1.0 INTRODUCTION AND GENERAL DESCRIPTION OF THE PLANT

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Note: The title blocks for the Chapter 1 figures refer to “Comanche Peak S.E.S.” The plant name has changed to CPNPP, and these figure title blocks will not be updated to reflect this.

1.1 INTRODUCTION

This updated Final Safety Analysis Report (FSAR) describes the design, construction and operation of a two-unit nuclear power plant designated as the Comanche Peak Nuclear Power Plant (CPNPP), Operating Licenses NPF-87 and NPF-89 for Units 1 and 2, respectively. All future references to the Final Safety Analysis Report (or FSAR) contained in this document refer to the updated FSAR. This report follows the format recommended by Regulatory Guide 1.70, Revision 2, Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants. The design information provided herein reflects the design in sufficient detail to enable a definitive evaluation as to whether CPNPP can be constructed and operated without undue risk to the health and safety of the public.

Vistra Operations Company LLC (Vistra OpCo) (operator) and Comanche Peak Power Company LLC (CP PowerCo) (owner) are subsidiaries of Vistra Corp. A simplified organizational diagram is shown in Figure 13.1-1.

Vistra OpCo - has corporate responsibility for the design, engineering, construction, licensing, operation, procurement, quality assurance and fuel management support to CPNPP.

1.1.1 STATION LOCATION

Located in Somervell County in North Central Texas, the CPNPP is about 65 miles southwest of the Dallas-Fort Worth Metropolitan Area. In addition, the Squaw Creek Reservoir (SCR), built for station cooling, extends northward into Hood County.

1.1.2 NUCLEAR STEAM SUPPLY SYSTEM

The Nuclear Steam Supply System (NSSS) for each unit is a pressurized water reactor, which along with the design and fabrication of the initial cores, is supplied by Westinghouse Electric Corporation.

1.1.3 CONTAINMENT

The containment for each unit is a steel-lined, reinforced concrete, cylindrical structure with a hemispherical dome designed by Gibbs & Hill, Inc. (Architect-Engineer).

1.1.4 CORE THERMAL POWER LEVEL

Each generating unit core was originally designed for a warranted power output of 3411 Megawatt thermal (MWt), which corresponds to a net electrical output of approximately 1150 Megawatt electric (MWe). This output, combined with the reactor coolant pump heat output of 14 MWt, gives a warranted NSSS output of 3425 MWt, which is the license application rating. A 4.5 percent increase in steam flow results in the maximum calculated NSSS output of 3579 MWt, and thus a maximum calculated core output of 3565 MWt. All safety systems, including the containment and engineered safety features, are designed for operation at the maximum calculated power output.

The thermal power output of Units 1 and 2 were subsequently increased to 3612 MWt.

1.1.5 SCHEDULE

CPNPP Unit 1 was declared for commercial operation on August 13, 1990. Unit 2 was declared for commercial operation on August 3, 1993.

1.2 GENERAL PLANT DESCRIPTION

1.2.1 SITE CHARACTERISTICS

The site of the Comanche Peak Nuclear Power Plant (CPNPP) is in an area of low population (520 persons within four miles of site) having a rural farm-ranch community setting. There are no residents within one mile of the site, and only 75 persons reside within two miles. There are approximately 12,000 persons within 10 miles of the site. The nearest communities, Glen Rose and Granbury (combined 1970 population about 4,000 persons), are 4 1/4 and 10 miles from the site, respectively. Fort Worth (1970 population about 400,000), from 33 to 50 miles from site, is the nearest major population center.

Land use in the site vicinity and surrounding area is predominantly agricultural. Dairy cattle are a minor portion of the total livestock with the closest commercial dairy herd nearly five (5) miles from the site. There is no commercial fishing in the site vicinity.

Although local population is projected to increase, no land use characteristics are projected to be drastically altered during station life. The site is well-suited for locating a nuclear generating installation because of a large exclusion area with a minimum boundary distance of 5,067 feet (1544 meters), a population center distance greater than 20 miles, a rock foundation for all seismic Category I structures, a lack of high surface water conditions, an available cooling water supply, and because of favorable hydrologic, geologic, seismic, and meteorologic conditions.

1.2.1.1 Geology and Seismology

The CPNPP site is located on the Comanche Plateau, a subdivision of the Central Texas section of the Great Plains physiographic province.

Gently dipping Lower Cretaceous limestone and sandstone directly underlie the site.

Structurally, the site occupies the south flank of the Fort Worth basin, a sedimentary depositional trough filled with Pennsylvanian and Permian sediments formed in mid-Pennsylvanian time. A regional unconformity separates the Paleozoic sediments from the lower Cretaceous sediments underlying the site.

Detailed investigation within five miles of the site has disclosed no evidence of surface faulting, thus eliminating the need for considering this factor as a design basis.

With regard to the stability of subsurface materials, there is no evidence in the site region indicating actual or potential uplift or subsidence, cavernous or karst terrain, tectonic warping or deformational zones pertinent to the site. Also not in evidence are zones of alteration, significant weathering, structural weakness, unrelieved residual stresses, or geologically hazardous materials.

Two major fault systems, the Balcones and the Luling-Mexia-Talco fault zones occur within 200 miles of the site. They are not regarded as active faults but are considered in establishing the Safe Shutdown Earthquake (SSE).

Seventeen seismic events have been reported with epicenters within 200 miles of the site. The closest large event was a modified Mercalli Intensity VII which occurred in 1882 near Bonham,

Texas, 155 miles northeast of the CPNPP. The closest known event to the site occurred 90 miles to the north in 1950 and exhibited an intensity of IV. Two events occurred 90 and 100 miles to the southeast of the site in 1932 (V-VI) and 1970 (IV), respectively. For the purpose of establishing the Safe Shutdown Earthquake (SSE), it has been assumed that a Bonham type event could occur at the closest approach of the Wichita-Ouachita tectonic subprovince to the site (70 miles). Based on these considerations, a conservative safe shutdown earthquake of 0.12g has been selected for the site. An acceleration of 0.06g has been selected for the site operating basis earthquake based on similar analysis.

In conclusion, the site is acceptable from a geologic and seismic standpoint for the facility designed for the SSE.

1.2.1.2 Hydrology

The CPNPP site is on Squaw Creek, a tributary of the Paluxy and thence Brazos rivers. Squaw Creek Reservoir (SCR) is impounded for station cooling by a dam constructed on Squaw Creek approximately 4 1/2 miles upstream of its confluence with the Paluxy and Brazos rivers.

The site grade is at elevation 810 feet, which provides 20.3 feet of freeboard above the Probable Maximum Flood (PMF) and superimposed wave runup on Squaw Creek Reservoir. The site grade is well above the maximum water levels conceivable on the Brazos River. Hence, the site will be unaffected by river flooding of any kind and will not be affected by tsunami, seiche, or ice flooding.

A continuous supply of service water is assured by means of a seismic Category I dam which encloses an arm of SCR. For the purpose of controlling the concentration of dissolved solids, SCR receives makeup water from Lake Granbury, on the Brazos River, and SCR blowdown is returned to Lake Granbury.

Most of the groundwater in the site region occurs in bedrock, although some does exist in shallow flood plain alluvium in stream valleys, but is not withdrawn for use. Bedrock aquifers pertinent to the CPNPP vicinity include, in order of increasing age, the Paluxy, Glen Rose, and Twin Mountains formations. Locally, the CPNPP and SCR are on the Glen Rose outcrop, which in turn is underlain by the Twin Mountains Formation. The Paluxy Formation is absent at the CPNPP location and is not found within the limits of the reservoir; the Glen Rose Formation is the upper most stratigraphic unit exposed in these areas.

In these formations, groundwater percolates slowly along bedrock joints and fractures, and through interstices in the rock fabric. The Twin Mountains Formation is the only moderately productive bedrock zone in the site vicinity. The Paluxy Formation has nominal pumpage near the site. The Glen Rose Formation yields very little water in the site area and is usually less productive than the other formations.

The station’s liquid waste processing system is designed so that radioactive liquid waste can be processed adequately prior to release to the environment as described in Section 11.2.

1.2.1.3 Meteorology

Located approximately 290 miles to the northwest of the Gulf of Mexico, the CPNPP site usually experiences a continental climate with marked temperature extremes both diurnally and

seasonally. Maritime tropical air masses almost completely dominate the weather in summer. During the winter, outbreaks of polar continental air are the most common frontal activity, while pacific maritime cold fronts are more frequent in the spring and fall. Wide variations in precipitation amounts occur from year to year, including both drought and persistent rains (occasionally induced by land-weakened, but rain-filled tropical cyclones from the Gulf of Mexico).

Tornado average return is about once in 320 years for the station site.

The accident dilution factors at the site are similar to those specified by Regulatory Guide 1.4 for tentative use in site evaluations. The highest frequency of poorer diffusion conditions generally occur with winds blowing toward the northwest, north-north-west, and north sectors. Some amelioration of the conditions in these sectors will be provided by the warmed waters of the SCR and by reduced topographic variation. The routine dilution factors were evaluated according to Regulatory Guide 1.111.

1.2.1.4 Environmental Radiation Monitoring

Pre-operational baseline radiological monitoring has been performed to determine the average background radiation level.

Conditions of radioactive releases are expected to be infrequent and thus contribute little dose to humans. Of primary concern are doses attributable to I-131 uptake and the submersion dose from noble gases. The primary emphasis of the operational environmental monitoring program will be to measure whole body dose from gaseous effluent and onsite sources, I-131 in air, and radionuclides (especially I-131) in milk. Radionuclides will also be measured in airborne particulate, surface water, well water, broadleaf vegetation, food products, fish, and milk. Frequency of data collection and location of measurements have been evaluated to maximize the data relating to the dose in humans.

1.2.2 PLANT DESCRIPTION

1.2.2.1 Design Criteria

The design of the CPNPP complies with Appendix A of 10 CFR Part 50, General Design Criteria for Nuclear Power Plants.

1.2.2.2 Nuclear Steam Supply System

The Nuclear Steam Supply system (NSSS) consists of a Westinghouse pressurized water reactor (PWR) and supporting auxiliary systems. The nuclear, mechanical, and thermal-hydraulic design of the reactor core is similar to the design of other Westinghouse units under construction. The rated thermal output of Unit 1 and Unit 2 is 3458 megawatt thermal (MWt). Heat balances, showing the major parameters of the plant for the original design rated power condition and maximum expected power condition and that provide bounding thermal/ hydraulic conditions for the core power uprates to 3458 MWt, are shown in Section 10.1.

Nominal performance of the NSSS based on 0 percent makeup is as follows:

		Unit 1[(a)]	Unit 2
1.	Thermal output of NSSS, MWt	3475	3474
2.	Thermal output of reactor core, MWt	3458	3458
3.	Steam flow from NSSS, lb/hr	15,334,487	15,359,775
4.	Steam pressure at a steam generator outlet, psia	967	995
5.	Maximum moisture content, percent	0.10	0.25
6.	Assumed feedwater temperature at steam generator inlet, °F	441.6 (HB) 443 (BE)	441.5

a) Based on nominal performance after 1RF12 [1 and 2].

1.2.2.2.1 Reactor Core

The reactor core is a multi-regions core composed of slightly enriched uranium dioxide pellets enclosed in pressurized, cold worked, sealed Zircaloy tubes. The fuel tubes are grouped and supported in fuel assemblies of the canless type. The basic fuel assembly consists of the control rod guide thimbles attached to the grids and the top and bottom nozzles. The fuel rods are held by the grids in the assembly to provide for very stiff support.

The initial loading of fuel into the core is designed so that fuel assemblies with the highest enrichment are placed in the outer region of the core while the two groups of fuel assemblies with lower enrichment are selectively arranged in the central region. During refueling operations, a portion of the fuel is discharged, and new fuel is loaded into the core. The fuel in the reactor core is arranged to achieve an acceptable power distribution.

Control rod assemblies are inserted into the guide thimbles of the fuel assemblies. The absorber sections of the control rods are fabricated from silver-indium-cadmium and sealed in stainless steel tubes. Neutron control for slow transients is provided by means of boric acid in solution in the Reactor Coolant System (RCS).

Each control rod cluster is attached to a spider connector and drive shaft. The drive mechanisms for the control rod assemblies are equipped with magnetic latches which are controlled by three magnetic coils and the latches are designed so that upon loss of power to the coils, the rod cluster control assembly (RCCA) is released and falls by gravity into the reactor core, effecting a shutdown of the reactor.

The reactor core rated thermal power for Unit 1 prior to 1RF09 is 3411 MWt; this results in a NSSS thermal power of approximately 3427 MWt with the additional 16 Mwt (approximate) of energy input to the RCS by nonreactor sources such as pump heat.

The reactor core thermal power for Unit 1 (after 1RF09) and Unit 2 is 3458 Mwt; this results in a NSSS thermal power of approximately 3474 Mwt with the additional 16 Mwt (approximate) of energy input to the RCS by nonreactor sources such as pump heat.

1.2.2.2.2 Reactor Coolant System

The Reactor Coolant System (RCS) consists of four parallel, similar, heat transfer loops, each consisting of a reactor coolant pump and steam generator, connected to the reactor vessel. In addition, the system includes a pressurizer, pressurizer relief tank, connecting piping, and the instrumentation necessary for operational control and protection.

The RCS transfers the heat developed in the reactor core to the steam generators, where steam is generated to drive the plant turbine- generator. Borated, demineralized light water is used as the heat transfer medium for the RCS and is circulated at a flow rate and temperature consistent with achieving the desired thermal-hydraulic performance for the reactor core. The water of the RCS also serves as neutron moderator, reflector, and solvent for the neutron absorber.

The reactor coolant pumps are Westinghouse vertical, single stage, centrifugal pumps of the shaft-seal type. The power supply systems for the pumps are designed to provide for adequate coolant flow to the reactor core under all required conditions.

The steam generators are Westinghouse vertical U-tube units containing Inconel tubes and equipped with integral moisture separation equipment to reduce the moisture content of the steam to 0.10 percent for Unit 1 and to 0.25 percent or less for Unit 2.

All reactor coolant piping as well as all pressure-containing and heat transfer surfaces in contact with the reactor water is stainless steel clad expect for the fuel tubes and steam generator tubing which are Zircaloy and Inconel, respectively. The reactor core internals, including the control rod drive shafts, are primarily stainless steel.

The electrically heated pressurizer, which is connected to one of the four parallel reactor coolant loops (RCLs), is designed to perform the following functions:

1. To maintain the pressure of the RCS during normal operation

2. To limit the pressure variations during plant loading transients

3. To maintain the RCS pressure within the design limits during abnormal occurrences.

1.2.2.3 Engineered Safety Features

The engineered safety features (ESF) systems are designed to limit the potential radiation exposure to the public as well as to plant personnel in the event of a postulated accidental release of radioactive fission products from the reactor system, particularly as the result of a loss-of-coolant accident (LOCA). These ESF function to localize, control, mitigate, and terminate all such postulated accidents to ensure that the guidelines of 10 CFR Part 100 for exposure limits are not exceeded.

The following is a list of ESF:

1. Containment

2. Containment Spray System

3. Containment spray chemical additive subsystem

4. Containment Isolation System

5. Combustible Gas Control Systems

6. Emergency Core Cooling System (ECCS)

7. Control Room habitability systems

8. Residual Heat Removal (RHR) System

Each of the ESF systems is missile-protected and designed to withstand all normal and accident loads including the Safe Shutdown Earthquake (SSE).

1.2.2.3.1 Containment

The containment is a steel-lined, reinforced concrete structure which consists of a vertical cylinder with a hemispherical dome supported on a foundation mat with a reactor cavity pit. The interior steel liner is constructed with carbon steel plate for leaktightness.

The Containment Building completely encloses the reactor and the RCS. An interior structure within the Containment Building supports and provides shielding for the reactor and other components of the NSSS.

The containment is designed to withstand the pressures and temperatures resulting from a spectrum of LOCAs and secondary system breaks.

1.2.2.3.2 Containment Spray System

The Containment Spray System, including the containment spray chemical additive subsystem, is designed to meet the following criteria:

1. To ensure that the containment pressure and temperature do not exceed the design parameters for all accidents including LOCAs

2. To ensure that offsite radiological consequences are within the limits of 10 CFR Part 100 by reducing the fission product concentration with containment

The Containment Spray System initially supplies the containment atmosphere with borated water drawn from the Refueling Water Storage Tank (RWST). When the water level in the RWST reach the low-level set point, water is drawn from the Containment sump and recirculated through the Containment spray heat exchangers to provide for the continued cooling of the containment atmosphere.

1.2.2.3.3 Containment Spray Chemical Additive Subsystem

The containment spray chemical additive subsystem is designed to insure that the concentration of the iodines within the Containment, following all postulated accidents, is reduced to limit the offsite radiological consequences.

The desired reduction in containment iodine concentration is accomplished by the injection of sodium hydroxide from the containment spray chemical additive subsystem into the containment spray water; this chemically-treated water absorbs the fission product iodines from the containment atmosphere and retains them in solution in the containment sump.

The containment spray chemical additive subsystem, together with the balance of the Containment Spray System, is designed as follows:

1. To be fully redundant

2. To meet the single failure criterion

3. To withstand all normal and accident loads

4. To be fully missile-protected

1.2.2.3.4 Containment Isolation System

In the event of postulated accidents, the Containment Isolation System is designed to minimize the leakage of radioactive materials through fluid lines penetrating Containment.

This design objective is achieved by the use of double isolation barriers. The use of double isolation barriers ensures that no single failure of any active or passive component renders the Containment Isolation System partially or wholly inoperable. The isolation valves are checked regularly during normal unit operation and are designed to assume a fail-safe position.

The Containment Isolation System ensures that the offsite radiological consequences of a main steam line rupture or LOCA are within the guidelines of 10 CFR Part 100.

1.2.2.3.5 Combustible Gas Control Systems

Combustible gas control systems are not required to be Engineered Safety Feature Systems.

The hydrogen purge system, a non-safety system, is designed to meet seismic category II requirements. This system purges the containment atmosphere through filters which reduce radioactive releases. See Section 6.2.5 for details.

1.2.2.3.6 Emergency Core Cooling System

The ECCS, with active and passive subsystems, is designed to perform the following functions:

1. To inject borated water into the RCS following a LOCA to minimize core-damage, metal-water reactions, and fission-product releases

2. To ensure adequate shutdown margin regardless of temperature

3. To provide for long-term postaccident cooling of the reactor core by recirculating borated water from the containment sump

1.2.2.3.7 Control Room Habitability Systems

The Control Room habitability systems are designed to ensure that the Control Room is habitable for a period not less than 30 days following any LOCA.

The Control Room air-conditioning system has a recirculation mode designed to maintain Control Room ambient conditions suitable for personnel occupancy during accident conditions. A set of emergency filtration units removes airborne activity from outside air during emergency pressurization and filtration modes of operation.

Communication and domestic facilities are provided to meet personnel needs during an extended stay in the Control Room area.

1.2.2.3.8 Residual Heat Removal System

The RHR System serves as a part of the ECCS during both the low pressure injection and recirculation phases following a LOCA. The RHR System is designed to provide an adequate supply of water for cooling of the reactor core; this is accomplished by the RHR pumps delivering water from the RWST to the RCLs. When the supply of refueling water is depleted, recirculation of containment sump water through a heat exchanger is initiated and maintained to ensure continued core cooling.

1.2.2.4 Plant Instrumentation and Control System

Instrumentation and controls are provided to monitor and maintain plant parameters within the prescribed operating ranges.

Reactor control is provided by the following:

1. Temperature coefficients of reactivity

2. Control rod cluster motion

3. Injection of neutron absorbing chemical shim in the form of boric acid

The control rod clusters provide for load-follow transients as well as for startup and shutdown requirements. The chemical shim is inserted during cold shutdown, partially removed during startup, and further adjusted during the lifetime of the core to compensate for such effects as fuel consumption and accumulation of fission products which act to decrease the excess reactivity of the core.

The reactor control system permits the plant to accept 10 percent step-load increases and ramp load increases of 5 percent per minute over the load-range of 15 to 100 percent of full power. Step-and ramp-load reductions of the same magnitude are also possible over the range of 15 to 100 percent of full power.

The non-nuclear-safety-related process and containment instrumentation measure temperatures, pressure, flows, and levels in the steam and auxiliary systems and in the Containment. Process variables required on a continuous basis for startup, operation, and shutdown of the unit are indicated, recorded, and controlled from the Control Room. The quantity and types of process instrumentation provided ensure safe and orderly operation of all systems and processes over the full operating range of the plant.

The nuclear-safety-related instrumentation and control systems provide automatic protection and exercise proper control to ensure safe reactor operation and to provide initiating signals to mitigate the consequences of design basis accidents (DBAs). Supervision of both nuclear and turbine generator plants is accomplished from the Control Room.

1.2.2.5 Power Conversion and Electrical Systems

The electrical systems transmit the output of the generator units to the utility system and provide power for plant auxiliary loads. Independent and redundant Class 1E standby power systems function during a DBA to ensure operation of necessary safety-related systems.

The main generator for each unit is rated 1410 MVA at 0.90 power factor, 22 kV, 60 Hz, three-phase, 1800 rpm. Power output from each generator is fed to its respective main step-up transformer bank through the isolated phase bus system. Generator voltage is stepped up to 345 kV and transmitted through overhead transmission lines to the 345 kV Switchyard where distribution to the utility system is accomplished. Seven 345 kV transmission lines connect the Switchyard to other switching stations in the Transmission Operator's transmission system. The AC station power is distributed through the 6900 V, 480 V, and 118 V uninterruptible AC auxiliary bus systems to the unit loads. The DC load requirements are satisfied by 125-V, 125⁄250-V, and 24⁄48-V systems.

During normal operation, the main generators supply power to all the auxiliary loads except for the loads connected to the emergency buses.

These loads are normally energized from the offsite power systems. Startup and shutdown power is derived from two independent offsite power circuits. One circuit is connected to the 345 kV Switchyard, and the second feeder is connected to the 138 kV system. Each high voltage circuit is transformed through startup transformers to 6900 V.

Under LOCA and loss-of-offsite power conditions, power to the safety-related loads is achieved through the standby diesel generator system. A total of four 100 percent diesel generators, two per unit, is provided. Each diesel generator is assigned to a specific safeguard bus. One diesel generator and its associated bus and loads can satisfy the safety load requirements for one unit. There is no sharing of diesel generators.

Each unit has two independent Class 1E 125-VDC systems to provide DC and uninterruptible AC power to essential plant instruments and controls. An independent 125/250-VDC system is provided for each unit to supply non-Class 1E power and control loads.

1.2.2.6 Fuel Handling and Storage System

The Fuel Handling System is designed to permit the handling of spent fuel under water from the time it leaves the reactor vessel during refueling until it is placed in a cask for offsite shipment.

The transfer of spent fuel under water provides both a transparent radiation shield and a reliable cooling source for the removal of residual and decay heat.

The Fuel Handling System is located in two separate areas. The first area is comprised of the refueling cavity inside the Containment Building, which is flooded during the refueling operations; the second area is the Fuel Building, which includes two spent fuel pools; a cask pit and connecting fuel transfer canal, and which is external to the Containment Building and accessible to authorized personnel. The two areas of the Fuel Handling System are connected by the Fuel Transfer System, which moves the fuel through an opening in the Containment. The Fuel Handling System provides for the safe handling of RCCA under all conditions and for the required assembly, disassembly, and storage of reactor internals. This system includes a refueling machine located inside the Containment above the refueling cavity Fuel Transfer System (which includes a fuel transfer tube), a fuel handling crane above the fuel storage racks area, the RCCA changing fixture, and lifting devices (which are used for handling the reactor vessel head and internals).

The new fuel assemblies can be stored in racks in the new fuel pit, in racks in the spent fuels pools, or in the in-containment storage racks. In each of the locations, the new fuel is separated by sufficient distance to preclude criticality. Spent fuel is placed in storage racks in the spent fuel pools where it will remain for a sufficient decay and cooling period before shipment in a cask to a fuel reprocessing facility or a storage facility.

1.2.2.7 Cooling Water Systems

Two separate closed-loop cooling water systems are provided to facilitate heat removal from ESF, safety-related, and non-safety- related equipment. These are the Component Cooling Water System (CCWS) and the Turbine Plant Cooling Water (TPCW) System. Cooling for the CCWS is provided by the Station Service Water System (SSWS) and the heat is ultimately rejected to the Safe Shutdown Impoundment (SSI). Cooling for the TPCW System is provided by the Circulating Water System and this resulting heat is ultimately rejected to the Squaw Creek Reservoir (SCR).

The design criteria for the CCWS are as follows:

1. To remove residual heat from the RCS by means of the RHR System during plant shutdown

2. To cool the letdown flow to the CVCS during power operation

3. To dissipate waste heat from containment and auxiliary components

4. To remove ESF heat loads after an accident

The CCWS transfers heat to the SSWS via the component cooling water heat exchangers.

The CCWS consists of two separate and independent full capacity, safeguards-related, redundant trains of pumps and heat exchangers to remove heat from the engineering safeguards components and a nonredundant, safety-related loop with an ANS non-safety-class-related portion. Corrosion inhibiting demineralized water is circulated through the CCWS. The CCWS is maintained at a higher pressure than the SSWS to prevent inleakage of potentially corrosive

station service water fluids. The CCWS provides an intermediate barrier between the RCS and the service water system to preclude direct leakage of radioactive material to the SSI.

An uninterrupted supply of water for the service water system is ensured by the SSI, a seismic Category I structure. Makeup water for the SSI is provided via a shallow canal connecting it with the SCR and can be provided via a bleed-off from the Circulating Water System.

The design criterion for the TPCW System is to remove waste heat from various Turbine Building auxiliaries.

The TPCW System passes its heat load via the TPCW heat exchanger to the Circulating Water System.

Cooling water for the main condensers is provided by the Circulating Water System. The Circulating Water System Intake Structure is located north of the plant on the SCR. The heated water of the Circulating Water System is discharged to the SCR via a discharge tunnel at a point southeast of the plant.

1.2.2.8 Auxiliary Systems

The auxiliary systems are designed to perform the following functions:

1. To supply RCS water requirements

2. To purify reactor coolant water

3. To introduce chemicals to inhibit corrosion

4. To introduce and remove chemicals for reactivity control

5. To cool system components

6. To remove residual heat during a portion of the reactor cooling period and when the reactor is shut down

7. To cool the spent fuel pool water

8. To permit sampling of reactor coolant

9. To provide for safety injection

10. To vent and drain the RCS and the auxiliary systems

11. To provide containment ventilation and cooling

12. To provide plant fire protection

The principal auxiliary systems are the following:

1. Chemical and Volume Control System (CVCS)

2. Sampling systems

3. Compressed air systems

4. Plant ventilation systems

5. Station Service Water System (SSWS)

6. Plant Fire Protection System

7. Spent Fuel Pool Cooling and Cleanup System

1.2.2.8.1 Chemical and Volume Control System

Performance of the following functions will establish the design criteria of the CVCS:

1. Purification of reactor coolant fluid

2. Corrosion control

3. Regulation of reactor coolant inventory

4. Reactivity shim

5. Seal water injection for reactor coolant pumps

During power operation, a continuous feed-and-bleed system is maintained to and from the RCS. The bypassed fluid is depressurized, cooled, purified, and stored by a series of valves, heat exchangers, demineralizers, and tanks. Furthermore, the amount of reactor coolant is automatically adjusted to compensate for changes in volume as a result of coolant temperature changes. The CVCS provides reactivity control by varying the boron concentration in the reactor coolant.

Water for the reactor coolant pump shaft seals is supplied from the charging pump.

The centrifugal charging pumps associated with the CVCS also serve as the high-head pumps for the ECCS. In the event of a LOCA, the CVCS is isolated except for the charging pumps, which inject borated water into the reactor core.

1.2.2.8.2 Sampling Systems

The CPNPP is equipped with three sampling systems: the Primary Sampling System, the Secondary Sampling System and the Post Accident Sampling System (PASS). The Primary Sampling System serves the RCS and its auxiliary systems while the Secondary Sampling System serves the feedwater and main steam systems. The PASS no longer functions, as described in FSAR Section II.B.3. These systems provide a determination of both the chemical and radiological makeup of various plant fluids. Samples drawn from radioactive sources are passed through sample coolers or delay coils, or both, as required.

1.2.2.8.3 Compressed Air Systems

Each of the two CPNPP units are equipped with oil-free air compressors, which discharge filtered and dried compressed air into the Instrument Air System. There are two instrument air compressors for each unit and two common instrument air compressors which can be aligned to either unit. An oil-flooded common compressor supplies both units with unfiltered compressed air into a separate Service Air System.

The Service Air System provides compressed air for routine maintenance at various stations throughout the plant.

The Instrument Air System provides compressed air, which is dried and filtered, for all air-operated instruments and valves.

1.2.2.8.4 Plant Ventilation System

To facilitate the independent control of the atmosphere in various plant areas, separate ventilation systems have been provided. The following major areas are served by separate ventilation system:

1. Containment Building.

2. Auxiliary, Safeguards, and Fuel Building.

3. Turbine Building.

4. Diesel Generator Building.

5. Control Room.

6. Uninterruptible Power Supply and Distribution Room.

In addition, a Containment Purge System and a Containment Preaccess Filtration System are provided for the containment atmosphere.

1.2.2.8.5 Station Service Water System

The SSWS removes heat from the CCWS to meet the cooling requirements of the plant as follows:

1. During normal operations

2. Following shutdown

3. During and after all postulated LOCAs

The SSWS is assured of a constant supply of water by the SSI Dam which is a seismic Category I structure founded on bedrock. The SSI maintains the necessary volume of water required by NRC Regulatory Guide 1.27. Makeup water is provided via a shallow canal to the SCR and can be provided via a bleed-off from the Circulating Water System.

The SSWS of each of the units is completely independent and redundant. Each unit has two fully independent trains, either of which can supply the required cooling waterflow. The pumps and heat exchangers of each train can be aligned with the other train in the event of a component failure.

1.2.2.8.6 Plant Fire Protection System

The Fire Protection system provides means for detecting, alarming, and extinguishing fires. The system is divided into two basic subsystems: the Fire Detection System and the fire extinguishing system. The Fire Detection System is a plantwide instrumentation system provided to detect fires in various areas of the plant and to alert the Control Room operators of a fire and its location. The fire extinguishing system includes such fire fighting equipment as sprinkler systems (wet pipe, deluge and preaction), water spray, standpipe and hose stations, Halon Systems, and portable extinguishers. The Fire Protection System is not required to ensure the integrity of the reactor coolant pressure boundary (RCPB), the capability to shut down the reactor, and the capability to prevent or mitigate the consequences of accidents which could result in potential offsite exposures as detailed in 10 CFR Part 100.

All fire protection piping in Category I structures is classified Class 5 as described in Section 3.2.2.

The station fire main system, including the associated pumps, piping, and valves, is shared by the two CPNPP units.

Noncombustible and fire-resistant materials are selected where practical for use throughout the CPNPP facility, particularly in controlled-access portions of the plant such as the Containment and Control Room.

1. Fire Detection System

   A Fire Detection System is provided throughout the plant. When a fire is detected, the respective zone and fire indication appears on the local zone panel and the main fire detection panel in the Control Room. An alarm indicating lamp illuminates in the base of the ionization detectors indicating the actuated detector, for other types of detectors a lamp illuminates at the local zone panel. The majority of detectors are placed overhead in their respective areas. Also, detectors located in areas protected by halon and automatic open head water spray systems actuate the fixed systems. Detectors sound an alarm condition via the Control Room fire detection panel.

   The Fire Detection System uses fire, smoke, and heat detection devices located throughout the entire plant; they include the following:

   a. Ionization smoke detectors

      Ionization detectors are of the two-chamber-type design. The first is a reference chamber to compensate against the sensitivity changes due to temperature, barometric pressure, and humidity variations. The second chamber is a sensing chamber open to the outside elements through a protective screening which permits combustion products to enter while preventing insects and foreign matter from entering and causing false alarms.

b. Thermal detectors

Spot-type thermal detectors are of the fixed-temperature, rate compensation type.

Continuous strip thermistor heat detectors are used to monitor the temperature of charcoal adsorber beds of the atmosphere cleanup units, and other areas where accurate measurement of temperatures for fire detection is required. The strip thermistors are of the negative coefficient of resistance type.

c. Flame detectors

Flame detectors are of the ultraviolet (UV) radiation sensing type. The detectors respond to the UV frequency of radiation given off by a fire. The UV flame detectors are installed with high ceilings or specific pieces of equipment which contain highly flammable materials.

The detectors are strategically positioned throughout the facility to detect fires, annunciate alarms in the Control Room and indicate the location of the fire on the Control Room fire detection panel and local zone panel.

The detection system is electrically supervised. Class A supervision is used in the cable spreading rooms. As a minimum, class B supervision is used in the remainder of the plant.

2. Fire Extinguishing System

The fire extinguishing system uses portable extinguishers, in conjunction with hose stations and fixed suppression systems, as primary and secondary means of suppression. Water is supplied for the standpipes and suppression systems from two dedicated above ground storage tanks via an underground piping distribution system and water supply lines for each building and transformer branch from the underground loop; every branch has a post indicator gate valve for isolation of the branch or building. The main loop is divided into sections by post indicator valves to allow isolation of the loop in case of a line break. There are three 50 percent capacity fire pumps to supply water to the system. One is an electric motor driven pump and the other two pumps are diesel engine driven. Water is supplied to the underground fire loop by the lead pump, the electric motor-driven pump, when the jockey pump cannot maintain the system pressure above a predetermined set point.

A siamese fire department connection is provided for emergency fill of the system by a fire truck or a portable auxiliary pump. This fill is used as a backup to the pumps. As required by National Fire Protection Association (NFPA) No. 24, a check valve and a ball drip valve are provided for the connection of the siamese to the main loop. The siamese connection is located adjacent to the Service Water Intake Structure.

The Safeguards, Fuel Handling, and Auxiliary buildings have internal loops to supply suppression and standpipes systems. These loops have multiple connections to the underground loop. Valves are available for isolation of sections of the loops as well as isolation of the internal loop from the underground loop. Each suppression and standpipe

system has a shutoff valve to facilitate work on the system. Approved control valves are located in the respective system in accordance with applicable NFPA standards.

The Turbine Buildings have an internal loop to supply standpipes. This internal loop has multiple connections to the underground loop in Unit 1 and Unit 2. Crosstie lines are provided between Unit 1 and Unit 2 Turbine Buildings to facilitate isolation of sections of either loop. Valves are provided in accordance with NFPA No. 14 to control the waterflow and isolate the system. The water spray and automatic sprinkler systems are connected to the outside loop via isolation valves located in the fire protection valve rooms, which are accessible from inside and outside the Turbine Buildings, as required, to control the waterflow to the suppression systems.

The suppression systems for the diesel generator day tanks are supplied from the main loop independent of each other. Each system has a deluge valve, a cutoff valve, and a detection system. Actuation of one system does not affect operation of the other diesel generator; each diesel generator compartment is provided with a watertight door to prevent flooding of the adjacent areas. The systems are actuated automatically based on detection of a fire or manually adjacent to the respective area. The systems can only be shut off manually at the valve.

The fire pump house structure is protected by an automatic wet-pipe sprinkler system. Water flow and valve tamper alarms are provided at the pump house location and in the Control Room.

The deluge water spray systems for the atmosphere cleanup units, except the containment preaccess units, are supplied by the interior building supply loop. The preaccess units are supplied by the Demineralized Water System. Should demineralized water be unavailable, the operator can manually route fire protection water through the demineralized water piping. Actuation of one system will not affect the operation of an adjacent atmosphere cleanup unit.

The halon system for each cable spreading room consists of a detection system, storage cylinders, manifold and header assembly, control valves, piping, nozzles, and local control panel. Each system is designed with two charges of halon. Halon is released automatically after receipt of a fire signal from detectors located in the cable spreading room. The Unit computer rooms are each provided with a manually actuated halon system, and ionization smoke detection provides control room personnel notification of a fire.

1.2.2.8.7 Spent Fuel Pool Cooling and Cleanup System

The Spent Fuel Pool Cooling and Cleanup System serves the spent fuel pools of both units.

The cooling portion of this system has two trains consisting of a pump, heat exchanger, and other associated equipment.

The purification portion of this system consists of two trains containing a filter and a demineralizer which can be operated in parallel with either of the two cooling trains.

The skimmer portion of this system consists of a single skimmer train and is shared between both pools.

1.2.2.9 Waste Processing Systems

The waste processing systems (WPS) are designed to process liquid, gaseous, and solid waste while achieving the lowest reasonable radioactive release to the environment available through current technology. Liquid and gaseous wastes to be recycled within the plant are first segregated from those to be processed or shipped offsite.

Segregation of wastes is consistently maintained in the subsystems to ensure proper handling.

The Liquid Waste Processing System (LWPS) is designed to perform the following functions:

1. To collect reactor grade water
2. To process reactor grade water
3. To recycle reactor grade water
4. To collect floor drains and laundry waste
5. To process floor drains and laundry waste
6. To recycle or discharge processed floor drain and laundry waste
7. To remove radioactive constituents
8. To concentrate the removed radioactive constituents
9. To process concentrated radioactive constituents for solidification and shipment offsite

In addition, all liquid wastes are sampled and activity levels recorded prior to release. All processed liquid effluents from the RCS are subjected to purification by the CVCS ion exchanger in addition to the components of the WPS. The limits established by 10 CFR Part 20 are met.

The Gaseous Waste Processing System (GWPS) is designed to perform the following functions:

1. To reduce fission product gas concentrations in the RCS during normal operation
2. To contain indefinitely the fission product gases removed from the RCS
3. To collect gases generated by other systems
4. To maintain a low level of hydrogen gas in the collected gases
5. To discharge radioactive effluents after monitoring for radioactivity via a controlled path to ensure that the expected offsite doses are as low as reasonably achievable (ALARA)

The waste gases generated during plant operations including anticipated operational occurrences are collected and processed and are stored in waste gas decay tanks.

The Solid Waste Management System is designed:

1. To provide a means of encapsulating or compacting radioactive solid wastes generated by reactor plant operations

2. To provide adequate equipment and storage area shielding for the protection of operating personnel pending shipment of waste to disposal facilities

3. To measure and record the radiation levels of the solid waste processed for shipment from the site to disposal facilities

4. To provide a 3- to 6-month storage capacity for the processed wastes depending upon plant operation.

Solid wastes are compacted into containers for offsite disposal.

The WPS, with the exception of the equipment associated with the reactor coolant drain tanks, are completely shared. The reactor coolant drain tanks and associated equipment are located inside their respective Containment structures.

1.2.2.10 Safe Shutdown Impoundment

The SSI is an enclosed body of water formed from a cove of the SCR and is retained by a seismic Category I dam. It is designed and constructed to withstand the most severe postulated natural phenomena. The water level of the SSI is maintained by a spillway between the SSI and the reservoir behind the Squaw Creek Dam.

Since the Squaw Creek Dam is not a seismic Category I structure, it is assumed that ground motion from the SSE results in catastrophic, total, and instantaneous drawdown of the water level on the downstream side of the SSI Dam. The spillway limits the low water level in the SSI to 769 ft 6 in, at which point the volume of water contained is approximately 300 acre-feet.

The SSI is designed to serve as the ultimate heat sink of the CPNPP and acts to dissipate heat rejected by the SSWS during postaccident shutdown and normal cooldown conditions. It has been sized to provide adequate cooling capacity for the CPNPP in accordance with the requirements of NRC Regulatory Guide 1.27.

The materials used in construction of the SSI Dam include limestone rock from excavation at the plant site, clay from a borrow area located upstream from the Squaw Creek embankment, and a granular filter zone material.

1.2.2.11 Shared Structures, Systems, and Components

1.2.2.11.1 Shared Structures

The following structures are shared by both units of the CPNPP:

1. Auxiliary Building, including Control Room

2. Fuel Building

3. Service Water Intake Structure

4. Circulating Water Intake Structure

5. Circulating Water Discharge Structure

1.2.2.11.2 Shared Systems

The following systems are shared (or have shared components) by both units of the CPNPP:

1. Chemical and Volume Control Systems (CVCS)

2. Boron Recycle System (BRS)

3. Compressed Air Systems (CASs)

4. Station Service Water System (SSWS)

5. Waste Processing Systems (WPS)

6. Fire Protection Systems

7. Hydrogen Purge System

8. Turbine Building ventilation system

9. Control Room ventilation system

10. Primary Plant ventilation system

11. UPS and Distribution Room Ventilation System

1.2.2.11.3 Shared Components

The following components are shared by both units of the CPNPP:

1. Fuel handling equipment (in Fuel Building only)

2. Startup transformers

3. 480-V motor control centers feeding to the common services systems (mechanical) load and lighting

4. Control Room panels where common services controls are located

1.2.2.12 General Arrangements

Figure 1.2-1 through Figure 1.2-46 provide sufficient detail of major structures and equipment to give an understanding of the general layout of the plant. Plan and elevation drawings are provided for the following structures:

1. Containment and Safeguards Buildings (Unit 1)

2. Containment and Safeguards Buildings (Unit 2)

3. Turbine Building (Unit 1)

4. Turbine Building (Unit 2)

5. Auxiliary and Electrical Control Building

6. Fuel Building

7. Circulating Water Intake Structure

8. Service Water Intake Structure

These figures are based on plant plan and elevation drawings. As such, the figures provide additional detail beyond what is necessary for general arrangements. These figures will be updated as necessary to provide a reasonable understanding of the general arrangement of major plant structures and equipment. The additional detail on these drawings is not updated.

1.2.2.13 References

1. TXU Letter CPSES-200401793, Transmittal of Contract Information - Best Estimate Feedwater Temperature, July 20, 2004.

2. Design Specification No. 418A23, Revision 2, Comanche Peak Nuclear Power Plant Unit 1, Delta 76 Replacement Steam Generator, September 2006.

1.2.3 OLD STEAM GENERATOR STORAGE FACILITY

The Old Steam Generator Storage Facility (OSGSF), shown in Figure 1.2-1, is a reinforced concrete structure supported on a reinforced concrete footing. The OSGSF consists of precast concrete wall closure panels and a reinforced concrete roof slab supported by structural steel beams. The OSGSF provides a secure long-term on-site storage facility for the four Unit 1 Old Steam Generators and Old Reactor Vessel Heads (and associated Control Rod Drive Mechanisms) for Units 1 and 2.

1.2.4 INDEPENDENT SPENT FUEL STORAGE INSTALLATION

The CPNPP site contains an Independent Spent Fuel Storage Installation (ISFSI) as shown on Figure 1.2-1. The ISFSI consists of a concrete pad with space for 84 natural convection air-cooled, HI-STORM shielded dry spent fuel storage casks, each capable of storing 32 spent nuclear fuel assemblies in a welded multiple purpose container. The ISFSI is inside a separate security protected area.

1.2.5 BEYOND DESIGN BASIS

1.2.5.1 NRC Order EA-12-049 Beyond Design Basis External Events

On March 12, 2012, the NRC issued Order EA-12-049, "Order to Modify Licenses with Regard to Requirements for Mitigation Strategies for Beyond-Design-Basis External Events." Strategies,

details, and programmatic controls for mitigating beyond-design-basis external events are contained in site specific program documents. The CPNPP site contains a FLEX Storage Building as shown on Figure 1.2-1. The FLEX Storage Building protects equipment credited for NRC Order EA-12-049. Modifications to a Structure, System, or Component (SSC) required by the strategies of NRC Order EA-12-049 and are reflected in the FSAR flow diagrams as required.

1.2.5.2 NRC Order EA-12-051 – Spent Fuel Pool Instrumentation

On March 12, 2012, the NRC issued Order EA-12-051, “Order to Modify Licenses with Regard to Reliable Spent Fuel Pool Instrumentation.” Details and programmatic controls are contained in site specific program documents. The instrument display for the wide range spent fuel pool instrumentation is installed in compliance with NRC Order EA-12-051 and is readily accessible during postulated events and allows for SFP level information to be promptly available to decision makers. As appropriate, Sections 9.1.3.2 and 9.1.3.5 and Tables 17A-1 and 17A-2 refer to the wide range Spent Fuel Pool Instrumentation installed in accordance with the NRC Order.

1.3 COMPARISON TABLES

1.3.1 DELETED.

1.3.2 COMPARISON OF FINAL AND PRELIMINARY DESIGNS

Table 1.3-2 details the significant design changes that have been made since the submittal of the PSAR.

The FSAR is complete and does not rely on information contained in the PSAR. The original intent of this table was to provide a comparison with the PSAR when the FSAR was initially submitted for NRC Staff review. The information in this table is considered historical and will not be updated.

TABLE 1.3-1
THIS TABLE HAS BEEN DELETED

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 1 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
I. Structures		
Category I Structures (other than containment)	3.8	Leak chase system behind liner of outdoor Category I tanks has been eliminated.
		The requirement for roof blow out panels above the main steam lines in the Safeguards Building has been deleted.
Control Room	3.11B	Limiting environmental conditions have been reduced to 80°F and 60 percent relative humidity.
	6.4 3.11B	Positive pressure of 1/2 inch water gauge changed to 0.1 inch water gauge during an accident.
Containment Systems	3.8	The containment external pressure design has been changed from a 3 psi differential to a 5 psi differential pressure.
		The containment internal structure through liner anchors was eliminated.
		The containment dome liner has been increased from 3/8 inch thickness to 1/2 inch.
		The containment liner paint has been changed.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 2 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
	3.9	The break flow in the reactor cavity analysis is limited to 144 in^2 by pipe whip restraints instead of the previous 150 in^2. (See Section 3.9N.1.4.6 for application of leak-before-break to GDC-4.) Cathodic protection not required for containment liner, reinforced steel and other such steel encased in concrete. Containment penetration sleeves changed from A-333-70, Grade 6 to: 1. SA-333, Grade 6 for sleeve sizes 20 in. and smaller 2. SA-516, Grade 70 or SA-537, Class 2 to sleeve sizes 20 in. and larger Inside weld no longer utilized on cold pipe penetration. Sleeve with inside weld and guard pipe on hot pipe penetration replaced with sleeve without inside welds or guard pipe.
	3.11B	Maximum operational temperature increased to 120°F.
	6.2.1	The reactor cavity analysis was redone using new support criteria and restraint design.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 3 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
	6.2.4	Changes in signals used to actuate the containment isolation system to reflect changes in WNES protection system. Addition of bellows expansion joints to the valve isolation tanks.
	6.2.5	The containment liner paint has been changed from a primer and top coat that was zinc free to inorganic zinc as a primer in phenolic topcoat. The use of aluminum and zinc inside containment is now permitted. Remote thermal conductivity hydrogen analyzers are replaced by in-containment electrochemical sensors and microprocessor based analyzers located in the control room. The containment free volume has been changed from 2.9×10^6 cubic feet to 2.985×10^6 cubic feet
	6.5	Containment sump pH limits changed from 9.0 to 8.6 and 9.3 to 9.5.
Missile Shield	3.5	The CRDM roll away missile shield is constructed of steel instead of reinforced concrete.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 4 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Safe Shutdown Impoundment	9.2.5	The presence of fractures in the SSI limestone foundation material was determined to be acceptable if the fractures occur above elevation 769.5 (spillway level), and if the fractures would not cause piping of core material. Clarification of the term “unweathered” to mean material that is firm, hard, and of appropriate strength parameters; the presence of slight oxidation resulting in some color change may not be an indication of weathering.
Shielding	12.1	Deleted the provisions for: 1. Local alarms at entries to Zone IV areas 2. Local and remote alarms at entries to Zone V areas has been deleted. The minimum density for ordinary concrete has been changed from 2.33 g/cc to 2.26 g/cc The secondary shield wall no longer surrounds the pressurizer and the thickness has been increased to 2 ft 9 in.
Tornado Characteristics	3.5	Tornado Missile Characteristics have been changed.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 5 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Piping Systems, Handling Equipment and Tornado Venting Components	3.2 and Appendix 17A	Portions of systems or components are designed as Seismic Category II components.
Non-nuclear Safety Piping Systems or Components	3.2 and Appendix 17A	Portions of systems or components are designed as Class 5 components.
Auxiliary Feedwater System	3.6B	The system is identified as high energy fluid system.
II. Reactor Systems		
Fuel	4.2	The reactors are fueled with 17 x 17 fuel assemblies in lieu of 15 x 15 fuel assemblies.
Reactor internals	3.1, 7.7	The use of part length control rods has been removed from the CPNPP design.
	4.5.2	The reactor internals have been modified to accept 17 x 17 fuel assemblies.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 6 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
RCS pressure boundary	5.2	Flux core welding on stainless steel piping components fabricated in the shop are permitted. The allowable heat input range is 15 to 100 KJ/in. The use of the following as valve materials is now permitted for valves 2.5 in. and larger: bodies: SA 351 grade CF8 and CF8M bonnets: SA 351 grade CF8M with nickel plating or stellite hardfacing closure bolting: SA 564 Type 630 and SA 193 grade B6 closure nuts: SA 193 grade B6 and SA 194 grade 8M The use of the following as valve materials is now permitted for valves 2.0 in. and smaller: bodies: SA-351 Gr. CF8M SA-182 F 316

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 7 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		bonnets: SA-182 F 316 discs: A-567 Gr. 1 (Stellite 21) stems: SA-564 Type 416 (Non-pressure retaining applications) The lower heat input range limit for welding processes of austenitic stainless steel have been deleted, and the use of automatic gas tungsten arc-cold wire processes is allowed. The maximum allowable heat input for this process is 45 KJ/in.
	7.6	An RCS cold overpressure control system is employed to provide for the mitigation of potential cold overpressurization transients, utilizing existing power operated relief valves with modifications to their actuation logic.
Steam Generators	5.4.3	CPNPP utilizes model D5 steam generators in Unit 2. Thermal sleeves in reactor coolant loop branch nozzles have been deleted.
RHR systems	3.6	The RHR system is no longer identified as a high-energy fluid system located outside the containment.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 8 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Safety Injection System	3.1	Change in SIS signal from coincident low pressurizer pressure and water level to low pressurizer pressure.
Reactor Coolant and Whip Restraint	3.1 3.6 3.9N 5.4	The leak-before-break technology has been applied to exclude from the design basis the dynamic effects of postulate ruptures in the reactor coolant loop piping.
III. Electrical Systems:		
Reactor Trip System	7.2	The reactor trip on low feedwater flow has been replaced by a low-low steam generator level trip. The RTD bypass line T hot and T cold measurements have been replaced by an N-16 power monitor and an inline T cold measurement CPNPP utilities 4 section power range neutron detector assemblies.
Engineered Safety Feature Systems	7.3	An improved steamline break protection system has been incorporated where safety injection and steamline isolation are initiated from low compensated steam line pressure.
CCW, SSW	7.3	Recirculation valves are now flow controlled, rather than pressure controlled

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 9 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Post-Accident Monitoring	7.5	SSW and CCW system parameters are not considered post-accident monitors
Hot Shutdown Panels	7.4	Indicators on the hot shutdown panel are no longer considered “Post accident monitors” and are treated as non-Class IE.
Electrical systems	8.1	Non-safety-related loads removed from Class 1E batteries and reassigned to ± 125-V non-Class 1E batteries. A static switch has been added to the BOP static uninterrupt. Power supply system inverter to accommodate the 120 V supply. 138 kV DeCordova line previously was directly fed from DeCordova substation. Now it is transferred through switching station located near DeCordova SES. Motor operated disconnect switch is added to the 138 kV line at CPNPP to isolate the startup transformer XST1 for maintenance. The condition of diesel generator trip on bus-fault is removed. Now diesel generator does not trip on bus fault condition. However, will trip on overspeed.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 10 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Some events which trip diesel generator excluding accident condition, are added and deleted as follows: 1. Bus fault signal to trip the generator breaker is deleted and following signal are included to trip the diesel generator. 2. Lube oil high temperature 3. Generator over current Diesel generator trips due to following events, during “no accident” conditions have been added 1. Generator Negative Sequence 2. Generator Ground 3. Generator Field Ground Trip Diesel generator rating modified from a 2000 hour rating to a 2 hour short term rating Plant computer removed from DC safeguards bus and added to 125/250 V bus. Both DC safeguard batteries are sized identically.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 11 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Back-up (Alternate) 120 V ac instrument power supply to the 118 V Class IE uninterruptible as bus is changed from 120 V single phase chassie emergency Lighting System to the 120 V ac supply from Class IE MCC via a 480/120 volt single phase Class 1E bypass transformer. Static switch is added in each BOP SUP system to allow automatic transfer from inverter to back-up (alternate) 120 V ac power source. The air circuit breakers are molded case type and fusible switches are provided on some switchboard feeders (instead of circuit breakers) for coordination purposes. Unit substation transformer was changed from 1500 kVA to 2000/1666 kVA. The backup supply for the 120-V NSSS instrument buses was originally taken from the 120-V single-phase emergency lighting system. These buses are now fed from the Class 1E motor control centers via a 480/120V single-phase Class 1E bypass transformer. Electric penetration centerline spacing changed from 2 ft 6 in. to 2 ft 2 in. Total integrated dose that electrical equipment is subjected to is increased to 2.0×10^8 rads (1.5 gamma and 0.5 beta).

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 12 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Added ±24 volt dc system for the turbine generator systems
IV. Mechanical Water Systems:		
Component Cooling	9.2	The following equipment is no longer served by the CCWS: 1. Safety injection pump seal cooler 2. Centrifugal charging pump seal cooler 3. SGBS heat exchanger The following equipment is now served by the CCWS: 1. Instrument air aftercoolers 2. Chilled water system condensers 3. Control room air-conditioning condensers 4. Pump added for component cooling water drain tank in safeguards building 5. Control room alarm added if makeup water added to surge tank

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 13 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		The CCWS heat exchangers will utilize Cu-Ni tubes instead of stainless steel.
		Component Cooling Water pump normal supply temperature is increased to 115^{o}F.
Condensate Storage Facility	9.2.6	Storage capacity increased from 480,000 gallons to 500,000 gallons.
		Secondary system water storage decreased from 240,000 gallons to 224,000 gallons.
		Reserve auxiliary feedwater capacity increased from 240,000 gallons to 276,000 gallons.
Containment Spray System	3.2	Containment spray system nozzles will not bear an “N” stamp
		Change from one spray header to seven.
	6.5	Minimum spray fall height changed from 117 feet to 115 feet 9 inches.
		Eductor calibrated flow changed from 37.5 gpm to 45 gpm.
		Containment spray delivery lag time changed from 40 seconds to 60 seconds.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 14 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Containment sprayed volume changed from 2.528 x 10^6 cubic feet to 1.725 x 10^6 cubic feet. (From 87% to 57.8%)
Piping systems	3.6	The pipe whip analyses are now performed by the ANSYS computer program of Swanson Analysis System, Inc.
	3.9	The addition of stress limits for seismic Category I piping.
Potable and sanitary water system	9.2.4	Sewage plant effluent is discharged to Squaw Creek Reservoir instead of the evaporation ponds. Domestic water storage tank is designed to ASME B&PV Code Section III Code Class 3 requirements. Capacity of the sewage treatment plant is increased from 5000 gpd (normal operation) to 10,000 gpd (normal operation) and to 30,000 gpd during the construction stage. Capacity of the hypochlorinator has been increased.
Station Service Water System	9.2.1	The following changes have been made: 1. The SSI study resulted in an increase of the accident inlet temperature of the service water to 105°F

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 15 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		2. Revision of Service Water Intake structure
		3. Service water pump design temperature increased from 95°F to 120°F
		4. Traveling screen spray piping changed from carbon steel to copper alloy
		5. The SSWS no longer removes heat from the control room air-conditioning system
		6. The SSWS has been added as a backup water supply to the Auxiliary Feedwater System
		7. The non-safety-related train of the SSWS has been deleted
		8. Automatic strainers have been deleted.
		9. Piping is constructed of either stainless steel (or other corrosion resistant material) or is plasite 7122 lined carbon steel as indicated in the appropriate drawings and/or specifications.
Valves: Feedwater Isolation Valves	6.2.4	The feedwater isolation valve will no longer fail closed upon loss of power. Closure will be upon energizing redundant solenoid valves.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 16 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Valves: General	3.1 6.2.4	Change from air-operated valves failing closed to air and solenoid-operated valves failing in the direction of greatest safety.
Materials	6.1B	The control of delta ferrite to reduce the susceptibility of stainless steel welds to hot cracking is no longer considered for ANS Safety Class 3 components. The requirement that the total leachable chloride and fluoride content of clean elastomers and plastics placed over all openings in components fabricated from austenitic stainless steel be limited to 15 and 10 ppm, respectively, has been deleted.
V. Fuel Storage and Handling Systems:		
Fuel storage and handling system	9.1	The following changes were made to the spent fuel storage handling system system: 1. An increase in total spent fuel storage space from 400 to 1166 spent fuel assemblies (1116 in spent fuel pools/25 in each Containment) 2. A decrease in center-to-center spacing from 21 to 16 in.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 17 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		3. An increase in Keff from 0.90 to 0.95 for spent fuel assemblies if immersed in unborated water. Purification loop was added to the refueling cavity. The number of dry storage racks has been increased from 129 to 132. An increase in Keff from 0.90 to 0.98 for new fuel assemblies if flooded with unborated water. The dry cask loading concept is eliminated. The design of the spent fuel pool cooling heat exchangers is changed from horizontal to vertical.
VI. Auxiliary Systems:		
Auxiliary Feedwater System	9.2.1	The SSWS has been added as a backup water supply.
	10.4.9	Manually controlled heating systems have been changed to automatic systems The Auxiliary Feedwater is supplied to the secondary side of the steam generator through a separate upper Auxiliary Feedwater nozzle.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 18 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Condensers	10.4.1	Design basis for auxiliary condensers was revised. Instrumentation for main condenser was modified to include high differential pressure alarm and temperature recorder in the control room. Isolation valves were added to the auxiliary condenser and turbine plant heat exchanger with change from a motor operator to a gear operator. The number of condensate pumps in the condenser hotwells has been changed from three to two half-capacity pumps.
Vacuum pumps	10.4.2	The number of mechanical vacuum pumps was changed from two to three. During startup, discharge is to the atmosphere; during normal operation, discharge is through the charcoal filter system.
Circulating water system	10.4.5	Design basis was revised. Pressure-differential alarm on the screen is annunciated in control room. Butterfly valves on the turbine plant cooling water heat exchangers were added. Circulating water pipes are made from coated carbon steel instead of concrete or plastic.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 19 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Condensate Cleanup System	10.4.6	Backwash recovery and powdered resin handling added to system design
Feedwater heaters	10.4.7	Separate external drain coolers have been added to the 6A and 6B heaters.
Steam Generator Blowdown System	10.4.8	Treatment capacity of the system has been increased from 50 to 640 gpm.
		Blowdown cooled by condensate of condensate/blowdown instead of component cooling water
		Blowdown now recycled to heater drain tank at high temperature
		The processing of spent resin has been changed from regeneration to flushing into the storage tank prior to drumming.
		Change in bed processing as a result of elimination of the regeneration mode.
		SGBS heat exchanger has been changed from ANS Safety Class 3 to NNS.
Fire Protection System	9.5.1	The Fire Protection System has been completely upgraded to address current NRC Regulatory Positions.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 20 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Lighting systems	9.5.3	Lighting system powered from localized motor control centers in lieu of a central power substation. Instrument room and primary plant egress routes have been added to the list of areas provided with DC emergency lighting. The AC Essential Lighting System and DC Emergency Lighting System are reclassified as non-Class 1E.
Diesel Generator Fuel Oil Day Tank	9.5.4	Fuel oil supply increased to 2160 gallons from 3 hours
The Diesel Generator Fuel Oil Storage and Transfer System	9.5.4	The diesel fuel oil storage tank is equipped with: 1. A fill line with a shutoff valve. 2. A perforated fill line which runs to within 2 feet of the bottom of the tank. 3. A dirt and water collector. 4. Hold down straps embedded in a concrete foundation. 5. A return line from the fuel oil day tank.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 21 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Diesel Generator Combustion Air Intake and Exhaust System	9.5.8	Exhaust relief valve discharge changed to vertical.
Main Steam Supply System	10.2	Design temperature reduced to 541.5°F from 544.6°F.
		A vibration trip and alarm have been added.
		A MSR high level trip and alarm have been added.
	10.3	The main steam relief valves are relocated upstream from the safety valves to gain accessibility to these valves.
		The main steam isolation valve materials have been changed.
		Each steam line of auxiliary feedwater pump turbine is provided with an air-operated stop valve instead of motorized stop check valve.
		Phosphate was deleted from use and condensate polishing added.
		Design pressure changed to 1200 psig.
		Safety class piping extended to the first moment restraint beyond the Main Steam Stop Valves.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 22 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Power operated relief valves now set to open at 1130 psia.
		The Main Steam Stop Valve integral bypass valve has been upgraded to code Class 1 from Code Class 2
		Lowest and highest safety valve set pressures changed to 1185 and 1235 psig.
Steam Generators	10.3	Steam generator safety valves rated to pass 105 percent rather than 110 percent of flow.
Compressed Air System	9.3.1	Air accumulators are upgraded to ANS Safety class 3 and designed to ASME B&PV Code Section III Class 3 Requirements.
VII. Ventilation Systems:		
Containment Ventilation Systems	6.2.6	The hydrogen purge system filter efficiencies have been increased from 90 to 95%.
	9.4	CRDM cooling is accomplished by two 100-percent CRDM exhaust fans per containment instead of three 50-percent exhaust fans.
		The following changes have been made in the neutron detector well cooling system:

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 23 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		1. Use of chilled water instead of component cooling water as the cooling medium of the neutron detector well cooling system 2. A reduction in airflow from 24,000 ft^3 per containment to 13,100 ft^3 per containment for the neutron detector well cooling system due to the increase in the temperature gradient across the cooling coils Heating, ventilating, and air-conditioning (HVAC) dampers are manufactured in accordance with ANS Safety Class 3, 10 CFR Part 50 Appendix B, and manufacturers' standards rather than ASME II. To provide for adequate cooling during accident conditions, centrifugal chillers are used for the emergency fan coil units instead of service water. Increase size of containment coolers to absorb heat load from the CRDM. Only the containment purge exhaust ductwork is classified ANS Safety Class 3.
Other ventilation systems		
	9.4.1	Roof-mounted fans changed to in-duct fans for improved missile protection Reduction in outside air intake quantity for control room

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 24 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Control room pressurization capacity increased from 150 cfm to 800 cfm.
		Control room air-conditioning units increased to 100 percent redundancy from 50 percent redundancy.
		Greater than 10 percent of the total air quantity is directed through the control room emergency filtration units.
		Four nuclear safety related vane-axial fans are utilized in each diesel generator room instead of one.
	9.4.1	The concept of lead system and standby system is deleted. Emergency recirculation will activate both Train A and Train B components.
	9.4.2	Auxiliary safeguards buildings ventilation supply distribution systems are now classified as seismic Category I
	9.4.3	Auxiliary building ventilation supply distribution system classified as seismic Category I
		Electrical area ventilation system contains only two 50-percent-capacity fans.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 25 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		Uncontrolled access area ventilation system contains two 50-percent-capacity fans.
		Battery rooms have in-duct fans instead of roof-mounted fans.
		Office and service area ventilation systems have 50-percent redundancy; these systems are not required to satisfy the single-failure criterion and are nonessential to the safe shutdown of the reactor.
	General	Auxiliary, safeguards, fuel-handling, and containment purge supply and exhaust are incorporated into primary plant ventilation system.
		Summer design condition of diesel generator building changed from 104°F to 122°F
		Filter bed thickness increased from 2 inches to 4 inches.
		Air inlet design temperature to the hydrogen purge exhaust unit changed from 120°F to 250°F.
		Instrument air accumulators are now provided for the control room HVAC system.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 26 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
VIII. Steam & Power Conversion Systems:		
Component and sub-system design	5.4	The steam generators for Unit 2 have been changed from D4's to D5's
Steam and power conversion system	10.1	The steam dump system to the condenser form the steam generator system is not a safety-related feature included in the steam and power conversion system.
	10.3	The Main Steam Isolation Valve (MSIV) Bypass Valve were converted form automatic to manual. The hydraulic actuactors were deleted and handwheels installed. The valves are capable of being locked in position.
Turbine-generator	10.2	Added Occupational Safety and Health Act (OSHA) to codes and standards. The worst case accident, a failure of the cast stage of the low pressure turbine rotor, is analyzed. The following have been added to the list of events that initiate a turbine trip: 1. Reactor trip 2. Steam generator high-high level

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 27 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
		3. Safety injection 4. Generator trip 5. Moisture separator high level (each MSR) 6. Excessive vibration during speed operation from 900 rpm until unit is synchronized. The following have been deleted from the list of events that initiate a turbine trip 1. Low flow of stator coolant The following have been added to the list of events that initiate a generator trip: 1. Loss of lube oil pressure 2. Low flow of stator coolant

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 28 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
IX. Waste Processing Systems:		
Liquid Waste Processing System	11.2	The following equipment has been added to the LWPS: 1. Laundry reverse osmosis system 2. Laundry holdup and monitor tanks 3. Laundry holdup and monitor tank pump 4. 10,000 gallon floor drain tank 2 5. 30,000 gallon floor drain tank 3
Solid Waste Processing System (SWPS)	11.4	Utilizes an ATCOR proprietary cement solidification system per ATCOR Topical Report No. ATC-132A.
X. Sampling & Monitoring Systems:		
Process Sampling System	9.3.2	Phosphate analyzers replaced by sodium ion analyzers. System changed to non-nuclear safety related and non-seismic down-stream from the external containment isolation valves.
	9.3.2, II.B.3	Inclusion of Post Accident Sampling System.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 29 of 30)

(See Note 1)

Systems or Components	CPNPP/FSAR Section	Changes
Containment Hydrogen Monitoring System	6.2.5	Remote thermal conductivity analyzers replaced by in-containment electrochemical sensors and microprocessor-based analyzers located in the control room.
Steam generator liquid sample monitors	11.5.2	Steam generator liquid sample monitors reduced from four to one to simplify design
Radiation Monitoring System	11.5	Changed from Analog to digital, microprocessor based system. Process monitors have been added to liquid waste processing and auxiliary condensate systems and to safeguards, auxiliary and fuel building vent ducts. Area monitors have been added to the fuel building to form a criticality alarm system. The auxiliary vent stack is no longer capable of being monitored by the containment air monitor Radiation monitors have been modified and have been added for accident monitoring.
	12.32	Change from analog system to digital microprocessor based system Radiation monitors have been added for accident monitoring.

TABLE 1.3-2
DESIGN CHANGES SINCE PSAR SUBMITTAL

(Sheet 30 of 30)

(See Note 1)

	Systems or Components	CPNPP/FSAR Section	Changes
XI.	Boron Recycle and Radioactive Waste Processing Systems		
1.	Boron Recycle	3.2	The code, code class, safety class, and seismic classification Table 17A-1 of system components were revised.
2.	Liquid Waste Processing System	3.2 Table 17A-1 Appendix 1A(B)	Same as for XI, item 1.
3.	Gaseous Waste Processing System	3.2 Table 17A-1 Appendix 1A(B)	Same as for XI, item 1.

NOTE 1: The FSAR is complete and does not rely on information contained in the PSAR. The original intent of this table was to provide a comparison with the PSAR when the FSAR was initially submitted for NRC Staff review. The information in this table is considered historical and will not be updated.

1.4 IDENTIFICATION OF AGENTS AND CONTRACTORS

1.4.1 OWNER AND OPERATORS

Vistra Operations Company LLC (Vistra OpCo) (operator) and Comanche Peak Power Company LLC (CP PowerCo) (owner) are subsidiaries of Vistra Corp. A simplified organizational diagram is shown in Figure 13.1-1.

1.4.2 CONTRACTORS

Vistra OpCo is responsible for the design and construction of CPNPP.

A historical list of major contractors during the design and construction of CPNPP is provided below.

- Gibbs & Hill, Inc. – original architect-engineer responsible for the design and engineering
- Stone and Webster Engineering Corporation (SWEC) – various engineering services
- Impell – various engineering services (e.g., fire protection, equipment qualification, cable tray hangers, conduit supports, startup assistance)
- Ebasco – various engineering services (e.g, HVAC, conduit supports, cable tray hangers, system interaction program)
- Westinghouse Electric Corporation – Nuclear Steam Supply System Manufacturer
- Dames & Moore – earth sciences engineer
- Brown & Root, Inc. – constructor
- Mason-Johnson & Associates – subsurface and soils investigation
- Freese and Nichols – reservoir design
- EDS Nuclear Incorporated – retained as a consultant for QA services to assist in the development of QA programs and to provide operations administrative support services

Responsibilities of the quality assurance program (e.g., audits, procurement) are described in Chapter 17.

TABLE 1.4-1
DELETED

1.5 REQUIREMENTS FOR FURTHER TECHNICAL INFORMATION

The activities described in Section 1.5 will be maintained as historical information and will not be updated.

Reference [1] presents descriptions of the safety-related research and development programs which are being carried out for, or by, or in conjunction with, Westinghouse Nuclear Energy Systems, and which are applicable to Westinghouse Pressurized Water Reactors (PWRs).

For each program still in progress the safety-related program is first introduced, followed, where appropriate, by background information. There is, then, a description of the program which relates the program objectives to the problem and presents pertinent recent results. Finally, a back up position may be given for programs (generally experimental rather than analytical) which have not yet reached a stage where it is reasonably certain that the results confirm the expectation. The back up position is one that might be used if the results are unfavorable; it is not necessarily the only course that might be taken.

The term “research and development”, as used in this report, is the same as that used by the NRC in 10CFR50.2, that is:

(n) ‘Research and development’ means (1) theoretical analysis, exploration or experimentation; or (2) the extension of investigative findings and theories of a scientific nature into practical application for experimental and demonstration purposes including the experimental production and testing of models, devices, equipment, materials, and processes.

The technical information generated by these research and development programs will be used either to demonstrate the safety of the design and more sharply define margins of conservatism, or will lead to design improvements.

1.5.1 BLOWDOWN HEAT TRANSFER TESTING

The Nuclear Regulatory Commission (NRC) acceptance criteria for Emergency Core Cooling Systems (ECCS) for light-water power reactors was issued in Section 50.46 of 10CFR50 on December 28, 1973. It defines the basis and conservative assumptions to be used in the evaluation of the performance of the ECCS. Westinghouse believes that some of the conservatism of the criteria is associated with the manner in which transient departure from nucleate boiling (DNB) phenomena are treated in the evaluation models. Transient critical heat flux data presented at the 1972 specialists meeting of the Committee on Reactor Safety Technology (CREST) indicated that the time to DNB can be delayed under transient conditions. To demonstrate the conservatism of the ECCS evaluation models, Westinghouse initiated a program to experimentally simulate the blowdown phase of a loss of coolant accident (LOCA). This testing is part of the Electric Power Research Institute (EPRI) sponsored Blowdown Heat Transfer Program, which was started early in 1976. Testing was completed in 1979. A DNB correlation will be developed by Westinghouse from these test results for use in the ECCS analyses.

1.5.1.1 Objective

The objective of the Blowdown Heat Transfer Test was to determine the time that DNB occurs under LOCA conditions. This information will be used to confirm the existing, or develop a new

Westinghouse transient DNB correlation. The steady state DNB data obtained from 15 x 15 and 17 x 17 test programs can be used to assure that the geometrical differences between the two fuel arrays can be correctly treated in the transient correlations.

1.5.1.2 Program

The program was divided into two phases. The Phase I tests started from steady state conditions, with sufficient power to maintain nucleate boiling throughout the bundle. Controlled ramps of decreasing test section pressure or flow initiated DNB. By applying a series of controlled conditions, investigation of the DNB was studied over a range of qualities and flows, and at pressures relevant to a PWR blowdown.

Phase I provided separate-effects data to permit heat transfer correlation development.

Typical parameters used for Phase I testing are shown in Table 1.5-1.

Phase II simulates PWR behavior during a LOCA to permit definition of the time delay associated with onset of DNB. Tests in this phase covered the large double ended guillotine cold leg break. All tests in Phase II were also started after establishment of typical steady state operating conditions. The fluid transient was then initiated, and the rod power decay programmed in such a manner as to simulate the actual heat input of fuel rods. The test was terminated when the heater rod temperatures reach a predetermined limit.

Typical parameters used for Phase II testing are shown in Table 1.5-2.

1.5.1.3 Test Description

The experimental program was conducted in the J-Loop at the Westinghouse Forest Hills Facility with a full length 5 x 5 rod bundle simulating a section of a 15 x 15 assembly to determine DNB occurrence under LOCA conditions.

The heater rod bundles used in this program were internally-heated rods, capable of a maximum power of 18.8 kW/ft, with a total power of 135 kW (for extended periods) over the 12 foot heated length of the rod. Heat was generated internally by means of a varying cross-sectional resistor which approximates a chopped cosine power distribution. Each rod was adequately instrumented with a total of 12 clad thermocouples.

1.5.1.4 Results

The experiments in the DDNB Facility resulted in cladding temperature and fluid properties measured as a function of time throughout the blowdown range from 0 to 20 seconds.

Facility modifications and installation of the initial test bundle were completed. A series of shakedown tests in the J-Loop were performed. These tests provided data for instrumentation calibration and check-out, and provided information regarding facility control and performance. Initial program tests were performed during the first half of 1975. Under the sponsorship of EPRI, testing was reinitiated during 1976 on the same test bundle. The testing was terminated in November and plans were made for a new test bundle and further testing during 1978-79. These tests were completed in December of 1979. A DNB correlation will be developed from these test results for use in the Westinghouse ECCS analyses.

REFERENCES

1. Eggleston, F. T., “Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries,” WCAP-8768, latest revision.

TABLE 1.5-1
DELAYED DEPARTURE FROM NUCLEATE BOILING PHASE I TEST PARAMETERS

Parameters	Nominal Value
Initial Steady State Conditions	
Pressure	1250 to 2250 psia
Test section mass velocity	12 to 2.5 x 10^6 lb/hr-ft^2
Core inlet temperature	550 to 600°F
Maximum heat flux	306,000 to 531,000 Btu/hr-ft^2
Transient Ramp Conditions	
Pressure decrease	0 to 350 psi/sec and subcooled depressurization from 2250 psia
Flow decrease	0 to 100%/sec
Inlet enthalpy	Constant

TABLE 1.5-2
DELAYED DEPARTURE FROM NUCLEATE BOILING PHASE II TEST PARAMETERS

Parameters	Nominal Value
Initial Steady State Conditions	
Pressure	2250 psia
Test section mass velocity	2.5×10^6 lb/hr-ft^2
Inlet temperature	545°F
Maximum heat flux	531,000 Btu/hr-ft^2
Transient Conditions	
Simulated break	Double ended cold leg guillotine breaks

1.6 MATERIAL INCORPORATED BY REFERENCE

Table 1.6-1 lists documents which provide information additional to that provided in this FSAR and have been filed separately with the Nuclear Regulatory Commission (NRC) in support of the CPNPP operating licenses. A change to any of the documents listed in this table is considered to be a change to the FSAR and is subject to the same processing requirements. It should be noted, however, that for documents with revisions and/or dates, only the specific revisions and/or dates listed are considered to be part of the FSAR. Later revisions to these documents are not part of the FSAR unless they are incorporated into this table (except for the Technical Requirements Manual (including TRM Bases) and Technical Specifications Bases).

Table 1.6-2 lists many of the topical reports and other licensing documents which were provided to the NRC in support of the CPNPP operating licenses as additional background and/or reference material for the FSAR. Material in this table is not considered to be "incorporated by reference" and changes to these documents are not considered to changes to the FSAR. Table 1.6-2 is historical and is not updated.

Documents listed in Table 1.6-2 (or are referenced elsewhere in the FSAR but are not listed in Table 1.6-1) are provided for information (e.g., references listed at the end of each FSAR section) or are documents to which CPNPP is committed (e.g., Regulatory Guides, Codes and Standards). Compliance with committed documents is required but the documents are not considered to be part of the FSAR.

TABLE 1.6-1
DOCUMENTS INCORPORATED BY REFERENCE

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Report	Reference Section(s)
"An Evaluation of the Rod Ejection Accident in Westinghouse Pressurized Water Reactors Using Spatial Kinetics Methods," WCAP-7588, Revision 1-A, January 1975	15.4
"Overpressure Protection for Westinghouse Pressurized Water Reactors," WCAP-7769, October 1971	5.2
"Pipe Breaks for the LOCA Analysis of the Westinghouse Primary Coolant Loop," WCAP-8082-P-A (Proprietary) and WCAP-8172-A (Non-Proprietary), January 1975 (Non-Proprietary), January 1975	3.6B
WCAP-10698-P-A, "SGTR Analysis Methodology to Determine the Margin to Steam Generator Overfill," August 1987.	5.2, 6.2, 15.1, 15.2, 15.3, 15.4, 15.5, 15.6
WCAP-14882-P-A, "RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses," April 1999.	
"VIPRE-01 Modeling and Qualification for Pressurized Water Reactor Non-LOCA Thermal-Hydraulic Safety Analysis," WCAP-14565-P-A, October 1999.	4.4, 15.1, 15.3, 15.4
"Fuel Densification Experimental Results and Model for Reactor Application," WCAP-8218-P-A (Proprietary) and WCAP-8219-A (Non-Proprietary), March 1975.	4.1, 4.2, 4.3, 4.4
"Westinghouse Mass and Energy Release Data For Containment Design," WCAP-8264-P-A, Rev. 1 July 2004 (Proprietary) and WCAP-8312-A, Revision 2, August 1975 (Non-Proprietary).	6.2
"Westinghouse Anticipated Transients Without Trip Analysis," WCAP-8330, August 1974	4.3, 4.6 15.1, 15.2 15.8
"Environmental Qualification of Westinghouse Class 1E Equipment," WCAP-8587, Rev. 6, March 1983	3.10N, 3.11N

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Report	Reference Section(s)
"General Method of Developing Multi-frequency Biaxial Test Inputs for Bistables," WCAP-8624 (Proprietary) September 1975 and WCAP-8695 (Non-Proprietary) August 1975.	3.10N
Response to Station Blackout Rule", TXX-92447 October 1, 1992	8B
TP-03143, "Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC," July 31, 2003.	3.5, 10.2
Technical Requirements Manual (TRM) for Comanche Peak Nuclear Power Plant Units 1 and 2	All Sections
Technical Specification Bases for Comanche Peak Nuclear Power Plant Units 1 and 2	All Sections
Foster, J.P., et. al., "Westinghouse Improved Performance Analysis and Design Model (PAD 4.0)," WCAP-15063-P-A, Revision 1, with Errata (Proprietary) and WCAP-15064-NP-A, Revision 1, with Errata (Non-Proprietary), July 2000.	4.2, 4.3, 4.4, 15.1
Davidson, S.L., Ryan, T. L., "VANTAGE+ Fuel Assembly Reference Core Report," WCAP-12610-P-A and Appendices, April 1995.	4.2, 4.3, 4.4, 15.1
Weiner, R. A., et. al., "Improved Fuel Performance Models for Westinghouse Fuel Rod Design and Safety Evaluations," WCAP-10851-P-A (Proprietary) and WCAP-11873-A (Non-Proprietary), August 1988.	4.2
Kersting, P. J., et. al., "Assessment of Clad Flattening and Densification Power Spike Factor Elimination in Westinghouse Nuclear Fuel," WCAP-13589-A, March 1995.	4.2
WCAP-13060-P-A, "Westinghouse Fuel Assembly Reconstitution Methodology," July 1993 (Technical Specification 5.6.5b Item 18).	4

TABLE 1.6-1
DOCUMENTS INCORPORATED BY REFERENCE

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Report	Reference Section(s)
WCAP-10444-P-A, "Reference Core Report VANTAGE 5, Fuel Assembly," September 1985 (Technical Specification 5.6.5b Item 16).	4.2, 4.3, 4.4, 15.1
Scherder, W. J. (ed.), et al. "Reactor Core Response to Excessive Secondary Steam Releases," WCAP-9226-P-A, Revision 1, February 1998.	4.4, 15.1
"Addendum 1 to WCAP-12488-A, Revision to Design Criteria," WCAP-12488-A, Addendum 1-A, Revision 1, January 2002.	4.2
Davidson, S. L. (ed.), et al., "Extended Burnup Evaluation of Westinghouse Fuel," WCAP-10125-P-A, December 1985.	4.2
WCAP-14040-NP-A, "Methodology used to Develop Cold Overpressure Mitigating System Setpoints and RCS Heatup and Cooldown Limit Curves."	5.2.2.11
Design Basis Document, DBD-ME-006, "Control of Heavy Loads at Nuclear Plants" (i.e., "CPSES Final Response to NUREG-0612", June 1983, as submitted by TXX-3659 dated June 8, 1983 / ML20072A945)	Appendix 1A(B), RG 1.13 and RG 1.104, 9.1.4, Table 17A-1
"RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses," WCAP-14882-P-A, April 1999.	5.2, 15.1, 15.2, 15.3, 15.4, 15.5, 15.6
"Revised Thermal Design Procedure," WCAP-11397-P-A, April 1989.	4.2, 4.3, 4.4, 15.0, 15.1, 15.2, 15.3, 15.4, 15.6
"TWINKLE - A Multi-dimensional Neutron Kinetics Computer Code," WCAP-7979-P-A, January 1975.	15.0, 15.1, 15.4
"FACTRAN - A FORTRAN-IV Code for Thermal Transients in a UO2 Fuel Rod," WCAP-7908-A, December 1989.	15.4
"LOFTRAN Code Description," WCAP-7907-P-A, April 1984.	15.4
"Methodology for the Analysis of the Dropped Rod Event," WCAP-11394-P-A, January 1990.	4.2, 4.3, 4.4, 15.1, 15.4

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DOCUMENTS INCORPORATED BY REFERENCE

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Report	Reference Section(s)
"ANC: A Westinghouse Advanced Nodal Computer Code," WCAP-10965-P-A, September 1986.	4.3, 15.4
WCAP-10079-P-A (Proprietary) and WCAP-10080-A (Non-proprietary), 1985. Meyer, P. E., "NOTRUMP, A Nodal Transient Small Break and General Network Code."	15.6
Report Reference Section(s) WCAP-10054-P-A, Addendum 2, Revision 1 (Proprietary), 1995. Thompson, C.M., et al, "Addendum to the Westinghouse Small Break ECCS Evaluation Model Using the NOTRUMP Code: Safety Injection into the Broken Loop and COSI Condensation Model."	15.6
WCAP-10054-P-A (Proprietary) and WCAP-10081-A (Non-proprietary), 1985. Lee, N. Rupprecht, S. D., Schwarz, W. R, and Tauche, W. D., "Westinghouse Small ECCS Evaluation Model Using the NOTRUMP Code."	15.6
Bajorek, S.M., et. al., 1998, "Code Qualification Document for Best-Estimate LOCA Analysis," WCAP-12945-P-A, Volume 1, Revision 2 and Volumes 2 through 5, Revision 1.	15.6
WCAP-16009-P-A, "Realistic Large-Break LOCA Evaluation Methodology Using the Automated Statistical Treatment of Uncertainty Method (ASTRUM)", (Proprietary), January 2005.	15.6
WCAP-8745, "Design Bases for the Thermal Overpower ΔT Trip Functions," September 1986.	15.1
WCAP-10325-P-A, (Proprietary) and WCAP-10326-A, (Nonproprietary), "Westinghouse LOCA Mass and Energy Release Model for Containment Design, March 1979 Version," May 1983.	6.2
WCAP-8264, Rev. 1, August 1975 and WCAP-8312, Rev. 2, "Topical Report Westinghouse Mass and Energy Release Data Containment Design."	6.2

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DOCUMENTS INCORPORATED BY REFERENCE

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Report	Reference Section(s)
WCAP-8822 and WCAP-8860, "Mass and Energy Releases Following a Steam Line Rupture," September 1976; WCAP-8822 and WCAP-8860, "Supplement 1 - Calculations of Steam Superheat in Mass/Energy Releases Following a Steam Line Rupture," September 1986; WCAP-8822 and WCAP-8860, "Supplement 2 - Impact of Steam Superheat in Mass/Energy Releases Following a Steam Line Rupture for Dry and Subatmospheric Containment Designs," September 1986.	6.2
WCAP-9272, "Westinghouse Reload Safety Evaluation Methodology," July 1985.	4.2, 4.3, 4.4, 15.1
RXE-94-001-A, "Safety Analysis of the Postulated Inadvertent Boron Dilution Event in Modes 3, 4, and 5," February 1994.	15.1
Skaritka, J., (ed.), "Fuel Rod Bow Evaluation," WCAP-8691, Rev. 1 (Proprietary) and WCAP-8692, Rev. 1 (Non-Proprietary), July 1979.	
WCAP-16045-P-A, "Qualification of the Two-Dimensional Transport Code PARAGON," August 2004.	4.3
WCAP-16045-P-A, Addendum 1, "Qualification of the NEXUS Nuclear Data Methodology," November 2005.	4.3
WCAP-10965-P-A, Addendum 2-A, "Qualification of the New Pin Power Recovery Methodology," September 2010.	4.3, 15.4

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Report	Reference Section(s)
"The Doppler Effect for a Non-Uniform Temperature Distribution in Reactor Fuel Elements," WCAP-2048, July 1962	4.3
"Single Phase Local Boiling and Bulk Boiling Pressure Drop Correlations," WCAP-2850 (Proprietary), April 1966 and WCAP-7916 (Non-Proprietary), June 1972	4.4
"In-Pile Measurement of UO_2 Thermal Conductivity," WCAP-2923, 1966	4.4
"Hydraulic Tests of the San Onofre Reactor Model," WCAP-3269-8, June 1964	4.4
"LEOPARD - A Spectrum Dependent Non-Spatial Depletion Code for the IBM - 7094," WCAP-3269-26, September 1963	4.3, 15.0, 15.4
"Saxton Core II Fuel Performance Evaluation," WCAP-3385-56, Part II, "Evaluation of Mass Spectrometric and Radio-chemical Analyses of Irradiated Saxton Plutonium Fuel," July 1970	4.3, 4.4
"Xenon-Induced Spatial Instabilities in Large PWRs," WCAP-3680-20, (EURAEC-1974) March 1968	4.3
"Control Procedures for Xenon-Induced X-Y Instabilities in Large PWR's" WCAP-3680-21, (EURAEC-2111) February 1969	4.3
"Xenon-Induced Spatial Instabilities in Three-Dimensions," WCAP-3680-22, (EURAEC-2116) September 1969	4.3
"Pressurized Water Reactor pH - Reactivity Effect Final Report," WCAP-3696-8, (EURAEC-2074) October 1968	4.3
"PUO_2 - UO_2 Fueled Critical Experiments," WCAP-3726-1, July 1967	4.3
"Melting Point of Irradiated UO_2," WCAP-6065, February 1965	4.2, 4.4

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Report	Reference Section(s)
"Burnup Physics of Heterogeneous Reactor Lattices," WCAP-6069, June 1965	4.4
"LASER - A depletion Program for Lattice Calculations Based on MUFT and THERMOS," WCAP-6073, April 1966	4.3
"Supplementary Report on Evaluation of Mass Spectrometric and Radio-chemical Analyses of Yankee Core I Spent Fuel, Including Isotopes of Elements Thorium Through Curium," WCAP-6086, August 1969	4.3
"Subchannel Thermal Anaylsis of Rod Bundle Cores," WCAP-7015, Revision 1, January 1969	4.4
"The PANDA Code," WCAP-7048-P-A (Proprietary) and WCAP-7757-A (Non-Proprietary), January 1975	4.3
"Evaluation of Protective Coatings for use in Reactor Containment," WCAP-7198-L (Proprietary), April 1969 and WCAP-7825 (Non-Proprietary), December 1971	6.1N
"The TURTLE 24.0 Diffusion Depletion Code," WCAP-7213-P-A (Proprietary) and WCAP-7758-A (Non-Proprietary), January 1975	4.3, 15.0 15.4
"Evaluation of Nuclear Hot Channel Factor Uncertainties," WCAP-7308-L (Proprietary) and WCAP-7810 (Non-Proprietary), December 1971	4.3
"Application of the THINC Program to PWR Design," WCAP-7359-L (Proprietary), August 1969 and WCAP-7838 (Non-Proprietary), January 1972	4.4
"Sensitized Stainless Steel in Westinghouse PWR Nuclear Steam Supply Systems," WCAP-7477-L (Proprietary), March 1970 and WCAP-7735 (Non-Proprietary), August 1971	5.2
"Solid State Logic Protection System Description," WCAP-7488-L (Proprietary), March 1971 and WCAP-7672 (Non-Proprietary), May 1971.	7.2, 7.3

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Report	Reference Section(s)
"Radiological Consequences of a Fuel Handling Accident," WCAP-7518-L (Proprietary) and WCAP-7828 (Non-Proprietary), June 1970	15.7
"Seismic Vibration Testing with Sine Beats," WCAP-7558, October 1971	3.10N
"Dynamic Fracture Toughness Properties of Heavy Section A533 Grade B Class 1 Steel Plate," WCAP-7623, December 1970	5.4
"Interchannel Thermal Mixing with Mixing Vane Grids," WCAP-7667-P-A (Proprietary) and WCAP-7755-A (Non-Proprietary), January 1975	4.4
"DNB Tests Results for New Mixing Vane Grids (R)," WCAP-7695-P-A (Proprietary) and WCAP-7958-A (Non-Proprietary), January 1975	4.4
"DNB Test Results for R Grid Thimble Cold Wall Cells," WCAP-7695-Addendum 1-P-A (Proprietary) and WCAP-7958-Addendum 1-A (Non-Proprietary), January 1975	4.4
"Testing of Engineering Safety Features Actuation System," WCAP-7705, Revision 2, January 1976.	7.3
"An Evaluation of Solid State Logic Reactor Protection in Anticipated Transients," WCAP-7706-L (Proprietary) and WCAP-7706 (Non-Proprietary), February 1973	4.6, 7.1, 7.2
"Electric Hydrogen Recombiner for PWR Containments," WCAP-7709-L, Supplements 1 through 7 (Proprietary) and WCAP-7820, Supplements 1 through 7 (Non-Proprietary), 1971 through 1977	3.11N, 6.2
"A Comprehensive Space-Time Dependent Anaylsis of Loss of Coolant (SATAN-IV Digital Code)," WCAP-7750, August 1971	3.6B

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Report	Reference Section(s)
"Behavior of Austenitic Stainless Steel in Post Hyporthetical Loss of Coolant Accident Environment," WCAP-7798-L (Proprietary) and WCAP-7803 (Non-Proprietary), January 1972	6.1N
"Nuclear Fuel Division Quality Assurance Program Plan," WCAP-7800, Revision 4-A, April 1975	4.2, 17
"Nuclear Design of Westinghouse Pressurized Water Reactors with Burnable Poison Rods," WCAP-7806, December 1971	4.3
"Power Distribution Control of Westinghouse Pressurized Water Reactors," WCAP-7811, December 1971	4.3
"Evaluation of Steam Generator Tube, Tubesheet and Divider Plate Under Combined LOCA Plus SSE Conditions," WCAP-7832, December 1973	5.4
"Inlet Orificing of Open PWR Cores," WCAP-7836, January 1972	4.4
"Neutron Shielding Pads," WCAP-7870, May 1972	3.9N
"LOFTRAN Code Description," WCAP-7907, June 1972	5.2, 15.0 15.1, 15.2 15.3, 15.4 15.5, 15.6
"FACTRAN - A FORTRAN-IV Code for Thermal Transients in a UO_2 Fuel Rod," WCAP-7908, June 1972	15.0, 15.2 15.3, 15.4
"MARVEL, A Digital Computer Code for Transient Analysis of a Multiloop PWR System," WCAP-7909, June 1972	6.2
"Power Peaking Factors," WCAP-7912-P-A (Proprietary) and WCAP-7912-A (Non-Proprietary), January 1975	4.3, 4.4
"Process Instrumentation for Westinghouse Nuclear Steam Supply Systems," WCAP-7913, January 1973.	7.2, 7.3

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Report	Reference Section(s)
"Damping Values of Nuclear Power Plant Components," WCAP-7921-AR, May 1974	1A(N), 3.7N
"Basis for Heatup and Cooldown Limit Curves," WCAP-7924-A, April 1975	5.3
"Effect of Axial Spacing on Interchannel Thermal Mixing with the R Mixing Vane Grid," WCAP-7941-P-A (Proprietary) and WCAP-7959-A (Non-Proprietary), January 1975	4.4
"THINC-IV An Improved Program for Thermal- Hydraulic Analysis of Rod Bundle Cores," WCAP-7956, June 1973	4.4
"Axial Xenon Transient Tests at the Rochester Gas and Electric Reactor," WCAP-7964, June 1971	4.3
"TWINKLE - A Multi-Dimensional Neutron Kinetics Computer Code," WCAP-7979-P-A (Proprietary) and WCAP-8028-A (Non-Proprietary), January 1975	15.0, 15.4
"Application of the THINC-IV Program to PWR Design," WCAP-8054 (Proprietary) and WCAP-8195 (Non-Proprietary), October 1973	4.4
A Summary Analysis of the April 30 Incident at the San Onofre Nuclear Generating Station Unit 1," WCAP-8099, April 1973	3.9N
"Reactor Coolant Pump Integrity in LOCA," WCAP-8163, September 1973	1A(N), 5.4
"Calculation Model for Core Reflooding After a Loss of Coolant Accident (WREFLOOD Code)," WCAP-8170 (Proprietary) and WCAP-8171 (Non-Proprietary), June 1974	15.6
"Operational Experience with Westinghouse Cores," WCAP-8183, Revision 6, June 1977	4.2, 11.1

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Report	Reference Section(s)
"WFLASH, A FORTRAN-IV Computer Program for Simulation of Transients in a Multi-Loop PWR," WCAP-8200, Revision 2 (Proprietary) and WCAP-8261, Revision 1 (Non-Proprietary), July 1974	15.6
"Safety Analysis of the 17x17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident," WCAP-8236 (Proprietary), December 1973 and WCAP-8288 (Non-Proprietary), January 1974	3.7N, 4.2
"Safety Analysis of the 8-Grid 17x17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident," WCAP-8236, Addendum 1(Proprietary), March 1974 and WCAP-8288, Addendum 1 (Non-Proprietary), April 1974	3.7N
"Documentation of Selected Westinghouse Structural Analysis Computer Codes," WCAP-8252, Revision 1, July 1977	3.6B, 3.9N
"Source Term Data for Westinghouse Pressurized Water Reactors," WCAP-8253, Amendment 1, July 1975	11.1, 12.3
"Nuclear Instrumentation System," WCAP-8255 January 1974.	7.2, 7.7
"Hydraulic Flow Test of the 17x17 Fuel Assembly," WCAP-8278 (Proprietary) and WCAP-8279 (Non-Proprietary), February 1974	4.2, 4.4
"The Effect of 17x17 Fuel Assembly Geometry on Interchannel Thermal Mixing," WCAP-8298-P-A (Proprietary) and WCAP-8299-A (Non-Proprietary), January 1975	4.4
"LOCTA-IV Program" Loss of Coolant Transient Analysis," WCAP-8301 (Proprietary) and WCAP-8305 (Non-Proprietary), June 1974	15.0, 15.6
"SATAN-IV Program: Comprehensive Space-Time Dependent Analysis of Loss of Coolant," WCAP-8302 (Proprietary) and WCAP-8306 (Non-Proprietary), June 1974	15.0, 15.6

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Report	Reference Section(s)
"Prediction of the Flow-Induced Vibration of Reactor Internals by Scale Model Tests," WCAP-8303-P-A (Proprietary) and WCAP-8317-A (Non-Proprietary), July 1975	3.9N
"Control of Delta Ferrite in Austenitic Stainless Steel Weldments," WCAP-8324-A, June 1975	1A(N), 5.2
"Containment Pressure Analysis Code (COCO)," WCAP-8327 (Proprietary) and WCAP-8326 (Non-Proprietary), June 1974	15.6
"Westinghouse ECCS Evaluation Model- Summary," WCAP-8339, July 1974	6.2, 15.6
"Westinghouse ECCS – Plant Sensitivity Studies," WCAP-8340 (Proprietary) and WCAP-8356 (Non-Proprietary), July 1974	15.6
"Westinghouse ECCS Evaluation Model Sensitivity Studies," WCAP-8341 (Proprietary) and WCAP-8342 (Non-Proprietary), July 1974	15.6
"Effects of Fuel Densification Power Spikes on Clad Thermal Transients," WCAP-8359, July 1974	4.3
"Westinghouse Nuclear Energy Systems Division Quality Assurance Plan," WCAP-8370, Revision 8A, September 1977	1A(N), 17
"Revised Clad Flattening Model," WCAP-8377 (Proprietary) and WCAP-8381 (Non-Proprietary), July 1974	4.2
"Power Distribution Control and Load Follow Procedures," WCAP-8385 (Proprietary) and WCAP-8403 (Non-Proprietary), September 1974	4.3, 4.4
"An Evaluation of Loss of Flow Accidents Caused by Power System Frequency Transients in Westinghouse PWRs," WCAP-8424, Revision 1, June 1975	15.3

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Report	Reference Section(s)
"17x17 Drive Line Components Tests - Phase IB, II, III, D-Loop Drop and Deflection," WCAP-8446 (Proprietary) and WCAP-8449 (Non-Proprietary), December 1974	3.9N
"Analysis of Data from the Zion (Unit 1) THINC Verification Test," WCAP-8453-A, May 1976	4.4
"Westinghouse ECCS Evaluation Model - Supplementary Information," WCAP-8471 (Proprietary) and WCAP-8472 (Non-Proprietary), April 1975	15.6
"Incore Power Distribution Determination in Westinghouse Pressurized Water Reactors," WCAP-8498, July 1975	4.3
"UHI Plant Internals Vibration Measurement Program and Pre and Post Hot Functional Examinations," WCAP-8516-P (Proprietary) and WCAP-8517 (Non-Proprietary), April 1975	3.9N
"Critical Heat Flux Testing of 17x17 Fuel Assembly Geometry with 22 Inch Grid Spacing," WCAP-8536 (Proprietary) and WCAP-8537 (Non-Proprietary), May 1975	4.4
"The Application of Preheat Temperatures after Welding Pressure Vessel Steels," WCAP-8577, February 1976	1A(N)
"Failure Mode and Effects Anaylsis (FMEA) of the Engineered Safeguard Features Actuation System," WCAP-8584 (Proprietary) and WCAP-8760 (Non-Proprietary), Revision 1, February 1980.	4.6, 7.3
"Westinghouse ECCS Evaluation Model - October 1975 Version," WCAP-8622 (Proprietary) and WCAP-8623 (Non-Proprietary), November 1975	15.6
"Experimental Verification of West Fuel Storage Criticality Analyses," WCAP-8682 (Proprietary) and WCAP-8683 (Non-Proprietary), December 1975	4.3

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Report	Reference Section(s)
"Fuel Rod Bowing," WCAP-8691 (Proprietary) and WCAP-8692 (Non-Proprietary), Revision 1, July 1979 as supplemented by NS-EPR-2515, October 9, 1981 and NS-EPR-2572, March 16, 1982	4.2
"Delta Ferrite in Production Austenitic Stainless Steel Weldments," WCAP-8693, January 1976	1A(N), 5.2
"MULTIFLEX - A FORTRAN-IV Computer Program for Analyzing Thermal-Hydraulic-Structure System Dynamics," WCAP-8708 (Proprietary) and WCAP-8709 (Non-Proprietary), February 1976	3.9N
"Improved Analytical Models Used in Westinghouse Fuel Rod Design Computations," WCAP-8720 (Proprietary) and WCAP-8785 (Non-Proprietary), October 1976	4.2
"Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries," WCAP-8768, latest revision	1.5, 4.2 4.3
"Verification of Neutron Pad and 17x17 Guide Tube Designs by Preoperational Tests on the Trojan 1 Power Plant," WCAP-8780, May 1976	3.9N
"Mass and Energy Releases Following a Steam Line Rupture," WCAP-8822 (Proprietary) and WCAP-8860 (Non-Proprietary), September 1976.	6.2
"Hybrid B_4C Absorber Control Rod Evaluation Report," WCAP-8846-A, October 1977	4.2, 15.3
"7300 Series Process Control System Noise Tests," WCAP-8892-A, April 1977	7.1
"Bench March Problem Solutions Employed for Verification of WECAN Computer Program," WCAP-8929, June 1977.	3.9N

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Report	Reference Section(s)
"Safety Analysis for the Revised Fuel Rod Internal Pressure Design Basis," WCAP-8963 (Proprietary), November 1976 and WCAP-8964 (Non-Proprietary), August 1977	4.2
"Dropped Rod Methodology for Negative Flux Rate Trip Plants", WCAP-10297-A (Proprietary)	15.4
"Evaluation of Mispositioned ECCS Valves," WCAP-8966 (Proprietary), September, 1977	6.3
"Westinghouse Emergency Core Cooling System Small Break October 1975 Model," WCAP-8970 (Proprietary) and WCAP-8971 (Non-Proprietary), April, 1977.	15.6
"Failure Mode and Effects Analysis (FMEA) of the Solid State full Length Rod Control System," WCAP-8976, September 1977	4.6
"Westinghouse Emergency Core Cooling System Evaluation Model for Analyzing Large LOCA's During Operation with One Loop Out of Service for Plants Without Loop Isolation Valves," WCAP-9166, February 1978	15.6
"Properties of Fuel and Core Component Materials," WCAP-9179 (Proprietary), September 1977	4.2
"N-16 Power Measuring System," WCAP-9190 (Proprietary) and WCAP-9191 (Non-Proprietary), December 1977	7.2
"Westinghouse ECCS Evaluation Model, February 1978 Version," WCAP-9220-P-A (Proprietary Version), WCAP-9221-P-A (Non-Proprietary Version), February 1978	15.6
"Westinghouse Emergency Core Cooling System Evaluation Model" – Modified October 1975 Version, WCAP-9168 (Proprietary) and WCAP-9169 (Non-Proprietary), September 1977	

TABLE 1.6-2 (HISTORICAL)
DOCUMENTS PROVIDED AS ADDITIONAL REFERENCE INFORMATION

(Sheet 11 of 12)

Report	Reference Section(s)
"Combination of Safe Shutdown Earthquake Loss of Coolant Accident Responses for Faulted Condition Evaluation of Nuclear Power Plants," WCAP-9279 (Non-Proprietary), March 1978	3.9N
"Integrity of the Primary Piping Systems of Westinghouse Nuclear Power Plants During Postulated Seismic Events," WCAP-9283 (Non-Proprietary), March 1978	3.9N
"Technical Bases for Eliminating Large Primary Loop Pipe Ruptures as the Structural Design Bases for Comanche Peak Units 1 and 2" WCAP-10527 (Proprietary) and WCAP-10528 (Non-Proprietary), April 1984	3.6B
"Evaluation Of Surveillance Frequencies And Out Of Service Times For The Reactor Protection Instrumentation System" WCAP-10271-P-A, May 1986	16.2
"Evaluation Of Surveillance Frequencies And Out Of Service Times For The Reactor Protection Instrumentation System" Supplement 1 WCAP-10271, Supplement 1-P-A, May 1986	16.2
"Evaluation Of Surveillance Frequencies And Out Of Service Times For The Engineered Safety Features Actuation System" WCAP-10271, Supplement 2, Revision 1, March 1987	16.2
"Reactor Trip Breaker Maintenance/Surveillance Optimization Program" WCAP-11312, April 1987	16.2
WCAP-8708-PA (Proprietary), WCAP-8709 (Non-Proprietary), "MULTIFLEX A FORTRAN Computer Program for Analyzing Thermal-Hydraulic Structure System Dynamics, "Takeuchi, K., et al., September 1977.	3.6B, 3.9N
WCAP-15029-P-A (Proprietary), WCAP-15030-NP-A, (Non-Proprietary) "Westinghouse Methodology for Evaluating the Acceptability of Baffle-Former-Barrel Bolting Distribution Under Faulted Load Conditions," E.R. Schwirian, et al., January 1999.	3.6B, 3.9N

TABLE 1.6-2 (HISTORICAL)
DOCUMENTS PROVIDED AS ADDITIONAL REFERENCE INFORMATION

(Sheet 12 of 12)

Report	Reference Section(s)
WCAP-15245 (Proprietary), WCAP-15246 (Non-Proprietary), “Control Rod Insertion Following a Cold Leg LBLOCA, D.C. Cook Units 1 and 2,” J.A. Barsic, et al., February 28, 1999.	3.6B, 3.9N
WCAP-11004-P (Proprietary), WCAP-11005 (Non-Proprietary), “Comparison of Data for Beaver Valley Power Station Unit 2 with WCAP-9735 Data, Prepared for NRC Review in Conjunction with Review of WCAP-9735, Docket No. 50-412,” D.R. Bhandari, et al., November 1985.	3.6B, 3.9N
WCAP-11522 (Proprietary), WCAP-11523 (Non-Proprietary), “Response to NRC Questions on the LOCA Hydraulic Forces Analysis of the Beaver Valley Power Station Unit 2, Prepared for NRC Review in Conjunction with Review of WCAP-9735, Docket No. 50-412,” D.C. Garner, et al., June 1987.	3.6B, 3.9N

1.7 ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS

A list of electrical, instrumentation, and control drawings is presented in Tables 1.7-1 and 1.7-2 for Unit 1 and Unit 2, respectively. These drawings include safety-related systems and non-safety-related systems which interact with the safety-related systems.

The fifth digit letter of the drawing number indicates type of drawing. E indicates Electrical drawing and M indicates Mechanical Instrumentation and Control (I&C) drawing.

The sixth digit numeral indicates Unit number. 1 indicates Unit 1 and 2 indicates Unit 2.

The revision and date indicated identify the drawing issue submitted to the NRC staff for review. Additional drawings or later revisions of these drawings will be provided upon request and the latest revision of all drawings are available for review on site.

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 1 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Schematics (Electrical)					
Main steam Reheat & Steam Dump System	2323-E1-0040	0[a] 00, 1-4, 17-28, 44-71, 92-95, 97	H (b)	01/08/80 (b)	
Aux. & Main Steam System	2323-E1-0039	0[a] 00, 41-48, 58-61, 63, 64	J (b)	01/08/80 (b)	
Steam generator feedwater	2323-E1-0038	0[a] 00, 3-11, 13, 15, 17, 19-30, 50, 51, 54-62, 64-99	M (b)	03/15/80 (b)	
Auxiliary feedwater	2323-E1-0037	0[a] 00, 1-16, 19-22, 24-38, 40, 41, 42, 46, 48, 60-68	S (b)	10/07/80 (b)	
Compressed air (Instrument Air)	2323-E1-0048	0[a] 00, 1, 4, 5, 7, 9, 12, 14, 16	I (b)	03/13/80 (b)	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 2 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Process sampling	2323-E1-0045	0[a] 01-14, 16, 17	E (b)	01/07/80 (b)	
Component cooling water	2323-E1-0050	2323_E1_0050 0[a] 00, 1, 3, 4, 6_19, 23, 26-29, 34-41, 43, 44, 47, 49, 50, 51, 53-55	M (b)	03/19/80 (b)	
Containment spray	2323-E1-0049	0[a] 00, 01-14, 18, 21	H (a)	02/07/80 (b)	
Reactor Building Safeguard Building sump pumps	2323-E1-0055	0[a] 00, 7, 8, 16-19	K (a)	01/28/80 (b)	
Demineralized water and reactor plant makeup	2323-E1-0044	0[a] 00, 6-9, 11, 12, 14, 16, 19	M (b)	07/15/80 (b)	
Containment ventilation	2323-E1-0059	0[a] 00, 1, 2, 13-16, 19-54, 58, 59, 62-70 85, 86, 89	L (b)	03/19/80 (b)	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 3 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Safeguard and electrical area ventilation	2323-E1-0053	0[a] 12, 13, 14, 16-55, 62, 63, 65-76, 90, 91	K (b)	03/10/80 (b)	
Auxiliary building and fuel handling area ventilation	2323-E1-0056	0[a] 00, 1, 2, 7-11, 14-53, 55-62, 64-67	I (b)	02/20/80 (b)	
Turbine Gen. Aux. & Misc. Syst.	2323-E1-0060	0[a] 59	N (b)	03/06/80 (b)	
Chilled water ventilation	2323-E1-0054	0[a] 00, 2-5, 7-23	G (b)	03/19/80 (b)	
Primary plant ventilation	2323-E1-0036	0[a] 1-32, 49, 50, 58, 59, 71-96	I (b)	03/19/80 (b)	
Service water and intake structure ventilation	2323-E1-0043	0[a] 00,1-23, 25-27, 40 41	L (b)	02/08/80 (b)	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 4 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
CVCS	2323-E1-0061	0[a] 00,1-6, 9, 10,14-20, 26-31, 36-41, 43-46, 65-69, 74-77, 80, 86	L (b)	03/18/80 (b)	
6.9 kV 3-line Normal Buses	2323-E1-0026	1, 2	2	12/13/79	
6.9 kV 3-line Safeguard Buses	2323-E1-0027	1, 2	3	10/15/79	
Safeguard 6.9-kV switchgear breakers	2323-E1-0031	0[a] 1-66	AA (b)	02/01/80 (b)	
Normal 6.9-kV switchgear breakers	2323-E1-0032	0[a] 1-72	AF (b)	02/28/80 (b)	
Safeguard 480-V switchgear breaker	2323-E1-0033	0[a] 1-42, 47-70	L (b)	01/30/80 (b)	
Normal 480-V switchgear breakers	2323-E1-0034	0[a] 1-43, 45-49, 51-57	K (b)	11/12/79 (b)	
Control Room and office Facil. HVAC	2323-E1-0035	0[a] 00, 1-30, 35-43, 46-53, 58-67, 69, 70, 71, 72-79, 88, 89, 92-97	S (b)	09/29/80 (b)	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 5 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
BOP Misc. System (Part of 2323-E1-0035)	2323-E1-0069	0[a] 1-7	B (b)	09/12/80 (b)	
BOP & NSSS Misc. Systems	2323-E1-0071	0[a] 14, 15, 24, 46-65 67, 68	L (b)	03/20/80 (b)	
Safety injection	2323-E1-0062	0[a] 00, 1, 2, 5-75	H (b)	03/13/80 (b)	
RHR	2323-E1-0063	0[a] 00, 1-9	G (b)	03/27/80 (b)	
RCS	2323-E1-0064	0[a] 00, 1, 2, 9-12, 16, 21-25, 40-43	H (b)	02/26/80 (b)	
Waste Processing	2323-E1-0065	0[a] 00, 13-16	Q (b)	03/03/80 (b)	
6.9-kV Switchgear Miscellaneous	2323-E1-0030	0[a] 9, 10, 19, 21, 22, 24, 26, 30, 35-37, 39, 43, 45, 47, 49, 51-54, 75, 58, 59, 60, 62, 63	AD (b)	08/12/80 (b)	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 6 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
BOP Misc. Systems DC Switchboard Battery Charger Inverter	2323-E1-0066	0[a] 00, 22, 23, 24-29, 66, 78-81, 84, 87	Z (b)	10/03/80 (b)	
Diesel Generator and Monitor Lights	2323-E1-0067	0[a] 1-36, 39, 40, 41-50, 52-65, 77-90	L (b)	09/05/80 (b)	
NSSS Misc. System	2323-E1-0070	0[a] 1-6, 8, 9, 21-26	F (b)	08/05/80 (b)	
Annunciator	2323-E1-0076	1, 2, 3-7, 9, 10, 13-20, 28-40, 42	(b)	(b)	
Ventilation System	2323-E1-0057	0[a] 00, 12-23	F (b)	01/11/80 (b)	
Radiation Monitoring System	2323-E1-0046	0[a] 62-65	G (b)	04/03/80 (b)	
Main 3-Line Meter Diagram	2323-E1-0025	1, 2	2	01/22/80	
Instrumentation and Control Diagram (ICD)					
Legend	2323-M1-2200-01		2	06/08/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 7 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2200-02		4	10/16/79	
	2323-M1-2200-03		2	06/08/79	
	2323-M1-2200-04		2	06/08/79	
	2323-M1-2200-05		2	06/08/79	
Main Steam Reheat and Steam Dump	2323-M1-2202-01		3	04/26/79	
	2323-M1-2202-02		4	12/14/79	
	2323-M1-2202-06		3	04/26/79	
	2323-M1-2202-07		3	04/26/79	
	2323-M1-2202-08		4	12/14/79	
	2323-M1-2202-10		3	04/26/79	
Steam Generator Feedwater	2323-M1-2203-02		3	07/03/79	
	2323-M1-2203-04		4	07/03/79	
	2323-M1-2203-05		3	07/03/79	
	2323-M1-2203-07		2	07/03/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 8 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2203-08		2	07/03/79	
Auxiliary feedwater	2323-M1-2206-01		3	08/27/79	
	2323-M1-2206-02		4	04/22/80	
	2323-M1-2206-03		3	08/27/79	
	2323-M1-2206-04		3	08/27/79	
	2323-M1-2206-05		3	08/27/79	
	2323-M1-2206-06		3	08/27/79	
	2323-M1-2206-07		3	08/27/79	
	2323-M1-2206-08		3	08/27/79	
	2323-M1-2206-09		3	08/27/79	
	2323-M1-2206-10		2	08/27/79	
Diesel Generator Auxiliary	2323-M1-2215-01		4	09/11/79	
	2323-M1-2215-02		3	05/18/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 9 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Compressed air	2323-M1-2216-03		4	11/06/79	
	2323-M1-2216-01		4	11/06/79	
Fire Protection	2323-M1-2225-01		3	07/13/79	Not Class 1
Process Sampling	2323-M1-2228-01		4	10/25/79	
	2323-M1-2228-02		4	10/25/79	
Component cooling water	2323-M1-2229-01		3	04/10/79	
	2323-M1-2229-02		3	04/10/79	
	2323-M1-2229-03		4	10/04/79	
	2323-M1-2229-04		3	04/10/79	
	2323-M1-2229-05		3	04/10/79	
	2323-M1-2229-06		4	03/17/80	
	2323-M1-2230-01		4	03/17/80	
	2323-M1-2230-02		5	03/17/80	
	2323-M1-2231-02		3	04/10/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 10 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2231-03		3	04/10/79	
	2323-M1-2231-05		3	04/10/79	
	2323-M1-2231-07		3	04/10/79	
Containment spray	2323-M1-2232-01		3	04/12/79	
	2323-M1-2232-02		3	04/12/79	
	2323-M1-2232-03		4	12/13/79	
	2323-M1-2232-04		3	04/12/79	
	2323-M1-2232-05		3	04/12/79	
	2323-M1-2232-06		3	04/12/79	
Station Service Water	2323-M1-2233-01		3	05/02/79	
	2323-M1-2233-02		3	05/02/79	
	2323-M1-2233-04		4	12/27/79	Not Class 1
	2323-M1-2233-05		3	05/02/79	
	2323-M1-2233-06		3	05/02/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 11 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2233-07		4	12/27/79	
	2323-M1-2233-08		4	12/27/79	
	2323-M1-2233-09		3	05/02/79	
	2323-M1-2234-01		3	05/02/79	Not Class 1
	2323-M1-2234-02		3	05/02/79	
Spent Fuel Pool Cooling and Cleanup	2323-M1-2235-01		3	02/09/79	Not Class 1
	2323-M1-2235-04		3	02/09/79	
Vents & Drains System	2323-M1-2236-01		3	02/01/79	
Safeguards and Auxiliary Building	2323-M1-2236-02		3	02/01/79	
Vents & Drains System Turbine and Fuel Handling Building	2323-M1-2237-02		3	02/26/79	Not Class 1
Vents & Drains System Containment Building	2323-M1-2238-02		4	02/08/79	
Steam Generator Blowdown Cleanup	2323-M1-2239-01		4	06/01/79	Not Class 1

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 12 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Demineralized and reactor makeup water	2323-M1-2241-03		3	02/08/79	
	2323-M1-2242-01		4	04/08/80	
Turbine Generator	2323-M1-2249-06		2	05/16/79	
Ventilation Containment	2323-M1-2300-04		3	02/07/79	
	2323-M1-2300-02		3	02/07/79	
	2323-M1-2300-01		3	02/07/79	
	2323-M1-2301-01		3	05/04/79	
	2323-M1-2301-02		4	05/04/79	
	2323-M1-2301-03		3	05/04/79	
	2323-M1-2301-04		3	05/04/79	
	2323-M1-2301-05		3	05/04/79	
	2323-M1-2301-06		3	05/04/79	
	2323-M1-2301-07		3	05/04/79	
	2323-M1-2301-08		4	05/04/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 13 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2301-09		3	05/04/79	
	2323-M1-2301-10		3	05/04/79	
	2323-M1-2301-11		2	05/04/79	
Ventilation safeguards and electrical area	2323-M1-2302-05		3	01/26/79	
	2323-M1-2302-06		3	01/26/79	
	2323-M1-2302-07		3	01/26/79	
	2323-M1-2302-08		3	01/26/79	
	2323-M1-2302-09		3	01/26/79	
	2323-M1-2302-10		2	01/26/79	
Ventilation Auxiliary building and fuel handling area	2323-M1-2303-01		3	01/26/79	
	2323-M1-2303-05		3	01/26/79	
	2323-M1-2303-05A		1	01/26/79	
	2323-M1-2303-06		3	01/26/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 14 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Control room air conditioning	2323-M1-2304-01		5	07/13/79	
	2323-M1-2304-02		5	07/13/79	
	2323-M1-2304-03		5	07/13/79	
	2323-M1-2304-04		5	07/13/79	
	2323-M1-2304-05		5	07/13-79	
	2323-M1-2304-06		5	07/13/79	
	2323-M1-2304-07		5	07/13/79	
	2323-M1-2304-08		5	07/13/79	
	2323-M1-2304-12		5	07/13/79	
	2323-M1-2304-13		5	07/13/79	
	2323-M1-2304-14		5	07/13/79	
Ventilation uncontrolled and miscellaneous areas	2323-M1-2305-04		3	03/09/79	
Ventilation chilled water	2323-M1-2307-01		3	02/09/79	
	2323-M1-2307-02		3	02/09/79	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 15 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2307-03		3	02/09/79	Not Class 1
	2323-M1-2307-04		3	02/09/79	
	2323-M1-2307-05		3	02/09/79	
	2323-M1-2307-06		3	02/09/79	
Ventilation equipment primary plant	2323-M1-2309-02		4	05/21/79	
	2323-M1-2309-04		4	05/21/79	
	2323-M1-2309-01		2	05/21/79	
	2323-M1-2309-05		4	05/21/79	Not Class 1
Ventilation safety chilled water	2323-M1-2311-01		3	02/07/79	
	2323-M1-2311-02		3	02/07/79	
	2323-M1-2311-03		2	02/07/79	
Ventilation service water intake structure and misc. buildings	2323-M1-2312-01		3	05/16/79	
Ventilation UPS & Distribution Room	2323-M1-2313-01		CP1	07/20/84	
Reactor Coolant	2323-M1-2250-03		1	12/28/79	Not Class 1

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 16 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M1-2251-07		1	08/11/81	
	2323-M1-2250-07A		1	08/11/81	
Boron Recycle	2323-M1-2259-02		1	11/19/79	Not Class 1
	2323-M1-2259-03		1	11/19/79	Not Class 1
Liquid Waste Processing Drain Channel A	2323-M1-2265-03		2	10/15/79	Not Class 1
Liquid Waste Processing Drain Channel B	2323-M1-2266-03		2	10/15/79	Not Class 1
	2323-M1-2266-04		2	10/15/79	Not Class 1
	2323-M1-2266-05		1	10/15/79	Not Class 1
Liquid Waste Processing Drain Channel C	2323-M1-2267-02		2	10/15/79	Not Class 1
	2323-M1-2267-03		2	10/15/79	Not Class 1
	2323-M1-2267-04		2	10/15/79	Not Class 1
	2323-M1-2267-05		2	10/15/79	Not Class 1

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 17 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Liquid Waste Processing Disposal System	2323-M1-2268-03		2	10/15/79	Not Class 1
	2323-M1-2268-04		2	10/15/79	Not Class 1
	2323-M1-2268-05		2	10/15/79	Not Class 1
	2323-M1-2268-06		2	10/15/79	Not Class 1
Waste Processing System(gas)	2323-M1-2270-05		1	10/15/79	Not Class 1
Others (Equipment Supplier Drawings)					
Process Control Block Diagram	8758D39	1[a] 2-14	3 [b]	--	
NIS Source Range Functional Block Diagram	5655D49	--	8	--	
NIS Intermediate Range Functional Block Diagram	5655D50	--	6	--	
NIS Power Range Functional Block Diagram	5655D51	--	9	--	
NIS Auxiliary Channels Functional Block Diagram	5655D52	--	6	--	

TABLE 1.7-1
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 1

(Sheet 18 of 18)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Solid State Protection System Functional Diagram	7247D05	1-16	(b)	--	
Pressurizer Heater Controller	413918	--	6	--	
Upgrade System Block Diagrams	9554085	1, 2	1	--	

a) Index Sheet

b) For Revision and Date see Individual Sheet

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 1 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Schematics (Electrical)					
Main steam	2323-E2-0040	0[a] 01-04, 18-28 44-47, 53-63	B (b)	03/15/80 (b)	
Steam generator feedwater	2323-E2-0038	Later	Later	Later	
Auxiliary feedwater	2323-E2-0037	Later	Later	Later	
Compressed air (Instrument air)	2323-E2-0048	Later	Later	Later	
Process sampling	2323-E2-0045	0[a] 01-14, 16, 17	A	05/13/80 (b)	
Component cooling water	2323-E2-0050	0[a] 01, 03, 04, 06-19, 23, 26-29, 34-41, 43, 44, 47, 49-51, 53-55	C (b)	09/2/80 (b)	
Containment spray	2323-E2-0049	0[a] 01-14, 18, 21	B (b)	10/31/80 (b)	
Extraction steam	2323-E2-0051	Later	Later	Later	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 2 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Auxiliary & main steam	2323-E2-0039	0[a] 41-48, 58-60, 63, 64	B (b)	07/16/80 (b)	
Safeguards and reactor building sump pumps	2323-E2-0055	0[a] 07, 08, 16-19	B (b)	03/17/80 (b)	
Demineralized water and and reactor plant makeup	2323-E2-0044	0[a] 06, 08, 09, 11, 14	A (b)	06/17/80 (b)	
Containment ventilation	2323-E2-0059	Later	Later	Later	
Safeguard and electrical area ventilation	2323-E2-0053	0[a] 65-76, 90, 91	B (b)	09/09/80 (b)	
Auxiliary building and fuel handling area ventilation	2323-E2-0056	0[a] 55-59	A (b)	07/28/80 (b)	
Turbine generator aux. and misc. system	2323-E2-0060	0[a] 59	D (b)	05/28/80 (b)	
Chilled water ventilation	2323-E2-0054	Later	Later	Later	
6.9-kV 3-line normal buses	2323-E2-0026	Later	Later	Later	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 3 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Service water and intake structure ventilation	2323-E2-0043	0[a] 01, 02, 05, 06, 13-20, 25-27	C (b)	09/15/80 (b)	
Diesel generator and monitor lights	2323-E2-0067	Later	Later	Later	
Vent system	2323-E2-0057	0[a] 12-23	A (b)	09/15/80 (b)	
Annunciator	2323-E2-0076	Later	Later	Later	
Radiation monitoring	2323-E2-0046	Later	Later	Later	
CVCS	2323-E2-0061	0[a] 01-06, 9, 10, 14-20, 26-31, 36-41, 43-46, 65-69, 74-77, 80, 86	B (b)	10/06/80 (b)	
Safety injection	2323-E2-0062	Later	Later	Later	
RHR	2323-E2-0063	0[a] 01-09	A (b)	11/10/80 (b)	
RCS	2323-E2-0064	0[a] 01, 02, 09-12, 16, 21-25	C (b)	10/01/80 (b)	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 4 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Waste processing	2323-E2-0065	0[a] 13-16	B (b)	07/10/80 (b)	
6.9-kV switchgear miscellaneous	2323-E2-0030	Later	Later	Later	
Safeguard 6.9-kV switchgear breakers	2323-E2-0031	Later	Later	Later	
Normal 6.9-kV switchgear breakers	2323-E2-0032	0[a] 05-38, 41-62, 67-72	B (b)	09/03/80 (b)	
480-V Safeguard switchgear breakers	2323-E2-0033	0[a] 01-40, 47-70	A (b)	12/08/80 (b)	
480-V Normal switchgear breakers	2323-E2-0034	Later	Later	Later	
6.9-kV 3-line Safeguard buses	2323-E2-0027	Later	Later	Later	
BOP NSS Misc. Systems	2323-E2-0071	Later	Later	Later	
BOP Misc. System DC Switchboard Battery Charger and Inverter	2323-E2-0066	Later	Later	Later	
NSSS Misc. System	2323-E2-0070	Later	Later	Later	
Main 3-Line Metering Diagram	2323-E1-0025	Later	Later	Later	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 5 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Instrumentation and Control Diagrams (ICD)					
Legend	2323-M2-2200-01		1	07/20/79	
	2323-M2-2200-02		3	10/16/79	
	2323-M2-2200-03		1	07/20/79	
	2323-M2-2200-04		1	07/20/79	
	2323-M2-2200-05		1	07/20/79	
Main Steam Reheat and Steam Dump	2323-M2-2202-01		2	12/21/79	
	2323-M2-2202-02		2	12/21/79	
	2323-M2-2202-06		2	12/21/79	
	2323-M2-2202-07		2	12/21/79	
	2323-M2-2202-08		2	12/21/79	
	2323-M2-2202-10		2	12/21/79	
Steam generator feedwater	2323-M2-2203-02		1	12/28/79	
	2323-M2-2203-04		1	12/28/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 6 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M2-2203-05		1	12/28/79	
	2323-M2-2203-07		1	12/28/79	
	2323-M2-2203-08		1	12/28/79	
Auxiliary feedwater	2323-M2-2206-01		1	08/27/79	
	2323-M2-2206-02		2	04/22/80	
	2323-M2-2206-03		1	08/27/79	
	2323-M2-2206-04		1	08/27/79	
	2323-M2-2206-05		1	08/27/79	
	2323-M2-2206-06		1	08/27/79	
	2323-M2-2206-07		1	08/27/79	
	2323-M2-2206-08		1	08/27/79	
	2323-M2-2206-09		1	08/27/79	
	2323-M2-2206-10		1	08/27/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 7 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Compressed air	2323-M2-2216-03		2	11/06/79	
	2323-M2-2216-01		2	11/06/79	
Condenser Vacuum and Waterbox Priming	2323-M2-2211-02		1	02/28/79	Not Class 1
Auxiliary Steam	2323-M2-2213-01		1	01/25/79	Not Class 1
Diesel Generator Auxiliary	2323-M2-2215-01		2	09/11/79	
	2323-M2-2215-02		1	05/18/79	
Fire protection	2323-M2-2225-01		1	08/30/79	Not Class 1
	2323-M2-2225-02		1	08/30/79	Not Class 1
Process sampling	2323-M2-2228-01		2	10/25/79	
	2323-M2-2228-02		2	10/25/79	
Spent fuel pool cooling and cleanup	2323-M2-2235-01		1	02/09/79	Not Class 1
Component cooling water	2323-M2-2229-01		1	04/10/79	
	2323-M2-2229-02		1	04/10/79	
	2323-M2-2229-03		2	10/04/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 8 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M2-2229-04		1	04/10/79	
	2323-M2-2229-05		1	04/10/79	
	2323-M2-2229-06		2	03/17/80	
	2323-M2-2229-07		2	12/10/79	Not Class 1
	2323-M2-2230-01		2	03/17/80	
	2323-M2-2230-02		3	03/17/80	
	2323-M2-2231-01		1	04/10/79	Not Class 1
	2323-M2-2231-02		1	04/10/79	
	2323-M2-2231-03		1	04/10/79	
	2323-M2-2231-04		1	04/10/79	Not Class 1
	2323-M2-2231-05		1	04/10/79	
	2323-M2-2231-06		1	04/10/79	Not Class 1
	2323-M2-2231-07		1	04/10/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 9 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Containment spray	2323-M2-2232-01		1	04/12/79	
	2323-M2-2232-02		1	04/12/79	
	2323-M2-2232-03		2	12/13/79	
	2323-M2-2232-04		1	04/12/79	
	2323-M2-2232-05		1	04/12/79	
	2323-M2-2232-06		1	04/12/79	
	2323-M2-2232-07		2	07/24/79	Not Class 1
Station Service water	2323-M2-2233-01		1	05/02/79	
	2323-M2-2233-02		1	05/02/79	
	2323-M2-2233-03		1	05/02/79	Not Class 1
	2323-M2-2233-04		2	12/27/79	
	2323-M2-2233-05		1	05/02/79	
	2323-M2-2233-06		1	05/02/79	Not Class 1
	2323-M2-2233-07		1	05/02/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 10 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M2-2234-02		1	05/02/79	
Vents and Drains System Safeguards and Auxiliary building	2323-M2-2236-01		1	02/01/79	
	2323-M2-2236-02		1	02/01/79	
Vents and Drains System Turbine and Fuel Handling Buildings	2323-M2-2237-02		2	03/26/80	Not Class 1
Vents and Drains System Containment building	2323-M2-2238-02		1	02/08/79	
Steam Generator Blowdown Cleanup	2323-M2-2239-01		2	06/01/79	Not Class 1
Demineralized and reactor makeup water	2323-M2-2241-01		1	02/08/79	
	2323-M2-2242-01		2	04/08/80	
Turbine Generator	2323-M2-2249-06		2	03/26/80	
Ventilation Containment	2323-M2-2300-01		1	02/07/79	
	2323-M2-2300-02		1	02/07/79	
	2323-M2-2300-04		1	02/07/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 11 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
	2323-M2-2301-01		1	05/04/79	
	2323-M2-2301-02		1	05/04/79	
	2323-M2-2301-03		1	05/04/79	
	2323-M2-2301-04		1	05/04/79	
	2323-M2-2301-05		1	05/04/79	
	2323-M2-2301-06		1	05/04/79	
	2323-M2-2301-07		1	05/04/79	
	2323-M2-2301-08		1	05/04/79	
Ventilation Safeguards and electrical area	2323-M2-2302-05		1	01/26/79	
	2323-M2-2302-06		1	01/26/79	
	2323-M2-2302-07		1	01/26/79	
	2323-M2-2302-08		1	01/26/79	
	2323-M2-2302-09		1	01/26/79	
	2323-M2-2302-10		1	01/26/79	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 12 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
Ventilation Auxiliary Building and fuel handling area	2323-M2-2303-01		1	01/26/79	
Ventilation Uncontrolled and miscellaneous areas	2323-M2-2305-02		1	03/09/79	
Ventilation Chilled water	2323-M2-2307-01		1	02/09/79	
	2323-M2-2307-02		1	02/09/79	
	2323-M2-2307-03		1	02/09/79	
Ventilation Equipment Primary plant	2323-M2-2309-01		1	05/21/79	Not Class 1
Ventilation Safety chilled water	2323-M2-2311-01		1	02/07/79	
	2323-M2-2311-02		1	02/07/79	
	2323-M2-2311-03		1	02/07/79	
Others (Equipment Supplier Drawings)					
Process Control Block Diagram	8758D39	1[a] 2-41	3 [b]	--	
NIS Source Range Functional Block Diagram	5655D49	--	8	--	

TABLE 1.7-2
CPNPP ELECTRICAL, INSTRUMENTATION, AND CONTROL DRAWINGS LIST FOR UNIT 2

(Sheet 13 of 13)

System or Title	Drawing No.	Sheet No.	Revision	Date	Remarks
NIS Intermediate Range Functional Block Diagram	5655D50	--	6	--	
NIS Power Range Functional Block Diagram	5655D51	--	9	--	
NIS Auxiliary Channels Functional Block Diagram	5655D52	--	6	--	
Solid State Protection System Functional Diagram	7247D05	1-16	(b)	--	
Pressurizer Heater Controller	413918	--	6	--	
Upgrade System Block Diagrams	9554D85	1, 2	1	--	

a) Index Sheet

b) For Revision and Date See Individual Sheet

APPENDIX 1A(N) - DISCUSSION OF REGULATORY GUIDES

This appendix discusses the CPNPP positions on, and compliance with, Division 1 regulatory guides as they apply to the Nuclear Steam Supply System scope of equipment and services. Appendix 1A(B) is provided separately for the balance of plant information.

Identification of specific issues of regulatory guides are provided in the discussions where appropriate.

Regulatory Guide 1.1

Net Positive Suction Head for Emergency Core Cooling and Containment Heat Removal System Pumps

Discussion

CPNPP complies with Safety Guide 1 (11/2/70) as discussed in Section 6.3.2.2.10.

Also refer to Appendix 1A(B) for further discussion.

Regulatory Guide 1.2

Thermal Shock to Reactor Pressure Vessels

Discussion

CPNPP follows all the recommendations of Safety Guide 2 (11/20/70). Regulatory Position C.1 is followed by Westinghouse's own analytical and experimental programs as well as by participation in the Heavy Section Steel Technology (HSST) program at the Oak Ridge National Laboratory.

Analytical techniques have been developed by Westinghouse to perform fracture evaluations of reactor vessels under thermal shock loadings.

Under the HSST program a number of 6 inch thick 39 inch outside diameter steel pressure vessels containing carefully prepared and sharpened surface cracks are being tested. Test conditions include both hydraulic internal pressure loadings and thermal shock loadings. The objective of this program is the validate analytical fracture mechanics techniques and demonstrate quantitatively the margin of safety inherent in reactor pressure vessels.

A number of vessels have been tested under hydraulic pressure loadings, and results have confirmed the validity of fracture analysis techniques. The results and implications of the hydraulic pressure tests are summarized in Oak Ridge National Laboratory report ORNL-TM-5090.

Three thermal shock experiments have been completed and are now being evaluated. Preliminary information indicates that the analytical techniques do agree favorably with experimental results.

Westinghouse is continuing to obtain fracture toughness data for reactor pressure vessel steels through internally funded programs as well as HSST sponsored work.

Fracture toughness testing of irradiated compact tension fracture toughness specimens has been completed. The complete post-irradiation data on 0.394, 2, and 4 inch thick specimens are now available from the HSST program. Both static and dynamic post-irradiation fracture toughness data have been obtained. Evaluation of the data obtained to date on material irradiated to fluences between 2.2 and 4.5 10^{19} n/cm^2 indicates that the reference toughness curve as contained in the American Society of Mechanical Engineers (ASME) Code, Section III, remains a conservative lower bound for toughness values for pressure vessel steels. Another fracture toughness program is now underway. This program involves the irradiation and testing of weld metal used in fabrication of operating pressure vessels of pre-1972 construction. These welds characteristically have high copper contents and low initial Charpy V-notch “shelf” energies. Results of this program are expected in 1977.

Details of progress and results obtained in the HSST program are available in the HSST program semiannual (quarterly beginning in 1974) progress reports, issued by Oak Ridge National Laboratory.

Regulatory Position C.2 is followed inasmuch as no significant changes have been made in approved core or reactor designs.

Regulatory Position C.3 is followed since the vessel design does not preclude the use of an engineering solution to assure adequate recovery of the fracture toughness properties of the vessel material.

If additional margin is needed, the reactor vessel can be annealed at any point in its service life. This solution is already feasible, in principle, and could be performed with the vessel in place.

Regulatory Guide 1.3

Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Boiling Water Reactors

Discussion

This regulatory guide is not applicable to the Comanche Peak Nuclear Power Plant (CPNPP).

Regulatory Guide 1.4

Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Pressurized Water Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.5

Assumptions Used for Evaluating Potential Radiological Consequences of a Steam Line Break Accident for Boiling Water Reactors

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.6

Independence Between Redundant Standby (Onsite) Power Sources and Between Their Distribution Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.7

Control of Combustible Gas Concentrations in Containment Following a Loss of Coolant Accident

Discussion

Based on a revision to 10CFR50.44, Regulatory Guide 1.7 no longer applies to CPNPP. See Section 6.2.5.

Regulatory Guide 1.8

Personnel Selection and Training

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.9

Selection of Diesel Generator Set Capacity for Standby Power Supplies

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.10

Mechanical (Cadweld) Splices in Reinforcing Bars of Category I Concrete Structures

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.11

Instrument Lines Penetrating Primary Reactor Containment

Discussion

CPNPP meets the recommendations of Safety Guide 11 (3/10/71) in accordance with the comments of Section 7.3.1.1.2.

Also refer to Appendix 1A(B) for further discussion.

Regulatory Guide 1.12

Instrumentation for Earthquakes

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.13

Spent Fuel Storage Facility Design Basis

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.14

Reactor Coolant Pump Flywheel Integrity

Discussion

The CPNPP design follows the recommendations of Revision 1 (8/75) of Regulatory Guide 1.14 except for the following:

1. Post-spin inspection

   Westinghouse has shown in Reference [4] that the flywheel would not fail at 290 percent of normal speed for a flywheel flaw of 1.15 inches or less in length. Results for a double ended guillotine break at the pump discharge with full separation of pipe ends assumed, show the maximum overspeed was to be less than 100 percent of normal speed. The maximum overspeed was calculated in Reference [4] to be about 280 percent of normal speed for the same postulated break, and an assumed instantaneous loss of power to the reactor coolant pump. In comparison with the overspeed presented above, the flywheel is tested at 125 percent of normal speed. Thus, the flywheel could withstand a speed up

to 2.3 times greater than the flywheel spin test speed 125 percent provided that no flaws greater than 1.15 inches are present. If the maximum speed were 125 percent of normal speed or less, the critical flaw size for failure would exceed 6 inches in length.

Nondestructive tests and critical dimension examinations are all performed before the spin tests. The inspection methods employed (described in Reference [4] provide assurance that flaws significantly smaller than the critical flaw size of 1.15 inches for 290 percent of normal speed would be detected. Flaws in the flywheel will be recorded in the pre-spin inspection program (see Reference [4]). Flaw growth attributable to the spin test (i.e., from a single reversal of stress, up to speed and back), under the most adverse conditions, is about three orders of magnitude smaller than what nondestructive inspection techniques are capable of detecting. For these reasons, Westinghouse performs no post- spin inspections and believes that pre-spin test inspections are adequate.

2. Interference fit stresses and excessive deformation

Much of Revision 1 to Regulatory Guide 1.14 deals with stresses in the flywheel resulting from the interference fit between the flywheel and the shaft. Because Westinghouse's design specifies a light interference fit between the flywheel and the shaft; at zero speed, the hoop stresses and radial stresses at the flywheel bore are negligible. Centering of the flywheel relative to the shaft is accomplished by means of keys and/or centering devices attached to the shaft, and at normal speed, the flywheel is not in contact with the shaft in the sense intended by Revision 1. Hence, the definition of "Excessive Deformation," as defined in Revision 1 of Regulatory Guide 1.14, is not applicable to the Westinghouse design since the enlargement of the bore and subsequent partial separation of the flywheel from the shaft does not cause unbalance of the flywheel. Extensive Westinghouse experience with reactor coolant pump flywheels installed in this fashion has verified the adequacy of the design.

Westinghouse's position is that combined primary stress levels, as defined in Revision 0 of Safety Guide 14 (C.2.a and C.2.c), are both conservative and proven and that no changes to these stress levels are necessary. Westinghouse designs to these stress limits and thus does not have permanent distortion of the flywheel bore at normal or spin test conditions.

3. Discussion B, cross rolling ratio of 1 to 3

Westinghouse's position is that specification of a cross rolling ratio is necessary since past evaluations have shown that ASME SA-533 Grade B Class 1 materials produced without this requirement have suitable toughness for typical flywheel applications. Proper material selection and specification of minimum material properties in the transverse direction adequately ensure flywheel integrity. An attempt to gain isotropy in the flywheel material by means of cross rolling is unnecessary since adequate margins of safety are provided by both flywheel material selection (ASME SA-533 Grade B Class 1) and by specifying minimum yield and tensile levels and toughness test values taken in the direction perpendicular to the maximum working direction of the material.

4. Regulatory Position C.1.a, relative to vacuum-melting and degassing process of the electroslag process

   The requirements for vacuum melting and degassing process or the electroslag process are not essential in meeting the balance of the Regulatory Position nor do they, in themselves, ensure compliance with the overall Regulatory Position. The initial Safety Guide 14 (10/27/71) stated that the "flywheel material should be produced by process that minimized flaws in the material and improves its fracture toughness properties." This is accomplished by using SA-533 material including vacuum treatment.

5. Regulatory Position C.2.b

   Westinghouse suggests that this paragraph be reworded as follows in order to remove the ambiguity of reference to an undefined overspeed transient.

   "Design speed should be 125 percent of normal speed or the speed to which the pump motor might be electrically driven by station turbine generator during anticipated transients, whichever is greater. Normal speed is defined as the synchronous speed of the alternating current drive motor at 60 hertz."

6. Inspection

   The reactor coolant pump flywheel inspection is in accordance with Technical Specification 5.5.7.

Regulatory Guide 1.15

Testing of Reinforcing Bars for Category I Concrete Structures

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.16

Reporting of Operating Information - Appendix A Technical Specifications

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.17

Protection of Nuclear Power Plants Against Industrial Sabotage

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.18

Structural Acceptance Test for Concrete Primary Reactor Containments

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.19

Nondestructive Examination of Primary Containment Liner Welds

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.20

Comprehensive Vibration Assessment Program for Reactor Internals During Preoperational and Initial Startup Testing

Discussion

The CPNPP position on Revision 2 (5/76) of Regulatory Guide 1.20 is discussed in Section 3.9N.2.4.

Regulatory Guide 1.21

Measuring, Evaluating, and Reporting Radioactivity in Solid Wastes and Releases of Radioactive Materials in Liquid and Gaseous Effluents from Light-Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.22

Periodic Testing of Protection System Actuation Functions

Discussion

CPNPP meets the recommendations of Safety Guide 22 (2/17/72) in accordance with the comments of Section 7.1.2.5.

Regulatory Guide 1.23

Onsite Meteorological Program

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.24

Assumptions Used for Evaluating the Potential Radiological Consequences of a Pressurized Water Reactor Radioactive Gas Storage Tank Failure

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.25

Assumptions Used for Evaluating the Potential Radiological Consequences of a Fuel Handling Accident in the Fuel Handling and Storage Facility for Boiling and Pressurized Water Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.26

Quality Group Classifications and Standards for Water, Steam and Radioactive-Waste-Containing Components of Nuclear Power Plants

Discussion

Quality group classification of fluid system equipment of the Westinghouse scope of supply for the CPNPP is as described in Section 3.2.

Nuclear Steam Supply System fluid system components important to safety are classified in accordance with the August 1970 Draft of American National Standard (ANSI) N18.2, “Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plant,” except for the deviation described in Section 3.2.2. In addition, components of the accumulator subsystem and the refueling water subsystem are placed in a higher classification. The effect of this higher classification is to produce the same quality levels for all components as is achieved by the application of the 1973 version of N18.2, as finally accepted by ANSI.

Classification by this means is an alternate acceptable method of meeting the intent of Revision 3 (2/76) of Regulatory Guide 1.26. The method of classifying is in accordance with NRC stated policy on regulatory guides that “methods and solutions different from those set out in the guides will be acceptable if they provide a basis for the findings requisite to the issuance or continuance of a permit or license by the Commission.”

Also refer to Appendix 1A(B) for further discussion.

Regulatory Guide 1.27

Ultimate Heat Sink for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.28

Quality Assurance Program Requirements (Design and Construction)

Discussion

The Westinghouse quality assurance program, including the Westinghouse position on Regulatory Guide 1.28, is presented in WCAP-8370 [5].

Also refer to Appendix 1A(B) for further discussion.

Regulatory Guide 1.29

Seismic Design Classification

Discussion

Seismic classification of fluid system equipment of Westinghouse scope of supply for the CPNPP is as described in Section 3.2.

The classification of components by safety class provides the means of establishing applicable aseismic design requirements of both components and system. At the time the CPNPP was designed, duplication by special seismic classification was unnecessary since ANSI N18.2, "Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants" was considered to establish seismic design requirements of systems having components classified as Safety Class 1, Safety Class 2, or Safety Class 3. Safety class fluid system components are designed to remain functional in the event of occurrence of the Design Basis Earthquake, defined in Section 2.1.5.4 of ANSI N18.2-1973 but now identified as the Safe Shutdown Earthquake by the NRC. The Reactor Protection System includes similar aseismic design features.

Classification by this means is an alternate acceptable method of meeting the intent of Revision 2 (2/76) of Regulatory Guide 1.29. The method of classifying is in accordance with NRC stated policy on regulatory guides that "methods and solutions different from those set out in the guides will be acceptable if they provide a basis for the findings requisite to the issuance or continuance of a permit or license by the Commission."

Also refer to Appendix 1A(B).

Regulatory Guide 1.30

Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment

Discussion

The design criteria Westinghouse applies to safety-related instrumentation and controls are consistent with ANSI N45.2.4-1972 (IEEE Standard 336-1971), “Installation, Inspection, and Testing Requirements of Instrumentation and Electrical Equipment During the Construction of Nuclear Power Generating Stations.” The major actions is complying with this standard fall within the scope of plant construction. Westinghouse functions in design and procurement of instrumentation and electrical equipment are supportive to the later fulfillment of the standard at the time of construction.

For those activities performed after September 1, 1972, Westinghouse follows the guidance of Safety Guide 30 (8/11/72). For those activities performed prior to September 1, 1972, the quality assurance procedures employed in the design of the CPNPP may vary in detail from the position of the regulatory guide, but meet its essential requirements.

Also refer to Appendix 1A(B).

Regulatory Guide 1.31

Control of Stainless Steel Welding

Discussion

The Westinghouse position concerning the control of delta ferrite in stainless steel welding is discussed in Section 5.2.3. The Westinghouse production weld verification program, as described in Reference [6], was approved as a satisfactory substitute for conformance with the NRC Interim Position on Revision 3 (4/78) of Regulatory Guide 1.31. The results of the verification program have been summarized and documented in Reference [7].

The welding of austenitic stainless steel is controlled to mitigate the occurrence of microfissuring or hot cracking in the weld. Although published data and experience have not confirmed that fissuring is detrimental to the quality of the weld, it is recognized that such fissuring is undesirable in a general sense. Also, it has been well documented in the technical literature that the presence of delta ferrite is one of the mechanisms for reducing the susceptibility of stainless steel welds to hot cracking. However, there are insufficient data to specify a minimum delta ferrite level below which the material will be prone to hot cracking. It is assumed that such a minimum lies somewhere between 0 and 3 percent delta ferrite.

The scope of these controls discussed herein encompasses welding processes used to join stainless steel parts in components designed, fabricated or stamped in accordance with ASME B&PV Code, Section III Class 1, 2, 3 and CS components. Delta ferrite control is appropriate for the above welding requirements except in the following cases where no filler metal is used (for example, in electron beam welding and in autogenous gas shielded tungsten arc welding), where stainless steel filler metal is used for weld metal cladding, explosive welding, and welding using fully austenitic welding materials.

The fabrication and installation specifications require welding procedure and welder qualification in accordance with Section III, and include the delta ferrite determinations for the austenitic stainless steel welding materials that are used for welding qualification testing and for production processing. Specifically, the undiluted weld deposits of the “starting” welding materials are

required to contain a minimum of 5 percent delta ferrite[a] as determined by chemical analysis and calculation using the appropriate weld metal constitution diagrams in Section III. When new welding procedure qualification tests are evaluated for these applications, including repair welding of raw materials, they are performed in accordance with Section III and Section IX. The results of all the destructive and non-destructive tests are reported in the procedure qualification record in addition to the information required by Section III.

The "starting" welding materials used for fabrication and installation welds of austenitic stainless steel materials and components meet the requirements of Section III. The austenitic stainless steel welding material conforms to ASME weld metal analysis A-7 (designated A-8 in the 1974 Edition of the ASME code), type 308 or 308L for all applications. Bare weld filler metal, including consumable inserts, used in inert gas welding processes conform to ASME SFA-5.9, and are procured to contain not less than 5 percent delta ferrite in the deposit according to Section III. Weld filler metal materials used in flux shielded welding processes conform to ASME SFA-5.4 or SFA-5.9 and are procured in a wire-flux combination to be capable of providing not less than 5 percent delta ferrite in the deposit according to Section III. Welding materials are tested using the welding energy inputs to be employed in production welding.

Combinations of approved heats and lots of "starting" welding materials are used for all welding processes. The welding quality assurance program includes identification and control of welding material by lots and heats as appropriate. All of the weld processing is monitored according to approved inspection programs which include review of "starting" materials, qualification records and welding parameters. Welding systems are also subject to quality assurance audit including calibration of gages and instruments, identification of "starting" and completed materials, welder and procedure qualifications, availability and use of approved welding and heat treating procedures, and documentary evidence of compliance with materials, welding parameters, and inspection requirements. Fabrication and installation welds are inspected using non-destructive examination methods according to Section III rules.

To assure the reliability of these controls, Westinghouse has performed a delta ferrite verification program, described in Reference [6]. The verification program has been approved as a valid approach to verify the Westinghouse hypothesis and is considered an acceptable alternative for conformance with the Interim Position on Regulatory Guide 1.31. The Regulatory Staff's acceptance letter and topical report evaluation were received on December 30, 1974. The program results, which support the hypothesis presented in Reference [6], are summarized in Reference [7].

Welds made in accordance with the criteria discussed herein have continually resulted in sound production welds, which are free from detrimental fissuring and consistently conform to Section III non-destructive acceptance standards.

Also refer to Appendix 1A(B).

a. The equivalent Ferrite Number may be substituted for percent delta ferrite.

Regulatory Guide 1.32

Criteria for Safety-Related Electric Power Systems from Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.33

Quality Assurance Program Requirements (Operation)

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.34

Control of Electroslag Weld Properties

Discussion

Where electroslag welding is used in fabricating nuclear plant components the Westinghouse procurement procedure requires vendors to meet the guidelines of Regulatory Guide 1.34 (12/28/72).

Regulatory Guide 1.35

Inservice Inspection of Ungrouted Tendons in Prestressed Concrete Containment Structures

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.36

Nonmetallic Thermal Insulation for Austenitic Stainless Steel

Discussion

The Westinghouse practice meets the recommendations of Regulatory Guide 1.36 (2/23/73) but is more stringent in several respects as discussed below.

The nonmetallic thermal insulation used on the reactor coolant pressure boundary is specified to be made of compounded materials which yield low leachable chloride and/or fluoride concentrations. The compounded materials in the form of blocks, boards, cloths, tapes, adhesives, cements, etc., are silicated to provide protection of austenitic stainless steels against stress corrosion which may result from accidental wetting of the insulation by spillage, minor leakage or other contamination from the environmental atmosphere. Each lot of insulation materials provides a compatible combination for the reactor coolant pressure boundary.

The tests for qualification specified by this regulatory guide (ASTM- C-692-71 or RDT M12-IT) allow use of the tested insulation materials if no more than one of the metallic test samples crack. Westinghouse rejects the tested insulation material if any of the test samples crack.

The Westinghouse procedure is more specific than the procedures suggested by this regulatory guide, in that the Westinghouse specification requires determination of leachable chloride and fluoride ions from a sample of the insulating material. The procedures in this regulatory guide (ASTM-D-512 and ASTM-D-1179) do not differentiate between leachable and unleachable halogen ions.

In addition, Westinghouse experience indicates that only one of the three methods allowed under ASTM-D-512 and ASTM-D-1179 for chloride and fluoride analysis is sufficiently accurate for reactor applications. This is the “referee” method, which is used by Westinghouse. These requirements are defined in Westinghouse Process Specification PS-83336KA.

Also refer to Appendix 1A(B).

Regulatory Guide 1.37

Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants

Discussion

This guide (dated March 16, 1973) endorses ANSI N45.2.1-1973, which applies to cleaning procedures at the construction site and is therefore not in the Westinghouse scope. Westinghouse procurement orders apply cleaning requirements during fabrication and packaging of safety-related components so that NSSS equipment is delivered to the site in a properly cleaned condition. A Westinghouse process specification provides detailed cleaning requirements for equipment manufacturers, and is included as a procurement requirement, where appropriate.

Also refer to Appendix 1A(B).

Regulatory Guide 1.38

Quality Assurance Requirements for Packaging, Shipping, Receiving, Storage, and Handling of Items for Water-Cooled Nuclear Power Plants

Discussion

For the CPNPP whose application was docketed prior to April 1, 1973, the quality assurance procedures employed in the design and construction phases may vary in detail from the current position of Revision 1 (10/76) of the regulatory guide but meet its essential requirements in that they follow good business practices as defined in the applicable Westinghouse process specifications.

For activities initiated after January 1, 1975, for the CPNPP, Reference [5] is applicable. This plan follows the guidance of ANSI N45.2.2-1978 (Unit 1) ANSI N45.2.2-1972 (Unit 2) (which is recognized by Regulatory Guide 1.38) in the design, procurement, fabrication and shipment of

safety-related NSSS equipment. Measures are applied, as appropriate, to apply packaging requirements to procurement orders, to review supplier packaging procedures, to apply proper cleaning requirements, to apply proper marking and identification, to provide protection to equipment from physical or weather damage, to apply special handling precautions and to define storage requirements. A Westinghouse process specification incorporates detailed packaging and handling requirements for equipment manufacturers, and is included as a procurement requirement where appropriate.

Also refer to Appendix 1A(B).

Regulatory Guide 1.39

Housekeeping Requirements for Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.40

Qualification Tests of Continuous-Duty Motors Installed Inside the Containment of Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.41

Preoperational Testing of Redundant On-Site Electric Power Systems to Verify Proper Load Group Assignments

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.42

Interim Licensing Policy on As Low As Practicable for Gaseous Radioiodine Releases from Light-Water-Cooled Nuclear Power Reactors

Discussion

This regulatory guide was withdrawn March 22, 1976.

Regulatory Guide 1.43

Control of Stainless Steel Weld Cladding of Low-Alloy Steel Components

Discussion

The reactor vessel bottom head (Unit 1) and heads (Unit 2), shell courses, and torus sections were constructed of SA-533, Grade B, Class 1, plate material made to a fine grain practice. This material was clad by the shielded metal arc, one-wire submerged arc, and the 4-inch strip submerged arc processes; the shielded metal arc and one-wire submerged arc processes are considered to be low heat input methods of cladding.

Although the plate material is SA-533, Grade B, Class 1, made to fine grain practice and not subject to qualification restrictions by Regulatory Guide 1.43 (5/73), the 4-inch strip clad process was qualified.

1. Unit 1

   The shielded metal arc and the one-wire submerged arc processes (both considered low heat input) were used to clad the vessel flanges and the closure head, which was forged of SA-508, Class 3. These processes were also used to clad the primary nozzles which were constructed of SA-508, Class 2, forging material.

2. Unit 2

   The shielded metal arc and the one-wire submerged arc processes (both considered low heat input) were used to clad the closure head and vessel flanges and the primary nozzles which were constructed of SA-508, Class 2, forging material.

The reactor vessel fabricator monitors and records the weld parameters to verify conformance with the parameters established by the procedure qualification as recommended by Regulatory Position C.3.

Consequently, the reactor vessel is in full compliance with Regulatory Guide 1.43.

The Unit 1 steam generator parts which are clad are constructed of SA-508, Grade 3a steel. The Unit 2 steam generator and pressurizer parts which are clad are constructed of SA-533, Grade A, Class 2, and SA-508, Class 2, steels. These materials are made to fine grain practices and all welding is done with low heat input techniques.

Regulatory Guide 1.44

Control of the Use of Sensitized Stainless Steel

Discussion

The Westinghouse position on Regulatory Guide 1.44 (5/73) is discussed in part in Section 5.2.3.4.

Westinghouse compliance with the separate positions of the guide are as follows:

The use of processing, packaging and shipping controls, and preoperational cleaning to preclude adverse effects of exposure to contaminants on all stainless steel materials are in accordance with Regulatory Position C.1.

Austenitic stainless steel starting materials are utilized in the final heat treated conditions required by the respective ASME Code, Section II, material specification for the particular type or grade of alloy in accordance with Regulatory Position C.2.

The Westinghouse position concerning material inspection programs and Regulatory Position C.3 is discussed in Section 5.2.3.4.

Westinghouse meets the intent of Regulatory Position C.4 in the manner discussed in detail in Section 5.2.3.4. However, the guide's exception (a) to Regulatory Position C.4 established 200°F as the upper limit for dissolved oxygen concentration above 0.1 parts per million (ppm). This temperature limit should be increased to 250°F which provides for much quicker reduction of the oxygen concentration by reaction with hydrazine. Startup operations provide for adding hydrazine after the temperature is at about 225°F. Oxygen scavenging at this temperature is rapid and complete; below 200°F, considerable time can be encountered in the oxygen removal operations.

The Westinghouse position is that the significant chemistry control associated with the above limits on oxygen is the control of chloride and fluoride ion concentrations at less than 0.15 ppm each at all times. The guide's exception (b) to Regulatory Position C.4 is covered in the discussion of delta ferrite in Section 5.2.3.4.

Westinghouse complies with Regulatory Position C.5 in the manner discussed in Section 5.2.3.4. The guide's exception (a) to Regulatory Position C.5 is covered in Section 5.2.3.4 on delta ferrite.

Westinghouse complies with Regulatory Position C.6 in the manner discussed in Section 5.2.3.4.

Also refer to Appendix 1A(B).

Regulatory Guide 1.45

Reactor Coolant Pressure Boundary Leakage Detection Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.46

Protection Against Pipe Whip Inside Containment

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.47

Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.48

Design Limits and Loading Combinations for Seismic Category I Fluid System Components

Discussion

Westinghouse supplied components are designed using the stress limits and loading combinations presented in Sections 3.9N.1 and 5.2 for Code Class 1 components and in Section 3.9N.3 for Code Class 2 and 3 components. The conservatism in these limits and the associated ASME design requirements precludes any component structural failure.

The operability of active Code Class 1, 2 and 3 valves and active Code Class 2 and 3 pumps (there are no active Class 1 pumps) will be verified by methods detailed in Sections 3.9N.1 and 5.2 for Code Class 1 components and in Section 3.9N.3 for Code Class 2 and 3 components.

The use of the above stated methods provides an acceptable alternate method to meeting the guidance of this regulatory guide dated May 1973.

Also refer to Appendix 1A(B).

Regulatory Guide 1.49

Power Levels of Nuclear Power Plants

Discussion

The CPNPP meets the recommendations of Revision 1 (12/73) of Regulatory Guide 1.49 since the projected initial power level is less than 3800 megawatts thermal (MWt) and analyses and evaluation are made at assumed core power levels less than the levels in this regulatory guide.

Regulatory Guide 1.50

Control of Preheat Temperature for Welding of Low-Alloy Steel

Discussion

Westinghouse considers that this regulatory guide (dated May 1973) applies to ASME Code, Section III, Class 1 components.

The Westinghouse practice for Class 1 components is in agreement with the recommendations of Regulatory Guide 1.50 except for Regulatory Positions C.1.b and C.1.c. For Class 2 and 3 components, Westinghouse does not apply any of Regulatory Guide 1.50 recommendations.

In the case of Regulatory Position C.1.b, the welding procedures are qualified within the preheat temperature ranges required by Section IX of the ASME Code. Westinghouse experience has shown excellent quality of welds using the ASME qualification procedures.

In the case of Regulatory Position C.2, the Westinghouse position is that this guide recommendation is both unnecessary and impractical. Code acceptance low-alloy steel welds have been and are being made under present Westinghouse specified procedures. It is not necessary to maintain the preheat temperature until a post-weld treatment has been performed as recommended by this regulatory guide, in the case of large components. In the case of reactor vessel main structural welds, the practice of maintaining preheat until the intermediate or final post-weld heat treatment has been followed by Westinghouse. In either case, the welds have shown high integrity. Westinghouse practices are documented in Reference [8], which has been accepted by the NRC.

Also refer to Appendix 1A(B).

Regulatory Guide 1.51

Inservice Inspection of ASME Code Class 2 and 3 Nuclear Power Plant Components

Discussion

This regulatory guide was withdrawn March 22, 1976.

Regulatory Guide 1.52

Design, Testing, and Maintenance Criteria for Engineered-Safety- Feature Atmosphere Cleanup System Air Filtration and Adsorption Units of Light-Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.53

Application of the Single-Failure Criterion to Nuclear Power Plant Protection Systems

Discussion

Westinghouse furnished systems meet the recommendations of this regulatory guide (dated June 1973) in accordance with the comments of Section 7.1.2.7.

Also refer to Appendix 1A(B).

Regulatory Guide 1.54

Quality Assurance Requirements for Protective Coatings Applied to Water-Cooled Nuclear Power Plants

Discussion

For the CPNPP, Westinghouse employs process specifications and the Westinghouse Quality Assurance Program, including quality assurance surveillance and auditing, to provide adequate confidence that coating work within Westinghouse scope will perform satisfactorily in service.

This regulatory guide (dated June 1973) recognizes ANSI N101.4-1972. Westinghouse uses alternate methods which have employed process specifications for protective coatings for several years. The Westinghouse process specifications cover the application of paint systems to structures and components in the reactor Containment which may come in contact with the fission product removal and/or Containment cooling spray. Regulatory Guide 1.54 provides that equipment covered with fabricated metal insulation does not require protective coatings, and thus, may not require the use of this Westinghouse process specification, e.g., steam generators. Applicable coating requirements are included in procurement orders.

Also refer to Appendix 1A(B).

Regulatory Guide 1.55

Concrete Placement in Category I Structures

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.56

Maintenance of Water Purity in Boiling Water Reactors

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.57

Design Limits and Loading Combinations for Metal Primary Reactor Containment System Components

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.58

Qualification of Nuclear Power Plant Inspection, Examination, and Testing Personnel

Discussion

For work within Westinghouse scope performed for the CPNPP, before January 1, 1975, the qualification of inspection, examination, and testing personnel was controlled by standard industry practice with the exception of nondestructive examination personnel, who were qualified in accordance with the requirements of SNT-TC-1A. The qualification of inspection, examination, and testing personnel was accomplished locally by Westinghouse suppliers through on-the-job training.

For activities initiated on the CPNPP after January 1, 1975, Westinghouse follows the guidance of this regulatory guide (dated August 1973) as defined in Reference [5] and as described below.

This guide recognizes ANSI N45.2.6-1973. Westinghouse policies and procedures for qualification of personnel engaged in inspection, examination and testing activities follow the guidance of this standard. Westinghouse uses demonstrated capability of performing the assigned tasks to predetermine standards or levels of proficiency as the primary basis for evaluating and certifying the personnel as an acceptable alternative to the specific years of education/experience.

Westinghouse applies the guidance of this standard to personnel who perform inspection, examination and testing activities including surveillance of these activities for safety-related equipment, materials and services in the Nuclear Energy Systems Division and at outside suppliers.

Also refer to Appendix 1A(B).

Regulatory Guide 1.59

Design Basis Floods for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.60

Design Response Spectra for Seismic Design of Nuclear Power Plants

Discussion

The design response spectra of Revision 1 (12/73) of Regulatory Guide 1.60 are acceptable to Westinghouse as long as the damping values discussed below under Regulatory Guide 1.61 are acceptable.

Also refer to Appendix 1A(B).

Regulatory Guide 1.61

Damping Values for Seismic Design of Nuclear Power Plants

Discussion

The damping values listed in Regulatory Guide 1.61 (dated October 1973) are acceptable to Westinghouse with the single exception of the large piping systems faulted condition value of 3 percent critical. Higher damping values when justified by documented test data have been provided for in Regulatory Position C.2. A conservative value of 4 percent critical has therefore been justified by testing for the Westinghouse reactor coolant loop configuration in Reference [9] and has been approved by the NRC Staff.

For piping systems analyzed by the response spectrum method, ASME Code Case N-411 damping values may be used in lieu of the damping values in this Regulatory Guide.

Also refer to Appendix 1A(B).

Regulatory Guide 1.62

Manual Initiation of Protective Actions

Discussion

Westinghouse furnished systems meet the recommendations of this regulatory guide (dated October 1973) in accordance with the comments of Section 7.3.2.2.7.

Regulatory Guide 1.63

Electric Penetration Assemblies in Containment Structures for Light- Water-Cooled Nuclear Power Plants

Discussion

The Reg. guide as well as IEEE Std. 317 are not applicable to the NSSS Scope of Supply.

Regulatory Guide 1.64

Quality Assurance Requirements for the Design of Nuclear Power Plants

Discussion

The Westinghouse commitment, as described in Chapter 17 of the CPNPP Preliminary Safety Analysis Report, was designed to meet the requirements of 10CFR50, Appendix B, which is an acceptable method for following the guidance of ANSI N45.2. For activity initiated by Westinghouse after January 1, 1975 for the CPNPP, the control measures described in Reference [5] are applicable. Reference [5] commits Westinghouse to meet the requirements of 10CFR50, Appendix B and follow the guidance of the NRC. Specifically, Westinghouse follows the guidance of ANSI/ASME NQA-1 for Unit 1 and ANSI N45.2.11-1974 for Unit 2, which is recognized by this standard in the design work for safety-related NSSS equipment. Westinghouse utilizes different but equivalent forms than those shown as samples in the standard. In regard to design verification, it is performed by individuals or groups other than those who performed the original design. However, in exceptional cases, when the designer's supervisor maintains a detailed expertise on the design, the supervisor will perform the design verification function, and document the reason for this action.

Also refer to Appendix 1A(B).

Regulatory Guide 1.65

Materials and Inspections for Reactor Vessel Closure Studs

Discussion

Westinghouse is in agreement with Regulatory Guide 1.65 (dated 10/73) except for material and tensile strength guidelines.

Westinghouse has specified both 45 ft-lb and 25 mils lateral expansion for control of fracture toughness determined by Charpy-V testing, required by the ASME Code, Section III, Summer 1973 Addenda and 10CFR50, Appendix G (July 17, 1973, Paragraph IV.A.4). These toughness requirements assure optimization of the stud bolt material tempering operation with the accompanying reduction of the tensile strength level when compared with previous ASME Code requirements.

The specification of both impact and maximum tensile strength as stated in the guide results in unnecessary hardship in procurement of material without any additional improvement in quality.

The closure stud bolting material is procured to a minimum yield strength of 130,000 psi and a minimum tensile strength of 145,000 psi. This strength level is compatible with the fracture toughness requirements of 10CFR50, Appendix G (July 17, 1973, Paragraph I.C), although higher strength level bolting materials are permitted by the code. Stress corrosion has not been observed in reactor vessel closure stud bolting manufactured from material of this strength level. Accelerated stress corrosion test data do exist for materials of 170,000 psi minimum yield strength exposed to marine water environments stressed to 75 percent of the yield strength (given in Reference 2 of the guide). These data are not considered applicable to Westinghouse reactor vessel closure stud bolting because of the specified yield strength differences and a less severe environment; this has been demonstrated by years of satisfactory service experience.

The ASME Code requirement for toughness for reactor vessel bolting has precluded the guide's additional recommendation for tensile strength limitation, since to obtain the required toughness levels, the tensile strength levels are reduced. Prior to 1972, the ASME Code required a 35 ft-lb toughness level which provided maximum tensile strength levels ranging from approximately 155 to 178 kpsi (Westinghouse review of limited data - 25 heats). After publication of the Summer 1973 Addenda to the ASME Code and 10CFR50, Appendix G, wherein the toughness requirements were modified to 45 ft-lb with 25 mils lateral expansion, all bolt material data reviewed on Westinghouse plants showed tensile strengths of less than 170 kpsi.

Additional protection against the possibility of incurring corrosion effects is assured by:

a. Decrease in level of tensile strength comparable with the requirement of fracture toughness as described above.

b. Design of the reactor vessel studs, nuts, and washers, allowing them to be completely removed during each refueling permitting visual and/or nondestructive inspection in parallel with refueling operations to assess protection against corrosion, as part of the inservice inspection program.

c. Design of the reactor vessel studs, nuts, and washers, providing protection against corrosion by allowing them to be completely removed during each refueling and placed in storage racks on the Containment operating deck, as required by Westinghouse refueling procedures. The stud holes in the reactor flange are sealed

with special plugs before removing the reactor closure. Thus, the bolting materials and stud holes are never exposed to the borated refueling cavity water.

Regulatory Guide 1.66

Nondestructive Examination of Tubular Products

Discussion

The guide (dated October 1973) states that "Nondestructive examination applied to tubular products used for components of the reactor coolant pressure boundary and other safety-related systems...should be capable of detecting unacceptable defects regardless of defect shape, orientation, or location in the product." To accomplish this, the regulatory guide suggests the addition of angle beam scanning in two axial directions.

Westinghouse considers the regulatory position regarding angle beam scanning in the axial direction as technically unnecessary since any flaws which might be developed by the processes employed in tubular product manufacture are invariably oriented in the axial direction, and the probability of developing metallurgical flaws of other than axial orientation is virtually nil. Flaws of transverse or circumferential orientation which might be developed would normally be mechanically induced surface defects which should be detected by surface nondestructive examination procedures.

The primary pressure boundary and safety-related tubular products within the Westinghouse scope of supply and the nondestructive examinations applied are tabulated in Table 1A(N)-1. In all cases the volumetric nondestructive examination is designed to detect the flaws inherent to the manufacturing process or processes employed. In those few cases where compliance with the guide is not indicated, Westinghouse believes that the nondestructive examinations performed in the normal procurement of the tubular products covered by the guide achieve the same purpose as the guide.

The hardwater items identified in Table 1A(N)-1 as not being in compliance with Regulatory Guide 1.66 are constructed in accordance with the applicable ASME Code, Section III rules as a minimum.

However, the ASME Code, Section III rules do not require the suggested provisions of the regulatory guide for tubular products which do not fall within the size range of 2.5 to 6 inches, outside diameter.

Tubular products for core support structures function only as structural beams. For the CPNPP these tubular products were procured prior to the effectivity of Section NG of the ASME Code but were purchased using the guidelines of draft Section NG-2000.

Also refer to Appendix 1A(B).

Regulatory Guide 1.67

Installation of Overpressure Protection Devices

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.68

Initial Test Programs for Water-Cooled Reactor Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.68.1

Preoperational and Initial Startup Testing of Feedwater and Condensate Systems for Boiling Water Reactor Power Plants

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.68.2

Initial Startup Test Program to Demonstrate Remote Shutdown Capability for Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.69

Concrete Radiation Shields for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.70

Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.71

Welder Qualification for Areas of Limited Accessibility

Discussion

Westinghouse practice does not require qualification or requalification of welders for areas of limited accessibility as described by the guide (dated December 1973) and has provided welds of high quality.

Westinghouse believes that limited accessibility qualification or requalification, which are additional to ASME Code, Section III and IX requirements, is an unduly restrictive requirement for shop fabrication, where the welders' physical position relative to the welds is controlled and does not present any significant problems. In addition, shop welds of limited accessibility are repetitive due to multiple production of similar components, and such welding is closely supervised.

For field application, the type of qualification should be considered on a case-by-case basis due to the great variety of circumstances encountered.

Also refer to Appendix 1A(B).

Regulatory Guide 1.72

Spray Pond Plastic Piping

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.73

Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants

Discussion

For safety-related motor operated valves inside Containment, Westinghouse complies with the guidance of Regulatory Guide 1.73 (dated January 1974 with the exception that stem mounted limit switches are tested separately to the requirements of IEEE Standard 382-1972.

Also refer to Appendix 1A(B).

Regulatory Guide 1.74

Quality Assurance Terms and Definitions

Discussion

This regulatory guide (dated February 1974) recognizes ANSI N45.2.10- 1973. For the CPNPP, Westinghouse follows the guidance of this standard, utilizing consistent terms and definitions in the description of the Westinghouse Quality Assurance Program.

Also refer to Appendix 1A(B).

Regulatory Guide 1.75

Physical Independence of Electric Systems

Discussion

Westinghouse furnished systems meet the recommendations of Revision 1 (1/75) of this regulatory guide in accordance with the comments of Section 7.1.2.2.1 and Reference [10].

Also refer to Appendix 1A(B).

Regulatory Guide 1.76

Design Basis Tornado for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.77

Assumptions Used for Evaluating a Control Rod Ejection Accident for Pressurized Water Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.78

Assumptions for Evaluating the Habitability of a Nuclear Power Plant Control Room During a Postulated Hazardous Chemical Release

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.79

Preoperational Testing of Emergency Core Cooling Systems for Pressurized Water Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.80

Preoperational Testing of Instrument Air Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.81

Shared Emergency and Shutdown Electric Systems for Multi-Unit Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.82

Sumps for Emergency Core Cooling and Containment Spray Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.83

Inservice Inspection of Pressurized Water Reactor Steam Generator Tubes

Discussion

Westinghouse steam generators are designed to permit access to tubes for inspection and/or plugging. The inservice inspection program and compliance with Revision 1 (7/75) of this guide is discussed in Section 5.4.2.2.

Also refer to Appendix 1A(B).

Regulatory Guide 1.84

Code Case Acceptability - ASME Section III Design and Fabrication

Discussion

Refer to the discussion of Regulatory Guide 1.85 and Appendix 1A(B).

Regulatory Guide 1.85

Code Case Acceptability - ASME Section III Materials

Discussion

The CPNPP had its second-round Preliminary Safety Analysis Report questions issued after the July 1, 1974 date on which the original versions of Regulatory Guides 1.84 and 1.85 were made effective. The major NSSS components that are Code Class 1, 2 or 3 were ordered before Westinghouse instituted procedures to control the use of code cases. The only examples of code

cases known to have been used that are not authorized by way of Regulatory Guides 1.84 and 1.85 are Code Case 1528 and Code Case 1637. Westinghouse is conducting a testing program to determine fracture toughness properties of the materials allowed by Code Case 1528 and is in the process of obtaining authorization. Authorization has been obtained from the Commission for use of Code Case 1637 in the purchase of tubing for Class 2 and 3 components.

Refer to Appendix 1A(B) for a further discussion of the CPNPP position.

Regulatory Guide 1.86

Termination of Operating Licenses for Nuclear Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.87

Guidance for Construction of Class 1 Components in Elevated- Temperature Reactors (Supplement to ASME Section III Code Cases 1592, 1593, 1594, 1595, and 1596)

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.88

Collection, Storage, and Maintenance of Nuclear Power Plant Quality Assurance Records

Discussion

This regulatory guide recognizes N45.2.9-1974. For the CPNPP, the Westinghouse Quality Assurance Records Program follows the guidance of this standard. Records are identified, indexed, stored, and protected in a manner consistent with Revision 2 (10/76) of Regulatory Guide 1.88.

For active files, Westinghouse maintains duplicate records in separate geographical locations as an alternative to the protective construction provisions of the standard.

Also refer to Appendix 1A(B) and 17.2.17.

Regulatory Guide 1.89

Qualification of Class 1E Equipment for Nuclear Power Plants

Discussion

The Westinghouse approach to satisfying the guidelines of Regulatory Guide 1.89 (dated November 1974) and IEEE Standard 323-1974 is documented in WCAP 8587, Revision 6-A which has been reviewed and approved by the NRC. However, as supported by the statements

of consideration for 10CFR50.49 (Federal Register, 48FR2731, January 21, 1983), the recommendations of this regulatory guide need not be applied for Class 1E equipment located in a mild environment area.

Also refer to Appendix 1A(B).

Regulatory Guide 1.90

Inservice Inspection of Prestressed Concrete Containment Structures with Grouted Tendons

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.91

Evaluation of Explosions Postulated to Occur on Transportation Routes Near Nuclear Power Plant Sites

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.92

Combining Modal Responses and Spatial Components in Seismic Response Analysis

Discussion

Westinghouse takes exception to several major positions in this regulatory guide (dated December 1974) as discussed below.

1. Combination of closely spaced modes

   In Section B.1 it is stated that, “Because of the likelihood that the response of closely spaced modes add directly, the values of the response of these modes should be combined by taking the absolute sum of the maximum values of the response of individual closely spaced modes, as explained in Regulatory Position C.2 of this guide.”

   The regulatory guide cited as its only reference in the closely spaced modes discussion one which did not propose to take the absolute sum of maximum values of the response of individual closely spaced modes. Instead, it has recommended that closely spaced modes be added using a coefficient E which is less than or equal to 1.0 and is a function of the frequency ratio, duration of the earthquake, and damping values. Similar formulas have been proposed by the ASME Dynamic Analysis Task Force which utilize the same E coefficient.

   The formula proposed in Regulatory Guide 1.92 has not been thoroughly discussed in the literature and does not realistically represent the dependence of the response of closely spaced modes on the proximity of frequencies and the modal damping values.

Westinghouse has presented an alternative approach in Section 3.7(N) which has factored in the E coefficient. This formula is more conservative than the referenced formula, since E is considered to be positive for all closely spaced modes, but is equal to or less conservative than the formula proposed in the regulatory guide.

2. Time-history dynamic analysis

Regulatory Position C.3 recommends that time-history dynamic analysis be conducted independently for each of the three directions and summed by the square root of the sum of the squares approach.

The three orthogonal earthquake components are statistically independent, and more importantly, they occur simultaneously. Furthermore, time-history analysis is most often conducted in order to properly account for either geometrical or material nonlinearities. All systems do not behave linearly in responding to the loading. Consequently, all three components of the input may need to be applied simultaneously, and therefore, simultaneous use of these components should be allowed by the regulatory guide in the time-history analysis.

For some systems, independent application of each time-history is justified and the methods given in the regulatory guide are applicable. Simultaneous application, however, must be included as an option.

For response spectra analyses, the option should be provided in the regulatory guide for the simultaneous application of the three load components of the earthquake rather than that method specified in the regulatory guide. Currently, the guide recommends independent application of the three components of the earthquake with subsequent combination by the square root of the sum of the squares. Westinghouse is currently preparing computer programs for the analyses of the Reactor Coolant System which will apply the three seismic load components simultaneously. In the seismic analysis, this method gives generally equivalent conservatisms in comparisons to the independent application of the loads and should be permitted by the regulatory guide.

Also refer to Appendix 1A(B).

Regulatory Guide 1.93

Availability of Electric Power Sources

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.94

Quality Assurance Requirements for Installation, Inspection, and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.95

Protection of Nuclear Power Plant Control Room Operators Against an Accidental Chlorine Release

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.96

Design of Main Steam Isolation Valve Leakage Control Systems for Boiling Water Reactor Nuclear Power Plants

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.97

Instrumentation for Light-Water-Cooled Nuclear Power Plants to Assess Plant Conditions During and Following an Accident

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.98

Assumptions Used for Evaluating the Potential Radiological Consequences of a Radioactive Offgas System Failure in a Boiling Water Reactor

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.99

Radiation Embrittlement of Reactor Vessel Materials

Discussion

CPNPP complies with Revision 2 (5/88) of this Regulatory Guide as discussed in Section 5.3.2.1.

Regulatory Guide 1.100

Seismic Qualification of Electric Equipment for Nuclear Power Plants

Discussion

As discussed in Section 3.10N, the CPNPP NSSS program for seismic qualification of safety-related electrical equipment to the recommendations of Regulatory Guide 1.100 is delineated in References 16 and 17. In summary, seismic qualification will be demonstrated by the following methods:

1. For equipment not subject to high energy line break conditions, which has been previously qualified by the single axis sine beat method (after demonstration of no resonant frequency below 33 hertz) as permitted by IEEE Standard 344-1971 "IEEE Guide for Seismic Qualification of Class 1 Electric Equipment for Nuclear Power Generating Stations" and included in the NRC seismic audit and, where required by the staff, the Supplemental Qualification Program, no additional qualification testing is required to demonstrate acceptability to IEEE 344-1975 provided that:

    a. It can be shown, by separate component testing and/or analysis, that there are no aging mechanisms that could prejudice the previously completed seismic qualification.

    b. Any design modifications made to the equipment do not significantly affect the seismic characteristics of the equipment.

    c. The adequacy of the original seismic test levels can be demonstrated as conservative by plant specific verification.

2. For new equipment, or equipment that cannot meet the provisions of item 1 above, seismic qualification will be performed in accordance with IEEE Standard 344-1975, "IEEE Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations." The method to be employed (i.e., test and/or analysis) is indicated, for the safety-related equipment in the Westinghouse scope of supply, in Reference 17. Where multifrequency biaxial inputs are employed for testing, the methodology described in Reference 18 will be employed. When flexible equipment size and weight precludes biaxial testing (e.g., enclosures), single axis testing with justification will be utilized to meet IEEE Standard 344-1975. For rigid equipment (i.e., no resonant frequency below 33 hertz), qualification will be by analysis in accordance with IEEE Standard 344-1975.

All Class 1E equipment located in a mild environment area will be seismically qualified as described above and in FSAR Sections 3.10N and 3.10B, except that the additional requirements imposed by IEEE Standard 323- 1974 do not apply. The procurement documents will specify that the effects of aging on seismic qualification be assessed and if there are aging effects, require pre-aging or analysis of aging effects as part of the seismic qualification.

Refer to FSAR Section 3.10N for further discussion of Regulatory Guide 1.100.

Also refer to Appendix 1A(B).

Regulatory Guide 1.101

Emergency Planning for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.102

Flood Protection for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.103

Post-Tensioned Prestressing Systems for Concrete Reactor Vessels and Containments

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.104

Overhead Crane Handling Systems for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.105

Instrument Setpoints

Discussion

CPNPP complies with the intent of Revision 2 (2/86) of this guide as discussed below.

Westinghouse Standard Technical Specifications provide the margin from the nominal trip setpoint to the allowable value to account for rack drift and calibration when measured during periodic testing. The allowances between the nominal trip setpoint technical specification limit and the safety analysis limit include a statistical combination of the following items:

1. the inaccuracies of the instrumentation,

2. non-instrument related effects, i.e., process measurement accuracy,

3. uncertainties in the calibration of both the transmitter and racks, and

4. Adverse environmental effects on transmitter accuracy caused by postulated or limiting postulated events (only for those systems required to mitigate consequences of an accident).

Potential transient overshoot is accounted for in the modeling of the event in the safety analyses.

Westinghouse chooses setpoints such that the accuracy of the instrumentation is sufficient to meet the assumptions of the safety analyses.

The range of the instrumentation is chosen based on the span necessary for the function. Narrow range instrumentation will be used where necessary. Instrumentation will be selected based on expected environmental and accident conditions. The need for qualification testing is evaluated and justified on a case-by-case basis.

Administrative procedures, coupled with the present cabinet alarms and/or locks, provide sufficient control over the setpoint adjustment mechanism such that no integral setpoint securing device is required. Integral setpoint locking devices are not supplied.

Also refer to Appendix 1A(B) for further discussion.

Regulatory Guide 1.106

Thermal Overload Protection for Electric Motors on Motor-Operated Valves

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.107

Qualifications for Cement Grouting for Prestressing Tendons in Containment Structures

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.108

Periodic Testing of Diesel Generators Used as Onsite Electric Power Systems at Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.109

Calculation of Annual Doses to Man from Routine Releases of Reactor Effluents for the Purpose of Evaluating Compliance with 10CFR Part 50, Appendix I

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.110

Cost-Benefit Analysis for Radwaste Systems for Light-Water-Cooled Nuclear Power Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.111

Methods for Estimating Atmospheric Transport and Dispersion of Gaseous Effluents in Routine Releases from Light-Water-Cooled Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.112

Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Light-Water-Cooled Power Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.113

Estimating Aquatic Dispersion of Effluents from Accidental and Routine Reactor Releases for the Purpose of Implementing Appendix I

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.114

Guidance on Being Operator at the Controls of a Nuclear Power Plant

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.115

Protection Against Low-Trajectory Turbine Missiles

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.116

Quality Assurance Requirements for Installation, Inspection, and Testing of Mechanical Equipment and Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.117

Tornado Design Classification

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.118

Periodic Testing of Electric Power and Protection Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.119

Surveillance Program for New Fuel Assembly Designs

Discussion

This regulatory guide was withdrawn June 23, 1977.

Regulatory Guide 1.120

Fire Protection Guidelines for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.121

Bases for Plugging Degraded PWR Steam Generator Tubes

Discussion

The plugging criteria for the steam generator U-tubes, as specified in the plant Technical Specifications, is based on the more conservative limit required by paragraph IWB-3521.1, Section XI of the ASME Boiler and Pressure Vessel Code, than the plugging criteria derived from the Regulatory Guide 1.121 (dated August, 1976) analysis. Alternate repair criteria which may exceed the plugging limit specified by IWB-3521.1 are defined in the plant Technical Specifications.

Regulatory Guide 1.122

Development of Floor Design Response Spectra for Seismic Design of Floor-Supported Equipment or Components

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.123

Quality Assurance Requirements for Control of Procurement of Items and Services for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.124

Design Limits and Loading Combinations for Class 1 Linear-Type Component Supports

Discussion

The CPNPP NSSS position on Revision 1 (1/78) of this guide is as follows:

1. Paragraph B.1

   In the regulatory guide, an increase in bolt allowable stress under emergency and faulted conditions is not permitted. Westinghouse believes that the present ASME Code rules are adequate for bolted connections.

   It is recognized after extensive experimental work by several researchers that the interaction curve between the shear and tension stress in bolts is more closely represented by an ellipse and not a line. This has been clearly recognized by the ASME. Code Case 1644-6 specifies stress limits for bolts and represents this tension/shear relationship as a non-linear interaction equation (incorporated into ASME III Appendix XVII via the Winter 77 Addenda) and has a built-in safety factor that ranges between 2 and 3 (depending on whether the bolt load is predominantly tension or shear) based on the actual strength of the bolt as determined by test. [19]

Study of three interaction curves of allowable tension and shear stress based on the ASME Code (emergency condition allowables per XVII-2110 and faulted condition allowables per F-1370) and the ultimate tensile and shear strength of bolts indicates that there is adequate safety margin between the emergency and faulted condition allowables and failure of the bolts. [20]

From this study it is observed that:

a. For the emergency condition, the safety factor (ratio of ultimate strength to allowable stress) varies between a minimum of 1.63 and a maximum of 2.73 depending upon the actual tensile stress/shear (T/S) ratio on the bolt.

b. For the faulted condition, the safety factor varies between a minimum of 1.36 to a maximum of 2.29, again depending upon actual T/S ratio on the bolt.

The stress limits used for Class 1 component bolting material are those given in ASME III Appendix F-1323.1(b), $Pm \leq 2.4\ Sm$ or $0.7\ S_u$ (whichever is less) using the material properties of Table I-1.3 taken at the appropriate temperature.

It is thus reasonable to allow an increase in these limits for the emergency and faulted conditions.

Based on the above discussion, for the emergency and faulted conditions, Westinghouse will use allowable bolt stresses specified in Code Case 1644-6, as increased according to the provisions of XVII-2100(a) and F-1370(a), respectively.

The increased design limit for the stress range identified in NF-3231.1(a) shall be limited to the smaller of $2\ S_y$ or S_u unless otherwise justified by shakedown analysis.

2. Paragraphs B.5 and C.8

The reduction of allowable stresses to no greater than level B limits (which in reality are design limits since design, level A and level B limits are the same for linear supports) for support structures in those systems with safety related functions occurring during emergency or faulted plant conditions is overly conservative. The primary concern is that the system remains capable of performing its safety function. For active components, this is accomplished through the operability program as discussed in Section 3.9N. In the case of Class 1 piping, maintaining the pipe stresses within level D limits assures that piping geometry is maintained and that required flow is not impeded. The selection of more restrictive stress limits for component supports is not necessary to assure the functional capability of the system.

3. Paragraph C.4

In the design of component support, member compressive axial loads shall be limited to 0.67 times the critical buckling strength. If, as a result of more detailed evaluation of the supports the member compressive axial loads can be shown to safely exceed 0.67 times the critical buckling strength for the faulted condition, verification of the support functional adequacy will be documented and submitted to the NRC for review. The member

compressive axial loads will not exceed 0.67 times the critical buckling strength without NRC acceptance. In no case shall the compressive load exceed 0.9 times the critical buckling strength.

This regulatory guide states that increases in Level A or B service limits does not apply to limits for bolted connections. The Westinghouse design of component supports restricts the use of bolting material to the following applications:

a. Westinghouse design uses bolting predominantly in tension. Oversized holes are generally provided and a mechanism other than the bolts is provided to take any shear loads. Shear or shear & tension interaction occur only in isolated locations;

b. Westinghouse bolts are limited to the following material A490, SA-354, SA-325, SA-540.

c. The diameters used range between 1/2" and 3".

These limitations on bolt usage are standard in the Westinghouse supports. We will limit tensile loads in the bolts to 0.7 S_u, but not to exceed in any case 0.9 S_y. The allowables are taken at temperature. In those few cases where bolts are used in shear or tension and shear, ASME Code Appendix XVII - 2460 Requirements will apply with an increase factor that is defined in Regulatory Guide 1.124 or in Appendix F-1370. This provides an adequate margin of safety for the Westinghouse design. If future revisions to the bolting criteria in ASME Section III modify the Westinghouse criteria listed above, we will review the criteria at the time.

4. Paragraph C.6(a)

Westinghouse will interpret this paragraph as follows: "The stress limits of XVII-2000 of Section III and Regulatory Position 3 increased according to the provisions of XVII-2110(a) of Section III and Regulatory Position 4, should not be exceeded for component supports designed by the linear elastic analysis method."

5. Paragraph C.7(b)

Westinghouse will use the provisions of F-1370(d) to determine service level D allowable loads for supports designed by the lead rating method. If future revisions to Appendix F modify this criteria, it will be reviewed further. If the load rating method is used, further details of its implementation will be provided at that time.

Also refer to Appendix 1A(B).

Regulatory Guide 1.125

Physical Models for Design and Operation of Hydraulic Structures and Systems for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.126

An Acceptable Model and Related Statistical Methods for the Analysis of Fuel Densification

Discussion

Fuel for CPNPP is provided by Westinghouse. The methodology for the analysis of fuel densification used by Westinghouse is described in Reference 12.

Regulatory Guide 1.127

Inspection of Water-Control Structures Associated with Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.128

Installation Design and Installation of Large Lead Storage Batteries for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.129

Maintenance, Testing, and Replacement of Large Lead Storage Batteries for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.130

Service Limits and Loading Combinations for Class 1 Plate- and Shell-Type Component Supports (Revision 1, October, 1978)

Discussion

The Applicant takes the following exceptions to this Regulatory Guide:

1. Regulatory Guide states in Paragraph B.1:

   "Allowable design limits for bolted connections are derived on a different basis that varies with the size of the bolt. For this reason, the increases permitted by NF-3224 and F-1323.1(a) of Section III are not directly applicable to bolts and bolted connections."

   It is the Applicant's position that it is reasonable to allow an increase in the limits for bolted connections for emergency and faulted conditions. Further justification of this position can be found in the discussion of Regulatory Guide 1.124 on Class 1 linear type supports.

2. Paragraphs C.2, C.4(a), and C.6(a) of the Regulatory Guide state that the allowable buckling strength should be calculated using a design margin of 2 for flat plates and 3 for shells for normal, upset and emergency conditions.

   In the design of plate-type supports, member compressive axial loads shall be limited per the requirements of Paragraph C.3 for normal, upset and emergency conditions.

3. In Paragraph C.7 of the Regulatory Guide, inclusion of the upset plant condition is unappropriate in the load combination under discussion. The Applicant will not include the upset plant condition in this combination.

4. In Paragraphs C.7(a) and B.1 of the Regulatory Guide, the stress limits of F-1370(c) are discussed. The criterion stated in F-1370(c), "...loads should be exceed 0.67 times the critical buckling strength of the support...".

   In the design of plate-type component supports, member compressive axial loads shall be limited to 0.67 times the critical buckling strength. If, as a result of a more detailed evaluation of the supports the member compressive axial loads can be shown to safely exceed 0.67 times the critical buckling for the faulted condition, verification of the support function adequacy will be documented and submitted to the NRC for review. The member compressive axial loads will not exceed 0.67 times the critical buckling strength without NRC acceptance.

5. In Paragraph C.7(b) of the Regulatory Guide, the limit based on the test load given in the Regulatory Guide, T.L. x 0.7 $S'_u/_{Su}$, is overly conservative and is inconsistent with ASME Code requirements presented in Appendix F.

   The Applicant will use the provisions of F-1370(c) to determine service level D allowable loads for supports designed by the load rating method.

Regulatory Guide 1.131

Qualification Tests of Electric Cables, Field Splices, and Connections for Light-Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.132

Site Investigations for Foundations of Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.133

Loose-Part Detection Program for the Primary System of Light-Water- Cooled Reactors

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.134

Medical Evaluation of Nuclear Power Plant Personnel Requiring Operator Licenses

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.135

Normal Water Level and Discharge at Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.136

Material for Concrete Containments

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.137

Fuel-Oil Systems for Standby Diesel Generators

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.138

Laboratory Investigations of Soils for Engineering Analysis and Design of Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.139

Guidance for Residual Heat Removal

Discussion

In lieu of Regulatory Guide 1.139, CPNPP addressed Branch Technical Position RSB 5-1 as provided in FSAR Chapter 5.

Regulatory Guide 1.140

Design, Testing, and Maintenance Criteria for Normal Ventilation Exhaust System Air Filtration and Adsorption Units of Light-Water- Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.141

Containment Isolation Provisions for Fluid Systems

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.142

Safety-Related Concrete Structures for Nuclear Power Plants (Other Than Reactor Vessels and Containments)

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.143

Design Guidance for Radioactive Waste Management Systems, Structures, and Components Installed in Light-Water-Cooled Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.144

Auditing of Quality Assurance Programs for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.145

Atmospheric Dispersion Models for Potential Accident Consequence Assessments at Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.146

Qualification of Quality Assurance Program Audit Personnel for Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.148

Functional Specification for Active Valve Assemblies in Systems Important to Safety in Nuclear Power Plants

Discussion

Refer to Appendix 1A(B).

Regulatory Guide 1.150

Ultrasonic Testing of Reactor Vessel Welds During Preservice and Inservice Examinations

Discussion

The CPNPP position on Revision 1 (2/83) of this guide is as follows:

Preservice Inspection

A partial R.G. 1.150 inspection was performed on Units 1 and 2 reactor vessels in accordance with Reference [21].

Inservice Inspection

CPNPP complies with revision 1 of this regulatory guide.

Regulatory Guide 1.155

Station Blackout

Discussion

Refer to Appendix 1A(B)

REFERENCES

1. Fletcher, W. D., Bell, M. J. and Picone, L. F., "Post LOCA Hydrogen Generation in PWR Containment", Nucl. Technol. 10, 420-427, 1971.

2. Zittel, H. E. and Row, T. H., "Radiation and Thermal Stability of Spray Solutions", Nucl. Technol. 10, 436-443, 1971.

3. Allen, A. O., "The Radiation Chemistry of Water and Aqueous Solutions", Princeton, N. J., Van Nostrand, 1961.

4. "Reactor Coolant Pump Integrity in LOCA", WCAP-8163, September 1973.

5. "Westinghouse Nuclear Energy Systems Division Quality Assurance Plan," WCAP-8370, Revision 8A and subsequent revisions.

6. Enrietto, J. F., "Control of Delta Ferrite in Austenitic Stainless Steel Weldments," WCAP-8324-A, June 1975.

7. Enrietto, J. F., "Delta Ferrite in Production Austenitic Stainless Steel Weldments," WCAP-8693, January 1976.

8. Caplan, J. S., "The Application of Preheat Temperatures after Welding Pressure Vessel Steels," WCAP-8577, February 1976.

9. "Damping Values of Nuclear Power Plant Components," WCAP-7921- AR, May 1974.

10. Letter NS-CE-604, dated March 31, 1975, C. Eicheldinger (Westinghouse) to the Secretary of the Nuclear Regulatory Commission.

11. "Environmental Qualification of Westinghouse Class 1E Equipment," WCAP-8587, October 1975.

12. Hellman, J. M. (Ed.), "Fuel Densification Experimental Results and Model for Reactor Operation," WCAP-8218-P-A (Proprietary) and WCAP-8219-A (Non-Proprietary), March 1975.

13. DELETED

14. Letter NS-CE-784, dated September 22, 1975, C. Eicheldinger (Westinghouse) to the Secretary of the Nuclear Regulatory Commission.

15. Letter NS-CE-1157 dated August 13, 1976, C. Eicheldinger (Westinghouse) to the Secretary of the Nuclear Regulatory Commission.

16. Butterworth, G. and Miller, R. B., "Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety-Related Electrical Equipment," WCAP-8587, Revision 2, February 1979.

17. "Equipment Qualification Data Packages," WCAP-8587, Supplement 1, November, 1978.

18. Jarecki, S. J., "General Method of Developing Multifrequency Biaxial Test Inputs for Bistables," WCAP-8624 (Proprietary), September 1975 and WCAP-8695 (Non-Proprietary), August 1975.

19. Fisher and Struik, Guide to the Design Criteria of Bolted & Riveted Joints, John Wiley & Sons, 1974, p. 54.

20. Chesson, E. Jr. Faustino, N. L., Munse, W. H., High Strength Bolts Subjected to Tension and Shear, Journal of the Structural Division, Proceedings of the American Society of Civil Engineer, October 1965, pp. 155-180.

21. Letter TXX-3632, dated March 1, 1983, H. C. Schmidt to B. J. Youngblood, NRC Licensing Branch No. 1.

TABLE 1A(N)-1
TUBULAR PRODUCTS WITHIN WESTINGHOUSE SCOPE RELATED TO REGULATORY GUIDE 1.66

Westinghouse Application	Tubular Product Form	UT[a]	ET[a]	RT[a]	PT[a]	Complies with Regulatory Guide 1.66
		Circumf. Axial				
Steam generator tubing	Seamless tube	X[b],[c]	X			Yes[d]
Pressurizer heater tube	Seamless tube	X[c]	X		O.D.	Yes
Heat exchanger tubing	Welded tube	Part.	X		O.D.	Yes
Vent pipe monitor tube	Seamless tube	X[c]	X		O.D.	Yes
Instrument tubing	Seamless tube	X[c]			O. D. ends	No[e]
Instrument nozzles	Bar	X[c]	X		O.D.	Yes
CRDM housings	Bar	X[c]	X		O.D., I.D.	Yes
Adaptor flange	Bar	X[c]	X		O.D., I.D.	Yes
Reactor coolant pipe wrought	Extrusions	X[c]		Welds	O.D., I.D.	No[e]
Surge line pipe wrought	Extrusions	X[c]		Welds	O.D., I.D.	No[e]

a) UT - Ultrasonic
ET - Eddy current
RT - Radiographic
PR - Dye penetrant

b) X indicates that the tests are performed.

c) Two directions.

d) Unit 2 only.

e) These items are outside the range of O.D. sizes covered by those sections of the ASME Code, Section III, which incorporate the ultrasonic testing provisions of the guide.

APPENDIX 1A(B) - DISCUSSION OF REGULATORY GUIDES

This appendix discusses the Comanche Peak Nuclear Power Plant (CPNPP) positions on, and compliance with, Division 1 regulatory guides as they apply to the balance of plant design and construction as well as start-up and operations. Appendix 1A(N) is provided separately for the NSSS related information.

Identification of specific issues of regulatory guides are provided in the discussions where appropriate.

Regulatory Guide 1.1

Net Positive Suction Head for Emergency Core Cooling and Containment Heat Removal System Pumps

Discussion

The CPNPP Containment Spray System design complies with the requirements of Safety Guide 1 (11/2/70) except as follows:

In the calculation of the net positive suction head (NPSH) for containment spray pumps it is assumed that the containment pressure is equal to the vapor pressure of the sump water.

The above assumption is conservative and consistent with the intent of the regulatory guide.

Also refer to Appendix 1A(N) and Section 6.2.2.3.4 for further discussion.

Regulatory Guide 1.2

Thermal Shock to Reactor Pressure Vessels

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.3

Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Boiling Water Reactors

Discussion

This regulatory guide is not applicable to the CPNPP which has pressurized water reactor steam supply systems.

Regulatory Guide 1.4

Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Pressurized Water Reactors.

Discussion

The analysis of the radiological consequences of the loss-of-coolant accident presented in Section 15.6.5 follows the guidance provided in Regulatory Guide 1.195 instead of Regulatory Guide 1.4.

Regulatory Guide 1.5

Assumptions Used for Evaluating Potential Radiological Consequences of a Steam Line Break Accident for Boiling Water Reactors.

Discussion

This regulatory guide is not applicable to the CPNPP which has pressurized water reactor steam supply systems.

Regulatory Guide 1.6

Independence Between Redundant Standby (Onsite) Power Sources and Between Their Distribution Systems

Discussion

The CPNPP design complies with the requirements of Safety Guide 6 (3/10/71). For details see Section 8.3.

Regulatory Guide 1.7

Control of Combustible Gas Concentrations in Containment Following a Loss of Coolant Accident

Discussion

Based on a revision to 10CFR50.44, Regulatory Guide 1.7 no longer applies to CPNPP. See Section 6.2.5.

Regulatory Guide 1.8

Personnel Selection and Training

Discussion

Minimum qualifications of unit staffs, with the exception of licensed Senior Reactor Operators and Reactor Operators, will be in accordance with Regulatory Guide 1.8, Revision 2. Minimum qualifications for licensed Senior Reactor Operators and Reactor Operators will be in accordance with Regulatory Guide 1.8, Revision 3.

The training requirements of Regulatory Guide 1.8, Revision 2 have been superseded by the provisions of 10CFR parts 50 and 55.

Regulatory Guide 1.9

Selection of diesel Generator Set Capacity for Standby Power Supplies

Discussion

The CPNPP Diesel generator sets comply with the requirements of Safety Guide 9 (3/10/71) with the following comment:

The voltage may dip below 75 percent of nominal voltage when the diesel generator breaker closes and energizes the two 2000/2666 kVA, 6.9 kV/480 V unit substation transformers supplied from each diesel generator. The dip is due to transformer magnetizing inrush current which exists for two to three cycles. The diesel generator sets are designed to recover to 80 percent of nominal voltage within 10 cycles for this transient. The effect on the first load groups (see Tables 8.3-1 and 8.3-2) therefore would be a maximum possible delay in motor starting of 12-13 cycles after closure of the diesel generator circuit breaker. However, the objective of the first load group and subsequent load groups is not affected. For details see Section 8.3.

Regulatory Guide 1.10

Mechanical (Cadweld) Splices in Reinforcing Bars of Category I Concrete Structures

Discussion

Testing and sampling of Mechanical (Cadweld) Splices in Reinforcing Bars of the CPNPP Concrete Containment Structure complies with the requirements of Revision 1 (1/2/73) of this regulatory guide. For other seismic Category I concrete structures, the testing and sampling of Mechanical (Cadweld) splices complies with the requirements of this guide except that the location of all splices are not recorded and shown in as-built drawings.

Also refer to Section 3.8.

Regulatory Guide 1.11

Instrument Lines Penetrating Primary Reactor Containment

Discussion

The CPNPP instrument lines penetrating primary reactor containment comply with the requirements of Safety Guide 11 (3/10/71), as described by Section 7.3.1.1.2 and 6.2.4.1.4.

Also refer to Appendix 1A(N).

Regulatory Guide 1.12

Instrumentation for Earthquakes

Discussion

The installation of instrumentation for earthquakes in the CPNPP plant meets the intent of Revision 1 (4/74) of this regulatory guide with respect to the ability to determine exceedance of the OBE in a timely manner. Seismic instrumentation includes only the free-field triaxial accelerometer installed in the Yard. Determination of OBE exceedance will be based on the methods of ANSI/ANS-2.10-2003, “Criteria for the Handling and Initial Evaluation of Records from Nuclear Power Plant Seismic Instrumentation.”

Also refer to Section 3.7B.4.

Regulatory Guide 1.13

Spent Fuel Storage Facility Design Basis

Discussion

The design of the CPNPP spent fuel storage facility complies with Revision 1 (12/75) of this regulatory guide except that the air filtration system is not actuated by a high radiation level alarm. Instead the normal plant exhaust filtration system operates continuously. Refer to Sections 3.5, 3.8.4, 9.1.2, 9.1.3, 9.1.4 and 9.4.2 for details.

Regulatory Guide 1.14

Reactor Coolant Pump Flywheel Integrity

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.15

Testing of Reinforcing Bars for Category I Concrete Structures

Discussion

The testing of reinforcing bars for CPNPP Category I concrete structures complies with the requirements of Revision 1 (12/28/72) of this regulatory guide.

Also refer to Section 3.8.

Regulatory Guide 1.16

Reporting of Operating Information - Appendix A Technical Specifications

Discussion

This regulatory guide was withdrawn on August 11, 2009.

Regulatory Guide 1.17

Protection of Nuclear Power Plants Against Industrial Sabotage

Discussion

Regulatory Guide 1.17 endorses the requirements and recommendations of ANSI N18.17-1973. This criteria does not reflect current regulations or industry standards and has been superseded by more recent guidelines. CPNPP takes exception to Regulatory Guide 1.17 and ANSI N18.17-1973 and instead implements a security program in accordance with the current applicable regulations and guidelines identified in FSAR section 13.6.

Regulatory Guide 1.18

Structural Acceptance Test for Concrete Primary Reactor Containments (Revision 1, December 28, 1972)

Discussion

The structural acceptance test for the CPNPP Concrete Containments are in accordance with paragraph CC-6000 of the ASME B&PV Code, Section III, Division 2, 1980 Edition with Summer 1980 Addenda (as applicable to non-prototype containments). As such, this test is in conformance with Revision 1 of this regulatory guide except as follows:

Radial deflections of the containment walls are measured along four azimuths only.

Vertical deflections along one azimuth are measured at two equally spaced intermediate points between the dome apex and the springline and along four azimuths at the dome springline. Vertical deflection is also measured at the dome apex.

Only radial deflections at 12 points around the largest opening (the equipment hatch) are measured. In addition, the increase in diameter of the opening is measured in two mutually perpendicular directions.

See Subsection 3.8.1.7.1 for description of the test.

Regulatory Guide 1.19

Nondestructive Examination of Primary Containment Liner Welds

Discussion

The requirements for nondestructive examination of the CPNPP containment liner welds are in conformance with the intent of Revision 1 (08/11/72) of this regulatory guide by use of ASME-ACI 359 Code applicable to this type of structure and the alternate requirements as discussed in Sections 3.8.1 and 3.8.2. Acceptance criteria are those provided in Subsection 3.8.1.6.5.

Regulatory Guide 1.20

Comprehensive Vibration Assessment Program for Reactor Internals During Preoperational and Initial Startup Testing

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.21

Measuring, Evaluating, and Reporting Radioactivity in Solid Wastes and Releases of Radioactive Materials in Liquid and Gaseous Effluents from Light-Water-Cooled Nuclear Power Plants

Discussion

Measuring, evaluating and reporting of radioactive materials released from CPNPP will meet the recommendations of Revision 1 (6/74) of this regulatory guide.

Also refer to Section 11.5.

Regulatory Guide 1.22

Periodic Testing of Protection System Actuation Functions

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.23

Onsite Meteorological Programs

Discussion

The meteorological monitoring program at CPNPP complies with the requirements and those applicable recommendations of the Second Proposed Revision 1 to Regulatory Guide 1.23 (April, 1986) as discussed in Section 2.3. Refer to the CPNPP Offsite Dose Calculation Manual (ODCM), Section I, 4.3.3.6 and Bases 3/4.3.3.6 for an exception.

Refer to Section 2.3 for a description of the design and siting of the primary meteorological tower. The quality assurance program for meteorological monitoring is identified in FSAR Table 17A-1 and Section 17.2.

Regulatory Guide 1.24

Assumptions Used for Evaluating the Potential Radiological Consequences of a Pressurized Water Reactor Radioactive Gas Storage Tank Failure

Discussion

The analysis of the radiological consequences of the radioactive gas storage tank failure accident presented in Section 15.7.1 complies with the requirements of Safety Guide 24 (3/23/72) except that only gamma radiation contribution is taken into account in the determination of whole body exposures and the dose calculation methodology is consistent with Regulatory Guide 1.195.

Regulatory Guide 1.25

Assumptions used for Evaluating the Potential Radiological Consequences of a Fuel Handling Accident in the Fuel Handling and Storage Facility for Boiling and Pressurized Water Reactors

Discussion

The analysis of the radiological consequences of the fuel handling accident presented in Section 15.7.4 follows the guidance provided in Regulatory Guide 1.195 instead of that in Regulatory Guide 1.25.

Regulatory Guide 1.26

Quality Group Classifications and Standards for Water-, Steam-, and Radioactive-Waste-Containing Components of Nuclear Power Plants

Discussion

The CPNPP quality group classification described in Section 3.2.2 is generally in accordance with ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973, which is an alternate acceptable method of meeting the intent of Revision 3 (2/76) of this regulatory guide. Deviations are described in Section 3.2.2.

Also refer to Appendix 1A(N).

Regulatory Guide 1.27

Ultimate Heat Sink for Nuclear Power Plants

Discussion

The CPNPP ultimate heat sink meets the requirements of Revision 2 (January 1976) of the Regulatory Guide 1.27 as described in Sections 2.3.1.2.10, 2.4.11.5, 2.4.11.6 and 9.2.5.

Regulatory Guide 1.28

Quality Assurance Program Requirements (Design and Construction)

Discussion

The quality assurance program (design and construction) for CPNPP complies with the requirements of Safety Guide 28 (6/7/72) except as stated in Appendix 1A(N). Revisions 1 (3/78) and 2 (2/79) of this guide are not addressed.

Also refer to Sections 17.1 and 17.2

Regulatory Guide 1.29

Seismic Design Classification

Discussion

The seismic Category I classifications of CPNPP structures, systems and components conforms to Revision 2 (2/76) of this regulatory guide.

Also refer to Appendix 1A(N) and Sections 3.1, 3.2 and 13.5.

Regulatory Guide 1.30

Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment

Discussion

Safety Guide 30 (8/11/72) is applicable to the installation, inspection, and testing of instrumentation and electric equipment at CPNPP. The quality assurance methods for operations phase activities will comply with the applicable guidance contained in this regulatory guide. The application of the requirements of ANSI N45.2.4 - 1972, as endorsed by this regulatory guide, will be in accordance with the guidance provided in ANSI N18.7 - 1976.

Also refer to ANSI/NCSL Z540-1-1994 “American National Standard for Calibration Laboratories and Measuring and Test Equipment General Requirements” for M&TE and reference standards traceability.

Also refer to Appendix 1A(N) and to Sections 17.1 and 17.2.

Regulatory Guide 1.31

Control of Ferrite Content in Stainless Steel Weld Metal

Discussion

The CPNPP design complies with the recommendations of Revision 3 (4/78) of this regulatory guide except as follows:

1. The acceptable alternative methods of FSAR Subsection 6.1B.1.1.3 are used.

2. The recommendations of Revision 1 (6/73) or Revision 2 (5/77) of this regulatory guide are used.

3. The exception described in Subsection 6.1B.1.1.4.

Also refer to Appendix 1A(N).

Regulatory Guide 1.32

Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants

Discussion

The CPNPP design complies with the requirements of Revision 2 (2/77) of this regulatory guide. For details see Section 8.3.

Regulatory Guide 1.33

Quality Assurance Program (Operation)

Discussion

The quality assurance requirements for the operations phase of CPNPP are in compliance with Revision 2 (2/78) of this regulatory guide as implemented by ANSI N18.7-1976, "Administrative Controls and Quality Assurance for Operational Phase of Nuclear Power Plants", with the following exceptions:

1. Audit Intervals (Section 17.2.18)

Audits shall be performed at the intervals designated herein for each audit area. Schedules shall be based on the month in which the audit starts. For audits, other than those audit areas with maximum frequency specified by regulation (Security and Emergency Plan), an interval extension grace period of up to 25% may be used for the scheduled commencement date when conditions, such as plant operational considerations or to accommodate performance based observations of activities related to the audit area, make meeting the specified schedule date impractical. When an audit interval extension greater than one month is used, the next audit for that particular audit area will be scheduled from the original anniversary month rather than from the month of the extended audit.

2. Regulatory Position C.4

The quality assurance program has specified alternatives to the program area audit frequencies stated in Regulatory Position C.4. The quality assurance program schedules the audits provided for in Regulatory Position C.4 at 24 month frequencies. The audit schedule is performance-based and additional audits may be scheduled based upon program or functional area performance or other factors that indicate the need for increased assessment.

3. Biennial Reviews of Plant Procedures

The intent of the biennial review is accomplished by CPNPP programmatic controls already in place. The following controls assure that procedures are appropriately reviewed and revised to incorporate information based on plant operations, design changes, regulatory requirements, industry experience and other conditions that may impact plant procedures.

- Site Modification Process
- Corrective Action Program
- Off-Normal Occurrence
- User Feedback and Procedure Compliance
- Operating Experience Review
- Vendor Technical Information
- Licensed Document Change/50.59 Evaluation
- Commitment Tracking System (CTS)
- Trending Activities
- Infrequently Performed Evolutions Control
- Requalification Training
- Quality Assurance Activities

Note: 10CFR50.59 has been revised and the terminology in Section 4.3.4 of ANSI N18.7-1976 is no longer current. Section 17.2 has been updated to reflect the revision to 10CFR50.59.

4. Identification of QA Program Requirements in Procurement Documents

Quality Assurance program requirements consistent with 10CFR50 Appendix B or ANSI N45.2 are not imposed in procurement documents for commercial grade calibration services from a National Voluntary Laboratory Accreditation Program (NVLAP) or American Association for Laboratory Accreditation (A2LA) accredited calibration laboratory evaluated in accordance with Section 17.2.7.

Regulatory Guide 1.34

Control of Electroslag Weld Properties

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.35

Inservice Inspection of Ungrouted Tendons in prestressed Concrete Containment Structures

Discussion

This regulatory guide is not applicable to the CPNPP which has a steel-lined, reinforced concrete containment structure.

Regulatory Guide 1.36

Nonmetallic Thermal Insulation for Austenitic Stainless Steel

Discussion

This regulatory guide is not applicable for components located inside CPNPP Containment Buildings, since only stainless steel metal reflective thermal insulation is used for austenitic stainless steel components located there. Nonmetallic thermal insulation used for austenitic stainless steel piping and components located outside the Containment Buildings meets the requirements of this regulatory guide dated February 23, 1973, with the exception of packaging and shipping requirements of paragraph C.1 of this guide. In lieu of controlled packaging and shipping, a receipt inspection is required, which consists of visual inspection for physical or water damage to shipping containers, to determine if insulation materials have been contaminated by external sources.

Also refer to Appendix 1A(N) and Section 6.1B, 6.2 and 5.2.3.

Regulatory Guide 1.37

Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants

Discussion

This regulatory guide is not applicable to the design and construction of CPNPP. The quality assurance methods for operating phase activities will comply with the applicable provisions of this regulatory guide, dated March 16, 1973, for those activities in this area which are similar to construction activities. The application of the requirements of ANSI N45.2.1 - 1973, as endorsed by this guide, will be in accordance with the guidance provided in ANSI 18.7 - 1976.

Also refer to Appendix 1A(N) and Section 17.2.

Regulatory Guide 1.38

Quality Assurance Requirements for Packaging, Shipping, Receiving, Storage, and Handling of Items for Water-Cooled Nuclear Power Plants

Discussion

This regulatory guide is not applicable to the design and construction of CPNPP. The quality assurance methods and administrative controls utilized in the area of packaging, shipping, receiving, storage and handling of items during the operations phase, as described in Section 17.2, will comply with the applicable guidance contained in Revision 1 (10/76) of this regulatory guide. The application of the provisions of ANSI N45.2.2 - 1972, as endorsed by this regulatory guide, will reflect the guidance provided in ANSI N18.7 - 1976.

Although ANSI N45.2.2 - 1972 is entitled, “Packaging, Shipping, Receiving, Storage, and Handling of Items for Nuclear Power Plants During the Construction Phase,” CPNPP meets the requirements of this standard during the operation phase with the clarification noted below:

Maintenance of items in-storage during operations will be in accordance with the preventive maintenance requirements of ANSI N18.7-1976 as supplemented by the EPRI guidelines for determining in-storage maintenance of items for nuclear facilities (NCIG-18).

For handling of items during operation phase, the following codes and standards shall apply.

ANSI B30.9 - 1971

ANSI B30.2 - 1976 (1967 for Design and Fabrication)

ANSI B30.5 - 1968

Regulatory Guide 1.38, Revision 2 (5/77) is not addressed.

Also refer to Appendix 1A(N) and Section 17.2.

Regulatory Guide 1.39

Housekeeping Requirements for Water-Cooled Nuclear Power Plants

Discussion

The administrative methods utilized to control station housekeeping practices will comply with ANSI N45.2.3 - 1973, as endorsed by Revision 2 (9/77) of this regulatory guide and as applied to the operations phase by the provisions of ANSI N18.7 - 1976.

Also refer to Section 17.2

Regulatory Guide 1.40

Qualification Tests of Continuous-Duty Motors Installed Inside the Containment of Water-Cooled Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP since there are no continuous-duty Class 1E motors inside the Containment. For details see Section 3.11B.

Regulatory Guide 1.41

Preoperational Testing of Redundant On Site Electric Power Systems to Verify Proper Load Group Assignments

Discussion

The CPNPP onsite electric power system design complies with the preoperational testing requirements of this guide dated March 1973. For details see Section 8.3. The Initial Test Program, as described in Section 14.2, will comply with the provisions of this regulatory guide.

Regulatory Guide 1.42

Interim Licensing Policy on As Low As Practicable for Gaseous Radioiodine Release from Light-Water-Cooled Nuclear Power Reactors

Discussion

This regulatory guide was withdrawn March 22, 1976.

Regulatory Guide 1.43

Control of Stainless Steel Weld Cladding of Low Alloy Steel Components

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.44

Control of the Use of Sensitized Stainless Steel

Discussion

The CPNPP design will comply with this regulatory guide, dated May 1973, except for Regulatory Position C6. The intergranular corrosion tests are eliminated because the weld arc heat input is controlled as described in Section 6.1B.1.1.2, Item 1e. Exception to this regulatory position is also taken for some equipment as described in Section 6.1B.11.2, Item 2.

Also refer to Appendix 1A(N) for further discussion.

Regulatory Guide 1.45

Reactor Coolant Pressure Boundary Leakage Detection Systems

Discussion

The reactor coolant pressure boundary leakage detection system meets the requirements of this guide, dated May 1973 with the exception of the radioactive gaseous monitor leak detection sensitivity (one gpm within one hour), as discussed in Section 5.2.5.3.

Regulatory Guide 1.46

Protection Against Pipe Whip Inside Containment

Discussion

In accordance with Standard Review Plan 3.6.2, protection against pipe whip inside the CPNPP Containment Buildings meets the requirements of Regulatory Guide 1.46, dated May 1973, or Branch Technical Position MEB 3-1 except as discussed in Section 3.6B.

Regulatory Guide 1.47

Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems

Discussion

The CPNPP bypassed and inoperable status indication system presented in Section 7.1 complies with the requirements of this regulatory guide dated May 1973.

Regulatory Guide 1.48

Design Limits and Loading Combinations for Seismic Category I Fluid System Components

Discussion

Balance of Plant components are designed using the stress limits and loading combinations presented in Section 3.9B.3. The conservatism in these limits and the associated ASME design requirements precludes any component structural failure.

The operability of active Code Class 2 and 3 valves and pumps will be verified by methods detailed in Section 3.9B.3.

Alternate provisions to meeting the guidance of this regulatory guide dated May 1973, have been utilized for valves and piping as provided in Section 3.9B.3.

Also refer to Appendix 1A(N).

Regulatory Guide 1.49

Power Levels of Nuclear Power Plants

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.50

Control of Preheat Temperature for Welding of Low-Alloy Steel

Discussion

This guide, dated May 1973, is considered applicable to ASME Section III Code Class 1 components only.

The CPNPP design of Code Class 1 components is in agreement with the requirements of this regulatory guide except for Regulatory Positions 1(a) and 1(b). The recommendations of this regulatory guide are not applied to Code Class 2 and 3 components.

In the case of Regulatory Position 2, the low alloy steel welds have been made to ASME code requirements, and therefore the regulatory guide requirements are considered unnecessary.

Also refer to Appendix 1A(N) and 6.1B for further discussion.

Regulatory Guide 1.51

Inservice Inspection of ASME Code Class 2 and 3 Nuclear Power Plant Components

Discussion

This regulatory guide was withdrawn July 21, 1975.

Regulatory Guide 1.52

Design, Testing, and Maintenance Criteria for Post Accident Engineered-Safety-Feature Atmosphere Cleanup System Air Filtration and Adsorption Units of Light-Water-Cooled Nuclear Power Plants

Discussion

The CPNPP design meets the requirements of Revision 1(07/76) of Regulatory Guide 1.52, as indicated in Table 6.5-1, except part C.2.g instrumentation (see Section 6.5.1.5); part C.3.h, water drains; Part C.3.n, ductwork design; Part C.4.c, ease of maintenance; part C.4.d, test probes; Part C.5.b, Air flow distribution; Parts C.5.c, and C.5.d, in-place testing, and part C.6, laboratory testing criteria for activated carbon.

1. Part C.2.g, instrumentation:

   The flow rates and pressure drops for the ESF filtration units are not recorded or monitored from the control room. Operability of the ESF filtration units is maintained in accordance with the Technical Specifications. For those units provided for postulated DBA conditions, alarm annunciators in the Control Room are utilized to monitor design limits (e.g., flow, pressure drops, temperature, etc.) as well as system malfunction (e.g., damper, fan). This design conforms to the intent of NRC Regulatory Guide 1.52, Revision 1 (07/76).

2. Part C.3.e:

   In lieu of ORNL-NSIC-65, ANSI N509 was used for design.

3. Part C.3.f:

   In lieu of ORNL-NSIC-65 or ERDA 76-21, filter banks are arranged in accordance with ANSI/ASME N509-1980.

4. Part C.3.g:

   In lieu of ORNL-NSIC-65, ANSI N509 was used for design.

5. Part C.3.h, Water drains:

   Check valves have been used on some drains in lieu of seals or traps. The drains for the demister compartments of the four Primary Plant Ventilation System ESF filtration units have not been provided with seals, traps, or check valves. Any minimal airflow through the demister drain will be filtered prior to discharge at the stack. This design conforms to the intent of NRC Regulatory Guide 1.52, Revision 1 (07/76).

6. Part C.3.n, Ductwork Design:

   In lieu of ORNL-NSIC-65 OR ANSI/ASME N509-1976, ANSI/ASME N509-1980 is used as noted below.

   The following exception is taken to the ANSI/ASME N509-1980 standard (NRC R.G. 1.52, Rev. 2, refers to ANSI/ASME N509-1976):

   Quantitative leak testing of ductwork is limited to:

   a. All Primary Plant Ventilation System ductwork located above el. 873'-6" designated as Safety Class 3.

   b. All ductwork from the fresh air intake up to the Emergency Pressurization Units of the Control Room Air Conditioning System (CRACS).

   c. All ductwork related to the make-up air system from the fresh air intakes up to the isolation dampers (CPX-VADPOU-17, 18, 20, & 21) of the CRACS.

   d. All ductwork penetrating the Control Room envelope.

7. Part C.4.c, Ease of Maintenance:

   The spacing provided between the components of the filtration units is, in some cases, less than the minimum required by this regulatory position. However, it has been determined by operation and maintenance personnel that the spacing provided is sufficient to perform all required maintenance operations. This design conforms to the intent of Reg Guide 1.52, Revision 1 (07/76).

8. Part C.4.d, Test Probes:

   Test ports will be provided in lieu of permanently installed test probes. Permanently installed probes are a convenience in the periodic testing of filters and absorbers.

Replacing them with test ports is of no safety significance. This design meets the intent of NRC Regulatory Guide 1.52, Revision 1 (07/76).

9. Part C.5.b, Air flow distribution:

The air flow distribution to the HEPA filters and iodine absorbers will be tested in place for uniformity initially and after maintenance affecting the flow distribution for all units except the control room pressurization units.

10. Parts C.5.c and C.5.d, In-Place Testing Criteria:

ANSI/ASME N509-1980 and ANSI/ASME N510-1980 will be issued for field testing activities in place of the older versions of these codes referenced in this Regulatory Guide.

In-place testing of the HEPA filter banks and adsorber will be required following any significant painting, fire or chemical release described in position C.5.c and C.5.d of this guide. This design conforms to the intent of NRC Regulatory Guide 1.52, Revision 2 (03/78).

HEPA filter and charcoal adsorber efficiency for the Primary Plant Filtration Units is 95% which corresponds to an acceptance criteria of less than 1% for in-place penetration and bypass leakage at rated flow.

11. Part C.6, Laboratory Testing Criteria for Activated Carbon:

Laboratory testing of activated carbon will be in accordance with ANSI/ASME N509-1980, ANSI/ASME N510-1980 and ASTM-D3803-1989. The parameters applicable to new and used charcoal at CPNPP shall be applied in the lab test as follows:

New Charcoal	Used Charcoal
4" Beds[(a)]	4" Beds[(a)]
40 ft/min	40 ft/min
30°C & 95% RH	30°C & 70% RH
Pre-equilibrated	Pre-equilibrated

a) When the carbon sample is less than 4" (e.g., as a result of settling during shipment) the correction as detailed in ASTM D3803-1989 Section 9.2 may be applied.

The allowable methyl iodide penetration for 99% efficiency units (control room ventilation filtration and pressurization units) shall be 0.5% instead of 0.175% as shown in Table 2 of the Regulatory Guide.

The allowable methyl iodide penetration for 95% efficiency units (primary plant ventilation filtration ESF units) shall be 2.5% instead of 1.0% as shown in Table 2 of the Regulatory Guide.

When all or most of the charcoal canisters in the filtration units have been used, a representative sample of the charcoal from the bed will be obtained by grain thief sampling in accordance with ANSI N509-1980, Appendix A. The representative samples obtained with the grain thief will be used for laboratory testing and/or refilling the charcoal canisters.

The carbon adsorber beds will be replaced when (1) testing in accordance with the frequency specified in Footnote c of Table 2 results in a representative sample failing to pass the applicable test in Table 2 or (2) no representative sample is available for testing, at a time when testing is required.

This design conforms to the intent of NRC Regulatory Guide 1.52, revision 2 (03/78).

Regulatory Guide 1.53

Application of the Single-Failure Criterion to Nuclear Power Plant Protection Systems

Discussion

BOP systems meet the recommendations of this regulatory guide (dated June 1973) in accordance with the comments of Section 7.1.2.7.

Also refer to Appendix 1A(N).

Regulatory Guide 1.54

Quality Assurance Requirements for Protective Coatings Applied to Water-Cooled Nuclear Power Plants

Discussion

Protective Coatings inside containment were declassified from Nuclear Safety Related to Non-safety Related during construction, prior to the receipt of the Operating License. The protective coatings program is subject to a specifically structured quality assurance program as described in Appendix 17A.

A detailed “suitability of application review” of the protective coatings inside containment for Units 1 and 2 has been performed based on the guidance in EPRI Report 1003102 “Guideline on Nuclear Safety-Related Coatings”, Revision 1 (formerly TR-109937) and ASTM D 5144-00, Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants. [ER-ME-124, Evaluation of CPNPP Protective Coatings] All of the protective coatings have been evaluated and classified as either “Acceptable” or “unqualified” coatings. This evaluation concluded that all of the applied protective coatings within both containment buildings are acceptable coatings with the exception of those coatings on the Coatings Exempt Log for each unit.

As of December 31, 2007, protective coatings inside containment are classified as Service Level I, coatings. The protective coatings program for ongoing Service Level I inspections, maintenance, repair and modifications is in accordance with the following guidance:

EPRI Report 1003102 “Guideline on Nuclear Safety-Related Coatings”, Revision1 (formerly TR-109937).

ASTM D 5144-00, Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants.

For the above reasons, Regulatory Guide 1.54, “Quality Assurance Requirements for Protective Coatings Applied to Water-Cooled Nuclear Power Plants”, does not apply to Comanche Peak.

Also refer to Appendix 1A(N) and Section 6.1B.2 for further discussion.

Regulatory Guide 1.55

Concrete Placement in Category I Structures

Discussion

The CPNPP design and construction requirements and procedures for concrete placements in Category I structures are in conformance with the requirements of this regulatory guide dated June 1973.

Also refer to Section 3.8.

Regulatory Guide 1.56

Maintenance of Water Purity in Boiling Water Reactors

Discussion

This regulatory guide is not applicable to the CPNPP which has pressurized water reactor steam supply systems.

Regulatory Guide 1.57

Design Limits and Loading Combinations for Metal Primary Reactor Containment System Components

Discussion

The following is applicable to structural systems, i.e. air locks, equipment hatch, etc.

The design limits and loading combinations utilized for the CPNPP metal containment system components conform to the requirements of this regulatory guide dated June 1973.

Also refer to Section 3.8.

Regulatory Guide 1.58

Qualification of Nuclear Power Plant Inspection, Examination, and Testing Personnel

Discussion

With reference to the subject Regulatory Guide as it applies to activities during the Construction phase:

1. For inspection activities within the scope of the ASME Code, inspectors are qualified in compliance with the requirements of Regulatory Guide 1.58, Revision 1.

2. For inspection activities outside the scope of the ASME Code, inspection personnel are qualified in general compliance with the requirements of Regulatory Guide 1.58, Revision 1 except as follows:

    a. Some inspection personnel qualifications are documented on a TUGCO form and not on the Constructor's form.

    b. Some of the qualification forms are signed by a TUGCO representative and not by a Constructor representative.

3. Qualification records are collected, stored and controlled in compliance with ANSI N45.2.9 Draft 11, Revision 0 dated January 17, 1973 as included in the "Gray Book".

For Operational Phase activities, CPNPP Vistra OpCo will comply with the provisions of Regulatory Guide 1.58, Revision 1, for the qualification of quality control inspection personnel.

Also refer to Appendix 1A(N) and Section 17.2.

Regulatory Guide 1.59

Design Basis Floods for Nuclear Power Plants

Discussion

The CPNPP probable maximum flood (PMF) analysis was done prior to the issuance of R.G. 1.59. However, a detailed examination has shown that it complies with revision 2 (8/77) of this regulatory guide except as noted below:

Revision 2 of this guide differs, in part, from revision 1 in that it refers to ANSI Standard N170-1976. The CPNPP PMF analysis differs slightly from the recommended methods in ANSI N170-1976 in three areas.

The probable maximum precipitation (PMP) used in the CPNPP analysis is based on Hydrometeorological Report (HMR) No. 33. ANSI N170 - 1976 refers to HMR 33 and also to a draft version of the later HMR 51. The use of HMR 33 PMP data instead of HMR 51 has only a small effect on the calculated high water levels.

The CPNPP PMF analysis used a rainfall time distribution slightly different from the time sequence recommended by ANSI N170 - 1976. This results in no significant difference in calculated maximum water height in either the reservoir or Safe Shutdown Impoundment.

ANSI N170 - 1976 recommends using an antecedent rainfall preceding or following the PMF. The CPNPP analysis assumed the reservoir is full to the top of the conservation storage. The assumption of antecedent rainfall results in slightly higher calculated maximum water levels.

The combined effect of the three computational differences above results in calculated maximum water elevations that are within design limits. Specifically, the resulting minimum freeboard values are in excess of the required freeboard heights for protection against wave action at the peak of the flood.

Regulatory Guide 1.60

Design Response Spectra for Seismic Design of Nuclear Power Plants

Discussion

The design response spectra for the seismic design of the CPNPP plant is in conformance with Revision 1 (12/73) of this regulatory guide except in the high frequency region greater than 33 Hz of vertical response spectra where the vertical response spectra are reduced to a maximum vertical ground acceleration of 0.08g at 50 Hz. This exception is in accordance with NRC Standard Review Plan, Section 3.7.1 and is also consistent with the recommendation of Newmark, Blume and Kapur (Reference 14 of Section 3.7B) from which the vertical design response spectra of this regulatory guide was developed. Similarly, vertical response spectrum frequency of 4.0 Hz was recommended by Newmark, Blume and Kapur and used here instead of 3.5 Hz adopted in Regulatory Guide 1.60. The effect of these deviations is insignificant.

Also refer to Appendix 1A(N).

Regulatory Guide 1.61

Damping Values for Seismic Design of Nuclear Power Plants

Discussion

The damping values utilized for the seismic design of the CPNPP are in conformance with this regulatory guide dated October 1973. For piping systems analyzed by the response spectrum method, ASME Code Case N-411 damping values may be used in lieu of the damping values in this Regulatory Guide.

Also refer to Appendix 1A(N) and Section 3.7B.

Regulatory Guide 1.62

Manual Initiation of Protective Actions

Discussion

Manual initiation controls for BOP systems meet the recommendations of this Regulatory Guide in accordance with the comments of Section 7.3.2.2.7. System level manual actuation of BOP components (such as the auxiliary feedwater pumps, containment spray pumps, component cooling water pumps, etc.) is accomplished through the system level actuation devices described in Section 7.3.2.2.7, for manual initiation of safety injection, spray actuation, or steam line isolation, as appropriate to the component.

Also refer to Appendix 1A(N).

Regulatory Guide 1.63

Electric Penetration Assemblies in Containment Structures for Light-Water-Cooled Nuclear Power Plants

Discussion

The CPNPP Electric Penetration Assemblies comply with the intent of Revision 2 (7/78) of this regulatory guide. For details see Section 8.3.

Regulatory Guide 1.64

Quality Assurance Requirements for the Design of Nuclear Power Plants

Discussion

The quality assurance program for design and construction at CPNPP incorporates the intended objectives of ANSI N45.2.11 (Draft 2, Revision 2 - 5/73). The quality assurance methods for operation phase activities will comply with applicable guidance contained in Revision 2 (6/76) of this regulatory guide. The application of the requirements of ANSI N45.2.11 - 1974, as endorsed by this regulatory guide, will be in accordance with the guidance provided by ANSI N18.7 - 1976.

Also refer to Appendix 1A(N) and Section 17.2.

Regulatory Guide 1.65

Materials and Inspections for Reactor Vessel Closure Studs

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.66

Nondestructive Testing of Tubular Products

Discussion

This regulatory guide was withdrawn September 28, 1977.

Regulatory Guide 1.67

Installation of Overpressure Protection Devices

Discussion

This regulatory guide was withdrawn by the NRC on April 15, 1983 because the Winter 1978 Addenda to the 1977 edition of the ASME Boiler and Pressure Vessel Code, Appendix O, Section III, Division 1, included requirements equivalent to the recommendations in this Reg. Guide. The NRC accepted these changes in the code by reference in 50.55a of 10CFR50 (46FR20153) on April 3, 1981.

Regulatory Guide 1.68

Preoperational and Initial Startup Test Programs for Water-Cooled Power Reactors

Discussion

The testing activities conducted as part of the Initial Test Program, as described in Section 14.2, and startup program comply with the intent of this Regulatory Guide, Revision 2, dated August 1978, with the following exceptions and justification.

1. Appendix A subparagraph 1.c

   As identified in Table 14.2-2, Sheet 42A of 60 of the FSAR, the response time acceptance criteria of the various logic channels will be consistent with Technical Specifications requirements. The Reactor Trip System Response Time is defined in the technical specifications as the time interval from when the monitored parameter exceeds its trip setpoint at the channel sensor until loss of stationary gripper coil voltage. The accident analysis accounts for conservative values for delay times, setpoint drift, etc. Therefore, it is not necessary to account these delay times in the test methods or acceptance criteria.

2. Appendix A subparagraph 1.d.1, 1.e.6, 1.e.g, 1.f.1, 1.j.2 & 1.j.17

   There are no safety related functions of the turbine control and bypass valves as discussed in FSAR Section 10.2.2.7.7. The circulating water system also has no safety related function as described in FSAR Section 10.4.5. Therefore these systems are not included in the preoperational program. However, as identified by Section 14.2.1 these systems are intended to be acceptance tested in accordance with applicable startup administrative procedures.

   The feedwater control system is not required for safety but does have an interface with the protective system. These interfaces are exclusively part of the feedwater isolation valves ability to close upon the actuation of the proper logic and is covered within preoperational test summary FSAR Table 14.2-2, Sheet 50.

3. Appendix A subparagraph 1.d.3

The accident analysis concerning the inadvertent depressurization of the reactor coolant system is discussed in FSAR Section 3.9N.1.1, Upset Condition number 5. Subsection 5.b lists the condition of the inadvertent opening of one pressurizer power operated relief valve (PORV). The analysis states that the limiting case is the actuation of the pressurizer safety valve. This is classified as an Upset Condition with no operational impairment. The design parameters listed in Table 5.4-16 of the FSAR indicate that the relieving capacity for the pressurizer power operated relief valve is one-half the capacity of the pressurizer safety valve, 210,000 lb/hr vs 420,000 lb/hr. Therefore there is no intention of performing capacity tests of the pressurizer power operated relief valves during the startup phase.

Testing of these valves is listed in Table 14.2-2, Sheet 53, of the FSAR as a preoperational test.

The accident analysis covering the opening of a steam generator power operated relief valve is similar. The analysis is described in FSAR Section 15.1.4. This accident analysis uses a value of 968,400 lbm/hr @ 1200 psia. In addition, for plant cooldown requirements, a minimum capacity of 62,150 lbm/hr @ 100 psia is also required. Table 10.3-3 in the FSAR describes these requirements. Because of extensive validation program conducted by the valve vendor, the uncertainties associated with a field test and the routine calibration of the valve stem travel stop positions, there is no need to test the capacities of the steam generator power-operated atmospheric relief valves during the startup program.

The testing of the steam generator power operated relief valves is presented in FSAR Table 14.2-2, Sheet 49.

4. Appendix A subparagraph 1.k.2,3

The equipment identified in the above paragraphs of Regulatory Guide 1.68 is calibrated and functionally tested as part of the instrument calibration program for the TUGCO Chemistry and Health Physics section. The calibration and functional testing is performed and documented in accordance with approved station calibration procedures. Therefore, TUGCO Startup will not perform additional testing, in the form of a preoperational test, on this equipment.

5. Appendix A subparagraph 1.n.11

Refer to the discussion of Regulatory Guide 1.68.3.

6. Appendix A subparagraph 1.o.1

The vendor has performed applicable load testing on the head lifting and internals lifting devices for 125% static loads.

7. Appendix A subparagraph 4.t

Natural circulation tests have been successfully completed at McGuire Unit 1, Salem 2, Sequoyah 1 and other Westinghouse plants similar to CPNPP. It is unnecessary for CPNPP to compare flow (without pumps) and temperature data to that of these plants

since no design differences exist which would significantly effect natural circulation capabilities. Typical natural circulation characteristics for 4 loop Westinghouse plants are given in WCAP-8460, “Natural Circulation Test Report for Zion Station Unit 1.” However, in order to verify natural circulation cooldown and boron mixing capability per requirements of Branch Technical Position RSB 5-1, CPNPP referenced test results from Diablo Canyon Unit 1 which were found to be acceptable. See Appendix 5A for further discussion.

8. Appendix A subparagraph 5.k.k

The most influential contributor for this transient is the value of moderator temperature coefficient of reactivity, which has a relatively low value at beginning of core life. Since this parameter is determined in other startup tests, thus validating the safety analysis, the performance of this test provides no new information needed to verify the plant design. The transient does introduce the potential for thermal stress damage to the steam generator feedwater inlet nozzles and it expends one of the analyzed thermal cycles. Therefore, we do not intend to perform a test to comply with this subparagraph.

9. Appendix A, Subparagraphs 4c, 5e, 5f and 5i

The pseudo ejected rod tests and pseudo dropped rod tests referenced by these sections of the Regulatory Guide have been successfully performed at plants of similar design to Comanche Peak. Previous tests on similar facilities indicate little new information is generated by the performance of these tests. Additionally, vendor predictions indicate that a violation of the F-delta-H Technical Specification may occur if the 50% power pseudo dropped rod test is performed at Comanche Peak. Recent industry experience indicates there is an increased potential for causing severe xenon transients by performing these tests. Based on these reasons, we do not intend on performing tests to comply with Appendix A, Subparagraphs 4c, 5e, 5f and 5i. A normal function of the Digital Rod Position Indication system is to detect misaligned control rods which satisfies subparagraph 5i. The Digital Rod Position Indication system is a diverse and independent method of detecting control rod positions.

10. Appendix A, subparagraph 5.d.

CPNPP plant design does not include part-length control rods. The ability to control core xenon transients is a design feature of the Westinghouse Nuclear Steam Supply System and has been demonstrated in numerous operating pressurized water reactors. In addition, compliance with Technical Specification 3/4.2.1, “Axial Flux Difference”, ensures proper power and flux distributions. On these bases, CPNPP does not intend to perform an Initial Startup Test to comply with subparagraph 5.d.

11. Appendix A, subparagraph 5.i.i.

The performance of this test provides no new information needed to verify the plant performance during design transients. Trip of the reactor coolant pumps result in a reactor trip with flow coastdown, as verified in the Reactor Coolant System Flow Coastdown Test, providing sufficient heat removal to ensure DNBR does not decrease below the limit value. Performing this test expends one of the analyzed transients and results in unnecessary cost and down time for the utility.

12. Appendix A, subparagraph 5.u.

Operability and response times of the main steam isolation valves will be verified in hot standby (mode 3) rather than at the recommended 25% power level. Testing at hot, zero power will result in more conservative results and will eliminate the unnecessary pressure and steam flow transients which would otherwise be induced.

13. Appendix A, subparagraph 5.m.m.

The performance of this test provides no new information needed to verify the plant performance during design transients. Closure of all Main Steam Isolation Valves from 100% power causes a turbine trip and a reactor trip. A turbine trip and reactor trip from 100% power will be performed during initial startup testing. Closure of the MSIV's may cause the operation of the pressurizer and steam generator power operated relief valves and/or safety valves which may then require repair and unnecessary down time for the utility. This test would expend one of the analyzed pressure transients for the reactor coolant system and steam generators and therefore will not be performed at CPNPP.

14. Appendix A, subparagraph 2.b

Hot no flow, cold no flow and cold full flow rod drops do not provide any additional useful data. By Technical Specifications, critical operations are only permitted when plant is hot and RCP operating and thus scram testing is not required for hot no flow, cold no flow and cold full flow conditions. We do not intend to take credit for cold no flow, cold full flow and hot no flow rod drops, although one or more of these tests may be optionally performed for diagnostic or demonstration purposes.

15. Appendix A, subparagraph 5.h

Rod drop times are measured during pre-critical testing at hot full-flow conditions. There is no provision in the design of CPNPP to allow for determination of rod scram times following normal plant trips. These tests meet the intent of subparagraph 5.h.

16. Appendix A, subparagraph 5.j

The CPNPP design does not include partial scram or rod runback features. Therefore, an Initial Startup Test will not be performed to comply with subparagraph 5.j.

17. Appendix A, subparagraph 5.l

The CPNPP residual heat removal (RHR) system does not include a "steam condensing mode" of operation. Also, the "reactor core isolation cooling (RCIC) system" pertains to Boiling Water Reactors. Since these features do not exist at CPNPP, they cannot be tested during the Initial Startup Test program. The remainder of paragraph 5.l will be complied with.

18. Appendix A, subparagraph 5.a

The power coefficient measurement test method recommended by Westinghouse cannot be used because it requires installed instrumentation (narrow range hot leg temperature)

which is not part of CPNPP design. Power coefficient measurement results, obtained during the startup of several Westinghouse plants, have verified consistency between design calculations and measurement data in plants having fuel similar to the Comanche Peak design. CPNPP's Initial Startup program includes measurement of the critical RCS boron concentration under the conditions of all rods out, hot full power, equilibrium xenon. From this, the core reactivity balance is verified. This test is considered sufficiently accurate to detect any significant differences between the designed and as-built core. This test is in addition to the minimum specifications of the Regulatory Guide.

19. Appendix C, subparagraph 4.h.

The power ascension program for CPNPP requires flux maps to be taken at each power plateau. The values for $F_Q(Z)$ (heat flux hot channel factor) and $F^N{}_H$ (nuclear enthalpy rise hot channel factor) obtained from these flux maps will be compared to and evaluated against Technical Specification limits. This evaluation will be used in place of extrapolating DNBR and linear heat rate values. This action is consistent with the Reference Startup Document supplied by the fuel vendor. Station Administrative procedures will require management review and approval of test results prior to ascending to the next power plateau.

20. Section C.8, Appendix A (Various)

Testing will not be performed at the Regulatory Guide suggested 25% (30%) power. Testing originally specified at 30% RTP will be conducted at 50% RTP. Where testing had been previously performed for both 30% and 50% RTP, only the 50% RTP test will be performed.

The tests that were previously prescribed to be conducted at the 30% rated thermal power (RTP) plateau include the following:

Core Performance Evaluation
Unit Load Transients (10% Step Load Change)
Automatic Reactor Control System Test

The core performance data that could be obtained at 30% RTP is utilized for gross calibration adjustments of the Nuclear Instrumentation System (NIS) prior to power escalation to 50% RTP. This activity will be performed at 25-30% RTP as a hold prior to escalation to 50% RTP. The flux distribution measurement at 30% RTP will not be performed unless the peaking factors measured at low power do not support escalation to 70% RTP, the NIS trip setpoint for the 50% RTP testing plateau. This is per the direction of RG-1.68 Appendix C, paragraph 4.h.

For Unit 2, one low power flux map will be taken prior to exceeding 30% RTP as a fulfillment of Regulatory Guide 1.68, Appendix A, subparagraph 4.e. At power, flux maps will be taken at 50%, 75% and 100% RTP to satisfy Regulatory Guide 1.68, Appendix A, subparagraph 5.b. Additional flux maps will be taken if required by Regulatory Guide 1.68, Appendix C, subparagraph 4.h.

For Unit 1, the unit load transient at approximately 30% RTP will be performed following completion of 50% RTP plateau testing to assure proper control system response. For Unit 2, the 30% RTP unit load transient will not be performed.

The Automatic Reactor Control System test is performed at 50% RTP and intended as a precursor to the first Unit Load Transient test. It is designed to ensure that the automatic rod control system can restore the Reactor Coolant System (RCS) temperature to within a +1.5 Deg-F deadband of the reference temperature. Prior to 50% RTP, proper operation of this function would be demonstrated by observation during the normal power escalation, where the control rods will be in automatic and already controlling the RCS temperature to within the deadband.

21. Appendix A, subparagraph 5.z

For Unit 2, each radiation monitor and detector will be calibrated in lieu of only verifying response to a radiation check source as part of the preoperational test program (Table 14.2-2, Sheet 24A). The calibration of the monitors and detectors meet the intent of RG 1.68, Revision 2, Appendix A, Sections 4.g and 5.z (see FSAR 11.5.2.11) and provides a more accurate method for demonstration of their proper operation.

22. Appendix A, subparagraph 1.j.(13).

The incore neutron instrumentation portion of this subparagraph cannot be fully tested as a preoperational test program item because the system is not fully installed until after initial fuel load. The fuel assemblies provide the structural support for the incore flux mapping system thimbles, the thimbles are not self-supporting and cannot be installed until after the fuel is all in place. For this reason, the only valid demonstration of proper system function over its full operational range must be performed after initial fuel load. Portions of this testing are also customarily performed at normal operating temperatures prior to initial reactor criticality to ensure that thimble thermal expansion does not create any system operation interferences.

23. Appendix A, subparagraph 1.j.(18).

The neutron response checks of the auxiliary startup instrumentation are not included as part of the preoperational test program. This testing is performed prior to the start of initial fuel loading, but is addressed as an initial startup test. The auxiliary startup instrumentation is typically received only a short time prior to the start of initial fuel load and is tested prior to the actual fuel movements. The neutron response check portion of this testing is required to be performed within 8 hours of the actual start of initial fuel loading. This time constraint makes it impractical to retain this testing in the preoperational testing program as it allows too little time to close out the preoperational testing program prior to fuel load.

24. Appendix A, subparagraph 5.q.

The failed fuel detection system consists of the failed fuel monitor with letdown chemistry samples used as both a backup and supplement. The failed fuel monitor, itself, is one of

the process radiation monitors that are tested elsewhere under subparagraph 5.z. Additional testing of this one monitor, beyond that already planned as part of the process monitor testing, would not be expected to provide additional useful test results. Normal plant operating surveillances provide sufficient assurance of failed fuel monitor operability and of the ability to detect any actual failed fuel.

Regulatory Guide 1.68.1

Preoperational and Initial Startup Testing of Feedwater and Condensate Systems for Boiling Water Reactor Power Plants

Discussion

This regulatory guide is not applicable to the CPNPP

Regulatory Guide 1.68.2

Initial Startup Test Program to Demonstrate Remote Shutdown Capability for Water Cooled Nuclear Power Plants

Discussion

The testing activities conducted as a part of the startup test program will comply with the applicable requirements of Revision 1 (7/78) of this regulatory guide.

Also refer to Section 14.2.

Regulatory Guide 1.68.3

Preoperational Testing of Instrument Air Systems

Discussion

The CPNPP Instrument Air System testing meets with the intent of the requirements of NRC Regulatory Guide 1.68.3, Regulatory Positions C.1 through C.11, as described below:

Position C.1: CPNPP meets the intent of position C.1 by performing preoperational tests on those aspects of the system which are important to safety as described in Section 14.2. The balance of the Instrument Air System testing is performed under the acceptance testing program at CPNPP.

Position C.7: Air operated components which are required to operate in order to perform a safety function are provided with safety related air accumulators as described in section 9.3.1.1. These accumulators and their associated check valves are, for the purposes of single failure analysis, considered an extension of the component being supplied. In no case are safety related air accumulators shared between two redundant components. Therefore, CPNPP meets the required single failure criterion. Tests on these accumulators are described in FSAR Section 14.2.

Position C.8: Preoperational tests for loss of Instrument Air are performed as described in FSAR Section 14.2. Prerequisite tests are performed to verify air operated component failure positions. This testing meets the intent of position C.8.

Position C.9: Regulatory Position C.9 is not applicable to CPNPP since there is no provision for connecting the plant Service Air System to the Instrument Air System.

Position C.10: Regulatory Position C.10 addresses air header pressure transients caused by large air users. The CPNPP Instrument Air Systems only bulk load is the Turbine Generator Gas Purge. This function is manually initiated and is only used while the plant is not operational. To date, operation of the Turbine Generator Gas Purge has not revealed any problems associated with pressure transients. Therefore, no testing will be performed.

Position C.11: Regulatory Position C.11 addresses Instrument Air System overpressure transients. The CPNPP Instrument Air System is protected from overpressurization by relief valves which are individually tested. No additional overpressure testing is performed.

Also refer to Section 14.2

Regulatory Guide 1.69

Concrete Radiation Shields for Nuclear Power Plants

Discussion

The regulatory guide states that ANSI N101.6-1972, Concrete Radiation Shields, is considered applicable to shielding structures for nuclear power plants, subject to certain conditions. These conditions are stated in Regulatory Positions Nos. 1-8 of the regulatory guide. The guidance provided in this regulatory guide, dated December 1973, is adhered to as follows:

1. Regulatory Position No. 1: The codes and standards of concrete shielding are those listed in subsection 3.8.4.2 (outside the Containment).

2. Regulatory Position No. 2: This position is complied with as described in Section 3.8.

3. Regulatory Position No. 3: In the design of concrete radiation shields, there has been consideration of steady state and transient thermal loads and loads due to postulated missiles, as described in Sections 3.3, 3.8.3.3, and 3.8.4.3.

4. Regulatory Position No. 4: This position is complied with as described in Sections 3.8.3 and 3.8.4.

5. The intent of Regulatory Position Nos. 5 and 6 is complied with.

6. Regulatory Position No. 7: Provision is made for adequate means by transferring all forces through construction joints. All loads described in Sections 3.3, 3.5, and 3.8 are considered in the design.

7. Regulatory Position No. 8: Vacuum box testing of pool liner welds, or an equivalent alternate method, are considered as an acceptable supplementary method of testing liner welds or leaktight integrity.

Also refer to Section 12.3.2.

Regulatory Guide 1.70

Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants, LWR Edition

Discussion

The CPNPP FSAR conforms to the format of Revision 2 (9/75) of this regulatory guide with minor variations where needed for clarity. In a number of sections the NSSS information is separated from the BOP (i.e. 1A(N) and 1A(B), 3.6N and 3.6B). Appendices are used appropriately and incorporated by reference.

Printing/reproduction for FSAR amendments may be performed mechanically or photographically, including “Xerox-type” copier quality reproduction.

Following clarification is provided for Revisions Section (Page V):

Amendment number and date of change is provided on each affected pages of Sections and Tables. For flow diagrams, it will be provided on the index pages in lieu of each flow diagram.

Regulatory Guide 1.71

Welder Qualification for Areas of Limited Accessibility

Discussion

The CPNPP design complies with the requirements of this regulatory guide, dated December 1973, for piping only.

Also refer to Appendix 1A(N).

Regulatory Guide 1.72

Spray Pond Plastic Piping

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.73

Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants

Discussion

Safety-related motor operated valves inside Containment comply with the guidance of Regulatory Guide 1.73, dated January 1974, with the exception that stem mounted limit switches are tested separately to the requirements of IEEE Standard 382-1972.

For details see Section 3.11B.

Also refer to Appendix 1A(N).

Regulatory Guide 1.74

Quality Assurance Terms and Definitions

Discussion

This guide is not applicable to CPNPP design and construction. The quality assurance provisions for operating phase activities are in accordance with the guidance of ANSI N45.2.10 - 1973, as endorsed by this regulatory guide dated February 1974.

Also refer to Section 17.2.

Regulatory Guide 1.75

Physical Independence of Electric Systems

Discussion

CPNPP design complies with the intent of Revision 1 (1/75) of this regulatory guide with the following comments:

Regulatory Position C.1 - Non-Class 1E power or control circuits may be isolated from their Class 1E power source by two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective device, and the circuit breakers periodically tested. Non-safety instrument circuits powered from distribution panels 1PC1, 1PC2, 1PC3, and 1PC4 will have a non-safety circuit breaker or fuse connected in series with the panel circuit breaker.

The non-Class 1E AC essential lighting circuits use interconnecting cable (i.e., from the distribution panel feeder breaker to the lighting load) routed in conduit. The routing of the circuits in conduit ensures the physical and electrical independence from Class 1E circuits beyond the isolation breaker.

The non-Class 1E DC emergency lighting circuits connected to dedicated batteries are routed in conduit. The routing of the circuits in conduit ensures physical and electrical independence from Class 1E circuits.

The lighting circuits routed in conduit meet the separation criteria of FSAR Section 8.3.1.4.

Electrically, Class 1E circuits are equal or superior to non-Class 1E circuits. Physically, Class 1E circuits are subject to more stringent analysis than non-Class 1E circuits. Therefore, devices qualified for non-Class 1E-to-Class 1E isolation applications provide more than adequate train-to-train isolation.

Separation between Radiax and Heliax radio antenna cables and Class 1E cables is not required. The analysis is provided in Section 8.3.1.4(4).

Separation between Thermistor and Class 1E cables is not required. The analysis is provided in Section 8.3.1.4(4).

Fiber optic cables used in non-Class 1E monitoring circuits carry no electrical energy by themselves and therefore are not required to maintain physical separation from Class 1E circuits.

Lesser internal wiring separation is being used between redundant safety systems and safety and non safety systems in BOP Analog Process Instrumentation Panels. This analysis is provided in Section 7.1.2.2.

The non-Class 1E diesel generator neutral grounding transformer is connected to the neutral of the Class 1E diesel generator. An analysis has been performed which demonstrates that a fault on the non-Class 1E portion of the circuit will not cause an unacceptable influence on the Class 1E system. In addition, the interconnecting cable is routed within the diesel generator room. The cable is routed in dedicated raceway and is inspected to Class 1E requirements.

Isolation between Non-Class 1E cables originating from Non-Class 1E, 6.9kV switchgear reactor coolant pump and potential transformer Class 1E fuse, is not required based on the analysis provided in Section 8.3.

Isolation between the Class 1E pressure indicating switch contact and Non-Class 1E time delay relay contact in Emergency Diesel Generator control panels CP1-ECCPEC-01 and 02 is not required based on the analysis provided in Section 8.3.

Isolation between Non-class 1E cables originating from the annuniciator in Spent Fuel Pool Panel CPX-EIPRLV-06 and Class 1E electronic boxes X-LY-4849A and B is not required based on the analysis provided in Section 8.3.

Isolation between the Non-Class 1E coaxial cable and the epoxy mica capacitor (EMC) which forms part of the Class 1E Partial Discharge Monitor Bus Coupler connected to each of the Unit 1 and Unit 2 Station Service Water Pump and Component Cooling Water motor feeders and Emergency Diesel Generators, is not required based on the analysis provided in Section 8.3.

Regulatory Position C.2 - For the purpose of electrical cable separation, acceptable barriers include rigid metal conduit, electrical metallic tubing (EMT), flexible metallic conduit, cable tray covers (both solid and ventilated types), cable bus enclosures, equipment and device enclosures, enclosed metal wireways inside equipment, a wrap of woven silicon dioxide and one hour fire rated materials (i.e. thermolag and one hour fire rated cable).

A wrap of woven silicon dioxide, thermolag and one hour fire rated cable are considered equivalent to a metal enclosed raceway with respect to protection from electrical failures.

Thermolag and one hour fire rated cable shall only be used as a Regulatory Guide 1.75 electrical separation barrier when installed to satisfy the requirements described in FSAR Section 9.5.1.2.

Metal Clad (MC) cables include copper sheathed (CS) cable, aluminum sheathed (ALS) cable and Galvanized Steel Sheathed Cable (GS). MC cables are considered the same as cable inside conduit for separation purposes. See Section 8.3 for the analysis and discussion on MC cables.

CS cable is constructed of continuous corrugated 16 mil thick copper tube with no outer jacket and 600V XHHW, 90°C insulation. CS cable will be used only inside the containment and only in the lighting system.

ALS cable is constructed of continuous corrugated 25 mil. thick seamless aluminum tube with an outer thermosetting chlorosulphonated polyethylene jacket and 600V XHHW, 90° C insulation. ALS cable will only be used outside of the containment building in the lighting, fire protection, heat tracing and communication systems.

The one hour fire rated cable provides a one hour fire rated barrier per ASTM standard E 119-1971. The cable is Class 1E qualified per IEEE 323-1974 and IEEE 383-1974 for flame retardancy (unaged cables only). See FSAR Section 1A(B), Regulatory Guide 1.131. The cable is constructed of a continuously welded corrugated 12 mil thick stainless steel sheath with high temperature nickel-clad copper conductors, glass braid cable jacket and silicone rubber insulation. This cable will be used to satisfy the fire safe shutdown requirements, as described in FSAR Section 9.5.1.2, in power and control circuits outside containment where the total radiation dose is less than or equal to 50 MRADS gamma. Cable sizes will be and smaller. The cable is considered equivalent to cable in conduit for electrical separation purposes.

Regulatory Position C.4 - The multiconductor cable between the Safety System Inoperable Indication (SSII) logic panel located in the control room and the termination cabinet in the cable spreading room meet the requirements of this position except for Class 1E environmental qualification. The cable materials meet IEEE Standard 383-1974 for the attributes of flame and radiation resistance. The cable is installed in a mild environment. Based on analysis, a fault at the SSII logic panel will not impact the availability of adjacent Class 1E cables with which these associated cables are routed. This analysis is provided in Section 8.3.

Regulatory Position C.6 - Lesser separation, based on RG 1.75 Rev. 3, is being used for interaction involving instrumentation cables only. This analysis is provided in Section 8.3.

Lesser separations are being used in several locations between Class 1E wiring and non-Class 1E Area Radiation Monitoring detector wiring and Public Address System speaker wiring based on analysis. This analysis is provided in Section 8.3.

Separation between Class 1E circuits and non-Class 1E Emergency Diesel Generator stator RTD circuits is not required based on analysis. This analysis is provided in Section 8.3.

Separation between Class 1E Cables and Non-Class 1E cables at the Electronics Boxes (X-LY-4849A-1, X-LY-4849A-2, X-LY-4849B-1 and X-LY-4849B-2) furnished by FCI is not required based on analysis. This analysis is provided in Section 8.3.

Separation between associated circuits and non-Class 1E circuits in fire panels CPX-EIPRLV-29, CPX-EIPRLV-29A, and CPX-EIPRLV-30 is not required based on analysis. This analysis is provided in Section 8.3.

Separation between associated cables and non-Class 1E cables at the Safety System Inoperable Indication panels is not required based on analysis. This analysis is provided in Section 8.3.

Separation between the output circuits of the spare inverters is not required based on the analysis provided in Section 8.3.

Separation between non-Class 1E cables originating from 6.9kV switchgear (Reactor Coolant Pump) and Class 1E PT fuses inside the Class 1E PT cabinet is not required based on the analysis. This analysis is provided in Section 8.3.

Separation between associated cables and non-Class 1E Regulating transformers TXEC1, TXEC2, TXEC3 and TXEC4 is not required based on analysis. This analysis is given in Section 8.3

Separation between associated cables and non-class 1E cable CP2-EIPRLV-48 is not required based on analysis. This analysis is given in section 8.3.

Minimum physical separation between class 1E cables and nonclass 1E differential relay (87/ST1 and 87/ST2) protection cables are not required inside the 6.9 kV class 1E switch gear based on analysis. This analysis is provided in Section 8.3.

Separation between non-Class 1E Train C cables connected to the line and load terminals of breaker CP1-BSDSEB-01, CP2-BSDSEB-01 and Class 1E Train B cables connected to the shunt trip coil are adequate to ensure that the non-Class 1E cabling will not degrade the Class 1E system based on analysis provided in Section 8.3.1.2.1.7.l.

When a cable, conduit or tray is protected by One-Hour Fire rated Thermo-lag installed to satisfy the requirements as described in FSAR section 9.5.1.2; there is no separation requirement from the Thermo-lag protected cable, conduit or tray, to the redundant cable, conduit or tray outside the Thermo-lag enclosure based on analysis. This analysis is provided in section 8.3.

Associated cables used to selectively connect the non-Class 1E alternate power diesel generators to the Class 1E 6.9kV switchgear are installed in outdoor raceway sections that are not seismically supported. These cables do not have Class 1E protection for over current and faults. This is based on the analysis provided in Section 8.3.

Separation/Isolation between associated control cables and non-Class 1E control cables at Instrument Air Compressor Termination Cabinet CP1-CICACO-01A is not required based on analysis. This analysis is provided in section 8.3.

Physical separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits inside the Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A/02A is not required based on the analysis provided in Section 8.3.1.4.5.

Regulatory Position C.9 - Splice type connections have been used to terminate field routed cables in raceways. Such splices are utilized in CPNPP design at:

1. Electric penetration assemblies (EPAs), Core Exit Thermocouple (CET) Integral Reference Junction (IRJ), Control Rod Drive Mechanism (CRDM) disconnect panel connectors and Electric Conductor Seal Assemblies (ECSAs) pigtail cables

2. Solenoid valves, limit switches, level switches, etc. (local mounted devices - LMDs)

3. Connection of LMDs to Electric Conductor Seal Assembly (ECSA) pigtails.

An analysis to justify cable splices in raceways is provided in Appendix 8A.

Regulatory Position C.10 - Cable jackets requiring field color coding are color coded at intervals not to exceed every 5 feet for exposed cable runs (cable not in raceway, i.e., cable not in cable tray, conduit or pull box) and at each end.

Regulatory Position C.12 - Power circuits for the following equipment located inside the Control Room complex, are routed in exclusive conduits within the Control Room complex:

Regulating transformers TXEC3 and TXEC4, and air conditioners CP1- VAACTC-01, CPX-VAACTC-01 and CP2-VAACTC-01.

The ratings of the above equipment do not categorize them as high energy equipment. Regulating transformers TXEC3 and TXEC4 and the air conditioners listed above are located in the Control Building Mezzanine Area, which is separated from the Control Room by a Seismic Category II partial height gypsum wall and concrete floor.

For details see Section 8.3.

Also refer to Appendix 1A(N).

Regulatory Guide 1.76

Design Basis Tornado for Nuclear Power Plants

Discussion

The CPNPP is designed to conform to the requirements of this regulatory guide, dated April 1974, except that it is designed to withstand the effects of a Design Basis Tornado having a maximum wind speed of 360 mph which is made up of a rotational speed of 300 mph and a translational speed of 60 mph. A simultaneous pressure drop of 3.0 psi at the rate of 1.0 psi per second is considered.

The Design Basis Tornado for CPNPP was determined prior to the issuance of this Regulatory Guide and was approved for use by the Atomic Energy Commission's Safety Evaluation Report Dated September 3, 1974.

Also refer to Section 3.3.

Regulatory Guide 1.77

Assumptions Used for Evaluating a Control Rod Ejection Accident for Pressurized Water Reactors

Discussion

The analysis of the radiological consequences of a control rod ejection accident presented in Section 15.4.8 follows the guidance provided in Regulatory Guide 1.195 instead of that in Regulatory Guide 1.77.

Regulatory Guide 1.78

Assumptions for Evaluating the Habitability of a Nuclear Power Plant Control Room During a Postulated Hazardous Chemical Release

Discussion

The CPNPP design meets the intent of this regulatory guide, dated June 1974, as discussed in Sections 2.2 and 6.4.

Regulatory Guide 1.79

Preoperational Testing of Emergency Core Cooling Systems for Pressurized Water Reactors

Discussion

The Initial Test Program, as described in Section 14.2, is in compliance with the provisions of Revision 1 (9/75) of this regulatory guide with the following exception.

1. Regulatory Position C.1.b(2)
   Recirculation Test - Cold Conditions

   A satisfactory in-plant test of the containment sump to demonstrate vortex control and acceptable pressure drops across screening and suction lines and valves was not practical for reasons which include physical limitations, difficulty of cleanliness control and the possibility of equipment damage.

   However, a full scale model of the Containment Recirc sumps, screens and surrounding area was used to demonstrate that unacceptable vortex formation in the sump area is precluded while simulating operation under various flow and pump combinations. In addition, the inlet loss coefficient across the sump screens and sump intake piping configuration was evaluated for comparison to analytically determined values and to verify the adequacy of new positive suction head at the pumps.

   The capability to realign valves and injection pumps to recirculate coolant from the containment floor was demonstrated. The full flow test was performed with the RHR pumps taking suction from the Refueling Water Storage Tank (RWST) and delivering to the RCS. The RHR pumps were not running when demonstrating the realignment capability. The net positive suction head was determined from the level in the RWST and

in the containment sump and shown to be greater than the required net positive suction head for the pump.

The lines from the containment sump to the RHR pumps were flushed and inspected to ensure that they are free from obstruction.

For details refer to Table 14.2-2, Sheet 15.

Regulatory Guide 1.80

Preoperational Testing of Instrument Air Systems

Discussion

Regulatory Guide 1.80 was superseded by Regulatory Guide 1.68.3.

Refer to the discussion of Regulatory Guide 1.68.3.

Regulatory Guide 1.81

Shared Emergency and Shutdown Electric Systems for Multi-Unit Nuclear Power Plants

Discussion

The CPNPP design complies with the provisions of Revision 1 (1/75) of this regulatory guide with an exception to Regulatory Position C1. For details see Section 8.3.1.2.1 Item 8.

Regulatory Guide 1.82

Sumps for Emergency Core Cooling and Containment Spray Systems

Discussion

The containment recirculation sump design in the CPNPP complies with the intent of the Regulatory Positions of this regulatory guide dated June 1974 with clarifications as noted below. The description and design evaluation of the containment recirculation sumps is provided in Section 6.2.2.

C.3 and C.8 - The screen design with (1) an outer trash rack and (2) a fine inner screen mounted on a structure with a solid top cover has been replaced by a strainer which incorporates the function of both the screen and the trash rack. The strainer is located on the lowest floor elevation in the containment exclusive of the reactor vessel cavity. The strainer is fully submerged prior to switchover of containment spray from injection to recirculation for LOCAs and secondary pipe breaks. This design meets the intent of the Reg. Guide.

C.4 and C.6 - The floor level in the vicinity of the sumps does not slope away from the sumps. However, the design meets the intent of the Reg. Guide by the provision of a one foot tall solid debris interceptor surrounding the sump strainers.

Regulatory Guide 1.83

Inservice Inspection of Pressurized Water Reactor Steam Generator Tube

Discussion

The inservice inspection of steam generator tubes complies with the intent of Revision 1 (7/75) of this regulatory guide.

Regulatory Guide 1.84

Design and Fabrication Code Case Acceptability ASME Section III Division 1

Discussion

Code cases listed in this regulatory guide will be used where appropriate.

By reference to ASME Section III requirements in the procurement specifications, the use of code cases by mechanical equipment suppliers requires mutual consent of the Owner or his agent and the manufacturer. The ASME Code Cases which are used for design and erection at CPNPP are identified in the appropriate mechanical design and erection specifications or the Brown & Root QA Manual; conditionally-approved Code Cases will show justification for their use, as required by NRC, in these documents. The application of ASME Code Cases is documented on the ASME Data Report Forms.

ASME Code Case N-31-1 regarding the use of elastomer diaphragm valves in ASME Class 2 and 3 applications has been applied to CPNPP. This Code Case was listed as having contingent approval in Regulatory Guide 1.84 Rev. 29 which was applicable when the Code Case was originally applied on April 15, 1994. The Code Case with the following conditions imposed by the Regulatory Guide should apply:

1. The service life of the elastomer should not exceed the manufacturer's recommended service life. This service life should not exceed 1/3 of the minimum cycle life established by the requirements of paragraph 3 of the Code Case. In addition, the service life of the elastomer diaphragm should not exceed 5 years.

2. The combined service life and storage life of the elastomer diaphragm should not exceed 10 years.

Also refer to Appendix 1A(N).

Regulatory Guide 1.85

Materials Code Case Acceptability ASME III Division 1

Discussion

Code cases listed in this regulatory guide will be used where appropriate.

By reference to ASME Section III requirements in the procurement specifications, the use of code cases by mechanical equipment suppliers requires mutual consent of the Owner or his agent and the manufacturer. The ASME Code Cases which are used for design and erection at CPNPP are identified in the appropriate mechanical design and erection specifications or the Brown & Root QA Manual; conditionally-approved Code Cases will show justification for their use, as required by NRC, in these documents. The application of ASME Code Cases is documented on the ASME Data Report Forms.

Also refer to Appendix 1A(N).

Regulatory Guide 1.86

Termination of Operating Licenses for Nuclear Reactors

Discussion

The termination of the operating license and subsequent decommissioning of the Comanche Peak Nuclear Power Plant will address the regulations in effect at that time.

Regulatory Guide 1.87

Guidance for Construction of Class 1 Components in Elevated-Temperature Reactors (Supplement to ASME Section III Code Cases 1592, 1593, 1594, 1595 and 1596)

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.88

Collection, Storage, and Maintenance of Nuclear Power Plant Quality Assurance Records

Discussion

The quality assurance program for design and construction at CPNPP incorporates the intended objectives of ANSI N45.2.9 (Draft 11, Revision 0, 1/73). During the operations phase, the quality assurance methods and administrative controls utilized for controlling and storing QA records, as described in Section 17.2, will comply with the applicable guidance provided in ANSI N45.2.9 - 1974, as endorsed by Revision 2 (10/76) of this regulatory guide; except for paragraph 3 of section 5.6. For this exception, CPNPP will comply with paragraph 3 of section 5.6.1 of ANSI N45.2.9-1979, which requires a minimum two hour rating for the structure, doors, frames and hardware of the storage facility. Application of the requirements of Revision 2 (10/76) of this regulatory guide will be in accordance with the guidance provided in ANSI N18.7 -1976.

To ensure adequate protection and timely processing of quality related documentation prior to transmittal of this documentation to the permanent records storage facility for retention, the following control measures shall be implemented in accordance with approved procedures:

1. Within 60 days from the time that the work activity was completed, quality related documentation shall be processed through any required post work or closure reviews and transmitted to the permanent records storage facility for retention;

2. Once the work activity has been completed, the associated quality related documentation shall be provided adequate protection from damage, deterioration, loss and unauthorized alteration.

Also refer to Section 17.2.17.

Regulatory Guide 1.89

Qualification of Class IE Equipment for Nuclear Power Plants

Discussion

The CPNPP Class 1E equipment design complies with the qualification program guidelines as delineated in this regulatory guide dated November 1974. However, as supported by the statements of consideration for 10CFR50.49 (Federal Register, 48FR2731, January 21, 1983), the recommendations of this regulatory guide need not be applied for Class 1E equipment located in a mild environment area.

Also refer to Appendix 1A(N).

Regulatory Guide 1.90

Inservice Inspection of Prestressed Concrete Containment Structures with Grouted Tendons

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.91

Evaluation of Explosions Postulated to Occur on Transportation Routes Near Nuclear Power Plant Sites

Discussion

No transportation routes in the vicinity of CPNPP are within the distances established by this regulatory guide dated January 1975.

Revision 1 (2/78) of this guide is addressed in Section 2.2.3.

Regulatory Guide 1.92

Combining Modal Responses and Spatial Components in Seismic Response Analysis

Discussion

Modal responses and spatial components are combined in the seismic response analysis of the CPNPP in conformance with Revision 1 (2/76) of this regulatory guide.

Also refer to Appendix 1A(N) and 3.7B.

Regulatory Guide 1.93

Availability of Electric Power Sources

Discussion

The CPNPP design complies with the requirements of this regulatory guide dated December 1974.

Also refer to Section 8.3.

Regulatory Guide 1.94

Quality Assurance Requirements for Installation, Inspection, and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP design and construction activities. The quality assurance methods for operations phase activities will comply with applicable guidance contained in Revision 1 (4/76) of this regulatory guide. The application of the requirements of ANSI N45.2.5 -1974, as endorsed by this regulatory guide, will be in accordance with the guidance provided in ANSI N18.7 - 1976.

Also refer to Section 17.2.

Regulatory Guide 1.95

Protection of Nuclear Power Plant Control Room Operators Against an Accidental Chlorine Release

Discussion

The CPNPP design complies with the intent of Revision 1 (1/77) of this regulatory guide as described in Section 2.2.

Regulatory Guide 1.96

Design of Main Steam Isolation Valve Leakage Control Systems for Boiling Water Reactor Nuclear Power Plants

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.97

Instrumentation for Light-Water-Cooled Nuclear Power Plants to Assess Plant Conditions During and Following an Accident

Discussion

A plant specific analysis of the information system requirements for CPNPP was conducted. This analysis was based on the design basis accident event scenarios and required operator actions, using the guidance provided in U.S. NRC Regulatory Guide 1.97, Revision 2. As a result, specific variables were selected and specific design and qualification criteria developed to assure the safety of CPNPP. These variables and criteria are described in detail in Section 7.5.

Regulatory Guide 1.98

Assumptions Used for Evaluating the Potential Radiological Consequences of a Radioactive Offgas System Failure in a Boiling Water Reactor

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.99

Effects of Residual Elements on Predicted Radiation Damage to Reactor Vessel Materials

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.100

Seismic Qualification of Electric Equipment for Nuclear Power Plants

Discussion

The CPNPP design for Class 1E Electrical Equipment complies with the requirements and recommendations of Revision 1 (8/77) of this regulatory guide with the following clarification:

The intent of Section C.1 is met by performing an analysis to verify the approach and/or by following the Equivalent Static Load Method.

Also refer to Section 3.10B and Appendix 1A(N).

Regulatory Guide 1.101

Emergency Planning for Nuclear Power Plants

Discussion

Section 13.3 Appendix A, “Emergency Planning,” is intended to comply with the contents of Revision 3 (8/92) of this regulatory guide with the following exception:

Vistra OpCo takes exception to the part of Section C which states licensees may use either NUREG 0654/FEMA-REP-1 or NUMARC/NESP-007 in developing their EAL scheme but may not use portions of both methodologies. CPNPP EALs were developed and implemented in accordance with the guidance in the NRC-endorsed NEI 99-01 Rev. 6, "Development of Emergency Action Levels for Non-Passive Reactors.”

Regulatory Guide 1.102

Flood Protection for Nuclear Power Plants

Discussion

Plant grade for the CPNPP is located above the PMF level, and Category I structures, with the exception of the Service Water Intake Structure and the Electrical and Control Building, are not subject to flooding.

The operating deck and safety-related equipment in the Service Water Intake Structure are located above the PMF level. Flooding of the Electrical and Control Building is prevented by the use of incorporated barriers (i.e., isolation valves and/or stop-gates in the Circulating Water System). Emergency operating procedures are discussed in Section 3.4 and 13.5. Therefore the CPNPP is in conformance with Revision 1 (9/76) of this regulatory guide.

Also refer to Sections 2.4 and 3.4.

Regulatory Guide 1.103

Post-Tensioned Prestressing Systems for Concrete Reactor Vessels and Containments

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.104

Overhead Crane Handling Systems for Nuclear Power Plants

Discussion

This regulatory guide was withdrawn August 16, 1979.

CPNPP cranes and hoists which are required to be single failure proof are described in “CPNPP Final Response to NUREG-0612”, dated June 1983. Updates to this response are made via DBD-ME-006 which is incorporated by reference (See Table 1.6-1). Also refer to sections 9.1.4.2 and 9.1.4.3.

Regulatory Guide 1.105

Instrument Setpoints

Discussion

The CPNPP design complies with the provisions of Revision 2 (2/86) of this regulatory guide. Setpoint securing devices do not have separate physical locks (requiring an administratively controlled key). The adjustment of setpoints and release/securing of securing devices is controlled by administrative procedures.

Also refer to Appendix 1A(N) and Section 7.1.

Regulatory Guide 1.106

Thermal Overload Protection for Electric Motors on Motor-Operated Valves

Discussion

The CPNPP design complies with the provisions of Revision 1 (3/77) of this regulatory guide as described in Section 8.3.1.1.11 Item 2.

Regulatory Guide 1.107

Qualifications for Cement Grouting for Prestressing Tendons in Containment Structures

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.108

Periodic Testing of Diesel Generators Used As Onsite Electric Power Systems at Nuclear Power Plants

Discussion

The CPNPP design complies with the intent of Revision 1 (8/77) of this regulatory guide.

1. The requirements of position C. 1b.4 are satisfied as described below:

    a. Diesel generator (D-G) “Ready Stand by” status indication in the Control Room can be acknowledged by the combination of a green light for output breaker and the absence of a “D-G disable” and a “D-G trouble” annunciator indication.

b. "D-G Lockout" status is indicated by means of a "D-G disable" annunciation in the Control Room. This status is also indicated by "DG PWR" indication on the safety system bypass and inoperable status indicating light box in the Control Room due to any of the following:

1. DC power unavailable

2. Remote-Local-Maintenance switch not in remote on the generator control panel.

c. "D-G undertest" status light is not provided in the CPNPP design.

d. A means of communication is provided between the diesel generator room and the Control Room.

2. The requirements of Regulatory Positions are clarified as described below:

Regulatory Position C.2.a.(4)

The largest single load is taken to be the largest single end load and not combined distribution system loads (e.g., distribution system transformers or load centers). This load is 783 kW equal to the nameplate rating of the Component Cooling Water pump.

Regulatory Position C.2.d

If the number of failures in the last 100 valid tests is seven or more for an individual diesel generator unit, the reliability of that diesel generator requires special evaluation (i.e., the reliability of both diesel generators is not impacted).

3. The CPNPP periodic testing program meets the intent of the regulatory guide with the following exceptions:

a. Regulatory Position C.2.a.3

During preoperational testing, testing will be conducted at the full-load-carrying capability for an interval of not less than 24 hours, of which 22 hours will be at the continuous rating capability of the diesel generator and 2 hours at a load equivalent to the two hour rating of the diesel generator. At a frequency specified in accordance with Technical Specification 5.5.21, "Surveillance Frequency Control Program", testing will be conducted to demonstrate full-load-carrying capability for an interval of not less than 24 hours, of which 22 hours will be at a load which exceeds the maximum expected diesel generator load requirements and 2 hours at a load which is approximately 110% of this maximum expected load.

b. Regulatory Position C.2.a.5

Demonstrate functional capability at full load temperature conditions by rerunning the test phase outlined in Regulatory Position C.2.a.1 and by demonstrating proper operation for shutdown-loading-sequence to shutdown-load requirements

immediately following the performance of C.2.a.3, except during preoperational testing when the proper operation for the design-accident-loading-sequence to design-load requirements will be demonstrated.

c. Regulatory Position C.2.a.9

Demonstrate the reliability of the Emergency Diesel Generator by performing at least a total of 69 (but no less than 35 per Emergency Diesel Generator) consecutive valid start and load tests on the Emergency Diesel Generators from cold ambient conditions to at least 50 percent continuous rating for at least 1 hour with no failures. These tests need not be performed as part of the pre-operational test program, but may be performed at any time prior to the required operability of the Emergency Diesel Generator.

d. Regulatory Position C.2.c.2

Periodic testing of the diesel generator units during normal plant operation to demonstrate full-load-carrying capability will be conducted at a load which exceeds the maximum expected diesel generator load requirements. During this test, loading to the diesel generator is accomplished in accordance with vendor recommendations.

e. Regulatory Position C.2.d

Diesel generators testing frequency is in accordance with the requirements of the technical specifications.

4. The requirements of Regulatory Position c.3.b is satisfied as described below:

Other existing reporting requirements adequately ensure that the NRC learns of significant problems with diesel generator performance. Reporting requirements for EDG failures provided in this section will not be submitted.

Regulatory Guide 1.109

Calculation of Annual Doses to Man from Routine Releases of Reactor Effluents for the Purpose of Evaluating Compliance with 10 CFR Part 50, Appendix I

Discussion

Compliance with Appendix I was evaluated in accordance with Revision 1 (10/77) of this regulatory guide.

Also refer to Appendix 11A.

Regulatory Guide 1.110

Cost-Benefit Analysis for Radwaste Systems for Light-Water-Cooled Nuclear Power Reactors

Discussion

This regulatory guide is not applicable to the CPNPP.

Regulatory Guide 1.111

Methods for Estimating Atmospheric Transport and Dispersion of Gaseous Effluents in Routine Releases from Light-Water-Cooled Reactors

Discussion

The methods of Revision 1 (7/77) of this regulatory guide were used in the evaluation of gaseous effluents as described in Appendix 11A.

Regulatory Guide 1.112

Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Light-Water-Cooled Power Reactors

Discussion

The radioactive source terms calculation presented in Section 11.1 complies with the requirements of Revision O-R (4/76 - reissued 5/77) of this regulatory guide.

Regulatory Guide 1.113

Estimating Aquatic Dispersion of Effluents from Accidental and Routine Reactor Releases for the Purpose of Implementing Appendix I

Discussion

The methods of Revision 1 (4/77) of this regulatory guide were used in the evaluation of effluents as described in Appendix 11A.

Regulatory Guide 1.114

Guidance on Being Operator at the Controls of a Nuclear Power Plant

Discussion

The CPNPP administrative operating procedures will incorporate the applicable guidance of Revision 1 (11/76) of this regulatory guide.

Also refer to Section 13.5.

Regulatory Guide 1.115

Protection Against Low Trajectory Turbine Missiles

Discussion

The turbines of the CPNPP are so oriented that low trajectory turbine missiles cannot strike any of the essential systems as discussed in Section 3.5. The CPNPP design is in compliance with Revision 1 (7/77) of this regulatory guide.

Regulatory Guide 1.116

Quality Assurance Requirements for Installation, Inspection, and Testing of Mechanical Equipment and Systems

Discussion

This regulatory guide is not applicable to CPNPP design and construction. The quality assurance methods for operations phase activities will comply with the applicable guidance contained in Revision 0-R (5/77) of this regulatory guide. Application of the requirements of ANSI N45.2.8 - 1975, as endorsed by this regulatory guide, will be in accordance with guidance contained in ANSI N18.7 - 1976.

Also refer to Section 17.2.

Regulatory Guide 1.117

Tornado Design Classification

Discussion

The CPNPP structures, systems, and components are designed to withstand the effects of a design basis tornado, including tornado missiles, in conformance with Revision 1 (4/78) of this regulatory guide.

Also refer to Section 3.5.

Regulatory Guide 1.118

Periodic Testing of Electric Power and Protection Systems

Discussion

Although the implementation of this regulatory guide, dated June 1976, does not include CPNPP, the CPNPP electrical power systems are testable to the extent described in FSAR Section 8.2.1.2.2, 8.3.1.1.11 item 3 and 8.3.2.1 item 6.

Regulatory Guide 1.119

Surveillance Program for New Fuel Assembly Designs

Discussion

This regulatory guide was withdrawn June 23, 1977.

Regulatory Guide 1.120

Fire Protection Guidelines for Nuclear Power Plants

Discussion

The fire protection system design for CPNPP is in accordance with the criteria outlines in Appendix A to Branch Technical Position APCSB 9.5-1, “Guidelines for Fire Protection for Nuclear Power Plants Docketed Prior to 7/1/76.” The fire protection system design complies with the requirements for plants under construction except as stated in Section 9.5.1 of the FSAR. Where the guidelines of Revision 1 (11/77) of Regulatory Guide 1.120 are comparable with those of BTP APCSB 9.5-1, the former criteria are employed to the extent practical for the CPNPP design.

The CPNPP Fire Protection Program is described in Section 9.5.1.6.1 and Appendix 13.3B.

Regulatory Guide 1.121

Bases for Plugging Degraded PWR Steam Generator Tubes

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.122

Development of Floor Design Response Spectra for Seismic Design of Floor Supported Equipment or Components

Discussion

Floor Design Response Spectra are developed in accordance with the methods described in Revision 1 (2/78) of this regulatory guide with the following conservative clarification to C.2:

In constructing instructure response spectra, uncertainties inherent to the analysis, such as the material properties of the foundation material and the structures, damping values, soil structure interaction, approximations in modeling techniques and computation of structural natural frequencies, are accounted for by parametric variations incorporated into the analysis and by broadening the peaks of the resulting envelope response spectra.

The procedure of parametric variations consists of evaluating and using in the dynamic analysis lower bound, best estimate, and upper bound values for the foundation spring constants in the case of all seismic Category I structures with the exception of the Fuel Building and the Service Water Intake Structure, where only lower bound and upper bound values are used. In addition, the analysis of the Containment Building is performed for each set of foundation spring constants by considering a cracked and an uncracked containment wall. Then, before smoothing of the envelope spectra, which envelopes the spectra developed for each parametric variation, a minimum frequency shift to the peaks of + 10 percent is also incorporated to account for other variations in the structural properties which are not considered in the described parametric

variation approach. This approach is more conservative than applying a peak shifting to spectral curves generated from a single dynamic model, without considering parametric variation.

The Fuel Building was re-analyzed to determine the effects of added mass due to the addition of high density spent fuel storage racks. Best estimate soil properties were used in the re-rack analysis in lieu of parametric variations involving upper and lower bound soil properties. The resulting spectrum peak responses were widened by at least +/- 15% in accordance with RG 1.122. The results from this approach remain conservative since results of previous studies show that variations in soil properties and building structure stiffness have minimal effect on the dynamic response of the Fuel Building.

Also see Section 3.7B.2.5.

Regulatory Guide 1.123

Quality Assurance Requirements for Control of Procurement of Items and Services for Nuclear Power Plants

Discussion

The quality assurance program for design and construction at CPNPP incorporates the intended objectives of ANSI N45.2.13 (Draft dated 5/73). The quality assurance methods and administrative controls utilized in the procurement of quality-related items and services, during the operations phase, as described in Section 17.2, will be consistent with the applicable guidance contained in Revision 1 (7/77) of this regulatory guide. The application of the provisions of ANSI N45.2.13 - 1976, as endorsed by this regulatory guide, will be in accordance with the guidance provided in ANSI N18.7 - 1976 except for imposing quality assurance program requirements consistent with 10CFR50 Appendix B or ANSI N45.2 in procurement documents for commercial grade calibration services from a NVLAP or A2LA accredited calibration laboratory evaluated in accordance with Section 17.2.7.

Regulatory Guide 1.124

Design Limits and Loading Combinations for Class 1 Linear-Type Component Supports

Discussion

All non-NSSS supplied Class I linear-type supports comply with Revision 1 (1/78) of this regulatory guide, with the following exceptions:

1. Paragraph C.4 (which is also referenced in Paragraphs C.5 and C.7)

   Service limits for bolts shall be increased for Service Level D (faulted) plant conditions. The increased allowable permitted for tensile stress in bolts shall not exceed the lesser of 0.7 Su or Sy at temperature. The increased allowable permitted for shear stress in bolts shall not exceed the lesser of 0.42 Su or 0.6 Sy. This exception is implemented by the use of ASME III Subsection NF paragraphs NF-3225 and NF-3324.6 of the 1983 Edition - Summer 1983 Addenda for the design of bolting.

The current industry position on the design of bolting is that the maximum safe increased allowables are achieved by limiting bolting tensile stress to the lesser of 0.7 Su or Sy at temperature and bolting shear stress to 0.42 Su or 0.6 Sy at temperature. The 0.7 Su limit is well recognized in Section III of the Code. The average shear strength of bolting material is about 0.62 Su according to test data, with a standard deviation of 0.033. Results indicate that the ratio of sear strength to tensile strength is independent of the bolt grade. Curves showing this appear on page 50 of "Guide to Design for Bolted and Riveted Joints," by J. W. Fisher. Test data are given in a paper by J. J. Wallaert and J. W. Fisher, "Shear strength of High-Strength Bolts," Journal of the Structural Division, ASCE, Volume 91, ST3, June 1965.

2. Paragraph C.8 (which is also referenced in Paragraphs C.6 and C.7)

As stated in Appendix 1A(N) of Section 1.0, system functional capability is provided for in Class 1 piping systems by maintaining the pipe stresses within Level D (faulted) limits. This assures that piping geometry is maintained and that required flow is not impaired. The selection of more restrictive stress limits for component supports is not necessary to assure the functional capability of the system.

However, to provide additional conservatism, ASME Class 1 component supports associated with the Safety Injection System and the Containment Spray system are designed to the limits described in Regulatory Position C-6 for Level C (emergency) conditions when subjected to the loadings associated with the Level D (faulted) plant condition.

Also refer to Appendix 1A(N) for a discussion of NSSS-supplied Class 1 supports.

Regulatory Guide 1.125

Physical Models for Design and Operation of Hydraulic Structures and Systems for Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP.

Regulatory Guide 1.126

An Acceptable Model and Related Statistical Methods for the Analysis of Fuel Densification

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.127

Inspection of Water-Control Structures Associated with Nuclear Power Plants

Discussion

CPNPP commits to comply with the requirements of Revision 1 (3/78) of this regulatory guide for the Safe Shutdown Impoundment in the manner described in Chapters 2.4 and 2.5 of the FSAR. Additionally, inspections addressed in the guide relative to Embankment Slope Stability, Unlined Saddle Spillways, the Reservoir proper and the Intake Structure and Discharge Channel will be accomplished.

This guide is not applicable to Squaw Creek Dam which is not safety related.

Regulatory Guide 1.128

Installation Design and Installation of Large Lead Storage Batteries for Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP.

Regulatory Guide 1.129

Maintenance, Testing, and Replacement of Large Lead Storage Batteries for Nuclear Power Plants

Discussion

The CPNPP design complies with the provisions of Revision 1 (2/78) of this regulatory guide, with the clarification that IEEE 450-1995 applies instead of IEEE 450-1975 as endorsed by the regulatory guide. Refer to Section 8.3 for further discussion.

Regulatory Guide 1.130

Service Limits and Loading Combinations for Class 1 Plate-and-Shell-Type Component Supports

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.131

Qualification Tests of Electric Cables, Field Splices, and Connections for Light-Water-Cooled Nuclear Power Plants

Discussion

Although the implementation of this regulatory guide, dated August 1977, does not include CPNPP, all Class 1E cables and field splices of the CPNPP meet most of the requirements of this regulatory guide with the few exceptions noted below:

1. All prefabricated cables of the CPNPP do not meet the following requirements:

a. Vertical tray flame test on the aged cables to determine their relative self-extinguishing tendencies. (Regulatory Guide position C.6),

b. Flame testing with the gas burner face in the front (Regulatory Guide position C.10). However, work is in progress by vendor on these items.

2. All 8kv cables, 600V power and lighting cables of the CPNPP do not meet the following requirements:

a. "Radiation Exposure Total" - per Regulatory Guide 1.89 Position C.5.

b. Vertical tray flame test on the aged cables to determine their relative self-extinguishing tendencies. (Regulatory Guide position C.6).

c. Flame testing with the gas burner face in the front (Regulatory Guide position C.10).

d. Gas and air flow requirements (Regulatory Guide position C.11).

3. All Raychem type WCSF-N field splices of the CPNPP have not been tested according to the following regulatory guide positions:

a. Vertical tray flame test on aged cable splices to determine their relative self-extinguishing tendencies. (Regulatory Guide Position C.6).

b. Flame testing with only natural grade propane and a release of approximately 70,000 BTU per hour at an air-gas ratio of 5 to 1. (Regulatory Guide Position C.9).

c. Flame testing with the gas burner face in the front. (Regulatory Guide Position C.10).

d. Gas and air flow requirements. (Regulatory Guide Position C.11).

e. Flame temperature requirements. (Regulatory Guide Position C.13).

4. The Partial Discharge Monitor Bus Coupler jumpers (15 kV rated) installed in the motor and emergency diesel generator termination boxes are not flame retardant and are not qualified to the requirements of Regulatory Guide 1.131.

Regulatory Guide 1.132

Site Investigations for Foundations of Nuclear Plants

Discussion

This regulatory guide is not applicable to CPNPP.

Regulatory Guide 1.133

Loose-Part Detection Program for the Primary System of Light-Water-Cooled Reactors

Discussion

The CPNPP O.L. Application was docketed prior to June 1, 1978, as provided by this guide dated September 1977.

A discussion of the CPNPP loose parts monitoring system is provided in Section 4.4.6.4.

Regulatory Guide 1.134

Medical Evaluation of Nuclear Power Plant Personnel Requiring Operator Licenses

Discussion

The medical certification and monitoring requirements of licensed personnel comply with the requirements of Revision 2 (4/87) of this regulatory guide.

Regulatory Guide 1.135

Normal Water Level and Discharge at Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP.

Regulatory Guide 1.136

Material for Concrete Containments

Discussion

This regulatory guide is not applicable to CPNPP; however, the application of ACI-359 is discussed in Section 3.8.1.2.1.

Regulatory Guide 1.137

Fuel-Oil Systems for Standby Diesel Generators

Discussion

The CPNPP design complies with the requirements of this regulatory guide, dated January 1978, with the following exception to portions of Parts C.1, C.1.b, C.1.d, C.1.g, C.2.b, C.2.c, C.2.d, and C.2.f:

1. Part C.1.b (Paragraph 7.1 of ANSI N195-1976 [49]):

Venting components for the storage and day tanks are not safety class 3. However, piping and components are seismically supported and missile protection requirements are satisfied for the vent path.

2. Part C.1.d (Paragraph 6.1 of ANSI N195-1876):

   The suction from the day tank is provided from the bottom of the tank.

3. Part C.1.g (Paragraph 7.5 of ANSI N195-1976):

   Fill lines to the storage and day tanks do not have strainers.

4. Part C.2.b:

   Prior to addition of new fuel into the storage tanks, the fuel is tested in accordance with tests specified in ASTM D975-1981 for:

   a. [a]API or Specific Gravity when tested in accordance with ASTM D1298-1980

   b. Kinematic viscosity

   c. Flash point

   d. Water and Sediment (Either a Clear and Bright Test, ASTM D4176-1982 or a water and sediment content of less than or equal to 0.05% volume when tested in accordance with ASTM D1796-1968)

   The remainder of the tests specified in Table-1 of ASTM D975-1981 will be completed within 31 days of the fuel addition and fuel oil will meet the requirements when tested in accordance with ASTM D975-1981 except that the analysis for sulphur may be performed in accordance with ASTM D1552-1979 [46], ASTM D2622-1982 [47], or ASTM D4294-2003.

5. Part C.2.c:

   New fuel sampling procedures for the fuel oil will be in accordance with ASTM D4057-1981, “Standard Practice for Manual Sampling of Petroleum and Petroleum Products”.

6. Part C.1 (Paragraph 6.3 of ANSI N195-1976):

   The strainers provided in the diesel fuel oil transfer system are not duplex strainers.

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a. If new fuel oil does not meet the diesel generator manufacturer’s requirements for absolute specific gravity at 60/60^{o}F of $\geq$ 0.8299 or for API gravity at 60^{o}F of $\leq$ 39^{o}, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer’s recommendations.

7. Part C.2.d:

   Accumulated condensate is checked for, and removed from storage tanks at least every 31 days.

8. C.1 (Section 4 of ANSI N195-1976):

   Section 4 of the standard includes requirements for security for the diesel-generator fuel-oil system. The requirements of this section are not considered applicable to the CPNPP Security Plan since this subject is addressed separately in more detail in other NRC documents. See Section 13.6 for a description of security requirements.

9. C.2.a

   If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60^{o}F of $\geq$ 0.8299 or for API gravity at 60^{o}F of $\leq$ 39^{o}, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations.

10. C.2.f

    The 10-year requirement to drain the supply tank and remove accumulated sediment in order to perform the ASME Section XI, Article IWD-2000, "Examination Requirements" is not valid. The test from ASME Section XI, Article IWD-2000 was a pressure test of the tank. The tanks are and have been vented to atmosphere. CPNPP has been draining and cleaning the tanks to perform ultrasonic wall thickness testing in lieu of the pressure test. Based on the following factors Emergency Diesel Generator (EDG) fuel oil storage tank testing is performed in accordance with the Preventative Maintenance Program:

    a. Results of prior tank cleaning and ultrasonic testing

    b. Fuel Oil Storage Tank (FOST) exterior coating

    c. Quality of FOST backfill material

    d. Cathodic protection of the FOST and associated piping

    e. All new fuel oil is sampled prior to admittance into the FOST in accordance with the Diesel Fuel Oil Testing Program

    f. FOST levels are recorded every 24 hours and after running the associated EDG

    g. FOST level is maintained greater than 88,000 gallons per daily logs with a Technical Specification minimum of 86,000 gallons

    h. TSTF-2-A, Revision 1, Relocate the 10 year sediment cleaning of the fuel oil storage tank to licensee control

Also refer to Section 9.5.4.

Regulatory Guide 1.138

Laboratory Investigations for Soils for Engineering Analysis and Design Of Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP; however, soils investigations are discussed in Section 2.5.

Regulatory Guide 1.139

Guidance for Residual Heat Removal

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.140

Design, Testing and Maintenance Criteria for Normal Ventilation Exhaust System Air Filtration and Absorption Units of Light-Water-Cooled Nuclear Power Plants

Discussion

The CPNPP design, maintenance and testing of the normal HVAC systems is in compliance with the requirements of Regulatory Guide 1.140 (03/78) except part C.3.h, water drains; part C.4.b, ease of maintenance; part C.4.c, test probes;, part C.5.b, air flow distribution; parts C.5.c and C.5.d, in-place testing; and part C.6, laboratory testing criteria for activated carbon.

1. Part C.3.h, Water drains:

   Check valves have been used on some drains in lieu of seals or traps. The drains for the demister compartments of the Hydrogen Purge Filtration units have not been provided with seals, traps, or check valves. Any minimal airflow through the demister drain will be filtered prior to discharge at the stack. This design conforms to the intent of NRC Regulatory Guide 1.140 (03/78).

2. Part C.4.b, Ease of Maintenance:

   The spacing provided between the components of the filtration units is, in some cases, less than the minimum required by this regulatory position. However, it has been determined by operation and maintenance personnel that the spacing provided is sufficient to perform all the required maintenance operations. This design conforms to the intent of Regulatory Guide 1.140 (03/78).

3. Part C.4.c, Test Probes:

   Test ports will be provided in lieu of permanently installed test probes. Permanently installed probes are a convenience in the periodic testing of filters and adsorbers.

Replacing them with test ports is of no safety significance. This design meets the intent of NRC Regulatory Guide 1.140 (03/78).

4. Part C.5.b, Air Flow Distribution:

The air flow distribution to the HEPA filters and iodine adsorbers will be tested in place for uniformity initially and after maintenance affecting the flow distribution for all units except the hydrogen purge units.

5. Parts C.5.c and C.5.d, In-Place Testing Criteria

ANSI/ASME N509-1980 and ANSI/ASME N510-1980 shall be used for field testing activities in place of the older versions of these codes referenced in this regulatory guide.

Atmospheric cleanup trains installed at CPNPP have two HEPA filter banks in series with a charcoal adsorber between them. In-place testing of the upstream HEPA bank only will be performed.

In-place testing of the HEPA filter banks and adsorber will not be required following painting, fire and chemical release described in position C.5.c and C.5.d of this guide. Only laboratory testing will be performed for carbon efficiencies. This design conforms to the intent of NRC Regulatory Guide 1.140 (03/78).

HEPA filter and charcoal adsorber efficiency for non-Engineered Safety Feature Filtration Units is 90% which corresponds to an acceptance criteria of less than 1% for in-place penetration and bypass leakage at rated flow.

The in-place testing of the HEPA filter and charcoal adsorber for the containment preaccess filtration units is performed in accordance with ANSI N510-1980, Table 1, NOTES 2 & 5. As discussed in FSAR section 9.4A, the preaccess filtration system is a recirculating system inside containment that reduces airborne contamination inside containment prior to personnel entry. Leakage through the HEPA or charcoal adsorber would not contribute to any increased radiological release to the environment. This design conforms to the intent of Reg. Guide 1.140 (3/78).

6. Part C.6, Laboratory Testing Criteria for Activated Carbon:

Laboratory testing of activated carbon will be in accordance with ANSI N509 and N510-1980 (which refers to ASTM D3803-79), except for the determination of the efficiency of charcoal to remove radioiodine. The parameters applicable to new and used charcoal at CPNPP shall be applied in the lab test as follows:

<u>New and Used Charcoal</u>
4" Beds
40 ft/min
30°C & 95% RH
Pre-equilibrated

The allowable methyl iodide penetration for these units shall be 10% as shown in Table 2 of the Regulatory Guide.

When all or most of the charcoal canisters in the filtration units have been used, a representative sample of the charcoal from the bed will be obtained by grain thief sampling in accordance with ANSI N509-1980, Appendix A. The representative samples obtained with the grain thief will be used for laboratory testing and/or refilling the charcoal canisters.

The carbon adsorber beds will be replaced when (1) testing in accordance with the frequency specified in Footnote c of Table 2 results in a representative sample failing to pass the applicable test in Table 2 or (2) no representative sample is available for testing, at a time when testing is required.

This design conforms to the intent of NRC Regulatory Guide 1.140 (03/78).

Regulatory Guide 1.141

Containment Isolation Provisions for Fluid Systems

Discussion

This regulatory guide is not applicable to CPNPP; however, the containment isolation is discussed in Section 6.2.

Regulatory Guide 1.142

Safety-Related Concrete Structures for Nuclear Power Plants (Other Than Reactor Vessels and Containments)

Discussion

This regulatory guide is not applicable to CPNPP; however the design of safety-related concrete structures is discussed in Section 3.8.

Regulatory Guide 1.143

Design Guidance for Radioactive Waste Management Systems, Structures, and Components Installed in Light-Water-Cooled Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP, however, the code, code class, safety class, and seismic classification of Liquid Waste Processing and Gaseous Waste Processing System components comply with Table 1 of Regulatory Guide 1.143, Rev. 1. Reference Section 11.2, 11.3 and 11.4 for related information.

Regulatory Guide 1.144

Auditing of Quality Assurance Programs for Nuclear Power Plants

Discussion

The program for auditing of Quality Assurance Programs for CPNPP complies with the requirements of Regulatory Guide 1.144 Revision 1 (9/80). The following suppliers are not audited by Vistra OpCo and are exempt from the requirements of this regulatory guide:

National or state calibration sources

Authorized Nuclear Inspection Agency

NVLAP and A2LA accredited commercial grade calibration suppliers evaluated in accordance with Section 17.2.7.

For triennial supplier audits (Section 17.2.7), the audit schedule shall be based on the month in which the audit starts. An interval extension grace period of up to 25% may be used for the scheduled commencement date when conditions, such as plant operational considerations or to accommodate supplier activities, make meeting the specified schedule date impractical. When an audit interval extension greater that one month is used, the next audit will be scheduled from the original anniversary month rather than from the month of the extended audit. The total combined time interval for any three consecutive audit intervals should not exceed 3.25 times the specified audit interval.

Regulatory Guide 1.145

Atmospheric Dispersion Models for Potential Accident Consequence Assessments at Nuclear Power Plants

Discussion

This regulatory guide is not applicable to CPNPP.

Regulatory Guide 1.146

Qualification of Quality Assurance Program Audit Personnel for Nuclear Power Plant

Discussion

Effective June 1, 1983, CPNPP Quality Assurance Audit Personnel are in compliance with regulatory guide 1.146, August 1980.

Regulatory Guide 1.148

Functional Specification for Active Valve Assemblies in Systems Important to Safety in Nuclear Power Plants

Discussion

CPNPP is not committed to this Regulatory Guide. The operability program for active valves is covered in Section 3.9N and 3.9B.

Regulatory Guide 1.150

Ultrasonic Testing of Reactor Vessel Welds During Preservice and Inservice Examinations.

Discussion

Refer to Appendix 1A(N).

Regulatory Guide 1.155

Station Blackout

Discussion

CPNPP complies with the guidance of Regulatory Guide 1.155 (August 1988) as described in Appendix 8B.

Regulatory Guide 1.163

Performance-Based Containment Leak-Test Program

Discussion

Effective by August 12, 1996, CPNPP complies with Regulatory Guide 1.163, September 1995.

Regulatory Guide 1.195

Methods and Assumptions for Evaluating Radiological Consequences of Design Basis Accidents at Light-Water Nuclear Power Reactors, May 2003.

Discussion

The radiological consequences analysis for the design basis accidents follow the guidance provided in Regulatory Guide 1.195 and CPNPP adopts the dose limits defined by Regulatory Guide 1.195. The CPNPP design for skin dose calculations DCFs uses DOE/EH-0070 instead of Figure 12 recommended in Regulatory Guide 1.195.

Regulatory Guide 1.196

Control Room Habitability at Light-Water Nuclear Power Reactors, May 2003.

Discussion

The Control Room Envelope Habitability Program in Technical Specification 5.5.20 is consistent with the guidance of TSTF-448, Revision 3, which incorporate the specific aspects of Regulatory Guides 1.196 and 1.197. CPNPP currently uses RG 1.52, Revision 1 for the control room design. CPNPP used Revision 2 for testing only. CPNPP uses RG 1.140 as information only for non-safety related air filtration systems.

Regulatory Guide 1.197

Demonstrating Control Room Envelope Integrity at Nuclear Power Reactors, May 2003.

Discussion

The Control Room Envelope Habitability Program in Technical Specification 5.5.20 is consistent with the guidance of TSTF-448, Revision 3, which incorporates the specific aspects of Regulatory Guide 1.197 with the following exceptions:

1. C. - Section 4.3.2 “Periodic CRH Assessment” from NEI 99-03 Revision 1 will be used as input to a site specificSelf assessment procedure.

2. C.1.2 - No peer reviews are required to be performed.

Regulatory Guide 8.10

Operating Philosophy for Maintaining Occupational Radiation Exposures as Low as Reasonably Achievable, May 1977.

Discussion

The administration of the As Low As Reasonably Achievable (ALARA) Program at CPNPP will be in accordance with Regulatory Guide 8.10 with the following exception:

1. The radiation workers at CPNPP will be tested at the frequency of 24 months nominal not to exceed 27 months for supplemental workers and 48 months not to exceed 51 months for utility workers. These frequencies are set as quarterly basis.

2.0 SITE CHARACTERISTICS

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2.1 GEOGRAPHY AND DEMOGRAPHY

2.1.1 SITE LOCATION AND DESCRIPTION

2.1.1.1 Specification of Location

The CPNPP site is located in Somervell County in North Central Texas. Squaw Creek Reservoir (SCR), established for station cooling, extends northward into Hood County. The site is owned by the Applicants. The site is situated along Squaw Creek, a tributary of the Paluxy River, which is a tributary of the Brazos River. The station site is over 30 miles southwest of the nearest portion of Fort Worth and approximately 4.5 miles north- northwest of Glen Rose, the nearest community (see Figure 2.1-1). Site coordinates are:

	Unit No. 1	Unit No. 2
Texas Grid (Feet)	Y = 229,723.96 X = 1,911,921.11	Y = 230,010.86 X = 1,911,951.27
U.T.M. Grid (Zone 14) (Meters)	N = 3,573,903 E = 614,393	N = 3,537,991 E = 614,401
Latitude	32º 17' 52.02"	32º 17' 54.85"
Longitude	97º 47' 06.15"	97º 47' 05.79"

2.1.1.2 Site Area

The site area map (Figure 2.1-2) shows the concurrent plant property and site boundary line, the Exclusion Area, and Squaw Creek Reservoir (SCR). The site area is approximately 7,700 acres. Site area access is by a plant railroad, which connects to the Atchison, Topeka and Santa Fe Railroad Company main line at Tolar, Texas, by a plant access road which connects to FM 56 (previously Highway 201) and by County road 213 (also known as Coates Rd) which connects to State Highway 144.

The plant railroad and access road are owned and controlled by the Applicants. There are no other highways, railways or navigable waterways which traverse or are immediately adjacent to the site. Vistra OpCo maintains Squaw Creek Park and controls access to the park and reservoir.

Principal plant structures are also shown on Figure 2.1-2.

2.1.1.3 Boundaries For Establishing Effluent Release Limits

The Exclusion Area consists of approximately 4,170 acres. Figure 2.1-2 depicts the Exclusion Area boundary. This boundary is used for establishing effluent release limits and enables the owners to fulfill their obligations with respect to the requirements of 10 CFR Parts 20 and 100 (see Section 2.1.2.1.5).

Figure 2.1-2 shows that the points of release for each of the two units are located closer to the southwest property line than any other segment of the property line. This southwesterly distance

coincides with the minimum Exclusion Area boundary distance, which is 5,067 feet (1544 meters) from the midpoint of the centerline between the Containment buildings.

2.1.2 EXCLUSION AREA AUTHORITY AND CONTROL

2.1.2.1 Authority

2.1.2.1.1 Surface Rights

CP PowerCo has acquired and will maintain surface ownership of all the land within the Exclusion Area (see Figure 2.1-2A). Accordingly, CP PowerCo has the authority to determine all activities within the Exclusion Area, except for certain improbable and de minimus mineral exploration activities as discussed in greater detail below.

That portion of Squaw Creek Reservoir which is within the Exclusion Area is subject to the waterway exclusion provided in 10 CFR Part 100.3(a). Consistent with that regulation, appropriate and effective arrangements will be made to control traffic on the reservoir to protect the public health and safety in case of emergency. The Exclusion Area is not traversed by any public highway or railroad. See Figure 2.1-2. The nearest primary public road, FM 56 (previously State Highway 201 as denoted on Figure 2.1-2), lies outside the Exclusion Area approximately 8900 feet southwest of the center line between the Containment buildings. CP PowerCo owns and operates the plant railroad from the plant to its junction with the Atchison, Topeka and Santa Fe Railroad at Tolar, Texas, approximately 11 miles from the site.

2.1.2.1.2 Mineral Rights

CP PowerCo has acquired mineral rights beneath all seismic Category I structures (see Figure 2.1-2C). Portions of the remainder of the Exclusion Area are subject to certain outstanding mineral rights. As noted above, CP PowerCo owns the surface rights for the entire Exclusion Area.

As to the mineral rights within the Exclusion Area not owned by CP PowerCo, CP PowerCo will assure that the exercise of such mineral rights will pose no health and safety threat during normal reactor operation or in the event of an accident. The only outstanding mineral rights in the Exclusion Area for CPNPP, and surrounding areas, relate to the exploration for and production of oil, gas, and other subsurface minerals. There are no outstanding rights which permit the production of surface minerals. As discussed in Section 2.5.1, the potential for commercial production of minerals at CPNPP and in the surrounding area is low. Thus, it is anticipated that the exercise of such outstanding mineral rights would involve only sporadic, exploratory activity, and little or no production.

Nevertheless, CP PowerCo will maintain sufficient authority and control over, and knowledge of, attempted ingress into the Exclusion Area to ensure that no unauthorized entry is allowed by use of a mobile patrol at random intervals. Ingress for the purpose of exercising mineral rights in any area within 2250 feet of a seismic Category I building (See Section 2.2.3.2.1) or within 2800 feet of either Containment Building (See Section 2.1.2.1.5) is prohibited by virtue of CP PowerCo ownership of the mineral rights or by deed restrictions on any reserved mineral rights. This area has been designated in Figure 2.1-2C as the “External Hazard Free Zone.” The distances of 2250 and 2800 feet are based upon the analysis of a postulated gas well in Section 2.2.3. The distance of 2250 feet from any seismic Category I building assures that any conceivable

detonation (of the maximum quantity of explosives that could be used within the exclusion area) would have no effect on the safety-related structures and components (see Regulatory Guide 1.91). The forbidden zone of 2800 feet in radius corresponds to the equivalent distance for the 2 hour dose limit of 10 CFR 100 and the maximum design basis accidents for CPNPP (See Section 15.6). Ingress to the remaining outer areas of the Exclusion Area will only be permitted pursuant to written agreements between CP PowerCo and the necessary parties which would provide that CP PowerCo has absolute authority to determine all activities within the Exclusion Area, including removal of personnel and equipment.

A mineral owner or lessee has no legal right to use physical force or to create a public disturbance to obtain access to the outer areas of the Exclusion Area for purposes of mineral exploration or extraction. If such access is sought as to areas outside the "External Hazard Free Zone", CP PowerCo likewise will refuse to allow access unless the written agreement discussed above has been executed by the mineral owner or lessee. In such case, legal remedy of a mineral owner or lessee to obtain access to the surface of the Exclusion Area after being so excluded would be to file a lawsuit in the State District Court for the county where the land is located (either Somervell or Hood County). Should such a suit be filed, CP PowerCo would then file an immediate cross-action to condemn the mineral rights of the party seeking ingress and thereby prevent the ingress. CP PowerCo has statutory authority to do so. Article 3269, V.A.T.S.

In this manner, CP PowerCo will have absolute authority to determine all activities within the Exclusion Area, including the authority to exclude or remove persons. Thus, the exercise of mineral rights in the Exclusion Area will pose no health and safety threat during normal reactor operation or in the event of an accident. In view of the unusual nature and limited scope of activities associated with the mineral rights and the plan and commitments by CP PowerCo to control all activities within the Exclusion Area, the present status of ownership is deemed to be of de minimis safety consequence.

As stated above, CP PowerCo has acquired the mineral rights beneath all seismic Category I structures as well as beneath the Squaw Creek Dam (See Figure 2.1-2C). No measurable subsidence due to mineral extraction is anticipated. See Section 2.5.1.2.6 for a discussion of the effects of mineral extraction in the area of the site.

2.1.2.1.3 Easements

A 6-inch natural gas pipeline and a 26-inch crude oil pipeline traverse the Exclusion Area about 4,900 feet southwest of the midpoint of the centerline between the Containment buildings as shown on Figure 2.1-2B. These pipelines are also described in Section 2.2.3. CP PowerCo has granted the pipeline owners easements which retain for CP PowerCo absolute control to determine all such activities within the Exclusion Area including ingress and egress for the purpose of maintaining the pipelines and their right-of-way.

2.1.2.1.4 Status of Ownership

CP PowerCo has acquired all of the land which will constitute the site property. One small tract of land within the site area has been excluded from purchase: the Hopewell cemetary east of the plant (see Figure 2.1-2). This tract is outside the Exclusion Area and is fenced off from the site property.

2.1.2.1.5 Minimum Exclusion Area Distance

The minimum distance to the Exclusion Area boundary from the midpoint of the centerline between the Containment buildings is 5,067 feet (1,544 meters) to the west-southwest. The minimum Exclusion Area boundary distance is substantially larger, and therefore more conservative, than the distance which literal compliance with 10 CFR 100 would dictate (See Section 15.6).

2.1.2.2 Control of Activities Unrelated to Plant Operation

Activities unrelated to plant operation which may be permitted within the Exclusion Area include the exercising of mineral rights and the maintenance of pipelines as described in Sections 2.1.2.1.2 and 2.1.2.1.3 above, and cattle grazing. Vistra OpCo will have the necessary control to determine these activities and will require that all persons involved in them report to the Plant Manager (See Section 13.1.1.2.1) or a designated representative prior to engaging in the activities.

Arrangements have been made (in coordination with the appropriate agencies) to control access to, activities on, and the removal of persons and property from the CPNPP owner controlled area in case of emergency. Arrangements for recreational use and emergency procedures governing such use have been completed. Vistra OpCo has the authority to exclude or remove any person from this area at any time.

Vistra OpCo conducts practices that provide knowledge of the approximate number and location of persons within the Exclusion Area engaged in such activities. Normal evacuation of persons within the Exclusion Area will take no more than two hours.

2.1.2.3 Arrangements for Traffic Control

In the event of an emergency, traffic on the plant access road will be controlled by Vistra OpCo.

If Squaw Creek Reservoir is opened to the public, arrangements will be made to control traffic in the event of an emergency (see Section 2.1.2.2, above).

2.1.2.4 Abandonment or Relocation of Roads

An unpaved county road which traversed the northeast corner of the Exclusion Area was abandoned in April 1975 because of the construction of Squaw Creek Dam and Reservoir. Arrangements for the closing of this section of road were made with the Somervell County Commissioner's Court in December 1974. No other public roads traverse the Exclusion Area.

2.1.3 POPULATION DISTRIBUTION

The purpose of this section is to provide detailed estimates of the present and projected size and distribution of population within a 50-miles radius of CPNPP. The population estimates provided in the PSAR have been reviewed, revised, and updated for purposes of the FSAR. Estimates of population distribution are provided for 1970 (most recent census year), 1976 (current year), and for census decades 1980 through 2020.

In reviewing and updating the population estimates in the PSAR, it was recognized that the actual centerline locations of the containment structures for Units 1 and 2 differ slightly (approximately 88 feet) from the locations as originally shown. In these revised population estimates, the actual centerline of the Unit 1 containment structure has been taken as the point of origin for the sector lines and concentric distance circles which form the sector-areas used in portraying population distribution within the 50-mile radius of CPNPP. While the 1970 county-by-county population base in the CPNPP area remains the same as shown in the PSAR, the slight difference in Unit 1 location causes some change in location of sector lines and distance circles. Thus, very small changes are found in this updated population study in the 1970 distribution of population by sector- areas.

The territory included within the 50-mile radius of CPNPP includes all or a part of 19 counties, all in Texas. The general location of CPNPP in Somervell County and the locations of the rest of the counties located within 50 miles of the plant site are shown in Figures 2.1-3 and 2.1-4. The population of these 19 counties is given in Table 2.1-1 for the census decades 1930 through 1970; in addition, the table provides an estimate of the 1976 population for each county and the projected future population for each census decade 1980 through 2020. Footnotes to the table provide brief comment regarding the sources of the historical data and the projections. Within the 50-mile radius of CPNPP, there is wide diversity in land use, urbanization, and population density. The plant site is located in Hood and Somervell counties, which are essentially rural, sparsely-populated areas. The entire population of Somervell County was 2,793 residents in 1970. Hood County, which is much larger in area than Somervell County (as may be seen in Figure 2.1-3), had a population of 6,368 inhabitants in 1970. In 1970, there were three small communities with a total of 4,339 people within the 10-mile area around the CPNPP site. The total population within the 10-mile area at that time was 5,353, or an overall population density of approximately 17 persons per square mile.

The sparsely-settled rural character extends well beyond the 10-mile radius, as indicated in Table 2.1-2, which lists all incorporated communities and all unincorporated settlements with over 1,000 inhabitants within the entire 50-mile radius of the site. The area extending from 10 to 20 miles out from the CPNPP site is even more sparsely populated than the 0 to 10-mile area. As the table shows, in 1970 there were two communities totaling 1,028 people within the entire 10 to 20-mile area around the CPNPP site, and the total population in this area was 7,532 (this is a 1970 population density of only 8 persons per square mile). Beyond the 20-mile radius there are more communities, and the 30 to 50-mile area to the northeast is dominated by the Fort Worth metropolitan complex.

In reviewing the county-by-county population projections which were given in the PSAR, it was found that most of the county population projections should be revised somewhat in accordance with current population estimates and the most recent projections prepared by the councils-of-government and the state agencies. Accordingly, in providing updated county population data (Table 2.1-1), current estimates for 1976 are given along with the revised projections for the years 1980 through 2020. It should be noted that the current estimates and the projections of expected population growth in Hood and Somervell counties are based upon an enumeration of housing units in the two counties in mid-1976 and an updated evaluation of recent and probable future trends in recreational development in the local area. Moreover, the estimates for Hood and Somervell counties include detailed consideration of the current and possible future impact of CPNPP construction and operation on population growth. As a general procedure in determining population distribution, the 1970 population of each county (wholly or partially within the 50-mile radius of the plan) was allocated to sector-areas within the county on

the basis of (1) the population of each community and enumeration district located wholly within a particular sector-area, and (2) a percentage share of the population of each enumeration district and community partially within a particular sector-area (the percentage share of population of an enumeration district to be allocated to a particular sector-area was generally assumed to be equal to the percentage portion of the area of the enumeration district within the particular sector-area).

In the case of Hood and Somervell counties, the county and enumeration district populations were allocated to sector-areas on the basis of (1) an actual count of housing units within each sector-area, and (2) the estimated number of residents per housing unit, considering available census data for the particular local area. The housing count for Hood and Somervell counties for the PSAR was made in 1973 utilizing a combination of available mapping, aerial photography, and field survey resources. The housing unit count made in mid-1976 for the FSAR was based on a comprehensive field survey and housing enumeration for all of Hood and Somervell counties. The estimates of population within individual sector-areas have been reconciled to the population totals for each county (or a portion of a county).

The percentage ratio of the total population of a county which was estimated to be within each individual sector-area in the county in 1970 (1976 in the case of Hood and Somervell counties) was assumed to remain the same in 1980 and beyond. This assumption was made after concluding that there is little possibility of a significant and radical change in the basic pattern of current population distribution within individual counties in the 50-mile area.

2.1.3.1 Population Within 10 Miles

The area within the 10-mile radius of CPNPP is predominantly a rural agricultural area. In 1970, there were three small communities having more than 100 residents. The total population of Hood and Somervell counties changed little over the period from 1930 to 1970 (see Table 2.1-1).

2.1.3.1.1 Current Population Within 10 Miles

Since 1970, Hood and Somervell counties have experienced increases in population, as indicated by the following:

	Population	
County (and Community)	1970 (census)	1976 (estimate)
Hood County	6,368	15,601
Granbury (county seat)	2,473	3,526
Tolar	312	435
Somervell County	2,793	5,216
Glen Rose (county seat)	1,554	2,790

The estimates for 1976 are based upon the July/August 1976 enumeration of housing units in the two counties (including the communities). The growth of Granbury may be understated in

comparing the data above because much of the urban growth of the community has taken place beyond the city limits.

The increase in population of Hood County since 1970 is largely related to the attractions of Lake Granbury for residential developments. These developments are attracting large numbers of permanent residents as well as vacation and recreational visitors. In 1970, there were 2,628 housing units in use as permanent residences in Hood County. By 1976, there were 5,566 housing units accounting for an estimated population of 15,601 full-time residents in Hood County and there were an additional 1,648 housing units in use primarily during weekends and vacations. A significant portion of the increase in housing units has been in mobile homes used both as vacation homes and permanent residences. The increase in permanent residents in Hood County is partially accounted for by attractive living conditions; many people that have moved into Hood County are commuting to jobs elsewhere. Significant numbers of retired people have established their permanent residence in Hood County. With the above, and the general stimulus of vacation and recreational activity, there has been an increase in local economic activity and job opportunity within the county.

The number of permanent housing units in Somervell County increased from 1,203 in 1970 (but only 1,035 households at that time) to an estimated 1,856 housing units in 1976. An additional 26 vacation homes were located in the housing enumeration, but as the data indicates, virtually the entire housing inventory of Somervell County (fixed structures, mobile homes, and apartments or courts) was permanently occupied in mid-1976. Somervell County has participated only peripherally in the Lake Granbury stimulus to housing development and recreational activity.

In addition to the foregoing, it should be noted that the influx of construction workers employed on CPNPP accounts for a significant portion of the population increase from 1970 to 1976, particularly in the case of Somervell County. For example, at the end of July 1976, there were 2,285 workers regularly employed at the CPNPP construction site. Of that number, 216 had moved into Somervell County specifically to work at CPNPP. In addition to the 2,285 man active work force as of the end of July 1976, there was a total of 1,968 workers who had been terminated since the start of construction on CPNPP in October 1974. Of these terminated employees, 177 had moved into Somervell County for the purpose of employment at CPNPP. Data on the CPNPP work force indicates that, on the average, each worker had 1.8 dependents. Thus, considering both active and terminated workers, a total of 1,100 workers and dependents had moved into Somervell County as of July 1976.

The tabulation below provides a summary showing the total number of CPNPP workers who have moved into Hood and Somervell counties through July 1976:

	Total No. of Workers as of July 1976	
	Active	Terminated
Total (on-site work force)	2,285	1,968
Residing in Somervell County:		
Local Hire	160	138

	Total No. of Workers as of July 1976	
	Active	Terminated
Relocated	216	177
Residing in Hood County:		
Local Hire	278	240
Relocated	201	173

The total number of workers (both active and terminated) and dependents relocating to Hood and Somervell counties through July 1976 is estimated as follows:

	Total Workers	Dependents	Total
Hood County	374	673	1,047
Somervell County	393	707	1,100

While 45 to 47 percent of these totals are represented by now terminated CPNPP employees and dependents, there are strong indications that a large portion of the terminated employees who originally moved with their dependents into the local area to work on CPNPP actually stayed in the area after termination of employment at CPNPP. The overall growth of population in both counties is creating increasing opportunities for employment in local service industries. Moreover, as indicated earlier, substantial numbers of new residents in Hood and Somervell counties are commuting to jobs outside the local area.

Table 2.1-3 shows estimated population distribution by sector and mileage zones within 10 miles of CPNPP for the years 1970 and 1976 and for the census decades 1980 through 2020. Figure 2.1-3 indicates the location of the plant in Somervell County and the sector areas out to a 10-mile radius. As may be seen from the data in the table, a substantial part of the increase in population in the two counties from 1970 to 1976 has occurred in or near the established communities. The population of the remainder of Somervell County has increased by an estimated 95 percent over the same period (1970-1976) and examination of the data in the table indicated that this growth was widely distributed in the county.

In some contrast to Somervell County, the population growth in Hood County has been more highly concentrated (in the areas near Lake Granbury). The population of the county seat of Granbury (within the city limits) increased by an estimated 42 percent from 1970 to 1976. The population of the balance of Hood County increased by well over 300 percent. Much of this increase has occurred in the north, north-north-east, and northwest sectors in the 5 to 10-mile area and the 10 to 20-mile area. These increased concentrations of population within particular sector-areas in Hood County are clearly evident in a comparison of the 1970 and 1976 data in Table 2.1-3.

2.1.3.1.2 Projected Population Within 10 Miles

The size and distribution of population within 10 miles of the CPNPP plant site as projected for the census decades 1980 through 2020 are shown in Table 2.1-3 (along with estimates for 1970 and 1976). These projections of future population distribution within the 10-mile radius of the site are fully reconciled to the overall population projections for Hood and Somervell counties (as described earlier). These projections take into account the size and residential distribution of the CPNPP construction work force and the expected phase-out of construction activity over the period 1980-1982.

It is expected that the continued growth of population and employment opportunities associated with lake-oriented residential community development and recreational activities in Hood County will tend to offset the sharp drop in construction employment on CPNPP. It is also expected that the completion of CPNPP will cause an actual decline in population in Somervell County after 1980 (for a period of two to three years), but by 1990 the loss will be regained due to the underlying slow rate of growth expected for the county.

2.1.3.1.3 Age Distribution of Population Within 10 Miles

To provide an estimate of the population by age groups, it was necessary first to compare the 1970 population distribution by age group for Somervell County and the U.S. as a whole. Because the percentage distribution by age groups 0 to 11, 12 to 18, and 19 and over for Somervell County and for the U.S. for 1970 differed by less than 10 percent, the distributions for the year 2000 for the project area were assumed to be the same as the Bureau of Census age-group distribution projections for the U.S. for the year 2000. If the difference had been greater than 10 percent, the U.S. percentages would have required adjustment based on projections for the North Central Texas region.

The U.S. projection for the year 2000 came from Table No. 3 of the “U.S. Statistical Abstract, 1975.” The data had to be modified slightly, however, to compensate for somewhat different age groupings. The population projection for the 10-mile area of CPNPP came from Table 2.1-3. The resulting population distribution by age group for the 10-mile area around CPNPP for the year 2000 (the mid-point of expected plant life) is given in Table 2.1-4.

2.1.3.2 Population Within 10 to 50 Miles

The population within the 10 to 50-mile radius of CPNPP includes the population of a large number of communities and cities including Fort Worth, as listed in Table 2.1-2. The location of all population centers is shown on Figure 2.1-4, which also shows sector lines and distance circles out to the 50-mile radius from CPNPP.

2.1.3.2.1 Current Population Within 10 to 50 Miles

The distribution of the 1970 population and the estimated 1976 population within the 10 to 50-mile radius of CPNPP is shown by sector-area in Table 2.1-5. The estimates of population distribution by sector-area for 1976 are correlated with the county-by-county projections, as described earlier. There was a 13.4 percent increase in population within the entire 10 to 50 mile area from 1970 to 1976 (compared with the 127 percent increase within the 10-mile area). Within the 10 to 50-mile area there are significant differences among counties with respect to population increase (or loss) from 1970 to 1976 as may be seen in reviewing Table 2.1-1. For

example, the population of Johnson County (to the east of the site) increased by 20 percent while Stephens County (to the west of the site) decreased by 3 percent. These differences in recent growth trends among the counties are reflected in the estimates of current population distribution by sector-area.

2.1.3.2.2 Projected Population Within 10 to 50 Miles

The present and projected population growth through the year 2020 is shown by county in Table 2.1-1 for all counties within the 50-mile radius of CPNPP. Table 2.1-5 provides estimates of the distribution of projected population by sector-area for the census decades 1980 through 2020 (along with comparable estimates for 1970 and 1976). As may be seen in a comparison of the cumulative estimates of population for 0 to 10 miles, 0 to 20 miles, etc., for each of the census years (summarized at the end of Table 2.1-5), population growth for the entire 0 to 50-mile area is projected to increase at a much slower rate than for the 0 to 10-mile area, as shown below:

Area (radius)	Population (000)		Percent Increase
	1976	2020	
0-10	12.1	31.1	156
0-20	24.7	64.9	163
0-30	83.8	198.5	137
0-40	438.8	1,024.8	134
0-50	894.0	2,090.5	134

2.1.3.2.3 Age Distribution of Population Within 10 to 50 Miles

The age distribution of the population of Somervell County (in which the plant is located) in 1970 differed by less than 10 percent from the 1970 age distribution for the populations of the United States as a whole. Accordingly, the projected percentage age distribution for the United States for the year 2000 (mid-point of plant life) was assumed for the population within the 10 to 50-mile area around CPNPP for the year 2000. The projected populated distribution by age group for the 10 to 50-mile area is provided in Table 2.1-6.

2.1.3.3 Transient Population

After consideration of the overall patterns of settlement, land use, and population distribution within the entire 50-mile radius of the site, it was concluded that transient population movements and daily and seasonal variations in population distribution and concentration within the 10-mile radius of CPNPP should be examined in some detail. It was also concluded that no such examination and projections should be made of transient movements in the 10 to 50-mile area. There are large-scale movements of daily commuters (in all directions) within the Fort Worth metropolitan complex. However, but the potential significance and meaning of such movements (compared with the potential significance of movements within the 10-mile area) does not appear to justify the inordinate effort that would be required to characterize, analyze, and project transient population movements within the 10 to 50-mile area. Accordingly, the discussions below pertain only to transient population within the 10-mile radius of CPNPP.

2.1.3.3.1 Seasonal Variation

This category of transient population is specifically concerned with overnight visitors coming into the 10-mile area. As will be shown, there are many more overnight visitors in the summer season than during other times of the year, but overnight visitors are found in the area throughout the year. The seasonal or overnight category of transients includes visitors staying for several days or weeks (this category excludes consideration of daily transients, regardless of the season). This category of transient population (seasonal or overnight visitors) includes visitors that would be found in hotel/motels, camp-grounds, recreational vehicle parks, organized camps (church groups, youth groups, etc.), mobile home-parks, and vacation homes. It also includes live-in students in a children's home despite the fact that their stays are for extended periods. These children are not included as part of the permanent population.

Because this category of transients is predominantly comprised of recreational and vacationing visitors, there is a distinct peaking in the summer season of total transients in the area at any one time. Moreover, normal and holiday weekend peaking is different and week- days differ from week-ends, both in summer and winter. See Table 2.1-7 for estimates of the length of stay of various types of transients.

In developing the estimates of seasonal transient population given in Table 2.1-8, consideration has been given to (1) the location and capacity of various facilities that accommodate overnight visitors, and (2) the actual patterns or levels of use which are typically experienced by the facilities at various time of the year. The table presents estimates of weekly, typical weekend and holiday week, and daily and overnite transient during summer and winter.

With respect to the projection of future levels of seasonal or overnight transients, different assumptions were made for the several different types of facilities. Sector-area estimates of vacation home and hotel/motel visitors were assumed to increase from 1980 through 2020 in proportion to the projected increase in population for the county as a whole (in which the sector-areas are located). Organized camp attendance is expected to remain at 1976 levels and the children's institution will remain stable from 1980 onward. Camping at various types of facilities along Lake Granbury is expected to reach its peak by 1990 (achieving levels similar to those at other older facilities in North Central Texas) and level off thereafter. Camping elsewhere in the 10-mile area (away from Lake Granbury) was estimated to grow in general accordance with the population growth projected for Dallas and Tarrant counties (major counties in the larger metroplex).

2.1.3.3.2 Daily Variation

This category of transient population is concerned with daily movements of population into and within the 10-mile area and peaking in the number of transients found in a particular sector-area at any given time. Estimates of daily transient movements and population concentration include consideration of movements to such facilities as public schools, private schools, urban and community shopping centers, and recreational facilities such as parks and lakes. Estimates have been made of daily visitor recreational use of Lake Granbury (in addition to use by overnight and vacationing visitors). Beginning in 1980, when it will have been filled, it is assumed that limited daily visitor recreational use will be permitted at a non-camping park facility on Squaw Creek Reservoir.

It is apparent from Table 2.1-8 that there are great differences in daily transient population movements depending on the season of the year and on the time of the week (weekday, normal weekend, and holiday weekend). It should be noted that the table does not provide estimates of the total number of daily visitors (of different types) but rather it provides estimates of the maximum number of visitors that might be expected in a sector-area during the peak hours of the day. It is also important to note that the estimates of peak daily transient population concentration in a particular sector-area are not accompanied by corresponding decreases in population in sector-area from which the transients originated. This is the situation where large numbers of school children (from numerous sector-areas) are concentrated daily in a single sector-area. This is simply a recognition that peak population concentration in different sector-areas may occur at different times of the day.

The general approach in estimating daily transient movements in 1976 was to utilize empirical data wherever possible, as in the case of school enrollments and the daytime use of some recreational and park facilities. The experience of state parks and other older reservoirs and water-oriented facilities was used as a basis for estimating current and projected daytime use of Lake Granbury and Squaw Creek Reservoir. The numbers and movement patterns of permanent and vacation residents in making use of shipping facilities in Granbury and Glen Rose was estimated on the basis of the geographical distribution of households and a number of working assumptions regarding the frequency and time of week for shoppers coming from various distances. It was assumed that the above communities served only the areas of their respective counties. Again, the estimates indicate the peak number of transients that would be in a sector-area at a particular time and not the total number of transients during a day.

With respect to the projections of daily transients for future years, several assumptions were made as in the case of the seasonal population estimates. Projections of daily transients associated with school enrollments and community shopping activity were related to basic population projections. Recreational use of Lake Granbury and Squaw Creek Reservoir was assumed to reach mature levels by 1990 and level off thereafter at levels found at similar, but older, water-related facilities elsewhere in North Central Texas. It is noted that Unit 2 of CPNPP will not be completed until 1982 or thereafter and that a construction work force will remain on site after completion and start of operations (at least in a test mode) of Unit 1 in late 1980.

2.1.3.3.3 Summary Effect of Transient Population Movements in 10-Mile Area

The most conservative estimate of area population, provided in Table 2.1-9, is the sum of the permanent population and the maximum transient population estimate (summer holiday weekend daily transients). Similar estimates for other transient periods may be obtained by simply summing the permanent population estimate (Table 2.1-4) and the appropriate transient population estimate. Comparison of Table 2.1-4 and 2.1-9 shows that inclusion of the maximum transient population increases total population by nearly 60 percent.

2.1.3.4 Low Population Zone

In accordance with 10 CFR Part 100 guidance, the low population zone shown in Figure 2.1-5 is defined as that area falling within a four-mile radius of the center of the station site. The present number of residents (approximately 500 persons according to the 1976 estimate) within this area is sufficiently small to ensure a reasonable probability that appropriate protective action could be taken in their behalf in the event of a serious accident as required by 10 CFR 100.3. Section 15.6 shows that this area is of sufficient size to preclude an individual located on its outer

boundary from receiving a total dose following a postulated accident in excess of requirements in 10 CFR Part 100.11. Resident and transient populations within the zone have been discussed in detail in Sections 2.1.3.1 through 2.1.3.3.

2.1.3.5 Population Center

The nearest population center (defined in 10 CFR Part 100 as more than 25,000 people) is Fort Worth, Texas. According to the 1970 census, Fort Worth has a population of 393,476 people (approximately 613,000 in metropolitan area). The growth rate was 10.4 percent between 1960 and 1970. The geographic center of Fort Worth, which is approximately 41 miles northeast from CPNPP, does not differ significantly from its population center. Dallas, with 844,401 inhabitants (1,555,950 in metropolitan area), is 67 miles northeast of the station. Cleburne, 23 miles east of the station site with a population of 16,015 people is the next largest community in the area. Cleburne is expected to reach a population of 25,000 by the mid 1980's, thus becoming the nearest population center as defined by 10 CFR Part 100.

2.1.3.6 Population Density

The cumulative resident population projected for the year of initial plant operation (1980) is compared with a cumulative population resulting from a uniform population density of 500 people/square mile in all directions from the plant in Figure 2.1-6. A similar comparison is made for the end of plant life (2020), but compared with a cumulative population resulting from a uniform population density of 1,000 people/square mile in Figure 2.1-7.

TABLE 2.1-1
HISTORICAL AND PROJECTED POPULATIONS FOR COUNTIES WITHIN
50 MILES OF CPNPP

(Sheet 1 of 3)

County	1930	1940	1950	1960	1970
Bosque	15,750	15,761	11,836	10,809	10,966
Comanche	18,430	19,245	15,516	11,865	11,898
Coryell	19,999	20,226	16,284	23,961	35,311
Dallas	325,691	398,564	614,799	951,527	1,327,321
Eastland	34,156	30,345	23,942	19,526	18,092
Ellis	53,936	47,733	45,645	43,395	46,638
Erath	20,804	20,760	18,434	16,236	18,141
Hamilton	13,523	13,303	10,660	8,488	7,198
Hill	43,036	38,355	31,282	23,650	22,596
Hood	6,779	6,674	5,287	5,443	6,368
Jack	9,046	10,206	7,755	7,418	6,711
Johnson	33,317	30,384	31,390	34,720	45,769
McLennan	98,682	101,898	130,194	150,091	147,553
Palo Pinto	17,576	18,456	17,154	20,516	28,962
Parker	18,759	20,482	21,528	22,880	33,888
Somervell	3,016	3,071	2,542	2,577	2,793
Stephens	16,560	12,356	10,597	8,885	8,414
Tarrant	197,178	225,521	361,253	538,495	716,317
Wise	19,178	19,074	16,141	17,012	19,687

TABLE 2.1-1
HISTORICAL AND PROJECTED POPULATIONS FOR COUNTIES WITHIN
50 MILES OF CPNPP

(Sheet 2 of 3)

County	1976	1980	Projections 1990	2000	2010	2020
Bosque	10,775	10,711	10,423	10,295	10,353	10,610
Comanche	11,850	11,899	12,046	12,428	13,015	13,780
Coryell	37,000	37,500	39,900	41,900	43,600	45,000
Dallas	1,520,000	1,678,100	2,126,100	2,658,100	3,294,900	4,056,700
Eastland	17,000	16,726	15,669	14,929	14,328	13,857
Ellis	52,100	55,500	63,000	69,600	76,000	82,000
Erath	19,200	19,900	21,900	23,800	25,600	27,400
Hamilton	6,750	6,400	5,600	4,900	4,300	3,700
Hill	21,900	21,861	21,641	22,008	22,816	24,055
Hood	15,601	22,000	27,500	33,577	40,796	49,118
Jack	6,600	6,500	6,300	6,000	5,700	5,400
Johnson	55,000	62,358	86,509	105,627	128,337	154,518
McLennan	152,900	156,600	166,600	174,700	181,700	187,700
Palo Pinto	23,650	24,000	25,000	25,750	26,250	26,500
Parker	36,600	42,589	50,105	56,881	63,055	68,975
Somervell	5,216	5,616	5,600	6,200	7,500	9,000
Stephens	8,200	8,077	7,922	7,964	8,130	8,424
Tarrant	825,000	892,000	1,114,300	1,370,900	1,673,800	2,029,900
Wise	21,600	22,900	26,600	30,500	34,700	39,300

Sources:

a. Historical data for 1930-1970 are from US Bureau of Census.

b. Estimates for 1976 for Hood and Somervell Counties are based on a comprehensive enumeration of housing units by Westwood Research, Inc.

TABLE 2.1-1
HISTORICAL AND PROJECTED POPULATIONS FOR COUNTIES WITHIN
50 MILES OF CPNPP

(Sheet 3 of 3)

c. Estimates for 1976 for counties other than Hood and Somervell, are derived from the long-term trend lines considering historical data through the 1970 Census and long-term projections described below.

d. The long-term projections of population growth in all counties are projections made by various local councils of governments and Texas state agencies. Where there are significant differences in projections made by different authorities for the same county, recent experience and current and short-term trends have been considered in selecting the long-term projections for inclusion in the table.

TABLE 2.1-2
CENTERS OF POPULATION WITHIN 50 MILES OF CPNPP

(Sheet 1 of 4)

Community[a]	1970 Population[b]	Distance from Site in Miles[c]	Direction
0-10 Miles			
Glen Rose	1,554	4.3-5.5	SE,SSE, &S
Granbury	2,473	9.5-10.9	N
Tolar	312	9.7-10.6	NW
10-20 Miles			
Granbury	(see Granbury, 0-10 Miles)		N
Godley	533	18.1-18.8	NE
Tolar	(see Tolar, 0-10 Miles)		NW
Walnut Springs	495	16.3-17.5	S
20-30 Miles			
Aledo	620	29	NNE
Benbrook	8,169	29-35	NNE & NE
Blum	382	25	ESE
Cleburne	16,015	21-25	ENE & E
Hico	975	26	SW & SSW
Iredell	316	22	SSW
Joshua	924	26	ENE
Keene	2,440	28	ENE
Lipan	333	22	NW
Meridian	1,162	27	SSE
Morgan	415	22	SSE
Rio Vista	370	24	E
Stephenville	9,277	24-27	WSW & W

TABLE 2.1-2
CENTERS OF POPULATION WITHIN 50 MILES OF CPNPP

(Sheet 2 of 4)

Community[a]	1970 Population[b]	Distance from Site in Miles[c]	Direction
30-40 Miles			
Alvarado	2,129	34	ENE & E
Benbrook	(see Benbrook, 20-30 Miles)		NNE & NE
Burelson	7,713	30-33	NE & ENE
Clifton	2,578	37	SSE
Crowley	2,662	32	NE
Dublin	2,810	36	WSW
Fort Worth	393,476[d]	33-50	NNE & NE
Edgecliff	1,143	36	NE
Everman	4,570	37	NE
Forest Hill	8,236	39-40	NE
Gordon	457	38	WNW
Lakeside	988	39-40	NNE
Mineral Wells	18,411	38-42	NNW
River Oaks	8,193	39-41	NNE & NE
Weatherford	11,750	31-33	N
Westover Hills	374	37	NE
Westworth Village	4,578	38	NNE & NE
White Settlement	13,449	35-38	NNE
Willow Park	230	31-34	NNE
40-50 Miles			
Arlington	90,643[d]	42-50	NE
Azle	4,493	42-44	NNE
Blue Mound	1,293	47	NE
Dallas	844,401[d]	47-50	ENE

TABLE 2.1-2
CENTERS OF POPULATION WITHIN 50 MILES OF CPNPP

(Sheet 3 of 4)

Community[a]	1970 Population[b]	Distance from Site in Miles[c]	Direction
Dalworthington Gardens	757	46	NE
De Leon	2,170	46	WSW
Fort Worth	(see Fort Worth, 30-40 Miles)[d]		NNE & NE
Grand Prairie	50,904[d]	49-50	ENE
Gustine	357	48	SW
40-50 Miles			
Haltom City	28,127	44-48	NE
Hamilton	2,760	44-46	SSW
Hillsboro	7,224	42-46	ESE
Hurst	27,215[d]	49-50	NE
Kennedale	3,076	40-41	NE
Lake Worth	4,958	40-41	NNE
Mansfield	3,658	41-42	ENE
Maypearl	462	45	E
Midlothian	2,322	48-49	ENE
Mineral Wells	(see Mineral Wells, 30-40 Miles)		NNW
Mingus	273	40-41	WNW
North Richland Hills	16,514[d]	47-50	NE
Pantego	1,168	46	NE
Reno	688	44-49	NNE
Richland Hills	8,865	47-50	NE
River Oaks	(see River Oaks, 30-40 Miles)		NNE & NE
Saginaw	2,382	45-47	NNE

TABLE 2.1-2
CENTERS OF POPULATION WITHIN 50 MILES OF CPNPP

(Sheet 4 of 4)

Community[a]	1970 Population[b]	Distance from Site in Miles[c]	Direction
Sansom Park	4,771	41-42	NNE
Springtown	1,194	46-47	N
Strawn	786	45	WNW
Valley Mills	1,022	49-50	SSE
Venus	414	41	ENE
Watauga	3,611[d]	49-50	NE

a) Communities include all unincorporated settlements with 1,000 or more inhabitants, and all incorporated communities.

b) Source is the US Bureau of Census (Ref. 2.1-1).

c) Distances are indicated to the nearest and most distant portion of the densely settled area of a community rather than to city limits, as such. Beyond 20 miles the distance is rounded to the nearest mile.

d) Population of the community is located partly outside of the 50 mile radius of CPNPP.

TABLE 2.1-3
POPULATION DISTRIBUTION BY SECTOR-AREA WITHIN 10 MILES OF CPNPP

(Sheet 1 of 2)

	0 to 1 Mile Zone								1 to 2 Mile Zone							
Sector	1970	1976	1980	1990	2000	2010	2020	Sector	1970	1976	1980	1990	2000	2010	2020
N								N	0	0	0	0	0	0	0
NNE								NNE	0	0	0	0	0	0	0
NE								NE	0	0	0	0	0	0	0
ENE								ENE	6	0	0	0	0	0	0
E								E	0	0	0	0	0	0	0
ESE								ESE	3	3	4	4	4	5	5
SE								SE	0	0	0	0	0	0	0
SSE	NO RESIDENT POPULATION							SSE	0	0	0	0	0	0	0
S								S	0	0	0	0	0	0	0
SSW								SSW	3	3	4	4	4	5	5
SW								SW	6	45	47	47	53	64	77
WSW								WSW	6	12	13	13	14	17	21
W								W	12	9	10	12	14	16	21
WNW								WNW	3	3	3	3	3	3	3
NW								NW	3	0	0	0	0	0	0
NNW								NNW	3	0	0	0	0	0	0
TOTAL								TOTAL	45	75	81	83	92	110	132

	2 to 3 Mile Zone								3 to 4 Mile Zone							
Sector	1970	1976	1980	1990	2000	2010	2020	Sector	1970	1976	1980	1990	2000	2010	2020
N	3	6	9	11	13	16	20	N	9	23	33	41	50	61	74
NNE	6	9	13	17	20	24	29	NNE	27	40	57	71	87	106	127
NE	15	42	58	73	91	111	132	NE	18	20	29	36	44	53	64
ENE	9	9	9	9	10	13	15	ENE	18	29	37	43	51	62	75
E	0	0	0	0	0	0	0	E	15	26	28	28	31	38	45
ESE	15	9	10	10	11	13	15	ESE	42	62	67	66	74	87	106
SE	3	6	6	6	7	8	10	SE	12	26	28	28	31	38	45
SSE	3	3	4	4	4	5	6	0	6	12	13	13	14	17	21
S	9	14	14	14	16	19	23	S	0	3	4	4	4	5	6
SSW	0	0	0	0	0	0	0	SSW	9	9	9	9	11	13	15
SW	3	12	12	13	14	17	21	SW	0	9	10	10	11	14	16
WSW	3	3	4	4	4	5	6	WSW	6	6	7	9	10	12	15
W	9	12	19	23	28	35	42	W	24	20	29	36	44	53	64
WNW	6	6	9	11	13	16	20	WNW	15	23	31	39	47	57	69
NW	0	0	0	0	0	0	0	NW	0	0	0	0	0	0	0
NNW	6	6	7	8	10	12	15	NNW	3	0	0	0	0	0	0
TOTAL	90	137	174	203	241	294	354	TOTAL	204	308	382	433	509	616	742

TABLE 2.1-3
POPULATION DISTRIBUTION BY SECTOR-AREA WITHIN 10 MILES OF CPNPP

(Sheet 2 of 2)

	4 to 5 Mile Zone								5 to 10 Mile Zone						
Sector	1970	1976	1980	1990	2000	2010	2020	Sector	1970	1976	1980	1990	2000	2010	2020
N	36	37	53	66	81	98	118	N	1,434	3,839	5,414	6,766	8,263	10,040	12,087
NNE	21	17	24	30	37	45	54	NNE	93	1,207	1,703	2,129	2,599	3,158	3,802
NE	27	104	144	181	221	268	324	NE	147	409	576	721	880	1,069	1,287
ENE	13	20	29	36	44	53	64	ENE	126	144	201	249	302	368	443
E	18	26	28	28	31	38	45	E	69	189	203	202	223	270	324
ESE	69	84	90	90	100	120	145	ESE	196	549	591	589	653	790	948
SE	90	76	82	82	91	110	131	SE	70	115	124	123	136	165	198
SSE	1,258	2,260	2,433	2,426	2,636	3,249	3,900	SSE	363	726	781	780	863	1,043	1,262
S	88	185	199	198	219	265	319	S	38	98	106	105	117	141	169
SSW	26	51	55	55	61	74	88	SSW	81	180	194	193	214	259	311
SW	18	45	48	48	53	65	77	SW	63	87	94	94	104	125	150
WSW	6	3	4	4	4	5	6	WSW	66	116	144	165	194	235	283
W	12	14	20	25	30	37	44	W	45	79	112	140	171	208	251
WNW	30	31	44	55	67	82	98	WNW	96	196	275	344	420	610	614
NW	3	0	0	0	0	0	0	NW	249	348	491	613	749	910	1,096
NNW	24	20	29	36	44	53	64	NNW	131	362	610	638	779	946	1,140
TOTAL	1,739	2,973	3,282	3,360	3,769	4,562	5,477	TOTAL	3,272	8,643	11,519	18,851	16,667	20,237	24,354

CUMULATIVE POPULATIONS
(Miles from Site)

Year	0-1	0-2	0-3	0-4	0-5	0-10
1970	3	48	138	342	2,081	5,353
1976	0	75	212	520	3,493	12,136
1980	0	81	255	637	3,919	15,438
1990	0	83	286	719	4,079	17,930
2000	0	92	333	842	4,611	21,278
2010	0	110	404	1,020	5,582	25,819
2020	0	132	486	1,228	6,705	31,059

TABLE 2.1-4
POPULATION DISTRIBUTION BY AGE GROUPS WITHIN 10 MILE RADIUS OF CPNPP FOR YEAR 2000

(Mid point of plant life)

	Population	
Age Groups	Number	Percentage[a]
0-11	4,807	22.6
12-18	2,843	13.4
19 and over	13,628	64.0
	21,278[b]	100.0

a) As specified in Appendix D of the Regulatory Guide 4.3, Revision 2, a “test of significance” was made comparing the 1970 population distribution by age groups of Somervell County with that of the United States as a whole. The test indicated that the two distributions differed less than 10 percent. Thus, the percentages in the table above are based on the unadjusted population distribution by age groups as projected for the United States in the year 2000. The United States projection of the population by age groups is found in Table No. 3 of the Statistical Abstract of the United States, 1975 (Ref. 2.1-3).

b) Population total of the area 0 to 10 miles for the year 2000 is from Table 2.1-3

TABLE 2.1-5
POPULATION DISTRIBUTION BY SECTOR-AREA WITH N 10 MILES OF CPNPP

	10 to 20 Mile Zone							20 to 30 Mile Zone							
Sector	1970	1976	1980	1990	2000	2010	2020	Sector	1970	1976	1980	1990	2000	2010	2020
N	1,779	3,935	5,517	6,883	8,392	10,173	12,223	N	1,437	1,552	1,806	2,124	2,412	2,674	2,925
NNE	819	2,213	3,072	3,850	4,693	5,691	6,838	NNE	2,065	2,241	2,601	3,069	3,497	3,893	4,283
NE	782	1,113	1,366	1,822	2,224	2,703	3,255	NE	1,675	1,925	2,123	2,762	3,381	4,113	4,965
ENE	529	642	730	1,012	1,235	1,500	1,806	ENE	13,042	15,675	17,771	24,655	30,104	36,578	44,037
E	312	377	428	592	722	878	1,056	E	14,148	16,996	19,269	26,729	32,636	39,650	47,738
ESE	280	334	374	502	606	733	880	ESE	1,958	2,038	2,131	2,424	2,682	3,014	3,415
SE	185	184	187	193	199	211	228	SE	463	452	451	443	445	455	474
SSE	201	211	212	207	209	217	230	SSE	1,916	1,882	1,871	1,821	1,799	1,808	1,853
S	647	660	659	641	640	653	679	S	235	231	229	223	220	222	227
SSW	150	175	182	193	201	221	245	SSW	1,204	1,150	1,111	1,016	939	878	827
SW	265	292	304	331	361	392	426	SW	918	913	906	898	894	896	897
WSW	338	369	388	429	470	509	549	WSW	7,895	8,355	8,659	9,631	10,358	11,142	11,923
W	290	315	339	379	419	461	504	W	2,987	3,160	3,275	3,605	3,917	4,214	4,510
WNW	298	351	402	462	526	595	671	WNW	465	490	508	558	605	651	696
NW	398	580	813	1,016	1,239	1,503	1,809	NW	781	1,045	1,377	1,678	2,002	2,383	2,815
NNW	259	803	1,123	1,401	1,706	2,068	2,484	NNW	975	1,053	1,230	1,448	1,645	1,828	2,003
TOTAL	7,532	12,554	16,096	19,913	23,842	28,508	33,883	TOTAL	52,164	59,158	65,318	82,984	97,536	114,339	133,588

	30 to 40 Mile Zone							40 to 50 Mile Zone							
Sector	1970	1976	1980	1990	2000	2010	2020	Sector	1970	1976	1980	1990	2000	2010	2020
N	14,448	15,610	18,165	21,371	24,251	26,892	29,418	N	4,108	4,440	5,130	6,029	6,848	7,605	8,340
NNE	35,194	40,381	43,806	54,574	66,911	81,386	98,337	NNE	32,037	36,712	39,926	49,632	60,687	73,594	88,663
NE	211,833	243,980	263,809	329,597	405,492	495,083	600,405	NE	313,754	361,432	390,785	488,176	600,592	733,291	889,299
ENE	14,675	17,463	19,604	26,641	32,576	39,616	47,765	ENE	13,818	15,844	17,154	21,181	25,544	30,615	36,474
E	2,718	3,178	3,555	4,791	5,778	6,956	8,319	E	2,258	2,537	2,742	3,323	3,808	4,351	4,947
ESE	3,013	2,919	2,914	2,885	2,934	3,042	3,208	ESE	8,761	8,497	8,486	8,409	8,558	8,877	9,361
SE	3,496	3,401	3,392	3,343	3,377	3,476	3,641	SE	1,969	1,948	1,958	1,978	2,029	2,101	2,195
SSE	3,296	3,240	3,220	3,134	3,096	3,113	3,189	SSE	2,135	2,118	2,121	2,110	2,118	2,149	2,208
S	768	750	741	711	693	687	694	S	911	922	917	930	939	951	957
SSW	472	442	420	368	322	283	243	SSW	3,533	3,313	3,142	2,747	2,405	2,109	1,815
SW	615	623	626	644	664	686	710	SW	1,065	1,052	1,048	1,044	1,058	1,091	1,138
WSW	3,682	3,897	4,038	4,443	4,830	5,196	5,560	WSW	3,506	3,496	3,512	3,560	3,675	3,851	4,078
W	736	780	808	889	966	1,039	1,112	W	1,283	1,243	1,237	1,212	1,206	1,210	1,226
WNW	733	647	654	693	725	753	774	WNW	1,343	1,131	1,131	1,168	1,201	1,222	1,232
NW	1,260	1,044	1,044	1,088	1,120	1,142	1,153	NW	1,138	943	943	982	1,012	1,032	1,041
NNW	18,748	16,566	17,292	18,708	19,907	20,908	21,752	NNW	10,769	9,562	10,008	10,852	11,571	12,172	12,685
TOTAL	315,687	354,921	384,088	473,880	573,652	690,258	826,280	TOTAL	402,388	455,190	490,240	603,333	733,251	886,221	1,065,659

CUMULATIVE POPULATIONS
(Miles from Site)

Year	0-10	0-20	0-30	0-40	0-50
1970	5,353	12,885	65,0493	80,736	783,124
1976	12,136	24,690	83,848	438,769	893,959
1980	15,438	31,534	96,852	480,940	971,180
1990	17,930	37,843	120,827	594,707	1,198,040
2000	21,278	45,120	142,656	716,308	1,449,559
2010	25,819	54,327	168,726	858,984	1,745,205
2020	31,059	64,942	198,530	1,024,810	2,090,469

TABLE 2.1-6
POPULATION DISTRIBUTION BY AGE GROUPS BETWEEN 10 AND 50 MILE RADIUS OF CPNPP FOR YEAR 2000

(Mid point of plant life)

	Population	
Age Groups	Number	Percentage[(a)]
0-11	322,791	22.6
12-18	191,390	13.4
19 and over	914,090	64.0
	1,428,271	100.0

a) As specified in Appendix D of the Regulatory Guide 4.3, Revision 2, a “test of significance” was made comparing the 1970 population distribution by age groups of Somervell County with that of the United States as a whole. The test indicated that the two distributions differed less than 10 percent. Thus, the percentages in the table above are based on the unadjusted population distribution by age groups as projected for the United States in the year 2000. The United States projection of the population by age groups is found in Table No. 3 of the Statistical Abstract of the United States, 1975 (Ref. 2.1-3).

b) Population total of total area between 10 miles and 50 miles for the year 2000 is from Table 2.1-5.

TABLE 2.1-7
TYPE AND TYPICAL LENGTH OF STAY OF TRANSIENTS WITHIN 10 MILE RADIUS OF CPNPP

Type of Transients	Typical Length of Stay
Overnight Transients (Seasonal Visitors)	
Camping (Campgrounds and RV Parks)	Winter visitors for weekends Summer visitors for one week
Childrens' Institution	Indefinite[(a)]
Hotels/Motels	Two days
Organizational Camps	Winter campers for weekends Summer campers for 1 to 2 weeks
Vacation Homes	Winter visitors for weekends Summer length of stay cannot be specified for lack of data
Daily Transients	
Childrens' Institution	Eight hours
Recreation (Parks and Lakes)	Four hours
Schools	Eight hours
Shopping Centers	Three hours

a) The children of the institution are listed as transients because they are not included in the estimate and projections of the permanent population. Some students are live-in and some attend day only.

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 1 of 24)

Daily Transient - Summer Weekday

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	3	10	10	10	10	N	0	38	143	143	143	143
NNE	0	7	27	27	27	27	NNE	0	23	88	88	88	88
NE	0	1	4	4	4	4	NE	0	80	295	295	295	295
ENE	0	1	4	4	4	4	ENE	0	14	52	52	52	52
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	1	1	1	1	NW	0	2	7	7	7	7
NNW	0	1	4	4	4	4	NNW	0	13	47	47	47	47
TOTAL	0	13	50	50	50	50	TOTAL	0	170	632	632	632	632

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	2	10	10	10	10	NE	0	0	0	0	0	0
ENE	0	2	5	5	5	5	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	100	110	138	171	211	260
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	3	3	3	3	NW	0	1	3	3	3	3
NNW	0	13	50	50	50	50	NNW	0	1	3	3	3	3
TOTAL	0	18	68	68	68	68	TOTAL	100	112	144	177	217	266

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 2 of 24)

Daily Transient - Summer Weekday

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	840	1161	1535	1807	2135	2505
NNE	0	0	0	0	0	0	NNE	176	219	380	380	380	380
NE	0	0	0	0	0	0	NE	171	213	369	369	369	369
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	285	308	309	346	419	504	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	285	308	309	346	419	504	TOTAL	1187	1593	2284	2556	2884	3254

CUMULATIVE TRANS ENT POPULATION (miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	100	385	1572
1980	13	183	201	313	621	2214
1990	50	682	750	894	1203	3487
2000	50	682	750	922	1268	3824
2010	50	682	750	967	1386	4270
2020	50	682	750	1016	1520	4774

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 3 of 24)

Daily Transient-Summer Weekend

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	6	23	23	23	23	N	0	85	319	319	319	319
NNE	0	17	61	61	61	61	NNE	0	52	195	195	195	195
NE	0	2	9	9	9	9	NE	0	177	660	660	660	660
ENE	0	2	9	9	9	9	ENE	0	31	115	115	115	115
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	2	2	2	2	NW	0	4	16	16	16	16
NNW	0	2	8	8	8	8	NNW	0	28	106	106	106	106
TOTAL	0	30	112	112	112	112	TOTAL	0	377	1411	1411	1411	1411

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	6	22	22	22	22	NE	0	0	0	0	0	0
ENE	0	3	11	11	11	11	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	800	878	1102	1370	1693	2076
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	2	6	6	6	6	NW	0	2	5	5	5	5
NNW	0	30	113	113	113	113	NNW	0	1	6	6	6	6
TOTAL	0	41	152	152	152	152	TOTAL	800	881	1113	1381	1704	2087

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 4 of 24)

Daily Transient-Summer Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	2075	2874	3778	4464	5291	6222
NNE	0	0	0	0	0	0	NNE	392	488	845	845	845	845
NE	0	0	0	0	0	0	NE	381	475	822	822	822	822
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	535	580	585	653	790	952	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	535	580	585	653	790	952	TOTAL	2848	3837	5445	6131	6958	7889

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	800	1335	4183
1980	30	407	448	1329	1909	5746
1990	112	1523	1675	2788	3373	8818
2000	112	1523	1675	3056	3709	9840
2010	112	1523	1675	3379	4169	11,127
2020	112	1523	1675	3762	4714	12,603

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 5 of 24)

Daily Transient-Summer Holiday Weekend

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	9	35	35	35	35	N	0	133	495	495	495	495
NNE	0	25	93	93	93	93	NNE	0	80	302	302	302	302
NE	0	4	14	14	14	14	NE	0	274	1022	1022	1022	1022
ENE	0	4	15	15	15	15	ENE	0	49	179	179	179	179
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	4	4	4	4	NW	0	6	24	24	24	24
NNW	0	3	13	13	13	13	NNW	0	44	164	164	164	164
TOTAL	0	46	174	174	174	174	TOTAL	0	586	2186	2186	2186	2186

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	8	33	33	33	33	NE	0	0	0	0	0	0
ENE	0	5	17	17	17	17	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	969	1060	1330	1658	2044	2506
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	3	10	10	10	10	NW	0	2	9	9	9	9
NNW	0	47	174	174	174	174	NNW	0	2	9	9	9	9
TOTAL	0	63	234	234	234	234	TOTAL	969	1064	1348	1676	2062	2524

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 6 of 24)

Daily Transient-Summer Holiday Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	3127	4328	5697	6725	7965	9362
NNE	0	0	0	0	0	0	NNE	608	757	1312	1312	1312	1312
NE	0	0	0	0	0	0	NE	590	735	1275	1275	1275	1275
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	1224	1339	1496	1766	2160	2625	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	1224	1339	1496	1766	2160	2625	TOTAL	4325	5820	8284	9312	10552	11949

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	969	2193	6518
1980	46	632	695	1759	3098	8918
1990	174	2360	2594	3942	5438	13,722
2000	174	2360	2594	4270	6036	15,348
2010	174	2360	2594	4656	6816	17,368
2020	174	2360	2594	5118	7743	19,692

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 7 of 24)

Daily Transient – Winter Weekday

0 to 1 Mile Zone							1 to 2 Mile Zone						
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	1	5	5	5	5	N	0	2	8	8	8	8
NNE	0	1	5	5	5	5	NNE	0	2	4	4	4	4
NE	0	1	4	4	4	4	NE	0	31	114	114	114	114
ENE	0	1	4	4	4	4	ENE	0	4	16	16	16	16
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	1	1	1	1	NW	0	2	7	7	7	7
NNW	0	1	4	4	4	4	NNW	0	4	13	13	13	13
TOTAL	0	5	23	23	23	23	TOTAL	0	45	162	162	162	162

2 to 3 Mile Zone							3 to 4 Mile Zone						
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	1	2	2	2	2	NE	0	0	0	0	0	0
ENE	0	1	5	5	5	5	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	17	87	87	87	87	87
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	8	8	11	14	17	21
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	O	O	O	O	O	O	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	3	3	3	3	NW	0	1	3	3	3	3
NNW	0	2	10	10	10	10	NNW	0	1	3	3	3	3
TOTAL	0	5	20	20	20	20	TOTAL	25	97	104	107	110	114

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 8 of 24)

Daily Transient – Winter Weekday

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	1215	1699	2160	2611	3156	3769
NNE	0	0	0	0	0	0	NNE	69	86	151	151	151	151
NE	0	0	0	0	0	0	NE	68	85	147	147	147	147
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	716	774	768	852	1031	1238	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	221	312	389	475	579	696
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	716	774	768	852	1031	1238	TOTAL	1573	2192	2847	3384	4033	4763

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	25	741	2314
1980	5	50	55	152	926	3118
1990	23	185	205	309	1077	3924
2000	23	185	205	312	1164	4548
2010	23	185	205	315	1346	5379
2020	23	185	205	319	1557	6320

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 9 of 24)

Daily Transient-Winter Weekend

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	2	8	8	8	8	N	0	3	13	13	13	13
NNE	0	2	9	9	9	9	NNE	0	2	6	6	6	6
NE	0	2	7	7	7	7	NE	0	51	187	187	187	187
ENE	0	2	7	7	7	7	ENE	0	7	27	27	27	27
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	2	2	2	2	NW	0	3	12	12	12	12
NNW	0	2	6	6	6	6	NNW	0	6	23	23	23	23
TOTAL	0	11	39	39	39	39	TOTAL	0	72	268	268	268	268

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	1	3	3	3	3	NE	0	0	0	0	0	0
ENE	0	2	9	9	9	9	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	20	22	27	34	42	51
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	5	5	5	5	NW	0	1	4	4	4	4
NNW	0	5	16	16	16	16	NNW	0	1	4	4	4	4
TOTAL	0	9	33	33	33	33	TOTAL	20	24	35	42	50	59

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 10 of 24)

Daily Transient-Winter Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	1631	2285	1209	3509	4231	5047
NNE	0	0	0	0	0	0	NNE	116	145	250	250	250	250
NE	0	0	0	0	0	0	NE	113	141	243	243	243	243
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	515	556	555	616	745	897	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	515	556	555	616	745	897	TOTAL	1860	2571	1702	4002	4724	5540

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	20	535	2395
1980	11	83	92	116	672	3243
1990	39	307	340	375	930	2632
2000	39	307	340	382	998	5000
2010	39	307	340	390	1135	5859
2020	39	307	340	399	1296	6836

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 11 of 24)

Daily Transient-Winter Holiday Weekend

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	4	16	16	16	16	N	0	7	27	27	27	27
NNE	0	4	17	17	17	17	NNE	0	4	12	12	12	12
NE	0	4	15	15	15	15	NE	0	104	390	390	390	390
ENE	0	4	16	16	16	16	ENE	0	15	55	55	55	55
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	1	4	4	4	4	NW	0	7	25	25	25	25
NNW	0	4	13	13	13	13	NNW	0	13	47	47	47	47
TOTAL	0	21	81	81	81	81	TOTAL	0	150	556	556	556	556

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	0	0	0	0	0	0
NNE	0	0	0	0	0	0	NNE	0	0	0	0	0	0
NE	0	2	7	7	7	7	NE	0	0	0	0	0	0
ENE	0	5	18	18	18	18	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	47	52	65	80	99	122
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	3	10	10	10	10	NW	0	2	9	9	9	9
NNW	0	9	34	34	34	34	NNW	0	2	9	9	9	9
TOTAL	0	19	69	69	69	69	TOTAL	47	56	83	98	117	140

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 12 of 24)

Daily Transient-Winter Holiday Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	2585	3612	4627	5560	6684	7951
NNE	0	0	0	0	0	0	NNE	241	300	520	520	520	520
NE	0	0	0	0	0	0	NE	234	292	505	505	505	505
ENE	0	0	0	0	0	0	ENE	0	0	0	0	0	0
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	681	736	740	824	998	1200	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	681	736	740	824	998	1200	TOTAL	3060	4204	5652	6585	7709	8976

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	47	728	3788
1980	21	171	190	246	982	5186
1990	81	637	706	789	1529	7181
2000	81	637	706	804	1628	8213
2010	81	637	706	823	1821	9530
2020	81	637	706	846	2046	11,022

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 13 of 24)

Overnight Transient-Summer Weekday

0 to 1 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE			NO TRANS ENT POPULATION			
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

1 to 2 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE			NO TRANSIENT POPULATION			
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

2 to 3 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE			NO TRANSIENT POPULATION			
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

3 to 4 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0
NNE	0	0	0	0	0	0
NE	0	0	0	0	0	0
ENE	0	0	0	0	0	0
E	13	70	70	70	70	70
ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0
S	0	0	0	0	0	0
SSW	80	88	110	138	170	208
SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0
W	0	0	0	0	0	0
WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0
TOTAL	93	158	180	208	240	278

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 14 of 24)

Overnight Transient-Summer Weekday

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	520	708	945	1107	1302	1525
NNE	0	0	0	0	0	0	NNE	748	974	1253	1438	1659	1913
NE	0	0	0	0	0	0	NE	241	320	472	519	576	641
ENE	0	0	0	0	0	0	ENE	26	36	45	55	66	80
E	275	275	275	275	275	275	E	24	26	26	28	34	41
ESE	0	0	0	0	0	0	ESE	183	184	184	185	187	189
SE	0	0	0	0	0	0	SE	450	450	450	450	450	450
SSE	240	297	314	341	379	423	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	70	70	70	70	70	70	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	585	642	659	686	724	768	TOTAL	2192	2698	3375	3782	4274	4839

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	93	678	2870
1980	0	0	0	158	800	3498
1990	0	0	0	180	839	4214
2000	0	0	0	208	894	4676
2010	0	0	0	240	964	5238
2020	0	0	0	278	1046	5885

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 15 of 24)

Overnight Transient-Summer Weekend

0 to 1 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE						
S		NO TRANSIENT POPULATION				
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

1 to 2 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE						
S		NO TRANSIENT POPULATION				
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

2 to 3 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE						
S						
SSW		NO TRANSIENT POPULATION				
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

3 to 4 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0
NNE	0	0	0	0	0	0
NE	0	0	0	0	0	0
ENE	0	0	0	0	0	0
E	13	70	70	70	70	70
ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0
S	0	0	0	0	0	0
SSW	161	175	220	274	337	413
SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0
W	0	0	0	0	0	0
WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0
TOTAL	174	245	290	344	407	483

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 16 of 24)

Overnight Transient-Summer Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	994	1346	1814	2113	2470	2879
NNE	0	0	0	0	0	0	NNE	1574	1832	2417	2788	3228	3737
NE	0	0	0	0	0	0	NE	449	590	896	967	1052	1150
ENE	0	0	0	0	0	0	ENE	52	73	90	109	133	160
E	275	275	275	275	275	275	E	48	52	51	57	69	82
ESE	0	0	0	0	0	0	ESE	191	192	192	194	198	203
SE	0	0	0	0	0	0	SE	450	450	450	450	450	450
SSE	306	373	409	459	524	601	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	70	70	70	70	70	70	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	651	718	754	804	869	946	TOTAL	3758	4535	5910	6678	7600	8661

CUMULATIVE TRANS ENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	174	825	4583
1980	0	0	0	245	963	5498
1990	0	0	0	290	1044	6954
2000	0	0	0	344	1148	7826
2010	0	0	0	407	1276	8876
2020	0	0	0	483	1429	10,090

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 17 of 24)

Overnight Transient-Summer Holiday Weekend

0 to 1 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E		NO TRANSIENT POPULATION				
ESE						
SE						
SSE						
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

1 to 2 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E		NO TRANSIENT POPULATION				
ESE						
SE						
SSE						
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

2 to 3 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E		NO TRANSIENT POPULATION				
ESE						
SE						
SSE						
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

3 to 4 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0
NNE	0	0	0	0	0	0
NE	0	0	0	0	0	0
ENE	0	0	0	0	0	0
E	13	70	70	70	70	70
ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0
S	0	0	0	0	0	0
SSW	161	175	220	274	337	413
SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0
W	0	0	0	0	0	0
WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0
TOTAL	174	245	290	344	407	483

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 18 of 24)

Overnight Transient-Summer Holiday Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	1457	1992	2695	3133	3652	4252
NNE	0	0	0	0	0	0	NNE	2096	2784	3670	4227	4886	5647
NE	0	0	0	0	0	0	NE	655	856	1316	1411	1524	1654
ENE	0	0	0	0	0	0	ENE	80	112	138	168	204	246
E	400	400	400	400	400	400	E	72	77	77	85	103	123
ESE	0	0	0	0	0	0	ESE	200	202	201	204	209	215
SE	0	0	0	0	0	0	SE	556	556	556	556	556	556
SSE	676	679	790	931	1105	1312	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	100	100	100	100	100	100	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	1176	1179	1290	1431	1605	1812	TOTAL	5116	6579	8653	9784	11134	12693

CUMULATIVE TRANS ENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	174	1350	6466
1980	0	0	0	245	1424	8003
1990	0	0	0	290	1580	10,233
2000	0	0	0	344	1775	11,559
2010	0	0	0	407	2012	13,146
2020	0	0	0	483	2295	14,988

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 19 of 24)

Overnight Transient-Winter Weekday

0 to 1 Mile Zone							1 to 2 Mile Zone						
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N							N						
NNE							NNE						
NE							NE						
ENE							ENE						
E							E						
ESE							ESE						
SE							SE						
SSE		NO TRANSIENT POPULATION					SSE		NO TRANSIENT POPULATION				
S							S						
SSW							SSW						
SW							SW						
WSW							WSW						
W							W						
WNW							WNW						
NW							NW						
NNW							NNW						
TOTAL							TOTAL						

2 to 3 Mile Zone							3 to 4 Mile Zone						
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N							N	0	0	0	0	0	0
NNE							NNE	0	0	0	0	0	0
NE							NE	0	0	0	0	0	0
ENE							ENE	0	0	0	0	0	0
E							E	13	70	70	70	70	70
ESE							ESE	0	0	0	0	0	0
SE							SE	0	0	0	0	0	0
SSE		NO TRANSIENT POPULATION					SSE	0	0	0	0	0	0
S							S	0	0	0	0	0	0
SSW							SSW	0	0	0	0	0	0
SW							SW	0	0	0	0	0	0
WSW							WSW	0	0	0	0	0	0
W							W	0	0	0	0	0	0
WNW							WNW	0	0	0	0	0	0
NW							NW	0	0	0	0	0	0
NNW							NNW	0	0	0	0	0	0
TOTAL							TOTAL	13	70	70	70	70	70

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 20 of 24)

Overnight Transient-Winter Weekday

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	9	9	0	9	0	0	N	200	280	357	431	517	617
NNE	0	0	0	0	0	0	NNE	206	288	369	443	530	631
NE	0	0	0	0	0	0	NE	48	65	92	104	119	136
ENE	0	0	0	0	0	0	ENE	10	14	17	21	26	31
E	0	0	0	0	0	0	E	10	11	11	12	14	17
ESE	0	0	0	0	0	0	ESE	3	3	3	4	4	5
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	28	28	31	38	45	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	0	28	28	31	38	45	TOTAL	477	661	849	1015	1210	1437

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	13	13	490
1980	0	0	0	70	98	759
1990	0	0	0	70	98	947
2000	0	0	0	70	101	1116
2010	0	0	0	70	108	1318
2020	0	0	0	70	115	1552

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 21 of 24)

Overnight Transient-Winter Weekend

0 to 1 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE		NO TRANSIENT POPULATION				
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

1 to 2 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE		NO TRANSIENT POPULATION				
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

2 to 3 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE						
S						
SSW		NO TRANSIENT POPULATION				
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

3 to 4 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0
NNE	0	0	0	0	0	0
NE	0	0	0	0	0	0
ENE	0	0	0	0	0	0
E	13	70	70	70	70	70
ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0
S	0	0	0	0	0	0
SSW	0	0	0	0	0	0
SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0
W	0	0	0	0	0	0
WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0
TOTAL	13	70	70	70	70	70

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 22 of 24)

Overnight Transient-Winter Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	266	369	475	569	682	812
NNE	0	0	0	0	0	0	NNE	363	467	575	667	776	902
ENE	0	0	0	0	0	0	ENE	12	17	21	25	31	37
E	100	100	100	100	100	100	E	12	13	13	14	17	21
ESE	25	25	25	25	25	25	ESE	179	179	179	180	181	182
SE	0	0	0	0	0	0	SE	235	235	235	235	235	235
SSE	240	285	303	330	366	406	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	70	70	70	70	70	70	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	435	480	498	525	561	601	TOTAL	1150	1392	1655	1870	2129	2426

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	13	448	1598
1980	0	0	0	70	550	1942
1990	0	0	0	70	568	2223
2000	0	0	0	70	595	2465
2010	0	0	0	70	631	2760
2020	0	0	0	70	671	3097

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 23 of 24)

Overnight Transient-Winter Holiday Weekend

0 to 1 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE		NO TRANSIENT POPULATION				
SSE						
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

1 to 2 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE		NO TRANSIENT POPULATION				
SSE						
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

2 to 3 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N						
NNE						
NE						
ENE						
E						
ESE						
SE						
SSE		NO TRANSIENT POPULATION				
S						
SSW						
SW						
WSW						
W						
WNW						
NW						
NNW						
TOTAL						

3 to 4 Mile Zone

Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0
NNE	0	0	0	0	0	0
NE	0	0	0	0	0	0
ENE	0	0	0	0	0	0
E	13	70	70	70	70	70
ESE	0	0	0	0	0	0
SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0
S	0	0	0	0	0	0
SSW	161	175	220	273	337	413
SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0
W	0	0	0	0	0	0
WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0
TOTAL	174	245	290	343	407	483

TABLE 2.1-8
TRANSIENT POPULATION PROJECTIONS

(Sheet 24 of 24)

Overnight Transient-Winter Holiday Weekend

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	0	0	0	0	0	N	546	781	1002	1202	1440	1715
NNE	0	0	0	0	0	0	NNE	788	1062	1337	1583	1875	2213
NE	0	0	0	0	0	0	NE	160	216	304	345	394	450
ENE	0	0	0	0	0	0	ENE	36	50	62	75	92	111
E	100	100	100	100	100	100	E	32	34	34	38	46	55
ESE	75	75	75	75	75	75	ESE	188	189	189	190	192	194
SE	0	0	0	0	0	0	SE	235	235	235	235	235	235
SSE	571	657	768	909	1078	1280	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	0	0	0	0	0	0
SW	70	70	70	70	70	70	SW	0	0	0	0	0	0
WSW	0	0	0	0	0	0	WSW	0	0	0	0	0	0
W	0	0	0	0	0	0	W	0	0	0	0	0	0
WNW	0	0	0	0	0	0	WNW	0	0	0	0	0	0
NW	0	0	0	0	0	0	NW	0	0	0	0	0	0
NNW	0	0	0	0	0	0	NNW	0	0	0	0	0	0
TOTAL	816	902	1013	1154	1323	1525	TOTAL	1985	2567	3163	3668	4274	4973

CUMULATIVE TRANSIENT POPULATION
(miles from site)

YEAR	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	0	0	174	990	2975
1980	0	0	0	245	1147	3714
1990	0	0	0	290	1303	4466
2000	0	0	0	343	1497	5165
2010	0	0	0	407	1730	6004
2020	0	0	0	483	2008	6981

TABLE 2.1-9
MAXIMUM PERMANENT AND TRANSIENT POPULATION DISTR BUTION BY SECTOR-AREA

(Sheet 1 of 2)

	0 to 1 Mile Zone							1 to 2 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	0	9	35	35	35	35	N	0	133	495	495	495	495
NNE	0	25	93	93	93	93	NNE	0	80	302	302	302	302
NE	0	4	14	14	14	14	NE	0	274	1022	1022	1022	1022
ENE	0	4	15	15	15	15	ENE	0	49	179	179	179	179
E	0	0	0	0	0	0	E	0	0	0	0	0	0
ESE	0	0	0	0	0	0	ESE	3	4	4	4	5	5
SE	0	0	0	0	0	0	SE	0	0	0	0	0	0
SSE	0	0	0	0	0	0	SSE	0	0	0	0	0	0
S	0	0	0	0	0	0	S	0	0	0	0	0	0
SSW	0	0	0	0	0	0	SSW	3	4	4	4	5	5
SW	0	0	0	0	0	0	SW	45	47	47	53	64	77
WSW	0	0	0	0	0	0	WSW	12	13	13	14	17	21
W	0	0	0	0	0	0	W	9	10	12	14	16	21
WNW	0	0	0	0	0	0	WNW	3	3	3	3	3	3
NW	0	1	4	4	4	4	NW	0	6	24	24	24	24
NNW	0	3	13	13	13	13	NNW	0	44	164	164	164	164
TOTAL	0	46	174	174	174	174	TOTAL	75	667	2269	2278	2296	2318

	2 to 3 Mile Zone							3 to 4 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	6	9	11	13	16	20	N	23	33	41	50	61	74
NNE	9	13	17	20	24	29	NNE	40	57	71	87	106	127
NE	42	66	106	124	144	165	NE	20	29	36	44	53	64
ENE	9	14	26	27	30	32	ENE	29	37	43	51	62	75
E	0	0	0	0	0	0	E	26	28	28	31	38	45
ESE	9	10	10	11	13	15	ESE	62	67	66	74	87	106
SE	6	6	6	7	8	10	SE	26	28	28	31	38	45
SSE	3	4	4	4	5	6	SSE	12	13	13	14	17	21
S	14	14	14	16	19	23	S	3	4	4	4	5	6
SSW	0	0	0	0	0	0	SSW	978	1069	1339	1669	2057	2521
SW	12	12	13	14	17	21	SW	9	10	10	11	14	16
WSW	3	4	4	4	5	6	WSW	6	7	9	10	12	15
W	12	19	23	28	35	42	W	20	29	36	44	53	64
WNW	6	9	11	13	16	20	WNW	23	31	39	47	57	69
NW	0	3	10	10	10	10	NW	0	2	9	9	9	9
NNW	6	54	182	184	186	189	NNW	0	2	9	9	9	9
TOTAL	137	237	437	475	528	588	TOTAL	1277	1446	1781	2185	2678	3266

TABLE 2.1-9
MAXIMUM PERMANENT AND TRANSIENT POPULATION DISTR BUTION BY SECTOR-AREA

(Sheet 2 of 2)

	4 to 5 Mile Zone							5 to 10 Mile Zone					
Sector	1976	1980	1990	2000	2010	2020	Sector	1976	1980	1990	2000	2010	2020
N	37	53	66	81	98	118	N	6966	9742	12,463	14,988	18,005	21,449
NNE	17	24	30	37	45	54	NNE	1815	2460	3,441	3,911	4,470	5,114
NE	104	144	181	221	268	324	NE	999	1311	1,996	2,155	2,344	2,562
ENE	20	29	36	44	53	64	ENE	144	201	249	302	368	443
E	26	28	28	31	38	45	E	189	203	202	223	270	324
ESE	84	90	90	100	120	145	ESE	549	591	589	653	790	948
SE	76	82	82	91	110	131	SE	115	124	123	136	165	198
SSE	3484	3772	3922	4452	5409	6525	SSE	726	781	780	863	1,043	1,252
S	185	199	198	219	265	319	S	98	106	105	117	141	169
SSW	51	55	55	61	74	88	SSW	180	194	193	214	259	311
SW	45	48	48	53	65	77	SW	87	94	94	104	125	150
WSW	3	4	4	4	5	6	WSW	115	144	165	194	235	283
W	14	20	25	30	37	44	W	79	112	140	171	208	251
WNW	31	44	55	67	82	98	WNW	196	275	344	420	510	614
NW	0	0	0	0	0	0	NW	348	491	613	749	910	1,095
NNW	20	29	36	44	53	64	NNW	362	510	638	779	946	1,140
TOTAL	4197	4621	4856	5535	6722	8102	TOTAL	12,968	17,339	22,135	25,979	30,789	36,303

CUMULATIVE POPULATIONS
(Miles from Site)

Year	0-1	0-2	0-3	0-4	0-5	0-10
1976	0	75	212	1489	5,686	18,654
1980	46	713	950	2396	7,017	24,356
1990	174	2443	2880	4661	9,517	31,652
2000	174	2452	2927	5112	10,647	36,626
2010	174	2470	2998	5676	12,398	43,187
2020	174	2492	3080	6346	14,448	50,751

2.2 NEARBY INDUSTRIAL, TRANSPORTATION AND MILITARY FACILITIES

2.2.1 LOCATIONS AND ROUTES

Investigation has been undertaken to locate all significant manufacturing plants; chemical plants; refineries; storage facilities; mining and quarrying operations; military bases; missile sites; transportation routes (air, land, and water); transportation facilities; oil and gas pipelines, drilling operations and wells; and underground gas storage facilities. The search included military firing ranges, and nearby airplane high and low-level flights and landing patterns (commercial, general aviation and military).

The CPNPP vicinity is essentially devoid of most of these features. Within the 10-mile area, there are no military bases, missile sites, military firing ranges, munitions facilities, airport approaches, chemical plants, and storage facilities, tank farms or upstream sources of corrosive or oil discharges. The nearest military bases are Carswell Air Force Base, approximately 38 miles north-northeast, and Fort Hood, approximately 65 miles south. There are no missile sites within 50 miles. The nearest manufacturing facility is a textile plant in Granbury, 10 miles north.

Principal transportation routes extending to within 10 miles of the Station site consist of two U. S. highways and one state highway, one railroad, eleven air routes (refueling tracks and Victor lanes), three gas transmission pipelines, and one crude oil transmission pipeline. Air routes out to more than 20 miles are shown on Figure 2.2-1; other transportation routes can be seen in Figure 2.1-5.

No heavily traveled highways pass close to the site. All nearby roads have one or two lanes and have relatively light traffic (Figure 2.1-5). The nearest highway is State Highway 144, 2.5 miles northeast. The closest approach to U. S. Highway 67 is approximately 4.5 miles south-southwest, and the nearest approach to U. S. Highway 377 is nine miles north-northeast. The Atchinson, Topeka and Santa Fe Railroad nearest approach is about 9.5 miles to the northwest.

2.2.2 DESCRIPTIONS

2.2.2.1 Description of Facilities

The only industrial, transportation, or military facilities within five miles of the site are the transportation corridors shown on Figures 2.1-5 and 2.2-1. Furthermore, there are no industrial, transportation or military facilities at greater distances which are significant to CPNPP from a safety standpoint.

2.2.2.2 Description of Products and Materials

The only hazardous materials (excluding local gas stations and materials not directly related to CPNPP) regularly manufactured, stored, used, or transported in the site vicinity are crude oil and natural gas transported through the pipelines described in Section 2.2.2.3.

2.2.2.3 Pipelines And Wells

The four pipelines traversing the site vicinity are described in Tables 2.2-1 and 2.2-2; their locations are shown in Figure 2.5.1-17. Two of the pipelines (the West Texas Gulf crude oil line

and the Lone Star Gas Company line) were relocated around Squaw Creek Reservoir to avoid the immediate Station and reservoir areas. The relocated routes are shown on Figure 2.1-2b.

Figure 2.5.1-17 also shows the location of gas and oil wells within five miles of CPNPP.

2.2.2.4 Waterways

Squaw Creek Reservoir (SCR) is a 3,272 acre cooling reservoir located on Squaw Creek (see Section 2.4), which is not used for navigation. A Safe Shutdown Impoundment (SSI) on the Panther branch of SCR impounds water for the Station Service Water System as described in Sections 2.4 and 9.2.5.

2.2.2.5 Aircraft

2.2.2.5.1 Airports

There are no airports within five miles of the Station. Furthermore, there are no airports with greater than 500 d^2 ("d" is the distance in miles from the Station) movements per year within 10 miles, nor are there any airports with projected operations in excess of 1000 d^2 movements per year within 50 miles.

The nearest landing field (Bar L Ranch) is a sod strip located six miles west-southwest of the site. There are several other sod strips within 20 miles of the site, but the only paved strips are Nassau Bay (private), Pecan Plantation (private), and the Granbury Municipal Airport. The Granbury Municipal Airport (11 miles north-northwest) has a lighted 2,000-foot paved runway, but there are no regularly scheduled commercial flights.

2.2.2.5.2 Air Routes

Eleven air routes pass within 10 miles of the site. Three of these routes are military refueling tracks (flight levels: 24,000 to 31,000 feet) having 10 nautical-mile wide paths (Figure 2.2-1). The outer limits of all three refueling tracks miss the Station site; their nearest approach to the site is the north edge of AR-102B (East), located 1.7 miles south. The north edge of AR-102B (West) is the next closest refueling track, 2.5 miles south. The closest approach of the south limit of AR-102 A (East) is 8.5 miles north of the Station.

The other air routes are Victor lanes which (Figure 2.2-1) are cleared for general, commercial and military aviation. Victor 163W and Victor 163 air lanes are the only routes which pass within 2 miles of the Station location.

The probability of an aircraft crashing into the plant (PFA) is estimated in the following manner:

PFA = C * N * X * (A/B)

where:

C = inflight crash rate (crashes per mile),

N = number of flights per year in the air sector,

X = flight distance in passing possible collision area,

A = effective area of plant, and

B = possible collision area

For CPNPP:

C = 1.14 x 10-9 crash/mile,

N = 130,104 flights/year,

X = 12.63 miles

A = 0.02 square miles,

B = 314.16 square miles.

Therefore, $P_{FA} = 1.91 \times 10^{-7}$. This probability is sufficiently low that an aircraft crash into the plant is not considered a design basis.

2.2.2.6 Projections of Industrial Growth

Industrial growth includes the Station and related pipelines, transmission lines, railroad spur, and access roads. With the exception of natural gas exploration, no industrial growth can reasonably be expected to occur in the site vicinity.

2.2.3 EVALUATION OF POTENTIAL ACCIDENTS

The nearby industrial, transportation, and military facilities described above were evaluated for their potential effects on the station in order to identify potential accidents to be considered as design basis events. In addition, the onsite storage of chlorine was evaluated for its effect on control room habitability as described in Section 6.4.

2.2.3.1 Determination of Design Basis Events

Events which potentially qualify as design basis events were categorized by the physical effects which accompany the accident.

2.2.3.1.1 Explosions

There are no offsite sources of explosive materials in the vicinity and transportation routes do not approach CPNPP within the distance prescribed by Reg. Guide 1.91 [8].

[redacted]

An explosion, therefore, is not considered a design basis event.

2.2.3.1.2 Flammable Vapor Clouds

The three natural gas pipelines in the site vicinity could be a source of flammable vapor clouds and were evaluated for their potential effects on the plant.

The gas wells in the site vicinity, described in Section 2.5.1, could also be a source of flammable vapor clouds and were evaluated for their potential effects on the plant. Since exploration in the area is still active, a hypothetical gas well was postulated using data available from the Texas Railroad Commission. A discussion of this hypothetical well is provided in Section 2.5.1.

2.2.3.1.3 Toxic Chemicals

Toxic chemicals which are stored and used onsite at CPNPP are evaluated in accordance with the criteria and guidance of Regulatory Guide 1.78 and NUREG 0570.

2.2.3.1.4 Fires

The only non-plant-related source of fire in the vicinity of CPNPP is the crude oil pipeline. The rupture of this pipeline was evaluated for its potential effect on the plant.

2.2.3.1.5 Collision with Intake Structure

The Service Water Intake Structure is located on the Safe Shutdown Impoundment (see Figure 1.2-1), which is not open to public transportation. Therefore, a significant collision with this structure is not considered a credible event.

2.2.3.1.6 Liquid Spills

The only source of liquid spills in the vicinity of CPNPP is the crude oil pipeline which was evaluated for its potential effect on the plant.

2.2.3.2 Effects of Design Basis Events

The design basis events identified above are:

1. Gas Pipeline and Gas Well Accidents

2. Accidental Release of Toxic Chemicals

3. Crude Oil Pipeline Rupture

Each of these is evaluated below.

2.2.3.2.1 Gas Pipeline and Gas Well Accidents

Potential accidents involving the release of natural gas from existing pipelines and postulated wells described in Section 2.2.2 do not pose a hazard to the plant. As shown below, in the event of an accident the concentration of gas at all plant air intakes is well below the lower flammability

limit. Also detonation of an unconfined natural gas-air mixture is not considered to be a credible event [1,2,3,4].

Each of the existing pipelines and a potential gas well were analyzed to determine the most limiting potential accident condition; the results of this analysis indicated that the most limiting release of natural gas would involve a break in the 36-inch Lo-Vaca natural gas pipeline. The analysis of this accident was performed using the following conservative assumptions:

1. [redacted]

2. Gas is released by a constant enthalpy process yielding a gas temperature for dispersion calculations of 40°F (for a 60°F initial gas temperature) due to the Joule-Thompson effect.

3. The flowrates out of the break is the maximum steady flow for single-ended break (572 m^3/sec) and a double-ended break (1144 m^3/sec).

4. Atmospheric dispersion factors were based on the 5-percentile meteorological conditions from onsite data analyses with a virtual source distance correction to account for initial finite source size.

5. A conservatively low wind speed of 1.0 m/sec was used.

6. Plume rise is in accordance with Briggs equation for stable (classes F and G) atmospheric conditions [5] and an air temperature of 100°F.

A perfect gas prior to its escape from the broken pipe will expand and accelerate toward the break. As the gas expands, it will tend to cool down from its original equilibrium temperature. In doing so, a temperature gradient between the pipe and the flowing gas will be established. In addition, friction between the pipe wall and the flowing gas will tend to heat the gas slightly above the temperature it would possess in a frictionless expansion. In the first instance, small amounts of heat will flow from the pipe wall into the expanding gas and in the other, energy already possessed by the gas is simply transformed into heat and causes the gas temperature to rise slightly more. Thus, the gas gains a small amount of energy from the pipe wall during this process.

In the second part of the blowdown process, the gas expands through the break at sonic velocity and, if still above atmospheric pressure, continues to expand until it reaches atmospheric pressure. During this phase, the process is essentially adiabatic. No work is performed either during the expansion or the slowing down period. Consequently, the net energy possessed by the gas just prior to reaching the break, and after it slows down in the atmosphere, remains unchanged.

Since (1) no work has been performed by the escaping natural gas, (2) small quantities of heat have been transferred from the pipe to the gas, and (3) a transformation of energy has occurred by virtue of the fact that the gas is at a substantially lower pressure at the end of the blowdown, the energy gain from (2) above and the energy transformation indicated in (1) above must be present in the form of heat. Therefore, the slowed down natural gas in the atmosphere would be at a slightly higher temperature than the original temperature. The blowdown is therefore, in

essence, a throttling or isenthalpic process. Since natural gas is a real gas, not a perfect gas, the Joule-Thompson effect will cause the natural gas to be about 20°F below that expected for a perfect gas after blowdown.

The above description of the process indicates that there is a tendency for a perfect gas after the blowdown and mixing phase to be at a slightly higher temperature than the original temperature. However, with real gases, this temperature increase is lessened or, perhaps, reversed slightly. Therefore, extra conservatism is introduced into the plume rise analysis by assuming that the natural gas does not mix thermally with air and is cooled isenthalpically 20°F below the temperature it possessed prior to the break.

The flowrate was found using the American Gas Association gas pipeline formula [6] using an initial upstream pressure of 1050 psig, a final (downstream or break) pressure of 14.7 psia, and pipe length of 70 miles. This length is the distance from the break to both the upstream and downstream compressors.

Dispersion of potentially explosive gas vapors resulting from a gas pipeline rupture is governed by meteorological conditions existing in the locality at the time. This analysis used design basis accident meteorological assumptions determined for a radioactive release in the analysis of a flammable and/or explosive structure. The design basis accident assumption doses were calculated using conditions which conservatively calculate radioactive cloud concentrations, i.e., those conditions which minimize diffusion of the cloud. The conditions assume five percentile dispersion factors as an analysis basis conservatism. The Dispersion calculations also conservatively assumed that there was no momentum flux in the plume rise calculation. In reality, because the pipe is below grade, the gas would be expected to have a significant vertical velocity component. This would tend to carry the plume higher than calculated. Atmospheric dispersion factors and the model used for virtual source correction are discussed as part of the description of the gas well blowout hazard analysis at the end of this section (2.2.3.2.1).

[redacted]

Though a sustained gas fire may occur at or near the gas well head, a gas well fire at the closest point where mineral exploration may be permitted has negligible heat effects on the nearest safety-related structure, and no effect on safety-related structures due to the release and delayed ignition of non-buoyant natural gas components such as ethane and propane. Conservatively estimating all the heat from a gas well blow out to be radiated isotropically, the resultant sustained incident energy on the surface of a sphere 2250 feet away gives a heat flux loading of about ten percent of the maximum solar insolation at the site. The heated air would rise clear of the plant at low wind speeds, while a strong wind is needed to keep the buoyant heated plume near the surface. These conditions imply ventilation and mixing sufficient to keep any temperature rise at the plant to a non-significant level.

Each Control Room fresh air intake is equipped with a smoke detector that will annunciate in the Control Room in the advent of elevated smoke levels. Their operation is discussed in Section 6.4.2.4 and 9.4.1.3. Thus, smoke effects from mineral explosion fires are responded to adequately in a manner similar to smoke from plant fires.

[redacted]

The atmosphere dispersion model used for the gas well blowout hazard analysis was the Gaussian plume model for centerline concentrations with ground level releases and the Pasquill-Gifford dispersion parameters for an assumed non-buoyant release. Atmosphere dispersion factors for an instantaneous release with a finite initial volume were calculated in accordance with Reg. Guide 1.78 [13]. Wind speed was selected to maximize the instantaneous concentration within the Control Room. Additional credit due to building wake effects was not included in the model. Also, the difference in height of the release point and Control Room inlet is not large enough to allow credit for an elevated release. Atmospheric dispersion factors for a continuous release are based on a 5-percentile meteorological conditions (such as Pasquill F at 1.0 meters/second) at the site. To establish these conditions, the 5-percentile x/Q's for each time period were first determined from the results of onsite dispersion modeling. See Section 2.3.4. Because approximately 30 percent of hourly observations at the plant site are in the Pasquill F

and G stability classes, a Pasquill F stability was assumed, and the wind speed was defined at the 5 percentile probability level of relative concentration. Having defined the 5-percentile wind speed and Pasquill Class for the three time periods of interest, x/Q values were then calculated at selected distances using these meteorological conditions. The parameters used were:

- Distance from plant to well: 2250 ft.
- Maximum postulated gas well blowout flow rate: 52.4×10^6 ft^3 day^{-1} (17.2 m^3 sec^{-1})
- Lower well gas flammability limit concentration: 4%
- Virtual source correction: 278 ft.

The assumption is that the concentration of the released gas at the break is limited to unity. To satisfy this requirement, the source is taken to be a virtual point located a distance away from the actual release point. From a table of the 5-percentile x/Q's for the plant, this value is found to occur at a distance of 278 ft. This then is effectively added to the plant-well separation distance as the virtual source correction term.

The flammable limit was calculated on a mole-fraction-weighted bases of all the components assuming they are mixed in the turbulent stream emitted from the well. The well gas composition used was:

Nitrogen	2.36 mole %
CO_2	0.55 mole %
Methane	73.97 mole %
Ethane	13.58 mole %
Propane	6.95 mole %
Iso-butane	0.88 mole %
N-butane	0.83 mole %
Iso-pentane	0.35 mole %
N-pentane	0.22 mole %
Hexanes plus	0.33 mole %

2.2.3.2.2 Accidental Release of Toxic Chemicals

A study has been performed to evaluate the potential for chlorine releases from offsite chlorine storage and transportation affecting control room habitability. This study revealed no chlorine storage locations or frequent chlorine transportation within a 5 mile radius of CPNPP in quantities that are large enough to require evaluation under the criteria of Regulatory Guide 1.78. Therefore, potential accidents involving the release of chlorine from offsite storage locations and transportation routes do not pose a significant hazard to control room habitability.

The circulating cooling water and service water systems will be chemically treated for control of biological growth with solutions of sodium hypochlorite and sodium bromide. The diluted sodium hypochlorite and sodium bromide solutions will not present a threat to control room habitability. No liquified chlorine will be stored within the site boundary in containers exceeding 150 lbs capacity. The only liquified chlorine that will be stored within the protected area will be in small quantities of 20 lbs or less to be used for laboratory purposes.

Other toxic chemicals which are stored and used onsite at CPNPP are evaluated for potential impact of their release upon control room habitability in accordance with the criteria and guidance of Regulatory Guide 1.78 and NUREG 0570. Toxic chemicals which are determined to be hazardous to control room habitability are controlled and detection instrumentation is provided, as appropriate.

2.2.3.2.3 Oil Pipeline Accidents

[redacted]

Gulf indicates that from experience the lower limit on immediate detection and isolation is 400 Bbl/hr.[10] Smaller leaks could be detected over a period of time and would result in the initiation of an immediate search for the leak. CPNPP personnel will monitor for oil accumulation in the three retaining ponds located between the pipeline and the westside of the SSI once per day ensuring that leaks smaller than 400 Bbl/hr do not go undetected for more than 32 hours.

[redacted]

[redacted]

Maximum Rate Case

[redacted]

Maximum Quantity Case

[redacted]

[redacted]

In order to prevent the spilled oil from reaching the SSI, three retaining ponds will be built between the pipeline and the SSI. These ponds will be designed to hold the maximum quantity of oil given above or 14,770 Barrels. Rain water accumulation in these ponds will be drained through an inverted weir to ensure capacity is available for the maximum oil spill quantity. These retaining ponds eliminate any potential hazards resulting directly or indirectly from the crude oil spilling onto the SSI surface.

With the crude oil being kept from the SSI by the retaining ponds, the only remaining potential impact on the plant would be from drifting vapor clouds. Potential toxic and flammable vapor effects have been considered as discussed below.

The crude oil carried in the pipe may contain hydrogen sulfide (H_2S) which is toxic gas. The pipeline operator has indicated that the maximum H_2S concentration is 80 ppm in air measured 2 3/4" above a crude oil sample at 60°F [9,11]. Further information from Gulf indicated the best available measurement of H_2S concentration in the crude is 250 ppm obtained by mass spectrographic techniques, while 400 ppm is obtained by other techniques.[11]

A conservative calculation of H_2S concentration in the CPNPP control room was made assuming:

1. All H_2S was released instantaneously from a leak of 13,500 Bbl/hr of crude oil containing 400 ppm H_2S.

2. This formed a continuous source of H_2S which is blown toward the plant with worst 5 percentile site atmospheric dispersion.

3. The control room air intake sees the resulting plume centerline concentration ($94 mg/m^3$) for 10 minutes (note that the pipeline leak flow only occurs for 2.5 minutes.)

4. Normal control room ventilation of 3000 CFM continues for the entire 10 minutes.

The resulting maximum concentration in the control room is 8.5 mg/m^3. This is less than the time weighted average threshold limit value of 15 mg/m^3 recommended by the American Conference of Governmental Industrial Hygienists.[12] The control room concentration and the outside air concentration are far below the R.G. 1.78[13] toxic limit of 750 mg/m^3.

Considering the conservative calculations and the adequate odor detection threshold of H_2S the oil pipeline is not a credible potential gas hazard.

Gulf Oil indicates that the crude oil in the pipeline may contain significant fractions of light hydrocarbons. The maximum light fraction concentrations are as follows:[14]

Methane	0.60%
Ethane	0.14%
Propane	0.92%
Butane	3.74%
Pentane	44.6%

As indicated above, the maximum oil spill quantity occurs for the 400 Bbl/hr spill rate. At this rate even if all the light fractions were vaporized instantaneously the maximum concentration at the nearest plant safety related structure would be 0.24% which is 15% of the mixture lower flammable limit of 1.6%. This result is for worst 5 percentile site meteorology and a distance of 2400 feet from the nearest retaining pond to the Service Water Intake Structure. This effectively assumes the spilled oil gets all the way to the retaining pond without losing any vapor, but when it does reach the pond all the volatile fraction is released immediately. This maximizes the vapor release rate and minimizes the source to plant distance.

For the maximum spill rate case two situations have been analyzed. The first is for vaporization from the oil accumulated in the retaining pond (conservatively assuming water in the pond). The second situation is for vaporization from the oil spill on land between the pipe and the retaining pond.

The rate of vaporization from the oil spill was determined from a heat balance equating the heat lost by evaporation to the heat gain from either the water or the soil plus heat from solar radiation.

The evaporation rate was determined from the mass transfer relationships of Bird, Stewart and Lightfoot.[15] The mass transfer coefficient was obtained from the heat/mass transfer analogy using forced convection flat plate correlations of Krieth.[16] The vapor phase composition was determined from the crude oil composition given above using two phase equilibrium ratios from Reference [17].

Heat transfer from water was determined assuming simple convection from the water at a maximum natural water temperature of 97ºF using a overall film coefficient of 100 Btu/hr ft^2 ºF. For a spill on land transient heat conduction from soil was considered utilizing a procedure which accounted for soil temperature variation with depth from a 135ºF initial surface temperature and

an initial oil temperature of 74°F. Solar heat flux of 365 Btu/hr ft^2 was used. These temperatures are for worst case daytime summer conditions. Temperatures at other times would be much less.

The results of the analysis indicate that on water the oil reaches a maximum temperature of 86°F with a maximum specific evaporation rate of 9.6 lb/hr ft^2. On land the oil reaches a maximum temperature of 77°F with a corresponding specific evaporation rate of 6.8 lb/hr ft^2. These values are based on the initial oil composition and do not reflect the reduction in evaporation rate which occurs as the more volatile fractions are evaporated. Also the maximum conditions on land last for only a very short time due to the continued cooling of the soil.

[redacted]

Pond Area (ft^2)	Distance from SSWS INTAKE (ft)	Concentration as fraction of LFL
1	[redacted]	[redacted]
2	[redacted]	[redacted]
3	[redacted]	[redacted]

[redacted]

From the above discussion it is concluded that vapors released from oil spilled as a result of failure of the Gulf Oil pipeline do not result in a credible potential hazard to CPNPP. This conclusion and the ability of the retaining ponds to prevent crude oil liquid from reaching the SSI indicate that the Gulf Oil crude oil pipeline does not present a credible hazard to the CPNPP.

REFERENCES

1. Foster, J. C., Jr., et al., "Detonatability of Some Natural Gas- Air Mixtures," Air Force Armament Laboratory, AFATE-TR-74-80, November 1970.

2. Kogarko, S. M., et al., "An Investigation of Spherical Detonations of Gas Mixtures." International Chemical Engineering, Vol. 6, No. 3, July 1966.

3. Loesch, F. C., "Thermal Radiation and Overpressure from Instantaneous LNG Release into the Atmosphere," TRW Report No. TRW-08072-4, April 26, 1968.

4. "Safety Evaluation Report, Tennessee Valley Authority, Hartsville Nuclear Plants A and B." U. S. Nuclear Regulatory Commission, NUREG-0014.

5. Briggs, G. A., "Plume Rise," TID 25075, Equations 4.19c, 4.32, and 5.7, November 1969.

6. Natural Gas Processors Suppliers Association, Engineering Data Book, Eighth Edition, p. 96, 1966.

7. Nuclear Power Plant Control Room Ventilation System Design for Meeting General Criterion 19, K. G. Murphy and Dr. K. G. Murphy and Dr. K. M. Campe, 13th AEC Air Cleaning Conference, August 1974.

8. NRC Regulatory Guide 1.91, “Evaluation of Explosions Postulated to Occur on Transportation Routes Near Nuclear Power Plant Sites,” Revision 1, February 1978

9. Letter from H. E. Evans (Gulf Refining Co.) to H. C. Schmidt (Texas Utilities Services, Inc.), dated October 2, 1978.

10. Telecon between R. W. Haskovec (Texas Utilities Services Inc.) and Mr. Newton Fincher (Gulf Refining Co.), August 24, 1979.

11. Telecon Between A. O. Smyth (Gulf Refining Co.) and E. R. Schmidt (NUS Corporation), February 12, 1980.

12. “Threshold Limit Values for Chemical Substances in Workroom Air,” American Conference of Governmental Industrial Hygienists, 1978.

13. NRC Regulatory Guide 1.78, “Assumptions for Evaluating the Habitability of a Nuclear Power Plant Control Room During a Postulated Hazardous Chemical Release”, June 1974.

14. Letter from H. E. Evans (Gulf Refining Co.) to H. C. Schmidt (Texas Utilities Services, Inc.) dated August 9, 1978.

15. Bird, R. B., Stewart, W. E., and Lightfoot, E. N., Transport Phenomena, John Wiley and Sons, New York (1942 pp 662-663).

16. Krieth, F., Principals Of Heat Transfer, Second ed., International Textbook Company, Scranton (1965) pp 296-314.

17. “Technical Data Book - Petroleum Refining”, American Petroleum Institute, Third Edition, 1976.

18. NRC Regulatory Guide 1.95, Protection of Nuclear Power Plant Control Room Operators Against An Accident Chlorine Release, Revision 1, January 1977.

19. NRC Regulatory Guide 1.111, Methods for Estimating Atmospheric Transport and Dispersion of Gaseous Effluents in Routine Releases from Light-Water-Cooled Reactors, Revision 1, July 1977.

20. NRC Regulatory Guide 1.145, Atmospheric Dispersion Models for Potential Accident Consequences Assessments at Nuclear Power Plants, Revision 1, November 1982.

TABLE 2.2-1
NATURAL GAS PIPELINE INFORMATION

Owners	Lone Star Gas Company	Lone Star Gas Company	Lovaca/Texas Utilities Fuel Company
[redacted]			

TABLE 2.2-2
WEST TEXAS GULF CRUDE OIL LINE

Owner	West Texas Gulf
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	[redacted]
[redacted]	

2.3 METEOROLOGY

2.3.1 REGIONAL METEOROLOGY

2.3.1.1 General Climate

The regional climatology of the site is described for the area bounded by 30 North, 100 West, 34 North, 96 West, shown in Figure 2.3-1. The Comanche Peak site is located approximately equidistant between Cleburne and Stephenville, Texas, west of the Brazos River. The site elevation is approximately 810 feet mean sea level (MSL). Over the region depicted in Figure 2.3-1, the terrain slopes gradually from 300 to 700 feet MSL in the southeast to 1,200 to 1,800 feet MSL in the northwest.

The climate of the region is continental and is characterized by rapid changes in temperature, marked extremes, and large daily and annual temperature ranges. The mean annual temperature decreases from southeast to northwest because of elevation and latitude changes. The general climate of the region is modified frequently by advancing warm moist air from the Gulf of Mexico, resulting in high humidities and cloudiness. Rainfall generally decreases from east to west and is heaviest in late spring and early summer.

In summer, the Bermuda High exerts a strong influence upon the weather of the region. It furnishes the tropical maritime air from the Gulf of Mexico which almost completely dominates the weather from May to September. This air mass is responsible for almost all of the thunderstorm activity in the region regardless of time of year. Occasionally, in summer, tropical continental air may move into West Texas from the high plateaus to the west. This air mass is characterized by very hot daytime temperatures and almost cloudless skies. Typically, summer has over 100 days with temperatures of 90°F or above and temperatures often exceed 100°F [1, 2, and 3]. Rainfall occurs during brief but sometimes intense showers and thunderstorms.

During the winter and early spring, outbreaks of polar continental air are the most common frontal activity. Although these fronts frequently have little weather associated with them, they often stall in Central and South Texas. Low ceiling and poor visibility may persist for three to four days or even longer over large areas [3]. On occasion, arctic air masses push through the region and cause some of the coldest temperatures. Cold spells, however, rarely last more than a few days. Normally, temperatures drop to 32°F or below about 30 days each year [1, 2]. Winter is the driest season, but one or two occurrences of snow and one or two occurrences of sleet or freezing rain may be expected in both January and February, the coldest months [3].

Pacific maritime cold fronts are more frequent in spring and fall than in winter or summer. These air masses usually bring clear skies to the region, although the weather along the leading edge of the front may be quite violent. Most of the dust storms of early spring and the violent thunderstorms during April, May, and June are associated with these frontal systems. Warm fronts are generally confined to the late fall and early spring months in this region. They are usually confined to the southern half of the region and move northward very slowly.

Spring is characterized by rapid changes of temperature, i.e., alternating periods of warm and cold conditions. On the average, thunderstorms are more frequent and more violent in the spring than any other season. Spring is normally the wettest season of the year. The fall is characterized by fair weather, low wind speeds, and moderate temperatures. It is the most pleasant season of the year.

Another major influence on the climate of this region is the so-called “dewpoint front,” which marks the westward extent of the moist air from the Gulf of Mexico. This trough of low pressure can exist at all times of the year. In summer, it marks the westward extension of the Bermuda High, while in winter it forms behind the migratory high moving across Central Texas following a cold front. The movement of this trough eastward results in a shift to west or southwest winds and dryer weather [3].

Annually, the prevailing surface winds in the region are from the south to southeast while the average wind speed is about 12 miles per hour (mph) [2]. In winter there is a secondary wind direction maximum from the north to northwest due to frequent outbreaks of polar air masses.

Annual average relative humidity is approximately 65 percent in the region while the mean percentage of possible sunshine is about 70 percent [1, 2].

Severe weather in the region is usually associated with heavy thunderstorms (including tornadoes) and tropical cyclones. Property damage occurs from flooding and high winds. Damaging hail also occurs in the site region.

The interplay between synoptic scale phenomena and topography is small in the region of interest. The effect of terrain features on synoptic scale flow can readily be ascertained when a larger area, which takes in the high country of West Texas and Eastern New Mexico, is included; i.e., the principal effect is that the high country forms a natural barrier to the flow of air. Consequently, moist tropical air from the Gulf of Mexico and air from the arctic or polar sources, which flows uninhibited through the site region, is effectively blocked from the areas to the west of the mountains. The net result is wide fluctuations in rainfall, humidities, and annual sunshine over the larger area.

2.3.1.2 Regional Meteorological Conditions for Design and Operating Bases

2.3.1.2.1 General

Meteorological data are presented in this section for severe weather phenomena such as hurricanes, tornadoes, thunderstorms, lightning, hail, high air pollution, and ice storms. Also presented are the meteorological data used for evaluating the performance of the ultimate heat sink and design basis tornado parameters.

2.3.1.2.2 Hurricanes

Tropical cyclones including hurricanes lose strength rapidly as they move inland, and the greatest concern is potential damage from winds or flooding due to excessive rainfall. The tropical cyclone season for Texas extends from June to October; storms are more frequent in August and September, and rarely occur after the first of October. The average frequency of tropical cyclones that affected Texas for the period 1931 to 1960 is approximately two per year; and of these about one in four were of hurricane force, i.e., winds greater than 74 mph [4].

The Probable Maximum Hurricane (PMH) is discussed in Section 2.4. For the Comanche Peak site, the PMH sustained (10-minute average) wind speed at 30 feet above ground is computed to be 81 mph [47].

2.3.1.2.3 Tornadoes

During the period 1955 through 1967, 58 tornadoes (4.46 mean annual frequency) occurred within a one-degree square centered near the Comanche Peak site [5]. It should be noted that statistical data on severe local storms, tornadoes particularly, are highly dependent on human observation. For example, as population density increases, the number of tornado occurrences observed and accurately reported generally increases.

The probability (P) of a tornado hitting a point in a given year is:

$$P = (2.8209 \times t)/A \qquad 2.3\text{-}1$$

where A is the area (in square miles) of a one-degree latitude-longitude square centered at the point, and t is the mean annual frequency [6]. The return period is the reciprocal of P. Table 2.3-1 summarizes the results for the site and nearby areas for the period 1955 through 1967. In the area north of about 31 North latitude, there is a greater frequency of large tornadoes with wide paths and long trajectories.

Based on the 13-year period of record from 1955 - 1967, the mean seasonal and annual number of tornado occurrences for Texas are [5]:

Winter	4.7	Summer	44.8
Spring	71.1	Autumn	23.7
	Annual	143.6	

During the period from 1968-1977 there were approximately 64 recorded tornado occurrences within about 40 miles of the plant site [40], but none of these occurrences were in Somervell County. Estimated characteristics of these tornados, expressed in terms of the Fujita- Pearson (FPP) tornado scale [4], are summarized below:

Scale No.	Maximum Windspeed[a]			Path Length		Path Width	
	F Scale (mph)	Expected Damage	No. of Occurrences	P Scale (miles)	No. of Occurrences	P Scale	No. of Occurrences
0	40-72	Light	12	<1.0[b]	33	<18 yd[b]	23
1	73-112	Moderate	34	1.0-3.1	9	18-55 yd	14
2	113-157	Considerable	14	3.2-9.9	8	56-175 yd	11
3	158-206	Severe	4	10-31	5	176-556 yd	5
4	207-260	Devastating	0	32-91	0	0.3-0.9 mile	0
5	261-318	Incredible	0	100-315	0	1.0-3.1 mile	1
				Unknown	9	Unknown	10

a) Subjective estimate based on observed damage given in the literature.

b) Tornados given in the literature as short and narrow are included in these categories.

For example, a tornado having a wind speed of 200 mph, a path length of 10 miles, and a path width of 100 yds would be expressed on the FPP scale as 3, 3, 2. The above data indicate that most of the tornados in this area have a path length of less than 20 miles, a path width of less than 180 yards, and a maximum wind speed of less than 160 mph.

The design basis tornado parameters used in the design and operation of CPNPP are listed below:

Translational Speed	60 mph
Rotational Speed	300 mph
Maximum Wind Speed (sum of the translational and rotational speed)	360 mph
Maximum Pressure Drop	3.0 psi
Rate of Pressure Drop	1.0 psi/sec

Compliance with Regulatory Guide 1.76 [7] is discussed in Appendix 1A(B). Tornado loadings are discussed in Section 3.3.2.

2.3.1.2.4 Thunderstorms

Thunderstorms, from which damaging local weather can develop (tornadoes, hail, high winds, and flooding), occur about 46 days each year based on Fort Worth data [8]. The maximum frequency of thunderstorms occurs from April to June, while months November through February have few thunderstorms. The monthly and seasonal distributions are displayed in Table 2.3-2.

2.3.1.2.5 Lightning

A typical thunderstorm is believed to produce one to three cloud-to-ground flashes each minute. The number of lightning strikes per square mile has been determined from photographs, records of strikes to power lines, and from electrical lightning counters. The combined results of several studies indicate that the number of flashes to ground per square mile per year is equal to between 0.05 to 0.8 times the number of thunderstorm days per year [9].

Applying these results to the thunderstorm data for Fort Worth in Table 2.3-2, the seasonal and annual estimates of lightning strikes are:

Winter	0.2 - 3.2	Summer	08 - 12.8
Spring	0.9 - 14.4	Autumn	0.4 - 6.4
	Annual	2.3 - 36.0	

Thus, the annual expectancy of lightning strikes for a square mile area in the site vicinity is between 2 and 36.

The expected probability of lightning strikes to the safety-related structures at the site on a seasonal and annual basis are presented in Table 2.3-2A. Included in this Table is the attractive area of each building and building dimensions used in the Table's development. The number of thunderstorm days for each season and annually, presented in Table 2.3-2, as well as the attractive area of each building, geographic latitude of the site and building dimensions were used in the development of Table 2.3-2A [42].

2.3.1.2.6 Hail

Almost all localities in Texas experience damage from hail. While the most commonly reported hailstones are 1/2 to 3/4 inch in diameter, hailstones 3 to 3-1/2 inches in diameter are reported in Texas several times a year.

During the period 1955 through 1967, there were 42 reports of large hail (3/4 inch diameter or larger) occurrences within a one-degree square centered near the Comanche Peak site [5]. This gives a mean annual frequency of 3.2 potentially damaging hailstorms per year for the entire one-degree square. Fortunately, recurrence of damaging hail at a specific location is very infrequent. The annual frequency of hail at Fort Worth is less than 0.05 percent of the hours observed [3].

The monthly and seasonal breakdown of large-hail occurrences (3/4 inch diameter or larger) for the state of Texas is given in Table 2.3-2 [5]. Damaging hailstorms are most frequent during April, May, and June, the period of severe-thunderstorm activity.

2.3.1.2.7 Air Pollution Potential

Conditions in the region generally favor turbulent mixing. Two conditions which reduce mixing, increasing the air pollution potential, are surface inversions and stable air layers aloft.

The surface inversion is generally a short-term effect and surface heating on most days creates a uniform mixing layer by mid-afternoon. On the other hand, if warming caused by subsiding air occurs, the second condition, namely a subsidence inversion, may result. Since both conditions usually occur in conjunction with light winds, the air pollution potential is amplified.

Holzworth [10] has computed mean morning and afternoon mixing depths and corresponding wind speeds for several stations in Texas. There is considerable variation in mixing depths among Texas stations; but the mixing depths all display similar seasonal variation, the depth being greatest during the warm months and shallowest during the cold months.

Since no data were presented by Holzworth for stations within the site region, data for Midland and San Antonio are presented in Table 2.3-3. Isopleths of the total number of forecast-days of high air pollution potential in five years are given in Figure 2.3-2. The data indicate that stable periods with light wind conditions are generally of short duration in the region.

Based on a 30-year period of record (1936-1965), Korshover [11] tabulated the number of times stagnating anticyclones persisted for four or more and seven or more days. Occurrences of stagnation were determined primarily on the basis of a surface pressure-gradient analysis. In the general area of the site, there were only three stagnation cases which persisted for at least four days during the 30-year period; the total number of stagnation days was 14. There were not any stagnation cases which persisted for seven or more days. The three stagnation cases which persisted for at least four days occurred during the fall.

2.3.1.2.8 Precipitation

Maximum rainfall, estimated by statistical analysis of regional precipitation data, is given in Table 2.3-4 for return periods of one to 100 years and for rainfall durations of from one-half hour to one day [12].

Probable maximum precipitation (PMP), sometimes called maximum possible precipitation, for a given area and duration is the depth which can be reached but not exceeded under known meteorological conditions. For the site area, the PMP for 6, 12, 24, and 48 hours is 25.5, 31.3, 34.7, and 39.1 inches, respectively [13]. These values are based on a 64 square-mile area.

The maximum recorded point rainfall for durations of 5 minutes to 24 hours at Abilene, Fort Worth, and Waco, are give in Table 2.3-5 [14].

On the other hand, drought conditions (extended periods of widespread meager precipitation) are known to occur in Texas. The most severe this century in Texas occurred during 1954-1956 (see Section 2.4.11). Fort Worth precipitation records, which illustrate the regional conditions, indicate that the average annual precipitation for this three-year period was 21.1 inches, with only 18.55 inches occurring in 1956 [15]. Although this period represents the worst drought in Texas, there have been three occurrences of annual precipitation less than 18.55 inches during the 81-year period from 1895 - 1975 at Fort Worth [8, 16]. The extreme minimum annual precipitation recorded, 17.91 inches, occurred in 1921.

Extreme snowfalls for representative stations in the area are given in Table 2.3-6 [17, 18].

Ice storms, precipitation in the form of freezing rain and/or sleet, occur occasionally in the region during the period December through March. The annual frequency of freezing rain at Carswell Air Force Base Fort Worth is only 0.3 percent of the hours observed, but the percentages for December, January, February, and March are 0.3, 2.5, 0.5, and 0.1 percent, respectively [3]. Moderate to heavy ice storms can be quite damaging to utility lines and trees as well as being a serious traffic hazard.

The worst ice storm on record in the Fort Worth-Dallas area occurred on January 6-9, 1937 [19]. As much as two inches of ice formed on wire and did not disappear until the 12th. Communications were disrupted and highway traffic was extremely hazardous. The estimated probability of ice thickness ≥ 2 inches occurring somewhere in North Texas, Arkansas, or Oklahoma in any one year is 0.06 [20].

The 100-year return period snow and ice pack for the area in which the plant is located, in terms of snow load on the ground and water equivalent, is listed below [21]:

Snow Load	= 7 lbs/ft^2
Water Equivalent	= 1.3 inches

The maximum 24-hour and seasonal snowfall recorded (1898-1970) in Fort Worth is 12 and 15.3 inches, respectively [18]. If the maximum seasonal snowfall (water equivalent of 1.5 inches) occurred in one storm on the 100-year return period snow and ice pack, the total water equivalent would be only 2.8 inches. In the period from 1970 - 1978, the maximum seasonal snowfall in Fort Worth was 17.6 inches (1977 - 1978 season). Even assuming the maximum winter snowfall occurred in one storm on top of an already present 100-year return period snowpack of 13 inches, the total weight would only be approximately 16 lbs/ft^2. This value is far below the design load on all nuclear safety-related buildings of an eight-inch maximum uniform depth of water (weight of about 42 lbs/ft^2) in addition to the regular live loads (see Section 2.4.1.1).

The 48-hour Probable Maximum Winter Precipitation (PMWP) for a 64 square-mile area is estimated to be 28.8 inches [13]. Because of the southern location of the site, almost all of this PMWP occurs as liquid. The roof drainage utilizes an exterior system, whereby the openings in the parapet walls are actually combined relief and drainage openings. These parapet walls relief openings insure that the eight-inch level is not exceeded during the PMP. They are specifically located at all roof low points and extend from the roof low point elevation to the top of the parapet (approximate height four feet). Their location is shown in Figure 1.2-2, 1.2-3, 1.2-5, 1.2-39 and 2.4-2. The length of an opening is six feet.

The size of these openings and their location preclude:

1. the possibility of ice and snow build-up blocking them and

2. roof ponding.

All design features of the relief and drainage opening are shown in Figure 2.4-42.

2.3.1.2.9 Dust Storms

Blowing dust or sand may occur occasionally in West Texas where strong winds are more frequent and vegetation is sparse. While blowing dust or sand may reduce visibility to less than five miles over an area of thousands of square miles, dust storms that reduce visibility to one mile or less are quite localized and depend on soil type, soil condition, and vegetation in the immediate area. At Fort Worth during the six year period from 1970-1975, only 0.02 percent of the observations had visibilities of one mile or less due to dust [22].

2.3.1.2.10 Ultimate Heat Sink

The performance of the ultimate heat sink is discussed in Section 9.2.5. The meteorological parameters used in the analysis are presented in Tables 2.3-7A, 2.3-7B and 2.3-7C.

Three meteorological records were examined in the analysis. These records are data from the onsite station and data from two offsite stations, Dallas-Fort Worth Regional Airport (DFW) and Waco Madison Cooper Airport. DFW is approximately 59 miles northeast of CPNPP, with a ground elevation of 551 ft. The DFW record available on magnetic tape begins with May 1953. Until the end of 1973, DFW observations were taken at Love Field. Waco is approximately 53 miles southeast, with ground elevation of 501 ft. The Waco record begins with July 1948. At the time of the analysis, data through December 1991 at both stations were available. The data consists of hourly (or three-hourly for the period 1965-1980) air and dew point temperatures, windspeed and direction, cloud cover and atmospheric pressure. In addition to these variables, surface heat exchange computations require solar radiation, which can be determined from cloud cover.

The onsite data set is not as complete as the offsite data. For this reason, no direct use of the onsite meteorological data set was made in the analysis. However, the 1974 onsite data set (Table 2.3-7A), previously identified as resulting in high Safe Shutdown Impoundment (SSI) temperatures for years preceding 1978, is presented here for comparison to the data sets selected from the offsite records for SSI performance analysis (Tables 2.3-7B and 7C). The day with the highest water temperature, as computed from the onsite meteorological record as a response to atmospheric heating or cooling, is July 15, 1974.

The two offsite records were used to compute SSI temperatures under normal operating conditions for comparison to SSI routinely monitored intake temperatures. These comparison showed that the DFW meteorological data more accurately represented onsite conditions than the Waco data. The DFW data were used for the SSI performance analysis.

The entire DFW record (39 years) was then used to compute fully-mixed water temperatures responding to atmospheric heating or cooling and steady heat load similar to those used in the SSI analysis. From this long record of response temperatures, 1-, 5-, 10-, 20-, and 30-day average response temperature were computed. Maximum values of these average temperatures were then identified and ranked by year. From this table, 1990 was identified as the year that would produce the highest temperatures in the SSI for all durations. The meteorological data from 1990 are shown in Table 2.3-7B for the period of maximum SSI temperatures. This period includes data for both for the 24 hour transient analysis and for the peak SSI intake temperature analysis. The day with the highest water temperatures computed from this meteorological record is August 31, 1990.

A similar search was conducted for the maximum 30-day evaporation. The fully-mixed computation was used to determine evaporation rates. The period of 30-day maximum evaporation was identified by examining the entire 39 year record using a moving sum procedure. The maximum natural evaporation occurred in 1980. The meteorological data for this period are shown in Table 2.3-7C. The 30-day period with the greatest potential for evaporation is June 25, 1980 to July 25, 1980. In order to be conservative, the evaporation analysis was continued for nine days beyond the 30 day minimum specified in R.G.1.27. The additional data are also shown in the Table.

2.3.1.2.11 Extreme Winds

Estimated extreme winds (fastest mile) for the general area based on the Frechet distribution are [24]:

Return Period (Years)	Wind Speed (Miles Per Hour)
2	51
10	61
50	71
100	76

Fastest mile winds are sustained winds, normalized to 30 feet above ground and include all meteorological phenomena except tornadoes.

Wind loadings for the site are discussed in Section 3.3.1.

2.3.2 LOCAL METEOROLOGY

2.3.2.1 Normal and Extreme Values of Meteorological Parameters

2.3.2.1.1 General

In this section, the normal and extreme statistics of wind, temperature, water vapor, precipitation, fog, and atmospheric stability are described. Long-term data from proximal weather stations (see Figure 2.3-3) have been used to supplement the shorter-term onsite data.

2.3.2.1.2 Surface Winds

Percentage frequencies of surface wind (wind roses) at Love Field, Dallas, for the years 1951-1960 are shown on an annual and monthly basis in Table 2.3-11 [25]. The symbol “+” indicates that the value is less than one-half of one percent but greater than zero. According to the annual table, surface wind directions at Dallas are from the southeast, south-southeast, and south 45 percent of the time. These directions predominate during the individual months also, but to a lesser extent during November through March. The annual average wind speed (not shown in Table 2.3-11) is 11.0 miles per hour. The maximum average wind speed occurs in the spring, while the minimum occurs in the fall.

Monthly and annual onsite wind frequency distributions at the 10-meter level are included in Table 2.3-27, presented in Section 2.3.3 [26]. Similar to the offsite distribution, the surface wind is from the southeast, south-southeast, and south 40 percent of the time. The annual average wind speed is lower onsite, averaging 8.5 miles per hour.

The “fastest mile” of record (long-term) at Fort Worth and Dallas for each month is presented in Table 2.3-12 [8, 27].

Frequency distributions of wind direction persistence, determined from observations at three-hour intervals over a five-year period (1969- 1973) from Fort Worth, Texas [28], are presented in Table 2.3-13. Persistence values are given for both one sector and three sectors (indicated by wind sector +22.5º in Table). Persistence was maintained through calm or missing observations if it was maintained subsequent to them. Because of these criteria, persistence as given by the number of consecutive three-hour observations tends to have a bias towards long durations.

Monthly and annual wind direction persistence, determined from hourly onsite observations at the 10-meter level [26] are presented in Table 2.3-14. These distributions, which are independent of stability, indicate that most cases of persistence are less than 12 hours in duration. During the four-year period of record, there were only eight cases of persistence greater than 24 hours, five of which occurred in the NNW sector. In the average wind speeds given in Table 2.3-14, 99.99 indicates that the wind speeds were invalid.

2.3.2.1.3 Temperatures

Monthly and annual values of daily mean temperature and average and extreme daily maximum and minimum temperatures are shown in Table 2.3-15, based on data records for Fort Worth [8]. From these data, the annual mean temperature in the site area is 66°F. The monthly averages indicate that July and August are the hottest months and January the coldest month.

The monthly and annual diurnal distributions of temperature at the site are shown in Table 2.3-16 [26]. Values of the mean, absolute maximum and minimum, and average daily maximum and minimum temperature for each month are also presented. The annual mean temperature from the onsite data is 17.1°C (63°F), which is some 3°F lower than the long-term mean as given by Fort Worth data.

Table 2.3-16 indicates that there were 93.8 percent valid primary ΔT measurements (10-60m) during the 4-year period of record (5/15/72 - 5/14/76). Table 2.3-23 shows that there were a total of 95.8 percent valid delta temperature (ΔT) measurements during the same 4-year period of record, including secondary ΔT measurements when primary ΔT measurements were invalid. Therefore, only two percent secondary ΔT measurements (10-30m) were substituted in the data record for invalid primary ΔT measurements.

In order to justify the direct substitution of secondary ΔT measurements during periods when the primary ΔT measurements were invalid, a simultaneous comparison of stability classes calculated from both ΔT measurements was made. Seven random days, encompassing all seasons, were selected from the 4-year period for the comparison. The comparison showed that there was a good correlation between the two ΔT samples with a higher frequency of both the most stable (F and G) and unstable (A and B) occurrences using the secondary ΔT measurements. Since the relative concentrations for CPNPP were calculated for an assumed

ground-level release, maximum relative concentrations (other conditions being equal), would occur during Pasquill G stabilities, the next highest relative concentrations would occur during Pasquill F stabilities, and so on. During the 7-day period of comparison, there were considerably more F and G occurrences for secondary ΔT measurements than for primary ΔT measurements. Therefore, substitution of secondary ΔT measurements during periods when primary ΔT measurements were invalid (two percent of the time) is quite conservative.

See Section 2.3.2.1.7 for more information on atmospheric stability.

2.3.2.1.4 Water Vapor

Monthly and annual average relative humidity for four different times of day are given in Table 2.3-17 from 10 years of record at the Fort Worth weather station [8]. Based on these data the annual average relative humidity is estimated to be about 68 percent. Monthly and annual average dewpoint temperatures and extreme maximum and minimum dewpoint temperatures are shown in Table 2.3-18, based on 1970-1975 data from Fort Worth [22].

The monthly and annual diurnal distributions of dewpoint temperature and relative humidity at the site are shown in Table 2.3-16, based on four years of onsite measurements [26]. Values of the mean, absolute maximum and minimum, and average daily maximum and minimum for both parameters are also presented. The annual average relative humidity is 63 percent, or five percent lower than the estimated long-term average at Fort Worth.

2.3.2.1.5 Precipitation

Monthly and annual precipitation normals and the mean number of days with precipitation equal to or greater than 0.10 and 0.50 inches, estimated for Comanche Peak by averaging data from Rainbow, Stephenville, Cleburne, Dublin, and Fort Worth are presented in Table 2.3-19 [15]. These data indicate that the highest monthly average rainfall occurs in May with an annual average of 31.3 inches in the area. The number of days with measurable precipitation ($\geq$ 0.01 inches) is also presented in Table 2.3-19 based on 1954-1973 data from Fort Worth [8]. Monthly precipitation extremes (maximum and minimum), presented in Table 2.3-20 for several stations in the area, indicate that the largest rainfalls occur during April and May [8, 15].

Snow and sleet occur from December through March with an occasional snow flurry in late November or early April. Monthly and annual average totals of snow and sleet from 20 years of record at Fort Worth [8] are shown in Table 2.3-19. These data give an annual expectancy of 2.9 inches of snow. Extremes of snowfall at selected stations in the area were previously presented in Table 2.3-5.

Monthly and annual precipitation wind roses are presented in Table 2.3-21. These data are based on four years of data at Fort Worth concurrent with the onsite data record [29]. These data show that of the 5.8 percent of the time that precipitation occurred, the maximum frequency of precipitation occurred with north winds. (Onsite data were not used because most of the hourly onsite precipitation data are invalid.)

2.3.2.1.6 Fog

Heavy fog is that which reduces visibility to one-quarter mile or less. Average monthly and annual number of heavy fog days based on 20 years of data at Fort Worth are presented in

Table 2.3-19 [8]. These data indicate that most of the heavy fog days occur in winter with a few occurrences during the remainder of the year.

2.3.2.1.7 Atmospheric Stability

Based on data for the period 1957-1971 at Fort Worth, the monthly and annual frequency distributions of stability classes are shown in Table 2.3-22 [30]. The stability classes are based on the Pasquill classification [31] and are defined in Table 2.3-22. These data indicate that the frequency of stable classes reaches a peak during the late summer and early fall.

Monthly summaries of the diurnal distribution of stability and stability persistence are presented in Table 2.3-23 for the onsite period of record [26]. Stability is determined from the 10-60 meter delta temperature measurements onsite and the class intervals specified in NRC Regulatory Guide 1.23 [32]. The annual percentages by stability class are as follows:

A	B	C	D	E	F	G
3.2	1.4	2.2	22.7	40.4	19.9	10.2

These frequencies are not directly comparable to the long-term frequencies at Fort Worth, since each distribution is based on a different criteria for determining Pasquill stability classes.

2.3.2.1.8 Mixing Heights

The frequencies of seasonal and annual mixing heights are included and discussed in Section 2.3.1.2.7. Since onsite measurements of mixing depth are neither required nor made, monthly mixing depths from upper air data at Carswell AFB, Fort Worth (5/72-10/73) and Stephenville, Texas (11/73-4/76), and surface observations from the National Weather Service (NWS) station in Fort Worth, concurrent with the onsite data record, are presented in Table 2.3.24 [33]. The method used for determining mixing depths is the same as described by Holzworth [10] with observations identified as P (precipitation), C (cold air advection), and M (missing) excluded from the record. Inclusion of P and C types would tend to increase the mean mixing depths given in Table 2.3-24.

2.3.2.1.9 Representativeness of the Onsite Data

To demonstrate that the onsite data sample is representative of a long term climatological average, concurrent data from Fort Worth (May 1972 through April 1976) were compared to a longer data record for Fort Worth (January through December, 1957-1971). The wind frequency distribution and the stability class frequency distribution and mean wind speeds are presented in Table 2.3-25.

Generally, the wind frequencies and stability distributions are in excellent agreement. The largest difference in wind frequency is only three percent and occurs in the south sector. One notable difference is the higher frequency of calms during the four-year period. The only significant differences in the stability class distributions occur in the “D” and “F” and “G” classes. For the four-year period, there are approximately five percent less “D” occurrences and five percent more “F” and “G” occurrences. The mean wind speeds are also in good agreement, averaging about one knot less during the four-year period.

Generally, the comparison implies that the Comanche Peak onsite data are representative of longer-term climatological conditions. The differences that do occur indicate that diffusion estimates will probably be higher than normal (conservative) for the four-year period of record.

2.3.2.2 Potential Influence of the Plant and Its Facilities on Local Meteorology

2.3.2.2.1 General

Potential modifications of the local meteorology at the site resulting from the construction and operation of the plant are believed to be small. The Reactor Complex is located approximately 450m west-northwest of the meteorological tower. The top of the dome is 69m above the level of the base of the meteorological tower. The Reactor Complex meets the requirements of the Standard Review Plan (46). Additionally wind was recorded from the west-northwest sector approximately 2.1% of all recordings; thus, any effect that the Reactor Complex does exhibit on our overall meteorological measurements program will be minimal. See Section 2.3.3.2 for a description of the Meteorological Instrumentation Building.

In addition, no other structures are in such proximity to the tower that will cause a significant alteration of the meteorological data. This is based upon the criteria that no structure equal to or greater than the height of a measuring level is closer than 10 times that measuring level to the tower, or that all sensors are at least 5 times the building height away from any building that is equal to or greater than the measuring levels.

The fill dirt pile located east-northeast of the meteorological towers is actually a mound of topsoil and will be distributed as required over the finish grade of the site. The topsoil mound is approximately 3000 feet from the primary meteorological tower location. Top of the pile is E1. 865'-9" approximate and the lower measure level on the tower E1. 870' approximate. Due to the linear distance away from the tower, and the relative height of the mound, the effects of the mound on meteorological measurements will be minimal. Additionally, wind was recorded from the ENE direction only 4% of all recorded winds (FSAR Table 2.3-21); therefore, for overall purposes, even if the topsoil mound did exhibit effects on measurements, the effect on cumulative total measurements would be minimal.

In sum, the Containment Building and associated facilities are expected to have some small influence on the local air flow; specifically, mechanical turbulence is expected downwind of the plant, due to building wake effects.

2.3.2.2.2 Impact of Squaw Creek Reservoir

Filling of Squaw Creek Reservoir commenced on February 15, 1977 with completion in May 1979. The potential impacts of the Squaw Creek Reservoir upon onsite meteorlogy are discussed in the Dames & Moore Report [34] which supports the conclusion that the reservoir will have a minimal effect upon local meteorology. Although no specific monitoring program is proposed to examine such effects, various meteorological measurements are sampled as discussed in 2.3.3.1.1.

The effects of the reservoir on wind speed, wind direction and vertical temperature gradient should be minimal. Small increases in the wind speed would be expected because the surface frictional coefficient of the air-water interface would be different from that which is now existent at elevation 770' MSL (reservoir elevation). Since Squaw Creek Reservoir has a maximum fetch of

about 4 1/2 miles, wind speeds may increase slightly for wind directions along the maximum fetch (southeast-northwest orientation). Small changes in the wind direction would also be expected due to variations of the surface frictional coefficient.

During winds from the northwest, northeast, and east-northeast, some modification of atmospheric conditions may occur: first as the trajectory moves over the reservoir surface, and subsequently as it returns to land, passes the tower and the reactor building, and then to the site boundary and some distance beyond. At all times with these wind directions the tower is likely to record higher wind speeds than prevail through the balance of a trajectory to the site boundary and beyond. As the trajectory returns to land, the reduction in wind speed due to the friction of land surfaces will be accompanied by increased turbulence. As a result of the offsetting process, the resultant dispersion values are not significantly altered.

On clear nights during light onshore winds from the above wind directions, the thermal stability of the air column may increase while the air column moves from the tower toward the site boundary and beyond. However, unless wind speeds are below about 3 mph, the overwater trajectory is not likely to have been of sufficient duration for significant modifications in its thermal stability to occur. In cases where the wind speed exceeds 3 mph, the tower will observe the air column stability near its overland equilibrium value and its readings will be representative of the thermal stability which will prevail throughout the trajectory to the site boundary and beyond.

With wind speeds of 3 mph or less in these directions, departures from overland stability values will be sensed at night by the tower, and the air column will move toward its more stable overland values during the trajectory from the reservoir shore to the site boundary and beyond. Conditions when the tower observed data may be somewhat unrepresentative and more favorable than the actual dispersion conditions downwind of the reactor building include the following:

1. When the wind direction is from the northwest, northeast, and east-northeast, and

2. The wind speed is 3 mph or less, and

3. Overland thermal stabilities have equilibrium values of Pasquill E, F, or G, and

4. During nighttime hours.

The onsite meteorological tower data for the 4-year period of record shows that conditions one through three occur concurrently only about 2.5 percent of the time. If daytime conditions were eliminated, the percentage would be somewhat smaller. By comparison, the combined frequency of occurrence of winds from these three directions is about 12 percent.

Thus, (1) 2.5 percent or less of the time, tower stability observations may be somewhat nonconservative when used to represent dispersion conditions at CPNPP, and (2) wind speeds may be higher than the representative overland values about 12 percent of the time, but downwind modifications to overland speeds are likely to be offset by compensatory increases in mechanically induced turbulence.

The effects of the reservoir on air temperatures in the area will be significant only when ventilation across the pond is at a minimum. The lack of ventilation would give air over the pond enough time to have its temperature measurably changed by the temperature of the surface

water for northwest or southeast winds. For low wind speeds, the maximum increase in air temperature would probably be large enough to have an effect upon thermal stability. It is estimated that the reservoir, due to heating of the air, would cause a decrease in the frequency of the more stable classifications and an increase in the frequency of the more unstable classifications. Thus, although overall changes are expected to be small, the Squaw Creek Reservoir should cause a slight improvement in diffusion meteorology due to enhanced ventilation and a reduction in the frequency of the very stable conditions.

Anticipated effects of the proposed Squaw Creek Reservoir on monthly average temperature, relative humidity, frequency of fogs for various visibility classes, and frequency of icing have been determined for the months of January and July (representing winter and summer conditions). Details of the methods used and the results are available in the Dames & Moore report [34] on the Squaw Creek Reservoir.

1. Temperature and Humidity Changes

   The night hours of both January and July experience the greatest modification in temperature and relative humidity. Largest modifications occur over the reservoir, with smaller effects over adjacent land area. The night hours of January and July show the greatest anticipated change of +1.2 degrees in temperature and the July nighttime hours have a 3.2 percent increase in relative humidity with 73 percent and 90 percent of the area affected, respectively. The daylight hours of July exhibit the smallest influence from the proposed reservoir.

2. Fogging - Frequency of Occurrence

   Conditional fog frequencies have been determined for various visibility classes for each of the following situations: (a) natural fogs unaffected by the reservoir, i.e., the baseline state, (b) natural fogs with visibilities altered by the reservoir, and (c) reservoir generated fogs.

   Natural fog (baseline) occurrences were determined from 3-hourly meteorological observations taken at the Dallas National Weather Service (NWS) station during the months of January and July over a 3-year period (1965-1967); occurrences were summarized by visibility classes. Natural fogs are not recorded by the NWS unless the visibility is seven miles or less. Onsite observations of fogs were not made.

   Data from the Dallas NWS station were selected to represent natural (baseline) fog occurrences at CPNPP on the basis of being the most representative data available. Fog occurrence at the Dallas NWS station is quite small [34]. Although these frequencies cannot be directly correlated to onsite fog frequencies, there are no known physical reasons that would contribute to markedly different baseline fog frequencies at CPNPP. The following meteorological characteristics are presented as evidence that baseline fog frequencies are similar or even less frequent onsite than at the Dallas NWS station:

   a. Pure radiation fog, which can be extremely localized, seldom occurs in the site area except for shallow patches [3].

b. Patchy or localized fog is more predominant, generally, in low lying area. Since CPNPP is located as much as 200 feet above adjacent terrain, baseline fog occurrences at the site should be less than at nearby low lying areas.

c. The site is not located adjacent to any major bodies of water (the Brazos River is over three miles away) that might increase local fog occurrences onsite.

d. Moisture levels decrease from East to West across much of Texas. Since CPNPP is located approximately 50 miles west of the Dallas NWS station, moisture conditions should be somewhat less favorable for fog formation onsite than at the Dallas NWS station.

Generally, except for July daylight hours, the presence of Squaw Creek Reservoir will decrease the visibility in naturally occurring fogs. In January, an increase of up to five percent is expected in the occurrence of fogs. July nighttime conditions show the least influence of the reservoir, with no change in the frequency of fog occurrence. During July daylight hours, a decrease of up to one percent in the occurrence of fog is expected.

3. Fogging - Visibility of One Mile or Less

Fog with visibility of one mile or less may be frequent enough to limit activities in the area. Daylight hours show the greatest increase of fog frequencies in this range of visibilities, +5 percent, and the largest area modified. During January daylight hours, the study also shows that Highway 144, East of the reservoir, should experience a three percent increase in occurrence of visibility restricted to one mile. Road 201, west of Squaw Creek Reservoir, should experience a one percent increase of fog visibility less than or equal to one mile. Highway 144 is expected to have an increase of one to two percent in fog frequency with visibility less than or equal to one mile during the night hours of January. This fog will extend for about 1.7 miles on the southeast side of the reservoir.

July daylight hours, for this visibility range, are expected to have a two percent increase in the maximum frequency of fogs. A one percent increase over natural fog frequencies is expected along Highway 144 north of the reservoir and along Roads 201, 204, and 51, west of Squaw Creek Reservoir. No modification is anticipated for the July night hours in this visibility class.

4. Icing

Potential icing conditions are specified by the occurrence of reservoir produced fog accompanied by subfreezing temperature. January night hours are anticipated to have a maximum of one percent increase in the frequency of icing. These conditions occur over or near the reservoir, and therefore have little effect on the area.

2.3.2.2.3 Topographical Description

A map of the Comanche Peak area for a distance of five miles from the site is shown in Figure 2.3-4. The eight radial lines correspond to the eight individual topographic cross-sections which appear in Figure 2.3-5 through 2.3-8. These figures indicate the maximum elevation versus distance from the plant in each sector. The site elevation is approximately 810 feet MSL.

The terrain varies from 600 to 1,000 feet MSL within five miles of the site, and is generally in this range out to 50 miles. General topographic features for a radius of fifty miles are shown in Figure 2.3-9.

Variable terrain has a potential to influence local diffusion characteristics. Terrain variations on the order of plus or minus 200 feet are not pronounced enough to cause any significant flow blocking. Two possible influences, though, cold air drainage and channeling, have been investigated. The occurrences of cold (more dense) air drainage down Squaw Creek was assessed by a comparison of wind direction frequencies between the 10-meter (850 ft. MSL) and 60-meter (1000 ft. MSL) levels (Figure 2.3-10) for a 131-day period. If drainage occurs, then marked increases of down-valley wind frequencies (ESE and SE) from the upper to the lower level would be expected. Marked changes in frequency do not appear in the data; therefore, it is concluded that cold air drainage along Squaw Creek is not significant. Though not significant, the occurrence of cold air drainage cannot be completely ruled out on the basis of a comparison of wind directions measured at approximately 850 and 1000 ft. MSL, since the Squaw Creek drainage basin lies below 700 ft. MSL. However, the Squaw Creek Reservoir has been filled since this comparison was made (see Section 2.3.2.2.2). This effectively modifies the topography over a large area surrounding CPNPP to a minimum elevation of 770 ft. MSL, or only about 40 feet less than site elevation. Thus, cold air drainage is unlikely.

Channeling of air flow, the other potential topographical effect, was studied by comparing 10-meter wind directions with nearby wind direction data from Dallas Love Field, where surroundings are relatively flat. A significant increase in wind direction frequencies for both up and down valley sectors (WNW, NW, NNW, ESE, and SE) should occur if channeling is an important influence. Approximately eight months of concurrent wind direction data, shown in Figure 2.3-11, indicate that channeling of the air along Squaw Creek is not a prominent effect.

The channeling and air-drainage study results are indicative of a relatively flat terrain. There is even less topographical variation after creation of the reservoir. This implies that there is less topographic effect on the local airflow and, therefore, a slight improvement in diffusion meteorology. In conclusion, the onsite data collected prior to, and after, the creation of the reservoir does not change appreciably.

2.3.2.3 Local Meteorological Conditions for Design and Operating Bases

Local meteorological data have not been used for design and operating basis considerations other than those conditions referred to in Sections 2.3.4 and 2.3.5. Design wind loadings, tornado loadings, and snow loadings are referred to under Regional Meteorology, Section 2.3.1.

2.3.3 ONSITE METEOROLOGICAL MEASUREMENTS PROGRAM

2.3.3.1 Pre-Operational Program

The pre-operational onsite meteorological program is designed to measure the parameters needed to evaluate the dispersive characteristics of the site for both the routine operational and the hypothetical accidental releases of radionuclides to the atmosphere.

2.3.3.1.1 Instrumentation and Recording Systems

The majority of the measurements are made on a 200-foot steel-framed tower which is located 1,500 feet east of the reactor complex (See Figure 2.3-12). The location of the tower in relation to the entire site area is shown in Figure 2.1-2. The terrain surrounding the tower is relatively hilly and is covered by widely scattered scrub brush, small mesquite, and cedar.

The following measurements are sampled on the tower:

1. Wind speed at two levels (10 and 60 meters)

2. Wind direction at two levels (10 and 60 meters)

3. Ambient temperature at three levels (10, 30, and 60 meters)

4. Ambient dewpoint temperature at two levels (10 and 60 meters)

5. Thermal stability, for two height intervals (10 to 30 meters, and 10 to 60 meters)

6. Wind direction variability (10 meters)

In addition to these measurements, precipitation and solar radiation (both total and net) are sampled near the base of the tower.

Descriptions of the instruments employed in these measurements are found in Tables 2.3-26.

To assure a high percentage of data recovery, duplicate recording systems are used. All variables are recorded in analog form on strip charts which record continuously, and digitally on magnetic tape which records ”one scan each minute.” The latter is the most common method of complying with criteria in Reg. Guide 1.23 for digital recording systems; that is, to record instantaneous values of each meteorological parameter once each minute. If the total number of minute observations during the hour for any parameter is less than 15, then that parameter is considered to be invalid for that hour. The magnetic tape is the primary recording system.

A comparison of analog and digital hourly average values was made to support the use of instantaneous values recorded once each minute to represent hourly average conditions. Comparisons of both wind speed and direction at the 10-meter level for a randomly selected 2-week period are summarized in Table 2.3-35. These data indicate overall agreement between the hourly average values from the analog and digital recording systems.

The digital recording system reduces each sensor output signal to a digital voltage form, with a range of 0 to 999 millivolts (mv) (except delta T, with a range of -999 to 999 mv), and records all variables on magnetic tape once per minute.

Data lost during outages or malfunctions of the digital recorder are reduced manually from the strip chart record. The hourly averages are logged for insertion into the meteorological record.

2.3.3.1.2 Maintenance and Calibration

To assure data quality and accuracy, a comprehensive calibration of the meteorological station components is performed at six month intervals. The procedure includes close visual inspection of all instrument sensors for wear, electronic component calibration, ambient temperature, and dewpoint comparison using mercury-type thermometers, and calibration of recorders.

Normal maintenance includes a comprehensive inspection of the station's electronic and mechanical equipment as part of an ongoing operation and maintenance program. Inspections are performed on a weekly frequency, as a minimum, but on average two operational inspections are performed per week. Station operating procedures call for, among other things, a manual check of the zero and full-scale positioning of the analog recorders, as well as a verification of the associated DC voltages displayed by the digital panel meter for the primary recording system.

The calibration and maintenance program is accomplished through the effort of experienced CPNPP technicians.

Pre-operational data recovery and analysis was initiated on May 15, 1972. For the four year period ending May 14, 1976, about 96 percent recovery of primary variables (wind speed, wind direction, and delta temperature) was accomplished.

2.3.3.1.3 Data Processing

With the exception of precipitation and wind direction, hourly averages of the digital minute-by-minute observations are calculated from the following scalar equation:

$$\bar{B}_j = \frac{r_j}{n} \sum_{i=1}^{n} B_{ji} \qquad 2.3\text{-}2$$

where

B_j = the average hourly value for the j^{th} variable (engineering units)

n = the total number of minute observations during the hour normally 60, but if n <15, data for that hour are considered to be missing)

B_{ji} = the i^{th} minute observation of the j^{th} variable (millivolts)

r_j = the conversion factor to change the j^{th} variable into engineering units

Whereas most of the averages are scalar inform, the average wind directions are determined by the following averaging technique:

Each minute observation of wind vector (speed and direction) is broken into its components, u and v, according to

$$u_i = S_i \times \sin(\theta i - \pi)$$
$$v_i = S_i \times \cos(\theta i - \pi) \qquad \text{2.3-3}$$

where

u_i = the east-west component of wind for the minute

v_i = the north-south component of wind for the minute

S_i = the scaler wind speed for the minute

θi = the wind direction for the minute

The u_i and v_i are added separately and the sums are divided by the total number of minute observations for the hour to establish the average components, u and v, i.e.:

$$\bar{u} = \frac{1}{n}\sum_{i=1}^{n} u_i \qquad \text{2.3-4}$$

$$\bar{v} = \frac{1}{n}\sum_{i=1}^{n} v_i$$

where

$\bar{u}$ = the average east-west component of wind for the hour

$\bar{v}$ = The average north-south component of wind for the hour

n = the number of valid minute observations for the hour

The average wind direction is found by converting the average components into a vector direction, i.e.,

$$\bar{\theta} = \tan^{-1}(\bar{u}/\bar{v} + \pi) \qquad \text{2.3-5}$$

where

$\bar{\theta}$ = the average vector wind direction during the hour

The precipitation accumulated during the hour is established by subtracting the amount of precipitation measured by the rain gauge at the beginning of the hour from the amount measured at the end of the hour.

Data analysis for both wind distribution and diffusion characteristics of the site require three basic atmospheric variables. These three variables together with the primary and secondary (back-up) measurements for each are as follows:

Horizontal wind speed:	primary, 10-meter level secondary, 60-meter level
Horizontal wind direction:	primary, 10-meter level secondary, 60-meter level
Delta temperature (ΔT)	primary T, 10 to 60 meters secondary T, 10 to 60 meters

The secondary measurement is needed only during periods of outage of the primary system, and is reduced to the appropriate level as follows: wind speed at 60 meters is converted to wind speed at 10 meters using the power law [35]:

$$V_{10} = V_{60}\left(\frac{10}{60}\right)^{S} \qquad 2.3\text{-}6$$

where

V_{10} = wind speed at 10 meters

V_{60} = wind speed at 60 meters

S = .26 for Pasquill classes A, B, C, and D or .50 for Pasquill classes E, F, and G

Wind direction at 60 meters is directly substituted for wind direction at 10 meters.

ΔT (redundant) between 10 and 60 meters is substituted for ΔT between 10 and 60 meters, as required, for classification of atmospheric stability in accordance with Table 2 of Regulatory Guide 1.23.

The final step in the data reduction program is the listing, in sequential order, of the concurrent, hourly-averaged values of the weather elements observed at the site. The data record provides the input data for all types of meteorological analyses needed to define the site atmospheric dispersive qualities. The hourly onsite data record used in the Section 2.3 analyses will be provided to the NRC separately from this report.

2.3.3.2 Operational Program

Prior to fuel load of Unit One, the meteorological measurements program of the Comanche Peak Nuclear Power Plant shall consist of the following:

1. A primary meteorological measurements program.

2. A backup meteorological measurements system.

3. A system for making near real-time predictions of the atmospheric effluent transport and diffusion.

4. A capability for remote interrogation, on demand, of the atmospheric measurements and prediction systems by the licensee, emergency response organizations, and the NRC Staff.

To accomplish these goals, “the pre-operational meteorological instrument system” has been modified. The current “operational meteorological measurements system” transmits the above mentioned meteorological parameters to a multi-channel digital recorder, METSYS Computer and the plant computer.

The operational program consists of a 60m multi-level primary meteorological tower located at plant grid coordinates N9519.44 ft. by E12111.75 ft. (el. 838 ft. - 9 in.) which is east of the Unit 1 and Unit 2 reactor buildings. The primary meteorological tower directly monitors or provides information to determine the following meteorological parameters:

- Wind speed at 10m and 60m;

- Wind direction at 10m and 60m;

- Ambient temperature at 10m;

- Delta-temperature between 10m and 60m (redundant channels at each level);

- Sigma theta at 10m;

- Precipitation near ground-level.

The primary tower is a 60m, guyed, open lattice type tower with an instrument elevator and instrumentation booms located at the 10m and 60m levels. These booms are oriented to the west. The aspirated temperature shields at 10m and 60m are oriented laterally to the north. The primary and backup tower instrument translators are located in an environmentally controlled building which is approximately 70 ft. West-Northwest of the primary tower.

The Meteorological Instrumentation Building was constructed near the base of the tower to support both primary and backup tower field instruments. It is 12 ft. wide by 16 ft. long by 9 ft. high with a steel floor and galvanized steel siding, roof and ceiling on a concrete slab. Environmental control in the summer is maintained by an air conditioning unit. In winter, environmental control is maintained by an electric heating system. The heating/AC unit is controlled by an automatic thermostat.

The operational program also consists of a separate 10m backup meteorological tower located approximately 75 ft. East-Northeast from the 60m primary tower. The backup tower monitors or determines the following parameters:

- Wind speed at 10m;
- Wind direction at 10m;
- Temperature at 10m;
- Sigma theta at 10m

The backup tower is a 10m free standing open lattice type tower with an instrumentation boom located on top of the tower at 10m. The aspirated temperature shield is orientated laterally to the north to further minimize the effects of direct sunlight on the measured temperature.

To ensure that meteorological data being collected by the backup tower at its location near the primary tower was representative of coincident data being recorded at the 10 meter level of the primary tower, a real-time data analysis consisting of a 3-month time period was performed. The real-time representativeness analysis consisted of statistical correlations for coincident wind speed, wind direction, and sigma-theta parameters from the two towers for the minimum time period specified above. The results of this analysis was used to demonstrate that meteorological data from the backup tower can be reliably substituted during loss-of-data periods from the primary tower.

The recording of meteorological data from both the primary and backup towers is accomplished by utilizing one digital and one auxiliary analog system. The recorder system is a digital paperless recorder on the meteorological instrument panel in the Control Room. All meteorological parameters from the primary tower, and wind speed, wind direction and sigma theta from the backup tower, are displayed on this recorder.

To ensure an acceptable data recovery of the precipitation measurements for the operational program, equipment was procured and is maintained to assure at least 90% annual data recovery. The primary system for recording precipitation data is checked at least once per week to assure its continued function.

Meteorological data from both towers is provided to the Met and plant computers. Signals from both meteorological towers are transmitted through shielded twisted pair cable to the digital to analog converter of the meteorological system receiver located in the Unit 1 Plant Computer Room.

Digital output of the Met and plant computers are located in the Unit 1 Plant Computer and Control Room. Digital output of the plant computers are also located in the Technical Support Center (TSC) and Emergency Operation Facility (EOF), where meteorological data consistent with the requirements of NUREG-0696 (45) are displayed.

The Met computer is designed to provide digital readout of meteorological data received from the primary and backup towers for all parameters. Both 15-minute and hourly-averaged data are generated. The 15-minute averaged data are derived from the meteorological parameters that are sampled every five seconds (except precipitation which is a totalized value). The hourly

averaged data are derived from the 15-minute averages. The 15-minute averaged data and the hourly-averaged data can be stored internally in the computer for a period of up to 10-years. The methodology for meteorological data processing as described in Section 2.3.3.1 was used on the meteorological data collected during the pre-operational monitoring program and for the data initially collected under the approved CPNPP operational meteorological monitoring program. The data averaging methodology has been updated with the installation of the Met computer. This updated data averaging methodology continues to meet the requirements of Regulatory Guide 1.23.

The plant computer is designed to provide digital readout of 15-minute averaged data from the primary meteorological tower. The 15-minute averaged data are based on a 5-second sampling rate. Trending data for the previous week are also available from the plant computer.

The following table shows the sampling time, averaging time and number of samples per averaging period for both the Met and plant computers:

Computer	Sampling Time (sec)	Averaging Time (min)	Number of Samples Per Averaging Period
Met	5	15	180
Met	(a)	60	(a)
Plant computer	5	15	180

a) The hourly average is derived from the 15-minute averages.

The operational program will be conducted in accordance with the requirements specified in Regulatory Guides 1.21 and 4.1 (36, 37), Second Proposed Revision 1 to Regulatory Guide 1.23 (44), and Revision 1 to NUREG-0654 (43). Descriptions of the instruments employed in the Operational meteorological measurements program are found in Table 2.3-34. Tables 2.3-34A and 2.3-34B provide information about meteorological instrumentation system accuracy.

2.3.3.3 Wind Roses By Pasquill Stability Classes

Monthly and annual wind roses for each Pasquill stability class (and all classes combined) for the four-year period of record onsite are shown in Table 2.3-27 [26]. These tables are based on wind distributions at the 10-meter level and the 10-60 meter ΔT measurements. During periods of outage of the primary variables, secondary wind and ΔT measurements were used as discussed in Section 2.3.3.1.3.

A discussion of the representativeness of the onsite data is provided in Section 2.3.2.1.9.

2.3.4 SHORT-TERM (ACCIDENT) DIFFUSION ESTIMATES

2.3.4.1 Objective

The onsite meteorological data record at the Comanche Peak site for the period May 15, 1972, through May 14, 1976, has been used to calculate dilution factors which can be anticipated in the event of an accidental release of radionuclides into the atmosphere. The one-hour dilution factors are calculated at the site boundary (exclusion area); for longer time periods the factors are calculated at the outer boundary of the low population zone (LPZ).

2.3.4.2 Calculations

Diffusion calculations for accidental or short-term releases of radionuclides were initially performed in accordance with the criteria provided in Regulatory Guide 1.4 [38]. Short term diffusion estimates were subsequently recalculated to incorporate the modeling features of NRC Draft Regulatory Guide 1.XXX [48]. It was assumed that the releases emanate from a point source at ground level; i.e., no advantage is realized from effluent emissions from elevated release points. During downwind transport, the effluent plume is assumed to spread according to a Gaussian dispersion model.

2.3.4.2.1 The Diffusion Model for Two Hours or Less

The analytical procedure for evaluating the 0-2 hour accident period is based on a revision of the model described in NRC Regulatory Guide 1.4; the revision incorporates the modeling features of NRC Draft Reg. Guide 1.XXX [48]. The changes reflect variations in atmospheric diffusion factors that occur as a function of wind direction and varying site boundary distance. Allowances are made for meandering plumes during light winds and stable atmospheric conditions. The model is distance and direction-dependent. The hourly χ/Q values used to represent the 0-2 hour accident period were determined in the following manner.

During neutral and stable conditions when the wind speed at the lower level is less than 6 meters per second (mps) the relative concentration is computed as:

$$\frac{\chi}{Q} = \frac{1}{\bar{u} \quad \pi\Sigma_y\sigma z} \qquad 2.3\text{-}7$$

provided it is less than the greater value calculated from either

$$\frac{\chi}{Q} = \frac{1}{\bar{u}(\pi\sigma_y\sigma_z + cA)} \qquad 2.3\text{-}8$$

or

$$\frac{\chi}{Q} = \frac{1}{\bar{u}(3\pi\sigma_y\sigma_z)} \qquad 2.3\text{-}9$$

where,

χ/Q = relative concentration at ground level (s/m^3)

$\bar{u}$ = hourly average wind speed at 10 m level (mps)

Σ_y = lateral plume spread (m) (a function of atmospheric stability, wind speed, and downwind distance from the release). For distances up to 800 meters, $\Sigma_y = M\sigma_y$; where M is a function of atmospheric stability and wind speed. For distances greater than 800 meters,

$$\Sigma_y = (M-1)\sigma_y(@800m) + \sigma_y$$

A = minimum cross-sectional area of the reactor containment structure (3200 m^2) assuming that:
width of containment building = 140 feet
height of spring line above grade = 190.5 feet
radius of dome circle = 70 feet

c = building shape factor (taken as 0.5)

σ_y = lateral plume spread (m) at a given distance and stability

σ_z = vertical plume spread (m) at a given distance and stability.

During all other atmospheric stability and/or wind speed conditions, χ/Q is the greater value calculated from Equations 2.3-8 and 2.3-9.

Plume meander was accounted for by modifying the lateral diffusivity, σ_y. The meander function (M) is evaluated as follows:

For Pasquill stabilities A-C at all wind speeds or all stabilities when wind speed > 6 mps; M = 1. For wind speed ≤2 mps;

Stab D; M = 2

Stab E; M = 3

Stab F; M = 4

Stab G; M = 6

For wind speed between 2 and 6 mps, M is evaluated by a curve fitting technique and multiplied by σ_y.

The direction dependent 5- and 50-percentile χ/Q's by sector were calculated by the following procedure. The 5- and 50-percentile values were determined by normalizing the frequency of wind into the sector of interest. The 5-percentile χ/Q value is the point in a given sector χ/Q

cumulative frequency distribution which is equalled or exceeded (5 x 6.25/f) percent of the time, where f is the frequency in percent of winds in the sector. Similary, the 50-percentile value is the point in a given sector χ/Q cumulative frequency distribution which is equalled or exceeded (50 x 6.25/f) percent of the time. Calm conditions were included in the calculations by setting the wind speed to the threshold value of the wind direction sensor (0.75 mph) and distributing them among the 16 direction sectors in proportion to the directional frequencies in each stability class of wind speeds of 2 mph or less. This is a slightly larger speed class interval than the 1.5 mph limit recommended by Reg. Guide 1.XXX [48], but is justified due to the relatively small number of observations in the lowest wind speed class (≤1 mph) at CPNPP.

2.3.4.2.2 Estimated Values of One-Hour Dilution Factors

The minimum Exclusion Area Boundary (EAB) distances in each sector are presented in Table 2.3-28. For each wind direction sector, the minimum EAB distance was assumed to be the minimum distance within a 45 degree direction sector, centered on the direction sector of interest. See Section 2.1.2 for a discussion of the exclusion area boundary parameters.

The highest 5-percentile concentration in any sector is 1.6×10^{-4} s/m^3 and occurs northwest of the plant. This maximum χ/Q was calculated at a distance of 2080m although the actual minimum EAB distance in the NW sector is 2106m. This is conservative. The highest 50-percentile concentration is 4.5×10^{-5} s/m^3 and occurs north-northwest of the plant.

2.3.4.2.3 Diffusion Estimates for Periods Greater Than Two Hours

Determination of χ/Q's for time periods of 0-8 hours, 8-24 hours, 1-4 days, and 4-30 days was accomplished by graphical interpolation between 2-hour χ/Q's in each sector and the annual average χ/Q's in each sector, both at the LPZ distance (4 miles). Hourly 5- and 50-percentile χ/Q's for each sector were computed at the LPZ using the same technique as described above for the 5- and 50-percentile χ/Q's by sector at the EAB (hourly χ/Q's were used to represent the 0-2 hour χ/Q's). The annual average χ/Q's used in the graphical interpolation are those given in Table 2.3-33 at the LPZ distance of 4 miles. For each sector, the hourly 5-percentile χ/Q's and the annual average χ/Q's, both at the distance of the LPZ, were plotted on a log-log coordinate graph with ordinate values representing χ/Q and abscissa values being hours.

These two points, (χ/Q_1, 2 hours) and (χ/Q annual, 8760 hours) were connected by a straight line and values of χ/Q for periods of 8, 16, 72 and 624 hours were extracted. These χ/Q's represent the 0-8 hour, 8-24 hour, 1-4 day, and the 4-30 day accident period 5-percentile χ/Q's per sector at the LPZ distance. Similar graphical interpolation was used to determine 50-percentile χ/Q's for each accident period.

The 5- and 50-percentile χ/Q's for each accident period at the LPZ are given in Tables 2.3-37 and 2.3-38 for each of the 16 meteorological sectors.

2.3.4.2.4 Representativeness and Topographic Effects

As discussed in Section 2.3.2.1.9, the onsite data sample was considered to be conservatively representative of meteorological conditions at the site. The notable differences observed during the onsite period of record were a higher frequency of calm wind speeds and a higher frequency

of stable conditions. These conditions have undoubtedly increased the magnitude of the χ/Q values given in Section 2.3.4 above those to be expected over a longer climatological interval.

Topographic effects at the site were discussed in Section 2.3.2.2.3. The results were indicative of a flat terrain with no appreciable effects on short-term diffusion estimates. After creation of the Squaw Creek Reservoir, a slight improvement is expected in diffusion meteorology due to both reduced topographic variation and the ameliorative effects of a heated reservoir on the frequency of very stable conditions.

2.3.5 LONG-TERM (ROUTINE) DIFFUSION ESTIMATES

2.3.5.1 Objective

The onsite meteorological record (5/15/72 to 5/14/76) is used to provide realistic estimates of annual average atmospheric dilution factors to a distance of 50 miles from the plant for use in calculating the dispersion through air pathways of radionuclides released in routine plant operations. As indicated in Section 2.3.3, the hourly onsite data record will be provided to the NRC separately from this report.

2.3.5.2 Calculations

The average annual dilution factors which are applicable to routine venting or other routine gaseous-effluent releases have been evaluated from the data record using the technique presented in Regulatory Guide 1.111 [39]. The equation used is a sector-spread equation, namely:

$$\chi/Q\ (i, D) = \frac{T}{mBD}\ \frac{\sqrt{2}}{\sqrt{n}}\ \sum_{1}^{n}\ \frac{1}{\overline{u}\ (i)\ (\sigma_z^{\,2}(p, D) + cV^2/\pi)^{1/2}} \qquad 2.3\text{-}10$$

where,

$\chi/Q(i,D)$	=	dilution factor (seconds/meter^3) at distance D, in affected direction Sector i
m	=	number of valid hourly observations in the data record
n	=	number of occurrences in the data record affecting direction Sector i
i	=	wind direction Sector index
p	=	Pasquill stability class index
$\sigma_z(p,D)$	=	hourly average vertical dispersion coefficient of the plume (meters) for a given Pasquill class, at distance D
c	=	building wake shape factor (taken as 0.5)

V	=	vertical height of the reactor containment structure (79.4 meters)
B	=	horizontal plume spread factor (taken as $\pi/8$ radians)
$\bar{u}(i)$	=	hourly average wind speed (meters/second) affecting direction Sector i
D	=	distance from reactor containment structure
T	=	terrain correction factor; open terrain, valley flow, or other factor for χ/Q adjustment (distance dependent)

The wake factor (cV^2/π) influence is limited such that the resultant χ/Q may not be reduced greater than a factor of $\sqrt{3}$, i.e., [π/Q] wake $\geq$ [$1/\sqrt{3}(\chi/Q)$ no wake]. Calm conditions are included in the calculations by setting the wind speed to one-half the threshold value of the speed or direction sensor and distributing them among the 16 direction sectors in proportion to the directional frequencies of the one and two mph speed class intervals in the appropriate stability class.

Equation 2.3-10 is the straight-line trajectory model defined in NRC Regulatory Guide 1.111 assuming a ground-level release mode; that is, the release occurs at an elevation less than or equal to the adjacent building height. Since the site is basically in open terrain with gently rolling hills, the T factor in Equation 2.3-10 is the open terrain correction factor given in Reg. Guide 1.111[39].

Annual average dilution factors to a distance of 50 miles from the plant are shown in Table 2.3-33. The maximum value at the actual EAB is 3.3×10^{-6} seconds/meter^3 and occurs north-northwest of the plant at a distance of 1.29 miles. There are no higher values beyond the site boundary since for ground level releases concentrations monontonically decrease from the release point to all locations downwind.

Long-term dilution factor estimates for distances out to five miles are displayed in Figure 2.3-18. Estimates out to 50 miles are illustrated in Figure 2.3-19.

The annual average dilution factors given in this section are likely to be quite conservative for the reasons given in Section 2.3.4.2.4.

REFERENCES

1. Visher, Stephen S., “Climatic Atlas of the United States,” Howard University Press, Cambridge, Mass., 1966.

2. “Climatic Atlas of the United States,” U.S. Department of Commerce, ESSA, Environmental Data Service, June 1968.

3. Orton, Robert B., “The Climate of Texas and Adjacent Gulf Waters,” U.S. Government Printing Office, Washington, D.C., 1965.

4. Cry, George W., “Effects of Tropical Cyclone Rainfall on the Distribution of Precipitation over the Eastern and Southern United States,” ESSA, Professional Paper 1, U. S. Department of Commerce, Washington, D. C., June 1967.

5. Pautz, M.E., “Severe Local Storm Occurrences, 1955-1967,” U. S. Department of Commerce, Weather Bureau, Office of Meteorological Operations, Weather Analysis and Predicting Division, Silver Springs, Md., September 1969.

6. Thom, H. C. S., “Tornado Probabilities,” Monthly Weather Review, October-December, 1963.

7. “Design Basis Tornado for Nuclear Power Plants,” Regulatory Guide 1.76, U. S. Nuclear Regulatory Commission, April 1974.

8. Local Climatological Data, Fort Worth, Texas - Annual Summary with Comparative Data, U. S. Department of Commerce, NOAA, Environmental Data Service 1973.

9. Uman, Martin A., “Understanding Lightning,” Westinghouse Research Laboratories, Bek Technical Publications, Carnegie, Pa., 1971.

10. Holzworth, G. C., “Mixing Heights, Wind Speeds, and Potential For Urban Air Pollution Throughout the Contiguous United States,” EPA, Research Triangle, N.C., January 1972.

11. Korshover, Julius, “Climatology of Stagnating Anticyclones East of the Rocky Mountains, 1936-1965,” U. S. Department of Health, Education and Welfare, National Center for Air Pollution Control, Cincinnati, Ohio, 1967.

12. Hershfield, David M., “Rainfall Frequency Atlas of the United States for Durations from 30 Minutes to 24 Hours and Return Periods From 1 to 100 Years,” U. S. Government Printing Office, Washington, D.C., May 1961.

13. “Seasonal Variation of the Probable Maximum Precipitation East of the 105th Meridian for Areas From 10 to 1000 Square Miles and Durations of 6, 12, 24 and 48 Hours,” Hydrometeorological Report No. 33, U. S. Department of Commerce, Weather Bureau, Washington, D.C., April 1956.

14. “Maximum Recorded United States Point Rainfall for 5 Minutes to 24 Hours for 296 First Order Stations.” Technical Paper No. 2, U. S. Weather Bureau, Washington, D.C., 1963.

15. Climatography of the United States No. 86-36, Climatic Summary of the United States - Supplement for 1951 through 1960, Texas, U. S. Department of Commerce, Washington, D. C., 1964.

16. Climatic Summary of the U. S., Section 32 - Northeast Texas, U.S. Department of Agriculture, Weather Bureau, 1931.

17. Climates of the States, Texas, U. S. Department of Commerce, NOAA, Environmental Data Source, 1969.

18. “Extremes of Snowfall: United States and Canada,” Weatherwise, December 1970.

19. Ludlum, David M., “Weather Record Book,” Weatherwise, Inc., Princeton, N. J., 1971.

20. Tattlemen, P. And Gringorten, I. I., “Estimated Glaze Ice And Wind Loads at the Earth’s Surface for the Continuous United States,” Air Force Cambridge Research Laboratories, L. G. Hanscom Field, Bedford, Mass., October 1975.

21. “Requirements for Minimum Design Loads in Buildings And Other Structures,” American National Standards Institute, ANSI A58.1, 1972.

22. Local Climatological Data, Fort Worth, Texas - Observations at 3- Hour Intervals, U. S. Department of Commerce, NOAA, Environmental Data Service, 1970-1975.

23. DELETED

24. Thom, H. C. S., “New Distributions of Extreme Winds in the United States,” Journal of the Structural Division, Proceedings of the ASCE, July 1968.

25. “Climatography of the United States No. 82-41, Summary of Hourly Observations, Dallas, Texas, 1951-1960,” U. S. Government Printing Office, Washington, D. C., 1962.

26. “On-site Local Meteorological Data - 5/72 to 5/76, Comanche Peak Steam Electric Station, Texas Utilities Generating Company,” Dames & Moore, Job #4486-025, Houston, Texas.

27. Local Climatological Data, Dallas, Texas - Annual Summary With Comparative Data, U. S. Department of Commerce, NOAA, Environmental Data Service, 1973.

28. “Persistence of Surface Direction by Wind Speed, 1969-1973, Fort Worth, Texas,” U. S. Department of Commerce, NOAA, Environmental Data Service, 1976.

29. “Monthly And Annual Special Wind Direction vs. Wind Speed With And Without Precipitation, Based on 8 obs/day, Fort Worth, Texas,” U. S. Department of Commerce, NOAA, Environmental Data Service, National Climatic Center, Asheville, N.C., October 1976.

30. “Monthly And Annual Wind Distribution by Pasquill Stability Classes, 1957-1971, Based on 8 obs/day, Fort Worth, Texas,” U. S. Department of Commerce, NOAA, Environmental Data Service, National Climatic Center, Asheville, N. C., June 1976.

31. Turner, D. B., “A Diffusion Model for an Urban Area,” Journal of Applied Meteorology, February 1964.

32. “Onsite Meteorological Programs,” Regulatory Guide 1.23, U. S. Nuclear Regulatory Commission, February 1972.

33. “Daily Mixing Height Study, Tables I & II, Fort Worth, Texas,”U.S. Department of Commerce, NOAA, Environmental Data Service, National Climatic Center, Asheville, N. C., October 1976.

34. “Potential Impact of the Release of Heat And Moisture from the Squaw Creek Reservoir,” for Texas Utilities Services Inc.,” by Dames & Moore, Houston, December 1973.

35. “Recommended Guide for the Prediction of the Dispersion of Airborne Effluents,” ASME, May 1968.

36. “Measuring, Evaluating, And Reporting Radioactivity in Solid Wastes And Releases of Radioactive Materials in Liquid And Gaseous Effluents from Light-Water-Cooled Nuclear Power Plants,” U. S. Regulatory Guide 1.21, U. S. Nuclear Regulatory Commission, Revision 1, June 1974.

37. “Program for Measuring Radioactivity in the Environs of Nuclear Power Plants,” Regulatory Guide 4.1, U. S. Nuclear Regulatory Commission, Revision 1, April 1975.

38. “Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Pressurized Water Reactors,” Regulatory Guide 1.4, U. S. Nuclear Regulatory Commission, Revision 1, June 1973.

39. “Methods for Estimating Atmospheric Transport And Dispersion of Gaseous Effluents in Routine Releases from Light-Water-Cooled Reactors,” Regulatory Guide 1.111, U. S. Nuclear Regulatory Commission, March 1976.

40. Storm Data - Monthly, U. S. Department of Commerce, NOAA Environmental Data Service, Asheville, N. C., January 1968 through December 1977.

41. Fujita, T. T., “Tornados Around the World,” Weatherwise, April 1973.

42. Marshall, J. L., “Lighting Protection,” John Wiley & Sons, New York, 1973.

43. “Criteria for Preparation And Evaluation of Radiological Emergency Response Plans And Preparedness in Support of Nuclear Power Plants,” NUREG-0654, U.S. Nuclear Regulatory Commission, (FEMA-REP-1, Federal Emergency Management Agency), November, 1980.

44. “Meteorological Measurement Programs for Nuclear Power Plant.” Second Proposed Revision 1 to Regulatory Guide 1.23, U.S. Nuclear Regulatory Commission, April, 1986.

45. “Functional Criteria for Emergency Response Facilities,” Final Report, NUREG-0696, U.S. Nuclear Regulatory Commission, February, 1981.

46. “Onsite Meteorological Measurements Programs,” Section 2.3.3, “Standard Review Plan for the Review of Safety Analysis Reports for Nuclear Power Plants,” LWR Edition, NUREG-0800, Rev. 1, May, 1978.

47. “Meteorological Characteristics of the Probable Maximum Hurricane, Atlantic and Gulf Coasts of the United States,” HUR 7-97, U.S. Department of Commerce, May 1968.

48. “Atmospheric Dispersion Models for Potential Accident Consequence Assessments at Nuclear Power Plants,” Draft Regulatory Guide 1.XXX, September 23, 1977.

TABLE 2.3-1
TORNADO POINT PROBABILITY WITHIN A GIVEN YEAR AT SELECTED LOCATIONS (1955-1967)[a]

Square Location	Significance	Annual Frequency, $\dot{t}$	Probability P	Return Period (Years)
Plant Site		4.46	0.00313	320
Adjacent (East)	Area Comparison	3.54	0.00248	403
Adjacent (South)	Area Comparison	3.23	0.00241	415
Adjacent (West)	Area Comparison	2.08	0.00146	685
Adjacent (Northwest)	Maximum in Texas	5.23	0.00371	269
North of Oklahoma City	Maximum in U.S.	8.00	0.00588	170

a) Period of record is in parentheses on this and succeeding tables when applicable.

TABLE 2.3-2
AVERAGE NUMBER OF THUNDERSTORM DAYS[a] AND LARGE-HAIL[b] DAYS

Period	Thunderstorms Fort Worth (1954-1973)	Large Hail Texas (1955-1967)
January	1	<1/2
February	2	2
March	4	7
April	7	25
May	7	30
June	6	18
July	5	4
August	5	1
September	4	2
October	3	3
November	2	1
December	1	<1/2
Winter	4	2
Spring	18	62
Summer	16	23
Autumn	9	6
Annual	46	93

a) Defined as day on which thunder is heard at station.

b) 3/4-inch diameter and larger.

TABLE 2.3-2A
SEASONAL AND ANNUAL ESTIMATES OF LIGHTNING STRIKES TO SAFETY-RELATED STRUCTURES

Name of Safety-Related Structure	Winter	Spring	Summer	Autumn	Annual
Auxiliary Building	0.019	0.084	0.074	0.042	0.214
Reactor Building (Each Unit)	0.074	0.332	0.295	0.166	0.848
Electrical and Control Building	0.011	0.051	0.046	0.026	0.131
Safeguards Building (Each Unit)	0.018	0.079	0.071	0.040	0.203
Diesel Building (Each Unit)	0.007	0.030	0.027	0.015	0.077
Fuel Building	0.018	0.082	0.073	0.041	0.209
Service Water Intake Structure	0.002	0.010	0.009	0.005	0.024
Condensate Storage Tank	0.003	0.015	0.014	0.008	0.039
Reactor Makeup Water Storage Tank	0.002	0.007	0.007	0.004	0.019
Refueling Water Storage Tank	0.003	0.015	0.014	0.008	0.039

(Attractive Areas and Building Dimensions.)

Name of Safety Related Structure	Height[a] (meters)	Length (meters)	Width (meters)	Attractive Area (kilometers)
Auxiliary Building	32.31	58.52	34.75	0.0272
Reactor Building	79.25	42.67	42.67	0.1078
Electrical and Control Building	23.16	50.60	36.58	0.0167
Safeguards Building[b]	32.92	40.01	40.01	0.0258
Diesel Building	20.73	39.01	9.75	0.0098
Fuel Building	32.92	43.28	41.45	0.0266
Service Water Intake Structure	8.84	32.46	14.63	0.0031
Condensate Storage Tank	15.24	15.24	15.24	0.0050
Reactor Makeup Water Storage Tank	10.97	9.14	9.14	0.0024
Refueling Water Storage Tank	15.24	15.24	15.24	0.0050

a) Maximum heights of each building were generally used.

b) Due to the shape of this building, it was partitioned into two sections. The values for length and width are adjusted values.

TABLE 2.3-3
MEAN SEASONAL AND ANNUAL MIXING DEPTHS AND WIND SPEEDS AT TEXAS STATIONS NEAREST THE REGION (1960-1964)

	Midland		San Antonio	
	Mixing Depth (meters)	Wind Speed (meters/sec)	Mixing Depth (meters)	Wind Speed (meters/sec)
I. Morning				
Winter	290	5.7	459	5.8
Spring	429	7.5	748	6.5
Summer	606	7.2	915	5.7
Autumn	419	6.0	654	5.5
Annual	436	6.6	694	5.9
II. Afternoon				
Winter	1276	7.8	1112	6.8
Spring	2449	9.0	1552	7.2
Summer	2744	6.7	2119	6.0
Autumn	1887	6.7	1572	6.1
Annual	2089	7.5	1589	6.5

TABLE 2.3-4
ESTIMATED RAINFALL (INCHES) FOR THE SITE AREA

	Average Return Period (Years)						
Duration	1	2	5	10	25	50	100
30 min	1.15	1.49	1.95	2.24	2.62	3.00	3.27
1 hr	1.50	1.82	2.40	2.84	3.34	3.75	4.20
2 hrs	1.74	2.18	2.95	3.45	4.06	4.60	5.10
3 hrs	1.93	2.43	3.24	3.85	4.47	5.15	5.64
6 hrs	2.35	2.90	3.88	4.60	5.45	6.05	6.85
12 hrs	2.95	3.38	4.60	5.45	6.45	7.25	8.20
24 hrs	3.10	3.90	5.30	6.30	7.45	8.45	9.45

TABLE 2.3-5
MAXIMUM RECORDED POINT RAINFALL FOR SELECTED STATIONS IN THE REGION

	Abilene (1906-1961)		Ft. Worth (1899-1961)		Waco (1941-1961)	
Duration	Amount (Inches)	Date	Amount (Inches)	Date	Amount (Inches)	Date
5 min.	0.68	5/18/42	0.71	5/20/28[a]	0.75	6/5/61[b]
10 min.	1.25	5/18/42	1.04	5/20/28[a]	1.15	6/5/61[b]
15 min.	1.76	5/18/42	1.40	8/22/16[a]	1.55	7/18/47[c]
30 min.	2.75	5/18/42	1.98	8/22/16[a]	2.35	7/18/47[c]
1 hr	3.47	7/31/11	3.35	9/05/32	3.15	6/19/47
2 hrs	4.42	5/22/08	5.59	9/05/32	4.20	6/19/47
3 hrs	4.53	5/11/28	5.99	9/04/32	4.20	6/19/47
6 hrs	6.26	5/11/28	6.93	9/04/32	4.44	5/12/53
12 hrs	6.56	5/11/28	9.04	9/04/32	4.64	5/12/53
24 hrs	6.78	5/22/08[d]	9.57	9/04/32	7.18	5/11/53[e]

a) Period of record is 1903 through 1961.

b) Period of record is 1953 through 1961.

c) Period of record is 1947 through 1961.

d) Period of record is 1886 through 1961.

e) Period of record is 1894 through 1961.

TABLE 2.3-6
EXTREME SNOWFALLS AT SELECTED STATIONS IN THE REGION

	Abilene (1886-1970)		Ft. Worth (1998-1970)		Waco (1899-1970)	
Period	Amount (Inches)	Date	Amount (Inches)	Date	Amount (Inches)	Date
24 hrs	8.0	1/15/19 1/16/19[a]	12.1	1/15/64 1/16/64	13.0	2/26/24
Calendar Month	9.5	2/90	12.1	1/64	13.0	12/29[a]
Season	18.4	1918-19	15.3	1963-64	15.0	1929-30

	(1940-1968)		(1954-1968)		(1943-1968)	
Month	Amount	Year	Amount	Year	Amount	Year
January	6.6	1966	12.1	1964	7.0	1949
February	8.4	1956	2.9	1966[a]	4.8	1966
March	5.7	1962	2.5	1962	1.0	1962[a]
April	T	1950[a]	0.0	--	0.0	--
October	T	1967	0.0	--	0.0	--
November	8.1	1968	T	1968[a]	T	1968[a]
December	4.3	1946	2.6	1963	2.0	1946

T, trace <01 in.

a) Amount also occurred on earlier date(s)

TABLE 2.3-7
THIS TABLE HAS BEEN DELETED

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 1 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
7/13/74	0	75.9	70.0	4.0	0
	3	73.9	69.1	0.9	0
	6	72.0	69.1	0.0	27
	9	78.1	71.1	2.5	3133
	12	87.1	68.0	4.7	6105
	15	91.9	64.0	5.6	5973
	18	93.9	61.0	4.7	2363
	21	87.1	63.0	4.0	0
7/14/74	0	82.0	63.0	4.0	0
	3	78.1	62.1	1.6	0
	6	75.0	63.0	1.6	16
	9	82.9	66.9	3.1	3053
	12	91.9	61.0	4.0	6307
	15	97.0	60.1	4.0	6174
	18	98.1	60.1	3.1	2389
	21	89.1	64.0	5.6	0
7/15/74	0	75.9	70.0	8.7	0
	3	73.0	70.0	4.0	0
	6	71.1	66.9	2.5	0
	9	80.1	66.9	4.0	3053
	12	88.0	64.9	4.7	6105
	15	93.9	63.0	6.3	5946
	18	93.0	63.0	7.8	2363
	21	87.1	64.9	4.0	0
7/16/74	0	80.1	66.9	4.0	0
	3	75.9	68.0	2.5	0
	6	73.0	68.0	0.9	0
	9	81.0	71.1	1.6	1487
	12	89.1	64.9	6.3	5707
	15	93.0	64.0	6.3	4995
	18	93.0	63.0	6.3	1911
	21	84.0	68.0	11.0	11

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 2 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
7/17/74	0	75.9	68.0	4.0	0
	3	73.0	68.0	1.6	0
	6	70.0	68.0	1.6	32
	9	78.1	70.0	5.6	2920
	12	87.1	64.9	5.6	5075
	15	91.0	62.1	8.7	5866
	18	88.0	64.9	11.9	2363
	21	82.0	66.9	5.6	0
7/18/74	0	75.9	61.0	2.5	0
	3	73.9	64.0	1.6	0
	6	70.0	64.9	1.6	32
	9	78.1	69.1	4.7	2989
	12	86.0	66.0	6.3	4969
	15	93.9	63.0	4.7	6017
	18	93.0	60.1	6.3	2654
	21	82.0	63.0	5.6	0
7/19/74	0	75.9	64.0	7.2	0
	3	73.9	64.0	2.5	0
	6	73.0	68.0	3.1	27
	9	80.1	66.9	6.3	2947
	12	90.0	64.0	5.6	6068
	15	95.0	57.0	6.3	6078
	18	97.0	55.9	5.6	2230
	21	89.1	61.0	6.3	16
7/20/74	0	81.0	60.1	9.4	0
	3	79.0	62.1	3.1	0
	6	78.1	63.0	2.5	16
	9	84.9	64.9	6.3	2947
	12	95.0	61.0	4.0	6366
	15	98.1	59.0	5.6	5707
	18	98.1	57.9	5.6	2256
	21	91.0	59.0	4.7	37

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 3 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
7/21/74	0	84.0	57.9	4.0	0
	3	82.0	61.0	4.7	0
	6	78.1	59.0	1.6	16
	9	88.0	60.1	4.7	2989
	12	98.1	55.9	4.7	6137
	15	100.9	53.1	6.3	5893
	18	100.9	52.0	6.3	2458
	21	93.0	55.9	4.0	0
7/22/74	0	84.9	61.0	4.0	53
	3	82.9	63.0	5.6	27
	6	79.0	62.1	2.5	27
	9	89.1	62.1	5.6	2845
	12	99.0	57.0	4.7	6158
	15	102.9	55.9	5.6	5866
	18	100.9	57.9	6.3	1539
	21	93.9	62.1	5.6	106
7/23/74	0	87.1	64.0	4.0	53
	3	82.9	64.0	4.7	53
	6	81.0	63.0	1.6	43
	9	90.0	64.9	5.6	2999
	12	99.0	61.0	4.7	6052
	15	102.9	57.0	7.2	5866
	18	102.0	57.9	7.8	2389
	21	93.9	61.0	4.0	133
7/24/74	0	88.0	64.9	4.7	95
	3	84.0	64.9	15.7	53
	6	81.0	64.9	8.7	43
	9	87.1	66.0	8.7	2580
	12	96.1	62.1	8.7	4979
	15	99.0	61.0	7.8	3849
	18	95.0	63.0	11.9	1794
	21	91.0	64.9	13.4	133

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 4 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
7/25/74	0	86.0	63.0	13.4	80
	3	84.0	64.9	7.8	53
	6	79.0	69.1	7.2	27
	9	78.1	73.0	6.3	1396
	12	93.0	68.0	12.5	6041
	15	100.0	62.1	9.4	6026
	18	102.0	57.9	7.2	1794
	21	93.9	62.1	13.4	133
7/26/74	0	88.0	62.1	11.0	122
	3	84.9	60.1	5.6	80
	6	82.0	62.1	9.4	106
	9	87.1	64.9	23.5	2856
	12	95.0	64.0	7.8	6371
	15	97.0	64.9	4.0	3053
	18	73.9	71.1	6.3	53
	21	77.0	71.1	4.7	0
7/27/74	0	75.9	72.0	0.0	0
	3	75.0	71.1	3.1	0
	6	73.0	71.1	0.9	0
	9	81.0	73.9	3.1	2999
	12	91.0	70.0	4.7	6052
	15	96.1	68.0	5.6	5378
	18	93.9	69.1	8.7	1991
	21	86.0	64.9	5.6	0
7/28/74	0	82.0	72.0	3.1	0
	3	79.0	72.0	3.1	0
	6	77.0	72.0	2.5	0
	9	87.1	75.9	1.6	2920
	12	95.0	75.0	2.5	6185
	15	98.1	75.0	4.7	5707
	18	98.1	73.9	5.6	1858
	21	90.0	71.1	6.3	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 5 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
7/29/74	0	84.0	71.1	4.7	0
	3	80.1	71.1	4.0	0
	6	75.9	70.0	3.1	0
	9	89.1	73.9	1.6	3053
	12	95.0	60.1	4.7	5458
	15	100.0	87.1	5.6	5909
	18	99.0	87.1	5.6	733
	21	77.0	66.9	11.0	0
7/30/74	0	75.0	68.0	3.1	0
	3	70.0	68.0	6.3	0
	6	70.0	70.0	2.5	0
	9	72.0	69.1	4.0	1062
	12	75.0	69.1	3.1	2527
	15	82.9	66.9	4.7	2325
	18	82.9	66.9	4.7	637
	21	80.1	68.0	4.0	0
7/31/74	0	77.0	68.0	0.0	0
	3	75.0	70.0	1.6	0
	6	72.0	69.1	1.6	0
	9	77.0	69.1	3.1	1327
	12	86.0	64.9	3.1	5707
	15	90.0	64.0	5.6	5707
	18	90.0	64.0	7.2	929
	21	82.0	69.1	7.2	0
8/1/74	0	75.0	70.0	7.2	0
	3	72.0	70.0	3.1	0
	6	70.0	70.0	3.1	0
	9	78.1	70.0	6.3	2150
	12	89.1	66.9	7.2	5197
	15	81.0	69.1	4.0	1630
	18	72.0	69.1	7.2	101
	21	72.0	70.0	7.2	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 6 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/2/74	0	70.0	70.0	4.7	0
	3	71.1	70.0	3.1	0
	6	71.1	70.0	2.5	0
	9	77.0	69.1	5.6	1858
	12	89.1	73.9	2.5	4247
	15	86.0	70.0	10.3	6238
	18	86.0	64.9	7.2	1168
	21	80.1	64.9	4.7	0
8/3/74	0	75.0	66.9	2.5	0
	3	73.9	70.0	4.7	0
	6	72.0	69.1	1.6	0
	9	75.9	71.1	4.0	1529
	12	80.1	60.1	7.8	1746
	15	84.9	55.0	8.7	2756
	18	84.0	52.0	11.0	1593
	21	77.0	50.0	7.2	0
8/4/74	0	73.0	46.9	0.0	0
	3	70.0	48.9	2.5	0
	6	68.0	51.1	0.0	0
	9	75.0	51.1	4.7	1991
	12	81.0	54.0	5.6	5176
	15	86.0	57.9	5.6	5644
	18	86.0	57.0	4.0	865
	21	82.0	61.0	4.0	0
8/5/74	0	77.0	64.9	4.0	0
	3	73.9	66.9	1.6	0
	6	72.0	68.0	2.5	0
	9	73.9	69.1	4.7	1301
	12	79.0	64.0	7.8	1248
	15	78.1	64.0	6.3	743
	18	69.1	66.9	7.8	207
	21	68.0	66.0	2.5	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 7 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/6/74	0	68.0	66.9	4.7	0
	3	66.9	66.9	3.1	0
	6	64.0	64.0	6.3	0
	9	66.9	64.9	2.5	345
	12	72.0	63.0	4.7	1555
	15	75.0	63.0	4.0	1497
	18	75.9	64.9	7.2	797
	21	72.0	69.1	4.0	0
8/7/74	0	69.1	69.1	4.7	0
	3	66.9	66.9	1.6	0
	6	66.9	66.9	0.9	0
	9	71.1	69.1	3.1	1762
	12	73.9	71.1	4.0	3483
	15	82.0	68.0	6.3	3971
	18	82.9	69.1	9.4	1741
	21	77.0	70.0	4.7	0
8/8/74	0	72.0	71.1	6.3	0
	3	71.1	71.1	5.6	0
	6	73.0	70.0	4.7	0
	9	71.1	71.1	7.2	547
	12	71.1	71.1	11.0	945
	15	79.0	72.0	13.4	1858
	18	82.9	72.0	14.1	1603
	21	79.0	75.0	11.9	0
8/9/74	0	75.0	75.0	7.8	0
	3	73.9	75.0	7.8	0
	6	75.0	73.9	8.7	0
	9	78.1	73.9	12.5	1757
	12	86.0	71.1	15.0	4587
	15	91.0	70.0	13.4	3823
	18	91.0	70.0	14.1	1858
	21	84.9	73.9	11.9	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 8 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/10/74	0	80.1	72.0	12.5	0
	3	77.0	73.9	7.8	0
	6	77.0	73.9	5.6	0
	9	78.1	75.0	9.4	1375
	12	82.9	73.9	11.0	2123
	15	70.0	69.1	7.8	239
	18	71.1	69.1	3.1	865
	21	68.0	68.0	6.3	0
8/11/74	0	68.0	68.0	6.3	0
	3	66.0	64.9	7.2	0
	6	66.0	64.0	2.5	0
	9	72.0	71.1	1.6	1401
	12	78.1	70.0	5.6	2458
	15	82.9	73.0	7.2	4066
	18	86.0	72.0	3.1	1491
	21	81.0	73.0	3.1	0
8/12/74	0	77.0	72.0	3.1	0
	3	75.0	75.0	3.1	0
	6	77.0	73.9	6.3	0
	9	75.9	77.0	4.7	765
	12	84.0	73.9	6.3	3982
	15	89.1	71.1	6.3	4895
	18	90.0	66.0	4.7	1885
	21	84.0	68.0	1.6	0
8/13/74	0	73.0	73.0	4.7	0
	3	71.1	72.0	2.5	0
	6	72.0	68.0	2.5	0
	9	79.0	78.1	5.6	2750
	12	87.1	75.0	6.3	5314
	15	91.0	70.0	4.7	4640
	18	87.1	73.9	4.7	823
	21	82.9	73.9	4.0	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 9 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/14/74	0	78.1	73.0	9.4	0
	3	73.9	73.9	3.1	0
	6	71.1	68.0	2.5	0
	9	80.1	77.0	5.6	2724
	12	89.1	73.0	7.8	6105
	15	91.9	70.0	4.7	4577
	18	93.0	70.0	7.2	1707
	21	86.0	71.1	6.3	0
8/15/74	0	79.0	73.0	4.0	0
	3	73.9	73.9	3.1	0
	6	71.1	68.0	4.0	0
	9	80.1	75.9	5.6	2788
	12	89.1	73.9	7.8	5973
	15	93.0	70.0	7.8	5776
	18	93.0	69.1	8.7	1848
	21	86.0	68.0	6.3	0
8/16/74	0	79.0	70.0	5.6	0
	3	77.0	75.0	6.3	0
	6	78.1	73.0	4.0	0
	9	80.1	75.9	7.2	2893
	12	88.0	73.0	7.2	6212
	15	93.0	69.1	6.3	5548
	18	95.0	66.0	15.7	1869
	21	86.0	68.0	11.0	0
8/17/74	0	81.0	70.0	7.2	0
	3	77.0	75.0	6.3	149
	6	73.9	75.0	2.5	133
	9	81.0	75.9	5.6	3079
	12	90.0	72.0	11.0	5999
	15	93.9	70.0	11.0	5309
	18	95.0	66.9	7.8	2060
	21	86.0	70.0	4.7	80

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 10 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/18/74	0	82.0	72.0	5.6	53
	3	79.0	73.0	6.3	186
	6	75.0	75.0	6.3	133
	9	82.0	77.0	11.9	3122
	12	91.0	72.0	10.3	6238
	15	95.0	69.1	9.4	5840
	18	96.1	66.9	5.6	2097
	21	87.1	69.1	3.1	175
8/19/74	0	82.0	69.1	4.7	133
	3	79.0	70.0	5.6	80
	6	75.0	73.9	9.4	80
	9	82.9	75.0	7.2	3026
	12	91.0	70.0	9.4	6105
	15	95.0	68.0	9.4	5574
	18	96.1	66.9	7.8	1746
	21	87.1	66.0	6.3	16
8/20/74	0	82.9	72.0	7.2	0
	3	79.0	77.0	7.2	16
	6	75.0	75.9	7.2	16
	9	82.0	73.9	8.7	3053
	12	91.0	70.0	11.0	6105
	15	95.0	68.0	10.3	5553
	18	93.9	68.0	9.4	1699
	21	87.1	71.1	7.2	0
8/21/74	0	80.1	71.1	4.7	0
	3	75.0	73.9	3.1	0
	6	73.0	66.9	2.5	0
	9	79.0	73.0	5.6	3026
	12	89.1	66.0	9.4	5978
	15	93.9	63.0	7.2	5335
	18	95.0	63.0	7.2	1673
	21	87.1	66.0	5.6	0

TABLE 2.3-7A
ONSITE METEOROLOGICAL DATA

(Sheet 11 of 11)

DATE	HOUR	AIR TEMP, F	DEWPOINT, TEMP, F	WINDSPEED, MPH	SOLAR RADIATION, BTU FT^2DAY^{-1}
8/22/74	0	80.1	66.9	3.1	0
	3	75.9	70.0	2.5	0
	6	73.0	72.0	0.0	0
	9	82.0	78.1	0.0	2813
	12	90.0	73.0	0.0	5707
	15	82.0	72.0	2.5	797
	18	82.0	73.9	5.6	133
	21	77.0	70.0	0.0	0
8/23/74	0	75.0	69.1	2.5	0
	3	72.0	66.9	2.5	0
	6	73.0	69.1	4.7	0
	9	81.0	73.0	0.0	2389
	12	88.0	73.9	3.1	5113
	15	95.0	75.0	3.1	4226
	18	93.0	73.0	7.2	1062
	21	81.0	66.9	7.8	0

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 1 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
8/26/90	0	82.9	66.9	0.0	0	0	29.38
	1	81.0	66.9	0.0	0	0	29.37
	2	81.0	66.9	0.0	0	0	29.37
	3	78.1	68.0	0.0	0	0	29.38
	4	78.1	68.0	0.0	0	0	29.38
	5	75.9	68.0	0.0	0	0	29.39
	6	75.0	66.9	0.0	0	0	29.41
	7	78.1	66.9	0.0	701	0	29.42
	8	84.9	68.0	6.7	2446	0	29.43
	9	88.0	69.1	6.7	4224	0	29.44
	10	91.9	70.0	0.0	5736	0	29.45
	11	95.0	69.1	0.0	6831	0	29.44
	12	98.1	68.0	4.9	6644	4	29.43
	13	99.0	68.0	4.3	6232	5	29.41
	14	99.0	66.0	0.0	5785	5	29.39
	15	100.0	66.0	4.9	5248	4	29.38
	16	100.0	63.0	0.0	4122	3	29.37
	17	100.0	64.0	0.0	2462	3	29.36
	18	98.1	66.0	3.4	792	3	29.36
	19	93.0	66.0	0.0	0	3	29.35
	20	90.0	64.9	0.0	0	0	29.36
	21	87.1	64.9	0.0	0	0	29.38
	22	86.0	64.9	0.0	0	0	29.38
	23	84.9	64.9	0.0	0	0	29.38
8/27/9	00	82.0	66.0	0.0	0	2	29.38
	1	82.0	66.0	0.0	0	2	29.38
	2	81.0	66.0	0.0	0	3	29.37
	3	81.0	64.9	0.0	0	7	29.37
	4	79.0	66.0	0.0	0	6	29.37
	5	81.0	66.0	0.0	0	6	29.38
	6	75.0	66.9	0.0	0	0	29.40

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 2 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	7	79.0	69.1	0.0	686	0	29.42
	8	87.1	69.1	0.0	2366	2	29.42
	9	91.9	64.0	0.0	4097	2	29.42
	10	95.0	63.0	0.0	5719	0	29.43
	11	99.0	62.1	0.0	6813	0	29.42
	12	100.0	62.1	0.0	7347	1	29.39
	13	100.0	64.0	0.0	7370	1	29.37
	14	102.0	63.0	0.0	6703	2	29.36
	15	102.0	64.0	0.0	5676	2	29.33
	16	102.0	64.0	0.0	4232	2	29.31
	17	99.0	64.9	0.0	2561	1	29.30
	18	99.0	64.0	3.4	810	0	29.31
	19	96.1	66.0	0.0	0	0	29.30
	20	93.0	66.9	0.0	0	0	29.31
	21	91.0	68.0	0.0	0	0	29.32
	22	87.1	69.1	0.0	0	0	29.33
	23	87.1	68.0	0.0	0	0	29.33
8/28/90	0	84.9	68.0	0.0	0	0	29.33
	1	84.9	69.1	0.0	0	0	29.33
	2	82.9	69.1	0.0	0	0	29.32
	3	82.0	68.0	0.0	0	0	29.32
	4	82.0	68.0	0.0	0	0	29.31
	5	78.1	69.1	0.0	0	0	29.32
	6	75.9	68.0	0.0	0	0	29.33
	7	78.1	69.1	0.0	671	0	29.34
	8	84.9	71.1	0.0	2411	0	29.35
	9	89.1	70.0	0.0	4190	0	29.35
	10	93.9	70.0	0.0	5701	0	29.36
	11	97.0	66.0	2.5	6794	0	29.35
	12	98.1	64.9	3.4	7374	0	29.34

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 3 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	13	98.1	63.0	0.0	7395	0	29.32
	14	99.0	64.9	4.9	6855	0	29.30
	15	100.0	66.0	3.4	5797	0	29.28
	16	100.0	66.0	6.7	4282	1	29.26
	17	99.0	62.1	4.3	2542	0	29.26
	18	97.0	64.0	3.4	778	0	29.25
	19	93.9	64.0	0.0	0	0	29.25
	20	89.1	66.0	0.0	0	0	29.27
	21	88.0	66.0	0.0	0	0	29.29
	22	86.0	66.0	0.0	0	0	29.30
	23	84.9	66.9	0.0	0	0	29.30
8/29/90	0	82.9	68.0	0.0	0	0	29.29
	1	82.9	69.1	2.5	0	0	29.29
	2	82.0	61.0	0.0	0	0	29.29
	3	81.0	69.1	0.0	0	0	29.29
	4	79.0	69.1	0.0	0	0	29.30
	5	77.0	70.0	0.0	0	0	29.30
	6	77.0	69.1	0.0	0	0	29.31
	7	78.1	70.0	0.0	657	0	29.32
	8	84.0	71.1	0.0	2394	0	29.33
	9	88.0	71.1	0.0	4172	0	29.34
	10	93.9	70.0	0.0	5684	0	29.35
	11	96.1	68.0	0.0	6775	0	29.35
	12	98.1	68.0	6.7	7305	1	29.33
	13	99.0	66.9	0.0	7371	0	29.30
	14	99.0	66.9	2.5	6827	0	29.28
	15	100.0	66.9	0.0	5728	1	29.26
	16	100.0	66.0	4.3	4164	2	29.24
	17	99.0	64.9	3.4	2489	1	29.24
	18	98.1	64.9	3.4	746	0	29.23

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 4 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	19	93.0	66.9	0.0	0	0	29.23
	20	89.1	68.0	0.0	0	0	29.25
	21	88.0	68.0	0.0	0	0	29.27
	22	87.1	68.0	3.4	0	0	29.28
	23	84.9	69.1	0.0	0	0	29.30
8/30/90	0	82.0	68.0	3.4	0	0	29.29
	1	82.0	69.1	3.4	0	0	29.29
	2	81.0	68.0	0.0	0	0	29.29
	3	82.0	68.0	0.0	0	0	29.29
	4	79.0	68.0	0.0	0	0	29.29
	5	77.0	68.0	0.0	0	0	29.30
	6	78.1	68.0	0.0	0	0	29.30
	7	79.0	69.1	0.0	641	0	29.31
	8	84.9	69.1	0.0	2376	0	29.32
	9	91.9	69.1	0.0	4155	0	29.32
	10	93.9	69.1	2.5	5666	0	29.33
	11	98.1	66.9	3.4	6756	0	29.33
	12	98.1	68.0	4.3	7284	1	29.31
	13	100.9	64.9	0.0	7155	2	29.29
	14	100.9	63.0	3.4	6755	1	29.27
	15	102.0	61.0	4.9	5733	0	29.25
	16	102.0	60.1	0.0	4239	0	29.23
	17	100.9	61.0	4.3	2468	0	29.22
	18	99.0	62.1	0.0	641	4	29.22
	19	93.0	62.1	0.0	0	4	29.22
	20	90.0	62.1	3.4	0	5	29.23
	21	89.1	64.0	4.3	0	4	29.24
	22	87.1	64.9	0.0	0	3	29.26
	23	84.9	66.9	4.9	0	2	29.27

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 5 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
8/31/90	0	82.9	66.9	4.9	0	3	29.27
	1	84.0	66.0	4.3	0	3	29.27
	2	82.0	66.0	0.0	0	2	29.26
	3	79.0	64.9	0.0	0	2	29.27
	4	80.1	66.9	0.0	0	2	29.27
	5	79.0	68.0	0.0	0	2	29.28
	6	78.1	66.0	0.0	0	1	29.29
	7	78.1	68.0	0.0	627	0	29.31
	8	87.1	69.1	0.0	2358	0	29.32
	9	91.9	66.0	4.3	4136	0	29.33
	10	95.0	68.0	3.4	5647	0	29.34
	11	99.0	66.9	7.6	6736	0	29.33
	12	100.9	64.9	4.3	7262	1	29.31
	13	102.9	64.9	6.7	7131	2	29.28
	14	106.0	61.0	4.3	6594	2	29.26
	15	105.1	63.0	3.4	5552	2	29.24
	16	102.9	61.0	4.3	3957	3	29.23
	17	104.0	59.0	4.9	2287	3	29.22
	18	100.9	60.1	3.4	523	6	29.22
	19	95.0	61.0	3.4	0	3	29.24
	20	89.1	64.0	3.4	0	0	29.25
	21	88.0	64.0	0.0	0	0	29.28
	22	87.1	62.1	0.0	0	2	29.30
	23	87.1	66.0	0.0	0	2	29.30
9/1/90	0	87.1	66.0	0.0	0	3	29.31
	1	86.0	66.9	4.3	0	2	29.30
	2	82.9	66.9	0.0	0	2	29.30
	3	80.1	66.0	6.7	0	2	29.30
	4	79.0	66.0	4.3	0	1	29.31
	5	77.0	64.9	4.3	0	1	29.32

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 6 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	6	77.0	66.0	0.0	0	2	29.33
	7	79.0	66.9	0.0	576	3	29.34
	8	84.0	69.1	4.3	1960	5	29.34
	9	86.0	70.0	0.0	3877	3	29.35
	10	90.0	69.1	6.7	4714	5	29.35
	11	91.0	68.0	6.7	6323	3	29.36
	12	91.9	68.0	0.0	4966	7	29.35
	13	93.0	66.0	4.3	5589	6	29.33
	14	99.0	64.9	4.3	4594	7	29.31
	15	98.1	60.1	10.7	5336	3	29.29
	16	97.0	57.0	13.4	4166	0	29.27
	17	97.0	55.0	14.1	2391	0	29.28
	18	93.9	51.1	11.6	583	4	29.30
	19	90.0	54.0	4.9	0	5	29.32
	20	84.9	55.9	0.0	0	4	29.34
	21	86.0	59.0	9.2	0	4	29.36
	22	82.0	66.0	10.1	0	4	29.38
	23	78.1	66.0	0.0	0	2	29.39
9/2/90	0	75.9	66.9	0.0	0	3	29.39
	1	73.9	66.9	0.0	0	5	29.39
	2	73.0	66.0	0.0	0	3	29.39
	3	72.0	66.0	0.0	0	3	29.40
	4	70.0	66.0	0.0	0	3	29.41
	5	70.0	66.0	0.0	0	3	29.43
	6	70.0	66.0	0.0	0	0	29.43
	7	72.0	66.9	0.0	597	0	29.44
	8	77.0	68.0	0.0	2322	0	29.45
	9	82.0	66.0	0.0	4100	0	29.47
	10	84.9	64.9	4.3	5609	0	29.48
	11	87.1	64.0	0.0	6695	0	29.48

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 7 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	12	91.0	63.0	6.7	7217	1	29.46
	13	91.0	62.1	3.4	7222	1	29.44
	14	93.0	63.0	8.3	6536	2	29.42
	15	93.0	61.0	7.6	5487	2	29.40
	16	93.9	62.1	8.3	4021	2	29.39
	17	93.9	62.1	9.2	2291	2	29.39
	18	91.0	62.1	4.9	603	2	29.39
	19	89.1	63.0	3.4	0	3	29.41
	20	86.0	64.0	0.0	0	4	29.43
	21	86.0	64.9	9.2	0	5	29.45
	22	82.9	66.0	4.3	0	2	29.46
	23	80.1	66.9	4.3	0	0	29.45
9/3/90	0	77.0	66.9	4.3	0	2	29.45
	1	75.9	66.9	0.0	0	3	29.44
	2	75.9	68.0	0.0	0	2	29.45
	3	75.9	66.9	0.0	0	2	29.46
	4	75.0	68.0	0.0	0	5	29.47
	5	75.9	68.0	0.0	0	6	29.48
	6	73.9	68.0	0.0	0	6	29.49
	7	75.9	70.0	0.0	548	3	29.51
	8	82.0	70.0	0.0	2289	1	29.52
	9	84.9	69.1	0.0	4054	1	29.53
	10	88.0	68.0	4.9	5554	1	29.54
	11	88.0	68.0	4.9	5980	4	29.53
	12	90.0	66.9	4.3	5546	6	29.52
	13	93.0	66.9	6.7	5548	6	29.50
	14	93.0	64.9	10.1	4553	7	29.48
	15	95.0	64.9	0.0	3816	7	29.46
	16	93.9	66.0	6.7	2788	7	29.45
	17	93.0	66.9	4.3	1937	5	29.44

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 8 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	18	93.0	64.9	4.9	527	4	29.45
	19	90.0	64.0	4.3	0	3	29.45
	20	87.1	66.9	3.4	0	3	29.48
	21	84.9	66.0	0.0	0	2	29.49
	22	82.9	66.9	3.4	0	0	29.49
	23	82.0	66.9	0.0	0	0	29.50
9/4/90	0	82.0	66.9	6.7	0	0	29.50
	1	80.1	66.9	6.7	0	0	29.51
	2	77.0	68.0	4.9	0	0	29.51
	3	75.9	66.9	0.0	0	0	29.53
	4	75.9	68.0	0.0	0	0	29.52
	5	75.0	68.0	0.0	0	0	29.54
	6	73.0	68.0	0.0	0	1	29.56
	7	77.0	68.0	0.0	568	0	29.58
	8	82.9	70.0	0.0	2286	0	29.60
	9	88.0	69.1	0.0	4062	0	29.60
	10	91.0	68.0	0.0	5570	0	29.60
	11	91.9	66.0	4.9	6609	1	29.60
	12	93.0	66.9	8.3	6794	3	29.58
	13	93.9	66.0	3.4	6465	4	29.56
	14	96.1	66.0	7.6	5957	4	29.54
	15	96.1	64.9	6.7	5238	3	29.52
	16	97.0	64.9	4.9	3816	3	29.50
	17	96.1	64.9	7.6	2258	1	29.48
	18	93.9	64.9	0.0	557	0	29.48
	19	91.0	64.0	0.0	0	1	29.48
	20	87.1	64.0	0.0	0	1	29.49
	21	86.0	64.9	0.0	0	2	29.50
	22	82.0	64.9	0.0	0	0	29.52
	23	81.0	66.0	0.0	0	2	29.53

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 9 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
9/5/90	0	81.0	66.0	0.0	0	2	29.52
	1	80.1	66.0	0.0	0	5	29.52
	2	80.1	66.0	0.0	0	5	29.53
	3	82.0	64.9	0.0	0	4	29.52
	4	75.9	66.0	0.0	0	3	29.52
	5	75.0	66.0	0.0	0	2	29.53
	6	75.0	64.9	0.0	0	1	29.53
	7	77.0	66.0	0.0	550	1	29.54
	8	84.0	66.9	0.0	2252	1	29.56
	9	88.0	68.0	4.3	4017	1	29.55
	10	91.9	68.0	2.5	5514	1	29.55
	11	97.0	64.9	2.5	6631	0	29.54
	12	98.1	64.9	0.0	7145	1	29.52
	13	99.0	66.0	4.9	7000	2	29.50
	14	99.0	64.0	4.3	6445	2	29.47
	15	100.0	64.0	0.0	5384	2	29.44
	16	98.1	64.0	0.0	3596	4	29.42
	17	99.0	62.1	0.0	2219	1	29.40
	18	97.0	61.0	0.0	494	3	29.40
	19	93.0	66.0	0.0	0	7	29.39
	20	88.0	64.9	0.0	0	3	29.40
	21	88.0	66.0	4.3	0	8	29.42
	22	84.9	66.9	0.0	0	10	29.42
	23	84.0	68.0	0.0	0	10	29.41
9/6/90	0	82.0	68.0	0.0	0	10	29.41
	1	81.0	68.0	0.0	0	10	29.40
	2	81.0	66.9	0.0	0	10	29.40
	3	79.0	68.0	0.0	0	8	29.39
	4	78.1	66.0	0.0	0	8	29.40
	5	75.0	66.9	0.0	0	8	29.40

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 10 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	6	75.0	66.9	0.0	0	7	29.40
	7	77.0	69.1	0.0	413	6	29.40
	8	82.0	68.0	0.0	2117	3	29.41
	9	88.0	68.0	0.0	3788	3	29.40
	10	93.0	68.0	0.0	5386	2	29.40
	11	96.1	66.9	0.0	6436	2	29.39
	12	98.1	64.9	4.3	6980	2	29.37
	13	99.0	64.9	4.3	6972	2	29.35
	14	100.0	64.9	6.7	6414	2	29.32
	15	100.0	64.0	7.6	5349	2	29.30
	16	100.0	64.0	3.4	3741	3	29.28
	17	99.0	64.0	6.7	2064	3	29.27
	18	97.0	64.0	8.3	414	5	29.26
	19	93.0	64.0	4.3	0	3	29.26
	20	89.1	64.9	2.5	0	2	29.27
	21	89.1	64.9	4.9	0	2	29.28
	22	87.1	64.9	3.4	0	5	29.29
	23	87.1	63.0	7.6	0	7	29.30
9/7/90	0	86.0	63.0	7.6	0	6	29.31
	1	84.0	63.0	7.6	0	4	29.31
	2	82.9	63.0	9.2	0	6	29.31
	3	80.1	64.0	4.9	0	6	29.32
	4	78.1	64.0	4.3	0	8	29.32
	5	78.1	64.9	4.3	0	8	29.33
	6	75.9	66.0	2.5	0	8	29.33
	7	77.0	68.0	0.0	306	8	29.33
	8	81.0	70.0	6.7	1302	8	29.33
	9	84.9	70.0	6.7	3067	6	29.34
	10	91.0	70.0	4.9	4936	4	29.34
	11	93.0	68.0	4.3	3846	8	29.35

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 11 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	12	96.1	66.9	3.4	2499	10	29.33
	13	98.1	64.0	4.9	2495	10	29.31
	14	99.0	63.0	7.6	2293	10	29.29
	15	98.1	61.0	4.9	3186	8	29.26
	16	98.1	62.1	7.6	2681	7	29.23
	17	97.0	61.0	10.7	1802	5	29.22
	18	95.0	62.1	8.3	437	3	29.23
	19	90.0	62.1	0.0	0	2	29.23
	20	88.0	63.0	0.0	0	10	29.26
	21	86.0	63.0	0.0	0	10	29.27
	22	86.0	64.0	0.0	0	10	29.28
	23	84.9	63.0	0.0	0	10	29.28
9/8/90	0	81.0	64.0	0.0	0	9	29.28
	1	82.9	64.0	0.0	0	9	29.28
	2	82.9	64.9	8.3	0	9	29.28
	3	84.0	66.9	4.3	0	9	29.29
	4	82.9	68.0	4.9	0	8	29.30
	5	82.0	69.1	2.5	0	9	29.30
	6	79.0	69.1	0.0	0	9	29.31
	7	79.0	69.1	0.0	178	10	29.32
	8	81.0	70.0	2.5	774	10	29.32
	9	82.9	70.0	6.7	1886	9	29.32
	10	89.1	71.1	6.7	2598	9	29.33
	11	91.0	70.0	6.7	2297	10	29.32
	12	93.0	68.0	6.7	2490	10	29.31
	13	88.0	64.9	4.9	2484	10	29.31
	14	82.0	70.0	0.0	2281	10	29.31
	15	79.0	71.1	7.6	1896	10	29.30
	16	75.9	73.0	7.6	1363	10	29.28
	17	75.9	73.9	0.0	739	10	29.28

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 12 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	18	77.0	73.9	4.3	152	10	29.29
	19	75.9	75.0	6.7	0	10	29.29
	20	77.0	73.9	0.0	0	10	29.30
	21	75.9	73.9	6.7	0	10	29.33
	22	73.9	72.0	4.9	0	10	29.33
	23	75.0	72.0	0.0	0	10	29.33
9/9/90	0	73.9	71.1	0.0	0	10	29.33
	1	73.9	71.1	0.0	0	9	29.33
	2	72.0	71.1	0.0	0	10	29.34
	3	73.0	71.1	0.0	0	10	29.34
	4	73.0	71.1	0.0	0	10	29.33
	5	73.0	72.0	0.0	0	10	29.33
	6	73.0	72.0	0.0	0	9	29.33
	7	73.9	73.0	0.0	235	9	29.34
	8	75.0	73.0	0.0	1280	8	29.35
	9	77.0	73.0	0.0	1877	9	29.37
	10	78.1	72.0	2.5	1913	10	29.38
	11	80.1	71.1	4.9	2289	10	29.40
	12	79.0	71.1	10.7	2481	10	29.38
	13	75.9	70.0	8.3	2474	10	29.37
	14	79.0	69.1	8.3	2270	10	29.36
	15	79.0	69.1	4.3	2547	9	29.34
	16	79.0	69.1	0.0	1348	10	29.33
	17	78.1	68.0	4.9	724	10	29.32
	18	77.0	69.1	0.0	142	10	29.33
	19	75.9	69.1	6.7	0	9	29.33
	20	73.9	70.0	4.3	0	9	29.35
	21	73.0	70.0	0.0	0	8	29.36
	22	72.0	70.0	0.0	0	7	29.36
	23	72.0	69.1	0.0	0	8	29.37

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 13 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
9/10/90	0	73.9	71.1	3.4	0	10	29.37
	1	73.9	71.1	0.0	0	10	29.37
	2	73.0	71.1	0.0	0	9	29.37
	3	72.0	70.0	0.0	0	9	29.36
	4	71.1	69.1	0.0	0	6	29.34
	5	72.0	70.0	0.0	0	10	29.36
	6	70.0	69.1	0.0	0	6	29.37
	7	71.1	70.0	0.0	328	7	29.38
	8	73.0	70.0	0.0	760	10	29.39
	9	73.9	69.1	4.3	1380	10	29.39
	10	72.0	70.0	4.9	1906	10	29.41
	11	72.0	71.1	8.3	2280	10	29.42
	12	73.9	71.1	8.3	2471	10	29.40
	13	79.0	70.0	6.7	2463	10	29.38
	14	81.0	69.1	4.3	3766	8	29.35
	15	77.0	71.1	0.0	2529	9	29.33
	16	80.1	66.9	6.7	2225	8	29.31
	17	81.0	66.9	13.4	1382	7	29.30
	18	78.1	66.9	7.6	288	6	29.31
	19	73.9	66.9	4.9	0	5	29.33
	20	72.0	66.9	0.0	0	3	29.34
	21	72.0	66.9	4.3	0	4	29.36
	22	71.1	68.0	0.0	0	5	29.36
	23	72.0	68.0	3.4	0	10	29.36
9/11/90	0	71.1	68.0	6.7	0	10	29.35
	1	71.1	68.0	4.9	0	10	29.34
	2	70.0	68.0	8.3	0	10	29.33
	3	70.0	68.0	8.3	0	10	29.31
	4	70.0	68.0	4.9	0	10	29.31
	5	70.0	68.0	8.3	0	10	29.32

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 14 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	6	69.1	69.1	7.6	0	10	29.33
	7	69.1	69.1	6.7	164	10	29.34
	8	69.1	69.1	9.2	753	10	29.34
	9	70.0	69.1	8.3	1372	10	29.35
	10	71.1	70.0	10.7	1897	10	29.36
	11	72.0	69.1	10.1	2271	10	29.36
	12	77.0	70.0	10.1	2462	10	29.35
	13	75.0	69.1	8.3	2452	10	29.34
	14	73.9	71.1	2.5	2244	10	29.32
	15	73.9	70.0	4.3	1855	10	29.31
	16	78.1	71.1	4.9	1784	9	29.28
	17	79.0	70.0	4.9	941	9	29.29
	18	78.1	70.0	0.0	165	9	29.29
	19	75.9	69.1	0.0	0	10	29.30
	20	75.0	71.1	0.0	0	10	29.31
	21	72.0	70.0	0.0	0	10	29.32
	22	72.0	70.0	0.0	0	10	29.33
	23	73.0	71.1	0.0	0	10	29.33
9/12/90	0	71.1	70.0	0.0	0	5	29.33
	1	71.1	70.0	0.0	0	8	29.32
	2	71.1	71.1	0.0	0	10	29.32
	3	71.1	71.1	0.0	0	8	29.31
	4	72.0	72.0	0.0	0	10	29.32
	5	72.0	71.1	0.0	0	10	29.33
	6	72.0	71.1	0.0	0	10	29.34
	7	73.0	72.0	0.0	158	10	29.36
	8	73.0	71.1	0.0	746	10	29.37
	9	73.9	71.1	0.0	1365	10	29.38
	10	75.9	71.1	0.0	2556	9	29.39
	11	81.0	71.1	2.5	3775	8	29.38

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 15 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	12	82.9	71.1	4.3	4091	8	29.37
	13	84.9	69.1	2.5	4073	8	29.35
	14	86.0	66.9	2.5	3725	8	29.33
	15	82.9	70.0	9.2	1842	10	29.31
	16	82.0	70.0	6.7	2177	8	29.31
	17	84.0	69.1	3.4	1324	7	29.30
	18	82.0	69.1	0.0	187	8	29.31
	19	80.1	71.1	0.0	0	8	29.31
	20	79.0	71.1	2.5	0	10	29.32
	21	77.0	72.0	0.0	0	8	29.33
	22	75.9	71.1	0.0	0	3	29.35
	23	73.9	70.0	0.0	0	3	29.35
9/13/90	0	73.9	70.0	0.0	0	3	29.35
	1	73.0	70.0	0.0	0	3	29.34
	2	71.1	69.1	0.0	0	3	29.34
	3	71.1	70.0	0.0	0	2	29.33
	4	71.1	70.0	0.0	0	6	29.33
	5	71.1	70.0	0.0	0	3	29.34
	6	71.1	70.0	0.0	0	4	29.35
	7	71.1	71.1	0.0	154	10	29.36
	8	73.9	72.0	0.0	740	10	29.37
	9	75.0	72.0	0.0	1357	10	29.38
	10	75.9	73.9	0.0	1881	10	29.38
	11	80.1	72.0	0.0	3049	9	29.38
	12	79.0	71.1	3.4	3303	9	29.36
	13	80.1	73.0	7.6	3287	9	29.35
	14	80.1	73.0	4.3	2219	10	29.33
	15	84.0	72.0	4.3	3050	8	29.30
	16	84.9	72.0	2.5	2152	8	29.28
	17	87.1	71.1	2.5	1296	7	29.28

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 16 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	18	84.0	70.0	2.5	263	4	29.28
	19	81.0	70.0	0.0	0	2	29.28
	20	78.1	71.1	4.3	0	5	29.30
	21	75.9	71.1	0.0	0	2	29.32
	22	75.9	71.1	4.3	0	2	29.33
	23	75.9	71.1	0.0	0	1	29.33
9/14/90	0	73.9	71.1	3.4	0	1	29.33
	1	73.0	71.1	0.0	0	1	29.34
	2	73.0	71.1	0.0	0	4	29.34
	3	73.9	73.0	2.5	0	10	29.34
	4	73.9	73.0	0.0	0	10	29.34
	5	75.0	73.0	0.0	0	10	29.35
	6	73.9	73.0	0.0	0	10	29.36
	7	75.0	72.0	0.0	202	9	29.37
	8	80.1	72.0	0.0	1427	7	29.38
	9	81.0	72.0	2.5	2252	8	29.40
	10	82.9	72.0	0.0	3647	7	29.40
	11	87.1	72.0	4.3	3036	9	29.40
	12	88.0	71.1	3.4	3289	9	29.38
	13	91.0	68.0	2.5	4035	8	29.36
	14	93.0	68.0	2.5	3682	8	29.34
	15	95.0	68.0	4.3	3026	8	29.32
	16	93.9	66.9	0.0	2126	8	29.31
	17	90.0	70.0	3.4	880	9	29.30
	18	89.1	70.0	4.3	126	9	29.30
	19	84.9	71.1	0.0	0	8	29.31
	20	82.0	72.0	0.0	0	3	29.32
	21	80.1	72.0	0.0	0	3	29.33
	22	78.1	71.1	6.7	0	3	29.34
	23	78.1	72.0	0.0	0	3	29.32

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 17 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
9/15/90	0	75.9	71.1	0.0	0	2	29.32
	1	75.0	71.1	0.0	0	2	29.32
	2	75.0	71.1	0.0	0	2	29.32
	3	73.9	71.1	0.0	0	2	29.31
	4	72.0	70.0	0.0	0	2	29.31
	5	71.1	70.0	0.0	0	2	29.32
	6	72.0	70.0	0.0	0	5	29.32
	7	73.0	71.1	0.0	195	9	29.35
	8	78.1	73.9	0.0	1210	8	29.35
	9	86.0	72.0	0.0	3212	5	29.35
	10	89.1	70.0	2.5	5015	3	29.34
	11	91.0	69.1	2.5	5721	4	29.33
	12	93.0	66.9	4.3	5792	5	29.31
	13	93.9	66.0	2.5	5759	5	29.29
	14	96.1	63.0	2.5	5249	5	29.26
	15	96.1	62.1	3.4	4606	4	29.23
	16	96.1	63.0	4.9	2452	7	29.21
	17	93.9	63.0	3.4	1061	8	29.20
	18	90.0	64.0	3.4	114	9	29.20
	19	84.9	64.9	0.0	0	10	29.20
	20	82.9	63.0	0.0	0	10	29.21
	21	82.0	64.9	0.0	0	10	29.22
	22	81.0	66.9	2.5	0	10	29.22
	23	80.1	68.0	3.4	0	10	29.22
9/16/90	0	77.0	69.1	2.5	0	10	29.22
	1	77.0	70.0	0.0	0	10	29.22
	2	75.9	70.0	0.0	0	10	29.22
	3	75.0	70.0	0.0	0	9	29.23
	4	75.9	71.1	0.0	0	10	29.22
	5	75.0	71.1	0.0	0	9	29.23

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 18 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	6	75.0	71.1	0.0	0	8	29.24
	7	73.9	71.1	0.0	271	7	29.25
	8	82.9	73.0	0.0	1398	7	29.26
	9	87.1	73.0	0.0	2920	6	29.27
	10	90.0	71.1	4.3	4062	6	29.27
	11	93.0	70.0	2.5	3713	8	29.27
	12	95.0	66.0	6.7	3260	9	29.24
	13	95.0	64.9	0.0	3240	9	29.22
	14	95.0	63.0	7.6	2950	9	29.20
	15	95.0	66.9	8.3	1785	10	29.18
	16	93.9	68.0	8.3	1244	10	29.17
	17	91.9	66.0	8.3	621	10	29.19
	18	91.0	66.9	8.3	76	10	29.20
	19	88.0	66.0	0.0	0	10	29.22
	20	87.1	69.1	6.7	0	10	29.26
	21	82.9	68.0	14.1	0	10	29.30
	22	79.0	68.0	4.9	0	10	29.26
	23	78.1	66.0	0.0	0	10	29.26
9/17/90	0	75.9	66.9	0.0	0	9	29.25
	1	73.9	66.9	0.0	0	10	29.27
	2	75.0	68.0	0.0	0	10	29.30
	3	73.9	66.9	0.0	0	10	29.30
	4	75.0	68.0	0.0	0	10	29.30
	5	73.9	68.0	0.0	0	9	29.31
	6	73.9	68.0	0.0	0	9	29.31
	7	75.0	69.1	2.5	182	9	29.33
	8	78.1	72.0	3.4	711	10	29.35
	9	80.1	73.9	6.7	1326	10	29.36
	10	82.0	73.0	13.4	2499	9	29.37
	11	84.9	73.0	9.2	2998	9	29.37

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 19 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	12	87.1	72.0	8.3	2399	10	29.36
	13	88.0	70.0	10.1	2382	10	29.34
	14	90.0	69.1	9.2	2931	9	29.32
	15	91.9	69.1	9.2	3447	7	29.30
	16	91.0	69.1	8.3	2050	8	29.30
	17	88.0	69.1	4.9	820	9	29.30
	18	88.0	69.1	7.6	68	10	29.31
	19	75.0	72.0	7.6	0	10	29.32
	20	77.0	75.0	0.0	0	10	29.35
	21	75.9	75.0	0.0	0	9	29.37
	22	75.9	73.0	0.0	0	10	29.38
	23	75.0	73.0	0.0	0	8	29.39
9/18/90	0	75.0	72.0	0.0	0	7	29.39
	1	73.9	73.0	0.0	0	3	29.39
	2	73.9	73.0	0.0	0	5	29.38
	3	73.0	72.0	0.0	0	4	29.38
	4	73.0	72.0	0.0	0	5	29.39
	5	73.0	73.0	0.0	0	9	29.39
	6	73.0	73.0	0.0	0	8	29.40
	7	73.9	73.0	0.0	216	8	29.42
	8	77.0	73.9	4.9	1541	6	29.43
	9	81.0	73.9	4.9	2567	7	29.44
	10	84.0	73.0	7.6	4023	6	29.44
	11	87.1	72.0	7.6	5647	4	29.43
	12	90.0	71.1	8.3	6424	3	29.41
	13	91.9	69.1	9.2	6376	3	29.39
	14	93.0	69.1	4.9	5793	3	29.36
	15	93.9	69.1	6.7	4494	4	29.34
	16	93.9	68.0	7.6	2903	5	29.32
	17	93.0	68.0	9.2	1514	4	29.31

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 20 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	18	91.0	68.0	9.2	168	2	29.31
	19	87.1	69.1	4.9	0	1	29.33
	20	84.0	70.0	0.0	0	4	29.35
	21	84.0	70.0	4.9	0	3	29.36
	22	82.0	72.0	0.0	0	3	29.38
	23	81.0	72.0	0.0	0	1	29.38
9/19/90	0	81.0	72.0	4.3	0	4	29.37
	1	79.0	72.0	4.3	0	2	29.36
	2	78.1	72.0	4.3	0	7	29.35
	3	78.1	72.0	6.7	0	5	29.35
	4	77.0	72.0	4.3	0	2	29.33
	5	75.9	73.0	0.0	0	0	29.35
	6	73.9	72.0	0.0	0	1	29.37
	7	75.9	73.0	0.0	355	1	29.39
	8	80.1	73.0	4.9	1977	1	29.40
	9	82.9	72.0	4.9	3646	2	29.40
	10	87.1	71.1	10.1	5091	2	29.40
	11	87.1	69.1	10.7	4276	7	29.40
	12	89.1	71.1	4.9	4628	7	29.39
	13	91.9	69.1	6.7	6343	3	29.36
	14	93.0	71.1	4.9	5756	3	29.31
	15	93.9	70.0	9.2	4457	4	29.28
	16	93.9	70.0	9.2	2332	7	29.26
	17	93.0	69.1	9.2	1123	7	29.26
	18	90.0	68.0	3.4	126	5	29.27
	19	86.0	66.9	0.0	0	7	29.29
	20	84.9	70.0	0.0	0	6	29.30
	21	82.9	70.0	3.4	0	5	29.29
	22	79.0	70.0	0.0	0	2	29.32
	23	81.0	71.1	7.6	0	4	29.35

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 21 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
9/20/90	0	80.1	71.1	4.3	0	9	29.35
	1	75.9	72.0	0.0	0	10	29.34
	2	75.9	72.0	0.0	0	9	29.34
	3	75.9	72.0	0.0	0	6	29.34
	4	75.0	71.1	3.4	0	7	29.34
	5	75.0	71.1	0.0	0	9	29.34
	6	73.9	71.1	0.0	0	9	29.35
	7	75.0	71.1	2.5	120	10	29.35
	8	77.0	72.0	2.5	689	10	29.35
	9	80.1	72.0	0.0	1302	10	29.36
	10	82.0	73.0	7.6	1820	10	29.36
	11	84.9	72.0	9.2	2186	10	29.35
	12	87.1	73.0	9.2	2366	10	29.34
	13	88.0	72.0	9.2	2345	10	29.32
	14	88.0	69.1	9.2	2126	10	29.30
	15	91.0	69.1	10.1	1726	10	29.27
	16	73.0	70.0	7.6	1182	10	29.31
	17	73.0	71.1	0.0	561	10	29.26
	18	75.0	72.0	0.0	46	10	29.27
	19	75.0	71.1	0.0	0	10	29.28
	20	75.0	72.0	0.0	0	10	29.30
	21	75.9	72.0	0.0	0	10	29.31
	22	75.0	71.1	0.0	0	8	29.31
	23	75.0	72.0	0.0	0	8	29.32
9/21/90	0	77.0	72.0	0.0	0	8	29.30
	1	75.0	72.0	0.0	0	6	29.31
	2	75.0	73.0	0.0	0	4	29.32
	3	75.9	73.0	0.0	0	9	29.32
	4	75.9	73.9	0.0	0	9	29.33
	5	75.9	73.9	4.9	0	6	29.33

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 22 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	6	75.9	73.9	4.9	0	9	29.33
	7	77.0	73.9	7.6	225	7	29.33
	8	79.0	75.0	10.1	1137	8	29.34
	9	82.0	73.9	10.1	2158	8	29.34
	10	82.9	73.9	11.6	2450	9	29.34
	11	84.0	73.9	10.7	2175	10	29.36
	12	82.9	73.0	7.6	2354	10	29.36
	13	82.9	73.0	0.0	2333	10	29.34
	14	82.0	71.1	7.6	2112	10	29.34
	15	82.0	71.1	7.6	1711	10	29.33
	16	82.0	71.1	4.9	1167	10	29.32
	17	81.0	70.0	7.6	547	10	29.33
	18	78.1	70.0	8.3	40	10	29.34
	19	75.9	71.1	4.3	0	10	29.34
	20	75.0	71.1	0.0	0	10	29.35
	21	75.0	71.1	0.0	0	10	29.37
	22	73.9	71.1	4.3	0	10	29.39
	23	75.9	73.0	0.0	0	10	29.39
9/22/90	0	73.0	71.1	0.0	0	8	29.40
	1	73.9	72.0	3.4	0	10	29.40
	2	73.9	72.0	0.0	0	10	29.40
	3	73.9	73.0	3.4	0	10	29.40
	4	73.9	73.0	4.3	0	10	29.41
	5	73.9	71.1	8.3	0	10	29.42
	6	73.9	71.1	6.7	0	10	29.43
	7	73.0	69.1	4.3	111	10	29.44
	8	73.9	68.0	6.7	912	9	29.46
	9	77.0	66.9	10.1	2502	7	29.47
	10	78.1	64.0	9.2	3507	7	29.48
	11	82.9	61.0	4.9	4738	6	29.48

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 23 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	12	84.9	57.9	7.6	5127	6	29.48
	13	87.1	57.0	7.6	4517	7	29.46
	14	84.9	57.0	8.3	4085	7	29.45
	15	88.0	55.0	4.9	2829	8	29.44
	16	87.1	53.1	10.7	1151	10	29.43
	17	84.9	55.0	15.0	532	10	29.45
	18	82.0	55.0	14.1	34	10	29.47
	19	80.1	57.0	0.0	0	10	29.47
	20	75.9	55.9	3.4	0	10	29.48
	21	73.0	57.0	0.0	0	10	29.50
	22	71.1	57.0	0.0	0	7	29.53
	23	70.0	57.0	0.0	0	7	29.54
9/23/90	0	68.0	59.0	0.0	0	7	29.55
	1	68.0	59.0	6.7	0	7	29.56
	2	70.0	55.0	6.7	0	8	29.58
	3	66.9	51.1	7.6	0	10	29.59
	4	66.0	48.0	10.1	0	5	29.61
	5	63.0	46.9	10.7	0	2	29.63
	6	62.1	46.0	11.6	0	1	29.65
	7	61.0	43.0	10.7	303	1	29.67
	8	62.1	42.1	15.0	1892	1	29.69
	9	64.9	37.9	14.1	3551	2	29.71
	10	66.9	37.9	14.1	5086	1	29.71
	11	69.1	37.9	10.7	6115	1	29.70
	12	71.1	37.0	14.1	6616	1	29.69
	13	72.0	37.0	11.6	6590	0	29.67
	14	73.0	37.9	13.4	5954	0	29.65
	15	75.0	37.0	11.6	4802	0	29.62
	16	73.9	37.9	13.4	3222	1	29.60
	17	73.0	37.9	10.7	1439	2	29.59

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 24 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
	18	70.0	37.9	8.3	75	3	29.60
	19	63.0	39.9	6.7	0	2	29.59
	20	59.0	42.1	4.3	0	3	29.60
	21	61.0	42.1	4.9	0	4	29.61
	22	57.9	44.1	4.3	0	10	29.62
	23	55.0	44.1	3.4	0	9	29.61
9/24/90	0	55.9	44.1	4.9	0	10	29.60
	1	57.0	42.1	4.3	0	10	29.60
	2	55.9	43.0	0.0	0	10	29.60
	3	54.0	43.0	3.4	0	9	29.60
	4	54.0	43.0	0.0	0	8	29.60
	5	55.0	43.0	4.3	0	9	29.60
	6	54.0	44.1	3.4	0	8	29.60
	7	57.0	46.0	0.0	171	8	29.62
	8	60.1	46.0	0.0	1099	8	29.62
	9	64.9	44.1	6.7	2115	8	29.62
	10	66.9	37.9	4.9	2973	8	29.62
	11	70.0	39.9	4.9	4174	7	29.60
	12	72.0	39.9	4.9	3869	8	29.58
	13	75.0	39.9	9.2	5020	6	29.55
	14	77.0	37.9	4.9	3453	8	29.52
	15	75.0	41.0	7.6	2253	9	29.50
	16	73.0	39.9	8.3	1120	10	29.48
	17	73.0	37.9	3.4	503	10	29.47
	18	72.0	41.0	3.4	31	9	29.46
	19	69.1	45.0	3.4	0	10	29.45
	20	68.0	45.0	3.4	0	8	29.45
	21	66.9	46.0	4.3	0	8	29.44
	22	66.9	46.9	4.3	0	10	29.44
	23	68.0	46.9	4.3	0	10	29.44

TABLE 2.3-7B
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 25 of 25)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE, IN Hg
9/25/90	0	66.9	48.0	4.9	0	8	29.42
	1	64.0	46.9	6.7	0	3	29.41
	2	61.0	48.9	4.9	0	2	29.41
	3	61.0	48.9	6.7	0	1	29.40
	4	62.1	48.9	7.6	0	1	29.40
	5	62.1	50.0	7.6	0	7	29.41
	6	61.0	50.0	7.6	0	4	29.40
	7	62.1	51.1	6.7	277	1	29.40
	8	66.9	50.0	13.4	1861	0	29.40
	9	72.0	51.1	8.3	3596	0	29.41
	10	75.9	50.0	8.3	5064	0	29.40
	11	81.0	50.0	8.3	6094	0	29.39
	12	82.9	51.1	10.1	6591	0	29.37
	13	84.0	53.1	8.3	6515	0	29.35
	14	84.9	54.0	8.3	5871	0	29.33
	15	86.0	55.9	6.7	4714	0	29.31
	16	86.0	55.9	7.6	3154	0	29.29
	17	86.0	55.0	7.6	1395	0	29.29
	18	82.9	54.0	8.3	53	0	29.29
	19	77.0	55.9	4.3	0	0	29.29
	20	77.0	55.9	8.3	0	0	29.30
	21	72.0	55.9	6.7	0	0	29.31
	22	71.1	57.0	8.3	0	0	29.32
	23	69.1	57.0	7.6	0	0	29.33

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 1 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
6/25/80	0	87.1	69.1	9.2	0	0	29.22
	3	84.0	73.0	10.1	0	0	29.22
	6	79.0	73.0	4.3	173	0	29.28
	9	91.9	73.9	9.2	4984	0	29.32
	12	100.9	72.0	13.4	7959	0	29.31
	15	108.0	66.0	10.1	6546	0	29.28
	18	105.1	64.0	9.2	1773	0	29.26
	21	93.0	61.0	9.2	0	0	29.27
6/26/80	0	84.0	64.0	6.7	0	0	29.29
	3	82.0	68.0	7.6	0	0	29.28
	6	81.0	66.9	9.2	168	1	29.29
	9	97.0	70.0	10.1	4976	0	29.32
	12	106.0	66.9	6.7	7956	0	29.32
	15	111.9	60.1	7.6	6548	0	29.28
	18	108.0	57.0	8.3	1776	0	29.27
	21	96.1	61.0	7.6	0	0	29.28
6/27/80	0	90.0	64.9	8.3	0	0	29.28
	3	86.0	66.0	8.3	0	0	29.27
	6	82.9	66.0	6.7	160	2	29.27
	9	98.1	66.9	7.6	4968	0	29.29
	12	108.0	64.9	4.9	7953	0	29.28
	15	111.9	64.0	7.6	6550	0	29.22
	18	109.0	62.1	10.7	1778	0	29.18
	21	97.0	63.0	8.3	0	0	29.21
6/28/80	0	91.0	66.0	10.7	0	0	29.23
	3	86.0	70.0	10.1	0	0	29.20
	6	82.0	70.0	6.7	159	0	29.23
	9	93.0	71.1	10.7	4960	0	29.27
	12	102.9	69.1	9.2	7950	0	29.26
	15	109.9	66.0	8.3	6551	0	29.22
	18	107.1	64.0	10.7	1780	0	29.19

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 2 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	21	93.9	64.0	10.1	0	0	29.23
6/29/80	0	89.1	69.1	11.6	0	0	29.26
	3	84.0	69.1	10.7	0	0	29.24
	6	79.0	71.1	4.9	138	4	29.29
	9	91.0	71.1	6.7	4661	3	29.34
	12	100.0	68.0	7.6	7740	2	29.34
	15	106.0	64.9	7.6	6510	1	29.29
	18	105.1	59.0	10.1	1781	0	29.27
	21	91.0	63.0	6.7	0	0	29.30
6/30/80	0	88.0	64.9	9.2	0	0	29.33
	3	82.0	68.0	4.9	0	0	29.35
	6	79.0	69.1	4.9	140	3	29.36
	9	91.0	70.0	10.7	4942	0	29.40
	12	102.0	66.0	10.7	7944	0	29.38
	15	106.0	62.1	6.7	6511	1	29.33
	18	104.0	61.0	9.2	1782	0	29.29
	21	91.9	64.0	7.6	0	0	29.31
7/1/80	0	87.1	66.9	7.6	0	0	29.32
	3	84.0	72.0	8.3	0	0	29.32
	6	81.0	72.0	6.7	144	0	29.35
	9	93.9	71.1	10.1	4922	0	29.39
	12	104.0	64.9	11.6	7941	0	29.36
	15	109.0	62.1	7.6	6554	0	29.30
	18	105.1	63.0	10.1	1783	0	29.28
	21	95.0	64.9	7.6	0	0	29.30
7/2/80	0	89.1	66.9	8.3	0	0	29.35
	3	86.0	69.1	8.3	0	0	29.34
	6	81.0	69.1	6.7	139	0	29.37
	9	95.0	68.0	7.6	4924	0	29.40
	12	106.0	66.9	9.2	7937	0	29.39

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 3 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	15	109.9	61.0	7.6	6554	0	29.32
	18	106.0	57.9	10.7	1783	0	29.29
	21	95.0	64.0	10.7	0	0	29.33
7/3/80	0	86.0	63.0	4.9	0	0	29.31
	3	75.9	61.0	0.0	0	0	29.31
	6	80.1	64.9	7.6	133	0	29.36
	9	93.0	66.9	6.7	4915	0	29.37
	12	102.0	64.9	6.7	7933	0	29.35
	15	109.0	59.0	9.2	6554	0	29.29
	18	104.0	61.0	10.1	1781	0	29.28
	21	95.0	69.1	10.1	0	0	29.30
7/4/80	0	88.0	70.0	4.3	0	0	29.34
	3	86.0	71.1	9.2	0	5	29.33
	6	82.0	68.0	8.3	126	1	29.36
	9	91.9	68.0	7.6	4905	0	29.39
	12	99.0	66.9	4.9	7929	0	29.38
	15	102.9	64.0	9.2	6554	0	29.33
	18	100.0	64.0	6.7	1781	0	29.31
	21	91.0	64.0	6.7	0	0	29.34
7/5/80	0	84.9	64.9	4.9	0	0	29.37
	3	82.9	66.0	9.2	0	0	29.36
	6	79.0	69.1	6.7	122	0	29.41
	9	90.0	71.1	8.3	4896	0	29.44
	12	97.0	66.0	9.2	7925	0	29.43
	15	100.9	64.9	9.2	6554	0	29.36
	18	100.9	60.1	8.3	1778	0	29.33
	21	91.0	68.0	8.3	0	0	29.36
7/6/80	0	87.1	66.9	4.3	0	0	29.39
	3	82.9	69.1	4.9	0	0	29.39
	6	81.0	70.0	4.9	117	0	29.43

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 4 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	9	91.0	69.1	8.3	4886	0	29.46
	12	97.0	68.0	9.2	6634	5	29.45
	15	104.0	66.0	10.1	5871	4	29.39
	18	100.0	64.0	9.2	1776	0	29.35
	21	91.0	66.0	6.7	0	0	29.37
7/7/80	0	86.0	66.0	4.3	0	0	29.40
	3	82.0	66.9	4.9	0	0	29.40
	6	82.0	70.0	7.6	110	0	29.42
	9	91.9	69.1	13.4	4876	0	29.47
	12	100.0	68.0	4.3	7866	1	29.46
	15	102.0	68.0	9.2	5871	4	29.40
	18	100.0	66.0	10.1	1727	2	29.35
	21	91.9	69.1	7.6	0	1	29.35
7/8/80	0	89.1	69.1	6.7	0	0	29.40
	3	84.9	70.0	6.7	0	0	29.43
	6	81.0	70.0	4.3	102	2	29.44
	9	89.1	71.1	6.7	4359	4	29.44
	12	100.9	68.0	13.4	7091	4	29.44
	15	105.1	64.9	10.7	6507	1	29.38
	18	100.9	64.0	10.7	1769	0	29.34
	21	91.9	64.0	4.3	0	0	29.36
7/9/80	0	87.1	66.0	6.7	0	0	29.38
	3	80.1	64.9	4.3	0	0	29.38
	6	79.0	66.0	6.7	99	0	29.42
	9	95.0	68.0	7.6	4855	0	29.44
	12	100.9	66.0	4.9	7703	2	29.43
	15	105.1	61.0	7.6	6378	2	29.36
	18	102.0	61.0	6.7	1478	5	29.33
	21	93.0	62.1	8.3	0	2	29.36

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 5 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
7/10/80	0	86.0	63.0	4.3	0	5	29.40
	3	80.1	63.0	4.3	0	0	29.39
	6	80.1	68.0	6.7	94	0	29.41
	9	93.9	70.0	9.2	4813	1	29.46
	12	100.0	66.0	6.7	7442	3	29.43
	15	102.0	62.1	3.4	6503	1	29.37
	18	102.0	61.0	4.3	1759	0	29.33
	21	91.9	63.0	6.7	0	0	29.36
7/11/80	0	87.1	63.0	4.9	0	0	29.37
	3	80.1	64.0	4.3	0	0	29.36
	6	79.0	66.9	4.9	88	0	29.39
	9	95.0	69.1	9.2	4834	0	29.41
	12	102.0	64.0	9.2	7695	2	29.38
	15	105.1	61.0	9.2	6500	1	29.32
	18	102.0	61.0	8.3	1754	0	29.28
	21	95.0	63.0	6.7	0	0	29.30
7/12/80	0	89.1	64.9	4.9	0	0	29.33
	3	84.0	64.9	9.2	0	0	29.32
	6	80.1	64.9	4.3	74	4	29.37
	9	93.9	69.1	8.3	4823	0	29.40
	12	100.9	64.9	6.7	7895	0	29.37
	15	106.0	64.0	10.7	6539	0	29.31
	18	102.9	61.0	8.3	1748	0	29.28
	21	90.0	59.0	4.3	0	0	29.29
7/13/80	0	84.0	61.0	4.3	0	0	29.32
	3	82.9	66.0	9.2	0	0	29.34
	6	79.0	66.0	4.9	77	0	29.36
	9	91.9	64.9	9.2	4812	0	29.40
	12	100.0	63.0	10.1	7890	0	29.39
	15	106.0	57.9	10.7	6536	0	29.31

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 6 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	18	104.0	57.0	9.2	1741	0	29.28
	21	91.0	57.0	4.9	0	0	29.30
7/14/80	0	82.0	62.1	4.3	0	0	29.33
	3	84.9	64.9	6.7	0	0	29.35
	6	80.1	66.0	9.2	72	0	29.37
	9	91.9	66.0	10.1	4801	0	29.40
	12	102.9	61.0	13.4	7885	0	29.39
	15	107.1	59.0	9.2	6532	1	29.33
	18	102.9	62.1	10.1	1733	0	29.31
	21	93.0	64.9	7.6	0	0	29.33
7/15/80	0	88.0	64.9	4.9	0	0	29.35
	3	84.9	68.0	7.6	0	0	29.37
	6	82.0	69.1	9.2	65	2	29.40
	9	91.0	69.1	9.2	4790	0	29.42
	12	100.0	68.0	9.2	7880	0	29.40
	15	106.0	63.0	13.4	6528	0	29.34
	18	102.0	61.0	11.6	1725	0	29.30
	21	91.9	62.1	6.7	0	0	29.32
7/16/80	0	88.0	63.0	6.7	0	0	29.34
	3	84.9	64.9	10.1	0	0	29.35
	6	80.1	66.9	4.9	62	0	29.36
	9	91.9	69.1	9.2	4778	1	29.38
	12	102.0	62.1	6.7	7874	0	29.37
	15	108.0	53.1	6.7	6522	1	29.30
	18	106.0	48.9	4.3	1716	0	29.25
	21	90.0	52.0	6.7	0	0	29.26
7/17/80	0	86.0	53.1	0.0	0	0	29.31
	3	80.1	55.0	4.9	0	0	29.29
	6	81.0	59.0	6.7	43	6	29.31
	9	95.0	57.0	6.7	2784	8	29.34

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 7 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	12	104.0	55.0	9.2	4595	8	29.32
	15	108.0	54.0	6.7	4442	7	29.27
	18	106.0	53.1	6.7	1607	3	29.22
	21	95.0	54.0	6.7	0	0	29.25
7/18/80	0	86.0	57.0	4.9	0	0	29.27
	3	81.0	57.0	4.9	0	0	29.26
	6	81.0	57.9	6.7	52	0	29.29
	9	95.0	61.0	9.2	4756	0	29.31
	12	104.0	60.1	3.4	7862	0	29.31
	15	109.0	59.0	10.1	5834	4	29.27
	18	105.1	57.9	8.3	1685	1	29.23
	21	95.0	59.0	6.7	0	0	29.28
7/19/80	0	89.1	60.1	4.3	0	0	29.31
	3	86.0	62.1	6.7	0	0	29.33
	6	84.0	64.0	4.3	47	0	29.35
	9	91.9	66.9	8.3	4744	0	29.38
	12	100.9	64.0	6.7	7856	0	29.36
	15	102.0	63.0	9.2	6463	1	29.31
	18	100.0	60.1	8.3	1685	0	29.28
	21	91.0	60.1	4.3	0	0	29.28
7/20/80	0	88.0	63.0	4.3	0	0	29.30
	3	82.0	62.1	4.3	0	0	29.30
	6	81.0	64.9	4.3	43	0	29.31
	9	89.1	66.0	4.3	4732	0	29.33
	12	98.1	64.0	8.3	7391	3	29.31
	15	102.9	63.0	4.3	5823	4	29.26
	18	100.9	60.1	4.9	1631	2	29.24
	21	90.0	60.1	3.4	0	1	29.27

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 8 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
7/21/80	0	86.0	63.0	2.5	0	4	29.29
	3	82.0	64.0	2.5	0	4	29.29
	6	80.1	64.9	2.5	37	2	29.31
	9	93.0	66.9	4.9	3953	5	29.36
	12	100.9	63.0	10.7	6569	5	29.32
	15	91.0	69.1	6.7	4423	7	29.29
	18	93.0	64.9	0.0	582	10	29.28
	21	82.9	69.1	0.0	0	9	29.31
7/22/80	0	82.9	68.0	4.3	0	5	29.33
	3	81.0	68.0	2.5	0	10	29.34
	6	78.1	68.0	3.4	20	8	29.37
	9	86.0	70.0	4.9	2749	8	29.40
	12	97.0	68.0	9.2	6564	5	29.38
	15	102.0	62.1	11.6	5430	5	29.30
	18	100.0	59.0	7.6	1606	2	29.28
	21	87.1	54.0	0.0	0	0	29.33
7/23/80	0	82.0	54.0	4.3	0	0	29.35
	3	75.9	53.1	6.7	0	0	29.33
	6	73.9	53.1	7.6	30	0	29.34
	9	87.1	55.0	4.3	4696	0	29.36
	12	97.0	43.0	10.7	7830	0	29.34
	15	102.9	46.0	13.4	6475	0	29.28
	18	100.0	53.1	9.2	1635	0	29.24
	21	89.1	59.0	6.7	0	0	29.27
7/24/80	0	81.0	57.9	0.0	0	0	29.29
	3	75.0	55.9	3.4	0	0	29.28
	6	73.0	59.0	4.3	27	1	29.30
	9	88.0	61.0	10.1	4410	3	29.34
	12	95.0	57.0	10.7	7772	1	29.32
	15	100.0	53.1	10.7	6424	1	29.28

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 9 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	18	97.0	51.1	8.3	1611	1	29.26
	21	89.1	51.1	4.9	0	0	29.28
7/25/80	0	79.0	55.0	2.5	0	0	29.30
	3	77.0	57.9	0.0	0	0	29.29
	6	77.0	62.1	2.5	22	3	29.31
	9	91.0	66.0	10.7	4550	2	29.35
	12	100.0	64.9	4.9	6545	5	29.33
	15	104.0	63.0	10.7	5785	4	29.28
	18	100.0	57.9	7.6	1564	2	29.26
	21	91.9	57.9	4.9	0	0	29.28
7/26/80	0	87.1	60.1	4.9	0	0	29.29
	3	84.9	64.9	7.6	0	0	29.29
	6	81.0	66.0	6.7	21	1	29.31
	9	91.0	66.9	7.6	4659	0	29.32
	12	100.9	63.0	7.6	7807	0	29.29
	15	105.1	60.1	9.2	3765	8	29.23
	18	88.0	66.9	4.9	557	10	29.22
	21	84.0	66.9	3.4	0	3	29.23
7/27/80	0	82.0	64.0	2.5	0	0	29.25
	3	80.1	64.0	0.0	0	6	29.26
	6	80.1	66.9	2.5	8	9	29.27
	9	88.0	70.0	6.7	2714	8	29.28
	12	97.0	70.0	4.3	6989	4	29.27
	15	88.0	71.1	4.9	4387	7	29.20
	18	86.0	71.1	9.2	1073	7	29.22
	21	75.9	66.9	2.5	0	0	29.27
7/28/80	0	75.0	68.0	6.7	0	5	29.30
	3	73.0	68.0	2.5	0	2	29.28
	6	73.0	66.9	0.0	14	0	29.30
	9	82.9	70.0	3.4	4634	0	29.33

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 10 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
	12	95.0	71.1	3.4	7740	1	29.32
	15	100.9	61.0	6.7	6384	1	29.28
	18	99.0	55.9	8.3	1466	3	29.28
	21	86.0	59.0	0.0	0	0	29.30
7/29/80	0	82.0	66.9	7.6	0	0	29.32
	3	75.0	66.9	2.5	0	0	29.33
	6	72.0	64.9	2.3	12	0	29.36
	9	88.0	69.1	2.5	4621	0	29.40
	12	98.1	57.0	6.7	7783	0	29.39
	15	102.0	57.0	9.2	6414	0	29.35
	18	99.0	57.9	7.6	1540	0	29.32
	21	84.9	60.1	4.3	0	0	29.33
7/30/80	0	80.1	61.0	4.3	0	0	29.36
	3	77.0	63.0	6.7	0	0	29.35
	6	77.0	66.0	7.6	10	0	29.37
	9	90.0	68.0	7.6	4608	0	29.39
	12	100.9	61.0	8.3	7774	0	29.37
	15	105.1	57.9	7.6	6402	0	29.30
	18	102.0	55.0	9.2	1521	0	29.28
	21	91.9	61.0	10.1	0	0	29.31
7/31/80	0	87.1	59.0	7.6	0	0	29.36
	3	81.0	63.0	6.7	0	0	29.36
	6	80.1	66.0	6.7	8	0	29.39
	9	91.9	63.0	9.2	4595	0	29.42
	12	102.9	50.0	13.4	7765	0	29.40
	15	106.0	48.9	7.6	6389	0	29.35
	18	102.0	46.9	10.1	1502	0	29.32
	21	86.0	50.0	4.9	0	0	29.35

TABLE 2.3-7C
OFFSITE METEOROLOGICAL DATA USED IN ULTIMATE HEAT SINK WATER SUPPLY PERFORMANCE EVALUATION

(Sheet 11 of 11)

DATE	HOUR	AIR TEMP.F	DEW-POINT TEMP.F	WIND-SPEED MPH	SOLAR RADIATION BTU FT^2DAY^{-1}	CLOUD COVER, TENTHS	PRESSURE IN Hg
8/01/80	0	80.1	57.0	4.9	0	0	29.38
	3	72.0	57.0	4.9	0	0	29.37
	6	75.9	59.0	4.9	7	0	29.39
	9	90.0	61.0	4.9	4581	0	29.43
	12	100.0	54.0	9.2	7756	0	29.40
	15	102.9	51.1	6.7	6375	0	29.33
	18	100.9	48.9	8.3	1473	1	29.28
	21	86.0	53.1	4.9	0	0	29.29
8/02/80	0	82.0	57.9	4.9	0	0	29.30
	3	78.1	55.9	4.9	0	0	29.28
	6	75.9	59.0	8.3	5	0	29.31
	9	90.0	66.0	13.4	4568	0	29.30
	12	99.0	63.0	11.6	7746	0	29.27
	15	104.0	57.0	13.4	6321	1	29.20
	18	102.0	57.9	10.7	1452	1	29.16
	21	93.9	60.1	8.3	0	0	29.17

TABLE 2.3-8
DELETED

TABLE 2.3-9
DELETED

TABLE 2.3-10
DELETED

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 1 of 13)

Annual

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	1	2	2	1	+	6
NNE	+	1	2	1	+	+	5
NE	1	2	2	1	+	+	5
ENE	+	1	1	1	+	+	4
E	+	1	2	+	+	+	4
ESE	+	1	2	1	+	+	5
SE	1	3	5	3	1	+	13
SSE	+	2	5	6	2	+	16
S	1	2	5	6	2	+	16
SSW	+	1	1	2	1	+	6
SW	1	1	1	1	+	+	4
WSW	+	+	+	+	+	+	1
W	+	1	+	+	+	+	2
WNW	+	1	1	+	+	+	2
NW	1	1	1	1	1	+	4
NNW	+	1	1	2	1	+	5
Calm	2						2
TOTAL	9	21	32	28	9	1	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 2 of 13)

January

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	3	3	3	1	+	10
NNE	1	2	2	1	+	0	6
NE	1	2	1	+	0	0	5
ENE	+	1	1	+	0	0	2
E	+	1	1	+	0	0	2
ESE	+	1	1	+	+	0	3
SE	1	2	4	2	+	0	9
SSE	+	2	5	4	1	+	12
S	1	1	5	6	2	+	14
SSW	+	1	1	2	1	+	5
SW	1	1	1	1	+	+	4
WSW	+	+	+	+	+	0	1
W	1	1	1	+	+	0	2
WNW	+	1	1	1	+	+	3
NW	1	2	2	2	1	+	8
NNW	1	2	3	3	1	+	10
Calm	3						3
TOTAL	13	22	31	26	8	1	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 3 of 13)

February

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	2	3	2	1	+	9
NNE	1	2	2	2	+	0	7
NE	1	3	2	1	+	0	6
ENE	+	1	1	+	+	0	3
E	1	1	1	+	+	0	3
ESE	+	1	2	1	+	0	4
SE	1	2	4	2	1	+	10
SSE	+	1	3	5	1	+	11
S	1	2	3	4	2	+	12
SSW	+	1	1	1	1	+	4
SW	1	1	1	1	+	+	5
WSW	+	+	+	+	+	+	1
W	1	1	+	+	+	+	3
WNW	+	1	1	1	1	+	3
NW	1	2	2	2	1	+	8
NNW	1	2	2	3	1	+	9
Calm	2						2
TOTAL	10	21	30	26	10	2	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 4 of 13)

March

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	2	3	2	1	+	8
NNE	+	2	2	2	+	0	6
NE	1	2	3	1	+	0	6
ENE	+	1	2	1	+	+	4
E	+	1	2	1	+	0	4
ESE	+	1	2	2	+	+	5
SE	+	2	4	3	1	0	9
SSE	+	1	3	6	3	+	13
S	+	1	2	5	3	1	12
SSW	+	+	1	2	1	+	5
SW	+	1	1	1	+	+	4
WSW	+	+	+	1	+	+	2
W	+	1	1	1	1	+	3
WNW	+	+	1	1	1	+	4
NW	+	1	1	2	1	1	6
NNW	+	1	2	3	2	+	8
Calm	1						1
TOTAL	6	17	29	31	14	3	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 5 of 13)

April

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	+	1	2	2	1	+	7
NNE	+	2	2	2	1	+	7
NE	1	2	2	1	+	+	6
ENE	+	1	1	1	+	+	3
E	+	1	1	1	+	0	3
ESE	+	1	1	1	+	+	4
SE	+	1	4	5	1	+	12
SSE	+	+	3	8	5	1	17
S	+	1	2	7	5	1	16
SSW	+	+	1	2	1	+	5
SW	+	1	1	1	+	+	3
WSW	+	+	+	1	+	+	2
W	+	1	1	+	+	+	2
WNW	+	+	+	1	1	+	2
NW	+	1	1	1	1	+	4
NNW	+	1	2	1	1	+	5
Calm	1						1
TOTAL	4	13	26	36	18	3	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 6 of 13)

May

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	1	2	1	+	+	5
NNE	+	1	1	1	+	0	4
NE	1	2	2	1	+	+	5
ENE	+	1	2	1	+	+	4
E	+	1	2	+	+	+	4
ESE	+	1	2	1	+	0	5
SE	1	2	5	5	1	+	14
SSE	+	2	4	9	4	1	18
S	+	1	4	9	5	1	19
SSW	+	1	1	2	1	+	6
SW	+	1	1	1	+	0	3
WSW	+	+	+	+	+	+	1
W	+	1	1	+	+	+	2
WNW	+	+	1	+	+	+	1
NW	+	1	1	1	+	+	3
NNW	+	1	1	1	+	+	4
Calm	1						1
TOTAL	6	18	29	32	13	2	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 7 of 13)

June

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	+	1	1	+	+	+	2
NNE	+	1	1	1	+	+	3
NE	+	1	1	+	+	+	4
ENE	+	1	2	1	+	0	4
E	+	1	2	1	+	0	4
ESE	+	1	3	2	+	0	6
SE	+	2	6	6	1	+	16
SSE	+	2	6	10	4	+	23
S	+	2	5	9	5	1	22
SSW	+	1	2	3	1	+	7
SW	+	1	1	1	+	+	4
WSW	+	+	+	+	+	0	1
W	+	+	+	+	0	+	1
WNW	+	+	+	+	+	+	1
NW	+	+	+	+	+	0	1
NNW	+	+	+	+	+	0	1
Calm	+						+
TOTAL	4	14	32	36	13	1	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 8 of 13)

July

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	+	1	1	+	+	0	2
NNE	+	1	1	1	+	+	2
NE	+	2	1	+	+	0	4
ENE	+	1	2	1	+	+	4
E	+	2	2	1	+	0	5
ESE	+	2	3	1	+	0	6
SE	1	4	6	4	1	+	16
SSE	+	2	7	8	1	+	19
S	+	3	8	8	2	+	21
SSW	+	1	3	3	1	+	9
SW	1	2	3	2	+	0	6
WSW	+	1	1	1	+	0	2
W	+	+	+	+	+	0	1
WNW	+	+	+	+	+	0	1
NW	+	+	+	+	+	0	1
NNW	+	+	+	+	+	+	1
Calm	1						1
TOTAL	5	21	39	29	6	+	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 9 of 13)

August

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	+	1	1	+	+	0	2
NNE	+	1	1	1	+	0	3
NE	1	2	1	+	+	+	4
ENE	+	2	2	1	+	0	5
E	1	2	2	1	+	0	5
ESE	+	2	3	1	+	+	7
SE	1	4	8	5	+	0	17
SSE	+	2	8	8	1	+	21
S	+	3	9	6	1	+	19
SSW	+	1	2	2	+	+	6
SW	+	1	2	1	+	+	5
WSW	+	+	+	+	0	0	1
W	+	+	+	+	0	0	1
WNW	+	+	+	+	+	0	1
NW	+	+	+	+	0	0	1
NNW	+	+	+	+	+	+	1
Calm	1						1
TOTAL	6	22	42	26	3	0	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 10 of 13)

September

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	1	1	1	+	0	4
NNE	1	2	2	1	+	+	6
NE	1	4	3	1	+	0	10
ENE	1	2	2	1	+	0	5
E	1	3	4	+	+	0	8
ESE	+	2	4	1	+	+	8
SE	1	4	8	3	+	0	16
SSE	+	2	6	5	1	+	15
S	1	2	5	5	1	+	14
SSW	+	1	1	2	+	0	4
SW	1	1	1	+	+	+	3
WSW	+	+	+	+	0	0	1
W	+	+	+	+	+	0	1
WNW	+	+	+	+	+	0	1
NW	+	1	+	+	+	0	2
NNW	+	1	1	+	+	0	2
Calm	2						2
TOTAL	9	26	40	22	3	+	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 11 of 13)

October

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	2	2	2	+	+	7
NNE	1	2	2	1	+	+	5
NE	1	3	1	+	+	0	6
ENE	+	2	1	+	+	0	4
E	1	2	1	+	+	0	4
ESE	1	2	3	+	+	0	6
SE	1	5	7	3	+	+	17
SSE	+	2	6	4	1	+	13
S	1	2	4	4	1	+	13
SSW	+	1	1	2	+	+	4
SW	1	1	1	1	+	0	4
WSW	+	+	+	+	+	0	1
W	+	1	+	+	+	0	2
WNW	+	1	+	+	+	+	2
NW	1	1	1	1	+	+	5
NNW	+	1	2	1	+	+	5
Calm	3						3
TOTAL	13	28	34	20	4	1	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 12 of 13)

November

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	2	2	2	1	+	9
NNE	1	2	2	1	+	+	6
NE	1	3	1	+	+	0	6
ENE	+	1	1	+	0	0	3
E	1	2	1	+	+	0	4
ESE	+	1	1	+	0	0	3
SE	1	4	3	1	+	0	9
SSE	+	2	4	5	1	0	12
S	1	2	5	5	2	+	15
SSW	+	1	1	2	1	+	5
SW	1	1	1	1	+	+	4
WSW	+	1	+	+	+	0	2
W	1	1	+	+	+	+	2
WNW	+	1	1	1	+	+	3
NW	1	2	1	1	1	+	6
NNW	1	2	2	2	1	+	7
Calm	4						4
TOTAL	15	26	27	23	7	1	100

TABLE 2.3-11
PERCENT FREQUENCY DISTRIBUTION OF WIND AT LOVE FIELD
(1951-1960)

(Sheet 13 of 13)

December

Wind Direction	Wind Speed (Miles per Hour)						
	0-3	4-7	8-12	13-18	19-24	≥25	Total
N	1	2	2	2	1	+	8
NNE	+	1	1	1	+	+	4
NE	1	1	1	1	+	0	4
ENE	+	1	1	+	+	0	2
E	+	1	1	+	+	0	2
ESE	+	1	1	1	+	+	3
SE	1	3	3	2	+	0	9
SSE	+	2	5	5	1	+	14
S	1	2	4	5	1	+	13
SSW	+	1	2	2	1	+	6
SW	1	2	1	1	+	0	5
WSW	+	1	+	+	+	+	2
W	1	1	1	1	+	+	4
WNW	1	1	1	1	+	+	5
NW	1	2	2	2	1	+	9
NNW	+	2	2	3	1	+	8
Calm	2						2
TOTAL	12	24	29	27	7	1	100

+ indicates percentages less than 0.5 but greater than zero.

TABLE 2.3-12
MONTHLY VARIATION OF EXTREME "FASTEST MILE" WINDS FOR FORT WORTH AND DALLAS

	Fort Worth (1954-1973)			Dallas (1941-1973)		
Month	Speed (MPH)	Direction	Year	Speed (MPH)	Direction[a]	Year
January	46	N	1957	50	W	1966
February	51	N	1962[b]	61	SW	1948
March	55	WNW	1954	59	W	1954
April	55	NW	1970	58	N	1961
May	55	SE	1955	51	S	1952
June	52	NW	1955	65	N	1954
July	65	N	1961	43	S	1948
August	73	N	1959	53	SE	1954
September	53	ESE	1961[b]	47	NW	1954
October	44	W	1957	61	NW	1960
November	50	NNW	1957	49	NW	1957
December	53	NW	1968	47	W	1957[b]
Year	73	N	1959	65	N	1954

a) Direction recorded to 8 compass points only.

b) Record also occurred in earlier year(s).

TABLE 2.3-13
WIND DIRECTION PERSISTENCE AT FORT WORTH (1969-1973)

(Sheet 1 of 2)

Wind Sector	Consecutive 3–Hourly Observations[(a)]											
	2	3	4	5	6	7	8	9	10	11	12	≥15
N	274	130	74	37	19	10	6	4	1	0	0	0
NNE	131	48	14	4	2	0	0	0	0	0	0	0
NE	64	21	5	4	0	0	0	0	0	0	0	0
ENE	40	4	1	1	1	0	0	0	0	0	0	0
E	121	33	12	3	1	0	0	0	0	0	0	0
ESE	126	37	8	2	1	1	1	0	0	0	0	0
SE	185	64	25	10	4	2	0	0	0	0	0	0
SSE	363	136	58	24	8	4	2	1	0	0	0	0
S	662	332	179	116	62	38	20	15	9	7	5	2
SSW	263	89	32	10	3	0	0	0	0	0	0	0
SW	68	12	4	1	0	0	0	0	0	0	0	0
WSW	30	8	1	1	1	0	0	0	0	0	0	0
W	56	22	8	3	2	0	0	0	0	0	0	0
WNW	35	11	3	2	0	0	0	0	0	0	0	0
NW	94	39	18	8	2	2	1	1	0	0	0	0
NNW	192	89	37	20	14	10	2	1	1	1	1	0
Total	2704	1075	479	246	120	67	32	22	11	8	6	2

TABLE 2.3-13
WIND DIRECTION PERSISTENCE AT FORT WORTH (1969-1973)

(Sheet 2 of 2)

Wind Sector ±22.5°	Consecutive 3–Hourly Observations[a]											
	2	3	4	5	6	7	10	15	20	25	30	≥40
N	276	182	131	95	65	44	13	3	1	1	0	0
NNE	118	76	49	41	28	19	6	1	0	0	0	0
NE	105	45	25	14	9	5	0	0	0	0	0	0
ENE	64	29	16	10	9	6	3	0	0	0	0	0
E	172	80	41	22	14	10	1	0	0	0	0	0
ESE	165	102	64	43	27	14	6	0	0	0	0	0
SE	235	157	99	67	42	32	14	1	1	0	0	0
SSE	348	259	201	148	117	90	43	14	5	2	2	0
S	323	253	200	166	141	111	77	45	23	9	3	1
SSW	263	183	133	97	76	49	32	13	6	3	1	0
SW	95	49	25	10	5	1	0	0	0	0	0	0
WSW	61	30	13	5	2	1	0	0	0	0	0	0
W	83	37	22	12	10	6	1	1	0	0	0	0
WNW	61	32	17	10	5	5	3	2	0	0	0	0
NW	91	67	48	31	28	22	8	2	0	0	0	0
NNW	191	111	56	37	23	18	5	0	0	0	0	0
Total	2651	1692	1140	808	601	433	212	82	36	15	6	1

a) Values in each sector are cumulative, beginning with the longest duration.

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 1 of 26)

JANUARY (1973 - 1976): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	12	10	6	6	1	0	1	0	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
NE	4	4	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	9	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	27	7	4	3	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	23	7	9	2	1	3	2	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S	31	14	7	3	2	1	3	2	0	2	0	0	0	0	0	1	0	0	0	0	0	0	0	0
SSW	21	11	8	4	2	2	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
SW	9	12	5	2	2	0	0	1	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
WSW	14	4	5	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	10	12	0	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	15	3	2	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	26	15	8	5	1	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	22	12	7	6	5	2	1	1	1	1	0	1	2	1	0	0	0	0	1	0	0	0	0	2
N	21	7	10	3	1	1	5	1	0	2	2	0	0	0	0	0	0	0	0	0	0	0	0	1

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

JANUARY (1973 - 1976): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3 39	3.35	4.20	3 60	99 99	0.	2 28	0.	3.56	0.	0.	0.	99 99	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2.72	3.05	1.68	3 20	3.56	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 05	2.38	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	1 05	0.	3.23	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2 31	2.28	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 57	2.50	2.15	1 88	2.63	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	3 24	3.72	3.16	3 93	2.44	3.44	5 09	5.20	0.	4 32	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	3.74	4.93	4.55	4 01	4.59	2.84	6.12	5.24	0.	5 62	0.	0.	0.	0.	0.	7.55	0.	0.	0.	0.	0.	0.	0.	0.
SSW	4.78	4.60	5.30	4 94	6.24	6.40	0.	7.24	0.	0.	0.	0.	7.21	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	4.78	4.18	3.99	6.12	4.34	0.	0.	6.70	6.43	0.	0.	5.25	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	3 05	3.80	4.66	2 04	0.	3.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	3 23	3.07	0.	7 32	1.74	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	3 63	2.36	4.00	2.42	0.	3.24	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 86	3.07	2.95	2.73	9.89	3.61	2 94	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3.64	3.96	3.82	4 57	5.21	3.76	6.48	99 99	3.91	6 04	0.	4.89	4 66	9.24	0.	0.	0.	0.	6.58	0.	0.	0.	0.	8.56
N	4 87	3.59	5.14	4 65	99 99	10.45	6 02	4.77	0.	5 25	4.58	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2976

TOTAL NO. OF INVALID OBSERVATIONS = 133 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

FEBRUARY (1973 - 1976): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	9	3	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	1	3	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	3	4	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	8	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	11	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	15	4	2	5	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	22	13	7	1	2	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
S	33	11	8	9	7	3	0	0	1	0	1	0	0	0	1	0	1	0	0	0	0	0	0	0
SSW	21	16	7	5	9	2	3	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
SW	12	9	2	3	2	0	0	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	11	1	1	2	2	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	9	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	14	4	3	3	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	25	18	6	2	1	3	2	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0
NNW	24	16	4	1	2	3	4	2	1	1	1	1	1	0	0	0	0	1	0	0	0	0	0	0
N	15	7	8	2	2	1	0	1	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0	1

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 4 of 26)

FEBRUARY (1973 - 1976): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	2 58	2.55	0.	0.	0.	3.96	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	6.48	2.44	2.74	2 02	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2.14	2.12	2.40	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	2 34	1.86	0.	0.	2.03	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	3.44	4.25	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	3 06	2.86	2.02	3 56	0.	1.60	0.	2.72	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	3 57	3.57	3.75	2 31	7.06	4.55	5 26	3.45	0.	0.	0.	0.	0.	0.	0.	5.57	0.	0.	0.	0.	0.	0.	0.	0.
S	3.49	5.55	4.59	5 56	5.77	6.56	0.	0.	8.53	0.	9.80	0.	0.	0.	4.73	0.	7.31	0.	0.	0.	0.	0.	0.	0.
SSW	4 59	4.25	5.70	5 58	7.09	5.51	7 05	8.76	0.	0.	0.	0.	4.15	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	4 31	5.69	4.38	6.10	4.87	0.	0.	6.16	0.	5.40	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	4.46	4.92	4.95	5 60	6.20	7.40	0.	0.	5.18	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	4.19	11.03	1.12	0.	3.00	0/	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	2 60	4.95	6.40	6 06	5.27	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	4 67	4.33	5.44	8 37	3.84	4.34	7 27	0.	0.	3 81	0.	0.	0.	0.	8 81	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	4 51	4.60	4.32	6 28	5.17	4.67	5.17	6.28	10.46	5 24	.82	4.03	8 37	0.	0.	0.	0.	8.13	0.	0.	0.	0.	0.	0.
N	4 35	4.69	5.71	3 28	6.31	5.10	0.	5.11	9.83	0.	0.	6.15	6.38	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	7.78

TOTAL NO. OF OBSERVATIONS = 2715

TOTAL NO. OF INVALID OBSERVATIONS = 124 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

MARCH (1973 - 1976): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	16	4	3	0	0	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	6	6	6	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	10	3	2	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
E	16	0	3	1	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	15	11	8	2	1	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	27	21	6	5	3	3	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	1	0	0
SSE	25	22	10	11	5	2	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
S	32	14	10	9	3	3	1	1	2	0	0	3	0	0	0	0	0	0	0	0	1	0	0	0
SSW	17	11	2	2	4	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	14	6	1	4	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	13	5	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	5	6	3	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	13	1	3	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	15	10	4	6	3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	14	6	8	1	0	3	3	1	1	2	1	1	0	0	0	1	0	0	0	0	0	0	0	0
N	12	7	6	2	2	1	1	0	1	0	1	0	0	0	2	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 6 of 26)

MARCH (1973 - 1976): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3.72	5.85	6.11	0.	0.	5.06	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2.45	3.42	4.03	3.76	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 35	2.81	3.75	0.	4.24	0.	0.	0.	0.	0.	4.58	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	3 08	0.	3.77	99.99	0.	2.55	0.	4.06	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2 91	3.49	4.27	4 68	7.47	4.07	4.41	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	3 33	4.42	2.90	5.41	6.37	6.03	0.	2.71	0.	0.	6.44	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	3.15	0.	0.
SSE	4 86	5.72	4.20	5 69	5.53	5.79	2 68	5.53	0.	0.	9.65	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	6 21	6.70	5.85	6 60	6.41	7.57	13.41	5.23	8.87	0.	0.	7 95.	0.	0.	0.	0.	0.	0.	0.	0.	8.76	0.	0.	0.
SSW	6 25	7.90	9.11	7 06	8.25	10.36	0.	0.	0.	9.17	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	4.17	5.51	5.58	4 95	10 91	0.	0.	0.	0.	0.	5.23	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	5 32	5.30	5.07	0.	8.35	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	3 56	5.87	5.61	5 53	6.26	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	5 80	1.48	5.48	3 38	0.	0.	0.	0.	6.64	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	4 69	4.66	5.54	6 24	6.40	0.	6 64	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	5 63	3.18	5.60	9.72	0.	6.13	8 69	6.81	5.45	6.71	7.50	7.68	0.	0.	0.	6.27	0.	0.	0.	0.	0.	0.	0.	0.
N	4 38	7.59	6.27	4 09	8.02	4.37	6 31	0.	6.24	0.	5.90	0.	0.	0.	7.72	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS =2976

TOTAL NO. OF INVALID OBSERVATIONS =125

99.99I NDICATES INVALID DATA PO NT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 7 of 26)

APR L(1973-1976):10-METERLEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	6	8	3	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	12	7	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	10	1	2	1	1	0	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	12	12	4	1	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	28	9	6	1	3	2	1	1	1	3	0	0	0	0	1	0	0	0	0	0	0	0	0	0
SE	21	18	10	6	5	3	2	2	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	26	17	9	7	1	5	2	2	1	1	2	0	0	0	0	0	1	0	0	0	0	0	0	1
S	16	12	8	4	3	1	3	3	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0
SSW	15	2	4	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	10	1	2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	6	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	5	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	4	4	2	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	13	7	6	3	2	1	1	1	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0
NNW	17	12	4	1	2	1	3	0	0	1	2	0	0	0	0	1	0	0	0	0	0	0	0	1
N	7	5	3	3	1	2	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

APRIL (1973-1976): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	2.47	3.68	2.31	5 83	6.61	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	3 23	3.28	0.	0.	0.	2.04	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 64	1.39	3.43	6 03	3.67	0.	4 00	0.	0.	4 80	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	4 80	4.17	3.18	2.15	6.73	4.24	4 89	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	3.70	4.52	5.93	2 66	4.84	4.93	6 07	7.66	8.31	8 64	0.	0.	0.	5.97	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	4 34	4.00	3.69	3 81	5.90	6.42	4 04	5.25	4.50	5 24	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	4 22	4.69	5.46	5 83	11.50	7.43	8.10	8.86	7.61	7 93	8.16	0.	0.	0.	0.	0.	7.51	0.	0.	0.	0.	0.	0.	7 63
S	4.46	6.00	6.05	7 30	4.80	7.18	8 52	7.15	0.	6 93	0.	0.	0.	0.	0.	0	0.	6.72	0.	0.	0.	0.	0.	0.
SSW	4.15	2.40	6.66	7 38	5.08	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	3.40	2.92	6.72	0.	0.	0.	0.	0.	6.55	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	5 82	5.27	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2.78	0.	8.10	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	2 05	3.72	6.26	3 64	0.	0.	7 20	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	3 81	4.00	5.35	2 83	4.29	9.76	3 93	6.04	0.	7.18	3.80	0.	0.	0.	0.	0.	0.	7.15	0.	0.	0.	0.	0.	0.
NNW	3 95	4.31	4.56	5 96	4.30	7.73	6 65	0.	0.	6.47	6.57	0.	0.	0.	0.	6.36	0.	0.	0.	0.	0.	0.	0.	7 57
N	2 64	3.78	3.29	3.18	7.81	4.45	0.	6.69	0.	99.99	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2880

TOTAL NO. OF INVALID OBSERVATIONS = 217 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 9 of 26)

MAY (1972 - 1976): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	9	3	3	3	0	0	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	6	4	3	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	16	7	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	10	5	3	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	24	13	4	0	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
SE	25	9	5	2	6	0	3	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	41	17	10	12	2	0	3	1	2	0	0	1	0	0	0	0	0	1	0	0	0	0	0	0
S	31	21	5	8	6	4	3	1	1	4	0	2	1	0	0	0	0	0	0	0	0	0	0	0
SSW	16	5	10	2	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	13	2	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	11	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	3	3	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	12	4	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	14	6	4	2	4	1	0	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0
NNW	18	4	2	2	0	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	16	3	3	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 10 of 26)

MAY (1972 - 1976): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3 01	3.10	3.64	3.16	0.	0.	5 09	1.77	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	3.10	3.20	2.93	6.48	.34	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2.13	2.02	4.17	0.	3.89	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	3 05	1.92	3.73	1.13	2.60	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	3 08	2.32	3.34	0.	2.15	0.	0.	2.61	0.	0.	0.	8.18	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2.71	3.71	3.68	2.79	3.82	0.	3 88	3.02	0.	3 21	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	3.13	4.26	3.41	3.76	3.45	0.	5.18	8.61	4.03	0.	0.	5.39	0.	0.	0.	0.	0.	3 28	0.	0.	0.	0.	0.	0.
S	3 54	4.00	4.39	5 39	3.44	3.97	4.78	4.42	4.00	5 02	0.	5.24	5.41	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	4 26	2.76	4.89	4 53	4.39	5.18	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	2 86	2.59	0.	4 92	7.00	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	2 93	0.	34	4 85	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	5 34	5.13	0.	4.70	0.	0.	0.	0.	0.	0.	4.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	3 68	2.28	4.13	4 36	0.	0.	0.	0.	0.	0.	0.	0.	0.	0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 36	3.26	5.02	4 20	3.92	6.00	0.	0.	0.	0.	5.10	5.23	0.	0.	0.	4.70	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3 68	3.25	4.39	12.84	0.	4.74	3 90	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
N	3.11	5.92	3.20	6 30	3.75	4.34	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2955

TOTAL NO. OF INVALID OBSERVATIONS = 292 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 11 of 26)

JUNE (1972 - 1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	6	2	2	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	11	5	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	7	3	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	16	9	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	26	8	7	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	49	24	9	5	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	51	27	13	4	3	3	3	3	0	0	1	0	0	2	0	0	1	0	0	0	0	0	0	0
S	44	20	11	13	7	5	4	3	2	3	2	1	2	1	0	2	1	0	0	0	0	0	0	0
SSW	22	13	6	6	2	5	1	1	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
SW	11	13	2	1	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	6	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	8	2	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	6	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	7	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	2	5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 12 of 26)

JUNE (1972 - 1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3.77	3.81	3.73	3.48	2.86	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	3 04	3.81	3.35	0.	0.	0.	0.	3.66	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 54	2.34	4.02	3 04	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	3 03	2.69	2.40	0.	0.	0.	0.	4.00	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2 57	3.03	2.74	2 90	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 83	3.20	2.73	2 28	5.30	0.	3 01	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	3.14	3.11	2.89	5 25	4.98	6.29	6.13	4.49	0.	0.	9.07	0.	0.	6.48	0.	0.	3.18	0.	0.	0.	0.	0.	0.	0.
S	3.78	4.46	5.37	4.42	4.06	3.91	6 20	5.25	4.81	5.14	5.71	8.94	5.70	5.33	0.	7.01	6.86	0.	0.	0.	0.	0.	0.	0.
SSW	4 85	5.46	4.39	5 39	5.25	4.88	5 92	5.40	0.	0.	0.	6.40	0.	0.	0.	0.	2.96	0.	0.	0.	0.	0.	0.	0.
SW	2 63	3.69	3.89	4 81	0.	4.05	4.75	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	2.15	0.	2.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2 31	4.54	1.68	5 36	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	1 56	0.	4.58	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	1 56	1.88	1.84	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3.15	0.	1.65	0.	0.	0.	0.	0.	0.	5 07	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
N	2 27	4.05	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2883

TOTAL NO. OF INVALID OBSERVATIONS = 142 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 13 of 26)

JULY (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	14	5	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	8	4	3	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	15	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	7	4	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	21	3	3	2	5	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	45	20	3	5	4	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	52	29	19	14	4	3	3	3	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
S	42	22	16	8	4	6	4	4	1	1	0	0	2	0	0	0	0	0	0	0	0	0	0	0
SSW	24	21	12	4	4	5	1	0	2	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
SW	18	2	4	3	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	12	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	11	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	6	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	8	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	4	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	8	1	4	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 14 of 26)

JULY (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	2 98	2.71	3.03	2 23	1.24	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2.43	3.22	2.99	3 69	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2.17	4.13	2.61	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	2 39	2.72	3.01	1 96	1.65	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2.16	3.57	3.22	4 86	3.56	2.84	0.	3.73	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 96	3.35	2.61	3 55	3.24	3.27	0.	0.	0.	0.	4.44	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	2 92	3.43	3.73	3 84	4.57	3.71	2.71	3.98	0.	4 39	5.39	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	3 24	3.83	3.96	5 06	4.79	3.94	6 60	4.80	4.30	3 94	0.	0.	7.05	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	3.76	4.35	3.97	3.19	3.91	3.89	8 63	0.	5.77	0.	0.	0.	4.98	0.	6 30	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	3 35	2.97	3.22	4.40	5.60	4.62	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	2 51	2.78	3.59	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	1 95	2.28	4.18	0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	2 69	2.47	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	1 57	2.50	1.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	2.42	2.10	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
N	2 62	2.94	4.30	1.79	0.	0.	3 06	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2976

TOTAL NO. OF INVALID OBSERVATIONS = 12 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 15 of 26)

AUGUST (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	8	1	0	0	1	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	13	5	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	7	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	15	2	1	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	29	17	7	5	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	49	19	14	4	3	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	50	27	18	12	2	2	0	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
S	48	34	15	11	7	4	0	1	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
SSW	36	22	9	3	5	0	1	1	1	2	2	1	0	0	0	0	0	0	0	0	0	0	0	0
SW	11	13	3	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	12	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	4	3	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	8	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	4	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 16 of 26)

AUGUST (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	2.46	3.01	0.	0.	4.24	0.	6 30	0.	3.71	4 08	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2.72	3.98	1.68	5 61	2.82	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 32	2.36	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	2.16	1.97	3.14	2 65	0.	3.35	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2.44	2.77	2.95	2 24	2.70	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 67	2.52	2.36	1.79	2.58	2.33	0.	2.46	3.63	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	2 68	3.17	3.85	3.10	4.32	5.30	0.	3.69	5.19	0.	0.	0.	0.	3.64	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	3 67	3.67	3.94	4 21	3.97	4.72	0.	4.81	4.80	0.	0.	0.	0.	0.	0.	5.58	0.	0.	0.	0.	0.	0.	0.	0.
SSW	3 68	3.87	4.02	4 58	3.53	0.	5 01	3.98	5.78	4.10	5.43	4.71	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	2.77	3.30	2.83	3 54	0.	0.	0.	3.80	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	1 86	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	1.44	0.	0.	0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	1 68	.48	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 58	1.97	2.10	2 39	0.	0.	1.73	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	2 57	3.66	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
N	3.12	3.32	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2976

TOTAL NO. OF INVALID OBSERVATIONS = 31 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 17 of 26)

SEPTEMBER (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	25	11	6	2	3	1	1	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
NE	21	8	2	5	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	11	5	2	3	2	0	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	13	6	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	26	5	7	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	26	20	6	4	5	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	30	15	13	8	2	2	1	1	1	1	0	0	0	1	0	0	0	2	0	0	0	0	0	0
S	27	16	4	5	6	5	2	1	1	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0
SSW	9	8	3	4	3	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	4	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	4	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	6	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	10	9	6	5	1	0	1	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
NNW	23	10	3	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
N	37	9	10	4	7	1	1	0	1	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

SEPTEMBER (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3 24	3.67	4.13	3 24	5.16	2.41	6 97	0.	4.86	0.	0.	0.	0.	0.	0.	0.	5.54	0.	0.	0.	0.	0.	0.	0.
NE	2 80	2.59	3.03	4 54	0.	3.17	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2.11	2.85	4.42	2 94	5.27	0.	4.14	0.	0.	2 93	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	2 21	2.00	2.25	1 98	2.81	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2.19	2.49	2.66	3.10	2.12	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 25	2.62	2.93	2 36	2.40	3.69	0.	3.99	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	2 95	3.22	3.88	3 88	4.63	3.01	3.10	4.78	3.39	3 90	0.	0.	0.	3.23	0.	0.	0.	5 25	0.	0.	0.	0.	0.	0.
S	3 86	3.97	4.25	3 98	5.31	4.39	5 53	3.60	7.01	4 31	0.	0.	3.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	3 32	4.86	5.17	5 86	5.40	0.	0.	5.66	5.22	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	4 22	4.07	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	2 23	0.	2.00	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	1 24	0.	0.	0	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	1 95	1.60	2.75	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 37	3.07	2.40	3.18	1.87	0.	2.40	3.09	0.	0.	0.	3.60	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3.18	3.06	6.76	0.	0.	1.98	0.	0.	0.	5 35	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
N	3 05	4.27	4.81	3 05	4.20	6.56	3 01	0.	7.01	0.	0.	0.	6.46	7.50	4 66	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2880

TOTAL NO. OF INVALID OBSERVATIONS = 35 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 19 of 26)

OCTOBER (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	8	6	6	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	12	3	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	6	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	10	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	22	7	2	4	0	2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	37	24	6	3	4	2	2	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
SSE	44	15	15	11	5	7	5	2	3	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
S	35	26	11	7	8	2	1	1	3	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0
SSW	19	10	10	2	3	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	8	6	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	6	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	7	3	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	5	2	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	25	6	3	3	4	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NNW	12	12	2	2	2	1	0	0	0	2	0	0	0	0	1	0	1	0	0	0	0	0	0	0
N	10	7	8	1	2	3	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

OCTOBER (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	2 60	3.07	3.29	6 31	3.10	0.	0.	0.	0.	7.16	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2 33	2.41	0.	2 29	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	1.49	1.74	1.71	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	1 64	2.34	3.65	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	1 91	2.82	2.07	2 22	0.	3.19	0.	0.	3.77	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2.16	2.58	2.27	1 93	2.27	3.32	2 68	0.	0.	0.	0.	0.	0.	3.10	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	2 51	3.24	3.58	3 39	4.20	4.42	4.47	4.01	4.94	0.	0.	5.20	0.	0.	0.	0.	4.22	0.	0.	0.	0.	0.	0.	0.
S	3.71	3.84	4.91	6 02	5.16	4.47	5 50	2.81	4.93	3.49	4.66	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	4.40	4.06	4.30	4 24	3.64	0.	1 24	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	2.79	2.21	4.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	1 96	2.87	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2 01	2.38	0.	0.	0.	0.	0.	1.43	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	1 38	1.79	3.34	1 98	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 50	2.49	2.54	2 64	2.45	0.	0.	3.43	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3.12	3.94	3.77	3 86	4.13	5.58	0.	0.	0.	4 96	0.	0.	0.	0.	4.16	0.	5.24	0.	0.	0.	0.	0.	0.	0.
N	2.13	5.51	3.47	3 33	3.67	4.63	7.15	0.	0.	2 07	5.70	7.49	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2976

TOTAL NO. OF INVALID OBSERVATIONS = 67 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

NOVEMBER (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	10	10	2	0	4	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	10	5	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	12	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	5	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	6	3	5	1	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	27	12	12	5	6	2	2	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	39	9	16	3	6	4	2	3	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
S	25	13	11	5	2	2	1	2	2	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0
SSW	7	15	5	4	1	1	4	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SW	6	6	2	2	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
WSW	5	6	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	11	2	3	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	16	6	7	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	20	17	5	3	7	3	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
NNW	20	10	1	9	2	2	1	2	1	0	0	0	1	0	0	0	1	0	1	0	0	0	0	1
N	12	6	5	3	4	1	1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

NOVEMBER (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3 02	2.11	1.16	0.	3.24	5.41	0.	4.39	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2 37	2.57	1.26	2 63	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 27	3.31	1.93	.62	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	1 89	1.77	0.	2 06	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	1 51	1.76	2.38	2.49	0.	2.31	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2 67	2.42	2.99	3 09	3.49	4.40	4 03	4.36	0.	5.45	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	3 65	4.45	3.33	3 33	5.44	3.90	2 39	4.67	2.93	0.	3.36	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	3 68	4.61	4.07	6 58	4.76	5.88	3 63	7.33	4.98	0.	4.54	0.	4.32	9.74	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	3 91	4.13	5.74	3 91	7.06	6.06	6 85	7.04	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	3 91	3.65	6.70	4.77	5.78	7.14	4 58	0.	0.	0.	0.	6.79	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	4 29	3.33	0.	1 84	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2 06	1.94	3.36	0.	0.	2.05	6 05	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	2.41	2.40	4.33	3.13	0.	4.81	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	3 51	3.81	6.11	4 50	2.82	6.35	4.43	9.21	0.	0.	0.	0.	0.	0.	0.	9.14	0.	0.	0.	0.	0.	0.	0.	0.
NNW	5 55	3.79	3.30	4 68	4.68	6.53	4 96	4.58	5.91	0.	0.	0.	4 90	0.	0.	0.	6.41	0.	7.74	0.	0.	0.	0.	7 00
N	3 97	3.84	3.43	3 58	3.96	4.72	4.44	5.84	4.98	3 60	0.	0.	8.52	6.87	0.	0.	0.	0.	0.	0.	0 99	0.	0.	0.

TOTAL NO. OF OBSERVATIONS = 2880

TOTAL NO. OF INVALID OBSERVATIONS = 21 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 23 of 26)

DECEMBER (1972-1975): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	9	5	1	5	2	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NE	4	2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	4	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
E	4	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	13	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SE	30	14	3	4	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SSE	27	15	9	2	3	3	1	0	0	2	0	0	0	0	1	0	0	0	0	0	0	0	0	0
S	26	22	7	3	5	4	2	3	2	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
SSW	17	8	4	5	2	5	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
SW	13	8	4	2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WSW	11	6	3	0	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	10	4	3	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	15	5	6	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	38	18	7	5	7	4	2	1	0	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0
NNW	25	18	11	7	1	2	2	0	1	0	0	1	1	1	0	0	1	0	0	0	0	0	0	1
N	18	3	7	2	2	2	2	0	0	0	0	0	0	2	0	0	0	0	0	0	1	0	0	0

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 24 of 26)

DECEMBER (1972-1975): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3.40	4.16	3.36	3.76	2.75	0.	0.	3.47	0.	2 34	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NE	2 27	2.90	99.99	4.72	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2.10	1.13	1.11	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	1.13	.58	0.	2 25	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2.17	0.	0.	1 97	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2.17	2.72	2.07	1 87	1.32	2.24	1 32	2.24	1.32	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSE	2 96	3.46	2.79	5 39	4.80	4.45	5.11	0.	0.	3 64	0.	0.	0.	0.	5 90	0.	0.	0.	0.	0.	0.	0.	0.	0.
S	4.75	4.60	5.27	4 06	4.49	5.44	4 97	6.32	5.12	0.	0.	0.	7.61	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
SSW	4 89	4.74	6.23	6.47	5.39	5.81	0.	0.	0.	0.	0.	0.	0.	0.	6 82	0.	0.	0.	0.	0.	0.	0.	0.	0.
SW	4 00	4.82	3.51	5 85	5.82	8.11	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	3 34	4.06	3.25	0.	6.12	0.	4 00	4.43	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2.10	2.85	3.25	2 23	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	2 97	3.30	2.50	5.42	0.	6.05	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	2 95	3.79	5.15	3 26	4.96	4.59	3 82	6.09	0.	3 25	4.08	7.01	0.	0.	6.15.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NNW	3 99	4.63	4.25	5.13	5.77	3.86	4 29	0.	8.28	0.	0.	7.06	7 38	7.37	0.	0.	7.05	0.	0.	0.	0.	0.	0.	8 80
N	4 31	3.11	4.74	6 55	5.75	6.62	7 30	0.	0.	0.	0.	0.	0.	5.67	0.	0.	0.	0.	0.	0.	5.79	0.	0.	0.

TOTAL NO. OFOBSERVATIONS = 2976

TOTAL NO. OF INVALID OBSERVATIONS = 74 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 25 of 26)

ANNUAL (5/72-5/76): 10-METER LEVEL

WIND DIRECTION PERSISTENCE - PASQUILL ALL

1 SECTOR PERSISTENCE

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	131	68	32	20	16	9	6	3	5	3	0	0	1	0	0	0	1	0	0	0	0	0	0	0
NE	108	56	23	17	3	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ENE	106	34	16	6	5	0	3	0	0	2	1	0	0	0	0	0	0	0	0	0	0	0	0	0
E	118	44	18	11	6	4	1	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ESE	230	81	49	21	12	10	3	3	2	3	0	1	0	0	1	0	0	0	0	0	0	0	0	0
SE	379	191	81	51	40	16	11	8	3	4	2	0	0	1	0	0	0	0	0	0	0	1	0	0
SSE	427	212	149	87	35	34	24	19	12	6	6	2	0	4	1	0	4	3	0	0	0	0	0	1
S	390	225	112	84	60	41	24	22	16	13	6	6	8	2	2	4	2	1	0	0	1	0	0	0
SSW	224	144	79	42	36	26	11	7	4	3	2	2	3	0	2	0	1	0	0	0	0	0	0	0
SW	128	80	26	21	9	6	2	4	3	1	1	2	0	0	0	0	0	0	0	0	0	0	0	0
WSW	110	30	17	5	4	4	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
W	85	34	20	9	3	1	1	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
WNW	112	35	27	12	2	3	2	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NW	204	114	53	35	30	15	10	6	0	4	3	3	0	0	2	2	0	1	0	0	0	0	0	0
NNW	194	104	43	29	14	16	16	6	5	9	4	4	5	2	1	2	3	1	2	0	0	0	0	5
N	162	64	63	22	22	12	12	4	4	6	4	2	3	4	3	0	0	0	0	0	1	0	0	2

TABLE 2.3-14
WIND DIRECTION PERSISTENCE AT CPNPP

(Sheet 26 of 26)

ANNUAL (5/72-5/76): 10-METER LEVEL

AVERAGE WIND SPEED (M/SEC)

CONSECUTIVE HOURS

SECTOR	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	>24
NNE	3 09	3.30	3.72	3 64	3.77	4.71	5 62	3.21	3.82	4 53	0.	0.	99.99	0.	0.	0.	5.54	0.	0.	0.	0.	0.	0.	0.
NE	2.75	3.11	3.04	3 87	2.24	2.60	0.	3.66	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ENE	2 22	2.44	3.07	3 08	4.47	0.	4 04	0.	0.	3 86	4.58	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
E	2 67	2.74	3.20	2 09	3.07	3.59	4 89	4.03	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
ESE	2 60	2.99	3.41	2 91	3.83	3.35	4 97	4.67	6.04	8 64	0.	8.18	0.	0.	5 97	0.	0.	0.	0.	0.	0.	0.	0.	0.
SE	2.76	3.13	2.81	3 01	3.65	4.35	3.41	3.72	4.21	4.79	5.44	0.	0.	3.10	0.	0.	0.	0.	0.	0.	0.	3.15	0.	0.
SSE	3 22	3.77	3.67	4.14	5.09	4.90	4 54	5.07	4.52	4 64	7.30	5.29	0.	4.96	5 90	0.	5.19	4 59	0.	0.	0.	0.	0.	7 63
S	3 94	4.45	4.72	5.14	4.74	4.97	6 37	5.52	5.69	4 96	5.85	7.21	5.84	7.54	5 54	6.79	7.08	6.72	0.	0.	8.76	0.	0.	0.
SSW	4.41	4.55	5.01	5.17	5.75	5.37	6 30	6.69	5.63	5.79	5.43	5.55	5.45	0.	6 56	0.	2.96	0.	0.	0.	0.	0.	0.	0.
SW	3 53	3.96	4.20	5 21	5.95	5.43	4 66	5.71	6.47	5.40	5.23	6.02	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WSW	3 36	3.84	3.95	3 98	6.72	5.66	4 00	4.43	5.18	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
W	2 63	3.89	4.42	5 85	3.67	2.05	6 05	1.43	0.	0.	4.91	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
WNW	3 09	2.75	4.22	4 03	5.27	4.70	5 86	0.	6.64	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.	0.
NW	3.15	3.61	4.29	4 02	4.11	5.13	4.42	5.16	0.	4 51	4.33	5.28	0.	0.	7.48	6.92	0.	7.15	0.	0.	0.	0.	0.	0.
NNW	3 99	4.00	4.54	5 58	4.89	5.05	5 91	5.70	6.80	5.72	5.36	5.91	6.10	8.31	4.16	6.31	6.23	8.13	7.19	0.	0.	0.	0.	8.14
N	3 62	4.66	4.64	3 89	4.97	5.57	5 69	5.60	7.01	4 20	5.19	6.82	7.12	6.43	6.70	0.	0.	0.	0.	0.	5.79	0.	0.	7.78

TOTAL NO. OF OBSERVATIONS = 35043

TOTAL NO. OF INVALID OBSERVATIONS = 1267 99.99 INDICATES INVALID DATA POINT

TABLE 2.3-15
VALUES OF MEAN, AVERAGE AND EXTREME DAILY MAXIMUM, AND AVERAGE AND EXTREME DAILY MINIMUM SURFACE TEMPERATURES (°F) AT FORT WORTH (1931 - 1960)

Month	Mean	Average Daily Maximum	Extreme Maximum[a]	Average Daily Minimum	Extreme Minimum[a]
January	46	56	88(1969)	35	4(1964)
February	49	60	87(1969)	39	12(1971)
March	56	67	91(1971)	44	19(1965)
April	65	76	95(1972)	54	30(1973)
May	73	83	96(1967)	63	42(1971)
June	82	92	105(1972)	71	51(1964)
July	85	96	105(1964)[b]	75	59(1972)[b]
August	85	96	108(1964)	75	56(1967)
September	78	89	102(1963)	68	46(1971)[b]
October	68	79	96(1963)	56	37(1966)[b]
November	55	66	88(1965)[b]	44	24(1970)
December	48	58	84(1966)	37	10(1963)
Annual	66	77	108(1964)	55	4(1964)

a) Period of record is July 1963 through 1973; year of occurrence given in parenthesis.

b) Record also occurred in earlier year(s).

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 1 of 13)

JANUARY (1973 - 1976)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	5.1	-.9	63.9	3 3	262	5.6	244	1.6	F
2	4.9	-.8	64.3	3.4	263	5.7	246	1.6	F
3	4.5	-.9	65.3	3 2	269	5.5	246	1.6	F
4	4.2	-1.0	66.4	3 2	281	5.6	255	1.6	F
5	4.0	-1.0	67.3	3.4	283	5.6	256	1.5	F
6	3.7	-1.2	67.5	3.4	296	5.8	266	1.6	F
7	3.5	-1.2	67.7	3.4	304	5.7	273	1.6	F
8	3.6	-.9	67.8	3 3	308	5.5	274	1.3	F
9	5.1	-.5	64.2	3 8	298	5.5	270	5	E
10	6.9	-.2	59.3	4 6	282	5.9	268	- 0	E
11	8.5	-.3	54.8	4 9	263	6.1	258	- 2	E
12	10 0	-.5	49.9	5 0	257	6.3	253	- 3	D
13	11.1	-.9	45.2	5 2	253	6.5	248	-.4	D
14	11.8	-.8	43.7	5 2	256	6.4	251	- 3	D
15	12.1	-.9	43.1	5 0	257	6.3	251	- 3	D
16	12 2	-.8	43.1	4 8	246	6.1	238	- 2	E
17	11.4	-.7	45.4	4 2	232	5.8	221	2	E
18	10.1	-.6	50.0	3 5	206	5.7	191	8	F
19	8.9	-.3	54.5	3 3	200	5.9	176	1.2	F
20	8.0	-.3	56.7	3 3	221	5.8	193	1.3	F
21	7.3	-.5	58.3	3 3	233	5.9	208	1.4	F
22	6.8	-.6	60.0	3 3	246	5.7	225	1.4	F
23	6.2	-.8	60.4	3 3	256	5.7	234	1.5	F
24	6.0	-.4	63.3	3.4	258	5.6	237	1.4	F
ABSOLUTE MAX	28 5	20.7	99.9	14.2		17.4			
AVG DAILY MAX	12 8	3.1	76.9	6.7		9.0			
MEAN	7.4	-.7	57.3	3 9	263	5.8	244	8	F
CL MATIC MEAN	7.4	- 6	58.3	4.1		5.8			
AVG DAILY MIN	2.0	-4.4	39.6	1 5		2.5			
ABSOLUTE M N	-14 3	-20.8	12.5	0 0		0.0			
STANDARD DEV	7.5	7.3	20.9	2.4		2.9			
VALID OBS	2677	2638	2520	2718	2642	2449	2447	2651	2651
INVALID OBS	299	338	456	258	334	527	529	325	325
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	90 0	88 6	84.7	91.3	88.8	82 3	82 2	89.1	89.1

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 2 of 13)

FEBRUARY (1973 - 1976)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	8.3	1.4	61.9	4 0	245	6.3	224	1.3	F
2	7.9	1.5	63.6	3 9	240	6.1	227	1.3	F
3	7.4	1.5	65.3	3 8	245	5.9	233	1.2	F
4	7.1	1.5	66.5	3.7	251	5.7	241	1.2	F
5	6.7	1.4	67.7	3 6	265	5.5	250	1.3	F
6	6.5	1.4	69.8	3.7	275	5.8	261	1.4	F
7	6.1	1.0	70.5	3.7	279	5.7	268	1.3	F
8	6.7	1.2	68.0	3.7	281	5.3	271	.7	E
9	8.3	1.5	64.0	4 5	270	5.5	263	.1	E
10	10 2	1.9	58.4	5 3	261	5.9	252	- 2	E
11	11.6	2.1	53.1	5 6	252	6.2	246	- 3	D
12	12.7	1.5	48.5	5 9	258	6.8	263	-.4	D
13	14 0	1.1	45.0	5 9	257	6.8	258	-.4	D
14	14 6	1.3	42.0	6 0	255	6.7	254	- 5	D
15	15 2	1.2	40.4	6 0	250	7.0	252	-.4	D
16	15 3	1.1	40.1	5 9	239	6.7	245	- 3	D
17	14 8	1.3	41.8	5.1	224	6.3	222	-.1	E
18	13 6	1.3	44.8	4 3	191	5.9	194	3	E
19	12 2	1.4	49.0	4 0	182	6.1	180	8	F
20	11.2	1.4	52.3	4 0	186	6.2	168	1.0	F
21	10 6	1.4	54.7	4.1	219	6.3	182	1.1	F
22	9.8	1.3	57.0	4.1	218	6.3	197	1.1	F
23	9.3	1.3	58.2	4 0	222	6.2	207	1.1	F
24	8.8	1.3	59.3	3 9	247	6.2	223	1.3	F
ABSOLUTE MAX	29 5	19.2	100 0	16.1		16.4			
AVG DAILY MAX	15 9	4.8	73.6	7 5		9.3			
MEAN	10.4	1.4	55.6	4 5	250	6.2	239	6	E
CL MATIC MEAN	10.7	1.3	56.2	4 6		6.3			
AVG DAILY MIN	5.4	-2.1	38.7	1 8		3.4			
ABSOLUTE M N	-6.2	-14 6	9.7	0 0		.4			
STANDARD DEV	7.2	7.0	21.3	2.7		2.8			
VALID OBS	2156	2124	2054	2619	2489	2125	2124	2362	2362
INVALID OBS	556	588	658	93	223	587	588	350	350
TOTAL OBS	2712	2712	2712	2712	2712	2712	2712	2712	2712
DATA RECOVERY	79 5	78 3	75.7	96.6	91.8	78.4	78.4	87.1	87.1

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 3 of 13)

MARCH (1973 - 1976)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	11.8	5.2	64.4	4 6	179	6.6	172	1.1	F
2	11.2	5.0	67.2	4 5	188	6.3	181	1.1	F
3	10 8	4.9	68.0	4 3	197	6.2	186	1.1	F
4	10.4	4.8	68.7	4 2	198	6.0	188	1.1	F
5	10.1	4.8	70.1	4.1	195	5.9	192	1.1	F
6	9.8	4.8	71.6	4 0	184	5.7	182	1.1	F
7	9.8	4.7	71.7	4 2	186	5.8	194	1.0	F
8	10 5	5.0	69.2	4.7	188	5.8	198	.4	E
9	11.8	5.3	64.5	5.4	186	6.3	199	.1	E
10	13 2	5.2	59.3	5.7	189	6.6	202	-.1	E
11	14 5	4.9	55.8	6 0	195	6.9	211	- 2	E
12	15 5	4.9	52.3	6.1	200	6.8	212	- 2	E
13	16 8	4.8	48.2	6 2	204	7.1	212	- 3	D
14	17.7	4.6	44.9	6 2	205	7.0	208	- 3	D
15	18.1	4.6	43.7	6.4	192	7.1	200	- 3	D
16	18.4	4.3	43.2	6 2	188	7.2	196	- 2	E
17	18 2	4.4	43.8	6.1	185	7.3	196	-.1	E
18	17.4	4.4	45.3	5 6	167	7.0	177	.1	E
19	16.1	4.8	49.9	5 0	156	6.8	150	5	E
20	15 0	5.0	53.5	4 8	156	7.0	144	8	F
21	14.1	5.3	57.5	4 8	163	7.2	157	1.0	F
22	13 3	5.2	59.8	4 6	166	6.8	157	1.1	F
23	12 6	5.2	61.4	4 6	181	6.7	174	1.2	F
24	11.9	4.7	63.3	5 0	181	6.8	175	1.2	F
ABSOLUTE MAX	36.4	22.4	100 0	18.3		17 9			
AVG DAILY MAX	19.4	9.4	78.7	8 3		10 0			
MEAN	13 8	4.9	57.9	5.1	186	6.6	186	5	E
CL MATIC MEAN	14 0	4.9	59.0	5 2		6.8			
AVG DAILY MIN	8.5	.4	39.2	2 0		3.5			
ABSOLUTE M N	-2.4	-12.1	11 0	0 0		0.0			
STANDARD DEV	6.6	6.9	22.6	2 8		2.8			
VALID OBS	2380	2380	2302	2806	2769	2338	2309	2569	2569
INVALID OBS	596	596	674	170	207	638	667	407	407
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	80 0	80 0	77.4	94.3	93.0	78 6	77 6	86.3	86.3

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 4 of 13)

APRIL (1973 - 1976)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	15 5	11 0	74.7	4 2	151	6.5	139	9	F
2	15 3	11.1	75.0	4.1	155	6.3	145	9	F
3	15 0	11.4	76.8	3 9	161	6.0	152	9	F
4	14 6	11 0	77.6	3.7	162	5.9	153	9	F
5	14.1	10.6	77.9	3 6	168	5.6	151	9	F
6	13.7	10.2	77.7	3.7	159	5.7	143	1.0	F
7	13 8	10.2	77.5	3 8	153	5.6	142	.7	E
8	14 8	10.6	75.2	4.1	143	5.4	144	2	E
9	16.1	11 0	71.2	4 9	145	6.0	144	- 0	E
10	17.4	11 0	65.8	5 2	148	6.3	143	- 2	E
11	18 5	10.6	61.8	5 6	152	6.8	150	- 2	E
12	19 3	10.5	58.8	5 6	149	6.7	145	- 3	D
13	20 0	10.2	56.9	5.7	144	6.8	140	-.4	D
14	20.7	10.1	54.4	6.1	141	7.1	141	-.4	D
15	21 2	10.3	52.8	6 0	142	7.2	140	-.4	D
16	21.4	10.7	52.8	6 0	142	7.2	136	- 3	D
17	21.4	10.9	53.3	5 9	135	7.2	131	- 2	E
18	20 9	10.8	55.3	5.7	129	7.2	125	-.1	E
19	19 9	10.9	58.6	4 9	127	6.6	121	2	E
20	19 0	11 0	61.8	4.4	124	6.5	116	5	E
21	18.1	11 0	64.6	4.4	132	6.7	119	6	E
22	17 3	10.8	66.2	4.4	134	6.7	122	8	F
23	16 6	10.9	69.9	4.4	142	6.7	132	8	F
24	15 5	11 2	74.9	4 8	144	6.3	125	9	F
ABSOLUTE MAX	32.1	24.8	100 0	15.2		18.1			
AVG DAILY MAX	21 9	13 9	85.4	7 8		9.6			
MEAN	17 6	10.7	65.6	4 8	143	6.5	136	3	E
CL MATIC MEAN	17 2	10.4	67.2	4 9		6.6			
AVG DAILY MIN	12.4	6.8	48.9	2.1		3.6			
ABSOLUTE M N	.7	-7.4	11.2	0 0		.5			
STANDARD DEV	5.3	7.4	22.7	2 6		2.9			
VALID OBS	2235	2181	2007	2619	2563	2124	2123	2611	2611
INVALID OBS	645	699	873	261	317	756	757	269	269
TOTAL OBS	2880	2880	2880	2880	2880	2880	2880	2880	2880
DATA RECOVERY	77 6	75.7	69.7	90.9	89.0	73.7	73.7	90.7	90.7

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 5 of 13)

MAY (1972 - 1976)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	19 2	17.0	83.9	3 0	173	5.2	167	1.2	F
2	18 8	16.7	85.0	2 9	183	5.2	173	1.3	F
3	18 5	16.3	85.8	3.1	190	5.3	180	1.3	F
4	18 0	16.3	86.7	2 8	190	4.9	184	1.2	F
5	17 6	16.2	86.5	2 8	185	4.9	179	1.3	F
6	17.4	16.0	86.0	2.7	196	4.7	181	1.3	F
7	17.4	16.1	85.3	2 8	206	4.6	188	1.1	F
8	18 5	17.0	84.4	3.1	208	4.6	206	3	E
9	19 9	17.1	80.5	3 8	187	5.0	197	-.1	E
10	21.1	17.2	77.2	4 2	177	5.5	183	-.4	D
11	22.4	17.2	71.9	4 2	176	5.4	181	- 5	D
12	23.7	16.3	66.3	4.1	171	5.3	180	- 5	D
13	24 3	15.6	61.3	4.1	163	5.3	172	- 6	D
14	25 0	16.6	60.1	4 2	158	5.6	171	- 6	D
15	25 6	16.5	58.6	4 3	150	5.7	158	- 6	D
16	25 8	16.5	56.1	4.4	153	6.0	159	- 5	D
17	25.7	16.7	57.4	4 3	146	5.8	156	-.4	D
18	25 3	16.9	60.0	4 2	141	5.8	147	- 2	E
19	24 3	17.1	63.9	3.7	132	5.5	140	- 0	E
20	23 0	17.2	69.7	3.1	140	5.3	144	5	E
21	21 9	17.4	75.3	3.1	137	5.5	144	8	F
22	21.1	17.6	79.3	3 0	142	5.5	142	1.0	F
23	20.4	17.5	82.3	3 0	156	5.3	153	1.2	F
24	17 9	14.8	79.4	3 2	163	5.2	123	1.1	F
ABSOLUTE MAX	33 8	25.2	100 0	12.0		15.4			
AVG DAILY MAX	25 8	19 2	93.0	5 9		8.2			
MEAN	21.4	16.7	73.1	3 5	163	5.3	166	.4	E
CL MATIC MEAN	21.4	16.4	73.4	3 6		5.6			
AVG DAILY MIN	17 0	13.6	53.9	1.4		3.0			
ABSOLUTE M N	9.6	-3.0	.7	0 0		.4			
STANDARD DEV	4.8	5.6	18.5	1 9		2.3			
VALID OBS	1476	1431	1261	2510	2399	2143	1533	2691	2691
INVALID OBS	1476	1521	1691	442	553	809	1419	261	261
TOTAL OBS	2952	2952	2952	2952	2952	2952	2952	2952	2952
DATA RECOVERY	50 0	48 5	42.7	85.0	81.3	72 6	51 9	91.2	91.2

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 6 of 13)

JUNE (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	23 6	21.0	84.9	3 2	172	5.9	160	9	F
2	23.1	21.1	86.1	3.1	174	5.4	154	9	F
3	22 6	21.0	88.6	3.1	178	5.2	163	1.0	F
4	22 3	21.1	90.1	3 0	178	4.7	160	1.0	F
5	22 0	21.0	91.3	2 8	180	4.5	151	1.1	F
6	21 8	20.9	92.3	2 6	168	4.3	144	1.1	F
7	22 0	21.2	92.2	2 6	169	4.1	144	1.0	F
8	23.1	21.6	88.0	3.1	179	4.4	157	2	E
9	24.4	21.7	84.7	3 8	182	5.3	167	- 2	E
10	25 8	21.5	77.3	4.1	183	5.8	166	- 5	D
11	27.1	21.2	69.7	4 2	179	5.8	165	- 6	D
12	28.1	21.1	64.7	4.4	172	6.1	162	- 5	D
13	28 9	20.8	60.3	4 5	167	6.1	162	- 5	D
14	29.7	20.6	58.4	4.7	166	6.1	160	-.7	D
15	29 9	20.4	57.2	4.7	165	6.3	162	- 6	D
16	30 2	19.5	54.7	4.7	165	6.1	155	- 6	D
17	30.4	20.8	56.8	4.7	159	6.0	152	- 5	D
18	30 2	20.9	58.0	4 6	160	6.3	153	- 3	D
19	29 0	20.9	61.2	4.4	157	6.1	146	-.1	E
20	27 8	21.3	68.6	3 8	157	5.7	143	6	E
21	26 6	21.5	73.2	3 5	159	5.8	141	.7	E
22	25.7	21.7	77.7	3 5	161	6.1	143	8	F
23	25 0	21.9	81.9	3.4	164	6.1	146	9	F
24	23 9	21.0	83.7	3.4	169	6.2	155	9	F
ABSOLUTE MAX	34 2	26.9	99.8	12.6		14 5			
AVG DAILY MAX	30.4	22.7	94.8	5 9		7.9			
MEAN	25 9	21.1	74.0	3.7	168	5.6	154	2	E
CL MATIC MEAN	25 9	20 3	73.4	3.7		5.9			
AVG DAILY MIN	21.4	17.9	51.9	1 5		3.9			
ABSOLUTE M N	14 2	-2.2	10.2	0 0		.3			
STANDARD DEV	4.0	3.5	16.4	2 2		2.6			
VALID OBS	714	698	638	2768	2649	904	697	2755	2755
INVALID OBS	2166	2182	2242	112	231	1976	2183	125	125
TOTAL OBS	2880	2880	2880	2880	2880	2880	2880	2880	2880
DATA RECOVERY	24 8	24 2	22.2	96.1	92.0	31.4	24 2	95.7	95.7

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 7 of 13)

JULY (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	25 0	20.8	75.3	2 8	174	4.8	176	8	F
2	24.7	20.9	78.5	2.7	179	4.7	182	8	F
3	24.1	21.1	81.0	2 6	181	4.4	183	8	F
4	23.7	21.2	82.7	2 5	188	4.3	189	8	F
5	23 5	21.0	82.8	2 3	189	4.2	197	8	F
6	23 2	20.9	83.6	2 0	184	3.9	198	9	F
7	23 3	21.1	84.3	2.1	184	3.5	199	6	E
8	24 5	21.8	82.4	2 8	191	3.4	201	-.1	E
9	26 3	21.9	76.5	3 5	199	4.1	206	-.4	D
10	27 9	21.7	69.5	3.7	197	4.2	201	- 5	D
11	29 0	21.4	63.8	3.7	190	4.4	196	- 6	D
12	30 0	20.9	57.6	3.7	184	4.3	185	-.7	D
13	30.7	20.4	53.5	3 8	172	4.5	171	-.7	D
14	31.4	20.0	50.05	4 5	164	4.8	162	-.7	D
15	31.7	19.6	49.0	4 5	155	4.8	154	-.7	D
16	31.7	19.7	48.5	4 5	151	5.0	149	- 6	D
17	31 6	19.5	48.9	4 2	149	5.0	145	- 5	D
18	30 9	19.7	50.0	4 2	149	5.2	148	- 3	D
19	30 0	20.2	53.6	4 0	151	5.1	149	-.1	E
20	28 8	20.5	59.3	3 5	150	5.2	147	.4	E
21	27 6	20.5	64.4	3 3	153	5.4	150	.7	E
22	26 6	20.6	68.5	3.1	159	5.3	157	8	F
23	26 0	20.7	70.9	3 0	165	5.1	166	8	F
24	25.4	20.8	73.3	3.1	170	4.9	174	9	F
ABSOLUTE MAX	38 2	26.2	99.8	14.2		14 5			
AVG DAILY MAX	32.4	22 6	88.7	6.1		7.4			
MEAN	27.4	20.7	66.3	3 3	169	4.6	172	.1	E
CL MATIC MEAN	27 6	20 5	67.4	3.7		4.9			
AVG DAILY MIN	22 9	18.4	46.1	1 3		2.4			
ABSOLUTE M N	17 9	10.2	19.6	0 0		0.0			
STANDARD DEV	4.0	2.9	19.1	2 2		1.9			
VALID OBS	2040	1966	1837	2950	2908	2114	2042	2916	2916
INVALID OBS	936	1010	1139	26	68	862	934	60	60
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	68 5	66.1	61.7	99.1	97.7	71 0	68 6	98.0	98.0

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 8 of 13)

AUGUST (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	24 9	20.5	76.8	2.4	171	4.7	174	8	F
2	24.4	20.6	78.9	2 3	174	4.5	180	8	F
3	23 9	20.7	81.0	2 2	176	4.3	184	8	F
4	23 6	20.8	83.3	2.1	179	4.1	189	9	F
5	23 2	20.8	84.2	2 0	179	4.0	191	1.0	F
6	22 9	20.8	85.5	1 8	172	3.7	188	1.0	F
7	22.7	21.0	86.5	1 8	164	3.6	186	9	F
8	23 9	21.7	84.8	2 2	169	3.3	181	0	E
9	25 8	21.9	78.8	3 0	189	3.8	196	-.4	D
10	27 8	21.5	68.9	3 6	190	4.4	193	- 5	D
11	29.1	21.1	61.4	3.7	183	4.5	186	-.7	D
12	30.4	20.5	55.5	3.7	176	4.5	177	-.7	D
13	31.1	20.0	51.8	3 8	167	4.6	164	- 8	C
14	31 6	19.6	48.9	3 9	156	4.8	154	-.7	D
15	31 9	19.4	48.3	4 0	158	5.1	160	- 6	D
16	32.1	19.1	45.9	3 8	152	4.8	156	- 6	D
17	32.1	19.1	46.4	4 0	152	5.1	155	-.4	D
18	31 3	19.4	49.6	4.1	151	5.5	151	- 2	E
19	30 3	19.5	52.9	3 8	152	5.5	150	2	E
20	28.7	19.7	58.7	3 2	152	5.5	152	6	E
21	27 6	19.9	62.9	3.1	156	5.5	154	8	F
22	26 6	20.1	68.2	2 8	160	5.3	159	8	F
23	26.1	20.2	70.4	2 5	165	4.9	162	8	F
24	25 5	20.5	74.3	2 5	164	5.0	168	8	F
ABSOLUTE MAX	38.7	26.1	99.9	12.3		12 2			
AVG DAILY MAX	32.7	22.4	89.8	5.4		7.3			
MEAN	27.4	20.4	66.3	3 0	166	4.6	170	2	E
CL MATIC MEAN	27 6	20 2	66.9	3 2		4.8			
AVG DAILY MIN	22 5	18.1	43.9	1.1		2.3			
ABSOLUTE M N	16 6	8.2	18.6	0 0		0.0			
STANDARD DEV	4.1	2.5	19.2	1 6		1.9			
VALID OBS	2370	2290	2183	2812	2794	2628	2423	2901	2901
INVALID OBS	606	686	793	164	182	348	553	75	75
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	79 6	76 9	73.4	94.5	93.9	88 3	81.4	97.5	97.5

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 9 of 13)

SEPTEMBER (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	22 3	18.5	77.5	2.7	136	4.9	154	8	F
2	21 8	18.5	80.4	2.7	142	4.8	162	8	F
3	21.4	18.5	82.2	2.7	141	4.8	161	8	F
4	20 9	18.5	83.8	2 5	138	4.5	167	9	F
5	20 5	18.5	85.7	2.4	112	4.3	165	9	F
6	20 2	18.6	86.4	2 3	56	4.1	160	9	F
7	20.1	18.5	87.0	2 3	48	4.0	157	9	F
8	20 6	19.1	86.9	2.4	98	3.9	164	3	E
9	22 2	19.5	83.3	3 2	153	4.1	182	- 2	E
10	23 9	19.4	74.7	3 8	165	4.9	187	-.4	D
11	25 3	19.1	68.2	3 9	156	5.1	184	- 5	D
12	26 6	18.5	61.7	4 0	154	5.2	180	- 6	D
13	27 5	18.2	57.4	4.1	137	5.2	171	- 6	D
14	28 3	18.0	53.5	4 3	115	5.3	150	- 6	D
15	28 8	17.7	51.4	4 2	81	5.4	131	- 5	D
16	29 0	17 2	50.2	4 3	81	5.5	122	- 5	D
17	28 8	16 9	49.8	4 2	83	5.5	122	-.4	D
18	28.1	17 2	51.8	4.1	83	5.7	118	- 2	E
19	26 8	17.3	55.9	3.4	94	5.5	121	2	E
20	25 5	17.5	61.1	3.1	100	5.8	123	6	E
21	24.4	17.5	65.1	3 0	97	5.7	122	.7	E
22	23 6	17.6	69.3	3 0	105	5.5	128	8	F
23	22.7	17.6	71.7	2 9	118	5.3	138	8	F
24	21.7	17.7	76.0	2 8	133	5.1	141	.7	E
ABSOLUTE MAX	37 0	26.3	100 0	11.4		14.7			
AVG DAILY MAX	29 6	20.7	92.2	5 5		7.3			
MEAN	24 2	18.2	68.6	3 3	110	5.0	147	2	E
CL MATIC MEAN	24 5	18 2	70.4	3 3		5.0			
AVG DAILY MIN	19 5	15.6	48.7	1 2		2.7			
ABSOLUTE M N	8.3	-2.0	19.8	0 0		0.0			
STANDARD DEV	5.5	4.8	18.8	1 9		2.2			
VALID OBS	1894	1846	1715	2812	2748	1929	1892	2836	2836
INVALID OBS	986	1034	1165	68	132	951	988	44	44
TOTAL OBS	2880	2880	2880	2880	2880	2880	2880	2880	2880
DATA RECOVERY	65 8	64.1	59.5	97.6	95.4	67 0	65.7	98.5	98.5

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 10 of 13)

OCTOBER (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	16.7	12.0	72.5	2 5	170	5.2	165	1.5	F
2	16 2	12.1	73.7	2 5	178	5.0	171	1.4	F
3	15 9	12.1	75.1	2.4	175	4.8	177	1.4	F
4	15.4	12.1	77.8	2.4	182	4.8	185	1.5	F
5	15.1	11 8	79.2	2 3	168	4.5	184	1.6	F
6	14 8	11 8	81.5	2 3	151	4.3	179	1.6	F
7	14 5	11 9	82.9	2 2	163	4.1	177	1.6	F
8	14 9	12.5	83.2	2 2	166	4.0	177	1.2	F
9	16 6	13.3	79.8	2 8	165	4.1	179	2	E
10	18 6	13.7	72.2	3.4	184	4.7	188	- 2	E
11	20 3	13.6	66.5	3 8	189	5.2	190	- 3	D
12	21.7	13.6	61.3	4.1	186	5.5	188	- 5	D
13	22.7	13.3	56.3	4.1	178	5.5	181	- 5	D
14	23 5	12.9	52.7	4 3	170	5.6	173	- 5	D
15	24.1	12.8	50.4	4.4	167	5.8	167	- 5	D
16	24 2	12.7	49.9	4 3	161	5.6	160	-.4	D
17	23 9	12.6	51.2	4.1	153	5.6	154	- 2	E
18	22 9	12.6	52.6	3 6	144	5.4	145	2	E
19	21 3	12.6	58.1	3 0	135	5.6	136	8	F
20	20.1	12.6	63.2	2 9	140	5.9	140	1.2	F
21	19.1	12.5	66.3	2 9	147	6.0	143	1.2	F
22	18 2	12.3	68.4	2 8	150	5.9	149	1.2	F
23	17.4	12.1	70.0	2 8	161	5.7	154	1.3	F
24	17 0	12.0	70.7	2 5	162	5.4	160	1.4	F
ABSOLUTE MAX	33.1	24.8	100 0	11.0		13.4			
AVG DAILY MAX	24 5	15 3	87.7	5.4		7.6			
MEAN	19 0	12.6	66.6	3.1	164	5.2	165	.7	E
CL MATIC MEAN	19 2	12 5	68.6	3 2		5.1			
AVG DAILY MIN	13 8	9.8	49.6	1 0		2.6			
ABSOLUTE M N	6.1	- 6	21.2	0 0		0.0			
STANDARD DEV	6.0	6.0	20.4	2 0		2.3			
VALID OBS	1935	1909	1753	2897	2787	1947	1935	2884	2884
INVALID OBS	1041	1067	1223	79	189	1029	1041	92	92
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	65 0	64.1	58.9	97.3	93.6	65.4	65.0	96.9	96.9

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 11 of 13)

NOVEMBER (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	10 2	5.0	69.9	3.4	244	5.7	207	1.2	F
2	9.7	4.7	71.0	3 3	245	5.5	202	1.3	F
3	9.5	4.7	71.7	3 3	257	5.6	206	1.2	F
4	9.2	4.5	71.9	3 2	272	5.4	230	1.3	F
5	9.1	4.7	72.9	3 3	257	5.7	225	1.3	F
6	9.0	4.6	74.4	3 2	271	5.4	218	1.3	F
7	8.9	4.8	75.0	3.1	267	5.4	231	1.1	F
8	9.9	5.1	72.4	3 5	272	5.3	241	5	E
9	11.7	5.5	66.3	4.4	260	5.9	249	- 0	E
10	13 3	5.3	58.6	4 8	234	6.4	229	- 2	E
11	14 9	5.4	53.9	5 0	229	6.6	216	- 3	D
12	16.1	5.0	51.1	5 0	231	6.6	213	-.4	D
13	16 9	5.3	49.6	5 0	222	6.9	208	- 3	D
14	17 2	4.7	46.8	5.1	225	6.6	213	- 3	D
15	17 0	4.4	46.7	5 0	222	6.4	213	- 2	E
16	16.7	4.7	48.3	4.4	215	6.0	192	- 0	E
17	15 6	4.8	50.8	3 6	182	5.5	163	.4	E
18	14.4	5.0	54.8	3 3	177	5.7	148	8	F
19	13 3	5.1	58.6	3 3	179	6.0	146	1.1	F
20	12 6	5.1	60.9	3 3	185	5.9	156	1.2	F
21	12 0	4.9	61.7	3 5	227	5.9	164	1.2	F
22	11.6	5.0	63.7	3 5	215	6.1	170	1.3	F
23	10 9	4.8	65.3	3.4	229	5.9	179	1.3	F
24	9.9	4.2	67.5	3.4	237	5.6	198	1.3	F
ABSOLUTE MAX	30.7	23.2	100 0	13.8		17 8			
AVG DAILY MAX	17.7	8.6	80.9	6 5		9.0			
MEAN	12 5	4.9	61.4	3 9	231	5.9	202	.7	E
CL MATIC MEAN	12 9	5.2	62.9	4 0		6.0			
AVG DAILY MIN	8.1	1.9	44.8	1.4		3.0			
ABSOLUTE M N	-4.2	-16 0	14.3	0 0		0.0			
STANDARD DEV	7.1	8.9	22.5	2 3		2.8			
VALID OBS	1902	1883	1798	2864	2784	1903	1903	2785	2785
INVALID OBS	978	997	1082	16	96	977	977	95	95
TOTAL OBS	2880	2880	2880	2880	2880	2880	2880	2880	2880
DATA RECOVERY	66 0	65.4	62.4	99.4	96.7	66.1	66.1	96.7	96.7

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 12 of 13)

DECEMBER (1972 - 1975)

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	5.5	-1.0	62.8	3.4	265	6.2	249	1.5	F
2	5.0	-1.1	64.6	3 3	269	6.0	242	1.6	F
3	4.6	-1.1	66.1	3 2	269	6.0	248	1.7	F
4	4.3	-1.1	67.4	3 2	260	6.0	243	1.7	F
5	3.9	-1.1	69.2	3 2	274	5.7	242	1.7	F
6	3.8	-1.1	69.0	3 2	282	5.7	255	1.7	F
7	3.6	-1.1	69.6	3 2	283	5.6	254	1.7	F
8	4.3	-.8	68.5	3.4	284	5.6	252	1.2	F
9	6.1	-.5	62.4	3 9	287	5.4	256	3	E
10	7.9	-.3	57.3	4 5	259	5.9	238	0	E
11	10 0	-.2	51.1	4 9	253	6.3	234	-.1	E
12	11.6	-.5	44.4	5 0	251	6.5	237	- 2	E
13	12 6	-.7	41.1	5.1	254	6.6	236	- 2	E
14	13.1	-1.0	39.6	5.1	256	6.6	243	- 2	E
15	13 5	-1.0	39.1	5 0	252	6.5	231	-.1	E
16	13.1	-1.0	40.0	4 6	236	6.3	218	0	E
17	11.8	-1.0	43.4	3 8	218	6.0	195	.4	E
18	10 2	-.8	47.7	3 5	222	6.3	174	1.0	F
19	9.1	-.8	51.2	3 3	239	6.4	183	1.1	F
20	8.2	-.8	53.8	3.4	276	6.5	194	1.3	F
21	7.5	-.8	56.3	3 5	267	6.4	191	1.4	F
22	7.1	-.7	58.1	3 6	259	6.5	210	1.4	F
23	6.7	-.7	59.4	3 6	261	6.4	219	1.4	F
24	6.4	-.8	60.1	3 3	272	6.4	230	1.5	F
ABSOLUTE MAX	28 5	17.5	99.9	12.7		17.7			
AVG DAILY MAX	13 8	2.5	74.7	6 5		9.3			
MEAN	7.9	-.8	55.8	3 8	262	6.2	233	9	F
CL MATIC MEAN	8.1	- 9	55.9	4 0		6.2			
AVG DAILY MIN	2.5	-4.2	37.1	1.4		3.1			
ABSOLUTE M N	-8.1	-16 0	11 3	0 0		0.0			
STANDARD DEV	7.1	6.8	20.0	2 3		2.9			
VALID OBS	2137	2141	2129	2801	2791	2093	2063	2890	2890
INVALID OBS	839	835	847	175	185	883	913	86	86
TOTAL OBS	2976	2976	2976	2976	2976	2976	2976	2976	2976
DATA RECOVERY	71 8	71.9	71.5	94.1	93.8	70 3	69.3	97.1	97.1

TABLE 2.3-16
STATISTICS AND DIURNAL VARIATION OF METEROLOGICAL PARAMETERS

(Sheet 13 of 13)

MAY 15 1972 TO MAY 14 1976

DATA SOURCE: ON-SITE
TABLE GENERATED: 10/04/76. 09 29.33.

COMANCHE PEAK STEAM ELECTRIC STATION
GLEN ROSE, TEXAS
TEXAS UTILIT ES GENERAT NG CO.
DAMES AND MOORE JOB NO: 4486-025-12

METEROLOGICAL PARAMETERS (HEIGHTS N METERS)

	DRY BULB	DEW PO NT	REL HUMID	WIND SPEED	WIND DIR.	WIND SPEED	W ND DIR.	DELTA TEMP	STAB CLASS
	10.00	10.00	10.00	10 00	10 00	60.00	60.00	10.00 60.00	10 00 60 00
HOUR	DEG C	DEG C	PCT	M/SEC	DEG	M/SEC	DEG	DEG C	
1	14 9	9.8	70.9	3 3	182	5.6	178	1.1	F
2	14 5	9.8	72.5	3 2	188	5.5	184	1.1	F
3	14.1	9.7	73.9	3.1	193	5.3	189	1.1	F
4	13.7	9.6	75.1	3 0	199	5.2	196	1.2	F
5	13.4	9.6	75.9	3 0	203	5.1	200	1.2	F
6	13.1	9.5	76.6	2 9	203	4.9	201	1.2	F
7	13 0	9.5	77.0	2 9	206	4.8	203	1.1	F
8	13 8	10.1	75.8	3 2	204	4.7	205	5	E
9	15 5	10.5	71.8	3 9	203	5.1	207	- 0	E
10	17.1	10.5	65.4	4.4	201	5.5	203	- 3	D
11	18 5	10.3	59.9	4 6	199	5.8	201	-.4	D
12	19.7	10.0	55.0	4.7	198	5.8	201	- 5	D
13	20 6	9.6	51.2	4 8	192	6.0	195	- 5	D
14	21 2	9.3	48.5	4 9	184	6.0	187	- 5	D
15	21 5	9.2	47.3	4 9	176	6.1	179	-.4	D
16	21 6	9.1	46.8	4 8	166	6.1	169	- 3	D
17	21.1	9.2	48.2	4 5	155	5.9	157	- 2	E
18	20 2	9.3	50.9	4 2	148	6.0	149	2	E
19	19 2	9.5	54.9	3 8	146	5.9	144	5	E
20	18.1	9.7	58.9	3 6	149	6.0	146	8	F
21	17.1	9.7	62.1	3 5	155	6.0	151	1.0	F
22	16 5	9.7	65.0	3 5	161	6.0	157	1.0	F
23	15.7	9.5	66.8	3.4	171	5.8	165	1.1	F
24	15.1	9.5	68.8	3 5	176	5.7	174	1.1	F
ABSOLUTE MAX	38.7	26.9	100 0	18.3		18.1			
AVG DAILY MAX	22.4	12 9	83.7	6 5		8.5			
MEAN	17.1	9.7	62.8	3 8	178	5.6	178	5	E
CL MATIC MEAN	17 3	9.7	64.0	4 0		5.7			
AVG DAILY MIN	12 2	6.6	44.2	1 5		2.9			
ABSOLUTE M N	-14 3	-20.8	.7	0 0		0.0			
STANDARD DEV	9.4	10.3	21.4	2.4		2.6			
VALID OBS	23916	23487	22197	33176	32323	24697	23491	32851	32851
INVALID OBS	11124	11553	12843	1864	2717	10343	11549	2189	2189
TOTAL OBS	35040	35040	35040	35040	35040	35040	35040	35040	35040
DATA RECOVERY	68 3	67 0	63.3	94.7	92.2	70 5	67 0	93.8	93.8

TABLE 2.3-17
MONTHLY AVERAGE RELATIVE HUMIDITY (PERCENT) AT FORT WORTH
(1964-1973)

	Hour (CST)			
Month	00	06	12	18
January	75	81	60	59
February	72	79	57	53
March	71	80	57	51
April	75	85	60	55
May	80	88	61	58
June	74	86	57	52
July	68	81	51	46
August	70	83	53	48
September	80	89	61	59
October	78	86	56	59
November	76	83	56	60
December	77	81	60	61
Annual	75	84	57	55

TABLE 2.3-18
MONTHLY AND ANNUAL MEAN, AND EXTREME MAXIMUM AND MINIMUM DEWPOINT TEMPERATURES (°F) AT FORT WORTH (1970-1975)

Month	Mean	Extreme Maximum	Extreme Minimum
January	34	67	01
February	36	63	08
March	44	70	12
April	52	75	08
May	61	76	36
June	67	79	44
July	68	78	50
August	68	78	51
September	66	77	40
October	57	75	30
November	43	72	02
December	37	68	05
Annual	53	79	01

TABLE 2.3-19
MONTHLY AND ANNUAL AVERAGE VALUES IN THE SITE AREA

Month	Precipitation (Inches)[a]	Number of Days With Precipitation			Snowfall (Inches)[b]	Number of Days With Heavy Fog[b]
		.01 In.[b]	.10 In.[c]	.50 In.[d]		
Jan	2.00	7	3	1	1.6	2
Feb	2.19	7	4	1	0.7	2
Mar	2.01	7	3	1	0.4	1
Apr	3.57	9	6	3	0.0	1
May	4.78	8	6	3	0.0	<1/2
June	2.84	6	4	2	0.0	<1/2
July	2.06	5	3	1	0.0	0
Aug	1.82	5	2	1	0.0	<1/2
Sept	2.80	7	4	1	0.0	<1/2
Oct	2.82	6	5	2	0.0	1
Nov	2.21	6	3	2	T	1
Dec	2.19	7	4	1	0.2	3
Annual	31.30	79	47	19	2.9	11

a) Based on 30 years of record (1931-1960) at Cleburne, Dublin, Fort Worth, Rainbow, and Stephenville (Period of record at Rainbow and Stephenville varies from 25-31 years).

b) Based on 20 years of record (1954-1973) at Fort Worth.

c) Based on 7 years of record (1954-1960) at Cleburne, Dublin, Fort Worth, Rainbow, and Stephenville.

d) Based on 10 years of record (1951-1960) at Cleburne, Dublin, Fort Worth, Rainbow, and Stephenville.

TABLE 2.3-20
MONTHLY PRECIPITATION EXTREMES (Inches) AT SELECTED STATIONS

Month	Fort Worth (1954-1973)		Cleburne (1956-1960)		Dublin (1951-1960)	
	Maximum[a]	Minimum[a]	Maximum[a]	Minimum[a]	Maximum[a]	Minimum[a]
January	3.60(1968)	0.19(1971)	2.65(1960)	0.11(1959)	3.34(1960)	0.04(1953)
February	6.20(1965)	0.15(1963)	3.34(1959)	0.78(1954)	2.16(1958)	0.52(1953)
March	6.39(1968)	0.10(1972)	4.50(1957)	0.15(1956)	2.63(1953)	0.12(1956)
April	12.19(1957)	0.92(1959)	12.19(1957)	1.38(1951)	8.42(1957)	1.62(1951)
May	12.64(1957)	1.06(1961)	8.60(1957)	2.50(1954)	10.89(1956)	2.18(1959)
June	6.94(1962)	0.40(1964)	7.55(1951)	0.03(1952)	7.08(1959)	0.10(1952)
July	11.13(1973)	0.09(1965)	3.22(1960)	0.29(1956)	7.88(1959)	0.22(1956)
August	6.85(1970)	0.01(1973)	7.85(1953)	0.02(1957)	2.54(1953)	0.02(1957)
September	9.25(1964)	0.23(1956)	6.71(1958)	0.15(1956)	3.71(1957)	0.07(1956)
October	9.22(1959)	0.20(1955)	9.60(1959)	0.05(1952)	9.42(1959)	0.00(1952)
November	6.23(1964)	0.20(1970)	8.73(1952)	0.12(1955)	6.27(1957)	T (1955)
December	6.99(1971)	0.21(1955)	5.85(1960)	0.00(1951)	4.11(1960)	0.12(1951)

a) Year of occurrence is given in parentheses.

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 1 of 13)

JANUARY (1973 - 1976)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	12	6	2					21	26.9	11.8
NNE		1	5			1				7	9.0	10.9
NE			3	1						4	5.1	11.0
ENE												
E		1	2	2						5	6.4	10.0
ESE	1	1								2	2.5	4.5
SE		1	2							3	3.8	8.7
SSE		1								1	1.3	5.0
S		2	3	1		1				7	9.0	10.9
SSW		1	1			1				3	3.8	11.7
SW												
WSW		2	1		1					4	5.1	9.8
W		1	1							2	2.6	7.5
WNW			2	1						3	3.8	10.3
NW		2		2						4	5.1	10.3
NNW		2	2	3		2				9	11.5	13.2
CALM	3									3	3.8	
TOTAL	4	16	34	16	3	5				78	100.0	10.4
PERCENT	5.1	20.5	43.6	20.5	3.8	6.4					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 2 of 13)

FEBRUARY (1973 - 1976)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	5	4	2	2				14	31.1	13.4
NNE		2	1							3	6.7	7.3
NE		2				1				3	6.7	11.0
ENE												
E	1									1	2.2	3.0
ESE												
SE		1								1	2.2	6.0
SSE		2	1							3	6.7	7.0
S	1	1	1		1					4	8.9	8.5
SSW												
SW			1							1	2.2	8.0
WSW												
W		2								2	4.4	5.5
WNW			1							1	2.2	9.0
NW				1		2				3	6.7	20.3
NNW	1			1	1	2	1			6	13.3	16.7
CALM	3									3	6.7	
TOTAL	6	11	10	6	4	7	1			45	100.0	11.0
PERCENT	13.3	24.4	22.2	13.3	8.9	15.6	2.2				100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 3 of 13)

MARCH (1973 - 1976)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	1	2	2	1				7	11.1	14.1
NNE			1	1						2	3.2	12.0
NE	1	1	2							4	6.3	6.5
ENE			5	2						7	11.1	11.4
E		2	1		1					4	6.3	9.5
ESE			1							1	1.6	9.0
SE	1	2	2	1	1	1				8	12.7	11.0
SSE		1	2	2	1	2				8	12.7	15.1
S		1	3	1	2	2				9	14.3	13.9
SSW				1						1	1.6	13.0
SW												
WSW												
W						1				1	1.6	20.0
WNW												
NW		1	1		1					3	4.8	10.3
NNW		1			1	2				4	6.3	16.5
CALM	4									4	6.3	
TOTAL	6	10	19	10	9	9				63	100.0	11.7
PERCENT	9.5	15.9	30.2	15.9	14.3	14.3					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 4 of 13)

APRIL (1973 - 1976)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	2	1						4	4.9	9.3
NNE		1	2	1						4	4.9	9.8
NE		2	2							4	4.9	7.5
ENE	1		1	1						3	3.7	9.3
E		2	2	2						6	7.3	9.7
ESE		2	2	1						5	6.1	9.4
SE			2			2				4	4.9	14.5
SSE	1	1	5	3	5	2				17	20.7	13.2
S		4	7	3	6	2				22	26.8	12.7
SSW		1								1	1.2	6.0
SW			1							1	1.2	12.0
WSW		1								1	1.2	5.0
W												
WNW				3						3	3.7	14.7
NW				1	1					2	2.4	15.5
NNW												
CALM	5									5	6.1	
TOTAL	7	15	26	16	12	6				82	100.0	11.0
PERCENT	8.5	18.3	31.7	19.5	14.6	7.3					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 5 of 13)

MAY (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N			3	1						4	9.3	11.8
NNE		1		1	1					3	7.0	12.7
NE		1		1						2	4.7	10.5
ENE			1		1					2	4.7	12.0
E	1									1	2.3	3.0
ESE		1	1							2	4.7	7.5
SE			1							1	2.3	10.0
SSE		1	1	2						4	9.3	11.3
S			4	1	3					8	18.6	12.6
SSW		1		3	1					5	11.6	13.0
SW												
WSW												
W		1	2							3	7.0	9.0
WNW												
NW			1			1				2	4.7	16.0
NNW		1	2	1						4	9.3	9.8
CALM	2									2	4.7	
TOTAL	3	7	16	10	6	1				43	100.0	10.9
PERCENT	7.0	16.3	37.2	23.3	14.0	2.3					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 6 of 13)

JUNE (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1								1	3.1	5.0
NNE												
NE		1								1	3.1	7.0
ENE		1								1	3.1	6.0
E		1	2			1				4	12.5	11.8
ESE		1	2	1		1				5	15.6	12.6
SE		1	4			2				7	21.9	12.0
SSE			1	1		1				3	9.4	14.3
S			1		1					2	6.3	12.5
SSW		1								1	3.1	7.0
SW							1			1	3.1	29.0
WSW		1								1	3.1	5.0
W		1	1	1						3	9.4	9.7
WNW												
NW												
NNW		1								1	3.1	7.0
CALM	1									1	3.1	
TOTAL	1	10	11	3	1	5	1			32	100.0	11.2
PERCENT	3.1	31.3	34.4	9.4	3.1	15.6	3.1				100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 7 of 13)

JULY (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	1							2	6.7	8.5
NNE	1	1	1	1	1					5	16.7	9.8
NE												
ENE												
E		1								1	3.3	5.0
ESE			1							1	3.3	8.0
SE	1									1	3.3	3.0
SSE	1		2	1		1				5	16.7	11.6
S		2	2							4	13.3	7.8
SSW		1								1	3.3	5.0
SW		1	1							2	6.7	7.5
WSW			1			1				2	6.7	14.0
W						1				1	3.3	21.0
WNW					1	1				2	6.7	18.5
NW			1							1	3.3	8.0
NNW												
CALM	2									2	6.7	
TOTAL	5	7	10	2	2	4				30	100.0	9.5
PERCENT	16.7	23.3	33.3	6.7	6.7	13.3					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 8 of 13)

AUGUST (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		2			1					3	9.7	10.3
NNE			2	1						3	9.7	11.3
NE			1							1	3.2	12.0
ENE			1							1	3.2	9.0
E		1	1	1						3	9.7	9.3
ESE	1									1	3.2	3.0
SE		3	2							5	16.1	7.2
SSE		1	2	1						4	12.9	10.3
S			2	1						3	9.7	12.7
SSW			1							1	3.2	8.0
SW												
WSW	1									1	3.2	3.0
W		1	1	1						3	9.7	11.0
WNW												
NW												
NNW		1				1				2	6.5	13.5
CALM												
TOTAL	2	9	13	5	1	1				31	100.0	9.8
PERCENT	6.5	29.0	41.9	16.1	3.2	3.2					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 9 of 13)

SEPTEMBER (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N	2	2	8	2	2	4				20	26.3	12.4
NNE			6	1						7	9.2	11.0
NE		3	3	1	2	1				10	13.2	11.2
ENE	1	1		1	1					4	5.3	9.8
E	1	3	2	1						7	9.2	8.1
ESE		2	1							3	3.9	8.3
SE	1	2	1							4	5.3	5.5
SSE		1	1	1						3	3.9	10.3
S			1	2	1					4	5.3	13.3
SSW		2	1							3	3.9	6.7
SW												
WSW			1							1	1.3	9.0
W		1								1	1.3	6.0
WNW		1	1							2	2.6	7.0
NW		2		1						3	3.9	9.3
NNW			3							3	3.9	9.3
CALM	1									1	1.3	
TOTAL	6	20	29	10	6	5				76	100.0	10.1
PERCENT	7.9	26.3	38.2	13.2	7.9	6.6					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 10 of 13)

OCTOBER (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N			2	1	1					4	6.8	13.3
NNE		1	2							3	5.1	8.3
NE		1	2							3	5.1	8.7
ENE					1					1	1.7	18.0
E		1	2	1						4	6.8	9.8
ESE		3	1							4	6.8	6.3
SE		2	2							4	6.8	8.0
SSE		1	4	2	1	1				9	15.3	11.9
S			5		1					6	10.2	10.8
SSW			1							1	1.7	8.0
SW	1									1	1.7	3.0
WSW			1							1	1.7	10.0
W			1							1	1.7	9.0
WNW	1		1	1	1					4	6.8	11.0
NW					1					1	1.7	16.0
NNW		4	3							7	11.9	7.1
CALM	5									5	8.5	
TOTAL	7	13	27	5	6	1				59	100.0	9.0
PERCENT	11.9	22.0	45.8	8.5	10.2	1.7					100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 11 of 13)

NOVEMBER (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N			1		1					2	3.3	14.0
NNE			1	1						2	3.3	12.0
NE		1								1	1.6	5.0
ENE		1	2	1						4	6.6	9.5
E		2								2	3.3	6.5
ESE	1	4								5	8.2	5.4
SE	1	2	2							5	8.2	7.2
SSE			8	3						11	18.0	10.5
S		3	1	1	3					8	13.1	11.5
SSW												
SW		1								1	1.6	6.0
WSW												
W		1		2						3	4.9	11.0
WNW		1		3						4	6.6	12.3
NW			2	1						3	4.9	12.3
NNW		3		1		1	1			6	9.8	13.3
CALM	4									4	6.6	
TOTAL	6	19	17	13	4	1	1			61	100.0	9.6
PERCENT	9.8	31.1	27.9	21.3	6.6	1.6	1.6				100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 12 of 13)

DECEMBER (1972 - 1975)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N		1	11	2	1	1				16	21.1	11.1
NNE	1	4	7							12	15.8	7.9
NE		4	3	1						8	10.5	7.9
ENE		2	1		1					4	5.3	10.0
E	1	2	2							5	6.6	6.2
ESE	1									1	1.3	3.0
SE		3	1							4	5.3	6.5
SSE		2	2	2	1					7	9.2	10.4
S				3	1					4	5.3	14.5
SSW												
SW												
WSW		1								1	1.3	5.0
W												
WNW	1		1							2	2.6	5.5
NW			2							2	2.6	10.5
NNW			4	1	2			1		8	10.5	14.9
CALM	2									2	2.6	
TOTAL	6	19	34	9	6	1		1		76	100.0	9.5
PERCENT	7.9	25.0	44.7	11.8	7.9	1.3		1.3			100.0	

TABLE 2.3-21
PRECIPITATION WIND ROSE FORT WORTH, TEXAS

(Sheet 13 of 13)

ANNUAL (5/72 - 4/76)

	SPEED GROUPS IN MPH											AVG
DIR	0-3	4-7	8-12	13-15	16-18	19-24	25-31	32-38	39 & GR	TOTAL	PERCENT	SPEED
N	2	11	46	19	12	8				98	14.5	12.0
NNE	2	11	28	7	2	1				51	7.5	9.9
NE	1	16	16	4	2	2				41	6.1	9.2
ENE	2	5	11	5	4					27	4.0	10.4
E	4	16	14	7	1	1				43	6.4	8.7
ESE	4	14	9	2		1				30	4.4	7.8
SE	4	17	19	1	1	5				47	7.0	9.1
SSE	2	11	29	18	8	7				75	11.1	11.8
S	1	13	30	13	19	5				81	12.0	12.1
SSW		7	4	4	1	1				17	2.5	9.8
SW	1	2	3				1			7	1.0	10.4
WSW	1	5	4		1	1				12	1.8	8.7
W		8	6	4		2				20	3.0	10.2
WNW	2	2	6	8	2	1				21	3.1	11.4
NW		5	7	6	3	3				24	3.6	12.8
NNW	1	13	14	7	4	8	2	1		50	7.4	12.7
CALM	32									32	4.7	
TOTAL	59	156	246	105	60	46	3	1		676	100.0	10.3
PERCENT	8.7	23.1	36.4	15.5	8.9	6.8	.4	.1			100.0	

TABLE 2.3-22
MONTHLY AND ANNUAL STABILITY CLASS PERCENT FREQUENCY
DISTRIBUTIONS AT FORT WORTH (1957-1971)

	Stability Class[a]					
Month	A	B	C	D	E	F and G
January	0.0	1.5	5.5	68.3	13.4	11.2
February	0.1	2.4	6.1	66.9	14.2	10.4
March	0.0	2.4	5.6	69.9	13.7	8.4
April	0.5	2.9	8.8	68.4	10.8	8.6
May	0.8	5.4	12.3	59.2	11.9	10.5
June	1.1	6.9	16.1	48.0	16.3	11.5
July	1.6	10.5	19.3	33.0	19.6	16.2
August	1.9	10.8	17.3	32.2	20.7	17.1
September	0.9	7.0	11.1	43.3	19.0	18.6
October	0.1	4.2	10.2	47.6	17.6	20.3
November	0.0	2.9	8.2	56.3	15.8	16.8
December	0.0	1.4	6.8	63.3	15.2	13.3
Annual	0.6	4.9	10.6	54.6	15.7	13.6

a) Definition of Pasquill Stability Classes

Pasquill Stability Class	Identified in Table as	Definition
1	A	Extremely Unstable
2	B	Unstable
3	C	Slightly Unstable
4	D	Neutral
5	E	Slightly Stable
6	F	Stable
7	G	Extremely Stable

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 1 of 13)

Stability Persistence Summary
January (1973 - 1976)

NUMBER OF CONSECUTIVE HOURS	NUMBER OF HOURS PASQU LL STAB LITY CLASS						
	-A-	-B-	-C-	-D-	-E-	-F-	-G-
2	6	4	14	456	851	327	436
3	3	0	5	361	681	217	361
4	1	0	1	292	563	146	301
5	0	0	0	247	466	100	246
6	0	0	0	211	393	71	203
7	0	0	0	181	330	47	168
8	0	0	0	160	277	30	136
9	0	0	0	140	230	17	106
10	0	0	0	131	190	8	79
11	0	0	0	125	159	2	58
12	0	0	0	119	134	0	40
13	0	0	0	114	112	0	24
14	0	0	0	109	94	0	11
15	0	0	0	105	79	0	3
16	0	0	0	102	66	0	0
17	0	0	0	100	55	0	0
18	0	0	0	98	47	0	0
19	0	0	0	96	41	0	0
20	0	0	0	94	35	0	0
21	0	0	0	92	30	0	0
22	0	0	0	90	26	0	0
23	0	0	0	88	23	0	0
24	0	0	0	86	21	0	0
>24	0	0	0	84	20	0	0

169 INVALID HOUR(S).

Diurnal Distribution of Stability
January (1973 - 1976)

T ME OF DAY (LST)[a]	NUMBER OF HOURS PASQU LL STAB LITY CLASS							
	-A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
0	0	0	0	6	41	31	32	110
1	0	0	0	5	45	29	38	117
2	0	0	0	6	45	27	40	118
3	0	0	0	8	41	30	38	117
4	0	0	0	10	40	29	38	117
5	0	0	0	10	42	26	39	117
6	0	0	0	13	42	26	36	117
7	0	0	0	13	39	25	39	116
8	0	0	0	16	46	22	32	116
9	0	0	0	31	58	9	16	114
10	1	1	2	49	51	8	3	115
11	2	2	7	59	41	2	2	115
12	3	7	7	61	33	3	1	115
13	3	6	10	61	33	2	1	116
14	2	1	11	66	34	2	1	117
15	1	0	6	64	44	3	0	118
16	0	0	1	49	65	4	0	119
17	0	0	0	17	95	7	0	119
18	0	0	0	9	55	46	9	119
19	0	0	0	7	38	50	24	119
20	0	0	0	7	36	44	32	119
21	0	0	0	7	38	39	35	119
22	0	0	0	7	39	32	41	119
23	0	0	0	7	43	30	39	119
ALL	12	17	44	588	1084	526	536	2807

169 NVALID HOUR(S).

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 2 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary February (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability February (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	25	5	10	374	871	404	242	0	1	0	0	3	39	40	21	104
3	16	1	4	271	704	284	191	1	1	0	0	2	42	36	29	110
4	11	0	1	196	595	199	151	2	0	0	0	2	46	36	26	110
5	8	0	0	137	508	137	116	3	0	0	0	4	45	38	23	110
6	5	0	0	89	438	96	88	4	0	1	0	3	44	38	24	110
7	2	0	0	52	382	67	65	5	0	0	0	6	43	36	25	110
8	1	0	0	31	330	46	45	6	0	0	0	3	41	40	26	110
9	0	0	0	18	285	31	31	7	0	0	0	5	38	42	25	110
10	0	0	0	13	247	19	20	8	0	0	0	8	61	25	16	110
11	0	0	0	9	210	10	13	9	0	1	0	27	64	10	8	110
12	0	0	0	7	176	5	6	10	2	1	1	42	57	6	1	110
13	0	0	0	6	146	2	3	11	3	2	6	60	33	6	0	110
14	0	0	0	5	123	0	1	12	5	4	10	59	30	0	1	109
15	0	0	0	4	104	0	0	13	7	6	7	61	24	3	0	108
16	0	0	0	3	88	0	0	14	7	4	4	65	26	2	0	108
17	0	0	0	2	75	0	0	15	6	3	4	63	31	1	0	108
18	0	0	0	1	65	0	0	16	2	3	2	53	47	2	0	109
19	0	0	0	0	56	0	0	17	1	1	0	21	85	1	0	109
20	0	0	0	0	48	0	0	18	1	1	0	6	79	20	2	109
21	0	0	0	0	42	0	0	19	1	0	0	1	60	34	13	109
22	0	0	0	0	37	0	0	20	0	0	0	4	45	43	17	109
23	0	0	0	0	32	0	0	21	0	1	0	3	42	46	17	109
24	0	0	0	0	27	0	0	22	0	0	0	2	42	43	22	109
>24	0	0	0	0	23	0	0	23	2	0	0	2	40	43	22	109
96 NVALID HOUR(S).								ALL	39	28	34	505	1104	591	318	2619
									96 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 3 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary March (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability March (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	28	4	10	381	1076	301	210	0	0	0	0	2	55	29	25	111
3	18	1	2	272	900	188	163	1	0	0	0	5	51	26	28	110
4	10	0	0	185	773	113	128	2	1	0	0	2	54	28	26	111
5	5	0	0	125	674	68	99	3	1	0	0	4	52	28	24	109
6	2	0	0	77	590	40	76	4	0	0	0	5	56	27	22	110
7	1	0	0	44	518	25	56	5	0	0	0	5	54	31	20	110
8	0	0	0	24	455	17	40	6	0	0	0	6	47	38	19	110
9	0	0	0	11	400	12	26	7	0	0	0	3	54	34	19	110
10	0	0	0	6	351	8	14	8	0	0	1	15	66	19	9	110
11	0	0	0	5	308	6	4	9	2	0	0	33	66	6	4	111
12	0	0	0	4	272	4	0	10	0	0	4	46	53	7	1	111
13	0	0	0	3	242	2	0	11	2	7	5	55	36	5	1	111
14	0	0	0	2	214	1	0	12	4	6	3	62	32	2	1	110
15	0	0	0	1	188	0	0	13	9	4	8	54	35	2	1	113
16	0	0	0	0	162	0	0	14	9	2	5	62	31	1	2	112
17	0	0	0	0	141	0	0	15	8	2	5	60	35	2	2	114
18	0	0	0	0	123	0	0	16	3	4	3	51	50	3	1	115
19	0	0	0	0	109	0	0	17	2	1	2	32	76	2	0	115
20	0	0	0	0	96	0	0	18	0	0	0	16	94	6	0	116
21	0	0	0	0	85	0	0	19	0	0	0	1	84	26	4	115
22	0	0	0	0	76	0	0	20	0	0	0	1	64	43	7	115
23	0	0	0	0	67	0	0	21	0	0	0	1	60	37	15	113
24	0	0	0	0	60	0	0	22	0	0	0	1	52	42	19	114
>24	0	0	0	0	54	0	0	23	0	0	0	1	55	32	25	113
	287 INVALID HOUR(S).							ALL	41	26	36	523	1312	476	275	2689
									287 NVALID HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 4 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary April (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability April (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	60	13	11	326	1154	273	103	0	0	0	0	4	60	28	15	107
3	38	4	1	229	1008	186	71	1	0	0	0	4	61	31	13	109
4	22	1	0	169	892	130	51	2	0	0	0	2	57	37	13	109
5	11	0	0	127	791	89	35	3	0	0	0	2	60	33	14	109
6	6	0	0	91	703	58	24	4	0	0	0	2	58	32	17	109
7	2	0	0	62	625	33	15	5	0	0	0	2	55	37	15	109
8	0	0	0	41	562	19	7	6	0	0	0	2	54	37	16	109
9	0	0	0	27	508	12	3	7	0	0	0	4	69	24	11	108
10	0	0	0	16	459	9	1	8	1	0	1	22	67	13	4	108
11	0	0	0	9	413	6	0	9	1	0	3	42	54	7	2	109
12	0	0	0	5	372	3	0	10	4	5	4	43	46	4	2	108
13	0	0	0	3	333	1	0	11	10	1	8	43	40	3	2	107
14	0	0	0	2	296	0	0	12	12	6	8	41	38	3	0	108
15	0	0	0	1	264	0	0	13	15	6	3	46	35	4	0	109
16	0	0	0	0	235	0	0	14	15	10	6	40	34	3	0	108
17	0	0	0	0	211	0	0	15	18	5	3	43	40	0	0	109
18	0	0	0	0	192	0	0	16	8	8	9	37	47	2	0	111
19	0	0	0	0	177	0	0	17	4	2	12	37	54	3	0	112
20	0	0	0	0	165	0	0	18	1	1	1	31	77	1	0	112
21	0	0	0	0	154	0	0	19	2	1	0	10	90	9	0	112
22	0	0	0	0	145	0	0	20	3	1	0	3	76	25	3	111
23	0	0	0	0	138	0	0	21	1	3	0	2	69	34	2	111
24	0	0	0	0	131	0	0	22	0	1	1	2	66	32	9	111
>24	0	0	0	0	124	0	0	23	0	0	0	4	63	30	12	109
262 INVALID HOUR(S).								ALL	95	50	59	468	1370	432	150	2624
									262 NVALID HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 5 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary May (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability May (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	37	21	30	519	832	320	159	0	0	0	0	0	38	26	13	77
3	25	9	8	368	656	212	111	1	0	0	0	0	52	43	20	115
4	16	4	0	263	534	139	77	2	0	0	0	0	50	40	26	116
5	9	2	0	183	435	92	48	3	0	0	0	0	48	43	25	116
6	4	0	0	129	359	65	29	4	0	1	0	1	49	38	26	115
7	1	0	0	85	295	43	16	5	0	0	0	0	48	41	25	114
8	0	0	0	56	245	25	8	6	0	0	0	0	52	34	28	114
9	0	0	0	32	204	16	4	7	0	0	0	2	52	39	20	113
10	0	0	0	12	173	9	1	8	1	0	0	17	74	21	1	114
11	0	0	0	2	143	4	0	9	1	0	2	47	59	4	1	114
12	0	0	0	0	121	1	0	10	3	5	3	67	36	0	0	114
13	0	0	0	0	101	0	0	11	4	7	7	74	22	0	0	114
14	0	0	0	0	84	0	0	12	5	10	9	73	15	0	0	112
15	0	0	0	0	72	0	0	13	11	10	13	64	14	0	0	112
16	0	0	0	0	61	0	0	14	13	7	20	58	14	0	0	112
17	0	0	0	0	52	0	0	15	12	11	17	60	14	0	0	114
18	0	0	0	0	45	0	0	16	6	7	15	68	18	0	0	114
19	0	0	0	0	40	0	0	17	3	5	9	71	28	0	0	116
20	0	0	0	0	36	0	0	18	1	0	1	64	49	1	0	116
21	0	0	0	0	33	0	0	19	0	0	1	33	77	5	0	116
22	0	0	0	0	31	0	0	20	0	0	0	0	87	20	4	111
23	0	0	0	0	29	0	0	21	0	0	0	1	67	43	5	116
24	0	0	0	0	27	0	0	22	0	0	0	0	55	47	14	116
>24	0	0	0	0	25	0	0	23	0	0	0	0	56	41	19	116
248 NVALID HOUR(S).								ALL	60	63	97	700	1074	486	227	2707
									248 INVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 6 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary June (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability June (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	91	7	24	615	782	401	84	0	1	0	0	1	50	52	9	113
3	72	0	9	464	612	293	61	1	0	0	0	3	53	51	8	115
4	58	0	3	343	492	223	43	2	0	1	0	1	53	52	9	116
5	47	0	0	253	408	167	30	3	0	2	0	0	50	51	13	116
6	38	0	0	182	345	122	20	4	2	0	0	1	54	43	15	115
7	31	0	0	131	288	83	12	5	2	0	0	0	53	44	16	115
8	24	0	0	87	235	53	7	6	1	0	1	1	49	43	20	115
9	19	0	0	53	187	32	4	7	2	0	0	3	65	33	11	114
10	16	0	0	27	147	15	2	8	2	0	0	29	66	10	3	110
11	13	0	0	8	116	8	0	9	2	0	0	63	46	4	0	115
12	11	0	0	2	90	5	0	10	7	1	3	82	21	1	0	115
13	9	0	0	0	68	3	0	11	8	6	7	77	18	0	0	116
14	7	0	0	0	48	2	0	12	12	5	12	75	8	1	1	114
15	5	0	0	0	35	1	0	13	12	5	15	71	10	1	1	115
16	3	0	0	0	26	0	0	14	14	11	11	70	8	1	0	115
17	1	0	0	0	18	0	0	15	16	1	14	71	12	1	0	115
18	0	0	0	0	12	0	0	16	12	3	9	70	18	1	0	113
19	0	0	0	0	10	0	0	17	9	3	7	69	27	0	0	115
20	0	0	0	0	8	0	0	18	7	2	1	64	38	3	0	115
21	0	0	0	0	6	0	0	19	5	1	0	43	59	8	0	116
22	0	0	0	0	4	0	0	20	3	0	0	4	92	15	1	115
23	0	0	0	0	2	0	0	21	1	0	0	2	70	39	3	115
24	0	0	0	0	1	0	0	22	1	0	0	1	58	49	7	116
>24	0	0	0	0	0	0	0	23	0	0	0	2	53	51	10	116
128 NVALID HOUR(S).								ALL	119	41	80	803	1031	554	127	2755
									128 INVALID HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 7 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary July (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability July (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	195	6	13	585	819	410	30	0	1	0	0	1	64	44	7	117
3	160	3	1	422	634	304	16	1	1	0	0	2	68	45	5	121
4	128	2	0	311	507	221	9	2	1	0	0	2	62	53	3	121
5	98	1	0	231	401	162	7	3	1	0	0	3	62	51	4	121
6	70	0	0	170	323	117	5	4	1	0	2	2	60	51	5	121
7	48	0	0	125	257	81	3	5	1	0	1	3	53	57	6	121
8	28	0	0	89	201	51	2	6	0	0	1	2	50	63	4	120
9	13	0	0	59	155	29	1	7	2	0	1	1	75	39	3	121
10	6	0	0	33	116	15	0	8	2	0	1	54	58	5	1	121
11	3	0	0	10	87	4	0	9	4	3	5	83	25	1	0	121
12	1	0	0	2	63	0	0	10	17	3	5	83	14	0	0	122
13	0	0	0	0	44	0	0	11	25	6	3	75	12	0	0	122
14	0	0	0	0	30	0	0	12	30	9	3	63	16	0	0	121
15	0	0	0	0	19	0	0	13	31	4	13	59	14	1	0	122
16	0	0	0	0	10	0	0	14	34	5	12	58	11	2	1	123
17	0	0	0	0	5	0	0	15	32	6	8	60	15	1	1	123
18	0	0	0	0	3	0	0	16	26	5	12	63	15	1	1	123
19	0	0	0	0	1	0	0	17	21	4	8	68	19	3	0	123
20	0	0	0	0	0	0	0	18	9	1	4	74	32	3	0	123
21	0	0	0	0	0	0	0	19	1	0	0	51	66	4	1	123
22	0	0	0	0	0	0	0	20	1	0	0	2	100	17	2	122
23	0	0	0	0	0	0	0	21	1	0	0	1	68	49	3	122
24	0	0	0	0	0	0	0	22	1	0	0	0	68	47	5	121
>24	0	0	0	0	0	0	0	23	1	0	0	0	70	43	7	121
60 INVAL D HOUR(S).								ALL	244	46	79	810	1097	580	60	2916
									60 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 8 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary August (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability August (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	188	17	19	464	847	486	47	0	2	0	0	1	68	46	4	121
3	147	3	5	324	691	355	25	1	1	1	0	3	65	47	7	124
4	114	0	1	221	580	258	14	2	1	0	0	4	62	53	4	124
5	87	0	0	165	487	179	7	3	0	0	0	5	54	59	6	124
6	61	0	0	123	408	123	3	4	0	0	2	2	59	52	9	124
7	41	0	0	91	341	81	2	5	0	0	1	3	52	53	15	124
8	23	0	0	64	289	54	1	6	1	1	0	2	53	54	13	124
9	11	0	0	41	243	33	0	7	2	0	0	2	48	65	7	124
10	4	0	0	19	207	19	0	8	2	0	0	38	73	10	1	124
11	2	0	0	4	174	10	0	9	12	2	4	68	36	2	0	124
12	1	0	0	1	145	4	0	10	18	1	6	78	19	0	0	122
13	0	0	0	0	121	0	0	11	27	9	17	53	16	0	0	122
14	0	0	0	0	101	0	0	12	27	19	13	48	15	0	0	122
15	0	0	0	0	87	0	0	13	39	7	15	48	10	2	0	121
16	0	0	0	0	75	0	0	14	34	11	14	42	15	7	0	123
17	0	0	0	0	69	0	0	15	27	10	14	48	17	5	1	122
18	0	0	0	0	65	0	0	16	23	6	6	62	19	6	0	122
19	0	0	0	0	61	0	0	17	17	2	4	70	27	2	1	123
20	0	0	0	0	58	0	0	18	6	4	2	65	43	2	2	124
21	0	0	0	0	55	0	0	19	2	0	0	22	90	6	3	123
22	0	0	0	0	53	0	0	20	1	0	0	1	84	36	1	123
23	0	0	0	0	52	0	0	21	0	0	0	2	64	54	4	124
24	0	0	0	0	51	0	0	22	0	0	0	1	65	51	6	124
>24	0	0	0	0	50	0	0	23	1	0	0	2	58	56	4	121
	29 INVAL D HOUR(S).							ALL	244	73	98	670	1112	668	88	2953
									29 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 9 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary September (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability September (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	82	12	13	483	1080	280	110	0	0	0	0	0	78	27	12	117
3	56	3	3	351	909	188	78	1	2	0	0	1	76	29	11	119
4	36	1	1	259	782	127	60	2	2	0	0	1	72	32	11	118
5	23	0	0	194	677	83	48	3	1	1	0	0	72	33	11	119
6	13	0	0	142	583	55	38	4	0	0	1	0	74	28	16	119
7	5	0	0	99	500	33	31	5	0	0	0	1	68	34	16	119
8	1	0	0	63	428	20	25	6	0	0	0	3	68	28	20	119
9	0	0	0	35	361	12	19	7	1	0	0	1	73	25	19	119
10	0	0	0	16	304	7	14	8	0	0	1	12	82	17	8	120
11	0	0	0	7	257	4	9	9	0	1	1	62	52	2	2	120
12	0	0	0	3	221	3	4	10	5	2	8	70	33	1	0	119
13	0	0	0	2	188	2	1	11	10	9	7	67	22	1	0	116
14	0	0	0	1	160	1	0	12	12	10	11	68	17	0	0	118
15	0	0	0	0	134	0	0	13	22	7	7	60	19	1	0	116
16	0	0	0	0	114	0	0	14	19	8	12	60	19	0	0	118
17	0	0	0	0	98	0	0	15	18	3	6	70	20	1	0	118
18	0	0	0	0	84	0	0	16	11	2	8	73	23	2	0	119
19	0	0	0	0	73	0	0	17	7	0	6	75	29	1	0	118
20	0	0	0	0	65	0	0	18	5	0	0	45	66	1	1	118
21	0	0	0	0	60	0	0	19	2	1	1	6	95	14	1	120
22	0	0	0	0	56	0	0	20	2	0	1	1	71	42	3	120
23	0	0	0	0	53	0	0	21	2	0	0	1	64	44	8	119
24	0	0	0	0	50	0	0	22	2	0	0	0	69	38	10	119
>24	0	0	0	0	48	0	0	23	1	0	0	1	71	37	10	120
36 INVAL D HOUR(S).								ALL	124	44	70	678	1333	438	160	2847
									36 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 10 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary October (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability October (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	16	8	18	582	717	371	395	0	0	0	0	6	38	40	34	118
3	8	1	7	452	568	237	319	1	0	0	0	6	41	36	37	120
4	2	0	2	347	467	149	257	2	0	0	0	5	41	30	44	120
5	0	0	1	264	387	96	205	3	0	0	0	5	41	33	41	120
6	0	0	0	198	324	59	161	4	0	0	0	7	36	35	43	121
7	0	0	0	143	273	36	126	5	0	0	0	7	40	26	48	121
8	0	0	0	99	231	18	93	6	0	0	0	4	43	28	46	121
9	0	0	0	67	196	10	67	7	0	0	0	5	41	31	44	121
10	0	0	0	56	166	7	44	8	0	0	0	13	50	20	38	121
11	0	0	0	51	139	4	28	9	0	0	0	39	61	19	2	121
12	0	0	0	46	115	2	15	10	0	0	8	66	41	5	1	121
13	0	0	0	41	94	0	6	11	0	8	11	67	32	2	1	121
14	0	0	0	37	74	0	2	12	8	6	8	73	22	2	0	119
15	0	0	0	33	56	0	0	13	9	5	12	76	18	0	0	120
16	0	0	0	29	44	0	0	14	7	9	9	79	17	0	0	121
17	0	0	0	26	34	0	0	15	2	7	9	91	13	0	0	122
18	0	0	0	23	27	0	0	16	2	1	10	84	25	0	0	122
19	0	0	0	20	21	0	0	17	1	0	0	72	47	2	0	122
20	0	0	0	17	18	0	0	18	0	0	0	16	97	9	0	122
21	0	0	0	14	15	0	0	19	0	0	0	4	64	49	5	122
22	0	0	0	12	13	0	0	20	0	0	0	5	40	57	20	122
23	0	0	0	10	11	0	0	21	0	0	0	5	38	52	27	122
24	0	0	0	8	9	0	0	22	1	0	0	7	38	46	29	121
>24	0	0	0	6	7	0	0	23	0	0	0	5	41	44	31	121
80 INVAL D HOUR(S).								ALL	30	36	67	747	965	566	491	2902

80 NVAL D HOUR(S).

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 11 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary November (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability November (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	10	7	14	542	806	439	316	0	0	0	0	14	32	39	28	113
3	6	2	6	418	646	328	254	1	0	0	0	15	38	36	31	120
4	4	0	4	319	524	242	200	2	0	0	0	14	38	37	31	120
5	2	0	2	248	435	182	158	3	0	0	0	11	42	39	28	120
6	1	0	1	192	365	141	124	4	0	0	0	10	41	40	29	120
7	0	0	0	146	304	109	97	5	0	0	0	10	43	36	29	118
8	0	0	0	119	249	85	75	6	0	0	0	10	41	38	30	119
9	0	0	0	101	201	65	56	7	0	0	0	14	44	33	28	119
10	0	0	0	87	168	50	41	8	1	0	0	18	64	23	12	118
11	0	0	0	77	139	39	29	9	1	0	1	50	54	11	2	119
12	0	0	0	67	115	34	18	10	2	5	5	63	37	5	1	118
13	0	0	0	58	94	31	9	11	3	4	9	67	29	3	1	116
14	0	0	0	49	76	28	3	12	6	3	8	68	25	3	0	113
15	0	0	0	42	63	25	0	13	3	4	4	74	24	2	1	112
16	0	0	0	37	50	22	0	14	1	1	6	81	22	2	0	113
17	0	0	0	33	40	19	0	15	0	1	0	70	42	4	0	117
18	0	0	0	29	33	18	0	16	0	1	0	28	86	4	1	120
19	0	0	0	25	28	17	0	17	0	0	0	12	81	25	2	120
20	0	0	0	23	23	16	0	18	0	0	0	10	52	49	9	120
21	0	0	0	22	18	15	0	19	0	0	0	12	40	50	18	120
22	0	0	0	21	15	14	0	20	0	0	0	10	41	43	26	120
23	0	0	0	20	12	13	0	21	0	0	0	10	43	38	29	120
24	0	0	0	19	9	12	0	22	0	0	0	9	46	36	29	120
>24	0	0	0	18	7	11	0	23	0	0	0	10	45	33	32	120
48 INVAL D HOUR(S).								ALL	17	19	33	690	1050	629	397	2835
									48 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 12 of 13)

NUMBER OF CONSECUTIVE HOURS	Stability Persistence Summary December (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	T ME OF DAY (LST)[a]	Diurnal Distribution of Stability December (1973 - 1976) NUMBER OF HOURS PASQU LL STAB LITY CLASS -A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
2	43	0	12	439	788	455	447	0	3	0	0	10	24	26	34	97
3	38	0	4	348	640	310	359	1	3	0	0	9	36	34	40	122
4	34	0	0	275	532	216	294	2	3	0	0	8	37	32	42	122
5	30	0	0	224	456	149	240	3	2	0	0	8	34	32	47	123
6	27	0	0	183	388	101	191	4	2	0	0	7	33	35	45	122
7	26	0	0	155	327	67	148	5	2	0	0	8	29	38	45	122
8	25	0	0	135	270	48	117	6	1	0	0	9	30	39	43	122
9	24	0	0	124	227	34	88	7	1	0	0	9	34	30	48	122
10	23	0	0	116	190	25	65	8	2	0	0	15	38	39	28	122
11	22	0	0	108	165	17	46	9	2	1	1	33	56	17	10	120
12	21	0	0	101	146	9	33	10	2	1	1	55	44	15	3	121
13	20	0	0	94	129	5	22	11	2	2	4	54	50	7	2	121
14	19	0	0	88	112	2	11	12	2	4	13	50	46	6	1	122
15	18	0	0	82	97	1	4	13	2	0	9	58	46	6	1	122
16	17	0	0	77	83	0	0	14	3	0	5	61	47	6	1	123
17	16	0	0	72	70	0	0	15	3	0	0	55	56	7	1	122
18	15	0	0	69	60	0	0	16	2	0	0	28	84	5	1	120
19	14	0	0	66	51	0	0	17	2	0	0	9	81	27	2	121
20	13	0	0	63	44	0	0	18	2	0	0	7	45	54	15	123
21	12	0	0	60	38	0	0	19	1	0	0	10	28	61	22	122
22	11	0	0	57	33	0	0	20	1	0	0	11	28	55	27	122
23	10	0	0	54	28	0	0	21	1	0	0	11	29	47	34	122
24	9	0	0	51	23	0	0	22	2	1	0	9	35	39	36	122
>24	8	0	0	48	20	0	0	23	3	0	0	8	31	35	45	122
	80 INVAL D HOUR(S).							ALL	49	9	33	542	1001	692	573	2899
									80 NVAL D HOUR(S).							

TABLE 2.3-23
STABILITY-PERSISTENCE AND DIURNAL DISTRIBUTION AT CPNPP (BASED ON 10-60 METER DELTA TEMPERATURE)

(Sheet 13 of 13)

Stability Persistence Summary
May 15 1972 - May 14 1976)

NUMBER OF CONSECUTIVE HOURS	NUMBER OF HOURS PASQU LL STAB LITY CLASS						
	-A-	-B-	-C-	-D-	-E-	-F-	-G-
2	781	104	188	5767	10633	4472	2586
3	587	27	55	4282	8669	3109	2019
4	436	8	13	3182	7270	2171	1598
5	320	3	3	2400	6160	1511	1251
6	227	0	1	1789	5259	1055	974
7	157	0	0	1316	4484	712	751
8	102	0	0	970	3818	473	568
9	67	0	0	710	3242	309	417
10	49	0	0	534	2757	194	291
11	40	0	0	417	2343	116	192
12	34	0	0	359	1997	71	118
13	29	0	0	323	1693	46	65
14	26	0	0	295	1428	34	28
15	23	0	0	270	1210	27	7
16	20	0	0	250	1022	22	0
17	17	0	0	234	873	19	0
18	15	0	0	220	758	18	0
19	14	0	0	207	668	17	0
20	13	0	0	197	596	16	0
21	12	0	0	188	536	15	0
22	11	0	0	180	489	14	0
23	10	0	0	172	447	13	0
24	9	0	0	164	409	12	0
>24	8	0	0	156	378	11	0
	1490 INVALID HOUR(S).						

Diurnal Distribution of Stability
May 15 1972 - May 14 1976)

T ME OF DAY (LST)[a]	NUMBER OF HOURS PASQU LL STAB LITY CLASS							
	-A-	-B-	-C-	-D-	-E-	-F-	-G-	ALL
0	8	0	0	48	587	428	234	1305
1	8	1	0	55	628	443	267	1402
2	8	1	0	47	617	457	275	1405
3	5	3	0	50	601	470	275	1404
4	5	2	5	50	604	448	289	1403
5	5	0	2	55	580	459	299	1400
6	3	1	2	55	570	468	301	1400
7	8	0	1	62	632	420	274	1397
8	11	0	4	257	745	224	153	1394
9	25	8	17	578	631	92	47	1398
10	61	25	50	744	452	52	12	1396
11	96	63	91	751	351	29	10	1391
12	126	89	105	741	297	20	5	1383
13	163	64	116	732	282	24	5	1386
14	158	69	115	742	278	26	5	1393
15	143	49	86	755	339	25	5	1402
16	95	40	75	666	497	30	4	1407
17	67	18	48	553	649	73	5	1413
18	32	9	9	407	727	195	38	1417
19	14	3	2	200	791	316	91	1417
20	11	1	1	49	764	440	143	1409
21	6	4	0	46	652	522	182	1412
22	8	2	1	39	633	502	227	1412
23	8	0	0	42	626	475	256	1407
ALL	1074	452	730	7724	*533	6638	3402	33553
	1490 INVALID HOUR(S).							

a) LOCAL STANDARD TIME

TABLE 2.3-24
MEAN MONTHLY MIXING DEPTHS FORT WORTH[a] (May 1972-April 1976)

	Morning[b] (meters)	Afternoon[c] (meters)
January	409	771
February	449	1023
March	545	1057
April	547	1203
May	519	1323
June	619	1515
July	689	2120
August	662	2042
September	678	1451
October	481	1355
November	578	995
December	414	804

a) Based on surface observations at Fort Worth and upper air observations at Carswell AFB, Fort Worth from 5/72-10/73 and at Stephenville from 11/73-4/76.

b) Determined from intersection of 1200 GMT RAOB temperature profile and the potential temperature based on the surface RAOB pressure at 1200 GMT and the minimum surface temperature between 0200 and 0600 LST plus 5.0°C (P, C, and M cases excluded).

c) Determined as in b) except using surface temperature at 1500 LST.

TABLE 2.3-25
COMPARISON OF 4- AND 15-YEAR WIND AND STABILITY CONDITIONS AT FORT WORTH

Wind Direction	Wind Frequency Distribution (%)	
	4-Year[a]	15-Year[b]
N	9.0	8.4
NNE	4.3	6.1
NE	2.6	3.9
ENE	2.1	3.0
E	4.0	4.4
ESE	3.9	6.0
SE	6.8	7.9
SSE	10.9	10.4
S	22.5	19.6
SSW	8.6	9.3
SW	3.1	3.4
WSW	2.0	2.5
W	2.3	2.2
WNW	2.1	1.8
NW	3.8	3.3
NNW	5.8	5.4
Calm	6.4	2.5

Pasquill Stability Class	Stability Class Frequency Distribution and Mean Speed			
	4-Year[a]		15-Year[b]	
	Frequency(%)	Mean Speed (knots)	Frequency(%)	Mean Speed (knots)
A	1.1	2.4	0.6	3.1
B	5.8	5.2	4.9	5.4
C	10.5	8.3	10.6	8.8
D	49.5	11.5	54.6	12.3
E	14.6	7.1	15.7	7.5
F & G	18.5	3.5	13.6	4.1
All Classes	100.0	8.6	100.0	9.7

a) May 1972 through April 1976, concurrent with onsite meterological record.

b) January 1957 through December 1971.

TABLE 2.3-26
METEOROLOGICAL INSTRUMENTATION COMANCHE PEAK PREOPERATIONAL METEOROLOGICAL PROGRAM

Measurement	Level (meters)	Instrument	Manufacturer	Model Number	Accuracy	Threshold	Instrument Range
Wind Speed	10 & 60	6 cup Anemometer	WeatherMeasure	W103/6L/A	±1% or .15 mph	.45 m/s[a]	1-100 mph
Wind Direction[b]	10 & 60	Wind Vane	WeatherMeasure	W104-2	±1%	.75 m/s[a]	0-540°
Temperature	10, 30 & 60	Thermistor Composite	WeatherMeasure	44018 X	±.1°C	N/A[c]	-20 to 120°F
		Aspirated Shield (Temp., Dewpoint, & ΔT)	WeatherMeasure	IS-6	N/A	N/A	N/A
Dewpoint	10 & 60	Lithium Chloride Dewcell	Foxboro	2711	±.5°C	N/A	-20 to 120°F
Delta Temperature (ΔT)	10-30 and 10-60	Thermistor Composite	WeatherMeasure	44018 X	.1°C	N/A	-20 to 20
Precipitation	Surface	Tipping Bucket Rain Gauge	WeatherMeasure	P 501	.01 in	.01 in	0 - 10 in
Total Solar Radiation	1	Pyranometer	WeatherMeasure	R 411	±2%	.05 ly/min	-.05 to 2.5 ly/min
Net Solar Radiation	1	Net Radiometer	WeatherMeasure	R 421	±1%	.03 ly/min	-.05 to 2.5 ly/min

a) m/s = meters per second

b) Wind direction variability at the 10-meter level uses this sensor

c) N/A - not applicable

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 1 of 52)

January (1973 - 1976): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
S	0	0	0	0	0	0	0	0	0	0	0	4	4	20.17
SSW	0	0	0	0	0	0	0	0	0	0	0	4	4	18.10
SW	0	0	0	0	1	0	0	0	0	0	0	1	2	12.65
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	2	2	12.35
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	0	1	0	0	0	0	0	0	0	11	12	16.92

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	1	0	0	0	0	0	1	7 00
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSW	0	0	0	0	0	0	0	0	0	0	1	2	3	15.43
SW	0	0	0	0	0	0	0	0	0	0	1	5	6	15.02
WSW	0	0	0	0	0	0	0	0	0	0	0	1	1	20.40
W	0	0	0	0	0	0	0	0	0	0	0	2	2	19.45
WNW	0	0	0	0	0	0	0	0	0	0	0	0	1	0.
NW	0	0	0	0	0	0	0	0	0	0	0	1	1	13.00
NNW	0	0	0	0	0	0	0	0	0	0	0	1	1	15.40
N	0	0	0	0	0	0	0	0	1	0	0	1	2	9 80
CALM													0	0.
TOTAL	0	0	0	0	0	0	1	0	1	0	2	13	17	14.75

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

January (1973 - 1976): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	1	0	0	0	0	1	1	1	4	9.45
NE	0	0	0	0	0	0	0	1	0	1	0	0	2	9 00
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	1	0	0	0	0	0	0	0	0	1	3 90
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	1	0	1	0	2	9.75
S	0	0	0	0	0	0	1	0	0	0	0	2	3	13.23
SSW	0	0	0	0	1	1	0	0	0	0	0	2	4	9 97
SW	0	0	0	0	0	1	0	0	0	0	1	2	4	11.30
WSW	0	0	0	2	2	0	0	0	0	0	0	3	7	9.14
W	0	0	0	0	0	0	0	0	0	1	0	2	3	16.57
WNW	0	0	0	0	0	0	0	0	0	0	0	1	1	17.60
NW	0	0	0	0	0	0	0	0	0	0	0	2	2	11.50
NNW	0	0	0	0	0	0	0	2	1	2	1	0	6	8 92
N	0	0	0	0	0	0	0	0	1	2	2	0	5	9.70
CALM													0	0.
TOTAL	0	0	0	3	4	2	1	3	3	7	6	15	44	10.46

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	2	2	4	5	6	8	6	6	10	49	8 80
NE	0	1	1	1	1	0	0	3	1	2	2	0	12	7 00
ENE	0	0	1	2	1	0	1	2	1	0	0	0	8	5 67
E	0	1	1	1	1	1	1	0	0	0	1	0	7	5.14
ESE	0	0	0	4	3	0	1	1	0	0	0	0	9	4.71
SE	0	0	1	0	2	4	3	2	1	0	1	0	14	6 28
SSE	0	0	0	0	1	3	3	3	6	1	4	10	31	9 90
S	0	0	0	0	1	0	6	2	2	4	1	20	36	11.23
SSW	0	0	1	2	2	1	0	0	2	3	0	34	45	13.20
SW	0	0	3	2	1	3	1	2	3	2	2	20	39	10.92
WSW	0	0	1	1	0	1	0	1	2	0	2	11	19	11.95
W	0	0	2	1	0	0	0	1	0	1	1	12	18	14.81
WNW	0	0	0	0	1	2	1	1	3	0	0	6	14	10.97
NW	0	0	1	4	2	0	1	2	1	5	3	4	23	8 05
NNW	0	1	4	2	6	1	2	5	4	14	18	45	102	11.79
N	0	1	1	4	2	0	3	2	5	6	8	27	59	12.37
CALM													5	0.
TOTAL	0	4	17	26	26	20	28	33	39	44	49	199	490	10.66

NUMBER OF NVAL D OBSERVATIONS = 98
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 3 of 52)

January (1973 - 1976): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	5	5	4	4	2	9	7	6	7	6	11	67	7.79
NE	0	1	3	8	3	2	0	5	4	1	0	1	28	5.72
ENE	0	1	2	3	3	3	0	0	0	0	0	0	12	4 09
E	1	3	1	2	1	1	0	1	1	0	0	0	11	3 88
ESE	0	0	1	6	6	5	1	1	0	0	0	0	20	4 56
SE	0	1	5	4	15	9	11	2	7	2	2	0	58	5 80
SSE	0	0	2	1	10	9	8	6	11	9	5	24	85	9 02
S	0	0	1	1	1	7	1	5	7	13	9	71	116	12.47
SSW	0	1	1	2	2	6	3	5	6	0	1	49	76	12.74
SW	0	0	1	3	2	1	6	4	3	4	2	37	63	11.87
WSW	0	0	1	0	0	1	5	0	0	3	0	7	17	10.69
W	0	0	2	1	1	3	1	2	2	2	1	13	28	11.61
WNW	0	0	0	3	4	3	3	2	6	0	2	4	27	7 93
NW	0	1	1	6	8	13	8	8	9	5	3	21	83	9.14
NNW	1	0	2	8	7	7	14	12	20	14	10	113	208	12.85
N	0	8	1	5	2	5	7	13	15	11	14	76	157	10.70
CALM													2	.10
TOTAL	3	21	29	57	69	77	77	73	97	71	55	427	1058	10.36

NUMBER OF NVAL D OBSERVATIONS = 26
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	2	1	1	3	1	0	1	1	0	0	11	5 03
NE	0	0	1	2	0	1	0	0	0	0	0	0	4	3 97
ENE	0	2	2	0	0	0	0	0	1	0	0	0	5	3.46
E	0	1	2	3	1	0	0	0	0	1	0	0	8	3 99
ESE	0	0	5	6	1	2	1	2	1	1	0	1	20	5.10
SE	0	0	3	12	6	5	7	5	3	3	0	0	44	5 57
SSE	0	1	5	4	7	9	5	11	11	6	3	7	69	7 21
S	0	3	5	7	5	10	3	9	11	7	10	28	98	9 22
SSW	1	1	5	4	0	4	1	5	2	6	11	23	63	9 61
SW	0	1	1	2	0	5	1	3	0	2	1	7	23	8.48
WSW	0	2	5	2	1	4	0	3	2	1	2	4	26	6.72
W	0	2	1	4	3	1	2	0	1	2	1	3	20	6 57
WNW	0	0	0	5	1	4	9	5	3	0	1	0	28	6 31
NW	1	1	4	6	8	8	8	4	9	0	2	1	52	5 94
NNW	0	3	0	7	4	4	0	5	3	3	2	5	36	6.71
N	0	1	2	3	2	1	1	3	1	1	1	3	19	6 88
CALM													0	0.
TOTAL	2	19	43	68	40	61	39	55	49	34	34	82	526	7 29

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 4 of 52)

January (1973 - 1976): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	2	2	1	0	1	0	0	1	0	2	10	5 83
NE	0	2	1	0	0	0	1	0	0	0	0	0	4	2 85
ENE	0	1	2	1	0	0	0	0	0	0	0	0	4	2.45
E	0	4	4	2	1	0	0	0	0	0	0	0	11	2 56
ESE	0	3	4	2	0	1	1	3	0	1	0	0	15	4 37
SE	0	7	11	16	9	2	3	4	2	1	0	0	55	4 09
SSE	1	4	11	17	11	5	8	0	1	0	0	0	58	4 01
S	1	9	5	7	8	9	5	2	2	4	0	0	52	4 68
SSW	0	3	3	0	6	4	4	2	2	2	0	1	27	5 67
SW	0	3	11	5	5	4	4	3	2	2	0	4	43	5 35
WSW	1	4	7	9	9	4	5	6	3	1	1	2	52	5.17
W	0	9	13	10	19	2	1	0	0	0	0	0	54	3.48
WNW	0	7	10	7	7	7	3	2	0	0	1	0	44	4 01
NW	1	4	10	7	8	10	8	12	1	2	3	0	66	5 28
NNW	2	5	5	1	2	4	1	2	2	1	1	1	27	4 65
N	0	2	1	1	2	0	0	0	0	0	0	0	6	2 92
CALM													8	.37
TOTAL	6	68	100	87	88	52	45	36	15	15	6	10	536	4.45

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	9	9	9	9	9	16	13	15	16	13	24	143	7.75
NE	0	5	6	12	4	3	2	9	5	4	2	1	53	5.70
ENE	0	4	7	6	4	3	1	2	2	0	0	0	29	4.19
E	1	9	8	9	4	2	1	1	1	1	1	0	38	3.76
ESE	0	3	12	8	10	8	4	8	1	2	0	1	67	4 68
SE	0	8	20	32	32	20	25	13	13	6	3	0	172	5 24
SSE	1	5	18	22	29	26	24	20	30	16	13	42	246	7.46
S	2	12	12	15	15	26	16	18	23	28	20	125	312	10.03
SSW	1	5	10	8	11	16	8	12	12	11	13	115	222	11.17
SW	0	4	17	13	8	14	12	12	8	10	7	76	181	9.73
WSW	1	7	14	15	12	12	10	10	7	5	5	28	126	7 56
W	0	14	19	16	23	6	4	3	3	6	3	32	129	7 80
WNW	0	7	11	15	13	16	16	10	12	0	4	11	115	6.44
NW	2	8	19	24	27	31	25	26	20	12	11	29	234	7 08
NNW	3	10	13	19	20	16	17	26	30	34	32	167	387	11.25
N	0	13	5	13	8	6	11	18	23	20	25	107	249	10.56
CALM													15	.21
TOTAL	12	123	200	246	229	214	192	201	205	171	152	758	2718	8 62

NUMBER OF NVAL D OBSERVATIONS = 258
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 5 of 52)

February (1973 - 1976): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
S	0	0	0	0	1	0	0	0	0	0	1	3	5	13.40
SSW	0	0	0	0	0	0	1	0	2	1	1	13	18	14.01
SW	0	0	0	1	0	0	0	0	0	0	1	8	10	12.51
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	1	1	15.50
NNW	0	0	0	0	0	0	0	0	0	0	0	3	3	15.37
N	0	0	0	0	0	0	0	0	0	0	0	2	2	16.25
CALM													0	0.
TOTAL	0	0	0	1	1	0	1	0	2	1	3	30	39	13.81

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2712

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	3	0	0	0	3	8 67
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	1	0	0	0	0	0	0	0	1	2	9 20
S	0	0	0	0	0	0	1	0	0	1	0	3	5	15.00
SSW	0	0	0	0	0	2	0	0	1	0	0	3	6	12.63
SW	0	0	0	0	0	0	0	0	1	0	0	2	3	13.17
WSW	0	0	0	0	0	0	0	0	0	0	2	0	2	10.35
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	3	3	12.43
NNW	0	0	0	0	0	0	0	0	0	0	1	1	2	12.30
N	0	0	0	0	0	0	0	0	0	0	0	2	2	16.65
CALM													0	0.
TOTAL	0	0	0	1	0	2	1	0	5	1	3	15	28	12.52

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2712

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 6 of 52)

February (1973 - 1976): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	1	0	0	1	10.00
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	1	1	14.00
SSE	0	0	0	0	0	0	0	0	0	0	0	3	3	14.70
S	0	0	0	1	0	1	0	2	1	1	1	0	7	7 54
SSW	0	0	0	0	0	0	0	0	1	0	0	0	1	8 20
SW	0	0	0	0	0	0	0	0	0	0	1	2	3	14.57
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	1	0	3	4	11.35
NNW	0	0	0	0	0	0	0	0	0	0	3	8	11	13.93
N	0	0	0	0	0	0	0	0	0	0	0	3	3	13.57
CALM													0	0.
TOTAL	0	0	0	1	0	1	0	2	2	3	5	20	34	12.12

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2712

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	3	1	6	1	0	3	1	1	1	1	18	6.10
NE	0	0	1	1	2	2	3	1	3	0	1	0	14	6 38
ENE	0	0	1	1	3	1	4	1	2	0	0	0	13	5 88
E	0	1	3	2	4	3	1	0	0	0	0	0	14	4 07
ESE	0	0	1	1	1	3	2	0	0	3	1	1	13	7 85
SE	0	0	0	1	3	2	5	1	1	1	1	4	19	8 38
SSE	0	0	0	3	2	1	2	0	3	3	3	11	28	10.51
S	0	1	0	1	5	2	4	3	7	7	3	31	64	12.14
SSW	0	1	0	1	2	1	3	4	4	6	1	50	73	13.85
SW	0	0	1	0	1	0	3	2	4	2	2	21	36	12.13
WSW	0	0	0	0	2	1	0	0	0	3	2	15	23	13.69
W	0	0	0	0	0	0	0	0	1	0	0	3	4	12.55
WNW	0	0	0	6	0	1	0	0	0	0	0	14	21	12.52
NW	0	0	1	1	3	2	0	0	1	3	3	43	57	14.14
NNW	0	0	1	2	0	2	0	0	1	4	8	33	51	13.08
N	0	0	1	1	1	3	1	2	1	2	8	32	52	13.17
CALM													5	0.
TOTAL	0	3	13	22	35	25	28	17	29	35	34	259	505	11.68

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2712

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 7 of 52)

February (1973 - 1976): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	5	5	2	1	4	0	0	1	2	22	5 84
NE	0	0	6	3	6	2	4	2	0	0	1	2	26	5 55
ENE	0	2	10	3	5	1	2	2	0	0	0	1	26	4.13
E	0	2	3	7	4	1	3	1	0	0	0	1	22	5 03
ESE	0	1	2	3	3	1	1	0	2	2	0	1	16	6 33
SE	0	0	4	4	4	11	5	4	6	2	1	2	43	6 67
SSE	0	1	2	5	5	4	4	2	11	10	8	34	86	10.58
S	0	0	0	7	3	8	2	7	5	6	11	92	141	13.93
SSW	0	0	0	2	3	4	4	6	4	6	5	69	103	14.61
SW	0	1	2	1	0	3	2	2	1	0	4	28	44	11.79
WSW	0	0	1	0	0	2	1	1	2	2	0	23	32	14.47
W	0	0	1	1	3	0	1	1	1	2	1	9	20	12.46
WNW	0	1	1	3	0	0	4	3	3	1	1	26	43	12.13
NW	0	0	2	5	7	6	9	8	5	2	7	55	106	12.37
NNW	0	2	1	2	6	11	8	18	19	22	14	107	210	12.43
N	0	1	0	2	5	3	5	7	7	13	9	87	139	13.53
CALM													14	0.
TOTAL	0	12	36	53	59	59	56	68	66	68	63	539	1093	11.72

NUMBER OF NVAL D OBSERVATIONS = 11
TOTAL NUMBER OF OBSERVATIONS = 2712

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	2	5	3	2	0	2	0	0	1	0	0	16	3 87
NE	0	2	3	2	5	0	0	0	0	0	0	0	12	3.48
ENE	0	1	2	0	1	2	0	0	0	0	0	1	7	6 59
E	0	0	1	2	1	0	0	0	0	0	0	2	6	7.12
ESE	0	3	4	2	3	1	0	3	0	0	0	2	18	5.44
SE	0	1	3	5	12	9	4	5	2	1	1	4	47	6 33
SSE	0	3	5	10	14	9	6	8	6	3	3	5	72	6.16
S	0	1	2	0	8	2	4	8	12	11	4	33	85	10.05
SSW	0	2	3	2	1	8	3	6	3	2	7	33	70	10.09
SW	1	1	2	2	0	0	0	4	3	2	1	16	32	9 84
WSW	0	1	0	3	1	1	1	2	1	0	2	5	17	8 28
W	0	0	0	1	0	3	0	3	1	3	1	1	13	8.45
WNW	0	1	1	2	2	2	3	9	2	4	2	7	35	8 39
NW	0	0	4	4	9	12	6	2	7	6	2	8	60	7.73
NNW	0	3	6	4	5	6	4	5	3	1	0	7	44	7.17
N	0	5	0	8	0	4	4	0	0	2	1	9	33	7 52
CALM													9	.22
TOTAL	2	26	41	50	64	59	37	55	40	36	24	133	576	7.78

NUMBER OF NVAL D OBSERVATIONS = 15
TOTAL NUMBER OF OBSERVATIONS = 2712

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

February (1973 - 1976): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	1	2	0	1	0	0	0	0	0	6	3 92
NE	1	0	0	0	0	0	0	0	0	0	0	0	1	1 00
ENE	0	1	1	1	0	0	0	0	0	0	0	0	3	2 53
E	1	1	2	0	0	0	0	0	0	0	0	0	4	1 95
ESE	1	2	3	2	2	0	0	0	0	0	0	1	11	4.11
SE	1	2	5	14	5	1	1	1	0	0	0	0	30	3.72
SSE	0	1	9	9	9	2	2	1	0	1	2	4	40	5 60
S	1	2	6	7	10	5	5	3	0	1	2	0	42	4 93
SSW	2	2	2	5	8	3	4	3	3	0	2	1	35	5.40
SW	2	4	4	6	2	1	0	0	0	1	1	2	23	4 34
WSW	0	0	5	1	1	1	3	2	1	2	0	1	17	5 91
W	0	5	4	6	2	4	2	0	0	0	1	0	24	4.10
WNW	0	4	5	4	2	4	2	0	1	0	0	1	23	4 50
NW	0	3	4	7	7	7	3	2	0	0	0	8	41	7.15
NNW	0	0	0	1	2	0	1	1	0	0	0	2	7	8 89
N	0	0	0	1	2	4	0	0	0	0	0	0	7	4 91
CALM													1	.20
TOTAL	9	28	51	65	54	32	24	13	5	5	8	20	315	5.11

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2712

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	4	10	10	15	3	4	7	4	3	2	3	66	5.45
NE	1	2	10	6	13	4	7	3	3	0	2	2	53	5 21
ENE	0	4	14	5	9	4	6	3	2	0	0	2	49	4 85
E	1	4	9	11	9	4	4	1	0	0	0	3	46	4.74
ESE	1	6	10	8	9	5	3	3	2	5	1	5	58	5 97
SE	1	3	12	24	24	23	15	11	9	4	3	11	140	6 21
SSE	0	5	16	28	30	16	14	11	20	17	16	58	231	8 37
S	1	4	8	16	27	18	16	23	25	27	22	162	349	11.45
SSW	2	5	5	10	14	18	15	19	18	15	16	169	306	12.25
SW	3	6	9	10	3	4	5	8	9	5	10	79	151	10.45
WSW	0	1	6	4	4	5	5	5	4	7	6	44	91	11.43
W	0	5	5	8	5	7	3	4	3	5	3	13	61	8 32
WNW	0	6	7	15	4	7	9	12	6	5	3	48	122	9 68
NW	0	3	11	17	26	27	18	12	13	12	12	121	272	10.93
NNW	0	5	8	9	13	19	13	24	23	27	26	161	328	11.83
N	0	6	1	12	8	14	10	9	8	17	18	135	238	12.41
CALM													29	.08
TOTAL	11	69	141	193	213	178	147	155	149	149	140	1016	2590	10.08

NUMBER OF NVAL D OBSERVATIONS = 122
TOTAL NUMBER OF OBSERVATIONS = 2712

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

March (1973 - 1976): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	1	1	1	0	3	9.47
NE	0	0	0	0	0	0	0	1	0	0	0	0	1	7.40
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	1	0	0	1	10.00
SSE	0	0	0	0	1	0	0	0	1	0	1	1	4	8.70
S	0	0	0	0	0	0	0	0	0	0	0	11	11	17.46
SSW	0	0	0	0	0	0	0	0	0	0	0	8	8	22.56
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	1	0	0	0	0	1	1	0	1	8	12	14.62
N	0	0	0	0	0	0	1	0	0	0	0	0	1	6.10
CALM													0	0.
TOTAL	0	0	1	0	1	0	1	2	3	2	3	28	41	15.48

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	1	0	0	1	9.10
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	1	0	0	2	3	10.80
SSE	0	0	0	0	0	0	0	0	1	0	0	1	2	10.70
S	0	0	0	0	0	0	0	0	0	0	0	2	2	15.45
SSW	0	0	0	0	0	0	0	0	0	0	0	6	6	16.90
SW	0	0	0	0	0	0	0	0	0	0	1	0	1	10.40
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	1	1	11.50
NNW	0	0	0	0	1	0	0	0	0	0	1	8	10	14.79
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	0	0	1	0	0	0	2	1	2	20	26	14.04

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

March (1973 - 1976): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	1	0	0	0	0	1	7 90
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	1	0	0	0	0	0	0	1	5.10
E	0	0	0	0	0	0	0	0	1	0	0	0	1	8.70
ESE	0	0	0	0	0	1	0	0	0	0	0	0	1	5 90
SE	0	0	0	0	0	0	0	0	0	0	2	0	2	10.20
SSE	0	0	0	0	1	0	0	0	1	0	1	2	3	11.13
S	0	0	0	0	0	0	0	0	0	0	0	5	5	17.80
SSW	0	0	0	0	0	0	0	0	0	0	1	3	4	13.17
SW	0	0	0	0	0	0	0	0	0	0	0	4	4	14.52
WSW	0	0	0	0	0	0	1	1	0	0	0	0	2	7 05
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	1	0	0	0	0	0	2	1	4	9 62
NNW	0	0	0	0	0	0	0	0	0	0	2	6	8	14.01
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	0	0	1	2	1	2	1	0	8	21	36	12.39

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	1	1	0	0	1	2	2	12	19	11.85
NE	0	0	1	1	0	1	1	1	0	2	5	1	13	8 67
ENE	0	0	2	4	4	0	0	1	2	1	0	1	15	5 69
E	0	1	0	1	3	1	0	0	0	0	3	0	9	6 20
ESE	0	0	2	1	3	4	3	1	1	5	3	6	29	8 52
SE	0	0	0	5	5	2	6	3	4	10	8	25	68	10.66
SSE	0	0	1	0	1	0	4	0	7	8	3	16	40	10.83
S	0	0	0	1	1	0	3	0	1	4	7	53	70	15.40
SSW	0	0	0	0	0	0	0	2	1	2	3	36	44	16.20
SW	0	0	0	2	1	3	1	2	2	4	2	25	42	12.89
WSW	0	0	0	0	0	0	1	1	0	1	6	14	23	14.16
W	0	0	0	0	0	0	1	0	0	0	1	18	20	14.69
WNW	0	1	0	0	1	0	1	0	0	1	0	9	13	12.24
NW	0	0	1	1	1	1	1	0	1	2	1	25	34	13.56
NNW	0	0	0	1	0	4	4	2	1	2	0	27	41	13.69
N	0	0	0	0	1	0	2	0	1	2	1	29	36	15.37
CALM													0	0.
TOTAL	0	2	7	17	22	17	28	13	22	46	45	297	516	12.73

NUMBER OF NVAL D OBSERVATIONS = 7
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 11 of 52)

March (1973 - 1976): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	3	2	2	2	3	2	1	6	8	29	58	10.87
NE	0	1	4	8	1	2	3	4	7	5	1	4	40	6 80
ENE	0	1	0	4	4	2	1	6	8	8	7	3	44	8 01
E	0	0	0	8	3	7	5	2	4	8	7	6	50	7 97
ESE	0	0	3	7	5	9	6	5	7	8	9	32	91	9 93
SE	0	0	6	5	6	6	5	1	6	7	10	57	109	10.97
SSE	0	0	1	3	4	3	8	4	6	8	6	121	164	14.22
S	0	0	2	2	3	2	3	3	10	6	11	154	196	16.39
SSW	0	0	0	0	1	1	2	0	3	1	5	49	62	17.51
SW	0	0	0	0	0	0	1	1	2	1	0	16	21	14.39
WSW	0	0	0	0	0	0	1	2	2	1	2	21	29	14.65
W	0	1	1	0	2	4	5	0	1	1	0	21	36	12.19
WNW	0	0	0	1	2	3	0	5	3	1	1	36	52	12.51
NW	0	1	2	3	0	6	4	5	2	6	5	61	95	12.92
NNW	0	2	0	2	4	4	7	2	6	5	6	97	135	14.17
N	2	0	0	2	4	2	9	3	4	3	8	74	111	13.78
CALM													1	0.
TOTAL	2	6	22	47	41	53	63	45	72	75	86	781	1294	13.04

NUMBER OF NVAL D OBSERVATIONS = 18
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	1	3	2	1	1	0	0	2	1	13	6 38
NE	0	1	3	2	3	0	0	0	0	0	2	1	12	5 34
ENE	0	1	1	5	1	2	1	2	0	1	0	0	14	4 91
E	0	1	4	6	1	1	1	2	2	2	0	2	22	5 82
ESE	0	1	5	11	11	3	2	2	1	0	1	0	37	4 64
SE	0	1	3	11	10	10	4	4	3	4	1	4	55	6 29
SSE	0	1	2	8	2	9	5	6	7	9	10	16	75	8 92
S	1	1	2	2	1	1	2	5	2	4	7	45	73	12.60
SSW	0	1	1	0	1	1	0	2	1	0	0	24	31	16.84
SW	0	0	2	1	0	1	2	1	1	1	1	10	20	10.08
WSW	0	0	0	0	2	0	2	5	2	1	0	2	14	8 38
W	0	1	0	1	1	3	4	1	0	0	0	1	12	6.11
WNW	0	1	1	3	3	2	2	4	0	1	1	1	19	6.11
NW	0	0	4	1	3	2	4	0	1	1	2	0	18	5 84
NNW	0	2	2	3	3	6	0	3	2	1	0	2	24	6 31
N	0	3	2	3	5	1	4	0	1	3	2	9	33	8.44
CALM													1	.10
TOTAL	1	16	33	58	50	44	34	38	23	28	29	118	473	8.49

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

March (1973 - 1976): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	0	0	1	0	0	0	1	1	2	2	8	8 57
NE	0	0	2	2	0	0	0	0	0	1	0	0	5	4 32
ENE	0	1	2	0	1	0	0	0	0	0	0	0	4	3 00
E	0	2	0	2	1	0	0	0	0	0	0	0	5	3.16
ESE	0	0	0	4	1	2	0	0	0	0	0	0	7	4 33
SE	0	2	3	7	5	8	1	2	0	0	0	3	31	5.76
SSE	0	1	6	4	3	1	3	3	2	0	0	6	29	7.77
S	0	1	1	3	2	2	1	4	1	0	2	7	24	9 50
SSW	0	5	0	4	1	0	0	1	1	0	1	9	22	8 83
SW	0	4	3	6	1	1	1	2	2	2	3	10	35	7.41
WSW	0	3	2	2	1	2	3	1	4	1	1	2	22	6.11
W	0	2	3	1	2	3	0	4	1	0	0	3	19	6 24
WNW	0	3	3	2	5	2	2	0	0	0	0	0	17	3 85
NW	0	4	0	2	0	2	4	0	0	2	0	0	14	5 08
NNW	0	2	1	3	1	0	0	2	1	0	0	7	17	10.88
N	0	2	1	0	0	0	0	0	0	3	1	8	15	13.03
CALM													1	.50
TOTAL	0	33	27	42	25	23	15	19	13	10	10	57	275	7 29

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	5	3	7	7	4	6	4	11	15	48	112	10.11
NE	0	2	10	13	4	5	5	7	9	14	8	7	84	7.12
ENE	0	3	5	13	11	5	2	10	10	10	7	4	80	6.70
E	0	4	4	18	10	11	9	5	8	11	10	8	98	6 90
ESE	0	2	10	25	21	21	13	10	12	14	14	41	183	8.19
SE	3	6	16	29	28	28	21	14	17	24	24	102	312	9.15
SSE	0	4	12	17	12	15	23	16	24	26	22	178	349	11.62
S	1	2	5	8	7	6	9	12	14	15	27	282	388	15.06
SSW	0	6	1	4	3	2	3	5	6	4	10	135	179	16.01
SW	0	4	5	9	2	5	5	6	7	8	7	65	123	11.16
WSW	0	3	2	2	3	2	8	10	8	5	9	39	91	11.27
W	0	4	4	2	5	10	10	5	2	1	1	43	87	10.63
WNW	0	5	4	6	11	7	5	9	3	3	2	47	102	9 83
NW	0	5	7	7	5	11	13	6	9	13	11	93	180	11.46
NNW	0	6	4	9	9	14	11	10	11	9	11	164	258	13.26
N	2	5	3	5	10	4	16	3	6	12	12	120	198	13.02
CALM													3	.20
TOTAL	6	63	97	170	148	153	157	134	150	180	190	1376	2827	11.48

NUMBER OF NVAL D OBSERVATIONS = 149
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

April (1973 - 1976): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	1	0	1	3	1	1	0	5	12	9 97
NE	0	0	0	0	0	1	1	1	0	0	0	0	3	6 30
ENE	0	0	0	0	1	1	1	0	0	0	0	0	3	5 60
E	0	0	0	0	0	1	0	0	0	1	0	0	2	7 65
ESE	0	0	0	0	0	0	0	0	0	2	0	3	5	13.84
SE	0	0	0	0	0	0	0	0	1	0	1	1	3	10.17
SSE	0	1	0	0	0	0	0	0	0	0	0	16	17	20.76
S	0	0	0	0	0	0	0	0	0	0	0	6	6	20.38
SSW	0	0	0	0	0	0	0	0	0	0	0	4	4	20.77
SW	0	0	0	0	0	0	0	0	0	0	0	3	3	18.33
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	3	3	17.73
WNW	0	0	0	0	0	0	0	0	1	0	0	3	4	14.10
NW	0	0	0	0	0	0	0	0	0	0	1	12	13	13.93
NNW	0	0	0	0	0	0	0	0	0	1	0	9	10	14.22
N	0	0	0	0	0	0	0	0	0	1	1	5	7	14.17
CALM													0	0.
TOTAL	0	1	0	0	2	3	3	4	3	6	3	70	95	14.90

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	1	1	0	0	0	1	4	7	12.91
NE	0	0	0	0	0	0	0	0	0	1	0	0	1	9 20
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	1	0	1	10.80
ESE	0	0	0	0	0	0	0	0	0	3	0	0	3	9 60
SE	0	0	0	0	0	0	0	0	1	1	0	7	9	12.86
SSE	0	0	0	0	0	0	0	0	0	0	0	7	7	16.29
S	0	0	0	0	0	0	0	0	0	0	0	7	7	19.34
SSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	5	5	18.66
W	0	0	0	0	0	0	0	0	0	0	0	1	1	21.70
WNW	0	0	0	0	0	0	0	1	0	1	0	1	3	9.73
NW	0	0	0	0	0	0	0	0	0	0	0	1	1	17.10
NNW	0	0	0	0	0	0	0	0	0	0	0	2	2	15.75
N	0	0	0	0	0	0	0	0	0	0	0	3	3	16.17
CALM													0	0.
TOTAL	0	0	0	0	0	1	1	1	1	6	2	38	50	14.91

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

April (1973 - 1976): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	1	1	2	14.05
NE	0	0	0	0	0	2	1	0	0	0	0	0	3	6.13
ENE	0	0	0	0	0	1	0	0	0	0	0	1	2	9.10
E	0	0	0	0	0	0	0	1	0	0	0	0	1	7 20
ESE	0	0	0	0	1	0	0	1	1	0	1	0	4	7 80
SE	0	0	0	0	0	0	0	0	1	3	0	5	9	12.68
SSE	0	0	0	0	0	0	0	0	0	0	0	7	7	17.50
S	0	0	0	0	0	0	0	0	0	0	0	5	5	19.72
SSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SW	0	0	0	0	0	0	0	0	0	0	1	5	6	13.95
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	2	2	25.00
WNW	0	0	0	0	0	1	0	0	0	0	0	2	3	12.30
NW	0	0	0	0	0	0	0	0	0	1	0	2	3	12.87
NNW	0	0	0	0	0	0	0	0	1	0	0	8	9	14.91
N	0	0	0	0	0	0	0	1	0	0	0	2	3	11.13
CALM													0	0.
TOTAL	0	0	0	0	1	4	1	3	3	4	3	40	59	13.82

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	1	5	1	2	1	1	1	3	5	20	8 33
NE	0	0	0	1	2	1	1	0	2	2	2	0	11	7.48
ENE	0	0	0	2	2	1	0	2	2	0	1	7	17	9 54
E	0	0	0	3	1	1	1	5	2	0	5	11	29	10.12
ESE	0	0	0	1	0	3	6	6	2	0	4	9	31	10.35
SE	0	0	1	0	0	1	3	3	6	9	11	36	70	11.75
SSE	0	0	0	3	1	0	2	1	3	1	2	50	63	14.73
S	0	0	0	1	3	0	0	1	3	1	2	48	59	16.16
SSW	0	0	0	0	0	1	1	0	1	0	1	15	19	14.39
SW	0	0	0	0	0	1	1	1	0	0	0	15	18	13.29
WSW	0	0	0	0	0	0	0	0	0	0	0	2	2	16.45
W	0	0	0	0	0	1	0	0	0	1	0	8	10	17.13
WNW	0	0	0	0	0	1	2	0	1	3	0	7	14	13.10
NW	0	0	0	3	1	0	2	0	2	1	1	17	27	14.91
NNW	0	0	0	1	0	1	1	2	2	2	5	42	56	14.73
N	0	0	0	0	0	0	1	2	1	2	0	13	19	12.68
CALM													0	0.
TOTAL	0	0	1	16	15	13	23	24	28	23	37	285	465	13.12

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 15 of 52)

April (1973 - 1976): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	4	1	2	8	4	2	4	0	0	2	27	6 61
NE	0	3	0	2	8	7	11	4	6	7	3	1	52	6.76
ENE	0	1	3	5	9	6	0	8	6	2	3	11	54	7 55
E	0	3	0	5	10	6	7	9	5	11	6	25	87	9 53
ESE	0	1	2	1	8	10	12	11	11	12	10	103	181	13.24
SE	0	1	2	2	10	8	16	12	12	17	16	78	174	10.77
SSE	0	0	4	1	5	5	6	5	12	12	16	182	248	14.32
S	0	0	0	6	6	5	7	8	4	8	8	107	159	13.16
SSW	0	0	2	2	3	3	1	5	1	2	2	13	34	10.28
SW	0	1	2	3	3	1	3	0	0	0	2	5	20	7.42
WSW	0	1	1	0	1	2	3	0	1	1	1	5	16	8 86
W	0	0	2	0	1	1	0	2	0	1	1	5	13	9.41
WNW	0	1	5	2	1	5	1	2	2	0	2	4	25	6 85
NW	0	0	3	3	6	6	2	3	6	3	3	43	78	11.67
NNW	1	0	1	7	7	6	5	8	11	5	15	62	128	11.86
N	0	2	0	4	5	5	10	5	7	5	4	7	54	8 03
CALM													2	0.
TOTAL	1	14	31	44	85	84	88	84	88	86	92	653	1352	11.45

NUMBER OF NVAL D OBSERVATIONS = 18
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	3	3	1	1	2	1	1	0	0	0	13	4 58
NE	0	2	4	3	0	1	1	0	0	1	0	1	13	4 51
ENE	0	2	5	6	1	3	1	0	3	0	1	1	23	4 86
E	0	1	6	7	4	2	1	2	2	1	4	2	32	5 96
ESE	0	3	7	10	6	6	4	4	1	4	2	7	54	6 22
SE	0	1	8	15	9	10	3	9	3	1	1	4	64	5 65
SSE	0	3	4	6	1	2	3	4	3	1	1	8	36	7.19
S	0	0	0	4	3	1	2	3	3	5	2	3	26	8 30
SSW	0	0	1	2	1	5	1	0	1	2	2	7	22	8.48
SW	0	1	1	1	2	2	0	0	0	0	0	1	8	5.10
WSW	0	0	4	0	0	1	0	1	0	0	0	0	6	4 03
W	0	2	0	2	3	1	1	0	0	0	0	1	10	4.73
WNW	1	1	3	3	3	4	2	1	0	1	0	0	19	4 62
NW	0	0	5	2	1	11	7	9	1	0	0	7	43	7 27
NNW	0	0	4	1	3	7	6	2	0	1	1	1	26	5 90
N	0	0	0	3	4	4	1	4	3	2	2	0	23	6.71
CALM													3	.43
TOTAL	1	17	55	68	42	61	35	40	21	19	16	43	421	6.18

NUMBER OF NVAL D OBSERVATIONS = 11
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 16 of 52)

April (1973 - 1976): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	0	0	0	0	0	0	0	0	0	0	1	1.70
NE	0	0	1	0	0	0	0	0	0	0	0	0	1	2 50
ENE	0	0	1	0	1	0	0	0	0	0	0	0	2	3 65
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	1	1	0	3	0	0	1	0	0	1	0	7	5.13
SE	0	2	2	8	2	3	4	0	0	0	0	2	23	4 97
SSE	1	0	1	4	4	2	4	0	0	0	0	0	16	4 67
S	0	2	2	2	5	4	1	0	1	0	0	0	17	4.42
SSW	0	2	0	2	4	1	2	1	1	1	0	0	14	5.41
SW	0	3	0	1	1	0	1	0	0	0	1	0	7	4 31
WSW	0	0	3	2	2	0	0	0	0	0	0	0	7	3 31
W	0	1	3	3	0	0	0	1	0	1	0	0	9	4 09
WNW	0	2	0	2	0	1	1	1	0	0	0	0	7	4.41
NW	1	0	2	2	0	4	4	6	7	1	0	2	29	7 03
NNW	0	0	1	0	0	2	2	1	1	0	0	0	7	6 21
N	0	1	0	0	0	0	1	0	0	0	1	0	3	6.13
CALM													0	0.
TOTAL	2	15	17	26	22	17	20	11	10	3	3	4	150	5.16

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	+	2	7	5	9	13	10	7	7	2	6	17	85	7 84
NE	0	5	5	6	11	12	15	5	8	11	5	2	85	6.42
ENE	0	3	9	13	15	12	2	10	11	2	5	20	102	7.14
E	0	4	6	15	15	10	9	17	9	13	16	38	152	8 86
ESE	0	5	10	12	18	19	22	23	16	21	18	122	286	11.28
SE	0	4	13	25	21	22	26	24	24	34	29	135	357	9.76
SSE	1	4	9	14	11	9	15	10	18	14	19	270	394	13.71
S	0	2	2	13	17	10	10	12	11	14	12	177	280	13.24
SSW	0	2	3	6	8	10	5	6	4	5	5	39	93	10.41
SW	0	5	3	5	6	4	5	1	0	0	4	29	62	9 63
WSW	0	1	8	2	3	3	3	1	1	1	1	12	36	8.76
W	0	3	5	5	4	3	1	3	0	3	1	20	48	10.47
WNW	1	4	8	7	4	12	6	5	4	5	2	17	75	7 95
NW	1	0	10	10	8	21	15	18	16	6	5	84	194	10.65
NNW	1	0	9	9	10	17	14	13	15	9	21	124	242	11.82
N	0	3	0	11	10	9	14	12	11	10	8	30	118	8 93
CALM													5	.26
TOTAL	4	47	107	158	170	186	172	167	155	150	157	1136	2614	10.74

NUMBER OF NVAL D OBSERVATIONS = 266
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May (1972 - 1976): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	4	1	1	1	2	0	1	10	7 93
NE	0	0	0	0	0	1	3	0	1	0	0	0	5	6 90
ENE	0	0	0	1	0	0	2	0	1	0	0	0	4	6.15
E	0	0	0	0	0	0	0	1	1	0	0	0	2	7 65
ESE	0	0	0	0	0	1	1	1	2	2	1	0	8	8.17
SE	0	0	0	0	0	0	1	1	2	1	0	3	8	9 85
SSE	0	0	0	0	0	0	2	1	2	0	1	0	6	8 23
S	0	0	0	0	0	0	0	2	1	0	0	0	3	8.13
SSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	1	0	0	0	1	8.70
WNW	0	0	0	0	0	0	2	0	1	0	0	0	3	7 30
NW	0	0	0	0	0	0	0	0	3	0	2	3	8	10.90
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	0	0	0	0	2	0	0	0	2	8.40
CALM													0	0.
TOTAL	0	0	0	1	0	6	12	7	18	5	4	7	60	8.44

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2955

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	2	0	2	0	0	1	0	0	0	5	5 34
NE	0	0	0	0	0	1	0	0	0	0	0	0	2	6 65
ENE	0	0	0	1	1	0	0	0	0	0	0	0	2	4 05
E	0	0	0	0	0	1	1	0	0	0	0	0	2	6 20
ESE	0	0	1	1	0	2	6	2	1	0	0	0	13	6 22
SE	0	0	0	0	1	0	2	2	1	0	0	0	6	6 92
SSE	0	0	0	0	0	1	0	0	5	1	3	1	11	9.43
S	0	0	0	0	0	1	1	0	2	2	2	1	9	9 04
SSW	0	0	0	0	0	0	0	1	1	2	0	0	4	8.77
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	1	0	0	0	0	1	8 00
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	1	0	0	0	1	1	2	5	10.58
NNW	0	0	0	0	0	0	0	1	0	0	0	0	1	8 00
N	0	0	0	0	0	1	1	0	0	0	0	0	2	6 05
CALM													0	0.
TOTAL	0	0	1	4	2	10	11	8	11	6	6	4	63	7 68

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2955

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May (1972 - 1976): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	2	1	3	2	0	0	1	0	0	9	5.72
NE	0	0	0	0	2	0	1	1	0	0	0	1	5	6.72
ENE	0	0	0	1	1	2	0	0	0	0	1	0	5	6 08
E	0	0	0	1	2	1	0	0	0	0	0	0	4	4 35
ESE	0	0	0	1	1	2	3	2	1	1	1	0	12	6 86
SE	0	0	1	0	0	2	2	2	0	0	0	1	10	7 05
SSE	0	0	0	1	1	2	1	3	2	2	1	3	16	8 29
S	0	0	0	0	0	0	1	3	3	3	2	3	15	9.41
SSW	0	0	0	0	2	0	0	1	1	1	2	2	9	9.14
SW	0	0	0	0	0	0	1	0	0	0	0	0	1	7 00
WSW	0	0	0	0	0	0	0	0	0	1	0	0	1	9 20
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	1	0	0	0	0	1	8 00
NW	0	0	0	0	0	0	0	0	0	0	1	1	2	11.50
NNW	0	0	0	0	1	0	0	0	2	0	0	0	3	7.40
N	0	0	0	1	0	1	0	0	0	0	0	0	2	4 90
CALM													0	0.
TOTAL	0	0	1	7	11	13	11	13	11	9	8	11	95	7 59

NUMBER OF NVAL D OBSERVATIONS = 2
TOTAL NUMBER OF OBSERVATIONS = 2955

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	0	2	5	5	4	3	6	1	2	1	30	6 93
NE	0	0	1	3	7	3	2	2	1	1	1	6	27	7.43
ENE	0	2	3	3	5	3	5	1	5	1	1	2	31	5 99
E	0	1	1	6	8	5	1	3	3	2	1	2	33	6 07
ESE	0	1	4	7	7	6	8	5	2	4	1	8	53	7 95
SE	0	0	1	1	8	7	8	10	12	7	4	14	72	8 53
SSE	0	0	4	4	7	13	9	5	8	6	8	25	89	9.41
S	0	0	1	2	6	5	6	2	17	13	7	60	119	10.57
SSW	0	0	0	2	2	0	1	2	5	2	10	32	56	11.09
SW	0	1	1	2	2	1	3	4	1	1	2	11	29	9 27
WSW	0	0	1	4	1	0	0	3	1	1	1	2	14	7 05
W	0	0	1	3	0	1	1	0	1	1	1	7	16	9 65
WNW	0	0	1	0	0	1	0	1	0	0	2	4	9	9 87
NW	0	0	1	0	0	1	1	1	3	1	2	11	21	11.19
NNW	0	1	0	3	2	1	1	6	5	5	4	14	42	9 85
N	0	0	0	3	8	3	1	6	6	5	2	10	44	8.42
CALM													1	.60
TOTAL	0	7	20	45	68	55	51	54	76	51	49	209	686	9 00

NUMBER OF NVAL D OBSERVATIONS = 14
TOTAL NUMBER OF OBSERVATIONS = 2955

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May (1972 - 1976): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	3	3	5	4	4	2	1	4	6	14	46	9 02
NE	0	3	2	7	5	1	5	2	5	1	2	4	37	6 64
ENE	0	1	4	5	4	0	3	3	2	3	0	1	26	5 65
E	2	2	0	5	7	6	1	3	0	2	2	5	35	6 57
ESE	0	5	5	5	9	5	4	3	4	1	1	19	61	8 00
SE	0	2	3	5	9	14	16	11	10	3	6	18	97	7 85
SSE	0	3	3	5	15	22	25	25	22	17	20	48	05	8 84
S	1	1	1	7	11	16	17	24	26	24	21	83	32	9 67
SSW	0	6	0	2	4	3	2	5	4	9	3	28	66	9 53
SW	0	2	2	3	2	1	2	2	0	2	0	5	21	6 99
WSW	0	1	0	2	1	0	0	2	0	0	2	2	10	7 66
W	0	0	2	1	1	1	0	0	1	0	3	4	13	8 33
WNW	0	0	2	1	1	0	0	2	4	1	1	9	21	9 69
NW	1	2	1	5	4	4	6	3	5	8	4	31	74	9 68
NNW	0	0	2	3	10	5	3	4	5	2	5	14	53	8 66
N	0	1	2	1	2	10	9	2	2	1	1	8	39	7 94
CALM													11	.24
TOTAL	4	29	32	60	90	92	97	93	91	78	77	293	1047	8 59

NUMBER OF NVAL D OBSERVATIONS = 27
TOTAL NUMBER OF OBSERVATIONS = 2955

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	0	2	1	1	0	0	2	0	1	0	0	8	4 82
NE	0	3	2	4	1	1	0	0	0	0	0	0	11	3 23
ENE	0	4	9	3	0	2	0	0	0	0	0	0	18	2 86
E	0	2	5	11	6	1	0	1	1	0	0	1	28	4 08
ESE	0	1	7	11	11	2	2	1	0	1	0	0	36	4 24
SE	0	3	7	16	14	9	1	2	3	1	0	0	56	4 51
SSE	0	5	4	10	12	8	10	3	5	2	1	5	65	5 82
S	0	5	7	7	10	12	8	4	4	6	2	3	68	5 96
SSW	1	1	3	4	1	1	1	5	0	3	1	4	25	7 81
SW	1	1	4	3	4	2	1	1	1	0	1	3	22	5.78
WSW	1	1	5	3	1	0	1	2	1	0	1	1	17	4 95
W	0	1	2	2	1	1	0	0	2	4	5	2	20	7 97
WNW	0	0	3	3	5	4	0	1	1	0	1	1	19	5 37
NW	1	1	3	4	7	7	5	0	0	4	1	7	40	6.77
NNW	0	0	3	3	7	1	3	3	0	0	0	0	20	4 84
N	0	1	2	1	3	3	2	2	2	2	0	1	19	6 22
CALM													5	.20
TOTAL	5	29	68	86	84	54	34	27	20	24	13	28	477	5.42

NUMBER OF NVAL D OBSERVATIONS = 9
TOTAL NUMBER OF OBSERVATIONS = 2955

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 20 of 52)

May (1972 - 1976): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	1	0	1	0	0	0	0	0	0	0	4	2.47
NE	0	0	2	0	0	0	0	0	0	0	0	1	3	6 00
ENE	0	0	0	3	1	0	1	0	0	1	0	0	6	4 98
E	0	2	2	3	3	0	0	2	0	0	0	0	12	4 02
ESE	0	5	1	3	2	1	0	1	0	0	0	0	13	3.45
SE	0	2	6	11	4	0	2	0	0	0	0	0	25	3 56
SSE	0	4	6	10	6	3	1	0	1	0	0	0	31	3 93
S	0	0	3	3	3	1	3	0	0	0	0	1	14	4 99
SSW	3	2	2	4	5	0	0	0	0	0	0	1	17	3.74
SW	1	2	2	3	1	1	0	0	0	0	0	1	11	3 85
WSW	1	2	3	2	2	1	2	0	1	0	0	0	14	4 03
W	0	3	4	1	0	0	0	0	0	0	0	1	9	3.12
WNW	2	1	3	5	2	0	0	0	0	0	0	0	13	3.19
NW	1	2	9	5	2	7	3	2	2	0	0	0	33	4.43
NNW	0	1	2	1	3	0	1	0	0	0	0	0	8	3 99
N	0	1	1	1	0	0	0	0	0	0	0	3	6	8 22
CALM													5	.24
TOTAL	8	29	47	55	35	14	13	5	4	1	0	8	224	3 98

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2955

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	3	6	10	13	18	11	8	9	9	8	16	112	7.40
NE	0	6	7	14	15	7	11	6	7	2	3	12	90	6.46
ENE	0	7	17	17	12	7	11	4	8	5	2	3	93	5.16
E	2	7	8	26	26	14	3	10	5	4	3	8	116	5 50
ESE	0	12	18	28	30	19	24	15	10	9	4	27	196	6 81
SE	0	7	18	33	36	32	32	28	30	12	10	36	274	6 96
SSE	0	12	17	30	41	49	48	37	45	28	34	82	423	8.12
S	1	6	12	19	30	36	36	35	53	49	34	151	462	9.18
SSW	4	9	5	12	14	4	4	14	11	17	16	67	177	9.19
SW	2	6	9	11	9	5	7	7	2	3	3	20	84	7 05
WSW	2	4	9	11	5	1	3	7	3	2	4	5	56	5 81
W	0	4	9	7	2	3	1	1	5	5	9	14	60	7.78
WNW	2	1	9	9	8	5	2	5	6	1	4	14	66	7 06
NW	3	5	14	14	13	20	15	6	13	14	11	55	183	8 37
NNW	0	2	7	10	23	7	8	14	12	7	9	28	127	8.12
N	0	3	5	7	13	18	13	10	12	8	3	22	114	7.77
CALM													22	.25
TOTAL	17	94	170	258	290	245	229	207	231	175	157	560	2655	7 67

NUMBER OF NVAL D OBSERVATIONS = 300
TOTAL NUMBER OF OBSERVATIONS = 2955

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

June (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	2	0	2	0	1	1	2	8	9.15
NE	0	0	0	1	0	1	0	0	1	2	2	3	10	10.03
ENE	0	0	0	0	0	0	1	0	1	0	0	0	2	7.15
E	0	0	0	0	1	1	1	0	0	2	0	0	5	7.42
ESE	0	0	0	0	0	0	0	1	2	0	0	1	4	9 30
SE	0	0	0	0	1	0	0	0	0	0	0	0	1	5 00
SSE	0	0	0	0	1	1	1	0	2	2	2	10	19	11.96
S	0	0	0	0	0	1	1	5	5	6	2	11	31	10.17
SSW	0	0	0	0	1	0	2	3	2	4	5	7	24	10.01
SW	0	0	0	1	2	0	0	1	0	0	0	2	6	8 67
WSW	0	0	0	0	1	3	0	0	1	0	0	0	5	6.40
W	0	0	0	0	1	0	0	0	0	0	0	0	1	5 00
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	1	0	0	0	0	0	1	7 00
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	1	0	0	0	0	0	0	0	1	4 60
CALM													0	0.
TOTAL	0	0	0	2	9	9	7	12	14	17	12	36	118	9.75

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	1	0	0	0	0	0	0	1	5 80
NE	0	0	0	0	0	0	2	0	0	1	0	0	3	7 93
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	2	0	2	0	0	0	0	4	6 65
ESE	0	0	0	0	1	0	1	1	1	1	0	1	6	9.13
SE	0	0	0	0	0	1	2	0	0	1	1	0	5	7 92
SSE	0	0	0	0	0	0	0	0	0	0	1	0	1	13.00
S	0	1	0	0	0	0	1	1	1	1	0	3	8	9 04
SSW	0	0	0	0	0	0	1	2	0	0	3	4	10	10.60
SW	0	0	0	0	0	0	0	1	0	1	1	0	3	9 50
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	1	0	0	1	4	7	7	2	5	5	9	41	9 03

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

June (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	2	2	14.10
NE	0	0	0	1	1	0	0	0	4	0	0	2	8	8 85
ENE	0	0	0	0	0	0	1	2	0	3	0	0	6	8 60
E	0	0	0	0	0	0	0	1	1	1	0	0	3	8 83
ESE	0	0	0	0	1	0	1	3	3	1	1	0	10	8 09
SE	0	0	0	0	1	2	1	0	2	1	0	1	8	7 87
SSE	0	0	0	0	0	0	2	1	4	2	0	0	9	8 56
S	0	0	0	0	0	1	5	0	2	0	1	4	13	9.18
SSW	0	0	0	0	0	0	1	1	0	2	2	5	11	10.69
SW	0	0	0	1	0	0	0	0	2	0	1	0	4	8 02
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	1	0	0	0	0	1	1	1	4	9 30
CALM													0	0.
TOTAL	0	0	0	2	4	3	11	8	18	11	6	15	78	9 03

NUMBER OF NVAL D OBSERVATIONS = 2
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	4	0	6	5	2	2	2	1	4	28	7 24
NE	1	0	1	4	4	0	4	5	3	1	0	2	25	6.72
ENE	0	1	1	5	2	2	6	4	2	2	0	0	25	6 26
E	0	2	4	7	5	7	7	3	6	3	3	2	49	6.45
ESE	1	0	8	3	5	8	13	5	9	4	3	3	62	6.77
SE	1	1	5	6	5	7	6	16	5	11	11	10	84	7 95
SSE	0	2	4	3	12	12	16	18	18	14	13	45	157	10.00
S	0	3	2	9	9	10	17	19	13	13	18	71	184	11.05
SSW	0	0	3	1	1	1	3	4	3	6	7	37	66	11.79
SW	0	2	2	4	1	1	4	4	3	7	8	13	49	9.16
WSW	0	1	2	1	1	0	0	1	0	0	0	1	7	5 23
W	0	2	3	2	3	2	0	1	1	0	0	0	14	4.49
WNW	0	0	0	0	1	0	0	0	0	0	0	0	1	5 00
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	1	0	2	0	1	1	0	4	0	0	6	15	8 99
N	0	1	2	4	1	4	1	1	1	1	0	0	16	5 26
CALM													0	0.
TOTAL	3	17	38	55	50	61	83	83	70	64	64	194	782	9 06

NUMBER OF NVAL D OBSERVATIONS = 21
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

June (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	0	1	1	3	5	5	0	0	0	2	18	6 92
NE	0	2	2	2	2	1	3	2	2	2	2	2	22	6 89
ENE	2	6	2	2	1	1	2	1	0	3	1	2	23	5.45
E	2	3	5	9	4	6	8	4	1	3	1	2	48	5 56
ESE	2	3	6	8	14	13	8	3	0	1	0	1	59	5.13
SE	1	7	19	21	15	18	16	13	12	9	6	11	148	6 35
SSE	3	8	8	20	17	21	20	12	17	14	16	57	213	9.14
S	1	6	8	5	10	11	14	12	19	24	15	148	273	11.90
SSW	1	2	3	4	2	5	6	8	8	8	10	67	124	11.47
SW	0	1	4	4	3	0	1	1	0	1	0	10	25	8 08
WSW	0	1	0	1	0	0	0	1	0	0	0	3	6	8.12
W	0	1	1	0	2	0	1	0	1	0	0	0	6	5 02
WNW	1	1	1	0	0	0	0	1	0	0	0	1	5	6 00
NW	2	0	4	2	3	1	0	1	1	0	0	1	15	4 65
NNW	0	1	0	3	1	0	0	1	1	1	1	6	15	9 87
N	0	1	2	0	3	3	1	0	1	1	2	4	18	8 05
CALM													1	.40.
TOTAL	15	44	65	82	78	83	85	65	63	67	54	317	1019	9 03

NUMBER OF NVAL D OBSERVATIONS = 12
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	1	0	1	1	3	1	2	0	1	0	12	6 04
NE	1	2	2	1	1	5	1	1	0	0	1	0	15	4 63
ENE	0	2	1	2	1	0	0	0	1	0	0	0	7	3 80
E	1	0	3	4	4	1	0	0	0	0	0	0	13	3.77
ESE	1	3	4	22	7	10	1	0	0	0	2	2	52	4 62
SE	1	7	14	16	12	10	9	4	4	0	1	3	81	4 95
SSE	2	7	12	11	21	14	7	7	6	6	2	5	100	5 66
S	0	7	7	10	8	14	16	16	12	10	8	34	142	8.41
SSW	1	3	3	3	4	4	4	3	1	3	3	6	38	7 07
SW	4	3	3	2	0	1	1	3	1	3	1	2	24	5 66
WSW	2	2	0	4	3	0	2	1	0	0	0	0	14	4 04
W	1	1	6	1	0	0	0	0	0	0	0	4	13	6 04
WNW	2	0	1	0	1	0	1	1	0	0	0	0	6	3 97
NW	3	0	2	2	4	1	0	0	0	0	0	0	12	3.42
NNW	2	1	2	2	1	0	0	0	0	0	1	1	10	4.41
N	2	0	0	0	2	0	1	0	0	1	0	0	6	4 63
CALM													1	0.
TOTAL	23	40	61	80	70	61	46	37	27	23	20	57	546	6 04

NUMBER OF NVAL D OBSERVATIONS = 8
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

June (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	0	0	0	0	0	0	0	0	0	0	2	1.70
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	1	1	1	1	0	0	0	0	0	0	0	0	4	2.15
E	1	2	0	0	0	0	0	0	0	0	0	0	3	1 53
ESE	0	1	0	0	0	0	0	0	0	0	0	0	1	1.70
SE	1	3	3	2	0	3	0	0	0	0	0	0	12	3.17
SSE	0	1	3	3	2	0	0	0	0	1	0	0	10	4 01
S	0	3	3	0	0	1	1	1	0	0	1	0	10	4 65
SSW	1	3	6	1	1	1	1	0	0	0	0	1	15	3 91
SW	0	1	1	3	1	0	0	2	0	0	0	0	8	4 69
WSW	0	4	3	1	2	0	0	0	0	0	0	0	10	3 00
W	2	4	6	2	0	0	0	0	0	0	0	0	14	2 34
WNW	0	4	3	0	0	0	0	0	0	0	0	0	7	2 26
NW	0	3	4	2	1	0	0	0	0	0	0	0	10	2 80
NNW	2	4	0	1	0	1	0	0	0	0	0	0	8	2 35
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													3	.53
TOTAL	8	36	33	16	7	6	2	3	0	1	1	1	117	3.13

NUMBER OF NVAL D OBSERVATIONS = 10
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	6	2	6	2	13	13	10	4	3	4	10	73	7 20
NE	2	5	5	9	8	7	10	8	10	6	5	9	84	6 97
ENE	3	10	6	10	4	3	10	7	4	8	1	2	68	5 68
E	4	7	12	20	14	17	16	10	8	9	4	4	125	5 81
ESE	4	7	18	33	28	31	24	13	15	7	6	8	194	5 86
SE	4	19	42	45	34	41	34	35	23	22	19	25	343	6 34
SSE	5	19	27	37	55	48	46	38	48	39	33	118	513	8.70
S	1	20	20	24	27	38	55	54	53	55	45	271	663	10.63
SSW	3	8	15	9	9	11	18	21	14	23	30	127	288	10.39
SW	4	7	10	15	7	2	6	12	6	12	11	27	119	7 87
WSW	2	8	5	8	7	5	2	3	1	0	0	6	47	5 34
W	3	8	17	5	6	3	1	1	2	1	0	12	59	5 93
WNW	3	5	5	0	2	0	1	3	1	0	0	4	24	5 61
NW	5	3	10	6	8	2	1	1	1	0	0	1	38	3 83
NNW	4	7	2	8	2	2	1	1	5	1	2	13	48	7 20
N	2	3	4	4	8	7	3	1	2	4	3	5	46	6 54
CALM													5	.40
TOTAL	49	142	200	239	221	230	241	218	197	190	163	642	2737	8 22

NUMBER OF NVAL D OBSERVATIONS = 143
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

July (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	2	1	2	3	1	0	1	0	10	7.16
NE	0	0	0	0	2	4	3	0	2	1	0	0	12	6.46
ENE	0	0	0	0	0	0	0	0	1	0	1	0	2	9 60
E	0	0	1	0	2	0	4	0	1	0	0	0	8	5 69
ESE	0	0	2	1	1	1	3	3	2	5	4	6	28	8 57
SE	0	0	0	0	1	0	5	3	4	6	4	3	26	8 86
SSE	0	0	2	1	1	1	5	4	8	4	4	4	34	8 84
S	0	0	0	1	0	1	6	2	9	11	4	29	63	11.68
SSW	0	0	0	0	1	0	0	2	1	7	1	32	44	13.95
SW	0	0	0	0	0	1	0	0	0	0	0	1	2	10.95
WSW	0	0	0	0	0	0	0	1	0	0	0	0	1	8 00
W	0	0	0	0	0	1	0	1	0	0	0	0	2	6 60
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	1	0	0	0	0	0	1	6 20
NNW	0	0	0	0	0	0	1	0	2	0	0	0	3	8.17
N	0	0	0	0	1	0	1	1	1	2	1	1	8	8 32
CALM													0	0.
TOTAL	0	0	5	3	11	10	31	20	32	36	20	76	244	10.14

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	1	1	0	1	2	0	0	0	0	0	5	5 24
NE	0	0	0	0	0	0	0	0	0	0	1	0	1	10.10
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	1	0	1	10.90
ESE	0	0	0	0	0	0	0	1	1	2	1	1	6	9 65
SE	0	0	0	0	0	0	1	2	0	1	1	2	7	10.13
SSE	0	0	0	0	0	0	0	1	0	2	1	2	6	10.55
S	0	0	0	0	0	0	0	0	0	1	2	4	7	11.49
SSW	0	0	0	0	0	0	0	0	0	0	2	2	4	12.30
SW	0	0	0	0	1	0	0	0	0	0	2	2	5	10.04
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	1	0	0	1	0	0	0	0	2	6.40
WNW	0	0	0	1	0	0	0	0	0	0	0	0	1	3.40
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	0	0	1	0	0	0	0	0	1	7 00
CALM													0	0.
TOTAL	0	0	1	2	2	1	4	5	1	6	11	13	46	9 62

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

July (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	1	0	1	1	0	0	0	3	7 30
NE	0	0	0	1	0	0	1	1	0	0	0	0	3	6.13
ENE	0	0	0	0	0	0	1	2	0	1	0	0	4	7 97
E	0	0	0	1	1	0	0	0	0	0	0	0	2	4.10
ESE	0	0	0	0	0	0	1	0	1	4	1	4	11	10.32
SE	0	0	0	1	1	0	0	1	1	1	1	3	9	8 32
SSE	0	0	0	0	0	1	3	1	1	0	1	4	11	9 80
S	0	0	0	0	0	0	0	1	2	1	2	4	10	11.18
SSW	0	0	0	0	1	0	2	0	1	0	4	4	12	10.03
SW	0	0	0	0	0	0	0	1	1	0	1	1	4	9.70
WSW	0	0	0	1	0	1	0	0	0	1	0	0	3	6 33
W	0	0	0	0	1	0	0	0	0	0	1	0	2	7.45
WNW	0	0	0	0	1	0	0	0	0	0	0	0	1	4 30
NW	0	0	0	0	0	0	0	0	0	1	0	0	1	9 20
NNW	0	0	0	1	0	0	0	0	0	0	0	0	1	3 60
N	0	0	0	0	0	1	0	0	0	0	0	0	1	6 00
CALM													0	0.
TOTAL	0	0	0	5	5	4	8	8	8	9	11	20	78	9 09

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	6	6	4	5	6	6	3	1	0	39	6 29
NE	0	1	1	0	2	1	3	2	8	2	4	4	28	8 29
ENE	0	0	3	2	0	0	2	8	3	2	1	2	23	7.43
E	0	0	0	5	0	2	8	6	0	1	1	0	23	6 50
ESE	0	0	3	3	0	7	12	6	9	2	3	5	50	7.49
SE	0	0	2	1	5	5	5	12	8	11	9	14	72	8 91
SSE	0	0	3	4	7	11	13	16	20	16	14	40	144	9 08
S	0	1	2	3	8	6	13	17	21	27	29	40	167	9 89
SSW	0	0	0	1	8	12	14	13	18	10	6	20	102	9.11
SW	0	0	0	3	4	7	7	4	3	4	2	6	40	7 62
WSW	0	0	0	1	5	2	4	4	3	1	3	0	23	7.12
W	0	0	3	7	2	2	4	0	6	2	2	4	32	6 95
WNW	0	0	0	0	0	1	1	4	3	2	1	0	12	8.12
NW	0	0	1	2	1	0	1	3	1	0	2	1	12	7 66
NNW	0	1	0	4	2	3	1	4	0	1	0	2	18	6 39
N	0	1	0	2	4	2	3	4	2	0	1	4	23	7.44
CALM													1	.70
TOTAL	0	5	19	44	54	65	96	109	111	84	79	142	809	8.49

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 27 of 52)

July (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	3	1	3	2	1	2	1	0	2	2	2	19	6 28
NE	0	8	5	4	5	6	4	1	1	0	1	0	35	4 27
ENE	0	3	1	5	3	3	5	2	0	0	0	1	23	5.12
E	1	5	6	9	7	6	2	0	1	0	0	0	37	3 90
ESE	0	8	7	1	14	9	3	6	2	0	1	0	51	4.76
SE	0	8	10	19	22	24	19	15	14	10	7	12	160	6 37
SSE	0	4	15	18	23	25	29	24	31	21	20	38	248	7.77
S	1	2	7	9	12	18	17	22	23	31	19	58	219	9.44
SSW	0	3	6	6	8	12	11	9	6	6	10	39	116	8 90
SW	0	4	3	3	6	6	2	5	5	5	4	14	57	8.40
WSW	0	1	3	1	3	2	1	2	1	1	0	2	17	6 56
W	0	1	4	2	1	4	2	0	0	0	0	0	14	4 29
WNW	0	0	3	6	3	2	2	1	0	1	0	1	19	5.11
NW	0	3	1	6	3	1	3	2	0	0	0	1	20	4 81
NNW	0	1	4	4	6	2	1	0	1	1	1	2	23	5 53
N	0	3	5	5	3	2	5	4	1	4	1	3	36	6.11
CALM													3	.27
TOTAL	2	57	81	101	121	123	108	94	86	82	66	173	1097	7 30

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	4	5	0	4	0	0	0	0	0	0	0	14	2.78
NE	0	2	3	1	2	0	2	0	1	0	0	0	11	3 97
ENE	2	8	2	2	3	2	0	0	0	0	0	0	19	2.74
E	2	1	2	4	2	1	0	0	0	0	0	0	12	3 02
ESE	0	10	11	7	1	1	2	0	1	0	0	0	33	2 98
SE	0	8	14	8	11	7	7	3	0	0	0	0	58	4.16
SSE	0	9	20	11	20	24	24	13	1	0	0	6	128	5 31
S	0	1	11	10	15	13	17	7	9	1	0	2	86	5 67
SSW	2	3	6	10	7	10	8	8	7	2	9	7	79	6 92
SW	0	6	7	4	5	4	7	0	3	1	2	1	40	5.12
WSW	0	3	9	5	5	2	1	1	1	0	0	0	27	3.74
W	0	3	8	1	0	0	2	0	0	0	0	0	14	2 93
WNW	0	3	6	6	0	0	0	0	0	1	0	0	16	3.17
NW	0	2	6	9	2	1	0	1	0	0	0	1	22	4 20
NNW	0	1	1	2	1	0	0	0	0	0	0	0	5	3 22
N	1	0	3	0	0	2	2	2	0	0	0	0	10	4 84
CALM													6	.37
TOTAL	8	64	114	80	78	67	72	35	23	5	11	17	580	4 80

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

July (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	1	0	0	0	0	0	0	0	0	0	0	1	1 90
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	1	2	1	0	0	0	0	0	0	0	0	4	2 37
SE	0	0	2	1	0	0	0	0	0	0	0	0	3	3 03
SSE	0	4	0	2	2	0	1	0	1	0	0	1	11	4 84
S	1	1	4	3	1	3	2	1	0	0	0	2	18	5 52
SSW	0	0	2	2	1	0	1	0	0	0	0	1	7	5 57
SW	0	1	0	0	0	1	0	0	0	0	0	1	3	7 53
WSW	0	0	1	1	0	0	0	0	0	0	0	1	3	7 53
W	1	1	0	0	0	0	1	0	0	0	0	0	3	3 23
WNW	0	0	0	0	0	0	0	0	0	1	0	0	1	10.00
NW	0	0	0	0	1	1	1	0	0	0	0	0	3	5.70
NNW	0	0	1	0	0	1	0	0	0	0	0	0	2	3 80
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													1	.70
TOTAL	2	9	12	10	5	6	6	1	1	1	0	6	60	5 04

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	9	8	10	14	8	11	11	8	5	4	2	91	5.77
NE	0	11	9	6	13	12	14	4	12	3	6	4	94	5 89
ENE	3	13	7	10	6	5	9	12	4	3	2	3	77	5 32
E	3	6	10	19	13	10	15	6	2	1	2	0	87	4.78
ESE	0	19	26	13	17	18	23	19	17	14	10	16	192	6 26
SE	0	16	30	31	42	36	39	37	27	29	22	34	343	6 81
SSE	0	18	41	37	53	62	75	59	62	43	40	95	585	7 60
S	2	6	24	28	36	41	55	50	64	72	56	139	573	9.15
SSW	2	8	14	19	26	34	36	32	33	26	32	106	368	9.13
SW	0	11	11	10	16	19	16	10	12	10	11	27	153	7.44
WSW	0	4	13	9	13	8	6	8	5	3	3	3	75	5.76
W	1	5	15	10	6	7	9	2	6	2	3	4	70	5.42
WNW	0	3	9	13	4	3	3	5	3	5	1	1	70	5 26
NW	0	6	9	17	7	3	6	6	1	1	2	3	61	5.19
NNW	0	4	6	11	9	6	3	4	3	2	1	4	53	5 57
N	1	4	8	7	8	7	12	11	4	6	3	8	79	6 57
CALM													12	.42
TOTAL	13	143	240	250	283	279	332	276	263	225	198	449	2963	7 37

NUMBER OF NVAL D OBSERVATIONS = 13
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 29 of 52)

August (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	0	2	0	2	0	0	0	0	0	6	4 50
NE	0	0	2	1	0	1	1	2	0	1	1	0	9	6 32
ENE	0	0	0	0	0	3	0	1	0	0	0	0	4	6.17
E	0	1	0	2	1	4	1	3	7	2	0	0	21	6 84
ESE	0	1	1	2	7	3	5	9	2	6	0	0	36	6 53
SE	0	0	0	2	4	2	7	11	6	6	1	0	39	7 27
SSE	0	1	0	1	3	2	4	6	6	6	2	1	32	7 64
S	0	0	0	3	1	5	6	4	2	2	2	3	28	7 34
SSW	0	0	1	1	1	4	4	3	4	4	7	4	33	8 81
SW	0	0	0	0	0	1	0	2	0	0	1	1	5	8 60
WSW	0	0	1	3	1	0	0	1	1	0	0	0	7	4.73
W	0	0	0	0	1	2	2	0	0	0	0	0	5	5.76
WNW	0	0	1	1	2	0	0	0	0	0	0	0	4	3 85
NW	0	0	0	0	0	0	0	1	0	0	0	0	1	7.40
NNW	0	1	0	1	2	1	0	0	2	0	0	0	7	5 39
N	0	0	0	0	0	2	2	1	0	1	0	0	6	7 22
CALM													1	.50
TOTAL	0	5	7	17	25	30	34	44	30	28	14	9	244	7 05

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	1	2	0	0	0	0	0	0	3	5.13
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	1	0	1	0	0	0	0	2	6 65
ESE	0	0	0	0	0	3	2	2	1	0	1	0	9	7 26
SE	0	0	0	0	1	1	2	2	1	1	0	0	8	7 05
SSE	0	0	0	0	1	0	1	2	3	5	2	2	16	9.12
S	0	0	0	0	0	0	1	0	3	4	5	4	17	10.22
SSW	0	0	0	0	0	0	1	0	2	2	1	4	10	10.58
SW	0	0	0	0	0	0	1	0	0	1	2	1	5	10.24
WSW	0	0	0	0	0	0	1	0	0	0	0	0	1	6.40
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
N	0	0	0	0	0	0	0	0	0	0	0	1	1	12.20
CALM													0	0.
TOTAL	0	0	0	0	3	7	9	7	10	13	11	12	72	8 97

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

August (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	1	0	0	0	1	0	1	0	3	6	11.37
NE	0	0	0	1	0	0	0	1	0	0	0	2	4	9 80
ENE	0	0	0	0	0	0	0	0	0	0	0	1	1	12.00
E	0	0	0	0	1	2	1	1	0	1	0	0	6	6 68
ESE	0	0	0	0	0	2	2	2	0	1	0	0	7	7 09
SE	0	0	0	0	1	2	3	3	0	0	1	0	10	6 91
SSE	0	0	0	0	0	1	2	0	1	3	1	3	11	9 20
S	0	0	0	0	1	0	3	0	4	1	4	6	19	9.76
SSW	0	0	0	0	0	3	2	2	3	2	2	7	21	9 82
SW	0	0	0	0	1	1	0	1	0	1	0	1	5	8.18
WSW	0	0	0	0	0	1	0	1	0	0	0	0	2	6 50
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	1	0	0	0	0	0	0	0	1	4 20
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	1	1	11.80
N	0	0	0	1	1	0	0	0	0	0	0	0	2	4 50
CALM													0	0.
TOTAL	0	0	0	3	6	12	13	12	8	10	8	24	96	8 85

NUMBER OF NVAL D OBSERVATIONS = 2
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	1	1	2	0	4	4	6	7	6	8	39	9 26
NE	0	0	4	0	3	5	7	2	2	5	1	9	38	8.13
ENE	0	0	0	2	4	3	2	3	2	1	2	1	20	6 80
E	0	1	3	3	5	5	0	5	2	0	1	0	25	5 51
ESE	0	0	1	2	4	12	9	11	8	1	0	1	49	6.76
SE	0	3	2	3	9	7	8	8	7	4	4	6	61	7 01
SSE	0	1	1	9	7	14	10	12	6	9	10	15	94	7.79
S	0	1	0	2	9	9	10	14	9	9	9	60	132	10.12
SSW	0	0	0	1	2	7	17	14	14	16	14	41	126	10.08
SW	1	1	0	2	3	5	6	7	4	4	3	6	42	7 58
WSW	0	0	2	3	2	2	1	1	0	0	0	0	11	4 60
W	0	0	0	2	2	0	1	0	0	0	0	0	5	4 66
WNW	0	0	0	0	0	2	0	0	0	0	0	0	2	5 95
NW	0	0	1	1	0	0	0	0	0	0	0	0	2	3 20
NNW	0	0	2	1	0	0	1	0	1	1	0	0	6	5 63
N	0	0	1	0	2	0	1	3	0	0	2	4	15	8.79
CALM													0	0.
TOTAL	1	7	18	32	54	71	77	84	61	59	52	151	667	8.42

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 31 of 52)

August (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	4	6	4	1	2	2	2	1	2	0	2	26	5 33
NE	1	2	9	3	5	1	1	1	2	1	0	0	26	4.10
ENE	2	1	6	3	4	1	1	0	1	0	0	0	19	3 60
E	0	5	6	8	10	4	0	1	1	1	0	2	38	4 66
ESE	0	5	12	15	16	12	11	6	4	3	1	2	87	5.15
SE	0	9	8	34	29	16	24	13	11	8	1	5	158	5 61
SSE	0	6	11	18	26	31	32	34	17	24	16	29	244	7 39
S	2	3	7	16	10	20	21	21	34	19	26	62	241	8 90
SSW	1	2	7	9	2	14	12	17	18	14	13	24	133	8 51
SW	0	2	4	4	7	9	5	8	3	4	1	4	51	6 67
WSW	0	3	3	2	3	1	1	0	1	0	0	1	15	4.45
W	0	2	2	1	0	1	0	0	0	1	0	0	7	3 96
WNW	0	3	0	1	2	2	0	0	2	0	0	0	10	4 58
NW	1	1	2	3	2	2	0	1	1	0	1	1	15	5 23
NNW	0	0	0	2	3	2	2	1	2	1	2	3	18	7 50
N	0	0	5	1	2	1	1	1	0	2	0	2	15	5 93
CALM													2	.45
TOTAL	7	48	88	124	122	119	113	106	98	80	61	137	1105	6 96

NUMBER OF NVAL D OBSERVATIONS = 7
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	5	3	2	2	0	1	0	0	0	0	0	0	13	2 01
NE	1	2	5	2	0	0	0	0	0	0	0	0	10	2 33
ENE	1	4	2	1	0	0	0	0	0	0	0	0	8	2 01
E	3	4	5	4	3	0	0	1	0	0	0	0	20	3 05
ESE	1	18	10	17	16	2	0	0	2	0	0	0	66	3 28
SE	1	12	29	28	21	12	5	2	0	0	0	1	111	3 81
SSE	0	11	12	18	24	16	13	12	5	0	1	0	112	4 80
S	4	11	8	19	16	17	18	6	7	1	1	1	109	4 96
SSW	1	7	13	8	9	8	7	6	2	4	1	6	72	5 52
SW	0	3	8	3	0	4	3	4	1	0	1	1	28	5 01
WSW	0	3	5	2	2	1	1	2	1	0	0	1	18	4 50
W	0	3	1	0	0	1	0	0	0	0	0	0	5	2.72
WNW	0	2	4	0	2	3	1	0	0	0	0	0	12	3 69
NW	0	3	3	7	2	2	4	0	1	0	0	0	22	4 24
NNW	1	3	4	0	1	0	1	0	0	0	0	0	10	2 67
N	2	4	4	4	0	1	1	0	0	0	0	0	16	2 84
CALM													29	.36
TOTAL	20	93	115	115	96	68	55	33	19	5	4	10	661	4 08

NUMBER OF NVAL D OBSERVATIONS = 7
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

August (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	1	0	0	0	0	0	0	0	0	0	0	1	1 20
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	1	1	0	0	0	0	0	0	0	0	0	2	1 90
SE	0	4	5	2	0	2	1	0	0	0	0	0	14	3.15
SSE	0	3	2	0	0	1	2	0	0	0	0	0	8	3 52
S	0	0	1	0	2	3	3	1	0	0	0	0	10	5.75
SSW	0	1	1	0	0	0	1	1	0	0	0	1	5	6 26
SW	0	1	2	1	0	2	1	1	0	0	0	0	8	4 52
WSW	1	1	1	0	1	0	0	0	0	0	0	0	4	2 60
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	4	2	0	0	0	0	0	0	0	0	0	6	1.77
NW	0	2	3	1	3	2	1	0	0	1	0	0	13	4.17
NNW	1	1	1	0	2	0	0	0	0	0	0	0	5	2 64
N	1	0	0	0	0	0	0	0	0	0	0	0	1	1 00
CALM													11	.44
TOTAL	3	19	19	4	8	10	9	3	0	1	0	1	88	3 37

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	5	8	10	8	5	3	8	7	7	10	6	13	90	6 90
NE	2	4	20	7	9	9	9	6	4	7	2	11	90	6.12
ENE	3	6	8	6	8	7	3	4	3	1	2	2	53	4 88
E	3	11	14	17	20	16	2	12	10	4	1	2	112	5.12
ESE	1	25	25	36	43	34	29	30	17	11	2	3	256	5 27
SE	1	28	44	69	65	42	50	39	25	19	7	12	401	5.46
SSE	0	22	26	46	61	65	64	66	40	50	32	50	522	6 97
S	6	15	16	40	39	54	62	46	59	36	48	136	557	8 36
SSW	2	10	22	19	14	36	44	43	44	42	38	87	401	8 58
SW	1	7	14	11	11	23	16	23	8	10	8	14	146	6.71
WSW	1	7	12	10	9	6	4	5	3	0	0	2	59	4 53
W	0	5	3	3	3	4	3	0	0	1	0	0	22	4 25
WNW	0	9	7	2	7	7	1	0	2	0	0	0	35	3.78
NW	1	6	9	12	7	6	5	2	2	1	1	1	53	4 52
NNW	2	5	7	4	8	3	4	1	5	2	2	4	47	5.49
N	3	4	10	6	5	4	5	5	0	5	2	7	56	5 92
CALM													43	.39
TOTAL	31	172	247	296	314	319	309	289	229	199	151	344	2943	6 66

NUMBER OF NVAL D OBSERVATIONS = 33
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

September (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	4	4	3	1	1	2	0	3	18	8.12
NE	0	0	0	2	1	0	0	0	0	0	0	0	3	3 97
ENE	0	1	1	2	2	0	0	0	0	1	0	1	8	5.14
E	0	0	0	2	3	0	2	1	1	0	0	0	9	5.43
ESE	0	0	1	3	3	7	7	0	1	2	0	0	24	5 84
SE	0	1	0	0	0	2	0	0	0	1	0	0	4	5 67
SSE	0	0	0	0	1	5	2	1	3	0	3	4	19	8.41
S	0	0	2	1	1	2	3	0	2	2	3	4	20	8 53
SSW	0	0	0	0	0	0	0	0	1	0	0	2	3	11.23
SW	0	0	0	0	1	0	2	0	0	0	0	0	3	6 07
WSW	0	0	0	1	1	0	0	0	0	0	0	0	2	4 35
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	1	0	0	0	0	0	0	0	0	1	3 50
NNW	0	0	1	1	0	0	0	0	0	0	0	0	2	2 90
N	0	0	0	2	2	0	1	0	0	1	0	1	7	6 39
CALM													1	.70
TOTAL	0	2	5	15	19	20	20	3	9	9	6	15	124	6 91

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	0	0	0	1	0	1	1	0	1	7	12	11.57
NE	0	0	0	0	0	0	0	0	0	0	1	0	1	10.80
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	1	0	0	0	1	0	1	0	0	3	6 87
SE	0	0	0	1	0	1	0	0	0	0	0	0	2	4 80
SSE	0	0	0	0	0	0	1	0	2	2	2	1	8	9 52
S	0	0	0	0	0	0	1	0	0	1	0	3	5	11.54
SSW	0	0	0	0	0	0	0	0	0	0	0	2	2	13.75
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	1	0	0	0	1	8.40
N	0	0	0	0	0	0	1	2	1	3	3	0	10	9 32
CALM													0	0.
TOTAL	0	1	0	2	0	2	3	4	5	7	7	13	44	10.07

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

September (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	1	0	0	0	0	3	0	3	7	11.73
NE	0	0	0	1	1	0	0	0	0	0	0	0	2	4 20
ENE	0	1	0	0	2	0	0	1	0	0	0	0	4	4 90
E	0	0	0	0	0	0	0	0	1	0	0	0	1	8.70
ESE	0	0	0	1	0	1	1	0	0	2	1	0	6	7.47
SE	0	0	0	0	0	0	0	0	1	0	1	0	2	9.15
SSE	0	0	0	0	1	0	3	2	1	2	0	3	12	8 61
S	0	0	0	1	0	0	0	1	0	0	1	3	6	9 85
SSW	0	1	0	0	0	0	0	0	0	0	1	2	4	9.72
SW	0	0	0	0	0	1	1	0	1	0	0	1	4	8.47
WSW	0	0	0	0	0	2	0	0	1	0	0	0	3	6 50
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	2	0	1	0	3	0	1	7	8 87
NNW	0	0	0	0	0	1	0	1	0	0	0	0	2	6 60
N	0	0	0	0	1	1	0	0	1	0	5	1	9	9 36
CALM													1	.40
TOTAL	0	2	0	3	6	8	5	6	6	10	9	14	70	8 52

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	1	2	3	2	3	2	5	2	3	3	20	47	9.45
NE	0	2	3	3	6	6	6	6	3	1	0	4	40	6.48
ENE	0	2	0	7	2	3	9	6	5	5	2	8	49	7.74
E	1	0	7	1	4	4	3	2	3	1	0	0	26	5 25
ESE	0	3	6	5	9	5	8	7	11	2	0	0	56	5 94
SE	0	1	2	7	4	9	7	3	3	6	5	3	50	6.76
SSE	0	1	0	4	2	6	5	10	6	12	6	22	74	9 33
S	0	3	1	5	1	3	5	10	10	5	12	55	110	10.83
SSW	0	0	1	0	1	1	6	5	1	8	8	31	62	11.61
SW	0	0	0	1	1	1	2	0	1	2	1	4	13	9.71
WSW	0	0	1	0	2	0	0	1	0	0	1	0	5	6 28
W	0	1	0	0	2	0	0	1	0	1	0	0	5	5.70
WNW	0	0	0	3	0	0	1	1	1	1	0	0	7	6.17
NW	0	0	0	3	3	3	0	2	4	3	2	4	24	8.16
NNW	0	0	1	2	5	0	0	1	5	4	1	20	39	11.23
N	0	4	2	3	3	4	6	3	4	7	2	28	66	10.63
CALM													5	.12
TOTAL	2	18	26	47	47	48	60	63	59	61	43	199	678	8 93

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

(Sheet 35 of 52)

September (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	4	8	13	14	4	18	9	16	10	11	28	135	8.18
NE	1	2	8	7	9	10	11	5	9	9	5	13	89	7 53
ENE	1	3	4	5	7	12	6	2	1	2	3	6	52	6.49
E	0	7	7	13	8	7	6	1	0	1	0	0	50	4 28
ESE	3	2	6	16	11	9	6	4	1	2	1	0	61	4.76
SE	1	2	9	12	22	23	18	18	11	5	3	5	129	6.14
SSE	2	3	7	10	14	26	30	20	19	23	13	43	210	8.14
S	2	4	4	8	4	11	6	12	18	16	22	46	153	9 29
SSW	1	1	1	2	0	2	4	2	6	5	6	30	60	10.43
SW	2	0	2	0	1	1	1	0	3	2	3	0	15	6.72
WSW	1	0	3	3	3	4	0	1	0	1	0	0	16	4.47
W	0	0	1	1	0	0	0	0	0	0	0	1	3	5 93
WNW	0	4	2	2	3	2	3	1	2	0	0	0	19	4 83
NW	1	3	5	7	2	9	11	5	3	1	2	3	52	5 98
NNW	1	3	11	7	3	7	5	5	5	7	1	17	72	7 82
N	2	6	7	11	16	13	16	20	16	9	9	75	200	9 65
CALM													14	.38
TOTAL	18	44	85	117	117	140	141	105	110	93	79	267	1330	7.71

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	6	0	0	2	1	1	2	0	0	2	0	15	4 53
NE	2	7	2	2	2	5	0	0	0	0	0	1	21	3 84
ENE	0	5	1	4	3	2	0	0	0	0	0	1	16	3 94
E	2	3	3	7	10	2	0	0	0	0	0	0	27	3 61
ESE	3	7	11	8	4	1	0	0	0	0	0	0	34	2 90
SE	0	5	13	13	11	19	6	2	2	0	0	0	71	4.43
SSE	1	4	9	4	6	19	13	6	2	0	1	0	65	5.15
S	0	2	5	6	6	2	4	7	7	6	2	3	50	6 81
SSW	1	2	1	1	0	0	0	2	1	2	1	0	11	5 68
SW	0	2	0	1	0	0	1	0	0	0	0	0	4	3 52
WSW	0	2	2	0	0	0	0	0	0	0	0	0	4	2.12
W	0	3	1	0	0	0	0	2	0	0	0	0	6	3.48
WNW	0	0	3	0	0	0	0	0	0	0	0	0	3	2.70
NW	1	3	7	3	4	2	5	4	1	2	0	0	32	4 82
NNW	1	5	7	6	6	7	2	0	0	1	0	1	36	4 34
N	4	4	4	3	3	3	0	1	0	0	0	0	22	3 28
CALM													20	.33
TOTAL	16	60	69	58	57	63	32	26	13	11	6	6	437	4 35

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

September (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	0	0	0	0	0	0	0	0	0	0	2	2 00
NE	0	1	0	0	0	0	0	0	0	0	0	0	1	1 80
ENE	0	0	0	0	2	0	0	0	0	0	0	0	2	5 00
E	0	0	1	0	1	1	0	1	0	0	0	0	4	5.15
ESE	0	0	0	1	1	0	0	0	0	0	0	0	2	4.40
SE	0	1	1	6	4	2	1	0	0	0	0	0	15	4 26
SSE	0	1	5	6	4	1	0	1	0	0	0	0	18	3.73
S	0	1	2	1	2	1	1	0	1	0	0	0	9	4 57
SSW	0	1	2	2	0	0	0	0	0	1	0	0	6	3.78
SW	0	0	0	0	0	0	0	1	0	0	0	0	1	7.70
WSW	0	1	1	0	0	0	0	0	0	0	0	0	2	2 25
W	0	2	4	1	0	0	0	0	0	0	0	0	7	2 56
WNW	3	6	2	0	2	0	0	0	0	0	0	0	13	2.18
NW	2	2	7	4	5	11	8	4	6	2	0	0	51	5 51
NNW	0	4	3	1	1	1	1	1	0	0	0	0	12	3 64
N	0	2	2	0	0	0	0	0	0	0	0	0	4	2 00
CALM													11	.18
TOTAL	5	24	30	22	22	17	11	8	7	3	0	0	160	3 96

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	2	14	10	16	23	13	24	18	20	18	17	61	236	8.42
NE	3	12	13	15	19	21	17	11	12	10	6	18	157	6 64
ENE	1	12	6	18	18	17	15	9	6	8	5	16	131	6.49
E	3	10	18	23	26	14	11	5	5	2	0	0	117	4 50
ESE	6	12	24	36	28	23	22	12	13	9	2	0	187	5 03
SE	1	10	25	39	41	56	32	23	17	12	9	8	273	5.71
SSE	3	9	21	24	28	57	54	40	33	39	25	73	406	7.74
S	2	10	14	22	14	19	20	30	38	30	40	114	353	9 30
SSW	2	5	5	5	1	3	10	9	9	16	16	67	148	10.34
SW	2	2	2	2	3	3	7	1	5	4	4	5	40	7 52
WSW	1	3	7	4	6	6	0	2	1	1	1	0	32	4 51
W	0	6	6	2	2	0	0	3	0	1	0	1	21	4 05
WNW	3	10	7	5	5	2	4	2	3	1	0	0	42	4 08
NW	4	8	19	18	14	27	24	16	14	11	4	8	167	6 03
NNW	2	12	23	17	15	16	8	8	11	12	3	38	165	7 51
N	6	16	15	19	25	21	24	26	22	20	19	105	318	9 23
CALM													52	.30
TOTAL	41	151	215	265	268	298	272	215	209	194	151	514	2845	7 30

NUMBER OF NVAL D OBSERVATIONS = 35
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

October (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	1	1	1	0	0	0	3	7.70
SSE	0	0	0	0	0	0	0	1	1	2	6	5	15	10.52
S	0	0	0	0	0	0	0	2	0	1	0	3	6	9.73
SSW	0	0	0	0	1	0	1	0	0	2	1	0	5	8 28
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	0	0	1	11	5.70
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	0	0	1	0	2	4	2	5	7	9	30	9 88

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	1	0	1	0	0	0	2	7 85
SE	0	0	0	0	0	1	0	0	0	0	0	0	1	5 30
SSE	0	0	0	0	2	2	0	0	0	1	2	5	12	9.41
S	0	0	0	0	0	0	0	1	3	1	1	2	8	9 82
SSW	0	0	0	0	0	1	0	0	0	0	3	2	6	10.35
SW	0	0	0	0	0	0	0	0	0	0	0	1	1	12.60
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	0	0	1	0	2	3	13.20
N	0	0	0	0	0	0	1	0	1	1	0	0	3	8 23
CALM													0	0.
TOTAL	0	0	0	0	2	4	2	1	5	4	6	12	36	9.76

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

October (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	2	1	1	0	0	0	0	4	6 30
NE	0	0	0	0	0	1	2	0	0	0	0	0	3	6 67
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	2	0	0	0	0	0	0	0	2	4 30
ESE	0	0	0	0	3	0	1	0	1	1	0	0	6	6 33
SE	0	0	0	0	1	0	0	1	0	0	0	0	2	5 90
SSE	0	0	0	0	1	0	1	0	1	1	3	6	13	10.32
S	0	0	0	0	2	1	1	2	2	1	0	0	9	6 92
SSW	0	0	0	1	0	0	2	0	0	0	1	3	7	8 86
SW	0	0	0	1	0	1	0	0	1	0	0	1	4	7.40
WSW	0	0	0	0	0	0	1	0	1	0	0	0	2	7 60
W	0	0	0	0	0	0	0	0	1	0	0	0	1	9 00
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NNW	0	0	0	0	0	0	0	1	0	0	1	7	9	13.04
N	0	0	0	0	0	0	1	0	1	1	2	0	5	8 94
CALM													0	0.
TOTAL	0	0	0	2	9	5	10	5	8	4	7	17	67	8 63

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	3	4	6	9	2	4	5	1	2	8	44	7 67
NE	0	0	5	4	5	6	1	0	0	0	1	7	29	6.76
ENE	1	0	4	4	5	3	2	2	0	0	0	0	21	4 50
E	0	6	6	4	3	3	1	3	1	1	0	0	28	4 30
ESE	0	2	5	3	9	6	7	4	8	2	0	0	46	5.73
SE	0	3	1	7	11	17	11	4	8	4	1	2	69	6.12
SSE	0	1	4	5	9	5	8	12	10	10	11	19	94	8 89
S	1	1	11	1	13	7	6	8	10	11	8	53	130	10.14
SSW	2	0	0	3	3	5	4	5	7	4	2	29	64	10.34
SW	0	1	3	0	2	1	2	3	1	2	1	5	21	7 62
WSW	0	1	0	2	5	1	1	3	0	0	1	1	15	6 07
W	0	2	1	0	4	2	1	3	0	0	0	0	13	5.19
WNW	0	1	3	0	1	0	0	1	1	1	1	1	10	6 36
NW	0	5	4	4	1	2	2	1	0	3	4	4	30	6 39
NNW	1	2	1	3	3	0	4	5	8	1	11	26	66	10.31
N	0	1	1	0	5	1	4	5	4	2	4	36	63	11.92
CALM													4	.35
TOTAL	5	26	52	44	85	68	56	63	63	43	47	191	747	8 38

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

October (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	3	5	2	5	0	9	2	0	0	2	18	47	9 29
NE	0	5	3	6	3	4	2	1	1	0	0	0	25	4.14
ENE	0	0	8	5	2	0	0	0	0	0	0	0	15	3 20
E	0	1	4	7	3	4	1	1	0	0	1	0	22	4 54
ESE	0	3	4	7	10	4	6	7	7	0	1	0	49	5 54
SE	3	5	9	8	14	22	13	15	12	7	3	4	115	6 25
SSE	1	9	1	8	6	12	8	27	18	20	16	61	187	9 54
S	0	3	4	1	8	2	16	19	9	13	6	87	168	11.24
SSW	0	0	1	2	5	6	7	2	7	1	0	20	51	10.01
SW	0	1	2	2	0	2	1	0	0	0	0	2	10	6 00
WSW	0	2	1	3	4	2	1	0	0	0	0	0	13	4 23
W	0	2	2	2	1	1	1	0	2	1	0	0	12	5 01
WNW	0	1	0	2	2	2	1	2	1	1	0	2	14	6.72
NW	1	2	5	7	6	5	1	7	3	4	3	6	50	6.77
NNW	0	2	5	8	8	10	7	10	12	7	5	21	95	8 08
N	0	2	7	4	3	10	9	11	4	3	2	25	80	9.17
CALM													11	.24
TOTAL	6	41	61	74	80	86	83	104	76	57	39	246	964	8 27

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	1	3	1	1	0	0	0	0	0	0	8	3 55
NE	1	5	3	0	0	0	0	0	0	0	0	0	9	1 86
ENE	3	7	6	0	0	0	0	0	0	0	0	0	16	1 85
E	1	2	3	4	2	0	0	0	0	0	0	0	12	3 00
ESE	1	7	11	17	9	4	1	1	2	0	1	0	54	4 03
SE	2	4	8	18	19	7	8	3	0	0	1	2	72	4 66
SSE	2	4	11	22	15	17	23	24	13	3	4	14	152	6 35
S	0	3	6	5	4	7	15	12	8	7	2	8	77	7 25
SSW	0	6	0	3	3	7	1	2	2	0	1	3	28	6 01
SW	1	3	4	0	1	3	1	0	0	0	0	0	13	3.46
WSW	0	1	2	0	1	1	0	0	0	0	0	0	5	3 30
W	0	3	1	1	1	0	0	0	0	0	0	0	6	2 60
WNW	1	0	3	0	1	0	0	0	0	0	0	0	5	2.70
NW	0	3	1	6	4	7	6	4	3	2	0	2	38	6 20
NNW	0	1	1	8	3	2	3	4	1	1	0	0	24	5 34
N	1	1	3	4	6	3	2	2	2	3	0	1	28	5 92
CALM													19	.36
TOTAL	13	52	64	91	70	59	60	52	31	16	9	30	566	5 27

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

October (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	3	0	1	0	0	0	0	0	0	0	0	4	2.15
NE	1	2	2	0	0	0	0	0	0	0	0	0	5	1 64
ENE	2	0	0	0	0	0	0	0	0	0	0	0	2	.90
E	1	2	3	0	0	0	0	0	0	0	0	0	6	2.15
ESE	1	9	7	4	1	2	2	0	0	0	0	0	26	2 94
SE	4	12	19	23	10	6	3	1	0	0	0	0	78	3 32
SSE	4	16	15	15	13	6	8	1	0	0	0	1	79	3 61
S	5	4	12	8	9	4	8	7	2	1	1	5	66	5 21
SSW	1	8	4	5	4	6	3	1	1	0	0	0	33	3 93
SW	0	4	0	2	2	1	0	0	1	0	0	0	10	3.75
WSW	1	3	3	2	0	0	0	0	0	0	0	0	9	2 22
W	1	7	10	4	2	0	0	0	0	0	0	0	24	2 50
WNW	5	6	9	2	4	1	3	0	0	0	0	0	30	2 94
NW	3	8	12	4	4	6	7	8	0	1	0	0	53	4 36
NNW	0	8	3	3	1	0	0	0	0	0	0	0	15	2 55
N	2	3	2	1	1	0	1	0	0	0	0	1	11	3.48
CALM													40	.31
TOTAL	31	95	101	74	51	32	35	18	4	2	1	7	491	3 37

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	8	9	10	12	12	12	7	5	1	4	26	107	7 81
NE	2	12	13	10	8	11	5	1	1	0	1	7	71	4 85
ENE	6	7	18	9	7	3	2	2	0	0	0	0	54	3 22
E	2	11	16	15	10	7	2	4	1	1	1	0	70	3 97
ESE	2	21	27	31	32	16	18	12	19	3	2	0	183	4 83
SE	9	24	37	57	56	53	36	25	21	11	5	8	342	5 21
SSE	7	30	32	50	47	42	48	66	43	37	42	113	557	7.76
S	6	11	33	15	36	21	46	51	34	35	18	158	464	9 28
SSW	3	14	5	14	16	25	18	10	17	7	8	57	194	8.43
SW	1	9	9	5	5	8	4	3	3	2	1	9	59	5 84
WSW	1	7	6	7	10	4	3	3	1	0	1	1	44	4.49
W	1	14	14	7	8	3	2	3	3	1	0	0	56	3.79
WNW	6	8	15	4	8	3	4	3	2	2	1	3	59	4.40
NW	4	18	22	21	15	20	16	20	6	10	7	12	171	5 83
NNW	1	13	10	22	15	12	14	20	21	11	17	57	213	8 59
N	3	7	13	9	15	14	18	18	12	10	8	63	190	9 25
CALM													74	.32
TOTAL	55	214	279	286	300	254	248	248	189	131	116	514	2908	6 93

NUMBER OF NVAL D OBSERVATIONS = 68
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

November (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	1	0	0	0	0	1	7.70
NE	0	0	0	0	1	3	0	0	0	1	0	0	5	6 36
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	1	0	0	0	0	0	0	1	5.70
SE	0	0	0	0	0	0	0	0	0	0	0	1	11	1 90
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
S	0	0	0	0	0	0	0	0	1	0	0	0	1	8.40
SSW	0	0	0	0	0	0	0	0	0	1	0	4	51	8 90
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	1	0	0	0	0	0	0	0	1	4 20
NNW	0	0	0	0	0	0	0	0	1	1	0	0	2	9.15
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	0	0	2	4	0	1	2	3	0	5	17	10.74L

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	1	0	0	0	0	0	0	1	5.70
NE	0	0	0	0	0	0	0	1	0	0	0	0	1	7 60
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	1	0	0	0	1	8.70
ESE	0	0	0	0	0	0	0	0	0	1	2	0	3	10.13
SE	0	0	0	0	0	0	0	0	0	1	0	2	3	12.10
SSE	0	0	0	0	0	0	0	0	0	0	1	0	1	10.50
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSW	0	0	0	0	0	0	0	0	0	0	0	1	1	18.00
SW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	0	0	1	0	1	11.00
NNW	0	0	0	0	0	0	0	0	0	1	0	5	6	11.38
N	0	0	0	0	1	0	0	0	0	0	0	0	1	4 50
CALM													0	0.
TOTAL	0	0	0	0	1	1	0	1	1	3	4	8	19	10.58

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

November (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	1	1	0	0	0	0	0	2	6.15
NE	0	0	0	0	0	2	1	1	0	0	0	0	4	6 32
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	2	0	0	0	0	0	0	2	5 55
SE	0	0	0	0	0	0	0	0	0	1	0	2	3	11.10
SSE	0	0	0	0	0	0	0	0	0	0	1	1	2	10.75
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSW	0	0	0	0	0	0	0	0	0	0	1	0	1	11.00
SW	0	0	0	0	0	0	0	0	1	1	1	1	4	10.00
WSW	0	0	0	0	0	0	0	1	0	1	0	1	3	9.77
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	1	0	0	1	0	0	2	8 00
NNW	0	0	0	0	0	0	0	0	0	1	4	3	8	10.61
N	0	0	0	0	0	0	0	0	0	1	1	0	2	9 95
CALM													0	0.
TOTAL	0	0	0	0	0	5	3	2	1	6	8	8	33	9 23

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	3	6	5	6	9	1	5	5	7	13	60	8 03
NE	0	0	4	5	8	6	6	5	0	2	1	0	37	5.48
ENE	0	1	3	0	2	3	4	4	3	1	0	0	21	6 09
E	0	2	1	2	4	2	2	0	0	1	0	0	14	4 81
ESE	1	2	0	2	2	5	5	2	1	1	1	0	22	5.72
SE	0	0	0	1	4	7	6	9	8	9	6	7	57	8 22
SSE	1	0	0	1	6	10	5	7	8	5	1	7	51	7 92
S	0	1	2	0	3	4	3	5	3	3	4	35	63	12.58
SSW	0	2	1	0	1	0	0	0	0	3	2	37	46	14.03
SW	0	0	1	0	1	1	3	0	5	3	4	19	37	11.53
WSW	0	0	1	0	1	1	3	3	1	0	1	7	18	9 54
W	0	2	2	0	0	1	4	2	4	0	2	8	25	9.11
WNW	0	0	1	1	1	6	3	1	1	2	0	3	19	7 87
NW	0	1	0	1	3	3	1	4	2	2	1	16	34	11.76
NNW	0	1	0	2	3	3	5	8	5	7	5	63	102	12.74
N	0	0	1	3	1	8	7	3	5	11	9	34	82	10.58
CALM													2	.60
TOTAL	2	12	20	24	45	66	66	54	51	55	44	249	690	9 94

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

November (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	2	5	4	7	5	4	2	2	2	2	4	39	6 29
NE	0	4	3	4	1	1	1	0	0	0	0	0	14	3.49
ENE	0	4	5	7	2	2	0	0	0	0	0	0	20	3 24
E	0	3	5	4	4	2	1	0	0	0	0	0	19	3 57
ESE	1	2	3	10	11	4	0	0	1	0	0	0	32	4 08
SE	0	5	3	8	12	14	12	13	12	11	2	18	110	7.44
SSE	1	3	0	7	13	14	22	7	7	5	11	56	146	9 62
S	0	1	3	1	1	3	4	6	11	11	17	81	139	12.94
SSW	0	0	2	1	0	4	1	5	3	4	7	41	68	12.57
SW	0	0	2	1	0	3	0	1	0	2	1	17	27	12.41
WSW	0	1	2	1	2	1	1	0	2	3	0	3	16	7 91
W	0	0	2	1	1	2	2	3	0	0	1	5	17	8.49
WNW	0	0	1	0	4	8	5	5	6	2	1	11	43	8 90
NW	0	1	1	4	4	4	5	7	7	10	12	76	131	13.30
NNW	1	0	2	3	2	4	6	7	9	19	13	74	140	12.57
N	0	0	2	5	7	3	2	8	6	8	5	39	85	11.14
CALM													4	.42
TOTAL	3	26	41	61	71	74	66	64	66	77	72	425	1050	10.36

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	5	3	2	3	4	1	0	1	0	0	0	19	3 85
NE	0	3	3	2	1	0	1	0	0	0	0	0	10	3 07
ENE	0	4	2	4	0	0	0	0	0	0	0	0	10	2 69
E	1	2	4	5	0	1	0	0	0	0	0	0	13	2 97
ESE	0	0	3	7	5	2	2	1	0	0	0	0	20	4 30
SE	0	4	6	10	18	7	5	5	7	7	4	5	78	6 36
SSE	2	7	6	6	5	15	16	14	15	11	3	26	126	7 91
S	1	0	2	1	4	11	12	12	9	3	7	12	74	8 28
SSW	0	1	6	2	5	5	4	5	4	5	3	8	48	7.47
SW	0	3	1	1	1	5	0	0	1	0	3	15	30	10.05
WSW	0	2	1	1	1	2	2	1	1	0	0	1	12	5 88
W	0	0	1	1	1	2	1	0	0	1	2	2	11	7 63
WNW	0	2	1	2	5	8	10	6	1	1	0	2	38	6.14
NW	0	1	2	1	3	9	14	14	12	4	3	6	69	7.72
NNW	0	3	0	2	3	6	4	5	2	3	3	7	38	7.72
N	0	2	2	0	6	0	3	3	1	2	0	2	21	6.16
CALM													12	.33
TOTAL	4	39	43	47	61	77	75	66	54	37	28	86	629	6 94

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

November (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	1	0	0	0	0	0	0	0	0	0	0	1	1 50
ENE	1	0	3	0	0	0	0	0	0	0	0	0	4	2.17
E	0	4	1	0	0	0	0	0	0	0	0	0	5	1.74
ESE	0	6	5	2	2	3	0	0	0	0	0	0	18	3.16
SE	2	4	13	10	13	6	1	2	2	1	1	0	55	4.13
SSE	1	2	8	9	12	8	9	4	2	3	0	3	61	5.45
S	0	2	8	10	5	8	5	2	4	1	0	0	45	4 84
SSW	1	4	1	1	4	0	0	3	6	1	0	0	21	5 61
SW	3	3	4	2	0	1	4	0	1	0	0	0	18	3 66
WSW	1	6	3	6	4	3	1	1	0	0	1	0	26	3 83
W	1	9	7	5	5	3	0	2	0	0	0	0	32	3 35
WNW	2	5	4	7	1	2	3	3	0	0	1	0	28	4.15
NW	0	2	8	8	11	6	1	3	3	3	1	0	46	5 03
NNW	1	0	3	1	0	0	2	0	0	0	0	1	8	5 04
N	0	1	0	1	0	0	2	0	0	0	0	0	4	4 80
CALM													22	.21
TOTAL	13	49	68	62	57	40	28	20	18	9	4	4	394	4 20

NUMBER OF NVAL D OBSERVATIONS = 3
TOTAL NUMBER OF OBSERVATIONS = 2880

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	7	11	12	15	17	15	4	8	7	9	17	122	6.77
NE	0	8	10	11	11	12	9	7	0	3	1	0	72	4 84
ENE	1	10	13	13	4	5	4	4	3	1	0	0	58	4.11
E	1	11	11	11	8	5	4	0	1	1	0	0	53	3.74
ESE	2	10	11	21	20	17	9	3	2	2	3	0	100	4 60
SE	2	13	22	29	47	34	24	30	29	31	14	35	310	6 84
SSE	5	12	14	23	36	47	52	32	32	24	17	96	390	8 22
S	1	4	15	13	14	26	24	25	28	18	28	128	324	10.61
SSW	1	7	10	5	10	10	5	13	13	14	13	91	192	10.98
SW	3	6	8	4	2	10	7	1	8	6	9	52	116	10.08
WSW	1	9	7	8	8	7	7	6	4	4	2	12	75	6 64
W	1	11	12	7	7	8	7	7	4	1	5	15	85	6 63
WNW	2	7	7	10	11	24	21	15	8	5	2	18	130	7 04
NW	0	5	11	14	22	22	22	29	24	20	18	98	285	10.33
NNW	2	4	5	8	8	15	17	21	17	32	25	153	307	11.67
N	0	3	5	10	17	12	14	15	12	22	15	75	20	10.08
CALM													40	.29
TOTAL	22	127	172	199	240	271	241	212	193	191	161	790	2859	8 62

NUMBER OF NVAL D OBSERVATIONS = 21
TOTAL NUMBER OF OBSERVATIONS = 2880

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

December (1972 - 1975): 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	2	2	15.50
S	0	0	1	0	0	2	0	0	0	2	0	2	7	8 89
SSW	0	0	1	0	0	0	0	1	0	0	1	9	12	11.86
SW	0	0	0	0	0	0	1	1	0	0	0	2	4	12.00
WSW	0	0	0	0	0	0	0	1	0	0	0	1	2	9 90
W	0	1	0	3	0	0	0	0	0	0	0	0	4	3.12
WNW	0	1	1	1	1	0	0	1	0	0	0	0	5	4 00
NW	0	0	1	0	2	1	0	1	1	4	1	0	11	7 89
NNW	0	0	0	0	0	1	1	0	0	0	0	0	2	6 50
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	2	4	4	3	4	2	5	1	6	2	16	49	8 89

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	1	0	0	0	0	0	0	1	0	1	3	8.13
NE	0	0	0	0	0	1	0	0	0	0	0	0	1	5 30
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0.	0.
S	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSW	0	0	0	0	0	0	0	0	0	0	0	0	0.	0.
SW	0	0	0	0	1	0	0	0	0	0	0	0	1	5 00
WSW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
W	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
WNW	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
NW	0	0	0	0	0	0	0	0	2	0	0	0	2	8 95
NNW	0	0	0	0	0	0	0	0	0	0	1	1	2	11.90
N	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
CALM													0	0.
TOTAL	0	0	1	0	1	1	0	0	2	1	1	2	9	8.49

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

December (1972 - 1975): 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	0	0	0	0	0	3	0	0	0	3	8 60
NE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ENE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
E	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
ESE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
SSE	0	0	0	0	0	0	0	0	0	0	0	0	0	0.
S	0	0	0	0	0	0	1	0	0	2	0	0	3	8.47
SSW	0	0	0	0	0	0	0	0	0	0	0	1	1	11.20
SW	0	0	0	0	0	0	0	0	1	0	0	0	1	8 60
WSW	0	0	0	0	0	0	0	1	0	0	0	0	1	7.70
W	0	0	0	0	0	0	0	1	0	0	0	0	1	7 60
WNW	0	0	0	0	0	0	0	2	0	0	0	1	3	9 30
NW	0	0	0	0	0	1	1	0	0	0	2	1	5	9.16
NNW	0	0	0	0	0	0	1	0	0	2	1	2	6	11.57
N	0	0	0	0	0	1	0	0	1	1	0	6	9	13.20
CALM													0	0.
TOTAL	0	0	0	0	0	2	3	4	5	5	3	11	33	10.55

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	1	4	6	9	3	4	3	6	1	5	42	7.45
NE	0	0	0	2	3	0	1	0	1	1	0	0	8	5.75
ENE	0	0	5	1	2	1	0	0	1	0	0	0	10	4 02
E	0	0	2	0	0	0	0	0	0	0	0	0	2	2 80
ESE	0	0	1	4	3	5	3	0	0	0	0	0	16	4.76
SE	0	0	0	1	1	8	6	2	0	2	0	0	20	6 20
SSE	0	1	2	5	1	4	6	2	3	4	1	10	39	7.78
S	0	0	0	0	4	2	3	1	1	2	3	29	45	11.90
SSW	0	0	0	2	2	0	0	1	1	1	2	35	44	13.72
SW	0	0	0	0	0	1	3	0	0	0	1	18	23	13.57
WSW	0	0	0	0	0	2	0	0	0	1	1	9	13	11.85
W	0	0	0	0	3	3	0	0	2	1	1	3	13	8 51
WNW	0	0	1	1	2	0	0	2	0	0	3	14	23	10.76
NW	0	0	1	1	3	0	4	2	4	2	3	16	36	10.40
NNW	0	0	0	0	2	8	2	3	7	2	4	63	91	14.77
N	0	0	0	1	0	1	2	2	5	4	5	34	54	13.34
CALM													0	0.
TOTAL	0	1	13	22	32	44	33	19	28	26	25	236	479	11.09

NUMBER OF NVAL D OBSERVATIONS = 63
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

December (1972 - 1975): 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	1	4	8	6	2	1	3	0	0	6	31	7 21
NE	0	0	3	3	2	3	3	1	0	1	0	1	16	5 07
ENE	0	4	4	4	3	1	1	1	0	0	0	0	18	3 52
E	0	2	6	7	0	1	1	0	0	1	0	0	18	3 61
ESE	1	0	5	3	5	4	1	1	1	0	0	1	22	4.75
SE	1	1	3	2	7	9	5	4	1	3	0	1	37	5.71
SSE	0	0	4	4	2	8	11	9	4	11	7	45	105	10.29
S	2	1	0	2	2	4	8	8	9	7	8	95	146	13.15
SSW	0	2	0	1	1	3	3	4	3	6	1	57	81	13.94
SW	0	0	1	1	2	1	4	0	3	6	1	19	38	10.98
WSW	0	0	1	1	3	0	0	2	3	0	2	18	30	10.52
W	0	1	4	2	1	0	4	1	2	2	0	5	22	7 26
WNW	0	0	2	2	1	7	5	3	9	3	2	7	41	8.10
NW	1	2	3	2	5	11	8	8	8	2	10	76	136	11.13
NNW	0	1	3	6	3	10	9	7	10	9	8	99	165	12.20
N	0	1	0	2	3	7	7	4	6	6	6	40	82	11.16
CALM													0	0.
TOTAL	5	15	40	46	48	75	72	54	62	57	45	469	988	10.67

NUMBER OF NVAL D OBSERVATIONS = 13
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	6	4	0	1	0	0	3	0	1	1	17	5 53
NE	0	4	0	0	1	1	0	2	0	3	0	2	13	6 50
ENE	1	1	3	0	0	1	0	0	0	0	0	1	7	4 34
E	0	5	2	1	3	2	1	0	1	0	0	0	15	3 83
ESE	0	6	7	6	1	3	0	0	0	0	0	0	23	2 95
SE	1	2	6	12	18	20	9	5	0	3	1	0	77	5 05
SSE	0	0	4	5	9	14	11	11	8	13	3	18	96	7 82
S	0	1	3	5	6	9	11	7	10	8	18	28	106	9.49
SSW	1	0	3	3	0	1	3	5	5	1	3	10	35	8 91
SW	0	0	3	1	0	2	3	0	0	3	1	11	24	9 90
WSW	0	3	2	3	0	2	6	3	2	4	2	2	29	7 04
W	0	4	0	4	5	1	1	4	1	0	0	0	20	4.71
WNW	1	2	4	1	10	4	2	1	0	3	1	1	30	5 29
NW	1	1	1	5	8	21	16	15	7	6	3	15	99	7.70
NNW	0	1	0	1	5	9	10	7	9	3	2	11	58	9 02
N	1	1	3	1	0	4	2	5	4	1	1	5	28	8 21
CALM													15	.44
TOTAL	6	32	47	52	66	95	75	65	50	48	36	105	692	7 24

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

December (1972 - 1975): 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	2	0	0	0	3	3	2	0	0	0	10	6.44
NE	0	2	0	1	0	1	1	0	0	0	0	0	5	3.74
ENE	0	4	1	2	0	1	1	0	1	0	0	0	10	3.60
E	2	5	0	0	0	0	0	0	0	0	0	0	7	1.16
ESE	1	3	4	6	7	1	0	0	0	0	0	0	22	3.54
SE	0	5	24	25	21	6	5	0	2	0	0	0	88	3.91
SSE	0	1	17	28	21	9	5	2	1	1	0	0	85	4.21
S	0	3	5	8	7	5	8	4	6	2	0	0	48	5.42
SSW	0	3	5	6	5	4	4	1	1	1	0	2	32	4.98
SW	0	5	4	8	4	2	5	2	1	2	0	0	33	4.61
WSW	0	2	6	8	7	3	3	1	2	0	1	0	33	4.58
W	0	6	7	6	6	3	0	0	2	1	0	0	31	3.91
WNW	1	4	2	5	13	4	4	2	0	0	0	0	35	4.51
NW	0	4	7	8	7	15	11	15	5	4	2	4	82	6.25
NNW	1	5	4	6	1	3	2	1	2	1	2	1	29	4.95
N	1	2	0	2	0	1	1	3	1	0	0	0	11	4.95
CALM													12	.43
TOTAL	6	54	88	119	99	58	53	34	26	12	5	7	573	4 58

NUMBER OF NVAL D OBSERVATIONS = 0
TOTAL NUMBER OF OBSERVATIONS = 2976

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	11	12	14	16	8	8	14	7	2	13	106	7.03
NE	0	6	3	6	6	6	5	3	1	5	0	2	43	5.48
ENE	1	10	13	7	5	4	2	1	2	0	0	1	46	3.73
E	2	12	10	8	3	3	2	0	1	1	0	0	42	3.24
ESE	2	9	17	19	16	13	4	1	1	0	0	1	83	3.93
SE	2	8	33	40	47	43	25	11	3	8	1	1	222	4.81
SSE	0	2	27	42	33	35	33	24	16	29	11	76	328	7.73
S	2	5	9	15	19	22	31	20	26	23	29	154	355	10.73
SSW	1	5	9	12	8	8	10	12	10	9	7	114	205	11.50
SW	0	5	8	10	7	6	16	3	5	11	3	50	124	9.52
WSW	0	5	9	12	10	7	9	8	7	5	6	30	108	7.89
W	0	12	11	15	15	7	5	6	7	4	1	8	91	5.56
WNW	2	7	10	10	27	15	11	11	9	6	6	23	137	6.89
NW	2	7	13	16	25	49	40	41	27	18	21	113	372	8.95
NNW	1	7	7	13	11	31	25	18	28	17	18	177	353	11.70
N	2	4	3	6	3	14	12	14	17	12	12	85	184	11.08
CALM													27	.44
TOTAL	17	105	193	243	249	279	238	181	174	155	117	848	2826	8.63

NUMBER OF NVAL D OBSERVATIONS = 150
TOTAL NUMBER OF OBSERVATIONS = 2976

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May 15, 1972 - May 14, 1976: 10-Meter Level

PASQU LL #A# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	1	0	9	11	9	11	5	7	3	11	68	8.13
NE	0	0	2	4	4	11	8	4	4	5	3	3	48	7 07
ENE	0	1	1	3	3	4	4	1	3	1	1	1	23	6.12
E	0	1	1	4	7	6	8	5	10	5	0	0	47	6 50
ESE	0	1	4	6	11	13	16	14	9	17	5	10	106	7.48
SE	0	1	0	2	6	4	14	16	14	15	6	8	86	8 09
SSE	0	2	2	2	7	9	14	13	23	14	19	43	148	10.53
S	0	0	3	5	3	11	16	15	20	24	12	76	185	11.04
SSW	0	0	2	1	4	4	8	9	10	19	16	87	160	12.78
SW	0	0	0	3	3	2	3	4	0	0	2	18	35	11.10
WSW	0	0	1	4	3	3	0	3	2	0	0	1	17	5 98
W	0	1	0	3	2	3	2	1	1	0	0	3	16	7 59
WNW	0	1	2	2	3	0	2	1	2	0	0	3	16	7.11
NW	0	0	1	1	3	1	2	2	4	4	4	16	38	10.50
NNW	0	1	2	2	2	2	2	1	6	2	1	23	44	11.44
N	0	0	0	2	4	2	5	2	3	5	2	9	34	9 23
CALM													2	.60
TOTAL	0	10	22	44	74	86	113	102	116	118	74	312	1073	9.71

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 35043

PASQU LL #B# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	1	2	3	0	7	3	1	5	2	2	12	38	9 29
NE	0	0	0	0	1	4	3	2	0	2	2	0	14	7 32
ENE	0	0	0	1	1	0	0	0	0	0	0	0	2	4 05
E	0	0	0	0	0	4	1	3	1	0	2	0	11	7 52
ESE	0	0	1	2	1	5	10	7	5	8	4	2	45	7 87
SE	0	0	0	1	2	4	7	6	4	5	2	13	44	9 27
SSE	0	0	0	1	3	3	2	3	11	11	11	21	66	10.29
S	0	1	0	0	0	1	5	2	9	11	10	29	68	11.55
SSW	0	0	0	0	0	3	2	3	4	4	10	26	52	12.06
SW	0	0	0	0	2	0	1	1	1	2	7	11	25	11.50
WSW	0	0	0	0	0	0	1	0	0	0	2	6	9	15.64
W	0	0	0	0	1	0	0	2	0	0	0	3	6	13.57
WNW	0	0	0	1	0	0	0	1	0	1	0	1	4	8.15
NW	0	0	0	0	0	1	0	0	2	1	2	8	14	11.48
NNW	0	0	0	0	1	0	0	1	1	2	3	20	28	13.12
N	0	0	0	0	1	1	4	2	3	4	3	7	25	10.20
CALM													0	0.
TOTAL	0	2	3	9	13	33	39	34	46	53	60	159	451	10.48

NUMBER OF NVAL D OBSERVATIONS = 1
TOTAL NUMBER OF OBSERVATIONS = 35043

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May 15, 1972 - May 14, 1976: 10-Meter Level

PASQU LL #C# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	0	0	3	3	7	4	4	4	7	2	10	44	9 07
NE	0	0	0	4	4	5	6	5	4	1	0	5	34	7.41
ENE	0	1	0	1	3	4	2	5	0	4	1	2	23	7 34
E	0	0	0	3	6	3	1	3	3	2	0	0	21	6.16
ESE	0	0	0	2	6	8	9	8	7	10	5	4	59	7.75
SE	0	0	1	1	4	6	6	7	7	6	5	13	56	8 82
SSE	0	0	0	1	3	4	12	7	11	10	9	32	89	10.08
S	0	0	0	2	3	3	12	9	14	9	11	32	95	10.37
SSW	0	1	0	1	4	4	7	4	6	5	14	29	75	10.01
SW	0	0	0	2	1	4	2	2	7	2	6	18	44	10.49
WSW	0	0	0	3	2	4	2	4	2	3	0	4	24	7 96
W	0	0	0	0	1	0	0	1	1	1	1	4	9	14.58
WNW	0	0	0	0	2	1	0	3	0	0	0	4	10	9 89
NW	0	0	0	0	1	3	2	1	0	7	5	11	30	10.05
NNW	0	0	0	1	1	1	1	4	4	5	12	36	65	12.88
N	0	0	0	2	3	4	1	1	4	6	11	13	45	10.05
CALM													1	.40
TOTAL	0	2	1	26	47	61	67	68	74	78	82	217	724	9.68

NUMBER OF NVAL D OBSERVATIONS = 6
TOTAL NUMBER OF OBSERVATIONS = 35043

PASQU LL #D# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	1	4	15	34	46	49	41	39	46	38	35	87	435	8.11
NE	1	4	22	25	43	31	35	27	24	19	18	33	282	7 03
ENE	1	6	23	33	32	20	35	34	28	13	7	21	253	6 56
E	1	15	28	35	38	34	25	27	17	9	15	15	259	6 08
ESE	2	8	31	36	46	64	77	48	51	24	16	13	436	7 01
SE	1	8	15	33	57	76	74	73	63	74	61	121	656	8 38
SSE	1	6	19	41	56	79	83	86	98	89	76	270	904	9 56
S	1	11	19	25	63	48	76	82	96	98	104	556	1179	11.38
SSW	2	3	6	13	24	29	49	50	57	61	56	397	747	11.81
SW	1	5	11	16	17	25	36	29	27	31	28	163	389	10.31
WSW	0	2	8	12	19	10	10	18	7	7	18	62	173	9 82
W	0	7	12	15	16	12	12	8	15	7	8	63	175	9 60
WNW	0	2	6	11	7	14	9	11	10	10	7	58	145	10.11
NW	0	6	11	21	18	12	13	15	19	22	22	141	300	11.17
NNW	1	7	9	23	23	24	22	36	43	44	56	343	631	12.42
N	0	8	9	21	28	26	32	33	35	44	42	251	529	11.36
CALM													23	.20
TOTAL	13	102	244	394	533	553	629	616	636	590	569	2614	7516	9 88

NUMBER OF NVAL D OBSERVATIONS = 208
TOTAL NUMBER OF OBSERVATIONS = 35043

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May 15, 1972 - May 14, 1976: 10-Meter Level

PASQU LL #E# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	2	23	42	46	56	39	63	39	34	33	38	120	535	7 98
NE	2	31	48	57	50	40	48	28	37	27	15	27	410	6 06
ENE	5	27	49	51	47	32	21	25	18	18	14	25	332	5 69
E	6	36	43	84	61	51	35	23	13	27	17	41	437	6 05
ESE	7	30	56	82	112	85	59	47	40	29	24	159	730	7 90
SE	6	41	81	124	165	174	160	121	114	84	57	211	1338	7 36
SSE	7	37	58	100	140	180	203	175	175	174	154	738	2141	9 82
S	9	21	37	65	71	107	116	147	175	178	173	1084	2183	11.66
SSW	3	17	23	33	31	63	56	68	69	62	63	486	974	11.54
SW	2	12	25	25	26	28	28	24	20	27	18	157	392	9 69
WSW	1	10	16	14	20	15	14	11	12	12	7	85	217	9 60
W	0	8	24	12	14	17	17	9	10	10	7	63	191	9.13
WNW	1	11	17	23	23	34	24	27	38	10	10	101	319	8 88
NW	7	16	30	53	50	68	57	58	50	41	50	375	855	10.61
NNW	4	12	31	55	60	68	67	75	101	93	81	615	1262	11.64
N	4	25	31	42	55	64	81	78	69	66	61	440	1016	10.64
CALM													65	.22
TOTAL	66	357	611	866	981	1065	1049	955	975	891	789	4727	13397	9 68

NUMBER OF NVAL D OBSERVATIONS = 136
TOTAL NUMBER OF OBSERVATIONS = 35043

PASQUILL #F# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	9	28	31	20	19	15	11	7	8	3	6	2	159	4.40
NE	5	33	31	21	16	14	5	3	1	4	3	5	141	4 01
ENE	7	41	36	27	10	14	2	2	5	1	1	4	150	3 60
E	10	22	40	58	37	11	3	6	6	4	4	7	208	4 25
ESE	6	59	85	124	75	37	15	14	8	6	6	12	447	4 22
SE	5	48	114	164	161	125	68	49	27	20	10	23	814	5 04
SSE	7	55	94	115	136	156	136	119	82	54	32	110	1096	6.46
S	6	35	58	76	86	99	112	96	94	68	63	201	994	8.18
SSW	8	27	45	42	32	54	33	49	29	30	42	131	522	8 29
SW	7	24	36	21	13	29	20	16	11	12	12	67	268	7 31
WSW	3	20	35	23	17	14	16	21	11	6	7	16	189	5.71
W	1	23	21	18	15	13	11	10	5	10	9	14	150	5.79
WNW	5	12	30	25	33	31	30	28	7	11	6	12	230	5 69
NW	7	15	42	50	55	83	75	53	42	25	13	47	507	6 65
NNW	4	23	30	39	42	48	33	34	20	14	9	35	331	6.48
N	11	22	25	30	31	26	23	22	14	17	7	30	258	6 39
CALM													120	.34
TOTAL	101	487	453	853	778	769	593	529	370	285	230	716	6584	6.17

NUMBER OF NVAL D OBSERVATIONS = 54
TOTAL NUMBER OF OBSERVATIONS = 35043

TABLE 2.3-27
WIND FREQUENCY DISTRIBUTIONS AT CPNPP

May 15, 1972 - May 14, 1976: 10-Meter Level

PASQUILL #G# (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	0	13	6	4	5	0	5	3	3	2	2	4	47	5.16
NE	2	8	8	3	0	1	2	0	0	1	0	1	26	3 26
ENE	4	10	11	8	5	1	2	0	1	1	0	0	43	3.13
E	5	22	13	7	6	1	0	3	0	0	0	0	57	2.72
ESE	3	32	28	25	19	10	3	5	0	1	1	1	128	3 57
SE	8	44	94	125	73	39	22	10	6	2	1	5	429	3 97
SSE	7	38	83	107	87	38	43	12	8	6	2	15	446	4 58
S	8	28	52	52	54	46	43	25	17	9	6	15	355	5 33
SSW	8	34	28	32	39	19	20	13	15	6	3	17	234	5 27
SW	6	31	31	37	17	14	16	11	7	7	5	18	200	5.10
WSW	5	26	38	34	29	14	17	11	11	4	4	6	199	4 63
W	5	49	61	39	36	15	4	7	3	2	1	4	226	3 62
WNW	13	46	43	34	36	21	18	8	1	1	2	1	224	3.77
NW	8	34	66	50	49	71	51	52	24	16	6	14	441	5.48
NNW	7	30	24	18	13	12	10	8	6	2	3	12	145	5 20
N	4	14	7	7	5	5	5	3	1	3	2	12	68	6.41
CALM													115	.32
TOTAL	93	459	593	582	473	307	261	171	103	63	38	125	3383	4.49

NUMBER OF NVAL D OBSERVATIONS = 19
TOTAL NUMBER OF OBSERVATIONS = 35043

PASQUILL ALL (FROM AEC/DELTA-T CRITERIA 10-60 METERS)
WINDS AT 10 METERS LEVEL

W ND FREQUENCY DISTR BUTION
(FREQUENCY N NUMBER OF OCCURRENCES)

W ND	UPPER CLASS INTERVALS OF W ND SPEED (MPH)													MEAN
DIRECTION	1	2	3	4	5	6	7	8	9	10	11	>11	TOTAL	SPEED
NNE	12	73	98	111	138	132	136	106	105	92	90	250	1343	7 57
NE	10	78	111	115	121	109	109	70	72	65	41	75	976	6.11
ENE	18	89	123	127	103	75	67	68	55	38	24	53	840	5.47
E	22	96	126	192	158	113	78	71	51	48	38	63	1056	5 56
ESE	18	131	208	280	272	224	195	149	125	97	62	224	1985	6 56
SE	23	146	312	453	473	430	359	290	238	212	146	407	3489	6 66
SSE	22	142	260	370	436	471	496	419	411	362	304	1251	4944	8 58
S	25	97	170	228	281	317	380	376	427	400	380	199	5080	10.41
SSW	21	84	104	123	134	177	176	196	191	189	204	1174	2773	10.50
SW	16	72	105	105	79	103	106	87	73	81	78	453	1358	8 80
WSW	9	59	98	92	90	66	60	68	45	33	38	182	840	7 50
W	6	91	120	87	86	61	46	38	35	31	26	162	789	7 09
WNW	19	72	99	96	104	10	183	80	59	33	25	186	957	7.11
NW	22	74	154	176	177	239	200	183	146	118	103	618	2210	8.72
NNW	16	75	101	139	143	158	135	160	181	163	167	1093	2531	10.81
N	19	71	72	109	130	130	152	142	129	146	128	762	1990	10.05
CALM													327	.30
TOTAL	278	1450	2261	2803	2925	2906	2778	2503	2343	2108	1854	8952	33488	8 50

NUMBER OF NVAL D OBSERVATIONS = 1555
TOTAL NUMBER OF OBSERVATIONS = 35043

TABLE 2.3-28
MINIMUM DISTANCE FROM THE REACTOR CONTAINMENT STRUCTURE TO THE EXCLUSION-AREA BOUNDARY

Direction From Plant	Minimum Distance (meters)
N	2148
NNE	2594
NE	2673
ENE	2588
E	2570
ESE	2199
SE	2234
SSE	2211
S	2040
SSW	1717
SW	1547
WSW	1544[a]
W	1631
WNW	2056
NW	2106
NNW	2083

a) Minimum Exclusion Distance

TABLE 2.3-29
THIS TABLE HAS BEEN DELETED.

TABLE 2.3-30
THIS TABLE HAS BEEN DELETED.

TABLE 2.3-31
THIS TABLE HAS BEEN DELETED.

TABLE 2.3-32
THIS TABLE HAS BEEN DELETED.

TABLE 2.3-33
AVERAGE ANNUAL RELATIVE CONCENTRATION AT CPNPP
(SEC./CUBIC METER) PERIOD OF RECORD: 5-15-72 TO 5-14-76

BASE DISTANCE IN MILES/KILOMETERS

AFTD SECT	DESIGN DIST MI	.50 .80	1.00 1.61	1.50 2.41	2.00 3.22	2.50 4.02	3.50 5.63	4.00 6.44	4.50 7.24
NNE	0.	1.3E-05	3.1E-06	1.2E-06	6.5E-07	4.2E-07	2.3E-07	1.8E-07	1.5E-07
NE	0.	9.2E-06	2.2E-06	8.5E-07	4.6E-07	2.9E-07	1.6E-07	1.3E-07	1.1E-07
ENE	0.	7.5E-06	1.8E-06	6.8E-07	3.7E-07	2.3E-07	1.3E-07	1.0E-07	8.4E-08
E	0.	8.9E-06	2.1E-06	8.0E-07	4.3E-07	2.7E-07	1.5E-07	1.2E-07	9.9E-08
ESE	0.	9.7E-06	2.3E-06	8.8E-07	4.7E-07	3.0E-07	1.6E-07	1.3E-07	1.1E-07
SE	0.	1.4E-05	3.4E-06	1.3E-06	7.1E-07	4.5E-07	2.5E-07	2.0E-07	1.6E-07
SSE	0.	1.1E-05	2.7E-06	1.1E-06	5.8E-07	3.7E-07	2.0E-07	1.6E-07	1.3E-07
S	0.	8.6E-06	2.2E-06	8.5E-07	4.7E-07	3.0E-07	1.6E-07	1.3E-07	1.1E-07
SSW	0.	7.1E-06	1.8E-06	6.9E-07	3.8E-07	2.4E-07	1.3E-07	1.0E-07	8.6E-08
SW	0.	6.5E-06	1.6E-06	6.4E-07	3.5E-07	2.3E-07	1.2E-07	9.7E-08	8.0E-08
WSW	0.	7.1E-06	1.8E-06	6.9E-07	3.8E-07	2.5E-07	1.3E-07	1.1E-07	8.7E-08
W	0.	8.6E-06	2.2E-06	8.4E-07	4.6E-07	2.9E-07	1.6E-07	1.3E-07	1.0E-07
WNW	0.	1.4E-05	3.4E-06	1.3E-06	7.3E-07	4.7E-07	2.6E-07	2.0E-07	1.7E-07
NW	0.	2.3E-05	5.7E-06	2.2E-06	1.2E-06	7.7E-07	4.1E-07	3.3E-07	2.7E-07
NNW	0.	2.5E-05	6.1E-06	2.4E-06	1.3E-06	8.4E-07	4.5E-07	3.6E-07	3.0E-07
N	0.	2.0E-05	4.9E-06	1.9E-06	1.1E-06	6.7E-07	3.6E-07	2.9E-07	2.4E-07

BASE DISTANCE IN MILES/KILOMETERS

AFTD SECT	DESIGN DIST MI	6.00 9.65	7.50 12.07	10.00 16.09	15.00 24.13	25.00 40.22	35.00 56.31	45.00 72.40	50.00 80.45
NNE	0.	9.6E-08	7.0E-08	4.6E-08	2.7E-08	1.4E-08	9.5E-09	7.0E-09	6.2E-09
NE	0.	6.8E-08	5.0E-08	3.3E-08	2.0E-08	1.0E-08	6.9E-09	5.1E-09	4.5E-09
ENE	0.	5.5E-08	4.1E-08	2.7E-08	1.6E-08	8.5E-09	5.7E-09	4.2E-09	3.7E-09
E	0.	6.5E-08	4.8E-08	3.1E-08	1.9E-08	1.0E-08	6.8E-09	5.0E-09	4.4E-09
ESE	0.	7.1E-08	5.2E-08	3.5E-08	2.1E-08	1.1E-08	7.4E-09	5.5E-09	4.9E-09
SE	0.	1.1E-07	7.8E-08	5.1E-08	3.1E-08	1.6E-08	1.1E-08	8.0E-09	7.1E-09
SSE	0.	8.4E-08	6.1E-08	4.0E-08	2.3E-08	1.2E-08	8.1E-09	6.0E-09	5.3E-09
S	0.	6.7E-08	4.9E-08	3.2E-08	1.9E-08	9.7E-09	6.4E-09	4.7E-09	4.1E-09
SSW	0.	5.5E-08	4.0E-08	2.6E-08	1.5E-08	7.8E-09	5.1E-09	3.8E-09	3.3E-09
SW	0.	5.1E-08	3.7E-08	2.4E-08	1.4E-08	7.3E-09	4.8E-09	3.5E-09	3.1E-09
WSW	0.	5.6E-08	4.0E-08	2.6E-08	1.5E-08	8.1E-09	5.3E-09	3.9E-09	3.5E-09
W	0.	6.7E-08	4.9E-08	3.2E-08	1.9E-08	9.7E-09	6.4E-09	4.7E-09	4.2E-09
WNW	0.	1.1E-07	7.9E-08	5.1E-08	3.0E-08	1.6E-08	1.1E-08	7.8E-09	6.9E-09
NW	0.	1.8E-07	1.3E-07	8.5E-08	5.0E-08	2.7E-08	1.8E-08	1.3E-08	1.2E-08
NNW	0.	1.9E-07	1.4E-07	9.2E-08	5.4E-08	2.9E-08	1.9E-08	1.4E-08	1.2E-08
N	0.	1.5E-07	1.1E-07	7.3E-08	4.3E-08	2.3E-08	1.5E-08	1.1E-08	9.7E-09

TABLE 2.3-34
METEOROLOGICAL INSTRUMENTATION COMANCHE PEAK OPERATIONAL METEOROLOGICAL PROGRAM

Measurement	Level (Meters)	Instrument Type	Threshold	Instrument Range
Wind Speed	10 & 60	3 Cup Anemometer	0.45 m/s[a]	0 to 100 mph[b]
Wind Direction	10 & 60	Wind Vane	0.45 m/s[a]	0 to 540°
Temperature	10	Platinum Temp. Sensor	N/A	-20° to +120°F
Delta-Temp	10 to 60	Platinum Temp. Sensor	N/A	-5° to +15°F
Precipitation	Surface	Tipping Bucket	0.01 in.	0-1 in.

a) m/s = meters per second

b) Envelops the range 0-67 mph as proposed by Reg. Guide 1.97 Rev. 2

TABLE 2.3-34A
METEOROLOGICAL INSTRUMENTATION COMANCHE PEAK OPERATIONAL METEOROLOGICAL PROGRAM OVERALL SYSTEM ACCURACIES[a]

Parameter	Recording Type	System Accuracy (ANSI/ANS-2.5-1984)[b]	Actual System Accuracy
Wind Speed	Digital	±0.5 mph, WS<5mph ±10%, otherwise	±0.39mph, WS<25mph ±1.10%, otherwise[c]
	Analog[d]	±0.75mph, WS<5mph ±15%, otherwise	±0.58mph, WS<25mph ±1.18%, otherwise[c]
Wind Direction	Digital	±5°	±3.4°[c]
	Analog[d]	±7.5°	±4.5°[c]
Temperature	Digital	±0.9°F	±0.6°F[c]
	Analog[d]	±0.9°F	±0.9°F[c]
Delta Temperature	Digital	±0.27°F	±0.17°F[c]
	Analog[d]	±0.27°F	±0.19°F[c]
Precipitation	Digital	Rain guage with ±0.01 in resolution ±10% measured value for total accumulated catch greater than 0.2 in	Rain guage with ±0.01 resolution ±0.011 in or ±1.1%[c]
	Analog[d]	Rain guage with ±0.01 in resolution +10% measured value for total accumulated catch greater than 0.2 in	Rain guage with ±0.01 resolution ±0.013 in or ±1.3%[c]

a) These values are based upon equipment being properly calibrated.

b) Endorsed by Reg. Guide 1.23 Second Proposed Revision 1, April 1986.

c) Accuracy values show were calculated for original system. Calculations made for subsequent equipment upgrades computed uncertainties equal or less. All uncertainties computed are below acceptance criteria of requirements documents.

d) Paperless digital recorder replaces previous analog recorders. Uncertainties calculated for paperless digital recorder do not exceed acceptance criteria requirements for digital channels.

TABLE 2.3-34B
METEOROLOGICAL INSTRUMENTATION COMANCHE PEAK
OPERATIONAL METEOROLOGICAL PROGRAM DELTA
TEMPERATURE SYSTEM ACCURACY[a]

	Instrument Accuracy	
1.	Sensor Accuracy Signal Conditioner Accuracy	±0.13°F
	Instrument Accuracy	±0.08°F
	Temperature Coefficient	±0.05°F
2.	Square Root of the Sum of the Squared Tolerances	±0.09°F
3.	Transmitter Accuracy	±0.04°F
4.	Receiver Accuracy	±0.04°F
5.	Current Driver Accuracy	±0.04°F
6.	Digital Recorder Accuracy	
	Input Resistor Accuracy	±0.05°F
	Input Accuracy	±0.05°F
7.	Square Root of the Sum of the Squared Tolerances	±0.071°F
8.	Analog Data Reduction Accuracy	±0.05°F
	System Accuracy[b]	
	Digital Recording	
	Square Root of the Sum of the Squared Tolerance of 1, 2, 3, 4, 5 and 6	±0.17°F

a) Measurement of delta temperature is made by comparing two identical sensors after having first calibrated out any difference between them at the reference temperature. The sensor error is then essentially the tracking error between the two sensors. This error includes accuracy and interchangeability and linearity. These errors plus those which are attributed to the signal conditioning electronics and recorders have been included in the analysis.

b) These values are well within the ±0.27°F criteria established by ANSI/ANS-2.5-1984, which is endorsed by Reg. Guide 1.23. Paperless digital recorder replaces previous analog recorders. Uncertainties calculated for paperless digital recorder do not exceed acceptance criteria requirements for digital channels. Calculations made for subsequent equipment upgrades computed uncertainties equal to or less than these uncertainties.

TABLE 2.3-35
COMPARISON OF ANALOG AND DIGITAL DATA

(Sheet 1 of 2)

I. W ND DIRECTION COMPARISON

DIGITAL OCCURRENCES

Wind Sector	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	N	TOTAL
NNE	6																6
NE	1	2															3
ENE		1	2	1													4
E			1	3													4
ESE				2	6	1											9
SE					1	25	2	1									29
SSE						4	32	1									37
S						1	4	47	1								53
SSW						1		4	19								24
SW									1	9							10
WSW										3	7						10
W											5	11	5				21
WNW												2	15	1			18
NW													3	38	1		42
NNW														2	29	1	32
N															2	23	25
Total	7	3	3	6	7	32	38	53	21	12	12	13	23	41	32	24	327

TABLE 2.3-35
COMPARISON OF ANALOG AND DIGITAL DATA

(Sheet 2 of 2)

II. WIND SPEED COMPARISON[a]

Wind speed (mph)	DIGITAL OCCURRENCES																	
	Calm	1	2	3	4	5	6	7	8	9	10	11	12	13	14-16	17-19	≥20	Total
Calm	2																	2
1		5	1															6
2	1	3	19															23
3			1	23														24
4				5	24	1												30
5					3	16												19
6						3	19	1										23
7							1	24	1									26
8								3	23	2	1							29
9									5	16								21
10										5	8							13
11											7	12	1					20
12												3	15					18
13													3	13	1			17
14-16														3	20			23
17-19															3	22		25
≥20																1	7	8
Total	3	8	21	28	27	20	20	28	29	23	16	15	19	16	24	23	7	327

a) All wind speed averages were rounded off to the nearest mph.

Note: In a perfect correlation, all occurrences would fall on the single diagonal indicated by the underlined values above.

TABLE 2.3-36
THIS TABLE HAS BEEN DELETED.

TABLE 2.3-37
5-PERCENTILE VALUES[(a)] OF THE ACCIDENT PERIOD DIFFUSION ESTIMATES AT THE LPZ

Sector Affected	1 Hour	0-8 Hours	8-24 Hours	1-4 Days	4-30 Days
N	42.6	18.7	12.4	5.05	1.40
NNE	41.2	16.8	10.7	4.04	1.00
NE	41.2	15.9	9.9	3.52	0.80
ENE	38.7	14.5	8.8	3.03	0.65
E	51.5	18.9	11.5	3.86	0.81
ESE	51.1	19.0	11.6	3.98	0.85
SE	47.4	19.2	12.2	4.58	1.12
SSE	36.3	14.8	9.5	3.57	0.88
S	25.7	10.7	6.9	2.68	0.69
SSW	23.7	9.6	6.1	2.29	0.56
SW	22.9	9.3	5.9	2.22	0.54
WSW	30.4	12.0	7.5	2.75	0.65
W	29.4	12.0	7.7	2.90	0.72
WNW	42.6	17.6	11.3	4.31	1.08
NW	55.6	23.8	15.6	6.22	1.66
NNW	51.1	22.5	15.0	6.15	1.72

a) 10^{-6} s/m^3

TABLE 2.3-38
50-PERCENTILE VALUES[a] OF THE ACCIDENT PERIOD DIFFUSION ESTIMATES AT THE LPZ

Sector Affected	1 Hour	0-8 Hours	8-24 Hours	1-4 Days	4-30 Days
N	9.50	5.34	4.00	2.14	0.87
NNE	4.31	2.55	1.96	1.11	0.49
NE	1.58	1.05	0.85	0.54	0.29
ENE	0.00	0.00	0.00	0.00	0.00
E	0.00	0.00	0.00	0.00	0.00
ESE	0.00	0.00	0.00	0.00	0.00
SE	7.04	3.91	2.91	1.54	0.61
SSE	3.56	2.13	1.65	0.95	0.43
S	2.93	1.75	1.35	0.77	0.35
SSW	1.46	0.94	0.75	0.46	0.23
SW	0.00	0.00	0.00	0.00	0.00
WSW	0.00	0.00	0.00	0.00	0.00
W	0.10	0.11	0.11	0.11	0.12
WNW	4.72	2.80	2.16	1.22	0.54
NW	12.40	6.81	5.04	2.63	1.03
NNW	12.80	7.09	5.28	2.78	1.11

a) 10^{-6} s/m^3

2.4 HYDROLOGIC ENGINEERING

2.4.1 HYDROLOGIC DESCRIPTION

2.4.1.1 Site and Facilities

The Comanche Peak Nuclear Power Plant (CPNPP) site is located approximately 65 miles southwest of the Dallas-Fort Worth, Texas, area, on a peninsula in Squaw Creek Reservoir (for a description of the Reservoir see Section 2.4.8.2). The Squaw Creek Reservoir (SCR) has a service spillway crest elevation of 775.0 feet. The calculated probable maximum flood level of Squaw Creek Reservoir is 789.7 feet. The station, which takes cooling water from one side of the peninsula and discharges to the other, has a site grade elevation of 810.0 feet.

The Safe Shutdown Impoundment (SSI) is formed by building a seismic Category I dam across an arm of the reservoir. The water level is maintained by an equalization channel between the SSI and SCR. Near the south end of the equalization channel, a non-intrusive, floating raft system has been added which bridges from the east bank to west bank of the channel. Details of the SSI Dam are presented in Section 2.5.6.

The Service Water Intake Structure is located on a bank of the SSI. The structure foundation is at elevation 749'0" with the top of slab at elevation 755'0". The operating deck of the structure is at elevation 796'0" with the pump discharge centerline at elevation 800'0". Access to the Service Water Intake Structure is provided by a personnel door and truck bay entrance at elevation 810'6".

The Containment Building is a completely closed, water-tight structure which cannot be affected by exterior flooding. Safety class equipment outside the Containment Building is located in the Safeguards, Fuel, Auxiliary, and Electrical and Control buildings. All exterior accesses to these buildings are above the 810 ft elevation with the exception of Turbine Building entrances to the Electrical and Control Building. This height is significantly above the maximum probable flood (see section 2.4.2). Protection against flooding of safety-related components and systems is discussed in Section 3.4.

A topographical map of the CPNPP site is shown in Figure 2.4-4, and changes made to the natural drainage features are indicated thereon.

The stability of the Squaw Creek Dam has been evaluated for construction and operating conditions, including conditions of maximum reservoir elevation and minimum tail water elevation, utilizing engineering properties of all embankment materials and all foundation materials, by normally recognized procedures such as that used by the Department of the Army, Corps of Engineers, as more fully described in Engineering Manual EM-1110-2-1902, “Stability of Earth and Rock-Fill Dams.” Squaw Creek Dam is not a seismic Category I structure.

Each building is equipped with a roof drainage system designed to effectively collect, pass and discharge the water volume resulting from a six-inch rainfall in one hour with a maximum intensity two inches in five minutes. The scuppers are in the parapet walls and the scupper invert elevation will not be more than three inches above the roof at the outside wall or not more than five inches above the low point of the roof. The roof drains and drain pipes are designed in accordance with “Roof Drain Design for Nuclear Project Safety Related Buildings” by Southern Services, Inc., dated December 8, 1972. The roofs of all nuclear-safety-related buildings are

designed to support an eight-inch maximum uniform depth of water in addition to the regular live loads considered (see Fig. 2.4-2 and 2.4-3). The parapet walls have relief openings to ensure that the eight-inch level will not be exceeded.

The onsite drainage system for the CPNPP site is a natural drainage system as outlined in Figure 2.4-4. As this figure shows, the safety-related buildings are located at a high point, with the surrounding grounds sloping towards the Squaw Creek Reservoir (SCR) to the north and the south including the Safe Shutdown Impoundment (SSI) to the south. The ground east and west of the buildings slopes towards the ditches as indicated on Figures 2.4-40 and 2.4-41. The onsite drainage system is further enhanced by the use of an underground storm drain system within the Protected Area/Vehicle Barrier System (PA/VBS) boundary which is supported by a series of catch basins, paved swales and rocked ditches. Surface runoff including that collected by the storm drain piping within the PA/VBS is discharged to the reservoirs on both sides of the peninsula either directly by a combination of open channel ditches and underground culverts or is collected at concrete designed Drainage Basins before being discharged to the applicable reservoirs via large underground pipe culvert systems. Although this level of detail is not shown on Figures 2.4-40 and 2.4-41, site plan drainage configuration changes and impact including that created by historical implementation and placement of new or modified design features, buildings, Security related features and other structures have been analyzed to address the adequacy of the onsite drainage system consistent with the design basis rainfall intensities and drainage requirements as described below. A possible clogging of any ditch or storm drain pipe within the PA/VBS boundary has been analyzed to not affect the system's water removal capacity.

In addition, the onsite drainage system is designed to remove the water resulting from a rainfall of six inches in one hour and 7.5 inches in two hours, in such a manner that the runoff is accomplished without ponds forming on the ground.

Further, the drainage system is designed to adequately drain a rainfall of 15 inches in one hour and 22 inches in two hours in such a way that there are no ponds which can back up into the structures and affect safety-related systems.

The design bases given above are consistent with the Probable Maximum Precipitation (PMP) as discussed in Section 2.4.3.1.

2.4.1.2 Hydrosphere

2.4.1.2.1 Regional Description

The Comanche Peak Nuclear Power Plant (CPNPP) site is located in the Squaw Creek catchment. Squaw Creek is a small tributary to the Paluxy River, which is a tributary to the Brazos River. The Brazos River and its tributaries constitute one of the principal river systems in the South-Central United States, extending from New Mexico, through Texas, to the Gulf of Mexico. The Texas portion of the Brazos River basin is illustrated in Figure 2.4-5.

The portion of the Brazos River catchment considered in this study is illustrated in Figure 2.4-6, a regional topographic map showing the site, existing and possible future reservoirs in the vicinity, and locations of the stream and rainfall gaging stations considered in this study. At gaging station 8-0910, east-southeast of the site, the Brazos River has a contributing catchment of 15,590 square miles [1].

For this study the most significant portion of the Brazos River catchment is that between Lakes Granbury and Whitney. In this reach, the Brazos River Channel is located in incised meanders formed by the river. These meanders may be the result of uplift of the area and/or sea level fluctuations after a mature meandering drainage pattern had been attained.

The meanders eroded through, and are flanked by, rock slopes confining the river within a relatively narrow channel. Immediately adjacent to the channel, within the meanders, is a narrow flood plain. Although accretion and erosion occur within the channel, as is typical of a meandering river, the well-defined meanders indicate that the channel location is closely confined. The geometry of the banks is governed closely by their location with respect to the meander pattern. The bank on the outside of a bend generally is steep, whereas the bank on the inside of the bend usually has a gentler slope.

2.4.1.2.2 Squaw Creek

Squaw Creek is a small, intermittent stream which drains parts of Hood and Somervell counties and empties into the Paluxy River, upstream of the confluence of the Brazos and Paluxy rivers. Squaw Creek frequently has no flow during dry periods. Squaw Creek Dam impounds SCR for CPNPP cooling water approximately 4.3 stream miles north of the creek's entrance into the Paluxy River. The principal features of the 64-square mile SCR catchment are illustrated in Figure 2.4-1. The Squaw Creek catchment is hydrologically similar to the Paluxy River basin.

The Squaw Creek watershed is underlain by sedimentary rocks of lower Cretaceous age (poorly cemented sandstone, limestone and shale) which dip gently to the east. The topography is influenced by the underlying geology, generally with steeper slopes in the limestone areas than in the shale and sandstone areas. Landforms are gently to steeply rolling. In the lower reaches of Squaw Creek, a small flood plain has developed.

Topographic maps prepared by the U.S. Geological Survey show a number of small man-made ponds in the catchment, some of which are in creek channels and others which are off-channel. The total volume of the on-channel and off-channel storage in these ponds has been estimated to be about 1150 acre-feet. There are three retaining ponds in the catchment for the purpose of mitigating oil spills. See FSAR Section 2.2.3.2.3. Other than these small ponds, there are no known control structures, weirs, or canals.

Tolar, with a 1970 population of 312 [2], is the only community in the catchment. The remainder of the catchment is largely ranchland with some cultivated areas.

2.4.1.2.3 Paluxy River

The Paluxy River basin is hydrologically similar to that of Squaw Creek and is used as a basis for developing hydrologic parameters for Squaw Creek and its subcatchments since Squaw Creek is ungaged. At gaging station 8-0915, near Glen Rose, the Paluxy River has a catchment of 410 square miles [1].

The Paluxy watershed also is underlain by sedimentary rocks of lower Cretaceous age (poorly cemented sandstone, limestone and shale) which dip gently to the east. The topography, governed somewhat by the geology, is generally stair-stepped in limestone areas and gently rolling in sandstone and shale areas.

There are no known control structures on the Paluxy River.

2.4.1.2.4 River Control Structures

The Brazos River is undergoing continuing development of reservoirs and control structures initiated for water supply, recreation and flow control. Table 2.4-1 lists important on-channel and off-channel reservoirs and dams upstream of Cox Bend. Cox Bend is shown on Figure 2.4-6. On-channel dams and reservoirs downstream of Cox Bend are identified on Table 2.4-2. Reservoir locations are represented on Figure 2.4-5.

Four potential future reservoirs are being considered on the Brazos between Possum Kingdom Reservoir and Lake Whitney [3]. These are described in Table 2.4-3 and their approximate locations are indicated in Figure 2.4-6.

2.4.1.2.5 Surface Water Users

No water rights have been issued on Squaw Creek, but two potential surface water users (irrigation) have filed claims to withdraw water. In due course, these claims will be evaluated by the Texas Water Rights Commission and either upheld or dismissed. The points of potential withdrawal are located within SCR limits, adjacent to the Hood-Somervell county line, and are listed in Table 2.4-4. No surface water users are known on the Paluxy River, downstream of the Squaw Creek confluence [4]. Cattle are watered from Squaw Creek and also probably from the Paluxy River.

There are numerous parties on the Brazos River downstream of the site vicinity who have applied for and/or received water allocations (hereafter called water users for non-irrigational and irrigational use). Non-irrigational users (municipal, industrial, mining and hydroelectric) are identified in Table 2.4-5 and on Figure 2.4-7.

Irrigational users are summarized by county in Table 2.4-6 and on Figure 2.4-7. The nearest irrigational water user below Squaw Creek Dam is approximately three stream miles downstream from the Paluxy and Brazos River confluence. The nearest extraction of water for public supply is at Waco, approximately 109 stream miles downstream of this confluence. There are no other known extractors of water for potable uses between Squaw Creek Dam and Waco.

2.4.1.2.6 Groundwater Environment

Most of the groundwater in the site region occurs in bedrock formations of very low permeability, although some does exist in shallow floodplain alluvium in stream valleys but is not withdrawn for use. The best producing formation is the Twin Mountains Formation which underlies the relatively impermeable Glen Rose Formation on which CPNPP and SCR are being constructed. Groundwater users near the site are described in Section 2.4.13.2 and presented in Tables 2.4-24 and 2.4-25.

2.4.2 FLOODS

2.4.2.1 Flood History

2.4.2.1.1 Brazos River

The maximum known flood on the Brazos River occurred in 1876 [1] well before flow monitoring began, and consequently, there is little quantitative data available on this flood. Flood records for the Brazos River at gaging station 8-0910, just upstream from the Paluxy River confluence, have been obtained since 1923 [1, 5, 6, 7]. Gage location is shown in Figure 2.4-6, and the annual maximum stage and discharge measurements for 1924 to 1975 are shown in Table 2.4-7. These records indicate that the highest water level recorded at gaging station 8-0910 was elevation 601.69 feet (May 27, 1957), corresponding to a discharge of 87,400 cubic feet per second (cfs).

For comparative purposes, Figure 2.4-8 shows the mean annual, the fifty-year, and the historical peak floods on the Brazos River. These correspond to flows of approximately 40,000 cfs, 110,000 cfs and 97,600 cfs, respectively, at gaging station 8-0910. The stage reached at station 8-0910 during the 97,600 cfs flood (May 18, 1935) was less than that of the 87,400 cfs flood due to backwater effects from the Paluxy River discharge.

The flood of record (elevation 601.69 feet) and the 1876 event are significantly lower than the CPNPP site grade (elevation 810 feet). Newly filled (1969) Lake Granbury and any subsequent flow-control facilities erected on the Brazos River watershed will further retard future flood levels.

2.4.2.1.2 Squaw Creek

There was no monitoring on the Squaw Creek watershed prior to 1966. A crest-stage partial-record station was installed in 1966 on Panter Branch, a tributary to Squaw Creek. Its location is shown in Figure 2.4-1, and the annual maximum stage and peak discharge measurements are presented in Table 2.4-8. In 1973, a continuous recording gage was installed on Squaw Creek at the State Highway 144 bridge, 2.1 miles upstream from the mouth of the creek. Annual peak discharges for the Squaw Creek gage during water years 1974 and 1975 are presented in Table 2.4-9.

2.4.2.1.3 Paluxy River

Periodic flow records were obtained in the 1920's and continuous monitoring began in 1947 at gaging station 8-0915 at Glen Rose [1, 5, 6, 7]. The gage location is shown on Figure 2.4-6, and the annual maximum stage and discharge measurements are provided in Table 2.4-10.

The maximum flood since 1876 is believed to have occurred on April 17, 1908. Information from a local resident has established the basis for an estimated stage at Elevation 637 feet at Glen Rose, corresponding to a discharge of 59,000 cfs for this event.

2.4.2.1.4 Flood Causes

All historical floods in the catchment area pertinent to the CPNPP site have been due to precipitation runoff into streams and rivers. The area is not subject to surges, tsunamis, or ice jams, and there have been no floods due to dam failure.

2.4.2.2 Flood Design Considerations

CPNPP safety-related facilities are designed to safely withstand all floods and flood waves which are remotely possible at the site. The crest of the service spillway, elevation 775 feet, has been utilized as the pre-flood condition. The planned design assures that the safety-related facilities of the CPNPP will not be adversely affected by this event. A summary of the consequences of flooding on all pertinent facilities is contained in Section 2.4.10. The PMF for Squaw Creek has been determined by imposing the 48-hour probable maximum precipitation (PMP) as obtained from Hydrometeorological Report No. 33 (HMR #33) [9] upon the catchment.

Studies show that the maximum sustained wind velocities in one general direction during major windstorms of record in most regions of the United States have averaged approximately 40 to 50 mph for a period of one hour [10]. In accordance with regional precedents set by the Corps of Engineers, wind tide and wave runup have been evaluated by procedures outlined in Engineering Technical Letter No. 1110-2-8 [10] using an overland wind velocity of 40-mph. Detailed analyses are described in Section 2.4.3.6.

Other events considered include the effect of domino-type failure of Morris Sheppard and DeCordova Bend Dams, and the effect of a 30-minute 80-mph wind coincident with the 10-year return period flood and with the maximum operating pool level. The domino-type failure is shown in Section 2.4.4.3 to have no effect on CPNPP or on Squaw Creek Dam. On SCR wave runup due to an 80-mph wind is shown in Section 2.4.3.6 to be less critical than for a 40-mph wind coincident with the PMF.

Events considered, but not applicable to CPNPP facilities include surge, tsunami, and ice flooding. Landsliding and/or reservoir slope failures are not an expected source of floodwaves. Natural slopes are 3:1 (Horizontal to Vertical), or flatter, except for occasional local portions of slope which are steeper (ranging to 2:1 or 1:1). Slope failures are improbable, but small localized sloughs conceivably might occur from erosion and/or weathering of the exposed edges of claystone seams from beneath overlying, more resistant rock zones. The small waves resulting from such failures will not adversely affect CPNPP facilities.

2.4.3 PROBABLE MAXIMUM FLOOD (PMF) ON STREAMS AND RIVERS

The PMF is defined by the Corps of Engineers as the “hypothetical flood characteristics that are considered to be most severe ‘reasonable possible’ at a particular location, based upon relative comprehensive hydrometeorological analyses of critical runoff producing precipitation and hydrologic factors favorable for maximum flood runoff.” The PMF for the SCR catchment has been derived synthetically, using procedures customarily applied by the U.S. Army Corps of Engineers. Results of the evaluations are summarized in Table 2.4-11 and Figure 2.4-9.

The flood routings for Squaw Creek Reservoir and Safe Shutdown Impoundment include allowance for rainfall on the surface of the reservoirs and the rainfall volumes are tabulated in Tables 2.4-11 and 2.4-15.

The nearby Brazos River has not been evaluated in detail for PMF because postulated upstream dam breaks are shown in Section 2.4.4 to be harmless to CPNPP facilities.

2.4.3.1 Probable Maximum Precipitation (PMP)

The PMP is the theoretically greatest precipitation over the applicable drainage area that would produce flood flows which have virtually no risk of being exceeded. From HMR No. 33 [9], it is found that the PMP is 25.5 inches in six hours on an area of 64 square miles in the CPNPP vicinity. Similarly, the PMP is 31.3 inches in 12 hours, 34.7 inches in 24 hours, and 39.1 inches in 48 hours. Table 2.4-11 lists the three hour rainfall increments, as divided for analyses.

For comparative purposes, a statistical method for estimating the largest rainfall likely to occur at a gaging station was applied to the nearby gaging stations shown on Figure 2.4-6. This procedure, developed by Hershfield [11], employs analyses of rainfall records to develop a statistical envelope defining the maximum precipitation occurrence. The method is applied to rainfall records from a single station, and a 24-hour estimate of maximum precipitation for that station is found. Results of analyses are compared to the 24-hour PMP and the maximum recorded rainfall in Table 2.4-12.

The amount of moisture present in the soil before the advent of the PMP, commonly referred to as the antecedent condition, is of importance in determining the amount of runoff produced by the PMP. The U.S. Bureau of Reclamation recommends “average” antecedent conditions [12]. Average conditions are defined as “the average of the conditions which have preceded the occurrence of the maximum annual flood on numerous watersheds.” The actual conditions assumed become relatively unimportant when considering 48-hour design storms because the initial 24-hour period of light rainfall saturates the ground prior to periods of intense rainfall.

The catchment area and site vicinity are gently to steeply rolling in character, with somewhat incised stream and river valleys. Relative relief is modest and does not require modification of the PMP value, or distribution, to account for orographic effects. Sub-basins within the SCR catchment are small; thus, catchment response to precipitation is relatively rapid. Therefore, postulation of storm centering is not of major importance, and the storm can be assumed to have uniform distribution and intensity over the entire catchment.

A comprehensive study of floods in Texas [5] indicates that snowmelt does not contribute significantly to river floods anywhere in the state. Similarly, no frozen-ground conditions are anticipated.

2.4.3.2 Precipitation Losses

Precipitation losses are estimated by evaluating the initial losses and the infiltration rates of the soils in a watershed. Evaluation of losses reveals that initial losses can be expected to be in the range of 0.5 inches with average antecedent conditions; and infiltration rates after presaturation will be in the range of 0.05 to 0.15 inches per hour. The PMF selected utilizes an initial loss of 0.5 inches and a conservative infiltration rate of 0.1 inch per hour as shown in Table 2.4-11.

These losses are based upon Corps of Engineers records for the Paluxy catchment shown below [13]:

			Losses (inches)			
Date of Flood	Rainfall (inches)	Duration (hours)	Initial (1st 3 hrs)	Balance	Total	Steady Loss (in/hr)
05/23/52	4.19	15	.85	2.24	3.09	.187
04/06/57	2.94	12	.47	1.36	1.83	.151

These values are indicative of the magnitude of precipitation losses expected on the SCR catchment due to its similarity to the Paluxy watershed in regard to general topography, geology, soil types and land usage. The values require evaluation, however, in light of some differences between the two watersheds, including: 1) the SCR catchment is generally not as steep as the Paluxy watershed; 2) the SCR watershed is much smaller; and 3) much of the relatively level, low-lying portion of the SCR catchment will be inundated by the reservoir.

Although the first difference described above tends to increase the precipitation losses on the SCR catchment in relation to that of the Paluxy River, the other two differences indicate a decrease in losses. Smaller drainage areas, such as the SCR catchment, generally have lower losses than larger areas, and submergence of much of the creek alluvium removes a section of the catchment which should have the most infiltration capacity. Thus, it is prudent to adopt lower estimates of losses on the SCR catchment. This conclusion is further supported by the relatively short duration of the available historical storms studied, in comparison to the 48-hour storm used in computing the PMF. In view of these factors, an initial loss estimate of 0.5 inches and an infiltration rate of about 0.1 inch per hour is considered appropriate for the catchment.

2.4.3.3 Runoff Model

2.4.3.3.1 Description of Squaw Creek Reservoir Catchment

Figure 2.4-1 illustrates the 64-square mile SCR catchment and the reservoir limits corresponding to a selected flood-stage at Elevation 780 feet.

The pre-flood condition reservoir level is assumed to be Elevation 775 feet.

During the PMF, pool level will rise from Elevation 775 feet (area 3,272 acres) to Elevation 789.7 feet (area 3,863 acres).

2.4.3.3.2 Synthetic Hydrographs

Synthetic methods for developing a runoff model have been undertaken. Three methods of synthetic hydrograph development were considered:

1. Use of Snyder's Unit hydrograph relations presented by the U.S. Army Corps of Engineers (USACE) [14].

2. Use of dimensionless hydrographs presented by the Soil Conservation Service (SCS) [14].

3. Use of triangular hydrograph techniques presented by the U.S. Bureau of Reclamation [12].

The empirical relations developed by Snyder have also been shown to be reliable through widespread usage. To employ this method, two coefficients which depend upon drainage basin characteristics are computed from hydrologic records for a representative portion of the drainage area under study, or for nearby catchment of similar characteristics [15]. Snyder's method was adopted for development of a runoff model for Squaw Creek.

2.4.3.3.3 Hydrograph Development

For purposes of analyses, the SCR catchment has been divided into three areas, as shown on Figure 2.4-1 and described below:

1. The Upper Squaw Creek catchment, which consists of about 38 square miles of land located above the reservoir area.

2. The Lower Squaw Creek catchment which consists of about 20.3 square miles of land located around the fringes of the reservoir. This area is actually comprised of a number of very small catchments, which are considered together for reasons of clarity and convenience.

3. The Squaw Creek Reservoir Area

The Corps of Engineers has developed unit hydrographs from gaging station records for two nearby watersheds, the Paluxy River and the Clear Fork of the Trinity River. The pertinent data from these studies [13] is summarized below:

	Drainage Area (Sq. Mi.)		
Paluxy River	361	.60	420
Clear Fork (upper portion)	210	1.50	480

The lower portion of the SCR catchment (see Figure 2.4-1) is generally similar to the Paluxy River watershed. The upper portion is considered to have characteristics falling between those of the Paluxy River and Clear Fork. These considerations lead to the adaptation of the following coefficients for the SCR catchment:

			Basis
Upper 38.0 Sq. Mi.	1.1	440	Combination of Paluxy & Clear Fork
Lower 20.3 Sq. Mi.	.3	420	Paluxy

All other information needed to apply Snyder's relationships (such as area, length, and center of gravity of the catchments) has been derived from U.S.G.S. quadrangle maps. These data are summarized in Table 2.4-13.

The unit hydrographs are shown in Figure 2.4-10.

2.4.3.3.4 Non-Linearity

High rainfall rates associated with intense storms may result in above normal concentrations of runoff (non-linearity) because of a more severe rainfall distribution and increased hydraulic efficiencies.

The USACE recommends increasing hydrograph ordinates by 25 to 50 percent during the period of most intense rainfall [14]. In view of this, the hydrograph for the upper SCR catchment was increased by 30 percent during the most intense period of rainfall (See Table 2.4-13). This magnitude of increase is considered applicable for a drainage basin of 38 square miles.

2.4.3.3.5 Translation of Inflows

The size and shape of the reservoir in relation to the catchment area suggest that there will be no significant translation time from inflow points to the dam and CPNPP site locations; accordingly, no delay is introduced into the computations.

2.4.3.4 Probable Maximum Flood Flow

The PMF inflow and outflow hydrographs to SCR resulting from the PMP are presented in Figure 2.4-9. As explained in Section 2.4.3.1, snowmelt was not considered in establishing the PMP.

Except for a few existing small farm ponds, there are no present or planned structures upstream of SCR; therefore, the effect of such structures was not considered in developing the PMF hydrograph.

The PMF was routed through the SCR assuming the reservoir level at the beginning of the PMF was at elevation 775, the service spillway crest elevation. All discharge was assumed to occur over the uncontrolled spillways of Squaw Creek Dam.

The current SCR operating practice (see Section 2.4.8.2.1) of pumping makeup water to cause a small flow over the service spillway may routinely raise the SCR level. This small SCR level increase and the resulting small flow over the service spillway is negligible when combined with the maximum postulated precipitation and flooding analysis results and has insignificant impact on the PMF peak reservoir level.

All streams in the SCR basin empty directly into SCR; therefore no channel routing coefficients were required. The applicability of the stream course response model to handle the PMF is discussed in Section 2.4.3.3.4. The ability of the SCR dam to withstand the PMF and coincident wave action is discussed in Section 2.4.3.6.

2.4.3.5 Water Level Determinations

The mass curve, the capacity-area-depth curves, and the spillway rating curves (Figure 2.4-9) are used in routing the PMF through the reservoir to evaluate water level. The resulting peak reservoir level is Elevation 789.7.

In routing, the reservoir water surface has been assumed to be nearly horizontal, and the volume of water in the reservoir has been assumed to be directly related to the reservoir elevation. These are reasonable assumptions in view of the shape and depth of the SCR. These assumptions allow the principle of continuity expressed as a storage equation $(\hat{I}\Delta t - \Delta s = \bar{\theta}\Delta t$, where $\hat{I}$ and $\bar{\theta}$ are the average rates of inflow and outflow for the time Δt, and Δs is the change in water volume during time Δt) to be applied directly to the routing problem [16].

2.4.3.6 Coincident Wind Wave Activity

The magnitude of the wind tide and wave runup are dependent upon the wind velocity, fetch and reservoir depth. The wind direction must coincide with the fetch direction. An overland wind velocity of 40 miles per hour has been approved by the USACE for use in determining freeboard requirements in the Fort Worth District. This 40 mph wind velocity is the highest that may reasonably be assumed to occur coincidentally with the probable maximum flood [17].

The effective fetch length for wave generation was determined for the center of Squaw Creek Dam (fetch of 1.28 miles) and for the exposed side of the CPNPP plant location (fetch of 1.25 miles). It was also determined for the Safe Shutdown Impoundment Dam and the protected side of the CPNPP plant location, but freeboard requirements were found to be less than two feet for these locations, so discussion is not included here. Computation of effective fetch considered radial lines at angles up to 42 degrees from the central or primary fetch line, as recommended by the USACE [10].

The average depth of the reservoir at PMF, Elevation 789.7 feet, is approximately 55 feet, and the longest theoretical deep-water wave length (for waves reaching the center of Squaw Creek Dam) is 46 feet, so the ratio of water depth to wave length is well over one-half, and the reservoir can be considered to have “deep water” [10].

Computation [10] utilizing data for wind velocity, fetch length, and reservoir depth yield the results shown in Table 2.4-14. The Table illustrates the maximum runup and setup of smooth and riprapped banks on the Squaw Creek Dam and at the exposed side of the CPNPP plant area. As

can be seen from the Table, wave runup and wind tide at the dam and plant are about 4 and 5.0 feet and elevations reached are 793.7 feet and 794.7 feet, respectively. Due to the much shorter fetch available around the area, water level elevation reached at the SSI is about 791.3 feet. All plant facilities are above the maximum wave runup and setup elevation of 794.7 feet. The Service Water Intake Structure is the only safety-related structure subject to wave action. The elevation at the operating deck is approximately 796, above the maximum expected wave runup. Section 2.4.10 discusses the effect of wave runup and wind tide on all pertinent safety-related facilities.

2.4.3.7 Flood Evaluations for Safe Shutdown Impoundment

[redacted] Water can pass freely between the SSI and SCR, and the water surface in both water bodies normally will be at the same elevation. Figure 2.4-11 is a graph of the discharge characteristics of the SSI spillway. Table 2.4-15 outlines predicted performance of the SSI during occurrence of simultaneous Probable Maximum Floods on the overall SCR watershed and the SSI watershed. The maximum level reached in the SSI during the PMF is computed to be 790.5 feet, leaving a freeboard in the SSI of 5.5 feet. Further details on the Safe Shutdown Impoundment Dam are given in Section 2.4.8.2.2. Table 2.4-16 gives the unit hydrograph parameters for the SSI watershed, and Figure 2.4-12 shows the unit hydrograph for the SSI watershed. A non-intrusive, floating raft system has been located near the south end of the equalization channel. This floating raft system bridges from the east bank to west bank of the channel. The raft system is retained at its designated position through the use of bollards on the banks of the channel. These bollards function as bumpers for the raft system. Should the equalization channel experience flooding conditions, the floating raft system is designed to freely float on the water surface, even if the floats are punctured. If the water level exceeds an elevation above approximately elevation 781 feet, the raft system will float over the bollards and, due to the direction of flow through the channel, the raft system will be pushed away from the SSI and into the SCR. As a result, this floating raft system will not affect the design function of the equalization channel to freely pass water to the SCR under flooding conditions up to a PMF.

2.4.4 POTENTIAL DAM FAILURES (SEISMICALLY INDUCED)

There are no impoundments other than small farm ponds on the SCR catchment; therefore, a postulated dam failure upstream on Squaw Creek is not appropriate for the CPNPP facilities. The farm ponds on the catchment have a combined volume which is less than one percent of the reservoir volume and are insignificant.

Failure of Squaw Creek Dam itself presents no danger of flooding the CPNPP, as the Station is above the PMF water level. The possibility of damage to Squaw Creek Dam by backwater due to flooding on the Brazos River in the event of a postulated domino-type failure of Morris Sheppard Dam and DeCordova Bend Dam is ruled out in Section 2.4.4.3.

2.4.4.1 Reservoir Description

Present and possible future reservoirs which might be considered to have an influence on the site from a safety or water-supply standpoint are described in Tables 2.4-1 through 2.4-3, and their locations are shown in Figures 2.4-5 and 2.4-6.

2.4.4.2 Dam Failure Permutations

Considering CPNPP safety, the most severe dam failure permutation conceivable is the failure of Morris Sheppard Dam and the subsequent domino-type failure of DeCordova Bend Dam. The 25-year floods of both the Brazos and Paluxy rivers can be added into the effects of the combined dam breaks without significantly intensifying the results. Failure of the Lake Whitney Dam, downstream, would not have an adverse effect on CPNPP.

The detailed analysis of the most severe dam failure permutation is presented in Section 2.4.4.3 and the effect of landslides into the reservoir is discussed in Section 2.4.2.2.

There will be no commercial water traffic on SCR, and no possible blockage of any water course in the site region could affect the plant.

2.4.4.3 Unsteady Flow Analysis of Potential Dam Failures

The possibility of damage to Squaw Creek Dam due to failure of Morris Sheppard Dam and DeCordova Bend Dam was examined by initiating a number of highly conservative simplified evaluations which would establish conservative water levels. The results of these studies clearly indicate that there is no possibility of flood damage to the CPNPP site and Squaw Creek Dam. The toe of Squaw Creek is above Elevation 640 feet, approximately 60 to 90 feet higher than the bankfull elevation of the Brazos River at the Brazos River - Paluxy River confluence (Elevation 560 feet). The maximum wave height that could be generated by complete and instantaneous removal of DeCordova Bend Dam is about 37 feet, according to dam break calculations described in Section 2.4.4.3.3.

Prior failure of Morris Sheppard Dam is not expected to increase this wave height significantly due to the modifying effects of 145 miles of river channel between Morris Sheppard and DeCordova Bend Dams. Although calculations (Sections 2.4.4.3.1) show that a wall of water over 60 feet high would be initiated by the complete and instantaneous removal of Morris Sheppard Dam, routing by the method of coefficients (Section 2.4.4.3.2) shows that the wall would have decayed during translation to DeCordova Bend Dam. Rather, there would be a flood, the peak of which would arrive some five hours after initial inflow to Lake Granbury reservoir had begun. The break in the DeCordova Bend Dam is postulated to occur when the reservoir level becomes full. Therefore, the reservoir inflow from an upstream dam break would serve to sustain the high rate of outflow through the break in DeCordova Bend Dam rather than to increase the initial outflow (height of water wall) significantly.

Even with the ultra-conservative assumption that the combined wave created by the dam breaks remained undiminished in traveling below DeCordova Bend to the Paluxy River - Brazos River confluence, the flood wave would fall more than 40 feet (elevation) short of reaching the toe of Squaw Creek Dam.

In view of the highly conservative assumptions made in the following flood analyses, the results are regarded as a very conservative upper limit on the river level in the event of the domino-type failure, rather than a refined estimate of the flood to be expected.

2.4.4.3.1 Morris Sheppard Dam Break

[redacted]

Evaluation assumptions:

1. Complete and instantaneous removal of dam.
2. [redacted]
3. Channel is frictionless, rectangular and horizontal.
4. No water downstream of dam.
5. Vertical accelerations are negligible

These highly conservative assumptions allow the following analysis to be made, using the method shown by Henderson [19] and Chow [20].

Crest Velocity

= $(gy)^{1/2}$ *

= $((32.2 \text{ ft/sec}^2) \times (145 \text{ ft}))^{1/2}$

= 46.5 mph

Trough Velocity (velocity of wave downstream)

= $2\ (gy)^{1/2}$

= 2 x (46.5 mph)

= 93 mph

Velocity of water through dam site

= $2/3\ (gy)^{1/2}$

= 2/3 (46.5 mph)

= 31 mph

Discharge

= $8/27\ (g)^{1/2}y^{3/2}W$

= $8/27\ (32.2\ ft/sec^2)^{1/2}\ (145\ ft)^{3/2}(1600)$

= 4,700,000 cfs

Time of constant discharge (assuming discharge will remain until lake is empty)

= 600,000 ac-ft/(.0826 ac-ft/cfs-hr) (4,700,000 cfs)

= 1.5 hours

where

t = length of time interval, in this case one hour.

X = a constant dependent upon channel characteristics.

g = acceleration of gravity

y = height of water surface exposed upon dam removal

2.4.4.3.2 Routing From Morris Sheppard Dam to DeCordova Bend Dam

Travel times for floods between Morris Sheppard Dam and DeCordova Bend Dam cannot be determined with accuracy because of such factors as incidence of rain and runoff accumulation within the reach. However, there are some data regarding water movement on the river. It has been found that it takes about two days for normal turbine releases from Morris Sheppard Dam to reach DeCordova Bend Dam [3]. This is equivalent to a velocity of under three mph. Flow travel time studies on the Brazos River between Lake Whitney and Richmond show that velocities vary between 1.6 mph and 3.8 mph. Travel time is greatest for low flows and least for small floods, and then increases again as flows become very large [21]. These data are inconclusive but they seem to indicate that the average velocity of flow from a dam break would be in the order of three or four mph.

The length of river channel required to reach a terminal velocity is not known; therefore, in the evaluations the flow from the Morris Sheppard Dam Break has been conservatively assumed to continue unbated at 31 mph, the theoretical velocity of flow through the dam section.

The distance traveled to Lake Granbury is 144.5 miles, and the lake is 33.5 miles long; therefore, travel time is indicated to be on the order of five hours, with one hour of this being flow along the reaches of Lake Granbury.

Routing was by the method of coefficients (Muskingum) [22] through a reach using total flood-wave travel equal to 4xt* and an X* of 0.3 (deemed appropriate for a wide rectangular channel where changes in discharges are small) and through a reach using total flood-wave travel equal to t and an X of zero (for reservoir-type storage routing). This evaluation yields an outflow hydrograph having a base time of over 12 hours and a peak flow of about 1,900,000 cfs, well under half of the theoretical initial discharge of 4,700,000 cfs.

2.4.4.3.3 DeCordova Bend Dam Break

[redacted]

Assumptions made:

1. DeCordova Bend Dam holds until lake level reaches top of dam.
2. Complete and instantaneous removal of dam.
3. [redacted]
4. Channel is frictionless, rectangular and horizontal.
5. No water downstream of dam.
6. Vertical accelerations are negligible.

These highly conservative assumptions allow the following analysis to be made:

Crest Velocity

$= (gy)^{1/2}$

$= (32.2 \text{ ft/sec}^2 (84 \text{ ft}))^{1/2}$

$= 52 \text{ ft/sec}$

$= 35 \text{ mph}$

Trough Velocity

$= 2 (gy)^{1/2}$

$= 2 (35 \text{ mph})$

$= 70 \text{ mph}$

Velocity of water through dam site

$= 2/3 (gy)^{1/2}$

$= 2/3\ 35$

$= 23 \text{ mph}$

Discharge

$= 8/27\ (g)^{1/2} y^{3/2} w$

$= 8/27\ (32.2\ ft/sec^2)^{1/2} (84\ ft)^{3/2} (2200\ ft)$

$= 2{,}800{,}000\ cfs$

Depth of water at dam site

$= 4/9\ Y$

$= 4/9\ (84\ ft)$

$= 37\ ft$

Theoretical time of discharge

$$= \frac{(2)(\text{length of lake})}{(\text{velocity of wave})}$$

$$= \frac{(2)(33.5\ mi)}{35\ mph}$$

$= 2\ hours$

Assuming no inflow, the 2,800,000 cfs discharge would empty Lake Granbury in less than a half hour. But the inflow from Possum Kingdom Reservoir would provide additional water to maintain the 2,800,000 cfs flow for two hours. After the two hour period, flow would diminish until the remaining 65,000 acre feet are discharged.

2.4.4.3.4 Routing from Decordova Bend

[redacted] Thus, there is no possibility of dam breaks on the Brazos River affecting the safety of the CPNPP.

In the event that other dams are constructed between Morris Sheppard Dam and DeCordova Bend Dam, their presence should in no way endanger the safety of the CPNPP, even in the case of a domino-type failure.

[redacted]

2.4.4.3.5 Consideration of Antecedent Flow

In the preceding description of analyses, river channels have been assumed to be dry. Inclusion of water in the river (antecedent flow) would not have a significant effect on the results of the dam break analyses. For example, the 25-year flood on the Brazos River would contribute only about 90,000 cfs (refer to Figure 2.4-13), which is only 3.2 percent of the theoretical flow from the dam breaks.

2.4.4.4 Water Level at CPNPP

The Station site can in no way be endangered by any dam breaks or series of dam breaks, as it is over 110 feet above postulated maximum water levels.

2.4.5 PROBABLE MAXIMUM SURGE AND SEICHE FLOODING

2.4.5.1 Probable Maximum Winds and Associated Meteorological Parameters

The probable maximum sustained over-land wind selected for evaluation of wave action of the maximum operated reservoir elevation is 81 mph. The wind has an estimated return period of 200 years [23].

2.4.5.2 Surge and Seiche History

There are no existing large bodies of water near the site that would allow development of either surge or seiche; therefore, there is no history of surge and seiches in the site vicinity.

2.4.5.3 Surge and Seiche

The small size, relatively shallow depth and irregular shape of Squaw Creek Reservoir indicates that there is a minimum probability of either surges or seiches occurring in the reservoir. Therefore, surge and seiche should not be considered significant at this site.

2.4.5.4 Wave Action

The effect of the maximum sustained wind on the reservoir surface has been evaluated in Section 2.4.5.1. This wind is considered coincident with a 10 year return period flood elevation in Squaw Creek (778.1 feet). Results of the wind wave activity calculation are presented in Table 2.4-14. With an effective fetch of 1.28 miles, computations indicate that the significant wave height will be approximately five feet with a period of 3.9 seconds. The maximum wave height will be about eight feet with a setup of 0.2 feet and runup of 6.8 feet [10].

This will occur at Squaw Creek dam as illustrated in Table 2.4-14.

The wind penetrated waves on the SSI are less than those on SCR due to the much shorter fetch available around the SSI area (see Figure 2.4-14, Figure 2.4-15 and Table 2.4-14).

2.4.5.5 Resonance

Due to the irregular shape and sloping sides of SCR and SSI, wave resonance will not have any significant effect on the maximum water elevation.

2.4.5.6 Runup

The maximum water elevation reached due to wave runup and setup at the plant site, Squaw Creek dam and SSI dam are 794.7 feet, 793.7 and 791.3 feet, respectively. All plant facilities will be above the maximum wave runup elevation of 794.7 feet.

The Service Water Intake Structure will be the only safety-related structure subject to wave action or wave runup. The operating deck will be approximately elevation 796', well above the maximum expected wave runup.

2.4.6 PROBABLE MAXIMUM TSUNAMI FLOODING

This site is nearly 300 miles from the Gulf of Mexico and the plant will be over 800 feet above sea level. Therefore, tsunami flooding will not occur.

2.4.7 ICE FLOODING

The Texas climate is too warm to allow the development of significant ice on any lake. Certainly there are no records of any major river in Texas freezing over at any time, so the possibility of ice flooding can be discounted.

2.4.8 COOLING WATER CANALS AND RESERVOIRS

2.4.8.1 Canals

No canals are involved.

2.4.8.2 Reservoirs

2.4.8.2.1 Squaw Creek Reservoir (SCR)

SCR is a cooling lake for CPNPP. The location and configuration of the reservoir are shown in Figures 2.4-5 and 2.4-6. Table 2.4-17 gives the area and capacity characteristics of the SCR site, based on planimeter measurements from U.S. Geological Survey quadrangle maps entitled Hill City, Texas, and Nemo, Texas, scale 1:24,000. The volumes and areas indicated are those of the entire reservoir, including the reserve storage within the Safe Shutdown Impoundment (SSI) described in Section 2.4.8.2.2 below. The performance capability of SCR as operational cooling pond was evaluated through mathematical modeling as documented in Reference [40].

During some periods and especially in summer months, the pumping of makeup water may be utilized to raise the level of SCR above the crest of the service spillway to keep a small flow of SCR water going over the crest of the service spillway even in the absence of rain. This practice enhances the cooling capacity of the reservoir.

1. Squaw Creek Dam

[redacted]

2. Spillways

[redacted]

Design bases for the service spillway at Squaw Creek Dam are as follows:

a. Spillway width: [redacted]

b. Design Head: [redacted]

c. Crest elevation: [redacted]

d. Depth of approach channel below crest: [redacted]

e. Slope of upstream face of weir: [redacted]

f. Type of spillway crest: [redacted]

g. Manning's "n" of approach channel: [redacted]
(Ref: EM 1110-2-1603)

h. End contraction effects from Engineering Manual Part CXVI

i. Entrance loss assumed to be 0.1 V**2/2g for approach channel

j. Coefficients of Discharge from Figure 251, Design of Small Dams, 1973 Edition

Design for rip-rap lining on the sides of the discharge channel of the service spillway was determined using U.S. Army Corps of Engineers EM 1110-2-1601 (1 July 1970) and

supplemental ETL 1110-2-120 (14 May 1971) along with U.S.A.E.W.E.S. Hydraulic Design Chart 712-1 (Revised September 1970). Unit weight used for rip-rap stone was 145 pcf. Average velocities of water flow were determined from backwater analyses starting at the confluence of the discharge channel with Squaw Creek and progressing through the channel to the stilling basin.

Velocities ranged from 16.2 fps in the lower reach of the channel to five fps just below the stilling basin. The termination of the rip-rap at the end of the discharge channel was designed based on the possibility that future stream bed degradation in Squaw Creek could cause critical depth of flow at the end of the rip-rap lining with a corresponding local flow velocity of 18.5 fps. Rip-rap bedding beneath the stone layer was designed as a filter to prevent migration of soil or bedding material through the slope protection layer. The bottom of the channel is sound limestone and did not require rip-rap protection.

The emergency spillway discharges into a tributary of Squaw Creek whose confluence with Squaw Creek is 7,000 feet downstream of the dam. A water surface profile in Squaw Creek showed that the water level at the toe of the dam will reach Elevation 656.0 during a Probable Maximum Flood. The tributary, through which the emergency spillway discharge will flow, is separated from the dam by a large hill composed of a thin overburden overlying limestone. The Squaw Creek Dam embankment will not be endangered by flows through the emergency spillway.

Design bases for the emergency spillway at Squaw Creek Dam are as follows:

a. Spillway width: [redacted]

b. Crest elevation [redacted]

c. Length of level section at elevation 783.0: [redacted]

d. Slope of spillway downstream from level section: [redacted]

e. Manning's "n" of spillway channel: [redacted]
(Ref: Chow: Open-Channel Hydraulics)

f. Side slopes: 1 vertical on 3 horizontal

g. Entrance loss assumed to be 0.1 V**2/2g.

The emergency spillway at Squaw Creek Dam is not lined. A concrete wall at Elevation 783.0 on the emergency spillway crest serves as the spillway's erosion control structure. Its purpose is to maintain a uniform elevation for the entire length of the crest. The wall is anchored in limestone. The maximum velocity along the emergency spillway during a Probable Maximum Flood is 10 fps. Velocities at the downstream edge of the spillway will be higher. These velocities will cause some erosion damage. The erosion will not be severe since the emergency spillway cut is into limestone. In places, surface materials (up to 2 feet thick) overlying the limestone consist of softer materials, graded for drainage. The frequency of operation of the emergency spillway is in excess of 100 years. The erosion damage will not endanger the dam or reduce its storage capacity.

Location of the spillways are indicated on Figure 2.1-2.

3. Flood Levels

Routing of the PMF through the reservoir has been discussed in detail under Section 2.4.3. The maximum high water level during the PMF would be Elevation 789.7, leaving a freeboard allowance of 6.3 feet, which is more than adequate to accommodate wave runup and wind tide on rip-rapped slopes as shown in Section 2.4.3.6.

In order to evaluate the probable frequency of flows through the emergency spillway cut, inflow hydrographs were also computed for floods having 50-year and 100-year probable recurrence intervals, using rainfall data developed for those frequencies by the U.S. Weather Bureau [24]. It was found that the 100-year flood would involve less total runoff than the volume of surcharge storage available between the crest of the service spillway and the crest of the emergency spillway, and that the 100-year storm would, therefore, not cause flow through the emergency spillway.

4. Brazos River Diversion System

Supplemental water from Lake Granbury on the Brazos River will be conveyed by pipeline to SCR when required. The pipeline is 48 inches in diameter, with a design delivery capability of 65.1 million gallons per day (MGD). To allow for one pumping unit being temporarily out of service, the station includes four pumps with 21.7 MGD of rated capacity each, for a total installed name-plate capacity of 86.8 MGD.

5. Service Outlet

Service outlet facilities are located at the right (south) abutment of the dam and consist of an intake tower and outlet conduit. The tower has ports at various levels in the reservoir to allow selective taking of water. Backup closure facilities are provided to protect against loss of storage due to inability to close a valve or a gate.

2.4.8.2.2 Safe Shutdown Impoundment (SSI)

A portion of the arm of the reservoir that is formed by the channel of Panther Branch is utilized as a Safe Shutdown Impoundment, holding water for normal and emergency cooling use. The area and capacity characteristics of this cooling water pond are given in Tables 2.4-18 and 2.4.19. The secondary reservoir is separated from the main body of the reservoir by a rock-fill dam. An open channel was excavated through the narrow ridge to the southwest of the SSI Dam to connect the SSI with the main body of the reservoir. The floor of the channel is at elevation 769.5, six inches below the normal minimum operating level, and under normal operating conditions water will pass back and forth to keep the large and small reservoir surfaces at the same elevation. If the level in the main reservoir should drop due to some emergency, the SSI Dam will hold back between 284 acre-feet and 367 acre-feet of reserve water to allow continued cooling and safe shutdown of the plant.

Details of the SSI Dam embankment are shown in Figure 2.4-21. The middle zone is of select, impervious material, wetted and rolled and carried down to impervious foundation material for effective cut-off.

[redacted]

Design bases for the spillway at the Safe Shutdown Impoundment are as follows:

a. Spillway width: [redacted]

b. Side Slopes [redacted]

c. Slope of Spillway channel: [redacted]

d. Manning's "n" of spillway channel: (Ref: Chow: Open-Channel Hydraulics) [redacted]

e. Entrance loss assumed to be 0.1 V**2/2g

f. Exit loss assumed to be V**2/2g

The spillway channel is in limestone and is unlined. The velocity through the spillway will reach 10.3 fps. A concrete erosion slab was constructed in the channel to maintain the control level at elevation 769.5.

The rock used to construct the outer zones of the SSI dam was selectively quarried and processed from planned plant site excavations to produce a reasonably well graded product conforming to the limits shown on Figure 2.5.6-12. Basic consideration in selecting the rock gradation were seismic performance and protection of the interior filters. Normal construction techniques for the rock fill resulted in larger rock sizes accumulating on the outer edges of the slopes. Analysis of wave protection requirements was based on an average over-water wind of 81 mph (probable maximum wind - 200 year frequency) over the effective fetch distance. The method of computing the effective fetch distance set forth in Department of Army, Office of the Chief of Engineers ETL 1110-2-221, 29 November 1976, was adopted. Comparison of the as constructed conditions with conventional rip rap protection requirements as set forth in EM-1110-2-2300 (1 MARCH 1971) indicates adequate protection. To assure continued performance of the SSI, the planned annual inspection (see Section 2.5.6.8.1) of the dam will specifically include observations of slope protection.

Estimated sediment production from the Panther Branch watershed above the SSI during the 40-yr projected service life of CPNPP was derived based on analytical procedures for small watersheds as described in Reference [38]. The anticipated reduction in storage capacity of the SSI during that period due to accumulation of sediment was found to be 91 ac-ft, of which 85 ac-ft would be below elevation 770.0 and the remaining 6 ac-ft between elevations 770.0 and 775.0. Comparative plots of area and capacity characteristics before and after sedimentation are shown in Figure 2.4-22. Table 2.4-19 outlines the predicted area and capacity values at the end of the 40-year period.

Inspections of the service water intake channel for sedimentation buildup will be performed at the frequencies stated in Regulatory Guide 1.127. This inspection will consist of a direct measurement of sediment depth at the bottom of the intake channel at a minimum of five

locations along its length. Measurements will be taken at approximately the same location each time the inspection is performed.

If it is determined that, as a result of these inspections, the sediment has accumulated enough to increase the bottom elevation one and one-half (1.5) feet over the entire length of the inspection area, then measures will be employed to remove the sediment from the intake channel. At that time, procedures will be developed and approved to accomplish this task.

A detailed layout of the SSI design features is given in Figure 2.4-23.

Seismic Design Criteria for the SSI are discussed in Section 3.7.

The ability of the SSI to meet criteria of Regulatory Guide 1.27 is discussed in Section 9.2.5.

2.4.9 CHANNEL DIVERSIONS

The SCR catchment has developed streams with distinct valleys and has sustained numerous farm ponds. Therefore, diversion of water from the catchment appears impossible. The reservoir is formed in the Glen Rose formation, a predominately limestone sequence. Information developed regarding this formation indicated it is relatively impermeable and free of sinkholes and solutioning. Thus, significant loss of water is improbable.

Lake Granbury, which is on the Brazos River, will be a major source of makeup cooling water. The loss of Lake Granbury makeup water due to the diversion of the Brazos River is highly improbable. Above Lake Granbury, the Brazos River channel is cut into bedrock which precludes any reasonable possibility of the river changing its channel significantly within the life of the CPNPP and thus affecting the supply of water.

Extraction of groundwater, oil, and gas from the region is relatively nominal. It is concluded that subsidence sometimes associated with these extractions will not occur in the vicinity of the CPNPP. The potential of subsidence at the site is discussed in detail in Section 2.5.1.2.6.

2.4.10 FLOODING PROTECTION REQUIREMENTS

Safety-related structures, except the Service Water Intake Structure and the Electrical and Control Building, are not subject to flooding, wave action, or wave runup and do not require flood or wave protection. The plant grade is at elevation 810 ft, and the peak SCR level is at elevation 789.7 ft for the probable maximum flood (PMF). (See Section 2.4.3.3.1)

The Service Water Intake Structure is protected from the effects of wind wave activity on the Squaw Creek Reservoir by the Safe Shutdown Impoundment (SSI) Dam. An overland 40-mph wind is assumed to occur coincidentally with the SSI PMF level of 790.5 ft. Since the wind wave activity at the Service Water Intake Structure is negligible, there are no significant dynamic forces to be considered. (See Section 2.4.5.) The design hydrostatic load includes the maximum runup elevation (see Section 2.4.3.6) caused by the 40-mph wind coincident with the PMF.

Design criteria for the Service Water Intake Structure are discussed in Section 3.8.4.1.4. The Electrical and Control Building would be subject to flooding if flood protection measures were not taken. These measures are described in Sections 3.4 and 13.5. It is not necessary to implement any other flood protection measures or emergency procedures.

2.4.11 LOW WATER CONSIDERATIONS

The SCR provides the cooling water for plant operation. In general, the combined amounts of natural and induced evaporation, plus releases for control of chemical quality, are more than the available runoff from Squaw Creek. Supplemental water will be brought from Lake Granbury on the Brazos River, in accordance with an agreement by and between Brazos River Authority and TP&L, TESCO, and DP&L, to hold the reservoir at the required operating elevation [39]. This agreement has no limitations which could be placed on the water that would have an adverse effect on safety-related makeup.

2.4.11.1 Low Flow In Rivers And Streams

Low flow in Squaw Creek or in the Brazos River is not of concern to plant safety because station cooling water is obtained from SCR. Thus, low flow considerations are significant in terms of successful over-all plant operations, but are not safety-related.

A study by the Texas Water Development Board indicated that from 1891 to 1960 Texas experienced 11 significant drought periods of varying severity and areal extent. Their ranking and time of occurrence are listed below:

1.	Most Severe	1954-1956
2.	Second	1916-1918
3.	Third	1909-1912
4.	Fourth	1901
5.	Fifth	1953
6.	Sixth	1933-1934
7.	Seventh	1950-1952
8.	Eighth	1924-1925
9.	Ninth	1891-1893
10.	Tenth	1937-1939
11.	Eleventh	1896-1899

The 1954-1956 drought was the most severe and was immediately preceded by the fifth and seventh ranked droughts, comprising a continuous series of years of rainfall deficiencies. This series, 1950 through 1956, comprises the most intense seven-year drought period that the State as a whole has experienced within the 70-year period of rainfall records. Dendrochronological studies in the southwestern part of the United States suggest that the 1950-56 drought period ranks among the most severe droughts of the past 400 years. Results of the studies indicate that records of stream flows during the 1950-56 period provide a basis for evaluation of the dependable amounts of water which can be obtained from existing and proposed reservoirs [37].

Detailed analysis of performance of SCR under the critical drought conditions of 1950 through 1956 [25] indicates a potential need for diversions from Lake Granbury ranging from 52,450 acre-feet per year. Of these amounts, an estimated 26,400 acre-feet per year would be returned at a steady rate to Lake Granbury in order to maintain proper balance of water quality in the two reservoirs. Thus, the net volumes of water taken from Lake Granbury during the drought would average 33,100 acre-feet per year and would range from a minimum of 26,050 acre-feet per year to a maximum of 38,260 acre-feet per year.

Substantially more than the required amounts will be available to SCR from Lake Granbury under terms of an existing agreement with the Brazos River Authority. This agreement covers total use from Lake Granbury and/or Lake Possum Kingdom of 70,000 acre-feet per year. The dependable yield of Lake Granbury [26] has been evaluated as at least 69,200 acre-feet per year, exclusive of the additional yield which could be made available by releases from Lake Possum Kingdom. The 70,000 acre-feet per year of potential supply is more than adequate to provide the necessary net diversions to SCR, plus anticipated requirements of other facilities which might also draw on Lake Granbury.

As a further check on the adequacy of the basic water supply for CPNPP, additional yield studies were carried out for Lake Granbury and Possum Kingdom Reservoir based on the years 1950-1957 and the following assumptions:

1. Historical inflows to Possum Kingdom Reservoir during the years in question were reduced to reflect:

    a. Total exclusion of runoff that originated above Hubbard Creek Dam, White River Dam, Salt Creek Dam, and Millers Creek Dam. These are the projects which either have been completed since the drought years or are expected to be completed in the foreseeable future.

    b. Maximum utilization by the City of Abilene of the increase in diversion rights on the Clear Fork of the Brazos, granted by the Texas Water Rights Commission since the drought years.

2. Releases of water from Possum Kingdom Dam for generation of hydroelectric power were reduced to a rate of 50,000 acre-feet per year, as compared with an average of approximately 350,000 acre-feet per year during the years of the study.

3. Runoff originating above Lake Palo Pinto was excluded from inflows to the Brazos between Possum Kingdom and Lake Granbury.

4. Historical losses of water in transit between Possum Kingdom and the site of Lake Granbury were assumed to still occur, although the releases from Possum Kingdom were substantially reduced.

5. Reservoir storage capacities were reduced to reflect predicted siltation to about the year 2006.

6. The studies were started with Lake Possum Kingdom assumed to be at the same elevation as actually experienced at the beginning of 1950, and Granbury was assumed to start at elevation 688.0, or five feet below the top of conservation storage.

7. The seasonal pattern of hydroelectric releases from Possum Kingdom was based on the actual pattern of such releases during the years of the study. The seasonal pattern of demand at Lake Granbury was based on the combined requirements of Squaw Creek Reservoir and the fossil-fueled power plants which will also derive their cooling water supply from the lake.

Under these conditions, which are believed to be highly conservative, the studies show that Lake Granbury would provide a firm yield of 70,000 acre-feet per year by utilizing only the upper 51 percent of the available conservation storage volume.

2.4.11.2 Low Water Resulting From Surges, Seiches or Tsunamis

Not applicable (See Sections 2.4.5 and 2.4.6).

2.4.11.3 Historical Low Water

The extreme variability of flow in the Brazos River is depicted through a flow probability curve (Figure 2.4-24), which shows that the average discharge of 1,555 cfs at gaging station 8-0910 is equalled or exceeded only about 17 percent of the time [27]. This flow is modified by regulation of water by upstream reservoirs which tend to decrease the variability. The impact that upstream control has had upon flow extremes is indicated from data which show that the Brazos River was known to dry up completely before construction of Possum Kingdom Reservoir and from Figure 2.4-25 which shows the lessening of annual flood events subsequent to Possum Kingdom Reservoir.

Squaw Creek has not been gaged long enough to allow a direct measure of flow variability, but indirect generalization of variability is gained by comparison with the Paluxy River. A flow probability curve for the Paluxy River is illustrated in Figure 2.4-25, which shows that the average discharge of 70.8 cfs was exceeded only about 11 percent of the time. Since the SCR catchment size is only about 16 percent that of the Paluxy watershed at gaging station 8-0915, this variability will be much more pronounced. Thus, inflow from Squaw Creek will be extremely variable.

Lake Granbury and SCR will serve to regulate the naturally variable flows and provide suitable minimum water levels on a dependable basis.

The Lake Granbury yield discussed in 2.4.11.1 above is based on not drawing that reservoir below elevation 675.0 [26]. The normal operating level in SCR will be held not less than elevation 770.0 feet.

2.4.11.4 Future Control

Development of additional control on the upriver sections of the Brazos River is possible; an initial study of potential reservoirs by the Brazos River Authority [40] identified three possible sites between Possum Kingdom Reservoir and Lake Granbury. These sites are briefly outlined on Table 2.4-3. Any one of these possible developments would increase storage availability and should strengthen water availability.

Downstream of DeCordova Bend Dam there is a possible reservoir site at Bee Mountain. The water at maximum level would be elevation 609 feet [40] and would impound the Brazos River

channel almost to DeCordova Bend Dam. It would be technically feasible to withdraw water from that lake also, offering an additional source of makeup water.

Under the Water Rights system of the State of Texas, issuance by the Texas Water Rights Commission of the permit to build and operate SCR precluded any further significant development and control upstream on Squaw Creek watershed itself.

2.4.11.5 Plant Requirements

The single source of safety-related cooling water and the ultimate plant heat sink for the Comanche Peak Nuclear Power Plant (CPNPP) is the SSI. This reservoir contains a volume of water, including evaporative contingency, that is sufficient to provide cooling for a period of over 30 days without makeup to safely limit the effects of an accident in one unit, to permit the safe shutdown of the other unit, or simultaneous shutdown of both units and maintaining them both in a safe shutdown condition. The available volume of water is determined during postulated 100-yr drought conditions as specified by NRC Regulatory Guide 1.27 and after a postulated 40 years of sedimentation. The minimum safety-related cooling water flow required for the two units during accident operation is 34,000 gpm, which represents the cooling water requirements of 17,000 gpm for post-LOCA cooling of one unit and 17,000 gpm for safe shutdown of the other unit. Expected safety-related cooling water flow is 51,000 gpm, which represents 17,000 gpm for post-LOCA cooling of one unit and 34,000 gpm for safe shutdown of the other unit. Plant water requirements for all modes of operation are given on Figures 2.4-26, 2.4-27, 2.4-28, and 2.4-29.

Cooling water for both units is withdrawn from the SSI and delivered by four 17,000-gpm-capacity station service water pumps enclosed in a seismic Category I structure, the Service Water Intake Structure. The Service Water Intake Structure sump descends to elevation 755 ft 0 in, and the service water pump impeller blades descend to elevation 758 ft 0 in. Each pump is designed to operate with a minimum submergence requirement of 4 feet - 6 inches above the bellmouth flare (El. 757 ft). As a result, a minimum water elevation of at least 761 ft 6 in is necessary for service water pump operation. The minimum water elevation, considering no makeup of water and the most severe period for evaporation described in Section 2.3.1.2.10, is 766 ft 4 in. at 30 days and 765 ft. 4 in. at 39 days. Therefore, substantial margin for submergence is provided.

Cooling water is returned to the SSI through the Service Water Discharge Structure. Water from this structure enters the SSI at a point remote enough from the Service Water Intake Structure and at a velocity high enough to ensure adequate mixing, dispersion, and evaporative cooling of the effluent. The station service water pumps, Service Water Intake Structure, and Service Water Discharge Structure are described in Section 9.2.1.

The SSI is formed from an inlet of the Squaw Creek Reservoir (SCR) and is separated from the reservoir by a seismic Category I dam. The surface elevation of both bodies of water is a minimum 770 ft 0 in during normal operation. The seismic Category I dam, however, will maintain the SSI at a surface elevation of 769 ft 6 inches in the event of an accident involving water loss from the SCR. The ultimate Heat Sink is described in Section 9.2.5.

The SSI is constructed specifically to serve as the ultimate heat sink for the CPNPP. Except for a backup source for auxiliary feedwater (Section 10.4.9.3), there are no other uses of this water during accident and safe shutdown operations. Water rights to the SSI are guaranteed by the Texas Water Rights Commission. For use of water near the plant site, see Subsection 2.4.13.

2.4.11.6 Heat Sink Dependability Requirements

The source of plant cooling water for the CPNPP is the SCR. This reservoir is formed by a dam structure extending across the Squaw Creek and creating a cooling water impoundment in the Squaw Creek Basin behind it (see Figure 2.4-4). The impoundment is normally supplied with makeup water from Lake Granbury. In addition, water is exchanged, as necessary, between Lake Granbury on the Brazos River and the SCR to reduce total dissolved solids levels.

The single source of safety-related cooling water is the SSI, which functions as the ultimate heat sink for the CPNPP. The SSI is formed from an inlet of the SCR and separated from it by a seismic Category I dam. A canal connects the SSI with the SCR and maintains an equal water level between the two bodies of water. The seismic Category I dam and canal maintain the water level of the SSI at 769 ft 6 inches in the event of accidental water loss from the SCR. Waterflow through the canal will stop at this elevation. A simultaneous failure of both the SCR dam and the SSI Dam is considered unlikely since the SSI Dam is a seismic Category I structure.

Cooling water for normal plant operation is withdrawn from the SCR by eight 275,000-gpm-capacity circulating water pumps. Each pump is a 25-percent-capacity unit, and all pumps are located in the Circulating Water Intake Structure. Each pump has impeller blades which extend to elevation 758 ft 0 in and to a minimum submergence requirement of 12 ft. The circulating water pumps are not required for plant shutdown.

Cooling water is returned to the SCR via the Circulating Water Discharge Structure. The structure discharge is located at an adequate distance from the Circulating Water Intake Structure to ensure sufficient water mixing and evaporative cooling.

Safety-related cooling water is withdrawn from the SSI by four 17,000-gpm-capacity service water pumps. All pumps are located in the Service Water Intake Structure, a seismic Category I building. Safety-related cooling water is returned to the SSI through the Service Water Discharge Structure. The discharge structure is located at a sufficient distance from the Service Water Intake Structure to ensure adequate water mixing and evaporative cooling.

The minimum water level of both the SCR and the SSI during normal plant operation is 770 ft 0 in. This water level is adequate for both circulating water pump and station service water pump operation. During postulated 100-year drought conditions, and after 40 years of sedimentation, the SSI is determined to have 284 acre-ft of water.

Figures 2.4-26 through 2.4-29 describe cooling water requirements for the various modes of plant operation. Section 9.2.5 discusses design bases and safety functions of the SSI as the CPNPP ultimate heat sink.

The anticipated loss from the SSI caused by evaporation is 92 acre-feet. The seepage and drift loss is negligible in comparison to the evaporation losses. Instrumentation, which is read in the Control Room, is provided to measure the SSI water level as shown on Figure 9.2-1. An alarm is provided in the Control Room for low SSI water level and high differential level across the travelling screens. The low level alarm is set at or above the normal low SSI water level as desired by the operator as an aid for lake level management. The SSI will be dredged as required.

The CPNPP design of the SSI as the single ultimate heat sink complies with the intent of NRC Regulatory Guide 1.27 as discussed in Chapter 1, Appendix 1A(B).

The SSI is the source of fire protection storage tank emergency fill water. SSI level changes caused of the Fire Protection System are insignificant since the system is used infrequently and for periods of short duration. Refer to Section 9.5.1 for a description of the Fire Protection System.

Cooling water is withdrawn from the SSI at the Service Water Intake Structure. Elevations of intake structure pump deck and channel floor are established with due consideration for pump minimum submergence requirements and reservoir level fluctuations. For elevations and layouts of Service Water Intake Structure components, see Section 9.2. Plant water requirements for various modes of operation are given in Figures 2.4-26, 2.4-27, 2.4-28, and 2.4-29.

2.4.12 DISPERSION, DILUTION, AND TRAVEL TIME OF ACCIDENTAL RELEASES OF LIQUID EFFLUENTS IN SURFACE WATERS

2.4.12.1 Introduction

This section provides a conservative analysis of a postulated accidental release of radioactive material in surface waters adjacent to the site. The postulated release was assumed to occur due to an accidental rupture of the waste holdup tank which is located in the Auxiliary Building near the Containment Building.

The volume of the tank is 30,000 gallons and at the time of rupture it was assumed that the tank was 80 percent full. The assumed concentrations of radionuclides in the tank at the time of rupture are given in Table 2.4-20.

It was conservatively assumed that all the liquid radwaste (24,000 gallons, or 7.36×10^{-2} acre-feet) is spilled into Squaw Creek Reservoir. Minimum dilution in Squaw Creek Reservoir would occur at minimum pool elevation 770.00 feet (msl), corresponding to a storage volume of 135,062 acre-feet. Assuming complete mixing, the minimum dilution factor is 135,062/(7.36 x 10-2) or 1.84 x 106.

The instantaneous concentrations in Squaw Creek Reservoir are calculated by dividing the concentrations in the tank by the dilution factor. Due to the decay characteristics of the radionuclides, most concentrations decrease with time, however, some radionuclides build up in the reservoir as decay products. A computer code, RADIOISOTOPE, described in Appendix 15B Section 15B.6, was used to calculated the decayed concentrations. The concentrations of each radionuclide in Squaw Creek Reservoir at the end of 1 day, 1 month and 1 year is shown in Table 2.4-21. Since the intent of the analysis is to demonstrate that the concentrations of radionuclides in the reservoir do not exceed the effluent concentrations for unrestricted areas given in 10CFR20 Appendix B, the total effluent concentration fraction is also given for each of the above time periods. The results demonstrate that the water volume of the reservoir provides sufficient dilution, and that most of the activity would decay away after one month.

The water usage pathways involving Squaw Creek Reservoir indicate that a portion of the reservoir inventory may be transferred to Lake Granbury and Lake Whitney. This transfer would

provide for additional dilution of radionuclides due to the added water volumes, and it is expected that the concentrations of radionuclides in Lake Granbury and Lake Whitney would be much lower than those in Squaw Creek Reservoir. The flow pattern which exists among these bodies of water, and the overall potential for further dilution of Squaw Creek Reservoir concentrations provided by those other two lakes is more fully addressed in Appendix 11A.

2.4.13 GROUNDWATER

The investigation of groundwater features included:

1. A review of groundwater information obtained from public agencies and from literature.

2. A field inventory of all wells used for water supply within two miles of the site and reservoir fringe.

3. A review of municipal, industrial, and irrigation wells within 20 miles of the site.

4. Measurements of water levels in several exploratory borings which were drilled to obtain geologic, geohydrologic, and foundation data.

5. The evaluation of the probable influence SCR and CPNPP will have on the quality and piezometric level of groundwater in the site vicinity.

[General References 29, 30, 31, 32].

2.4.13.1 Description and On-Site Use

2.4.13.1.1 General Groundwater Features

The CPNPP site is located within the Great Plains Physiographic Province, about 4 1/2 miles north of Glen Rose, Texas, and approximately four miles west of the Brazos River. The power plant site is on a relatively narrow ridge which trends east-southeast and is flanked on the north and south by the SCR (Figure 2.1-2). Topography in the site vicinity ranges from slightly undulating to stair-stepped. Ground surface near the site ranges from elevations 600 to 700 feet (Mean Sea Level Datum) in valleys to elevations 900 to 1000 feet on ridges. The Brazos River channel a few miles east of the site is at approximate elevation 560.

The site is underlain by a sedimentary rock sequence which, at the surface, has been weathered to a clayey, silty, sandy overburden soil with some rock fragments. Overburden on the ridges and slopes ranges in thickness from a few inches to a few feet and rock outcrops are common. The soils and much of the rock are relatively impermeable. In valley bottoms, rock is overlain by a combined thickness of alluvial sediments and underlying residual soil. Total overburden ranges from 10 to 15 feet in depth. The sediments are of minor thickness and vary from slightly permeable to essentially impermeable.

In the site area, including the floodplain, infiltration into subsurface formation is retarded because of low permeability of the strata. Most precipitation flows across the surface and drains away as surface runoff or returns to the atmosphere by evaporation and transpiration. Water-bearing strata in the area are mainly recharged in their outcrops areas.

Data from a nearby gaging station on the Paluxy River indicate that the net surface runoff and base flow (which is derived from shallow groundwater) together total some two inches annually. Evaporation from a free water surface, such as a lake, is approximately 72 inches per year [33].

Most of the groundwater in the site region occurs in bedrock. Some water does exist in shallow floodplain alluvium in stream valleys, but is not withdrawn for use. Regional bedrock aquifers in order of increasing age are the Paluxy, Glen Rose and Twin Mountains formations, of the Comanche series, Cretaceous age [34]. Locally, CPNPP and SCR are on the Glen Rose outcrop, which, in turn, is underlain by the Twin Mountains Formation. The Paluxy Formation is absent at the CPNPP location, and within the limits of the reservoir; the Glen Rose Formation is the upper stratigraphic unit exposed in these areas.

The structure, stratigraphy, and composition of the three strata are described in detail in Section 2.5.1. A geologic cross section representing the approximate orientation of these strata near the site is shown on Figure 2.5.1.2-5 (see upper right portion of cross section). The approximate limits of formation outcrops in the counties surrounding the site are shown on Figures 2.4-30, 2.4-31 and 2.4-32.

The Paluxy, Glen Rose, and Twin Mountains formations are large in area. Their outcrops form a strip of land tens of miles wide that extends south from Central Oklahoma, strikes westward in Central Texas, and extends into Mexico [35]. The formations in the site region dip eastward.

The Twin Mountains and Paluxy formations are principally sandstone, but also have shale, limestone, claystone, and siltstone inclusions. Limestone is the dominant rock type within the Glen Rose Formation, but it also contains significant quantities of shale, siltstone and claystone. In these formations, groundwater percolates slowly along bedrock joints and fractures, and through interstices in the rock fabric.

The Twin Mountains is the only relatively productive bedrock zone in the site vicinity. The Paluxy Formation has nominal pumpage near the site and the Glen Rose Formation yields very little water in the site area and is usually less productive than the other formations.

At distances of 20 to 50 miles downdip from the outcrop, the groundwater becomes saline and the formations lose their importance as sources of fresh water [36]. The three formations are discussed individually in succeeding sections.

2.4.13.1.2 Twin Mountains Formation

A detailed description of the lithology, stratigraphy, and structure of the Twin Mountains Formation is given in Section 2.5.1. The formation is predominantly sandstone with subsidiary shale, claystone, siltstone, and limestone.

The principal origins of the groundwater within the Twin Mountains Formation are rainfall and streamflow occurring in the outcrop area (Figure 2.4-32).

Downdip from the outcrop, groundwater in the Twin Mountains Formation is confined by fine-grained materials in the overlying Glen Rose Formation. Hydrostatic pressure in the Twin Mountains is great enough to create static water levels which rise above the formation and, sometimes, great enough to cause flowing wells. The piezometric level measured in this

formation at the site in Boring P-10 is approximately elevation 670, about 60 feet above the formation surface.

Groundwater is discharged in the outcrop area by evapotranspiration and localized springs and seeps along drainages incised below the water table. Downdip (where the formation is confined, such as at CPNPP site), natural discharge is limited to a minimal upward movement into overlying formations.

Although the Twin Mountains Formation is a moderately productive stratum in the site vicinity, two packer pressure tests of 60 feet of this rock in a boring at the CPNPP site did not result in water take. This test indicates that there are essentially impermeable rock zones within this formation.

2.4.13.1.3 Glen Rose Formation

This formation is predominantly limestone, but significant amounts of shale, siltstone and claystone are also present. A full description of the geology of the Glen Rose Formation is presented in Section 2.5.1.

The principal origins of groundwater in the Glen Rose Formation are rainfall on the outcrop area and minor seepage from the overlying Paluxy Formation and underlying Twin Mountains Formation. The Glen Rose Formation outcrop area is shown on Figure 2.4-31.

CPNPP and SCR is constructed on the Glen Rose Formation; thus the character and rate of groundwater movement in this formation is of special interest. The Glen Rose limestones are essentially impermeable due to slight amounts of argillaceous impurity and are thus resistant to solution effects. Open voids, caverns, joints, collapse features, and frequent fractures common to some limestones are notably absent in the Glen Rose Formation at the site; therefore, the groundwater moves very slowly through and into the Glen Rose Formation, principally through the joints and fractures that do exist. Occasional isolated sand lenses also contain groundwater.

Detailed examination of cores from borings revealed no solutioning features and minimal fractures. Packer pressure tests in the Glen Rose Formation performed in most borings at the station, SSI Dam and spillway, and Squaw Creek dam and spillway incurred essentially no “water take” in rock beneath the upper, usually thin, weathered zone. Pressure tests were performed within the rock in many borings to evaluate jointing and/or the absence of solutioning features and to evaluate general permeability characteristics. Tests employed single and double packers and gage pressure ranging to 80 psi. The results of packer tests are presented on Table 2.5.6-1. Only zones at a depth range of 194 feet to 214 feet (elevation 649.04 to 629.04 feet) recorded any water loss. Upon review of the Log of Boring for Boring P-10, the cause of the water losses can be attributed to a zone of sandstone and sand lenses.

Drill water occasionally was lost while drilling through the upper weathered zone. A relatively thick pervious zone was encountered in Borings DI-8 and DI-9, approximately 1 1/4 miles southeast of the station. These borings are at the middle to outer extremity of a peninsula which will form the south abutment for Squaw Creek Dam. There, a large amount of drill water was lost while advancing Boring DI-8 to 56 ft and DI-9 to shallower depth through some zones of weathered, fractured rock. Indications were that water loss occurred principally within the upper 20 to 30 feet of each boring. There is no evidence of solutioning in the zone of water loss.

Boring DI-10, drilled farthest from the peninsula point, incurred no drill water loss, thereby indicating that conditions improve southward along the peninsula.

Northwest of the site, where the Glen Rose Formation is covered by outliers of the Paluxy Formation, a few domestic water wells are completed in the Glen Rose Formation. These wells produce potable water and they are reliable during droughts; this reliability is due to the slow release of groundwater to the Glen Rose Formation from the overlying Paluxy Formation. Elsewhere, wells completed in the Glen Rose Formation are often unreliable during droughts.

In its outcrop areas, the Glen Rose Formation discharges water naturally through springs and seeps. In the confined portions of the formation, when differential hydraulic pressures occur, there is very minor transfer of water into overlying and underlying formations from the Glen Rose Formation.

2.4.13.1.4 Paluxy Formation

This formation is predominately sandstone, but shale, siltstone, claystone and limestone are also present. A detailed description of the lithology, stratigraphy and structure of the Paluxy Formation is given in Section 2.5.1. The Paluxy Formation is not present at the CPNPP and SCR locations.

Recharge to the Paluxy Formation occurs in the outcrop areas as a result of infiltration of rainfall and of seepage from streams.

It also receives water from water-bearing units under greater hydraulic head which adjoin the Paluxy Formation. Figure 2.4-30 shows the outcrop area of the Paluxy Formation. South of the CPNPP site, across the Paluxy River, the formation is confined by overlying fine-grained bedrock strata. These strata are not of significance to CPNPP.

Groundwater discharges from the Paluxy Formation as springs and seeps in some outcrop areas. Where the formation is confined, there is limited water movement into overlying confining units if those units are at lower hydraulic head.

2.4.13.1.5 Onsite Water Table

Following the subsurface exploration program, a number of the borings were used to determine water levels. Of these borings, P-10 was completed in the Twin Mountains aquifer; the piezometric water level in that boring is elevation 670. The remainder of the borings monitored for groundwater were completed in the Glen Rose Formation. Static water levels observed in these borings are presented in Figure 2.5.5-77 and range from elevation 749 to 830.

As indicated in Section 3.8.5.1.5, groundwater is not expected to reach higher than elevation 775.0 feet. All piezometric levels recorded on Figure 2.55-77 were measures of perched water in the upper zone of the Glen Rose Formation in the immediate area of each piezometer. The resulting elevation range of 749 to 830 feet is attributed to surface run-off and not a true measure of permanent groundwater in the Formation.

Water levels in the Glen Rose Formation are expected to show some variation in response to seasonal climatic changes; those in the Twin Mountains Formation will be much less influenced by seasonal conditions because of the distance from the recharge area.

Piezometers which measured the levels recorded in Figure 2.5.5-77 were installed during preliminary design work at the site, before the site was excavated to plant grade (elevation 810). A permanent system of piezometors has been installed in order to monitor ground water levels at the site. This program is described in Section 2.5.4.13. A Groundwater Monitoring Program is established and incorporates several sentinel wells in various locations throughout the plant site for prompt identification of potential radiological release source locations.

2.4.13.1.6 Water Quality

Potable groundwater occurs in the Twin Mountains, Glen Rose and Paluxy formations. The results of chemical analyses of groundwater obtained from wells drawing from these formations are summarized in Table 2.4-22. (Well locations are shown on Figure 2.4-33).

Water in the Twin Mountains Formation is a sodium bicarbonate type with a dissolved solids content varying generally from 200 to 900 mg/1. In and near the outcrop areas, Twin Mountains water is used for irrigation. At the site, however, the water is unsuitable for irrigation due to the local soil conditions and the increased sodium content of the water. The results of physical and chemical analyses performed on the samples taken from the production and observation wells during the years of 1975-76 are presented and discussed in the Environmental Report/Operating License Stage (ER/OLS). The sodium content of the water samples ranges from 100 to 150 mg/1, with dissolved solids content varying from 300 to 500 mg/1. The temperature of groundwater follows the seasonal atmospheric average temperature values, and ranges from 20° to 26°C (68° to 79°F). The conductivity values vary between 550 to 1300 mhos.

The quality of water obtained from the Glen Rose Formation is variable; in localized areas it is not potable. Northwest of the site, water is produced from the Glen Rose Formation where it is capped by an outlier of Paluxy Formation. The analysis from well No. 48 is an example of the water quality obtained from this area. The chemical analysis from well No. 48 is presented in Table 2.4-22.

The Paluxy Formation is tapped by some domestic water wells south of the Paluxy River. The chemical analysis of water from a well five miles east of Walnut Springs in Bosque County is shown in Table 2.4-22. This well is located close to the outcrop area of the Paluxy Formation, where the water is typically a hard calcium bicarbonate type. Further downdip, the water becomes a progressively softer, sodium bicarbonate type.

2.4.13.1.7 Onsite Water Uses

Groundwater usage on the site is presented in Table 2.4-23. Continuous usage will be about 127 gpm and the peak plant requirement is estimated to be 330 gpm. Alternate source of supply of water for this usage will be from the surface water pre-treatment system.

2.4.13.2 Sources

2.4.13.2.1 Regional Use

1. Paluxy and Glen Rose Formations

   Groundwater is pumped from the Paluxy and Glen Rose formations by small-capacity wells for livestock and rural domestic use. Pumpage is not metered within the region.

Withdrawals from both formations, however, are estimated to be less than 100 acre-feet per year. Water extraction from these formations has no identifiable effect on regional piezometric levels. Groundwater use is not expected to increase significantly in the future because these formations are poor aquifers and will probably not be developed for water supply by either cities or industries or for large scale irrigation. More favorable water supplies are available from surface sources or from the Twin Mountains Formation.

The regional movement of water in the Paluxy and Glen Rose formations is downdip to the east. The hydraulic gradient in the Paluxy is about 20 feet per mile in downdip areas. The rate of regional groundwater movement is estimated to be on the order of one foot per day or less in the Paluxy Formation.

Because groundwater in the Paluxy is used only for domestic and livestock purposes in rural areas, wells are usually of small pumping capacity, typically under 10 gallons per minute or less. It is estimated that maximum potential well yield in the region is probably under 50 gallons per minute.

There are few wells withdrawing water from the Glen Rose Formation. This formation typically is not capable of supplying more than about 10 gallons per minute to wells. Water levels in the Glen Rose fluctuate in response to precipitation, and some wells in the Glen Rose are reported to have failed during certain droughts.

Generally, water use from the Paluxy and Glen Rose formations is small and individual wells are of very limited capacity. The aquifers are variable in their hydraulic characteristics and also in the quality of water they yield. As a result, no significant development of these sources of groundwater is anticipated.

2. Twin Mountains Formation

Prior to 1880 there was relatively little groundwater development from this formation. Since the early 1900's pumpage has increased to include wells from municipal, industrial and irrigation purposes.

The Twin Mountains Formation is the primary source of groundwater used in the region although this use is not extensive.

Principal wells in the area, i.e., public supply, irrigation and industrial, which are within 20 miles of the site, are listed on Table 2.4-24 and locations are shown on Figure 2.4-33. Texas Water Development Board studies [37] indicate that current annual pumpage from the Twin Mountains Formation is about 100 acre-feet per year within a 20-mile radius of the site. By the year 2020, the total pumpage is projected to be not more than 200 acre-feet per year.

The groundwater movement is downdip to the east (Figure 2.4-34), at a rate of approximately two feet per day. The current piezometric gradient is about 20 feet per mile. Permeability of the formation ranges from 90 to 240 gallons per day per square foot [34].

The recharge area (outcrop area) is shown on Figure 2.4-32. Because the site is near the recharge area and because of the relatively small projected amount of future pumping, no

significant change in groundwater level is expected in the site vicinity during the life of the plant.

2.4.13.2.2 Local Use

All water wells within a two-mile radius of the site were inventoried. In addition, information was gathered on water wells in and adjacent to the planned impoundment. The locations of wells in the vicinity are shown on Figure 2.4-35. The well data are listed on Table 2.4-25. All wells in the site vicinity are of small capacity and are used for individual rural home and livestock requirements. With few exceptions, all are deep wells, completed in the Twin Mountains aquifer.

To the north of the site, a few wells are completed in the Glen Rose Formation. The presence of adequate and reliable supplies of groundwater in the Glen Rose Formation is dependent on a sandstone cover (the Paluxy Formation). This sandstone cap results in prolonged percolation to the Glen Rose and results in relatively high water levels, even during drought conditions.

Data collected on local water levels has permitted mapping of the Twin Mountains piezometric surface in the site area; this map is shown on Figure 2.4-36, which is more detailed than the regional map, Figure 2.4-33.

Two production wells (PW-1 and 2), and three observation wells (OB-1, 2 and 4) are constructed onsite and their locations are shown in Figure 2.4-37. Observation well No. 3 was an existing well in the plant property. All these wells tap water from Twin Mountain Formation. Pumping tests were run PW-2 as the pumped well and PW-1 as observation well. From these tests a coefficient of transmissibility of 12,000 gpd/ft and a storage coefficient of 0.00005 were calculated. The two production wells (PW-1 and 2) along with two others on-site were plugged and capped in 2013.

Using these aquifer characteristics and a pumping rate of Q = 100 gpm, a series of drawn-down curves in relation to distance and time are plotted (Figure 2.4-38).

To satisfy water demands during construction, pumpage has occurred in the production wells. The pumpage rates for 1975 are presented in Table 2.4-26. These figures show that the average pumping rate was about 150 to 160 gpm. The hydrographs of observation wells during this pumpage are shown in Figure 2.4-39. The early measurements made in OB-4 indicate an abnormally high water level and sudden change in the level from August to September. This abnormality and change may be due to incomplete development of the well after completion. Each of these wells is sampled by pumping once a month as part of the quality monitoring program with the exception of Production Well Numbers 1 and 2 which were no longer sampled after being plugged and capped in 2013. Therefore, OB-4 may have been further developed by the pumpage and by the removal of any obstruction.

The average effect on the water levels of the observation wells due to plant pumpage is approximately nine feet. The hydrograph of the observation wells indicates that water levels from September to December have remained fairly constant. This constancy begins after approximately 200 days of pumpage, which conforms to the distance draw-down curves where the greatest amount of draw-down occurs in the first 180 days.

Operational pumpage of the plant will be 127 gpm on continuous basis. The peak plant requirement is estimated to be 330 gpm for a short time period. The alternate source of water

supply is the surface water pre-treatment system taking raw water from Squaw Creek Reservoir. This minimizes the effects of operational pumpage on draw-down. The nearest boundary line of the project area to the production wells is more than one mile. From the draw-down curves presented in Figure 2.4-38, it can be estimated that draw-down due to 34.5 gpm continuous pumpage during the life of the plant (40 years) will not be more than 2.5 feet at the boundary line. In case of 330 gpm peak pumpage rate, the temporary draw-down will be 3.3 feet for one day pumpage and 7.25 feet for three days pumpage at the same location.

Operational pumpage from wells in the owner controlled area followed a stepwise trend. The first six years 1975 to 1980 pumpage averaged 145 gpm. From 1981 to 1990 (years leading up to and including Unit 1 start-up) pumpage averaged 16 gpm. From 1991 to 2011 production well pumpage averaged 23.2 gpm. Since 2012, operational pumpage from groundwater dropped to less than 1 gpm when Somervell County Water District became the sole supplier of potable water.

Historical groundwater levels around the plant site can be estimated from the records of four Texas Water Development Board observation wells in Somervell County. The locations of these wells are also shown in Figure 2.4-37 and their records are presented in Table 2.4-27. The records indicate a fluctuation of levels and also localized cones of depression.

Based upon the geohydrologic characteristics at the site, it is estimated that the piezometric level in the Twin Mountains Formation will be depressed locally due to pumpage from the production wells, but without adverse effect on the station or the existing wells withdrawing water from the formation. Within the Glen Rose Formation, water levels will not be affected due to this pumpage.

All wells and borings in the impoundment area are grouted and sealed to insure no direct intrusion of reservoir water into subsurface aquifers.

2.4.13.3 Accident Effects

2.4.13.3.1 Introduction

This section provides a conservative analysis of a postulated accidental release of radioactive liquid at the surface of the site. The postulated release is assumed to occur due to an accidental rupture of the floor drain tank which is located in the Auxiliary Building near the Containment Building. This tank is located inside a Category I building with a drain collection system.

The volume of the tank is 30,000 gallons and at the time of rupture it was assumed that the tank was 80 percent full. The assumed quantities of radionuclides in the tank at the time of rupture are given in Table 2.4-20A. It is further assumed that all the liquid gets outside the Auxiliary Building unimpeded and no action is taken to contain or clean up the water spill.

The Auxiliary Building is founded in the Glen Rose limestone which is an essentially impermeable formation (see section 2.4.13.1.3). No ground water was encountered during excavation into the Glen Rose limestone for the plant foundations. However, during design validation efforts, it was determined that perched water may exist at elevations higher than 775'-0". This condition was created by the excavation and backfill process associated with buried piping ductbanks and the service water intake structure when backfill was placed using pervious material. This arrangement allows surface runoff to be trapped in the pervious zone

(backfilled area) thus raising the local water level (called perched water level) above elevation 775'-0". This ground water therefore, is not hydraulically connected to the Twin Mountains aquifer which is approximately 200 feet below plant grade. Design Basis groundwater level for lateral forces on the plant structures is conservatively taken as 810'-0" except at the Service Water Intake Structure. It is expected that the ground water gradient will be from the reservoir toward the plant site, thus restricting the movement of any released liquids above elevation 810'-0" to predominantly vertical flow. A buildup in the ground water level through percolation of rainfall over the plant site into the essentially impervious limestone is considered unlikely. Therefore, because mounding due to percolation is highly unlikely, it is correspondingly improbable that recharge in the plant area would increase the potential for outward horizontal flow toward the reservoir. If the released liquid could travel horizontally through the Glen Rose limestone, it would enter Squaw Creek reservoir. The ability of the surface water regime to dilute and disperse such releases is discussed in Section 2.4.12.

In order to evaluate the effect of a postulated accident on the Twin Mountains aquifer, a conservative mathematical model with simplifying assumptions was used to model the dispersion of the liquid through the Glen Rose limestone down to the top of the Twin Mountains aquifer.

2.4.13.3.2 Description of the Model

The dispersion and decay of the radionuclides within the Glen Rose limestone was evaluated using the model described briefly below.

The governing equation employed in this one-dimensional dispersive and convective model may be written as follows:

$$\frac{\partial}{\partial t}(R^j C_j) + \frac{\partial}{\partial x}(UC_j) = S^j + \frac{\partial}{\partial x}\left\{ Dx^j \frac{\partial C_j}{\partial x} \right\} - R\lambda^j jC_j$$

where

C_j is the concentration of the chemical species j;

R^j is the retardation factor for the species j, comprising adsorption, ion exchange, precipitation, colloid filtration and irreversible mineralization;

U is the groundwater velocity in the x-direction;

S^j is the source term for chemical species j;

D_x^j is the dispersion coefficient for species j is the x direction;

λ^j is the decay coefficient for species j (also given by $1n2/T^{1/2}$ where $T^{1/2}$ is the radionuclide half-life);

x is the distance in the x-direction; and

t is the elapsed time.

This governing equation is solved by a numerical model which employs a partly implicit solution procedure to predict chemical species concentrations as a function of space and time.

The discretized form of the governing equation is obtained by integration over control volumes defined by a numerical grid set up for the region of interest.

2.4.13.3.3 Selection of Parameters

1. Initial and Boundary Conditions

Initial and boundary conditions must be supplied to the numerical model in order to complete the mathematical specification of the problem. The initial conditions designate the distribution of radionuclides over the entire solution domain at the commencement of the analysis. In this problem, zero concentrations were specified everywhere initially.

Boundary conditions represent variations of the concentrations, the concentration fluxes, or a combination of these, at the boundaries of the domain. In the case being considered, the lower boundary condition was specified as a zero gradient boundary. That is, the radionuclide concentrations just within the upper part of the Twin Mountains aquifer were

considered to be identical to those just within the lower part of the Glen Rose limestone; i.e., the radionuclides are transported by convection alone. The upper boundary coincides with the base of the storage tank. The upper boundary condition was therefore the radionuclide source of the concentration contained in the tank.

2. Reduction Factor

The reduction factor was conservatively assumed to be equal to 1, implying that no retardation in movement or interaction between radionuclides and soil occurs.

3. Initial Flow Velocity

The initial flow velocity of radionuclides out of the base of the tank is limited by the capacity of the soil to absorb the flow; i.e., the saturated vertical permeability of the Glen Rose limestone. The permeability value was estimated from packer-test data to be:

$k = 2.8 \times 10^{-2}$ ft/day.

4. Dispersion Coefficient

The equation given by Fried and Combarnous (43) can be utilized in the calculations:

$Dx = D_o [0.67 + 0.5(U_x d_{50}/D_o)1.2]$

where d_{50} is the mean particle size (ft), U_x is the groundwater velocity (ft/day), and D_o is the molecular diffusion in bulk water (ft^2/day). D_o is equal to approximately 9.7×10^{-4} ft^2/day. Letting the groundwater velocity go the zero, and changing units yields:

$D_x = 0.0004$ cm^2/minute

Another value reported for the soils similar in dispersion characteristics is the value given in Reference 44, which is:

$D_x = 0.0028$ cm^2/minute

Taking the geometric average of these two values as a conservative approximation of the dispersion coefficient yields:

$D_x = 0.001$ cm^2/minute = 1.6×10^{-3} ft^2/day.

5. Decay Coefficient

The decay coefficient, λ^j, is calculated from known half-life values for each of the radionuclides of concern.

6. Source Term

The source term, S^j, is calculated from the known curie content of the tank, and the volume of fluid within the tank, assuming that the entire base of the tank ruptures instantaneously.

2.4.13.3.4 Computation Results

The analysis showed that only one radionuclide, CS-137, penetrated the entire 150-foot depth of the Glen Rose Limestone to reach the Twin Mountains aquifer below. To illustrate the dispersion of the major radionuclides of concern, the results for those with the greatest release concentrations and longest half-lifes have been tabulated in Table 2.4-28. This table illustrates the maximum depth to which significant concentrations penetrate. Concentrations of less than 10^{-13} μ Ci/ml (2.83×10^{-15} Ci/ft^3) were considered insignificant.

CS-137 is shown to penetrate the entire 150 feet in 400 years; therefore, more detailed results are presented for this radionuclide in Table 2.4-29. This table shows the concentrations of CS-137 throughout the limestone as a function of time. This conservative analysis calculates the maximum concentration of any radionuclide anywhere in the groundwater to be 4.8×10^{-12} μ Ci/ml as a result of the postulated accident.

2.4.13.4 Monitoring or Safeguard Requirements

No planned releases to the ground water environment will take place at the plantsite. However a Groundwater Monitoring Program exists for early detection of inadvertent releases of radioactive material. Pertinent information is provided in Section 6.1 of the Environmental Report.

2.4.13.5 Design Bases for Subsurface Hydrostatic Loading

The lateral pressure (σ) caused by the groundwater at a given point is equal to the unit weight of water (γ) times the vertical distance from the water table to the point at which the pressure is computed (H):

$$\sigma = \gamma H$$

Uplift pressures are similarly computed as H, where H is the vertical distance from the water table to the surface on which the uplift is computed.

No ground water was encountered during excavation and construction of the plant structures. However, during design validation efforts it was determined that ground water or perched water may exist at elevations higher than 775'-0". Therefore, safety-related plant structures are conservatively designed for hydrostatic loads with the design basis ground water level at elevation 810'-0", except for the Service Water Intake Structure. Hydrostatic loads on the SWIS are determined with a design basis ground water level at elevation 793'-0".

There is no dewatering at the site during or after construction.

2.4.14 TECHNICAL SPECIFICATION AND EMERGENCY OPERATION REQUIREMENTS

The most adverse hydrological conditions detailed in the preceding sections do not affect the operation of safety-related facilities. Emergency procedures are required, however, if Squaw Creek Reservoir elevation exceeds 778 feet to ensure that the Electrical and Control Building is not flooded via the Circulating Water System. See Section 13.5 for a discussion of these procedures.

The maximum predicted normal SSI temperature for two unit full power operation is 102°F. Therefore, this is the initial condition assumed in the Station Service Water System (Section 9.2.1), Component Cooling Water System (Section 9.2.2), and Containment Heat Removal (Section 6.2.2) analyses and specified in the Technical Specifications.

The lowest normal SSI elevation is 770 ft. 0 in. as described in Section 2.4.11.5. This elevation is the initial condition assumed in the ultimate heat sink analysis described in Section 9.2.5.

REFERENCES

1. Water Resources Data for Texas," Part 1, Surface Water Records, U. S. Geological Survey, annual publications, water years 1966-70.

2. "Number of Inhabitants, 1970 Census of Population, Texas," U. S. Department of Commerce, Bureau of the Census, 1971.

3. Letters from Burke G. Bryan, Chief Engineer, Brazos River Authority, dtd, 22 May 1972.

4. Digital Computer Printouts entitled, "River Order List of The Water Rights Master File," Texas Water Rights Commission, 1972.

5. "Bulletin 6311, Floods in Texas, Magnitude and Frequency of Peak Flows," Texas Water Commission, 1973.

6. "Geological Survey Water-Supply Paper 1682, Magnitude and Frequency of Floods in the United States," Part 8, U. S. Geological Survey, 1969.

7. "Geological Survey Water-Supply Paper 1922, Surface Water Supply of the United States 1961-1965," Part 8, Volume 1, U. S. Geological Survey, 1970.

8. "Flood Stages and Discharges for Small Streams in Texas," U. S. Geological Survey Texas District Open File Report, 1971.

9. "Hydrometeorological Report No. 33, Seasonal Variation of the Probable Maximum Precipitation East of the 105th Meridian for Areas of 10 to 1000 Square Miles and Durations of 6, 12, 24, and 48 Hours." Hydrometeorological Section, Hydraulic Services Division, U. S. Weather Bureau in Cooperation with the U. S. Corps of Engineers, 1956.

10. "Engineering Technical Letter No. 1110-2-8, Computation of Free- board Allowances for Waves in Reservoirs," Department of the Army, Office of the Chief of Engineers, Washington, D.C., 1966.

11. Hershfield, D. M., “Extreme Rainfall Relationships,” Journal of the Hydraulics Division, Proceedings of the American Society of Civil Engineers, November, 1962.

12. “Design of Small Dams,” U. S. Department of Interior, Bureau of Reclamation, 1960.

13. Corps of Engineers data provided by Freese, Nichols and Endress, Fort Worth, Texas.

14. “EM 1110-2-140S Flood-Hydrograph Analyses and Computations,” Corps of Engineers, U. S. Army, August, 1959.

15. “Report 33, Symposium on Consideration of Some Aspects of Storms and Floods in Water Planning,” Texas Water Development Board, 1966.

16. Linsley, R. K., and Franzini, J. B., Water Resources Engineering, McGraw-Hill, 1964.

17. Letter from Fort Worth Corps of Engineers Office, dated 1972.

18. “Report 48, Dams and Reservoirs in Texas, Historical and Descriptive Information,” Texas Water Development Board, 1967.

19. Henderson, R. M. Open Channel Flow, MacMillan, 1966.

20. Chow, Ven Te, Open-Channel Hydraulics, McGraw-Hill, 1969.

21. 21.“Report 115, Time of Travel of Translatory Waves on the Brazos, Leon and Little Rivers, Texas,” Texas Water Development Board, 1970.

22. “EM 1110-2-1408, Routing of Floods through River Channels,” Corps of Engineers, U. S. Army, 1960.

23. Thom, H. C. S., New Distributions of Extreme Winds of the United States, published in the Journal of Structural Division, Proceedings of the ASCE, July 1968.

24. U. S. Department of Commerce, Weather Bureau: Technical Paper No. 40, “Rainfall Frequency Atlas of the United States For Durations From 30 Minutes to 24 Hours and Return Periods From 1 to 100 Years,” Washington May 1961.

25. Freese, Nichols and Endress: “Engineering Report on Squaw Creek Reservoir,” prepared for Texas Utilities Services Inc., 1972.

26. Forrest and Cotton, Inc.: “Report on Yield of De Cordova Bend Reservoir if Operated for Water Supply Purposes,” prepared for Texas Electric Service Company, February 1965.

27. “Streamflow Characteristics of the Brazos River Basin, Texas,” U. S. Geological Survey, Open File 87, 1965.

28. “The Texas Water Plan,” Texas Development Board, 1968.

29. Interview with Mr. Paul R. Gillette, Chief, Flood Plain Management Branch, Texas Water Development Board, Austin, Texas.

30. Interview with Mr. Robert L. Bluntzer, Director, Water Availability Division, Texas Water Development Board, Austin, Texas.

31. Interview with Mr. Herbert Wolf, Publications Editor, Texas Water Development Board, Austin, Texas.

32. Interview with Mrs. Beth Willard, Librarian, Water Resources Research Center Archives, University of California, Los Angeles.

33. Texas Board of Water Engineers (1960), "Monthly Reservoir Evaporation Rates for Texas, 1940 through 1957," Bulletin 6006.

34. Texas Water Development Board (1974), "Ground-Water Resources of Part of Central Texas with Emphasis on the Antlers and Travis Peak Formations".

35. Sellards, E.H., Adkins, W. S., and Plummer, F. B., (1932), "The Geology of Texas, Volume I, Stratigraphy," University of Texas Bulletin 3232, Bureau of Economic Geology.

36. Cronin, J. C., Follett, C. R., Shafer, G. H., and Bettman, P. L., (1963), "Reconnaissance Investigation of the Ground-Water Resources of the Brazos River Basin, Texas," Texas Water Commission Bulletin 6310.

37. Texas Water Development Board (1968), "The Texas Water Plan".

38. U. S. Department of Agriculture, Soil Conservation Service: "Guide to Sedimentation Investigations," South Regional Technical Service Area, Fort Worth, Texas, March 1965, Revised October 1965 and July 1968.

39. Agreement By and Between Brazos River Authority and Texas Power & Light Company and Texas Electric Service Company.

40. Ambursen Engineering Corporation: "Plan of Development by Brazos River Authority for Water Conservation, Flood Control and Power, Brazos River, Texas," October 1955.

41. Nuclear Regulatory Guide, 1.113, May, 1976.

42. Deleted.

43. Fried, J.J. and Combarnous, M.A., "Dispersion in Porous Media," Advances in Hydroscience, edited by Ven Te Chow, Academic Press, 1971.

44. Ahlstrom, S.W. et al., Methods for Estimating Transport Model Parameters for Regional Groundwater Systems, Bettelle Northwest Laboratories - 1717, 1974.

TABLE 2.4-1
PRINCIPAL DAMS AND RESERVOIRS UPSTREAM OF COX BEND

(Sheet 1 of 4)

White River Dam and White River Reservoir

Location - on White River in Crosby County

Catchment - 172 sq. Mi.

Capacity - 38,600 ac-ft.

Owner- White River Municipal Water District

Use - municipal, industrial, mining

[redacted]

Sweetwater Dam and Lake Sweetwater

Location - on Bitter and Cottonwood Creeks in Noland County

Catchment - 104 sq. miles

Capacity – 11,900 ac-ft.

Owner - City of Sweetwater

Use - municipal and industrial

[redacted]

Fort Phantom Hill Dam and Fort Phantom Hill Reservoir

Location - on Elm Creek in Taylor County

Catchment - 478 sq. mi.

Capacity - 74,310 ac-ft.

Owner - City of Abilene

Use - municipal and recreational

[redacted]

TABLE 2.4-1
PRINCIPAL DAMS AND RESERVOIRS UPSTREAM OF COX BEND

(Sheet 2 of 4)

Stamford Dam and Lake Stamford

Location - on Paint Creek in Haskell County

Catchment - 360 sq. mi.

Capacity - 53,070 ac-ft.

Owner - City of Stamford

Use - municipal and industrial

[redacted]

Hubbard Creek Dam and Hubbard Creek Reservoir

Location - on Hubbard Creek in Stephens County

Catchment - 1,107 sq. mi.

Capacity - 317,800 ac-ft.

Owner - West Central Texas Municipal Water District

Use - municipal, mining and industrial

[redacted]

Gonzales Creek Dam and Lake Daniel

Location - on Gonzales Creek in Stephens County

Catchment - 115 sq. mi.

Capacity - 10,000 ac-ft.

Owner - City of Breckenridge

Use - municipal and industrial

[redacted]

TABLE 2.4-1
PRINCIPAL DAMS AND RESERVOIRS UPSTREAM OF COX BEND

(Sheet 3 of 4)

Graham Dam and Lake Graham

Location - on Salt Creek in Young County

Catchment - 205 sq. mi.

Capacity - 53,680 ac-ft.

Owner - City of Graham

Use - municipal and industrial

[redacted]

Morris Sheppard Dam and Possum Kingdom Reservoir

Location - on Brazos River at River Mile 687 in Palo Pinto County

Catchment - 13,310 sq. mi. contributing area.

Capacity - 600,000 ac-ft. (original capacity was 724,700 ac-ft., but sedimentation had reduced capacity to 600,000 ac-ft. by 1966.)

Owner - Brazos River Authority

Use - municipal, mining, irrigation, power and recreation

[redacted]

Palo Pinto Creek Dam and Lake Palo Pinto

Location - on Palo Pinto Creek in Palo Pinto County

Catchment - 471 sq. mi.

Capacity - 44,100 ac-ft.

Owner - Palo Pinto County Municipal District No. 1

Use - municipal and industrial

TABLE 2.4-1
PRINCIPAL DAMS AND RESERVOIRS UPSTREAM OF COX BEND

(Sheet 4 of 4)

[redacted]

DE Cordova Bend Dam and Lake Granbury

Location - on Brazos River at River Mile 542.5 in Hood County

Catchment - 15,451 sq. mi. contributing area.

Capacity - 155,000 ac-ft.

Owner - Brazos River Authority

Use - municipal, industrial, irrigational and recreational.

[redacted]

Reference: "Report 48, Dams and Reservoirs in Texas, Historical and Descriptive Information," Texas Water Development Board, 1967.

TABLE 2.4-2
PRINCIPAL DAMS AND RESERVOIRS DOWNSTREAM OF COX BEND

Whitney Dam and Whitney Reservoir (Lake Whitney)

Location - on Brazos River at River Mile 442.4 in Hill and Bosque Counties

Catchment - 16,930 sq. mi. contributing area

Capacity - 435,000 ac-ft.

Owner - U.S. Army Corps of Engineers

Use - Power and Flood Control

[redacted]

Note: Although there are many reservoirs downstream of Cox Bend on tributaries of the Brazos River, there are no other structures on the Brazos River.

Reference: "Report 48, Dams and Reservoirs in Texas, Historical and Descriptive Information," Texas Water Development Board, 1967.

TABLE 2.4-3
POSSIBLE FUTURE RESERVOIRS ON BRAZOS RIVER BETWEEN POSSUM KINGDOM RESERVOIR AND LAKE WHITNEY

Name	Location	Water Elevation When Filled (feet)	Storage Volume (acre-feet)
Turkey Creek	Downstream from Possum Kingdom Reservoir	884	159,000
Inspiration Point	Between Turkey Creek and Hightower	820	203,000
Hightower	Upstream of Lake Granbury	752	520,000
Bee Mountain	Between Lake Granbury and Lake Whitney	609	306,000

TABLE 2.4-4
SURFACE WATER CLAIMS FILED (IRRIGATIONAL) - SQUAW CREEK[a]

Applicant	Location of Applicant	Amount Requested (Ac-ft/yr)
Dean Williams	Within Squaw Creek Reservoir in Somervell County near Somervell-Hood County line	30
Carlisle Cravens	Within Squaw Creek Reservoir in Hood County near Somervell-Hood County Line	150

a) This list was derived from the “Water Rights Master File” of the Texas Water Rights Commission. There are no permits for non-irrigational water users on Squaw Creek.

TABLE 2.4-5
MUNICIPAL, INDUSTRIAL, MINING AND HYDROELECTRIC WATER USERS
ON THE BRAZOS RIVER[a]

(Sheet 1 of 2)

User	County	Amount Allocated (ac-ft/yr)	User[b]
Brazos River Authority	Palo Pinto	1,500,000	H
Brazos River Authority	Hood	10,000	M
Brazos River Authority	Hood	70,000	I
General Portland Inc.	Johnson	250	Mi
Harles Williams	Johnson	75	Mi
Texas Power & Light Co.	McLennan	12,000	I
Young Brothers, Inc.	McLennan	20	I
City of Waco	McLennan	5,510	M
City of Waco	McLennan	3,537	R
City of Marlin	Falls	3,500	M
City of Marlin	Falls	2,000	I
Texas A&M University	Burleson	500	I
Richmond Rice Assoc.	Ft. Bend	12,000	I
Brazos River Authority	Ft. Bend	50,000	I
Brazos River Authority	Ft. Bend	25,000	M
General Crude Oil, and Dow Chemical	Ft. Bend	20,000	I
Turner Lumber Co.	Brazoria	265	R
Dow Chemical Co.	Brazoria	260,000	I

TABLE 2.4-5
MUNICIPAL, INDUSTRIAL, MINING AND HYDROELECTRIC WATER USERS
ON THE BRAZOS RIVER[a]

(Sheet 2 of 2)

User	County	Amount Allocated (ac-ft/yr)	User[b]
Troy Dacos	Brazoria	160	R
Robert P. York	Brazoria	1,500	R
City of Freeport	Brazoria	31,360	M

a) This list includes all applications, claims and certified filings which appeared on the “Water Right Master File” of the Texas Water Rights Commission in 1976.

b) Explanation of symbols:

H - Hydroelectric, Mi - Mining, I - Industrial, R - Recreational, M - Municipal.

TABLE 2.4-6
IRRIGATIONAL WATER USAGE FROM BRAZOS RIVER BY COUNTY[a]

County	Amount Allocated (ac-ft/yr)
Palo Pinto	4,400
Parker	3,000
Hood: Brazos River Authority	20,000
Other	2,300
Somervell	500
Bosque	6,900
Johnson	30
Hill	1,400
McLennan	9,100
Falls	6,200
Robertson	13,600
Milan	1,500
Burleson	6,200
Brazos	7,200
Waller	1,300
Grimes	6,000
Washington	50
Austin	90
Ft. Bend	258,000
Brazoria	2,600

a) This list includes applications, claims and certified filings derived from the “Water Right Master File” of the Texas Water Rights Commission during 1976.

TABLE 2.4-7
ANNUAL FLOODS AT BRAZOS RIVER GAGING STATION 8-0910
(NEAR GLEN ROSE)

(Sheet 1 of 3)

Location. 4.4 miles upstream from Paluxy River confluence. 6 miles northeast of Glen Rose at Brazos River Mile 511.

Catchment area. 24,830 sq. mi., approximately; of which about 15,590 sq. Mi. contributes directly to surface runoff.

Gage. Nonrecording at site 2.5 miles downstream prior to May 8, 1931; recording thereafter. At site 2.4 miles downstream May 9, 1931, to Sept. 30, 1957. Present gage used as supplementary flood gage for 1957. Datum of gage is 567.82 ft. above mean sea level, datum of 1929.

Stage-discharge relation. Defined by current-meter measurements.

Bankfull Stage. 20 ft.

Remarks. Stages affected by backwater from Paluxy River at times and, since 1945, by medium-water bridge 2.5 miles downstream. Peaks partly retarded by Possum Kingdom Reservoir since March, 1941.

Average Discharge. 1,555 cfs

Water Year	Date	Gage Height (Feet)	Discharge (cfs)
1924	Oct. 17, 1923	13.0	37,500
1925	May 8, 1925	15.1	45,700
1926	June 21, 1926	13.2	38,300
1927	Oct. 19, 1926	14.0	41,400
1928	May 20, 1928	10.4	27,700
1928	Sept. 12, 1929	13.42	38,400
1930	June 17, 1930	19.6	68,300
1931	Oct. 7, 1930	12.18	31,700
1932	Sept. 10, 1932	16.37	49,300
1933	May 27, 1933	13.19	36,600
1934	Mar. 4, 1934	4.11	5,240
1935	May 18, 1935	23.68	97,600
1936	Sept. 27, 1936	19.42	67,300
1937	June 9, 1937	9.93	22,200
1938	Mar. 29, 1938	15.12	45,200
1939	June 23, 1939	9.85	22,600

TABLE 2.4-7
ANNUAL FLOODS AT BRAZOS RIVER GAGING STATION 8-0910
(NEAR GLEN ROSE)

(Sheet 2 of 3)

Water Year	Date	Gage Height (Feet)	Discharge (cfs)
1940	Aug. 19, 1940	13.62	38,300
1941	Nov. 25, 1940	14.90	44,200
1942	Apr. 26, 1942	19.23	66,400
1943	Oct. 18, 1942	17.47	54,100
1944	May 2, 1944	10.21	24,100
1945	Mar. 30, 1945	13.85	39,200
1946	Sept. 27, 1946	8.24	11,500
1947	Dec. 12, 1946	16.89	38,900
1948	Feb. 25, 1948	8.68	12,500
1949	May 17, 1949	26.7	74,000
1950	July 28, 1950	11.92	20,700
1951	June 18, 1951	5.05	5,680
1952	May 24, 1952	14.19	27,900
1953	May 17, 1953	5.21	5,920
1954	May 15, 1954	17.34	25,600
1955	Sept. 30, 1955	19.74	42,300
1956	Oct. 9, 1955	15.78	30,600
1957	May 27, 1957	33.87[a]	87,400
1958	May 2, 1958	21.00	36,100
1959	July 8, 1959	11.50	8,900
1960	Oct. 5, 1959	28.10	65,500
1961	June 19, 1961	16.80	21,700
1962	July 29, 1962	25.32	50,500
1963	Apr. 30, 1963	13.37	13,100
1964	Sept. 22, 1964	11.01	8,110
1965	May 20, 1965	17.43	23,500
1966	May 3, 1966	25.90	49,800
1967	July 22, 1967	14.19	15,000

TABLE 2.4-7
ANNUAL FLOODS AT BRAZOS RIVER GAGING STATION 8-0910
(NEAR GLEN ROSE)

(Sheet 3 of 3)

Water Year	Date	Gage Height (Feet)	Discharge (cfs)
1968	Mar. 21, 1968	19.01	28,400
1969	May 9, 1969	21.20	35,700
1970	Dec. 30, 1970	16.65	21,300
1971	Sept. 2, 1971	12.66	11,400
1972	Oct. 20, 1971	13.05	12,200
1973	Apr. 23, 1973	13.61	13,600
1974	Oct. 15, 1973	11.94	9,190
1975	Nov. 1, 1974	25.42	46,800

Reference: “Bulletin 6311, Floods in Texas, Magnitude and Frequency of Peak Floods”, Texas Water Commission in 1963.

“Geological Survey Water Supply Paper 1682, Magnitude and Frequency of Floods in the United States”, Part 8, U.S. Geological Survey.

“Geological Survey Water Supply Paper 1922, Surface Water Supply of the United States 1961-65”, Part 8, Volume 1, U.S. Geological Survey.

“Water Resources Data for Texas”, Part 1, Surface Water Records, U.S. Geological Survey, annual publications, water years 1966-70.

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a) Peak flood level of record (Elevation 601.69 feet). Peak flood occurred in 1876; no data are available for this flood.

TABLE 2.4-8
ANNUAL FLOODS AT GAGING STATION 8-0917 ON PANTER BRANCH OF SQUAW CREEK

Location[a] -- 4.6 miles southeast of Tolar at culvert on State Highway 51.

Catchment area. -- 7.82 sq. mi.

Type of gage. --Crest --Stage

Water Year	Date	Max. Discharge (cfs)
1966	April 29, 1966	880
1967	May 20, 1967	1,650
1968	May 9, 1968	3,800
1969	May 7, 1969	610

Reference: "Flood Stages and Discharges for Small Streams in Texas" by E. E. Schroeder, a U.S. Geological Survey Texas District Open-File Report.

a) Location shown on Figure 2.4-1.

TABLE 2.4-9
ANNUAL FLOODS AT GAGING STATION 08091750 ON SQUAW CREEK NEAR GLEN ROSE

Location - Lat. 32°16'12", long. 97°43'56", Somervell County, on left bank at downstream side of bridge on State Highway 144, 2.1 miles upstream from mouth, and 2.8 miles northeast of Glen Rose

Catchment Area - 62.5 $mi.^2$

Type of Gage - Water-stape recorder. Datum of gage is 569.02 ft. above sea level.

Water Year	Date	Max. Discharge (cfs)
1974	October 12, 1973	730
1975	April 8, 1975	9,030

Reference: U.S. Geological Survey.

TABLE 2.4-10
ANNUAL FLOODS AT GAGING STATION 8-0915 ON PALUXY RIVER NEAR GLEN ROSE

(Sheet 1 of 2)

Location. -- On left bank of Paluxy River at downstream side of pier of bridge on U.S. Highway 67, 1 mile upstream from Cross Branch, 1.2 miles southwest of Glen Rose, Somervell County, and 4.7 miles upstream from the mouth.

Catchment area. -- 410 sq. mi.

Gage. -- Nonrecording at site 1.8 miles downstream at datum 13.62 ft. lower prior to May 14, 1947; recording thereafter. Datum of gage is 609.66 ft. above mean sea level, datum of 1929, Fort Worth supplementary adjustment of 1942.

Stage-discharge relation. -- Defined by current-meter measurements below 150 cfs 1923-25; defined by current-meter measurements below 32,000 cfs and extended to 59,000 cfs by logarithmic plotting subsequent to May, 1947.

Historical Data. -- Flood of April 17, 1908, was the greatest since at least 1877, from information by local resident.

Average Discharge. -- 70.8 cfs

Water Year	Date	Gage Height (Feet)	Discharge (cfs)
1908	Apr. 17, 1908	27.2 Est.[(a)]	59,000 est.
1919	Nov. , 1918	26	53,000
1922	May 21, 1922	26	53,000
1947	Sept. 15, 1947	7.16	2,580
1948	Feb. 25, 1948	13.92	11,000
1949	May 17, 1949	25.1	48,500
1950	Oct. 24, 1949	9.18	4,570
1951	June 3, 1951	8.80	4,130
1952	May 23, 1952	22.3	36,200
1953	May 15, 1953	8.64	3,930
1954	April 12, 1954	10.0	5,510
1955	May 19, 1955	22.5	37,000
1956	May 1, 1956	16.6	17,300
1957	Apr. 26, 1957	24.12	44,000
1958	July 6, 1958	9.5	4,900
1959	Apr. 19, 1959	8.22	3,530

TABLE 2.4-10
ANNUAL FLOODS AT GAGING STATION 8-0915 ON PALUXY RIVER NEAR GLEN ROSE

(Sheet 2 of 2)

Water Year	Date	Gage Height (Feet)	Discharge (cfs)
1960	Oct. 4, 1959	25.4	50,000
1961	July 17, 1961	8.63	4,100
1962	Oct. 9, 1961	17.48	19,800
1963	Oct. 8, 1962	18.23	21,900
1964	Apr. 21, 1964	13.13	9,960
1965	May 10, 1965	12.79	9,480
1966	Apr. 30, 1966	12.38	8,840
1967	July 19, 1967	11.39	7,240
1968	May 20, 1968	15.92	15,500
1969	Apr. 17, 1969	14.05	12,700
1970	Oct. 12, 1970	11.97	8,150
1971	May 29, 1971	8.14	3,740
1972	Oct. 19, 1971	14.49	12,500
1973	Apr. 24, 1973	19.05	24,600
1974	Sept. 20, 1974	7.09	2,820
1975	Oct. 31, 1974	10.75	6,450

References: Same as for Table 2.4-7.

a) Maximum flood, since 1877 (based upon information from local resident), corresponding to Elevation 637 feet

TABLE 2.4-11
SQUAW CREEK RESERVOIR PROBABLE MAXIMUM FLOOD

(Sheet 1 of 2)

Hours	Probable Maximum Storm Rainfall (Inches)	Losses (Inches)	Rainfall Excess (Inches)	Probable Maximum Flood Hydrograph		Rainfall on Reservoir (ac-ft)	Cumulative Inflow (ac-ft)
				(cfs)	(ac-ft)		
3	.5	.5	.0	0	0	158	158
6	.5	.3	.2	951	118	158	434
9	.6	.3	.3	2,358	410	189	1,033
12	.7	.3	.4	3,774	760	221	2,014
15	.8	.3	.5	5,140	1,105	253	3,372
18	.8	.3	.5	6,002	1,382	253	5,007
21	.9	.3	.6	6,864	1,595	284	6,886
24	.9	.3	.6	7,473	1,778	284	8,948
27	2.3	.3	2.0	14,424	2,585	726	12,259
30	5.5	.3	5.2	34,370	5,665	1,737	19,661
33	20.0	.3	19.7	121,907	14,958	6,317	40,936
36	3.5	.3	3.2	142,576	33,962	1,105	76,003
39	.6	.3	.3	66,914	25,258	189	101,450

TABLE 2.4-11
SQUAW CREEK RESERVOIR PROBABLE MAXIMUM FLOOD

(Sheet 2 of 2)

Hours	Probable Maximum Storm Rainfall (Inches)	Losses (Inches)	Rainfall Excess (Inches)	Probable Maximum Flood Hydrograph (cfs)	Probable Maximum Flood Hydrograph (ac-ft)	Rainfall on Reservoir (ac-ft)	Cumulative Inflow (ac-ft)
42	.5	.3	.2	21,545	10,402	158	112,010
45	.5	.3	.2	7,796	3,281	158	115,449
48	.5	.3	.2	4,079	1,389	158	116,996
51				2,118	768		117,764
54				907	375		118,139
57				329	153		118,292
60				115	55		118,347
63				40	19		118,366
66				15	7		118,373
69				4	2		118,375
72				2	1		118,376
Totals:	39.1	5.0	34.1				

TABLE 2.4-12
COMPARISON OF PROBABLE MAXIMUM PRECIPITATION MAXIMUM RECORDED RAINFALL AND MAXIMUM RAINFALL LIKELY TO OCCUR AT SELECTED GAGING STATIONS NEAR CPNPP

Station	Period of Record Analyzed	PMP[a] (Inches)	Maximum Recorded in 24 Hours (Inches)	Maximum Likely to Occur in 24 Hours[b] (Inches)
Iredell	1964-1971	39.5	4.45	22.4
Kopperl	1952-1971	39.5	4.90	20.2
Rainbow	1936-1971	39.5	6.00	23.6
Stephenville	1952-1971	39.5	7.75	29.6

a) 24-hour point PMP as obtained from HMR #33 (Reference 9) is approximately 39.5 inches for all gaging stations. 24-hour point rainfall was converted to 48-hour 64-square mile rainfall (38.7 inches) for PMF determination.

b) 24-hour maximum rainfall likely to occur as computed through use of a statistical envelope technique developed by Hershfield (Reference 11).

TABLE 2.4-13
UNIT HYDROGRAPH CHARACTERISTICS

		Lower SCR Catchment	Upper SCR Catchment (Except Maximum Six Hours)	Upper SCR Catchment (During Maximum Six Hours)
1.	A (Sq. Mi.)	20.3	38.0	38.0
2.	L (Mi.)	4.4[(a)]	13.1	13.1
3.	Lca (Mi.)	2.7[(a)]	6.0	6.0
4.	$(LLca)^{.3}$	2.1[(a)]	3.7	3.7
5.	C_t	.6	11.1	1.1
6.	$t_p=C_t(LLca)^{.3}$ (hours)	1.26	4.07	4.07
7.	$t_a=t_p/5.5$ (hours)	.23	.74	.74
8.	t_r (hours)	3.00	3.00	3.00
9.	$t_{pr}=t_p+.25(t_r-t_a)$ (hours)	1.95	4.64	4.64
10.	$t_{pr}+.5t_r$ (hours)	3.45	6.14	6.14
11.	C_p640	420	440	440 x 1.3[(b)]
12.	$\frac{Q_{pr} = C_p640xA}{t_{pr}(cfs)}$	4,370	3,600	4,680[(b)]

a) Typical values for adjoining areas.

b) For the most intense 6-hour rainfall period, the unitgraph ordinate is increased by 30 percent of the Upper SCR Catchment.

TABLE 2.4-14
COINCIDENT WIND WAVE ACTIVITIES

WIND & WAVE CHARACTERISTICS	PMF (elev. 789.7) and 40 mph Overland Wind			10 Year Return Period Flood (elev. 778.1) and PMW (81 mph Overland Wind)		
	PLANT SITE	SQUAW CREEK DAM	SAFE SHUTDOWN IMPOUNDMENT	PLANT SITE	SQUAW CREEK DAM	SAFE SHUTDOWN IMPOUNDMENT
Effective Fetch	1.25 mi	1.28 mi	0.36 mi	1.25 mi	1.28 mi	0.36 mi
Average Depth	88 ft.	88 ft.	68 ft.	78 ft.	78 ft.	58 ft.
Wind Ratio	1.155	1.18	1.05	1.155	1.158	1.05
Set-up	0.04 ft.	0.06 ft.	0.01 ft.	0.20 ft.	0.21 ft.	0.06 ft.
Significant Wave	2.25 ft.	2.5 ft.	1.1 ft.	4.7 ft.	4.8 ft.	2.3 ft.
Maximum Wave	3.76 ft.	4.17 ft.	1.84 ft.	7.85 ft.	8.01 ft.	3.84 ft.
Wave Period	2.8 sec.	3.0 sec.	1.9 sec.	3.8 sed.	3.85 sec.	2.6 sec.
Wave Length	40.14 ft.	46.1 ft.	18.48 ft.	74.71 ft.	75.9 ft.	34.6 ft.
Wave Steepness	0.094	0.09	0.1	0.105	0.106	0.11
Relative Runup	1.3 (Smooth 1:3 Slope)	0.95 (rip-rap 1:2 Slope)	0.85 (rip-rap 1:2 1/2 Slope)	1.3 (Smooth 1:3 Slope)	0.85 (rip-rap 1:2 Slope)	0.85 (rip-rap 1:2 1/2 Slope)
Runup	4.9 ft.	3.96 ft.	1.56 ft.	10.20 ft.	6.81 ft.	3.26 ft.
Runup + Setup	5.0 ft.	4.0 ft.	1.6 ft.	10.40 ft.	7.0 ft.	3.3 ft.
Elevation Reached	794.7 ft.	793.7 ft.	791.3 ft.	788.5 ft.	785.1 ft.	781.4 ft.

TABLE 2.4-15
SAFE SHUTDOWN IMPOUNDMENT PROBABLE MAXIMUM FLOOD

(Sheet 1 of 2)

Time in Hours	Incremental Rainfall (Inches)	Incremental Rainfall Excess (Inches)	PMF Hydrograph (cfs)	PMF Hydrograph (ac-ft)	Rainfall on SSI Surface (ac-ft)	Cumulative Inflow (ac-ft)	SSI Surface Elevation	SSI Spillway Flow (cfs)
3	.5	.0	0	0	1	1	775.0	0
6	.5	.2	111	14	2	17	775.3	60
9	.6	.3	200	39	2	58	775.5	130
12	.7	.4	285	60	3	121	775.7	200
15	.8	.5	354	79	3	203	776.0	270
18	.8	.5	368	89	3	295	776.5	320
21	.9	.6	431	99	3	397	777.0	350
24	.9	.6	442	108	4	509	777.5	370
25	.6	.5	504	39	4	552	777.8	380
26	.8	.7	851	55	4	611	778.1	380
27	1.2	1.1	1,311	88	4	703	778.6	830
28	2.2	2.1	2,103	140	10	853	779.3	1,900
29	2.6	2.5	3,586	231	11	1,095	780.3	3,000
30	2.9	2.8	4,683	340	11	1,446	781.5	4,200
30-1/2	2.4	2.35	5,151	203	10	1,659	782.1	4,600
31	2.6	2.55	5,883	227	11	1,897	782.9	4,900
31-1/2	5.9	5.85	7,451	275	25	2,197	784.2	6,200
32	9.3	9.25	10,079	361	40	2,598	785.5	8,400

TABLE 2.4-15
SAFE SHUTDOWN IMPOUNDMENT PROBABLE MAXIMUM FLOOD

(Sheet 2 of 2)

Time in Hours	Incremental Rainfall (Inches)	Incremental Rainfall Excess (Inches)	PMF Hydrograph (cfs)	PMF Hydrograph (ac-ft)	Rainfall on SSI Surface (ac-ft)	Cumulative Inflow (ac-ft)	SSI Surface Elevation	SSI Spillway Flow (cfs)
32-1/2	1.6	1.55	15,828	526	7	3,131	787.5	11,600
33	1.5	1.45	20,849	799	6	3,936	790.1	17,600
33-1/2	1.0	.95	17,310	798	4	4,738	790.5	18,400
34	.9	.85	13,321	633	4	5,375	789.8	15,400
34-1/2	.7	.65	10,124	483	3	5,861	789.2	10,200
35	.4	.35	7,783	370	2	6,233	789.5	7,200
35-1/2	.3	.25	6,107	285	1	6,519	789.7	5,600
36	.2	.15	4,787	225	1	6,745	789.8	4,600
37	.2	.1	2,832	314	1	7,060	789.7	3,500
38	.2	.1	1,538	178	1	7,239	789.2	2,000
39	.2	.1	838	97	1	7,337	788.5	1,500
42	.5	.2	178	101	2	7,440	786.4	600
45	.5	.2	145	40	2	7,482	784.9	300
48	.5	.2	146	36	1	7,519	783.9	100
51	.0	.0	28	22		7,541	783.3	100
54	.0	.0	3	4		7,545	782.9	100
	44.9	39.9						

TABLE 2.4-16
UNIT HYDROGRAPH CHARACTERISTICS FOR THE PANTHER BRANCH WATERSHED ABOVE THE SAFE SHUTDOWN IMPOUNDMENT

1. A = 3.47 Square Miles
2. L = 4.15 Miles
3. L_{ca} = 2.05 Miles
4. $(LL_{ca})^{.3}$ = 1.90
5. C_t = .6
6. $t_p = C_t(LL_{ca})^{.3}$ = 1.14 Hours
7. $t_r = t_p/5.5$ = .21 Hour
8. t_R = .5 Hour
9. $t_{pR} = t_p + .25(t_R - t_r)$ = 1.21 Hours
10. $t_{pR} = .5t_R$ = 1.46 Hours
11. C_p640 = 420
12. $Q_{pR} = C_p640 \times A / t_{pR}$ = 1,200 cfs
13. $q_{pR} = C_p640/t_{pR}$ = 347 cfs per Square Mile
14. W-50 = 1.4 Hours
15. W-75 = .8 Hour

TABLE 2.4-17
SQUAW CREEK RESERVOIR AREA AND CAPACITY CHARACTERISTICS

(Sheet 1 of 3)

Elev.	0	1	2	3	4	5	6	7	8	9	
650	-	-	-	-	-	0	5	11	17	22	Acres
	-	-	-	-	-	0	3	11	25	45	Ac-ft
660	28	34	38	43	51	59	69	79	91	104	Acres
	70	101	137	177	224	279	343	417	502	599	Ac-ft
670	118	132	146	158	172	188	204	216	232	250	Acres
	710	835	974	1,126	1,291	1,471	1,667	1,877	2,101	2,342	Ac-ft
680	268	292	316	344	368	396	420	448	472	500	Acres
	2,601	2,881	3,185	3,515	3,871	4,253	4,661	5,095	5,555	6,041	Ac-ft
690	524	536	548	562	573	586	598	611	624	637	Acres
	6,553	7,083	7,625	8,180	8,748	9,328	9,920	10,525	11,143	11,773	Ac-ft

TABLE 2.4-17
SQUAW CREEK RESERVOIR AREA AND CAPACITY CHARACTERISTICS

(Sheet 2 of 3)

Elev.	0	1	2	3	4	5	6	7	8	9	
700	652	670	691	712	733	755	776	797	820	842	Acres
	12,417	13,078	13,758	14,460	15,182	15,926	16,692	17,478	18,286	19,117	Ac-ft
710	866	892	924	956	988	1,018	1,050	1,081	1,114	1,147	Acres
	19,971	20,850	21,758	22,698	23,670	24,673	25,707	26,773	27,871	29,002	Ac-ft
720	1,180	1,210	1,244	1,276	1,310	1,344	1,378	1,414	1,447	1,482	Acres
	30,165	31,360	32,587	33,847	35,140	36,467	37,828	39,224	40,655	42,120	Ac-ft
730	1,516	1,554	1,592	1,630	1,967	1,705	1,743	1,781	1,819	1,857	Acres
	43,619	45,154	46,727	48,338	49,985	51,672	53,396	55,158	56,958	58,796	Ac-ft
740	1,895	1,933	1,969	2,007	2,044	2,080	2,117	2,154	2,192	2,230	Acres
	60,672	62,586	64,537	66,525	68,551	70,613	72,712	74,848	77,021	79,232	Ac-ft
750	2,268	2,308	2,348	2,390	2,431	2,473	2,516	2,558	2,600	2,642	Acres
	81,481	83,769	86,097	88,466	90,876	93,328	95,822	98,359	100,938	103,559	Ac-ft

TABLE 2.4-17
SQUAW CREEK RESERVOIR AREA AND CAPACITY CHARACTERISTICS

(Sheet 3 of 3)

Elev.	0	1	2	3	4	5	6	7	8	9	
760	2,684	2,724	2,764	2,803	2,843	2,884	2,924	2,964	3,004	3,045	Acres
	106,222	108,926	11,670	114,454	117,277	120,141	123,045	125,989	128,973	131,998	Ac-ft
770	3,084	3,122	3,162	3,195	3,234	3,272	3,308	3,329	3,354	3,380	Acres
	135,062	138,165	141,307	144,485	147,700	150,953	154,243	157,562	160,903	164,270	Ac-ft
780	3,411	3,445	3,489	3,534	3,578	3,624	3,672	3,722	3,772	3,823	Acres
	165,665	171,093	174,560	178,072	181,628	185,229	188,877	192,574	196,321	200,119	Ac-ft
790	3,874	3,925	3,976	4,030	4,079	4,130	4,082	4,235	4,286	4,339	Acres
	203,967	207,867	211,818	215,821	219,875	223,980	228,136	232,345	236,606	240,919	Ac-ft
800	4,391	-	-	-	-	-	-	-	-	-	Acres
	245,284	-	-	-	-	-	-	-	-	-	Ac-ft

Note: Based on sedimentation survey range sestablished pursuant to the requirements of the Texas Water Rights Commission.

TABLE 2.4-18
SAFE SHUTDOWN IMPOUNDMENT AREA AND CAPACITY CHARACTERISTICS

Elev.	0	1	2	3	4	5	6	7	8	9	
730	-	-	-	-	-	-	0.0	0.1	0.2	0.4	Acres
	-	-	-	-	-	-	0.0	0.0	0.2	0.5	5Ac-ft
740	0.7	1.1	1.6	2.2	2.8	3.4	4.0	4.6	5.2	5.9	Acres
	1	2	3	5	8	11	15	19	24	29	Ac-ft
750	6.6	7.4	8.2	9.1	10.0	11.0	12.1	13.2	14.3	15.5	Acres
	36	43	50	59	69	79	91	103	117	132	Ac-ft
760	16.7	17.9	19.2	20.5	21.8	23.2	24.7	26.2	27.7	29.2	Acres
	148	165	184	204	225	247	271	297	324	352	Ac-ft
770	30.8	32.5	34.3	36.1	37.9	39.8	41.7	43.6	45.6	47.6	Acres
	382	414	447	482	519	558	599	642	686	733	Ac-ft
780	49.7	51.8	53.9	56.0	58.1	60.3	62.5	64.8	67.1	69.4	Acres
	781	832	885	940	997	1,056	1,118	1,181	1,247	1,316	Ac-ft
790	71.7	-	-	-	-	-	-	-	-	-	Acres
	1,386	-	-	-	-	-	-	-	-	-	Ac-Ft.

TABLE 2.4-19
PREDICTED AREA AND CAPACITY CHARACTERISTICS OF THE SAFE SHUTDOWN IMPOUNDMENT AFTER FORTY YEARS OF SEDIMENTATION

Elev	0	1	2	3	4	5	6	7	8	9	
740					0	0.5	1.1	1.7	2.3	3.0	Acres
					0	0.3	1	2	4	6	Ac-Ft
750	3.7	4.5	5.3	6.2	7.1	8.1	9.2	10.3	11.4	12.6	Acres
	9	13.1	18.0	24	30	38	46	56	67	79	Ac-ft
760	13.8	15.0	16.3	17.6	18.9	20.3	21.8	23.3	24.8	26.3	Acres
	92	107	122	139	157	177	198	220	244	270	Ac-ft
770	27.9	30.5	33.1	35.4	37.8	39.8	41.7	43.6	45.6	47.6	Acres
	297	325	357	391	428	467	508	550	595	642	Ac-ft
780	49.7	51.8	53.9	56.0	58.1	60.3	62.5	64.8	67.1	69.4	Acres
	690	741	794	849	906	965	1027	1090	1156	1225	Ac-ft
790	71.7										Acres
	1295										Ac-ft

TABLE 2.4-20
CONCENTRATIONS OF RADIONUCLIDES IN POSTULATED ACCIDENTAL RELEASE[a]

(Sheet 1 of 3)

Isotope	Floor Drain Tank Concentration uCi/cc	Half Life (Year)
Br-83	5.4-3	2.73-4
Br-84	3.0-3	6.05-5
Br-85	3.4-4	5.45-6
I-130	2.3-3	1.41-3
I-131	2.8-1	2.20-2
I-132	1.1-1	2.63-4
I-133	4.0-1	2.37-3
I-134	5.3-2	9.99-5
I-135	2.1-1	7.55-4
Rb-86	8.9-5	5.11-2
Rb-88	2.3-1	3.39-5
Sr-89	3.6-4	1.38-1
Sr-90	1.0-5	2.92+1
Sr-91	7.0-4	1.08-3
Y-90	1.2-6	7.30-3
Y-91M	4.1-4	9.47-5
Y-91	6.5-5	1.60-1
Y-93	3.7-5	1.15-3
Zr-95	6.1-5	1.76-1
Nb-95	5.1-5	9.64-2
Mo-99	8.7-2	7.53-3

TABLE 2.4-20
CONCENTRATIONS OF RADIONUCLIDES IN POSTULATED ACCIDENTAL RELEASE[a]

(Sheet 2 of 3)

Isotope	Floor Drain Tank Concentration uCi/cc	Half Life (Year)
Tc-99M	5.2-2	6.87-4
Ru-103	4.6-5	1.08-1
Ru-106	1.0-5	1.01+0
Rh-103M	5.1-5	1.07-4
Rh-106	1.1-5	9.47-7
Te-125M	2.9-5	1.59-1
Te-127M	2.8-4	2.99-1
Te-127	9.2-4	1.07-3
Te-129M	1.4-3	9.20-2
Te-129	1.8-3	1.32-4
Te-131M	2.6-3	3.42-3
Te-131	1.2-3	4.76-5
Te-132	2.8-2	8.93-3
Cs-134	2.6-2	2.05+0
Cs-136	1.4-2	3.59-2
Cs-137	1.9-2	3.00+1
Ba-137M	1.8-2	4.85-6
Ba-140	2.2-4	3.51-2
La-140	1.6-4	4.59-3
Ce-141	7.1-5	8.90-2
Ce-143	4.2-5	3.76-3

TABLE 2.4-20
CONCENTRATIONS OF RADIONUCLIDES IN POSTULATED ACCIDENTAL RELEASE[a]

(Sheet 3 of 3)

Isotope	Floor Drain Tank Concentration uCi/cc	Half Life (Year)
Ce-144	3.3-5	7.79-1
Pr-143	5.1-5	3.72-2
Pr-144	3.8-5	3.29-5
H-3	1.0+0	1.23+1
Cr-51	1.9-3	7.58-2
Mn-54	3.1-4	8.55-1
Fe-55	1.6-3	2.60+0
Fe-59	1.0-3	1.23-1
Co-58	1.6-2	1.95-1
Co-60	2.0-3	5.27+0
Np-239	1.2-3	6.45-3

a) Note: 5.4-3 = 5.4×10^{-3}

TABLE 2.4-20A
CONCENTRATIONS OF RADIONUCLIDES IN POSTULATED ACCIDENTAL RELEASE[a]

(Sheet 1 of 2)

Radionuclide	Release (Curies)	Half Life (Years)
H-3	3.18 E+2	1.23 E+1
Br-84	3.91 E+0	6.10 E-5
Rb-88	3.36 E+2	3.40 E-5
RB-89	9.99 E+0	3.00 E-5
Sr-89	3.00 E-1	1.40 E-1
Sr-90	1.54 E-2	2.60 E+1
Sr-91	1.73 E-1	1.10 E-3
Sr-92	6.72 E-2	2.97 E-4
Y-90	1.82 E-2	7.33 E-3
Y-91	5.54 E-1	1.61 E-1
Y-92	6.54 E-2	4.11 E-4
Zr-95	6.36 E-2	1.78 E-1
Nb-95	6.27 E-2	9.59 E-2
Mo-99	4.81 E+2	7.64 E-3
I-131	2.27 E+2	2.20 E-2
I-132	8.18 E+1	2.62 E-4
I-133	3.63 E+2	2.40 E-3
I-134	5.45 E+1	9.99 E-5
I-135	2.00 E+2	7.64 E-4
Te-132	2.36 E+1	8.90 E-3
Te-134	2.63 E+0	8.00 E-5
Cs-134	2.73 E+1	2.05 E+0

TABLE 2.4-20A
CONCENTRATIONS OF RADIONUCLIDES IN POSTULATED ACCIDENTAL RELEASE[a]

(Sheet 2 of 2)

Radionuclide	Release (Curies)	Half Life (Years)
Cs-136	1.36 E+1	3.56 E-2
Cs-137	1.36 E+2	3.01 E+1
Cs-138	8.90 E+1	6.10 E-5
Ba-140	3.91 E-1	3.51 E-2
La-140	1.36 E-1	4.60 E-3
Ce-144	3.09 E-2	7.78 E-1
Pr-144	3.09 E-2	3.30 E-5
Cr-51	8.64 E-2	7.62 E-2
Mn-54	7.18 E-2	8.30 E-1
Mn-56	2.73 E+0	3.00 E-4
Co-58	2.36 E+0	1.95 E-1
Co-60	6.99 E-2	5.26 E+0
Fe-59	1.00 E-2	1.24 E-1

a) E-5 = 10^{-5}

TABLE 2.4-21
MAXIMUM CONCENTRATIONS IN SURFACE WATER DUE TO POSTULATED RELEASES FROM THE LIQUID RADIOACTIVE WASTE STORAGE TANK

(Sheet 1 of 3)

Concentration in Squaw Creek Reservoir (uCi/cc) Based on Time After Tank Rupture

Isotope	Instantaneous	One Day	One Month	One Year
Br-83	2.94-09[a]	2.78-12	0.00+00	0.00+00
Br-84	1.64-09	3.92-23	0.00+00	0.00+00
Br-85	1.85-10	0.00+00	0.00+00	0.00+00
I-129	0.00+00	4.03-20	7.39-19	1.58-18
I-130	1.25-09	3.26-10	3.45-27	0.00+00
I-131	1.53-07	1.40-07	1.15-08	3.28-21
I-132	6.00-08	1.28-08	2.68-11	3.20-42
I-133	2.18-07	9.80-08	8.22-18	0.00+00
I-134	2.89-08	1.61-16	0.00+00	0.00+00
I-135	1.15-07	9.27-09	2.00-40	0.00+00
Rb-86	4.85-11	4.68-11	1.59-11	6.26-17
Rb-88	1.25-07	5.57-32	0.00+00	0.00+00
Sr-89	1.96-10	1.94-10	1.30-10	1.30-12
Sr-90	5.45-12	5.45-12	5.44-12	5.32-12
Sr-91	3.82-10	6.61-11	5.37-33	0.00+00
Y-90	6.54-13	1.75-12	5.44-12	5.33-12
Y-91M	2.24-10	4.20-11	3.42-33	0.00+00
Y-91	3.54-11	3.73-11	2.68-11	5.07-13
Y-93	2.02-11	3.87-12	6.37-33	0.00+00
Zr-95	3.33-11	3.29-11	2.41-11	6.46-13
Nb-95M	0.00+00	4.04-14	1.78-13	4.79-15

TABLE 2.4-21
MAXIMUM CONCENTRATIONS IN SURFACE WATER DUE TO POSTULATED RELEASES FROM THE LIQUID RADIOACTIVE WASTE STORAGE TANK

(Sheet 2 of 3)

Concentration in Squaw Creek Reservoir (uCi/cc) Based on Time After Tank Rupture

Isotope	Instantaneous	One Day	One Month	One Year
Nb-95	2.78-11	2.79-11	2.79-11	1.40-12
Mo-99	4.74-08	3.69-08	2.45-11	4.83-48
Tc-99M	2.84-08	3.44-08	2.36-11	4.66-48
Ru-103	2.51-11	2.46-11	1.48-11	4.03-14
Ru-106	5.45-12	5.44-12	5.15-12	2.74-12
Rh-103M	2.78-11	2.22-11	1.33-11	3.64-14
Rh-106	6.00-12	5.44-12	5.15-12	2.74-12
Te-125M	1.58-11	1.56-11	1.11-11	2.04-13
Te-127M	1.53-10	1.52-10	1.26-10	1.50-11
Te-127	5.02-10	2.11-10	1.27-10	1.50-11
Te-129M	7.63-10	7.48-10	4.11-10	4.07-13
Te-129	9.82-10	4.87-10	2.67-10	2.65-13
Te-131M	1.42-09	8.14-10	8.41-17	0.00+00
Te-131	6.54-10	1.82-10	1.88-17	0.00+00
Te-132	1.53-08	1.23-08	2.60-11	3.10-42
Cs-134	1.42-08	1.42-08	1.38-08	1.01-08
Cs-136	7.63-09	7.24-09	1.56-09	3.17-17
Cs-137	1.04-08	1.04-08	1.03-08	1.01-08
Ba-137M	9.82-09	9.80-09	9.78-09	9.58-09
Ba-140	1.20-10	1.14-10	2.36-11	3.10-19
La-140	8.72-11	9.72-11	2.72-11	3.57-19

TABLE 2.4-21
MAXIMUM CONCENTRATIONS IN SURFACE WATER DUE TO POSTULATED RELEASES FROM THE LIQUID RADIOACTIVE WASTE STORAGE TANK

(Sheet 3 of 3)

Concentration in Squaw Creek Reservoir (uCi/cc) Based on Time After Tank Rupture

Isotope	Instantaneous	One Day	One Month	One Year
Ce-141	3.87-11	3.79-11	2.04-11	1.60-14
Ce-143	2.29-11	1.38-11	6.11-18	0.00+00
Ce-144	1.80-11	1.80-11	1.67-11	7.39-12
Pr-143	2.78-11	2.73-11	6.57-12	2.45-19
Pr-144	2.07-11	1.80-11	1.67-11	7.39-12
H-3	5.45-07	5.45-07	5.43-07	5.16-07
Cr-51	1.04-09	1.01-09	4.89-10	1.11-13
Mn-54	1.69-10	1.69-10	1.58-10	7.52-11
Fe-55	8.72-10	8.72-10	8.54-10	6.68-10
Fe-59	5.45-10	5.37-10	3.44-10	1.99-12
Co-58	8.72-09	8.64-09	6.51-09	2.47-10
Co-60	1.09-09	1.09-09	1.08-09	9.56-10
Np-239	6.54-10	4.87-10	9.49-14	1.30-56
Pu-239	0.00+00	4.47-17	1.75-16	1.75-16
Total Effluent Concentration Fraction (10CFR20)	0.22	0.19	0.04	0.02

a) Note: 2.94-09 = 2.94×10^{-9}

TABLE 2.4-22
GROUNDWATER CHEMICAL ANALYSES

Well Number[a] Or Location	Silica (SiO_2)	Calcium (Ca)	Magnesium (Mg)	Sodium[a] (Na)	Potassium (K)	Bicarbonate (HCO_3)	Sulfate (SO_4)	Chloride (Cl)	Fluoride (F)	Nitrate (NO_3)	Dissolved Solids	Total Hardness $CaCO_3$
9	15	13.5	8.2	142	5.9	372	28.0	35.0	0.2	0.2	620	67.5
11	15	16.9	7.2	161	6.0	360	56.0	52.5	0.4	0.2	675	71.7
13	15	13.5	10.3	173	6.0	302	61.3	105	0.4	0.1	686	75.9
14	15	18.6	9.5	194	7.5	354	74.9	105	0.4	0.1	779	85.3
37	15	16.9	7.2	204	6.8	250	90.1	158	1.0	0.05	749	71.1
39	15	11.8	5.1	170	5.5	400	25.5	52.5	0.3	0.01	504	50.4
42	10	11.8	8.2	201	6.7	326	64.2	123	0.4	0.07	751	63.1
48	15	107	8.2	84	1.9	363	36.2	105	0.3	0.5	721	301
City of Glen Rose	--	21	18	107	--	390	21.0	17.0	0.3	0.4	377	129
State Well 3259501 5 miles East of Walnut Springs (Paluxy Aquifer)	13	46	29	60	--	379	17.0	22.0	0.1	6.8	386	234

a) See Table 2.4-25

Values are milligrams per liter

TABLE 2.4-23
ESTIMATED WATER USAGE AT CPNPP[a]

	Annual Usage (gallons)	Avg. Annual flow (gpm)	Max. Daily flow (gpm)
Makeup water required	55,713,600	106	305
Potable and Sanitary	11,037,600	21	25
Total water required	66,751,200	127	330

a) To be supplied by water wells or surface water pre-treatment system or both.

TABLE 2.4-24
PUBLIC SUPPLY, INDUSTRIAL AND IRRIGATION WELLS, 0-20 MILES

(Sheet 1 of 2)

Well Number[a]	Piezometric Elevation (ft) and Date		Yield (gpm)	Drawdown[b] (feet)	Use[c]	
1	701	3/15/68	100	90	PS	City of Walnut Springs
2	-	--	135	90	PS	City of Walnut Springs
3	628	8/24/66	-	110	Irr	James Smith
4	890	10/14/65	69	0	Irr	Lee Manning
5	974	3/27/69	150	0	Irr	J. W. Waldie
6	992	10/14/65	150	0	Irr	J. W. Waldie
7	-	--	550	0	Irr	Triangle Ranch
8	-	--	-	0	Irr	Triangle Ranch
9	845	10/26/65	120	0	Irr	Stanley Allen
10	834	3/27/69	46	0	Irr	Stanley Allen
11	868	3/26/69	-	30	Irr	E. L. Huffman
12	855	10/15/65	-	30	Irr	E. L. Huffman
13	615	9/13/60	-	40	Irr	Roy Kenedy
14	620	1960	614	100	PS	City of Glen Rose
15	627	9/21/60	250	100	PS	City of Glen Rose

TABLE 2.4-24
PUBLIC SUPPLY, INDUSTRIAL AND IRRIGATION WELLS, 0-20 MILES

(Sheet 2 of 2)

Well Number[a]	Piezometric Elevation (ft) and Date		Yield (gpm)	Drawdown[b] (feet)	Use[c]	
16	620	6/19/30	50	100	PS	Young Women's Christian Ass.
17	611	9/14/60	-	90	Irr	Squaw Creek Cemetery Ass.
18	508	7/20/66	222	190	Ind	Texas Lime Company
19	575	9/15/60	100	140	Ind	Texas Cedar Oil Company
20	481	6/06/68	-	210	PS	U.S. Army Corps of Engineers
21	-	--	-	0	PS	City of Tolar
22	-	--	-	0	PS	City of Tolar
23	-	--	65	0	Irr	City of Granbury (9 wells)
24	-	--	-	25	Irr	L. L. Williams
25	-	--	-	100	PS	Camp El Jesom

a) Well locations are shown on Figure 2.4-33.

b) Estimated drawdown, based on original static, piezometric level, before 1900.

c) Use: Ind, Industrial; Irr, Irrigation; PS, Public Supply.

TABLE 2.4-25
LOCAL WATER WELLS

(Sheet 1 of 3)

Well	Owner	Driller	Date Completed	Depth of well (ft)	Casing		Water-bearing Unit	Altitude of land surface (ft)	Water Level		Method of Lift	Use of Water
					Diameter (in)	Depth (ft)			Below land surface datum (ft)	Date of Measure-ment		
1	John Stufflebeme	Morris Pollack	__	__	__	__	Twin Mtn	660	__	__	E	D
2	do well 2	do	__	__	__	__	do	700	__	__	W	S
3	do well 3	do	__	__	6	__	do	825	__	__	C,E	S
4	do well 4	do	__	__	6	__	do	680	__	__	W	S
5	do well 5	do	__	__	6	__	do	760	__	__	W	S
6	do well 6	do	__	__	6	__	do	700	__	__	W	S
7	do well 7	do	__	__	6	__	do	850	__	__	W	S
8	do well 8	do	__	__	6	__	do	690	__	__	W	S
9	Mary Lou Strawn	Morris Pollack	10/68	292 (?)	__	__	do	670	__	__	S,E	D,S
10	-- Miller	__	5/72	201	__	__	do	710	76	5/31/72	S,E	D,S
11	Charles Branham Well 1	Morris Pollack	__	200 (?)	__	__	Twin Mtn	710	__	__	J,E	D,S
12	Unknown	__	__	__	__	__	Twin Mtn	690	__	__	E	S
13	A. H. Weeks	Morris Pollack	8/68	360	4	364.5	Twin Mtn	905	250	8/24/68	S,E	D,S
14	Carlisle Cravens Well 1	__	1920 (?)	138	__	__	Twin Mtn	750	__	__	S,E	D
15	do well 2	__	1950 (?)	320	__	__	Twin Mtn	810	__	__	C,W	S
16	do well 3	__	1920 (?)	320	__	__	Twin Mtn	885	__	__	W	S
17	do well 4	__	1920 (?)	320	__	__	Twin Mtn	915	__	__	W	S
18	do well 5	__	1920 (?)	150	__	__	Twin Mtn	760	__	__	W	S
19	Carlisle Cravens Well 6	__	1920 (?)	320	__	__	Twin Mtn	925	__	__	W	S
20	do well 7	__	do	150	__	__	Twin Mtn	925	__	__	W	S

TABLE 2.4-25
LOCAL WATER WELLS

(Sheet 2 of 3)

Well	Owner	Driller	Date Completed	Depth of well (ft)	Casing		Water-bearing Unit	Altitude of land surface (ft)	Water Level		Method of Lift	Use of Water
					Diameter (in)	Depth (ft)			Below land surface datum (ft)	Date of Measure-ment		
21	do well 8	__	do	320	__	__	Twin Mtn	930	__	__	W	S
22	John Stufflebeme Well 9	Morris Pollack	__	__	__	__	Twin Mtn	740	__	__	W	S
23	Cowl Martin Well 1	Morris Pollack	1963 (?)	__	__	__	Twin Mtn	850	__	__	W	D,S
24	do well 2	do	1963 (?)	__	__	__	Twin Mtn	835	__	__	C,E	S
25	Jack Bridges	do	__	300 (?)	__	__	Twin Mtn	870	__	__	C,W	D,S
26	Gilbert Daniel	Morris Pollack	11/68	363	__	__	Twin Mtn	905	__	__	S,E	D,S
27	-- O'Connor	__	__	__	__	__	Twin Mtn	880	__	__	E	D,S
28	R. L. Pendleton	__	__	__	__	__	Twin Mtn	890	__	__	C,E	D
29	Henry A. Roach	__	__	__	__	__	Twin Mtn	900	__	__	C,E	D,S
30	Arthur Hankins	Morris Pollack	1950 (?)	__	__	__	Twin Mtn	880	__	__	C,E	D,S
31	Jake Kincaid	__	__	__	__	__	Twin Mtn	900	__	__	W	D,S
32	Joseph C. Ice Well 1	Morris Pollack	__	350 (?)	__	__	Twin Mtn	875	__	__	C,E	S
33	do well 2	do	6/69	352	4	352	Twin Mtn	900	212	6/14/69	S,E	D
34	Sydney Cruce	__	1946 (?)	327	__	__	Twin Mtn	950	270	8/09/72	C,E	D
35	Carlisle Cravens Well 9	__	1920 (?)	150	__	__	Twin Mtn	780	__	__	W	S
36	do well 10	__	__	320	__	__	Twin Mtn	950	__	__	W	S
37	V. J. Moloney, Jr. Well 1	__	1930 (?)	136	__	__	Twin Mtn	760	__	__	C,E	D,S
38	do well 2	__	1930 (?)	130 (?)	__	__	Twin Mtn	760	__	__	H	N
39	William L. Riley	Morris Pollack	1968	219	__	__	Twin Mtn	710	__	__	S,E	D,S
40	Charles Branham Well 2	do	1958	200	__	__	Twin Mtn	725	__	__	S,E	D,S

TABLE 2.4-25
LOCAL WATER WELLS

(Sheet 3 of 3)

Well	Owner	Driller	Date Completed	Depth of well (ft)	Casing		Water-bearing Unit	Altitude of land surface (ft)	Water Level		Method of Lift	Use of Water
					Diameter (in)	Depth (ft)			Below land surface datum (ft)	Date of Measure-ment		
41	__	__	__	__	__	__	Twin Mtn	820	__	__	S,E	D
42	R. N. Rawls	__	__	300 (?)	__	__	Twin Mtn	890	__	__	C,E	D,S
43	Walter Potts	__	__	__	__	__	Twin Mtn	840	__	__	S,E	D
44	Vernon Wheeler	__	__	__	__	__	Twin Mtn	830	__	__	S,E	D
45	Carlisle Cravens Well 11	__	1920 (?)	150	__	__	Twin Mtn	770	__	__	W	N
46	Ray Umphress Well 1	__	1910 (?)	400 (?)	4	__	Twin Mtn	865	217	10/13/72	C,E	D,S
47	do well 2	__	1910 (?)	164	__	__	Twin Mtn	750	78	10/13/72	C,G	S
48	-- McGuire	__	__	__	__	__	Glen Rose	912	14.4	10/13/72	J,E	D
49	-- Rogers	__	__	__	__	__	Glen Rose	990	52	10/13/72	J,E	D
50	R. Thomas, Jr.	__	__	325	__	__	Twin Mtn	930	191	10/13/72	S,E	D
51	Burton Wiggins	__	__	350 (?)	__	__	Twin Mtn	880	__	__	W	D
52	__	__	__	__	__	__	Glen Rose	885	120	10/13/72	C,E	S
53	J. T. Rumfield	__	1965	385	__	__	Twin Mtn	930	302	1965	S,E	D
54	__	__	__	__	__	__	Twin Mtn	810	__	__	J,E	D

Key:

Method of lift and type of pump: C, Cylinder; E, Electric; G, Gasoline, Butane or Diesel Engine; H, Hand; J, Jet; N, None; W, Windmill.

Use of Water: D, Domestic; S, Livestock

TABLE 2.4-26
SUMMARY OF PUMPAGE DURING CONSTRUCTION IN 1975

(Gallons)

Month	PW No. 1	PW No. 2	Total
Dec. 9-18	3,191,250		3,191,250
Jan. 26-31	1,822,500		1,822,500
Feb. 1-28	2,707,500		2,707,500
Mar. 1-31	5,385,000	Mar. 31 start	5,385,000
Apr. 1-30	4,347,300	958,900	5,306,200
May 1-31	5,048,200	501,100	5,549,300
June 1-30	5,414,500	1,079,100	6,493,600
July 1-31	5,728,400	838,300	6,566,700
Aug. 1-31	5,805,800	1,980,300	7,786,100
Sept. 1-30	3,032,100	3,449,600	6,481,700
Oct. 1-31	4,157,200	3,706,500	7,863,700
Nov. 1-30	5,027,800	3,270,500	8,298,300
Dec. 1-31	4,747,800	2,040,200	6,788,000
TOTALS	56,415,350	17,824,500	74,239,850

TABLE 2.4-27
HISTORICAL GROUNDWATER LEVELS AROUND THE PLANT SITE

State Well No. 41-902 Land Surface Datum Elevation 760 ft. Depth of Well 288 ft. Completion Interval 169-288 ft.		State Well No. 43-805 Land Surface Datum Elevation 640 ft. Depth of Well 464 ft. Completion Interval 442-464 ft.		State Well No. 50-402 Land Surface Datum Elevation 980 ft. Depth of Well 372 ft. Completion Interval 330-372 ft.		State Well No. 51-501 Land Surface Datum Elevation 730 ft. Depth of Well 370 ft. Completion Interval 35-370 ft.	
Date of Measurement	Depth of Water (ft.)	Date of Measurement	Depth of Water (ft.)	Date of Measurement	Depth of Water (ft.)	Date of Measurement	Depth of Water (ft.)
09-16-1950	+4.00	08-14-1950	- 8.98	07-06-1964	-230.00	03-03-1971	- 96.30
09-13-1960	-2.42	11-22-1950	- 8.39	04-02-1970	-234.25	03-07-1972	- 69.60
05-14-1963	-3.73	02-22-1951	- 9.69	03-03-1971	-238.52	04-17-1975	-116.42
05-14-1964	-3.28	05-01-1951	- 8.70	03-08-1972	-239.81		
05-03-1965	-3.92	06-04-1951	- 8.94	03-19-1973	-239.90		
04-05-1966	-6.16	05-13-1953	-11.13	02-22-1974	-239.98		
05-10-1966	-3.97	12-09-1954	-13.73	04-17-1975	-237.69		
03-13-1967	-4.00	06-14-1955	-15.14				
04-24-1967	-3.39	12-11-1956	-19.62				
03-27-1968	-3.37	11-21-1957	-20.52				
04-02-1968	-3.41	09-14-1960	-26.32				
03-26-1969	-4.59	05-23-1962	-28.77				
04-08-1969	-3.21	05-14-1963	-28.80				
03-04-1970	-4.70	05-14-1964	-32.62				
03-04-1971	-7.56	05-03-1965	-34.98				
03-07-1972	-6.20	04-05-1966	-37.31				
03-19-1973	-8.95	05-10-1966	-36.78				
03-22-1974	-9.64	03-15-1967	-36.78				
		04-24-1967	-36.94				
		03-27-1968	-36.96				
		04-02-1968	-36.86				
		03-21-1969	-36.58				
		04-08-1969	-38.20				
		03-05-1970	-40.80				
		03-03-1971	-45.78				
		03-07-1972	-48.55				
		03-23-1973	-56.58				
		02-20-1974	-62.14				
		04-17-1975	-70.37				

TABLE 2.4-28
MAXIMUM DEPTH OF SIGNIFICANT CONCENTRATIONS OF RADIONUCLIDES IN THE GLEN ROSE LIMESTONE DUE TO POSTULATED ACCIDENTAL RELEASES

Radionuclide	H-3	SR-90	I-131	CS-134	CS-137	CO-60
Release (Curies)	3.18E2	1.54E-2	2.27E2	2.73E1	1.36E2	6.99E-2
Half-Life (Years)	1.23E1	2.60E1	2.20E-2	2.05E0	3.01E1	5.26E0
Depth, Z, to which significant concentrations penetrate (ft)	127	132	47	112	150	92
Concentration at depth Z (Ci/ft^3)	5.50E-15	5.68E-15	7.36E-15	3.72E-15	1.36E-13	1.53E-14
Time to reach this depth (years)	100	300	1.3	1.3	400	1.3

NOTE: E-13 = x 10^{-13}

TABLE 2.4-29

CONCENTRATIONS (Ci/ft^3) OF Cs-137 IN THE GLEN ROSE LIMESTONE DUE TO POSTULATED ACCIDENTAL RELEASES

Time After Rupture (Years)	DEPTH BELOW BASE OF RUPTURED TANK (FEET)						
	0.0	22.5	47.5	72.5	97.5	122.5	150.0
1	3.53E-2	2.37E-3	4.95E-6	1.75E-9	2.14E-13	1.30E-17	4.82E-22
5	2.44E-2	4.52E-3	2.74E-5	2.67E-8	8.29E-12	1.17E-15	9.92E-20
10	1.70E-2	4.21E-3	2.94E-5	3.27E-8	1.14E-11	1.77E-15	1.70E-19
50	2.02E-3	2.10E-3	3.76E-5	1.07E-7	8.90E-11	2.99E-14	5.94E-18
100	3.17E-4	6.87E-4	3.01E-5	2.19E-7	4.60E-10	3.80E-13	1.80E-16
400	8.77E-8	4.76E-7	1.83E-7	1.80E-8	6.66E-10	1.12E-11	1.36E-13
500	6.85E-9	4.06E-8	2.07E-8	3.09E-9	1.86E-10	5.39E-15	1.18E-13
1000	4.15E-14	2.98E-13	2.90E-13	1.19E-13	2.61E-14	3.33E-15	4.09E-16

2.5 GEOLOGY AND SEISMOLOGY

2.5.1 BASIC GEOLOGICAL AND SEISMIC INFORMATION

The site of the Comanche Peak Nuclear Power Plant (CPNPP) is located on the Comanche plateau, a subdivision of the Central Texas section of the Great Plains physiographic province (F2.5.1-1). Gently dipping Lower Cretaceous limestone and sandstone directly underlie the site.

Structurally, the site is located on the southern flank of the Fort Worth Basin (F2.5.1-3), a sedimentary depositional trough formed in mid-Pennsylvanian time. The trough is filled with Pennsylvanian and Permian sediments. A regional unconformity separates these Paleozoic sediments from the Lower Cretaceous sediments underlying the site (F2.5.1-5).

Two major fault systems, the Balcones and the Luling-Mexia-Talco fault zones, occur within 200 miles of the site (F2.5.1-3 and F2.5.1-4). Both of these fault systems can be observed on the surface. Subsurface faults have been identified within seven miles of the site. These faults appear to die out in sediments over 270 million years old. (See Section 2.5.1.2.4.)

Seventeen seismic events have been reported with epicenters within 200 miles of the site (F2.5.2-2). The closest large event was in Intensity VII and occurred in 1882 near Paris, Texas, 155 miles northeast of the CPNPP (F2.5.2-3). The nearest event was the Wortham-Mexia Intensity V-VI shock of 1932, occurring 90 miles southeast of the site (F2.5.2-6). The region within 200 miles of the CPNPP site has been divided, based on geologic structure, into tectonic provinces and subprovinces (F2.5.2-8). There have been no reported epicenters within the subprovince in which the site is located. An evaluation of the seismic history of the surrounding provinces as been made and a conservative maximum acceleration of 0.12 g has been selected for the Safe Shutdown Earthquake (SSE) (see Section 2.5.2.6).

Detailed investigation within five miles of the site has disclosed no evidence of surface faulting, hence eliminating the need for taking this factor into account for design purposes. The absence of surface faulting was confirmed in the detailed mapping of the excavations (see Section 2.5.4.5).

With regard to the stability of subsurface materials, there is no evidence in the site region indicating actual or potential uplift or subsidence, cavernous or karst terrain, tectonic warping or deformational zones pertinent to the site. Zones of alteration, weathering, structural weakness, unrelieved residual stresses or geologically hazardous materials are not in evidence.

Evaluations undertaken indicate that natural and man-made slopes at the Station site and on the reservoir rim will be stable.

The site investigation program consisted of the following:

1. Geological investigation and mapping of the site and surrounding area;

2. Test borings, in situ packer and infiltration tests and water level monitoring;

3. Geophysical surveys; and

4. Detailed geological mapping of all Category I excavations.

Studies pertinent to Section 2.5 have been performed by Dames & Moore, Consultants in the Environmental and Applied Earth Sciences, and Mason-Johnston & Associates, Inc., Consulting Geologists and Engineers. Information regarding these firms is provided in Section 1.4.

Studies and analyses pertinent to petroleum geology in Section 2.5.1.2.6 were supplemented by H. J. Gruy and Associates, Inc. of Dallas, Texas.

Plot Plans presented on Figures 2.5.5-5 and 2.5.5-6 illustrate the layout of facilities and the locations of borings. The location of the geophysical surveys is presented on Figure 2.5.4-14. Cross sections illustrating general features of subsurface conditions and foundation grades of Category I Station facilities are presented on Figures 2.5.4-40, 2.5.4-41, 2.5.6-10 and 2.5.6-11. The section locations are shown on Figure 2.5.5-5.

In 1989, and prior to CPNPP Unit 1 licensing, ongoing research funded by the NRC identified a possible capable fault in Southwestern Oklahoma. Field reconnisance by NRC contract seismologists identified the Criner fault in Southwest Oklahoma as possibly capable. Investigation of the Criner fault by the NRC seismologists and TU Electric seismologists allowed the NRC to conclude that the Criner fault did not pose a safety concern to CPNPP. In 1994, a review of the seismicity of the North Texas and Southwestern Oklahoma was performed. The review concluded that:

> CPNPP is located near the center of one of the world's major seismologically and geologically stable continental interiors. During the 300 year historical period no large earthquake has been recorded within 200 miles of the site. During this same period, no large earthquake which cannot be attributed directly to geologic structures known earlier through ongoing seismicity has been documented within this stable continental interior.
>
> The site is underlain by undeformed rocks of Cretaceous age more than 80 million years old. No geological faults have been found in these rocks near the site. Their sub-horizontal stratigraphy across hundreds of miles of exposure and as found in countless oil, gas, water, and investigation wells in Texas, Louisiana, Oklahoma and the adjacent states testifies to the tectonic quiescence of the region for many millions of years. In this region, excepting deformation associated with the Meers fault in Oklahoma, no evidence has been found of tectonic activity at the earth's surface younger than late Miocene age, or about 11 millions years ago.
>
> Prior to the completion of the PSAR for CPNPP in the early 1970's, only 78 earthquakes in the region had been large enough to be reported as felt events during the period of colonization. Since that time in the early 70's, because of increased sophistication of instrumental monitoring, an additional 225 mainly very small events have been recorded. All are spatially associated with earthquakes reported prior to that time. Therefore, even with the large increase in the number of recorded events afforded by sophisticated instrumental monitoring, there is no suggestion of a need to identify new earthquake sources. None of these newly recorded events are closer than 50 miles from the site. No earthquakes have been felt at the site since the beginning of site selection activities in the 1960s'.
>
> A total of 84 publications on the geology and seismology of the region, published since 1972, have been found in an intensive literature survey in support of this FSAR review. These publications have been studied carefully, and none suggest a need to consider any

more conservative evaluation of earthquake parameters than postulated in the original FSAR. In fact, there is much to suggest these original values for base ground acceleration of 0.12g horizontal and 0.08g vertical may have been overly conservative and that designs based on the minimum of 0.10g required by 10CFR 100, Appendix A, Paragraph V(2)(1)A, would be sufficiently conservative.

It is concluded that the site is acceptable from a geological and seismic standpoint for the facility designed for the Safe Shutdown Earthquake.

2.5.1.1 Regional Geology

2.5.1.1.1 Physiography and Geologic Setting

The site of the Comanche Peak Nuclear Power Plant (CPNPP) is located on the Comanche plateau, a subdivision of the Great Plains physiographic province [1]. The relationship of the site to these features and to other physiographic units in the region is shown on Figure 2.5.1-1. The Great Plains province is bordered on the east by the Coastal Plain, Ouachita and Ozark Provinces. The western and northern boundary of the Coastal Plain province coincides in general with the limit of the Upper/Lower Cretaceous sedimentary boundary. The boundary extent in common with the Great Plains (Comanche Plateau area) is reasonably sharp and marked by the contrast between the harder limestones (Lower Cretaceous) and the softer shale-chalk-sandstone (Upper Cretaceous) to the east. North of the Brazos river, the Central Lowlands-Coastal Plain boundary is less distinct and the Coastal Plain limit is arbitrarily continued at the same geologic position by placement at the base of Upper Cretaceous.

The closest point of the Coastal Plain Province is 25 miles east of the site. The Coastal Plain is comprised of semiconsolidated and unconsolidated sediments of upper Cretaceous, Teriary, and Quaternary age. The geological materials become progressively younger and are generally softer towards the Gulf of Mexico; all the sediments dip gently seaward.

Near the CPNPP, the Great Plains province of Texas is subdivided on the degree of erosion of the resistant Lower Cretaceous limestone cover and on the nature of the older rocks thereby exposed. The Comanche plateau subdivision in which the Station site is located is a submaturely dissected area which slopes eastward at a gradient conforming the dip of the lower Cretaceous rocks. The eastern boundary of the Comanche plateau, from a point near Waco southward, is formed by the Balcones escarpment.

The Palo Pinto Country adjoins the Comanche Plateau to the Northwest. Palo Pinto country, named by Hill [1], is the locality in which Pennsylvanian rocks have been exposed by stripping away of the Edwards limestone (Upper Cretaceous). It is characterized by steep sided mesas cut by canyons, the mesas being remnants of strong sandstone beds. Between the mesas, on lower-lying shales, are rolling, mesquite-covered plains.

The formations forming the Comanche Peak Plateau and the outlier remnants of the Callahan Divide to the west are principally limestones of Lower Cretaceous age. These more resistant rocks are grouped with associated sands and calcareous clay or marl units into three subdivisions, the Trinity, Fredericksburg, and Washita groups (Figure 2.5.1-2). South and west of the Brazos River, the youngest (Washita) rocks are thin and have a small extent of a real outcrop. This group is absent in the site vicinity. The Fredericksburg group of formations (Edwards and Comanche Peak Limestones and underlying Walnut Clay) are confined to the major drainage

divides; the only complete section in the site vicinity is at Comanche Peak, the prominent landmark a little more than five miles north of the plant site. The Trinity rocks are roughly equal in a real extent to those of the Fredericksburg. They crop out in the western marginal area of the plateau and in the valley areas projecting southeastward. Classic exposures are present in the valleys of the Brazos River, Paluxy river, and Squaw Creek in the site area. The plant dams and reservoirs are all within the Glen Rose bedrock outcrop with the overlying Paluxy Sand on the adjacent divides. The underlying basal Trinity sands (Twin Mountains Formation), unexposed in the Squaw Creek drainage, crop out about eight miles to the southwest of the site in the Paluxy River Valley.

The Trinity formations exhibit characteristic terrain aspects. The outcrop area of the Paluxy Sand is confined to the summit regions of the drainage divides and forms gently rolling hills of red, sandy soil which supports deciduous trees and native grasses. Areas underlain by the Glen Rose Formation are typically prairies having relatively steep, stairstepped slopes developed on limestone alternating principally with claystone, siltone and/or shale. The residual soil derived from the Glen Rose beds supports cedar and sparse grassland vegetation.

A flat, broadly undulating plain lying to the west of the site characterizes the outcrop area of the Twin Mountains formation, and its vegetation cover closely resembles that described for the Paluxy Sand.

Topographic elevations in the site region range from about 550 feet to 1000 feet above sea level.

2.5.1.1.2 Geologic History

Figure 2.5.1-3 shows the regional tectonic structures within a two hundred mile radius of the CPNPP site. The major features shown on this figure are discussed in this section. Each of the major structural features has undergone a distinct geologic history. By far, the greatest tectonic activity on a regional scale occurred prior to Mesozoic time.

Seven tectonic provinces have been delineated within a two hundred mile radius of the CPNPP site. These provincial boundaries are shown on Figure 2.5.2-2 and their use in developing the Safe Shutdown Earthquake is discussed in Section 2.5.2.6.

1. Central Texas

   The Llano Uplift (Figure 2.5.1-3) is post-Early Ordovician and pre Late Pennsylvanian in age. In late Pennsylvanian time, the Precambrian surface was 10,000 feet higher in the western part of the Llano uplift than beneath the flanking Fort Worth Basin. Erosional losses indicate that uplift accounted for one-third and basinal subsidence for two-thirds of the vertical movement in these structural adjustments [2].

   The northeast trending horsts (Figure 2.5.1-4) of the region developed at the close of the mid-Pennsylvanian contemporaneously with the Ouachita Orogeny which intensely folded the equivalent and older beds in Oklahoma and Arkansas.

   The Llano Uplift remained stable during Mesozoic and Cenozoic time. Continental elevation during the Cenozoic resulted in the removal of the relatively thin cover of Mesozoic rocks, exhuming rocks ranging from Precambrian to Pennsylvanian in age

which had been overlapped by Lower Cretaceous limestone and shale in the Llano region.

2. North of the Llano Uplift

North of the Llano Uplift are a series of arches (Bend, Lampasas, Red River, Muenster) and basins (Fort Worth or Strawn). Most of the tectonic activity on these structures took place in mid-Pennsylvanian time and just preceded the Ouachita Orogeny (see below). The Strawn (Fort Worth) Basin was initiated at the same time as neighboring uplifts and represents differential warping of the basement. The Strawn Basin continued to receive sediments through upper middle-Pennsylvanian time and was progressively deepened.

The Red River Uplift and Muenster Arch (Figure 2.5.1-3) likewise were initiated either during or at the close of the Mississippian. Their history in Mississippian time appears to coincide with that of the Llano region (as a positive element but not pronounced uplift). Both Llano and Red River uplifts and Muenster Arch were submerged by the lower Pennsylvanian sea.

3. Ouachita Folded Belt

In the middle Cambrian a trough developed in the area which became the site of continuing sedimentation from late Cambrian through early Pennsylvanian time [3]. Compressional forces from the south during later Pennsylvanian time (Ouachita Orogeny) metamorphosed these sediments and elevated them into a chain of mountains. Major crustal movements terminated near the end of the Paleozoic era. Erosional denudation of the mountain belt occurred simultaneously with its development and continued through early Mesozoic time. The eroded surface and the younger, unaltered Paleozoic strata served as the foundation for the Jurassic and younger sediments of the Gulf Coastal Plain.

4. Southern Oklahoma Uplifts

Orogeny in the Wichita mountains (Figure 2.5.1-3) began no later than in early Pennsylvanian time. Numerous pulses of folding doubtless affected the Wichita throughout middle and late Pennsylvanian time and continued into the Permian. The last strong pulse was in the early Permian [4].

The Arbuckle mountain region experienced volcanic activity in Cambrian time and sedimentation throughout Paleozoic time, culminating in a late Paleozoic orogeny [5]. During the orogeny, there was close folding of the Arbuckle anticline and Ardmore basin, overturing and thrusting of some 30,000 feet of geosynclinal sediments, and collapse of the eastern segment of the Southern Oklahoma geosyncline. Two orogenic pulses occurred. Slight additional folding persisted into early Permian time.

Major deformation of the Ouachita mountains occurred in early Pennsylvanian time, after more than 35,000 feet of Paleozoic sediments had accumulated in the gradually downwarping basin.

5. Gulf Coast Plain

Subsidence of the Coastal Plain contemporaneous with tilting along its coastline during each Tertiary epoch of deposition resulted in a thickening of these strata to form prisms of sediment. Elevation of the upland area and sinking of the Gulf floor resulted in southeastward development of offlapping, successively younger continental shelves [6].

The Sabine Uplift (Figure 2.5.1-3) on the Coastal Plain has experienced periods of growth, dormancy and rejuvenation.

In Mesozoic time, the East Texas embayment, a synclinal depression, developed over a large portion of the Coastal Plain of northeast Texas. Salt deposits underlie most of this basin [7]. Numerous salt domes have been found in this area and comprise the inner salt dome province of the Texas Coastal Plain. The embayment persisted until near the close of the Eocene. Its original northwest axis was then modified by the Sabine uplift so that the south portion of the embayment now trends north while the northern part is deflected to the northeast [8].

Several prominent fault zones exist in the Coastal Plain. Because of their particular relevance to the site, their history of movement is discussed in Section 2.5.2.3.

2.5.1.1.3 Stratigraphy

A geologic map of the region is presented on Figure 2.5.1-4. Figure 2.5.1-2 summarizes the stratigraphy for the region around the site. The cross-sections provided in Figures 2.5.1-5 and 2.5.1-6 illustrate the stratigraphic relationships in the region along east-west and north-south lines, respectively. Sedimentary rocks of the Lower Cretaceous Comanche series (maximum age 136 million years) crop out in the Station site vicinity. Because of the easterly dip of these strata, progressively younger beds extending through the Upper Cretaceous Gulf series sediments (age 65-100 million year) outcrop to the east.

The Comanche series thins westward from the Station site [9] and is underlain by westerly dipping pre-Cretaceous sediments of Permian and earlier age (225 million years and older). Exposures of these older rocks become more abundant in a westerly direction where the Comanche series has been removed by erosion.

The Gulf series consists of the Navarro, Taylor, Austin, Eagle Ford and Woodbine formations. Total Thickness of the series ranges up to about 2,800 feet. The sequence consists of marl, limestone, limey shale, chalk, shale, sand sandstone and clay, along with minor lignite and gypsum.

A regional unconformity separates the Comanche series from the overlying Gulf series. However, here are numerous facies changes along both strike and dip caused by changing depositional environments [9] [10]. The Washita group, at the top of the Comanche series, consists of the Buda Formation (limestone), the Del Rio Formation (shale) and the Georgetown Formation (limestone). The Washita group ranges up to approximately 300 feet in thickness.

The Fredericksburg group, occurring stratigraphically below the Washita, has a maximum thickness of about 600 feet in the region and consists of the Kiamichi Formation (limestone), the

Edwards Formation (limestone), the Comanche Peak Formation (limestone), and the Walnut Formation (shale).

Below the Fredericksburg group lies the Trinity group, which forms the base of the Cretaceous sequence. Three formations of the Trinity group comprise bedrock in the site region and are approximately equal in thickness. In descending order, they are the Paluxy, the Glen Rose and the Twin Mountains formations. The Paluxy Formation at the top of the group is a sand which thins southward and eastward (downdip). The Paluxy Formation has been eroded from the immediate Station and reservoir area. However, it does crop out on the periphery of the site area and there consists of fine-to very fine-grained, well-sorted, poorly-cemented friable sandstone with occasional siltstone and claystone interbeds. The Paluxy Formation is mined locally for industrial uses including glass making, foundry molds and mineral filler.

The Glen Rose Formation, which underlies the Paluxy, constitutes the principal bedrock formation in the Station site area. It pinches out to the west (updip) and to the north and is absent in northern Wise County [9]. Eastward the Glen Rose Formation thickens substantially in the subsurface and contains a massive evaporite bed. Alternating beds of resistant limestone and weak claystone form stair-step topography where the Glen Rose is exposed. The Paluxy-Glen Rose contact in this area is abrupt, distinct and conformable [10].

Below the Glen Rose unit lies the Twin Mountains Formation. This stratum forms a gradational contact with the Glen Rose formation and is composed principally of sandstone, limestone, and claystone [2]. The sandstones are water bearing and serve as a source for water supplies in the site vicinity [11]. The Twin Mountains Formation is not exposed in the immediate site area.

In the subsurface to the east, rocks of Jurassic age are present. There strata range in age from 136 to 195 million years and include the Cotton Valley, Buckner, Smackover and Louann formations. The Jurassic sequence may be underlain by the Eagle Mills Formation of Triassic age (at least 195 million year old) and/or lower Paleozoic sedimentary and metamorphic rocks.

Generally speaking, the rocks in the above section became more dense with increasing age or depth in the stratigraphic column [12] as indicated on Figure 2.5.1-7.

West of the Station site and beneath the veneer of Lower Cretaceous formations (Figure 2.5.1-5) lies a succession of Permian, Pennsylvanian, Mississippian and Cambro-Ordovician strata consisting predominantly of resistant limestone and shale with some sandstone. Basement granite likely underlies this Paleozoic section.

North of the Station site, in southern Oklahoma, the Cretaceous section is similarly composed of the Gulf and Comanchean series (Figure 2.5.1-6). However, the Gulf series here is thinner. In Oklahoma, the Comanchean is comprised of the same three groups as in Texas: the Washita, Fredericksburg and Trinity groups. The Washita is thicker and contains more sand in this region. The Paleozoic section in southern Oklahoma consists mainly of limestone, dolomite, marl and shale dating from the Cambrian through the Permian. One sandstone unit appears in the lower Cambrian. The Precambrian basement (over 570 million years in age) consists mainly of quartzite, granite, and gabbro.

2.5.1.1.4 Structure

1. General

A large number of varied regional geologic structures are present within 200 miles of the site. These are described in the following paragraphs, subsection 2.5.2.3 and shown on Figures 2.5.1-3, -4, -4a, -5 and -6. Tables 2.5.1-1 through 2.5.1-4 list the principal fold and fault structures in Texas and in the adjoining states of Oklahoma, Arkansas and Louisiana.

a. North Central Texas (Great Plains Physiographic Province)

The Llano Uplift, located approximately 100 miles southwest of the site, is structurally a large dome. Precambrian rocks dating more than 570 million years in age are exposed in the center of the dome over an area roughly 40 to 70 miles and are surrounded by formations of Paleozoic age (225 to 570 million years old) and Cretaceous rocks deposited 65 to 136 million years ago. The Precambian of the uplift consists of metamorphic rocks (schist, gneiss and marble), batholithic intrusions (granite) and late dike intrusions (felsite). Extensive faulting is associated with the Llano Uplift (Figures 2.5.1-3, -4, -4a and -6) and extends to the northeast and southwest under a covering of late Pennsylvanian and Cretaceous sediments. Evidence for extension of this faulting into the site vicinity could not be found in the field through stereoscopic study of areal photographs, however, data from recent hydrocarbon exploration in the site vicinity has indicated this faulting is present in the subsurface to the west of the site. (See Section 2.5.1.2.4.) Several major structural arches also originate in the Llano Uplift, and they, too, extend under the Cretaceous and Upper Pennsylvania formations [8]. One of these features, the San Marcos arch, plunges southeastward into the Coastal Plain and coincides in position and trend with known Precambrian folding. Other structures, namely the Lampasas Arch trending northeast, the Edwards Arch trending southwest and the Bend Arch trending north, possess an axial orientation approximately at right angles to the lines of Precambrian folding.

North of the Llano Uplift, differential warping and uplift associated with faulting and subsidence in Pennsylvanian time formed the Fort Worth Basin and associated structurally high areas. The structural highs include: the Bend Arch to the west, the Red River Uplift and its continuation, the Muenster Arch to the north, and the Lampasas Arch to the south.

The Forth Worth Basin or Syncline has a northwest-southeast axis and extends as far to the northwest as southern Clay County. Because most of the strata in the basin were deposited during the Strawn time of the Pennsylvanian Period, the basin is commonly referred to as the “Strawn Basin.” The full thickness of Pennsylvanian sediments in the Forth Worth Syncline is not known. No faults are known or suggested in the Basin. The Pennsylvanian section is comparatively thinner over the Bend Arch, Red River Uplift, and Muenster Arch structural highs.

The Bend Arch is a northward plunging anticline and is the most pronounced of the arches originating from the Llano Uplift. Because it originates from the north

side of the uplift, there is a possibility that its southern extension may be connected with the southwest trending Edwards Arch. The Bend Arch has probably resulted from two earth movements: a homoclinal tilt to the southwest in upper Pennsylvanian time, and a Post-Permian tilt of the entire region to the northwest, resulting in the northwest dip of the Pennsylvanian and Permian sediments.

The Muenster Arch, together with the Wichita Mountains Uplift comprise an uplift some 350 miles long. The trend has a Precambrian core and is flanked with truncated lower and middle Paleozoic strata. Geophysical evidence indicates a structural relationship between the Muenster Arch and the Wichita Mountain Uplift (Figure 2.5.1-8). The arch is completely blanketed by Cretaceous sediments and is shown in Figure 2.5.1-6.

The Lampasas Arch is a broad arch originating in the northeast portion of the Llano Uplift and is shown on Figure 2.5.1-6.

The Hardeman Basin is a depression at the western end of the Red River Uplift, filled with Pennsylvanian and Permian sediments.

b. Southern Oklahoma (Wichita-Arbuckle-Ouachita Mts.)

North of the Red River Uplift, in Oklahoma, are several major uplifts. From west to east, they are the Wichita Mountains, the Arbuckle Mountains and the Ouachita Mountains (shown on Figure 2.5.1-3 as the Ouachita Mountains uplift).

The topographic relief of the Wichitas averages generally less than 500 feet but ranges up to approximately 1,400 feet at Mount Scott (Altitude 2,840 feet) [4]. The Wichitas form hills, low mountains and ridges that emerge from a nearly flat plain. Forming the core and the principal part of the mountains are northwestward-trending igneous rocks (granite, anorthosite, rhyolite porphyry and gabbro) of Cambrian age. On the plunging east end of the uplift, Paleozoic rocks are now exposed in the form of ridges and isolated hills. These ridges are approximately as they existed when they were covered by early Permian sediments (shale and conglomerate) that form the extensive plains surrounding the Wichita Mountains.

The Arbuckle Mountains, 100 miles east of the Wichitas, are mostly gently rolling plains, with their major relief (607 feet) in the western portion. They are formed by a large inlier of folded and faulted Paleozoic and Precambrian rocks. It is covered on the east, north and west by gently westward-dipping Pennsylvanian and Permian strata and on the south by gently southwest-dipping early Cretaceous sediments of the Gulf Coastal Plain [5].

Immediately to the east of the Arbuckles are the Ouachita Mountains which are principally a flysch sequence quite unlike the Arbuckles in stratigraphy and structural development. There are three areas where pre-Mississippian sedimentary rocks are exposed. The Broken Bow-Benton anticlinorium is the one major structural component of the Ouachita Mountains that can be projected with

confidence into the subsurface in North Texas (Lamar and Red River counties, approximately 150 and 175 miles east-northeast of the site).

Each of the above uplifts is accompanied by extensive faulting, as shown on Figure 2.5.1-3. The major faults in Southern Oklahoma are listed in Table 2.5.1-3; those of the Arbuckles are mainly normal faults and those of the Ouachitas are mainly thrust faults. These faults all pass beneath the Cretaceous overlap of sediments without apparent evidence of offset.

c. Eastern Texas (Gulf Coastal Plain)

The principal features of the Gulf Coastal Plain are included in Figure 2.5.1-3. The Ouachita folded belt (synonymous with the Ouachita structural front and the Ouachita tectonic belt) is the buried southernly extension of the Ouachita Mountain uplift [14]. The fold belt has a minimum length of 1,300 miles and a width of 90 miles and lies under the inner margin of the Coastal Plain, ranking with the Appalachian belt in order of magnitude and in significance as a major continental structural feature [3]. Their subsurface southwestward extension from the Ouachita Mountains south passes along the east side of the Fort Worth basin, curving around the Llano uplift and trending westward as far as the Marathon uplift and thence southward, probably into Mexico. Deformation, including thrusting, is considerable in the belt. Sediments of the Ouachita facies are metamorphosed to varying degrees.

The structure of the Gulf Coastal Plain itself assumes a broad homocline dipping gently gulfward. Regional structural features that alter the general structure and stratigraphy of the plain are the Sabine Uplift, the East Texas Embayment, the San Marcos Arch and the Rio Grande Embayment.

A complex system of normal faults follows the margin of the Coastal Plain coastward of the Ouachita structural front. Three individual systems of the faults are recognized: the Balcones, the Luling-Mexia-Talco and the Charlotte-Fashing fault zones. The surface trace of the Balcones fault probably overlies the Ouachita folded belt along its considerable length [8]. The zone probably extends as far north as Kaufman County (90 miles east-northeast of the site) and as far south as Uvalde County (240 miles southwest of site). Displacement along the fault decreases markedly both north and south of Bexar County (200 miles south-southwest of the site).

The Luling-Mexia-Talco fault zone lies coastward of the Balcones zone and is comprised of individual faults largely antithetic to the southeastward throws of the Balcones.

Within the Luling-Mexia-Talco fault system, grabens of varying widths are common. An en echelon pattern to the individual faults within the system is normal [15].

The Charlotte-Fashing fault system is coastward of the Luling Fault system and may be considered an echelon to it. Major grabens comprise part of the system.

The location of the fault system is approximately 200 miles from the site and is shown on Figure 2.5.1-3.

The Sabine Uplift is a large, relatively flat-topped dome with gently sloping flanks in eastern Texas and northwestern Louisiana. It is reflected in the surficial Tertiary strata. Faulting in the Mt. Enterprise and Rodessa fault systems, Figure 2.5.2-2, has somewhat modified the structure.

Flanking the west side of the Sabine Uplift is the East Texas Embayment. The Embayment is situated in the bend of the Ouachita fold belt as its trend changes from west to south in northeast Texas and southeast Oklahoma. Numerous piercement salt domes and some faults appreciably modify the general simplicity of the basin [16].

The final Coastal Plain structure to be considered is the San Marcos Arch. It is a broad, gentle, structural nose extending southeast from the Llano Uplift. It separates the East Texas and Rio Grande embayments and plunges southeast along its trend.

2. Geophysical Surveys

Two types of geophysical investigation have been utilized to examine regional geologic structure and assist the identification of tectonic provinces. These methods include gravity and natural gamma aeroradioactivity surveys.

a. Gravity Survey

The regional Bouguer gravity map (Figure 2.5.1-8) depicts large gravity anomlies, providing an indication of the rock densities associated with the known regional tectonic features and some insight into less well known basement features. The Bouguer reduction commonly yields negative values for areas which are located in continental interiors. Positive Bouguer anomaly values (observed Bouguer gravity minum regional Bouguer gravity) generally indicate emplacement of high-density rocks near the surface [68] such as Llano Uplift.

Major lithologic-structural divisions of the basement agree well with the regional Bouguer gravity trends and regional gravity configuration is controlled largely by basement phenomena. In Texas, the general westward decrease in Bouguer values is caused by the isostatic effect of the High Plains.

The site, as shown on the regional Bouguer gravity map is located within the Fort Worth (Strawn) Basin and west of the Ouachita tectonic belt [19]. The features are shown on Figure 2.5.2-2. Basic data for construction of the map came from field measurement of the total gravitational force (commonly called absolute gravity), using the pendulum, and from conventional gravity-meter surveys which have been correlated with the absolute gravity determinations.

The updip limit of the Ouachita facies coincides with a series of gravitational maxima trending in the same general direction and also with both the

Luling-Mexia-Talco fault system and the Choctaw fault system farther north (Figure 2.5.1-8, 19).

Conspicuous positive Bouguer gravity anomalies are apparent from a series of maxima which coincide with the Muenster and Wichita uplifts and the Amarillo Uplift (Figure 2.5.2-2). These maxima begin rather abruptly in northwest Collin County, Texas, (immediately north of Dallas) and extend northwest through Oklahoma. The maximum of +35 milligals on the Wichita trend in Kiowa and Greer counties of Oklahoma (approximately 200 miles north-northwest of the site) is contrasted to a -81 milligal minimum to the south in Wilbarger County, Texas (approximately 150 miles northwest of the site). The amount of basement uplift and the density contrast between granite and flanking sedimentary rocks is inadequate to account for the magnitude of the anomaly. Either there is a greater amount of gabbroic rock in the Precambrian of the Wichita-Amarillo trend than is apparent from examination of surface exposures and drilling records, or the answer lies in the distribution of mafic material in the subcrust.

A crustal model for Central Texas [65, 66] is shown on Figure 2.5.1-8. Calculated and observed Bouguer gravity anomalies show good correlation and agreement is also found with crustal seismic models proposed by others for Texas. Crustal thickness in the site vicinity is between 40 and 50 km. (25 and 31 miles) according to the model.

All of the major structural uplifts of the southern Mid-Continent correspond to regional isostatic maxima and the basins with gravity minima [67]. The overall isostatic anomaly pattern (see Figure 2.5.1-8, inset) is broadly coincidental with basement structure; however, some broad-wavelength anomalies, one of which is also of high amplitude, apparently result from density variations deep within the crust rather than from structural relief. The most prominent of these anomalies is a 200-mile-long minimum which is aligned east-west across Central Texas from near the New Mexico border to near the Ouachita tectonic belt [67].

The structural grain of the Marathon-Ouachita orogenic belt (stable Ouachita, see Figure 2.5.2-2) is represented by a series of disconnected gravity minima and a parallel nearly continuous, lineal gravity maximum. This elongated gravity maximum can be traced for about 500 miles as a continuous feature from the Ouachita Mountains to near the Marathon area and corresponds closely with the axis of greatest metamorphism in the Ouachita trend.

A major 90 milligal Hayford-Bowie isostatic gravity minimum is centered in the Ouachita Mountains and includes the adjacent Arkoma basin.

Isostatic and Bouguer gravity anomalies produced by the Waco Uplift are superimposed on the broader rimming gravity maximum. This structure has been shown to be faulted and anticlinal, and it probably represents a crustal block partially involved in late Paleozoic orogenic thrusting related to the Ouachita belt [67].

The site is situated on the central stable continental land mass and exhibits a Bouguer gravity value averaging -48 miligals [19] with low lateral gradients.

These gravitational trends indicate that the site is far removed from tectonic boundaries and confirm the regional tectonic provinces developed in Section 2.5.2.2.

b. Radioactivity Survey

Airborne measurement of terrestrial radioactivity serves as an aid to geologic mapping and as general background data for later environmental monitoring. Varying amounts of uranium, thorium, and potassium in the surface rock and soil generate measurable gamma radiation [17]. A regional natural gamma aeroradioactivity map on Figure 2.5.1-9 shows the gamma ray intensity measured from an aircraft at 500 feet above the ground surface. Flight-line spacings were one mile and two miles. The Upper Cretaceous units possess individual gross lithologic homogeneity which permits association of aeroradioactivity patters with particular units [18]. The Lower Cretaceous outcrop area is characterized by lithologic variation and consequent radioactive mineral variations but most stratigraphic units are too limited in a real extent to permit their differentiation by the scintillation detector. The strata are so nearly horizontal that geologic contacts approximate the topographic contours. In this terrain, few correlations are possible.

2.5.1.1.5 Stability of Subsurface Materials

Information relating to the stability of subsurface materials in the site vicinity are provided in Section 2.5.1.2.6.

2.5.1.1.6 Groundwater

A detailed description of groundwater is provided in Section 2.4.13.

2.5.1.2 Site Geology

2.5.1.2.1 Physiography

The Station site is located in the Squaw Creek drainage area immediately west of the Brazos River as shown on Figure 2.4-6. The Station area will be prepared for construction by leveling a ridge between Squaw Creek and Panther Branch, one of its tributaries. Figure 2.4-1 depicts the locality. The site for the main dam, Squaw Creek Dam, is on Squaw Creek just below the mouth of Panther Branch. A smaller impoundment dam (Safe Shutdown Impoundment) will be built on Panther Branch about 3,000 feet upstream from Squaw Creek. Location of the principal Station components are presented on Figure 1.2-1 and detailed topography also is illustrated on Figure 2.4-1.

The site is situated on the indurated and consolidated Lower Cretaceous sediments typical of the Comanche Plateau subdivision of the Great Plains physiographic province. Maximum relief in the immediate site area is approximately 220 feet, with grades ranging as low as elevation 640 feet in creek floodplains to elevation 860 feet in uplands. The floodplains of Panther Branch and Squaw Creek vary in width to approximately 1,000 and 3,000 feet, respectively. Surface materials in the floodplains are principally fine-grained clayey soils.

Valley slopes rising above the floodplain are relatively gentle to moderately steep, ranging from 15 to 30 degrees or more, and generally exhibit a somewhat stair-stepped appearance. Rock outcrops of limestone and claystone comprise approximately 40 to 60 percent of the slopes. In other areas, including flat-topped plateau remnants, bedrock frequently crops out and/or is mantled by a thin cover of residual soil which generally consists of clay silt and/or sandy clay material.

Vegetation in the site area includes cedar, scrub oak, juniper, mesquite, cactus and various types of grass. Large deciduous trees grow in the valley bottoms.

2.5.1.2.2 Geologic History

The history of geologic events in the site area subsequent to the development of an erosional surface on the Precambrian basement is one of crustal downwarping, alternate marine invasion and retreat, and related accumulation and erosion of sediments.

Throughout Paleozoic time, marine submergence alternated within periods of land uplift and emergence. During these pulsations, sedimentation in the site region varied from deep-water marine limestones and shales to sandstones and shales carried into shallower seas from emergent sourcelands. Cycles of deposition alternated with periods of erosion and partial removal of these materials. Major interruptions in the depositional history can be identified by significant unconformities at the contact zones between the Cambro- Ordovician limestone and Mississipian shales and limestones, and between the Pennsylvanian shales/sandstones and the lower Cretaceous formations immediately underlying the site.

Extensive faulting is associated with the Llano uplift and extends to the northeast (toward site) and under a covering of late Pennsylvanian and Cretaceous sediments. This faulting is associated with uplift of Llano Uplift and Lampasas Arch (Figures 2.5.1-3, -4, -4a, and 6) and downwarping to the west (Figure 2.5.1-5). The major time of fault activity occurred in mid-Pennsylvanian time just prior to the Ouachita Orogeny.

Mesozoic history began with much of North-Central Texas emergent and this condition continued through the early Mesozoic (Triassic and Early Jurassic). Following regional tilting to the south and east, the ancestral Gulf of Mexico came into being and the great submergence of Cretaceous time succeeded the initial Late Jurassic marine invasion. The Balcones Fault Zone is generally considered as a basement hingeline with increased rate of submergence and depositional thickening to the south and east of this perpherial fault zone. The Jurassic seas of the East Texas embayment extended toward this hinge zone and then lapped out against the terrain of the Ouachita Tectonic Belt. The site area is about 75 miles west of the nearest inferred Jurassic shoreline.

The early Cretaceous seas overlapped the Jurassic strata and transgressed the peneplained, eastward sloping surface developed on the generally westward-dipping Paleozoic rocks of the site area (Figure 2.5.1-5). Cretaceous history is marked by a series of depositional cycles, i.e., progressive transgressions with stillstands and/or minor regressions. Except for marginal coastal areas dominated by sand deposition, these cycles are basically couplets of clastic deposits (sand or shale) followed by carborate deposits. In the site area, the basal Cretaceous sands (Trinity or Twin Mountains) and the Glen Rose Limestone, laterally replaced by the lower part of the Paluxy Sand, represent the initial major cycle. The upper part of the Paluxy Sand and the overlying Walnut Clay plus the Comanche Peak and Edwards Limestones constitute the middle Lower

Cretaceous cycle. The last Lower Cretaceous cycle (Washita) has been removed by erosion in the site vicinity. These three cycles are shown on Figure 2.5.1-2.

Deposition and erosion continued through the Upper Cretaceous and into the Tertiary with final continental uplift occurring in late Tertiary time accompanied by renewed erosion and removal of the Cenozoic sediments as well as portions of the underlying Cretaceous rocks. This erosional cycle persists into the present.

2.5.1.2.3 Stratigraphy

Within the general 5-mile radius area of the site vicinity, the stratigraphic units exposed consist of Quaternary fluvial deposits exposed in the drainage lowlands and Lower Cretaceous strata of the Edwards, Comanche Peak, Walnut, Paluxy, and Glen Rose formations exposed on the adjacent uplands. The Site vicinity Geologic Map, Figure 2.5.1-10, illustrates the outcrop area of each formation. The geologic column, figure 2.5.1-11, illustrates the formation sequence.

The Paleozoic section is not exposed within a five-mile radius of the site. A typical Paleozoic section as obtained from regional mapping and recent gas exploration wells in the site vicinity is also presented on Figure 2.5.1-11. This Paleozoic section consists of predominantly sandstone and shale of the Strawn and Atoka Stoke Series (Pennsylvanian); limestones of the Marble Falls Formation (Pennsylvanian) and Ellenburger Formation (Ordovician); and sandstone of the Hickory Formation (Cambrian).

These sediments overlie Precambrian granite basement. Because the Paleozoic sediments are not exposed in the site vicinity, these have not been included in the detailed stratigraphy given below.

Quarternary deposits in the site areas are recent floodplain alluvium (Qa1) and Pleistocene fluvial terrace sediments (Qt). These deposits consist of gravel, sand, silt, silty clay, and organic material and are confined to the bottoms of the Squaw Creek, Paluxy River, and Brazos River valleys. The scattered patches of Pleistocene fluvial terrace remnants are adjacent to the recent floodplain alluvium but at a slightly higher elevation.

The areal distribution of the modern floodplain alluvium and the terrace remnants of older floodplains conforms with that of the 1:250,000 scale, Dallas Sheet, Geologic Atlas of Texas (Univ. Texas Bur. Econ. Geol., 1973). Regional treatments of Brazos River terrace deposits [50, 51] document three terrace levels of Pleistocene age in the general area under consideration. The separate levels, as recognized in the Waco (McLennan County) area downstream are referred to by elevation above the base of the modern floodplain of the Brazos. The first terrace is about 30 feet above, the second terrace 45 to 60 feet above, and third terrace 90 to 120 feet above. Those terraces mapped in the site area represent the first and second and possibly the third. No scarps were noted that would assist in defining the separate terraces. Data on age in years B. P. (Before Present) are not available. Typical Pleistocene vertebrate fossils - mammouth, mastodon, camel, horse, etc., have been collected from all three terraces [50, 51, 52, 53, 54]. At higher elevations outside the area, from the tops of buttes in the Lampasas Cut Plain, lag gravels considered to represent the Uvalde terrace gravel (Micocene-Pliocene) have been reported [54].

The Edwards, Comanche Peak, and Walnut formations are of little relevance to the immediate site vicinity but are typically developed at nearby Comanche Peak to the north (Figure 2.5.1-10). Descriptive features of each unit are summarized below:

Edwards Limestone (Ked): massive, light gray, scarp-forming high purity limestone with reef-rock organisms (typically rudists), locally with abundant chert nodules in thinner-bedded, dense, lime mudstones; forms upper scarp slopes of high areas (Lampasas Cut Plains); thickness 16 - 60 feet in Waco Geologic Atlas Sheet, about 30 feet in site vicinity.

Comanche Peak Limestone (Kcp): nodular, gray to white, fairly hard but slope-and-bench forming, low purity limestone with numerous clay partings and filled burrows; distinct calcareous shale beds in he upper part and a few, harder, bench-forming oyster-bearing beds near the base; about 90-100 feet thick in site vicinity.

Walnut Clay (Kwa): dark grey to brown, fossiliferous, calcareous clay with hard massive beds of oysters in the lower part; thick, chalky, marly, nodular limestone interbedded with thin shale in upper part; about 70 feet thick in site area, thins to north and thickens southward by facies replacement of upper Paluxy Sand.

The Glen Rose/Paluxy contact as mapped (Fig. 2.5.1-10) marks the abrupt lithologic change from the gray-to-white ledge forming limestones of the Glen Rose to the overlying reddish, unconsolidated, sandy deposits of the Paluxy. Associated changes occur in soil type, vegetation, cultural use, and topographic nature. These distinctions are quite evident on the ground, are reflected clearly on areal photos, and the cartographic boundary was transferred without difficulty to 7.5 minute topographic quadrangles, scale 1:24,000, covering the area.

Standard field mapping procedures employed involved initial ground check points, by car mileage, on all country roads and on selected permissible private roads. The logged data, recorded on county road maps (scale 1 inch = 1 mile), controlled the boundary trace as extended on photo-index sheets or photomosaics. The basic reconnaissance data were refined by use of stereo areal photos for inaccessible or critical areas or for secondary marker beds.

In a similar manner, other formation boundaries within the 5-mile radius were mapped and extended a mile or more beyond. The detailed areally geology of the study area was compiled on a composite topographic base map of scale 1 inch = 3000 feet. Figure 2.5.1-10 is a reduced and simplified version of this base map.

The Paluxy Sand (Kpx) is a weathered, friable, well-sorted, fine- grained, poorly-cemented sandstone with some soft claystone and hard siltstone remains in the immediate site vicinity. The Paluxy Sand is a really extensive and cultivated on the drainage uplands.

The Glen Rose Limestone (Kgr) constitutes the bedrock of the Station site and reservoir area. The outcrop of this formation, its general topography and location of principal station facilities are shown on Figure 2.5.1-12. The Glen Rose Limestone was originally called the “Alternating Beds” because the outcrop pattern is characterized by stair-step topography resulting from differential weathering of impure, nodular limestones, softer claystone beds, and resistant, sparry-cemented, medium-to-thick bedded, hard limestones. The local order of occurrence or dominance of these lithologic types, plus other physical or paleontologic aspects, permits locally useful zonation as

shown on Figure 2.5.1-13 and on the lithologic cross section of the station vicinity shown on Figure 2.5.1-14.

The graphic lithologic log and the gamma-ray/resistivity log of boring P10 (Figure 2.5.1-13) illustrate a zonation of strata based on cores of all site boring logs, and related field information in the site vicinity.

Zone I, about 80-90 feet thick, is the bench-and-slope, uppermost outcrop unit indicative of major soft claystone beds alternating with hard, ledge-forming limestones. The basal claystone in contact with the underlying massive-bedded limestone of Zone II is the most prominent field contact visible throughout the reservoir area.

Zone II, 60-70 feet thick, is a massive unit of mostly nodular limestone with thin claystone partings best developed in the basal portion.

Zone III, 60-65 feet thick, is the most consistent unit recognizable in electric logs in the region. The upper part, IIIa, is an alternating claystone-limestone section transitional with the more massive units above and below; the lower part, IIIb, is the massive-bedded, bluff-forming Thorp Spring Member which is well exposed in the Paluxy River Bed at Glen Rose and further upstream.

Zone IV, at the base of the Glen Rose is about 30-40 feet thick; and consists of an alternation of claystone and limestone, both sandy or with thin sand lenses. The unit is transitional with the fine grained, calcareous, silty sandstone of the underlying Twin Mountains Formation.

The regional persistence of the Glen Rose zonation is indicated on the 25-mile correlation cross section (Figure 2.5.1-14) from the plant site east to Cleburne, Texas.

Where the Glen Rose beds are not exposed at the surface, data obtained from borings indicate that it is generally manteled by a few feet (range of a few inches to 15 feet) of surficial soils. These soils consist of mixtures of clay, silt and sand with some gravel and cobble-size rock fragments. In valley bottoms, the Glen Rose beds are overlain by alluvial sediments and residual soils which range to approximately 10 to 15 feet in thickness. As evidenced in many of the boring locations, weathering processes have produced a weathering zone on the Glen Rose up to several feet in thickness. This zone has been chemically altered, partially oxidized, and in some areas over 40 top 60 percent of the interval decomposed to a soil consistency.

The upper limit of bedrock, below which no soil-like inclusions were encountered in borings, is indicated on the vicinity Bedrock Contour Map, Figure 2.5.1-15. Bedrock contours in the immediate plant site are shown on Figure 2.5.1-16. Borings logs are presented in Section 2.5.5.3.

The conditions encountered at Category I facilities including the overburden and top of rock and moderately to severely weathered rock zone where it overlies the better rock are shown on geologic cross sections in Figures 2.5.4-40, 2.5.4-41, and 2.5.6-10. The section locations are shown on Figure 2.5.5-5. Pertinent information regarding foundation grades, existing groundwater conditions and characteristics of subsurface materials are summarized on these figures. The surficial rock generally is of a tan to orange-tan limestone which is occasionally dolomitic and fossiliferous. The remainder of the Glen Rose in this area consists of light to dark

gray argillaceous limestone with lenses and/or zones of gray to greenish-gray calcareous claystone. It is sporadically fossiliferous, dolomitic, pyritic and chrystalline. The unweathered Glen Rose is consolidated rock, medium hard to soft and thin-bedded to massive, becoming generally massive with fewer claystone zones below at approximately 770 feet in elevation. Oolitic zones are common in the massive portion of the formation. There are occasional zones which consist of limestone fragments enclosed in a claystone matrix, indicative of slump brecciation during deposition. In outcrops, the claystone lenses weather out to produce a thin-bedded appearance.

The Twin Mountains Formation underlies the Glen Rose Formation with a gradational contact. It was encountered in borings in the Station area at approximately elevation 610 to 615 feet. The sandstone beds of the Twin Mountains Formation constitute the aquifer for domestic water supplies and are collectively referred to in the region as the “Trinity Sand” [20].

In the postconstruction S/L cluster borings (Fig. 2.5.1-16), drilled for seismic crosshole survey, the Twin Mountain Formation was found to be about 250-ft thick and unconformably overlying the Mineral Wells Formation of the Strawn Series (Pennsylvanian age) at elevation 366 ft (Figure 2.5.4-30A). The Twin Mountains Formation is composed of sandstone and claystone, with occasional limestone layers [131]. The upper portion of this formation (above elevation 495 ft), as encountered in the borings, consists of interbedded sandstone, claystone, and argillaceous limestone. The sandstone is the dominant lithology. It is very soft to hard, friable in places, poorly cemented, light brown to light gray, and fine to medium-grained. The claystone and argillaceous limestone are medium hard to hard, dark gray to greenish-gray, occasionally green or maroon, and silty or sandy in places. The middle portion of the Twin Mountains formation (between elevation 495 and 458 ft) consists primarily of generally hard and brittle, reddish-brown, and highly slickensided claystone. The lower portion (below elevation 458 ft) is composed of interbedded sandstone and claystone. The sandstone is generally coarse-grained and occasionally conglomeratic, with infrequent fractures and common clay-rich intervals. Hard sandstone and claystone with up to 3-ft thick beds comprise the lowermost 20 ft of the formation. The presence of relatively unconsolidated sandstone and soft claystone in the middle and lower portions of the formation was indicated by the caliper logs of the S/L clusterly at elevation 489 ft (middle portion) and elevation 428 ft (lower portion). Poorly cemented sandstone was also encountered in borings P-9 and P-10 in which the drill rods rapidly penetrated some sandy zones due to easy cutting and/or jetting erosion by high-velocity drilling fluid.

The Mineral Wells Formation [130] was encountered in the S/L cluster Borings at elevation 366 ft, and was penetrated for 65 ft, exposing predominantly claystone beds. These beds, being of Paleozoic age, are distinguished from the similar overlying Lower Cretaceous beds by a higher degree of induration manifested in greater hardness and slickenside frequency, and by the occasional presence of vertical fractures filled with limestone. They are also characterized by their massive nature, dark color, and intercollated limestone seams and layers [131].

Figure 2.5.1-14 illustrates the approximate relationship of stratigraphy at the site to the stratigraphy found in borings within approximately five to 25 miles of the site.

The extensive limestone strata which are present at the site and vicinity are not subject to solutioning because of argillaceous impurities. There is no evidence of solutioning observed in the core obtained from the numerous borings. Fractures or joints occur infrequently.

A more detailed description of subsurface conditions is presented in Section 2.5.4.

2.5.1.2.4 Structure

Dip of the sediments in the site vicinity is to the east at approximately 25 feet per mile. Slight, gently warping of the sediments is occasionally evidenced in road cuts and natural exposures in the area surrounding the site. These are minor local features which are very limited in a real extent and degree of flexure. The infrequent slicken sides encountered in the Glen Rose Formation appear to be the result of differential settlement which occurred during compaction and diagenesis of the formation. The extensive investigations conducted at the site vicinity revealed no evidence of faults, shear zones or other anomalies.

Examinations of electric logs from recent gas exploration wells in the site vicinity suggests the presence of subsurface fault. The locations of these wells are presented on Figure 2.5.1-17. The faults locations are presented on Figure 2.5.1-23. Larger scale maps of the fault nearest the site are presented in Figures 2.5.1-23A, 24 and 26. This fault has no well cuts and is inferred from trends in subsurface data. The fault exhibits normal movement and is approximately 65 miles long (Figure 2.5.1-23). The throw on the fault appears to decrease upward until normal conditions are reached at a subsea datum of approximately 2,000 feet.

The faulting is associated with basement faulting. The structure model which accommodates this faulting is one of older faulting from basement movement which faulted the Ellenburger (see Section 2.5.1.1.4).

Later movement renewed the fault to cut the Marble Falls and Big Saline formations and continued upward into the Strawn but died out as the Strawn formation was deposited. There is no evidence of renewed movements after middle to late Strawn time (over 270 million years ago).

Figures 2.5.1-23A, 24A and 26 are the revised structure maps of the Ellenburger, Marble Falls and Big Saline formations. They are based on more data than was available for the maps filed with the initial FSAR. The current information does not indicate any faulting within five miles of the site. A structural nose trending southwest to northeast is present on the Ellenburger, Marble Falls and Big Saline horizons. No faulting is interpreted with this nose nor is any anticipated within five miles of the plant site. If by further hydrocarbon exploration any faulting is found within the five-mile radius, it is not expected that it will extend above a datum of approximately 2,000 feet subsea (approximately the middle or upper portion of the Strawn formation) [124]. On the Strawn marker structural map (Figure 2.5.1-28), no faulting is depicted within five miles of the plant site. Where a fault is mapped on older beds to the northwest of the plant (approximately seven or eight miles from the site) there is no faulting at the level of the Strawn marker. This indicates the faulting has died out in the Strawn Formation between the Big Saline Formation and the Strawn marker. Cross section A-A' (Figure 2.5.1-30) illustrates this upward limitation of the faulting. Cross section B-B' (Figure 2.5.1-31) illustrates the absence of faulting near the site.

Cross Section A-A^1 (Figure 2.5.1-30) is essentially a strike section on the Big Saline and older units as shown on Figure 2.5.1-23A. Cross Section B-B^1 (Figure 2.5.1-31) is a dip section of the Big Saline and older units (Figure 2.5.1-23A). In contrast, cross section A-A^1 (Figure 2.5.1-30) is a dip section on the Strawn marker and cross section B-B^1 (Figure 2.5.1-31) is a strike section (See Figure 2.5.1-28). The erosional top of the Ellenburger dips at about 100 ft/mi (1°) to the east and the Strawn dips at about 200 ft/mi (2°) to the northwest. These gradients reflect typical regional gradiens related to development of the Fort Worth Basin within the Texas platform. The gradients of 1-2° are not considered steeply dipping.

Figure 2.5.1-32 (Cretaceous Structural Map) presents the structural contours on the Paluxy Sand-Glen Rose Limestone contact over the 5-mile radius area and, over a smaller area, on the prominent marker bed about 75 feet below the top of the Glen Rose. Contouring on the marker bed is confined to the central portion of the site area in a 2- to 3-mile strip adjacent to the NW-SE trending valley of Squaw Creek. Elevations for datum control of contouring were taken from published USGS topographic quadrangles. The principal points of the Paluxy-Glen Rose contact were established in the field through road traverses; the upper Glen Rose marker bed points were in large part transferred directly from areal photographs to the topographic base (Figure 2.5.1-10, Vicinity Geologic Map).

Figure 2.5.1-32 shows a consistent north-south regional strike and constant dip to the east at a rate of about 25 feet per mile for both of these horizons. The mappable continuity of the Paluxy-Glen Rose contact and of the upper Glen Rose marker bed plus the structural contouring indicate no surface anomaly in the site area.

Structural cross sections C-C' and D-D' (Figures 2.5.1-34 and -35) supplement the structural maps and likewise indicate no evidence of post-Paleozoic faulting, known or inferred, west or northwest of the 5-mile radius area. The most significant subsurface Glen Rose datum is the massive Thorp Spring Limestone Member. The top of this high resistivity unit marks the boundary between Glen Rose zones IIIa and IIIb of the lithologic section at the plan site (Figure 2.5.1-13). All correlation lines extended from well to well, based on gamma-ray logs updip and self-potential/resistivity logs downdip, are essentially parallel to the Zone IIIa - Zone IIIb boundary (= top of Thorp spring Member). No faulting is indicated in either of the cross sections within the Cretaceous rocks.

2.5.1.2.5 Groundwater

Detailed information regarding site groundwater is described in Section 2.4.13.

2.5.1.2.6 Engineering Geological Evaluation

1. General

   Detailed discussion of engineering geological considerations in relation to specific seismic Category I structures is presented in Section 2.5.4.

2. Site Behavior During Past Earthquakes

   The response of the site to past seismic activity is discussed in Section 2.5.2.1, Seismicity. Although a few distant large earthquakes (1811-1812, 1882, 1931) have been, or could have been felt at the site, it is unlikely that the CPNPP site has experienced greater than Intensity III (less than 0.09 g) in historic time. Table 2.5.2-1 describes the earthquake intensity scale. Detailed onsite studies have not disclosed any physical evidence to indicate that the site has experienced major seismic activity at any time. These studies included a detailed survey of the site vicinity in order to detect any features that could be related to major seismic activity such as sand dikes, intrusions, scarps, or dissimilar materials placed into juxtaposition. No such features were found during the preliminary investigations and no such features were found during the excavation of the site.

3. Stability of Subsurface Materials

3.1 Uplift and Subsidence

3.1.1 Uplift

Tectonic tilting or warping is not characteristic of this region; hence, significant uplift or subsidence due to such phenomena will not occur. No evidence of actual or potential uplift or subsidence was revealed at the site.

3.1.2 Subsidence

3.1.2.1 Solution Potential

The limestone of the Glen Rose Formation is the only calcareous material in lateral or vertical proximity to the site. The thin-bedded and impure nature of this rock renders it unfavorable for the development of solution cavities. Study of topographic maps and areal photography and examination of rock cores shows no indication of karst formation or other solution activity in the area. In addition, detailed onsite reconnaissance has failed to find any evidence of sink holes or solution cavities. The absence of such features was confirmed in interviews with federal civil construction agencies.

Many of these agencies have had experience with the Edwards Limestone, but this data can be applied to the Glen Rose formation because the Edwards is purer limestone than the Glen Rose and is more susceptible to solution activity. If there has been no reported solution activity in the Edwards, it can be conservatively assumed that the Glen Rose will have no solution activity.

The chief geologist of the Fort Worth District of the Corps of Engineers reports that solution voids have never been encountered in the corps drilling in the Edwards Limestone near the Balcones Fault Zone (Wallis Samuelson, personal communication). The Texas Water Development Board and United States Department of Agriculture Soil Conservation Service have no record of solution activity in the Cretaceous limestones in the site vicinity (Gunner Brune, personal communications). The Texas State Board of Economic Geology reports no accounts of solution problems in the Glen Rose Limestone [43].

The results of the petrographic and chemical analysis of the Glen Rose Formation show that the limestone has very low porosity, poor sorting and slight to moderate compaction, all of which are indicators of low permeability. In addition, there was no evidence of styolites, pressure solution of grains, or collapse structures of any scale noted in the thin section analysis. The rarity of dolomite as reported from petrographic and X-ray analysis suggests that collapse structures have not occurred from the effect of volume change in the transformation of Mg-calcite to dolomite

and calcite. The low amount of magnesium shown in the chemical analysis indicates that the magnesium is already in the dolomite phase and that future dolomitization as an isochemical reaction is impossible.

These laboratory tests indicate that the Glen Rose Limestone should have a very low permeability, an indication which is confirmed by the packer test data presented in Table 2.5.6-1. The laboratory data also indicates that future solutioning of the limestone should not occur.

Considering the various material types, chemistry, distribution and site environment, it is concluded that solutioning activity in the limestones beneath the site does not exist. Results of detailed petrographic and chemical analysis of the limestone below the site indicate that the potential for solutioning is very low. Detailed results of these tests are given in Section 2.5.4.2.3.

At a considerable distance east of the site vicinity, anhydrite and gypsum and present in Glen Rose Formation [10]. Under certain conditions such material could be subject to solutioning. However, these conditions are not known to occur near the site.

3.1.2.2 Oil and Gas Wells

The Paluxy and Glen Rose formations are major lower Cretaceous hydrocarbon producing reservoirs in the East Texas Embayment. These reservoirs are mostly east and south of the Luling-Mexia-Talco Fault Zone [44] and are nearly 100 miles east of the site. Erratic sands in the Pennsylvanian Strawn, and the Ordovocian Ellenburger limestone are major producing zones in the Bend Arch area 25 to 50 miles or more to the west.

No exploratory or producing oil or gas wells existed within five miles of CPNPP prior to 1974. At that time, three exploratory wells have been drilled in that area and all three were reported as dry and abandoned. The first area production was reported early in 1977. In light of new data, a study of mineral potential was performed by H. J. Gruy and Associates, Inc. and is presented here.

Using the subsurface geologic terminology established by Gulf Energy and Minerals Co. -U. S. No. 1 Otis Rollins well in the Louis Boatwright Survey A-33, Hood County, there are four Paleozoic formations having the potential for hydrocarbon production in the area of the CPNPP. In ascending order, these are the Ellenburger Formation, the Marble Falls Formation, the Big Saline sandstone and the Strawn sandstones. The size of a possible blowout from the above listed formations is in the order they are listed (i.e., maximum in the Ellenburger and minimum in the Strawn sandstones). The probability of finding hydrocarbons is in the reverse order as listed. In the following discussion, wells which fall within a five-mile radius of the plant site are listed with a well identification number keyed to Tables 2.5.1-5 through 2.5.1-10 and to Figure 2.5.1-17

entitled “Natural Gas and Oil Pipelines and Wells Within Five Miles of CPNPP”. As noted on this Table, all hydrocarbons produced within five (5) miles of the site are believed to be from stratigraphic traps not structural traps [124]. The outline on the net hydrocarbon isopachous (Figures 2.5.1-25, 27 and 29) maps which represent the size of the hydrocarbon accumulations do not appear to be structurally controlled.

A structure map (Figure 2.5.1-23) on the top of the Ellenburger formation was constructed for Somervell, Erath, Hood, Johnson, Hill Bosque and Hamilton counties. All wells (for which there were public records) that reached the Ellenburger Formation were used in this study. There is no structural high on the Ellenburger under or adjacent to CPNPP.

Within the studied area, only two wells produced from the Ellenburger Formation. The Cities Service No. 1 Foss, approximately 23.5 miles northwest of CPNPP in northeastern Erath County, is indicated by completion card information to have an initial potential flowing of 196 barrels of oil per day from Ellenburger perforations at 5,065 to 5,076 feet. No state production records exist confirming this production. The second well is the HNG Oil Company No. 1 Ryan well in the northeastern corner of Hamilton County, approximately 25.2 miles to the southwest of the plant site. This well reported an initial potential pumping of 28 barrels of oil and 41 barrels of water per day from Ellenburger perforations located at a depth of 4,423 to 4,434 feet. The total cumulative production of this well from state records is 1,571 barrels of oil, and the well has been shut in since June 1976.

Electric well log analysis was performed for the Ellenburger Formation in seven wells. A summary of the calculated petrophysical properties is presented in Table 2.5.1-7. The nearest well testing the Ellenburger Formation (approximately 2.4 miles northeast of the plant) is the Dallas Production Company No. 1 Hubbard (Well I. D. No. 14) in Hood County. The No. 1 Hubbard tested salt water from the Ellenburger Formation. A well approximately 5.9 miles southwest of the plant site, the Dorchester Exploration No. 1 Davis-Bonney well, calculated 45 feet of net hydrocarbon pay by electric well log analysis in the Ellenburger Formation. This well was not tested nor was completion attempted in the Ellenburger Formation.

The Dorchester Exploration No. 1 Davis-Bonney well located approximately 5.9 miles southwest of the plant site is currently plugged and abandoned. Records from the Oil and Gas Division of the Texas Railroad Commission show that it was plugged on March 6, 1975.

The Ellenburger does not possess hydrocarbon potential under or adjacent to the CPNPP. This is supported by the test of salt water in the closest well to the north and the location of the plant site in a structurally low position as mapped on the top of the Ellenburger Formation.

The structure map (Figure 2.5.1-24) of the top of the Marble Falls Formation indicates no structural high under the site. There is Marble Falls gas production reported to the west and northwest of the plant site with the nearest gas well, the Ben J. Taylor No. 1 Cravens “B” (Well I. D. No. 22), located 2.5 miles to the northwest of the plant site. Marble Falls gas production is associated with a structural high, trending from southwest to northeast, to the west of the plant site. A net hydrocarbon isopachous map (Figure 2.5.1-25) has been prepared for the Marble Falls Formation. An area of thick net hydrocarbon exists approximately 4.3 miles to the west of the plant site. The maximum net hydrocarbon thickness calculated from electric well log analysis is 62 feet. This net hydrocarbon isopachous map indicates a rapid thinning to the east, and we postulate that under CPNPP there will be zero feet of net hydrocarbon porosity found in the Marble Falls Formation.

The structure map (Figure 2.5.1-26) on the top of the Big Saline sandstone portrays a structural high, trending from southwest to northeast, to the west of CPNPP. The Comanche Peak site is low on the east flank of this trend and on the east flank of the regional dip into the deeper portion of the Fort Worth Basin. The nearest producing well reportedly completed in the Big Saline sandstone is the Tri-Venture II No. 1 Nelon (Well I. D. No. 18), located approximately 3.2. Miles to the west-northwest of the plant site. A net hydrocarbon isopachous map (Figure 2.5.1-27) of the Big Saline sandstone was prepared. Fourteen feet of net hydrocarbon-bearing porosity was calculated by electric well log analysis for this zone in a well approximately 4.4 Miles to the west-northwest of the plant site. The net hydrocarbon isopachous map of the Big Saline sandstone indicates a north-northeast to south-southwest trending thick area approximately 3.8 Miles to the west of CPNPP. The Big Saline sandstone zone thins and loses porosity to the east and southwest. This loss of porosity lessens the probability of any production from the Big Saline sandstone under CPNPP.

A structure map (Figure 2.5.1-28) of the top of the Strawn Marker indicates a northwestward dip at a uniform rate interrupted by minor anomalies. An isochore map from the top of the Strawn Marker to the top of the Big Saline sandstone indicates an eastward thickening of the Strawn Formation into the deeper part of the Fort Worth Basin. As the overall zone thickens, there is a loss of both total number of sandstones and porosity. This results in a reduction in the calculated net hydrocarbon pay sand in the Strawn Formation.

All sand lenses in the Strawn Formation have been combined into one net hydrocarbon isopachous map (Figure 2.5.1-29) because of the lenticularity and limited drainage of any individual sand within the Strawn Formation. The maximum net hydrocarbon pay thickness calculated in a well from electric well log analysis was 43 feet. The net hydrocarbon isopachous map of the Strawn Formation indicates a north-south trending area of hydrocarbon-bearing sandstone approximately 3.8 miles west of the plant site. The nearest well completed in the Strawn Formation is the Gulf Oil No. 5 Bratten Ranch (Well I. D. No. 15), 4.4 Miles to the west of CPNPP.

Towards the plant site the Strawn sandstones thin, become tighter and undergo a facies change into a shaley equivalent. Some Strawn sandstone gas production may be achieved adjacent to the boundaries of the Exclusion Area and even near or under the plant site. However, it is evident that less than 20 feet of net hydrocarbon Strawn will exist within the Exclusion Area and any gas well would be marginal to subcommercial.

A well was drilled within the boundaries of the Squaw Creek Reservoir. The data on this well can be found under Well I. D. No. 24 listed in Table 2.5.1-5. This Well was plugged under the supervision of a Texas Railroad Commission representative in June 1971. Table 2.5.1-5 lists pertinent data for the wells within a five-mile radius of the plant site.

Eighteen wells in the Strawn sandstones, three wells in the Big Saline sandstone, eight wells in the Marble Falls Formation and ten wells in the Ellenburger Formation were selected for study of petrophysical properties. The distance of the studied wells was from 2.4. to 33.3 Miles from the plant site. The results are summarized in Table 2.5.1-6. Water saturations, porosites and net hydrocarbon thicknesses were calculated from electric well logs. The reservoir pressures, temperatures and gas gravities are known only for a few wells.

Table 2.5.1-7 is derived from Table 2.5.1-6 and is a statistical summary of reservoir properties. For each zone, the minimum, the maximum and the average values of net hydrocarbon thickness, porosity, water saturation and permeability are listed. The permeability values are obtained from radial steady-state flow calculations on tested wells.

For each possible producing zone, a maximum hypothetical well is postulated with maximum bottom-hole pressure and optimum valves for net hydrocarbon feet, porosity and water saturation. The absolute open flow potential is calculated for each zone using radial steady-state gas flow calculation. The results are shown in Table 2.5.1-8.

The maximum absolute open flow potential for an out-of-control well (blowout) of 43,800 thousand cubic feet of gas per day is calculated for a hypothetical Ellenburger well using 1.5. times the maximum permeability. This rate is 3.21 times the rate of 13,360 thousand cubic feet of gas per day calculated for the average Ellenburger well. The maximum subsidence of 0.54 feet is calculated for the hypothetical Ellenburger well at depletion. This subsidence extends outward approximately 2,250 feet from the wellbore. In this subsidence calculation, it is assumed that the zone above the producing Ellenburger is not compressible.

The gas gravity data on six wells (Table 2.5.1-6) indicate a range of 0.69 to 0.70. Data was available only on these six wells within the study area.

Reservoir data on some selected wells within the five-mile radius from the plant site are shown in Table 2.5.1-9. Table 2.5.1-10 presents calculations of the gas in-place for a hypothetical well in each formation.

The Marble Falls and the Big Saline formations produce by depletion drive. Depletion drive is associated with minimal to no water production. The Strawn sandstones do have some associated water production. Gulf Energy & Minerals Co.-U.S. (the major operator in the area) does have a water disposal well located seven miles northwest of the plant site. This well is the Gulf Energy & Minerals Co.-U.S. No. WD-2 T. H. Nix located in the Lewis Boatwright Survey A-33, southern Hood County, Texas. Water is being injected for disposal in the Ellenburger formation from 5,550 to 5,796 feet. This well was completed on October 10, 1977.

Water production associated with the hydrocarbon production from the Strawn sandstones within five miles of the plant is minimal. The Strawn sandstones are lenticular with low porosities and, although small amounts of water are produced, the reservoir drive mechanism is primarily a depletion drive, not a water drive. The Strawn sandstones are competent sandstones, and fracturing is usually necessary to achieve gas production. There is little or no potential subsidence due to production of hydrocarbons and water from the Strawn sandstones.

The water production as reported by Gulf Energy & Minerals Co.-U.S. to the Railroad Commission of Texas on Form G-10 for the test period of October, November and December, 1977 is summarized as follows:

Field	Formation	Number of Wells	Average Water Production in Barrels per Day per Well
Bratten Ranch	Marble Falls	3	8
Bratten Ranch	Big Saline	1	4
Bratten Ranch	Strawn	4	7
McIntosh	Big Saline	1	35
McIntosh	Strawn	5	8
Otis Rollins	Marble Falls	1	1

The hydrocarbon production to the northwest of the plant site is from the above-listed fields.

In summary, the new data on mineral potential support the data available prior to 1974. On the basis of calculations and the consolidated nature and integrity of the rock, no measurable subsidence is anticipated from depletion of the recorded or future (postulated) gas reservoirs.

3.1.2.3 Groundwater and Geothermal

Groundwater withdrawals, in the event they would be greatly accelerated from any of the aquifers underlying the site, should not lead to a subsidence problem. The formations are sufficiently well consolidated and structurally of a nature that the potential for surface subsidence is virtually nil.

Sedimentary strata under the site are indurated or semi-indurated. The semi-indurated strata are composed of poorly graded granular materials which are not subject to change in volume upon dewatering. The United States Geological Survey in Austin, Texas, reports that subsidence of Cretaceous sandstones in the site vicinity has never been observed and is highly unlikely to ever occur (Al Winslow, personal communication). The United States Geological Survey in Fort Worth, Texas, also has no knowledge of subsidence in Cretaceous sandstone in the site vicinity. (John Montgomery, personal communication).

An estimate, based upon conservative consolidation characteristics, has been made of the potential subsidence of the site area due to a conservative drip of 220 feet in the piezometric level in the Twin Mountains Formation.

This estimate of potential subsidence was based upon a consolidation test performed on a sample of the Twin Mountains Formation, the results of which are presented on Figure 2.5.1-21. This sample was taken from uncemented sandstone zone with one of the lowest RQD (Rock Quality Designation) values and which is considered to be more compressible than the other strata within the Twin Mountains Formation. The method and results of the settlement calculation are presented on Figure 2.5.1-22. The settlement obtained from this calculation is very conservative due to the assumption made that the entire Twin Mountains formation has the same compressibility as the sampled tested. Based upon an evaluation of boring logs, it is estimated that actual subsidence under the conservatively assumed 220 foot drawdown would be less than one-half the value calculated or about six inches.

Considering the site geology and the regional potential for groundwater withdrawal, it is unlikely that the site will experience any subsidence at all during the life of the plant. Even if subsidence did occur, it would be limited to a few inches, be regional in nature (i.e., no differential subsidence) and have no effect on the plant operations.

There are no known existing or potential geothermal energy locations near the site. Geothermal energy can be inferred by existence of hot springs, or the distribution of young volcanic rocks [45]. Neither of these are present near the site; the nearest known higher-than-average thermal gradient is at Marlin, Texas, nearly 100 miles to the southeast. Here water at a temperature of 147°F has been found in deep wells. The geothermal

gradient in the site vicinity is insufficient for any consideration to be given to the use of geothermal energy.

3.1.2.4 Reservoir Effects

The increase in earthquake hazards at the site due to the creation of the SCR is negligible. Most theories on the cause of reservoir-created earthquakes attribute the cause to lubrication of preexisting faults under the reservoir sites [46 and 47]. Because there are no known faults within five miles of the site, the potential for reservoir induced earthquakes is considered nil. This conclusion is reinforced by the fact that no reservoir-induced earthquakes have been reported for the two existing reservoirs which are in close proximity to the site and in similar geologic systems. These are Lake Granbury and Lake Whitney whose locations are shown on Figure 2.4-5.

A subsurface fault has been identified approximately 7 miles NNW of the Class I structures as identified on Figures 2.5.1-23, 23A, 24, and 26. A cross section through this fault is shown on Figure 2.5.1-30. This fault is mapped at the Ellenburger, Marble Falls and Big Saline horizons; however, it cannot be traced upward beyond the middle portion of the Strawn formation. Based on all available data, it appears that this fault dies out upward in the middle to upper portion of the Strawn formation (over 270 million years old) [124]. No actual cut of the fault has been found in any well, and the faulting is inferred from the subsurface data established by the well control.

Other known faults in the site area exhibit this same dying out upward in the middle to upper portion of the Strawn formation [124]. Based on this evidence, any other presently unknown faulting which might exist under the plant site or under the Squaw Creek reservoir would occur only in beds older than middle to upper Strawn [124]. There appears to be no potential for hydraulic interconnection through any existing or future wells with the faulted strata. The only formations with any significant aquifer potential are the Ellenburger and Hickory. Faulting in the area does not have sufficient displacement to place the Ellenburger or Hickory in communication with younger porous formations (see Figure 2.5.1-30). The formations above the Ellenburger have discontinuous lenticular porosity. Therefore, the potential for any hydraulic interconnection by faulting with other porous and permeable strata is minimal.

Table 2.5.1-11 has been constructed from a representative sampling of the wells on which drill-stem tests and production test data are available. These data illustrate that no geopressures (only normal pressures) are found in the Ellenburger, Marble Falls, Big Saline formations or Strawn sandstones.

Due to the indurated and consolidated nature of the Cretaceous rock at the site, subsidence of any significance due to the reservoir load is not considered possible. Minor elastic-deformations of the rock may occur;

however, due to the relatively shallow depths of the reservoir these deformations will be negligible.

Due to the low permeability of the limestones and the lack of solution potential of the Creataceous deposits in the site vicinity, the potential for solutioning due to reservoir seepage is considered negligible.

Result of petrographic and chemical analyses of the Twin Mountains Formation indicate that the solution potential of the sandstone is very low. Results of these tests are presented in Section 2.5.4.2.3.

Piezometric data indicate that the Glen Rose Formation contains a perched water table and that the Twin Mountain Formation will not be hydraulically connected to the new reservoirs.

Seepage from the reservoirs should be minimal because the jointing in the Glen Rose limestone is not continuous (see Section 2.5.1.2.6); also the Glen Rose has been shown to maintain a separate piezometric level and the solutioning potential of the limestone has been demonstrated to be very low.

3.2 Deformational Zones

Joint patterns in bedrock have been observed both on areal photographs and in the field. No evidence for movement along joints (i.e., faulting) has been observed, however.

Strikes of the primary joint systems are N20 E and N78 W, as measured in river bottom exposures. The degree of weathering in even the more resistant rock types obscures joint patterns both in road cuts and in natural exposures away from the river channels. The surface joints are generally spaced several feet apart and either are opened or filled with soil, vegetation and small weathered rock fragments.

The joints in the limestone do not extend into the overlying and underlying claystone beds (O. T. Hayward, personal communications). Residual boulders of the massive Thorp Springs limestone near the base of the Glen Rose Formation are present in the Paluxy River bed near Glen Rose. These boulders are the combined result of rock structure joint sets intersecting at right angles and weathering processes. These perched boulders are strictly a surface phenomenon [55].

Part (b) of Figure 2.5.1-18 shows a rosette diagram of the strikes and frequencies of observed structural trends in the lower half of the Paluxy basin. To varying extents over the region these alignments influence the courses of stream channels, as illustrated in part (A) of the figure. Emphasis was placed on field checking of alignments in the site vicinity to determine whether evidence existed for faulting.

No evidence for faults or shear zones was revealed within a five-mile radius of the site, either in the regional or the site studies.

The origin of the joints is not a question easily resolved. The north-northeast trending system coincides with directions of the Balcones and Luling-Mexia-Talco fault systems to the east so that the possibility of a relationship could not be completely dismissed.

Considerable topographic relief was established on the pre-Cretaceous (Wichita) peneplain surface (Figure 2.5.1-19) prior to burial by Cretaceous sediments. It is probable that subsequent differential consolidation of the covering sediments resulted in some stresses which led to the formation of joints and occasional slickensides which were observed in borings at the site. The same phenomenon of compaction and consolidation probably warped the regional dips slightly into very gentle folds which are known to occur sparingly throughout the vicinity.

On a broader scale, the crossing of tectonic axes resulting from conflicting compressional forces created areas of fracture and distortion throughout Texas and Oklahoma [34]. This is best illustrated in the Arbuckle Mountain area where the Wichita axis intersects the Ouachita axis. The site is distantly removed from intersections of tectonic axes.

3.3 Alteration, Weathering and Structurally Weak Zones

Results of detailed site investigations revealed no zones of alteration, irregular weathering profiles or zones of structural weakness which would adversely affect the stability of the Station facilities. General excavation to the Station grade of elevation 810 feet MSL will result in removal of essentially all weathered material.

3.4 Unrelieved Residual Stresses in Bedrock

The tectonics of the area indicate that there should be no abnormal stresses due to active tectonic forces. It is not felt that stress measurements are required to confirm this opinion.

3.5 Geologically Hazardous Materials

The occurrence of anhydrite in certain geologic formations in Central Texas has led to violent explosions and uplifts due to the hydration of the mineral to gypsum. The reaction is accompanied by great pressure and an increase in volume. Figure 2.5.2-10 shows the location of outcrops which contain critical amounts of anhydrite and/or gypsum.

The deposits, generally of Permian age, usually take the form of anhydrite at depths greater than 500 ft. Between the depths of 100 and 500 feet, either anhydrite or gypsum can occur, while gypsum normally is found at depths less than 100 feet. Erosion reduces the overburden, permitting groundwater to reach the anhydrite and convert it to gypsum. Mean annual precipitation is less than 30 inches where outcrops of gypsum and gypsiferous material occur. The site is distantly removed from all historic uplifts of this type.

The areal distribution of expansive soils in the region is illustrated on Figure 2.5.1-20. It is not anticipated that materials of a swelling character will have any adverse effect upon structures at the site whose foundations are constructed below the soil zone.

3.6 Effects of Man's Activities

Except for the removal of minor quantities of sand, gravel and dimension stone in the site vicinity, no mining has occurred. In connection with plant operation, it is anticipated that a water supply will be developed within wells on the order of 600 gallons per minute from the Twin Mountains Formation. This pumpage may eventually reduce the artesian pressure in the area, but withdrawal of groundwater will have no deleterious effect on rock deformation or stability. Extraction of subsurface fluids is discussed in Section 2.5.1.2.6.

4. Physical Properties of Geologic Materials

The results of onsite geophysical surveys and laboratory testing are provided in Section 2.5.4.4.

5. Safety Criteria

Safety criteria are discussed in detail in Sections 2.5.4.10 and 3.7.

2.5.2 VIBRATORY GROUND MOTION

Vibratory ground motion and the design earthquake (Safe Shutdown Earthquake) are evaluated in consideration of:

1. A study of the seismic history of the region;

2. A study of the geology and tectonics of the region;

3. The relationship of seismicity to geology; and,

4. The response of the site to both the Safe Shutdown Earthquake and Operating Basis Earthquake.

Regional and site geology are describe in Sections 2.5.1.1 and 2.5.1.2

2.5.2.1 Seismicity

The earthquake history of Texas has been poorly documented in the past. This result has been due, in part, to the recognized low seismicity of the area, the lack of seismograph instrumentation for recording events, a generally sparse population and insufficient interest in minor seismic activity. The new Texas earthquakes of consequence occurred in areas where there was little damage to the works of man. Except for the interest of the academic community and petroleum geologists, all but the larger events probably would have remained unreported.

Improved instrumentation in recent years and the increased awareness of geotechnical professionals of geologic and other environmental hazards has resulted in an apparent increase in reports of seismic activity. Consequently the increase in reports probably do not reflect an actual increase in seismic events. Some reports may be due to blasting, sonic booms, water flood operations in oil fields, or reservoir loadings of large dams.

Central and East Texas lie within the zone of least seismic activity in the United States. According to Algermissen and Perkins' (1973) Seismic Risk Map of the United States, the region lies within a Zone 0 (no damage) area [64] as indicated on Figure 2.5.2-1. King, using Woollard's [24] data, indicates an average of less than one epicenter per 10,000 square kilometers. This evaluation is based on reported and instrumentally detected epicenters for about 200 years and relies, necessarily, on incomplete data.

Texas was sparsely populated before 1820, but by the time Texas became a Republic, there were widely scattered ranches and settlements which would have reported larger events such as the Bonham, Texas earthquake of 1882. It is likely that all Modified Mercalli Intensity VII earthquakes in Texas have been reported for the past 150 years. The complete record for smaller events is shorter. A list of the earthquakes in Texas and adjacent areas of interest is presented in Table 2.5.2-2. Figure 2.5.2-2 shows the epicenters with respect to regional tectonic provinces.

The literature of Texas seismicity reflects a scarcity of recent damaging shocks, or even widely-felt shocks. No Texas earthquake with an intensity greater than VI has been reported east of the 100th meridian since 1882. Sporadic damaging events have occurred in Texas through the years; the most recent was in 1969 (Intensity VI, in West Texas. Sketchy field reports are available for most of the larger events. The few large West Texas shocks in the 1930's warranted field investigations and reports.

Seventeen seismic events (or series of events) have been reported with epicenters within 200 miles of the site (1882-1975). Three events (or series of events) are within 100 miles of the site. The closes known event to the site occurred 90 miles to the north in 1950 and exhibited an intensity of IV. Two events occurred 90 to 100 miles to the southeast of the site in 1932 (V-VI) and 1970 (IV) respectively. The highest reported intensity within 200 miles of the site was a Modified Mercalli Intensity VII exhibited by both the 1882 Paris and 1891 Rusk events located 155 and 160 respectively from the site.

The localized areas where seismic activity is observed to cluster are apparent (Figure 2.5.2-2). The El Reno, Oklahoma area located 200 miles from the site has been the locus of significant seismic activity. The most important event in this area occurred in 1952 with an intensity of VII and produced an intensity I-III at the site [27].

The central and northern Panhandle region of Texas 280 miles northwest of the site is also the locus of significant clustering of seismic activity and the 1936 Borger intensity VI earthquake [114].

Earthquakes of the south central region of the United States generally occur as single major shocks usually with associated smaller aftershocks and foreshocks. Occasionally, seismic energy release takes place in the form of earthquake swarms such as the earthquake series observed for the Sabine Uplift in 1964 or the recent 1974 Arkansas swarm. Swarm activity is not unusual in the eastern and central United States.

Both Rossi-Forel and Modified Mercalli intensities are used to describe the effects of historic earthquakes (see Table 2.5.2-1). A comparison of these intensity values with other commonly used intensity scales follows.

U.S.A. Modified Mericalli (M.M.) 1931	Japanese (JMA) 1950	Rossi-Forel (RF) 1873	European (Mercalli-Cancani-Sieberg) 1917
I	0	I	I
II	I	I-II	II
III	II	III	III
IV	II-III	IV-V	IV
V	III	V-VI	V
VI	IV	VI-VII	VI
VII	IV-V	VIII-	VII
VIII	V	VIII+-IX	VIII
IX	V-VI	IX+	IX
X	VI	X	X
XI	VII	X	XI
XII	VII	X	XII

A detailed description of individual earthquakes which have occurred in the area and which are considered to be relevant to this study for the purpose either establishing the SSE or to further illustrating possible tectonic relationships is presented below.

2.5.2.1.1 1882, October 22 - Bonham (Paris), Texas Earthquake

Intensity VII [26]: “About 4:15 P.M. an earthquake was felt in northern Texas, western Arkansas, and eastern Kansas, and presumably in the intervening portions of the Indian Territory. The region affected extended from Greenville and Paris, Texas and Little Rock, Arkansas northeasterly to Wichita and Leavenworth, Kansas, a distance of some 300 miles. The shock was reported from numerious places with these boundaries, and also, as a light shock, from Warrenton, Missouri, which is father eastward...no damage was done other than overturning movable articles and knocking bricks from chimney tops.” The original epicentral location was assigned at 35°N, 94°W [28]. Dockekal [69] has recently revaluated the effects and characteristics of this earthquake. Based on a reinterpretation of the original references, he relocates the epicenter 107 miles to the southsouthwest in the vicinity of Bonham and Paris, Texas. Accordingly, this event will be called the Bonham, Texas earthquake in further discussions. Intensities of MM VI-VII and MM VII for this earthquake are given by the USC & GS [70] and Docekal [69] respectively.

2.5.2.1.2 1891, January 8 - Rusk, Texas Earthquake

The original listing of this event was apparently by Milne [28], who used the London Times of January 10, 1891 [71] as his source.

> The Times account read:
>
> Rusk, Cherokee, Co., Texas -
>
> "Two sharp earthquake shocks were felt here yesterday. Chimneys were thrown to the ground and people awakened by the violence of the oscillation." - Reuter

Another account was the following from the Dallas Morning News, January 9, 1891 [72]

> Rusk, Texas - "Last night at 12:00 this town and immediate vicinity experienced two well defined shocks of a seismic nature. Each paraxym was accompanied by a detonation loud and long as thunder rolling from south to north. Several chimneys were levelled to the earth and sleepers in various portions of town were awakened. J.W. McCord, in charge of the county jail, a very strong structure, declares that for one full minute he apprehended a collapse of the building. Thos. Miller said the Acme Hotel was shaken to its foundation. During these disturbances there was a slight rainfall and some electrical force prevailing, but not of sufficient strength to produce the shock. A few parties here who were through the Charleston shocks of 1886 pronounced the phenomenon last night a genuine earthquake."

The above account, with some deletion, was also found in the San Antonio Express, January 9, 1891 [73]; the Galveston Daily News, January 9, 1891 [94]; the Houston Daily Post, January 9, 1891 [75]; the Fort Worth Daily Gazette, January 9, 1891 [76]; and the Terrell Times-Star, January 17, 1891 [77].

The only other original account seems to be the following from the January, 1891 edition of the Victoria Advocate: "It was for Rusk to furnish a long felt want in Texas, which it did on the 8th last. There was a genuine earthquake there which overturned chimneys, rocked buildings and shook up the city generally. Texas will not be outdone".

Eppley [78] has assigned an intensity of MM VII to this event based on an interpretation of Milne's Catalog of Destructive Earthquakes [28], and on the reports of chimneys being thrown down, which is one criterion of MM VII intensity. However, the original accounts are generally inconclusive and completely confined to the immediate vicinity of Rusk, which had an overwhelming lack of supportive data from the surrounding area and contained over 60,000 residents (22,975 in Cherokee County, 19,360 in Houston County, and 20,923 in Anderson County - 1890 census figures). This is contrasted with the felt area and damage reports from the October 22, 1882 Bonham (Paris) Earthquake (MM VII) (see Figure 2.5.2-3) which attracted accounts from numerous surrounding communities and was felt over an area of 135,000 to 195,000 square miles [78]. Given the felt area for the MM V quakes, reports of an MM VII quake would be expected from many surrounding towns rather than just from the town of Rusk.

The 15 January 1891 edition of the weekly Jacksonville Banner stated "Rusk claims to have felt an earthquake last week." In spite of the fact that Jacksonville is only 12 miles north-northwest of Rusk, the town obviously felt nothing.

Eastern Texas experienced very violent weather conditions during the period January 6-8, 1891. Tornadoes were reported in Queen City, Cass County and Halletsville, Lavaca County, and severe thunderstorms, heavy rainfall, and flash flooding was common. The damage reports in Rusk included damage to chimneys and two tall buildings, the Acme Hotel and the jail, which were one block apart. The damage reports are suggestive of a localized tornado or funnel-cloud that possibly did not quite touch down and furthermore, such a phenomenon would not have been observed at midnight.

The January 8, 1891 Rusk event probably was not an earthquake, but a localized tornado, which would be consistent with the recorded regional weather patterns of that two day period [120]. For the purpose of safe shutdown earthquake determination, however, the event will be treated as an earthquake.

2.5.2.1.3 1930, October 19 - Donaldsonville, Louisiana Earthquake

The October 19, 1930, Donaldsonville Earthquake is the largest reported earthquake to have occurred in the Gulf Coast. The earthquake was not recorded instrumentally in the area. Both the New Orleans, Louisiana and Springhill, Alabama seismograph stations were not recording at the time. The Georgetown University Observatory in Washington, D.C., recorded an earthquake at 6:22 CST. This may have been the Donaldsonville event.

The following sources have been examined regarding this earthquake:

Source	Origin Time	Epicenter Location	Felt Area (Sq. Mi.)	Maximum Intensity
U.S.C.G.S.- Memo to H.F. Reid [24]			12,000	V-VI (RF)
Neumann Ethq. Notes [80]			900	V-VI (RF)
			V (RF) Area	
Reid, H.F. (Notes, File) [24]	06:15:20	30N, 91W	V-VI (RF)	
Neumann & Bodle U.S.C.G.S. [8]		30N, 91W	15,000	V-VII (RF)
Branner & Howell Arkansas Geolgoical Survey [82]			12,000	
Docekal, [69]	06:17	30.08N 90.97W	18,500	VI (MM)
Coffman & Von Hake [83]	06:17	30.0N 91.0W	15,000	VI (MM)
	(RF) Rossi-Forel		(MM) Modified Mercalli	

Regarding this earthquake, Neumann [24] states, “The outstanding feature of the Louisiana shock was the irregular distribution of intensity in the Central zone and the relatively large area of the zone.” Neumann states that these features would probably indicate a deep-seated origin.

Neumann considered the region of greatest disturbance to be an area of about 900 sq. mi. including parts of Assumption, Saint Martin, Iberville, Ascension, St. James, and Lafourche Parishes. Neumann’s isoseismal map is shown in Figure 2.5.2-4. As can be seen, the intensity V (RF) isoseismal includes towns recording intensities of III (RF). An area of 12,000 sq. mi. is included in the IV (RF) isoseismal. Here also are found towns reporting intensities of III (RF) to VI (RF).

A possible explanation for this irregularity of distribution in ratings may be that parts of the affected area contained buildings constructed of poor masonry. Many areas in southern Louisiana are underlain by relatively weak alluvial soils with high ground water conditions. It is interesting to note the sequence of intensity ratings north and south of the intensity V (RF) isoseismal along the Mississippi River. The higher values are found along the river to the north.

The maximum intensity assigned to the Donaldsonville Earthquake has increased over the years. Initially, Neumann [24] assigned a maximum intensity of VI (RF). This corresponds roughly to a MM V-VI. Reid also recorded the maximum intensity as VI (RF) [24]. The official USCGS publication, “U.S. Earthquakes of 1930” by Neumann and Bodle [81], does not give a maximum intensity rating per se but lists the towns of Donaldsonville, Napoleonville, White Castle, Franklin, Morgan City, and Allemands as intensity V (RF) to VII (RF). Docekal [69] and the most recent catalog of NOAA (83), give the value of MM VI.

Damage reports appearing in USCGS, the “Donaldsonville Chief” (84), the “Baton Rouge Times” [85], and the AP wire of October 20, 1930 have been reviewed. The greatest damage occurred in the towns of Gonzales and Napoleonville. These two towns are north and south of the Mississippi (a distance of 12 miles, respectively. At Gonzales some chimneys were cracked almost from top to bottom. Parts of others above the roofline were knocked off. This was reported by Assessor J. Paul Bourgeois. At Napoleonville, a few chimneys were damaged also and a few windows were broken and some furniture moved. These facts suggest a MM VI rating for Gonzales and Napoleonville. At Allemends - 37 miles southeast of Donaldsonville - some pictures fell indicating a MM VI. However, the earthquake itself was reported felt by only a few people. The town of Morgan City, Donaldsonville, Franklin, and White Castel felt the earthquake at MM V level.

2.5.2.1.4 1931, August 16 - Valentine, Texas Earthquake

The Intensity VIII (MM) Valentine event, affecting 450,000 square miles is Texas’ largest historic earthquake. Damage was most severe at Valentine, as the epicenter was nearby at Mount Livermove.

Isoseismal mapping (Figure 2.5.2-5) indicates that damage could have been more widespread in a populated area. Good documentation has been provided by Sellards [30], and by Byerly [31, 32]. Byerly has computed the Richer magnitude of this event as 6.4. The event was felt in Waco, as well as most northern and central Texas cities. For instance, a Waco newspaper reported this earthquake without reference to its origin, giving rise to erroneous suggestion of a Waco earthquake. The epicenter of the earthquake was approximately 380 miles from the CPNPP site

and produced a Rossi-Forel Intensity II-III (Modified Mercalli Intensity II-III) at the site. This event is not associated with structures near the site.

2.5.2.1.5 1932, April 9 - Wortham-Mexia, Texas Earthquake

Regarding the Mexia earthquake, Frank Neumann of the USCGS states [86]:

> "April 9; 4:15. Mexia and Wortham, Texas, MM V. Loose bricks thrown down. Plaster cracked. Damage slight. Also felt at Cooledge, Currie, Groesbeck, Hillsboro, Teague, and Richland. A detailed report by E.H. Sellards was published in University of Texas Bulletin issued in February, 1933."

The report appearing in the Bulletin of the Seismological Society of America (BSSA) stated [87]:

> "Richland, Texas. April 9, 1932. --Windows were rattled and houses shaken by several slight earth tremors felt in Richland, Currie, Wortham, College Station, Mexia, Groesbeck, Teague, and other small towns in the vicinity shortly after 4 A.M. on April 9th. Damage was limited to fallen chimneys."

The report in the Houston Chronicle for Sunday, April 10, 1932 read:

> "Light Earth Tremor Felt Around Mexia"---"More than Seven towns in that Vicinity Feel Shock; Little Damage is Caused by Disturbance, Associated Press. Corsicana, April 9.---Earth tremors were felt shortly after 4 A.M. today at Richland, Coolidge, Mexia, Groesbeck, Teague, and other towns in that vicinity. People were awakened by the rattling of windows and the shaking of buildings. Little damage was reported. The disturbance seemed to center at a point between Wortham and Mexia, where the tremors were most pronounced. The tremors decreased in intensity away from this point. Apparently the disturbance affected an area about 25 miles in diameter. A check was made to determine whether blasting operations could have been responsible, but nothing of this nature was found, so it was decided the tremors were caused by some subsurface disturbance. The affected area was situated along the Balcones fault, well known to oil men. Renewed faulting was believed to have been a possible cause. A report from Mexia said that the tremors were felt at 4:13 A.M. Wortham reported it as occurring at 4:17 A.M. The shocks were but a few seconds duration. Several brick chimneys toppled at Wortham and Mexia, but otherwise damage was negligible."

By far the most detailed account of this earthquake is given by Sellards based on observations he made in the area from May 2-5, 1932. No seismograph recorded this earthquake. Sellards states that the seismographs at Denton, 110 miles northwest of Mexia, and at Austin, 120 miles southwest of Mexia, failed to register the earthquake [33].

Sellards found that most of the damage caused by the earthquake occurred at Wortham. The effect of the earthquake at Wortham "was only to shake bricks from a few chimneys and to enlarge or develop cracks, possibly already present as incipent breaks, in one brick veneer building." He further states that "the most pronounced of these cracks is at the northeast side of the house and runs in a zigzag course from the roof to a window and from the bottom of the window to the concrete baseboard and does not pass through the brick. Some cracks in this building had formed previous to the earthquake."

At the home of a Mrs. Turner, in the southwestern part of town, Sellards found that “four top layers of brick were loosened from the chimney. Twenty four bricks fell to the ground, and the other of these four layers were more or less displaced.” A few bricks fell from other chimneys in the southern and southwestern parts of Wortham. Sellards says that “the bricks in the chimneys affected were probably not firmly held by the mortar, as indicated by the fact that the bricks separated in falling.”

Sellards states that almost everyone in Wortham was awakened and more or less alarmed. “Many ran out of their houses or made examinations to determine the cause of the disturbance.” He ranked the intensity as a V on the Rossi-Forel scale.

Sellards states that outside Wortham the effects were less perceptible. He says that “at Mexia only a few people were awakened. Many were more or less alarmed, but no damage of any kind resulted. At this place the intensity was not above IV (RF).” A Mr. R.A. Liddle gave Sellards a detailed account of his recollecting the event. Mr. Liddle stated that the tremor came from a southeasterly direction; that no sound was heard other than the creaking of the building; and that dishes and stoves rattles. Mr. Liddle further states that “the second movement was several seconds after the first, and was barely noticeable.”

Sellards states that in all directions from Mexia and Wortham, the tremors died out very rapidly. “At Richland, 10 miles northeast of Wortham, the earthquake was felt by only a few persons who were already awake. At Teague, 12 miles east, and at Groesbeck, 11 miles south of Mexia, a slight shock was felt by only a few persons. To the west as far as Prairie Hill, 18 miles, the shock was felt by several persons some whom were awakened.” Finally, Sellards found that the maximum distances at which the shock was at all perceptible to persons awake were Corsicana, 21 miles north of Wortham, Thornton, 19 miles south of Mexia, and Watts, 23 miles west of Mexia. At Fairfield, 19 miles east of Mexia, the shock was not perceptible and at Streetman, 10 miles northeast of Wortham, the shock was “scarcely perceptible.” Sellards’ outline of the sensible area is shown in Figure 2.5.2-6.

Sellards located the earthquake on the Mexia-Wortham faults west of Mexia. The quake was located here because at Tehuacana the shock was fairly strong and because the tremor was sensible for a greater distance west of Mexia than east. The main inconsistency with this location is that Wortham, where the shock was the strongest, is not central to the area in which the earthquake was felt. Possibly if faulting occurred from south to north along fault, the greater intensity at Wortham may be explained by a focusing effect. The USCGS [83] locates the epicenter about eight kilometers east and slightly north of Mexia which is indicated in Figure 2.5.2-6. The only other epicenter location that has been estimated was by Docekal [69], who puts the epicenter 16-17 km almost due south of Mexia, as shown in Figure 2.5.2-6. This location is probably too far south of Wortham and too close to Groesbeck which did not experience any damage from the quake. The USCGS location would more or less agree with Mr. Liddle’s feeling that the shock originated southeast of Mexia [83].

The most likely explanation for the damage experienced at Wortham and the small perceptiple area is that the event occurred at shallow depth. Local geologic conditions may have added to the problem by locally amplyifying or at least failing to attenuate the ground motion.

The most recent intensity assigned to this earthquake is MM V-VI by Coffman and Von Hake [70]. The felt area given is 1,000 square miles in agreement with the area contained within Sellards’ “limit of felt area” of radius 17 miles. A higher intensity of MM VII is given to this earthquake by

Docekal [69]. It apprears that the Docekal value is based on chimney damage alone (see previous description of chimney damage). The felt area given by Docekal is "2,000 sq. mi." [69]. According to Brazee's [88] (Figure 2.5.2-7), and Nuttli and Zollweg's [117] recent work on felt areas for earthquakes in the eastern and central U.S., both of these assigned intensities are too large for the Wortham-Mexia Earthquake. Even a MM V maximum intensity for this earthquake yields a felt area five times that given by Sellards.

The possibility exists that this event may be related to petroleum withdrawal in the area. The first major Texas oil discovery occurred in June, 1894, when the city of Corsicana was drilling a water well. The Mexia Field was discovered in 1920, and the Wortham Field in 1924- 25. The Mexia field is shown in Figure 2.5.2-6. The Mexia Field covers an area of about 3800 acres. Figure 2.5.2-7 shows the annual amounts of crude oil taken from the Mexia field from 1920 up to and including 1932 (data for the years 1927-1929 could not be obtained). For each barrel of crude oil taken out, there are roughly two and one- half barrels of saltwater removed [89].

Two years earlier, Sellards [90] had studied subsidence in the Goose Creek Oil Field in Harris County, Texas. Sellards writes that, unlike the Goose Creek Field where "the geological section includes only relatively incoherent strata, under conditions as at Mexia where the section includes resistant strata, as the limestone of the Midway and Austin formations, subsidence, if such occurs might result in a jar locally producing a tremor of considerable force." Sellards then calculates the total volume of fluid withdrawn from the Mexia and Wortham Fields. This amounted to some four billion cubic feet.

Sellards concludes by saying: "Whether this tremor therefore proves continued activity in the Mexia-Wortham line of faulting or records merely local subsidence in these oil fields incident to the removal of oil is at present undetermined."

In addition to the Corsicana and Mexia-Wortham Fields, there are a number of other oil fields along the Mexia-Talco Fault System. Some of these fields still producing are listed below:

Cumulative Oil Withdrawn
From Discovery to 1/73
by Counties

Discovery Date	Field	County	Area (Acres)	(106 BBLS)
1894	Corsicana	Navarro	4000	192
	Darst Creek	Guadalupe	1900	157
1925	Lytton Springs	Caldwell	2000	
1928	Salt Flat	Caldwell	7000	227
1936	Talco	Titus & Franklin	9500	298

The combined total area of these fields is about 25,000 acres. The counties in which the above and smaller Mexia-Talco Fault Zone fields are located have produced about 875 million barrels of oil to date compared with the 110 million barrels withdrawn from the Mexia-Wortham Fields as of

January 1932, as calculated by Sellards [33]. If the Wortham-Mexia event of 1932 was caused by petroleum withdrawal, it was certainly anomalous in light of the far greater volumes withdrawn from other fields along the fault zone (underlain by essentially similar geological conditions) without any recorded or reported seismic activity.

Sellards [33] states that as of January 1, 1932, 704 x 106 cubic feet of gas had been taken out of the Mexia and Wortham Fields. While the report in the Houston Chronicle [19] states that no blasting operations were in progress at the time of the earthquake, the possibility of an underground gas explosion cannot be ruled out as a possible cause of the disturbance. If faulting were the cause, according to Brazee [88], an area within a radius of some 35 miles should have felt the earthquake at intensity V level if the maximum intensity of the Mexia quake is assigned an MM V rating.

2.5.2.1.6 1952, April 9 - El Reno, Oklahoma Earthquake

The El Reno, Oklahoma event in one of the largest historic earthquakes in Oklahoma. Cloud and Murphy [27] indicate that the event was felt over Oklahoma (except for the panhandle), eastern Kansas, central and northern Texas, western Iowa, Missouri, Arkansas and the southern tip of Nebraska. Slippage along the Nemaha fault about five miles southwest of Oklahoma City is credited by Cloud and Murphy as the source of the event and the felt data conforms to the known trace of the fault. Figure 2.5.2-8 outlines the areas in which the various intensity levels were observed. Maximum intensity was VII although damage was not extensive. Portions of chimneys fell in El Reno and Ponca City and bricks loosened from a building wall and tile facing of commercial buildings bulged at Oklahoma City.

Intensity I-III was reported from Dallas and Fort Worth, Texas for this event. Aftershocks were observed on April 11, 16; July 16, and August 14.

2.5.2.1.7 1959, June 17 - Lawton, Oklahoma Earthquake

Eppley and Cloud [93] report that this shock was felt over an area of about 12,000 square miles in Oklahoma and Texas. Maximum intensity for this event was VI and slight damage consisting of a few reports of cracked plaster and minor pavement cracking was reported.

Intensity levels of I to IV were reported from Dallas and Wichita Falls. The configuration of isoseismals for this event (Fig. 2.5.2-9) exhibit a marked attenuation transverse to geologic structure to the south, in the direction of the site.

2.5.2.2 Geologic Structure and Tectonic Activity

In the following sections all tectonic provinces within 200 miles of the site are identified and described. These provinces are delineated on the basis of 1) basement type; 2) relative isostatic stability of each type; and 3) specific history of stress application within the same crust and tectonic fabric. These major tectonic provinces are further subdivided on the basis of magnitude and mode of crustal deformation, and timing of stress application. Both the timing and magnitude of crustal activity is recorded in the stratigraphic thicknesses and facies distribution throughout the region.

The term tectonic province as used in this study is a region of the North American continent characterized by a relative consistency of the geologic structural features contained therein.

Principal geologic structures and faulting are described with regard to development of the tectonic provinces and relation to existing seismicity.

2.5.2.2.1 Regional Tectonic Provinces

Earthquakes are generated by the release of strain which has accumulated due to regional or local stresses. Sbar and Sykes [95] state that: "In eastern North America intercratonic earthquakes appear to occur in areas of high stress along unhealed fault zones of late Paleozoic or younger age," thus emphasizing the need for a detailed geologic investigation. They describe two types of contemporary stress that are possible along these faults. Remanant stress is defined as "that stress remaining in the rock after the stress applied at some earlier time has been removed." Tectonic stress, on the other hand, can be caused by (1) glacial unloading such as in eastern North American (Woolard [96] (2) sinking of oceanic crust (Sclater, et al. [97], (3) lithostatic loading, such as the Gulf of Mexico (Hubbert and Willis [98], Voight [99]; (4) differences in thickness and density distribution in the continental and oceanic crust at the continental margin (Bott and Dean, [100]); and (5) convective flow in the upper mantle (Voight [99], Morgan [101], Sbar and Sykes [95], among others).

Given these origins of stress and possible location of earthquakes on unhealed faults, it is not possible to make meaningful delineations of tectonic provinces without a detailed study of the geologic history of a given region. In addition, the state of the art geologic concepts learned from detailed studies in other areas of similar geologic fabric and history must be considered. Seven tectonic provinces are delineated within two hundred miles of the Comanche Peak Nuclear Power Plant as indicated on Figure 2.5.2-2. Clockwise from the site these are: 1) the West Central Texas Province; 2) the Wichita province; 3) the Seminole Arch-Nemaha Uplift Province; 4) the Ouachita Province 5) the Balcones-Mexia-Talco Province; 6) the Mesozoic Shelf Province; and 7) the Gulf Coast Province.

These provinces can be grouped according to the history and origin of the underlying crust: (1) Precambrian continental crust which can be undeformed or deformed; (2) deformed Paleozoic oceanic crust; and (3) downwarped Precambrian or Paleozoic oceanic crust. The West Central Texas and Seminole Arch-Nemaha Uplift Provinces are underlain by relatively undeformed Precambrian crust. The Wichita province is underlain by Precambrian crust that has been deformed by the Wichita Orogeny. The Wichita Orogeny occurred during Upper Mississippian through about Mid-Pennsylvanian time and resulted in a series of generally northwest to southeast trending structural features reflecting southwesterly to northeasterly compressional forces.

This major group of Provinces involving Precambrian crust was partially overridden during the Ouachita Orogeny. This orogeny, extending to slightly more recent time than the Wichita and involving compressional forces from the south and southeast, consolidated Paleozoic oceanic crust and its overlying sediments forming the crust that now underlies the Mesozoic Shelf Province [56].

The Balcones-Mexia-Talco Province represents the response to the collapse of the Ouachita Belt perhaps at the Precambrian continental margin [47]. To the south of the relatively stable Mesozoic Shelf Province is the Gulf coast Province. The crust underlying this province has undergone continuous subsidence since its origin [57, 58], receiving more than 60,000 feet of sediments.

The classification of tectonic provinces presented above represents regional interpretation based on gross structural geologic characteristics, crustal structure and geologic history. Each of the major provinces can be subdivided into tectonic subprovinces based on an understanding of more definitive structural criteria and the interaction of structural elements throughout geologic time. Although genetically related to the major province, each subprovince can be defined by its unique structural characteristics.

The provinces and subprovinces shown on Figure 2.5.2-2 are considered sufficiently detailed to allow development of the Safe Shutdown Earthquake for the CPNPP site and are identified along with their principal tectonic components in the following section.

1. West Central Texas Tectonic Province

   The West Central Texas Province is bounded to the north by the Wichita Province and to the southeast by the Ouachita Province.

   The backbone of the province (the Precambrian Crust) crops out in the Llano Uplift subprovince. The other subprovinces are delineated using Upper Paleozoic geography. The paleography includes basins: 1) Delaware; 2) Val Verde; and 3) Midland, making up the basin subprovince; and the Central Basin Platform making up the other subprovince. The area not included in these subprovinces is called the Central Texas Shelf subprovince [59].

2. Wichita Tectonic Province

   The Wichita Tectonic Province lies to the north of the West Central Texas Province and is differentiated from it by structures formed during the Wichita Orogeny. Major structures, folds, and high angle reverse faults strike predominantly northwest-southeast throughout the extent of this province. This province is bounded to the northeast by the Seminole Arch-Nemaha Uplift Province and to the east by the Ouachita Province.

   The Wichita Orogeny formed four subprovinces which include two uplifts and two depressions. The southern boundary is the Matador-Red River-Muenster Uplift subprovince. This is separated from the Amarillo- Wichita-Arbuckle Uplift subprovince by the Palo Duro Basin subprovince. The Anadarko Basin subprovince forms the northern boundary of the Wichita Province.

3. Seminole Arch-Uplift Tectonic Province

   The Seminole Arch-Nemaha Uplift Tectonic Province is a relatively stable area bounded on the south and west by the Wichita Province and to the southeast by the Ouachita Province. This province remained stable while over 28,000 feet of sediments were being deposited in the Anadarko Basin (a subprovince of the Wichita Tectonic Province) and approximately 17,000 feet of sediments were being deposited in the Arkoma Basin to the east the northernmost subprovince of the Ouachita Province. Preconsolidation deformation near Bartlesville, Oklahoma suggests significant past earthquake activity in this area [118].

4. Ouachita Tectonic Province

The Ouachita Tectonic Province represents he inland extent of the complex zone of deformation formed during the late Paleozoic Ouachita Orogeny. This orogenic pulse, which featured compressional forces from the south and southeast, consolidated Paleozoic oceanic crust and it overlying sediments and produced overthrusting along the contact with the Precambrian continental landmass. The Province is bounded on the west by the stable West Central Texas Tectonic Province, on the northwest by the Wichita and Seminole-Nemaha Uplift Tectonic Provinces, and in a gulfward direction by the peripheral fault systems of the Balcones-Mexia-Talco Tectonic Province. It can be differentiated into three subdivisions based upon the relationship of the tectonic belt to underlying structural trends and/or to subsequent structural behavior.

4.1 Exposed Ouachitas

This subprovince contains uplifted and exposed deformed Paleozoic sediments that overrode the Precambrian continental crust from the south. Its northern portion, the Arkoma Basin, contains east-west trending gently folds and faults. The Ouachita Mountains, in the southern portion, feature tight, closely spaced folds and numerous east-west trending high-angle reverse faults and thrust sheets.

4.2 Wichita Ouachita

This subprovince is a transitional zone, where the predominant structural trends change from east-west to northeast-southwest. It also represents the region where the Ouachita Orogenic Belt overrode the southeastward-plunging structures of the Arbuckle Uplift and Muenster Arch portions of the Wichita Tectonic Province which were formed during the earlier Wichita Orogeny.

4.3 Buried Ouachita

This subprovince represents the portion of the Ouachita Tectonic Belt that was placed on the Precambrian continental crust of central Texas then subsequently overlapped and buried by extensive Cretaceous sediments.

5. Balcones-Mexia-Talco Tectonic Province

The Balcones-Mexia-Talco tectonic province and eastern counterpart, which includes the South Arkansas and Pickens-Quitman-Gilbertown Fault Systems, represent a complex system of normal faults that developed at the periphery of the Gulf Coastal Plain in response to the collapse of the Ouachita Orogene. Tensional forces developed during a relaxation that followed the compressional phase of the orogenic cycle, causing the formation of classic post-orogenic block faulting. The inner boundary of this province is defined by the innermost faults of the Balcones system; the gulfward boundary, with the Mesozoic Shelf Tectonic Province, is defined by the outermost faults of the Luling and Mexia-Talco fault systems. Although the deformed Paleozoics of the Ouachita Orogene undoubtedly extend gulfward of the limits of the Ouachita Tectonic Province and probably served as the foundation upon which the Mesozoic Shelf Tectonic Province was developed, it is considered that subsequent to the compressional phase of the Ouachita

Orogeny, tectonic activity was largely confined to the regional zone of weakness defined by he peripheral fault systems. This long, linear province then serves as a logical buffer zone or tectonic boundary between the older Ouachita Tectonic Province and the younger Mesozoic Shelf Tectonic Province.

6. Mesozoic Shelf Tectonic Province

The Mesozoic Shelf Tectonic Province is defined as a geologically stable shelf area that developed on the relatively thick subcontinental crust of Paleozoic sediments deformed during the Ouachita Orogeny. Its inner boundary is defined by the peripheral fault systems that were formed over the Ouachita Tectonic Belt and which are herein called the Balcones-Mexia-Talco Tectonic Province. Movement along these basement-related faults commenced in the late Paleozoic or early Mesozoic [102] and continued intermittently until at least early Tertiary time [15, 63, 103], although some faults of this province seem to displace formations as young as Pliocene [15].

The Mesozoic Shelf Tectonic Province is bounded in a gulfward direction by the Stuart City Reef which marked the lower Cretaceous shelf edge and is located above the zone of crustal transition from relatively thick subcontinental crust in a landward direction to thin oceanic crust in a gulfward direction [104, 105].

The above province and boundary descriptions apply for the overall Mesozoic Shelf Tectonic province. The province, however, contains several major subdivisions.

6.1 East Texas Embayment

The East Texas Embayment is at least as old as the Jurassic and was well developed by the beginning of lower Cretaceous time. The total thickness of Jurassic sediments is unknown, but the maximum thickness of Cretaceous and lower Teritiary sediments (predominantly marine) probably exceeds 15,000 feet [106]. The Houston Embayment is the downdip counterpart of the East Texas Embayment, and the Tyler, or Mineola Basin, is the deepest part.

The axis of the embayment was originally essentially north-south, but movement of the Sabine Uplift during upper Cretaceous time deflected the axis so that the northernmost portion now trends northeasterly [102].

The East Texas Basin contains a series of shallow piercement salt domes which have developed from the Jurassic parent salt unit in response to loading by the thick sedimentary over burden.

6.2 Sabine Uplift

The Sabine Uplift is a broad domal feature that persisted as a stable structural element throughout the Cenozoic [107] as is clearly indicated by the sediments that thin across it. A series of short, discontinuous faults has developed across the uplift and along its flanks. The Sabine Arch is a southward extension of the uplift that persisted throughout the Cenozoic as a stable positive structural element.

6.3 Mount Enterprise

The extent of the Mt. Enterprise subprovince is defined by faults of the Mt. Enterprise System. As defined by Murray [102], these structures bear no presently discernible relationship to other known systems of faulting. Faults of this region typically form graben structures with dips ranging between 35° and 60°. Activity may have commenced as early as Comanchean (lower Cretaceous) time but movements reached a maximum during Gulfian (upper Cretaceous) time and continued at least into the Claibornian (middle Eocene).

The Stuart City Reef marked the Comanchean shelf edge, separating the lagoonal depositional environments of the Mesozoic Shelf from the deep water environment of the ancestral Gulf of Mexico.

In Texas, this reef trend is now buried under more than 10,000 feet of younger sediments but its younger equivalent in Mexico (El Abra) is exposed in outcrop.

7. Gulf Coast Tectonic Province

The Gulf Coast Tectonic Province is divided into an east and west division by the Sabine Arch similar to the Mesozoic Shelf Province. This division separates the two major depocenters of the Gulf Coast Tertiary geosyncline. During the Eocene and Oligocene, the area of maximum deposition (over 50,000 feet) was central and southern Texas but, during the Miocene and Pliocene-Pleistocene, the depocenter (over 40,000 feet) had shifted to a position off southern Louisiana [61].

2.5.2.2.2 Geologic Structures and Faults

In this section the locations of geologic structures and descriptions of the significant faulting in each tectonic province are presented. The significant geologic structures and faults are shown in relation to the tectonic provinces on Figure 2.5.2-2.

1. West Central Texas Tectonic Province

The West Central Texas province is part of the central stable region of the United States founded upon Precambrian crystalline rock and mantled by a thin veneer of sedimentary rock. The crystalline basement is comprised of both igneous and metamorphic rocks similar to rocks of the Canadian Shield.

The central stable region has undergone vertical movements resulting in the formation of broad basins and arches. The arches, basins and other structures of this region formed principally during the Paleozoic era. These structures are rippled and checked by numerous folds, faults and unconformities which constitute important structures for oil and gas accumulation.

The Concho arch, which rose in late Devonian time, is the principal structure of the West Central Texas Basin [108]. The arch consists of a broad gentle uplift in central Texas that extends from the present Llano Uplift northwestward at least 150 miles and may have connected with uplifts in northeastern New Mexico and southern Colorado. West of the arch is the Reagan Range, a small sharp uplift. Immediately west of the Reagan Range

is the Pecos Range which extends northward into Southeastern New Mexico. During Permian time the Pecos Range was known as the central basin platform. Farther to the west is the Diablo Range which parallels the Rio Grande and extends from the west Texas Davis Mountains northwestward to New Mexico.

The Llano Uplift and its northeastward extension, the Bend Arch came into existence in Cretaceous time but had their beginnings toward the end of the Pennsylvanian.

Major basins developed between uplifts at different times. The Fort Worth (Strawn) and Kerr basins are chiefly Pennsylvanian depressions on either side of the Concho arch. The basins and platforms of west Texas formed in Permian time.

Faults of the Llano Uplift are known from surface mapping [111]. Well and geophysical data have extended the known length of these faults more than 100 miles to the north-northeast into the Strawn basin. The faults are probably high-angle and normal and have blocked out narrow grabens and horsts. These high blocks from east to west are the Richland Springs, Pototac, San Saba and Lampasas arches. Geophysical data from the Fort Worth basin has shown that some of these faults have displaced the Smithwick (early Pennsylvanian) formation at 1100 feet. However, middle Strawn beds only slightly younger in age are only slightly disturbed along a major fault near Regency in the Colorado River area [108].

No evidence of this faulting is known in younger Pennsylvanian beds. Therefore the faulting in the Llano Uplift and the Strawn Basin occurred in early Pennsylvanian time or earlier [2,108].

2. Wichita Tectonic Province

Geologic structures of the Wichita Tectonic Province are the Wichita Mountains, the Amarillo Uplift, the Criner Hills, the Red River Uplift, the Arbuckle Mountains, and the Anadarko and Ardmore Basins (Figure 2.5.2-2).

The Amarillo Uplift, the Wichita Mountains and Criner Hills comprise a single tectonic unit extending some 330 miles from Southwestern Oklahoma into the northwestern Panhandle of Texas. Precambrian granites form the cores of the Wichita mountains. The Amarillo Uplift is a series of buried hills without surface expression also composed of granite and crystalline rocks overlain by Permian and Pennsylvanian strata and bounded by an echelon faults. The Criner Hills are a complexly-faulted horst exposed at the surface and considered a southeasterly continuation of the Wichitas.

The Red River uplift is comprised of narrow east-west buried granite hills which extend from the New Mexico line across the Llano Estacado to Wichita Falls and then to the southeast. Well-expressed faults and folds trend obliquely across the uplift in a northwest direction producing an echelon offsets of the buried granite ridges. The western portion of the Red River Uplift is known as the Matador Arch, the east central part as the Electra Range, and the eastern end as the Muenster Range.

The Arbuckle Mountains are the surface expression of a complex anticline between Davis and Ardmore. The core of the anticline consists of two prominent peaks of Precambrian rocks with folded sequences of Precambrian to late Pennsylvanian rocks exposed in

range. Major features of the series of northwest-southeast-trending faults and folds comprising the Arbuckles from north to south include the Hunton Arch, the Mill Creek syncline, the Tishomingo anticline, the Wichita syncline, and the Arbuckle anticline.

The Ardmore basin is a west-northwestward-trending trough between the Criner Hills on the south and the Arbuckle Mountains on the north and contains over 30,000 feet of Paleozoic sediments.

North of the Wichita Mountains lies the Anadarko basin which occupies much of western Oklahoma. The axis trends west-northwest and approximately parallels the Wichita-Amarillo Ranges.

The early Paleozoic Wichita geosyncline was compressed into northwest fault blocks and folds of the Wichita and Red River uplifts in early Pennsylvanian time; and then again in late Pennsylvanian time the Arbuckle Mountains were formed by north-south compression. This complex of related chains is known as the Wichita system [108].

3. Seminole Arch-Nemaha Uplift Tectonic Province

The Nemaha Range, a sharp uplift, trends south-southwest from Omaha, Nebraska into northern Oklahoma. The range is buried beneath a considerable thickness of Paleozoic strata. Uplift and dissection were sufficient to expose Precambrian crystalline rocks before burial. The east flank is bounded by the Nemaha which extends into the Wichita trend to the south [83].

The Seminole Arch extends northward from the Arbuckle Mountains. Both the Seminole and Nemaha structures appear to be block-faulted. All of these geologic structures are located over 170 miles from the site.

4. Ouachita Tectonic Province

Four independent thrust sheets form the principal tectonic structures of the Ouachita Tectonic Province. From northwest to southeast these are the Choctau, the Pine Mountain, the Ti Valley fault and the Windingstair faults. Each of these sheets has been thrust from south to north and has been broken by numerous smaller high-angle reverse faults that probably join the main thrusts at depth. Larger cross faults are occasionally present with numerous minor cross faults. The major thrusts of the Ouachita system have been identified in eastern Oklahoma but appear to die out to the east in the shale sequences. None of these thrust faults are presently known to be active. A zone of open folds lies to the north and comprises the Arkoma Valley basin.

5. Balcones-Mexia-Talco Province

The Balcones-Mexia-Talco faults are a complex assembly of faults that follow approximately the border of the Tertiary and Cretaceous formations of the Gulf Coastal Plain and are probably related to collapse of the continental margin on the seaward side of Paleozoic orogenic belts [47, 113].

It is believed that the fault systems, generally categorized as the Balcones, Luling-Mexia-Talco, and Charlotte-Fashing fault zones, have basement control or deep

tectonic involvement, and that thege fault systems are formed over the original zone of weakness that allowed the Ouachita Geosyncline to form along the edge of the Texas Craton [112].

Wells drilled in the general area of the Balcones fault system encountered beneath the Cretaceous a sequence of steeply-dipping clastic sedimentary rocks showing varying degrees of weak metamorphism [3]. Near the beginning of the Miocene period (about 22.5 million years ago) the Coastal Plain strata faulted and slumped towards the Gulf at an increased rate of speed [15]. This was due to a readjustment of the earth's crust caused by sinking of the Gulf Coast region under the heavy load of Cretaceous and Tertiary sediments. The Balcones fault acted as the hinge over which the sediments attempted to stretch [63]. The movement on the Luling-Mexia-Talco fault system was likely due to readjustment incidental to the Balcones faulting and the consequent dip towards the Balcones zone.

The Balcones fault zone forms a great arc, convex gulfward. Its strike is easterly in Kinney, Uvalde and Medina Counties and changes to northeasterly and then north-northeasterly across Bexar, Comal, Hays, Travis, Williamson, Bell, and McLennan Counties.

Faults in the zone are all normal and trend mostly down to the coast, with antihetic faults that form grabens. Distinct faults within the zone are partially an echelon. The average dip of the fault is 45 degrees which generally decreases with depth [15, 47]. It is believed that these normal faults penetrate the basement, beneath Cretaceous and Tertiary sediments. A typical cross section through the Balcones Fault Zone is shown on Figure 2.5.2-11.

Displacements of up to 1,500 to 1,700 feet occur across the Balcones fault zone. The latter figures are for Bexar County in which maximum displacement has been reported. Northeast and southwest of Bexar County the displacement diminshes in magnitude. Individual faults may have any amount of displacement up to 700 feet. Topographic relief along the fault also varies from extensive to none. A prominent escarpment exists throughout most of the area where more resistant Lower Cretaceous strata have been uplifted [47].

The Luling-Mexia-Talco Fault Zone probably represents complimentary (antithetic) up-and-down-to-the-coast movements that developed in response to the down-to-the-coast movement along the Balcones Fault Zone [64]. A typical cross-section through the fault zone is shown in Figure 2.5.2-12. The Balcones and Luling-Mexia-Talco fault zones, although related to the action of the Gulf Coast Geosyncline, are not considered growth faults as are the younger more coastward faults. Although they exhibit some of the same characteristics as growth faults (i.e., decreasing dip with depth) they are believed to have basement control as opposed to the more shallow and younger faults in the thicker sediments of the Gulf Coast Geosyncline [47].

Both the Balcones and the Luling-Mexia-Talco fault systems pass within 200 miles of the site. The Balcones system, the nearest, passes within 50 miles of the site and is situated just east of the Ouachita Tectonic front. The Luling-Mexia-Talco Fault system passes within 80 miles of the site to the east and generally is parallel to the trend of the Balcones system; both fault systems are shown on Figure 2.5.2-13.

The possibility of the Balcones and the Luling-Mexia-Talco fault systems being active according to 10CFR100, Appendix A, has been carefully considered. Details in the literature, field observations by prominent earth scientists, the opinions of other experts and our own assessment of field conditions have all lent appropriate weight in making a judgement. Insufficient evidence exists to warrant classification of these fault systems as active along any portion of their known limits.

The 300-mile-lone Balcones fault zone passes closest to the site in Hill County at which location it is 50 miles southeast of the site.

An examination both of areal photographs and field conditions was conducted in order to evaluate the possibility of activity along the nearest portion of the fault zone. These efforts revealed no sound evidence indicating any degree of activity. In many cases, faults within the zone have brought earth materials of contrasting physical properties into juxtaposition. Problems of civil structures stem from this cause or others, rather than from displacement along the failure (fault) plane.

The investigation of possible activity included a thorough field inspection of both the closest approach of the fault system to the site and the closest municipalities to these points along the fault. The purpose of the latter investigation was to obtain evidence that would not be apparent in an undeveloped area; evidence such as distressed buildings and streets, utility line maintenance records, etc. This field investigation also included interviews with individuals who had knowledge of the fault system, including academic personnel (retired and active), petroleum companies, government and private consulting geologists. In addition, Soil Conservation Service personnel were questioned about any anomalous field observations they might have observed over the years. Public works employees in Waco, Hillsboro and Austin were interviewed, including water, sewer, and street maintenance personnel. County and state highway maintenance personnel were also interviewed. The list of personnel contacted is provided in the references. This investigation effort failed to reveal any evidence of fault activity. In addition to the intensive investigation of the fault in the site region an investigation was made to obtain evidence of activity along the entire length of the fault. Again, no evidence of activity was found.

Activity along the Balcones Fault Zone may have started as early as Jurassic time (195 million years ago) and persisted through the Tertiary. Nowhere is there evidence that terraces of Pleistocene age (1.85 million years) or younger have been displaced through faulting.

Accounts do appear in the geologic literature that appear to indicate historic movements in the ground surface, across highways, archaeological digs, and pipelines.

Bryan [36 and 37] indicated that sewer and water mains in McLennan County were severed where they crossed faults in the Balcones system. Investigation of this reported occurrence conducted in Hill, McLennan and Travis Counties and interviews with a geologist familiar with other portions of the Balcones fault system failed to disclose evidence for movements across highways or pipelines along faults within the system.

Weeks [15] cities “breastworks” near Lexington in northern Lee County as possible evidence for relatively recent movements along a portion of the Luling-Mexia-Talco Fault

Zone. He describes these “breastworks” by saying that the upthrown side of the fault stands a few feet higher at the surface than the downthrown side. However, in retrospect, Weeks says that the features he observed by chance in the field in 1945 could very likely be attributed to sources other than fault breakage, such as differential settlement or differential weathering along the fault zone (Weeks, personal communications).

Wiggins [39], in describing the geology of the Ham-Gosset Oil Field, noted disturbance of roads in the Luling-Mexia-Talco fault zone, and suggested that this represented surface fault rupture. During recent conversations with the author (Wiggins, personal communication) he indicated that several times since publication of the paper he has endeavored to confirm his initial hypothesis, but has not been able to do so.

Bell and Brill [38] reported a persistent crack in sandy soil in eastern Lavaca County over the Mirando-Provident City fault zone, and Sellards [33] reported a crack near Wortham; however, Lavaca County is over 200 miles from the site and it was not possible to establish the validity of the observations as clear proof of active faulting; or of oil field subsidence.

Seismic activity of any significance is not normally associated with faults in the region.

An earthquake had been reported in the literature [36] as having occurred along the Balcones fault in or near the city of Waco on August 16, 1931. However, neither of the Waco newspapers of the time (Waco News Tribune and Waco Times Herald, August 17, 1931) reported any local tremors, although the August 16, 1931 earthquake of Valentine, Texas received abundant press coverage in Waco. The Valentine event is the largest to have been recorded in the state, and it was felt in Waco. See the dicussion of important earthquakes in Section 2.5.2.1.

Slow, continuous, viscous movement of soil or rock masses without seismic disburbance is commonly referred to as “creep” or “tectonic creep” if it can be demonstrated that releases of tectonic stress have been the cause rather than other stresses associated with activities of man.

In the vicinity of the city of Houston it is known that creep is occurring now along normal down-to-the-coast faults. The question therefore arises as to whether this coastward activity can be related to major tectonic structures far removed from the coast as are the Balcones and Luling-Mexia-Talco fault zones [40, 41, and 42].

As stated previously, creep is probably caused by shearing and plastic deformation along the Louann Salt. From Figure 2.5.2-10 it is evident that the geographic distribution of the Louann Salt is such that it plays no part in the tectonics of the Balcones and Luling-Mexia-Talco fault systems. For these structures, rather, it is likely that there is basement control.

6. Mesozoic Shelf Tectonic Province

The Mount Enterprise Fault System refers to a restricted series of faults in the southern part of the East Texas Embayment in Anderson, Cherokee, Rusk and Panola Counties. This includes the Mt. Enterprise, Holly Grove, Jacksonville, Elkhart, Medford, Elkins and other un-named faults lying along this trend in the counties mentioned.

The Mount Enterprise Fault system lies at the southern rim of the East Texas Basin on the western edge of the Sabine Uplift. The system represents a series of narrow graben structures that developed in repsonse to tensional forces caused by the arching of sediments across a buried spur of the Sabine Uplift and occupies the southern limit of the East Texas Embayment. Aeromagnetic data reveals a basement ridge beneath the Mt. Enterprise Fault System.

The time frame of the faulting is consistant, in that post-Claibornian (Eocene) faulting took place after the rapid downwarping of Claibornian time during which the sediments reached maximum thickness. The Mt. Enterprise faults therefore are actually related to downwarping of the Tyler Basin and independent of other regional fault systems.

The closest approach of the Mt. Enterprise Fault system to the site is 125 miles.

7. Gulf Coast Tectonic Province

Coastward of the Balcones-Mexia-Talco system of faults lies a series of major down-to-the-coast faults with compensating up-to-the-coast faults that trend roughly parallel to the Gulf of Mexico. Included in this system are the Wilcox (Mirando-Provident City), Vicksburg (San Fordyce-Vanderbilt), McAllen and Williaman fault zones of Texas; the Bancroft Fault Zone of Texas and Louisiana; and the Mamou, Tempetate- Baton Rouge, Lake Arthur and Grand Chenier fault zones of Louisiana.

In contrast to the older fault systems, the younger faults in the Gulf Coast Tectonic province are not believed to have basement control, but are believed to have decreasing dip with depth and die out in incompetent strata which underlie the coastal sediments. These faults are normal down-to-the-coast faults and many have been termed “growth faults”.

The Coastal Plain is a wedge of sediments which thickens outwards towards the coast. This column of material is likely to be still undergoing consolidation and, therefore, subsidence. At depth there are incompetent strata which dip towards the coast in conformity with the rest of the sedimentary strata.

Perhaps the weakest of such beds is the Louann Salt of Jurassic Age, the distribution of which is shown on Figure 2.5.2-10. The Coastal Plain sediments are believed to move toward the Sigsbee Deep in the center of the Gulf of Mexico under the influence of gravity. The rate of flow of the incompetent bed, being greater than that of the overlying beds, creates tensional stresses in the overburden. Shearing as well as plastic deformation takes place in these beds. This leads to movement along what are termed “growth faults”.

There is clear field evidence to indicate that the growth faults increase in amount of displacement and decrease in dip with depth. Evidence strongly suggests that they flatten to a bedding-plane fault where they reach a sizeable incompetent bed. They do not appear to extend into the basement complex. Detailed seismic work recently undertaken offshore by the major oil companies has disclosed the presence of parallel ridges which in all likelihood represents the “toeing out” of these large mobilized blocks which may be likened to immense landslides.

This “shallow tectonic” activity does not lend itself readily to earthquakes because of the great thickness of soft unconsolidated sediments which behave plastically, absorbing the strain.

The closest approach of the Gulf Coast Tectonic province to the site is 180 miles.

2.5.2.3 Correlation of Epicenters with Geologic Structures or Tectonic Provinces

The relationship of epicentral locations of earthquakes to tectonic features is discussed in this section. An attempt is made to associate historic earthquakes with active tectonic structures, such as faults or orogenic belts. The procedure adopted for establishing the distance of possible future events from a site is to assume that past events associated with an active structure can occur anywhere along that structure, including its point of nearest approach to the site. In the absence of such relationships earthquakes are regarded as random within the tectonic province with which they are associated.

Figure 2.5.2-2 was prepared primarily to relate historical seismic events to tectonic structures or tectonic provinces based on careful consideration of geologic evidence including geologic history. The classification of tectonic provinces represents regional interpretation based on gross structural geologic characteristics, crustal structure and geologic history. Each of the major provinces can be subdivided into tectonic subprovinces based on an understanding of more definitive structural criteria and the interaction of structural elements throughout geological time. Although genetically related to the major province, each subprovince can be defined by its unique structural characteristics.

Detailed discussion of the tectonic provinces and their subdivisions which are pertinent to the development of the SSE for the CPNPP site are presented in Section 2.5.2.2.

The locations of all recorded earthquakes in Texas and adjacent areas (Modified Mercalli intensity IV or greater) are shown on Figure 2.5.2-2.

2.5.2.3.1 West Central Texas Tectonic Province

All recorded events, except one, have occurred in the Central Basin Platform subprovince. These events are all over 200 miles from the site and in each case appear to be related to local basin formation.

One Intensity IV-V has been reported in the Basin subprovince. This event occurred in an area of recurrent seismic activity and block faulting on the Central Basin Platform. This activity may be associated with petroleum production. No events have been reported in the Central Texas Shelf subprovince, in which the site is located.

2.5.2.3.2 Wichita Tectonic Province

There has been one Intenisty VII event within the Wichita Province and several V to VI events. The largest event occurred in the Anadarko Basin subprovince near the border of the Seminole-Nemaha Uplift Province. This event has been attributed to the Nemaha fault [27] but may well be more characteristic of the Anadarko basin. Recurrent seismic activity is associated with this area as revealed by earthquakes in 1918, 1929, 1933, 1952, and 1953.

The concentration of events in the Amarillo Uplift area is likely attributable to petroleum production, or associated water flooding [25]. At least one event, however, near Lake Meredith, may be the result of crustal loading by this body of water [25].

There is also some possibility that mountain building in the Wichita mountains is still continuing.

The nearest event to the site in the Wichita Province is an Intensity IV, 70 miles from the site and located on the Munester Arch.

2.5.2.3.3 Seminole Arch-Nemaha Uplift Province

This province is just over 190 miles from the site and has experienced no event larger than Intensity VII. Recurrent seismic activity appears to be confined principally to buried block faults in the Precambrian basement along the Nemaha and Seminole arches.

The closest event to the site occurred in 1959 with an intensity of V.

2.5.2.3.4 Ouachita Tectonic Province

This province has experienced twelve events of Intensity V, two of Intensity VI, and one of Intensity VII.

The 1882 Intensity VII event near Paris, Texas has been associated with the buried Arbuckle Mountains under the Ouachita folded belt in the northern portion of he Wichita-Ouachita subprovince [65]. The Intensity VII event occurred approximately 155 miles from the CPNPP site.

Minor seismicity appears to be spatially associated with the thrust faults of southeastern Oklahoma. South central Arkansas also has experienced a significant amount of seismic activity which suggests a continuation of the thrusts of southeast Oklahoma to the east in Arkansas.

2.5.2.3.5 Mesozoic Shelf Tectonic Province

Several earthquakes have been reported in the Mesozoic Shelf Province, with the largest being an Intensity VII near Rusk, Texas. The Rusk event occurred in the Mt. Enterprise subprovince which contains a discontinuous series of narrow grabens representing response to tensional forces generated by a draping of sediments across the southern rim of the East Texas Embayment.

Several earthquakes have occurred in or bordering on the area of the Sabine Uplift. The closest to the site is over 150 miles. These include a 1940 MM IV event in extreme northwestern Louisiana and a series of four events in 1957 in northeastern Texas, all of which are located in the proximity of the Rodessa Fault Zone which is located in the northwestern flank of the uplift. Other events associated with the Sabine Uplift are the series of quakes ranging from MM IV-V that occurred in the Pineland-Hemphill area along the Texas-Louisiana border in 1964. These events are located on the south flank of the Sabine Uplift and at the juxtaposition of the southward trending Sabine Arch with the eastward trending gulfward edge of the Mesozoic Shelf Tectonic Province.

2.5.2.3.6 Balcones-Mexia-Talco Tectonic Province

This province has experienced very little activity, the largest event being an Intensity V-VI near Mexia, Texas in 1932. This event was felt in an area of only 1,000 square miles indicating a very shallow focus [33,61]. Sellards [33] states that the focus for the event was in the Mexia-Talco Fault Zone, west of the town of Mexia. There is also the possibility that the event was caused by oil and gas in the Mexia Field.

2.5.2.3.7 Gulf Coast Tectonic Province

The Gulf Coast Province has experienced several events of low intensity (IV-VI) with the larger events occurring over 200 miles from the site. No major activity has been reported in this province and, due to its distance from the site (approximately 183 miles), it is not significant to the CPNPP site.

The largest historic event (MM VI) occurred near Donaldsonville, La. at a distance of approximately 425 miles from the site. This event is believed to be associated with crustal loading resulting from the deposition of vast quantities of Mississippi River sediments gulfward of that area.

2.5.2.3.8 Far West Texas

In far west Texas, several earthquakes have been recorded, including the highest intensity disturbance experienced in the state (Valentine earthquake of Intensity VIII in 1931). These earthquakes occurred in a heavily faulted area along the southwest margin of the Marathon Basin. Structures there have a northwest orientation, normal to the usual strike of Balcones-Luling-Mexia fault systems. This northwest trending structure has recently been termed the “Texas Lineament.” The structure has not been precisely defined, but possibly represents a late Paleozoic fault zone along which significant right lateral displacement has occurred. The presence of several thermal springs in the vicinity of the Lineament may be noteworthy as they are the only occurrences of thermal springs in Texas [34, 35]. Frequently, thermal springs are associated with active tectonic structures. Their absence over the remainder of the state may be significant. The Texas Lineament is nowhere presumed to be closer than 250 miles to the site.

2.5.2.4 Maximum Earthquake Potential

The historic earthquakes of greatest intensities applicable to site seismic design criteria are the 1882 Intensity VII Paris earthquake, the 1891 Intensity VII Rusk earthquake, and the 1932 Intensity V-VI Mexia earthquake. The only larger event in Texas is the 1931 Intensity VIII Valentine earthquake in west Texas; however, the epicenter is 400 miles from the site and not within a tectonic province within 200 miles of the site. A recurrence of the Valentine event would not result in significant ground motion at the site.

In order to evaluate the effect of earthquakes in surrounding provinces on site ground motion, conservative attenuation curves (based on past Texas and Oklahoma earthquakes) were developed. Attenuation curves were developed by using the maximum epicenter-isoseismal distance for each of three historic earthquakes. The curves are based on the 1882 Paris, the 1931 Valentine, and the 1952 El Reno earthquakes. These curves are shown on

Figure 2.5.2-14. The Bonham, Texas and El Reno, Oklahoma events were used for the Intensity VII curve and the Valentine event for the intensity VIII.

No preferential attenuation or areal perceptibility considerations were employed to increase the attenuation characteristics observed in order that the curves be adequately conservative. Comparison of the attenuation-distance relations shown in Figure 2.5.2-14 with similar relations proposed by Gupta and Nuttli [121], Howell and Schultz [122], and Cornell and Merz [123] indicates that in each case the observed attenuation does not exceed that recognized by the authors.

Using these curves and the conservative acceleration versus intensity relation proposed by Trifunac and Brady [62], the effect of earthquake activity in each tectonic province and subprovince was evaluated. Mean values of peak acceleration [62] were used for evaluation of accelerations from each earthquake intensity after conservative attenuation to the site. Resulting acceleration values, presented in Table 2.5.2-3, are considered very conservative.

Where events cannot be assigned to a specific structure in a subprovince, they are assumed to occur at the nearest point of that subprovince to the site. Using this procedure, the highest intensity possible at the site is a moderate IV resulting from a Modified Mercalli Intensity VII assumed to occur at the closest approach (70 miles NE of the site) of the Wichita-Ouachita Subprovince to the West Central Texas Tectonic Province. This results in a ground acceleration of 0.10 g at the site.

Good agreement is found using the method described by Howell [109] in which an Average Regional Seismic Hazard Index (ARSHI) for the area is defined between 5.22 and 5.70. This method is based on seismic energy release. Since ARSHI values are identical to intensity for any single earthquake a measure of conservatism is therefore demonstrated by choosing a moderate intensity VI as the maximum potential earthquake.

Consideration also has been given to on-site ground motion related to larger historical events originating beyond 200 miles and extrapolated to the site. The largest known events would not have shaken the site at intensities higher than III or IV. This would be the level of the historical events of a New Madrid shock [115, 116] migrated closer along related structures southward, or a Valentine event migrated closer southeastward along related structure.

In summary, the maximum ground motion which conservatively should be postulated for a Maximum Potential Earthquake would result from the occurrence of an earthquake equal to the 1882 Intensity VII shock near Bonham, Texas. Ground motion from this event, minor local events, or large distant shocks would produce horizontal ground motion at the site less than or equal to 0.10 g.

2.5.2.5 Seismic Wave Transmission Properties of the Site

To determine seismic wave transmission and other physical properties of the subsurface material, the following geophysical studies were conducted at the Station and Safe Shutdown Impoundment Dam Areas:

1. Seismic refraction surveys to evaluate the compressional waver velocities of bedrock and the materials overlying bedrock and to evaluate the depth to bedrock;

2. Uphole velocity surveys to provide additional data regarding compressional wave velocities of bedrock and materials overlying bedrock;

3. A surface wave survey to evaluate types of surface waves and their characteristics;

4. Preexcavation crosshole shear wave surveys to evaluate shear and compressional wave velocities of bedrock and the materials overlying bedrock;

5. Postconstruction crosshole seismic wave survey to evaluate shear and compressional wave velocities of a 500-ft thick bedrock interval below plant-grade, with a particular emphasis on claystone interbeds;

6. Ambient noise studies to evaluate the predominant characteristics of groundmotion due to background noise level; and,

7. Gamma ray and electrical resistivity logs in the Squaw Creek Dam site borings, and gamma-ray, gamma-gamma (density), electrical resistivity, and caliper logs in borings S-1A, L-1, and L-2 at the Station site to evaluate strategicgraphic features and obtain rock physical properties at various depths.

Details of the geophysical surveys and results are tabulated and discussed in Section 2.5.4.4.

The properties of the materials underlying the site are described in Section 2.5.4.2. All major structures will be founded on sound bedrock. The low level of anticipated as well as past seismic activity at the site is further improved by the conservative assumptions applied to identify the maximum potential earthquake for design purposes. Therefore, detailed analysis of the seismic wave transmission characteristics of the site is not considered necessary.

2.5.2.6 Safe Shutdown Earthquake (SSE)

A conservative Safe Shutdown Earthquake having a peak horizontal ground acceleration at the top of bedrock of 0.12 g has been selected for design.

Design response spectra and artificial time-history records are described in Section 3.7.1. SSE selection has been based on a conservative evaluation of the level of ground motion that is remotely possible considering seismic and tectonic history and geologic structure.

2.5.2.7 Operating Basis Earthquake (OBE)

In accordance with 10CFR100, Appendix A, the Operating Basis Earthquake is defined to be equal to 1/2 the SSE (i.e. the horizontal ground acceleration at the top of the bedrock has been selected to be 0.06 g, 1/2 the SSE acceleration).

Using the seismic source areas proposed by Algermissen and Perkins [110] the horizontal acceleration in rock with ninety percent probability of not being exceeded in fifty years is five percent of gravity (0.05 g) for southeast central Oklahoma and correspondingly less for the site vicinity. Based on the probabilistic assessment and the historical record it is unlikely that any acceleration corresponding to the OBE would be experienced during the life of the facility.

The Seismic System Analysis (Refer to Section 3.7.2) utilized OBE=1/2 SSE for all stress and deformation criteria.

2.5.3 SURFACE FAULTING

No evidence of surface faulting was found within five miles of the site. For a detailed description of the site geology see Section 2.5.1.

2.5.4 STABILITY OF SUBSURFACE MATERIALS & FOUNDATIONS

2.5.4.1 Geologic Features

Geologic considerations relating to stability of subsurface materials are described in Section 2.5.1.2.6.3. In support of the discussion presented in Section 2.5.1.2.6.3, no evidence of solutioning or past solution activity has been observed in any excavations made in the unweathered Glen Rose Limestone.

The geological history of the site area is presented in Section 2.5.1.2.2.

The information pertaining to the site foundation medium, particularly the data generated from the postconstruction S/L cluster borings (S-1, S-1A, L-1, and L-2) by a detailed seismic crosshole survey, geophysical logging, continuous coring, and dynamic testing of representative core samples [131, 132], was used to divide the subsurface materials underlying the site to a depth of 500 ft into intervals on the basis of their lithologic characteristics and, consequently, engineering properties. These intervals are identified in Table 2.5.4-5E by stratigraphic units, prevalent lithologies, and ranges of thickness, depth and elevation. The depths and elevations shown on Table 2.5.4-5E apply to the vicinity of the power block area. Due to the slight eastward dip of the strata (Section 2.5.1.2.4), these depths and elevations differ slightly for other locations.

2.5.4.2 Properties of Subsurface Materials

The subsurface materials of Cretaceous age, which underlie the site to a depth of apporximately 444 ft (elevation 366 ft) have been sampled and laboratory tested for a wide range of physical and chemical properties. The materials of the Glen Rose Formation, extend from elevation 810 ft (plant grade) to elevation 610 ft and consist of light-to-dark gray argillaceous limestone with lenses and zones of gray to greenish-gray calcareous claystone. The materials of the Twin Mountains Formation, extend from elevation 610 ft to elevation 366 ft and consist of interbedded claystone and sandstone sequences.

The static engineering properties of the subsurface materials underlying the site to a depth of about 50 ft (from plant grade at elevation 810 ft to elevation 760 ft) were determined in the preexcavation stage of the site geotechnical investigations. These properties are discussed in Section 2.5.4.2.1. The dynamic engineering properties of the subsurface materials underlying the site were studied in both the preexcavation and the postconstruction stages of the site geotechnical investigations. The preexcavation investigations were conducted at a number of locations, and were effectively limited to a depth of 200 ft (although some boreholes were drilled to a maximum depth of 400 ft). The postconstruction investigations reached a depth of 500 ft at the location represented by the S/L cluster borings. The dynamic engineering properties are discussed in Section 2.5.4.2.2.

2.5.4.2.1 Static Engineering Properties

The static strength and other physical properties of the upper portion of the Glen Rose Formation (Zone I and the top of Zone II) were determined in the laboratory by means of the following types of tests:

1. Unconfined compression tests.

2. Bulk modulus of elasticity determination.

3. Double ring shear test.

4. Brazil tension test.

5. Unit dry weight determinations.

6. Poisson's Ratio determination.

A considerable number of selected representative specimens of the rock formation were subjected to unconfined compression tests. The unconfined compression tests were conducted in accordance with standard ASTM Procedures. On a number of the samples, vertical strain was measured by means of a dial micrometer in order that the stress-strain (bulk modulus) relationship could be determined. The results of the unconfined compression tests are shown on Table 2.5.4-1. The results of the stress-strain (Young's modulus) observations made on the selected specimens during the unconfined compression test procedure are shown in graphical form on Figures 2.5.4-1 (Sheets 1 thru 25).

Triaxial compression tests were conducted on selected claystone samples. A listing of the samples tested is given below:

Boring No.	Sample No.	Depth, Feet	Lab No.	Length, Inches	Diameter Inches
P-25	W-34	76.0-77.0	1	4.465	2.010
P-25	W-34	76.0-77.0	2	4.350	2.018
P-25	W-28	64.0-65.0	3	4.500	2.045
P-25	W-28	64.0-65.0	4	4.395	2.048
P-25	W-73	154.0-156.0	5	4.345	2.065
P-25	W-73	154.0-156.0	6	4.638	2.065

Each sample was subjected to a back pressure of 200 PSI and the claystone was essentially saturated as full pressure response was noticed during the test. Higher saturation pressure (greater than 200 PSI) was not desirable since such greater pressures accelerate consolidation and reduce permeability. A typical stress-strain curve (Lab Number 4) is shown on Figure 2.5.4-2 and a Mohr Diagram for each of the samples tested is shown as Figure 2.5.4-3.

The ratio of lateral strain to vertical strain, Poisson's Ratio, was determined on two selected typical samples of claystone and three selected typical samples of limestone. Both directions of strain were measured using SR-4 Resistant Wire Strain Gauges attached to the surface of the sample by means of appropriate adhesives. As many as four cycles of axial load application were made in order to fully investigate stress-strain characteristics of the typical materials. The results, in graphical form, of these determinations are shown by Figures 2.5.4-4 through 2.5.4-13, inclusive.

The double ring shear strength of selected typical rock samples was determined in pre-chosen direction essentially parallel to the bedding planes. The double ring shear test apparatus consists of three parallel steel plates with openings of a size very slightly greater than the sample diameter. Two of these steel plates are supported on horizontal base and, in turn, support the rock specimen. The third plate or ring, which is sandwiched between the outer plates, acts as a loading device and is allowed to react with the specimen. The results of these determinations are shown on Table 2.5.4-2.

The tensile strength of selected and typical rock materials were determined by the Brazil tension test. These tests are accomplished by placing a rock specimen horizontally between two bearing plates and loading in compression until failure occurs in tension. The results of these determinations are shown in summary form on Table 2.5.4.-3.

Table 2.5.4-4 is a summary of the static rock properties for the upper portion of the Glen Rose Formation on which the major plant structures were found.

2.5.4.2.2 Dynamic Properties of Subsurface Materials

The dynamic properties of the subsurface materials were determined by field geophysical surveys conducted in both the preexcavation and the postconstruction stages of the site geotechnical investigations, and by laboratory testing of rock core samples obtained from deep postconstruction borings S-1, S-1A, and L-2 which were drilled together with boring L-1 for a detailed crosshole seismic survey. The scope and the results of the geophysical surveys are presented in Section 2.5.4.4. Discussion of the laboratory testing is included in this section.

The laboratory testing program included measurement of unit weights and performance of resonant column testing (which provided shear moduli and damping properties). Out of 79 core pieces delivered to the laboratory, 64 were tested; all of them for unit weight, and 17 were subjected to resonant column testing. These samples were taken from the S/L cluster borings between elevation 809 ft and elevation 428 ft. Within this depth interval of 381 ft, 30 samples were obtained from the Glen Rose Formation (17 claystone, 12 limestone, and 1 sandstone sample), and 34 samples were obtained from the Twin Mountains Formation (11 claystone, 3 limestone, and 20 sandstone samples). A summary of the laboratory test results [132] is shown on Table 2.5.4-5G. Reference 132 discusses the details of the test program and presents data on the variation of shear modulus and critical damping with strain levels.

Considering the ranges of stiffness indicated by Tables 3.7B-24 through 29, and by inspection of the sensitivity of the elastic expressions listed in Tables 3.7B-5 and 6 to variation in shear modulus (G) and Poisson's ratio (ν), it is evident that the profile stiffness represented on Table 2.5.4-5H lies within the ranges of stiffness indicated by Tables 3.7B-25 through 29. On this basis, the additional information provided by the postconstruction geophysical survey generally confirms the validity of the profile stiffness represented on Table 2.5.4-5.

2.5.4.2.3 Rock Core Compositional Analyses

Petrographic, solubility, x-ray and chemical analyses were performed on ten cores of Glen Rose and Twin Mountains formations. These tests indicate that both the Glen Rose and Twin Mountains formations are not susceptible to solutioning which verifies field observations and experience.

1. Petrographic Analysis

The cores were slabbed lengthwise and one slice was kept intact and prepared for observation. It was ground on 220, 400, and 600 grit grinding powder, washed, then etched for 3 seconds in 10% HC1 solution, washed, and dried. The procedure was completed for all cores as well as for the rock chips remaining from the thin sections that were prepared. The etched slabs were examined under binocular microscope.

Thin sections were made for each core and were ground to a thickness of approximately 18 microns, after which petrographic modal analyses were completed for each thin section. All work was done on Leitz Dialux-Pol polarizing research microscopes. The description terminology is standard Fold Carbonate Classification with the textural packing classification of R. Dunham added as a supplement. Both classifications are described in detail in American Association of Petroleum Geologists Memoir 1 on the Classification of Carbonate Rocks (ca 1962). Three thin sections were prepared from the cores from borings P-9 and P-10. Single thin sections were prepared from all other cores. Detailed results of the petrographic analyses are given in the following sections.

1.1 Boring P-9 Depth 124 to 124.5 Feet

Core Description: Limestone, calcitic mud; stratification very irregular, lensing, very thinly laminated (fraction of mm) to very thin bedded (circa 2 cm). Lenses are flattened parallel to stratificated plans, are 1x2 mm; are made of light gray lime mud (5Y 7/1) bounded by darker very thin laminae (gray-5Y 5/1) which are laterally discontinuous.

Porosity: low, 1-3% prior to slabbing. Discontinuous pockets, a maximum of 1/4 inch in size, of lightly cemented material are present which disintegrate upon contact with water. These are isolated from each other and are apparent organic burrow fillings. Thus, porosity created by opening of these pockets is not effective.

Grain Orientation: Generally poor due to the moderate to strong bioturbation of sediment. Post-depositional compaction has improved grain alignment parallel to stratification. Most of the lenses were originally round or oval-shaped animal burrows; some were simply sea-floor depressions that were filled by mud and later flattened somewhat.

Thin section Analyses - Carbonates - (Summary of three sections)

I. Complete Lithic Name (Folk-Dunham): Slightly sandy, disturbed, ostracod/mollusca, bio-pelmicrite, wackestone

II. A. Allochem Total: 45%

Pellets: 21%

Biogenic: 23%

Others: Carbonaceous plant detritus

Phosphatic fish remains (less than 2%)

B. Matrix: 45% (incl. 5% secondary dolomite)

Sparry Cement: less than 1%

Porosity: less than 2%

Terrigenous Material: approximately 10-12%

III. Composition (detailed description of items under “II”):

Biogenic: Ostracods, Mollusca, Foraminifera Plant, Phosphate

Pellets: mean size = 0.110-0.115 mm, more concentrated in burrows\

Terrigenous Material: visible quartz, opaque minerals. Qtz = med. silt to v.f. sand, mean size = 0.070mm, sub- angular to angular. Opaque minerals = pyrite, iron oxide.

Matrix: micritic calcite predominately; dolomite euhedral rhombs (0.020-0.025mm) average about 5% throughout all thin sections of P-9, reaches 20%+ within individual burrows.

Cement: Sparry calcite, primarily within tests of organisms

IV. Texture:

A. Grain size (mean, sorting): Mean grain size = medium silt; sorting poor (0.02-0.03mm)

B. Orientation/Bioturbation: Medium to highly biogenically disturbed. Burrowed zones sharply defined by disturbances of grain orientation.

C. Compaction: Moderate compaction - burrows shells compressed.

D. Resedimentation/Abrasion: None except for movement during bioturbation.

E. Diagenesis: Dolomite is secondary, common in sediments with high content of organic matter initially (especially in muds)

V. Depositional/Environmental Interpretation:

Low energy, carbonate mudflat, probably lagoonal/restricted circulation. Fine terrigenous material probably introduced during storms, then reworked by burrowing infauna (e.g. worms, crustaceans)

VI. Other Notes: Most of the pore space in the thin sections due to plucking of softer clay-rich material during section grinding, i.e., this is not original porosity.

1.2 Boring P-10 Depth 114 to 114.5 Feet

Core Description: Limestone, muddy carbonate (calcitic) sand; stratification very irregular, thin to medium bedded, very discontinuous laterally. Patches of finer calcitic mud, internally laminated, that appear identical to the general P-9 lithology (confirmed in thin section examination). These patches are renants that were not removed by the bioturbation which dominates in the rest of the slab. That other P-9 type sediment has been reworked is indicated by the lithoclasts that are present in the carbonate sand of the slab and in thin section. Some of the reworking was by mechanical wave action, some by burrowing organisms.

Porosity: Low, probably less than 5%. Some discontinuous pockets of coarse-grained carbonate sand are weakly consolidated and with some porosity; this porosity is not effective. Compaction of sediment is moderate.

Grain Orientation: Generally poor due to moderate bioturbation of sediment. The bulk of the common sandsize carbonate was deposited under conditions of relatively high wave and/or current energy with probably very good initial grain orientation. Subsequent reworking by organisms has destroyed this. Sorting is poor with both mud and calcite spar-cemented patches of grains.

Thin Section Analysis - Carbonates - (Summary of three sections)

I. Complete Lithic Name (Folk-Dunham): Disturbed, foraminiferal molluscan biosparite/biomicrite-Packstone

II. A. Alochem Total: 80%

Intraclasts: 5%

Pellets: 10%

Biogenic: 65%

B. Matrix: 7-12%

Sparry Cement: 12-7%

Porosity: less than 5%

Terrigenous Material: less than 3%

III. Composition (detailed description of items under “II”:

Biogenic: Foraminifer (Milliolid-types) Mollusca (pelec., gast.) Echinodermata Ostracods

Pellets: very fine-fine grain sand size

Intraclasts: micritic, to 2.5mm in diameter. Often terrigenous, with P-9 lithic type.

Terrigenous Material: Very fine quartz, pyrite, ferrug. opaques. Pyrite generally filling some foram chambers.

Matrix; Cement: Calcite, only trace of dolomite seen. Both are quite variable in distribution due to the bioturbation; cement is not only in voids of tests, between grains, but also within micrite envelopes developed on molluscan debris.

IV. Texture

A. Grain size (mean, sorting): Mean grain size. 0.35- 0.375mm, sorting poor; patchy distribution of micrite, both within tests and as intersititial matrix.

B. Orientation/Bioturbation: Generally random grain orientation due to extensive bioturbation.

C. Compaction: Moderate, no severe crushing of shells

D. Resedimentation/Abrasion: Local resedimentation, indicated by presence of intraclasts, commonly of P-9 type. Abrasion neglible.

E. Diagenesis: Micrite envelopes due to algal boring (of fungal activity) are extensively developed on most pelecypod material; internal aragonite is dissolved and replaced by calcite. Calcite spar cement looks like vadose zone development.

V. Depositional/Environmental Interpretation:

Higher energy situation than for P-9. Possible intertidal zone on well circulated carbonate mudbank. Could be in same setting as that for P-9, but with better water circulation and higher energy.

Large patches of P-9 type sediment (silty mud) are present in upper part of core P-10 and were verified in thin section P-10C.

1.3 Boring P-10A Depth 70.8 - 71.2 feet

Core Description: Mudstone, dolomitic, silty laminated. Laminae disturbed by burrowing organisms. Sample light gray (5Y7/1 dry).

Porosity: 3%. Burrowed zones may have slightly higher porosites due to concentration of dolomite and silt size material.

Grain Orientation: Good, evidenced by laminations attributed to alignment of clay minerals.

Thin Section Analysis - Clastics

I. Complete Lithic Name: Dolomitic silty mudstone.

II. Composition:

Quartz: 15%

Organic: 1%

Pyrite: 3%

Matrix: 80%

Matrix primarily clay minerals, silt size quartz and 15-20T dolomite ranging in size from 0.004 mm to 0.025 mm, mean grain size 0.015 mm.

III. Texture:

A. Size Range: 0.048 coarse silt to clay sized material. Mean grain size is in the clay size range.

B. Sorting: well sorted

C. Orientation: Good. Attributed to clay minerals.

D. Porosity: 3%

E. Compaction: Fair amount of compaction.

F. Diagenesis: Secondary dolomite

IV. Depositional/Environmental Interpretation:

Low energy, probably a lagoonal deposit.

1.4 Boring P-10A Depth 141.0 - 141.8 feet

Core Description: Limestone, calcitic mud; thinly laminated to thinly bedded. Stratification very irregular with laminae laterally discontinuous. Lenses common and flattened parallel to the stratification plane. Lenses range up to 1 x 4 cm. and are composed of light gray to light olive gray lime mud (5Y6/1.5) bounded by darker laminae light gray (5Y7/1) which are laterally discontinuous. Most lenses appear to be burrows flattened by compaction.

Porosity: Low, 3%. The darker laminae are easily disintegrated upon contact with water due to the presence of expandable clays. These laminae are cut by lenses (burrows) which are little affected by water contact.

Grain Orientation: Poor due to moderate bioturbation of the sediment. Post-depositional compaction has flattened the lenses and aligned some grains parallel to stratification.

Thin Section Analysis - Carbonates:

I. Complete Lithic Name (Folk-Dunham): Disturbed, silty, dolomitic Molluscan Biomicrite-Wackestone.

II. A. Allochem Total: 25%

Intraclasts: 2%

Pellets: 6%

Biogenic: 17%

B. Matrix: 58%

Sparry Cement: --%

Porosity: 3%

Terrigenous Material: 8%

Pyrite: 5%

III. Composition (detailed description of items under "II"):

Biogenic: Molluscan (concentrated in burrows)

Intraclasts: Micritic to 2.4mm in diameter

Terrigenous Material: Visible quartz, medium silt to very fine sand size (concentrated in burrows)

Matrix: Fine dolomite crystals abundant in matrix. Size range 0.021 mm - 0.002 mm. Mean grain size = 0.008 mm.

IV. Texture:

A. Grain size (mean, sorting); Poor sorting within burrows, mean grain size in burrows 0.30 mm.

B. Orientation/Bioturbation: Random grain orientation within burrows.

C. Compaction: Moderate -- no severe crushing of shell debris.

D. Resedimentation/Abrasion: None except for bioturbation.

E. Diagenesis: Diagenetic formation of dolomite.

V. Depositional/Environmental Interpretation:

Low energy, carbonate mudflat deposit, restricted circulation. Dolomite burrowed by infauna and terrigenous material as well as molluscan fragments and intraclasts introduced in burrows.

1.5 Boring P-10A Depth 223.5 - 223.8 feet

Core Description: Limestone, calcitic mud; thinly laminated to thinly bedded. Stratification very irregular with laminae laterally discontinuous. Lenses are common and flattened parallel to the stratification plane. Lenses observed range in size from 1 x 2 mm to 1 x 2 cm. and are composed of white to light gray lime mud (5Y8/1 - 5Y7/1) bounded by darker laminae gray (5Y6/1). Most of the lenses appear to be burrows which were originally round or oval shaped and since flattened by compaction.

Porosity: low, 2%. A few pockets of expandable clay are present which disintegrate upon contact with water. These pockets or lenses apparently isolated animal burrows. Porosity is not improved by solutioning of these isolated lenses.

Grain Orientation: Poor due to moderate to intense bioturbation of the sediment. Post-depositional compaction has realigned some grains parallel to stratification.

Thin Section Analysis - Carbonates:

I. Complete Lithic Name (Folk-Dunham): Disturbed, slightly silty, obstracod/Mollusca biomicrite-Packstone.

II. A. Alochem Total: 80%

Intraclasts: 2%

Pellets: 15%

Biogenic: 43%

Bryozoan coated grains: 20%

Organic and phosphatic remains: 1%

B. Matrix: 7%

Sparry Cement: 5-8%

Porosity: 2%

Terrigenous Material: 5%

III. Composition (detailed description of items under “II”):

Biogenic: Ostracod-foraminifers Mollusca Echinodermata Bryozoa

Terrigenous Material: Quartz, medium silt to fine sand.

Matrix: Predominantly micritic calcite, minor amount of dolomite.

Cement: Sparry calcite primarily within tests of organisms.

IV. Texture:

A. Grain size (mean, sorting): Poor sorting due to bioturbation. Mean grain size 0.21 mm to 0.30 mm.

B. Orientation/bioturbation: Medium to highly bioturbated, random grain orientation.

C. Compaction: Moderate -- Burrows and shells compressed.

D. Resedimentation/Abrasion: None except for bioturbation. Burrows commonly mud and/or spar rich.

E. Diagenesis: Secondary dolomite.

V. Depositional/Environmental Interpretation:

Low energy, carbonate mudflat, restricted circulation, possibly lagoonal. Material reworked by burrowing infauna. Terrigenous material possibly introduced and concentrated by burrowing organisms from adjacent strata.

1.6 Boring P.10A Depth 242.3 - 242.6 feet

Core Description: Sandstone, fine grained quartz sand, gray (5Y6/1 dry). Cross stratified non-cemented, very friable although well compacted.

Porosity: 10-12%. Moderate to well compacted sandstone accounting for relatively low porosity.

Grain Orientation: Poor grain orientation. Locally organic material concentrated along stratification planes.

Thin Section Analysis - Clastics:

I. Complete Lithic Name: Fine-grained quartz arenite (Orthoquartize)

II. Composition:

Quartz: 90%

Chert: 2%

Feldspar: 1%

Rock Fragments: 1

Pyrite: 2%

Organic: 2%

Matrix: 2%

Carbonate and clay matrix (carbonate 1%)

III. Texture:

A. Size Range: 0.39mm (medium sand) to 0.02 mm (medium silt). Mean grain size 0.127mm (fine sand).

B. Sorting: Well sorted.

C. Textural maturity: Mature

D. Roundness: Subangular to rounded.

E. Orientation: Weak

F. Porosity: 10-12%

G. Compaction: Moderate to well compacted - pressure solutioning minimal.

H. Diagenesis: Minor overgrowths.

IV. Depositional/Environmental Interpretation: High energy, possibly a barrier bar or beach deposit.

1.7 Boring P-10A Depth 264 - 264.3 feet

Core Description: Sandstone, fine grained quartz sand, light gray (5Y7/1 dry). Stratified, non-cemented, very friable, although well compacted.

Porosity: 12-15% Moderate to well compacted sandstone accounting for relatively low porosity.

Grain Orientation: Weak or poor grain orientation. Sediment slightly bioturbated with burrows higher in clay content.

Thin Section Analysis - Clastics:

I. Complete Lithic Name: Fine-grained Quartz arenite (orthoquartzite).

II. Composition:

Quartz: 91%

Feldspar: 1%

Organic: 1%

Chert: 2%

Rock Fragments: 1%

Pyrite: 1%

Matrix:

Carbonate: 2%

Cement: 0%

III. Texture:

A. Size Range: 0.30mm (medium sand) to 0.06 (silt, very fine sand). Mean grain size 0.012 (fine sand).

B. Sorting: well sorted

C. Textural Maturity: Mature

D. Roundness: Subangular to rounded.

E. Orientation: Weak.

F. Porosity: 12-15%

G. Compaction: Moderate to well compacted - pressure solutioning minimal

H. Diagenesis: Minor quartz overgrowths.

IV. Depositional/Environmental Interpretation:

High energy, possibly beach or barrier bar deposit.

1.8 Boring P-10A Depth 282.2 - 282.5 Feet

Core Description: Sandstone, very fine grained, poorly sorted quartz sand. Stratification disturbed by burrowing organisms. Burrows light gray (5Y6.5/1) dry), sand white (5Y8/1 dry).

Porosity: 7-10T Moderate to well compacted sand. Burrows have poor porosity due to concentration of clay minerals.

Grain Orientation: Weak or poor grain orientation due to bioturbation.

Thin Section Analysis - Clastics:

I. Complete Lithic Name: Very fine grained Quartzwacke.

II. Composition:

Quartz: 67%

Feldspar: 1%

Rock Fragments: 1%

Organic: 1%

Matrix: 31% Predominantly silt size quartz and clay.

III. Texture:

A. Size Range: 0.204mm (fine sand) to clay sized material. Mean grain size 0.102mm (very fine sand).

B. Sorting: Poor sorting.

C. Textural Maturity: Immature.

D. Roundness: Subangular to rounded.

E. Orientation: Weak

F. Porosity: 7-10%

G. Compaction: Moderate.

H. Diagenesis: Bioturbated with possible concentration of clay along burrows.

IV. Depositional/Environmental Interpretation

Initial sorting of quartz due to high energy but final deposition probably occurred at low energy due to mixing of clay and sand. Possible beach or dune sand deposited in lagoonal mud by storm winds or waves.

2. Solubility Tests

Eight core samples were tested for solubility. Part of each core was ground to less than 1450 mesh and dried at 105°C. Five grams of ground rock were then agitated for two hours in one liter or distilled water. The resulting solubilities are maximum values due to the slight losses of rock resulting from removal of the water before weighing. The results are presented in Table 2.5.4-8.

In addition, five grams of the cores were treated with 5 percent acetic acid until no more effervescence resulted. The results are presented in Table 2.5.4-8.

The residue from samples from boring P-9 and P-10 were heated to 800°C for four hours which caused a color change.

In addition, five grams of the cores were treated with 5 percent acetic acid until no more effervescence resulted. The results are presented in Table 2.5.4-8.

The residue from samples from boring P-9 and P-10 were heated to 800°C for four hours which caused a color change from black to light brown and weight losses of 11.6 and 12.3 percent respectively. These correspond to a net loss of 1.6 and 0.5 percent of the total sample.

3. X-Ray Analysis

Core samples were run on the x-ray diffractometer. The results are presented in Table 2.5.4-9.

The acetic acid residue of samples were x-rayed with the following results:

SAMPLE	DEPTH	RESULTS
P-9	124	Quartz, dolomite
P-10	114	Quartz, dolomite, pyrite
P-10A	141.1	Quartz, minor illite, montmorillonite
P-10A	223.5	Quartz, minor montmorillonite, illite

Additional x-ray study was conducted on P-9. The finest fraction (2 microns) was found to be mainly calcite with minor quartz. The small, discontinuous, porous portions of the rock are calcite, dolomite, and minor quartz.

Two specimens from the claystone section of the Glen Rose Limestone (core P-10A, 70.8 feet; core P-14, 64 feet) were analyzed chemically (Table 2.5.4-10) and by x-ray diffraction (Table 2.5.4-9). Combined interpretation of x-ray and chemicals data indicates the following approximate clay mineral contents:

	P10A, 70.8 feet		P-14, 64 feet	
Clay Mineral	% of Clay	% of Rock	% of Clay	% of Rock
Kaolinite	45	25	13	5
Illite	55	30	87	35
Montmorillonite	0	0	0	0

Averaging results on the two specimens, the claystone section has the following approximate composition:

Clay Mineral	% of Clay	% of Rock
Kaolinite	29	15
Illite	71	32
Montmorillonite		None Detected

Kaolinite was not detected in core P-14, 64 feet, but its presence is suggested by the ration of A1203/K20 (=4.17) in the chemical analysis. This ratio is higher than the 3.4 A1203/K20 ratio which would be expected in a pure illite.

Montmorillonite was not detected in either specimen. However it could be present below the X-ray diffraction detection limit (5-10% of rock).

If this were the case some of the reported kaolinite would be montmorillonite since both are primarily aluminum silicates.

Montmorillonite was detected in two specimens of Glen Rose Limestone at deeper levels. Approximate clay mineral contents of these specimens are:

	P10A, 141.1 feet		P10A, 223.5 feet	
Clay Mineral	% of Clay	% of Rock	% of Clay	% of Rock
Kaolinite	0	0	0	0
Illite	70	28	50	5
Montmorillonite	30	12	50	5

4. Chemical Analyses

   The core samples were analyzed for metal oxides with an atomic absorption spectrometer. The results are presented in Table 2.5.4-10.

2.5.4.3 Exploration

A comprehensive investigation has been completed to evaluate the subsurface conditions beneath all structures. This exploration consisted, in part, of drilling borings, performing geophysical surveys, installing piezometers and wells, and mapping foundation excavations.

The locations of the borings, wells and piezometers are shown on Figures 2.5.5-5 and 2.5.5-6. The methods used in drilling the borings are discussed in Section 2.5.5.3. The description of the geophysical surveys is presented in Section 2.5.4.4.

Geological cross sections are presented in Figures 2.5.4-40, 2.5.4-41, and 2.5.6-7 through 2.5.6-11. The limits of excavations are shown in Figure 2.5.4-27.

2.5.4.4 Geophysical Surveys

In the preexcavation period, the following geophysical surveys were conducted at the Station and Safe Shutdown Impoundment Dam areas, providing data to depths of 200-ft and 100-ft, respectively:

1. Seismic refraction surveys to evaluate the compressional wave velocities of bedrock and the materials overlying bedrock and to evaluate the depth to bedrock;

2. Uphole velocity surveys to provide additional data regarding compressional wave velocities of bedrock and materials overlying bedrock;

3. A surface wave survey to evaluate types of surface waves and their characteristics;

4. Crosshole shear wave surveys to evaluate compressional and shear wave velocities of bedrock and the materials overlying bedrock;

5. Ambient noise studies to evaluate the predominant characteristics of ground motion due to background noise level; and,

6. Gamma ray and resistivity logs (at Squaw Creek Dam also) to evaluate stratigraphic features.

The locations of surveys 1 through 5 are shown on Figure 2.5.4-14. Locations of borings P-10, M-1, D1-2, D1-8A and D1-9 in which survey 6 was performed are shown on Figures 2.5.5-5 and 2.5.5-6. In the postconstruction period, two methods of geophysical exploration were conducted in the deep S/L cluster borings. Both provided detailed information on rock properties from a 500-ft-thick sedimentary section. These methods were:

- Crosshole measurement of seismic wave velocities on which rock shear modulus and Poisson's ratio were based; and

- Complex borehole logging including gamma-ray, electrical resistivity, gamma-gamma, and caliper surveys. The geophysical logs contributed to the straitigraphic refinement of the drilled column. The gamma-gamma logs also provided in situ densities.

The summary of geophysical test data including pre-excavation survey data and ambient ground motion measurements are provided in tables 2.5.4-5A through 2.5.4-5D.

The geophysical explorations are discussed in detail in the following sections.

2.5.4.4.1 Seismic Refraction Survey

A seismic refraction survey, 4,000 lineal feet in length, was conducted at the Station location along two seismic profiles. A seismic refraction survey (1,500 lineal feet) also was performed at the Safe Shutdown Impoundment (SSI) Dam site. Adjustments were made to seismic line locations where pipelines were present. Seismic energy was produced by detonation of explosive charges (five to 15 pounds of Nitromon-S; a Du Pont product) placed in drilled holes. These holes ranged in depth from five to 15 feet. The energy released by the charges was detected by vertically-oriented geophones spaced at 50-foot intervals along the seismic profiles. The geophones, manufactured by Electro-Tech Labs, have a natural frequency of 14 Hertz (cycles per second) and are fitted with a spike or metal plate to obtain proper coupling with the site materials. The energy impulse received by the geophones was recorded by a Dresser SIE seismic amplifier system coupled with a Dresser SIE R6 recording oscillograph.

The compressional wave velocities and the corresponding depths to bedrock were evaluated by plotting first arrival times of seismic energy at each geophone against the distance of each geophone from the seismic energy source. The time-distance data for each profile and the corresponding subsurface cross sections of the profiles are shown on Figures 2.5.4-15 through 2.5.4-18. Representative compressional wave velocity data determined from the seismic refraction survey (and the uphole survey) are summarized on Tables 2.5.4-5B and 2.5.4-5C. The accuracy of the calculated depths to the bedrock surface is considered to be 10 to 20 percent. This estimate is based on a number of factors. The velocity and characteristics of the near surface materials can introduce some error into interpreted depths. Uphole surveys, an examination of soil and core samples, and an examination of the boring logs were used to reduce this type or error. The timing accuracies of the equipment and the seismic caps also lead to the selection of tolerance which acts as a guide in determining the accuracy of the estimate.

2.5.4.4.2 Uphole Velocity Survey

Uphole velocity surveys were performed at the station and SSI dam locations. These provided a comparison with compressional wave velocities obtained by the seismic refraction surveys at each location. Borings P-1 at the Station location and M-5 on the SSI dam were selected for these surveys. A geophone array was placed on the ground surface around each boring. Small explosive charges were fired at 10-foot intervals from the bottom to the top in each boring. The seismic energy released by the explosives was detected by the geophones in each array. This energy was recorded by the same system that was used in the seismic refraction surveys. The results of the uphole velocity surveys are presented on Figures 2.5.4-19 and 2.5.4-20, as well as on Tables 2.5.4-5B and 2.5.4-5C.

2.5.4.4.3 Surface Wave Survey

A surface wave survey was conducted at the station and SSI dam locations to evaluate the types and characteristics of any observable surface waves.

Two Sprengnether Engineering Seismograph VS-1200, three-component geophones were placed 300 to 350 feet apart on the ground. The output of these geophones was fed into a VS-1100 amplifier with a gain characteristic of 100, and from the amplifier into a special attenuator circuit. The resultant output was recorded on the Dresser SIE R6 recording oscillograph.

Explosive charges were placed in drill holes at varying distances from, but in-line to, the geophones. Explosive charges used in this survey varied from five to 40 pounds of Nitromon-S.

In addition to this procedure, recordings were made of the energy produced by horizontal impacts of an eight-pound sledgehammer. This energy was detected by six one-component (horizontal) geophones spaced five to 20 feet apart.

The compressional and shear wave velocities developed from the surface wave survey are presented on Tables 2.5.4-5B and 2.5.4-5C.

2.5.4.4.4 Crosshole Shear Wave Surveys

Crosshole shear wave surveys were performed at two plant locations and at the SSI dam site utilizing borings drilled into bedrock. These surveys were conducted in two separate time periods; initially, prior to excavation, and subsequently, after construction was essentially completed. The preexcavation survey was made in somewhat different manner than the postconstruction survey. The difference included: the scope and the location of the survey, the techniques applied, and the thickness of the stratigraphic column involved.

During the preexcavation crosshole survey, stationary subsurface shots were used as a source of the seismic energy. Wave propagation from two shot points was recorded in borings P-1 and P-2 at distances ranging from 370 ft to 1,175 ft, and from a single shot point in two preconstruction SSI dam borings M-5 and M-6 at distances of 400 ft and 700 ft, respectively. Compressional and shear wave arrival times were recorded in each of these borings by using 4 to 6 geophones vertically arranged within the upper three zones of the Glen Rose Formation between elevation 650 ft and elevation 818 ft.

For the preexcavation survey, two three-component, low-frequency geophones (Mark Products L-1-3DS) were placed at corresponding elevations deep in each recording borehole. Explosive charges were detonated in drilled holes at fixed distances from each of the borings. The resultant seismic energy was recorded by the Dresser SIE system. After each shot (energy detonation), each geophone was raised 25 feet. Details of the crosshole shear wave survey at both the plant site and the Safe Shutdown Impoundment Dam are presented in Table 2.5.4-5A. In addition to this data, seismic energy developed by a technique of producing horizontally-generated waves was also recorded by each of the geophones at 25-foot intervals. The technique for producing horizontally-generated waves was developed by the Dames & Moore geophysical staff. It involves the use of explosives (Primacord) placed in trenches at the ground surface adjacent to a boring. Two trenches are used, one on either side of a boring. Depending on the site conditions, the trenches vary in length from three to eight feet long, and in depth from six inches to two feet deep. The trenches used on this site were five feet long, and one foot deep. The trenches are “v” shaped with one face cut vertical and the other sloped at some angle to the vertical. This angle will vary for different conditions, but for this survey, an angle of 60 degrees was used. The vertical face of the trenches on either side of a boring are opposed to each other as shown in Figure 2.5.4-21. Primacord is placed in the bottom of each trench and fired simultaneously by the use of seismic caps. The trenches are not backfilled, allowing for a backblast. The resultant detonations produce a horizontal torque, centered around the boring, which is recorded in the boring by use of a three-component geophone.

The results of this survey are plotted as an uphole shear survey, with proper calculations for the shot offset and depth. Primary and secondary arrivals are plotted on a time-depth plot. The plot for the survey at boring M-5 is shown on Figure 2.5.4-21.

Shear wave arrivals often are found to be masked by relatively large-amplitude motions caused by multiple reflected and refracted compressional waves and the surface waves. To overcome this difficulty high-energy and low-energy recordings were made at corresponding depths in each of the borings.

The compressional and shear wave velocities, determined in the initial survey by the crosshole techniques and by the surface wave surveys, are presented on Tables 2.5.4-5B and 2.5.4-5C.

During the postconstruction crosshole survey [131] conducted in four closely spaced, up to 500-ft-deep S/L cluster borings (Figure 2.5.4-14), seismic waves were generated by a vertically mobile hammer inserted in “source” boring S-1A, and these waves were detected by three horizontally separated geophones installed at the same elevation in each of the three “listening” borings: S-1, L-1, and L-2. These three “listening” borings were 5 ft, 8 ft, and 9.5 ft, respectively, away from the “source” boring. The measuring points were located within the interval from elevation 797 ft (12 ft below plant grade), which is within Zone I of the Glen Rose Formation, to elevation 317 ft (492 ft below plant grade), which is beneath the unconformity separating the Cretaceous Twin Mountains Formation from the underlying Paleozoic Mineral Wells Formation.

The postconstruction survey was performed at a location immediately west of the Unit 2 turbine building area (Figure 2.5.4-14A). The survey equipment consisted of a four-channel Nicolet Oscilloscope (Model 4094-2), a Bison shear-wave hammer (Model 1465-1), and three Geospace Geophones (Model HS-J-LP3D). The Bison downhole shear-wave hammer was the energy source. This hammer consists of a stationary body with a sliding mass, and is hydraulically locked in the “source” boring. Using a hoist cable, the mass is lifted or dropped to produce impacts against the stationary hammer body. The hammer, designed to permit either upward or

downward impacts, generates vertically polarized shear waves of opposing polarities. These polarities allow identification of the shear wave arrivals in the presence of refracted or reflected seismic arrivals in layered formations. Crosshole measurements were normally taken at 3-ft intervals from 0 to 45 ft, at 5-ft intervals from 45 to 300 ft, and at 10-ft intervals thereafter. Additional measurements were performed to determine the seismic wave velocities in claystone layers at depths of approximately 20 to 30 ft and 100 ft. Crosshole measurements were taken at a total of 90 elevations, resulting in a total of about 440 individual records. The survey procedures are described below:

- The Bison hammer was suspended in boring S-1A ("source boring") at the desired depth. The hammer shoes were hydraulically extended to clamp the hammer in-place at the elevation.

- Geophones were suspended in borings S-1, L-1, and L-2 ("listening" borings) at the same elevation, and the pneumatic tubes were inflated to couple the geophones to the casing wall.

- Typically, 10 impacts (upward or downward) were performed with the shear-wave hammer.

- The wave motion produced with each individual impact was stored in the Nicolet Model 4094-2 oscilloscope. The motions produced by 10 impacts were averaged to yield one record for that combination of depth, impact direction, and "listening" boring.

- Upon completion of each series of impacts, the data were written onto a floppy disk as a permanent record and labeled to indicate depth, impact direction, and "listening" boring.

- This procedure was repeated such that two records for upward impacts and two records for downward impacts were obtained for each "listening" boring at each elevation.

- Upon completion of the recording sequence, the hammer and geophones were released and advanced to the next recording elevation where the procedure was repeated.

The borings used for the postconstruction crosshole geophysical surveys were cored either continuously (S-1, and S-1A below 325 ft) or partially (L-1 and L-2). They were logged geophysically (except for S-1), cased by PVC pipe which was grouted (except for S-1), and subjected to a verticality survey to allow accurate determination of interhole separation at all depth levels where crosshole seismic velocity measurements were made.

Generalized subsurface conditions at the location of the S/L cluster borings are shown in Figure 2.5.4-30A which includes a generalized geologic column, the geophysical logs of boring S-1A, and a profile of seismic wave velocities. The mean compressional and shear wave velocities, was well as the Poisson's ratio for various depth intervals between plant grade and the bottom of the S/L cluster boreholes (at a depth of 503 ft), are listed on Table 2.5.4-5F.

2.5.4.4.5 Ambient Vibration Measurement

Measurements of ground motion due to background (ambient) vibrations were made at each site when drilling rigs or other equipment were not operating.

The three-component VS-1200 Sprengnether Engineering seismograph was used to record ambient ground motions. This seismograph has gain characteristics of 20 inches per inch per second in the velocity mode, 12 inches per inch per second in the acceleration mode and 200 inches per inch in the displacement mode. A VS-1100 amplifier with a gain characteristic of 100 was used for all recordings. The resultant maximum gain level is 2,000 for the velocity mode, 1,200 for the acceleration mode and 20,000 for the displacement mode.

The three components of ground motion measured were radial, vertical and transverse. The seismometer was oriented facing either north or east. Table 2.5.4-5D presents the results of the ambient ground motion measurements.

2.5.4.4.6 Borehole Geophysical Logs

Gamma ray and resistivity surveys were made in preexcavation borings P-10, DI-2, DI-9, M-1 and DI-8A to aid in evaluating the correlating on-site stratigraphy. These boring locations are shown on Figures 2.5.5-5 and 2.5.5-6. The survey results are illustrated on Figures 2.5.4-22 through 2.5.4-26.

Gamma-ray, electrical resistivity, gamma-gamma, and caliper surveys were made in postconstruction boreholes S-1A, L-1, and L-2 as an aid to correlation of the borings and for refinement of the site stratigraphy. The gamma-gamma survey also provided a continuous record of in situ densities. The location of the borings are shown in Figures 2.5.4-14A and 2.5.5-5. The geophysical logs of these borings are almost identical due to closeness of the boreholes; the survey results are well represented by the logs of boreholes S-1A, which are shown in Figure 2.5.4-30A.

2.5.4.4.7 Interpretation of Geophysical Data

The refraction survey data indicated slight variations in the compressional wave velocity of the bedrock at the Station location. This variation in compressional wave velocities is most prevalent at shallow depth near the edges of the peninsula on which the Station is located.

The compressional wave velocities from the uphole velocity survey are different from the compressional wave velocities from the refraction survey. This is because a refraction survey develops an average compressional wave velocity over a large lateral distance, whereas the uphole velocity survey records the compressional wave velocity around the isolated point (i.e., the boring). It is felt that the compressional wave velocities obtained by the seismic refraction survey are a more appropriate measure of the actual dynamic properties of the site materials.

In the preexcavation stage of the site geophysical investigation, the geophysical surveys did not penetrate into the Twin Mountains Formation. However, indirect geophysical information from the surface and shear wave surveys indicates that both the compressional and shear wave velocities of the Twin Mountains Formation are lower than the corresponding velocities of the Glen Rose Formation. This was supported by the geologic borings which obtained basically 100 percent core recovery within the Glen Rose Formation and a much lower percentage recovery and Rock Quality Determination (RQD) in the underlying Twin Mountains Formation. The velocity inversion between the Glen Rose and Twin Mountains Formations was assumed to result in a seismic energy response characterized by the Glen Rose Formation acting as a seismic energy channel, or in essence, a plate of limestone. All these conclusions and assumptions have been

substantiated by the detailed postconstruction crosshole seismic survey that penetrated through the Twin Mountains Formation into the uppermost part of the underlying Mineral Wells Formation.

Surface waves at this site have unusual characteristics because of the seismic velocity inversion. The uppermost higher velocity Glen Rose Formation which acts as a plate or seismic channel, produces several discrete, but overlapping Rayleigh wave trains. These wave trains vary from 15 to 25 hertz, and from 5,000 to 2,500 feet per second (respectively) apparent velocity. The wave trains are essentially trapped waves within the plate of limestone and indicate an average shear wave velocity greater than 5,000 feet per second in the limestone. These waves are apparently second of higher mode vibrations within the plate. The maximum amplitudes and longest wave train have a frequency of 25 hertz. The maximum amplification of seismic energy, from an explosive source detonated at shallow depth is, therefore, at 25 hertz.

Some of the seismic records show a low-frequency wave train of about five to seven hertz. This wave is not well defined, but is probably the first mode Rayleigh wave vibration in the limestone plate.

The refraction and surface wave surveys are considered to give reasonably accurate data for the determination of compressional wave velocities while equally reliable data for the determination of the shear wave velocities was obtained by the Primacord technique. The surface wave surveys also produced a set of plausible values, with the initial crosshole surveys producing the least reliable set of values. However, the postconstruction crosshole survey produced more detailed measurements of both compressional and shear wave velocities. In comparision, the preexcavation geophysical explorations were not so detailed, systematic, and deeply penetrating. The postconstruction velocity measurements, together with the gamma-gamma log-derived densities, were the basis for the final design values of shear modulus and Poisson's ratio for the generalized profile (Table 2.5.4-5E). The initially selected values of the parameters based on the preexcavation geophysical surveys, are shown on Table 2.5.4-5. The dynamic rock properties based on both the preexcavation and postconstruction geophysical surveys are presented in Table 2.5.4-5H.

It should be noted that in the final determinations of compressional and shear wave velocities the information from all the techniques are integrated and used, with an understanding of the reliabilities, to produce the reported results.

2.5.4.5 Excavations and Backfill

The limits of required excavation for the power plant is shown on Figure 2.5.4-27. Figure 2.5.4-28 shows the location of the plant site and all cut and fill areas that occur adjacent to the plant site.

Figure 2.5.4-30 shows a 43-ft thick geologic section of the rock encountered in the excavations for Category I structures. This section is typical for the Category I portion of the power plant and related adjacent Category I support facility excavations. The limestone bed identified as bed "R" in Figure 2.5.4-30 is a reference bed in the plant site. The elevation of the top of bed "R" in Unit 1 reactor excavation was 782 feet 7 inches and 780 feet 9 inches in the Service Water Intake Structure excavation. This difference is due to the undulating nature of the beds and also because the Service Water Intake is downdip of Unit 1 Containment. Figure 2.5.4-30A is a general subsurface profile which includes a 500-ft thick geologic section of the materials constituting the foundation base of all plant structures.

Structural backfill was required on the north side of the Service Water Intake Structure. Figure 2.5.4-31 is a cross section of this area.

All Seismic Category I Electrical Duct Banks and the Service Water Pipe Trench require Category I backfill and bedding and are delineated in Figure 2.5.4-27. Figure 2.5.4-32 shows typical sections of the Service Water Pipe Trench and a Seismic Category I Electrical Duct Bank.

Geologic sections and blast-fracture maps were constructed as each excavation was completed for Category I structures. Figures 2.5.4-33, 2.5.4-34 and 2.5.4-35 are the fracture maps developed for the Unit 1 Containment and reactor excavation. Figure 2.5.4-36 is the photo grid for photographs taken of the Unit I Containment and Reactor excavation walls and Figure 2.5.4-37 (21 sheets) are the photographs depicted in Figure 2.5.4-36.

Excavation methods consisted of blasting, utilizing the presplit, primacord method, or heavy equipment demolition tools where blasting was not an acceptable method. Removal of the broken rock was accomplished by heavy equipment.

Protection of the exposed rock against weathering was required due to the argillaceous nature of some of the limestone beds and the claystone beds which are present. The wall portion of the excavations was protected by the application of shotcrete. The floor portion was protected by a minimum three inch thick concrete seal slab.

No groundwater was encountered in the primary Glen Rose Limestone, and therefore only normal pumping equipment and procedures were required to remove storm runoff and concrete curing water which collected in the open excavations.

Upon completion of removal of material from an excavation, a preliminary inspection was performed by the site geologist. After the final cleanup, just prior to placement of shotcrete or seal slab, photographs were taken and blast fracture maps were developed by the site geologist. At the time the geologist approved an excavation, the QA/QC program became involved, assuring cleanliness prior to application, curing and testing of protective coatings, and the inspection of these protective coatings prior to being covered by structural concrete.

To measure the rebound of the foundation rock due to the excavation on the plant site, two extensometers were installed. Figure 2.5.4-28 shows the locations of the extensometers. Monitoring results are presented in section 2.5.4.13.

The primary source for Category I backfill material was on-site quarries which are shown on Figure 2.5.4-29. A limited quantity of Category I backfill was obtained from Bridgeport, Texas.

During the preliminary grading and excavation of the plant site area, down to elevation 810', a bed of very hard, tan, fossiliferous, crystalline limestone, was intersected at an elevation of 825'. This particular bed is present throughout the site area and an initial quarry site northwest of the plant (Pit “A”, Figure 2.5.4-29) was selected for sampling and testing. Material was quarried from Pit “A” and processed through the crusher plant on site, a sample pulled and graded to the proper gradation, and then subjected to cyclic laboratory testing to verify that the material met specification requirements. Also sampled for testing was the concrete agregate sand which is being used as a bedding material around the pipe in the Service Water Pipe Trench and under the Seismic Category I Electrical Manholes. A complete discussion of this testing is covered in Section 2.5.4.7.

The compaction requirement for the backfill materials, placed against the Service Water Intake Structure, is a minimum of 95% of the maximum dry density as determined in accordance with AASHTO Designation T-99, Method D. In-place density was determined in accordance with ASTM D1556 (Sand Cone), ASTM D2167 (Balloon), or ASTM D3017 and D2922 (Nuclear). These density requirements apply to all Category I backfill material. Figure 2.5.4-38 shows the gradation requirements for the backfill material.

Based on the results of in-place density tests for Category I backfill, the following average backfill densities were obtained:

γDry = 130 pcf

γWet = 135 pcf

γSat = 140 pcf

The compaction requirement for the bedding material placed in both the Seismic Category I Electrical Duct Banks and Service Water Pipe Trench, is a minimum density of not less than 80% of the relative density as determined by ASTM Test Designation D2049, latest revision effective prior to September 4, 1975. In-place density was determined in accordance with ASTM D1556 (Sand Cone), ASTM D2167 (Balloon), or ASTM D2922 (Nuclear). Figure 2.5.4-38 shows the gradation requirements for the bedding material. The gradation, relative density, and percent compaction summaries are presented in Figures 2.5.4-44 thru 53.

Based on the results of in-place density tests for Category I bedding, the following average densities were obtained:

γDry = 120 pcf

γWet = 130 pcf

γSat = 135 pcf

2.5.4.6 Groundwater Conditions

A detailed description of groundwater is presented in Section 2.4.13. No groundwater was encountered during excavation for the plant foundation. Groundwater observations from piezometers installed at the site are provided on Figure 2.5.5-77.

2.5.4.7 Response of Soil to Dynamic Loading

The plant structures are founded on rock, however, granular material was required as backfill, pipe bedding and for the dam filters. The response of these materials to dynamic loading was assumed during the initial design phase. Later, an investigation of the actual dynamic strengths of these materials was performed by means of a cyclic triaxial testing program.

During design, adopted values for the cyclic shear strength for Category I backfill and bedding material were based on the published data for granular soils. A discussion of the published data is presented in Section 2.5.6.4.3.4. Since the D50 of the bedding material was approximately the same as for Filter “A” for the SSI Dam, the material was assumed to have essentially the same

cyclic shear strength characteristics. Category I backfill material has a D_{50} of approximately 15mm and since this material is coarser than the published data utilized, the cyclic shear strength characteristics of this material should be greater. For design, the cyclic shear strength criteria was assumed based on the data presented on Figure 2.5.6-48. The specific criteria was that the material with the gradation limits as shown on Figure 2.5.4-38 shall not allow development of double amplitude strain larger than 5% under specific corresponding stress conditions shown on Table No. 2.5.4-11.

In order to verify the adopted cyclic shear strength criteria for Category I bedding, backfill and filter materials, a test program utilizing cyclic triaxial tests was developed. A minimum of four cyclic triaxial tests were performed for each type of material utilized; one at each confining pressure as listed on Table 2.5.4-11. The applied stress given for the development of 5% double amplitude strain in five cycles was utilized since this stress is more critical than the stress combination specified for ten cycles of loading. Thus any specimen that developed 5% strain at ten cycles of load under a higher stress level automatically meets the criteria specified at ten cycles.

Four (4) different materials have been tested to determine if they meet the cyclic shear strength criteria as described above. The materials which have been tested are as follows:

Sample No.	Material
1	Filter A Material
2	ASTM C-33, Fine Aggregate
3	Glen Rose “Gray” Limestone Crushed Stone
4	Glen Rose Limestone “Crowder Quarry” Crushed Stone

The gradation limits of these materials, as well as the actual gradation curve of the tested materials, are shown are Figures 2.5.4-38A through 2.5.4-38D, respectively. The densities of the four (4) materials tested are summarized on Table 2.5.4-12.

For conventional laboratory triaxial testing of soils, it is generally recognized that the diameter of the test speciman should be at least six times the maximum particle size to avoid possible effects of sample geometry on test data. For Samples 3 and 4, the maximum size for the crushed stone was on the order of 2 inches, which would require triaxial cells capable of taking 12-inch test specimens. Such large cells, especially for dynamic testing, are not available for cyclic triaxial testing; therefore, modeling techniques have been utilized to model the field gradation of Samples 3 and 4.

One technique recommended by John Lowe [134] describes a technique of preparing a model soil with a grain size distribution curve parallel to the field material. Research on laboratory modeling of soils utilizing the parallel grading curve method has shown that the method can be used to predict the static and dynamic strength and deformation characteristics of soils. For Samples 3 and 4, a laboratory gradation range parallel to the field gradation requirements of the material has been drawn and an artificially blended material has been prepared (within these

limits) and utilized for the cyclic triaxial testing. (See Figures 2.5.4-38 C & D). For Samples 1 and 2, samples falling within the field gradation curves were obtained and were utilized for the cyclic triaxial testing. The test program followed guidelines set up in Report No. NUREG-31 entitled "Laboratory Triaxial Testing Procedures to Determine the Cyclic Strength of Soils", which was prepared under Contract No. NRC-E (11- 1) 2433 for the U.S. Nuclear Regulatory Commission.

The results of the initial tests are summarized in Table 2.5.4-13. One plot for each material tested at various confining pressures, relating double amplitude axial strain to the number of cycles of constant cyclic shear stress, are shown on Figures 2.5.4-39, Sheets 1 through 4. All of the materials tested were of sufficient strength to meet the cyclic strain criteria at 5 cycles of load. The ASTM C- 33, Fine Aggregate material (Sample 2) and the "Crowder Quarry" Crushed Stone (Sample 4) also met the strain criteria at 10 cycles of applied load. The Glen Rose Limestone Crushed Stone (Sample 3) met the strain criteria at 10 cycles for confining pressures of 2000 and 4000 pounds per square foot. However, the test results showed that the Glen Rose crushed limestone at 6000 and 8000 psf confining pressure, (Sample 3) and Filter "A" (Sample 1) did not meet the strain criteria in 10 cycles at the stress ratios utilized in the test program. Therefore, additional tests were run on Sample 1 and Sample 3 at reduced stress levels corresponding to stress ratios required to develop 5 percent strain in 10 cycles of load. The results of these tests are summarized in Table 2.5.4-13A.

For Sample 3, additional tests at reduced stress levels showed that the material did not meet the cyclic strain criteria at effective normal pressure of 6000 and 8000 psf in 10 cycles of load. The figures relating double amplitude strain to number of cycles for Sample 3 show that the cyclic strains were relatively low until about 6 to 8 cycles, after which large strains developed quite rapidly in only a few additional cycles and failure developed due to necking of the specimen.

For Sample 1, Filter "A" material, additional tests at reduced stress levels showed that the material did not meet the specification requirements at 10 cycles of loading. Utilizing all the test results for Filter "A" material, a liquefaction curve has been developed for Filter "A" material and is shown on Figure 2.5.1-6. From this liquefaction curve, a new curve has been developed relating the cyclic shear stress required to cause 5 percent strain to normal effective stress and is presented as Figure 2.5A-17.

The cyclic shear strength criteria of Filter "A" material for 10 cycles of load should be as follows:

Effective Normal Pressure Kips per Sq. Ft.	Cyclic Shear Stress Required to Cause 5×10^{-2} strain
0	0
2.0	0.90
4.0	1.60
6.0	2.20
8.0	2.80

In summary, the following points should be made:

1. Filter "A" material met the cyclic shear criteria for 5 cycles of load. The criteria for 10 cycles of load described in the previous paragraph has been adopted for design of pipe bedding material, where Filter "A" material is utilized in construction as a bedding material. How these results affect the stability of the SSI Dam is discussed in Appendix 2.5A.

2. ASTM C-33, Fine Aggregate met all of the cyclic shear strength criteria as shown in Table 2.5.4-11.

3. Glen Rose Limestone Crushed Stone met all the cyclic shear strength criteria as shown in Table 2.5.4-11 except at normal effective stresses of 6 and 8 KSF at 10 cycles of load. This material can be utilized as backfill material up to a confining pressure of 4 KSY (i.e., backfill to approximately 33 feet in depth.).

4. "Crowder Quarry" Crushed Stone met all the cyclic shear strength criteria as shown in Table 2.5.4-11.

The majority of Category I backfill has been "Crowder Quarry" crushed stone which meets all specifications for cyclic testing when compacted to the required density and the gradation of the materials meets the gradation requirements. Bedding material for Category I pipe has been ASTM C-33 fine aggregate sand which also meets all cyclic shear strength criteria. As stated previously the effect of the cyclic strength tests results on the stability of the SSI Dam are discussed in Appendix 2.5A.

In 1985 after the on-site sources of backfill material were no longer available, another source of material was investigated. Samples of crushed limestone from a quarry in Bridgeport, Texas were tested in order to determine their grain size, compaction and cyclic shear strength characteristics. The results of the test program are presented in Reference 144. Figure 2.5.4-57 shows the results of the cyclic strength testing program in terms of stress ratio versus number of cycles for 5% double amplitude strain. Four cyclic triaxial tests were performed on samples with a maximum grain size of 0.75-inch and a confining stress of 4000psf. At a stress ratio of 0.29, no significant strain developed; however, at a stress ratio of 0.42, 5% double amplitude strain developed in slightly less than 10 cycles. Figure 2.5.4-57 shows that the exact criteria for cyclic strength are not met for test #2; however, the test results are conservative since the in-place backfill contains grain sizes up to 2 inches. The conclusion presented in Reference 144 was that the material met the intent of the specification and was appropriate for use as Category I backfill.

2.5.4.8 Liquefaction Potential

The entire Nuclear Power Plant foundation consists of firm, unweathered, Glen Rose Limestone with no liquefaction susceptible soils present.

The cyclic shear strength of all Category I backfill and bedding materials used show that there is no liquefaction potential; all materials used meet or exceed the design criteria for cyclic strain for the ground acceleration of 0.12 g adopted for the Safe Shutdown Earthquake.

2.5.4.9 Earthquake Design Basis

A conservative Safe Shutdown Earthquake (SSE) having a peak horizontal ground acceleration at the top of bedrock of 0.12 g has been selected for design (Section 2.5.2.6). The Operating Basis Earthquake (OBE) is equal to 1/2 the SSE. Design response spectra and artificial time-history records are described in Section 3.7B.

2.5.4.10 Static Stability

2.5.4.10.1 Bearing Capacity

The major structures at the plant have been constructed on individual mat foundations (except individual spread footings under the Turbine Building columns) founded in the Glen Rose Limestone Formation. Properties of the limestone as well as the claystone material were determined and are presented in Section 2.5.4.2.

The ultimate bearing capacity of the Glen Rose Formation located under each structure has been determined from equations derived for local shear failure from the theory of elasticity [133] as follows:

q (ultimate) = (shear strength) (3.14)

The shear strength has been taken as one-half of the unconfined compressive strength. The above equation does not take into account the effects of overburden or the size of the foundation which would increase the safety factor against a bearing capacity failure. A study of the founding elevations indicated that some upper structures are founded on the claystone. However, when considering the anticipated pressure bulbs due to the applied loads, it is unrealistic to assume that claystone alone is subject to the applied stress. In fact, any shear failure occurring in the claystone zones would also have to pass through the thick underlying limestone layers.

From the results of the unconfined compression tests presented in Table 2.5.4-1, a reasonable average value of unconfined compressive strength for the claystone below elevation 810 is 40 ksf. This results in a very conservative value of 60 ksf for the ultimate bearing capacity of rock beneath Category I structures. The factor of safety against a local bearing capacity failure is the ultimate bearing capacity divided by the maximum foundation edge pressure. Table 2.5.4-6 summarizes the static bearing capacity factors of safety for the Category I structures. The safety factors against local shear failure exceed the minimum allowable of higher than the values presented in Table 2.5.4-6 due to the effects of embedment and foundation size.

The dynamic bearing capacity factors of safety were calculated for the maximum loads caused by the SSE. Table 2.5.4-6a presents the results of this analysis. The safety factors exceed the minimum allowable of 2.

The maximum edge pressure resulting from the overturning effects of earthquake or tornado loadings is 24.9 Kips per square foot under the Containment structure. The safety factor against overturning of the Containment is 2.0. It is calculated by taking moments about the lower edge of the foundation mat using the overturning forces generated by the safe shutdown earthquake. Vertical pressure distribution under the Containment during the SSE is shown in Figure 2.5.4-42.

2.5.4.10.2 Settlement of Foundations

The foundations of all structures in the plant area are founded on the Glen Rose formation. The removal of the overburden to elevation 810 resulted in a slight elastic rebound measured by extensometers to be 0.02 inches (Section 2.5.4.13). Due to the nature of the supporting rock and plant loads, a very small elastic settlement of the structures is anticipated. Time-dependent consolidation settlement of the claystone layers will not be considered due to the fact that these layers are heavily overconsolidated and are minor in proportion to the limestone over an approximate 200 feet thick depth of influence.

The settlements of the Category I structures were calculated using the theory of elasticity for a uniformly loaded area on an elastic half-space. The following equation [138] was utilized:

$$s = C_s \, p \, B\left(\frac{1-\nu^2}{E}\right)$$

where:

s = elastic settlement

C = shape and rigidity factor

p = uniform foundation pressure

B = width of rectangular footing or diameter of a circular footing

υ = static Poisson's ratio, 0.25

E = static elastic modulus, 4.8 x 105 psi.

The rock mass deformation modulus was estimated based on site seismic surveys (Table 2.5.4-5) with a correction for rock mass quality (RQD) [139]. The modulus was also checked using the results of moduli determined by laboratory methods (Table 2.5.4-4). Uncertainty of the effect of blasting and construction activities on the in-situ modulus was considered when evaluating an appropriate modulus for the use in the analysis. The modulus used for the settlement analysis was 4.8 x 105 psi.

A summary of the settlement of the Category I structures is presented in Table 2.5.4-7. The maximum predicted settlement is 0.26 inches at the center of the reactor containment. Since the settlement is elastic, it should occur immediately after final load application.

Expansive characteristics of the claystone were measured by conducting an absorption-pressure-swell test. The results of the test indicate that introduction of water to the sample could generate a pressure of 1140 pounds per square foot in the upward direction when swell is prevented. Therefore, is the applied load equals or exceeds this value, no swell should be anticipated. However, if the loads on the claystone layer are less than 1140 pounds per square foot, some swelling will take place. Figure 2.5.4-43 relates the total potential swell to the thickness of a claystone layer. The interpretation of this graph is shown by use of a hypothetical example on the same sheet. The small amount of swell calculated indicates no problems are anticipated due to the swell potential of the claystone.

During construction, additional samples of the claystone layers were obtained and tested to evaluate the swell potential of the claystone beneath Category I Pipe Structure.

The tests showed a maximum of free-swell of 3.0 percent and ranged from 0.7 to 3.0 percent. The maximum pressure to restrain swelling was 1940 pounds per square foot and ranged from 517 to 1940 pounds per square foot. These values are directly comparable with values reported during design (3.0 percent free-swell and 1875 pounds per square inch absorption pressure).

Along the alignment of the Category I Piping, an analysis shows the swell potential to be in the range of 0.01 inches, which is considered negligible. Therefore, in areas where intact rock has been exposed above the 15 foot line, the removal of claystone beds was not warranted. In areas where pre-splitting of the rock had already been carried down to the 15 foot depth, all fractured rock has been removed to reach the intact sound Glen Rose Limestone and Category I backfill material has been placed. Figure 2.5.4-37A is a plan view of the Surface Water Intake and Discharge pipe trench excavation showing the intact rock elevation along the alignment.

2.5.4.10.3 Lateral Forces

The magnitude and distribution of lateral earth pressures is a function of the allowable yielding of the wall, the backfill or rock characteristics, water pressure, surcharge loads from adjacent structures, and, for seismically designed structures, the earthquake loading. The concrete foundation walls were conservatively assumed to be rigid, unyielding walls. Therefore, the coefficient of earth pressure at rest, K_o, has been used in evaluating lateral soil pressures on these walls. A value of $K_o = 0.47$ was used based on an assumed friction angle of 32° for backfill at the site.

It was conservatively assumed that the groundwater table is at plant grade (elevation 810) except at the service water intake structure where plantside water level was taken as elevation 793. The actual groundwater level conditions have been verified by piezometric data (Section 2.5.4.13).

For structures cast against excavated rock, lateral pressure against the walls has been taken as:

$$\sigma h = \sigma v\left(\frac{v}{1-v}\right) = 0.33v$$

where:

σh = horizontal stress or pressure, psf

σv = vertical stress or pressure, psf

υ = Poisson's Ratio - 0.25

The vertical stress used for the individual structures should also include any surcharge load which exists.

This is an estimate of rock pressure which does not take into account the relative stiffness of the wall and the rock mass. In some cases, a more detailed evaluation of rock pressures based on the unique geological discontinuities was performed.

Dynamic loadings include pressures due to the soil or rock mass, water, and surcharge, accelerated in the vertical and horizontal directions. Figure 2.5.4-54 graphically depicts the static and dynamic lateral earth pressures.

The terms presented on Figure 2.5.4-54 are defined as follows:

γ_t = total unit weight of soil/rock (pcf)

γw = unit weight of water (pcf)

κ_o = coefficient of at-rest earth pressure

q = surcharge load

K_{AE} = seismic earth pressure coefficient

α_h = horizontal seismic coefficient

σ_c = compaction-induced earth pressure

These terms are more clearly defined in Reference 145, 146 and 147.

2.5.4.11 Design Criteria

Methods used to evaluate bearing capacity, settlement and lateral earth pressures are discussed in Section 2.5.4.10. Soil and rock properties used in the analyses are provided in Sections 2.5.4.4 and 2.5.4.5. The slope stability of the SSI and Squaw Creek dams is discussed in Section 2.5.6.

The minimum acceptable design factor of safety for bearing capacity is 3.0 under static loading conditions and 2.0 for dynamic conditions. The minimum allowable factor of safety for the SSI dam slope stability is 1.5 for static conditions and 1.1 for dynamic conditions.

2.5.4.12 Techniques to Improve Subsurface Conditions

No special techniques were required to improve foundation conditions except where blasting shattered or fractured the rock adjacent to or within the plant foundation.

In order to improve these areas of exception, dental concrete was used to replace fractured rock and to fill overexcavated areas. In areas where fractures not requiring removal were encountered, grout pipes were installed. Grouting operations are to be performed at the appropriate time, i.e., when the foundation affected was completed to the point that the grout was confined.

2.5.4.13 Subsurface Instrumentation

2.5.4.13.1 Extensometers

Two extensometers were installed in locations shown on Figure 2.5.4-28. Extensometers were installed during removal of common material from the plant site to measure the rebound of the foundation material and were monitored during the excavation phase of plant construction. At the end of the excavation phase the extensometers were terminated and the data evaluated. The measured rebound was 0.02 inches which is well within the anticipated amount of rebound.

2.5.4.13.2 Post-construction Groundwater Monitoring

A system of 17 piezometers has been installed in the plant area in order to monitor groundwater conditions. The location of the piezometers is shown in Fig. 2.5.4-55 and the typical piezometer detail is shown in Fig. 2.5.4-56.

Design of plant structures initially assumed groundwater at elevation 775 ft. (Section 2.4.13). Installation and observation of these piezometers has verified the actual groundwater condition at the site. The piezometer boreholes were completed as test borings, with continuous NX rock core and water pressure tests throughout the cored interval. Boring logs are presented in Figures 2.5.5-78 through 94. Upon completion of piezometer installation, groundwater data was collected for a period of one year to determine seasonal variations. The groundwater level was also correlated with weekly rainfall.

2.5.4.14 Construction Notes

Required excavation operations, including blasting, at times resulted in fracturing or shattering of the foundation rock adjacent to or within the plant foundation. Such excavation and subsequent backfilling with pervious material of the duct banks, piping, etc. throughout the plant site has changed the rock formation from an impervious material to a fractured and pervious material. The horizontal and vertical extent of the fractures and backfilling varies over the site.

Section 2.5.4.12 describes the treatment of shattered or fractured foundation rock.

Frequent changes in construction techniques were made in the early stages of excavation in the plant area. All such changes were within specification requirements, were made to reduce fracturing of rock, and included the following techniques:

1. Line drilling to isolate the proposed excavation area

2. Pre-splitting

3. Rock sawing (isolation)

4. Blast hole spacing

5. Amount and type of explosives

6. Time-delay of multiple blasts

No significant construction problems, other than rock fracturing, have been encountered.

During construction small volumes of water in-leakage at various structures was observed. The in-leakage problem exists at the turbine building and various Category I structures. This condition was solved at the turbine building by installing a permanent drainage system.

The in-leakage is caused by perched water collected at the foundation exteriors. The perched water source is mainly the result of surface water, such as rainfall runoff or construction and maintenance runoff. The water migrates to the foundation exterior surfaces through the utility trenches, filled with permeable bedding and backfill material, that intersect the structures.

Seventeen piezometers (Section 2.5.4.13.2) have been installed at the locations shown in Figure 2.5.4-55 and were monitored for a period of one year. The perched water condition around the plant structures has been evaluated and no remedial actions were necessary.

2.5.5 STABILITY OF SLOPES

2.5.5.1 Slope Characteristics

There are no slopes, either natural or man-made, in the vicinity of the plant, whose failure could adversely effect the plant.

The existing slopes along the creek bed that form the reservoir behind the Safe Shutdown Impoundment Dam are comparatively gently slopes and are characterized by a very thin mantle of soil underlain by limestone of the Glen Rose Formation. In many areas, all soil cover has been eroded and the rock outcrops. The steepest slope present along the reservoir boundary occurs on the south side of the Safe Shutdown Impoundment and approximately 600 feet upstream from the axis of the SSI Dam (Figure 2.5.5-2). Current topographic data indicates that at this location, the ground elevation increases from elevation 750 to elevation 800 over a horizontal distance of approximately 150 feet, or a slope of approximately 3 horizontal to 1 vertical. All other slopes along the total impoundment shoreline are in the range of 4 horizontal to 1 vertical, or flatter.

Topography of a part of the reservoir in the plant site vicinity, shown on a 10 foot contour interval is presented on Figure 2.5.5-5 together with the location of Sections Numbered 1-1, 2-2, 3-3, and 4- 4. These sections include the steepest slope condition along the shoreline of the SSI reservoir.

Subsurface profiles at each location were obtained by trench excavation. A wedge-type stability analysis using moist, saturated and submerged unit weights, appropriate strength properties for the materials present, and a seismic coefficient of 0.12 g was made for Sections 1-1, 2-2, 3-3 and 4-4. The results indicate a minimum safety factor against a slope slide of greater than 3. The Sections, including the results of the wedge-type stability analysis, are shown on Figures 2.5.5-1, 2.5.5-2, 2.5.5-3, and 2.5.5-4.

Some degree of weathering of the rock outcrops has occurred, and if one assumes that the weathering has extended to such a degree as to be the equivalent of a granular material with a 0 angle of 35 degrees, the existing slope will have a safety factor against a shallow near- surface slide greater than 2.0, a value considered entirely adequate. Any slide deeper than that of the weathered rock will have a larger safety factor.

2.5.5.2 Design Criteria and Analyses

The method of dynamic analysis of the SSI Dam is described in Section 3.7B.2.13 and the details of the analysis are presented in Section 2.5.6.5.

2.5.5.3 Log of Borings

The locations of borings made for the plant and dam explorations are shown in Figures 2.5.5-5 and 2.5.5-6. The logs of these borings are presented in Figures 2.5.5-7 through 2.5.5-76.

Borings were drilled with truck-mounted rotary-wash or auger drilling equipment. Overburden soil samples suitable for testing were obtained from some of the borings, and the underlying rock has been continuously cored utilizing standard NX coring equipment. The soils encountered have been classified in accordance with the Unified Soil Classification System. Geologic rock classifications have been based on macroscopic and hand lens examination of cores. Percent core recovery and R.Q.D. (Rock Quality Designation) information are presented on the boring logs.

The borings were drilled in the overburden soils by advancing a 3 or 4 inch diameter hole to the desired sampling depth. Undisturbed soil samples were obtained by hydraulically pressing a 2-inch Shelby tube or a 2.5-inch diameter thinwall sampler. Some disturbed samples were obtained from the auger flight, and disturbed Standard Penetration Test samples were also taken.

Drilling mud and casing were used when necessary during the drilling of overburden soils. Prior to coring the underlying rock, each boring was flushed thoroughly with water.

Two-inch diameter plastic pipes were installed in several borings to enable observation of groundwater levels (Figure 2.5.5-5). Because the Glen Rose formation is essentially impermeable, water levels reach static equilibrium slowly. Therefore, observations were continued for an extended period to confirm static levels. The observations made in borings at Category I facilities are summarized in Figure 2.5.5-77.

The static water level in the Twin Mountains formation was observed in Boring P-10, one of two borings penetrating the stratum. The piezometric level is indicated to be Elevation 670.

2.5.5.4 Compacted Fill

A detailed discussion of the areas requiring fill and the sources of fill materials required is given in Section 2.5.4.5. The extent of these fill areas are shown on Figure 2.5.4-28.

2.5.6 EMBANKMENTS AND DAMS

2.5.6.1 General

2.5.6.1.1 Squaw Creek Dam

The purpose of Squaw Creek Dam (SCD) is to impound a cooling lake for CPNPP. The location and configuration of the reservoir are shown in Figures 2.5.6-1 and 2.4-1. Under normal operating conditions, the reservoir will remain in the five-ft range between a minimum operating

level at elevation 770.0 and the crest of the service spillway at elevation 775.0. Rainfall or pumping of makeup water can result in flow over the Service Spillway Crest.

The layout of Squaw Creek Dam is shown in Figure 2.4-16. A typical cross-section of the embankment is shown in Figure 2.4-17. The dam is designed to pass the Probable Maximum Flood with sufficient freeboard to prevent overtopping by wave action generated by a sustained 40 mph overland wind. The top of the dam is a elevation 796.0 with overbuild to elevation 798.4 to compensate for estimated settlement. The central section is constructed of select, impervious material, with a cutoff extending down to impervious foundation material. The outer zones of the embankment are of less select material with the outer portion of the downstream shell being rock fill. A filter system separates the impervious central zone from the less select outer zone on the downstream side and extends outward to the downstream toe to provide drainage and piping protection for the core.

The reservoir side of the dam is protected by rock riprap and a gravel blanket from the top of the embankment to elevation 760.0, which is 10.0 ft. below the minimum operating level. The top width of the embankment is 20 ft., exclusive of riprap and blanket. Design of the riprap was based on an average over-water wind of 95 mph.

The service spillway is an uncontrolled structure 100 ft wide, with an ogee crest at elevation 775.0 Additional discharge capacity for protection from extreme floods is provided by a broadcrest emergency spillway, 2,200 ft. wide, excavated in rock of the east abutment at elevation 783.0. The sides of the discharge channel from the stilling basin to Squaw Creek are protected with rock riprap and blanket. The floor of the channel is native limestone. A 12-inch makeup water pipeline and 36-inch diameter return pipeline crosses the emergency spillway along its crest. Both lines are placed in a trench cut in limestone and is covered with a concrete cap. The top of this cap is at elevation 783.0.

Service outlet facilities are located near the right abutment of the dam and consist of an intake tower and outlet conduit. The tower has a 6 ft. by 6 ft. control gate at its base leading to a 6 ft. diameter conduit through the dam that discharges to Squaw Creek. The tower has ports at various levels to allow selective taking of water from the reservoir. Backup closure facilities are provided to protect against loss of storage due to inability to close a valve or gate.

2.5.6.1.2 Safe Shutdown Impoundment Dam

A portion of the arm of the reservoir that is formed by the channel of Panther Branch is utilized as a Safe Shutdown Impoundment, which holds water for emergency cooling use. The secondary reservoir is separated from the main body of the reservoir by a rock fill dam. An open spillway channel was excavated through the narrow ridge to the southwest of the SSI Dam, to connect the SSI with the main body of the reservoir. The floor of the channel is at elevation 769.5, six in. below the normal minimum operating level, and water will pass back and forth to keep the large and small reservoir surfaces at the same elevation under normal operating conditions. If the level in the main reservoir should drop due to some emergency, the SSI Dam will hold back 367 acre-ft of reserve water to allow continued cooling and safe shutdown of the plant.

The location and layout of the SSI Dam are shown in Figures 2.5.6-1 and 2.4-23. Details of the SSI Dam embankment are shown in Figure 2.4-21. The middle zone is of select, impervious material, wetted and rolled, and carried down to impervious foundation material for effective cut-off. The outer zones are rock fill. Two-stage filters were placed between the core and the

rock fill sections. The embankment crest is 40 feet wide at elevation 796.0. The spillway channel is 40 feet wide, with side slopes of 1-on-3 and a channel slope of 0.003 upstream and downstream from a concrete wall approximately flush with the channel bottom.

The SSI Dam is designed to pass the Probable Maximum Flood with or without the presence of SCR, with sufficient freeboard to prevent overtopping by waves generated in either direction, i.e. from the SSI or SCR, by a sustained 40 mph overland wind.

The embankment is designed to be stable during the rapid drawdown caused by the sudden and total loss of the SCR.

Seismic Design Criteria for the SSI are discussed in Section 3.7.

The ability of the SSI to meet the criteria of Regulatory Guide 1.27 is discussed in Section 9.2.5.

2.5.6.2 Exploration

The subsurface investigation conducted for Squaw Creek Dam and SSI Dam during design consisted of the undisturbed sampling of alluvial overburden materials as well as the undisturbed sampling of the primary sediments present.

The cohesive soils present along the centerline of the dam site were sampled at vertical intervals of 5.0 ft or at a change in material type, whichever was the lesser distance. These samples of cohesive materials were obtained by means of a thin-walled, seamless, Shelby tube sampler forced into the soil by means of a rapid thrust from two balanced hydraulic rams. The samples were then field extracted, preserved in moisture-proof polyethylene plastic and sealed to prevent changes in moisture content and accompanying changes in physical characteristics, identified as to boring number and depth, and packaged for transport to the laboratory in Dallas, Texas, for further tests and analyses.

Primary sediments encountered were sampled continuously utilizing a double tube, bottom discharge, conventional coring tool fitted with an appropriate cutting bit, using water as a drilling fluid.

Field infiltration tests were conducted in selected zones of the alluvial material in order to establish the permeability of the soil. In addition, pressure testing of the primary sediments was accomplished by the insertion of an expandable packer at the location selected while water pressure inside the boring, between the packer and the bottom of the boring, was regulated by a water pump at the ground surface. The depth of the tests pressure in pounds per square in., and the quantity of water consumed during the tests are recorded, where applicable, on the individual field boring logs, and are summarized on Table 2.5.6-1. In addition, selected samples of primary sediments penetrated at the site were preserved in polyethylene plastic, in the manner described above, and all samples were transferred to the laboratory for further evaluation and tests.

Detailed descriptions of materials penetrated as well as all field testing conducted in conjunction with this investigative program are shown, where appropriate, on the boring logs which are discussed in Section 2.5.5.3.

2.5.6.2.1 Squaw Creek Dam

A series of NX-size vertical core borings were drilled in connection with the exploration for the foundations of the Squaw Creek Dam and appurtenances. Figure 2.5.5-6 shows the location of core borings in relationship to the centerline of the dam.

Borings identified as the D-I series and the D-II series, a total of 28 NX-size core borings, represent the foundation exploration along the centerline of the dam. A total of nine NX-size core borings, identified as Series ES borings, have been utilized to explore the emergency spillway location. Borings identified as SO-1 through SO-11 were drilled for the purpose of exploring two potential alignments for the service outlet works to be located in the valley section of the proposed construction. The service spillway, located on the right abutment of the proposed dam site, has been explored by a series of 16 NX-size core borings numbered SS-1 through SS-16. Additional foundation data was obtained through the drilling for two, four-inch diameter core borings located within the proposed spillway and identified herein as Borings Numbered SS-17 and SS-18.

In addition, further investigation of the service spillway area was accomplished by means of four NX-size, inclined, core borings drilled at the locations shown on the Figure 2.5.5-6 and identified as core borings numbered SSA-1 through SSA-4.

Geologic Profiles representing cross-sections along the Embankment Centerline, the Service Outlet Works, and the Service Spillway are shown on Figure 2.5.6-7 through 2.5.6-9, respectively.

2.5.6.2.2 Safe Shutdown Impoundment Dam (SSI Dam)

A series of NX-size vertical and angle core borings were drilled in connection with the exploration for the foundation of the SSI Dam, the equalization channel and rock excavation areas. Figure 2.5.5-5 shows the location of the core borings in relation to the centerline of the dam and equalization channel.

Borings identified as the M Series, a total of ten NX-size core borings, represent the foundation excavation along the alignment of the SSI Dam. Borings MS-7 and MS-8 are vertical, NX-sized core borings drilled in the equalization channel area. Borings BA-1 and 2 are angle borings drilled in the balancing channel area.

A geological profile along the centerline of the SSI Dam is presented in Figure 2.5.6-10. A geological section was developed for the equalization channel using the information from MS-7 and 8 and BA-1 and 2. This geological section is shown on Figure 2.5.6-11.

2.5.6.2.3 Borrow Material

Several potential borrow areas were explored and evaluated for use as potential impervious core and random fill material for Squaw Creek Dam and SSI Dam. Figure 2.5.4-29 shows the approximate locations of the different borrow areas investigated. Each of the borrow areas was investigated by a series of six-inch diameter continuous-flight auger borings or excavated pits. Laboratory analysis and engineering evaluation have been conducted on the materials lifted from the auger borings in order to develop design parameters utilizing such materials. Additional core

borings with field testing were drilled during construction in the area of the Service Water Outlet alignment and in the West Abutment to help verify depths to competent rock.

2.5.6.3 Foundation and Abutment Treatment

2.5.6.3.1 Squaw Creek Dam

Squaw Creek Dam is an earth and rockfill embankment structure founded on existing valley alluvium. To minimize leakage through the foundation, a core trench was excavated through the alluvium and weathered rock to impervious unweathered limestone and claystone. This core trench extends well into the abutments. Figure 2.5.6-2 is a profile of the completed excavation.

All transverse cracks were removed by excavation or intersected by a keyway trench, except in the vicinity of the service outlet where removal would undermine the structure. Here a rock saw was employed to cut an 11-in.-wide trench across the cracks to the depth of the crack. The trench was then backfilled with dental concrete. Water was encountered in vertical cracks exposed in the keyway trench, especially on the right side of the creek and at the horizontal contact of the alluvium and limestone. This water could cause the exposed claystone seams to weather rapidly causing overhangs of the limestone layer above the seams. To prevent this weathering, pneumatically placed concrete (shotcrete) was used to seal the claystone seams. The inflow of water from the vertical cracks was controlled by inserting a steel pipe at the base of the crack and then sealing the crack with shotcrete. When the shotcrete had gained strength and it was desired to begin backfill, the pipes were sealed with wood plugs. Potholes and overhangs were filled with dental concrete. Dental concrete and shotcrete were used to cover rough rock surfaces to provide a smooth surface to place backfill against. Figure 2.5.6-5C(1) is a photograph showing shotcreting activities.

Prior to excavation, the foundation for the random fill and rockfill zones of the embankment was cleared and grubbed of trees and roots and then stripped to rock or a maximum depth of 9 in. to remove topsoil, rubbish and vegetation. The foundations for the service spillway an service outlet structures were then excavated to solid limestone.

Claystone seams were sealed with a coating of shotcrete. Over excavation was corrected with 1500 psi concrete. Anchor bars were installed as required and the foundation drains were drilled after placement of the structural concrete. Water encountered in the excavation of the foundation for the service outlet intake tower was controlled by sumping and pumping.

The effectiveness of the cutoff trench was judged visually. Mason- Johnston & Associates, Inc., Geotechnical Consultants, determined when the weathered zone had been penetrated. The effectiveness of the saw cutoff trench was tested by observing the depth of the crack and, in some cases, the stoppage of the flow of water.

Weathered material extended to a greater depth in the right abutment than anticipated, causing the cutoff trench to be extended some 250 ft. into this abutment. The trench was terminated prior to reaching the service spillway. The water tightness of the formation between the service spillway structure and the termination of the deep core trench was verified by drilling both vertical and horizontal holes and pressure testing. These hole were filled with grout.

Construction procedures used for the foundation and abutment treatment were as follows:

Soil material was removed by scrapers.

Limestone and claystone were removed by blasting which was prededed by pre-splitting the limits of the excavation. Spacing and loading of the blast holes were adjusted as conditions dictated. Blasted material was removed by front end loaders and end dump trucks. Final shaping of the excavation was accomplished with demolition tools and hand-operated pavement breakers.

Dental concrete had a 28 day design strength of 3000 psi. The concrete was centrally batched and placed in accordance with the procedures of Section 8, ACI 301-72.

The pneumatically-placed concrete was a wet pre-mixed mixture conveyed by air slugs through a flexible tube and deposited in place by air pressure. The material had a minimum 7 day strength of 2400 psi and a minimum 28 day strength of 3000 psi.

All excavations were pumped dry, cleaned with high velocity jets of air and/or water and inspected prior to backfill.

Estimates of the construction quantities involved are included in Section 2.5.6.9.1.

2.5.6.3.2 Safe Shutdown Impoundment Dam

Safe Shutdown Impoundment Dam is a rockfill dam with an impervious earth core. The entire foundation was excavated to unweathered limestone to provide suitable support for the rockfill shells and to provide an impervious base for the core. (The term unweathered is interpreted with respect to engineering properties and not in a purely geological sense.) Figure 2.5.6-3 is a profile of the completed excavation.

All transverse cracks under the core zone were removed by excavation or sealed by making a 10-in.-wide saw cut across the crack for its full depth and backfilling with dental concrete. Some seepage was encountered on the upstream side of the excavation in rock strata above the foundation base. Pneumatically placed concrete (shotcrete) was used to seal claystone seams to prevent weathering and possible overhangs. This shotcrete was removed just prior to placement of core and filter material in their respective zones.

The abutments against which the core material would be place were cut to a slope not steeper than one on one in order to provide “wedging” as the core consolidated, thus increasing the watertightness of the contact between core and abutment. Vertical faces not in excess of 30 in. (the largest size rock in rockfill) were allowed under the rockfill zone. Verticals in excess of 30 in. were brought to a slope of one on one with fill concrete. Potholes were leveled with dental concrete.

The spillway is an open cut channel between SCR and the SSI. This channel was bottomed on limestone. The crest elevation is maintained by a concrete wall across the channel that is excavated into the limestone foundation.

The effectiveness of the impervious cutoff was further assured by excavating a trench 2 ft. wide, and ranging in depth form 6 to 8 ft., across the base of the foundation on the centerline of the dam. This trench was visually inspected by the Mason-Johnston Geotechnical Consultant to assure that all cracks had been intercepted prior to its backfill with dental concrete. All

excavation to unweathered limestone was accepted by the Engineering Geologist prior to covering with shotcrete, dental or fill concrete. Upon completion of all excavation, the entire foundation was jointly inspected and accepted by the Freese and Nichols Project Engineer and the Mason-Johnston Project Geotechnical Consultant.

Construction procedures used for the foundation and abutment treatment were the same as described for the SCD in Section 2.5.6.3.1. Estimates of the construction quantities involved are included in Section 2.5.6.9.2.

2.5.6.4 Embankment

2.5.6.4.1 Squaw Creek Dam

A typical cross-section of Squaw Creek Dam is shown on Figure 2.4-17. The top of the dam is at elevation 796.0. The central core is constructed of select, impervious material wetted and rolled, with a cutoff trench extending down into the impervious rock foundation material. The outer zone of the embankment are of a less select material with the outer zone of the downstream side being a rock fill. The filter system separates the impervious central zone from the less select outer zone on the downstream side and extends outward to the downstream toe to provide drainage and protection of the core.

The upstream side of the dam is protected by riprap and gravel blanket from the top of the embankment to elevation 760.0, which is ten feet below the minimum operating level. The crest width of the embankment is twenty feet.

The sides slopes of the dam as shown on Figure 2.5.1-7 are three horizontal to one vertical on the downstream slope and two horizontal to one vertical to elevation 775 on the upstream slope and below 775 a three horizontal to one vertical slope. An upstream berm is located at elevation 760.0.

2.5.6.4.2 Safe Shutdown Impoundment Dam

A typical section of the SSI Dam is shown on Figure 2.4-21. The middle zone of the SSI Dam is composed of select impervious material, wetted and rolled, and carried down to the underlying firm, unweathered Glen Rose Limestone Formation. The outer zones of the SSI Dam consist entirely of the firm Glen Rose Limestone selectively quarried and processed to remove all claystone particles from the rock shell. The two zones of the SSI Dam are separated by two-stage filters. All components of the SSI embankment extend vertically downward to contact with the firm, unweathered, Glen Rose Limestone.

There are many aspects of the design construction procedure of this Class I Dam which make it much more conservative than most other existing dams. Primarily, these consist of the following:

1. The outside slope of the SSI Dam is flatter than slopes commonly used for a fill rock dam. Dams considerably higher than the SSI have been constructed in more seismically active areas with outside slopes in the range of 1.75 horizontal to 1 vertical.

2. The rock shell fill of the SSI Dam was placed in layers and compacted with multiple passes of a vibratory 10 ton roller. This compaction of the fill rock will provide a material with great strength and low compressibility.

3. The foundation of the entire SSI embankment is supported by the firm, unweathered Glen Rose Limestone.

4. The crest width of the dam (elevation 796) is 40 ft., which is considerably greater than most existing dams of 70 ft. height.

5. The width of the core at normal water surface is 20 feet.

6. The core material, while considered practically impervious, is not highly plastic. (CL in the Unified Soil Classification System).

7. Two stage filter, both upstream and downstream of the core, are present to control seepage, prevent piping, and control the effects of any cracks that may develop during the occurrence of the SSE.

A comparison of the SSI Dam with other existing fill rock dams in areas much more seismically active than the CPNPP site is presented in Table 2.5.6-2. Each of the dams in Table 2.5.6-2 has been constructed with slopes, both upstream and downstream, steeper than those for the SSI Dam at the CPNPP Project and constructed in a much more seismically active area. In summary, this SSI Dam is considered most conservative in design and construction and is conservatively appropriate for its intended purpose of providing a reservoir to function as the Ultimate Heat Sink during SSE conditions.

2.5.6.4.3 Material Properties Used for Design

A discussion of the laboratory procedures and tests used for adoption of both static and dynamic material properties for the rock shell, filter and clay core materials is included in the following subsections.

1. Shell Material

The laboratory modeling technique used during design for shell material consisted of the following sequential steps:

a. A probable range of gradations resulting from drilling, blasting, loading, placing, and compacting of the Glen Rose Limestone was estimated from prior experience, taking into account the known gradation of core material. The limits of probable design gradation of the shell material are shown on Figure 2.5.6-12.

b. Laboratory model techniques were used to obtain engineering properties of the rockfill. Actual NX (2 1/8" diameter) cores of the limestone from the plant site exploration borings, including a statistical average amount of claystone seams, were crushed in the laboratory to conform to the estimated gradation on a fraction-by- fraction basis; i. e., ultimate 10 inch diameter stones were reduced in the laboratory to particles 3 mm in diameter; one-quarter inch particles were reduced to 0.2 mm particles, etc. The gradation curve of the shell gradation model is shown on Figure 2.5.6-13.

c. The material representing the shell was then remolded, placed in a triaxial chamber, and a multi-stage “Q” test was conducted. The Mohr Diagram obtained

from this type of test is shown as Figure 2.5.6-14. Engineering properties of the shell material utilized for design purpose, are as follows:

Cohesion = 0

$\varphi = 37°$

Tan $\varphi = 0.754$

Unit Dry/Moist Weight = 115 pcf

Unit Saturated Weight = 135 pcf

2. Core Material

Borrow areas upstream of Squaw Creek Dam and downstream from the SSI Dam were investigated as potential borrow to form the impervious clay core of the two dams. Materials available for use in the core are CL in classification (Unified Soil Classification System) and have Liquid Limits ranging from 31 to 44 percent, Plastic indices ranging from 11 to 22, and Specific Gravities ranging from 2.65 to 2.70. Strength properties were determined on remolded materials prepared at moisture contents ranging from one percent below optimum to 3 percent above optimum and at density ranges equal to approximately 95 percent of the maximum density by the Standard AASHO Compaction Procedure.

Strength properties were determined by means of the Direct Shear test, where the rate of strain was computed to avoid a build-up of pore pressure and by triaxial testing in the form of Q test and R tests where complete saturation was obtained using back pressure and where pore pressures were measured.

Settlement and drainage characteristics of the impervious clay core were estimated from data obtained during consolidation testing with the coefficient of permeability, in the range of 10^{-8} centimeters per second, being computed from consolidation characteristics.

Typical results of laboratory tests conducted during design analysis of the material that has been used in the clay core of the SSI Dam are represented by Figures 2.5.6-15 through 2.5.6-17.

Cyclic strength determinations were not an important consideration in the dynamic analysis as explained in Section 2.5.6.5.2.

Based on a detailed analysis of the clay material, the following properties have been adopted for design of the core material:

Liquid Limit = 30-40%

Plasticity Index = 12-20

Optimum Moisture = 12%

Embankment Unit Dry Weight = 108 pcf

Embankment Unit Saturated Weight = 128 pcf

Strength Properties, Considolidated-Undrained (CU) Cohesion = 432 psf

$\varphi = 18°$

Tan $\varphi = 0.325$

3. Filter Zones

Materials that were used to form the upstream and downstream filters, separating the core from the shell, conformed to the gradation requirement of both the shell and core materials and was obtained offsite from Brazos River alluvial deposits. A two stage filter system has been utilized with the gradation limits of each zone shown on Figure 2.5.6-12.

4. Dynamic Strength Properties of the Filters and Rock Shell

For design purposes the seismic shear strength of the filter materials and rock shell for the dam has been estimated based on published data for granular soil, field embankment tests, and adopted criteria for construction. Factors considered in the assessment of the seismic strength characteristics included; (a) relative density; (b) grain size; (c) gradation; and (d) effect of initial static shear stresses.

Published data on the effect of relative density on seismic strength of filter materials have shown an increase in strength with increase in relative density. Based on this data, a relative density of at least 80 percent has been specified for the filters.

The effect of mean grain size on the cyclic shear strength of granular soils have been studied by Lee and Fitton [136], Seed and Peacock [137], and Wong (1970). The results of Lee and Fitton indicate a substantial increase in seismic strength as D_{50} increases from about 0.1 to 4 mm, and a very large increase in strength as D_{50} increases above 4 mm. The data presented by Seed and Peacock confirms Lee and Fitton's results. The results of Wong indicate only a slight increase in strength as the mean grain size increases from 0.6 to 10 mm, but a rapid increase in strength for mean grain sizes larger than 10mm.

The mean grain size of the fine filter is 0.6 mm (average) and the coarse filter 8.0 mm (average) as shown by Figure 2.5.6-12. Although the available published data would indicate a slight to substantial increase in strength as the mean grain size increases, this effect has been conservatively ignored in assessing the cyclic shear strength characteristics of the filter. Rather, the cyclic shear strength of the filter has been assessed on the basis of data for Sacramento River sand [135]. The data on this particular sand (a uniformly graded fine sand with $D_{50} = 0.2$ mm) has been utilized because of the comprehensive cyclic test data published, covering a wide range of confining pressures, relative densities, and seismic stress-strain levels.

Although the filter material is a well-graded material, whereas Sacramento River sand is a uniformly-graded sand, it is felt that little, if any, difference in the strength can be detected between well-graded and uniformly graded material for the same mean grain size. Research by Lee and-Fitton [136] confirm this conclusion. Wong (1970) tested a well-graded sandy gravel and found that the strengths were lower than uniformly graded gravels at low strains (approximately 5×10^{-2}); however, the well-graded gravels were more resistant to the development of large strain; in other words, 10×10^{-2}. Therefore, the published cyclic triaxial tests data for Sacramento River sand provide a reasonable basis for assessing the cyclic strength characteristics of the filters. Figure 2.5.6-48, prepared from this data, shows the cyclic shear stresses required to cause 5×10^{-2} strain in 5 and 10 cycles for filter materials compacted at a relative density of 80%.

The dynamic strength of the rock shell has been estimated in a similar manner as for the filters. Based on data by Wong (1970), the effect of mean grain sizes larger than 10 mm results in a rapid increase in cyclic strength. For a mean grain size of 100 mm, an increase in the cyclic strength of 83% may be expected and for a mean grain size of 152 mm at least 100% increase may be expected. The gradation of the rock in the actual test embankment had a mean grain size of 152 mm and the average of the rock shell in the actual dam is estimated to be no lower than 100 mm. Thus, the cyclic strength may be expected to be increased between 83% and 100%. To be conservative in analysis of the SSI Dam, the relationship shown in Figure 2.5.6-47 was derived. This figure presents the cyclic shear stresses required to cause 5×10^{-2} strain in 10 cycles for the compacted rock shell.

Initial static shear stress effects which occur in the filters and rock shell, and would result in an increase in material strength, have been ignored.

In developing the cyclic shear strength characteristics for design, it was assumed that pore pressures do not dissipate during earthquake motions. Because of the large particle size and high permeability of the rockfill, it is certain that any pore pressures would dissipate to some degree during the ground motions, and this would result in higher rock shell strengths than are being used in this analysis.

Lee and Fitton [136] found that clay materials have cyclic shear strengths greater than cyclic shear strengths for sands; however, the clay core of the SSI Dam has not been evaluated for cyclic shear strength but has been analyzed using the conservative assumption of liquefaction as described in Section 2.5.6.5.2.

During construction a test program was set up to evaluate the cyclic shear strengths of backfill material used at CPNPP to verify design assumptions. Included in the specimens tested were the fine filter material, Filter “A”. The details of the investigation are discussed in Section 2.5.4.7. How the results of Filter “A” tests affect the stability of the SSI Dam is discussed in Appendix 2.5A.

2.5.6.4.4 Rock Field Tests

A series of field tests for fill rock properties was conducted during design at the site of an abandoned quarry located at the CPNPP site. The location was selected to ensure that materials tested would be representative of those actually used in the construction of the SSI

Dam. Core borings were made at the proposed quarry site and a careful study was made of all cores to insure that the test quarry rock was representative of that which would be used in construction.

The primary objectives of the rock field tests included:

1. Determination of excavation procedures.

2. Evaluation of several wall-protection schemes.

3. Determination of compacted fill rock properties by construction and evaluation of the test embankment section.

Excavation operations at the quarry site consisted of the removal of all residual soils by the use of suitable-equipped crawler equipment, the drilling of blast holes to depths considered necessary to provide the desired quantity of materials, presplitting holes and blasting so as to provide a near vertical quarry face, and the removal of all excavated materials by loading equipment. The material was then transported to the test-embankment site and compacted in various layer thicknesses with various roller passes. The completed test-embankment was evaluated by means of a test trench. The test trench showed the presence of claystone particles. Since the test-embankment obtained with this type of quarry operation was considered totally unsuitable for high quality, pervious, rock fill section due to the presence of claystone particles, this source of material was abandoned.

Making full use of the results obtained from the first test quarry program, a new area immediately adjacent to the first area was prepared for excavation by removing all residual soils, drilling and shooting pre-splitting holes to establish a good quarry wall surface, then drilling and shooting blast holes on several patterns. Blast hole depths were controlled so as to bottom immediately above the elevation of known claystone layers. Shot hole patterns of 7 ft. by 7 ft. and 9 ft. by 9 ft. were used; in each case, holes were 3.5 in. in diameter. All holes were bottom loaded with DuPont Pour-vex, column loaded with HD-1 primer, and shot with primer cord. Water gel was used when some of the holes were noted to contain water.

Shot hole spacing, depths, and powder were varied to provide proper maximum size rock, well-graded rock with a minimum of spalls, and little or no claystone particles. With the controlled blasting operation, the limestone portion of the quarry rock was loaded out to the processing area and the remaining claystone areas and zones, readily identified, were removed by crawler equipment and wasted.

Quarried rock was processed by use of a vibrating grizzly and a vibrating screen; the size of openings in the vibrating screen were altered, as required, until material of a desired gradation was obtained.

The test embankment construction consisted of the following basic steps:

1. An area was selected for the test embankment and stripped of all overburden soil and weathered rock until a smooth, hard, and level unweathered Glen Rose Limestone surface was obtained. Corners of a 50 x 100 foot rectangle were marked.

2. Processed quarried rock was then hauled to the test embankment site, dumped, and smoothed to an approximately two-foot thick lift with a dozer. A ten-ton smooth wheel vibratory roller with the vibration mechanism disengaged was then allowed to make one pass to further smoother the surface of the lift. A grid pattern of approximately 10 by 7 ft. was established (21 points) on the surface of the rock layer and the elevation at each grid point determined and recorded.

3. Each layer was then compacted with a self-propelled vibratory roller weighing 10 tons and creating a dynamic force of 40,000 pounds, vibrating at 1400 cycles per minute, and having a drum diameter of 60 inches with a drum length of 104 inches. The elevation of the marked grid pattern points was determined after each pass of the vibratory roller; the number of passes varied up to six.

4. After the construction of the embankment was complete, a trench was excavated in the central portion of the test embankment and at right angles to the direction of roller passes to determine conditions of the embankment cross section.

Following the completion of the embankment, a designated area of the embankment was sluiced with water to visually determine the permeability and drainage characteristics of he rock fill and to provide an approximation of the moisture content retained in the fill.

Two large-scale density tests were made on the embankment in the following manner:

1. Following the placement of a particular lift and the application of a particular number of passes of a vibratory roller, a four-foot diameter steel ring was placed on the rock surface and secured in place by sand bags on the perimeter flange of the rings.

2. The rock material inside the ring and for a depth equal to the lift thickness, in the range of two feet, was hand excavated and saved for further use.

3. The excavation was lined with polyethylene film very loosely, including the bottom and sides.

4. Water levels from calibrated barrels were noted and water from these barrels was introduced into the polyethylene lines excavation until the level was within one in. of the steel ring.

5. The steel ring was then removed.

6. Additional water was added until the excavation was full; water levels in the calibrated barrel were noted and the volume of he excavation determined by the volume of water used.

7. The weight of the rock excavated was determined.

8. The “moist” density was determined by dividing the weight of the rock by the volume of the hole.

9. The rock from the excavation was “quartered”, piece-by-piece by hand, to obtain a smaller but representative sample and the moisture content was determined.

10. The “dry” density was then determined.

The results of all field observations and compaction tests, including grain size distribution, have been analyzed and are provided in Figure 2.5.6-18 through 2.5.6-24 and Table 2.5.6-8.

After removal of the desired quantity of rock from the test quarry, the existing floor was cleaned and the previously obtained vertical wall resulting from the pre-splitting operation was washed clean with a water nozzle. Shot crete panels were then constructed in order to produce a protecting surface against weathering on the vertical rock face. Two panels were selected; on the first panel a shot crete thickness of approximately one in. was constructed while on the second panel a surface approximately two in. in thickness was constructed. A seven sack cement mix was utilized in the construction of both panels.

Two shear tests were made in the field. The first shear test was run to determine the angle of internal friction between the rockfill embankment material and the in-place, firm, unweathered Glen Rose Limestone material. After determination of the friction angle between the rockfill embankment material and the underlying relatively smooth rock surface, a second direct shear test was made of the rockfill embankment material itself.

The shear box consisted of a reinforced, metal wall, six-ft cube, open on top and bottom, and separated on the bottom from the underlying relatively smooth limestone base by a system of small diameter rollers. Loads were applied to the box by means of a calibrated hydraulic ram acting through a suitable yoke arrangement and a ball-type joint. Deflections, or movements, were measured by means of dial micrometers placed at appropriate locations along the bottom and top edge of the shear box. Schematic details of the physical arrangement are illustrated on Figure 2.5.6-25.

Boulders larger than 12 in. in diameter were removed from the rockfill material so as to be compatible with the physical dimensions of the large scale shear box; all materials were weighed prior to placing in the shear box and then compacted with a “jumping jack” compactor to a density equivalent of that previously obtained from large scale density tests made after completion of the test embankment. The gravitational weight of the rock produced the normal force for four successive layer thicknesses.

The results of these tests are shown on Figures 2.5.6-21 and 2.5.6-22. The angle of sliding friction of rockfill material on in place limestone of approximately 42.9 degrees was determined as shown by Figure 2.5.6-23.

At the completion of the sliding resistance shear test, a strip of metal 4 in. in height was taken out of the shear box at about mid- height of the box and the hydraulic ram was removed so that the upper part of the box could be sheared (relative to the fixed lower portion of the box) to determine the angle of internal friction of the rockfill embankment material. The rock in the zone of the shear plane was limited to slightly less than 4 in. in size to correspond with the mechanics of the shear box. The results of these tests are shown on Figures 2.5.6-21 and 2.5.6-22 and indicate an angle of internal friction of 48 degrees.

Rockfill embankment material removed from the large scale field density test location was screened to determine the grain size distribution. The results of this determination are shown on

the attached Figure 2.5.6-24. The results of the field density test are shown in summary form on Table 2.5.6-3.

2.5.6.5 Slope Stability

2.5.6.5.1 Squaw Creek Dam

Embankment stability studies for the Squaw Creek Embankment have been made for the condition immediately after construction of the embankment but prior to the filling of the reservoir; and for a condition after construction with the reservoir present and a steady seepage condition established. In addition, an upstream reservoir drawdown case has been analyzed. Selected cases have been analyzed for static conditions and for a pseudo-dynamic condition wherein an acceleration of 0.12g is applied to the driving forces.

Appropriate soil strength parameters, described in detail in the following sub-sections, were used in embankment slope stability analysis. By means of the Swedish method of slices, a series of circular slides was analyzed until the minimum safety factor representing the critical slide surface was obtained. Full use of a computer program was made for this analysis. The computer program used for this analysis is described in Reference 134. After the location of the critical sliding surface was determined, hand computational methods were used to further evaluate the effects of pore pressure build-up during construction, steady seepage conditions, and the effect of an upstream reservoir drawdown. A series of wedge type analyses were made, by hand computational methods, to further determine the stability of the embankments under both static and pseudo-dynamic conditions. The results of these studies are shown on Figure 2.5.6-26. The safety factors presented on the table on Figure 2.5.6-26 are considered adequate and acceptable for the stability of the embankment under static loading conditions, wherein the minimum safety factor is 1.49, and under dynamic conditions, wherein the minimum safety factor is 1.04.

1. Strength Parameters

   The primary formation present at the site, as previously described, is that of the Glen Rose Limestone and strength and physical properties of these materials were evaluated by means of unconfined compression tests, Brazil Tensile Tests, and double ring shear tests.

   A portion of the embankment was constructed of rock obtained from required excavations on the site and this rock was quarried and run through a grizzly crusher so as to be well-graded, quarry run, with a maximum rock size limited to thirty inches in diameter. Strength and physical properties of this material were determined in the laboratory by a modeling technique. The modeling technique is described in Section 2.5.6.4.3. The strength properties of the impervious core and random fill material were determined by an evaluation of borrow material as described in Section 2.5.6.4.3.

   The design strength parameters of the various materials that will comprise the embankment section used in the slope stability analysis are as follows:

   Core:

γ = 117 pcf (95%-100% compaction)

C_u = 720 psf

φ_u = 15.5°

tan φ_u = 0.277

C = 2300 psf

φ = 16°

tan φ = 0.287

Random:

γ = 122 pcf (95%-100%)

C_u = 432 psf

φ_u = 19°

tan φ_u = 0.344

C = 2020 psf

φ = 19°

tan φ = 0.34

Rock:

γd = 115 pcf; $_s$= 135 pcf

$\overline{C}$ = 0

φ = 37.5°

tan φ = 0.767

Verification of design parameters is discussed in Appendix 2.5B.

2.5.6.5.2 Safe Shutdown Impoundment Dam (SSI Dam)

Stability of the SSI Dam was evaluated for operating conditions when the embankment and foundation are subject to ground motion from the SSE and when there is a total and instantaneous drawdown of the downstream side of the SSI Embankment resulting from the catastrophic loss of the Squaw Creek Dam. The dynamic response of the dam was determined by the procedure stated in Section 3.7.2.13. The details of the analysis are presented in the following subsections.

The static slope stability analysis of the embankment was performed and is shown on Figure 2.5.6-27. During the initial investigation, uniform upstream and downstream slope of 1.75 (horizontal) to 1 (vertical) and 2.5 (horizontal) to 1 (vertical) were evaluated.

The circular failure surface analysis utilized design strength properties given in Section 2.5.6.4.3. The failure surface which cuts through the core, filter and rock shell was found to provide a safety factor of 1.73 for the 1.75:1 slope condition. However, the most critical failure surface corresponds to a relatively shallow shear surface near the outer face of the shell where the safety factor was found to be 1.32. When the condition of rapid drawdown with negligible drainage of the rock slopes was considered, the safety factor was found to be 0.91.

Since the effects of the postulated seismic loading will be applied simultaneously to the condition of rapid drawdown, the result in outer slope stability would be reduced to a still lower value. The results of the stability studies, using design strength properties, for the uniform slope of 2.5:1 provided a safety factor of 2.43 for a critical sliding surface passing through the core, filter and rock shell. This factor decreased to 1.89 for a relatively shallow failure surface near the outer face of the rock shell. When rapid drawdown conditions were considered, again with negligible drainage of the shell, the safety factor was found to be 1.38. The effects of postulated seismic loading reduce the safety factors from those of static and total drawdown conditions; however, it appeared likely that the outer shell slopes in the range 2.5 to 1 would be adequate. The dynamic stability analysis was conducted utilizing the 2.5 to 1 slopes.

The liquefaction potential of the rock shell material and of the clay core material is considered practically non-existent; that of the filter zone materials is considered low. However, the effects of any possible liquefaction of the filter zone or core on the stability of the embankment has been analyzed by replacing the filter and core with a heavy liquid (static) and making a wedge type failure surface analysis where a seismic coefficient is applied. Filter zone gradation requirements are dependent on the gradation of the core material and shell material. The gradations for the rock shell material as well as the coarse and fine filters for the SSI Dam are shown on Figure 2.5.6-12. The properties of the impervious core filters and rock shell are discussed in Section 2.5.6.4.3.

1. Finite Element Model

   The structural integrity of the SSI Dam during the SSE was analyzed using a finite element method similar to that described by Clough and Chopra [135] and Chopra, Clough, Seed, et al. [136]. The finite element model consisted of a grid of 15 x 12 (180) nodal points (see Figure 2.5.6-28) connected in a regular fashion with 14 x 11 (154) quadrilateral elements. The general quadrilateral elements were formed by assembling the stiffness matrices of four constant triangles (CST). For the rockfill materials, the formulation of the stiffness matrix was the standard plain strain CST as described by Clough [137]. The mass matrix was formed by lumping one-quarter of the mass of each quadrilateral at the corner nodes in a manner consistent with Subsection 3.7.2.3. Static condensation was then used on the two center node displacements to reduce the element to 8 degrees of freedom. The clay core was modeled in a similar manner except that an additional hydrostatic variable was added to improve the accuracy in treating the nearly incompressible clay using the method described in Herrman [138]. The reformulated quadrilateral has two center node displacements and one pressure variable that are statically condensed to again give an 8 degree of freedom element. The clay core represents only a small portion of the cross-section and is modeled by the two central columns of elements in Figure 2.5.6-28.

2. Reservoir Dam Interaction

The hydrodynamic model of the SSI reservoir was accomplished using the added mass matrix described by Zienkiewicz [139] for the upstream and downstream faces of the SSI Dam. This model neglects the compressibility of the water but is a good approximation for dams below 100 feet in height [140]. The added mass coefficients were formed by solution of Laplace's equation in a semi-infinite plane reservoir and lead to a banded (non-diagonal) mass matrix. The added mass matrix coefficients are in the form of influence coefficients and decay rapidly away from the diagonal of the mass matrix. The band of the mass matrix was limited to 5 nodes.

3. Material Properties

The Shear moduli for the seismic analysis were based on the recommendations of Hardin & Drnevich [141], the results of the laboratory tests, the direct shear test, the average number of cycles expected during the duration of the SSE, and the average maximum shear strains predicted by the maximum spectral response for the lowest fundamental mode of vibration. The following properties were used:

ROCK FILL:

N	=	10 cycles
e	=	0.46 (void ratio)
γ	=	0.05 (Shear Strain)
γDRY	=	115 pcf
γSAT	=	135 pcf
ν	=	0.35 (Poisson's Ratio)
λavg.	=	0.08 (critical damping ratio)
G	=	45000 j psf

where is the effective vertical stress

CLAY CORE:

γDRY	=	115 pcf
ν	=	0.495 (Poisson's Ratio)
λavg.	=	0.08 (critical damping ratio)
G	=	500,000 psf

4. Natural Frequency and Loads of Vibration

The natural frequencies and nodes shape were determined using the subspace interaction procedure described in Bathe & Wison [142]. The first 10 natural frequencies were found to be in the range of 2.5 to 10 cycles per second.

5. Ground Motion History

An artificial ground motion history consistent with AEC Regulatory Guide 1.60 (revised) and the Newmark, Blume, and Kapur spectral envelope was used to perform both a step-by-step time motion history analysis and a spectral analysis of the seismic response of the SSI Dam. The artificial history was selected consistent with an anticipated structural damping of 8 percent. (The closest artificial history was for 7 percent structural damping). The 8 percent damping was determined from the elevation of the maximum shear strain together with the average vertical stress of a number of cycles using the Hardin & Drnevich formula.

The spectral envelope and artificial ground motion acceleration history are shown in Figures 3.7-4 and 3.7-9 for the horizontal component and Figures 3.7-18 and 3.7-23 for the vertical component. The ground motions were normalized so that the maximum horizontal acceleration was 0.12 g (at time = 3.52 seconds) and the maximum vertical acceleration was 0.08 g (at time = 3.44 seconds). It is important to note that although the peak vertical acceleration is 2/3 of the peak horizontal, as given by the Newmark, Blume, Kapur formula, it lies outside and is as severe as that for the horizontal component for all but the lowest fundamental frequency for the dam. Table 2.5.6-4 summarizes the Spectral Velocities for horizontal and vertical ground acceleration is 2/3 the horizontal, the maximum vertical accelerations in pseudo one degree of freedom systems of comparable frequency content of the dam are equal to the maximum horizontal accelerations. This is a very conservative modeling of an earthquake. A further comparison of the artificial acceleration history (Figure 3.7-9) with an actual 1925 earthquake at the site of Sheffield Dam [143] reveals that the artificial history has many more severe peaks (a > 0.05 g) than that of the Sheffield Earthquake.

The artificial time histories were developed and the dam analyzed for a duration of 10.24 seconds. Considering the very severe nature of the horizontal and vertical ground motion histories and since the number of cycles is not as significant a parameter for rockfill as for earth dams, this duration was deemed adequate for design purposes.

6. Step-By-Step Mode Superposition

Based on the natural frequencies and modes determined from the finite element model, a step-by-step mode superposition analysis was performed using the artificial histories for the combined horizontal and vertical components of the ground motion. The full 10.24 second duration was analyzed using a time increment of 0.01 seconds. The first 10 modes were analyzed using the Newmark generalized acceleration method. Figure 2.5.6-29 is a computer plot of the history of the acceleration of the nodal point at the downstream crest of the dam (“15-12”). The maximum acceleration of this node is 0.7 g and occurs at 5.85 seconds.

The response of the SSI Dam at times of 5.61 seconds, 5.70, 5.76, 5.85 (the maximum acceleration), and 5.91 seconds was selected for detailed study. At each of the stated time intervals, three drawings were prepared showing the nodal point acceleration vectors, the nodal point displacements, and the ratio of shear stress divided by the overburden stress. For time interval 5.61 seconds through 5.91 seconds, these relationships are shown on Figures 2.5.6-30 through 2.5.6-44, inclusive. The figures illustrating the acceleration vectors and the displacements at each nodal point and for each selected time interval also contain a vertical profile of acceleration or displacement from the top of the dam to the base of the dam along line 9.

Final displacements of the SSI Dam were estimated and have been plotted at a distorted scale to emphasize the permanent deflections and are shown on Figure 2.5.6-45. The procedure of determining permanent displacements consisted of the following:

a. Nodal point displacements were determined at the time of the maximum acceleration of nodal point “15-12”, the time of 5.85 seconds.

b. It was assumed that the artificial time history (Figure 2.5.6-29) for duration of 10.24 seconds may contain 15 peaks (all less than the maximum occurring at 5.85 seconds).

c. It was assumed that the number of such peaks over a 30 second time interval would be three times that occurring in the 10 second interval.

d. It was assumed that the computed displacements at each nodal point were in error by 100 percent and were multiplied by two (doubled).

Thus, the computed displacements at a time of 5.85 seconds were multiplied by a factor of 90 (15 times 3 times 2 = 90) to arrive at the final displacement configuration as shown on Figure 2.5.6-45. These displacements are considered extremely unlikely to actually occur in view of the pyramiding of the very conservative assumptions involved.

The procedure used in evaluating the seismic stability of the SSI Dam at local points and on horizontal planes consists of the following steps:

a. From the artificial ground motion study of the dam, the induced shear stresses at the various locations throughout the dam were evaluated.

b. The normal effective stresses at the various locations throughout the dam were determined.

c. Determinations of the cyclic shear stresses required to cause 5×10^{-2} strain were made at the various locations throughout the dam by use of Figures 2.5.6-47 and 2.5.6-48.

d. An evaluation was made of the ration of cyclic shear stresses required to cause 5×10^{-2} strain (f) with the shear stresses induced by the artificial ground motion,

The ratio, $\tau f/\tau d$ has been considered to represent a local factor of safety against the development of 5×10^{-2} strain. Figure 2.5.6-46 provides a graphic representation of the local factors of safety analyzed at Time=5.85 seconds (the time of the maximum acceleration). Table 2.5.6-5 presents the average of the local safety factors along the various horizontal planes. In this table, the average safety factors along horizontal planes decrease as the planes approach the crest elevation of the dam, as expected.

Local or point safety factors on or near the outer slopes near the crest of the dam have values less than one; these are not considered significant.

During construction, a test program to evaluate the cyclic shear strength of Class 1 backfill material was conducted. The results modified the curves on Figure 2.5.6-47 and Figure 2.5.6-48 as well as Figure 2.5.6-46. The effect of these changes on the stability of the SSI Dam are discussed in Appendix 2.5A.

7. Determination of Safety Factors

The evaluation of the effects of the SSE on the stability of the SSI rockfill dam has been made using two basic approaches. The first consisted of determining the dynamic response of the dam, the determination of the time of maximum acceleration, the magnitude of shear stresses at nodal points at this time, and the determination of the stress ration ($\tau d/\tau f$) or point safety factors. The details of this procedure are more fully described in Section 2.5.6.5.2.6.

The second procedure of determining the stability of the SSI rockfill dam during the conditions of the SSE consisted of a series of wedge analyses wherein the ultimate assumption was made that the core would liquefy, regardless of the value of acceleration, and would be resisted by the rock shell when the shell itself is subjected to the maximum value of acceleration as determined by the finite element procedure described in Section 2.5.6.5.2.6 for the time of 5.85 seconds.

The equation for determining the mean acceleration, a, for any distance, z, below the top of the dam, as related to the height of the dam (H) and the magnitude of the peak acceleration at the top of the dam (Ψ), was rigorously determined as:

$$\bar{a} = \Psi\left(1 - \frac{2z}{3H}\right)$$

The derivation of the equation is shown on Figure 2.5.6-49.

The equation of forces for a wedge type of analysis at any horizontal plane through the SSI rockfill shell when the shell itself was subjected to acceleration was rigorously derived and, in terms of safety factor, is expressed as:

$$F = \frac{\gamma r \cot\beta \tan\varphi}{K^{\gamma} r \cot\beta \tan\varphi + \gamma c}$$

The derivation of this equation is shown on Figure 2.5.6-50.

Thus, for any horizontal plane through the SSI rockfill dam at a depth z below the crest of the dam, the mean acceleration over the vertical distance H was determined using the maximum crest acceleration at T=5.85 seconds; this mean acceleration was redefined as k and the resulting safety factor determined for each of two cases. Case I represents the condition after loss of Squaw Creek Reservoir, with no tail water, and uses the saturated weight of rock shell material together with a angle of 37 degrees as determined on the laboratory model of the rock shell material, a value less than the field sliding resistance and direct shear tests which gave values in the range of 42 degrees. Case II represents a condition when Squaw Creek Reservoir

is present and the water surface on both sides of the SSI rockfill dam are equal at elevation 775 (normal water surface). Again, the angle of the rockfill shell remains as in Case I at 37 degrees.

Table 2.5.6-6 presents the safety factors resulting from this analysis for each horizontal plane. That table readily shows that the safety factor decreases as the plane analyzed approaches the crest of the dam which, of course, is the location of the greatest acceleration resulting from the occurrence of the SSE.

A comparison of the Safety Factor values obtained by the ultimate assumption of liquefaction of the core and wedge element analysis, as shown on Figure 2.5.6-51 show, without exception, that the core liquefaction and wedge analysis results in a lower safety factor than the average safety factor along the same horizontal plane in the finite element analysis. In both types of analyses, the safety factors along horizontal planes decrease as the planes approach the crest elevation of the dam, as expected.

In the case of the finite element analysis, Figure 2.5.6-46 shows local values less than one for points on the outer slope surfaces near the crest of the dam. These values result from the presence of little or no normal stress and are interpreted as meaning that, during the occurrence of the SSE, an occasional rock particle on the outer slope may be displaced enough to tumble or roll some short distance down the slope.

To further investigate this condition, a study was made to determine the magnitude of acceleration required to displace a rock lying adjacent to but not on the outer slope line. The details of this study are shown on Figure 2.5.6-52 and indicate that the acceleration must approach 4 g in magnitude for one rock to be displaced over another, an acceleration magnitude approximately six times greater than indicated to be present by the response of the dam to the artificial ground motion which, in itself, is considered most conservative.

Design verification of material properties and stability analysis are discussed in Appendix 2.5A.

2.5.6.6 Seepage Control

2.5.6.6.1 Squaw Creek Dam

Squaw Creek Dam is provided with a central impervious core extending to impervious foundation material. No seepage analysis was made since this core is considered to be a positive cutoff. The dam is provided with an 8 ft. wide sand filter, between the impervious core and the downstream random zone. The sand filter also extends horizontally with a thickness of 3 ft. between Stations 60+00 and 73+40. A 4-ft. wide vertical sand drain is located 370 ft from the centerline adjacent to the downstream toe of the dam between Stations 60+00 and 73+40. The sand drain penetrates the alluvium and bottoms on the foundation limestone, and is contiguous with the horizontal filter blanket. The entire drainage system exits at Squaw Creek where the volume of seepage may be observed. The purpose of the drain is to remove the possibility of a high phreatic level in the downstream random zone and to protect against piping of the core material in the event of its fracture. The vertical toe drain acts as a relief well for any hydrostatic pressures that may develop in the alluvial foundation. Material underlying the horizontal drainage blanket consists of existing in-place alluvial soils having a gradation essentially equal to that of the random zone. Immediately overlying the horizontal filter blanket is a 5 ft thick layer of random zone material.

The gradation of the filter is within the following limits:

Sieve Size	Percent Passing
1"	100
1/2"	65 - 100
1/4"	55 - 85
10 m	30 - 65
40 m	10 - 40
100 m	5 - 20
200 m	0 - 5

This gradation is designed to prevent piping of the core and random embankment zones. The design was developed using the procedures given in Reference [125].

2.5.6.6.2 Safe Shutdown Impoundment Dam

The Safe Shutdown Impoundment Dam is provided with a central impervious core founded on an impervious foundation. The shells are free-draining rockfill. No seepage analysis was made since the core is considered to be a positive cutoff. During normal operation there will be no differential head across the core.

Dual filters, each 5 ft thick, are provided on each side of the core between the core material and the rock fill shells. Filters on the downstream side are designed to prevent piping of the impervious material into the rock fill shell in the event of fracture of the core. Filters on the upstream side are designed to fill fractures in the core, should they occur. The gradations of the filter materials are based on design procedures in Reference [126] and are as follows:

Filter A

Sieve Size	Percent Passing
1/2"	100
4 m	78 - 100
10 m	50 - 85
40 m	33 - 60
100 m	8 - 27
200 m	0 - 10

Filter B

Sieve Size	Percent Passing
6"	100
3"	88 - 100
3/4"	60 - 78
4 m	32 - 45
10 m	15 - 26
40 m	0 - 10

Filter B is adjacent to the rock shells.

2.5.6.7 Diversion and Closure

2.5.6.7.1 Squaw Creek Dam

Squaw Creek is an intermittent stream. The U.S.G.S. gage “Squaw Creek near Glen Rose, Texas”, located on the State Highway 144 bridge approximately 2-1/2 miles downstream from the dam, showed in 1974 a maximum flow of 730 cfs and a minimum of 0.02 cfs. The average flow is 1.81 cfs.

The program for caring for water during construction was to leave an opening in the embankment until the remaining embankment and appurtenant structures were completed, then close the opening. This way, the natural flow of the stream would be passed on downstream.

The drawings required a 200 ft minimum opening in the embankment at elevation 655.0, centered on Squaw Creek. The maximum allowable side slopes of the embankment in this closure section were four (H) on one (V).

The core trench was completed on either side of Squaw Creek and the embankments were brought above natural ground prior to excavating the core trench across the creek. A cofferdam was constructed upstream and downstream of the excavation to stop the flow of the creek and to keep the excavation dry. The flow of the creek was pumped downstream from the cofferdams into Squaw Creek. Upon completion of the excavation and foundation treatment, the core trench was backfilled and the embankment was constructed to elevation 655.0.

During construction of the adjacent embankment and prior to closure, the normal flow of the creek was pumped over the embankment in the stream bed. Flood flows were passed over this embankment resulting in some embankment loss on several occasions. The lost embankment and any saturated embankment were replaced.

The closure of the embankment was not started until the following three items had been completed:

1. Embankment, including riprap and blanket, constructed to elevation 793.

2. Service spillway substantially compete including discharge channel with slope protection.

3. Service outlet complete except for gates and operators, but including service and emergency gate frames. The Service bridge was not complete.

Just prior to closure, all water impounded in Squaw Creek was drained through the service outlet by breaching the cofferdam in its inlet channel, and all undesirable materials were removed from the closure section. Closure was started on July 28, 1976, and was completed on May 14, 1977. Figure 2.5.6-5D(2) is a photograph of the SCD closure section.

Closure began by constructing the upstream random zone to elevation 673.0 to protect the downstream zones from flooding. Once this was accomplished, the filter blanket was placed and all zones of embankment brought to the same elevation. After completion of the filter blanket and placement of protective fill over it, a cofferdam was temporarily placed in the service outlet channel so that the service and emergency gates could be installed in their frames. The cofferdam was then removed to allow the flow of Squaw Creek to pass through the service outlet conduit.

The service outlet was closed and filling started on February 15, 1977. The following conditions were met prior to commencing impoundment:

1. Makeup Pump Station was operational.

2. Forty-eight in. portion of makeup pipeline was operational.

3. Embankment in the closure section was completed to at least elevation 729.

4. Service Spillway and Outlet Works construction was completed, including all mechanical and electrical activities and power supply to it.

5. SSI Dam was completed to an elevation which assured that SSI Dam construction would progress ahead of filling.

6. Rock shell and borrow material necessary for the remainder of the SSI Dam were stockpiled above elevation 775.

7. Riprap and bedding material for Squaw Creek Dam were stockpiled above elevation 775.

8. Flexible base material for dam roads was screened through a grizzly and stockpiled in an area accessible to the dam after the reservoir area was inundated.

9. Construction of the following structures was completed, or assurance established that construction could continue if the lake should fill:

   a. Circulating Water Intake Structure

   b. Circulating Water Discharge Structure

c. Service Water Intake Structure

d. Circulating Water Discharge Training Dike

Since the impoundment would make upstream borrow sources inaccessible, provisions were made for alternate downstream sources of material to complete the closure. Closure continued until the entire embankment was topped out.

2.5.6.7.2 Safe Shutdown Impoundment Dam

The flow in Panther Branch at the site of the SSI Dam is intermittent, like Squaw Creek, but subject to flood flows. It was not feasible to leave a closure section in the SSI Dam because the foundation excavation would be excavated 8 feet below the creek bottom, and it would be necessary to pump out this excavation after each flow. It was also believed that better construction quality would be obtained by constructing the entire length of this relatively short embankment level. Similarly, it was not desirable to allow flows to pass over completed embankment since material carried by the flows would contaminate the filters and possibly the rockfill zones. The dam does not have an outlet works through which Panther Branch could be diverted.

A cofferdam was constructed immediately upstream of the SSI Dam to contain the flow of the creek and to discharge the creek's flow through the interconnecting spillway channel into SCR below the SSI. This dam was constructed of alluvial material excavated from the floor of the SSI. It had a core trench to a impervious foundation which extended into each abutment as well as the valley floor. The side slopes were 3 (H) on 1 (V) and the 20-foot wide crown was at elevation 780-10.5 feet above the interconnecting spillway channel crest.

Drainage between the cofferdam and the SSI Dam foundation excavation was diverted to a pond and pumped over the cofferdam along with seepage and runoff trapped in the foundation excavation. All excavations were dewatered by sumping and pumping.

The cofferdam was lowered to elevation 755, then breached at three places when the SSI Dam was completed to elevation 780 and the Service Water Intake Structure was complete. Water was pumped over the cofferdam to equalize the SSI on either side of it prior to breaching.

2.5.6.8 Instrumentation

2.5.6.8.1 SSI Instrumentation

In order to verify the design parameters utilized and to evaluate the post-construction surface deformation, surface alignment and piezometric levels with in the SSI Dam, a system of permanent instrumentation has been installed as shown in Figure 2.5.6-53 and 2.5.6-54. The installed permanent instrumentation consists of surface movements, settlement plates to monitor internal vertical deformations, and wellpoint piezometers to monitor pore-water pressures within the embankment.

The settlement plates installed within the core of the SSI Dam are located as follows:

Station	Tip Elevation
6 + 50 C_L	770
8 + 50 C_L	770
10 + 50 C_L	770

Each settlement plate consists of a rectangular piece of steel conforming to ASTM A-36 of the following dimensions: 2.5'+ 0.25" X 2.5'+0.25" with a thickness of 1/2"+0.125". Each settlement plate was installed a minimum of one ft below the current working surface, at the time of installation, and backfilled and hand compacted in conformance with F&N specification SSI-1 (latest revision). All settlement plates were contractor-installed. At a later date if monitoring is needed, the settlement plates can be located using rotary core-drilling equipment. After the boring is advanced, a permanent observation casing can be installed, so that the settlement plates can be periodically probed in order to determine the interior movements taking place within the core of the SSI Dam.

Surface alignments monuments have been installed on the crest of the SSI Dam at the following locations:

No.	Station	Offset
SSI-SM-1	4+50	15.0' D.S.
SSI-SM-2	4+50	15.3' U.S.
SSI-SM-3	6+50	14.8' D.S.
SSI-SM-4	8+50	15.4' U.S.
SSI-SM-5	8+50	14.2' D.S.
SSI-SM-6	8+50	15.0' U.S.
SSI-SM-7	10+50	15.0' D.S.
SSI-SM-8	10+50	15.0' U.S.

All surface monuments were installed in accordance with MJA QAP-19. Installation of the surface monuments was accomplished by advancing a hole into the rockfill approximately 5 ft in depth. The hole was somewhat conical as it shown on Detail A of Figure 2.5.6-53. The surface monuments consisted of a No. 8 reinforcing bar approximately five ft in total length. The reinforcing bars were set into the bottom of the conical-shaped hole and encased in grout up to the approximate surface elevation. After the grout achieved a proper cure (28 days), the final elevations and offsets of the surface monuments were determined by the contractor in accordance with F&N SSI-1 Special Conditions, Item 9A and 9B (REV. 11).

The horizontal and vertical coordinates of the alignment monuments will be determined annually and compared with the original coordinates to determine the vertical and horizontal movements of the crest of the SSI Dam.

Wellpoint-type piezometers have been installed in the core of the SSI Dam at the following locations:

No.	Station	Tip Elevation
P-I-1	6+60 C	746.95'
P-I-2	6+70 C	766.71'
P-II-2	9+50 C	746.90'
P-II-2	9+60 C	767.03'

All piezometers have been installed in accordance with MJ QAP-18.

The wellpoint-type piezometers consist of industrially accepted stainless steel jacket wellpoints approximately 30" X 1.25"; a #60 mesh wire screen forms the outside of the wellpoint. Each wellpoint is coupled to a 1-in. standard galvanized riser pipe which extends from the tip elevation to the crest elevation. At the crest, the riser pipe has been fitted with a cap which is vented to allow pressure equalization.

Installation of the piezometers was accomplished by advancing an NX- size core boring down to a predetermined depth utilizing the rotary drilling process. After the boring was completed and bailed, the bottom portion of the boring was filled with approximately 1 ft of Ottawa sand. This sand formed a cushion for the wellpoint piezometer which was placed on the sand. After the wellpoint was seated, the sand column was extended around the wellpoint to approximately 3 ft above the screen. The remainder of the boring annular space, extending through the clay core of the dam, was filled to approximately elevation 786 with a dry bentonitic-type drilling additive. The final step in the installation process was the application of contractor-furnished grout through the remainder of the rockfill section up to the surface. The grout was formed into a conical shape at the surface to provide a positive anchor for the riser pipe. The uppermost part of the riser was recessed within this grout cone so as to be slightly below the existing crest elevation.

The piezometers installed in the SSI Dam will be monitored periodically during reservoir impoundment and operation in order to gain a measure of the piezometric level within the core of the dam.

The SSI dam will be inspected and piezometer readings taken annually. Results will be evaluated for significant or abnormal deviations from previously observed data trends. Unusual readings will be cause for increased surveillance of the SSI dam and determination and/or correction of the conditions which caused these readings.

2.5.6.8.2 Squaw Creek Dam Instrumentation

In order to verify design parameters, and to evaluate post construction surface deformations, alignments, piezometric levels and internal settlements within Squaw Creek Dam, the system of

permanent instrumentation as shown on the attached Figure 2.5.6-55 and 2.5.6-56 has been installed. Permanent instrumentation has consisted of surface alignment monuments to monitor the horizontal and vertical surface movements at the dam crest, settlement plates installed within the interior of the embankment core in order to monitor internal vertical deformations and wellpoint and pneumatic piezometers in order to monitor porewater pressures within the embankment section.

Surface monuments have been installed along the crest of Squaw Creek Dam at the following locations:

No.	Station	Offset	Elevation
SCD-SM-1	39+00	12.0' D.S.	796.7
SCD-SM-2	39+00	12.0' U.S.	796.8
SCD-SM-3	41+00	12.3' D.S.	796.6
SCD-SM-4	41+00	11.8' U.S.	797.0
SCD-SM-5	43+00	12.0' D.S.	796.9
SCD-SM-6	43+00	11.9' U.S.	797.1
SCD-SM-7	45+00	12.2' D.S.	797.0
SCD-SM-8	47+00	12.2' D.S.	797.3
SCD-SM-10	47+00	11.9' U.S.	797.8
SCD-SM-11	49+00	11.8' D.S.	798.3
SCD-SM-12	49+00	11.5' U.S.	798.5
SCD-SM-13	51+00	12.0' D.S.	798.5
SCD-SM-14	51+00	11.6' U.S.	799.0
SCD-SM-15	53+00	12.0' D.S.	798.6
SCD-SM-16	53+00	12.2' U.S.	799.3
SCD-SM-17	55+00	12.2' D.S.	798.5
SCD-SM-18	55+00	11.9' U.S.	799.3
SCD-SM-19	57+00	12.1' D.S.	798.7
SCD-SM-20	57+00	12.0' U.S.	798.2
SCD-SM-21	59+00	12.4' D.S.	798.3
SCD-SM-22	59+00	11.8' U.S.	799.0

No.	Station	Offset	Elevation
SCD-SM-23	61+00	12.2' D.S.	798.3
SCD-SM-24	61+00	11.9' U.S.	799.0
SCD-SM-25	63+00	12.1' D.S.	798.8
SCD-SM-26	63+00	12.2' U.S.	799.3
SCD-SM-27	65+00	11.7' D.S.	798.9
SCD-SM-28	65+00	11.7' U.S.	798.8
SCD-SM-29	67+00	12.9' D.S.	798.8
SCD-SM-30	67+00	11.9' U.S.	799.1
SCD-SM-31	69+00	12.2' D.S.	798.7
SCD-SM-32	69+00	12.2' U.S.	799.0
SCD-SM-33	71+00	12.4' D.S.	798.7
SCD-SM-34	71+00	11.6' U.S.	798.9
SCD-SM-35	73+00	12.3' D.S.	797.6
SCS-SM-36	73+00	12.2' U.S.	797.8
SCD-SM-37	75+00	12.1' D.S.	797.7
SCD-SM-38	75+00	11.9' U.S.	796.5

Installation of the surface monuments was accomplished in general accordance with MJ QAP-19. The installation procedure is similar to that described in Subsection 2.5.6.8.1.

The surface monuments installed on the crest of Squaw Creek Dam will be periodically observed for horizontal and vertical movements throughout the course of reservoir impoundment and operation.

Settlement place have been installed in the core of Squaw Creek Dam at the following locations:

Station	Offset	Elevation
50+49.2	0.6 Left of C	724.43
60+40.0	C	710.05
70+50.0	C	674.53

Each settlement plate consists of a rectangular piece of steel conforming to ASTM-A-36 with the following dimensions: 2.5' X 2.5' X 0.375'. Installation of the settlement plates was accomplished in the same general manner as that outlined in Subsection 2.5.6.8.1.

If settlement readings are desired, a core boring will be advanced to penetrate through the core of the dam down to the elevation of the settlement plates; and a permanent observation casing will be installed rising from the elevation of the settlement plate up to the crest of the dam. Utilizing the casing as access to the settlement plate, elevations could be taken periodically on the settlement plates and monitored to observe the internal vertical deformations.

Various combinations of wellpoint and pneumatic transducer piezometers have been installed in Squaw Creek Dam as shown on Table 2.5.6-7.

The piezometers were grouped in three ranges. Typically, each range of piezometers consists of a set of piezometers upstream of the centerline spread uniformly throughout the core of the dam, a set of piezometers downstream of the centerline spread uniformly throughout the core, and a set of piezometers near the toe of the dam with piezometers in the random section and in the alluvial section.

In order to facilitate the ongoing construction process, the piezometers contained within Range II were installed during the course of constructions. Two different types of piezometers were installed at Range II. Standard industrially-accepted stainless steel jacket wellpoint piezometers were installed at all locations for Range II. The wellpoints were nominally 30 inches in length by 1.25 inches and were encased by a No. 670 mesh screen. Additionally, companion pneumatic transducers were installed at the same elevation as the two deepest piezometers for each set (i.e., upstream, downstream, and near the toe). The pneumatic piezometer operates on a null balance principle whereby external pore pressure exerted on the transducer tip is exactly balanced b by an equal and opposite internal gas pressure supplied when reading the piezometer via an interconnecting tubing which connects the transducer to the surface readout station. By installing the pneumatic piezometers as companions to wellpoint piezometers at equal depths, their pore pressure readings may be compared to those obtained by the standard wellpoint type piezometers.

Similarly, wellpoint and pneumatic piezometers were installed near the toe of the dam at Station 46 and Station 69. The remainder of the piezometers, that is, the upstream (wellpoint) and near downstream (pneumatic) sets at Stations 46 and 69, were installed after completion of the Squaw Creek Dam.

The upstream set of piezometers for both Stations 46 and 69 were installed by drilling down through the completed crest of the dam. In order to avoid any disturbance to the finished grades on the downstream slope, it was impossible to install the downstream set of piezometers for Stations 46 and 49 by using vertical rotary drilling. Therefore, it as decided to install these piezometers by angle-core drilling from the existing completed crest of the dam down to the appropriate elevation and offset as previously determined. By using the angle-hole method of installation, standard wellpoint piezometers could not be utilized. Therefore, pneumatic transducers were installed for the near downstream sets at Stations 46 and 69. Even though the hole where the piezometers has been installed is drilled at an angle, the pore pressure reading at the tip elevation is in no way affected by the angularity of the hole.

The installation of the wellpoint piezometers in Squaw Creek Dam was essentially similar to that utilized for the SSI Dam as described in Paragraph 2.5.6.8.1, except that grout was not utilized to backfill the holes. Instead, the holes were filled with a bentonitic drilling additive. Installation of the pneumatic transducers was accomplished in a similar manner to that of the wellpoints. In this case, the riser pipe was utilized to protect the readout tubings leading from the pneumatic transducers to the ground surface.

Regular readings of the wellpoint and pneumatic piezometers are being made during impoundment and will continue during operation of the reservoir. These readings will be used to verify design parameters and to alert the geotechnical consultants to any extraordinary rise in pore pressure within the core of the dam or to any blockage of the filter on the downstream toe of the dam.

2.5.6.9 Construction Notes

2.5.6.9.1 Squaw Creek Dam

Estimated construction quantities for Squaw Creek Dam are as follows:

1.	Core Trench Excavation - Common	141,166 C.Y.
2.	Core Trench Excavation - Rock	137,488 C.Y.
3.	Emergency Spillway Excavation	1,205,165 C.Y.
4.	Service Outlet Excavation	104,282 C.Y.
5.	Service Spillway Excavation - Common	191,177 C.Y.
6.	Service Spillway Excavation - Rock	205,452 C.Y.
7.	Impervious Fill	1,138,935 C.Y.
8.	Random Fill	3,733,191 C.Y.
9.	Rock Fill	526,149 C.Y.
10.	Filter Sand	178,555 C.Y.
11.	Riprap	41,274 C.Y.
12.	Riprap Bedding	13,150 C.Y.

Figure 2.5.6-5A(1) is a photograph of the SCD site after stripping activities.

Construction of Squaw Creek Dam commenced on November 17,1 974, with the clearing and grubbing of the dam's foundation. Details of and problems encountered during the foundation and abutment excavation and treatment are discussed in Section 2.5.6.3.1. Figure 2.5.6-5B(1) shows a photograph of the foundation preparation activities. Soil material removed from the core trench was placed in the random embankment zones. Rock material excavated was wasted since it was generally too large for the rockfill zone.

On February 14, 1975, excavation of the east core trench was completed. The core trench was backfilled in section; that is, the east side of Squaw Creek was excavated to the abutment then backfilled. The same procedure was used on the west side of the creek which was completed on March 25, 1975. The excavation of the core trench in the closure section was completed on November 3, 1975.

Upon completing the backfill of the east core trench, placement of material in the east embankment began. As mentioned before, soil material excavated from the core had been placed in the random zone.

Select and random materials were obtained from upstream borrow pits. These pits were irrigated prior to use to bring the moisture content close to optimum. Rockfill material was obtained from the emergency spillway excavation. The east embankment was completed to elevation 791 on March 4, 1976, and the west embankment on July 19, 1976, when both embankments were brought to elevation 793. Figure 2.5.6-5A through Figure 2.5.6-5J are progress photographs of the construction of SCD.

Completion of the west embankment was delayed due to the time required to construct the service outlet conduit and backfill adjacent to it. Additional delay was encountered due to the need to extend the core excavation deep into the right abutment to unfractured material. The embankment could not be placed higher than the backfill in the abutment without forming a pocket that would pond water.

The specifications required that the select zone have a minimum liquid limit of 35. This requirement was relaxed to a minimum of 30 for fill placed above elevation 780 due to a shortage of material with a higher liquid limit.

It was originally planned to place all material excavated from the emergency spillway in the rock zone of the embankment. However, more random material containing weathered rock was encountered in the emergency spillway than anticipated. This material contained too much rock to be used as random fill and too much soil to be used as rockfill. It was elected to waste it rather than try to process it.

Later, a grizzly was used to separate the rock from the soil. The rock was processed for riprap and the soil stockpiled for use in the random zone. The remaining excavation was divided into two classifications. The upper zone contained weathered crystalline limestone of suitable quality to be used as riprap. The lower zone contained unweathered limestone and claystone seams not suitable for riprap, but adequate for rockfill. These zones were selectively excavated and used as stated.

The rockfill zone of the dam now required more material than would be available from the emergency spillway excavation. Rather than quarry additional rockfill, it was decided to reduce the rockfill zone to be compatible with the volume of material available. A minimum horizontal thickness of 16 ft. was maintained for workability. The purpose of this zone is to use materials required to be excavated from the emergency spillway and to act as slope protection for the back of the dam. Specifications were modified to allow quarry run material not exceeding 24 in. in size.

Difficulties with compacting material adjacent to piezometer clusters at Station 50+00 and 60+00 are discussed in Appendix 2.5B along with corrective action taken.

Closure of the stream began on July 28, 1976. The details of this operation are described in Section 2.5.6.7.1. The dam was completed on June 16, 1977.

2.5.6.9.2 Safe Shutdown Impoundment Dam

Estimated construction quantities for the Safe Shutdown Impoundment Dam are as follows:

1.	Excavation	447,800 C.Y.
2.	Impervious Fill	63,900 C.Y.
3.	Filter A	22,800 C.Y.
4.	Filter B	22,800 C.Y.
5.	Rock Fill - Graded	450,700 C.Y.
6.	Rock Fill - Quarry Run	97,600 C.Y.

Construction of the SSI Dam began on February 10, 1975, with the clearing and grubbing of the dam's foundation. Details of and problems encountered during the foundation and abutment excavation and treatment are discussed in Section 2.5.6.3.2. Figures 2.5.6-4A thru 2.5.6-4G are a series of photographs of the dam during construction.

The most significant problem in the foundation excavation was that weathered and unsound rock was encountered deeper than expected. It was originally estimated that 305,000 C.Y. of material would have to be removed to secure a proper foundation; however, 548,300 C.Y. were removed.

Specifications required all slopes in the abutments to be 3 (H) on 1(V). This was to be accomplished with dental concrete where necessary. Excavation of the rock ledges tended to break vertical, and the claystone ledges had to be excavated vertical. Sloping there verticals to 3 (H) on 1 (V) slope would require a considerable amount of concrete; therefore, the specifications were modified to allow a 1 on 1 slope, with a maximum of 30 inches of vertical rock face when in rock. This applied to the rockfill and filter zones only. In the core zone, the faces were hand trimmed to a 1 on 1 slope and all claystone layers were protected from weathering with 1-1/2 inches of pneumatically-placed concrete. The specifications for dental concrete required that it be consolidated to a dense impervious mass. When placed on a 1 on 1 slope, it could not be vibrated without flowing down slope. Since this material is used only under the rockfill and filter zones where being impervious is not important, the specifications were changed to allow the use of fill concrete. Fill concrete was consolidated by vibration insofar as possible then by tamping. Core test samples demonstrated that the compressive strength of this material was adequate to support the rockfill.

On April 13, 1976, a joint inspection of the foundation was made by the Freese and Nichols Project Engineer and the Mason-Johnston Project Geotechnical Consultant, resulting in permission being given to proceed with backfill. The presence of weathered (Fractured) layers of limestone was noted in the right abutment. Attempts to remove this material were unsuccessful and it was believed that this material extended through the abutment. Removal of this material was not required since it was structurally sound and located above elevation 769.5 where water

tightness is not important. The fractures were judged to be tight enough not to cause piping of the core material during its limited exposure to flood water.

Sources of embankment materials were as follow:

1. Rock Fill - Borrow “B” and “C” quarries as defined in Appendix 2.5A, “Construction Records and Design Verification of SSI Dam.”

2. Impervious Core - Selected borrow pits in Panther Branch valley below the SSI, and in Squaw Creek Valley above SCD.

3. Filter Materials - Commercial sources.

The time required to carry out effectively selective quarry operation in the plant site to obtain sufficient material to construct the SSI Dam was prohibitive when compared to the excavation schedule for the plant. Quarries of satisfactory quality were located alone Squaw Creek, two miles upstream from SCD.

Rock fill was selectively quarried from these sources to minimize claystone content and brought to a stockpile near the crusher. Sufficient material was stockpiled before impoundment of water in SCR since the quarry would be one of the first areas flooded. The rock material was hen run through the crusher and either placed as embankment or stockpiled.

Rock fill was dumped from end-dump trucks and spread into a two-ft- thick layer with a dozer. Segregation problems caused by spreading were corrected through modifying the dozer blade by adding “wings” to keep the larger stones from rolling to the side. Compaction was accomplished by four passes with a 10 ton vibrating roller.

The preirrigated impervious core material was processed with a mixer prior to loading in dump trucks and being transported to the fill. The material was dumped and spread into a loose lift not greater than 8 inches thick and compacted by a minimum of 8 passes of a sheepsfoot roller.

Originally up to a 12-inch loose lift thickness was allowed. Considerable difficulty was experienced in compacting the filter zones to the 80 percent relative density required. This problem was solved by reducing the lift thickness from 12 inches to 8 inches and compacting with hand operated vibrating sleds.

In August 1976, it was determined that a portion of the filter material did not comply with gradation requirements. At this time, the top of the embankment was at approximately elevation 756. An investigation of the location and extent of the undesirable material was jointly conducted by Freese and Nichols and Mason-Johnston. It was determined that acceptable filter material was present in all locations below elevation 747.5. All of the undesirable filter material was removed down to elevation 747.5. The mechanical requirements of the removal process also required the removal of limited quantities of acceptable filter, rock shell and core material. The excavated areas were then rebuilt with materials meeting the specifications.

The presence of the undesirable filter material in the dam embankment and the subsequent removal of the material and its replacement with desired material were determined to constitute a reportable deficiency as defined by paragraph 10 CFR 50.55(e) (1) (iii). A complete report of the

deficiency and the corrective actions taken was made to the Nuclear Regulatory Commission [127 and 128].

The embankment was 50 percent complete on August 20, 1976, a hold point to core drill the impervious material for geotechnical evaluation. The SSI Dam was completed on May 12, 1977, also a hold point to core drill the impervious material for geotechnical evaluation. The findings and details of the geotechnical evaluations, performed by Mason-Johnston, Inc., are included in Appendix 2.5A entitled "Construction Records and Design Verification of SSI Dam.:"

2.5.6.10 Operational Notes

The Safe Shutdown Impoundment Dam and the Squaw Creek Dam have been monitored as discussed in Section 2.5.6.8 and the performance has been satisfactory. The performance of the Safe Shutdown Impoundment Dam and the Squaw Creek Dam will continue to be monitored and any unsatisfactory conditions will be evaluated and corrected.

REFERENCES

1. Hill, R.T., Physical Geography of the Texas Region, 21st Annual Report, United States Geological Survey, pt. 7, p. 131, 1901.

2. Cheney, M. G., "Tectonics of Central Texas": AAPG Bulletin, Vol. 36, No. 12, pp. 2237-2265, 1950.

3. Flawn, P. T., Tectonics: The Ouachita System, Texas Bur., Econ. Geol., Pub. 6120, pp. 162-173, 1961.

4. Chase, B. W., Et al., "Resume of Geology of the Wichita Mountains, Oklahoma," Ardmore Geological Society, Petrol. Geology of Southern Oklahoma, Vol. 1, AAPG, Tulsa, pp. 36-55, 1956.

5. Hamm, W. E., "Regional Geology of the Arbuckle Mountains, Oklahoma," Oklahoma Geological Survey Guide Book XVII, 52 pp., 1969.

6. Bornhauser, M., "Gulf Coast Tectonics": AAPG Bulletin, Vol. 42, No. 2, pp. 339-370, 1958.

7. Eaton, R., "Resume of Subsurface Geology of Northeast Texas with Emphasis on Salt Structures," G.C. Assoc. Geol. Socs., Vol. 6, pp. 79-84, 1956.

8. Sellards, E.H., 1935, Balcones zone of faulting and folding, in The Geol. of Texas, Vol. II, Structural and Economic Geol., Pt. 1. Texas Bur. Econ., Geol. Bull., No. 3401, pp. 49-62, 1935.

9. Fisher, W. L. and Peter U. Rodda, "Lower Cretaceous Sands of Texas: Stratigraphy and Resources," Bureau of Economic Geology, University of Texas, Austin, R. of I. No. 59, 1967.

10. Nagle, S., "Glen Rose Cycles and Facies, Paluxy River Valley, Somervell County, Texas," Bureau of Economic Geology, Geol. Circular 68-1, 25 pp., 1968.

11. Fielder, A. G. “Artesian Water in Somervell County, Texas,” United States Geological Survey, W.S.P. 660, 86 pp., 1934.

12. Crosby, G. W., “Gravity and Mechanical Study of the Great Bend in the Mexia-Talco Fault Zone, Texas,” J. of Geophysical Research, Vol. 76, No. 11, pp. 2690-2705, 1971.

13. Sellards, E. H., and Leo Hendricks, Structural Map of Texas, 1:500,000, N. E. Quad., 1951.

14. Howell, J. V. and Lyons, P. L., “Oil and Gas Possibilities of the Ouachita Province,” in the Geol. of the Ouachita Mts., Symposium: Dallas Geol. Soc. and Ardmore Geol. Soc. pp. 57-61, 1959.

15. Weeks, A. W., “Balcones, Luling and Mexia Fault Zones in Texas,” AAPG Bull., Vol. 29, No. 12, pp. 1733-1737, 1945.

16. Coon, L. A., “Tertiary-Cretaceous Growth of the East Basin,” Gulf Coast Assoc. Geol. Socs., Trans., Vol. 6, pp. 85-90, 1956.

17. Pitkin, J. A., Airborne Measurements of Terrestrial Radioactivity as an Aid to Geological Mapping U.S.G.S. Prof. Paper 516-F, pp. F1-F29, 1968.

18. Pitkin, J. A., Natural Gamma Aeroradioactivity Map of Fort Worth-Dallas Area, Texas 1:250,000., U.S.G.S. Geophysical Investigations Map GP-696, 1970.

19. American Geophysical Union and U.S.G.S., Bouguer Gravity Anomaly Map of the U.S., 1964.

20. Boone, P., “Basal Trinity Sands,” Baylor Geological Studies, Bulletin No. 15, 16 pp. 1965.

21. United States Geological Survey, Texas - Oil and Gas Fields, Pipelines and Exposed Basement, 1:1,000,000 Oil and Gas Invest. Map: OM-214, 1964.

22. Cram, E. H., Editor, “Future Petroleum Provinces of the United States,” AAPG Memoir 15, 1971.

23. King, Philip B., Tectonics of Quaternary Time in Middle North America, H. E. Wright, Jr. and D. G. Frey, Editors, The Quarternary of the United States, Princeton University Press, pp. 831-870, 1965.

24. Woollard, G. P., A Catalogue of Earthquakes in the United States Prior to 1925, based on unpublished data completed by Harry Fielding Reid. Hawaiian Institute of Geophysics, Data Report 10, HIG 68-9, University of Hawaii, Honolulu, 1968.

25. Shurbet, D. H., “Increased Seismicity in Texas,” Texas Journal of Science, Vol. 21, No. 1, pp. 37-41, 1969.

26. Rockwood, C. G. “Notes of American Earthquakes,” American Journal of Science, Vol. 125, p. 359, 1883.

27. Murphy, L. M., Cloud, W. K., “United States Earthquakes, 1952,” United States Department of Commerce Coast and Geodetic Survey, No. 770, 1954.

28. Milne, J., “Catalogue of Destructive Earthquakes, 1891,” Report of the 81st Meeting of the British Association for the Advancement of Science, Protsmouth, p. 736, 1911.

29. Eppley, R. A. Earthquake History of the United States, Part 1, Exclusive of California and Western Nevada, United States Department of Commerce, Environmental Science Services Administration, Coast and Geodetic Survey, No. 41-1, Revised Edition through 1963, Washington, 1965.

30. Sellards, E. H., “The Valentine, Texas, Earthquake,” The University of Texas Bulletin, No. 3201, Austin, 1932.

31. Byerly, P., “The Texas Earthquake of August 16, 1931,” Bulletin of the Seismology Society of America, Vol. 2, No. 2, 1934.

32. Byerly, P., “The Texas Earthquake of August 16, 1931,” Bulletin of the Seismology Society of America, Vol. 24, No. 3, 1934.

33. Sellards, E. H., “The Wortham-Mexia, Texas Earthquake,” The University of Texas Bulletin No. 3201, Austin, 1932.

34. United States Geological Survey, “Thermal Springs in United States,” W. S. P. 679-B, pp. 59-206, 1937.

35. United States Geological Survey, “Thermal Springs in United States,” pp. 492, 383 pp., 1965.

36. Bryan, Frank, “Evidence of Recent Movements along Faults of the Balcones System in Central Texas, “AAPG Bulletin, Vo. 20, No. 10, pp. 1357-1371, 1936.

37. Bryan, Frank, “Recent Movements on a Fault of the Balcones System, McLennan County, Texas, “AAPG Bulletin, Vol. 17, No. 4, pp. 439-442, 1933.

38. Bell, D. E., and Brill, V. A., “Active Faulting in Lavaca County, Texas, AAPG Bulletin, Vol. 22, No. 1, pp. 104-106, 1938.

39. Wiggins, P. N., III, “Geology of the Ham Gossett Oil Field, Kaufman County, Texas,” AAPG Bulletin Vol. 38, No. 2, pp. 306- 318, 1954.

40. Van Siclen, D. C., “The Houston Fault Problem”: American Institute of Professional Geologists, Proceedings, 3rd Annual Meeting, Texas Section - Dallas, pp. 9-31, 1967.

41. Sheets, M. M., “Active Surface Faulting in the Houston Area, Texas,” Bulletin, Houston Geological Society, Vol. 13, No. 7, pp. 24-33, 1971.

42. Imlay, R., “Jurassic Formations of the Gulf Region,” AAPG Bulletin, Vol. 27, pp. 1407-1533, 1943.

43. Baker, C. L., “Geology of Dam Sites in Comachean Cretaceous Rocks on the Brazos River in Johnson, Hood, Bosque and Hill Counties, Texas,” Bureau of Economic Geology Open Site Report.

44. U. of Texas, publication 5116, Oil and Gas in Northeast Texas.

45. Classification of Public Lands Valuable for Geothermal Steam and Associated Geothermal Resources,” Department of Interior Geological Survey Circular 647.

46. Carder, D. S., “Reservoir Loading and Local Earthquakes,” Proceedings of the Sixth Annual Engineering Geology and Soils Engineering Symposium, Boise, Idaho, 1968.

47. Murray, G. E., Geology of the Atlantic and Gulf Coast Provinces of North America: Harper and Row, New York, 1961.

48. Granata, W. H., Jr., Cretaceous Structural Development of the Sabine Uplift Area, G. C. Assoc. Geol. Society, Trans., Vol. 12, pp. 117-119, 1962.

49. Barosh, P. J., “Use of Seismic Intensity Data to Predict the Effects of Earthquakes and Underground Nuclear Explosions in Various Geologic Settings,” Geological Survey Bulletin 1279, 1969.

50. Stricklin, F. L., Jr. 1961. Degradational Stream Deposits of the Brazos River, Central Texas, Geol. Soc. American Bull., Vol. 72, No. 1.

51. Epps, L. W., 1973. A Geologic History of the Brazos River. Baylor Geological Studies, Bull. No. 24.

52. Adkins, W. S., 1924. Geology and Mineral Resources of McLennan County; Univ. Texas Bull. 2340.

53. Bronaugh, R. L., 1950. Geology of the Brazos River Terraces in McLennan County, Texas. Masters thesis, Univ. Texas.

54. Byrd, C. L., 1971. Origin and History of the Uvalda Gravel of Central Texas. Baylor Geological Studies. Bull. No. 10.

55. Southwestern Association Student Geological Society, 1970. Fall Field Trip Guidebook, Baylor Geological Society.

56. Wilhelm, S. and M. Erving (1972), “Geology and History of the Gulf of Mexico,” G.S.A. Bull., Vol. 83, p. 575-599.

57. Freeland, George, L., and Dietz, Robert S., 1971, Plate Tectonic Evolution of the Carribean - Gulf of Mexico region (abstr): ESO (Am. Geophys. Union, Trans.), Vol. 52, No. 4, p. 351-352.

58. Carey, S. W. Ed, 1958, “A Tectonic Approach to Continental Drift”, in A Symposium on Continental Drift,” University of Tasmania, Hobart, 375 pp.

59. Wermund, E. G., and Jenkins, W. A., Jr., 1969, “Late Pennsylvanian Series in North-Central Texas, in a Guidebook to the Late Pennsylvania, Shelf Sediments North-Central Texas”: Dallas, Texas, Dallas Geol. Soc., p. 1-11.

60. Hardin, G. C., Jr., 1962, “Notes on Cenozoic Sedimentation in the Gulf Coast Geosyncline, U.S.A.” in Rainwater, E. H. and Zingula, R. P., Eds., Geology of the Gulf Coast and Central Texas, field trip Guidebook for 1961, Ann. mtg., Geol. Soc. American: Houston Geol. Soc., p. 1-15.

61. Woodbury, H. O., Murray I. B., Jr., Plickford, P. J. and Alsers, W. H., 1973, “Pliocene and Pleistocene Depocenters Outer Continental Shelf, Louisiana and Texas,” AAPG, Vol. 57, p. 2428- 2439.

62. Trifunac, M. D. and A. G. Brady, On the Correlation of Seismic Intensity Scales with the Peaks of Recorded Strong Ground Motion, Bull. Seis. Soc. Am., Vol. 65, No. 1, pp. 139-162, 1975.

63. Young, K., Mesozoic History, Llano Region in Geology of the Gulf Coast and Central Texas and Guidebook of Excursions: GSA, 1962 Annual Meeting, Houston, Texas, Houston, Geol. Soc., pp. 98-106, 1962.

64. Algermissen, S. T., and Perkins, Some Problems in Seismic Zoning, Earthquake Research in NOAA, 1971-1972; NOAA Tech. Report ERL 256-ESL 28, 1973.

65. Keller, G. R. Jr., Curstal Structure of the Texas Gulf Coast, Doctoral, Texas Tech, 1973.

66. Keller, G. R. and S. E. Cebull, Plate tectonics and the Ouachita System in Texas, Oklahoma and Arkansas, Geol. Soc., Am. Bull., V. 84. No. 5, p. 1659-1665, 1973.

67. Nicholas, R. L. and R. A. Rozendal, Subsurface Positive Elements in the Ouachita Fold Belt and their relation to Paleozoic Cratonic Margin, AAPG Bull., V. 59, No. 2, p. 193-216, 1975.

68. Woolard, G. P., Crustal Structure from Gravity and Seismic Measurements, J. Geophys. Res., V. 64, No. 10, 1959.

69. Docekal, J., “Earthquakes of the Stable Interior, with Emphasis on the Midcontinent,” Dissertation, Lincoln, Nebraska, 1970.

70. Coffman, J. L., and Von Hake, Carl A., Earthquake History of the United States: Pub. 41-1, Rev. Ed. (through 1970), U. S. Dept of Commerce, NOAA, Environmental Data Serv., 208 pp., 1973.

71. London Times, Page 5, Col. F, from British Museum Library, Colindale, London, January 10, 1891.

72. Dallas Morning News, Jan. 9, 1891, p. 1.

73. San Antonio Express, Jan. 9, 1891.

74. Galveston Daily News, Jan. 9, 1891.

75. Houston Daily Post, Jan. 9, 1891.

76. Fort Worth Daily Gazette, Jan. 9, 1891.

77. Terrell Times-Star, Jan. 17, 1891.

78. Eppley, R. A., Earthquake History of the United States, Part 1, Exclusive of California and Western Nevada, United States Department of Commerce, Environmental Science Services Administration, Coast and Geodetic Survey, No. 41-1, Revised Edition through 1963, Washington.

79. Dallas Morning News, 1974, Texas Almanac and State Industrial Guide, Dallas, Texas 704, p. 1974-1975.

80. Neumann, F., 1930, The Louisiana Earthquake of October 19, 1930, Earthquake Notes, Seismological Society of America, Vol. 2, No. 3, p. 1.

81. Neumann, F. and Bodle, R. R., United States Earthquakes 1930, United States Department of Commerce, coast and Geodetic Survey No. 528, 1932.

82. Branner, G. C., and M. M. Howell, Earthquake Risks in Arkansas: Ark. Geol. Sur. Info. Circ. 4, 13 pp., 1933.

83. Oetking, P., D. E. Feray and H. B. Renfro, Geological Highway Map of the Midcontinent Region, AAPG-USGS, 1966.

84. Donaldsonville Chief, Nov. 1930.

85. Baton Rouge State Times, Oct. 20, 1930.

86. Neuman, F., 1934, United States Earthquakes, United States Department of Commerce, Coast and Geodetic Survey, No. 563, 1932.

87. Seismological Notes, Bulletin of the Seismological Society of America, Vol. 22, p. 173, 1932.

88. Brazee, R. J., Attenuation of Modified Mercalli Intensities with Distance for the United States East of 106oW: Earthquake Notes, Seismological Society of America, Vol. 43, pp. 41-52, 1972.

89. Flawn, P. T., Environmental Geology: Conservation, Land Use Planning and Resource Management, Harper and Row, New York, New York, 313 p., 1970.

90. Sellards, E. H., Subsidence at Goose Creek Oil Field, Univ. Texas, Bull. 3001, p. 29-36, 1930.

91. Houston Chronical, April 10, 1932.

92. Neumann, F. and Bodle, R. R., United States Earthquakes, United States Department of Commerce, Coast and Geodetic Survey, 1961.

93. Eppley, R. A. and W. K. Cloud, United States Earthquakes 1959, United States Department of Commerce, Coast and Geodetic Survey, 1961.

94. Coffman, J. L. and W. K. Cloud, United States Earthquakes, United States Department of Commerce, Environmental Science Service Administration, 1970.

95. Sbar, M. L. and L. R. Sykes, Contemporary Compressive Stress and Seismicity in Eastern North America: An Example of Intraplate tectonics: GSA, Vol. 84, pp. 1871-1882, 1973.

96. Woollard, G. P., Tectonic Activity in North America as indicated by Earthquakes, pp. 125-133, in The Earth's Crust and Upper Mantle, P. J. Hart (ed.), Am. Geophys. Union Geophysical Monograph 13, 735, p., 1969.

97. Sclater, J. G., R. N. Anderson and M. L. Bell, Elevation of Ridges and Evolution of the Central Eastern Pacific; Jour. Geophys. Research, Vol. 76, pp. 7888-7923, 1971.

98. Hubbert, M. K., and D. G. Willis, Mechanics of Hydraulic Fracturing; Am. Inst. Mining Engineers Trans., Vol. 210, pp. 153-168, 1957.

99. Voight, B., Evolution of North Atlantic Ocean: Relevance of Rock-Pressure Measurements, in North Atlantic-Geology and Continental Drift; Am. Assoc. Petroleum Geologists Mem. 12, pp. 955, 962, 1969.

100. Bott, M. H. P. and D. S. Dean, Stress Systems at Young continental Margins, Nature, vol. 235, pp. 23-25, 1972.

101. Morgan, W. J., Deep Mantle Convection Plumes and Plate Motions: Am. Assoc. Petroleum Geologists Bull., Vol. 56, pp. 203-213, 1972.

102. Murray, G. E., Geology of the Atlantic and Gulf Coast Province of North America, Harper & Row, Publishers, New York, New York, 1961.

103. Hager, D. S. and Burnett, C. M. Mexia-Talco Fault Line in Hopkins and Delta Counties, Texas: AAPG, Vol. 44, pp. 316-356, 1960.

104. Shelton, J. W., Role of Contemporaneous Faulting During Basinal Subsidence: AAPG, Vol. 52, pp. 399-413, 1968.

105. Worzel, J. L. and J. S. Watkins, Evolution of the Northern Gulf Coast Deduced from Geophysical Data: GCAGS, Vol. 23, pp. 84-91, 1973.

106. Muehlberger, W. R., P. S. Clabaugh and M. L. Hightower, Palestine and Grand Saline Salt Domes, Eastern Texas, Field Excursion Number 6, in Geology of the Gulf Coast and Central Texas and Guidebook of Excursion: GSA 1962, Annual Meeting, Houston, Texas, Houston Geol. Soc., pp. 226-277, 1962.

107. Hardin, G. C., Jr., Notes on Cenozoic Sedimentation in the Gulf Coast Geosyncline, U.S.A. in Rainwater, E. H. and Zingula, R. P., eds., Geology of the Gulf Coast and Central Texas, field trip guidebook for 1962 Ann. Mtg., Geol. Soc. America: Houston Geol. Soc., pp. 1-15, 1962.

108. Eardley, A. J., Structural Geology of North America, Harper and Row, New York, New York 1962.

109. Howell, B. F., Average Regional Seismic Hazard Index (ARSHI) in the United States, in Geology, Seismicity, and Environmental Impact, edited by D. E. Moran, J. E. Slosson, R. O. Stone and C. A. Yelverton, University Publishers, Los Angeles, Ca., 1974.

110. Algermissen, S. T. and D. M. Perkins, A probabilistic estimate of maximum acceleration in rock in the contiguious United States, U. S. Geological Survey open file report 76-416, 1976.

111. Freeman, T., and Garner L. Wilde, Age and Stratigraphic implication of major fault in Llano region, Central Texas: AAPG Bull., V. 48, No. 5, pp. 714-718, 1964.

112. Foley, L. L., Mechanics of the Balcones and Mexia Faulting: AAPG Bull., V. 10, pp. 1261-1269, 1926.

113. Hage, D. S. and C. M. Burnett, Mexia-Talco Fault Line in Hopkins and Delta Counties, Texas, AAPG Bull., V. 44, No. 3, pp. 316- 356, 1960.

114. Northrop, S. A. and A. R. Sanford, Earthquakes of Northeastern New Mexico and the Texas Panhandle, N. M. Geol. Soc., Annu. Field Conf. Guide, No. 23, pp. 148-160, 1972.

115. Nuttli, O. W., Mississippi Valley Earthquakes of 1811 and 1812; Intensities, Ground Motion and Magnitudes, Bull. Seis. Soc. Am., V. 63, No. 1, pp. 227-248, Feb. 1973.

116. Nuttli, O. W., Seismic Wave Attenuation and Magnitude Relations for Eastern North America, J. Geophys. Re., No. 78, pp. 876-885, 1973.

117. Nuttli, O. W. and J. E. Zollweg, The Relation Between Fault Area and Magnitude for Central United States Earthquakes, Bull, Seis. Soc. Am., V. 64, p. 73-85, 1974.

118. Roscoe, Bailey, Jr., Tectonic Origin of Preconsolidation Deformation in Upper Pennsylvanian Rocks near Bartlesville, Oklahoma, AAPG Bull., V. 59, No. 9, pp. 1626-1638, 1975.

119. Coffman, J. C. and C. A. VonHake, Earthquake History of the United States: Pub. 41-1, Rev. Ed, (through 1970), U. S. Dept. of Commerce, NOAA< Environmental Data Serv., 208 p., 1973.

120. Kilbourne, R. T., W. Lutschak and R. H. Rice, The Rusk, Texas “Earthquake” of January 9, 1891 (abst.) Geol. Soc. Am. Annual Meeting, 1974, Miami, Florida.

121. Gupta, I. N. and O. W. Nuttli, Spatial Attenuation of Intensities for Central U. S. Earthquakes, Bull. Seis. Soc. Am., v. 66, No. 3, pp. 743-761, 1976.

122. Howell, B. F. and T. R. Schultz, Attenuation of Modified Mercalli Intensity with Distance from the Epicenter, Bull, Seis. Soc. Am., V. 65, pp. 651, 665, 1975.

123. Cornell, C. A. and H. A. Merz, A seismic risk analysis of Boston, ASCE National Structural Engineering Meeting, April, 1974, Cincinnati, Ohio.

124. H. S. Gruy and Associates, Inc., Letter to Texas utilities Service, Inc. regarding responses to NRC 361.0 Seismology - Geology Branch Questions June 28, 1978.

125. Department of the Army, Corps of Engineers, Engineering Manual Em-1110-2-1901, February 1952, “Soil Mechanics Design Seepage Control, Section 1-09, Filter Design.”

126. “Comanche Peak Steam Electric Station, Safe Shutdown Impoundment Dam, Foundation and Embankment Criteria,” Mason-Johnston & Associates, Inc., Dallas, Texas, July 12, 1974.

127. Texas Utilities Generating Company, Letter to Mr. W. C. Seidle, Chief, Reactor Construction & Operations Division, U. S. Nuclear Regulatory Commission, Office of Inspection & Enforcement, December 10, 1976.

128. Texas Utilities Generating Company, Letter to Mr. W. C. Seidle, Chief, Reactor Construction & Operations Division, U.S. Nuclear Regulatory Commission, Office of Inspection & Enforcement, January 17, 1977.

129. Turner, G. L. , 1957, Paleozoic Stratigraphy of the Fort Worth Basin; in Abilene and Fort Worth Geological Societies 1957 Joint Field Trip Guide Book, p. 57-77.

130. Hall, W.D., 1976, Hydrogeologic Significance of Depositional Systems and Facies in Lower Cretaceous Sandstones, North Central Texas, University of Texas, Bureau of Economic Geology, Geological Circular 76-1.

131. Dames and Moore, Crosshole Geophysical Survey, Comanche Peak Steam Electric Station, Glen Rose, Texas, Vos. 1, 2, 3, and 4, July 1986.

132. Dames and Moore, Geotechnical Laboratory Testing, Comanche Peak Steam Electric Station, Glen Rose, Texas, October 1986.

133. Jurgenson, L., 1934, “The Application of Theories of Elasticity and Plasticity to Foundation Problems”, Journal of the Boston Society of Civil Engineers, July (reprinted in Contributions to Soil Mechanics 1925-1940, Boston Society of Civil Engineers, 1940, pp. 148-183).

134. Lowe III, John, “Shear Strength of Course Embankment Dam Material”, Proceedings of Eighth International Congress on Large Dams, held May 4-8, 1964, Edinburgh, Great Britian.

135. Lee, Kenneth L. and Seed, H. Bolton, “Cyclic Stress Conditions Causing Liquefaction of Sands; ASCE Journal of the Soil Mechanics and Foundation Division, Vol. 93, No. SMI, Jan. 1967, pp. 47-70.

136. Lee, K.L. and Fitton, J.A., “Factors Affecting the Cyclic Loading Strength of Soil,” Vibration Effects of Earthquakes on Soils and Foundations, ASTM STP-450, American Society for Testing Materials, 1969.

137. Seed, H.B. and Peacock, W.H., “Applicability of Laboratory Test Procedures for Measuring Soil Liquefaction Characteristics Under Cyclic Loading.” Earthquake Engineering Research Center, Report No. EERC 70-8, November 1970.

138. Perloff, W., Pressure Distribution and Settlement, in: Foundation Engineering Handbook, edited by Winterkorn, H.F. and Fang, H., Van Nostrand Reinhold Company, New York, 1975, pp. 148-196.

139. Hendron, Jr., A.J., Mechanical Properties of Rock, in: Rock Mechanics in Engineering Practice, edited by Stagg, K.G. and Zienkiewicz, O.C., John Wiley and Sons, London, 1969, pp. 21- 53.

140. Chopra. Earthquake Behavior of Reservoir Dam Systems. Proceedings ASCE EM-1, 1969.

141. Hardin and Drnevich. Shear Modulus and Damping in Soils: Design Equations and Curves. Proc. ASCE SM7, July 1972

142. Bathe & Wilson. Large Eigenvalue Problems in Dynamic Analysis. Proc. ASCE EM6, 1972.

143. Seed and Lysmer. Applications of Finite Element Analysis in Soil Dynamics. Proc. Symposium on Application of the Finite Element Method in Geotechnical Engineering, Vicksburg, 1972.

144. Dames & Moore, Report, Cyclic Triaxial Testing, Crushed Limestone Samples, Comanche Peak Steam Electric Station, Glen Rose, Texas, January 20, 1986.

145. Bowles, J. E., 1977, Foundation Analysis & Design, McGraw-Hill Book Company, New York, N. Y.

146. Seed, H. B. and Whitman, R. V., 1970, Design of Earth Retaining Structures for Dynamic Loading, Specialty Conference on Lateral Stresses and Design of Earth Retaining Structures, ASLE, New York, N. Y.

147. Broms, B., 1971, Lateral Earth Pressures due to Compaction of Cohesionless Soils, Proceedings, Fourth Conference on Soil Mechanics, Budapest, pages 373-384.

General Reference - Geology

Blank, H. R., Waco-Blacklands Exper. Watershed, Texas Bur. Econ. Geol., Rept, Invest. No. 12, 46 pp., 1952.

Brunstein, Jules, Editor, Bibliography of Gulf coast Geology, Spec. Pub. 1, Vols. 1 and 2, Gulf Coast Association of Geological Societies, 1045 pp., 1970.

Collier, J., Tectonic History of Bosque County, in Popular Geology of Central Texas: Baylor Geol. Soc. Field Trip, Valley Mills to Walnut Springs, Texas, Spring 1961 Guidebook, pp. 12-13, 1961.

Foley, L. L., Mechanics of the Balcones and Mexia Faulting: AAPG Bull., Vol. 10, pp. 1261-1269, 1926.

Fowler, Philip, Faults and Folds of South-Cenral Texas: Gulf coast Assoc. of Geol. Soc. Trans., Vol. 6, pp. 37-46, 1956.

Granata, W. H., Jr., Cretaceous Structural Development of the Sabine Uplift Area, G. C. Assoc. Geol. Socs., Trans., Vol. 12, pp. 117-119, 1962.

Green, J. H., The Effect of Artesian Pressure Decline on Confined Aquifer Systems and Its Relation to Land Subsidence: USGS W.S.P. 1779-T, pp. T1-T11, 1964.

Hayward, O. T., Structural Significance of the Bosque Escarpment, McLennan County, Texas (abs): Dissert. Abs., Vol. 17, No. 9, p. 1982, 1957.

Jones, R. A., Mechanics of the Balcones Fault System: Oil Weekly, Vol. 51, p. 307, Oct. 19, 1928.

Logue, L. L., Gravity Anomalies of Texas, Oklahoma and the U.S.: Oil and Gas Journal, Vol. 52, No. 50, pp. 132-135 and insert map, 1954.

Lyncy, S. A. and Paul Weaver, “Pavement Disruption by Rencent Earth Movement,” 11th Ann. Symp. of Highway Engineering Geologist, Tallahassee, Florida, 1960.

Melton, F. A., Central Fracture Systems, Bull. 3401, Vol. II, Texas Bureau Econ. Geology, 1934.

Muehlberger, W. R., and Kurie, A. E., Fracture Stude of Central Travis County, Texas, A Preliminary Statement: Gulf Coast Assoc. of Geol. Soc., Trans., Vol. 6, pp. 43-49, 1956.

Murray, G. E. Geology of the Atlantic and Gulf Coastal provinces of North America; Harper and Brothers, New York, 692, pp., 1961.

Perry, Lee, Jr., The Balcones Fault Zone of Texas; Compass, Vol. 20, No. 3, pp. 158-161, 1940.

Pratt, W. E., Some Questions on the Cause of the Subsidence of the Surface in the Goose Creek Field, Texas: AAPG Bulletin, Vol. 11, pp. 887-889, 1927.

Reasor, D. D., Balcones Fault System - Its Northeast Extent: AAPG Bulletin, Vol. 45, No. 10, pp. 1759-1762, 1961.

Taylor to Glen Rose, Geol. Section: South Texas Geol. Soc., Guidebook 19th Field Trip, San Antonio, 20 pp., 1953.

Texas Bur. of Econ. Geology 1970, Geol. Atlas of Texas, Waco Sheet.

General Reference - Earthquakes

Austin Statesman, Panhandle Has Light Tremor, July 20, 1966.

Austin Statesman, Weak Earthquakes Rumble El Paso, May 13, 1969.

Austin Statesman, Earthquake In Austin? Could Be, February 5, 1970.

Bowie, William, The Relation of Isostasy to Seismology, Bulletin of The Seismological Society of America, p. 158, 1925.

Byerly, P., The Texas Earthquake of August 16, 1931, Bulletin of The Seismological Society of America, Vol. 24, No. 3, 1934.

Coffman, Jerry L. & Von Hake, Carl A., Earthquake History of the United States, Pub. 41-1, U. S. Department of Commerce, National Oceanic and Atmospheric Administration, 1973.

De Montessue De Ballore, F., Tremblements De Terre Et Eruptions Volcaniques All Centre-Amerique, Societe Des Sciences Naturelles De Saone-Ft. Loire, Dijon, No. 210, 1888.

Eppley, R. A., Earthquake History of the United States, Part I, Exclusive of California and Western Nevada, United States Department of Commerce, Environmental Science Services Administration, Coast and Geodetic Survey No. 41-1, Revised Edition through 1963, Washington 1965, 1965.

Henley, A. D., Seisic Activity Near the Texas Gulf Coast, Engineering Geology, Bulletin, Association of Engineering Geologists, Bulletin Vol. 3, No. 1-2 and pre-print memo Nat. Convention - Association of Engineering Geologists, Denver, October 1965, January - July 1966.

Henley, Aubrey D., Engineering Geology, Seismic Activity Near the Texas

Gulf Coast, Bulletin Association of Engineering Geologists, Vol. 3, No. 2, pp. 33-39, January - July 1966.

Herrin, Eugene, Seismological Bulletin, Dallas, Seismological Observatory, Vol, 1, 1954, 18pp., Southern Methodist University, Dallas, Texas.

Houston Post, The, Houston Records First Earthquake, June 13, 1969.

Houston Post, The, Houston has Second Tremor, June 19, 1969.

Houston Post, The Earth Shocks Here Baffling, Not Dangerous, Expert Says, June 20, 1969.

Hull, L. V., New Dallas Seismological Observatory at Southern Methodist University, Bulletin of Seismological Society of America, Vol. 46, No. 4, pp. 321-330, October 1956.

Humphreys, W. J., Seismological Reports for October, November and December 1913, United States Weather Bureau, Monthly Weather Review, U. S. Dept. of Commerce, Washington, D. C., Vol. 42, pp. 687-689, December 1914.

Jenneman, V. F., Microseisms at Corpus Christi (Abstract), Union Geod. Et. Geophy. Int. Assoc. Seismo., Compte Rerdus No. 9, BIS P. 24, Geophysical Abstracts, 139, p. 278, October - December 1949.

King, Philip, B., Tectonics of Quaternary Time in Middle North America, H. E. Wright, Jr. and D. G. Frey, Editors, the Quaternary of the United States, Princeton University Press, pp. 831-970, 1965.

London times, page 5, Col. F, from British Museum Newspaper Library, Colindale, London, January 10, 1891.

Miller, W. D. and Reddell, D. L., Alaskan Earthquake Damages - Texas High Plains - Water Wells Trans. August, Vol. 45, No. 4, pp. 659-663, December 1964.

Milne, J., Catalogue of Destructive Earthquakes, Report of the 81st Meeting of the British Association for the Advancement of Science Portsmouth 1911, p. 736, 1911.

Neumann, F., Quarterly Seismological Report 1924 to 1927, United States Coast and Geodetic Survey.

Neumann, F., Seismological Report, U. S. Coast & Geodetic Survey No. 363, 1926.

Neumann, Frank, The Texas Earthquakes of July 30, 1925 (Abstract) Bulletin of Seismological Society of America Vol. 16, No. 2, p. 158, 1926.

Pratt, W., An Earthquake in the Panhandle of Texas, Bulletin of Seismological Society of America, Vol. 16, No. 2, June, 1926.

Roberts, E. B. and Ulrich, F. P., Seismological Activity of the United States Coast and Geodetic Survey in 1948, Earthquake of March 11, 1948, Bulletin of Seismological Society of America, Vol. 40.3, p. 201- 203, July 1950.

Rockwood, C. G., Notes on American Earthquakes, American Journal of Science, Vol. 125, p. 359, 1883.

Seismological Notes, Bulletin of the Seismological Society of America, Vol. 3, p. 80, 1913.

Vol. 13, No. 78, 1923.

Vol. 41, No. 4, p. 392, October 1951.

Sellards, E. H., The Valentine, Texas Earthquake, The University of Texas Bulletin No. 3201, pp. 113-138, Contributions to Geology 1932, January 1, 1932.

Sellards, E. H., The Wortham-Mexia, Texas Earthquake, The University of Texas Bulletin No. 3201, January 1, 1932.

Sellards, E. H., The Texas-Oklahoma Earthquake of April 11, 1934, The University of Texas Bulletin No. 3501, February 1936.

Sellards, E. H., The Borger, Texas Earthquake of June 19, 1936, The University of Texas Publication No. 3945, pp. 699-704, Contributions to Geology 1939, June 1940.

Sheets, Martin, M., Gulf Coast Earthquakes Diatrophism During Historic Time in Gulf Coastal Plain, Bulletin AAPG Vol. 31, No. 2, pp. 221-226 (see p. 201-209), February 1947.

Shurbet, D. H., Increased Seismicity in Texas, Texas Journal of Science, Vol. 21, No. 1, pp. 37-41, October 1969.

Tondorff, Francis, A., The Registration of Earthquakes and Press Dispatches on Earthquakes, Georgetown University.

UED, Texas-Lousiana Earthquake 24 April, 1964. Earth Sciences Division - Teledyne Systems Company, Seismic Data Laboratory Report No. 114, Long Range Seismic Measurements, project 8.4, ARPA Project Vela-Uniform, February 9, 1965.

Udden, J. A., The Southwest Earthquake of July 30, 1925, The University of Texas Bulletin No. 2609, March 1, 1926.

U. S. Coast & Geodetic Survey, Preliminary Report of the Southwest Texas Earthquake of August 16, 1931, Earthquake Notes, Vol. III, No. 3, p. 1-3, December 1931.

U. S. National Oceanic and Atmospheric Administration, National Survey, National Earthquake Information Center, Preliminary Determination of Epicenters Monthly Listing.

U. S. National Oceanic and Atmospheric Administration, National Ocean Survey, National Earthquake Information Center Hypocenter Data File, June 11, 1638 - Apr. 30, 1975.

United States Weather bureau, Monthly Weather Review, Washington, D. C., pp. 22-23, October 1882.

United States Weather Bureau, Monthly Weather Review, Washington, D. C., Section V - Seismology, p. 689, December 1914.

Wollard, G. P., A Catalogue of Earthquakes in the United States prior to 1925, Based on unpublished data completed by Harry Fielding Reid, Hawaiian Institute of Geophysics, Data Report No. HIG 68-69, University of Hawaii, Honolulu, June 27, 1968.

United States Department of the Interior, Geological Survey, Earthquake Information Bulletin, 1969, 1976.

United States Earthquakes (1928-1972), U. S. Department of Commerce, Coast and Geodetic Survey, published yearly.

U. S. National Oceanic and Atmospheric Administration, National Ocean Survey, National Earthquake Information Center, Hypocenter Data File (1928 - April, 1975).

Dewey, J. F. and J. M. Bird, Mountain belts and new global tectonics, J. Geophy., Res., V. 75, p. 2626-2647, 1970.

Durham, C. O., Jr., and Grover E. Murray, Tectonics of Atlantic and Gulf Coastal province in symposium on the chronology of tectonic movements in the United States; Am. Jour. Sci., V. 265, No. 5, p. 428- 441, 1967.

Keller, G. R. and S. E. Cebull, Plate Tectonics and the Ouachita System in Texas, Oklahoma and Arkansas, Geol. Soc. Am. Bull., V. 84, No. 5, p. 1659-1665, 1973.

Neumann, F., The Texas Earthquakes of July 30, 1925 (abst.) Bull. Seis. Soc. Am., V. 16, No. 3, p. 158, 1926.

Neumann, F., The Louisiana Earthquake of October 19, 1930, Earthquake Notes, Bull. Seis. Soc. Am., V. 2, No. 3, p. 1, 1930.

Seed, H. B., C. Ugas and J. Lysmer, Site Department spectra for earthquake resistant design, Bull., Seis. Soc. Am., V. 66, No. 1, p. 221-244.

Seismological Notes, Bulletin of the Seismological Society of America, V. 64, No. 4, p. 1605, V. 65, No. 4, p. 1038, V. 65, No. 6, p. 1887.

Shurbet, D. H. Excitation of Rayleigh Waves, Geophys. Res., V. 74, No. 22, p. 5339-5342, 1969.

Street, R. L., R. B. Hermann and O. W. Nuttli, Earthquake mechanics in the Central United States, Science, V. 184, No. 4143, p. 1285-1827, 1974.

Viele, G. W., Structure and Tectonic History of the Ouachita Mountains, Arkansas in Gravity and Tectonics, p. 361-377 illus., John Wiley and Sons, New York, 1973.

Wiggins, J. H., construction of strong motion response spectra from magnitude and distance data, Bull. Seis. Soc. Am., V. 54, No. 5, 1257- 1269, 1964.

Woollard, G. P., Tectonic activity in North America as indicted by earthquakes in The Earth's Crust and Upper Mantle edited by P. J. Hart, Am. Geophys, Union Monograph 13, p. 25-133, 1969.

Yamamoto, J., A method for estimation of intensity of an earthquake from its magnitudes, Bull. of the International Institutue of Seismology and Earthquake Engineering, II, 67-72, 1973.

Gupta, I. N. and O. W. Nuttli, Spatial Attenuation of Intensities for Central U.S. Earthquakes, Bull. Seis. Soc. Amer., V. 66, No. 3, pp. 743-753, 1976.

Trifunas, M. D. and A. G. Brady, On the correlation of seismic intensity scales with the peaks of recorded strong ground motion, Bull. Seis. Soc. Am., Vol. 65, No. 1, pp. 139-162, 1975.

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TABLE 2.5.1-1
TABULATION OF PRINCIPAL FAULTS - TEXAS

Structure Name (County)	Basis of Identification			Age of Movement		Characteristics	Distance From Site (miles)
	Surface	Borehole	Geophysics	Major	Last		
Balcones Fault Zone (Hill), McLennan) 15,13	X	X		Oligocene, Beginning in Cretaceous	Miocene?	Strikes NNE; SE side down; 1200-1700' maximum displacement	44
Luling-Darst Creek Fault Zone 15,47	X	X	X	Oligocene, Beginning in Cretaceous	Quarter-nary	Strikes NNE; NW side down; 1500-2000 feet maximum displacement; 450' average displacement	150
Luling-Mexia-Talco Fault Zone (Bowie, Morris, Titus, Franklin, Delta, Hunt, Kaufman, Navarro, Limestone, Ellis, Southwest) 13	X	X	X	Oligocene, Beginning in Cretaceous	Eocene?	Graben, str kes NNE to EW, 600' displacement; 500 mi. long; NW and SE sides down; 30°-55° dips.	30
Fort Chadbourne Fault (Schleicher, Tom Green, Runnels, Taylor) 13			X			Strikes N-S; E side down	40
Rodessa Fault Zone (Marion, Cass) 13,47		X				Strikes NE; related to Sabine Uplift	180
Mt. Enterprise Fault Zone (Anderson Chero-kee, Rusk, Shelby) 47	X	X	X	mid-Cretaceous		Strikes WSW; Graben; 30°-60° dips. Related to Sabine uplift.	140

_____ Numbers refer to references in text

TABLE 2.5.1-2
TABULATION OF PRINCIPAL FAULTS - TEXAS

Structure Name (County)	Basis of Identification			Age of Movement		Characteristics	Distance From Site (miles)
	Surface	Borehole	Geophysics	Major	Last		
Muenster Arch (Montague, Cooke, Denton) 13		X		Late Paleozoic		Strikes NW; related to Red River uplift	60
Fort Worth Basin (Bosque, Johnson, Tarrant, Denton, Westward) 13		X		Pennsylvanian		Strikes NW	0
Bend Arch (McCulloch, Coleman Eastland, Stephens, Young) 13		X		Late Paleozoic	Post Permian	Strikes N-S; Northward plunge; related to Llano uplift	30
East Texas Embayment (Axis in Houston, Anderson Smith, Wood, Camp, Cass) 16,47	X	X	X	Early Mesozoic	Cenozoic	Strikes NNE	90
Llano uplift (Mason, Llano, San Saba, Burnet, Blanco, Gillespie) 13,47	X			Post Early Ordovician	Pre-Late Penn.	Dome; 40 x 70 miles	70
Fort Worth Syncline (Tarrant, Wise, Jack, Clay) 13		X				Strikes ESE	60
Red River Uplift (Denton to Cottle) 13		X	X	Mid-Paleozoic		Strikes E-W; 200 miles long, few miles wide	120
Edwards Arch (Kimble, Edwards) 13		X				Strikes NE; related to Llano Uplift	180
San Marcos Arch (Axis in Blanco, Hays, Caldwell, Gonzales) 13,47	X	X		Late Paleozoic	Early Cenozoic	Strikes NW; Se plunge; related to Llano Uplift	170
Lampasas Arch (Lampasas) 13		X				Strikes NE; related to Llano Uplift	40
Marathon-Quachita Tectonic Belt (Mexico, Texas, S.E. Oklahoma, Arkansas, Miss., S.W. Alabama) 13,47		X		Mid to late Paleozoic		Strikes NNE nearest site 1300 miles long, minimum 90 miles wide.	35

____ Numbers refer to references in text

TABLE 2.5.1-3
TABULATION OF PRINCIPAL FAULTS - OKLAHOMA, ARKANSAS, LOUISIANA

Structure Name (County)	Basis of Identification			Age of Movement		Characteristics	Distance From Site (miles)
	Surface	Borehole	Geophysics	Major	Last		
Boktukola Thrust Fault (Oklahoma-Choctaw, Pusnmataha, McCurtain; Arkansas) 3	X			Pennsylvanian		Str kes NE; thrust toward NW; related to Quachita uplift	180
Winding-stair Thrust Fault (Oklahoma-Octavia, Latimer, Pittsburgh, Atoka; Arkansas) 3	X			Pennsylvanian		Str kes NE to E; thrust toward NW; related to Quachita uplift	175
Ti Valley Thrust Fault (Oklahoma-Octavia, Latimer, Pittsburgh, Atoka; Arkansas) 3	X			Pennsylvanian		Str kes NE to E; thrust toward NW; related to Quachita uplift	160
Chocktaw Thrust Fault (Oklahoma-Latimer, Pittsburgh, Atoka; Arkansas) 3	X			Pennsylvanian		Str kes NE to E; thrust toward NW; related to Quachita uplift	160
Pine Mountain Thrust Fault (Oklahoma-Latimer, Pittsburgh, Atoka) 3	X			Pennsylvanian		Str kes NE to E, thrust toward NW; related to Quachita uplift	180
Jackfork Mountain Thrust Fault (Oklahoma-Latimer, Pusnmataha, Atoka) 3	X			Pennsylvanian		Str kes NE to E; thrust toward NW; related to Quachita uplift	170
Octavia Thrust Fault (Oklahoma-Pushmataha) 3	X			Pennsylvanian		Str kes NE to E; thrust toward NW; related to Quachita uplift	180
Bancroft Fault Zone (Louisiana and Texas) 3	X			Tertiary		Str kes NE to E, normal down to coast faults	180
Reagan Fault (Oklahoma) 3	X			Pennsylvanian		Str kes NW; down to SE; related to Arbuckle uplift	150
Millcreek Fault (Oklahoma) 3	X			Pennsylvanian		Str kes NW; down to SW; related to Arbuckle uplift	150
Sulphur Fault (Oklahoma) 3	X			Pennsylvanian		Str kes NE; down to NE; related to Arbuckle uplift	160
Washita Valley Fault (Oklahoma) 3	X			Pennsylvanian		Str kes NW; down to NE Related to Arbuckle uplift	140

____ Numbers refer to references in text

TABLE 2.5.1-4
TABULATION OF PRINCIPAL FAULTS - OKLAHOMA, ARKANSAS, LOUISIANA

Structure Name	Basis of Identification			Age of Movement		Characteristics	Distance From Site (miles)
	Surface	Borehole	Geophysics	Major	Last		
Arbuckle Uplift (Oklahoma) 5	X	X	X	Late Pennsylvanian	Permian	Strikes NW	175
Broken Bow Uplift-Anticlinorium (Oklahoma, Arkansas) 3	X			Late Pennsylvanian	Permian	Str kes NE	190
Quachita Mountians Uplift (Oklahoma, Arkansas) 3	X	X		Early Pennsylvanian	Late Paleozoic	Str kes E-W; 200 miles long; 50-60 miles wide	185
Wichita Uplift (Oklahoma) 4	X			Late Pennsylvanian	Late Permian	Str kes NW; eastward plunge	160
Sabine Uplift 48	X	X	X	Mid-Cretaceous	Early Tertiary	100-mile diameter	160
Nemaha Arch (Oklahoma-Kansas) 108		X	X	Paleozoic	Late Pennsylvanian	Str kes N-S	180
Seminol Arch (Southern Oklahoma) 83		X	X	Paleozoic	Late Pennsylvanian	Str kes N-S	175

____ Numbers refer to references in text

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 1 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
1	Hood	Ben J. Taylor/Cravens No. 4 2,000' FNWL & 2,000' FNEL H. H. Meyers Survey A-580 7 miles northeast of Paluxy	Gas well 9-30-77	5,010 ft. Marble Falls	4,740-4,750	Big Saline	2,074	Stratigraphic
1-A	Hood	Ben J. Taylor/Cravens No. 5 2,256' FSEL & 660' FNEL H. H. Meyers Survey A-580 3 miles northeast of Paluxy	Gas well 12-14-77	4,784 ft. Big Saline	2,975-2,985	Strawn	505	Stratigraphic
1-B	Hood	Lone Star Petroleum/Comanche Peak “D” No. 1 4,030' FSL & 2,092' FWL Felix Cattonet Survey A-92 4 miles south and 1 mile west of Granbury	Suspended operations 8-13-79	4,885 ft.				
1-C	Hood	Lone Star Petroleum/Angelia No. 12,279' FSEL & 2,312' FSWL Felix Cattonet Survey A-92 4 miles south and 2 miles west of Granbury	Suspended operations 8-20-79	4,998 ft.				
2	Hood	Mid-Continent Pet./Squaw Creek Cattle Co. No. 1 1,263' FSL & 1,263' FEL P. S. Hughlett Survey A-256 10 miles south of Granbury	Dry 5-8-50	5,576 ft. Ellenburger	none	none	none	
2-A	Hood	Dallas Production, Inc./Williams No. 1-C 2,250' FSEL & 3,950' FME'y NEL F. C. Cattonet Survey A-92 6.3 miles south of Granbury	Dry 2-25-80	5,260 ft.	none	none	none	
2-B	Hood	Dallas Production, Inc./Cook No. 1-C2,600' FNEL & 3,875' FSEL John C. Grace Survey A-1906 miles south of Granbury	Dry 2-25-80	5,275 ft.	none	none	none	
2-C	Hood	Lone Star Petroleum/Comanche Peak No. 1-B 467' FSL & 467' FWL Felix Cattonet Survey A-92 5 miles southwest of Paluxy	Gas well 12-8-78	5,022 ft Marble Falls.	4,214-4,438	Atoka Conglomerate	4,100	Stratigraphic
2-D	Hood	Franke Basin Exploration Co./ Nedelkovich No. 1 4,024' FSL & 700' FWL John C. Grace Survey A-1905 miles south and 1 mile west of Granbury	Dry 6-9-80	365 ft.	none	none	none	

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 2 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
2-E	Hood	Ben J. Taylor/Cravens No. 8 500' FNWL & 500' FNEL Richard Disney Survey A-144 5 miles south of Granbury	Gas well 8-16-79	5,571 ft. Ellenburger	4,690-4,704	Big Saline	950	Stratigraphic
2-F	Hood	Taylor Operating Co./Rawls No. 1 1,800' FSL & 2,300' FWL J. C. Grace Survey A-190 1 mile north and 7 miles east of Paluxy	Gas well 4-11-79	5,100 ft. Marble Falls	4,303-4,839	Strawn and Big Saline	98	Stratigraphic
2-G	Hood	Dallas Production, Inc./Sanders “A” No. 1 2,950' FNWL & 4,050' FNEL E. J. Johnson Survey A-292 5 miles south of Granbury	Suspended operations 8-18-80	5,300 ft.				
3	Hood	Fitzgerald/Van Morrison No. 1 330' FNL & 1,388' FWL Alexander O'Brien Survey A-433 7 miles southeast of Granbury	Dry 8-10-51	4,508 ft. Strawn	none	none	none	
3-A	Hood	Dallas Production, Inc./Carter No. 1-A900' FSEL & 4,600' FSWL T. E. Tucker Survey A-566 2.8 miles southeast of Mambrino	Proposed location 12-30-77					
4	Hood	Tri-Venture II/Trotter No. 1 545' FNWL& 660' FNEL J. Doane Survey A-145 13 miles southwest of Granbury	Gas well 11-5-77	5,218 ft. Barnett Shale[(a)]	4,770-4,780	Big Saline	378	Stratigraphic
4-A	Hood	Energy Fund of America/Henry P. Maj No. 1 1,434' FNWL & 1,291' FNEL J. Doane Survey A-145 3 miles northeast of Paluxy	Dual gas well 6-8-78	5,059 ft. Marble Falls	4,719-4,725 4,965-4,969 3,148-3,160	Big Saline Marble Falls Strawn	} 90 260	Stratigraphic
4-B	Hood	Louisiana Land & Expl./Rodney Samson G. U. No. 1 499' FNL & 747' FWL W. R. Carey Survey A-85 1.5 miles south of Hill City	Dry 4-27-79	5,356 ft. Barnett Shale[(a)]	none	none	none	
5	Hood	Energy Fund of America/O. Coker No. 1731' FEL & 6214' FSL J. Doane Survey A-145 5.6 miles south of Tolar	Dry 6-18-79	5,133 ft. Marble Falls	none	none	none	
6	Hood	Scates Exploration/Lowdon No. 1480' FNWL & 4,900' FSWL William R. Carey Survey A-85 1.8 miles north of Hill City	Dry 10-15-79	5,275 ft.	none	none	none	

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 3 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
7	Hood	Ben J. Taylor/Cravens No. 3-A 1,964' FNWL & 122' FNEL William R. Carey Survey A-85 7 miles northeast of Paluxy	Abandoned location 8-1-77					
7-A	Hood	Louisiana Land & Expl./Lowden No. 13,600' FNWL & 1,508' FNEL William R. Carey Survey A-85 3 miles north and 4 miles east of Paluxy	Gas well 7-13-78	5,366 ft. Barnett Shale[a]	4,445-4,477	Strawn	1,100	Stratigraphic
8	Hood	Ben J. Taylor/Cravens No. 3 2,104' FNWL & 222' FNEL William R. Carey Survey A-85 5 miles south and 2 miles east of Tolar	Gas well 9-5-78	5,150 ft. Marble Falls	4,748-4,760	Big Saline	950	Stratigraphic
9	Hood	Gulf/Bratton Ranch No. 4 2,500' FSEL & 4,300' FSWL J. Doane Survey A-145 5 miles northeast of Paluxy	Gas well 5-16-77	5,100 ft. Barnett Shale[a]	4,636-4,642	Big Saline	3,650	Stratigraphic
9-A	Hood	Wayman Buchanan/Swaim No. 1 4,735' FNWL & 5,000' FSWL J. Doane Survey A-145 4 miles northeast of Paluxy	Gas well 5-1-78	5,169 ft. Barnett Shale[a]	4,671-5,110	Marble Falls	975	Stratigraphic
10	Hood	Energy Fund of America/Moore No. 12,204' FSEL & 1,776' FNEL J. Doane Survey A-145 8 miles southeast of Tolar	Dry 9-17-77	5,094 ft. Marble Falls	none	none	none	
10-A	Hood	Energy Fund of America/Argo Peek No. 1 724' FEL & 1,028' FSL J. Doane Survey A-145 6.2 miles south of Tolar	Abandoned location 2-6-78					
11	Hood	Tri-Venture II/Sullivan No. 1 500' FSWL & 708' FSEL William R. Carey Survey A-85 9 miles southwest of Granbury	Gas well 11-1-77	5,085 ft. Marble Falls	4,466-4,486	Big Saline	355	Stratigraphic
11-A	Hood	Sirius Energy Co./J. T. Parker No.2467' FNWL & 900' FSWL Cyrus T. Egery Survey A-169 2 miles north and 3 miles east of Paluxy	Gas well 7-27-78	5,285 ft Barnett Shale[a]	4,445-4,763	Strawn	259	Stratigraphic
12	Hood	Ben J. Taylor/Wiggins No. 1 4,081' FNEL & 500' FSEL William R. Carey Survey A-85 6 miles northeast of Paluxy	Gas well 9-7-78	5,170 ft.	3,943-3,957	Strawn	287	Stratigraphic

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 4 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
12-A	Hood	Ben J. Taylor/Cravens No. 7 2,230' FSEL & 660' FSWL Richard Disney Survey A-144 5 miles south and 4 miles east of Tolar	Gas well 9-5-78	5,093 ft. Marble Falls	4,663-4,673	Big Saline	1,500	Stratigraphic
12-B	Hood	Lone Star Petroleum/Squaw Creek No. 15,433' FNL & 664' FEL Richard Disney Survey A-144 10 miles south and 8 miles west of Granbury	Dry9-8-80	4,850 ft.	none	none	none	
13	Hood	The Baron Co./Randall No. 2 5,880' FWL & 470' FNL Cyrus T. Egery Survey A-169 10 miles south of Granbury	Abandoned location 1-30-78					
13-A	Hood	Ben J. Taylor/Wiggins No. 2 900' FNWL & 2,000' FNEL Cyrus W. Egery Survey A-169 8 miles south of Granbury	Gas well 7-25-79	4,950 ft. Marble Falls	4,719-4,738	Big Saline	3,900	Stratigraphic
14	Hood	Dallas Production, Inc./Hubbard No. 11,490' FNWL & 3,367' FSWL J. C. Grace Survey A-204 10 miles south and 1 mile east of Granbury	Dry 2-12-74	5,965 ft. Ellenburger	none	none	none	
14-A	Hood	Dallas Production, Inc./Williams “B” No. 1 2,650' FNEL & 3,030 FSEL J. C. Grace Survey A-203 4 miles southwest of Mambrino	Abandoned location 9-25-78					
14-B	Hood	Vanguard Exploration Co./Cleveland No. 2 467' FNL & 1,050' FEL J. Grace Survey A-203 3 miles northeast of Hill City	Location 11-3-80					
15	Hood	Gulf/Bratten Ranch No. 5 587' FNL & 3,000' FEL Thomas J. Stansbury A-498 7 miles southeast of Tolar	Gas well 6-20-77	5,100 ft. Barnett Shale[a]	2,359-2,369	Strawn	1,080	Stratigraphic
16	Hood	Gulf/Bratten Ranch No. 8 1,980' FNWL & 5,500' FSWL Thomas J. Stansbury Survey A-498 3 miles east of Paluxy	Gas well Recompleted 12-13-78	4,950 ft. Marble Falls	4,126-4,277	Strawn	360	Stratigraphic
17	Hood	Gulf/Bratten Ranch No. 6 3,000' FSL & 6,700' FWL Thomas J. Stansbury Survey A-498 3 miles east of Paluxy	Gas well 7-18-77	5,000 ft. Barnett Shale[a]	4,686-4,696	Marble Falls	1,930	Stratigraphic

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 5 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
17-A	Hood	O.D.P. Ltd./Roach No. 1 3,875' FNWL & 1,955' FNEL Thomas J. Stansbury Survey A-498 7.5 miles southeast of Tolar	Dry 4-9-79	5,084 ft.	none	none	none	
17-B	Somervell	Dallas Production, Inc./Davis No. 1-A 660' FNWL & 15,600' FNEL Galveston County School Land Survey A-37 2.3 miles southwest of Hill City	Gas well 3-24-80	5,107 ft.	3,942-3,954	Strawn	2,350	
17-C	Somervell	Dallas Production, Inc./Nystel No. 1660' FNWL & 6,950' FNEL Galveston County School Land Survey A-37 5.8 miles northwest of Glen Rose	Abandoned location 7-24-78					
17-D	Somervell	Gulf/H. W. McDowell No. 1-A 7,500' FNL & 16,200' FEL (approximately) Galveston County School Land Survey A-37 5.9 miles northwest of Glen Rose	Gas well 3-13-78	5,400 ft. Ellenburger	4,112-4,130	Strawn	4,800	Stratigraphic
18	Hood	Tri-Venture II/Nelon No. 1 1,797' FSWL & 3,186' FNWL Cyrus W. Egery Survey A-169 10 miles southwest of Granbury	Gas well 10-31-77	5,254 ft. Marble Falls	4,742-4,750	Big Saline	275	Stratigraphic
18-A	Hood	Sirius Energy/J. T. Parker No. 1 2,904' FNL & 735' FWL Cyrus W. Egery Survey A-169 8 miles southeast of Tolar	Gas well 3-13-78	5,164 Marble Falls	4,730-4,746	Big Saline	174	Stratigraphic
19	Hood	The Baron Co./Randall No. 1 5,190' NE & 1,680' SE of NW/C Cyrus W. Egery Survey A-169 9 miles southwest of Granbury	Gas well 9-4-78	5,654 ft. Ellenburger	4,732-4,744	Big Saline	1,200	Stratigraphic
20	Hood	The Baron Co./Nelon No. 2-B 4,830' FEL & 3,296' FSL Cyrus W. Egery Survey A-169 10 miles southwest of Granbury	Abandoned location 1-30-78					
21	Hood	The Baron Co./Nelson No. 1-B 3,040' FNEL & 3,296' FSEL Cyrus W. Egery Survey A-169 2 miles southeast of Hill City	Gas well 10-19-78	5,625 ft. Ellenburger	4,946-5,095	Big Saline	328	Stratigraphic
22	Hood	Ben J. Taylor/Cravens “B” No. 1 1,939' FE'yL & 4,669' FS'yL Cyrus W. Egery Survey A-169 9 miles southwest of Granbury	Gas well 5-5-77	5,030 ft. Marble Falls	4,835-5,030	Marble Falls	1,700	Stratigraphic

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 6 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
22-A	Hood	S. W. Gas Pipeline/M. Peveler No. 1500' FWL & 2,850' FNL James D. Elliot Survey A-170 2.5 miles east of Hill City	Gas well 6-27-78	5,350 ft.	3,816-3,842	Pennsylvanian Conglomerate	13,000	Stratigraphic
22-B	Hood	Ben J. Taylor/Cravens No. 2-B 3,500' FSL & 468' FWL James D. Elliot Survey A-170 10 miles southwest of Granbury	Gas well 1-3-79	5,136 ft. Barnett Shale[a]	3,790-3,822	Strawn	909	Stratigraphic
22-C	Hood	S. W. Gas Pipeline/M. Peveler No. 2 825' FNWL & 2,050' FSWL James D. Elliot Survey A-170 1 mile north and 2 miles east of Hill City	Dry 12-8-80	5,374 ft. Barnett Shale[a]	none	none	none	
22-D	Hood	S. W. Gas Pipeline/M. Peveler No. 3467' FNL & 4,950' FWL James D. Elliot Survey A-170 3 miles east of Hill City	Gas well 2-21-79	5,330 ft. Barnett Shale[a]	3,797-4,254	Strawn	430	Stratigraphic
22-E	Hood	S. W. Gas Exploration/M. Peveler No. 51,340' FNL & 550' FEL James D. Elliot Survey A-170 5 miles east of Hill City	Gas well 7-17-79	5,369 ft. Barnett Shale[a]	4,734-4,763	Big Saline	1,450	Stratigraphic
22-F	Hood	Taylor Operating Co./Rawls No. 2 330' FNL & 1,650' FWL J. Grace Survey A-203 4 miles northeast of Hill City	Gas well 4-11-79	5,010 ft. Marble Falls	4,800-4,820	Big Saline	1,050	Stratigraphic
22-G	Hood	S. W. Gas Pipeline/M. Peveler No. 4 5,165' FNL & 2,740' FWL James D. Elliot Survey A-170 3 miles east of Hill City	Gas well 2-21-79	5,370 ft. Barnett Shale[a]	4,851-5,142	Marble Falls	2,700	Stratigraphic
23	Somervell	Dallas Production, Inc./Kinnard No. 1467' FNWL & 467' FSWL William B. Smith Survey A-91 6 miles northwest of Glen Rose	Dry [b] 10-8-79	5,605 ft. Ellenburger	none	none	none	
23-A	Somervell	Dallas Production, Inc./O'Connor No. 1 3,850' FNWL & 1,320' FNEL Galveston County School Land Survey A-37 4.7 miles northwest of Glen Rose	Suspended operations 11-27-78					
24	Somervell	McPherson/Williams No. 1 1,916' FEL & 2,825' FNL William B. Smith Survey A-90 6 miles north of Glen Rose	Dry 6-29-71	800 ft.	none	none	none	

TABLE 2.5.1-5
WELL DATA ON ALL WELLS WITHIN FIVE-MILE RADIUS OF COMANCHE PEAK S.E.S AS OF DECEMBER 31, 1980

(Sheet 7 of 7)

Well I.D. No.	County	Operator/Lease/Well Number Location	Formation Status	Total Depth and Deepest Perforations Reached	Depth of Producing (ft)	Initial Potential Formation	Reservoir Trapping (Mcf/day)	Mechanism
24-A	Somervell	Dallas Production, Inc./McConal No. 1 660' FNWL & 6,900' FNEL J. A. Hernandez Survey A-42 4.5 miles north of Glen Rose	Abandoned location 7-24-78					
25	Somervell	W. H. Price/Martin No. 1 6,200' FSL & 800' FEL Galveston County School Land Survey A-37 2.5 miles northwest of Glen Rose	Dry 10-26-77	5,560 ft. Ellenburger	none	none	none	
25-A	Somervell	Dallas Production, Inc./Millican No. 1 1,980' FSEL & 11,800' FSWL Milam County School Land Survey A-135 1.8 miles northeast of Glen Rose	Abandoned location 7-24-78					

a) The Barnett Shale is a non-productive unit located between the Ellenburger and Marble Falls formations.

b) This status is from Petroleum Information Corporation. No status is on file with the Texas Railroad Commission.

TABLE 2.5.1-6
RESERVOIR DATA SUMMARY

(Sheet 1 of 4)

Operator/Lease	County	Interval (ft)	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom- Hole Temperature (°F)	Shut-in Wellhead Pressure (psig)	Shut-in Bottom-Hole Pressure (psig)	Gas Gravity (air=1.0)	Absolute Open Flow (Mcf/day)	Distance From Plant (miles)
Strawn											
Gulf/McIntosh No. 1	Hood	4,356-4,380	52.2	9.3	50	162	1,036	1,183	0.700	3,090	7.5
Gulf/McIntosh No. 3	Hood	2,916-2,926	-	-	10[a]	140	488	762	0.690	14,000	7.0
Gulf/Otis Rollins No. 1	Hood	2,700-4,882	55.8	10.0	20	162	-	-	-	-	7.2
Mid Continent Pet./ Cattle Co. No. 1	Hood	-	-	-	60[b]	-	-	-	-	-	4.3
Fitzgerald/Van Morrison No. 1	Hood	-	-	-	65[b]	-	-	-	-	-	4.6
Gulf/Bratten Ranch No. 3	Hood	4,054-4,064	-	-	10[a]	140	1,052	1,181	0.700	742	6.0
Gulf/Bratten Ranch No. 7	Hood	4,188-4,198	-	-	10[a]	142	921	1,042	-	850	5.5
Gulf/Bratten Ranch No. 1-A	Hood	4,181-4,188	-	-	-	142	853	535	0.700	900	5.1
Gulf/Bratten Ranch No. 8	Hood	2,849-4,525	-	11.2	85[b]	-	-	-	-	-	4.9
Gulf/Bratten Ranch No. 6	Hood	2,740-4,505	52.1	10.3	25	-	-	-	-	-	4.6
Tri-Venture II/ Trotter No. 1	Hood	2,700-4,700	47.6	11.4	33	-	-	-	-	-	4.7

TABLE 2.5.1-6
RESERVOIR DATA SUMMARY

(Sheet 2 of 4)

Operator/Lease	County	Interval (ft)	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom- Hole Temperature (°F)	Shut-in Wellhead Pressure (psig)	Shut-in Bottom-Hole Pressure (psig)	Gas Gravity (air=1.0)	Absolute Open Flow (Mcf/day)	Distance From Plant (miles)
Energy Fund/ O. Coker No. 1	Hood	-	-	-	24[(b)]	-	-	-	-	-	4.7
Ben J. Taylor/ Wiggins No. 1	Hood	2,780-4,740	51.6	9.9	43	-	-	-	-	-	3.2
Dallas Production/ Hubbard No. 1	Hood	2,710-4,796	42.2	10.5	15	-	-	-	-	-	2.4
Dorchester/Davis-Bonney No. 1	Somervell	2,440-4,325	43.2	8.6	22	-	-	-	-	-	6.1
Mitchell/Ogle No. 1	Erath	-	-	-	20[(b)]	-	-	-	-	-	33.3
HNG Oil/Ryan No. 1	Hamilton	3,722-3,776	49.4	11.4	5	140	-	-	-	-	25.2
Gulf/O. Gardner No. 1	Hood	4,182-4,190	-	-	-	142	856	821	-	1,025	7.2
Big Saline											
Gulf/Bratten Ranch No. 5	Hood	4,605-4,636	-	6.0	14[(b)]	-	-	-	-	-	4.3
Tri-Venture II/ Trotter No. 1	Hood	4,756-4,772	30.6	10.4	4	-	-	-	-	-	4.7
Ben J. Taylor/ Wiggins No. 1	Hood	4,744-4,790	45.2	6.0	5	147	-	-	-	-	3.2

TABLE 2.5.1-6
RESERVOIR DATA SUMMARY

(Sheet 3 of 4)

Operator/Lease	County	Interval (ft)	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom- Hole Temperature (°F)	Shut-in Wellhead Pressure (psig)	Shut-in Bottom-Hole Pressure (psig)	Gas Gravity (air=1.0)	Absolute Open Flow (Mcf/day)	Distance From Plant (miles)
Marble Falls											
Dorchester/ Davis-Bonney No. 1	Somervell	4,472-4,684	46.6	7.7	9	-	-	-	-	-	6.1
Gulf/Otis Rollins No. 1	Hood	5,130-5,140	-	-	46[a]	162	1,440	1,678	0.690	1,200	7.2
Gulf/Bratten Ranch No. 8	Hood	4,700-4,800	-	8.0	20[b]	156	1,512	1,762	0.700	1,900	4.9
Gulf/Bratten Ranch No. 6	Hood	4,576-4,933	44.3	7.0	3	150	1,478	1,807	-	1,950	4.6
Gulf/Bratten Ranch No. 5	Hood	4,683-5,050	-	4.8	62[b]	150[a]	-	-	-	-	5.0
Tri-Venture II/ Trotter No. 1	Hood	4,830-5,100	40.4	5.9	25	-	-	-	-	-	4.7
Ben J. Taylor/ Wiggins No. 1	Hood	4,834-5,110	29.5	5.0	15	150[a]	-	-	-	-	3.2
Dallas Production/ Hubbard No. 1	Hood	4,960-5,034	57.1	10.0	6	-	-	-	-	-	2.3
Ellenburger											
Texas Pacific/Fee Deed No. 1 Well No. 2	Erath	3,844-3,878	34.3	9.2	8	111	-	-	-	-	46.4
Mitchell/T.G. Ogle No. 1	Erath	4,388-4,680	27.2	8.2	4	127	-	-	-	-	33.0

TABLE 2.5.1-6
RESERVOIR DATA SUMMARY

(Sheet 4 of 4)

Operator/Lease	County	Interval (ft)	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom- Hole Temperature (°F)	Shut-in Wellhead Pressure (psig)	Shut-in Bottom-Hole Pressure (psig)	Gas Gravity (air=1.0)	Absolute Open Flow (Mcf/day)	Distance From Plant (miles)
Dorchester/ Young No. 1-7	Erath	4,941-4,972	17.1	5.1	2	146	-	-	-	-	25.8
Dallas Production/ Hubbard No. 1	Hood	(Poor Logs)	-	-	-	-	-	-	-	-	2.3
Dorchester/ Davis-Bonney No. 1	Somervell	5,024-5,144	27.5	6.6	45	169	-	-	-	-	6.1
HNG Oil/Ryan No. 1	Hamilton	4,414-4,438	42.9	10.3	6	-	-	-	-	-	25.2
T. J. Nunley/ Maxine Brooks No. 1	Hood	5,147-5,198	100.0	7.1	0	-	-	-	-	-	21.2
Cities Service/ Allen ”B” No. 1	Hood	4,173-5,501	37.6	8.3	8	128	-	-	-	-	20.9
Caliph Resources/ Millington No. 7	Hood	5,457-5,499	-	6.2	4(b)	125	-	-	-	-	14.5
Sunray Oil/ John R. Black No. 1	Hood	5,220-5,400	-	-	34(b)	129	-	-	-	-	13.6

a) Estimated

b) Porosity thickness

TABLE 2.5.1-7
STATISTICAL SUMMARY OF RESERVOIR PROPERTIES

Sand	Number of Wells Used for Log Analysis	Net Hydrocarbon, feet			Porosity, percent			Water Saturation, percent			Permeability, millidarcies		
		Maximum	Minimum	Average	Maximum	Minimum	Average	Maximum	Minimum	Average	Maximum	Minimum	Average
Strawn	8	50.0	5.0	27.0	11.4	8.6	10.2	55.8	42.2	49.2	6.0[a]	4.0[a]	5.0[a]
Big Saline	2	5.0	4.0	4.5	10.4	6.0	8.2	45.2	30.6	37.9	10.0[b]	-	-
Marble Falls	7	25.0	3.0	11.6	10.0	5.0	7.8	57.1	29.5	43.6	17.0[c]	1.0[c]	7.0[c]
Ellenburger	7	45.0	2.0	12.2	10.3	5.1	7.6	42.9	17.1	31.1	15.0[b]	-	-

a) Only two wells used to calculate permeability in this zone.

b) Estimated.

c) Only three wells used to calculate permeability in this zone.

TABLE 2.5.1-8
ABSOLUTE OPEN FLOW POTENTIAL CALCULATIONS

Sand	Maximum Bottom-Hole Pressure (psia)	Temperature (°R)	Gas Viscosity (cp)	Compressi-bility (Z)	Drainage Radius (ft)	Maximum Net Hydrocarbon (ft)	Permeability, md		Flow Rate, MMcf/d			Percent[a] over Average	Subsidence (ft)
							Maximum	Average	Maximum	(Permeability) 1.5 times Maximum	Average		
Strawn	1,200	607	0.0138	0.8673	1,053	50	6	5	5.8	8.7	4.80	181	0.36
Big Saline	1,500[b]	610	0.0146	0.8450	1,290	5	10[b]	5[b]	1.4	2.1	0.71	300	0.04
Marble Falls	1,900	615	0.0159	0.8230	1,489	25	17	7	17.8	26.7	7.33	364	0.28
Ellenburger	2,000[b]	620[b]	0.0163	0.8189	2,106	45	15[b]	7[b]	29.2	43.8	13.63	321	0.54

a) The percent that the 1.5 times maximum flow rate is over the average well flow rate.

b) Estimated.

TABLE 2.5.1-9
RESERVOIR DATA ON SELECTED WELLS WITHIN FIVE-MILE RADIUS OF CPNPP[a]

(Sheet 1 of 2)

Well I.D. No.	Operator/Lease	Interval	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom-Hole Temperature (F°)
Strawn						
2	Mid-Continent Pet./Cattle Co. No. 1	-	-	-	60[b]	-
3	Fitzgerald/Van Morrison No. 1	-	-	-	65[b]	-
4	Tri-Venture II/Trotter No. 1	2,700-4,700	47.6	11.4	33	151
5	Energy Fund/O. Coker No. 1	-	-	-	24[b]	-
12	Ben J. Taylor/Wiggins No. 1	2,780-4,740	51.6	9.9	43	153
14	Dallas Production/Hubbard No. 1	2,710-4,796	42.2	10.5	15	-
16	Gulf/Bratten Ranch No. 8	2,849-4,525	-	-	85[b]	143
17	Gulf/Bratten Ranch No. 6	2,740-4,505	52.1	10.3	25	150
Big Saline						
2	Mid-Continent Pet./Cattle Co. No. 1	-	-	-	15[b]	-
4	Tri-Venture II/Trotter No. 1	4,756-4,772	30.6	10.4	4	151
12	Ben J. Taylor/Wiggins No. 1	4,744-4,790	45.2	6.0	5	153

TABLE 2.5.1-9
RESERVOIR DATA ON SELECTED WELLS WITHIN FIVE-MILE RADIUS OF CPNPP[a]

(Sheet 2 of 2)

Well I.D. No.	Operator/Lease	Interval	Water Saturation (%)	Porosity (%)	Net Hydrocarbon (ft)	Bottom-Hole Temperature (F°)
15	Gulf/Bratten Ranch No. 5	4,605-4,636	-	6.7	14[b]	147
Marble Falls						
4	Tri-Venture II/Trotter No. 1	4,830-5,100	40.4	5.9	25	151
12	Ben J. Taylor/Wiggins No. 1	4,834-5,110	29.5	5.0	15	153
14	Dallas Production/Hubbard No. 1	-	57.1	10.0	6	-
15	Gulf/Bratten Ranch No. 5	4,683-5,050	-	4.8	62[b]	147
16	Gulf/Bratten Ranch No. 8	4,700-4,800	-	8.0	20[b]	147
17	Gulf/Bratten Ranch No. 6	4,756-4,933	44.3	7.0	3	150
Ellenburger						
14	Dallas Production/Hubbard No. 1	-	-	-	18[b]	-

a) Wells with adequate data for calculations presented above.

b) Indicates gross porosity thickness.

TABLE 2.5.1-10
GAS IN-PLACE ASSOCIATED WITH HYPOTHETICAL WELL

Sand	Maximum Porosity (%)	Minimum Water Saturation (%)	Temperature (^{O}R)	Gas Deviation Factor (Z)	Maximum Net Hydrocarbon (ft)	Maximum Pressure (psia)	Drainage Area (acres)	Gas In-Piece (MMef)
Strawn	11.4	42.2	607	0.8673	50	1,200	80	929
Big Saline	10.4	30.6	610	0.8450	5	1,500	120	195
Marble Falls	10.0	29.5	615	0.8230	25	1,900	160	1,636
Ellenburger	10.3	17.1	620	0.8189	45	2,000	320	7,488

TABLE 2.5.1-11
SURVEY OF REPORTED BOTTOM-HOLE PRESSURE IN HOOD AND SOMERVELL COUNTIES, TEXAS

(Sheet 1 of 2)

Well I.D. No.	Well Name and Location	County	Formation	Depth, ft.	Shut-in Pressure	Gradient psi/ft.	Remarks
	Delray Oil, Inc. No. 1 G. P. Snyder "R" J. Echols Survey A-24 3 mi. south of Glen Rose	Somervell	Ellenburger	6,049-6,114	2,761# 1 hr. 30 min.	.45643	Recovered 552' muddy salt water (chlorides 86,000 ppm). Dry and abandoned 10-16-75.
	Cities Service No. 1-N Wilson James Cole Survey A-91 2 mi. North of Lipan	Hood	Ellenburger	5,182-5,285	2,318# 30 min. (Probable not Stabilized.)	.44732	Open hole drill-stem test recovered 180' gas in drill pipe; 120' salty, gas-cut mud; 180' gas-cut muddy salt water; 1,260' gas-cut salt water. Dry and abandoned 9-22-61.
	Sunray No. 1 John Black Lower Thomas Robinson Survey 3.5 mi. west of Thorp Springs	Hood	Ellenburger	5,438-5,448	2,550# Time not reported.	.46892	Recovered 4,200' salt water in 2-hr. open hole drill-stem test. Dry and abandoned 7-7-51.
17	Gulf Oil Corp. No. 6 Bratten Ranch T. J. Stansbury Survey A-498 3 mi. east of Paluxy	Hood	Marble Falls	4,686-4,696	1,807# Time not reported. (After treatment.)	.38520	Gas well (CAOF 1,950 Mcfgpd). Completed 7-18-77
16	Gulf Oil Corp. No. 8 Bratten Ranch Thomas Stansbury Survey A-498 3 mi. east of Paluxy	Hood	Marble Falls	4,696-4,776	1,746# Time not reported.	.36866	Gas well (CAOF 2,725 Mcfgpd). Completed 6-16-77
	Cities Service No. 1-N Wilson James Cole Survey A-91 2 mi. north of Lipan	Hood	Big Saline	4,141-4,201	1,983# 30 min.	.47887	Open hole drill-stem test flow estimated 4.4 million cubic feet of gas per day; recovered 70' slightly gas and salty cut mud, 60' gassy and heavy salt water-cut mud, 30' muddy salt water.
4	Tri-Venture, II No. 1 Trotter J. Doan Survey 13 mi. southwest of Granbury	Hood	Big Saline	4,770-4780	1,861# Time not reported. (After treatment.)	.38973	Gas well (CAOF 378 Mcfgpd). Completed 11-7-77.
18A	Sirius Energy Co. No. 1 J. T. Parker Cyrus T. Egery Survey A-169 1 mi. north and 4 mi. east of Paluxy	Hood	Big Saline	4,730-4,746	1,822# Time not reported. (After fracturing formation for production)	.38455	Gas Well (CAOF 174 Mcfgpd). Completed 1-23-78.

TABLE 2.5.1-11
SURVEY OF REPORTED BOTTOM-HOLE PRESSURE IN HOOD AND SOMERVELL COUNTIES, TEXAS

(Sheet 2 of 2)

Well I.D. No.	Well Name and Location	County	Formation	Depth, ft.	Shut-in Pressure	Gradient psi/ft.	Remarks
	Delray Oil, Inc. No. 1 Bramlett Isacc Addison Survey A-2 4 miles south of Glen Rose	Somervell	Strawn	2,610-2,708	855# 1 hr. 30 min.	.32759	Open hole drill-stem test recovered 60' drilling mud. Dry and abandoned 5-22-76.
			Strawn	2,590-2,621	370# 2 hrs.	.14286	Open hole drill-stem test recovered 25' drilling mud.
	Delray Oil, Inc. No. 1 G. P. Snyder "R" J. Echols Survey A-24 3 mi. south of Glen Rose	Somervell	Strawn	2,964-2,995	1,064# 3 hrs.	.35897	Recovered 15' drilling mud.
			Strawn	2,428-2,464	837# 2 hrs.	.34473	Recovered 35' drilling mud.
	Gulf Oil Corp. No. 3 Bratten Ranch J. B. White Survey A-576 10 mi. southeast of Tolar	Hood	Strawn	4,054-4,064	1,181# Time not reported.	.29095	Gas well (CAOF 742 Mcfgpd). Completed 3-31-77.
	Gulf Oil Corp. No. 2 T. H. Nix Lewis Boatwright Survey A-33 3 mi. south of Tolar	Hood	Strawn	2,802-2,812	599# Time not reported.	.21339	Gas well (CAOF 161 Mcfgpd). Completed 2-21-78.
	Gulf Oil Corp. No. 1 V. Swain Lewis Boatwright Survey A-33 3 mi. south of Tolar	Hood	Strawn	2,762-2,768	689# Time not reported.	.24918	Gas well (CAOF 3,200 Mcfgpd). Completed 9-27-77.
	Gulf Oil Corp. No. 1 D. McIntosh T. W. Dundick Survey A-23 4 mi. southeast of Tolar	Hood	Strawn	4,356-4,380	1183# Time not reported.	.27083	Gas well (CAOF 3,090 Mcfgpd). Completed 11-18-76.

TABLE 2.5.2-1
MODIFIED MERCALLI INTENSITY SCALE OF 1931 (Abridged)

I. Not felt except by a very few under especially favorable circumstances. (I Rossi-Forel Scale.)

II. Felt only by a few persons at rest, especially on upper floors of buildings. Delicately suspended objects may swing. (I to II Rossi-Forel Scale.)

III. Felt quite noticeably indoors, especially on upper floors of buildings, but many people do not recognize it as an earthquake. Standing motorcars may rock slightly. Vibration like passing of truck. Duration estimated. (III Rossi-Forel Scale.)

IV. During the day felt indoors by many, outdoors by few. At night some awakened. Dishes, windows, doors disturbed; walls make creaking sound. Sensation like heavy truck striking building. Standing motorcars rocked noticeably. (IV to V Rossi-Forel Scale.)

V. Felt by nearly everyone, many awakened. Some dishes, windows, etc., broken; a few instances of cracked plaster; unstable objects overturned. Disturbances of trees, poles, and other tall objects sometimes noticed. Pendulum clocks may stop. (V to VI Rossi-Forel Scale.)

VI. Felt by all, many frightened and run outdoors. Some heavy furniture moved; a few instances of fallen plaster or damaged chimneys. Damage slight. (VI to VII Rossi-Forel Scale.)

VII. Everybody runs outdoors. Damage negligible in buildings of good design and construction; slight to moderate in well-built ordinary structures; considerable in poorly built or badly designed structures; some chimneys broken. Noticed by persons driving motor cars.

VIII. Damage slight in specially designed structures; considerable in ordinary substantial buildings with partial collapse; great in poorly built structures. Panel walls thrown out of frame structures. Fall of chimneys, factory stacks, columns, monuments, walls. Heavy furniture overturned. Sand and mud ejected in small amounts. Changes in well water. Persons driving motorcars disturbed. (VIII+ to IX- Rossi-Forel Scale.)

IX. Damage considerable in specially designed structures; well-designed frame structures thrown out of plumb; great in substantial building with partial collapse. Buildings shifted off foundations. Ground cracked conspicuously. Underground pipes broken. (IX+ Rossi-Forel Scale.)

X. Some well-built wooden structures destroyed; most masonry and frame structures destroyed with foundations; ground badly cracked. Rails bent. Landslides considerable from river banks and steep slopes. Shifted sand and mud. Water splashed (slopped) over banks. (X Rossi-Forel Scale.)

XI. Few, if any (masonry) structures remain standing. Bridges destroyed. Broad fissures in ground. Underground pipelines completely out of service. Earth slumps and land slips in soft ground. Rails bent greatly.

XII. Damage total. Waves seen on ground surface. Lines of sight and level distorted. Objects thrown upward into the air

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 1 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References		
1843 Feb 14	New Orleans, LA	29.9	90.1					Docekal		1970
1843 Feb 15	New Orleans, LA	29.9	90.1					Docekal		1970
1852 Jan 17	British Honduras (felt in Galveston)							De Montessus		1888
1873 Apr 30	Manor, Texas	30.3	97.7		III-IV			Docekal		1970
1882 Apr 11	New Orleans, LA	29.9	90.1					Docekal		1970
1882 Oct 22	NE Texas (felt in Paris and Greenville Northwestward to Kansas)	33.6	95.6		VII	(5.5)	195,000	Byerly Eppley Henley Rockwood Shurbet U.S. Weather Bureau	July/	1934 1965 1966 1883 1969 1882
1886 Jan 22	Grand Coteau, LA	30.3	92.0					Docekal		1970
1887 May 3	W Texas (Sonora, MX)	31.0	109.0		VIII-IX			Eppley Woollard		1965 1968
1891 Jan 8	Rusk, Cherokee County, Texas	31.7	95.2		VII			Eppley Henley London Times Milne		1965 1966 1891 1911
1907 Apr	Panhandle, Texas	(35.5	101.2)					Udden		1926
1908 July 19	El Reno & Guthrie, Okla.	35.7	97.7		III			Docekal		1970

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 2 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1910 May 8	Hempstead, Waller County, Texas	30.1	96.0		IV			Seis. Notes	1913
1910 May 11	Waller & Washington Counties, Texas	30.1	96.0		IV		1,000	Seis. Notes Docekal	1913 1970
1911 Mar 31	Pine Bluff, Ark.	34.22	92.0		VI		20,000	Docekal	1970
1914 Dec 29	Anderson County, Texas	30.5	95.9		IV-V			Humphreys Docekal	1914 1970
1914 Dec 30	Anderson County, TX	30.5	95.9		V			U.S. Weather Bureau	1914
1915 Oct 8	Muskogee, Okla.	35.7	95.4		III		3,000	Docekal	1970
1917 Jan 28	Panhandle, Texas	35.4	101.3		II-III			Tondorf	
1917 Mar 24	Panhandle, Texas	(35.3	101.2)		V			Henley Udden	1966 1926
1917 Mar 28	Panhandle, Texas	35.3	101.3		VI		2,000	Eppley Pratt Docekal U.S. Hypo. Data File	1965 1926 1970
1918 Sept 10	El Reno, Okla.	35.5	98.0		V			Docekal	1970
1918 Sept 11	El Reno, Okla. (2 shocks)	35.5	98.0					Docekal	1970
1918	Yukon, Okla.	35.5	97.7		III-IV			Docekal	1970
1923 Mar 6	El Paso, Texas	31.7	106.5		IV-V		20,000	Docekal	1970

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 3 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1923 May 6	Felt Sierra Blanco and SE El Paso, Texas	31.7	106.5		IV-V		20,000	Seis. Notes Docekal	1923 1970
1924 June 2	Cleveland, Okla.	36.3	96.5		III		20,000	Docekal	1970
1924 Aug 12	N. Mexico	36.0	104.5					U.S. Hypo. Data File	
1925 July 8	Harrison and Alpena, Ark.	36.3	93.2		IV		4,000	Docekal	1970
1925 Jul 29	Silverton, Texas	34.5	101.2		IV			Neumann/Quarterly Report Neumann (Seis. Rep.) Docekal	 1926 1970
1925 Jul 30	Childress, Texas (02:00)	(34.5	100.3)		V			Neumann/Quarterly Report Neumann (Tex. Earth) Udden	 1926 1926
1925 Jul 30	Panhandle, Texas (06:00)	35.4	101.3		VI		200,000	Bowie Eppley Henley Neumann/Quarterly Report Pratt Udden Docekal	1925 1965 1966 1926 1926 1970
1925 Jul 30	Panhandle, Texas (06:00 - 08:00)				II-VI			Neumann (Seis. Rep.)	1926
1926 June 20	Okla. - Ark. border	35.6	94.9		V		18,000	Docekal	1970
1927 Dec 14	Burrwood, LA	28.9	89.4		IV		4,000	Docekal	1970
1928 Oct 31	Mexico, Felt in Texas	26.0	106.0		(VI)			Eppley USE	1965 1928

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 4 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References		
1929 July 28	Burrwood, LA.	28.9	89.4		IV		3,000	Docekal		1970
1929 Nov 16	Malvern, Ark.	34.3	92.7		V		350	Docekal		1970
1929 Nov 26	Ashland, Kansas	37.2	99.7		IV			Docekal		1970
1929 Dec 27	El Reno, Okla.	35.5	98.0		VI		8,000	U.S. Hypo. Data File		
								Docekal		1970
1930 Oct 19	Donaldsonville, LA.	30.0	91.0		VI		18,500	Docekal		1970
								USE		1930
								U.S. Hypo. Data File		
1931 Aug 16	Valentine, TX (2 shocks)	30.6	104.1		VIII	6.4	450,000	Byerly	April/	1934
								Byerly	July/	1934
								Eppley		1965
								Henley		1966
								Sellards (Val., Tex)		1932
								USC & CS	Dec/	1931
								U.S. Hypo. Data File		
								Docekal		1970
								USE		1931
1931 Aug 18	Valentine, Texas (aftershock of 8/16/31)	30.6	104.1		V		Uncertain	Eppley		1965
								Sellards (Val., Tex)		1932
								USE		1931
1931 Aug 26	Valentine, Texas	30.6	104.1		III			Sellards (Val., Tex)		1932
1931 Oct 2	El Paso, Texas	31.7	106.5		II-III			Sellards (Val., Tex)		1932
								Docekal		1970
								USE		1931

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 5 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1931 Nov 3	Valentine, Texas	29.9	104.2		II			Shurbet U.S. Hypo. Data File USE	1969 1931
1932 Apr 9	Mexia-Wortham, Texas	31.5	96.5		V-VI		1,000	Eppley Henley Sellards (Mexia) Docekal USE	1965 1966 1932 1970 1932
1933 Aug 19	El Reno, Okla.	35.4	97.8		V			USE Docekal	1932 1970
1934 Apr 11	Trout Switch, Lamar County, Texas (Texas- Oklahoma Border)	33.9	95.5		V		3,000	Eppley Henley Sellards Docekal	1965 1966 1936 1970
1936 Jan 8	Carlsbad, New Mexico	32.4	104.2		II-III			Docekal	1970
1936 Mar 14	Wright City, Oklahoma	34.0	95.0		V		800	Eppley Docekal USE	1965 1970 1936
1936 Jun 19	Borger, Texas (Texas Panhandle – two foreshocks)	35.7	101.4		VI		44,000	Eppley Henley Sellards U.S. Hypo. Data File Docekal USE	1965 1966 1940 1970 1936
1936 July 11	Kenton, Okla.	36.9	102.9		III-IV		200	Docekal	1970

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 6 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1936 Aug 7	El Paso, Texas	31.7	106.5		III			Henley Shurbet Docekal USE	1966 1969 1970 1936
1936 Oct 15	El Paso, Texas	31.7	106.5		IV			Docekal USE	1970 1936
1937 Mar 31	El Paso, Texas	31.7	106.4		III			Henley Shurbet Docekal USE	1966 1969 1970 1937
1937 June 8	Tecumsen, Okla.	35.3	96.9		IV		1,000	Docekal	1970
1937 Aug 7	El Paso, Texas	31.7	106.5		III			Docekal	1970
1937 Sept 30	Fort Stanton, New Mexico	33.5	105.5		IV			Docekal	1970
1938 Apr 25	Fendley, Ark.	34.3	93.5		IV			Docekal USE	1970 1938
1939 June 1	Holdenville, Okla.	35.0	96.0		IV		25,000	Docekal USE	1970 1939
1939 June 19	Arkadelphia, Ark.	34.1	92.6		IV		25,000	Eppley Docekal USE	1965 1970 1939
1940 Dec 2	Rodessa, LA	33.0	94.0		IV			USE	1940
1941 Oct 18	Bessie, Okla.	35.4	99.0		V		100	Docekal USE	1970 1941

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 7 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1941 June 28	Vicksburg, Miss.	32.3	90.8		III-IV			Docekal USE	1970 1941
1942 June 11	Enid, Okla.	36.4	97.9		III		1,500	Docekal USE	1970 1942
1947 Sept 20	Winfield, LA	31.9	92.7		IV-V			Docekal USE	1970 1949
1943 Mar 11	Dalhart, TX	36.0	102.5		VI		115,000	Eppley Robetrs/Ulrich Shurbet Docekal USE	1965 1950 1969 1970 1948
1949 Feb 2	Carlsbad, New Mexico	32.4	104.2		IV		50,000	Docekal USE	1970 1949
1949 May 23	East Vaughn, N. Mexico	34.6	105.2		IV			Docekal USE	1970 1949
1950 Mar 20	Chico, TX	33.5	97.1		IV			Docekal USE	1970 1950
1951 Jun 20	Amarillo, TX	35.5	103.0		VI		25,000	Eppley Seis. Notes Shurbet U.S. Hypo. Data File Docekal USE	1965 1951 1969 1970 1951

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 8 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1952 Apr 9	El Reno, Okla.	35.4	97.8		VII	5.5	247,000	Eppley Hull U.S. Hypo. Data File Docekal USE	1965 1956 1970 1952
1952 Apr 11	El Reno, Okla. (2 shocks)	35.4	97.8		IV		3,000	Docekal USE	1970 1952
1952 Apr 15	El Reno, Okla.	35.4	97.8		II-III		3,000	Docekal USE	1970 1952
1952 Apr 16	El Reno, Okla.	35.4	97.8		V		3,000	Docekal USE	1970 1952
1952 July 16	El Reno, Okla.	35.4	97.8		III-IV			Docekal USE	1970 1952
1952 Aug 14	El Reno, Okla.	35.4	97.8		IV			Docekal	1970
1952 Oct 7	Wewoka, Okla.	35.2	96.5		V			USE Docekal	1952 1970
1952 Oct 17	Orange, TX	30.1	93.7		IV			Docekal USE	1970 1952
1953 Mar 16	Union City, Okla.	35.4	98.0		III			USE Docekal	1953 1970
1953 Mar 17	Concho, Okla. (2 shocks)	35.6	98.0		V-VI		2,700	USE Docekal	1953 1970
1953 June 6	Ada, Okla.	34.8	96.7		IV			USE Docekal	1953 1970

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 9 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1954 Apr 11	Holdenville (2 shocks)	35.0	96.4		IV			USE Docekal	1954 1970
1954 Apr 12	Holdenville, Okla.	35.0	96.4		IV			USE Docekal	1954 1970
1954 Apr 13	Holdenville, Okla.	35.0	96.4		IV			USE Docekal	1954 1970
1954 June 23	210 km from Dallas, TX				*			Herrin	1954
1955 Jan 26	Valentine, TX	30.6	104.5		IV			Shurbet USE Docekal	1969 1955 1970
1955 Feb 1	Gulfport, Mississippi	30.4	89.1		V			USE Docekal	1955 1970
1956 Jan 6	S. Kansas	37.3	98.8		VI		22,500	Docekal	1970
1956 Jan 7	Galveston Island, TX	29.3	94.8		IV			USE Docekal	1956 1970
1956 Feb 16	Edmond, Okla.	35.6	97.5		VI		5,000	USE Docekal	1956 1970
1956 Apr 2	Antlers, Okla.	35.2	95.6		V		2,000	USE Docekal	1956 1970
1956 Oct 30	Catoosa, Okla.	36.2	95.8		VII		9,500	Docekal USE	1970 1956

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 10 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1957 Mar 19	Gladewater, TX (10:38)	32.6	94.7		V		18,300	Eppley Shurbet Docekal USE	1965 1969 1970 1957
1957 Mar 19	Gladewater, TX (11:41)	32.2	94.8		V		10,100	USE	1957
1957 Mar 19	Gladewater, TX (16:36)	32.2	94.8		V		10,100	USE	1957
1957 Mar 19	Gladewater, TX (16:45)	32.2	94.8		V		10,000	USE	1957
1958 May 24	Gulf of Mexico	27.0	95.0		*			U.S. Hypo. Data File	
1958 Nov 6	New Orleans, LA	30.0	90.1		IV			Docekal USE	1970 1958
1958 Nov 19	Baton Rouge, LA	30.5	91.2		V		300	Eppley Docekal USE	1965 1970 1958
1959 Feb 2	Texas Panhandle	35.5	100.9		V		40,000	Docekal	1970
1959 June 15	Ada, Okla.	34.8	96.7		V		5,000	USE Docekal	1959 1970
1959 Jun 17	Lawton, Okla.	34.6	98.4		VI		14,250	U.S. Hypo. Data File Docekal USE	 1970 1959
1959 Oct 15	Creole, LA	29.8	93.1		IV		2,500	Docekal	1970
1959 Dec 9	Mexico	28.5	105.0					U.S. Hypo. Data File	

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 11 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1960 May 4	Pine Bluff, Ark.	34.2	92.0		IV			USE Docekal	1960 1970
1961 June 10	Adamson, Okla.	34.9	95.6		V		2,500	USE Docekal	1961 1970
1961 May 26	Antlers, Okla. (3 shocks)	34.6	95.0		III		2,500	USE Docekal	1961 1970
1961 May 27	Antlers, Okla.	34.6	95.0		V		8,000	USE Docekal	1961 1970
1963 June 6	New Mexico	36.5	104.3	(33)**		3.8		U.S. Hypo. Data File	
1963 Sept 10	Mexico	28.9	104.0	(33)**		4.2		U.S. Hypo. Data File	
1963 Sept 13	Mexico	29.1	105.6	(33)**		4.7		U.S. Hypo. Data File	
1964 Feb 2	Western Okla.	35.1	99.7	(33)**	V			U.S. Hypo. Data File Docekal	 1970
1964 Mar 27	Epicenter: Alaska (Damage to water wells and fluctuations in surface water and groundwater wells occurred in the southern High Plains Region of Texas.)						22,000	Miller/Reddel	1964
1964 Apr 23	Texas-Louisiana Border	31.5	93.8		IV	3.7		Docekal USE	1970 1964

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 12 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1964 Apr 24	Texas-Louisiana	31.6	93.8	12-33	IV	3.7-4.36		Henley	1966
								UED	1965
								U.S. Hypo. Data File	
								Docekal	1970
								USE	1964
1964 Apr 24	Texas-Louisiana Border	31.6	93.8	(33)**	IV-V	3.7		U.S. Hypo. Data File	
								USE	1966
1964 Apr 27	Texas-Louisiana Border	31.5	93.8	(33)**	IV	3.4	300	U.S. Hypo. Data File	
								Docekal	1970
								USE	1964
1964 Apr 28	Hemphill, TX	37.7	93.6	(33)**	V	4.4		U.S. Hypo. Data File	
								Docekal	1970
								USE	1964
1964 Apr 30	Pineland, TX	31.2	94.0					Docekal	1970
								USE	1964
1964 May 7	Pineland, TX	31.2	94.0		VI			Docekal	1970
								USE	1964
1964 June 2	Texas-Louisiana Border (2)	31.3	94.0	30	IV	4.2		Henley	1966
								U.S. Hypo. Data File	
								USE	1964
1964 June 2	Pineland, TX	(31.3	94.0)		Light	4.2		USE	1964
1964 Aug 16	Hemphill, TX	(31.3	94.0)		V			USE	1964
1965 Feb 3	Logan, New Mexico	35.5	103.6		IV		1,200	USE	1965
								Docekal	1970

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 13 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1965 Apr 13	Mexico	30.3	105.1	(33)**		4.2		U.S. Hypo. Data File	
1965 Aug 30	W Central Texas	32.1	102.3	(33)**	IV	3.5		Shurbet U.S. Hypo. Data File	1969
1966 Jan 15	Texas-Louisiana Border (Sonic Booms)	30.3	93.7		III-IV			USE Docekal	1968 1970
1966 Feb 1	Texas-Louisiana Border (Sonic Booms)	30.3	93.7		III			USE Docekal	1968 1970
1966 Feb 2	Texas-Louisiana Border (Sonic Booms)	30.3	93.7		IV-V		6,000	USE Docekal	1968 1970
1966 Feb 3	Texas-Louisiana Border (Sonic Booms)	30.3	93.7		IV			USE Docekal	1968 1970
1966 Mar 24	Saline, TX	30.0	94.0		III-IV			Docekal	1970
1966 Jul 20	Borger, TX	35.6	101.4	(33)**	IV-V	3.9-4.8	12,000	Austin Statesman Shurbet U.S. Hypo. Data File Docekal USE	7/20/66 1969 1970 1966
1966 July	Amarillo, TX							Shurbet	1969
1966 Aug 14	Kermit, TX (11 aftershocks)	31.7	103.1	(33)**	IV-V	3.4	20,000	Shurbet U.S. Hypo. Data File Docekal USE	1969 1970 1966
1966 Aug 19	Mexico (04:15)	30.3	105.6	(33)**		4.1		U.S. Hypo. Data File	

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 14 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1966 Aug 19	Mexico (07:38)	30.2	105.7	(33)**		3.9		U.S. Hypo. Data File	
1966 Aug 19	Mexico (08:38)	30.3	105.6	(33)**		4.0		U.S. Hypo. Data File	
1966 Aug 20	Mexico	30.1	105.5	(33)**		4.3		U.S. Hypo. Data File	
1966 Aug 21	Mexico	30.1	105.6	(33)**		4.1		U.S. Hypo. Data File	
1966 Nov 28	Mexico	30.4	105.4	(33)**		3.8		U.S. Hypo. Data File	
1966 Dec 5	Mexico	30.4	105.4	(33)**		4.2		U.S. Hypo. Data File	
1967 June 4	Greenville, Miss.	33.6	90.9		VI	3.8	25,000	USE Docekal	1967 1970
1967 June 29	Greenville, Miss.	33.6	90.9		V	3.4		USE Docekal	1967 1970
1968 Jan 4	Hartshorne, Okla.	34.9	95.5		IV			USE	1968
1968 Mar 9	New Mexico	32.5	106.0	(33)**				U.S. Hypo. Data File	
1966 Oct 10-11	Durant, Okla. (2 shocks)	34.0	96.0					USE	1969
1968 Oct 14	Durant, Okla.	34.0	96.3		VI			Docekal USE	1970 1969
1969 Jan 1	W Central Arkansas	34.8	92.6		VI	4.2	23,000	U.S. Hypo. Data File USE	 1969
1969 May 2	Wewoka, Okla.	35.2	96.3	(33)**	V	4.6	13,000	U.S. Hypo. Data File USE	 1969

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 15 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1969 May 12	Mexico-Texas Border (4 shocks - 2:26, 2:49, 2:51, 4:39)	31.8	106.4	19	VI	3.4 3.3		Austin Statesman U.S. Hypo. Data File USE	5/13/69 1969
1969 June 12	Houston, TX (Hobby Airport)	29.7	95.3		I-II	1.5-2		Houston Post Houston Post	6/13/69 6/20/69
1969 June 18	Houston, TX					1.5		Houston Post Houston Post	6/19/69 6/20/76
1969 June 19	Houston, TX					1.5		Houston Post	6/20/69
1969 Oct 19	Mexico	30.8	105.7	(33)**		3.8		U.S. Hypo. Data File	
1970 Jan 12	New Mexico (near Texas border)	36.1	103.2	(33)**	VI	3.5	3,700	U.S. Hypo. Data File USE	 1970
1970 Feb 3	Travis Peak, TX	(31	97)		(IV)			News Release	
1970 Feb 5	Travis Peak, TX							Austin Statesman	2/5/70
1970 Jul 30	West Texas	31.7	103.0	10	(III)	3.0		U.S. Hypo. Data File	
1971 Jul 31	West Texas	31.7	103.1	10	(IV)	3.4		U.S. Hypo. Data File	
1972 Mar 10	Oklahoma City, Okla.	35.5	97.5		(II-III)			USE	1972
1974 Feb 15	Texas Panhandle	36.5	100.7	24	V	4.6		U.S. Hypo. Data File	
1974 Feb 15	Whelen Springs, Ark.	34.0	93.0	1	(III)	3.6		U.S. Hypo. Data File	
1974 Feb 15	Whelen Springs, Ark.	34.0	93.1	1	III	3.6		U.S. Hypo. Data File	

TABLE 2.5.2-2
REGIONAL EARTHQUAKES

(Sheet 16 of 16)

Date	Location	Latitude N°	Longitude W°	Depth km	Maximum Intensity MM1931	Magnetic	Felt Area Square Mile	References	
1974 Feb 15	(swarm)	34.0	93.0	1	V	4.0		U.S. Hypo. Data File	
1974 Feb 15	(swarm)	34.0	93.0	1	III	2.8		U.S. Hypo. Data File	
1974 Feb 16	(swarm)	34.0	93.0	1		1.6		U.S. Hypo. Data File	
1974 Feb 16	(swarm)	34.0	93.0	1		1.8		U.S. Hypo. Data File	
1974 Feb 16	(swarm)	34.0	93.1	1		2.3		U.S. Hypo. Data File	
1974 Nov 28	Carlsbad, New Mexico	32.3	104.1	5	(V)	3.9		Person	1976
1974 Dec 13	Tucker-Coy, Ark.	34.7	91.9	5	V	3.4		Person	1976
1974 Dec 16	Oklahoma City, Okla.	35.3	97.5	10	IV	2.3		Person	1976
1975 Sept 9	Bay St. Louis, Miss.	30.4	89.2		III	3		Person	1976
1975 Sept 13	Wilson, Okla.	34.2	97.4		IV	3.4		Person	1976

TABLE 2.5.2-3
EARTHQUAKE ATTENUATION

Tectonic Subprovince	Maximum Earthquake Intensity In Subprovince And Date		Closest Point Of Subprovince To Site (Miles)	Distance of Maximum Earthquake From Site (Miles)	Distance Used for Attenuation Of Earthquake (Miles)	Intensity Used For Attenuation	Intensity At Site	Maximum Ground Acceleration At Site
Wichita Province								
Anadarko Basin	VII	1952	163	213	163	VII	IV +	<0.05
Amarillo Uplift								
Wichita Mtns-								
Arbuckle Uplift	VI	1968	91	141	91	VI	<VI	0.06
Palo Duro Basin-								
Hardeman Basin	VI	1951	153	373	153	VI	<VI	0.06
Matador Arch-								
Red River Uplift								
Meunster Arch	IV	1950	70	91	70	IV	<IV	<0.06
Seminole-Nemana Uplift Province	VII	1952	172	213	172	VII	IV +	<0.05
Ouachita Province								
Exposed Ouachita	VI	1969	162	345	162	VI	<VI	<0.06
Wichita-Ouachita	VII	1882	68	155	70	VII	VI +	0.10
Stable Ouachita	0		40					
West Central Texas Province								
Basin Subprovince	IV	1965	187	266	187	IV	<IV	<0.05
Blano Uplift	0		82					
Central Texas Shelf	0		0					
Balcones Mexia-Talco	IV	1970	50	104	50	IV	<IV	<0.05
Province	V-VI	1932	50	90	50	VI	<VI	0.06
Mesozoic Shelf Province								
East Texas Empayment	0		106					
Mt. Enterprise	VII	1981	128	160	128	VII	V +	0.06
West Subprovince	V	1914	88	190	88	V	<V	<0.05
Sabine Uplift	V	1957	168	186	168	V	<V	<0.05
Gulf Coast Province	VI	1930	183	438	183	VI	<VI	0.06

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 1 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-1	40.0 - 40.5	796.7 - 796.2	18.0[(a)]	Claystone, gray
P-1	69.5 - 70.5	767.2 - 766.2	405.8	Limestone, gray
P-1	90.0 - 90.7	746.7 - 746.0	214.8	Limestone, gray
P-3	39.8 - 40.7	784.9 - 784.0	11.2[(a)]	Claystone, gray
P-3	59.9 - 60.8	764.8 - 763.9	349.0	Limestone, gray
P-3	70.0 - 70.6	754.7 - 754.1	318.5	Limestone, gray
P-4	59.6 - 60.3	789.7 - 789.0	70.0	Limestone, gray, shaly
P-4	80.3 - 80.8	769.0 - 768.5	335.9	Limestone, gray
P-4	100.0 - 100.9	749.3 - 748.4	305.4	Limestone, gray
P-9	7.5 - 8.0	826.2 - 825.7	3.8	Claystone, tan, weathered
P-9	12.9 - 14.0	820.8 - 819.7	69.8	Claystone, tan, weathered
P-9	20.0 - 22.0	813.7 - 811.7	178.9	Limestone, gray
P-9	24.3 - 25.0	809.4 - 808.7	5.0[(a)]	Claystone, black
P-9	28.7 - 29.5	805.0 - 804.2	192.0	Limestone, gray
P-9	33.5 - 34.9	800.2 - 798.8	225.0	Limestone, gray

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 2 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-9	36.1 - 36.5	797.6 - 797.2	121.2	Limestone, gray
P-9	39.0 - 39.5	794.7 - 794.2	9.9[(a)]	Claystone, gray
P-9	42.2 - 42.5	791.5 - 791.2	124.7	Limestone, gray
P-9	50.7 - 51.8	783.0 - 781.9	12.1[(a)]	Claystone, gray
P-9	52.4 - 52.9	781.3 - 780.8	278.4	Limestone, gray
P-9	58.2 - 58.7	775.5 - 775.0	183.2	Limestone, gray
P-9	60.0 - 60.4	773.7 - 773.3	31.4[(a)]	Claystone, gray
P-9	64.6 - 64.5	769.1 - 769.2	88.6	Limestone, gray, shaly
P-9	70.0 - 70.8	763.7 - 762.9	155.4	Limestone, gray
P-9	81.1 - 81.5	752.6 - 752.2	192.0	Limestone, gray
P-9	91.1 - 92.4	742.6 - 741.3	606.0	Limestone, gray
P-9	106.0 - 107.2	727.7 - 726.5	253.0	Limestone, gray
P-9	112.1 - 113.5	721.6 - 720.2	331.6	Limestone, gray
P-9	128.3 - 129.0	705.4 - 704.7	530.1	Limestone, gray
P-9	134.9 - 136.8	698.8 - 696.9	88.1[(a)]	Claystone, dark, gray
P-10	11.2 - 11.7	831.8 - 831.3	20.5	Limestone, tan, weathered

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 3 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-10	14.1 - 15.0	828.9 - 828.0	87.3	Limestone, tan, weathered
P-10	16.6 - 17.3	826.4 - 825.7	325.0	Limestone, gray, (band)
P-10	22.0 - 22.8	821.0 - 820.2	133.1	Claystone, gray, calcareous
P-10	28.2 - 28.6	814.8 - 814.4	582.5	Limestone, white
P-10	31.5 - 33.5	811.5 - 809.5	152.7	Limestone, gray
P-10	43.3 - 44.0	799.7 - 799.0	222.5	Limestone, gray
P-10	45.7 - 46.2	797.3 - 796.8	240.0	Limestone, gray
P-10	48.7 - 49.3	794.4 - 793.8	15.0[(a)]	Claystone, gray
P-10	54.6 - 55.3	788.5 - 787.8	121.2	Limestone, gray
P-10	58.5 - 59.0	784.6 - 784.1	21.5[(a)]	Claystone, gray
P-10	60.9 - 62.0	782.2 - 781.1	7.1[(a)]	Claystone, gray
P-10	63.1 - 63.5	780.0 - 779.6	15.2[(a)]	Claystone, gray
P-10	70.0 - 70.8	773.1 - 772.3	48.5[(a)]	Claystone, gray (band)
P-10	70.8 - 71.6	772.3 - 771.5	139.6	Limestone, gray
P-10	74.6 - 76.0	768.5 - 767.1	329.2	Limestone, gray

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 4 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-10	77.3 - 78.0	765.8 - 765.1	220.2	Limestone, gray
P-10	81.3 - 82.0	761.8 - 761.1	270.5	Limestone, gray
P-10	88.8 - 89.7	754.3 - 753.4	200.7	Limestone, gray
P-10	97.6 - 99.4	745.5 - 743.7	235.6	Limestone, gray
P-10	107.0 - 108.1	736.1 - 735.0	445.0	Limestone, gray
P-10	115.5 - 116.1	727.6 - 727.0	196.3	Limestone, gray
P-10	143.6 - 144.7	699.5 - 698.4	261.8	Limestone, gray
P-11	20.5 - 21.2	827.5 - 826.8	149.8	Limestone, gray
P-11	26.2 - 27.0	821.8 - 821.0	69.7	Limestone, gray, shaly
P-11	30.0 - 31.4	818 - 816.6	198.5	Limestone, gray
P-11	36.3 - 36.7	811.7 - 811.3	147.3	Limestone, gray
P-11	47.0 - 48.0	801.0 - 800.0	261.8	Limestone, gray
P-11	54.3 - 54.8	793.7 - 793.2	381.3	Limestone, gray
P-11	61.2 - 61.6	786.8 - 786.4	57.9[(a)]	Claystone, gray
P-11	72.3 - 73.2	775.7 - 774.8	27.1[(a)]	Claystone, gray
P-11	73.9 - 74.5	774.1 - 773.5	161.4	Limestone, gray

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 5 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-11	82.7 - 83.7	765.3 - 764.3	266.1	Limestone, gray
P-11	91.0 - 92.0	757.0 - 756.0	161.4	Limestone, gray
P-13	13.5 - 14.1	837.3 - 836.7	6.3	Claystone, tan, weathered
P-13	24.0 - 25.5	826.8 - 825.3	48.5	Claystone, gray
P-13	28.5 - 29.1	822.3 - 821.7	39.0	Claystone, gray
P-13	39.5 - 40.1	811.3 - 810.7	327.2	Limestone, gray
P-13	50.1 - 51.0	800.7 - 799.8	301.0	Limestone, gray
P-14	12.6 - 13.2	837.8 - 837.2	10.9	Limestone, tan, weathered
P-14	30.6 - 31.1	819.8 - 819.3	698.1	Limestone, gray
P-14	34.7 - 35.2	815.7 - 815.2	130.9	Limestone, gray
P-14	54.9 - 55.7	795.5 - 794.7	15.7[(a)]	Claystone, gray
P-14	68.0 - 68.7	782.4 - 781.7	17.4[(a)]	Claystone, gray
P-14	75.7 - 77.1	774.7 - 773.3	114.6	Limestone, gray (band)
P-14	83.3 - 83.9	767.1 - 766.5	240.0	Limestone, gray
P-16	8.2 - 9.0	833.3 - 832.5	15.2	Claystone, tan, weathered
P-16	15.2 - 15.9	826.3 - 825.6	89.0	Limestone, gray, shaly

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 6 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-16	25.2 - 26.0	816.3 - 815.5	179.9	Limestone, gray
P-16	39.2 - 40.0	802.3 - 801.5	117.8	Limestone, gray
P-16	44.6 - 45.1	796.9 - 796.4	20.6[(a)]	Claystone, dark gray
P-16	56.7 - 57.6	784.8 - 783.9	9.2[(a)]	Claystone, dark gray
P-16	64.5 - 65.6	777.0 - 775.9	15.6[(a)]	Claystone, dark gray
P-16	74.5 - 75.0	767.0 - 766.5	200.7	Limestone, gray
P-17	6.1 - 7.5	830.9 - 829.5	71.2	Limestone, tan, weathered
P-17	24.3 - 24.8	812.7 - 812.2	220.3	Limestone, gray
P-17	37.9 - 38.4	799.1 - 798.6	353.4	Limestone, gray
P-17	45.3 - 45.9	791.7 - 791.1	123.4	Limestone, gray
P-17	48.4 - 48.9	788.6 - 788.1	186.9	Limestone, gray
P-17	64.6 - 66.0	772.4 - 771.0	5.3[(a)]	Claystone, dark gray
P-17	73.6 - 74.6	763.4 - 762.4	280.0	Limestone, gray
P-25	43.0 - 43.8	807.7 - 806.9	112.0[(a)]	Claystone
P-25	49.3 - 50.0	801.4 - 800.7	183.2[(a)]	Claystone

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 7 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-25	56.0 - 56.9	794.7 - 793.8	77.1[(a)]	Claystone
P-26	42.2 - 42.8	809.0 - 808.4	5.5[(a)]	Claystone
P-26	69.5 - 70.4	781.7 - 780.8	21.2[(a)]	Claystone
P-26	70.4 - 71.3	780.8 - 779.9	21.2[(a)]	Claystone
P-26	75.0 - 76.0	776.2 - 775.2	21.6[(a)]	Claystone
P-27	34.4 - 35.6	809.7 - 808.5	2.1[(a)]	Claystone
P-27	37.4 - 39.4	806.7 - 804.7	54.6[(a)]	Claystone
P-27	49.0 - 49.9	795.1 - 794.2	23.5[(a)]	Claystone
P-27	59.0 - 60.5	785.1 - 783.6	9.6[(a)]	Claystone
P-27	64.3 - 65.6	779.8 - 778.5	15.6[(a)]	Claystone
P-27	69.0 - 69.7	775.1 - 774.4	23.8[(a)]	Claystone
P-27	71.7 - 72.6	772.4 - 771.5	12.4[(a)]	Claystone
P-28	31.0 - 31.5	813.1 - 812.6	73.5	Claystone
P-28	57.0 - 58.3	787.1 - 785.8	19.6[(a)]	Claystone

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 8 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-28	62.5 - 63.0	781.6 - 781.1	13.0[a]	Claystone
P-28	69.3 - 70.5	774.8 - 773.6	33.3[a]	Claystone
P-29	25.2 - 26.8	810.2 - 808.6	53.4[a]	Claystone
P-29	46.0 - 46.8	789.4 - 788.6	36.7[a]	Claystone
P-29	49.1 - 49.9	786.3 - 785.5	15.3[a]	Claystone
P-29	53.5 - 54.9	781.9 - 780.5	17.9[a]	Claystone
P-29	60.7 - 61.4	774.7 - 774.0	15.5[a]	Claystone
P-30	41.8 - 43.3	810.7 - 809.2	46.3	Claystone
P-30	52.1 - 53.0	800.4 - 799.5	14.6[a]	Claystone
P-30	55.0 - 56.0	797.5 - 796.5	10.4[a]	Claystone
P-30	63.9 - 64.8	788.6 - 787.7	31.4[a]	Claystone
P-30	74.1 - 76.0	778.4 - 776.5	35.9[a]	Claystone
P-31	0 - 1.5	832.4 - 830.9	1.7	Sandy clay
P-31	5.5 - 6.3	826.9 - 826.1	49.4	Limestone, weathered

TABLE 2.5.4-1
SUMMARY OF UNCONFINED COMPRESSION TESTS

(Sheet 9 of 9)

BORING NUMBER	DEPTH FT.	ELEVATION FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-31	11.9 - 13.3	820.5 - 819.1	41.3	Claystone
P-31	16.2 - 17.0	816.2 - 815.4	167.8	Limestone with claystone
P-31	19.2 - 20.0	813.2 - 812.4	113.3	Limestone with claystone
P-31	22.8 - 23.9	809.6 - 808.5	4.9[a]	Claystone
P-31	30.1 - 30.7	802.3 - 801.7	104.6	Limestone with claystone
P-31	43.2 - 44.5	789.2 - 787.9	85.7	Limestone with claystone
P-33	8.3 - 10.0	797.9 - 796.2	166.9	Limestone with claystone
P-33	13.5 - 14.0	792.7 - 792.2	15.9[a]	Claystone with limestone
P-33	18.3 - 19.0	787.9 - 787.2	121.6	Limestone
P-33	37.5 - 38.3	768.7 - 767.9	72.9	Limestone with claystone
P-33	45.8 - 46.7	760.4 - 759.5	162.2	Limestone with claystone
P-33	48.3 - 49.2	757.9 - 757.0	176.6	Limestone with claystone
P-33	51.2 - 52.5	755.0 - 753.7	232.7	Limestone with claystone
P-33	58.6 - 59.4	747.6 - 746.8	149.2	Limestone with claystone

a) Denotes claystone below elevation 810

TABLE 2.5.4-2
SUMMARY OF DOUBLE RING SHEAR TESTS

BORING NUMBER	DEPTH FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-9	36.5-36.8	22.9	Limestone, gray
P-9	42.5-42.8	26.4	Limestone, gray
P-9	60.4-60.9	2.9	Claystone, gray
P-9	81.5-82.0	12.8	Limestone, gray
P-10	17.3-17.9	43.6	Limestone, gray, (band)
P-10	44.0-44.6	22.0	Limestone, gray
P-10	49.3-49.8	3.4	Claystone, gray
P-10	59.0-59.4	4.5	Claystone, gray
P-11	36.7-37.1	31.6	Limestone, gray
P-11	61.6-62.0	1.4	Claystone, gray
P-11	74.5-75.0	12.8	Limestone, gray
P-14	31.1-31-7	61.1	Limestone, gray
P-16	45.1-45.6	3.1	Claystone, dark gray
P-17	24.8-25.3	44.0	Limestone, gray
P-17	49.2-49.8	11.4	Limestone, gray

TABLE 2.5.4-3
SUMMARY OF BRAZIL TENSILE TESTS

BORING NUMBER	DEPTH FT.	ULTIMATE STRENGTH TSF	DESCRIPTION
P-9	52.9-53.4	53.8	Limestone, gray
P-9	58.7-59.3	22.4	Limestone, gray
P-9	70.8-71.6	51.6	Limestone, gray
P-10	28.6-29.0	76.5	Limestone, white
P-10	46.2-46.7	34.4	Limestone, gray
P-10	55.3-56.1	16.8	Limestone, gray
P-11	27.0-27.7	11.7	Limestone, gray with claystone
P-11	54.8-55.3	53.8	Limestone, gray
P-14	35.2-35.9	23.0	Limestone, gray
P-14	83.9-84.5	16.5	Limestone, gray
P-17	38.4-38.8	58.7	Limestone, gray

TABLE 2.5.4-4
STATIC ROCK PROPERTIES

LOCATION	DEPTH[(a)] FT.	FOUNDATION MATERIAL	UNIT WEIGHT PCF	ELASTIC MODULUS PSI	SHEAR MODULU PSI	POISSON'S RATIO
Containment Structure (Boring P-9)	32.6-33.5	Hard Limestone	156.0	$4.0X10^6$	$2.09X10^6$	0.17
Containment Structure (Boring P-9)	60.9-62.7	Claystone	135.5	$5.6X10^4$	$2.18X10^4$	0.282
Containment Structure Boring P-10)	80.0-81.3	Hard Limestone	152.9	$1.6X10^6$	$6.6X10^5$	0.218
Safeguards Bldg. (Boring P-11)	73.0-75.0	Claystone	-	$8.9X10^4$	$3.48X10^4$	0.278
Safeguards Bldg. (Boring P-11)	81.3-81.8	Limestone with Claystone	-	$1.3X10^6$	$5.2X10^5$	0.248

a) Depth from existing ground surface.

TABLE 2.5.4-5
INITIALLY SELECTED VALUES OF PREEXCAVATION-DYNAMIC FOUNDATION DESIGN PARAMETERS

Foundation Material	Compressional Wave Velocity, Vc (ft/sec)	Shear Wave Velocity (ft/sec)	Wave Density* (lbs/cu ft)	Insitu Wet Poisson's Ratio (ν)	Shear Modulus (G) or Modulus of Rigidity (lbs/sq in)
Weathered Rock	2,600	1,000	144	.35	4.5×10^4
Moderately Weathered to Fresh Rock (Glen Rose Limestone above approximately elevation 770)	9,500 to 11,000	5,500 to 6,000	150	.30	8.0×10^5
Underlying Massive Fresh Rock (Glen Rose Limestone below approximately elevation 770)	11,000 to 12,500	6,000 to 6,500	155	.27	1.2×10^6
Underlying Twin Mountains Rock	7,000 to 8,000	3,200	135	.32	3.0×10^5

Young's Modulus, $E = 2(1 + \nu) G$.

$$\text{Mass Density} = \rho = \frac{\text{Insitu Density}}{32.2}$$

$$E = V_C^2 \frac{(1+\nu)(1-2\nu)}{(1-\nu)}$$

TABLE 2.5.4-5A
DETAILS OF PREEXCAVATION CROSSHOLE SURVEYS

Plant Site

Shot Station	Shot Station Ground Elevation	Shot Elevation	Distance From Boring P-1	Distance From Boring P-2	Geophone Elevations Boring P-1	Geophone Elevations Boring P-2
A	848	841	870	1175	693, 718, 743 768, 793, 818	743, 768, 793 818
B	846	839	370	675	Same as A	Same as A

Safe Shutdown Impoundment Dam Site

Shot Station	Shot Station Ground Elevation	Shot Elevation	Distance From Boring M-5	Distance From Boring M-6	Geophone Elevations Boring M-5	Geophone Elevations Boring M-6
A	775	748	400	700	650, 675, 700 725	650, 675, 700 725

NOTE: All elevations and distances are in feet

Boring	Surface Elevation
P-1	837
P-2	827
M-5	738
M-6	730

TABLE 2.5.4-5B
REPRESENTATIVE GEOPHYSICAL DATA[a] FROM PREEXCAVATION SURVEYS AT STTION LOCATION

Approx. Depth (Feet)	Materials Description	Compressional Wave Velocity		Shear Wave Velocity		"Dynamic" Poisson's Ratio[b]
		Ft/Sec	Method	Ft/Sec	Method	
0 to 12	Clayey, sandy silt (soil derived by rock weathering) and weathered limestone	2,700 2,500	Refraction Surface Wave	1,000-1,100	Surface Wave	0.42
12 to 200 (bottom of boring)	Light to dark gray argillaceous limestone (Glen Rose Formation)	12,500 9,600 11,500 11,200	Refraction Uphole Crosshole Surface Wave	5,500 5,600-6,200	Surface Wave Crosshole	0.35 to 0.38

a) Data from uphole and crosshole surveys at borings P-1 and P-2 and refraction survey in vicinity.

b) Values are derived by computation using compressional and shear wave data.

TABLE 2.5.4-5C
REPRESENTATIVE GEOPHYSICAL DATA[a] FROM SAFE SHUTDOWN IMPOUNDMENT DAM LOCATION

Approximate Depth (Feet)	Materials Description	Compressional Wave Velocity		Shear Wave Velocity		"Dynamic" Poisson's Ratio[b]
		Ft/Sec	Method	Ft/Sec	Method	
0 to 10	Clayey, Sandy silt and weathered limestone	2,000	Refraction	650	Surface Wave	0.43
10 to 17	Weathered to harder limestone	5,000	Surface Wave	1,200-1,300	Surface Wave	0.46
17 to 100 (bottom of boring)	Light to dark gray argillaceous limestone (Glen Rose Formation)	11,800 10,600 11,000 11,500-12,000	Refraction Uphole Crosshole Surface Wave	5,300 4,000 4,200	Surface Wave Crosshole Uphole Shear[c]	0.40 to 0.43

a) Data from uphole and crosshole surveys at boring M-5 and refraction survey at Safe Shutdown Impoundment data location.

b) Values are derived by computation using compression and shear wave data.

c) Obtained from primacord technique described in Section 2.7.2.6.4.

TABLE 2.5.4-5D
AMBIENT GROUND MOTION MEASUREMENTS

Location (Boring)	Frequency (Hertz)	Gain x100	Ground Motion in Inches x 10^{-3}		
			Radial	Vertical	Transverse
P-8	33-1/3	Dx20	(a)	(a)	.004
	50, 66-2/3	Ax5	.040	(a)	.080
	50, 100	Vx5	.040	.020	.060
P-1	100	Dx20	(a)	(a)	.0005
	40, 50, 66-2/3	Ax20	.040	.025	.008
	50, 100	Vx20	.025	.010	.025
M-5	Not Observable	Dx5	(a)	(a)	(a)
	33-1/3, 40 66-2/3	Ax5	.050	.033	.050
	30, 33-1/3, 100	Vx5	.035	.010	.025

Key:

D=Displacement (inches)
V=Velocity (inches/second)
A=Acceleration (inches/second/second)

a) Amplitude of record trace too small to distinguish magnitude

TABLE 2.5.4-5E
LITHO-STRATIGRAPHY OF SUBSURFACE MATERIALS TO A DEPTH OF 500 FT

Geologic Characterization					
Geologic Units			Superpositional Boundaries		
Zone	Formation	Prevalent Lithologies	Depth Range (Ft)	Elevation Range	Thickness
I Upper	Glen Rose	Hard limestone with soft claystone interbeds	0 - 23	810 - 787	23
I Lower	Glen Rose	Soft claystone alternating with hard limestone	23 - 39	787 - 771	16
II	Glen Rose	Massive nodular limestone with occasional claystone partings	39 - 102	771 - 708	63
IIIA	Glen Rose	Alternating limestone and claystone	102 - 110	708 - 700	8
IIIB	Glen Rose	Massive limestone	110 - 163	700 - 647	53
IV	Glen Rose	Alternating limestone and claystone with sand lenses	163 - 200	647 - 610	37
Upper	Twin Mountains	Soft-to-hard sandstone with interbedded medium-to-hard claystone	200 - 315	610 - 495	115
Middle	Twin Mountains	Hard claystone	315 - 352	495 - 458	37
Lower	Twin Mountains	Alternating hard sandstone and claystone	352 - 444	458 - 366	92
---	Mineral Wells	Massive and hard claystone	444 - 504[(a)]	366 - 306[(a)]	60[(a)]

a) Maximum depth of borings.

TABLE 2.5.4-5F
SUMMARY OF THE RESULTS OF THE POSTCONSTRUCTION CROSSHOLE SURVEY

Formation	Depth (ft)	Elevation (ft)	Compressional Wave (ft/sec)		Shear Wave (ft/sec)		Poisson's Ratio
			No. of Tests	Mean	No. of Tests	Mean	
GLEN ROSE	0 - 194	809 - 615					
- Limestone (upper)	0 - 36	809 - 773	26	8,631	30	3901	0.37
- Claystone	19 - 36	790 - 773	16	5,766	31	2380	0.40
	99 - 105	710 - 704					
- Limestone	36 - 99	773 - 710	137	10,802	161	5066	0.36
	105 - 194	704 - 615					
TWIN MOUNTAINS	194 - 443	615 - 366					
- Sandstone	varies	varies	71	7,495	67	3073	0.40
- Claystone	varies	varies	38	7,861	38	2884	0.42
- Limestone	211 - 217	598 - 592	4	10,904	4	5210	0.35
- Sandstone/Claystone	varies	varies	23	10,452	24	4691	0.37
MINERAL WELLS	443 - 503+	366 - 306	20	10,162	19	4339	0.39

TABLE 2.5.4-5G
DYNAMIC ROCK PROPERTIES DETERMINED BY LABORATORY TESTING OF CORE SAMPLES

		Unit Weight (PCF)		Shear Modulus (KSI) and Damping Ratio (%)		
Lithology	Formation	No. of Tests	Mean	No. of Tests	Mean Shear Modulus	Mean Damping Ratio
	Glen Rose	17	137.8	4	69	4.4
Claystone	Twin Mountains	11	137.9	2	110	5.0
	All	28	137.9	6	83	4.6
	Glen Rose	12	154.4	4	727	1.9
Limestone	Twin Mountains	3	156.4	1	729	2.3
	All	15	154.8	5	727	2.0
	Glen Rose	1	144.4	-	-	-
Sandstone	Twin Mountains	20	135.0	6	166	4.7
	All	21	135.4	6	166	4.7
	Total	64	Total	17		

NOTE: Samples were obtained from S/L cluster borings S-1, S-1A, L-1. L-2 core samples were not tested.

TABLE 2.5.4-5H
DYNAMIC FOUNDATION DESIGN PARAMETERS BASED ON PREEXCAVATION AND POSTCONSTRUCTION GEOPHYSICAL SURVEYS

Zone Description			Seismic Velocities (Ft/Sec)		Unit Weight (PCF)	Shear Modulus[a] (KSI)	Poisson's Ratio
Depth (ft)	Elevation (ft)	Thickness (ft)	Compressional Vp	Shear Vs			
0[b]	810	23[c]	9400	4500	150	650	0.35
23	787	16	5600	2700	145	230	0.35
39	771	161	11400	5500	155	1000	0.35
200	610	160	6400	3100	135	280	0.35
360	450		9400	4500	150	650	0.35

a) Shear Modulus = Mass Density $(Vs)^2$

b) Plant Grade

c) Some boring logs indicate substantial claystone strata within this zone. Such strata are treated as subzones and properties of the zone extending from elevation 787 to elevation 771 are applied.

TABLE 2.5.4-6
SUMMARY OF STATIC BEARING CAPACITY ANALYSIS OF CATEGORY I STRUCTURES

Structure	Founding Elevation Ft.	Maximum Pressure Ksf	Ultimate Bearing Capacity Ksf	Factor of Safety
Reactor Containments	795.5/769.2	12.2	60.0	4.9
Safeguard Structures	805/767.5/781.75	5.7	60.0	10.5
Fuel Building	805.3/795.2/780.5	10.7	60.0	5.6
Auxiliary Building	785/773	6.9	60.0	8.7
Service Water Intake Structure	749	12.1	60.0	5.0
Condensate Storage Tank	805.5	5.1	60.0	11.7
Refueling Water Storage Tank	805.5	5.1	60.0	11.7
Reactor Makeup Water Storage Tank	806.6	4.7	60.0	12.7

TABLE 2.5.4-6A
SUMMARY OF DYNAMIC BEARING CAPACITY ANALYSIS OF CATEGORY I STRUCTURES

Structure	Founding Elevation Ft.	Maximum Pressure Ksf	Ultimate Bearing Capacity Ksf	Factor of Safety
Reactor Containments	795.5/769.2	24.9	60.0	2.4
Safeguard Structures	805/767.5/781.75	9.6	60.0	6.3
Fuel Building	805.3/795.2/780.5	13.9	60.0	4.3
Auxiliary Building	785/773	13.4	60.0	4.5
Service Water Intake Structure	749	18.1	60.0	3.3
Condensate Storage Tank	805.5	11.9	60.0	5.0
Refueling Water Storage Tank	805.5	11.9	60.0	5.0
Reactor Makeup Water Storage Tank	806.6	11.0	60.0	5.4

TABLE 2.5.4-7
SUMMARY OF SETTLEMENT ANALYSIS FOR CATEGORY I STRUCTURES

Building	Maximum Settlement (in.) Center	Maximum Settlement (in.) Edge	Differential Settlement (in.)
Containment Structures (2)	0.26	0.16	0.10
Safeguard Buildings (2)	0.12	0.04	0.08
Fuel Building	0.18	0.09	0.09
Auxiliary Building	0.18	0.09	0.09
Service Water Intake Structure	0.08	0.04	0.04
Condensate Storage Tank	0.04	0.02	0.02
Refueling Water Storage Tank	0.04	0.02	0.02
Reactor Makeup Water Storage Tank	0.02	0.01	0.01

TABLE 2.5.4-8
SOLUBILITY ANALYSES RESULTS

Boring	Depth	Percent Soluble In Distilled Water	Percent Residue After Treating With 5% Acetic Acid
P- 9 (Glen Rose)	124	2.7	13.6
P-10 (Glen Rose)	114	1.0	4.4
P-10A (Twin Mountains)	70.8	0.7	90.3
P-10A (Twin Mountains)	141.1	0.5	72.4
P-10A (Twin Mountains)	223.5	1.4	20.7
P-10A (Twin Mountains)	242.3	1.3	98.1
P-10A (Twin Mountains)	264	1.1	98.5
P-10A (Twin Mountains)	282.2	0.0	98.5

TABLE 2.5.4-9
RESULTS OF X-RAY DIFFRACTOMETER ANALYSIS

Sample	Depth	Results
P-9 (Glen Rose)	124	Calcite (somewhat impure), minor quartz
P-10 (Glen Rose)	114	Calcite (somewhat impure)
P-10A (Glen Rose)	70.8	Quartz, illite, minor dolomite
P-10A (Glen Rose)	141.1	Calcite, dolomite, quartz
P-10A (Twin Mountains)	223.5	Calcite, dolomite, minor quartz
P-10A (Twin Mountains)	242.3	Quartz, minor calcite
P-10A (Twin Mountains)	264.0	Quartz, minor calcite
P-10A (Twin Mountains)	282.2	Quartz, minor calcite
P-14 (Glen Rose)	64	Quartz, illite, minor dolmite
P-16 (Glen Rose)	116	Calcite, illite, dolmite, minor quartz

TABLE 2.5.4-10
RESULTS OF CHEMICAL ANALYSES

Boring	Depth	Percentage of Metal Oxides[a] CaO	MgO	FeO[b]	MnO	SrO	S_1O_2	Al_2O_3	Na_2O	K_2O
P-9	124	47.2	1.47	0.52	0.02	0.04	7.57	1.71	0.13	0.37
P-10	114	52.1	0.72	0.64	0.04	0.05	2.35	0.47	0.08	0.15
P-10A	70.8	3.24	3.73	6.4	0.04	[c]	55.86	16.1	0.58	0.38
P-10A	141.1	14.76	4.33	3.75	0.03	[c]	47.07	11.81	0.40	2.64
P-10A	223.5	42.77	1.58	0.83	0.05	[c]	12.82	3.98	0.11	0.60
P-10A	242.3	0.84	0.16	1.48	0.01	[c]	91.33	1.43	0.18	0.78
P-10A	264	0.68	0.13	0.41	0.01	[c]	94.13	0.75	0.15	0.58
P-10A	282.2	0.60	1.04	1.17	0.01	[c]	84.66	6.21	0.48	1.67
P-14	64	1.33	2.18	4.38	0.06	[c]	68.13	13.96	0.69	3.35
P-16	116	43.85	1.42	2.4	0.03	[c]	10.83	3.35	0.15	0.52

a) Remainder of each sample is mainly CO_2, H_2O and organic carbon.

b) Total Fe as FeO.

c) Sample not tested for SrO.

TABLE 2.5.4-11
CYCLIC SHEAR STRENGTH CRITERIA

Normal Effective Stress $\bar{\sigma}_o$ (Kips/ft^2)	Cyclic Shear Stress to 0.05 Strain $\sigma_a/2$ (Kips/ft^2)		Cyclic Stress Ratio to 0.05 Strain $\sigma_a/2\ \bar{\sigma}_o$	
	5 cycles	10 cycles	5 cycles	10 cycles
0	0	0	0	0
2	1.20	1.05	0.600	0.525
4	2.10	1.85	0.525	0.463
6	2.90	2.60	0.483	0.433
8	3.65	3.25	0.456	0.406

This table was developed based on Figure 2.5.6-48.

See Figure 2.5A-17 for Filter “A” cyclic shear strength criteria.

TABLE 2.5.4-12
CYCLIC TRIAXIAL TEST PROGRAM SUMMARY OF DENSITIES

Material	Specific Gravity	γ_d min (pcf)	γ_d max (pcf)	Density Criteria	Required γ_d test (pcf)	Average γ_d test (pcf)
Filter A (Sample #1)	2.65	96.8[(a)]	127.8[(a)]	80% relative density	120.1	120.6
Fine Aggregate (Sample #2)	2.62	95.6[(a)]	121.6[(a)]	80% relative density	115.3	115.7
Glen Rose Crushed Stone (Sample #3)	2.51		123.6[(b)]	95% of standard Proctor maximum density	117.4	118.6
Crowder Quarry Crushed Stone (Sample #4)	2.60		134.6[(b)]	95% of standard Proctor maximum density	127.9	128.6

a) Relative density test ASTM D2049

b) Maximum density AASHTO T99, Method D

TABLE 2.5.4-13
SUMMARY OF CYCLIC STRENGTH VALUES

0.05 Double Amplitude Axial Strain Failure Criteria

Soil Type	Confining Pressure	Density Required	Density Tested	Max Shear Stress Required	Max Shear Stress Tested	Double Amplitude Axial Strain Developed in/in		Dynamic Criteria Achieved 0.05 D. A. Axial Strain	
	psf	pcf	pcf	psf	psf	5 cycles	10 cycles	5 cycles	10 cycles
Sample #1									
Filter “A”	2000	120.1	120.6	1200	1260	0.0222	0.0953	Yes	No
	4000		120.5	2100	2120	0.0167	0.0657	Yes	No
	6000		120.6	2900	2940	0.0271	0.1138	Yes	No
	8000		120.6	3650	3680	0.0294	0.105	Yes	No
Sample #2									
C-33, Fine	2000	115.3	114.6	1200	1220	0.0131	0.0225	Yes	Yes
Aggregate	4000		115.7	2100	2140	0.0121	0.0199	Yes	Yes
	6000		116.1	2900	2970	0.0108	0.0173	Yes	yes
	8000		116.4	3650	3700	0.0131	0.0216	Yes	Yes
Sample #3									
Glen Rose	2000	117.4	118.1	1200	1230	0.0084	0.0127	Yes	Yes
Crushed	4000		118.6	2100	2160	0.0117	0.0237	Yes	Yes
Stone	6000		118.7	2900	2980	0.0182	>.20	Yes	No
	8000		119.1	3650	3740	0.0270	>.20	Yes	No
Sample #4									
Crowder	2000	127.9	127.9	1200	1220	0.0050	0.0062	Yes	Yes
Quarry	4000		127.9	2100	2140	0.0050	0.0068	Yes	Yes
Crushed	6000		129.7	2900	2960	0.0060	0.0081	Yes	Yes
Stone	8000		129.0	3650	3720	0.0070	0.0084	Yes	Yes

TABLE 2.5.4-13A
SUMMARY OF ADDITIONAL CYCLIC SHEAR STRENGTH TESTS

SOIL TYPE	CONFINING PRESSURE (psf)	DENSITY TESTED (pcf)	STRESS RATIO	DOUBLE AMPLITUDE AXIAL STRAIN DEVELOPED (IN/IN)		DYNAMIC CRITERIA ACHIEVED 0.05 D. A. AXIAL STRAIN	
				5 CYCLES	10 CYCLES	5 CYCLES	10 CYCLES
Sample #1	2000	119.8	0.51	.013	.073	Yes	No
Filter “A”	2000	119.3	0.36	.004	.012	Yes	Yes
	4000	120.1	0.46	.016	.083	Yes	No
	6000	120.0	0.43	.031	.127	Yes	No
	8000	120.8	0.41	.062	.211	No	No
Sample #3	6000	118.3	0.43	.017	.141	Yes	No
Glen Rose	8000	117.8	0.40	.045	>.20	Yes	No
Crushed							
Store							

TABLE 2.5.6-1
PACKER TEST DATA

(Sheet 1 of 2)

Boring	Depth Interval Tested (Feet)	Gage Pressure (Psi)	Time Interval (Minutes)	Water Loss (Gallons)
P-1	20 - 200	20	5	-0-
P-1	40 - 200	40	5	-0-
P-4	16 - 200	25	5	-0-
P-8	20 - 200	40	5	-0-
P-8	10' intervals	40	5	0.03 in one 10' zone
P-9	20 - 200 in 10' intervals	25 - 70	5	-0-
P-10	18 - 229 in 10' intervals	18 - 80	2 - 5	2.6 @ 204'-214' 0.6 @ 194'-204'
P-10	220 - 294	72	5	-0-
P-10	215 - 294	35	5	-0-
P-11	27 - 198	25	5	-0-
P-12	18 - 198	30	5	-0-
P-13	28 - 200	25	5	-0-
P-14	28 - 200	25	5	-0-
P-15	20 - 188	25	5	-0-
P-16	20 - 192	20	5	-0-
P-17	21 - 187	25	4	-0-
P-18	23 - 150	30	5	-0-
P-19	25 - 150	25	5	-0-
P-20	16 - 150	25	5	-0-
M-1	26 - 150	25	5	-0-
M-2	13 - 73	45	5	-0-

TABLE 2.5.6-1
PACKER TEST DATA

(Sheet 2 of 2)

Boring	Depth Interval Tested (Feet)	Gage Pressure (Psi)	Time Interval (Minutes)	Water Loss (Gallons)
M-3	24 - 95	45	5	-0-
M-4	24 - 168	25	5	-0-
M-5	14 - 100	25	5	-0-
M-6	14 - 100	25	5	-0-
MS-7	20 - 75	45	5	-0-
MS-8	21 - 75	25	5	-0-
DI-1	14 - 165	25	5	-0-
DI-2	19 - 165	25	5	-0-
DI-3	14 - 200	25	5	-0-
DI-4	12 - 50	25	5	-0-
DI-5	22 - 69	25	5	-0-
DI-6	21 - 100	25	5	-0-
DI-7	36 - 75	25	5	-0-
DI-8	56 - 130	25	5	-0-
DI-9	28 - 220	25	5	-0-
DI-10	18 - 160	25	5	-0-
DI-8A	20 - 150	25	5	-0-

TABLE 2.5.6-2
TABULATION OF SELECTED ROCK-FILL DAMS STATE OF CALIFORNIA

NAME	DATE COMP.	HEIGHT FT.	SLOPE U.S.	SLOPE D.S.
Salt Springs	1931	315	1.3:1	1.4:1
New Exchequer	1967	479	1.4:1	1.4:1
Morena	1895	177	0.9:1	1.4:1
Lower Hell Hole	1966	410	1.4:1 2.5:1	angle of repose
Hillside	1910	81	0.8:1	1.3:1
Cogswell[a]	1935	265	1.2:1 1.4:1	1.4:1 1.7:1
SSI DAM		70	2.5:1	2.5:1
Proposed				

a) (apx. 100 yds. from San Gabriel Fault)

This data was furnished by personal communication from Mr. Wilferd W. Peak, Chief Engineering Geologist, California Division of Safety of Dams, 1416 Ninth Street, Sacramento, California 95814.

TABLE 2.5.6-3
TEST EMBANKMENT SECTION LARGE SCALE FIELD DENSITY TESTS

Test No.	Moist Weight Lb.	Volume Ft.	Moist Unit Weight Pcf		Dry Unit Weight Pcf After 2 passes	Saturated[a] Unit Weight Pcf After 2 passes
			5 passes	2 passes		
1	2125.3	17.16	123.8	120.95	116.2	132.9
2	2386.7	19.28	123.8	120.95	116.2	132.9

a) These values were computed using specific gravity of 2.54.

TABLE 2.5.6-4
SPECTRAL VELOCITIES AS A FUNCTION OF FREQUENCY

Frequency	Horizontal	Vertical
w (cps)	Sv (in/sec)	Sv (in/sec)
3	6.6	5.2
4	4.7	4.8
5	3.7	3.7
6	3.0	3.0

TABLE 2.5.6-5
AVERAGE LOCAL SAFETY FACTORS

Elevation of Horizontal Plane	Safety Factor
739	3.56
754	2.32
767	1.75
782	1.45

TABLE 2.5.6-6
SAFETY FACTORS FOR WEDGE ANALYSIS

Elevation of Base Of Sliding Wedge	Safety Factors	
	Case II	Case I
725[a]	1.48	1.70
739	2.33	1.51
754	1.20	1.35
767	1.10	1.22
782[b]	1.04	1.04

a) This elevation represents the total height of the SSI Rockfill Dam.

b) This elevation is above the reservoir elevation and both cases are equal.

TABLE 2.5.6-7
SQUAW CREEK DAM

(Sheet 1 of 2)

Piezometer Locations

Range I

Station	Piezometer No.	Tip Offset from	Nominal Tip Elevation	Type
46+00	P-I-1	6.7' U.S.	715	Wellpoint
46+05	P-I-2	7.0' U.S.	735	Wellpoint
46+11	P-I-3	7.0' U.S.	755	Wellpoint
46+15	P-I-4	7.0' U.S.	770	Wellpoint
46+01	P-I-5	30' D.S.	715	Pneumatic
46+06	P-I-6	30' D.S.	735	Pneumatic
46+11	P-I-7	30' D.S.	755	Pneumatic
46+14	P-I-8	30' D.S.	770	Pneumatic
46+02	P-I-9	140.3' D.S.	710	Wellpoint
46+08	P-I-9a	139.9' D.S.	710	Pneumatic
46+12	P-I-10	139.7' D.S.	725	Wellpoint
46+17	P-I-10a	140.4' D.S.	725	Pneumatic (Abandoned)

Range II

Station	Piezometer No.	Tip Offset from	Nominal Tip Elevation	Type
55+95	P-II-OaCB	20.3' U.S.	650	Pneumatic (Abandoned)
56+05	P-II-1	18.4' U.S.	650	Wellpoint (Replaced)
56+10	P-II-2	20.5' U.S	690	Wellpoint
56+16	P-II-2a	20.4' U.S	690	Pneumatic
56+20	P-II-3	19.8' U.S	730	Wellpoint
56+25	P-II-4	21.4' U.S.	770	Wellpoint

TABLE 2.5.6-7
SQUAW CREEK DAM

(Sheet 2 of 2)

Station	Piezometer No.	Tip Offset from	Nominal Tip Elevation	Type
56+02	P-II-5	35.9' D.S.	650	Wellpoint (Replaced)
56+07	P-II-5a	35.6' D.S.	650	Pneumatic
56+12	P-II-6	35.8' D.S.	690	Wellpoint
56+17	P-II-6a	35.8' D.S.	690	Pneumatic
56+22	P-II-7	35.6' D.S.	730	Wellpoint (Replaced)
56+27	P-II-8	35.4' D.S.	770	Wellpoint
56+92	P-II-9	183.5' D.S.	680	Wellpoint (Replaced)
55+96	P-II-9a	183.6' D.S.	680	Pneumatic
56+02	P-II-10	184.4' D.S.	695	Wellpoint
56+06	P-II-10a	185.4' D.S.	695	Pneumatic (Abandoned)

Range III

Station	Piezometer No.	TIp Offset from	Nominal Tip Elevation	Type
69+00	P-III-1	6.2' U.S.	650	Wellpoint (Replaced)
69+05	P-III-2	5.9' U.S.	690	Wellpoint (Abandoned)
69+10	P-III-3	5.9' U.S.	730	Wellpoint
69+15	P-III-4	5.7' U.S.	770	Wellpoint (Abandoned)
69+00	P-III-5	45' D.S.	650	Pneumatic
69+05	P-III-6	45' D.D.	690	Pneumatic
69+10	P-III-7	45' D.S.	730	Pneumatic
69+15	P-III-8	45' D.S.	770	Pneumatic
69+03	P-III-9	284.6' D.S.	652	Wellpoint
69+09	P-III-9a	284.4' D.S.	652	Pneumatic (Abandoned)
69+10	P-III-10	283.8' D.S.	667	Wellpoint
69+20	P-III-10a	285.5' D.S.	667	Pneumatic (Abandoned)

TABLE 2.5.6-8
TEST EMBANKMENT SECTION LARGE SCALE FIELD DENSITY TESTS

Test No.	Moist Weight Lb.	Volume $Ft.^3$	Moist Unit Weight Pcf		Dry Unit Weight Pcf		Saturated[(a)] Unit Weight Pcf	
			5 passes	2 passes	5 passes	2 passes	5 passes	2 passes
1	2125.3	17.16	123.8	121.0	119.0	116.2	134.5	132.9
2	2386.7	19.28	123.8	121.0	119.0	116.2	134.5	132.9

a) These values were computed using specific gravity of 2.54.

APPENDIX 2.5A - CONSTRUCTION RECORDS AND DESIGN VERIFICATION OF SSI DAM

2.5A.1 DESIGN VERIFICATION AT FIFTY PERCENT (50%) CONSTRUCTION COMPLETION

In order to verify that the SSI Dam was constructed according to design criteria, two borings were drilled through the impervious core when the embankment was approximately 50% complete. The location of the borings were as follows:

Boring No.	Station	Ground Surface Elevation
SSI-1	6+00	756.4
SSI-2	10+00	756.3

Continuous samples of the impervious core were taken utilizing a Shelby-tube sampler. A detailed log of each boring is shown on the attached Figures 2.5A-1 & -2.

The following laboratory tests were conducted on samples obtained from the impervious core: unit dry weight determinations, Atterberg limit determinations and hand vane shear tests. The results of these tests are summarized on Table 2.5A-1, -2 for Borings SSI-1 & 2, respectively. The results of the Atterberg limit tests confirmed that the impervious core met the liquid limit specification requirement. The hand vane shear tests were conducted as a relative indicator of the uniformity of the layer compaction. For each specimen tested, a vane shear reading was made in both the top and bottom of the sample. Differences in the readings are interpreted to represent the relative variation in the compaction of the embankment. Only slight variations were noted in the vane shear readings.

A series of hydrometer and sieve analyses were conducted on representative samples and are shown on Figure 2.5A-3. All the tested specimens are classified as CL material.

The strength parameters of the impervious core were evaluated by a series of unconfined compression tests as well as three (3) multi-stage triaxial compression tests. Summarized on Table 2.5A-3 are the results of the unconfined compression tests. Figure 2.5A-4 is a typical stress-strain curve developed for one of the unconfined compression tests. The results of the triaxial tests are summarized on Figure 2.5A-5. The results confirmed that the material actually possessed a strength higher than assumed during design.

In summary, the results of these tests confirmed that all design criteria and specification requirements had been met at 50 percent completion of the embankment.

2.5A.2 DESIGN VERIFICATION AT ONE HUNDRED PERCENT (100%) CONSTRUCTION COMPLETE

Upon completion of the SSI dam, a core boring was drilled through the entire embankment at centerline station 9+75 to obtain undisturbed Shelby-tube samples throughout the entire depth of the impervious core section. A detailed “Log of Boring” is attached as Figure 2.5A-6.

Utilizing all the samples obtained of the impervious core, a series of laboratory tests, including Atterberg Limits, unit dry weight determinations, and hand vane shear tests was conducted. The results are summarized on Table 2.5A-4. The results of the Atterberg limit tests confirmed that the impervious core met the liquid limit specification requirement. The results of the vane shear readings show the material to be relatively uniform in compaction.

To evaluate the shear strength of the impervious core, a series of consolidated undrained triaxial tests were conducted with the results plotted on Figure 2.5A-7. This is a plot of the Mohr Stress circles developed from the individual tests with the design criteria failure envelope drawn showing its relationship to the test results. As can be seen on this Figure, the results of the CU triaxial tests fit very closely to the adopted design strengths.

On Figures 2.5A-8 through 10 are results of consolidation tests to the impervious core.

The results of the testing shows that the impervious core met all specification requirements and design criteria.

2.5A.3 FIELD DENSITY TEST DISTRIBUTION

To further show the adequacy of the completed embankment, a plot showing the distribution of density control tests has been developed and is shown on Figures 2.5A-11, Sheets 1 through 15. These plots show the distribution of the density tests conducted for the impervious core, filter zone and rock fill material at 5-foot intervals through the embankment. In addition, the test results have been shown in histogram plots. These histogram plots are shown as Figures 2.5A-12 through 15. In summary, the results showed the following:

Data Base	No. of Tests	Mean	Standard Deviation	Design Criteria
% Compaction of Impervious Core	309	102%	2.3	>95%
Moisture Content Variation, W_o-W_f	309	-0.59%	1.1	-3%+1%
Percent Relative Density-Filter	585	96.7%	9.3	>80%
Rock Fill Density	75	125.0pcf	9.8	115pcf

Histograms of the grain size distributions for the rock fill, filter A and filter B are presented in Figures 2.5A-19 thru 2.5A-21.

2.5A.4 DYNAMIC STABILITY RE-EVALUATION

During construction a series of cyclic triaxial tests were conducted on filter “A” material to verify whether or not his material met the adopted design criteria as discussed in Section 2.5.6.4.3.4. According to the test results presented in Section 2.5.4.7, filter “A” material failed to meet the adopted design criteria. Based on the cyclic triaxial test results for the filter “A” material, Figure 2.5A-16 was developed which relates the cyclic strength of the material based on a failure criteria for both 5% and 10% of double amplitude strain. From this figure, new cyclic design criteria for 10 cycles of loading and a failure criteria of 5% double amplitude strain was developed for the

filter material and is shown on Figure 2.5A-17. Utilizing this new design criteria, a new seismic stability cross-section of the SSI Dam was developed. Figure 2.5A-18 relates shear stress induced by the artificial ground motion, τ_d, to the cyclic shear stress required to cause 5% double amplitude strain, τ_f. As was the cause in Section 2.5.6.5.2.6, the ratio τ_d/τ_f represents a local safety factor against the development of 5% strain in 10 cycles of loading. Figure 2.5A-18 is the plot showing the calculated local safety factors at the nodal points of the finite element grid. Several points had values less than one (1) for the rock shell at points on the outer slope surfaces near the crest of the dam. These values result from the presence of little or no normal stress. An interpretation of these results would be that, during the occurrence of the SSE, an occasional rock particulate on the outer slope may be displaced enough to tumble or roll some short distance down the slope.

Near the top of the filter zone, two nodal points have safety factors calculates as 0.94 and 0.97. This is interpreted as meaning that, during the occurrence of the SSE, the upper zone of the filter material will experience 5% double amplitude strains. This will not result in the liquefaction of the filter material and will not impair the function of the dam.

TABLE 2.5A-1
SSI DAM SUMMARY OF CLASSIFICATION TESTS BORING SSI-1

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
0.0-1.5	47	24	90	124+	21.3	105.0	CL
1.5-3.0	43	19	124+	124+	21.2	106.7	CL
3.0-4.5	42	20	124+	124+	22.3	103.8	CL
4.5-6.0	37	16	124+	124+	19.4	110.2	CL
6.0-7.5	37	17	104	107	21.2	107.6	CL
7.5-9.0	40	20	90	114	19.8	106.0	CL
9.0-10.5	39	19	(a)	(a)	20.5	106.7	CL
10.5-12.0	37	17	124+	105	18.1	110.6	CL
12.0-13.5	34	15	124+	122	19.3	110.2	CL
13.5-15.0	39	19	94	94	21.9	105.5	CL
15.0-16.5	36	16	124+	100	19.4	111.1	CL
16.5-18.0	43	20	118	124+	21.5	104.6	CL
18.0-19.5	40	18	124+	124+	19.3	107.3	CL
19.5-21.0	40	17	104	112	22.8	101.7	CL
21.0-22.5	39	18	110	97	20.9	105.8	CL
22.5-24.0	38	18	124+	124+	19.7	109.0	CL
24.0-25.5	35	15	104	92	21.7	104.9	CL
25.5-27.0	42	20	(a)	(a)	20.6	108.5	CL
27.0-28.5	37	18	124+	124+	17.0	113.4	CL
28.5-30.0	38	18	124+	124+	20.0	109.5	CL
30.0-31.5	37	16	124+	124+	20.5	104.4	CL
31.5-33.0	46	23	124+	124+	23.1	101.9	CL
33.0-33.8	42	21	89	88	23.2	102.6	CL

a) Insufficient Material to run test.

Note:Vane reading in units of kilopascals.

TABLE 2.5A-2
SSI DAM SUMMARY OF CLASSIFICATION TESTS BORING SSI-2

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
0.0-1.5	46	22	124+	124+	20.8	105.6	CL
1.5-3.0	42	20	124+	124+	21.5	103.9	CL
3.0-4.5	41	20	124+	124+	21.1	106.1	CL
4.5-6.0	37	17	105	104	20.9	104.7	CL
6.0-7.5	35	16	92	97	20.6	109.6	CL
7.5-9.0	34	17	124+	124+	19.3	110.4	CL
9.0-10.5	39	19	82	78	21.1	103.0	CL
10.5-12.0	39	20	(a)	(a)	21.7	108.5	CL
12.0-13.5	41	19	124+	124+	21.7	104.4	CL
13.5-15.0	40	19	109	72	22.5	98.0	CL
15.0-16.5	41	20	88	98	21.3	104.6	CL
16.5-18.0	40	19	124+	124+	19.1	109.0	CL
18.0-19.5	40	17	108	124	20.9	105.7	CL
19.5-21.0	39	19	124+	114	21.3	103.7	CL
21.0-22.5	38	17	124+	100	20.7	107.1	CL
22.5-24.0	35	16	124+	124+	15.2	117.2	CL
24.0-25.5	36	17	124+	124+	19.2	108.8	CL
25.5-27.0	34	14	124+	124+	19.5	107.2	CL
27.0-28.5	33	14	124+	124+	19.4	107.6	CL
28.5-30.0	36	16	124+	124+	18.5	107.5	CL
30.0-30.6	36	15	(a)	(a)	(a)	(a)	CL

a) Insufficient Material to Conduct Test.

Note: Vane reading in units of kilopascals.

TABLE 2.5A-3
SSI DAM SUMMARY OF UNCONFINED COMPRESSIVE STRENGTH TESTS

Boring No.	Depth Ft.	Qu T.S.F.	Strain %	Moisture %	U.D.W. pcf	Description
SSI-1	4.5-6.0	2.4	13.2	19.4	104.5	CLAY, sandy, very stiff w/pbls., light brown
SSI-1	9.0-10.5	3.1	13.1	20.5	106.7	CLAY, sandy, very stiff w/pbls., light brown
SSI-1	21.0-22.5	4.2	12.1	20.9	107.2	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-1	25.5-27.0	4.2	16.8	20.6	108.5	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-2	1.5-3.0	3.7	14.0	21.5	106.8	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-2	6.0-7.5	3.9	10.9	20.6	107.7	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-2	10.5-12.0	3.6	16.1	21.7	108.5	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-2	16.5-18.0	3.3	16.9	19.1	107.6	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.
SSI-2	25.5-27.0	3.6	15.6	19.5	109.4	CLAY, sandy, very stiff w/pbls., dk. brn. to brn.

TABLE 2.5A-4
SSI DAM SUMMARY OF CLASSIFICATION TESTS BORING IC-3

(Sheet 1 of 2)

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
12.5-14.0	40	19	124	116	22.9	101.4	CL
14.0-15.5	52	31	106	83	23.4	100.9	CH
15.5-17.0	49	27	106	120	22.9	101.3	CL
17.0-18.5	44	22	99	124	21.9	102.7	CL
18.5-20.0	53	33	98	124	21.7	103.7	CH
20.0-21.5	45	23	113	110	22.7	103.4	CL
21.5-23.0	53	31	115	118	21.9	103.2	CH
23.0-24.5	51	29	(a)	(a)	20.0	106.4	CH
24.5-26.0	48	28	(a)	(a)	20.0	105.1	CL
26.0-27.5	50	27	124	124	21.2	105.3	CL
27.5-29.0	45	24	120	124	21.8	103.6	CL
29.0-30.5	46	26	124	124	18.9	107.6	CL
30.5-32.0	39	24	124	124	17.9	109.0	CL
32.0-33.0	44	25	124	124	19.6	106.9	CL
33.0-34.5	45	24	113	103	21.4	104.0	CL
34.5-36.0	42	22	114	124	18.8	105.4	CL
36.0-37.5	45	26	124+	120	18.8	106.3	CL
37.5-39.0	47	28	124+	124+	19.4	106.6	CL
39.0-40.5	44	25	124+	124+	17.9	111.5	CL
40.5-42.0	36	18	(a)	(a)	17.2	111.1	CL
42.0-43.0	39	20	(a)	(a)	16.8	109.1	CL
43.0-44.5	49	27	124+	124+	20.2	108.7	CL
44.5-46.0	40	20	(a)	(a)	19.5	106.3	CL

TABLE 2.5A-4
SSI DAM SUMMARY OF CLASSIFICATION TESTS BORING IC-3

(Sheet 2 of 2)

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
46.0-47.5	40	20	124+	124+	17.3	111.8	CL
47.5-49.0	45	26	110	109	21.4	104.7	CL
49.0-50.5	42	23	98	124	18.1	109.3	CL
50.5-52.0	42	23	(a)	(a)	19.7	106.6	CL
53.0-54.5	41	21	124+	124+	19.9	107.8	CL
54.5-56.0	45	24	(a)	(a)	21.3	105.8	CL
56.0-57.5	45	26	116	116	21.0	103.6	CL
57.5-59.0	44	25	124	124	21.2	106.5	CL
59.0-60.5	43	22	124	110	20.2	105.0	CL
60.5-62.0	46	23	124	124	22.2	102.5	CL
62.0-63.0	47	27	(a)	(a)	(a)	(a)	CL
63.0-63.4	43	24	(a)	(a)	(a)	(a)	CL
63.4-65.0	41	21	(a)	(a)	18.4	111.9	CL
65.0-66.5	41	21	124	124	19.3	104.9	CL
66.5-68.0	(a)	(a)	124	124	21.4	104.9	CL
68.0-69.5	41	22	124	124	19.4	106.0	CL
69.5-70.6	42	22	124	124	19.6	105.1	CL

a) Test Not Run.

Note: Vane reading in units of kilopascals.

APPENDIX 2.5B - CONSTRUCTION RECORDS AND DESIGN VERIFICATION OF SQUAW CREEK DAM

2.5B.1 DESIGN VERIFICATION AT 50% CONSTRUCTION COMPLETION.

In order to verify that the Squaw Creek Dam embankment met the design criteria at 50% construction completion, four (4) borings were drilled through the embankment as follows:

Boring No.	Station	Embankment Zone	Ground Elevation
SCD-1	60+00,60'D.S.	Random	701.0
SCD-2	59+50,20'D.S.	Random	702.0
SCD-2	59+50,20'U.S.	Impervious Core	680.0
SCD-4	58+00,20'U.S.	Impervious Core	709.0

Near continuous sampling of the earth embankment was achieved by the utilization of a Shelby-tube sampler. A detailed log of each boring is shown on Figures 2.5B-1 thru 2.5B-4, respectively.

From each boring, a series of Atterberg Limits, Unit Dry Weight Determinations and Hand Vane Shear Tests were conducted. The test results for each Boring are summarized on Table 2.5B-1 thru 2.5B-4, respectively. The results of the classification tests confirm that the impervious core and random consist of CL soils, with the higher plastic clays in the impervious core. The Hand Vane Shear tests were conducted to provide a relative indicator of the uniformity of the embankment layer compaction. For each specimen tested, a Vane Shear reading was made on both the top and bottom of the samples. Differences in the readings represent the relative variation in the compaction of the embankment layers. Several soft zones were detected in Boring SCD-2 and SCD-4; however, these zones were isolated and the overall consistency of the material is considered stiff.

The strength parameters of the embankment fill were evaluated by a series of fourteen (14) Unconfined Compression Tests and six (6) Multistage, Q, Triaxial Compression Tests. The results of the Unconfined Compression Tests are summarized on Table 2.5B-5, with a typical stress-strain curve developed for one of the tested specimen shown on Figure 2.5B-5. The results of the Triaxial tests are shown on Figure 2.5B-6. The test results confirm that the materials actually possess a higher cohesive strength than assumed during design.

2.5B.2 FIELD DENSITY TEST DISTRIBUTION

To further show the adequacy of the completed embankment, plots showing the distribution of density control tests have been developed and are shown on Figures 2.5B-7, Sheets 1 through 33. These plots show the distribution of the density tests conducted for the impervious core random zone and filter zone at 5-foot intervals through the embankment. In addition, the tests

results have been shown in histogram plots, which are shown as Figures 2.5B-8 through 10, respectively. In summary, the results showed the following:

Data Base	No. of Tests	Mean	Standard Deviation	Design Criteria
% Compaction of Impervious Core	2959	100.3%	2.7	95%
Moisture Content Variation, W_o-W_f	3128	-0.59%	1.3	-3%+1%
Percent Relative Density-Filter	148	102.3%	13.5	80%

TABLE 2.5B-1
SQUAW CREEK DAM BORING SCD-1 SUMMARY OF CLASSIFICATION AND VANE SHEAR TESTS

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
0.0-1.0	32	18	124+	124+	9.8	127.1	CL
1.0-2.0	34	17	124+	124+	13.3	123.5	CL
2.0-3.5	47	31	124+	124+	12.7	122.3	CL
3.5-5.0	34	18	124+	124+	14.2	118.1	CL
5.0-6.5	39	24	124+	124+	13.9	119.4	CL
6.5-8.0	(a)	(a)	124+	124+	10.6	130.4	CL
8.0-9.5	30	18	124+	124+	11.5	127.2	CL
9.5-11.0	(a)	(a)	(a)	(a)	13.5	(a)	CL
11.0-12.0	(a)	(a)	124+	124+	14.2	113.5	CL
12.0-13.5	42	27	124+	124+	15.2	112.9	CL
13.5-15.0	36	19	120	124+	18.3	113.2	CL
15.0-16.5	40	27	119	124	16.7	115.0	CL
16.5-18.0	33	18	(a)	(a)	16.8	106.8	CL
18.0-19.5	29	15	98	109	16.1	115.3	CL
19.5-21.0	31	16	60	70	14.7	121.6	CL
21.0-22.5	(a)	(a)	71	96	16.6	113.3	CL
22.5-24.0	34	17	124+	124+	12.9	122.3	CL
24.0-25.5	(a)	(a)	(a)	(a)	9.9	(a)	CL
25.5-27.0	35	17	(a)	(a)	19.7	(a)	CL
27.0-28.5	35	19	97	85	16.2	114.0	CL
28.5-30.0	28	13	(a)	(a)	13.1	(a)	CL
30.0-31.5	24	9	(a)	(a)	12.9	(a)	CL
31.5-33.0	31	18	(a)	(a)	15.9	(a)	CL
33.0-34.5	31	15	124+	121	15.3	117.8	CL
34.5-36.0	30	15	124+	124+	15.3	117.1	CL
36.0-37.5	26	12	124+	124+	12.2	124.6	CL
37.5-39.0	32	16	124+	(a)	13.5	120.8	CL
39.0-40.5	26	12	124+	124+	11.0	125.9	CL
40.5-42.0	25	12	124+	124+	11.5	123.1	CL
42.0-43.5	25	12	(a)	(a)	(a)	(a)	CL

a) Insufficient Material to run Test

TABLE 2.5B-2
SQUAW CREEK DAM BORING SCD-2 SUMMARY OF CLASSIFICATION AND VANE SHEAR TESTS

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
0.0-1.5	46	24	103	124+	18.2	109.7	CL
1.5-3.0	46	24	124+	118	19.1	107.1	CL
3.0-4.5	45	25	124+	123	20.1	103.3	CL
4.5-6.0	48	26	124+	124+	18.7	109.3	CL
6.0-7.5	43	24	84	92	19.5	107.9	CL
7.5-9.0	46	26	106	112	19.6	107.7	CL
9.0-10.5	43	24	45	15	27.2	95.0	CL
10.5-12.0	46	25	68	116	21.9	105.0	CL
12.0-13.5	50	26	55	92	24.0	100.1	CL
13.5-15.0	46	25	124+	124+	20.3	108.4	CL
15.0-16.5	47	26	124+	124+	19.5	112.3	CL
16.5-18.0	48	26	124	124+	21.1	107.8	CL
18.0-19.5	50	27	100	90	21.8	105.7	CL
19.5-21.0	40	20	96	106	20.4	106.8	CL
21.0-22.5	43	23	92	100	19.4	109.4	CL
22.5-24.0	44	23	76	107	20.7	105.4	CL
24.0-25.5	31	16	111	117	15.3	118.9	CL
25.5-27.0	39	21	93	67	20.5	109.9	CL
27.0-28.5	42	22	101	114	19.1	110.7	CL
28.5-29.5	43	22	82	98	20.4	107.9	CL
29.5-30.5	43	22	90	102	24.0	101.9	CL
30.5-31.5	33	14	124+	124+	17.3	114.3	CL
31.5-33.0	40	20	72	80	21.0	106.0	CL
33.0-34.0	46	24	94	94	21.8	106.0	CL
34.0-35.0	40	20	80	102	21.0	105.7	CL
35.0-36.0	40	20	(a)	(a)	20.7	(a)	CL
36.0-37.0	43	24	73	102	20.4	108.9	CL
45.0-46.0	34	19	81	76	18.5	108.6	CL
46.0-47.0	38	20	80	110	20.5	107.0	CL
47.0-48.0	41	18	69	78	23.3	102.2	CL
48.0-49.0	44	26	63	66	25.3	95.4	CL
51.1-52.2	41	25	112	113	20.5	109.2	CL
52.2-53.1	(a)	(a)	(a)	(a)	23.8	104.8	CL
53.1-54.6	43	21	40	81	27.1	94.9	CL
54.6-56.2	33	14	124+	124+	18.8	106.5	CL
56.2-57.5	41	23	103	107	21.3	104.5	CL

a) Insufficient Material to run Test

TABLE 2.5B-3
SQUAW CREEK DAM BORING SCD-3 SUMMARY OF CLASSIFICATION AND VANE SHEAR TESTS

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	CLASS.
0.0 - 1.5	39	20	117	124+	18.6	108.7	CL
1.5 - 3.0	46	26	(a)	(a)	(a)	(a)	CL
3.0 - 4.5	51	28	103	113	21.4	105.3	CH
4.5 - 6.0	41	25	124+	124+	19.4	106.3	CL
6.0 - 7.5	41	19	124+	119	20.9	103.2	CL
7.5 - 9.0	45	29	108	103	20.4	106.4	CL
9.0 - 10.5	(a)	(a)	101	113	20.1	101.5	CL
10.5 - 12.0	48	33	60	70	23.0	102.1	CL
12.0 - 13.5	30	17	124+	124+	14.0	120.5	CL
13.5 - 15.0	45	29	113	102	20.5	108.3	CL
15.0 - 16.5	40	26	121	105	20.7	105.1	CL
16.5 - 18.0	37	22	124+	124+	17.6	106.3	CL
18.0 - 19.5	44	23	(a)	(a)	(a)	(a)	CL
19.5 - 21.0	40	24	80	99	22.1	104.3	CL
21.0 - 22.5	35	19	88	124+	19.2	108.6	CL
22.5 - 24.0	36	22	70	124+	16.9	114.0	CL
24.0 - 25.5	(a)	(a)	124+	124+	18.1	111.8	CL
25.5 - 27.0	34	19	124+	124+	15.4	116.9	CL
27.0 - 28.5	45	24	105	97	21.6	106.0	CL
28.5 - 30.0	41	21	124+	124+	17.5	114.1	CL
30.0 - 31.5	40	23	124+	124+	18.5	111.6	CL
31.5 - 33.0	42	25	102	94	20.2	106.4	CL
33.0 - 34.5	39	17	90	92	22.5	101.5	CL
34.5 - 36.0	38	17	98	76	21.7	103.8	CL
36.0 - 36.5	41	19	90	94	23.2	99.8	CL

a) Insufficient Material to run Test

TABLE 2.5B-4
SQUAW CREEK DAM BORING SCD-4
SUMMARY OF CLASSIFICATION AND VANE SHEAR TESTS

(Sheet 1 of 2)

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
0.0-1.5	45	23	124+	124+	21.2	106.1	CL
1.5-3.0	44	21	124+	70	20.6	102.9	CL
3.0-4.5	44	23	124+	124+	17.8	112.4	CL
4.5-6.0	43	21	124+	124+	19.1	106.5	CL
6.0-7.5	43	21	95	124+	20.7	105.8	CL
7.5-9.0	49	25	114	124+	21.8	103.3	CL
9.0-10.5	44	22	103	124+	19.4	106.7	CL
10.5-12.0	39	19	124+	97	17.0	111.4	CL
12.0-13.5	37	19	102	98	18.4	109.0	CL
13.5-15.0	36	19	73	124+	18.0	108.8	CL
15.0-16.5	36	17	77	50	18.2	109.3	CL
16.5-18.0	42	22	115	124+	19.9	106.8	CL
18.0-19.5	49	28	124+	105	21.9	103.1	CL
19.5-21.0	45	23	124+	124+	17.3	111.7	CL
21.0-22.5	46	24	124+	124+	19.3	108.1	CL
22.5-24.0	48	28	124+	124+	17.4	110.7	CL
24.0-25.5	45	24	116	124+	19.0	108.0	CL
25.5-27.0	42	23	124+	124+	13.6	117.7	CL
27.0-28.5	41	23	(a)	(a)	14.6	(a)	CL
28.5-30.0	38	20	124+	120	15.2	117.0	CL
30.0-31.5	38	19	124+	124+	15.0	117.8	CL
31.5-33.0	42	23	96	72	20.0	106.9	CL
33.0-34.5	42	25	115	73	18.5	108.5	CL
34.5-36.0	38	20	115	97	17.8	110.6	CL

TABLE 2.5B-4
SQUAW CREEK DAM BORING SCD-4
SUMMARY OF CLASSIFICATION AND VANE SHEAR TESTS

(Sheet 2 of 2)

Depth Ft.	L.L. %	P.I.	Vane Reading Top	Vane Reading Bot	M.C. %	UDW pcf	Class.
36.0-37.5	40	20	123	116	17.7	112.1	CL
37.5-39.0	34	16	74	80	19.1	106.2	CL
39.0-40.5	43	23	99	104	19.0	112.2	CL
40.5-42.0	44	23	66	69	22.0	106.2	CL
42.0-43.5	41	20	114	110	18.4	112.5	CL
43.5-45.0	41	20	(a)	(a)	21.7	(a)	CL
45.0-46.5	48	26	96	92	22.0	105.0	CL
46.5-48.0	46	27	84	94	21.0	108.1	CL
48.0-49.5	42	23	92	122	18.3	111.0	CL
49.5-51.0	32	16	(a)	(a)	16.9	113.9	CL
51.0-52.5	34	16	84	88	18.4	111.5	CL
52.5-54.0	31	15	108	103	16.8	115.0	CL
54.0-55.5	44	22	80	95	20.9	105.7	CL
55.5-57.0	45	26	77	85	20.1	105.2	CL
57.0-58.5	41	21	124+	90	19.0	107.9	CL
58.5-60.0	40	21	124+	124+	18.0	107.9	CL
60.0-61.5	45	23	90	100	20.0	106.0	CL
61.5-63.0	38	19	114	88	19.8	108.2	CL
63.0-64.5	47	24	90	62	22.3	100.7	CL
64.5-65.5	43	22	100	124+	19.3	106.8	CL

a) Insufficient Material to run Test

TABLE 2.5B-5
SQUAW CREEK DAM SUMMARY OF UNCONFINED COMPRESSIVE STRENGTH TESTS

Boring No.	Depth Ft.	Qu T.S.F.	Strain %	U.D.W. pcf	Moisture %	Description
SCD-1	34.5-36.0	3.2	13.4	117.0	15.3	CLAY, sandy, lt. tan
SCD-2	1.5- 3.0	7.6	1.2	106.5	19.1	CLAY, sandy, brown
SCD-2	10.5-12.0	3.5	5.9	106.5	21.9	CLAY, lt. brown
SCD-2	18.0-19.5	2.9	10.3	107.6	21.8	CLAY, w/L.S. part., lt. brown
SCD-2	31.5-33.0	2.0	6.0	102.5	21.0	CLAY, sandy, w/L.S. part, reddish brown
SCD-2	35.0-36.5	2.3	12.1	117.7	11.8	CLAY, sli. sandy, dark gray
SCD-2	45.0-46.0	2.8	16.8	110.5	18.5	CLAY, sandy, brown
SCD-3	10.5-12.0	2.2	2.2	105.7	23.0	CLAY, brown
SCD-4	1.5-3.0	4.0	8.1	108.6	20.6	CLAY, brown
SCD-4	27.0-28.5	7.5	5.5	116.1	14.6	CLAY, sandy, w/L.S. gravel, brown
SCD-4	34.5-36.0	4.8	5.7	113.8	17.8	CLAY, w/some L.S. part., brown
SCD-4	40.5-42.0	2.8	14.4	105.2	22.0	CLAY, brown
SCD-4	49.5-51.0	2.3	11.1	113.9	16.9	CLAY, sandy, brown
SCD-4	54.0-55.5	2.7	17.1	107.0	20.9	CLAY, sandy, brown

3.0 DESIGN OF STRUCTURES, COMPONENTS, EQUIPMENT AND SYSTEMS

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3.0 DESIGN OF STRUCTURES, COMPONENTS, EQUIPMENT AND SYSTEMS

3.1 CONFORMANCE WITH NUCLEAR REGULATORY COMMISSION (NRC) GENERAL DESIGN CRITERIA

This section briefly discusses the extent to which the design criteria for the plant structures, systems, and components important to safety comply with Title 10, Code of Federal Regulations, Part 50 (10 CFR Part 50), Appendix A, General Design Criteria (GDC) for Nuclear Power Plants. Each criterion is first quoted directly from 10 CFR Part 50 and then discussed. In the discussion of each criterion, it is shown how the CPNPP design meets the criterion, and the sections of the FSAR, where more detailed information is presented to demonstrate compliance with the criterion, are referenced.

3.1.1 OVERALL REQUIREMENTS

3.1.1.1 Criterion 1 - Quality Standards and Records

"Structures, systems, and components important to safety shall be designed, fabricated, erected, and tested to quality standards commensurate with the importance of the safety functions to be performed. Where generally recognized codes and standards are used, they shall be identified, evaluated to determine their applicability, adequacy, and sufficiency, and shall be supplemented or modified as necessary to assure a quality product in keeping with the required safety function. A quality assurance program shall be established and implemented in order to provide adequate assurance that these structures, systems, and components will satisfactorily perform their safety functions. Appropriate records of the design, fabrication, erection, and testing of structures, systems, and components important to safety shall be maintained by or under the control of the nuclear power unit licensee throughout the life of the unit [1]."

Discussion

Structures are identified and classified in accordance with the requirement that they be designed to withstand the effects of earthquakes, as delineated in Section 3.2.

The systems and components of the facility are classified according to their importance in the prevention and mitigation of accidents. Reactor components use the classification system developed by American National Standards Institute (ANSI) N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants. Classifications and any deviations are described in Section 3.2. Each component is given a safety class designation.

The codes, standards, and quality control applicable to each type of component are identified in pertinent equipment specifications. Where applicable, design and fabrication are in accordance with the codes specified in 10 CFR Part 50, Section 55a. See Section 5.2.1.1. Alternative requirements, as provided by ASME Code Cases, are utilized at CPNPP in accordance with 10 CFR Part 50, Section 55a(a)(3). By reference to ASME Section III requirements in the procurement specifications, the use code cases by mechanical equipment suppliers requires mutual consent of the Owner or his agent and the manufacturer. The ASME Code Cases which are used for design and erection at CPNPP are identified in the appropriate mechanical design and erection specifications or the Brown & Root QA Manual; conditionally-approved Code Cases will show justification for their use, as required by NRC, in these documents. This application of ASME Code Cases is documented on the ASME Data Report Forms. For further discussion, see

the text concerning Regulatory Guide 1.85 in Section 1A(N). The quality assurance (QA) program that is implemented to ensure that safety-related systems, components, and structures perform as intended is discussed in Chapter 17. The QA program conforms to the intent of 10 CFR Part 50, Appendix B, Quality Assurance Criteria for Nuclear Power Plants and Fuel Reprocessing Plants. Those items for which the requirements of 10 CFR Part 50, Appendix B, are met are listed in the list of quality assured items in Appendix 17A.

Chapter 14 describes initial tests to ensure performance of installed equipment commensurate with the importance of the safety function.

The component safety classifications are also shown on the flow diagrams presented with their respective sections. Records for the design, fabrication, erection, and testing of safety-related systems, components, and structures are maintained as described in Chapter 17.

Westinghouse, in accordance with the Westinghouse Nuclear Energy System Division's (WNESD's) Quality Assurance Plan, maintains (either in its possession or under its control) a complete set of records of the design, fabrication, construction, and testing of safety components which it supplies.

The Westinghouse QA program conforms to the requirements of 10 CFR Part 50, Appendix B, and is discussed in Westinghouse Commercial Atomic Power (WCAP) 8370, Revision 7A.

3.1.1.2 Criterion 2 - Design Bases for Protection Against Natural Phenomena

"Structures, systems, and components important to safety shall be designed to withstand the effects of natural phenomena such as earthquakes, tornadoes, hurricanes, floods, tsunami, and seiches without loss of capability to perform their safety functions. The design bases for these structures, systems, and components shall reflect: (1) Appropriate consideration of the most severe of the natural phenomena that have been historically reported for the site and surrounding area, with sufficient margin for the limited accuracy, quantity, and period of time in which the historical data have been accumulated, (2) appropriate combinations of the effects of normal and accident conditions with the effects of the natural phenomena, (3) the importance of the safety functions to be performed [1]."

Discussion

The natural phenomena and their magnitude are selected in accordance with their probability of occurrence at the Comanche Peak site. The design is based on the most severe of the natural phenomena recorded for the site, with an appropriate margin to account for uncertainties in the historical data. The natural phenomena postulated in the design are presented in Sections 2.3, 2.4, 2.5, and 3.3. The design criteria for the structures, systems, and components affected by each natural phenomenon are presented in Sections 3.2, 3.3, 3.4, 3.5, 3.7, and 3.8.

Combinations of natural phenomena and plant originated accidents considered in the design are identified in Sections 3.8, 3.9, and 3.10. The importance of the safety functions is identified with the classification system developed by the American Nuclear Society (ANS) and is generally in accordance with ANSI N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants. This identification and any deviations are included in Section 3.2.

3.1.1.3 Criterion 3 - Fire Protection

"Structures, systems, and components important to safety shall be designed and located to minimize, consistent with other safety requirements, the probability and effect of fires and explosions. Noncombustible and heat resistant materials shall be used wherever practical throughout the unit, particularly in locations such as the containment and control room. Fire detection and fighting systems of appropriate capacity and capability shall be provided and designed to minimize the adverse effects of fires on structures, systems, and components important to safety. Fire fighting systems shall be designed to assure that their rupture or inadvertent operation does not significantly impair the safety capability of the structures, systems, and components [1]."

Discussion

Comanche Peak Nuclear Power Plant is designed to minimize the probability and effect of fires and explosions. Noncombustible and fire resistant materials are used in the Containment Building, in the Control Room, and wherever practical throughout the station. Atmospheric conditions inside the Containment Building and Control Room are not of an explosive nature. Alarm and detection devices are located throughout the station and connected to the annunciation system in the Control Room.

To protect both plant equipment and personnel from fire normal fire protection is provided by water and halon fire suppression systems, hose stations, and portable fire extinguishers. The fire fighting systems are designed to ensure that rupture or inadvertent operation does not significantly impair systems important to safety.

In the design and installation of fire protection equipment, the requirements of the National Fire Protection Association (NFPA), American Nuclear Insurers (ANI), and applicable codes and regulations have been followed. The applicable codes, standards, and regulations used in the design of the Fire Protection System and plant equipment are further discussed in Section 9.5.1 and the Fire Protection Report.

Finally, fire protection and detection system reliability is ensured by periodic tests and inspections.

3.1.1.4 Criterion 4 - Environmental and Dynamic Effects Design Bases

"Structures, systems, and components important to safety shall be designed to accommodate the effects of and to be compatible with the environmental conditions associated with normal operation, maintenance, testing, and postulated accidents, including loss-of-coolant accidents. These structures, systems, and components shall be appropriately protected against dynamic effects, including the effects of missiles, pipe whipping, and discharging fluids, that may result from equipment failures and from events and conditions outside the nuclear power unit. However, the dynamic effects associated with postulated pipe ruptures in nuclear power units may be excluded from the design basis when analyses reviewed and approved by the Commission demonstrate that the probability of fluid system piping rupture is extremely low under basis conditions fluid system piping.[1]"

Discussion

The station's structures, systems, and components important to safety are designed to accommodate the effects of, and to be compatible with, the environmental conditions associated with normal operation, maintenance, testing, and postulated accidents, including loss-of-coolant accident (LOCA). Environmental conditions are described in Section 3.11.

These structures, systems, and components are appropriately protected against dynamic effects, including the effects of missiles, pipe whipping, and discharging fluids, that can result from equipment failures and from events and conditions outside the nuclear power unit.

Details of the design, environmental testing, and construction of these systems, structures, and components are included in Chapters 3, 5, 6, 7, 8, 9 and 10. Evaluation of the performance of safety features is contained in Chapter 15.

The leak before break methodology demonstrates that the probability of fluid system piping rupture is extremely low under conditions consistent with the design basis for the piping. It has been applied to CPNPP Units 1 and 2 to exclude from the design basis the dynamic effects of postulated pipe ruptures in the primary coolant loop piping and 10 inch and larger reactor coolant loop branch lines, as discussed in Sections 3.6B.2.5.1 and 3.6B.2.5. Implementation of this technology eliminates the need for respectively pipe whip restraints and jet impingement barriers. Containment design, emergency core cooling and environmental qualification requirements are not influenced by this modification.

3.1.1.5 Criterion 5 - Sharing of Structures, Systems, and Components

"Structures, systems, and components important to safety shall not be shared between nuclear power units unless it can be shown that such sharing will not significantly impair their ability to perform their safety functions including, in the event of an accident in one unit, an orderly shutdown and cooldown of the remaining unit [1]."

Discussion

The facilities which have shared systems or components are tabulated in Section 1.2 along with references to the appropriate sections containing specific design details. Construction of the second unit will not impair the safety functions of any shared system or component.

The sharing of systems or components does not impair system capability to perform safety functions, e.g., the orderly shutdown and cooldown of one unit in the event of an accident in the other unit.

3.1.2 PROTECTION BY MULTIPLE FISSION PRODUCT BARRIERS

3.1.2.1 Criterion 10 - Reactor Design

"The reactor core and associated coolant, control, and protection systems shall be designed with appropriate margin to assure that specified acceptable fuel design limits are not exceeded during any condition of normal operation, including the effects of anticipated operational occurrences [1]."

Discussion

The reactor core and associated coolant, control, and protection systems are designed with adequate margins to do the following:

1. To preclude significant fuel damage during normal core operation and operational transients (Condition I)[a] or during transient conditions arising from occurrences of moderate frequency (Condition II).[a]

2. To ensure return of the reactor to a safe state following infrequent faults (Condition III)[a] with only a small fraction of fuel rods damaged, although sufficient fuel damage can occur to preclude resumption of operation without considerable outage time.

3. To ensure that the core is intact, with acceptable heat transfer geometry, following transients arising from occurrences of limiting faults (Condition IV).[a]

Chapter 4 discusses the design bases and design evaluation of reactor components, including the fuel, reactor vessel internals, and reactivity control systems. Details of the control and protection systems instrumentation design and logic are discussed in Chapter 7. This information supports the accident analyses of Chapter 15, which show that the acceptable fuel design limits are not exceeded for Condition I and II occurrences.

3.1.2.2 Criterion 11 - Reactor Inherent Protection

“The reactor core and associated coolant systems shall be designed so that in the power operating range the net effect of the prompt inherent nuclear feedback characteristics tends to compensate for a rapid increase in reactivity [1].”

Discussion

When the reactor is critical, prompt compensatory reactivity feedback effects are ensured by the negative fuel temperature effect (Doppler effect) and by the operational limit on moderator temperature coefficient of reactivity. The negative Doppler coefficient of reactivity is ensured by the inherent design using low-enrichment fuel; the limit on moderator temperature coefficient of reactivity is ensured by administratively controlling the dissolved absorber concentration or by burnable poison.

These reactivity coefficients are discussed in Section 4.3.

3.1.2.3 Criterion 12 - Suppression of Reactor Power Oscillations

“The reactor core and associated coolant, control, and protection systems shall be designed to assure that power oscillations which can result in conditions exceeding specified acceptable fuel design limits are not possible or can be reliably and readily detected and suppressed [1].”

(a.) Defined by ANSI N18.2-1973

Discussion

Power oscillations of the fundamental mode are inherently eliminated by the negative power coefficient of reactivity.

In addition, oscillations, due to xenon spatial effects, in the radial, diametral, and azimuthal overtone modes are heavily damped because of the inherent design and because of the negative power coefficient of reactivity.

Also, oscillations, caused by xenon spatial effects, in the axial first overtone mode may occur; however, assurance that fuel design limits are not exceeded by xenon axial oscillations is provided by reactor trip functions using the measured axial power imbalance as an input.

Finally, oscillations, due to xenon spatial effects, in axial modes higher than the first overtone are heavily damped because of the inherent design and because of the negative Doppler coefficient of reactivity.

Xenon stability control is discussed in Section 4.3.

3.1.2.4 Criterion 13 - Instrumentation and Control

“Instrumentation and control shall be provided to monitor variables and systems over their anticipated ranges for normal operation, for anticipated operational occurrences, and for accident conditions as appropriate to assure adequate safety, including those variables and systems that can affect the fission process, the integrity of the reactor core, the reactor coolant pressure boundary, and the containment and its associated systems. Appropriate controls shall be provided to maintain these variables and systems within prescribed operating ranges [1].”

Discussion

To ensure adequate safety, instrumentation and control systems are provided to monitor and control significant variables over their anticipated range for all conditions in the Reactor Core, Reactor Coolant System (RCS), steam and power conversion system, radioactive waste systems, engineered safety features (ESF) systems, and the Containment Building. Parameters that must be provided for operator use under normal operating and accident conditions are indicated in the Control Room in close proximity to the controls which maintain the indicated parameters in the proper range.

The quantity and types of process instrumentation provided ensures safe and orderly operation of all systems over the full design range of the plant. These systems are described in Chapters 6, 7, 8, 9, 10, 11, and 12.

3.1.2.5 Criterion 14 - Reactor Coolant Pressure Boundary

“The reactor coolant pressure boundary shall be designed, fabricated, erected, and tested so as to have an extremely low probability of abnormal leakage, of rapidly propagating failure, and of gross rupture [1].”

Discussion

The RCS pressure boundary is designed to accommodate the system pressures and temperatures attained under the expected modes of plant operation, including anticipated transients, and to maintain the stresses within applicable stress limits (Section 5.2.). Also, reactor coolant pressure boundary (RCPB) materials and selection and fabrication techniques ensure a low probability of gross rupture of significant leakage.

In addition to the loads imposed on the system under normal operating conditions, consideration is also given to abnormal loading conditions, such as seismic and pipe rupture, which are discussed in Sections 3.6 and 3.7.

The leak before break methodology demonstrates that the probability of rupturing primary coolant piping is extremely low under design basis conditions. It has been applied to CPNPP Units 1 and 2 to exclude from the design basis the dynamic effects of postulated ruptures in the primary coolant loop piping, as discussed in Section 3.6B.2.5.1. Implementation of this technology eliminates the need for primary coolant loop piping whip restraints and jet impingement barriers. Containment design, emergency core cooling and environmental qualification requirements are not influenced by this modification.

The system is protected from overpressure by means of pressure-relieving devices as required by applicable codes.

In conclusion, the RCS boundary has provisions for inspection, testing, and surveillance of critical areas to assess the structural and leaktight integrity (Section 5.2). For the reactor vessel, a material surveillance program conforming to applicable codes is provided (Section 5.3).

3.1.2.6 Criterion 15 - Reactor Coolant System Design

“The reactor coolant system and associated auxiliary, control, and protection systems shall be designed with sufficient margin to assure that the design conditions of the reactor coolant pressure boundary are not exceeded during any condition of normal operation, including anticipated operational occurrences [1].”

Discussion

The design pressure and temperature for each component in the RCS and associated auxiliary, control, and protection systems are selected to be above the maximum coolant pressure and temperature under all normal and anticipated transient load conditions.

In addition, RCPB components achieve a large margin of safety by the use of proven American Society of Mechanical Engineers (ASME) materials and design codes, use of proven fabrication techniques, nondestructive shop testing, and of integrated hydrostatic testing of assembled components.

Chapter 5 discusses the RCS design.

3.1.2.7 Criterion 16 - Containment Design

“Reactor containment and associated systems shall be provided to establish an essentially leak-tight barrier against the uncontrolled release of radioactivity to the environment and to assure that the containment design conditions important to safety are not exceeded for as long as postulated accident conditions require [1].”

Discussion

A steel-lined, reinforced concrete containment structure encloses the entire RCS and is designed to withstand the pressures and temperatures resulting from a spectrum of postulated LOCAs and secondary system breaks.

The Emergency Core Cooling System cools the reactor core and limits the release of radioactive materials to the environment.

Next, to ensure its integrity, the Containment Spray System is incorporated in the containment design.

The Containment Spray System is designed to function after a LOCA to reduce the pressure inside the containment to near atmospheric pressure and to remove fission product activity from the containment atmosphere.

To sum up, the Containment structure and ESF systems are designed to safely sustain internal and external environmental conditions that may reasonably be expected to occur during the life of the plant, including both short- and long-term effects following a LOCA (See Sections 6.2, 6.5, 15.6, and 3.8.1).

3.1.2.8 Criterion 17 - Electric Power Systems

“An onsite electric power system and an offsite electric power system shall be provided to permit functioning of structures, systems, and components important to safety. The safety function for each system (assuming the other system is not functioning) shall be to provide sufficient capacity and capability to assure that (1) specified acceptable fuel design limits and design conditions of the reactor coolant pressure boundary are not exceeded as a result of anticipated operational occurrences, and (2) the core is cooled and containment integrity and other vital functions are maintained in the event of postulated accidents.

“The onsite electric power sources, including the batteries, and the onsite electrical distribution system, shall have sufficient independence, redundancy, and testability to perform their safety functions, assuming a single failure.

“Electric power from the transmission network to the onsite electric distribution system shall be supplied by two physically independent circuits (not necessarily on separate rights of way) designed and located so as to minimize to the extent practical the likelihood of their simultaneous failure under operating and postulated accident and environmental conditions. A switchyard common to both circuits is acceptable. Each of these circuits shall be designed to be available in sufficient time following a loss of all onsite alternating current power supplies and the other offsite electrical power circuit, to assure that specified acceptable fuel design limits and design conditions of the reactor coolant pressure boundary are not exceeded. One of these circuits

shall be designed to be available within a few seconds following a loss-of-coolant accident to assure that core cooling, containment integrity, and other vital safety functions are maintained.

"Provisions shall be included to minimize the probability of losing electric power from any of the remaining supplies as a result of, or coincident with, the loss of power generated by the nuclear power unit, the loss of power from the transmission network, or the loss of power from the onsite electrical power supplies [1]."

Discussion

To ensure the functioning of safety-related systems, an onsite electrical power system and an offsite electrical power system are provided and are designed with adequate independence, capacity, redundancy, and testability.

Independence is provided for both systems by physical separation of components and cables to minimize vulnerability of redundant systems to single credible accidents. For details of separation, see Chapter 8.

Offsite power is provided to the safety-related systems through the startup transformers. The startup transformers are independent of each other and are each capable of supplying the Class 1E systems of each unit. One transformer is connected to the 345 kilovolts (kV) Switchyard and the other to the 138 kV system, so that a single component failure will not prevent the safety-related systems from performing their function (Section 8.2).

The onsite AC source of electrical power consists of four diesel generators, one connected and used exclusively for Unit 1 train A, one connected and used exclusively for Unit 1 train B, one connected and used exclusively for Unit 2 train A, and one connected and used exclusively for Unit 2 train B. One diesel generator is capable of supplying sufficient power for the operation of the minimum safety features required for the unit during a postulated LOCA. However, during a postulated LOCA, both diesel generators start automatically, connect to the two safety features buses of the unit, and sequentially start the safety-related loads.

The two safety features buses and their associated diesel generators are so arranged that a failure of a single component will not prevent the power supply systems from performing their function (Section 8.3).

In conclusion, for each unit, two DC battery systems are provided in physically separated rooms; each one is adequate to supply the DC control power required for the engineered safety features. Failure of a single component in this system does not impair control of the minimum safety features required to maintain each unit in a safe condition. See Section 8.3 for failure mode analysis.

3.1.2.9 Criterion 18 - Inspection and Testing of Electric Power System

"Electric power systems important to safety shall be designed to permit appropriate periodic inspection and testing of important areas and features, such as wiring, insulation, connections, and switchboards, to assess the continuity of the systems and the condition of their components. The systems shall be designed with a capability to test periodically (1) the operability and functional performance of the components of the systems, such as onsite power sources, relays, switches, and buses and (2) the operability of the systems as a whole and, under conditions as

close to design as practical, the full operational sequence that brings the systems into operation, including operation of applicable portions of the protection system and the transfer of power among the nuclear power unit, the offsite power system, and the onsite power system [1]."

Discussion

All Class 1E power systems are designed to permit periodic inspection and testing. The testing procedure simulates a loss of bus voltage to start a diesel generator and connect it to the bus. Full load testing of the diesel generator is performed by manually synchronizing the generator to the normal power supply. Under conditions as close to design as practical, these tests, therefore, prove the capability of the power supply system to assess the continuity of the systems and the conditions of components.

Inspections and tests which constitute compliance with this criterion are included in Section 8.3.

3.1.2.10 Criterion 19 Control Room

"A control room shall be provided from which actions can be taken to operate the nuclear power unit safely under normal conditions and to maintain it in a safe condition under accident conditions, including loss-of-coolant accidents. Adequate radiation protection shall be provided to permit access and occupancy of the control room under accident conditions without personnel receiving radiation exposures in excess of 5 rem whole body, or its equivalent to any part of the body, for the duration of the accident."

"Equipment at appropriate locations outside the control room shall be provided (1) with a design capability for prompt hot shutdown of the reactor, including necessary instrumentation and controls to maintain the unit in a safe condition during hot shutdown, and (2) with a potential capability for subsequent cold shutdown of the reactor through the use of suitable procedures [1]."

Discussion

Safe occupancy of the Control Room during abnormal conditions is provided for in the design. The Control Room is in a seismic Category I structure (See Figures 1.2-33 and 13.5-1). Adequate shielding is provided for the Control Room in the event of a design basis accident (DBA), as detailed in Sections 12.1 and 15.6. The Control Room Air-Conditioning System features redundant equipment, and radiation and smoke detectors, with appropriate alarms and interlocks. Provisions are made for Control Room air to be recirculated through high-efficiency particulate air (HEPA) and charcoal filters, as discussed in Sections 6.4, 6.5 and 9.4.

The Control Room is continuously occupied by qualified operating personnel under all operating and accident conditions.

In the unlikely event that access to the Control Room is restricted, the hot shutdown panel, local control stations, or manual operation of critical components can be used to effect hot shutdown from outside the Control Room for an extended period.

Before evacuation takes place the reactor can be manually tripped and neutron level and control rod position can be verified. By use of appropriate procedures and equipment, the unit can also be brought to cold shutdown conditions. (See Section 7.4.)

3.1.3 PROTECTION AND REACTIVITY CONTROL SYSTEMS

3.1.3.1 Criterion 20 - Protection System Functions

"The protection systems shall be designed (1) to initiate automatically the operation of appropriate systems, including the reactivity control systems, and to assure that specified acceptable fuel design limits are not exceeded as a result of anticipated operational occurrences and (2) to sense accident conditions and to initiate the operation of systems and components important to safety [1]."

Discussion

A fully automatic protection system with appropriate redundant channels is provided to cope with transients where insufficient time is available for manual corrective action. The design basis for all protection systems is in accordance with the intent of Institute of Electrical and Electronic Engineers (IEEE) 279-1971 and IEEE 379-1972. The Reactor Protection System automatically initiates a reactor trip when any variable monitored by the system or combination of monitored variables exceeds the normal operating range. Set points are designed to provide an envelope of safe operating conditions with adequate margin for uncertainties to ensure that fuel design limits are not exceeded.

Reactor trip is initiated by removing power to the control rod drive mechanisms (CRDM) of all full-length rod cluster control assemblies (RCCA). This causes the rods to insert by the force of gravity, rapidly reducing the reactor power output. The response and adequacy of the protection system has been verified by analysis of anticipated transients.

Refer to Chapter 7, Instrumentation and Controls, for additional information regarding actuating devices to the protection system.

The ESF Actuation System automatically initiates emergency core cooling and other safeguards functions by sensing accident conditions using redundant analog channels measuring diverse variables. In addition, manual actuation of safeguards can be performed where ample time is available for operator action. In either case, the ESF Actuation System automatically trips the reactor on manual or automatic safety injection signal generation.

3.1.3.2 Criterion 21 - Protection Systems Reliability and Testability

"The protection system shall be designed for high functional reliability and inservice testability commensurate with the safety functions to be performed. Redundancy and independence designed into the protection system shall be sufficient to assure that (1) no single failure results in loss of the protection function and (2) removal from service of any component or channel does not result in loss of the required minimum redundancy unless the acceptable reliability of operation of the protection system can be otherwise demonstrated. The protection system shall be designed to permit periodic testing of its functioning when the reactor is in operation, including a capability to test channels independently to determine failures and losses of redundancy that may have occurred [1]."

Discussion

The protection system is designed for high functional reliability and inservice testability. It is designed with sufficient redundancy and independence to permit acceptable reliability of operation. System design also includes the capability to perform periodic testing of channels independently while the reactor is in operation.

Compliance with this criterion is discussed in detail in Sections 7.2.2.2.3 and 7.3.2.2.5.

3.1.3.3 Criterion 22 - Protection System Independence

"The protection system shall be designed to assure that the effects of natural phenomena, and of normal operating, maintenance, testing, and postulated accident conditions on redundant channels do not result in loss of the protection function, or shall be demonstrated to be acceptable on some other defined basis. Design techniques, such as functional diversity or diversity in component design and principles of operation shall be used to the extent practical to prevent loss of the protection function [1]."

Discussion

Protection system components are designed and arranged so that there is no loss of safety function in the environment accompanying any emergency situation in which the components are required to function. Various means are used to accomplish this. Functional diversity has been designed into the system. The extent of this functional diversity has been evaluated for a wide variety of postulated accidents. Also, diverse protection functions will automatically terminate an accident before intolerable consequences can occur.

Automatic reactor trips are based upon neutron flux measurements, reactor coolant loop (RCL) temperature measurements, pressurizer pressure and level measurements, and reactor coolant pump power underfrequency and undervoltage measurements. Trips can also be initiated manually or by safety injection signal (Section 7.2).

With regard to the ESF Actuation System, a LOCA safety injection signal can be obtained manually or by automatic initiation from two diverse sets of signals:

1. Low pressurizer pressure

2. High containment pressure

For a steam line break accident, diversity of safety injection signal for automatic actuation is provided by:

1. Low compensated steamline pressure

2. High containment pressure (for a steam line break inside the Containment Building)

For information regarding the independence of the protection system logic, see Section 7.3.

High quality components, conservative design, applicable quality control, inspection, calibration, and tests are used to guard against common mode failure. Qualification testing is performed on

the various safety systems to demonstrate functional operation at normal and post-accident conditions of temperature, humidity, pressure, chemistry, and radiation for specified periods, if required. See Section 3.11 for further details. Typical protection system equipment is subjected to tests under simulated seismic conditions using conservatively large accelerations and applicable frequencies. The test results indicate no loss of the protection function. See Section 3.10 for further details.

3.1.3.4 Criterion 23 - Protection System Failure Modes

"The protection system shall be designed to fail into a safe state or into a state demonstrated to be acceptable on some other defined basis if conditions such as disconnection of the system, loss of energy (e.g., electric power, instrument air), or postulated adverse environments (e.g., extreme heat or cold, fire, pressure, steam, water, and radiation) are experienced [1]."

Discussion

The protection system is designed with due consideration of the most probable failure modes of the components under various perturbations of the environment and energy sources. Each reactor trip channel is designed on the de-energize-to-trip principle so that loss of power, disconnection, open-channel faults, and the majority of internal channel short-circuit faults cause the channel to go into its tripped mode.

The protection system is discussed in Sections 7.2 and 7.3.

3.1.3.5 Criterion 24 - Separation of Protection and Control Systems

"The protection system shall be separated from control systems to the extent that failure of a single control system component or channel, or failure or removal from service of any single protection system component or channel which is common to the control and protection systems leaves intact a system satisfying all reliability, redundancy, and independence requirements of the protection system. Interconnection of the protection and control systems shall be limited so as to assure that safety is not significantly impaired [1]."

Discussion

The protection system is separate and distinct from the control systems. Control systems may be dependent on the protection system because control signals are derived from protection system measurements where applicable. These signals are transferred to the control system by isolation amplifiers, which are classified as protection components. The adequacy of system isolation has been verified by testing under conditions of postulated credible faults. The failure of a single control system component or channel, or failure or removal from service of any single protection system component or channel which is common to the control and protection systems, leaves intact a system satisfying all reliability, redundancy, and independence requirements of the protection system. See Chapter 7 for details. Distinction between channel and train is made in this discussion. The removal of a train from service is allowed only during testing of the train.

3.1.3.6 Criterion 25 - Protection System Requirements For Reactivity Control Malfunctions

"The protection system shall be designed to assure that specified acceptable fuel design limits are not exceeded for any single malfunction of the reactivity control systems, such as accidental withdrawal (not ejection or dropout) of control rods [1]."

Discussion

The protection system is designed to limit reactivity transients so that fuel design limits are not exceeded. Reactor shutdown by full-length rod insertion is completely independent of the normal control function since the trip breakers interrupt power to the rod mechanisms regardless of existing control signals. In the postulated accidental withdrawal (assumed to be initiated by a control malfunction), flux, temperature, pressure, level, and flow signals are independently generated. Any of these signals (trip demands) can operate the breakers to trip the reactor.

Analyses of the effects of possible malfunctions are discussed in Chapter 15. These analyses show that for postulated dilution during refueling, cold shutdown, hot standby, startup, or manual or automatic operation at power, the operator has ample time to determine the cause of dilution, and to initiate reboration before the shutdown margin is lost. The analyses also show that acceptable fuel damage limits are not exceeded even in the event of a single malfunction of either system.

3.1.3.7 Criterion 26 - Reactivity Control System Redundancy and Capability

"Two independent reactivity control systems of different design principles shall be provided. One of the systems shall use control rods, preferably including a positive means for inserting the rods, and shall be capable of reliably controlling reactivity changes to assure that under conditions of normal operation, including anticipated operational occurrences, and with appropriate margin for malfunctions such as stuck rods, specified acceptable fuel design limits are not exceeded. The second reactivity control system shall be capable of reliably controlling the rate of reactivity changes resulting from planned, normal power changes (including xenon burnout) to assure acceptable fuel design limits are not exceeded. One of the systems shall be capable of holding the reactor core subcritical under cold conditions [1]."

Discussion

Two reactivity control methods are provided. These are rod control cluster assemblies (RCCA) and chemical shim (boric acid). The RCCAs are inserted into the core by the force of gravity.

During operation the shutdown rod banks are fully withdrawn. The full-length control rod system automatically maintains a programmed average reactor temperature compensating for reactivity effects associated with scheduled and transient load changes. The shutdown rod banks, along with the full-length control banks, are designed to shut down the reactor with adequate margin under conditions of normal operation and anticipated operational occurrences, thereby ensuring that specified fuel design limits are not exceeded. The most restrictive period in core life is assumed in all analyses and the most reactive rod cluster is assumed to be in the fully withdrawn position.

The boron system maintains the reactor in the cold shutdown state independent of the position of the control rods and can compensate for xenon burnout transients.

Details of the construction of the RCCA are presented in Chapter 4; the operation is discussed in Chapter 7. The means of controlling the boric acid concentration are described in Chapter 9; performance analyses under accident conditions are included in Chapter 15.

3.1.3.8 Criterion 27 - Combined Reactivity Control Systems Capability

"The reactivity control systems shall be designed to have a combined capability, in conjunction with poison addition by the emergency core cooling system, of reliably controlling reactivity changes to assure that under postulated accident conditions and with appropriate margin for stuck rods the capability to cool the core is maintained [1]."

Discussion

The facility is provided with means of making the core subcritical and maintaining it at that level under any anticipated conditions and with an appropriate margin for contingencies. These means are discussed in detail in Chapters 4 and 9. Combined use of the rod cluster control assemblies and the chemical shim permits the necessary shutdown margin to be maintained during long-term xenon decay and plant cooldown. Upon trip for this determination, the single highest worth control cluster is assumed to be stuck full-out upon trip.

3.1.3.9 Criterion 28 - Reactivity Limits

"The reactivity control systems shall be designed with appropriate limits on the potential amount and rate of reactivity increase to assure that the effects of postulated reactivity accidents can neither (1) result in damage to the reactor coolant pressure boundary greater than limited local yielding nor (2) sufficiently disturb the core, its support structures or other reactor pressure vessel internals to impair significantly the capability to cool the core. These postulated reactivity accidents shall include consideration of rod ejection (unless prevented by positive means), rod dropout, steam line rupture, changes in reactor coolant temperature and pressure, and cold water addition [1]."

Discussion

The maximum reactivity worth of control rods and the maximum rates of reactivity insertion employing control rods are limited to values that prevent rupture of the RCS boundary or disruptions of the core or vessel internals to a degree that could impair the effectiveness of emergency core cooling.

The maximum positive reactivity insertion rates for the withdrawal of RCCA and the dilution of the boric acid in the RCS are limited by the physical design characteristics of the RCCA and of the Chemical and Volume Control System (CVCS). Technical specifications on shutdown margin and on RCCA insertion limits and bank overlaps as functions of power provide additional assurance that the consequences of the postulated accidents are no more severe than those presented in the analyses of Chapter 15. Reactivity insertion rates, dilution, and withdrawal limits are also discussed in Section 4.3. The capability of the CVCS to avoid an inadvertent excessive rate of boron dilution is discussed in Chapter 15.

Assurance of core cooling capability following Condition IV accidents, such as rod ejections, steam line break, and similar accidents, is given by keeping the RCPB stresses within faulted condition limits as specified by applicable ASME codes. Structural deformations are checked also and limited to values that do not jeopardize the operation of necessary safety features.

3.1.3.10 Criterion 29 - Protection Against Anticipated Operational Occurrences

"The protection and reactivity control systems shall be designed to assure an extremely high probability of accomplishing their safety functions in the event of anticipated operational occurrences [1]."

Discussion

The protection and reactivity control systems are designed to ensure extremely high probability of performing their required safety functions in any anticipated operational occurrences. Likely failure modes of system components are designed to be safe modes. Equipment used in these systems is designed, constructed, operated, and maintained with a high level of reliability. In addition, loss of power to the protection system results in a reactor trip. Details of system design are covered in Chapter 7; also refer to the discussions of GDC 20 through 25.

3.1.4 FLUID SYSTEMS

3.1.4.1 Criterion 30 - Quality of Reactor Coolant Pressure Boundary

"Components which are part of the reactor coolant pressure boundary shall be designed, fabricated, erected, and tested to the highest quality standards practical. Means shall be provided for detecting and, to the extent practical, identifying the location of the source of reactor coolant leakage [1]."

Discussion

RCPB components are designed, fabricated, inspected, and tested in conformance with ASME Boiler and Pressure Vessel (B&PV) Code, Section III. All components are classified according to ANSI N18.2-1973 except as noted in Sections 3.2 and 5.2 and are accorded the quality measures appropriate to the classification. The design bases and evaluations of RCPB components are discussed in Chapter 5.

Leakage from components is detected by an increase in the amount of makeup of water required to maintain a normal level in the pressurizer.

The reactor vessel closure joint is provided with a temperature-monitored leakoff between double gaskets. Leakage into the containment is drained to the Containment Building sump.

Leakage is also detected by measuring both the airborne activity and the rate at which condensate is drained from the Containment Building recirculation units. Monitoring the inventory of reactor coolant in the system at the pressurizer, volume control tank, and coolant drain collection tanks makes available an accurate indication of integrated leakage. The RCPB leakage detection system is discussed in Section 5.2.5.

3.1.4.2 Criterion 31 - Fracture Prevention of Reactor Coolant Pressure Boundary

"The reactor coolant pressure boundary shall be designed with sufficient margin to assure that when stressed under operating, maintenance, testing, and postulated accident conditions (1) the boundary behaves in a nonbrittle manner and (2) the probability of rapidly propagating fracture is minimized. The design shall reflect consideration of service temperatures and other conditions of the boundary material under operating, maintenance, testing, and postulated accident conditions and the uncertainties in determining (1) material properties, (2) the effects of irradiation on material properties, (3) residual, steady state, and transient stresses, and (4) size of flaws [1]."

Discussion

Close control is maintained over material selection and fabrication for the RCS to ensure that the boundary behaves in a nonbrittle manner. Those RCS materials which are exposed to the coolant are corrosion resistant, stainless steel, or Inconel. The reference temperature (RT_{NDT}) of the reactor vessel structural steel is established by Charpy V-notch and drop weight tests in accordance with 10 CFR Part 50, Appendix G.

As part of the reactor vessel specification, certain requirements which are not specified by the applicable ASME codes are performed, as follows:

1. Ultrasonic Testing

   In addition to code requirements, the performance of a 100 percent volumetric ultrasonic test of reactor vessel plate for shear wave and a post-hydro-test ultrasonic map of all welds in the pressure vessel are required. Cladding bond ultrasonic inspection to more restrictive requirements than those specified in the codes is also required to preclude interpretation problems during inservice inspection.

2. Radiation Surveillance Program

   In the surveillance programs, the evaluation of the radiation damage is based on pre-irradiation and post-irradiation testing of Charpy V-notch and tensile specimens. These programs evaluate the effect of radiation on the fracture toughness of reactor vessel steels based on the measured change in reference transition temperature and fracture mechanics measurements. These measurements are performed in accordance with American Society Testing Material (ASTM) E 185-1982, Recommended Practice for Surveillance Tests for Nuclear Reactor Vessels, and the requirements of 10 CFR Part 50, Appendix H.

3. Material chemistry (copper, nickel, phosphorous, sulfur, and vanadium) of the reactor vessel core region is controlled to reduce sensitivity to embrittlement, which is caused by irradiation over the life of the plant.

The fabrication and quality control techniques used in the fabrication of the RCS are equivalent to those used for the reactor vessel. The inspections of reactor vessel, pressurizer, piping, pumps, and steam generator are governed by the requirements of ASME Codes (see Chapter 5.0).

Allowable pressure-temperature relationships for plant heatup and cooldown rates are calculated using methods derived from the ASME B&PV Code, Section III, Appendix G, Protection Against Non-Ductile Failure. The approach specifies that allowed stress intensity factors for all vessel operating conditions shall not exceed the reference stress intensity factor (KIR) for the metal temperature at any time. Operating specifications include conservative margins for predicted changes in the material (RT_{NDT}) due to irradiation.

3.1.4.3 Criterion 32 - Inspection of Reactor Coolant Pressure Boundary

"Components which are part of the reactor coolant pressure boundary shall be designed to permit (1) periodic inspection and testing of important areas and features to assess their structural and leaktight integrity, and (2) an appropriate material surveillance program for the reactor pressure vessel [1]."

Discussion

The design of the RCPB provides the capability for accessibility during service life to the entire internal surfaces of the reactor vessel, certain external zones of the vessel (including the nozzle to reactor coolant piping welds and the top and bottom heads), and external surfaces of the reactor coolant piping, except for the area of pipe within the primary shielding concrete. The inspection capability complements the leakage detection systems in assessing pressure boundary component integrity. The RCPB is periodically inspected under the provisions of ASME B&PV Code, Section XI.

Monitoring of changes in the fracture toughness properties of the reactor vessel core region plates forging, weldments, and associated heat-treated zones are performed in accordance with 10 CFR Part 50, Appendix H, Reactor Vessel Material Surveillance Program Requirements. Samples of reactor vessel plate materials are retained and cataloged in case future engineering development shows the need for further testing.

The material properties surveillance program includes not only the conventional tensile and impact tests, but also fracture mechanics specimens. The observed shifts in RT_{NDT} of the core region materials with irradiation are used to confirm the allowable limits calculated for all operational transients.

3.1.4.4 Criterion 33 - Reactor Coolant Makeup

"A system to supply reactor coolant makeup for protection against small breaks in the reactor coolant pressure boundary shall be provided. The system safety function shall be to assure that specified acceptable fuel design limits are not exceeded as a result of reactor coolant loss due to leakage from the reactor coolant pressure boundary and rupture of small piping or other small components which are part of the boundary. The system shall be designed to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished using the piping, pumps, and valves used to maintain coolant inventory during reactor operation [1]."

Discussion

The CVCS provides a means of reactor coolant makeup and adjustment of the boric acid concentration. Makeup is added automatically if the level in the volume control tank falls below the preset level. The high pressure centrifugal charging pumps provided are capable of supplying the required makeup and reactor coolant seal injection flow when power is available from either onsite or offsite electric power systems. These pumps also serve as high head safety injection pumps. Functional reliability is ensured by the provision of standby components ensuring a safe response to probable modes of failure. Details of system design are included in Sections 6.3 and 9.3; details of the electric power system are included in Chapter 8.

3.1.4.5 Criterion 34 - Residual Heat Removal

"A system to remove residual heat shall be provided. The system safety function shall be to transfer fission product decay heat and other residual heat from the reactor core at a rate such that specified acceptable fuel design limits and the design conditions of the reactor coolant pressure boundary are not exceeded.

"Suitable redundancy in components and features, and suitable interconnections, leak detection, and isolation capabilities shall be provided to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure [1]."

Discussion

The Residual Heat Removal (RHR) System in conjunction with the steam and power conversion system is designed to transfer the fission production decay heat and other residual heat from the reactor core within acceptable limits. The Auxiliary Feedwater System provides backup for the steam and power conversion system in this function. The Auxiliary Feedwater System is described in Section 10.4.9.

The crossover from the steam and power conversion system to the RHR System occurs at approximately 350 degrees Fahrenheit (°F) and 425 pounds per square inch gauge (psig).

Suitable redundancy at temperatures below approximately 350°F is accomplished with the two RHR pumps (located in separate compartments with means available for draining and monitoring of leakage), the two heat exchangers, and the associated piping, cabling, and electric power sources. The RHR System is capable of operating on either onsite or offsite electrical power systems.

Suitable redundancy at temperatures above approximately 350°F is provided by the four steam generators and associated piping systems.

Details of system design are in Section 5.4.7.

3.1.4.6 Criterion 35 - Emergency Core Cooling

"A system to provide abundant emergency core cooling shall be provided. The system safety function shall be to transfer heat from the reactor core following any loss of reactor coolant at a

rate such that (1) fuel and clad damage that could interfere with continued effective core cooling is prevented and (2) clad metal water reaction is limited to negligible amounts.

“Suitable redundancy in components and features, and suitable interconnections, leak detection, isolation, and containment capabilities shall be provided to assure that for onsite electrical power system operation (assuming offsite power is not available) and for offsite electrical power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure [1].”

Discussion

An Emergency Core Cooling System (ECCS) is provided to cope with any LOCA in the plant design basis. Abundant cooling water is available in an emergency to transfer heat from the core at a rate sufficient to maintain the core in a coolable geometry and to ensure that clad metal-water reaction is limited to less than 1 percent. Adequate design provisions are made to ensure performance of the required safety functions even with a single failure.

Details of the capability of the systems are included in Section 6.3. An evaluation of the adequacy of the system functions is included in Chapter 15. Performance evaluations are conducted in accordance with 10 CFR Part 50, Section 50.46, and Appendix K.

3.1.4.7 Criterion 36 - Inspection of Emergency Core Cooling System

“The emergency core cooling system shall be designed to permit appropriate periodic inspection of important components, such as spray rings in the reactor pressure vessel, water injection nozzles, and piping, to assure the integrity and capability of the system [1].”

Discussion

Design provisions facilitate access to the critical parts of the injection nozzles, pipes, and valves for visual inspection and for nondestructive inspection where such techniques are desirable and appropriate. The design is in accordance with the requirements of ASME B&PV Code, Section XI, Subarticle IWA-1500.

The components outside the Containment Building are accessible for leaktightness inspection during operation of the reactor.

Details of the inspection program of the ECCS are discussed in Section 6.3 and in the technical specifications.

3.1.4.8 Criterion 37 - Testing of Emergency Core Cooling System

“The emergency core cooling system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leaktight integrity of its components, (2) the operability and performance of the active components of the system, and (3) the operability of the system as a whole and, under conditions as close to design as practical, the performance of the full operational sequence that brings the system into operation, including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of the associated cooling water system [1].”

Discussion

Each active component of the ECCS can be individually actuated on the normal power source and can be transferred to the emergency power source at any time during appropriate plant periodic tests.

Tests can also be performed during shutdown to demonstrate proper automatic operation of the ECCS.

The details of the ECCS testing program are included in Section 6.3.

3.1.4.9 Criterion 38 - Containment Heat Removal

"A system to remove heat from the reactor containment shall be provided. The system safety function shall be to reduce rapidly, consistent with the functioning of other associated systems, the containment pressure and temperature following any loss-of-coolant accident and maintain them at acceptably low levels.

"Suitable redundancy in components and features, and suitable interconnections, leak detection, isolation, and containment capabilities shall be provided to assure that for onsite electric power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure [1]."

Discussion

Two separate containment spray recirculation trains described in Section 6.2.2, each with 100 percent capacity, remove heat from the Containment Building following certain design basis accidents. Each train contains two separate pumps, one heat exchanger, and seven spray headers, and each system is fed from its individual electrical Class 1E bus. Each Class 1E bus is connected to a separate offsite power source and is also connected to its individual onsite power source as described in Section 8.3. Containment isolation valves separate all components from the containment penetrations.

Finally, the containment heat removal system is designed to ensure that the failure of any single active component, assuming the availability of either onsite or offsite power exclusively, does not prevent the system from accomplishing its planned safety function.

3.1.4.10 Criterion 39 - Inspection of Containment Heat Removal System

"The containment heat removal system shall be designed to permit appropriate periodic inspection of important components, such as the torus, sumps, spray nozzles, and piping to assure the integrity and capability of the system.[1]"

Discussion

The containment heat removal system design permits appropriate periodic inspection of the components as described in Section 6.2.2. The essential equipment of the Containment Spray System is outside the Containment Building, except for risers, distribution header piping, spray nozzles, and the containment sump.

The containment sump, spray piping, and nozzles can be inspected during shutdowns. Those portions of the containment spray suction piping between the sump and the Refueling Water Storage Tank as well as associated equipment outside the Containment Building are accessible for inspection.

3.1.4.11 Criterion 40 - Testing of Containment Heat Removal System

"The containment heat removal system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leaktight integrity of its components, (2) the operability and performance of the active components of the system, and (3) the operability of the system as a whole, and, under conditions as close to the design as practical, the performance of the full operational sequence that brings the system into operation, including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of the associated cooling water system [1]."

Discussion

System piping, valves, pumps, heat exchangers, and other components of the containment heat removal system are arranged so that each component can be tested periodically for operability, including transfer to emergency power sources. The delivery capability of the Containment Spray System is tested periodically, to the extent practical, up to the last isolation valve before the spray nozzles. The delivery capability of the spray nozzles is tested periodically by blowing low pressure air through the nozzles and verifying the flow. The Containment Spray Systems are tested for operational sequence as close to the design as practical (see Section 6.2.2).

3.1.4.12 Criterion 41 - Containment Atmosphere Cleanup

"Systems to control fission products, hydrogen, oxygen, and other substances which may be released into the reactor containment shall be provided as necessary to reduce, consistent with the functioning of other associated systems, the concentration and quantity of fission products released to the environment following postulated accidents, and to control the concentration of hydrogen or oxygen and other substances in the containment atmosphere following postulated accidents to assure that containment integrity is maintained.

"Each system shall have suitable redundancy in components and features, and suitable interconnections, leak detection, isolation, and containment capabilities to assure that for onsite electrical power system operation (assuming offsite power is not available) and for offsite electric power system operation (assuming onsite power is not available) its safety function can be accomplished, assuming a single failure [1]."

Discussion

The Containment Spray System includes a chemical additive subsystem using a basic sodium hydroxide solution to enhance post-accident fission product removal efficiency as described in Section 6.5.2. The unit is equipped with two independent spray systems supplied from separate buses, as described in Chapter 8, and either system alone can provide the iodine removal capacity for which credit is taken as described in Section 15.6.

Post-accident combustible gas control is ensured by a Hydrogen Purge System described in Section 6.2.5.

3.1.4.13 Criterion 42 - Inspection of Containment Atmosphere Cleanup Systems

"The containment atmosphere cleanup systems shall be designed to permit appropriate periodic inspection of important components, such as filter frames, ducts, and piping to assure the integrity and capability of the systems [1]."

Discussion

The containment atmosphere cleanup systems are designed and located so that they can be inspected periodically, as detailed in Section 6.5.

3.1.4.14 Criterion 43 - Testing of Containment Atmosphere Cleanup Systems

"The containment atmosphere cleanup systems shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leaktight integrity of its components, (2) the operability and performance of the active components of the systems such as fans, filters, dampers, pumps, and valves and (3) the operability of the systems as a whole and, under conditions as close to design as practical, the performance of the full operational sequence that brings the systems into operation, including operation of applicable portions of the protection system, the transfer between normal and emergency power sources, and the operation of associated systems [1]."

Discussion

The containment atmosphere cleanup system can be tested as follows:

The operation of the spray pumps can be tested by recirculation through a test line to the Refueling Water Storage Tank (RWST). The system valves can be operated through their full travel, and the system can be checked for leaktightness during testing. See Section 6.5 for details. Power transfer is described in Chapter 8.

3.1.4.15 Criterion 44 - Cooling Water

"A system to transfer heat from structures, systems, and components important to safety to an ultimate heat sink shall be provided. The system safety function shall be to transfer the combined heat load of these structures, systems, and components under normal operating and accident conditions.

"Suitable redundancy in components and features, and suitable interconnections, leak detection, and isolation capabilities shall be provided to assure that for onsite electrical power system operation (assuming offsite power is not available) and for offsite electrical power system operation (assuming onsite power is not available) the system safety function can be accomplished, assuming a single failure [1]."

Discussion

The cooling water system for safety-related functions consists of the Station Service Water System (SSWS) and the Component Cooling Water System (CCWS).

The SSWS removes heat from the component cooling heat exchangers.

The CCWS is a closed system. It is designed to remove residual heat from the RCS, cool the letdown flow to the CVCS, cool safety-feature heat loads, and dissipate rejected heat from various plant components. The ultimate heat sink used to dissipate rejected heat from the reactor facility during normal and emergency shutdown conditions is the Safe Shutdown Impoundment (SSI).

Both the SWS and CCWS have two flow loops with redundant pumps, heat exchangers, and piping arrangements. The system is designed to meet the required safety function so that no single failure impairs cooling of essential equipment. See Section 9.2 for details.

The ultimate heat sink has the capability to ensure either the simultaneous shutdown and cooldown of both units or the shutdown and cooldown of one unit simultaneously with the dissipation of post- accident heat from the other unit. Both systems are operable from either the offsite power system or the onsite diesel generators.

3.1.4.16 Criterion 45 - Inspection of Cooling Water System

“The cooling water system shall be designed to permit appropriate periodic inspection of important components, such as heat exchangers and piping, to assure the integrity and capability of the system [1].”

Discussion

The essential components are located in accessible areas and are interchanged periodically to enable inspection to be performed. Appropriate design and layout features to allow periodic visual inspection are also incorporated. The system (the Station Service Water System and the Component Cooling Water System), therefore, can be inspected completely to find and correct incipient malfunctions. See Section 9.2 for details.

3.1.4.17 Criterion 46 - Testing of Cooling Water System

“The cooling water system shall be designed to permit appropriate periodic pressure and functional testing to assure (1) the structural and leaktight integrity of its components, (2) the operability and the performance of the active components of the system, and (3) the operability of the system as a whole, and, under conditions as close to design as practical, the performance of the full operational sequence that brings the system into operation for reactor shutdown and for loss-of-coolant accidents, including operation of applicable portions of the protection system and the transfer between normal and emergency power sources [1].”

Discussion

The design provides for testing of operating system components, which are interchanged periodically, for operability and functional performance.

Redundancy and isolation are provided to allow periodic pressure and functional testing of the system as a whole, including the functional sequence that initiates system operation. At any time during reactor operation, each component of the system can be mechanically connected to the preferred power source to demonstrate operability.

3.1.5 REACTOR CONTAINMENT

3.1.5.1 Criterion 50 - Containment Design Basis

"The reactor containment structure, including access openings, penetrations, and the containment heat removal system shall be designed so that the containment structure and its internal compartments can accommodate, without exceeding the design leakage rate and, with sufficient margin, the calculated pressure and temperature conditions resulting from any loss-of-coolant accident. This margin shall reflect consideration of (1) the effects of potential energy sources which have not been included in the determination of the peak conditions, such as energy in steam generators and energy from metal water and other chemical reactions that may result from degraded emergency core cooling functioning, (2) the limited experience and experimental data available for defining accident phenomena and containment responses, and (3) the conservatism of the calculational model and input parameters [1]."

Discussion

The Containment Building is designed to withstand the pressures and temperatures resulting from a spectrum of LOCAs and secondary system ruptures.

Vital Containment Building subcompartments, such as the steam generator compartment, the reactor cavity, and the pressurizer compartment, are designed to withstand, with a safe margin, peak differential pressures resulting from postulated hot-leg, cold-leg, and pressurizer line breaks.

The assumptions used, other details of the containment pressure temperature transient analysis, and the subcompartmental differential pressure analysis are presented in Section 6.2.1. The structural details are described in Section 3.8. See Section 6.2.2 for containment heat removal; Section 3.8 for access openings; and Sections 3.8, 6.2.4, and 8.3 for penetrations.

3.1.5.2 Criterion 51 - Fracture Prevention of Containment Pressure Boundary

"The reactor containment boundary shall be designed with sufficient margin to assure that under operating, maintenance, testing, and postulated accident conditions (1) its ferritic materials behave in a nonbrittle manner and (2) the probability of rapidly propagating fracture is minimized.

"The design shall reflect consideration of service temperatures and other conditions of the containment boundary material during operation, maintenance, testing, and postulated accident conditions, and the uncertainties in determining (1) material properties, (2) residual, steady state, and transient stresses, and (3) size of flaws [1]."

Discussion

The containment liner material meets the requirements of the Proposed Standard Code for Concrete Reactor Vessels and Containment (April 1973) developed by the joint American Concrete Institute (ACI)-ASME Technical Committee on Concrete Pressure Components for Nuclear Service, which is made up of ACI Committee 359 and the ASME B&PV Code, Section III, Division 2, Subgroup on Concrete Components.

Piping penetrating the containment and forming part of the containment pressure boundary meets the requirements of the ASME B&PV Code, Section III, Division 1, particularly paragraph NE 1131 and Subarticle NE 2300.

These requirements ensure conformance to GDC 51.

For additional details, refer to Section 3.8.

NOTE: Inspection of attachment welds during hydrostatic testing, as required by the ASME B&PV Code was not performed on all type MV containment penetrations for Unit 1.

3.1.5.3 Criterion 52 - Capability For Containment Leakage Rate Testing

"The reactor containment and other equipment which may be subjected to containment test conditions shall be designed so that periodic integrated leakage rate testing can be conducted at containment design pressure [1]."

Discussion

The containment is designed and constructed, and the necessary equipment is provided, to permit periodic integrated leakage-rate tests during plant lifetime, in accordance with reduced pressure-test program requirements of 10 CFR 50, Appendix J. For additional details, see Section 6.2.6 and the technical specifications.

3.1.5.4 Criterion 53 - Provisions for Containment Testing and Inspection

"The reactor containment shall be designed to permit (1) appropriate periodic inspection of all important areas, such as penetrations, (2) an appropriate surveillance program, and (3) periodic testing at containment design pressure of the leaktightness of penetrations which have resilient seals and expansion bellows [1]"

Discussion

In accordance with the requirements stated in Section 6.2.6, the design of the containment permits the periodic inspection, surveillance, and testing of the leaktightness of the containment and its penetrations which have resilient seals and expansion bellows.

3.1.5.5 Criterion 54 - Piping Systems Penetrating Containment

"Piping systems penetrating primary reactor containment shall be provided with leak detection, isolation, and containment capabilities having redundancy, reliability, and performance capabilities which reflect the importance to safety of isolating these piping systems. Such piping systems shall be designed with a capability to test periodically the operability of the isolation valves and associated apparatus and to determine if valve leakage is within acceptable limits [1]."

Discussion

Piping systems penetrating the primary reactor containment are provided with containment isolation valves.

Fittings are provided to permit periodic leakage-rate testing of isolation valves to ensure that leakage is within the acceptable limit. (see the technical specifications).

Penetrations that must be closed for containment isolation have redundant valving and associated apparatus as described in Section 6.2.4. Each valve is tested periodically during normal operation or during shutdown conditions to ensure its operability when needed.

3.1.5.6 Criterion 55 - Reactor Coolant Pressure Boundary Penetrating Containment

"Each line that is part of the reactor coolant pressure boundary and that penetrates primary reactor containment shall be provided with containment isolation valves as follows, unless it can be demonstrated that the containment isolation provisions for a specific class of lines, such as instrument lines, are acceptable on some other defined basis:

1. One locked closed isolation valve inside and one locked closed isolation valve outside containment; or

2. One automatic isolation valve inside and one locked closed isolation valve outside containment; or

3. One locked closed isolation valve inside and one automatic isolation valve outside containment. A simple check valve may not be used as the automatic isolation valve outside containment; or

4. One automatic isolation valve inside and one automatic isolation valve outside containment. A simple check valve may not be used as the automatic isolation valve outside containment.

"Isolation valves outside containment shall be located as close to the containment as practical and, upon loss of actuating power, automatic isolation valves shall be designed to take the position that provides greater safety."

"Other appropriate requirements to minimize the probability or consequences of an accidental rupture of these lines or of lines connected to them shall be provided as necessary to assure adequate safety. Determination of the appropriateness of these requirements, such as higher quality in design, fabrication, and testing, additional provisions for inservice inspection, protection against more severe natural phenomena, and additional isolation valves and containment, shall include consideration of the population density, use characteristics, and physical characteristics of the site environs [1]."

Discussion

Each line that is part of the RCPB and penetrates the containment is provided with isolation valves meeting this criterion. For a detailed description of the Containment Isolation System, see Section 6.2.4. Instrument lines are designed in accordance with the requirements of NRC Regulatory Guide 1.11, Instrument Lines Penetrating Primary Reactor Containment.

3.1.5.7 Criterion 56 - Primary Containment Isolation

“Each line that connects directly to the containment atmosphere and penetrates primary reactor containment shall be provided with containment isolation valves as follows, unless it can be demonstrated that the containment isolation provisions for a specific class of lines, such as instrument lines, are acceptable on some other defined basis:

1. One locked closed isolation valve inside and one locked closed isolation valve outside containment; or

2. One automatic isolation valve inside and one locked closed isolation valve outside containment; or

3. One locked closed isolation valve inside and one automatic isolation valve outside containment. A simple check valve may not be used as the automatic isolation valve outside containment; or

4. One automatic isolation valve inside and one automatic isolation valve outside containment. A simple check valve may not be used as the automatic isolation valve outside containment.

“Isolation valves outside containment shall be located as close to the containment as practical and upon loss of actuating power, automatic isolation valves shall be designed to take the position that provides greater safety [1].”

Discussion

Each line that connects directly to the containment atmosphere and penetrates the containment is provided with redundant containment isolation barriers, except where it is demonstrated that the containment isolation provisions for a specific class of lines, such as instrument lines, are acceptable. The failure of one barrier does not result in the loss of isolation capability. Details are provided in Section 6.2.4 and on the corresponding figures listed in Table 6.2.4-1.

The isolation system for each line is designed to fail in a safe mode by designing the air-operated valves and solenoid-operated valves to fail in the direction of greatest safety.

Motor-operated valves, on the other hand, fail in the mode in which they are when failure occurs. Different power sources for each valve in series ensures that isolation is not defeated by a single failure.

3.1.5.8 Criterion 57 - Closed System Isolation Valves

“Each line that penetrates primary reactor containment and is neither part of the reactor coolant pressure boundary nor connected directly to the containment atmosphere shall have at least one containment isolation valve which shall be either automatic, or locked closed, or capable of remote manual operation. This valve shall be outside the containment and located as close to the containment as practical. A simple check valve may not be used as the automatic isolation valve [1].”

Discussion

Each line that penetrates containment and is not connected directly to the containment atmosphere or is not part of the RCPB has at least one isolation valve located outside the Containment Building near the penetration. Details are provided in Section 6.2.4, and on the corresponding figures listed in Table 6.2.4-1.

3.1.6 FUEL AND RADIOACTIVITY CONTROL

3.1.6.1 Criterion 60 - Control of Releases of Radioactive Materials to the Environment

"The nuclear power unit design shall include means to control suitably the release of radioactive materials in gaseous and liquid effluents and to handle radioactive solid wastes produced during normal reactor operation, including anticipated operational occurrences. Sufficient holdup capacity shall be provided for retention of gaseous and liquid effluents containing radioactive materials, particularly where unfavorable site environmental conditions can be expected to impose unusual operational limitations upon the release of such effluents to the environment [1]."

Discussion

Waste handling systems are incorporated in the facility design for processing and/or retention of radioactive wastes for normal operation and anticipated operational occurrences. Controls and monitoring are provided to ensure that releases during normal operation do not exceed a few percent of the limits of 10 CFR Part 20 and yield offsite doses within the numerical guides for design objectives and limiting conditions of operation set forth in 10 CFR Part 50, Appendix I.

Section 9.4 describes the Primary Plant Ventilation System and non-ESF exhaust units which satisfy GDC-60.

Chapter 11 describes the radioactive waste processing systems' design criteria, holdup capacities, and estimated releases of radioactive effluents to the environment. Compliance with 10 CFR Part 50, Appendix I, is described in Appendix 11A.

3.1.6.2 Criterion 61 - Fuel Storage and Handling and Radioactivity Control

"The fuel storage and handling, radioactive waste, and other systems which may contain radioactivity shall be designed (1) to assure adequate safety under normal and postulated accident conditions. These systems shall be designed with a capability to permit appropriate periodic inspection and testing of components important to safety, (2) with suitable shielding for radiation protection, (3) with appropriate containment, confinement, and filtering systems, (4) with a residual heat removal capability having reliability and testability that reflects the importance to safety of decay heat and other residual heat removal, and (5) to prevent significant reduction in fuel storage coolant inventory under accident conditions [1]."

Discussion

The spent fuel pool, Spent Fuel Pool Cooling and Cleanup System, fuel storage and handling system, radioactive waste processing systems, and other systems that contain radioactivity are designed as follows to ensure adequate safety under normal and postulated accident conditions:

1. Components are designed and located so that appropriate periodic inspection and testing can be performed.

2. All areas of the plant are designed with suitable shielding for radiation protection based on anticipated radiation dose rates and occupancy as discussed in Section 12.3.

3. Individual components that contain significant radioactivity are located in confined areas which are adequately ventilated through appropriate filtering systems. Radioactive waste management is discussed in detail in Chapter 11.

4. The Spent Fuel Pool Cooling and Cleanup System provides cooling to remove residual decay heat from the fuel stored in the spent fuel pool and is designed with redundancy and testability to ensure continued heat removal. A purification loop is provided to remove fission product activity. The Spent Fuel Pool Cooling and Cleanup System is described in Section 9.1.

5. The spent fuel pool is designed so that no postulated accident can cause excessive loss-of-coolant inventory.

6. The Primary Plant Ventilation System is designed to filter the exhaust from the fuel storage and handling, radioactive waste and other systems which may contain radioactivity. The non-ESF exhaust units which satisfy GDC-61 are described in Section 9.4.

The piping connected to the fuel pool is designed so that a significant loss of fuel pool water does not occur because of a pipe rupture. Level instrumentation indicates a reduction in fuel pool water level, and redundant sources of fuel pool water are available.

3.1.6.3 Criterion 62 - Prevention of Criticality in Fuel Storage and Handling

“Criticality in the fuel storage and handling system shall be prevented by physical systems or processes, preferably by use of geometrically safe configurations [1].”

Discussion

Criticality in new fuel and spent fuel storage areas is prevented by both physical separation of fuel assemblies and the presence of borated water in the spent fuel storage pool. For the high density Region I racks in Spent Fuel Pool 1 (SFP1) and Spent Fuel Pool 2 (SFP2), the space between storage positions within the rack module and between individual rack modules is blocked to prevent insertion of fuel assemblies. With respect to criticality, there are no administrative restrictions on placement of spent fuel within the high density Region I racks in SFP1 and SFP2. For the high density Region II racks in SFP1 and SFP2, there is no space between storage positions within the rack module and the space between individual rack modules is blocked to prevent insertion of fuel assemblies. Placement of fuel in the high density Region II racks in SFP1 and SFP2 is administratively controlled based on initial enrichment and minimum burnup requirements. Placement of fuel in the spent fuel pool outside of a rack module, such as a gap between the rack module and pool wall or within the oversized inspection cells, is administratively controlled. A discussion of the SFP Region I and Region II oversized inspection cells is contained in the Technical Specification Bases 3.7.17. For high density Region I racks (any storage configuration) and high density Region II racks (1 out of 4 storage

configuration and 2 out of 4 storage configuration), fuel assembly spacing is such that subcriticality is ensured ($k_{eff} \leq 0.95$) even if assemblies are immersed in unborated water. For high density Region II racks (3 out of 4 storage configuration and 4 out of 4 storage configuration) fuel assembly spacing is such that subcriticality is ensured ($k_{eff} \leq 0.95$ taking credit for soluble boron and $k_{eff} < 1.0$ assuming unborated water). Criticality prevention and criticality considerations are discussed in Sections 9.1 and 4.3, respectively. CPNPP complies with 10CFR50.68(b) in lieu of maintaining a system capable of detecting a criticality event as described in 10 CFR 70.24.

The containment refueling cavity of each unit has additional interim storage space for one (1) low density rack. For the low density rack in the refueling cavity, the space between storage positions within the rack module is blocked to prevent insertion of fuel assemblies. There are no administrative restrictions on placement of spent fuel within the low density rack. Fuel assembly spacing is such that subcriticality is ensured ($k_{eff} \leq 0.95$) even if assemblies are immersed in unborated water.

3.1.6.4 Criterion 63 - Monitoring Fuel and Waste Storage

"Appropriate systems shall be provided in fuel storage and radioactive waste systems and associated handling areas (1) to detect conditions that may result in loss of residual heat removal capability and excessive radiation levels and (2) to initiate appropriate safety actions [1]."

Discussion

Monitoring systems are provided to alarm on excessive temperature or low water level in the spent fuel pool so that appropriate safety actions can be initiated by operator action.

Radiation monitors and alarms are also provided to warn personnel of impending excessive levels of radiation or airborne activity. The Area Radiation Monitoring System is described in Section 12.3.4, and the Process Radiation Monitoring System is detailed in Section 11.5.

3.1.6.5 Criterion 64 - Monitoring Radioactivity Releases

"Means shall be provided for monitoring the reactor containment atmosphere, spaces containing components for recirculation of loss-of-coolant accidents fluids, effluent discharge paths, and the plant environs for radioactivity that may be released from normal operations, including anticipated operational occurrences, and from postulated accidents [1]."

Discussion

The Containment Building atmosphere is monitored continuously during normal and transient operations, using particulate, iodine, and gaseous monitors. Under post-accident conditions, samples of the containment atmosphere provide data of existing airborne radioactive concentrations within the Containment Building. Radioactivity levels contained in the facility effluent discharge paths and in the plant environs are continuously monitored during normal and accident conditions by the Process Radiation Monitoring System described in Section 11.5. Provisions for monitoring the plant areas for radioactivity are included in the Area Radiation Monitoring System described in Section 12.3.4. In addition to the installed detectors, periodic surveys are conducted using portable equipment as discussed in Section 12.5.

REFERENCES

1. Title 10, Code of Federal Regulations, Part 50, Appendix A, General Design Criteria (GDC) for Nuclear Power Plants.

3.2 CLASSIFICATION OF STRUCTURES, COMPONENTS, AND SYSTEMS

3.2.1 SEISMIC CLASSIFICATION

3.2.1.1 Seismic Category I

The plant structures, Reactor Coolant System, engineered safety features, and safety-related systems and components are identified and classified in accordance with the seismic requirements of General Design Criterion 2 of Appendix A to 10 CFR Part 50, General Design Criteria for Nuclear Power Plants [5]. NRC Regulatory Guide 1.29 [2] designates those structures, systems, and components which must remain functional during the safe shutdown earthquake (SSE) as seismic Category I items.

Specifically, all seismic Category I structures, systems, and components must withstand the effects of the SSE and ensure the following:

1. The integrity of the reactor coolant pressure boundary

2. The capability to shut down the reactor and maintain it in a safe shutdown condition, or

3. The capability to prevent or mitigate the consequences of incidents which can result in potential offsite exposures comparable to the guideline exposures of 10 CFR Part 100

The following structures, systems, and components of the Comanche Peak Nuclear Power Plant (CPNPP), including their foundations and supports, are designated as seismic Category I and are designed to withstand the effects of the SSE.

The pertinent quality assurance requirements of Appendix B to 10 CFR Part 50 are applied to all activities affecting the safety-related functions of these structures, systems, and components as shown in Table 17A-1.

3.2.1.1.1 Seismic Category I Structures

1. Containment Buildings including internal structures

2. Safeguards Buildings including diesel generator room and emergency switchgear room

3. Auxiliary Building

4. Electrical and Control Building

5. Fuel Building

6. Service Water Intake Structure

7. Safe Shutdown Impoundment Dam

8. Refueling Water Storage Tanks and Associated Piping Tunnels

9. Reactor Makeup Water Storage Tanks and Associated Piping Tunnels

10. Condensate Storage Tanks and Associated Piping Tunnels

3.2.1.1.2 Seismic Category I Mechanical Systems and Components

All, or portions, of the following mechanical systems or components are seismic Category I as described in Appendix 17A and Table 17A-1:

17A SYSTEM

NO.	SYSTEM
1.	Reactor Coolant System (RCS)
2.	Chemical and Volume Control System (CVCS)
3.	Boron Thermal Regeneration Subsystem (BTRS)
4.	Safety Injection System (SIS)
5.	Residual Heat Removal (RHR) System
6.	Boron Recycle System (BRS)
7.	Containment Spray System (CSS)
8.	Containment Isolation System
9.	Combustible Gas Control System
10.	Component Cooling Water System (CCWS)
11.	Station Service Water System (SSWS)
12.	Main Steam, Reheat and Steam Dump System
13.	Auxiliary Feedwater System
14.	Steam Generator Feedwater System
15.	Diesel Generator, Fuel Oil, and Auxiliary Systems
16.	Spent Fuel Pool Cooling and Cleanup System
17.	Liquid Waste Processing System (LWPS)
18.	Gaseous Waste Processing System (GWPS)
20.	Demineralized Water Makeup System
21.	Vents and Drains System
22.	Containment Ventilation Systems

17A SYSTEM

NO.	SYSTEM
23.	Control Room Air-Conditioning System
24.	Safeguards Building HVAC System
25.	Fuel Building Ventilation System
26.	Diesel Generator Building Ventilation System
27.	Uncontrolled Access Area Ventilation System
28.	Primary Plant Ventilation System
29.	Auxiliary Building HVAC System
30.	Service Water Intake Structure Ventilation System
31.	Chilled Water Systems
32.	Process Sampling System
32 a.	Post Accident Sample System
33.	Fuel Handling Equipment
34.	Containment Equipment
35.	Miscellaneous Handling Equipment
39.	Fire Protection System
40.	Plant Gas System
42.	Tornado Venting Components
43.	Compressed Air Systems
45.	Potable and Sanitary Water System
49.	Pipe Whip Restraints
51.	Uninterruptible Power Supply (UPS) Area Air-Conditioning System

3.2.1.1.3 Seismic Category I Electrical Systems and Components

All, or portions, of the following electrical systems or components are seismic Category I as described in Appendix 17A and Table 17A-1:

17A SYSTEM

NO.	SYSTEM
37.	Electrical Equipment
38.	Radiation Monitoring System
41.	Instrumentation and Control Equipment

3.2.1.1.4 Structures and Systems of Mixed Category

None of the plant structures are classified as partially seismic Category I; however, certain structural items within seismic Category I structures are classified as seismic Category II or non-seismic as appropriate. See Table 17A-1, item 36, for typical structural classifications. The boundaries of seismic Category I portions of systems are shown on the piping and instrumentation diagrams in appropriate sections of the FSAR. Seismic category II piping segments located inside a non-seismic building are described in Section 3.2.2.d.

3.2.1.2 Seismic Category II

Those portions of structures, systems, or components whose continued function is not required but whose failure could reduce the functioning of any seismic Category I system or component required to satisfy the requirements of C.1.a through C.1.q of Regulatory Guide 1.29 to an unacceptable safety level or could result in incapacitating injury to occupants of the control room (See Figure 13.5-1) are designated seismic Category II and are designed and constructed so that the SSE would not cause such failure. The pertinent quality assurance requirements of Appendix B to 10CFR50 are applied to all activities affecting the seismic requirements of those systems and components.

Insulation (for in-Containment piping) and/or some non-nuclear-safety portions of the fluid systems listed in 3.2.1.1.2 are designated seismic Category II in Table 17A-1. NNS portions of the HVAC systems listed in 3.2.1.1.2 are also designated seismic Category II in Table 17A-1.

In addition, all, or portions, of the following mechanical systems or components are seismic Category II as described in Appendix 17A and Table 17A-1:

17A SYSTEM

NO.	SYSTEM
6.	Boron Recycle System (BRS)
9.	Combustible Gas Control System
17.	Liquid Waste Processing System (LWPS)
18.	Gaseous Waste Processing System (GWPS)
19.	Solid Waste Management System (SWMS)

17A SYSTEM

NO.	SYSTEM
25.	Fuel Building Ventilation System
28.	Primary Plant Ventilation System
29	Auxiliary Building HVAC System
32.	Process Sampling System
32.a	Post Accident Sample System
32.b	Secondary Sampling System
33.	Fuel Handling Equipment
34.	Containment Building Miscellaneous Equipment
35.	Miscellaneous Handling Equipment
39.	Fire Protection System
42.	Tornado Venting Components
43.	Compressed Air Systems
46.	Condensate System
47.	Auxiliary Steam System
48.	Steam Generator Blowdown & Cleanup System
49.	Pipe Whip Restraints
50.	Meteorological Instrumentation
51.	Uninterruptible Power Supply (UPS) And Distribution Rooms System
52.	Turbine Plant Cooling Water System
53.	Condensate Polishing System
54.	Condenser Vacuum and Waterbox Priming System
55.	Heater Drains System
56.	Chemical Feed System
58.	Chlorination Systems

3.2.2 SYSTEM QUALITY GROUP CLASSIFICATION

1. General Classification

The system quality group classification applies to fluid systems which are directly depended on to prevent or mitigate the consequences of accidents and malfunctions originating within the reactor coolant pressure boundary or which are relied on to permit reactor shutdown and to maintain the reactor in the safe shutdown condition. These fluid systems are also depended on to contain radioactive material.

Fluid system components important to safety are classified in accordance with the ANSI N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants [4] classification except as described below. This classification system is compatible with requirements of NRC Regulatory Guide 1.26 [1] and is submitted as an alternate acceptable method of meeting the intent of NRC Regulatory Guide 1.26. Quality standards generally applicable to the safety classifications are given in Table 3.2-1. Clarification of the application of quality standards on a system basis is provided in Appendix 17A.

2. Safety Class Definitions

Fluid system components are classified as Safety Class 1, Safety Class 2, Safety Class 3, and non-nuclear-safety (NNS) in accordance with their importance to nuclear safety. This importance, as established by the assigned safety class, shall be applied in the design, materials, manufacture or fabrication, assembly, erection, construction, and operation. A single system may have components in more than one safety class.

The definitions of safety classes listed apply to fluid pressure boundary components and the reactor Containment. Supports that have a nuclear safety function shall be the same safety class as the components that they support. Selection of loading combinations and design methods for supports is the responsibility of the designer.

a. Safety Class 1

Safety Class 1 applies to components whose failure could cause a Condition III or Condition IV loss of reactor coolant as defined in ANSI N18.2 [4]. Condition III occurrences include incidents any one of which may occur during the lifetime of a particular plant. Condition IV occurrences are faults that are not expected to occur but are postulated because their consequences would include the potential for release of significant amounts of radioactive material. Condition IV faults are the most drastic which must be designed against and thus represent the limiting design case. All components located within the reactor coolant pressure boundary (as defined by 10CFR50.2) are Safety Class 1 as required by 10CFR50.55a with the exception of the 3/4” instrument lines and other 3/4” piping connected to the pressurizer above the normal water level and the pressurizer relief and safety valve discharge line. See Section 5.2.1.1 for additional information.

b. Safety Class 2

Safety Class 2 applies to the reactor containment and to the following components:

1. Components of the reactor coolant pressure boundary not in Safety Class 1

2. Components of safety systems that are necessary to remove heat directly from the reactor or reactor containment, to circulate reactor coolant of any safety system purpose, to control within the reactor containment radioactivity released, or to control hydrogen in the reactor containment. A safety system (in this context) is any system that is necessary to shut down the reactor, cool the core or cool another safety system or the reactor containment (after an accident), or it is any system that contains, controls, or reduces radioactivity released in an accident. Only those portions of a system that are designed primarily to accomplish one of those functions, or the failure of which could prevent accomplishing one of those functions, are included.

c. Safety Class 3

Safety Class 3 is applied to those components not in Safety Class 1 or Safety Class 2:

1. The failure of which would result in release to the environment of radioactive gases normally required to be held for decay, or that are necessary to:

   (a) Provide or support any safety system function

   (b) Control outside the reactor containment airborne radioactivity released in an accident, or

   (c) Remove decay heat from spent fuel

d. Non-Nuclear-Safety

Non-nuclear-safety (NNS) applies to portions of the nuclear power plant not covered by Safety Classes 1, 2, or 3 that can influence safe, normal operation or that may contain radioactive fluids. Design of non-nuclear-safety components shall be to applicable industry codes and standards.

The piping Class G designation is used to identify those non-nuclear safety related (NNS) piping and plumbing lines which are not located in Seismic Category I structures.

The piping Class 5 designation is used to identify those non-nuclear safety (NNS) piping and plumbing lines which are located in Seismic Category I structures. Class 5 piping is designed as seismic Category II or non-seismic. Based on

specific routing, all non-seismic Class 5 lines larger than 2" (larger than 4" for air filled copper tubing) are evaluated for their capability to reduce the functioning of Seismic Category I systems and components as defined in position C.1.a through C.1.q of Regulatory Guide 1.29 to an unacceptable level as the result of an SSE and are seismically supported where required. In some special cases as noted in Table 17A-1, Class 5 lines 2" and smaller are designated as NNS, seismic Category II and seismically supported. As such, all activities affecting the design and construction of Seismic Category II systems are subject to the pertinent quality assurance requirements of Appendix B to 10CFR50.

Sleeve-coupled fittings by LOKRING Technology have been qualified for, and can be used in, non-safety, non-critical B31.1 piping application NPS ¼ to 4 for CS and NPS ¼ to 3 for SS, and including piping whose only design function requirement is to remain in place during and following a Seismic Event, that is mounted Seismic Category II inside Seismic Category I Buildings (Reference FDA-2020-000077-01).

High energy piping segments, which are listed in FSAR Section 3.7B.2.8 as located in the Turbine Building, are designated class 5 piping and classified as seismic Category II. These piping segments are seismically analyzed for break postulation and seismic qualification. Additional analyses are performed to demonstrate that unacceptable interactions of these piping segments with non-Category I structures/components will not occur during a seismic event.

Class 5 lines, which are determined as not reducing the functioning of the systems and components described above to an unacceptable degree, and Class G lines are fabricated and installed in accordance with applicable industry codes and standards.

All, or portions of the systems or components that are Seismic Category II are described in Appendix 17A and Table 17A-1.

3. Radioactive Waste Management System (RWMS)

The RWMS designation is used to identify the boundaries of the radioactive waste management system on applicable CPNPP flow diagrams, as the RWMS does not match the functional system boundaries. The RWMS is considered to begin at the interface valve(s) in each line from other systems provided for collecting wastes that may contain radioactive materials and to terminate at the point of controlled discharge to the environment, at the point of recycle back to storage for reuse in the reactor, or at the point of storage of packaged solid wastes prior to shipment offsite to a licensed burial ground. The RWMS portion of the steam generator blowdown system begins at, but does not include, the outermost containment isolation valve on the blowdown line and terminates at the point of controlled discharge to the environment, at the point of the interface with other liquid waste systems, or at the point of recycle back to the secondary system.

The radioactive waste management system (RWMS) includes portions of the following systems and system which interface with them:

LWPS	Liquid Waste Processing System
GWPS	Gaseous Waste Processing System
SWMS	Solid Waste Management System
CVCS	Chemical and Volume Control System
BRS	Boron Recycle System
SGBS	Steam Generator Blowdown System
SFPCCS	Spent Fuel Pool Cooling and Cleanup System

The piping and components comprising the RWMS are classified seismic category I, II or NONE, and ANS Safety Class 2, 3 or NNS based on their functional requirements as defined in Chapter 11 and as shown in Appendix 17A, Table 17A-1.

3.2.3 PLANT DRAWINGS

Tables 3.2-3 and 3.2-4 correlate FSAR figure numbers to plant drawings. In the electronic FSAR, figures are not located after the tables and can only be accessed/viewed by the use of links. These links are located 1) in the List of Figures portion for each chapter Table of Contents, 2) in FSAR text/tables which reference the specific figure and, 3) in Table 3.2-4 (only for FSAR figures that are also Station drawings (i.e., “E” and “M” series drawings)). FSAR Tables 3.2-3 and 3.2-4 provide a cross reference of station drawing numbers to FSAR figure numbers (only for station drawings that are also FSAR figures.) Links to FSAR figures that have multiple sheets will take you to the first sheet. Once you are linked to the first sheet you can step through the remaining sheets of the figure.

3.2.3.1 Flow Diagrams

The safety classes and their interfaces are depicted in the flow diagrams. Coordinates are provided to all piping and instrument drawings (P&ID’s) for locating equipment and interconnections from figure to figure. For meaning of mechanical symbols and abbreviations, see Figure 3.2-1.

Table 3.2-3 include listing of Unit 1 and Common flow diagrams and of Unit 2 flow diagrams and their respective FSAR Figure Numbers. These drawings provide the details required by Regulatory Guide 1.70, Revision 2, for their respective system sections and for FSAR Sections 3.9 (active pumps and valves), 6.2.4 (containment isolation), Section 6.4 (control room habitability) and for Chapter 7 (instrumentation & controls) which refer to them. The flow diagrams also provide the correlation for ANSI Safety Class 1, 2, and 3 and for NNS fluid system components in Table 17A-1.

Table 3.2-4 provides a numerical listing of FSAR Figure Numbers and their respective flow diagrams.

The flow diagrams contain the following significant information for their respective system:

1. All major components, piping, points where connecting systems and subsystems tie together, and instrumentation associated with the system and subsystems.

2. Safety Classes and their interfaces.

3. Tag numbers for equipment referenced in text and tables throughout the FSAR.

4. The extent of systems located in Containment and the extent of system located Seismic Category I buildings.

5. Other details required by RG 1.70, Revision 2.

The flow diagrams contain the following information which is not significant:

1. Line numbers, manual valve/damper numbers (not referenced in text or tables), and air flow rates on HVAC drawings (except for fan capacity and control room ESF filtration air flows (Figure 9.4-1, Sheets 1 & 2)).

2. Drawing notes not related to system operation or function and not related to RG 1.70, Revision 2 content requirements.

3. Reducer sizes and/or their use attached directly to equipment.

4. Other trivial details which satisfy regulatory requirements and commitments, have no potential safety impact, and was not the basis for the NRC safety review as documented in the SER/SSERs.

3.2.3.2 One Line Diagrams

The safety classification of electrical systems is depicted on plant one line diagrams. Table 3.2-3 include listing of Unit 1 and common one line diagrams and of Unit 2 one line diagrams and their respective FSAR figure numbers.

One line diagrams provide the details required by Regulatory Guide 1.70, Revision 2, for AC and DC power distribution systems.

The one line diagrams contain the following significant information for AC and DC power distribution system.

1. Power supply feeders.

2. Loads supplied from each bus.

3. Interconnections between buses, buses and loads, and buses and supplies.

4. Interconnection between safety-related and non safety-related buses.

5. Power distribution equipment capacities.

6. Safety related equipment identification.

7. Power distribution equipment and buses shared between units.

8. Power distribution circuit protective devices.

9. Other details required by RG 1.70, Revision 2.

The one line diagrams contain the following information which is not significant for the purposes of FSAR figures.

1. Cable numbers.

2. Circuit breaker/fuse numbers

3. Protective device, circuit breaker/fuse/relay, manufacturer type and part numbers.

4. Bus compartment and circuit numbers.

5. Control circuit details showing control and indicating devices, e.g., control switches, elector switches, alarms, status lights, and indicating lights.

6. Schematic and wiring drawing references.

7. Notes not related to power distribution system operation/function and not related to RG 1.70 content requirements.

8. Other trivial details which satisfy regulatory requirements and commitments, have no potential safety impact, and was not the basis for the NRC safety review as documented in the SER/SSERs.

3.2.3.3 Radiation Zone and Shield Thickness Drawings

These drawings depict the radiation protection design features for the entire plant. The information depicted in these drawings provides the details required by Regulatory Guide 1.70, Revision 2 with respect to radiation protection design features. These figures are used to trigger engineering ALARA evaluations of proposed plant changes and thus help to ensure that collective station radiation exposure as well as individual radiation exposure remains ALARA throughout the life of the plant.

Table 3.2-3 includes a listing of Unit 1 and Common radiation zone and shield thickness drawings and of Unit 2 radiation zone and shield thickness drawings and their respective FSAR figure numbers.

Table 3.2-4 provides a numerical listing of FSAR Figure Numbers and their respective radiation zone and shield thickness drawings.

The radiation zone and shield thickness drawings contain the following significant information to describe the CPNPP radiation protection design features.

1. Room and Area Design Radiation Zones

2. Shield Wall Thickness

3. Room and Equipment Layouts

4. Room and Area Access Control Features

5. Area and Process Radiation Monitor Locations

REFERENCES

1. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam-, and Radioactive-Waste-Containing Components of Nuclear Power Plants, Revision 3, February 1976, U.S. Nuclear Regulatory Commission.

2. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976, U.S. Nuclear Regulatory Commission.

3. ASME B&PV Code, Section III, Nuclear Power Plant Components, American Society of Mechanical Engineers, 1974.

4. ANSI N18.2, Nuclear Safety Criteria For The Design of Stationary Pressurized Water Reactor Plants, 1973.

5. Code of Federal Regulations, 10 CFR Part 50, Appendix A, General Design Criteria.

6. ASME B&PV Code, Section VIII, Pressure Vessels, Division 1 and Division 2, July 1974.

7. ANSI B31.1-1973, Power Piping, American National Standards Institute, 1973.

8. API Standard 650, including Revision 1, Welded Steel Tanks for Oil Storage, February 1984.

9. AWWA D-100, Welded Steel Tanks for Water Storage, 1984.

TABLE 3.2-1
QUALITY STANDARDS

(Sheet 1 of 2)

Components	Safety Class 1 (Note 1)	Safety Class 2 (Notes 2 and 5)	Safety Class 3 (Notes 2 and 5)	NNS Class (Notes 3 and 4)
Pressure Vessels	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 1, Components NB-3300	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 2, Components NC-3300, NC-3200	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 3, Components ND-3300 and ND-3800	ASME B&PV Code, Section VIII, Division 1
Atmospheric Storage Tanks	N/A	N/A	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 3, Components Article ND-3800	ASME B&PV Code, Section VIII, Division 1; API 620; or AWWA D-100.
0-15 PSIG Tanks	N/A	N/A	N/A	API Standard API-620 or 650
Steel-lined Concrete Storage Tanks	N/A	ACI-318	ACI-318	N/A
Supports	ASME B&PV Code Section III, Nuclear Power Plant Components, Class 1, Components Article NF	ASME B&PV Code Section III, Nuclear Power Plant Components, Class 2, Article NF	ASME B&PV CODE Section III, Nuclear Power Plant Components, Class 3, Article NF, ANSI B31.1	ANSI B31.
Piping	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 1, Components NB-3600	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 2, Components NC-3600	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 3, Components ND-3600	ANSI B31.1, Power Piping

TABLE 3.2-1
QUALITY STANDARDS

(Sheet 2 of 2)

Components	Safety Class 1 (Note 1)	Safety Class 2 (Notes 2 and 5)	Safety Class 3 (Notes 2 and 5)	NNS Class (Notes 3 and 4)
Pumps	ASME B&P V Code, Section III, Nuclear Power Plant Components, Class 1, Components NB-3400	ASME B&PV Code, Section III, Nuclear Power Plant Components Class 2, Components NC-3400	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 3, Components ND-3400	Mfrs. Standards
Valves	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 1, Components NB-3500	ASME B&PV Code, Section III, Nuclear Power Plant Components Class 2, Components NC-3500	ASME B&PV Code, Section III, Nuclear Power Plant Components, Class 3, Components ND-3500	ANSI B31.1

Notes:

1. Pressure vessels which are part of the reactor coolant pressure boundary meet the requirements of 1971 Version of the ASME B&PV Code, with application of all Addenda through and including the Summer 1972 Addenda. Pumps, valves and piping which are part of the reactor coolant pressure boundary meet the requirements of the 1971 Version, with application of all Addenda through and including the Winter 1972 Addenda. Later Code revisions may be used optionally in accordance with requirements of 10 CFR 50.55a.

2. See Reference 3. Code Class 2 and 3 components purchased prior to the effective date of Reference 3 meet the requirements of the Code in effect at the time of purchase (earliest Code specified was the Code in effect on May 1972). Later Code versions, including Code Cases, may be used optionally in accordance with ASME Code implementation requirements.

3. See Reference 6. Later Code versions may be used optionally.

4. See Reference 7, 8, 9. Later Standard versions may be used optionally.

5. See Section 3.8 and Table 17A-1 (for clarification and exceptions).

TABLE 3.2-2
EQUIPMENT CODE AND CLASSIFICATION LIST FLUID SYSTEM COMPONENTS

This table has been superseded by Table 17A-1 in FSAR Appendix 17A. All information formerly provided in this table is provided there.

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 1 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
a) ONE LINE DIAGRAM						
E1-0001	CP-35	110				8.3-1
E1-0001-A	CP-18	103	E2-0001-A	CP-4	103	8.3.13
E1-0002	CP-27	109				8.3.2
E1-0002-A	CP-25	106	E2-0002-A	CP-22	110	8.3-2
E1-0002-B	CP-17	109	E2-0002-B	CP-9	108	8.3-2
E1-0003	CP-30	107	E2-0003	CP-27	107	8.3-5
E1-0003-A	CP-33	108	E2-0003-A	CP-24	107	8.3-5
E1-0003-B	CP-14	107				8.3-5
E1-0004	CP-44	111	E2-0004	CP-32	111	8.3-6
E1-0004-A	CP-32	110	E2-0004-A	CP-27	107	8.3-6
E1-0005	CP-27	106	E2-0005	CP-15	102	8.3-8
E1-0005-A	CP-23	105	E2-0005-A	CP-17	104	8.3-8
E1-0006	CP-15	96	E2-0006	CP-11	107	8.3-7
E1-0006-A	CP-27	105	E2-0006-A	CP-15	107	8.3-7
E1-0007	CP-36	108	E2-0007	CP-27	107	8.3-9
E1-0007-A	CP-35	102	E2-0007-A	CP-27	107	8.3-9
E1-0007-B	CP-37	107	E2-0007-B	CP-21	107	8.3-9
E1-0007-C	CP-42	107	E2-0007-C	CP-29	107	8.3-9
E1-0009	CP-27	107	E2-0009	CP-17	107	8.3-10
E1-0009-A	CP-24	104	E2-0009-A	CP-18	110	8.3.10
E1-0009-B	CP-10	111	E2-0009-B	CP-16	110	8.3-10
E1-0010	CP-45	110				8.3-11
E1-0010-A	CP-40	108				8.3-11
E1-0010-B	CP-42	101				8.3-11
E1-0014	CP-33	110	E2-0014	CP-18	111	8.3-12
E1-0014-A	CP-21	101	E2-0014-A	CP-13	106	8.3-12
E1-0014-B	CP-10	109				8.3-12
E1-0014-C	CP-9	101				8.3-12

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 2 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
E1-0018	CP-25	110	E2-0018	CP-19	107	8.3-15
E1-0018-A	CP-22	107	E2-0018-A	CP-13	105	8.3-15
E1-0018-B	CP-16	104	E2-0018-B	CP-12	106	8.3-15
E1-0018-C	CP-15	107	E2-0018-C	CP-13	107	8.3-15
E1-0018-D	CP-26	105	E2-0018-D	CP-18	105	8.3-15
E1-0018-E	CP-18	102	E2-0018-E	CP-11	101	8.3-15
E1-0018-F	CP-19	105	E2-0018-F	CP-14	105	8.3-15
E1-0018-G	CP-17	102	E2-0018-G	CP-10	104	8.3-15
E1-0018-H	CP-9	99	E2-0018-H	CP-8	99	8.3-15
E1-0018-J	CP-3	96	E2-0018-J	CP-4	96	8.3-15A
E1-0018-01	CP-51	109				8.3-15A
E1-0018-01A	CP-17	109				8.3-15A
E1-0018-01B	CP-32	108				8.3-15A
E1-0018-01C	CP-18	111				8.3-15A
E1-0018-01D	CP-30	99				8.3-15A
E1-0018-01E	CP-33	107				8.3-15A
E1-0018-01F	CP-15	111	E2-0018-01	CP-13	110	8.3-15A
E1-0018-01G	CP-15	111	E2-0018-01A	CP-12	111	8.3-15A
E1-0018-01H	CP-16	101	E2-0018-01B	CP-12	100	8.3-15A
E1-0018-02	CP-19	109	E2-0018-02	CP-10	109	8.3-15B
E1-0018-02A	CP-16	109	E2-0018-02A	CP-8	109	8.3-15B
E1-0018-02B	CP-4	96	E2-0018-02B	CP-3	96	8.3-15B
E1-0018-02C	CP-16	104	E2-0018-02C	CP-9	104	8.3-15B
E1-0018-02D	CP-15	109	E2-0018-02D	CP-10	104	8.3-15B
E1-0018-03	CP-25	105	E2-0018-03	CP-15	108	8.3-15C
E1-0019	CP-27	102	E2-0019	CP-18	102	8.3-14A
E1-0019-A	CP-16	107	E2-0019-A	CP-11	107	8.3-14A
E1-0020	CP-23	110	E2-0020	CP-14	110	8.3-14
E1-0020-A	CP-17	110	E2-0020-A	CP-9	110	8.3-14

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 3 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
E1-0020-B	CP-18	102	E2-0020-B	CP-7	110	8.3-14
E1-0020-C	CP-8	102	E2-0020-C	CP-8	101	8.3-14
E1-0020-D	CP-18	101	E2-0020-D	CP-9	101	8.3-14
E1-0020-E	CP-11	101	E2-0020-E	CP-5	101	8.3-14
E1-0020-F	CP-17	102	E2-0020-F	CP-10	104	8.3-14
E1-0020-G	CP-9	107	E2-0020-G	CP-7	101	8.3-14
E1-0020-H	CP-17	106	E2-0020-H	CP-13	100	8.3-14
E1-0020-J	CP-12	109	E2-0020-J	CP-5	101	8.3-14
E1-0020-K	CP-24	102				8.3-14
E1-0020-L	CP-23	103				8.3-14
E1-0022-02	CP-1	100				8.3-3
E1-0022-04	CP-1	100	E2-0022-04	CP-5	101	8.3-18
E1-0022-05	CP-2	102				8.3-4
E1-0024-03	CP-16	103	E2-0024-03	CP-8	108	8.3-15
E1-0024-03A	CP-9	100	E2-0024-03A	CP-3	96	8.3-15
b) FLOW DIAGRAM						
M1-0200	CP-28	109				3.2-1
M1-0202	CP-34	108	M2-0202	CP-23	108	10.3-1
M1-0202-01	CP-23	111	M2-0202-01	CP-18	111	10.3-1
M1-0202-01A	CP-24	109	M2-0202-01A	CP-14	109	10.3-1
M1-0202-01B	CP-12	102	M2-0202-01B	CP-8	102	10.3-1
M1-0202-02	CP-21	102	M2-0202-02	CP-16	101	10.3-1
M1-0202-03	CP-2	97	M2-0202-03	CP-2	97	10.3-1
M1-0203	CP-29	101	M2-0203	CP-15	108	10.4-9
M1-0203-01	CP-25	102	M2-0203-01	CP-15	99	10.4-9
M1-0203-01A	CP-15	102	M2-0203-01A	CP-8	96	10.4-9
M1-0204	CP-39	105	M2-0204	CP-30	110	10.4-8
M1-0204-02	CP-36	103	M2-0204-02	CP-31	104	10.4-8
M1-0204-03	CP-20	103	M2-0204-03	CP-14	104	10.4-8

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 4 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0205	CP-20	102	M2-0205	CP-13	103	10.4-13
M1-0205-A	CP-21	106	M2-0205-A	CP-14	105	10.4-13
M1-0206	CP-21	107	M2-0206	CP-15	107	10.4-11
M1-0206-01	CP-19	107	M2-0206-01	CP-14	107	10.4-11
M1-0206-02	CP-22	107	M2-0206-02	CP-12	107	10.4-11
M1-0207	CP-28	107	M2-0207	CP-19	107	10.4-14
M1-0207-A	CP-14	106	M2-0207-A	CP-13	105	10.4-14
M1-0207-B	CP-6	104	M2-0207-B	CP-3	105	10.4-14
M1-0208	CP-13	103	M2-0208	CP-10	103	10.4-14
M1-0209	CP-33	103	M2-0209	CP-19	103	10.4-14
M1-0209-A	CP-27	109	M2-0209-A	CP-18	103	10.4-14
M1-0210	CP-32	108	M2-0210	CP-34	111	10.4-5
M1-0210-A	CP-13	104	M2-0210-A	CP-8	104	10.4-5
M1-0210-B	CP-17	106	M2-0210-B	CP-16	102	10.4-5
			M2-0210-C	CP-10	108	10.4-5
M1-0210-01	CP-40	108				10.4-5
M1-0211	CP-44	111	M2-0211	CP-29	103	10.4-3
M1-0211-01	CP-22	103	M2-0211-01	CP-15	108	10.4-3
M1-0211-02	CP-7	103	M2-0211-02	CP-6	108	10.4-3
M1-0212	CP-32	105	M2-0212	CP-18	101	10.4-15
M1-0212-A	CP-28	105	M2-0212-A	CP-24	110	10.4-15
M1-0212-B	CP-38	110	M2-0212-B	CP-17	108	10.4-15
M1-0213	CP-23	101				10.4-16
M1-0213-A	CP-11	105				10.4-16
M1-0213-B	CP-13	108				10.4-16
M1-0213-C	CP-8	95				10.4-16
M1-0213-01	CP-20	99				10.4-16
M1-0214	CP-24	100	M2-0214	CP-13	98	10.4-17
M1-0214-A	CP-23	104	M2-0214-A	CP-22	99	10.4-17

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 5 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0214-B	CP-15	102	M2-0214-B	CP-14	102	10.4-17
M1-0214-01	CP-12	101	M2-0214-C	CP-14	101	10.4-17
M1-0215	CP-18	97	M2-0215	CP-9	97	9.5-57
M1-0215-A	CP-13	97	M2-0215-A	CP-6	97	9.5-57
M1-0215-B	CP-12	95	M2-0215-B	CP-8	95	9.5-56
M1-0215-C	CP-13	95	M2-0215-C	CP-8	95	9.5-56
M1-0215-D	CP-27	110	M2-0215-D	CP-19	111	9.5-55
M1-0215-E	CP-29	111	M2-0215-E	CP-22	110	9.5-55
M1-0215-F	CP-8	106	M2-0215-F	CP-9	106	9.5-52
M1-0215-G	CP-9	106	M2-0215-G	CP-12	106	9.5-52
M1-0215-H	CP-12	109	M2-0215-H	CP-12	111	9.5-54
M1-0215-J	CP-12	109	M2-0215-J	CP-11	111	9.5-54
M1-0216	CP-47	111	M2-0216	CP-21	101	9.3-1
M1-0216-A	CP-42	109	M2-0216-A	CP-10	100	9.3-1
M1-0216-B	CP-21	102	M2-0216-B	CP-20	102	9.3-1
M1-0216-C	CP-11	103				9.3-1
M1-0216-01	CP-28	106				9.3-1
M1-0217	CP-19	100	M2-0217	CP-11	96	9.3-2
M1-0217-A	CP-19	107	M2-0217-A	CP-5	95	9.3-2
M1-0218	CP-14	94	M2-0218	CP-8	94	9.3-1
M1-0218-01	CP-20	102	M2-0218-01	CP-19	101	9.3-1
M1-0218-01A	CP-20	101				9.3-1
M1-0218-02	CP-17	102	M2-0218-02	CP-12	99	9.3-1
			M2-0218-04	CP-10	99	9.3-1
			M2-0218-04A	CP-8	98	9.3-1
M1-0219	CP-21	107	M2-0219	CP-15	103	9.3-1
M1-0219-01	CP-15	94				9.3-1
M1-0219-02	CP-20	104				9.3-1
M1-0219-02A	CP-15	104				9.3-1

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 6 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0219-03	CP-25	101				9.3-1
M1-0220	CP-52	105				9.3-1
M1-0220-01	CP-34	109	M2-0220-01	CP-12	102	9.3-1
M1-0220-01A	CP-28	107	M2-0220-01A	CP-10	110	9.3-1
			M2-0220-02	CP-25	110	9.3-1
M1-0222	CP-23	105	M2-0222	CP-21	103	10.4-20
M1-0222-A	CP-14	105				10.4-20
M1-0222-B	CP-16	103				10.4-20
M1-0222-C	CP-22	103				10.4-20
			M2-0222-01	CP-22	107	10.4-20
M1-0223	CP-18	101	M2-0223	CP-13	101	10.4-4
M1-0223-A	CP-10	109	M2-0223-A	CP-9	110	10.4-4
M1-0224	CP-17	107				9.2-4
M1-0224-A	CP-7	96				9.2-4
M1-0224-01	CP-21	97				9.2-4A
M1-0224-01A	CP-13	97				9.2-4A
M1-0224-01B	CP-18	107				9.2-4A
M1-0224-01C	CP-21	109				9.2-4A
M1-0224-01D	CP-17	106				9.2-4A
M1-0224-01E	CP-15	106				9.2-4A
M1-0224-01F	CP-8	109				9.2-4A
M1-0224-01G	CP-9	109				9.2-4A
M1-0224-01H	CP-10	106				9.2-4A
M1-0224-01J	CP-4	97				9.2-4A
M1-0224-01K	CP-2	94				9.2-4A
M1-0224-01L	CP-7	100				9.2-4A
M1-0224-01M	CP-7	103				9.2-4A
M1-0224-01N	CP-4	109				9.2-4A
M1-0225	CP-23	103				9.5-43

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 7 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0225-01	CP-14	108	M2-0225-01	CP-17	108	9.5-44
M1-0225-02	CP-24	110				9.5-45
M1-0225-02A	CP-15	104				9.5-45
M1-0225-03	CP-18	104	M2-0225-03	CP-13	110	9.5-46
M1-0225-03A	CP-8	97	M2-0225-03A	CP-5	94	9.5-46
M1-0225-04	CP-12	97				9.5-47
M1-0225-04A	CP-16	110				9.5-47
M1-0225-05	CP-12	94	M2-0225-05	CP-6	96	9.5-48
M1-0225-06	CP-35	111				9.5-61
MX-0225-07	CP-19	111				9.5-62
MX-0225-08	CP-11	101				9.5-62
MX-0225-09	CP-11	97				9.5.62
M1-0226	CP-25	107				9.2-7
M1-0227	CP-61	108				9.2-6
M1-0227-01	CP-15	105				9.2-6
M1-0227-03	CP-7	106				9.2-6
M1-0228	CP-32	101	M2-0228	CP-18	111	9.3-4
M1-0228-A	CP-22	102	M2-0228-A	CP-15	102	9.3-4
M1-0228-B	CP-14	100	M2-0228-B	CP-11	100	9.3-4
M1-0228-C	CP-10	99	M2-0228-C	CP-9	98	9.3-4
M1-0228-01	CP-20	100	M2-0228-01	CP-10	100	9.3-4
M1-0228-02	CP-7	95				9.3-4
M1-0229	CP-23	105	M2-0229	CP-19	100	9.2-3
M1-0229-A	CP-22	111	M2-0229-A	CP-15	111	9.2-3
M1-0229-B	CP-26	111	M2-0229-B	CP-16	111	9.2-3
M1-0230	CP-26	98	M2-0230	CP-21	111	9.2-3
M1-0230-A	CP-22	106	M2-0230-A	CP-7	111	9.2-3
M1-0230-B	CP-23	107				9.2-3
M1-0230-C	CP-8	103				9.2-3

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 8 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0231	CP-24	98	M2-0231	CP-16	102	9.2-3
M1-0231-A	CP-15	109	M2-0231-A	CP-15	98	9.2-3
M1-0232	CP-32	111	M2-0232	CP-23	111	6.2.2-1
M1-0232-A	CP-26	109	M2-0232-A	CP-20	107	6.2.2-1
M1-0233	CP-45	111	M2-0233	CP-20	105	9.2-1
M1-0233-A	CP-21	108	M2-0233-A	CP-9	108	9.2-1
M1-0234	CP-31	108	M2-0234	CP-13	108	9.2-1
M1-0235	CP-23	106	M2-0235	CP-17	107	9.1-13
M1-0235-01	CP-20	109				9.1-13
M1-0235-02	CP-26	109				9.1-13
M1-0236	CP-21	106	M2-0236	CP-20	103	9.3-6
M1-0236-A	CP-21	106	M2-0236-A	CP-15	103	9.3-6
M1-0236-B	CP-3	103	M2-0236-B	CP-4	106	9.3-6
M1-0236-01	CP-18	109				9.3-7
M1-0236-01A	CP-13	103				9.3-7
M1-0236-02	CP-17	97				9.3-7
M1-0236-02A	CP-13	104				9.3-7
M1-0236-03	CP-15	103	M2-0236-03	CP-17	109	9.3-7
M1-0236-04	CP-15	106				9.3-7
M1-0237	CP-50	109	M2-0237	CP-21	110	9.3-8
M1-0237-A	CP-14	95	M2-0237-A	CP-9	97	9.3-8
M1-0237-01	CP-17	109	M2-0237-01	CP-9	103	9.3-9
M1-0237-01A	CP-12	109				9.3-9
M1-0238	CP-22	107	M2-0238	CP-14	103	9.3-5
M1-0238-A	CP-12	101	M2-0238-A	CP-12	102	9.3-5
M1-0239	CP-27	101	M2-0239	CP-18	107	10.4-10
M1-0239-01	CP-17	108	M2-0239-01	CP-6	108	10.4-10
M1-0240	CP-17	109				9.2-1
M1-0240-01	CP-13	105				10.4-6

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

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U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0241	CP-32	106	M2-0241	CP-22	107	9.2-5
M1-0241-A	CP-31	105				9.2-5
M1-0241-01	CP-29	109				9.2-5
M1-0242	CP-33	108	M2-0242	CP-18	110	9.2-5
M1-0242-A	CP-20	110				9.2-5
M1-0242-B	CP-16	111				9.2-5
M1-0243	CP-26	99	M2-0243	CP-9	94	10.4-18
M1-0243-A	CP-26	101	M2-0243-A	CP-11	105	10.4-18
M1-0243-B	CP-11	99				10.4-18
M1-0243-02	CP-24	108	M2-0243-02	CP-14	103	10.4-19
M1-0244	CP-15	99	M2-0244	CP-13	110	10.4-7
M1-0244-A	CP-8	99	M2-0244-A	CP-10	110	10.4-7
M1-0244-B	CP-11	104	M2-0244-B	CP-9	100	10.4-7
M1-0244-01	CP-26	110	M2-0244-01	CP-24	98	10.4-7
M1-0244-01A	CP-14	103	M2-0244-01A	CP-15	103	10.4-7
M1-0245	CP-12	108	M2-0245	CP-12	108	3.8-22
			M2-0245-A	CP-6	109	3.8-23
M1-0250	CP-36	111	M2-0250	CP-19	110	5.1-1
M1-0251	CP-34	108	M2-0251	CP-21	103	5.1-1
M1-0253	CP-22	109	M2-0253	CP-7	102	9.3-10
M1-0253-A	CP-10	98	M2-0253-A	CP-16	103	9.3-10
M1-0254	CP-25	102	M2-0254	CP-33	111	9.3-10
M1-0255	CP-30	109	M2-0255	CP-14	107	9.3-10
M1-0255-01	CP-37	110	M2-0255-01	CP-7	96	9.3-10
M1-0255-02	CP-15	108	M2-0255-02	CP-15	107	9.3-10
M1-0256	CP-11	97				9.3-10
M1-0256-A	CP-18	102	M2-0256-A	CP-14	102	9.3-10
			M2-0256-B	CP-6	96	9.3-10
M1-0257	CP-33	109				9.3-10

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

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U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0258	CP-31	106				9.3-11
M1-0259	CP-23	99				9.3-11
M1-0259-A	CP-8	103				9.3-11
M1-0259-01	CP-10	94				9.3-11
M1-0260	CP-37	106	M2-0260	CP-21	103	5.4-6
M1-0261	CP-24	109	M2-0261	CP-16	107	6.3-1
M1-0262	CP-27	107	M2-0262	CP-22	109	6.3-1
M1-0263	CP-17	107	M2-0263	CP-17	107	6.3-1
M1-0263-A	CP-20	107	M2-0263-A	CP-7	95	6.3-1
M1-0263-B	CP-14	103	M2-0263-B	CP-13	103	6.3-1
			M2-0263-C	CP-7	105	6.3-1
M1-0264	CP-19	101	M2-0264	CP-11	101	11.2-2
M1-0265	CP-32	110				11.2-3
M1-0265-01	CP-11	98				11.2-3
M1-0266	CP-43	104				11.2-4
M1-0266-A	CP-18	107				11.2-4
M1-0266-B	CP-5	104				11.2-9
M1-0266-01	CP-12	98				11.2-4
M1-0267	CP-29	99				11.2-5
M1-0268	CP-26	106				11.2.6
M1-0268-01	CP-16	94				11.2-7
M1-0268-02	CP-25	109				11.2-8
M1-0269	CP-24	105				11.3-1
M1-0269-A	CP-19	107				11.3-1
M1-0269-B	CP-22	111				11.3-1
M1-0269-C	CP-1	107				11.3-1
M1-0269-D	CP-2	107				11.3-1
M1-0269-01	CP-22	105				11.3-1
M1-0270	CP-20	105				11.3-1

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

(Sheet 11 of 13)

U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0270-A	CP-6	94				11.3-1
M1-0280	CP-27	107				9.2-15
M1-0280-A	CP-16	108				9.2-15
M1-0281	CP-17	105				9.2-16
M1-0281-A	CP-14	96				9.2-16
M1-0281-B	CP-10	99				9.2-16
M1-0300	CP-12	107	M2-0300	CP-7	109	9.4-5
M1-0300-A	CP-10	105	M2-0300-A	CP-11	101	9.4-5
M1-0301	CP-21	98	M2-0301	CP-13	103	9.4-6
M1-0301-A	CP-18	110	M2-0301-A	CP-15	106	9.4-6
M1-0302	CP-23	107	M2-0302	CP-13	101	9.4-4
M1-0302-A	CP-13	101	M2-0302-A	CP-4	100	9.4-4
M1-0302-B	CP-17	101	M2-0302-B	CP-11	101	9.4-4
M1-0302-C	CP-8	100	M2-0302-C	CP-7	100	9.4-4
M1-0303	CP-19	109				9.4-2
M1-0303-A	CP-8	100				9.4-2
M1-0303-B	CP-15	100				9.4-2
M1-0303-C	CP-16	106				9.4-2
M1-0303-01	CP-21	104				9.4-2
M1-0304	CP-34	101				9.4-1
M1-0304-A	CP-14	99				9.4-1
M1-0304-B	CP-19	101				9.4-1
M1-0304-C	CP-4	102				9.4-1
M1-0304-D	CP-4	101				9.4-1
M1-0304-01	CP-14	105				9.4-14
M1-0304-01A	CP-8	105				9.4-14
M1-0304-01B	CP-13	105				9.4-14
M1-0305	CP-20	108				9.4-8
M1-0305-A	CP-11	108				9.4-8

TABLE 3.2-3
PLANT DRAWING VERSUS FIGURE CROSS REFERENCE

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U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-0306	CP-30	104	M2-0306	CP-23	100	9.4-3
M1-306-A	CP-4	102				9.4-3
M1-0307	CP-46	108	M2-0307	CP-14	96	9.4-11
M1-0307-A	CP-8	102	M2-0307-A	CP-30	111	9.4-11
M1-0307-B	CP-8	105				9.4-11
M1-0307-C	CP-4	94				9.4-11
M1-0308	CP-12	101				9.4-1
M1-0308-A	CP-5	101				9.4-1
M1-0309	CP-30	101				9.4-9
M1-0309-A	CP-20	101				9.4-9
M1-0309-B	CP-11	101				9.4-9
M1-0311	CP-30	101	M2-0311	CP-19	105	9.4-12
M1-0311-A	CP-11	98	M2-0311-A	CP-6	100	9.4.12
M1-0311-B	CP-9	98	M2-0311-B	CP-7	100	9.4-12
M1-0312	CP-18	109				9.4-7
M1-0313	CP-21	104				9.4-15
c) RADIATION ZONES & SHIELD THICKNESS						
M1-3500-001	CP-01	100				12.3-4
M1-3500-002	CP-01	100				12.3-5
M1-3500-003	CP-01	100				12.3-5.1
M1-3500-004	CP-02	102				12.3-6
M1-3500-005	CP-01	100				12.3-7
M1-3500-006	CP-01	100				12.3-8
M1-3500-007	CP-01	100				12.3-9
M1-3500-008	CP-01	100				12.3-10
M1-3500-009	CP-01	100				12.3-11
M1-3500-010	CP-02	100				12.3-12
M1-3500-011	CP-01	100				12.3-13
M1-3500-012	CP-01	100				12.3-14

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U 1 & X Drawing	Rev.	AM	U2 Drawing	Rev.	AM	FSAR Figure No.
M1-3500-013	CP-01	100				12.3-14.1
M1-3500-014	CP-01	100				12.3-15
M1-3500-015	CP-01	100				12.3-16
M1-3500-016	CP-01	100				12.3-17
M1-3500-017	CP-01	100				12.3-18
M1-3500-018	CP-01	100				12.3-19
M1-3500-019	CP-01	100				12.3-20
M1-3500-020	CP-01	100				12.3-21.1
M1-3500-021	CP-01	100				12.3-21.2
M1-3500-022	CP-01	100				12.3-23
M1-3500-023	CP-02	100				12.3-21
			M2-3500-001	CP-01	100	12.3-23.2
			M2-3500-002	CP-01	100	12.3-23.3
			M2-3500-003	CP-02	102	12.3-23.4
			M2-3500-004	CP-01	100	12.3-23.5
			M2-3500-005	CP-01	100	12.3-23.6
			M2-3500-006	CP-01	100	12.3-23.7
			M2-3500-007	CP-01	100	12.3-23.8
			M2-3500-008	CP-01	100	12.3-22.1
			M2-3500-009	CP-01	100	12.3-22.2
			M2-3500-010	CP-01	100	12.3-23.1
			M2-3500-011	CP-01	100	12.3-22

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 1 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
3.2-1	M1-0200	
3.8-22	M1-0245	M2-0245
3.8-23		M2-0245-A
5.1-1	M1-0250	M2-0250
5.1-1	M1-0251	M2-0251
5.4-6	M1-0260	M2-0260
6.2.2-1	M1-0232	M2-0232
6.2.2-1	M1-0232-A	M2-0232-A
6.3-1	M1-0261	M2-0261
6.3-1	M1-0262	M2-0262
6.3-1	M1-0263	M2-0263
6.3-1	M1-0263-A	M2-0263-A
6.3-1	M1-0263-B	M2-0263-B
6.3-1		M2-0263-C
8.3-1	E1-0001	
8.3-2	E1-0002	
8.3-2	E1-0002-A	E2-0002-A
8.3-2	E1-0002-B	E2-0002-B
8.3-3	E1-0022-02	
8.3-4	E1-0022-05	
8.3-5	E1-0003-A	E2-0003-A
8.3-5	E1-0003	E2-0003
8.3-5	E1-0003-B	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 2 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
8.3.6	E1-0004	E2-0004
8.3-6	E1-0004-A	E2-0004-A
8.3-7	E1-0006	E2-0006
8.3-7	E1-0006-A	E2-0006-A
8.3-8	E1-0005	E2-0005
8.3-8	E1-0005-A	E2-0005-A
8.3-9	E1-0007	E2-0007
8.3-9	E1-0007-A	E2-0007-A
8.3-9	E1-0007-B	E2-0007-B
8.3-9	E1-0007-C	E2-0007-C
8.3-10	E1-0009	E2-0009
8.3-10	E1-0009-A	E2-0009-A
8.3-10	E1-0009-B	E2-0009-B
8.3-11	E1-0010	
8.3-11	E1-0010-A	
8.3-11	E1-0010-B	
8.3-12	E1-0014	E2-0014
8.3-12	E1-0014-A	E2-0014-A
8.3-12	E1-0014-B	
8.3-12	E1-0014-C	
8.3-13	E1-0001-A	E2-0001-A
8.3-14	E1-0020	E2-0020
8.3-14	E1-0020-A	E2-0020-A

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 3 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
8.3-14	E1-0020-B	E2-0020-B
8.3-14	E1-0020-C	E2-0020-C
8.3-14	E1-0020-D	E2-0020-D
8.3-14	E1-0020-E	E2-0020-E
8.3-14	E1-0020-F	E2-0020-F
8.3-14	E1-0020-G	E2-0020-G
8.3-14	E1-0020-H	E2-0020-H
8.3-14	E1-0020-J	E2-0020-J
8.3-14	E1-0020-K	
8.3-14	E1-0020-L	
8.3-14A	E1-0019	E2-0019
8.3-14A	E1-0019-A	E2-0019-A
8.3-15	E1-0018	E2-0018
8.3-15	E1-0018-A	E2-0018-A
8.3-15	E1-0018-B	E2-0018-B
8.3-15	E1-0018-C	E2-0018-C
8.3-15	E1-0018-D	E2-0018-D
8.3-15	E1-0018-E	E2-0018-E
8.3-15	E1-0018-F	E2-0018-F
8.3-15	E1-0018-G	E2-0018-G
8.3-15	E1-0018-H	E2-0018-H
8.3-15	E1-0024-03	E2-0024-03
8.3-15	E1-0024-03A	E2-0024-03A

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 4 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
8.3-15A	E1-0018-J	E2-0018-J
8.3-15A	E1-0018-01	
8.3-15A	E1-0018-01A	
8.3-15A	E1-0018-01B	
8.3-15A	E1-0018-01C	
8.3-15A	E1-0018-01D	
8.3-15A	E1-0018-01E	
8.3-15A	E1-0018-01F	E2-0018-01
8.3-15A	E1-0018-01G	E2-0018-01A
8.3-15A	E1-0018-01H	E2-0018-01B
8.3-15B	E1-0018-02	E2-0018-02
8.3-15B	E1-0018-2A	E2-0018-02A
8.3-15B	E1-0018-2B	E2-0018-02B
8.3-15B	E1-0018-2C	E2-0018-02C
8.3-15B	E1-0018-2D	E2-0018-02D
8.3-15C	E1-0018-03	E2-0018-03
8.3-18	E1-0022-04	E2-0022-04
9.1-13	M1-0235	M2-0235
9.1-13	M1-0235-01	
9.1-13	M1-0235-02	
9.2-1	M1-0233	M2-0233
9.2-1	M1-0233-A	M2-0233-A
9.2-1	M1-0234	M2-0234

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 5 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.2-1	M1-0240	
9.2-3	M1-0229	M2-0229
9.2-3	M1-0229-A	M2-0229-A
9.2-3	M1-0229-B	M2-0229-B
9.2-3	M1-0230	M2-0230
9.2-3	M1-0230-A	M2-0230-A
9.2-3	M1-0230-B	
9.2-3	M1-0230-C	
9.2-3	M1-0231	M2-0231
9.2-3	M1-0231-A	
9.2-4	M1-0224	
9.2-4	M1-0224-A	
9.2-4A	M1-0224-01	
9.2-4A	M1-0224-01A	
9.2-4A	M1-0224-01B	
9.2-4A	M1-0224-01C	
9.2-4A	M1-0224-01D	
9.2-4A	M1-0224-01E	
9.2-4A	M1-0224-01F	
9.2-4A	M1-0224-01G	
9.2-4A	M1-0224-01H	
9.2-4A	M1-0224-01J	
9.2-4A	M1-0224-01K	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 6 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.2-4A	M1-0224-01L	
9.2-4A	M1-0224-01M	
9.2-4A	M1-0224-01N	
9.2-5	M1-0241	M2-0241
9.2-5	M1-0241-A	
9.2-5	M1-0241-01	
9.2-5	M1-0242	M2-0242
9.2-5	M1-0242-A	
9.2-5	M1-0242-B	
9.2-6	M1-0227	
9.2-6	M1-0227-01	
9.2-6	M1-0227-03	
9.2-7	M1-0226	
9.2-15	M1-0280	
9.2-15	M1-0280-A	
9.2-16	M1-0281	
9.2-16	M1-0281-A	
9.2-16	M1-0281-B	
9.3-1	M1-0216	M2-0216
9.3-1	M1-0216-A	M2-0216-A
9.3-1	M1-0216-B	M2-0216-B
9.3-1	M1-0216-C	
9.3-1	M1-0216-01	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 7 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.3-1	M1-0218	M2-0218
9.3-1	M1-0218-01	M2-0218-01
9.3-1	M1-0218-01A	
9.3-1	M1-0218-02	M2-0218-02
9.3-1		M2-0218-04
9.3-1		M2-0218-04A
9.3-1	M1-0219	M2-0219
9.3-1	M1-0219-01	
9.3-1	M1-0219-02	
9.3-1	M1-0219-02A	
9.3-1	M1-0219-03	
9.3-1	M1-0220	
9.3-1	M1-0220-01	M2-0220-01
9.3-1	M1-0220-01A	M2-0220-01A
9.3-1		M2-0220-02
9.3-2	M1-0217	M2-0217
9.3-2	M1-0217-A	M2-0217-A
9.3-4	M1-0228	M2-0228
9.3-4	M1-0228-A	M2-0228-A
9.3-4	M1-0228-B	M2-0228-B
9.3-4	M1-0228-C	M2-0228-C
9.3-4	M1-0228-01	M2-0228-01
9.3-4	M1-0228-02	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 8 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.3-5	M1-0238	M2-0238
9.3-5	M1-0238-A	M2-0238-A
9.3-6	M1-0236	M2-0236
9.3-6	M1-0236-A	M2-0236-A
9.3-6	M1-0236-B	M2-0236-B
9.3-7	M1-0236-01	
9.3-7	M1-0236-01A	
9.3-7	M1-0236-02	
9.3-7	M1-0236-02A	
9.3-7	M1-0236-03	M2-0236-03
9.3-7	M1-0236-04	
9.3-8	M1-0237	M2-0237
9.3-8	M1-0237-A	M2-0237-A
9.3-9	M1-0237-01	M2-0237-01
9.3-9	M1-0237-01A	
9.3-10	M1-0253	M2-0253
9.3-10	M1-0253-A	M2-0253-A
9.3-10	M1-0254	M2-0254
9.3-10	M1-0255	M2-0255
9.3-10	M1-0255-01	M2-0255-01
9.3-10	M1-0255-02	M2-0255-02
9.3-10	M1-0256	
9.3-10	M1-0256-A	M2-0256-A

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 9 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
		M2-0256-B
9.3-10	M1-0257	
9.3-11	M1-0258	
9.3-11	M1-0259	
9.3-11	M1-0259-A	
9.3-11	M1-0259-01	
9.4-1	M1-0304	
9.4-1	M1-0304-A	
9.4-1	M1-0304-B	
9.4-1	M1-0304-C	
9.4-1	M1-0304-D	
9.4-1	M1-0308	
9.4-1	M1-0308-A	
9.4-2	M1-0303	
9.4-2	M1-0303-A	
9.4-2	M1-0303-B	
9.4-2	M1-0303-C	
9.4-2	M1-0303-01	
9.4-3	M1-0306	M2-0306
9.4-3	M1-0306-A	
9.4-4	M1-0302	M2-0302
9.4-4	M1-0302-A	M2-0302-A
9.4-4	M1-0302-B	M2-0302-B

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 10 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.4-4	M1-0302-C	M2-0302-C
9.4-5	M1-0300	M2-0300
9.4-5	M1-0300-A	M2-0300-A
9.4-6	M1-0301	M2-0301
9.4-6	M1-0301-A	M2-0301-A
9.4-7	M1-0312	
9.4-8	M1-0305	
9.4-8	M1-0305-A	
9.4-9	M1-0309	
9.4-9	M1-0309-A	
9.4-9	M1-0309-B	
9.4-11	M1-0307	M2-0307
9.4-11	M1-0307-A	M2-0307-A
9.4-11	M1-0307-B	
9.4-11	M1-0307-C	
9.4-12	M1-0311	M2-0311
9.4-12	M1-0311-A	M2-0311-A
9.4-12	M1-0311-B	M2-0311-B
9.4-14	M1-0304-01	
9.4-14	M1-0304-01A	
9.4-14	M1-0304-01B	
9.4-15	M1-0313	
9.5-43	M1-0225	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 11 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
9.5-44	M1-0225-01	M2-0225-01
9.5-45	M1-0225-02	
9.5-45	M1-0225-02A	
9.5-46	M1-0225-03	M2-0225-03
9.5-46	M1-0225-03A	M2-0225-03A
9.5-47	M1-0225-04	
9.5-47	M1-0225-04A	
9.5-48	M1-0225-05	M2-0225-05
9.5-52	M1-0215-F	M2-0215-F
9.5-52	M1-0215-G	M2-0215-G
9.5-54	M1-0215-H	M2-0215-H
9.5-54	M1-0215-J	M2-0215-J
9.5-55	M1-0215-D	M2-0215-D
9.5-55	M1-0215-E	M2-0215-E
9.5-56	M1-0215-B	M2-0215-B
9.5-56	M1-0215-C	M2-0215-C
9.5-57	M1-0215	M2-0215
9.5-57	M1-0215-A	M2-0215-A
9.5-61	M1-0225-06	
9.5-62	MX-0225-07	
9.5-62	MX-0225-08	
9.5.62	MX-0225-09	
10.3-1	M1-0202	M2-0202

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 12 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
10.3-1	M1-0202-01	M2-0202-01
10.3-1	M1-0202-01A	M2-0202-01A
10.3-1	M1-0202-01B	M2-0202-01B
10.3-1	M1-0202-02	M2-0202-02
10.3-1	M1-0202-3	M2-0202-03
10.4-3	M1-0211	M2-0211
10.4-3	M1-0211-01	M2-0211-01
10.4.3	M1-0211-02	M2-0211-02
10.4-4	M1-0223	M2-0223
10.4-4	M1-0223-A	M2-0223-A
10.4-5	M1-0210	M2-0210
10.4-5	M1-0210-A	M2-0210-A
10.4-5	M1-0210-B	M2-0210-B
10.4-5		M2-0210-C
10.4-5	M1-0210-01	
10.4-6	M1-0240-01	
10.4-7	M1-0244	M2-0244
10.4-7	M1-0244-A	M2-0244-A
10.4-7	M1-0244-B	M2-0244-B
10.4-7	M1-0244-01	M2-0244-01
10.4-7	M1-0244-01A	M2-0244-01A
10.4-8	M1-0204	M2-0204
10.4-8	M1-0204-02	M2-0204-02

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 13 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
10.4-8	M1-0204-03	M2-0204-03
10.4-9	M1-0203	M2-0203
10.4-9	M1-0203-01	M2-0203-01
10.4-9	M1-0203-01A	M2-0203-01A
10.4-10	M1-0239	M2-0239
10.4-10	M1-0239-01	M2-0239-01
10.4-11	M1-0206	M2-0206
10.4-11	M1-0206-01	M2-0206-01
10.4-11	M1-0206-02	M2-0206-02
10.4-13	M1-0205	M2-0205
10.4-13	M1-0205-A	M2-0205-A
10.4-14	M1-0207	M2-0207
10.4-14	M1-0207-A	M2-0207-A
10.4-14	M1-0207-B	M2-0207-B
10.4-14	M1-0208	M2-0208
10.4-14	M1-0209	M2-0209
10.4-14	M1-0209-A	M2-0209-A
10.4-15	M1-0212	M2-0212
10.4-15	M1-0212-A	M2-0212-A
10.4-15	M1-0212-B	M2-0212-B
10.4-16	M1-0213	
10.4-16	M1-0213-A	
10.4-16	M1-0213-B	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 14 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
10.4-16	M1-0213-C	
10.4-16	M1-0213-01	
10.4-17	M1-0214	M2-0214
10.4-17	M1-0214-A	M2-0214-A
10.4-17	M1-0214-B	M2-0214-B
10.4-17	M1-0214-01	M2-0214-C
10.4-18	M1-0243	M2-0243
10.4-18	M1-0243-A	M2-0243-A
10.4-18	M1-0243-B	
10.4-19	M1-0243-02	M2-0243-02
10.4-20	M1-0222	M2-0222
10.4-20	M1-222-A	
10.4-20	M1-0222-B	
10.4-20	M1-0222-C	
10.4-20		M2-0222-01
11.2-2	M1-0264	M2-0264
11.2-3	M1-0265	
11.2-3	M1-0265-01	
11.2-4	M1-0266	
11.2-4	M1-0266-A	
11.2-4	M1-0266-01	
11.2-5	M1-0267	
11.2-6	M1-0268	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 15 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
11.2-7	M1-0268-01	
11.2-8	M1-0268-02	
11.2-9	M1-0266-B	
11.3-1	M1-0269	
11.3-1	M1-0269-A	
11.3-1	M1-0269-B	
11.3-1	M1-0269-C	
11.3-1	M1-0269-D	
11.3-1	M1-0269-01	
11.3-1	M1-0270	
11.3-1	M1-0270-A	
12.3-4	M1-3500-001	
12.3-5	M1-3500-002	
12.2-5.1	M1-3500-003	
12.3-6	M1-3500-004	
12.3-7	M1-3500-005	
12.3-8	M1-3500-006	
12.3-9	M1-3500-007	
12.3-10	M1-3500-008	
12.3-11	M1-3500-009	
12.3-12	M1-3500-010	
12.3-13	M1-3500-011	
12.3-14	M1-3500-012	

TABLE 3.2-4
FIGURE v/s PLANT DRAWING CROSS REFERENCE

(Sheet 16 of 16)

FSAR Figure No.	U1&X Drawing	U2 Drawing
12.3-14.1	M1-3500-013	
12.3-15	M1-3500-014	
12.3-16	M1-3500-015	
12.3-17	M1-3500-016	
12.3-18	M1-3500-017	
12.3-19	M1-3500-018	
12.3-20	M1-3500-019	
12.3-21	M1-3500-023	
12.3-21.1	M1-3500-020	
12.3-21.2	M1-3500-021	
12.3-22		M2-3500-011
12.3-22.1		M2-3500-008
12.3-22.2		M2-3500-009
12.3-23	M1-3500-022	
12.3-23.1		M2-3500-010
12.3-23.2		M2-3500-001
12.3-23.3		M2-3500-002
12.3-23.4		M2-3500-003
12.3-23.5		M2-3500-004
12.3-23.6		M2-3500-005
12.3-23.7		M2-3500-006
12.3-23.8		M2-3500-007

3.3 WIND AND TORNADO LOADINGS

3.3.1 WIND LOADINGS

3.3.1.1 Design Wind Velocity

Seismic Category I structures are designed for a basic maximum wind velocity of 80 mph at 30 ft above ground based on a 100-year period of recurrence.

The design wind velocity is selected on the basis of U.S. Weather Bureau data for the general area, as indicated in Figure 1(b) of ASCE Paper No. 3269, “Wind Forces on Structures”.

Wind velocity distribution for various height zones above the ground is in accordance with Table 1 of ASCE Paper No. 3269, “Wind Forces on Structures”, and is as follows:

Height Above Ground (ft)	Wind Velocity (mph)
0 to 50	80
50 to 150	95
150 to 400	110
400 to 700	120

A gust factor of 1.1 is applied to the above wind velocities for all seismic Category 1 structures except the yard tanks for which a gust factor of 1.3 is used. The adequacy of a 1.1 gust factor is confirmed by calculations in accordance with Reference 6.

3.3.1.2 Determination of Applied Forces

The dynamic wind pressure q is defined as $q = 0.00256V^2$, where q is in psf and V, the wind velocity (including the gust factor), is in mph. ASCE Paper No. 3269 is used as a guide in determining wind pressure distribution on the structures. For the containment structure and cylindrical tanks, the coefficients are in accordance with Table 4(f) of ASCE paper 3269. For rectangular structures, the shape coefficient is 1.3. The total wind pressure load on the structure is 1.3q; of the 1.3q, 0.8q is applied as positive pressure on the windward walls, and 0.5q is applied as negative pressure on the leeward walls. Concurrently, the end walls receive 0.8q negative pressure, and the roof receives 0.5q uplift.

Any wind shielding effect that one structure can provide to another is neglected.

Wind loads are applied to the structures as static loads. Only dead loads are considered to resist uplift caused by wind loads. Horizontal wind loads are distributed by the walls to roof and floor diaphragms, and they are distributed from these to the foundations by the lateral load-carrying elements of the structural system.

Load combinations that include wind are described in Section 3.8.

3.3.2 TORNADO LOADINGS

All essential systems and components are protected from tornado induced loads in order to maintain the following:

1. The ability to achieve and maintain a safe shutdown in both units.

2. The integrity of the reactor-coolant pressure boundary.

3. The ability to prevent failures that could lead to radioactive releases resulting in offsite exposures greater than 25 percent of the guideline exposures of 10 CFR Part 100.

These systems and components are located within Seismic Category I buildings as listed in Section 3.2.1.1. The arrangement of these structures can be found on the plot plan, Fig. 1.2-1.

Seismic Category I buildings are vented to the atmosphere in the event of a tornado. These buildings are designed to withstand the loadings due to wind, depressurization and repressurization, and tornado generated missiles. These buildings are also designed to provide protection from the tornado and its effects to Seismic Category I systems and components within these structures.

Tornado impingement is not considered coincident with other low probability events such as a Safe Shutdown Earthquake (SSE), loss of coolant accident (LOCA), or a fire. Because of the potential switchyard damage, a trip of the turbine-generators and loss of offsite power are assumed to result from the design basis tornado.

3.3.2.1 Applicable Design Parameters

The design basis tornado has a peripheral tangential velocity of 300 mph. and a translational velocity of 60 mph.

The design basis tornado drops the local atmospheric pressure 3 psi at a constant rate of 1.0 psi/sec. The pressure is then held at -3 psig for 10 seconds at which time repressurization occurs at a rate of 1.0 psi/sec., for a total duration of 16 seconds. The number 10 seconds for the time held at the negative pressure is chosen for analysis purposes to ensure that complete depressurization is considered.

The spectrum and pertinent characteristics of tornado generated missiles are described in Section 3.5 on Table 3.5-8. Missile loads are considered to act simultaneously with wind and pressure loads.

3.3.2.2 Tornado Protection Design Features

Venting from the building interior compartments to the exterior is provided by blow-out door F-4EX, tornado pressure relief dampers, and tornado pressure relief blowout panels.

1. Blow-out door F-4EX

   The Fuel Building is primarily vented by an opening covered by blow-out door F-4EX. This door is designed to blowout when subjected to tornado induced loading.

2. Tornado Pressure Relief Dampers in Exterior Walls

All exterior tornado dampers are specified to remain closed during winds with speeds up to 119 mph. Tornado dampers in exterior walls are specified to open in either direction when the differential pressure across the damper reaches 0.25 psi in either direction. Concrete missile shields, as shown on Figure 3.3-1, protect the dampers and the building interior from tornado generated missiles.

3. Tornado Pressure Relief Blowout Panels

Airtight blowout panels are used in the venting of the Auxiliary Building, the Control Room and the Safeguards Building. All blowout panels are specified to remain airtight during winds with speeds up to 119 mph. Blowout panels are specified to open when the differential pressure across the panel reaches 0.25 psi. Where a missile can impact a blowout panel, a Seismic Category I building is protected by reinforced concrete missile resisting walls and roof so arranged as to stop a missile as shown in Figure 3.3-2 and Figure 3.3-3. Where a missile is stopped by a concrete barrier and could then enter by gravity into a Category I building protective grating is provided as shown in Figure 3.3-3.

The Diesel Generator Building is primarily vented through the Diesel Generator air intakes.

Venting between interior compartments that are separate fire areas is provided by fire rated architectural door openings, tornado pressure relief dampers (in series with fire dampers), and fire rated HVAC air transfer grilles.

1. Fire Rated Architectural Door Openings

Door E-3AX is specified to blowout of the E&C building into the turbine building. The door is designed to blowout when subjected to tornado induced loading.

Several standard hollow metal doors are held open with fusible link arm-holders designed to close the door during a fire. Wire-mesh doors are also provided for radiation protection (access control) as required. Where other considerations, such as HVAC integrity, require that a hollow metal door be closed, and venting is still required the door is modified in such a way as to allow it to release during a tornado. These doors are also restrained during release to prevent destruction of the door. Fire rated doors are also provided in the same doorway and are also held open with fusible link arm-holders designed to close the door during a fire.

2. Tornado Pressure Relief Dampers in Interior Fire Walls

These dampers are normally closed, where required to provide a ventilation or other environmental boundary, and are specified to open in either direction when the differential pressure across the damper reaches 3" wg. except for selected dampers located in the Safeguards Buildings which have been modified by permanently opening a portion of the modules of each damper to provide an airflow path through areas served by the Electrical Area Ventilation System to the Main Steam and Feedwater Area Ventilation System air inlet duct in the Mechanical Equipment Room (1(2)-104).

The tornado dampers located in the UPS HVAC chase are also normally closed and open in either direction when the differential pressure reaches 7.5" wg.

3. Fire Rated Air Transfer Grilles

Several fire-rated HVAC air transfer grilles are considered for tornado venting. These vents are not attached to ductwork which would potentially block the vent. These vents are normally open.

Venting between interior compartments that are not separate fire areas is provided by access gratings, excess area in wall penetrations, tornado pressure relief dampers and architectural door openings.

1. Tornado Pressure Relief Dampers in Interior Walls

These dampers operate in the same manner as the interior dampers in the fire-rated openings.

2. Architectural Door Openings

Several wire-mesh doors are provided for tornado venting.

Several hollow-metal doors are provided with air transfer louvers. In cases where doors are required by other considerations to be in closed positions, some doors are designed to blow open when subjected to tornado induced loading.

Since the tornado venting characteristics of the buildings are dependent on the positions of architectural doors, these positions must be considered. Applicable plant design documents show all doors that have a required open position for purposes of tornado venting. Seismic Category I structures are designed to withstand the impact loads from the spectrum of tornado generated missiles as shown in Table 3.5-8. Structures are of sufficient thickness to prevent perforation or the generation of secondary missiles by tornado generated missiles to which they can be exposed.

Non-Seismic Category I structures or structural components whose failure under tornado loading could possibly impair the function of Seismic Category I structures, equipment, systems or components are designed to withstand the effects of the tornado loading to the extent that the functional requirements of Seismic Category I structures, equipment, systems and components are not impaired.

3.3.2.3 Method of Analysis

The dynamic wind pressure, q, is calculated from the following equation:

$$q = 0.00256\ V^2$$

where:

q = wind pressure [psf]

v = wind velocity [mph]

The tornado wind velocity is not assumed to vary with height above the ground. The tornado wind pressure is applied as a static load. The pressure distribution and shape coefficients for tornado loadings are the same as the coefficients used for wind, as described in Subsection 3.3.1.2. For all Seismic Category I structures a 360 mph wind velocity is applied uniformly. A gust factor of 1.0 shall be used.

For Seismic Category I buildings other than the Containment the effects of venting are considered in determining the maximum pressure differential across building wall, floor and roof systems. Calculations of differential pressure loads are made utilizing an improved version of the COMPARE - Mod 1 computer code. The subcompartments are represented as volumes that are connected by pressure-dependent variable area junctions or vents. The volume thermodynamics are based on a homogeneous gas, assumed to undergo a quasi-static process. Flow between volumes is based on a one dimensional solution of the momentum equation that includes an accounting for the effects of inertia. Air is assumed to be an ideal gas. The relative humidity is assumed to be zero for conservatism in terms of pressure. For analysis purposes the turbine building is considered to exist at atmospheric environmental conditions. This is justified by the nature of the building design. No flow is considered available through HVAC ductwork due to the potential for damage to it.

Volumes in the flow model represent separate rooms, groups of rooms, or sections of rooms so that the maximum attainable differential pressure across any structure inside a volume is analytically shown to be 0.25 psi.

Tornado missile impact velocities and barrier requirements are presented in Section 3.5. The procedures used for transforming tornado-generated missile loadings, which are considered impactive dynamic loads, into effective loads are described in Section 3.5. Load combinations that include tornados are described on Section 3.8. A load factor of 1.0 is used for tornado loads in the load combination equations. The load factor is justified by the low probability of occurrence at a specific point, by the short duration of associated loading, and by the conservative nature of the maximum tornado wind velocity selected. Non-Seismic Category I equipment that is not in a Seismic category I building and hence can possibly become a tornado generated missile is evaluated as a potential missile. Because the damage potential of such a missile is not greater than the effect of the spectrum of tornado-generated missiles listed in Table 3.5-8 the equipment is not protected from tornado loads.

3.3.2.4 Design Evaluation

The various combinations of tornado effects which the structures will withstand are as follows:

$W_t = W_w$

$W_t = W_p$

$W_t = W_m$

$W_t = W_w + 0.5\ W_p$

$W_t = W_w + W_m$

$W_t = W_w + 0.5\ W_p + W_m$

where:

W_t = total tornado load

W_w = tornado wind load

W_p = tornado differential pressure load

W_m = equivalent tornado missile load

The local design of the exterior walls and roofs of each Seismic Category I structure provides each element with the capability to withstand the most adverse of the six load combinations in each direction. All interior structural components in Seismic Category I structures are verified to withstand the pressure load only. This is justified by the low air velocities induced by the relatively large flow areas inside the rooms. These interior structures include non-load bearing partitions such as fire-walls, windows, and accoustic ceilings which could potentially become missiles if damaged. The same loading combinations are considered to act on all doors that are not specified as a pressure relief door or not required to be open and which could potentially become a missile if damaged. The structural capabilities of hollow metal doors are verified by the ASTM test method.

Since the interior of the building depressurizes, the electrical and I&C cabinets are subject to pressure loadings in the same manner as the buildings. Available vents are measured and pressure loadings are determined in the same manners as the buildings. The pressure loadings are transformed into member stresses and displacements by a stress analysis. All cabinets containing equipment required to achieve and maintain a safe shutdown during and after a tornado, and to operate tornado protection devices are verified to maintain the structural integrity and the operability of the equipment when subjected to tornado pressure loads.

The Safe Shutdown Impoundment (SSI) Dam has outer surfaces or shells of limestone rock with a minimum thickness of 10 feet. This minimum thickness occurs only at the crest of the dam; the rock thickness increases rapidly below the crest to a maximum of more than 100 ft. (each surface) at the base of the embankment. The penetration of any tornado missile from the spectrum presented in Table 3.5-8 into such a surface is far less than the minimum thickness of the limestone. However, even if total penetration of the longest missile of the spectrum considered can somehow occur, and even if the direction of penetration is parallel to and located at the SSI water level, a limestone rock and filter thickness that exceeds the total missile length by 56 percent is present on either dam surface. The longest missile is considered to be a 35-ft. (10.68-m) long utility pole.

Squaw Creek Dam is not a Category I structure and the effect of a tornado missile is not pertinent.

REFERENCES

1. American Society of Civil Engineers, Wind Forces on Structures, Transactions of the American Society of Civil Engineers, Volume 126, Part II, Paper No. 3269, 1961.

2. R. G. Gido, G. J. E. Willcutt, Jr., J. L. Lunsford, and J. S. Gilbert, COMPARE-MOD 1 Code Addendum, NUREG/CR-1185, Addendum 1 (LA-7199-MS, Addendum), June 1980, and Attachment September, 1980.

3. K. C. Mehata, et al. The Kalamazoo Tornado of May 13, 1980. Committee on Natural Disasters, Commission on Sociotechnial Systems. National Research Council, National Academy Press 1981.

4. Amercian Society for Testing and Materials. Standard Test Method for Structural Performance of Exterior Windows, Curtain-Walls, and Doors by Uniform Static Air Pressure Difference, ANSI/ASTM E 330-79.

5. American Society of Civil Engineers, Nuclear Power Plant Tornado Design Considerations Journal of the Power Division, Proc. ASCE, Vol. 97, No. P02.

6. American Nuclear Society, Building Code Requirements for Minimum Design Loads in Buildings and Other Structures, ANSI A58.1-1972.

TABLE 3.3-1
HAS BEEN DELETED.

3.4 WATER LEVEL (FLOOD) DESIGN

3.4.1 FLOOD PROTECTION

Seismic Category I equipment, systems, and components in the Service Water Intake Structure are located above the probable maximum flood (PMF) level and do not require protection from flood. The PMF is considered in the design of the submerged pump shafts.

The following seismic Category I systems and components are located below the PMF level of 789.7 ft:

Safeguards Building:

1. Containment Spray System - containment spray pumps

2. Residual Heat Removal System - RHR pumps

3. Safety Injection System - SIS pumps

4. Vents and Drains System - floor drain sumps and pumps

Grade elevation for the structure in which these seismic Category I system components are located is 810.0 ft. No doors or entries to these structures are located below 810.0 ft. There is, then, no possibility of flooding these components from the PMF condition.

Water stops are used in all construction joints below grade even though the ground water elevation as outlined in Section 2.5.4.6 is no higher than 775.0 ft. Sump pumps are provided to handle the minute amounts of water which could leak into the structures.

Since there is no possibility of flooding any of these seismic Category I components as a result of PMF or ground flooding, no specific flood protection of the above equipment is required.

Electrical and Control Building:

1. Safety Chilled Water System - Chillers and pumps

2. UPS and Distribution Room Air Conditioning System - Air Conditioning Units

Grade elevation for the structure in which these seismic Category I system components are located is 810.0 feet. Doors to this structure are located above PMF level with the exception of three doors into the Turbine Building at Elevation 778 feet (see Figures 1.2-31, 22, and 27).

Flooding of the Turbine Building due to PMF is prevented because the Circulating Water and its interconnecting system are closed systems (See Section 10.4). In addition, if Squaw Creek Reservoir elevation rises above 778 feet, the operators will verify that any Circulating Water System equipment which is to be opened for maintenance will be isolated from Squaw Creek Reservoir by valves and/or stop-gates. Due to the access opening in the operate deck of the circulating water discharge structure, the stop gates isolate the CW Tunnels for lake levels below 778 feet only. Any open CW system at elevation below the PMF level of 789.7 feet must be closed or isolated in order to isolate the lake levels above 778 from the open pathway that could

lead to flooding of the Turbine Building and lower level of the Electrical & Control Building on Elevation 778.

Therefore, it is not necessary to bring the reactor to a cold shutdown for flood conditions.

3.4.2 ANALYSIS PROCEDURES

Grade elevation of safety-related structures, except the Service Water Intake Structure, is 810.0 ft. The PMF level is 789.7 ft. Thus no direct static or dynamic effects of the design basis flood conditions are considered for these structures. Initially, the static hydrostatic pressure that results from the ground water elevation of 775.0 ft. was considered in the design of structures located below this elevation. The design-basis groundwater elevations were later changed to 793 ft-0 in for the Service Water Intake Structure (SWIS) and 810 ft-0 in for other safety-related structures. All safety-related structures have been designed for these groundwater levels. Hydrostatic pressure p is determined from the following equation:

$$p = \gamma h$$

where

γ = the unit weight of water

h = the depth below the groundwater elevation

The Service Water Intake Structure is protected from the effects of wind-wave activity on the Squaw Creek Reservoir by the Safe Shutdown Impoundment Dam, but it is subject to wind-wave activity on the Safe Shutdown Impoundment. A 40-mph wind is assumed coincidental with the PMF (789.7 ft). The wind-wave activity at the Service Water Intake is negligible; therefore, there are no significant dynamic forces to be considered. The maximum water level used in design, as a hydrostatic load, takes into consideration the maximum runup elevation caused by a 40-mph wind coincident with the PMF. An 80-mph sustained wind has been superimposed on a normal water level of 775 ft which results in a lower combined setup and runup elevation than the still water elevation that occurs during the PMF. Therefore, this is not a critical design condition with regard to maximum water level at the Service Water Intake Structure.

The Service Water Intake Structure is not subject to flood currents; therefore, dynamic water force as a result of flood currents is not a design load.

3.4.3 FLOODING FROM TANK RUPTURE

Table 3.4-1 lists all the non-seismic Category I liquid storage tanks larger than 1500 gallons located in the Containment, Safeguards and Auxiliary buildings. The effects of flooding on shutdown capability of the plant were analyzed assuming that the release of tank contents is instantaneous and the sump pumps will not be operational. In all cases, tank rupture results in local flooding only, with no effect on plant safety or the plant safe shutdown capability. Tanks less than or equal to 1500 gallons are bounded by the room design described in Section 9.3.3.3.

TABLE 3.4-1
EFFECT OF LIQUID STORAGE TANKS FAILURE ON THE SAFE SHUTDOWN OF THE PLANT

(Sheet 1 of 3)

ITEM	TANK	SYSTEM	CAPACITY	LOCATION BLDG.	EL	POINT OF COLLECTION OF SPILLS	EFFECTS OF FLOODING SHUTDOWN CAPABILITY OF THE PLANT	COMMENTS
1	Pressurizer relief tank	RC	1800 ft^3	Containment	822'-9"	Sump No. 1 via floor drains	None	
2	DELETED							
3	DELETED							
4	DELETED							
5	DELETED							
6	Laundry and hot shower tank	WP	10,000 gal.	Auxiliary	790'-6"	Sump No. 7 via floor drains	None	
7	Spent resin storage tank	WP	4,100 gal.	Auxiliary	810'-6"	Floor drain tank No. 1 via floor drains	None	
8	Waste holdup tank	WP	10,000 gal.	Auxiliary	790'-6"	Waste holdup tank compartment	None	The tank is located in a watertight compartment which drains to sump No. 7 through normally locked-closed valves
9	DELETED							
10	Recycle holdup tanks	BR	112,000 gal.	Auxiliary	790'-6"	Recycle holdup tanks compartment	None	The tank is located in a watertight compartment which drains to sump No.8 through normally locked-closed valves. Backflow thru floor drains was evaluated.
11	Steam generator blowdown spent resin storage tank	S.G. blow-and cleanup	3700 gal	Auxiliary	790'-6"	Steam generator blowdown tank compartment and AB Sump No. 1 via drains	None	The tank is located in a watertight compartment, the spill will be confined to the tank compartment and AB Sump No. 1
12	DELETED							

TABLE 3.4-1
EFFECT OF LIQUID STORAGE TANKS FAILURE ON THE SAFE SHUTDOWN OF THE PLANT

(Sheet 2 of 3)

ITEM	TANK	SYSTEM	CAPACITY	LOCATION BLDG.	EL	POINT OF COLLECTION OF SPILLS	EFFECTS OF FLOODING SHUTDOWN CAPABILITY OF THE PLANT	COMMENTS
13	Laundry holdup and monitor tank	WP	5,000 gal.	Auxiliary	790'-6"	Sump No. 7 via floor drains	None	
14	Waste evaporator condensate tank	WP	5,000 gal	Auxiliary	790'-6"	Waste evaporator condensate tank compartment and AB Sump No. 6	None	The spill will be confined to the tank compartment and sump
15	Floor drains Waste Monitor tanks	WP	5,000 gal.	Auxiliary	790'-6"	Waste monitor tanks compartment and AB Sump No. 5	None	The spill will be primarily confined to the tank compartment and sump
16	DELETED							
17	DELETED							
18	DELETED							
19	Laundry water head tank	WP	5,000 gal.	Auxiliary	852'-6"	Floor drain tank No. 1 via floor drains	None	
20	Floor drain tanks	WP	10,000 gal.	Safeguard	773'-0"	Floor drain tank compartment	None	The tank is located in a watertight compartment which drains to sump No.1 via through normally locked-closed valves
			30,000 gal.	Auxiliary	790'-6"	Floor drain tank compartment and AB Sump No. 4	None	The contents of the tank will be retained partially inside the compartment, the balance will be disposed via the floor drain system
21	DELETED							
22	Component cooling water drain tank	CCW	2,300 gal.	Safeguards	773'-0"	Component cooling water drain tank compartment and SB Sumps Nos. 1 and 3	None	The spill will be contained inside the tank compartment and sumps

TABLE 3.4-1
EFFECT OF LIQUID STORAGE TANKS FAILURE ON THE SAFE SHUTDOWN OF THE PLANT

(Sheet 3 of 3)

ITEM	TANK	SYSTEM	CAPACITY	LOCATION BLDG.	EL	POINT OF COLLECTION OF SPILLS	EFFECTS OF FLOODING SHUTDOWN CAPABILITY OF THE PLANT	COMMENTS
23	DELETED							
24	DELETED							
25	DELETED							
26	Hot phase separator	CP	11,420 gal.	Fuel	860'-0"	FB Sump No. 2 via floor drains	None	
27	Waste conditioning tank	WP	2,000 gal.	Fuel	841'-0"	FB Sump No. 2 via floor drains	None	
28	Tanks ≤ 1,500 gallons	Various	≤ 1,500 gal	Various	Various	Local room and floor drains	None	See Section 9.3.3.3

3.5 MISSILE PROTECTION

3.5.1 MISSILE SELECTION AND DESCRIPTION

The basic approach to ensure missile protection of systems and components both inside and outside of Containment involves:

1. Examination of systems in order to identify and classify potential missiles

2. Postulation of the generation of potential missiles and the provision of protection against them

3. Insurance of the design adequacy of equipment against the generation of missiles.

3.5.1.1 Internally Generated Missiles (Outside Containment)

The principal design bases are that missiles generated outside of Containment but internal to the plantsite do not cause damage to a system or component, (which is provided for either continued safe operation or shutdown during operating conditions, operational transients, and postulated accident conditions) which are associated with the effects of missile formation. The seismic category and quality group classifications for these systems are identified in Section 3.2.

Equipment outside Containment (see Section 3.5.2) has been evaluated for potential missile sources. The information in this section concerning potential missile sources and systems which require protection from internally generated missiles outside Containment is provided as a result of that evaluation.

The specific systems which require protection from missiles are a function of the missile source analysis and are listed in Table 3.5-1. Systems outside Containment which are required for safe shutdown of the reactor are protected regardless of the missile source.

3.5.1.1.1 Missile Selection

In order to conservatively establish the protection and redundancy requirements for safety-related equipment, missile generation is postulated to occur from high energy systems and rotating equipment. Only one source of a primary missiles is postulated for each event.

3.5.1.1.2 Missile Protection

To prevent a total loss of safety-related system integrity due to internally generated missiles, general design provides two redundant trains for each safety-related system. These trains are separated by missile barriers so that one missile, if it is generated in a system, cannot damage a redundant train of any system.

In addition, systems and components which are identified as potential missile sources are, wherever possible, arranged and oriented so that the tangent structure or component is capable of withstanding the impact, and the design criteria are not violated.

In order to determine the effects of a postulated missile, safety-related equipment is considered and protection is only provided for essential safety-related equipment (as defined in 3.6B.1.2.1) on a case by case basis.

In general, safety-related pumps, tanks, heat exchangers, and other large components are placed in separate compartments surrounded by missile barriers. Piping is run in common pipe chases, and valves are located in common valve rooms. For safety-related equipment, the pipe chases and valve rooms for different trains in general are isolated from each other so that a missile in one pipe chase or valve room cannot affect any equipment in the other chase or room.

Information on postulated turbine missiles is provided in Section 3.5.1.3.

3.5.1.1.3 Design Adequacy of Equipment

Equipment within Westinghouse scope has been reviewed for design adequacy against missile generation.

1. Valves

   Valves in high-pressure systems have been reviewed. As a result of this review, it is concluded that there are no credible sources of missiles associated with valves because there is no single failure associated with any potential valve parts that can result in the generation of a missile. Therefore, there are no missiles associated with valves within Westinghouse scope outside the Containment.

2. Pumps

   Pumps located (within Westinghouse scope) outside the Containment have been evaluated for missiles associated with overspeed failure. The maximum no-load speed of these pumps is equivalent to the operating speed of their motors. Consequently, no pipe break or single failure in the suction line can increase pump speed over that of the no-load condition. Furthermore, there are no pipe break plus single failure combinations which can result in a significant increase in pump suction or discharge head. Therefore, no overspeed is expected, and missiles associated with pumps within Westinghouse scope outside Containment are not credible.

3. Generator Flywheel

   The fabrication specifications of the motor generator set flywheels control the material, and are in accordance with ASTM A 533, 7D, Grade B, Class I, with inspections in accordance with MIL-I-45208A; flame cutting and machining operations are governed to prevent flaws in the material. Nondestructive testing for nil ductility (ASTM E208) and Charpy V-notch (ASTM A 593), as well as ultrasonic testing (ASTM A 578 and A 579) and magnetic particle testing (ASME B&PV Code, Section III, Paragraph NB 2545) are performed on each flywheel material lot. In addition to these requirements, stress calculations are performed in conformance with guidelines of the ASME B&PV Code, Section III, Appendix A, to show that the combined primary stresses caused by centrifugal forces and the shaft interference fit do not exceed one-third of the yield strength at normal operating speeds (1800 rpm) and do not exceed two-thirds of the yield strength at 25 percent overspeed. However, no overspeed is expected because the

flywheel weighs approximately 1300 lb and has dimensions of 35.26 in. in diameter by 4.76 in. in width. The flywheel mounted on the generator shaft, which is directly coupled to the motor shaft, is driven by a 200 hp, 1800 rpm induction motor. The torque developed by the motor is insufficient for overspeed. Therefore, there are no credible missiles from the motor generator sets.

3.5.1.2 Internally Generated Missiles (Inside Containment)

The principal design bases are that missiles generated within the Containment do not cause loss of function of any structure, system or component whose failure could lead to offsite radiological consequences or which is required for safe plant shutdown to a cold condition assuming an additional single failure. The seismic category and quality group classifications for nuclear steam supply systems and components are in Section 3.2.

Equipment (outside of and within Westinghouse scope) inside Containment (see Section 3.5.2) has been evaluated for potential missile sources. As a result of this review, the following information concerning potential missile sources and systems which require protection from internally generated missiles inside Containment is provided:

3.5.1.2.1 Missile Selection

1. Equipment Within Westinghouse Scope

The possibility of a catastrophic failure of the reactor vessel, steam generators, pressurizer, reactor coolant pump casings, and piping that can lead to generation of missiles is not considered credible. Massive and rapid failure of these components is not credible because of the material characteristics; inspection; quality control during fabrication, erection, and operation; conservative design; and prudent operation, as applied to the particular component. The reactor coolant pump flywheel is not considered a source of missiles for the reasons discussed in Section 5.4.1. Nuts and bolts are of negligible concern because of their small amount of stored elastic energy.

NSSS components which are considered to have a potential for Missile Generation inside the reactor containment are as follows:

a. control rod drive mechanism (CRDM) housing plug, drive shaft, and the drive shaft and drive mechanism latched together

b. Certain valves

c. Temperatures and pressure sensor assemblies

d. Pressurizer heaters

Gross failure of a CRDM housing sufficient to allow a control rod to be rapidly ejected from the core is not considered credible for the following reasons:

a. Full-length CRDMs are shop tested at 3107 psig (Unit 1) or 3450 psig and 4105 psig, respectively (Unit 2).

b. Unit 1

The mechanism housings are inspected and leak tested as they are installed on the reactor vessel to the head adapters.

Unit 2

The mechanism housings are individually hydrotested to 3107 psig as they are installed on the reactor vessel to the head adapters, and checked during the hydrotest of the completed RCS.

c. Stress levels in the mechanism are not affected by system transients at power, or by thermal movement of the coolant loops.

d. The mechanism housings are made of Type F304LN (Unit 1) or Type 304 (Unit 2) stainless steel. This material exhibits excellent notch toughness at all temperatures encountered.

However, it is postulated that the top plug on the CRDM becomes loose and it is forced upward by the water jet (Unit 2). The following sequence of events is assumed: the drive shaft and control rod cluster are forced out of the core by the differential pressure of 2500 psi across the drive shaft. The drive shaft and control rod cluster, latched together, are assumed to be fully inserted when the accident starts. After approximately 12 ft of travel, the rod cluster control spider hits the underside of the upper support plate. Upon impact, the flexure arms in the coupling, which join the drive shaft and control cluster, fracture and completely free the drive shaft from the control rod cluster. The control cluster is completely stopped by the upper support plate; however, the drive shaft continues to be accelerated upward to hit the missile shield provided. A description of the missile shield is provided in Section 3.5.1.2.4.

a. Valves

Valve stems are not considered credible sources of missiles.

All the isolation valves installed in the RCS have stems with a back seat. This effectively eliminates the possibility of ejecting valve stems even if the stem threads fail. Stems are threaded into a yoke and are locked in place with a locking bar. The actuator assembly around the yoke consists of other structural components as the frame which surrounds the spring adjustment and spring. Failure of the stem below the threaded region would be blocked by these components which are directly above the stem. The stem cannot be ejected sideways due to the fact it would be guided by the bonnet, packing and/or the trim depending on the location of the failure. In addition, the valves have a backseat or stop which will provide another mechanism against the stem being ejected. The backseat or stop would prevent the lower portion of the stem from exiting the valve [12].

Additional interference is encountered with air- and motor-operated valves.

Valves with a nominal diameter larger than 2 in. have been designed against bonnet body connection failure and subsequent bonnet ejection by the following means:

1. By using the design practice of the ASME B&PV Code, Section VIII, which limits the allowable stress of bolting material to less than 20 percent of its yield strength

2. By using the design practice of the ASME B&PV Code, Section VIII for flange design

3. By controlling the load during the bonnet body connection stud tightening process

The pressure-containing parts are designed in accordance with Class 1, 2, or 3 requirements established by the ASME B&PV Code, Section III.

The proper stud-torquing procedures and the use of torque wrench, with indication of the applied torque, restrict the stress of the studs to the allowable limits established in the ASME B&PV Code, Section III. This stress level is far below the material yield. The complete valves are hydrotested in accordance with the ASME B&PV Code, Section III. The stainless steel bodies and bonnets are volumeterically tested and surface-tested to verify soundness.

Valves with a nominal diameter of 2 in. or smaller are forged and have a screwed bonnet with a canopy seal. The canopy seal is the pressure boundary, while the bonnet threads are designed to withstand the hydrostatic and force. The pressure-containing parts are designed in conformance with the criteria established by the ASME B&PV Code, Section III.

While valve missiles are not generally postulated (the reasons are as previously mentioned), the valves in the region where the pressurizer extends above the operating deck are exceptions. Valves in this region are the pressurizer safety valves, the motor-operated isolation valves in the relief line, the air-operated relief valves, and the air-operated spray valves. Although failure of these valves is also considered not credible, failure of the valve bonnet body bolts is nevertheless postulated, and provisions are made to assure integrity of the Containment liner from the resultant bonnet missile.

To the extent practical, all valves are also oriented so that a missile will strike a barrier.

All BOP valves in high energy systems located inside the Containment Building are evaluated.

b. Thermowells

The only credible source of jet-propelled missiles from the reactor coolant piping and piping systems connected to the RCS is that represented by the temperature and pressure sensor assemblies. The resistance temperature sensor assemblies

can be of two types: with well and without well. Two rupture locations have been postulated: around the weld (or thread) between the temperature element assembly and the boss for the without well element, and the weld (or thread) between the well and boss for the with well element.

A temperature sensor is installed on the reactor coolant pumps close to the radial bearing assembly. A hole is drilled in the gasket and sealed on the internal end of the steel plate. In evaluating missile potential, it is assumed that this plate can break and the pipe plug on the external end of the hole can become a missile.

In addition, it is assumed that the welding between the instrumentation well and the pressurizer wall can fail and that the well and sensor assembly can become a jet-propelled missile.

c. Pressurizer Heaters

Finally, it is assumed that the pressurizer heaters can become loose and become jet-propelled missiles.

2. Equipment Outside of Westinghouse Scope

In order to conservatively establish the protection and redundancy requirements for safety-related equipment, credible missile generation from the following sources are evaluated:

a. Valves in high energy systems;

b. Temperature and Sensor Assemblies in high energy systems;

c. Vessels and Compressed Gas Cylinders in high energy systems; and

d. Dynamic Equipment.

3.5.1.2.2 Missile Description

The CRDM Missiles are summarized in Table 3.5-2.

The missile characteristics of the valve bonnets in the region where the pressurizer extends above the operating deck are given in Table 3.5-3 while the missile characteristics of the piping temperature sensor assemblies are given in Table 3.5-4. A 10-degree expansion, half angle water jet is assumed. The missile characteristics of the piping pressure element assemblies are less severe than those of Table 3.5-4.

The missile characteristics of the reactor coolant pump temperature sensor, the instrumentation well of the pressurizer, and the pressurizer heaters are given in Table 3.5-5. A 10-degree expansion, half-angle water jet is assumed.

The missile characteristics of equipment that are in Westinghouse scope are given in Table 3.5-6.

3.5.1.2.3 Missile Protection

Internal missiles which can be generated from pressure-containing components that are part of the RCS or main steam and feedwater systems are considered in the design of the Containment. The entire RCS and parts of the main steam and feedwater system are surrounded by the secondary shield wall and their components, arranged so that a missile generated from one component does not damage any other component whose failure can lead to an offsite radiation dose in excess of that prescribed in 10 CFR Part 100. Equipment that is not in the Westinghouse scope is either surrounded by concrete barriers, provided with redundant counterparts or designed to adequately resist missile generation or external penetration.

Missile barriers and missile resistant structures are of sufficient thickness and strength to prevent emission of secondary missiles when struck by a primary missile.

3.5.1.2.4 The Control Rod Drive Shaft Missile Shield

Unit 1 Missile Shield

Damage to equipment inside the Containment or to the Containment itself caused by missiles generated by the failure of a CRDM housing (as described in Section 3.5.1.2.1) is prevented by an integral steel missile shield. This missile shield is circular with an approximate outside diameter of 134 inches, constructed of an array of W6 x 16 carbon steel beams welded to the top of a 1.5 inch thick carbon steel plate and built to seismic Category I requirements. The missile shield is part of the CRDM cooling plenum assembly and is approximately 5 feet above the top of the CRDM pressure housings. The missile shield's array of W6 x 16 carbon steel beams are bolted to the plenum's carbon steel support column assembilies. These support columns are bolted to plenum's upper ring girder and lower ring girder. The plenum is mounted to the head assembly and is positioned above the seismic support platform and the CRDM and DRPI cable connections. The plenum assembly is attached to the head lift rig via the lift rig extension sections. The plenum missile shield assembly remains installed of the reactor vessel head assembly during refueling.

Unit 2 Missile Shield

Damage to equipment inside the Containment or to the Containment itself caused by missiles generated by the failure of a CRDM housing (as described in Section 3.5.1.2.1) is prevented by a roll-away steel missile shield. (See Figure 3.5-1) This missile shield is approximately 16 ft. square and constructed of 2-inch-thick carbon steel plate and built to seismic Category I requirements (no function after the LOCA is required). The shield which is located over the refueling cavity and is approximately 2 feet above the top of the CRDM housings is mounted on the manipulator rails for ease of removal during refueling. (See Figure 3.5-2.)

The CRDM cooling fans, along with cable trays containing cables which are connected to the CRDMs, are mounted on the missile shield. Neither the fans nor the cables are nuclear-safety-related and therefore do no have to be protected from damage resulting from missile impact on the shield.

During refueling, the cables are removed from the cable trays and the shield is rolled to the far end of the manipulator rail where it is stored throughout refueling. The fans are not removed

from the shield during refueling. When refueling is completed, the shield is positioned over the reactor and the cables are reinstalled in the cable trays.

3.5.1.2.5 Secondary Missiles

As stated in Section 3.5.1 the basic approach to ensure missile protection of safety systems and equipment inside the Containment Building has been the assurance of the design adequacy of equipment against the generation of missiles rather than to try to contain the missile. The precautions taken to prevent missile generation are as described in Section 3.5. In spite of the design precautions, the generation of potential missiles is postulated and the provision for protection against them is described in Section 3.5. It has not been postulated that secondary missiles are generated by impingement of these missiles.

3.5.1.3 Turbine Missiles

3.5.1.3.1 Turbine Placement and Orientation

Turbine missiles have been evaluated not to be a credible threat for the CPNPP design basis.

3.5.1.3.2 Turbine Overspeed Protection

For a detailed description of the operational mechanism for the overspeed protection components of the turbine, refer to Section 10.2.2. The components reliability analysis has been based on the following assumptions:

1. Failure rates for the turbine valves and controls derived primarily from actual operating experience and calculated with a statistical confidence level of 95 percent

2. The turbine valves are tested per the Preventative Maintenance Program. The overspeed protection system is automatically tested periodically during each shift and is verified operational per the Preventative Maintenance Program. The turbine trip block is tested per the Preventative Maintenance Program.

3. An average of one true load rejection per year

4. Consideration of the turbine extraction of common mode failure possibilities in addition to random failures

Thus, the overspeed failure probability has been estimated to be very small (1.37×10^{-5} per turbine-year for a four flow turbine generator) and per the Preventative Maintenance Program turbine valve test interval [15], and subsequently a high degree of component reliability exists.

3.5.1.3.3 Turbine Valve Testing

Periodic testing of the stop and control valves is necessary to insure reliability and continuity of service; therefore, valve testing for all nuclear units is recommended with a manual test or with the aid of an automatic turbine tester (ATT) per the Preventative Maintenance Program.

For high reliability of the electrohydraulic control (EHC), one primary AC power source for valve testing is provided from safeguard bus; two separate DC sources for control circuits are provided from one +24 VDC system with common neutral. For more details of inservice inspection, refer to Section 10.2.3.6.

3.5.1.3.4 Turbine Characteristics

The turbine is a multicasing, tandem compound, four flow, reaction type, 1800-rpm unit with 46-in. last stage blades. For more detailed characteristics, refer to Section 10.2.2.

For major components, rotor materials and their properties, steam environment, and other properties related to the turbine disk integrity, refer to Section 10.2.3.

Turbine operational and transient characteristics are analytically described in Section 10.2.1 (Design Bases).

3.5.1.4 Missiles Generated by Natural Phenomena

The seismic Category I structures are designed for the effects of tornado-generated missiles. The missile characteristics are shown in Table 3.5-8.

Missiles, generated by floods, are not a design consideration as the PMF is lower than grade elevation. The Service Water Intake Structure is designed for tornado-generated missiles, among other loads. This provides adequate protection for any low velocity waterborne object.

The safety-related systems/components and non-safety related 1E components not contained within reinforced concrete buildings or structures are:

1. Service Water traveling screens

2. Diesel fuel oil storage tanks

3. Instruments located in the turbine building

4. Feedwater control and bypass valve appurtenances

5. Auxiliary Feedwater Pump (AFWPT) Exhaust Stack

The safety related Service Water traveling screens are intended for long term protection of the Service Water System from accumulation of debris (ie. CCW Heat Exchanger tube plates) and for short term protection against floating debris, small enough to pass through the trash racks positioned ahead of the screens, but large enough to potentially damage the Service Water Pumps.

In the event that the exposed parts of both screens, are struck by missiles, the only mechanical consequence is that the screens could lose their ability to rotate. The screens are positioned in channels which prevent the screen from collapsing following a postulated missile strike on the non-protected portion.

Local essential lighting for the Service Water System is located within the reinforced concrete service water structure but adjacent to a door not designed to resist damage from a tornado missile. The essential lighting fixtures, fixture primary supports, lamps and bulbs are classified non-nuclear safety grade. Essential lighting systems are discussed in Section 9.5.3.2.

The diesel fuel oil storage tanks and associated piping are located underground adjacent to the diesel building and are protected by a 1 foot 9 inch thick reinforced concrete slab at grade level. Manholes for the tanks are protected by steel plate covers with a minimum thickness of 1 3/4 inches.

Instruments that actuate anticipatory reactor trip on turbine trip are located in the turbine building as shown on Figure 7.1-3, sheet 16. They are included in the design as good engineering practice. The instruments are procured as Class 1E but are not Nuclear Safety Related and are not protected from tornado missiles because their function is anticipatory, not essential. The anticipatory reactor trip on turbine trip is discussed in Section 7.2.1.1.2 item 6.

The electrical appurtances on the Feedwater control and bypass valves, and the cables and junction boxes serving them, are non-safety related, but designated as Class 1E. Credit is taken for these valves in the analysis of the Feedwater Line Breaks and Main Steam Line Breaks inside Containment as non-safety related backup to the Nuclear Safety Related Feedwater Isolation Valves. This equipment is not protected against tornado missiles because these pipe rupture events are not postulated to occur simultaneous with a tornado.

The Turbine Driven AFW Pump steam supply and exhaust stack has been evaluated for tornado-generated missile strikes resulting from missiles as described in Table 3.5-8 originating from all potential directions. Missile strikes on the steam exhaust stack and the flash tank vent pipes were analyzed using the explicit time-history finite element method. The analysis assumed concurrent missile strikes on the exhaust stack and on the two flash tank vents. The analysis determined that the exhaust stack and the two flash tank vents would not crimp shut in a manner that would prevent the turbine driven pump from performing its safety function nor would it adversely impact any plant SSCs required for safe shutdown.

All other safety related equipment is located within reinforced concrete structures.

The SSW discharge piping which extends from the discharge structure is classified as NNS (see Figure 9.2-1) and is not protected from tornado missiles. An analysis of tornado missile impacts on the exposed piping shows that the flow area reduction is acceptable and SSW would be able to perform its safety functions during and after a tornado.

The Refueling Water Storage Tank vents described in Section 6.3.3.7 and the Condensate Storage tank vents described in Section 9.2.6 are classified as non-Nuclear Safety related and, in accordance with ANSI N18.2a, are designed to break away if struck by a tornado missile. The exterior walls and roofs of all building housing safety-related equipment are of reinforced concrete and are designed to protect safety-related equipment against damage from the effects, including tornado missile effects, of the design basis tornado of Region I.

The buildings and structures housing safety-related equipment are shown on FSAR Plot Plan Figure 1.2-1.

Barriers are located at all exterior wall and roof openings of buildings or structures housing safety related equipment to prevent tornado missiles from damaging the contained safety related equipment.

The types of barriers used are:

1. Door openings - All safety related equipment is protected against damage from tornado missiles by missile resistant steel doors or steel plates completely covering the opening.

2. Ventilation Pressure Barriers - Where openings are provided for blow-out panels, tornado dampers, or air intake louvers, all safety related equipment is protected against damage from tornado missiles either by exterior missile resistant reinforced concrete barriers completely surrounding the opening or by missile resistant concrete barriers inside the building. In cases where a postulated missile is stopped by an interior concrete barrier and could then fall into the structure, a grating is provided.

3. Hatch Openings - All safety related equipment is protected against damage from tornado missiles by missile resistant steel plates completely covering the roof openings. Manhole covers on top of all outdoor seismic Category I tanks are steel plate with a thickness of no less than 1 3/4 inches.

   Tornado missiles are postulated to impact normal to the plant structure with maximum mass and velocity as defined in Table 3.5-8 of the FSAR.

3.5.1.5 Missiles Generated by Events Near the Site

As described in Section 2.2, potential missiles generated by events near the site will not endanger any safety-related facilities.

3.5.1.6 Aircraft Hazards

As discussed in Section 2.2, aircraft missiles are not considered a design basis.

3.5.2 SYSTEMS TO BE PROTECTED

All plant structures, systems, and components whose failure can lead to offsite radiological consequences or which are required to shut down the reactor and maintain it in a safe condition, assuming an additional single failure, are listed in Section 3.2. In general, reinforce concrete floors and internal compartment walls are used as barriers for protection of all essential components and systems against postulated missiles. For arrangement of floors and internal walls, see Figures 3.8-1 through 3.8-4. For general arrangement, see Figures 1.2-1 through 1.2-46.

3.5.3 BARRIER DESIGN PROCEDURES

Only one missile is postulated at a time and is assumed to strike the barrier end-on.

The missiles have two effects on structures, walls, or any other barrier: local effects and overall response. The local effects include penetration, perforation, and spalling or scabbing. The overall response includes the flexural and shear effects.

Reinforced concrete external roofs and walls of seismic Category I structures form barriers against tornado-generated missiles. Buried underground, safety-related duct runs and piping are also protected.

3.5.3.1 Local Effects

The estimate of missile penetration (D) in concrete barriers is based on the Modified Petry Formula [1]. Sufficient thickness of concrete is provided to prevent perforation, and when required, to prevent backface spalling or scabbing. To prevent perforation, a concrete thickness of at least twice the penetration thickness determined for an infinitely thick slab is provided. The thickness (T) of the concrete barrier, to prevent backface spalling or secondary missiles which are injurious to safety-related equipment, is determined by the following equation [1]:

$$T = 2.4D$$

D is the maximum penetration in an infinitely thick slab. A minimum concrete barrier thickness of 30 inches for exterior walls and 21 inches for roofs is provided for protection from tornado-generated missiles. See Section 3.8.4.1.7 for a discussion of the properties of the concrete and typical reinforcement details.

Determination of penetration by missiles into steel plates is in accordance with the Stanford Research Institute Formula [2], or the Ballistic Research Lab Formula [10].

3.5.3.2 Overall Barrier Response

The overall response of structural barriers to missile impact depends upon the available ductility. This ductility is a function of the controlling nature of the structural behavior. Thus, for concrete beams, walls, and slabs, where flexure controls design, the permissible ductility ratio is taken as 0.05/(p - p'), p being the ratio of tension reinforcement and p', the ratio of compression reinforcement. However, the ductility ratio does not exceed 10.0 for beams, walls, and slabs. For beam, walls, and slabs where shear may control design, the ductility ratio is 1.0 when shear is carried by concrete alone, and 1.3 when shear is carried by concrete and stirrups or bent bars. Flexural strength is determined based on ultimate strength theory with the limitations on ductility as follows:

1. For beam-column members where the compressive load does not exceed either 0.1 f'C Ag or one-third of that which would produce balanced conditions, whichever is smaller, the allowable ductility ratio is 10.

2. For beam-column members where the design is controlled by compression, the allowable ductility is 1.3.

3. For members which are between these two values, ductility shall be taken as decreasing linearly from 10 to 1.3.

For steel barriers, the ductility ratio does not exceed 10.0 for flexure, compression or shear. For steel columns proportioned to preclude elastic buckling, (l/r equal to or less than 20) the limiting ductility ratio is 1.3. For steel columns not proportioned to preclude elastic buckling (l/r greater than 20), the limiting ductility ratio is 1.0, where l is the effective length of the member and r is the

least radius of gyration. For steel tension members, the limiting ductility ratio is taken as 0.5 Eu/Ey, where Eu is the ultimate strain and Ey is the yield strain.

The design load, caused by the missile, is based on the absorption of the kinetic energy of the missile by the target at its maximum Deflection [4]. It is limited by the yielding, buckling, crushing, or local failure of the missile.

REFERENCES

1. Amirikian, A., "Design of Protective Structures," NAVDOCKS P-51, Bureau of Yards and Docks, Department of the Navy, August, 1950.

2. Cottrell, W. B., and Savolainen, A. W., "U. S. Reactor Containment Technology," ORNL NSIC 5, Vol. 1, Chapter 6, Oak Ridge National Laboratory.

3. Stevenson, J. D., "2nd Annual Short Course on Structural Design, Analysis and Testing of Nuclear Plant Equipment and Structures," Case Western Reserve University, Cleveland, Ohio, June, 1974.

4. Williamson, R. A., and Alvy, R. R., "Impact Effect of Fragments Striking Structural Elements," Holmes & Narver, Inc., Anaheim, California, Revised, November, 1973.

5. Deleted.

6. Bush, S. H., "Probability of Damage to Nuclear Components Due to Turbine Failure" presented at Topical Meeting on Water Reactor Safety, March 26-28, 1973, CONF-730304, USAEC

7. Not Used.

8. Allis-Chalmer P. S. Inc., Engineering Report No. ER-503. Turbine Missile Analysis for 1800 r/min Nuclear Steam Turbine-Generators with 44-inch Last Stage Blades," July 1975.

9. USNRC, Regulatory Guide 1.115, "Protection Against Low-Trajectory Turbine Missiles," Rev. 1, July 1977.

10. Bechtel Corporation Topical Report, BC-TOP-9-A "Design of Structures for Missile Impact," Revision 2, September 1974.

11. Deleted.

12. WPT-16042, "Missile Sources in Westinghouse Supplied Valves"

13. Deleted.

14. Deleted.

15. CT-27331, Revision 8, “Missile Probability Analysis Methodology for Luminant Generation Company LLC, Comanche Peak Units 1 and 2 with Siemens Retrofit Turbines,” August 23, 2016.

16. Letter C. E. Rossi (NRC) to J. A. Martin (Westinghouse Electric Corp.) dated February 2, 1987 (NRC Safety Evaluation Report for WSTG-1-P, May 1981, “Analysis of the Probability of the Generation and Strike of Missile From a Nuclear Turbine”, March 1974, WSTG-2-P, May, 1981 and WSTG-3-P, July 1984)

17. TP-03143, “Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC,” July 31, 2003.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 1 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Containment Spray System				
Containment spray pumps	Safeguards building, floor elevation 773 ft 0 in	Section 6.2 Figure 1.2-10 Figure 1.2-16	Valve bonnet or part from rotating equipment	Two pumps for each train are located in an enclosed cubicle Pumps cannot be damaged by failed components from other systems.
Spray additive tank	Safeguards building, elevation 790 ft 6 in.	Sections 6.2.2, 6.5 Figure 1.2-10 Figure 1.2-16	Valve bonnet or part from rotating equipment	Tank is located in isolated compartment.
Containment spray heat exchangers	Safeguards building, floor elevation 790 ft 6 in.	Section 6.2.2 Figure 1.2-10 Figure 1.2-16	Missile from valve in adjacent valve room	Heat exchanger is separated from valve room by wall.
Piping and valves	Safeguards building	Section 6.2.2	Valve bonnet from other pipes in pipe chase	Redundant train
Chemical eductor	Safeguards building, elevation 786 ft 7 in.	Section 6.5	Valve bonnet or parts from rotating equipment	Redundant train
Containment sumps valve isolation tanks	Safeguards building, elevation 790 ft 6 in.	Section 6.2 Figure 1.2-10 Figure 1.2-16	None postulated	Tanks for redundant trains are separated from each other by concrete barriers.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 2 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Service Water System				
Pumps	Pump house	Section 9.2 Figure 1.2-45	Missile from service water pump	Redundant and separated pumps
Piping and valves	Pump house, auxiliary and safe guards buildings	Section 9.2	Valve missile from other piping in pipe chase	Redundant and separated trains
Spent Fuel Pool Cooling System (not needed for reactor shutdown)				
Spent fuel cooling pumps	Fuel building, floor elevation 810 ft 6 in.	Section 9.1 Figure 1.2-38	Missile from spent fuel cooling pump	Heat exchanger and pump of one train share a common cubicle which is separated from the other train by a wall.
Spent fuel cooling heat exchangers	Fuel building, floor elevation 810 ft 6 in.	Section 9.1 Figure 1.2-38	Missile from spent fuel cooling pump	Only one train is needed to cool both pools except during abnormal design conditions
Piping and valves	fuel building	Section 9.1	Missile from pump	Pumps are enclosed in concrete cubicle with piping associated only with that train.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 3 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Component Cooling Water System				
Pumps	Auxiliary building, floor elevation 810 ft 6 in.	Section 9.2 Figure 1.2-32	Missile from other CCWS pump	Pumps from redundant trains in separate cubicles are separated by corridors. Pump from same train of different units is separated by wall.
Heat exchangers	Auxiliary building, floor elevation 790 ft 6 in.	Section 9.2 Figure 1.2-31	None postulated	Four heat exchangers (two per unit) are in a common compartment. No pumps or high pressure valves are in area.
Surge tank		Figure 1.2-35		
Piping and valves	Auxiliary building	Section 9.2	Valve bonnet	Redundant and separated trains
Auxiliary Feedwater System				
Condensate storage tank	Yard	Section 10.4.9 Figure 1.2-1	None postulated	Tank is shielded against tornado-generated missiles.
Pumps	Safeguards building, floor elevation 790 ft 6 in.	Section 10.4.9 Figure 1.2-10 Figure 1.2-16	Missiles from other auxiliary feedwater pumps	Each pump is enclosed in a separate cubicle.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 4 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Piping and valves	Yard and safeguards building	Section 10.4.9	Missiles from other auxiliary feedwater pumps	Each pump is enclosed in a separate cubical with its train associated piping.
Main Steam Supply System				
Piping and valves	Safeguards building, floor elevation 873 ft 6 in.	Section 10.3 Figure 1.2-14 Figure 1.2-20	Valve bonnets from safety and relief valves	Each steam line is located in separate compartment. Each compartment is divided to separate relief and safety valves from isolation valve.
Power-operated relief valves	Safeguards building, floor elevation 885 ft	Section 10.3 Figure 1.2-14 Figure 1.2-20	Valve bonnets from safety of same steam line	Safety valves are pointed away from relief valve and each other.
Feedwater System				
Piping and valves	Safeguards building	Section 10.4.7.2	Valve bonnets from isolation valves	Each line is in own separate cubicle. Valve bonnet from one isolation valve cannot damage any other feedline.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 5 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Safety Injection System				
Refueling water storage tank	Yard	Section 6.3 Figure 1.2-1	No postulated internal source	Tank is shielded against tornado-generated missiles.
Piping and valves	Safeguards building and yard	Section 6.3	Valve parts from other systems	Redundant trains
High-head SIS pumps	Safeguards building, floor elevation 773 ft 0 in	Section 6.3 Figure 1.2-10 Figure 1.2-16	No postulated internal source	Each safety injection pump is in its own cubicle on opposite sides of the safeguards building
Boron injection tank	Safeguards building, floor elevation 810 ft 6 in.	Section 6.3.2	N/A	Tank has been deleted from the CPNPP design.
BIT recirculation pump	Safeguards building, floor elevation 810 ft 6 in.	Section 6.3.2	N/A	Pump has been deleted from the CPNPP design.
Boron injection surge tank	Safeguards building, elevation 852 ft 6 in.	Section 6.3.2	N/A	Surge tank has been deleted from the CPNPP design.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 6 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
CVCS and Liquid Poison System				
Centrifugal charging pumps	Auxiliary building, floor elevation 810 ft 6 in.	Section 9.3.4 Figure 1.2-32	No postulated internal missile per discussion on Westinghouse-supplied pumps	Charging pumps are in separate cubicles with walls between them.
Boric acid tanks	Auxiliary building, floor elevation 810 ft 6 in.	Section 9.3.4 Figure 1.2-32	No postulated missile	Two tanks per unit are located in common cubicle with no missile sources in or near cubicle.
Piping and valves	Auxiliary and safeguards parts buildings	Section 9.3.4	Missile from valve or pump shutdown.	Redundant trains for parts of system are needed for safe
Residual Heat Removal System				
Containment sumps valve isolation tanks	Safeguards building floor elevation 790 ft 6 in.	Section 5.4.7 Figure 1.2-10 Figure 1.2-16	Part from containment spray valve	Spray valve located in separate tank on opposite side of column and redundant trains.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 7 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Pump	Safeguards building, floor elevation 773 ft 0 in.	Section 5.4.7 Figure 1.2-10 Figure 1.2-16	Missile from containment spray pump in next room	Each RHR pump is in a separate cubicle with walls separating pumps from each other and from containment spray pumps.
Heat exchanger	Safeguards building, floor elevation 90 ft 6 in.	Section 5.4.7 Figure 1.2-10 Figure 1.2-16	Missile from valve in adjacent valve room	RHR heat exchangers are separated from the valve room by a wall. Two redundant trains are provided.
Hydrogen Purge System				
Filters (charcoal, HEPA, roughing)	Auxiliary building, elevation 873 ft 6 in.	Section 6.2.5 Figure 1.2-35	No postulated missiles from other internal sources other than fan in same train	Trains A and B are isolated.
Fans	Auxiliary building, elevation 873 ft 6 in.	Section 6.2.5 Figure 1.2-35	No postulated missiles from external sources	Trains A and B are isolated.
Piping and valves	Auxiliary building	Section 6.2.5	No postulated missile	None provided; there are no high-pressure piping or pumps nearby.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 8 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Containment Isolation System				
Discussed in Section 6.2.4				
Control Room HVAC				
Air conditioning units (roughing filter, fan, heating and chilled water cooling coils)	Control building, elevation 854 ft 4 in.	Section 9.4.1 Figure 1.2-34	Fan wheel from fan in same train	Fan housing and separation wall between trains; one train can be damaged but other is protected
Emergency filtering unit (roughing, charcoal and HEPA filter and fans)	Control building, elevation 854 ft 4 in.	Section 9.4.1 Figure 1.2-34	Fan wheel from fan in same train	Fan housing and separation wall between trains; one train can be damaged but other is protected.
Piping, valves, and dampers	Control building, elevation 854 ft 4 in.	Section 9.4.1	Fan wheel from fan in same train	Fan housing and separation wall between trains; one train can be damaged but other is protected.
Fans	Control building, elevation 854 ft 4 in.	Section 9.4.1 Figure 1.2-34	Wheel in fan	Fan housing and separation wall between trains; one train can be damaged but other is protected.
Safeguards building emergency closed loop air cooling system (cooling coils, fans)	Inside of various pump rooms	Section 9.4	Missile from pump which the unit cools	Separate cooling units for redundant pumps.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 9 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Fuel Building Ventilation System (For Fuel Accident Only)				
Filters (roughing, charcoal, HEPA)	Auxiliary building, elevation 873 ft 6 in.	Section 9.4.2 Figure 1.2-35	Fan blade	Complete redundancy and separation of trains
Fans (exhaust)	Auxiliary building, elevation 873 ft 6 in.	Section 9.4.2 Figure 1.2-35	Fan blade	Complete redundancy and separation of trains
Ducts (exhaust)	Fuel and auxiliary buildings	Section 9.4.2	Fan blade	Complete redundancy and separation of trains
Standby Power System				
Diesel generators	Diesel generator building, floor elevation 810 ft 6 in.	Section 8.3 Figure 1.2-11 Figure 1.2-17	Piston from one diesel generator	Each diesel generator is located in separate room with missile barrier between generators.
Diesel generator supportive subsystems (including lubrication, intercooling, and jacket water systems)	Diesel generator building, floor elevation 810 ft 6 in.	Section 9.5.5 Figure 1.2-11 Figure 1.2-17	Piston from one diesel generator	Supportive subsystems are in room with generator they service; redundant subsystem are for each generator.

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 10 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Diesel Generator Starting System				
Air compressors	Diesel generator, floor elevation 844'-0"	Section 9.5.6 Figure 1.2-12 Figure 1.2-18	Piston from diesel generator	Separate starting system for each diesel in same room as diesel.
Air receivers	Diesel generator floor elevation 844'-0"	Section 9.5.6 Figure 1.2-12 Figure 1.2-18	Piston from diesel generator	Separate starting system for each diesel in same room as diesel.
Piping and valves	Diesel generator floor elevation	Section 9.5.6	Piston from diesel generator	Separate starting system for each diesel in same room as diesel.
Emergency Diesel Fuel Oil Storage and Transfer System				
Transfer pumps	Diesel generator building, floor elevation 844’-0”	Section 9.5.4 Figure 1.2-12 Figure 1.2-18	Diesel generator piston	Located in pit in room of diesel generator that it services; each generator has separate pumps.
Storage tanks	Buried in yard	Section 9.5.4 Figure 1.2-1	None postulated	Tank is buried in ground. Piping is separated and buried underground.
Day tanks	Diesel generator building, floor elevation 844'-0"	Section 9.5.4 Figure 1.2-12 Figure 1.2-18	Diesel generator piston	Separate tank for each diesel generator

TABLE 3.5-1
INTERNALLY GENERATED MISSILES OUTSIDE CONTAINMENT

(Sheet 11 of 11)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Diesel fuel filter	Diesel generator building, floor elevation	Section 9.5.4	Diesel generator piston	Separate filter for each unit
Piping and valves (required for continued operation of diesel generators)	Yard and diesel generator building	Section 9.5.4	Diesel generator piston (only for piping in building)	Interior piping and valves are confined to the cubical containing the diesel which they serve.

TABLE 3.5-2A
SUMMARY OF THE UNIT 1 CONTROL ROD DRIVE MECHANISM MISSILE ANALYSIS

	Calculation Data			
Postulated Missiles	Missile Weight (lb)	Impact Velocity (ft/sec)	Kinetic Energy (ft-lb)	Assumptions
Drive shaft	133	140	40478	------

TABLE 3.5-2B
SUMMARY OF THE UNIT 2 CONTROL ROD DRIVE MECHANISM MISSILE ANALYSIS

Typical Examples of Postulated Missiles	Calculation Data: Missile Weight (lb)	Calculation Data: Impact Velocity (ft/sec)	Calculation Data: Kinetic Energy (ft-lb)	Calculation Data: Penetration (in.)	Assumptions
CRDM housing plug becomes loose and accelerated by the	11	380	24700	$\leq$1.69	Plug is water jet
Drive Shaft (Drive Rod)	120	173	55800	$\leq$1.69	------
CRDM and drive shaft latched together	1500	25.7	15400	$\leq$1.69	------

TABLE 3.5-3
VALVE MISSILE CHARACTERISTICS

Missile Description	Weight (lb)	Flow Discharge Area (in^2)	Thrust Area (in^2)	To Impact Area (in^2)	Impact Area Ratio (psi)	Velocity (ft/sec)
Safety relief valve bonnet (3 in. x 6 in. or 6 in. x 6 in.)	350	2.86	80	24	15.6	110
3-in. motor-operated isolation valve bonnet, plus motor and stem (3 in.)	400	5.5	113	28	14.1	135
2-in. air-operated relief valve bonnet, plus stem	75	1.8	20	20	3.75	115
3-in. air-operated spray valve bonnet, plus stem	120	5.5	50	50	2.4	190
4-in. air-operated spray valve	200	9.3	50	50	4	190

TABLE 3.5-4
PIPING TEMPERATURE ELEMENT ASSEMBLY MISSILE CHARACTERISTICS

1. For a Tear Around the Weld Between the Boss and the Pipe

	Without Well	With Well
Flow discharge area, in^2	0.11	0.60
Thrust area, in^2	7.1	9.6
Missile weight, lb	11.0	15.2
Area of impact, in^2	3.14	3.14
Missile Weight		
Impact Area, psi	3.5	4.84
Velocity, ft/sec	20	120

2. For a Tear at the Junction Between the Temperature Element Assembly and the Boss for the Without Well Element and at the Junction Between the Boss and the Well for the With Well Element

	Without Well	With Well
Flow discharge area, in^2	0.11	0.60
Thrust area, in^2	3.14	3.14
Missile weight, lb	4.0	6.1
Area of impact, in^2	3.14	3.14
Missile Weight		
Impact Area, psi	1.27	1.94
Velocity, ft/sec	75	120

TABLE 3.5-5
CHARACTERISTICS OF OTHER MISSILES POSTULATED WITHIN REACTOR CONTAINMENT

	Reactor Coolant Pump Temperature Element	Instrument Wall of Pressurizer	Pressurizer Heaters
Weight, lb	0.25	55	15
Discharge area, in^2	0.50	0.442	0.80
Thrust area, in^2	0.50	1.35	2.4
Impact area, in^2	0.50	1.35	2.4
Missile Weight			
Impact area, psi	0.5	4.1	6.25
Velocity, ft/sec	260	100	55

TABLE 3.5-6
INTERNALLY GENERATED MISSILES (INSIDE CONTAINMENT)

(Sheet 1 of 2)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Control Rod Drive Mechanism (CRDM)	Containment Building	Section 5.3 Figure 1.2-12	Unit 1 Missile from CRDM drive shaft Unit 2 Missile from CRDM housing plug, drive shaft and drive mechanism latched together	Unit 1 Integral steel missile shield above CDRM housing Unit 2 Roll away steel missile shield above CRDM housing
Valves (Pressurizer) – pressurizer safety valves - motor operated isolation valves in the relief line - air operated relief valves - air operated spray valves	Containment Building	Section 5. 4	failure of valve body bonnet bolts	Valves orientated so that a missile will strike a missile barrier
Pressurizer Heaters	Containment building El. 853'-6"	Section 5.4 Figure 1.2-13	pressurizer heaters become loose	located in a concrete cubicle
Pressurizer	Containment Building El. 853'-6"	Section 5.4 Figure 1.2-13	Instrumentation well	located in a concrete cubicle

TABLE 3.5-6
INTERNALLY GENERATED MISSILES (INSIDE CONTAINMENT)

(Sheet 2 of 2)

System and Component	Location	Section for Description and Reference Drawings	Identification of Missiles	Missile Protection Provided
Reactor Coolant Pump (RCP)	Containment Building El. 832'-6"	Section 5.4 Figure 1.2-13	RCP temperature sensor pipe plug	oriented to strike a concrete barrier
Reactor Coolant System (RCS) Piping	Containment Building	Section 5.4	Temperature sensors	RCS is surrounded by the secondary shield wall and components are arranged so that a missile does not damage a component whose failure can lead to an offsite radiation dose in excess of that prescribed in 10 CFR 100

TABLE 3.5-7
HIGH TRAJECTORY MISSILE TARGETS AND TARGET AREAS

Target Building	Target Components	Target Building Area (Ft^2)	Target Components Area (Ft^2)
Unit 1 Safeguard Building	Main steam piping including main steam isolation valves, safety relief valves and main steam moments restraints	4,030	732
Unit 1 Diesel Generator Building	Diesel generator exhaust piping	4,160	712
Unit 1 Reactor Containment	Steam generators, reactor coolant pumps, reactor vessel and portion on reactor coolant piping, main steam piping	14,314	1,146
Auxiliary Building	Component Cooling water surge tanks, chilled water surge tanks	22,080	552
Fuel Building	Spent fuel pool	15,857	1,987
Unit 2 Reactor Containment	(Same as in Unit 1)	14,314	1,146
Unit 2 Safeguard Building	(Same as in Unit 1)	4,030	732
Unit 2 Diesel Generator Building	(Same as in Unit 1)	4,160	712
Electrical and Control Building	Entire building area included for conservatism	16,355	16,355
Service Water Intake Structure	Service Water System Components (Pumps and Piping)	4,686	562

TABLE 3.5-8
TORNADO-GENERATED MISSILES (HORIZONTAL MODE)

Missile		Mass (kg)	Dimensions (m)	Velocity (m/sec)	Kinetic Energy (kg-m)
A.	Wood plank	52	.092 x .289 x 3.66	83	18,300
B.	6-in. Schedule 40 pipe	130	.168D x 4.58	52	17,900
C	1-in. steel rod	4	.0254D x .915	51	530
D.	Utility pole	510	.343D x 10.68	55	78,600
E.	12-in. Schedule 40 pipe	340	.32D x 4.58	47	38,300
F.	Automobile	1810	5 x 2 x 1.3	59	321,000

Note:

Vertical velocities of 70 percent of the postulated horizontal velocities are used except for Missile C, which is used to test barrier openings. Missile C is assumed at the same speed in all directions. Kinetic energy for vertical impact is reduced according to missile velocity.

TABLE 3.5-9
TABLE 3.5-9 HAS BEEN DELETED

TABLE 3.5-10
TABLE 3.5-10 HAS BEEN DELETED

APPENDIX 3.5A - THEORY OF MISSILE ENERGY [8]

3.5A HAS BEEN DELETED

APPENDIX 3.5B - TURBINE MISSILE STRIKE PROBABILITY DISTRIBUTIONS [8]

3.5B HAS BEEN DELETED

3.6N PROTECTION AGAINST DYNAMIC EFFECTS ASSOCIATED WITH THE POSTULATED RUPTURE OF PIPING

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3.6B PROTECTION AGAINST DYNAMIC EFFECTS ASSOCIATED WITH THE POSTULATED RUPTURE OF PIPING

This section describes the design bases and protective measures used to ensure that all essential structures, systems and components required for a safe shutdown and to maintain the reactor in a cold shutdown condition are adequately protected from the dynamic effects associated with postulated pipe ruptures located inside and outside of the containment. The criteria for protection against pipe ruptures conforms to the requirements of Appendix A of 10 CFR 50 General Design Criterion 1, 2, 4 and 14. The pipe rupture protection criteria also conforms to the guidelines of NRC Regulatory Guide 1.46 [1], Standard Review Plans 3.6.1 and 3.6.2 [2], and Branch Technical Positions APCSB 3-1 [3] and MEB 3-1 [2].

3.6B.1 POSTULATED PIPING FAILURES IN FLUID SYSTEMS

This section sets forth the design bases, description, and safety evaluation for protection and determination of the dynamic effects associated with the postulated rupture of piping in fluid systems both inside and outside containment.

3.6B.1.1 Design Basis

The following design bases are utilized in determining the consequences of pipe failures on essential systems or components important to plant safety or shutdown (see Section 3.6B.1.2) which are located in the vicinity of high or moderate energy piping:

1. Piping systems, valves and components required to achieve a safe shutdown will be protected.

2. Required redundancy will be maintained in the protection system (IEEE Standard 279), Class 1E electric systems (IEEE Standard 308), ESF equipment, cable penetrations and their interconnecting cables.

3. HVAC equipment required for safe shutdown will be protected. Portions of Primary Plant Ventilation and associated chilled water are credited after 72 hours to mitigate the long term effects of HELBs outside containment. This is sufficient time to effect required repairs.

4. Instrumentation required for post accident monitoring will be protected as described in Section 7.5.1.3.1.3.

5. Containment leaktightness will be maintained.

6. LOCA breaks will not propagate to steam and/or feed water line breaks and vice versa.

7. A non-LOCA break will not be allowed to propagate into a LOCA.

8. LOCA break propagation to the unaffected Reactor Coolant System loops will be prevented.

9. LOCA propagation in the affected loop is permitted, but is limited as discussed in Section 3.6B.1.2.2.

10. In-core instrumentation lines will be protected.

The criteria for determining the location of pipe breaks is given in section 3.6B.2.1. A discussion of the effects of pipe failure on each system and typical piping runs with the location of failure points shown on drawings is given in subsection 3.6B.2.5 for high energy systems.

A list of high energy lines that are considered for pipe rupture analysis is included as Table 3.6B-1.

All large bore High Energy Lines in safety related structures are Seismic Category I and II. Small bore High Energy Lines in safety related structures are Seismic Category I or II except for the main steam line drip pot drain lines in the Safeguards Building, Main Steam and Feedwater Penetration Area. See Table 3.6B-1 for applicable line numbers.

3.6B.1.2 Description

Essential systems are defined as those systems that are needed to shut down the reactor and mitigate the consequences of the pipe break for a given postulated piping break.

3.6B.1.2.1 Protection Criteria

Depending upon the type and location of the postulated pipe break, certain safety equipment may not be classified as essential for the particular event. Some safety equipment will be essential for almost all cases. This category includes service water to the ultimate heat sink and the pressurizer level instrumentation. The containment integrity and leak tightness will be maintained for any LOCA break. The effects of a postulated piping failure, including environmental conditions resulting from the escape of contained fluids, will not preclude habitability of the control room or access to surrounding areas important to the safe control of reactor operations needed to cope with the consequences of the piping failure. Accordingly, protection from the effects of pipe rupture will be provided for only that safety-related equipment considered as essential on a case-by-case basis.

The systems or portions of systems and equipment for which protection against postulated pipe failures is required are identified below. However, in general, protection from pipe failure need not be provided if any of the following conditions exists:

1. The piping is physically separated (or isolated) from structures, systems, or components important to safety by protective barriers, or is restrained from whipping.

2. Following a single break, the unrestrained pipe movement of either end of the ruptured pipe about a plastic hinge formed at the location determined by calculation cannot impact any structure, system, or component important to safety.

3. The internal energy level associated with the whipping pipe can be demonstrated to be insufficient to impair the safety function of any structure, system, or component to an unacceptable level.

The following systems or portions of these systems are required to mitigate the consequences of a postulated pipe failure:

1. Auxiliary Feedwater System
2. Chemical and Volume Control System
3. Feedwater System
4. Main Steam System
5. Reactor Coolant System
6. Residual Heat Removal System
7. Safety Injection System
8. Safety Chilled Water System
9. Hydrogen and Nitrogen System
10. Containment Spray System
11. Diesel Generator System
12. Component Cooling Water System
13. Service Water System
14. HVAC Systems required for the operation of other systems
15. Containment Isolation System
16. Control Room Ventilation System
17. Class 1E electrical systems, ac and dc (including switchgear, cables, batteries and distribution systems)

The instrumentation required to mitigate the consequences of a postulated pipe break and/or for accident monitoring is discussed in Sections 7.4, 7.5 and 7.6.12.

The results of the environmental analysis, which discuss the effects of pressure, temperature, humidity, and flooding on safety related equipment, are provided in Sections 3.11 and 3.6B.2.5. The only high-energy lines located near the Control Room are the main steam and feedwater lines which are located outside the Control Room adjacent to the north and south walls. Accordingly, protection is provided as described in Section 3.6B.2.5 so that habitability of the Control Room is not jeopardized by postulated pipe failures.

3.6B.1.2.2 LOCA Break Propagation Criteria

A loss of reactor coolant accident is assumed to occur for a branch line break down to the restraint of the second normally open automatic isolation valve (Case II in Figure 3.6B-10) on outgoing lines and down to and including the second check valve (Case III in Figure 3.6B-10) on incoming lines normally with flow. A pipe break beyond the restraint or second check valve will not result in an uncontrolled loss of reactor coolant if either of the two valves in the line closes.

Accordingly, both of the automatic isolation valves are suitably protected and restrained as close to the valves as possible so that a pipe break beyond the restraint will not jeopardize the integrity and operability of the valves. Further, periodic testing capability of the valves to perform their intended function is essential. This criterion takes credit for only one of the two valves performing its intended function. For normally closed isolation or incoming check valves (Cases I and IV in Figure 3.6B-10) a loss of reactor coolant accident is assumed to occur for pipe breaks on the reactor side of the valve.

Branch lines connected to the Reactor Coolant System are defined as “large” for the purpose of these criteria if they have an inside diameter greater than 4 inches. Rupture of these lines results in a rapid blowdown from the Reactor Coolant System and protection is basically provided by the accumulators and the low head safety injection pumps (residual heat removal pumps).

Branch lines connected to the Reactor Coolant System are defined as “small” if they have an inside diameter equal to or less than 4 inches. This size is such that Emergency Core Cooling System analyses using realistic assumptions show that no clad damage is expected for a break area of up to 12.5 square inches.

In order to assure the design function of essential systems in the event of a LOCA, break propagation within the affected loop shall be limited as follows:

1. Reactor Coolant Loop Piping

   Propagation of damage resulting from rupture of the main reactor coolant loop is permitted to occur but must not exceed the design basis for the containment, environmental qualification of equipment and Emergency Core Cooling System performance.

   The application of Leak-Before-Break technology to CPNPP Unit 1 and 2, discussed in Section 3.6B.2.1.1, has shown that the dynamic effects of main reactor coolant loop rupture, such as loop hydraulic forces, reactor internals reaction loads and primary equipment support loads, can be excluded from the design basis, as allowed by General Design Criterion 4 (Section 3.1.1.4).

2. Large Branch Lines

   In the event of a rupture of a large branch line resulting in a LOCA, propagation of the break in the affected loop must not exceed 20 percent of the flow area of the line which initially ruptured.

3. Small Branch Lines

In the event of a rupture of a small branch line resulting in a LOCA:

a. Break propagation must be limited to the affected leg of the affected loop.

b. Propagation of the break in the affected leg must be limited to a total break area of 12.5 square inches. However, no propagation is permitted when the initiating small break is the high head safety injection line.

c. Propagation of the break to a high head safety injection line connected to the affected leg is prevented if the line break results in a loss of core cooling capability due to a spilling injection line.

3.6B.1.2.3 Environmental Analysis to Determine HELB Temperature Transients for Purposes of Equipment Qualification (Outside Containment)

1. Design Bases

Environmental analyses are performed to model the thermal effects of HELBs on safety related areas and equipment outside containment that are needed for safe shutdown of the plant. This information is used to determine that the plant can be safely shutdown following a high energy line break.

The environmental effects of HELBs are analyzed for those systems where breaks or cracks are required to be postulated that have the potential of releasing steam or flashing water. These systems include Main Steam, Feedwater, Chemical and Volume Control, Steam Generator Blowdown and Auxiliary Steam.

2. Analytical Techniques

The COMPARE/MOD1A computer code is used to model the effects of HELBs. COMPARE is a thermal-hydraulic modeling code that is used for subcompartment transient analyses. It models the physical plant arrangement as a series of volumes and junctions. The volumes are made up of rooms, parts of rooms or groups of rooms that are combined together to form one volume. The junctions are naturally occurring flow paths based on the geometry of the plant.

The COMPARE code has the ability to calculate the transient conditions in up to 100 volumes that are connected by up to 200 junctions. Each volume is assumed to contain a homogeneous mixture of air, steam and water. The volumes may contain heat conduction elements. Mass and energy addition (blowdown) may also be added to a volume.

Mass and energy accounting is performed on each volume for each discrete time step. This then describes the thermodynamic conditions, which are assumed to be constant, from which the new junction flows are calculated. Several options are available to calculate the junction flows. These are a homogeneous equilibrium gas mixture, a two-phase Moody critical flow model or an inertial flow model based on the solution of the

one-dimensional momentum equation with critical flow based on one of the first two methods listed. The third is the option used for the C.P.S.E.S.

Doors whose angular positions are a function of pressure, spring forces, gravity forces or inertia forces can be represented to model the vent flow area.

Heat sinks can be modeled into the volumes and the associated parameters can include density, specific heat, initial conditions, boundary conditions, changes in materials and changes in heat transfer coefficients.

3. Plant Models

Because of the locations of the postulated high energy line breaks outside containment, several detailed computer models of the plant are developed. These models are used to calculate the effects of breaks occurring in the main steam and feedwater penetration areas, the safeguards building, the auxiliary building and the electrical and controls building. The models are based on the physical arrangement of the as-built plant and include door positions, tornado dampers, blowout panels, blowout doors, equipment removal gratings, ventilation transfer grills and physically isolated areas. The door positions are in accordance with those specified by the Tornado Analysis.

For each of the plant models many different kinds of input information is calculated. Some of the calculated input parameters include the room volumes corrected by subtracting the volume occupied by large piping and major equipment, the specific surface areas of the walls, floor and ceiling for input as a heat sink and junction inertial values (L/A) specific to each of the flow paths of the as-built plant. Also the exact free area of the tornado dampers, blowout panels and blowout doors are used, along with the geometry of each to accurately model the swing of each as a function of time and differential pressure.

Sensitivity analyses are performed by changing several input parameters, one at a time, to determine the adequacy of the parameters used in the analysis.

A survey is made of all the compartments in the plant that contain HELBs and the governing worst case break for each of the compartments is determined. Using the plant models mentioned above, each of the individual breaks that is the governing case for its compartment is then analyzed to determine the plant environment.

Using the results of the above analyses, a study is then performed to determine equipment affected by the environment. The affected equipment is then evaluated, on a case by case basis, to determine the equipment required for safe shutdown.

4. The governing results for each room and piece of equipment in terms of temperature, pressures and relative humidities are used as input to Section 3.11.

This input is then used to equate whether the existing demonstrated (maximum) qualification parameters envelope the short duration HELB parameters for temperature, pressure and relative humidity. Where the HELB parameters are not enveloped by the existing environmental qualification parameters, an Engineering Analysis is performed. This analysis is performed in order to justify continued operability during (where required)

and after the HELB event, with no appreciable degradation in performance or qualified life of the affected safety or non-safety related electrical equipment.

5. Break Exclusion Areas

For the main steam piping, the break exclusion region extends to the moment restraint downstream of the main steam isolation valve. The restraint is located next to the main steam isolation valve (at the outside wall of the Safeguards Building). It is designed to withstand the loadings resulting from a postulated piping failure beyond the break exclusion region so that neither isolation valve operability nor the leaktight integrity of the containment will be impaired.

For the main feedwater piping, break exclusion region extends to the moment restraint upstream of the main feedwater control valve. The moment restraint is designed to withstand the loadings resulting from a postulated piping failure beyond the break exclusion region. In addition to the main steam and main feedwater piping, about 10 ft of main feedwater bypass piping connected to each of the four main feedwater piping is also designated as a break exclusion region (Unit 2 only).

The piping in the break exclusion regions is designed to meet the requirements of the ASME B & PV Code, Section III, Subarticle NE-1120 and the additional design requirements specified in Branch Technical Position MEB 3-1.

The combination low stress and fatigue, in combination with augmented inservice inspection, precludes the possibility of pipe breaks or cracks in the break exclusion regions. However, for the purpose of determining environmental qualification requirements for safety related equipment in the break exclusion region, a non-mechanistic one square foot crack is postulated.

Method of Analysis

1. Introduction

A subcompartment multinode pressurization analysis has been performed using the COMPARE - MOD 1A computer code which is described in Reference [27]. The analysis results provide information on environmental effects of cracks in the 32-inch main steam pipe or the 18-inch main feedwater pipe within their respective compartments in the Safeguards Building. In addition, flooding due to the above has also been evaluated.

In accordance with the criteria stated in Section 3.6B.2.1.2, no specific pipe breaks are postulated in the above lines within the Safeguards Building. However, to provide an additional level of assurance of safety-related equipment operability in these compartments, the building structure and safety related equipment were reviewed for the environmental conditions (pressure, temperature and flooding), that would result from a non-mechanistic crack, equal in area to one square foot of either a main steam line or feedwater line. The mass and energy release data assumes that the non-mechanistic crack is independent of the location of the MSIV; that is, the open MSIV on the line with the postulated crack is not required to close on steam line isolation to limit the mass and energy release. This conservative assumption is the only failure of equipment outside the main steam and feedwater compartments or qualified equipment inside the compartment

for this scenario. Thus, no single failure or loss of offsite power need be assumed. In addition, the scenario need not assume that the event causes a loss of offsite power since FSAR Section 8.2.2 provides the results of analyses which show that loss of offsite power due to a reactor trip/turbine trip from 100% power is not probable. It is assumed that jets are not generated and hence jet loads are not considered.

Peak compartment temperatures were found to occur following the onset of a postulated MSLB using blowdown data that includes superheated steam effects. This analysis is detailed in FSAR Section 3.6B.2.5.2 and Tables 3.6B-4A (Unit 1), 3.6B-4B (Unit 2) and 3.6B-5.

The method used for the pressurization analysis of the structural compartments in the Safeguards Building is similar to the method used in the subcompartment analyses of the containment structure (Section 6.2.1.2) except that: (1) no asymmetric loading has been analyzed and (2) the type and size of break is limited to a one square foot non-mechanistic crack.

Analyses have been performed for each unit.

2. Compartment Nodalization

The affected compartments in the Safeguards Building containing the high energy lines are represented in a nodalization scheme by a series of volume nodes. The compartment boundaries are shown in Figure 3.6B-207. The line cracks were postulated in room numbers 108A through 108H for main steam and in room numbers 100A through 100H for feedwater, respectively.

In addition, flooding level calculations were based on a feedwater line crack in room numbers 100A, 100B, 100E and 100F.

The choice of nodal subdivisions was governed by the actual physical subdivision of the structure (natural boundaries) which form flow restrictions.

3. Vent Openings and Flow Obstructions

In determining the maximum levels of pressure and temperature resulting from the postulated cracks, a conservative approach was taken in that no credit was taken for utilizing the various building structures and equipment as heat sinks that would contribute to lowering the calculated pressure and temperature.

Venting between the affected compartments is obtained by the following:

a. Floor openings in main steam compartments nearest the Containment Building (room numbers 108A, 108B, 108C, 108D).

b. The seven-feet by three-feet openings in walls of all the compartments, each opening being located seven-feet above the compartment floor.

c. Pipe sleeve openings.

The total area considered for venting from the main steam and feedwater compartments consists of the following:

a. Pipe sleeve openings through the west wall of the Safeguards Building from room numbers 108E, 108F, 108G, 108H, 100E, 100F, 100G and 100H.

b. Pipe sleeve openings for the safety and relief valve vent stack piping through the roof of room numbers 108A, 108B, 108C and 108D.

Seals for these pipe sleeves are provided for weather protection. They have little strength and will "blow out" at a very low differential pressure. For the purpose of analysis, the seals are discounted. The vent openings are modeled as flow paths (or junctions) between each compartment.

For the purpose of calculating the temperature and pressure inside the main steam and feedwater compartments, flow obstruction is assumed to occur as a result of the failure of HVAC ducts passing between the various main steam and feedwater compartments from the compartments to outside areas. These ducts are assumed to fail in such a way as to block any flow that normally would pass through them. Also, the equipment hatches located on the roof of the main steam isolation valve compartments and the seals provided in the gaps between the Safeguards and Containment Buildings are assumed to remain in place.

Structural Design Margins

The main steam line and feedwater line compartments in the Safeguards Building are designed to withstand the environmental effects (pressure, temperature, humidity and flooding) of a non-mechanistic crack in a main steam or feedwater line with a flow area of one square foot. These effects are not combined with seismic effects and are not factored (the load factor is taken as unity). In addition it is assumed that jets are not generated and hence jet loads are not considered.

Mass and Energy Release Rates

1. Main Steam Line Break

The mass and energy release rates were obtained from Westinghouse. Data for the limiting case (1 ft^2 steam line split at 90% including power uncertainty) is shown in Table 3.6B-4A (Unit 1) and Table 3.6B-4B (Unit 2).

2. Main Feedwater Line Break

The worst case crack in the main feedwater line occurs downstream of the check valve. For Unit 1, the mass and energy release rates were developed using RELAP5/MOD 3 [36].

For Unit 2, the mass and energy release rates were developed based on methodology consistent with ANSI/ANS 58.2, Appendix E (see Table 3.6B-6).

Design Data

The main steam and feedwater isolation valve compartments are located on the two top floors of the Safeguard Building which is situated between the Containment Building and the Electrical and Control Building. Figure 3.6B-207 provides plan and elevation drawings of the Safeguard Building.

The main steam and feedwater piping in this area consists of straight piping runs approximately 40-feet long, extending from the containment penetrations, through the compartments to moment limiting restraints mounted in the Safeguards Building wall through which these lines exit to the outside. The compartments contain the main steam line isolation valves, main steam safety valves, main steam power-operated relief valves, main steam isolation bypass valves, steam supply valves to the auxiliary feedwater turbine pump, feedwater isolation valves, feedwater isolation bypass valves and auxiliary feedwater containment isolation valves. The piping arrangement is shown in Figure 3.6B-208.

Also in the subcompartments are various instruments and controls, cables and branch piping lines for the main steam drain and secondary sampling system.

Environmental Effects

The ability to safely shutdown the plant will not be affected by the release of steam and water in the main steam and feedwater compartments. Plant design allows for loss of one main steam and feedwater loop and associated equipment without jeopardizing the safe shutdown of the plant. The following discussion addresses design, operation and consequences of failure of equipment located inside the main steam and feedwater compartments (See Table 3.6B-5 for detail of equipment qualification for MSLB).

1. Main Steam Isolation Valves (MSIVs) and Bypass Valves

   The MSIVs and their electrical supplies are designed to operate at the maximum environmental conditions resulting from the postulated main steam or feedwater pipe crack. These valves except for the valve in the compartment where the crack is located, are required to close following a main steam line or feedwater line crack in order to limit the blowdown through the crack to one steam generator. The signal to close is derived from steam line pressure transducers located on the main steam line inside the compartment. These instruments are qualified for the above environment.

   The MSIV bypass valves are not required for safe-shutdown. These valves are locked closed manual valves and no failure could cause them to open.

2. Power-Operated Main Steam Relief Valves (PORV's)

   These valves are air-to-open, fail closed. The pressure transducers which provide the signal to open these valves must not fail in a mode which could cause the PORV on an intact steam generator to inadvertently open. The PORVs for the steam generator associated with the crack may fail in either the open or closed position without increasing the severity of the event beyond the present analysis or decreasing the ability to mitigate the event and safely shutdown the unit. The PORVs and pressure transducers have a

demonstrated operability beyond that required for their safety function (See Table 3.6B-5).

3. Auxiliary Feedwater Turbine Steam Supply Valves

The failure mode of the valve in the compartment with the crack due to the accident environment does not prevent safe shutdown of the unit. The valve in the compartments without the crack have a demonstrated operability beyond that required for its safety function.

4. Auxiliary Feedwater Containment Isolation Valves

These valves are motor operated, fail-as-is and are normally open and de-energized. Consequently, the accident environment cannot cause these valves to close. It is not postulated that there is a requirement to close the valve supplying auxiliary feedwater to the faulted steam generator because the auxiliary feedwater pump and flow control valves which are isolated from the break environment would be available to perform this function. The valves have a demonstrated operability beyond that required for their safety function.

5. Auxiliary Feedwater Flow Indicators

These instruments indicate flow in the control room for accident monitoring, but have no control function. Although other indication could perform this function (eg. steam generator level) these instruments must not fail in a manner that could lead operators to defeat or fail to accomplish a required safety function. They have a demonstrated operability beyond that required for the safe plant shutdown.

Flooding Evaluation

The worst flooding will occur as a result of a feedwater line break in any one of the feedwater piping compartments 100A through 100D. The water spills over an opening 7' high and 3' wide into compartments 100E through 100H. The maximum flood height in the break compartment is 9.56’. The maximum flood level in the spill over compartment is 8.06’. The structural integrity of the building will not be impaired as a result of these flood levels concurrent with the maximum pressure developed.

Chronology of Events

During the following events, the maximum temperature occurs inside the compartments during the main steam line crack and the maximum pressure occurs during the feedwater line crack.

1. Main Steam Line Crack

Maximum temperature inside the compartments containing essential equipment is reached within 155 seconds into the event. The time sequence of events for the MSLB with superheat is described in Table 3.6B-4A (Unit 1) and Table 3.6B-4B (Unit 2).

2. Feedwater Line Crack

   Low water level in the affected steam generator causes reactor trip and auxiliary feedwater pump start. Feedwater pumps will continue to deliver feedwater to the crack for approximately two minutes when low water level in the condenser hot wells causes the condensate pumps to trip. Low suction pressure will trip the main feed pumps and consequently pumped flow to the crack will be terminated. At that time steam from all steam generators and from the auxiliary feedwater being added to the faulted loop steam generator will be emitted from the crack. Main steam isolation occurs on low steam pressure or on operator action. Operator action terminates auxiliary feedwater addition to the faulted steam generator. The steam leakage phase of the feedwater line crack accident will increase the environmental temperature to a level which is less than that reached during the main steam line crack accident. The maximum pressure occurs during the water leakage phase of the feedwater line crack accident.

3.6B.1.3 Safety Evaluation

The results of the pipe whip and jet impingement analyses, which verify that the consequences of failures of high energy lines do not affect the ability to safely shut down the plant, are provided in the corresponding FSAR sections for the systems listed in Section 3.6B.1.2.1. Incorporated in the analyses are the results of the failure modes and effects analysis, which are described in the safety evaluation section for each system.

Specific design features utilized for the protection of essential systems are identified in Section 3.6B.2.5. Protection for essential structures and components to ensure that their minimum required function can be accomplished following a postulated pipe rupture is provided by one or more of the following methods:

1. Separation and remote location of piping from essential structures and components. Separation is achieved by physical plant arrangements that provided sufficient distances between essential structures, components and piping such that the effects of postulated pipe ruptures cannot impair the structural integrity or operability of the essential structure or component.

2. Barriers, shields and enclosures. Barriers, shields or enclosures are provided for piping or, alternatively, protection of components within structures or compartments is provided. These elements are designed to withstand and contain the effects of the postulated pipe ruptures.

3. Pipe whip restraints. Where physical separation, shields or enclosures are not feasible, protection of essential systems and components is attained by the use of pipe whip restraints and barriers. Alternatively, essential systems and components may be designed to withstand the effects of the postulated pipe rupture.

The design criteria for the restraints are given in Section 3.6B.2.3.

No protection or restraints are provided when it is determined that the piping failure will not cause unacceptable damage to essential systems.

Each longitudinal or circumferential break in high-energy fluid system piping is considered separately as a single postulated initial event occurring during normal plant conditions.

In analyzing the effects of postulated piping failures, the following assumptions are considered with regard to the operability of systems and components:

1. Offsite power is assumed to be unavailable if a trip of the turbine-generator system or reactor protection system is a direct consequence of the postulated piping failure.

2. A single active failure is assumed in systems used to mitigate consequences of the postulated piping failure and to shut down the reactor, except as noted in (3) below. The single active component failure is assumed to occur in addition to the postulated piping failure and any direct consequences of the piping failure, such as unit trip and loss of offsite power.

3. Where the postulated piping failure is assumed to occur in one of two or more redundant trains of a dual-purpose moderate-energy essential system, (i.e., one required to operate during normal plant conditions as well as to shut down the reactor and mitigate the consequences of the piping failure), single failures in the other train or trains of that system are not assumed provided the following conditions are met:

   a. The system is designed to seismic Category I Standards.

   b. The system is powered from both offsite and onsite sources.

   c. The system is constructed, operated, and inspected to quality assurance, testing, and in-service inspection standards appropriate for nuclear safety systems.

4. All available systems, including those actuated by operator actions, are employed to mitigate the consequences of a postulated piping failure. In judging the availability of systems, account is taken of the postulated failure and its direct consequences such as unit trip and loss of offsite power, and of the assumed single active failure and its direct consequences.

5. Operator action credited to mitigate the consequences of the postulated pipe failure is analyzed for each specific event. In general, no operator action is considered to mitigate the consequences of a postulated pipe failure for 30 minutes.

   However, if detection and termination of blowdown can be accomplished in the control room, then an operator action of 10 minutes is credited.

6. A whipping pipe has insufficient energy to cause loss of structural (pressure retaining) integrity in target piping systems of equal or larger nominal pipe size and wall thickness and has sufficient energy to cause through-wall leakage cracks in target piping systems of equal or larger nominal pipe size but with less wall thickness. Any whipping pipe has sufficient energy to cause loss of function of active components such as valves, pumps and motors.

7. Unless it is determined by appropriate calculations or tests, a jet emanating from either a circumferential or a longitudinal break of a high energy system has sufficient energy to

cause loss of structural integrity in piping systems of smaller diameter, through wall leakage cracks in larger pipes with lesser wall thickness, and loss of function in active components such as valves, pumps and motors. Check valves that have equal or larger nominal pipe size and wall thickness than the pipe containing the break shall remain operable.

8. Credit is taken, where possible for automatic valves which fail in the safe position.

9. Essential components which are qualified or can be demonstrated to operate in an adverse environment due to a pipe failure are not protected.

The protective measures utilized, as described in this section, will not prevent the access required to conduct the inservice examinations specified in the ASME Boiler and Pressure Vessel Code, Section XI, Division 1, “Rules for Inspection and Testing of Components in Light-Water Cooled Plants.”

All essential systems, components, piping, component enclosures, protective structural barriers and structures supporting pipe whip restraints are designed to seismic Category I requirements, with the following exceptions. The moderate energy spray shields have structural supports that are designed to Seismic Category I, however, the shielding is constructed of a plastic type material and as such has no seismic pedigree. Architectural features (e.g. doors), tornado dampers, HELB mitigation dampers, ductwork, penetration seals, and backwater valves of the floor drain system are non-seismic category I. (Although some backwater valves are credited to operate for the mitigation of postulated piping failures, a seismic event is not assumed in the analysis of these failures.) Primary plant ventilation (Section 9.4.3) supply (Figure 9.4-9, Sh. 3) and associated chilled water (Section 9.4E) are non-seismic category I. Also see Section 7.6.12 for detection and mitigation equipment exceptions.

3.6B.2 DETERMINATION OF BREAK LOCATIONS AND DYNAMIC EFFECTS ASSOCIATED WITH THE POSTULATED RUPTURE OF PIPING (INSIDE AND OUTSIDE CONTAINMENT)

The design bases for locating postulated breaks and cracks in piping inside and outside of the Containment, the procedure used to define the jet thrust reaction at the break or crack location and the jet impingement loading on adjacent structures, equipment, systems, and components are described as follows.

3.6B.2.1 Criteria Used To Define Break And Crack Location And Configuration

This section provides criteria for the location and configuration of postulated pipe breaks in high energy piping systems.

3.6B.2.1.1 Reactor Coolant System (RCS) Main Loop Piping

The generic Leak-Before-Break technology described in NUREG-1061 Volume 3 [18] has been applied to the CPNPP Units 1 and 2 RCS main loop piping.

This application of Leak-Before-Break methodology, allowed under the modified GDC-4, is discussed in reference [19].

The analyses show that the probability of RCS main loop piping breaks is extremely low, thus the dynamic effects of these breaks are not considered in the design basis of CPNPP. CPNPP leak-before-break was developed from detailed analysis of the following factors [19]:

1. The loads, material properties, transients, and geometry for CPNPP Units 1 and 2 RCS primary loop are enveloped by the Westinghouse generic leak-before-break analyses.

2. Stress corrosion cracking is precluded by use of fracture resistant materials in the piping system and controls on reactor coolant chemistry, temperature, pressure, and flow during normal operation.

3. Water hammer is precluded in the RCS primary loop piping because of system design, testing, and operational considerations.

4. The effects of low and high cycle fatigue on the integrity of the primary piping are negligible.

5. Ample margin exists between the leak rate of the reference flaw and the criteria of Reg. Guide 1.45. (See FSAR Sections 1A(B) and 5.2.5).

6. Ample margin exists between the reference flaw chosen for leak detectability and the “critical” flaw.

7. Ample margin exists in the material properties used to demonstrate end-of-life (relative to aging) stability of the reference flaw.

The reference flaw will be stable throughout reactor life because of the ample margins in e, f, and g, above and will leak at a detectable rate which will assure a safe plant shutdown.

Although the dynamic effects of the RCS main loop piping breaks are not considered in the design basis, GDC 4 (FSAR Section 3.1.1.4) requires that primary coolant piping breaks be included in the design of the reactor containment, the ECCS, and in environmental qualification. In order to comply with this requirement, main loop breaks are postulated as discussed in Sections 6.2, 6.3 and 3.11 respectively.

The new Leak Before Break (LBB) leak rate calculation methodology for Alloy 82/182 welds with Structural Weld Overlay (SWOL), has been approved by the Nuclear Regulatory Commission for use in Pressurizer Surge Line (PSL) LBB analysis at the Waterford nuclear power plant. This methodology is applicable for CPNPP and has been utilized for the LBB analysis for Alloy 82/182 welds with SWOL for the pressurize surge line nozzle.

3.6B.2.1.2 High Energy Piping Other Than The RCS Main Loop

High energy fluid systems are those systems that during normal plant conditions are either in operation or maintained pressurized under conditions where either or both of the following are met:

Maximum operating temperature exceeds 200°F

Maximum operating pressure exceeds 275 psig.

It is noted that systems which operate within the pressure and temperature conditions specified for high energy fluid systems for less than two percent (2%) of the time are considered moderate energy. Breaks for these systems are postulated as described in Section 3.6B.2.1.4. A list of high energy lines is given in Table 3.6B-1.

Design basis break locations and types are postulated in accordance with NRC Regulatory Guide 1.46 and Branch Technical Position MEB 3-1. Where required, postulated pipe breaks are selected as described below and analyzed to demonstrate the capability for a safe shutdown of the plant.

Application of leak before break methodology justifies the elimination of postulated pipe ruptures in 10 inch and larger RCS branch lines. This analysis demonstrates the piping integrity and serves as the basis for excluding from consideration the dynamic effects associated with postulated pipe ruptures.

1. ASME Section III, Code Class 1 Piping (Excluding Primary Reactor Coolant Loop Piping)

   For high energy piping in the RC, CVC, RHR, and SI systems, the 1977 Edition up to and including Summer 1979 Addenda of ASME B&PV Code Section III, Code Class 1 will be used to postulate pipe breaks. For all other high energy piping the 1974 Edition up to and including Winter 1975 Addenda is applicable. The pipe breaks are postulated to occur at terminal ends and at all intermediate locations in the piping system where none of the following criteria is met:

   a. The primary plus secondary stress intensity range (equation (10) of ASME III Subparagraph NB-3653) derived on an elastically calculated basis under loadings associated with the OBE and normal and upset plant conditions does not exceed 2.4 S_m or

   b. The primary plus secondary stress intensity range derived on an elastically calculated basis under loadings associated with the OBE and normal and upset plant conditions exceeds 2.4 S_m but is less than 3.0 S_m, and the cumulative usage factor is less than 0.1 or

   c. The primary plus secondary stress intensity range derived on an elastically calculated basis under loadings associated with the OBE and normal and upset plant conditions exceeds 3.0 S_m but the stress ranges computed by Equations (12) and (13) of Subparagraph NB-3653 of ASME III are less than 2.4 S_m and the cumulative usage factor is less than 0.1.

2. ASME Section III, Code Class 2 and 3 Piping

   ASME B&PV Code, Section III, Class 2 and Class 3 piping breaks are postulated to occur at terminal ends and intermediate locations in each piping run or branch run. Breaks at intermediate locations are selected by either of the following criteria:

   a. At each location where the stresses associated with normal and upset plant conditions and an OBE event, calculated by the sum of Equations (9) and (10) of paragraph NC-3652 of the ASME B&PV Code, Section III, exceed 0.8(1.2 S_h+S_A).

b. Certain pipe runs contain Class 2 piping extensions to Class 1 lines up to the first anchor point beyond the Class 1/Class 2 boundary. For these pipe runs a break is postulated at the Class 2 terminal end. Breaks are not postulated in the Class 2 portion if the stresses in the Class 1 portion, calculated using Equation (10) of NB-3653, are above 2.4 S_m and the sum of the stresses in the Class 2 portion calculated using Equations (9) and (10) of NC- 3652, are below 0.8 (1.2 S_h + S_A).

3. Non-Nuclear Piping

Breaks are postulated to occur in non-nuclear piping systems at the locations as specified for ASME Section III, Class 2 and 3 piping in accordance with Section B criteria above when a seismic stress analysis is performed.

Breaks in non-nuclear piping systems are postulated at terminal ends and at intermediate points, such as fittings (elbows, tees, reducers, etc.), welded attachments and valves in each run or branch run where a stress analysis is not performed.

4. Fluid System Piping Between Containment Isolation Valves

There are no ASME Section III, Class I piping penetrations. The piping between containment Isolation valves is ASME B&PV Code, Section III, Class 2 piping.

Breaks are not postulated in the break exclusion areas of high energy fluid system piping from the inside containment process pipe-to-penetration weld to and including the outside containment isolation valve moment restraint forging as shown on Figures 3.6B-15, 16, 17, 18, 23, 25, 26, 27, 28, 29, 30, 42, 82, 85, 87 and 88. Moment restraints/forgings are provided to protect the containment isolation valves, process pipes and penetrations within the break exclusion areas. A typical moment restraint/forging, including the boundaries of the break exclusion area, is shown in Figure 3.6B-5.

The following stress limits of Branch Technical Positions APCSB 3-1 [3] and MEB 3-1 [2] for ASME B&PV Code, Section III, Class 2 piping are met within the break exclusion areas:

a. The maximum stress ranges as calculated by the sum of Equations (9) and (10), paragraph NC-3652 of the ASME B&PV Code, Section III, considering normal and upset plant conditions (i.e., sustained loads, occasional loads, and thermal expansion), and an OBE event do not exceed 0.8 ($1.2S_h$+ S_A).

b. The maximum stress, as calculated by Equation (9), paragraph NC-3652 of the ASME B&PV Code, Section III, under the loadings resulting from a postulated piping failure beyond the break exclusion area, does not exceed 1.8 S_h. The moment restraint forging is permitted higher stresses outboard of the restrained locations provided a plastic hinge is not formed.

Welded attachments, for pipe supports or other purposes, to these portions of piping are avoided except where detailed stress analyses or tests are performed to demonstrate compliance with the limits defined in 4.a and 4.b above. In

addition, the number of circumferential and longitudinal piping welds and branch connections is minimized. Guard pipes are not used.

The length of these portions of piping is reduced to the minimum length practical. The design of pipe anchors or restraints (e.g., connections to Containment penetrations and pipe whip restraints) does not require welding directly to the outer surface of the piping (e.g., flued-integrally forged pipe fittings are used). If welded attachments are required, such welds are 100-percent volumetrically examinable in service, and a detailed stress analysis shall be performed to demonstrate compliance with the limits defined in 4.a and 4.b above.

3.6B.2.1.3 Type of Breaks Postulated in Fluid System Piping Other Than the RCS Main Loop

Circumferential and longitudinal breaks are postulated in piping systems at locations previously discussed. The following are definitions and criteria used to determine the type of break to be considered:

1. Circumferential Pipe Breaks

Circumferential breaks are defined as a full cross section area break, with at least one diameter lateral displacement of the ruptured pipe. Circumferential breaks are postulated in piping with nominal size greater than 1", unless the break separation is physically limited by piping restraints, by structural members, or by piping stiffness as demonstrated by inelastic limit analysis. Circumferential breaks are perpendicular to the pipe axis, and the break area is equivalent to the internal cross-sectional area of the ruptured pipe. Dynamic forces at the break location are based on a calculated fluid pressure as modified by an analytically or experimentally determined thrust coefficient. These forces separate the piping axially causing the ruptured pipes to move in the direction of the thrust force. The ruptured pipe is assumed to whip in the plane defined by the piping geometry and configuration causing the pipe to move in the direction of the jet reaction.

Only circumferential breaks are postulated at terminal points, whether or not the piping is seamless. For a terminal point at an equipment or vessel nozzle, a circumferential break is postulated at the pipe-to-nozzle weld. If the terminal point is at a normally closed valve pressurized on one side, the circumferential break is at the pipe-to-valve weld on the high energy side. For a terminal point where a fitting is installed, a circumferential pipe break is postulated only at terminal-to-fitting weld. The pipe-to-fitting weld (the second weld on fitting) may be used as an intermediate pipe break if required by stress level.

2. Longitudinal Pipe Breaks

Longitudinal breaks are assumed to result in an axial split of the pipe without severance. Splits are oriented (but not concurrently) at two diametrically opposed points on the piping circumference so that a jet reaction causing out-of-plane bending of the piping configuration results. Alternatively, a single split may be assumed at the section of highest tensile stress as determined by an elastic stress analysis.

Longitudinal breaks in high energy fluid system piping are postulated in nominal pipe sizes 4-inches and larger. Longitudinal breaks are not postulated at terminal ends. The provision for longitudinal breaks at terminal points applies not only to anchored points, but

also to terminal points at branch connections and at the point immediately outboard of the break exclusion area. Longitudinal breaks are also not postulated at intermediate locations where the criterion for the minimum number of breaks as described in Sections 3.6B.2.1.2.1 and 3.6B.2.1.2.1 is satisfied.

If a detailed stress analysis or test is performed the results may be used to predict the most probable rupture location(s) for the longitudinal breaks. Without the benefit of the stress analysis report, longitudinal breaks will be postulated to occur on each side of a fitting.

Only one type of break need be postulated at a location based on the stresses as identified by a stress analysis. If the primary plus secondary stress in the axial direction is found to be at least 1.5 times that in the circumferential direction for the most severe normal plus upset loading combination, then only a circumferential break need be postulated. Conversely, if the primary plus secondary stress in the circumferential direction is found to be at least 1.5 times that in the axial direction for the most severe loading combination noted above, then only a longitudinal break need be postulated.

The dynamic force of the fluid jet discharge is based on a circular break area equal to the effective cross-sectional flow area of the pipe at the break location and on a calculated fluid pressure modified by an analytically or experimentally determined thrust coefficient as determined for a circumferential break at the same location.

Piping movement is assumed to occur in the direction of the jet reaction unless limited by structural members, piping restraints, or piping stiffness as demonstrated by inelastic limit analysis.

3.6B.2.1.4 Moderate Energy Piping

Moderate-energy fluid systems are the fluid systems that during normal plant conditions are either in normal operation or maintained pressurized under conditions where both of the following are met:

Maximum operating temperature is 200°F or less

Maximum operating pressure is 275 psig or less

Through-wall leakage cracks are postulated for moderate energy piping systems inside and outside the containment in accordance with the following criteria:

1. Through wall leakage cracks are postulated either at any location in moderate energy piping larger than 1 inch, where essential equipment may be affected by the environmental effects from fluid spray and flooding or only where the stresses associated with normal and upset plant conditions and an OBE event as calculated by the sum of equations (9) and (10) of paragraph NC-3652 of the ASME B&PV Code, Section III exceed .4 ($1.2\ S_h + S_a$).

2. Fluid flow from a crack is based on a circular opening of area equal to that of a rectangle which is one-half pipe internal diameter in length and one-half pipe wall thickness in width.

3. Cracks instead of breaks are postulated in the piping of those fluid systems that qualify as high energy fluid systems for only short operational periods, but qualify as moderate energy fluid systems for the major operational period.

   An operational period is considered short if the fraction of time that the system operates within the pressure-temperature conditions specified for high energy fluid systems is less than two percent (2%) of the time that the system operates as a moderate energy fluid system.

   Where through wall leakage cracks are postulated in one of two or more redundant trains of a dual-purpose moderate-energy essential system (i.e., one train required to operate during normal plant conditions as well as to shut down the reactor and mitigate the consequences of the piping failure), single failures of components in the other train or trains of that system are not assumed provided the system is designed to seismic Category I standards, is powered from both offsite and onsite power sources, and is constructed, operated, and inspected to quality assurance, testing, and inservice inspection standards appropriate for nuclear safety systems.

3.6B.2.1.5 Definition of Plant Operating Conditions

Plant operating conditions which are used in calculating the stresses for ASME Section III piping to establish pipe break locations are as follows:

1. Normal Plant Conditions

   Plant operating conditions during reactor startup, operation at power, hot standby, or reactor cooldown to cold shutdown condition.

2. Upset Plant Conditions

   Plant operating conditions during system transients that may occur with moderate frequency during plant service life and are anticipated operational occurrences, but not during system testing.

3.6B.2.2 Analytical Methods to Define Forcing Functions and Response Models

The analytical methods used to define the forcing functions and the response models are described in the following sections.

3.6B.2.2.1 Reactor Coolant System Main Loop Piping

In order to determine the thrust and reactive force loads to be applied to the reactor coolant loops during the postulated loss of coolant accident (LOCA), it is necessary to have a detailed description of the hydraulic transient. Hydraulic forcing functions are calculated for the ruptured and intact reactor coolant loops as a result of a postulated LOCA. These forces result from the transient flow and pressure histories in the RCS. The calculation is performed in two steps. The

first step is to calculate the transient pressure, mass flow rates, and thermodynamic properties as a function of time. The second step uses the results obtained from the hydraulic analysis along with input of areas and direction coordinates and calculates the time history of forces at appropriate locations in the reactor coolant loops.

The hydraulic model represents the behavior of the coolant fluid within the entire RCS. Key parameters calculated by the hydraulic model are pressure, mass flow rate, and fluid density. These are input to the thrust calculation, with appropriate plant layout information, to determine the time-dependent loads exerted by the fluid on the loops. In evaluating the hydraulic forcing functions during a postulated LOCA, the pressure and momentum flux terms are dominant. The inertia and gravitational terms are taken into account in evaluation of the local fluid conditions in the hydraulic mode.

The blowdown hydraulic analysis is required to provide the basic information concerning the dynamic behavior of the reactor core environment for the loop forces, reactor kinetics and core cooling analysis. This requires the ability to predict the flow, quality, and pressure of the fluid throughout the reactor system. The SATAN-IV code [11] was developed with a capability to provide this information.

1. Unit 1

   The MULTIFLEX 3.0 [42] computer code was developed with this capability, which is an enhancement and extension of MULTIFLEX 1.0 [37] NRC reviewed and approved computer code developed for the space-time dependent analysis of nuclear power plants. The MULTIFLEX 3.0 features which differ from MULTIFLEX 1.0 are primarily related to vessel forces. The loop forcing functions do not differ significantly from those generated using the NRC approved MULTIFLEX 1.0 model. MULTIFLEX 3.0 has been accepted by the NRC for several other applications [38, 39, 40 and 41] and also has been extensively used for the LOCA analyses of reactor internals components of numerous other 2, 3, and 4 loop nuclear power plants.

   MULTIFLEX is a digital computer program for calculation of pressure, velocity and force transients in reactor primary coolant systems during the subcooled, transition and the early saturation portion of blowdown caused by a LOCA. During this phase of the accident, large amplitude rarefaction waves are propagated through the system with the velocity of sound causing large differences in local pressures. As local pressures drop below saturation, causing the formation of steam, the amplitudes and velocities of these waves drastically decrease. Therefore, the largest forces across the loop piping due to wave propagation occur during the subcooled portions of the blowdown transient. MULTIFLEX includes mechanical structure models and their interaction with the thermal-hydraulic system, although these features are only involved in the vessel and steam generator modeling.

2. Unit 2

The SATAN-IV code performs a comprehensive space time-dependent analysis of a LOCA and is designed to treat all phases of the blowdown. The stages are:

1. A subcooled stage where the rapidly changing pressure gradients in the subcooled fluid exerts an influence upon the RCS internals and support structures

2. A two phase depressurization stage

3. The saturated stage.

The code employs a one dimensional analysis in which the entire RCS is divided into control volumes. The fluid properties are considered uniform and thermodynamic equilibrium is assumed in each element. Pump characteristics, pump coastdown and cavitation, core and steam generator heat transfer, including the W-3 departure from nucleate boiling (DNB) correlation, and reactor kinetics are incorporated in the code.

The THRUST (Unit 1) or STHRUST (Unit 2) computer programs [12] were developed to compute the transient (blowdown) hydraulic loads resulting from a LOCA.

The blowdown hydraulic loads on primary loop components are computed from the equation

$$F = 144A\left[\frac{(P - 14.7) + (m^2)}{144\rho\ g\ A^2 m}\right]$$

which includes both the static and dynamic effects. The symbols and units are:

F = force (lb_f)

A = aperture area (ft^2)

P = system pressure (psig)

m = mass flow rate (lbm/sec)

ρ = density (lb_m/ft^3)

g = gravitational constant (32.174 ft-$lb_m/lb_f sec^{2)}$

A_m = mass flow area (ft^2)

In the model to compute forcing functions the reactor coolant loop system is represented by a similar model as employed in the blowdown analysis. The entire loop layout is described in a global coordinate system. Each node is fully described by blowdown hydraulic information and the orientation of the streamlines of the force nodes in the system. These include flow area and projection coefficients along the three axes of the global coordinate system. Each node is modeled as a separate control volume, with one or two flow apertures associated with it. Two apertures are used to simulate a change in flow direction and area. Each force is divided into its x, y, and z components using the projection coefficients. The force components are then summed over the total number of apertures in any one node to give a total x force, total y force, and total z force. These thrust forces serve as input to the piping/restraint dynamic analysis described in Section 3.9N.14.

3.6B.2.2.2 High-Energy Piping Other Than RCS Main Loop

The time dependent function representing the thrust forces caused by the jet flow from a postulated pipe break or crack includes the combined effects of the thrust impulse resulting from the sudden pressure drop at the initial moment of pipe rupture, the thrust transient resulting from wave propagation and reflection, and the blowdown thrust resulting from buildup of the discharge flow rate which may reach steady state if there is a fluid energy reservoir having sufficient capacity to develop a steady jet for a significant interval. Alternatively, in a simplified method, the jet thrust force is represented by a steady state function. This function, representing the force, would have a magnitude not less than:

$$F_{ss} = C_t PA$$

where:

F_{ss} = steady state thrust force (lb_f)

P = system pressure prior to pipe break (lb_f/in^2)

A = pipe break area (in^2)

C_t = steady state thrust coefficient

The steady state thrust coefficient C_t is dependent on the fluid state and the frictional loss terms. The value of steady state thrust coefficient and the time to reach steady state flow conditions are calculated from references [15], [16] and [22].

The rigorous time dependent blowdown forces resulting from a postulated pipe rupture are determined using the RELAP-5 computer code [6 or 34] except where STEHAM [28] is used as part of the alternative dynamic methodology for postulated main steam line breaks described later in this section. RELAP-5 is a thermal/hydraulic program commonly used in the nuclear industry to evaluate the behavior of water cooled reactor systems during postulated accidents such as pipe ruptures. The program is acceptable (see Reference [7]) as a means of determining the hydraulic forcing function at the pipe break. CALPLOTF-III [20] or REPIPE [29] or other post processor programs to the RELAP program are used to develop the break force time-history plots.

The RELAP-5 program solves the transient energy, momentum, and fluid state equations to determine the system flow, pressure, and thermodynamic conditions. The break force is computed using the one-dimensional momentum equation and the appropriate density, internal energy, and pressure values. The rupture load is the summation of the pressure, momentum, and change in momentum terms at the time interval in question.

RELAP-5 has the capability of solving the fluid state equation for subcooled water, flashing water, two-phase steam/water mixtures, and superheated steam. The ASME steam tables [9] have been incorporated into RELAP-5 so that the fluid state properties are accurately determined. RELAP-5 has a provision for modeling components such as valves, check valves, pumps, heat exchangers, and reactors along with the associated piping.

Transients can be initiated by the control card added to the program which is used to describe leaks (pipe breaks), valves opening and closing, check valve pressure drop-flow-characteristics, pump coastdowns, and so forth.

The flow system is described as a series of volumes connected by flow paths or junctions. RELAP-5 requires input data that completely describe the thermodynamic conditions and physical data of the system being analyzed. Pressure, temperature, and flow conditions along with physical dimensions, flow areas, friction characteristics must all be specified as initial conditions. The break area can be reduced by an analytically or experimentally derived discharge coefficient. However, in lieu of such data it is conservatively assumed that the discharge coefficient is 1.0 for both longitudinal and circumferential breaks. In a similar manner, the break area is assumed to open within one millisecond (0.001 second).

Rigorous time dependent blowdown forces and pipe segment forces resulting from the postulated breaks were also determined by using the STEHAM computer program. The analysis is based on the method of characteristics with finite difference approximations both in space and in time. STEHAM calculates the one-dimensional transient flow responses and the flow-induced forcing functions in a piping system caused by rapid operational changing of piping components, such as the stop valve and the safety/relief valve which are used to represent pipe break. Flow characteristics of piping components are mathematically formulated as boundary condition and frictional effects are appropriately taken into consideration in this computer program. Output from STEHAM consists of time-histories of flow pressures, flow densities, flow velocities, inertia, and momentum functions. CHPLOT [32] program is used to plot the force time-history data developed by STEHAM. These data are used as input to the LIMITA3 structural computer program.

The piping dynamic responses resulting from a postulated pipe rupture are determined using the PIPERUP [13], SHPLAST 2267 [24], ABAQUS [21], ANSYS [30 or 35] or LIMITA3 [31] computer codes. The programs are adaptations of the finite element method to the requirements of pipe rupture analyses. They perform a dynamic, nonlinear, elastic-plastic analysis of piping systems subjected to time-history forcing functions. These forces result from fluid jet thrust at the location of a postulated longitudinal or circumferential rupture of high energy piping and ensuing acoustic disturbances within the piping.

The piping is mathematically modeled in the PIPERUP, SHPLAST 2267, ABAQUS, ANSYS or LIMITA3 program as an assembly of weightless structural members connecting discrete nodal points. A typical pipe whip mathematical model is shown in Figure 3.6B-6. Weight of the system, including distributed weight of the piping and concentrated weights (e.g., valves), is lumped at selected mass points (lumped parameter analysis model). Nodal points are placed in such a manner as to isolate particular types of piping elements such as straight runs of pipe, valves, elbows, etc. for which force-deformation characteristics may be determined. Nodal points are also placed at all discontinuities such as piping restraints, branch lines, and changes in cross-section. Piping restraints are modeled with an initial gap and in PIPERUP with a bilinear stiffness curve, or, in SHPLAST 2267, ABAQUS, ANSYS and LIMITA3 with multilinear stiffness curve. A typical piping stress-strain curve is shown in Figure 3.6B-7. The first stiffness represents linear elastic behavior and the second stiffness models linear strain hardening behavior. All five programs utilize a direct step-by-step integration method to determine the time history response of the ruptured piping system. A typical restraint impact curve is shown in Figure 3.6B-8. An incremental procedure is used to account for the nonlinear deformation and elastic-plastic effect of the pipe and restraints.

3.6B.2.3 Dynamic Analysis Methods to Verify Integrity and Operability

3.6B.2.3.1 Reactor Coolant System Main Loop

The leak-before-break technology has been applied to CPNPP Units 1 and 2 to exclude from the design basis the dynamic effects of postulated ruptures in the RCS main loop piping. This applies, in particular, to jet impingement loads on components and supports.

Jet loads from large branch nozzle breaks are addressed in Section 3.6B.2.3.2.

3.6B.2.3.2 High-Energy Piping Other than the RCS Main Loop

Pipe beaks are postulated in high-energy piping in accordance with the criteria in Section 3.6B.2.1.2. The analyses for determining the dynamic effects of pipe break are as follows:

1. Jet Impingement

   A circumferential or longitudinal break in a high energy line results in a jet of fluid emanating from the break point. For subcooled high energy lines where the fluid temperature is less than its saturation temperature at the surrounding environmental pressure, the discharge jet is characterized by a nearly constant diameter jet approximately equal to the break diameter. Since the fluid temperature is below saturation it will not flash but instead will form an incompressible fluid jet.

   In general, most of the high energy line breaks result in a two-phase choked (critical) flow at the break exit plane. Fluid pressure at the exit plane is in general at some pressure greater than ambient. As the fluid leaves the pipe break area, it expands as the jet pressure decreases from the higher exit (break) plane pressure to the atmospheric pressure surrounding the jet.

   A jet discharging from a saturated steam line will accelerate and expand due to the pressure differential, and it will partially condense to a low-moisture wet steam with the liquid phase in the form of dispersed, entrained water droplets. A jet discharging from a subcooled or saturated hot water line (greater than 212°F) will flash to a low quality wet steam. The flashing will cause the jet diameter to expand very rapidly.

   ANSI/ANS-58.2-1988 [22] provides an acceptable basis (including conservative analytical models) for the evaluation of jet impingement loads. The CPNPP jet impingement methodologies and models are consistent with ANSI/ANS 58.2-1988, as briefly described in the following sections:

   a. Jet Category and Geometry

      The area of the break is assumed to be equal to the flow area of the ruptured pipe. All the high energy line break jets can be summarized into the following three categories:

      1. Category I Jets - Non-Expanding Jets

For the liquid jets whose temperature is below the saturation temperature at ambient pressure, the initial free expansion does not occur. Incompressible liquid jets are assumed to travel with no increase in jet area. However, for target identification a conservative zone of influence of two diameters is utilized. The pressure is assumed to be uniform throughout the jet area.

2. Category II Jets - Steam and Flashing Water Jets which meet the Criteria of NUREG/CR-2913[25]:

The high energy two-phase jet is a complicated multidimensional flow phenomena. The high pressure and high temperature fluid that exits the break expands with supersonic velocities downstream of the break. Upon encountering a target (or obstacle) a shock wave forms in the flow field, and it is the thermodynamic properties downstream of this shock that determine the pressure field and load on the target. A multidimensional analysis, such as demonstrated in NUREG/CR-2913 [25], more realistically evaluates the thermodynamic properties of these jets. The jet cone shape for these Category II jets is a function of the break initial conditions (i.e., temperature and pressure). These jet shapes are determined by using the NUREG/CR-2913 contour curves and are used to establish the jet zone of influence for target identification. The NUREG/CR-2913 model provides a method for calculating target loads for initial pipe rupture fluid conditions of pressure between 60 and 170 bars (870 psia -2466 psia) and with subcooling of 0°C (0°F) to 70°C (126°F). Note that, for Category II jets, the jet shape from the break exit to the jet core region is not defined in NUREG/CR-2913 [25] and should be computed from ANSI/ANS-58.2-1988 [22] (see next section).

3. Category III Jets - All other steam and flashing water jets:

Category III jets are assumed to expand as a three region cone defined in Figure 3.6B-96A for circumferential breaks and Figure 3.6B-96B for longitudinal breaks [22].

(a) Circumferential Break with Full Separation:

Jet Region 1 ($L < L_c$). Region 1 includes a cone-shaped region containing the jet core and the remainder of the jet. This geometry is shown in Figure 3.6B-96C.

The jet core length is related to the jet subcooling at the jet break plane and has been correlated using the following expression

$$L_c/D_e = 0.26 \quad (\Delta T_{sub})^{1/2} + 0.5 \qquad (1)$$

where:

L_c = core length

D_e = pipe inside diameter

T_{sub} = jet subcooling at stagnation conditions in °F at the break plane

Figure 3.6B-96D can be used to relate jet stagnation subcooling at the break plane to stagnation conditions in the vessel supplying the jet flow, accounting for irreversible losses in the blowdown line.

In Region 1, for $0 < L < L_c$, the jet core diameter, D_c, is given by

$$D_c/D_e = (C^*_{Te})^{1/2} (1-L/L_c) \quad (2)$$

where:

C^*_{Te} = 2.0 for $\Delta T_{sub} > 0$

1.26 for $\Delta T_{sub} = 0$

L = distance from break plane to target

Jet Region 2 ($L_c < L < L_a$). In Region 2, the jet expands to its asymptotic area which can be calculated as:

$$A_a/A_e = G_e^2/(g_c \rho P_{ma} C_T P_o) \quad (3)$$

where:

A_a = jet area at the asymptotic plane

A_e = break plane area

C_T = steady-state thrust coefficient (as defined in ANSI/ANS-58.2 [22])

G_e = mass flow rate per unit area from the break plane

g_c = gravitational constant

P_o = initial total (stagnation) pressure in the vessel

ρ_{ma} = asymptotic plane density. If two-phase, density will be given by

ρ_{ma} = $1/\lfloor x/\rho_g + (1-x)/\rho_f \rfloor$

x = mixture vapor mass fraction i.e. quality, at the asymptotic plane pressure, Pa, and stagnation enthalpy

ρ_f = saturated liquid density at the asymptotic plane pressure

ρ_g = saturated vapor density at the asymptotic plane pressure

[Note: Figure 3.6B-96E may be used in place of Equation (3)]

The jet pressure at the asymptotic plane, Pa can be expressed as the following expression

$$(Pa)/P_{amb} = 1 - 0.5\,(1 - 2P_{amb}/P_o) f(h_o) \qquad (4)$$

where:

P_{amb} = ambient pressure

Pa = asymptotic plane static pressure

$$f(h_o) = \begin{cases} [0.1 + (h_o - h_f)/h_{fg}]^{1/2} & \text{for } (h_o - h_f)/h_{fg} > -0.1 \\ 0 & \text{for } (h_o - h_f)/h_{fg} < -0.1 \end{cases}$$

h_o = stagnation enthalpy in the vessel[(a)]

h_f, h_{fg} = saturated liquid enthalpy and heat of vaporization in the vessel

---

a) h_o in the vessel and at the break plane are assumed to be equal.

The distance from the break plane to the asymptotic plane is defined by:

$$L_a/D_e = 1/2\left[(A_a/A_e)^{1/2} - 1\right] \qquad (5)$$

The jet area at any location from the break plane to the asymptotic plane (Regions 1 and 2) may be calculated from the following relationship:

$$A_j/A_{je} = \left[1 + L/L_a\,(A_a/A_{je} - 1)\right], \qquad (6)$$

where:

A_j = jet area

A_{je} = jet area at break plane

Jet Region 3 ($L \geq L_a$). In Region 3, the jet area is given by

$$A_j A_a = (1 + (2(L - L_a)/D_a)(\tan 10°))^2 \quad (7)$$

where:

D_a = jet diameter at the asymptotic plane

(b) Longitudinal Break

The jet shape for longitudinal breaks, as shown in Figure 3.6B-96B shall be assumed to be the same as the circumferential break defined in 3.6B.2.3.2.1.a.3.a. A jet diameter for a circular break of the same area may be used and the jet direction taken to be perpendicular to the axis of the pipe.

2. Effective Target Distance

a. Category II Jets

For steam and flashing water jets within the limits of NUREG/CR-2913 [i.e., stagnation pressure from 60 bars (870 psia) to 170 bars (2466 psia) and subcooling of 0°C (0°F) to 70°C (126°F)] the effective target distance is taken as ten (10) times the inside diameter of the ruptured pipes [25].

b. Category I & III Jets

For all other high energy line break jets, jets are assumed to travel until impact with a target or a barrier.

3. Jet Force and Pressure

a. Category I Jets

The generalized momentum equation that describes the jet blowdown force, F_b, is;

$$F_b = G^2{}_e A_e/\rho_e g_c + A_e(P_e - P_{amb}) \quad (8)$$

where:

P_e = fluid pressure at the break flow area

ρ_e = fluid density at the break flow area

And for calculating target loads a conservative quasi-steady-state jet force, F_j, is used:

$$F_j = A_e (C_T P_o - P_{amb}) - C_T P_o A_e \quad (9)$$

However, the above equation for F_j is modified as follows:

1. For jets where $\Delta T_{sub} > 0$, C_T is multiplied by $(2.0/C_{Te})$ in region 1, where $C_{Te} = C_T$ based on FL/D=O and the break plane stagnation condition.

2. Unless otherwise justified, F_j is not to be less than the initial jet force based on equation (8).

For liquid jets whose temperature is below the saturation temperature at ambient pressure, a uniform pressure over the jet cross-section can be assumed, which is consistent with the jet area and the total jet force as defined by equation (8) or (9).

b. Category II Jets

In NUREG/CR-2913 [25], an advanced two-dimensional computer code (CSQ) was used to develop a two-phase jet load data base. The results showed that the jet force/pressure can be expressed as functions of four independent variables:

$F_j = F_j(r/D_e, L/D_e, P_o, T_o, \text{or } X_o)$ and

$P_j = P_j(r/D_e, L/D_e, P_o, T_o, \text{or } X_o)$

where:

r = radial distance from the jet center line

T_o = stagnation temperature in the vessel

X_o = stagnation quality in the vessel

P_j = radial pressure distribution on and an area perpendicular to the jet centerline

The values of F_j and P_j corresponding to some specific conditions of r/D_e, L/D_e, P_o, T_o or X_o can be found in NUREG/CR-2913 [25].

c. Category III Jets

The total jet force defined by Equations (8) and (9) can also be applied to Category III Jets. However, it should be recognized that typical Category III Jets (except low pressure jets which are discussed later) are under expanded, which means that the jet pressure at the break plane is greater than the ambient pressure. The jet expands rapidly as it leaves the break and comes to equilibrium with the surrounding pressure. Therefore, the jet impingement pressure decreases in a radial direction, away from the jet centerline. According to ANSI/ ANS-58.2-1988, the radial pressure distributions in the different jet regions are given below.

1. For Region 1, defined as $0 < L < L_c$, the jet pressure in the core and outside the core is given by

   Jet Core; for $0 < r < D_c/2$,

$$P_j = P_{oe} = \frac{F_j}{A_{je}} \quad = \text{stagnation pressure at the break} \tag{11}$$

   Outside Core; for $D_c/2 < r < D_j/2$,

$$\frac{P_j}{P_{oc}} = \left(\frac{D_j - 2r}{D_j - D_c}\right)\left[1 - \frac{2[D_j^2 + D_j D_c + D_c^2 - 3D_c^2\ C^*_{Te}\ ]}{(D_j^2 - D_c^2\ )}\right] \tag{12a}$$

$$\text{if}\left[(D_j/D_e)^2 + 2(D_j\ D_c/D_e^2\ ) + 3(D_c/D_e)^2\right] > 6C^*_{T_e} \text{ then}$$
$$\frac{P_j}{P_{oe}} = \left(\frac{D_j - 2r}{D_j - D_c}\right)^2\left[\frac{6(D_e^2\ C^*_{T_e} - D_c{}^2)}{(D_j - D_c)(D_j + 3\ D_c)}\right] \tag{12b}$$

2. For Region 2, defined as $L_c < L < L_a$, the jet pressure is given by

$$\frac{P_j}{P_{jc}} = \left(1 - \frac{2r}{D_j}\right)\left[1 - 2\left(\frac{2r}{D_j}\right)\left[1 - 3C_{Te}\left(\frac{D_e}{D_j}\right)^2\left(\frac{P_{oe}}{P_{jc}}\right)\right]\right] \tag{13}$$

   where

$$\frac{P_{jc}}{P_{oe}} = \left[F_c - 3C_{Te}\left(\frac{D_e}{D_a}\right)^2\right]\frac{L_a(L - L_c)}{L(L_a - L_c)} \quad \begin{array}{l}\text{= Jet centerline}\\ \text{pressure for}\\ (L_c<L<L_a)\end{array} \tag{14}$$

   F_c = 1.0 for $(D_j/D_e)^2 < 6C_{Te}^*$ at $L=L_c$

F_c = 6 CTe* $(D_e/D_j)^2$ for $(D_j/D_e)^2 > 6$ C_{Te}* at $L=L_c$

C_{Te} = Steady-state thrust coefficient defined at break plane conditions

3. For Region 3, defined as $L \geq L_a$, the jet pressure is given by

$$\frac{P_j}{P_{jc}} = \frac{D_j - 2r}{D_j} \quad (15)$$

where

Pjc = 3 F_j/A_j, jet centerline pressure for ($L > L_a$) (16)

It should be noted that the above pressure distribution can only be applied to large targets on the jet center line and underexpanded jets. For a small target which is not on the jet center line, judgement should be made to decide if the impingement load can be computed using the jet radial pressure distribution. This is particularly true at the outer edge of the jet where the jet flow may be away from the jet center line and not parallel to it. Additionally, if the stagnation pressure at the break flow area is sufficiently close to the ambient pressure (low pressure jets), the core length, L_c, calculated by equation (1) may be greater than the distance to the asymptotic surface La, calculated by equation (5). For this bounding case, the core length may be set to zero (L_c =0) and the jet pressure distribution assumed to be uniform over the jet cross section and equal to F_j/A_j.

4. Jet Impingement Force

The jet impingement force which is applied to a given target is a function of the fraction of the jet which is intercepted by the target. If the entire jet is intercepted, then the entire jet force is applied to the target.

$$F_{jt} = F_j \quad (17)$$

If the target intercepts a fraction of the jet, but not the entire jet, the jet pressure distribution over the target must be integrated to obtain the jet force.

$$F_{jt} = \int P_j \, dA_t \quad (18)$$

where

P_j = radial jet pressure distribution at the impingement plane, as described in 3.6B.2.3.2.3 above.

A_t = Target area

The impingement load may be estimated from the jet axial force and an approximate correction factor,

$$F_{imp} = K\varphi\ F_{jt}(DLF) \quad (19)$$

where

F_{imp} = impingement force on the target, as a function of time

$K\varphi$ = the shape factor, a measure of the target's potential for changing the momentum of the jet, as described in Appendix D of Reference 22

DLF = Dynamic Load Factor [26]

5. Pipe Whip Dynamic Analysis Criteria

An analysis of the pipe run or branch is performed for each longitudinal and/or circumferential postulated rupture at the break locations determined in accordance with the criteria of Section 3.6B.2.1.2. The loading condition of a pipe run or branch prior to postulated rupture, in terms of internal pressure, temperature and stress state, is assumed to be the condition associated with the normal plant operating condition.

For a circumferential rupture, pipe whip dynamic analyses are only performed for that end (or ends) of the pipe or branch that is connected to a contained fluid energy reservoir having sufficient capacity to develop a jet stream. Dynamic analytical methods used for calculating the piping and piping/restraint system response to the jet thrust developed after a postulated rupture adequately account for the effects of the following:

a. Translational masses (and rotational masses for major components) and stiffness properties of the piping system, restraint system, major components and support walls.

b. Transient forcing function(s) acting on the piping system and jet thrusts on affected structures.

c. Elastic and inelastic deformation of piping and/or restraint.

A 10 percent increase of minimum specified design yield strength (S_y) is used to account for strain rate effects in inelastic nonlinear analyses.

3.6B.2.3.3 Pipe Whip Restraint Design Criteria

1. Design Bases

Pipe whip restraints function primarily as a load carrying member for the low probability occurrence of a pipe break. The restraints are designed for one time use only and function to control the movement of the ruptured pipe. The design basis for the pipe

break event is that of a faulted condition. The restraints and the structure which supports them are analyzed accordingly.

2. Functional Requirements

High-energy pipe whip restraints are designed to ensure that the pipe whip will be eliminated or minimized. On the other hand, the restraints are designed to permit the predicted thermal and seismic movements of the pipes.

3. Design Parameters

After the pipe restraint locations are identified, the following design parameters are determined:

a. Jet thrust force

b. Pipe seismic displacements

c. Pipe thermal displacements

d. Pipe insulation thickness

e. Minimum allowable tolerance between restraint and pipe insulation.

f. Maximum allowable pipe movement.

The jet thrust force and maximum allowable pipe movement are used in the analysis process. Tolerance, insulation, and seismic and thermal movements are used in determining the minimum gap between the restraint and pipe surfaces.

The pipe whip restraints used for the CPNPP project are U-bar, crushable pipe, crushable pad (honeycomb), and elastic hard restraints as shown in Figures 3.6B-1 through 3.6B-4. Existing pipe supports may also be considered as additional hardware capable of providing protection when their use can be analytically justified.

4. Analysis and Design

The maximum allowable design limits for the restraints are as follows:

a. The permanent strain in the metallic ductile materials is limited to fifty percent of the minimum ultimate uniform strain (strain corresponding to the maximum stress point on the appropriate engineering stress-strain curve) based on restraint material tests and/or ASME Code [4].

b. The design limit for crushable pipe restraints longer than three outside pipe diameters is 50 percent of energy absorbing capacity (crushed to 70 percent of inside diameter of pipe).

c. The permissible diameter deformations limit for crushable pipe restraints shorter than two outside pipe diameters is the lesser of D/2 or ASTM where;

D = initial outside diameter of the crushable pipe

ASTM = maximum flattening limit for A106 Gr. B carbon steel pipe as prescribed by ASTM A 530-76.

d. If a crushable material, such as honeycomb, is used, the allowable capacity of crushable material shall be limited to 80% of its rated energy dissipating capacity as determined by dynamic testing, at loading rates within + 50% of the specified design loading rate. The rated energy dissipating capacity shall be taken as not greater than the area under the load-deflection curve. The portion of the curve in which the value of load vs. deflection has departed from the essentially horizontal portion shall not be used.

Typical characteristics of pipe whip restraint components are as follows:

a. Energy absorption members of yielding type restraints are those which, under the influence of the whipping pipe, absorb energy by significant plastic deformation. U-bars, crushable pipe, and crushable pad assemblies, designed with materials having high ultimate strain and relatively high energy absorption capacity, are used in CPNPP.

b. Elastic pipe whip restraints are also used. They are essentially plane frames or space frames. Upon rupture, the energy of the whipping pipe is completely absorbed by the steel restraint structure. Movement of the ruptured pipe is controlled by permitting deformation of the elements of the structural restraints within the elastic range of the material. Existing pipe supports activated during pipe rupture are governed by the criteria applicable to their faulted design rating or other acceptable limits which assures protection is provided such that the effects of the break are mitigated.

c. Restraint connecting members are components which form a direct link between the plastic yielding restraint members and the structure (e.g., clevises, brackets, and pins of a U-bar restraint).

d. Restraint connecting member structural attachments are fasteners which provide the means of securing the restraint connecting members of the structure (e.g., weld attachments and bolts).

e. Structural and civil components are steel and concrete structures which ultimately carry the restraint load (e.g., walls, frames, columns, and beams). A dynamic load factor is considered in the design of these structural and civil components. This factor depends on the natural frequency of these components and the restraint force time history. It is evaluated for each postulated break affecting a specific structural or civil component.

f. The design of the pipe whip restraint is for one time usage.

In designing a restraint, the following three loading conditions are considered:

a. An in-plane (the plane of the U-bar) impact loading at an angle up to 45 degrees from the axis of the U-bar.

b. An out-of-plane (the plane of the U-bar) impact loading at an angle up to 10 degrees from the axis of the U-bar.

c. An impact loading at an angle up to 10 degrees from the normal to the crushable pad/pipe axis.

The materials used in restraint design are selected to ensure ductile behavior. The U-bars are made of SA 479 type 304 annealed stainless steel. The crushable pipes are made of SA 106 Grade B carbon steel. The crushable pads are metallurgically bonded type 304 stainless steel sheets. The elastic restraints are made of ASTM A-588 Grade 50 steel. The other restraint components, such as pins, bolts, and anchors, are designed to remain within their elastic limits.

3.6B.2.4 Guard Pipe Assembly Design Criteria

The CPNPP containment is of single barrier design. Guard pipes are not used in the penetration design.

3.6B.2.5 Material to be Submitted for the Operating License Review

This section presents a summary of the dynamic analyses applicable to high-energy piping systems resulting from postulated pipe breaks. The following information is provided for the various high energy systems in the subparagraphs of this Section:

1. Implementation of the stress criteria as outlined in Section 3.6B.2.1

2. The type, number, and location of postulated breaks on which the dynamic analyses are based.

3. The number and locations of pipe whip restraints required to protect essential systems.

4. The results of the jet thrust, impingement functions, and pipe break analysis as described in Section 3.6B.2.2 and 3.6B.2.3 consistent with Reference [5], where the stress intensity ranges and/or usage factors exceed the criteria of 2.4 Sm and 0.2, respectively.

5. The design adequacy of essential systems and components to ensure that their design-intended functions will not be impaired to an unacceptable level of integrity or operability as the result of high-energy pipe breaks.

6. Description of protective enclosures provided to protect safety-related equipment from the effects of a possible rupture in a high energy fluid piping system, including openings in these enclosures.

The implementation of criteria for inservice inspection is discussed in Section 6.6.8.

3.6B.2.5.1 Reactor Coolant System Main Loop Piping

Table 3.6B-2 and Figure 3.6B-9 identify the RCS main loop break locations. The eight main loop piping and three large branch nozzle break locations (breaks 1 to 11 in Table 3.6B-2 and Figure 3.6B-9) are included in the CPNPP design basis for containment design, ECCS and environmental qualification requirements. These eleven postulated breaks are not part of the design basis for dynamic effects, as discussed in Sections 3.6B.2.1.1 and 3.6B.2.1.2. The two remaining branch nozzle breaks (breaks 12 and 13 in Table 3.6B-2 and Figure 3.6B-9) are included in the design basis for dynamic effects.

Design loading combinations and applicable criteria for ASME Class 1 components and supports are provided in Section 3.9N. The forces associated with rupture of the branch nozzles (12 and 13, Table 3.6B-2) to the reactor coolant loop piping systems are considered in combination with normal operating loads and earthquake loads for the reactor coolant loop design in order to assure continued integrity of vital components and engineered safety features. Pipe rupture loads include not only the jet thrust forces acting on the piping but also jet impingement loads on the primary equipment supports.

Barriers and layout are used to provide protection from pipe whip, blowdown jet and reactive forces. Some of the barriers utilized for protection against pipe whip are as follows. The steam generator compartment walls serve as a barrier between the reactor coolant loops and the containment liner. In addition, the refueling cavity walls, various structural beams, the operating floor, and the steam generator compartment walls enclosed each reactor coolant loop into a separate compartment, thereby preventing an accident, which may occur in any loop, from affecting another loop or the containment liner. The portion of the steam and feedwater lines within the containment have been routed behind barriers which separate these lines from all reactor coolant piping. The barriers described above will withstand loadings caused by jet forces and pipe whip impact forces.

Other than for the Emergency Core Cooling System lines, which must circulate cooling water to the vessel, the engineered safety features are located outside of the steam generator compartment walls. The Emergency Core Cooling System lines which penetrate the steam generator compartment walls are routed around and outside the walls to penetrate the walls in the vicinity of the loop to which they are attached.

3.6B.2.5.2 High-Energy Piping Other Than RCS Main Loop

In this section, a summary is presented giving the results of the detailed stress analysis and, describing methods of protection employed to protect essential equipment against the effects of pipe breaks for the high energy systems outlined in Section 3.6B.1.2.1.

1. Main Steam System

   a. General Description

      The main steam piping inside containment is carbon steel ASME SA-155, Grade KCF 70 material designed in accordance with the ASME Code, Section III, Class 2 criteria. The main steam system inside containment consists of four 32 inch OD (1.25 inch minimum wall thickness) lines running from each steam generator to the containment penetrations.

The main steam piping outside containment from the containment penetrations to the main steam isolation valve moment restraints is the same as the main steam piping inside containment. The piping from the MSIV moment restraints to the high pressure turbine is carbon steel ASME SA-155 Grade KC 70 material designed in accordance with ANSI B31.1 as non-nuclear class piping. These lines are 34 inch OD (1.25 inch minimum wall thickness). The piping connected to the main steam drip pots are of carbon steel material with portions designed in accordance with ANSI B31.1 and ASME Code, Section III, Class 2 criteria. The Class 2 portion of the system consists of ASME SA-333 Grade 6, two inch schedule 80 pipe and the non-nuclear portion of the system consists of ASME SA-106 Grade B, two inch schedule 80 pipe. The location and configuration of the main steam lines with respect to structures, equipment, and other piping are shown on Figures 1.2-8, 1.2-14 and 1.2-25. The criteria described above and as follows is applicable for both Units 1 and 2.

b. Pipe Whip Analysis

Isometrics of the main steam lines inside Containment indicating the location of the highest stress node points, postulated breakpoints, and restraints are provided in Figures 3.6B-11 through 3.6B-14 and 3.6B-100 through 3.6B-103. The systems and equipment necessary to mitigate the consequences of a main steam line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at inter-mediate locations in accordance with the criteria out-lined in Section 3.6B.2.1. The steam generator nozzles and the flued heads at the containment penetrations are considered terminal ends. A circumferential break is postulated to occur at any one of these points. Restraints are provided on each line to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

Isometrics of the main steam lines outside containment indicating the location of postulated breakpoints and restraints are provided in Figures 3.6B-15 through 3.6B-18, and 3.6B-104 through 3.6B-107. Since these lines consist of non-nuclear piping, pipe breaks are postulated at each fitting, valve, or welded attachment. Since these lines are greater than 4 inches in diameter, circumferential or longitudinal breaks are postulated. Pipe whip restraints are provided as necessary that are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects. A portion of the main steam lines between the containment penetrations and isolation valves, are designed in accordance with the criteria in Section 3.6B.2.1.2. Therefore, circumferential pipe breaks are not postulated in the regions between the penetrations and the moment restraints after the isolation valves. However, a 1.0 ft^2 crack is postulated and evaluated for environmental effects, in accordance with the criteria in BTP ASB 3-1. As shown on Figures 3.6B-25 and 3.6B-119 through 3.6B-122, pipe breaks are postulated in the main steam blowdown lines outside containment. Since these lines consist of non-nuclear piping, break points are postulated at terminal ends and at each intermediate fitting, valve or welded attachment with the exception of the four break exclusion areas as shown.

Review of the piping layout and plant arrangement showed that breaks in portions of the main steam lines could adversely impact the safeguards, switchgear, electrical and controls and turbine buildings. Accordingly, these lines are restrained as necessary to protect the structures. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects on the structures. Since there are no essential systems or components in the turbine building, protection is primarily provided to protect the structures noted above from a main steam pipe break.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

In piping tunnels that contain break exclusion regions of main steam lines, the safety-related equipment in these tunnels are designed to withstand the environmental effects of a non-mechanistic crack with a flow area of 1.0 ft^2. Jets are not generated, and hence jet Loads are not considered in break exclusion regions (superpipe areas).

A small quantity of essential cables is located in the superpipe areas; however, these cables service equipment that because of their function, are appropriately located in these areas. Other essential cables are not located in and do not transit through these areas. The low concentrations of the essential cables located in these areas does not violate the NRC staff's pipe break separation goal for break exclusion areas.

d. Environmental Analysis

The safety-related systems required to mitigate the consequences of a main steam line break inside Containment are designed to perform their safety function under the environmental conditions resulting from a LOCA or MSLB as discussed in Section 3.11.

The penetration area is isolated from the rest of the Safeguards Building by the seismic gap seal between the Safeguards Building and the containment wall, fast closing isolation dampers in the interconnecting HVAC ducts and the pressure resisting watertight doors. These features preclude this crack from producing any detrimental environmental effects on the rest of the plant.

Any one of the main steam lines is postulated to develop a 1 ft^2 crack in the penetration area outside Containment. This crack is postulated to be the most severe break for all rooms, based on the impact to equipment in rooms outside the

break area, the amount of energy released, and the generation of the highest peak. The postulation of this crack is in accordance with Branch Technical Position ASB 3-1.

The only equipment that could be required to operate and that could be in the room where the break occurs are the ventilation dampers. These dampers close on pressure differential; pressure helps keep these dampers closed. The temperature spike will not degrade the ability of these dampers to close and remain closed.

Mass and energy release data were calculated for 100% including power uncertainty, 90%, 80%, 70% and 50% power. Full power including power uncertainty with the 1.0 ft^2 break provided the worst environmental results, i.e., the highest peak temperature. Table 3.6B-4A (Unit 1) and Table 3.6B-4B (Unit 2) provide the mass and energy release data. Mass/energy releases were calculated using a modified version of the LOFTRAN code [33], and it and the related blowdown analysis are plant-specific to CPNPP. LOFTRAN was modified to model the heat transfer which may occur in the uncovered portion of the tube bundle of a steam generator and to calculate the resulting superheated steam mass and energy release. Conservative assumptions were made in order to lead to early tube bundle uncovery, and as a result the earliest superheat initiation time. Primary system temperatures were maximized to increase the primary-to-secondary heat transfer, promoting earlier tube bundle uncovery and maximizing the superheat steam enthalpy following tube bundle uncovery. Table 3.6B-4A and Table 3.6B-4B also provides the time sequence of events based on a 1.0 ft^2 break and an initial power level of 90% including power uncertainty. The environmental analysis used to determine the temperature transient due to this break is described in Section 3.6B.1.2.3.

An Equipment Qualification review of Class 1E equipment in the affected compartment has been performed and Class 1E items have been determined to be qualified for the environment. The Class 1E cables located in the areas affected by the crack are those cables required to support the Class 1E equipment in the same areas. All of these cables are qualified for the LOCA/MSLB inside containment. As such, these cables will remain fully operational throughout this event. Cable separation in these areas meets the requirements of Regulatory Guide 1.75. See Table 3.6B-5 for the remaining safety-related equipment located in the area affected by the steam line break with superheat.

2. Feedwater System

a. General Description

The main feedwater lines inside containment are carbon steel ASME SA-333, Grade 6 material designed in accordance with the ASME Code, Section III, Class 2 criteria. Due to the Unit RSG installation the 18” x 16” reducing elbow material at the RCS nozzle inlet is now SA-508, Grade 2, Class 1. Each of the lines consists of an 18 inch schedule 80 seamless pipe running from the Containment penetration to each steam generator. The main feedwater piping

outside Containment from the containment penetration to the feedwater containment isolation valve is the same as the feedwater piping inside Containment. The piping from the feedwater containment isolation valve to the feedwater control valve moment restraint is also the same as the feedwater piping inside containment, except that these lines are 18 inch schedule 140 seamless pipe. The main feedwater lines from the feedwater control valve moment restraint connect to the main feedwater header combining into one main feedwater line which originates from the feedwater heater in the turbine building. The feedwater lines to the main feedwater header and the feedwater control valve by-pass lines are carbon steel ASME SA-106, Grade B material designed in accordance with ANSI B31.1 as non-nuclear class piping. These lines consist of 18 inch schedule 140 and eight inch schedule 120 pipe. The main feedwater header and feedwater line in the turbine building are carbon steel ASTM A155 Gr KC 60 material designed in accordance with ANSI B31.1 as non-nuclear class piping. These lines are 30 inch OD (2.125 inch minimum wall thickness) pipe. The location and configuration of the feedwater lines with respect to structures, equipment, and other piping are shown in Figures 1.2-8, 1.2-13 and 1.2-25.

The criteria described above and as follows is applicable for both Units 1 and 2.

b. Pipe Whip Analysis

Isometrics of the main feedwater lines inside Containment indicating the location of the highest stress node points, postulated breakpoints, and restraints are provided in Figures 3.6B-19 through 3.6B-22 and 3.6B-108 through 3.6B-111. The systems and equipment necessary to mitigate the consequences of a main feedwater line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. The flued heads at the containment penetrations and the steam generator nozzles are considered terminal ends. A circumferential break is postulated to occur at any one of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects. Existing pipe supports were also activated in the pipe whip analysis as additional existing hardware capable of mitigating the effects of the break.

Isometrics of the main feedwater lines outside containment, indicating the location of postulated breakpoints and restraints, are provided in Figures 3.6B-23, 3.6B-24 and 3.6B-112 through 3.6B-118. Since these lines consist largely of non-nuclear piping, pipe breaks are postulated at each fitting, valve, or welded attachment. For all lines, circumferential or longitudinal breaks are postulated. Pipe whip restraints are provided to prevent unacceptable damage to essential components and building structures. The pipe whip restraints are designed to prevent plastic hinge formation to preclude adverse pipe whip effects. A portion of the main feedwater lines between the containment penetrations and the feedwater control valves, is designed in accordance with the criteria in Section 3.6B.2.1.2. Therefore, pipe breaks are not postulated in the regions between the penetrations and the moment restraints after the control valves. However, a one square foot

crack is postulated and evaluated for environmental effects in accordance with the criteria in BTP ASB 3-1.

Review of the piping layout and plant arrangement showed that breaks in portions of the main feedwater lines could adversely impact the safeguards, switchgear, electrical and controls and turbine buildings. Accordingly, these lines are restrained as necessary to protect the structures. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects on the structures. Since there are no essential systems or components in the turbine building, protection is primarily provided to protect the structures noted above from a feedwater line break.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

Jet Loads are not considered in break exclusion regions (superpipe areas). A small quantity of essential cables is located in the superpipe areas; however, these cables service equipment that because of their function, are appropriately located in these areas. Other essential cables are not located in and do not transit through these areas. The low concentrations of the essential cables located in these areas does not violate the NRC staff's pipe break separation goal for break exclusion areas.

d. Environmental Analysis

The safety-related systems required to mitigate the consequences of a main feedwater line break inside Containment are designed to perform their safety function under the environmental conditions resulting from a LOCA or MSLB as discussed in Section 3.11.

Any one of the main feedwater lines is postulated to develop a 1 ft^2 crack in the penetration area outside Containment. The postulation of this crack is in accordance with Branch Technical Position ASB 3-1.

The penetration area is isolated from the rest of the safeguards building by the seismic gap seal between the safeguards building and the containment wall, fast closing isolation dampers in the interconnecting HVAC ducts and the pressure resisting watertight doors. These features preclude this crack from producing any detrimental environmental effects on the rest of the plant. High energy flooding is evaluated in the same manner as moderate energy flooding per Section 3.6B.2.5.3.

3. Auxiliary Feedwater System

a. General Description

The Auxiliary Feedwater System lines are carbon steel, ASME SA 106, Grade B and SA 333, Grade 6 material designed in accordance with ASME Code Section III, Class 2 or 3 criteria as applicable. The suction lines of the auxiliary feedwater pumps consist of 10, 8 and 6 inch category 152 piping which is schedule 40, with design pressure of 150 psig. These lines run from the condensate storage tank and service water piping to the suction of each auxiliary feedwater pump. The discharge lines of the auxiliary feedwater pumps consist of 6, 4 and 3 inch category 2002 (category 2003, directly after the isolation valve) piping which is schedule 160 up to 3 inches, and schedule 120 for sizes above 3 inches. The design pressure is 1800 psig. The piping that connects to the main feedwater piping after the last auxiliary feedwater isolation valves is category 1303 which is schedule 80, with a design pressure of 1200 psig to match the main feedwater piping. The discharge lines of the auxiliary feedwater pumps connect to the main feedwater lines. The turbine driven auxiliary feedwater pump discharge piping up to isolation check valves AF-078, AF-106, AF-086, and AF-096 is classified as moderate energy piping since operation as a high energy fluid system is for a limited period as prescribed by 3.6B.2.1.4. The location and configuration of the auxiliary feedwater lines with respect to structures, equipment and other piping are shown in Figure 1.2-10. The criteria described above and as follows are applicable for both Units 1 and 2.

b. Pipe Whip Analysis

Isometrics of the Auxiliary Feedwater System lines indicating the location of the highest stress node points, postulated breakpoints and restraints are provided in Figures 3.6B-26 through 3.6B-33 and 3.6B-124 through 3.6B-130. The systems and equipment necessary to mitigate the consequences of a Auxiliary Feedwater System line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. Intermediate breaks are postulated as shown. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

d. Environmental Analysis

The environmental conditions resulting from a break in an Auxiliary Feedwater System line do not affect the operation of safety-related systems required to mitigate the consequences of the accident.

The Auxiliary Feedwater System is high energy only on the basis of the system pressure. During the AFW System operation, for breaks upstream of isolating check valves 1AF-093, 1AF-101, 1AF-083, 1AF-075, 2AF-093, 2AF-101, 2AF-083 and 2AF-075, cold water jets are developed. For breaks downstream of these check valves there exist jets that can produce a steam atmosphere in the feedwater tunnel area. These high temperature jets originate from feedwater during plant warm-up (up to 250°F maximum) (Unit 2 only). The resultant pressure and temperature from these breaks are less severe than those analyzed for this area due to the non-mechanistic one square foot crack of the feedwater piping.

High energy flooding is evaluated in the same manner as moderate energy flooding per Section 3.6B.2.5.3.

4. Auxiliary Steam System

a. General Description

The Auxiliary Steam System Piping is comprised of non-nuclear safety related carbon steel, ASME SA-106 Grade B material. The auxiliary steam system supplies steam to both Units 1 and 2 components.

The steam supply piping to the floor drain waste evaporator package, WPS waste evaporator package, BRS recycle evaporator package and CVCS boric acid batching tank, all located in the Auxiliary Building, are 10, 6, 4 and 2 inch, schedule 40 category 152 Class 5 piping with a design pressure of 150 psig. These lines run from the auxiliary steam header of the components described above and are seismically supported. The return lines of the above components are 2, 1-1/2 and 3/4 inch schedule 40, category 152 Class 5 piping with a design pressure of 150 psig. The return lines are connected to the auxiliary steam drain tank. The location and configuration of the auxiliary steam lines with respect to structures, equipment and other piping are shown in Figures 1.2-31 and 1.2-32.

b. Pipe Whip Analysis

Isometric drawings of the Auxiliary Steam System Piping indicating the locations of the highest stresses, postulated breakpoints and restraints are provided in Figures 3.6B-58 through 3.6B-63. The systems and equipment necessary to mitigate the consequences of an Auxiliary Steam System line break are described in Section 3.6B.1. Break locations were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on

essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

d. Environmental Analysis

The environmental conditions resulting from a break in an Auxiliary Steam System line do not affect the operation of safety-related systems required to mitigate the consequences of the accident.

The most significant environmental conditions generated by breaks in the auxiliary steam system are the various elevated temperatures. The pressure transients are very slight and do not pose a threat to the safe operation or the structure of the plant. The temperature transients caused by any of the auxiliary system breaks do not affect the safe shutdown of the plant because: 1) equipment required for safe shutdown is located in an area not affected by the break; 2) the equipment is qualified to parameters higher than those experienced during the break; 3) the equipment fails in a safe position; or 4) the equipment has been analyzed to show that the short-duration increased temperatures will not cause appreciable degradation in performance or qualified life.

Blowdown from any of the postulated auxiliary steam system line breaks is terminated automatically. The transients were analyzed for the full period of the blowdown.

Details of the instrumentation required to mitigate the blowdown is in Section 7.6.12.

5. Steam Generator Blowdown Cleanup System

a. General Description

The Steam Generator Blowdown Cleanup System Piping is comprised of non-nuclear safety related, carbon steel ASME SA-106 Grade B material, or non-nuclear safety related stainless steel ASTM A-312 TP 304 or 316 material.

The blowdown lines from each steam generator up to the isolation valve moment restraints are 2", 3" or 4" safety class 2, Schedule 80, category 1303 piping with a design pressure of 1200 psig. The piping connecting the 8-inch header of the steam generator blowdown heat exchanger to the pressure reducing

valve PV-5180 is schedule 80, category 1302, Class 5, with a design pressure of 1200 psig. The portion of the blowdown piping on the Switchgear Building roof is category 1302G, and not seismically supported. The piping from valve PV-5180 up to valve SB-170 is 8-inch, category 302, Class 5, schedule 40, with a design pressure of 450 psig. The piping from valve SB-170 to the filters and demineralizers is 6-inch, category 301, Class 5, schedule 40S with a design pressure of 370 psig. The piping that connects the discharge of relief valve SB020 to the condenser is 4 and 8 inch, category 302, Class 5, Schedule 40, with a design pressure of 450 psig. The piping downstream of the filters is moderate energy piping. The location of the steam generator blowdown cleanup system with respect to structures and equipment are shown in Figures 1.2-31 and 1.2-35.

b. Pipe Whip Analysis

Isometrics of the Steam Generator Blowdown Cleanup System Piping indicating the locations of the highest stresses postulated breakpoints and restraints are provided in Figures 3.6B-38 through 3.6B-45 and 3.6B-136 through 3.6B-144. The systems and equipment necessary to mitigate the consequences of a Steam Generator Blowdown Cleanup System line break are described in Section 3.6B.1. Break locations were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1 except for category 1302G piping, where breaks are postulated at terminal ends and at each intermediate fitting, valve or welded attachment. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

d. Environmental Analysis

The environmental conditions resulting from a break in a Steam Generator Blowdown System line do not affect the operation of safety-related systems required to mitigate the consequences of the accident.

All equipment required for mitigating the consequences of the accident are located in areas that are isolated from the pipe break event by the automatic closure of HVAC dampers. The leakage that occurs prior to the closure of the dampers has been analyzed and the effects of temperature, pressure and

humidity intrusion evaluated. A total of 20 dampers, 2 in each HVAC duct, are installed and designed to close in a short time after the pipe break occurrence (less than 5 seconds). The dampers are qualified for the transient environmental conditions of the break.

Blowdown from any of the postulated blowdown line breaks is terminated automatically. The transients were analyzed for the full period of the blowdown.

Details of the instrumentation required to mitigate the blowdown is in Section 7.6.12.

High energy flooding is evaluated in the same manner as moderate energy flooding per Section 3.6B.2.5.3.

6. Chemical & Volume Control System

a. General Description

The Chemical and Volume Control System lines are of stainless steel ASME SA-312, TP 304 material or stainless steel ASME SA-312 TP 304 material or stainless steel ASME SA-376 TP 304 material designed in accordance with ASME Code Section III, Class 1, 2 or 3 criteria as applicable. The letdown line from the cold leg of the RCS to the letdown orifices consists of 3 and 2 in category 2501 schedule 160 piping with a design pressure of 2485 psig. The letdown orifices on the low pressure letdown valve PCV-131 is 2 and 3 inch category 601, schedule 40S piping with a design pressure of 700 psig. The letdown line from the low pressure letdown valve to the demineralizers and RHT of the BRS is 3 inch category 301, Schedule 40S, with a design pressure of 370 psig. The suction lines of the charging pumps consist of 1, 2, 3, 4 and 6 inch category 151, schedule 40S piping with a design pressure of 150 psig. These lines run from the volume control tank, CVCS boric acid tank and chemical mixing tank to the suction of the charging pumps. The discharge lines of the charging pumps consist of 4, 3 and 2 inch category 2501 schedule 160 piping, with a design pressure of 2485 psig. These lines run from the charging pumps to the cold leg of the RCS and the RCP seals. The location and configuration of the CVCS lines with respect to structures and equipment are shown in Figures 1.2-11, 1.2-12, 1.2-17, 1.2-18 and 1.2-32.

b. Pipe Whip Analysis

Isometrics of the CVS system piping indicating location of the highest stresses, postulated breakpoints and re-straints are provided in Figures 3.6B-70 through 3.6B-88 and 3.6B-183 through 3.6B-206. The systems and equipment necessary to mitigate the consequences of a CVC system line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shields or barriers to protect the target.

d. Environmental Analysis

The environmental conditions resulting from a break in an CVC System line do not affect the operation of safety-related systems required to mitigate the consequences of the accident.

The most significant environmental conditions generated by breaks in the CVC system are the various elevated temperatures. The pressure transients are small and do not pose a threat to the safe operation or the structure of the plant. The temperature transients caused by any of the CVCS system breaks do not affect the safe shutdown of the plant because: 1) equipment required for safe shutdown is located in an area not affected by the break; 2) the equipment is qualified to parameters higher than those experienced during the break; 3) the equipment fails in a safe position; or 4) the equipment has been analyzed to show that the short-duration increased temperatures will not cause appreciable degradation in performance or qualified life.

Details of the instrumentation required to mitigate the blowdown is in Section 7.6.12.

High energy flooding is evaluated in the same manner as moderate energy flooding per Section 3.6B.2.5.3.

7. Residual Heat Removal System

a. General Description

The Residual Heat Removal System piping from the RCS hot leg of loop 1 and 4 to the second isolation valve are of stainless steel, ASME SA-376 TP 316 material designed in accordance with ASME Code Section III, Class 1 criteria. These lines consist of 12 inch category 2501, schedule 140 piping with a design pressure of 2485 psig. The piping from the second isolation valve to the containment penetrations is of stainless steel ASME SA-312 TP 304 material designed in accordance with ASME Code section III, Class 2 criteria. These lines are 12 inch category 601, schedule 40S piping with a design pressure of 600 psig. The piping from the containment penetration to the suction of the RHR pumps is of stainless steel ASME SA-358, Class 1. These lines are 16 inch (0.5 inch minimum wall thickness) category 601 with a design pressure of 600 psig. The discharge lines of the Residual Heat Removal pumps are of stainless steel, ASME SA-312

TP 304 material designed in accordance with ASME Code Section III, Class 2 criteria. These lines consist of 8, 10 and 12 inch category 601, schedule 40S piping with a design pressure of 600 psig. The discharge lines of the residual heat removal pumps connect to safety injection system cold leg injection headers. The location of the Residual Heat Removal System piping with respect to structures and equipment is shown in Figures 1.2-10 and 1.2-16.

b. Pipe Whip Analysis

Isometric drawings of the RHR System piping indicating the locations of the highest stresses postulated breakpoints and restraints are provided in Figures 3.6B-64, 3.6B-172 and 3.6B-173. The systems and equipment necessary to mitigate the consequences of a RHR system line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components (as defined in Section 3.6B.1.2.1), associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

d. Environmental Analysis

The RHR System is considered a moderate energy system outside containment in accordance with Branch Technical Position MEB 3-1.

Moderate energy flooding is evaluated in accordance with Section 3.6B.2.5.3.

8. Safety Injection System

a. General Description

The Safety Injection System Piping from the discharge header of the charging pumps to the Reactor Coolant System loop cold legs are stainless steel, ASME SA-376 TP 304 or TP 316 material, designed in accordance with ASME Code Section III, Class 1 or 2 criteria, as applicable. These lines consist of 1, 1-1/2, 2, 3, 4 and 6 inch category 2501, schedule 160 piping with a design pressure of 2485 psig. The piping from the accumulators to the Reactor Coolant System loop cold legs is stainless steel, ASME SA-376 TP 304 or TP 316 material, designed in accordance with ASME Code Section III, Class 1 or 2

criteria, as applicable. These lines are 10 inch category 2501 schedule 140 piping with a design pressure of 2485 psig.

The safety injection pump discharge piping up to the isolation valves outside the containment are stainless steel ASME SA-312 TP 304 or TP 316 material, designed in accordance with ASME Code Section III, Class 2 criteria. These lines consist of 3 and 4 inch category 1501, Schedule 80S piping with a design pressure of 1860 psig. The safety injection pump discharge piping from the isolation valves outside the containment to the Reactor Coolant System hot legs and Safety Injection System cold legs is stainless steel ASME SA-376 TP 304 or TP 316 material, designed in accordance with ASME Code Section III, Class 1 or 2 criteria as applicable. These lines consist of 3/4, 2, 4, 6, 8 and 10 inch category 2501, schedule 160 or 140 piping, as applicable, with a design pressure of 2485 psig. The safety injection pump suction piping is of stainless steel, ASME SA-312, TP 304 or TP 316 material, designed in accordance with ASME Code Section III, Class 2 criteria. These lines consist of 6 and 8 inch category 151 schedule 40S piping with a design pressure of 150 psig. The location of the Safety Injection System piping with respect to structures and equipment are shown in Figures 1.2-10, 1.2-11, 1.2-16 and 1.2-17.

b. Pipe Whip Analysis

Isometric drawings of the Safety Injection System lines indicating the locations of the highest stresses postulated breakpoints and restraints are provided in Figures 3.6B-48 through 3.6B-57 and 3.6B-147 through 3.6B-169. The systems and equipment necessary to mitigate the consequences of a Safety Injection System line break are described in Section 3.6B.1. Breakpoints were postulated at the terminal ends of the piping run and at intermediate locations in accordance with the criteria outlined in Section 3.6B.2.1. A circumferential break is postulated to occur at each of these points. Pipe whip restraints are provided as required to prevent impact on essential components. The pipe whip restraints are designed to prevent plastic hinge formation and thereby preclude adverse pipe whip effects.

c. Jet Impingement Analysis

The jet impingement analysis for this system is performed to determine the effects of jet impingement loading on essential components, associated supports and building structures.

In those cases where the analysis shows that the component or structure is not capable of withstanding the load then protection is required. Protection consists of either relocating the target, or installing jet shield or barriers to protect the target.

d. Environmental Analysis

The environmental conditions resulting from a break in an SI System line do not affect the operation of safety-related systems required to mitigate the consequences of the accident.

Those portions of the Safety Injection System which are considered high energy, outside the containment, are based on the system pressure. The piping is considered as containing cold water so a break in a line cannot generate a steam atmosphere.

High energy flooding is evaluated in the same manner as moderate energy flooding per Section 3.6B.2.5.3.

3.6B.2.5.3 Moderate Energy Piping

In evaluating the effects of a moderate energy system piping failure, the postulated failure is a crack which results in neither pipe whip nor jet impingement but rather in spraying water streams. As such, the consequences are of an environmental/flooding nature. The effects of cracks, as postulated in Section 3.6B.2.1.4, are evaluated for all essential equipment on a case by case basis.

If it is determined that an essential component is not qualified or cannot be demonstrated to operate under the adverse conditions caused by the crack then the essential component is protected. Protection is accomplished either by relocating the component, installing a barrier or curb or by designing a shield.

REFERENCES

1. NRC Regulatory Guide 1.46, Protection Against Pipe Whip Inside Containment, U.S. Nuclear Regulatory Commission, 5/73.

2. Acceptance Criteria of NRC Standard Review Plan, Section 3.6.1, Plant Design For Protection Against Postulated Piping Failures in Fluid Systems Outside Containment and Section 3.6.2, Determination of Break Locations and Dynamic Effects Associated with the Postulated Rupture of Piping. Also Branch Technical Position MEB 3-1, Postulated Breaks and Leakage Locations in Fluid System Piping Outside Containment.

3. Branch Technical Position APCSB 3-1, Protection Against Postulated Piping Failures in Fluid Systems Outside Containment.

4. ASME Boiler and Pressure Vessel Code, Section III, Nuclear Power Plant Components, 1974 with Addenda.

5. WCAP-8082-P-A (Proprietary), January 1975, and WCAP 8172 A (Nonproprietary), January 1975, Pipe Breaks for the LOCA Analysis of the Westinghouse Primary Coolant Loop.

6. NUREG/CR-1826, RELAP-5/MOD 1, Rev 2 Sept. 1981.

7. Interim Acceptance Criteria for Emergency Core Cooling System for Light-Water Power Reactors, Interim Policy Statement, Atomic Energy Commission Federal Register, June 20, 1971, p. 12217.

8. Lintner, M.A., A Plotting Routine for RELAP Data Tapes, Idaho Nuclear Corporation, January 1970.

9. 1967 ASME Steam Tables - Thermodynamics and Transport Properties of Steam, American Society of Mechanical Engineers, 1967.

10. Maximum Flow Rate of a Single Component, Two Phase Mixture, Journal of Heat Transfer, ASME pp. 134 142, February 1965.

11. Bordelon, F.M., “A Comprehensive Space-Time Dependent Analysis of Loss of Coolant (SATAN-IV Digital Code),” WCAP-7750, August 1971.

12. “Documentation of Selected Westinghouse Structural Analysis Computer Codes,” WCAP-8252, Revision 1, May 1977.

13. Nuclear Services Corporation, “PIPERUP; A Computer Program For Analysis of Piping Systems Subject to Pipe Rupture Loads,” Revision 2, December 1974.

14. N.M. Newmark, “A Method of Computation for Structural Dynamics,” JEM Div, Vol., 85, No. EM3, 1959.

15. Moody, F. J., “Fluid Reaction and Impingement Loads”, Vol. 1, ASCE Specialty Conference on Structural Design of Nuclear Plant Facilities, pp. 219-262, December 1973.

16. Webb, S. W., “Evaluation of Subcooled Water Thrust Forces” Nuclear Technology, Vol. 31, October 1976.

17. Regulatory Guide, G.J.E. Willcutt, Jr., J.L. Lunsford and J.S. Gilbert, COMPARE-MOD1: “A Code for the Transient Analysis of Volumes with Heat Sinks, Flowing Vents, and Doors, LA-7199-MS, March 1978.

18. NUREG - 1061 Volume 3, “Report of the USNRC Piping Review Committee. Evaluation of Potential Pipe Breaks”, November, 1984.

19. WCAP-10527 (Proprietary), WCAP-10528 (Non-Proprietary) “Technical Bases for Eliminating Large Primary Loop Pipe Ruptures as the Structural Design Bases for Comanche Peak Units 1 and 2”, April, 1984.

20. EBASCO Services Inc., “CALPLOTF-III” computer code to calculate blowdown loads using output from RELAP-5 Rev. 0, Nov. 1985.

21. Hibbitt and Karlsson and Sorensen Inc., “ABAQUS-ND” A Finite Element Code for Nonlinear Dynamic Analysis.

22. Design Basis for Protection of Light Water Nuclear Power Plants Against Effects of Pipe Rupture, American National Standard ANSI/ANS-58.2-1988.

23. Federal register 12502 Vol. 51, No. 70, April 11, 1986 “Modification of General Design Criterion 4 Requirements for Protection Against Dynamic Effects of Postulated Pipe Ruptures”.

24. EBASCO Services Inc., “SHPLAST 2267, Dynamic Pipe Whip Analysis” version 0 Rev. 27, June 1983.

25. NUREG/CR-2913 “Two-Phase Jet Loads”, G.G. Weigand, S. L. Thompson, and D. Tomasco January 1983.

26. Biggs, J.M., et.al, “Structural Design For Dynamic Loads”, McGraw-Hill, 1959.

27. TU Electric, “COMPARE MOD1A”, 1991 (Los Alamos LA-7199-MS informal report NRC-4)

28. Stone & Webster Engineering Corporation, “STEHAM”, Steamhammer Analysis for Piping Systems, ME-167, Version 02, level 03, 83.167.

29. “REPIPE”, CDC Application Reference Manual, post processor for RELAP CDC.

30. Stone & Webster Engineering Corporation, “ANSYS”, Computer Program: Engineering Analysis System, version 04, level 02, ST-384.

31. Stone & Webster Engineering Corporation “LIMITA3”, Nonlinear Dynamic Analysis of Frame Structures, version 03, level 00, ST-225.

32. Stone & Webster Engineering Corporation, “CHPLOT”, Time-History Plotting; ME-179, version 02, level 02, 86.009.

33. Burnett, T.W.T, et al., “LOFTRAN Code Description,” WCAP-7907, June 1972.

34. RELAP5/MOD3.2, NUREG/CR-5535, June 1995.

35. ANSYS, Version 8.1.

36. NUREG/CR-5535, RELAP5/MOD3 Code Manual.

37. WCAP-8708-PA (Proprietary), WCAP-8709 (Non-Proprietary), “MULTIFLEX A FORTRAN Computer Program for Analyzing Thermal-Hydraulic Structure System Dynamics,” Takeuchi, K., et al., September 1977.

38. WCAP-15029-P-A (Proprietary), WCAP-15030-NP-A, (Non-Proprietary) “Westinghouse Methodology for Evaluating the Acceptability of Baffle-Former-Barrel Bolting Distribution Under Faulted Load Conditions,” E.R. Schwirian, et al., January 1999.

39. WCAP-11004-P (Proprietary), WCAP-11005 (Non-Proprietary), “Comparison of Data for Beaver Valley Power Station Unit 2 with WCAP-9735 Data, Prepared for NRC Review in Conjunction with Review of WAP-9735, Docket No. 50-412,” D.R. Bhandari, et al., November 1985.

40. WCAP-11522 (Proprietary), WCAP-11523 (Non-Proprietary), “Response to NRC Questions on the LOCA Hydraulic Forces Analysis of the Beaver Valley Power Station Unit 2, Prepared for NRC Review in Conjunction with Review of WCAP-9735, Docket No. 50-412,” D.C. Garner, et al., June 1987.

41. WCAP-9735, Revision 2 (Proprietary) and WCAP-9735, Revision 1 (Non-Proprietary) “MULTIFLEX 3.0 A FORTRAN IV Computer Program for Analyzing Thermal-Hydraulic-Structural System Dynamics Advanced Beam Model,” K. Takeuchi, et al., February 1998.

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 1 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
SYSTEM AUXILIARY FEEDWATER (AF)					
4AF-1-49-2002-3	SB	X		MOTOR DRIVEN AF PUMP TRAIN A (DISCHARGE)	10.4-11
4AF-1-36-2002-3	SB	X		AF PUMP TRAIN A DISCH. (TEST LINE INCLUDED)	10.4-11
3AF-1-38-2002-3	SB	X		AF PUMP TRAIN A MINIMUM FLOW RECIRC. LINE	10.4-11
2AF-1-65-2002-3	SB	X		AF PUMP TRAIN A MINIMUM FLOW RECIRC. LINE	10.4-11
3AF-1-28-2002-3 (FROM 2X3 REDUCER TO BREAKDOWN ORIFICE CP1-AFORBO-01 ONLY)	SB	X		AF PUMP TRAIN A MINIMUM FLOW RECIRC.	10.4-11
6AF-1-10-2002-3	SB	X		AF TRAIN A HEADER TO SG-01 AND SG-02	10.4-11
4AF-1-50-2002-3 (THRU VALVE 1AF-074 TO 4X3 REDUCER)	SB	X		AF TRAIN A TO SG-01	
3AF-1-67-2002-3	SB	X		AF TRAIN A TO SG-01	10.4-11
3AF-1-75-2002-3	SB	X		AF TRAIN A TO SG-01	10.4-11
4AF-1-97-2002-3 (FROM 4X3 REDUCER THRU VALVE 1-AF-0121 AND CP1-AFORFR-08 TO VALVE 1-AF-0075)	SB	X		AF TRAIN A TO SG-01	
4AF-1-105-2002-3		X		AF TRAIN A TO SG-01	10.4-114
4AF-1-52-2003-2	RB	X		AF TRAIN A TO SG-01 (NOTE 1)	10.4-11
4AF-1-17-1303-2	SB	X		AF TRAIN A TO SG-01 (NOTE 1)	10.4-11
4AF-1-104-2002-3	SB	X		TURBINE DRIVEN AF PUMP TO SG-01	10.4-11
4AF-1-21-2003-3	SB	X		TURBINE DRIVEN AF PUMP TO SG-01 (NOTE 1)	10.4-11

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 2 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
4AF-1-61-2002-3	SB	X		AF TRAIN A TO SG-02	10.4-11
3AF-1-69-2002-3	SB	X		AF TRAIN A TO SG-02	10.4-11
3AF-1-77-2002-3	SB	X		AF TRAIN A TO SG-02	10.4-11
4AF-1-99-2002-3 (THRU VALVE 1AF-123 AND ORIFICE CP1-AFORFR-06 TO VALVE 1AF-083)	SB	X		AF TRAIN A TO SG-02	10.4-11
4AF-1-107-2002-3	SB	X		AF TRAIN A TO SG-02	10.4-11
4AF-1-53-2003-2	SB	X		AF TRAIN A TO SG-02 (NOTE 1)	10.4-11
4AF-1-18-1303-2	SB	X		AF TRAIN A TO SG-02 (NOTE 1)	10.4-11
4AF-1-22-2003-2	SB	X		TURBINE DRIVEN AF PUMP TO SG-02 (NOTE 1)	10.4-11
4AF-1-106-2002-3	SB	X		TURBINE DRIVEN AF PUMP TO SG-02	10.4-11
4AF-1-42-2002-3	SB	X		MOTOR DRIVEN AF PUMP, TRAIN B DISCHARGE TO SG-03 AND SG-04	10.4-11
6AF-1-09-2002-3	SB	X		MOTOR DRIVEN AF PUMP, TRAIN B DISCHARGE TO SG-03 AND SG-04	10.4-11
3AF-1-31-2002-3	SB	X		AF PUMP TRAIN B MINIMUM FLOW RECIRC LINE	10.4-11
2AF-1-66-2002-3	SB	X		AF PUMP TRAIN B MINIMUM FLOW RECIRC LINE	10.4-11
3AF-1-30-2002-3 (FROM 2X3 REDUCER TO BREAK DOWN ORIFICE CP1-AFORBO-02 ONLY)	SB	X		AF PUMP TRAIN B MINIMUM FLOW RECIRC LINE	10.4-11
6AF-1-33-2002-3	SB	X		TEST LINE, TRAIN B	10.4-11
4AF-1-62-2002-3	SB	X		TRAIN B TO SG-03	10.4-11
3AF-1-71-2002-3	SB	X		AF TRAIN B TO SG-03	10.4-11

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 3 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
3AF-1-79-2002-3	SB	X		AF TRAIN B TO SG-03	10.4-11
4AF-1-101-2002-2 (FROM 4X3 REDUCER THRU VALVE 1AF-125 AND ORIFICE CP1-AFORFR-04 TO VALVE 1AF-093	SB	X		AF TRAIN B TO SG-03	10.4-11
4AF-1-109-2002-3	SB	X		AF TRAIN B TO SG-3	10.4-11
4AF-1-54-2003-2	SB	X		AF TRAIN B TO SG-3 (NOTE 1)	10.4-11
4AF-1-019-1303-2	SB	X		AF TRAIN B TO SG-3 (NOTE 1)	10.4-11
4AF-1-23-2003-2	SB	X		TURBINE DRIVEN AF PUMP TO SG-3 (NOTE 1)	10.4-11
4AF-1-108-2002-3	SB	X		TURBINE DRIVEN AF PUMP TO SG-03	10.4-11
4AF-1-51-2002-3	SB	X		AF TRAIN B TO SG-04	10.4-11
3AF-1-73-2002-3	SB	X		AF TRAIN B TO SG-04	10.4-11
3AF-1-81-2002-3	SB	X		AF TRAIN B TO SG-04	10.4-11
4AF-1-103-2002-3 (FROM 4X3 REDUCER THRU VALVE 1AF-127 AND ORIFICE CP1-AFORFR-10 TO VALVE 1AF-101)	SB	X		AF TRAIN B TO SG-04	10.4-11
4AF-1-111-2002-3	SB	X		AF TRAIN B TO SG-04 (NOTE 1)	10.4-11
4AF-1-55-2003-2	SB	X		AF TRAIN B TO SG-04 (NOTE 1)	10.4-11
4AF-1-20-1303-2	SB	X		AF TRAIN B TO SG-04 (NOTE 1)	10.4-11
4AF-1-24-2003-2	SB	X		TURBINE DRIVEN AF PUMP TO SG-04 (NOTE 1)	10.4-11
4AF-1-110-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-04	10.4-11

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
8AF-1-11-2002-3	SB	X		TURBINE DRIVEN AF FEED PUMP TO ALL STEAM GENERATORS (NOTE 2)	10.4-11
3AF-1-68-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-01 (NOTE 2)	10.4-11
3AF-1-76-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-01 (NOTE 2)	10.4-11
4AF-1-96-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-01 (NOTE 2)	10.4-11
3AF-1-70-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-02 (NOTE 2)	10.4-11
3AF-1-78-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-02 (NOTE 2)	10.4-11
4AF-1-98-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-02 (NOTE 2)	10.4-11
3AF-1-72-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-03 (NOTE 2)	10.4-11
3AF-1-80-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-03 (NOTE 2)	10.4-11
4AF-1-100-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-03 (NOTE 2)	10.4-11
3AF-1-74-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-04 (NOTE 2)	10.4-11
6AF-1-25-2002-3	SB	X		TURBINE DRIVEN AF FEED PUMP TO ALL STEAM GENERATORS (NOTE 2)	10.4-11
3AF-1-26-2002-3	SB	X		TURBINE DRIVEN PUMP MIN. FLOW RECIRC LINE (NOTE 2)	10.4-11
3AF-1-82-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-04 (NOTE 2)	10.4-11
4AF-1-102-2002-3	SB	X		TURBINE DRIVEN PUMP TO SG-04 (NOTE 2)	10.4-11
6AF-1-29-2002-3	SB	X		TURBINE DRIVEN AUX. FEEDWATER PUMP TEST LINE (NOTE 2)	10.4-11

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6AF-1-88-2002-3	SB	X		MOTOR DRIVEN PUMPS (TRAIN A + B CROSS-TIE LINE EXCLUDING THE SEGMENT BETWEEN VALVES 1AF-090 AND 1AF-091	10.4-11
SYSTEM CHEMICAL AND VOLUME CONTROL (CS)					
3CS-1-235-2501R-1	RB		X	RCS LETDOWN LINE	9.3-10
3CS-1-001-2501R-2	RB		X	LETDOWN LINE PRIMARY SIDE H. EX. INLET	9.3-10
3CS-1-002-2501R-2	X		X	LETDOWN LINE PRIMARY SIDE H. EX OUTLET	9.3-10
2CS-1-005-2501R-2	RB		X	LETDOWN ORIFICE INLET	9.3-10
2CS-1-008-2501R-2	RB		X	LETDOWN ORIFICE DISCHARGE	9.3-10
2CS-1-004-2501R-2	RB		X	LETDOWN ORIFICE INLET	9.3-10
2CS-1-007-2501R-2	RB		X	LETDOWN ORIFICE DISCHARGE	9.3-10
2CS-1-003-2501R-2	RB		X	LETDOWN ORIFICE	9.3-10
2CS-1-006-2501R-2	RB		X	LETDOW ORIFICE DISCHARGE	9.3-10
2CS-1-009-601R-2	RB		X	LETDOWN ORIFICE DISCHARGE	9.3-10
2CS-1-010-601R-2	RB		X	LETDOWN ORIFICE DISCHARGE	9.3-10
2CS-1-011-601R-2	RB		X	LETDOWN ORIFICE DISCHARGE	9.3-10
3CS-1-012-601R-2	SB RB		X	LETDOWN LINE (NOTE 1)	9.3-10
2CS-1-546-601R-2	RB		X	LETDOWN LINE RELIEF VALVE LINE	9.3-10
3CS-1-013-601R-2	SB		X	CONNECTION TO LETDOWN REHEAT H EX	9.3-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
4CS-1-549-601R-2	SB		X	LETDOWN REHEAT HX INLET	9.3-10
4CS-1-550-601R-2	SB		X	LETDOWN REHEAT HX OUTLET	9.3-10
3CS-1-014-601R-2	SB		X	LETDOWN REHEAT HX OUTLET	9.3-10
3CS-1-354-601R-2	SB		X	VALVES TCV-381B BY PASS LINE	9.3-10
2CS-1-348-601R-2	SB		X	RHRS CLEAN UP CROSS CONNECTION	9.3-10
3CS-1-015-601R-2	SB	X		LETDOWN LINE	9.3-10
2CS-1-017-601R-2	AB	X		CHARGING DISCHARGE FROM TBX-CSAPPD-01	9.3-10
4CS-1-076-2501R-2	AB	X		CHARGING PUMP DISCHARGE FROM TBX-CSAPCH-01	9.3-10
4CS-1-344-2501R-2	AB	X		CHARGING PUMP DISCHARGE FROM TBX-CSAPCH-01	9.3-10
4CS-1-085-2501R-2	AB	X		CHARGING PUMP DISCHARGE FROM TBX-CSAPCH-02	9.3-10
4CS-1-346-2501R-2	AB	X		CHARGING PUMP DISCHARGE FROM TBX-CSAPCH-02	9.3-10
4CS-1-110-2501R-2	AB	X		CHARGING PUMP DISCHARGE HEADER	9.3-10
3CS-1-075-2501R-2	AB	X		CHARGING PUMP DISCHARGE HEADER	9.3-10
3CS-1-343-2501R-2	AB	X		CENTRIFUGAL CHARGING PUMP TRAIN A, BY PASS	9.3-10
2CS-1-345-2501R-2	AB	X		CENTRIGUGAL CHARGING PUMP TRAIN B, BY PASS	9.3-10
2CS-1-483-2501R-2	AB	X		CENTRIFUGAL CHARGING PUMP TRAIN A/B CROSS CONNECTIONS (NOTE 20)	9.3-10
2CS-1-482-2501R-2	AB	X		FCV-0121 BYPASS (NOTE 19)	9.3-10
2CS-1-338-2501R-2	AB	X		POS. DISPL. PUMP BYPASS VALVE LINE (NOTE 4)	9.3-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
3CS-1-074-2501R-2	AB SB	X		CHARGING LINE	9.3-10
2CS-1-349-2501R-2	AB	X		FLOW CONTROL VALVE BY PASS LINE	9.3-10
3CS-1-077-2501R-2	RB SB	X		CHARGING LINE (REGEN H. EX. INLET) (NOTE 1)	9.3-10
3CS-1-547-2501R-2	RB		X	CHARGING LINE (REGEN. H. EX. OUTLET)	9.3-10
3CS-1-079-2501R-2	RB		X	CHARGING LINE	9.3-10
3CS-1-081-2501R-2	RB		X	CHARGING LINE	9.3-10
3CS-1-078-2501R-2	RB		X	ALTERNATE CHARGING LINE (NOTE 7)	9.3-10
3CS-1-0802501R-2	RB	X		ALTERNATE CHARGING LINE (NOTE 8)	9.3-10
2CS-1-111-2501R-2	RB		X	PRESSURIZER AUX SPRAY LINE	9.3-10
2CS-1-112-2501R-2	RB	X		PRESSURIZER AUX SPRAY LINE (NOTES 4, 6)	9.3-10
2CS-1-087-2501R-2	AB SB	X		RCP SEAL WATER INJECTION	9.3-10
2CS-1-088-2501R-2	AB	X		SEAL WATER INJECTION FILTER	9.3-10
2CS-1-089-2501R-2	AB	X		SEAL WATER INJECTION FILTER OUTLET	9.3-10
2CS-1-090-2501R-2	AB SB	X		SEAL INJECTION SUPPLY HEADER	9.3-10
2CS-1-092-2501R-2	SB	X		SEAL INJECTION LINE RCP, LOOP 1 (NOTE 1)	9.3-10
2CS-1-093-2501R-2	SB	X		SEAL INJECTION LINE RCP, LOOP 1	9.3-10
2CS-1-102-2501R-2	RB	X		SEAL INJECTION LINE RCP, LOOP 1	9.3-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 8 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2CS-1-103-2501R-1	RB	X		SEAL INJECTION LINE RCP, LOOP 1	9.3-10
		(A JET)	(B JET)		
1 1/2CS-1-248-2501R-1	RB	X	X	SEAL INJECTION RCP, LOOP 1 (NOTE 5)	9.3-10
2CS-1-095-2501R-2	RB	X		SEAL INJECTION LINE RCP, LOOP 2 (NOTE 1)	9.3-10
2CS-1-104-2501R-2	RB	X		SEAL INJECTION RCP, LOOP 2	9.3-10
2CS-1-105-2501R-2	RB	X		SEAL INJECTION RCP, LOOP 2	9.3-10
		(A JET)	(B JET)		
1 1/2CS-1-249-2501R-2	RB	X	X	SEAL INJECTION RCP, LOOP 2 (NOTE 5)	9.3-10
2CS-1-096-2501R-2	SB	X		SEAL INJECTION RCP, LOOP 3	9.3-10
2CS-1-098-2501R-2	RB SB	X		SEAL INJECTION RCP, LOOP 3 (NOTE 1)	9.3-10
2CS-1-106-2501R-2	RB	X		SEAL INJECTION RCP, LOOP 3	9.3-10
2CS-1-107-2501R-1	RB	X		SEAL INJECTION RCP, LOOP 3	9.3-10
		(A JET)	(B JET)		
1 1/2 RCS-1-243-2501R-2	RB	X	X	SEAL INJECTION RCP LOOP 3 (NOTE 5)	9.3-10
2CS-1-099-2501R-2	SB	X		SEAL INJECTION LINE RCP, LOOP 4	9.3-10
2CS-1-101-2501R-2	SB	X		SEAL INJECTION RCP LOOP 4 (NOTE 1)	9.3-10
2CS-1-108-2501R-1	RB	X		SEAL INJECTION RCP, LOOP 4	9.3-10
2CS-1-109-2501R-1	RB	X		SEAL INJECTION RCP, LOOP 4	9.3-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21)		REMARKS	FLOW DIAGRAM FIGURE NO.
		SUB COOLED (A JET)	2 PHASE FLOW (B JET)		
1 1/2CS-244-2501R-1	RB	X	X	SEAL INJECTION RCP LOOP 4 (NOTE 5)	9.3-10
2CS-X-008-152-5	AB		X	BORIC ACID BATCHING TANK	9.3-10
2CS-1-165-2501R-2	RB	X		EXCESS LETDOWN HX OUTLET	9.3-10
3CS-1-628-2501R-2	AB	X		CHARGING PUMP A BYPASS	9.3-10
3CS-1-901-2501R-2	SB	X		LETDOWN LINE	9.3-10
3CS-1-902-2501R-2	SB		X	TCV BYPASS	9.3-10
3CS-1-903-2501R-2	SB	X		LETDOWN LINE	9.3-10
3CS-1-904-601R-2	SB		X	LETDOWN LINE	9.3-10
3CS-1-906-601R-2	SB		X	LETDOWN LINE	9.3-10
3CS-1-907-601R-2	SB		X	LETDOWN LINE	9.3-10
3CS-1-908-2501R-2	SB		X	LETDOWN LINE	9.3-10
3CS-1-910-2501R-2	SB		X	LETDOWN REHEAT HX INLET	9.3-10
3CS-1-911-601R-2	SB		X	LETDOWN REHEAT HX INLET	9.3-10
3CS-1-912-2501R-2	SB		X	LETDOWN REHEAT HX OUTLET	9.3-10
3CS-1-913-601R	SB		X	LETDOWN REHEAT HX OUTLET	9.3-10
3CS-1-922-601R-2	SB	X		LETDOWN REHEAT HX OUTLET	9.3-10
3CS-2-921-601R-2 (UNIT 2)	SB	X		LETDOWN REHEAT HX OUTLET	9.3-10
2CS-X-09-152-5	AB		X	AUX. STEAM TO BORIC ACID BATCHING TANK	9.3-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2CS-1-579-2501R-2	RB		X	LETDOWN HX INLET	9.3-10
2CS-1-945-2501R-2	AB		X	CENTRIFUGAL CHARGING PUMP MINIMUM FLOW BYPASS (NOTE 17)	9.3-10
2CS-2-946-2501R-2 (UNIT 2)	AB	X		CENTRIFUGAL CHARGING PUMP MINIMUM FLOW BYPASS (NOTE 17)	9.3-10
2CS-1-946-2501R-2	AB	X		CENTRIFUGAL CHARGING PUMP MINIMUM FLOW BYPASS (NOTE 18)	9.3-10
2CS-2-947-25014-2 (UNIT 2)	AB	X		CENTRIFUGAL CHARGING PUMP MINIMUM FLOW BYPASS (NOTE 18)	9.3-10
SYSTEM FEEDWATER SYSTEM (FW)					
3CS-1-629-2501R-2	AB	X		CHARGING PUMP B BYPASS	9.3-10
3CS-1-960-601R-2	RB		X	LETDOWN LINE RV LINE	9.3-10
18FW-1-13-2002-5	SB (Roof)		X	SG NUMBER 3 FEED LINE	10.4-9
18FW-1-34-2003-2 AND 2" DR. CONN. TO 1FW-230	SB		X	SG NUMBER 3 FEED LINE (NOTE 1)	10.4-0
18FW-1-17-1303-2 AND 2" CONN TO VALVES 1FW-111 (DRAIN) 1FW-127 (VENT)	RB, SB SB RB	X		SG NUMBER 3 FEED LINE (NOTE 1)	10.4-9
16FW-1-73-1303-2	RB		X	SG NUMBER 3	10.4-9
8FW-1-54-2002-5	SB (Roof)		X	SG NUMBER 3 FEED FLOW CONTROL VALVE BYPASS CONNECTION	10.4-9
8FW-1-42-2002-5 AND 2" CONN TO VENT VALVE 1FW-067	SB (Roof)		X	CONTROL VALVE BYPASS	10.4-9

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
3FW-1-105-2003-2	SB		X	SG NUMBER 3 FWIV BYPASS (NOTE 1)	10.4-9
3FW-1-114-1303-2	SB		X	(NOTE 1)	
6FW-2-93-203-2	SB		X	SG NUMBER 3 BYPASS LINE (NOTES 1 and 25)	10.4-9
6FW-1-97-1303-2 AND 2" LINES TO VALVES 1FW-247 (VENT) 1FW-264 (DRAIN) 1FW-265 (DRAIN)	RB, SB SB RB RB		X	SG NUMBER 3 PREHEATER BYPASS LINE (NOTE 26)	10.4-9
1 1/2 FW-1-30-1303-2	SB		X	CONNECTION FROM PREHEATER BYPASS (NOTES 1 and 26)	10.4-9
6FW-2-101-1303-2	RB		X	FW SPLIT FLOW BYPASS FOR SG NUMBER 3 (NOTE 25)	10.4-9
18FW-1-14-2002-5	SB (Roof)		X	SG NUMBER 2 FEED LINE	10.4-9
18FW-1-35-2003-2 AND 2" DR. CONN. TO IFW-228	SB		X	SG NUMBER 2 FEED LINE	10.4-9
18FW-1-18-1303-2 AND 2" CONNS TO VALVES 1FW-114 (DRAIN) 1FW-129 (VENT)	SB, RB SB RB		X	SG NUMBER 2 FEED LINE	10.4-9
16FW-1-74-1303-2	RB		X	SG NUMBER 2 FEED LINE	10.4-9
8FW-1-55-2002-5	SB (Roof)	X		SG NUMBER 2 FEED FLOW CONTROL VALVE BYPASS	10.4-9
8FW-1-43-2002-5 AND 2" CONN TO VENT VALVE 1FW-073	SB (Roof)		X	SG NUMBER 2 FEED FLOW CONTROL VALVE BYPASS	10.4-9
3FW-1-104-2003-2	SB		X	SG NUMBER 2 FWIV BYPASS (NOTE 2)	10.4-9
2FW-1-113-1303-2	SB		X	SG NUMBER FWIV BYPASS (NOTE 2)	10.4-9

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6FW-2-92-2003-2	SB		X	SG NUMBER 2 PREHEATER BYPASS LINE (NOTES 1 and 25)	10.4-9
6FW-1-96-1303-2 AND 2" LINES TO	RB, SB		X	SG NUMBER 2 PREHEATER BYPASS LINE	10.4-9
1FW-246 (VENT)	SB			(NOTES 1 and 26)	10.4-9
1FW-261 (DRAIN)	RB				10.4-9
1FW-262 (DRAIN)	RB				10.4-9
1 1/2FW-1-31-1303-2	SB		X	CONNECTION FROM PREHEATER BYPASS LINE (NOTES 1 and 26)	10.4-9
6FW-2-100-1303-2	RB		X	FW SPLIT FLOW BYPASS FOR SG NUMBER 2 (NOTE 25)	10.4-9
18FW-1-15-2002-5	SB (Roof)		X	SG NUMBER 1 FEED LINE	10.4-9
18FW-1-36-2003-2 AND 2" DR CONN TO 1FW-220	SB		X	SG NUMBER 1 FEED LINE (NOTE 1)	10.4-9
18FW-1-19-1303-2 AND 2" CONN TO VALVES	RB, SB		X	SG NUMBER 1 FEED LINE (NOTE 1)	10.4-9
1FW-117 (VENT)	SB				
16FW-1-75-1303-2	RB		X	SG NUMBER 1 FEED LINE	10.4-9
8FW-1-56-2002-5	SB (Roof)		X	SG NUMBER 1 FEED FLOW CONTROL VALVE BYPASS CONNECTION	10.4-9
8FW-1-44-2002-5 AND 2" CONN TO VENT	SB (Roof)		X	SG NUMBER 1 FEED FLOW CONTROL VALVE BYPASS	10.4-9
VALVE 1FW-079	SB			CONNECTION	
3FW-1-103-2003-2	SB		X	SG NUMBER 1 FWIV BYPASS (NOTE 1)	10.4-9
3FW-1-112-1303-2	SB		X	SG NUMBER 1 FWIV BYPASS (NOTE 1)	10.4-9
6FW-2-91-2003-2	SB		X	SG NUMBER 1 PREHEATER BYPASS LINE (NOTES 1 and 25)	10.4-0

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6FW-1-95-1303-2 AND 2" LINES TO VALVES 1FW-245 (VENT) 1FW-258 (DRAIN) 1FW-259 (DRAIN)	RB SB SB RB RB		XX	SG NUMBER PREHEATER BYPASS LINE (NOTES 1 and 25)	10.4-9
1 1/2FW-1-32-1303-2	SB		X	CONNECTION FROM CHEM PREHEATER BYPASS LINE (NOTES 1 and 26)	10.4-9
6FW-2-99-1303-2	RB		X	SPLIT FLOW BYPASS LINE FOR SG NUMBER 1 (NOTE 25)	10.4-9
18FW-1-16-2002-5	SB (Roof)		X	SG NUMBER 4 FEED LINE	10.4-9
18FW-1-37-2003-2 AND 2" DR CONN TO 1FW-226	SB		X	SG NUMBER 4 FEED LINE (NOTE 1)	10.4-9
18FW-1-20-1303-2 AND 2" CONN TO VALVES 1FW-120 (VENT)	RB, SB SB		X	SG NUMBER FEED LINE (NOTE 1)	10.4-9
16FW-1-76-1303-2	RB		X	SG NUMBER 4 FEED LINE	10.4-9
8FW-1-57-2002-5	SB (Roof)		X	SG NUMBER 4 FEED FLOW CONTROL VALVE BYPASS	10.4-9
8FW-1-45-2002-5 AND 2" DR. CONN TO VALVE 1FW-185	SB (Roof) SB		X	SG NUMBER 4 FEED FLOW CONTROL VALVE BYPASS	10.4-9
3FW-1-106-2003-2	SB		X	SG NUMBER 4 FWIV BYPASS (NOTE 1)	10.4-9
3FW-1-111-1303-2 BYPASS (NOTE 1)	SB		X	SG NUMBER 4 FWIV	10.4-9
6FW-2-94-2003-2 LINE (NOTE 1)	SB		X	SG NUMBER 4 PREHEATER BYPASS (NOTE 25)	10.4-9

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6FW-2-98-1303-2 AND 2" LINES TO VALVES 1FW-248 (VENT) 1FW-255 (DRAIN) 1FW-256 (DRAIN)	RB, SB SB RB RB		X	(NOTES 1 and 26)	10.4-9
1 1/2FW-1-33-1303-2	SB		X	CONNECTION FROM PREHEATER BYPASS LINE (NOTES 1 and 26)	10.4-9
6FW-2-102-1303-2	RB		X	FW SPLIT FLOW BYPASS FOR SG NUMBER 4 (Unit 25)	10.4-9
30FW-1-12-2002-G	TB		X	FEEDWATER HEATERS OUTLET HEADER	10.4-9
30FW-1-011-2002G	TB		X	FEEDWATER HEADERS OUTLET	10.4-9
2FW-2-905-1303-2	RB		X	CHEMICAL FEED SG NUMBER 3	10.4-9
2FW-2-906-1303-2	RB		X	CHEMICAL FEED SG NUMBER 4	10.4-9
2FW-2-907-1303-2	RB		X	CHEMICAL FEED SG NUMBER 2	10.4-9
2FW-2-900-1303-2	RB		X	CHEMICAL FEED SG NUMBER 2	10.4-9
2FW-2-901-1303-2	RB		X	CHEMICAL FEED SG NUMBER 4	10.4-9
2FW-2-902-1303-2	RB		X	CHEMICAL FEED SG NUMBER 1	10.4-9
2FW-2-903-1303-2	RB		X	CHEMICAL FEED SG NUMBER 1	10.4-9
2FW-2-904-1303-2	RB		X	CHEMICAL FEED SG NUMBER 3	10.4-9
SYSTEM MAIN STEAM (MS)					
32MS-1-01-1303-2 AND 2" V. CON TO VALVE 1MS-615	RB SB		X	MAIN STEAM PIPES & INCLUDING SSV CONNECTIONS (NOTE 1)	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
32MS-1-02-1303-2 AND 2" V. CON TO VALVE 1MS-614	RB SB		X	MAIN STEAM PIPES & INCLUDING SSV CONNECTIONS (NOTE 1)	10.3-1
32MS-1-03-1303-2 AND 2" V. CON TO VALVE 1MS-616	RB SB		X	MAIN STEAM PIPES & INCLUDING SSG CONNECTIONS (NOTE 1)	10.3-1
32MS-1-04-1303-2 AND 2" V CON TO VALVE 1MS-617	RB SB		X	MAIN STEAM PIPES & INCLUDING SSV CONNECTIONS (NOTE 1)	10.3-1
2MS-1-05-1303-2	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 1)	10.3-1
2MS-1-166-1302-5	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 1) (NOTE 24)	10.3-1
2MS-1-10-1303-2 POT DRAIN LINES (NOTE 1)	SB		X	MAIN STEAM LINE DRAIN	10.3-1
2MS-1-164-1302-5	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 24)	10.3-1
2MS-1-15-1303-2	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 1)	10.3-1
2MS-1-162-1302-5	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 24)	10.3-1
2MS-1-20-1303-2	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 1)	10.3-1
2MS-1-168-1302-5	SB		X	MAIN STEAM LINE DRAIN POT DRAIN LINES (NOTE 24)	10.3-1
34MS-1-07-1302-5 34MS-1-07-1302-G	SB (Roof) TB		X	MAIN STEAM PIPES DOWN STREAM OF MSIV'S	10.3
34MS-1-12-1302-5 34MS-1-12-1302-G	SB (Roof) TB		X	MAIN STEAM PIPES DOWNSTREAM OF MSIVs	10.3-1
34MS-1-17-1302-5 34MS-1-17-1302-G	SB (Roof) TB		X	MAIN STEAM PIPES DOWNSTREAM OF MSIVs	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
34MS-1-22-1302-5 34MS-1-22-1302-G	SB (Roof) TB		X	MAIN STEAM PIPES DOWNSTREAM OF MSIVs	10.3-1
4MS-1-25-1303-2	SB		X	MS TO AF PUMP TURBINE (NOTE 1)	10.3-1
4MS-1-26-1303-2	SB		X	MS TO AF PUMP TURBINE (NOTE 1)	10.3-1
8MS-1-223-1303-2	SB		X	ATMOSPHERIC STEAM DUMPS (NOTE 1)	10.3-1
8MS-1-240-1303-2	SB		X	ATMOSPHERIC STEAM DUMPS (NOTE 1)	10.3-1
8MS-1-257-1303-2	SB		X	ATMOSPHERIC STEAM DUMPS (NOTE 1)	10.3-1
8MS-1-274-1303-2	SB		X	ATMOSPHERIC STEAM DUMPS (NOTE 1)	10.3-1
2MS-1-206-1303-2	RB		X	SG NUMBER 1 BLOWDOWN SYSTEM	10.3-1
3MS-1-74-1303-2	RB SB		X	SG NUMBER 1 BLOWDOWN SYSTEM (NOTE 1)	10.3-1
2MS-1-203-1303-2	RB		X	SG NUMBER 1 BLOWDOWN SYSTEM	10.3-1
2MS-1-341-1303-2	RB		X	SG NUMBER 1 BLOWDOWN SYSTEM	10.3-1
3MS-1-342-1303-2	RB		X	SG NUMBER 1 BLOWDOWN SYSTEM	10.3-1
3MS-78-1302-5	SB		X	SG NUMBER 1 BLOWDOWN SYSTEM	10.3-1
2MS-199-1303-2	RB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1
3MS-1-75-1303-2	RB SB		X	SG NUMBER 2 BLOWDOWN SYSTEM (NOTE 1)	10.3-1
4MS-1-151-1303-2	RB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1
2MS-1-195-1303-2	RB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2MS-1-339-1303-2	RB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1
4MS-1-340-1303-2	RB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1
3MS-1-79-1302-5	SB		X	SG NUMBER 2 BLOWDOWN SYSTEM	10.3-1
2MS-1-210-1303-2	RB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
4MS-1-150-1303-2	RB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
3MS-1-76-1303-2	RB SB		X	SG NUMBER 3 BLOWDOWN SYSTEM (NOTE 1)	10.3-1
3MS-1-80-1302-5	SB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
2MS-1-213-1303-2	RB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
2MS-1-343-1303-2	RB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
4MS-1-344-1303-2	RB		X	SG NUMBER 3 BLOWDOWN SYSTEM	10.3-1
2MS-1-218-1303-2	RB		X	SG NUMBER 4 BLOWDOWN SYSTEM	10.3-1
3MS-1-73-1303-2	RB SB		X	SG NUMBER 4 BLOWDOWN SYSTEM (NOTE 1)	10.3-1
3MS-1-77-1302-5	SB		X	SG NUMBER 4 BLOWDOWN SYSTEM	10.3-1
2MS-1-221-1303-2	RB		X	SG NUMBER 4 BLOWDOWN SYSTEM	10.3-1
2-MS-1-346-1303-2	RB		X	SG NUMBER 4 BLOWDOWN SYSTEM	10.3-1
3-MS-1-345-1303-2	RB		X	SG NUMBER 4 BLOWDOWN SYSTEM	10.3-1
2MS-1-163-1302-5	SB		X	MAIN STEAM DRAIN POT DR. LINES, DOWN STREAM OF FLOW CONTROL ORIFICES (Note 24)	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2MS-1-165-1302-5	SB		X	MAIN STEAM DRAIN POT DR. LINES, DOWN STREAM OF FLOW CONTROL ORIFICES (Note 24)	10.3-1
2MS-1-167-1302-5	SB		X	MAIN STEAM DRAIN POT DR. LINES, DOWN STREAM OF FLOW CONTROL ORIFICES (Note 24)	10.3-1
2MS-1-169-1302-5	SB		X	MAIN STEAM DRAIN POT DR. LINES, DOWN STREAM OF FLOW CONTROL ORIFICES (Note 24)	10.3-1
4MS-1-27-1302-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
4MS-1-28-1302-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
4MS-1-29-1302-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
8MS-1-415-152-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
16MS-1-416-152-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
10MS-1-425-152-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
1 1/2MS-1-419-152-3	SB	X		(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
1 1/2MS-1-420-152-3	SB	X		(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
4-MS-1-437-152-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2-MS-1-438-152-3	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2-MS-1-439-152-5	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2-MS-1-440-152-5	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2-MS-1-441-151-5	SB		X	(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
6-MS-1-442-152-5	SB		X	(NOTE 3) MS TO AF PMP TURBINE	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
4MS-1-914-152-3 4MS-2-907-152-3 (U2)	SB	X		(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2MS-1-915-152-3 2MS-2-908-152-3 (U2)	SB	X		(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2MS-916-152-3	SB	X		(NOTE 3) MS TO AF PUMP TURBINE	10.3-1
2MS-1-918-1303-2	RB		X	CHEM. FD TO BLOWDOWN LINE	10.3-1
2MS-1-919-1303-2	RB		X	CHEM. FD TO BLOWDOWN LINE	10.3-1
2MS-1-920-1303-2	RB		X	CHEM. FD TO BLOWDOWN LINE	10.3-1
2MS-1-921-1303-2	RB		X	CHEM. TD TO BLOWDOWN LINE	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-03 TO SAFETY RELIEF VALVE 1MS-0093 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-03 TO SAFETY RELIEF VALVE 1MS-0094 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-03 TO SAFETY RELIEF VALVE 1MS-0095	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-03 TO SAFETY RELIEF VALVE 1MS-0096 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-03 TO SAFETY RELIEF VALVE 1MS-0097 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-58 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-59 (NOTE 1)	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-60 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-61 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-62 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-02 TO SAFETY RELIEF VALVE 1MS-21 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-01 TO SAFETY RELIEF VALVE 1MS-22 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-01 TO SAFETY RELIEF VALVE 1MS-23 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-01 TO SAFETY RELIEF VALVE 1MS-24 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-01 TO SAFETY RELIEF VALVE 1MS-25 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-04 TO SAFETY RELIEF VALVE 1MS-129 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-04 TO SAFETY RELIEF VALVE 1MS-130	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-04 TO SAFETY RELIEF VALVE 1MS-131 (NOTE 1)	10.3-1
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-04 TO SAFETY RELIEF VALVE 1MS-132 (NOTE 1)	10.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 21 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6-2003-2	SB		X	6 INCH LINE FROM 32-MS-1-04 TO SAFETY RELIEF VALVE 1MS-133 (NOTE 1)	10.3-1
SYSTEM REACTOR COOLANT (RC)					
29RC-1-001-2501R-1	RB		X	HOT LEG	5.1-1
31RC-1-002-2501R-1	RB		X	CROSS OVER	5.1-1
27.5RC-1-003-2501R-1	RB		X	COLD LEG	5.1-1
12RC-1-007-2501R-1	RB		X	RHR TRAIN A CONNECTION	5.1-1
6RC-1-008-2501R-1	RB		X	SIS INJ. PUMP CONNECTION	5.1-1
2RC-1-015-2501R-1	RB		X	TO WPS (L) VIA RCDT	5.1-1
4RC-1-018-2501R-1	RB		X	TO PRESSURIZER SPRAY LINE	5.1-1
3RC-1-019-2501R-1	RB		X	ALTERNATE CH. LINE	5.1-1
10RC-1-021-2501R-1	RB		X	SIS ACCUMULATOR & RHR INJECTION	5.1-1
1 1/2 RC-1-020-2501-1	RB		X	SIS BIT CONN. RC LOOP #01	5.1-1
29RC-1-058-2501R-1	RB		X	HOT LEG	5.1-1
31RC-1-059-2501R-1	RB		X	CROSS OVER	5.1-1
27.5RC-1-060-2501R-1	RB		X	COLD LEG	5.1-1
12RC-1-069-2501R-1	RB		X	RHR TRAIN B CONN.	5.1-1
6RC-1-070-2501R-1	RB		X	SIS-INJ. PUMP CONN.	5.1-1
14RC-1-135-25O1R-1	RB		X	PRESSURIZER SURGE LINE	5.1-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 22 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
4RC-1-075-2501R-1	RB		X	TO PRESSURIZER SPRAY LINE	5.1-1
3RC-1-076-2501R-1	RB		X	CHARGING LINE	5.1-1
10RC-1-078-2501R-1	RB		X	SIS ACCUMULATOR RHRS INJ	5.1-1
1 1/2RC-1-079-2501R-1	RB		X	SIS BIT CONN.	5.1-1
2RC-1-072-2501R-1	RB		X	TO WPS (L) VIA RDT	5.1-1
29RC-1-040-2501R-1	RB		X	HOT LEG	5.1-1
31RC-1-041-2501R-1	RB		X	CROSS OVER	5.1-1
27.5RC-1-042-2501R-1	RB		X	COLD LEG	5.1-1
6RC-1-029-2501R-1	RB		X	SIS INJ. PUMP CONN.	5.1-1
2RC-1-035-2501R-1	RB		X	TO WPS (L) VIA RCDT	5.1-1
10RC-1-037-2501R-1	RB		X	SIS ACCUMULATOR & RHRS INJ.	5.1-1
1 1/2RC-1-039-2501R-1	RB		X	SIS BIT CONN. RC LOOP #02	
29RC-1-023-2501R-1	RB		X	HOT LEG	5.1-1
31RC-1-024-2501R-1	RB		X	CROSS OVER	5.1-1
27.5RC-1-025-2501R-1	RB		X	COLD LEG	5.1-1
3RC-1-052-2501R-1	RB		X	RCS LETDOWN LINE	5.1-1
2RC-053-2501R-1	RB		X	TO WPS (L) VIA RCDT	5.1-1
6RC-1-046-2501R-1	RB		X	SIS INJ. PUMP CONN.	5.1-1
1 1/2RC-1-057-2501R-1	RB		X	SIS BIT CONN.	5.1-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 23 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
10RC-1-055-2501R-1	RB		X	SIS ACCUMULATOR & RHRS INJ.	5.1-1
4RC-1-087-2501R-1	RB		X	PRESSURIZER SPRAY LINE FROM RC LOOP 1	5.1-1
4RC-1-088-2501R-1	RB		X	PRESSURIZER SPRAY LINE FROM RC LOOP 4	5.1-1
6RC-1-147-2501R-1	RB		X	PRESSURIZER SPRAY LINE	5.1-1
4RC-1-091-2501R-1	RB		X	PRESSURIZER SPRAY LINE	5.1-1
2RC-1-132-2501R-1	RB		X	AUX. SPRAY LINE FROM CVCS	5.1-1
6RC-1-096-2501R-1	RB		X	PRESSURIZER SAFETY LINES	5.1-1
6RC-1-098-2501R-1	RB		X	PRESSURIZER SAFETY LINES	5.1-1
6RC-1-100-2501R-1	RB		X	PRESSURIZER SAFETY LINES	5.1-1
6RC-1-108-2502R-1	RB		X	PRESSURIZER RELIEF LINE	5.1-1
3RC-1-146-2501R-1	RB		X	RELIEF VALVE INLET	5.1-1
3RC-1-111-2501R-1	RB		X	RELIEF VALVE INLET	5.1-1
6RC-1-161-2501R-1	RB		X	PRESSURIZER SPRAY	5.1-1
4RC-1-162-2501R-1	RB		X	PRESSURIZER SPRAY	5.1-1
6RC-1-163-2501R-1	RB		X	PRESSURIZER SPRAY	5.1-1
4RC-1-164-2501R-1	RB		X	PRESSURIZER SPRAY	5.1-1
SYSTEM RESIDUAL HEAT REMOVAL (RHR)					
12-RH-1-001-2501R-1	RB	X		FROM RCS HL LOOP 4 (NOTE 16)	5.4-6
12-RH-1-002-2501R-1	RB	X		FROM RCS HL LOOP 4 (NOTE 16)	5.4-6

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 24 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
SYSTEM AUXILIARY STEAM (SA)					
6SA-X-18-152-5	AB EC TB		X	BORON RECYCLE EVAPORATOR AND BORIC ACID BATCHING TANK SUPPLY HEADER	10.4-16
2SA-X-63-152-5	AB		X	BORIC ACID BATCHING TANK STEAM SUPPLY	10.4-16
2SA-X-54-152-5	AB		X	DEMIN. WATER “FLUSHING CONN.” (NOTE 9)	10.4-16
4SA-X-57-152-5	AB		X	BRS RECYCLE EVAPORATOR SUPPLY	10.4-16
4SA-X-58-152-5	AB		X	BRS RECYCLE EVAPORATOR FEED PREHEATER STEAM	10.4-16
2SA-X-59-152-5	AB		X	BRS RECYCLE EVAPORATOR FEED PREHEATER STEAM	10.4-16
1-1/2SA-X-84-152-5	AB		X	BRS RECYCLE EVAPORATOR CONDENSATE	10.4-16
1-1/2SA-X-85-152-5	AB		X	BORON RECYCLE EVAPORATOR CONDENSATE TRIP BYPASS	10.4-16
2SA-X-48-152-5	AB		X	BRS RECYCLE EVAPORATOR CONDENSATE	10.4-16
8SA-X-115-152-5	AB		X	AUXILIARY STEAM CONDENSATE COOLER INLET HEADER	10.4-16
2SA-X-93-152-5	AB		X	BORIC ACID BATCHING TANK CONDENSATE	10.4-16
2SA-X-120-152-5	AB		X	BORIC ACID BATCHING TANK CONDENSATE TRIP BYPASS	10.4-16
10SA-X-19-152-5	AB EC TB		X	FLOOR DRAIN WASTE EVAPORATOR AND WPS WASTE EVAPORATOR STEAM SUPPLY HEADER	10.4-16

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 25 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2SA-X-38-152-5	AB		X	DEMIN. WATER FLUSHING CONN. (NOTE 9)	10.4-16
6SA-X-39-152-5	AB		X	FLOOR DRAIN WASTE EVAPORATOR STEAM SUPPLY	10.4-16
4SA-X-40-152-5	AB		X	FLOOR DRAIN WASTE EVAPORATOR STEAM SUPPLY	10.4-16
2SA-X-42-152-5	AB		X	FLOOR DRAIN WASTE EVAPORATOR FEED PREHEATER STEAM SUPPLY	10.4-16
1-1/2SA-X-65-152-5	AB		X	FLOOR DRAIN WASTE EVAPORATOR CONDENSATE	10.4-16
1-1/2SA-X-66-152-5	AB		X	FLOOR DRAIN WASTE EVAPORATOR CONDENSATE STRN. BYPASS	10.4-16
2SA-X-52-152-5	AB		X	FLOOR DRAIN WASTE EVAP. CONDENSATE	10.4-16
6SA-X-46-152-5	AB		X	WPS WASTE EVAPORATOR STEAM SUPPLY	10.4-16
2SA-X-53-152-5	AB		X	DEMIN. WATER FLUSHING CONN. (NOTE 9)	10.4-16
4SA-X-47-152-5	AB		X	WPS WASTE EVAPORATOR STEAM SUPPLY	10.4-16
4SA-X-41-152-5	AB		X	WPS WASTE EVAPORATOR FEED PREHEATER STEAM SUPPLY	10.4-16
2SA-X-49-152-5	AB	X		WPS WASTE EVAPORATOR FEED PREHEATER STEAM SUPPLY	10.4-16
1-1/2SA-X-74-152-5	AB		X	WPS WASTE EVAPORATOR CONDENSATE	10.4-16
1-1/2SA-X-75-152-5	AB		X	WPS WASTE EVAP. CONDENSATE STRN. BYPASS	10.4-16
2SA-X-50-152-5	AB		X	WPS WASTE EVAP CONDENSATE	10.4-16
2SA-X-129-152-5	EC TB		X	DRAIN POT X-SA-7 TO MAIN CONDENSER	10.4-16

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 26 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
2SA-X-130-152-5	EC		X	DRAIN POT X-SA-7 TO MAIN CONDENSER	10.4-16
6SA-X-906-152-5	AB		X	BRS RECYCLE EVAPORATOR SUPPLY	10.4-16
SYSTEM STEAM GENERATOR BLOWDOWN (SB)					
3SB-1-01-1302-5	SB		X	STEAM GENERATOR BLOWDOWN SG. 01	10.4-10
3SB-1-03-1302	SB		X	STEAM GENERATOR BLOWDOWN, SG. 02	10.4-10
3SB-1-09-1302-5	SB		X	STEAM GENERATOR BLOWDOWN SG. 03	10.4-10
3SB-1-18-1302-5	SB		X	STEAM GENERATOR BLOWDOWN, SG. 04	10.4-10
8SB-1-60-1302-5	SB TB EC		X	STEAM GENERATOR BLOWDOWN, HEADER TO SG. BLOWDOEN HT. EXCHANGE CP1-SBAHSB-01	10.4-10
2SB-1-72-1302-5	SB		X	N2 GAS SUPPLY CONNECTION (NOTE 10)	10.4-10
8SB-1-25-1302-5	EC		X	SG. HT. EXCH. BYPASS	10.4-10
8SB-1-48-1302-5	EC	X		SG. HT. EXCH. DISCHARGE	10.4-10
3SB-1-109-1302-5	EC	X		SG. HT. EXCH. DISCHARGE	10.4-10
3SB-1-110-1302-5	EC	X		SG. HT. EXCH. DISCHARGE	10.4-10
8SB-1-37-302-5	EC AB	X		SG. HT. EXCH. DISCHARGE	10.4-10
3SB-1-40-302-5	EC	X		RELIEF VALVE CONNECTION	10.4-10
8SB-1-68-301-5	AB	X		SG. BLOWDOWN FILTER HEADER	10.4-10
6SB-1-101-301-5	AB	X		SG. BLOWDOWN FILTER CP1-SBFLSB-01 INLET CONNECTION	10.4-10

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6SB-1-52-301-5	AB	X		FILTER BYPASS (NOTE 11)	10.4-10
6SB-1-24-301-5	AB	X		SG. BLOWDOWN FILTER CP1-SBFLSB-02 INLET CONNECTION	10.4-10
8SB-1-30-302-5	EC		X	SG. BLOWDOWN HX COND. OUTLET	10.4-10
SYSTEM SAFETY INJECTION SYSTEM (SI)					
10SI-1-119-2501R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 1	6.3-1
10SI-1-103-2501R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 1 (NOTE 12)	6.3-1
10SI-1-179-2501R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 1 (NOTE 12)	6.3-1
2SI-1-123-601R-2	RB	X		TO WPS VIA RCDT PUMPS (NOTE 15)	6.3-1
2" LEVEL TRANSMITTER PIPING	RB	X		LEVEL TRANSMITTER 1-LT-950 AND 1-LT-951 ON ACCUMULATOR INJ. TANK 1 (NOTE 13)	6.3-1
10SI-1-120-601R-2	RB	X		ACCUMULATOR INJ. TANK 2 TO RCS COLD LEG LOOP 2	6.3-1
10SI-1-104-2501R-2	RB	X		ACCUMULATOR INJ. TANK 2 TO RCS COLD LEG LOOP 2 (NOTE 12)	6.3-1
10SI-1-180-2501R-1	RB	X		ACCUMULATOR INJ. TANK 2 TO RCS COLD LEG LOOP 2 (NOTE 12)	6.3-1
2SI-1-124-601R-2	RB	X		TO WPS VIA RCDT PUMPS (NOTE 15)	6.3-1
2" LEVEL TRANSMITTER PIPING	RB	X		LEVEL TRANSMITTER 1-LT-952 AND 1-LT-953 ON ACCUMULATOR INJ. TANK 2 (NOTE 13)	6.3-1
10SI-1-121-601R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 3	6.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
10SI-1-105-2501R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 3 (NOTE 12)	6.3-1
10SI-1-181-2501R-1	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 3 (NOTE 12)	6.3-1
2SI-1-125-601R-2	RB	X		TO WPS VIA RCDT PUMPS (NOTE 15)	6.3-1
2" LEVEL TRANSMITTER PIPING	RB	X		LEVEL TRANSMITTER 1-LT-954 AND 1-LT-955 ON ACCUMULATOR INJ. TANK 3 (NOTE 13)	6.3-1
10SI-1-122-601R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 4	6.3-1
10SI-1-106-2501R-2	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 4 (NOTE 12)	6.3-1
10SI-1-182-2501R-1	RB	X		ACCUMULATOR INJ. TO RCS COLD LEG LOOP 4 (NOTE 12)	6.3-1
2SI-1-126-601R-2	RB	X		TO WPS VIA RCDT PUMPS (NOTE 15)	6.3-1
2" LEVEL TRANSMITTER	RB	X		LEVEL TRANSMITTER 1-LT-956 AND 1-LT-957 ON ACCUMULATOR INJ. TANK 4 (NOTE 13)	6.3-3
2SI-1-063-2501R-1	TB	X		SI TO CL LOOP (NOTE 14)	6.3-1
6SI-1-089-2501R-1	RB	X		SI/RHR TO CL LOOP 1 (NOTE 14)	6.3-1
2SI-1-065-2501R-1	RB	X		SI TO CL LOOP 2	6.3-1
6SI-1-328-2501R-1	RB	X		SI/RHR TO CL LOOP 2 (NOTE 14)	6.3-1
8SI-1-090-2501R-1	RB	X		SI/RHR TO CL LOOP 2 (NOTE 14)	6.3-1
2SI-1-067-2501R-1	RB	X		SI TO CL LOOP 3 (NOTE 14)	6.3-1

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 29 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.
6SI-1-330-2501R-1	RB	X		SI/RHR TO CL LOOP 3 (NOTE 14)	6.3-1
8SI-1-091-2501R-1	RB	X		SI/RHR TO CL LOOP 3 (NOTE 14)	6.3-1
2SI-1-069-2501R-1	RB	X		SI TO CL LOOP 4 (NOTE 14)	6.3-1
6SI-1-092-2501R-1	RB	X		SI/RHR TO CL LOOP 4 (NOTE 14)	6.3-1
4SI-1-001-2501R-1	AB	X		ECCS FROM CHG PUMPs	6.3-1
4SI-1-008-2501R-2	SB	X		ECCS FROM CHG PUMPs	6.3-1

NOTES TO THE HIGH ENERGY LINE LIST

LEGEND:

SB - SAFEGUARDS BUILDING

EC - ELECTRICAL AND CONTROLS BUILDING

RB - CONTAINMENT BUILDING

AB - AUXILIARY BUILDING

TB - TURBINE BUILDING

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 30 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.

1. This piping is within a Break Exclusion Area. This includes the piping outside containment from the containment penetration to the containment isolation valve and including the applicable moment restraint(s).

2. These Steam Turbine Driven AF Pump discharge lines are classified as moderate energy piping since operation as a high energy fluid system is for a limited period as prescribed by 3.6B.2.1.4. Breaks in these lines are not postulated because they are not operational during normal plant conditions and because of insufficient stored energy due to back leakage of any of the following flow check valves:

   1AF-078 1AF-098 1AF-086 1AF-106

   This leakage will place the lines under pressure back to valve 1AF-038 in line 8-AF-1-011-2002-3. This latter valve is assumed to hold.

3. These MS lines are classified as moderate energy piping since operation as a high energy fluid system is for a limited period as prescribed by 3.6B.2.1.4.

4. Not used.

5. This line is downstream of the last check valve (closest to the RCP). Therefore the back flow jet from the downstream side (B Jet) of any break (i.e. flow of RC water from the pump body) will be at 2192 psig & 559^{o}F. The flow from this source will be 2-phase flow.

6. This line (2-CS-1-112-2501R-1) is between a check valve (1CS-8377) and a flow control valve (1-8145), both normally closed during power operation. Pressurization of this line is assumed due to poss ble valve leakage. Temperature in this line, however, approaches ambient. Therefore, a break of this line will result in a subcooled (solid) jet of water.

7. This line (3-CS-1-078-2501R-2) is high energy up to normally closed valve 1-8147.

8 This line (3-CS-1-080-2501R-1) is between check valves 1-8379A and 1-8379B. Pressurization of this line is assumed due to possible valve leakage.

9. This flushing connection will be subject to SA operating line pressure and temperature of 50 psig and 297^{o}F during normal operation of the SA System up to the check valve.

10. Valve 1NG-026 (see 2323-M1-0243 Sheet A, Zone E-4) is closed and sealed during normal plant operation at power. Therefore, a break in the N2 gas supply line (2-SB-1-072-1302-5) would result in a high energy jet consisting of two phase back flow from the Steam Blowdown System header.

11. This line is considered to be High Energy up to normally closed valve 1SB-054.

12. Emergency supply of borated water from Accumulator Injection Tanks 1 through 4 to RCS Cold Leg Loops 1 through 4. It is assumed that check valves 1-8956 A,B,C, and D hold and 1-8948 A,B,C, and D leak.

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

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LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.

13. This line connects to nitrogen overpressure space in the accumulator tank.

14. To SIS Cold Legs Loops 1 through 4, it is assumed that check valves 1SI-8819 A,B,C, and D and 1SI-8818 A,B,C, and C, all hold.

15. These lines are assumed to be high energy only up to normally closed valves 1SI-8955 A<B<C< and D.

16. For normally closed isolation valves a loss of reactor coolant accident is assumed to occur for pipe breaks on the reactor side of the first valve.

17. This line (2CS-1-945-2501R-2) is high energy up to normally closed valve 1-8511A.

18. This line (2CS-1-946-2501R-2) is high energy up to normally closed valve 1-8511B.

19. This line (2CS-1-482-2501R-2) is high energy up to normally closed valve 1-CS-8387A.

20. This line (2CS-1-483-2501R-2) is high energy up to normally closed valve 1-CS-08387B.

21. The categorization of the break jets (subcooled liquid, two phase flow) is arrived at by consideration of contr buting fluid inventory upstream and downstream of the break (i.e. piping, pressure vessels, heat exchangers, pressurizer, steam generator). The plant operating conditions for the jet selection are defined in Section 6.2.2 of ANS 58.2 (1988 final draft issue) as stated below:

    (a) For those portions of piping systems which are normally pressurized during operation, the thermodynamic state in the pipe and the associated reservoirs shall be that corresponding to 100 percent power.

    (b) For those portions of high energy piping systems which are normally pressurized only during plant conditions other that 100 percent power, the thermodynamic state and associated operating conditions shall be determined using the most severe mode. The most severe jet was selected for the purposes of calculating fluid reaction forces and plant environmental conditions.

22. Vent and drain lines (1 1/2 inches and below) branching off high energy lines are also considered high energy lines up to and including the closed valve.

23. Unit 1 lines listed; Unit 2 lines similar unless otherwise noted.

24. This high energy line is located is safety related area; however, no detailed seismic design was performed. All essential equipment required for mitigation are seismically designed and/or supported.

25. Unit 2 only - Line eliminated on Unit 1 with RSG installation.

TABLE 3.6B-1
HIGH ENERGY LINE LIST (See Notes 22 and 23)

(Sheet 32 of 32)

LINE NUMBER	BUILDING	CONDITION OF JET (SEE NOTE 21) SUB COOLED	CONDITION OF JET (SEE NOTE 21) 2 PHASE FLOW	REMARKS	FLOW DIAGRAM FIGURE NO.

26. Preheater Bypass Line Unit 2 only. For Unit 1, line partially removed from connection to 18” main feedwater line to connection with 4” auxiliary feedwater line. Remaining portion of line is for auxiliary feedwater flow only.

TABLE 3.6B-2
POSTULATED BREAK LOCATIONS FOR THE LOCA ANALYSIS OF THE PRIMARY COOLANT LOOP[a]

Location of Postulated Rupture [b]	Type	Break Opening Area[c]
1. Reactor vessel inlet nozzle	Guillotine	Effective cross-sectional flow area of the loop pipe
2. Reactor vessel outlet nozzle	Guillotine	Effective cross-sectional flow area of the loop pipe
3. Steam generator inlet nozzle	Guillotine	Cross-sectional flow area of the loop pipe
4. Steam generator outlet nozzle	Guillotine	Cross-sectional flow area of the loop pipe
5. Reactor coolant pump inlet nozzle	Guillotine	Cross-sectional flow area of the loop pipe
6. Reactor coolant pump outlet nozzle	Guillotine	Cross-sectional flow area of the loop pipe
7. 50° elbow on the intrados	Longitudinal	Cross-sectional flow area of the loop pipe
8. Loop closure weld in crossover leg	Guillotine	Cross-sectional flow area of the loop pipe
9. Residual heat removal (RHR) line/primary coolant loop connection	Guillotine (viewed from the RHR Line)	Cross-sectional flow area of the RHR line
10. Accumulator (ACC) line/primary Coolant loop connection	Guillotine (viewed from the ACC line)	Cross-sectional flow area of the ACC line
11. Pressurizer surge (PS) line/primary coolant loop connection	Guillotine (viewed from the PS line)	Cross-sectional flow area of the PS line
12. Six inch (6") safety injection (SI) line primary coolant loop connection	Guillotine (Viewed from the SI line)	Cross-Sectional flow area of the SI line
13. Four inch (4") Pressurizer Spray line (Spray Line) primary coolant loop	Guillotine (Viewed from the Spray Line)	Cross-Sectional Flow area of the Spray line

a) Rupture locations 1 through 11 are postulated for containment design, ECCS and Environmental Qualification. These breaks are not postulated for dynamic effects (Sections 3.6B.2.1.1 and 3.6B.2.1.2)

b) Refer to Figure 3.6B-9 for location of postulated breaks in reactor coolant loop.

c) Less break opening area will be used if justified by analysis, experiments, or considerations of physical restraints such as concrete walls or structural steel.

TABLE 3.6B-3
THIS TABLE HAS BEEN DELETED

TABLE 3.6B-4A
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) - MODERATOR DENSITY COEFFICIENT = 0.38 Δk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 1)
(Sheet 1 of 2)

TIME (sec)	FLOW (lbm/sec)	ENTHALPY (Btu/lbm)
0.0	0	0
0.2	2196.38	1188.307
6.0	2009.046	1191.674
12.0	1898.802	1193.597
18.0	1836.559	1194.632
24.0	1975.015	1193.251
29.6	2180.119	1189.614
30.0	2179.208	1189.639
36.0	2072.859	1191.627
42.0	2002.969	1192.88
48.0	1927.116	1194.198
54.0	1852.632	1195.434
60.0	1781.938	1196.541
66.0	1715.045	1197.543
72.0	1651.719	1198.437
78.0	1592.69	1199.241
84.0	1539.075	1199.937
90.0	1490.214	1200.549
96.0	1442.744	1201.115
102.0	1396.781	1201.633
108.0	1357.275	1202.05
114.0	1324.872	1202.372
120.0	1300.142	1202.604
126.0	1283.228	1202.756
132.0	1272.949	1202.846
138.0	1267.352	1202.894

TABLE 3.6B-4A
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) - MODERATOR DENSITY COEFFICIENT = 0.38 Δk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 1)
(Sheet 2 of 2)

TIME (sec)	FLOW (lbm/sec)	ENTHALPY (Btu/lbm)
144.0	1264.349	1202.919
150.0	1263.092	1202.93
162.0	1262.974	1202.931
174.0	1262.364	1202.936
186.0	1255.459	1203.546
198.0	1243.922	1204.636
210.0	1228.394	1206.321
218.0	1209.594	1209.435
226.0	1185.714	1212.072
234.0	1155.766	1214.435
242.0	1119.07	1216.633
250.0	1078.283	1218.95
258.0	1036.129	1221.4
266.0	993.7287	1223.977
274.0	949.1581	1226.713
282.0	900.4384	1229.79
290.0	846.1362	1233.41
298.0	785.2863	1237.589
306.0	358.9854	1266.399
314.0	165.2518	1277.871
322.0	96.78773	1281.54
330.0	86.57766	1281.399
1800.0	99.79793	1275.065
1815.0	0	0

TABLE 3.6B-4A
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) - MODERATOR DENSITY COEFFICIENT = 0.38 Δk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 1)
(Sheet 3 of 3)

Reactor Trip on Overpower N-16 Setpoint at 22.5 sec.

Feedwater Isolation at 32.2 sec.

Safety Injection on Low Pressurizer Pressure at 53.3 sec.

Low SG Water Level Setpoint reached at 71.5 sec.

Tube Bundle Uncovery at 182.2 sec.

Steam Line Isolation at 297.8 sec.

AFW Isolation assumed at 138.3 sec.

TABLE 3.6B-4B
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) MODERATOR DENSITY COEFFICIENT = 0.38 Dk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 2)
(Sheet 1 of 3)

TIME (sec)	FLOW (lbm/sec)	ENTHALPY (Btu/lbm)
0.0	0	0
0.2	2166.222	1188.911
6.0	1965.62	1192.472
12.0	1853.074	1194.388
18.0	1795.158	1195.318
24.0	2103.635	1190.959
29.6	2167.3	1189.858
30.0	2101.651	1191.124
36.0	2011.238	1192.735
42.0	1933.979	1194.085
48.0	1853.376	1195.426
54.0	1777.098	1196.618
60.0	1705.335	1197.686
66.0	1638.036	1198.628
72.0	1576.008	1199.464
78.0	1519.357	1200.193
84.0	1468.629	1200.814
90.0	1425.646	1201.31
96.0	1390.138	1201.703
102.0	1358.259	1202.038
108.0	1331.529	1202.306
114.0	1310.011	1202.512
120.0	1293.449	1202.665
126.0	1281.434	1202.773
132.0	1273.271	1202.844
138.0	1268.199	1202.887

TABLE 3.6B-4B
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) MODERATOR DENSITY COEFFICIENT = 0.38 Dk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 2)
(Sheet 2 of 3)

144.0	1265.266	1202.912	
150.0	1264.22	1202.921	
162.0	1262.706	1203.782	
174.0	1258.14	1204.527	
186.0	1240.455	1208.688	
198.0	1208.474	1212.962	
210.0	1158.5	1218.177	
218.0	1110.042	1222.49	
226.0	1043.983	1228.022	
234.0	955.1543	1235.031	
242.0	843.4539	1243.472	
250.0	543.1462	1260.558	
258.0	200.8643	1278.106	
266.0	104.4548	1283.656	
274.0	87.92005	1283.82	
282.0	92.18698	1282.319	
290.0	97.37733	1281.058	
298.0	99.62243	1280.323	
306.0	99.99.89142	1279.861	
314.0	99.76444	1279.483	
322.0	99.6874	1279.141	
330.0	99.67809	1278.837	
1800.0	99.80119	1268.596	
1815.0	0	0	

TABLE 3.6B-4B
MASS/ENERGY RELEASES 90% POWER (INCLUDING POWER UNCERTAINTY) MODERATOR DENSITY COEFFICIENT = 0.38 Dk/gm/cc FOR 1.0 FT^2 MAIN STEAM LINE BREAK (UNIT 2)
(Sheet 3 of 3)

Reactor Trip on High Neutron Flux Setpoint at 19.8 sec.

Feedwater Isolation at 31.1 sec.

Safety Injection on Low Pressurizer Pressure at 55.4 sec.

Low SG Water Level Setpoint reached at 177.6 sec.

Tube Bundle Uncovery at 152.0 sec.

Steam Line Isolation at 247.1 sec.

AFW Isolation assumed at 140.4 sec

TABLE 3.6B-5
EQUIPMENT AFFECTED BY MSLB SUPERHEAT CONDITION

(Sheet 1 of 4)

EQUIPMENT	REQUIRED OPERABILITY TIME	POSTULATED TEMPERATURE PEAK	DEMONSTRATED TEMPERATURE PEAK (3)	DEMONSTRATED OPERABILITY TIME	TIME MARGIN	NOTES
MSIV	Not required (in room with break)	<334°F	355°F	N/A	N/A	(1)
	72 hours (not in room with break)	<334°F	355°F	30 days	27 days	(2 and 5)
MSIV Bypass (MSIVBP)	Not required	<334°F	N/A	N/A	N/A	(13)
Main Steam Drain Pot Isolation Valves (MSDPIV)	Not required (in room with break)	<334°F	346°F	N/A	N/A	(1)
	30 minutes (not in room with break)	<334°F	346°F	100 days	>99 days	(4)
Feedwater Isolation Valves (FIVs)	72 hours	<334°F	390°F	30 days	27 days	(5 and 14)
Feedwater Bypass Valves (FIBVS)	30 minutes	<334°F	346°F	100 days	>99 days	(4)
Feedwater Sample Isolation Valves (FSIVs)	30 minutes	<334°F	346°F	100 days	>99 days	(4)
Turbine Driven Auxiliary Feedwater Pump Turbine Steam Supply Valve (TDAFPTSUP)	Not required (in room with break)	<334°F	345°F	N/A	N/A	(1)
	72 hours (not in room with break)	<334°F	345°F	30 days	27 days	(5)
Main Steam Power Operated Relief Valves (MSPORVs)	Not required (in room with break)	<334°F	334°F	N/A	N/A	(1)
	72 hours (not in room with break)	<334°F	334°F	>30 days	>27 days	(5)

TABLE 3.6B-5
EQUIPMENT AFFECTED BY MSLB SUPERHEAT CONDITION

(Sheet 2 of 4)

EQUIPMENT	REQUIRED OPERABILITY TIME	POSTULATED TEMPERATURE PEAK	DEMONSTRATED TEMPERATURE PEAK (3)	DEMONSTRATED OPERABILITY TIME	TIME MARGIN	NOTES
Barton Steam Line Pressure Transmitters	72 hours	<334°F	400°F	100 days	97 days	(5, 6 and 12)
Rosemount Steam Line Pressure Transmitters	Not required (in room with break)	N/A	N/A	N/A	N/A	(7)
	72 hours (not in room with break)	<334°F	350°F	100 days	97 days	(5 and 15)
Watertight Doors	N/A	N/A	N/A	N/A	N/A	(8)
HVAC Isolation Dampers	72 hours	<334°F	336°F	100 days	>97 days	(5 and 9)
Main Steam Safety Valve (MSSVs)	Not required (in room with break)	<334°F	600°F	N/A	N/A	(1)
	72 hours (not in room with break)	<334°F	600°F	100 days	97 days	(5 and 11)
Accessory Limit Switches for MSIV and MSIVBP (on intact steam generators)	72 hours	<334°F	380°F	100 days	97 days	(5 and 12)
Accessory Limit Switches for MSDPIV, TDAFPTSUP, SPORV, FIV and FIBV (on intact steam generators)	72 hours	<334°F	372°F	100 days	97 days	(5 and 12)
Accessory Limit Switches for FSIVs and FIBV	Not required	<334°F	N/A	N/A	N/A	(16)
Auxiliary Feedwater Flow Transmitters (on intact steam generators)	72 hours	<334°F	411°F	100 days	97 days	(5 and 12)

TABLE 3.6B-5
EQUIPMENT AFFECTED BY MSLB SUPERHEAT CONDITION

(Sheet 3 of 4)

EQUIPMENT	REQUIRED OPERABILITY TIME	POSTULATED TEMPERATURE PEAK	DEMONSTRATED TEMPERATURE PEAK (3)	DEMONSTRATED OPERABILITY TIME	TIME MARGIN	NOTES
Main Steam Safety Valve Position Indicator (on intact steam generators)	72 hours	<334°F	510°F	100 days	97 days	(5 and 12)
Auxiliary Feedwater Containment Isolation MOVs	Not Required	<334°F	385°F	100 days	100 days	(10)

Notes:

(1) These valves may fail in either the open or closed position without increasing the severity of the event beyond the present analysis or decreasing the ability to mitigate the event and safely shut down the unit.

(2) These valves will operate to close within 30 minutes. Beyond this time (for up to 72 hours while proceeding to cold shutdown) these valves must remain closed. Since the Class 1E solenoid valves of the MSIV operator are energized to close and are only qualified for 90 minutes, the non Class 1E solenoid on each MSIV (which prevent reopening of the valve should the Class 1E solenoids fail) are qualified for greater than 72 hours to ensure that the MSIV remains closed. Should offsite power not be available, guidance is provided in the applicable emergency operating procedure for early restoration of power to the battery charger for the batteries which power these non-1E solenoids. See the Table 17A-1 for the MSIV Train C solenoids.

(3) For no equipment where operability is required does the postulated accident profile exceed the demonstrated qualification profile.

(4) These valves operate to close within 30 minutes. Operability is not required beyond then for the valves will remain in a safe position.

(5) 72 hour operability time is based on achieving cold shutdown within 72 hours such that the operability of these items would no longer be required.

(6) The low steamline pressure signal or high negative pressure rate signal is provided by these protection channel transmitters. Thirty minute operability time is based on providing steam line isolation signal and providing required instrumentation to identify and isolate the faulted steam line. Operability after 30 minutes is based on control of cooldown rates.

7) This transmitter provides an analog signal for the operation of the Main Steam PORV associated with the break. This transmitter may fail without increasing the severity of the event beyond the present analysis or decreasing the ability to mitigate the event and safely shutdown the unit.

(8) These are mechanical devices that do not have an active function. The materials have been analyzed to show that the doors will not lose their leak tightness.

TABLE 3.6B-5
EQUIPMENT AFFECTED BY MSLB SUPERHEAT CONDITION

(Sheet 4 of 4)

EQUIPMENT	REQUIRED OPERABILITY TIME	POSTULATED TEMPERATURE PEAK	DEMONSTRATED TEMPERATURE PEAK (3)	DEMONSTRATED OPERABILITY TIME	TIME MARGIN	NOTES

(9) The dampers are mechanical devices. They are mechanically closed by the pressure buildup from the break and are latched in the closed position (as well as being held closed by compartment pressure). Failure of the environmentally sensitive components due to this event will not prevent closure of the damper nor cause the damper to reopen. Therefore, the required operability is early in the event, but the dampers remain continuously operable in spite of the postulated environment. The data shown is for damper gasket material which ensures damper leak tightness.

(10) This equipment is not required to isolate auxiliary feedwater supply to the faulted steam generator. Isolation is achieved by valves not exposed to the high energy environment.

(11) The Main Steam safety valve is required to operate to maintain hot standby conditions. No Class 1E cables/components are associated with the Main Steam safety valve's function.

(12) This equipment is required to indicate the status of valve positions or process variables for accident monitoring.

(13) These valves are locked closed manual valves and no failure could cause them to open.

(14) These valves will isolate to close within 30 minutes. Beyond this time (for up to 72 hours while proceeding to cold shutdown) these valves must remain fully closed. Since the Class 1E solenoid valves of the FIV are energized to close and are only qualified for 80 minutes, the valve is maintained closed by automatic de-energization of the electric pump which provides hydraulic fluid for opening.

(15) These transmitters provide analog signals for the control operations of the main steam PORV's. Although they are not required to function, they must not fail in a manner which would open the PORVs.

(16) Valve Position Indication of these normally closed valves is not essential for cracks in break exclusion piping.

TABLE 3.6B-6
MASS AND ENERGY RELEASE RATES 1 FT^2 BREAK - FEED WATER LINE
(Unit 1 and Unit 2 with original steam generators)

TIME (SEC.)	FLOW (LB/SEC.)	ENTHALPY FLOW (BTU/SEC)
0.0	6200	2.64×10^6
8.281	5100	2.437×10^6
108.6	0	0

Note: For Unit 1 with replacement steam generators, mass and energy release data were generated for each loop utilizing RELAP5/MOD3.

TABLE 3.6B-7
THIS TABLE HAS BEEN DELETED

TABLE 3.6B-8
THIS TABLE HAS BEEN DELETED

3.7N SEISMIC DESIGN

In addition to the steady state loads imposed on the system under normal operating conditions, the design of equipment and equipment supports requires that conditions also be given to abnormal loading conditions such as earthquakes. Seismic loadings are considered for earthquakes of two magnitudes: Safe Shutdown Earthquake (SSE) and Operating Basis Earthquake (OBE). The SSE is defined as the maximum vibratory ground motion at the plant site that can reasonably be predicted from geologic and seismic evidence. The OBE is that earthquake which, considering the local geology and seismology, can be reasonably expected to occur during the plant life.

For the OBE loading condition, the Nuclear Steam Supply System is designed to be capable of continued safe operation. The design for the SSE is intended to assure:

1. That the integrity of the reactor coolant pressure boundary is not comprised;

2. That the capability to shutdown the reactor and maintain it in a safe condition is not comprised; and

3. That the capability to prevent or mitigate the consequences of accidents which could result in potential offsite exposures comparable to the guideline exposures of 10CFR100 is not comprised.

It is necessary to ensure that required critical structures and components do not loose their capability to perform their safety function. Not all critical components have the same functional safety requirements. For example, a safety injection pump must retain its capability to function normally during the SSE. Therefore, the deformation in the pump must be restricted to appropriate limits in order to assure its ability to function. On the other hand, many components can experience significant permanent deformation without loss of function. Piping and vessels are examples of the latter where the principal requirement is that they retain their contents and allow fluid flow.

The seismic requirements for safety-related instrumentation and electrical equipment are covered in Section 3.10. The safety class definitions, classification lists, operating condition categories and the methods used for seismic qualification of mechanical equipment are given in Section 3.2.

3.7N.1 SEISMIC INPUT

3.7N.1.1 Design Response Spectra

Refer to Section 3.7B.1.1.

3.7N.1.2 Design Time History

Refer to Section 3.7B.1.2.

3.7N.1.3 Critical Damping Values

The damping values given in Table 3.7N-1 are used for the systems analysis of Westinghouse equipment and the original Westinghouse analyses. These are consistent with the damping values recommended in Regulatory Guide 1.61 except in the case of the primary coolant loop system components and large piping (excluding reactor pressure vessel internals) for which the damping values of 2 percent and 4 percent are used as established in testing programs reported in Reference [1]. As an alternative and as defined in Table 3.7N-1, piping systems analyzed by the response spectrum method may use ASME Code Case N-411 damping values. Qualification analyses of the primary coolant loop equipment (steam generators; reactor pressure vessel and reactor coolant pumps) incorporate equipment damping values as defined in Table 3.7N-1. ASME Code Case N-411 damping values are used for the reconciliation analyses of all NSSS scope piping stress analysis packages, including the Primary Coolant Loop piping for Unit 2. The time history method of seismic analysis and composite modal damping are used as discussed in Section 3.9N.1.4.3 for the reactor coolant loops for Unit 1. ASME Code Case N-397 is not used for NSSS scope piping analysis but any subsequent usage will be detailed in the FSAR and in compliance with reference [8].

The damping values for control rod drive mechanisms (CRDM's) and the fuel assemblies of the Nuclear Steam Supply System, when used in seismic system analysis, are in conformance with the values for welded and/or bolted steel structures (as appropriate) listed in Regulatory Guide 1.61.

Tests on fuel assembly bundles justified conservative component damping values of 7 percent for OBE and 10 percent for SSE to be used in the fuel assembly component qualification. Documentation of the fuel assembly tests is found in Reference [2].

The damping values used in component analysis of CRDM's and their seismic supports were developed by testing programs performed by Westinghouse. These tests were performed during the design of the CRDM support; the support was designed so that the damping in Table 3.7N-1 could be conservatively used in the seismic analysis. The CRDM support system is designed with plates at the top of the mechanism and gaps between mechanisms. These are encircled by a box section frame which is attached by tie rods to the refueling cavity wall. The test conducted was on a full size CRDM complete with rod position indicator coils, attachment to a simulated vessel head, and variable gap between the top of the pressure housing support plate and a rigid bumper representing the support. The internal pressure of the CRDM was 2250 pounds per square inch (psi) and the temperature on the outside of the pressure housing was 400°F.

The program consisted of transient vibration tests in which the CRDM was deflected a specified initial amount and suddenly released. A logarithmic decrement analysis of the decaying transient provides the effective damping of the assembly. The effect on damping of variations in the drive shaft axial position, upper seismic support clearance and initial deflection amplitude was investigated.

The upper support clearance had the largest effect on the CRDM damping with the damping increasing with increasing clearance. With an upper clearance of 0.06 inches, the measured damping was approximately 8 percent. The clearances in a typical upper seismic CRDM support is a minimum of 0.10 inches. The increasing damping with increasing clearances trend from the test results indicated that the damping would be greater than 8 percent for both the OBE and the SSE based on a comparison between typical deflections during these seismic events to the initial

deflections of the mechanisms in the test. Component damping values of 5 percent are therefore conservative for both the OBE and the SSE. The CRDM component damping applies to both Units 1 and 2 since the upper support clearances are the same and the L106A and L106C models are structurally similar.

These damping values are used and applied to CRDM component analyses by response spectra and time history techniques.

3.7N.1.4 Supporting Media for Seismic Category I Structures

Refer to Section 3.7B.1.4.

3.7N.2 SEISMIC SYSTEM ANALYSIS

This section describes the methods of seismic analysis performed for safety-related components and systems within Westinghouse's scope.

3.7N.2.1 Seismic Analysis Methods

Those components and systems that must remain functional in the event of the SSE (Seismic Category I) are identified by applying the criteria of Section 3.2.1. This equipment is classified into three types according to its dynamic characteristics. The analysis methods used for this equipment also depended on these classifications.

The first type is flexible equipment. This equipment is characterized by several modes in the frequency range that could produce amplification of the base input motion. In addition, more than one mode could be subjected to the peak response. Because of these reasons, dynamic analyses were performed for these components using modal analysis techniques with either the response spectrum method or integration of the uncoupled modal equations of motion, or by direct integration of the coupled differential equation of motion. Details of the methods used for these analyses are described in Section 3.7N.2.1.1 through 3.7N.2.1.5.

The second classification is rigid equipment. This equipment has a fundamental natural frequency that is sufficiently high (greater than 33 Hz) so that it is not excited by input motion at the base. Such equipment is particularly suitable for static analysis as described in Section 3.7N.2.1.6.

Finally, the third type of equipment is classified as limited flexible, with only one predominate mode in the frequency range subject to possible amplification of the input motion. The fundamental mode of this type of equipment is basically a translational bending mode at a frequency less than 33 Hz. The second mode is usually a rocking mode with a frequency greater than 33 Hz. Because of the simple response characteristics of the equipment, dynamic analysis techniques that account for multiple mode effects and closely spaced modes are not required. Therefore, this equipment was evaluated using static analysis methods as described in Section 3.7N.2.1.6.

3.7N.2.1.1 Dynamic Analysis - Mathematical Model

The first step in any dynamic analysis is to model the structure or component, i.e., convert the real structure or component into a system of masses, springs, and dashpots suitable for

mathematical analysis. The essence of this step is to select a model so that the displacements obtained will be a good representation of the motion of the structure or component. Stated differently, the true inertia forces should not be altered so as to appreciably affect the internal stresses in the structure of component. Some typical modeling techniques are presented in Reference [3].

Equations of Motion

Consider the multi-degree of freedom system shown in Figure 3.7N-1. Making a force balance on each mass point r, the equations of motion can be written in the form:

$$m_r \ddot{y}_r + \sum^{i} c_{ri} \dot{u}_i + \sum^{i} k_{ri} u_i = 0 \qquad (3.7N\text{-}1)$$

where

m_r = the value of the mass or mass moment of rotational inertia at mass point r

$\ddot{y}_r$ = absolute translational or angular acceleration of mass point r

c_{ri} = damping coefficient - external force or moment required at mass point r to produce a unit translational or angular velocity at mass point i, maintaining zero translational or angular velocity at all other mass points. Force or moment is positive in the direction of positive translational or angular velocity

$\dot{u}_i$ = translational or angular velocity of mass point i relative to the base

k_{ri} = stiffness coefficient - the external force (moment) required at mass point r to produce a unit deflection (rotation) at mass point i, maintaining zero displacement (rotation) at all other mass points

Force (moment) is positive in the direction of the displacement (rotation)

u_i = displacement (rotation) of mass point i relative to the base

As an example, note that Figure 3.7N-1 does not attempt to show all of the springs (and none of the dashpots) which are presented in Equation (3.7N-1).

Since:

$$\ddot{y}_r = \ddot{u}_r + \ddot{y}_s \qquad (3.7N\text{-}2)$$

where

$\ddot{y}_s$ = absolute translational (angular) acceleration of the base

$\ddot{u}_r$ = translational (angular) acceleration of mass point r relative to the base

Equation (3.7N-1) can be written as:

$$m_r\ddot{u}_r + \sum^{i} c_{ri}\dot{u}_i + \sum^{i} k_{ri}u_i = -m_r\ddot{y}_s \qquad (3.7N\text{-}3)$$

For a single degree of freedom system with displacement u, mass m, damping c, and stiffness k, the corresponding equation of motion is:

$$m\ddot{u} + c\dot{u} + ku = -m\ddot{y}_s \qquad (3.7N\text{-}4)$$

3.7N.2.1.2 Modal Analysis

Natural Frequencies and Mode Shapes

The first step in the modal analysis method is to establish the normal modes, which were determined by eigen solution of Equation (3.7N-3). The right hand side and the damping term are set equal to zero for this purpose as illustrated in Reference [4] (pages 83 through 111). Thus, Equation (3.7N-3) becomes:

$$m_r\ddot{u}_r + \sum^{i} k_{ri}u_i = 0 \qquad (3.7N\text{-}5)$$

The equation given for each mass point r in Equation (3.7N-5) can be written as a system of equations in matrix form as:

$$[M]\{\ddot{\Delta}\} + [K]\{\Delta\} = 0 \qquad (3.7N\text{-}6)$$

where

[M] = mass and rotational inertia matrix

{Δ} = column matrix of the general displacement and rotation at each mass point relative to the base

[K] = square stiffness matrix

$\{\ddot{\Delta}\}$ = column matrix of general translational and angular accelerations at each mass point relative to the base, $d^2\{\Delta\}/dt^2$

Harmonic motion is assumed and the $\{\Delta\}$ is expressed as:

$$\{\Delta\} = \{\delta\}\sin_{\omega}t \qquad (3.7N\text{-}7)$$

where

$\{\delta\}$ = column matrix of the spatial displacement and rotation at each mass point relative to the base

ω = natural frequency of harmonic motion in radians per second

The displacement function and its second derivative are substituted into Equation (3.7N-6) and yield:

$$[K]\{\delta\} = \omega^2[M]\{\delta\} \qquad (3.7N\text{-}8)$$

The determinant [K] - ω^2 [M] is set equal to zero and is then solved for the natural frequencies. The associated mode shapes are then obtained from Equation (3.7N-8). This yields n natural frequencies and mode shapes where n equals the number of dynamic degrees of freedom of the system. The mode shapes are all orthogonal to each other and are sometimes referred to as normal mode vibrations. For a single degree of freedom system, the stiffness matrix and mass matrix are single terms and the determinant $[K]^{-2}$ [M] when set equal to zero yields simply:

$$k - \omega^2 m = 0$$

or:

$$\omega = \sqrt{\frac{k}{m}} \qquad (3.7N\text{-}9)$$

where ω is the natural angular frequency in radians per second.

The natural frequency in cycles per second is therefore:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \qquad (3.7N\text{-}10)$$

To find the mode shapes, the natural frequency corresponding to a particular mode, ω_n, can be substituted in Equation (3.7N-8).

Modal Equations

The response of a structure or component is always some combination of its normal modes. Good accuracy can usually be obtained by using only the first few modes of vibration. In the normal mode method, the mode shapes are used as principal coordinates to reduce the equations of motion to a set of uncoupled differential equations that describe the motion of each mode n. These equations may be written as (Reference [4], pages 116 through 125):

$$\ddot{A}_n + 2_{\omega n} p_n \dot{A}_n = -\Gamma n \ddot{y}_s \qquad (3.7N\text{-}11)$$

where the modal displacement or rotation, A_n, is related to the displacement or rotation of mass point r in mode n, u_{rn}, by the equation:

$$u_{rn} = A_n \phi_{rn} \qquad (3.7N\text{-}12)$$

where

ϕ_{rn} = the modal displacement for mode n at mass point r

ω_n = natural frequency of mode n in radians per second

p_n = critical damping ratio of mode n

Γn = modal participation factor of mode n given by:

$$\Gamma n = \frac{\sum^{n} m_r \varphi'_{rn}}{\sum^{n} m^2_{r\varphi} rn} \qquad (3.7N\text{-}13)$$

where

φ'_{rn} = value of φ_{rn} in the direction of the earthquake

The essence of the modal analysis lies in the fact that Equation (3.7N-11) is analogous to the equation of motion for a single degree of freedom system that will be developed from Equation (3.7N-4). Dividing Equation (3.7N-4) by m gives:

$$\ddot{u} + \frac{c}{m}\dot{u} + \frac{k}{m}u = -\ddot{y}_s \qquad (3.7N\text{-}14)$$

The critical damping ratio of a single degree of freedom system, p, is defined by the equation:

$$p \equiv \frac{c}{c_c} \qquad (3.7N\text{-}15)$$

where the critical damping coefficient is given by the expression:

$$c_c = 2m\omega \qquad (3.7N\text{-}16)$$

Substituting Equation (3.7N-16) into Equation (3.7N-15) and solving for c/m gives:

$$\frac{c}{m} = 2\omega\rho \qquad (3.7N\text{-}17)$$

Substituting this expression and the expression for k/m given by Equation (3.7N-9) into Equation (3.7N-14) gives:

$$\ddot{u} + 2\omega\rho\dot{u} + \omega^2 u = -\ddot{y}_s \qquad (3.7N\text{-}18)$$

Note the similarity of Equations (3.7N-11) and (3.7N-18). Thus each mode may be analyzed as though it were a single degree of freedom system and all modes are independent of each other. By this method a fraction of critical damping, i.e., c/c_c, may be assigned to each mode and it is not necessary to identify or evaluate individual damping coefficients, i.e., c. However, assigning only a single damping ratio to each mode has a drawback. Normally, there are two ways used to overcome this limitation when considering a slightly damped structure (e.g., steel supported by a massive moderately damped structure (e.g., concrete)).

The first method is to develop and analyze separate mathematical models for both structures using their respective damping values. The massive moderately damped support structure is analyzed first. The calculated response at the support points for the slightly damped structures are used as forcing functions for their subsequent detailed analysis. The second method is to inspect the mode shapes to determine which modes correspond to the slightly damped structure and then use the damping associated with the structure having predominant motion.

3.7N.2.1.3 Response Spectrum Analysis

The response spectrum is a plot showing the variation in the maximum response (Reference [5], pages 24 through 51) (displacement, velocity, and acceleration) of a single degree of freedom system versus its natural frequency of vibration when subjected to a time history motion of its base. Examples of response spectra are shown in Figures 3.7N-2 and 3.7N-3.

The response spectrum concept can be best explained by outlining the steps involved in developing a spectrum curve. Determination of a single point on the curve requires that the response (displacement, velocity, and acceleration) of a single degree of freedom system with a given damping and natural frequency is calculated for a given base motion. The variations in

response are established and the maximum absolute value of each is plotted as an ordinate with the natural frequency used as the abscissa. The process is repeated for other assumed values of frequency is sufficient detail to establish the complete curve. Other curves corresponding to different fractions of critical damping are obtained in a similar fashion. Thus, the determination of each point of the curve requires a complete dynamic response analysis, and the determination of a complete spectrum may involve hundreds of such analyses. However, once a response spectrum plot is generated for the particular base motion, it may be used to analyze each structure and component with that base motion. The spectral acceleration, velocity, and displacement are related by the equation:

$$S_{a_n} = \omega_n S_{v_n} = \omega_n^2 S_{d_n} \qquad (3.7N\text{-}19)$$

There are two types of response spectra that must be considered. If a given building is shown to be rigid and to have a hard foundation, the ground response spectrum or ground time history is used. It is referred to as a ground response spectrum. If the building is flexible and/or has a soft foundation, the ground response spectrum is modified to include these effects. The response spectrum at various support points must be developed. These are called floor response spectra. The specific response spectra used are discussed in Sections 3.7B.1 and 3.7B.2.5.

3.7N.2.1.4 Integration of Modal Equations

This method can be separated into the following two basic parts:

1. Integration procedure for the uncoupled modal Equation (3.7N-11) to obtain the modal displacements and accelerations as a function of time.

2. Using these modal displacements and accelerations to obtain the total displacements, accelerations, forces, and stresses.

Integration Procedure

Integration of these uncoupled modal equations is done by step-by-step numerical integration. The step-by-step numerical integration procedure consists of selecting a suitable time interval, Δt, and calculating modal acceleration, $\ddot{A}_n$, modal velocity, $\dot{A}_n$, and modal displacement, A_n, at discrete time stations Δt apart, starting at t = 0 and continuing through the range of interest for a given time history of base acceleration.

Total Displacements, Accelerations, Forces and Stresses

From the modal displacements and accelerations, the total displacements, accelerations, forces and stresses can be determined as follows:

1. Displacement of mass point r in mode n as a function of time is given by Equation (3.7N-12) as:

$$u_{rn} = An\varphi rn \qquad (3.7N\text{-}20)$$

with the corresponding acceleration of mass point r in mode n as:

$$\ddot{u}_{rn} = \dot{A}_n \varphi_{rn} \qquad (3.7N\text{-}21)$$

2. The displacement and acceleration values obtained for the various modes are superimposed algebraically to give the total displacement and acceleration at each time interval.

3. The total acceleration at each time interval is multiplied by the mass to give an equivalent static force. Stresses are calculated by applying these forces to the model or from the deflections at each time interval.

3.7N.2.1.5 Integration of Coupled Equations of Motion

The dynamic transient analysis is a time history solution of the response of a given structure to know forces and/or displacement forcing functions. The structure may include linear or nonlinear elements, gaps, interfaces, plastic elements, and viscous and Coulomb dampers. Nodal displacements, nodal forces, pressure, and/or temperatures may be considered as forcing functions. Nodal displacements and elemental stresses for the complete structure are calculated as functions of time.

The basic equations for the dynamic analysis are as follows:

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{F(t)\} \qquad (3.7N\text{-}22)$$

where the terms are as defined earlier and {F(t)} may include the effects of applied displacements, forces, pressures, temperatures, or nonlinear effects such as plasticity and dynamic elements with gaps. Options of translational accelerations input to a structural system and the inclusion of static deformation and/or preload may be considered in the nonlinear dynamic transient analysis. The option of translational input such as uniform base motion to a structural system is considered by introducing an inertia force term of $-[M]\{\ddot{z}\}$ to the right hand side of the basic Equation (3.7N-22), i.e.,

$$[M]\{\ddot{x}\} + [C]\{\dot{x}\} + [K]\{x\} = \{f\} - [M]\{\ddot{z}\} \qquad (3.7N\text{-}23)$$

The vector $\{\ddot{z}\}$ is defined by its components $\ddot{z}_i$ where i refers to each degree of freedom of the system. $\ddot{z}_i$ is equal to a1, a2, or a3 if the i-th degree of freedom is aligned with the direction of the system translational acceleration a1, a2, or a3, respectively. $\ddot{z}_i$ = 0 if the i-th degree of freedom is not aligned with any direction of the system translational acceleration. Typical application of this option is a structural system subjected to a seismic excitation of a given ground acceleration record. The displacement {x} obtained from the solution of Equation (3.7N-23) is the displacement relative to the ground.

The option of the inclusion of initial static deformation or preload in a nonlinear transient dynamic structural analysis is considered by solving the static problem prior to the dynamic analysis. At each stage of integration in transient analysis, the portion of internal forces due to static deformation is always balanced by the portion of the forces which are statically applied. Hence, only the portion of the forces which deviate from the static loads will produce dynamic effects. The output of this analysis is the total result due to static and dynamic applied loads.

One available method for the numerical integration of Equations (3.7N- 22) and (3.7N-23) is the Newmark Beta integration scheme preposed by Chan, Cox, and Benfield (Reference [6]). In this integration scheme, Equations (3.7N-22) and (3.7N-23) are replaced by:

$$\begin{aligned}\frac{1}{(\Delta t)^2}[M]\{x_{n+2}-2x_{n+1}+x_n\}+\frac{1}{2(\Delta t)}\{x_{n+2}-x_n\}[C]\\ +[K]\{\beta x_{n+2}+(1-2\beta)x_{n+1}+\beta x_n\}\\ =\{\beta F_{n+2}+(1-2\beta)F_{n+1}+\beta F_n\}\end{aligned} \qquad \text{(3.7N-24)}$$

where

n, n+1, n+2 = past, present, and future (updated) values of the variables

β = parameter to be selected on the basis of numerical stability and accuracy

F = the total right hand side of the equation of motion (Equation (3.7N-22) or (3.7N-23))

$\Delta t = t_{n+2} - t_{n+1} = t_{n+1} - t_n$

The value of β is chosen equal to 1/3 in order to provide a margin of numerical stability for nonlinear problems. Since the numerical stability of Equation (3.7N-24) is mostly determined by the left hand side terms of that equation, the right hand side terms were replaced by F_{n+2}. Furthermore, since the time increment may vary between two successive time substeps, Equation (3.7N-24) may be modified as follows:

$$\begin{aligned}\frac{2}{(\Delta t+\Delta t_1)}[M]\left\{\frac{x_{n+2}-x_{n+1}}{\Delta t}-\frac{x_{n+1}-x_n}{\Delta t_1}\right\}+\frac{1}{(\Delta t+\Delta t_1)}[C]\{x_{n+2}-x_n\}\\ +\frac{1}{3}[K]\{x_{n+2}+x_{n+1}+x_n\}=\{F_{n+2}\}\end{aligned} \qquad \text{(3.7N-25)}$$

By factoring x_{n+2}, x_{n+1}, and x, and rearranging terms, Equation (3.7N-26) is obtained as follows:

$$\begin{aligned}\{C_5[M]+C_3[C]+(1/3)[K]\}\{x_{n+2}\}=\{F_{n+2}\}\\ +\{C_7[M]-(1/3)[K]\}\{x_{n+1}\}\\ +\{-C_2[M]+C_3[C]-(1/3)K\}\{x_n\}\end{aligned} \qquad \text{(3.7N-26)}$$

Where

$$C_2 = \frac{2}{\Delta t_1(\Delta t + \Delta t_1)}$$

$$C_3 = \frac{1}{\Delta t + \Delta t_1}$$

$$C_5 = \frac{2}{\Delta t_1(\Delta t + \Delta t_1)}$$

$$C_7 = C_2 + C_5$$

The above set of simultaneous linear equations is solved to obtain the present values of nodal displacements x_t in terms of the previous (known) values of the nodal displacements. Since [M], [C], and [K] are included in the equation, they can also be time or displacement dependent.

3.7N.2.1.6 Static Analysis - Rigid and Limited Flexible Equipment

Rigid equipment and limited flexible equipment as defined in Section 3.7N.2.1 are generally analyzed using the static analysis method. This technique involves the multiplication of the total weight of the equipment or component member by a specified seismic acceleration coefficient. The magnitude of the seismic acceleration coefficient was established on the basis of the excitation level that the component was expected to experience in the plant.

For rigid equipment, the seismic acceleration coefficients were compared with the high frequency (greater than 33 Hz) acceleration levels for the applicable response spectra developed for the plant to confirm the design analysis. The seismic acceleration coefficients for limited flexible equipment are compared with the acceleration levels from the applicable response spectra at the calculated fundamental natural frequency of the component. If the design seismic acceleration coefficients for either rigid or limited flexible equipment are exceeded by the actual plant acceleration levels, the design analyses is performed again at the actual level to confirm the equipment adequacy.

3.7N.2.2 Natural Frequencies and Response Loads

Refer to Section 3.7B.2.2.

3.7N.2.3 Procedures Used for Modeling

Procedures used for modeling are discussed in Section 3.7N.2.1.1.

The mathematical model used for the dynamic analyses of the Reactor Coolant System is shown in Figure 3.9N-1. Figure 3.9N-2 shows the mathematical model used for the reactor pressure vessel.

3.7N.2.4 Soil/Structure Interaction

Refer to Section 3.7B.2.4.

3.7N.2.5 Development of Floor Response Spectra

Refer to Section 3.7B.2.5.

3.7N.2.6 Three Components of Earthquake Motion

The seismic design of the piping and equipment includes the effect of the seismic response of the supports, equipment, structures and components. The system and equipment response is determined using three earthquake components, two horizontal and one vertical. The design ground response spectra, specified in Section 3.7B.1, are the bases for generating these three input components. Floor response spectra are generated for two perpendicular horizontal directions, (i.e., N-S, E-W) and the vertical direction. System and equipment analysis is performed with these input components applied in the N-S, E-W and vertical directions. The damping values used in the analysis are those given in Table 3.7N-1.

In computing the system and equipment response by response spectrum modal analysis the methods of Section 3.7N.2.7 are used to combine all significant modal responses to obtain the combined unidirectional responses.

The combined total response is then calculated using the square root of the sum of the squares formula applied to the resultant unidirectional responses. For instance, for each item of interest such as displacement, force, stresses, etc., the total response is obtained by applying the above described method. The mathematical expression for this method (with R as the item of interest) is:

$$R_C = \left[\sum_{T=1}^{3} R_T^2 \right]^{1/2} \qquad (3.7N\text{-}27)$$

where

$$R_T = \left[\sum_{i=1}^{N} R_{Ti}^2 \right]^{1/2} \qquad (3.7N\text{-}28)$$

where

R_C = total combined response at a point

R_T = value of combined response of direction T

R_{Ti} = absolute value of response for direction T, mode i

N = total number of modes considered

The subscripts can be reversed without changing the results of the combination.

Again, for the case of closely spaced modes, R_T in Equation (3.7N-28) shall be replaced with R_T as given by Equation (3.7N-29) in Section 3.7N.2.7.

3.7N.2.7 Combination of Modal Responses

The total unidirectional seismic response is obtained by combining the individual modal responses utilizing the square root of the sum of the squares method. For systems having modes with closely spaced frequencies, this method is modified to include the possible effect of these modes. The groups of closely spaced modes are chosen such that the difference between the frequencies of the first mode and the last mode in the group does not exceed 10 percent of the lower frequency. Combined total response for systems which have such closely spaced modal frequencies is obtained by adding to the square root of the sum of the squares of all modes the product of the responses of the modes in each group of closely spaced modes and a coupling factor. This can be represented mathematically as:

$$R_T^2 = \sum_{i=1}^{N} R_i^2 + 2 \sum_{j=1}^{S} \sum_{K=M_j}^{N_j - 1} \sum_{\ell = K+1}^{N_j} R_K R_\ell \varepsilon_{K\ell} \qquad \text{(3.7N-29)}$$

where

R_T = total unidirectional response

R_i = absolute value of response of mode i

N = total number of modes considered

S = number of groups of closely spaced modes

M_j = lowest modal number associated with group j of closely spaced modes

N_j = highest modal number associated with group j of closely spaced modes

$\varepsilon K\ell$ = coupled factor with

$$\varepsilon K\ell = \left\{ 1 + \left(\frac{\omega_K^i - \omega_\ell^i}{\beta_K^i \omega_K + \beta_\ell^i \omega_\ell} \right)^2 \right\}^{-1} \qquad \text{(3.7N-30)}$$

and

$$\omega_{K'} = \omega_K [1 - (\beta K')^2]^{1/2} \qquad \text{(3.7N-31)}$$

$$\beta' K = \beta K + \frac{2}{\omega_K t_d} \qquad \text{(3.7N-32)}$$

where

ω_K	=	frequency of closely spaced mode K
βK	=	fraction of critical damping in closely spaced mode K
t_d	=	duration of the earthquake

An example of this equation applied to a system can be supplied with the following considerations. Assume that the predominant contributing modes have frequencies as given below:

Mode	1	2	3	4	5	6	7	8
Frequency	5.0	8.0	8.3	8.6	11.0	15.5	16.0	20

There are two groups of closely spaced modes, namely with modes {2, 3, 4} and {6, 7}. Therefore:

S	=	2 number of groups of closely spaced modes
M_1	=	2 lowest modal number associated with group 1
N_1	=	4 highest modal number associated with group 1
M_2	=	6 lowest modal number associated with group 2
N_2	=	7 highest modal number associated with group 2
N	=	8 total number of modes considered

The total response for this system is, as derived form the expansion of Equation (3.7N-29):

$$R_T^2 = [R_1^2 + R_2^2 + R_3^2 + \ldots + R_8^2] + 2\,R_2\,R_3\,\varepsilon_{23} + 2\,R_2\,R_4\,\varepsilon_{24} + 2\,R_3\,R_4\,\varepsilon_{34} + 2\,R_6\,R_7\,\varepsilon_{67} \qquad \text{(3.7N-33)}$$

3.7N.2.8 Interaction of Non-Category I Structures with Seismic Category I Structures

Refer to Section 3.7B.2.8.

3.7N.2.9 Effects of Parameter Variations on Floor Response Spectra

Refer to Section 3.7B.2.9.

3.7N.2.10 Use of Constant Vertical Static Factors

Constant vertical static factors are not used as the vertical floor response load for the seismic design of safety classed systems and components within Westinghouse's scope of responsibility. All such systems and components are analyzed in the vertical direction.

3.7N.2.11 Methods Used to Account for Torsional Effects

Refer to Section 3.7B.2.11.

3.7N.2.12 Comparison of Responses

Refer to Section 3.7B.2.12.

3.7N.2.13 Methods for Seismic Analysis of Dams

Refer to Section 3.7B.2.13.

3.7N.2.14 Determination of Seismic Category I Structure Overturning Moments

Refer to Section 3.7B.2.14.

3.7N.2.15 Analysis Procedure for Damping

In instances under the standard scope of Westinghouse supply and analysis, either the lowest damping value associated with the elements of the system are used for all modes, or an equivalent modal damping value is determined by testing programs such as was done for the reactor coolant loop (Reference [1]).

3.7N.3 SEISMIC SUBSYSTEM ANALYSIS

This section describes the seismic analysis performed on subsystems within Westinghouse's scope of responsibility.

3.7N.3.1 Seismic Analysis Methods

Seismic analysis methods for subsystems within Westinghouse's scope of responsibility are given in Section 3.7N.2.1.

3.7N.3.2 Determination of Number of Earthquake Cycles

For each OBE the system and component will have a maximum response corresponding to the maximum induced stresses. The effect of these maximum stresses for the total number of OBE's must be evaluated to assure resistance to cyclic loading.

The OBE is conservatively assumed to occur 20 times over the life of the plant. The number of maximum stress cycles for each occurrence depends on the system and component damping values, complexity of the system and component, duration and frequency contents of the input earthquake. A precise determination of the number of maximum stress cycles can only be made using time history analysis for each item which is not feasible. Instead, a time history study has

been conducted to arrive at a realistic number of maximum stress cycles for all Westinghouse systems and components.

To determine the conservative equivalent number of cycles of maximum stress associated with each occurrence, an evaluation was performed considering both equipment and its supporting building structure as single degree of freedom systems. The natural frequencies of the building and the equipment are conservatively chosen to coincide. The damping in the equipment and building are equivalent to the damping values in Table 3.7N-1.

The results of this study indicate that the total number of maximum stress cycles in the equipment having peak acceleration above 90 percent of the maximum absolute acceleration did not exceed eight cycles.

If the equipment was assumed to be rigid in a flexible building, the number of cycles exceeding 90 percent of the maximum stress was not greater than three cycles.

This study was conservative since it was performed with single degree of freedom models which tend to produce a more uniform and unattenuated response than a complex interacted system. The conclusions indicate that 10 maximum stress cycles for flexible equipment (natural frequencies less than 33 Hz) and 5 maximum stress cycles for rigid equipment (natural frequencies greater than 33 Hz) for each of 20 OBE occurrences should be used for fatigue evaluation of Westinghouse systems and components.

3.7N.3.3 Procedures Used for Modeling

Refer to Section 3.7N.2.3 for modeling procedures for subsystems in Westinghouse's scope of responsibility.

3.7N.3.4 Basis for Selection of Frequencies

The analysis of equipment subjected to seismic loading involves several basic steps, the first of which is the establishment of the intensity of the seismic loading. Considering that the seismic input originates at the point of support, the response of the equipment and its associated supports based upon the mass and stiffness characteristics of the system, will determine the seismic accelerations which the equipment must withstand.

Three ranges of equipment/support behavior which affect the magnitude of the seismic acceleration are possible:

1. If the equipment is rigid relative to the structure, the maximum acceleration of the equipment mass approaches that of the structure at the point of equipment support. The equipment acceleration value in this case corresponds to the low period region of the floor response spectra.

2. If the equipment is very flexible relative to the structure, the internal distortion of the structure is unimportant and the equipment behaves as though supported on the ground.

3. If the periods of the equipment and supporting structure are nearly equal, resonance occurs and must be taken into account.

Also, as noted in Section 3.7N.3.2, rigid equipment/support systems have natural frequencies greater than 33 Hz.

3.7N.3.5 Use of Equivalent Static Load Method of Analysis

The equivalent static load or static analysis method involves the multiplication of the total weight of the equipment or component member by the specified seismic acceleration coefficient. The magnitude of the seismic acceleration coefficient is established on the basis of the expected dynamic response characteristics of the component. Components which can be adequately characterized as single degree of freedom systems are considered to have a modal participation factor of one. Seismic acceleration coefficients for multi-degree of freedom system which may be in the resonance region of the amplified response spectra curves are increased by 50 percent to account conservatively for the increased modal participation.

3.7N.3.6 Three Components for Earthquake Motion

Methods used to account for three components of earthquake motion for subsystems in Westinghouse's scope of responsibility are given in Section 3.7N.2.6.

3.7N.3.7 Combination of Modal Responses

Methods used to combine modal responses for subsystems in Westinghouse's scope of responsibility are given in Section 3.7N.2.7.

3.7N.3.8 Analytical Procedures for Piping

The Class 1 piping systems are analyzed to the rules of the American Society of Mechanical Engineers (ASME) Code, Section III, NB-3650. When response spectrum methods are used to evaluate piping systems supported at different elevations, the following procedures are used. The effect of differential seismic movement of piping supports is included in the piping analysis according to the rules of the ASME Code, Section III, NB-3653. According to ASME definitions, these displacements cause secondary stresses in the piping system. The response quantity of interest induced by differential seismic motion of the support is computed statically and addresses the relative phase of the seismic motion at each support.

In the response spectrum dynamic analysis for evaluation of piping systems supported at different elevations, the envelope floor response spectrum corresponding to the support locations is used. Westinghouse does not have in their scope of analysis any piping systems interconnected between buildings.

3.7N.3.9 Multiply Supported Equipment Components with Distinct Inputs

When response spectrum methods are used to evaluate Reactor Coolant System primary components interconnected between floors, the procedures in the following paragraphs are used. There are no components in Westinghouse scope of analysis which are connected between buildings. The primary components of the Reactor Coolant System are supported at no more than two floor elevations.

A dynamic response spectrum analysis is first made assuming no relative displacement between support points. The response spectra used in this analysis is the most severe floor response spectra.

Secondly, the effect of differential seismic movement of components interconnected between floors is considered statically in the integrated system analysis and in the detailed component analysis. The results of the building analysis are reviewed on a mode-by-mode basis to determine the differential motion in each mode. Per ASME Code rules, the stress caused by differential seismic motion is clearly secondary for piping (NB-3650) and component supports (NF-3231). For components, the differential motion is evaluated as a free end displacement, since, per NB-3213.19, examples of a free end displacement are motions "that would occur because of relative thermal expansion of piping, equipment, and equipment supports, or because of rotations imposed upon the equipment by sources other than the piping." The effect of the differential motion is to impose a rotation on the component from the building. This motion, then, being a free end displacement and being similar to thermal expansion loads, will cause stresses which will be evaluated with ASME Code methods including the rules of NB-3227.5 used for stresses originating from restrained free end displacements.

The results of these two steps, the dynamic inertia analysis and the static differential motion analysis, are combined absolutely with due consideration for the ASME classification of the stresses.

3.7N.3.10 Use of Constant Vertical Static Factors

Constant vertical load factors are not used as the vertical floor response load for the seismic design of safety-related components and equipment within Westinghouse's scope of responsibility.

3.7N.3.11 Torsional Effects of Eccentric Masses

The effect of eccentric masses, such as valves and valve operators, is considered in the seismic piping analyses. These eccentric masses are modeled in the system analysis and the torsional effects caused by them are evaluated and included in the total system response. The total response must meet the limits of the criteria applicable to the safety class of the piping.

3.7N.3.12 Buried Seismic Category I Piping Systems and Tunnels

Refer to Section 3.7B.3.12.

3.7N.3.13 Interaction of Other Piping with Seismic Category I Piping

Refer to Section 3.7B.3.13.

3.7N.3.14 Seismic Analyses for Reactor Internals

The seismic analysis of the reactor internals is conducted in accordance with the guidelines specified in Regulatory Guide 1.92. The seismic analysis determines the response of the reactor internals to Operational Basis Earthquake (OBE) and Safe Shutdown Earthquake (SSE) vertical and horizontal seismic shock components.

The input response spectra used in the reactor internals analysis are based on the accelerations at the reactor vessel supports taken either from the plant specific spectra or from Westinghouse generic E-spec spectra which envelop the applicable plant specific spectra. The horizontal and vertical seismic analysis use the modal response spectrum method and the WECAN general purpose finite element program to determine the internals response. The method used to obtain the combined response of the modal spectral responses is square-root-of-the-sum-of-the squares (SRSS).

The effect of closely spaced modes is considered using the Ten Percent Method (Regulatory Guide 1.92, Paragraph 1.2.2); however, the effect has been shown to be insignificant. The maximum, or total, seismic response value of the reactor internals is obtained by taking the SRSS of the maximum values of the co-directional responses due to the three components of earthquake motion. In general, this combination is made in the Stress Analysis section of the particular structural component.

The horizontal and vertical seismic models contain 118 and 23 active dynamic degrees of freedom, respectively. Results from the modal analysis of the horizontal and vertical systems indicates, in general, 12 and 3 modes present with frequencies less than 33Hz.

In developing the seismic model of the reactor vessel and internals, a systematic approach was used to ensure that basic fundamental frequencies, i.e., both component and system frequencies, are described and inherent in the mathematical models. The approach used to verify the mathematical modeling of reactor vessel and internals was to compare and require that the system frequencies and mode shapes, from the mathematical models, be in agreement with plant test and scale model test data.

In determining the seismic response of the reactor system, due to the excitation of unidirectional shock spectrum, only those modes contributing to the first 80-90 percent of total system mass were considered in the solution.

Hydrodynamic mass effects, for horizontal and vertical directions, was included in the reactor vessel-internals system models. The numerical values for the various hydrodynamic mass effects within the reactor system were based on scale model and plant tests and applicable analytical expressions; e.g., Fritz, Fritz & Kiss, etc.

The effect of significant nonlinearities in the reactor system, i.e., gaps between reactor vessel and internals, on the seismic response is considered in the system analysis. The nonlinearities due to the gaps are included by determining an effective stiffness at the gap location. The validity of this approach has been investigated and found to be conservative for the frequency response range of the reactor internals.

The structural damping values used in the system seismic analysis are in accordance with Regulatory Guide 1.61 (i.e. 2 and 4 percent for OBE and SSE respectively.)

In addition, the stiffness of the primary piping and the stiffness of reactor vessel supports are considered in the analysis. Coupling effects between the horizontal and vertical directions are small and are not included in the analysis. Diagrams of the internals model are shown in Figures 3.7N-4 and 3.7N-5 for the horizontal and vertical analysis respectively.

3.7N.3.15 Analysis Procedure for Damping

Analysis procedures for damping for subsystems in Westinghouse's scope of responsibility are given in Section 3.7N.2.15.

3.7N.4 SEISMIC INSTRUMENTATION

Refer to Section 3.7B.4.

REFERENCES

1. "Damping values of Nuclear Plant Components," WCAP-7921-AR, May 1974.

2. Gesinski, T. L. and Chiang, D., "Safety Analysis of the 17x17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident," WCAP-8236 (Proprietary), December 1973 abd WCAP-8288 (Non-Proprietary), January 1974.

3. Lin, C. W., "How to Lump the Masses - A Guide to the Piping Seismic Analysis," ASME Paper 74-NE-7 presented at the Pressure Vessels and Piping Conference, Miami, Florida, June 1974.

4. Biggs, J. M., Introduction to Structural Dynamics, McGraw-Hill, New York, 1964.

5. Thomas, T. H., et al., "Nuclear Reactors and Earthquakes," TID- 7024, U. S. Atomic Energy Commission, Washington, D. C., August 1963.

6. Chan, S. P., Cox, H. L. and Benfield, W. A., "Transient Analysis of Forced Vibration of Complex Structural-Mechanical Systems," J. Royal Aeronautical Society, July 1962.

7. "Safety Analysis of the 8-Gird 17 x 17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident," WCAP-8236, Addendum 1 (Proprietary), March 1974 and WCAP-8288, Addendum 1 (Non-Proprietary), April 1974.

8. U.S. Nuclear Regulatory Commission Evaluation of Request for Use of ASME Code Cases N-397 and N-411, March 13, 1985.

TABLE 3.7N-1
DAMPING VALUES USED FOR SEISMIC SYSTEMS ANALYSIS FOR
WESTINGHOUSE SUPPLIED EQUIPMENT

Item	Damping (Percent of Critical)	
	Upset Conditions (OBE)	Faulted Condition (SSE, DBA)
Primary coolant loop system components and large piping[a],[b]	2	4
Small piping[b]	1	2
Welded steel structures	2	4
Bolted and/or riveted steel structures	4	7

a) Applicable to 12 inch or larger diameter piping.

b) For piping analyzed by the response spectrum method, ASME Code Case N-411 damping values may also be used in lieu ofthe damping provided in this table.

3.7B SEISMIC DESIGN

3.7B.1 SEISMIC INPUT

3.7B.1.1 Design Response Spectra

Design response spectra for both horizontal and vertical ground motion for the SSE are shown in Figures 3.7B-1 and 3.7B-7, respectively. Response spectra for 2, 5, 7, 10, and 15 percent of critical damping are provided for both the horizontal and vertical motions and are scaled to the maximum ground accelerations of 0.12g and 0.08g selected for the SSE. For the OBE, a scaling factor of 0.5 is applied to the SSE design spectra.

The response spectra are constructed on the basis of the recommendations of Newmark, Blume, and Kapur [14] and conform to the requirements of NRC Regulatory Guide 1.60, Revision 1, with the exception of the 33 Hz to 50 Hz frequency range. In that range, the vertical response spectrum of NRC Regulatory Guide 1.60, Revision 1, differs from the vertical response spectrum of Reference [14]. The effects of this deviation on the results of the analyses of structures and systems are negligible because they only affect the modes which have low amplification. Similarly, the method recommended in Reference [14] for the construction of vertical response spectra leads to a slight deviation from NRC Regulatory Guide 1.60, Revision 1 recommendations for accelerations corresponding to 3.5 Hz. The magnitudes of these differences are negligible.

The response spectra indicate the estimated response of a structure subject to significant nearby earthquake ground motion. The spectra are presented over a range of frequencies corresponding to natural frequencies of structural elements, and they represent the maximum amplitude of motion in structural elements for typical degrees of structural damping. Because the design response spectra have been developed from a large number of real records, following the procedures recommended by Newmark, the effect of strong motion duration and distance of focal depth are included [29].

There are, of course, general associations between duration of strong motion and the size of an event. Longer durations of strong motion are expected with greater-sized earthquakes. Higher frequency accelerations are attenuated with greater distance from the epicenter of the earthquake. These conditions are inherent in the strong motion records which are the source of Newmark's work. In no case are the amplification factors less than one.

3.7B.1.2 Design Time History

One horizontal and one vertical SSE artificial time history have been developed for the design response spectra requirements presented in this section and Section 3.7B.1.1.

As an alternative to a site-dependent analysis, these artificial time history records are suitable for use as base excitations for the dynamic structural analysis.

The mathematical procedures used to generate these artificial time history records can be briefly summarized as follows:

1. The spectral characteristics of the selected site SSE design response spectra are extracted to construct a frequency response function with proper phase factor modification.

2. A fast Fourier transform of the frequency response function is performed to obtain a filter impulse response function.

3. The filter impulse response function is then integrated with a set of pseudorandom numbers to obtain an artificial time history record.

4. A comparison is made between the response spectrum derived from the artificial time history and the site SSE design response spectrum. Any unacceptable deviations are corrected by adding a series of sinusoidal impulses with proper amplitude and phase angles until the desirable fit is achieved.

5. The artificial time history records meet the minimum acceptance criteria given by Table 3.7.1-1 in Section 3.7.1 of the Standard Review Plan.

The response spectra derived from the horizontal artificial time history record and the selected site SSE design response spectra are presented in Figures 3.7B-2 through 3.7B-6, for five structural damping values. The corresponding artificial time history is presented in Figure 3.7B-14. The response spectra from the vertical artificial time history record and the SSE design response spectra are presented on Figures 3.7B-8 through 3.7B-12, and the corresponding artificial time history is presented on Figure 3.7B-19.

Time history durations of approximately 10 sec have been found necessary to allow the modifications of the time histories to match response spectra values at periods of three to four sec. A 10 sec record allows two to three cycles for modification by sinusoidal impulses. A record length of 10.24 sec is obtained because the fast Fourier transform used for this purpose operates on sets of numbers which are as powers to time; i.e., 1024 is equal to two raised to the tenth power.

The artificial time history records are generated at 0.01 sec equal time intervals with a time duration of 10.24 sec. They are in the digitized form of 1024 acceleration values.

New ground motion time history accelerations were generated for the re-analysis of the Fuel Building due to the addition of high density spent fuel storage racks. Three statistically independent acceleration time histories were developed, two in the horizontal direction used as East-West and North-South ground motions and one for the Vertical direction. An iterative method was used to compute ground motion time history accelerations using the design response spectra in Figures 3.7B-3 and 3.7B-9 as target spectra. In each iteration the response spectrum was recomputed from the generated time history accelerations and compared to the target response spectra. Iterations were repeated until a satisfactory match was achieved. The resulting artificial ground acceleration time history motions are 20 seconds long and the time size is 0.005 seconds.

The response spectra resulting from the horizontal artificial time history records and the selected site SSE design response spectra are presented in Figures 3.7B-3A and 3.7B-3B for 5% damping, Figures 3.7B-4A and 3.7B-4B for 7% damping, and Figures 3.7B-5A and 3.7B-5B for 10% damping. The corresponding artificial acceleration time histories are presented in

Figures 3.7B-14A and 3.7B-14B. The response spectra from the vertical acceleration time history record and the SSE design response spectra are presented in Figures 3.7B-9A, 3.7B-10A, and 3.7B-11A for 5%, 7%, and 10% damping, respectively. The corresponding artificial acceleration time history is shown in Figure 3.7B-19A. The artificial acceleration time histories are normalized to 1.0g and 0.67g for the horizontal and vertical directions, respectively. Scaling factors of 0.12 and 0.06 were applied to represent the design maximum ground accelerations described in Section 3.7B.1.1 (i.e., 0.12 times 1.0g and 0.67g results in maximum ground accelerations of 0.12g and 0.08g horizontal and vertical, respectively, for the SSE and 0.06 times 1.0g and 0.67g results in maximum ground accelerations of 0.06g and 0.04g horizontal and vertical, respectively, for the OBE).

Three checks were made to show the validity of the new time history accelerations:

1. Statistical independence of the three artificial acceleration time histories was verified via the normalized correlation coefficients recommended by Clough and Penzien [52].

2. The power spectral density of the generated acceleration time history motions were computed and shown to meet the criteria of Appendix A of Standard Review Plan Section 3.7.1 [46].

   The computed power spectral density function was smoothed in accordance with Standard Review Plan Section 3.7.1 by averaging the computed values in the frequency band of ±20% around the frequency point.

   The computed smoothed power spectral density functions are shown in Figures 3.7B-20 and 3.7B-21 and exceed 80% of the target PSD throughout the frequency range from 0.3 Hz to 24 Hz.

3. Response spectra for each ground motion acceleration were computed at damping levels of 5, 7, and 10%. Response spectra computations were carried out using the method recommended by Nigam and Jennings [48] at 400 equally spaced intervals from 0.8 Hz to 40 Hz.

   The computed spectra envelop the target spectra in the entire frequency range from 0.8 Hz to 33 Hz with only a few minor exceptions. In very few cases the computed spectra dip below the target values. However, these points are acceptable based on the provisions of Standard Review Plan Section 3.7.1. None of the points are below the prescribed allowable of 10 percent below the design response spectra and the number of points below the design response spectra is less than the prescribed maximum of five.

3.7B.1.3 Critical Damping Values

The specific percentages of critical damping values used for Category I structures, systems, and components are based on the materials, stress levels, and type of connections of the particular structure or component. They are determined in accordance with the recommendations of NRC Regulatory Guide 1.61 and Reference [14]. For piping systems analyzed by the response spectrum method, ASME Code Case N-411 damping values may also be used in lieu of the damping values in Regulatory Guide 1.61.

Structure and component damping values used in the response spectrum and time history analyses are given in Table 3.7B-1. With the exception of piping stress analysis problem number 1-081, all non-NSSS piping stress analysis of safety related piping at CPNPP utilizes Code Case N-411. This includes all analysis of new piping and support optimization work, as well as all pipe stress reconciliation work. Damping factors associated with foundation springs are discussed in Section 3.7B.2.4. Damping values used for qualifying Westinghouse piping and equipment are shown in Section 3.7N.

3.7B.1.4 Supporting Media for Seismic Category I Structures

All seismic Category I structures are founded on the firm, unweathered Glen Rose Limestone which constitutes the principal bedrock formation on the site.

Below the Glen Rose unit lies the Twin Mountains Formation, which forms a gradational contact with the Glen Rose unit and is composed principally of sandstone, limestone, and clay stone. The portion of the Glen Rose unit which provides the founding material for the Category I structures consists of argillaceous limestone with lenses and zones of calcereous clay stone. Approximately 150 to 160 ft of this formation is present beneath the lowermost foundation. The upper portion of the Glen Rose unit consists of weathered rock and a soil cover of a few feet. Prevailing rock characteristics are presented in Table 3.7B-3.

The soil cover and the upper 40 ft (approximately) of the Glen Rose Limestone are totally removed by foundation excavation. Thus, all of the moderately-to-severely weathered rock present at the site is removed.

With the exception of the Service Water Intake Structure, no structural backfill is used under or against Category I structures.

More detailed description of the site geology, the subsurface conditions, and the engineering properties of site materials are included in Section 2.5.

Foundation elevations, depths of embedment, total structural heights, and foundation plan dimensions for the Category I structures are presented in Table 3.7B-4.

3.7B.2 SEISMIC SYSTEM ANALYSIS

3.7B.2.1 Seismic Analysis Methods

Methods of seismic analysis used for seismic Category I structures, systems, and components, as well as applicable stress and deformation criteria, and mathematical models, are described in this section.

Seismic analysis of seismic Category I structures, systems, and components is performed by the use of the response spectrum or the time history concept of analysis, or both [28], [30], [35]. The use of the response spectrum concept provides a convenient procedure for seismic analysis. Spectrum analysis uses the natural frequencies, mode shapes, and appropriate modal dampings as a fraction of critical damping, and is an approximate method for determining the seismic response of linear elastic multidegree-of-freedom systems with lumped masses and elastic properties in discrete parts.

In a time history analysis, there are two basic ways of using the time history for linear elastic systems, namely, by a modal analysis time history, which uses the same free vibration characteristics and damping factors as the spectrum analysis, or by solving a system of coupled differential equations of motion by direct numerical integration. In the latter case, the numerical integration using a suitable technique must be performed simultaneously for all of the coupled equations. This procedure is cumbersome, requiring a large amount of computations, and is susceptible to computational difficulties. For example, it is difficult to know how small the time intervals should be to avoid mathematical instability. Furthermore, there is no really satisfactory way to determine all of the damping coefficients in these coupled differential equations of motion. Because of these difficulties, the modal method of analysis is used. Only in the case of nonlinear behavior when structures, systems, and components cannot be regarded as linear elastic, such as springs with nonlinear restoring-force functions and nonlinear elastic properties of materials, is the method of direct numerical integration of coupled differential equations of motion used.

Where the aforementioned methods do not provide reliable results, or where analysis appears impractical, dynamic testing of equipment is performed to ensure functional integrity.

The methods used for seismic analysis of particular seismic Category I structures, systems, and components are summarized in Table 3.7B-2.

It should be noted that the modal analysis time history method is used to generate responses at selected locations, such as the ones required for the development of instructure response spectra. Responses at selected locations resulting from both response spectrum concept and time history are compared. Static loads resulting from a dynamic analysis are used in the design of some structural components such as foundation mats, floors, and shear walls [34].

3.7B.2.1.1 Idealization of Seismic Category I Structures, Systems, and Components

A most important part of seismic analysis is devising a mathematical model that satisfactorily represents the dynamic behavior of a seismic Category I structure, system, or component. The modeling technique used results in mathematical models composed of a network of lumped masses and elastic properties in discrete parts. Normally, characteristic points or nodes are selected so that they coincide with concentrations of mass, e.g., at floors, changes of cross sectional area, or at locations which are important for stiffness. The characteristic points for lumping of the masses of an axisymmetric shell-type structure are selected at the centroids of horizontal cross-sections through individual components of the structure. These centroids lie on the vertical centerline of the structure. Each mass has six degrees of freedom, namely, three translations in the three principal orthogonal directions and three rotations about the three principal orthogonal axes. In general, responses associated with all of these degrees of freedom can be coupled and excited by each direction of seismic motion. Bending and shearing effects are considered in determining the discrete rigidities between the lumped masses.

For all seismic Category I structures except the Service Water Intake Structure, Seismic Category I Tanks, and re-analysis of the Fuel Building for addition of high density spent fuel storage racks, finite element techniques that simulate floor slabs and shear wall assemblies are used to generate the reduced stiffness matrix associated with the number of dynamic degrees of freedom required for the dynamic analysis. The mathematical model for which this reduced matrix is generated consists of lumped masses, viscous dashpots, and elastic properties in discrete parts. For the Service Water Intake Structure, Seismic Category I Tanks, and re-analysis of the Fuel Building, stiffness properties are calculated for the structural elements between

lumped mass elevations using standard structural techniques. The mathematical models representing the seismic Category I structures and the method chosen for the selection of the number of masses are described in Subsection 3.7B.2.1.6.

For ease of computation, the mathematical model is reduced to contain as few dynamic degrees of freedom as feasible so that it can be analyzed successfully by means of algorithms adopted for today's high-speed digital computers.

Foundation structure interaction is represented by decoupled springs, dashpots, and effective masses generally associated with the six degrees of freedom in a global orthogonal system. The methods used to determine the foundation parameters related to torsion, rocking, and translation are described in Subsection 3.7B.2.4.

The re-analysis of the Fuel Building for added mass due to the addition of high density spent fuel storage racks uses the computer code CLASSI to perform the soil structure interaction analysis. The soil was modeled as a horizontally layered viscoelastic system overlying a half-space. The method used to perform this analysis is described in Subsection 3.7B.2.4.

3.7B.2.1.2 Analytical Approach

In order to analyze the response of a linear elastic lumped mass system, the natural frequencies and corresponding mode shapes are first determined. This determination is accomplished by extracting the eigenvalues and associated eigenvectors from a homogeneous system of equations which result from undamped free vibration and are comprised of stiffness or flexibility and mass matrices developed from the mathematical model. These free vibration characteristics are calculated by using any one of the suitable algorithms coded into the computer programs, such as the diagonalization method originated by Jacobi, Householder's tridiagonalization method combined with the Sturm sequence method, and methods such as those used in computer programs presented in Section 3.7B(A). After establishing the free vibration characteristics, such as natural frequencies and associated mode shapes, the next step consists of response computations obtained by using the response spectrum approach or time history analysis or both [28], [30], [31], [35], [38].

1. Response Spectrum Analysis

   The response spectrum analysis is performed using various computer programs consisting of different subroutines developed by Vistra OpCo or it's engineering services contractor as described in Section 3.7B(A).

   The analysis of the structures founded on bedrock uses spectral values from the free-field horizontal and vertical ground response spectra developed for this site. Spectral values associated with modal dampings and natural frequencies are obtained for each mode. Then the maximum absolute accelerations, inertia forces, shears, moments, and relative displacements are obtained in each mode. The maximum modal responses of all the modes are combined by the square root of the sum of the squares (SRSS), by absolute sum, and by combinations thereof, as discussed in Subsection 3.7B.2.7.

   A separate analysis is made on the model representing the structure founded on bedrock for each of the three orthogonal principal directions of input ground excitations.

Vertical and two horizontal ground excitations are assumed to act simultaneously. Hence, the combined effects of earthquakes on structures, components, or elements are computed by taking the SRSS of the particular effects at any particular point, caused by each of the three components of earthquake motion (two horizontal motions at right angles and one vertical motion).

In the case of shell structures when shell theory is used, maximum stress resultants (membrane shears, moments, and forces), as well as unit stresses and displacements, are obtained. This is accomplished by applying distributed inertia forces and using a suitable computer program.

The total overturning moment at the base of a structure is obtained. The maximum dynamic foundation pressure is evaluated to ensure that it is within permissible limits.

The analysis is performed for both the SSE and OBE unless it is apparent that one of these controls the design.

2. Time History Analysis and Instructure Response Spectra

After the mathematical models of structures are analyzed for their characteristics of free vibration, the time history responses at selected mass points are obtained using the artificial time history ground motion [30], [31], [38]. Derivation of the appropriate time history ground motion is discussed in Subsection 3.7B.1.2. Once the time history response of a selected mass point is generated, the next step is to subject a single-degree-of-freedom system, with the natural frequency range of interest and various damping ratios, to this time history motion. A spectrum response curve is obtained by plotting the maximum acceleration response as ordinates and the corresponding natural periods of the single-degree-of-freedom system as abscissa. The enveloping technique used for the construction of instructure response spectra consists of enveloping the maximum peaks. Since the frequencies of the structures can only be computed approximately because of the linear and nonlinear deformability, the energy dissipation, variation in elastic properties of both structure and foundation, and the idealization of structure with lumped masses and elastic properties in discrete parts, parametric studies are performed in order to take into account these effects for the construction of instructure response spectra. These effects result in the shifting of the resonance peaks of the instructure response spectra. The peaks are widened by at least ±10 percent of the resonance frequencies to account for these effects. The widening exceeds ±10 percent if the parametric studies indicate that such widening is necessary to achieve conservative results. The ground design response spectra and design time history are discussed in Section 3.7B.1.1 and 3.7B.1.2 respectively. Parametric studies and spectra widening are discussed in Subsection 3.7B.2.9.

The Fuel Building was re-analyzed to determine the effects of added mass due to the addition of high density spent fuel storage racks. These analyses were based on best estimate soil properties and the resulting spectrum peak responses were widened by at least ± 15% in accordance with RG 1.122. Best estimate soil properties were used in the re-rack analysis in lieu of parametric variations involving upper and lower bound soil properties. Results of previous studies show that variations in soil properties and building structure stiffness have minimal effect on dynamic response (less than 7%).

Results of the Fuel Building re-analysis demonstrate that the original instructure response spectra largely envelope instructure response spectra considering the effects of high density spent fuel storage. Where the original instructure response spectra were not enveloped, new design instructure response spectra was defined as the envelope of the two as discussed in Section 3.7B.2.5.

The preceding analyses are accomplished by using suitable computer programs as presented in Section 3.7B(A) and in accordance with References [30], [31], [36], and [38].

3.7B.2.1.3 Testing and Analysis for Equipment

Seismic Category I equipment, equipment supports, and components are designed to ensure functional operability during and after an earthquake of magnitude up to and including the SSE (refer to Section 3.2 for the list of seismic Category I mechanical and electrical equipment). The capability of all seismic Category I electrical and mechanical equipment and equipment supports to satisfy this requirement is verified by testing or analysis, or both.

The seismic qualification of seismic Category I electrical equipment and equipment supports is in accordance with IEEE 344-1975. Seismic Category I mechanical equipment and equipment supports are qualified in accordance with the ASME B&PV Code, Section III, Division 1, and the applicable NRC regulatory guides.

When dynamic analysis or testing is employed, the equipment specification or other controlled means furnishes the vendor with appropriate instructure response spectra or acceleration requirements for both SSE and OBE with instructions on their use. In the case where a supporting structure is required to be modeled as an integral part of the equipment, the vendor is furnished with the mathematical model representing the structure consisting of lumped masses and elastic properties in discrete parts. The lumped masses, flexibility or stiffness characteristics, or both, can be presented in matrix form. Any additional information such as the floor motion time history is furnished only upon request.

Where applicable, the response spectrum technique is employed in the dynamic analysis of seismic Category I equipment and components. The time history modal analysis of seismic Category I structures, as previously explained, generates time histories at various support elevations and instructure response spectra for use in analysis of systems and equipment.

In general, at each level of the supporting structure where vital items are located, two horizontal and one vertical response spectra corresponding to coupled translational motions of the supporting structure in the three orthogonal principal directions and the coupled rotational motions of the structure about the three axes for each direction of ground excitation are developed. The instructure response spectra are smoothed so that the response curve is an upper bound envelope of all the acceleration points. Parametric studies and spectra widening are discussed in Subsection 3.7B.2.9.

Generally, multimass presentation of seismic Category I equipment and components, except piping, is used in accordance with the lumped parameter modeling techniques and normal mode theory, as described in the references listed in Section 3.7B. Piping systems are analyzed as described in Subsection 3.7B.3.8.

Simplified analytical models such as one or two degree-of-freedom models are used where they provide a suitable representation of the systems.

Equipment whose lowest dominant natural frequency is 33 Hz or higher is considered rigid. In this case the acceleration of the equipment is assumed to be the same as the zero period acceleration of the appropriate response spectrum.

Where dynamic testing is used to ensure functional integrity, test performance data and results reflect the following:

1. Performance data for equipment which, under the specified conditions, has been subjected to equal or greater dynamic loads than those to be experienced under the specified seismic conditions

2. Test data from previously tested comparable equipment which, under similar conditions, has been subjected to equal or greater dynamic loads than those specified

Mechanical and electrical equipment and components which are required to maintain functional integrity and operability during and after a postulated seismic event are described in Subsection 3.9B.2 and Section 3.10B. Actual testing of such equipment and components to demonstrate seismic adequacy is performed by the suppliers using a shake table, test bed, or other appropriate device.

1. Seismic testing for equipment operability conforms to the following:

    a. A test required to confirm the functional operability of seismic Category I electrical and mechanical equipment and instrumentation during and after an earthquake of magnitude up to and including the SSE is performed. Analysis without testing may be performed only if structural integrity alone can ensure the design intended function. When a complete seismic testing is impracticable, a combination of test and analysis is performed.

    b. The characteristics of the required input motion are specified by one of the following:

        1. Response spectrum

        2. Time history

        Such characteristics, as derived from the structures or systems seismic analysis, are representative of the input motion at the equipment mounting locations.

    c. Where practicable, equipment which is required to function during and/or after an earthquake is tested in the operational condition. Operability is verified during and/or after the testing.

    d. The actual input motion is characterized in the same manner as the required input motion and the conservatism in amplitude and frequency content is demonstrated. The frequency spectrum covers the range from 1 through 33 Hz.

e. Seismic excitations generally have a broad frequency content. Therefore, random vibration input motion is generally used. Single frequency input, such as sine beats, can be applicable provided one of the following conditions is met:

1. The characteristics of the required input motion indicate that the motion is dominated by one frequency, i.e., by structural filtering effects.

2. The anticipated response of the equipment is adequately represented by one mode.

3. The input has sufficient intensity and duration to excite all modes to the required magnitude so that the testing response spectra envelope the corresponding response spectra of the individual modes.

f. To account for the simultaneous occurrence of random earthquake motions, multiaxis testing is performed. Test input motions are applied simultaneously in the vertical and each of the horizontal principal directions. Single-axis testing is accepted in accordance with the provisions of IEEE 344-1975, i.e., where justified by demonstrating that earthquake motions at the mounting locations are amplified in one direction, or that the equipment is constrained to respond in one direction, or that coupling between the responses is negligible. In the case of single-frequency input, the time phasing of the inputs in the vertical and horizontal directions is such that a purely rectilinear resultant input is avoided.

The acceptable alternate to this procedure is to test the equipment with vertical and horizontal inputs in-phase, and then repeat the test with inputs 180 degrees out-of-phase.

In addition, it is required that the test be repeated with the equipment rotated 90 degrees horizontally.

g. The fixture design meets the following requirements:

1. Simulates the actual service mounting

2. Causes no dynamic coupling to the test item

h. The in situ application of vibratory devices to superimpose the seismic vibratory loadings on the complex active device to provide data for operability testing is acceptable when application is justifiable.

i. The test program may be based upon selectively testing a representative number of mechanical components according to type, load level, size, and similar items on a prototype basis.

j. Where a random vibration input motion is used, the amplitudes of the excitations are controlled to provide a test response spectrum which meets or exceeds the applicable required response spectra.

2. Seismic design adequacy of supports conforms to the following:

   a. Analyses or tests are performed for all supports of electrical and mechanical equipment and instrumentation to ensure their structural capability to withstand seismic excitation.

      In accordance with the intent of IEEE 344-1975, if supports are similar and justified as such, or if based on engineering investigation, the worst case support is chosen from a group of supports to be qualified and is justified, only one of the similar supports or the worst case support requires a complete dynamic seismic analysis or full scale test, or a combination of both.

      Justification of this procedure is based upon comparison analysis or past experience that the supports to be qualified are similar or that the worst case has been chosen. Upon such justification and dynamic analysis or full scale testing, or a combination of both, of the similar or worst case support, the group of supports being investigated is accepted as seismically qualified.

   b. The analytical results include the following:

      1. The required input motions to the mounted equipment are obtained and characterized in the manner stated in Item 1.b of this subsection.

      2. The combined stresses of the support structures are within the limits of the ASME B&PV Code, Section III, Division 1, Subsection NF, Component Support Structures, or other comparable stress limits.

   c. Supports are tested with equipment installed. If the equipment is inoperative during the support test, the response at the equipment mounting locations is monitored and characterized in the manner stated in Item 1.b. In such a case, equipment is tested separately and the actual input to the equipment is made more conservative in amplitude and frequency content than the monitored response.

   d. The requirements of Items 1.b, 1.d, 1.e, 1.f, and 1.g are applicable when tests are conducted on the equipment supports.

Equipment, when under test, is mounted in a manner that simulates the intended service mounting in its operating condition.

When sinusoidal beat input motion is applicable as per Items 1.e.1), 1.e.2), and 1.e.3), testing is normally performed in two phases as follows:

1. Phase 1 consists of a low-amplitude continuous sweep frequency search, with a sinusoidal steady-state input of at least 0.2g over a minimum frequency range of 1.0 to 33 Hz in order to determine potential resonant regions.

2. Phase 2 consists of sinusoidal beat testing at resonant frequencies determined in Phase 1 with amplitudes correlated to the maximum acceleration of the equipment support. In addition to sinusoidal beat testing at the known resonant frequencies, the test

is also performed at a number of preselected frequencies in the 1.0 to 33 Hz band. This ensures that any shifts in resonant frequencies caused by nonlinear effects are adequately evaluated.

A sinusoidal beat consists of 10 cycles at the frequency being tested with amplitudes of each cycle varying as a sine function from zero to a maximum at the fifth cycle and then decreasing to zero at the tenth cycle. The maximum sine beat amplitude is made to correspond to the maximum zero period SSE acceleration of the structure at the appropriate floor.

A test at any frequency consists of five beats with a pause between beats of approximately two sec or more, as required to ensure no superposition of motion.

Sinusoidal steadystate testing at resonant frequencies rather than sinusoidal beat testing is performed at the option of the equipment supplier.

When sinusoidal beat testing or sinusoidal steady-state testing is used, the equipment is subjected to the input motion for at least the entire duration of an earthquake of 30 sec.

All tests are performed independently for each of the two horizontal directions and the vertical direction. Three-axes simultaneous testing is also required. Satisfactory alternatives as recommended in Section 6.6.6 of IEEE 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations, are allowed where three-axis testing is impractical.

Equipment when under test is required to demonstrate a capability to perform its intended function, and sufficient monitoring equipment is used to evaluate performance during test.

Other test procedures proposed by equipment suppliers equivalent to the methods described previously are reviewed by Vistra OpCo or its engineering services contractor, and used for seismic qualification of equipment when formally approved by Vistra OpCo or its engineering services contractor.

The equipment supplier is required to document his previous experience in analysis or testing of seismic Category I equipment similar in nature to that being provided.

The equipment supplier is required to furnish documentation to demonstrate that the equipment meets the seismic design criteria established in the specification in both performance and structural integrity.

If proof of performance is obtained by analytical means, the equipment supplier is required to present documentation in a step-by-step form which is readily auditable by persons skilled in such analyses.

If proof of performance is obtained by testing, the test data include detailed information on the following:

1. Equipment identification

2. Test facility (including location)

3. Test equipment

4. Test method

5. Test data (including unsuccessful test of components and subsequent remedial measures)

6. Results and conclusions

7. Attestation

3.7B.2.1.4 Differential Seismic Movement of Interconnected Components

The seismic analysis of seismic Category I subsystems and equipment subjected to differential support motion is performed in three parts using lumped mass mathematical models as follows:

1. Modal response spectrum analysis is performed for all three principal orthogonal directions of support motion for each direction of ground excitation using appropriate instructure response spectra constructed on the basis of superimposing the spectra for all support points and enveloping them as stated in Subsection 3.7B.2.5. The analysis of these subsystems or components follows the same considerations as those described in Subsection 3.7B.2.1 for seismic Category I structures. The vertical analysis is combined with both horizontals, according to the statement in Subsection 3.7B.2.1.2, to produce basic dynamic loading conditions.

2. The same multimass lumped parameter model is subjected to a stress analysis due to differential displacements of the support points. The displacements used are consistent with the directions of structural excitation being considered in the spectrum analysis. This results in basic differential displacement loading conditions.

3. The results obtained from the spectrum analysis and differential displacement analysis are then combined directly. The effects of these loading conditions on the components and the supporting structures are determined.

3.7B.2.1.5 Stress and Deformation Criteria

The ground design response spectra and design time history are discussed in Section 3.7B.1.1 and 3.7B.1.2 respectively.

Primary steady-state stresses including the effects of the normal operating loads plus the OBE loads are maintained well within the elastic limit of the material affected.

For systems and equipment, self-limiting secondary stresses may exceed allowable primary stress to the extent permitted by the appropriate codes. For the OBE, the equipment function is performed without permanent deformation.

Primary steady-state stresses, including the effects of the normal operating loads plus the SSE loads, are limited to prevent loss of function of the equipment. For the purpose of calculation, the no-loss-of-function stresses are limited to 90 percent of the yield strength of the material, except when valid plastic analysis demonstrates structural integrity or when the stress limits are

specifically controlled by an applicable code or standard as committed to by CPNPP for example, see Section 3.9B. Local, self-limiting, secondary stresses may exceed yield stress levels to the extent set forth in the appropriate design standards and codes.

Deformations resulting from the combined influence of normal operating loads and the loads from the SSE are investigated to verify that they do not impair the functional performance required for a safe and orderly shutdown of the plant.

For fatigue analysis required by some codes, the number of expected earthquakes, the duration of strong motion vibration, and the number of cycles the equipment or system is exposed to are evaluated for the OBE, and specified as 600 load cycles.

Because the earthquake transients are part of mechanical loading conditions, the origin of their determination is separate and distinct from those transients which result from fluid pressure and temperature. Therefore, the superposition of the earthquake cycles on the fluid pressure and thermal transients is not considered.

3.7B.2.1.6 Descriptions of Mathematical Models

The mathematical models used for the dynamic analysis of the major structures are described as follows:

1. Model for Containment and Internal Structures

   The mathematical model for the Containment and Internal Structures consists of lumped masses, elastic properties, and viscous dashpots in discrete parts. The number of masses and associated degrees of freedom is kept to a minimum to reduce unnecessary complexity.

   Plan and elevation views of the Containment and Internal Structures are shown on Figures 3.7B-23 and 3.7B-24. The corresponding mathematical models are shown on Figure 3.7B-34. The coordinates of the lumped masses are listed in Table 3.7B-9. The values of the masses and mass moments of inertia associated with the lumped masses of the structures and foundation are presented in Table 3.7B-14. The degrees of freedom assigned to the lumped masses of the structures are identified in Table 3.7B-19.

   The dome and the cylinder of the Containment Structure are simulated by five masses. The Internal Structure, made up of concrete walls, columns, slabs, and beams, is represented by three masses lumped at floor levels. The mat is simulated by a single mass. Each mass is assumed to have six dynamic degrees of freedom, namely, three translations in the three orthogonal principal directions and three rotations about these three axes.

   Two parametric analyses are performed for the Containment Structure where the full thickness (uncracked) and the half thickness (cracked) of the cylindrical wall are considered to be effective.

   Foundation-structure interaction, such as the foundation spring constants, damping ratios associated with each foundation spring, and effective masses of the foundation material, is based on the theory of a circular base resting on elastic half space as described in

Subsection 3.7B.2.4 [1], [2], [23], [32], [37]. These values are determined for all six degrees of freedom of the global orthogonal system. The effect of the embedment is also evaluated. Estimated values, upper bound values, and lower bound values of foundation spring constants used in the parametric analyses described in Subsection 3.7B.2.4 are presented in Table 3.7B-24.

The stiffness or the flexibility matrices of the Containment Structure and Internal Structure are generated using suitable computer programs based on finite element techniques. The Containment Structure wall and dome and Internal Structure walls, columns, and floors are modeled with finite elements. The stiffness matrix which corresponds to the finite element model is then reduced to the number of dynamic degrees of freedom required for the dynamic analysis [3], [23], [30], [38].

Past experience shows that the mathematical model described previously conservatively predicts the dynamic behavior of the actual structure subjected to seismic disturbances.

2. Models for the Safeguards Building, Electrical and Auxiliary Buildings, Fuel Building, Service Water Intake Structure and Category I Tanks.

The mathematical models for these structures are comprised of lumped masses, elastic properties, and dashpots in discrete parts.

The locations of the mass points are chosen at floor levels and points considered of critical interest, such as equipment support levels. Because the structures cannot be considered symmetrical and the torsional modes of vibration can be excited by ground motions, each mass is assumed to have six degrees of freedom, namely, three translations in the orthogonal principal directions and three rotations about the three principal axes which account for the torsional and rocking modes of vibration.

Plan and elevation views of the Safeguards Building, Electrical and Auxiliary Buildings, Fuel Building, and Service Water Intake Structure are shown on Figures 3.7B-25 through 3.7B-33. The corresponding mathematical models are shown on Figures 3.7B-35 through 3.7B-38. The coordinates of the lumped masses are listed in Tables 3.7B-10 through 3.7B-13. The values of the masses and mass moments of inertia associated with the lumped masses of the structure and foundation are presented in Tables 3.7B-15 through 3.7B-18. The degrees of freedom assigned to the lumped masses of the structures are identified in Tables 3.7B-20 through 3.7B-23.

The Fuel Building was re-analyzed to determine the effects of added mass due to the addition of high density spent fuel storage racks. The refined lumped mass model incorporates a new node at the bottom of the spent fuel pools (elevation 819.56 ft) to allow finer distribution of the building mass and more accurate determination of the input motion to the new fuel racks. With the additional node at the 819.56 ft elevation, the fluid mass is distributed between the 819.56 ft, 838.75 ft, and 860 ft elevation. A portion of the fluid is modeled as a sloshing mass at the 806 ft elevation in accordance with TID-7024. As a further refinement, at each floor, four (4) nodes are added at extreme locations and connected rigidly with the floor center of mass. The model incorporates rigid links between the center of rigidity and center of mass at each floor elevation to account for the effects of torsion. The math model used for the Fuel Building re-analysis is shown in

Figure 3.7B-37A. The coordinates of the lumped masses are shown in Table 3.7B-12A and the values of the masses are presented in Table 3.7B-17A.

Foundation spring constants associated with three orthogonal principal translations and two rocking motions about the two horizontal orthogonal axes are determined on the basis of a rectangular or circular base resting on an elastic half space [1], [2], [23], [37]. Torsional foundation spring constants, damping ratios, and effective masses and rotary inertias for foundation below the vibrating mat associated with the foundation springs are determined on the basis of the equivalent radius for the rectangular base of dimensions 2c by 2d using the theory of the elastic half space for a circular footing according to Subsection 3.7B.2.4 [1], [2], [23], [32], [37]. The effects of the embedment of the structures are evaluated and taken into consideration in the analysis. Values of foundation spring constants used in the parametric analyses described in Subsection 3.7B.2.4 are presented in Tables 3.7B-25 through 3.7B-29.

The Fuel Building re-analysis was based on soil-structure interaction analysis using the computer program CLASSI. Soil properties and impedances used in the soil-structure interaction analysis are discussed in Subsection 3.7B.2.4. The effects of embedment are evaluated and taken into consideration in the analysis.

The stiffness matrices of the buildings are generated using suitable computer programs based on finite element techniques or on stiffness properties calculated for the structural elements between lumped mass elevations. For unsymmetric structures the stiffness matrices include the effects of torsional rigidities of shear wall assemblies between floors. The stiffness matrices obtained for finite element models are reduced to conform to the number of degrees-of-freedom of the dynamic models which are used in the dynamic analysis [3], [23], [30], [38].

3.7B.2.2 Natural Frequencies and Response Loads

The natural frequencies and modal participation factors for all modes resulting from the parametric analyses of representative seismic Category I structures are presented in Tables 3.7B-30 through 3.7B-45. Response loads for these structures obtained by the square root of the sum of the squares method (SRSS) are summarized in Tables 3.7B-46 through 3.7B-50 in the form of modal accelerations.

For comparison, envelope values of time history analysis results for the Electrical and Auxiliary Buildings and for the Fuel Building are presented in Tables 3.7B-51 and 3.7B-52.

Seismic loads used for the design of seismic Category I structures are obtained by multiplying the response accelerations with the appropriate masses.

Response spectra at all floors are developed for all seismic Category I structures as indicated in Subsection 3.7B.2.5.

3.7B.2.3 Procedure Used for Modeling

The structures and their contents possess mass which contributes to the inertia loading of the structure. The complexity of the spatial distribution makes it necessary to concentrate the mass at characteristic points or nodes. These points are selected so that they coincide with

concentrations of mass, e.g., at the floors, or with locations which are important for stiffness. In some instances, the nodes are selected at intermediate points of structures and equipment that can be regarded as being of uniform construction. This discretization into characteristic points permits a more accurate prediction of the dynamic behavior of actual structures and equipment.

At each node, the structure or system is given six degrees of freedom (three translation components and three rotation components).

No simplifications aimed at reducing the total number of degrees-of-freedom considered in the analysis are made. All six degrees-of-freedom of each node are treated as generalized displacements for all seismic Category I structures.

The idealization of the mass is performed on the basis of relative displacements. If the horizontal cross-section of the structural component, for example, does not deform significantly, and the contents undergo essentially the same displacement as the structure, all mass in a given place can be represented by a point mass placed at the centroid.

It is not feasible to formulate a mathematical model which would include, in addition to the primary structure, all of the equipment, piping systems, and other lightweight structures. These subsystems are therefore uncoupled from the primary structures and are analyzed by the response spectrum approach procedure. In order to use the spectrum analysis for secondary systems, floor response spectra are developed as described in Subsection 3.7B.2.5.

The criteria employed for system/subsystem decoupling are consistent with the provisions of USNRC Standard Review Plan, Subsection 3.7.2, June 1975. They are based on the mass ratio, R_m of the supported subsystem mass to the corresponding support mass, and the frequency ratio, R_f of the supported subsystem fundamental frequency to the corresponding supporting system dominant frequency such that:

1. If $R_m < 0.01$, decoupling can be done for any R_f

2. If $0.01 < R_m < 0.1$, decoupling can be done if $R_f < 0.8$ or $R_f > 1.25$

3. If $R_m > 0.1$, an approximate model of the subsystem should be included in the primary system model.

where:

$$R_m = \frac{\text{Total mass of supported subsystem}}{\text{Mass of supporting structure}}$$

$$R_f = \frac{\text{Fundamental frequency of the supported system}}{\text{Frequency of the dominant support motion}}$$

The floor response spectra are generated using the mathematical models which consist of the lumped masses computed from tributary structure dead loads, a portion of live loads, and fixed equipment loads. In some cases, the uncoupled mathematical models, with lumped masses

representing the equipment, include the effective masses and flexibility of the supporting structure.

3.7B.2.4 Soil-Structure Interaction

The mathematical model for performing the dynamic analysis of seismic Category I structures supported on the ground is comprised of lumped masses and elastic properties in discrete parts. Because these structures are founded on sound bedrock (Glen Rose Limestone) with shear wave velocities of 5500 to 6000 ft/sec, the foundation-structure interaction is evaluated using the conventional elastic half-space theory in accordance with References [1], [2], [23], [32], and [37]. The justification for the use of this theory is based on the fact that sound bedrock is much closer to being a truly elastic material than any other common foundation material. Using the half-space theory, foundation spring constants with associated effective masses of the rock and damping ratios caused by radiation damping are determined.

Principles of radiation damping apply more correctly for this case than for foundations on other materials. The radiation damping is associated with the energy which is carried away from the immediate vicinity of the foundation by stress waves. In addition to the radiation damping, the internal damping is also determined. The internal damping within the foundation material arises primarily because of a nonlinear effect known as hysteresis; i.e., the stress-strain relationship during unloading is not the same as that during loading. The combined effects of radiation and internal damping are evaluated in accordance with References [32] and [37]. However, damping values used in the analysis are limited to a maximum of 10 percent associated with horizontal and vertical translational foundation springs and five percent for rocking and torsional foundation springs.

Because of the possible variations in measuring the properties of the foundation material, parametric studies are performed on the basis of the best estimate as well as the upper and lower bound estimates of the foundation spring constants with the exception of the Fuel Building and Service Water Intake Structure where only lower bound and upper bound values are used. The range of foundation spring constants considered in the parametric studies accounts for the variations in the properties of the foundation material encountered at the site, as well as the anticipated variations in measuring these properties. These variations are considered to be 10 percent in measuring the mass density and 15 percent in measuring the shear-wave velocity.

The Fuel Building was re-analyzed to determine the effects of added mass due to high density spent fuel storage. The re-analysis incorporates soil-structure interaction using the computer code CLASSI. In the SSI analyses, the soil was modeled as a horizontally layered viscoelastic system overlying a half-space. The solution utilizes soil impedances derived from elastic half-space theory and based on best estimate soil properties.

CLASSI requires as input the modal properties of the building (mode shapes, frequencies, and modal damping ratios) with fixed base. Modal damping ratios for the fixed base building are in accordance with Table 3.7B-1. Modes with frequencies higher than 35 Hz do not need to be considered, however, the number of modes used in the CLASSI analyses account for at least 90% of the total mass.

3.7B.2.4.1 Evaluation of Soil-Structure Interaction Parameters

The foundation-structure interaction in the mathematical models represented by foundation spring constants, associated damping factors, and effective foundation material masses and rotary inertias (in general, corresponding to all six degrees-of-freedom) is evaluated from the theory of a rigid base resting on an elastic half space in accordance with References [1], [2], [23], [32], and [37]. The solution of this problem was originally given by Boussinesq [8].

Based on the preceding theory, the foundation spring constants are obtained from Tables 3.7B-5 and 3.7B-6 in conjunction with Figure 3.7B-39 for rigid circular and rectangular bases resting on the surface of an elastic half-space. An approximate evaluation is made by using an equivalent circular base having the same second moment about its centroidal axis for the contact area. On this basis, the equivalent radius for the rectangular base of dimensions 2c by 2d is determined by the following equation:

$$r_0 = \sqrt[4]{\frac{8dc(c^2 + d^2)}{3\pi}} \qquad (3.7B\ 1)$$

The determination of radiation damping (see Table 3.7B-7), occasionally called geometric damping, caused by foundation structure interaction is also based on vibrations of a rigid circular base on the elastic half-space.

The internal damping caused by energy loss during stress reversals is evaluated on an empirical basis.

Combined effects of radiation and internal damping are taken into account by direct addition of the two values of damping. For rotational motions, the radiation damping is low, and the internal damping can be a significant part of the total damping. However, for translational motions, radiation damping is much greater than internal damping.

There are practical reasons why the damping shown by the theory may not be realistic in actual mass foundation systems. Therefore, damping values that are used in the dynamic analysis are limited to the maximum values set forth in Subsection 3.7B.2.4.

The effective masses and rotary inertias of the foundation material are estimated on the basis of the theory for the elastic half-space.

Effective mass and mass moment of inertia for foundation material below a vibrating footing are estimated from the following formulas:

$$M_{eff} = \propto \bullet \rho \bullet r_0^{\,3} \qquad (3.7B\text{-}2)$$

and

$$I_{eff} = \beta \bullet \rho \bullet r_0^{\,5} \qquad (3.7B\text{-}3)$$

where

M_{eff}	denotes the effective mass of the foundation material.
I_{eff}	is the effective rotary inertia of the foundation material.
ρ	is the mass density of the foundation material.
$\propto$ and β	are the coefficients given by theory and shown in Table 3.7B-8 [2].

For rectangular footings of dimensions 2c by 2d, the equivalent radii are obtained for torsion from Equation 3.7B-1 and for translation and rocking from the following equations:

For translation:

$$r_0 \sqrt{\frac{4cd}{\pi}} \qquad (3.7B\text{-}4)$$

For rocking:

$$r_0 = \sqrt[4]{\frac{16cd^3}{3\pi}} \qquad (3.7B\text{-}5)$$

The values of effective foundation masses and mass moments of inertia determined for representative seismic Category I structures are presented in Tables 3.7B-14 through 3.7B-18. They are calculated based upon the assumption that the centers of gravity of the effective soil masses are located on the top of the soil at the centroid of the soil mat contact areas. These values are then added to the lumped mass points representing the building foundations.

Foundation embedment depths, including a description of the foundation medium and its properties, are given in Subsection 3.7B.1.4.

The effect of embedment on foundation rigidities is also determined approximately from the theory of elastic half-space. For example, the torsional rigidity and the rigidity in horizontal translation are determined on the basis of the assumption that the vertical contact area, engaging the elastic half-space, is an imaginary area of rectangular shape whose vertical dimension is twice that of the actual contact area of embedment. The rigidities obtained by the foregoing procedure are divided in half to account for the actual vertical contact area of embedment.

The equations used for evaluating translational and rotational spring constants for embedment are presented in Subsection 3.7B.2.4.2.

The additional nomenclature used in Tables 3.7B-5, 3.7B-6, 3.7B-7, and 3.7B-8 is as follows:

E and G are the modulus of elasticity and the shear modulus of the foundation material, respectively. Nu is the Poisson's ratio of the foundation material.

The shear modulus is obtained from the field-measured, shear-wave velocity C and mass density of the foundation material from the equation

$$G = C_s^2 \cdot \rho \qquad (3.7B\text{-}6)$$

The methods of determining the shear-wave velocity are discussed in Section 2.5.4.4. Strains developed by these methods are in the order of 10^{-5} to 10^{-3} percent. Based on the shear modulus of the competent rock on which the structure is founded (8×10^5 lb/in^2), the estimated strains caused by a seismic event are of the same order of magnitude. In this range of strains, the shear modulus of rock is relatively independent of the strain levels.

The analysis of all CPNPP seismic Category I structures is based upon a rigid mat approach, and the spring constants of the foundation material are determined by means of the formulas listed in Table 3.7B-5 and Table 3.7B-6 and by means of the coefficients in Figure 3.7B-39.

The CLASSI analyses used for the Fuel Building re-analysis were carried out based on a surface founded structure, rigid foundation, layered soil over a half-space, and input motions applies at the free field surface level.

CLASSI is based on a specialized form of substructuring which uses the finite element method to perform the detailed analysis of the superstructure and uses the continuum mechanics method to calculate the interaction of the foundation with the soil medium and with incident seismic waves. These substructuring procedures are made possible by balancing the forces and moments at the foundation, which serves as the common ground for both the superstructure and the soil medium.

The CLASSI substructure approach divides the SSI problem into the following three steps:

1. Determination of the foundation scattering matrices.

2. Determination of the frequency-dependent impedance functions.

3. Analysis of the coupled soil-structure system, using results from steps 1 and 2 and the dynamic properties of the structure.

In the first step, CLASSI evaluates the harmonic response of the rigid, massless foundation bonded to the soil and subjected to a given incident seismic wave in the absence of the superstructure. The free field motion is then used in conjunction with the complex, frequency-dependent scattering matrix in order to determine the foundation input motion. The foundation input motion corresponds to the response of the rigid, massless foundation to the seismic environment described by the free-field in the absence of the superstructure. The response of the rigid massless foundation to the seismic excitation can be described by the six component vector:

$$U^*_o = (\Delta^*_X, \Delta^*_Y, \Delta^*_Z, \theta^*_X, \theta^*_Y, \theta^*_Z)^T$$

In which $\Delta^*_X, \Delta^*_Y, \Delta^*_Z$ represents the translational components of the response and $\theta^*_X, \theta^*_Y, \theta^*_Z$ represent the rotational components of the response.

The foundation input motion U^*_o is related to the free-field ground motion by means of the complex-valued, frequency dependent scattering matrix [S(w)]:

$U^*_o = [S(w)]\ f(w)$

Where the vector f(w) is the complex Fourier transform of the free-field ground motion. At a given frequency, w, each complex number in f(w) corresponds to the amplitude and phase of a wave component of the free-field motion. Each column of the scattering matrix [S(w)] represents the response of a massless rigid foundation to a given incident wave of unit amplitude. The matrix product [S(w)] f(w) is therefore the response of the rigid massless foundation to a particular free-field motion.

For a surface-founded rigid foundation subjected to vertically propagating shear or compressional waves, the response of the foundation includes only translational components with amplitudes equal to those of the free-field motion on the ground surface. However, if the foundation is embedded, a horizontal component of the control motion consisting of vertically propagating shear waves produces both a horizontal translation and a rocking motion of the massless foundation. This is primarily due to the scattering waves from the soil-foundation interface and the kinematic constraints imposed on the soil by the rigid foundation.

Input for the first step in the SSI analysis consists of the properties of the underlying soil layers and the frequency points for Green's function. Soil properties used in the SSI analysis are best estimate and are shown in Table 3.7B-3A.

In the second step, the foundation impedances corresponding to a rigid foundation on a uniform or layered viscoelastic media are developed. The foundation impedances are complex-valued, frequency-dependent functions which relate the dynamic forces that the foundation exerts on the soil to the resulting soil displacements, i.e.:

$F_S(w) = [K(w)]\ U_S$

Where $F_S(w)$ represents the generalized forces, [K(w)] is the complex impedance matrix, and U_S represents the generalized displacements. The real part of the complex impedance matrix represents the stiffness of the soil and the imaginary part represents the energy dissipation of the soil, including both radiation and material damping.

In the third step, analysis of the coupled soil-structure system is carried out in the frequency domain. The impedances and scattering matrices are used to solve the equations of the coupled soil-structure system. For this step, the dynamic characteristics of the structure are used to reduce the effects of the superstructure to six dynamic inertial parameters for each mode and a rigid body mass matrix of the structure about a reference point on the foundation where the SSI response is determined. Once the motion of the foundation has been obtained, the time history response at any level of the structure is computed using Fourier transform techniques.

The Fuel Building has an irregular basemat which exhibits a range of embedment from five (5) to thirty 30) feet. Soil properties were adjusted to account for embedment as described in Subsection 3.7B.2.4.2. Vertical and horizontal time histories were generated by vertically propagating waves; compressional (P) waves for the vertical earthquake and shear waves for the

horizontal earthquake. Translational acceleration time histories were calculated for the extreme location nodes at each floor.

3.7B.2.4.2 Effect of Embedment on Foundation Rigidities

The effect of the embedment on foundation rigidities is based on the conservative assumption that there is no friction between the vertical contact surfaces of the structures and the foundation medium. Therefore, the effect of the embedment on vertical translational rigidity and torsional rigidity, in the case of circular mats and axisymmetric shells of revolution structures, is neglected. The effect of the embedment on rocking rigidities, about the two horizontal axes located at the base, is normally small and is also neglected, unless a significant contribution to the total rocking rigidities due to this cause is expected to play an important role in the analysis and design.

The effect of the embedment on foundation torsional rigidity due to the two opposite, vertical contact surfaces of a structure having a rectangular shape is obtained, according to References [23] and [33], from the following formula:

$$k_\theta = \frac{G}{1-\nu}\beta\psi 4hd^2 \qquad (3.7B\text{-}13)$$

where

h	=	the depth of the embedment
2d	=	the length of the contact surface
G	=	the rigidity modulus
ν	=	the Poisson's ratio
$\beta\Psi$	=	the value given in Figure 3.7B-39 for various values of d/h.

For example, for h = 20 ft and d = 100 ft, d/h = 5; this corresponds to $\beta\Psi$ which equals 0.9 in Figure 3.7B-39. Then, the foundation torsional rigidity constant is:

$$k_e = \frac{G}{1-\nu}0.9x4x20x100^2 = 720{,}000\frac{G}{1-\nu}$$

This torsional rigidity is added to the torsional rigidity associated with the other two opposite, vertical surfaces and the torsional rigidity associated with the horizontal contact surface obtained in accordance with Subsection 3.7B.2.4.

The effect of the embedment on horizontal translational rigidity is given by the following formula:

$$k_x = \frac{G}{1-\nu}\beta_z\sqrt{hd} \qquad (3.7B\text{-}14)$$

where β_z is the value given in Figure 3.7B-39 for various values of d/h.

Taking the same example as for torsional rigidity, the following horizontal translational rigidity constant is obtained for the value of β_z = 2.5 resulting from Figure 3.7B-39:

$$k_\Psi = \frac{G}{1-\nu}(2.5\sqrt{20x100}) = 112\frac{G}{1-\nu}$$

This horizontal translational rigidity is added to the horizontal translational rigidity corresponding to the horizontal contact surface obtained as presented in Subsection 3.7B.2.4.

A matter of practical interest is that the rigidities in torsion and horizontal translation due to the effects of embedment, obtained on the basis of the elastic half-space theory, are essentially in agreement with the rigidities obtained using References [39] and [40] when the sloping planes of the effective foundation zone are assumed to slope at an angle of 30 degrees. It should be noted that these values are also consistent for practical purposes with the values obtained from Reference [7].

Where the effect of embedment on rocking rigidities plays an important role in the analysis and design of seismic Category I structures, equipment, systems, and components, the rocking rigidity constants for embedment are based on the conservative assumption that the side pressure distribution is uniform due to horizontal load. In this case, the horizontal translational spring constant k_x is obtained from the formula presented in Table 3.7B-6. Then, the rocking rigidity constant due to two opposite vertical contact surfaces of the embedment is obtained using Equation 3.7B-9 of Subsection 3.7B.2.4 as follows:

$$k_\Psi = \frac{1}{12}\frac{G}{1-\nu}\beta_z h^2\sqrt{2hd} \qquad (3.7B\text{-}15)$$

where β_z is a value given in Figure 3.7B-39 for various values of 2d/h.

For example, the rocking rigidity constant corresponding to a length of embedment 2d = 200 ft and a depth of embedment h = 20 ft (β_z = 2.85) is:

$$k_\Psi = \frac{1}{12}\frac{G}{1-\nu}(2.85x400\sqrt{4{,}000}) = 6000\frac{G}{1-\nu}$$

It should be noted that this value can be insignificant compared to the rocking rigidity constant associated with the horizontal contact surface of the base with dimensions 2c by 2d. For example, assume that the width of the base mat is 2c = 100 ft (c/d = 0.5); then, the rocking rigidity constant corresponding to the horizontal contact surface is:

$$k_{\Psi b} = \frac{G}{1-\nu}(0.42x8x100x50^2) = 840{,}000\frac{G}{1-\nu}$$

This value is much larger than the value of rocking rigidity constant obtained for the effect of embedment. Therefore, in this case, the effect of the embedment on rocking rigidity is neglected.

In reality, the rocking rigidity constant for embedment is higher in value than the one obtained here. Perhaps a more realistic value can be obtained by assuming that the vertical contact surface of the embedment with the depth h has a mirror image surface with the depth of 2h. Then half of the value for rocking rigidity constant based on the elastic half space theory seems to be more appropriate when the ratio of the actual depth of embedment to the length of embedment is less than unity. For example, using this approach, the following value for rocking rigidity constant for embedment is obtained:

$$k_{\Psi} = \frac{G}{1-\nu} 0.4 \cdot 8 \cdot 100 \cdot 20^2 x \frac{1}{2} = 64{,}000 \frac{G}{1-\nu}$$

Incidentally, this value and the values obtained for the ratios of the depth to the length of embedment less than one are in close agreement with the values obtained on the basis of the approach to the problem for cohesive soils as presented in References [39] and [40]. These values also compare well for practical purposes with the ones obtained using formulation presented in Reference [7].

For the dynamic analysis of seismic Category I structures which have relatively shallow depths of embedment (such as the Safeguards, Electrical and Auxiliary, and Fuel buildings), the effect of embedment on rotational foundation rigidities (torsion and rocking) is negligible. The Service Water Intake Structure (SWIS), which has a greater depth of embedment, is analyzed by including the effects of embedment according to procedures recommended in References 42, 43, 44 and 45. The Service Water Intake Structure is basically socketed into rock with soil backfill on three sides above the top of rock. The embedment effects were calculated for all rock and then for all soil above the founding levels and an average effect representative of the actual soil/rock profile was selected. Although the Fuel Building has a relatively shallow embedment, the affects of embedment were considered in the re-analysis for addition of high density spent fuel storage racks. The procedures recommended in References 42, 43, 44, and 45 were used to evaluate the effects of embedment for the re-analysis.

3.7B.2.5 Development of Floor Response Spectra

The methods of seismic analysis are covered in Subsection 3.7B.2.1. The response spectrum method for the development of instructure response spectra is not used.

Instructure response spectra at selected locations of interest are developed on the basis of computed responses to an artificial time history input of ground motion. The time history of the simulated earthquake ground motion is developed to be compatible to the given ground response spectra. Having established the time history of the ground motion, the lumped mass mathematical models of seismic Category I structures are analyzed and time histories at desired masses lumped at floor levels or any other location of interest are generated. Once the time history of the floor motion is obtained, the next step consists of subjecting a single degree-of-freedom system with the natural frequency range of interest and various damping ratios to the floor time history motion. The maximum acceleration responses obtained are then plotted as ordinates and the corresponding natural periods of the single oscillators are plotted as abscissa. The envelope of maximum peaks is used for the construction of instructure response spectra.

In constructing instructure response spectra, uncertainties inherent to the analysis, such as the material properties of the foundation material and the structures, damping values, soil structure interaction, approximations in the modeling techniques, and computation of structure natural frequencies, are accounted for by parametric variations incorporated into the analysis and by broadening of the peaks of the resulting envelope response spectra as described in Subsection 3.7B.2.9.

The procedure of parametric variations consists of evaluating and using in the dynamic analysis lower bound, best estimate, and upper bound values for the foundation spring constants in the case of all seismic Category I structures with the exception of the Fuel Building and the Service Water Intake Structure where only lower bound and upper bound values are used. In addition, the analysis of the Containment Building is performed for each set of foundation spring constants by considering a cracked and an uncracked Containment wall.

Responses including translational and rotational effects are calculated for each structure at each center of mass, and transferred to the points farthest from the center of mass. Responses in each direction from each of the three directions of earthquake input are calculated separately, and resulting response spectra are combined by the square root of the sum of the squares (SRSS) method to calculate the total response.

Typical response spectra for the Category I Buildings are presented in Figures 3.7B-41 through 3.7B-50C. Curves Ax, Ay, and Az represent the spectra in the X, Y, and Z directions for the combined effect of the three simultaneous earthquakes. The coupling effects of the nonsymmetric structure are included. These design spectra were generated and peak broadened by computer and are therefore labelled refined. Some other design spectra were generated by computer but not peak broadened by computer and therefore have extra conservatism due to the hand smoothing technique.

The Fuel Building was re-analyzed to determine the effects of added mass due to the addition of high density spent fuel storage racks. The acceleration response spectra of the CLASSI output time histories were calculated using the computer program RESPEC. The program uses the procedure recommended by Nigam and Jennings [48]. The model for the Fuel Building re-analysis incorporated node points at extreme locations at each floor level. At each extreme floor location, the generated spectra from the soil structure interaction analysis for each response direction were combined by the square root of the sum of the squares (SRSS) method. This process combined the “X” direction response due to the “X” direction input motion along with the “X” direction responses from the “Y” and “Z” direction input motions, and was repeated for the “Y” and “Z” direction responses. This method is acceptable because of the statistical independence of the ground motion time history accelerations.

Best estimate soil properties were used in the re-analysis in lieu of parametric variations involving upper and lower bound soil properties. The resulting instructure response spectra was combined with the original instructure response spectra to define new enveloped design response spectra for the Fuel Building. The resulting response spectra is dominated by the original analysis as shown in Figures 3.7B-50AA and 3.7B-50AB. The response spectra shown in Figure 3.7B-50A completely envelopes the re-rack spectra and is not affected.

For certain special subsystems such as the RCL subsystem, response spectra at the exact locations of the subsystems considered (e.g., at the steam generator support or the reactor nozzle) are developed as follows: Floor time histories for the three translational and three

rotational degrees-of-freedom and for each earthquake excitation (SSE and OBE) are derived at the nodes corresponding to the floors which contain the selected locations. Response spectra are developed at these nodes by subjecting a single-degree-of-freedom system with the natural frequency range of interest and various damping ratios to the floor time history motions obtained. The response spectra at the selected points are then developed by rigid body transformations.

Figures 3.7B-51, 3.7B-52, and 3.7B-53 represent the response spectra of translational accelerations in three orthogonal directions at the location of the outermost support of the steam generator for two percent equipment damping and for SSE excitations in X, Y, and Z directions, respectively.

3.7B.2.6 Three Components of Earthquake Motion

The three orthogonal components of the design earthquake motion are assumed to act simultaneously. The combined responses (shears, moments, deflections, and so forth) of structures, components, and elements to the simultaneous application of the two horizontal and one vertical ground excitations are obtained by means of the SRSS method because it is considered unlikely that the peak values of the responses from ground excitations in different directions can coincide. This procedure is in conformance with the recommendations of NRC Regulatory Guide 1.92.

3.7B.2.7 Combination of Modal Responses

When the response spectrum concept of analysis is used, only the maximum modal responses are known and the phasing of modes cannot be determined as in the time history analysis. Therefore, the total response at a point in the multi-degree-of-freedom system can only be approximated. The maximum modal responses are normally combined by SRSS, by absolute sum, or by combinations thereof.

The method of combining maximum modal responses is not straightforward. When frequencies of the modes are closely spaced (differences of 10 percent in frequency of the lower mode), the absolute sum procedure of combining the responses in these modes is used.

When the absolute sum procedure for combining some of the modal responses is used, the total maximum response is obtained by treating the responses resulting from the absolute sum as pseudomodal responses and combining them with all other modal responses in an SRSS manner. This procedure conforms to the recommendations of NRC Regulatory Guide 1.92. When additional conservatism is desired, the total maximum response is obtained by adding the values of the responses resulting from the absolute sum to the SRSS value of the rest of the modal responses.

3.7B.2.8 Interaction of Non-Category I Structures with Seismic Category I Structures

A number of structures such as the Turbine Building, the Switchgear Buildings, the Circulating Water Intake and Discharge Structures, the Maintenance Building, and the Administration Building are designated as non-Category I.

The only non-Category I structures which are adjacent to any seismic Category I structure are the Turbine Building and the Switchgear Buildings. These structures do not share a common mat with the adjacent seismic Category I structure, and all structures are founded on firm rock.

Therefore, there is no possible interaction of non-Category I structures with seismic Category I structures resulting from seismic motion. Sufficient space is provided between the Turbine and Switchgear Buildings and the adjacent seismic Category I structure so as to prevent contact because of deformations occurring in the structures during a seismic event.

The possibility of structural failure during a seismic event is considered for the Turbine Building. Structural failure in the direction of the adjacent seismic Category I structure is prevented by the bearing of the mezzanine and operating floor slabs on the concrete turbine generator pedestal. The Switchgear Buildings are design to withstand a seismic event equal to the SSE.

The seismic Category II high energy piping segments which are located inside the Turbine Building, and attached to a seismic Category I structure are shown by analysis to remain undamaged by non-Category I structures and components during a seismic event. The piping segments involved are:

- portion of the Steam Generator Blowdown line (8-SB-2-060-1302-5)

- portion of the Heater Drain line (8-HD-2-069-302-5)

Non-Category I equipment and components located in seismic Category I buildings are investigated by analysis or testing, or both, to ensure that under the prescribed earthquake loading, structural integrity is maintained, or the non-Category I equipment and components do not adversely affect the integrity or operability, or both, of any designated seismic Category I structure, equipment, or component to the extent that these seismic Category I items cannot perform their required functions.

Sleeve-coupled fittings by LOKRING Technology have been qualified for, and can be used in, non-safety, non-critical B31.1 piping application NPS ¼ to 4 for CS and NPS ¼ to 3 for SS, and including piping whose only design function requirement is to remain in place during and following a Seismic Event, that is mounted Seismic Category II inside Seismic Category I Buildings (Reference FDA-2020-000077-01).

3.7B.2.9 Effects of Parameter Variations on Floor Response Spectra

The instructure response spectra are developed using the time history of the instructure motion resulting from the ground motion time history. The instructure response is evaluated by performing time history modal analysis on a lumped-mass mathematical model which simulates the structure and foundation-structure interaction. Because of the uncertainties associated with the energy dissipation, the variation in elastic properties of both structure and foundation, the idealization of structure with lumped masses and elastic properties in discrete parts, and the frequency content and amplitude modulation of simulated ground motion, the free vibration characteristics and the response of structures can only be approximately computed. Parametric studies and conservative assumptions are made to take into account these uncertainties in construction of instructure response spectra.

The amplifications at resonance peaks of the instructure response spectra are generally not sensitive to frequency-shifting because the eigenvalues and eigenvectors of the structure do not

change appreciably with small to moderate changes of mass or flexibility and the amplification region of the free field is very wide at dominant structure modes.

Damping factors play an important role in determining amplification of the structure; the smaller the damping, the greater the amplification. Therefore, the damping values are determined conservatively as discussed and presented in Subsections 3.7B.1.3, 3.7B.2.4, and 3.7B.2.15. The conservatism of the amplification is also reflected in the chosen artificial ground motion.

The enveloping technique used for the construction of instructure response spectra consists of enveloping the maximum peaks. Since the frequencies of the structures can only be computed approximately because of the linear and nonlinear deformability, the energy dissipation, variation in elastic properties of both structure and foundation, and the idealization of structure with lumped masses and elastic properties in discrete parts, parametric studies are performed in order to take into account these effects for the construction of instructure response spectra. These effects result in the shifting of the resonance peaks of the instructure response spectra. The peaks are widened by at least ±10 percent of the resonance frequencies to account for these effects. The widening exceeds ±10 percent if the parametric studies indicate that such widening is necessary to achieve conservative results. The ground design response spectra and design time history are discussed in Section 3.7B.1.1 and 3.7B.1.2 respectively.

The Fuel Building was re-analyzed to determine the effects of added mass due to the addition of high density spent fuel storage racks. These analyses were based on best estimate soil properties and the resulting spectrum peak responses were widened by at least ± 15% in accordance with RG 1.122. Best estimate soil properties were used in the re-rack analysis in lieu of parametric variations involving upper and lower bound soil properties. Results of previous studies show that variations in soil properties and building structure stiffness have minimal effect on dynamic response (less than 7%). Widening the peaks of instructure spectra based on best estimate soil properties by ± 15% provides additional margin in the design.

The preceding analysis are accomplished by using suitable computer programs as presented in Section 3.7B(A) and in accordance with References [30], [31], [36], and [38].

3.7B.2.10 Use of Constant Vertical Static Factors

Constant static factors such as vertical response loads for the seismic design of seismic Category I structures, systems, and components are not used. Instead, multimass dynamic analysis for both horizontal and vertical directions of excitation is performed as described in Subsection 3.7B.2.1.

3.7B.2.11 Method Used to Account for Torsional Effects

The methods of seismic analysis of structures are presented in Subsection 3.7B.2.1. The torsional effects on structures depend on the geometric configuration of the structure, the elastic properties of the material of the structure, and the foundation structure interaction.

Torsional effects in unsymmetric structures are induced primarily because the centroids of masses which simulate floor slabs and portions of the shear walls between the adjacent floors do not coincide with the shear centers of the shear wall assemblies.

A finite element computer program is used for the seismic Category I structures except the Service Water Intake Structure, seismic Category I Tanks, and re-analysis of the Fuel Building for the addition of high density spent fuel storage racks. The structure walls and floors are modeled with finite elements. The stiffness matrix corresponding to the finite element model is reduced to the number of dynamic degrees-of-freedom required for the dynamic analysis in accordance with References [3], [23], and [38]. The approach used in the seismic analysis accounts for all torsional effects. For the Service Water Intake Structure, seismic Category I tanks, and re-analysis of the Fuel Building, stiffness properties are calculated for the structural elements between lumped mass elevations using standard structural techniques. Static factors are not used nor are any other approximate methods used in lieu of a combined vertical, horizontal, and torsional system dynamic analysis to account for torsional accelerations in the seismic design of seismic Category I structures.

In the design of the Seismic Category I structures, shears resulting from torsion were computed using the larger of either the actual computed eccentricity at each floor elevation or 5% of the maximum building dimension (normal to the direction of excitation).

For seismic Category I structures, the mathematical model is normally composed of masses lumped at floor levels. Each mass is assumed to have six degrees-of-freedom; namely, three translations in the three orthogonal directions and three rotations about these axes which account for the rotational modes of vibration. Torsional rigidities of shear wall assemblies between the floors are determined and taken into account in the analysis. Torsional spring constants of the foundation, associated damping ratios, and effective inertia of the foundation are determined on the basis presented in Subsection 3.7B.2.4. The model used for the re-analysis of the Fuel Building incorporates offsets between the center of mass and the center of rigidity to account for torsional effects.

3.7B.2.12 Comparison of Responses

The results of the response spectrum analysis are used for the design of all seismic Category I structures. The procedure for combining modal responses is presented in Subsection 3.7B.2.7.

The time history method of analysis is used to generate time histories of the instructure support motion at selected critical locations of interest. The maximum responses attained from the time history analysis are compared with the ones resulting from the spectrum analysis. This comparison is used to check both analyses and to verify the conservatism of the procedure for combining modal responses in the spectrum analysis. Responses at selected points in the seismic Category I structures resulting from both response spectrum concept and the time history analysis are listed in Tables 3.7B-46 through 3.7B-52.

3.7B.2.13 Methods for Seismic Analysis of Dams

The Safe Shutdown Impoundment (SSI) Dam impounds water which serves as the ultimate heat sink for the power plant.

The SSI Dam is analyzed using a two-dimensional finite element model in which the dam is assumed to behave elastically and the elastic continuum is modeled as an assemblage of discrete elements connected at a set of common nodal points [25]. The elastic properties are selected based on predictions of the strain distribution in the dam during an earthquake.

Hydro-dynamic forces are included, using one-dimensional elements to model the water [26]. The effect of dam-foundation interaction is studied by extending two-dimensional elements into the foundation. Vertical as well as horizontal ground motion is considered.

The finite element representation yields the stiffness and mass matrices for the structure. The resulting eigenvalue problem for undamped free vibration is solved using conventional eigenvalue techniques [27] to give the natural frequencies and mode shapes for the dam. The mode shapes and spectral response curves are used in a conventional modal superposition method of analysis in which damping coefficients are introduced into the various modes.

It should be noted that the main dam is not considered as a seismic Category I structure. Therefore, no seismic analysis of the main dam is performed.

3.7B.2.14 Determination of Seismic Category I Structure Overturning Moments

A description of the dynamic methods and procedures used to determine seismic Category I structure overturning moments is provided in this subsection. A description of the procedures used to account for foundation reactions and vertical earthquake effects is also included.

All dynamic analyses performed for seismic Category I structures are based on the assumption that the structures can be simulated by mathematical models corresponding to linear elastic lumped systems.

A coefficient of stability C_{st} and an eccentricity e shown on Figure 3.7B-40, which govern the evaluation of the dynamic contribution to the foundation pressure beneath the base, are determined.

The degree of stability is dependent on the coefficient of stability C_{st}, which is defined as the ratio of the resisting moment to the overturning moment taken about the extreme point A of the foundation (see Figure 3.7B-40) as expressed in the following equation:

$$C_{st} = \frac{Wa}{M_o} \qquad (3.7B\text{-}22)$$

where

W	=	the weight of the entire structure
a	=	the horizontal distance from the center of gravity of the entire structure to the point of rotation designated as A on Figure 3.7B-40
M_o	=	the total overturning moment about point A induced by seismic disturbances.

The overturning moment in each significant structure mode is given as:

$$M_i = \sum_{n=1}^{N} M_{i,n} + \sum_{n=1}^{N} H_{i,n}\bar{Y}_n + \sum_{n=1}^{N} V_{i,n}\bar{X}_n \qquad (3.7B\text{-}23)$$

where

N	=	the total number of nodes in the dynamic model of the structure
$M_{i,n}$	=	the modal inertia moment at each node
$H_{i,n}$	=	the horizontal modal inertia force at each node
$V_{i,n}$	=	the vertical modal inertia force at each node
$\bar{X}_n$	=	the horizontal distances from the lines of action of the vertical inertia forces to the point of rotation A
$\bar{Y}_n$	=	the vertical distances from the lines of action of the horizontal inertia forces to the point of rotation A

The total overturning moment M_o in the stability Equation 3.7B-22 is obtained from the combined effect of all dominant structure modes. The modal overturning moments for each direction of ground excitation are combined, as discussed in Subsection 3.7B.2.7. The combined overturning moments resulting from the vertical and the two horizontal ground motions are calculated by the SRSS combination. In addition, the overturning moment resulting from the vertical ground motion is combined with the larger of the two overturning moments resulting from the two horizontal ground motions by absolute sum combination. Both results are compared and the stability of the structures is checked based upon the most critical of the two cases. For a conservative evaluation of stability against overturning, the vertical seismic forces acting on the seismic Category I systems are considered to act in an upward direction and thus combined with the lateral forces.

An alternate method for calculating the total overturning moment M_o is as follows:

Horizontal and vertical inertia forces are obtained by multiplying the building mass at each story level by the corresponding building acceleration. The building acceleration used considers the three directions of earthquake motion, (i.e., the square root of the sum of the squares of the maximum representative accelerations of the codirectional accelerations caused by each of the three components of earthquake motion). The overturning moment due to horizontal inertia forces is determined by multiplying the individual horizontal inertia force by the vertical distance from it's center of mass to the point of rotation. Similarly, the overturning moment due to vertical inertia is found by multiplying the individual vertical inertia force by the horizontal distance from its center of mass to the point of rotation. The total overturning for the structure, in a particular direction, is found by summing the individual overturning moments due to the horizontal inertia force in the corresponding direction and the overturning moment due to the vertical inertia force.

The criteria used to select an acceptable structural configuration are as follows:

1. The stability coefficient C_{st} is not less than 1.10 for the SSE.

2. Maximum static and dynamic foundation pressures remain within the allowable ultimate capacity.

To calculate dynamic foundation pressures, the resultant reaction R and the eccentricity e shown on Figure 3.7B-40 are first determined. The resultant reaction is a function of the weight of the entire structure and the total vertical seismic load. The total vertical seismic load is determined by the combined effects of all dominant structure modes induced by horizontal and vertical seismic excitations; i.e., the resultant modal vertical earthquake loads for each direction of ground motion and the simultaneous effect of three earthquakes are combined as discussed in Subsection 3.7B.2.7.

3.7B.2.15 Analysis Procedure for Damping

Equivalent modal damping is evaluated using the concept of weighted modal damping [13]. This is an approximate rule for determining modal damping by weighing the damping associated with the individual components according to the energy stored in each component. Concrete structures, steel structures and systems, and foundation materials have inherently different damping properties; the effective damping in any vibration mode of the total system depends on the degree of participation of these components in the modal response.

The weighted modal damping is evaluated as follows:

$$D_n = \frac{\Sigma E_{nc} D_c}{\Sigma E_{nc}} \qquad (3.7B\text{-}24)$$

where

D_n = the weighted damping expressed as a fraction of critical damping in mode n

D_c = the damping expressed as a fraction of critical damping for component c

E_{nc} = the modal energy stored in component c

If a dynamic analysis is performed in a coupled system of equations of motion using a direct integration technique, a damping matrix [C] is computed and used in the dynamic solution. The damping matrix is expressed as a function of critical modal damping [23], [28], [37], [38] in the following equation in matrix form:

$$[C] = 2[m][\Phi][M]^{-1}[\omega]\,[\beta]\,[\Phi]^{T}[m] \qquad (3.7B\text{-}25)$$

where

$[m]$	=	mass matrix
$[\Phi]$	=	mode shape matrix
$[\Phi]^T$	=	transpose of mode shape matrix
$[M] = [\Phi]^T[m][\Phi]$	=	generalized mass matrix
$[\omega]$	=	diagonal matrix of natural circular frequencies
$[\beta]$	=	diagonal matrix of modal damping expressed in fractions of critical damping

Equation 3.7B-25 can be used in the solution of the coupled differential equations of motion presented in matrix form as follows:

$$[m]\{\ddot{u}\} + [c]\{\dot{u}\} + [k]\{u\} = \{f(t)\} \qquad (3.7B\text{-}26)$$

where

$[k]$	=	stiffness matrix
$\{\ddot{u}\}, \{\dot{u}\}, \{u\}$	=	column matrices of relative accelerations, velocities, and displacements, respectively
$\{f(t)\}$	=	column matrix of time dependent forcing functions

3.7B.3 SEISMIC SUBSYSTEM ANALYSIS

3.7B.3.1 Seismic Analysis Methods

The seismic Category I subsystems which comprise all seismic Category I mechanical and electrical systems and components and their supports, are analyzed by the response spectrum method, or the equivalent static load method. The seismic excitation used for the analysis of subsystems is obtained from the appropriate floor response spectra as described in Section 3.7B.2.

Subsystems and equipment are generally analyzed by the response spectrum method which consists of the following steps:

1. The system is modeled using a multi-degree-of-freedom representation. The size of the model is reviewed to assure that an adequate and sufficient number of masses or degrees of freedom are used to compute the response of the system. A sufficient number is considered adequate provided additional degrees of freedom do not result in more than a 10% increase in response, or the number of degrees of freedom equals or exceeds twice the number of modes with frequencies less than 33 Hz, or based upon engineering

judgement that the number of degrees of freedom chosen will meet or exceed the intent of the prior two guidelines. The mathematical model consists of a network of elastic springs and lumped masses which are selected to coincide with actual mass concentrations, or to provide adequate representation of mass distribution. The degrees of freedom assigned to each mass of the model correspond to translations along three orthogonal axes, and rotations about these axes, completely defining the motion of the mass point in space. At points of support the total number of degrees of freedom is reduced by the number of constraints at these points. Reduced degrees of freedom are acceptable provided the analysis adequately and conservatively predicts the dynamic response of the equipment.

2. A dynamic analysis is performed to determine the natural frequencies and mode shapes of the mathematical model.

3. The participation factors for each direction of support motion are calculated.

4. For each significant mode a spectral response value corresponding to the modal natural frequency is determined from the applicable floor response spectra.

5. The modal responses consisting of modal absolute accelerations, modal relative displacements, and modal inertia loads are calculated for a sufficient number of modes. The results are combined by the SRSS method as described in paragraph 3.7B.3.7 or other methods in conformance with USNRC Regulatory Guide 1.92.

The number of modes chosen is considered adequate provided the inclusion of additional modes does not result in more than a 10% increase in responses, or based upon evaluation of the dynamic participation factors to assure that all significant modes have been included.

An analytical technique, developed in accordance with NUREG/CR-1161 (Reference 41), is used for piping systems to account for the modal contribution above the cutoff frequency.

Structurally simple equipment and systems, which can be represented either by a single degree-of-freedom model or a simple mathematical model, and equipment and subsystems which have been found to have no natural frequencies below 33 Hz are generally analyzed by the equivalent static load method as described in Section 3.7B.3.5.

The seismic response loads obtained by either the modal response analysis or equivalent static load method are combined with all other external loads such as operating loads, hydrodynamic loads, and piping interaction loads for design purposes. Non-linear responses of subsystems are considered on an individual basis where such phenomena are identified as existing, and are accounted for by analysis. Such an analysis was performed to account for the predetermined support clearance tolerances of the Service Water Intake Structure pumps.

For further details on seismic analysis methods, see Section 3.7B.2.1.

3.7B.3.2 Determination of Number of Earthquake Cycles

The number of maximum amplitude loading cycles is specified for seismic Category I structures, systems, and components as a minimum of 600 loading cycles for the OBE, and 120 loading cycles for the SSE.

For ASME Code Class 2 and 3 piping systems including supports for ASME Code Class 1, 2, and 3 piping a minimum of 50 loading cycles for the OBE and 10 loading cycles for the SSE is specified.

3.7B.3.3 Procedure Used for Modeling

The dynamic analysis of any complex system requires the discretization of its mass and elastic properties. This is accomplished by concentrating the mass of the system at distinct characteristic points or nodes, and interconnecting them by a network of elastic springs representing the stiffness properties of the systems, which are computed either by hand calculations for simple systems, or by finite element methods for more complex systems. Nodes are located at all mass concentrations and at additional points within the system, selected in such a way as to provide an adequate representation of the mass distribution of the system. At each node, degrees of freedom corresponding to translations along three orthogonal axes, and rotations about these axes are assigned. The number of degrees of freedom is reduced by the number of constraints, where applicable. For equipment qualification, reduced degrees of freedom are acceptable provided the analysis adequately and conservatively predicts the response of the equipment.

For additional description of modeling procedures see Sections 3.7B.2.3 and 3.7B.3.1.

3.7B.3.4 Basis for Selection of Frequencies

In general, subsystems can be analyzed using time history or model response spectrum methods, or both.

Mathematical models representing subsystems are subjected to their support motions, which reflect the seismic environment of the free- field and structural amplifications. Therefore, when these support motions are used as input to the dynamic system, each mode responds according to the amplification which has been predetermined in the time history analysis of the supporting structure.

Elimination of resonance condition is considered good practice in the design of subsystems. The resonance peaks are readily identified from the appropriate response spectra. Elimination of resonance is the principal aim of the design. To eliminate this resonance condition, the fundamental frequency of seismic Category I subsystems can be selected to be greater than twice or less than one-half the dominant frequency of the supporting system or some modification of the dominant natural frequencies can be achieved by providing stiffer or more flexible supports and smaller or bigger mass characteristics of the subsystem. When this becomes impossible or impractical, the subsystem is analyzed and designed for amplified responses, including the resonance condition.

3.7B.3.5 Use of Equivalent Static Load Method of Analysis

Where a subsystem can be adequately and realistically represented as a one-degree-of-freedom system, and no determination of natural frequency is made, the response of the subsystem is assumed to be the peak acceleration of the appropriate floor response spectra curves at the appropriate value of damping.

For a subsystem which can be adequately and realistically represented as a simple model, similar to the guidelines of NRC Regulatory Guide 1.100, Rev. 1, and produce conservative analysis results, and no determination of natural frequencies is made, the response of the subsystem is assumed to be the peak of the appropriate floor response spectra at the appropriate value of damping multiplied by a factor of 1.5. A factor less than 1.5, but not less than 1.0 may be used, provided conservative results are obtained and proper justification provided.

Equipment having a minimum natural frequency equal to or greater than 33 Hz is also sometimes designed by the equivalent static load method, in which case the applied seismic loads correspond to accelerations equal to at least the zero-period accelerations of the appropriate floor response spectra.

3.7B.3.6 Three Components of Earthquake Motion

The combined effect of the three components of earthquake motion on seismic Category I subsystems is determined by the SRSS method as described in Subsection 3.7B.2.6.

3.7B.3.7 Combination of Modal Responses

When the response spectrum concept of analysis is used, only the maximum modal responses are known, and the phasing of modes cannot be determined. Therefore, the total response at a point in the multi-degree-of freedom system can only be approximated. The maximum modal responses are combined by the methods of NRC Regulatory Guide 1.92, Revision 1. For equipment and subsystem analyses, the methods presented in the Regulatory Guide paragraphs 1.1, 1.2.1, 1.2.2, or 1.2.3 are acceptable methods for vendor qualification.

3.7B.3.8 Analytical Procedures for Piping

3.7B.3.8.1 Design Criteria

Piping design criteria for Code Class 1 piping are in accordance with NB-3000 of the ASME B&PV Code, Section III. For Code Class 2 and Code Class 3 piping, see Section 3.9B.3.

Piping is anchored so that the total movements caused by relative building motion plus thermal growth do not overstress the system.

Critical areas of valve and piping inside the Containment are affected by relative motion between the Containment Building and the internal structure. Similar criteria are followed in these areas, especially at elevations where relative movements between Containment wall and internal structure are greater.

Piping is analyzed as an elastic system subject to thermal loadings and given displacements at anchor points.

Two analyses are made to determine the following:

1. Stresses imposed by thermal movements between equipment and anchors and by anchor movements between structures

2. Dynamic stresses imposed by seismic loading as a result of relative motion of buildings

Each piping system is idealized as a mathematical model consisting of lumped masses connected by elastic members. In order to adequately represent the dynamic and elastic characteristics of the piping system, lumped masses are located at carefully selected points. Sufficient mass points are located to ensure that all modes with frequencies less than 33 Hz are considered in the analysis. The number of degrees of freedom is verified to be equal to or greater than twice the number of modes with frequencies less than 33 Hz. In the modeling of the piping system, valves, reducers, tee and branch connections attached to the pipe are included. The location, type and stiffness of supports provided are reviewed and included in the analysis. Class 2 and 3 piping systems, whose nominal pipe diameter is 2 inches or less and whose operating temperature is less than 200°F may be analyzed by using section 3.7B.3.5 for seismic inertial effect.

Anchors with all six degrees restrained have thermal movement included in the analysis (i.e., anchors at equipment nozzles, containment penetrations, or embedded pipes).

There are three (3) categories of displacement for each direction of earthquake. Two of these categories represent rigid body motion of the structure, motions that are common to all points on the structure. The third category represents deformation of the structure, that is relative displacements of points on the structure.

When all of the points of fixity are located on a single structure, the rigid body motions of the structure, translation and rotation, do not result in relative motion of the points of fixity. Since the third category of displacement, deformation of the structure, represents a small portion of the total displacement profile, the effects of this displacement on the points of fixity are neglected.

For piping passing between buildings or equipment mounted on individual structures or foundations (such as big tanks), the relative displacement of support points located in different structures is considered in piping stress analysis.

Maximum relative displacements in two horizontal and the vertical direction between piping supports and anchor points between buildings are used as equivalent static displacement boundary conditions in order to calculate the secondary stresses of the piping system. Relative seismic displacements used are obtained from a dynamic analysis of the structures, and are always considered to be out-of-phase between different buildings and the equipment if applicable to obtain the most conservative piping responses.

3.7B.3.8.2 Basis for Computing Combined Responses

For the seismic design of piping, the horizontal and vertical loadings are obtained from the instructure response spectra that have been generated for the appropriate structures and elevations as outlined in Subsection 3.7B.2.1.2, and References [30], [31], and [36].

The combined effect of the three components of earthquake motion on the seismic design of piping is determined by the SRSS method (section 3.7B.2.6). The maximum modal responses are combined by the methods of NRC Regulatory Guide 1.92, Revision 1. The methods presented in Regulatory Guide paragraphs 1.1, 1.2.1, 1.2.2 or 1.2.3 are acceptable methods for vendor qualification.

Restraints are designed for loadings that are obtained from the piping analysis.

3.7B.3.8.3 Amplified Seismic Responses

For the seismic design of piping, input loading is obtained from the vertical and two horizontal modal response spectra curves for the appropriate damping of the building and/or structure.

Where a piping system is subjected to more than one amplified response spectrum, such as support points located in different structures or different elevations of the same structure, either of the following two methods shall be used, to generate loads at the pipe supports of the piping system.

1. Envelope all the amplified response spectra and apply to the piping system

2. Utilize independent support motion by applying the applicable amplified response spectra to each pipe support of the piping system.

3.7B.3.9 Multiple Supported Equipment Components with Distinct Inputs

The seismic analysis of multiply supported seismic Category I subsystems and equipment subjected to differential support motion within a building or between two buildings is performed in three parts, using lumped mass mathematical models, as follows:

1. Modal response spectrum analysis is performed for all three principal orthogonal directions of support motion for each direction of ground excitation using appropriate in-structure response spectra, constructed on the basis of superimposing the spectra for all support points and enveloping them as stated in Subsection 3.7B.2.5. The vertical analysis is combined with both horizontals as described in Subsection 3.7B.2.1.2, Item 1.

2. The same multimass lumped parameter model is subjected to a static analysis for the differential displacements of the support points. The displacements used are consistent with the directions of structural excitation considered in the spectrum analysis. This results in basic differential displacement loading conditions.

3. The results obtained from the spectrum analysis and differential displacement analysis are then combined absolutely. The effects of these loading conditions on the components and the supporting structures are determined.

3.7B.3.10 Use of Constant Vertical Static Factors

Constant static factors are used in some cases for the design of seismic Category I subsystems and equipment. The criteria for using this method are presented in Subsection 3.7B.3.5.

3.7B.3.11 Torsional Effects of Eccentric Masses

The criteria used to account for the torsional effects of valves and other eccentric masses (e.g., valve operators) in the seismic piping analyses are as follows:

1. When valves and other eccentric masses are considered rigid, the mass of the operator and valve body or other eccentric mass will be located at their respective center of gravity. The eccentric components (i.e., yoke, valve body) will be modeled as rigid members.

2. When valves and other eccentric masses are not considered rigid, the dynamic models are simulated by the lumped masses in discrete locations (i.e., center of gravity of valve body and valve operator), coupled by elastic members with properties of the eccentric components.

3.7B.3.12 Buried Seismic Category I Piping Systems and Tunnels

For seismic Category I piping systems outside the Containment structure, including those placed in underground concrete conduits but excluding those directly buried underground, the same design criteria and analytical procedures described in Subsection 3.7B.3.8 are used to ascertain that allowable piping and structural stresses are not exceeded at Containment penetrations and at entry points into other structures.

Some seismic Category I piping systems are comprised of segments which are completely buried underground and which interface with the Auxiliary Building or the Service Water Intake Structure, or with other seismic Category I structures. Other seismic Category I piping segments are enclosed in concrete conduits which are buried underground and are connected to the conduit walls by appropriate restraints and supports.

All seismic Category I buried piping and concrete conduits are encased in a lean concrete fill or located in compacted backfill with a density sufficient to ensure that the backfill does not lose its integrity as a result of liquefaction during an SSE. If required, the effects of small settlements of structures on adjacent piping are reduced by providing flexible joints, split sleeves, and similar devices. Consolidation of the backfill is expected to be negligible under the pipe and conduit weights. Shearing distortions assumed for the design of the piping and conduits are based on consideration of the elastic properties of the compacted backfill or concrete fill, as well as those of the surrounding natural ground.

The following procedures are considered in the design of seismic Category I buried piping and concrete conduits.

3.7B.3.12.1 Stresses Caused by Free-Field Seismic Wave Propagation

The basic concept governing the response of those portions of buried underground pipes and concrete conduits away from the interaction effects of any external support except the surrounding soil is that, compared to the pipes and conduits, the soil is stiff; therefore, the earthquake deformation of the soil is imposed on the pipes and conduits which must conform to this deformation. Thus, the complex dynamic analysis of a discrete system involving soil-pipe or soil-conduit interaction that may be required for relatively soft soils is not considered. The analysis of seismic Category I tunnels is consistent with the recommendations of both References 15 and 29. The procedures outlined in paragraph 10.6 of Reference 15 are employed in analyzing buried piping.

1. The maximum bending stress resulting from shear wave is obtained by assuming that the shear wave travels parallel to the axis of the pipe or conduit. The curvature of the pipe or conduit is then determined as the ratio of the maximum ground acceleration

perpendicular to the axis of the pipe or conduit and the shear wave velocity squared [15]. The product of the curvature and the flexural rigidity of the pipe or conduit yields the bending moment from which the unit stresses are obtained.

2. The maximum axial stress resulting from compressional wave is obtained by assuming that the compressional wave propagates in the soil along the pipe axis. The axial strain is then determined as the ratio of the maximum ground velocity in the direction of the axis of the pipe or conduit and the compressional wave velocity [15], [16], [17], [21], [22]. The maximum axial stress is obtained as a product of the axial strain and the modulus of elasticity of the pipe material. Slippage occurs if the total induced axial force over a straight pipe exceeds the friction force between the soil and the pipe surface.

   Then, the maximum axial stress in a buried pipe or concrete conduit under free-field conditions corresponds to the limiting frictional force.

3. The maximum axial stress caused by a shear wave occurs when the shear wave propagates at the incident angle of 45 degrees from the longitudinal axis of the pipe or concrete conduit [18], [21], [22]. This subjects different parts of the pipe or concrete conduit to out-of-phase displacements and results in a compressional wave propagating along the axis of the pipe or concrete conduit. The maximum induced axial stress is computed as the product of the strain, which can be obtained in accordance with References [18], [21], and [22] and the modulus of elasticity of the pipe material.

4. The maximum combined axial stress caused by stretching and bending strains induced by a single shear wave is computed following the studies concerning subways [18], [21]. It should be noted that it is impossible to have a single shear wave that simultaneously induces the maximum stresses described in Subsection 3.7B.3.12.1, Items 1 and 3. The axial effect of the shear wave propagation assumes its maximum value at the incident angle of 45 degrees measured from the longitudinal axis of the pipe or conduit, while the maximum bending effect value is at the incident angle of shear wave propagation of 0 degrees that is measured in the same way. The maximum strain is obtained by maximizing the expression of the sum of the two effects as presented in References [18], [21], and [22], the results of which are extended to buried pipe applications as in Reference [19].

3.7B.3.12.2 Stresses Caused by Differential Displacements Between Soil and Structure

As a result of soil-structure interaction, differential displacements during seismic disturbance are usually experienced between the structure and the soil at the entry points of buried pipes. The maximum horizontal and vertical differential displacements are obtained by performing the seismic spectrum analysis of each seismic Category I structure. These displacements are used in obtaining additional stresses in buried pipes. For pipes extending from one structure to another, an out-of-phase assumption is made to account for the possible phase differences of the seismic ground waves.

1. Bending and shearing stresses caused by differential displacements perpendicular to the pipe axis are obtained from the studies concerning elastic pile theory involving coefficients of subgrade reaction [4], [5], [6], [23]. When the soil surrounding the pipe can be assumed to be a homogeneous isotropic medium, solutions for beams on an elastic foundation such as the ones presented in Reference [24] are used.

2. The maximum axial stresses resulting from differential displacement along the pipe or conduit axis are computed from the consideration of load transfer from the pipe or conduit to the surrounding soil by friction to accommodate axial differential displacement at the location where the pipe is entering a structure, as well as from the elastic deformation of the soil at the other end of the pipe or conduit. However, a conservative estimate of this maximum axial stress can be obtained as the product of the axial displacement, the coefficient of horizontal subgrade reaction, and the ratio of the moduli of elasticity of the pipe or conduit material and the soil. The procedure is based on the assumption that the strain in the pipe or conduit is the same as that of the surrounding soil.

If the computed combined stresses, which include stresses resulting from earthquake, internal pressure, thermal expansion, and other operating loads, exceed the allowable limits at the penetrations, one or more of the following devices are used to relieve the stresses caused by the differential displacements:

1. The portions of the pipe at the entry points are protected from soil pressure by providing a concentric split sleeve.

2. The stresses resulting from differential displacements are reduced by replacing the compacted backfill soil or concrete fill around the pipe near the penetrations by another softer soil material.

3.7B.3.13 Interaction of Other Piping with Seismic Category I Piping

3.7B.3.13.1 Seismic Category I Piping with Connecting Non-Category I Piping

Interaction of seismic Category I piping with non-Category I piping connected to it is considered in the following two respects:

1. The loads transmitted under seismic excitation between the two systems locally at the point of their connection

2. The effect of the non-Category I system on the dynamic characteristics and the seismic response of the seismic Category I system

Consideration of both effects is achieved by incorporating into the analysis of the seismic Category I system a length of pipe that represents the actual dynamic behavior of the complete run of the non-Category I system. The length considered extends, but is not limited to, the first anchor point beyond the point of change from seismic Category I to non-Category I. Whenever possible, an anchor is located at the intersection of the seismic Category I piping with the non-Category I piping. In cases where location of the anchor or restraint is not possible at the category change, it is placed on the non-Category pipe, and that portion of the line up to the anchor or restraint is analyzed according to seismic Category I criteria. In either case, the non-Category I piping is always isolated from the Category I piping by anchors or seismic restraints.

3.7B.3.13.2 Seismic Category I Piping with Adjacent Non-Category I Piping

Non-Category I Piping Systems whose failure is not acceptable, adjacent to Seismic Category I piping, are analyzed by the nomograph method, other simplified dynamic analysis methods, or

enveloping methods that will limit stress levels to assure structural integrity and prevent any unacceptable physical interaction with adjacent seismic Category I piping and components. The nomograph method provides seismic restraint spacing based on the natural frequency of the supported piping. This support spacing assures that the first natural frequency of the non-Category I piping is beyond that value which is twice the resonant frequency.

3.7B.3.14 Seismic Analyses for Reactor Internals

Seismic analyses for the reactor internals are presented in Section 3.7N.

3.7B.3.15 Analysis Procedure for Damping

Damping values expressed as percents of critical damping are determined for the type of material and fabrication of subsystems in accordance with the recommendations of NRC Regulatory Guide 1.61.

Typical damping values are presented in Table 3.7B-1. For the analysis of multidegree-of-freedom systems, equivalent modal dampings are determined according to the concept of weighted modal damping as described in Subsection 3.7B.2.15 and in Reference [13].

3.7B.3.16 Analysis Procedure for High Density Spent Fuel Storage Racks

The analysis undertaken to confirm the structural integrity of the high density spent fuel storage racks was performed in compliance with the USNRC Standard Review Plan [46, 49, 50, and 51] and the OT Position Paper [47]. The primary objective of the analysis is to determine the nature of the dynamic response of the SFP rack configuration to plant-specific seismic excitation and to establish the factor of safety. Both single rack and Whole-Pool-Multi-Rack (WPMR) analyses were performed. In the WPMR analysis, the entire assemblage of rack modules was modeled in one comprehensive simulation.

The response of a freestanding rack module to seismic inputs is highly nonlinear and involves a complex combination of motions (sliding, rocking, twisting, and turning), resulting in impacts and friction effects. Some of the unique attributes of the rack dynamic behavior include a large fraction of the total structural mass in a confined rattling motion, friction support of rack pedestals against lateral motion, and large fluid coupling effects due to deep submergence and motion of closely spaced adjacent structures.

Linear methods, such as modal analysis and response spectrum techniques, cannot accurately simulate the structural response of such a highly nonlinear structure to seismic excitation. Therefore, evaluation of the high density spent fuel storage racks is based on direct integration of the nonlinear equations of motion with the pool slab acceleration time-histories applied as the forcing functions.

The three-dimensional single rack dynamic model used in the rack analyses has the capability to effect momentum transfers which occur due to rattling of fuel assemblies inside storage cells and the capability to simulate lift-off and subsequent impact of support pedestals with the pool liner (or bearing pad). The contribution of the water mass in the interstitial spaces around and between the rack modules and around fuel assemblies was also included. This fluid mass significantly alters the dynamic behavior of fuel and rack masses during a seismic event.

Despite the versatility of the three dimensional seismic model, the accuracy of the single rack simulations has been suspect due to one key element; hydrodynamic participation of water around the racks. During dynamic rack motion, hydraulic energy is either drawn from or added to the moving rack, modifying its submerged motion in a significant manner. Therefore, the dynamics of one rack affects the motion of all others in the pool.

For closely spaced racks, demonstration of kinematic compliance was verified by including all modules in one comprehensive simulation using a WPMR model. In the WPMR analysis, all rack modules were modeled simultaneously and the coupling effect due to this multi-body motion was included. The worst case loads and stresses that result from either the three-dimensional single rack dynamic model and the WPMR model were used to determine the structural adequacy of the racks.

3.7B.4 SEISMIC INSTRUMENTATION

Seismic instrumentation is provided within the plant so that in case of an earthquake, sufficient data is generated to permit verification of the dynamic analysis of the plant and evaluation of the safety of continued operation.

3.7B.4.1 Comparison with Regulatory Guide 1.12

The seismic monitoring system meets the intent of RG 1.12, Rev. 1, with respect to the ability to determining exceedance of the OBE in a timely manner. Seismic instrumentation includes only the free-field triaxial accelerometer installed in the Yard. For the purpose of determining whether or not the OBE has been exceeded, ANSI/ANS-2.10-2003, “Criteria for the Handling and Initial Evaluation of Records from Nuclear Power Plant Seismic Instrumentation,” uses the free-field accelerometer data as input to address 10CFR100 Appendix A requirements in paragraph (V)(a)(2). Exceedance above the OBE ground motion level will require structural response analyses to be performed (ANSI/ANS-2.23-2002, “Nuclear Power Plant Response to an Earthquake”). The seismic monitoring system used to record a seismic event is in accordance with the requirements described in ANSI/ANS-2.2-2002, “Earthquake Instrumentation Criteria for Nuclear Power Plants.” The seismic monitoring system has the following components:

1. A triaxial accelerometer located in the free-field. The function of the triaxial accelerometer is to provide the acceleration time-history responses where the affects associated with surface features, buildings, and components will be insignificant.

2. A digital recorder that continuously monitors the free-field accelerometers with a solid state archival system. The function of the recorder is to provide a recording of the seismic event for post-event evaluation. The recorder will capture the 3-seconds preceding the exceedance of the seismic trigger threshold, and will continue to record data for a minimum of 5-seconds beyond the last recorded exceedance of the seismic trigger threshold.

3. A controller that will evaluate the recorded data in a timely manner. The function of the controller is to determine whether the OBE has been exceeded.

4. An uninterrupted power supply battery backup. The function of the battery backup is to ensure a minimum of 25-minutes of recorded data will be collected.

5. The free-field accelerometers will also function as the seismic switch. The recorder will continuously monitor the free-field accelerometers. The seismic trigger is a threshold ground acceleration value designed to initiate the event recording process at a value significantly below that of the OBE.

A schematic diagram of the seismic monitoring system is presented on Figure 3.7B-54. The free-field triaxial accelerometer described previously is provided for CPNPP Unit 1 as allowed by Section 4.5 of ANSI/ANS-2.2-2002. The seismic monitoring system conforms to the requirements of ANSI/ANS-2.2-2002, as described above, including calibration and channel checks.

3.7B.4.2 Location and Description of Instrumentation

The recorder, controller, computer screen display, printer, and uninterruptible power system are located in an instrument rack in the Control Room. In case of any seismic activity of sufficient intensity to activate the seismic monitoring system, the Control Room operator is alerted by means of the seismic annunciation system, which consists of visual and audible alarms. The first notification will be that an event is in progress by the exceedance of the seismic trigger threshold and activation of the recorder. The second notification will be provided by the controller to indicate that the OBE has been exceeded and that shut down is required. A redundant set of notification indications is provided on the front of the seismic monitoring system in the form of light-emitting diodes (LEDs).

3.7B.4.3 Control Room Operator Notification

This equipment computes and displays the free-field response spectra automatically on the computer screen following the event and prints a hardcopy on the system's printer. No Operator action is required to compute and print the response spectra. The seismic monitoring instrumentation facilitates the Operators ability to rapidly determine whether or not the OBE has been exceeded. In the event that the system fails to provide annunciation to one or both Control Rooms, local LED indication is provided on the face of the controller. The Control Room Staff will be alerted to the seismic event through perception of the building motions and external news sources. In addition, the Control Room Operator is provided various screen displays that present the results of the analysis of the ground motion data. Therefore, failure to fully annunciate will not prevent the Control Room Staff from being alerted to the situation. The operator is required to request maintenance and engineering support to evaluate the validity of any alarm or indication of a seismic event. In response to a seismic event, the operator is required to perform focused walkdowns to assess plant damage and the availability of safe shutdown equipment. The operator is required to shutdown both units if the Seismic Monitoring System determines that the OBE has been exceeded. In the event that the System is not available or functioning, the operator is required to request engineering to determine whether or not the OBE was exceeded.

3.7B.4.4 Comparison of Measured and Predicted Responses

Recorded actual time histories from a significant seismic event that has occurred at the site are used to determine whether the OBE has been exceeded. The acceleration response spectrum from the actual time histories will be compared to the design spectra in the FSAR Figures 3.7B-1 and 3.7B-6. Due to large conservatism in the generation of in-structure responses, a detailed comparison of measured to predicted responses is not required.

REFERENCES

1. Richart, F.E., Jr., Hall, J.R., Jr., and Woods, R.D., 1970, Vibrations of Soils and Foundations, Prentice-Hall, Inc., Englewood Cliffs, New Jersey.

2. Whitman, R.V., and Richart, F.E., Jr., November 1967, Design Procedures for Dynamically Loaded Foundations, Journal of the Soil Mechanics and Foundation Division, Proceedings of the American Society of Civil Engineers.

3. Pestel, E.C., and Leckie, F.A., 1963, Matrix Methods in Elastomechanics, McGraw-Hill Book Company, Inc., New York.

4. Vesic, A.B., Bending of Beams Resting on an Isotropic Elastic Solid, Proceeding of the American Society of Civil Engineers, April 1961, vol. 87, no. EM2.

5. Terzaghi, K., Evaluation of Coefficients of Subgrade Reaction, Geotechnique Institution of Civil Engineers, London, 1955, vol. V, pp. 297-326.

6. Matlock, H., and Reese, L.C., 1962, Generalized Solutions for Laterally Loaded Piles, American Society of Civil Engineers Transactions, paper no. 3370, vol. 127, Part I.

7. Novak, M., Vibrations of Embedded Footings and Structures, American Society of Civil Engineers National Structural Engineering Meeting, San Francisco, California, April 9-13, 1973.

8. Timoshenko, S., and Goodier, J.N., 1951, Theory of Elasticity, 2nd ed., McGraw-Hill Book Company, Inc., New York.

9. Benjamin, J.R., 1959, Statically Indeterminate Structures, McGraw-Hill Book Company, Inc., New York.

10. Terzaghi, K., 1965, Theoretical Soil Mechanics, John Wiley and Sons, Inc., New York.

11. Marks' Mechanical Engineers' Handbook; Baumeister, T., ed., 1958, McGraw-Hill Book Company, Inc., New York, 6th ed.

12. Dunham, C.W., 1962, Foundations of Structures, McGraw-Hill Book Company, Inc., New York.

13. Whitman, R.V., 1970, Soil-Structure Interaction, in Hansen, R.J., ed., Seismic Design for Nuclear Power Plants, The M.I.T. Press, Cambridge, Massachusetts.

14. Newmark, N.M., Blume, J.A., and Kapur, K.K, Design Response Spectra for Nuclear Power Plants, American Society of Civil Engineers National Structural Engineering Meeting, San Francisco, April 1973.

15. Newmark, N.M., and Rosenblueth, E., 1971, Fundamentals of Earthquake Engineering, Prentice-Hall, Inc., Englewood Cliffs, N.J.

16. Sakurai, A., and Takahashi, T., Dynamic Stresses of Underground Pipe Lines During Earthquakes, Proceedings--The Fourth World Conference on Earthquake Engineering, Santiago, Chile, 1969, vol.II.

17. Newmark, N.M., Problems in Wave Propagation in Soil and Rock, Proceedings-International Symposium on Wave Propagation and Dynamic Properties of Earth Material, Albuquerque, New Mexico, 1967.

18. Kuesel, T.R., Earthquake Design Criteria for Subways, Proceedings-American Society of Civil Engineers, June 1969, vol. 95, no. ST6, pp. 1213-1231.

19. Hadjian, A.H., Discussion of the Paper, Earthquake Design Criteria for Subways by T.R. Kuesel, Proceedings-American Society of Civil Engineers, January 1970, vol. 96, no. ST1, pp. 159-160.

20. Jacobsen, L.S., and Ayre, R.S., 1958, Engineering Vibrations, McGraw-Hill Book Company, Inc., New York, p. 110.

21. Yeh, G.C.K., October 1974, Seismic Analysis of Slender Buried Beams, Bulletin of the Seismological Society of America, vol. 64, no. 5, pp. 1551-1562.

22. Iqbal, M.A., and Goodling, E.C., Jr., Seismic Design of Buried Piping, Second American Society of Civil Engineers Specialty Conference-Structural Design of Nuclear Plant Facilities, New Orleans, Louisiana, December 8 10, 1975.

23. Stoykovich, M., 1973, Evaluation of Torsional Effects in the Seismic Analysis of Symmetric and Unsymmetric Nuclear Power Plant Structures, Nuclear Engineering and Design, North-Holland Publishing Company, Amsterdam, vol. 24, no. 1, pp. 110-124.

24. Timoshenko, S., Strength of Materials, Part II, An Advanced Theory and Problems, D. Van Nostrand Company, Inc., New York, 2nd ed.

25. Clough, R.W., and Chopra, A.K., 1966, Earthquake Stress Analysis in Earth Dams, Journal of the Engineering Mechanics Division, American Society of Civil Engineers, vol. 92, no. EM 2.

26. Chopra, A.K., and others, Earthquake Analysis of Reservoir Dam Systems, Proceedings-Fourth World Conference on Earthquakes, Chile, 1969.

27. Bathe, K.J., and Wilson, E.L., 1972, Large Eigenvalue Problems In Dynamic Analysis, Journal of the Engineering Mechanics Division, American Society of Civil Engineers, vol. 98, no. EM 6.

28. Stoykovich, M., Use of Time History Analysis, Proceedings-Symposium on Structural Design of Nuclear Power Plant Facilities, Department of Civil Engineering, University of Pittsburgh, December 1972, pp. 576-614.

29. Newmark, N.M., Earthquake Response Analysis of Reactor Structures, First International Conference on Structural Mechanics in Reactor Technology, Berlin, September 1971;

also published in 1972 in Nuclear Engineering and Design, North-Holland Publishing Company, pp. 303-322.

30. Stoykovich, M., March 1974, Criteria for Seismic Analysis of Nuclear Plant Structures and Substructures, Nuclear Engineering and Design, North-Holland Publishing Company, vol. 27, no. 1, pp. 106-120.

31. Stoykovich, M., Development and Use of Seismic Instructure Response Spectra in Nuclear Plants, published (abridged version) in the book of preprints of the Third International Conference on Structural Mechanics in Reactor Technology, held at Imperial College, London, England, September 1-5, 1975, paper code no. K5/4; published as an expanded version in the book of preprints of the International Seminar on Extreme Load Conditions and Limit Analysis Procedures for Structural Reactor Safeguards and Containment Structures, held in Berlin, Germany; September 8-11, 1975, paper code no. U1/5; and published as a modified version, August 1976, in Nuclear Engineering and Design, North-Holland Publishing Company, Amsterdam, vol. 38, no. 2.

32. Stoykovich, M., Damping Associated with Foundation Structure Interaction in Seismic Analysis of Nuclear Power Plants, Proceedings-the Second American Society of Civil Engineers Specialty Conference on Structural Design of Nuclear Plant Facilities, New Orleans, Louisiana, December 8-10, 1975.

33. Stoykovich, M., Finite Element Versus Elastic Half-Space Representation of Soil Structure Interaction in Seismic Design of Nuclear Facilities, Panel Presentation, American Society of Civil Engineers Specialty Conference on Structural Design of Nuclear Plant Facilities, Chicago, Illinois, December 17-18, 1973, vol. III.

34. Ho, P.K., and Stoykovich, M., Analysis and Design of Nuclear Power Plant Structures, American Society of Civil Engineers Proceedings-Specialty Conference on Structural Design of Nuclear Plant Facilities, Chicago, Illinois, December 17-18, 1973, vol II.

35. Stoykovich, M., Use of Floor or Ground Response Spectrum, Proceedings-Symposium on Structural Design of Nuclear Power Plant Facilities, Department of Civil Engineering, University of Pittsburgh, December 1972.

36. Stoykovich, M., Methods of Determining Floor Design Response Spectra, Proceedings-Symposium on Structural Design of Nuclear Power Plant Facilities, Department of Civil Engineering, University of Pittsburgh, December 1972.

37. Stoykovich, M., Variation of Input Parameters Considered in Determining Floor Response Spectra, Proceedings-Symposium on Structural Design of Nuclear Power Plant Facilities, Department of Civil Engineering, University of Pittsburgh, December 1972.

38. Stoykovich, M., Seismic Design and Analysis of Nuclear Plant Components, American Society of Civil Engineers Proceedings-Specialty Conference on Structural Design of Nuclear Plant Facilities, Chicago, Illinois, December 17-18, 1973, vol. I, pp. 1-28.

39. Pauw, A., 1953, A Dynamic Analogy for Foundation - Soil Systems, Symposium-Dynamic Testing of Soils, American Society for Testing and Materials Special Technical Publication no.156.

40. Leonards, G.A., ed., 1962, Foundation Engineering, McGraw-Hill Book Company, Inc., New York.

41. U.S. Nuclear Regulatory Commission, NUREG/CR-1161, Recommended Revision to Nuclear Regulatory Commission Seismic Design Criteria, December 1979.

42. Johnson, G. R., et al., “Stiffness Coefficients for Embedded Footings,” ASCE Journal of the Geotechnical Engineering Division, 1975, GTS, pp. 789-800.

43. Analyses for Soil-Structure Interaction Effects for Nuclear Power Plants, ASCE Publication, 1979, p. 155.

44. Elsabee, F., et al., “Dynamic Stiffness of Embedded Foundations,” in Advances in Civil Engineering Through Engineering Mechanics, ASCE Publication, 1977, pp. 40-43.

45. Kaldjian, M. J., “Torsional Stiffness of Embedded Footings,” ASCE Journal of the Soil Mechanics and Foundations Division, 1971, SM7, pp. 969-980.

46. USNRC NUREG-0800, Standard Review Plan, Section 3.7.1, “Seismic Design Parameters,” Revision 2.

47. USNRC Office of Technology, “OT Position for Review and Acceptance of Spent Fuel Storage and Handing Applications,” dated April 14, 1978, as amended January 18, 1979.

48. Nigam, Navin C. and Jennings, Paul C., “Digital Calculation of Response Spectra from Strong Motion Earthquake Records,” California Institute of Technology, Earthquake Engineering Research Laboratory, June 1968.

49. USNRC NUREG-0800, Standard Review Plan, Section 3.7.2, “Seismic System Analysis,” Revision 2, November 1989.

50. USNRC NUREG-0800, Standard Review Plan, Section 3.7.3, “Seismic Subsystem Analysis,” Revision 2.

51. USNRC NUREG-0800, Standard Review Plan, Section 3.8.4, “Other Seismic Category I Structures,” Revision 1.

52. Clough, Ray W. and Penzien, Joseph, “Dynamics of Structures,” McGraw Hill, Inc., 1975.

TABLE 3.7B-1
DAMPING VALUES

Item, Equipment, or Structure	Damping, Percent Critical[a] OBE	SSE
Equipment	2	3
Welded steel structures	2	4
Bolted steel structures	4	7
Reinforced concrete structures	4	7
Piping [b]	variable	

a) NRC Regulatory Guide 1.61

b) Damping used in piping analysis is in accordance with the recommendations of NRC Regulatory Guide 1.61 or as specified in Code Case N-411 of the ASME Boiler and Pressure Vessel Code.

TABLE 3.7B-2
METHODS OF SEISMIC ANALYSIS USED FOR SEISMIC CATEGORY I
STRUCTURES, SYSTEMS, AND COMPONENTS

Structure System, or Component	Equivalent Static Load	Response Spectrum Analysis	Modal Analysis Time History	Remarks
1. Structures				
Containment and internal structures	X	X	X	
Safeguards Building	X	X	X	
Electrical and Auxiliary Building	X	X	X	
Fuel Building	X	X	X	
Condensate storage tank	X	X	X	
RWST	X	X	X	
Reactor makeup Water storage tank	X	X	X	
Service Water Intake Structure	X	X	X	
2. Mechanical Components and Systems				
Westinghouse equipment				See Section 3.7N
High density spent fuel storage racks				See Section 3.7B.3.16
Other equipment	X	X		
3. Electrical Components and Systems				
Westinghouse equipment				See Section 3.7N
Other equipment	X	X		

TABLE 3.7B-3
DYNAMIC FOUNDATION DESIGN PARAMETERS

Foundation Material	Compressional Wave Velocity, V_c (ft/sec)	Shear Wave Velocity (ft/sec)	In-Situ Wet Density[a] (lb/ft^3)	Poisson's Ratio (ν)	Shear Modulus G or Modulus of Rigidity (lb/in^2)
Weathered rock	2,600	1,000	144	0.35	4.5×10^4
Moderately weathered to fresh rock (Glen Rose limestone above approximate elevation 770 ft)	9,500 to 11,000	5,500 to 6,000	150	0.30	8.0×10^5
Underlying massive fresh rock (Glen Rose limestone below approximate elevation 770 ft)	11,000 to 12,500	6,000 to 6,500	155	0.27	1.2×10^6
Underlying Twin Mountains rock	7,000 to 8,000	3,200	135	0.32	3.0×10^5

Young's modulus, $E = 2(1 + \nu) G$

$$E = pV_c^2 \frac{(1+\nu)(1-2\nu)}{(1-\nu)}$$

a) Mass density = $p = \frac{\text{in-situ density}}{32.2}$

TABLE 3.7B-3A
SOIL PROPERTIES FOR SOIL STRUCTURE INTERACTION ANALYSIS OF FUEL BUILDING

Layer No.	Elevation From	Elevation To	Thickness (ft)	Unit Weight (K/ft^3)	Mass Density ($K\text{-}s^2/ft^4$)	Shear Modulus (Ksf)	Shear Wave Velocity (ft/s)	Poisson's Ratio
A	796 ft	787 ft	9	0.150	0.00466	93,600	4482	0.35
B	787 ft	771 ft	16	0.145	0.00450	33,120	2713	0.35
C	771 ft	610 ft	161	0.155	0.00481	144,000	5472	0.35
D	610 ft	450 ft	160	0.135	0.00419	40,320	3102	0.35
Halfspace	450 ft	-200 ft	α	0.150	0.00466	93,600	4482	0.35
E								

TABLE 3.7B-4
SEISMIC CATEGORY I STRUCTURES FOUNDATION CHARACTERISTICS

Structure	Bottom of Mat Elevation	Plan Dimensions	Depth of Embedment	Structural Height
Containment Building	769 ft 2 in.	73 ft 6 in. radius	40 ft 10 in.	Top of dome elevation, 1070 ft 6 in.
Internal structure	793 ft 6 in.	72 ft 0 in. radius	-	Top floor elevation, 905 ft 9 in.
Safeguards Building	767 ft 4 in.	62 ft 6 in. by 98 ft 0 in.	42 ft 8 in.	Top of roof elevation, 918 ft 4 in.
Electrical Building	772 ft 10 in.	165 ft 6 in. by 101 ft 0 in.	37 ft 2 in.	Top of roof elevation, 873 ft 4 in.
Auxiliary Building	784 ft 10 in.	136 ft 0 in. by 192 ft 6 in.	25 ft 2 in.	Top of roof elevation, 900 ft 3 in.
Fuel Building	780 ft 6 in.	143 ft 6 in. by 137 ft 5 in.	29 ft 6 in.	Top of roof elevation, 918 ft 0 in.
Service Water Intake Structure	749 ft 0 in.	106 ft 6 in. by 74 ft 9 in.	Approx. 41 ft (varies)	Top of roof elevation, 838 ft 0 in.
Condensate storage tank	805 ft 6 in.	26 ft 6 in. radius	Approx. 4 ft 6 in.	Top of roof elevation, 860 ft 0 in.
RWST	805 ft 6 in.	26 ft 6 in. radius	Approx. 4 ft 6 in.	Top of roof elevation 860 ft 0 in.
Reactor makeup water storage tank	806 ft 6 in.	16 ft 6 in. radius	Approx. 3 ft 6 in.	Top of roof elevation, 846 ft 0 in.

TABLE 3.7B-5
SPRING CONSTANTS FOR RIGID CIRCULAR FOOTINGS RESTING ON ELASTIC HALF-SPACE

Motion	Spring Constant
Vertical	$k_z = \frac{4GR_o}{(1-\nu)}$
Horizontal	$k_x = \frac{32(1-\nu)GR_o}{7-8\nu}$
Rocking	$k_\psi = \frac{8GR_o^3}{3(1-\nu)}$
Torsion	$k_\theta = \frac{16GR_o^3}{3}$

Note:

$$G = \frac{E}{2(1+\nu)}$$

TABLE 3.7B-6
SPRING CONSTANTS FOR RIGID RECTANGULAR FOOTINGS RESTING ON ELASTIC HALF-SPACE

Motion	Spring Constant
Vertical	$k_z = \frac{G}{1-\nu}\beta_z\sqrt{4cd}$
Horizontal	$k_x = 4(1+\nu)G\beta_x\sqrt{cd}$
Rocking	$k_\psi = \frac{G}{1-\nu}\beta_\psi 8cd^2$

Note:

Values for $B_{z,}B_{x,}$ and B_ψ are given on Figure 3.7B-39 for various values of d/c.

TABLE 3.7B-7
PROPERTIES OF MASS-FOUNDATION SYSTEM THEORY OF RIGID CIRCULAR FOOTINGS ON ELASTIC HALF-SPACE

Mode of Vibration	Mass or Inertia Patio	Damping Ratio $D = C/C_c$
Vertical translation	$B_z = \frac{(1-\nu)m}{4pr_o^3}$	$D_z = \frac{0.425}{\sqrt{B_z}}$
Horizontal translation	$B_x = \frac{(7-8\nu)m}{32(1-\nu)pr_o^3}$	$D_x = \frac{0.288}{\sqrt{B_x}}$
Rocking	$B_\psi = \frac{3(1-\nu)I_\psi}{8pr_o^5}$	$D_\psi = \frac{0.15}{(1+B_\psi)\sqrt{B_\psi}}$
Torsion (about vertical axis)	$B_\theta = \frac{I_\theta}{pr_o^5}$	$D_\theta = \frac{0.5}{1+2B_\theta}$

Note:

- m = mass of the mat
- I_Ψ = moment of inertia about horizontal axis located at base
- I_θ = moment of inertia about vertical axis
- p = mass density of foundation medium
- ν = Poisson's Ratio of foundation medium
- C = coefficient of viscous damping
- C_c = coefficient of viscous critical damping

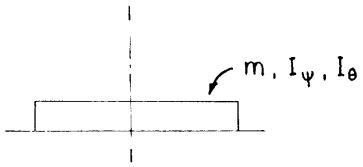

TABLE 3.7B-8
COEFFICIENTS FOR ESTIMATING EFFECTIVE MASS AND MASS MOMENT OF INERTIA OBTAINED FROM THEORY OF ELASTIC HALF-SPACE

Mode of Vibration	Coefficient		
	$\nu = 0$	$\nu = 0.25$	$\nu = 0.5$
Vertical translation	$\alpha = 0.5$	$\alpha = 1.0$	$\alpha = 2.0$
Horizontal translation	$\alpha = 0.2$	$\alpha = 0.2$	$\alpha = 0.1$
Rocking	$\beta = 0.4$	Not computed	
Torsion (about vertical axis)	$\beta = 0.3$	$\beta = 0.3$	$\beta = 0.3$

TABLE 3.7B-9
CONTAINMENT AND INTERNAL STRUCTURES MASS POINT NODAL COORDINATES

Mass Point	X (ft)	Y (ft)	Z (ft)
1	0.00	246.60	0.00
2	0.00	197.00	0.00
3	0.00	145.00	0.00
4	0.00	87.75	0.00
5	0.00	29.25	0.00
6	9.45	89.83	4.44
7	-0.28	55.94	8.27
8	-5.87	26.98	2.42
9	2.40	- 7.75	0.38
Foundation spring	0.00	-18.85	0.00

Notes:

a. The origin of coordinates is at elevation 805 ft 6 in.

b. For orientation of coordinate axes, see Figure 3.7B-23

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-34.

TABLE 3.7B-10
SAFEGUARDS BUILDING MASS POINT NODAL COORDINATES

Mass Point	X (ft)	Y (ft)	Z (ft)
1	48.35	893.49	35.64
2	52.66	872.40	70.80
3	54.11	851.80	82.06
4	58.60	831.34	90.09
5	61.01	809.47	103.87
6	44.15	789.68	69.87
7	31.69	772.53	49.74
Foundation spring		(See Table 3.7B-25.)	

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-25.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-35.

TABLE 3.7B-11
ELECTRICAL AND AUXILIARY BUILDINGS MASS POINT NODAL COORDINATES

Mass Point	X (ft)	Y (ft)	Z (ft)
1	59.31	871.29	-1.12
2	55.47	852.50	-0.18
3	46.31	829.55	0.15
4	53.78	805.63	0.75
5	57.20	776.79	-0.14
6	179.37	893.85	-0.22
7	166.49	872.32	3.47
8	172.92	849.55	0.40
9	174.87	831.15	-2.09
10	178.11	811.46	1.58
11	162.22	791.36	-3.30
Foundation spring		(See Tables 3.7B-26 and 3.7B-27.)	

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-28

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-36.

TABLE 3.7B-12
FUEL BUILDING MASS POINT NODAL COORDINATES

Mass Point	X (ft)	Y (ft)	Z (ft)
1	68.33	916.17	0 .00
2	68.21	894.27	-46.68
3	54.86	859.10	-41.12
4	55.91	840.06	-41.15
5	54.24	824.69	-40.52
6	65.72	894.67	48.60
7	60.53	859.37	44.54
8	52.83	840.41	38.59
9	52.07	824.65	40.55
10	51.09	812.53	6.47
Foundation spring	65.40	805.50	-2.17

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-30.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-37.

TABLE 3.7B-12A
FUEL BUILDING RE-ANALYSIS MODEL NODAL COORDINATES

(Sheet 1 of 2)

Mass Points

Mass Point	X (ft)	Y (ft)	Z (ft)
10	67.46	805.33	-0.59
20	48.60	819.56	0.20
1	50.34	838.75	-0.64
2	56.21	860.00	3.82
3	67.74	899.50	1.61
4	69.37	918.25	0.64
2030 Spent Fuel/Racks	40.13	827.06	0.00
220 Sloshing	40.13	860.00	0.00
109	35.13	805.33	0.00
107	51.68	810.50	0.00
108	35.13	810.50	0.00
208	35.13	819.56	0.00
209	51.68	819.56	-1.28
19	51.68	838.75	-1.28
18	45.80	838.75	-0.35
28	45.80	860.00	-0.35
29	67.18	860.00	0.00
39	67.18	899.50	0.00
38	68.88	899.50	0.00
48	68.88	918.25	0.00
210	56.21	860.00	0.00
2010	48.60	819.56	0.00
2020	48.60	827.06	0.00
1002	-18.08	805.33	-0.59
1003	153.00	805.33	-0.59
1004	67.46	805.33	-0.59

TABLE 3.7B-12A
FUEL BUILDING RE-ANALYSIS MODEL NODAL COORDINATES

(Sheet 2 of 2)

Mass Points

Mass Point	X (ft)	Y (ft)	Z (ft)
1005	67.46	805.33	-98.61
2002	-40.55	819.56	0.20
2003	137.75	819.56	0.20
2004	48.60	819.56	97.83
2005	48.60	819.56	-97.43
102	-37.07	838.75	-0.64
103	137.75	838.75	-0.64
104	50.34	838.75	97.43
105	50.34	838.75	-98.71
202	-25.33	860.00	3.82
203	137.75	860.00	3.82
204	56.21	860.00	105.07
205	56.21	860.00	-97.43
302	-2.27	899.50	1.61
303	137.75	899.50	1.61
304	67.74	899.50	100.65
305	67.74	899.50	-97.43
402	0.00	918.25	0.64
403	138.74	918.25	0.64
404	69.37	918.25	39.28
405	69.37	918.25	-38.00

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes and structure dynamic model showing the mass points, see Figure 3.7B-37A.

TABLE 3.7B-13
SERVICE WATER INTAKE STRUCTURE MASS POINT NODAL COORDINATES

Mass Point	X (ft)	Y (ft)	Z (ft)
1	29.65	0	54.86
2	21.91	20	60.81
25	21.92	41	55.56
3	19.96	62	53.36

Notes:

a. The origin of coordinates is at elevation 776 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-32

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-38.

TABLE 3.7B-14
CONTAINMENT AND INTERNAL STRUCTURES MASS POINT MASS DATA

Mass Point	Mass (kip-sec^2-ft)			Mass Moment of Inertia (kip-sec^2-ft)		
	Mx	My	Mz	Ix	Iy	Iz
1	173	173	173	186,400	340,500	186,400
2	375	375	375	966,000	1,734,000	966,000
3	542	555	542	1,395,000	2,510,000	1,395,000
4	538	538	538	1,463,000	2,618,000	1,463,000
5	538	538	538	1,463,000	2,618,000	1,463,000
6	440	440	440	599,800	982,500	429,600
7	708	531	710	816,400	1,112,000	623,600
8	602	640	602	440,900	984,800	588,200
9	1821	1959	1821	2,033,000	3,888,000	2,097,000
9A	322	2146	322	3,864,000	2,898,000	3,864,000

Notes:

a. Mass points 1 through 9 are for the structures. Mass point 9A is for the foundation material.

b. The origin of coordinates is at elevation 805 ft 6 in. For orientation of coordinate axes, see Figure 3.7B-23.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-34.

TABLE 3.7B-15
SAFEGUARDS BUILDING MASS POINT MASS DATA

Mass Point	Mass (kip-sec^2-ft)			Mass Moment of Inertia (kip-sec^2-ft)		
	Mx	My	Mz	Ix	Iy	Iz
1	171	171	171	138,564	218,891	132,264
2	323	323	323	719,229	1,104,267	399,863
3	393	393	393	1,014,757	1,560,267	565,170
4	479	479	479	1,353,810	2,020,784	698,972
5	569	569	569	1,728,231	2,663,410	957,458
6	470	470	470	844,598	1,154,650	336,715
7	254	254	254	216,360	303,136	94,036
5A	106	703	106	356,800	727,400	1,679,000
6A	65	432	65	1,190,000	607,900	464,400
7A	64	430	64	489,500	237,200	160,700

Notes:

a. Mass points 1 through 7 are for the structure. Mass points 5A, 6A and 7A are for the foundation material.

b. The origin of coordinates is at elevation 0 ft 0 in. For orientation of coordinate axes, see Figure 3.7B-25.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-35.

TABLE 3.7B-16
ELECTRICAL AND AUXILIARY BUILDINGS MASS POINT MASS DATA

Mass Point	Mass (kip-sec^2-ft)			Mass Moment of Inertia (kip-sec^2-ft)		
	Mx	My	Mz	Ix	Iy	Iz
1	191	191	191	352,578	621,762	273,288
2	371	371	371	1,029,267	1,540,460	525,576
3	335	335	335	935,606	1,329,372	413,593
4	404	404	404	1,232,237	1,789,889	585,062
5	570	570	570	1,475,049	2,636,970	1,184,607
6	373	373	373	1,170,341	1,756,542	611,452
7	530	530	530	1,543,640	2,372,764	859,910
8	860	860	860	2,390,334	3,484,584	1,151,946
9	710	710	710	2,110,568	3,065,440	989,810
10	663	663	663	2,193,210	2,996,694	859,967
11	1022	1022	1022	2,877,228	5,305,467	2,585,200
5A	310	2065	310	7,211,000	3,355,000	2,046,000
11A	623	4153	623	18,830,000	9,913,000	8,046,000

Notes:

a. Mass points 1 through 11 are for the structures. Mass points 5A and 11A are for the foundation material.

b. The origin of coordinates is at elevation 0 ft 0 in. For orientation of coordinate axes, see Figure 3.7B-28.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-36.

TABLE 3.7B-17
FUEL BUILDING MASS POINT MASS DATA

Mass Point	Mass (kip-sec^2-ft)			Mass Moment of Inertia (kip-sec^2-ft)		
	Mx	My	Mz	Ix	Iy	Iz
1	156	156	156	97,540	370,520	274,990
2	144	144	144	62,892	329,880	282,850
3	234	234	234	148,060	548,670	432,090
4	304	304	304	149,840	563,140	425,110
5	209	209	209	107,960	429,550	328,460
6	167	167	167	73,818	392,660	338,010
7	261	261	261	140,530	562,750	455,640
8	342	342	342	191,730	623,910	445,600
9	188	188	188	105,060	358,800	259,940
10	1434	1434	1434	3,032,500	4,287,500	2,147,000
10A	445	2574	445	7,134,826	4,677,522	5,362,199

Notes:

a. Mass points 1 through 10 are for the structure. Mass point 10A is for the foundation material.

b. The origin of coordinates is at elevation 0 ft 0 in. For orientation of coordinate axes, see Figure 3.7B-30.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-37.

TABLE 3.7B-17A
FUEL BUILDING RE-ANALYSIS MASS POINT DATA

Mass Point	Mass ($kip\text{-}sec^2\text{-}ft$)			Mass Moment of Inertia ($kip\text{-}sec^2\text{-}ft$)		
	Mx	My	Mz	Ix	Iy	Iz
10	1124	1124	1124	1,996,564	1,535,468	3,351,743
20	701	701	701	1,497,815	2,356,604	893,774
1	907	907	907	1,854,838	3,049,616	1,226,955
2	577	577	577	1,325,568	2,288,451	1,030,240
3	407	407	407	981,658	1,700,226	756,403
4	211	211	211	132,334	510,951	380,890
2030 Spent Fuel/Racks	66.8	66.8	66.8	178,310	186,076	10,271
220 Sloshing	45.6	45.6	35.8	125,015	127,022	10,306

TABLE 3.7B-18
SERVICE WATER INTAKE STRUCTURE MASS POINT MASS DATA

Mass Point	Mass ($kip\text{-}sec^2\text{-}ft$)			Mass Moment of Inertia ($kip\text{-}sec^2\text{-}ft$)		
	Mx	My	Mz	Ix	Iy	Iz
1	736	1446	736	1.964×10^6	1.308×10^6	5.79×10^5
2	280	280	280	3.59×10^5	4.17×10^5	1.08×10^5
25	98	98	98	1.57×10^5	1.83×10^5	$.335 \times 10^5$
3	89	89	89	0.99×10^5	1.21×10^5	$.25 \times 10^5$

Notes:

a. Mass 1 includes the equivalent soil mass.

b. The origin of coordinates is at elevation 776 ft, 0 in. For orientation of coordinate axes, see Figure 3.7B-32.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-38.

TABLE 3.7B-19
CONTAINMENT AND INTERNAL STRUCTURES DEGREES OF FREEDOM

(Sheet 1 of 2)

Mass Point	Degree of Freedom	Direction of Motion	
1	1	Translation	X
1	2	Translation	Y
1	3	Translation	Z
1	4	Rotation	θ_x
1	5	Rotation	θ_y
1	6	Rotation	θ_z
2	7	Translation	X
2	8	Translation	Y
2	9	Translation	Z
2	10	Rotation	θ_x
2	11	Rotation	θ_y
2	12	Rotation	θ_z
3	13	Translation	X
3	14	Translation	Y
3	15	Translation	Z
3	16	Rotation	θ_x
3	17	Rotation	θ_y
3	18	Rotation	θ_z
4	19	Translation	X
4	20	Translation	Y
4	21	Translation	Z
4	22	Rotation	θ_x
4	23	Rotation	θ_y
4	24	Rotation	θ_z
5	25	Translation	X
5	26	Translation	Y
5	27	Translation	Z
5	28	Rotation	θ_x
5	29	Rotation	θ_y
5	30	Rotation	θ_z
6	31	Translation	X

TABLE 3.7B-19
CONTAINMENT AND INTERNAL STRUCTURES DEGREES OF FREEDOM

(Sheet 2 of 2)

Mass Point	Degree of Freedom	Direction of Motion	
6	32	Translation	Y
6	33	Translation	Z
6	34	Rotation	θ_x
6	35	Rotation	θ_y
6	36	Rotation	θ_z
7	37	Translation	X
7	38	Translation	Y
7	39	Translation	Z
7	40	Rotation	θ_x
7	41	Rotation	θ_y
7	42	Rotation	θ_z
8	43	Translation	X
8	44	Translation	Y
8	45	Translation	Z
8	46	Rotation	θ_x
8	47	Rotation	θ_y
8	48	Rotation	θ_z
9	49	Translation	X
9	50	Translation	Y
9	51	Translation	Z
9	52	Rotation	θ_x
9	53	Rotation	θ_y
9	54	Rotation	θ_z

Notes:

a. The origin of coordinates is at elevation 805 ft 6 in.

b. For orientation of coordinate axes, see Figure 3.7B-23.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-34.

TABLE 3.7B-20
SAFEGUARDS BUILDING DEGREES OF FREEDOM

(Sheet 1 of 2)

Mass Point	Degree of Freedom	Direction of Motion	
1	1	Translation	X
1	2	Translation	Y
1	3	Translation	Z
1	4	Rotation	θ_x
1	5	Rotation	θ_y
1	6	Rotation	θ_z
2	7	Translation	X
2	8	Translation	Y
2	9	Translation	Z
2	10	Rotation	θ_x
2	11	Rotation	θ_y
2	12	Rotation	θ_z
3	13	Translation	X
3	14	Translation	Y
3	15	Translation	Z
3	16	Rotation	θ_x
3	17	Rotation	θ_y
3	18	Rotation	θ_z
4	19	Translation	X
4	20	Translation	Y
4	21	Translation	Z
4	22	Rotation	θ_x
4	23	Rotation	θ_y
4	24	Rotation	θ_z

TABLE 3.7B-20
SAFEGUARDS BUILDING DEGREES OF FREEDOM

(Sheet 2 of 2)

Mass Point	Degree of Freedom	Direction of Motion	
5	25	Translation	X
5	26	Translation	Y
5	27	Translation	Z
5	28	Rotation	θ_x
5	29	Rotation	θ_y
5	30	Rotation	θ_z
6	31	Translation	X
6	32	Translation	Y
6	33	Translation	Z
6	34	Rotation	θ_x
6	35	Rotation	θ_y
6	36	Rotation	θ_z
7	37	Translation	X
7	38	Translation	Y
7	39	Translation	Z
7	40	Rotation	θ_x
7	41	Rotation	θ_y
7	42	Rotation	θ_z

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of ccordinate axes, see Figure 3.7B-25.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-35.

TABLE 3.7B-21
ELECTRICAL AND AUXILIARY BUILDINGS DEGREES OF FREEDOM

(Sheet 1 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
1	1	Translation	X
1	2	Translation	Y
1	3	Translation	Z
1	4	Rotation	θ_x
1	5	Rotation	θ_y
1	6	Rotation	θ_z
2	7	Translation	X
2	8	Translation	Y
2	9	Translation	Z
2	10	Rotation	θ_x
2	11	Rotation	θ_y
2	12	Rotation	θ_z
3	13	Translation	X
3	14	Translation	Y
3	15	Translation	Z
3	16	Rotation	θ_x
3	17	Rotation	θ_y
3	18	Rotation	θ_z
4	19	Translation	X
4	20	Translation	Y
4	21	Translation	Z
4	22	Rotation	θ_x
4	23	Rotation	θ_y
4	24	Rotation	θ_z

TABLE 3.7B-21
ELECTRICAL AND AUXILIARY BUILDINGS DEGREES OF FREEDOM

(Sheet 2 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
5	25	Translation	X
5	26	Translation	Y
5	27	Translation	Z
5	28	Rotation	θ_x
5	29	Rotation	θ_y
5	30	Rotation	θ_z
6	31	Translation	X
6	32	Translation	Y
6	33	Translation	Z
6	34	Rotation	θ_x
6	35	Rotation	θ_y
6	36	Rotation	θ_z
7	37	Translation	X
7	38	Translation	Y
7	39	Translation	Z
7	40	Rotation	θ_x
7	41	Rotation	θ_y
7	42	Rotation	θ_z
8	43	Translation	X
8	44	Translation	Y
8	45	Translation	Z
8	46	Rotation	θ_x
8	47	Rotation	θ_y
8	48	Rotation	θ_z

TABLE 3.7B-21
ELECTRICAL AND AUXILIARY BUILDINGS DEGREES OF FREEDOM

(Sheet 3 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
9	49	Translation	X
9	50	Translation	Y
9	51	Translation	Z
9	52	Rotation	θ_x
9	53	Rotation	θ_y
9	54	Rotation	θ_z
10	55	Translation	X
10	56	Translation	Y
10	57	Translation	Z
10	58	Rotation	θ_x
10	59	Rotation	θy
10	60	Rotation	θ_z
11	61	Translation	X
11	62	Translation	Y
11	63	Translation	Z
11	64	Rotation	θ_x
11	65	Rotation	θ_y
11	66	Rotation	θ_z

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-28.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-36.

TABLE 3.7B-22
FUEL BUILDING DEGREES OF FREEDOM

(Sheet 1 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
1	1	Translation	X
1	2	Translation	Y
1	3	Translation	Z
1	4	Rotation	θ_x
1	5	Rotation	θ_y
1	6	Rotation	θ_z
2	7	Translation	X
2	8	Translation	Y
2	9	Translation	Z
2	10	Rotation	θ_x
2	11	Rotation	θ_y
2	12	Rotation	θ_z
3	13	Translation	X
3	14	Translation	Y
3	15	Translation	Z
3	16	Rotation	θ_x
3	17	Rotation	θ_y
3	18	Rotation	θ_z
4	19	Translation	X
4	20	Translation	Y
4	21	Translation	Z
4	22	Rotation	θ_x

TABLE 3.7B-22
FUEL BUILDING DEGREES OF FREEDOM

(Sheet 2 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
4	23	Rotation	θ_y
4	24	Rotation	θ_z
5	25	Translation	X
5	26	Translation	Y
5	27	Translation	Z
5	28	Rotation	θ_x
5	29	Rotation	θ_y
5	30	Rotation	θ_z
6	31	Translation	X
6	32	Translation	Y
6	33	Translation	Z
6	34	Rotation	θ_x
6	35	Rotation	θ_y
6	36	Rotation	θ_z
7	37	Translation	X
7	38	Translation	Y
7	39	Translation	Z
7	40	Rotation	θ_x
7	41	Rotation	θ_y
7	42	Rotation	θ_z
8	43	Translation	X
8	44	Translation	Y

TABLE 3.7B-22
FUEL BUILDING DEGREES OF FREEDOM

(Sheet 3 of 3)

Mass Point	Degree of Freedom	Direction of Motion	
8	45	Translation	Z
8	46	Rotation	θ_x
8	47	Rotation	θ_y
8	48	Rotation	θ_z
9	49	Translation	X
9	50	Translation	Y
9	51	Translation	Z
9	52	Rotation	θ_x
9	53	Rotation	θ_y
9	54	Rotation	θ_z
10	55	Translation	X
10	56	Translation	Y
10	57	Translation	Z
10	58	Rotation	θ_x
10	59	Rotation	θ_y
10	60	Rotation	θ_z

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-30.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-37.

TABLE 3.7B-23
SERVICE WATER INTAKE STRUCTURE DEGREES OF FREEDOM

Mass Point	Degree of Freedom	Direction of Motion	
1	1	Translation	X
1	2	Translation	Y
1	3	Translation	Z
1	4	Rotation	θ_x
1	5	Rotation	θ_y
1	6	Rotation	θ_z
2	7	Translation	X
2	8	Translation	Y
2	9	Translation	Z
2	10	Rotation	θ_x
2	11	Rotation	θ_y
2	12	Rotation	θ_z
25	13	Translation	X
25	14	Translation	Y
25	15	Translation	Z
25	16	Rotation	θ_x
25	17	Rotation	θ_y
25	18	Rotation	θ_z
3	19	Translation	X
3	20	Translation	Y
3	21	Translation	Z
3	22	Rotation	θ_x
3	23	Rotation	θ_y
3	24	Rotation	θ_z

Notes:

a. The origin of coordinates is at elevation 776 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-32.

c. For structure dynamic model showing the mass point numbers, see Figure 3.7B-38.

TABLE 3.7B-24
CONTAINMENT AND INTERNAL STRUCTURES FOUNDATION SPRING CONSTANTS

Direction	Lower Pound	Best Estimate	Upper Pound
Translation along X axis (kip/ft)	0.116×10^{8}	0.465×10^{8}	0.930×10^{8}
Translation along Y axis (kip/ft)	0.125×10^{8}	0.499×10^{8}	0.998×10^{8}
Translation along Z axis (kip/ft)	0.116×10^{8}	0.465×10^{8}	0.930×10^{8}
Rotation along X axis (kip-ft/rad)	0.390×10^{11}	0.156×10^{12}	0.312×10^{12}
Rotation along Y axis (kip-ft/rad)	0.600×10^{11}	0.240×10^{12}	0.480×10^{12}
Rotation along Z axis (kip-ft/rad)	0.390×10^{11}	0.156×10^{12}	0.312×10^{12}

Notes:

a. The origin of coordinates is at elevation 805 ft 6 in.

b. For orientation of coordinate axes, see Figure 3.7B-23.

TABLE 3.7B-25
SAFEGUARDS BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 1 of 3)

Mass No.	Spring Location X (ft)	Spring Location Y (ft)	Spring Location Z (ft)	Lower Bound Foundation Spring Constants: Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	92.22	806.50	149.62	5565×10^{7}	7033×10^{7}	$.5565 \times 10^{7}$	$.2970 \times 10^{10}$	$.1047 \times 10^{11}$	$.1026 \times 10^{11}$
5	58.50	801.125	127.50	4338×10^{6}	0.0	$.5245 \times 10^{6}$	$.2020 \times 10^{8}$	$.2329 \times 10^{9}$	$.1672 \times 10^{8}$
5	60.75	801.125	125.625	4775×10^{6}	0.0	$.3760 \times 10^{6}$	$.1445 \times 10^{8}$	$.1466 \times 10^{9}$	$.4595 \times 10^{7}$
5	53.93	785.00	90.71	$.4730 \times 10^{7}$	$.6554 \times 10^{7}$	$.4730 \times 10^{7}$	$.9880 \times 10^{10}$	$.1067 \times 10^{11}$	$.4655 \times 10^{10}$
6	31.25	780.44	0.00	$.3415 \times 10^{6}$	0.0	$.4348 \times 10^{6}$	$.1088 \times 10^{8}$	$.1648 \times 10^{9}$	$.8470 \times 10^{7}$
6	62.50	780.625	47.625	$.5650 \times 10^{6}$	0.0	$.4475 \times 10^{6}$	$.1141 \times 10^{8}$	$.4273 \times 10^{9}$	$.1442 \times 10^{8}$
6	31.25	780.44	95.25	$.3415 \times 10^{6}$	0.0	$.4348 \times 10^{6}$	$.1088 \times 10^{8}$	$.1648 \times 10^{9}$	$.8470 \times 10^{7}$
6	58.50	790.375	127.50	$.4338 \times 10^{6}$	0.0	$.5245 \times 10^{6}$	$.2020 \times 10^{8}$	$.2329 \times 10^{9}$	$.1672 \times 10^{8}$
6	60.75	790.375	125.625	$.4775 \times 10^{6}$	0.0	$.3760 \times 10^{6}$	$.1445 \times 10^{8}$	$.1466 \times 10^{9}$	$.4595 \times 10^{7}$
7	30.50	767.50	47.625	$.4723 \times 10^{7}$	$.5708 \times 10^{7}$	$.4723 \times 10^{7}$	$.4315 \times 10^{10}$	$.5035 \times 10^{10}$	$.1770 \times 10^{10}$
7	31.25	771.81	0.00	$.3415 \times 10^{6}$	0.0	$.4348 \times 10^{6}$	$.1088 \times 10^{6}$	$.1648 \times 10^{9}$	$.8470 \times 10^{7}$
7	62.50	771.875	47.625	$.5650 \times 10^{6}$	0.0	$.4475 \times 10^{6}$	$.1141 \times 10^{8}$	$.4273 \times 10^{9}$	$.1442 \times 10^{8}$
7	31.25	771.81	95.25	$.3415 \times 10^{6}$	0.0	$.4348 \times 10^{6}$	$.1088 \times 10^{8}$	$.1648 \times 10^{9}$	$.8470 \times 10^{7}$

TABLE 3.7B-25
SAFEGUARDS BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 2 of 3)

Mass No.	Spring Location X (ft)	Spring Location Y (ft)	Spring Location Z (ft)	Best Estimate Foundation Spring Constants: Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	92.22	806.50	149.62	$.2226 \times 10^{8}$	$.2813 \times 10^{8}$	$.2226 \times 10^{8}$	$.1188 \times 10^{11}$	$.4187 \times 10^{11}$	$.4102 \times 10^{11}$
5	58.50	801.125	127.50	$.1735 \times 10^{7}$	0.0	$.2098 \times 10^{7}$	$.8080 \times 10^{8}$	$.9315 \times 10^{9}$	$.6687 \times 10^{8}$
5	60.75	801.125	125.625	$.1910 \times 10^{7}$	0.0	$.1504 \times 10^{7}$	$.5780 \times 10^{8}$	$.5865 \times 10^{9}$	$.1838 \times 10^{8}$
6	53.93	785.00	90.71	$.1892 \times 10^{8}$	$.2578 \times 10^{8}$	$.1892 \times 10^{8}$	$.3952 \times 10^{11}$	$.4267 \times 10^{11}$	$.1862 \times 10^{11}$
6	31.25	780.44	0.0	$.1366 \times 10^{7}$	0.0	$.1739 \times 10^{7}$	$.4351 \times 10^{8}$	$.6590 \times 10^{9}$	$.3388 \times 10^{8}$
6	62.50	780.625	47.625	$.2260 \times 10^{7}$	0.0	$.1790 \times 10^{7}$	$.4565 \times 10^{8}$	$.1709 \times 10^{10}$	$.5768 \times 10^{8}$
6	31.25	780.44	95.25	$.1366 \times 10^{7}$	0.0	$.1739 \times 10^{7}$	$.4351 \times 10^{8}$	$.6590 \times 10^{9}$	$.3388 \times 10^{8}$
6	58.50	790.375	127.50	$.1735 \times 10^{7}$	0.0	$.2098 \times 10^{7}$	$.8080 \times 10^{8}$	$.9315 \times 10^{9}$	$.6687 \times 10^{8}$
6	60.75	790.375	125.625	$.1910 \times 10^{7}$	0.0	$.1504 \times 10^{7}$	$.5780 \times 10^{8}$	$.5865 \times 10^{9}$	$.1838 \times 10^{8}$
7	30.50	767.50	47.625	$.1889 \times 10^{8}$	$.2283 \times 10^{8}$	$.1889 \times 10^{8}$	$.1726 \times 10^{11}$	$.2014 \times 10^{11}$	$.7079 \times 10^{10}$
7	31.25	771.81	0.0	$.1366 \times 10^{7}$	0.0	$.1739 \times 10^{7}$	$.4351 \times 10^{8}$	$.6590 \times 10^{9}$	$.3388 \times 10^{8}$
7	62.50	771.875	47.625	$.2260 \times 10^{7}$	0.0	$.1790 \times 10^{7}$	$.4565 \times 10^{8}$	$.1709 \times 10^{10}$	$.5768 \times 10^{8}$
7	31.25	771.81	95.25	$.1366 \times 10^{7}$	0.0	$.1739 \times 10^{7}$	$.4351 \times 10^{8}$	$.6590 \times 10^{9}$	$.3388 \times 10^{8}$

TABLE 3.7B-25
SAFEGUARDS BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 3 of 3)

Mass No.	Spring Location X (ft)	Spring Location Y (ft)	Spring Location Z (ft)	Upper Bound Foundation Spring Constants: Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	92.22	806.50	149.62	$.4452 \times 10^{8}$	$.5626 \times 10^{8}$	$.4452 \times 10^{8}$	$.2376 \times 10^{11}$	$.8374 \times 10^{11}$	$.8204 \times 10^{11}$
5	58.50	801.125	127.5	$.3470 \times 10^{7}$	0.0	$.4196 \times 10^{7}$	$.1616 \times 10^{9}$	$.1863 \times 10^{10}$	$.1337 \times 10^{9}$
5	60.75	801.125	125.625	$.3820 \times 10^{7}$	0.0	$.3008 \times 10^{7}$	$.1156 \times 10^{9}$	$.1173 \times 10^{10}$	$.3676 \times 10^{8}$
6	53.93	785.00	90.71	$.3784 \times 10^{8}$	$.5156 \times 10^{8}$	$.3784 \times 10^{8}$	$.7904 \times 10^{11}$	$.8534 \times 10^{11}$	$.3724 \times 10^{11}$
6	31.25	780.44	0.0	$.2732 \times 10^{7}$	0.0	$.3478 \times 10^{7}$	$.8702 \times 10^{8}$	$.1318 \times 10^{10}$	$.6776 \times 10^{8}$
6	62.50	780.625	47.625	$.4520 \times 10^{7}$	0.0	$.3580 \times 10^{7}$	$.9130 \times 10^{8}$	$.3418 \times 10^{10}$	$.1154 \times 10^{9}$
6	31.25	780.44	95.25	$.2732 \times 10^{7}$	0.0	$.3478 \times 10^{7}$	$.8702 \times 10^{8}$	$.1318 \times 10^{10}$	$.6776 \times 10^{8}$
6	58.50	790.375	127.50	$.3470 \times 10^{7}$	0.0	$.4196 \times 10^{7}$	$.1616 \times 10^{9}$	$.1863 \times 10^{10}$	$.1337 \times 10^{9}$
6	60.75	790.375	125.625	$.3820 \times 10^{7}$	0.0	$.3008 \times 10^{7}$	$.1156 \times 10^{9}$	$.1173 \times 10^{10}$	$.3676 \times 10^{8}$
7	30.50	767.50	47.625	$.3778 \times 10^{8}$	$.4566 \times 10^{8}$	$.3778 \times 10^{8}$	$.3452 \times 10^{11}$	$.4028 \times 10^{11}$	$.1416 \times 10^{11}$
7	31.25	771.81	0.0	$.2732 \times 10^{7}$	0.0	$.3478 \times 10^{7}$	$.8702 \times 10^{8}$	$.1318 \times 10^{10}$	$.6776 \times 10^{8}$
7	62.50	771.875	47.625	$.4520 \times 10^{7}$	0.0	$.3580 \times 10^{7}$	$.9130 \times 10^{8}$	$.3418 \times 10^{10}$	$.1154 \times 10^{9}$
7	31.25	771.81	95.25	$.2732 \times 10^{7}$	0.0	$.3478 \times 10^{7}$	$.8702 \times 10^{8}$	$.1318 \times 10^{10}$	$.6776 \times 10^{8}$

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate area see Figure 3.7B-25.

TABLE 3.7B-26
ELECTRICAL BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 1 of 2)

Lower Bound Foundation Spring Constants

Mass No.	Spring Location X (ft)	Y (ft)	Z (ft)	Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	98.75	778.50	0.0	0.1587×10^7	0.0	0.1195×10^7	0.0	0.3626×10^{10}	0.0
5	48.75	772.25	0.0	0.7970×10^7	$.9632 \times 10^7$	0.7970×10^7	2.1985×10^{10}	2.4830×10^{10}	0.8027×10^{10}
5	48.75	787.875	-82.75	0.1225×10^7	0.0	0.1615×10^7	0.0	0.1345×10^{10}	0.0
5	48.75	787.875	82.75	0.1225×10^7	0.0	0.1615×10^7	0.0	0.1345×10^{10}	0.0

Best Estimate Foundation Spring Constants

Mass No.	Spring Location X (ft)	Y (ft)	Z (ft)	Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	98.75	778.50	0.0	0.6355×10^7	0.0	0.4782×10^7	0.0	1.4506×10^{10}	0.0
5	48.75	772.25	0.0	3.1880×10^7	3.8530×10^7	3.1880×10^7	8.7940×10^{10}	9.9320×10^{10}	3.2110×10^{10}
5	48.75	787.875	-82.75	0.4900×10^7	0.0	0.6460×10^7	0.0	0.5382×10^{10}	0.0
5	48.75	787.875	82.75	0.4900×10^7	0.0	0.6460×10^7	0.0	0.5382×10^{10}	0.0

TABLE 3.7B-26
ELECTRICAL BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 2 of 2)

Mass No.	Spring Location X (ft)	Spring Location Y (ft)	Spring Location Z (ft)	Upper Bound Foundation Spring Constants: Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
5	98.75	778.50	0.0	1.2700×10^{7}	0.0	0.9564×10^{7}	0.0	2.9012×10^{10}	0.0
5	48.75	772.25	0.0	6.3760×10^{7}	7.7060×10^{7}	6.3760×10^{7}	1.7588×10^{11}	19.8640×10^{10}	6.4220×10^{10}
5	48.75	787.875	-82.75	0.9800×10^{7}	0.0	1.2920×10^{7}	0.0	1.0764×10^{10}	0.0
5	48.75	787.875	82.75	0.9800×10^{7}	0.0	1.2920×10^{7}	0.0	1.0764×10^{10}	0.0

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate area see Figure 3.7B-28.

TABLE 3.7B-27
ELECTRICAL BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 1 of 2)

Lower Bound Foundation Spring Constants

Mass No.	Spring Location X (ft)	Y (ft)	Z (ft)	Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
11	167.25	784.75	0.0	1.0060×10^{7}	1.2160×10^{7}	1.0060×10^{10}	3.7550×10^{10}	4.6800×10^{10}	1.9015×10^{10}
11	235.75	795.00	0.0	0.2117×10^{7}	0.0	0.1650×10^{7}	0.0	0.6535×10^{10}	0.0

Best Estimate Foundation Spring Constants

Mass No.	Spring Location X (ft)	Y (ft)	Z (ft)	Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
11	167.25	784.75	0.0	4.0240×10^{7}	4.8640×10^{7}	4.0240×10^{7}	15.0200×10^{10}	1.8720×10^{10}	7.6060×10^{10}
11	235.75	795.00	0.0	0.8466×10^{7}	0.0	0.6604×10^{7}	0.0	2.6139×10^{10}	0.0

TABLE 3.7B-27
ELECTRICAL BUILDING FOUNDATION SPRING CONSTANTS

(Sheet 2 of 2)

Mass No.	Spring Location X (ft)	Spring Location Y (ft)	Spring Location Z (ft)	Upper Bound Foundation Spring Constants: Translation Along X Axis (kip/ft)	Translation Along Y Axis (kip/ft)	Translation Along Z Axis (kip/ft)	Rotation About X Axis (kip-ft/rad)	Rotation About Y Axis (kip-ft/rad)	Rotation About Z Axis (kip-ft/rad)
11	167.25	784.75	0.0	8.0480×10^{7}	9.7280×10^{7}	8.0480×10^{7}	3.0040×10^{11}	3.7440×10^{11}	1.5212×10^{11}
11	235.75	795.00	0.0	1.6940×10^{7}	0.0	1.3200×10^{7}	0.0	5.2280×10^{10}	0.0

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate area see Figure 3.7B-28.

TABLE 3.7B-28
FUEL BUILDING FOUNDATION SPRING CONSTANTS

Direction	Upper Bound	Lower Bound
Translation along X axis (kip/ft)	3.9913×10^{7}	1.99565×10^{7}
Translation along Y axis (kip/ft)	5.4593×10^{7}	2.72965×10^{7}
Translation along Z axis (kip/ft)	3.9913×10^{7}	1.99565×10^{7}
Rotation along X axis (kip-ft/rad)	2.4524×10^{11}	1.22620×10^{11}
Rotation along Y axis (kip-ft/rad)	1.3182×10^{11}	6.59100×10^{10}
Rotation along Z axis (kip-ft/rad)	2.1453×10^{11}	1.07265×10^{11}

Notes:

a. The origin of coordinates is at elevation 0 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-30.

TABLE 3.7B-29
SERVICE WATER INTAKE STRUCTURE FOUNDATION SPRING CONSTANTS

Direction	Upper Bound	Lower Bound
Translation along X axis (kip/ft)	5.21×10^{7}	2.60×10^{7}
Translation along Y axis (kip/ft)	4.14×10^{7}	2.07×10^{7}
Translation along Z axis (kip/ft)	5.21×10^{7}	2.60×10^{7}
Rotation along X axis (kip-ft/rad)	1.28×10^{11}	0.64×10^{11}
Rotation along Y axis (kip-ft/rad)	1.94×10^{11}	0.97×10^{11}
Rotation along Z axis (kip-ft/rad)	7.72×10^{10}	3.86×10^{10}

Notes:

a. The origin of coordinates is at elevation 776 ft 0 in.

b. For orientation of coordinate axes, see Figure 3.7B-32.

c. Spring constants include embedment effects.

TABLE 3.7B-30
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, LOWER BOUND, SSE

(Sheet 1 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	2.574	4.03	-0.74	52.06
2	2.577	51.87	-0.71	-4.03
3	5.664	0.16	-6.38	35.01
4	5.965	-25.94	63.51	1.65
5	6.068	28.56	60.37	3.26
6	6.369	-9.68	-2.02	7.42
7	8.575	-1.41	-0.18	24.63
8	9.182	25.84	-1.14	18.95
9	9.256	-20.83	0.36	20.85
10	12.254	5.70	1.37	22.12
11	12.422	22.37	2.52	-6.72
12	13.000	8.31	-15.72	1.95
13	15.056	-0.51	-0.75	1.57
14	15.633	0.15	0.96	-3.05
15	16.338	1.72	0.79	0.89
16	17.835	0.11	0.22	6.31
17	18.685	-3.98	-2.30	-0.51
18	20.265	-1.05	-0.38	1.04
19	20.523	-1.20	-0.61	-3.00
20	20.944	3.31	1.35	3.18
21	21.087	-3.46	-1.27	3.81
22	21.629	-0.76	0.97	-0.43
23	22.904	0.57	-0.85	0.47
24	23.099	-2.14	1.81	0.18
25	24.515	0.05	-0.01	-0.32
26	25.593	0.30	0.77	-0.03
27	26.181	0.17	-0.08	2.66
28	26.345	-2.90	0.13	0.19

TABLE 3.7B-30
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, LOWER BOUND, SSE

(Sheet 2 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
29	28.101	1.12	-1.84	-0.35
30	28.280	-0.07	0.03	-0.79
31	28.455	-0.81	-0.76	-0.07
32	29.221	0.08	-0.01	-0.25
33	29.602	0.22	0.88	0.23
34	32.176	0.01	0.05	-0.73
35	32.491	0.18	-0.13	1.60
36	35.039	-1.35	-0.06	0.06
37	37.042	0.11	0.02	-0.14
38	38.465	-0.02	-0.64	-0.02
39	38.730	0.05	0.14	-0.19
40	39.202	-0.21	0.14	-0.06
41	39.440	-0.05	0.11	0.11
42	40.519	-0.00	-0.00	0.04
43	41.729	0.06	0.02	-0.00
44	44.048	0.01	-0.27	0.04
45	49.042	0.03	-0.38	-0.00
46	52.598	0.05	-0.00	0.11
47	52.664	-0.11	0.02	0.05
48	53.887	0.02	-0.34	-0.00
49	56.683	-0.01	0.00	-0.06
50	56.705	0.07	-0.03	-0.02
51	59.069	-0.00	0.18	-0.00
52	69.662	-0.03	-0.07	-0.02
53	73.651	-0.00	-0.04	0.04
54	81.116	0.01	-0.06	-0.00

TABLE 3.7B-31
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, BEST ESTIMATE, SSE

(Sheet 1 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	3.289	3.26	-0.26	44.06
2	3.290	43.86	-0.26	-3.25
3	6.317	0.91	-1.21	34.84
4	7.006	38.38	-3.67	-0.62
5	7.462	-2.89	0.23	2.68
6	9.019	-2.22	-0.51	13.02
7	9.672	2.44	72.63	0.71
8	10.238	7.31	-0.75	23.27
9	10.251	22.35	-1.39	-6.96
10	14.686	12.12	40.48	6.38
11	15.224	-1.46	-8.75	28.55
12	16.676	29.58	-21.94	-0.57
13	18.522	1.30	2.13	9.06
14	18.765	4.09	7.17	-2.95
15	19.173	1.01	-0.91	-5.77
16	19.583	2.12	1.70	10.20
17	20.124	7.92	1.55	1.13
18	20.632	1.16	0.89	2.04
19	20.961	-0.74	-0.40	2.82
20	22.412	8.65	12.58	1.78
21	23.357	-0.53	-0.60	22.20
22	23.837	-18.69	0.95	-0.24
23	24.885	3.23	4.20	-9.53
24	26.140	-3.54	-6.77	-0.04
25	26.241	0.13	2.19	0.56
26	27.501	-9.33	-3.83	9.80
27	27.615	9.06	0.97	12.21
28	28.316	0.26	-0.32	-1.45

TABLE 3.7B-31
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, BEST ESTIMATE, SSE

(Sheet 2 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
29	28.682	-0.75	-6.47	1.05
30	28.817	4.15	0.74	8.58
31	29.251	-15.59	4.42	2.89
32	30.440	1.34	0.02	-7.45
33	31.276	0.40	-4.13	-3.58
34	33.304	-0.64	-0.11	3.35
35	33.645	1.25	-0.55	10.75
36	35.739	8.76	0.16	-0.24
37	37.197	-1.09	-0.09	1.34
38	38.610	-0.27	-3.03	0.09
39	39.486	1.74	-0.48	0.27
40	39.981	-0.55	0.74	0.65
41	40.158	-0.03	0.00	0.76
42	41.576	-0.00	-0.20	-1.11
43	42.263	0.03	-0.13	-0.09
44	44.258	0.03	-1.20	0.30
45	49.154	0.16	-1.67	-0.04
46	52.958	0.32	-0.01	0.60
47	53.039	-0.60	-0.00	0.33
48	54.006	0.20	-1.50	-0.01
49	56.811	0.10	-0.05	0.38
50	56.850	-0.41	0.17	0.11
51	59.109	-0.06	0.81	-0.00
52	69.706	-0.18	-0.33	-0.12
53	73.707	-0.04	-0.20	0.21
54	81.140	-0.08	0.25	0.04

TABLE 3.7B-32
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, UPPER BOUND, SSE

(Sheet 1 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	3.455	2.93	-0.12	41.23
2	3.456	41.09	-0.07	42.93
3	6.523	0.87	-0.88	34.22
4	7.339	37.32	-2.83	-0.47
5	7.642	-2.27	0.15	1.47
6	9.138	-2.39	-0.12	12.39
7	10.372	6.98	0.57	31.44
8	10.376	20.94	1.26	-6.79
9	10.628	0.31	51.84	0.13
10	15.806	4.48	6.72	23.53
11	16.363	18.71	25.50	4.80
12	18.818	17.93	-18.08	0.96
13	18.974	-1.22	1.00	11.52
14	19.477	7.39	-27.49	-0.32
15	19.678	-1.54	-1.86	3.60
16	20.271	0.68	3.17	-4.27
17	20.537	3.97	0.59	0.31
18	20.662	-0.42	-1.11	-2.25
19	21.491	0.17	0.23	-0.92
20	24.138	3.19	19.73	0.94
21	24.737	-0.03	-0.39	15.31
22	24.812	13.43	-1.62	-0.01
23	25.476	1.02	-2.86	-0.33
24	27.638	0.19	0.17	6.03
25	27.704	4.30	-1.46	-0.32
26	28.566	1.66	0.77	1.02
27	28.570	1.35	1.09	-1.05
28	29.988	0.58	-1.43	14.24

TABLE 3.7B-32
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS UNCRACKED MODEL, UPPER BOUND, SSE

(Sheet 2 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
29	30.654	0.61	-0.29	2.82
30	32.704	18.02	5.99	-0.60
31	34.011	-3.42	21.82	0.40
32	35.289	-0.95	0.58	0.50
33	36.745	1.85	0.70	-0.66
34	36.745	0.59	-0.71	-3.04
35	37.206	-1.42	-0.14	4.46
36	39.727	-13.63	2.10	-3.79
37	40.369	0.07	-0.27	-3.39
38	40.558	-0.17	9.57	17.27
39	40.909	-2.65	-7.10	24.17
40	41.954	24.56	1.65	1.44
41	43.860	3.08	-7.11	-2.29
42	48.424	1.94	2.43	0.11
43	48.600	0.00	1.49	-2.03
44	50.527	-0.24	5.49	0.00
45	55.746	0.71	2.44	0.06
46	55.809	0.05	0.35	-0.70
47	56.278	-0.04	5.92	-0.01
48	60.814	-0.40	-6.43	-0.03
49	62.444	-0.09	-0.10	1.27
50	68.033	0.40	-1.90	1.53
51	70.793	1.03	-2.65	-1.23
52	76.486	2.61	0.50	0.04
53	80.807	-0.89	1.14	-2.28
54	83.946	1.21	-1.30	-1.20

TABLE 3.7B-33
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, LOWER BOUND, SSE

(Sheet 1 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	2.124	3.81	-0.45	47.31
2	2.125	47.21	-0.42	-3.81
3	5.013	3.06	0.15	-6.51
4	5.260	2.96	-6.09	42.73
5	5.512	44.36	-23.07	-3.45
6	5.671	13.36	80.54	2.59
7	7.117	0.29	-0.34	21.11
8	7.223	15.39	-0.69	-0.05
9	8.365	-0.64	-0.06	5.46
10	9.746	4.51	28.16	2.62
11	10.913	3.71	-2.37	35.41
12	11.138	35.00	-3.61	-3.83
13	13.008	-0.33	0.87	-5.29
14	13.257	4.86	0.08	-1.39
15	13.356	-2.59	0.22	-2.25
16	14.355	-0.34	0.02	-4.69
17	14.555	-6.71	1.00	0.63
18	16.117	-0.55	-0.70	3.48
19	16.806	1.99	0.56	0.76
20	17.325	-0.64	-0.35	-2.76
21	17.445	-0.78	-0.14	0.80
22	17.614	-0.41	0.14	-0.08
23	17.850	-0.78	1.76	0.02
24	19.269	-0.53	-0.16	5.78
25	19.593	3.84	1.34	1.05
26	20.024	-0.82	-0.33	1.09
27	20.219	-3.05	-0.88	-3.49
28	20.319	-2.35	-0.50	3.82

TABLE 3.7B-33
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, LOWER BOUND, SSE

(Sheet 2 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
29	20.689	3.59	1.73	3.23
30	21.174	3.12	1.47	-3.60
31	22.172	0.21	0.43	0.42
32	23.212	-3.89	1.88	-0.17
33	23.301	-0.87	0.31	1.06
34	26.483	-0.06	0.09	0.12
35	27.299	0.09	0.01	-0.56
36	27.692	0.06	-1.38	-0.01
37	28.018	-0.02	0.08	-0.54
38	28.661	0.03	-0.05	-0.14
39	29.212	-0.34	1.88	0.34
40	32.234	-0.16	0.17	-1.50
41	34.961	-1.23	-0.07	0.05
42	37.119	0.35	-0.02	0.07
43	37.208	-0.10	0.02	0.15
44	37.857	0.08	-0.24	-0.23
45	37.953	-0.01	-0.51	0.12
46	39.945	-0.16	0.02	-0.04
47	40.061	-0.05	0.04	0.12
48	41.664	-0.00	0.27	0.01
49	41.796	-0.00	-0.06	0.02
50	44.157	-0.00	-0.31	0.07
51	49.352	0.06	-0.48	-0.02
52	69.633	-0.03	-0.07	-0.02
53	73.620	-0.00	-0.04	0.04
54	81.101	0.01	-0.06	-0.00

TABLE 3.7B-34
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, BEST ESTIMATE, SSE

(Sheet 1 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	2.441	3.26	-0.13	41.00
2	2.441	40.95	-0.12	-3.26
3	5.399	0.38	0.00	-0.67
4	6.243	1.21	-1.37	39.72
5	6.856	45.55	-4.27	-1.14
6	7.320	0.25	0.65	18.99
7	7.387	6.37	9.39	-0.00
8	7.461	0.83	56.16	0.13
9	8.990	-1.79	-0.14	10.23
10	13.254	4.48	4.18	18.19
11	13.278	16.68	14.63	-3.70
12	13.655	3.14	60.53	5.18
13	14.451	1.38	2.58	-7.09
14	14.458	6.27	-2.44	1.10
15	15.104	-0.00	-0.35	-1.80
16	15.293	1.10	-4.02	26.09
17	16.428	27.84	-14.08	-1.04
18	17.528	-0.64	-0.17	-8.01
19	17.600	-3.79	0.14	0.86
20	18.040	6.14	-12.29	0.01
21	19.073	1.99	2.09	-1.81
22	19.257	-0.66	-1.95	1.61
23	19.435	-1.94	0.33	3.40
24	19.831	-1.01	-0.81	4.08
25	19.901	2.63	0.63	0.45
26	20.155	-1.55	-1.28	1.97
27	21.618	0.53	0.57	-2.22
28	22.379	5.07	12.10	1.29

TABLE 3.7B-34
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, BEST ESTIMATE, SSE

(Sheet 2 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
29	24.264	10.65	7.58	-0.37
30	24.635	1.05	-0.06	-12.99
31	24.980	-0.77	0.31	5.72
32	25.651	1.92	-0.71	26.71
33	26.519	-7.36	-0.06	-2.31
34	26.613	27.71	0.28	-1.48
35	28.514	-1.03	7.28	-0.34
36	28.521	-1.11	6.05	0.56
37	38.126	-0.65	1.02	-8.52
38	31.134	2.92	-5.71	-4.38
39	32.955	-0.14	0.75	-9.49
40	33.507	-2.06	0.41	-3.68
41	35.547	-7.46	-0.31	0.39
42	37.687	3.03	0.32	0.52
43	37.824	0.39	0.87	-1.05
44	38.107	1.06	-2.53	-0.54
45	38.725	-0.22	-0.34	1.87
46	40.028	-1.30	0.18	-0.38
47	40.334	-0.55	0.36	1.21
48	41.703	0.05	-1.30	-0.12
49	42.475	-0.70	0.25	-0.33
50	44.475	-0.10	-1.36	0.68
51	49.594	0.41	-2.17	-0.12
52	69.670	-0.17	-0.31	-0.11
53	73.669	-0.04	-0.19	0.20
54	81.123	-0.08	0.25	0.04

TABLE 3.7B-35
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, UPPER BOUND, SSE

(Sheet 1 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	2.502	2.95	-0.05	39.49
2	2.502	39.46	-0.03	-2.95
3	5.459	0.16	-0.00	-0.27
4	6.501	0.99	-0.94	37.09
5	7.229	43.28	-2.52	-0.73
6	7.352	0.23	0.11	19.62
7	7.438	7.95	-2.92	-0.15
8	7.712	0.94	46.33	0.09
9	9.131	-2.13	-0.08	11.00
10	13.439	1.56	0.12	13.85
11	13.446	13.03	0.95	-1.40
12	14.541	-0.11	-0.44	-3.93
13	14.543	3.92	0.04	-0.02
14	15.329	0.13	0.32	-0.08
15	15.808	-5.32	10.81	22.44
16	16.235	16.43	32.13	-6.41
17	17.394	-11.81	14.47	0.11
18	17.620	1.46	-0.15	-8.70
19	17.638	6.53	-1.40	-1.47
20	19.330	12.88	-26.39	0.10
21	19.484	-0.86	2.15	2.37
22	19.549	-4.04	15.52	-0.06
23	19.707	-1.64	-2.61	3.76
24	19.942	-0.02	0.98	-1.69
25	19.950	1.41	0.84	0.16
26	20.398	0.77	3.02	-1.34
27	21.904	0.09	0.13	-0.53
28	24.254	-1.26	-20.44	-0.72

TABLE 3.7B-35
CONTAINMENT AND INTERNAL STRUCTURES MODAL FREQUENCIES AND PARTICIPATION FACTORS CRACKED MODEL, UPPER BOUND, SSE

(Sheet 2 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
29	25.308	0.07	-0.01	-0.68
30	26.257	1.23	4.68	-0.45
31	26.554	-0.27	0.00	0.52
32	26.666	0.52	1.03	0.48
33	28.548	0.02	0.01	-0.38
34	29.675	0.83	-1.09	19.54
35	32.420	22.28	3.06	-0.86
36	34.720	-2.14	18.34	0.06
37	35.223	0.53	3.22	-1.50
38	35.377	0.88	-7.93	-1.77
39	37.005	-2.25	0.34	6.43
40	38.961	-15.73	3.07	-5.54
41	39.470	0.74	6.96	15.07
42	39.648	-1.35	-4.78	-15.72
43	39.689	-7.42	3.51	-6.53
44	39.952	-9.23	-6.92	20.32
45	41.471	20.52	-0.00	2.78
46	42.092	4.75	5.26	-2.40
47	44.362	2.01	-8.31	-2.04
48	53.338	1.16	12.28	-0.14
49	59.789	-0.07	-0.06	1.44
50	65.451	-0.72	1.23	-3.18
51	68.199	2.87	-2.84	-1.29
52	73.305	-2.08	-0.75	-0.67
53	78.154	0.24	-0.94	2.32
54	82.853	0.96	-1.36	-0.74

TABLE 3.7B-36
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS LOWER BOUND, SSE

(Sheet 1 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	4.768	31.13	-2.71	-16.53
2	6.696	19.08	20.59	30.70
3	9.581	-3.96	56.19	-15.67
4	10.594	2.74	-12.57	17.20
5	13.228	27.50	-12.15	-16.98
6	15.762	19.26	15.99	18.70
7	18.075	11.04	-1.47	-7.10
8	18.526	1.83	-5.70	15.37
9	23.246	6.57	-1.94	3.00
10	24.216	-6.99	-3.07	13.24
11	24.669	11.63	2.24	5.36
12	26.456	-6.46	5.25	0.72
13	27.883	-1.63	1.26	-0.27
14	30.224	0.44	2.80	4.56
15	32.850	2.16	1.01	-2.59
16	33.823	1.12	1.41	-0.58
17	34.718	0.98	-1.98	-0.44
18	37.609	-0.47	-0.35	0.17
19	40.132	-1.58	-0.12	0.71
20	41.165	0.23	0.16	-1.07
21	41.993	-0.69	-0.21	0.06

TABLE 3.7B-36
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS LOWER BOUND, SSE

(Sheet 2 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
22	44.005	0.17	-0.11	-0.08
23	44.050	0.66	-0.51	-0.19
24	45.674	0.01	0.26	-0.38
25	48.003	-0.43	0.21	-0.52
26	48.878	0.94	0.08	-0.30
27	51.920	-0.02	1.05	0.05
28	52.807	-0.04	1.09	0.12
29	55.348	0.50	0.20	0.04
30	59.281	0.08	-0.68	0.01
31	60.856	-0.12	0.22	0.00
32	63.802	0.14	0.03	0.00
33	65.907	0.07	-0.05	0.00
34	68.573	-0.00	-0.46	-0.01
35	74.424	0.02	0.11	0.01
36	77.527	-0.03	0.01	0.01
37	86.736	-0.02	-0.02	-0.00
38	91.109	-0.00	0.32	0.00
39	97.830	0.00	-0.08	-0.00
40	06.868	0.00	-0.00	-0.00
41	117.617	-0.00	-0.18	-0.00
42	137.447	-0.00	-0.06	-0.00

TABLE 3.7B-37
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 1 of 2)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	7.082	33.05	-1.63	-12.02
2	8.662	11.20	8.72	32.89
3	14.099	9.35	12.96	6.86
4	14.916	2.79	48.62	-12.85
5	18.540	-19.12	24.17	5.70
6	21.540	0.35	4.63	17.32
7	22.725	14.22	10.02	3.09
8	27.266	-1.76	13.72	3.47
9	29.366	3.64	6.73	13.11
10	30.466	-4.91	-19.58	9.45
11	31.796	-0.85	4.76	-3.19
12	34.137	17.68	-8.78	1.58
13	36.238	-11.75	-4.07	-3.65
14	38.271	-3.70	-4.86	-16.85
15	39.362	-1.43	8.94	-11.32
16	41.648	7.15	-3.70	-3.46
17	42.641	-11.17	-4.21	-11.94
18	43.723	6.58	1.62	1.03
19	44.481	-13.22	0.19	13.95
20	46.026	1.95	-0.06	-5.03
21	46.838	5.60	0.03	0.70

TABLE 3.7B-37
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 2 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
22	49.260	1.64	0.71	2.24
23	49.949	2.50	-0.37	0.41
24	51.666	-5.20	-0.80	2.40
25	52.897	-3.03	0.72	-0.38
26	54.688	-1.40	6.02	0.11
27	56.262	3.68	1.99	-2.70
28	58.654	3.27	0.12	0.80
29	60.134	-0.29	-0.41	-2.25
30	60.400	-1.14	2.44	-0.24
31	63.725	0.48	-2.07	0.22
32	64.124	-0.82	-1.25	0.05
33	69.447	-0.33	-1.10	-0.07
34	69.987	0.41	-1.75	-0.05
35	75.545	0.11	0.78	0.06
36	77.734	0.18	0.02	-0.06
37	87.496	0.10	0.24	0.01
38	92.059	-0.02	1.35	0.03
39	97.915	-0.02	0.40	0.02
40	106.878	-0.00	0.00	0.00
41	118.232	-0.02	-0.81	-0.04
42	137.557	-0.01	-0.27	-0.01

TABLE 3.7B-38
SERVICE WATER INTAKE STRUCTURE MODAL FREQUENCIES AND
PARTICIPATION FACTORS LOWER BOUND, SSE

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	11.313	-21.293	-1.791	0.213
2	14.626	-.5070001	3.043	-25.66
3	16.241	-1.043	47.581	1.991
4	17.707	-3.707	0.337	2.373
5	23.975	-28.005	-0.264	0.01
6	24.767	-0.155	0.998	-22.391
7	29.819	-0.363	0.288	13.161
8	33.756	-3.922	-2.451	-0.89
9	34.974	-9.597	0.324	-1.413
10	35.444	-3.913	-1.371	1.243
11	41.936	-1.891	0.861	8.776
12	43.327	-3.692	2.796	-2.289
13	46.842	-4.94	-0.888	0.158
14	49.277	-1.391	-0.141	-3.584
15	52.073	-1.658	-0.116	0.915
16	54.557	-0.759	0.269	-0.205
17	57.317	-1.173	0.076	0.382
18	59.172	-0.04	-0.063	-0.677

TABLE 3.7B-39
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS UPPER BOUND, SSE

(Sheet 1 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	7.882	33.22	-1.11	-8.20
2	9.169	6.52	6.53	33.16
3	15.142	11.04	5.77	6.88
4	16.759	4.77	42.63	-10.11
5	20.776	15.97	-25.10	-5.36
6	22.414	3.32	3.36	14.15
7	25.931	-10.50	14.08	0.25
8	30.545	-2.03	8.49	11.39
9	31.660	-2.44	6.69	-6.53
10	35.141	-4.87	-22.00	3.52
11	35.591	4.09	5.04	1.63
12	37.550	-10.23	17.18	-0.27
13	39.709	8.97	4.45	-6.01
14	40.757	4.73	9.78	6.53
15	43.854	1.43	6.41	1.62
16	47.031	-15.06	8.15	-8.62
17	48.584	5.14	7.19	-3.43
18	50.038	2.51	4.55	4.31
19	52.349	4.75	-0.86	14.50
20	52.780	-3.39	-4.50	2.12
21	53.786	-4.84	2.33	-15.83

TABLE 3.7B-39
SAFEGUARDS BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS UPPER BOUND, SSE

(Sheet 2 of 2)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
22	55.488	-0.78	-8.67	-15.75
23	58.298	-7.77	-7.76	1.99
24	59.792	18.14	-1.69	-1.57
25	61.324	-4.96	-4.87	9.78
26	61.574	-0.36	-4.31	-8.73
27	63.237	-5.66	2.00	2.20
28	64.686	-12.00	0.35	5.60
29	66.737	1.46	-0.69	-2.61
30	68.534	-0.13	-7.23	-0.10
31	70.712	6.23	1.88	-0.84
32	71.898	2.00	-2.34	3.37
33	73.001	-0.38	-0.57	-5.12
34	75.630	3.23	-2.04	0.43
35	77.047	-0.22	2.50	0.06
36	78.285	-1.04	-0.60	0.35
37	88.803	-0.26	-0.96	-0.00
38	93.0569	-0.05	2.82	0.09
39	98.114	0.07	-1.14	-0.08
40	106.897	0.00	0.00	-0.00
41	119.263	0.04	1.84	0.09
42	137.744	0.02	0.63	0.03

TABLE 3.7B-40
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS LOWER BOUND, SSE

(Sheet 1 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	4.309	-3.45	3.42	53.63
2	4.615	57.29	-22.12	3.79
3	6.360	21.59	103.60	-2.22
4	6.937	-1.20	-5.31	-8.91
5	8.086	39.73	-17.46	2.02
6	8.431	1.44	-2.44	-55.24
7	9.539	13.16	-24.79	0.46
8	12.920	-28.79	-1.82	0.85
9	14.204	0.00	0.22	-0.37
10	15.470	0.60	-0.47	10.77
11	15.919	-2.65	0.19	-16.86
12	18.125	14.90	0.46	-10.31
13	18.212	8.14	0.68	15.11
14	19.347	4.92	-1.56	2.34
15	19.983	7.87	-2.15	-2.12
16	21.023	-9.06	-0.81	-0.48
17	22.377	-0.58	0.54	-9.27
18	22.525	2.64	5.71	0.92
19	22.940	5.84	1.64	-1.20
20	24.146	-0.42	0.16	-4.27
21	24.566	5.43	0.59	-0.27
22	26.007	-7.92	2.39	0.97
23	26.276	1.45	-0.42	4.49

TABLE 3.7B-40
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS LOWER BOUND, SSE

(Sheet 2 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
24	27.308	0.79	-0.34	2.68
25	27.842	0.24	-0.38	-2.45
26	29.259	3.32	1.52	0.47
27	30.170	0.51	0.13	-5.83
28	32.358	-0.07	-0.05	2.35
29	32.848	-0.63	-0.26	-0.58
30	34.566	0.09	-0.06	0.01
31	36.090	-0.06	0.02	-0.74
32	37.811	0.59	-0.04	-0.08
33	38.080	-1.36	0.07	0.18
34	39.431	0.28	-0.00	0.91
35	41.089	-0.37	-0.00	-0.71
36	41.336	-0.48	-0.00	0.58
37	43.906	-0.11	-0.02	-0.24
38	44.187	0.59	0.05	-0.10
39	44.989	0.03	0.35	0.01
40	47.079	-0.13	0.05	0.11
41	48.701	-0.01	0.01	-0.23
42	48.954	0.02	-0.16	0.02
43	50.440	0.06	-0.01	0.11
44	51.466	0.00	0.08	0.02
45	51.779	-0.01	0.03	0.07

TABLE 3.7B-40
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS LOWER BOUND, SSE

(Sheet 3 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
46	52.841	0.03	0.00	0.08
47	56.324	-0.02	-0.00	-0.05
48	58.716	0.01	-0.28	-0.00
49	60.178	0.00	0.00	0.02
50	64.398	-0.00	0.00	0.02
51	65.300	0.01	-0.05	0.00
52	69.281	0.03	0.00	-0.00
53	69.974	0.01	0.14	0.00
54	78.774	0.00	-0.02	0.01
55	85.197	-0.00	0.04	0.00
56	85.517	-0.01	0.00	-0.00
57	91.319	-0.00	-0.05	0.00
58	93.209	0.00	-0.01	-0.00
59	94.674	-0.00	0.02	-0.00
60	95.244	-0.00	0.00	0.00
61	100.430	0.00	0.00	0.00
62	101.711	0.00	0.00	0.00
63	106.190	-0.00	0.01	-0.00
64	110.577	-0.00	0.00	0.00
65	113.414	-0.00	0.00	0.00
66	113.800	-0.00	0.00	-0.00

TABLE 3.7B-41
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 1 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	6.070	61.27	-3.63	-5.41
2	6.478	5.97	-0.17	61.42
3	8.708	1.74	-1.17	-6.78
4	11.194	7.76	94.79	0.01
5	12.456	23.34	-21.79	-3.03
6	12.680	-0.92	3.06	-30.31
7	14.694	11.01	36.54	0.23
8	15.613	18.85	-22.12	-0.42
9	16.403	-3.29	1.61	-6.98
10	18.671	1.94	4.63	-4.39
11	19.891	-8.71	6.27	1.07
12	20.783	0.77	0.23	18.60
13	22.109	1.39	-1.90	-11.75
14	22.413	18.45	-9.59	0.41
15	22.897	3.26	-0.36	-0.18
16	23.177	-10.82	8.85	-0.01
17	24.474	5.16	2.41	-0.70
18	24.639	13.75	22.96	-0.85
19	26.736	0.49	-0.43	-23.66
20	27.260	0.66	-0.13	5.27
21	28.337	5.02	-1.32	-3.81
22	29.000	-11.27	6.30	-9.05
23	29.040	9.16	-7.05	-7.30

TABLE 3.7B-41
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 2 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
24	30.076	-0.26	0.53	10.86
25	31.354	-5.93	-8.24	0.39
26	32.998	-19.66	0.25	-0.16
27	34.047	1.39	0.01	5.08
28	36.599	2.64	0.72	-1.51
29	36.969	0.19	-0.10	18.44
30	38.039	-0.80	0.20	6.29
31	38.630	-1.42	-0.54	0.75
32	39.407	20.56	0.04	-1.62
33	39.851	5.70	-0.05	2.97
34	41.640	-5.31	-0.00	-6.36
35	41.904	11.92	0.07	-0.41
36	43.084	0.94	-0.03	-15.90
37	44.515	-3.09	-0.19	1.41
38	44.777	0.26	-0.01	7.39
39	45.109	0.28	1.58	0.08
40	47.194	-0.30	0.24	0.68
41	48.735	0.03	-0.07	1.53
42	49.017	0.21	-0.69	0.05
43	50.510	-0.29	0.09	-0.37
44	51.523	0.00	0.33	-0.07
45	51.799	-0.11	0.14	0.51

TABLE 3.7B-41
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 3 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
46	52.891	0.15	0.01	0.78
47	56.336	0.10	0.03	0.26
48	58.819	0.06	-1.25	-0.00
49	60.184	-0.00	0.00	0.15
50	64.433	-0.01	0.02	0.12
51	65.364	-0.09	0.26	-0.01
52	69.300	-0.15	-0.03	0.03
53	70.023	0.07	0.61	0.03
54	78.789	0.00	-0.10	0.06
55	85.204	-0.00	0.19	0.00
56	88.522	-0.06	0.00	-0.00
57	91.331	0.00	0.21	-0.00
58	93.212	0.00	-0.05	-0.01
59	94.681	0.01	-0.10	0.00
60	95.247	0.01	-0.00	0.02
61	100.430	0.00	0.03	0.00
62	101.713	0.03	0.03	0.00
63	106.195	-0.00	0.05	-0.00
64	110.577	0.00	-0.00	-0.00
65	113.415	-0.01	0.02	-0.01
66	113.801	0.00	0.01	0.00

TABLE 3.7B-42
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 1 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	6.385	60.53	-1.01	-3.62
2	7.114	4.19	-0.43	62.20
3	9.070	2.05	-0.53	-8.38
4	13.825	-9.86	76.26	4.55
5	13.862	23.69	-36.36	-1.37
6	15.231	-1.87	14.58	-22.50
7	16.628	-0.79	4.18	-2.39
8	16.874	5.39	37.73	0.82
9	18.410	10.03	-24.13	-0.12
10	19.974	-8.55	-8.20	-0.70
11	20.459	4.72	2.23	0.68
12	22.594	0.01	-1.99	-17.19
13	23.065	8.36	-6.25	-0.92
14	23.357	0.46	-0.98	2.19
15	24.502	15.40	-14.13	0.63
16	25.142	2.31	-3.73	-0.41
17	26.162	-7.27	-14.28	-0.10
18	27.292	5.37	41.10	-1.38
19	28.190	1.18	1.95	-2.21
20	29.172	0.67	-2.32	-16.59
21	31.297	4.76	-8.08	-0.08
22	31.631	5.33	-4.43	-10.71
23	31.910	4.53	-2.47	13.98

TABLE 3.7B-42
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 2 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
24	32.948	-1.93	1.19	1.02
25	34.683	6.60	-21.69	0.23
26	35.956	-18.66	6.63	0.05
27	36.731	-1.47	0.24	-0.20
28	38.904	2.52	-0.44	12.43
29	39.227	2.06	2.76	1.06
30	40.951	-6.96	0.06	6.93
31	41.102	-8.02	0.12	-3.23
32	41.557	-15.43	1.27	1.09
33	42.411	0.37	-0.09	12.35
34	44.175	-0.76	-0.06	-1.61
35	45.267	2.33	-3.39	0.65
36	45.969	13.10	1.58	-3.62
37	46.418	6.57	0.43	22.52
38	47.822	15.22	-0.18	-6.82
39	48.952	-0.67	0.19	7.62
40	49.135	-0.37	1.40	-3.33
41	50.695	1.98	-0.25	6.80
42	51.594	0.08	0.67	2.69
43	51.855	-0.19	0.39	1.39
44	52.892	2.02	0.04	-1.84
45	54.565	26.38	0.00	0.13

TABLE 3.7B-42
ELECTRICAL AND AUXILIARY BUILDINGS MODAL FREQUENCIES AND PARTICIPATION FACTORS BEST ESTIMATE, SSE

(Sheet 3 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
46	56.360	0.24	0.08	2.47
47	57.441	-0.09	-0.01	22.34
48	58.991	-0.07	-2.82	-0.05
49	60.220	-0.04	-0.00	2.16
50	64.486	-0.07	0.05	0.48
51	65.468	-0.42	0.65	-0.05
52	69.329	-0.39	-0.06	0.08
53	70.094	-0.19	-1.30	-0.07
54	78.811	-0.01	0.21	-0.14
55	85.214	0.00	0.40	0.00
56	88.530	-0.14	0.02	-0.01
57	91.345	-0.01	-0.43	0.00
58	93.218	0.02	-0.10	-0.04
59	94.693	0.04	-0.22	0.01
60	95.254	-0.04	0.00	0.06
61	100.431	0.01	0.06	0.00
62	101.717	0.07	0.06	0.00
63	106.197	0.01	-0.11	0.00
64	110.579	0.00	-0.00	-0.01
65	113.418	-0.01	0.05	-0.02
66	113.801	0.00	-0.03	0.00

TABLE 3.7B-43
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
LOWER BOUND, SSE

(Sheet 1 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	8.014	46.79	12.94	4.23
2	8.395	-2.53	-12.17	49.62
3	9.260	-4.86	0.61	-6.76
4	10.372	-11.69	74.34	10.36
5	14.203	38.39	5.27	0.12
6	14.551	-0.07	-6.66	33.83
7	18.252	2.96	7.07	-2.86
8	18.578	-3.47	-2.13	-5.19
9	19.580	0.60	3.02	5.48
10	29.180	0.74	-4.43	-0.05
11	31.674	-2.72	-3.18	2.29
12	32.240	1.03	1.39	4.78
13	33.654	0.46	0.26	-0.26
14	34.636	-1.93	0.37	-1.12
15	36.847	-1.54	0.97	1.78
16	37.046	-3.90	0.55	-0.13
17	38.128	0.89	-0.25	2.12
18	40.202	-0.03	-0.36	-1.56
19	42.014	0.23	0.16	-0.22
20	43.311	-0.16	-0.75	0.48
21	46.039	0.51	0.14	0.37

TABLE 3.7B-43
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
LOWER BOUND, SSE

(Sheet 2 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
22	46.562	0.33	-0.02	-0.30
23	51.890	-0.18	0.29	-0.10
24	52.565	0.04	-0.06	0.00
25	52.759	0.12	0.17	-0.21
26	55.343	0.04	0.00	0.04
27	56.413	-0.19	0.90	-0.20
28	57.616	-0.30	0.40	0.30
29	59.398	0.09	-0.29	-0.43
30	62.495	-0.51	-0.08	-0.42
31	63.225	-0.30	-0.34	0.30
32	65.080	-0.15	-0.62	0.08
33	68.111	0.02	-0.23	-0.01
34	71.964	0.17	-0.00	-0.08
35	73.321	-0.20	0.09	-0.10
36	74.423	-0.20	0.07	0.04
37	78.191	-0.00	-0.05	-0.01
38	38.532	0.01	-0.02	-0.12
39	87.897	0.01	0.01	0.02
40	101.105	-0.01	-0.05	0.01
41	102.827	-0.02	-0.00	-0.12
42	105.013	0.01	0.01	-0.09

TABLE 3.7B-43
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
LOWER BOUND, SSE

(Sheet 3 of 3)

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
43	106.400	-0.10	-0.05	0.01
44	107.610	-0.00	-0.02	-0.07
45	109.281	-0.11	0.00	-0.00
46	111.256	-0.01	0.00	-0.02
47	113.420	-0.02	-0.02	-2.02
48	114.764	0.02	0.00	-0.00
49	115.340	0.02	0.04	-0.00
50	118.027	0.04	-0.07	-0.00
51	122.716	-0.00	0.00	-0.00
52	133.602	0.00	0.00	0.00
53	137.483	0.00	0.00	-0.00
54	139.213	-0.00	0.04	-0.00
55	172.564	0.00	0.00	0.00
56	185.389	-0.00	-0.00	-0.00
57	189.382	-0.00	-0.03	0.00
58	197.406	-0.00	0.01	-0.00
59	200.907	-0.00	-0.00	0.00
60	269.036	0.00	-0.00	0.00

TABLE 3.7B-44
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
UPPER BOUND, SSE

(Sheet 1 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
1	9.482	41.46	5.61	0.94
2	10.341	0.04	-6.13	46.06
3	11.725	1.12	-0.99	7.91
4	14.219	-2.81	75.50	6.00
5	17.529	42.37	-3.00	2.61
6	18.283	-6.85	-3.18	32.46
7	20.578	-3.12	2.56	-20.52
8	22.398	12.35	4.80	1.49
9	22.826	2.48	6.12	-1.61
10	29.562	-1.88	8.96	0.10
11	33.147	3.61	3.48	-9.13
12	33.712	1.19	1.15	4.34
13	33.822	-7.08	-6.22	-3.23
14	35.564	1.41	-1.64	1.25
15	37.509	-0.92	-1.50	1.10
16	38.558	-3.80	1.66	5.06
17	39.105	7.30	-2.01	2.85
18	40.455	-0.16	-0.97	-4.51
19	42.050	-0.72	-0.30	0.58
20	43.400	-0.29	-1.73	1.25
21	46.100	1.21	0.37	0.97
22	46.616	0.79	0.05	-0.72
23	51.913	0.41	-0.58	0.22

TABLE 3.7B-44
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
UPPER BOUND, SSE

(Sheet 2 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
24	52.598	-0.07	0.15	-0.04
25	52.781	0.28	0.35	-0.01
26	55.346	0.09	0.01	0.10
27	56.566	0.40	-1.85	0.47
28	57.734	-0.64	0.85	0.61
29	59.521	0.24	-0.65	-0.96
30	62.721	-1.03	-0.10	-1.01
31	63.381	-0.76	-0.72	0.56
32	65.320	-0.36	-1.32	0.19
33	68.151	-0.06	0.52	0.02
34	72.008	0.28	0.00	-0.16
35	73.365	0.40	-0.19	0.22
36	74.483	-0.43	0.17	0.08
37	78.203	-0.01	-0.11	-0.02
38	83.553	0.03	-0.05	-0.24
39	87.899	-0.03	-0.02	-0.05
40	101.110	-0.03	-0.11	0.03
41	102.847	0.05	0.01	0.25
42	105.024	0.02	0.02	-0.19
43	106.426	0.21	0.12	-0.03
44	107.621	0.00	0.04	0.15
45	109.302	-0.24	0.01	-0.01

TABLE 3.7B-44
FUEL BUILDING MODAL FREQUENCIES AND PARTICIPATION FACTORS
UPPER BOUND, SSE

(Sheet 3 of 3)

Mode	Frequency (Hz)	Participation Factors		
		X-Earthquake	Y-Earthquake	Z-Earthquake
46	111.259	-0.04	0.00	-0.05
47	113.426	-0.05	-0.05	-0.04
48	114.772	0.05	0.00	-0.01
49	115.350	0.04	0.09	-0.01
50	118.048	0.08	-0.15	0.00
51	122.723	0.00	0.03	-0.00
52	133.608	0.00	0.00	0.00
53	137.490	0.00	0.00	-0.00
54	139.228	0.01	-0.09	0.00
55	172.565	0.00	0.00	0.00
56	185.393	-0.00	-0.00	-0.00
57	189.393	-0.00	-0.06	0.00
58	197.410	-0.00	0.03	-0.00
59	200.912	0.00	0.00	-0.00
60	269.038	0.00	-0.00	0.00

TABLE 3.7B-45
SERVICE WATER INTAKE STRUCTURE MODAL FREQUENCIES AND PARTICIPATION FACTORS UPPER BOUND, SSE

		Participation Factors		
Mode	Frequency (Hz)	X-Earthquake	Y-Earthquake	Z-Earthquake
1	11.808	-19.204	-0.995	0.186
2	16.07	-0.466	1.042	-23.067
3	18.161	-2.899	-0.288	2.952
4	22.329	-0.714	47.003	1.004
5	27.753	-23.194	-0.41	-0.318
6	31.033	-0.3	-1.656	21.343
7	35.528	-3.489	0.61	-5.757
8	35.606	-4.602	2.56	6.371
9	39.008	-10.54	1.926	-1.433
10	41.556	-8.903	-7.544	-0.517
11	47.207	-10.72	3.002	6.139
12	47.285	-3.796	1.4	-14.066
13	51.626	-12.732	-0.8	0.785
14	52.885	-1.152	-0.052	-11.36
15	55.16	-0.838	-0.038	0.066
16	59.357	-0.354	0.197	2.053
17	61.053	-3.046	0.465	-0.466
18	66.59	-1.022	0.073	0.668

TABLE 3.7B-46
CONTAINMENT AND INTERNAL STRUCTURES SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 1 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.67	0.11	0.09
2	0.11	0.57	0.03
3	0.09	0.04	0.68
4	0.11×10^{-2}	0.13×10^{-2}	0.36×10^{-2}
5	0.13×10^{-2}	0.03×10^{-2}	0.14×10^{-2}
6	0.39×10^{-2}	0.45×10^{-2}	0.12×10^{-2}
7	0.51	0.07	0.06
8	0.09	0.44	0.05
9	0.06	0.02	0.51
10	0.08×10^{-2}	0.05×10^{-2}	0.33×10^{-2}
11	0.13×10^{-2}	0.03×10^{-2}	0.12×10^{-2}
12	0.32×10^{-2}	0.11×10^{-2}	0.08×10^{-2}
13	0.39	0.07	0.05
14	0.08	0.40	0.05
15	0.05	0.02	0.40
16	0.06×10^{-2}	0.04×10^{-2}	0.29×10^{-2}
17	0.12×10^{-2}	0.03×10^{-2}	0.11×10^{-2}
18	0.28×10^{-2}	0.08×10^{-2}	0.06×10^{-2}
19	0.29	0.09	0.06
20	0.07	0.32	0.03
21	0.06	0.03	0.31
22	0.04×10^{-2}	0.03×10^{-2}	0.22×10^{-2}
23	0.10×10^{-2}	0.02×10^{-2}	0.08×10^{-2}
24	0.21×10^{-2}	0.10×10^{-2}	0.04×10^{-2}
25	0.17	0.08	0.04
26	0.05	0.26	0.02
27	0.04	0.03	0.18
28	0.02×10^{-2}	0.05×10^{-2}	0.15×10^{-2}
29	0.06×10^{-2}	0.01×10^{-2}	0.07×10^{-2}
30	0.15×10^{-2}	0.08×10^{-2}	0.03×10^{-2}

TABLE 3.7B-46
CONTAINMENT AND INTERNAL STRUCTURES SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 2 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
31	0.41	0.11	0.04
32	0.11	0.27	0.05
33	0.04	0.07	0.42
34	0.09×10^{-2}	0.12×10^{-2}	0.34×10^{-2}
35	0.09×10^{-2}	0.04×10^{-2}	0.34×10^{-2}
36	0.42×10^{-2}	0.44×10^{-2}	0.14×10^{-2}
37	0.26	0.10	0.03
38	0.06	0.25	0.03
39	0.03	0.06	0.25
40	0.07×10^{-2}	0.08×10^{-2}	0.27×10^{-2}
41	0.08×10^{-2}	0.05×10^{-2}	0.20×10^{-2}
42	0.27×10^{-2}	0.19×10^{-2}	0.09×10^{-2}
43	0.20	0.10	0.02
44	0.04	0.24	0.03
45	0.03	0.05	0.17
46	0.06×10^{-2}	0.06×10^{-2}	0.20×10^{-2}
47	0.06×10^{-2}	0.04×10^{-2}	0.14×10^{-2}
48	0.22×10^{-2}	0.10×10^{-2}	0.06×10^{-2}
49	0.14	0.04	0.02
50	0.04	0.23	0.02
51	0.02	0.02	0.14
52	0.02×10^{-2}	0.02×10^{-2}	0.13×10^{-2}
53	0.04×10^{-2}	0.01×10^{-2}	0.03×10^{-2}
54	0.13×10^{-2}	0.03×10^{-2}	0.02×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-19.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

TABLE 3.7B-47
SAFEGUARDS BUILDING SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 1 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.38	0.13	0.23
2	0.09	0.30	0.13
3	0.22	0.22	0.37
4	0.20×10^{-2}	0.30×10^{-2}	0.18×10^{-2}
5	0.28×10^{-2}	0.27×10^{-2}	0.37×10^{-2}
6	0.27×10^{-2}	0.32×10^{-2}	0.19×10^{-2}
7	0.28	0.08	0.17
8	0.05	0.22	0.09
9	0.18	0.19	0.28
10	0.19×10^{-2}	0.25×10^{-2}	0.16×10^{-2}
11	0.20×10^{-2}	0.23×10^{-2}	0.19×10^{-2}
12	0.25×10^{-2}	0.25×10^{-2}	0.18×10^{-2}
13	0.21	0.08	0.14
14	0.04	0.20	0.07
15	0.14	0.16	0.21
16	0.18×10^{-2}	0.20×10^{-2}	0.16×10^{-2}
17	0.18×10^{-2}	0.20×10^{-2}	0.17×10^{-2}
18	0.24×10^{-2}	0.20×10^{-2}	0.17×10^{-2}
19	0.15	0.09	0.10
20	0.04	0.18	0.07
21	0.10	0.11	0.13
22	0.17×10^{-2}	0.17×10^{-2}	0.14×10^{-2}
23	0.14×10^{-2}	0.14×10^{-2}	0.13×10^{-2}

TABLE 3.7B-47
SAFEGUARDS BUILDING SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 2 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
24	0.21×10^{-2}	0.14×10^{-2}	0.15×10^{-2}
25	0.12	0.05	0.07
26	0.05	0.15	0.06
27	0.06	0.07	0.12
28	0.16×10^{-2}	0.13×10^{-2}	0.14×10^{-2}
29	0.08×10^{-2}	0.09×10^{-2}	0.09×10^{-2}
30	0.20×10^{-2}	0.13×10^{-2}	0.14×10^{-2}
31	0.12	0.04	0.05
32	0.03	0.17	0.07
33	0.05	0.04	0.12
34	0.15×10^{-2}	0.11×10^{-2}	0.13×10^{-2}
35	0.08×10^{-2}	0.06×10^{-2}	0.10×10^{-2}
36	0.19×10^{-2}	0.11×10^{-2}	0.12×10^{-2}
37	0.12	0.05	0.05
38	0.04	0.18	0.08
39	0.05	0.04	0.12
40	0.14×10^{-2}	0.10×10^{-2}	0.13×10^{-2}
41	0.21×10^{-2}	0.05×10^{-2}	0.10×10^{-2}
42	0.18×10^{-2}	0.09×10^{-2}	0.12×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-20.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

TABLE 3.7B-48
ELECTRICAL AND AUXILIARY BUILDING SRSS ACCELERATIONS FOR SSE
FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 1 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.33	0.15	0.05
2	0.12	0.26	0.02
3	0.05	0.03	0.30
4	0.05×10^{-2}	0.05×10^{-2}	0.24×10^{-2}
5	0.05×10^{-2}	0.03×10^{-2}	0.10×10^{-2}
6	0.23×10^{-2}	0.28×10^{-2}	0.04×10^{-2}
7	0.28	0.14	0.04
8	0.12	0.24	0.01
9	0.04	0.03	0.28
10	0.05×10^{-2}	0.05×10^{-2}	0.23×10^{-2}
11	0.05×10^{-2}	0.03×10^{-2}	0.07×10^{-2}
12	0.22×10^{-2}	0.27×10^{-2}	0.03×10^{-2}
13	0.20	0.11	0.03
14	0.11	0.20	0.01
15	0.03	0.03	0.24
16	0.04×10^{-2}	0.04×10^{-2}	0.23×10^{-2}
17	0.04×10^{-2}	0.03×10^{-2}	0.06×10^{-2}
18	0.19×10^{-2}	0.24×10^{-2}	0.03×10^{-2}
19	0.14	0.08	0.02
20	0.11	0.19	0.01
21	0.03	0.02	0.17
22	0.03×10^{-2}	0.03×10^{-2}	0.22×10^{-2}
23	0.04×10^{-2}	0.02×10^{-2}	0.06×10^{-2}
24	0.18×10^{-2}	0.24×10^{-2}	0.02×10^{-2}

TABLE 3.7B-48
ELECTRICAL AND AUXILIARY BUILDING SRSS ACCELERATIONS FOR SSE
FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 2 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
25	0.12	0.04	0.01
26	0.10	0.18	0.01
27	0.02	0.01	0.12
28	0.03×10^{-2}	0.02×10^{-2}	0.21×10^{-2}
29	0.03×10^{-2}	0.01×10^{-2}	0.05×10^{-2}
30	0.17×10^{-2}	0.25×10^{-2}	0.02×10^{-2}
31	0.46	0.22	0.06
32	0.12	0.35	0.02
33	0.06	0.04	0.41
34	0.04×10^{-2}	0.12×10^{-2}	0.27×10^{-2}
35	0.06×10^{-2}	0.05×10^{-2}	0.14×10^{-2}
36	0.29×10^{-2}	0.20×10^{-2}	0.04×10^{-2}
37	0.34	0.17	0.04
38	0.09	0.32	0.02
39	0.05	0.03	0.35
40	0.04×10^{-2}	0.11×10^{-2}	0.26×10^{-2}
41	0.04×10^{-2}	0.04×10^{-2}	0.11×10^{-2}
42	0.28×10^{-2}	0.17×10^{-2}	0.04×10^{-2}
43	0.27	0.14	0.04
44	0.10	0.29	0.02
45	0.04	0.03	0.29
46	0.03×10^{-2}	0.10×10^{-2}	0.26×10^{-2}
47	0.03×10^{-2}	0.03×10^{-2}	0.09×10^{-2}
48	0.27×10^{-2}	0.15×10^{-2}	0.03×10^{-2}

TABLE 3.7B-48
ELECTRICAL AND AUXILIARY BUILDING SRSS ACCELERATIONS FOR SSE
FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 3 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
49	0.20	0.11	0.03
50	0.10	0.27	0.02
51	0.03	0.04	0.23
52	0.03×10^{-2}	0.08×10^{-2}	0.25×10^{-2}
53	0.03×10^{-2}	0.03×10^{-2}	0.07×10^{-2}
54	0.25×10^{-2}	0.12×10^{-2}	0.03×10^{-2}
55	0.14	0.08	0.03
56	0.10	0.24	0.01
57	0.02	0.03	0.16
58	0.02×10^{-2}	0.05×10^{-2}	0.24×10^{-2}
59	0.03×10^{-2}	0.02×10^{-2}	0.05×10^{-2}
60	0.23×10^{-2}	0.11×10^{-2}	0.02×10^{-2}
61	0.12	0.05	0.02
62	0.07	0.23	0.01
63	0.02	0.01	0.12
64	0.02×10^{-2}	0.03×10^{-2}	0.23×10^{-2}
65	0.02×10^{-2}	0.02×10^{-2}	0.04×10^{-2}
66	0.23×10^{-2}	0.11×10^{-2}	0.02×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-21.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

TABLE 3.7B-49
FUEL BUILDING SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 1 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.50	0.18	0.07
2	0.05	0.25	0.04
3	0.05	0.13	0.48
4	0.07×10^{-2}	0.06×10^{-2}	0.25×10^{-2}
5	0.16×10^{-2}	0.07×10^{-2}	0.27×10^{-2}
6	0.26×10^{-2}	0.09×10^{-2}	0.05×10^{-2}
7	0.40	0.17	0.16
8	0.05	0.23	0.12
9	0.03	0.11	0.37
10	0.05×10^{-2}	0.09×10^{-2}	0.24×10^{-2}
11	0.14×10^{-2}	0.06×10^{-2}	0.20×10^{-2}
12	0.25×10^{-2}	0.08×10^{-2}	0.04×10^{-2}
13	0.19	0.08	0.07
14	0.04	0.21	0.08
15	0.02	0.08	0.22
16	0.03×10^{-2}	0.05×10^{-2}	0.20×10^{-2}
17	0.08×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
18	0.19×10^{-2}	0.06×10^{-2}	0.04×10^{-2}
19	0.16	0.06	0.06
20	0.04	0.20	0.08
21	0.02	0.07	0.19
22	0.02×10^{-2}	0.04×10^{-2}	0.17×10^{-2}
23	0.07×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
24	0.16×10^{-2}	0.06×10^{-2}	0.03×10^{-2}
25	0.15	0.06	0.05
26	0.04	0.19	0.07
27	0.02	0.06	0.16
28	0.02×10^{-2}	0.04×10^{-2}	0.16×10^{-2}
29	0.07×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
30	0.14×10^{-2}	0.06×10^{-2}	0.03×10^{-2}
31	0.46	0.17	0.11
32	0.04	0.25	0.13
33	0.04	0.11	0.37
34	0.06×10^{-2}	0.07×10^{-2}	0.25×10^{-2}

TABLE 3.7B-49
FUEL BUILDING SRSS ACCELERATIONS FOR SSE FROM RESPONSE SPECTRUM ANALYSIS

(Sheet 2 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
35	0.12×10^{-2}	0.06×10^{-2}	0.21×10^{-2}
36	0.25×10^{-2}	0.09×10^{-2}	0.05×10^{-2}
37	0.23	0.08	0.04
38	0.04	0.22	0.10
39	0.02	0.08	0.23
40	0.04×10^{-2}	0.05×10^{-2}	0.22×10^{-2}
41	0.08×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
42	0.20×10^{-2}	0.06×10^{-2}	0.03×10^{-2}
43	0.20	0.07	0.03
44	0.04	0.21	0.09
45	0.02	0.07	0.19
46	0.04×10^{-2}	0.05×10^{-2}	0.19×10^{-2}
47	0.07×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
48	0.17×10^{-2}	0.06×10^{-2}	0.03×10^{-2}
49	0.17	0.06	0.03
50	0.04	0.20	0.08
51	0.02	0.06	0.17
52	0.02×10^{-2}	0.04×10^{-2}	0.17×10^{-2}
53	0.07×10^{-2}	0.04×10^{-2}	0.10×10^{-2}
54	0.15×10^{-2}	0.06×10^{-2}	0.03×10^{-2}
55	0.14	0.05	0.01
56	0.04	0.19	0.04
57	0.02	0.05	0.14
58	0.02×10^{-2}	0.04×10^{-2}	0.15×10^{-2}
59	0.06×10^{-2}	0.04×10^{-2}	0.09×10^{-2}
60	0.12×10^{-2}	0.06×10^{-2}	0.03×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-22.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

TABLE 3.7B-50
SERVICE WATER INTAKE STRUCTURE SRSS ACCELERATIONS FOR SSE
FROM RESPONSE SPECTRUM ANALYSIS UPPER BOUND SOIL CASE

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	2.19	0.41	0.75
2	0.52	4.16	0.32
3	0.75	0.25	2.01
4	0.0136	0.0050	0.0317
5	0.0105	0.0019	0.0049
6	0.0285	0.0127	0.0187
7	4.74	0.33	0.85
8	0.32	5.18	0.36
9	0.68	0.41	4.26
10	0.0127	0.0095	0.0447
11	0.0577	0.0120	0.0237
12	0.0588	0.0282	0.0340
13	7.98	1.14	2.28
14	2.09	6.58	1.25
15	1.08	0.60	7.29
16	0.0302	0.0128	0.0969
17	0.0370	0.0033	0.0316
18	0.1745	0.0455	0.0748
19	12.00	1.03	1.20
20	3.05	7.40	2.04
21	1.03	0.71	10.25
22	0.0440	0.0165	0.1271
23	0.0776	0.0126	0.0472
24	0.2329	0.0571	0.0973

a) Degrees of freedom are identified in Table 3.7B-23.

b) Tabulated accelerations are in units of ft/sec./sec. for transitional degrees of freedom and radians/sec./sec. for rotational degrees of freedom.

TABLE 3.7B-51
ELECTRICAL AND AUXILIARY BUILDINGS PEAK ACCELERATIONS FOR SSE
FROM TIME-HISTORY ANALYSIS

(Sheet 1 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.34	0.15	0.02
2	0.12	0.26	0.01
3	0.02	0.02	0.32
4	0.01×10^{-2}	0.03×10^{-2}	0.23×10^{-2}
5	0.02×10^{-2}	0.02×10^{-2}	0.06×10^{-2}
6	0.19×10^{-2}	0.21×10^{-2}	0.01×10^{-2}
7	0.29	0.13	0.02
8	0.12	0.26	0.01
9	0.02	0.02	0.28
10	0.01×10^{-2}	0.03×10^{-2}	0.23×10^{-2}
11	0.01×10^{-2}	0.02×10^{-2}	0.05×10^{-2}
12	0.19×10^{-2}	0.21×10^{-2}	0.01×10^{-2}
13	0.24	0.10	0.01
14	0.12	0.26	0.01
15	0.02	0.02	0.24
16	0.01×10^{-2}	0.03×10^{-2}	0.23×10^{-2}
17	0.01×10^{-2}	0.01×10^{-2}	0.05×10^{-2}
18	0.18×10^{-2}	0.21×10^{-2}	0.01×10^{-2}
19	0.20	0.08	0.01
20	0.11	0.25	0.01
21	0.01	0.01	0.22
22	0.01×10^{-2}	0.02×10^{-2}	0.22×10^{-2}
23	0.01×10^{-2}	0.01×10^{-2}	0.04×10^{-2}
24	0.16×10^{-2}	0.23×10^{-2}	0.01×10^{-2}

TABLE 3.7B-51
ELECTRICAL AND AUXILIARY BUILDINGS PEAK ACCELERATIONS FOR SSE
FROM TIME-HISTORY ANALYSIS

(Sheet 2 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
25	0.15	0.04	0.04
26	0.10	0.23	0.01
27	0.00	0.00	0.17
28	0.01×10^{-2}	0.01×10^{-2}	0.21×10^{-2}
29	0.00×10^{-2}	0.01×10^{-2}	0.03×10^{-2}
30	0.15×10^{-2}	0.24×10^{-2}	0.01×10^{-2}
31	0.52	0.22	0.03
32	0.13	0.34	0.01
33	0.04	0.02	0.43
34	0.02×10^{-2}	0.06×10^{-2}	0.27×10^{-2}
35	0.02×10^{-2}	0.02×10^{-2}	0.12×10^{-2}
36	0.29×10^{-2}	0.15×10^{-2}	0.02×10^{-2}
37	0.37	0.17	0.02
38	0.10	0.34	0.02
39	0.03	0.02	0.37
40	0.02×10^{-2}	0.06×10^{-2}	0.26×10^{-2}
41	0.02×10^{-2}	0.02×10^{-2}	0.10×10^{-2}
42	0.28×10^{-2}	0.15×10^{-2}	0.02×10^{-2}
43	0.29	0.14	0.02
44	0.11	0.32	0.01
45	0.02	0.02	0.31
46	0.02×10^{-2}	0.05×10^{-2}	0.26×10^{-2}
47	0.03×10^{-2}	0.02×10^{-2}	0.08×10^{-2}
48	0.27×10^{-2}	0.14×10^{-2}	0.01×10^{-2}

TABLE 3.7B-51
ELECTRICAL AND AUXILIARY BUILDINGS PEAK ACCELERATIONS FOR SSE
FROM TIME-HISTORY ANALYSIS

(Sheet 3 of 3)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
49	0.24	0.11	0.01
50	0.11	0.30	0.01
51	0.02	0.02	0.28
52	0.02×10^{-2}	0.04×10^{-2}	0.25×10^{-2}
53	0.01×10^{-2}	0.02×10^{-2}	0.06×10^{-2}
54	0.26×10^{-2}	0.13×10^{-2}	0.01×10^{-2}
55	0.19	0.08	0.01
56	0.12	0.26	0.01
57	0.01	0.01	0.20
58	0.01×10^{-2}	0.03×10^{-2}	0.24×10^{-2}
59	0.01×10^{-2}	0.01×10^{-2}	0.04×10^{-2}
60	0.17×10^{-2}	0.11×10^{-2}	0.01×10^{-2}
61	0.16	0.05	0.01
62	0.08	0.25	0.01
63	0.00	0.01	0.18
64	0.01×10^{-2}	0.02×10^{-2}	0.23×10^{-2}
65	0.01×10^{-2}	0.01×10^{-2}	0.03×10^{-2}
66	0.23×10^{-2}	0.10×10^{-2}	0.01×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-21.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

TABLE 3.7B-52
FUEL BUILDING PEAK ACCELERATIONS FOR SSE FROM TIME-HISTORY ANALYSIS

(Sheet 1 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
1	0.55	0.16	0.02
2	0.04	0.27	0.03
3	0.01	0.11	0.51
4	0.01×10^{-2}	0.05×10^{-2}	0.24×10^{-2}
5	0.08×10^{-2}	0.02×10^{-2}	0.12×10^{-2}
6	0.25×10^{-2}	0.08×10^{-2}	0.01×10^{-2}
7	0.44	0.14	0.07
8	0.03	0.26	0.08
9	0.01	0.09	0.42
10	0.01×10^{-2}	0.06×10^{-2}	0.23×10^{-2}
11	0.08×10^{-2}	0.02×10^{-2}	0.10×10^{-2}
12	0.24×10^{-2}	0.07×10^{-2}	0.01×10^{-2}
13	0.23	0.06	0.03
14	0.03	0.24	0.07
15	0.01	0.05	0.27
16	0.01×10^{-2}	0.04×10^{-2}	0.20×10^{-2}
17	0.05×10^{-2}	0.01×10^{-2}	0.06×10^{-2}
18	0.18×10^{-2}	0.07×10^{-2}	0.02×10^{-2}
19	0.21	0.05	0.03
20	0.03	0.23	0.07
21	0.01	0.04	0.22
22	0.01×10^{-2}	0.04×10^{-2}	0.17×10^{-2}
23	0.05×10^{-2}	0.01×10^{-2}	0.06×10^{-2}
24	0.16×10^{-2}	0.06×10^{-2}	0.01×10^{-2}
25	0.19	0.04	0.03
26	0.03	0.23	0.06
27	0.01	0.04	0.19
28	0.01×10^{-2}	0.04×10^{-2}	0.16×10^{-2}
29	0.05×10^{-2}	0.01×10^{-2}	0.07×10^{-2}
30	0.14×10^{-2}	0.06×10^{-2}	0.01×10^{-2}
31	0.49	0.14	0.06
32	0.04	0.26	0.12
33	0.01	0.09	0.42
34	0.01×10^{-2}	0.06×10^{-2}	0.25×10^{-2}

TABLE 3.7B-52
FUEL BUILDING PEAK ACCELERATIONS FOR SSE FROM TIME-HISTORY ANALYSIS

(Sheet 2 of 2)

Degree of Freedom[a]	X-Earthquake[b]	Y-Earthquake[b]	Z-Earthquake[b]
35	0.06×10^{-2}	0.01×10^{-2}	0.10×10^{-2}
36	0.25×10^{-2}	0.07×10^{-2}	0.01×10^{-2}
37	0.24	0.07	0.04
38	0.03	0.25	0.09
39	0.01	0.05	0.27
40	0.01×10^{-2}	0.05×10^{-2}	0.21×10^{-2}
41	0.06×10^{-2}	0.01×10^{-2}	0.06×10^{-2}
42	0.19×10^{-2}	0.06×10^{-2}	0.01×10^{-2}
43	0.23	0.05	0.03
44	0.04	0.25	0.07
45	0.01	0.05	0.22
46	0.01×10^{-2}	0.04×10^{-2}	0.19×10^{-2}
47	0.05×10^{-2}	0.01×10^{-2}	0.06×10^{-2}
48	0.17×10^{-2}	0.06×10^{-2}	0.01×10^{-2}
49	0.21	0.04	0.03
50	0.04	0.24	0.07
51	0.01	0.04	0.19
52	0.01×10^{-2}	0.04×10^{-2}	0.17×10^{-2}
53	0.04×10^{-2}	0.01×10^{-2}	0.07×10^{-2}
54	0.14×10^{-2}	0.06×10^{-2}	0.01×10^{-2}
55	0.19	0.04	0.01
56	0.03	0.23	0.03
57	0.01	0.03	0.18
58	0.01×10^{-2}	0.04×10^{-2}	0.15×10^{-2}
59	0.04×10^{-2}	0.01×10^{-2}	0.07×10^{-2}
60	0.12×10^{-2}	0.06×10^{-2}	0.01×10^{-2}

a) Degrees of freedom corresponding to translations in the x, y, and z directions, and rotations about x, y, and z axes are identified in Table 3.7B-22.

b) Tabulated accelerations are in units of (g) for translational degrees of freedom, and (g/ft) for rotational degrees of freedom.

APPENDIX 3.7B(A) - COMPUTER PROGRAMS USED IN DYNAMIC AND STATIC ANALYSES

3.7B(A).1 INTRODUCTION

Computer programs used in the dynamic and static analyses of seismic Category I structures, systems, and equipment are described herein and listed in Table 3.7B(A)-1. Program version, software or operating system, computer hardware, and status of recognition in the public domain are also specified in Table 3.7B(A)-1. The recognized computer programs in the public domain presented in Section 3.7B(A) have sufficient history of use to justify their applicability and validity without further verification. Other computer programs presented in this section were developed by Gibbs & Hill. These programs are occasionally linked and modified by G&H to suit a particular need for the solution of a problem. They are made operational on the CDC 6600 system. The applicability and validity of these programs are demonstrated by comparing the results obtained from each computer program with the results derived from a similar program available in the public domain. Verification of these programs is completed and maintained at G&H. See also Section 3.9B.1.2

Additional computer programs used in the analysis of ASME Code Class 2 and Class 3 piping systems, including supports for ASME Code Class 1, 2, and 3 piping are provided in Appendix 3B.

3.7B(A).2 COMPUTER PROGRAM (QUAKE)

The QUAKE program performs the dynamic analysis of a lumped mass system. The input includes mass data, structural stiffness, foundation spring constants, structure and foundation damping values, and other data related to the dynamic analysis model. The output provides modal shears and moments of statically determinate structures and combines them by the square root of the sum of the squares of modal values, by absolute sum, and by algebraic sum. QUAKE is comprised of selected subroutines presented in Reference [1]; the program is verified by using the STARDYNE computer program described in Subsection 3.7B(A).5 and by hand calculations. The QUAKE program has the following capabilities:

1. Extracting the eigenvalues and the corresponding eigenvectors from the following equation as described in Reference [35] of Section 3.7B:

$$\left[\frac{1}{\omega^2} [I] - [\delta][m] \right] \{\varnothing\} = 0 \qquad \text{(3.7B(A)-1)}$$

where

[I] = the unit matrix

ω = the natural circular frequency

[m] = the mass matrix

$[\delta]$ = the flexibility matrix

$\{\varnothing\}$ = the column matrix of eigenvectors

Equation 3.7B(A)-1 can be written in terms of the stiffness matrix [k] as follows:

$$[[k] - \omega^2 [m]] \{\varnothing\} = 0 \qquad (3.7B(A)\text{-}2)$$

To extract eigenvalues and eigenvectors, Equation 3.7B(A)-1 is converted by using the following substitution:

$$[m] = [U]^T [U] \qquad (3.7B(A)\text{-}3)$$

In general, matrix [U] consists of only diagonal and upper diagonal elements.

Substitution of [m] from Equation 3.7B(A)-3 and $\{\varnothing\}$ from Equation $\{\varnothing\} = [U]^{-1} \{\Phi\}$ into Equation 3.7B(A)-2 yields the following:

$$[[k] [U]^{-1} - \omega^2 [U]^T [U] [U]^{-1}] \{\Phi\} = 0 \qquad (3.7B(A)\text{-}4)$$

Premultiplication of Equation 3.7B(A)-4 by $\frac{[u][k]^{-1}}{w^2}$ yields the equation:

$$[{}_{\omega}1_2[I] - [U][\delta][U]^T] \{\Phi\} = 0 \qquad (3.7B(A)\text{-}5)$$

The matrix product $[U] [\delta] [U]^T$ is a symmetric matrix. The eigenvalues and eigenvectors are extracted from Equation 3.7B(A)-5 using a subroutine called EIGEN [1]. This subroutine uses an algorithm known as the Jacobi Diagonalization method.

The mode shapes $[\varnothing]$ as corresponding eigenvectors of Equation 3.7B(A)1 are obtained from the following equation:

$$[\varnothing] = [U]^{-1} [\Phi] \qquad (3.7B(A)\text{-}6)$$

2. Computing participation factors Γ by using a matrix manipulation in the following manner:

$$\{\Gamma\}_j = [M]^{-1} [\varnothing]^T [m] \{D\}_j \qquad (3.7B(A)\text{-}7)$$

where

$$[M] = [\varnothing]^T [m] [\varnothing] \qquad (3.7B(A)\text{-}8)$$

$[\varnothing]$ = the matrix of mode shapes

$[\varnothing]^T$ = the transpose of the mode shape matrix

$[M]$ = the diagonal generalized mass matrix

$\{\Gamma\}_j$ = the column matrix of participation factors for seismic motion in the jth direction

$[D]_j$ = the column matrix governed by the seismic motion in the jth direction

The normalization and orthogonality conditions are represented by matrix Equation 3.7B(A)-8.

3. Computing equivalent modal damping (or composite modal damping) according to the energy stored in each component and in each vibration mode.

   Each component, such as concrete structures, steel structures and systems, and foundation materials, can have different damping properties. The effective damping in any vibration mode of the total system depends upon the degree of participation of these materials in the modal response.

The stiffness matrix [k] of the entire system is obtained by the summation of the stiffness matrices [k] for each component r that has inherently different damping properties as expressed by the following equation:

$$[k] = \sum_{r=1}^{N} [k]r \qquad (3.7B(A)\text{-}9)$$

Where

N = total number of components with different damping properties

The corresponding flexibility matrix is obtained by inverting the stiffness matrix as follows:

$$[\delta] = [k]^{-1} \qquad (3.7B(A)\text{-}10)$$

The fictitious force matrix [f] is obtained as a product of the stiffness matrix and the mode shape matrix $[\Theta]^r$ as follows:

$$[f] = [k] [\varnothing] \qquad (3.7B(A)\text{-}11)$$

The mode shape matrix $[\Theta]_r$ for each component r is given by the following equation:

$$[\varnothing]_r = [k]_r^{-1} [f] \qquad (3.7B(A)\text{-}12)$$

The associated modal energy component matrix, which comprises diagonal terms only, is given by the following equation:

$$[E]_r = \frac{1}{2}[\varnothing]_r^T [k]_r [\varnothing]_r \qquad (3.7B(A)\text{-}13)$$

The total energy for the entire system is given by the following equation:

$$[E] = \tfrac{1}{2}[M] [\omega^2] \qquad (3.7B(A)\text{-}14)$$

The fraction of modal energy components is given by the following equation:

$$[MEC] = [E]^{-1} [E]_r \qquad (3.7B(A)\text{-}15)$$

so that $$\sum_{r=1}^{N} [E]_r = [E] \qquad (3.7B(A)\text{-}16)$$

and $$\sum_{r=1}^{N} [MEC]_r = [1] \qquad (3.7B(A)\text{-}17)$$

The column matrix of weighted modal damping ratios is found by using the following equation:

$$[D] = [E]^{-1} \sum_{r=1}^{N} [E]_r \quad \{D\}_r \qquad (3.7B(A)\text{-}18)$$

For a system comprised of the foundation and structure components, the modal energy matrix for the foundation is first generated by means of Equation 3.7B(A)-13. The modal energy matrix for the structure is obtained by using Equations 3.7B(A)- 16. The column matrix of weighted modal damping ratios for the entire system is obtained as indicated in Equation 3.7B(A)-18.

4. Computing absolute modal accelerations, relative displacements, and inertia loads using the spectrum approach; also, the modal responses are combined as described in Section 3.7B.2.7.

The matrix of maximum modal absolute accelerations [W] is obtained from the following equation in matrix form:

$$[\ddot{W}] = [\varnothing][Sa][r] \qquad (3.7B(A)\text{-}19)$$

where

[Sa] = the diagonal matrix of maximum modal spectral accelerations

[Γ] = the diagonal matrix of modal participation factors

The matrix of maximum modal relative displacements [d] is computed by the following equation:

$$[d] = [\ddot{W}][{}_{\omega}1_2] \qquad (3.7B(A)\text{-}20)$$

where

$[{}_{\omega}1_2]$ = the diagonal matrix of eigenvalues, if the flexibility matrix is used in the equation for the extraction of eigenvalues and eigenvectors

The matrix of maximum modal inertia forces [F] is the product of mass matrix [m] and maximum modal acceleration matrix $[\ddot{W}]$.

$$[F] = [m][\ddot{W}] \qquad (3.7B(A)\text{-}21)$$

The maximum modal inertia forces [F] are combined by the square root of the sum of the squares, by absolute sum, and by algebraic sum, and in accordance with the provisions of the NRC Regulatory Guide 1.92.

5. Calculating the modal shears and moments of structures and combining them by the square root of the sum of the squares (SRSS) of modal values, by absolute sum, and by algebraic sum, and in accordance with the provisions of the NRC Regulatory Guide 1.92.

   The matrix of modal shears and bending moments [Q] is computed from the following equation:

$$[Q]^T = [F]^{T\,[J]} \qquad (3.7B(A)\text{-}22)$$

where

$[F]^T$ = the transpose of the matrix of inertia forces and inertia moments

[J] = the summation matrix

3.7B(A).3 FINITE ELEMENT COMPUTER PROGRAM (NASTRAN)

This is a large computer program for dynamic and static analysis of linear elastic systems by the finite element approach. The program can handle large problems with many degrees-of-freedom. The only limitations on problem size are those imposed by practical considerations of running time and by the ultimate capacity of auxiliary storage devices. The program contains various types of finite elements. A manual of this program is given in Reference [2]. The official documentation consists of manuals presented in References [3], [4], [5], and [6]. These documents are available from COSMIC, Barrow Hall, University of Georgia, Athens, Georgia 30601.

3.7B(A).4 COMPUTER PROGRAM (CSMP)

The CSMP program is an IBM system/360 program for the simulation of continuous systems [7]. It provides an application-oriented input language that accepts problems expressed in the form of either an analog block diagram or a system of ordinary differential equations. Data input and output are facilitated by means of application oriented-control statements.

The manual presented in Reference [7] contains a general description of the program, detailed programming information, and a description of the inputs and outputs.

Several different types of routines are available to perform the integration operation. They include both fixed integration step-size routines and variable step-size routines. Five fixed step-size routines are available: fixed Runge-Kutta, Simpson's, trapezoidal, rectangular, and second order Adams. Two variable step-size routines are available: fifth-order Milne predictor-corrector and fourth-order Runge-Kutta.

3.7B(A).5 FINITE ELEMENT COMPUTER PROGRAM (STARDYNE)

The STARDYNE program consists of a series of compatible digital computer programs designed to analyze linear elastic structural models. It encompasses the full range of static and dynamic analyses.

This program, whose user's manual is presented in Reference [8], is used to evaluate a wide variety of static and dynamic problems. It contains various types of finite elements.

The static capability includes the computation of structural deformations and member loads and stresses caused by an arbitrary set of thermal, applied loads or prescribed displacements, or both.

Dynamic response analysis is performed for a wide range of loading conditions using the normal mode technique.

Transient, steady-state harmonic, random, and shock spectra excitation type inputs are applied. Different plotting capabilities are available.

3.7B(A).6 FINITE ELEMENT COMPUTER PROGRAM (ICES STRUDL-II)

The ICES STRUDL-II program was developed at the Massachusetts Institute of Technology. It is a part of the Integrated Civil Engineering System (ICES), which is a project of cooperation between government, industry, and university groups interested in the development of a large-scale, computer-based system [9]. The program has been updated and debugged by McDonnell-Douglas Automation Company [10].

This program provides the design engineer with an integrated set of facilities for the selection of structural members; these facilities are structured so that knowledgeable users can easily modify and expand the current capabilities. A user can add his own selection procedures and design codes, as well as change information such as assumed values for optional design data.

The system is organized for the easy addition of new codes and procedures.

This program contains various types of finite elements, such as triangular, rectangular, and quadrilateral plate elements representing or bending behavior, or both three dimensional solid elements, and beam elements.

3.7B(A).7 STRUCTURAL ENGINEERING SYSTEM SOLVER (STRESS) COMPUTER PROGRAM

The STRESS program is designed specifically for the solution of structural engineering problems as presented in Reference [11]. It uses a problem-oriented input language that enables the structural engineer to communicate with the computer even without previous programming experience.

The program can analyze structures with prismatic members in two or three dimensions, with either pinned or rigid joints, and subjected to concentrated or distributed loads, displacements, or temperature effects.

3.7B(A).8 COMPUTER PROGRAM FOR THE ANALYSIS OF SHELLS OF REVOLUTION (KALNINS)

This program calculates the stresses and displacements in thin-walled elastic, orthotropic shells of revolution when subjected to symmetrical and nonsymmetrical loads by means of the method of analysis presented in Reference [12]. The program is applicable to rotationally symmetric shells to which any number of axisymmetric branches are attached. The only restriction on the shell is that shell geometry, boundaries, and material properties are symmetric about one axis; that is, each section of the shell perpendicular to the axis must be circular. The geometry and elastic parameters, however, can vary in an arbitrary manner along the meridian of the shell. The circumferential loads are expressed in terms of Fourier harmonics.

This program uses a numerical method which combines the direct integration and the finite difference techniques for the solution of general shell equations.

3.7B(A).9 COMPUTER PROGRAM (SCONV)

The SCONV program uses convolution integration to solve a general dynamic problem for a linear elastic system with the support time history excitation RS(t) where R is the scaling factor and S(t) is the time-dependent support acceleration. For any linear elastic system, the governing equation for motions in matrix form is as shown in References [30] and [38] of Section 3.7B:

$$[m]\{\ddot{u}\} + [c]\{\dot{u}\} + [k]\{u\} = -[m]\{D\}R\ddot{S}(t) \qquad \text{3.7B(A)-23}$$

can be decoupled into a system of second order differential equations as follows:

$$\{\ddot{q}\} + 2[\beta][\omega]\{\dot{q}\} + \lfloor\omega^2\rfloor\{q\} = -\{\gamma\}R\ddot{S}(t) \qquad \text{(3.7B(A)-24)}$$

where

[q] = the normal coordinate, i.e., {u} = [ϕ] {q}

and [β], = the matrices of coefficients of critical damping,

[ω],{γ}, circular frequencies, participation factors, direction

[D] and [ϕ] vector, and eigenvectors, respectively

The general solution of Equation 3.7B(A)-24 for any system is as follows:

$$q_n(t) = C_n(t) - \frac{\gamma n}{\omega_n^2} A_n(t) \qquad \text{(3.7B(A)-25)}$$

where

$$C_n(t) = e^{-\beta_n\omega_n t}\left(\frac{\dot{q}_0 + \beta_n\omega_n q_0}{\bar{\omega}_n}\sin\bar{\omega}_n t + q_0\cos\bar{\omega}_n t\right)$$

$$A_n(t) = \int_0^t \frac{\omega_n^2}{\bar{\omega}_n} e^{-\beta_n\omega_n(t-\tau)} R\ddot{S}(\tau)\sin[\bar{\omega}_n(t-\tau)]d\tau \qquad \text{(3.7B(A)-26)}$$

$$\text{and } \bar{\omega}_n = \sqrt{\omega_n^2\ 1-\beta_n^2}$$

$q_n(t)$ is the relative displacement of the nth mode at time t, and q_o and q_o are the initial velocity and displacement.

By differentiating Equation 3.7B(A)-25 once and twice, the relative velocity and acceleration are obtained.

$$\dot{q}_n(t) = \beta_n\omega_n C_n(t) + D_n(t) + \frac{\gamma_n}{\omega_n^2}(\beta_n\omega_n A_n(t) - \beta_n(t)) \qquad \text{(3.7B(A)-27)}$$

$$\ddot{q}_n(t) = -(\omega_n^{-2} - \beta_n^2\ \omega_n^2\ )C_n(t) - 2\beta_n\omega_n D_n(t) + \frac{\gamma_n}{\omega_n^2}[(\omega_n^2\ - 2\beta_n^2\ \omega_n^2\ )A_n(t) + 2\beta_n\omega_n B_n(t)] - \gamma_n R\ddot{S}(t) \qquad \text{(3.7B(A)-28)}$$

where

$$B_n(t) = \int_0^t \omega_n^2\ R\ddot{S}(\tau)e^{-\beta_n\omega_n(t-\tau)}\cos[\bar{\omega}_n(t-\tau)]d\tau$$

and

$$D_n(t) = e^{-\beta_n\omega_n t}[(\dot{q}_0 + \beta_n\omega_n q_0)\cos\bar{\omega}_n t - q_0\bar{\omega}_n\ \sin\bar{\omega}_n t] \qquad \text{(3.7B(A)-29)}$$

The solution procedure is based on the step-by-step method. The total time duration is divided into small time intervals or steps. Displacement and velocity from the previous step are used as initial conditions for the next step. At each step, the relative displacement, velocity, and acceleration are calculated. The absolute acceleration is then combined by modal superposition as follows:

$$[\ddot{W}] = [\varphi]\{\ddot{q}\} + \{D\}R\ddot{S}(t) \qquad \text{(3.7B(A)-30)}$$

where

$\{\ddot{W}\}$ = the column matrix of absolute accelerations

The numerical method used for integration of Equations 3.7B(A)-26 and 3.7B(A)-29 is Simpson's rule.

Different output options are available. The user can ask for any or all of the following output

1. Time history of the absolute acceleration of any mass
2. Maximum absolute acceleration of all masses
3. Maximum load effects, such as shears, moments, and similar load effects at any mass point of the system

4. All load effects of the system corresponding to the inertia forces at any time instant

This program was verified by using the IBM program, CSMP, described in Section 3.7B(A).4.

3.7B(A).10 STORING OF DESIGN RESPONSE SPECTRUM ON THE COMPUTER (SPECTRA)

Dynamic analysis of structures by the spectrum approach necessitates reading the spectral acceleration values at discrete pairs of frequency and damping values. Design response spectra provided are usually in log-log scale and are for a few damping values only (10 percent, 7 percent, 5 percent, 2 percent, and 0.5 percent, normally). Calculated modal damping can fall in between these values or beyond; if so, it becomes necessary to interpolate or extrapolate. All this can be done visually, but the values thus read off the graphs are susceptible to errors. To achieve satisfactory accuracy, to avoid the tedium of visual reading, and to expedite the computer solution of the problem at hand, a subroutine is developed to simulate the reading process by certain measured parameters. A single call of this subroutine yields the spectral accelerations corresponding to a set of specified pairs of frequencies and the associated modal damping values.

Essentially, the method involves finding a polynomial expression for the maximum amplification as a function of damping. This is done by using the least square method, described in References [13], [14], [15], and [16], to fit a curve of a frequency control point that goes through the available points determined by amplification factors and damping ratios as shown in Figure 3.7B(A)-1. The maximum amplification occurs over a range of frequencies, beyond which the amplification is found to decay linearly in the log field. The decaying segments may be subsequently followed by other segments of different slopes. The points of discontinuities of the spectral curves, consisting of segments, are assumed to correspond to the same frequencies, irrespective of damping. The slopes of the decaying segments are, however, allowed to vary, i.e., the spectral curves need not be parallel to each other for each damping. The slopes of these segments for various damping values are again described by a least square polynomial fit as a function of damping. With this amount of information it is possible to find out the spectral accelerations corresponding to any frequency and the associated damping.

3.7B(A).11 FLOOR RESPONSE SPECTRA PROGRAM (TIME)

This program, developed by G&H, is used to obtain floor response spectra for equipment from the floor time histories of the building. Response spectra for different equipment dampings can be obtained; the output data are used in plotting response spectra curves.

TIME is used to produce response spectra for equipment mounted on flexible, i.e., low, natural-frequency floors.

The program uses an exact solution for the second order differential equation, assuming the forcing function (i.e., floor time history) to be linear between each time step; this solution is based on the formulations given in the dynamic analysis section of Reference [17].

3.7B(A).12 COMPUTER PROGRAM (ADLPIPE)

This computer program provides analyses of Class 1, 2 and 3 piping systems, which are classified as, Seismic Category I, for piping flexibility due to thermal expansion, as well as static

and dynamic analyses of elastic multibranch piping systems with flexible or rigid anchors and intermediate restraints. The piping system can contain a number of sections between network nodal points comprised of straight or curved members. A network nodal point is a branching point of pipes, an anchor, or any point at which motion is prescribed, or a point of lumped mass. Input preparation is described in Reference [19].

This program computes at each point within the piping system, the forces, moments, and displacements resulting from thermal expansion, dead loads, static loads, and displacements applied in the directions associated with all six degrees-of-freedom of the global orthogonal system and earthquake disturbances.

The normal mode technique in conjunction with the three-dimensional response spectra is used for obtaining seismic response. The resultant internal forces and moments are computed from the SRSS of their modal values. Algorithms used in this program for the extraction of eigenvalues and associated eigenvectors are the Jacobi rotation scheme [20] and the Givens-Householder scheme with modification as suggested in Reference [18]. The program is based on the systematic use of transfer matrices.

The program computes stresses within piping systems in accordance with ANSI B31.1-1973 and the ASME B&PV Code, Section III, Nuclear Power Plant Components. The combination of load conditions to compute usage factors in accordance with the latter code is included in this version of the program.

Typical benchmark calculations for the verification of the ADLPIPE program are presented in Reference [21].

3.7B(A).12.1 Description of the Analytical Techniques

The following documents describe the mathematical techniques that are used:

1. Generalized Piping System Response to Ground Shock Spectra by I.W. Dingwell, Arthur D. Little, Inc.

2. A Method of Computing Stress Range and Fatigue Damage in a Nuclear Piping System by W.B. Wright and E.C. Rodabaugh, Nuclear Engineering and Design, vol. 22, no. 2, October 1972, pp. 318- 325.

3. Method of Calculating Static and Dynamic Moments for Stress Evaluation at Tees and Branches, Arthur D. Little, Inc., May 1973.

4. Method of Calculating Thermal Stress Range for dT1, dT2, dTa, and dTb Terms, Arthur D. Little, Inc., May 1973.

5. Mathematical Analysis and Logical Procedure by I.W. Dingwell and R.T. Bradshaw, Arthur D. Little, Inc., 1970.

3.7B(A).13 FINITE ELEMENT COMPUTER PROGRAM (ANSYS)

This is a general purpose three-dimensional computer program for the solution of a large class of problems in engineering. This computer program, identified by the acronym ANSYS for

Engineering Analysis System [22], provides a flexible framework for implementation of the finite element analysis technology. The program is capable of taking practically any type of loading conditions and boundary conditions and has the capabilities for static and dynamic, elastic and plastic, fluid flow, and transient heat transfer analyses.

The matrix displacement method of analysis, based on finite element idealization, is used throughout. The library of finite elements available numbers more than twenty for static and dynamic analyses.

These elements are of various types, e.g., frames, plane stress and axisymmetric triangles, three-dimensional solids, springs, masses, dampers, plates, axisymmetric shells, general shells, and friction interface elements.

The program uses a direct solution, developed by the matrix displacement method, for the system of simultaneous linear equations.

Plotting subroutines are available.

3.7B(A).14 COMPUTER PROGRAM (SUPERPIPE)

See Section 3.9B.1.2.3.

3.7B(A).15 STRUDL-SW (STRUCTURAL DESIGN LANGUAGE)

The STURDL-SW computer program [24] is a general finite element program which uses the stiffness analysis method to analyze a wide range of structural problems. It handles two- and three-dimensional trusses and frames, having linear elastic members with either static or dynamic loadings.

STRUDL-SW has been documented by bench marking procedures against the GTSTRUDL computer code. GTSTURDL is a recognized program in the public domain.

3.7B(A).16 SHELL-1 (THIN SHELL OF REVOLUTION UNDER ARBITRARY LOADING)

The SHELL-1 computer program [25] is a general shell program. This program is based upon the general numerical procedure proposed by B. Budiansky and P. P. Radowski, and Greenbaum to analyze a shell of revolution subjected to arbitrary loadings. The program is applicable only for axisymmetric structures such as containment shells; however, the program has the capacity of taking non-symmetric loads and boundary conditions through the technique of Fourier harmonic decomposition.

This program uses a finite difference stress analysis solution. It can be used to determine the forces, moments, shears, displacements, rotations, and stresses in a shell of revolution subject to arbitrary loads expanded in a Fourier series of up to 150 terms. Single layer shells with up to 30 simply connected branches may be analyzed. Poisson’s ratios may change at discontinuity points, and Young’s modulus and the thermal coefficient of expansion may be different at each point. The allowed types of loading include elastic restraints, pressures in three orthogonal directions, temperature changes which may have a gradient through the shell thickness, and simplified input for weight of the shell or earthquake forces.

This computer program has been qualified by comparing it's results against those of hand calculations.

3.7B(A).17 TIMHIS6 (TIME HISTORY ANALYSIS)

The TIMHIS6 Program [26] computes time history response and amplified response spectra (ARS) at any mass location of a lumped mass system due to a synthetic earthquake input. The responses are computed by integration of the modal equations of the system by exact methods. The program's main application is the generation of ARS.

This program has been documented by bench marking procedures against the STARDYNE computer program. STARDYNE is a recognized program in the public domain.

3.7B(A).18 SBMMI (SINGLE BARRIER MASS MISSILE IMPACT)

THE SBMMI computer program [27] computes the elastic-plastic structural response of a barrier due to the following loads: (a) static loads; (b) suddenly applied constant dynamic loads which remain permanently on the structure; (c) suddenly applied constant dynamic loads representing missile impact with a finite force and specific momentum; and (d) suddenly applied dynamic loads of zero time duration and specific momentum representing missile impact. The barrier is modelled as a single barrier mass and a non-linear spring, with the above loads applied. The equation of motion is integrated in time assuming constant acceleration in each time step.

This computer program has been qualified by comparing it's results against those of hand calculations.

3.7B(A).19 CLASSI (CONTINUUM LINEAR ANALYSIS OF SOIL STRUCTURE INTERACTION)

The program CLASSI is a computer code generated to calculate the three-dimensional dynamic response of structures including soil-structure interaction effects. CLASSI is based on a specialized form of substructuring which uses the finite element method to perform the detailed analysis of the superstructure and uses the continuum mechanics method to calculate the interaction of the foundation with the soil medium and with incident seismic waves. These substructuring procedures are made possible by balancing the forces and moments at the foundation, which serves as the common ground for both the superstructure and the soil medium.

The capabilities of CLASSI in modeling systems which combine the soil, the foundation, and the structure are:

1. The soil profile may consist of multiple layers lying over a homogenous halfspace. The layers may have different shear moduli, Poisson's ratio, density, depth, and material damping characteristics.

2. The foundation can be of arbitrary geometry but must be surface mounted. The foundation can be discretized into regular (square or rectangular) segments. The stress distribution between the foundation and the soil depends on the degree of discretization of the foundation since constant stress is assumed for each segment.

3. The structural model can be developed using any standard finite element program. The fixed base structural modal properties obtained from the finite element program are subsequently used as input to CLASSI. In this way,the structure can have any degree of complexity.

The CLASSI substructure approach divides the SSI problem into the following three steps.

1. Determination of the foundation scattering matrices.

2. Determination of the frequency-dependent impedance functions.

3. Analysis of the coupled soil-structure system, using results from steps 1 and 2 and the dynamic properties of the structure.

In the first step, CLASSI evaluates the harmonic response of the rigid, massless foundation bonded to the soil and subjected to a given incident seismic wave in the absence of the superstructure. The free field motion is then used in conjunction with the complex, frequency-dependent scattering matrix in order to determine the foundation input motion.

In the second step, the foundation impedances corresponding to a rigid foundation on a uniform or layered viscoelastic media are developed.

In the third step, analysis of the coupled soil-structure system is carried out by CLASSI in the frequency domain. Time history of responses are obtained by inverse Fourier transform techniques.

3.7B(A).20 RESPEC (RESPONSE SPECTRA GENERATOR)

The computer program RESPEC computes the response spectra of acceleration time histories digitized at equal intervals. Spectral accelerations are computed for frequencies and damping ratios selected by the user. Each spectral value is obtained by computing the maximum response of a damped single-degree-of-freedom oscillator subjected to the specified acceleration time history. The response is computed by explicit integration of the differential equations of motion, assuming “at rest” initial conditions. The program uses the numerical method set forth by Nigam and Jennings in Earthquake Engineering Research Laboratory report “Diginal Calculation of Response Spectra from Strong-Motion Earthquake Records.” The spectral velocity and spectral displacement may also be calculated.

3.7B(A).21 LS-DYNA

The computer program LS-DYNA (developed by Lawrency Livermore Laboratories) is used to perform an elasto-plastic finite element analysis of a spent fuel handling accident. The program simulates the transient collision event with full consideration of plastic, large deformation, wave propagation, and elastic/plastic buckling modes.

3.7B(A).22 MR216 (DYNARACK)

The computer program DYNARACK is used to perform non-linear, direct integration finite element analyses of the Whole Pool Multi-Rack (WPMR) configuration for the spent fuel storage racks. The program uses a step-by-step solution in time employing a central difference algorithm

to evolve to a converged solution. Using the structural model for every set of 22-DOF rack models that comprise a Whole Pool Multi-Rack simulation, equations of motion corresponding to each degree-of-freedom are obtained using Lagrange's Formulation of the dynamic equations of motion. The system kinetic evergy includes contributions from solid structures and from trapped and surrounding fluid. The final system of equations obtained have the matrix form:

$$[M] \{d^2 q / dt^2 \} = \{Q\} + \{G\}$$

where:

- [M] = total mass matrix (including structural and fluid mass contributions). The size of this matrix will be (22 x NOR) x (22 x NOR). NOR = number of racks in the spent fuel pool.
- {q} = the nodal displacement vector relative to the pool slab displacement.
- {G} = a vector dependent on the given ground acceleration.
- {Q} = a vector dependent on the spring forces (linear and nonlinear) and the coupling between degrees-of-freedom.

The equation can be rewritten as:

$$\{d^2 q / dt^2\} = [M]^{-1} \{Q\} + [M]^{-1} \{G\}$$

This equation set is mass uncoupled, displacement coupled at each instant in time. The numerical solution uses a central difference scheme build into the proprietary computer program. Results are archived at appropriate time intervals for permanent record and for performing subsequent post-processing for structural integrity evaluation.

This computer program has the capability to execute concurrent sliding, rocking, bending, twisting, and other motion forms compatible with the freestanding rack design. The program has the capability to effect momentum transfers which occur due to rattling of fuel assemblies inside storage cells and the capability to simulate lift-off and subsequent impact of support pedestals.

3.7B(A).23 OTHER COMPUTER PROGRAMS

There are other small programs not described above. These programs are developed for the purpose of bulk data manipulation to replace regular manual calculations. From time to time, additional small programs of this kind are developed to suit particular needs. The verification of these programs is generally accomplished by sample calculations.

REFERENCES

1. IBM Application Program, 1130 Scientific Subroutine Package, Programmer's Manual, Program Number 1130-CM-02X.

2. McCormick, Caleb, W., October 11, 1971, NASTRAN Basic Training Manual for CDC 6000 Series, the MacNeal Schwendler Corporation, 7442 No. Figueroa, Los Angeles, California 90041.

3. NASTRAN Theoretical Manual (NASA SP-221).

4. NASTRAN User's Manual (NASA SP-222).

5. NASTRAN Programmer's Manual (NASA SP-223).

6. NASTRAN Demonstration Problem Manual (NASA SP-224).

7. IBM Application Program, System/360 Continuous System Modeling Program, User's Manual, Program Number 360A-CX-16X.

8. MRI/STARDYNE, Static and Dynamic Structural Analysis System, User's Information Manual, developed by and proprietary to Mechanics Research, Inc.

9. ICES STRUDL-II, June 1971, The Structural Design Language, Engineering User's Manual, Additional Design and Analysis Facilities, MIT, Department of Civil Engineering, School of Engineering, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139, vol. 2, 2nd ed., Research Report R70-77.

10. ICES STRUDL DYNAL User's Manual, McDonnell-Douglas Automation Company, St. Louis, Missouri.

11. IBM Application Program, Structural Engineering System Solver (STRESS), for the IBM 1130, Version 2, User's Manual, Program Number 1130-EC-03X.

12. Kalnins, A., September 1964, Analysis of Shells of Revolution Subjected to Symmetrical and Nonsymmetrical Loads, Journal of Applied Mechanics.

13. Kelly, L.G., 1967, Handbook of Numerical Methods and Applications, Addison Wesley.

14. Forsythe, G.E., 1957, SIAM 5, Society for Industrial and Applied Mathematics, pgs. 74-88.

15. Library of Mathematical Subprograms, Reference Manual, Control Data Company.

16. IBM Application Program, 1130 Statistical System, Program Number 1130-CA-06X, Application Description, H20-0341-1.

17. MRI/STARDYNE Theoretical Manual, Control Data Corporation CYBERNET SERVICES.

18. Wilkinson, J.H., 1965, The Algebraic Eigenvalue Problem, Oxford University Press, London.

19. I.W. Dingwell of Arthur D. Little, Inc., April 1977, ADLPIPE, Static and Dynamic Pipe Design and Pipe Stress Analysis, Input Preparation Manual, Cambridge, Massachusetts.

20. Greenstadt, J., 1959, The Determination of the Characteristic Roots of a Matrix by the Jacobi Method, Chapter 7, in Mathematical Methods for Digital Computers, John Wiley, New York.

21. Documentation of ADLPIPE for Static and Dynamic Loads and Stress Evaluation, September 1973, Arthur D. Little, Inc., Cambridge, Massachusetts.

22. ANSYS, Engineering Analysis System, December 1, 1971, User's Manual, John A. Swanson, Ph.D., Swanson Analysis System, Inc., 870 Pine View Drive, Elizabeth, Pennsylvania 15037.

23. SUPERPIPE, Static and Dynamic Loads and Stress Evaluation of Piping Systems, Program Version Number 15c, Impell Corp., San Francisco, Calif.

24. STRUDL-SW, "Structural Design Language," Stone and Webster Engineering Corporation, April 1980.

25. SHELL-1, "Thin Shell of Revolution Under Arbitrary Loading," Stone and Webster Engineering Corporation, September 1971.

26. TIMHIS6, "Time History Analysis," Stone and Webster Engineering Corporation, March 1978.

27. SBMMI, "Single Barriers Mass Missile Impacts," Stone and Webster Engineering Corporation, January 1981.

28. CLASSI (Continuum Linear Analysis of Soil Structure Interaction), "A Computer Program for Soil-Structure Interaction Using a Substructuring Technique," Version v.0.0, Impell Corporation.

29. RESPEC, "A Computer Program for the Generation of Response Spectra," Version v.6/10/75, Impell Corporation.

30. LS-DYNA, Version 950, Liversomre Software Technology Corporation, 1999.

31. MR216, "Multi-Rack Transient Analysis Code", Holtec International, v 2.0 Code.

TABLE 3.7B(A)-1
LIST OF COMPUTER PROGRAMS USED

(Sheet 1 of 2)

Computer Program Version	Software or Operating System	Computer Hardware	Note
Program Number 1130-CM-02X	IBM Application Program Scientific Subroutine Package	IBM 1130 and IBM 370	a
Program Number 360A-CX-16X	IBM Application Program, System/360, Continuous System Modeling Program (CSMP)	IBM 360	
Version 3 April, 1974	MRI/STARDYNE	CDC 6600	a
December 1971	ANSYS	CDC 6600 Series and IBM 360/65	a
May 1974	McAuto/ICES-STRUDL ICES-STRUDL-II	IBM 370	
Version II Program Number 1130-EC-03X	IBM Application Program Structural Engineering System Solver (STRESS)	IBM 1130 and IBM 370	a
February 1977	ADLPIPE	IBM 1130, IBM 360/370, and CDC 6600	a
Program Number 2D17	ADLPIPE	CYBER 175/176	a
October 1971	NASTRAN	CDC 6600	a
1974	QUAKE	CDC 6600	b
April 1971	KALNINS	CDC 6600	a
Program Number 15c	SUPERPIPE	CYBER 170 CYBER 175 CYBER 176	a

TABLE 3.7B(A)-1
LIST OF COMPUTER PROGRAMS USED

(Sheet 2 of 2)

Computer Program Version	Software or Operating System	Computer Hardware	Note
April 1973	SCONV	CDC 6600	b
1974	SPECTRA	CDC 6600	b
May 1974	TIME	CDC 6600	b
V03/L03	STRUDL-SW	IBM 3081	c
V01/L07	SHELL-1	IBM 3081	c
V00/L02	TIMHIS6	IBM 3081	c
V00/L00	SBMMI	IBM 3081	c
4.2A	ANSYS	IBM 3081	a
5.4	ANSYS	Compac/DEC Alphaserver 1200	a
12.0	ANSYS	Personal Computer	e
12.1	ANSYS	Personal Computer	e
13.0	ANSYS	Personal Computer	e
14.0	ANSYS	Personal Computer	e
0.0	CLASSI	CYBER	d
June 1975	RESPEC	CYBER	d
v2.0	DYNARACK	Personal Computer	e
950	LS-DYNA	Personal Computer	a

Notes:

a. Recognized computer program in public domain.

b. Computer program developed, verified, and documented by G&H.

c. Computer program developed, verified and documented by SWEC.

d. Computer program verified and documented by Impell.

e. Computer program verified and documented by Holtec International.

3.8 DESIGN OF CATEGORY I STRUCTURES

3.8.1 CONCRETE CONTAINMENT

3.8.1.1 Description of the Containment

3.8.1.1.1 General Description

The Reactor Containment structure is a fully continuous, steel-lined, reinforced concrete structure. It consists of a vertical cylinder and a hemispherical dome and is supported on an essentially flat foundation mat with a reactor cavity pit projection. The Containment superstructure is independent of the adjacent interior and exterior structures. Sufficient space is provided between the Containment and the adjacent structures to prevent contact under all combinations of loadings.

3.8.1.1.2 Dimensions of Containment

The dimensions of the Containment are as follows:

1. Inside diameter (ID): 135 ft 0 in.

2. Height of cylinder (top of foundation mat to dome spring line): 195 ft 0 in.

3. Inside radius of hemispherical dome: 67 ft 6 in.

4. Thickness of cylindrical walls: 4 ft 6 in.

5. Thickness of dome: 2 ft 6 in.

6. Foundation mat thickness: 12 ft 0 in.

7. Top of the foundation mat: approximately 4 ft 6 in. below grade

3.8.1.1.3 Containment Function

The Containment structure is designed to serve the following functions:

1. Provide vapor containment and limit the leakage from the Containment following an accident within the Containment

2. Isolate the RCS from postulated extreme environmental conditions, including tornadoes and external missiles

3. Biological shielding

3.8.1.1.4 Arrangement of Main Reinforcing Steel

The principal reinforcement used in the mat, walls, and dome are No. 18 bars, made continuous at splices by the use of Cadweld mechanical connectors produced by the Erico Corporation.

The reinforcing steel pattern in the cylinder wall consists of vertical bars at each face, horizontal hoop bars at each face, and 45-degree diagonal bars, in each direction, near the outside face.

The foundation mat is reinforced with top and bottom layers of bars placed as shown on Figure 3.8-12.

The dome reinforcement consists of top and bottom meridional layers of rebars, extending from the vertical bars of the cylindrical wall and top and bottom layers of circumferential hoop bars, as shown on Figure 3.8-11.

The meridional reinforcement terminated in the dome is anchored by the use of a positive mechanical anchor, such as a bearing plate cadwelded to the end of the bar, and satisfies the other anchorage requirements in accordance with CC-3531.1.2 of the ASME-ACI 359 document.

At penetration openings, reinforcing steel is generally curved around the openings where practical, and supplemental bars are provided around the opening as required. At large major penetrations such as the personnel lock and the equipment hatch some of the wall reinforcement is terminated at the opening by cadwelding steel plates on the end of the bar. Additional reinforcing is provided around these openings to carry stress concentrations and redistributions at these discontinuities. For details, see Figures 3.8-13 and 3.8-14.

3.8.1.1.5 Steel Liner

The entire inside face of the Containment (mat, walls, and dome) is lined with a continuous welded steel liner plate, attached with anchors to the reinforced concrete, to ensure a high degree of leaktightness. The thickness of the liner in the wall is 3/8 in. and in the dome is 1/2 in.; a 1/4 in. thick plate is used on top of the foundation mat and covered with a layer of concrete. Local thickened liner plate sections are provided at penetrations, at major pipe and duct support attachments and at crane and rotating platform girder brackets, and at the bottom of the cylindrical wall's steel liner. Overlay plates and/or structural shapes may be used on the interior side of the liner for support of minor pipes and ducts, conduits, cable trays, and equipment.

Leak-chase channels are provided at liner seams which, after construction, are inaccessible for other means of leaktightness examination. For typical liner details, see Figures 3.8-5 and 3.8-6.

3.8.1.1.6 Containment Penetrations and Attachments

Access to the Containment structure is provided by a personnel airlock, an emergency airlock, and an equipment hatch. Containment airlocks are tested in accordance with 10CFR Part 50 Appendix J, Option B. A constant pressure of P_a is used to pressurize the volume between the airlock seals.

1. Personnel Airlock

   The personnel airlock is an approximately 9 ft inside diameter electro-hydraulically operated double-door assembly. Each door is hinged and gasketed, with leakage test taps aligned to the annulus between the gasket sealing surfaces. Both doors are interlocked so that if one door is open, the other cannot be activated. Both doors are also furnished with hydraulic actuated as well as manual pressure equalizing valves which can

be operated by persons leaving or entering the personnel hatch. Plan and elevation drawings of the personnel airlock are shown in Figures 3.8-20 and 3.8-21. The configuration of the airlock seal test provisions is shown in Figure 3.8-21. Figure 3.8-22 identifies all personnel airlock mechanical and electrical penetrations. The personnel airlock has provisions for test pressurization at a pressure of P_a of the space between the two grooves at both ends of the airlock as well as provisions for pressurization at a pressure of P_a of the volume between the airlock doors. The doors are designed to maintain their functional capability during testing with no additional requirements for blocking beyond normal locking procedure.

2. Emergency Airlock

The emergency airlock is an approximately 5-ft 9-in. inside diameter manually operated double-door assembly, with 2-ft 6-in. diameter doors. Both doors of the emergency airlock are furnished with manually operated pressure-equalizing connection and valves which are interlocked with the door operating mechanism and serve to equalize differential pressure across locked doors. The reactor building to airlock door (interior) requires installation of strongbacks for the performance of the overall leakage check. Other operating features are similar to those of the personnel airlock described previously. Figure 3.8-23 identifies all emergency airlock mechanical and electrical penetrations.

3. Equipment Hatch

The equipment hatch is a 16-ft 0-in. ID single closure penetration. The bolted hatch cover is mounted on the inside of the Containment, and is double-gasketed with a leakage test tap between the gaskets. The hatch cover is provided with a hoist for handling. For details of the airlocks and equipment hatch, see Figure 3.8-9.

4. Pipe Penetrations

Other smaller penetrations through the Containment include the main steam and feedwater lines, hot and cold pipes, the fuel transfer tube, and electrical conductors. All penetration sleeves are welded to the liner and anchored into the reinforced concrete Containment wall. For typical details, see Figures 3.8-7 and 3.8-8.

5. Fuel Transfer Tube Penetration

A fuel transfer tube penetration is provided for fuel transfer between the refueling canal in the Containment structure and the spent fuel pools in the Fuel Building. The penetration consists of a 20 in. stainless steel pipe inside a carbon steel sleeve. The inner pipe acts as the transfer tube; the outer tube is welded to the Containment liner. Bellows expansion joints are provided to permit differential movements. The fuel transfer tube is equipped with a bolted blind flange with double O-ring seals inside the containment.

6. Electrical Penetrations

Header plate type penetrations are used for electrical conductors passing through the Containment. The penetration header plate with double O-ring gaskets is bolted to a

weld neck flange which is welded to a steel penetration sleeve. The steel penetration sleeves are welded to the Containment vessel liner. See Section 8.3 for additional information.

7. Liner Attachments

Major pipe and duct supports, and crane and Containment access rotating platform girder support brackets are welded to a thickened section of the liner plate, and anchored into the reinforced concrete Containment wall, as shown on Figure 3.8-6. Overlay plates and/or structural shapes may be used from the interior side of the liner, for support of minor pipes and ducts, conduits, cable trays and equipment.

8. Containment Alternate Access for the Steam Generator and Reactor Pressure Vessel Head Replacement (Unit 1)

The Steam Generator (SG) and Reactor Vessel Head (RVH) Replacement Project created and restored a construction alternate access in the Containment Building (Containment Alternate Access) in accordance with administrative procedures and the design control program. The alternate access was used to facilitate the movement of original and replacement SGs and RVH out of and into the Containment Building. In accordance with the ASME Section XI repair/replacement program, the alternate access was restored consistent with the original containment specifications with any exceptions reconciled to the original specification.

Codes and Specifications

Restoration of the Containment Alternate Access was performed as a repair/replacement activity in accordance with the requirements of ASME Section XI, 1998 edition, 1999 and 2000 addenda.

The basic code for the restored Containment Building structure are appropriate portions of the Proposed Standard Code for Concrete Reactor Vessels and Containments (April 1973); ASME-ACI 359. The restored structure meets all applicable design loads and load combinations required by ASME-ACI 359.

Concrete placement, curing, and repair were in accordance with ACI 301-05. Concrete mix proportioning was per ACI 211.1-91 (reapproved 2002).

Project specification for restoration of the Containment Alternate Access address:

- Reinforcing steel procurement, testing and placement
- Cadweld® reinforcing steel splices procurement, testing and installation
- Concrete mix design, testing and placement
- Structural steel and materials procurement

Liner Restoration

The cut section of the Containment Building liner plate was rewelded to the liner plate with a full penetration weld. The new liner plate seam welds were examined using NDE methods specified within CC-5520. Liner weld was leak tested by vacuum box test method to satisfy leaktightness requirements of NRC Regulatory Guide 1.19.

Replacement material was purchased for Nelson studs in accordance with the requirements of the original plant specification for the Unit 1 liner plate.

Reinforcing Steel Restoration

The reinforcing steel bars cut during the creation of the Containment Alternate Access were reinstalled using Cadweld® splices or welding, as required. Reinforcing steel bars that were damaged during the creation of the access were repaired in accordance with AWS D1.4-98 or were replaced with reinforcing steel procured in accordance with the project specification. New No. 6 and No. 18 reinforcing steel used for the Containment Building wall restoration conform to ASTM A615 Grade 60 or ASTM A706 and meet or exceed the additional physical and chemical composition requirements described in UFSAR Section 3.8.1.6.2 for the Containment Building structure existing reinforcing steel.

Concrete Restoration

The concrete removed from within the Containment Alternate Access was restored with fresh concrete with a specified 28-day compressive strength of 4000 psi. Fresh concrete was qualified, tested, mixed, and placed in accordance with the project specification.

3.8.1.1.7 Drawings

For various Containment structure details described in Subsection 3.8.1.1, see Figures 3.8-1 through 3.8-15.

3.8.1.2 Applicable Codes, Standards, and Specifications

3.8.1.2.1 Basic Code

The basic code used for the materials, design, fabrication, construction, examination, testing, and surveillance of the Containment are the appropriate portions of the Proposed Standard Code for Concrete Reactor Vessels and Containments (April 1973), which was issued for trial use and comments. This basic code was developed by the joint ACI-ASME Technical Committee on Concrete Pressure Components for Nuclear Service, which is made up of ACI Committee 359 and the ASME B&PV Code, Section III, Division 2, Subgroup on Concrete Components. The specific portions of this document that apply (except where otherwise specifically indicated in this FSAR) are as follows:

1. Subsection CA (General Requirements)

   Article CA-4000 Quality Assurance

2. Subsection CC (Concrete Containment)

Article CC-1000	Introduction
Article CC-2000	Materials (except CC-2232.4, CC-2240, and CC-2400, for Preplaced Aggregate Concrete, Grout, and Materials for Prestressing Systems; and except portions of CC-2231.2 for testing for: coefficient of thermal expansion - CRD-C-39 thermal conductivity - CRD-C-44 creep - ASTM C-512 shrinkage coefficient - ASTM C-157 aggregates for radiation - shielding concrete - ASTM C-637)
Article CC-3000	Design (except CC-3830 for Transitions from Concrete Containment to Steel Containment Vessels and Prestressed Concrete sections)
Article CC-4000	Fabrication, Construction, and Installation (except CC-4230 for Preplaced Aggregate Concrete, and CC-4400 for Fabrication and Installation of Prestressing Systems)
Article CC-5000	Construction, Testing and Examination (except CC-5235 for Preplaced Aggregate Concrete and CC-5400, for Examination of Prestressing Systems)
Article CC-7000	Concrete Containment Structures Protection Against Overpressure
Article CC-9000	Inservice Surveillance (except CC-9230 for Structural Integrity of Prestressed Concrete Containments)
Appendix I	Tables of Prestressing and Liner Materials (except that materials for prestressing do not apply)
Appendix III	Glossary of Terms and Symbols
Appendix VI	Porosity Charts
Appendix IX	Nondestructive Examination Methods
Appendix D	Nonmandatory Preheat Procedures

The procedures used in the design of this facility are consistent with the requirements contained in the applicable portions of the ASME-ACI 359 document as described in the following paragraphs.

The code (hereinafter referred to as the ASME-ACI 359 document) is used as the basic code for this facility because it is by far the most complete, comprehensive document available for the design, construction, inspection and testing of a concrete containment structure. Another advantage is that this document is virtually complete, in itself, without reference to portions of other codes or standards. Reference to other documents such as ASTM specifications or ACI standards are made only where the total document is applicable.

The previously listed portions of the ASME-ACI document that apply are all of the applicable requirements of Subsection CC (which comprises the technical requirements for a concrete containment), the applicable appendices, and some aspects of the Quality Assurance Article (CA-4000). The exceptions taken concern items which are not applicable (preplaced aggregate, grout, and prestress systems). Also, in answer to AEC (presently NRC) Question 3.6 of Amendment No. 1, conservative modifications are made regarding some load combinations and allowable stresses. (See Subsections 3.8.1.2.1, 3.8.1.2.5, 3.8.1.3.1, 3.8.1.3.2, 3.8.1.4.5, 3.8.1.5.1, 3.8.1.5.2, and 3.8.3.2.4.)

Requirements for punching shear are in accordance with Section 11.10.3 of the 1971 edition of the ACI 318 Code.

Specific references to the articles in Subsection CA, General Requirements, which are of a legal nature rather than a technical nature (Articles CA-1000, CA-2000, CA-3000, CA-4000, CA-5000, and CA-8000) have been omitted. These articles include requirements for such items as code jurisdiction, effective dates of code edition and addenda, certificates of authorization, responsibility of parties, stamping of containment, inspector's certification, authorized inspection agency, and so forth. These legal requirements are not applicable to the Comanche Peak Nuclear Power Plant (CPNPP) since the Code edition in force for this project is the trial use and comments issue.

3.8.1.2.2 ACI Committee 349

Since the ASME-ACI 359 document used as the basic code is co-sponsored by the ACI, it supersedes the ACI Committee 349 document, Criteria For Reinforced Concrete Nuclear Power Containment Structures; therefore, the ACI 349 criteria, published in the ACI Journal, January 1972, is not used as the basic containment criteria. However, the following sections of the ACI 349 criteria complement the requirements of the ASME-ACI 359 document and are used as a reference in the design of this facility:

1. Section 2.2.1 — Concerning strain limitations on self-limiting-type bending moments (used in conjunction with CC-3110(b) of the ASME-ACI 359 document)
2. Appendix C — method of calculating stresses and strains when a thermal gradient is combined with other loads.
3. The ASME-ACI 359 document does not provide guidance in determining thermal stresses. Therefore, the guidance provided in ACI 349-76, "Code Requirements for Nuclear Safety Related Concrete Structures", Appendix A, "Thermal Considerations" is used.

3.8.1.2.3 Additional Specifications and Standards

The following is a list of specifications and standards that are referred to in the applicable portions of the ASME-ACI 359 document described in Subsection 3.8.1.2.1 and which are applicable to this facility.

1. Liner, Penetrations, Containment Vessel Metal Components, and Attachments:

ASME B&PV Code, Section III, Division I, Subsection NE, 1971 through and including the 1973 Summer addenda (for the electrical penetration sleeves, fuel transfer tube penetration sleeve, emergency and personnel air locks, and equipment hatch) and 1974 through and including the Summer 1976 addenda (for process piping penetrations subjected to pressure-induced stresses and unsupported by concrete for load-carrying purposes).

ASME B&PV Code, Section V, 1974 (for liner radiographic examinations and electrical penetration ultrasonic examinations)

ASME B&PV Code, Section IX, 1971 through and including the Summer 1973 addenda (for welding qualifications)

ASME B&PV Code, Section II, 1971 through and including the Summer 1973 addenda, Part A (ferrous materials) and Part C (welding rods, electrodes, and filler materials)

AISC Specification for the Design, Fabrication and Erection of Structural Steel For Buildings, 1969 including Supplement Numbers 1, 2, and 3 hereafter referred to as AISC Specification.

Except: when supported by an engineering analysis, connections using A325 or A490 high strength bolts need not be pretensioned to the values required by AISC Specification, Table 1.23.5 (for steel brackets and attachments)

ASTM A 20-72a	Specification for General Requirements for Delivery of Steel Plates for Pressure Vessels
SA 370-74	Specification for Methods and Definitions for Mechanical Testing of Steel Products
ASTM A 578-71b	Specification for Straight-Beam Ultrasonic Examination of Plain and Clad Steel Plates for Special Applications
SA 537-74	Specification for Carbon Manganese Silicon Steel Plates, Heat Treated for Pressure Vessels
SA 333-74	Specification for Seamless and Welded Steel Pipe for Low-Temperature Service
SA 182-74	Specification for Forged or Rolled Alloy Steel Pipe Flanges, Forged Fittings, and Valve and Parts for High-Temperature Service
SA 350-74	Specification for Forged or Rolled Carbon and Alloy Steel Flanges, Forged Fittings, and Valves and Parts for Low-Temperature Service
SA-320-74	Specification for Alloys Steel Bolting Materials for Low-Temperature Service
SA-105-74	Specifications for Forged or Rolled Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service

SA-354-74	Specification for Quenched and Tempered Alloy Steel Bolts and Studs with Suitable Nuts
ASTM A 36-74	Specification for Structural Steel (for miscellaneous attachments)
ASTM A 108-73	Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality
SA 516-74	Specification for Carbon Steel Plates for Pressure Vessel for Moderate and Lower Temperature Services

2. Reinforcing Steel

ASTM A 615-72	Specification for Deformed and Plain Billet Steel Bars for Concrete Reinforcement
ASME SFA 5.1	Specification for Mild Steel Covered Arc-Welding Electrodes

3. Concrete

ASTM C 150-74	Specification for Portland Cement
ASTM C 33-74	Specification for Concrete Aggregates
ASTM C 131-69	Test for Resistance to Abrasion of Small Size Coarse Aggregate by Use of the Los Angeles Machine
ASTM C 142-71	Test for Clay Lumps and Friable Particles in Aggregates
ASTM C 117-69	Test for Materials Finer Than No. 200 Sieve in Mineral Aggregates by Washing
ASTM C 87-69	Test for Effect of Organic Impurities in Fine Aggregate on Strength of Mortar
ASTM C 40-73	Test for Organic Impurities in Sands for Concrete
ASTM C 289-71	Test for Potential Reactivity of Aggregates (Chemical Method)
ASTM C 136-71	Test for Sieve or Screen Analysis of Fine and Coarse Aggregates
ASTM C 88-73	Test for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C 127-73	Test for Specific Gravity and Absorption of Coarse Aggregate
ASTM C 295-65	Recommended Practice for Petrographic Examination of Aggregates for Concrete
ASTM D 512-67	Tests for Chloride Ion in Water and Waste Water

ASTM C 151-74a	Test for Autoclave Expansion of Portland Cement
ASTM C 191-74	Test for Time of Setting of Hydraulic Cement by Vicat Needle
ASTM C 260-74	Specification for Air-Entraining Admixtures for Concrete
ASTM C 494-71	Specification for Chemical Admixtures for Concrete
ACI 211.1-74	Recommended Practice for Selecting Proportions for Normal and Heavy-Weight Concrete
ACI 304-73	Recommended Practice for Measuring, Mixing, Transporting, and Placing Concrete
ASTM C 143-74	Test for Slump of Portland Cement Concrete
ASTM C 172-71	Sampling Fresh Concrete
ASTM C 192-69	Making and Curing Concrete Test Specimens in the Laboratory
ASTM C 31-69	Making and Curing Concrete Test Specimens in the Field
ASTM C 39-72	Test for Compressive Strength of Cylindrical Concrete Specimens
ASTM C 109-73	Test for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. (50 mm) cube specimens)
ASTM C 231-74	Test for Air Content of Freshly Mixed Concrete by the Pressure Method
ACI 214-65	Recommended Practice for Evaluation of Compression Test Results of Field Concrete
ASTM C 78-64	Test for Flexural Strength of Concrete
ASTM C 496-71	Test for Splitting Tensile Strength of Cylindrical Concrete Specimens
ASTM C 469-65	Test for Static Modulus of Elasticity and Poisson’s Ratio of Concrete in Compression
ASTM C 642-69T	Test for Specific Gravity, Absorption, and Voids in Hardened Concrete
ASTM C 94-74	Specification for Ready-Mixed Concrete
ACI 347-68	Recommended Practice for Concrete Formwork
ACI 305-72	Recommended Practice for Hot Weather Concreting
ACI 306-66	Recommended Practice for Cold Weather Concreting

4. Testing and Surveillance

10 CFR Part 50, Appendix J, Primary Reactor Containment Leakage Testing for Water-Cooled Power Reactors (2/5/73) ANSI N45.4-1972 Leakage-Rate Testing of Containment Structures for Nuclear Reactors

3.8.1.2.4 Specifications and Standards Not Referred to in ASME-ACI 359

AWS D12.1-61 Recommended Practices for Welding Reinforcing Steel, Metal Inserts, and Connections in Reinforced Concrete Construction

AWS D1.4-1998 Structural Welding Code (Unit 1)

3.8.1.2.5 Summary of Principal Plant Specifications

1. The principal plant specifications for the steel-lined, reinforced concrete containment are the fabrication and construction specifications for the following:

   a. Liner (including penetrations and attachments)

   b. Reinforcing steel

   c. Splicing of reinforcing steel

   d. Concrete

2. The applicable portions of the ASME-ACI 359 document, as described in Subsection 3.8.1.2.1, are included in the plant construction specifications in regard to materials, construction techniques, fabrication, welding, examination, testing, and so forth. The following are the principal appropriate portions of the ASME-ACI 359 document which are incorporated into the plant specifications:

   a. Liner Specifications

CC-2500	Materials for Liners
CC-2520	Special Materials Testing
CC-2530	Examination and Repair of Liner Materials
CC-2540	Marking of Liner Materials
CC-2600	Welding Materials
CC-2612	Weld Metal Tests NB-2432 may be used in lieu of CC-2612.2 for chemical analysis of filler metal or weld deposit.
CC-2620	Stud Welding Materials Except, 1100 Aluminum may be used as a flux for stud welding without a chemical analysis of each batch as stipulated by CC-2623.2.
CC-2630	Identification of Welding Materials

CC-2700	Materials Manufacturers' Quality Assurance Programs
CC-3840	Design of Welded Construction
CC-4120	Certification of Materials and Fabrication by Component Manufacturer and/or Installer and/or Constructor
CC-4520	Forming, Fitting, and Aligning
	Except for the following: When qualifying the procedure for the forming and bending process, the Charpy V-notch impact test temperatures, of the specimens used to establish a transition curve, may be conducted at a minimum of five different temperatures distributed throughout the transition region, in lieu of CC-4521.3.2(e) requirement for conducting tests at each temperature increment of 10°F.
	When thermal cutting is performed to prepare weld joints or edges, to remove attachments or defective material or any other purpose, preheating may be in accordance with the applicable section of the ASME III B&PV Code in lieu CC-4521.1.1.
	Engineers may approve on a case by case basis other stud welding equipment in addition to the requirements of CC-4543.5(a).
	Engineers may review post-weld heat treatment records in lieu of CC-4552.2.2 requirement of review by inspector.
CC-4530	Welding Qualifications
CC-4540	Rules Governing Making, Examining, and Repairing Welds

Material for non-pressure parts or pads which are permanently attached by welding to liners may use the following acceptance criteria in lieu of CC-4543.1:

The allowable loadings for attachment materials may be determined by the principles of fracture mechanics based on calculated normal tensile stresses in a defect free stress model at the surface nearest the location of an assumed defect or charpy impact testing may be waived in accordance with the provisions of the 1986 edition of ASME III, Division 2, Subsection CC4543.1.

CC-4550	Heat Treatment
CC-5520	Required Examination of Welds

The criteria for extent and frequency of radiographic examination may be as follows in lieu of CC-5521.1.1.(a) (b) (c) (d) (e) and (f):

1. For each welder and welding position (flat, vertical, horizontal and overhead), the first 10 feet of weld shall be examined radiographically. If this radiograph meets the acceptance standards that 10 feet of weld shall be accepted.

2. For the first three shell rings only, for each welder and welding position (flat, vertical, horizontal and overhead) the first 10 feet of weld shall be performed on a representative mock-up. These mock-ups shall simulate as close as practicable the actual conditions that the welder will experience during [Aduction. These welds shall be 100 percent radiographically examined. Should a question of interpretation arise as to the acceptance of the weld in accordance with the radiographic acceptance standard, a cross-sectional coupon can be cut from the weld to visually verify or refute the film interpretation.

   Further, all production welds on the first three shell rings are to be 100 percent examined by magnetic particle inspection and vacuum box tested, in lieu of radiographic examination.

3. If the radiography in “1” above does not meet the acceptance standards, the portions of the 10 foot increment of weld which do not meet the performance requirements shall be repaired and re-radiographed.

4. Welders who have satisfactorily welded the first 10 feet of weld as described above shall have one 12-inch-long radiograph made of each subsequent 50 feet of weld or fraction thereof which he produces. If the first radiograph in each 50 foot increment meets the acceptance standards, the welder shall be permitted to continue welding the next 50-foot increment of production weld. A minimum of 2 percent of all liner seam welds shall be examined by radiography.

5. If the 12-inch radiograph in the 50-foot-long increment of weld does not meet the acceptance standards, two 12-inch films shall be taken at least 1 ft removed on each side from the original spot within the 50-foot-long increment. If these radiographs meet the radiographic acceptance standards, the 50 feet of weld represented shall be accepted. The defective areas shown in the first radiograph shall be repaired and re-radiographed.

6. If either of the second radiographs does not meet the acceptance standards, the entire weld test unit is unacceptable. The remaining portion of the 50-foot increment of this weld shall be radiographed. The portions of the 50-foot increment of weld which do not meet the acceptance standards are to be repaired and re-radiographed.

The criteria for magnetic-particle (MT or MPE) or liquid-penetrant (PT or LPE) examination in lieu of radiographic examination may be used as follows in lieu of CC-5521.2.1:

Where radiographic examination of liner seam welds is not feasible or where the weld is located in areas which will not be accessible after construction, the entire length of weld shall be examined by the magnetic-particle or liquid-penetrant method. Where magnetic-particle or liquid-penetrant inspection discloses welding that does not meet the magnetic-particle or liquid-penetrant acceptance standards, additional testing shall be performed to the same extent as required for radiography in CC-5520. Unacceptable indications of the weld shall be eliminated or repaired by welding as required.

In addition to the required radiography of CC-5520, all seam welds at abrupt changes in liner configuration (i.e., cylinder to sphere) shall be examined by the magnetic particle method for their entire length.

The following criteria may be used in lieu of CC-5523.1 for magnetic-particle (MT or MPE) or liquid-penetrant (PT or LPE) examination of full penetration attachment welds to containment liner inserts for welding and examinations performed prior to and including June, 1989:

Full penetration attachment welds shall be examined by examining a portion of the welds using the magnetic-particle (MT or MPE) or liquid penetrant (PT or LPE) method with a linear indication of greater than 1/2" being unacceptable. The accessible portion of the welds below the springline (El. 1000'-6") will be inspected to establish the overall acceptability of the welds inspected in this manner.

This exception does not include the insert plates for the polar crane and the rotating platform. Modifications/additions to full penetration attachment welds to the containment liner performed after June, 1989, will use the CC-5523.1 criteria.

CC-5530 Acceptance Standards

The acceptance standards for the liner butt welds examined by radiography may be in accordance with ASME B&PV Code Section VIII, Division 1, paragraph UW-51 and AEC Regulatory Guide 1.19 in lieu of CC-5532 Radiography Acceptance Standards.

Non-full penetration attachment welds to the reactor containment liner may use the following acceptance standards:

The fillet welds of the attachment to the reactor containment liner shall be examined by either the magnetic particle method or liquid penetrant method. The acceptance standards shall be in accordance with the following criteria.

For attachment welds, fillet weld size specified on the drawings are the minimum size required, except as permitted below, along the full length of the weld joint. Additional welding is acceptable provided it does not distort the items being joined together.

A fillet weld in any single continuous weld may be less than the fillet weld dimension by not more than 1/16 inch provided that the undersize portion of the weld does not exceed 10 percent of the length of weld.

Liquid Penetrant Acceptance Standard

1. Linear indications in which the length is more than three times the width.

2. Round indications are indications which are circular or elliptical with the length less than three times the width.

3. Only indications with major dimensions greater than 1/16 inch are considered relevant.

4. Unless otherwise specified, the following relevant indications are unacceptable.

   (a) Any cracks and linear indications.

   (b) Rounded indications with dimensions greater than 3/16 inch.

   (c) Four or more rounded indications aligned and separated by 1/16 inch or less, edge to edge.

   (d) Ten or more rounded indications in any 6 square inch of surface, the major dimension of this area not exceeding 6 inches, with the area taken in the most unfavorable location relative to the indications being evaluated.

Magnetic Particle Acceptance Standards

1. Only indications with major dimensions greater than 1/16 inch are considered relevant.

2. Unless otherwise specified, the following relevant indications are unacceptable:

   (a) Any cracks or linear indications.

   (b) Rounded indications with dimensions greater than 3/16 inch.

   (c) Four or more rounded indications aligned and separated by 1/16 inch or less, edge to edge.

   (d) Ten or more rounded indications in any 6 square inch of surface, the major dimension of this area not exceeding 6 inches, with the area taken in the most unfavorable location relative to the indications being evaluated.

CC-5540 Examination of Stud Welds

(For protective coatings on liner and other steel, see Subsection 3.8.1.6.5, Item 2.g.)

b. Reinforcing Steel Specification

CC-2300	Materials for Reinforcing Systems
CC-2320	Material Identification
CC-2330	Special Materials Testing
CC-2700	Materials Manufacturers' Quality Assurance Programs
CC-3430	Concrete Temperatures
CC-3533	Reinforcing Steel Cover and Spacing Requirements
CC-4120	Certification of Materials and Fabrication by Component Manufacturer and/or Installer and/or Constructor
CC-4320	Bending of Reinforcing Bars
CC-4340	Placing Reinforcement
CC-4350	Spacing of Reinforcement
CC-4360	Surface Condition

c. Specification for Mechanical Butt Splicing (Cadwelds) of Reinforcing Steel

CC-2700	Materials Manufacturers' Quality Assurance Programs
CC-4120	Certification of Materials and Fabrication by Component Manufacturer and/or Installer and/or Constructor
CC-4333	Mechanical Butt Splices Using Sleeve and Filler Material
CC-4335	Mechanical Joints
CC-5320	Examination of Sleeve with Filler Metal Connection

d. Concrete Specification

CC-2220	Materials for Concrete
CC-2230	Concrete Mix Design
CC-2250	Marking and Identification of Concrete Materials
CC-2700	Materials Manufacturers' Quality Assurance Programs
CC-4220	Batching, Mixing, and Transporting
CC-4225	Depositing
CC-4240	Curing

CC-4250	Formwork and Construction Joints
CC-4260	Cold and Hot Weather Conditions
CC-4270	Repairs to Concrete
CC-5220	Concrete Materials (testing and examination of)
CC-5230	Concrete (testing and examination of)

3.8.1.2.6 Applicable NRC Regulatory Guides

The following NRC (formerly AEC) Regulatory Guides are applicable to this Containment and are complied with:

NRC Regulatory Guide 1.10	Mechanical (Cadweld) Splices in Reinforcing Bars of Category I Concrete Structures (Revision 1, 1-2-73, of former Safety Guide 10)
NRC Regulatory Guide 1.15	Testing of Reinforcing Bars for Category I Concrete Structures (Revision 1, 12-28-72 of former Safety Guide 15)
NRC Regulatory Guide 1.18	Structural Acceptance Test for Concrete Primary Reactor Containments (Revision 1, 12-28-72 of former Safety Guide 18)
NRC Regulatory Guide 1.19	Nondestructive Examination of Primary Containment Liner Welds (Revision 1, 8-11-72 of former Safety Guide 19)
NRC Regulatory Guide 1.28	Quality Assurance Program Requirements (Design and Construction) (6-7-72 of former Safety Guide 28)
NRC Regulatory Guide 1.29	Seismic Design Classification (Revision 2, 2-76 of former Safety Guide 29)
NRC Regulatory Guide 1.55	Concrete Placement in Category I Structures (6-73)

3.8.1.3 Loads and Load Combinations

3.8.1.3.1 Loads

The following loads are considered in the design of the steel-lined, reinforced concrete Containment structure (essentially in accordance with the ASME-ACI 359 document):

1. D = dead load of the Containment, and all superimposed permanent loads

2. L = live loads, comprising conventional floor and roof live loads, movable equipment loads, cables, and lateral soil pressure

3. Pa = Containment pressure load due to the DBA, at 50 psig

4. T = thermal effect

   a. To = thermal loads during normal operating conditions, including liner expansion and temperature gradients in the wall

      1. Normal operating temperature range inside the Containment is 50°F to 120°F.

      2. Ambient temperature range at the outside face of the Containment wall is 20°F to 110°F.

   b. Ta = added thermal loads (over and above operating thermal loads), exerted by the liner, which may occur during an accident and which correspond to the factored accident pressure (i.e., 1.0 Pa, 1.25 Pa, or 1.5 Pa); the accident temperature causes an almost instantaneous increase in the liner temperature, with little initial effect on the temperature of the relatively thick concrete wall. This sudden increase in liner temperature creates compressive stresses and strains in the liner which thrusts against the reinforced concrete section, having an effect on the reinforcing steel similar to an added internal pressure.

      1. The design temperature corresponding to an accident pressure of 50 psig is 280°F.

      2. The design temperature corresponding to an accident pressure of 1.25 x 50 psig is 295°F.

      3. The design temperature corresponding to an accident pressure of 1.5 x 50 psig is 305°F.

   c. Tt = Thermal loads during pressure test, including liner expansion and a temperature gradient in the wall; a maximum gradient of 40°F is assumed during test.

5. Seismic loads representing two magnitudes of earthquake are considered, as follows:

   a. E' = SSE

   b. E = 1/2 of SSE=OBE

   The vertical and horizontal earthquake accelerations are assumed to act simultaneously.

   The earthquake forces acting on the Containment structure are taken from the results of dynamic analyses, based on seismic input described in Section 3.7.

6. W = design wind load (See Section 3.3.)

7. Wt = tornado load including wind, differential pressure and missiles (See Sections 3.3 and 3.5.)

8. Ro = piping loads acting on the Containment during operating conditions

9. Ra = piping loads acting on the Containment, due to increased temperature resulting from the design accident

10. Yr = equivalent static load on structure or penetration generated by the reaction on the broken high-energy pipe during the postulated break, and including an appropriate dynamic load factor to account for the dynamic nature of the load.

11. Pv = negative internal pressure during operation; maximum Pv equals 5 psig.

12. Yj = equivalent static jet impingement load on a structure generated by the postulated break, and including an appropriate dynamic load factor to account for the dynamic nature of the load.

13. Ym = equivalent static missile impact load on a structure generated by or during the postulated break, as from pipe whipping, and including an appropriate dynamic load factor to account for the dynamic nature of the load.

14. Ha = represents the load on the Containment resulting from post-LOCA internal flooding.

3.8.1.3.2 Load Combinations

The design of the reinforced concrete Containment structure incorporates the service load combination requirements and the factored load combination requirements, as follows (in accordance with the ASME-ACI 359 document):

1. Service Load Combinations

   a. Construction Category

      1.0 D + 1.0 L + 1.0 To

   b. Test Category

      1.0 D + 1.0 L + 1.15 Pa + 1.0 Tt

   c. Normal Category

      1. 1.0 D + 1.0 L + 1.0 To + 1.0 E + 1.0 Ro + 1.0 Pv

      2. 1.0 D + 1.0 L + 1.0 To + 1.0 W + 1.0 Ro + 1.0 Pv

2. Factored Load Combinations

   a. Abnormal Category

      1.0 D + 1.0 L + 1.5 Pa + 1.0 (To + Ta) + 1.0 Ra

b. Extreme Environmental Category

1. 1.0 D + 1.0 L + 1.0 To + 1.0 Wt + 1.0 Ro + 1.0 Pv

2. 1.0 D + 1.0 L + 1.0 To + 1.0 E' + 1.0 Ro + 1.0 Pv

c. Abnormal Severe Environmental Category

1. 1.0 D + 1.0 L + 1.25 Pa + 1.0 (To + Ta) + 1.25 E + 1.0 Ra + 1.0 (Yr+Yj+Ym)

2. 1.0 D + 1.0 L + 1.25 Pa + 1.0 (To + Ta) + 1.25 W + 1.0 Ra + 1.0 (Yr+Yj+Ym)

3. 1.0 D + 1.0 L + 1.0 To + 1.0 E + 1.0 Ha

4. 1.0 D + 1.0 L + 1.0 To + 1.0 W + 1.0 Ha

d. Abnormal-Extreme Environmental Category

1.0 D + 1.0 L + 1.0 Pa + 1.0 (To + Ta) + E' + 1.0 Ra + 1.0 (Yr+Yj+Ym)

3. Variable and Interrelated Loads

For loads which may vary, the values (within the possible range) which produce the most critical combination of loading are used in design. For loads which are interrelated as a function of time, the maximum values of these effects do not necessarily occur simultaneously. Recognition is given to the time increments associated with such conditions.

4. Allowable Stresses

The allowable stresses associated with the service load combinations and the factored load combinations are given in Subsection 3.8.1.5.

5. Load Combinations for Localized Areas

The design load combinations used to examine the effects on localized areas such as penetrations, shell discontinuities, crane girder brackets, and local areas of high thermal gradients are the same load combinations used for the general Containment structure, as previously described.

6. Time-Dependent Loads

Time-dependent loads such as the effects of creep, shrinkage, and other related effects are ignored in the design of the reinforced concrete Containment structure. (See Subsection 3.8.1.4.1, Subsection 2).

7. Explanation of the Use of an Ultimate Strength Approach With a Load Factor of 1.0

Factored load combinations that include extreme environmental effects (SSE or tornado effects) incorporate a load factor of 1.0 using an ultimate strength approach with stresses within the range of general yield. This approach is justified based on the fact that the extreme environmental effects that are considered are of an upper bound conservative magnitude and have an extremely low probability of occurrence. In addition, the maximum SSE is assumed to occur concurrently with the DBA, an extremely unlikely occurrence. Additional margin of safety is provided by the fact that, under these factored load combinations, the average stress in the reinforcing steel is limited to 90 percent of yield, rather than full yield.

3.8.1.4 Design and Analysis Procedures

3.8.1.4.1 General Analysis of Entire Containment Structure

The Containment structure, including the foundation mat, is analyzed and designed for all load combinations as described in Subsection 3.8.1.3. Table 3.8-1 shows critical loading combinations, type of stress and computed and corresponding allowable stresses at key locations in the Containment structure.

1. Foundation Mat Analysis

The Containment structure foundation mat is analyzed by a finite element method of analysis using the ANSYS computer program. (See Appendix 3.7A for a description of the program.) The program uses the stiffness method of structural analysis and contains the various types of finite elements, i.e., triangular, rectangular, and quadrilateral plate elements representing membrane or bending behavior, or both, and beam elements. The model used for the foundation mat analysis includes the mat and the Containment cylindrical wall to a height of approximately 76 ft above the mat. This height of Containment wall is sufficient to represent the effect of the Containment wall stiffness on the mat behavior under the various loading conditions. The rock beneath the foundation mat is represented by appropriate linear springs. The rock mat contact area under various loading combinations is discussed in Subsection 3.8.5.4.1, Item 1.

The input to this program consists of the geometry of the structure, the material properties, the appropriate boundary conditions, and the loadings. The boundary conditions at the cut Containment wall section (approximately 76 ft above the mat) are represented by the equivalent load reactions at this point for each type of loading (dead load, pressure, thermal gradient, seismic, and so forth). The loads acting on the mat from the concrete internal structure are represented in the model as equivalent pressures at the interface surface of the mat.

The output of this program contains the displacements, rotations, forces, shears, moments, and stresses throughout the structure. This output is used for the design of the foundation mat. In regard to the Containment wall design, the output from this analysis is used only to check the design of the wall at the junction with the mat. The Containment superstructure design (walls and dome) is based on a supplementary analysis as described in Subsection 3.8.1.4.1, Item 2.

2. Containment Superstructure Analysis

The Containment shell is analyzed using the SHELL-1 computer program. (See Appendix 3.7A for a description of the program.) This program uses a numerical method which combines the direct integration and the finite difference techniques for solving general shell equations. The model consists of the hemispherical dome and the cylindrical wall down to the top of the mat. The input to this model consists of the geometry of the structure, the material properties, and the loadings. Fixed boundary conditions are used at the junction of the bottom of the wall and the top of the mat. The results of the mat analysis are used to verify the design at the bottom of the wall.

Creep and shrinkage of concrete are important considerations in the analysis and design of a prestressed concrete containment. However, for the nonprestressed reinforced concrete containment being used on this project, the effects of creep and shrinkage are not significant and can be safely ignored. Shrinkage in a reinforced concrete containment results in meridional and radial displacements which are the opposite of the displacements caused by the principal loadings, internal pressure, and temperature. Since it is not additive to these major loads, it can be ignored. The amount of cracking in a reinforced concrete containment is dependent on the tensile stresses in the shell. For load combinations which do not include internal pressure, the containment is assumed to be completely uncracked. For load combinations which include internal pressures, the analysis uses a variably cracked model, in which the concrete is completely cracked in the membrane regions, with the stiffness of the concrete ignored in both directions and only the properties of the reinforcing steel considered. The stiffness is increased in the vertical direction of the cylinder due to the decrease in the net tensile force when the dead weight of the containment is included. Also the stiffness is adjusted at the discontinuities to account for the increased stiffness when compression and/or high moment exist.

The overall Containment shell is analyzed, neglecting the presence of the penetrations. The analysis of the portions of the containment shell where the stress pattern is influenced by the major penetrations (airlocks and equipment hatch) is performed as described in Subsection 3.8.1.4.2.

Below the dome spring line, radial shear forces occur in the containment wall as a result of the design loadings, due to the discontinuity between the wall and the dome. In the original design, reinforcement was provided for the Unit 2 containment wall between elevations 993'-1 1/2" and 996'-9 1/2" to conservatively resist maximum envelope shear forces, for the most critical loading combinations using the model which gave the maximum forces. No attempt was made to reduce the reinforcement by investigating the applicability of the model used to the critical loading condition.

Some of the shear reinforcement indicated on the construction drawings for the Unit 2 containment wall at the elevations indicated above were inadvertently omitted. When this omission was discovered, after the concrete had been poured, additional shear reinforcement was provided to that called for on the construction drawings between the construction joint at elevation 997'-6" and the dome spring line.

Due to the effects of the high vertical and circumferential tensile membrane forces in the wall resulting from the critical LOCA loading combination, the following model was used to determine shear forces and to verify the as-built shear reinforcement: stiffness of the concrete was ignored in the stiffness calculations and the stiffness of the reinforcing steel only was used in both horizontal and vertical directions. All shear load in the subject area

was assumed to be resisted by the shear tie reinforcing only and no reliance was placed on the shear capacity of the concrete. The as-built condition was found to be more than adequate. Though ignored, the concrete would resist some of the shear load; in fact, the total shear load would be resisted by the concrete together with the as-built reinforcing steel shear ties. All code allowables were met for the model assumptions applicable to this structural modification of Unit 2.

3. Check of Model Validity and Analysis Results

For methods of checking the validity of the model and the results of the analysis, see Subsection 3.8.3.4.1.

3.8.1.4.2 Analysis at Major Penetrations

The effect of the major penetrations (airlocks and equipment hatch) on the Containment wall is analyzed using a finite element model. The computer program used ANSYS (See Appendix 3.7A for description.) For the purpose of this analysis, a segment of the Containment wall containing the penetration is isolated and analyzed for the same loading conditions as those for which the entire Containment shell is analyzed. The boundary of the segment is approximately three times the penetration diameter from the center of the penetrations, except at the boundary with the mat which is approximately two times the penetration diameter from the center of the penetration. The finite element model of the segment consists of solid finite elements or plate elements, or both connected at their nodes. The boundary conditions applied to this model are obtained from the analysis of the entire Containment shell, as described in Subsection 3.8.1.4.1, Item 2. The model considers the various degrees of cracking as described in Subsection 3.8.1.4.1, Item 2. The program used in this analysis considers, in addition to the strains in the plane of the wall, strains in the orthogonal direction.

The output of this analysis includes the displacements, rotations, forces, shears, and moments which are used for the design of the reinforced concrete ring girder around the penetration.

3.8.1.4.3 General Design Criteria and Procedures

The General Design Criteria (GDC) are in accordance with CC-3000 of the ASME-ACI 359 document, except as otherwise specifically indicated.

1. Concrete Tensile Strength

Concrete tensile strength is not relied upon to resist flexural and membrane tension.

2. Interaction of Liner Plate and Reinforced Concrete

The SHELL-1 analysis considers the interaction of the liner plate and the concrete structure under all conditions of loading. However, if the action of the liner results in lower concrete or reinforcement stresses, the presence of the liner is disregarded for that particular case.

In considering the interaction between the liner, when subjected to the hot accident temperature, and the reinforcing steel within the cracked concrete section, strain compatibility between the liner and the reinforcing steel is considered. In the equations of

strain compatibility for the design of the reinforcing steel for this hot liner condition, the effect of the concrete is ignored, since it is assumed to be cracked and incapable of carrying any of the tensile loads.

The interaction of the liner plate is only used to increase the effective internal pressure during an accident case. It is not used to reduce the internal pressure acting on the concrete.

3. Evaluation of Effect of Variations in Assumptions and Materials

The fact that reinforced concrete is not a homogeneous material is accounted for in the design; stiffness properties are altered where the section is assumed to crack. Properties of materials are known with sufficient accuracy, and assumptions made are sufficiently conservative so that other variations need not be considered.

4. Temperature Effects

The temperature gradient through the containment wall during operation is essentially linear and is a function of the internal operating temperature and the average external ambient temperature. The accident temperature primarily affects the liner, rather than the concrete and reinforcing steel, due to insulating properties of the concrete. By the time the temperature of the concrete within the interior of the concrete shell begins to rise significantly, the internal accident pressure within the Containment has fallen off to a point below the peak values. Therefore, it is not necessary to consider peak accident temperature in the concrete coincident with peak pressures in the Containment. (The thrust caused by the instantaneously hot liner against the reinforced concrete wall is considered simultaneously with the peak pressure.) Also, temperature stresses of the reinforcing steel in the Containment shell caused by the maximum thermal gradient do not significantly influence the capacity of the structure to resist membrane forces. Temperature gradients induce stresses in the structure which are internal in nature, causing tension on one face and compression on the other face; the resultant membrane force is zero. If loading combinations concurrent with these temperature gradient effects cause local stresses in the horizontal and vertical bars of one face to reach the yield point, any further load is transferred to the unyielding elements on the other face of the wall. Because of the self-limiting nature of stresses resulting from a thermal gradient, the reinforcing bars across a horizontal or vertical section have a magnitude of final membrane load resistance essentially equal to that which would be carried if temperature gradient effects were neglected. This design approach is basically in accordance with CC-3110 of the ASME-ACI 359 document. However, for factored load combinations which include a thermal gradient, the maximum strain in the reinforcing steel is limited to approximately 1.5 times the yield strain (in accordance with ACI 349, Section 2.2.1). The total reinforcement across any section for any factored load combination has an average tensile stress not more than 0.9 times the yield stress.

3.8.1.4.4 Design of Reinforced Concrete at Penetrations

1. Major Penetrations (Airlocks and Equipment Hatch)

The Containment wall at the major penetrations is designed as a ring beam around the openings and is thickened around these penetrations. The results of the analyses

performed at these openings (see Subsection 3.8.1.4.2) are used to design the ring beams. The ring beams are designed to resist biaxial bending moments, axial tension, torsion, and biaxial shear, resulting from all the load combinations listed in Subsection 3.8.1.3.2. At the openings, some of the typical wall reinforcement close to the outside of the opening is curved around the openings. The remainder of the typical wall reinforcement is terminated at the opening by cadwelding steel plates on the end of the bars.

Additional reinforcement is provided around the opening, principally in circumferential and radial directions relative to the centerline of the opening, to limit stresses to the allowable values (see Subsection 3.8.1.5). For the typical arrangement of reinforcing steel at these major openings, see Figure 3.8-14.

2. Smaller Penetrations (for Pipe Lines)

Penetration sleeves are anchored into the reinforced concrete Containment wall by means of steel lugs welded to the sleeve. The anchors and local wall reinforcement are designed to resist the torsion, bending, and shear that the pipe is capable of exerting on the penetration, as limited by the full plastic strength of the pipe and an axial load based on the maximum possible pipe jet reaction. All possible combinations of loadings are considered to act simultaneously. The typical wall reinforcement is curved around these penetrations and kept continuous wherever possible. Additional local reinforcement is provided around the opening, as required to resist the loads imposed by the pipe, as described in the preceding paragraphs. This additional reinforcement is provided in vertical, horizontal, and radial directions, relative to the centerline of the opening. For typical arrangement of reinforcing steel at these penetrations, see Figure 3.8-13.

3.8.1.4.5 Analysis and Design for Shear Effects

1. Tangential Shear

Tangential shear is due principally to horizontal seismic motion. The maximum intensity of concrete tangential shear stress, Vu (in psi), is obtained from the results of the analysis in accordance with Subsection 3.8.1.4.1, Item 2, for all load combinations.

The factored load design for tangential shear is in accordance with CC-3411.5 of the ASME-ACI 359 document, except that the maximum allowable tangential shear stress carried by the concrete, Vc, does not exceed 60 psi. In regard to this section of the ASME-ACI 359 document, this Containment complies with the requirements CC-3411.5 (a) through (d), and it does not support principal equipment laterally; therefore, the maximum allowable tangential shear stress which the concrete can be assumed to safely resist is 60 psi.

Where the maximum tangential shear stress, Vu, exceeds the concrete allowable shear, Vc, the excess (Vu-Vc) is resisted by inclined reinforcement placed near the outside of the wall at 45 degrees in each direction. Design of the reinforcing steel for tangential shear is in accordance with CC-3521.1.1 of the ASME-ACI 359 document for factored load design. In calculating the stresses in the reinforcing steel caused by tangential shear, compatibility of strains between the inclined steel and the vertical and horizontal steel is considered in accordance with M.J. Holley's Provision of Required Seismic

Resistance, included in Seismic Design for Nuclear Power Plants by the MIT Press. For arrangement of reinforcing steel, see Figure 3.8-10. (Service load combinations are checked in accordance with CC-3420 and CC-3522 of the ASME-ACI 359 document.)

2. Radial Shear

The maximum radial shear occurs at the junction of the bottom of the Containment wall and the top of the foundation mat, under the pressure loading. The values of radial shear are obtained from the results of the analysis described in Subsection 3.8.1.4.1, Item 2, for all load combinations. Design for radial shear is in accordance with CC-3411, CC-3421, and CC-3521 of the ASME-ACI 359 document. Radial shear loads are resisted by radial bars inclined at 45 degrees with the horizontal and extending between the vertical bars near the inside surface and the outside surface of the cylinder wall. Above the mat, where the radial shear is maximum, plate bars 4 in by 1 in are welded to the vertical reinforcing steel. See Figure 3.8 10 for arrangement of radial shear reinforcement.

3.8.1.4.6 Analysis and Design of Liner and Anchorage

The liner for the cylindrical walls is 3/8 in-thick steel plate and for the dome 1/2 in-thick steel plate, each anchored into the concrete with 5/8 in. by 6 3/8 in. long, headed, welded studs, Type H4, as produced by the Nelson Stud Welding Co., or engineer-approved equal. Studs are spaced to satisfy the criteria described in Subsection 3.8.1.5.3. The approximate spacing of the anchor studs in the cylindrical wall and dome is 12 in. each way. The wall and dome liner serve as the inside formwork for placing of concrete. The liner on top of the mat is 1/4-in. thick. This bottom liner is installed after foundation mat construction by welding at seams to structural members that have been embedded in the top of the mat. These embedded structural anchors are approximately 8 to 10 ft apart. The liner on top of the mat is covered with approximately 30 in. of concrete. The vertical wall liner is anchored at the foundation mat; this end anchor is designed to resist the maximum compression and tension to which the liner plate is subjected. See Figure 3.8-5 for liner anchorage details.

The analysis and design of the liner, anchors, and attachments are in accordance with CC-3120, CC-3600, CC-3700, and CC-3800 of the ASME-ACI 359 document.

The liner and anchors are designed to withstand the effects of all load combinations as described in Subsection 3.8.1.3.2, using load factors equal to 1.0.

The stability of the liner is ensured by anchorage to the reinforced concrete. The anchorage system prevents distortions sufficient to impair leaktightness. The liner plate anchorage system is designed to withstand without rupture all in-plane (shear) loads or deformations exerted by the liner plate and also to resist all loads applied normal to the liner surface. The anchors are designed to elastically carry the forces resulting from the various load combinations, or to have sufficient ductility to relieve the forces, or to bring necessary additional anchors into action without rupture of the liner or anchor. The anchorage is designed so that if any one anchor fails, successive failure of adjacent anchors does not occur in the manner of a chain reaction. See ACI 349, Section 2.6.5.5.

In general the maximum load affecting the design of the liner and anchors, is that caused by the maximum temperature rise due to an accident. This temperature increase causes the liner, which is restrained by the reinforced concrete wall, to be stressed in compression. The

compressive stress is calculated by equating the strains between the liner and the reinforced wall. The resulting stresses and strains in the liner are less than allowable as stated in Subsection 3.8.1.5.3. The maximum load in an anchor stud is an in-plane shear load which can occur if the plate on one side of an anchor bows inward in a flexural mode, causing a reduction of membrane compression on one side of the anchor. This inward bowing of the plate can be caused by initial construction deformity, variation of liner plate curvature, loss of an anchor, and similar occurrences.

The resulting unbalanced plate stress imparts a shear load and corresponding displacement on the adjacent anchor, and, to a lesser degree, on each successive anchor further away from the bowed-in plate. The shear load versus displacement of the anchor stud is based on test data developed by the Nelson Stud Welding Co. The analysis to determine the load and displacement on each stud is performed by making a series of successive approximations using the test curves for anchor shear load versus displacement. The analysis is based on maintaining equilibrium of loads in the plate and anchors for any free body cut through a section of plate and compatibility between the strain in the plate and displacement of the studs. The resulting maximum loads and displacements in the studs are less than allowable as stated in Subsection 3.8.1.5.3.

The anchor design and analysis consider the effects of the following (in accordance with CC-3800 of the ASME-ACI 359 document):

1. Variation of liner plate curvature: the anchors are designed for possible inward bowing of the plate, as described previously.

2. Variation in liner plate thickness due to rolling tolerances: the range of maximum and minimum plate thickness is assumed in the design of the liner plate and anchors.

3. Variation of liner plate yield strength: the anchors are analyzed assuming the liner remains elastic under all conditions, i.e., the liner strains are converted to stress using Hooke's Law with the modulus of elasticity and Poisson's Ratio below yield. See CC-3630 of the ASME-ACI 359 document.

4. Liner plate seam offset: stresses due to maximum allowable seam offset, as stated in the construction specification, are considered in design.

5. Variation in anchor spacing: maximum range of anchor spacing, as allowed in the construction specification, are considered in design.

6. Variation in anchor stiffness due to a variation of the concrete modulus: this variation is considered by modifying the load versus displacement test data for the stud anchors and considering in the analysis a range of minimum and maximum possible values of concrete moduli.

7. Local concrete crushing in the anchor zone: such crushing is reflected in the anchor stud test data.

The liner plate is thickened at penetrations in accordance with the requirements of Subsection NE of the ASME B&PV Code, Section III.

3.8.1.4.7 Design of ASME B&PV Code, Section III, Division 1, Class MC Components

For the design of ASME B&PV Code, Section III, Division 1, Class MC Steel Components, such as the airlocks, equipment hatch, and portions of penetration sleeves subject to pressure induced stresses, see Subsection 3.8.2.

3.8.1.5 Structural Acceptance Criteria

3.8.1.5.1 Reinforcing Steel - Allowable Stresses and Strains

1. Based on factored load combinations, as described in Subsection 3.8.1.3.2, Item 2, the allowable average tensile stress in reinforcing steel at any section in the Containment is 90 percent of the yield stress (0.90 fy) in accordance with CC-3412 of the ASME-ACI 359 document. When considering load combinations which include a self-limiting thermal gradient, the maximum allowable strain in the reinforcing steel in one face may reach 1.5 times the yield strain, provided that the average stress in the reinforcing steel across the entire section does not exceed 0.9 fy. For additional discussion on this criterion, see Subsection 3.8.1.4.3, Item 4.

2. Based on service load combinations, as described in Subsection 3.8.1.3.2, Item 1, the allowable tensile stress in reinforcing steel is 50 percent of the yield stress (0.50 fy). However, this value is increased by 33 1/3 percent when considering load combinations which include any one (or more) of the following temporary loads:

    a. Temporary pressure loads during the test condition

    b. Temperature effects

This criterion is in accordance with CC-3422 of the ASME-ACI 359 document, except that the allowable stress may not be increased 33 1/3 percent for earthquake or wind loads.

3.8.1.5.2 Concrete - Allowable Stresses and Strains

1. Based on factored load combinations, as described in Subsection 3.8.1.3.2, Item 2, the allowable concrete stresses are in accordance with CC-3411 of the ASME-ACI 359 document, as follows:

    a. Membrane compression = 0.60 f'c

    b. Membrane plus bending compression = 0.75 f'c

    c. Local compression at discontinuities and at the inside face due to temperature gradients from accident conditions = 0.90 f'c.

    d. Concrete tensile strength is not relied upon to resist flexural and membrane tension.

    e. Radial shear is in accordance with CC-3411.4.1 of the ASME-ACI 359 document. (Where the calculated shear is greater than the allowable concrete shear, steel

shear reinforcement is provided in accordance with CC-3521 of the ASME-ACI 359 document.)

f. Tangential shear is in accordance with CC-3411.5.1 of the ASME-ACI 359, as follows, except that the maximum allowable tangential shear stress carried by the concrete, Vc, does not exceed 60 psi.

1. Provisions (a) and (d) of CC-3411.5.1 are satisfied, and this Containment does not support principal equipment laterally; therefore, the allowable tangential shear stress that can be resisted by the concrete, Vc, equals 60 psi.

2. Where the maximum tangential shear stress, Vu, exceeds the concrete allowable shear, Vc, inclined reinforcement is provided to resist the excess (Vu - Vc).

3. Design of the reinforcing steel for tangential shear is in accordance with CC-3521.1.1 of the ASME-ACI 359 document.

g. Requirements for punching shear are in accordance with Section 11.10.3 of the 1971 edition of the ACI 318 Code.

2. Based on service load combinations as described in Subsection 3.8.1.3.2, Item 1, the allowable concrete stresses are in accordance with CC-3421 of the ASME-ACI 359 document, as follows:

a. Membrane compression = 0.30 f'c

b. Membrane plus bending compression = 0.45 f'c

Note: Allowable stresses indicated in Subsection 3.8.1.5.2, Items 2.a and 2.b are increased by 33 1/3 percent when considering load combinations which include thermal. (The allowable increase when considering wind or earthquake, as permitted by the ASME-ACI 359 document, is not applied.)

c. Local compression at discontinuities and in the vicinity of liner anchors = 0.60 f'c

d. Concrete tensile strength is not relied upon to resist flexural and membrane tension.

e. Allowable stress in shear is 50 percent of the values indicated for factored loads, except that 67 percent of the factored load stresses are allowed for load combinations which include the pressure loads during the test condition. Additional requirements are in accordance with ASME-ACI 359 document, CC-3421.3 and CC-3522, except that the tangential shear stress carried by the concrete, Vc, does not exceed 40 psi.

3.8.1.5.3 Liners, Anchors, and Attachments - Allowable Stresses and Strains

1. The allowable stresses and strains in the liner plate are in accordance with Table CC-3700 1 of the ASME-ACI 359 document, as follows:

   a. Considering Calculated Membrane Stresses and Strains Only

      1. Construction category: tensile or compressive stress = 2/3 x yield stress

      2. Test category: compressive strain = .002 in./in.; tensile strain = .001 in./in.

      3. Normal category: same as test category

      4. Severe environmental category: same as test category

      5. Extreme environmental category: same as test category

      6. Abnormal category: compressive strain = .005 in./in.; tensile strain = .003 in./in.

      7. Abnormal-severe environmental category: same as abnormal category

      8. Abnormal-extreme environmental category: same as abnormal category

   b. Considering Combined Membrane and Bending Stresses and Strains

      1. Construction category: tensile and compressive stress = 2/3 x yield stress

      2. Test category: compressive strain = .004 in./in.; tensile strain = .002 in./in.

      3. Normal category: same as test category

      4. Severe environmental category: same as test category

      5. Extreme environmental category: same as test category

      6. Abnormal category: compressive strain = .014 in./in.; tensile strain = .010 in./in.

      7. Abnormal-severe environmental category: same as abnormal category

      8. Abnormal-extreme environmental category: same as abnormal category

   c. The load categories indicated previously include loads as defined in Subsection 3.8.1.3.2, except that load factors for all load cases are equal to 1.0.

2. The allowable forces and displacements of the liner anchors are in accordance with Table CC-3700-2 of the ASME-ACI 359 document, as follows:

   a. Considering Mechanical Loads Only

      1. Test category: applied load equals the lesser of 0.67 x yield load or 0.33 x ultimate load

      2. Normal category: same as test category

      3. Severe environmental category: same as test category

      4. Extreme environmental category: same as test category

      5. Abnormal category: applied load equals the lesser of 0.9 x yield load or 0.50 x ultimate load

      6. Abnormal-severe environmental category: same as abnormal category

      7. Abnormal-extreme environmental category: same as abnormal category

   b. Considering Displacement Limited Loads

      1. Test category: displacement equals 0.25 x ultimate displacement

      2. Normal category: same as test category

      3. Severe environmental category: same as test category

      4. Extreme environmental category: same as test category

      5. Abnormal category: displacement equals 0.50 x ultimate displacement

      6. Abnormal-severe environmental category: same as abnormal category

      7. Abnormal-extreme environmental category: same as abnormal category

   c. The load categories previously indicated include loads as defined in Subsection 3.8.1.3.2, except that load factors for all cases are equal to 1.0. Mechanical loads are defined as those which are not self-limiting or self-relieving with load application. Displacement limited loads are those resulting from constraint of the structure or constraint of adjacent material and are self-limiting or self-relieving. The yield and ultimate capacity of the liner stud anchors are based on the results of tests performed by the Nelson Stud Welding Company.

3. Allowable stresses and strains in penetration assemblies are in accordance with CC-3740 of the ASME-ACI 359 document. The design allowables for the penetration nozzles are the same as those used for metal containments (ASME B&PV Code, Section III, Division I), as discussed in Subsection 3.8.2. For additional criteria for ASME B&PV

Code, Section III, Class MC steel components such as the airlocks and the equipment hatch, see Subsection 3.8.2.

4. The design allowables for overlay plates, brackets, and attachments are the same as those given in AISC Specification for resisting mechanical loads in the construction, test, and normal categories. For all other categories indicated in Subsection 3.8.1.3.2, the allowable can be increased by a factor of 1.5.

   For overlay plates, brackets, and attachments which resist external mechanical loads and are not continuous through the liner plate, the allowable strength of the liner plate in the through-the-thickness direction will be taken as one-half of that in the transverse direction.

5. The liner is investigated for fatigue using the methods and limits established by ASME B&PV Code, Section III, Division I, Subsection NE.

3.8.1.5.4 Effect of Two- and Three-Dimensional Stress Strain Fields on the Behavior of the Structure

1. Liner

   The computed stresses and strains in the liner consider the effect of the two dimensional stress-strain field by the use of Poisson's Ratio in stress and strain determination. The liner anchor design considers biaxial liner loading by not relying on liner plate yielding to limit the forces applied to the anchor; liner strains are converted to stress and membrane forces assuming the material remains elastic. Because of the use of this conservative design approach, biaxial yield test values are not required.

2. Major Penetrations

   The analysis performed at the major penetrations, as described in Subsection 3.8.1.4.2, considers the three dimensional stress-strain field. Allowable stresses are the same as indicated in Subsections 3.8.1.5.1, 3.8.1.5.2, and 3.8.1.5.3.

3. Reinforced Concrete Section

   Based on the conservative assumption of fully cracked concrete, in both directions, used in the design of the reinforcing steel, the effect of a two- or three-dimensional stress-strain field is not a consideration in the design of the typical reinforced concrete section in the wall and dome of the Containment. However, the effect of the two-dimensional-stress strain field in the liner, in calculating the hot liner thrust against the reinforced concrete section, is considered.

3.8.1.5.5 Effects of Repeated Reactor Shutdowns and Startup During the Life of the Plant

1. The number of assumed reactor shutdowns and startups during the life of the plant is assumed to be 200 cycles over a period of 40 years.

2. The cycled stresses and strains in the reinforced concrete sections caused by reactor shutdowns and startups are minor compared to the stresses caused by the critical design

loading based on the abnormal (accident pressure and temperature) and extreme environmental (SSE) conditions. Therefore, the cycled stresses and strains caused by reactor shutdowns and startups do not degrade the margin of safety in the reinforced concrete.

3. The effect of cycled stresses and strains in the liner is considered by performing a fatigue analysis, in accordance with Subsection 3.8.1.5.3, Item 5, which includes the reactor shutdown and startup cycles.

3.8.1.5.6 Connections and Joints

The connections and joints of the various elements of the Containment, such as the crane girder bracket, liner end anchorage at bottom of wall, and similar elements, are designed using the same appropriate stress and strain allowables described elsewhere in Subsection 3.8.1.5.

3.8.1.5.7 Conditions at End of Service Life of Structure

Since a conventionally reinforced concrete containment is essentially a passive structure, as compared to a prestressed structure which relies on active prestress forces to meet its design function requirements, the margins of safety against all loading conditions are essentially the same throughout the life of this structure.

3.8.1.6 Materials, Quality Control, and Special Construction Techniques

3.8.1.6.1 Concrete

1. Materials

a. Cement

Cement is in conformance with the requirements of ASTM C-150-74, Specification for Portland Cement, Type II.

b. Aggregates

Aggregates are in conformance with the requirements of ASTM C 33-74, Specification for Concrete Aggregates, with the following additional requirements:

1. Gradations 357 or 467 are not furnished as one graded aggregate, but are obtained by combining at least two separate gradation sizes.

2. The potential reactivity of the aggregate is established by the methods described in the Appendix to ASTM C 33-74.

3. Aggregate shapes and sizes are in accordance with CC-2222.1.1 of the ASME-ACI 359 document.

c. Mixing Water

Mixing water is clean and free from injurious amounts of oils, acids, alkalis, salts, and organic materials, or other substances which could be deleterious to concrete or steel. Quality control tests are in accordance with the requirements of CC-2223 of the ASME-ACI 359 document.

d. Admixtures

1. Air-entraining admixtures conform to the requirements of ASTM C 260-74, Specification for Air-Entraining Admixtures for Concrete.

2. Chemical admixtures conform to the requirements of ASTM C 494-71, Specification for Chemical Admixtures for Concrete

2. Concrete Strength

Concrete has a minimum compressive strength of 4000 psi, in 28 days, when tested in accordance with ASTM C 39-72.

3. Other Concrete Properties

The following concrete properties are determined in accordance with the noted ASTM standards:

ASTM C 78-64 for flexural strength

ASTM C 496-71 for splitting tensile strength

ASTM C 469-65 for static modulus of elasticity

ASTM C 642-69T for specific gravity of concrete

4. Selection of Concrete Mix Proportions

Concrete mix proportions are established on the basis of laboratory trial batches, in accordance with the requirements of CC-2232 of the ASME-ACI 359 document. The following industry standards are referred to in this document:

ACI 211.1-74	Recommended Practice for Selecting Proportions for Normal and Heavy Weight Concrete
ACI 214-65	Recommended Practice for Evaluation of Compression Test Results of Field Concrete
ASTM C 39-72	Test for Compressive Strength of Cylindrical Concrete Specimens
ASTM C 192-69	Making and Curing Concrete Test Specimens in the Laboratory

5. Construction of Concrete

Concrete construction, including the stockpiling, batching, mixing, conveying, depositing, consolidation, curing, and construction joint preparation, is in accordance with CC-4200 of the ASME-ACI 359 document. The following industry standards are referred to in this document:

ACI 304-73	Recommended Practice for Measuring, Mixing, Transporting, and Placing Concrete
ASTM C 94-74	Specification for Ready-Mixed Concrete
ACI 347-68	Recommended Practice for Concrete Formwork
ACI 305-72	Recommended Practice for Hot Weather Concreting
ACI 306-66	Recommended Practice for Cold Weather Concreting

6. Examination, Testing and Other Quality Control Procedures for Concrete

a. Quality Assurance

The QA procedures are in accordance with the requirements, in general, as described throughout the ASME-ACI 359 document.

The quality control (QC) for concrete begins with the selection and testing of the ingredients of the mix and extends through proportioning, batching, mixing, transporting, placing, and curing.

b. Testing and Examination of Concrete Ingredients

1. Cement

QC testing is in accordance with ASTM C 109-73, Test for Compressive Strength of Hydraulic Cement Mortars. Other QA requirements, including testing frequency, are in accordance with CC-5221 of the ASME-ACI 359 document.

2. Aggregates

QC testing is in accordance with CC-5224 of the ASME-ACI 359 document with the exception that the frequency of tests in accordance with ASTM C289 can be one test for each two month period provided low alkali cement is used. The following industry standards are called for in testing of aggregates to ascertain conformance to ASTM C 33-74, Specification for Concrete Aggregates:

ASTM C 131-69	Test for Resistance to Abrasion of 11 Size Coarse Aggregate by Use of the Los Angeles Machine

ASTM C 142-71	Test for Clay Lumps and Friable Particles in Aggregates
ASTM C 117-69	Test for Materials Finer Than No. 200 Sieve in Mineral Aggregates by Washing
ASTM C 87-69	Test for Effect of Organic Impurities in Fine Aggregate on Strength of Mortar
ASTM C 40-73	Test for Organic Impurities in Sands for Concrete
ASTM C 289-71	Test for Potential Reactivity of Aggregates (Chemical Method)
ASTM C 136-71	Test for Sieve or Screen Analysis of Fine and Coarse Aggregates
ASTM C 88-73	Test for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C 127-73	Test for Specific Gravity and Absorption of Coarse Aggregate
ASTM C 295-65	Recommended Practice for Petrographic Examination of Aggregates for Concrete

3. Mixing Water

   Quality control testing requirements, including testing frequency, are in accordance with CC-5225 of the ASME-ACI 359 document.

c. Testing and Examination of Concrete

Concrete is tested in accordance with CC-5230 of the ASME-ACI 359 document. The following is a summary of the major testing requirements, as stated in this document:

1. Slump Test

   Testing is performed in accordance with ASTM C 172-71, Sampling Fresh Concrete, and ASTM C 143-74, Test for Slump of Portland Cement Concrete, CC-5232 of the ASME-ACI 359 document.

2. Air Content

   Testing is performed in accordance with ASTM C 231-74, Test for Air Content of Freshly Mixed Concrete by the Pressure Method, CC-5233 of the ASME-ACI 359 document.

3. Mechanical Properties

Testing is performed in accordance with ASTM C 39-72, Test for Compressive Strength of Cylindrical Concrete Specimens, and ASTM C 31-69, Making and Curing Concrete Test Cylindrical Specimens in the Field. The samples for strength tests are taken in accordance with ASTM C 172-71, Sampling Fresh Concrete. Testing frequency, and the acceptance criteria, are in accordance with CC-5234.2 of the ASME-ACI 359 document.

d. Certification, Marking, and Identification of Materials

Certified materials test reports are prepared in accordance with the requirements of CC-2130 of the ASME-ACI 359 document. Marking and identification of materials are in accordance with CC-2250 of the ASME-ACI 359 document.

e. Certification and Tests and Examinations

Certification of tests and examinations is provided in accordance with the requirements of CC-4120 of the ASME-ACI 359 document.

3.8.1.6.2 Reinforcing Steel

1. Material Specification

Reinforcing steel conforms to the requirements of ASTM A 615-72 Grade 60 or ASTM A706 (Unit 1).

2. Physical Properties

The specified minimum yield strength is 60,000 psi, and the specified minimum ultimate strength is 90,000 psi. The minimum elongation is 7 percent in 8 inches.

3. Chemical Properties

Some arc-welding is performed on the reinforcing steel, such as the welded shear bars at the Containment wall (See Figure 3.8-10) and restored reinforcing steel for the Containment Alternate Access (Unit 1); the reinforcing steel which is welded has a limited chemistry as defined in CC-2333 of the ASME-ACI 359 document, or carbon equivalency determined and reinforcing steel welded in accordance with the requirements of AWS D1.4-1998 (Unit 1).

4. Fabrication and Installation of Reinforcing Steel

Fabrication and installation of reinforcing steel are in accordance with CC-4300 of the ASME-ACI 359 document.

5. Examination and Testing of Reinforcing Steel

Special materials testing is performed in accordance with CC-2330 of the ASME-ACI 359 document. This section requires that one full diameter tensile test bar from each bar size be tested for each 50 tons, or fraction thereof, of reinforcing bars produced from each heat of steel. Bend tests also are performed in accordance with CC-2332 of the ASME-ACI 359 document. This section requires that, for No. 14 and No. 18 bars, bend testing of bars be conducted at the rate of one test bar for each bar size from each heat. NRC Regulatory Guide 1.15 is also complied with.

6. Certification, Marking, and Identification of Materials

Certified materials test reports are furnished in accordance with the requirements of CC-2130 of the ASME-ACI 359 document. Marking and identification of reinforcing steel are in accordance with the requirements of CC-2320 and CC-4122 of the ASME-ACI 359 document.

7. Certification of Tests and Examinations

Certification of tests and examinations is provided in accordance with CC-4120 of the ASME-ACI 359 document.

3.8.1.6.3 Mechanical Butt Splices (Cadwelds)

No. 14 and No. 18 reinforcing bars are spliced by use of Cadweld connections, as described in CC-4333 of the ASME-ACI 359 document. Such splices develop the tensile limits shown in Table CC-4330 of that document. Crew qualification is in accordance with CC-5321. Nondestructive testing is in accordance with CC-5322. Tensile testing and test frequency conform to the applicable requirements CC-5323 and CC-5324. The procedure for substandard test results conforms to CC-5325. Certified mill test reports are furnished in accordance with CC-2131. Marking and identification conform to CC-4122. Tests and examinations are certified in accordance with CC-4120. The requirements of the NRC Regulatory Guide 1.10 are also complied with.

3.8.1.6.4 Welding to Reinforcing Bars

Limited welding to reinforcing bars, such as required for welding shear bars to vertical reinforcing at the base of the Containment wall and restoration of the Containment Alternate Access (Unit 1), conforms to the requirements of CC-4334 of the ASME-ACI 359 document or chemistry composition determined to ensure proper welding (Unit 1).

3.8.1.6.5 Liner and Attachments

1. Materials

a. Liner, including thickened plates at penetrations and crane girder brackets, is in accordance with SA 537-74 Class 2.

b. Penetration sleeves are in accordance with SA 333-74 Grade 6 (seamless), SA 537-74 Class 2, or SA 516-74 Grade 70.

c. Stud anchors are in accordance with ASTM A 108-73.

d. Embedded steel members in mat are in accordance with ASTM A 36-74.

e. Penetration caps are in accordance with SA 105-74, SA 350-74 Grade LF1 or LF2, SA 516-74 Grade 60, 65, or 70 or SA 333-74 Grade 6.

f. Forgings, including penetration forgings, are in accordance with SA 350-74 Grade LF1 or LF2 or SA 182-74 Type F316.

g. Penetration anchorage studs and lugs are in accordance with 108-73 Grades 1015, 1016, or 1018 or ASTM A36-74 or SA 516-74 Grades 60, 65, or 70 or SA 537-74 Class 2.

h. Welding materials are in accordance with CC-2600 of the ASME-ACI 359 document, with the exceptions as specified in Section 3.8.1.2.5.2.a.

2. Special Materials Testing and Examination

a. Notch Toughness Testing

Notch toughness testing is performed on liner materials in accordance with the requirements of CC-2520 of the ASME-ACI 359 document.

b. Ultrasonic Testing

Liner plate materials which must transmit orthogonal loads in the through-the-thickness direction, such as the thickened liner plate at the crane girder brackets, are examined by the straight-beam ultrasonic method in accordance with ASTM A 578-71b. This examination, the acceptance standards, and related procedures are in accordance with CC-2530 of the ASME-ACI 359 document.

Liner plate in the area of overlay plates and/or structural attachments are not ultrasonic tested. The liner plate design, discussed in Subsection 3.8.1.5.3.4, will consider through-the-thickness properties.

c. Fabrication, Installation, and Welding of Liner

The following is a list of the major portions of CC-4500 of the ASME-ACI 359 document, covering the requirements for fabrication, installation, and welding of the liner, with the exceptions as specified in Section 3.8.1.2.5.2.a.

CC-4520	Forming, Fitting, and Aligning Exemptions and clarifications are discussed in Subsection 3.8.1.2.5.2.a.
CC-4522	Forming Tolerances
CC-4530	Welding Qualifications

CC-4540	Rules Governing Making, Examining, and Repairing Welds Clarifications are described in 3.8.1.2.5.2.a
CC-4550	Heat Treatment Clarifications are described in 3.8.1.2.5.2.a

d. Examination of Liners and Attachments

Examination of liners and attachments are in accordance with CC-5500 of the ASME-ACI 359 document as described in 3.8.1.2.5.2.a. Leak chase systems are installed at inaccessible welds. NRC Regulatory Guide 1.19 is also complied with.

e. Certification, Marking, and Identification of Materials

Certified materials test reports are prepared in accordance with the requirements of CC-2130 of the ASME-ACI 359 document. Marking and identification of liner materials are in accordance with the requirements of CC-2541 of the ASME-ACI 359 document.

f. Certification of Tests and Examinations

Certification of tests and examinations are provided in accordance with the requirements of CC-4120 of the ASME-ACI 359 document.

g. Protective Coatings

Suitable protective coatings are applied to the interior surfaces of the Containment liner as described in 6.1B.2.

3.8.1.6.6 Personnel Airlock, Equipment Hatch, and Emergency Airlock

1. Materials

See Subsection 3.8.2.6.

2. Code Requirements

The personnel airlock, equipment hatch, and emergency airlock satisfy all the requirements (materials, fabricating, welding, examination, testing, and other requirements) of the ASME B&PV Code, Section III, Division 1, Subsection NE, Class MC components. (See Subsection 3.8.2.) The personnel and emergency airlocks meet the requirements to obtain an N-stamp.

3. Outage Equipment Hatch

During Mode 5, 6 or defueled, an outage equipment hatch may be installed to allow access to containment. The outage equipment hatch is designed to provide a fission

product barrier in the event of a design basis Fuel Handling accident, and provides the same level of protection as the permanent inner equipment hatch in Mode 5, 6 or defueled. The outage equipment hatch will not replace the permanent inner equipment hatch during reduced inventory operation.

3.8.1.6.7 Quality Assurance Program

The documentation and maintenance of a QA program in the construction of the Containment are in accordance with CC-2700 of the ASME-ACI 359 document and in accordance with Chapter 17 of this FSAR. NRC Regulatory Guide No. 1.28 is also complied with.

3.8.1.7 Testing and Inservice Inspection Requirements

3.8.1.7.1 Structural Acceptance Test of Containment

The completed Containment structure is subjected to an acceptance test by which the internal pressure is increased from atmospheric pressure to a value of 1.15 times the Containment design pressure (1.15 x 50 psig = 57.5 psig) in five or more approximately equal pressure increments. The Containment is depressurized in the same number of increments. The purpose of the structural acceptance test is to demonstrate that the structure responds satisfactorily to the required internal pressure loads by making measurements of deflections and deformations under load to provide correlation with the theoretically predicted response.

Since this Containment is not a prototype, the measurements are limited to gross deformations and crack mapping. Strain measurements are not taken. The prototype containment on which strain measurements are correlated with deflection measurements is Consolidated Edison's Indian Point No. 2, NRC Docket No. 50247-47.

To the extent feasible, the test is conducted during a period of stable ambient temperature, atmospheric pressure, and humidity. Inside and outside of the Containment, atmospheric temperature, pressure, and humidity are monitored continuously during each test. The test will not be conducted under extreme weather conditions such as snow, heavy rain, or strong wind.

Deflection measurements are recorded at atmospheric pressure and at each level of the pressurization and depressurization cycle. At each level, the pressure is held constant for at least one hour before the deflections are recorded. Radial displacements of the containment cylinder are measured at five approximately equally spaced elevations between the base slab and dome springline and at the dome springline. These measurements are made along four azimuths spaced approximately equally around the containment. Radial deflections of the containment wall adjacent to the largest opening (the equipment hatch) are measured at 12 points as described in paragraph CC-6232(b) of the ASME B&PV Code, Section III, Division 2 (1980 Edition). The increase in diameter of the opening is measured in two mutually perpendicular directions. Vertical displacements of the cylinder at the top relative to the base are measured along four azimuths as described above. Vertical deflections of the containment dome at the apex and at two other equally spaced intermediate points, between a point near the apex and the springline, are measured along one azimuth. A listing of the numerical values, in the form of acceptance criteria for the measurements that are taken during the structural acceptance test, are evaluated in accordance with the ASME B&PV Code referenced below.

Crack patterns that exceed 0.01 inches in width before, during or after the test are mapped at locations described in paragraph CC-6233 of the ASME B&PV Code, Section III, Division 2 (1980 Edition).

The analysis of data and preparation of a test results report will be in accordance with paragraph CC-6260 of the ASME B&PV Code, Section III, Division 2 (1980 Edition).

All aspects of the structural integrity test, including the acceptance criteria are in accordance with paragraph CC-6000 of the ASME B&PV Code, Section III, Division 2, 1980 Edition with Summer 1980 Addenda (except that paragraph CC-6212.1, CC-6234 and CC-6236 do not apply since the CPNPP Containment is not a prototype containment).

3.8.1.7.2 Initial Leakage Rate Tests

Initial Containment leakage testing was in accordance with all the requirements of 10 CFR Part 50, Appendix J, Primary Reactor Containment Leakage Testing for Water-Cooled Power Reactors (2/5/73).

A preoperational Type A integrated leakage-rate test was performed on both units at the calculated peak Containment internal pressure. In addition, a reduced pressure test of not less than 50% of the peak pressure was performed for Unit 1. Type B tests of components and Type C tests of Containment isolation valves were performed in accordance with 10 CFR Part 50, Appendix J. For calculated peak containment internal pressure, see Section 6.2.1.

The maximum allowable leakage-rate (La as defined in 10 CFR Part 50, Appendix J), related to the maximum Containment leakage under design basis pressurization accident conditions, is 0.10 percent of the weight of contained air, at the calculated peak Containment internal pressure per 24 hour period.

For a discussion of the test objectives and the acceptance criteria, see the Technical Specifications for Containment Tests.

Initial test methods were in accordance with ANSI N45.4-1972, Leakage Rate Testing of Containment Structures for Nuclear Reactors (03/26/73), with the exceptions of the calculation of leakage rates using the mass-point method of ANSI/ANS 56.8-1987 and the isolation of penetrations, as stated in Table 14.2-2 (Sheet 59).

3.8.1.7.3 Inservice Inspection Program

The inservice inspection program conforms to the requirements of 10 CFR Part 50, Appendix J, Option B. Periodic testing, test objectives, and acceptance criteria are in accordance with 10CFR50 Appendix J, Option B.

3.8.2 STEEL CONTAINMENT

This section, as outlined in the NRC Regulatory Guide 1.70, Rev. 2., Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants, regarding a steel containment, is not applicable to the Comanche Peak Containment structure because a steel-lined reinforced concrete containment is being used, as described in Subsection 3.8.1. Certain steel components in the Containment system are classified in accordance with ASME B&PV Code, Section III, as

Class MC components, as described in Subsection 3.8.1. These are the personnel airlock, the equipment hatch, the emergency airlock, and other penetrations subject to pressure-induced stresses.

This section addresses itself to the requirements of these ASME B&PV Code, Section III, Class MC steel components.

3.8.2.1 Description of the Containment

For the description of the components (airlocks, equipment hatch, and other penetrations), see Subsection 3.8.1.1.6. The Containment itself is a steel-lined reinforced concrete containment.

3.8.2.2 Applicable Codes, Standards, and Specifications

3.8.2.2.1 Basic Code

The basic code for these steel components is the ASME B&PV Code, Section III, Division 1, Subsection NE, for Class MC components. This code is applicable for all the requirements of the components, including materials, design, fabrication, examination, testing, and so forth, except the equipment hatch and the sleeves for the electrical penetrations, fuel transfer tube, and process piping penetrations are not code stamped and are pressure tested as part of the reinforced concrete containment structure as discussed in Section 3.8.1.7. The 1971 edition of the code, through and including the summer 1973 addenda, is used for the electrical penetration sleeves, fuel transfer tube penetration sleeve, emergency and personnel airlocks, and equipment hatch. The 1974 edition through and including the Summer 1976 Addenda is used for the process piping penetrations.

3.8.2.2.2 Other Applicable Codes, Specifications, and Standards

See Section 3.8.1.2.3 and 3.8.2.6.6 of this FSAR.

3.8.2.2.3 Applicable NRC Regulatory Guide

NRC Regulatory Guide 1.57 Design Limits and loading Combinations for Metal Primary Reactor Containment System Components (6-73) (Applicable only to appropriate containment components constructed in accordance with Subsection NE of the ASME B&PV Code, Section III, Class MC.)

3.8.2.3 Loads and Load Combinations

The applicable loads stated in Subsection 3.8.1.3 are considered in the design of the ASME B&PV Code, Section III, Class MC steel components. For the various load combinations used in the design of Class M components and the related allowable stresses and strains, see Subsection 3.8.2.5.

3.8.2.4 Design and Analysis Procedures

The design and analysis of the Class MC components are in accordance with all the requirements of Subsection NE of the ASME B&PV Code, Section III, including the applicable

portions of Appendix A. Analysis procedures also use published formulas such as Roark's Formulas for Stress and Strain and Timoshenko's Theory of Plates and Shells.

3.8.2.5 Structural Acceptance Criteria

3.8.2.5.1 General Criteria

The design is such that all the stress and strain limits, as defined in NE-3000 of the ASME B&PV Code, Section III, are satisfied for pressure loads in combination with all mechanical loads and thermal loads, as discussed in Subsection 3.8.2.5.2.

3.8.2.5.2 Design Load Combination Stress Limits

1. The ASME B&PV Code, Section III, design criteria for Class MC components are based on establishing stress and strain limits which vary depending on the following factors:

   a. Types of Stress

      As defined by ASME B & PV Code Section III:

      P_m - General Primary Membrane Stress

      P_l - Local Primary Membrane Stress

      P_b - Primary Bending Stress

      P_e - Secondary Expansion Stress

      Q - Secondary Membrane plus Bending Stress

      F - Peak Stress

      S_y - Yield Strength

      S_u - Tensile Strength

      S_m - Design Stress Intensity Value

   b. Types of Loads

      The loads considered in the load combinations in 3.8.2.5.2.2 are as defined in Section 3.8.1.3.1 and as follows:

      P_e - Design External Pressure

      T_e - Thermal loads under thermal conditions during event causing external pressure

      R_e - Pipe reaction under thermal conditions during event causing external pressure

2. Load Combinations and Acceptance Criteria

LOAD COMBINATION	ACCEPTANCE CRITERIA
$D+L+1.15\ P_a+T_t$	$P_m < 0.9\ S_y$
	P_l or $P_l + P_b \leq 1.25\ S_y$
	$P_l+P_b+P_e+Q \leq 3\ S_m$
	$P_l+P_b+P_e+Q+F$ Fatigue Analysis
$D+L+T_o+R_o$	$P_m \leq S_m$
$D+L+T_o+R_o+E$	P_l or $P_l+Pb \leq 1.5\ S_m$
	$P_l+P_b+P_e+Q \leq 3\ S_m$
	$P_l+P_b+P+Q+F$ Fatigue Analysis
$D+L+T_a+R_a+P_a+E$	$P_m \leq S_m$
$D+L+T_e+R_e+P_e+E$	P_l or $P_l + P_b \leq 1.5\ S_m$
$D+L+T_a+R_a+P_a+E$	$P_m \leq$ Larger of $1.2\ S_m$ or S_y
$D+L+T_e+R_e+P_e+E$	P_l or $P_l + P_b \leq$ Larger of
	$1.8\ S_m$ or $1.5\ S_y$
$D+L+T_a+R_a+P_a+Y_r+Y_j+Y_m+ E$	$P_m \leq 0.85$ times smaller
	of $0.7\ S_u$
	or $\left(S_y + \frac{S_u - S_y}{3}\right)$
	P_l or $P_l+P_b \leq (1.5)(0.85)$
	times smaller of $0.7\ S_\upsilon$ or
	$\left(S_y + \frac{S_u - S_y}{3}\right)$

3. Compressive Stresses

In areas of compressive stress, buckling criteria are considered in accordance with the applicable sections of NE-3000 of the ASME B&PV Code, Section III.

4. Fatigue Analysis

   The requirements for an analysis for cyclic operation is investigated in accordance with NE-3131 (d) and the referenced portions therein.

3.8.2.6 Materials, Quality Control, and Special Construction Techniques

The Containment is a steel-lined reinforced concrete structure. The airlocks, equipment hatch, and other penetrations are fabricated from the following materials in accordance with Section III of ASME-ACI 359, Proposed Standard Code for Concrete Reactor Vessels and Containments (April 1973):

1. Plate is in accordance with ASME SA-537 Class 2 OR SA-616 Grade 70, or equivalent.

2. Penetrations are in accordance with ASME SA-333 Grade 6, SA-537 Class 2, or SA-516 Grade 70, or equivalent.

   Penetration caps are in accordance with ASME SA-105, SA-350 Grade LF1 or LF2, SA-516 Grade 60,65 or 70 or SA-333 Grade 6, or equivalent.

3. Forgings including penetration forgings are in accordance with ASME SA-350 Grade LF1 or LF2 or SA-182 Type F316, or equivalent.

4. Bolting is in accordance with ASME SA-320, or equivalent.

   The materials meet all the requirements in the ASME B&PV Code, Section III, Division 1, Subsection NE, for Class MC components, including the Charpy impact test requirements.

   The QA program for fabrication and erection is in accordance with the requirements of Subsection NE of the ASME B&PV Code, Section III, Division 1.

   The full penetration butt weld between the sleeve and weld neck flange of the electrical penetration assemblies may be examined by the ultrasonic examination method plus the magnetic particle or liquid penetrant method per the special exceptions of NE-5231.1 in lieu of the radiographic examination method of NE-5211.

3.8.2.7 Testing and Inservice Inspection Requirements

Testing of Class MC components is described in Section 6.2.6. Overall Containment testing is in accordance with Subsection 3.8.1.7.

3.8.3 CONCRETE AND STEEL INTERNAL STRUCTURES OF STEEL OR CONCRETE CONTAINMENTS

3.8.3.1 Description of the Internal Structures

3.8.3.1.1 General Description

The Containment internal structures are primarily of reinforced concrete and consist of the following major elements:

1. Primary Shield Wall (Reactor Cavity)

   The primary shield wall, a heavily reinforced concrete cylinder, is situated at the approximate center of the Containment vessel, and extends up from the interior base slab to surround the reactor vessel. This reactor cavity structure provides support for the reactor vessel. The vessel supports consist of support pads and shoes which are mounted on support members within the concrete cavity structure. During normal operation, the primary shield wall provides biological shielding for maintenance inspection. Under seismic loading, this structure serves to provide seismic shear resistance and stiffens the Containment internal structure.

2. Primary Loop Compartment Walls (Steam Generator Compartment)

   The compartments are formed by the secondary shielding walls on the exterior and by the reactor and refueling canal walls on the interior. These walls extend from the interior base slab up to the operating floor. The compartment houses the steam generator, reactor coolant pumps, and the RCLs. The compartment walls provide radiation shielding, isolation of the RCS, and lateral restraint for the steam generator, pump, and pressurizer.

3. Operating Floor

   The operating floor is supported by the primary loop compartment walls and concrete columns adjacent to the containment shell which extend down to the interior base slab. The operating floor provides a working and access floor during refueling, maintenance, and repair operations. Vent areas are provided where required.

4. Refueling Cavity

   The refueling cavity provides shielded access for transport of spent fuel and new fuel between the reactor vessel and the fuel transfer penetration. It also provides shielding storage space for the reactor vessel internals during refueling or maintenance. The cavity is lined with stainless steel.

5. Interior Base Slab

   The interior base slab is placed on top of the foundation mat liner plate. This slab provides lateral and flexural restraint at the base of the primary loop compartment walls and the primary shield wall. The slab ties the primary loop compartment walls to the primary shield walls and provides a diaphragm for seismic shear distribution at the bottom

of the internal structure. It also protects the foundation mat liner from any missiles generated in the primary loop compartments and from the effects of accident temperatures.

6. Missile Shields

The primary loop compartment walls and the operating deck provide missile protection for the RCS from potential missiles outside the primary loop compartment. Conversely, the walls and interior base slab provide missile protection to the Containment and to the safeguard and auxiliary systems located outside the primary loop compartment from postulated missiles originating inside the primary loop compartments. The missile-shielding function of the primary loop compartment walls is supplemented by a control rod drive missile shield positioned over the reactor vessel head, which is designed to contain any postulated ejected control rods from the reactor vessel. The Unit 1 missile shield is removed with the reactor vessel head assembly and the Unit 2 missile shield is removed during refueling operations.

7. Intermediate Floors

Intermediate floors are provided at several elevations, including a principal floor at elevation 860 ft for miscellaneous equipment supports, access, maintenance, and similar items.

8. Removable Slabs and Walls

Removable slabs and walls are provided, where required, for maintenance access, e.g., the missile shield over the reactor and in the walls around the regenerative heat exchanger and reactor coolant drain pumps.

3.8.3.1.2 Polar Crane and Telescopic Jib Crane

A Nuclear Safety Related polar crane is provided inside the Containment, on a circular steel runway girder which is supported by brackets from the cylindrical Containment wall. The polar crane will remain stable and not become derailed when subjected to the specified load combinations (See FSAR Sections 3.8.3.3.3 and 9.1.4).

The primary load path from the crane wheels to the runway girders is maintained within allowable stress limits.

A telescopic jib crane is provided inside Containment. It is supported by a structural steel support mounted on east-west divider wall between S.G. compartments at elevation 905'-9" (see Figures 1.2-8 and 1.2-15). Two such supports are provided-one each between S.G. compartments 1 & 3 and 2 & 4. The telescopic jib crane can be located as required at either of these two supports. This crane will only be used to handle loads during plant shutdown (e.g., MODES 5, 6, and defueled). The telescopic jib crane and it's support are Non-Nuclear Safety Related, Seismic Category II.

Both the Polar Crane and Telescopic Jib Crane operating at the same time constitutes Multi-crane Operation which requires special administrative controls to reduce the likelihood of

unplanned crane interactions. These administrative controls include a qualified multi-crane coordinator located in a position of good visibility to observe both cranes.

A Containment Anti-Collision Control System is also installed on the telescopic jib crane to reduce the likelihood of unplanned crane interactions with the polar crane bridge. The function of the Programmable Logic Controller (PLC) and associated components on the telescopic jib crane is to keep the jib crane boom below the Polar Crane bridge. When an administratively controlled jib crane key is placed in the Jib Crane controls, the boom height limit is bypassed.

In addition, the Polar Crane Radio Control system will restrict Bridge movement unless a different and administratively controlled polar crane key is placed into the Polar Crane Radio Control transmitter. Operation of the trolley and hoists from the polar crane radio controls and full operation from the polar crane cab is still available under the administrative controls for multi-crane operation.

The PLC is removed from containment during power operation to protect it from radiation.

3.8.3.1.3 Supports for Reactor Pressure Vessel, Steam Generator, Reactor Coolant Pump, Pressurizer, and Loop Piping

Descriptions of the supports for the reactor pressure vessel, steam generator, reactor coolant pump, pressurizer, and loop piping are presented in Section 5.4.14.

3.8.3.1.4 Drawings

For various details of internal structures, see Figure 3.8-15.

3.8.3.2 Applicable Codes, Standards, and Specifications

3.8.3.2.1 Basic Codes

Except where specifically noted otherwise, the basic codes used for the materials, design, construction, and so forth of the Containment internal structures are the applicable sections of the following:

1. Reinforced concrete is in accordance with ACI 318-71 Building Code Requirements for Reinforced Concrete.

2. Structural steel is in accordance with AISC Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings (1969) including Supplement Numbers 1, 2, and 3 hereafter referred to as AISC Specification. Except:

   a. When supported by an engineering analysis, connections using A325 or A490 high strength bolts need not be pretensioned to the values required by AISC Specification, Table 1.23.5.

   b. Welded construction is in accordance with “Structural Welding Code, AWS D1.1-75” (except paragraph 5.23.2.4) instead of “AWS D1.1 - Rev. 1-73” as required by AISC, section 1.17.

Welder qualification for partial groove and fillet welds on tubular joints (T, K and Y connections) is in accordance with paragraph 5.23.2.4 and Table 10.5 Columns 9, 10 and 11 “Minimum Welder Qualifications for all Positions” of “Structural Welding Code, AWS D1.1-90”.

c. For prequalified joint details and base metals, “AWSD1.1 - Rev. 1-73” and later editions may be used instead of only “AWSD1.1-Rev. 1-73” as required by AISC, section 1.17.

Visual inspection of structural welds will be in accordance with Visual Weld Acceptance Criteria for Structural Welding at Nuclear Power Plants (VWAC) Revision 2 dated May 7, 1985. (SEE Appendix 3.8A)

The acceptance criteria contained in VWAC Revision 2 are applicable to all structural steel welds at CPNPP other than ASME-class structural welds.

3.8.3.2.2 Supplementary Codes

Where specifically referred to in this section on internal structures, certain portions of the ASME-ACI 359 document apply, as described in Subsection 3.8.1.2.1.

ACI 349-76, Appendix A, is used to determine stresses resulting from thermal gradients.

3.8.3.2.3 Additional Specifications and Standards

The following is a list of additional specifications and standards which are applicable to the internal structures:

1. Concrete

(Same as Subsection 3.8.1.2.3, Item 3.)

2. Steel

ASTM A 36-74	Specification for Structural Steel
ASTM A 307-74	Specification for Carbon Steel Externally and Internally Threaded Standard Fasteners
ASTM A 325-74	Specification for High-Strength Bolts for Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers
ASTM A 240-74a	Specification for Heat Resisting Chromium and Chromium Nickel Stainless Steel Plate, Sheet, and Strip for Fusion Welded Unfired Pressure Vessels
ASTM A 370-74	Methods and Definitions for Mechanical Testing of Steel Products
ASTM A 578-71b	Specification for Straight-Beam Ultrasonic Examination of Plain and Clad Steel Plates for Special Applications
ASTM A 6-74	Specification for General Requirements for Delivery of Rolled Steel Plates, Shapes, Sheet Piling, and Bars for Structural Use
RCRBSJ - 1974	Specification for Structural Joints Using ASTM A325 or A490 Bolts, AISC 1974 Edition. RCRBSJ - Research Council on Rivited and Bolted Structural Joints.

3. Reinforcing Steel

ASTM A 615-72	Specification for Deformed and Plain Billet Steel Bars for Concrete Reinforcement (Grade 60)

4. Polar Crane

Crane Manufacturer's Association of America, Inc. (CMAA) Specification No. 70

3.8.3.2.4 Summary of Principal Plant Specifications

The principal plant specifications for the internal structures are primarily those involving reinforced concrete construction. The applicable portions of the ASME-ACI 359 document, as described in Subsection 3.8.1.2.1, are included in the plant construction specifications in regard to materials, construction techniques, examination, quality control, and so forth. The following principal portions of the ASME-ACI 359 document are incorporated into the plant specifications:

1. Reinforcing Steel Specification Same as Subsection 3.8.1.2.5, Item 2.b.

2. Concrete Specifications

Same as Subsection 3.8.1.2.5, Item 2.d.

3. Cadweld Splices

Generally, large-size bars requiring Cadweld splices are not used in the internal concrete structures. Where such splices are used, the requirements indicated in Subsection 3.8.1.2.5, Item 2.c are complied with.

4. Structural Steel

The plant specification for structural steel includes the requirements in AISC Specification.

3.8.3.2.5 Applicable NRC Regulatory Guides

The following NRC Regulatory Guides are applicable to the internal structures and are complied with:

NRC Regulatory Guide 1.15	Testing of Reinforcing Bars for Category I Concrete Structures (Revision 1, 12-28- 72 of former Safety Guide 15)
NRC Regulatory Guide 1.28	Quality Assurance Program Requirements (Design and Construction) (6-7-72 of former Safety Guide 28)

3.8.3.3 Loads and Load Combinations

3.8.3.3.1 Loads

The loads and load combinations for supports which are supplied by Westinghouse are provided in Section 3.9N.1.4. The following loads are considered in the design of the internal structures of the Containment:

1. Normal Loads

Normal loads are those loads which are encountered during normal plant operation and shutdown. They include the following:

a. D = dead loads, including any permanent equipment loads, and their related moments and forces

b. L = live loads, including any movable equipment loads and other loads which vary in intensity and occurrence such as soil and hydrostatic pressures, pressure differences caused by variation in heating, cooling, and their related moments and forces

c. To = thermal effects and loads during normal operating or shutdown conditions based on the most critical transient or steady-state condition

d. Ro = pipe reactions during normal operating or shutdown conditions based on the most critical transient or steady-state condition

2. Severe Environmental Loads

Severe environmental loads are those loads that could be encountered infrequently during the plant life. This category includes the following:

Feqo = loads generated by 1/2 the SSE

= OBE

3. Extreme Environmental Loads

Extreme environmental loads are those loads which are credible but highly improbable. They include the following:

Feqs = loads generated by the SSE

4. Abnormal Loads

Abnormal loads are loads generated by a postulated high energy pipe break accident within the Containment or compartment thereof. This category includes the following:

a. Pa = pressure equivalent static load within or across a compartment generated by the postulated break, including an appropriate dynamic factor to account for the dynamic nature of the load

b. Ta = thermal loads under thermal conditions generated by the postulated break, including To

c. Ra = pipe reactions under thermal conditions generated by the postulated break, including Ro

d. Yr = equivalent static load on the structure generated by the reaction on the broken high energy pipe during the postulated break, including an appropriate dynamic factor to account for the dynamic nature of the load

e. Yj = jet impingement equivalent static load on the structure generated by the postulated break, including an appropriate dynamic factor to account for the dynamic nature of the load

f. Ym = missile impact equivalent static load on the structure generated by or during the postulated break, such as pipe whipping, including an appropriate dynamic factor to account for the dynamic nature of the load

In determining an appropriate equivalent static load for Yr, Yj, and Ym, elastoplastic behavior is assumed with appropriate ductility ratios as long as excessive deflections do not result in loss of function. For concrete structures, the ductility ratios are described in Section 3.5.3.2.

5. Other Definitions

a. For structural steel, S is the required section strength based on the elastic design methods and the allowable stresses defined in AISC Specification.

A 33-percent increase in allowable stresses for concrete and steel because of seismic or wind loadings is not permitted.

b. For concrete structures, U is the section strength required to resist design loads and is based on methods described in ACI 318-71.

c. For structural steel, Y is the section strength required to resist design loads and is based on plastic design methods described in Part 2 of AISC Specification.

3.8.3.3.2 Load Combinations and Acceptance Criteria for Internal Concrete Structures of the Containment

1. Load Combinations for Service Load Conditions

a. U = 1.4 D + 1.7 L

b. U = 1.4 D + 1.7 L + 1.9 Feqo

If thermal stresses due to To and Ro are present the following combinations also apply:

c. U = .75 (1.4 D + 1.7 L + 1.7 To + 1.7 Ro)

d. U = .75 (1.4 D + 1.7 L + 1.9 Feqo + 1.7 To + 1.7 Ro)

L is considered for its full value or its complete absence.

2. Load Combinations for Factored Load Conditions

For conditions that represent extreme environmental, abnormal, abnormal/severe environmental, and abnormal/extreme environmental conditions, respectively, the following load combinations are satisfied:

a. U = D + L + To + Ro + Feqs

b. U = D + L + Ta + Ra + 1.5 Pa

c. U = D + L + Ta + Ra + 1.25 Pa + 1.0 (Yr + Yj + Ym) + 1.25 Feqo

d. U = D + L + Ta + Ra + 1.0 Pa + 1.0 (Yr + Yj + Ym) + 1.0 Feqs

In combinations b, c, and d, the maximum values of Pa, Ta, Ra, Yj, Yr, and Ym, including an appropriate dynamic factor, are used unless a time history analysis is performed to justify otherwise. Combinations c and d and the corresponding structural acceptance criteria shall be first satisfied without Yr, Yj, and Ym. When considering these loads, local

section strength capacities may be exceeded under these concentrated loads, provided there will be no loss of function of any safety-related system.

L is considered for its full value or its complete absence.

3.8.3.3.3 Load Combinations and Acceptance Criteria for Internal Steel Structures of the Containment

1. Load Combinations for Service Load Conditions

Either the elastic working stress design methods of Part 1 of AISC Specification or the plastic design methods of Part 2 of AISC Specification are used.

a. When the elastic working stress design methods are used, the following apply:

1. $S = D + L$

2. $S = D + L + Feqo$

When thermal stresses caused by To and Ro are present the following combinations are satisfied:

3. $1.5\ S = D + L + To + Ro$

4. $1.5\ S = D + L + To + Ro + Feqo$

L is considered for its full value or its complete absence.

b. When plastic design methods are used, the following apply:

1. $Y = 1.7D + 1.7L$

2. $Y = 1.7D + 1.7L + 1.7\ Feqo$

When thermal stresses caused by To and Ro are present, the following combinations are satisfied:

3. $Y = 1.3\ D + 1.3\ L + 1.3\ To + 1.3\ Ro$

4. $Y = 1.3\ D + 1.3\ L + 1.3\ To + 1.3\ Ro + 1.3\ Feqo$

L is considered for its full value or its complete absence.

2. Load Combinations for Factored Load Conditions

a. If elastic working stress design methods are used, the following load combinations are satisfied:

1. 1.6 S = D + L + To + Ro + Feqs

2. 1.6 S = D + L + Ta + Ra + Pa

3. 1.6 S = D + L + Ta + Ra + Pa + 1.0 (Yj + Yr + Ym) + Feqo

4. 1.7 S = D + L + Ta + Ra + Pa + 1.0 (Yj + Yr + Ym) + Feqs

b. If plastic design methods are used, the following load combinations are satisfied:

1. .90 Y = D + L + To + Ro + Feqs

2. .90 Y = D + L + Ta + Ra + 1.5 Pa

3. .90 Y = D + L + Ta + Ra + 1.25 Pa
+ 1.0 (Yj + Yr + Ym) + 1.25 Feqo

4. .90 Y = D + L + Ta + Ra + 1.0 Pa
+ 1.0 (Yj + Yr + Ym) + 1.0 Feqs

In these combinations, thermal loads are neglected when they are secondary and self-limiting in nature and when the material is ductile.

In combinations shown in Items 2.a.2), 3), and 4), and in Items 2.b.2) 3), and 4), the maximum values of Pa, Ta, Ra, Yj, Yr, and Ym, including an appropriate dynamic factor, are used unless a time history analysis is performed to justify otherwise.

In determining the equivalent static load for the differential pressure Pa, the impulsive nature of the load is taken into account by considering the time history of the applied pressure and the natural frequencies of the structures to which the pressure is applied (including the secondary shield walls and operating and intermediate floors). The steel is designed so that the maximum stress for any load combination, which includes differential pressure, is less than the yield stress, thus assuring elastic behavior.

For combinations shown in Subsection 3.8.3.3.3, Items 2.a.3) and 4), and in Subsection 3.8.3.3.3, Items 2.b.3) and 4), local stresses caused by the concentrated loads Yr, Yj, and Ym may exceed the allowables when there is no loss of function of any safety-related system. Furthermore, in computing the required section strength, the plastic section modulus of steel shapes is used.

3.8.3.3.4 Variable Loads

For loads which vary, the values (within the possible range) which produce the most critical combination of loading are used in design.

3.8.3.3.5 Interrelated Loads

For loads which are interrelated as a function of time, such as accident-induced pressure and jet and thermal effects, the maximum values of these effects do not necessarily occur

simultaneously. Recognition is given to the time increments associated with these postulated failure conditions.

3.8.3.3.6 Live Loads

The live loads used in design for each category of loading are the live loads consistent with the conditions assumed for that particular category of loading.

3.8.3.3.7 Load Combinations for Localized Areas

The design load combinations which are used to examine the effects on localized areas, such as loads transferred from support structures, are the same load combinations used for the general internal structure, as described above.

3.8.3.3.8 Time-Dependent Loads

Where significant, time-dependent loads such as the effects of creep, shrinkage, and other related effects are included with dead load effects as described in Section 9.3.7 of ACI 318-71.

3.8.3.4 Design and Analysis Procedures

3.8.3.4.1 General Analysis of Internal Structure

The internal structure is analyzed and designed for all load combinations as described in Subsection 3.8.3.3. Table 3.8-1 shows critical loading combinations, type of stress and computed and corresponding allowable stresses at key location in the Containment internal structures.

1. General Analysis Procedure

   The Containment internal structure is analyzed by hand calculations and finite element methods of analysis using the ANSYS computer program (see Appendix 3.7B (A) for description of program), unless noted otherwise. The Unit 1 Steam Generator compartments are analyzed, using hand calculations and the STRUDL computer program (see Appendix 3.7B(A) for description of program.) The ANSYS program uses the stiffness method of structural analysis and contains plate elements and beam elements to represent membrane and bending behavior. The models used for the analysis included the interior base slab and the internal structure components described in Subsection 3.8.3.1, such as the primary shield wall, steam generator compartments, refueling canal walls, and intermediate floors.

   The input to these analyses consist of the geometry of the structure, the material properties, the appropriate boundary conditions, and the loadings.

   All loading combinations specified in Section 3.8.3.3.2 have been addressed in these analyses. Equivalent static loads were developed to include appropriate dynamic load factors to account for the dynamic nature of these loads (e.g., subcompartment pressurization due to postulated high-energy line pipe break accidents). Thermal effects and loadings during normal operating (To) and postulated high-energy line pipe break conditions were addressed.

The output of these analyses contain the displacements, rotations, forces, shears, moments, and stresses throughout the structure.

This output is used for the general design of the internal structure.

2. Geometry Check of Model

The finite element model of the structure consists of numerous nodes and elements. During the preparation of the finite element models, it is necessary to identify and eliminate any input errors in the coordinates of the nodes and the connectivity of the elements to the nodes. Computerized plotting is used in preparing plans, sections, and isometric views of the finite element model from which input errors are readily detected.

3. Check of Validity of Computer Program

The validity of the computer program used in the structural analysis is verified before the program is used. Appendix 3.7B(A) lists computer programs used in this project.

4. Check of Analysis Results

Results of the finite element analyses are checked before they are used for design. The checks include the following:

a. Force and moment equilibrium check.

b. Global equilibrium check.

3.8.3.4.2 Additional Analysis at Local Areas

Analysis is performed in local critical areas, such as wall openings, or areas subject to localized accident loads such as jet forces or missile forces.

3.8.3.4.3 Interaction with Nuclear Steam Supply System Equipment Supports

The seismic dynamic analysis of the major NSSS equipment supports (reactor vessel, steam generators, reactor coolant pumps, and pressurizer) takes into consideration the interaction between the equipment and supports and the internal structure. The internal structures supporting the equipment are designed for the resulting seismic loads transmitted by the equipment supports. Loads transmitted from the equipment supports to the internal structures as a result of a pipe rupture accident are also considered in the design.

3.8.3.4.4 Lateral Load Transfer at Foundation Mat

Lateral loads such as seismic forces or LOCA forces are transmitted down to the foundation mat, primarily by means of shear wall action, through the primary and secondary shield walls. At the base of these walls, these lateral loads are transferred to the interior base slab by shear keys. This interior base slab acts as a diaphragm, transferring the lateral load to the key formed by the reactor cavity pit, and from there to the foundation mat.

3.8.3.4.5 Design Variables

The general analysis performed according to Subsection 3.8.3.4.1 assumes a linear elastic response with uncracked concrete section properties. The local analysis of critical areas, described in Subsection 3.8.3.4.2, considers, for loads where significant cracking is probable, the effects of cracked section properties on the analysis. Properties of materials are known with sufficient accuracy, and assumptions made are sufficiently conservative, so that other variables need not be considered in design.

3.8.3.4.6 Temperature Effects Analysis

When subjected to service load combinations described in Subsection 3.8.3.3.2, stresses caused by thermal gradients are calculated in accordance with ACI 349-76, Appendix A and conform to allowable stresses as described in Subsection 3.8.3.5. For factored load combinations described in Subsection 3.8.3.3.2, subsection 2, thermal gradient effects are determined by analysis as recommended by ACI 349-76, Appendix A, the maximum strain in the reinforcing steel does not exceed 1.5 times the yield strain, as described in Subsection 3.8.1.4.3, Item 4.

3.8.3.4.7 Reinforced Concrete Design

1. Geometry of Reinforcing Steel

   In general, all walls and slabs are reinforced in two orthogonal directions at each face. Shear reinforcement is provided in accordance with ACI 318-71. Beams and girders are conventionally reinforced using top and bottom longitudinal bars and vertical stirrups. Most of the bars are No. 11 size or smaller, permitting the use of lapped splices. In some areas where large size No. 14 or No. 18 bars are required, Cadweld splices are used.

2. Proportioning of Reinforcing Steel

   The results of the analysis under all load combinations include moments, axial forces, and shears at each section of the walls, slabs, and beams. Sufficient reinforcing steel is provided to resist the moments, axial forces, and shears as required to satisfy the requirements of ACI 318-71.

3. Bond and Anchorage Requirements of Reinforcing Steel

   Chapter 12 of ACI 318-71, Development of Reinforcement, is complied with in determining bond and anchorage requirements or they are based on available test data.

3.8.3.4.8 Evaluation of Radiation-Generated Heat Effects

Concrete temperatures do not exceed the values indicated in the ASME- ACI 359 document, CC-3430(a) for long-term loading and CC-3430(b) for accident or other short-term loading. If required, insulation or cooling systems, or both, are provided to limit the temperature of the concrete to an acceptable level.

3.8.3.4.9 Structural Steel Design

The methods of designing structural steel components are in accordance with the AISC Specification, including design for bending moments, tension, compression, connections, buckling criteria, and so forth.

3.8.3.4.10 Polar Crane and Telescopic Jib Crane

The polar crane is designed in accordance with the criteria in CMAA Specification No. 70.

The telescopic jib crane is designed in accordance with requirements of API Spec. 2C, ASME NOG-1 and ASME NUM-1.

3.8.3.4.11 Structural Response of Missile Barriers

The methods for predicting overall structural response of the missile barriers and other seismic Category I structures, subject to missile impact, are described in Section 3.5.3.

3.8.3.4.12 Supports for Reactor Coolant System Components

The models and methods of analysis for the RCS component supports are presented in Section 3.9N.1.4.4.

3.8.3.5 Structural Acceptance Criteria

3.8.3.5.1 Functional Criteria

The Containment internal structure is designed to satisfy the following functional requirements:

1. To support (under all load conditions) and provide access to the NSSS and auxiliary and safeguard systems located within the Containment

2. To provide shielded access for spent fuel and new fuel between the reactor vessel and the fuel transfer penetration

3. To provide isolation of the Containment liner from postulated missile effects resulting from the loads associated with rupture of the RCS

4. To provide isolation of the RCS from the effects of postulated failures in systems and components located inside the Containment, but outside the primary loop compartments

5. To provide biological shielding

3.8.3.5.2 Reinforced Concrete Acceptance Criteria

The design of the reinforced concrete structures is based on an ultimate strength concept in accordance with the requirements of ACI 318-71. In this ultimate strength method of design, the factor of safety is applied to the individual loads (as load factors) rather than to the allowable stresses. The ultimate load on the reinforced concrete member is based on the factored load combinations described in Subsection 3.8.3.3.2 (for both service load combinations and factored

load combinations). The ultimate capacity of the reinforced concrete member, as defined in ACI 318-71 is based on the ultimate concrete strength (in compression or shear) and the yield strength of the reinforcing steel. The ultimate capacity of the reinforced concrete member is equal to, or greater than, the required capacity based on the factored load combinations. A further factor of safety, provided by ACI 318-71 or is that the calculated ultimate capacity of the member is reduced by a capacity reduction factor, as follows:

1. 0 = 0.90 for flexure, with or without axial tension

2. 0 = 0.90 for axial tension

3. 0 = 0.85 for shear and torsion

4. 0 = 0.75 for spirally reinforced concrete compression members

5. 0 = 0.70 for other members in compression

The magnitude of the load factors applied to each type of load varies depending on the nature of the applied load, the frequency or probability of occurrence, the accuracy in predicting the magnitude of load, and the degree of conservatism in establishing the unfactored load.

3.8.3.5.3 Thermal Gradient Effects

For factored load combinations, as described in Subsection 3.8.3.3.2, Item 2, which include thermal gradient effects, the permissible maximum strain in the reinforcing steel does not exceed 1.5 times the strain corresponding to the yield stress, as described in Subsection 3.8.3.4.6.

3.8.3.5.4 Structural Steel Allowable Stresses and Strains

1. Service Load Conditions

   For service load combinations (see Subsection 3.8.3.3.3, Item 1) the allowable steel stresses, fs, are equal to the normal allowable stresses as stated in AISC Specification, except as noted for thermal stresses in Subsection 3.8.3.3.3, Item 1.

2. Factored Load Conditions

   For factored load combinations, as indicated in Subsection 3.8.3.3.3, Item 2, the allowable steel stresses are equal to 1.60 times the normal allowable stresses, with a maximum of 0.9 x Fy in tension, compression, or bending and a maximum of 0.50 Fy in shear.

3.8.3.5.5 Earthquake Response of Interior Structure Related to the Requirements of Attached Equipment

The seismic dynamic analysis of the NSSS equipment considers the interaction between the equipment and supports and the internal structure. Deformations of the structure and equipment are checked and, when required, are limited to ensure that no loss of function of any component can occur.

3.8.3.5.6 Shear Response of Internal Structures

The concrete shear capacity, including shear reinforcement where required, is in accordance with the requirements of ACI 318-71.

3.8.3.5.7 Missile Load and Pipe Break Criteria at Local Areas

When subjected to impact loads by missiles and forces caused by a pipe rupture, localized yielding is permitted when it is demonstrated that the deflections or deformations of the structures and supports are within the ductility limits (Section 3.5.3.2) necessary to ensure that functional requirements are not impaired.

3.8.3.5.8 Criteria for Reactor Coolant System Supports

The stress criteria for the RCS supports are presented in Section 3.9N.1.4.8.

3.8.3.6 Materials, Quality Control, and Special Construction Techniques

3.8.3.6.1 Concrete

1. Materials

   a. Cement is in conformance with the requirements of ASTM C 150-74, Specification for Portland Cement, Type II.

   b. Aggregates are in conformance with the requirements of ASTM C 33-74, Specification for Concrete Aggregates.

   c. Mixing water is potable or nonpotable, but is clean and free from injurious amounts of oils, acids, alkalis, salts, and organic materials or other substances which are deleterious to concrete or steel. Tests are in accordance with the requirements of CC-2223 of the ASME-ACI 359 document.

   d. Air-entraining admixtures conform to the requirements of ASTM C 260-74, Specification for Air-Entraining Admixtures for Concrete.

   e. Chemical admixtures conform to the requirements of ASTM C 494-71, Specification for Chemical Admixtures for Concrete.

2. Concrete Strength

   Concrete, when tested in accordance with ASTM C39-72, has a minimum compressive strength of 4000 psi in 28 days.

3. Other Concrete Properties

   See Subsection 3.8.1.6.1, Item 3.

4. Selection of Concrete Mix Proportions

   See Subsection 3.8.1.6.1, Item 4.

5. Construction of Concrete

   See Subsection 3.8.1.6.1, Item 5.

6. Examination, Testing, and Other Quality Control Procedures for Concrete

   See Subsection 3.8.1.6.1, Item 6.

3.8.3.6.2 Reinforcing Steel

See Subsection 3.8.1.6.2.

3.8.3.6.3 Mechanical Butt Splices (Cadwelds)

Refer to Subsection 3.8.1.6.3 and to Appendix 1A(B).

3.8.3.6.4 Structural and Miscellaneous Steel

1. Materials

   Listed below are specifications for structural and miscellaneous steel generally used. Other ASTM, conforming materials may be used as specified by project specifications/ design drawings.

ASTM A 36-74,	Specification for Structural Steel
ASTM A 537-74a,	Specification for Pressure Vessel Plates, Heat Treated, Carbon-Manganese-Silicon
ASTM A 307-74,	Specification for Carbon Steel Externally and Internally Threaded Standard Fasteners
ASTM A 325-74,	Specification for High-Strength Bolts for Structural Steel Joints, Including Suitable Nuts and Plain Hardened Washers
ASTM A 540-70,	Specification for Alloy Steel Bolting Materials for Special Applications
ASTM A 240-74a,	Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Fusion Welded Unfired Pressure Vessels-Type 304 L

2. Fabrication and Erection of Structural Steel

   Fabrication and erection of structural and miscellaneous steel are in accordance with the requirements of the AISC Specification.

3. Examination and Testing of Structural Steel

Structural and miscellaneous steel are examined and tested in accordance with the AISC Specification and the material specifications.

4. Certification, Marking, and Identification of Materials

All materials used will be provided with one of the following certifications which will become a part of the permanent project records:

a. Certified mill test reports giving chemical and physical properties,

b. Certified Materials Test Reports (independent party testing),

c. Certificates of Compliance which specify material specification, grade, class, and heat treatment as applicable.

Marking and identification are in accordance with the material specifications.

5. Certification of Tests and Examinations

Certification of all necessary tests and examinations are obtained from the mill, fabricator, or erector, as required.

3.8.3.6.5 Effect of Radiation on Structural Materials

The effects of radiation for the life of the plant do not degrade the structural integrity of the internal structures.

3.8.3.6.6 Quality Assurance Program

The documentation and maintenance of a QA program in the construction of the internal structures are in accordance with Chapter 17. NRC Regulatory Guide 1.28 is also complied with.

3.8.3.7 Testing and Inservice Inspection Requirements

3.8.3.7.1 Stainless Steel Liner

The seam welds of the stainless steel liner of the fuel transfer canal are vacuum box tested for leaktightness prior to filling with water. In addition, the stainless steel liner is provided with a leak chase system behind the liner, which is subject to periodic inservice monitoring for leaks.

3.8.3.7.2 Testing of Internal Structure

Except for the pneumatic pressure test of the entire Containment interior, as described in Subsection 3.8.1.7, there is no testing of the interior structures, such as differential pressure testing of individual compartments.

3.8.3.7.3 Testing of Polar Crane

The polar crane is tested to 125 percent of the rated capacity, in accordance with the requirements of ANSI B 30.2.0 (referenced in CMAA Specification No. 70).

3.8.3.7.4 Inservice Inspection Requirements

There is no inservice inspection program for the internal structures.

3.8.3.7.5 Tests and Inspections for Reactor Coolant System Supports

The tests and inspections for RCS supports are presented in Section 5.4.14.

3.8.4 OTHER SEISMIC CATEGORY I STRUCTURES

Other seismic Category I structures are as listed in Section 3.2.1.1. The design bases for these structures for normal operating conditions are governed by the applicable codes and standards. These structures are also designed so that there will be no loss of function caused by a rupture of any single pipe and a SSE if that function is related to public safety. Table 3.8-1 shows critical loading combinations, type of stress and computed and corresponding allowable stresses at key locations in other Seismic Category I structures.

3.8.4.1 Description of the Structures

3.8.4.1.1 Safeguards Building

The Safeguards Building is a multistory, reinforced concrete structure. There are two stories below grade and four stories above grade. Floor systems, columns, interior walls, and exterior walls are of reinforced concrete and are designed to support the vertical loads. In addition, the reinforced concrete floor systems and walls are designed to function as diaphragms and shear walls to resist the horizontal loads.

Adequate space is provided between the Safeguards Building and adjacent structures to avoid the contact of structures during SSE-induced motions. Piping and other systems crossing from the Safeguards and Electrical Buildings to other buildings or structures are designed with sufficient flexibility to maintain their function within applicable code allowable stress limits during the SSE.

The floors below grade house the safety injection pumps, RHR pumps and coolers, Containment spray pumps and coolers, and the auxiliary feedwater pumps. Floors above grade house the emergency diesel generators, electrical switchgear, motor control centers, and CRDM controls. Equipment locations are shown on Figures 1.2-1 through 1.2-11.

3.8.4.1.2 Auxiliary Building/Electrical and Control Building

The Auxiliary Building is a multi-story, reinforced concrete structure with one story below grade and four stories above grade. Floor systems, columns, and walls are of reinforced concrete and function to support vertical and horizontal loads in the same manner as those of the Safeguards Building. As with the Safeguards Building, space is provided to avoid the contact of structures, and systems are designed to accommodate the differential movements of adjacent buildings.

The Electrical and Control Building has five internal masonry block walls at Elevation 830'-0". Masonry block walls within Seismic Category I structures shall be designed to the load combinations and acceptance criteria specified in the NRC Structural Engineering Branch Interim Criteria for Masonry Walls. Upon evaluation, four of the walls were found to be acceptable based on no adverse seismic interaction with safety-related equipment; the fifth was found to be acceptable upon modification to prevent seismic interaction with ductwork. An additional masonry block wall is located in the Service Water Intake Structure; see Section 3.8.4.1.4. General information on masonry walls and removable precast block walls are contained within Table 17A-1.

The Auxiliary Building/Electrical and Control Building houses the Control Room, battery room, compressors, ventilating equipment, waste treatment equipment, and other fluid auxiliary systems. Equipment locations are shown on Figures 1.2-1 through 1.2-11.

3.8.4.1.3 Fuel Building

The Fuel Building is a reinforced concrete structure whose principal function is to house the new fuel storage area and the two spent fuel storage pools.

Both new and spent fuel bundles are stored in stainless steel racks which are located in the spent fuel pools which are filled with borated water. The spent fuel pools have thick concrete floors and walls and are lined with stainless steel plates for leaktightness.

The Fuel Building has an overhead electric crane capable of handling the fuel shipping cask. The crane is located so that it does not pass over either of the spent fuel pools. In addition, interlocks are provided to prevent movement of the shipping cask over the new fuel storage area. A fuel handling crane is mounted on the operating floor to transport new and spent fuel assemblies.

For the Fuel Building arrangement and equipment locations, see Figures 1.2-1 through 1.2-11.

3.8.4.1.4 Service Water Intake Structure

The Service Water Intake Structure is a reinforced concrete building located at the SSI. It houses the service water pumps and fire pumps and is equipped with trash racks, travelling screens, and screen wash pumps.

The base mat, walls, and floors are of reinforced concrete. Floor systems and walls are designed to support both vertical and horizontal loads, functioning as diaphragms and shear walls for horizontal loads.

Seismic loads used in the design of the Service Water Intake Structure include the hydrodynamic loads caused by seismic effects from the water contained in the SSI. The safety equipment contained within the building is suitably protected from tornado wind loads and tornado-generated missiles.

The Service Water Intake Structure has one internal masonry block wall. Conduits and an MCC are located in its vicinity, but the conduits and MCC were evaluated and found to provide no safety-related function. Therefore, the wall construction was found to be acceptable. General information on masonry walls and removable precast block walls is contained within Table 17A-1.

3.8.4.1.5 Drawings

For various details of other seismic Category I structures, see Figures 3.8-1 through 3.8-4 and Figure 3.8-16.

3.8.4.1.6 Outdoor Seismic Category I Tanks (Refueling Water Storage, Condensate Storage, and Reactor Makeup Water Storage)

The outdoor seismic Category I tanks are reinforced concrete structures, cylindrical in shape, with stainless steel liners to provide leaktightness and prevent absorption of radioactive material by the concrete (Refueling Water Storage Tank (RWST) only).

REFUELING WATER STORAGE AND CONDENSATE STORAGE TANKS:

Outside diameter of wall	50'-0"
Outside diameter of mat	53'-0"
Concrete wall thickness	2'-6"
Concrete mat thickness	5'-0"
Concrete roof thickness	1'-9"
Total height	54'-6"

REACTOR MAKE UP WATER STORAGE TANK

Outside diameter of wall	30'-0"
Outside diameter of mat	33'-0"
Concrete wall thickness	2'-6"
Concrete mat thickness	4'-0"
Concrete roof thickness	1'-9"
Total height	39'-6"

The tanks are designed to withstand all credible loadings and to maintain their integrity during operation. These loadings include both normal operating loads such as structure weight, hydrostatic pressure of the contained fluid, live loads on the roof, thermal loads and environmental loads such as the 1/2 SSE, SSE, normal wind and tornados (wind, differential pressure and missiles), and hydrodynamic forces caused by seismic effects on the contained fluid in accordance with methods as shown in Reference 21.

The load combinations given in Subsection 3.8.4.3 are used for the design of the structures, using design methods and strength requirements in accordance with ACI 318-71. Flexural

tensile cracking is permitted but is controlled by reinforcing steel. A minimum of 0.25 percent reinforcing steel is provided in the tank walls in both directions, vertical and hoop.

The structural analysis of the tanks is performed using the finite difference method. The SHELL-1 computer program is used for the analysis of axisymmetric static loads and nonaxisymmetric loads caused by wind, tornados, and earthquakes. This program is discussed in Section 3.7B(A). The motion of the contained fluid during an earthquake is accounted for by incorporating fluid masses into the mathematical model of the tank. The nonaxisymmetrical cases of wind, tornado, and seismic load (individually applied) are superimposed on the axisymmetric load cases to produce the highest design values, without regard to the direction of the nonaxisymmetric loads.

3.8.4.1.7 Concrete Barriers Provided for Category I Structures

In general, reinforced concrete external roofs and walls of seismic Category I structures form barriers against tornado generated missiles.

The compressive strength of the concrete used in the Category I structures which must resist the effects of postulated tornado winds and missiles, f'c is 4000 psi at 28 days.

Reinforcement steel is provided at each face in both directions with typical wall reinforcement #8 @ 8 each face each way (p = 0.004 E.F., E.W.) and typical roof reinforcement #10 @ 10 each face, each way (p = 0.006 E.F., E.W.).

3.8.4.2 Applicable Codes, Standards, and Specifications

Unless otherwise indicated, the design and construction of other seismic Category I structures are based upon the appropriate sections of the applicable codes, standards, specifications, recommendations, and references listed in Subsection 3.8.3.2. For anchorage requirements of reinforcing steel see Section 3.8.3.4.7(3).

3.8.4.3 Loads and Load Combinations

3.8.4.3.1 Loads

The following loads are considered in the design of other seismic Category I structures:

1. Normal Loads

   Normal loads are those which are encountered during normal plant operation and shutdown. They include the following:

   a. D = dead loads and their related moments and forces, including any permanent equipment loads

b. L = live loads and their related moments and forces, including any movable equipment loads and other loads which vary with intensity and occurrence such as soil and hydrostatic pressures and pressure differences caused by variation in heating and cooling and outside atmospheric changes. Hydrostatic pressures are included in this category. Hydrostatic loads from the water contained in the spent fuel pools and transfer canal may be treated as dead load (D) since they are maintained filled and the water level is maintained within specified tolerances. Ground water level is discussed in Subsection 3.8.5.1.5

c. To = thermal effects and loads during normal operating or shutdown conditions, based on the most critical transient or steady-state condition

d. Ro = pipe reactions during normal operating or shutdown conditions, based on the most critical transient or steady-state condition

2. Severe Environmental Loads

Severe environmental loads are those loads that could be encountered infrequently during the plant life. They include the following:

a. Feqo = loads generated by half the SSE = OBE

b. W = loads generated by the design wind specified for the plant

3. Extreme Environmental Loads

Extreme environmental loads are those loads which are credible but highly improbable. They include the following:

a. Feqs = loads generated by the SSE

b. Wt = loads generated by the design tornado specified for the plant: the loads include those caused by the tornado wind pressure, tornado-created differential pressures, and tornado-generated missiles.

c. Fd = Impact effects and loads resulting from a fuel handling accident in the Spent Fuel Pool.

4. Abnormal Loads

Abnormal loads are those loads generated by a postulated high-energy pipe break accident within a building or compartment thereof, or both. They include the following:

a. Pa = pressure-equivalent static load within or across a compartment or building, or both, generated by the postulated break; it includes an appropriate dynamic factor to account for the dynamic nature of the load

b. Ta = thermal loads under thermal conditions generated by the postulated break and including To

c. Ra = pipe reactions under thermal conditions generated by the postulated break and including Ro

d. Yr = equivalent static load on the structure generated by the reaction on the broken high-energy pipe during the postulated break; it includes an appropriate dynamic factor to account for the dynamic nature of the load.

e. Yj = jet impingement equivalent static load on a structure generated by the postulated break; it includes an appropriate dynamic factor to account for the dynamic nature of the load.

f. Ym = Missile impact equivalent static load on a structure generated by or during the postulated break, e.g., pipe whipping; it includes an appropriate dynamic factor to account for the dynamic nature of the load

For evaluation of the Fuel Building Spent Fuel Pool walls and slabs the effects of gamma heating shall be included along with To and Ta.

In determining an appropriate equivalent static load for Yr, Yj, and Ym, elastoplastic behavior is assumed with appropriate ductility ratios as long as excessive deflections do not result in loss of function.

5. Other Definitions

a. For structural steel, S is the required section strength based on the elastic design methods and the allowable stresses defined in Part 1 of AISC Specification.

The 33-percent increase in allowable stresses for concrete and steel because of seismic or wind loadings is not permitted.

b. For concrete structures, U is the section strength required to resist design loads based on methods described in ACI 318-71.

c. For structural steel, Y is the section strength required to resist design loads based on plastic design methods described in Part 2 of the AISC Specification.

3.8.4.3.2 Load Combinations and Acceptance Criteria for Other Seismic Category I Concrete Structures

1. Load Combinations for Service Load Conditions

a. U = 1.4 D + 1.7 L

b. U = 1.4 D + 1.7 L + 1.9 Feqo

c. U = 1.4 D + 1.7 L + 1.7 W

If thermal stresses due to To and Ro are present, the following also apply:

d. U = .75 (1.4 D + 1.7 L + 1.7 To + 1.7 Ro)

e. U = .75 (1.4 D + 1.7 L + 1.9 Feqo + 1.7 To + 1.7 Ro)

f. U = .75 (1.4 D + 1.7 L + 1.7 W + 1.7 To + 1.7 Ro)

L is considered for its full value or for its complete absence, and the following combinations are also satisfied:

g. U = 1.2 D + 1.9 Feqo

h. U = 1.2 D + 1.7 W

2. Load Combinations for Factored Load Conditions

For these conditions, which represent extreme environmental, abnormal environmental, abnormal/severe environmental, and abnormal/extreme environmental conditions, respectively, the following load combinations are satisfied:

a. U = D + L + To + Ro + Feqs

b. U = D + L + To + Ro + Wt

c. U = D + L + Ta + Ra + 1.5 Pa

d. U = D + L + Ta + Ra + 1.25 Pa
+ 1.0 (Yr + Yj + Ym) + 1.25 Feqo

e. U = D + L + Ta + Ra + 1.0 Pa
+ 1.0 (Yr + Yj + Ym) + 1.0 Feqs

See Section 3.6B.1.2.3 “Structural Design Margins” for additional requirements in the MS/FW penetration areas inside the Safeguards Building.

For the evaluation of local bearing effects in the spent fuel pools resulting from postulated fuel handling accidents, the following load combination shall be used:

f. U^a = D + L + To + Fd

In combinations shown in Items c, d, and e, the maximum values of Pa, Ta, Ra, Yj, Yr, and Ym, including an appropriate dynamic factor, are used, unless a time-history analysis is performed to justify otherwise. Combinations (b), (d), and (e) and the corresponding structural acceptance criteria should be satisfied first without the tornado missile load in (b) and without Yr, Ym, Yj, in

a. U is determined per the tri-axial compressive strength methodology described in NUREG/CR-6608

(d) and (e). For combinations shown in Items b, d and e, local stresses caused by the concentrated loads Wt, Yr, Yj, and Ym may exceed the allowables when there is no loss of function of any safety-related system.

L is considered for its full value or for its absence.

3.8.4.3.3 Load Combinations and Acceptance Criteria for other Seismic Category I Steel Structures

1. Load Combinations for Service Load Conditions

The AISC Specification is used for either the elastic working stress design methods of Part 1 or the plastic design methods of Part 2.

a. If the elastic working stress design methods are used, the following combinations apply:

1. S = D + L

2. S = D + L + Feqo

3. S = D + L + W

If thermal stresses due to To and Ro are present, the following combinations are satisfied:

4. 1.5 S = D + L + To + Ro

5. 1.5 S = D + L + To + Ro + Feqo

6. 1.5 S = D + L + To + Ro + W

L is considered for its full value or for its absence.

b. If plastic design methods are used, the following apply:

1. Y = 1.7 D + 1.7 L

2. Y = 1.7 D + 1.7 L + 1.7 Feqo

3. Y = 1.7 D + 1.7 L + 1.7 W

If thermal stresses due to To and Ro are present, the following also apply:

1a. Y = 1.3 D + 1.3 L + 1.3 To + 1.3 Ro

2a. Y = 1.3 D + 1.3 L + 1.3 Feqo + 1.3 To + 1.3 Ro

3a. Y = 1.3 D + 1.3 L + 1.3 W + 1.3 To + 1.3 Ro

L is considered for its full value or for its complete absence.

2. Load Combinations for Factored Load Conditions

a. If elastic working stress design methods are used, the following load combinations are satisfied:

1. 1.6 S = D + L + To + Ro + Feqs

2. 1.6 S = D + L + To + Ro + Wt

3. 1.6 S = D + L + Ta + Ra + Pa

4. 1.6 S = D + L + Ta + Ra +Pa
+ 1.0 (Yj + Yr + Ym) + Feqo

5. 1.7 S = D + L + Ta + Ra + Pa
+ 1.0 (Yj + Yr + Ym) + Feqs

b. If plastic design methods are used, the following load combinations are satisfied:

1. .90 Y = D + L + To + Ro + Feqs

2. .90 Y = D + L + To + Ro + Wt

3. .90 Y = D + L + Ta + Ra + 1.5 Pa

4. .90 Y = D + L + Ta + Ra + 1.25 Pa
+ 1.0 (Yj + Yr + Ym) + 1.25 Feqo

5. .90 Y = D + L + Ta + Ra + 1.0 Pa
+ 1.0 (Yj + Yr + Ym) + 1.0 Feqs

In these combinations, thermal loads are neglected when they are secondary and self-limiting in nature and when the material is ductile.

See Section 3.6B.1.2.3 “Structural Design Margins” for additional requirements in the MS/FW penetration areas inside the Safeguards Building.

In combinations shown in Items 2.a.3), 4), and 5), and in Items 2.b.3), 4), and 5) the maximum values of Pa, Ta, Ra, Yj, Yr, and Ym, including an appropriate dynamic factor, are used, unless a time history analysis is performed to justify otherwise. For combinations shown in Items 2.b.2), 4), and 5), local stresses caused by concentrated loads Yr, Yj, Ym, and Wt may exceed the allowables when there is no loss of function of any safety-related system. Furthermore, in computing the required section strength, the plastic section modulus of steel shapes is used.

3.8.4.3.4 Variable Loads

For loads which may vary, the values (within the possible range) which produce the most critical combination of loading are used in design.

3.8.4.3.5 Time Dependent Loads

Time dependent loads, such as the effects of creep, shrinkage, and other related effects, are included with dead load effects, as described in Section 9.3.7 of ACI 318-71 building code, where such loads affect design.

3.8.4.4 Design and Analysis Procedures

The static analysis of the other Seismic Category I structures is performed utilizing a combination of hand calculations and computerized analysis to determine the distribution of member forces and moments. These analyses use conventional methods of frame, shell, or shear wall analysis depending on the type of structural element under consideration.

Base mats of the other Seismic Category I structures are analyzed utilizing either computerized finite element analyses or hand calculations to determine the distribution of forces and moments within the mats. The mat analyses account for the stiffness of the supporting rock and do not consider any resistance to uplift from the rock (i.e., the interface between the mat and rock can not transfer any tension forces). A range of rock stiffness is considered in the analysis to account for uncertainties in the determination of the rock properties.

3.8.4.5 Structural Acceptance Criteria

3.8.4.5.1 General Criteria

The structural design criteria for other seismic Category I structures are established to provide structures that withstand all normal operating loads and also the extreme environmental loads of an earthquake or a tornado. The structures are designed to meet the requirements of the codes and standards listed in Subsection 3.8.3.2. The design of reinforced concrete structures conforms to the requirements of ACI 318-71, and structural steel design is in accord with AISC Specification.

The structural systems of these other seismic Category I buildings, except the outdoor seismic Category I tanks, consist of reinforced concrete floor slabs, beams, girders, walls, columns, and raft type foundation mats. Floor systems and walls are designed for vertical and lateral loads. Seismic, tornado, and other lateral loads applied to the total structure are resisted by the diaphragm action of the floors and shear walls and are transmitted to the foundation mats. Columns, because of their relatively low lateral stiffness, are not assumed to participate in resisting these lateral loads.

The seismic Category I structures other than the Containment are separated from each other and from the Containment to prevent contact caused by earthquake-induced motion. The widths of separations are determined from the deflections calculated in the dynamic analyses of the various structures.

The criteria for the outdoor seismic Category I tanks are described in Subsection 3.8.4.1.6.

3.8.4.5.2 Reinforced Concrete Acceptance Criteria

The design of the reinforced concrete structures is based on an ultimate strength concept, in accordance with the requirements of ACI 318-71. In this ultimate strength method of design, as

compared to previous working stress methods of design, the factor of safety is applied to the individual loads (as load factors) rather than to the allowable stresses. The ultimate load on the reinforced concrete member is based on the factored load combinations, as described in Subsection 3.8.4.3.2, Item 2. The ultimate capacity of the reinforced concrete member, as defined in ACI 318-71, is based on the ultimate concrete strength (in compression or shear) and on the yield strength of the reinforcing steel. The ultimate capacity of the reinforced concrete member is equal to or greater than the capacity required, based on the factored load combinations. A further factor of safety is provided by ACI 318-71 in that the calculated ultimate capacity of the member is reduced by a capacity reduction factor, as indicated in Subsection 3.8.3.5.2.

The magnitude of the load factors applied to each type of load varies, depending on the factors discussed in Subsection 3.8.3.5.2.

3.8.4.5.3 Missile Load and Pipe Break Criteria at Local Areas

For local areas subjected to loads, such as missiles, and to forces caused by pipe rupture, localized yielding is permitted when the deflections or deformations of the structures and supports are within the (ductility) limits (Section 3.5.3.2) necessary to ensure that functional requirements are not impaired.

3.8.4.5.4 Bracket or Corbel Criteria at Local Areas

The fuel building crane corbel supports at elevation 831 are designed in accordance with the PCI Design Handbook Second Edition 1978.

3.8.4.6 Materials, Quality Control, and Special Construction Techniques

The materials and QC procedures used in the construction of other seismic Category I structures are as discussed in Subsection 3.8.3.6. No special construction techniques are required for these structures.

3.8.4.7 Testing and Inservice Inspection Requirements

With the exception of the stainless steel liners for the spent fuel pool and the outdoor seismic Category I tanks, no special testing of the completed structure or inservice inspection is required for seismic Category I structures other than the Containment. The seam welds of the stainless steel liners are tested for leaktightness prior to filling with water by the vacuum box method. In addition, the spent fuel pools are provided with a leak-chase system behind their liners which is subject to periodic inservice monitoring for leaks.

3.8.5 FOUNDATIONS

3.8.5.1 Description of the Foundations

3.8.5.1.1 Foundation Mat and Supports for Containment and Internal Structure

The foundation mat for the Containment and internal structure is described in Subsection 3.8.1.1.1. It is essentially a flat mat, approximately 12 ft thick, with a reactor cavity pit projection. The foundation mat is covered with a steel liner that is an integral part of the

Containment liner system, which is then covered with a 30-in.- thick interior base slab, as described in Subsection 3.8.3.1.1. The arrangement of reinforcing steel at the junction of the bottom of the Containment wall and the top of the foundation mat is shown on Figure 3.8-10. A discussion of the lateral load transfer to the mat from the vertical members of the internal structure is provided in Subsection 3.8.3.4.4. The geometry of the reinforcing steel in the mat is described in Subsection 3.8.1.1.4. The principal reinforcing in the mat consists of No. 18 bars, cadwelded at splices.

The steel supports for the steam generators, the reactor pressure vessel, the reactor coolant pump, and the pressurizer are discussed in Section 5.4.14 and shown on Figures 5.5-1 to 5.5-5. Embedded steel, suitably anchored, is provided in the concrete internal structure to support the NSSS equipment supports. Typical details, illustrating the manner in which the embedded steel used to support NSSS equipment is anchored to the internal structures, are shown on Figures 3.8-17 and 3.8-18.

Reinforced concrete supports are approximately equally spaced around the periphery of the reactor vessel. Initial drawings for the Unit 2 reactor pressure vessel support were prepared with the supports oriented as a mirror image of the Unit 1 supports. The reactor vessel supplied for Unit 2, however, is a duplicate for the Unit 1 reactor vessel. As a result, the Unit 2 reactor vessel supports were misoriented by about 45 degrees from their proper location. See Figure 3.8-24. Since concrete had been poured to elevation 819'-0", vertical holes were drilled, anchor bolts installed in the holes and then grouted as a substitute for the anchor bolts with anchor plates, which had been improperly located. In place of the #18 horizontal rebars used in the wrong location, #11 rebars of equal total area were installed and grouted in drilled holes as shown in Figure 3.8-25. The procedure and materials used in the installation of the rebars in drilled holes in concrete supports which were then grouted is similar to the procedures used in the drilling and grouting described in Reference [23]. Design and analysis procedures for these supports is discussed in Section 3.8.5.4.1.

3.8.5.1.2 Foundation For Other Seismic Category I Structures

The foundations for other seismic Category I structures are generally flat slabs that vary in thickness from approximately 3 to 6 ft, depending on the requirements for shear and bending moment. Reinforcing steel is provided in two mutually perpendicular directions, at both the top and bottom faces. Shear reinforcement is provided where required by the provisions of ACI 318-71. The principal reinforcing in the foundations for other seismic Category I structures consists of No. 11 bars, lap spliced in accordance with ACI 318-71. For transfer of base shear, see Subsection 3.8.5.3.3.

3.8.5.1.3 Relationship Between Various Seismic Category I Structures

For the physical relationship between the various foundations, see Figures 3.8-1 through 3.8-4. Sufficient space, not less than one inch, is provided between foundations to prevent contact during SSE. However, during construction, it was observed that some debris had fallen into the gap between foundation mats, which could not be removed. Since all the foundations are on bed rock and are very stiff, it was determined that the presence of debris in the foundation gaps will not result in significant load transfer from one building to another such that the seismic response is altered during SSE.

3.8.5.1.4 Underlying Foundation Material

All seismic Category I foundations are founded on competent rock, as described in Sections 2.5 and 3.7.

3.8.5.1.5 Effects of Ground Water

Although some of the Category I foundations are below the PMF level of the reservoir (elevation 789.7 ft), ground water is not expected to reach higher than elevation 775.0 ft because of the impermeable nature of the rock. See Section 2.4.13.1.3 for a discussion of the level of permeability of the Glen Rose Formation, and Section 2.4.13.1.5 for a discussion of the onsite water table.

Elevation 775.0 ft. is only slightly above the deepest of the seismic Category I foundations (Safeguards Building, floor elevation 773'-0). In the event of in-leakage into the building, the small amount will be collected by the floor drains and routed to the floor drain sumps. Each of the two floor drain sumps is equipped with duplex sump pumps cable of pumping 50 gpm of fluid per pump. Therefore waterproofing is not required. Also refer to Section 9.3.3.

3.8.5.1.6 Cathodic Protection

Substrata and water resistivity tests have been performed, and where resistivity is less than approximately 5000 ohms cm, a suitable cathodic protection system is provided to protect buried and submerged metallic structures such as piping, fuel oil tanks, and intake structures.

3.8.5.1.7 Drawings

For various details of foundations, see Figures 3.8-1 through 3.8-4 and Figure 3.8-12.

3.8.5.2 Applicable Codes, Standards, and Specifications

The applicable codes, standards, and specifications for foundations and concrete supports are listed in the following applicable sections:

1. For Containment foundation and internal structure, see Subsections 3.8.1.2 and 3.8.3.2, respectively.

2. For other seismic Category I structures foundations, see Subsection 3.8.4.2.

3. For the additional standard ACI 436, see Subsection 3.8.5.4.2, Item 2.

3.8.5.3 Loads and Load Combinations

3.8.5.3.1 Foundation Design Loads and Combinations

The loads and load combinations used in the design of the foundations are the same as the loads and load combinations used in the design of the superstructures. Following is a list of the referenced sections that apply:

1. For Containment foundation and internal structure, see Subsections 3.8.1.3 and 3.8.3.3.

2. For other seismic Category I structures' foundations, see Subsection 3.8.4.3.

3.8.5.3.2 Various Effects at Foundations

1. The analysis performed for each structure considers the effect of torsional moments.

2. The factor of safety against overturning, for each structure, is greater than 1.1 when considering any load combinations (see Subsection 3.8.5.5.3).

3. The base shear is transferred from the foundation to rock through bond and surface friction and is not a problem for structures founded on competent rock, such as that which exists at the Comanche Peak site.

3.8.5.3.3 Loads Transferred from Supported Components to Foundations

The load combinations considered in the determination of the total loads transferred from supported components, such as the NSSS equipment, to the foundations are the same load combinations as those described in Subsection 3.8.5.3.1. For additional discussion, see Subsection 3.8.3.4.3.

3.8.5.4 Design and Analysis Procedures

3.8.5.4.1 Foundation and Supports for Containment and Internal Structure

The analysis of the foundation mat for the Containment and internal structure is described in Subsection 3.8.1.4.1, Item 1. The output of this analysis includes the displacements, rotations, forces, shears, moments, and stresses which are used for the design of the foundation mat.

1. Determination of Rock Contact Area Under the Foundation

   For load combinations which include the overturning effects of earthquake or tornado forces, some lift-off may occur, resulting in a condition where only a portion of the foundation is in contact with the rock. The area of contact between the bottom surface of the foundation and the rock is dependent on the resultant forces applied to the entire structure, based on the load combination being considered. The rock reaction is simulated by attaching appropriate springs to the nodes on the foundation mat that are within the area of contact. The predicted amount of foundation contact area affects the results of the analysis described in Subsection 3.8.1.4.1, Item 1.

   The following procedures are used to determine the contact area. Before proceeding with the analysis of the structure under a combined loading, which includes earthquake loads or tornado loads, an area of rock contact is postulated. This postulated contact area is determined by first assuming that the structure is rigid with respect to the foundation springs. Then, by considering equilibrium of the applied force and the rock-spring reactions, the postulated contact area is determined by trial and error through a systematic search process. After the analysis based on the postulated contact area is performed, the resulting contact area is checked. If the postulated and resulting contact areas are significantly different, a new postulated contact area is determined based on

analysis results, and the structure is reanalyzed. The checking cycle is terminated when the postulated contact area and the resulting contact area of the same analysis converge within a tolerable limit (approximately 5 percent difference).

2. NSSS Equipment Concrete Supports

The concrete supports for the NSSS equipment are designed for all the loading combinations (listed in Subsection 3.8.3.3) which include seismic and blowdown effects. A discussion of the blowdown effects as a result of a LOCA is contained in Sections 3.6 and 3.9. The dynamic analysis under seismic loading is described in Section 3.7.

Shear resistance capability through the use of rebar reinforcement (described in Section 3.8.5.1.1) is assumed to be developed by means of shear friction. The shear friction design procedure is described in ACI 318-77 [Reference 24]. Effectiveness of the grout used in the installation of the rebars in drilled holes in concrete supports is confirmed by Reference [23].

3.8.5.4.2 Foundations for Other Seismic Category I Structures

1. The analysis of foundations for other seismic Category I structures is described in Subsection 3.8.4.4. The output of the analysis includes the displacements, rotations, forces, shears, moments, and stresses in the foundation mat which are used for the design of the mat.

The determination of the rock contact area under the foundation when the structure is subjected to lateral forces, such as earthquake or tornado, is as described in Subsection 3.8.5.4.1, Item 1.

2. For the other seismic Category I structures that are not analyzed by using computer programs, such as the Fuel Building and the Service Water Intake Structure (see Subsection 3.8.4.4), the foundations are designed using techniques similar to those described in the report of ACI Committee 436, Suggested Design Procedures for Combined Footings and Mats, ACI Manual of Concrete Practice, Part 2, 1968.

3.8.5.4.3 Descriptions of Computer Programs

For the descriptions of the computer programs used in the referenced Subsections of 3.8.5, see Appendix 3.7B(A).

3.8.5.4.4 Computer Program Check

For the validity of the computer programs, the analytical models, and the results of the analyses, see Subsection 3.8.3.4.1.

3.8.5.4.5 Design Variables

For a general discussion of design variables, see Subsection 3.8.3.4.5. In addition, a minimum and maximum value of rock springs are considered for foundation design to account for a realistic range of probable values. Upper and lower bound estimates of the rock spring values are based on possible variations in measuring the material proportion of the rock. The variations

estimated on the basis of past experience are five percent in the mass density of the rock and 15 percent in the measurement of shear wave velocity.

3.8.5.4.6 Lateral Load Transfer to Soil

All seismic Category I structures are founded on bedrock. The transfer of lateral loads from the bottom of the foundation to the rock is primarily through bond and surface friction between concrete and rock. In addition, for the Containment foundation mat, the reactor cavity pit provides keying action.

3.8.5.4.7 Effects of Settlement

The nature of the bedrock on which all the seismic Category I structures are founded is such that both the gross and differential settlements are insignificant.

3.8.5.5 Structural Acceptance Criteria

3.8.5.5.1 Allowable Stresses and Strains

The allowable stresses and strains for the foundations are essentially the same as the allowables for the superstructures which are supported thereon, considering the same load combinations. The applicable Subsections are 3.8.1.5 and 3.8.3.5 (for foundation for Containment and internal structure) and 3.8.4.5 (for foundation for other seismic Category I structures).

3.8.5.5.2 Foundation Settlements

Considering the nature of the rock on which all seismic Category I structures are founded, gross and differential settlements are insignificant.

3.8.5.5.3 Factors of Safety Against Overturning, Sliding, and Floatation

The factors of safety against overturning, sliding, and floatation, when considering all load combinations including the SSE or tornado loading, are as follows:

	Minimum Factors of Safety		
Load Combination	Overturning	Sliding	Floatation
D + H + Feqo	1.5	1.5	-
D + H + W	1.5	1.5	-
D + H + Feqs	1.1	1.1	-
D + H + W_t	1.1	1.1	-
D + F'	-	-	1.5

D, Feqo, Feqs, W, and W_t are loads defined in Subsection 3.8.4.3. H is lateral earth pressure load. F' is the buoyant force due to the PMF. For earthquake loads Feqo and Feqs, three components of earthquakes are combined as described in Section 3.7B.

3.8.5.5.4 Effects of Rock Structure Interaction

The rock structure interaction effects are considered in the seismic dynamic analysis discussed in Section 3.7B. In regard to the static analysis of the foundations, the interaction between the foundation and the rock is reflected in the use of appropriate rock springs. The maximum pressure of the foundation on the rock is determined for each load combination, and is within the allowable limits as defined in Section 2.5.4.

3.8.5.6 Materials, Quality Control, and Special Construction Techniques

The materials, quality control, and special construction techniques for the foundations are essentially the same as those required for the structures which are supported thereon. The following sections apply:

1. For Containment foundation Subsection 3.8.1.6

2. For other seismic Category I structures foundations Subsection 3.8.4.6

The QC procedures for the concrete, including the type of tests required, the frequency and location of sampling, and the test requirements are described in the referenced sections cited in Section 3.8.1.6.1.

3.8.5.7 Testing and Inservice Inspection Requirements

1. The structural acceptance test of the Containment, as described in Subsection 3.8.1.7.1, subjects the Containment foundation to a test pressure load which demonstrates that the foundation responds satisfactorily to the maximum internal pressure.

2. Leak chase channels for leaktightness examination are provided at seams in the Containment mat liner that are inaccessible for other means of inspection.

3. Other than a visual inspection after construction, no other special testing or inservice inspection is required for seismic Category I foundations.

REFERENCES

1. ASME-ACI 359 document, Proposed Standard Code for Concrete Reactor Vessels And Containments, issued for trial use and comments, April 1973.

2. ACI Committee 349, Criteria For Reinforced Concrete Nuclear Power Containment Structures, ACI Journal, January 1972.

3. 10 CFR Part 50, Appendix J, Primary Reactor Containment Leakage Testing for Water-Cooled Power Reactors, February 5, 1973.

4. ANSI N45.4-1972, Leakage-Rate Testing of Containment Structures For Nuclear Reactors, March 16, 1972.

5. NRC Regulatory Guide 1.10, Mechanical (Caldweld) Splices in Reinforcing Bars of Category I Concrete Structures, Revision I, January 2, 1973, U.S. Nuclear Regulatory Commission.

6. NRC Regulatory Guide 1.15, Testing of Reinforcing Bars for Category I Concrete Structures, Revision 1, December 28, 1972, U.S. Nuclear Regulatory Commission.

7. NRC Regulatory Guide 1.18, Structural Acceptance Test for Concrete Primary Reactor Containments, Revision 1, December 28, 1972, U.S. Nuclear Regulatory Commission.

8. NRC Regulatory Guide 1.19, Nondestructive Examination of Primary Containment Liner Welds, Revision 1, August 11, 1972, U.S. Nuclear Regulatory Commission.

9. NRC Regulatory Guide 1.28, Quality Assurance Program Requirements (Design and Construction), June 7, 1972, U.S. Nuclear Regulatory Commission.

10. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 1, August 1973, U.S. Nuclear Regulatory Commission.

11. ASME Boiler and Pressure Vessel Code For Nuclear Power Plant Components, Section III, Division I, Subsection NE, 1971.

12. ACI 318-71 Building Code Requirements For Reinforced Concrete, 1971.

13. AISC Specification For the Design, Fabrication and Erection of Structural Steel For Buildings, 1969 including Supplement Numbers 1, 2, and 3.

14. Roark, R.J., Formulas For Stress and Strain, McGraw-Hill, 4th ed., 1965.

15. Timoshenko, S., Theory of Plates and Shells, McGraw-Hill, 1940.

16. NRC letter, Visual Weld Acceptance Criteria for Structural Welding at Nuclear Power Plants (VWAC), Revision 2 (letter from James P. Knight of NRC to Douglas E. Dutton of Nuclear Construction Group dated June 26, 1985.

17. ACI 349-76, Code Requirements for Nuclear Safety Related Concrete Structures, Appendix A, 1976.

18. PCI Design Handbook, precast and prestressed concrete Second Edition 1978.

19. AWS D1.1-75 Structural Welding Code requirements for Structural Welded Construction.

20. AWS D1.1- Rev. 1-73 and later edition for prequalified joint details and base materials.

21. U. S. Atomic Energy Comm., “Nuclear Reactors and Earthquakes”, TID-7024, Office of Technical Service, Wash., D. C. 1963 p. 183-195 and 367-390.

22. AWS D1.1-90 “Structural Welding Code” for Welder Qualifications of partial penetration groove and fillet welds on tubular T, K and Y connections.

23. Report of pull-out tests and grouped rebars, Robert W. Hunt Company, December 1, 1997.

24. ACI 318-77, Building Code Requirements for Reinforced Concrete, 1977.

25. 10CFR Part 50, Appendix J, Primary Reactor Containment Leakage Testing for Water-Cooled Power Reactors, Option B, Performance Based Requirements, effective October 26, 1995.

TABLE 3.8-1
SUMMARY TABLE FOR CATEGORY 'I' STRUCTURES

(Sheet 1 of 3)

TYPE OF STRUCTURE	KEY LOCATIONS[a]	CRITICAL LOADING COMBINATION	TYPE OF STRESS	ACTUAL STRESS (KSi)	ALLOW STRESS (KSi)
CONTAINMENT BLDG.	Bottom of Containment Wall @ EL.805'-6"	D+L+1.25 Pa+To+Ta+1.25E+ Ra+Yr+Yj+Ym	Tension	49.3	54.0
			Shear	13.6	50.0[b]
	Top of Containment Wall @ EL.1000'-6"	D+L+1.25 Pa+To+Ta+1.25E+ Ra+Yr+Yj+Ym	Tension	54.0	54.0
	Inside Face of Equipment Hatch at = 225° for Unit 1, 315° for Unit 2	D+L+1.5 Pa+To+Ta+Ra	Tension	54.0	54.0
REACTOR BLDG. (Unit 1)	Steam Generator Compartment #3. Northeast Wall EL 897	D+L+To+Ra+Pa+(Yr+Yj+Ym) + Feqs	Axial Tension and Flexure	54.0	54.0
			Transverse Shear	0.202	0.203
REACTOR BLDG. (Unit 2)	Steam Generator Compartment #1.West Cavity Wall EL 863	D+L+Ta+Ra+Pa+(Yr+Yj+Ym) + Feqs	Axial Tension and Flexure	54.0	54.0
			Transverse Shear	0.087	0.101[c]
	Steam Generator Upper Lateral Support Beam (Struct. Stl.)	D+L+Ta+Ra+Pa+(Yr+Yj+Ym) + Feqs	Axial Compression & Flexure Shear	Unit 1 25.1 8.43	Unit 1 45.0 25.0
				Unit 2 20.6 6.72	Unit 2 45.0 20.0

TABLE 3.8-1
SUMMARY TABLE FOR CATEGORY 'I' STRUCTURES

(Sheet 2 of 3)

TYPE OF STRUCTURE	KEY LOCATIONS[a]	CRITICAL LOADING COMBINATION	TYPE OF STRESS	ACTUAL STRESS (KSi)	ALLOW STRESS (KSi)
AUXILIARY BLDG.	5-A Wall East of F-A Above EL 790'-6"	0.75[1.4D+1.7L+1.9Feqo+ 1.7To+1.7Ro]	Flexure Shear[d]	54 25.5	54 51
FUEL BLDG.	C-F Wall Above EL 860'-0" South of col. Line 1-F	D+L+To+Ro+W_t	Flexure Shear[d]	37.8 37.2	54 51
	Bottom of A-F Wall Between 4-F and 2-F	D+L+Ta+Ra+Pa+ (Yr+Yj+Ym) + Feqs	Flexure Shear	35.3 0.063	54 0.183[c]
SAFEGUARDS BLDG.	4-S Wall Below EL 810'-6" for Unit 1, 13-S Wall Below EL 810'-6" for Unit 2	0.75[1.4D+1.7L+1.9Feqo+ 1.7To+1.7Ro]	Flexure Shear[d]	48.6(U1) 38.9(U2) 20.6(U1) 29.2(U2)	54 51
	8-S Wall EL 810'-6" for Unit 1, 9-S Wall Between EL 773'-0" and 790'-6" for Unit 2	0.75[1.4D+1.7L+1.9Feqo+ 1.7To + 1.7Ro] 1.4D+1.7L+1.9 Feqo	Flexure Shear[d]	54 (U1) 44.3(U2) 39.0(U1) 41.8(U2)	54 51
SERVICE WATER INTAKE STRUCTURE	Bottom of North Exterior Wall EL 755'-0"	1.4D+1.7L+1.9Feqo	Flexure	53.2	54.0

TABLE 3.8-1
SUMMARY TABLE FOR CATEGORY 'I' STRUCTURES

(Sheet 3 of 3)

TYPE OF STRUCTURE	KEY LOCATIONS[a]	CRITICAL LOADING COMBINATION	TYPE OF STRESS	ACTUAL STRESS (KSi)	ALLOW STRESS (KSi)
	Bottom of West Exterior Wall EL 755'-0"			53.7	
REFUELING WATER STOR. TANKS & CONDENSATE STOR. TANKS	Base of the Circumferential Exterior Wall	1.4D+1.7L+1.7+Hydro+ 1.9Feqo	Flexure Shear	52.4 .122	54 .126[c]
REACTOR MAKE-UP WATER STOR. TANK	Base of the Circumferential Exterior Wall	0.75[1.4D+1.7L+1.9Feqo+ 1.7To+1.7Ro]	Flexure Shear	50.0 .068	54 .090[c]

Notes:

a) Unit 2 stesses are the same as Unit 1 stresses except as noted.

b) 1" x 4" Bar Stock

c) Concrete Shear Stress

d) Stress shown is for shear reinforcing steel.

VISUAL WELD ACCEPTANCE CRITERIA

FOR STRUCTURAL WELDING

AT NUCLEAR POWER PLANTS

NOTICE

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PREFACE

This document has been prepared and issued under the auspices of the NUCLEAR CONSTRUCTION ISSUES GROUP (NCIG). NCIG was formed by several utilities for the purpose of developing a common approach to the resolution of issues at nuclear power plant construction sites. The first issue considered by NCIG was in the area of acceptance criteria for and inspection of welds in certain supports and structures. This document contains NCIG's resolutions of that issue.

SECTION 1.0

INTRODUCTION

1.0 INTRODUCTION

1.1 This document provides Acceptance Criteria for visual inspection of structural welds in nuclear power plants. The development of such acceptance criteria by the owner and the Engineer[a] falls within the provisions of the AISC Specification[b] and AWS D1.1[c]. This provision is clarified in the 1985 edition of AWS D1.1. A new paragraph 1.1.1.1 has been added which states:

> "1.1.1.1 The fundamental premise of the Code is to provide general stipulations adequate to cover any situation. Acceptance criteria for production welds different from those specified in the Code may be used for a particular application provided they are suitably documented by the proposer and approved by the Engineer. These alternate acceptance criteria can be based upon evaluation of suitability for service using past experience, experimental evidence or engineering analysis considering material type, service load effects, and environmental factors."

In addition, the commentary for this new paragraph reads:

> "C1.1.1.1 The workmanship criteria provided in Section 3 of the Code are based upon knowledgeable judgement of what is achievable by a qualified welder. The criteria in Section 3 should not be considered as a boundary of suitability for service. Suitability for service analysis would lead to widely varying workmanship criteria unsuitable for a standard code. Furthermore, in some cases, the criteria would be more liberal than what is desirable and producible by a qualified welder. In general, the appropriate quality acceptance criteria and whether or not a deviation is harmful to the end use of the product should be the Engineer's decision. When modifications are approved, evaluation of suitability for service using modern fracture mechanics techniques, a history of satisfactory service, or experimental evidence is recognized as a suitable basis for alternate acceptance criteria for welds."

1.2 Engineer's Responsibility: The Engineer, as used in this document, is the individual or the organization designated by the owner as being responsible for the design of the structure being welded or inspected. The owner may fulfill the role of the Engineer for the design and inspection of any structure, in accordance with the provisions of the owner's Quality Assurance Program.

(a). See paragraph 1.2 for a definition of the Engineer, as used in this document.

(b). American Institute of Steel Construction, "Specification for Design, Fabrication and Erection of Structural Steel for Buildings."

(c). American Welding Society, "Structural Welding Code - Steel, D1.1."

Before the provisions of this document are used for structures which have already been designed, the Engineer shall review this document to assure applicability to the structures for which it will be used. Before application, the Engineer shall determine that the provisions of this document are consistent with the engineering considerations used for the original design.

The Engineer shall specify the structures to which these Acceptance Criteria will be applied.

It is not the intent of this document to preclude the development and use of other visual weld acceptance criteria. The Engineer may specify alternative acceptance criteria; this may be done based on the provisions of paragraph 1.1.1.1 of the 1985 edition of AWS D1.1, with documented justification and appropriate project approvals.

1.3 After the Engineer accepts this document for use, it may be used to supplement specifications for design and construction as appropriate.

1.4 The application and distribution of these Acceptance Criteria and Inspection Guidelines shall be controlled in accordance with applicable document control procedures.

1.5 The workmanship provisions of AWS D1.1 are not modified by the Acceptance Criteria presented in this document and are to be implemented by the fabricator or erector during production.

1.6 Section 2.0 of this document contains a Scope statement, specifies applicable materials, weld Acceptance Criteria, and measurement units; Section 3.0 provides Inspection Guidelines for use in applying these Acceptance Criteria; Section 4.0 contains engineering evaluations for Acceptance Criteria.

SECTION 2.0

ACCEPTANCE CRITERIA

2.0 ACCEPTANCE CRITERIA

2.1 Introduction

2.1.1 This Section contains Acceptance Criteria for use when inspecting welds in nuclear power plant structures and supports that have been designed and fabricated to the requirements of the AISC Specification and AWS D1.1.

2.1.2 The provisions of this document do not apply to ASME Code[(d)] stamped work.

2.2 Scope

2.2.1 This document is intended to be used with design and construction specifications for nuclear power plants where the structures have already been designed, fabricated, and/or erected, and for new work.

2.2.2 This document is applicable to welded construction for safety and non-safety related structures, including seismically loaded structures, where fatigue[(e)] is not the governing design consideration. Examples of typical structures to which these criteria apply include, but are not necessarily limited to, steel components such as:

2.2.2.1 Main building framing members and connecting members;

2.2.2.2 Supports for equipment, components and piping[(f)], cable trays and conduit, and HVAC ducts;

2.2.2.3 Miscellaneous steel including bracing and stiffeners; embedments; stairways and handrails, doors and door frames, windows and window frames, gratings, covers, etc.

(d). American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section III, Nuclear Components.

(e). For the purposes of this document, structures whose design is governed by fatigue are those structures for which an analysis is required for cyclic service and whose endurance limit must be considered in the design. At a nuclear power plant, most supports and pipe hangers, which might be designed considering seismic loads, are not governed by fatigue design limits.

(f). Excluding component supports stamped in accordance with the ASME Code Section III, Subsection NF.

2.2.3 This document and its provisions may be used for the inspection and acceptance of welds in other structures and systems not specifically addressed in 2.2.2, provided the provisions of Sections 1.2 and 2.3 are met.

2.3 Materials

2.3.1 This document is applicable to the inspection of the following structural steels:

Material Number 1, Groups Number 1 and 2, as listed in Table C2 of AWS B2.1. (These materials include those structural steels also listed in Table 4.1.1 of AWS D1.1 as Groups I and II; P Number 1, Groups 1 and 2, as listed in Table QW-422 of Section IX of the ASME Code; and S Number 1, Groups 1 and 2 as listed in Table 1 of ASME Code Case N-71-12.)

2.4 Measurement Units

Table 2-1 identifies the smallest measurement units the Inspector will use when inspecting the listed weld attributes. When measuring and recording dimensions of the weld attributes listed in Table 2-1, these dimensions shall be rounded off to the nearest significant unit.

Table 2-1

Weld Attribute	Reference Section	Smallest Measurement Unit (Significant Unit - Inches)
Fillet Weld Size	2.5.2.2	1/16
Incomplete Fusion	2.5.2.3	1/8
Weld Overlap	2.5.2.4	1/8
Undercut Depth	2.5.2.7	1/32
Surface Porosity	2.5.2.8	1/16
Weld Length	2.5.2.9	1/8 or 1/4
Weld Location	2.5.2.9	1
Slag	2.5.2.11	1/4

2.5 Acceptance Criteria

2.5.1 These Acceptance Criteria are to be used for the acceptance inspection of welds in the uncoated condition. These criteria may also be used for subsequent inspections after the welds have been coated, with the concurrence of the Engineer.

2.5.2 A weld shall be acceptable by visual inspection, subject to the following:

2.5.2.1 Weld Cracks: The weld shall have no cracks.

2.5.2.2 Fillet Weld Size: A fillet weld shall be permitted to be less than the size specified by 1/16 inch for 1/4 the length of the weld. Oversized fillet welds shall be acceptable if the oversized weld does not interfere with mating parts.

2.5.2.3 Incomplete Fusion: In fillet welds, incomplete fusion of 3/8 inch in any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable. For groove welds, incomplete fusion is not acceptable. For fillet and groove welds, rounded end conditions that occur in welding (starts and stops) shall not be considered indications of incomplete fusion and are irrelevant.

2.5.2.4 Weld Overlap: Overlap is acceptable provided the criteria for weld size and fusion can be satisfied. When fusion in the overlap length cannot be verified, an overlap length of 3/8 inch in any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable.

2.5.2.5 Underfilled Craters: Underfilled craters shall be acceptable provided the criteria for weld size are met. Craters which occur outside the specified weld length are irrelevant provided there are no cracks.

2.5.2.6 Weld Profiles

2.5.2.6.1 The faces of fillet welds may be convex, flat, or concave, provided the criteria for weld size are met.

2.5.2.6.2 The faces of groove welds may be flat or convex.

2.5.2.6.3 Convexity of fillet and groove welds are not criteria for acceptance and need not be measured.

2.5.2.6.4 The thickness of groove welds is permitted to be a maximum of 1/32 inch less than the thinner member being joined.

2.5.2.7 Undercut

2.5.2.7.1 For material 3/8 inch and less nominal thickness, undercut depth of 1/32 inch on one side for the full length of the weld, or 1/32 inch on one side for 1/2 the length of the weld and 1/16 inch for 1/4 the length of the weld on the same side of the member, is acceptable. For members welded on both sides where undercut exists in the same plane of a member, the cumulative lengths of undercut shall be limited to the lengths of undercut allowed on one side. Melt-through that results in a hole in the base metal is unacceptable.

2.5.2.7.2 For materials greater than 3/8 inch nominal thickness, undercut depth of 1/32 inch for the full length of the weld and 1/16 inch for 1/4 the length of the weld on both sides of the member is acceptable. When either welds or undercut exist only on one side of the member or are not in the same plane, the allowable undercut depth of 1/32 inch may be increased to 1/16 inch for the full length of the weld.

2.5.2.8 Surface Porosity: Only surface porosity whose major surface dimension exceeds 1/16 inch shall be considered relevant. Fillet and groove welds which contain surface porosity shall be considered unacceptable if:

2.5.2.8.1 The sum of diameters of random porosity exceeds 3/8 inch in any linear inch of weld or 3/4 inch in any 12 inches of weld; or

2.5.2.8.2 Four or more pores are aligned and the pores are separated by 1/16 inch or less, edge to edge.

2.5.2.9 Weld Length and Location: The length and location of welds shall be as specified on the detail drawing, except that weld lengths may be longer than specified. For weld lengths less than 3 inches, the permissible underlength is 1/8 inch and for welds 3 inches and longer the permissible underlength is 1/4 inch. Intermittent welds shall be spaced within 1 inch of the specified location.

2.5.2.10 Arc Strikes: Arc strikes and associated blemishes are acceptable provided no cracking is visually detected.

2.5.2.11 Surface Slag and Weld Spatter: Slag whose major surface dimension is 1/8 inch or less is irrelevant. Isolated surface slag that remains after weld cleaning and which does not exceed 1/4 inch in its major surface dimension, is acceptable. (Slag is considered to be isolated when it does not occur more frequently than once per weld or more than once in a 3 inch weld segment.) Spatter remaining after the cleaning operation is acceptable.

SECTION 3.0

INSPECTION GUIDELINES

3.0 INSPECTION GUIDELINES

These Inspection Guidelines are to be used for visual inspection of structural welds made in accordance with the provisions of AWS D1.1. These guidelines provide background information and instructions to assist the Inspector in evaluating weld attributes. Measuring techniques and guidance on the accuracy, frequency, and locations for measuring welds are discussed. It is important for the Inspector to understand weld size tolerances and significant measurement units in order to preclude rejection of adequate welds.

Effective implementation of the Acceptance Criteria requires that all Inspectors evaluating any weld use the same inspection techniques, acceptance criteria, and measurement accuracy. The Inspector's duties, inspection philosophy, and inspection guidelines are presented with this goal in mind.

3.1 The Inspector

3.1.1 As used herein, the Inspector is that person performing acceptance inspections of completed welds.

3.1.2 The Inspector using these Acceptance Criteria is to perform inspections in accordance with the guidelines provided in this document. Section 6 of AWS D1.1 addresses some inspection provisions for structural welding, but they are applicable to the in-process welding control of the project.

3.2 General Inspection Requirements

3.2.1 Weld inspection and tests are performed as necessary prior to assembly, during assembly, during welding, and after welding to ensure that materials and workmanship meet the requirements of contract documents. It is the responsibility of the contractor to assure that the workmanship standards and in-process controls of AWS D1.1 are met as appropriate.

3.2.2 In nuclear construction, in-process workmanship may be monitored and audited by the Quality Control group as required by the constructor's Quality Assurance Program. The Acceptance Criteria of this document are to be used to determine the adequacy of completed structural welds.

3.2.3 The inspection of structural welds to the Acceptance Criteria of this document shall be documented in accordance with project requirements. The marking of welds during in-process work is not required as part of the acceptance inspections addressed by this document.

3.3 Inspection Philosophy

3.3.1 The workmanship provisions of Section 3 of AWS D1.1 are necessary for in-process quality control purposes. The determination of whether corrective action is necessary to assure good workmanship during

production shall be made in accordance with the in-process welding control of the project.

3.3.2 Inspections subsequent to the acceptance inspection are intended to evaluate and verify that previous inspections were competently performed and to demonstrate compliance with design requirements. They are not intended to upgrade or downgrade the level of workmanship or impose more stringent criteria or examination methods.

3.4 Inspection Principles

The following identifies general inspection principles that will enable Inspectors to consistently inspect welds to verify compliance with the Acceptance Criteria.

3.4.1 Each weld attribute shall be inspected and evaluated independently. It is not necessary to consider cumulative effects.

3.4.2 Acceptance inspections shall be performed promptly after welding has been completed so that deficiencies, if any, may be identified and resolved in a timely manner.

3.4.3 Visual inspection of welds is normally performed on the as-welded surface after the weld cleaning operation.

3.4.4 Measuring devices should be graduated in increments compatible with the applicable significant unit, e.g., 1/16 inch increments, rather than decimals. When taking or recording measurements, measurements should be rounded off to the nearest significant unit.

3.4.5 Lighting, natural or artificial, shall be of sufficient intensity and placement to illuminate the area being examined. Lighting shall be considered adequate when, for example, the Inspector can resolve a black line 1/32 inch wide or less on an 18 percent neutral gray card placed on the surface to be inspected. Backlighting is not required. Visual inspection may be aided by using a flashlight.

3.4.6 Applicable drawings for the structure shall be reviewed prior to inspection, and referred to during inspection, as necessary.

3.4.7 These Acceptance Criteria are to be used for the acceptance inspection of welds in the uncoated condition. These criteria may also be used for subsequent inspection after the welds have been coated, with the concurrence of the Engineer. Subsequent inspections related to suspected weldment cracking may require the removal of the coating or the use of appropriate magnetic particle inspection.

3.5 Weld Attribute Descriptions, Acceptance Criteria and Inspection Guidelines

This Section includes a definition or description of each weld attribute to be inspected, identifies the weld Acceptance Criteria to be applied, and provides specific inspection guidelines.

3.5.1 Weld Cracks

3.5.1.1 Description of Weld Cracks: Weld cracks are discontinuities characterized by a sharp tip and high ratio of length and depth to opening displacement.

3.5.1.2 Acceptance Criteria: The weld shall have no cracks.

3.5.1.3 Inspection Guidelines: Weld cracks shall be identified by visually examining the weld and heat affected zone. Visually detected cracks shall be identified for repair. The use of magnification devices for detection of cracks by visual examination is not required. However, additional lighting, such as a flashlight, or magnifiers may be appropriate for further investigation of suspected cracks.

3.5.2 Fillet Weld Size

3.5.2.1 Description of Fillet Weld Undersize: Fillet weld undersize occurs when the amount of weld metal deposited results in a size that is significantly less than specified.

The undersize condition may occur in the legs or the throat of fillet welds, as shown in Figure 3-1.

3.5.2.2 Acceptance Criteria: A fillet weld shall be permitted to be less than the size specified by 1/16 inch for 1/4 the length of the weld. Oversized fillet welds shall be acceptable if the oversized weld does not interfere with mating parts.

3.5.2.3 Inspection Guidelines: The Inspector shall measure weld size using appropriate measuring devices. Potentially undersized areas are the most appropriate locations for measuring weld size. Such areas can usually be determined by viewing the weld length. Areas of the weld that appear to be significantly undersized should be examined based on the requirements of the Acceptance Criteria.

Scales or fillet weld gages may be used for measuring the size of welds. Measurement increments of 1/16 inch are appropriate for these gages. The Inspector has the responsibility for assuring that the weld meets the Acceptance Criteria for size, but

continuous measurement of size over the full length of the weld is not mandatory and weld size measurements should be rounded off to the nearest 1/16 inch.

3.5.3 Incomplete Fusion

3.5.3.1 Description of Incompletion Fusion: Incomplete fusion is a condition in which coalescence of weld metal and base metal (or previous passes of filler metal) did not occur.

3.5.3.2 Acceptance Criteria: In fillet welds, incomplete fusion of 3/8 inch in any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable. For groove welds, incomplete fusion is not acceptable. For fillet and groove welds, rounded end conditions that occur in welding (starts and stops) shall not be considered indications of incomplete fusion and are irrelevant.

3.5.3.3 Inspection Guidelines: Inspection for incomplete fusion shall be performed by visually examining the weld. The wetting and flow of weld metal at the fusion line is the best indication of fusion. Incomplete fusion in fillet welds which does not exceed 3/8 inch in any 4 inch segment of weld or 1/4 inch in welds less than 4 inches long is acceptable. Visually detected incomplete fusion in groove welds is not acceptable.

Indications of incomplete fusion at the start and stop of a weld which are observed only at the weld root are not indicative of the fusion in the main fun of the weld. (See Figure 3-2.) This apparent incomplete fusion is the result of transient conditions and should be considered acceptable. These brief transients affect only a very short portion of the weld and are not a cause for rejection because the main run of the weld is not affected. Measurements of incomplete fusion length should be rounded off to the nearest 1/8 inch.

3.5.4 Weld Overlap

3.5.4.1 Description of Weld Overlap: Weld overlap is the protrusion or rollover of weld metal over the adjacent base metal at the toe of the weld.

Fillet and Groove Weld overlap is shown in Figure 3-3.

3.5.4.2 Acceptance Criteria: Overlap is acceptable provided the criteria for weld size and fusion can be satisfied. When fusion in the overlap length cannot be verified, an overlap length of 3/8 inch in

any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable.

3.5.4.3 Inspection Guidelines: The area of any overlap should be inspected to assure weld size and fusion requirements are met. Generally, in an overlap condition, if incomplete fusion occurs, it will be short in length. The wetting and flow of weld metal at the fusion line is the best indication of fusion. Portions of individual weld ripples or weave patterns which overlap are not a concern when there is fusion on each side. Where fusion cannot be verified, the unverified length should not exceed 3/8 inch in any 4 inch segment or 1/4 inch in welds less than 4 inches long. Overlap in excess of 3/8 inch (or 1/4 inch) length is acceptable when both fusion and weld size can be verified. Measurements of overlap length should be rounded off to the nearest 1/8 inch.

3.5.5 Underfilled Craters

3.5.5.1 Description of Underfilled Craters: A crater is a depression at the termination of a weld bead. Underfilled craters are small areas in which the nominal weld size has not been achieved.

3.5.5.2 Acceptance Criteria: Underfilled craters shall be acceptable provided the criteria for weld size are met. Craters which occur outside the specified weld length are irrelevant provided there are no cracks.

3.5.5.3 Inspection Guidelines: The length of the weld should be visually examined to locate the craters. Craters are to be evaluated using the acceptance criteria for undersized welds. Craters should be visually inspected for cracks.

3.5.6 Weld Profiles

3.5.6.1 Descriptions Applicable to Weld Profiles:

Description of Convexity and Concavity:

Convexity is the maximum distance from the face of a convex fillet weld perpendicular to a line joining the weld toes. Concavity is the maximum distance from the face of a concave fillet weld perpendicular to a line joining the weld toes.

Description of Groove Weld Reinforcement:

Groove weld reinforcement is the weld metal added in excess of the quantity needed to fill the weld joint. It increases the joint thickness.

3.5.6.2 Acceptance Criteria:

3.5.6.2.1 The faces of fillet welds may be convex, flat, or concave provided the criteria for weld size are met.

3.5.6.2.2 The faces of groove welds may be flat or convex.

3.5.6.2.3 Convexity of fillet and groove welds are not criteria for acceptance and need not be measured.

3.5.6.2.4 The thickness of groove welds is permitted to be a maximum of 1/32 inch less than the thinner member being joined.

3.5.6.3 Inspection Guidelines: Fillet weld convexity is controlled during the welding process and need not be measured as part of the acceptance inspection. Concavity is acceptable provided the requirements for weld size are met.

Groove weld reinforcement is controlled during the welding process and need not be measured as part of the acceptance inspection.

When evaluating groove weld thickness, the 1/32 inch undersize allowance is a maximum value. It is equivalent to rounding off to the nearest 1/16 inch; that is, if the thickness of a groove weld is 1/32 inch or less than the specified thickness, it is acceptable; if it is more than 1/32 inch less than the specified thickness, it is to be rejected.

3.5.7 Undercut

3.5.7.1 Description of Undercut: Undercut appears as a groove melted into the base metal adjacent to the toe or root of a weld and is an area left unfilled by weld metal. Undercut may occur in any length from a fraction of an inch to the full length of the weld.

Fillet and groove weld undercut is shown in Figure 3-4.

3.5.7.2 Acceptance Criteria:

3.5.7.2.1 For material 3/8 inch and less nominal thickness, undercut depth of 1/32 inch on one side for the full length of the weld, or 1/32 inch on one side for 1/2 the length of the weld and 1/16 inch for 1/4 the length of the weld on the same side of the member, is acceptable. For members welded on both sides where undercut exists in the same plane of a

member, the cumulative lengths of undercut shall be limited to the lengths of undercut allowed on one side. Melt-through that results in a hole in the base metal is unacceptable.

3.5.7.2.2 For materials greater than 3/8 inch nominal thickness, undercut depth of 1/32 inch for the full length of the weld and 1/16 inch for 1/4 the length of the weld on both sides of the member is acceptable. When either welds or undercut exist only on one side of the member or are not in the same plane, the allowable undercut depth of 1/32 inch may be increased to 1/16 inch for the full length of the weld.

3.5.7.3 Inspection Guidelines: The criteria for determining the acceptance of undercut are depth and length. The method to be used for determining acceptable depth is primarily visual. Potentially rejectable areas may be evaluated using a scale, comparative sample, or suitable gage to determine the depth of undercut. Melt-through is unacceptable.

These Acceptance Criteria allow undercut for the full length of the weld on one side of a member. However if undercut exists for the full length, it may be indicative of a welding process control problem, and should be brought to the attention of appropriate personnel.

The acceptance criteria refer to situations where undercut may exist on both sides of the member and in the same plane. Sketch A of Figure 3-4 illustrates undercut on only one side of a member. Sketch B shows undercut on both sides of a member and in the same plane. Sketch C is an example of undercut on both sides of a member but not in the same plane. In Sketch B, the total length of undercut is determined by adding together the length of undercut on each side of the member; for material 3/8 inch and less nominal thickness the total length of undercut should not exceed the length of undercut that would be allowed if there was undercut on only one side of the member. In Sketch C, the undercut on one side is not in the same plane as the undercut on the opposite side of the member and therefore, the length of undercut at each location is to be evaluated independently.

Undercut depth is to be estimated to the nearest 1/32 inch and undercut length is to be estimated to determine compliance with the length of undercut criteria. There are significant margins in these criteria so that it is not necessary to sum isolated intermittent undercut (i.e., 1/4 inch here, 5/8 inch there) to develop the total length of undercut.

3.5.8 Surface Porosity

3.5.8.1 Description of Surface Porosity: Porosity consists of discontinuities formed by gas entrapment during solidification.

Porosity appears on the surface of welds as circular voids. The voids are caused by gas in the molten weld material which collects in discrete bubbles. The bubbles, moving toward the weld surface while it is still molten, form voids which are trapped as the metal solidifies. Gas bubbles merge and varying void sizes result. This type of porosity is frequently referred to as “pin holes” when the porosity extends to the weld surface.

3.5.8.2 Acceptance Criteria: Only surface porosity whose major surface dimension exceeds 1/16 inch shall be considered relevant. Fillet and groove welds which contain surface porosity shall be considered unacceptable if:

3.5.8.2.1 The sum of diameters of random porosity exceeds 3/8 inch in any linear inch of weld or 3/4 inch in any 12 inches of weld; or

3.5.8.2.2 Four or more pores are aligned and the pores are separated by 1/16 inch or less, edge to edge.

3.5.8.3 Inspection Guidelines: Porosity voids are generally less than 1/16 inch in diameter. Thus, relatively few are relevant. Very seldom should it be necessary to use any measuring device, but when there is concert, a 1/16 inch wire and a tape or scale are adequate.

Estimating the number of random voids which appear to be over 1/16 inch in diameter within one inch and twelve inch spans will usually provide adequate information for weld acceptance.

When evaluating potentially rejectable aligned porosity, a 1/16 inch diameter wire can be used to measure both the size of the pores and the edge to edge spacing of the pores. If the size of a pore is 1/16 inch or less it is irrelevant and need not be counted; if the spacing between any two relevant pores exceeds 1/16 inch, those pores are irrelevant for the purpose of evaluating aligned porosity.

3.5.9 Weld Length and Location

3.5.9.1 Description of Weld Length and Location: Weld length and location are an expression of the design and detail drawing

requirements specifying the length of welds and the physical location of welds on the structure.

3.5.9.2 Acceptance Criteria: The length and location of welds shall be as specified on the detail drawing, except that weld lengths may be longer than specified. For weld lengths less than 3 inches, the permissible underlength is 1/8 inch and for welds 3 inches and longer the permissible underlength is 1/4 inch. Intermittent welds shall be spaced within 1 inch of the specified location.

3.5.9.3 Inspection Guidelines: Welds should be visually inspected to determine that they are in the locations shown on the design drawings. Weld length should be measured to the tolerances designated in the drawings. When the design drawings do not provide tolerances for weld length and location, an underlength of 1/8 inch is acceptable for welds less than 3 inches and 1/4 inch for welds 3 inches or longer. Welds may be longer than specified.

Welds are usually designated as full length of the member or continuous. When the length of a weld is specified, it usually applies to intermittent welding, such as, “3 inches long at 12 inches center-to-center,” or “6 inches long at 12 inches center-to-center.” The total sum of the lengths of the intermittent welds is more important than the actual length of the individual intermittent welds. Either way, a 1/8 inch or 1/4 inch tolerance, as applicable to segment length is acceptable if no other tolerance is given.

Intermittent welds may be located within 1 inch of the designated location.

Measurements shall be rounded off to the nearest 1/16 inch for size; to the nearest 1/8 or 1/4 inch for length; and to the nearest inch for location of intermittent welds.

3.5.10 Arc Strikes

3.5.10.1 Description of Arc Strikes: An arc strike is an area of base metal struck by inadvertent arcing from the welding electrode, which may cause a small amount of base metal melting in the impingement region. Arc strikes sometimes cause a discoloration of the base metal.

3.5.10.2 Acceptance Criteria: Arc strikes and associated blemishes are acceptable provided no cracking is visually detected.

3.5.10.3 Inspection Guidelines: No special inspection is required for detection of arc strikes. Arc strikes are usually found near the weld and the Inspector should review them while performing

inspections for other weld attributes. Arc strikes and the associated blemishes should be visually examined for cracks.

3.5.11 Surface Slag and Weld Spatter

3.5.11.1 Description of Surface Slag and Weld Spatter: Surface slag is the nonmetallic residue remaining from the reactions occurring in the arc and weld pool during welding. Spatter consists of metal particles expelled during fusion welding that do not form a part of the weld. Slag and spatter are removed during the cleaning operation by brushing, grinding, chipping, or other suitable means.

3.5.11.2 Acceptance Criteria: Slag whose major surface dimension is 1/8 inch or less is irrelevant. Isolated surface slag that remains after weld cleaning and which does not exceed 1/4 inch in its major surface dimension, is acceptable. (Slag is considered to be isolated when it does not occur more frequently than once per weld or more than once in a 3 inch weld segment.) Spatter remaining after the cleaning operations is acceptable.

3.5.11.3 Inspection Guidelines: Inspection for surface slag and weld spatter should be performed by visually examining the length of the weld and adjacent base metal areas. Any areas not meeting the acceptance criteria should be cleaned again and re-examined. (Cleaning required for subsequent processing, such as painting or NDE, is not addressed here.)

3.6 Significant Measurement Units

3.6.1 Measurement units are identified in Table 2-1.

3.6.2 It is not appropriate to mix or interchange fractional and decimal dimensions or tolerances. For example, undercut must not be measured in terms of hundredths or thousandths of an inch. When acceptance limits are given in fractions of an inch, all measuring tools must also be graduated in fractions of an inch. Decimal gages are only to be used when dimensions or tolerances are explicitly specified in decimals.

3.7 Equivalency

These Acceptance Criteria have been developed based on the effect that discontinuities may have on the reduction of cross-sectional area. The Acceptance Criteria for some discontinuities (such as weld size, incomplete fusion, weld overlap, undercut and surface porosity) provide limits based on both depth or surface dimensions and length. It is intended that the Engineer may provide for acceptance of welds on an equivalency basis; for example, in material 3/8 inch or less, in nominal thickness, undercut depth that exceeds 1/32 inch may

be acceptable up to 1/16, if the length of undercut is proportionately less. Equivalency is meant as a means for the Engineer to evaluate a weld condition documented by the Inspector as not meeting the acceptance criteria.

The Inspector must base his acceptance of welds on the criteria specified and provided by the Engineer.

SECTION 4.0

COMMENTARY

4.0 COMMENTARY

4.1 Criteria Evaluation

This Section contains engineering evaluations of the Acceptance Criteria given in Section 2.0. In general, it is important to recognize that the engineering design of structures and weld connections is based on conservatively estimated loads. The allowable stresses for materials given in Codes are also conservative. Code allowable stresses have been chosen by applying a "safety factor" or "design factor" to compensate for design and analysis techniques and assumptions, fabrication tolerances, inspection techniques, variables in material properties, and other unknowns. For these reasons, there is no need to modify allowable stresses or design techniques when implementing the Acceptance Criteria of this document.

Because fatigue is not a concern for the structures addressed in this document, the evaluation of the Acceptance Criteria is primarily based on an allowable reduction of area consistent with construction Codes and Standards. Welding Research Council Bulletin 222, "The Significance of Weld Discontinuities" reports that geometrical discontinuities are influential in static behavior only to the extent they reduce cross-sectional area.

4.1.1 Weld Cracks

4.1.1.1 Acceptance Criteria: The weld shall have no cracks.

4.1.1.2 Discussion: Visually detected cracks in welds are unacceptable

4.1.2 Fillet Weld Size

4.1.2.1 Acceptance Criteria: A fillet weld shall be permitted to be less than the size specified by 1/16 inch for 1/4 the length of the weld. Oversized fillet welds shall be acceptable if the oversized weld does not interfere with mating parts.

4.1.2.2 Discussion: Undersized fillet welds could cause increased stress in the throat of the weld and a subsequent reduction of the load carrying capacity of the weld. Therefore it is appropriate to review the increase in stress due to the allowable reduction of weld size.

Evaluation of the Acceptance Criteria for a 3/16 inch fillet weld results in a potential reduction in shear area of about 8%. (See paragraph 4.3 for a sample calculation of the effect on shear area due to undersized welds.)

This evaluation is based on the assumption that both legs are reduced equally, whereas the condition generally found is that

one leg is reduced and the other is not. In addition, no consideration is given for either convexity or penetration, both of which may increase the weld shear area. Also, no credit is taken for actual weld metal strength, which is generally about 10% greater than specified.

4.1.3 Incomplete Fusion

4.1.3.1 Acceptance Criteria: In fillet welds, incomplete fusion of 3/8 inch in any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable. For groove welds, incomplete fusion is not acceptable. For fillet and groove welds, rounded end conditions that occur in welding (starts and stops) shall not be considered indications of incomplete fusion and are irrelevant.

4.1.3.2 Discussion: Incomplete fusion describes a weld zone region within which the desired coalescence of base metal and weld metal did not take place due to the absence of complete melting. An important aspect of structural welding is the welding procedure and the performance test which all welders must execute satisfactorily. These are the industry accepted primary safeguards for prevention of unacceptable incomplete fusion.

The use of an acceptance criterion which allows an incomplete fusion length of 3/8 inch in any 4 inch segment is equivalent to a maximum potential reduction of load carrying area of less than 10% (9.4%). Furthermore, the 9.4% calculated reduction in the strength of any 4 inch segment of a fillet weld that contains a 3/8 inch length of incomplete fusion (where fusion cannot be readily verified) is based on the assumption of no weld fusion for the width of the indication and the full depth of the weld. This is considered to be an extremely conservative assumption because the condition is local and usually affects only a minor portion of one weld bead; that is, the condition does not affect the full depth of the weld. Such indications (i.e., short lengths of visually detected incomplete fusion), when further investigated, generally exhibit a high percentage of complete fusion.

Indications of incomplete fusion at the starts and stops of welds are usually observed only at the weld root and are not indicative of the fusion in the main run of the weld, provided fusion is evident at the weld toes. (See Figure 3-2.) This apparent incomplete fusion is the result of transient conditions and is acceptable. These brief transients affect only a very short portion of the weld at the start and stop, and are not a cause for rejection because the main run of the weld is not affected.

4.1.4 Weld Overlap

4.1.4.1 Acceptance Criteria: Overlap is acceptable provided the criteria for weld size and fusion can be satisfied. When fusion in the overlap length cannot be verified, an overlap length of 3/8 inch in any 4 inch segment, and 1/4 inch in welds less than 4 inches long, is acceptable.

4.1.4.2 Discussion: Weld overlap or rollover may occur intermittently or as a continuous condition. Short lengths of localized or intermittent overlap at weld ripples, weld weaves or at bead starts is not a concern when the adjacent portions of a weld are fused to the base material. Fusion to the base metal is a function of the arc energy and the heat content of the molten weld pool. Neither of these factors change rapidly enough to adversely affect fusion in a short length. A short length of overlap is evidence of a localized variation in welding technique that alters the weld surface appearance, but not the underlying weld deposit.

Overlap of 3/8 inch in any 4 inch segment results in a maximum potential reduction of load carrying capacity of the weld of less than 10% (9.4%). The basis and assumptions for this calculated reduction of 9.4% due to overlap are the same as those for incomplete fusion. (See paragraph 4.1.3.2.) Thus, the evaluation of the Acceptance Criteria for overlap is consistent with the evaluation for incomplete fusion. This is desirable, because at times there is difficulty in distinguishing between overlap and incomplete fusion. With equivalent acceptance criteria, any difference is immaterial.

4.1.5 Underfilled Craters

4.1.5.1 Acceptance Criteria: Underfilled craters shall be acceptable provided the criteria for weld size are met. Craters which occur outside the specified weld length are irrelevant provided there are no cracks.

4.1.5.2 Discussion: Weld craters are slightly underfilled areas sometimes found at weld stops. A crater may constitute a short length of weld undersize. The justification for undersized fillet welds is given in Section 4.1.2.

4.1.6 Weld Profiles

4.1.6.1 Acceptance Criteria:

4.1.6.1.1 The faces of fillet welds may be convex, flat, or concave provided the criteria for weld size are met.

4.1.6.1.2 The faces of groove welds may be flat or convex.

4.1.6.1.3 Convexity of fillet and groove welds are not criteria for acceptance need not be measured.

4.1.6.1.4 The thickness of groove welds is permitted to be a maximum of 1/32 inch less than the thinner member being joined.

4.1.6.2 Discussion: Fillet weld convexity and groove weld reinforcement do not adversely affect weldments for these structural applications, but removal is not necessarily beneficial to suitability for service, and could create other problems.

The undersize allowances of up to 1/32 inch for groove welds is consistent with the requirements of AWS D1.1, paragraph 3.6.3 for flush butt joints and is equivalent to rounding off to the nearest 1/16 inch.

4.1.7 Undercut

4.1.7.1 Acceptance Criteria:

4.1.7.1.1 For material 3/8 inch and less nominal thickness, undercut depth of 1/32 inch on one side for the full length of the weld, or 1/32 inch on one side for 1/2 the length of the weld and 1/16 inch for 1/4 the length of the weld on the same side of the member, is acceptable. For members welded on both sides where undercut exists in the same plane of a member, the cumulative lengths of undercut shall be limited to the lengths of undercut allowed on one side. Melt-through that results in a hole in the base metal is unacceptable.

4.1.7.1.2 For materials greater than 3/8 inch nominal thickness, undercut depth of 1/32 inch for the full length of the weld and 1/16 inch for 1/4 the length of the weld on both sides of the member is acceptable. When either welds or undercut exist only on one side of the member or are not in the same plane, the allowable

undercut depth of 1/32 inch may be increases to 1/16 inch for the full length of the weld.

4.1.7.2 Discussion: Weld undercut is caused by melting away of base metal at the toe of a fillet weld or groove weld. Thus, undercut is best characterized as a broad shallow discontinuity rather than a crack-like defect. Undercut is a base metal condition rather than weld metal condition.

For some structures, undercut limitations have been based upon concern for fatigue loading and other factors which by scope definition are not a concern for the service conditions being covered by these Acceptance Criteria. In some cases, because of plant conditions, the Engineer may specify the alternative criteria.

Undercut develops when base metal is melted and flows under the effects of gravity and/or surface tension, without being replaced by weld filler metal. Melting and flow cause the defect to have a shallow, broad contour, as limited by the molten metal surface tension and viscosity. The broad shallow nature of undercut will permit stress redistribution in that area.

Because fatigue is not a concern for the structures addressed in this document, undercut allowance is properly based on potential reduction of area. There is significant technical support for this approach in addressing the acceptability of undercut. Principal reference in this regard is Welding Research Council Bulletin 222, “The Significance of Weld Discontinuities.” Bulletin 222 reports that geometrical discontinuities are influential in static behavior only to the extent that they reduce the cross-sectional area.

The NCIG undercut criteria were evaluated considering the effect of undercut on the net load carrying base metal area. Table 4-1 shows the effect of the undercut allowed by the Acceptance Criteria of this document on area reduction for several thicknesses of material. (See Section 4.3 for a sample calculation for the effect of undercut.)

Figure 8.15.1.5 of AWS D1.1 specifies permissible undercut values for various thicknesses of materials and principal tensile stress conditions. Table 4-1A provides a comparison of the percent reduction in area allowed based on the NCIG Acceptance Criteria and the requirements of AWS D1.1 for various material thicknesses.

TABLE 4-1

EFFECT OF UNDERCUT ON REDUCTION OF CROSS SECTION

Material Thickness (Inches)	Allowable Undercut (Depth and Length	Maximum Area Change (%)
3/16	1/32 for L (one side) or 1/32 for L/2 + 1/16 for L/4 (one side)	16.7
1/4	(same as for 3/16")	12.5
5/16	(same as for 3/16")	10.0
3/18	(same as for 3/16")	8.3
7/16	1/16 for L/4 + 1/32 for 3L/4 (both sides)	17.9
5/8	(same as for 7/16")	12.5

L is the full weld length.

TABLE 4-1A

COMPARISON OF THE EFFECT OF UNDERCUT ON PERCENT REDUCTION OF AREA

Material Thickness	NCIG	AWS D1.1		
		No Calculated Stress	Stress Parallel to Undercut	Stress Perpendicular to Undercut
3/16"	16.7	24.9	12.4	12.4
1/4"	12.5	28.0	18.7	9.3
5/16"	10.0	28.0	22.4	7.5
3/8"	8.3	27.8	18.7	6.2
7/16"	17.9	23.0	16.0	5.3
5/8"	12.5	20.0	11.2	3.7

From the above comparison, the reduction of area based on the NCIG acceptance criteria for undercut is comparable to the reduction of area permitted by AWS D1.1.

4.1.8 Surface Porosity

4.1.8.1 Acceptance Criteria: Only surface porosity whose major surface dimension exceeds 1/16 inch shall be considered relevant. Fillet and groove welds which contain surface porosity shall be considered unacceptable if:

4.1.8.1.1 The sum of diameters of random porosity exceeds 3/8 inch in any linear inch of weld or 3/4 inch in any 12 inches of weld; or

4.1.8.1.2 Four or more pores are aligned and the pores are separated by 1/16 inch or less, edge to edge.

4.1.8.2 Discussion: Steel weld metal porosity is usually caused by improper cleaning of the base metal or contamination of the weld metal, and additionally, by inadequate shielding of the welding arc. Porosity is created when the molten steel solidifies before all the gas has issued from the liquid weld metal.

Numerous publications indicate that the presence of porosity in amounts of 5% to 7% has an insignificant influence on weld strength in non-fatigue stress applications, and that 10% porosity does not diminish weld metal strength below the minimum specified values. If the porosity is assumed to extend through the full depth of the weld, the 3/4 inch in 12 inches represents 6% of a weld, therefore there is a significant margin before porosity becomes a concern. Since this is surface porosity, the effect is really insignificant.

The conclusion that porosity less than 1/16 inch in diameter is irrelevant is consistent with the provisions of ASME Section III, Subsection NF, paragraph NF-5360(a); the criteria for aligned porosity is consistent with paragraph NF 5360(a)(4).

4.1.9 Weld Length and Location

4.1.9.1 Acceptance Criteria: The length and location of welds shall be as specified on the detail drawing, except that weld lengths may be longer than specified. For weld lengths less than 3 inches, the permissible underlength is 1/8 inch and for welds 3 inches and longer the permissible underlength is 1/4 inch. Intermittent welds shall be spaced within 1 inch of the specified location.

4.1.9.2 Discussion: Welds have been rejected when their length is slightly more or less than specified on construction drawings. The purpose of verifying that welds are of proper length and location is to assure that the welds are of proper length and

location is to assure that the welds are capable of transmitting loads through the structure in accordance with the design.

Welds are most often designated on drawings as being the full length of the member or continuous around the member. Specific weld lengths are designated for intermittent welding. The total sum of the lengths of the intermittent welds is more important than the actual length oft he individual intermittent welds. Intermittent welds that are longer than specified increase total weld length, reduce stress, and are acceptable.

For the minimum allowable intermittent fillet weld length of 1-1/2 inch, the 1/8 inch permitted underlength results in an area reduction of about 8%. A similar reduction, about 8% occurs for the 1/4 inch permitted underlength applied to a 3 inch weld. This effect decreases with increasing weld length, being less than 5% for welds over 5 inches in length. This reduction of area is consistent with the reductions of area allowed for other weld attributes. (See Section 4.3 for a sample calculation of the effect of weld underlength.)

In general, welds should be located as shown on drawings. However, for welds of specific length, such as intermittent welds, a linear offset in location by as much as 1 inch is not considered to be significant because the load will still be properly transmitted through the structure.

4.1.10 Arc Strikes

4.1.10.1 Acceptance Criteria: Arc strikes and associated blemishes are acceptable provided no cracking is visually detected.

4.1.10.2 Discussion: Arc strikes are caused by inadvertent arcing between the welding electrode and the base metal, causing a small amount of base metal melting in the arc impingement region. More than ordinary attention is sometimes given arc strike areas because of potential hardness and the possibility of cracking.

For the structural steels to which this document applies, the maximum hardness of the arc strike region is limited by the relatively low carbon equivalents of the materials. Recently published work(g) has shown that cracking does not occur in arc strikes on materials of this class, and concludes that since

(g). Van Malssen, S.H., "The Effects of Arc Strikes on Steels Used in Nuclear Construction," Welding Journal, July 1984.

uncorrected arc strikes do not affect the strength of the material, they should not be of concern when the design is not governed by fatigue. The article states one of the problems suspected to be associated with arc strikes is the possibility of cracking occurring as a result of the rapid heating and cooling of the material. The results documented in the article demonstrate that the suspected cracking does not occur.

The article also discusses potential encroachment on required thickness by arc strikes and concludes that such encroachment on the required thickness by arc strikes is unlikely.

4.1.11 Surface Slag and Weld Spatter

4.1.11.1 Acceptance Criteria: Slag whose major surface dimension is 1/8 inch or less is irrelevant. Isolated surface slag that remains after weld cleaning and which does not exceed 1/4 inch in its major dimension, is acceptable. (Slag is considered to be isolated when it does not occur more frequently than once per weld or more than once in a 3 inch weld segment.) Spatter remaining after the cleaning operation is acceptable.

4.1.11.2 Discussion: Sometimes removal of all slag from completed welds requires extraordinary effort since occasional tightly adhering slag in small areas requires the use of pneumatic chisels or grinders to completely remove the slag. This usually results in a less presentable surface and removal may cause more harm than good. It is good practice to remove surface slag by wire brushing or chipping hammers, but there are limits as to what can be accomplished with such tools. It is not considered a necessity to remove small tightly adherent areas of slag since it has no effect on the weld. The amount of slag permitted by the Acceptance Criteria should not mask any significant condition of concern.

The basic purpose for removing weld spatter is to clean the area so it can be properly inspected and painted. Tightly adhering weld spatter is not harmful to the weld or base metal.

4.2 Cumulative Effects

4.2.1 Eleven attributes are considered in this document. The cumulative effect that deficiencies in any one of these attributes may, or may not have, on any other attribute is summarized in Table 4-2. It is concluded that it is not necessary to consider cumulative effects when evaluating welds in these

types of structures and therefore each weld attribute should be inspected and evaluated independently.

4.2.2 The justification for not considering the cumulative effects of any of the potentially negative combinations that are possible, follows:

4.2.2.1 Both base metal and weld metal strength are significantly higher than minimum specified values.

4.2.2.2 Small size welds are rarely loaded to allowable stress limits.

4.2.2.3 Convexity and reinforcement do not reduce the load carrying capacity of the weld.

4.2.2.4 These structures are designed using conservative load definitions, load combinations, analytical techniques, and design methods.

4.2.2.5 Designers round-up to the next larger fillet weld size in 1/16 in. increments.

4.2.2.6 Member sizes and material thicknesses are selected by the designer based on manufacturer's standards or material availability. As a result, these structures are usually fabricated using material that exceeds the size or thickness needed to satisfy design requirements for area or load carrying capability.

4.2.3 Weld Underlength and Fillet Weld Undersize: It is possible that these attributes could combine to decrease the load capacity of a weld. However, underlength is a consideration that is generally applicable only to intermittent welds and such welds are not common in these types of structures.

TABLE 4-2

COMMENTS APPLICABLE TO CUMULATIVE EFFECTS

Weld Attributes	Comment
Weld Cracks	Not allowed, therefore not cumulative.
Fillet Weld Undersize	Could combine with weld underlength (refer to Section 4.2.3 for Discussion).
Incomplete Fustion	Local condition which does not produce a general reduction in strength.
Weld Overlap	Local condition which does not produce a general reduction in strength.
Underfilled Craters	Weld size restrictions must be met; therefore, not cumulative.
Weld Profiles:	
Convexity and Groove Weld Reinforcement	Do not result in a reduction in weld or material strength or area.
Concavity	Weld size restrictions must be met; therefore not cumulative.
Undercut	This is a base metal condition which does not affect the weld.
Surface Porosity	Porosity does not adversely affect weld strength and therefore is not cumulative.
Weld Length	Could combine with fillet weld undersize (refer to Section 4.2.3 for Discussion).
Arc Strikes	Could result in minor reduction in base metal area, but is not a cumulative effect with other base metal considerations.
Surface Slag and Weld Spatter	Do not reduce weld strength or area.

On a worst case basis, if a fillet weld specified to be 3/16 inch by 3 inches long was undersized and underlength but met the Acceptance Criteria for both of these attributes, the total reduction in throat area compared to the throat area of the weld as specified, would be approximately 16 percent, which is reasonable on the basis of the conservatisms stated in 4.2.2. (See Section 4.3 for a sample calculation of percent reduction in throat area.)

4.2.4 Fillet Weld Undersize and Undercut: Fillet weld undersize is a condition which may affect the load carrying capability of the weld and undercut is a condition in the base material adjacent to the weld. These two effects do not occur at the same point and are not cumulative. The failure plane for a structure with fillet welds is through the weld. This failure plane is not in line with nor in close proximity to any plane affected by undercut. Thus the two attributes cannot combine nor bear upon one another.

4.2.5 Incomplete Fusion or Weld Overlap and Other Weld Attributes: The 9.4% calculated reduction in the strength of a 4 inch segment of a fillet weld that contains a 3/8 inch length of incomplete fusion or weld overlap (where fusion cannot be readily verified) is based on the assumption of no weld fusion for the width of the indication and the full depth of the weld. This is considered to be an extremely conservative assumption because the condition is local and usually affects only a minor portion of one weld bead; that is, the condition does not affect the full depth of the weld. Such indications (i.e., short lengths of visually detected incomplete fusion or weld overlap), when further investigated, generally exhibit complete fusion. Because the assumed condition is extremely rare, it is not necessary to consider the cumulative effect of either incomplete fusion or weld overlap with any other deficiency (such as weld underlength or weld undersize) when inspecting welds in these types of structures.

4.2.6 Other Weld Attributes: The other attributes are either not related or are clearly independent of each other.

4.3 Sample Calculations

4.3.1 Sample Calculation for the Combined Effect of Weld Underlength and Undersize

For a 3/16” X 3” long fillet weld that meets NCIG criteria for length and size:

Theoretical throat area:

$$= (S)\,(L)\,(.707)$$

$$= (3/16)\,(3)\,(0.707) = 0.398 \text{ in.}^2$$

Actual throat area:

$$= (S)\,(L')\,(.707) - (s)\,(L')\,(\%L')\,(.707)$$

$$= (3/16)\,(2\text{-}3/4)\,(.707) - (1/16)\,(2\text{-}3/4)\,(0.25)\,(.707)$$

$$= 0.365 - 0.030$$

$= 0.335 \text{ in.}^2$

$$\text{Area reduction} = \frac{0.398 - 0.335}{0.398}$$

$= 0.158$ or approximately 16%(h)

Where:

S = Specified Weld Size
L = Specified Weld Length
L^1 = Permissible Weld Length = L-1/4”
s = Permissible Undersize = 1/16”
$\%L^1$ = % of Weld Length that may be Undersize = 25%

4.3.2 Sample Calculation for the Effect of Undercut in 5/8 Inch Material

$$\text{Area Reduction} = \frac{(2)\,(u)\,(L/4) + (2)\,(u')\,(3L/4)}{(t)\,(L)}$$

$$= \frac{(2)\,(1/16)\,(L/4) + (2)\,(1/32)\,(3L/4)}{(5/16)\,(L)}$$

$$= \frac{0.031 + 0.047}{0.625}$$

$= 0.125$ or 12.5%

Where:

t = Base Metal Thickness

L = Specified Weld Length

u = Permissible undercut depth for L/4

u’ = Permissible undercut depth for 3L/4

4.3.3 Sample Calculation for the Effect of Undersize in a 3/16 Inch Fillet Weld

$$\text{Area Reduction} = \frac{(s)\,(L/4)\,(.707)}{(S)\,(L)\,(.707)}$$

(h). A calculation to determine the reduction in weld metal area along the heat affected zone (weld leg) would obviously yield an identical result. This same observation can be made for the results given in 4.3.3 and 4.3.4.

$$= \frac{(1/16)\ (L/4)\ (.707)}{(3/16)\ (L)\ (.707)}$$

$$= \frac{(0.0156)}{(0.1875)}$$

$$= 0.083 \text{ or } 8.3\%$$

Where:

s = Permissible Undersize

S = Specified Weld Size

L = Specified Weld Length

4.3.4 Sample Calculations for the Effect of Weld Underlength

For an intermittent fillet weld specified to be 1-1/2 inches long that meets the NCIG criteria for length:

Theoretical throat area:

$$= (S)\ (L)\ (.707)$$

$$= (S)\ (1.5)\ (.707)$$

$$= 1.061\ (S)\ \text{in.}^2$$

Actual throat area:

$$= (S)\ (L')\ (.707)$$

$$= (S)\ (1.5 - 0.125)\ (.707)$$

$$= 0.972\ (S)\ \text{in.}^2$$

$$\text{Area reduction} = \frac{1.061 - 0.972}{1.061}$$

$$= 0.083 \text{ or } 8.3\%$$

Where:

S = Specified Weld Size

L = Specified Weld Length

L' = Permissible Weld Length

For an intermittent weld specified to be 5 inches long that meets the NCIG criteria for length:

Theoretical throat area:

= (S) (5) (.707)

= 3.535 (S) $in.^2$

Actual throat area:

= (S) (5 - 0.25) (.707)

= 3.358 (S) $in.^2$

Area reduction = $\frac{3.535 - 3.358}{3.535}$

= 0.050 or 5%

3.9N MECHANICAL SYSTEMS AND COMPONENTS

3.9N.1 SPECIAL TOPICS FOR MECHANICAL COMPONENTS

3.9N.1.1 Design Transients

The following five operating conditions as defined in Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code are considered in the design of the Reactor Coolant System (RCS), RCS component supports, and reactor internals.

1. Normal conditions

   Any condition in the course of startup, operation in the design power range, hot standby and system shutdown, other than upset, emergency, faulted or testing conditions.

2. Upset conditions (incidents of moderate frequency)

   Any deviations from normal conditions anticipated to occur often enough that design should include a capability to withstand the conditions without operational impairment. The upset conditions include those transients which result from any single operator error or control malfunction, transients caused by a fault in a system component requiring its isolation from the system and transients due to loss of load or power. Upset conditions include any abnormal incidents not resulting in a forced outage and also forced outages for which the corrective action does not include any repair of mechanical damage. The estimated duration of an upset condition shall be included in the design specifications.

3. Emergency conditions (infrequent incidents)

   Those deviations from normal conditions which require shutdown for correction of the conditions or repair of damage in the system. The conditions have a low probability of occurrence but are included to provide assurance that no gross loss of structural integrity will result as a concomitant effect of any damage developed in the system. The total number of postulated occurrences for such events shall not cause more than 25 stress cycles having an Sa value greater than that for 10^6 cycles from the applicable fatigue design curves of the ASME Code, Section III.

4. Faulted conditions (limiting faults)

   Those combinations of conditions associated with extremely low probability, postulated events whose consequences are such that the integrity and operability of the nuclear energy system may be impaired to the extent that consideration of public health and safety are involved. Such considerations require compliance with safety criteria as may be specified by jurisdictional authorities.

5. Testing conditions

   Testing conditions are those pressure overload tests including hydrostatic tests, pneumatic tests, and leak tests specified. Other types of tests shall be classified under normal, upset, emergency or faulted conditions.

To provide the necessary high degree of integrity for the equipment in the RCS the transient conditions selected for equipment fatigue evaluation are based upon a conservative estimate of the magnitude and frequency of the temperature and pressure transients resulting from various operating conditions in the plant. To a large extent, the specific transient operating conditions to be considered for equipment fatigue analyses are based upon engineering judgement and experience. The transients selected are representative of operating conditions which prudently should be considered to occur during plant operation and are sufficiently severe or frequent to be of possible significance to component cycle behavior. The transients selected may be regarded as a conservative representation of transients which, used as a basis for component fatigue evaluation, provide confidence that the component is appropriate for its application over the design life of the plant.

The following design conditions are given in the equipment specification for RCS components.

The design transients and the number of cycles of each that is normally used for fatigue evaluations are shown in Table 3.9N-1. In accordance with the ASME Code, Section III, emergency and faulted conditions are not included in fatigue evaluations.

The Components Cyclic or Transient Limits Program provides controls to track the FSAR Table 3.9N-1A, cyclic and transient occurrences to ensure that components are maintained within the design limits.

Normal Conditions

The following primary system transients are considered normal conditions:

1. Heatup and cooldown at 100°F per hour.
2. Unit loading and unloading at 5 percent of full power per minute.
3. Step load increase and decrease of 10 percent of full power.
4. Large step load decrease with steam dump.
5. Steady state fluctuations.
6. Feedwater cycling at hot shutdown.
7. Loop out of service.
8. Unit loading and unloading between 0 and 15 percent of full power.
9. Boron concentration equalization.
10. Refueling.

1. Heatup and cooldown at 100°F per hour

The design heatup and cooldown cases are conservatively represented by continuous operations performed at a uniform temperature rate of 100°F per hour. (These operations

can take place at lower rates approaching the minimum of 0°F per hour. The expected normal rates are 50°F per hour).

For these cases, the heatup occurs from ambient (assumed to be 120°F) to the no-load temperature and pressure condition and the cooldown represents the reverse situation. In actual practice, the rate of temperature change of 100°F per hour will not be attained because of other limitations such as:

a. Material ductility considerations which establish maximum permissible temperature rates of change, as a function of plant pressure and temperature, which are below the design rate of 100°F per hour.

b. Slower initial heatup rates when using pump energy only.

c. Interruptions in the heatup and cooldown cycles due to such factors as drawing a pressurizer steam bubble, rod withdrawal, sampling, water chemistry and gas adjustments.

The number of such complete heatup and cooldown operations is specified as 200 each, which corresponds to five such occurrences per year for the 40 year plant design life.

2. Unit loading and unloading at 5 percent of full power per minute

The unit loading and unloading cases are conservatively represented by a continuous and uniform ramp power change of 5 percent per minute between 15 percent load and full load. This load swing is the maximum possible, consistent with operation under automatic reactor control. The reactor temperature will vary with load as prescribed by the Reactor Control System. The number of loading and unloading operations is defined as 13,200. One loading operation per day yields 14,600 such operations during the 40 year design life of the plant. By assuming a 90 percent availability factor, this number is reduced to 13,200.

3. Step load increase and decrease of 10 percent of full power

The ±10 percent step change in load demand is a transient which is assumed to be a change in turbine control valve opening due to disturbances in the electrical network into which the plant output is tied. The Reactor Control System is designed to restore plant equilibrium without reactor trip following a ±10 percent step change in turbine load demand initiated from nuclear plant equilibrium conditions in the range between 15 percent and 100 percent full load, the power range for automatic reactor control. In effect, during load change conditions, the Reactor Control System attempts to match turbine and reactor outputs in such a manner that peak reactor coolant temperature is minimized and reactor coolant temperature is restored to its programmed setpoint at a sufficiently slow rate to prevent excessive pressurizer pressure decrease.

Following a step decrease in turbine load, the secondary side steam pressure and temperature initially increase since the decrease in nuclear power lags behind the step decrease in turbine load. During the same increment of time, the RCS average temperature and pressurizer pressure also initially increase. Because of the power

mismatch between the turbine and reactor and the increase in reactor coolant temperature, the control system automatically inserts the control rods to reduce core power. With the load decrease, the reactor coolant temperature will ultimately be reduced from its peak value to a value below its initial equilibrium value at the inception of the transient. The reactor coolant average temperature setpoint change is made as a function of turbine-generator load as determined by first stage turbine pressure measurement. The pressurizer pressure will also decrease from its peak pressure value and follow the reactor coolant decreasing temperature trend. At some point during the decreasing pressure transient, the saturated water in the pressurizer begins to flash which reduces the rate of pressure decrease. Subsequently the pressurizer heaters come on to restore the plant pressure to its normal value.

Following a step increase in turbine load, the reverse situation occurs, i.e., the secondary side steam pressure and temperature initially decrease and the reactor coolant average temperature and pressure initially decrease. The control system automatically withdraws the control rods to increase core power. The decreasing pressure transient is reversed by actuation of the pressurizer heaters and eventually the system pressure is restored to its normal value. The reactor coolant average temperature will be raised to a value above its initial equilibrium value at the beginning of the transient.

The number of each operation is specified at 2000 times or 50 per year for the 40 year plant design life.

4. Large step load decrease with steam dump

This transient applies to a step decrease in turbine load from full power, of such magnitude that the resultant rapid increase in reactor coolant average temperature and secondary side steam pressure and temperature will automatically initiate a secondary side steam dump that will prevent both reactor trip and lifting of steam generator safety valves. Thus, since the Comanche Peak Nuclear Power Plant (CPNPP) is designed to accept a step decrease of 50 percent from full power the Steam Dump System provides the heat sink to accept 40 percent of the turbine load. The remaining 10 percent of the total step change is assumed by the Reactor Control System (control rods). If a Steam Dump System was not provided to cope with this transient, there would be such a strong mismatch between what the turbine is asking for and what the reactor is delivering that a reactor trip and lifting of steam generator safety valves would occur.

The number of occurrences of this transient is specified at 200 times or 5 per year for the 40 year plant design life.

5. Steady state fluctuations

It is assumed that the reactor coolant temperature and pressure at any point in the system vary around the nominal (steady state) values. For design purposes two cases are considered:

a. Initial fluctuations

These are due to control rod cycling during the first 20 full power months of reactor operation. Temperature is assumed to vary ±3°F and pressure by

±25 pounds per square inch (psi), once during each 2 minute period. The total number of occurrences is limited to 1.5×10^5. These fluctuations are assumed to occur consecutively, and not simultaneously with the random fluctuations.

b. Random fluctuations

Temperature is assumed to vary by ±0.5°F and pressure by ±6 psi, once every 6 minutes. With a 6 minute period, the total number of occurrences during the plant design life does not exceed 3.0×10^6.

6. Feedwater cycling at hot shutdown

These transients can occur when the plant is at no-load conditions, during which intermittent feeding of 32°F feedwater into the steam generators is assumed. Due to fluctuations arising from this mode of operation, the reactor coolant average temperature decreases to a lower value and then immediately begins to return to normal no-load temperature. This transient is assumed to occur 2000 times over the life of the plant.

7. Loop out of service

The plant may be operated at a reduced power level with a single loop out of service for limited periods of time. This is accomplished by reducing power level and tripping a single reactor coolant pump.

It is assumed that this transient occurs twice per year or 80 times in the life of the plant. Conservatively, it is assumed that all 80 occurrences can occur in the same loop. In other words it must be assumed that the whole RCS is subjected to 80 transients while each loop is also subjected to 80 inactive loop transients.

When an inactive loop is brought back into service, the power level is reduced to approximately 10 percent and the pump is started. It is assumed that an inactive loop is inadvertently started up at maximum allowable power level 10 times over the life of the plant. (This transient is covered under Upset Conditions.) Thus, the normal startup of an inactive loop is assumed to occur 70 times during the life of the plant.

8. Unit loading and unloading between 0 and 15 percent of full power

The unit loading and unloading cases between 0 and 15 percent power are represented by continuous and uniform ramp power changes, requiring 30 minutes for loading and 5 minutes for unloading. During loading, reactor coolant temperatures are increased from the no-load value to the normal load program temperatures at the 15 percent power level. The reverse temperature change occurs during unloading.

Prior to loading, it is assumed that the plant is at hot shutdown conditions, with 32°F feedwater cycling. During the 2 hour period following the beginning of loading, the feedwater temperature increases from 32°F to 300°F due to steam dump and turbine startup heat input to the feedwater. Subsequent to unloading, feedwater heating is terminated, steam dump is reduced to residual heat removal requirements, and feedwater temperature decays from 300°F to 32°F.

The number of these loading and unloading transients is assumed to be 500 each during the 40 year plant design life, which is equivalent to about one occurrence per month.

9. Boron concentration equalization

Following any large change in boron concentration in the RCS, spray is initiated in order to equalize concentration between the loops and the pressurizer. This can be done by manually operating the pressurizer backup heaters, thus causing a pressure increase, which will initiate spray at a compensated pressurizer pressure of approximately 2275 psia. The proportional sprays return the pressure to 2250 psia and maintain this pressure by matching the heat input from the backup heater until the concentration is equalized. For design purposes, it is assumed that this operation is performed once after each load change in the design load follow cycle. With two changes per day and a 90 percent plant availability factor over the 40 year design life, the total number of occurrences is 26,400.

10. Refueling

At the end of plant cooldown the fluid in the RCS is at 140°F. At this time the vessel head is removed and the refueling canal is filled. This is done by pumping water from the refueling water storage tank, which is outside and conservatively assumed to be at 32°F, into the loops by means of the residual heat removal pumps. It is conservatively assumed that the cold water is replaced with the colder water within 10 minutes.

This operation is assumed to occur twice per year or 80 times over the life of the plant.

Upset Conditions

The following primary system transients are considered upset conditions.

1. Loss of load (without immediate reactor trip).

2. Loss of power.

3. Partial loss of flow.

4. Reactor trip from full power.

5. Inadvertent Reactor Coolant System depressurization.

6. Inadvertent startup of an inactive loop.

7. Control rod drop.

8. Inadvertent Emergency Core Cooling System actuation.

9. Operating Basis Earthquake.

10. Excessive Feedwater Flow

1. Loss of Load (without immediate reactor trip)

   This transient applies to a step decrease in turbine load from full power (turbine trip) without immediately initiating a reactor trip and represents the most severe pressure transient on the RCS under upset conditions. The reactor eventually trips as a consequence of a high pressurizer level trip initiated by the Reactor Protection System (RPS). Since redundant means of tripping the reactor are provided as a part of the RPS, transients of this nature are not expected, but are included to ensure a conservative design.

   The number of occurrences of this transient is specified at 80 times or 2 times per year for the 40 year plant design life.

2. Loss of power

   This transient applies to a blackout situation involving the loss of outside electrical power to the station, assumed to be operating initially at 100 percent power, followed by reactor and turbine trips. Under these circumstances, the reactor coolant pumps are de-energized and, following coastdown of the reactor coolant pumps, natural circulation builds up in the system to some equilibrium value. This condition permits removal of core residual heat through the steam generators which at this time are receiving feedwater, assumed to be at 32°F, from the Auxiliary Feedwater System operating from diesel generator power. Steam is removed for reactor cooldown through atmospheric relief valves provided for this purpose.

   The number of occurrences of this transient is specified at 40 times or 1 per year for the 40 year plant design life.

3. Partial loss of flow

   This transient applies to a partial loss of flow from full power, in which a reactor coolant pump is tripped out of service as the result of a loss of power to that pump. The consequences of such an accident are a reactor and turbine trip, on low reactor coolant flow, followed by automatic opening of the Steam Dump System and flow reversal in the affected loop. The flow reversal causes reactor coolant at cold leg temperature to pass through the steam generator and be cooled still further. This cooler water then flows through the hot leg piping and enters the reactor vessel outlet nozzles. The net result of the flow reversal is a sizable reduction in the hot leg coolant temperature of the affected loop.

   The number of occurrences of this transient is specified at 80 times or 2 times per year for the 40 year plant design life.

4. Reactor trip from full power

   A reactor trip from full power may occur from a variety of causes resulting in temperature and pressure transients in the RCS and in the secondary side of the steam generator. This is the result of continued heat transfer from the reactor coolant in the steam generator. The transient continues until the reactor coolant and steam generator secondary side temperatures are in equilibrium at zero power conditions. A continued

supply of feedwater and controlled dumping of steam remove the core residual heat and prevent the steam generator safety valves from lifting. The reactor coolant temperature and pressure undergo a rapid decrease from full power values as the RPS causes the control rods to move into the core.

Various moderator cooldown transients associated with reactor trips can occur as a result of excessive feed or steam dump after trip or large load increase. For design purposes, reactor trip is assumed to occur a total of 400 times or 10 times per year over the life of the plant. The various types of trips and the number of occurrences for each are as follows:

a. Reactor trip with no inadvertent cooldown - 230 occurrences.

b. Reactor trip with cooldown but no safety injection - 160 occurrences.

c. Reactor trip with cooldown actuating safety injection - 10 occurrences.

5. Inadvertent Reactor Coolant System depressurization

Several events can be postulated as occurring during normal plant operation which will cause rapid depressurization of the RCS. These include:

a. Actuation of a single pressurizer safety valve.

b. Inadvertent opening of one pressurizer power operated relief valve due either to equipment malfunction or operator error.

c. Malfunction of a single pressurizer pressure controller causing one power operated relief valve and two pressurizer spray valves to open.

d. Inadvertent opening of one pressurizer spray valve, due either to equipment malfunction or operator error.

e. Inadvertent auxiliary spray.

Of these events, the pressurizer safety valve actuation causes the most severe transients, and is used as an “umbrella” case to conservatively represent the reactor coolant pressure and temperature variations arising from any of them.

When a pressurizer safety valve opens, and remains open, the system rapidly depressurizes, the reactor trips, and the Emergency Core Cooling System (ECCS) is actuated. Also, the passive accumulators of the ECCS are actuated when pressure decreases by approximately 1600 psi, about 12 minutes after the depressurization begins. The depressurization and cooldown are eventually terminated by operator action. All of these effects are completed within approximately 18 minutes. It is conservatively assumed that none of the pressurizer heaters are energized.

With pressure constant and safety injection in operation, boiloff of hot leg liquid through the pressurizer and open safety valve will continue.

For design purposes this transient is assumed to occur 20 times during the 40 year design life of the plant.

6. Inadvertent startup of an inactive loop

This transient can occur when a loop is out of service. With the plant operating at maximum allowable level the reactor coolant pump in the inactive loop is started as a result of operator error. Reactor trip occurs on high nuclear flux. This transient is assumed to occur 10 times during the life of the plant.

7. Control rod drop

This transient occurs if a bank of control rods drop into the fully inserted position due to a single component failure. The reactor is tripped on low pressurizer pressure. It is assumed that this transient occurs 80 times over the life of the plant.

8. Inadvertent Emergency Core Cooling System Actuation

A spurious safety injection signal results in an immediate reactor trip followed by actuation of the high head centrifugal charging pumps. These pumps deliver cold water from Refueling Water Storage Tank to the RCS cold legs. The initial portion of this transient is similar to the reactor trip from full power with no cooldown. Controlled steam dump and feedwater flow after trip removes core residual heat. Reactor coolant temperature and pressure decreases as the control rods move into the core.

Later in the transient, the injected water causes the RCS pressure to increase to the pressurizer power operated relief valve setpoint and the primary and secondary temperatures to decrease gradually. The transient continues until the operator stops the charging pumps. It is assumed that the plant is then returned to no-load conditions, with pressure and temperature changes controlled within normal limits.

For design purposes this transient is assumed to occur 60 times over the life of the plant.

9. Operating Basis Earthquake

The mechanical stresses resulting from the Operating Basis Earthquake are considered on a component basis. Fatigue analysis, where required by the codes, if performed by the supplier as part of the stress anal report. The earthquake loads are a part of the mechanical loading conditions specified in the equipment specifications. The origin of their determination is separate and distinct from those transients resulting from fluid pressure and temperature. They are, however, considered in the design analysis. The number of occurrences for fatigue evaluation is assumed to be 20 earthquakes at 10 cycles each (200 cycles total).

10. Excessive Feedwater Flow

The pressure and temperature variations associated with this transient are considered in connection with analyzing the primary and secondary sides of the Steam Generator, the Reactor Coolant System, and the Pressurizer.

This transient is conservatively defined to cover the occurrence of several events of the same general nature. These include:

- Inadvertent opening of a feedwater control valve
- Turbine overspeed (110 percent) with an open feedwater control valve.
- Small steam break with an open feedwater control valve.

The excessive feedwater flow transient results from inadvertent opening of a feedwater control valve when the plant is at hot shutdown and the steam generator is in the no-load condition. The feedwater, condensate, and heater drain systems are in operation. The stem of a feedwater control valve has been assumed to fail, with the valve immediately reaching the full-open position. The feedwater flow to the affected loop is assumed to stop from essentially zero flow to the value determined by the steam flow is assumed to remain at zero, and the temperature of the feedwater entering the steam generator is conservatively assumed to be 32°F. Feedwater flow is isolated on a safety injection signal actuated by low steam line pressure or low pressurizer pressure signal.

Auxiliary feedwater flow, initiated by the safety injection signal, is assumed to continue with all pumps discharging into the affected steam generator. It is also assumed, for conservatism in the secondary side analysis, that auxiliary feedwater flows to the steam generators not affected by the malfunctioned valve, in the so-called “unfailed loops.” Plant conditions stabilize at the values reached in 600 seconds, at which time auxiliary feedwater flow is terminated. The plant is then either taken to cold shutdown, or returned to the no-load condition at a normal heatup rate with the auxiliary feedwater system under manual control.

For design purposes, this transient is assumed to occur 30 times during the life of the plant.

Emergency Conditions

The following primary system transients are considered emergency conditions:

1. Small loss of coolant accident.
2. Small steam break.
3. Complete loss of flow.

1. Small loss of coolant accident

For design transient purposes the small loss of coolant accident is defined as a break equivalent to the severance of a 1 inch inside diameter branch connection. (Breaks smaller than 0.375 inch inside diameter can be handled by the normal makeup system and produce no significant fluid systems transients.) Breaks which are much larger than 1 inch will cause accumulator injection soon after the accident and are regarded as faulted conditions. For design purposes it is assumed that this transient occurs five times during the life of the plant. It should be assumed that the ECCS is actuated immediately

after the break occurs and subsequently delivers water at a minimum temperature of 32°F to the RCS.

2. Small steam break

   For design transient purposes, a small steam break is defined as a break equivalent in effect to a steam safety valve opening and remaining open. This transient is assumed to occur five times during the life of the plant. The following conservative assumptions are used in defining the transients:

   a. The reactor is initially in a hot, zero-power condition.

   b. The small steam break results in immediate reactor trip and ECCS actuation.

   c. A large shutdown margin, coupled with no feedback or decay heat, prevents heat generation during the transient.

   d. The ECCS operates at a design capacity and repressurizes the RCS within a relatively short time.

3. Complete loss of flow

   This accident involves a complete loss of flow from full power resulting from simultaneous loss of power to all reactor coolant pumps. The consequences of this incident are a reactor trip and turbine trip on undervoltage followed by automatic opening of the Steam Dump System. For design purposes this transient is assumed to occur five times during the plant lifetime.

Faulted Conditions

The following primary system transients are considered faulted conditions. Each of the following accidents should be evaluated for one occurrence:

1. Reactor coolant pipe break (large loss of coolant accident).

2. Large steam line break.

3. Feedwater line break.

4. Reactor coolant pump locked rotor.

5. Control rod ejection.

6. Steam generator tube rupture.

7. Safe Shutdown Earthquake.

1. Reactor coolant pipe break (large loss of coolant accident)

   Following rupture of a reactor coolant pipe resulting in a large loss of coolant, the primary system pressure decreases causing the primary system temperature to decrease. Because of the rapid blowdown of coolant from the system and the comparatively large heat capacity of the metal sections of the components, it is likely that the metal will still be at or near the operating temperature by the end of blowdown. It is conservatively assumed that the ECCS is actuated to introduce water at a minimum temperature of 32°F into the RCS. The safety injection signal will also result in reactor and turbine trips.

2. Large steam line break

   This transient is based on the complete severance of the largest steam line. The following conservative assumptions were made:

   a. The reactor is initially in a hot, zero-power condition.

   b. The steam line break results in immediate reactor trip and ECCS actuation.

   c. A large shutdown margin, coupled with no feedback or decay heat, prevents heat generation during the transient.

   d. The ECCS operates at design capacity and repressurizes the RCS within a relatively short time.

   The above conditions result in the most severe temperature and pressure variations which the primary system will encounter during a steam break accident.

3. Feedwater line break

   This accident involves a double ended rupture of the main feed-water piping from full power, resulting in the rapid blowdown of one steam generator and the termination of main feedwater flow to the others. The blowdown is completed in approximately 27 seconds. Conditions were conservatively chosen to give the most severe primary side and secondary side transients. All auxiliary feedwater flow exits at the break. The incident is terminated when the operator manually realigns the Auxiliary Feedwater System to isolate the break and to deliver auxiliary feedwater to the intact steam generators.

4. Reactor coolant pump locked rotor

   This accident is based on the instantaneous seizure of a reactor coolant pump with the plant operating at full power. The locked rotor can occur in any loop. Reactor trip occurs almost immediately, as the result of low coolant flow in the affected loop.

5. Control rod ejection

   This accident is based on the single most reactive control rod being instantaneously ejected from the core. This reactivity insertion in a particular region of the core causes a severe pressure increase in the RCS such that the pressurizer safety valves will lift and

also causes a more severe temperature transient in the loop associated with the affected region than in the other loops. For conservatism the analysis is based on the reactivity insertion and does not include the mitigating effects (on the pressure transient) of coolant blowdown through the hole in the vessel head vacated by the ejected rod.

6. Steam generator tube rupture

   This accident postulates the double ended rupture of a steam generator tube resulting in a decrease in pressurizer level and reaction coolant pressure. Reactor trip will occur due to the resulting safety injection signal. In addition, safety injection actuation automatically isolates the feedwater lines, by tripping all feedwater pumps and closing the feedwater isolation valves. When this accident occurs, some of the reactor coolant blows down into the affected steam generator causing the shell side level to rise. The primary system pressure is reduced below the secondary safety valve setting. Subsequent recovery procedures call for isolation of the steam line leading from the affected steam generator. This accident will result in a transient which is no more severe than that associated with a reactor trip from full power. Therefore it requires no special treatment insofar as fatigue evaluation is concerned, and no specific number of occurrences is postulated.

7. Safe Shutdown Earthquake

   The mechanical dynamic or static equivalent loads due to the vibratory motion of the Safe Shutdown Earthquake are considered on a component basis.

Test Conditions

The following primary system transients under test conditions are discussed:

1. Turbine roll test.
2. Primary side hydrostatic test.
3. Secondary side hydrostatic test.
4. Primary side leakage test.
5. Secondary side leakage test.
6. Tube leakage test.

Pressure tests are performed in accordance with either the: ASME Code, Section III, Article NB-6000, 1980 edition and Addenda through Summer 1981 or ASME Code, Section XI, as described in FSAR Section 5.2.4.7.

1. Turbine roll test

   This transient is imposed upon the plant during the hot functional test period for turbine cycle checkout. Reactor coolant pump power will be used to heat the reactor coolant to operating temperature (no-load conditions) and the steam generated will be used to

perform a turbine roll test. However, the plant cooldown during this test will exceed the 100°F per hour design rate.

The number of such test cycles is specified at 20 times, to be performed at the beginning of plant operating life prior to irradiation. This transient occurs before plant startup and the number of cycles is therefore independent of other operating transients.

2. Primary side hydrostatic test

The pressure tests include both stop and field hydrostatic tests which occur as a result of component or system testing. This hydro test is performed at a water temperature which is compatible with reactor vessel material ductility requirements and a test pressure of 3107 psig (1.25 times design pressure). In this test, the RCS is pressurized to 3107 psig coincident with steam generator secondary side pressure of 0 psig. The RCS is designed for 10 cycles of these hydrostatic tests, which are performed prior to plant startup. The number of cycles is independent of other operating transients.

Additional hydrostatic tests will be performed to meet the inservice inspection requirements of the ASME Code, Section XI, subarticle IS5-20 (Winter 1972 Addenda). A total of four such tests is expected. The increase in the fatigue usage factor caused by these tests is easily covered by the conservative number (200) of primary side leakage tests that are considered for design.

3. Secondary side hyrostatic test

The secondary side of the steam generator is pressurized to 1.25 design pressure with a minimum water temperature of 120°F coincident with the primary side at 0 psig.

For design purposes it is assumed that the steam generator will experience 10 cycles of this test.

These tests may be performed either prior to plant startup, or subsequently following shutdown for major repairs or both. The number of cycles is therefore independent of other operating transients.

4. Primary side leakage test

For design purposes the following is considered:

1) Subsequent to each time the primary system has been opened, a leakage test will be performed, 2) During this test the primary system pressure is, for design purposes, raised to 2500 psia, with the system temperature above the minimum temperature imposed by reactor vessel material ductility requirements, while the system is checked for leaks. 3) It is assumed that 200 cycles of this test will occur during the 40 year life of the plant. In actual practice the system leakage test is performed in accordance with ASME Section XI following each refueling outage.

5. Secondary side leakage test

During the life of the plant it may be necessary to check the secondary side of the steam generator (particularly, the manway closure) for leakage. For design purposes it is assumed that the steam generator secondary side is pressurized to just below its design pressure, to prevent the safety valves from lifting. In order not to exceed a secondary side to primary side pressure differential of 670 psi, the primary side must also be pressurized. The primary system must be above the minimum temperature imposed by reactor vessel material ductility requirements. It is assumed that this test is performed 80 times during the 40 year life of the plant.

6. Tube leakage test

During the life of the plant it may be necessary to check the steam generator for tube leakage and tube-to-tubesheet leakage. This is done by visual inspection of the underside (channel head side) of the tubesheet for water leakage, with the secondary side pressurized. Tube leakage tests are performed during plant cold shutdowns.

For these tests the secondary side of the steam generator is pressurized with water, initially at a relatively low pressure, and the primary system remains depressurized. The underside of the tubesheet is examined visually for leaks. If any are observed, the secondary side is then depressurized and repairs made by tube plugging. The secondary side is then repressurized (to a higher pressure) and the underside of the tubesheet is again checked for leaks. This process is repeated until all the leaks are repaired. The maximum (final) secondary side test pressure reached is 840 psig.

The total number of tube leakage test cycles is defined as 800 during the 40 year life of the plant. Following is a breakdown of the anticipated number of occurrences at each secondary side test pressure:

Test Pressure (psig)	Occurrences
200	400
400	200
600	120
840	80

Both the primary and secondary sides of the steam generators will be at ambient temperatures during these tests.

3.9N.1.2 Computer Programs Used in Analyses

The following computer programs have been used in dynamic and static analyses to determine mechanical loads, stresses, and deformations of Seismic Category I components and equipment. These are described and verified in Reference [1] and [13]. The description and verification procedure for STRUDL and NASTRAN is presented in reference [16] and [17].

1. WESTDYN - static and dynamic analysis of piping systems.

2. FIXFM - time history response of three dimensional structures; also called FIXFM3; now a subroutine combined into WESTDYN.

3. WESDYN-2 - piping system stress analysis from time history displacement data; also called WESTDYN; now a subroutine combined into WESTDYN.

4. STHRUST - hydraulic loads on loop components from blowdown information.

5. WECAN - finite element structural analysis, includes dynamic analysis of piping systems.

6. DARI-WOSTAS - Dynamic transient response analysis of reactor vessel internals.

7. STRUDL - linear elastic structural analysis.

8. NASTRAN - finite element structural analysis.

9. PC-PREPS - Linear Elastic Structural Analysis Computer Code for Pipe Supports.

3.9N.1.3 Experimental Stress Analysis

No experimental stress analysis methods are used for Category I systems or components. However, Westinghouse makes extensive use of measured results from prototype plants and various scale model tests as discussed in Section 3.9N.2.

3.9N.1.4 Considerations for the Evaluation of the Faulted Condition

3.9N.1.4.1 Loading Conditions

The structural stress analyses performed on the RCS consider the loadings specified as shown in Table 3.9N-2. These loads result from thermal expansion, pressure, weight, Operating Basis Earthquake (OBE), Safe Shutdown Earthquake (SSE), design basis loss of coolant accident, and plant operational thermal and pressure transients.

3.9N.1.4.2 Analysis of the Reactor Coolant Loop and Supports

The reactor coolant loop piping is evaluated in accordance with the criteria of ASME III, NB-3650 and Appendix F. The loads included in the evaluation result from the SSE, deadweight, pressure, and LOCA loadings (loop hydraulic forces, asymmetric subcompartment pressurization forces, and reactor vessel motion including uplift and lateral displacement). The results of the stress analysis of the reactor coolant loop piping are given in Table 3.9N-21.

The loads used in the analysis of the reactor coolant loop/supports System are described in detail below.

Pressure

Pressure loading is identified as either membrane design pressure or general operating pressure, depending upon its application. The membrane design pressure is used in connection

with the longitudinal pressure stress and minimum wall thickness calculations in accordance with the ASME Code.

The term operating pressure is used in connection with determination of the system deflections and support forces. The steady state operating hydraulic forces based on the system initial pressure are applied as general operating pressure loads to the reactor coolant loop model at change in direction or flow area.

Weight

A dead weight analysis is performed to meet ASME Code requirements by applying a 1.0 g load downward on the complete piping system. The piping is assigned a distributed mass or weight as a function of its properties. This method provides a distributed loading to the piping system as a function of the weight of the pipe and contained fluid during normal operating conditions.

Seismic for Unit 1

For the complete elimination of all the snubbers from the upper supports of the steam generators on the reactor coolant loops, the piping stress analysis of the reactor coolant loops for seismic loading is done using the time history method of seismic analysis. The response spectrum method of seismic analysis had been used in the piping stress analysis done previous to the elimination of the snubbers. The time history method of seismic analysis is now used in the piping stress analysis done for the elimination of the snubbers. Because the structural model includes the reactor coolant loops coupled to the internal concrete structure and the foundation spring, the time history accelerations at the foundation spring are applied to the model as the input seismic loading.

Two types of seismic moments are required in the calculation of piping stresses according to the equations found in Section NB-3650 of the ASME Code. The first type, for equations 9 and 13, includes the effects from just inertia. The second type, for equations 10 and 11, includes the effects from both inertia and anchor motions. The moments resulting from the time history method of seismic analysis inherently include the effects from both inertia and anchor motions. Therefore, these moments are used appropriately in equations 10 and 11, but they are also used conservatively in equations 9 and 13.

Seismic for Unit 2

The forcing functions for the reactor coolant loop seismic piping analyses are derived from dynamic response analyses of the Containment Building subjected to seismic ground motion. Input is in the form of floor response spectrum curves at various elevations within the Containment Building.

For the OBE and SSE seismic analyses Code Case N-411 damping is used in analysis of the reactor coolant loop/supports system.

In the response spectrum method of analysis, the total response loading obtained from the seismic analysis consists of two parts: the inertia response loading of the piping system and the differential anchor movements loading. Two sets of seismic moments are required to perform an ASME Code analysis. The first set includes only the moments resulting from inertia effects and these moments are used in the resultant moment (Mi) value for equations 9 and 13 of NB-3650.

The second set includes the moments resulting from seismic anchor motions and are used in equations 10 and 11 of NB-3650. Differential anchor movement is discussed in Section 3.7N.

Loss of Coolant Accident

Blowdown loads are developed in the broken and unbroken reactor coolant loops as a result of transient flow and pressure fluctuations following a postulated pipe break to the large branch nozzles of the reactor coolant loop (breaks 12 and 13, Table 3.6B-2). Structural consideration of dynamic effects of postulated pipe breaks requires postulation of a finite number of break locations. Postulated pipe break locations are given in Section 3.6B.

Broken loop time history dynamic analysis is performed for these postulated break cases. Hydraulic models are used to generate time- dependent hydraulic forcing functions used in the analysis of the reactor coolant loop for each break case. For a further description of the hydraulic forcing functions, refer to Section 3.6B.2.2.1.

Transients

The ASME Code requires satisfaction of certain requirements relative to operating transient conditions. Operating transients are tabulated in Section 3.9N.1.1.

The vertical thermal growth of the reactor pressure vessel nozzle centerlines is considered in the thermal analysis to account for equipment nozzle displacement as an external movement.

The hot moduli of elasticity E, the coefficient of thermal expansion at the metal temperature, the external movements transmitted to the piping due to vessel growth, and the temperature rise above the ambient temperature T, define the required input data to perform the flexibility analysis for thermal expansion.

To provide the necessary high degree of integrity for the RCS, the transient conditions selected for fatigue evaluation are based on conservative estimates of the magnitude and anticipated frequency of occurrence of the temperature and pressure transients resulting from various plant operation conditions.

3.9N.1.4.3 Reactor Coolant Loop Models and Methods

The analytical methods used in obtaining the solution consists of the transfer matrix method and stiffness matrix formulation for the static structural analysis, the response spectra method for seismic dynamic analysis for Unit 2, the time history method of seismic analysis for Unit 1, and time history integration method for the loss of coolant accident dynamic analysis.

The integrated reactor coolant loop/supports system model is the basic system model used to compute loadings on components, component supports, and piping. The system model includes the stiffness and mass characteristics of the reactor coolant loop piping and components, the stiffness of supports, the stiffnesses of auxiliary line piping which affect the system and the stiffness of piping restraints. The deflection solution of the entire system is obtained for the various loading cases from which the internal member forces and piping stresses are calculated.

Static

The reactor coolant loop/supports system model, constructed for the WESTDYN computer program, is represented by an ordered set of data which numerically describes the physical system. Figure 3.9N-1 shows an isometric line schematic of this mathematical model. The steam generator and reactor coolant pump vertical and lateral support members are described in Section 5.4.14.

The spatial geometric description of the reactor coolant loop model is based upon the reactor coolant loop piping layout and equipment drawings. The node point coordinates and incremental lengths of the members are determined from these drawings. Geometrical properties of the piping and elbows along with the modulus of elasticity E, the coefficient of thermal expansion, the average temperature change from ambient temperature T, and the weight per unit length are specified for each element. The primary equipment supports are represented by stiffness matrics which define restraint characteristics of the supports. Due to the symmetry of the static loadings, the reactor pressure vessel centerline is represented by a fixed boundary in the system mathematical model. The horizontal thermal growth of the reactor pressure vessel (RPV) is included in the loop model by modeling the RPV as an equivalent pipe from the RPV centerline to the RPV nozzle. The vertical thermal growth at the RPV nozzle location includes the thermal growth of the RPV support structure and RPV nozzle and is applied to the loop model as an external displacement.

The model is made-up of a number of sections, each having an overall transfer relationship formed from its group of elements. The linear elastic properties of the section are used to define the stiffness matrix for the section. Using the transfer relationship for a section, the loads required to suppress all deflections at the ends of the section arising from the thermal boundary forces for the section are obtained. These loads are incorporated into the overall load vector.

After all the sections have been defined in this matter, the overall stiffness matrix and associated load vector to suppress the deflection of all the network points is determined. By inverting the stiffness matrix, the flexibility matrix is determined. The flexibility matrix is multiplied by the negative of the load vector to determine the net-work point deflections due to the thermal and boundary force effects. Using the general transfer relationship, the deflections and internal forces are then determined at all node points in the system.

The static solutions for deadweight, thermal, and general pressure loading conditions are obtained by using the WESTDYN-7 computer program. The derivation of the hydraulic loads for the loss coolant accident analysis of the loop is covered in Section 3.6N.2.

Seismic for Unit 1

For the complete elimination of all the snubbers from the upper supports of the steam generators on the reactor coolant loops, the piping stress analysis of the reactor coolant loops for seismic loading is done using the time history method of seismic analysis. More specifically, the modal superposition method of time history analysis for nonlinear structures in the WECAN computer program is used. The analysis accounts for the replacement steam generator in addition to the elimination of the snubbers.

The reactor pressure vessel; the hot leg piping, replacement steam generator, crossover leg piping, reactor coolant pump, and cold leg piping for each of the four reactor coolant loops; the

internal structure, containment building, and foundation spring, as shown in Figure 3.7B-34; and the supports connecting the reactor coolant loops to the internal structure are all included in the structural model created for WECAN. The stiffness and mass characteristics of each of these items are accounted for in the structural model. Pipe, mass, stiffness matrix, and mass elements are used in the modeling of the equipment; pipe, elbow, mass, stiffness matrix, and mass elements are used in the modeling of the piping; stiffness matrix and mass elements are used in the modeling of the building and foundation spring; and stiffness matrix and nonlinear impact elements are used in the modeling of the supports.

The reactor pressure vessel is represented by nine discrete masses distributed along the vertical centerline of its shell and the vertical centerline of its core internals, as shown in Figure 3.9N-2. The replacement steam generator is represented by eight discrete masses distributed along the vertical centerline of its shell. The reactor coolant pump is represented by two discrete masses, one located at the intersection of the axial centerlines of the suction and discharge nozzles and the other located at the center of gravity of the motor.

The reactor pressure vessel supports, the steam generator columns, and the reactor coolant pump columns are active for all operating conditions of the plant. they are represented by stiffness matrices in the model. The horizontal component of the steam generator lower support, the steam generator upper support, and the reactor coolant pump tie rods are inactive for heatup, cooldown, and normal operating conditions of the plant, but they become active for seismic loading. They are represented by stiffness matrices and/or nonlinear impact elements in the model. Seismic loading is assumed to occur at the full power condition.

The modeling of the internal structure, containment building, and foundation spring is discussed in Section 3.7B.2.1.6. Two sets of stiffness properties for the containment building, uncracked and cracked, and three sets of stiffness values for the foundation spring, best-estimate, lower-bound, and upper-bound, are discussed. Six different models result from the various combinations of these stiffness characteristics. A separate time history analysis is made for each of the six models, and then the results for the six models are enveloped.

The time history accelerations for the safe shutdown earthquake are presented in Section 3.7B.1.2. The time history accelerations are applied to the model at the foundation spring as the input seismic loading. The horizontal time history acceleration shown in Figure 3.7B-14 is applied in the north-south and east-west directions. The vertical time history acceleration shown in Figure 3.7B-19 is applied in the vertical direction. A separate time history analysis is made for each of the three directions of shock, north-south, east-west, and vertical, and then the results for the three directions are combined by the square-root-sum-of-the-squares method. More specifically, to calculate the total seismic result for an item such as a piping load, support load, displacement, or acceleration, the maximum value of the item over time is determined for each of the three directions, and then the three maximum values are combined by the square-root-sum-of-the-squares method.

Analysis is also done for the operational basis earthquake. The input accelerations for the operational basis earthquake are one half the input accelerations for the safe shutdown earthquake, as noted in Section 3.7B.1.1.

The uncertainties in modeling discussed in Section 3.7B.2.9 are addressed by varying the time increment of the input time history acceleration. Three cases are considered. In the first case, the time increment is used as is. In the second case, the time increment is decreased by 10% to

raise the effective frequency content of the input time history acceleration by about 10%. In the third case, the time increment is increased by 10% to lower the effective frequency content by about 10%. A separate time history analysis is made for each of the three cases, and then the results for the three cases are enveloped.

Composite modal damping is used. The damping values presented in Sections 3.7B.1.3 and 3.7B.2.4 and Table 3.7B-1 are used, with 2% of critical damping being used for the piping for the operational basis earthquake, and 3% for the safe shutdown earthquake.

The modal superposition method of time history analysis is used. The modal analysis of the model is done with all the nonlinear impact elements inactive, but then the modal results are corrected to account for the activity of the nonlinear impact elements in the modal superposition analysis.

Seismic for Unit 2

The model used in the static analysis is modified for the dynamic analysis by including the mass characteristics of the piping and equipment. All of the piping loops are included in the system model. The effect of the equipment motion on the reactor coolant/supports system is obtained by modeling the mass and the stiffness characteristics of the equipment in the overall system model.

The steam generator is typically represented by three discrete masses. The lower mass is located at the intersection of the centerlines of the inlet and outlet nozzles of the steam generator. The middle mass is located at the steam generator upper support elevation, and the third mass is located at the top of the steam generator.

The reactor coolant pump is typically represented by a two discrete mass model. The lower mass is located at the intersection of the centerlines of the pump suction and discharge nozzles. The upper mass is located near the center of gravity of the motor.

The reactor vessel and core internals are typically represented by approximately 10 discrete masses. The masses are lumped at various locations along the length of the vessel and along the length of the representation of the core internals, as shown in Figure 3.9N-2.

The component upper and lower lateral supports are inactive during plant heatup, cooldown and normal plant operating conditions. However, these restraints become active when the plant is at power and under the rapid motions of the reactor coolant loop components that occur from the dynamic loadings, and are represented by stiffness matrices and/or individual tension or compression spring members in the dynamic model. The analyses are performed at the full power condition.

The response spectra method employs the lumped mass technique, linear elastic properties, and the principal of modal superposition. The floor response spectra are applied along both horizontal axes and the vertical axis simultaneously.

From the mathematical description of the system, the overall stiffness matrix [K] is developed from the individual element stiffness matrices using the transfer matrix method. After deleting the rows and columns representing rigid restraints, the stiffness matrix is revised to obtain a reduced

stiffness matrix [K_R] associated with mass degree of freedom only. From the mass matrix and the reduced stiffness matrix, the natural frequencies and the normal modes are determined. The modal participation factor matrix is computed and combined with the appropriate response spectra value to give the modal amplitude for each mode. The total modal amplitude is obtained by taking the square root of the sum of the squares of the contributions for each direction.

The modal amplitudes are then converted to displacements in the global coordinate system and applied to the corresponding mass point. From these data the forces, moments, deflections, rotations, support reactions and piping stresses are calculated for all significant modes.

The total seismic response is computed by combining the contributions of the significant modes by the square root of the sum of the squares method and accounting for closely spaced modes as described in Section 3.7N.

Loss of Coolant Accident

The mathematical model used in the static analyses is modified for the loss of coolant accident analyses to represent the severance of the reactor coolant loop piping nozzles at the postulated break location. Modifications include addition of the mass characteristic of the piping and equipment.

The time-history hydraulic forces at the node points are combined to obtain the forces and moments acting at the corresponding structural lumped mass node points.

The dynamic structural solution for the full power loss of coolant accident and steam line break is obtained by using a modified-predictor-corrector-integration technique and normal mode theory.

When elements of the system can be represented as single acting members (tension or compression members), they are considered as nonlinear elements, which are represented mathematically by the combination of a gap, a spring, and a viscous damper. The force in this nonlinear element is treated as an externally applied force in the overall normal mode solution. Multiplied nonlinear elements can be applied at the same node, if necessary.

The time-history solution is performed in subprogram FIXFM3. The input to this subprogram consists of the natural frequencies, normal modes, applied forces and nonlinear elements. The natural frequencies and normal modes for the modified reactor coolant loop dynamic model are determined with the WESTDYN program. To properly simulate the release of the strain energy in the pipe, the internal forces in the system at the postulated break location due to the initial steady state hydraulic forces, thermal forces, and weight forces are determined. The release of the strain energy is accounted for by applying the negative of these internal forces as a step function loading. The initial conditions are equal to zero because the solution is only for the transient problem (the dynamic response of the system from the static equilibrium position). The time-history displacement solution of all dynamic degrees of freedom is obtained using subprogram FIXFM and employing 4 percent critical damping.

The loss of coolant accident displacements of the reactor vessel are applied in time-history form as input to the dynamic analysis of the reactor coolant loop. The loss of coolant accident analysis of the reactor vessel includes all the forces acting on the vessel including internals

reactions, cavity pressure loads, and loop mechanical loads. The reactor vessel analysis is described in Section 3.9N.1.4.6.

The main loop piping breaks and, therefore, the vessel cavity pressurization effects are not part of the design basis (because they are excluded as dynamic effects section 3.1.1.4). However, they are included as a conservative estimate of vessel motion due to LOCA. If required, vessel motion for the postulated branch nozzle breaks can be used in place of the conservative main loop piping breaks discussed in this section.

The resultant asymmetric external pressure loads on the RCP and steam generator resulting from a postulated pipe rupture and pressure buildup in the loop compartments are applied to the same integrated RCL/supports system model used to compute loadings on the components, component supports, and RCL piping as discussed above. The response of the entire system is obtained for the various external pressure loading cases from which the internal member forces and piping stresses are calculated. For each pipe break case considered, the equipment support loads and piping stresses resulting from the external pressure loading are added to the support loads and piping stresses calculated using the loop LOCA hydraulic forces and RPV motion.

The break locations considered for subcompartment pressurization are those postulated for the RCL LOCA analysis as discussed in Section 3.6N and WCAP-8172 (Reference [1] of Section 3.6N). The asymmetric subcompartment pressure loads are provided to Westinghouse. The analysis to determine these loads is discussed in Section 6.2.

The time-history displacement response of the loop is used in computing support loads and in performing stress evaluation of the reactor coolant loop piping.

The time-history displacements of the FIXFM3 program are used as input to program WESDYN to determine the internal forces, deflections, and stresses at each end of the piping elements. For this calculation the displacements are treated as imposed deflections on the reactor coolant loop masses.

Transient

Operating transients in a nuclear power plant cause thermal and/or pressure fluctuations in the reactor coolant fluid. The thermal transients cause time varying temperature distributions across the pipe wall. These temperature distributions resulting in pipe wall stresses may be further subdivided in accordance with the Code into three parts, a uniform, a linear, and a nonlinear portion. The uniform portion results in general expansion loads. The linear portion causes a bending moment across the wall and the nonlinear portion causes a skin stress.

The transients as defined in Section 3.9N.1.1 are used to define the fluctuations in plant parameters. A one dimensional finite difference heat conduction program is used to solve the thermal transient problem. The pipe is represented by at least 50 elements through the thickness of the pipe. The convective heat transfer coefficient employed in this program represents the time varying heat transfer due to free and forced convection. The outer surface is assumed to be adiabatic while the inner surface boundary experiences the temperature of the coolant fluid. Fluctuations in the temperature of the coolant fluid produce a temperature distribution through the pipe wall thickness which varies with time. An arbitrary temperature distribution across the wall is shown in Figure 3.9N-3.

The average through-wall temperature, T_A, is calculated by integrating the temperature distribution across the wall. This integration is performed for all time steps so that T_A is determined as a function of time.

$$T_A(t) = \frac{1}{H}\int_0^H T(X, t)\ dX$$

The range of temperature between the largest and smallest value of T_A is used in the flexibility analysis to generate the moment loadings caused by the associated temperature changes.

The thermal moment about the mid-thickness of the wall caused by the temperature distribution through the wall is equal to:

$$M = E\alpha\int_0^H \left(X - \frac{H}{2}\right) T(X, t)dX$$

The equivalent thermal moment produced by the linear thermal gradient as shown in Figure 3.9N-3 about the mid-wall thickness is equal to:

$$M_L = E\alpha_{12}\frac{\Delta T_1}{}H^2$$

Equating M_L and M, the solution for T_1 as a function of time is:

$$\Delta T_1(t) = \frac{12}{M^2}\int_0^H \left(X - \frac{H}{2}\right) T(X, t)dX$$

The maximum nonlinear thermal gradient, T_2, will occur on the inside surface and can be determined as the difference between the actual metal temperature on this surface and half of the average linear thermal gradient plus the average temperature.

$$\Delta T_{21}(t) = \left|T(0, t) - T_A(t)\right| - \frac{\left|\Delta T_1(t)\right|}{2}$$

Load Set Generation

A load set is defined as a set of pressure loads, moment loads, and through-wall thermal effects at a given location and time in each transient. The method of load set generation is based on Reference [2]. The through-wall thermal effects are functions of time and can be sub-divided into four parts:

1. Average temperature (T_A) is the average temperature through-wall of the pipe which contributes to general expansion loads.

2. Radial linear thermal gradient which contributes to the through-wall bending moment (ΔT_1).

3. Radial nonlinear thermal gradient (ΔT_2) which contributes to a peak stress associated with shearing of the surface.

4. Discontinuity temperature ($T_A - T_B$) represents the difference in average temperature at the cross sections on each side of a discontinuity.

Each transient is described by at least two load sets representing the maximum and minimum stress state during each transient. The construction of the load sets is accomplished by combining the following to yield the maximum (minimum) stress state during each transient.

1. ΔT_1

2. ΔT_2

3. $\alpha_A T_A - \alpha_B T_B$

4. Moment loads due to T_A.

5. Pressure loads.

This procedure produces at least twice as many load sets as transients for each point.

As a result of the normal mode spectral technique employed in the seismic analysis the, load components cannot be given signed values. Seismic loads are considered in the fatique analysis as having either a positive or negative sign, thus ensuring the most conservative combination of seismic loads are used in the stress evaluation.

For all possible load set combinations, the primary plus secondary and peak stress intensities, fatigue reduction factors (Ke) and cumulative usage factors, U, are calculated. The WESTDYN program is used to perform this analysis in accordance with the ASME Code, Section III, Subsection NB-3650. Alternatively, detailed finite element stress analyses may be used to determine primary plus secondary and peak stress intensities, for the load set combinations. Since it is impossible to predict the order of occurrence of the transients over a 40 year life it is assumed that the transients can occur in any sequence. This is a very conservative assumption.

The combination of load sets yielding the highest alternating stress intensity range is used to calculate the incremental usage factor. The next most severe combination is then determined and the incremental usage factor calculated. This procedure is repeated until all combinations having allowable cycles $<10^6$ are formed. The total cumulative usage factor at a point is the summation of the incremental usage factors.

3.9N.1.4.4 Primary Component Supports Models and Methods

The static and dynamic structural analyses employ the matrix method and normal mode theory for the solution of lumped parameter, multimass structural models. The equipment support

structure models are dual purpose since they are required: 1) to quantitatively represent the elastic restraints which the supports impose upon the loop, and 2) to evaluate the individual support member stresses due to the forces imposed upon the supports by the loop.

A description of the supports is found in Section 5.4.14. Detailed models are developed using beam elements and plate elements, where applicable.

The reactor vessel supports are modeled using the NASTRAN computer program. Structure geometry, topology and member properties are used in the modeling. NASTRAN is a general purpose finite element computer program. Its capability, broad in scope, permits the solution of large, complicated structural problems. Static, dynamic, modal, plastic and thermal analyses are possible. NASTRAN employs the matrix displacement method for each finite element in the idealized structure. A “wave front” direct-solution technique is employed to give accurate results in a minimum of computer time. The analysis solution output includes geometry plots, nodal displacements, element stresses and nodal forces. Steam generator and reactor coolant pump supports are modeled as linear or non-linear springs.

The steam generator lower beam was modeled using the STRUDL Program. STRUDL, part of the integrated civil engineering computer system (ICES), is widely used for the analysis and design of structures. It is applicable to linear elastic two and three-dimensional frame or truss structures, employs the stiffness formulation, and is valid only for small displacements.

Structure geometry, topology, element orientation and cross-section properties are described in free format. Member and support joint releases, such as pins and rollers, are specified. Otherwise, six restraint components are assumed at each end of each member and at each support joint.

The STRUDL system performs structural stability and equilibrium checks during the solution process and prints error messages if these conditions are violated. Type, location, and magnitude of applied loads or displacements are specified for any number of loading conditions.

One important feature of STRUDL is that any desired changes, deletions, or additions can be made to the structural model during the solution process. This produces results for a number of structure configurations, each with any number of loading conditions.

Printed output content, specified by input commands, includes member forces and distortions, joint displacements, support joint reactions, and member stresses.

The steam generator upper lateral supports are modeled/analyzed using the STRUDL computer program. The following discussion about snubbers applies only for Unit 2. The snubbers have been eliminated for Unit 1. The average snubber stiffness is combined with the stiffness of the entire upper support and used in the seismic analysis. Non-linear tension and compression spring rates are used in the LOCA model. The spring rates are 5400 k/in compression and 4300 k/in tension. Entrapped air is bled from the system during installation and therefore has no effect on the snubber operation. The characteristics of the fluid over a temperature range of 50°F to 500°F were considered, and it was determined the temperature of the fluid has no effect on snubber operation over its operating temperature range (<300°F). Load conditions/transients analyzed include normal operating and seismic (OBE), and SSE seismic combined with LOCA. The load ratios of actual load/rated capacity are: normal & OBE, 0.460; SSE seismic combined with LOCA, 0.265.

For each operating condition, the loads (obtained from the RCL analysis) acting on the support structures are appropriately combined. The main loop piping breaks are not part of the design basis (excluded as dynamic effects, Section 3.1.1.4). Reactor coolant loop normal and upset condition thermal expansion loads are treated as primary loadings for the primary component supports. The adequacy of each member of the steam generator supports, reactor coolant pump supports, and piping restraints is verified by solving the ASME III Subsection NF stress and interaction equations. The adequacy of the RPV support structure is verified using the NASTRAN computer program and comparing the resultant stresses to the criteria given in ASME III Subsection NF.

Tables 3.9N-14 through 3.9N-17 present maximum stresses in each member of the steam generator, reactor coolant pump, and pressurizer support structures expressed as a percentage of maximum permissible values for all operating condition loadings. The loads on the reactor vessel supports and the resulting stresses are shown in Table 3.9N-19. The above loads and stresses include the effects of loads resulting from asymmetric subcompartment pressurization.

3.9N.1.4.5 Analysis of Primary Components

Equipment which serves as part of the pressure boundary in the reactor coolant loop includes the steam generators, the reactor coolant pumps, the pressurizer, and the reactor vessel. This equipment is evaluated for the loading combinations outlined in Table 3.9N-2. The equipment is analyzed for: 1) the normal loads of deadweight, pressure and thermal, 2) mechanical transients of OBE, SSE, and pipe ruptures, including the effects of asymmetric subcompartment pressurization and 3) pressure and temperature transients outlined in Section 3.9N.1.1.

The results of the reactor coolant loop analysis are used to determine the loads acting on the nozzles and the support/component interface locations. These loads are supplied for all loading conditions on an “umbrella” load basis. That is, on the basis of previous plant analyses a set of loads are determined which should be larger than those seen in any single plant analysis. The umbrella loads represent a conservative means of allowing detailed component analysis prior to the completion of the system analysis. Upon completion of the system analysis, conformance is demonstrated between the actual plant loads and the loads used in the analyses of the components. Any deviations where the actual load is larger than the umbrella load will be handled by individualized analysis.

Seismic analyses are performed individually for the reactor coolant pump, the pressurizer, and the steam generator. Detailed and complex dynamic models are used for the dynamic analyses. The response spectra corresponding to the building elevation at the highest component/building attachment elevation is used for the component analysis. Seismic analyses for the steam generator and reactor coolant pump are performed using 2 percent damping for the OBE and 4 percent damping for the SSE. The reactor pressure vessel is qualified by static stress analysis based on loads that have been derived from dynamic analysis.

Reactor coolant pressure boundary components are further qualified to ensure against unstable crack growth under faulted conditions by performing detailed fracture analyses of the critical areas of this boundary. Actuation of the ECCS produces relatively high thermal stresses in the system. Regions of the pressure boundary which come into contact with ECCS water are given primary consideration. These regions include the reactor vessel beltline region, the reactor vessel inlet nozzles, and the safety injection nozzles in the piping system.

Two methods of analysis are used to evaluate thermal effects in the regions of interest. The first method is linear elastic fracture mechanics (LEFM). The LEFM approach to the design against failure is basically a stress intensity consideration in which criteria are established for fracture instability in the presence of a crack. Constantly, a basic assumption employed in LEFM is that a crack or crack-like defect exists in the structure. The essence of the approach is to relate the stress field developed in the vicinity of the crack tip to the applied stress on the structure, the material properties, and the size of defect necessary to cause failure.

The elastic stress field at the crack tip in any cracked body can be described by a single parameter designated as the stress intensity factor, K. The magnitude of the stress intensity factor K is a function of the geometry of the body containing the crack, the size and location of the crack, and the magnitude and distribution of the stress.

The criterion for failure in the presence of a crack is that failure will occur whenever the stress intensity factor exceeds some critical value. For the opening mode of loading (stresses perpendicular to the major plane of the crack) the stress intensity factor is designated as K_I and the critical stress intensity factor is designated K_{IC}. Commonly called the fracture toughness, K_{IC} is an inherent material property which is a function of temperature and strain rate. Any combination of applied load, structural configuration, crack geometry and size which yields a stress intensity factor K_{IC} for the material will result in crack instability.

The criterion of the applicability of LEFM is based on plasticity considerations at the postulated crack tip. Strict applicability (as defined by ASTM) of LEFM to large structures where plane strain conditions prevail requires that the plastic zone developed at the tip of the crack does not exceed 2.25 percent of the crack depth. In the present analysis, the plastic zone at the tip of the postulated crack can reach 20 percent of the crack depth. However, LEFM has been successfully used quite often to provide conservative brittle fracture prevention evaluations, even in cases where strict applicability of the theory is not permitted due to excessive plasticity. Recently, experimental results from Heavy Section Steel Technology (HSST) Program intermediate pressure vessel tests have shown that LEFM can be applied conservatively as long as the pressure component of the stress does not exceed the yield strength of the material. The addition of the thermal stresses, calculated elastically, which results in total stresses in excess of the yield strength does not affect the conservatism of the results, provided that these thermal stresses are included in the evaluation of the stress intensity factors. Therefore, for faulted condition analyses, LEFM is considered applicable for the evaluation of the vessel inlet nozzle and beltline region.

In addition, it has been well established that the crack propagation of existing flaws in a structure subjected to cyclic loading can be defined in terms of fracture mechanics parameters. Thus, the principles of LEFM are also applicable to fatigue growth of a postulated flaw at the vessel inlet nozzle and beltline region.

For the safety injection and charging line nozzles, which are fabricated from Type 304 stainless steel, LEFM is not applicable because of extreme ductility of the material. For these nozzles, the thermal effects are evaluated using the principles of Miner’s hypothesis of linear cumulative damage in conjunction with fatigue data from constant stress or strain fatigue tests. The cumulative usage fatigue defined as the sum of the ratios of the number of cycles of each transient (n) to the allowable number of cycles for the stress range associated with the transient (N) must not exceed 1.0.

An example of a faulted condition evaluation carried out according to the procedure discussed above in given in Reference [3]. This report discusses the evaluation procedure in detail as applied to a severe faulted condition (a postulated loss of coolant accident), and concludes that the integrity of the reactor coolant pressure boundary would be maintained in the event of such an accident.

The pressure boundary portions of Class 1 valves in the RCS are designed and analyzed according to the requirements of NB-3500 of the ASME Code, Section III. These valves are identified in Section 3.9N.3.2.

Valves in sample lines connected to the RCS are not considered to be ANS Safety Class 1 nor ASME Class 1. This is because the nozzles where the line connect to the primary system piping are orificed to a 3/8 inch hole. This hole restricts the flow such that loss through a severance of one of these lines can be made up by normal charging flow.

3.9N.1.4.6 Dynamic Analysis of Reactor Pressure Vessel for Postulated Loss of Coolant Accident

1. Introduction

This section presents the method of computing the reactor pressure vessel response to a postulated loss of coolant accident (LOCA). The dynamic analysis of the reactor vessel was performed prior to the leak-before-break update to GDC-4 and therefore included the effects of main loop piping breaks and reactor vessel cavity pressurization. These effects, no longer required for the dynamic analysis, are conservative loadings for the postulated LOCA analysis of the reactor vessel and internals discussed in this section and identified in Table 3.6B-2. The structural analysis considers simultaneous application of the time-history loads on the reactor vessel resulting from the reactor coolant loop mechanical loads, internal hydraulic pressure transients, and reactor cavity pressurization (for postulated breaks in the reactor coolant pipe at the vessel nozzles). The vessel is restrained by reactor vessel support pads and shoes beneath four of the reactor vessel nozzles and the reactor coolant loops with the primary supports of the steam generators and the reactor coolant pumps.

Pipe displacement restraints installed in the primary shield wall limit the break opening area of the vessel nozzle pipe breaks to less than 144 square inches. This break area was determined to be an upper bound by using worst case vessel and pipe relative motions based on similar plant analyses. Detailed studies have shown that pipe breaks at the hot or cold leg reactor vessel nozzles, even with a limited break area, would give the highest reactor vessel support loads and the highest vessel displacements, primarily due to the influence of reactor cavity pressurization. The dynamic analysis of the reactor vessel was performed prior to the leak-before-break update to GDC-4 and therefore included the effects of main loop piping breaks and reactor vessel cavity pressurization. These effects, no longer required for the dynamic analysis, are conservative loadings for the postulated LOCA analysis of the reactor vessel and internals discussed in this section and identified in Table 3.6B-2. By considering these breaks, the most severe reactor vessel support loads are determined. For completeness, an additional break outside the shield wall, for which there is no cavity pressurization, was also analyzed, specifically, the pump outlet nozzle pipe break.

2. Interface Information

Asymmetric reactor cavity pressurization loads were provided to Westinghouse by Gibbs and Hill, Inc.

All other input information was developed within Westinghouse. This information includes: reactor internals properties, loop mechanical loads and loop stiffness, internal hydraulic pressure transients, and reactor support stiffnesses. These inputs allowed formulation of the mathematical models and performance of the analyses, as will be described.

3. Loading Conditions

The dynamic analysis of the reactor vessel was performed prior to the leak-before-break update to GDC-4 and therefore included the effects of main loop piping breaks and reactor vessel cavity pressurization. These effects, no longer required for the dynamic analysis, are conservative loadings for the postulated LOCA analysis of the reactor vessel and internals discussed in this section and identified in Table 3.6B-2.

Following a postulated pipe rupture at the reactor vessel nozzle, the reactor vessel is excited by time-history forces. As previously mentioned, these forces are the combined effect of three phenomena: (1) reactor coolant loop mechanical loads, (2) reactor cavity pressurization forces and (3) reactor internal hydraulic forces.

The reactor coolant loop mechanical forces are derived from the elastic analysis of the loop piping for the postulated break. This analysis is described in Section 3.9N.1.4.3. The loop mechanical forces which are released at the broken nozzle are applied to the vessel in the RPV blowdown analysis.

Reactor cavity pressurization forces arise for the pipe breaks at the vessel nozzles from the steam and water which is released into the reactor cavity through the annulus around the broken pipe. The reactor cavity is pressurized asymmetrically with higher pressure on the side of the broken pipe resulting in horizontal forces applied to the reactor vessel. Smaller vertical forces arising from pressure on the bottom of the vessel and the vessel flanges are also applied to the reactor vessel. The cavity pressure analysis is described in Section 6.2.

The internals reaction forces develop from asymmetric pressure distributions inside the reactor vessel. For a vessel inlet nozzle break and pump outlet nozzle break, the depressurization wave path is through the broken loop inlet nozzle and into the region between the core barrel and reactor vessel. This region is called the downcomer annulus. The initial waves propagate up, down and around the downcomer annulus and up through the fuel. In the case of an RPV outlet nozzle break the wave passes through the RPV outlet nozzle and directly into the upper internals region, depressurizes the core, and enters the downcomer annulus from the bottom of the vessel. Thus, for an outlet nozzle break, the downcomer annulus is depressurized with much smaller differences in pressure horizontally across the core barrel than for the inlet break. For both the inlet and outlet nozzle breaks, the depressurization waves continue their propagation by reflection and translation through the reactor vessel fluid but the initial depressurization wave has the greatest effect on the loads.

The reactor internals hydraulic pressure transients were calculated including the assumption that the structural motion is coupled with the pressure transients. This phenomena has been referred to as hydroelastic coupling or fluid-structure interaction. The hydraulic analysis considers the fluid-structure interaction of the core barrel by accounting for the deflections of constraining boundaries which are represented by masses and springs. The dynamic response of the core barrel in its beam bending mode responding to blowdown forces compensates for internal pressure variation by increasing the volume of the more highly pressurized regions. The analytical methods used to develop the reactor internals hydraulics are described in WCAP- 8708[8].

4. Reactor Vessel and Internals Modeling

The reactor vessel is restrained by two mechanisms: (1) the four attached reactor coolant loops with the steam generator and reactor coolant pump primary supports and (2) four reactor vessel supports, two beneath reactor vessel inlet nozzles and two beneath reactor vessel outlet nozzles. The reactor vessel supports are described in Section 5.4.14 and are shown in Figures 5.4-12, and 3.8-17. The support shoe provides restraint in the horizontal directions and for downward reactor vessel motion.

The reactor vessel model consists of two non-linear elastic models connected at a common node. One model represents the dynamic vertical characteristics of the vessel and its internals, and the other model represents the translational and rotational characteristics of the structure. These two models are combined in the DARI-WOSTAS code[1] to represent motion of the reactor vessel and its internals in the plane of the vessel centerline and the broken pipe centerline.

The model for horizontal motion is shown in Figure 3.9N-12. Each node has one translational and one rotational degree of freedom in the vertical plane containing the centerline of the nozzle attached to the broken pipe and the centerline of the vessel. A combination of beam elements and concentrated masses are used to represent the components including the vessel, core barrel, neutron panels, fuel assemblies, and upper support columns. Connections between the various components are either pin-pin rigid links, translational impact springs with damping or rotational springs.

The model for vertical motion is shown in Figure 3.9N-13. Each mass node has one translational degree of freedom. The structure is represented by concentrated masses, springs, dampers, gaps, and frictional elements. The model includes the core barrel, lower support columns, bottom nozzles, fuel rods, top nozzles, upper support structure, and reactor vessel.

The horizontal and vertical models are coupled at the elevation of the primary nozzle centerlines. Node 1 of the horizontal model is coupled with node 2 of the vertical model at the reactor vessel nozzle elevation. This coupled node has external restraints characterized by a 3 x 3 matrix which represents the reactor coolant loop stiffness characteristics, by linear horizontal springs which describe the tangential resistance of the supports, and by individual non-linear vertical stiffness elements which provide downward restraint only. The supports as represented in the horizontal and vertical models (Figures 3.9N-12 and 3.9N-13) are not indicative of the complexity of the support system used in the analysis. The individual supports are located at the actual support pad locations and accurately represent the independent non-linear behavior of each support.

5. Analytical Methods

The time-history effects of the cavity pressurization loads, internals loads and loops mechanical loads are combined and applied simultaneously to the appropriate nodes of the mathematical model of the reactor vessel and internals. The analysis is performed by numerically integrating the differential equations of motion to obtain the transient response. The output of the analysis includes the displacements of the reactor vessel and the loads in the reactor vessel supports which are combined with other applicable faulted condition loads and subsequently used to calculate the stresses in the supports. Also, the reactor vessel displacements are applied as a time-history input to the dynamic reactor coolant loop blowdown analysis. The resulting loads and stresses in the piping components and supports include both loop blowdown loads and reactor vessel displacements. Thus, the effect of vessel displacements upon loop response and the effect of loop blowdown upon vessel displacements are both evaluated.

6. Results of the Analysis

As described, the reactor vessel and internals were analyzed for three postulated break locations. Table 3.9N-12 summarizes the displacements and rotations of and about a point representing the intersection of the centerline of the nozzle attached to the leg in which the break was postulated to occur and the vertical centerline of the reactor vessel. Positive vertical displacement is up and positive horizontal displacement is away from and along the centerline of the vessel nozzle in the loop in which the break was postulated to occur. These displacements were calculated using an assumed break opening area for the postulated pipe ruptures at the vessel nozzles of 144 in^2 and a double-ended rupture at the pump outlet nozzle. These areas are estimated prior to performing the analysis. Following the reactor coolant system structural analysis, the relative motions of the broken pipe ends are obtained from the reactor coolant loop blowdown analysis. The actual break opening area is then verified to be less than the estimated area used in the analysis and assures that the analysis is conservative.

The maximum loads induced in the vessel supports due to the postulated pipe break are given in Table 3.9N-13. These loads are per vessel support and are applied at the vessel nozzle pad. It is conservatively assumed that the maximum horizontal and vertical loads occur simultaneously and on the same support, even though the time-history results show that these loads occur neither simultaneously nor on the same support. The largest vertical loads are produced on the support opposite the broken nozzle. The largest horizontal loads are produced on the supports which are perpendicular to the broken nozzle horizontal centerline. Peak loads are based on a dynamic analysis of breaks 12 and 13 as defined in Table 3.6B-2 and Figure 3.6B-9. Analysis results are detailed in the Westinghouse Topical Report WCAP-12106 [15].

3.9N.1.4.7 Stress Criteria for Class 1 Components and Component Supports

All Class 1 components and supports are designed and analyzed for the design, normal, upset, and emergency conditions to the rules and requirements of the ASME Code, Section III. The design analysis or test methods and associated stress or load allowable limits that will be used to evaluation of faulted condition are those that are defined in Appendix F of the ASME code with the following supplementary option:

1. Elastic/inelastic system analysis and component/test load method

The test load method given in F 1370(d) is an acceptable method of qualifying components in lieu of satisfying the stress/load limits established for the component analysis.

The test load option was used to qualify the reactor pressure vessel nozzle support pads. To duplicate the loads that act on the pads during faulted conditions, the tests, which utilized a one-eighth linear scale model, were performed by applying a unidirectional static load to the nozzle pad. The load on the nozzle pad was reacted by a support shoe which was mounted to the test fixture.

This modeling and application of load thus allows the maximum load capacity of the support pads to be accurately established. The test load, LT, was then determined by multiplying the maximum collapse load by sixty-four (ratio of prototype area to model area) and including temperature effects in accordance with the rules of the ASME Code, Section III.

The loads on the reactor vessel support pads, as calculated in the system analysis for faulted conditions are limited to the value of .80 LT. The tests performed and the limits established for test load method ensure that the experimentally obtained value for LT is accurate and that the support pad design is adequate for its intended function.

Loading combinations and allowable stresses for ASME Code, Section III, Class 1 components and supports are given in Table 3.9N-2 and 3.9N-3. For component support bolting materials, the stress limits and increase factors applicable for the faulted condition are provided in the discussion of Regulatory Guide 1.124 in FSAR Appendix 1A(N). For faulted condition evaluations, the effects of the SSE and loss of coolant accident (LOCA) are combined using the square root of the sum of the squares method. Information supporting this methodology is provided in References [18], [19], [20] and [21].

For faulted condition analysis of Class 1 branch piping attached to the reactor coolant loop, Equation (9) of ASME III Subsection NB-3652 is applied with a stress limit of 3.0 Sm. This criterion provides sufficient assurance that the piping will not collapse or experience cross distortion such that the function of the system would be impaired. The basis for this position is described in the Westinghouse response to NRC Question 110.34 on the RESAR-414 application (Docket No. STN 50-572), which subsequently received a Preliminary Design Approval (PDA) in November 1978.

3.9N.1.4.8 Analytical Methods for RCS Class 1 Branch Lines

The analytical methods used to obtain the solution consist of the transfer matrix method and stiffness matrix formulation for the static structural analysis, the response spectrum method for seismic dynamic analysis, and dynamic structural analysis for the effect of a reactor coolant loop pipe break.

The integrated Class 1 piping/supports system model is the basic system model used to compute loadings on components, component and piping supports, and piping. The system models include the stiffness and mass characteristics of the Class 1 piping components, the reactor coolant loop, and the stiffness of supports which affect the system response. The deflection

solution of the entire system is obtained for the various loading cases from which the internal member forces and piping stresses are calculated.

Static

The Class 1 piping system models are constructed for the WESTDYN computer program, which numerically describes the physical system. A network model is made up of a number of sections, each having an overall transfer relationship formed from its group of elements. The linear elastic properties of the section are used to define the characteristic stiffness matrix for the section. Using the transfer relationship for a section, the loads required to suppress all deflections at the ends of the section arising from the thermal and boundary forces for the section are obtained.

After all the sections have been defined in this manner, the overall stiffness matrix and associated load vector to suppress the deflection of all the network points is determined. By inverting the stiffness matrix, the flexibility matrix is determined. The flexibility matrix is multiplied by the negative of the load vector to determine the network point deflections due to the thermal and boundary force effects. Using the general transfer relationship, the deflections and internal forces are then determined at all node points in the system. The support loads are also computed by multiplying the stiffness matrix by the displacement vector at the support point.

Seismic

The models used in the static analyses are modified for use in the dynamic analyses by including the mass characteristics of the piping and equipment.

The lumping of the distributed mass of the piping systems is accomplished by locating the total mass at points in the system which will appropriately represent the response of the distributed system. Effects of the primary equipment motion, that is, reactor vessel, steam generator, reactor coolant pump, and pressurizer, on the Class 1 piping system are obtained by modeling the mass and the stiffness characteristics of the primary equipment and loop piping in the overall system model.

The supports are represented by stiffness matrices in the system model for the dynamic analysis. Shock suppressors which resist rapid motions are also included in the analysis. The solution for the seismic disturbance employs the response spectra method. This method employs the lumped mass technique, linear elastic properties, and the principle of model superposition.

The total response obtained from the seismic analysis consists of two parts: the inertia response of the piping system and the response from differential anchor motions. The stresses resulting from the anchor motions are considered to be secondary and, therefore, are included in the fatigue evaluation.

Loss of Coolant Accident

The mathematical models used in the seismic analyses of the Class 1 lines are also used for RCL pipe break effect analysis. To obtain the proper dynamic solution both for lines attached to the unbroken loops and lines attached to the broken loop, the time history deflections from the analysis of the reactor coolant loop are applied at branch nozzle connections.

Fatigue

A thermal transient heat transfer analysis is performed for each different piping component on all the Class 1 branch lines. The normal, upset, and test condition transients identified in Section 3.9.1.1 are considered in the fatigue evaluation.

The thermal quantities T_1, T_2 and a_aT_a, $-a_bT_b$ are calculated on a time history basis, using a one-dimensional finite difference heat transfer computer program. Stresses due to these quantities were calculated for each time increment using the methods of NB-3650 of ASME III.

For each thermal transient, two loadsets are defined, representing the maximum and minimum stress states for that transient.

As a result of the normal mode spectral technique employed in the seismic analysis, the load components cannot be given signed values. Eight load sets are used to represent all possible sign permutations of the seismic moments at each point, thus insuring the most conservative combinations of seismic loads are used in the stress evaluation.

The WESTDYN computer program is used to calculate the primary-plus-secondary and peak stress intensity ranges, fatigue reduction factors and cumulative usage factors for all possible load set combinations. It is conservatively assumed that the transients can occur in any sequence, thus resulting in the most conservative and restrictive combinations of transients.

The combination of load sets yielding the highest alternating stress intensity range is determined and the incremental usage factor calculated. Likewise, the next most severe combination is then determined and the incremental usage factor calculated. This procedure is repeated until all combinations having allowable cycles $<10^6$ are formed. The total cumulative usage factor at a point is the summation of the incremental usage factors.

3.9N.2 DYNAMIC TESTING AND ANALYSIS

3.9N.2.1 Preoperational Vibration and Dynamic Effects Testing on Piping

This section is discussed in 3.9B.2.1.

3.9N.2.2 Seismic Qualification Testing of Safety-Related Mechanical Equipment

The operability of Category I mechanical equipment must be demonstrated if the equipment is determined to be active, i.e., mechanical operation is relied on to perform a safety function. The operability of active Class 2 and 3 pumps, active Class 1, 2, or 3 valves, and their respective drives, operators and vital auxiliary equipment will be shown by satisfying the criteria given in Section 3.9N.3.2. Other active mechanical equipment will be shown operable by either testing, analysis or a combination of testing and analysis. The operability programs implemented on this other active equipment will be similar to the program described in Section 3.9N.3.2 for pumps and valves. Testing procedures similar to the procedures outlined in Section 3.10N for electrical equipment will be used to demonstrate operability if the component is mechanically or structurally complex such that its response cannot be adequately predicted analytically. Analysis may be used if the equipment is amenable to modeling and dynamic analysis.

Inactive Seismic Category I equipment will be shown to have structural integrity during all plant conditions in one of the following manners: 1) by analysis satisfying the stress criteria applicable to the particular piece of equipment, or 2) by test showing that the equipment retains its structural integrity under the simulated test environment.

A list of Seismic Category I equipment and the method of qualification used is provided in Table 3.2-2.

3.9N.2.3 Dynamic Response Analysis of Reactor Internals Under Operational Flow Transients and Steady State Conditions

The vibration characteristics and behavior due to flow induced excitation are very complex and not readily ascertained by analytical means alone. Reactor components are excited by the flowing coolant which causes oscillatory pressures on the surfaces. The integration of these pressures over the applied area should provide the forcing functions to be used in the dynamic analysis of the structures. In view of the complexity of the geometries and the random character of the pressure oscillations, a closed form solution of the vibratory problem by integration of the differential equation of motion is not always practical and realistic. The determination of the forcing functions as a direct correlation of pressure oscillations cannot be practically performed independently of the dynamic characteristics of the structure. The main objective is to establish the characteristics of the forcing functions that essentially determine the response of the structures. By studying the dynamic properties of the structure from previous analytical and experimental work, the characteristics of the forcing function can be deduced. These studies indicate that the most important forcing functions are flow turbulence, and pump related excitation. The relevance of such excitations depends on many factors such as type and location of component and flow conditions.

The effects of these forcing functions have been studied from test runs on models, prototype plants and in component tests, References [4], [5], and [6].

The Indian Point No. 2 plant has been established as the prototype for a four-loop plant internals verification program and was fully instrumented and tested during hot functional testing. In addition, the Trojan plant and the Sequoyah No. 1 plant have provided prototype data applicable to CPNPP, Reference [5], [6], [7] and [14].

The CPNPP is similar to Indian Point No. 2; the only significant differences are the modifications resulting from the use of 17 x 17 fuel, replacement of the annular thermal shield with neutron shielding panels, and the change to the UHI-style inverted top hat support structure configuration. These differences are addressed below.

1. 17 x 17 fuel

   The only structural changes in the internals resulting from the design change from the 15 x 15 to the 17 x 17 fuel assembly are the guide tube and control rod drive line. The new 17 x 17 guide tubes are stronger and more rigid, hence they are less susceptible to flow induced vibration. The fuel assembly itself is relatively unchanged in mass and spring rate, and thus no significant deviation of internals vibration is expected from the vibration with the 15 x 15 fuel assemblies.

2. Neutron shielding pads lower internals

   The primary cause of core barrel excitation is flow turbulence, which is not affected by the upper internals [5]. The vibration levels due to core barrel excitation for Trojan and CPNPP both having neutron shielding pads are expected to be similar. The coolant inlet temperature of CPNPP is slightly higher than Trojan 1 and the flowrate is slightly higher. Scale model tests show that core barrel vibration varies as velocity raised to a small power [4]. The difference in fluid density and flowrate result in an approximately 6% higher core barrel vibration for CPNPP than for Trojan 1. However, scale model test results [4] and results from Trojan [7] show that core barrel vibration of plants with neutron shielding pads is significantly less than that of plants with thermal shields. This information and the fact that low core barrel stresses with large safety margins were measured at Indian Point No. 2 (thermal shield configuration) lead to the conclusion that stresses approximately equal to or less than those of Indian Point No. 2 result on the CPNPP internals with the attendant large safety margins.

3. UHI-style inverted top hat upper support configuration

   The components of the upper internals are excited by turbulent forces due to axial and cross flows in the upper plenum and by pump related excitations [5], [7]. Sequoyah and CPNPP have the same basic upper internals configuration, therefore, the general vibration behavior is not changed. Comanche Peak upper internals adequacy has been determined from data from the instrumented plant test at Sequoyah 1, scale model tests and numerous operation plants. The results of testing at Sequoyah 1, [14] showed that the components are excited by flow induced and pump related excitations. Analyses of the data indicate that the instrumented components have adequate factors of safety, that random flow-induced responses are adequately predicted by scale models and that the margins are higher with the core in place than during hot functional testing.

   In addition, the Comanche Peak upper internals configuration was tested in a scale model using the same modeling techniques as for the scale model tests of the UHI configuration. The responses of the Comanche Peak upper internals have been calculated using the Sequoyah 1 and scale model information. The results show adequate factors of safety for all components.

   The original test and analysis of the four-loop configuration is augmented by References [4], [5], [6], [7] and [14] to cover the effects of successive hardwater modifications.

3.9N.2.4 Preoperational Flow-Induced Vibration Testing of Reactor Internals

Because of CPNPP reactor internals design configuration is well characterized, as was discussed in Section 3.9N.2.3, it is not considered necessary to conduct instrumented tests of the CPNPP hardware. The requirements of Regulatory Guide 1.20 will be met by conducting the confirmatory preoperational testing examination for integrity per Section D, of Regulatory Guide 1.20, “Regulations for Reactor Internals Similar to the Prototype Design.” This examination will include some 35 points with special emphasis on the following areas.

1. All major load-bearing elements of the reactor internals relied upon to retain the core structure in place.

2. The lateral, vertical and torsional restraints provided within the vessel.

3. Those locking and bolting devices whose failure could adversely affect the structural integrity of the internals.

4. Those others locations on the reactor internal components which are similar to those which were examined on the prototype Indian Point No. 2 and on Trojan 1 and Sequoyah 1.

5. The inside of the vessel will be inspected before and after the hot functional test, with all the internals removed, to verify that no loose parts or foreign material are in evidence.

A particularly close inspection will be made on the following items or areas using a 5X or 10X magnifying glass where applicable.

1. Lower internals

    a. Upper barrel to flange girth weld.

    b. Upper barrel to lower barrel girth weld.

    c. Upper core plate aligning pin. Examine bearing surfaces for any shadow marks, burnishing, buffing or scoring. Inspect welds for integrity.

    d. Irradiation specimen guide screw locking devices and dowel pins. Check for lockweld integrity.

    e. Baffle assembly locking devices. Check for lockweld integrity.

    f. Lower barrel to core support girth weld.

    g. Neutron shield panel screw locking devices and dowel pin welds. Examine the interface surfaces for evidence of tightness and for lockweld integrity.

    h. Radial support key welds.

    i. Insert screw locking devices. Examine soundness of lockwelds.

    j. Core support columns and instrumentation guides. Check joints for tightness and soundness of the locking devices.

    k. Secondary core support assembly weld integrity.

    l. Lower radial support keys and inserts. Examine for shadow marks, burnishing, buffing or scoring. Check the integrity of the lockwelds. These members supply the radial and torsional constraint of the internals at the bottom relative to the reactor vessel while permitting axial and radial growth between the two. One would expect to see, on the bearing surfaces of the key and keyway, burnishing, buffing or shadow marks which would indicate pressure loading and relative

motion between the two parts. Some scoring of engaging surfaces is also possible and acceptable.

m. Check gaps between baffle plate joints.

2. Upper internals

a. Thermocouple conduits, clamps and couplings.

b. Guide tube, support column, orifice plate, and thermocouple assembly locking devices.

c. Support column and thermocouple conduit assembly clamp welds.

d. Upper core plate alignment inserts. Examine for shadow marks, burnishing, buffing or scoring. Check the locking devices for integrity of lockwelds.

e. Thermocouple conduit gusset and clamp welds (where applicable).

f. Thermocouple conduit end-plugs. Check for tightness.

g. Guide tube enclosure welds, tube-transition plate welds and card welds.

Acceptance standards are the same as required in the shop by the original design drawings and specifications.

During the hot functional test, the internals will be subjected to a total operating time at greater than normal full-flow conditions (four pumps operating) of at least 240 hours. This provides a cyclic loading of approximately 10^7 cycles on the main structural elements of the internals. In addition there will be some operating time with only one, two and three pumps operating.

Pre- and post-hot functional inspection results serve to confirm predictions that the internals are well behaved. When no signs of abnormal wear, no harmful vibrations are detected and no apparent structural changes take place, the four-loop core support structures are considered to be structurally adequate and sound for operation.

3.9N.2.5 Dynamic System Analysis of the Reactor Internals Under Faulted Conditions

Analysis of the reactor internals for blowdown loads resulting from a loss of coolant accident is based on the time-history response of the internals to simultaneously applied blowdown forcing functions.

The reactor internals hydraulic pressure transients were calculated including the assumptions that the structural motion is coupled with the pressure transients. This is referred to as fluid-structure-interaction. The hydraulic analysis considers the fluid structure interaction of the core barrel by accounting for the deflections of the constraining boundaries which are represented by masses and springs. The dynamic response of the core barrel in its beam bending mode responding to blowdown forces compensates for internal pressure variation by increasing the volume of the more highly pressurized regions. The analytical methods used to

develop the reactor internals hydraulics are described in WCAP-9735 [27] for (Unit 1) and WCAP-8708 [8] (Unit 2).

Unit 1

The horizontal and vertical forces exerted on reactor internals and the core, following a LOCA, are computed by employing the MULTIFLEX 3.0 [27], which is an enhancement and extension of the MULTIFLEX 1.0 [22], NRC reviewed and approved computer code developed for the space-time dependent analysis of nuclear power plants. MULTIFLEX 3.0 has been accepted by the NRC for several other applications [23, 24, 25 and 26] and also has been extensively used for the LOCA analyses of reactor internals components of numerous other 2, 3, and 4 loop nuclear power plants.

Unit 2

The horizontal and vertical models are coupled at the elevation of the primary nozzle centerlines. Node 1 of the horizontal model is coupled with node 2 of the vertical model at the reactor vessel nozzle elevation. This coupled node has external restraints characterized by a 3 x 3 matrix which represents the reactor coolant loop stiffness characteristics, by linear horizontal springs which describe the tangential resistance of the supports, and by individual non-linear vertical stiffness elements which provide downward restraint only. The supports are represented in the horizontal and vertical models (Figures 3.9N-12 and 3.9N-13) are not indicative of the complexity of the support system used in the analysis. The individual supports are located at the actual support pad locations and accurately represent the independent non-linear behavior of each support.

System blowdown analysis of the reactor internals carried out using DARI-WOSTAS computer code (Reference 10) is a non-linear elastic analysis, wherein the nonlinearities are due to system mechanical gaps. ASME Code Section III, Subsection NG does allow the system analyses to be performed on elastic basis.

In the mathematical models of the Blowdown System Analysis, the values used for structural damping are within the Regulatory guide 1.61 (i.e., 4 percent of critical damping) value. For impact damping a coefficient of restitution C_R value of about 55 percent was selected to consider impact losses in the analysis. This C_R value is considered to be conservative based on some published results. The gaps considered in the analysis are the physical gaps based on nominal dimensions obtained from assembly drawings.

In addition, because of the complexity of the system and the components, it is necessary to use finite element stress analysis codes to provide more detailed information at various points.

A blowdown digital computer program [8] which was developed for the purpose of calculating local fluid pressure, flow, and density transients that occur in Pressurized Water Reactor coolant systems during a loss of coolant accident is applied to the subcooled, transition, and saturated two-phase blowdown regimes. This is in contrast to programs such as WHAM [9] which are applicable only to the subcooled region and which, due to their method of solution, could not be extended into the region in which large changes in the sonic velocities and fluid densities take place. This blowdown code is based on the method of characteristics wherein the resulting set of

ordinary differential equations, obtained from the laws of conservation of mass, momentum, and energy, are solved numerically using a fixed mesh in both space and time.

Although spatially one dimension conservation laws are employed, the code can be applied to describe three dimensional system geometries by use of the equivalent piping networks. Such piping networks may contain any number of pipes of channels of various diameters, dead ends, branches (with up to six pipes connected to each branch), contractions, expansions, orifices, pumps and free surfaces (such as in the pressurizer). System losses such as friction, contraction, expansion, etc., are considered.

The blowdown code evaluates the pressure and velocity transients for a maximum of 2400 locations throughout the system. Theses pressure and velocity transients are stored as a permanent tape file and are made available to the program FORCE 2 (Reference 8) which utilizes a detailed geometric description in evaluating the loadings on the reactor internals.

Each reactor component for which FORCE 2 (Reference 8) calculations are required is designated as an element and assigned an element number. Forces acting upon each of the elements are calculated summing the effects of:

1. The pressure differential across the element.

2. Flow stagnation on, and unrecovered orifice losses across the element.

3. Friction losses along the element.

Input to the code, in addition to the blowdown pressure and velocity transients, includes the effective area of each element on which the force acts due to the pressure differential across the element, a coefficient to account for flow stagnation and unrecovered orifice losses, and the total area of the element along which the shear forces act.

The mechanical analysis [10] has been performed using conservative assumptions. Some of the more significant assumptions are:

1. The mechanical and hydraulic analyses have considered the effect of hydroelasticity.

2. The reactor internals are represented by a multi-mass system connected with springs and dashpots simulating the elastic response and the viscous damping of the components. The modeling is conducted in such as way that uniform masses are lumped into easily identifiable discrete masses while elastic elements are represented by springs.

The model described is considered to have a sufficient number of degrees of freedom to represent the most important modes of vibration in the vertical direction. This model is conservative in the sense that further mass-spring resolution of the system would lead to further attenuation of the shock effects obtained with the present model.

The pressure waves generated within the reactor are highly dependent on the location and nature of the postulated pipe failure. In general, the more rapid the severance of the pipe, the more severe the imposed loadings on the components. A one millisecond severance time is taken as the limiting case.

In the case of the hot leg break, the vertical hydraulic forces produce an initial upward lift of the core. A rarefraction wave propagates through the reactor hot leg nozzle into the interior of the upper core barrel. Since the wave has not reached the flow annulus on the outside of the barrel, the upper barrel is subjected to an impulsive compressive wave. Thus, dynamic instability (buckling) or large deflections of the upper core barrel, or both, is a possible response of the barrel during hot leg break results in transverse loading on the upper core components as the fluid exits the hot leg nozzle.

In the case of the cold leg break, a rarefraction wave propagates along a reactor inlet pipe, arriving first at the core barrel at the inlet nozzle of the broken loop. The upper barrel is then subjected to a non-axisymmetric expansion radial impulse which changes at the rarefaction wave propagates both around the barrel and down the outer flow annulus between vessel and barrel. After the cold leg break, the initial steady state hydraulic lift forces (upward) decreases rapidly (within a few milliseconds) and the increase in the downward direction. These cause the reactor core and the lower support structure to move initially downward.

If the simultaneous seismic event with the intensity of the SSE is postulated with the loss of coolant accident, the imposed loading on the internals component may be additive in certain cases and therefore the combined loading must be considered. In general, however, the loading imposed by the earthquake is small compared to the blowdown loading.

The summary of the mechanical analysis follows:

Vertical Excitation Model for Blowdown

For the vertical excitation, the reactor internals are represented by a multi-mass system connected with springs and dashpots simulating the elastic response and the viscous damping of the components. Also incorporated in the multi-mass system is a representation of the motion of the fuel elements relative to the fuel assembly grids. The fuel elements in the fuel assemblies are kept in position by friction forces originating from the preloaded fuel assembly grid fingers. Coulomb type friction is assumed in the event that sliding between the rods and the grid fingers occurs. In order to obtain an accurate simulation of the reactor internals response, the effects of the internal damping, clearances between various internals, snubbing action caused by solid impact, Coulomb friction induced by fuel rod motion relative to the grids, and preloads in hold down springs have been incorporated in the analytical model. The modeling is conducted in such a way that uniform masses are lumped into easily identifiable discrete masses while elastic elements are represented by springs.

The appropriate dynamic differential equations for the multi-mass model describing the aforementioned phenomena are formulated and the results obtained using a digital computer program which computes the response of the multi-mass model when excited by a set of time dependent forcing functions. The appropriate forcing functions are applied simultaneously and independently to each of the masses in the system. The results from the program give the forces, displacements and deflections as functions of time for all the reactor internals components (lumped masses). Reactor internals response to both hot and cold leg pipe ruptures were analyzed.

Transverse Excitation Model for Blowdown

Various reactor internal components are subjected to transverse excitation during blowdown. Specifically, the barrel, guide tubes, and upper support columns are analyzed to determine their response to this excitation.

Core Barrel - For the hydraulic analysis of the pressure transients during hot leg blowdown, the maximum pressure drop across the barrel is a uniform radial compressive impulse.

The barrel is then analyzed for dynamic buckling using the following conservative assumptions:

1. The effect of the fluid environment is neglected.

2. The shell is treated as simply supported.

During cold leg blowdown, the upper barrel is subjected to a non-axisymmetric expansion radial impulse which changes as the rarefaction wave propagates both around the barrel and down the outer flow annulus between vessel and barrel.

The analysis of transverse barrel response to cold leg blowdown is performed as follows:

1. The core barrel is treated as a simply supported cylindrical shell of constant thickness between the upper flange weldment and the lower support plate weldment. No credit is taken for the supports at the barrel midspan offered by the outlet nozzles. This assumption leads to conservative deflection estimates of the upper core barrel.

2. The core barrel is analyzed as a shell with two variable sections to model the support flange and core barrel.

3. The barrel with the core and thermal shielding pads is analyzed as a beam fixed at the top and elastically supported at the lower radial support and the dynamic response is obtained.

Guide Tubes - The guide tubes in closest proximity to the outlet nozzle of the ruptured loop are the most severely loaded during a blowdown. The transverse guide tube forces decrease with increasing distance from the ruptured nozzle location.

All of the guide tubes are designed to maintain the function of the control rods for a break size of 144 $in.^2$ and smaller. No credit for the function of the control rods is assumed for break size areas above 144 $in.^2$. However, the design of the guide tube will permit control rod operation in all but four control rod positions, which is sufficient to maintain the core in a subcritical configuration, for break sizes up to a double-ended hot leg break. This double-ended hot leg break imposes the limiting lateral guide tube loading.

Upper Support Columns - Upper support columns located close to the broken nozzle hot leg break will be subjected to transverse loads due to cross flow. The loads applied to the columns are computed with a method similar to the one used for the guide tubes, i.e., by taking into consideration the increase in flow across the column during the accident. The columns are

studied as beams with variable section and the resulting stresses are obtained using the reduced section modulus and appropriate stress risers for the various section.

The effects of the gaps that could exist between vessel and barrel, between fuel assemblies, and between fuel assemblies and baffle plates are considered in the analysis. Linear analysis will not provide information about the impact forces generated when components impinge each other, but can, and is, applied prior to gap closure. Reference [10] provides further details of the blowdown method used in the analysis of the reactor internals.

The stresses due to the Safe Shutdown Earthquake (vertical and horizontal components) are combined with the blowdown stresses in order to obtain the largest principal stress and deflection.

All reactor internals components were found to be within acceptable stress and deflection limits for both hot leg and cold leg loss of coolant accidents occurring simultaneously with the Safe Shutdown Earthquake.

The results obtained from the linear analysis indicate that during blowdown, the relative displacement between the components will close the gaps and consequently the structures will impinge on each other, making the linear analysis unrealistic and forcing the application of nonlinear methods to study the problem. Although linear analysis will not provide information about the impact forces generated when components impinge on each other, it can, and is, applied prior to gap closure. The effects of the gaps that could exist between vessel and barrel, between fuel assemblies, between fuel assemblies and baffle plates, and between the control rods and their guide paths were considered in the analysis. Both static and dynamic stress intensities are within acceptable limits. In addition, the cumulative fatigue usage factor is also within the allowable usage factor of unity.

The stresses due to the Safe Shutdown Earthquake (vertical and horizontal components) were combined in the most unfavorable manner with the blowdown stresses in order to obtain the largest principal stress and deflection.

The results indicate that the maximum deflections and stress in the critical structures are below the established allowable limits. For the transverse excitation, it is shown that the upper barrel does not buckle during a hot leg break and that it has an allowable stress distribution during a cold leg break.

Even though control rod insertion is not required for plant shutdown, this analysis shows that most of the guide tubes will deform within the limits established experimental to assure control rod insertion. For the guide tubes deflected above the no loss of function limit, it must be assumed that the rods will not drop. However, the core will still shut down due to the negative reactivity insertion in the form of core voiding. Shutdown will be aided by the great majority of rods that do drop. Seismic deflections of the guide tubes are generally negligible by comparison with the no loss of function limit.

3.9N.2.6 Correlations of Reactor Internals Vibration Tests with the Analytical Results

As stated in Section 3.9N.2.3, it is not considered necessary to conduct instrumented tests of the CPNPP reactor vessel internals. Adequacy of these internals will be verified by use of the Sequoyah and Trojan results. References [5] and [6] describe predicted vibration behavior

based on studies performed prior to the plant tests. These studies, which utilize analytical models, scale model test results, component tests, and results of previous plant tests, are used to characterize the forcing functions and establish component structural characteristics so that the flow induced vibratory behavior and response levels for CPNPP are estimated. These estimates are then compared to values deduced from plant test data obtained from the Sequoyah and Trojan internals vibration measurements programs.

3.9N.3 ASME CODE CLASS 1, 2 AND 3 COMPONENTS, COMPONENT SUPPORTS AND CORE SUPPORT STRUCTURES

The ASME Code Class components are built to accepted ASME Code, Section III requirements. For Code Class 1 components, very stringent requirements are imposed and are met. For Code Class 2 and 3 components, the requirements are less stringent but adequate, in accordance with the lower classification.

See Section 3.9N.1 for more detailed discussions on ASME Code Class 1 components.

3.9N.3.1 Loading Combinations Design Transients, and Stress Limits

Design pressure, temperature, and other loading conditions that provide the bases for design of fluid systems Code Class 2 and 3 components are presented in the sections which describe the systems.

3.9N.3.1.1 Design Loading Combinations

The design loading combinations for ASME Code Class 2 and 3 components and supports are given in Table 3.9N-4. The design loading combinations are categorized with respect to normal, upset, emergency, and faulted conditions. Loads for each plant operating condition are combined by the absolute sum method. Stress limits for each of the loading combinations are component oriented and are presented in Tables 3.9N-5, 3.9N-6, 3.9N-7, and 3.9N-8 for tanks, inactive[(a)] pumps, active pumps, and valves, respectively. Active[(b)] pumps and valves are discussed in Section 3.9N.3.2. The component supports are designed in accordance with the ASME Code, Section III, Subsection NF. Details regarding the application and usages of Code Case N-318 are as listed in Table 3.9B-1F.

For core support structures, design loading conditions are given in Section 4.2.2. Loading conditions are discussed in Section 4.2.3.

In general, for reactor internals components and for core support structures the criteria for acceptability in regard to mechanical integrity analyses are that adequate core cooling and core

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a. Inactive components are those whose operability are not relied upon to perform a safety function during the transients or events considered in the respective operating condition category.

b. Active components are those whose operability is relied upon to perform a safety function (as well as reactor shutdown function) during the transients or events considered in the respective operating condition categories.

shutdown must be assured. This implies that the deformation of the reactor internals must be sufficiently small so that the geometry remains substantially intact. Consequently, the limitations established on the internals are concerned principally with the maximum allowable deflections and stability of the parts in addition to a stress criterion to assure integrity of the components.

For the loss of coolant plus the Safe Shutdown Earthquake condition, deflections of critical internal structures are limited. In a hypothesized downward vertical displacement of the internals, energy absorbing devices limit the displacement after contacting the vessel bottom head, ensuring that the geometry of the core remains intact.

The following mechanical functional performance criteria apply:

1. Following the design basis accident, the functional criterion to be met for the reactor internals is that the plant shall be shutdown and cooled in an orderly fashion so that fuel cladding temperature is kept within specified limits. This criterion implies that the deformation of critical components must be kept sufficiently small to allow core cooling.

2. For large breaks, the reduction in water density reduces the reactivity of the core, thereby shutting down the core whether the rods are tripped or not. The subsequent refilling of the core by the Emergency Core Cooling System uses borated water to maintain the core in a subcritical state. Therefore, the main requirement is to assure effectiveness of the Emergency Core Cooling System. Insertion of the control rods, although not needed, gives further assurance of ability to shut the plant down and keep it in a safe shutdown condition.

3. The inward upper barrel deflections are controlled to ensure no contacting of the nearest rod cluster control guide tube. The outward upper barrel deflections are controlled in order to maintain an adequate annulus for the coolant between the vessel inner diameter and core barrel outer diameter.

4. The rod cluster control guide tube deflections are limited to ensure operability of the control rods.

5. To ensure no column loading of rod cluster control guide tubes, the upper core plate deflection is limited.

Methods of analysis of testing for core support structures are discussed in Sections 3.9N.1.3, 3.9N.1.4.1, 3.9N.2.3, 3.9N.2.5, and 3.9N.2.6. Stress limits and deformation criteria are discussed in Section 4.2.1.

3.9N.3.1.2 Design Stress Limits

Design stress limits established for the components are sufficiently low to assure that violation of the pressure retaining boundary will not occur. Limits, for each of the loading combinations, are component oriented and are presented in Tables 3.9N-5 through 3.9N-8.

Specifically, maximum calculated support stresses for each of the loading combinations, described in Table 3.9N-4, are limited by the stress criteria presented in Table 3.9N-7. These criteria assure that stresses in the pump supports remain essentially elastic; thus, a pump/motor misalignment will not occur. The pump support stress criteria, along with the balance of the

operability program described in Section 3.9N.3.2, demonstrates that pump operability would not be impaired during or following any of the loading conditions postulated to occur.

In addition, design procedures for Level C and D stress limits, as delinated in Section III of the ASME Code, provide adequate assurance that structural discontiuities in piping, tanks and vessels will retain their specified geometric configuration during the improbable emergency and faulted condition events. These procedures provide adequate margins to assure the primary pressure boundary of components and the functions of component supports. Conservative stress indices and intensification factors based upon analytical and experimental results as specified in the Code are used for analysis in the areas of structural discontiuity of piping, tanks and vessels. The use of these proven procedures and conformance with ASME III requirements provides an acceptable basis to assure the functional capability of these components.

3.9N.3.2 Pump and Valve Operability Assurance

Mechanical equipment classified as safety-related must be shown capable of performing its function during the life of the plant under postulated plant conditions. Equipment with faulted condition function requirements include active pumps and valves in fluid systems important to safety. Seismic analysis is presented in Section 3.7N and covers all safety-related mechanical equipment. A list of all active pumps supplied by Westinghouse is presented in Table 3.9N-9. As can be seen on this table, CPNPP has no Class 1 active pumps. Active valves supplied by Westinghouse are listed in Table 3.9N-10.

The methods used to demonstrate the operability of active pumps and valves are described in this section. Equipment specifications include requirements for operability under the specified plant conditions and define appropriate acceptance criteria to ensure that the program requirements defined below are satisfied.

All active pumps are qualified for operability by first being subjected to rigid tests both prior to installation in the plant and after installation in the plant. The in-shops tests include: 1) hydrostatic tests of pressure-retaining parts to 150 percent times the design pressure times the ratio of material allowable stress at room temperature to the allowable stress value at the design temperature, 2) seal leakage tests at the same pressure used in the hydrostatic tests, 3) performance tests, while the pump is operated with flow, to determine total developed head, minimum and maximum head, net positive suction head (NPSH) requirements and other pump/motor parameters. Also monitored during these operating tests are bearing temperatures and vibration levels. Bearing temperature limits are determined, by the manufacturer, based on the bearing material, clearances, oil type, and rotational speed. These limits are approved by Westinghouse. A comparison of the Westinghouse specified bearing vibration limits to industry standards is presented in Figure 3.9N-4. Both limits are below appropriate limits specified to the manufacturer for the design of each active pump. After the pump is installed in the plant, it undergoes the cold hydro tests, hot functional tests, and the required periodic inservice inspection and operation. These tests demonstrate that the pump will function as required during all normal operating conditions for the design life of the plant.

In addition to these tests, the safety-related active pumps are qualified for operability during SSE condition by assuring that 1) the pump will not be damaged during the seismic event, and 2) the pump will continued operating despite the SSE loads.

The pump manufacturer is required to show that the pump will operate normally (i.e., perform their safety function) when subjected to the maximum seismic accelerations and maximum faulted nozzle loads. Maximum nozzle loads imposed by the piping systems and the seismic accelerations imposed due to building locations and/or piping system design are confirmed in this process to be less than the maximum nozzle loads and seismic loads used for component design. Thus, active pumps are qualified for loads which are at least as severe as the maximum loads which are expected to occur as a result of faulted condition loadings.

It is required that test or dynamic analysis be used to show the lowest natural frequency of the pump is greater than 33 Hz. The pump, when having a natural frequency above 33 Hz, is considered essentially rigid. This frequency is considered sufficiently high to avoid problems with amplification between the component and structure for all seismic areas. A static shaft deflection analysis of the rotor is performed with the conservative SSE accelerations of 2.1 g in two orthogonal horizontal directions and 2.1 g vertical acting simultaneously. The deflections determined from the static shaft analysis are compared to the allowable rotor clearances. The nature of seismic disturbances dictates that the maximum contact (if it occurs) will be of short duration. In order to avoid damage during the faulted plant condition, the stresses caused by the combination of normal operating loads, SSE, and dynamic system loads will be limited to the material elastic limit, as indicated in Table 3.9N-7. Stress limits which are applied to active pumps, in Table 3.9N-7, are only nominally higher than Level B stress limits and less than Level C, as delineated in Section III of the ASME Code. This assures that the pumps will not experience permanent deformation or otherwise be damaged during the short duration of the faulted condition event.

The average membrane stresses (σ_m) for the faulted condition loads are maintained at 1.2S, or approximately 0.75 σ_y (σ_y = yield stress) and the maximum stress in local fibers (σ_m + bending stress σ_b) are limited to 1.8S, or approximately 1.1 σ_y. In addition, the pump casing stresses caused by the maximum seismic nozzle loads are limited to stresses outlined in Table 3.9N-7. The maximum seismic nozzle loads are also considered in an analysis of the pump supports to assure that a system misalignment cannot occur.

Performing these analyses with the conservative loads stated and with the restrictive stress limits of Table 3.9N-7 as allowables assures that critical parts of the pump will not be damaged during the faulted condition and that, therefore, the reliability of the pump for post-faulted condition operation will not be impaired by the seismic event.

If the natural frequency is found to be below 33 Hz, an analysis will be performed to determine the amplified input accelerations necessary to perform the static analysis. The adjusted accelerations will be determined using the same conservatisms contained in the 2.1 g horizontal and 2.1 g vertical accelerations used for “rigid” structures. The static analysis will be performed using the adjusted accelerations; the stress limits stated in Table 3.9N-7 must still be satisfied.

The second criterion necessary to assure operability is that the pump will function throughout the SSE. The pump/motor combination is designed to rotate at a constant speed under all conditions unless the rotor becomes completely sized, i.e., with no rotation. Typically, the rotor can be seized 5 full seconds before a circuit-breaker, to prevent damage to the motor, shuts down the pump. However, the high rotary inertia in the operating pump rotor, and the nature of the random, short duration loading characteristics of the seismic event, will prevent the rotor from losing its function. In actuality, the seismic loadings will cause only a slight increase, if any, in the

torque (i.e., motor current) necessary to drive the pump at the constant design speed. Therefore, the pump will not shutdown during the SSE and will operate at the design speed despite the SSE loads.

To complete the seismic qualification procedures, the pump motor is independently qualified for operation during the maximum seismic event. Any auxiliary equipment which is identified to be vital to the operation of the pump or pump motor, and which is not qualified for operation during the pump analysis or motor qualifications, is also separately qualified for operation at the accelerations it would see at its mounting. The pump motor and vital auxiliary equipment is qualified by meeting the requirements of IEEE Standard 344-1975. If the testing option is chosen, sine-beat testing for electrical equipment will be justified by satisfying one or more of the following requirements to demonstrate that multi-frequency response is negligible to the sinebeat input of sufficient magnitude to conservatively account for this effect.

1. The equipment response is basically due to one mode.

2. The sine-beat response spectra envelopes the floor response spectra in the region of significant response.

3. The floor response spectra consists of one dominant mode and has a narrow peak at this frequency.

The degree of coupling in the equipment will, in general, determine if a single or multi-axis test is required. Multi-axis testing will be required if there is considerable cross coupling. If coupling is very light, then single axis testing is justified. Or, if the degree of coupling can be determined, then single axis testing can be used with the input sufficiently increased to include the effect of coupling on the response of the equipment.

From the previous arguments, the safety-related pump/motor assemblies will not be damaged and will continue operating under SSE loadings, and therefore, will perform their intended functions. These proposed requirements take into account the complex characteristics of the pump and are sufficient to demonstrate and assure the seismic operability of the active pumps.

The functional ability of active pumps after a faulted condition is assured since only normal operating loads and steady state nozzle loads will then exist. Since it is demonstrated that the pumps are not damaged during the faulted condition, the post-faulted condition operating loads will be identical to the normal plant operating loads. This is assured by requiring that the imposed nozzle loads (steady-state loads) for normal conditions and post-faulted conditions are limited by the magnitudes of the normal condition nozzle loads. The post-faulted condition ability of the pumps to function under these applied loads is proven during the normal operating plant conditions for active pumps.

Safety-related active valves must perform their mechanical motion in times of an accident. Assurance is supplied that these valves will operate during a seismic event. Qualification tests accompanied by analyses are conducted for all active valves.

The safety-related valves are subjected to a series of stringent tests prior to service and during the plant life. Prior to installation, the following tests are performed: shell hydrostatic test to ASME Section III requirements, backseat and main seat leakage tests, disc hydrostatic test, functional tests to verify that the valve will open and close within the specified time limits when

subjected to the design differential pressure, operability qualification of motor operators for the environmental conditions over the installed life (i.e., aging radiation, accident environment simulation, etc.) according to IEEE Standard 382. Cold hydro qualification tests, hot functional qualification tests, periodic inservice inspections, and periodic inservice operation are performed in-situ to verify and assure the functional ability of the valve. These tests guarantee reliability of the valve for the design life of the plant. The valves are designed using either stress analyses or the pressure containing minimum wall thickness requirements. On all active valves, an analysis of the extended structure is also performed for static equivalent seismic SSE loads applied at the center of gravity of the extended structure. Stress limits imposed on the non-ASME Code extended structures of active valves assure that the extended structures do not experience permanent deformation or damage and that the functional capability of the valve is not impaired.

Such analyses verify that active valves will perform their safety function when subjected to the most severe loads which would be imposed by the SSE coincident with the maximum faulted plant condition nozzle loads. The maximum nozzle loads imposed by the piping systems and the seismic accelerations imposed due to building locations and/or piping system design are confirmed in this process to be less than the maximum nozzle loads and seismic loads used for component design. Thus, active valves are qualified for loads which are at least as severe as the maximum loads which are expected to occur as a result of faulted condition loadings. Stress limits for active valves are presented in Tables 3.9N-3 (Class 1) and 3.9N-8 (Class 2 & 3). The maximum stress limits allowed in these analyses show structural integrity and are the limits recommended by the ASME for the particular ASME class of valve analyzed.

In addition to these tests and analyses, representative valves of each design type will be tested for verification of operability during a simulated seismic event by demonstrating operational capabilities within the specified limits. The testing procedures are described below.

The valve will be mounted in a manner which will conservatively represent typical valve installations. In general, active valves are supported by the pipe attached to the valve. The valve will include the operator and all accessories normally attached to the valve in service. The operability of the valve during SSE is demonstrated by satisfying the following criteria:

1. All the active valves are designed to have a first natural frequency which is greater than 33 Hz. This is shown by suitable test or analysis.

2. The extended structure of the valve system will be statically deflected an amount equal to that determined by an analysis as representing SSE accelerations applied at the center of gravity of the operator along the direction of the weakest axis of the yoke. The design pressure of the valve will be simultaneously applied to the valve during the static deflection tests.

3. The valve will then be operated while in the deflected position. The valve must perform its safety-related function within the specified operating time limits.

4. Motor operators, pilot solenoid valves and limit switches necessary for operation will be qualified as operable during SSE by appropriate IEEE Seismic Qualification Standards, prior to their installation on the valve.

The accelerations which are used for the static valve qualification are equivalent, as justified by analysis, to the simultaneous application of not less than 2.1 g in two orthogonal horizontal

directions and not less than 2.1 g in the vertical direction. Accelerations imparted to the valve assemblies by the piping are not greater than the accelerations used for the static valve qualification.

If the frequency of the valve is less than 33 Hz, a dynamic analysis of the valve will be performed to determine the equivalent acceleration which will be applied during the static test. The analysis will provide the amplification of the input acceleration considering the natural frequency of the valve and the frequency content of the applicable plant floor response spectra. The adjusted accelerations will be determined using the same conservatisms contained in the 2.1 g horizontal and 2.1 g vertical accelerations used for “rigid” valves. The adjusted accelerations will then be used in the static analysis and the valve operability will be assured by the methods outlined in steps (2) to (4) above using the modified acceleration input.

The above testing program applies to valves with extended structures. The testing will be conducted on a representative number of valves. Valves from each of the primary safety-related design types will be tested. Valve sizes which cover the range of sizes in service will be qualified by the tests and the results will be used to qualify all valves within the intermediate range of sizes.

Valves which are safety-related but can be classified as not having an extended structure, such as check valves and safety-relief valves, will be considered separately.

The check valves are characteristically simple in design and their operation will not be affected by seismic accelerations or the maximum applied nozzle loads. The check valve design is compact and there are no extended structures or masses whose motion could cause distortions which could restrict operation of the valve. The nozzle loads due to maximum seismic excitation will not affect the functional ability of the valve since the valve designed to be isolated from the casing wall. The clearance supplied by the design around the disc will prevent the disc from becoming bound or restricted due to any casing disortions caused by nozzle loads. Therefore, the design of these valves is such that once the structural integrity of the valve is assured using standard design or analysis methods, the ability of the valve to operate is assured by the design features. In addition to these design considerations, the valve will also undergo 1) in-shop hydrostatic tests, 2) in-shop seat leakage test, and 3) periodic in-situ valve exercising and inspection to assure the functional ability of the valve.

The pressurizer safety valves are qualified by the following procedures. These valves are also subjected to tests and analysis similar to check valves: stress and deformation analyses for SSE loads, in-shop hydrostatic and seat leakage tests, and periodic in situ valve inspection. In addition to these tests, a static load equivalent to the SSE is applied at the top of the bonnet and the pressure is increased until the valve mechanism actuates. Successful actuation within the design requirements of the valve assures its overpressurization safety capabilities during a seismic event.

Using the methods described, all the safety-related valves in the systems are qualified for operability during a seismic event. These methods conservatively simulate the seismic event and ensure that the active valves will perform their safety-related function when necessary.

3.9N.3.3 Design and Installation Details in Mounting of Pressure Relief Devices (Pressurizer Safety and Relief System)

The pressurizer safety and relief valve discharge piping systems provide overpressure protection for the RCS. The three spring-loaded safety valves, located on top of the pressurizer, are designed to prevent system pressure from exceeding design pressure by more than 10 percent. The two power-operated relief valves, also located on top of the pressurizer, are designed to prevent system pressure from exceeding the normal operating pressure by more than 100 psi. A water seal is maintained upstream of each valve to minimize leakage. Condensate accumulation on the inlet side of each valve prevents any leakage of hydrogen gas or steam through the valves. The valve outlet side is sloped to prevent the formulation of additional water pockets.

If the pressure exceeds the set point and the valves open, the water slug from the loop seal discharges. The water slug, driven by high system pressure, generates transient thrust forces at each location where a change in flow direction or area occurs. The valve discharge conditions considered in the analysis of the PSARV piping systems are as follows: 1) the three safety valves are assumed to open simultaneously while the relief valves remain closed, and 2) the two relief valves open simultaneously while the safety valves are closed.

For each pressurizer safety and relief piping system, an analytical hydraulic mode is developed. The piping from the pressurizer nozzle to the relief tank nozzle is modeled as a series of control volumes and flow paths. The pressurizer is modeled as a reservoir which contains steam at constant pressure (2500 psia for safety system and 2350 psia for relief system) and at constant temperature of 680°F. The pressurizer relief tank is modeled as a sink which contains steam and water mixture.

Fluid acceleration inside the pipe generates reaction forces on all segments of the line which are bounded at either end by an elbow or bend. Reactor forces resulting from fluid pressure and momentum variations are calculated. These forces can be expressed in terms of the fluid properties available for the transient hydraulic analysis. The momentum equation can be expressed in vector form as:

$$\Sigma \vec{F}_{CV} = \frac{1}{g_C}\frac{\partial}{\partial t}\int_v \vec{\gamma} u dv + \frac{1}{g_C}\int_A \gamma \vec{u}(u \cdot \vec{n} dA)$$

From which the total force on the pipe can be derived:

$$\Sigma F_{pipe} = F_1 - F_S - F_2 - f_{s1} r_1 \frac{(1-\cos\alpha_1)}{\sin\alpha_1} - f_{s2} r_2 \frac{(1-\cos\alpha_2)}{\sin\alpha_2}$$

$$= \frac{r_1}{g_C}\frac{(1-\cos\alpha_1)}{\sin\alpha_1}\frac{\partial W}{\partial t}\bigg|_{Bend1} + \frac{r_2}{g_C}\frac{(1-\cos\alpha_2)}{\sin\alpha_2}\frac{\partial W}{\partial t}\bigg|_{Bend2}$$

$$+ \frac{1}{g_C}\int_{pipe}^{straight}\frac{\partial W}{\partial t} d\ell$$

Where $W = \frac{\gamma \vec{u}}{g_C}$ weight density. g_C = gravitational constant

The terms are indicated in Figure 3.9N-4a.

Unbalanced forces are calculated for each straight segment of pipe from the pressurizer to the relief tank. The hydraulic analysis includes the effects of water slug discharge and two phase flow. The time histories of these forces are used for the subsequent structural analysis of the pressurizer safety and relief lines.

The structural model used in the seismic analysis of the safety and relief lines is modified for the valve thrust analysis to represent the safety and relief valve discharge. The time-history hydraulic forces are applied to the piping system lump mass points. The dynamic solution for the valve thrust is obtained by using a modified predictor-corrector-integration technique and normal mode theory.

The time-history solution is performed in subprogram FIXFM. The input to this subprogram consists of the natural frequencies and normal modes, applied forces, and nonlinear elements. The natural frequencies and normal modes for the modified pressurizer safety and relief line dynamic model are determined with the WESTDYN program. The support loads are computed by multiplying the support stiffness matrix and the displacement vector at each support point. The time- history displacements of the FIXFM subprogram are used as input to the WESDYN2 subprogram to determine the internal forces, deflections, and stresses at each end of the piping elements.

The loading combinations considered in the analysis of the PSARV piping are given in Table 3.9N.2. Additional load combinations for consideration of valve thrust are indicated in Table 3.9N-22, as well as the applicable stress criteria.

3.9N.3.4 Component Supports

See Section 3.9N.1 for ASME Code Class 1 component supports.

Class 2 and 3 supports are designed as follows:

Class 2 and 3 component supports for the NSSS are generally of linear or plate and shell type; however, standard component supports may be used. The mandatory date for compliance with ASME B&PV Code, Section III, Subsection NF is July 1, 1974. Therefore, the standard component supports purchased prior to July 1, 1974 satisfy the requirements for the linear type presented in this Section. Those procured after July 1, 1974 comply with the requirements of Subsection NF.

1. Linear

   a. Normal - The allowable stresses of Appendix XVII of ASME Section III, as referenced in Subsection NF, are employed for normal condition limits.

   b. Upset - Stress limits for upset conditions are the same as normal condition stress limits. This is consistent with Subsection NF of ASME Section III (see NF-3320).

c. Emergency - For emergency conditions, the allowable stresses or load ratings are 33 percent higher than those specified for normal conditions. This is consistent with Subsection NF of ASME Section III in which (see NF-3231) limits for emergency conditions are 33 percent greater than the normal condition limits.

d. Faulted - Section 3.9N.1 specifies limits which assure that no large plastic deformations will occur (Stress <1.2 Sy). If any inelastic behavior is considered in the design, detailed justification will be provided for this limit. Otherwise the supports for active components will be designed so that stresses are less than or equal to S_y. Thus the operability of active components will not be endangered by the supports during faulted conditions.

2. Plates and shells

a. Normal - Normal condition limits are those specified in Subsection NF of ASME Section III (see NF-3320).

b. Upset - Limits for upset conditions equal normal condition limits and are consistent with Subsection NF of ASME Section III (see NF-3320).

c. Emergency - For emergency conditions, the allowable stresses or load ratings are 20 percent higher than those specified for normal conditions.

d. Faulted - Same as faulted limits for linear supports.

For active Class 2 or 3 pumps, support adequacy will be proven by satisfying the criteria in Section 3.9N.3.2. The requirements consist of both stress analysis and an evaluation of pump/motor support misalignment.

Active valves are, in general, supported only by the pipe attached to the valve. Exterior supports on the valve are generally not used.

3.9N.4 CONTROL ROD DRIVE SYSTEM (CRDS)

3.9N.4.1 Descriptive Information of CRDS

Full Length Control Rod Drive Mechanism

Control rod drive mechanisms are located on the dome of the reactor vessel. They are coupled to rod control clusters which have absorber material over the entire length of the control rods and derive their name from this feature. The full length control rod drive mechanism is shown in Figure 3.9N-5 and schematically in Figure 3.9N-6.

The primary function of the full length control rod drive mechanism is to insert or withdraw rod cluster control assemblies within the core to control average core temperature and to shutdown the reactor.

The full length control rod drive mechanism is a magnetically operated jack. A magnetic jack is an arrangement of three electromagnets which are energized in a controlled sequence by a

power cycler to inset or withdraw rod cluster control assemblies in the reactor core in discrete steps. Rapid insertion of the rod cluster control assemblies occurs when electrical power is interrupted.

The control rod drive mechanism consists of four separate subassemblies. They are the pressure vessel, coil stack assembly, latch assembly, and the drive rod assembly.

1. Unit 1

   For Unit 1, the pressure vessel is a one-piece housing which is connected to the reactor vessel had adapter by a full penetration weld. The lower portion of the pressure vessel contains the latch assembly. The upper portion of the pressure vessel provides space for the drive rod during its upward movement as the control rods are withdrawn from the core.

   Unit 2

   For Unit 2, the pressure vessel includes a latch housing and a rod travel housing which are connected by a threaded, seal welded, maintenance joint which facilitates replacement of the latch assembly. The closure at the top of the rod travel housing is a threaded plug for pressure integrity with a canopy seal weld. Mechanical canopy seal clamp assemblies may be used to contain or prevent leaks in the canopy seal.

   The latch housing is the lower portion of the vessel and contains the latch assembly. The rod travel housing is the upper portion of the vessel and provides space for the drive rod during its upward movement as the control rods are withdrawn from the core.

2. The coil stack assembly includes the coil housings, and electrical conduit and connector, and three operating coils: 1) the stationary gripper coil, 2) the movable gripper coil, and 3) the lift coil.

   The coil stack assembly is a separate unit which is installed on the drive mechanism by sliding it over the outside of the latch housing/pressure vessel. It rests on the base of the latch housing/pressure vessel without mechanical attachment.

   Energizing the operating coils causes movement of the pole pieces and latches in the latch assembly.

3. The latch assembly includes the guide tube, stationary pole pieces, movable pole pieces, and two sets of latches: 1) the movable gripper latches and 2) the stationary gripper latches.

   The latches engage grooves in the drive rod assembly. The movable gripper latches are moved up or down in 5/8 inch steps by the lift pole to raise or lower the drive rod. The stationary gripper latches hold the drive rod assembly while the movable gripper latches are repositioned for the next 5/8 inch step.

4. The drive rod assembly includes a flexible coupling, a drive rod, a disconnect button, a disconnect rod, and a locking button.

The drive rod has 5/8 inch grooves which receive the latches during holding or moving of the drive rod. The flexible coupling is attached to the drive rod and provides the means for coupling to the rod cluster control assembly.

The disconnect button, disconnect rod, and locking button provide positive locking of the coupling to the rod cluster control assembly and permits remote disconnection of the drive rod.

The control rod drive mechanism is a trip design. Tripping can occur during any part of the power cycler sequencing if electrical power to the coils is interrupted.

Unit 1

The Unit 1 control rod drive mechanism is welded with a full penetration weld to an adaptor on top of the reactor vessel and is coupled to the rod cluster control assembly directly below.

Unit 2

The Unit 2 control rod drive mechanism is threaded and seal welded on an adaptor on top of the reactor vessel and is coupled to the rod cluster control assembly directly below.

The mechanism is capable of raising or lowering a 360 pound load, (which includes the drive rod weight) at a rate of 45 inches/minute. Withdrawal of the rod cluster control assembly is accomplished by magnetic forces while insertion is by gravity.

The mechanism internals are designed to operate in 650°F reactor coolant. The pressure vessel is designed to contain reactor coolant at 650°F and 2500 psia. The three operating coils are designed to operate at 392°F with forced air cooling required to maintain that temperature.

The full length control rod drive mechanism shown schematically in Figure 3.9N-6 withdraws and inserts a rod cluster control assembly as shaped electrical pulses are received by the operating coils. An ON or OFF sequence, repeated by silicon controlled rectifiers in the power programmer, causes either withdrawal or insertion of the control rod. Position of the control rod is measured by 42 discrete coils mounted on the position indicator assembly surrounding the rod travel housing/pressure vessel. Each coil magnetically senses the entry and presence of the top of the ferromagnetic drive rod assembly as it moves through the coil center line.

During plant operation the stationary gripper coil of the drive mechanism holds the rod cluster control assembly in a static position until a stepping sequence is initiated at which time the movable gripper coil and lift coil is energized sequentially.

Rod Cluster Control Assembly Withdrawal

The rod cluster control assembly is withdrawn by repetition of the following sequence of events (refer to Figure 3.9N-6).

1. Movable gripper coil (B) - ON

The latch locking plunger raises and swings the movable gripper latches into the drive rod assembly groove. A 0.047 inch axial clearance exists between the latch teeth and the drive rod.

2. Stationary gripper coil (A) - OFF

   The force of gravity, acting upon the drive rod assembly and attached control rod, causes the stationary gripper latches and plunger to move downward 0.047 inch until the load of the drive rod assembly and attached control rod is transferred to the movable gripper latches. The plunger continues to move downward and swings the stationary gripper latches out of the drive rod assembly groove.

3. Lift coil (C) - ON

   The 5⁄8 inch gap between the movable gripper pole and the lift pole closes and the drive rod assembly raises one step length (5/8 inch).

4. Stationary gripper coil (A) - ON

   The plunger raises and closes the gap below the stationary gripper pole. The three links, pinned to the plunger, swing and the stationary gripper latches into a drive rod assembly groove. The latches contact the drive rod assembly and lift it (and the attached control rod) 0.047 inch. The 0.047 inch vertical drive rod assembly movement transfer the drive rod assembly load from the movable gripper latches to the stationary gripper latches.

5. Movable gripper coil (B) - OFF

   The latch locking plunger separates from the movable gripper pole under the force of a spring and gravity. Three links, pinned to the plunger, swing the three movable gripper latches out of the drive rod assembly groove.

6. Lift coil (C) - OFF

   The gap between the movable gripper pole and lift pole opens. The movable gripper latches drop 5⁄8 inch to a position adjacent to a drive rod assembly groove.

7. Repeat step 1

   The sequence described above (items 1 through 6) is termed as one step or one cycle. The rod cluster control assembly moves 5⁄8 inch for each step or cycle. The sequence is repeated at a rate of up to 72 steps per minute and the drive rod assembly (which has a 5⁄8 inch groove pitch) is raised 72 grooves per minute. The rod cluster control assembly is thus withdrawn at a rate up to 45 inches per minute.

Rod Cluster Control Assembly Insertion

The sequence for rod cluster control assembly insertion is similar to that for control rod withdrawal, except the timing of lift coil (C) ON and OFF is changed to permit lowering the control assembly.

1. Lift coil (C) - ON

   The 5/8 inch gap between the movable gripper and lift pole closes. The movable gripper latches are raised to a position adjacent to a drive rod assembly groove.

2. Movable gripper coil (B) - ON

   The latch locking plunger raises and swings the movable gripper latches into a drive rod assembly groove. A 0.047 inch axial clearance exists between the latch teeth and the drive rod assembly.

3. Stationary gripper coil (A) - OFF

   The force of gravity, acting upon the drive rod assembly and attached rod cluster control assembly, causes the stationary gripper latches and plunger to move downward 0.047 inch until the load of the drive rod assembly and attached rod cluster control assembly is transferred to the movable gripper latches. The plunger continues to move downward and swings the stationary gripper latches out of the drive rod assembly groove.

4. Lift coil (C) - OFF

   The force of gravity and spring force separates the movable gripper pole from pole from the lift pole and the drive rod assembly and attached rod cluster control drop down 5⁄8 inch.

5. Stationary gripper (A) - ON

   The plunger raises and closes the gap below the stationary gripper pole. The three links, pinned to the plunger, swing the three stationary gripper latches into a drive rod assembly groove. The latches contact the drive rod assembly and lift it (and the attached control rod) 0.047 inch. The 0.047 inch vertical drive rod assembly movement transfers the drive rod assembly load from the movable gripper latches to the stationary gripper latches.

6. Movable gripper coil (B) - OFF

   The latch locking plunger separates from the movable gripper pole under the force of a spring and gravity. Three links, pinned to the plunger, swing the three movable gripper latches out of the drive rod assembly groove.

7. Repeat step 1

   The sequence is repeated, as for rod cluster control assembly withdrawal, up to 72 times per minute which gives a insertion rate of 45 inches per minute.

Holding and Tripping of the Control Rods

During most of the plant operating time, the control rod drive mechanisms hold the rod cluster control assemblies withdrawn from the core in a static position. In the holding mode, only one coil, the stationary gripper coil (A), is energized on each mechanism. The drive rod assembly and attached rod cluster control assemblies hang suspended from the three latches.

If power to the stationary gripper coil is cut off, the combined weight of the drive rod assembly and the rod cluster control assembly plus the stationary gripper return spring is sufficient to move latches out of the drive rod assembly groove. The control rod falls by gravity into the core. The trip occurs as the magnetic field, holding the stationary gripper plunger half against the stationary gripper pole, collapses and the stationary gripper plunger half is forced down by the weight stationary gripper return spring and weight acting upon the latches. After the rod cluster control assembly is released by the mechanism, it falls freely until the control rods enter the dashpot section of the thimble tubes in the fuel assembly.

3.9N.4.2 Applicable CRDS Design Specifications

For those components in the CRDS comprising portions of the reactor coolant pressure boundary, conformance with the General Design Criteria and 10CFR50, Section 50.55a is discussed in Sections 3.1 and 5.2. Conformance with pertaining regulatory guides is discussed in Sections 4.5 and 5.2.3 and Appendix 1A(N).

Design Bases

Bases for temperature, stress on structural members, and material compatibility are imposed on the design of the reactivity control components.

Design Stresses

The CRDS is designed to withstand stresses originating from various operating conditions as summarized in Table 3.9N-1.

Allowable stresses: For normal operating conditions Section III of the ASME Code is used. All pressure boundary components are analyzed as Class 1 components.

Dynamic analysis: The cyclic stresses due to dynamic loads and deflections are combined with the stresses imposed by loads from component weights, hydraulic forces and thermal gradients for the determination of the total stresses of the CRDS.

Control Rod Drive Mechanisms

The control rod drive mechanisms (CRDM's) pressure housings are Class 1 components designed to meet the stress requirements for normal operating conditions of Section III of the ASME Code. Both static and alternating stress intensities are considered. The stresses originating from the required design transients are included in the analysis.

A dynamic seismic analysis is required on the CRDM's when a seismic disturbance has been postulated to confirm the ability of the pressure housing to meet ASME Code, Section III allowable stresses and to confirm its ability to trip when subjected to the seismic disturbance.

Full Length Control Rod Drive Mechanism Operational Requirements

The basic operational requirements for the full length CRDM's are:

1. 5/8 inch step.

2. 144 inch travel.

3. 360 pound maximum load.

4. Step in or out at 45 inches/minute (72 steps/minute).

5. Electrical power interruption shall initiate release of drive rod assembly.

6. Trip delay time of less than 150 milliseconds - Free fall of drive rod assembly shall begin less than 150 milliseconds after power interruption no matter what holding or stepping action is being executed with any load and coolant temperature of 100°F to 550°F.

7. 40 year design life with normal refurbishment.

3.9N.4.3 Design Loads, Stress Limits, and Allowable Deformations

3.9N.4.3.1 Pressure Vessel

The pressure retaining components are analyzed for loads corresponding to normal, upset, emergency and faulted conditions. The analysis performed depends on the mode of operation under consideration.

The scope of the analysis requires many different techniques and methods, both static and dynamic.

Some of the loads that are considered on each component where applicable are as follows:

1. Control rod trip (equivalent static load).

2. Differential pressure.

3. Spring preloads.

4. Coolant flow forces (static).

5. Temperature gradients.

6. Differences in thermal expansion

   a. Due to temperature differences.

   b. Due to expansion of different materials.

7. Interference between components.

8. Vibration (mechanically or hydraulically induced).

9. All operational transients listed in Table 3.9N-1.

10. Pump overspeed.

11. Seismic loads (Operational Basis Earthquake and Safe Shutdown Earthquake).

12. Blowdown forces (due to cold and hot leg break).

The main objective of the analysis is to satisfy allowable stress limits, to assure an adequate design margin, and to establish deformation limits which are concerned primarily with the functioning of the components. The stress limits are established not only to assure that peak stresses will not reach unacceptable values, but also to limit the amplitude of the oscillatory stress component in consideration of fatigue characteristics of the materials. Standard methods of strength of materials are used to establish the stresses and deflections of these components. The dynamic behavior of the reactivity control components has been studied using experimental test data and experience from operating reactors.

3.9N.4.3.2 Drive Rod Assembly

All postulated failures of the full length drive rod assemblies either by fracture or uncoupling lead to are reduction in reactivity. If the drive rod assembly fractures at any elevation, that portion remaining coupled falls with and is guided by the rod cluster control assembly. This always results in reactivity decrease for full length control rods.

3.9N.4.3.3 Latch Assembly and Coil Stack Assembly

Results of Dimensional and Tolerance Analysis

With respect to the CRDM system as a whole, critical clearances are present in the following areas:

1. Latch assembly (diametral clearances).

2. Latch arm-drive rod clearances.

3. Coil stack assembly-thermal clearances.

4. Coil fit in coil housing.

The following write-up defines clearances that are designed to provide reliable operation in the CRDM in these four critical areas. These clearances have been proven by life tests and actual field performance at operating plants.

Latch Assembly - Thermal Clearances

The magnetic jack has several clearances where parts made of Type 410 stainless steel fit over parts made from Type F304LN (Unit 1) or Type 304 (Unit 2) stainless steel. Differential thermal expansion is therefore important. Minimum clearances of these parts at 68°F is 0.011 inches. At the maximum design temperature of 650°F minimum clearances is 0.0045 inches and at the maximum expected operating temperatures of 550°F is 0.0057 inches.

Latch Arm - Drive Rod Clearances

The CRDM incorporates a load transfer action. The movable or stationary gripper latch are not under load during engagement, as previously explained, due to load transfer action.

Figure 3.9N-7 shows latch clearance variation with the drive rod as a result of minimum and maximum temperatures. Figure 3.9N-8 shows clearance variations over the design temperature range.

Coil Stack Assembly - Thermal Clearances

The assembly clearances of the coil stack assembly over the latch housing/pressure vessel was selected so that the assembly could be removed under all anticipated conditions of thermal expansion.

At 70°F the inside diameter of the coil stack is 7.308/7.298 inches. The outside diameter of the latch housing/pressure vessel is 7.260/7.270 inches.

Thermal expansion of the mechanism due to operating temperature of the CRDM result in minimum inside diameter of the coil stack being 7.310 inches at 222°F and the maximum latch housing/pressure vessel diameter being 7.302 inches at 532°F.

Under the extreme tolerance conditions listed above it is necessary to allow time for a 70°F coil housing to heat during a replacement operation.

Four coil stack assemblies were removed for four hot CRDMs mounted on 11.035 inch centers on a 550°F test loop, allowed to cool, and then placed without incident as a test to prove the preceding.

Coil Fit in Core Housing

CRDM and coil housing clearances are selected so that coil heat up results in a close to tight fit. This is done to facilitate thermal transfer and coil cooling in a hot CRDM.

3.9N.4.3.4 Evaluation of Control Rod Drive Mechanisms and Supports

The control rod drive mechanisms (CRDMs) and CRDM support structure are evaluated for the loading combinations outlined in Table 3.9(N)-2.

Unit 1: A detailed finite element model of the CRDMs and CRDM supports is constructed using the WECAN (seismic analysis) or STRUDL (LOCA) analysis) computer programs. The models consist of beam, pipe, and spring elements. For the seismic analysis, nonlinearities in the structure are represented. The time history is derived from an envelope of the seismic response spectra at the reactor vessel head center of gravity and the tie rod containment building elevation. Maximum forces and moments in the CRDMs and support structure are then determined. For the LOCA analysis, the structural model is linearized and the reactor vessel head center of gravity LOCA response spectra is applied to determine the maximum forces and moments in the structure.

Unit 2: A detailed finite element model of the CRDMs and CRDM supports is constructed using the WECAN computer program with beam, pipe, and spring elements. For the LOCA analysis, nonlinearities in the structure are represented. These include RPI plate impact, tie rods, and lifting leg clevis/RPV head interface. The time history motion of the reactor vessel head, obtained from the RPV analysis described in 3.9(N).1.4.6, is input to the dynamic model. Maximum forces and moments in the CRDMs and support structure are then determined. For the seismic analysis, the structural model is linearized and the floor response spectra corresponding to the CRDM tie rod elevation is applied to determine the maximum forces and moments in the structure.

The bending moments calculated for the CRDMs for the various loading conditions are compared with maximum allowable moments determined from a detailed finite element stress evaluation of the CRDMs. Adequacy of the CRDM support structure is verified by comparing the calculated stresses to the criteria given in ASME III, Subsection NF.

The highest loads occur at the head adaptor, the location where the mechanisms penetrate the vessel head. The bending moments at this location are presented in Table 3.9N-20 for the longest and shortest CRDM.

3.9N.4.4 CRDMS Performance Assurance Program

Evaluation of Material's Adequacy

The ability of the pressure housing components to perform throughout the design lifetime as defined in the equipment specification is confirmed by the stress analysis report required by the ASME Code, Section III.

Internal components subject to wear will withstand a minimum of 3,000,000 steps without refurbishment as confirmed by life tests (Reference [11]). Latch assembly inspection is recommended after 2.5×10^6 steps have been accumulated on a single CRDM.

To confirm the mechanical adequacy of the fuel assembly, the CRDM, and full length rod cluster control assembly, functional test programs have been conducted on a full scale 12 foot control rod. The 12 foot prototype assembly was tested under simulated conditions of reactor temperature, pressure, and flow for approximately 1000 hours. The prototype mechanism accumulated about 3,000,000 steps and 600 trips. At the end of the test the CRDM was still operating satisfactorily. A correlation was developed to predict the amplitude of flow excited vibration of individual fuel rods and fuel assemblies. Inspection of the drive line components did not reveal significant fretting.

These tests include verification that the trip time achieved by the full length CRDMs meet the design requirements of 2.4 seconds from start of rod cluster control assembly motion to dashpot entry. This trip time requirements will be confirmed for each CRDM prior to initial reactor operation and at periodic intervals after initial reactor operation as required by the proposed Technical Specifications.

There are no significant differences between the prototype CRDMs and the production units. Design materials, tolerances and fabrication techniques (Section 4.2.3.3.2) are the same.

These tests have been reported in Reference [11].

In addition, dynamic testing programs have been conducted by Westinghouse and Westinghouse Licensees to demonstrate that control rod scram time is not adversely affected by postulated seismic events. Acceptable scram performance is assured by also including the effects of the allowable displacements of the driveline components in the evaluation of the test results.

It is expected that all CRDMs will meet specified operating requirements for the duration of plant life with normal refurbishment. However, a Technical Specification pertaining to an inoperable rod cluster control assembly has been set. Latch assembly inspection is recommended after 2.5×10^6 steps have been accumulated on a single CRDM.

If a rod cluster control assembly cannot be moved by its mechanism, adjustments in the boron concentration ensure that adequate shutdown margin would be achieved following a trip. Thus, inability to move one rod cluster control assembly can be tolerated. More than one inoperable rod cluster control assembly could be tolerated, but would impose additional demands on the plant operator. Therefore, the number of inoperable rod cluster control assemblies has been limited to one as discussed in the Technical Specifications.

In order to demonstrate proper operation of the CRDM and to ensure acceptable core power distributions during rod cluster control assembly partial-movement checks are performed on the rod cluster control assemblies (refer to the Technical Specifications). In addition, periodic drop tests of the full length rod cluster control assemblies are performed at each refueling shutdown to demonstrate continued ability to meet trip time requirements, to ensure core subcriticality after reactor trip, and to limit potential reactivity insertions from a hypothetical rod cluster control assembly ejection. During these tests the acceptable drop time of each assembly is not greater than 2.4 seconds, at full flow and operating temperature, from the beginning of motion to dashpot entry.

Actual experience in operating Westinghouse plants indicates excellent performances of CRDMs.

All units are production tested prior to shipment to confirm ability of the CRDM to meet design specification-operation requirements.

Each production full length CRDM undergoes a production tests as listed below:

Test	Acceptance Criteria
Cold (ambient) hydrostatic	ASME Section III
Confirm step length and load transfer (stationary gripper movable gripper or movable gripper to stationary gripper)	Step Length 5/8 ± 0.015 inches axial movement to Load Transfer 0.047 inches nominal axial movement
Cold (ambient) performance Test at design load – 45 5 full travel excursions	Operating Speed 45 inches/minute Trip Delay Free fall of drive rod to begin within 150 milliseconds

3.9N.5 REACTOR VESSEL INTERNALS

3.9N.5.1 Design Arrangements

The reactor vessel internals are described as follows:

The components of the reactor internals are divided into three parts consisting of the lower core support structure (including the entire core barrel and neutron shield pad assembly), the upper core support structure and the incore instrumentation support structure. The reactor internals support the core, maintain fuel alignment, limit fuel assembly movement, maintain alignment between fuel assemblies and control rod drive mechanisms, direct coolant flow past the fuel elements, direct coolant flow to the pressure vessel head, provide gamma and neutron shielding, and guides for the incore instrumentation. The coolant flows from the vessel inlet nozzles down the annulus between the core barrel and the vessel wall and then into a plenum at the bottom of the vessel. It then reverses and flows up through the core support and through the lower core plate. The lower core plate is sized to provide the described inlet flow distribution to the core. After passing through the core, the coolant enters the region of the upper nozzles and directly through the vessel outlet nozzles. A small portion of the coolant flows between the baffle plates and the core barrel to provide additional cooling of the barrel. Similarly, a small amount of the entering flow is directed into the vessel head plenum and exits through the vessel outlet nozzles.

Lower Core Support Structures

Bottom Mounted Instrumentation wear reduction sleeves are installed in the reactor vessel lower internals instrumentation columns of Unit 1. The wear reduction sleeves are completely contained in the instrumentation columns with the flange resting on the top of the lower core plate. The major containment and support member of the reactor internals is the lower core support structure, shown in Figure 3.9N-9. This support structure assembly consists of the core barrel, the core baffle, the lower core plate and support columns, the neutron shield pads, and the core support which is welded to the core barrel. All the major material for this structure is Type 304 stainless steel. The lower core support structure is supported at its upper flange from a ledge in the reactor vessel head flange and its lower end is restrained in its transverse movement

by a radial support system attached to the vessel wall. Within the core barrel are an axial baffle and a lower core plate, both of which are attached to the core barrel wall and form the enclosure periphery of the assembled core. The lower core support structure and principally the core barrel serve to provide passageways and control for the coolant flow. The lower core plate is positioned at the bottom level of the core below the baffle plates and provides support and orientation for the fuel assemblies.

The lower core plate is a member through which the necessary flow distribution holes for each fuel assembly are machined. Fuel assembly locating pins (two for each assembly) are also inserted into this plate. Columns are placed between this plate and the core support of the core barrel in order to provide stiffness and to transmit the core load to the core support. Adequate coolant distribution is obtained through the use of the lower core plate and core support.

The neutron shield pad assembly consists of four pads that are bolted and pinned to the outside of the core barrel. These pads are constructed of Type 304 stainless steel and are approximately 48 inches wide by 148 inches long by 2.8 inches thick. The pads are located azimuthally to provide the required degree of vessel protection. Specimen guides in which material surveillance samples can be inserted and irradiated during reactor operation are attached to the pads. The samples are held in the guides by a preloaded spring device at the top and bottom to prevent sample movement. Additional details of the neutron shield pads and irradiation specimen holders are given in Reference [12].

Vertically downward loads from weight, fuel assembly preload, control rod dynamic loading, hydraulic loads and earthquake acceleration are carried by the lower core plate partially into the lower core plate support flange on the core barrel shell and partially through the lower support columns to the core support and thence through the core barrel shell to the core barrel flange supported by the vessel head flange. Transverse loads from earthquake acceleration, coolant cross flow, and vibration are carried by the core barrel shell and distributed between the lower radial support to the vessel wall, and to the vessel flange. Transverse loads of the fuel assemblies are transmitted to the core barrel shell by direct connection of the lower core plate to the barrel wall and by upper core plate alignment pins which are welded into the core barrel.

The main radial support system of the lower end of the core barrel is accomplished by “key” and “keyway” joints to the reactor vessel wall. At equally spaced points around the circumference, an Inconel clevis block is welded to the vessel inner diameter. Another Inconel insert block is bolted to each of these blocks and has a “keyway” geometry. Opposite each of these is a “key” which is attached to the internals. At assembly, as the internals are lowered into the vessel, the keys engage the keyways in the axial direction. With this design, the internals are provided with a support at the furthest extremity, and may be viewed as a beam fixed at the top and simply supported at the bottom.

Radial and axial expansions of the core barrel are accommodated but transverse movement of the core barrel is restricted by this design. With this system, cyclic stresses in the internal structures are within the ASME Code, Section III limits. In the event of an abnormal downward vertical displacement of the internals following a hypothetical failure, energy absorbing devices limit the displacement after contacting the vessel bottom head. The load is then transferred through the energy absorbing devices of the internals to the vessel.

The energy absorbers, cylindrical in shape, are contoured on their bottom surface to the reactor vessel bottom head geometry. Assuming a downward vertical displacement the potential energy of the system is absorbed mostly by the strain energy of the energy absorbing devices.

Upper Core Support Assembly

The upper core support assembly, shown in Figures 3.9N-10 and 3.9N-11 consists of the top support plate, assembly, and the upper core plate between which are contained support columns and guide tube assemblies. The support columns establish the spacing between the top support plate assembly and the upper core plate and are fastened at top and bottom to these plates. The support columns transmit the mechanical loadings between the two plates and serve the supplementary function of supporting thermocouple guide tubes. The guide tube assemblies, sheath and guide the control rod drive shafts and control rods. They are fastened to the top support plate and are restrained by pins in the upper core plate for proper orientation and support. Additional guidance for the control rod drive shafts is provided by the upper guide tube which is attached to the upper support plate and guide tube.

The upper core support assembly is positioned in its proper orientation with respect to the lower support structure by flat-sided pins pressed into the core barrel which is turn engage in slots in the upper core plate. At an elevation in the core barrel where the upper core plate is positioned, the flat-sided pins are located at angular positions of 90 degrees from each other. Four slots are milled into the core plate at the same positions. As the upper support structure is lowered into the main intenals, the slots in the plant engage the flat-sided pins in the axial direction. Lateral displacement of the plate and of the upper support assembly is restricted by this design. Fuel assembly locating pins protrude from the bottom of the upper core plate and engage the fuel assemblies as the upper assembly is lowered into place. Proper alignment of the lower core support structure, the upper core support assembly, the fuel assemblies and control rods are thereby assured by this system of locating pins and guidance arrangement. The upper core support assembly is restrained from any axial movements by a large circumferential spring which rests between the upper barrel flange and the upper core support assembly and is compressed by the reactor vessel head flange.

Vertical loads from weight, earthquake acceleration, hydraulic loads and fuel assembly preload are transmitted through the upper core plate via the support columns to the top support plate assembly and then the reactor vessel head. Transverse loads from coolant cross flow, earthquake acceleration, and possible vibrations are distributed by the support columns to the top support plate and upper core plate. The top support plate is particularly stiff to minimize deflection.

Incore Instrumentation Support Structures

The incore instrumentation support structures consist of an upper system to convey and support thermocouples penetrating the vessel through the head and a lower system to convey and support flux thimbles penetrating the vessel through the bottom (Figure 7.7-9 shows the basic flux-mapping system).

The upper system utilizes the reactor vessel head penetrations. Instrumentation port columns are slip-connected to inline columns that are in turn fastened to the upper support plate. These port columns protrude through the head penetrations. The thermocouples are carried through these port columns and the upper support plate at positions above their readout locations. The

thermocouple conduits are supported from the columns of the upper core support system. The thermocouple conduits are sealed stainless steel tubes.

In addition to the upper incore instrumentation, there are reactor vessel bottom port columns which carry the retractable, cold worked stainless steel flux thimbles that are pushed upward into the reactor core. Conduits extend from the bottom of the reactor vessel down through the concrete shield area and up to a thimble seal line. The minimum bend radii are about 144 inches and the training ends of the thimbles (at the seal line) are extracted approximately 15 feet during refueling of the reactor in order to avoid interference within the core. The thimbles are closed at the leading ends and serve as the pressure barrier between the reactor pressurized water and the containment atmosphere.

Mechanical seals between the retractable thimbles and conduits are provided at the seal line. During normal operation, the retractable thimbles are stationary and move only during refueling or for maintenance, at which time a space of approximately 15 feet above the seal line is cleared for the retraction operation.

The incore instrumentation support structure is designed for adequate support of instrumentation during reactor operation and is rugged enough to resist damage or distortion under the conditions imposed by handling during the refueling sequence. These are the only conditions which affect the incore instrumentation support structure.

3.9N.5.2 Design Loading Conditions

The design loading conditions that provide the basis for the design of the reactor internals are:

1. Fuel assembly weight.

2. Fuel assembly spring forces

3. Internals weight.

4. Control rod trip (equivalent static load).

5. Differential pressure.

6. Spring preloads.

7. Coolant flow forces (static).

8. Temperature gradients.

9. Differences in thermal expansion

    a. Due to temperature differences.

    b. Due to expansion of different materials.

10. Interface between components.

11. Vibration (mechanically or hydraulically induced).

12. One or more loops out of service

13. All operational transients listed in Table 3.9N-1.

14. Pump overspeed.

15. Seismic loads (Operation Basis Earthquake and Safe Shutdown Earthquake).

16. Blowdown forces (due to cold and hot leg break).

The main objective of the design analysis is to satisfy allowable stress limits, to assure an adequate design margin, and to establish deformation limits which are concerned primarily with the functioning of the components. The stress limits are established not only to assure that peak stresses will not reach unacceptable values, but also to limit the amplitude of the oscillatory stress component in consideration of fatigue characteristics of the materials. Both low and high cycle fatigue stresses are considered when the allowable amplitude of oscillation is established. Dynamic analysis on the reactor internals are provided in Section 3.9N.2.

As part of the evaluation of design loading conditions, extensive testing and inspections are performed from the initial selection of raw materials up to and including component installation and plant operation. Among these tests and inspections are those performed during component fabrication, plant construction, startup and checkout, and during plant operation.

3.9N.5.3 Design Loading Categories

The combination of design loadings fit into the normal, upset, emergency or faulted conditions as defined in the ASME Code, Section III.

Loads and deflections imposed on components due to shock and vibration are determined analytically and experimentally in both scaled models and operating reactors. The cyclic stresses due to these dynamic loads and deflections are combined with the stresses imposed by loads from component weights, hydraulic forces and thermal gradients for the determination of the total stresses of the internals.

The reactor internals are designed to withstand stresses originating from various operating conditions as summarized in Table 3.9N-1.

The scope of the stress analysis problem is very large requiring many different techniques and methods, both static and dynamic. The analysis performed depends on the mode of operation under consideration.

Allowable Deflections

For normal operating conditions, downward vertical deflection of the lower core support plate is negligible.

For the loss of coolant accident plus the Safe Shutdown Earthquake condition, the deflection criteria of critical internal structures are the limiting values given in Table 3.9N-11. The

corresponding no loss of function limits are included in Table 3.9N-11 for comparison purposes with the allowed criteria.

The criteria for the core drop accident is based upon analyses which have to determine the total downward displacement of the internal structures following a hypothesized core drop resulting from loss of the normal core barrel supports. The initial clearance between the secondary core support structures and the reactor vessel lower head in the hot condition is approximately one half inch. Additional displacement of approximately 3/4 inch would occur due to strain of the energy absorbing devices of the secondary core support; thus the total drop distance is about 1-1/4 inch which is insufficient to permit the trips of the rod cluster control assembly to come out of the guide thimble in the fuel assemblies.

Specifically, the secondary core support is a device which will never be used, except during a hypothetical accident of the core support (core barrel, barrel flange, etc.). There are 4 supports in each reactor. This device limits the fall of the core and absorbs much of the energy of the fall which otherwise would be imparted to the vessel. The energy of the fall is calculated assuming a complete and instantaneous failure of the primary core support and is absorbed during the plastic deformation of the controlled volume of stainless steel, loaded in tension. The maximum deformation of this austenitic stainless piece is limited to approximately 15 percent, after which a positive stop is provided to ensure support.

3.9N.5.4 Design Bases

The design bases for the mechanical design of the reactor vessel internals components are as follows:

1. The reactor internals in conjunction with the fuel assemblies shall direct reactor coolant through the core to achieve acceptable flow distribution and to restrict bypass flow so that the heat transfer performance requirements are met for all modes of operation. In addition, required cooling for the pressure vessel head shall be provided so that the temperature differences between the vessel flange and head do not result in leakage from the flange during reactor operation.

2. In addition to neutron shielding provided by the reactor coolant, a separate neutron pad assembly is provided to limit the exposure of the pressure vessel in order to maintain the required ductility of the material for all modes of operation.

3. Provisions shall be made for installing incore instrumentation useful for the plant operation and vessel material test specimens required for a pressure vessel irradiation surveillance programs.

4. The core internals are designed to withstand mechanical loads arising from Operating Basis Earthquake, Safe Shutdown Earthquake and pipe ruptures and meet the requirement of Item 5 below.

5. The reactor shall have mechanical provisions which are sufficient to adequately support the core and internals and to assure that the core is intact with acceptable heat transfer geometry following transients arising from abnormal operating conditions.

6. Following the design basis accident, the plant shall be capable of being shutdown and cooled in an orderly fashion so that fuel cladding temperature is kept within specified limits. This implies that the deformation of certain critical reactor internals must be kept sufficiently small to allow core cooling.

The functional limitations for the core structures during the design basis accident are shown in Table 3.9N-11. To ensure no column loading or rod cluster control guide tubes, the upper core plate deflection is limited to not exceed the value shown in Table 3.9N-11.

Details of the dynamic analyses, input forcing functions, and response loadings are presented in Section 3.9N.2.

The basis for the design stress and deflection criteria is identified below:

Allowable Stresses

For normal operating conditions Section III Subsection NG of the ASME Code is used as a basis for evaluating acceptability of calculated stresses. Both static and alternating stress intensities are considered.

It should be noted that the allowable stresses in Section III of the ASME Code are based on unirradiated material properties. In view of the fact that irradiation increases the strength of the Type 304 stainless steel used for the internals, although decreasing its elongation, it is considered that use of the allowable stresses in Section III is appropriate and conservative for irradiated internal structures.

The allowable stress limits during the design basis accident used for the core support structures are based on the 1974 Edition of the ASME Code for Core Support Structures, Subsection NG, and the Criteria for Faulted Conditions.

In summary, the design and construction of the CPNPP core support structures conforms to the requirements of the Subsection NG of Section III of the ASME code, except that a) the internals are not “stamped” and b) a specific stress report is note required.

3.9N.6 INSERVICE TESTING OF PUMPS AND VALVES

Refer to Section 3.9B.6.

REFERENCES

1. “Documentation of Selected Westinghouse Structural Analysis Computer Codes,” WCAP-8252, Revision 1, July 1977.

2. “Sample Analysis of a Class 1 Nuclear Piping System,” prepared by ASME Working Group on Piping, ASME Publication, 1972.

3. Buchalet, C. and Mager, T. R., “A Summary Analysis of the April 30 Incident at the San Onofre Nuclear Generating Station Unit 1,” WCAP-8099, April 1973.

4. “Prediction of the Flow-Induced Vibration of Reactor Internals by Scale Model Tests,” WCAP-8303-P-A (Proprietary) and WCAP- 8317-A (Non-Proprietary), July 1975.

5. Bloyd, C. N. and Singleton, N. R., “UHI Plant Internals Vibration Measurement Program and Pre and Post Hot Functional Examinations,” WCAP-8516-P (Proprietary) and WCAP-8517 (Non- Proprietary), April 1975.

6. Trojan Final Safety Analysis Report, Appendix A-12 (Docket No. 50-344).

7. Bloyd, C. N., Ciarametaro, W. and Singleton, N. R., “Verification of Neutron Pad and 17 x 17 Guide Tube Designs by Preoperational Tests on the Trojan 1 Power Plant,” WCAP-8780, May 1976.

8. Takeuchi, K., et al., “MULTIFLEX-A Fortran-IV Computer Program for Analyzing Thermal-Hydraulic-Structure System Dynamics,” WCAP-8708, (Proprietary) and WCAP-8079 (Non-Proprietary), February 1976.

9. Fabic, S., “Computer Program WHAM for Calculation of Pressure, Velocity, and Force Transients in Liquid Filled Piping Networks,” Kaiser Engineers Report No. 67-49-R, November 1967.

10. Bohm, G. J. and LaFaille, J. P., “Reactor Internals Response Under a Blowdown Accident, Procedures, First Intl. Conf. on Structural Mech. in Reactor Technology,” Berlin, September 20- 24, 1971.

11. Cooper, F. W. Jr., “17 x 17 Drive Line Components Tests - Phase IB, II, III, D-Loop Drop and Deflection,” WCAP-8446 (Proprietary) and WCAP-8449 (Non-Proprietary), December 1974.

12. Kraus, S., “Neutron Shielding Pads,” WCAP-7870, May 2972.

13. “Bench Mark Problem Solutions Employed for Verification of WECAN Computer Program,” WCAP-8929, June 1977.

14. Altman, D. A., et. al, “Verification of Upper Head Injection Reactor Vessel Internals by Preoperational Tests on the Sequoyah/Power Plant”, WCAP-9646, March, 1981.

15. “Dynamic Analysis of Reactor Pressure Vessel System for Postulated Loss-of-Coolant Accident of Auxiliary Line Breaks, WCAP-12106, January 1989.

16. “Structural Analysis of Reactor Coolant Loop for Comanche Peak Nuclear Power Plant Units No. 1 and 2,” WCAP-10197 Volume 2, Appendix A.

17. NATD Quality Assurance Program, WCAP-9565.

18. “Combination of Safe Sutdown Earthquake and Loss-of-Coolant Accident Responses for Faulted Condition Evaluation of Nuclear Power Plants,” WCAP-9279 (Non-proprietary).

19. “Integrity of the Primary Piping Systems of Westinghouse Nuclear Power Plants During Postulated Seismic Events,” WCAP-9283 (Non-Proprietary).

20. Westinghouse Letter NS-CE-1728 from C. Eicheldinger to John F. Stolz, dated March 17, 1978, Subject: WCAP Synopses.

21. Westinghouse Letter NS-CE-1737 from C. Eicheldinger to John F. Stolz, dated March 21, 1978.

22. WCAP-8708-PA (Proprietary), WCAP-8709 (Non-Proprietary), "MULTIFLEX A FORTRAN Computer Program for Analyzing Thermal-Hydraulic Structure System Dynamics," Takeuchi, K., et al., September 1977.

23. WCAP-15029-P-A (Proprietary), WCAP-15030-NP-A, (Non-Proprietary), "Westinghouse Methodology for Evaluating the Acceptability of Baffle-Former-Barrel Bolting Distribution Under Faulted Load Conditions," E.R. Schwirian, et al., January 1999.

24. WCAP-15245 (Proprietary), WCAP-15246 (Non-Proprietary), "Control Rod Insertion Following a Cold Leg LBLOCA, D.C. Cook Units 1 and 2," J.A. Barsic, et al., February 28, 1999.

25. WCAP-11004-P (Proprietary), WCAP-11005 (Non-Proprietary), "Comparison of Data for Beaver Valley Power Station Unit 2 with WCAP-9735 Data, Prepared for NRC Review in Conjunction with Review of WCAP-9735, Docket No. 50-412," D.R. Bhandari, et al., November 1985.

26. WCAP-11522 (Proprietary), WCAP-11523 (Non-Proprietary), "Response to NRC Questions on the LOCA Hydraulic Forces Analysis of the Beaver Valley Power Station Unit 2, Prepared for NRC Review in Conjunction with Review of WCAP-9735, Docket No. 50-412," D.C. Garner, et al., June 1987.

27. WCAP-9735, Revision 2 (Proprietary) and WCAP-9735, Revision 1 (Non-Proprietary), "MULTIFLEX 3.0 A FORTRAN IV Computer Program for Analyzing Thermal-Hydraulic-Structural System Dynamics Advanced Beam Model," K. Takeuchi, et al., February 1998.

TABLE 3.9N-1
SUMMARY OF REACTOR COOLANT SYSTEM DESIGN TRANSIENTS

(Sheet 1 of 3)

Normal Conditions		Occurrences
1.	Heatup and cooldown at 100°F/hr (pressurizer cooldown 200°F/hr)	200 (each)
2.	Unit loading and unloading at 5% of full power/min	13,200 (each)
3.	Step load increase and decrease of 10% of full power	2,000 (each)
4.	Large step load decrease with steam dump	200
5.	Steady state fluctuations	
	a. Initial fluctuations	1.5×10^5
	b. Random fluctuations	3.0×10^6
6.	Feedwater cycling at hot shutdown	2000
7.	Loop out of service	
	a. Normal loop shutdown	80
	b. Normal loop startup	70
8.	Unit loading and unloading between 0 to 15% of full power	500 (each)
9.	Boron concentration equalization	26,400
10.	Refueling	80

Upset Conditions		Occurrences
1.	Loss of load, without immediate reactor trip	80
2.	Loss of power (blackout with natural circulation in the Reactor Coolant System)	40
3.	Partial loss of flow (loss of one pump)	80

TABLE 3.9N-1
SUMMARY OF REACTOR COOLANT SYSTEM DESIGN TRANSIENTS

(Sheet 2 of 3)

4.	Reactor trip from full power	
	a. Without cooldown	230
	b. With cooldown, without safety injection	160
	c. With cooldown and safety injection	10
5.	Inadvertent reactor coolant depressurization	20
6.	Inadvertent startup of an inactive loop	10
7.	Control rod drop	80
8.	Inadvertent Emergency Core Cooling System actuation	60
9.	Operating Basis Earthquake (20 earthquakes of 10 cycles each)	200 cycles
10.	Excessive Feedwater Flow	30

Emergency Conditions[(a)]		Occurrences
1.	Small loss of coolant accident	5
2.	Small steam break	5
3.	Complete loss of flow	5

Faulted Conditions[(a)]		Occurrences
1.	Main reactor coolant pipe break (dynamic effects, resulting from large loss of coolant accident at the RHR, surge and accumulator nozzles, Section 3.6)	1
2.	Large steam break	1
3.	Feedwater line break	1
4.	Reactor coolant pump locked rotor	1
5.	Control rod ejection	1

TABLE 3.9N-1
SUMMARY OF REACTOR COOLANT SYSTEM DESIGN TRANSIENTS

(Sheet 3 of 3)

6.	Steam generator tube rupture	(included under upset conditions, reactor trip from full power with safety injection)
7.	Safe Shutdown Earthquake	1

Test Conditions		Occurrences
1.	Turbine roll test	20
2.	Primary side hydrostatic test	10
3.	Secondary side hydrostatic test	10
4.	Primary side leak test	200
5.	Secondary side leak test	80
6.	Tube leakage test	800

a) In accordance with the ASME Nuclear Power Plant Components Code, emergency and faulted conditions are not included in fatigue evaluation.

TABLE 3.9N-1A
COMPONENT CYCLIC OR TRANSIENT LIMITS

COMPONENT	CYCLIC OR TRANSIENT LIMIT	DESIGN CYCLE OR TRANSIENT
Reactor Coolant System	200 heatup cycles at ≤ 100°F/h and 200 cooldown cycles at ≤ 100°F/h.	Heatup cycle -T_{avg} from ≤ 200°F to ≥ 500°F. Cooldown cycle - T_{avg} from ≥ 550°F to ≤ 200°F.
	200 pressurizer cooldown cycles at ≤ 200°F/h.	Pressurizer cooldown cycle temperatures from ≥ 650°F to ≤ 200°F.
	80 loss of load cycles, without immediate Turbine or Reactor Trip	≥ 15% of RATED THERMAL POWER to 0% of RATED THERMAL POWER.
	40 cycles of loss-of-offsite A.C. electrical power	Loss-of-offsite A.C. electrical ESF Electrical System.
	80 cycles of loss of flow in one reactor coolant loop.	Loss of only one reactor coolant pump.
	400 Reactor trip cycles.	100% to 0% of RATED THERMAL POWER.
	10 auxiliary spray actuation cycles.	Spray water temperature differential > 320°F, but ≤ 625°F
	200 leak tests.	Pressurized to ≥ 2485 psig.
	10 hydrostatic pressure tests.	Pressurized to ≥ 3107 psig.
Secondary Coolant System	1 steam line break.	Break in a > 6-inch steam line.
	10 hydrostatic pressure tests.	Pressurized to ≥ 1481 psig.

TABLE 3.9N-2
LOADING COMBINATIONS FOR ASME CLASS 1 COMPONENTS AND COMPONENT SUPPORTS (EXCLUDING PIPE SUPPORTS)

Condition Classification	Loading Combination
Design	Design Pressure, Design Temperature, Deadweight, Operating Basis Earthquake[a]
Normal	Normal Condition Transient, Deadweight
Upset	Upset Condition Transients, Deadweight, Operating Basis Earthquake
Emergency	Emergency Condition Transients, Deadweight
Faulted	Faulted Condition Transients, Deadweight, Safe Shutdown Earthquake or (Safe Shutdown Earthquake and Pipe Rupture Loads)

a) The Operating Basis Earthquake is not considered a design condition for NSSS Class 1 valves and piping. The primary stresses are calculated for the Operating Basis Earthquake loads and compared with the upset condition allowable stresses.

TABLE 3.9N-3
ALLOWABLE STRESSES FOR ASME SECTION III CLASS 1 COMPONENTS[a]

(Sheet 1 of 2)

Operating Condition Classification	Vessels/Tanks	Piping	Pumps	Valves	Component Supports
Normal	ASME Section III	ASME Section III	ASME Section III	ASME Section III	ASME Section III Subsection NF
Upset	ASME Section III	ASME Section III	ASME Section III	ASME Section III	ASME Section III Subsection NF
Emergency	ASME Section III	ASME Section III	ASME Section III	ASME Section III	ASME Section III Subsection NF
Faulted	ASME Section III see Section 3.9N.1.4	ASME Section III see Section 3.9N.1.4	ASME Section III See Section 3.9N.1.4 (No active class 1 pump used)	See Note 1	ASME Section III Subsection NF See Section 3.9N.1.4

P_e, P_m, P_b, Q_t, C_p, S_n & S_m as defined by Section II of the ASME Code

TABLE 3.9N-3
ALLOWABLE STRESSES FOR ASME SECTION III CLASS 1 COMPONENTS[a]

(Sheet 2 of 2)

NOTE 1 TO TABLE 3.9N-3

CLASS 1 VALVE FAULTED CONDITION CRITERIA

	ACTIVE		INACTIVE
a)	Calculate Pm from para. NB3545.1 with Internal Pressure Ps = 1.25Ps Pm ≤ 1.5Sm	a)	Calculate Pm from para. NB3545.1 with Internal Pressure Ps = 1.50Ps Pm ≤ 2.4 Sm or 0.7Su
b)	Calculate Sn from para. NB3545.2 with Cp = 1.5 Ps = 1.25Ps Qt2 = 0 Ped = 1.3X value of Ped from equations of 3545.2(b) (1) Sn ≤ 3Sm	b)	Calculate Sn from para. NB3545.2 with Cp = 1.5 Ps = 1.50Ps Qt2 = 0 Ped = 1.3X value of Ped from equations of NB3545.2(b) (1) Sn ≤ 3Sm

a) A test of the components may be performed in lieu of analysis.

TABLE 3.9N-4
DESIGN LOADING COMBINATIONS FOR ASME CODE CLASS 2 AND COMPONENTS AND COMPONENT SUPPORTS (EXCLUDING PIPING AND PIPE SUPPORTS)[a]

Condition Classification	Loading Combination
Design and Normal	Design pressure design temperature,[b] deadweight, nozzle loads[c]
Upset	Upset condition pressure, upset condition metal temperature,[b] deadweight, OBE, nozzle loads[c]
Emergency	Emergency condition pressure, emergency condition metal temperature,[b] deadweight, nozzle loads[c]
Faulted	Faulted condition pressure, faulted condition metal temperature,[b] deadweight, SSE, nozzle loads[c]

a) For loading combinations and stress limits for ASME Class 2 and 3 piping, refer to Table 3.9B-1B and Table 3.9B-1C for Class 2 and 3 supports.

b) Temperature is used to determine allowable stress only.

c) Nozzle loads are those loads associated with the particular plant operating conditions for the component under consideration.

TABLE 3.9N-5
STRESS CRITERIA FOR SAFETY RELATED ASME CLASS 2 AND CLASS 3 TANKS

Condition	Stress Limits
Design and Normal	$\sigma_m \leq 1.0\ S$ $(\sigma_m \text{ or } \sigma_L) +$ $\sigma_b \leq 1.5\ S$
Upset	$\sigma_m \leq 1.1\ S$ $(\sigma_m \text{ or } \sigma_L) +$ $\sigma_b \leq 1.65\ S$
Emergency	$\sigma_m \leq 1.5\ S$ $(\sigma_m \text{ or } \sigma_L) +$ $\sigma_b \leq 1.80\ S$
Faulted	$\sigma_m \leq 2.0\ S$ $(\sigma_m \text{ or } \sigma_L) +$ $\sigma_b \leq 2.4\ S$

TABLE 3.9N-6
STRESS CRITERIA FOR ASME CODE CLASS 2 AND CLASS 3 INACTIVE PUMPS

Condition	Stress Limits	P_{max}[(a)]
Design and Normal	The pump shall conform to the requirements of ASME Section III, NC-3400 (or ND-3400)	
Upset	$\sigma_m \leq 1.1\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 1.65\ S$	1.1
Emergency	$\sigma_m \leq 1.5\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 1.80\ S$	1.2
Faulted	$\sigma_m \leq 2.0\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 2.4\ S$	1.5

a) The maximum pressure shall not exceed the tabulated factors listed under P_{max} times the design pressure.

TABLE 3.9N-7
DESIGN CRITERIA FOR ACTIVE PUMPS

Condition	Design Criteria
Normal	ASME Section III Subsection NC-3400 and ND-3400
Upset	$\sigma_m \leq 1.0\ S$ $\sigma_m + \sigma_b \leq 1.5\ S$
Emergency	$\sigma_m \leq 1.2\ S$ $\sigma_m + \sigma_b \leq 1.65\ S$
Faulted	$\sigma_m \leq 1.2\ S$ $\sigma_m + \sigma_b \leq 1.8\ S$

TABLE 3.9N-8
STRESS CRITERIA FOR SAFETY RELATED ASME CODE CLASS 2 AND CLASS 3 VALVES

Condition	Stress Limits (Notes 1-5)	P_{max} (Note 6)
Design and Normal	Valve bodies shall conform to the requirements of ASME Section III, NC-3500 (or ND-3500)	
Upset	$\sigma_m \leq 1.1\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 1.65\ S$	1.1
Emergency	$\sigma_m \leq 1.5\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 1.80\ S$	1.2
Faulted	$\sigma_m \leq 2.0\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 2.4\ S$	1.5

Notes:

1. Valve nozzle (piping load) stress analysis is not required when both the following conditions are satisfied by calculation: 1) section modulus and area of a plane, normal to the flow, through the region of valve body crotch is at least 10% greater than the piping connected (or joined) to the valve body inlet and outlet nozzles; and, 2) code allowable stress, S, for valve body material is equal to or greater than the code allowable stress, S, of connecting piping material. If the valve body material allowable stress is less than of the connected piping, the valve section modulus and area as calculated in (1) above shall be multiplied by the ratio (S_{pipe}/S_{valve}). If unable to comply with this requirement, the design by analysis procedure of NB-3545.2 is an acceptable alternate method.

2. Casting quality factor of 1.0 shall be used.

3. These stress limits are applicable to the pressure retaining boundary, and include the effects of loads transmitted by the extended structures, when applicable.

4. Design requirements listed in this Table are not applicable to valve discs, stems, seat rings, or other parts of valves which are contained within the confines of the body and bonnet, or otherwise not part of the pressure boundary.

5. These rules do not apply to Class 2 and 3 safety relief valves. Safety relief valves will be designed in accordance with ASME Section III requirements.

6. The maximum pressure resulting from upset, emergency or faulted conditions shall not exceed the tabulated factors listed under P_{max} times the design pressure or the rated pressure at the applicable operating condition temperature. If the pressure rating limits are met at the operating conditions, the stress limits in Table 3.9N-4 are considered to be satisfied.

TABLE 3.9N-9
ACTIVE PUMPS

Pump	Item No.	System	ANS Safety Class	Normal Mode	Post LOCA Mode	Basis
Centrifugal charging pump no. 1	APCH	CVCS	2a	ON/OFF	ON	High head safety injection
Centrifugal Charging pump no. 2	APCH	CVCS	2a	ON/OFF	ON	High head safety injection
Boric acid transfer pumps	APBA	CVCS	2b	ON/OFF	ON/OFF	Required for emergency boration safe shutdown
Residual heat removal pump no. 1 and 2	APRH	RHRS	2a	OFF	ON	Required for safety injection and plant cooldown
Safety injection pump no. 1 and 2	APSI	SIS	2a	OFF	ON	Required for safety injection

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 1 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
HV-3607	RCS	Solenoid	1"	Globe/2	Closed	RCS high point vents
HV-3608	RCS	Solenoid	1"	Globe/2	Closed	RCS high point vents
HV-3609	RCS	Solenoid	1"	Globe/2	Closed	Pressurizer high point vents
HV-3610	RCS	Solenoid	1"	Globe/2	Closed	Pressurizer high point vents
PCV-455A	RCS	Nitrogen	3"	Globe/1	Closed	Primary Pressure Control after SGTR only. (Open/Close)
PCV-456	RCS	Nitrogen	3"	Globe/1	Closed	Primary Pressure Control after SGTR only. (Open/Close)
8000A/B	RCS	Motor	3"	Gate/1	Open/Closed	Unblock Pressurizer PORV after SGTR (Open); Block Stuck Open Pressurizer PORV after SGTR (Close)
8010A/B/C	RCS	Self-actuated	6"	Safety/1	Closed	Pressurizer code safety valves
8026	RCS	Air	1"	Diaphragm/2	Open if sample is being taken or PRT is being vented	Containment isolation
8027	RCS	Air	1"	Diaphragm/2	Open if sample is being taken or PRT is being vented	Containment isolation
8046	RCS	Self-actuated	3"	Check/2	N/A	Containment isolation

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 2 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
8047	RCS	Air	3"	Diaphragm/2	Open if PRT spray is actuated	Containment isolation
RC-036	RCS	Self-actuated	3/4"	Relief/2	Closed	Pressure relief, Containment isolation
XCS-0037 (Note 1)	CVCS	Self-actuated	3/4"	Check/3	Closed	Pump Protection
XCS-0039 (Note 1)	CVCS	Self-actuated	3/4"	Check/3	Closed	Pump Protection
XCS-0041 (Note 1)	CVCS	Self-actuated	3/4"	Check/3	Closed	Pump Protection
XCS-0044 (Note 1)	CVCS	Self-actuated	3/4"	Check/3	Closed	Pump Protection
8100	CVCS	Motor	2"	Globe/2	Open	Containment isolation
8104	CVCS	Motor	2"	Globe/2	Closed	Required for safe shutdown redundancy
8105	CVCS	Motor	3"	Gate/2	Open	ECCS operation and containment isolation
8106	CVCS	Motor	3"	Gate/2	Open	ECCS operation
8110	CVCS	Motor	2"	Globe/2	Open	ECCS operation
8111	CVCS	Motor	2"	Globe/2	Open	ECCS operation
8112	CVCS	Motor	2"	Globe/2	Open	Containment isolation
8145	CVCS	Air	2"	Globe/1	Closed	RCS pressure boundary
8152	CVCS	Air	3"	Globe/2	Open	Containment isolation
8153	CVCS	Air	1"	Globe/1	Closed	RCS pressure boundary
8154	CVCS	Air	1"	Globe/1	Closed	RCS pressure boundary

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 3 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
1-8121	CVCS	Self-actuated	2”	Relief/2	Closed	Pressure relief during LOCA
2CS-8000	CVCS	Self-actuated	1”	Relief/2	Closed	Pressure relief during LOCA
8160	CVCS	Air	3"	Globe/2	Open	Containment isolation
CS-8180	CVCS	Self-actuated	3/4"	Check/2	N/A	Pressure relief, Containment isolation
8202A/B (Note 1)	CVCS	Solenoid	1"	Globe/2	N/A	ECCS operation
8210A/B (Note 1)	CVCS	Solenoid	1"	Globe/2	N/A	ECCS operation
HV-8220 (Note 1)	CVCS	Air	1"	Ball/2	Open	ECCS operation
HV-8221 (Note 1)	CVCS	Air	1"	Ball/2	Open	ECCS operation
CS-8350A/B/C/D	CVCS	Self-actuated	2"	Check/1	N/A	RCS pressure boundary
8351A/B/C/D	CVCS	Motor	2"	Globe/2	Open	Containment isolation
CS-8367A/B/C/D	CVCS	Self-actuated	2"	Check/1	N/A	RCS pressure boundary
CS-8368A/B/C/D	CVCS	Self-actuated	2"	Check/2	N/A	Containment isolation
CS-8377	CVCS	Self-actuated	2"	Check/1	N/A	RCS pressure boundary
8378A/B	CVCS	Self-actuated	3"	Check/1	N/A	RCS pressure boundary, Boration
8379A/B	CVCS	Self-actuated	3"	Check/1	N/A	RCS pressure boundary, Boration
8381	CVCS	Self-actuated	3"	Check/2	N/A	Containment isolation, Boration
CS-8442	CVCS	Self-actuated	2"	Check/2	N/A	Required for safe shutdown redundancy
CS-8473	CVCS	Self-actuated	2"	Check/3	N/A	Boric acid flowpath

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 4 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
CS-8480A/B (Note 1)	CVCS	Self-actuated	2"	Check/2	N/A	Minimum flow path, ECCS flowpath
8481A/B	CVCS	Self-actuated	4"	Check/2	N/A	ECCS operation, Boration
CS-8487	CVCS	Self-actuated	2"	Check/3	N/A	Boric acid flowpath
8497	CVCS	Self-actuated	3"	Check/2	N/A	ECCS flowpath boundary
8510A/B	CVCS	Self-actuated	1-1/2"	Relief/2	Closed	Pump protection x 2"
8511A/B	CVCS	Motor	2"	Globe/2	Closed	Pump protection
8512A/B	CVCS	Motor	2"	Globe/2	Open	ECCS operation
8546	CVCS	Self-actuated	8"	Check/2	N/A	ECCS operation, Boration
LCV-112B/C	CVCS	Motor	4"	Gate/2	Open	ECCS operation
LCV-112D/E	CVCS	Motor	8"	Gate/2	Closed	ECCS operation
LCV-459	CVCS	Air	3"	Globe/1	Open	RCS pressure boundary (Remote manual isolation)
LCV-460	CVCS	Air	3"	Globe/1	Open	RCS pressure boundary (Remote manual isolation)
8701A/B	RHRS	Motor	12"	Gate/1	Closed	RCS pressuure boundary, Containment isolation Open for normal plant cooldown
8702A/B	RHRS	Motor	12"	Gate/1	Closed	RCS pressure boundary, Open for normal plant cooldown
8716A/B	RHRS	Motor	10"	Gate/2	Open	ECCS operation

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 5 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
RH-0033	RHRS	Self	3/4” x 1”	Relief/2	Closed	Thermal Relief for Bonnet of 8716A
RH-0034	RHRS	Self	3/4” x 1”	Relief/2	Closed	Thermal Relief for Bonnet of 8716B
8730A/B	RHRS	Self-actuated	10"	Check/2	N/A	ECCS operation and normal cooldown
FCV-610	RHRS	Motor	3"	Globe/2	Open/Closed	RHR pump miniflow provides pump protection
FCV-611	RHRS	Motor	3"	Globe/2	Open/Closed	Control RHRS miniflow, provides pump protection
SI-166 (Note 1)	SIS	Self-actuated	3/4	Check/3	Closed	Nitrogen Accumulator isolation
SI-167 (Note 1)	SIS	Self-actuated	3/4	Check/3	Closed	Nitrogen Accumulator isolation
SI-168 (Note 1)	SIS	Self-actuated	3/4	Check/3	Closed	Nitrogen Accumulator isolation
SI-169 (Note 1)	SIS	Self-actuated	3/4	Check/3	Closed	Nitrogen Accumulator isolation
8800A/B	SIS	Air	3"	Globe/2	Closed	Closes on “S” signal if Open
8801A/B	SIS	Motor	4”	Gate/2	Closed	ECCS flowpath, Containment isolation
8802A/B	SIS	Motor	4"	Gate/2	Closed	ECCS flowpath cold leg to hot leg switchover sequence, Containment isolation
8804A/B	SIS	Motor	8"	Gate/2	Closed	ECCS flowpath, recirculation
8806	SIS	Motor	8"	Gate/2	Open	ECCS flowpath, recirculation boundary
8807A/B	SIS	Motor	6"	Gate/2	Closed	ECCS recirculation, operation, safety injection charging pump cross connection

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 6 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
8809A/B	SIS	Motor	10"	Gate/2	Open	ECCS and plant cooldown flowpath, Containment isolation
8811A/B	SIS	Motor	14"	Gate/2	Closed	ECCS flowpath from Containment sump, Containment isolation
SI-0182	SIS	Self-actuated	3/4"	Relief/2	Closed	Thermal relief, Containment isolation
SI-0183	SIS	Self-actuated	3/4"	Relief/2	Closed	Thermal relief, Containment isolation
8812A/B	SIS	Motor	14"	Gate/2	Open	ECCS operation
8813	SIS	Motor	2"	Globe/2	Open	Safety injection pump miniflow and RWST isolation during recirculation
8814A/B	SIS	Motor	1-1/2"	Globe/2	Open	Safety injection pump miniflow and RWST isolation during recirculation
8815	SIS	Self-actuated	3"	Check/1	N/A	ECCS flowpath and RCS pressure boundary, Containment isolation
8818A/B/C/D	SIS	Self-actuated	6"	Check/1	N/A	ECCS flowpath and RCS pressure boundary, Containment isolation
SI-8819A/B/C/D	SIS	Self-actuated	2"	Check/1	N/A	ECCS flowpath and RCS pressure boundary, Containment isolation
8821A/B	SIS	Motor	4"	Gate/2	Open	Cold leg to hot leg recirculation switchover during ECCS sequence
8823	SIS	Air	3/4"	Globe/2	Closed	Containment isolation

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 7 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
8824	SIS	Air	3/4"	Globe/2	Closed	Containment isolation
8825	SIS	Air	3/4"	Globe/2	Closed	Containment isolation
8835	SIS	Motor	4"	Gate/2	Open	Cold leg to hot leg recirculation switchover during ECCS sequence, Containment isolation
8840	SIS	Motor	10"	Gate/2	Closed	ECCS flowpath, hot leg recirculation, Containment isolation
8841A/B	SIS	Self-actuated	6"	Check/1	N/A	ECCS flowpath, RCS pressure boundary, Containment isolation
8843	SIS	Air	3/4"	Globe/2	Closed	Containment isolation, Must Close if Open on Phase A signal
8871	SIS	Air	3/4"	Globe/2	Closed	Containment isolation, Closes on Phase A signal if Open
8880	SIS	Air	1"	Globe/2	Open	Containment isolation, Closes on Phase A signal if Open
8881	SIS	Air	3/4"	Globe/2	Closed	Containment isolation
8888	SIS	Air	3/4"	Globe/2	Closed	Containment isolation
8890A/B	SIS	Air	3/4"	Globe/2	Closed	Containment isolation
SI-8900A/B/C/D	SIS	Self-actuated	1-1/2"	Check/1	N/A	ECCS operation, RCS pressure boundary

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 8 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
SI-8905A/B/C/D	SIS	Self-actuated	2"	Check/1	N/A	ECCS operation, RCS pressure boundary, Containment isolation
SI-8919A/B	SIS	Self-actuated	1-1/2"	Check/2	Closed	Recirculation flow path
8922A/B	SIS	Self-actuated	4"	Check/2	N/A	ECCS flowpath
8923A/B	SIS	Motor	6"	Gate/2	Open	Provide train isolation in case of passive failure
8924	SIS	Motor	6"	Gate/2	Open	Provides redundant train isolation of safety injection charging pump cross connection (passive failure protection)
8926	SIS	Self-actuated	8"	Check/2	N/A	ECCS flowpath
8948A/B/C/D	SIS	Self-actuated	10"	Check/1	N/A	ECCS operation, RCS pressure boundary
8949A/B/C/D	SIS	Self-actuated	6"	Check/1	N/A	ECCS operation, RCS pressure boundary
8956A/B/C/D	SIS	Self-actuated	10"	Check/1	N/A	ECCS operation, RCS pressure boundary
8958A/B	SIS	Self-actuated	14"	Check/2	N/A	ECCS flowpath
8964	SIS	Air	3/4"	Globe/2	Closed	Containment isolation, Closes on Phase A signal if Open
SI-8968	SIS	Self Actuated	1"	Check/2	N/A	Containment isolation
8969A/B	SIS	Self-actuated	8"	Check/2	N/A	ECCS operation
1SI-8972 (2SI-8983) (Note 1)	SIS	Self-actuated	3/4"	Relief/2	Closed	Pressure relief, Containment isolation

TABLE 3.9N-10
UNIT 1 & 2 ACTIVE VALVES (NOTE 2, 3, 4)

(Sheet 9 of 9)

Valve Location Number	System	Actuated By	Size	Type/ANS Safety Class	Normal Position	Basis
7126	LWPS	Air	3/4"	Diaphragm/2	Open	Containment isolation
7136	LWPS	Air	3"	Diaphragm/2	Open	Containment isolation
7150	LWPS	Air	3/4"	Diaphragm/2	Open/Closed	Containment isolation
WP-7176 (Note 1)	LWPS	Self-actuated	3/4"	Relief/2	Closed	Pressure relief, Containment isolation
WP-7177 (Note 1)	LWPS	Self-actuated	3/4"	Relief/2	Closed	Pressure relief, Containment isolation
LCV-1003	LWPS	Air	3"	Globe/2	Open	Containment isolation

Notes:

1) Valves procured under BOP criteria. See Section 3.9B.

2) Unit 1 and Unit 2 valve tag numbers are the same except for the prefix, as follows:
Examples:
1-8027 is Unit 1
2-8027 is Unit 2
8027 indicates both Units
XCS-0037 indicates a common valve which serves both Units.

3) See Section 3.9B, Table 3.9B-10 for Balance-of-Plant Active Valve List.

4) See Section 6.2.4 for containment isolation details.

TABLE 3.9N-11
MAXIMUM DEFLECTIONS ALLOWED FOR REACTOR INTERNAL SUPPORT STRUCTURES

Component	Allowable Deflections (in)	No-Loss-of Function Deflections (in)
Upper Barrel		
Radial Inward	4.1	8.2
Radial outward	1.0	1.0
Upper package	0.10	0.15
Rod cluster guide tubes	1.00	1.75

TABLE 3.9N-12
MAXIMUM REACTOR VESSEL DISPLACEMENTS AT REACTOR VESSEL CENTERLINE[a]

	Maximum Horizontal Displacement (inches)	Maximum Vertical Displacement (inches)	Maximum Rotation (radians)
144 Square Inch	+0.116	+0.095	+0.00057
RPV Inlet	0.0	-0.087	-0.00066
144 Square Inch	0.0	+0.0014	0.0
RPV Outlet	-0.0081	-0.046	-0.0029
Double Ended	+0.074	+0.0029	+0.0031
Pump Outlet	-0.00064	-0.06	-0.00047
6" Safety Injection	+0.02365	-0.01194	+0.00011
Line Break	-0.02539	-0.02229	-0.00013
4" Spray Line	0.01986	-0.01468	0.0001
Break	-0.02255	-0.02182	-0.000084

a) Although the main loop piping breaks are not part of the design basis (excluded as dynamic effects, Section 3.1.1.4), they are conservatively included in the design basis for Reactor Vessel Displacements. The load and stress evaluation tables for the primary component supports are based on the results obtained from the effects of the 4 and 6 inch auxiliary nozzle breaks (numbers 12 and 13 provided on Table 3.6B-2 and Figure 3.6B.9).

TABLE 3.9N-13
MAXIMUM REACTOR VESSEL SUPPORT LOADS FOR POSTULATED PIPE RUPTURE CONDITIONS

LOCA Maximum Vertical Load Per Support Including Deadweight	LOCA Maximum Horizontal Load Per Support
(Kips)	(Kips)
1113	606

TABLE 3.9N-14A
STEAM GENERATOR LOWER SUPPORT MEMBER STRESSES FOR UNIT 1

Member Stresses
(Percent of Allowable/Loading Conditions)
Units (kips)

(Sheet 1 of 2)

	Normal		Upset		Faulted		
	Load[a]	Percent Stressed	Load[a]	Percent Stressed	Load[a]	Break	Percent Stressed
LS-1 (Bumper)	176	7.4	525	22.1	1205	envelope	44.7
LS-2 (Beam)	0	0	379	80.0[d]	691	envelope	86.0[d]
LS-3 (Bumper)	11	0.5	301	12.7	547	envelope	20.3
Column 1	0	0.0	+332	27.9	+542	envelope	26.3
	-255	17.2	-616	41.5	-1597	envelope	94.7
Column 2	0	0.0	+129	10.8	+102	envelope	5.0
	-332	22.4	-686	46.2	-1207	envelope	71.7

TABLE 3.9N-14A
STEAM GENERATOR LOWER SUPPORT MEMBER STRESSES FOR UNIT 1

Member Stresses
(Percent of Allowable/Loading Conditions)
Units (kips)

(Sheet 2 of 2)

	Normal		Upset		Faulted		
	Load[(a)]	Percent Stressed	Load[(a)]	Percent Stressed	Load[(a)]	Break	Percent Stressed
Column 3	0	0.0	+171	14.4	+145	envelope	7.0
	-338	22.8	-746	50.3	-1251	envelope	74.3
Column 4	0	0.0	+253	21.3	+393	envelope	19.1
	-267	18.0	-567	38.2	-1223	envelope	72.7

Notes:

a) (+) = tensions and (-) = compressions for column loadings.

b) The load table does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

c) The bumpers, beam, and columns are located as shown in Figure 304 of revision 5 of volume 2 of Westinghouse report no. WCAP-10197.

d) These percentages are conservative for the loads reported, as noted in Table 6-1 of revision 5 of volume 2 of Westinghouse report no. WCAP-10197. The 80.0 percent is actually based on a load of 623 kips, and the 86.0 percent is actually based on a load of 1048 kips.

TABLE 3.9N-14B
STEAM GENERATOR LOWER SUPPORT MEMBER STRESSES FOR UNIT 2

Member Stresses
(Percent of Allowable/Loading Conditions)
Units (kips)

(Sheet 1 of 2)

	Normal		Upset		Faulted		
	Load[(a)]	Percent Stressed	Load[(a)]	Percent Stressed	Load[(a)]	Break	Percent Stressed
LS-1 (Bumper)	+192	8.1	+423	17.8	+1311	SI	48.7
LS-2 (Beam)	0	--	+379	38.0	+625	SPRAY	61.3
	--	--	--	--	--		--
LS-3 (Bumper)	+16	0.7	+395	16.6	+925	FW	34.3
Column 1	+7	0.6	+194	16.3	+347	SPRAY	16.8
	-234	15.8	-421	28.4	-898	MS	53.4
Column 2	0	0.0	0	0.0	+171	MS	8.3
	-327	22.0	-512	34.5	-1016	MS	59.7

TABLE 3.9N-14B
STEAM GENERATOR LOWER SUPPORT MEMBER STRESSES FOR UNIT 2

Member Stresses
(Percent of Allowable/Loading Conditions)
Units (kips)

(Sheet 2 of 2)

	Normal		Upset		Faulted		
	Load[(a)]	Percent Stressed	Load[(a)]	Percent Stressed	Load[(a)]	Break	Percent Stressed
Column 3	0	0.0	0	0.0	+247	SPRAY	12.0
	-322	21.7	-501	33.8	-1004	MS	59.7
Column 4	0	0.0	+171	14.4	+310	MS	15.0
	-244	16.4	-417	28.1	-908	MS	54.0

Notes:

a) (+) = tensions and (-) = compressions for column loadings.

b) The load table does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

TABLE 3.9N-15A
STEAM GENERATOR UPPER SUPPORT MEMBER STRESSES FOR UNIT 1

Member Stresses
(Percent of Allowable/Loading Condition)
Units (kips)

	Normal		Upset		Faulted		
	Load	Percent Stressed	Load	Percent Stressed	Load	Break	Percent Stressed
US-1 (Snubbers)[b]							
US-2 (Bumper)	0	0.0	785	22.9	1445	envelope	37.2
US-3 (Bumper)	0	0.0	833	24.3	1726	envelope	44.4
US-4 (Beam)	0	0.0	727	21.2	1030	envelope	26.5

Notes:

a) The bupers and beam are located as shown in Figure 3-5 of revision 5 of volume 2 of Westinghouse Report no. WCAP-10197.

b) The snubbers have been eliminated for Unit 1.

c) The load table does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

TABLE 3.9N-15B
STEAM GENERATOR UPPER SUPPORT MEMBER STRESSES FOR UNIT 2

Member Stresses
(Percent of Allowable/Loading Condition)
Units (kips)

	Normal		Upset		Faulted		
	Load	Percent Stressed	Load	Percent Stressed	Load	Break	Percent Stressed
US-1 (Snubbers)[b]	0	0.0	+131 -623	14.6 69.2	+238 -1170	MS	11.9 58.5
US-2 (Bumper)	0	0.0	511	14.9	1043	MS	26.9
US-3 (Bumper)	0	0.0	618	18.0	1065	MS	27.4
US-4 (Beam)	0	(a)	622	(a)	1035	FW	(a)

Notes:

a) Scope of evaluation by others.

b) The snubbers were rated at 450 kips for normal and OBE seismic operation, and at 1000 kips for the faulted condition, per snubber in accordance with the snubber stress report.[1]

c) The load table does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

[1] Paul-Monroe Hydraulics Inc., Report A-690623, Revision O, entitled “Multiplant II 100 kip Snubber Stress Report”.

TABLE 3.9N-16A
REACTOR COOLANT PUMP SUPPORT MEMBER STRESSES FOR UNIT 1

Member Stresses
(Percent of Allowable/Loading Condition)
Units (kips)

	Normal		Upset		Faulted		
	Load[(a)]	Percent Stressed	Load[(a)]	Percent Stressed	Load[(a)]	Break	Percent Stressed
Tie Rod-A	0	0.0	255	22.7	406	envelope	25.8
Tie Rod-B	264	23.5	424	37.8	1356	envelope	86.3
Tie Rod-C	161	14.3	234	20.8	847	envelope	53.9
Column-1	+0	0.0	+69	5.8	+169	envelope	8.2
	-258	16.8	-407	26.4	-502	envelope	28.3
Column-2	+96	8.1	+331	27.8	+450	envelope	21.8
	-25	1.6	-260	16.9	-383	envelope	21.6
Column-3	+0	0.0	+224	18.8	+299	envelope	14.5
	-136	8.8	-370	24.0	-547	envelope	30.8

Notes:

a) (+) = tensions and (-) = compression for column loadings.
b) The load tables does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9)
c) The tie rods and columns are located as shown in figure 3-7 revision 5 of volume 2 of Westinghouse Report no. WCAP-101097.

TABLE 3.9N-16B
REACTOR COOLANT PUMP SUPPORT MEMBER STRESSES FOR UNIT 2

Member Stresses
(Percent of Allowable/Loading Condition)
Units (kips)

	Normal		Upset		Faulted		
	Load[(a)]	Percent Stressed	Load[(a)]	Percent Stressed	Load[(a)]	Break	Percent Stressed
Tie Rod-A	0	0.0	151	13.4	334	SPRAY	21.3
Tie Rod-B	270	24.0	434	38.6	1376	SPRAY	87.5
Tie Rod-C	170	15.1	334	29.7	879	SPRAY	55.9
Column-1	+0	0.0	+204	17.1	+437	SPRAY	21.2
	-275	17.9	-560	36.4	-788	SPRAY	45.2
Column-2	+115	9.7	+404	33.9	+617	SPRAY	29.9
	-19	1.2	-308	20.0	-567	FW	32.5
Column-3	+0	0.0	+282	23.7	+479	SPRAY	23.2
	-142	9.2	-427	27.7	-677	SPRAY	38.8

Notes:

a) (+) = tensions and (-) = compression for column loadings.

b) The load tables does not reflect the dynamic effects of breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9)

TABLE 3.9N-17
PRESSURIZER UPPER LATERAL SUPPORT MEMBER STRESSES

(Sheet 1 of 2)

Member Stresses
(Percent of Allowable/Loading Condition)
Units (kips)

Member	Normal		Upset		Faulted[(a)]	
	Load	Percent Stressed	Load	Percent Stressed	Load	Percent Stressed
S-1 (Strut)	N = -7	2.4	N = -119	46.1	N = -226	70.5
	V = ±3	2.4	V = ±45	46.1	V = ±96	70.5
S-2 (Strut)	N = -4	1.8	N = -75	32.2	N = -159	43.7
	V = ±2	1.8	V = ±32	32.2	V = ±60	43.7
S-3 (Strut)	N = -7	1.3	N = -119	26.2	N = -226	39.3
	V = ±2	1.3	V = ±17	26.2	V = ±36	39.3
S-4 (Strut)	N = -4	1.2	N = - 75	23.2	N = -159	31.8
	V = ±1	1.2	V = ±13	23.2	V = ±24	31.8

TABLE 3.9N-17
PRESSURIZER UPPER LATERAL SUPPORT MEMBER STRESSES

(Sheet 2 of 2)

PRESSURIZER LOWER SUPPORT LOADS

LOADING CONDITIONS	AXIAL LOAD (kips)	SHEAR LOAD (kips)	OVERTURNING MOMENT (in.-kips)	ANCHOR BOLT Stress Ratio
Normal	- 385	± 29	±2320	3.2
Upset	- 439	±102	±11070	11.1
Faulted	+ 626	±195	±18880	12.4

a) A pressurizer safety and relief line break is controlling

TABLE 3.9N-18
PRIMARY PIPE RESTRAINT LOADS AND STRESSES

Restraints	Loading Condition	Stresses	
		Load (kips)	(Pct of Allowable)
Crossover leg bumper, SG side	Upset Faulted	(a)	
Crossover leg bumper, RCP side	Upset Faulted	(a)	
Crossover leg vertical run	Upset Faulted	(a)	
Steam generator inlet	Upset Faulted	(a)	
Primary shield wall (hot leg)	Faulted	(a)	
Primary shield wall (cold leg)	Faulted	(a)	

a) The postulated break dynamic effects associated with the primary pipe whip restraints are no longer required to be considered as justified by leak-before-break application to the primary coolant loop piping.

TABLE 3.9N-19A
REACTOR VESSEL SUPPORT LOADS AND STRESSES FOR UNIT 1
SUPPORT LOADS[a]

(Sheet 1 of 2)

Loading Condition	Load (kips)	
	Vertical	Tangential
	Hot Leg / Cold Leg	Hot Leg / Cold Leg
Pressure + Weight	586 / 582	7 / 12
Normal Thermal	442 / 204	28 / 18
Upset	497 / 247	46 / 13
Over-temperature Thermal		
Faulted	800 / 443	126 / 148
Over-temperature Thermal		
OBE	407 / 306	283 / 412
SSE	582 / 428	435 / 655
LOCA	1098 / 1132	314 / 461
Normal[b]	1028 / 786	35 / 6
Upset[c]	1435 / 1092	318 / 418
Faulted[d]	1285	897

TABLE 3.9N-19A
REACTOR VESSEL SUPPORT LOADS AND STRESSES FOR UNIT 1
SUPPORT LOADS[a]

(Sheet 2 of 2)

REACTOR VESSEL SUPPORT STRESSES

Loading Condition	Stress Ratio (pct of Allowable)
Normal	28.0
Upset	57.6
Faulted	64.6

a) The load table does not reflect the dynamic effects of the breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

b) Pressure + Weight + Normal Thermal.

c) Pressure + Weight + Upset Thermal + OBE.

d) Higher of (1) Pressure + Weight + $(SSE^2 + LOCA^2)^{1/2}$ or (2) Pressure + Weight + Faulted Thermal.

TABLE 3.9N-19B
REACTOR VESSEL SUPPORT LOADS AND STRESSES FOR UNIT 2
SUPPORT LOADS[a]

(Sheet 1 of 2)

	Load (kips)	
Loading Condition	Vertical	Tangential
Dead Weight	540	41
Thermal	465	41
O. T. Thermal	338	114
Pressure	2	10
OBE	404	405
SSE	742	648
LOCA-1[b],[c]	1113	370
LOCA-2[b],[d]	590	606
Normal [e]	1345	165
Upset [f]	1749	570
Faulted-1 [g]	1480	756
Faulted-2 [g]	1285	897

TABLE 3.9N-19B
REACTOR VESSEL SUPPORT LOADS AND STRESSES FOR UNIT 2
SUPPORT LOADS[a]

(Sheet 2 of 2)

REACTOR VESSEL SUPPORT STRESSES

Loading Condition	Actual Stress (ksi)	Allowable Stress (ksi)	Stress Ratio (Pct of Allowable)
Normal	P_M = 6.71 $P_M + P_B$ = 6.87	S_m = 23.3 1.5SM = 34.95	28.8 19.7
Upset	P_M = 13.7 $P_M + P_B$ = 17.1	S_m = 23.3 1.5 S_m = 34.95	58.6 48.8
Faulted[g]	P_M = 43.5 $P_M + P_B$ = 48.8	0.70 S_u = 49.0 1.05 S_u = 73.	88.9 66.5

a) The load table does not reflect the dynamic effects of the breaks 1 through 11 (Table 3.6B-2 and Figure 3.6B-9).

b) Includes dead weight

c) LOCA-1: Maximum Vertical Load with corresponding Tangential Load

d) LOCA-2: Maximum Tangential Load with corresponding Vertical Load

e) Dead weight + Thermal + O.T. Thermal + Pressure

f) Normal + OBE

g) Deadweight + Pressure + $(SSE^2 + LOCA^2)^{1/2}$

TABLE 3.9N-20
CRDM HEAD ADAPTOR BENDING MOMENTS

	LOCA (in-kip)	COMBINATION OF SSE AND LOCA (in-kip)	% OF ALLOWABLE
Unit 1			
Longest CRDM	32	109	45%
Shortest CRDM	86	141	58.8%
Unit 2			
Longest CRDM	101.30	108.67	45.3%
Shortest CRDM	143.125	166.69	69.5%

TABLE 3.9N-21A
MAXIMUM STRESSES IN THE REACTOR COOLANT LOOP PIPING FOR UNIT 1

	Hot Leg		Crossover Leg		Cold Leg	
Evaluation	Maximum	Allowable[a]	Maximum	Allowable[a]	Maximum	Allowable[a]
Eq 9 design stress (ksi) (DW, P, OBE)	25.4	28.35	24.4	28.35	22.2	28.35
Eq 9 faulted stress (ksi) (DW, P, SSE, LOCA)[b]	53.5	56.7	30.7	56.7	26.1	56.7
Eq 12 stress (ksi)	31.3	56.7	6.5	56.7	20.2	56.7
Eq 13 stress (ksi)	56.8	57.46[c]	55.6	56.7	57.1	57.46[c]
Fatigue usage factor	0.85	1.0	.012	1.0	0.22	1.0

Notes:

[a]: Allowable stress based on Material SA-351 CF8A @ 650°F

[b]: Reactor Coolant Loop piping stresses in this table are based on postulated breaks to the 6" SI, 4" Spray, SG Feedwater, SG Auxiliary Feedwater and Main Steam Line.

[c]: Allowable stress based on Material SA-351 CF8A @ 618°F

TABLE 3.9N-21B
MAXIMUM STRESSES IN THE REACTOR COOLANT LOOP PIPING FOR UNIT 2

	Hot Leg		Crossover Leg		Cold Leg	
Evaluation	Maximum	Allowable[a]	Maximum	Allowable[a]	Maximum	Allowable[a]
Eq 9 design stress (ksi) (DW, P, OBE)	20.7	28.35	22.7	28.35	22.3	28.35
Eq 9 faulted stress (ksi) (DW, P, SSE, LOCA)[b]	26.4	56.7	29.8	56.7	28.6	56.7
Eq 12 stress (ksi)	31.4	56.7	6.6	56.7	20.2	56.7
Eq 13 stress (ksi)	57.45	57.46[c]	55.6	56.7	57.1	57.46[c]
Fatigue usage factor	.85	1.0	.10	1.0	.20	1.0

Notes:

[a]: Allowable stress based on Material SA-351 CF8A @ 650°F

[b]: Reactor Coolant Loop piping stresses in this table are based on postulated breaks to the 6" SI, 4" Spray, SG Feedwater, SG Auxiliary Feedwater and Main Steam Line.

[c]: Allowable stress based on Material SA-351 CF8A @ 618°F

TABLE 3.9N-22
ADDITIONAL LOADING COMBINATIONS FOR SAFETY AND POWER RELIEF VALVE LINES

Condition Classification	Loading Combination	Applicable Stress Criteria
Design	Design Pressure Deadweight Valve Thrust	Design
Normal and Upset (Fatigue)	Normal Transient Upset Transients Deadweight Valve Thrust	Levels A and B (Fatigure Evaluation)
Faulted	Pressure Deadweight Valve Thrust	Level D

3.9B MECHANICAL SYSTEMS AND COMPONENTS

The text marked “later” will be included in future amendments after the required analysis has been completed in a timely fashion according to licensing requirements.

3.9B.1 SPECIAL TOPICS FOR MECHANICAL COMPONENTS

3.9B.1.1 Plant Conditions & Component Operating Conditions

3.9B.1.1.1 Plant Conditions

Plant conditions categorized as Normal, Upset, Emergency, Faulted and Test are defined in the ASME B&PV Code Section III, paragraph NB 3113, and are discussed below.

The following five plant conditions as defined in Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code are considered in the design of mechanical systems and system supports.

1. Normal conditions

   Any condition in the course of startup, operation in the design power range, hot standby and system shutdown, other than upset, emergency, faulted or testing conditions.

2. Upset conditions (incidents of moderate frequency)

   Any deviations from normal conditions anticipated to occur often enough that design should include a capability to withstand the conditions without operational impairment. The upset conditions include those transients which result from any single operator error or control malfunction, transients caused by a fault in a system component requiring its isolation from the system and transients due to loss of load or power. Upset conditions include any abnormal incidents not resulting in a forced outage and also forced outages for which the corrective action does not include any repair of mechanical damage.

3. Emergency conditions (infrequent incidents)

   Those deviations from normal conditions which require shutdown for correction of the conditions or repair of damage. The conditions have a low probability of occurrence.

4. Faulted conditions (limiting faults)

   Those combinations of conditions associated with extremely low probability, postulated events whose consequences are such that the integrity and operability of the nuclear energy system may be impaired to the extent that consideration of public health and safety are involved. Such considerations require compliance with safety criteria as may be specified by jurisdictional authorities.

5. Testing conditions

   Testing conditions are those pressure tests including hydrostatic tests, pneumatic tests, and leak tests specified. Other type of tests shall be classified under normal, upset, emergency or faulted conditions.

Normal Conditions

The following primary system transients are considered normal conditions:

1. Heatup and cooldown at 100°F per hour.
2. Unit loading and unloading at 5 percent of full power per minute.
3. Step load increase and decrease of 10 percent of full power.
4. Large step load decrease with steam dump.
5. Steady state fluctuations.
6. Feedwater cycling at hot shutdown.
7. Loop out of service.
8. Unit loading and unloading between 0 and 15 percent of full power.
9. Boron concentration equalization.
10. Refueling.

Upset Conditions

The following primary system transients are considered upset conditions:

1. Loss of load (without immediate reactor trip).
2. Loss of power.
3. Partial loss of flow.
4. Reactor trip from full power.
5. Inadvertent Reactor Coolant System depressurization.
6. Inadvertent startup of an inactive loop.
7. Control rod drop.

8. Inadvertent Emergency Core Cooling System actuation.

9. Operating Basis Earthquake.

Emergency Conditions

The following primary system transients are considered emergency conditions:

1. Small loss of coolant accident.

2. Small steam break.

3. Complete loss of flow.

4. High Energy Line Break (except as noted in items 1, 2, and 3 in the Faulted Condition)

Faulted Conditions

The following primary system transients are considered faulted conditions. Each of the following accidents should be evaluated for one occurrence:

1. Reactor coolant pipe break (large loss of coolant accident).

2. Large steam line break.

3. Feedwater line break.

4. Reactor coolant pump locked rotor.

5. Control rod ejection.

6. Steam generator tube rupture.

7. Safe Shutdown Earthquake.

3.9B.1.1.2 Component Operating Conditions

Component operating conditions are categorized under the same categories as plant conditions (Normal, Upset, Emergency, Faulted, and Test). These component operating conditions do not necessarily correspond to the plant conditions. Combinations of loadings associated with various plant conditions are categorized and specified as one or more component operating conditions.

All combinations of loads imposed on active components, which are relied upon to operate normally in the course of accomplishing a safety function under all plant conditions, are specified as normal component operating conditions and as design conditions for ASME B&PV Code, Section III design of the component.

For inactive components which accomplish a safety-related function by virtue of their pressure retaining integrity and are not required to perform a mechanical motion, some loading

combinations are to be specified as upset, emergency, or faulted component operating conditions in addition to normal conditions.

3.9B.1.2 Computer Programs Used in Analyses

Computer programs that can be used in the dynamic and static analyses of seismic Category I piping systems are described herein. Developed by G&H and others, these programs are also occasionally linked and modified to suit the need for a solution to a particular dynamic problem. See also Section 3.7B(A). Additional computer programs used in the analysis of ASME Code Class 2 and Class 3 piping systems, including supports for ASME Code Class 1, 2, and 3 piping, are provided in Appendix 3B.

1. Finite Element Computer Program (ANSYS)

   This is a general purpose computer program for the solution of a large class of problems in engineering. This computer program, identified by the acronym ANSYS for engineering analysis system [1], provides a flexible framework for implementation of the finite element analysis technology. The program has the capabilities for static and dynamic, elastic and plastic, fluid flow, and transient heat transfer analyses.

   The matrix displacement method of analysis, based on finite element idealization, is used throughout. A library of more than 40 finite elements is available for static and dynamic analyses. There are various types of these elements, e.g., plane stress, axisymmetric triangles, three dimensional solids, springs, masses, dampers, plates, axisymmetric shells, general shells, and friction interface elements.

   The program uses a direct solution, developed by the matrix displacement method, for the system of simultaneous linear equations.

   Plotting subroutines are available.

2. Computer Program (ADLPIPE)

   The ADLPIPE [2] computer program performs static and dynamic analyses of complex nuclear safety class 1, 2 & 3 piping systems. All safety-related piping systems that require analysis are analyzed using the ADLPIPE program. The input data is preprocessed and plots are made for input and model evaluation. The output automatically includes a stress analysis in accordance with requirements of ANSI B31.1 (1967), ANSI B31.1 (1973), and ASME B&PV Code, Section III, Classes 1, 2, and 3.

   The analysis in ASME B&PV Code, Section III, Class 1 includes fatigue usage and simplified elastic-plastic analyses. All forces, moments, deflections, and a summary stress report are included in the output. Additionally, the program has orthographic, isometric, and stereoscopic plotting capabilities to aid in checking input and interpreting computed results.

   The static loads on the piping systems are thermal internal pressure deadweight, static equivalent of seismic forces, externally applied forces and moments, and wind effects.

The dynamic seismic loads are computed using normal mode theory and seismic response spectra, or time history forcing functions.

Normal mode technique in conjunction with the three-dimensional response spectra is used for obtaining seismic response. The resultant internal forces and moments are computed from the SRSS of their modal values and for closely spaced modes, ADLPIPE has an option to take absolute sum of forces and moments. Algorithms used in this program for the extraction of eigenvalues and associated eigenvectors are the Jacobi rotation scheme and the Givens-Householder scheme with modification [3]. The program is based on the systematic use of transfer matrices.

One of the techniques used is documentation by benchmark calculations. This type of documentation transcends the complex mathematics, programming, and computer systems that are involved in program solution and allows direct comparison of computed results. The benchmark calculation is the principal form of documentation of ADLPIPE, and a number of these benchmarks are presented in part in this section.

Two auxiliary methods of problem solution evaluation are also included in ADLPIPE. The first is a detailed input error check routine, and the second is an intermediate printout of mathematical manipulations and calculations. There are three types of documentation of ADLPIPE. The first is the multitude of hand checks that are made during the development and change of the program. The second is by the many user groups, who have their own method of evaluation and documentation, both analytical and experimental; these groups have contributed immeasurably to the current state of ADLPIPE reliability.

The third type is the documentation and internal checks that Arthur D. Little, Inc., has generated. This type of documentation and internal checks is in four forms as follows: 52 common errors are checked for and automatically reported; all internal program data can be printed during problem solution; sample problems (benchmarks) are compared to other solutions; there is a description of the mathematical techniques that are used.

a. Input Check

   An automatic message for 52 different types of input error is provided [4].

b. Intermediate Data

   1. Force vectors are printed prior to inversion of the stiffness matrix.

   2. Deflection vector is printed after stiffness matrix inversion.

   3. Member data is printed out after input is read.

   4. Contracted stiffness matrix is printed prior to inversion.

   5. Eigenvectors or dynamical matrix are printed after eigenvalue routine.

   6. Eigenvalues of dynamical matrix are printed after eigenvalue routine.

7. Dynamical matrix is printed after formation from stiffness matrix.

8. Flexibility matrix is printed after inversion of stiffness matrix.

9. Reduced stiffness matrix and mass vector are printed after reduction of stiffness matrix to order of dynamical matrix.

10. Flags, properties, stress coefficients, and moments for each member are provided.

11. Modal effective mass for dynamic model/solution evaluation are provided.

c. Typical Benchmark Calculations

Table 3.9B-9 defines and references eight benchmark calculations typical of the verification that has been done with ADLPIPE. The first four are illustrated in Appendix A of Reference [4], where each problem is briefly defined and solutions from other sources are compared to the ADLPIPE solution.

d. Description of the Analytical Techniques

The following documents describe the mathematical techniques that are used:

1. Generalized Piping System Response to Ground Shock Spectra by I.W. Dingwell, Arthur D. Little, Inc.

2. A Method of Computing Stress Range and Fatigue Damage in a Nuclear Piping System by W.B. Wright and E.C. Rodabaugh, Nuclear Engineering and Design, Volume 22, 1972.

3. Method of Calculating Static and Dynamic Moments for Stress Evaluation at Tees and Branches, Arthur D. Little, Inc., May 1973.

4. Method of Calculating Thermal Stress Range for T1, T2, Ta, and Tb Terms, Arthur D. Little, Inc., May 1973.

5. Mathematical Analysis and Logical Procedure, I.W. Dingwell and R.T. Bradshaw, Arthur D. Little, Inc., 1970.

3.9B.1.2.3 Computer Program (SUPERPIPE)

SUPERPIPE is a general purpose piping program which performs comprehensive structural analyses of linear elastic piping systems for dead weight, thermal expansion, seismic spectra or time history, arbitrary force time history and other loading conditions. Analyses are performed to ASME requirements for Class 1, 2 and 3 systems. See Reference [23] of Section 3.7B(A).

A piping system is idealized as a mathematical model consisting of lumped masses connected by massless elastic members. The location of lumped masses is chosen to accurately represent the dynamic characteristics of the system for a dynamic analysis, and to adequately represent the weight distribution of the system for dead load analysis. Static or dynamic equilibrium

equations are formulated using the direct stiffness method, in which element stiffness matrices are formed according to virtual work principles and assembled to form a global stiffness matrix for the system, relating external forces and moment to joint displacements and rotations. Appropriate stiffness modifications for curved components are included. Diagonal mass and damping matrices are assumed.

Static equilibrium equations are solved using Gaussian reduction techniques on the global stiffness matrix. For dynamic problems, the equilibrium equations may be solved using either step-by-step direct integration of the coupled equations of motion, or by first calculating natural frequencies and mode shapes and transforming the system into a set of uncoupled equations of motion. For seismic analysis of piping systems the latter approach is typically used in the dynamic analysis technique known as the response spectrum mode super-position method. In this technique, the earthquake excitation is characterized by acceleration response spectra, and the total response of the system is evaluated as a combination of the individual responses of the significant natural modes of vibration of the system. Natural frequencies and mode shapes are calculated using the determinant search technique. The method of combination of model responses can be selected from any one of those specified in Regulatory Guide 1.92. Earthquakes acting in all three directions simultaneously may be computed.

SUPERPIPE has been verified for a comprehensive set of sample problems. This has included bench marking against the ASME Sample Problems 1 and 6 contained in ASME publication “Pressure Vessel and Piping 1972, Computer Program Verification”, and against a Class 1 sample problem contained in ASME publication “Sample Analysis of a Piping System, Class 1 Nuclear”, 1972. Extensive bench marking has also been performed against the programs, PISOL1A and PISOL3A which are well recognized and utilized throughout the industry. Additionally, the program has been bench marked against the programs such as HUPIPE, ADLPIPE, PIPESD and EDSGAP. SUPERPIPE has been used on a number of domestic and foreign nuclear plants.

3.9B.1.2.4 LEAP, (ME101) Linear Elastic Analysis of Piping

Author: Bechtel Corporation, San Francisco, California

Source: Bechtel Corporation, San Francisco, California

Description

ME101 is a finite element computer program which performs linear elastic analysis of piping systems, transient thermal analysis, local stress analysis, and response spectra merging.

The input data format is specifically designed for pipe stress analysis. ME101 performs a thorough check of the input prior to performing analysis. The program rearranges the geometry automatically to optimize the finite element model. The coordinate and keyword data can be specified in English, Metric or S1 units.

ME101 performs stataic and dynamic analysis of piping systems in accordance with Section III of the ASME Boiler and Pressure Vessel Code, ANSI B31.1 Power Piping Code and ANSI B31.3 Chemical Plant and Petroleum Refinery Piping Code. The flexibility factors, stress intensification factors (SIF) and stress indices (B1,B2) are incorporated in the stress evaluations.

Static analysis capabilities include thermal expansion, deadweight, seismic and uniformly distributed loads, and externally applied forces, moments, displacements and rotations. For dead weight and thermal expansion analysis, supports with gaps can be considered.

Response spectrum analysis is based upon standard modal superposition techniques. The input excitation may be in the form of single or multiple seismic response spectra or time dependent loading functions. In the response spectrum analysis, the user may request modal combinations by various alternate methods typical of the industry. ME101 can further consider differential damping according to NRC RG 1.61 and PVRC Code Case N-411 within a single response spectrum analysis.

ME101 uses an out-of-core solution and active column techniques for both static and dynamic analyses. It has no limit to the number of equations or bandwidth.

ME101 considers zero period accelerations with or without missing mass corrections in the seismic response spectrum anayses. In the time history analysis (modal or direct integration), the excitation may be in the form of nodal forces, support displacements or support accelerations. ME101 can also compute the response to dynamic loads using a direct integration time history method considering nonlinear kinematic hardening supports (such as the Bechtel energy absorber) with or without gaps. Transient analyses and response spectrum analyses of this nonlinear type support are also available.

ME101 compares stresses vs. allowables according to ASME/ANSI code equations. ME101 has load combination capabilities that allow the results of several load cases to be combined according to certain algebraic rules to form additional load cases. The additional load cases resulting from the combination may be utilized in stress comparisons or restraint load summaries. ME101 also provides a customized report writer that is designed to create report or table used in equipment nozzle evaluation, piping support/hanger guidance, valve acceleration summary, penetration load summary, special piping component qualification, pipe movement summary etc. This report is construed, directly from data stored in the internal ME101 Data Base (“MASTER” file). This report enables user to manually input the allowable side by side with the extracted value, or allows the user to uniquely define the interaction equations. The report writer module can also screen or filter results according to allowable criteria.

ME101 can generate isometric plots of the piping configuration with optional node numbering. It can also generate plots for deflections and mode shapes.

The program is validated utilizing fifty-nine validation problems, including the eleven NUREG/CR 1677 benchmark problems. Validation is documented in the ME101 program validation manual.

3.9B.1.2.5 LSAPS (ME214) Local Stress Analysis for Piping Systems

Author: Bechtel Corporation, San Francisco, California

Source: Bechtel Corporation, San Francisco, California

ME214 is an interactive computer program developed to perform local stress analysis due to integral welded attachments and non-integral welded attachments such as line bearing and point contact loads, restraint of local pressure and thermal expansion of the pipe. The program is capable of calculating local pipe stresses based on Welding Research Council Bulletin No. 107

and ASME Code Case N-318 and N-392. The program also calculates local pipe stresses due to non-integral welded attachments using analytical solutions as provided in Comanche Peak Design criteria documents.

The program is developed in accordance with ASME Section III Subsection NC and ND. The loading conditions and allowables are in accordance with Subsection NC and ND piping stress requirements. Code case allowables are used if the local pipe stresses are calculated based on the methodology provided in the code cases.

The program has been verified using Bechtel standard computer program ME210 and hand calculation.

Validation is documented in the program validation manual.

3.9B.1.2.6 SAPCAS (ME 215) (ME101FE) Stress Analysis for Pipe Component and Pipe Support Using Finite Element Method

Author: Bechtel Corporation, San Francisco, California

Source: Bechtel Corporation, San Francisco, California

ME 215 (ME101FE) is a special purpose finite element computer program for computing membrane and membrane-plus-bending stress intensities at pipes, pads, attachments, and welds. The piping component can be a circular run pipe, elbow, or square tubular steel. The attachment can be circular pipe, rectangular tube, or rectangular solid Lug. The program is capable of computing local stress on pipe components and supports such as stanchion, trunnion, lug, etc. This program is more refined and will provide a realistic solution for local piping stress as compared to, conservative ME214. The program is capable of calculating local pipe stresses where the results may exceed the limitation of the welding research council (WRC) Bulletin No. 107, ASME code cases, and/or other design documents such as CPNPP 2EP-5.12 and 2EP-5.13. The program is developed based on well established SAP computer program.

The program has been design verified using ANSYS Version 4.4 and ASME Bench Mark Problems. Additional Comparisons have been made with conservative ME214 results and a recent PVRC Technical Paper by D. C. Foster.

3.9B.1.3 Experimental Stress Analysis

Experimental stress analysis methods will not be used in the design of code or non-code components for the faulted plant condition.

3.9B.1.4 Considerations for the Evaluation of the Faulted Condition

1. ASME Code Class 2 and 3 Components

   The ASME Code Class 2 and 3 components are constructed in accordance with the ASME B&PV Code, Section III, Subsections NA, NC, and ND, respectively. The methods for the dynamic analysis for the Class 2 and 3 components are defined in Section 3.9B.

In the event that the design stress limits permitted in the design criteria exceed the yield strength of the material, analysis is performed in accordance with the ASME B&PV Code, Section III, Article NB-3000. The following types of analyses are used:

a. Elastic analysis for the design of Class 2 and 3 components in accordance with subparagraph NB 3227.6 of ASME B&PV Code, Section III.

b. Limit analysis is done at a specific location when the limits on local membrane stress intensity and primary bending stress intensity do not satisfy the limits of ASME B&PV Code, Section III, subparagraph NB 3228.2.

2. Bolts

For the Code Class 2 and 3 components not in Westinghouse's scope, the following are the criteria for the design of bolting:

a. Bolts for Component Supports (Other than Pipe Supports)

Normal and Upset Conditions - The design limits are as per ASME B&PV Code, Section III, Article XVII-2460 and NRC approved ASME Code Cases as indicated in the specification.

Emergency Condition - The bolt allowables are 1.33 times the allowable used in the normal and upset conditions.

Faulted Condition - The bolt allowables for normal and upset conditions are increased by a factor which is the lower of 1.2 (Sy/Ft) or 0.7 (Su/Ft). However, the total allowable stress shall not exceed 0.9 Sy in any case.

Sy = Yield stress

Ft = Allowable tensile stress

Su = Ultimate tensile stress

b. Bolts for Pipe Support Joints

The design limits for bolted pipe support joints are in accordance with the ASME B&PV Code, Section III, 1983 Edition including the Summer 1983 Addenda, paragraphs NF 3225 and NF 3324.6.

c. Bolts for Flange Joints

The design limits for bolted flange joints connecting pipe to pipe and/or pipe to equipment are in accordance with the ASME B&PV Code, Section III, 1983 Edition, subparagraphs NC/ND 3658.2 and NC/ND 3658.3.

d. Bolts and Components

Bolts for components do not exceed the Code allowable stress limits.

3. Non-ASME Code Class Components

The supplier of the mechanical components not covered by the ASME B&PV Code must demonstrate by experimental testing or calculation that the design stress limits (with use of applicable codes and standards) are sufficiently below the elastic limits to ensure operability.

3.9B.2 DYNAMIC TESTING AND ANALYSIS

3.9B.2.1 Preoperational Vibration and Dynamic Effects Testing on Piping

The purpose of preoperational vibration and dynamic effects tests of the piping systems and their supports, as described below, is to confirm that these piping systems, restraints, components and supports have been adequately designed to withstand flow induced dynamic loadings under steady state and operational transient conditions anticipated during the life of the plant, and that normal thermal motion is not restrained.

3.9B.2.1.1 Thermal Expansion Test

Thermal expansion (hot deflections) test will be conducted on the following piping systems:

Reactor Coolant System

Main Steam

Steam Supply to Auxiliary Feedwater Pump Turbine

Main Feedwater

Pressurizer Relief Line

RHR in Shutdown Cooling Mode

Auxiliary Steam (within seismic Category I Structure)

Steam Generator Blowdown

Safety Injection System (those line adjoining RCS which experience temp. > 200°F)

Auxiliary Feedwater

CVCS (Charging line from Regen. Hx to RCS, Letdown Line from RCS to Letdown Hx)

During the thermal expansion test, pipe deflections will be measured or observed at various locations based on the location of snubber, hangers, and expected large displacement. One complete thermal cycle, (i.e., cold position to hot position to cold position) will be monitored. For most systems, the thermal expansion will be monitored at cold conditions and at normal operating temperature. Intermediate temperatures are generally not practical due to the short time during which the normal operating temperature is reached. For the Reactor Coolant System

and the Main Steam System, measurements will be made at cold, 250°F, 350°F, 450°F and normal operating temperatures.

Acceptance criteria for the thermal expansion test verify that the piping system is free to expand thermally (i.e., piping does not bind or lock at spring hangers and snubbers nor interfere with structures or other piping), and to confirm that piping displacements do not exceed design limits, as described by ASME Section III, (i.e., the induced stresses do not exceed the sum of the basic material allowable stress at design temperature and the allowable stress range for expansion stresses).

3.9B.2.1.2 Dynamic Transient Response Testing

Instrumented dynamic transient response testing augmented by visual observation for the specification flow modes analyzed will be performed on the following piping systems:

	System	Transient Tested
1.	Main Steam System	Main Turbine Trip at 100% steam flow
2.	Safety Injection System	Safety Injection Pump Trip During Injection
3.	Main Feedwater System	Main Feedwater Pump Trip
4.	Auxiliary Feedwater System	Auxiliary Feedwater Pump (including Steam Supply Line) Trip
5.	Pressurizer Relief Valve	Pressurizer Relief Valve Discharge Piping of Reactor Actuation Coolant System
6.	Service Water System	Service Water Pump Trip & Restart

During the dynamic transient response testing for the specified piping systems, the piping will be instrumented to measure transient loads at selected locations and visually observed at other selected locations to ensure that severe vibrations do not exist. The instrument data will be compared with the analytically predicted values.

Instrumented measurements will also be conducted as needed for other systems and flow modes. The acceptance criteria for dynamic transients is based on the allowable design stress limits for occasional loads, as described by ASME Section III, such that the induced stress does not exceed allowables at design temperature as specified in ASME Section III Subsection NB, NC or ND, as applicable, Paragraph 3650.

If evidence of excessive piping motion is noted, the line(s) will be reviewed to determine what remedial action, if any, is necessary. Repetition of these transients may be required to make the determination of acceptability.

3.9B.2.1.3 Steady State Vibrations Tests

Steady State vibration tests will be conducted on (1) ASME Code Class 1, 2, 3 systems (2) other high-energy piping systems inside Seismic Category I Structures, (3) high-energy portions of

systems whose failure could reduce the functioning of any Seismic Category I Plant Feature to an unacceptable level.

The steady state tests are performed for the systems listed below at normal operating conditions consistent with the requirements of Chapter 14.2.

Reactor Coolant System

Residual Heat Removal System

Safety Injection System

Chemical and Volume Control System

Boron Thermal Regeneration System

Boron Recycle System

Containment Spray System (Minimum flow and test lines only)

Liquid Waste Processing System

Diesel Generator Fuel Oil System

Service Water System

Auxiliary Feedwater System

Spent Fuel Cooling and Cleanup System

Main Feedwater System

Main Steam Reheat and Steam Dump System

Component Cooling Water System

Condensate System

Steam Generator Blowdown System

Vents and Drain System

Chilled Water System

Demineralized and Reactor Makeup Water System

Process Sampling System

Auxiliary Steam System

During normal operating conditions, qualified engineers will observe the lines where accessible, to determine the acceptability of the steady state vibrations. One of the following resolutions will be applied if observed vibrations are excessive: (1) the piping will be monitored by suitable instrumentation at or near the locations where vibrations appear to be excessive to demonstrate that the vibration level does not cause ASME Code stress and fatigue allowables to be exceeded, (2) the cause of the vibration will be eliminated, or (3) the pipe routing and/or the support system will be modified to reduce the vibrations to an acceptable level.

3.9B.2.1.4 Test Performance and Acceptance

Where practical, the above tests will be performed during the preoperational testing phase of the startup program. Those tests that cannot be performed as part of the preoperational testing phase due to required plant conditions will be performed as part of the initial startup and power escalation phase.

For those piping systems in which corrective action is required, the vibration and/or thermal expansion observation shall be repeated to determine that the thermal movements and/or vibration have been reduced to satisfy the acceptance criteria .

Detailed test procedures will be prepared and approved as described by Chapter 14.2.

Design personnel will review the test procedures and acceptance criteria for technical adequacy and evaluate test results. The test results will be compared with the predicted results and acceptance criteria to determine acceptability.

3.9B.2.2 Seismic Qualification Testing of Safety-Related Mechanical Equipment

All safety-related mechanical equipment which is required to retain structural integrity or structural integrity and operability during and after a postulated earthquake is subject to seismic qualification.

Active pumps and valves which must perform a mechanical motion during the course of accomplishing a system safety function include the active ASME Boiler and Pressure Vessel (B&PV) Code Class 2 and Class 3 pumps, Code Class 1, Class 2, and Class 3 valves, and their respective drives, operators, and vital auxiliary equipment. This equipment is qualified by testing, or analysis, or both, in accordance with the criteria given in Subsection 3.9B.3.2 and the recommendations of NRC Regulatory Guide 1.48 and as described in the following paragraphs.

Analysis without testing is accepted if it can be conservatively demonstrated that structural integrity alone can ensure operability of the seismic Category I equipment. When a complete seismic test is impracticable, combinations of testing and analysis are performed.

Seismic qualification by analysis is applicable to mechanical equipment which has relatively simple configurations and which can be modeled accurately. When analytical modeling is used, the equipment is modeled as a network of lumped masses and elastic springs in discrete parts. The response spectrum method is applied to calculate stresses and deformations resulting from the base excitations characterized by the required OBE and SSE in-structure floor response spectra of the seismic Category I buildings, and for the seismic analysis and testing of all seismic Category I subsystems and equipment located in the seismic Category I buildings as described in Section 3.7B.

The calculated seismic stresses are combined with the design load and thermal stresses for the various plant conditions defined in the ASME B&PV Code, Section III. It is ascertained that for each condition the resulting stresses are within the limits specified by the code and the recommendations of NRC Regulatory Guide 1.48.

When structural integrity alone cannot ensure operability for mechanically or structurally complex equipment not amenable to modeling and dynamic analysis, structural integrity and operability during and after a postulated earthquake are ensured by testing. This method consists of mounting the equipment to be qualified on a shake table, which is vibrated in such a way as to equal or exceed the required OBE and SSE in-structure floor response spectra applicable at the equipment locations in the seismic Category I buildings. A minimum of five OBE tests and one SSE test are performed. Equipment is tested in its operational condition; and, when possible, during the tests, operating, thermal and seismic loads are applied simultaneously. Operability is verified both during and after the tests.

Multi-directional seismic loading effects and dynamic coupling of the equipment are considered through the use of multi-axis testing techniques as recommended by IEEE 344-1975, IEEE Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations, and described in Section 3.7B.2.1.3. When dynamic analysis is used for seismic qualification, the dynamic coupling effect is considered by modeling the equipment with masses with a sufficient number of degrees of freedom and elastic properties representing the multi-directional stiffness of its various interconnecting parts. Multi-directional seismic loading is accomplished by performing the response spectrum analysis for each of the three orthogonal directions of earthquake excitation and combining the results by the square root of the sum of the squares (SRSS) technique.

A general classification of safety-related mechanical equipment and applicable quality standards is given in Table 3.2-1. Fluid system components and the applicable codes are classified in Table 17A-1.

A detailed description of seismic analysis and testing procedures is given in Section 3.7B.2.1.3.

The seismic qualification program for safety-related mechanical equipment is conducted by the equipment vendor. The procedures proposed for qualification and the results obtained by analysis or testing, or both, are submitted to the Applicant or his Agent for review.

All supports of seismic Category I mechanical equipment are seismically qualified to ensure their structural capability to withstand seismic excitation. The seismic qualification is accomplished by analysis, testing, or a combination of both for a particular support or a support representative of a group of supports.

If supports are similar and justified as such or if the worst case support determined by consideration of dynamic response (stiffness, structural strength, supported load) is chosen from a group of supports to be qualified and justified as such, only one of the similar supports or the worst case support requires a complete dynamic seismic analysis or a full-scale test, or a combination of both.

Justification of this procedure is based upon a simplified comparison analysis or by past experience indicating that the supports to be qualified are similar or that the worst case has been chosen. Upon such justification and dynamic analysis, or full scale testing, or a combination of

both of the similar or worst case support, the group of supports being investigated is accepted as seismically qualified.

The criteria governing the analysis or testing methods for seismic qualification of supports are presented in Section 3.7B.2.1.3

3.9B.2.3 Dynamic Response Analysis of Reactor Internals Under Operational Flow Transients and Steady-State Conditions

See Section 3.9N.2.3

3.9B.2.4 Preoperational Flow-Induced Vibration Testing of Reactor Internals

See Section 3.9N.2.4

3.9B.2.5 Dynamic System Analysis of the Reactor Internals Under Faulted Condition

See Section 3.9N.2.5

3.9B.2.6 Correlations of Reactor Internals Vibration Tests with the Analytical Results

See Section 3.9N.2.6

3.9B.3 ASME CODE CLASS 2 AND 3 COMPONENTS AND COMPONENT SUPPORTS

The ASME Code class components are designed and fabricated in accordance with ASME B&PV Code, Section III requirements.

Note: Inspection of Attachment welds during Hydrostatic Testing, as required by the ASME B&PV Code, was not performed on all type MV containment penetrations for Unit 1.

The qualification of ASME Section III Code Class 2 and 3 piping systems and their supports are in accordance with:

ASME Boiler and Pressure Vessel Code, Section III, Division 1 Nuclear Power Plant Components, 1974 Edition including the Summer 1974 Addenda Subsection NC and ND, and 1974 Edition including the Winter 1974 Addenda Subsection NF.

In addition to the above, as permitted by paragraph NA-1140 of the 1974 Edition of the Code, specific paragraphs in more recent editions and addenda of the ASME Code have been invoked.

Specific ASME Code Edition, Addenda and Cases utilized for the qualification of ASME Section III Code Class 2 and 3 piping systems and supports are documented in design, purchase and construction specifications. The use of Code Case N-397 is not utilized for non-NSSS piping stress analysis for Unit 1 but any subsequent usage will be detailed in the FSAR and in compliance with reference [9]. Details regarding the application and usages of N-318 are as listed in Table 3.9B-1F.

3.9B.3.1 Loading Combinations, Design Transients, and Stress Limits

Design pressure, temperature, and other loading conditions that provide the bases for design of fluid systems for ASME B&PV Code Class 2 and Class 3 components are presented in the sections which describe the systems.

3.9B.3.1.1 Design Loading Combinations

Design loading combinations and stress limits for ASME B&PV components, piping and piping supports are provided in the following tables:

	Loading Combinations	Stress Limits
Components	3.9B-1A	Note 1
Piping	3.9B-1B	3.9B-1B
Piping Supports	3.9B-1C	3.9B-1D 3.9B-1E

Note 1 Stress limits for each of the loading combinations are presented in Tables 3.9B-2, 3.9B-3, 3.9B-4, and 3.9B-5 for vessels, inactive[(a)] pumps, active pumps, and valves respectively.

The design loading combinations are categorized with respect to normal, upset, emergency, and faulted plant conditions. (Refer to Section 3.9B.1.1 for definition of plant conditions). Peak dynamic responses from loadings shown in Tables 3.9B-1A and 3.9B-1B are combined using the Square Root of the Sums of the Squares (SRSS) technique. This method of combining dynamic responses is consistent with the position outlined in NUREG-0484, "Methodology for Combining Dynamic Responses," Revision 1 dated May, 1980. Active[(b)] pumps and valves are discussed in Subsection 3.9B.3.2. Table 3.9B-8 lists all non-NSSS active pumps by system. Table 3.9B-10 lists all non-NSSS active valves including their design parameter and safety function.

For the design of supports for Class 2 and 3 active pump and valves, load combinations are also given in the component design specification. Stress limits conform to the requirements of Regulatory Guide 1.48 and Subsection NF of the ASME B&PV Code, Section III. In addition, component supports are included in the analysis of the component to ensure that deformations do not preclude operability. Operability requirements are presented in Section 3.9B.3.2.

a. Inactive components are those whose operability is not relied upon to perform a safety function while being subjected to the loading combinations associated with the respective plant operating condition categories.

b. Active components are those whose operability is relied upon to perform a safety function while being subjected to the loading combinations associated with the respective plant operating condition categories.

Systems which are required to operate during and after a postulated plant accident condition comply with the functional capability requirements delineated in References [5], [6], [7], and [8] in addition to the ASME Code requirements.

This requirement will ensure that the piping system will maintain its capability to deliver the rated flow and retain its dimensional stability under events specified above.

References [7] and [8] provides an alternative functional capability evaluation for stainless steel elbows and bends.

3.9B.3.2 Pump and Valve Operability Assurance

A list that identifies all active Code Class 2 and 3 pumps and valves is presented in Tables 3.9.B-8 and 3.9B-10.

Design specifications for active pumps and valves include the requirements for operability under the specified plant conditions. The design specifications define the design loads and the corresponding stress limits as discussed in Section 3.9B.3.1.1, relevant environmental conditions and operability requirements.

Active pumps and valves are qualified for operability in accordance with the requirements of Regulatory Guide 1.48. Pump and valve supports are designed in accordance with ASME B&PV Code Section III, Subsection NF. The following qualification methodology is used in the acceptance review of a vendor's operability program to ensure active valve or pump operability.

All active pumps are qualified for operability by first being subjected to rigid tests both prior to and after installation in the plant. The in-shop tests include the following:

1. Hydrostatic tests of pressure retaining parts to 150 percent times the design pressure.

2. Performance tests, while the pump is operated with flow, to determine total developed head, minimum and maximum head, net positive suction head (NPSH), and other pump/motor parameters.

Also monitored during these operating tests are bearing temperatures and vibration levels. Bearing temperature limits are determined by the manufacturer based on the bearing material, clearances, oil type, and rotational speed.

In addition to these tests, the safety-related active pumps are qualified for operability during an SSE condition by ensuring that the pump is not damaged during the seismic event and that the pump continues operating after being subjected to the SSE loads.

The pump manufacturer is required to show that the pump operates normally when subjected to the maximum seismic accelerations and maximum nozzle loads associated with the plant faulted condition. It is required that a test or a dynamic analysis be used to show that the lowest natural frequency of the pump is greater than 33 Hz. When having a natural frequency above 33 Hz, the pump is considered essentially rigid. This frequency is considered sufficiently high to avoid problems with amplification between the component and structure for all seismic areas. A static shaft deflection analysis of the rotor is performed with conservative SSE accelerations. The deflections determined from the static shaft analysis are compared to the allowable rotor

clearances. The nature of seismic disturbances dictates that the maximum contact (if it occurs) is of short duration. To avoid damage during the faulted plant condition, the stresses caused by the combination of normal operating loads, SSE, and dynamic system loads are limited to below the material elastic limit(s), as indicated in Table 3.9B-4. The average membrane stresses (σm) for the faulted condition loads are maintained at 1.0 S, and the maximum stress in local fibers (σm + bending stress σb) is limited to 1.5 S. In addition, the pump casing stresses caused by the maximum seismic nozzle loads are limited to stresses outlined in Table 3.9B-4. The maximum seismic nozzle loads are also considered in an analysis of the pump supports to ensure that a system misalignment cannot occur.

Performing these analyses with the conservative loads stated and with the restrictive stress limits of Table 3.9B-4 as allowables can ensure that critical parts of the pump are not damaged during the faulted condition and that the reliability of the pump for postfaulted condition operation is not impaired by the seismic event.

If the natural frequency is found to be below 33 Hz, an analysis is performed to determine the amplified input accelerations necessary to perform the static analysis. The adjusted accelerations are determined using the same conservatisms contained in the accelerations used for rigid structures. The static analysis is performed using the adjusted accelerations; the stress limits stated in Table 3.9B-4 must still be satisfied.

The second criterion necessary to ensure operability is that the pump functions throughout the SSE. The pump/motor combination is designed to rotate at a constant speed under all conditions unless the rotor becomes completely seized; i.e., there is no rotation. Typically, the rotor can be seized five full seconds before a circuit breaker opens to prevent damage to the motor. However, the high rotary inertia in the operating pump rotor, and the nature of the random, short duration loading characteristics of the seismic event prevents the rotor from losing its function. In actuality, the seismic loadings cause only a slight increase, if any, in the torque (i.e., motor current) necessary to drive the pump at the constant design speed. Therefore, the pump does not shut down during the SSE and it operates at the design speed despite the SSE loads.

To complete the seismic qualification procedures, the pump motor is independently qualified for operation during the maximum seismic event. Any auxiliary equipment which is identified to be vital to the operation of the pump or pump motor and which is not qualified for operation during the pump analysis or motor qualifications is also separately qualified for operation at the accelerations it will be subjected to where it is mounted. The pump motor and vital auxiliary equipment are qualified by meeting the requirements of IEEE 344-1975. If the testing option is chosen, sine-beat testing for electrical equipment will be justified by satisfying one or more of the following requirements to demonstrate that multi-frequency response is negligible or the sine-beat input is of sufficient magnitude to conservatively account for this effect.

1. The equipment response is basically due to one mode.

2. The sine-beat response spectra envelop the floor response spectra in the region of significant response.

3. The floor response spectra consist of one dominant mode and have a narrow peak at this frequency.

The degree of coupling in the equipment does, in general, determine if a single or multi-axis test is required. Multi-axis testing is required if there is considerable cross-coupling. If coupling is very light, then single-axis testing is justified; or, if the degree of coupling can be determined, single-axis testing can be used with the input sufficiently increased to include the effect of coupling on the response of the equipment.

From the previous arguments, the safety-related pump/motor assemblies are not damaged and continue operating under SSE loadings, and therefore, they perform their intended functions.

These proposed requirements take into account the complex characteristics of the pump and are sufficient to demonstrate, and ensure, the seismic operability of the active pumps.

The functional capability of active pumps after a faulted condition is ensured because only normal operating loads and steady-state nozzle loads then exist. Because the pumps are not damaged during the faulted condition, the postfaulted condition operating loads are identical to the normal plant operating loads. This is ensured by requiring that the imposed nozzle loads (steady-state loads) for normal conditions and postfaulted conditions be limited to the magnitudes of the normal condition nozzle loads. The postfaulted condition capability of the pumps to function under these applied loads is proven during the normal operating plant conditions for active pumps.

Safety-related active valves must perform their mechanical motion during the course of accomplishing a system safety function. When appropriate, assurance is supplied that these valves operate during a seismic event. When full scale tests are not performed, tests accompanied by analyses are conducted for these active valves.

The safety-related valves are subjected to a series of stringent tests prior to service and during the plant life. Prior to installation, the following tests are performed: shell hydrostatic test to ASME B&PV Code, Section III requirements, backseat and main seat leakage tests, disc hydrostatic test, functional tests to verify that the valve opens and closes within the specified time limits when subjected to the design differential pressure, operability qualification of motor operators for the environmental conditions over the installed life (i.e., aging, radiation, accident environment simulation, and so forth) according to procedures specified in IEEE 382. Cold hydro qualification tests, hot functional qualification tests, periodic inservice inspections, and periodic inservice operation are performed in situ to verify and ensure the functional capability of the valve. These tests guarantee reliability of the valve for the design life of the plant. The valves are designed using either stress analyses or the pressure containing minimum wall thickness requirements. On active valves not subject to dynamic testing for SSE conditions, an analysis of the extended structure is also performed for static equivalent seismic SSE loads applied at the center of gravity of the extended structure; the maximum stress limits allowed in these analyses show structural integrity and are the limits recommended by ASME for the particular ASME class of valve analyzed.

In addition to these tests and analyses, representative valves of each design type are tested for verification of operability during a simulated seismic event by demonstrating operational capabilities within the specified limits. The testing procedures are described as follows:

The valve is mounted in a manner which conservatively represents typical valve installations. The valve includes the operator and all accessories normally attached to the valve in service. The operability of the valve during SSE is demonstrated by satisfying the following criteria:

1. Active valves are typically designed to have a first natural frequency which is greater than 33 Hz. This can be shown by a suitable test or analysis.

2. The extended structure of the valve system is statically deflected an amount equal to that determined by an analysis as representing SSE accelerations applied at the center of gravity of the operator alone in the direction of the weakest axis of the yoke. The design pressure of the valve is simultaneously applied to the valve during the static deflection tests.

3. The valve is then operated while in the deflected position. The valve must perform its safety-related function within the specified operating time limits.

4. Motor operators, pilot solenoid valves, and limit switches necessary for operation are qualified as operable during SSE by appropriate IEEE seismic qualification standards, prior to their installation on the valve.

5. Operability may be demonstrated on a shaketable, with the valve body at design pressure, and temperature, in lieu of the static deflection tests.

6. The accelerations which are used for the valve qualification are equivalent, as justified by analysis, to (3.0g) in two orthogonal horizontal directions and (2.0g) in a vertical direction. The analyst will verify that valve seismic accelerations do not exceed the design g valves above.

7. If the frequency of the valve is less than 33 Hz, a dynamic analysis of the valve is performed to determine the equivalent deflection which is applied during the static test. The analysis provides the amplification of the input acceleration with consideration of the natural frequency of the valve and the frequency content of the applicable plant floor response spectra. The adjusted accelerations are determined using the same conservatisms contained in the (3.0g) horizontal and (2.0g) vertical accelerations used for rigid valves. The adjusted accelerations are then used in the static analysis, and valve operability is ensured by the methods previously outlined in steps 2 through 4, using the modified acceleration input.

This testing program applies to valves with powered operators which have extended structures. The testing is conducted on a representative number of valves. Valves from each of the primary safety-related design types are tested. Valve sizes which cover the range of sized in service are qualified by the test; the results are used to qualify all valves within the intermediate range of sizes.

Valves which are safety-related but can be classified as not having an extended structure, such as check valves and safety relief valves, are considered separately.

The check valves are characteristically simple in design and their operation is not affected by seismic accelerations or the maximum applied nozzle loads. The check valve design is compact and there are no extended structures or masses whose motion could cause distortions which could restrict operation of the valve. The nozzle loads caused by maximum seismic excitation do not affect the functional capability of the valve since the valve disc is designed to be isolated from the casing wall. The clearance supplied by the design around the disc prevents the disc from becoming bound or restricted because of any casing distortions caused by nozzle loads.

Therefore, the design of the valves is such that once the structural integrity of the valve is ensured, using standard design or analysis methods, the capability of the valve to operate is ensured by the design features. In addition to these design considerations, the valve also undergoes in-shop hydrostatic tests, and in-shop seat leakage test, and periodic in situ valve exercising and inspection to ensure the functional capability of the valve.

Manual valves which are Active are those for which credit is taken for post-accident operator action to perform a Nuclear Safety Function. The utilization of a manual Active valve is such that these valves are not required to perform a mechanical motion while being subjected to seismic loads. Therefore, a demonstration of operability of the valve during SSE is not required. Once the structural integrity of the valve is ensured, by compliance with stress limits using standard design or analysis methods, the capability of the valve to operate is assured by the design features. In addition to these design considerations, the valve also undergoes in-shop hydrostatic tests, in-shop seat leakage tests, and periodic in-situ valve exercising and inspection to assure the functional capability of the valve.

The functional capability of the feedwater check valves to operate following a postulated feedwater line break are also demonstrated.

Feedwater check valve maximum disc impact velocity and differential pressure across the valve disc are determined. A stress analysis of the valve considers the impact and the seismic inertia loads, which demonstrates the valve design for the postulated break.

Using the methods described, all the safety-related active valves in the systems are qualified for operability during a seismic/dynamic event. These methods conservatively simulate the seismic/ dynamic event and ensure that the active valves perform their safety-related function when necessary.

3.9B.3.3 Design and Installation of Pressure Relief Devices

The piping systems are designed to accommodate the effects of weight, dynamic blowdown thrust loads, and bending torsional and axial loads. Deflections resulting from these loading conditions are calculated in the stress analysis. Nozzle size and corresponding wall thickness are selected to ensure that the requirements of the design pressure and temperature are met and that the allowable stress limits are not exceeded.

The relief/safety valve stations in seismic Category I piping systems are shown on the piping and instrumentation diagrams in appropriate sections of the FSAR.

1. Installation Criteria

   The relief/safety valve nozzles, the exhaust stacks, and the piping on which the valves are installed are designed to withstand the thrusts that are imposed on the piping system when the relief/safety valve opens.

   The requirements set herein are in addition to all requirements of the applicable specifications, codes, and standards.

Piping systems subject to relief/safety valve reaction forces are provided with shock absorbing devices if necessary. This does not interfere with normal or intended thermal movement of the piping system.

The discharge piping is suspended, supported, restrained, guided, or anchored to avoid interference with structural members or equipment as well as to accommodate insulation limitations.

2. Design Criteria

a. Open Systems

For relief valves which are vented to the atmosphere and mounted in seismic Category I piping systems, Equation (9) of paragraph NC 3652, of the ASME B&PV Code, Section III, is used with the MB term to include the reaction force moment. In addition, the criteria of ASME Code, Section III, Appendix O are met.

Total resultant forces on the outlet elbow of pressure relieving devices are determined as follows:

$$\text{Total resultant forces} = \frac{WV}{g} + P\,A = Fv + Fp$$

where: W = lb/sec

A = area of pipe, in^2

V = velocity, ft/sec

P = pressure, psig

$g = 32.2\ ft/sec^2$

Fv = thrust due to velocity, lb

Fp = thrust due to pressure, lb

b. Closed Systems

A closed relief system may be either a system in which fluid discharges into a closed vessel or an open discharge system with a long discharge pipe. Of particular concern in closed relief systems are the large forces that may occur on piping that contains water seals (slug flow), two-phase flow, or if there is a water column in the discharge piping.

To establish the forcing functions necessary to perform a structural analysis of the piping, thermal/hydrodynamic models of the piping system are constructed. These models consist of one-dimensional representation of the piping system divided into reservoirs, pumps, valves, lengths of piped segments, branch connections, and other special piping components. Effects such as flow

restrictions and frictional resistance are considered. The time dependent pressure, temperature, density, velocity, and momentum are computed. Unbalanced segment forces are then obtained as a function of time.

The forcing functions are them applied to the piping structural model and system responses are determined by performing a time-history dynamic analysis.

As an alternative to using dynamic analysis to generate fluid transient forcing functions, conservative hand calculations may be performed to develop bounding pipe segment forces. These forces then may be analyzed statically, with a Dynamic Load Factor (DLF) or 2.0, or dynamically to obtain piping structural responses.

3.9B.3.4 Component Supports

3.9B.3.4.1 Nuclear piping (ASME Class 1)

Class 1 pipe support are designed in accordance with FSAR Table 3.9B-1C and 3.9B-1D.

3.9B.3.4.2 Nuclear Piping (ASME Class 2 and 3)

Plant conditions and load combinations for Class 2 and Class 3 components are shown in Table 3.9B-1A. Safety-related component supports are designated with the same safety class as their respective components, and are subject to the same plant conditions and loading combinations.

Class 2 and 3 supports are designed as follows:

1. Component Standard Supports

Normal Condition	Load Rating = T.L. x $1.0(S \text{ or } F_{All})/S_u$
Upset Condition	Load Rating = T.L. x $1.0(S \text{ or } F_{All})/S_u$
Emergency Condition	Load Rating = T.L. x $1.33\ (F_{All}/S_u)$
Emergency Condition (Plate and Shell Type)	Load Rating = T.L. x $1.2\ (S/S_u)$

where

T.L. = support test load equal to less than the load under which the component support fails to perform its specified support function

F_{All} = allowable value for the type of stress in XVII-1100 of the ASME III Code

S = allowable stress value at the design temperature (NF- 3112.2) from the applicable table of Appendix I of the ASME III Code

S_u = specified minimum ultimate tensile strength of the material used in the support as given in the applicable table of Appendix I of the ASME III Code

Faulted Condition - Analyzed in accordance with Tables 3.9B-1D and 3.9B-1E.

2. Linear Supports

a. Normal

The allowable stresses of Appendix XVII, as referenced in subsection NF, of the ASME B&PV Code, Section III, are used for normal condition limits.

b. Upset

Stress limits for upset conditions are the same as normal condition stress limits. This is consistent with subsection NF of the ASME B&PV Code, Section III (see Subarticle NF-3230).

c. Emergency

For emergency conditions, the allowable stresses or load ratings are 33 percent higher than those specified for normal conditions. This is consistent with subsection NF of ASME B&PV Code, Section III (see Subarticle NF-3231.lb), in which limits for emergency conditions are 33 percent greater than the normal condition limits.

d. Faulted

The allowable stresses of NF-3231.1c subsection NF, of the ASME Code Section III are used for the faulted condition.

3. Plate and Shell Supports

a. Normal

Normal condition limits are those specified in subsection NF of the ASME B&PV Code, Section III (see Subarticle NF- 3320).

b. Upset

Limits for upset conditions equal normal condition limits and are consistent with subsection NF of the ASME B&PV Code, Section III (see Subarticle NF-3320).

c. Emergency

For emergency conditions, the allowable stresses or load ratings are 20 percent higher than those specified for normal conditions.

d. Faulted

Limits for faulted conditions are the same as those for linear supports.

For active Class 2 and Class 3 pumps, support adequacy is proven by satisfying the criteria in Subsection 3.9B.3.2. In general, active valves are supported only by the pipe attached to the valve. Exterior supports on the valve are generally not used.

4. Mechanical Snubbers

Mechanical snubbers are utilized in the Containment Building, Safeguards Building, Auxiliary Building and Fuel Building as indicated on Table 3.9B-11, and are used for both shock and/or vibration. The rated load of the various models used is provided in Table 3.9-12.

The snubber design specification establishes the design, performance, fabrication, inspection, testing and installation requirements for these mechanical snubbers. It is the intent of the specification to provide nuclear quality equipment and be in compliance with ASME Boiler and Pressure Code Section III, including Subsection NF.

Performance testing consists of calculating the breakaway friction force, lost motion, and an acceleration/load test. In addition, the snubbers are tested for operation for the faulted load.

The average of the minimum tension and compression spring constants is used as input to the program. Load conditions for snubber design is SSE seismic for all safety-related systems. When applicable, loads due to relief valve discharge, turbine trip and valve closure are included in the analysis. This information is available via computer printout upon completion of the code analysis.

3.9B.3.4.3 Instrument Impulse Tubing Supports for ASME III Class 2 and Class 3 Safety Related Applications

1. All instrument impulse tubing, valves and fittings connecting instruments to ASME III Class 2 and Class 3 piping root valves will be seismically supported.

2. All instrument impulse tubing, valves and fittings connecting nuclear safety related instruments to Non-ASME piping or ducting in seismic category I structures will be seismically supported.

3. The support design will consider gravity, seismic and thermal loading combinations and will be consistent with the design requirements of AISC.

4. Any welding of the support system will be per AWS specifications consistent with that performed on other nuclear safety related, Non-ASME supports.

5. Subsection NF supports will be employed on the ASME III main line piping including the instrument root valve.

6. The material used for fabrication of the tubing supports will be purchased with certificates of compliance to applicable ASTM standards.

7. Other seismically designed support systems such as cable tray supports, pipe supports or conduit supports may be used for tubing if paragraph c above is met. Any necessary re-analysis will be performed to justify the additional loads.

8. AISC, "Specification for the Design, Fabrication and Erection of Structural Steel for Buildings" Nov. 1, 1978 including Supplement 1 March 11, 1986 may be utilized in the analysis of structural tubing.

9. The tubing stress analysis will conform to the loading combinations and stress limits of ASME Section III class 2 and 3 equations 8 thru 11.

3.9B.4 CONTROL ROD DRIVE SYSTEM (CRDS)

Refer to Section 3.9N.4

3.9B.5 REACTOR VESSEL INTERNALS

Refer to Section 3.9N.5

3.9B.6 INSERVICE TESTING OF PUMPS & VALVES

3.9B.6.1 Inservice Testing of Pumps

3.9B.6.1.1 Scope

Inservice testing of pumps shall be in accordance with the Inservice Testing Plan.

3.9B.6.1.2 Test Program

Establishment of reference values and a periodic testing schedule shall be in accordance with the Inservice Testing Plan.

3.9B.6.1.3 Test Frequencies and Durations

Test frequencies and durations shall be in accordance with the Inservice Testing Plan.

3.9B.6.1.4 Methods of Measurement

Methods of measurements shall be in accordance with the Inservice Testing Plan.

3.9B.6.2 Inservice Testing of Valves

3.9B.6.2.1 Scope

Inservice testing of valves shall be in accordance with the Inservice Testing Plan.

3.9B.6.2.2 Valve Test List

The scope of the valves tested and format for the valve tables are provided in the Inservice Testing Plan.

3.9B.6.2.3 Test Procedures

Valve test provisions in Technical Specifications shall meet any additional conditions of the Inservice Testing Plan.

REFERENCES

1. Swanson, John A., Engineering Analysis System, User's Manual (ANSYS), Swanson Analysis System, Inc., Elizabeth, Pa.

2. Dignwell, I.W., Static, Thermal, Dynamic Pipe Stress Analysis, Input Preparation IBM 1130 for Gibbs & Hill, Inc., (ADLPIPE), Arthur D. Little, Inc., Cambridge, Mass., Sept. 1972.

3. Greenstadt, J., "The Determination of the Characteristic Roots of a Matrix by Jacobi Method," Mathematical Methods for Digital Computers, John Wiley & Sons, Inc., New York, N.Y., 1959.

4. Documentation of ADLPIPE for Static and Dynamic Loads and Stress Evaluation, Arthur D. Little, Inc., Cambridge, Mass., Sept., 1973.

5. NEDO-21985, Functional Capability Criteria for Essential Mark II Piping, September 1978, prepared by Battelle Columbus Laboratories for General Electric Company.

6. NRC Evaluation of Topical Report, Piping Functional Capability Criteria, May 1980.

7. Stress Criteria for Demonstrating Functional Capability of ASME Class 2 and 3 Stainless Steel Elbows, Westinghouse Letter Number NS-LT-9447 TBX/TCS-4705, September 1981.

8. TUGCO Letter to NRC Proposing Stress Limits, TXX-3414, October 6, 1981.

9. U. S. Nuclear Regulatory Commission Evaluation of Request for Use of ASME Code Cases N-397 and N-411, March 13, 1985.

TABLE 3.9B-1A
DESIGN LOAD COMBINATIONS FOR ASME CODE CLASS 2 AND CLASS 3 COMPONENTS (EXCLUSIVE OF PIPING AND PIPING SUPPORTS)

Plant Conditions(1)	Loading Combinations
Normal	Design pressure + design temperature(2) + deadweight + thermal expansion + sustained loads + nozzle loads(3) + occasional loads, as applicable.
Upset	Design pressure + deadweight + thermal expansion + sustained loads + occasional loads + 0.5 SSE + transients + nozzle loads(3) + dynamic upset events(5)
Emergency	Design pressure + deadweight + sustained loads + thermal expansion + nozzle loads(3) + occasional load, as applicable
Faulted(4)	Design pressure + deadweight + sustained loads + SSE + pipe rupture and/or impingement effects where applicable + nozzle loads(3) + thermal expansion + occasional load + dynamic faulted events(5), as applicable.

Notes:

1. For active pumps the plant conditions listed are all considered normal component operating conditions and are specified as design conditions. For valves and inactive components, the plant conditions listed may be specified as component normal, upset, emergency or faulted conditions, depending on the function to be performed by the component.

2. Design temperature is used to determine allowable stress only.

3. Nozzle loads are those loads associated with the particular plant operating conditions for the component under consideration.

4. For the faulted condition. For the purpose of analysis, a pipe rupture shall be considered as required by section 3.6 of the FSAR for any pipe connected to the component. The results of the analysis shall demonstrate that the effects of a pipe rupture, if any, will not cause any other pipe connected to the component to rupture, or cause the component to move from its foundation.

5. Other dynamic events include relief valve discharge/waterhammer, etc.

TABLE 3.9B-1B
LOADING COMBINATION AND STRESS LIMITS FOR ASME III CLASS 2 & 3 PIPING

(Sheet 1 of 3)

Plant Condition	Loading Combination	Load Definition	Allowable Stresses (Paragraphs NC and ND-3650)	
Normal/ Testing	Design pressure deadweight	Sustained	S_h (eq 8)	$S_h + S_A$ (eq 11)
	Temperature (6) Thermal Anchor Movements Containment Displacements (10)	Thermal expansion Anchor Movements	S_A (eq 10)	
Upset	Design press. (3) Deadweight	Sustained	S_h (eq 8)	$S_h + S_h$ (eq 11) (5)
	1/2 SSE inertia 1/2 SSE anchor movement (2) Dynamic Upset Events (4)	Occasional	1.2 S_h (eq 9)	
	Temperature (6) Thermal Anchor Movements 1/2 SSE Anchor Movement (2)	Thermal expansion Anchor movements	S_A (eq 10)	
Emergency	Design pressure (3) Deadweight Jet impingement loads (7) Dynamic Emergency Events Pipe Impact Loads (7)	Sustained Occasional	1.8 S_h (eq 9)	

TABLE 3.9B-1B
LOADING COMBINATION AND STRESS LIMITS FOR ASME III CLASS 2 & 3 PIPING

(Sheet 2 of 3)

Plant Condition	Loading Combination	Load Definition	Allowable Stresses (Paragraphs NC and ND-3650)	
Faulted	Design pressure (3) Deadweight	Sustained		$S_h + S_A$ (8) (eq 11)
	SSE Inertia Jet impingement loads (7) Dynamic Faulted Events Pipe Impact Loads (7) LOCA	Occasional	2.4 S_h (eq 9)	
	Temperature (12) Thermal Anchor Movements Containment Displacements (11) SSE Anchor Movements	Thermal expansion Anchor Movements	S_A (eq 10) (8)	

Notes:

1. Based on ASME Code Case 1606-1 “Stress Criteria Section III, Class 2 & 3 piping subject to upset, emergency and faulted operating conditions.”
2. Anchor movement included in Equation (10) if omitted from Equation (9).
3. Design pressure is used since peak pressure and earthquake are not taken to be acting concurrently.
4. Dynamic upset events include: Relief Valve Actuation Steam/Water Hammer, etc.

TABLE 3.9B-1B
LOADING COMBINATION AND STRESS LIMITS FOR ASME III CLASS 2 & 3 PIPING

(Sheet 3 of 3)

Plant Condition	Loading Combination	Load Definition	Allowable Stresses (Paragraphs NC and ND-3650)

5. Does not include occasional loads. Equation (11) = Equation (8) + Equation (10).
6. Temperatures used correspond to Normal/Upset system operating conditions.
7. For essential piping only, as identified in Damage Study Analyses performed in accordance with FSAR Section 3.6B.
8. Applicable to those piping systems whose normal function is to prevent or mitigate the consequences of events associated with a plant faulted condition (i.e., containment spray and safety injection systems).
9. Deleted.
10. Containment displacements due to Preoperational Containment Structural Integrity Pressure Test and subsequent Containment Integrated Leak Rate Pressure Test.
11. Containment displacements due to pressure and temperature response of the containment during a faulted plant condition (LOCA).
12. Temperatures used to correspond to faulted plant conditions.

TABLE 3.9B-1C
LOADING COMBINATIONS ASME SECTION III CLASS 1, 2, AND 3 PIPING SUPPORTS

SYSTEM CONDITION	LOADING COMBINATION
Testing	Deadweight, Thermal, Containment Pressurization
Normal	Thermal Expansion Deadweight Equipment and Insulation Weights
Upset	Thermal Expansion Deadweight Equipment and Insulation Weights 1/2 SSE (Inertia and Anchor Movements) Plant Upset Dynamic Events (2)
Emergency	Thermal Expansion Deadweight Equipment and Insulation Weights Pipe Impact Loads (1) (2) Jet Impingement Loads (1) (2) Plant Emergency Dynamic Events (2)
Faulted	Thermal Expansion Deadweight Equipment and Insulation Weights 1.0 SSE (Inertia and Anchor Movements) Pipe Impact Loads (1) (2) Jet Impingement Loads (1) (2) Plant Faulted Dynamic Events (2) Containment Anchor Movements

Notes:

1. Not to be considered simultaneously

2. When applicable

TABLE 3.9B-1D
ALLOWABLE STRESSES ASME B&PV CODE SECTION III CLASS 1 PIPING SUPPORTS

		Allowable Stresses and Component Classification			
		Component Non-Standard Supports		Component Standard Supports	
Type Analysis	System Condition	Plates & Shell	Linear	Plate & Shell	Linear
	Design Normal Upset	Fig. NF-3221.1	NF-3231.1(a)	Fig. NF-3221.1	NF-3231.1(a)
Elastic	Emergency	Fig. NF-3221.1	NF-3231.1(b)	Fig. NF-3221.1	NF-3231.1(b)
	Faulted (4)	Appendix F (NA-2140)	NF-3231.1(c)	Appendix F (NA-2140)	NF-3231.1(c)
Load Rating	Normal Upset Emergency	NF-3262.2	NF-3262.3	NF-3262.4	NF-3262.4
	Faulted (4)	Note (2)	Note (2)	Note (2)	Note (2)

Notes:

1. Limit Analyses per Appendix XVII, Article 4000 may be used for elastic analysis of linear supports.

2. Guidelines not available. Faulted condition analysis to be performed with elastic analysis.

3. Loading combinations provided in Table 3.9B-1C.

4. Piping supports in systems whose normal function is to prevent or mitigate the consequences of events associated with a plant faulted condition (i.e., containment spray and safety injection systems) are designed within the limits specified for emergency condition.

TABLE 3.9B-1E
ALLOWABLE STRESSES ASME B&PV CODE SECTION III CLASS 2 AND 3 PIPING SUPPORTS

		Allowable Stresses and Component Classification			
		Component Non-Standard Supports		Component Standard Supports	
Type Analysis	System Condition	Plates & Shell	Linear	Plate & Shell	Linear
	Design Normal Upset	NF-3321.1 NF-3321.2(a) NF-3321.2(b)	NF-3231.1(a)	NF-3321.1 NF-3321.2(a) NF-3321.2(b)	NF-3231.1(a)
Elastic	Emergency	NF-3321.2(c)	NF-3231.1(b)	NF-3321.2(c)	NF-3231.1(b)
	Faulted (4)	NF-3321.2(d)	NF-3231.1(c)	NF-3321.2(d)	NF-3231.1(c)
Load Rating	Normal Upset Emergency	NF-3262.2	NF-3262.3	NF-3262.4	NF-3262.4
	Faulted (4)	Note (2)	Note (2)	Note (2)	Note (2)

Notes:

1. Limit Analyses per Appendix XVII, Article 4000 may be used for elastic analysis of linear supports.

2. Guidelines not available. Faulted condition analysis to be performed with elastic analysis.

3. Loading combinations provided in Table 3.9B-1C.

4. Piping supports in systems whose normal function is to prevent or mitigate the consequences of events associated with a plant faulted condition (i.e., containment spray and safety injection systems) are designed within the limits specified for emergency condition.

TABLE 3.9B-1F
USES OF CODE CASE N-318

(Sheet 1 of 3)

PIPE SUPPORT MARK NUMBER	TYPE OF WELD	NUMBER OF SIDES WELDED
CC-1-034-016-C82R	FILLET	THREE
CC-1-106-001-A43R	FILLET	THREE
CC-1-234-019-C53R	FILLET	TWO
CH-1-269-705-A43R	FILLET	TWO
CT-1-049-432-C82S	FILLET	TWO
CT-1-135-417-C72R	FILLET	TWO
CT-1-077-407-C72R	FILLET	TWO
CT-1-075-403-C72S	FILLET	TWO
CC-1-008-001-A43R	FILLET	THREE
CT-1-013-402-C52S	FILLET	THREE
SI-1-196-004-S32R	FILLET	THREE
CC-X-039-001-F43S	FILLET	THREE
CC-X-043-700-F43S	FILLET	THREE
CC-X-019-009-A43R	PARTIAL PENETRATION	THREE
CC-X-910-708-E23R	FILLET	TWO
CT-1-075-409-C62S	FILLET	TWO
VA-X-002-711-A75R	FILLET	THREE
CC-1-031-010-S43S	FILLET	FOUR (ALL ROUND)
CC-1-040-019-E33S	FILLET	THREE
CC-1-123-001-A43R	FILLET	THREE
FW-1-017-709-C72K	FILLET	THREE
SI-1-031-709-A32R	FILLET	THREE
CC-1-195-006-C42R	FILLET	THREE
SW-1-129-736-A43R	FILLET	THREE
SB-1-037-008-A45R	FILLET	THREE
AF-1-102-107-S33R	FILLET	THREE
FW-1-017-717-C52S	FILLET	THREE

TABLE 3.9B-1F
USES OF CODE CASE N-318

(Sheet 2 of 3)

PIPE SUPPORT MARK NUMBER	TYPE OF WELD	NUMBER OF SIDES WELDED
CT-1-013-405-C82S	FILLET	THREE
CC-1-050-701-A43R	PARTIAL PENETRATION	THREE
CT-1-074-411-C82R	FILLET	TWO
CT-1-039-443-C42K	FILLET	TWO
FW-2-018-441-C72R	FULL PENETRATION	TWO
AF-2-006-402-S33R	FILLET	THREE
CT-2-046-015-C92K	FILLET	THREE
CT-2-013-432-C52S	FILLET	THREE
CT-2-013-403-C72S	FILLET	THREE
CT-2-013-442-C82S	FILLET	THREE
CC-2-195-408-C52R	FILLET	TWO
CC-2-077-403-S43S	FILLET	TWO
SI-2-095-403-S42R	FILLET	TWO
H-FW-2-SB-023-003-5	FILLET	TWO
H-PS-2-RB-007-011-2	FILLET	TWO
H-MS-2-RB-011-005-2	FILLET	TWO
SI-2-051-415-C42K	FILLET	TWO
SI-2-051-421-C42R	FILLET	THREE
SI-2-095-415-C42R	FILLET	THREE
SI-2-087-416-C42K	FILLET	TWO
SI-2-306-404-C42R	FILLET	TWO
SI-2-306-416-C42R	FILLET	TWO
SI-2-306-420-C42R	FILLET	THREE
CS-2-001-454-C42S	FILLET	TWO
CS-2-079-408-C42S	FILLET	TWO
CS-2-079-409-C42S	FILLET	TWO

TABLE 3.9B-1F
USES OF CODE CASE N-318

(Sheet 3 of 3)

PIPE SUPPORT MARK NUMBER	TYPE OF WELD	NUMBER OF SIDES WELDED
CS-2-RB-022-703-2	FILLET	TWO
CS-2-079-431-C42K	PARTIAL PENETRATION	TWO
CS-2-RB-003-709-2	FILLET	TWO
PS-2-RB-024-003-2	FILLET	TWO
H-CH-2-SB-051-007-3	FILLET	THREE
H-CH-2-SB-051-008-3	FILLET	THREE
H-CH-2-SB-041-017-E	FILLET	THREE
H-CH-2-SB-042-010-3	FILLET	THREE
H-CH-2-SB-045-009-3	FILLET	THREE
H-CH-2-SB-039-001-3	FILLET	THREE
H-CH-2-SB-043-011-3	FILLET	TWO
H-CH-2-SB-044-006-3	FILLET	TWO
H-CH-2-SB-044-007-3	FILLET	TWO
H-CH-2-SB-046-003-3	FILLET	TWO
H-CH-2-SB-046-009-3	FILLET	TWO
H-PS-2-RB-015-007-2	FILLET	TWO
H-CH-2-SB-043-012-3	FILLET	TWO
H-CH-2-SB-044-003-3	FILLET	TWO
H-CH-2-SB-046-007-3	FILLET	TWO
H-PS-2-RB-010-003-2	FILLET	TWO
H-PS-2-RB-010-014-2	FILLET	TWO
H-PS-2-RB-010-016-2	FILLET	TWO
RC-2-097-403-C86S	FILLET	TWO
RC-2-115-417-C76S	FILLET	TWO
RC-2-115-419-C66K	FILLET	THREE
H-RC-2-SB-007-001-3	FILLET	TWO
H-RC-2-AB-001-016-3	FILLET	TWO

TABLE 3.9B-2
STRESS CRITERIA FOR SAFETY-RELATED ASME CODE CLASS 2 AND CLASS 3 VESSELS

Plant Condition[a]	Stress Limits
Normal	The vessel shall conform to the requirements of ASME B&PV Code, Section VIII, Division 1.
Upset	$\sigma_m \leq 1.1s \geq \frac{\sigma_m + \sigma_b}{1.5}$
Emergency	$\sigma_m \leq 1.1s \geq \frac{\sigma_m + \sigma_b}{1.5}$
Faulted	$\sigma_m \leq 1.5s \geq \frac{\sigma_m + \sigma_b}{1.5}$

a) See Note 1 on Table 3.9B-1A.

TABLE 3.9B-3
STRESS CRITERIA FOR ASME CODE CLASS 2 AND CLASS 3 INACTIVE PUMPS

Plant Condition[a]	Stress Limits
Normal	The pump shall conform to the requirement of ASME B&PV Code, Section III, NC-3400 or ND-3400.
Upset	$\sigma_m < 1.1\ S$
	$\sigma_m + \sigma_b < 1.65\ S$
Emergency	$\sigma_m < 1.1\ S$
	$\sigma_m + \sigma_b < 1.65\ S$
Faulted	$\sigma_m < 1.2\ S$
	$\sigma_m + \sigma_b < 1.8\ S$

a) See note 1 on Table 3.9B-1A.

TABLE 3.9B-4
DESIGN CRITERIA FOR ACTIVE PUMPS

Plant Condition[a]	Design Criteria
Normal	ASME B&PV Code, Section III, NC-3400 and ND-3400
Upset	$\sigma_m < 1.0\ S$
	$\sigma_m + \sigma_b < 1.5\ S$
Emergency	$\sigma_m < 1.0\ S$
	$\sigma_m + \sigma_b < 1.5\ S$
Faulted	$\sigma_m < 1.0\ S$
	$\sigma_m + \sigma_b < 1.5\ S$

a) See Note 1 on Table 3.9B-1A.

TABLE 3.9B-5
STRESS CRITERIA FOR SAFETY RELATED ASME CODE CLASS 2 AND 3 VALVES

Condition	Stress Limits (Notes 1-5)	P_{max}(Note 6)
Design & Normal	Valve bodies shall conform to the requirements of ASME Section III, NC-3500 (or ND-3500)	
Upset	$\sigma_m \leq 1.1\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b < 1.65\ S$	1.1
Emergency	$\sigma_m \leq 1.5\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 1.80\ S$	1.2
Faulted	$\sigma_m \leq 2.0\ S$ $(\sigma_m \text{ or } \sigma_L) + \sigma_b \leq 2.4\ S$	1.5

Notes:

1. Valve nozzle (piping load) stress analysis is not required when both the following conditions are satisfied by calculation: 1) section modulus and area of a plane, normal to the flow, through the region of valve body crotch is at least 10% greater than the piping connected (or joined) to the valve body inlet and outlet nozzles; and, 2) code allowable stress, S, for valve body material is equal to or greater than the code allowable stress, S, of connected piping material.

   If the valve body material allowable stress is less than that of the connected piping, the valve section modulus and area as calculated in (1) above shall be multipled by the ratio (S_{pipe}/S_{valve}). If unable to comply with this requirement, the design by analysis procedure of NB-3545.2 is an acceptable alternate method.

2. Casting quality factor of 1.0 shall be used.

3. These stress limits are applicable to the pressure retaining boundary, and include the effects of loads transmitted by the extended structures, when applicable.

4. Design requirements listed in this table are not applicable to valve discs, stems, seat rings, or other parts of valves which are contained within the confines of the body and bonnet, or otherwise not part of the pressure boundary.

5. These rules do not apply to Class 2 and 3 safety relief valves. Safety relief valves will be designed in accordance with ASME Section III requirements.

6. The maximum pressure resulting from upset, emergency or faulted conditions shall not exceed the tabulated factors listed under P_{max} times the design pressure or the rated pressure at the applicable operating condition temperature.

TABLE 3.9B-6
THIS TABLE IS INTENTIONALLY NOT USED

TABLE 3.9B-7
THIS TABLE IS INTENTIONALLY NOT USED

TABLE 3.9B-8
ACTIVE PUMPS

(Sheet 1 of 2)

PUMP	TAG NUMBER	SYSTEM	CLASS	NORMAL MODE	ACCIDENT MODE	ACTIVE FUNCTION
Auxiliary Feedwater (Motor Driven)	CP1&CP2-AFAPMD-01, 02	AF	3	ON/OFF	ON	Required for Removing Reactor Decay Heat (Safe Hot Shutdown)
Auxiliary Feedwater (Turbine Driven)	CP1&CP2-AFAPTD-01	AF	3	OFF	ON	Required for Removing Reactor Decay Heat (Safe Hot Shutdown)
Service Water	CP1&CP2-SWAPSW-01, 02	SW	3	ON/OFF	ON	Required for Transferring Heat from Primary Plant Safeguards Components to the Ultimate Heat Sink
Component Cooling Water	CP1&CP2-CCAPCC-01, 02	CC	3	ON/OFF	ON	Required for Transferring Heat from Components to the Service Water System
Containment Spray	CP1&CP2-CTAPCS-01, 02, 03, 04	CT	2	OFF	ON	Required for Containment Heat Removal
Spent Fuel Pool	CPX-SFAPSF-01, 02	SF	3	ON/OFF	ON	Required for Cooling of Spent Fuel
Reactor Makeup Water	CP1&CP2-DDAPRM-01	DD	3	ON/OFF	ON/OFF	Required to Provide Seismic Category I Makeup for Spent Fuel Pool, Component Cooling Water, and Safety Chilled Water

TABLE 3.9B-8
ACTIVE PUMPS

(Sheet 2 of 2)

PUMP	TAG NUMBER	SYSTEM	CLASS	NORMAL MODE	ACCIDENT MODE	ACTIVE FUNCTION
Reactor Makeup Water	CPX-DDAPRM-01	DD	3	ON/OFF	ON/OFF	Required to Provide Seismic Category I Makeup for Spent Fuel Pool, Component Cooling Water, and Safety Chilled Water
Safety Chilled Water	CP1&CP2-CHAPCP-05, 06	CH	3	ON/OFF	ON	Required to Transfer the Heat Rejected by Engineering Safety Feature Pump Motors and Electrical Switchgear to CCW
Diesel Generator Fuel Oil Transfer Pump	CP1&CP2-DOAPFT-01, 02, 03, 04	DO	3	OFF	ON/OFF	Required to Transfer Fuel Oil to Day Tank from the Storage Tank
Safeguard Bldg Floor Drain Sump Pumps	CP1&CP2-WPAPSS-01, 02, 03, 04	VD	3	ON/OFF	ON/OFF	Required to Detect and Mitigate passive failures in the ECCS and Containment Spray System post-LOCA and to prevent Flooding of Safety Related Equipment

TABLE 3.9B-9
BENCHMARK CALCULATIONS FOR ADLPIPE COMPUTER PROGRAM

Type of Analysis	Checks	Reference
Thermal and deadweight combined	Forces, moments, and deflections through the system	Pressure Vessel and Piping/1972 Computer Program Verification ASME Chapter 6, p. 1
Dynamic	Natural frequencies of a three-dimensional structure; mode shapes are checked (not published)	Pressure Vessel and Piping/1972 Computer Program Verification ASME Chapter 6, p. 1
Stress and usage factor	Stress range calculation and fatigue usage factor	Sample Analysis of a Piping System-Class 1 Nuclear, ASME
Thermal and deadweight (separate)	Forces, defelctions, stresses in accordance with ANSI B31.1	Stress in Three Dimensional Pipe Bends by W. Hovgaard Trans. ASME, Volume 57, 1935 pp. 401-456
Thermal	Thermal stress in accordance with ANSI B31.1, anchor reactions	Design of Piping System, M. W. Kellogg Company, p. 47
Dynamic	Natural frequencies, mode shapes, and response spectra, deflections and moments	Shock and Vibration by Young, ASME
Stress	All stress coefficients (product of stress indexes and geometry) used in Section III Class 1 Piping Analysis, that is specified in footnotes to stress indexes	Hand calculations
Stress	All stress components and their sum on selected piping	Hand calculations

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 1 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-2333A	MS	Globe/Gas-Hydraulic	32	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2333B	MS	Globe/Handwheel	4	2	Local Manual	Closed	Steam Line Isolation, Containment Isolation
HV-2409	MS	Globe/Air	2	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2452-1	MS	Globe/Air	4	2	Auto Trip/ Remote Manual	Closed	Turbine Driven AFW Pump Steam Supply, Containment Isolation
PV-2325	MS	Globe/Air	8	2	Remote Manual	Closed	Containment Isolation, Cooldown
HV-2334A	MS	Globe/Gas-Hydraulic	32	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2334B	MS	Globe/Handwheel	4	2	Local Manual	Closed	Steam Line Isolation, Containment Isolation
HV-2410	MS	Globe/Air	2	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
PV-2326	MS	Globe/Air	8	2	Remote Manual	Closed	Containment Isolation, Cooldown
HV-2335A	MS	Globe/Gas-Hydraulic	32	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2335B	MS	Globe/Handwheel	4	2	Local Manual	Closed	Steam Line Isolation, Containment Isolation
HV-2411	MS	Globe/Air	2	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
MS-0101	MS	Gate/Handwheel	4	2	Local Manual	Open	TDAFWP Steam Supply, Containment Isolation
MS-0128	MS	Gate/Handwheel	4	2	Local Manual	Open	TDAFWP Steam Supply Containment Isolation
PV-2327	MS	Globe/Air	8	2	Remote Manual	Closed	Containment Isolation, Cooldown
HV-2336A	MS	Globe/Gas-Hydraulic	32	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2336B	MS	Globe/Handwheel	4	2	Local Manual	Closed	Steam Line Isolation, Containment Isolation
HV-2412	MS	Globe/Air	2	2	Auto Trip	Open	Steam Line Isolation, Containment Isolation
HV-2452-2	MS	Globe/Air	4	2	Auto Trip/ Remote Manual	Closed	Turbine Driven AFW Pump Steam Supply, Containment Isolation
PV-2328	MS	Globe/Air	8	2	Remote Manual	Closed	Containment Isolation Cooldown
HV-2397	MS	Globe/Air	3	2	Auto Trip	Open	Auxiliary Feedwater System Actuation, Containment Isolation & Steam Generator Isolation

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 2 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-2398	MS	Globe/Air	3	2	Auto Trip	Open	Auxiliary Feedwater System Actuation, Containment Isolation & Steam Generator Isolation
HV-2399	MS	Globe/Air	3	2	Auto Trip	Open	Auxiliary Feedwater System Actuation, Containment Isolation & Steam Generator Isolation
HV-2400	MS	Globe/Air	3	2	Auto Trip	Open	Auxiliary Feedwater System Actuation, Containment Isolation & Steam Generator Isolation
HV-2401A/B	MS	Globe/Air	3/4	2	Auto Trip	Open	Auxiliary Feedwater System Actuation
HV-2402A/B	MS	Globe/Air	3/4	2	Auto Trip	Open	Auxiliary Feedwater System Actuation
HV-2403A/B	MS	Globe/Air	3/4	2	Auto Trip	Open	Auxiliary Feedwater System Actuation
HV-2404A/B	MS	Globe/Air	3/4	2	Auto Trip	Open	Auxiliary Feedwater System Actuation
HV-2405	MS	Globe/Air	3/4	2	Auto Trip	Open	Containment Isolation
HV-2406	MS	Globe/Air	3/4	2	Auto Trip	Open	Containment Isolation
HV-2407	MS	Globe/Air	3/4	2	Auto Trip	Open	Containment Isolation
HV-2408	MS	Globe/Air	3/4	2	Auto Trip	Open	Containment Isolation
MS-0021	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, Decay Heat Removal, CODE Safety Valve
MS-0022	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation CODE Safety Valve
MS-0023	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0024	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0025	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0026	MS	Gate/Gear/Extension Stem	8	2	Local Manual	Open	Atmospheric Relief Valve Isolation
MS-0058	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, Decay Heat Removal, CODE Safety Valve
MS-0059	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Value

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 3 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
MS-0060	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0061	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0062	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0063	MS	Gate/Gear/Extension Stem	8	2	Local Manual	Open	Atmospheric Relief Valve Isolation
MS-0093	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, Decay Heat Removal, CODE Safety Valve
MS-0094	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0095	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0096	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0097	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0098	MS	Gate/Gear/Extension Stem	8	2	Local Manual	Open	Atmospheric Relief Valve Isolation
MS-0129	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, Decay Heat Removal, CODE Safety Valve
MS-0130	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0131	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0132	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0133	MS	Safety	6	2	Self-Actuated	Closed	Steam Line Isolation, CODE Safety Valve
MS-0134	MS	Gate/Gear/Extension Stem	8	2	Local Manual	Open	Atmospheric Relief Valve Isolation
MS-0142	MS	Check	4	3	Self-Actuated	Closed	Turbine Driven AFW Pump Steam Supply Path and Boundary
MS-0143	MS	Check	4	3	Self-Actuated	Closed	Turbine Driven AFW Pump Steam Supply Path and Boundary

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 4 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
1MS-0680 (2MS-0664)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0681 (2MS-0665)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0682 (2MS-0666)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0683 (2MS-0665)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0684 (2MS-0668)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0685 (2MS-0667)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0686 (2MS-0670)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
1MS-0687 (2MS-0669)	MS	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply To S.G. PORV After Loss of Instrument Air
HV-2397A	MS	Globe/Air	3	2	Auto Trip	Open	Steam Generator Isolation
HV-2398A	MS	Globe/Air	3	2	Auto Trip	Open	Steam Generator Isolation
HV-2399A	MS	Globe/Air	3	2	Auto Trip	Open	Steam Generator Isolation
HV-2400A	MS	Globe/Air	3	2	Auto Trip	Open	Steam Generator Isolation
HV-4075B/C	FP	Gate/Motor	4	2	Auto Trip	Closed	Containment Isolation
HV-2134	FW	Gate/Gas-Hydraulic	18	2	Auto Trip	Open	Feedwater Isolation, Containment Isolation
HV-2135	FW	Gate/Gas-Hydraulic	18	2	Auto Trip	Open	Feedwater Isolation, Containment Isolation
HV-2136	FW	Gate/Gas-Hydraulic	18	2	Auto Trip	Open	Feedwater Isolation Containment Isolation
HV-2137	FW	Gate/Gas-Hydraulic	18	2	Auto Trip	Open	Feedwater Isolation, Containment Isolation
HV-2185	FW	Globe/Air	3	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 5 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-2186	FW	Globe/Air	3	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
HV-2187	FW	Globe/Air	3	2	Auto Close	Closed	Feedwater Isolation Containment Isolation
HV-2188	FW	Globe/Air	3	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
2-FV-2193	FW	Globe/Air	6	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
2-FV-2194	FW	Globe/Air	6	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
2-FV-2195	FW	Globe/Air	6	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
2-FV-2196	FW	Globe/Air	6	2	Auto Close	Closed	Feedwater Isolation, Containment Isolation
FW-0070	FW	Check	18	2	Self-Actuated	Open	Feedwater Isolation
FW-0076	FW	Check	18	2	Self-Actuated	Open	Feedwater Isolation
FW-0082	FW	Check	18	2	Self-Actuated	Open	Feedwater Isolation
FW-0088	FW	Check	18	2	Self-Actuated	Open	Feedwater Isolation
FW-0195	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
FW-0196	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
FW-0197	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
FW-0198	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
1-FW-0199	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
1-FW-0200	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
1-FW-0201	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
1-FW-0202	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
2-FW-0199	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
2-FW-0200	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
2-FW-0201	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below
2-FW-0202	FW	Check	6	2	Self-Actuated	Open/Closed	See Note 9, below

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 6 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
2-FW-0191	FW	Check	6	2	Self-Actuated	Closed	AFW Flow Path Boundary
2-FW-0192	FW	Check	6	2	Self-Actuated	Closed	AFW Flow Path Boundary
2-FW-0193	FW	Check	6	2	Self-Actuated	Closed	AFW Flow Path Boundary
2-FW-0194	FW	Check	6	2	Self-Actuated	Closed	AFW Flow Path Boundary
HV-2491A/B	AF	Gate/Motor	4	2	Remote Manual	Open	Containment Isolation
2-FV-2181	FW	Butterfly/Air	6	2	Auto Trip	Open	AFW Flow Path Boundary
2-FV-2182	FW	Butterfly/Air	6	2	Auto Trip	Open	AFW Flow Path Boundary
2-FV-2183	FW	Butterfly/Air	6	2	Auto Trip	Open	AFW Flow Path Boundary
2-FV-2184	FW	Butterfly/Air	6	2	Auto Trip	Open	AFW Flow Path Boundary
HV-2492A/B	AF	Gate Motor	4	2	Remote Manual	Open	Containment Isolation
HV-2493A/B	AF	Gate Motor	4	2	Remote Manual	Open	Containment Isolation
HV-2494A/B	AF	Gate Motor	4	2	Remote Manual	Open	Containment Isolation
HV-2480	AF	Gate Motor	6	3	Remote Manual	Closed	SW Flow Path to Suction
HV-2481	AF	Gate Motor	6	3	Remote Manual	Closed	SW Flow Path to Suction
HV-2482	AF	Gate Motor	8	3	Remote Manual	Closed	SW Flow Path to Suction
FV-2456	AF	Globe/Air	2	3	Auto Trip	Open	Recirculation Flow Path/ Closes to ensure adequate flow for accident mitigation
FV-2457	AF	Globe/Air	2	3	Auto Trip	Open	Recirculation Flow Path/ Closes to ensure adequate flow for accident mitigation
AF-0009	AF	Check	3	3	Self-Actuated	Open	Condensate Storage Tank Safety Class Isolation
AF-0350	AF	Check	2	3	Self-Actuated	Open	Condensate Storage Tank Safety Class Isolation
AF-0351	AF	Check	3	3	Self-Actuated	Closed	Condensate Storage Tank Safety Class Isolation
AF-0014	AF	Check	6	3	Self-Actuated	Closed	Suction from Condensate Storage Tank
AF-0024	AF	Check	6	3	Self-Actuated	Closed	Suction from Condensate Storage Tank

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 7 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
AF-0032	AF	Check	8	3	Self-Actuated	Closed	Suction from Condensate Storage Tank
AF-0065	AF	Check	6	3	Self-Actuated	Closed	Discharge Flow Path
AF-0051	AF	Check	6	3	Self-Actuated	Closed	Discharge Flow Path
AF-0038	AF	Check	8	3	Self-Actuated	Closed	Discharge Flow Path
AF-0093	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0098	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0083	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0086	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0075	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0078	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0101	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
AF-0106	AF	Check	4	3	Self-Actuated	Closed	Discharge Flow Path
1AF-0215 (2AF-0291)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0216 (2AF-0236)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0217 (2AF-0237)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0218 (2AF-0238)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0219 (2AF-0239)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0220 (2AF-0240)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
AF-0221	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 8 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
AF-0222	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0223 (2AF-0224)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0224 (2AF-0223)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0226 (2AF-0227)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0227 (2AF-0226)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
AF-0228	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
AF-0229	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0230 (2AF-0231)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
1AF-0231 (2AF-0230)	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Aux. F. W. Reg Valve After Loss of Instrument Air
AF-0232	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to AFW Turbine Isolation Valve After Loss of Instrument Air
AF-0233	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to AFW Turbine Isolation Valve After Loss of Instrument Air
AF-0234	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to AFW Turbine Isolation Valve After Loss of Instrument Air
AF-0235	AF	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to AFW Turbine Isolation Valve after Loss of Instrument Air
PV-2453A/B	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
PV-2454A/B	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control
HV-2459	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control
HV-2460	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control
HV-2461	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control
HV-2462	AF	Globe/Air/Handwheel	3	3	Remote Manual/ Local Manual	Open	AFW Flow Path, Steam Generator Isolation, Manually Modulated for Flow Control
HV-2484	AF	Butterfly/Motor	12	3	Auto Trip	Open	Condensate Storage Tank Isolation
HV-2485	AF	Butterfly/Motor	12	3	Auto Trip	Open	Condensate Storage Tank Isolation
AF-0167	AF	Check	8	3	Self-Actuated	Closed	Pump Recirculation Flow Path
HV-4777	CT	Gate/Motor	16	2	Auto Trip	Closed	Discharge Path Flow, Containment Isolation
CT-0145	CT	Check	16	2	Self-Actuated	Closed	Discharge Path Flow, Containment Isolation
HV-4776	CT	Gate/Motor	16	2	Auto Trip	Closed	Discharge Path Flow, Containment Isolation
CT-0142	CT	Check	16	2	Self-Actuated	Closed	Discharge Path Flow, Containment Isolation
HV-4782	CT	Gate/Motor	16	2	Remote Manual	Closed	Recirculation Flow Path, Containment Isolation
HV-4783	CT	Gate/Motor	16	2	Remote Manual	Closed	Recirculation Flow Path, Containment Isolation
CT-0309	CT	Relief	3/4	2	Self-Actuated	Closed	Thermal Relief, Containment Isolation
CT-0310	CT	Relief	3/4	2	Self-Actuated	Closed	Thermal Relief, Containment Isolation
CT-0047	CT	Check	4	2	Self-Actuated	Closed	Pump Recirculation Flow Path
CT-0048	CT	Check	4	2	Self-Actuated	Closed	Pump Recirculation Flow Path
CT-0063	CT	Check	4	2	Self-Actuated	Closed	Pump Recirculation Flow Path
CT-0064	CT	Check	4	2	Self-Actuated	Closed	Pump Recirculation Flow Path

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-4758	CT	Butterfly/Motor	16	2	Remote Manual	Open	CT Flow Path from RWST, Train A, Recirculation Boundary
HV-4759	CT	Butterfly/Motor	16	2	Remote Manual	Open	CT Flow Path from RWST, Train B, Recirculation Boundary
LV-4755	CT	Diaphragm/Motor	3	3	Auto Trip	Closed	Chemical Additive Flow to Suction Train B
LV-4754	CT	Diaphragm/Motor	3	3	Auto Trip	Closed	Chemical Additive Flow to Suction Train A
CT-0072	CT	Check	2	2	Self-Actuated	Closed	Chemical Additive Flowpath Train A
CT-0082	CT	Check	2	2	Self-Actuated	Closed	Chemical Additive Flowpath Train A
CT-0020	CT	Check	2	2	Self-Actuated	Closed	Chemical Additive Flowpath Train B
CT-0031	CT	Check	2	2	Self-Actuated	Closed	Chemical Additive Flowpath Train B
CT-0065	CT	Check	10	2	Self-Actuated	Closed	Containment Spray Discharge Flow, Train A
CT-0094	CT	Check	10	2	Self-Actuated	Closed	Containment Spray Discharge Flow, Train A
CT-0013	CT	Check	10	2	Self-Actuated	Closed	Containment Spray Pump Discharge Flow, Train B
CT-0042	CT	Check	10	2	Self-Actuated	Closed	Containment Spray Pump Discharge Flow, Train B
CT-0025	CT	Check	16	2	Self-Actuated	Closed	Containment Spray Pump Suction Flow, RWST Isolation, Train B
CT-0148	CT	Check	16	2	Self-Actuated	Closed	Containment Spray Pump Suction Flow Path, Train B
CT-0077	CT	Check	16	2	Self-Actuated	Closed	Containment Spray Pump Suction Flow, RWST Isolation, Train A
CT-0149	CT	Check	16	2	Self-Actuated	Closed	Containment Spray Pump Suction Flow Path, Train A
FV-4772-1	CT	Globe/Motor	4	2	Auto Trip	Open	Pump Recirculation Flow Path
FV-4772-2	CT	Globe/Motor	4	2	Auto Trip	Open	Pump Recirculation Flow Path
FV-4773-1	CT	Globe/Motor	4	2	Auto Trip	Open	Pump Recirculation Flow Path
FV-4773-2	CT	Globe/Motor	4	2	Auto Trip	Open	Pump Recirculation Flow Path
CP1/CP2-CTVBCA-01	CT	Vacuum Bkr	2	3	Self-Actuated	Closed	Vacuum relief (CAT)

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
CP1/CP2-CTVBCA-02	CT	Vacuum Bkr	2	3	Self-Actuated	Closed	Vacuum relief (CAT)
HV-4710	CC	Globe/Air	4	2	Auto Trip	Open	Containment Isolation
HV-4711	CC	Globe/Air	4	2	Auto Trip	Open	Containment Isolation
HV-4725	CC	Globe/Air	2	2	Auto Trip	Open	Containment Isolation
HV-4726	CC	Globe/Air	2	2	Auto Trip	Open	Containment Isolation
1CC-1067 (2CC-1090)	CC	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
CC-0611	CC	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief During LOCA
CC-0618	CC	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief During LOCA
HV-4708	CC	Gate/Motor	8	2	Auto Trip	Open	Containment Isolation
HV-4701	CC	Gate/Motor	8	2	Auto Trip	Open	Containment Isolation
CC-0629	CC	Check	2	2	Self-Actuated	Closed	Containment Isolation
HV-4700	CC	Gate/Motor	8	2	Auto Trip	Open	Containment Isolation
CC-0713	CC	Check	8	2	Self-Actuated	Open	Containment Isolation
HV-4699	CC	Gate/Motor	8	2	Auto Trip	Open	CCW System Isolation
HV-4709	CC	Gate/Motor	4	2	Auto Trip	Open	Containment Isolation
HV-4696	CC	Gate/Motor	4	2	Auto Trip	Open	Containment Isolation
CC-0831	CC	Check	1	2	Self-Actuated	Closed	Containment Isolation
HV-4514	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Loop Isolation, Train A
HV-4515	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Loop Isolation, Train B
HV-4572	CC	Butterfly/Motor	18	3	Auto Trip	Closed	CCW Flow Path RHR Loop, Train A
CC-0109	CC	Butterfly/Handwheel	18	3	Local Manual	Closed	CCW Flow Path RHR Loop, Train A
HV-4573	CC	Butterfly/Motor	18	3	Auto Trip	Closed	CCW Flow Path RHR Loop, Train B
CC-0157	CC	Butterfly/Handwheel	18	3	Local Manual	Closed	CCW Flow Path RHR Loop, Train B

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ACTIVE VALVES

(Sheet 12 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-4574	CC	Butterfly/Motor	18	3	Auto Trip	Closed	CCW Flow Path Containment Spray Loop, Train A
HV-4575	CC	Butterfly/Motor	18	3	Auto Trip	Closed	CCW Flow Path Containment Spray Loop, Train B
HV-4512	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Loop Isolation, Train A
HV-4513	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Loop Isolation, Train B
HV-4631A	CC	Globe/Air	2	3	Auto Trip	Open	Primary Sampling System Loop Isolation
HV-4631B	CC	Globe/Air	2	3	Auto Trip	Open	Primary Sampling System Loop Isolation
FV-4536	CC	Butterfly/Air	10	3	Auto Trip	Open	Recirculation Loop Isolation
FV-4537	CC	Butterfly/Air	10	3	Auto Trip	Open	Recirculation Loop Isolation
HV-4524	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Non-safeguard Loop Isolation
HV-4525	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Non-safeguard Loop Isolation
HV-4526	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Non-safeguard Loop Isolation
HV-4527	CC	Butterfly/Motor	24	3	Auto Trip	Open	CCW Non-safeguard Loop Isolation
LV-4500	CC	Globe/Air/Handwheel	3	3	Note 8	Closed	Emergency Makeup Water Path
LV-4500-01	CC	Globe/Air	3	3	Note 8	Open/Closed	Emergency Makeup Water Path
LV-4501	CC	Globe/Air/Handwheel	3	3	Note 8	Closed	Emergency Makeup Water Path
CC-0003	CC	Check	3	3	Self-Actuated	Closed	Emergency Makeup Water Path
CC-0004	CC	Check	3	3	Self-Actuated	Open	Emergency Makeup Water Path Boundary
CC-0031	CC	Check	24	3	Self-Actuated	Open	Discharge Flow Path
CC-0061	CC	Check	24	3	Self-Actuated	Open	Discharge Flow Path
FV-4650A	CC	Butterfly/Air	10	3	Auto Trip	Open	Ventilation Chiller Isolation
FV-4650B	CC	Butterfly/Air	10	3	Auto Trip	Open	Ventilation Chiller Isolation
X-PV-3583	CC	Plug/Motor	3	3	Auto Pressure Control	Open	Control Room A/C Condensor CCW Regulating Valve
X-PV-3584	CC	Plug/Motor	3	3	Auto Pressure Control	Open	Control Room A/C Condensor CCW Regulating Valve

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ACTIVE VALVES

(Sheet 13 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
X-PV-3585	CC	Plug/Motor	3	3	Auto Pressure Control	Open	Control Room A/C Condensor CCW Regulating Valve
X-PV-3586	CC	Plug/Motor	3	3	Auto Pressure Control	Open	Control Room A/C Condensor CCW Regulating Valve
PV-4552	CC	Ball/Air/Handwheel	3	3	Auto/Local Manual	Open	Safety Chiller Condenser CCW regulating valve
PV-4553	CC	Ball/Air/Handwheel	3	3	Auto/Local Manual	Open	Safety Chiller Condenser CCW regulating valve
X-PCV-H116A	CC	Plug/Air	1	3	Auto Pressure Control	Open	UPS A/C Condenser CCW Regulating Valve
X-PCV-H116B	CC	Plug/Air	1	3	Auto Pressure Control	Open	UPS A/C Condenser CCW Regulating Valve
CC-0646	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
CC-0657	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
CC-0687	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
CC-0694	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
1CC-1075 (2CC-0371)	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
1CC-1076 (2CC-0372)	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
1CC-1077 (2CC-0373)	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
1CC-1078 (2CC-0374)	CC	Check	2	3	Self-Actuated	Open	RCP Thermal Barrier Hx Back Flow Prevention
1CC-1079 (2CC-1091)	CC	Check	1/2	3	Self-Actuated	Closed	Inst. Air Safety Class Isolation
1CC-1080 (2CC-1092)	CC	Check	1/2	3	Self-Actuated	Open	Inst. Air Safety Class Isolation
1CC-1081 (2CC-1093)	CC	Check	1/2	3	Self-Actuated	Open	Inst. Air Safety Class Isolation
1CC-1082 (2CC-1094)	CC	Check	1/2	3	Self-Actuated	Open	Inst. Air Safety Class Isolation
HV-4286	SW	Butterfly/Motor	24	3	Auto Trip	Open	SW Flow Path, Train A
HV-4287	SW	Butterfly/Motor	24	3	Auto Trip	Open	SW Flow Path, Train B
HV-4393	SW	Butterfly/Motor	10	3	Auto Trip	Open	SW Flow Path, Train A
HV-4394	SW	Butterfly/Motor	10	3	Auto Trip	Open	SW Flow Path, Train B
HV-4395	SW	Butterfly/Motor	10	3	Remote Manual	Closed	Alternate AFW Flow Path, Train A

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-4396	SW	Butterfly/Motor	10	3	Remote Manual	Closed	Alternate AFW Flow Path, Train B
SW-0373	SW	Check	24	3	Self-Actuated	Open	Discharge Flow Path
SW-0374	SW	Check	24	3	Self-Actuated	Open	Discharge Flow Path
CP1(CP2)-SWVAVB-01	SW	Vacuum Relief	2	3	Self-Actuated	Closed	Mitigates Water Hammer
CP1(CP2)-SWVAVB-02	SW	Vacuum Relief	2	3	Self-Actuated	Closed	Mitigates Water Hammer
XSF-0003	SF	Check	10	3	Self-Actuated	Open	Discharge Flow Path
XSF-0004	SF	Check	10	3	Self-Actuated	Open	Discharge Flow Path
XSF-0160	SF	Check	3	3	Self-Actuated	Closed	Makeup Path
XSF-0161	SF	Diaphram/Handwheel	3	3	Local Manual	Closed	Spent Fuel Pool Makeup
XSF-0179	SF	Diaphram/Handwheel	3	3	Local Manual	Closed	Spent Fuel Pool Makeup
XSF-0180	SF	Check	3	3	Self-Actuated	Closed	Makeup Path
SI-0166	SI	Check	3/4	3	Self-Actuated	Closed/Open	Close on Non-Safety Line Break to maintain Accumulator nitrogen pressure.
SI-0167	SI	Check	3/4	3	Self-Actuated	Closed/Open	Close on Non-Safety Line Break to maintain Accumulator nitrogen pressure.
SI-0168	SI	Check	3/4	3	Self-Actuated	Closed/Open	Close on Non-Safety Line Break to maintain Accumulator nitrogen pressure.
SI-0169	SI	Check	3/4	3	Self-Actuated	Closed/Open	Close on Non-Safety Line Break to maintain Accumulator nitrogen pressure.
HV-5365	DD	Globe/Air	3	2	Auto Trip	Closed	Containment Isolation
HV-5366	DD	Globe/Air	3	2	Auto Trip	Closed	Containment Isolation
DD-0006	DD	Check	3	3	Self-Actuated	Open	Safety Class Isolation
1DD-0064 (2DD-0008)	DD	Check	2	3	Self-Actuated	Open	Safety Class Isolation
1DD-0065 (2DD-0002)	DD	Check	3	3	Self-Actuated	Open	Safety Class Isolation
1DD-0066 (2DD-0009)	DD	Check	2	3	Self-Actuated	Open	Safety Class Isolation

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
DD-0016	DD	Check	2	3	Self-Actuated	Open	Recirculation Flow Path
DD-0018	DD	Check	3	3	Self-Actuated	Open	Discharge Flow Path
1DD-0020	DD	Ball/Handwheel	3	3	Local Manual	Open	Safety Class Isolation
XDD-0103	DD	Ball/Handwheel	3	3	Local Manual	Open	Safety Class Isolation
DD-0430	DD	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
XDD-0044	DD	Check	2	3	Self-Actuated	Open	Recirculation Flow Path
XDD-0048	DD	Check	3	3	Self-Actuated	Closed	Discharge Flow Path
DO-0005	DO	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
DO-0004	DO	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
DO-0016	DO	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
DO-0017	DO	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
DO-0049	DO	Check	2	3	Self-Actuated	Closed	Fuel Oil Flow Path
1DO-0050 (2DO-0052)	DO	Check	2	3	Self-Actuated	Closed	Fuel Oil Flow Path
DO-0111	DO	Relief	1-1/2	3	Self Actuated	Closed	Recirculation Flow Path (miniflow)
DO-0187	DO	Relief	1-1/2	3	Self Actuated	Closed	Recirculation Flow Path (miniflow)
DO-0211	DO	Relief	1-1/2	3	Self Actuated	Closed	Recirculation Flow Path (miniflow)
DO-0287	DO	Relief	1-1/2	3	Self Actuated	Closed	Recirculation Flow Path (miniflow)
HV-3487	CA	Globe/Air	3	2	Auto Trip	Open	Containment Isolation
CI-0030	CA	Check	3	2	Self-Actuated	Open	Containment Isolation
1CI-0644 (Note 4)	CI	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Control Room Damper After A Non-Safety Air Line Break.
1CI-0645 (Note 4)	CI	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Control Room Damper After A Non-Safety Air Line Break.

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
1CI-0646 (Note 4)	CI	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Control Room Damper After A Non-Safety Air Line Break.
1CI-0647 (Note 4)	CI	Check	1/2	3	Self-Actuated	Closed/Open	Prevent Air Loss From Accumulator Air Supply to Control Room Damper After A Non-Safety Air Line Break.
HV-4168	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4169	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4170	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4167	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4166	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
1DO-0062	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
1DO-0063	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
1DO-0064	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
1DO-0065	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
2DO-0074	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
2DO-0075	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
2DO-0076	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
2DO-0077	DG	Check	1 1/2	3	Self-Actuated	Closed	Starting Air Pressure Boundary
DO-0104	DG	Check	1	3	Self-Activated	Open	Jacket Water Flow Path Boundary
DO-0204	DG	Check	1	3	Salf-Activated	Open	Jacket Water Flow Path Boundary
DO-0107	DG	3-Way	8	3	Self-Activated	Open	Jacket Water Temperature Control
DO-0207	DG	3-Way	8	3	Self-Activated	Open	Jacket Water Temperature Control
DO-0157	DG	Check	6	3	Self-Activated	Closed	Lube Oil Flow Path
DO-0257	DG	Check	6	3	Self-Activated	Closed	Lube Oil Flow Path

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
DO-0158	DG	Check	6	3	Self-Activated	Closed	Lube Oil Flow Path Boundary
DO-0258	DG	Check	6	3	Self-Activated	Closed	Lube Oil Flow Path Boundary
HV-4176	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4165	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4175	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4171	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
PS-0500	PS	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
PS-0501	PS	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
PS-0502	PS	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief Containment Isolation
PS-0503	PS	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
HV-4172	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4173	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4174	PS	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-4178	PS	Angle/Air	3/4	2	Auto Trip	Closed	RHR Loop to Primary Sampling System
HV-4179	PS	Angle/Air	3/4	2	Auto Trip	Closed	RHR Loop to Primary Sampling System
HV-4182	PAS	Angle/Air	3/4	2	Auto Trip	Closed	ECCS Operation
CA-0016	CA	Check	3	2	Self-Actuated	Closed	Containment Isolation
HV-3486	CA	Globe/Air	3	2	Auto Trip	Closed	Containment Isolation
HV-6084	CH	Gate/Motor	6	2	Auto Trip	Open	Containment Isolation
CH-0024	CH	Check	6	2	Self-Actuated	Open	Containment Isolation
HV-6082	CH	Gate/Motor	6	2	Auto Trip	Open	Containment Isolation
HV-6083	CH	Gate/Motor	6	2	Auto Trip	Open	Containment Isolation
1CH-0271 (2CH-0281)	CH	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation

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ACTIVE VALVES

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Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
1CH-0272 (2CH-0282)	CH	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
HV-6720	CH	Globe/Air	1	3	Remote Manual	Closed	Emergency Makeup Water Path
CH-0300	CH	Check	1	3	Self-Actuated	Closed	Emergency Makeup Water Path
CH-0301	CH	Check	1	3	Self-Actuated	Open	Emergency Makeup Water Path Boundary
CH-0302	CH	Globe/Handwheel	1	3	Local Manual	Closed	Safety Chilled Water Surge Tank Makeup
CH-0305	CH	Globe/Handwheel	1	3	Local Manual	Closed	Safety Chilled Water Surge Tank Makeup
HV-5542	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5543	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5563	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5540	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5541	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5562	VA	Butterfly/Motor	12	2	Auto Trip	Closed	Containment Isolation
HV-5556	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5557	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5544	RM	Globe/Solenoid	1	2	Auto Trip	Open	Containment Isolation
HV-5545	RM	Globe/Solenoid	1	2	Auto Trip	Open	Containment Isolation
HV-5558	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5559	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5560	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5561	PAS	Globe/Solenoid	1	2	Auto Trip	Closed	Containment Isolation
HV-5546	RM	Globe/Solenoid	1	2	Auto Trip	Open	Containment Isolation
HV-5547	RM	Globe/Solenoid	1	2	Auto Trip	Open	Containment Isolation
HV-5548	VA	Butterfly/Air	18	2	Auto Trip	Closed	Containment Isolation

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 19 of 22)

Valve Identification or Location No. (Note 1)	System	Valve Type And Actuator	Size In.	ANS Safety Class	Method of Actuation	Normal Position	Function (Note 5)
HV-5549	VA	Butterfly/Air	18	2	Auto Trip	Closed	Containment Isolation
HV-5157	VD	Diaphragm/Air	4	2	Auto Trip	Open	Containment Isolation
HV-5158	VD	Diaphragm/Air	4	2	Auto Trip	Open	Containment Isolation
VD-0003	VD	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
VD-0004	VD	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
VD-0011	VD	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
VD-0012	VD	Check	2	3	Self-Actuated	Closed	Discharge Flow Path
HV-7311	WP	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
HV-7312	WP	Angle/Air	3/4	2	Auto Trip	Closed	Containment Isolation
1VD-0907 (2VD-0896)	VD	Relief	3/4	2	Self-Actuated	Closed	Pressure Relief, Containment Isolation
1BS-0064 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA
1BS-0065 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA
1BS-0066 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA
1BS-0067 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA
1BS-0068 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA
1BS-0069 (Note 6)	BS	Relief	1/4	2 (Note 7)	Self-Actuated	Closed	Pressure Relief, during LOCA

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 20 of 22)

SYMBOL	SYSTEM
AF	AF Auxiliary Feedwater
BS	Buildings and Structures (Containment Pressure Airlock)
CA	Compressed Air-Service Air
CC	Component Cooling Water
CH	Chilled Water
CI	Compressed Air - Instrument Air
CS	Chemical and Volume Control System
CT	Containment Spray
DD	Demineralized and Reactor Makeup
DG	Diesel Generator and Auxiliary Systems
DO	Diesel Generator Fuel Oil
FP	Fire Protection
FW	Steam Generator Feedwater
MS	Main Steam, Reheat and Dump
PAS	Post Accident Sampling
PS	Process Sampling Primary Plant
RHR	Residual Heat Removal
SF	Spent Fuel Pool Cooling and Clean-Up
SW	Service Water
VA	Heating and Ventilation
VD	Vents and Drains
WP	Waste Processing
RM	Radiation Monitoring

TABLE 3.9B-10
ACTIVE VALVES

(Sheet 21 of 22)

GENERAL

RMWST	Reactor makeup water storage tank
RWST	Refueling Water Storage Tank
ATM	Atmospheric
N/A	Not Applicable

TABLE 3.9B-10
ACTIVE VALVES

Notes:

1) See Section 3.9N, Table 3.9N-10 for NSSS Active Valve List.

2) Unit 1 and Unit 2 Tag Numbers are generally the same except for the prefix. Any Unit 2 difference will be identified in a future amendment.

   Examples:

   1-HV-2333A is Unit 1
   2-HV-2333A is Unit 2
   HV-2333A indicates both Units
   CP1-CTVBCA-01 is Unit 1
   CP2-CTVBVCA-01 is Unit 2
   CP1/CP2-CTVBCA-01 indicates both Units
   X-PV-3533 indicates a common valve which serves both Units
   1AF-230 indicates Unit 1
   (2AF-231) indicates equivalent Unit 2 valves

3) Deleted.

4) Common valve; function is XCI.

5) See Section 6.2.4 for containment isolation details.

   Note: Steamline Isolation and Feedwater Isolation indicates valves which also function as Containment Isolation valves.

6) This equipment is unit specific. There is no corresponding equipment for the opposite unit.

7) These valves are not ASME Code and are not included in the Inservice Test Program. However, they are included in another program to assure their operability (as allowed by NUREG-1482, Appendix A, Position 11). Also, these valves are not included in the containment isolation tables of Section 6.2.4. These valves are within a closed system and satisfy the requirements of NUREG-0800, Section 6.2.4.11.6 paragraph o.

8) Valves LV-4500 and LV-4501 are fail closed valves to prevent overfill of the CCW surge tank on loss of power or air. Their nuclear safety function is to be manually opened via their hand wheels to provide seismic Category I makeup via LV-4500-1 which is a fail open valve. LV-4500-1 may be either open or closed in normal operation.

9) The feedwater check valves in the auxiliary feedwater lines to the steam generator open to provide the AFW flowpath during design basis events. These valves must close in the event of a line break upstream of the outboard check valve in order to terminate a Condition III loss of feedwater event. In addition, these valves also must close post-LOCA to protect the cold containment penetration during intermittent operation of AFW.

TABLE 3.9B-11
LOCATION OF SNUBBERS IN THE CONTAINMENT, SAFEGUARDS,
AUXILIARY AND FUEL BUILDINGS

System	Containment Building	Safeguards Building	Auxiliary Building	Fuel Building
Main Steam	X	X		
Feedwater	X	X		
Auxiliary Feedwater		X		
Safety Injection	X	X		
Component Cooling Water	X	X	X	X
Residual Heat Removal	X	X		
Containment Spray	X	X		
Service Water		X	X	
Reactor Coolant System	X			
Boron Recycle System	(Only Rigid Restraints Planned)			
Diesel Oil System	(Only Rigid Restraints Planned)			
Chemical and Volume Control	X	X	X	
Spent Fuel Pool Cooling and Cleanup System	X	X		X
Boron Thermal Regeneration Sub-system	(Only Rigid Restraints Planned)			
Demineralized Water Makeup System	(Only Rigid Restraints Planned)			

TABLE 3.9B-12
RATED LOAD OF MECHANICAL SNUBBERS

The snubbers are manufactured by Pacific Scientific and are rated as follows:

Model	Rated Loads (lbs)
PSA-1/2	650
PSA-1/4	350
PSA-100	120,000
PSA-35	50,000
PSA-10	15,000
PSA-3	6,000
PSA-1	1,500

3.10N SEISMIC QUALIFICATION OF SEISMIC CATEGORY I INSTRUMENTATION AND ELECTRICAL EQUIPMENT

This section presents information to demonstrate that instrumentation and electrical equipment classified as seismic Category I is capable of performing designated safety related functions in the event of an earthquake. The information presented includes identification of the Category I instrumentation and electrical equipment that are within the scope of the Westinghouse Nuclear Steam Supply System (NSSS), the qualification criteria employed and for each item of equipment; the designated safety related functional requirements, definition of the applicable seismic environment and documentation of the qualification process employed to demonstrate the required seismic capability.

3.10N.1 SEISMIC QUALIFICATION CRITERIA

3.10N.1.1 Qualification Standards

The methods of meeting the general requirements for seismic qualification of Category I instrumentation and electrical equipment as described by General Design Criteria (GDC) 1, 2, and 23 are described in Section 3.1. The general methods of implementing the requirements of Appendix B to 10CFR Part 50 are described in Chapter 17.

The Commission's recommendations concerning the methods to be employed for seismic qualification of electrical equipment are contained in Regulatory Guide 1.100, which endorses IEEE-344-1975. Westinghouse will meet this standard, as modified by Reg. Guide 1.100, by either type test, analysis, or an appropriate combination of these methods. Westinghouse will meet this commitment employing the methodology described in Reference [1], and the qualification program plans and procedures described in Reference [2].

3.10N.1.2 Performance Requirements for Seismic Qualification

Reference 2 contains an equipment qualification data package (EQDP) for every item of instrumentation and electrical equipment classified as seismic Category I within the Westinghouse NSSS scope of supply. Table 3.10N.1 identifies the Category I equipment supplied by Westinghouse for this application and references the applicable EQDP contained in Supplement 1. Each EQDP in Supplement 1 contains a section entitled "Performance Specifications". This specification establishes the safety related functional requirements of the equipment to be demonstrated during and after a seismic event. The required response spectrum (RRS) employed by Westinghouse for generic seismic qualification is also identified in the specification, as applicable. The spectra employed envelopes the plant specific spectra defined in Section 3.7.

3.10N.1.3 Acceptance Criteria

Seismic qualification must demonstrate that Category I instrumentation and electrical equipment is capable of performing designated safety related functions during and after an earthquake of magnitude up to and including the Operating Basis Earthquake (OBE) and Safe Shutdown Earthquake (SSE) without the initiation of undesired spurious actuation which might result in consequences adverse to safety. The qualification must also demonstrate the structural integrity of mechanical supports and structures at the OBE level. Some permanent mechanical

deformation of supports and structures is acceptable at the SSE level providing that the ability to perform the designated safety related functions is not impaired.

3.10N.2 METHODS AND PROCEDURES FOR QUALIFYING ELECTRICAL EQUIPMENT AND INSTRUMENTATION

In accordance with IEEE 344-1975, seismic qualification of safety related electrical equipment is demonstrated by either type testing, analysis or a combination of these methods. The choice of qualification method employed by Westinghouse for a particular item of equipment is based upon many factors including; practicability, complexity of equipment, economics, availability of previous seismic qualification to earlier standards, etc. The qualification method employed for a particular item of equipment is identified in the individual Equipment Qualification Data Packages (EQDP's) of Reference 2.

All NSSS safety-related electrical equipment located in a mild environment area will be seismically qualified as described in FSAR Section 3.10N, except that the additional requirements imposed by IEEE Standard 323-1974 do not apply. The procurement documents will specify that the effects of aging on seismic qualification be assessed and if there are aging effects, require pre-aging or analysis of aging effects as part of the seismic qualification.

3.10N.2.1 Seismic Qualification by Type Test

From 1969 to mid-1974 Westinghouse seismic test procedures employed single axis sine beat inputs in accordance with IEEE 344-1971 to seismically qualify equipment. The input form selected by Westinghouse was chosen following an investigation of building responses to seismic events as reported in Reference 3. In addition, Westinghouse has conducted seismic retesting of certain items of equipment as part of the Supplemental Qualification Program (Reference 4). This retesting was performed at the request of the NRC staff on agreed selected items of equipment employing multi-frequency, multi-axis test inputs (Reference 5) to demonstrate the conservatism of the original sine-beat test method with respect to the modified methods of testing for complex equipment recommended by IEEE 344-1975.

The original single axis sine beat testing and the additional retesting completed under the Supplemental Test Program has been the subject of generic review by the Staff. For equipment which has been previously qualified by the single axis sine beat method and included in the NRC seismic audit and, where required by the Staff, the Supplemental Qualification Program (Reference 4), no additional qualification testing is required to demonstrate acceptability to IEEE 344-1975 provided that:

1. The Westinghouse aging evaluation program for aging effects on complex electronic equipmet located outside containment demonstrates there are no deleterious aging phenomena. In the event that the aging evaluation program identifies materials that are marginal, either the materials will be replaced or the projected qualified life will be adjusted.

2. Any changes made to the equipment due to 1. above or due to design modifications do not significantly affect the seismic characteristics of the equipment.

3. The previously employed test inputs can be shown to be conservative with respect to applicable plant specific response spectra.

This equipment is identified in Reference 1, Table 7.1 and the test results in the applicable EQDP's of Reference 2.

For equipment tests after July, 1974 (i.e. new designs, equipment not previously qualified or previously qualified that does not meet 1, 2 and 3 above) seismic qualification by test is performed in accordance with IEEE 344-1975. Where testing is utilized, multi-frequency multi-axis inputs are developed by the general procedures outlined in Reference 5. The test results contained in the individual EQDP's of Reference 2 demonstrate that the measured Test Response Spectrum envelopes the applicable Required Response Spectrum (RRS) defined for generic testing as specified in Section 1 of the EQDP (Reference 2). Qualification for plant specific use is established by verification that the generic RRS specified by Westinghouse envelopes the applicable plant specific response spectrum. Alternative test methods, such as single frequency, single axis inputs, are used in selected cases as permitted by IEEE 344-1975 and Regulatory Guide 1.100.

3.10N.2.2 Seismic Qualification by Analysis

The structural integrity of safety related motors (Table 3.10N.1 EQDP- AE-2 and 3) is demonstrated by a static seismic analysis in accordance with IEEE 344-1975 with justification. Should analysis fail to show the resonant frequency to be significantly greater than 33 Hz, a test is performed to establish the motor resonant frequency. Motor operability during a seismic event is demonstrated by calculating critical deflections, loads and stresses under various combinations of seismic, gravitational and operational loads. The worst case (maximum) values calculated are tabulated against the allowable values. On combining these stresses, the most unfavorable possibilities are considered in the following areas; 1) maximum rotor deflection, 2) maximum shaft stresses, 3) maximum bearing load and shaft slope at the bearings, 4) maximum stresses in the stator core welds, 5) maximum stresses in the stator core to frame welds, 6) maximum stresses in the motor mounting bolts and, 7) maximum stresses in the motor feet.

The analytical models employed and the results of the analysis are described in Section 4 of the applicable EQDP's (Reference 2).

3.10N.3 METHOD AND PROCEDURES FOR QUALIFYING SUPPORTS OF ELECTRICAL EQUIPMENT AND INSTRUMENTATION

Where supports for the electrical equipment and instrumentation are within the Westinghouse NSSS scope of supply, the seismic qualification tests and/or analysis are conducted including the supplied supports. The EQDP's contained in Reference 2 identify the equipment mounting employed for qualification purposes and establish interface requirements for the equipment to ensure subsequent in-plant installation does not prejudice the qualification established by Westinghouse.

3.10N.4 OPERATING LICENSE REVIEW

The results of tests and analyses that ensure that the criteria established in Section 3.10N.1 have been satisfied employing the qualification methods described in Sections 3.10N.2 and 3.10N.3 are included in the individual EQDP's contained in Reference 2 as they become available.

REFERENCES

1. Butterworth, G. and Miller, R. B., “Methodology for Qualifying Westinghouse WRD Supplied NSSS Safety Related Electrical Equipment,” WCAP-8587, Revision 6, March 1983.

2. “Equipment Qualification Data Packages,” Supplement 1 to WCAP- 8587.

3. Morrone, A., “Seismic Vibration Testing with Sine Beats,” WCAP- 7558, October 1971.

4. NS-CE-692, Letter dated July 10, 1975 from C. Eicheldinger (Westinghouse) to D. B. Vasello (NRC).

5. Jarecki, S. J., “General Method of Developing Multi-frequency Biaxial Test Inputs for Bistables,” WCAP-8624 (Proprietary) September 1975 and WCAP-8695 (Non-Proprietary) August 1975.

TABLE 3.10N-1
SEISMIC CATEGORY I INSTRUMENTATION AND ELECTRICAL EQUIPMENT IN WESTINGHOUSE NSSS SCOPE OF SUPPLY

EQUIPMENT	EQDP[a]
SAFETY RELATED VALVE ELECTRIC MOTOR OPERATORS	EQDP-HE-1 AND 4
SAFETY RELATED PILOT SOLENOID VALVES	EQDP-HE-2
SAFETY RELATED EXTERNALLY MOUNTED LIMIT SWITCHES	EQDP-HE-3
RCS SAFETY VALVE LIMIT SWITCHES	EQDP-HE-7
CONAX CONNECTORS	EQDP-HE-8
SOLENOID VALVES - RCS VENT	EQDP-HE-10
LARGE PUMP MOTORS (OUTSIDE CONTAINMENT)	EQDP-AE-2
CANNED PUMP MOTORS (OUTSIDE CONTAINMENT)	EQDP-AE-3
PRESSURE TRANSMITTERS	EQDP-ESE-1 AND 2
DIFFERENTIAL PRESSURE TRANSMITTERS	EQDP-ESE-3 AND 4
RESISTANCE TEMPERATURE DETECTORS	EQDP-ESE-6 AND 7
NUCLEAR INSTRUMENTATION SYSTEM (NIS)	EQDP-ESE-10
PROCESS PROTECTION SETS	EQDP-ESE-13
INDICATORS, POST-ACCIDENT MONITORING	EQDP-ESE-14
RECORDERS, POST-ACCIDENT MONITORING	EQDP-ESE-15
SOLID-STATE PROTECTION SYSTEM & SAFEGUARD TEST CABINET (2 TRAIN)	EQDP-ESE-16
REACTOR TRIP SWITCHGEAR	EQDP-ESE-20
CONTAINMENT PRESSURE SENSOR	EQDP-ESE-21
EXCORE NEUTRON DETECTORS (POWER RANGE)	EQDP-ESE-22
NITROGEN-16 DETECTOR	EQDP-ESE-27
DIFFERENTIAL PRESSURE INDICATING SWITCHES	EQDP-ESE-40
GROUP B	
INCORE THERMOCOUPLE CONNECTORS	EQDP-ESE-43
THERMOCOUPLE JUNCTION BOX	EQDP-ESE-44
EXCORE NEUTRON DETECTORS (SOURCE RANGE), PRE-AMPLIFIERS AND FLUX DOUBLING CIRCUIT	EQDP-ESE-47

a) Equipment Qualification Data Package

3.10B SEISMIC QUALIFICATION OF SEISMIC CATEGORY I INSTRUMENTATION AND ELECTRICAL EQUIPMENT

3.10B.1 SEISMIC QUALIFICATION CRITERIA

The seismic Category I instrumentation and electrical equipment which require seismic qualification are as follows:

1. 6900-V switchgear (nuclear-safety-related)
2. 6900-V to 480-V transformers (associated with nuclear-safety-related buses)
3. 480-V switchgear (nuclear-safety-related)
4. 480-V motor control and motor control centers (nuclear-safety- related)
5. 125-V station batteries and racks
6. 480-VAC to 125-VDC battery chargers
7. 125-VDC panels and switchboards (nuclear-safety-related)
8. 125-VDC to 120-VAC inverters (nuclear-safety-related instrument buses)
9. Nuclear-safety-related instrument bus panels
10. Containment penetration assemblies
11. Not Used
12. Diesel generator and accessories
13. Diesel generator control panels
14. Relay boards and racks (nuclear-safety-related)
15. Instrument racks (nuclear-safety-related)
16. Hot shutdown panel (used in event of Control Room evacuation)
17. Main Control boards
18. Vertical panels (including solid state sequencer and isolation equipment)
19. Wire and cable raceway system (nuclear-safety-related)
20. Electrical supports (nuclear-safety-related)
21. Containment particulate, iodine, and gas radiation monitors

22. Motors (nuclear-safety-related)

23. 120/208 VAC power distribution and lighting panels

24. Miscellaneous 3-phase 480-108/120v and 1-phase 480-120v transformers

25. Control Room ventilation radiation monitors

26. Accident Monitoring System including sensors (see Section 7.5)

27. Electronic Transmitters-pressure and differential pressure (nuclear-safety-related).

28. Analog Control Cabinets (Train A and Train B)

29. Resistance Temperature Detectors (nuclear-safety-related)

30. Level switches (nuclear-safety-related)

31. Pressure switches (nuclear-safety-related)

32. Differential pressure switches (nuclear-safety-related)

The criteria for establishing seismic design adequacy, by testing or analysis, are described in Section 3.7B.2.

Seismic Category I instrumentation and electrical equipment including standby power systems are designed to maintain structural integrity and functional operability during and after an earthquake of magnitude up to and including the SSE. Seismic design of the Reactor Protection System and ESF circuits is discussed in Section 3.10N.

Horizontal and vertical ground accelerations during an SSE are used to formulate floor response spectra at each equipment location as described in Section 3.7B.2. Equipment and system specifications include the appropriate response spectra used for the analysis or testing of the equipment. The seismic qualification program for electrical equipment is as described in Section 3.7B.2.1.3. Analysis is used for the qualification of equipment of relatively simple geometry which can be modeled accurately.

Suppliers are required to furnish documentation of actual test results or detailed computations, if an analytical approach is used, to substantiate the capability of the equipment to perform its intended function under the specified conditions.

Seismic Category I instrumentation and electrical equipment are seismically qualified in accordance with the procedures and documentation requirements specified in IEEE 344-1975," Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations." For equipment in mild environment areas, the procurement documents will require that the effects of aging on seismic qualification be assessed and, if there are aging effects, will require pre-aging or analysis of aging effects as part of the seismic qualification requirements.

3.10B.2 METHODS AND PROCEDURES FOR QUALIFYING INSTRUMENTATION AND ELECTRICAL EQUIPMENT

All seismic Category I instrumentation and electrical equipment are qualified by analysis or testing, or both. Seismic analysis without testing is used if it is demonstrated that the performance of the equipment subjected to earthquake motion can be conservatively predicted and if the functional operability of the instrumentation or equipment is assured by its structural integrity alone.

The seismic analysis methods (response spectrum, time history) and the testing procedures used are described in Section 3.7B.2.1.

Where analysis is used for qualifying the seismic Category I instrumentation and electrical equipment, it is required that the maximum stresses and deformations in the equipment including the effects of the normal operating loads plus the SSE be limited to prevent loss of function of the equipment. To ensure structural integrity and functional operability of the equipment after several occurrences of the OBE, it is required that the maximum stresses in the equipment, including the effects of the normal operating loads plus the OBE, be maintained within the normal allowable material working stress limits set forth in the appropriate design standards and codes, and that the equipment operate and maintain structural integrity without permanent deformation.

When testing is used for the seismic qualification of seismic Category I instrumentation and electrical equipment, tests at the SSE level are performed to qualify the equipment for operation or structural integrity, or both, during and after the SSE. Each test at the SSE level is preceded by a minimum of five tests at the OBE level to verify the structural and functional integrity of the equipment after several occurrences of the OBE.

3.10B.3 METHODS AND PROCEDURES OF ANALYSIS OR TESTING OF SUPPORTS OF INSTRUMENTATION AND ELECTRICAL EQUIPMENT

Supports of instrumentation and electrical equipment such as battery racks, instrument racks, control consoles, cabinets, and panels are analyzed or tested, or both, by their suppliers in accordance with the methods and procedures described in Section 3.7B.2.1.3.

Such supports are generally required to have overall natural frequencies greater than 33 Hz. Where this requirement cannot be met, suppliers are required to qualify their products by performing full dynamic analysis or testing, or both, to demonstrate their structural integrity during and after the SSE, and to generate response spectra or derive maximum amplification factors at all equipment and instrument mounting locations. The equipment and instruments to be mounted on these supports are then analyzed or tested by their suppliers on the basis of the response spectra or maximum accelerations furnished by support manufacturers.

The supplier accounts for possible amplification through equipment supports by analysis or testing, as described in Section 3.7B.2. Verification is by documentation based on either actual tests or analytical methods.

Documentation pertaining to the seismic qualification of all seismic Category I electrical equipment, instrumentation, and supports is reviewed for compliance with the requirements set forth in Section 3.7B.2, NRC regulatory guides, and applicable codes and standards.

The following is a description of the analysis and testing procedures used for qualification of Seismic Category I cable trays:

The types of cable trays used are: ladder type with or without tray covers and solid bottom type with or without covers. Trays are supported on structural steel frames anchored to the ceilings, walls, or floors, and braced transversely and longitudinally. The configuration of the tray and support system consists of tray segments with bolted splice connections, tray attachment to the respective supports by bolted connections and support attachment to the concrete super-structures generally by bolted connections (or by welded connections if embedded plates are available).

Under normal conditions of loading and support spacing, trays and their supports may have natural frequencies which are lower than 33 Hz; therefore, they are required to sustain amplified accelerations, and are designed for the peak values of the floor response spectra at the applicable support elevation, unless a frequency analysis is performed to justify the use of lower acceleration values.

For the qualification of trays, maximum equivalent static loads for SSE are determined from the peaks of floor response spectra curves corresponding to 7% damping, unless a frequency analysis is performed to justify the use of lower acceleration values and are based on the full mass of the tray (weight of the tray plus cable). Suppliers are required to determine by static testing on representative trays of each type being used, the strength of the trays in each of the three principal directions of load application, and to compare the equivalent static loads in each direction with the respective strengths of the trays. The results are combined using an interaction formula to account for the simultaneous application of dead load and earthquake excitation in the three principal directions.

Supports for the Seismic Category I electrical cable tray and conduit system are designed to resist the gravity and seismic loads imposed on them by the cable trays and/or conduits. The design of the supports is generally based on the peak values of the applicable floor response spectra, unless a frequency analysis is performed to justify the use of lower acceleration values. Floor response spectra corresponding to 7% and 4% dampings for the SSE and OBE, respectively, are used for the design of cable tray/support systems, consistent with Regulatory Guide 1.61 for bolted steel structures. For the design of conduit supports, floor response spectra corresponding to 3% and 2% dampings for the SSE and OBE, respectively are used. For Thermolag upgrade to safety related conduits, damping values of 4% and 7% for the OBE and SSE are used, also consistent with Regulatory Guide 1.61 for bolted structures.

Supports for cable trays are made of relatively light weight structural steel shapes, welded together such as to provide a structural element to support the cable tray raceway. Supports for conduits are made of structural steel or unistrut members. The cable tray or conduit raceway support systems are adequately braced in the transverse and longitudinal directions to account for the loadings in these directions.

The design of the cable tray supports is based on elastic working stress design methods. Upon evaluation, there were found no conditions under SSE loading combinations that would exceed the yield strength of the materials used in the supports except as permitted by Reference [1].

The factor of safety on expansion anchors used for all Seismic Category I cable tray supports (Hilti Kwik-Bolt) has been established to be a minimum of 4 for both OBE and SSE loading

conditions. Steel structures subject to this loading condition are designed based on elastic working stress design methods, as noted above, and meet the requirements of AISC working stress allowables. The expansion anchors which are used with these steel structures have been designed in accordance with the manufacturer's recommendation of a minimum factor of safety of 4 for the maximum working load conditions (OBE) and maximum faulted (extreme) load conditions (SSE) for Seismic Category I cable tray hangers.

The factor of safety on expansion anchors used for Seismic Category II cable tray supports when subjected to full SSE loading conditions is a minimum of 3. The steel structures which are subjected to this loading condition when designed using working stress design methods are allowed to be stressed to a level of 1.6 times the stress of OBE loading conditions. This increased allowable in stress levels is partly attributed to the high improbability of the occurrence of such SSEloading combinations. This same rationale is also applicable to the associated factor of safety of 3 for expansion anchors. Since the loading condition in question is one of extreme environmental conditions and highly improbable, the lower factor of safety is applicable. Also, Seismic Category II hangers are not required to be functional for a safe shutdown of the plant. During a SSE event, their major function is not to adversely affect other components which are required for safe shutdown of the plant. In addition to manufacturer's recommendations, technical information supporting this decision is found in Appendix B of Reference [2] as well as Reference [3].

3.10B.4 OPERATING LICENSE REVIEW

The available seismic qualification documentation for seismic Category I instrumentation and electrical equipment is retained in the plant records.

REFERENCES

1. AISC Specification For the Design, Fabrication and Erection of Structural Steel for Buildings, 1969 including Supplement Numbers 1, 2, and 3.

2. ACI Committee 349, Criteria for Reinforced Concrete Nuclear Power Containment Structures, ACI Journal, January 1972.

3. NRC IE Information Notice No. 79-14, Safety Classification of Electrical Cable Support Systems, June 11, 1979.

TABLE 3.10B-1
DELETED

3.11N ENVIRONMENTAL DESIGN OF MECHANICAL AND ELECTRICAL EQUIPMENT

The mechanical and electrical portions of the Engineered Safety Features and the Reactor Protection Systems are designed to ensure acceptable performance in all environments anticipated under normal, test, and design basis accident conditions. This section presents information on the design basis and qualification verifications for mechanical and electrical equipment in the Engineered Safety Features and the Reactor Protection System that are within the scope of the Westinghouse Nuclear Steam Supply System (NSSS). Also, this section refers to equipment located in a potentially harsh environment, unless otherwise noted. Section 3.7N presents the seismic design requirements and Section 3.10N presents the seismic qualification of electrical equipment.

3.11N.1 EQUIPMENT IDENTIFICATION AND ENVIRONMENTAL CONDITIONS

A list of Class 1E equipment located in potentially harsh environments and their postulated environmental extremes are reflected in controlled documents, (e.g., EEQSPs, EQML). The definition of potentially harsh and mild environments at CPNPP is provided in Section 3.11B.1.

The safety-related mechanical equipment at CPNPP has been designed to withstand environmental effects as required by GDC-4 and Appendix B of 10 CFR Part 50. These requirements are satisfied through the design, specification, procurement, and quality assurance procedures at CPNPP as supplemented by the pump and valve operability programs and the CPNPP maintenance and surveillance programs.

The energy and environmental variations to which the Westinghouse supplied protection system equipment will be qualified are defined in the applicable equipment qualification data package (Supplement 1 to WCAP-8587).

The CPNPP POST-LOCA long-term operability duration is 100 days, per TXX-88742, letter from W. G. Counsil to USNRC [4].

3.11N.2 QUALIFICATION TESTS AND ANALYSIS

For Westinghouse NSSS Class 1E equipment located in a potentially harsh environment, Westinghouse meets the Institute of Electrical and Electronics Engineers (IEEE) Standard 323-1974, “IEEE Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations,” including IEEE Standard 323a-1975, the Nuclear Power Engineering Committee (NPEC) Position Statement of July 24, 1975, by an appropriate combination of any or all of the following: type testing, operating experience, qualification by analysis, and ongoing qualification. The Westinghouse approach to satisfying IEEE-Standard 323-1974 is documented in WCAP-8587 [1] which has been accepted by the NRC Staff. Controlled documents (e.g., EEQSPs, EQML) identify the Westinghouse supplied Class 1E equipment located in a potentially harsh environment and the corresponding Equipment Qualification Data Packages.

For NSSS Class 1E equipment located in a mild environment area, CPNPP demonstrates qualification through procurement documents, which include the most stringent requirements for the equipment’s end use. The maintenance and surveillance programs in conjunction with a trending program provide the assurance that equipment which meets the requirements of the procurement documents is maintained throughout the equipment’s installed life (see Appendix 3A Section 2.4).

All seismic Category I Class 1E equipment located in a mild environment area is seismically qualified to IEEE Standard 344-1975 and Regulatory Guide 1.100 using the methods, procedures, and documentation described in FSAR Sections 3.7 and 3.10N.

The overall Class 1E Westinghouse equipment qualification program includes generic environmental conditions, e.g., temperature, pressure, relative humidity, chemistry, radiation, which are established for the various pieces of Westinghouse supplied Class 1E equipment. The conditions vary according to location of the equipment. The generic environmental conditions for which the equipment is qualified are reported in the specific Equipment Qualification Data Package and the Environmental Equipment Qualification Summary Packages. The postulated environmental extremes used for equipment qualification are provided in the EEQSPs.

The peak calculated containment vapor temperature for the design basis MSLB can be found in Table 6.2.1-2A. Component thermal analysis is required only if the environmental qualification test conditions are less than those calculated from the containment pressure temperature transient response analysis. Transmitters and valves/valve operators inside containment and electrical cables are qualified to the required parameter by testing or by a combination of testing/ analysis in accordance with IEEE 323-1974 (and the daughter standards such as IEEE 383-1974 for the electrical cables) if these are required to operate for design basis MSLB.

How the requirements of the General Design Criteria (GDC) 1, 4, 23, and 50 are met is addressed in Section 3.1. Specific information concerning how GDC 1 and 4 are met is reported in the applicable Equipment Qualification Data Packages [1, 3]. Specific information concerning how GDC 23 is met can be found in Section 7.2.2.2. Specific information concerning how GDC 50 is met is provided in Section 6.2. Specific information concerning how Appendix B of 10 CFR Part 50 is met is located in Chapter 17. Regulatory Guides 1.30, 1.40, 1.73 and 1.89 are addressed in Appendix 1A(N).

3.11N.3 QUALIFICATION TEST RESULTS

Qualification test results are provided in the Equipment Qualification Data Packages and in the Environmental Equipment Qualification Summary Packages.

3.11N.4 LOSS OF VENTILATION

Refer to Section 3.11B.4

3.11N.5 ESTIMATED CHEMICAL AND RADIATION ENVIRONMENT

Chemical environments in the primary containment as result of an accident condition are shown in the EEQSPs and discussed in Section 3.11B.5.1. The postulated integrated radiation environments for normal and accident conditions are shown in the EEQSPs and discussed in Section 3.11B.5.2.

REFERENCES

1. Jordan, W. G., Lorentz, D. G. and Miller, R. B., “Methodology For Qualifying Westinghouse PWR-SD Supplied NSSS Safety Related Electrical Equipment”, WCAP-8587, Revision 1, September, 1977.

2. Wilson, J. F., “Electric Hydrogen Recombiner for PWR Containments,” WCAP-7709L, Supplements 1 through 7 (Proprietary) and WCAP-7820, Supplements 1 through 7 (Non-Proprietary), 1971 through 1977.

3. “Equipment Qualification Data Packages”, Supplement to WCAP-8587, Revision 1, April 1978.

4. TXX-88742, Letter from W. G. Counsil to USNRC dated October 24, 1988.

TABLE 3.11N-1

Table 3.11N-1 is deleted and the specific environmental details are reflected in control documents (EEQSPs, EQML, DBDs).

TABLE 3.11N-2

Table 3.11N-2 is deleted and the specific environmental details are reflected in control documents (EEQSPs, EQML, DBDs).

TABLE 3.11N-3

Table 3.11N-3 is deleted and the specific environmental details are reflected in control documents (EEQSPs, EQML, DBDs).

3.11B ENVIRONMENTAL DESIGN OF MECHANICAL AND ELECTRICAL EQUIPMENT

To ensure proper performance during normal, abnormal, and design basis accident conditions, instrumentation and electrical components of the Engineered Safety Features (ESF) and other safety-related systems are designed for the environmental conditions and bases described in Section 3.11B.1. Specific information is also included in Sections 3.11B.2 and 3.11B.5 to demonstrate that the safety-related components have the capability to function as required in the combined temperature, pressure, humidity, chemistry, and radiation dose of the post-accident environment. In addition, an evaluation is made in Section 3.11B.4 of the environmental effects that would follow the loss of plant ventilation system used for cooling Class 1E electrical equipment. This Section refers to equipment located in a potentially harsh environment, unless otherwise noted.

Certain Class 1E equipment is furnished with the same train environmental support system (See Sections 9.4.5, 9.4C.1, 9.4C.3, 9.4C.4 and 9.4F). Redundant environmental support systems are not required for any Class 1E equipment because Class 1E equipment is already redundant. Redundant environmental support systems are provided where both trains of Class 1E equipment are located and emergency ventilation is required (See Sections 9.4.1, 9.4B and 9.4C.8). This type of design enables the safety related environmental systems to sustain a single active component failure without total loss of function (See Section 9.4).

As described in Section 7.5, certain Non-1E accident monitoring instruments are also included in the scope of the environmental qualification program.

The accident (limiting) environmental conditions are discussed in Section 3.11B.4.

Section 3.7B presents the seismic design requirements, and Section 3.10B presents the seismic qualification of electrical equipment.

LOCAs and steam line breaks are design basis accidents that will cause enveloping environmental changes within the Containment. Pressure and temperature transients for these design basis accidents are presented in Section 6.2. Loss of ventilation and High Energy Line Breaks (HELBs) outside containment are postulated events that can cause environmental changes in areas outside the containment. The environmental analysis performed to determine HELB environmental consequences for purposes of qualifying equipment located outside containment is described in Section 3.6B.1.2.3. Loss of ventilation is discussed in Section 3.11B.4 below.

In addition to HELBs as described in Section 3.6B, non-mechanistic cracks in break exclusion piping (i.e., superpipe) in the main steam and feedwater penetration area outside Containment are postulated and evaluated for environmental effects. (See Section 3.6B.2.5.2 for details).

3.11B.1 EQUIPMENT IDENTIFICATION AND ENVIRONMENTAL CONDITIONS

Safety-related equipment and components are designed to function during and subsequent to the DBA and are located within the Containment as well as in other seismic Category I structures (Safeguards buildings, Auxiliary Building, Fuel Building, Electrical and Control Building and Service Water Intake Structure).

The environmental qualification program at CPNPP includes all Class 1E equipment located in potentially harsh environment areas. Class 1E and certain Non-1E accident monitoring equipment located in a mild environment area is included in the maintenance and surveillance programs discussed in Appendix 3A, Section 2.4. For CPNPP, mild environment areas are defined as areas, outside the containment, that are not potentially harsh due to a design basis event. In addition, a mild environment is an environment that would at no time be significantly more severe than the environment that would occur during normal plant operation, including anticipated operational occurrences.

A potentially harsh environment is defined as an environment where safety-related equipment would experience, due to the direct effects of a design basis event (Loss of Coolant Accident, Main Steam Line Break, High Energy Line Break, Superpipe Cracks) any of the following parameters:

1. An ambient pressure increase greater than two pounds per square inch (2psi) above atmospheric, or

2. An ambient temperature increase greater than five degrees centigrade (5°C or 9°F) above the postulated maximum temperature based on normal and anticipated operational occurrences, or

3. A total integrated radiation exposure dose of $1x10^4$ rads Gamma.

NOTE: For postulated radiation environments between $1x10^3$ and $1x10^4$ rads Gamma, electronic equipment including semiconductors are evaluated to demonstrate qualification to the postulated radiation dose.

4. A relative humidity value of 100%

5. Flooding

   Class 1E equipment subject to flooding due to a design basis accident (as described above) is qualified to operate for the flooding environment. If the equipment cannot be qualified, an evaluation is performed to determine if it is required to operate for mitigation of the accident. If the equipment is not required to operate, an evaluation is performed to demonstrate that failure of the equipment is not detrimental to the safe shutdown of the plant.

   For Class 1E equipment which experiences flooding due to a Moderate Energy Line Crack, the impact of that flooding is addressed by the Systems Interaction Program as described in Section 3.6 of the FSAR. Wherever reasonable, CPNPP has been designed to locate Class 1E equipment such that the equipment does not experience flooding.

   As part of this process, all safety and non-safety-related electrical equipment, instrumentation and control circuits, and components located inside containment below elevation 816'-10", which may become submerged as a result of LOCA, have been identified and evaluated. The analysis concluded that:

a. There is no safety significance from the failure of this electrical equipment (e.g., spurious actuation or loss of actuation function) as a result of flooding.

b. There will be no degradation of Class 1E electrical power sources serving this electrical equipment as a result of such submergence.

c. The design changes for the required safety-related equipment, conduit and/or cable routing subject to submergence have been identified to resolve any unacceptable environmental interaction with required safety-related equipment.

A list of Class 1E equipment located in potentially harsh environments and their postulated environmental extremes are provided in Controlled Documents (e.g., EEQSPs, EQML). Environmental qualification documentation for Class 1E equipment is documented in Environmental Equipment Qualification Summary Packages (EEQSPs) and is retained in the plant records.

The ESF and other safety-related equipment which must remain operable during and after the DBA are further discussed in appropriate chapters of the FSAR as follows:

1. ESF and other Safety-Related Equipment Located Inside Containment

   a. Mechanical equipment (which contains Class 1E electrical components) in Chapter 6.

   b. Class 1E equipment in Chapter 8.

   c. Instrumentation and controls in Chapter 7.

2. ESF and other Safety-Related Equipment Located Outside Containment

   a. Mechanical equipment (which contains Class 1E electrical components) is described in Chapters 6, 9 and 10.

   b. Instrumentation and controls are described in Chapter 7.

   c. Class 1E equipment is described in Chapter 8.

The safety-related mechanical equipment at CPNPP has been designed to withstand environmental effects as required by GDC-4 and Appendix B of 10 CFR Part 50. These requirements are satisfied through the design, specification, procurement, and quality assurance procedures used at CPNPP as supplemented by the pump and valve operability programs and the CPNPP maintenance and surveillance programs.

3.11B.2 QUALIFICATION TESTS AND ANALYSES

Qualification tests and analyses are performed on all ESF equipment and components located in a potentially harsh environment as necessary to ensure their availability during and after a DBA. These tests consist of simulation of actual physical conditions on the equipment or a prototype on a generic basis, or analysis, or a combination of prototype tests and analysis as applicable. Qualification testing is performed under simulated conditions of temperature, pressure, relative

humidity, chemistry, and radiation dose in excess of those expected for post-DBA conditions. The testing period is sufficient to ensure the capability to function during and after a DBA.

In order to provide assurance that all ESF and other Safety Related equipment has the capability to meet environmental conditions as required the appropriate quality assurance programs are established and implemented.

Class 1E instrumentation and electrical equipment is capable of operating in the worst expected environmental conditions as required for each component and its location. The Class 1E electrical equipment is specified for manufacture in accordance with the criteria listed in Section 8.3. All Class 1E equipment located in a potentially harsh environment will be qualified per IEEE 323-1974 [11] and other applicable standards per Section 8.3 and below.

IEEE-323-1974 requirements include the need to establish the qualified life of Class 1E equipment. This requirement has, in many cases, represented a state-of-the-art challenge in assessing the longevity of equipment under normal and accident environments. Many of the equipment qualification test reports have identified relatively short lives of certain components. In some cases, these estimated, seemingly short, qualified lives may be the result of testing or accelerated aging limitations, rather than due to intrinsic equipment limitations. Vistra OpCo, as the first operating license applicant with an IEEE-323-1974 commitment has been confronted with an enormous developmental program in meeting these new requirements.

The CPNPP POST-LOCA long-term operability duration is 100 days, per TXX-88742, letter from W. G. Counsil to USNRC [20].

Consistent with Vistra OpCo's commitment to meet the requirements of IEEE-323-1974 subject to state-of-the-art limitations, Vistra OpCo will modify any of the existing qualified lives as additional information and better testing and analytical techniques are developed. Therefore, the existing qualified lives currently listed in the Environmental Equipment Qualification Summary Packages should be regarded as decision points with regard to an ongoing aging evaluation, rather than a fixed component replacement interval. Changes from those qualified lives currently indicated will be documented by revisions to the appropriate summary packages.

Instrumentation and control equipment mounted in the Control Room area is not subject to damaging vibration in either normal or postaccident modes, since no rotating equipment or fluid system components which could induce vibration are in proximity to the Control Room areas.

For Class 1E instrumentation and control equipment, including sensors, mounted in seismic Category I areas outside of the Control Room, the following design practices ensure that non-seismic vibration does not degrade the equipment:

1. Pressure and differential pressure transmitters and switches are mounted "off line," on secure and rigid (fn>33 hertz) instrument racks and supports. The instruments are supported independently of their connection to the process.

2. In-line temperature switches (thermostats) are not used.

3. In-line flow meters (such as target meters, rotameters, turbine meters, or paddle switches) are not used.

4. For equipment such as RTDs, level switches and valve limit switches, which are subject to non-seismic vibration, the qualification reports account for expected vibration, including the OBE tests required by IEEE 344-1975.

In addition to these design features, assurance that damaging vibration effects do not occur in service are provided by the preoperational tests and inspections as described in Section 14.2 as well as by the periodic on-line testing performed in accordance with Technical Specifications.

Electrical equipment and components located in potentially harsh environments outside of the Containment are qualified to meet the plant environmental conditions as listed in the EEQSPs.

Qualification tests for all safety related electric valve operators outside as well as inside containment are designed in accordance with References [8, 11, 13 and 15] to demonstrate their capability to function during and after accident environmental conditions. In addition, safety related electric valve operators installed inside the containment are also qualified in accordance with R.G. 1.73 [6].

The Class 1E instrumentation and control equipment in a potentially harsh environment have been qualified in accordance with IEEE 344-1975 and IEEE 323-1974 criteria. Designated non-Class 1E Accident Monitoring equipment located in a potentially harsh environment (see FSAR Section 7.5) have also been environmentally qualified.

Because the Containment has no Class 1E motors which are required to operate on a continuous basis during or following a DBA, NRC Regulatory Guide 1.40 [3] is not applicable to CPNPP.

Safety related equipment located inside the Containment that must remain operable during a DBA are identified in controlled documents. As indicated in the list referenced previously, the only motors in the Containment required to operate in the event of a DBA are those motor-operated isolation valves which operate at the onset of a DBA.

The peak calculated containment vapor temperature for the design basis MSLB can be found in Table 6.2.1-2A. Component thermal analysis is required only if the environmental qualification test conditions are less than those calculated from the containment pressure-temperature transient response analysis. Transmitters and valves/valve operators inside containment, containment electrical penetrations/header plate assemblies and electrical cables are qualified to the required parameter by testing or by a combination of testing/analysis in accordance with IEEE 323-1974 (and the daughter standards such as IEEE 317-1976 for the penetration assembly and IEEE 383-1974 for the electrical cables) if the devices are required to operate for design basis MSLB.

The Class 1E motors located outside the Containment in a potentially harsh environment are type-tested in accordance with the intent of IEEE 334-1974 [12].

Class 1E splices and cable connectors inside containment are environmentally qualified to meet the requirements of IEEE 323-1974 and IEEE 383-1974. The splices are listed and their use justified in Section 8.3 and Appendix 8A, “Analysis to Justify Cable Splices in Raceway.” There are no class 1E termination cabinets or terminal blocks inside containment other than class 1E terminal blocks which are part of class 1E equipment (such as Limitorque) which are qualified to IEEE-323-1974 requirements.

The initial preoperational test procedures incorporate the recommendations of NRC Regulatory Guide 1.41 [4] for verifying proper load group assignment.

Provisions are integrated into the design criteria for required test circuitry, undervoltage-sensing relays, and simulated accident test signals to permit testing as outlined in Regulatory Guide 1.41.

Subsequent to the initial service date, the station standard maintenance and test procedures require that any major modification or repair to the onsite power systems or redundant load groups necessitates retesting of these systems to verify independence among redundant systems.

The design basis for the maintenance, testing, and replacement of plant batteries is in compliance with the recommendations set forth in IEEE 450-1995 [17], as discussed in Section 8.3.2.

Inspection of the batteries are described in Section 8.3.

It is not proposed to test equipment of a passive nature, such as primary and secondary shielding.

Qualification tests for electric penetration assemblies are based on References [10 and 11].

Qualification tests for electric cables are based on References [11 and 16]. Test temperature profiles used for environmental qualification and included in the applicable EEQSP.

For Class 1E and certain Non-1E accident monitoring equipment located in a mild environment area, CPNPP will demonstrate qualification through procurement documents, which include the most stringent requirements for the equipment's end use. The maintenance and surveillance programs in conjunction with a trending program provide the assurance that equipment which meets the requirements of the procurement documents is maintained throughout the equipment's installed life (see Appendix 3A, Section 2.4).

All seismic Category I Class 1E equipment located in a mild environment will be seismically qualified to IEEE Standard 344-1975 and Regulatory Guide 1.100 [19] using the methods, procedures, and documentation described in FSAR Sections 3.7 and 3.10B.

How the requirements of GDC 1, 4, 23 and 50 are met is addressed in Section 3.1. Specific information concerning how GDC 1 and 4 are met is reported in Appendix A of Reference [18]. Specific information concerning how GDC 23 is met can be found in Sections 7.2 and 7.3.

The quality assurance program to be applied to the design, fabrication and testing of all BOP safety related equipment conform to the requirements of 10CFR part 50 Appendix B. The QA program is described in Chapter 17.

Specific information concerning how Appendix B of 10 CFR Part 50 is met is discussed in Chapter 17. Level of compliance with NRC Regulatory Guides 1.30, 1.40, 1.63, 1.73, 1.89, 1.100 and 1.131 is addressed in Appendix 1A(B).

3.11B.3 QUALIFICATION TEST RESULTS

All safety-related equipment and components in a potentially harsh environment are demonstrated to perform their designed safety function under all normal, abnormal and accident conditions, by appropriate testing and analyses.

The detailed qualification information and test results are documented in the Environmental Equipment Qualification Summary Packages.

3.11B.4 LOSS OF VENTILATION

3.11B.4.1 Environmental Design Basis

The plant design features ensure that room ambient temperatures do not exceed the maximum operational temperature limit for instrumentation and electrical equipment. The design conditions for normal plant operation are given in Table 9.4-2 for each room/area.

1. Control Room Ventilation System

   The Control Room air-conditioning system is designed to maintain its ambient temperature at 75°F (±5°F) and 35-50 percent relative humidity. The system is of seismic Category I design and includes the following design features:

   a. Sufficient redundancy in equipment and power supplies is provided to enable the system to sustain a single active component failure without loss of safety function.

   b. Redundant fans are connected to separate Class 1E buses as described in Section 8.3.

   c. Instrumentation and controls which incorporate audible and visual alarms enable the operator to continuously monitor system performance and alert him in the event of system malfunction.

   d. Failure modes for isolation valves and dampers (as described in Section 9.4.1) are set so that their failure does not render the system inoperable.

2. Primary Plant Ventilation

   a. Areas with Emergency Fan Coil Units.

      Certain areas with safety related equipment are provided with Class 1E emergency fan coil units which are powered from the same train as the equipment being cooled.

   b. Areas without Emergency Fan Coil Units.

Class 1E equipment in these areas are qualified for the temperature resulting from loss of ventilation (following a LOCA with Loss of Offsite Power).

3. Ventilation Systems of Other Areas

Other ventilation systems which serve areas containing safety-related equipment are described in Section 9.4.

3.11B.4.2 Limiting Environmental Conditions

1. UPPER LIMITS

The normal upper limit environmental conditions for the Control Room are as follows:

Temperature, °F	80
Relative humidity %	50
Chemistry	N/A
Radiation environment	Refer to plant drawings and calculations

All other areas upper limit environmental conditions for safety related equipment are shown in plant drawings and calculations.

2. LOWER LIMITS

The lower limit operating environment that Class 1E equipment is expected to operate in is specified in Table 9.4-2 and augmented by the following:

a. The lower limit operating temperature:
40°F for all areas outside containment except the Control Room and Battery Room;
70°F for the Control Room and Battery Room;
50°F for all areas inside containment.

The IE Devices outside of a permanent building or in an unheated enclosure are qualified for the Outdoor Design Conditions, protected or analyzed as acceptable as is. The Extreme Outdoor Temperatures have also been analyzed to have no effect on these components.

b. The lower limit operating relative humidity:
uncontrolled

c. The lower limit operating pressure:
atmospheric for all areas except inside containment;
-0.5 psig inside containment.

3.11B.4.3 Testing of Control and Electrical Equipment

Factory testing of protective system equipment is performed as stated in Section 3.11B.2. Periodic testing is provided in accordance with IEEE 338, “Criteria for the Periodic Testing of Nuclear Power Generation Station Protection Systems” [14], and IEEE 279, “Criteria for Protection Systems” [8] as described in Chapters 7 and 8.

The previously stated criteria for onsite testing are applied to all components of protection systems and are performed under ambient environmental conditions.

Overall testing activities are described in quality assurance programs identified in Chapter 17.

3.11B.5 ESTIMATED CHEMICAL AND RADIATION ENVIRONMENT

3.11B.5.1 Chemical Environment

The majority of pressure boundary components are constructed principally of stainless steel. Containment structures are principally constructed of concrete and carbon steel (galvanized or coated). These materials are compatible with, and do not suffer significant degradation in the containment spray environment. Other pressure boundary component (i.e. gaskets) materials are also selected for their compatibility with the spray environment.

Details concerning the chemicals used in and the chemistry of the containment spray system is presented in Sections 6.2.2 and 6.5.2. Accident chemistry qualification of safety-related electrical and instrumentation components in a potentially harsh environment (i.e. inside containment) is shown in the EEQSPs.

The atmosphere inside the containment building after a LOCA consists of steam-air-water spray mixture. The water spray solution is borated water with a maximum of 2600 ppm boron buffered with a sodium hydroxide solution (i.e. 30% wt. NaOH in water) to a final spray solution pH between 8.25 to 10.5.

3.11B.5.2 Radiation Environment

The radiation environment for normal operation and for the DBA for each safety-related component has been determined. The accident source terms are based on the TID-14844 [18] release model and, accordingly, are consistent with NRC Regulatory Guide 1.4 [1].

The postulated integrated normal/accident dose is given in the EEQSPs.

REFERENCES

1. NRC Regulatory Guide 1.4, Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Pressurized Water Reactors, Revision 2, 6/74.

2. NRC Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection and Testing of Instrumentation and Electric Equipment, 8/11/72.

3. NRC Regulatory Guide 1.40, Qualification Tests of Continuous- Duty Motors Installed Inside the Containment of Water-Cooled Nuclear Power Plants, 3/16/73.

4. NRC Regulatory Guide 1.41, Preoperational Testing of Redundant Onsite Electric Power Systems to Verify Proper Load Group Assignments, 3/16/73.

5. NRC Regulatory Guide 1.63, Electric Penetration Assemblies in Containment Structures for Water-Cooled Nuclear Power Plants, Revision 2, 7/78.

6. NRC Regulatory Guide 1.73, Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants, 1/74.

7. NRC Regulatory Guide 1.89, Qualification of Class 1E Equipment for Nuclear Power Plants, 11/74.

8. IEEE 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations, sponsored by the IEEE Joint Committee on Nuclear Power Standards.

9. IEEE 308-1971, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Stations.

10. IEEE 317-1976, Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations.

11. IEEE 323-1974, Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations.

12. IEEE 334-1974, Standard for Type Tests of Continuous Duty Class 1E Motors for Nuclear Power Generating Stations.

13. IEEE 336-1971, Installation, Inspection, and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations.

14. IEEE 338-1971, Trial Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection Systems, prepared by Subcommittee 3 of the IEEE Joint Committee on Nuclear Power Standards, Revision No. 15, May 15, 1971.

15. IEEE 382-1972, IEEE Trial-Use Guide for Type Test of Class 1E Electric Valve Operators for Nuclear Power Generating Stations.

16. IEEE 383-1974, Standard for Type Test of Class 1E Electric Cables, Field Splices and Connections for Nuclear Power Generating Stations.

17. IEEE 450-1995, IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead Acid Batteries for Stationary Applications.

18. TID-14844, Calculation of Distance Factors for Power and Test Reactor Sites, by J.J. DiNunno, F.D. Anderson, R.E. Bauer, and R.L. Waterfield, Division of Licensing and Regulation, United States Nuclear Regulatory Commission.

19. NRC Regulatory Guide 1.100, Seismic Qualification of Electric Equipment for Nuclear Power Plants, 8/77.

20. TXX-88742, Letter from W. G. Counsil to USNRC dated October 24, 1988.

TABLE 3.11B-1
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TABLE 3.11B-2
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3A ENVIRONMENTAL QUALIFICATION REPORT

3A.1 INTRODUCTION

This Environmental Qualification Report (EQR) supports the operating license application for Comanche Peak Nuclear Power Plant (CPNPP). This report provides design information in sufficient detail to allow a definitive evaluation of the equipment environmental qualification program for CPNPP.

The CPNPP environmental qualification program is described in Section 3A.2 and a detailed comparison to NUREG-0588 is provided in Section 3A.3. A summary of Class 1E equipment types located in a potentially harsh environment and their postulated environmental extremes are provided in controlled documents (e.g., EEQSPs, EQML).

The basic scope of the program includes all Class 1E equipment at CPNPP located in a potentially harsh environment. The definition of a potentially harsh environment at CPNPP is provided in Section 3.11B.1. Detailed system and equipment descriptions are included in the appropriate sections of the FSAR; in particular, those systems required for Hot Standby and Cold Shutdown are described in FSAR Section 7.4. Section 7.1 identifies all safety-related systems.

The basic scope of the CPNPP environmental qualification program which will qualify equipment to the standards of IEEE 344-1975 and IEEE 323- 1974, includes all safety related Class 1E equipment located in a potentially harsh environment, designated Accident Monitoring equipment (see FSAR Section 7.5), and any other electrical equipment which CPNPP has committed to qualify in accordance with these criteria. Certain non-safety related equipment, if subjected to adverse environmental conditions (such as high energy line breaks), could impact safety analyses and the adequacy of the protection functions performed for by safety grade equipment. Such equipment (e.g., Automatic rod control system, the pressurizer PORV control system, main feedwater control system and the steam generator PORV control system) may be required to be qualified to operate in this environment or be located in areas not subject to those environments. All safety-related systems are identified in FSAR Section 7.1. Detailed system and equipment descriptions are included in the appropriate sections of the FSAR; in particular, the equipment required for Hot Standby and Cold Shutdown is described in FSAR Section 7.4. A list of major equipment which is classified as IEEE Class 1E is provided in FSAR Table 17A-1.

A review has been conducted of the CPNPP Emergency Operating procedures. The review identified all display instrumentation referenced by the emergency operating procedures and listed the safety classification for these instruments. This review verified that the monitoring instrumentation as described in FSAR Section 7.5 is adequate for the execution of the emergency operating procedures.

Modifications to the CPNPP design due to the incorporation of TMI lessons learned have been included into the Environmental Qualification Program, as applicable.

3A.2 CPNPP QUALIFICATION PROGRAM

The equipment qualification program for CPNPP can be discussed as two separate programs. The first program covers the Balance of Plant (BOP) equipment and the second program covers the Nuclear Steam Supply System (NSSS) equipment. Of course, many aspects of these two programs are similar and will be discussed only once under the general category of the CPNPP

qualification program. Where it is better to present the information separately, the two programs shall be referred to as the NSSS qualification program and the BOP qualification program. The final review of all equipment qualification programs is referred to as the CPNPP qualification review program and is described in Section 3A.2.3 below.

3A.2.1 NSSS QUALIFICATION PROGRAM

The NSSS qualification program covers the Class 1E Electrical Equipment located in a potentially harsh environment supplied to CPNPP by the Westinghouse Electric Corporation under the CPNPP NSSS contract. This program is basically described in WCAP-8587.

3A.2.2 BOP QUALIFICATION PROGRAM

The BOP qualification program covers all Class 1E Electrical Equipment located in a potentially harsh environment not included in the NSSS qualification program.

The BOP and NSSS qualification programs are comprised of approximately 70 individual qualification packages. In each of the individual packages, the functional requirements and environmental conditions for the Class 1E electrical equipment are defined. The equipment vendor used the functional requirements and environmental conditions specified in procurement documents to develop the individual qualification program for his equipment. The vendor's program is reconciled in the individual qualification packages.

Some individual programs were "generic" programs. These programs cover a range of environmental conditions and functional requirements and were not conducted specifically for CPNPP. For these generic programs, the vendor usually provides the generic report along with documentation to show that the generic report adequately envelopes the CPNPP requirements and that the report applies to the equipment supplied to CPNPP. This documentation is submitted to Vistra OpCo for review, comment and acceptance.

Many individual qualification programs were conducted specifically for CPNPP. For these programs, a qualification program is developed by the vendor. These plans are submitted to Vistra OpCo for review, comment and acceptance. Using these plans as a basis, the testing and/or analysis is conducted and a final report is submitted for acceptance. These programs should normally result in an accepted qualification plan and an accepted qualification report. In some cases, the qualification report will contain all the required information in the plan and thus become a combined plan/report. A combined plan/report will usually supercede all previous plans.

3A.2.3 CPNPP QUALIFICATION REVIEW PROGRAM

The CPNPP qualification review program provides a final review of the BOP and NSSS programs in order to accept these programs for CPNPP. The review process consists of an overall review of the programs as well as a specific review of the environmental qualification for each type of Class 1E electrical equipment located in a potentially harsh environment.

The overall reviews of the programs are documented in Section 3A.3 where each program (BOP and NSSS) is compared to NUREG-0588. Each item by item comparison describes how the respective program meets each requirement of NUREG-0588.

All environmental qualification data are contained in specific Environmental Equipment Qualification Summary Packages (EEQSPs) which are maintained in the Environmental Qualification Files. These auditable files are updated and kept current as equipment is qualification tested, replaced or modified. Each EEQSP is reviewed and approved by Vistra OpCo.

3A.2.4 CPNPP QUALIFICATION MAINTENANCE PROGRAM

The Qualification Maintenance Program at CPNPP is designed to be integrated into the existing maintenance program while assuring that all qualified life contingencies and qualification requirements are fulfilled. The Qualification Maintenance Program consists of maintenance activities which in many cases, occur normally at a power plant, but that are formalized in the case of Class 1E electrical equipment located in a potentially harsh environment.

All maintenance activities that are required by the equipment qualification reports for equipment located in a potentially harsh environment shall be performed within the time intervals specified. In addition, this equipment shall be replaced before their qualified lives expire. As an alternative to replacement, it may be possible to extend qualified lives through additional testing or analysis. Any requalification of equipment will be controlled and documented in a process similar to the original qualification effort.

As described in Sections 3.11B and 7.5, in addition to the equipment in the harsh equipment qualification program (Section 3A.1), Class 1E equipment and certain Non-1E accident monitoring instruments located in a mild environment are also included in the scope of the environmental qualification maintenance program.

A maintenance and surveillance program shall be implemented to identify and prevent significant age related degradation in electrical and mechanical equipment. Provisions for preventing or detecting age related degradation in safety grade equipment are specified and include the following:

1. Development of a maintenance program to maintain safety related equipment at the quality required for it to perform its intended function.

2. Evaluation of causes of equipment failures and review of experience with failed equipment and similar components to determine whether a replacement component of the same type can be expected to perform its function reliably.

3. Inspection and performance testing to ensure an appropriate quality level.

4. Development of preventive maintenance schedules that describe the frequency and type of maintenance to be performed. A preliminary schedule should be developed early in plant life and it should be refined and changed as experience is gained with the equipment.

5. Surveillance testing related to the results of reliability analyses, frequency and type of service, or age of the item or system, as appropriate.

Whenever possible, direct replacement spare parts and replacements shall be specified for Class 1E equipment. A direct replacement part shall be considered to be one that has been

evaluated by engineering to confirm that the form, fit and functional characteristics satisfy the original design basis. Parts with expired shelf lives will not be used on Class 1E equipment at CPNPP.

The beginning of life date for Class 1E electrical equipment located in a potentially harsh environment at CPNPP shall be considered to be the date of the initial fuel load. For equipment with short qualified lives, an engineering analysis shall be performed to determine if a portion of a qualified life has been used before the initial fuel load date. If this is determined, the initial qualified life of this equipment shall be adjusted to account for this age related degradation or the equipment shall be refurbished.

3A.3 CPNPP COMPARISON TO NUREG-0588

The CPNPP environmental qualification program has been reviewed using the criteria provided in NUREG-0588. The results of this review are provided in Tables 3A.3-1 and 3A.3-2. These tables describe the BOP and NSSS qualification programs, respectively, relative to each criteria in NUREG-0588. As defined in Section 3A.1, the scope of this program for equipment located in a harsh environment does not include equipment which is located in a mild environment area.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 1 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
1.	ESTABLISHMENT OF THE QUALIFICATION PARAMETER FOR DESIGN BASIS EVENTS	1.	
1.1)	Temperature and Pressure Conditions Inside Containment - Loss-of-Coolant Accident (LOCA)	1.1	
(1)	The time-dependent temperature and pressure, established for the design of the containment structure and found acceptable by the staff, may be used for environmental qualification of equipment.	(1)	A spectrum of high energy line breaks in the primary and secondary systems was considered in developing the maximum containment and temperature envelopes. The temperature and pressure envelopes which were used for the qualification equipment are provided in design documents. The development of the individual temperature and pressure profiles for containment (for specific accident scenarios) is discussed in FSAR Section 6.2.
(2)	Acceptable methods for calculating and establishing the containment pressure and temperature envelopes to which equipment should be qualified are summarized below. Acceptable methods for calculating mass and energy release rates are summarized in Appendix A.	(2)	See (1) above.
Pressurized Water Reactors (PWRs)			
Dry Containment - Calculate LOCA containment environment using CONTEMPT-LT or equivalent industry codes. Additional guidance is provided in Standard Review Plan (SRP) Section 6.2.1.1.A, NUREG-75/087.			

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 2 of 21)

NUREG-0588 Category I Position	CPNPP STATUS
Ice Condenser Containment - Calculate LOCA containment environment using LOTIC or equivalent industry codes. Additional guidance is provided in SRP Section 6.2.1.1.A, NUREG-75/087.	
Boiling Water Reactors (BWRs)	
Mark I, II and III Containment - Calculate environment using methods of GESSAR Appendix 3B or equivalent industry codes. Additional guidance is provided in SRP Section 6.2.1.1.C, NUREG-75/087.	
(3) In lieu of using the plant-specific containment temperature and pressure design profiles for BWR and ice condenser types of plants, the generic envelope shown in Appendix C may be used for qualification testing.	(3) Not applicable
(4) The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4) See (1) above
1.2 Temperature and Pressure Conditions Inside Containment-Main Steam Line Break (MSLB)	1.2
(1) The environmental parameters used for equipment qualification should be calculated with a plant-specific model reviewed and approved by the staff.	(1) See 1.1(1) above.
(2) Models that are acceptable for calculating containment parameters are listed in Section 1.1(2).	(2) See 1.1(1) above.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 3 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
(3)	In lieu of using the plant-specific containment temperature and pressure design profiles for BWR and ice condenser plants, the generic envelope shown in Appendix C may be used.	(3)	Not applicable
(4)	The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4)	See 1.1(1) above.
(5)	Where qualifications has been completed but only LOCA conditions were considered, it must be demonstrated that the LOCA qualification conditions exceed or are equivalent to the maximum calculated MSLB conditions. The following technique is acceptable:	(5)	The methods of 1.1(1) were used in determining the specified environment
(a)	Calculate the peak temperature envelope from an MSLB using a model based on the staff's approved assumptions defined in Section 1.1(2)		
(b)	Show that the peak surface temperature of the component to be qualified does not exceed the LOCA qualification temperature by the method discussed in item 2 of Appendix B.		

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 4 of 21)

NUREG-0588 Category I Position	CPNPP STATUS
(c) If the calculated surface temperature exceeds the qualification temperature, the staff requires that (i) requalification testing be performed with appropriate margins, or (ii) qualified physical protection be provided to assure that the surface temperature will not exceed the actual qualification temperature. For plants that are currently being reviewed, or will be submitted for an operating license review within six months from issue date of this report, compliance with items (i) or (ii) above may represent a substantial impact. For those plants, the staff will consider additional information submitted by the applicant to demonstrate that the equipment can maintain its functional operability if its surface temperature rises to the value calculated.	
1.3 Effects of Chemical Spray The effects of caustic spray should be addressed for the equipment qualification. The concentration of cuastics used for qualification should be equivalent to or more severe than those used in the plant containment spray system. If the chemical composition of the caustic spray can be affected by equipment malfunctions, the most severe caustic spray environment that results from a single failure in the spray system should be assumed. See SRP Section 6.5.2 (NUREG-75/087), paragraph II, item (e) for caustic spray solution guidelines.	1.3 The effect of caustic spray was addressed for the qualification of Class 1E equipment inside containment which is required to function during or after an accident that can initiate containment spray. See FSAR Section 3.11B.5.1.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
1.4 Radiation Conditions Inside and Outside Containment The radiation environment for qualification of equipment should be based on the normally expected radiation environment over the equipment qualified life, plus that associated with the most severe design basis accident (DBA) during or following which that equipment must remain functional. It should be assumed that the DBA related environmental conditions occur at the end of the equipment qualified life. The sample calculations in Appendix D and the following positions provide an acceptable approach for establishing radiation limits for Qualification. Additional radiation margins identified in Section 6.3.1.5 of IEEE Std. 323-1974 for qualification type testing are not required if these methods are used.	1.4 The specified radiation environment includes the expected normal dose for the qualified life of the equipment and the most severe DBA dose following which that equipment is required to remain functional. The accident dose is assumed to occur at the end of the qualified life of the equipment.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 6 of 21)

NUREG-0588 Category I Position	CPNPP STATUS
(1) The source term to be used in determining the radiation environment associated with the design basis LOCA should be taken as an instantaneous release from the fuel to the atmosphere of 100 percent of the noble gases, 50 percent of the iodines, and 1 percent of the remaining fission products. For all other non-LOCA design basis accident conditions, a source term involving an instantaneous release from the fuel to the atmosphere of 10 percent of the noble gases (except Kr-85 for which a release of 30 percent should be assumed) and 10 percent of the iodines is acceptable.	(1) The same conservative source terms, namely 100 percent of noble gases, and 50 percent of the iodines are assumed to be instantaneously released from the fuel to the containment atmosphere for the design basis LOCA. The 1 percent of the remaining fission products are assumed to be released from the fuel and carried with the primary coolant to the containment sump, as given in Appendix D NUREG-0588. These assumptions are used for all DDA's for which core damage must be assumed, assumed, except fuel handling accidents. For fuel handling accidents, the guidance of Regulatory Guide 1.25 is followed.
(2) The calculation of the radiation environment associated with design basis accidents should take into account the time-dependent transport of released fission products within various regions of containment and auxiliary structures	(2) The time dependent transport of released fission products has been considered in calculation of the radiation environment associated with DBAs
(3) The initial distribution of activity within the containment should be based on a mechanistically rational assumption. Hence, for compartmented containments, such as in a BWR, a large portion of the source should be assumed to be initially contained in the drywell. The assumption of uniform distribution of activity throughout the containment at time zero is not appropriate.	(3) Both models of assuming uniform distribution in the entire containment free volume and uniform distribution in the containment volume excluding the relatively isolated compartments at time zero after LOCA are analyzed. The latter generates more conservative doses and the results are used as the required radiation environment.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(4)	Effects of ESF systems, such as containment sprays and containment ventilation and filtration systems, which act to remove airborne activity and redistribute activity within containment, should be calculated using the same assumptions used in the calculation of offsite dose. See SRP Section 15.6.5 (NUREG-75/087 and the related sections referenced in the Appendices to that section.	(4)	A two-region spray model is utilized to calculate the airborne iodine activity in containment atmosphere. The assumption and limits of spray removal credit are consistent with Appendix D of NUREG-0588. Effects of ventilation and filtration have not been considered.
(5)	Natural deposition (i.e., plate-out) of airborne activity should be determined using a mechanistic model and best estimates for the model parameters. The assumption of 50 percent instantaneous plate-out of the iodine released from the core should not be made. Removal of iodine from surfaces by steam condensate flow or washoff by the containment spray may be assumed if such effects can be justified and quantified by analysis or experiment.	(5)	Plateout of airborne iodines on containment heat-sink surfaces is modeled as first-order rate removal process using a calculated plate-out removal constant. Once plated out, the iodines remain as the surface source, reduced only by the radioactive decay process.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 8 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
(6)	For unshielded equipment located in the containment, the gamma dose and dose rate should be equal to the dose and dose rate at the centerpoint of the containment plus the contribution from location dependent sources such as the sump water and plate-out, unless it can be shown by analyses that location and shielding of the equipment reduces the dose and dose rate.	(6)	Gamma doses and dose rates have been calculated for the center point of the containment, at the surface of containment sump water and as a function of distance from the plateout sources. However, in shielded compartments or other areas where the dose rate is reduced due to location of shielding effects, computations have been performed for specific configurations to determine the post-accident integrated dose. The sources used in these computations include the airborne source, plateout source and sump water source in the containment sump as appropriate in addition to the recirculated emergency core cooling system.
(7)	For unshielded equipment, the beta doses at the surface of the equipment should be the sum of the airborne and plate-out sources. The airborne beta dose should be taken as the beta dose calculated for a point at the containment center.	(7)	The beta doses have been calculated at containment center point, sump water surface and as a function of distance from the plateout source. The doses from airborne, plateout source, and/or sump source are summed. The airborne beta dose at containment center is calculated using an infinite medium model. The semi-finite medium approximation is used for the airborne beta dose at sump water surface. For a dose point close to the containment wall. The airborne beta source includes a semi-infinite medium plus an infinite slab source of finite thickness.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(8)	Shielded components need be qualified only to the gamma radiation levels required, provided an analysis or test shows that the sensitive portions of the component or equipment are not exposed to beta radiation or that the effects of beta radiation heating and ionization have no deleterious effects on component performance.	(8)	Equipment is qualified to the gamma and beta dose specified unless a specific test or analysis is provided to show that the beta dose has no deleterious effects on the equipment performance.
(9)	Cables arranged in cable trays in the containment should be assumed to be exposed to half the beta radiation dose calculated for a point at the center of the containment plus the gamma ray dose calculated in accordance with Section 1.4(6). This reduction in beta dose is allowed because of the localized shielding by other cables plus the cable tray itself.	(9)	A semi-infinite medium has been used to calculate the airborne beta dose where justified.
(10)	Paints and coatings should be assumed to be exposed to both beta and gamma rays in assessing their resistance to radiation. Plate-out activity should be assumed to remain on the equipment surface unless the effects of the removal mechanisms, such as spray wash-off or steam condensate flow, can be justified and quantified by analysis or experiment.	(10)	Failure of the paint or coating on class 1E equipment will not effect the ability of class 1E equipment to perform its required function and the CPNPP design is such that the failure of these coating's will not effect the operation of safety systems inside containment; therefore, no additional special calculations were performed relative to the coatings on class 1E equipment.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(11)	Components of the emergency core cooling system (ECCS located outside containment (e.g., pumps, valves, seals and electrical equipment) should be qualified to withstand the radiation equivalent to that penetrating the containment, plus the exposure from the sump fluid using assumptions consistent with the Appendix K to 10 CFR Part 50.	(11)	All safety related Class 1E electrical equipment, including ECCS Components, located in a harsh environment outside containment are qualified to a total radiation level which included radiation penetrating the containment wall or streaming through containment openings, from sump fluid contained in recirculation piping outside containment and from 40 year normal power operation.
(12)	Equipment that may be exposed to radiation doses below 10^4 rads should not be considered to be exempt from radiation qualification, unless analysis supported by test data is provided to verify that these levels will not degrade the operability of the equipment below acceptable values.	(12)	For each piece of Class 1E equipment regardless of environment, the specified radiation dose is addressed.
(13)	The staff will accept a given component to be qualified provided it can be shown that the component has been qualified to integrated beta and gamma doses which are equal to or higher than those levels resulting from an analysis similar in nature and scope to that included in Appendix D (which uses the source term given in item (1) above), and that the component incorporates appropriate factors pertinent to the plant design and operating characteristics, as given in these general guidelines.	(13)	Class 1E equipment for CPNPP is qualified to integrated radiation levels greater than or equal to the levels resulting from analyses similar to Appendix D

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(14)	When a conservative analysis has not been provided by the applicant for staff review, the staff will use the radiation environment guidelines contained in Appendix D, suitably corrected for the differences in reactor power level, type, containment size, and other appropriate factors	(14)	A conservative analysis similar to Appendix D has been performed to determine the required radiation environment.
1.5	Environmental Conditions for Outside Containment	1.5	
(1)	Equipment located outside containment that could be pipe breaks should be qualified to the conditions resulting from the accident for the duration required. The techniques to calculate the environmental parameters described in Sections 1.1 through 1.4 above should be applied.	(1)	HELB's outside containment are described in FSAR Section resulting form the HELB's will be incorporated into the specified environment for the Class 1E equipment per Section 5.
(2)	Equipment located in general plant areas outside containment where equipment is not subjected to a design basis accident environment should be qualified to the normal and abnormal range of environmental conditions postulated to occur at the equipment location.	(2)	Class 1E equipment which is not subject to a design basis environment is qualified to the normal and abnormal range of environmental conditions postulated for that equipment.
(3)	Equipment not served by Class 1E environmental support systems, or served by Class 1E support systems that may be secured during plant operation or shutdown, should be qualified to the limiting environmental conditions that are postulated for that location, assuming a loss of the environmental support system.	(3)	Class 1E equipment which is supported by non-Class 1E environmental support systems (or secured Class 1E systems) is qualified to the limiting environmental conditions that are postulated assuming the loss of those support systems (provided that the function or integrity of this Class 1E equipment is required under these conditions).

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
2.	QUALIFICATION METHODS	2.	
2.1	Selection of Methods	2.1	
(1)	Qualification methods should conform to the requirements defined in IEEE Std. 323-1974.	(1)	The qualification methods used conform to IEEE Standard 323-1974.
(2)	The choice of the methods selected is largely a matter of technical judgment and availability of information that supports the conclusions reached. Experience has shown that qualification of equipment subjected to an accident Environment without test data is not adequate to demonstrate functional operability. In general, the staff will not accept analysis in lieu of test data unless (a) testing of the component is impractical due to size limitations, and (b) partial type test data is provided to support the analytical assumptions and conclusions reached.	(2)	Equipment subjected to an accident environment Is qualified using test data or at least partial type test data to support the analytical assumptions and conclusions.
(3)	The environmental qualification of equipment exposed to DBA environments should conform to the following positions. The basis should be provided for the time interval required for operability of this equipment. The operability and failure criteria should be specified and the safety margins defined.	(3)	The time interval required for operability was based on the operability assumptions used in the various applicable accident analysis. The operability requirements and safety margins are provided in various locations including the appropriate sections of the FSAR, purchase specifications, and qualification documentation. CPNPP uses failure to meet operability requirements as its failure criteria, unless specific failure criteria is provided in the qualification documentation.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 13 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
(a)	Equipment that must function in order to mitigate any accident should be qualified by test to demonstrate its operability for the time required in the environmental conditions resulting from that accident.	(a)	Class 1E equipment is qualified to function in the environment resulting from the accident it is designed to mitigate.
(b)	Any equipment (safety-related or non-safety-related) that need not function in order to mitigate any accident but that must not fail in a manner detrimental to plant safety should be qualified by test to demonstrate its capability to withstand any accident environment for the time during which it must not fail.	(b)	Class 1E equipment that need not function but must not fail during an accident is qualified not to fail due to the environment conditions resulting from that accident, per paragraph (1) above.
(c)	Equipment that need not function in order to mitigate any accident and whose failure in any mode in any accident environment is not detrimental to plant safety need only be qualified for its non-accident service environment. Although actual type testing is preferred, other methods when justified may be found acceptable. The bases should be provided for concluding that such equipment is not required to function in order to mitigate any accident, and that its failure in any mode in any accident environment is not detrimental to plant safety.	(c)	Class 1E equipment whose function is only required in non-accident environment conditions and whose failure in any mode in an accident environment is not detrimental to plant safety is only analyzed to its non-accident environments.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 14 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
(4)	For environment qualification of equipment subject to events other than a DBA, which result in abnormal environmental conditions, actual type testing is preferred. However, analysis or operating history, or any applicable combination thereof, coupled with partial type test data may be found acceptable, subject to the applicability and detail of information provided.	(4)	For Class 1E equipment which is not subjected to environmental conditions resulting directly from a DBA, type testing is normally used to demonstrate qualification. When alternate or combination methods are used, the methods are justified.
2.2	Qualification by Test	2.2	
(1)	The failure criteria should be established prior to Testing.	(1)	Failure criteria may be specifically specified or implied as not meeting the acceptance criteria.
(2)	Test results should demonstrate that the equipment can perform its required function for all service conditions postulated (with margin) during its installed life.	(2)	Test results demonstrate that the equipment can perform its required function during its qualified life and during all required service conditions.
(3)	The items described in Section 6.3 of IEEE Std. 323-1974 supplemented by items (4) through (12) below constitute acceptable test procedures.	(3)	The guidelines of Section 6.3 of IEEE Std. 323-1974 were used in establishing test procedures.
(4)	When establishing the simulated environmental profile for qualifying equipment located inside containment, it is preferred that a single profile be used that envelopes the environmental conditions resulting from any design basis event during any mode of plant operation (e.g., a profile that envelopes the conditions produced by the main steamline break and loss-of-coolant accidents).	(4)	A single composite profile was used for the simulation of DBA (LOCA/ MSLB) environmental conditions with adequate margin.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 15 of 21)

NUREG-0588 Category I Position		CPNPP STATUS	
(5)	Equipment should be located above flood level or protected against submergence by locating the equipment in qualified watertight enclosures. Where equipment is located in watertight enclosures, qualification by test or analysis should be used to demonstrate the adequacy of such protection. Where equipment could be submerged, it should be identified and demonstrated to be qualified by test for the duration required.	(5)	See FSAR Section 3.11B.1.
(6)	The temperature to which equipment is qualified, when exposed to the simulated accident environment, should be defined by thermocouple readings on or as close as practical to the surface of the component being qualified.	(6)	During the simulation of DBA conditions, temperature reading were taken to ensure as best as possible that the environment in the vicinity of the test component is the same as the environment required by the test envelope.
(7)	Performance characteristics of equipment should be verified before, after, and periodically during testing throughout its range of required operability.	(7)	Functional testing is performed at various points to verify the required operability of the equipment.
(8)	Caustic spray should be incorporated during simulated event testing at the maximum pressure and at the temperature conditions that would occur when the onsite spray systems actuate.	(8)	Caustic spray is incorporated in the testing during the portion of the accident envelope that best simulates DBA conditions.
(9)	The operability status of equipment should be monitored continuously during testing. For long-term testing, however monitoring at discrete intervals should be justified if used.	(9)	The operability status of equipment is monitored as necessary to verify the required operability of the equipment as defined by its Class 1E function.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(10)	Expected extremes in power supply voltage range and frequency should be applied during simulated event environmental testing.	(10)	Characteristics of the equipment under voltage and frequency extremes is demonstrated during the performance test of the equipment and not necessarily during the simulated event test. These characteristics are re-examined during simulated event testing if there is reason to believe or test evidence that the ability of the equipment to perform under these extremes are changed.
(11)	Dust environments should be addressed when establishing qualification service conditions.	(11)	The dust environment at CPNPP was considered in the drafting of the specifications of Class 1E equipment for CPNPP. The dust condition at CPNPP by design are not considered a significant factor in the qualification of the CPNPP Class 1E equipment.
(12)	Cobalt-60 is an acceptable gamma radiation source for environmental qualification.	(12)	Cobalt-60 and Cesium 137 are used as the radiation sources for environmental qualification.
2.3	Test Sequence	2.3	
(1)	The test sequence should conform fully to the guidelines established in Section 6.3.2 of IEEE Std. 323-1974. The test procedures should insure that the same piece of equipment is used throughout the test sequence, and that the test simulates as closely as practicable the postulated accident environment.	(1)	The test sequence provided in Section 6.3.2 of IEEE Std. 323-1974 was followed. The testing was performed on the same piece of equipment (with the exception that some performance tests may have been run on identical or essentially similar equipment as allowed by Section 6.3.2(3). Where other test sequences are used, suitable justification is provided.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 17 of 21)

NUREG-0588 Category I Position	CPNPP STATUS
2.4 Other Qualification Methods Qualification by analysis operating experience implemented, as described in IEEE Std. 323-1974 and other ancillary standards, may be found acceptable. The adequacy of these methods will be evaluated on the basis of the quality and detail of the information submitted in support of the assumptions made and the specific function and location of the equipment. These methods are most suitable for equipment where testing is precluded by physical size of the equipment being qualified. It is required that, when these methods are employed, some partical type tests on vital components of the equipment be provided in support of these methods.	2.4 Then qualification is by methods other than type test, adequate support information have been provided to find these methods acceptable. On some equipment, some partial type tests of vital components have been performed to support the overall qualification methods.
3. MARGINS	3.
(1) Quantified margins should be applied to the design parameters discussed in Section 1 to assure that the postulated accident conditions have been enveloped during testing. These margins should be applied in addition to any margins (conservatism) applied during the derivation of the specified plant parameters.	(1) Adequate margin has been applied to the design parameters in the simulation of postulated DBA conditions during type tests.
(2) In lieu of other proposed margins that may be found acceptable, the suggested values indicated in IEEE Std. 323-1974, Section 6.3.1.5, should be used as a guide. (Note exceptions stated in Section 1.4.)	(2) The margins suggested in Sections 6.3.1.5 of IEEE Std. 323-1974 have been met or exceeded (except as allowed by Section 1.4).

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(3)	When the qualification envelope in Appendix C is used, the only required margins are those accounting for the inaccuracies in the test equipment. Sufficient conservatism has already been included to account for uncertainties such as production errors and errors associated with defining satisfactory performance (e.g., when only a small number of units are tested).	(3)	Not applicable to CPNPP
(4)	Some equipment may be required by the design to only perform its safety function within a short time period into the event (i.e., within seconds or minutes), and, once its function is complete, subsequent failures are shown not to be detrimental to plant safety. Other equipment may not be required to perform a safety function but must not fail within a short time period into the event, and subsequent failures are also shown not to be detrimental to plant safety. Equipment in these categories is required to remain functional in the accident environment for a period of at least 1 hour in excess of the time assumed in the accident. For all other equipment (e.g., post-accident monitoring, recombiners, etc.), the 10 percent time margin identified in Section 6.3.1.5 of IEEE Std. 323-1974 may be used.	(4)	No BOP equipment falls into this category.
4.	<u>AGING</u>	4.	
(1)	Aging effects on all equipment, regardless of its location in the plant, should be considered and included in the qualification program.	(1)	Aging effects were considered for all Class 1E equipment located in a harsh environment at CPNPP. All qualification programs for CPNPP Class 1E equipment address aging.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(2)	The degrading influences discussed in Sections 6.3.3, 6.3.4 and 6.3.5 of IEEE Std. 323-1974 and the electrical and mechanical stresses associated with cyclic operation of equipment should be considered and included as part of the aging programs.	(2)	Aging effects (including cyclic operation, integrated radiation dose, and vibrations) were included in the aging programs where these degrading influences could effect the required operability of the equipment.
(3)	Synergistic effects should be considered in the accelerated aging programs. Investigation should be performed to assure that no known synergistic effects have been identified on materials that are included in the equipment being qualified. Where synergistic effects have been identified, they should be accounted for in the qualification programs. Refer to NUREG/CR-0276 (SAND 78-0799) and NUREG/CR-0401 (SAND 78-1452), “Qualification Testing Evaluation Quarterly Reports,” for additional information.	(3)	Synergistic effects have been considered in the review of the qualification programs and the aging performed. No known synergistic effects have been defined that could have a significant impact on the qualification of any of the equipment qualified for CPNPP.
(4)	The Arrhenius methodology is considered an acceptable method of addressing accelerated aging. Other aging methods that can be supported by type tests will be evaluated on a case-by-case basis.	(4)	Arrhenius methodology was used in addressing thermal aging.
(5)	Known material phase changes and reactions should be defined to insure that no known changes occur within the extrapolation limits.	(5)	The accelerated aging temperatures and extrapolation limits used in the qualification programs were reviewed. No known material phase changes of concern could be identified for the programs accepted for CPNPP.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(6)	The aging acceleration rate used during qualification testing and the basis upon which the rate was established should be described and justified.	(6)	The methodologies for the accelerated aging rates complies with (4) above.
(7)	Periodic surveillance testing under normal service conditions is not considered an acceptable method for on-going qualification, unless the plant design includes provisions for subjecting the equipment to the limiting service environment conditions (specified in Section 3(7) of IEEE Std. 279-1971) during such testing	(7)	Periodic surveillance testing is not used as a method of on-going qualification.
(8)	Effects of relative humidity need not be considered in the aging of electrical cable insulation.	(8)	_____
(9)	The qualified life of the equipment (and/ or component as applicable) and the basis for its selection should be defined	(9)	The qualified life for all Class 1E equipment at CPNPP is provided and its basis is defined.
(10)	Qualified life should be established on the basis of the severity of the testing performed, the conservatisms employed in the extrapolation of data, the operating history, and in other methods that may be reasonably assumed, coupled with good engineering judgement.	(10)	The qualified life is normally based on the accelerated aging (and other aging) performed during testing but may have as a partial basis other factors such as, operating history, stress test performed, analysis of failure modes and aging mechanisms, etc, - all applied with good engineering judgement.

TABLE 3A.3-1
CPNPP COMPLIANCE WITH NUREG-0588 FOR BOP CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
5.	QUALIFICATION DOCUMENTATION	5.	
(1)	The staff endorses the requirements stated in IEEE Std. 323-1974 that, “The qualification documentation shall verify that each type of electrical equipment is qualified for its application and meets its specified performance requirements. The basis of qualification shall be explained to show the relationship of all facets of proof needed to support adequacy of the complete equipment. Data used to demonstrate the qualification of the equipment shall be pertinent to the application and organized in an auditable form.”	(1)	Proper qualification documentation is provided for all CPNPP Class 1E equipment located in a harsh environment
(2)	The guidelines for documentation in IEEE Std. 323-1974 when fully implemented are acceptable. The documentation should include sufficient information to address the required information identified in Appendix E. A certificate of conformance by itself is not acceptable unless it is accompanied by test data and information on the qualification program.	(2)	The CPNPP documentation complies with these requirements.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
1.	ESTABLISHMENT OF THE QUALIFICATION PARAMETER FOR DESIGN BASIS EVENTS	1.	See Table 3A.3-1 for all information on this portion of NUREG-0588.
1.1)	Temperature and Pressure Conditions Inside Containment - Loss-of-Coolant Accident (LOCA)	1.1	See Table 3A.3-1.
(1)	The time-dependent temperature and pressure, established for the design of the containment structure and found acceptable by the staff, may be used for environmental qualification of equipment.	(1)	See Table 3A.3-1.
(2)	Acceptable methods for calculating and establishing the containment pressure and temperature envelopes to which equipment should be qualified are summarized below. Acceptable methods for calculating mass and energy release rates are summarized in Appendix A.	(2)	See Table 3A.3-1.

Pressurized Water Reactors (PWRs)

Dry Containment - Calculate LOCA containment environment using CONTEMPT-LT or equivalent industry codes. Additional guidance is provided in Standard Review Plan (SRP) Section 6.2.1.1.A, NUREG-75/087.

Ice Condenser Containment - Calculate LOCA containment environment using LOTIC or equivalent industry codes. Additional guidance is provided in SRP Section 6.2.1.1.B, NUREG-75/087.

Boiling Water Reactors (BWRs)

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
Mark I, II and III Containment - Calculate LOCA environment using methods of GESSAR Appendix 3B or equivalent industry codes. Additional guidance is provided in SRP Section 6.2.1.1.C, NUREG-75/087.			
(3)	In lieu of using the plant-specific containment temperature and pressure design profiles for BWR and ice condenser types of plants, the generic envelope shown in Appendix C may be used for qualification testing.	(3)	See Table 3A.3-1.
(4)	The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4)	See Table 3A.3-1.
1.2	Temperature and Pressure Conditions Inside Containment-Main Steam Line Break (MSLB)	1.2	
(1)	The environmental parameters used for equipment qualification should be calculated with a plant-specific model reviewed and approved by the staff.	(1)	See Table 3A.3-1.
(2)	Models that are acceptable for calculating containment parameters are listed in Section 1.1(2).	(2)	See Table 3A.3-1.
(3)	In lieu of using the plant-specific containment temperature and pressure design profiles for BWR and ice condenser plants, the generic envelope shown in Appendix C may be used.	(3)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(4)	The test profiles included in Appendix A to IEEE Std. 323-1974 should not be considered an acceptable alternative in lieu of using plant-specific containment temperature and pressure design profiles unless plant-specific analysis is provided to verify the adequacy of those profiles.	(4)	See Table 3A.3-1.
(5)	Where qualifications has been completed but only LOCA conditions were considered, it must be demonstrated that the LOCA qualification conditions exceed or are equivalent to the maximum calculated MSLB conditions. The following technique is acceptable:	(5)	See Table 3A.3-1.
(a)	Calculate the peak temperature envelope from an MSLB using a model based on the staff's approved assumptions defined in Section 1.1(2).		
(b)	Show that the peak surface temperature of the component to be qualified does not exceed the LOCA qualification temperature by the method discussed in item 2 of Appendix B.		

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
(c) If the calculated surface temperature exceeds the qualification temperature, the staff requires that (i) requalification testing be performed with appropriate margins, or (ii) qualified physical protection be provided to assure that the surface temperature will not exceed the actual qualification temperature. For plants that are currently being reviewed, or will be submitted for an operating license review within six months from issue date of this report, compliance with items (i) or (ii) above may represent a substantial impact. For those plants, the staff will consider additional information submitted by the applicant to demonstrate that the equipment can maintain its functional operability if its surface temperature rises to the value calculated.	

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
1.3 Effects of Chemical Spray The effects of caustic spray should be addressed for the equipment qualification. The concentration of caustics used for qualification should be equivalent to or more severe than those used in the plant containment spray system. If the chemical composition of the caustic spray can be affected by equipment malfunctions, the most severe caustic spray environment that results from a single failure in the spray system should be assumed. See SRP Section 6.5.2 (NUREG-75/087), paragraph II, item (e) for caustic spray solution guidelines.	1.3 The maximum concentration of boron employed for containment spray is 2600 ppm and the maximum permitted pH of the final spray solution is 10.5. For qualification testing Westinghouse specifies a chemical spray of 2500 ppm boron buffered with 0.9 dissolved sodium hydroxide to maintain a pH of approximately 10.7, starting at time zero and terminating after 24 hours. This spray concentration results in an increase in alkalinity of at least 10% compared to the maximum concentration defined by the specification and significantly exceeds the range of sump pH values permitted long-term by the technical specification. See also Table 3A.3-1.
1.4 Radiation Conditions Inside and Outside Containment The radiation environment for qualification of equipment should be based on the normally expected radiation environment over the equipment qualified life, plus that associated with the most severe design basis accident (DBA) during or following which that equipment must remain functional. It should be assumed that the DBA related environmental conditions occur at the end of the equipment qualified life.	1.4 See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
	The sample calculations in Appendix D and the following positions provide an acceptable approach for establishing radiation limits for qualification. Additional radiation margins identified in Section 6.3.1.5 of IEEE Std. 323-1974 for qualification type testing are not required if these methods are used.		
(1)	The source term to be used in determining the radiation environment associated with the design basis LOCA should be taken as an instantaneous release from the fuel to the atmosphere of 100 percent of the noble gases, 50 percent of the iodines, and 1 percent of the remaining fission products. For all other non-LOCA design basis accident conditions, a source term involving an instantaneous release from the fuel to the atmosphere of 10 percent of the noble gases (except Kr-85 for which a release of 30 percent should be assumed) and 10 percent of the iodines is acceptable.	(1)	See Table 3A.3-1.
(2)	The calculation of the radiation environment associated with design basis accidents should take into account the time-dependent transport of released fission products within various regions of containment and auxiliary structures.	(2)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(3)	The initial distribution of activity within the containment should be based on a mechanistically rational assumption. Hence, for compartmented containments, such as in a BWR, a large portion of the source should be assumed to be initially contained in the drywell. The assumption of uniform distribution of activity throughout the containment at time zero is not appropriate.	(3)	See Table 3A.3-1.
(4)	Effects of ESF systems, such as containment sprays and containment ventilation and filtration systems, which act to remove airborne activity and redistribute activity within containment, should be calculated using the same assumptions used in the calculation of offsite dose. See SRP Section 15.6.5 (NUREG-75/087) and the related sections referenced in the Appendices to that section.	(4)	See Table 3A.3-1.
(5)	Natural deposition (i.e., plate-out) of airborne activity should be determined using a mechanistic model and best estimates for the model parameters. The assumption of 50 percent instantaneous plate-out of the iodine released from the core should not be made. Removal of iodine from surfaces by steam condensate flow or washoff by the containment spray may be assumed if such effects can be justified and quantified by analysis or experiment.	(5)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(6)	For unshielded equipment located in the containment, the gamma dose and dose rate should be equal to the dose and dose rate at the centerpoint of the containment plus the contribution from location dependent sources such as the sump water and plate-out, unless it can be shown by analyses that location and shielding of the equipment reduces the dose and dose rate.	(6)	See Table 3A.3-1.
(7)	For unshielded equipment, the beta doses at the surface of the equipment should be the sum of the airborne and plate-out sources. The airborne beta dose should be taken as the beta dose calculated for a point at the containment center.	(7)	See Table 3A.3-1.
(8)	Shielded components need be qualified only to the gamma radiation levels required, provided an analysis or test shows that the sensitive portions of the component or equipment are not exposed to beta radiation or that the effects of beta radiation heating and ionization have no deleterious effects on component performance.	(8)	See Table 3A.3-1.
(9)	Cables arranged in cable trays in the containment should be assumed to be exposed to half the beta radiation dose calculated for a point at the center of the containment plus the gamma ray dose calculated in accordance with Section 1.4(6). This reduction in beta dose is allowed because of the localized shielding by other cables plus the cable tray itself.	(9)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(10)	Paints and coatings should be assumed to be exposed to both beta and gamma rays in assessing their resistance to radiation. Plate-out activity should be assumed to remain on the equipment surface unless the effects of the removal mechanisms, such as spray wash-off or steam condensate flow, can be justified and quantified by analysis or experiment.	(10)	See Table 3A.3-1.
(11)	Components of the emergency core cooling system (ECCS) located outside containment (e.g., pumps, valves, seals and electrical equipment) should be qualified to withstand the radiation equivalent to that penetrating the containment, plus the exposure from the sump fluid using assumptions consistent with the Appendix K to 10 CFR Part 50.	(11)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
(12) Equipment that may be exposed to radiation doses below 10^4 rads should not be considered to be exempt from radiation qualification, unless analysis supported by test data is provided to verify that these levels will not degrade the operability of the equipment below acceptable values.	(12) For safety-related electrical equipment that is not required to operate in a high energy line break (HELB) environment and for which the anticipated qualified life integrated radiation dose is 10^4 Rads or less, Westinghouse did not include a radiation aging simulation as part of any qualification testing. For such equipment and components, a supplement to WCAP-8587 demonstrates, based on available test information on materials and components, that up to approximately 10^5 Rads there is no detectable effect on the structural characteristics of materials and components that would affect the capability of equipment to perform during a seismic event. See also Table 3A.3-1.
(13) The staff will accept a given component to be qualified provided it can be shown that the component has been qualified to integrated beta and gamma doses which are equal to or higher than those levels resulting from an analysis similar in nature and scope to that included in Appendix D (which uses the source term given in item (1) above), and that the component incorporates appropriate factors pertinent to the plant design and operating characteristics, as given in these general guidelines.	(13) See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(14)	When a conservative analysis has not been provided by the applicant for staff review, the staff will use the radiation environment guidelines contained in Appendix D, suitably corrected for the differences in reactor power level, type, containment size, and other appropriate factors	(14)	See Table 3A.3-1.
1.5	Environmental Conditions for Outside Containment	1.5	
(1)	Equipment located outside containment that could be subjected to high-energy pipe breaks should be qualified to the conditions resulting from the accident for the duration required. The techniques to calculate the environmental parameters described in Sections 1.1 through 1.4 above should be applied.	(1)	See Table 3A.3-1.
(2)	Equipment located in general plant areas outside containment where equipment is not subjected to a design basis accident environment should be qualified to the normal and abnormal range of environmental conditions postulated to occur at the equipment location.	(2)	See Table 3A.3-1.
(3)	Equipment not served by Class 1E environmental support systems, or served by Class 1E support systems that may be secured during plant operation or shutdown, should be qualified to the limiting environmental conditions that are postulated for that location, assuming a loss of the environmental support system.	(3)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
2.	QUALIFICATION METHODS	2.	
2.1	Selection of Methods	2.1	
(1)	Qualification methods should conform to the requirements defined in IEEE Std. 323-1974.	(1)	The methodology employed by Westinghouse to qualify safety-related electrical equipment is described in WCAP-8587 and conforms to the requirements of IEEE Std. 323-1974.
(2)	The choice of the methods selected is largely a matter of technical judgment and availability of information that supports the conclusions reached. Experience has shown that qualification of equipment subjected to an accident environment without test data is not adequate to demonstrate functional operability. In general, the staff will not accept analysis in lieu of test data unless (a) testing of the component is impractical due to size limitations, and (b) partial type test data is provided to support the analytical assumptions and conclusions reached.	(2)	Westinghouse qualifies equipment that is required to perform a safety function in a high energy line break environment by test.
(3)	The environmental qualification of equipment exposed to DBA environments should conform to the following positions. The bases should be provided for the time interval required for operability of this equipment. The operability and failure criteria should be specified and the safety margins defined.	(3)	The required and demonstrated duration of the safety function of equipment subject to high energy line break conditions is defined for equipment supplied by Westinghouse. The specified required duration is selected to envelope the range of operability assumptions contained in the Accident Analyses reported in the FSAR.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
			The primary purpose of equipment qualification is to reduce the potential for common-mode failures due to postulated environmental conditions. A test unit will therefore be considered to have failed the test if the functional requirements identified cannot be met, unless an investigation can establish that the failure mechanism is not of common-mode origin or that plant specific analyses can demonstrate that the reduced capability is acceptable. Margins are discussed under 3 below.
(a)	Equipment that must function in order to mitigate any accident should be qualified by test to demonstrate its operability for the time required in the environmental conditions resulting from that accident.	(a)	When Westinghouse employs testing to qualify electrical equipment, that must function in order to mitigate any accident, the acceptance criterion for the test is that the safety related function must be demonstrated for the specified duration while the equipment is exposed to the simulated environmental conditions resulting from the accident.
(b)	Any equipment (safety-related or non-safety-related) that need not function in order to mitigate any accident but that must not fail in a manner detrimental to plant safety should be qualified by test to demonstrate its capability to withstand any accident environment for the time during which it must not fail.	(b)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(c)	Equipment that need not function in order to mitigate any accident and whose failure in any mode in any accident environment is not detrimental to plant safety need only be qualified for its non-accident service environment. Although actual type testing is preferred, other methods when justified may be found acceptable. The bases should be provided for concluding that such equipment is not required to function in order to mitigate any accident, and that its failure in any mode in any accident environment is not detrimental to plant safety.	(c)	Where Westinghouse supplies an item of safety-related electrical equipment that is located in an area where it can experience the environment resulting from a high energy line break, but is not required to perform any safety-function, Westinghouse has verified that any consequential failure of such equipment, due to the adverse environment, does not prejudice the safety related functions of other equipment claimed in the accident analysis.
(4)	For environment qualification of equipment subject to events other than a DBA, which result in abnormal environmental conditions, actual type testing is preferred. However, analysis or operating history, or any applicable combination thereof, coupled with partial type test data may be found acceptable, subject to the applicability and detail of information provided.	(4)	See Table 3A.3-1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
2.2	Qualification by Test	2.2	
(1)	The failure criteria should be established prior to Testing.	(1)	In Supplement 1 to WCAP-8587 Westinghouse has identified, for each item of safety-related equipment, the safety functions to be performed for all normal, abnormal or accident conditions during or after which the equipment is required to provide a protective function. As stated, a test unit will be considered to have failed the test if the safety related functional requirements cannot be met, unless an investigation can establish that the failure mechanism is not of common-mode origin or that plant specific analyses can demonstrate that the reduced capability is acceptable.
(2)	Test results should demonstrate that the equipment can perform its required function for all service conditions postulated (with margin) during its installed life.	(2)	Westinghouse qualification programs are designed to demonstrate that the equipment can perform its required safety function(s) for all postulated service conditions.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
			Via the treatment of aging, as described in the response to position 4. below, a generic qualified life is established by Westinghouse for each item of equipment. This generic qualified life may be extended on a plant specific basis by employing less conservative plant specific assumptions concerning the plant normal operating environmental conditions or by employing less conservative aging assumptions (i.e., less conservative activation energy for equipment). The qualified life established for the equipment on a specific plant will ultimately define the permitted installed life of the equipment. The subject of margin is discussed in the response to position 3. below.
(3)	The items described in Section 6.3 of IEEE Std. 323-1974 supplemented by items (4) through (12) below constitute acceptable guidelines for establishing test procedures.	(3)	When testing is the selected methodology for qualifying equipment, Westinghouse has established the test program in conformance with Section 6.3 of IEEE 323-1974.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
(4) When establishing the simulated environmental profile for qualifying equipment located inside containment, it is preferred that a single profile be used that envelopes the environmental conditions resulting from any design basis event during any mode of plant operation (e.g., a profile that envelopes the conditions produced by the main steamline break and loss-of-coolant accidents).	(4) Westinghouse prefers to use a single profile, enveloping both MSLB and LOCA, for qualification of equipment located inside containment which is required to perform a safety function to mitigate both High Energy Line Breaks (HELB). This approach is optimum in terms of schedule, manpower and materials. However, there is no technical justification for making this a requirement for all equipment inside containment. The exceptions to the use of a single qualification envelope for LOCA and MSLB are identified. In general, these occur when: - A component is only claimed to mitigate against one of the HELB's. In such a case, qualification has been completed to conditions enveloping the possible consequences inside containment from the single HELB and Westinghouse has verified that failure of the component in any other more limiting HELB environment will not prejudice any safety-related function. - The test conditions are found to be unacceptably conservative.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(5)	Equipment should be located above flood level or protected against submergence by locating the equipment in qualified watertight enclosures. Where equipment is located in watertight enclosures, qualification by test or analysis should be used to demonstrate the adequacy of such protection. Where equipment could be submerged, it should be identified and demonstrated to be qualified by test for the duration required.	(5)	See Table 3A.3-1.
(6)	The temperature to which equipment is qualified, when exposed to the simulated accident environment, should be defined by thermocouple readings on or as close as practical to the surface of the component being qualified.	(6)	In performing qualification tests for high energy line break environments, Westinghouse requires that the external environment temperature be measured as close to the test unit surface as practicable.
(7)	Performance characteristics of equipment should be verified before, after, and periodically during testing throughout its range of required operability.	(7)	Where the safety-related function of the equipment requires operation during the HELB, Westinghouse verifies the equipment performance before, during and after the simulated event and verifies that the safety-related function is demonstrated for the specified required duration of the function.
(8)	Caustic spray should be incorporated during simulated event testing at the maximum pressure and at the temperature conditions that would occur when the onsite spray systems actuate.	(8)	The response to item 1.3(1) is applicable for equipment located inside containment and qualified by test to operate in a high energy line break environment.
(9)	The operability status of equipment should be monitored continuously during testing. For long-term testing, however, monitoring at discrete intervals should be justified if used.	(9)	The response to item 2.2(7) is applicable.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position		CPNPP STATUS	
(10)	Expected extremes in power supply voltage range and frequency should be applied during simulated event environmental testing.	(10)	Westinghouse does not require that power supply voltage and frequency be varied during the HELB simulation. Most of the Class 1E equipment to be qualified in Westinghouse scope is suppled by a guaranteed stabilized power supply. As a consequence, the range of electrical parameters employed is extremely small and variations within the permitted range are considered insignificant. Exceptions to this generic position are those items of equipment which could experience a significant change in power supply voltage and frequency at the time they are called upon to perform the specified safety function, and a justification is provided for the adequacy of the qualification of these items to HELB conditions.
(11)	Dust environments should be addressed when establishing qualification service conditions.	(11)	See Table 3A.3-1.
(12)	Cobalt-60 is an acceptable gamma radiation source for environmental qualification.	(12)	Westinghouse employs Cobalt-60 sources to simulate the effects of gamma and, in some cases, beta radiation for equipment qualified by test to operate in a HELB environment.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
2.3 Test Sequence	2.3
(1) The test sequence should conform fully to the guidelines established in Section 6.3.2 of IEEE Std. 323-1974. The test procedures should insure that the same piece of equipment is used throughout the test sequence, and that the test simulates as closely as practicable the postulated accident environment.	(1) Section 6.3.2 of IEEE 323-1974 neither mandates a single unique test sequence or requires that the same piece of equipment be used through the test sequence: - the standard identifies a test sequence that is thought to be the most conservative for most equipment, however, alternative sequences are clearly permitted with justification. - Section 6.3.2(3) specifically permits the performance test at extremes of the normal ambient to be performed on another, essentially similar, piece of equipment. The following basic test sequences are employed by Westinghouse: (a.) Equipment required to operate in a HELB environment The preferred test sequence is that recommended by IEEE 323-1974: - All production units were subjected to a calibration and/or verification test at ambient conditions. This test included verification of all safety-related functions. - No specific abnormal tests were completed since the HELB environment envelopes the abnormal condition with ample margin.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
	- The selected test unit(s) were aged employing the methodology described in Subprogram A of Appendix B to WCAP-8587. - The same production unit was tested to verify equipment capability during a simulated seismic event. - The same production unit was tested under applicable simulated HELB and post-HELB conditions. Exceptions to the above generic position are identified and justified.
	(b.) <u>Equipment not required to operated in a HELB environment</u> The following test sequence is employed: - Electronic production units were, in general subject to a burn-in period. - All production units were subjected to a calibration and/or verification test at ambient conditions. This test included verification of all safety-related functions. - A performance test was completed on a sample production unit at the specified abnormal environmental extremes. - A similar production unit was tested to verify the capability of the equipment during and after a simulated seismic event.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 22 of 31)

NUREG-0588 Category I Position	CPNPP STATUS
	As indicated by this test sequence Westinghouse performs seismic testing on new equipment. Aging is addressed by a generic test program on a representative sample of aged components as described in Subprogram C of Appendix B to WCAP-8587. Westinghouse has deliberately selected this approach due to the advantages stated in Section 6 of Appendix B of WCAP-8587. The test sequence to be employed for component test program requires seismic testing of aged components, in conformance with IEEE 323-1974. The objective of the component test program is to demonstrate that there is no in-service aging mechanism capable of reducing electrical component capability to perform during a seismic event. Thus, success of this program verifies the validity of completing seismic tests on un-aged equipment. Exceptions to the above generic position are identified and justified. The subject of margin is discussed in the response to Position 3 above.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 23 of 31)

NUREG-0588 Category I Position		CPNPP STATUS	
2.4	Other Qualification Methods	2.4	
	Qualification by analysis or operating experience implemented, as described in IEEE Std. 323-1974 and other ancillary standards, may be found acceptable. The adequacy of these methods will be evaluated on the basis of the quality and detail of the information submitted in support of the assumptions made and the specific function and location of the equipment. These methods are most suitable for equipment where testing is precluded by physical size of the equipment being qualified. It is required that, when these methods are employed, some partial type tests on vital components of the equipment be provided in support of these methods.		See Table 3A.3-1.
3.	MARGINS	3.	
(1)	Quantified margins should be applied to the design parameters discussed in Section 1 to assure that the postulated accident conditions have been enveloped during testing. These margins should be applied in addition to any margins (conservatism) applied during the derivation of the specified plant parameters.	(1)	In general, Westinghouse has applied margin with respect to the design postulated accident conditions defined for each item of equipment, in Section 1 of the corresponding EQDP.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 24 of 31)

NUREG-0588 Category I Position		CPNPP STATUS	
(2)	In lieu of other proposed margins that may be found acceptable, the suggested values indicated in IEEE Std. 323-1974, Section 6.3.1.5, should be used as a guide. (Note exceptions stated in Section 1.4.)	(2)	Westinghouse has applied specific margin to design parameters, in deriving type test parameters, as described in Section 7.1 of WCAP-8587. This method of applying margin is in accordance with Section 6.3.1.5 of IEEE 323-1974, which recognizes increasing test duration as methods of incorporating margin in the test plan.
(3)	When the qualification envelope in Appendix C is used, the only required margins are those accounting for the inaccuracies in the test equipment. Sufficient conservatism has already been included to account for uncertainties such as production errors and errors associated with defining satisfactory performance (e.g., when only a small number of units are tested).	(3)	Westinghouse does not employ the envelope identified in Appendix C for qualification purposes.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

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NUREG-0588 Category I Position	CPNPP STATUS
(4) Some equipment may be required by the design to only perform its safety function within a short time period into the event (i.e., within seconds or minutes), and, once its function is complete, subsequent failures are shown not to be detrimental to plant safety. Other equipment may not be required to perform a safety function but must not fail within a short time period into the event, and subsequent failures are also shown not to be detrimental to plant safety. Equipment in these categories is required to remain functional in the accident environment for a period of at least 1 hour in excess of the time assumed in the accident analysis. For all other equipment (e.g., post-accident monitoring, recombiners, etc.), the 10 percent time margin identified in Section 6.3.1.5 of IEEE Std. 323-1974 may be used.	(4) In general, equipment required to operate in a high energy line break environment is qualified to perform its safety function over a considerable period of time (weeks/ months). The exceptions to this generic position are identified and justified. In qualifying equipment, Westinghouse did not always include any systematic margin on the specified duration of the safety function. Margin was included in qualification testing, as described in the response to position 3.(2). In addition to those areas where margin has been specifically included, Westinghouse has employed a sequence of testing that adds additional conservatism to the qualification program. Some of the areas where margin is usually implicit in the test sequence is as follows: - The full radiation dose, simulating the effects of in-service and high energy line break (HELB) applications, is applied in a single step prior to seismic and HELB test simulations. - When the aging is applied, the sequence in which the postulated aging mechanisms are simulated is designed to ensure a conservative simulation as discussed in the response to Staff position 4.1.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 26 of 31)

NUREG-0588 Category I Position	CPNPP STATUS
	- The seismic event simulation applies significant mechanical stress to the equipment prior to the HELB simulation. - The single envelope normally employed for HELB simulation, not only encompasses the effects of LOCA and MSLB accidents, but a whole spectrum of break sizes and locations within these accident definitions. As a consequence, the envelope employed invariably contains significant margin with respect to the transient for any single break size and location. - The single HELB simulation normally employed combines the high irradiation dose associated with the LOCA with the high temperature associated with the MSLB.
4. <u>AGING</u>	4.
(1) Aging effects on all equipment, regardless of its location in the plant, should be considered and included in the qualification program.	(1) Westinghouse considers and includes in the equipment qualification programs, the effects of aging, as applicable, irrespective of the location of the equipment in the plant.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 27 of 31)

NUREG-0588 Category I Position		CPNPP STATUS	
(2)	The degrading influences discussed in Sections 6.3.3, 6.3.4 and 6.3.5 of IEEE Std. 323-1974 and the electrical and mechanical stresses associated with cyclic operation of equipment should be considered and included as part of the aging programs.	(2)	Appendix B to WCAP-8587 describes the aging mechanisms to be considered and the methodology to be employed in simulating the aging effects. The mechanisms to be considered in the Westinghouse program include the degrading influences discussed in Sections 6.3.3, 6.3.4 and 6.3.5 of IEEE 323-1974 including electrical and mechanical stresses associated with cyclic operation of equipment. The order in which these aging mechanisms are applied has been selected so that no aging mechanism, when simulated, reduces the impact of any previously applied mechanisms. On the contrary, the order selected tends to enhance the effects of any simulated aging mechanism as described in paragraph 40 of Appendix B to WCAP-8587.
(3)	Synergistic effects should be considered in the accelerated aging programs. Investigation should be performed to assure that no known synergistic effects have been identified on materials that are included in the equipment being qualified. Where synergistic effects have been identified, they should be accounted for in the qualification programs. Refer to (SAND 78-0799) and (SAND 78-1452), “Qualification Testing Evaluation Quarterly Reports,” for additional information.	(3)	Westinghouse has not identified any synergistic effects involving the materials and components comprising the equipment to be qualified in this program.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 28 of 31)

NUREG-0588 Category I Position		CPNPP STATUS	
(4)	The Arrhenius methodology is considered an acceptable method of addressing accelerated aging. Other aging methods that can be supported by type tests will be evaluated on a case-by-case basis.	(4)	The Arrhenius equation is only employed by Westinghouse in calculating appropriate temperature and duration parameters to accelerate the effects of thermal aging.
(5)	Known material phase changes and reactions should be defined to insure that no known changes occur within the extrapolation limits.	(5)	Only when such changes are in a non-conservative direction would such mechanisms be a concern. Westinghouse will identify any know phase change or reaction which reduces the effects produced by any accelerated aging process and thereby makes extrapolating the results non-conservative. At this time no such mechanisms are known.
(6)	The aging acceleration rate used during qualification testing and the basis upon which the rate was established should be described and justified.	(6)	A supplement to WCAP-8587 justifies the acceleration parameters and rates employed for the aging program described in Appendix B to WCAP-8587.
(7)	Periodic surveillance testing under normal service conditions is not considered an acceptable method for on-going qualification, unless the plant design includes provisions for subjecting the equipment to the limiting service environment conditions (specified in Section 3(7) of IEEE Std. 279-1971) during such testing.	(7)	Westinghouse does not employ periodic surveillance testing or any form of on-going qualification in the program described in WCAP-8587.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 29 of 31)

NUREG-0588 Category I Position		CPNPP STATUS	
(8)	Effects of relative humidity need not be considered in the aging of electrical cable insulation.	(8)	For equipment subjected to High Energy Line Break (HELB) environments, the aging effects due to humidity during normal operation are judged to be insignificant compared to the effects of the high temperature steam accident simulation and therefore no additional humidity aging simulation is required. For equipment not subjected to HELB environments, the use of materials and components know to be significantly affected by humidity has been avoided.
(9)	The qualified life of the equipment (and/or component as applicable) and the basis for its selection should be defined	(9)	On completing the qualification program for a particular item of equipment, Westinghouse completed the corresponding Equipment Qualification Data Package (EQDP). The final version of the EQDP identifies the demonstrated qualified life and justifies the value selected based on the aging mechanisms that have been simulated.
(10)	Qualified life should be established on the basis of the severity of the testing performed, the conservatisms employed in the extrapolation of data, the operating history, and in other methods that may be reasonably assumed, coupled with good engineering judgement.	(10)	Westinghouse conforms to this position in establishing the demonstrated qualified life of WRD supplied equipment.

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 30 of 31)

NUREG-0588 Category I Position	CPNPP STATUS
5. QUALIFICATION DOCUMENTATION	5.
(1) The staff endorses the requirements stated in IEEE Std. 323-1974 that, "The qualification documentation shall verify that each type of electrical equipment is qualified for its application and meets its specified performance requirements. The basis of qualification shall be explained to show the relationship of all facets of proof needed to support adequacy of the complete equipment. Data used to demonstrate the qualification of the equipment shall be pertinent to the application and organized in an auditable form."	(1) See Table 3A.3-1. In support of this effort Westinghouse has supplied: - identification of the limits to any environmental qualification - copies of all referenced Westinghouse topical reports - a summary of the Westinghouse review to establish the adequacy of its qualification programs with respect to NUREG-0588 In addition, Westinghouse will maintain the available raw test data which supports the referenced qualification tests on equipment subject to high energy line break and available information concerning the performance testing of production units.
(2) The guidelines for documentation in IEEE Std. 323-1974 when fully implemented are acceptable. The documentation should include sufficient information to address the required information identified in Appendix E. A certificate of conformance by itself is not acceptable unless it is accompanied by test data and information on the qualification program.	(2) The Westinghouse qualification test reports meet the requirements of Section 5 to IEEE 323-1974 by providing the following essential information as a minimum: - safety related functional requirements to be demonstrated - range of applicable environmental parameters to be considered - identification of the test unit - description of the test facility and monitoring instrumentation

TABLE 3A.3-2
CPNPP COMPLIANCE WITH NUREG-0588 FOR NSSS CLASS 1E ELECTRICAL EQUIPMENT

(Sheet 31 of 31)

NUREG-0588 Category I Position	CPNPP STATUS
	- description of test unit mounting and interfaces - summary of the test procedures - summary of the test results Westinghouse maintains the supporting raw test data, detailed procedures employed etc. on file and available for audit. Significant exceptions to this generic position are identified and justified.

TABLE 3A.4-1
IS DELETED AND THE SPECIFIC ENVIRONMENTAL DETAILS ARE
REFLECTED IN CONTROL DOCUMENTS (EEQSPs, EQML, DBDs)

APPENDIX 3B

COMPUTER PROGRAMS USED IN THE SYNAMIC AND STATIC ANALYSIS OF ASME CODE CLASS 2 AND 3 PIPING SYSTEMS INCLUDING SUPPORTS FOR ASME CODE CLASS 1, 2 AND 3 PIPING

INTRODUCTION

The following computer programs are used for the analysis of ASME Code Class 2 and 3 piping systems, including supports for ASME Code Class 1, 2, and 3 piping:

1. NUPIPE-SW
2. BAP
3. BSPLT
4. STARDYNE
5. PITRUST
6. PILUG
7. PITRIFE
8. STEHAM
9. WATHAM
10. WATSLUG
11. ELBOW
12. PSPECTRA
13. STRUDL-SW
14. STRUDAT AND SANDUL
15. BASEPLATE-II
16. BIP
17. APE
18. CHPLOT
19. RELAP5
20. NUDL
21. CCW-318
22. CCW-392
23. LOTUS
24. ANSYS
25. PSAP
26. REPIPE
27. SCAP
28. ETA
29. HYDTRAN
30. ME101, LEAP
31. ME150, FAPPS
32. ME035, BASEPLATE
33. ME153, MAPPS
34. ME149, SIGNIT
35. ME148, CAPPS
36. ME214, LSAPS

37. *WESTDYN and Associated Pre- and Post-processors
38. *PIPSAN
39. *WESPLAT
40. *LOCAL
41. *PCLUG
42. *Stress Calculation Spreadsheets
43. *WECAN
44. ME215 (ME101FE), SAPCAS
45. NE820, HSTA
46. PC-PREPS

For each computer program there is a brief description of the program's theoretical basis, the assumptions, and the references used in the program, the extent of its application, and a summary of manual or comparison qualification.

* Unit 2 analyses only

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APPENDIX 3B

3B.1 NUPIPE - SW

General Description

The NUPIPE-SW piping program performs a linear elastic analysis of three dimensional piping systems subjected to thermal, static, and dynamic loads. It utilizes the finite element method of analysis.

NUPIPE-SW handles all loading conditions required for complete nuclear piping analyses. A given piping configuration may be analyzed successively for a number of static and dynamic load conditions in a single computer run. Separate load cases, such as thermal expansion and anchor displacements may be combined to form additional analysis cases. The piping dead load analysis considers both distributed weight properties of the piping and any added concentrated weights.

A lumped mass model of the system is used for all dynamic analysis; both translational and rotational degrees of freedom may be considered. Location of lumped masses and degrees of freedom at each mass point are pre-selected by the analyst. The program automatically computes values of translational lumped masses.

Program input consists basically of program control, piping configuration description, and load specification information. As part of the program input process a “missing mass” corrective feature is utilized in the response spectra analysis. This provides an approximation to the response of modes above the specified cutoff frequency in calculating the support loads. The output for each loading condition analyzed consists of support reactions, internal forces and moments, deflections, rotations, and member stresses. Output from seismic analysis includes system normal mode information. Several reports may be generated based on report specification. These reports include pipe stress summaries, pipe support tabulations and piping isometric plots.

The NUPIPE-SW program performs analysis in accordance with ASME Section III, Nuclear Power Plant Components (Code). Features ensuring code conformance include use of accepted analysis methods, incorporation of specified stress intensification factors, stress indices and flexibility factors, proper combination of moment resultants, and provision to (automatically) generate results of combined loading cases. A program option is available to specify among Class 1 analysis in accordance with NB-3600 of the Code, Analysis per ANSI B31.1.0 power piping code and combined Class 1 and Class 2 analysis per Articles NB-3600 and NC-3600 of the Code.

Program Verification

The NUPIPE-SW program has been verified against the NRC Benchmark problems for response spectrum seismic analysis. The model shown in Figure 3B.1-1 is a three dimensional piping system consisting of straight and curved elements starting and terminating at anchors. The seismic analysis used a standard SRSS combination. The second model shown in Figure 3B.1-2 is a multi branched piping system which resembles a table. For the seismic analysis a 10 percent grouping method utilizing a standard SRSS combination is selected. The results of

comparisons from both programs are presented in Tables 3B.1-1 through 3B.1-4. The models used are presented in Figures 3B.1-1 and 3B.1-2.

The NUPIPE-SW missing mass option has been verified against NUPIPE-II. The mathematical model in Figure 3B.1-3 has a total of 87 translational dynamic degrees of freedom. The NUPIPE-SW and NUPIPE-II cutoff mode is mode 19 with a frequency of 44 hertz. The ANCHOR stiffnesses input are the same for the NUPIPE-SW and NUPIPE-II analyses. The results of comparison are in Table 3B.1-5 with the model appearing in Figure 3B.1-3.

The additional model used in the missing mass verification is shown in Figure 3B.1-4. It is a free-ended piping system with three equally spaced lumped mass points. The three lumped mass points have a translational dynamic degree of freedom in the X-direction only. This results in a maximum of three modes being generated. Several computer runs were made using NUPIPE-SW. The first computer run had the missing mass option with a cutoff mode set at 2. The second run and no missing mass option and a cutoff mode equal to 3. The results of comparison are in Table 3B.1-6.

The NUPIPE-SW program has been verified with ADLPIPE (A.D. Little Corp.) for thermal analysis The results from both programs are presented in Tables 3B.1-7 through 3B.1-10. The model used for this comparison is shown in Figure 3B.1-5 with the operating conditions indicated on the figure.

3B.2 BAP

The Baseplate Analysis Processor (BAP) computer program is a preprocessor/postprocessor that works in conjunction with the program ANSYS. The purpose of BAP is to generate the ANSYS input necessary for the static, non-linear analysis of baseplates subjected to out-of-plane loads, to distribute in-plane loads to the anchor bolts assuming an infinitely rigid baseplate, and to post process the ANSYS results into a report-style format.

BAP has been documented by benchmarking procedures against the Baseplate Investigation Processor computer code, which is a recognized program in the public domain.

3B.3 BSPLT

BSPLT was developed by SWEC and is a fully documented computer program. BSPLT is used for the qualification of baseplates with four anchor bolts. This program computes bolt tension, shear, tension-shear interaction factors, and plate-bending stresses for various load conditions based on loads applied at the surface of the plate.

The BSPLT computer program has been verified by a comparison with a hand calculation. The sample problem was a 19-in. by 21-in. four-bolt baseplate with a plate thickness of 1 in. and a bolt size of 1 in. The results are summarized in Table 3B.3-1. The computer program results are within acceptable limits.

3B.4 STARDYNE

The STARDYNE Structural Analysis System, written by Mechanics Research, Inc., of Los Angeles, California, is a fully warranted and documented computer program available at Control Data Corporation.

The MRI STARDYNE Systems consists of a series of compatible, digital computer programs designed to analyze linear and non-linear elastic structural methods. The system encompasses the full range of static and dynamic analyses.

The static capability includes the computation of structural deformations and member loads and stresses caused by an arbitrary set of thermal, nodal-applied loads, and prescribed displacements.

Using the normal mode technique, linear dynamic response analyses can be performed for a wide range of loading conditions, including transient, steady-state harmonic, random, and shock spectra excitation types. Dynamic response results can be presented as structural deformations, internal member loads and stresses, and statistical data.

The non-linear dynamic analysis program is integrated in the rest of the STARDYNE system. The equations of motion for the linear portion of the structural model are generated and modified to account for the non-linear springs. The resulting non-linear equations of motion are directly integrated, using either the Newmark or Wilson implicit integration operators. The user may enter sets of structural loadings, which vary with time, and specify time points at which the program is to output the structural response.

This computer program is considered verified by constant use and by the vendor's original documentation and qualification.

3B.5 PITRUST

PITRUST is a program to calculate local stresses in the pipe caused by cylindrical welded attachments under external loadings. This program uses the Bijlaard method to calculate local stresses in the pipe wall caused by cylindrical welded attachments under external loadings, including pressure, dead load, thermal load, and combinations of maximum dynamic loads.[1]

PITRUST has been verified by comparing its solution of a test problem to the solution of the same problem by an independently written piping local stress program, CYLNOZ, in the public domain. The CYLNOZ piping local stress program was written by Franklin Institute (Philadelphia, PA) and is presently used by engineering companies. The test program is of a 72.375-in. outside diameter by 0.375-in. thick run pipe, reacting under an external loading condition of 1,000 lb. force (normal and shear) and 1,000 in.-lb. bending and torsional moments transmitted by a 16-in. outside diameter nozzle. A comparison of results is tabulated in Table 3B.5-1. PITRUST also has been verified by comparing its solution of the test problem to the experimental results.[2] A comparison of these results is tabulated in Table 3B.5-2.

3B.6 PILUG

PILUG is a program to calculate local stresses in the pipe wall caused by rectangular welded attachments under external loadings. This program uses the Bijlaard method to calculate local stresses in pipe wall caused by rectangular welded attachments under external loadings, including pressure, dead load, thermal load, and combinations of maximum dynamic loads.[3]

PILUG has been verified by comparing its solution of test problem to results obtained by hand calculations using the formulations of Reference 3. A comparison of results is tabulated in Table 3B.6-1.

3B.7 PITRIFE

General Description

PITRIFE (SWEC 1982) is a computer program for calculating the local discontinuity stresses in a pipe at the intersection with a circular trunnion due to loads applied to the trunnion. It is a post-processor program that uses the results of a finite element model of two intersecting cylinders. Based upon the stresses calculated with the finite element model, nondimensional stress coefficients are computed for a size-on-size pipe-trunnion configuration for three different values of average pipe radius to wall thickness (R/t = 5, 10, 20). Additionally, non-dimensional stress coefficients are computed for a trunnion radius equal to 0.707 times the pipe radius (0.707 size-on-size) for the three values of R/t. To facilitate the determination of non-dimensional stress coefficients for other values of R/t, a rotated parabola curve that fits the three R/t data points is generated for both the size-on-size and the 0.707 size-on-size data. The PITRIFE program reconstructs these curves and uses them to interpolate and extrapolate for stress coefficients for different values of R/t. The finite element models are analyzed using the STRUDL-II computer program (ICES) SWEC 1977.

Program Verification

The PITRIFE computer program has been verified by demonstrating that the maximum stress intensities as given by PITRIFE equal the values given by the finite element analysis for specific size-on-size and 0.707 size-on-size models. A comparison of these results is tabulated in Table 3B.7-1. The program was verified for other ratios of trunnion to pipe radius by demonstrating that the stress coefficients and maximum stress intensities derived by hand calculation equal the coefficients used in the program to calculate maximum stress intensity. A comparison of these results is given Table 3B.7-2.

3B.8 STEHAM

General Description

STEHAM is a computer program which is used to determine the steam hammer transients of piping systems. This program uses the method of characteristics with finite difference approximations both in space and in time[4,5,6]. It calculates the one-dimensional transient flow responses and the flow-induced forcing functions in a piping system caused by rapid operational changes of piping components, such as the stop valve and the safety relief valve. Flow characteristics of piping components are mathematically formulated as boundary conditions in the program. These components include the flow control valve, the stop valve, the safety relief valve, the steam manifold, and the steam reservoir. Frictional effects are taken into consideration.

This program accepts the following as input:

a. The flow network representation of the piping system.

b. The initial flow conditions along the piping system.
c. Time-dependent flow characteristics of piping components.

Output consists of time-histories of flow pressures, flow densities, flow velocities, inertia, and momentum functions.

Program Verification

STEHAM is verified by comparing its solutions of a test problem (Figures 3B.8-1 and 3B.8-2) to the results of the same problem obtained by an independent analytical approach, as well as an experimental measurement, as published in Reference 7 and 8. A comparison of results for time-history pressure responses is plotted on Figures 3B.8-3, 3B.8-4, and 3B.8-5. The forcing functions developed for nodal points of the piping system calculated from the relation $F = (p + PV^2/g) A - p_a A$ also have been checked by hand calculation as tabulated in Table 3B.8-1.

3B.9 WATHAM

General Description

WATHAM is a computer program which is used to determine the flow-induced forcing functions acting on piping systems due to water hammer. These forcing functions may then be used as input to a structural dynamic analysis, such as a NUPIPE program run.

WATHAM is applicable to a water hammer problem or, more generally, any unsteady, incompressible fluid flow. These events may be caused by normal or abnormal operational changes of piping components, such as the start up and trip of pumps or the rapid opening and closing of valves.

The analysis is based upon the method of characteristics with finite-difference approximations, both in time and space for the solution of one-dimensional liquid flows. Influences of piping components, including flow valves, pipe connection, reservoirs, and pumps have been considered in the analysis.

WATHAM input requires the geometry of the piping system, pipe properties, water properties, operational characteristics of pump and valve, flow frictional coefficients, and the initial water flow conditions. The output provides the time-history functions of piezometric heads, velocities, and nodal forces for all nodes and the inertial unbalanced force for each segment. It also gives the maximum value of all the preceding functions and their occurring time in the process of flow-transient.

Program Verification

Figure 3B.9-1 depicts a flow network with nine pipes, its geometrical properties, and steady-state flow conditions. The flow transient mode analyzed is the sudden closure of a valve at the down steam end. Figure 3.9B-2 shows the hydraulic network for WATHAM. Table 3B.9-1 illustrates the input data needed for WATHAM run. Figures 3B.9-3 and 3B.9-4 show a comparison of head-time curves[9,10] with WATHAM. Table 3B.9-2 presents the comparison of nodal forces between hand calculation and WATHAM computation.

In general, WATHAM results are in agreement with Streeter's results[9]. The small discrepancy is attributed to the modeling of reservoir boundary condition. In WATHAM, the energy equation between the reservoir is used, rather than assuming that the head of pipe entrance is the same as that of the reservoir.

3B.10 WATSLUG

General Description

The purpose of WATSLUG is to determine forcing functions on piping systems during water slug discharge events for subsequent input to piping dynamic analysis.

The analysis is based upon rigid body motion of the generally sub-cooled water slug and ideal gas representations of the steam or air using rigid column theory to facilitate tracking the several water-steam or water-air interfaces. The driving force is the steam pressure between the valve and the slug, less friction and other losses, and back pressure. Density changes due to possible local flashing of the water slug are considered. Having recourse to the control volume theory, the subsequent segment-forced calculation is carried out.

The input consists of complete piping system geometry, pipe dimensions (Table 3B.10-1), valve flow characteristics, valve opening time, detail upstream steam conditions, and initial downstream steam or air conditions (Table 3B.10-2), while the output contains forcing functions for each piping segment based upon flow velocities, pressures, and densities during the water slug discharge event. Forces were written on tape for direct input to NUPIPE-SW.

Program Verification

The WATSLUG model of the test problem is diagrammed on Figure 3B.10-1, while the NUPIPE-SW model is diagrammed on Figure 3B.10-2. WATSLUG is verified by comparing the solution of this test problem to the results for the same problem obtained by an independent analytical approach (RELAP5/MOD 1), as shown on Figures 3B.10-3 and 3B.10-4, and the comparison of predicted-versus-measured support reactions. NUPIPE-SW generated support reactions due to WATSLUG forcing functions were compared with experimental measurements from a test run of this problem, EPRI Test 980 (RELAP/MOD 1) shown on Figures 3B.10-5 and 3B.10-6.

The WATSLUG forcing functions and the resultant NUPIPE-SW support reactions compare favorably with the RELAP/MOD 1 predicted forcing functions and the EPRI-measured support reactions, respectively.

3B.11 ELBOW

General Description

ELBOW calculates the circumferential and longitudinal stresses on the inside and outside surfaces of an elbow subjected to internal pressure, in-plane bending, out-of-plane bending, torsion, and linear temperature gradient through the wall. Stress indexes and flexibility factors for the elbows are also calculated. Results can be used directly for design and analysis of elbows in accordance with Article NB-3600 of ASME Section III.

The solution method uses Table NB-3685.1-1 relative to internal pressure and Table NB-3685.1-2, with modifications as indicated by Dodge and Moore (1972) relative to moment loadings and flexibility factors.

The complete analysis of Rodabough and George (1957), based on the minimum potential energy method, was written in terms of infinite series. A modified version of the analytical method by Rodabough and George (1957), which considered both in-plane and out-of-plan bending as well as the influence of internal pressure, was selected by ORNL as the most appropriate basis in determining the stresses and flexibility for elbows. The analysis is a generation of the work done by Von Karman. The modification to the analysis method of Rodabough and George (1957) include a generalization of the “correction for transverse compression” recommended by Gross. The ORNL computer program ORNL-ELBOW was written by Dodge and Moore (1973) to implement this analysis procedure.

The SWEC computer program ELBOW uses the same theoretical considerations to obtain the flexibility factor and detailed stresses in the elbows.

When this program is used, it is considered to be a detailed analysis. For elbows free from local discontinuities, ELBOW solutions are the detailed solutions to Equation 10, Table NB-3653, including the consideration of C_1 and C_2, but without $|\alpha_a T_a - \alpha_b T_b|$ term. The solutions are also the detailed solutions to Equation 11 if $|\Delta T_2|$ term is negligible.

The program does not take into account the effects of discontinuities on the elbows. The influence length of a concentrated force or moment in a shells structure is about 2.5 $\sqrt{rt}$, where r is the radius of curvature of the shell surface; or for a pipe, r is the mean radius and t is the thickness of the pipe. For the portion of elbow at a distance of 2.5 $\sqrt{rt}$ away from local discontinuities, detailed stresses can be obtained by using this program.

In general, for an elbow welded to tangent pipe of the same thickness, the effects of straight tangent pipe on the elbow can be neglected (Table NB-3683.2). However, if two elbows are welded together or joined by a piece of straight pipe that is less than one pipe diameter in length, some intensification effects may have to be considered.

This program is an efficient and easy-to-use program for determining stresses, stress indexes, and flexibility factors for elbows. Comparison with experimental results indicates that the results accurately represent the maximum stresses which occur at the center of the bend. Since end effects are not included in the analytical solution on which this program is based, the calculated stresses and flexibility of the elbow may be larger than the actual values.

Required input data include elbow dimensions, material properties, applied moments and forces, internal pressure, and linear temperature gradients.

Output includes stress indexes, flexibility factor for the elbow, and circumferential and longitudinal stresses on the inside and outside surfaces at specific locations.

Program Verification

Sample problems were selected for solution by ELBOW, and these results were compared with those obtained from hand calculations. The following cases were selected for purposes of verification:

Case 1 - Elbow is subjected to internal pressures. Results are given in Table 3B.11-2.

Case 2 - Elbow is subjected to a linear temperature gradient through the pipe wall. Results are given in Table 3B.11-3.

Case 3 - Elbow is subjected to combined loadings at one end. Results are given in Table 3B.11-4.

Elbow properties used for the analyses are given in Table 3B.11-1.

3B.12 PSPECTRA

General Description

PSPECTRA (ME-164) is a data-generating program written and fully documented by SWEC for in-house use. It is used to combine amplified response spectra of seismic and other dynamic events. The methods of spectrum combination include absolute summation, square root of the sum of the squares (SRSS), and maximum value enveloping. PSPECTRA is also used to generate required response spectra which are in accordance with Regulatory Guide 1.122, Revision 1. This involves spreading the peak accelerations and sloping the sides parallel to the original peaks of the input amplified response spectra. The output curves can be generated in terms of accelerations (g's) and either period (sec) or frequency (Hz).

Program Verification

A comparison of a generated response spectrum versus the two input response spectra that were combined by absolute summation is provided on Figure 3B.12-1. Figure 3B.12-2 provides a generated required response spectrum with spread peaks and parallel sloped sides superimposed on the input amplified response spectrum (ARS). The ARS is generated by the time-history method. These figures demonstrate the function and adequacy of the program.

3B.13 STRUDL (STRUCTURAL DESIGN LANGUAGE)

General Description

The STRUDL computer code used within SWEC was developed from Version 2, Modification 2 (June 1972) of the Integrated Civil Engineering Systems (ICES) STRUDL II program which was design and formulated by the Department of Civil Engineering at the Massachusetts Institute of Technology. STRUDL II is a recognized program in the public domain. The software system is IBM-MVS Release 3.8. The hardware configuration is IBM-3033.

The finite element method provides for the solution of a wide range of solid mechanics problems. Its implementation within the context of the STRUDL analysis facilities expands these problems

for the treatment of plane stress, plane strain, plate bending, shallow shell, and three-dimensional stress analysis problems.

The three-dimensional finite element capability of STRUDL is used to analyze the drywell at the region of the equipment hatch and personnel door assembly and other regions of interest.

Seismic Category I structures are analyzed for seismic effect using the dynamic analysis capability of STRUDL. The analysis yields frequencies of vibration, mode shapes, displacements, velocities, accelerations, and forces.

Program Verification

comparisons of results for five test problems performed by both STRUDL and GT-STRUDL are provided herein. GT-STRUDL is a recognized program in the public domain, developed by the GT-ICES Systems Laboratory, School of Civil Engineering, Georgia Institute of Technology, Atlanta, Georgia. In all cases, there is excellent agreement of results between STRUDL and GT-STRUDL.

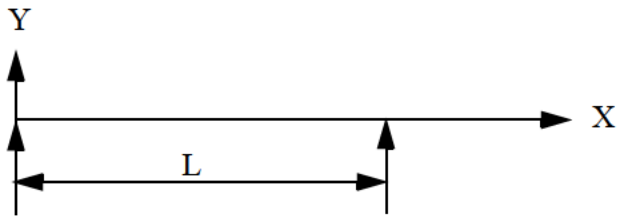
COMPARISON OF STRUDL VERSUS GT-STRUDL RESULTS
FOR DYNAMIC ANALYSIS CAPABILITY OF STRUDL

Problem No. 1

Find the natural frequencies F (I) of vibration for an I-beam with simply supported end, vibrating in the plane of its web.

The pertinent parameters of the beam are as follows:

Length	=	30 ft.
Modulus of Elasticity	=	30×10^6 psi
Moment of Inertia	=	3021 $in.^4$
Weight per foot	=	100 lb.



The theoretical results can be verified from Vibration Problems in Engineering, Fourth Edition, S. Timoshenko, D. W. Young, and W. Weaver, page 423, Problem 1.

Results: Natural Frequency F(I)

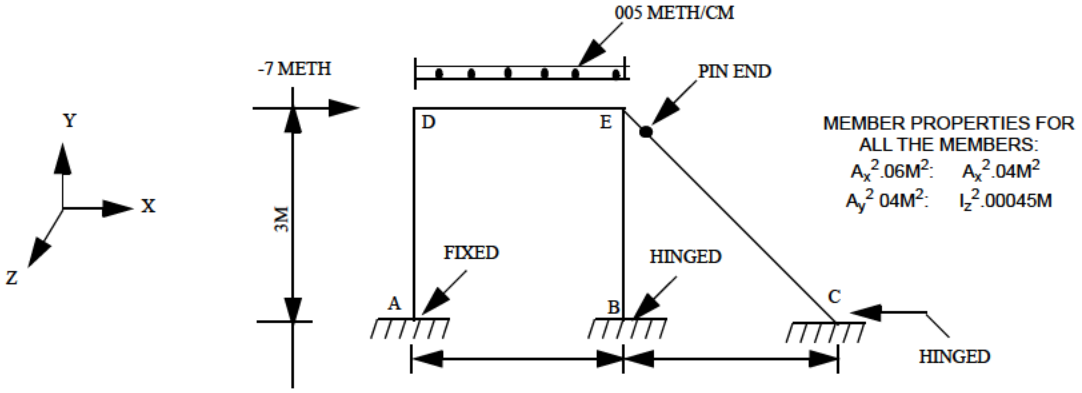
where:

I = mode number
= $24.8\ (I)^2$ cycles/sec
= $155.82\ (I)^2$ rad/sec

The comparison of results (i.e., eigenvalues and eigenvectors) of the theoretical values, STRUDL and GT-STRUDL, is tabulated in Tables 3B.13-1 and 3B.13-2. The eigenvalues for STRUDL and GT-STRUDL agree with each other (Table 3B.13-1). The eigenvectors for STRUDL and GT-STRUDL agree with each other (Table 3B.13-2).

COMPARISON OF STRUDL VERSUS GT-STRUDL RESULTS
FOR STATIC ANALYSIS CAPABILITY OF STRUDL

Problem No. 2



The frame as shown on the sketch is tested for the loads as shown on the sketch. Also, the frame was tested for joint displacement of joints A and B in the Y direction and also the joint displacement of joint A in the X direction. The member forces and the joint forces of the STRUDL run agreed with the GT-STRUDL run. The comparison of the results is tabulated in Table 3B.13-3 and 3B.13-4.

Loading Condition 1

Member DE force Y uniform W-0.005 Metn/cm

Joint D load force X-0.7 Metn

Loading Condition 2

Joint A displaced Y -0.8 cm

Joint B displaced Y -0.3 cm

Loading Condition 3

Joint A displaced X -0.2 cm

Problem No. 3

A foundation mat was analyzed using the finite element capability of STRUDL for a variety of loading combinations. A comparison check is performed for a loading condition which combines the self weight of the substructure and superstructure, dead load of 2.5 ft. of soil above the mat, and east-west tornado loading, by using the finite element capability of GT-STRUDL, a computer program in the public domain. A finite element model is provided in Figure 3B.13-1. Sign convention details are provided in Figure 3B.13-2. Refer to Table 3B.13-5 and 3B.13-6 for comparison between the results obtained from STRUDL and GT-STRUDL.

Problem No. 4

A comparison check is performed for suspended ceiling design using the static analysis capability of STRUDL versus GT-STRUDL. A model is provided in Figure 3B.13-3. The loading condition accounts for the dead loads of ceiling. Refer to Tables 3B.13-7 and 3B.13-8 for comparison between the results obtained from STRUDL and GT-STRUDL.

Problem No. 5

A comparison check is performed for suspended ceiling design using the dynamic analysis capability of STRUDL versus GT-STRUDL. A model is provided in Figure 3B.13-3. The loading condition accounts for the dynamic seismic loads resulting from ceiling dead load. Refer to Tables 3B.13-9, 3B.13-10, and 3B.13-11 for comparison between the results obtained from STRUDL and GT-STRUDL.

3B.14 STRUDAT AND SANDUL

STRUDAT and SANDUL were developed by SWEC and are fully qualified and documented. The computer programs perform ASME III, Appendix XVII and AISC code checks and size fillet welds for pipe support frames and anchors for various loading conditions and combinations.

STRUDAT is a post-processor for STRUDL-SW and translates STRUDL-SW output for applied unit loads into a protected disk file, subsequently accessed by SANDUL. This access is fully traceable.

SANDUL combines the load components for each applied load into the appropriate load conditions, then applies the appropriate signs to each applied load for each loading condition. This load condition matrix is multiplied by the unit load member force and unit load reaction matrixes from STRUDAT to generate the computed member forces and reactions. This is based on the principle of superposition. These member forces are used to calculate the required fillet weld sizes and member stresses. Allowable stresses are also computed for each member. The output provides a ratio of the member stresses and required weld size for each member and weld, respectively. In addition, a summary of the support reactions is provided in the output.

The STRUDAT and SANDUL computer programs have been verified by comparison with STRUDL-SW and hand computations. These results are presented in Table 3B.14-1 through 3B.14-3. The sample model is shown on Figure 3B.14-1. This comparison provides results within an acceptable range.

3B.15 CDC - BASEPLATE II

General Description

BASEPLATE II is a combination of a pre and post-processor to the STARDYNE program for the purpose of analyzing flexible baseplates on a geometrically non-linear foundation. The program employs an automatic mesh generation technique with the user in control of mesh size and element configuration.

Input includes plate geometry, nonstandard and standard (library) attachments, anchor locations, anchor and concrete stiffnesses, material properties, anchor allowables, and up to 50 loading conditions.

Output consists of a printer plot of the baseplate showing attachment and anchor locations, plate deformations and principal stresses, anchor bolt tension, and resultant shear load for each anchor, together with calculated tension/shear interaction and factor of safety.

Program Verification

BASEPLATE II is verified and qualified through Control Data Corporation quality assurance programs, which are periodically evaluated by SWEC and which have been found to be satisfactory.

3B.16 BIP

General Description

BIP (Baseplate Information Processor) is a set of two programs, BIP1 and BIP2, which are pre and post-processors to ANSYS. These programs reduce the time required for baseplate analysis. Baseplates are analyzed for in-plane and out-of-plane loads that are transferred through the attachments. The baseplate may be treated as infinitely rigid for in-plant loads, resulting in a statically determinate solution for anchor bolt shear loads. Out-of-plane loads are analyzed by ANSYS to account for plate flexibility as well as gaps between the baseplate and concrete and interference fit with anchors.

Input to BIP includes plate and attachment geometry, anchor locations, anchor and concrete stiffness values, anchor and concrete gaps, material properties, anchor allowables with reduction factors, and up to 10 loading conditions.

Output consists of an input echo, a printer plot of the baseplate showing attachment and anchor locations, resultant shear at each anchor together with reduced tension allowables based on tension-shear interaction, plate deformations and stresses, and reactions (including bolt pullout loads).

Program Verification

The BIP program is a publicly available program and is verified and qualified through Boeing Computer Services Quality Assurance Programs. These are periodically evaluated by SWEC and have been found to be satisfactory.

3B.17 APE

General Description

Computer program APE (Anchor Plate Evaluation) calculates the shear and tension loads for each anchor of a group of drilled-in anchors of a baseplate subjected to in-plane and out-of-plane loads. A reduced tension allowable, based on the calculated shear load and tension shear interaction, is also calculated.

Program input includes plate and attachment geometry, anchor bolt locations and allowables, and applied loads.

Output consists of anchor bolt pattern center of gravity and polar moment inertia and resultant shear and allowable tension load for each anchor. Load factors for out-of-plane loading, and anchor tension loads, including load factors, for each anchor.

Program Verification

Verification of the APE program (Version 01, Level 00) was performed by comparing results of APE analyses with similar results obtained from Boeing Computer Services Program BIP (refer to Section 3B.16). Comparison of results from APE (Version 01, Level 00) and Boeing Computer Services Program BIP are shown in Table 3B.17-1. APE results are shown to be conservative or comparable with respect to BIP. All results are based on loadings at the critical anchor.

3B.18 CHPLOT

CHPLOT is a program which will plot any number of data values (variables) versus time. Although the plot input data file can be in the form of card data, the more appropriate application of this program is to be used in conjunction with a program that creates a plot data file (on disk or tape) having the format required for input to this program.

Plots are available in two sizes: one with axes of 5 in. (Ordinate) by 8 in. (Absicissa) which fits the standard 8 1/2 in. by 11-in. page, and the other is 8 in. by 12 in. for fitting an 11-in. by 15-in. page. Plots are normally one data value versus time per graph, although up to 14 data values (plots) can be plotted on one graph.

3B.19 RELAP5

General Description

RELAP5 is a public-domain computer program which was written by Idaho National Engineering Laboratory and is available from several sources including Control Data Corporation (CDC). The program may be used to evaluate the behavior of a system subjected to postulated transients such as loss of coolant from large or small pipe breaks, pump failures, etc., or may be used to evaluate the thermal-hydraulic response of fluid transients in piping systems.

RELAP5 uses a five equation two-phase flow hydrodynamic model consisting of two phase continuity equations, two-phase momentum equations and an overall energy equation augmented by the requirement that one of the phases is assumed saturated. In this model, only two interphase constitutive relations are required, those for interphase drag and interphase mass exchange. Models are included for abrupt area changes, choking, mass transfer interphase drag, wall friction and branching.

For hydrodynamics, the space approximation uses a staggered mesh where integral forms of the continuity and energy equations are approximated over control volumes and line integral forms of the momentum equations are applied from the midpoint of one control volume to the midpoint of the adjoining control volume. Hydrodynamic equations are advanced in time using a semi-implicit, linearized method. Head conduction is approximated by finite differences and advanced by the Crank-Nicolson scheme. A modified Runge-Kutta technique for stiff equations is used to

solve the reactor kinetics equations. The interaction among hydrodynamics, heat conduction, trips, reactor kinetics and the control system is explicit.

The program requires numerical input data that completely describe the initial fluid conditions and geometry of the system being analyzed. The input data include physical characteristics such as fluid volume geometry and pump characteristics, range of time step size, output variables, output frequency and trips. The piping network is described to RELAP5 as a set of connected control volumes. RELAP5 computes fluid properties within these volumes and in the connecting flow junctions at discrete time points (steps).

In general the program provides, as output, variables necessary to describe the transient state of the system being analyzed. These results include, as applicable, time varying pressure, momentum flux and energy states throughout a fluid system containing water, steam, and/or a two-phase mixture.

Program Verification

RELAP5 is a recognized program in the public domain.

3B.20 NUDL (ME-268)

This computer program is a pipe support load combination generator that serves as an aid in the design of pipe supports. The program reads the pipe support loads from a file created by NUPIPE-SW, combines them according to a set of specified load combination equations, and outputs them to the printer in tabular format and to a disk file in STRUDL command format.

This computer program has been qualified by SWEC in accordance with SWEC Engineering Assurance Procedures.

3B.21 CCW-318

This program's purpose is to evaluate the design of rectangular cross section welded attachments on ASME Code Class 2 and 3 and ANSI B31.1 piping using Code Case N-318-1, -2, -3.

This program has been qualified by comparing the results against those of hand calculations.

3B.22 CCW-392

This program's purpose is to evaluate the design of hollow circular cross section integral welded attachments to ASME Code Class 2 and 3 and ANSI B-31.1 piping using Code Case N-392.

This program has been qualified by comparing the results against those of hand calculations.

3B.23 LOTUS

ME-277 is a collection of worksheets which have been converted to LOTUS 1-2-3 templates closely replicating the printed forms, allowing the assistance of an IBM PC in performing these tasks.

The ME-277 templates cover calculations on weld stresses, local and bearing stresses, code checks, and other stress calculations for pipe support evaluation. These evaluations are primarily to confirm the Subsection NF or AISC code requirements.

This program has been qualified by comparing the results against those of hand calculations.

3B.24 ANSYS (ST-384)

ANSYS is a general purpose finite element analysis program. It can be used to evaluate problems which are not solvable through standard project methods. Among the available options are static and dynamic finite element analysis for both linear and nonlinear behavior. Various types of elements are available for optimal solution to the particular problem.

This program has been qualified by SWEC in accordance with SWEC Engineering Assurance Procedures.

3B.25 PSAP (HY-066)

This program performs a comprehensive hydraulic analysis and design of a network piping system. It provides energy gradients, pressures, and flow rate distribution for the entire system under steady state flow balance condition.

This program has been qualified by SWEC in accordance with SWEC Engineering Assurance Procedures.

3B.26 REPIPE (ME-279)

This program computes the loading time-history on a piping network based upon results from computer program RELAP5 hydrodynamic analysis of the contained fluid. The time-varying pressure, momentum flus, and energy states throughout a fluid system containing water, steam, and/or a two-phase mixture from RELAP5 output are used as an input to the REPIPE program to produce force time-histories for input to the piping stress analysis program.

This program has been qualified by SWEC in accordance with SWEC Engineering Assurance Procedures.

3B.27 SCAP

The SCAP Computer Program calculates the resulting forces and pipe local stresses at a stiff clamp U-bolt pipe support caused by all possible load conditions. The methodology used is based on CPPP-7, Rev. 4, Attach. 4-6C, D & E.

The SCAP program is applicable only to the analysis of a stiff clamp attached to a single strut or snubber.

The SCAP program computes the resulting forces and local stresses in the pipe. The local pipe stresses are checked against the ASME Boiler and Pressure Vessel Code, Section III, Division 1, Class 2 and 3 piping allowables, and ANSI B31.1 code allowables for different load conditions.

This program has been qualified by comparing the results against those of hand calculations.

3B.28 ETA

This program's purpose is to evaluate the design of hollow circular cross section integral welded attachments to ASME Code Class 2 and 3 and ANSI B31.1 piping elbows using Code Class N-392 and SWEC analyzed stress intensification factors, per CPPP-7, Rev. 4, Attach. 4-6H.

This program has been qualified by comparing the results against those of hand calculations.

3B.29 HYDTRAN (HY-001)

This program simulates the hydraulic transient response in a piping system resulting from pump startup, pump shutdown, pump trip, and/or valve operation and was developed for use with a generalized complex hydraulic system. It is capable of analyzing various combinations of hydraulic components normally encountered in power plant systems. The method of characteristics is used for the solution of these transients which produces pressures, flows, and flow-induced forcing functions which may be input to a dynamic analysis such as a NUPIPE program run.

This program has been qualified by SWEC in accordance with SWEC Engineering Assurance Procedures.

3B.30 LEAP, (ME-101) LINEAR ELASTIC ANALYSIS OF PIPING

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

Description

ME101 is a finite element computer program which performs linear elastic analysis of piping systems, transient thermal analysis, local stress analysis, and response spectra merging.

The input data format is specifically designed for pipe stress analysis. ME-101 performs a thorough check of the input prior to performing analysis. The program rearranges the geometry automatically to optimize the finite element model. The coordinate and keyword data can be specified in English, Metric or S1 units. ME101 performs static and dynamic analysis of piping systems in accordance with Section III of the ASME Boiler and Pressure Vessel Code, ANSI B31.1 Power Piping Code and ANSI B31.3 Chemical Plant and Petroleum Refinery Piping Code. The flexibility factors, stress intensification factors (SIF) and stress indices (B1, B2) are incorporated in the stress evaluations.

Static analysis capabilities include thermal expansion, deadweight, seismic and uniformly distributed loads, and externally applied forces, moments, displacements and rotations. For dead weight and thermal expansion analysis, supports with gaps can be considered.

Response spectrum analysis is based upon standard modal superposition techniques. The input excitation may be in the form of a single or multiple seismic response spectra or time dependent loading functions. In the response spectrum analysis, the user may request modal combinations by various alternate methods typical of the industry. ME101 can further consider differential damping according to NRC RG 1.61 and PVRC Code Case N-411 within a single response spectrum analysis.

ME101 uses an out-of-core solution and active column techniques for both static and dynamic analyses. It has no limit to the number of equations or bandwidth.

ME101 considers zero period accelerations with or without missing mass corrections in the seismic response spectrum analyses. In the time history analysis (modal or direct integration), the excitation may be in the form of nodal forces, support displacements or support accelerations. ME101 can also compute the response to dynamic loads using a direct integration time history method considering nonlinear kinematic hardening supports (such as the Bechtel energy absorber) with or without gaps. Transient analyses and response spectrum analyses of this nonlinear type support are also available. ME101 compares stresses vs. allowables according to ASME/ANSI code equations. ME101 has load combination capabilities that allow the results of several load cases to be combined according to certain algebraic rules to form additional load cases. The additional load cases resulting from the combination may be utilized in stress comparisons or restraint load summaries. ME101 also provides a customized report writer that is designed to create report or table use in equipment nozzle evaluation, piping support/hanger guidance, valve acceleration summary, penetration load summary etc. This report is construed directly from data stored in the internal ME101 Data Base (“MASTER” file). This report enables user to manually input the allowable side by side with the extracted value, or allows the user to uniquely define the interaction equations. The report writer module can also screen or filter results according to allowable criteria.

ME101 can generate isometric plots of the piping configuration with optional node numbering. It can also generate plots for deflections and mode shapes.

The program is validated utilizing fifty-nine validation problems, including the eleven NUREG/CR 1677 benchmark problems. Validation is documented in the ME101 program validation manual.

3B.31 FAPPS (ME150), FRAME ANALYSIS PROGRAM FOR PIPE SUPPORTS

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

FAPPS is an interactive program specifically developed for the analysis of pipe support frames, associated welds, baseplates, embedments and local effects such as punching shear, web crippling and local flange bending. It performs static linear analyses of elastic structures. The program performs code checks for AISC, ASME III subsection NF and AIJ codes for “normal,” “upset,” “emergency” and “faulted” load conditions. The program addresses recent NRC and industry issues. It provides margin factors for all analyzed components.

Validation is documented in the program validation manual.

3B.32 BASEPLATE (ME035), NONLINEAR BASE PLATE ANALYSIS PROGRAM

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

ME035 is a finite-element program for the evaluation of baseplates. It has the capability of analyzing flexible baseplates on a geometrically nonlinear foundation. It has a library of standard attachments and accepts nonstandard and multiple attachments. It accommodates welded and bolted conditions of the baseplate.

Validation is documented in the program validation manual.

3B.33 MAPPS (ME153), MISCELLANEOUS APPLICATION PROGRAM FOR PIPE SUPPORTS

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

MAPPS is an interactive program that performs various pipe support related analyses. A description of the several analytical modules of MAPPS is provided below:

a. UNIFORM WELD ANALYSIS:

This module presents a method of determining weld size for the connecting structural members based on an approach described in Design of Welded Structures and Design of Weldments, both by O.W. Blodgett. It accepts five different types of shapes and analyzes from two to sixteen weld configurations. The program computes weld properties, minimum weld requirements, and adjusts weld effective throats to account for skewed angles. It also computes weld stresses and margin factors.

b. NONUNIFORM WELD ANALYSIS:

This module provides analysis of any two dimensional, nonuniform weld pattern with variable effective throat. It permits the user to model the weld as a series of joints and elements for welds without a constant effective throat over the entire weld configuration. Actual “as-built,” skewed T-joint and pipe-to-pipe connections are examples where effective throat over the entire weld length is not always achieved. The program gives the user the choice of either algebraic or absolute combinations of stress components and computes the combined stresses at every joint of the weld.

c. CLIP ANGLE ANALYSIS:

This module analyses stresses in clip angles used in pin-ended connections. The program also analyzes welds between clip angles and framed members and clip angles and supporting members. The program analyzes clip angles for three forces and for axial moment. It also permits variations of the orientation angle between the framed member and supporting structure.

d. BETA ANGLE CHECK:

The program determines “beta” angles used in programs like STRUDL and FAPPS. Beta angles are measured in a plane normal to the longitudinal axis of a structural member and serve to define the orientation of the member’s cross-section.

e. LOCAL EFFECT EVALUATION:

The program analyzes W-beam connections for local effects due to local flange bending and web crippling. It also performs a punching shear and web crippling analysis for tube and pipe members.

Validation is documented in the program validation manual.

3B.34 SIGNIT (ME149), SIGNATURE READY ENGINEERED CALCULATION SYSTEM FOR PIPE SUPPORTS

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

SIGNIT provides a system for generating “signature-ready” calculations for pipe supports that integrates all necessary program outputs together, develops the necessary portions of the calculations such as objective, methodology, conclusions, summary of results, check list, assumptions, sources of data and references and organizes the calculation package.

Validation is documented in the program validation manual.

3B.35 CAPPS (ME148), COMANCHE PEAK APPLICATIONS PROGRAMS FOR PIPE SUPPORTS

CAPPS is an interactive program which serves as a special application program for analysis of various pipe support components to their specific requirements. The various modules that cover these special application are listed below. This program has been specifically developed to address unique requirements applicable to pipe support design at Comanche Peak. The program provides to the user instructions regarding key points to be considered in the design and key data to be used in the form of a technical advisor. It calculates allowable stresses for plates and welds for all load conditions and all pipe support class in various buildings. It also provides footprint loads in project specific form to be utilized for evaluation of building structures. The various modules which are included in this program are:

Friction clamp Anchor, Intermediate Plate, Closed End Tubing, Richmond Inserts, Nuts on tubing, Flare bevel weld, Pipe weld to pipe/elbow and Strut/Snubber Spring/U-bolt/U-strap.

The program has been verified using hand calculations, validation is documented in the program validation manual.

3B.36 LSAPS (ME214) LOCAL STRESS ANALYSIS FOR PIPING SYSTEMS

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

ME214 is an interactive computer program developed to perform local stress analysis due to integral welded attachments and non-integral welded attachments such as line bearing and point contact loads, restraint of local pressure and thermal expansion of the pipe. The program is capable of calculation local pipe stresses based on Welding Research Council Bulletin No. 107 and ASME Code Case N-318 and N-392. The program also calculates local pipe stresses due to non-integral welded attachments using analytical solutions as provided in Comanche Peak design criteria documents.

The program is developed in accordance with ASME Section III Subsection NC and ND. The loading conditions and allowables are in accordance with Subsection NC and ND piping stress

requirements. Code case allowables are used if the local pipe stresses are calculated based on the methodology provided in the code cases.

3B.37 WESTDYN AND ASSOCIATED PRE- AND POST-PROCESSORS

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

WESTDYN is used for the structural analysis of piping system in Unit 2 only. WESTDYN, (described in WCAP-8252), calculates displacement, forces, and stresses distributions in three-dimensional piping models subjected to static and dynamic loads.

The dynamic analysis incudes seismic or hydrodynamic response spectra and time-history dynamic analysis. The time-history dynamic analysis incudes options for nonwear support, support gaps, and unidirectional single-acting restraints.

The following standard post-processing programs are provided as a part of the WESTDYN system: SELCOMB, STRSCHK, WESFOR, VALVE and CONFOR. In addition the following time history pre- and post-processing programs are provided: MERGE, HYDFO, SUPFOR, FORCE and DEFLOUT.

The WESTDYN post-processors provide for the stress analysis of ASME Class 1, 2, 3, or ANSI B31.1 code. The Post-processors are also used for generating support load summary sheets, and equipment and component qualification input data.

This family of programs has been verified in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.38 PIPSAN

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

PIPSAN (PIPe Support ANalysis) is an interactive program utilizing finite beam elements to model, analyze, and evaluate three-dimensional linear elastic structures in Unit 2 only. The required input consists of model geometry; connectivity of members; cross-sectional and material properties; external restraints; and location, direction, and magnitude of applied loads. Loading conditions (i.e., normal, upset, emergency or faulted) are required for stress evaluation.

This program has been verified in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.39 WESPLAT

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

WESPHALT is used for the analysis of any rectangular plate attached by anchors to a rigid foundation in Unit 2 only. WESPLAT utilizes a non-linear plate bending finite element approach.

The baseplate is modeled using an assembly of plate elements at node points. The plate is divided into elements dependent upon the user-specified problem.

This program has been verified in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.40 LOCAL

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

Program LOCAL is a PC based computer program which evaluates pipe primary and secondary local stresses that are due to support attachments in Unit 2 only. It also computes fatigue local stress effects due to upset and thermal loading for Class 1 pipe support attachments. The types of attachments considered include U-bolts, anchors, super stiff clamps, straps, lugs, trunnions, pads, and structural beams in contact with the pipe. The analysis method is in accordance with the procedures outlined in

- WRC Bulletin #107
- ASME B&PV Code, Section III, 1977 Edition through Summer, 1979 Addenda.
- Code Case N-391, 11/28/83
- Code Case N-392, 11/28/83
- Code Case N-122, 1/21/82

Program verification has been performed in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.41 PCLUG

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

PCLUG is a PC based program used to evaluate secondary stresses due to integral structural attachments to straight piping in Unit 2 only. The evaluations are made as per the ASME Code Case N-318-2, for Class 2 or 3 piping.

Program verification has been performed in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.42 STRESS CALCULATION SPREADSHEETS

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

Lotus 123 release 2.2 with the IMPRESS add-in is used to perform several repetitive stress calculations in Unit 2 only. Use of a spreadsheet assures consistency of calculations and permits engineers to focus on the engineering process vs. hand calculator arithmetic. IMPRESS add-in prints the resulting calculations in a neat, legible form.

The following spreadsheets are in use:

1. Swing Angle - calculates swing angles, binding angles, component extension/retraction (as applicable) for springes, snubbers, and struts.

2. Tube Steel Local Stress - analyzes acceptability of local stress on tube steel connections. This is actually a family of four spreadsheets. One is simply a menu allowing the operator to select one of the three analysis spreadsheets. Each spreadsheet addresses one of the following weld configurations: two-sided weld normal to main member, two-sided weld parallel to main member, and three/four sided welds. TUBSTL01, 02, and 03.WK1.

3. Tube Steel Welds - analyzes forces on specific weld configurations for a tube steel-to-tube steel weld, tube steel-to-base plate weld, and finally, flexibility was included to allow analysis of tube steel-to-rear bracket welds.
WELDTS01.WK1.

Additional spreadsheets will be developed as the need arises based on the approach stated above and the QA requirements as stated below.

The spreadsheet quality control complies with the NATD Quality Assurance Program, WCAP-9565, which defines the requirements for the development, verification, configuration management, and error reporting of software related nuclear Basic Components.

3B.43 WECAN

Author: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania
Source: Westinghouse Electrical Corporation, Pittsburgh, Pennsylvania

WECAN is a general purpose finite element program with the capability to solve a wide variety of structural analysis problems in Unit 2 only. These problems can be one, two, or three-dimensional in nature. Capabilities include:

- Static elastic and inelastic analysis
- Steady state and transient heat conduction
- Steady state hydraulic analysis
- Full and reduced modal analysis
- Harmonic response analysis
- Dynamic transient analysis
- Linear buckling analysis

Bench mark problem solutions for the verification of WECAN are contained in WCAP-8929. WECAN verified is performed in accordance with the Westinghouse QA program as specified in WCAP-9565.

3B.44 ME215 (ME101FE), SAPCAS

Author: Bechtel Corporation, San Francisco, California
Source: Bechtel Corporation, San Francisco, California

ME215 (ME101FE) is a special purpose finite element computer program for computing membrane and membrane-plus-bending stress intensities at pipes, pads, attachments, and welds. The piping components can be a circular run pipe, rectangular tube, or rectangular solid lug. The program is capable of computing local stress on pipe components and supports such as stanchion, trunnion, lug, etc. This program is more refined and will provide a realistic solution for local pipe stress as compared to, conservative ME214. The program is capable of calculating local pipe stresses where the results may exceed the limitation of the welding research council (WRC) Bulletin No. 107, ASME code cases, and/or other design documents such as CPNPP 2EP-5.12 and 2EP-5.13. The program is developed based on well established SAP computer program.

3B.45 HSTA (NE820) HYDRAULIC SYSTEM TRANSIENT ANALYSIS

Author: Bechtel Software, Inc., San Francisco, California
Source: Bechtel Software, Inc., San Francisco, California

Description

The Hydraulic System Transient Analysis (HSTA) is a PC based computer code that analyzes flow transients in liquid piping systems. The program generates piping segment time history forcing functions, as water hammer events, to be input into piping stress dynamic analysis codes such as ME101 for overall piping and support stress evaluations. The HSTA program solves one-dimensional unsteady flow problems utilizing the mass and momentum conservation equations (along the pipe axis), to obtain pressure and velocity as functions of time and distance. The solution is based on the finite difference solution of the method of characteristics (MOC), where the differential equations of flow are linearized to analyze transients resulting from valve opening and closure, pump startup or seizure, rapid decompression, column separation, and filling of a voided line. The code can also be applied to steam or gaseous systems where no significant changes to fluid density occurs during the transients.

HSTA input includes the piping modeled as a network in the form of links connected to each other through junctions. Hydraulic devices are located at these junctions and treated as boundary conditions. Piping frictional losses are also considered and the code enables modeling changes in elevation. The output features has the capability of providing snapshots of head or velocities at all nodes of selected links viewed on computer monitor. Other features including echoing of input date, printouts of pressure and velocities, and plotting of time-history pipe segment forces in a format compatible for input to ME101 piping stress code. A Windows compatible post-processor program (PLOTHG2) is used for plotting the results that are computed by HSTA as provided in the Code's user manual. The HSTA Code Validation & Verification process, that included the control and execution of a Quality Software application used by Engineering for Piping Stress and Support Analysis, was performed in accordance with the requirements of Station Procedures. Once the HSTA program is executed, the output file "input.o" is generated. There are additional files geneated depending on the options selected in the input file. These output files are identified by various file extensions as follows:

File Extension	Type
Input .o	Normal Output File
.b	Bubble Size file
.m	ME101 - format force time history file

.t	Columnar - format force time history file
.h	Head plot file
.v	Velocity plot file
.f	Pipe-run force plot file
.a	Surge Vessel plot file
.r	Restart file
.s	Save file

Another executable file is used to run the plotting program by placing the file in the same directory where the input files are located.

Furthermore, the Validation Report by Bechtel Software addresses limitations and comparisons regarding the validation between the program output results and either measured or independently calculated data for valve closure, branching, pump failure, open surge vessel (with one-way check valve), air tank (for suppression of pressure surges), positive displacement pump, liquid (water) column separation in a horizontal line or a siphon system, and a line filling case.

The HSTA program is compatible with other industry recognized codes utilized in CPNPP piping analysis, namely, “WATHAM” and “HYDTRAN”. Nonetheless, for a given analytical model previously analyzed by another equivalent code and is to be executed under the HSTA Code, it is the responsibility oft he user to ensure acceptable comparisons are obtained and consistency in the input/output results is maintained and in accordance with the applicable design criteria. This is typical practice for the execution of a compatible code performing similar functions when the code of the analysis of record is no longer available. Also, to ensure the safety of the program and related data files, other areas that are also checked is the virus detection and protection software included in the booting process activated whenever the computer is turned on. Any error reporting and/or corrective action are addressed. Any interference between such software and the Bechtel computer applications would be detected during the validation procedures described above.

3B.46 PC-PREPS

Description

PC-PREPS is a fully qualified structural analysis computer code which uses linear elastic theory to analyze and qualify structural members, welds, local tube steel stresses, anchor bolts and surface mounted baseplates in accordance with AISC, AWS and/or ASME criteria. The program may also be used to compare vendor items with their allowable loads and movements. PC-PREPS has several code check options available to qualify structural members.

REFERENCES

1. Local Stress in Spherical and Cylindrical Shells Due to External Loading. Welding Research Council Bulletin, WRC-107, 1965.

2. Corum, J. M. and Greenstreet, W. L., Experimental Elastic Stress Analysis of Cylinder to Cylinder Shell Models and Comparison with Theoretical Predictions. First International Conference on Structural Mechanics in Reactor Technology (Berlin, Preprints Volume 3, Part G, 1971).

3. Local Stress in Spherical and Cylindrical Shells Due to External Loading. Welding Research Council Bulletin, WRC-107, 1965.

4. Jonsson, V. K., Matthews, L., and Spalding, D. B.; Numerical Solution Procedure for Calculating the Unsteady One-Dimensional Flow of Compressible Fluid. ASME Paper No. 73-FE-30.

5. Luk, C. H., Effects of the Steam Chest on Steam Hammer Analysis for Nuclear Piping Systems. ASME Paper No. 75-PVP-61.

6. Moody, F. J., Time-Dependent Pipe Forces Caused By Blowdown and Flow Stoppage. ASME Paper No. 73-FE-23.

7. Progelhof, R. C. and Owczarek, J. A., The Rapid Discharge of a Gas from a Cylindrical Vessel Through a Nozzle. AIAA Journal, Volume 1, No. 9, September 1963, pp. 2182 to 2184.

8. Progelhof, R. C. and Owczarek, J. A., The Rapid Discharge of a Gas from a Cylindrical Vessel Through an Orifice. ASME Paper No. 63-WA-10.

9. Streeter, V. L. and Wylie, E. G., Hydraulic Transients, McGraw-Hill Book Company, New York, NY, 1967.

10. Fabic, S. Computer Program WHAM for Calculation of Pressure, Velocity, and Force Transients in Liquid Filled Piping Networks. Report No. 67-49-R, Kaiser Engineers, November 1967.

11. ICES STRUDL - II Structural Design Language Engineering User's Manual, Volume I Frame Analysis, November 1968, Volume II Addition Design and Analysis Facilities (Chapters III and IV), June 1971.

12. Zienkiewicz, O. C. and Cheung, Y. K. The Finite Element Method. McGraw-Hill Book Company, Inc., New York, NY, 1967.

13. GT-STRUDL - GT-ICES Systems Laboratory, School of Civil Engineering, Georgia Institute of Technology, Atlanta, GA.

14. RELAP5/MOD1/Code Manual, Volumes 1, 2, and 3. Prepared by Energy Technology Center of Control Data Corporation, Minneapolis, MN, based on NUREG/CR-1826 EGG-2070, 1984.

15. ICES STRUDL - II Structural Design Language Engineering User's Manual, Volume I Frame Analysis, November 1968, Volume II Addition Design and Analysis Facilities (Chapters III and IV), June 1971.

16. Zienkiewicz, O. C. and Cheung, Y. K. The Finite Element Method. McGraw-Hill Book Company, Inc., New York, NY, 1967.

17. GT-STRUDL - GT-ICES Systems Laboratory, School of Civil Engineering, Georgia Institute of Technology, Atlanta, GA.

18. “Documentation of Selected Westinghouse Structural Analysis Computer Codes”, WCAP-8252, Revision 1, July 1977.

19. “Bench Mark Problem Solutions Employed for Verification of WECAN Computer Program”, WCAP-8929, June 1977.

20. NATD Quality Assurance Program, WCAP-9565.

TABLE 3B.1-1
COMPARISON OF NATURAL FREQUENCIES FOR FIGURE 3B.1-1

	Frequency (Hz)	
Mode	NUPIPE-SW	NRC
1	28.510	28.53
2	55.698	55.77
3	81.411	81.50
4	141.618	141.7
5	162.633	162.8

TABLE 3B.1-2
COMPARISON OF INTERNAL MEMBER LOADS AND NODAL DEFLECTIONS COMBINED RESULTS FOR MODES 1-5 FOR FIGURE 1.0

	Member	Forces			Moments			Deflections		
Source	Ends	Fx	Fy	Fz	Mx	My	Mz	Dx	Dy	Dz
NUPIPE-SW	1	5.	18.	36.	53.	269.	116.	.000	.000	.000
	2	5.	18.	36.	53.	106.	39.	.002	.000	.005
NRC	1	5.	18.	36.	52.	269.	116.	.00000	.00000	.00000
	2	5.	18.	36.	52.	105.	39.	.00200	.00000	.00480
NUPIPE-SW	3	5.	7.	8.	53.	34.	29.	.006	.000	.015
	4	6.	7.	8.	12.	69.	24.	.007	.001	.017
NRC	3	5.	7.	8.	52.	34.	29.	.00580	.00000	.0146
	4	6.	7.	8.	12.	69.	24.	.00740	.00060	.017400
NUPIPE-SW	4	9.	9.	11.	12.	69.	24.	.007	.001	.017
	5	12.	4.	11.	35.	45.	22.	.008	.002	.014
NRC	4	9.	9.	11.	12.	69.	24.	.00740	.00060	.01740
	5	12.	4.	11.	35.	45.	22.	.00780	.00160	.01420
NUPIPE-SW	7	28.	24.	4.	35.	21.	41.	.008	.003	.007
	8	25.	27.	4.	35.	18.	73.	.006	.002	.002
NRC	7	28.	24.	4.	35.	20.	41.	.00780	.00249	.00680
	8	25.	27.	4.	35.	18.	73.	.00580	.00199	.00168
NUPIPE-SW	8	26.	30.	6.	35.	18.	73.	.006	.002	.002
	9	24.	31.	6.	9.	45.	115.	.002	.001	.000
NRC	8	26.	30.	6.	36.	18.	73.	.00580	.00199	.00168
	9	24.	31.	6.	9.	45.	115.	.00220	.00074	.00000
NUPIPE-SW	10	24.	7.	35.	9.	156.	54.	.001	.000	.000
	11	24.	7.	35.	9.	207.	65.	.000	.000	.000
NRC	10	24.	7.	35.	9.	156.	54.	.00062	.00019	.00000
	11	24.	7.	35.	9.	206.	65.	.00000	.00000	.00000

TABLE 3B.1-3
COMPARISON OF NATURAL FREQUENCIES FOR FIGURE 3B.1-2

	Frequency (Hz)	
Mode	NUPIPE-SW	NRC
1	8.711	8.712
2	8.805	8.806
3	17.507	17.51
4	40.364	40.37
5	41.624	41.63

TABLE 3B.1-4
COMPARISON OF INTERNAL MEMBER LOADS AND NODAL DEFLECTIONS COMBINED RESULTS FOR MODES 1-5 FOR FIGURE 3B.1-2

	Member	Forces			Moments			Deflections		
Source	Ends	Fx	Fy	Fz	Mx	My	Mz	Dx	Dy	Dz
NUPIPE-SW	15	767.	109.	108.	0.	428.	436.	.000	.000	.000
	1	757.	109.	108.	0.	23.	20.	.230	.001	.225
NRC	15	766.	109.	108.	0.	428.	436.	.00000	.00000	.00000
	1	766.	109.	108.	0.	23.	20.	.23000	.00127	.22500
NUPIPE-SW	1	766.	78.	77.	0.	23.	20.	.230	.001	.225
	7	766.	78.	77.	0.	337.	337.	.462	.003	.447
NRC	1	766.	78.	77.	0.	23.	20.	.23000	.00127	.22500
	7	766.	78.	77.	0.	337.	337.	.46180	.00253	.44620
NUPIPE-SW	7	7.	469.	13.	0.	6.	337.	.462	.003	.447
	6	7.	469.	13.	0.	3.	0.	.462	.001	.447
NRC	7	7.	469.	13.	0.	6.	337.	.46180	.00253	.44620
	6	7.	469.	13.	0.	3.	0.	.46180	.00097	.44620
NUPIPE-SW	5	19.	297.	13.	0.	6.	337.	.462	.003	.447
	14	19.	297.	13.	0.	4.	134.	.462	.003	.447
NRC	5	19.	297.	13.	0.	6.	337.	.46180	.00252	.44620
	14	19.	297.	13.	0.	4.	124.	.46180	.00332	.44630
NUPIPE-SW	14	7.	297.	1.	0.	4.	124.	.462	.003	.447
	13	7.	297.	1.	0.	4.	124.	.462	.003	.447
NRC	14	7.	297.	1.	0.	4.	124.	.46180	.00332	.44630
	13	7.	297.	1.	0.	4.	124.	.46180	.00332	.44630

TABLE 3B.1-5
COMPARISON OF SUPPORT LOADS BETWEEN NUPIPE II AND NUPIPE-SW
FOR FIGURE 3B.1-3

Program	Node	FX	FY	FZ	MX	MY	MZ
NUPIPE II	5520	556.	437.	961.	1102.	2202.	530.
NUPIPE SW		560.	436.	966.	1097.	2189.	533.
NUPIPE II	158		1361.				
NUPIPE SW			1344.				
NUPIPE II	159		1380.				
NUPIPE SW			1384.				
NUPIPE II	178	2019.	1060.	966.	7853.	21812.	3588.
NUPIPE SW		2020.	1059.	965.	7838.	21794.	3581.

TABLE 3B.1-6
COMPARISON OF SUPPORT LOAD FOR FIGURE 3B.1-4

	Mode	Node Pt.	FX(lb)	MX(ft-lb)
NUPIPE-SW with missing mass cutoff mode = 2.0	1,2, PSEUDO MODE	5	2072	42114
NUPIPE-SW with no missing mass cutoff mode = 3.0	1-3 SRSS	5	2072	42114

TABLE 3B.1-7
COMPARISON OF SUPPORT REACTIONS DUE TO THERMAL EXPANSION
FOR FIGURE 3B.1-5

Node	Program	Force (lb)			Moments (ft-lb)		
		FX	FY	FZ	MX	MY	MZ
1	NUPIPE-SW	-3275	-2934	-1214	-9307	2858	22302
1	ADLPIPE	-3274	-2934	-1214	-9307	2858	22302
11	NUPIPE-SW	3275	2934	1214	15281	-14855	-9415
11	ADLPIPE	3274	2934	1214	15281	-14855	-9415

TABLE 3B.1-8
COMPARISON OF DEFLECTIONS DUE TO THERMAL EXPANSION FOR
FIGURE 3B.1-5

Node	Program	Deflection (in.)		
		DX	DY	DZ
2	NUPIPE-SW	-0.074	0.135	-0.032
2	ADLPIPE	-0.074	0.135	-0.032
5	NUPIPE-SW	-0.132	0.259	-0.093
5	ADLPIPE	-0.132	0.259	-0.093
7	NUPIPE-SW	0.014	0.084	-0.157
7	ALDPIPE	0.014	0.084	-0.157

TABLE 3B.1-9
COMPARISON OF STRESS DUE TO THERMAL EXPANSION FOR
FIGURE 3B.1-5

Node	NUPIPE-SW (psi)	ADLPIPE (psi)
1	6411	6410
2	1158	1158
3	6622	6619
5	6938	6936
7	4908	4906
9	2725	2724
11	6138	6137

TABLE 3B.1-10
COMPARISON OF INTERNAL FORCES AND MOMENTS DUE TO THERMAL EXPANSION FOR FIGURE 3B.1-5

Node	Program	Force (lb)			Moments (ft-lb)		
		FX	FY	FZ	MX	MY	MZ
2	NUPIPE-SW	-3275	-2934	-1214	-2024	2858	2655
2	ADLPIPE	-3274	-2934	-1214	-2024	2858	2655
5	NUPIPE-SW	-3275	-2934	-1214	5258	1340	-13324
5	ADLPIPE	-3274	-2934	-1214	5258	1340	-13324
7	NUPIPE-SW	-3275	-2934	-1214	5258	-6549	5748
7	ALDPIPE	-3274	-2934	-1214	5258	-6549	5748

TABLE 3B.3-1
COMPARISON OF BSPLT COMPUTER PROGRAM RESULTS WITH HAND CALCULATION RESULTS

Input Data

Loads

F_X = 4201 lb

F_Y = 3785 lb

F_Z = 0

M_X = 0

M_Y = 0

M_Z = 7857 in.-lb

Output Data	Hand Calculation	BSPLT
Bolt Tension	2188 lb	2175 lb
Bolt Shear	946 lb	946 lb
Tension-Shear		
Interaction Factor	0.66	0.65
Plate Bending Stress	8637 psi	8534 psi

TABLE 3B.5-1
COMPARISON OF PITRUST WITH FRANKLIN INSTITUTE PROGRAM, CYLNOZ, AND HAND CALCULATION

Source of Stress	Stress (psi)		
	Franklin Institute Corrected Values	Output From PITRUST	Hand Calculation
Circumferential			
P (normal)	395	399	399.99
P (bending)	1,875	1,833	1,877.3
M_c (normal)	35.85	35.57	36.06
M_c (bending)	364.7	366.6	354.3
M_L (normal)	79.05	79.66	79.54
M_L (bending)	90.52	80.57	79.42
Axial			
P (normal)	813	812	814.8
P (bending)	812.3	827	810.6
M_c (normal)	91.79	105	95.45
M_c (bending)	158.8	158.8	160
M_L (normal)	37.06	37	37.12
M_L (bending)	117.9	105	103.85
Shear Stress by M_T	6.63	6.63	6.63
Shear Stress by V_c	106.1	106.1	106.1
Shear Stress by V_L	106.1	106.1	106.1

TABLE 3B.5-2
COMPARISON OF PITRUST WITH REFERENCE 2 RESULTS

Location and Cause	PITRUST Results (psi)	Experimental Results (psi)
Element A Longitudinal Moment, M_L		
Circumferential Stress	20,438.9	20,000
Axial Stress	26,292.6	25,000
Element B Longitudinal Moment, M_C		
Circumferential Stress	22,016.2	24,000
Axial Stress	13,105.8	13,000

TABLE 3B.6-1
COMPARISON OF PILUG COMPUTER PROGRAM OUTPUT WITH HAND CALCULATIONS

(Sheet 1 of 2)

Test Problem: Run Pipe Outside Diameter = 17 in.

Run Pipe Thickness = 0.812 in.

Axial Length of LUG = 12 in.

Width of LUG along Circumference = 3 in.

Loads: P = 3300 lb

V_c = 1788 lb

V_L = 2478 lb

M_c = 81834 in.-lb

M_L = 103320 in.-lb

M_T = 76284 in.-lb

Stress in Circumferential Direction (psi)

		Stress from		
Figure[a]	B	Hand Calculation	Computer Output	Remarks
3C	0.5485	387	330	Membrane stress due to P
1C	0.326	2,165	2,160	Bending stress due to P
3A	0.294	671	629	Membrane stress due to M_c
1A	0.388	18,976	19,904	Bending stress due to M_c
3B	0.467	3,014	2,961	Membrane stress due to M_L
1B	0.416	6,143	5,969	Bending stress due to M_L

TABLE 3B.6-1
COMPARISON OF PILUG COMPUTER PROGRAM OUTPUT WITH HAND CALCULATIONS

(Sheet 2 of 2)

		Stress from		
Figure[a]	B	Hand Calculation	Computer Output	Remarks
Stress in Axial Direction (psi)				
4C	0.4447	683	690	Membrane stress due to P
2C	0.4632	773	792	Bending stress due to P
4A	0.294	1,897	1,864	Membrane stress due to M_c
2A	0.550	6,357	5,942	Bending stress due to M_c
4B	0.467	2,365	2,328	Membrane stress due to M_L
2B	0.582	4,989.7	4,842	Bending stress due to M_L
Shear Stress (psi)				
---	---	1,304.8	1,304.8	Shear stress due to M_T
---	---	-366.99	-366.99	Shear stress due to V_L
---	---	127.15	127.16	Shear stress due to V_c

a) Local Stress in Spherical and Cylindrical Shells due to External Loading. Welding Research Council Bulletin, WRC-107, 1965

TABLE 3B.7-1
COMPARISON OF PITRIFE COMPUTER PROGRAM OUTPUT WITH STUDL-II OUTPUT

Test Problem	Size-On-Size	0.707 Size-on-Size
Average Pipe Radius (in.)	3.00	3.00
Average Trunnion Radius (in.)	3.00	2.12
Pipe Wall Thickness (in.)	0.30	0.30
Trunnion Wall Thickness (in.)	0.30	0.21

SIZE-ON-SIZE
MAXIMUM STRESS INTENSITY, psi (α = 30°)

Load	PITRIFE Output	STRUDL-II Output
FX = 10,000 lb	5,763	5,768
FY = 10,000 lb	7,844	7,846
FZ = 10,000 lb	6,507	6,506
MX = 10,000 in.-lb	1,329	1,329
MY = 10,000 in.-lb	1,688	1,687
MY = 10,000 in.-lb	4,068	4,066

0.707 SIZE-ON-SIZE
MAXIMUM STRESS INTENSITY, psi (α = 30°)

Load	PITRIFE Output	STRUDL-II Output
FX = 10,000 lb	13,471	13,458
FY = 10,000 lb	9,616	9,611
FZ = 10,000 lb	20,105	20,030
MX = 10,000 in.-lb	4,371	4,368
MY = 10,000 in.-lb	2,467	2,467
MZ = 10,000 in.-lb	6,178	6,176

TABLE 3B.7-2
COMPARISON OF PITRIFE COMPUTER PROGRAM OUTPUT WITH HAND CALCULATIONS

Test Problem

Average Pipe Radius	= 1.5 in.
Average Trunnion Radius	= 1.35 in.
Trunnion Wall Thickness	= 0.27 in.
Pipe Wall Thickness	= 0.30 in.

LOADS FOR EACH LOAD TYPE COMBINED (DL, OBEI, THER, OCCU, ETC)

FX = FY = FZ = 10,000 lb

MX = MY = MZ = 10,000 in.-lb

MNS = 200 psi

Internal Pressure = 100 psi

STRESS COEFFICIENTS - 0.9 SIZE-ON-SIZE - FX LOADING (α = 30°C)

Stress Type	Coefficient by Hand Calculation	Coefficient From PITRIFE
Longitudinal - Inside Fiber	-1.2652	-1.2652
Circumferential - Inside Fiber	-0.2764	-0.2764
Shear - Inside Fiber	0.2041	0.2041
Longitudinal - Outside Fiber	0.7454	0.7454
Circumferential - Outside Fiber	1.3509	1.3509
Shear - Outside Fiber	0.2041	0.2041

MAXIMUM STRESS INTENSITY - 0.9 SIZE-ON-SIZE (α = 30°C)

	Maximum Stress Intensity, psi	
Load Condition	Hand Calculation	PITRIFE
P + DL + MNS_1	28,181	28,182
P + DL + SRSS (OBEI, OCCU) + MNS_2	73,220	73,220
P + DL + OBEA + THER + MNS_3	88,216	88,216
P + DL + OCCE + MNS_4	59,853	59,853
P + DL + SRSS (SSEI, OCCF) + MNS_5	73,220	73,220

TABLE 3B.8-1
NODAL FORCE COMPARISON

Diameter D = 0.25 ft

Area A = $\pi D^2/_4$ = 0.0490874 ft^2

p = pressure lb/ft^2

= density lb/ft^3

V = velocity ft/sec

g = gravitational constant 32.2 ft/sec^2

p_a = ambient pressure (14.7 x 144 lb/ft^2)

at time t = 0.00650 sec

Node No.	Pressure (psia)	Velocity (fps)	Density (lb/ft^3)	Force	
				STEHAM (lb)	Hand Calculation (lb)
1	42.523	0.0	0.23954	186.57	196.67
5	42.785	5.7843	0.24076	198.43	198.53
10	44.231	31.219	0.24647	209.00	209.11
15	47.003	78.172	0.25737	230.62	230.73
20	50.214	129.89	0.26979	257.84	257.97
25	52.095	159.43	0.27697	274.93	275.06
30	52.209	161.97	0.27742	276.09	276.23
35	52.168	162.21	0.27731	275.83	275.97

TABLE 3B.9-1
INPUT DATA FOR WATHAM

Pipe No.	Total Length (ft)	Inside Diameter (ft)	Friction Factor	No. of Nodes	Nodal Span (ft)	Thickness (in.)	Velocity (fps)
1	2,000	3.0	0.03	7	333.33	0.30824	4.24413
2	3,000	2.5	0.28	9	375	0.44	2.92132
3	2,000	2.0	0.024	6	400	0.50026	4.98473
4	1,800	1.5	0.02	7	300	0.11108	3.59336
5	1,500	1.5	0.022[(a)]	5	375	0.264	4.52142
6	1,600	1.5	0.025	6	320	0.13796	2.29183
7	2,200	2.5	0.04	8	314.29	0.21534	3.65878
8	1,500	2.0	0.03	6	300	0.14811	3.83245
9	2,000	3.0	0.024	7	333.33	0.30824	4.24413

a) Friction factor in Pipe 5, 0.022, differs slightly from that of hand calculation, 0.020.

Note:

The initial heads of all nodes are calculated by using the Darcy-Weisbach equation.

TABLE 3B.9-2
COMPARISON OF NODAL FORCE CALCULATION AT TIME = 2.34 sec

Pipe No.	Node No.	Force (kip)	
		WATHAM	Hand Calculation
1	1	276.34	276.48
1	2	300.46	300.62
1	3	317.78	317.94
1	4	329.59	329.76
1	5	341.39	341.56
1	6	355.31	355.49
1	7	369.52	369.71

Nodal force calculation is based on the following equation:

$$F = A\left(\rho H + \frac{\rho V^2}{g}\right)$$

where:

F = nodal force, lb

ρ = density, lb/cu ft

H = nodal head, ft

g = 32.2 ft/sec^2

V = nodal velocity, fps

A = pipe area, sq ft

TABLE 3B.10-1
INPUT DATA FOR WATSLUG

Cutoff Mode	Cutoff Frequency	Time Step	Integration Time	Damping Ratio
53	433 Hz	0.0009 sec	0.5 sec	10 percent

Pipe Section	Total Length (ft)	Outside Diameter (in.)	Thickness (in.)	Weight (lb/ft)
1	4.73	8.625	0.906	74.71
2	12.31	6.625	0.864	53.16
3	12.43	6.625	0.28	18.97
4	69.0	12.75	0.688	88.60
5	1.1	12.75	1.5	-----
6	1.0	8.625	0.322	28.55
7	0.83	6.625	0.432	28.57

$E_{hot} = E_{cold}$ = Young's Modulus of pipe = 28.3×10^6 psi

TABLE 3B.10-2
INPUT DATA FOR WATSLUG

Pipe No.	Total Length (ft)	Inside Diameter (ft)	Friction Factor
1	16.125	0.408	0.015
2	12.563	0.5054	0.015
3	63.562	0.948	0.013

Valve Characteristrics

Orifice Area (ft^2)	Opening Time (sec)	Discharge Coefficient	Flow Rate (lbm/sec)
0.0253	0.015	0.805	120.83

Upstream Steam Conditions

Pressure (psia)	Temperature	Density lbm/ft^3	Pressure Rise Rate psi/sec
2690	679°F (1139°R)	8.862	-40*

Downstream Gas Conditions

Pressure (psia)	Temperature	Density lbm/ft^3
15	80°F (540°R)	0.09975

Waterslug Weight = 69.8 lb

Notes:

* Pressure is decreasing after valve opens

TABLE 3B.11-1
ELBOW PROGRAM – ELBOW PROPERTIES USED FOR VERIFICATION PROBLEMS

Outside diameter	30.0 in.
Minimum wall thichness	0.5239 in.
Bend radius	44.214 in.
Pipe radius	14.735 in.
Young's modulus	28.3×10^{6} psi
Poisson's ratio	0.3
Coefficient of thermal expansion	9.11×10^{-6} in./in. °F

TABLE 3B.11-2
ELBOW PROGRAM – CASE 1 RESULTS

Internal Pressure Equals 413.58 psi

	Circumferential Stresses, psi		Longitudinal Stresses, psi		Stress Intensities, psi	
	Inside	Outside	Inside	Outside	Inside	Outside
ELBOW Program	11,676	11,676	5,714	5,714	12,090	11,676
Hand Calculation[(a)]	11,676	11,676	5,714	5,714	12,090	11,676

Note:

a) Hand calculation is based on Article NB-3685.1 of ASME Section III, 1974.

TABLE 3B.11-3
ELBOW PROGRAM – CASE 2 RESULTS

Liner Temperature Gradient Through Wall Equals 100ºF

	Circumferential Stresses, psi		Longitudinal Stresses, psi		Stress Intensities, psi	
	Inside	Outside	Inside	Outside	Inside	Outside
ELBOW Program	18,415	-18,415	18,415	-18,415	18,415	18,415
Hand Calculation[(a)]	18,415	-18,415	18,415	-18,415	18,415	18,415

Note:

a) Hand calculation of Timoshenko and Goodier 1970.

TABLE 3B.11-4
ELBOW PROGRAM – CASE 3 RESULTS

Combined loadings as follows:

1. Internal pressure equals 413.58 psi
2. Linear Temperature gradient equals 100°F
3. Axial Force equal to 60,000 lb
4. Torsional moment equal to 3,500,000

	Circumferential Stresses, psi		Longitudinal Stresses, psi		Stress Intensities, psi	
	Inside	Outside	Inside	Outside	Inside	Outside
ELBOW Program	30,091	-6,739	24,129	-12,701	32,165	14,362
Hand Calculation*	30,091	-6,739	24,129	-12,701	32,165	14,362

TABLE 3B.13-1
COMPARISON OF EIGENVALUES FROM THEORETICAL RESULTS, STRUDL RESULTS, AND GT-STRUDL RESULTS (PROBLEM NO. 1)

"I" Mode No.	Theoretical Results: Frequency cycles/sec	STRUDL Results: Frequency cycles/sec	GT-STRUDL Results: Frequency cycles/sec
1	24.8	24.84	24.84
2	99.2	99.33	99.33
3	223.2	223.37	223.38
4	396.8	396.39	396.40

Note:

For comparison purposes, the results of four modes have been tabulated.

TABLE 3B.13-2
COMPARISON OF EIGENVECTORS FROM STRUDL AND GT-STRUDL
(PROBLEM NO. 1)

(Sheet 1 of 2)

		Y-Displacement	
Mode	Joint	STRUDL	GT-STRUDL
1	1	0.0	0.0
	2	0.309	0.309
	3	0.588	0.588
	4	0.809	0.809
	5	0.951	0.951
	6	1.000	1.000
	7	0.951	0.951
	8	0.809	0.809
	9	0.588	0.588
	10	0.309	0.309
	11	0.0	0.0
2	1	0.000	0.000
	2	0.618	0.618
	3	1.000	1.000
	4	1.000	1.000
	5	0.618	0.618
	6	0.000	0.000
	7	-0.618	-0.618
	8	-1.000	-1.000
	9	-1.000	-1.000
	10	-0.618	-0.618
	11	0.000	0.000

TABLE 3B.13-2
COMPARISON OF EIGENVECTORS FROM STRUDL AND GT-STRUDL
(PROBLEM NO. 1)

(Sheet 2 of 2)

		Y-Displacement	
Mode	Joint	STRUDL	GT-STRUDL
3	1	0.0	0.0
	2	-0.809	-0.809
	3	-0.951	-0.951
	4	-0.309	-0.309
	5	0.588	0.588
	6	1.000	1.000
	7	0.588	0.588
	8	-0.309	-0.309
	9	-0.951	-0.951
	10	-0.809	-0.809
	11	0.0	0.0
4	1	0.0	0.0
	2	1.0	1.0
	3	0.618	0.618
	4	-0.618	-0.618
	5	-1.0	-1.0
	6	0.000	0.000
	7	-1.0	-1.0
	8	0.618	0.618
	9	-0.618	-0.618
	10	-1.0	-1.0
	11	0.0	0.0

Note:

For comparison purposes, the results of four modes have been tabulated.

TABLE 3B.13-3
THE MEMBER FORCES FROM STRUDL AND GT-STRUDL COMPUTER RUNS FOR DIFFERENT LOADING CONDITIONS
(PROBLEM NO. 2)

(Sheet 1 of 2)

			STRUDL			GT-STRUDL		
Member	Loading Condition	Joint	Axial	Shear Y	Mom Z	Axial	Shear Y	Mom Z
AD	1	A	1652.25	-237.33	-7958.75	1652.25	-237.33	-7958.75
		D	-1652.25	237.33	-20071.84	-1652.25	237.33	-20071.86
	2	A	-583.75	-675.01	-104030.94	-583.75	-675.01	-104031.4
		D	583.75	675.01	24305.86	583.75	675.01	24305.9
	3	A	-694.20	1480.59	107811.12	-694.20	1480.59	170811.6
		D	694.20	-1480.59	67060.81	694.20	-1480.59	67061.1
DE	1	D	1780.56	1652.25	20071.84	1780.56	1652.24	20071.8
		E	-1780.56	1654.68	-20215.77	-1780.56	1654.69	-20215.9
	2	D	675.01	-583.75	-24305.86	675.01	-583.75	-24305.9
		E	-675.01	583.75	-44640.70	-675.01	583.75	-44640.8
	3	D	-1480.59	-694.20	-67060.81	-1480.59	-694.20	-67061.1
		E	1480.59	694.20	-14930.68	1480.59	694.20	-14930.7

TABLE 3B.13-3
THE MEMBER FORCES FROM STRUDL AND GT-STRUDL COMPUTER RUNS FOR DIFFERENT LOADING CONDITIONS
(PROBLEM NO. 2)

(Sheet 2 of 2)

			STRUDL			GT-STRUDL		
Member	Loading Condition	Joint	Axial	Shear Y	Mom Z	Axial	Shear Y	Mom Z
BE	1	B	45.28	171.16	0.00	45.29	171.16	0.00
		E	-45.28	-171.16	20215.80	-45.29	-171.16	20216.93
	2	B	286.70	377.96	-0.01	286.70	377.96	0.00
		E	-286.70	-377.96	44640.70	-286.70	-377.96	44640.84
	3	B	2301.20	126.41	0.00	2301.20	126.41	0.00
		E	-2301.20	-126.41	-14930.65	-2301.20	-126.41	14930.73
EC	1	E	2276.03	0.0	0.0	2276.04	0.0	0.0
		C	-2276.03	0.0	0.0	-2276.04	0.0	0.0
	2	E	420.09	0.0	0.0	420.09	0.0	0.0
		C	-420.09	0.0	0.0	-420.09	0.0	0.0
	3	E	-2272.64	0.0	0.0	-2272.65	0.0	0.0
		C	2272.64	0.0	0.0	2272.65	0.0	0.0

TABLE 3B.13-4
THE JOINT LOADS (AT SUPPORTS) FROM STRUDL AND GT-STRUDL COMPUTER RUNS FOR DIFFERENT LOADING CONDITIONS (PROBLEM NO. 2)

		STRUDL			GT-STRUDL		
Joint	Loading	X Force	Y Force	Z Mom	X Force	Y Force	Z Mom
A	1	237.33	1652.25	-7958.75	237.33	1652.25	-7958.75
	2	675.01	-583.75	-104030.94	675.01	-583.75	-104031.48
	3	-1480.59	-694.20	107811.12	-1480.59	-694.20	107811.62
B	1	-171.16	45.28	0.00	-171.16	45.29	0.00
	2	-377.96	286.70	-0.01	-377.96	286.70	0.00
	3	-126.41	2301.20	0.00	-126.41	2301.20	0.00
C	1	-1609.40	1609.40	0.00	-1609.40	1609.40	0.00
	2	-297.05	297.05	0.00	-297.05	297.05	0.00
	3	1607.00	-1607.00	0.00	1607.00	-1607.00	0.00

TABLE 3B.13-5
COMPARISON OF ELEMENT (RANDOMLY SELECTED) STRESSES (PROBLEM NO. 3)

(Sheet 1 of 3)

		Mxx		Myy		Mxy		Vxx		Vyy	
Element	Node	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	1	-12.546	-12.543	-78.334	-78.354	19.545	19.547	13.417	13.412	3.617	3.618
	2	-4.098	-4.109	-18.611	-18.642	19.107	19.109	13.417	13.412	-8.769	-8.756
	10	-13.852	-13.831	-69.887	-69.860	46.293	46.298	-3.549	-3.538	-8.769	-8.756
	9	-8.936	-8.952	-56.769	-56.764	46.730	46.735	-3.549	-3.538	-3.617	3.618
6	6	-2.145	-2.146	-16.671	-16.678	13.396	13.400	-0.595	-0.597	-0.158	-0.158
	7	-2.876	-2.879	-19.390	-19.409	10.630	10.635	-0.595	-0.597	0.916	0.924
	15	-3.956	-3.940	-11.933	-11.916	9.555	9.558	0.694	0.701	0.916	0.924
	14	-6.974	-6.986	-12.939	-12.935	12.320	12.323	0.694	0.701	-0.158	-0.158
10	11	-10.414	-10.417	-4.335	-4.336	65.119	65.122	1.027	1.026	-13.922	-13.924
	12	-15.984	-15.983	7.190	7.180	13.283	13.285	1.027	1.026	-9.953	-9.948
	20	-74.387	-74.369	-3.683	-3.672	22.101	22.102	5.789	5.798	-9.953	-9.948
	19	-99.338	-99.307	-12.307	-12.310	73.937	73.939	5.789	5.798	-13.922	-13.924

TABLE 3B.13-5
COMPARISON OF ELEMENT (RANDOMLY SELECTED) STRESSES (PROBLEM NO. 3)

(Sheet 2 of 3)

		Mxx		Myy		Mxy		Vxx		Vyy	
Element	Node	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
16	18	3.765	3.747	26.738	26.721	94.743	94.475	-15.849	-15.844	0.332	0.335
	19	-56.181	-56.179	-5.239	-5.250	71.624	71.624	-15.849	-15.844	-9.083	-9.078
	27	-117.077	-117.062	-7.560	-7.547	49.705	49.704	-27.147	-27.139	-9.083	-9.078
	26	1.424	1.396	31.393	31.403	72.554	72.555	-27.147	-27.139	0.332	0.335
22	25	-2.301	-2.323	56.058	56.053	74.477	74.481	-3.966	-3.962	-1.428	-1.429
	26	0.446	0.451	33.162	33.148	71.634	71.637	-3.966	-3.962	-2.312	-2.305
	34	11.675	-11.652	29.188	29.207	49.249	49.251	-5.178	-5.163	-2.312	-2.305
	33	-1.561	-1.594	45.381	45.381	52.093	52.094	-5.178	-5.178	-1.428	-1.429
45	51	-149.275	-149.281	-2.222	-2.223	29.027	29.028	6.244	6.243	0.660	0.658
	52	-113.469	-113.472	-1.815	-1.825	26.628	26.629	6.244	6.243	-1.319	-1.314
	60	-124.368	-124.354	-0.097	-0.085	23.243	23.244	3.869	3.877	-1.319	-1.314
	59	-144.766	-144.787	-2.139	-2.142	25.642	25.643	3.869	3.877	0.660	0.658
52	59	-144.756	-144.773	-2.047	-2.055	24.031	24.030	3.874	3.879	2.303	2.305
	60	-124.346	-124.339	0.0146	0.0123	22.659	22.658	3.874	3.879	-2.621	-2.620

TABLE 3B.13-5
COMPARISON OF ELEMENT (RANDOMLY SELECTED) STRESSES (PROBLEM NO. 3)

(Sheet 3 of 3)

		Mxx		Myy		Mxy		Vxx		Vyy	
Element	Node	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
	68	-143.165	-143.158	0.591	0.595	24.111	24.111	-2.034	-2.030	-2.621	-2.620
	67	-129.535	-129.548	-1.240	-1.237	25.483	25.482	-2.034	-2.030	2.303	2.305
60	68	-130.197	-130.198	2.811	2.810	22.296	22.298	22.805	22.805	-4.389	-4.388
	69	-20.373	-20.368	25.257	25.246	11.012	11.013	22.805	22.805	5.429	5.433
	77	-19.106	-19.097	61.775	61.791	34.229	34.230	34.586	34.591	5.429	5.433
	76	-157.545	-157.547	-0.386	-0.386	45.512	45.512	34.586	34.591	-4.389	-4.388
68	77	-24.513	-24.517	54.034	54.028	35.968	35.968	14.908	14.911	0.413	0.414
	78	11.657	11.666	104.330	104.326	39.457	39.457	14.908	14.911	1.380	1.384
	86	21.811	21.826	103.782	103.797	82.856	82.858	16.069	16.074	1.380	1.384
	85	-24.572	-24.578	56.964	56.973	79.367	79.369	16.069	16.074	0.413	0.414
80	91	-36.410	-36.423	2.194	2.186	30.708	30.708	1.185	1.190	7.587	7.593
	92	35.756	-35.747	8.412	8.143	65.545	65.546	1.185	1.190	6.573	6.572
	100	-7.093	-7.089	8.341	8.342	69.355	69.356	0.424	0.424	6.573	6.572
	99	-4.321	-4.324	3.110	3.110	34.518	34.518	0.424	0.424	7.587	7.593

TABLE 3B.13-6
COMPARISON OF RESULTANT (RANDOMLY SELECTED) JOINT DISP. SUPPORTS (GLOBAL) (PROBLEM NO. 3)

	Z Displacement		X Rotation		Y Rotation	
Joint	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	-0.0447149	-0.0447119	0.0004818	0.0004817	0.0001927	0.0001925
5	-0.0484141	-0.0484130	0.0003599	0.0003598	0.0001907	0.0001908
15	-0.0483233	-0.0483236	0.0003430	0.0003429	0.0002060	0.0002060
20	-0.0427901	-0.0427890	0.0003791	0.0003790	0.0002582	0.0002580
30	-0.0435345	-0.0435547	0.0002832	0.0002832	0.0002928	0.0002920
40	-0.0442976	-0.0442990	0.0002317	0.0002317	0.0002164	0.0002163
70	-0.0354678	-0.354688	0.0002588	0.0002586	0.0004367	0.0004369
100	-0.0259728	-0.0259738	0.0001780	0.0001779	0.0008314	0.0008313
112	-0.0426962	-0.0426994	0.0001656	0.0001656	0.0008937	0.0008930

TABLE 3B.13-7
COMPARISON OF MEMBERS (RANDOMLY SELECTED) FORCES AND MOMENTS (PROBLEM NO.4)

		Axial		Shear Y		Shear Z		Torsional Moment		Bending Y Moment		Bending Z Moment	
Member	Joint	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	1	0.0	0.0	0.0	0.0	0.527	0.527	0.0	0.0	0.0	0.0	0.0	0.0.
	2	0.0	0.0	0.0	0.0	-0.371	-0.371	0.0	0.0	-21.567	-21.568	0.0	0.0
4	4	0.0	0.0	0.0	0.0	-0.713	-0.713	0.0	0.0	5.158	5.158	0.0	0.0
	5	0.0	0.0	0.0	0.0	0.869	0.869	0.0	0.0	32.818	32.818	0.0	0.0
10	10	0.0	0.0	0.0	0.0	0.226	0.226	0.0	0.0	5.627	5.627	0.0	0.0
	11	0.0	0.0	0.0	0.0	-0.070	-0.070	0.0	0.0	-12.738	-12.738	0.0	0.0
20	3	0.0	0.0	0.0	0.0	0.158	0.158	0.0	0.0	0.0	0.0	0.0	0.0
	23	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-5.670	-5.670	0.0	0.0
25	14	0.0	0.0	0.0	0.0	0.117	0.117	0.0	0.0	1.818	1.818	0.0	0.0
	17	0.0	0.0	0.0	0.0	-0.093	-0.093	0.0	0.0	-4.338	-4.338	0.0	0.0
30	36	0.0	0.0	0.0	0.0	-0.093	-0.093	0.0	0.0	4.338	4.338	0.0	0.0
	39	0.0	0.0	0.0	0.0	0.117	0.117	0.0	0.0	-1.818	-1.818	0.0	0.0
50	12	0.0	0.0	0.0	0.0	0.426	0.426	0.0	0.0	0.0	0.0	0.0	0.0
	54	0.0	0.0	0.0	0.0	0.000	0.000	0.0	0.0	-15.336	-15.336	0.0	0.0
60	47	0.0	0.0	0.0	0.0	-0.069	-0.069	0.0	0.0	12.621	12.621	0.0	0.0
	48	0.0	0.0	0.0	0.0	0.225	0.225	0.0	0.0	-5.569	-5.569	0.0	0.0

TABLE 3B.13-8
COMPARISON OF RESULTANT JOINT LOADS - SUPPORTS (PROBLEM NO. 4) (GLOBAL)

	X Force		Y Force		Z Force		X Mom		Y Mom		Z Mom	
Joint	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	0.0	0.0	0.0	0.0	0.953	0.95	30.0	0.0	0.0	0.0	0.0	0.0
5	0.0	0.0	0.0	0.0	1.885	1.885	0.0	0.0	0.0	0.0	0.0	0.0
9	0.0	0.0	0.0	0.0	1.536	1.536	0.0	0.0	0.0	0.0	0.0	0.0
12	0.0	0.0	0.0	0.0	0.769	0.769	0.0	0.0	0.0	0.0	0.0	0.0
41	0.0	0.0	0.0	0.0	0.953	0.953	0.0	0.0	0.0	0.0	0.0	0.0
45	0.0	0.0	0.0	0.0	1.885	1.885	0.0	0.0	0.0	0.0	0.0	0.0
49	0.0	0.0	0.0	0.0	1.536	1.536	0.0	0.0	0.0	0.0	0.0	0.0
52	0.0	0.0	0.0	0.0	0.769	0.769	0.0	0.0	0.0	0.0	0.0	0.0

TABLE 3B.13-9
COMPARISON OF EIGENVALUES, FREQUENCIES, AND PERIODS (PROBLEM NO. 5)

(Sheet 1 of 2)

	Eigenvalue		Frequency (Cycles/Time Unit)		Period (Time Unit/Cycle)	
MODE	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	1.791123D 03	1.791171D+03	6.735703D 00	6.735791D+00	1.484626D-01	1.484607D-01
2	1.892565D 03	1.892596D+03	6.923816D 00	6.923873D+00	1.444290D-01	1.444278D-01
3	1.925683D 03	1.925715D+03	6.934134D 00	6.984191D+00	1.431817D-01	1.431805D-01
4	1.935117D 03	1.938148D+03	7.006645D 00	7.006702D+00	1.427217D-01	1.427205D-01
5	1.947453D 03	1.947485D+03	7.023501D 00	7.023558D+00	1.423791D-01	1.423780D-01
6	1.949031D 03	1.949063D+03	7.026403D 00	7.026403D+00	1.423215D-01	1.423203D-01
7	1.949454D 03	1.949486D+03	7.027108D 00	7.027166D+00	1.423060D-01	1.423049D-01
8	1.949909D 03	1.949940D+03	7.027927D 00	7.027984D+00	1.422895D-01	1.422883D-01
9	2.894955D 03	2.895024D+03	8.563297D 00	8.563400D+00	1.167775D-01	1.167760D-01
10	2.935117D 03	2.935188D+03	8.622493D 00	8.622597D+00	1.159757D-01	1.159743D-01
11	2.935117D 03	2.935188D+03	8.622493S 00	8.622597D+00	1.159757D-01	1.159743D-01
12	3.566743D 03	3.566836D+03	9.505086D 00	9.505210D+00	1.052068D-01	1.052055D-01
13	3.941438D 03	3.941542D+03	9.991886D 00	9.992018D+00	1.000812D-01	1.000799D-01
14	8.732367D 03	8.732600D+03	1.487257D 01	1.487277D+00	6.723786D-02	6.723696D-92
15	1.345767D 04	1.345793D+04	1.846330D 01	1.846330D+01	5.416204D-02	5.416151D-02

TABLE 3B.13-9
COMPARISON OF EIGENVALUES, FREQUENCIES, AND PERIODS (PROBLEM NO. 5)

(Sheet 2 of 2)

	Eigenvalue		Frequency (Cycles/Time Unit)		Period (Time Unit/Cycle)	
MODE	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
16	1.346419D 04	1.346445D+04	1.846759D 01	1.846777D+01	5.414892D-02	5.414839D-02
17	1.346597D 04	1.346623D+04	1.846881D 01	1.846899D+01	5.414534D-02	5.414481D-02
18	2.573237D 04	2.573314D+04	2.553054D 01	2.553092D+01	3.916877D-02	3.916819D-02
19	2.650393D 04	2.650463D+04	2.591047D 01	2.591081D+01	3.859444D-02	3.859393D-02
20	2.745771D 04	2.745854D+04	2.637256D 01	2.637296D+01	3.791820D-02	3.791763D-02
21	3.177105D 04	3.177167D+04	2.836847D 01	2.836875D+01	3.525040D-02	3.525006D-02
22	3.248699D 04	3.248762D+04	2.868632D 01	2.868660D+01	3.485952D-02	3.485948D-02
23	3.525845D 04	3.525916D+04	2.988489D 01	2.988520D+01	3.346172D-02	3.346138D-02
24	3.953962D 04	3.954066D+04	3.164728D 01	3.164769D+01	3.159829D-02	3.159788D-02
25	5.170876D 04	5.171001D+04	3.619113D 01	3.619157D+01	2.763108D-02	2.763074D-02
26	5.592949D 04	5.593058D+04	3.763922D 01	3.763958D+01	2.656804D-02	2.656778D-02
27	5.592949D 04	5.593058D+04	3.763922D 01	3.763958D+01	2.656804D-02	2.656778D-02
28	5.959983D 04	5.960165D+04	3.885462D 01	3.885521D+01	2.573697D-02	2.573657D-02
29	6.538255D 04	6.538428D+04	4.069594D 01	4.069648D+01	2.457248D-02	2.457215D-02
30	8.206529D 04	8.206758D+04	4.559318D 01	4.559382D+01	2.193311D-02	2.193280D-02

TABLE 3B.13-10
COMPARISON OF EIGENVECTORS FOR FEW RANDOMLY SELECTED MODES AND JOINTS (PROBLEM NO. 5)

Eigenvectors (Global)

		X-Displacement		Z-Displacement	
Mode	Joint	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
1	1	0.0	0.0	0.0	0.0
	2	0.721	-0.721	0.000	0.000
	13	0.721	-0.721	0.000	0.000
	23	1.00	-1.00	0.000	0.000
	43	0.999	-0.999	0.000	0.000
6	1	0.0	0.0	0.0	0.0
	2	0.000	0.000	-0.0009	-0.0009
	13	0.000	0.000	-0.0014	-0.0014
	23	0.000	0.000	-0.567	-0.567
	43	0.000	0.000	-0.0005	-0.0005
3	1	0.0	0.0	0.0	0.0
	2	0.000	0.000	0.0095	0.0095
	13	0.000	0.000	0.0153	0.0153
	23	0.000	0.000	1.000	1.000
	43	0.000	0.000	0.013	0.013

Y-displacements are approximately 0.000 for all three modes.

TABLE 3B.13-11
COMPARISON OF JOINT DISPLACEMENTS AT THE FREE JOINTS FOR RANDOMLY SELECTED JOINTS (GLOBAL)
(PROBLEM NO. 5)

Joint	Response Type	X Displacement		Y Displacement		Z Displacement	
		STRUDL	GT-STRUDL	STRUDL	GT-STRUDL	STRUDL	GT-STRUDL
7	RMS	0.000	0.000	0.000	0.000	0.00099	0.00099
	ABS SUM	0.000	0.000	0.000	0.000	0.00213	0.00212
	CMS	0.000	0.000	0.000	0.000	0.00117	0.00116
15	RMS	0.000	0.000	0.000	0.000	0.0105	0.0104
	ABS SUM	0.000	0.000	0.000	0.000	0.0105	0.0104
	CSM	0.000	0.000	0.000	0.000	0.0105	0.0104
22	RMS	0.000	0.000	0.000	0.000	0.0424	0.0424
	ABS SUM	0.000	0.000	0.000	0.000	0.0449	0.0449
	CSM	0.000	0.000	0.000	0.000	0.0424	0.0424
36	RMS	0.000	0.000	0.000	0.000	0.0287	0.0288
	ABS SUM	0.000	0.000	0.000	0.000	0.0287	0.02920
	CSM	0.000	0.000	0.000	0.000	0.0287	0.02920

RMS = Root Mean Square

ABS SUM = Absolute Sum

CSM = Closely Spaced Mode

TABLE 3B.14-1
COMPARISON OF SANDUL LOAD COMBINATIONS WITH HAND COMPUTATION

Load Condition 3

Load	Hand	SANDUL
FX	1424	1424
	-2102	-2102
FY	0	0
	-834	-834
FZ	978	978
	-652	-652
MX	2758	2758
	0	0
MY	3830	3830
	-5542	-5542
MZ	216	216
	-114	-114

TABLE 3B.14-2
COMPARISON OF MEMBER FORCES BETWEEN SANDUL AND STRUDL-SW

Load Condition 3
Loading Combination 49
Member 4, Joint 5

Member Force	STRUDL-SW	SANDUL
Axial (lb)	600	600
Shear Y-direction (lb)	298	298
Shear Z-direction (lb)	535	535
Torsion (in.-lb)	-3707	-3707
Bending about Y axis (in.-lb)	5872	5872
Bending about Z axis (in.-lb)	-2025	-2025

TABLE 3B.14-3
COMPARISON OF STRESSES AND WELD SIZE BETWEEN SANDUL AND HAND COMPUTATION

Same case as Table 3B.14-2

Item	Hand	SANDUL
Axial Stress (psi)	34	34
Normal Stress (psi)	483	483
Allowable Normal Stress (psi)	27,720	27,620
Shear Stress (psi)	147	147
Allowable Shear Stress (psi)	22,260	22,399
Fillet Weld Size (in.)	0.013	0.013

TABLE 3B.17-1
APE PROGRAM VERIFICATION PROBLEM COMPARISON OF DRILLED-IN ANCHOR ROAD

(Sheet 1 of 3)

			BIP (ST 361)		APE (ST 378, V01L00)	
Attachment	Attachment Location	Loading	Anchor Shear	Anchor Tension	Anchor Shear	Anchor Tension
TS 4 x 4	Corner of Surface-Mounted Plate	F_z = 3,000 lb	0	2,269.36	0	2,296.87
TS 4 x 4	Corner of Surface-Mounted Plate	M_x = 12,000 in.-lb	0	1,256.61	0	2,625.00
TS 4 x 4	Corner of Surface-Mounted Plate	M_x = 8,000 in.-lb M_y = 8,000 in.-lb	0	1,204.13	0	3,500.00
L 4 x 4	Corner of Surface-Mounted Plate	F_z = 3,000 lb	0	2,116.39	0	2,268.52
L 4 x 4	Corner of Surface-Mounted Plate	M_x = 12,000 in.-lb	0	1,418.15	0	2,333.33
L 4 x 4	Corner of Surface-Mounted Plate	M_x = 8,000 in.-lb M_y = 8,000 in.-lb	0	1,506.74	0	3,111.11
TS 4 x 4	Center of Embedded Plate	F_z = 20,000 lb	0	12,742.60	0	12,752.10
TS 4 x 4	Center of Embedded Plate	M_x = 50,000 in.-lb	0	4,285.70	0	5,934.86

TABLE 3B.17-1
APE PROGRAM VERIFICATION PROBLEM COMPARISON OF DRILLED-IN ANCHOR ROAD

(Sheet 2 of 3)

			BIP (ST 361)		APE (ST 378, V01L00)	
Attachment	Attachment Location	Loading	Anchor Shear	Anchor Tension	Anchor Shear	Anchor Tension
TS 4 x 4	Center of Embedded Plate	M_x = 25,000 in.-lb M_y = 25,000 in.-lb	0	3,756.76	0	5,934.86
TS 4 x 4	Corner of Embedded Plate	F_z = 20,000 lb	0	17,591.00	0	17,607.22
TS 4 x 4	Corner of Embedded Plate	M_x = 50,000 lb	0	9,735.84	0	10,937.50
TS 4 x 4	Corner of Embedded Plate	M_x = 25,000 in.-lb M_y = 25,000 in.-lb	0	5,575.90	0	10,937.50
L 4 x 4	Center of Embedded Plate	F_z = 20,000 lb	0	14,585.30	0	15,940.12
L 4 x 4	Center of Embedded Plate	M_x = 50,000 in.-lb	0	4,959.99	0	5,934.86
L 4 x 4	Center of Embedded Plate	M_x = 25,000 in.-lb M_y = 25,000 in.-lb	0	4,344.17	0	5,934.86
L 4 x 4	Corner of Embedded Plate	F_z = 20,000 lb	0	18,596.50	0	22,009.02

TABLE 3B.17-1
APE PROGRAM VERIFICATION PROBLEM COMPARISON OF DRILLED-IN ANCHOR ROAD

(Sheet 3 of 3)

			BIP (ST 361)		APE (ST 378, V01L00)	
Attachment	Attachment Location	Loading	Anchor Shear	Anchor Tension	Anchor Shear	Anchor Tension
L 4 x 4	Corner of Embedded Plate	M_x = 50,000 in.-lb	0	10,559.60	0	10,937.50
L 4 x 4	Corner of Embedded Plate	M_x = 25,000 in.-lb M_y = 25,000 in.-lb	0	8,648.15	0	10,937.50
C 4 x 5.4	Corner of Embedded Plate	$F_x = F_y$ = 2,800 lb F_z = 1,970 lb M_x = 30,430 in.-lb M_y = 27,390 in.-lb M_z = 0	1,763.35	11,067	1,763.35	32,351.37
TS 4 x 4	Along Edge of Surface-Mounted Plate	F_z = 3,000 lb	0	1,237.02	0	1,458.33
TS 4 x 4	Along Edge of Surface-Mounted Plate	M_x = 12,000 in.-lb	0	786.89	0	1,250.00
TS 4 x 4	Along Edge of Surface-Mounted Plate	M_x = 8,000 in.-lb M_y = 8,000 in.-lb	0	856.54	0	1,833.33

4.0 REACTOR

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4.1 SUMMARY DESCRIPTION

This chapter describes: 1) the mechanical components of the reactor and reactor core including the fuel rods and fuel assemblies, 2) the nuclear design, and 3) the thermal-hydraulic design. The initial reactor core is composed of an array of fuel assemblies that are identical in mechanical design but different in fuel enrichment. Within each fuel assembly all rods are of the same enrichment. Three different enrichments are employed in the first core: 2.10 (region 1), 2.60 (region 2), and 3.10 (region 3) weight percents. It was required that the initial core loading maximum enrichment not exceed 3.2 weight percent U235. For subsequent reloads, the target maximum enrichment is up to 5.0 weight percent. Reload fuel shall be similar in physical design to the initial core loading and shall have a maximum enrichment not to exceed 5.0 weight percent U235.

A fuel assembly is composed of 264 fuel rods in a 17x17 square array. The center position in the fuel assembly is reserved for incore instrumentation. The remaining 24 positions in the fuel assembly have guide thimbles which are joined to the top and bottom nozzles of the fuel assembly and serve to support the fuel grids. A fuel assembly may have limited substitution of zirconium alloy or stainless steel filler rods in place of fuel rods, in accordance with NRC approved applications of fuel rod configurations. The fuel grids consist of an egg crate arrangement of interlocked straps that maintain lateral spacing between the rods. The grid straps have spring fingers and dimples which grip and support the fuel rods. The middle grids also have coolant-mixing vanes. The flow mixer grid straps contain only support dimples and coolant mixing vanes.

The reactor core is comprised of an array of fuel assemblies which have different fuel enrichments. Fuel cycle times of 6 months to 24 months are expected and may be employed with the core described herein.

Commencing with Unit 1 Cycle 12 and Unit 2 Cycle 10, the Westinghouse fuel assemblies are of the VANTAGE + design with Intermediate Flow Mixing grids (IFMs) and include a thin zirconium oxide coating applied to the bottom seven inches of each fuel rod to provide additional debris fretting protection in this area. All fuel assemblies are equipped with the “Small Hole” debris filtering bottom nozzle, an alternate protective grid (P-grid), and long solid end plugs on the fuel rods.

The IFM grids are designed to improve CPNPP fuel performance in the following areas:

- enhanced DNB margin as a result of increased flow mixing;
- mitigate Crud Induced Power Shift (CIPS); and
- stabilize fuel assemblies to decrease susceptibility to Incomplete Rod Insertion (IRI) events.

Commencing with Unit 2 Cycle 13 and Unit 1 Cycle 16, the Westinghouse fuel assemblies utilize the Standardized Debris Filter Bottom Nozzle (SDFBN) and the Robust Protective Grid (RPG). These features are discussed in detail in Section 4.2.2.

The Westinghouse fuel assemblies consist of 17x17 fuel rods with 0.360" nominal outer diameter fuel rods. The Westinghouse assemblies typically have 2.60 w/o enriched U235 axial blankets in the top and bottom six inch zones of each fuel rod.

Wet annular burnable absorber (WABA) rodlets consisting of B4C - Al2O3 pellets can be used in selected fresh Westinghouse fuel for the reload core design to shape the power distribution and achieve a desirable moderator temperature coefficient. Additionally, selected fresh Westinghouse fuel will contain Integral Fuel-Burnable Absorber (IFBA) pellets, which also act as a burnable absorber. These fuel pellets are coated with a thin layer of ZrB2. Both the WABA and IFBA rods used typically contain 120 inches of absorber material, axially centered in the active fuel region, but the length can vary depending on the specific core design requirements.

The core is cooled and moderated by light water at a pressure of 2250 pounds per square inch absolute (psia) in the Reactor Coolant System. The moderator coolant contains boron as a neutron poison. The concentration of boron in the coolant is varied as required to control relatively slow reactivity changes including the effects of fuel burnup. Additional boron, in the form of burnable poison rods, is employed in the core to compensate for excess reactivity, control of radical flux distributions, and to maintain the required moderator temperature coefficient.

The fuel rods are supported at intervals along their length by grid assemblies which maintain the lateral spacing between the rods throughout the design life of the assembly. The fuel rods consist of slightly enriched uranium dioxide ceramic cylindrical pellets contained in slightly cold worked Zircaloy-4 or ZIRLO® High Performance Fuel Cladding Material which is plugged and seal welded at the ends to encapsulate the fuel. All fuel rods are pressurized with helium during fabrication.

The center position in the assembly is reserved for the incore instrumentation, while the remaining 24 positions in the array are equipped with guide thimbles joined to the grids and the top and bottom nozzles. Depending upon the position of the assembly in the core, the guide thimbles are used as core locations for rod cluster control assemblies (RCCAs), neutron source assemblies, and burnable poison assemblies. Otherwise, the guide thimbles are fitted with plugging devices to limit bypass flow.

The bottom nozzle is a box-like structure which serves as a bottom structural element of the fuel assembly and directs the coolant flow distribution to the assembly.

The top nozzle assembly functions as the upper structural element of the fuel assembly in addition to providing a partial protective housing for the RCCA or other components.

The RCCAs each consist of a group of individual absorber rods fastened at the top end to a common hub or spider assembly. These assemblies contain full length absorber material to control the reactivity of the core under operating conditions, to control axial power distribution and to provide a mechanism for reactor shutdown.

The nuclear design analyses and evaluation establish physical locations for control rods and burnable poisons and physical parameters such as fuel enrichments and boron concentration in the coolant. The nuclear design evaluation established that the reactor core has inherent characteristics which together with corrective actions of the reactor control, and protective

systems provide adequate reactivity control even if the highest reactivity worth RCCA is stuck in the fully withdrawn position.

The design also provides for inherent stability against diametral and azimuthal power oscillations and for control of induced axial power oscillation through the use of the control rods.

The thermal-hydraulic design analyses and evaluation establish coolant flow parameters which assure that adequate heat transfer is provided between the fuel clad and the reactor coolant. The thermal design takes into account local variations in dimensions, power generation, flow distribution and mixing. The fuel assembly spacer grid design induces additional flow mixing between the various flow channels within a fuel assembly as well as between adjacent assemblies.

Instrumentation is provided in and out of the core to monitor the nuclear, thermal-hydraulic, and mechanical performance of the reactor and to provide inputs to automatic control functions. Table 4.1-1 presents a comparison of the principal nuclear, thermal-hydraulic, and mechanical design parameters of the Comanche Peak units.

The analysis techniques employed in the core design are addressed in Table 4.1-2. The loading conditions considered in general for the core internals and components are addressed in Section 4.2 and Table 4.1-3.

TABLE 4.1-1
REACTOR DESIGN COMPARISION TABLE

(Sheet 1 of 3)

Design Parameters[a]		CPNPP Pre-Uprate Design	CPNPP Uprate Design
1.	Reactor core heat output (MW_t)[b]	3458	3612
2.	Reactor core heat output (10^6) BTU/hr)[b]	11,799	12,325
3.	Heat generated in fuel (%)	97.4	97.4
4.	System pressure, nominal (psia)	2250	2250
5.	Minimum DNBR for design transients		
	Typical flow channel		
	WRB-1	1.23	1.23
	WRB-2	1.23	1.23
	Thimble (cold wall) flow channel		
	WRB-1	1.23	1.23
	WRB-2	1.22	1.22
6.	DNB correlation[c]	WRB-2/WRB-1 W-3	WRB-2/WRB-1 W-3
Coolant flow(e)			
7.	Total vessel flow-rate (10^6 lb_m/hr) (based on thermal design flow - TDF, including bypass)	142.04	142.30
8.	Effective flow-rate for heat transfer (10^6 lb_m/hr) (based on TDF, excluding bypass)[d]	133.80	134.04
9.	Effective flow area for heat transfer (ft^2)	54.13	54.13
10.	Average mass velocity along fuel rods (ft/sec) (based on TDF, excluding bypass)[d]	14.84	14.84
11.	Average mass velocity (10^6 lb_m/hr-ft^2) (based on TDF, excluding bypass)[d]	2.472	2.476
Coolant temperature			
12.	Nominal inlet (^oF)	559.2	558.0
13.	Average rise in vessel (^oF)	60.0	62.4
14.	Average rise in core (^oF)	63.3	65.8
15.	Average in core (^oF)	592.6	592.8
16.	Average in vessel (^oF)	589.2	589.2

TABLE 4.1-1
REACTOR DESIGN COMPARISION TABLE

(Sheet 2 of 3)

Design Parameters[a]		CPNPP Pre-Uprate Design	CPNPP Uprate Design
Heat transfer			
17.	Active heat transfer surface area (ft^2)[e]	57,505	57,505
18.	Average heat flux (BTU/hr-ft^2)[e]	199,900	208,802
19.	Maximum heat flux for normal operation (BTU/hr-ft^2)[e][f]	483,758	522,005
20.	Average linear power (kW/ft)[e]	5.520	5.766
21.	Peak linear power for normal operation (kW/ft)[e][f]	13.36	14.41
22.	Peak linear power resulting from overpower transients/ operator errors, assuming a maximum overpower of 118.5% (kW/ft)[e]	$\leq$22.4	$\leq$22.4
23.	Heat flux hot channel factor (FQ)	2.42	2.50
24.	Peak fuel central temperature for prevention of centerline melt (°F)	4700	4700
25.	Design	RCC canless 17 x 17	RCC canless 17 x 17
26.	Number of fuel assemblies	193	193
27.	UO_2 rods per assembly	264	264
28.	Rod pitch (in.)	0.496	0.496
29.	Overall dimensions (in.)	8.246 x 8.426	8.246 x 8.426
30.	Fuel weight, as UO_2 (lb)	205,352	205,352
31.	Zircaloy/ZIRLO weight (lb) (active core)	50,759	50,759
32.	Number of grids per assembly	2 - R type 6 - Z type 3 - IFM 1 - Alt P-Grid	2 - R type 6 - Z type 3 - IFM 1 - Alt P-Grid or RPG
Fuel rods			
33.	Number	50,952	50,952

TABLE 4.1-1
REACTOR DESIGN COMPARISION TABLE

(Sheet 3 of 3)

Design Parameters[a]		CPNPP Pre-Uprate Design	CPNPP Uprate Design
34.	Outside diameter (in.)	0.360	0.360
35.	Diametral gap (in.)	0.0062	0.0062
36.	Clad thickness (in.)	0.0225	0.0225
37.	Clad material	ZIRLO	ZIRLO
Fuel pellets			
38.	Material	UO_2 sintered	UO_2 sintered
39.	Density (% of theoretical)	95	95
40.	Diameter (in.)	0.3088	0.3088
41.	Length (in.) (standard pellet/blanket pellet)	0.370/0.500	0.370/0.500
Rod cluster control assemblies			
42.	Neutron absorber	Ag-In-Cd	Ag-In-Cd
43.	Cladding material	Type 304 SS cold-worked	Type 304 SS cold-worked
44.	Clad thickness (in.)	0.0185	0.0185
45.	Number of clusters	53	53
46.	Number of absorber rods per cluster	24	24
47.	Core height, active fuel (in.)[e]	143.7	143.7

a) Unit 1 flow rates are used for comparison. The lower Unit 1 flow rates are more limiting with respect to DNB. Also, these flows are to be used for both units for uprate conditions. Unit 2 coolant temperatures are used for comparison. The higher Unit 2 temperatures are more limiting with respect to DNB.

b) The proposed power level of 3612 MW_t has been used for thermal-hydraulic design analyses.

c) See paragraph 4.4.2.1.1 for the use of the W-3 correlation.

d) Based on thimble plugs inserted.

e) Based on densified active fuel length.

f) Based on maximum FQ of 2.42 for pre-uprate design and 2.50 for uprate design.

TABLE 4.1-2
ANALYTICAL TECHNIQUES IN CORE DESIGN

(Sheet 1 of 2)

Analysis	Technique	Computer Code Referenced	Section
Mechanical design of core internals loads, deflections, and stress analysis	Static and dynamic modeling structural analysis code, and others	Blowdown code, FORCE, finite element	3.7.2.1 3.9.2 3.9.3
Fuel rod design Full performance characteristics (temperature, internal pressure, clad stress, etc.)	Semi-empirical thermal model of fuel rod with consideration of fuel density changes, heat transfer, fission gas release, etc.	Westinghouse fuel rod design model	4.2.1.1 4.2.3.2 4.2.3.3 4.3.3.1 4.4.2.8
Nuclear design Cross-sections and group constants	Microscopic data; macroscopic constants for homogenized core regions	Modified ENDF/B library LEOPARD/CINDER type PHOENIX-P NEXUS/PARAGON	4.3.3.2 4.3.3.2
	Group constants for control rods with selfshielding	HAMMER-AIM PHOENIX-P NEXUS/PARAGON	4.3.3.2 4.3.3.2
X-Y and X-Y-Z power distributions, fuel depletion, critical boron concentrations, X-Y and X-Y-Z xenon distributions, reactivity coefficients	2-group diffusion theory	TURTLE (2-D) or ANC (2-D or 3-D) NEXUS/PARAGON/ANC	4.3.3.3 4.3.3.3
Axial power distributions, control rod worths, and axial xenon distribution	1-D, 2-group diffusion theory 3D, Diffusion Theory-Based Nodal Method	PANDA NEXUS/PARAGON/ANC	4.3.3.3 4.3.3.3
Fuel rod power	Integral transport theory	LASER	4.3.3.1

TABLE 4.1-2
ANALYTICAL TECHNIQUES IN CORE DESIGN

(Sheet 2 of 2)

Analysis	Technique	Computer Code Referenced	Section
Effective resonance temperature	Monte Carlo weighting function	REPAD	4.3.3.1
Criticality of reactor and fuel assemblies	2-D, 2-group diffusion theory	APX system of codes, KENO-IV, SCALE 5.1, PARAGON	4.3.2.6
Vessel irradiation	Multigroup spatial dependent transport theory	DOT	4.3.2.8
Thermal-hydraulic design Steady state	Sub-channel analysis of local fluid conditions in rod bundles, including inertial and cross-flow resistance terms; solution progresses from core-wide to hot assembly to hot channel.	VIPRE-01	4.4.4.5.1
Transient departure from nucleate boiling	Sub-channel analysis of local fluid conditions in rod bundles during transients by including accumulation terms in conservative equations; solution progresses from core-wide to hot assembly to hot channel.	VIPRE-01	4.4.5.4

TABLE 4.1-3
DESIGN LOADING CONDITIONS FOR REACTOR CORE COMPONENTS

1. Fuel assembly weight and core component weights (BAs, sources, plugging devices)
2. Fuel assembly spring forces and core component spring forces
3. Internals weight
4. Control rod trip (equivalent static load)
5. Differential pressure
6. Spring preloads
7. Coolant flow forces (static)
8. Temperature gradients
9. Differences in thermal expansion
    a. Due to temperature differences
    b. Due to expansion of different materials
10. Interference between components
11. Vibration (mechanically or hydraulically induced)
12. One or more loops out of service
13. All operational transients listed in table 3.9.N.1-1
14. Pump overspeed
15. Seismic loads (operating basis earthquake and safe shutdown earthquake)
16. Blowdown forces (due to cold and hot leg break)

4.2 FUEL SYSTEM DESIGN

The plant conditions for design are divided into four categories in accordance with their anticipated frequency of occurrence and risk to the public: Condition I - Normal Operation; Condition II - Incidents of Moderate Frequency; Condition III - Infrequent Incidents; Condition IV - Limiting Faults. The bases and description of plant operation and events involving each condition are given in the accident analysis, Chapter 15.

The reactor is designed so that its components meet the following performance and safety criteria:

1. The mechanical design of the reactor core components and their physical arrangement, together with corrective actions of the reactor control, protection and emergency cooling systems (when applicable) assure that:

   a. Fuel damage[a] is not expected during Condition I and Condition II events. It is not possible, however, to preclude a very small number of rod failures. These are within the capability of the plant cleanup system and are consistent with plant design bases.

   b. The reactor can be brought to a safe state following a Condition III event with only a small fraction of fuel rods damaged[a] although sufficient fuel damage might occur to preclude immediate resumption of operation.

   c. The reactor can be brought to a safe state and the core can be kept subcritical with acceptable heat transfer geometry following transients arising from Condition IV events.

2. The fuel assemblies are designed to withstand, without exceeding the criteria of Section 4.2.1.5, loads induced during shipping, handling and core loading.

3. The fuel assemblies are designed to accept control rod insertions in order to provide the required reactivity control for power operations and reactivity shutdown conditions.

4. All fuel assemblies have provisions for the insertion of incore instrumentation necessary for plant operation.

5. The reactor internals in conjunction with the fuel assemblies and incore control components direct reactor coolant through the core. This achieves acceptable flow distribution and restricts bypass flow so that the heat transfer performance requirements can be met for all modes of operation.

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a. Fuel damage as used here is defined as penetration of the fission barrier (i.e., the fuel rod clad).

4.2.1 DESIGN BASES

The fuel rod and fuel assembly design bases are established to satisfy the general performance and safety criteria presented in Section 4.2.

The detailed fuel rod design establishes such parameters as pellet size and density, clad-pellet diametral gap, gas plenum size, and helium pre-pressurization level. The design also considers effects such as fuel density changes, fission gas release, clad creep, and other physical properties which vary with burnup. The integrity of the fuel rods is ensured by designing to prevent excessive fuel temperatures, excessive internal rod gas pressures due to fission gas releases, and excessive cladding stresses and strains. This is achieved by designing the fuel rods so that the conservative design basis in the following subsections are satisfied during Condition I and Condition II events over the fuel lifetime. For each design basis, the performance of the limiting fuel rod must not exceed the limits specified by the design basis.

Structural integrity of the fuel assembly structure is assured by setting limits on stresses and deformations due to various loads and by determining that the assembly structure does not interfere with the functioning of other components. Two types of loads are considered.

1. Normal and abnormal loads which are defined for Conditions I and II.

2. Abnormal loads which are defined for Conditions III and IV.

The design bases for the incore control components are described in Section 4.2.1.6.

4.2.1.1 Cladding

1. The desired fuel rod clad is a material which has a superior combination of neutron economy (low absorption cross section), high strength to resist deformation due to differential pressures and mechanical interaction between fuel and clad, high corrosion resistance to coolant, fuel and fission products, and high reliability. Zircaloy-4 and ZIRLO have this desired combination of clad properties. There is considerable PWR operating experience on the capability of Zirconium alloys as a clad material. Information of the material chemical and mechanical properties of the cladding is given in Reference 2 and Reference 9 with due consideration of temperature and irradiation effects.

   Other fuel rod criteria are specified in Reference 32.

2. Stress-strain limits

   a. Clad stress

      The von Mises criterion is used to calculate the effective stresses. The cladding stresses under Condition I and II events are less than the Zircaloy 0.2% offset yield stress with due consideration of temperature and irradiation effects. While the clad has some capability for accommodating plastic strain, the yield stress has been accepted as a conservative design basis.

   b. Clad tensile strain

The total tensile creep strain is less than 1 percent from the un-irradiated condition. The elastic tensile strain during a transient is less than 1 percent from the pre-transient value. This limit is consistent with proven practice.

3. Vibration and fatigue

a. Strain fatigue

The cumulative strain fatigue cycles are less than the design strain fatigue life. This basis is consistent with proven practice (Ref. [1]).

b. Vibration

Potential fretting wear due to vibration is prevented by design of the fuel assembly grid springs and dimples assuring that the stress-strain limits are not exceeded during design life. Fretting of the clad surface can occur due to flow-induced vibration between the fuel rods and fuel assembly grid springs. Vibration and fretting forces vary during the fuel life due to clad diameter creepdown combined with grid spring relaxation.

4.2.1.2 Fuel Material

1. Thermal-physical properties

Fuel pellet temperatures - The center temperature of the hottest pellet is to be below the melting temperature of the UO_2 (melting point of 5080ºF [3] unirradiated and decreasing by 58ºF per 10,000 MWD/MTU). While a limited amount of center melting can be tolerated, the design conservatively precludes center melting. A calculated fuel centerline temperature of 4700ºF has been selected as an overpower limit to assure no fuel melting. This provides sufficient margin for uncertainties as described in Section 4.4.2.9.

The normal design density of the fuel is 95 percent of theoretical.

2. Fuel densification and fission product swelling

The design bases and models used for fuel densification and swelling are provided in Reference 5.

3. Chemical properties

The mechanical and chemical characteristics of the cladding is discussed in Reference 9.

4.2.1.3 Fuel Rod Performance

1. Fuel rod models

The basic fuel rod models and the ability to predict operating characteristics are given in References 5 and 9.

2. Mechanical design limits

Cladding collapse shall be precluded during the fuel rod design lifetime. The models described in References 6 and 29 are used for this evaluation.

The rod internal gas pressure shall remain below the value which causes the fuel-clad diametral gap to increase due to outward cladding creep during steady state operation. Rod pressure is also limited such that extensive departure from nucleate boiling (DNB) propagation shall not occur during normal operation and accident events.

The fuel rod internal pressure design criteria will permit the internal pressure to exceed the reactor coolant system pressure. Consideration has been given to the effects of a postulated propagation process wherein fuel rods are simultaneously in DNB and exceed system pressure, causing subsequent ballooning and touching of adjacent rods resulting in more rods reaching DNB and failing.

Even with a very conservative assessment of this process, the number of fuel rods which could fail in Condition III and IV events increased by only a small percentage. The dose consequences of the accidents evaluated in Chapter 15 would remain essentially unchanged considering this increase.

4.2.1.4 Spacer Grids

1. Mechanical limits and materials properties

The grid component strength criteria are based on experimental tests. The limit is established at 0.9 Pc, where Pc is the experimental collapse load. This limit is sufficient to assure that under worst-case combined seismic and blowdown loads the core will maintain a geometry amenable to cooling.

2. Vibration and fatigue

The grids shall provide sufficient fuel rod support to limit fuel rod vibration and maintain clad fretting wear to within acceptable limits (defined in Section 4.2.1.1).

4.2.1.5 Fuel Assembly

1. Structural design

As previously discussed in Section 4.2.1, the structural integrity of the fuel assemblies is assured by setting design limits on stresses and deformations due to various operational and accident loads.

These limits are applied to the design and evaluation of the top and bottom nozzles, guide thimbles, grids, and the thimble joints.

The design bases for evaluating the structural integrity of the fuel assemblies are:

For the normal operating and upset conditions, the fuel assembly component structural design criteria are established for the two primary material categories, namely austenitic

steels and Zircaloy/ZIRLO. The stress categories and strength theory presented in the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, Section III, are used as a general guide. The maximum shear-theory (Tresca criterion) for combined stresses is used to determine the stress intensities for the austenitic steel components. The stress intensity is defined as the numerically largest difference between the various principal stresses in a three dimensional field. The allowable stress intensity value for austenitic steels, such as nickel-chromium-iron alloys, is given by the lowest of the following:

a. One-third of the specified minimum tensile strength or 2/3 of the specified minimum yield strength at room temperature;

b. One-third of the tensile strength or 90 percent of the yield strength at temperature but not to exceed 2/3 of the specified minimum yield strength at room temperature.

The stress limits for the austenitic steel components are given below. All stress nomenclature is per the ASME Boiler and Pressure Vessel Code, Section III.

Stress Intensity Limits

Categories	Limit
General Primary Membrane Stress Intensity	S_m
Local Primary Membrane Stress Intensity	1.5 S_m
Primary Membrane plus Bending Stress Intensity	1.5 S_m
Total Primary plus Secondary Stress Intensity	3.0 S_m

The Zircaloy-4 or ZIRLO structural components which consist of guide thimbles, spacer grids, Intermediate Flow Mixing (IFM) grids and fuel tubes are in turn subdivided into two categories because of material differences and functional requirements. The fuel tube and grid design criteria is covered separately in Section 4.2.1.1 and 4.2.1.4, respectively. The maximum shear theory is used to evaluate the guide thimble design. For conservative purposes, the Zircaloy/ZIRLO unirradiated properties are used to define the stress limits.

c. Abnormal loads during Conditions III or IV.

Westinghouse (W) fuel assemblies (except W Optimized Fuel Assemblies): worst case represented by combined seismic and blowdown loads.

W Optimized Fuel Assemblies: worst case represented by seismic loads, or blowdown loads during a LOCA event.

1. Deflections or failures of components cannot interfere with the reactor shutdown or emergency cooling of the fuel rods.

2. The fuel assembly structural component stresses under faulted conditions are evaluated using primarily the methods outlined in Appendix F of the ASME Boiler and Pressure Vessel Code, Section III. Since the current analytical methods utilize elastic analysis, the stress allowables are defined as the smaller value of 2.4 S_m or 0.70 S_u for primary membrane and 3.6 S_m or 1.05 S_u for primary membrane plus primary bending, where Su is the material ultimate tinsels strength. For the austenitic steel fuel assembly components, the stress intensity is defined in accordance with the rules described in the previous section for normal operating conditions. For the Zircaloy-4 or ZIRLO components the stress intensity limits are set at 2/3 of the material yield strength, S_y, at reactor operating temperature. This results in Zircaloy stress limits being the smaller of 1.6 S_y or 0.70 S_u for primary membrane and 2.4 S_y or 1.05 S_u for primary membrane plus bending. For conservative purposes, the Zircaloy/ZIRLO unirradiated properties are used to define the stress limits.

   Typical material and chemical properties of the fuel assembly components are given in References [2] and [9] for Zircaloy-4 and ZIRLO, respectively.

2. Thermal-hydraulic design

   This topic is covered in Section 4.4.

4.2.1.6 Incore Control Components

The core components consist of the rod cluster control assemblies (RCCAs), the primary and secondary source assemblies, the thimble plug assemblies and burnable absorber assemblies. A description of these components is provided in subsection 4.2.2.

1. Mechanical Design Basis

   a. Control Rod Assemblies

      The mechanical design bases for the control rods are consistent with the loading conditions under Article NB-3000 of the ASME Boiler and Pressure Vessel Code, Section III.

      - External pressure equal to the Reactor Coolant System operating pressure with appropriate allowance for overpressure transients.
      - Wear allowance equivalent to 1000 reactor trips.
      - Bending of the rod due to a misalignment in the guide tube.
      - Forces imposed on the rods during rod drop.
      - Loads imposed by the accelerations of the control rod drive mechanism.
      - Radiation exposure during maximum core life.

- Temperature effects from room to operating conditions.

b. Burnable Absorber, Thimble Plug and Source Rod Assemblies

The burnable absorber assemblies, thimble plug assemblies and source assemblies are static core components.

The mechanical design of these components satisfy the following:

1. Accommodate the differential thermal expansion between the fuel assembly and the core internals,

2. Maintain positive contact with the fuel assembly and the core internals.

3. Limits the flow through each occupied thimble to acceptable design values.

2. Thermal-Physical Properties of the Absorber Material

The absorber material for the RCCAs is Ag-In-Cd with nominal proportions silver 80%, Indium, 15% and Cadmium 5%.

The burnable absorber material is non-structural with nominal lengths between 120 and 142 inches depending on the cycle specific core design. The structural elements of the burnable poison rod are designed to maintain the absorber geometry even if the absorber material is fractured. The rods are designed so that the absorber material is below its softening temperature. In addition, the structural elements are designed to prevent excessive slumping.

3. Compatibility of the Absorber and Cladding Materials

The control rod and source rod cladding is cold drawn type 304 stainless steel tubing. Extensive in-reactor experience and available quantitative information show that reaction rates between 304 stainless steel and water, or other in-core metals are negligible at operational temperatures (Reference [2]).

4.2.1.7 Surveillance Program

A testing and fuel surveillance operational experience program has been and is being conducted to verify the adequacy of the fuel performance and design bases. For W-supplied fuel, the program for standard fuel is discussed in Section 4.2.4.5. Reference [7] provides a description of the tests performed and a summary of the results for the W fuel. Fuel surveillance and testing results, as they become available, are used to improve fuel rod design and manufacturing processes and assure that the design bases and safety criteria are satisfied.

4.2.2 DESIGN DESCRIPTION

Commencing with Unit 1 Cycle 12 and Unit 2 Cycle 10, the Westinghouse fuel assembly design is the VANTAGE + fuel design. The fuel assemblies each contain 264 fuel rods of 0.360" nominal outer diameter, twenty-four guide thimble tubes, and one instrumentation tube in a 17x17 array

supported by eight spacer grids, three Intermediate Flow Mixer (IFM) grids and one debris filtering protective grid (P-grid) in the fuel assembly structure. The instrumentation tube is located in the center position of the fuel assembly and provides a channel for insertion of an incore neutron detector, if the fuel assembly is located in an instrumented core position. The guide thimble tubes provide channels for insertion of either a rod cluster control assembly (RCCA), a neutron source assembly (primary or secondary), a burnable poison assembly (WABA) or a plugging device, depending on the position of the particular fuel assembly in the core. The assembly mid-span grids and the IFM grids are made of ZIRLO and the top and bottom grids and the P-grid are made of Inconel. Each fuel assembly contains a reconstitutable top nozzle (RTN) that has a unique serial number engraved on the nozzle for assembly ID. Each fuel assembly also contains a debris filter bottom nozzle (DFBN). The DFBN, P-grid, debris mitigation long bottom end-plugs on the fuel rods and a protective oxide coating applied to approximately the bottom 7" of each fuel rod provides the maximum protection for debris-induced fretting. The zirconium oxide pre-coating provides a hardened surface on the fuel cladding making the clad more resistant to debris-induced fretting during the first few months of operation before the oxide coating develops naturally. Since zirconium oxide forms naturally on the surface of the fuel rod clad during normal fuel operations, the pre-coating of the clad does not create any new potential adverse impacts to safety analyses or fuel operations. The central fuel pellets are 5 w/o enriched U^{235}, dished and chamfered on the ends. Typically, a nominal 6” of pellets at the top and bottom of the fuel rods are axial blanket pellets which are 2.6 w/o enriched U^{235}. The central fuel pellets may be coated with zirconium di-boride (ZrB_2) as a burnable absorber. These pellets are normally referred to as Integral Fuel Burnable Absorber (IFBA) pellets and can have varying loading of boron to support the fuel management design of the core. The IFBA fuel rods are typically pressurized to 100 psig and utilize annular blanket pellets to allow for the extra gas production associated with the helium release from the ZrB_2. Non-IFBA fuel rods are typically pressurized to 275 psig. The annular blankets and reduced diameter plenum springs (variable pitch plenum spring) also provide a larger internal plenum to mitigate the effects of the increased gas production to ensure the fuel rods meet their design criterion for rod internal pressure. All of the fuel pellets are sintered to a nominal 95.5% of theoretical density. Figure 4.2-1 shows a cross section of a typical fuel assembly array, and Figure 4.2-2 shows a typical fuel assembly full length view. Figure 4.2-3 shows a typical fuel rod configuration.

If a fuel rod becomes damaged during operations, such as debris fretting failures, the rod can be removed and a stainless steel filler rod can be inserted to allow continued use of the fuel assembly in subsequent cycles.

The fuel rods are loaded into the fuel assembly structure so that there is clearance between the fuel rod ends and the top and bottom nozzles. All fuel assemblies in the core are functionally identical.

Each fuel assembly is installed vertically in the reactor vessel and stands upright on the lower core plate, which is fitted with alignment pins to locate and orient the assembly. After all fuel assemblies are set in place, the upper support structure is installed. Alignment pins, built into the upper core plate, engage and locate the upper ends of the fuel assemblies. The upper core plate then bears downward against the fuel assembly top nozzle via the holddown springs to hold the fuel assemblies in place.

Improper orientation of fuel assemblies within the core is prevented by the use of an indexing hole in one corner of the top nozzle top plate. The assembly is oriented with respect to the

handling tool and the core by means of a pin which is inserted into this indexing hole. Additionally, visual confirmation of orientation is provided by an engraved identification number on the opposite corner clamp.

Several optional design enhancements have been made to the W optimized fuel design being used at Comanche Peak. These enhancements include the following:

- An alternate protective grid is included at the bottom of the assembly, to provide an additional debris barrier, thereby improving fuel reliability. The protective grid will also provide additional grid/rod fretting resistance by supporting the bottom of the fuel rod. Beginning with Unit 2 Cycle 13 and Unit 1 Cycle 16, the Westinghouse fuel assemblies utilize the Robust Protective Grid (RPG). This feature is discussed in detail in Section 4.2.2.2.4.

- A lengthened, debris-mitigating fuel rod bottom end plug, longer than prior design, is used which extends up through the protective grid, limiting the span for fretting to the solid end plug.

- A lengthened external grip top end plug is also used for reconstitution capability.

- A smaller gap between the rods and the bottom nozzle exists as a result of the longer end plug while maintaining the fuel stack at the same elevation.

- The mid-grids and IFM grids are fabricated from the ZIRLO alloy for improved corrosion performance margin to high burnup.

- The top and bottom nozzles and top grid assembly sleeve are fabricated of a low cobalt Type 304 Stainless Steel for dose reduction.

- Extended burnup bottom grid design.

- An oxide coating at the bottom 7 inches of the fuel rod for increased resistance to debris in this area.

4.2.2.1 Fuel Rods

The fuel rods consist of uranium dioxide ceramic pellets contained in Zircaloy-4 or ZIRLO tubing which is plugged and seal welded at the ends to encapsulate the fuel (the reconstituted rods are made of stainless steel). A schematic of a typical fuel rod is shown in Figure 4.2-3. The fuel pellets are right circular cylinders consisting of slightly enriched uranium dioxide powder which has been compacted by cold pressing and then sintered to the required density. The ends of each pellet are dished slightly to allow greater axial expansion at the center of the pellets.

To avoid overstressing of the clad or seal welds, void volume and clearances are provided within the rods to accommodate fission gases released from the fuel, differential thermal expansion between the clad and the fuel, and fuel density changes during irradiation.

Shifting of the fuel within the clad during handling or shipping prior to core loading is prevented by a stainless steel or Inconel helical spring which bears on top of the fuel. At assembly the pellets are stacked in the clad to the required fuel height, the spring is then inserted into the top end of

the fuel tube and the end plugs pressed into the ends of the tube and welded. All fuel rods are internally pressurized with helium during the welding process in order to minimize compressive clad stresses and prevent clad flattening due to coolant operating pressures.

The fuel rods are presently being pre-pressurized and designed so that: 1) the internal gas pressure mechanical design limit given in Section 4.2.1.3 is not exceeded, 2) the cladding stress-strain limits (Section 4.2.1.1) are not exceeded for Condition I and II events, and 3) clad flattening will not occur during the fuel core life.

Integral Fuel Burnable Absorber (IFBA)

The IFBA coated fuel pellets are identical to the enriched uranium dioxide pellets except for the addition of a thin zirconium diboride (ZrB_2) coating on the pellet cylindrical surface. Coated pellets occupy the central portion of the fuel column. The number and pattern of IFBA rods within an assembly may vary depending on specific application. The ends of the enriched coated pellets and enriched uncoated pellets are dished to allow for greater axial expansion at the pellet centerline and to increase the void volume for fission gas release. Analysis of IFBA rods includes any geometry changes necessary to model the presence of burnable absorber, and conservatively models the gas release from the coating.

Variation of IFBA loading is allowed according to the needs presented for power distribution control in designing the core each cycle. Enriched IFBA, containing a greater presence of the neutron absorbing isotope B-10, can be specified in terms of multiples of the natural boron IFBA rod. IFBA loadings of 1.5X or 2.0X have been applied in various regions, for example. This is not to preclude other loadings (1.25X, 1.6X) according to need. Also, variations in the length of the span of the fuel stack that include IFBA pellets is at the discretion of the core designer, according to requirements imposed. The acceptability of a proposed IFBA configuration is dependent on meeting fuel rod design criteria. The principle criteria which come to bear are the fuel rod internal pressure criterion and the clad stress criterion.

The rod internal pressure design criterion is impacted by the IFBA loading because some additional amount of helium is produced from the “burnout” of the ZrB_2 pellet coating over and above the inventory of fission product gases. Different IFBA loadings will produce varying increases of rod internal pressure. To offset this extra helium within the rod, the initial helium prepressurization of the IFBA rod (100 psig backfill pressure) is reduced as compared to the non-IFBA rod (275 psig backfill pressure). Also, for greater IFBA loadings, further pressure reduction measures are employed, such as providing additional void volume in the rod by inserting annular axial blankets. The annular axial blanket is formed by inserting pellets with an annulus (and void of approximately 25% of a solid pellet) for several inches in the top and bottom ends of the fuel stack. Regardless of the region specific IFBA load, continued compliance to the rod internal pressure design criterion is assured by reload analysis and evaluation.

With respect to the clad stress design criterion, generic analyses and significant design experience have demonstrated that changes in boron loadings and pre-pressurization levels consistent with current W non-IFBA and IFBA designs do not significantly impact the ability of the fuel to satisfy clad stress limits. Furthermore, for Comanche Peak, assuming 2X IFBA design (with annular axial blankets), the significant design parameters which can impact clad stress, including overall IFBA pellet diameter and rod internal pressure (which impacts the rate of clad creepdown onto the pellet surface), have been shown to satisfy all design criteria.

Axial Blankets

The axial blankets are a section of fuel pellets at each end of the fuel rod pellet stack. Axial blankets reduce neutron leakage and improve fuel utilization (thus reducing the fuel costs). Neutrons born near the top or bottom of the assembly have a high probability of migrating into the surrounding water/core support structure without causing fission. By implementing axial blankets, a larger proportion of enriched fuel is located in higher flux regions of the core, thus reducing the probability of neutron leakage. The axial blankets utilize chamfered pellets which are physically different (length) than the other fuel pellets to help prevent accidental mixing during manufacturing.

4.2.2.2 Fuel Assembly Structure

The fuel assembly structure consists of a bottom nozzle, top nozzle, guide thimbles and grids. A typical fuel assembly structure is shown in Figure 4.2-2.

4.2.2.2.1 Bottom Nozzle

The bottom nozzle serves as a bottom structural element of the fuel assembly and directs the coolant flow distribution to the assembly. The square nozzle is fabricated from Type 304 stainless steel and consists of a structural support with flow channels and four angle legs with bearing plates. The legs form a plenum for the inlet coolant flow to the fuel assembly. The structural support also prevents accidental downward ejection of the fuel rods from the fuel assembly. The bottom nozzle is fastened to the fuel assembly guide tubes by locked screws which penetrate through the nozzle and mate with a threaded plug in each guide tube.

Coolant flow through the fuel assembly is directed from the plenum in the bottom nozzle upward through the penetrations in the plate to the channels between the fuel rods.

Axial loads (holddown) imposed on the fuel assembly and the weight of the fuel assembly are transmitted through the bottom nozzle to the lower core plate. Indexing and positioning of the fuel assembly is controlled by alignment holes in two diagonally opposite bearing plates which mate with locating pins in the lower core plate. Any lateral loads on the fuel assembly are transmitted to the lower core plate through the locating pins.

The bottom nozzle design has a reconstitution feature which allows the nozzle to be easily removed. A locking cup is used to lock the guide thimble screw in place at assembly or during reconstitution and nozzle reattachment if required.

The Comanche Peak fuel also has a debris-filter bottom nozzle (DFBN) feature. The DFBN is a revised version of the 17x17 nozzle design. The revised design includes an improved pattern of flow holes which:

1. Reduces the passage of debris into the fuel assembly,

2. Maintains the structural integrity of the nozzle design,

3. Maintains the hydraulic performance of the design.

Westinghouse has developed the Standardized Debris Filter Bottom Nozzle (SDFBN) for Westinghouse 17x17 fuel which is designed to have a loss coefficient that is the same independent of supplier. The SDFBN has eliminated the side skirt communication flow holes as a means of improving the debris mitigation performance of the bottom nozzle. This nozzle has been evaluated and meets all of the applicable mechanical design criteria. In addition, there is no adverse affect on the thermal hydraulic performance of the SDFBN either with respect to the pressure drop or with respect to DNB. The SDFBN was implemented at Comanche Peak beginning with Unit 2 Cycle 13 and Unit 1 Cycle 16.

4.2.2.2.2 Top Nozzle

The top nozzle assembly functions as the upper structural element of the fuel assembly in addition to providing a partial protective housing for the rod cluster control assembly or other components. It consists of an adapter plate, enclosure, top plate, and pads. The assembly has holddown springs mounted on the top nozzle. The springs are made of Inconel-718 and spring screws are made of Inconel-600 or Inconel-718, whereas other components are made of Type 304 stainless steel.

The square adapter plate is provided with penetrations to permit the flow of coolant upward through the top nozzle. Other round holes are provided to accept thimble tubes. The ligaments in the plate cover the tops of the fuel rods and prevent their upward ejection from the fuel assembly. The enclosure is a box-like structure which sets the distance between the adapter plate and the top plate. The top plate has a large square hole in the center to permit access for the control rods and the control rod spiders. Holddown springs are mounted on the top plate and are fastened in place by bolts and clamps located at two diagonally opposite corners. On the other two corners integral pads are positioned which contain alignment holes for locating the upper end of the fuel assembly. The top nozzle is designed to allow reconstitution of the assembly.

The Westinghouse Integral Nozzle (WIN) top nozzle design, first introduced in Unit 2 Cycle 12 and Unit 1 Cycle 15, is a direct replacement for the removable top nozzle design. The WIN design eliminates the spring screw for attachment of the hold-down springs, which are now assembled into the nozzle pad and pinned in place. The flow plate, thermal characteristics, and method of attachment of the nozzle are all unchanged from the removable top nozzle design.

4.2.2.2.3 Guide and Instrument Thimbles

The guide thimbles are structural members which also provide channels for the neutron absorber rods, burnable poison rods, neutron source or thimble plug assemblies. Each thimble is fabricated from Zircaloy-4 or ZIRLO tubing having two different diameters. The tube diameter at the top section provides the annular area necessary to permit rapid control rod insertion during a reactor trip. Holes are provided on the thimble tube above the dashpot to reduce the rod drop time. The lower portion of the guide thimble is swaged to a smaller diameter to reduce diametral clearances and produce a dashpot action near the end of the control rod travel during normal trip operation. The dashpot is closed at the bottom by means of an end plug which is provided with a small flow port to avoid fluid stagnation in the dashpot volume during normal operation. The top end of the guide thimble is fastened to a sleeve by a mechanism which fits into the top nozzle adapter plate. The lower end of the guide thimble is fitted with an end plug which is then fastened into the bottom nozzle by a locked thimble screw. Each mid-grid is fastened to the guide thimble assemblies to create an integrated structure. The described methods of grid fastening are

standard and have been used successfully since the introduction of Zircaloy guide thimbles in 1969.

The central instrumentation thimble of each fuel assembly is constrained by seating in counterbores in each nozzle. This tube is a constant diameter and guides the incore neutron detectors.

4.2.2.2.4 Grid Assemblies

The fuel rods are supported at intervals along their length by grid assemblies which maintain the lateral spacing between the rods. Each fuel rod is supported within each grid by the combination of support dimples and springs. The grid assembly consists of individual slotted straps interlocked in an “egg-crate” arrangement. The straps contain spring fingers, support dimples and mixing vanes. The magnitude of the grid restraining force on the fuel rod is set high enough to minimize possible fretting, without overstressing the cladding at the points of contact between the grids and fuel rods. The grid assemblies also allow axial thermal expansion of the fuel rods without imposing restraint sufficient to develop buckling or distortion of the fuel rods.

For the W optimized fuel, the grid material is Inconel-718 or ZIRLO. The magnitude of the grid-restraining force on the fuel rod is set high enough to minimize possible fretting without overstressing the cladding at the points of contact between the grids and fuel rods. The grid assemblies also allow axial thermal expansion of the fuel rods without imposing restraint sufficient to develop buckling or distortion of the fuel rods. The chemical composition of the IFM and mid-grids fabricated with ZIRLO alloy is similar to Zircaloy-4 except for a slight reduction in the content of tin (Sn) and iron (Fe) and the elimination of chromium (Cr). The ZIRLO alloy also contains a nominal amount of niobium (Nb). These changes, although small, are responsible for the improved corrosion resistance of ZIRLO compared to Zircaloy-4. The ZIRLO mid-grids and IFM grids have shorter sleeves.

Three types of grid assemblies are used in the W optimized fuel assembly. Six ZIRLO grids, with mixing vanes projecting from the edges of the straps into the coolant stream, are used in the high heat flux elevations of the fuel assemblies to promote mixing of the coolant. Two grids, one at each end of the assembly, do not contain mixing vanes on the internal straps. The top and bottom non-mixing vane grids are Inconel. Three ZIRLO IFM grids promote flow mixing and are non-structural components. The outside straps on all grids contain mixing vanes which, in addition to their mixing function, aid in guiding the grids and fuel assemblies past adjacent surfaces during handling or loading and unloading of the core. The Inconel and ZIRLO grids have been sized appropriately to assure sufficient structural strength for the assembly.

The IFM grids are located in the three uppermost spans between the ZIRLO mixing vane structural grids and incorporate a similar mixing vane array. Their prime function is mid-span flow mixing in the hottest fuel assembly spans. Each IFM grid cell contains four dimples which are designed to prevent mid-span channel closure in the spans containing IFMs and fuel rod contact with the mixing vanes. This simplified cell arrangement allows short grid cells so that the IFM grid can accomplish its flow mixing objective with minimal pressure drop.

The IFM grids are not intended to be structural members. The outer strap configuration was designed similar to current fuel designs to preclude grid hang-up and damage during fuel handling. Additionally, the grid outer dimensions are smaller which further minimizes the potential for damage and reduces calculated forces during seismic/LOCA events. A coolable

geometry is, therefore, assured of the IFM grid elevation, as well as at the structural grid elevation.

The protective grid is made of Inconel 718. The protective grid is not intended to be a structural member.

Later, the “alternate” protective grid as used at Comanche Peak was developed to address plant specific concerns of Regulatory Guide 1.82, which requires that no passageway in the fuel assembly be less than the size of the mesh in the containment sump filter. The alternate protective grid design meets the regulatory guide by using an inner strap that is 0.1 inch shorter than the previous protective grid. The 0.1 inch is taken off the bottom of the inner straps and has the effect of raising the protective grid an additional 0.1 inch off the bottom nozzle.

Westinghouse has developed the Robust Protective Grid (RPG) as a result of observed failures in the field as noted in Post Irradiation Exams (PIE) performed at several different plants. It was determined that observed failures were the result of two primary issues; 1) fatigue failure within the protective grid itself at the top of the end strap and 2) stress corrosion cracking (SCC) primarily within the rod support dimples. The RPG implemented design changes such as increasing the maximum nominal height of the grid, increasing the ligament length and the radii of the ligament cutouts, and the use of four additional spacers or inserts to help strengthen the grid. The nominal height of the grid was increased to allow “V-notch” window cutouts to be added to help minimum flow-induced vibration caused by vortex shedding at the trailing edge of the inner grid straps. The design changes incorporated into the RPG design helped address the issues of fatigue failures and failures due to SCC. It was determined that the above changes do not impact the thermal hydraulic performance of the RPG as there is no change to the loss coefficient. In addition, the RPG retains the original protective grid function as a debris mitigation feature. The RPG was implemented at Comanche Peak beginning with Unit 2 Cycle 13 and Unit 1 Cycle 16.

4.2.2.3 Incore Control Components

Reactivity control is provided by neutron absorbing rods and a soluble chemical neutron absorber (boric acid). The boric acid concentration is varied to control long term reactivity changes such as:

1. Fuel depletion and fission product buildup.

2. Cold to hot, zero power reactivity change.

3. Reactivity change produced by intermediate term fission products such as xenon and samarium.

4. Burnable poison depletion.

Chemical and volume control is discussed in Chapter 9.

The rod cluster control assemblies provide reactivity control for:

1. Shutdown

2. Reactivity changes due to coolant temperature changes in the power range.

3. Reactivity changes associated with the power coefficient or reactivity.

4. Reactivity changes due to void formation.

Burnable absorbers may also be used for reactivity control. As the boron concentrations in the reactor coolant system is increased, the moderator temperature coefficient becomes more positive. The use of a soluble absorber alone could result in an unacceptably positive modertor temperature coefficient at beginning-of-life. Therefore, burnable absorbers may be used to reduce the soluble boron concentration sufficiently to ensure that the moderator temperature coefficient is acceptable for power operating conditions.

The most effective reactivity control components are the full length rod cluster control assemblies and their corresponding control rod drive mechanisms (CRDMs) which are the only moving parts in the reactor. Figure 4.2-8 identifies the full length rod cluster control and CRDM assembly, in addition to the arrangement of these components in the reactor relative to the interfacing fuel assembly and guide tubes. In the following paragraphs, each reactivity control component is described in detail. The CRDM assembly is described in Section 3.9N.4.

The neutron source assemblies provide a means of monitoring the core during periods of low neutron activity. The thimble plug assemblies limit bypass flow through those fuel assembly thimbles which do not contain control rods, burnable poison rods, or neutron source rods.

4.2.2.3.1 Full Length Rod Cluster Control Assembly

The full length rod cluster control assemblies are divided into two categories: control and shutdown. The control groups compensate for reactivity changes due to variations in operating conditions of the reactor, i.e., power and temperature variations. Two nuclear design criteria have been employed for selection of the control group. First the total reactivity worth must be adequate to meet the nuclear requirements of the reactor. Second, in view of the fact that these rods may be partially inserted at power operation, the total power peaking factor should be low enough to ensure that the power capability is met. The control and shutdown group provides adequate shutdown margin.

The rod cluster control assembly comprises a group of individual neutron absorber rods fastened at the top end to a common spider assembly, as illustrated in Figure 4.2-9.

The absorber material used in the control rod is silver-indium-cadium alloy. The absorber materials are essentially “black” to thermal neutrons and have sufficient additional resonance absorption to significantly increase their worth. The absorber is in the form of rods which are sealed in cold-worked, chromium plated, stainless steel tubes to prevent the absorber materials from coming in direct contact with the coolant (Figure 4.2-10). Sufficient diametral and end clearance is provided to accommodate relative thermal expansions and material swelling, as shown in Section 4.2.3.6.

The bottom plugs are made bullet-nosed to reduce the hydraulic drag during reactor trip and to guide smoothly into the dashpot section of the fuel assembly guide thimbles.

The material used in the absorber rod end plugs is Type 308 stainless steel. The design stresses used for the Type 308 material are the same as those defined in the ASME Code, Section III, for Type 304 stainless steel. At room temperature the yield and ultimate stresses per ASTM-580 are exactly the same for the two alloys. In view of the similarity of the alloy composition, the temperature dependence of strength for the two materials is also assumed to be the same.

The allowable stresses used as a function of temperature are listed in Table 1-1.2 of Section III of the ASME Boiler and Pressure Vessel Code. The fatigue strength for the Type 308 material is based on the S-N curve for austenitic stainless steels in Figure 1-9.2 of Section III.

The spider assembly is in the form of a central hub with radial vanes containing cylindrical fingers from which the absorber rods are suspended. Handling detents and detents for connection to the drive rod assembly are machined into the upper end of the hub. A coil spring inside the spider body absorbs the impact energy at the end of a trip insertion. The radial vanes are joined to the hub by tack weld and braze and the fingers are joined to the vanes by brazing. A centerpost which holds the spring and its retainer is threaded into the hub within the skirt and welded to prevent loosening in service. All components of the spider assembly are made from Types 304 and 308 stainless steel except for the retainer which is fabricated of 17-4 pH material and the springs which are Inconel-718 alloy.

The absorber rods are fastened securely to the spider to assure trouble free service. The rods are first threaded into the spider fingers and then pinned to maintain joint tightness, after which the pins are tack welded in place. The end plug below the pin position is designed with a reduced section to permit flexing of the rods to correct for small operating or assembly misalignments.

The overall length is such that when the assembly is withdrawn through its full travel the tips of the absorber rods remain engaged in the guide thimbles so that alignment between rods and thimbles is always maintained. Since the rods are long and slender, they are relatively free to conform to any small misalignments with the guide thimble.

4.2.2.3.2 Burnable Poison Assembly

The poison rods in each fuel assembly are grouped and attached together at the top end of the rods to a hold down assembly by a flat perforated retaining plate which fits within the fuel assembly top nozzle and rests on the adaptor plate. The retaining plate and the poison rods are held down and restrained against vertical motion through a spring pack which is attached to the plate and is compressed by the upper core plate when the reactor upper internals assembly is lowered into the reactor. This arrangement ensures that the poison rods cannot be ejected from the core by flow forces. Each rod is permanently attached to the base plate by a nut which is secured into place.

Wet annular burnable absorber (WABA) rodlets consisting of $B_4C - Al_2O_3$ pellets are used in selected fresh Westinghouse fuel depending upon the fuel management requirements for the cycle to shape the power distribution and achieve a desirable moderator temperature coefficient. Additionally, selected fresh Westinghouse fuel will contain Integral Fuel-Burnable Absorber (IFBA) pellets, which also act as a burnable absorber. These fuel pellets are coated with a thin layer of ZrB_2. The WABA and IFBA rods typically contain 120 inches of absorber material, axially

centered in the active fuel region, but the length may vary depending on the fuel management requirements of the cycle. Each burnable poison assembly consists of burnable poison rods attached to a hold down assembly. A typical burnable poison assembly is shown in Figure 4.2-12. When needed for nuclear considerations, burnable poison assemblies are inserted into selected thimbles within fuel assemblies.

The poison rods consist of absorber material contained within Zircaloy tubular cladding which is plugged and seal welded at the ends to encapsulate the absorber.

The absorber provides sufficient boron content to meet the criteria discussed in Section 4.3.1.

4.2.2.3.3 Neutron Source Assembly

The purpose of the neutron source assembly is to provide a base neutron level to ensure that the detectors are operational and responding to core multiplication neutrons. Since there is very little neutron activity during loading, refueling, shutdown, and approach to criticality, a neutron source is placed in the reactor to provide a positive neutron count of at least 2 counts per second on the source range detectors attributable to core neutrons. The detectors, called source range detectors, are used primarily when the core is subcritical and during special subcritical modes of operations.

The source assembly also permits detection of changes in the core multiplication factor during core loading refueling, and approach to criticality. This can be done since the multiplication factor is related to an inverse function of the detector count rate. Therefore a change in the multiplication factor can be detected during addition of fuel assemblies while loading the core, a change in control rod positions, and changes in boron concentration.

Both primary and secondary neutron source rods can be used. A primary source rod, containing a radioactive material, spontaneously emits neutrons during initial core loading and reactor startup. After the primary source rod decays beyond the desired neutron flux level, neutrons are then supplied by the secondary source rod. The secondary source rod contains a stable material, which must be activated by neutron bombardment during reactor operation. The activation results in the subsequent release of neutrons. This becomes a source of neutrons during periods of low neutron flux, such as during refueling and subsequent startups.

The reactor cores employ a sufficient number of secondary source assemblies (typically two) to ensure that the required base neutron level is provided.

Each secondary source assembly contains a symmetrical grouping of secondary source rods. Locations not filled with a source rod contain a thimble plug. A typical source assembly is shown in Figure 4.2-15.

Neutron source assemblies are employed at opposite sides of the core. The assemblies are inserted into the rod cluster control guide thimbles in fuel assemblies at selected unrodded locations.

As shown in Figure 4.2-15, the source assemblies contain a holddown assembly similar to that of the burnable poison assembly.

The primary and secondary source rods utilize the same cladding material as the absorber control rods. The secondary source rods contain Sb-Be pellets stacked to a height of approximately 88 inches. The primary source rods contain capsules of californium (Pu-Be possible alternate) source material and alumina spacer pellets to position the source material within the cladding. The rods in each assembly are permanently fastened at the top end to a holddown assembly.

The other structural members are constructed of Type 304 stainless steel except for the springs. The springs exposed to the reactor coolant are Inconel 718.

4.2.2.3.4 Thimble Plug Assembly

In order to limit bypass flow through the rod cluster control guide thimbles in fuel assemblies which do not contain either control rods, source rods, or burnable poison rods, the fuel assemblies are fitted with thimble plug assemblies at those locations.

A typical thimble plug assembly is shown in Figure 4.2-16 and consists of a flat base plate with short rods suspended from the bottom surface and a spring pack assembly. The 24 short rods, called thimble plugs, project into the upper ends of the guide thimbles to reduce the bypass flow.

Each thimble plug is permanently attached to the base plate by a nut which is secured to the threaded end of the plug. Similar short rods are also used on the source assemblies and burnable poison assemblies to plug the ends of all vacant fuel assembly guide thimbles. At installation in core, the thimble plug assemblies interface with both the upper core plate and with the fuel assembly top nozzles by resting on the adaptor plate. The spring pack is compressed by the upper core plate when the upper internals assembly is lowered into place.

All components in the thimble plug assembly, except for the springs, are constructed from Type 304 stainless steel. The springs are Inconel-718.

4.2.3 DESIGN EVALUATION

The fuel assemblies, fuel rods and incore control components are designed to satisfy the performance and safety criteria of Section 4.2, the mechanical design bases of Section 4.2.1, and other interfacing nuclear and thermal-hydraulic design bases specified in Sections 4.3 and 4.4. Effects of Conditions II, III, IV or anticipated transients without trip on fuel integrity are presented in Chapter 15 or the supporting topical reports.

A representative Fuel System Design Evaluation for fuel and components supplied by Westinghouse is provided in this section.

The initial step in fuel rod design evaluation for a region of fuel is to determine the limiting rod(s). Limiting rods are defined as those rod(s) whose predicted performance provides the minimum margin to each of the design criteria. For a number of design criteria the limiting rod is the lead burnup rod of a fuel region. In other instances it may be the maximum power or the minimum burnup rod. For the most part, no single rod will be limiting with respect to all design criteria.

After identifying the limiting rod(s), a worst-case evaluation is made which utilizes the limiting rod design basis power history and considers the effects of model uncertainties and dimensional variations. Furthermore, to verify adherence to the design criteria, the conservative case

evaluation also considers the effects of postulated transient power increases which are achievable during operation consistent with Conditions I and II. These transient power increases can affect both rod average and local power levels. The analytical methods used in the evaluation result in performance parameters which demonstrate the fuel rod behavior. Examples of parameters considered include rod internal pressure, fuel temperature, clad stress, and clad strain. In fuel rod design analyses these performance parameters provide the basis for comparison between expected fuel rod behavior and the corresponding design criteria limits.

Fuel rod and assembly models used for the various evaluations are documented and maintained under an appropriate control system. Materials properties used in the design evaluations are given in Reference [2] and [9].

4.2.3.1 Cladding

1. Vibration and wear

   Fuel rod vibrations are flow induced. The effect of the vibration on the fuel assembly and individual fuel rods is minimal. The cyclic stress range associated with deflections of such small magnitude is insignificant and has no effect on the structural integrity of the fuel rod.

   The reaction force on the grid supports due to rod vibration motions is also small and is much less than the spring preload. Firm fuel clad spring contact is maintained. No significant wear of the clad or grid supports is expected during the life of the fuel assembly based on out-of-pile flow tests performance of similarly designed fuel in operating reactors, and design analysis.

   Clad fretting and fuel vibration have been experimentally investigated as shown in Reference 20 for the Westinghouse optimized fuel.

2. Fuel rod internal pressure and cladding stresses

   The burnup dependent fission gas release model (Reference 5) is used in determining the internal gas pressures as a function of irradiation time. The plenum height of the fuel rod has been designed to ensure that the maximum internal pressure of the fuel rod will not exceed the value which would cause the fuel-clad diametral gap to increase during steady state operation.

   The clad stresses at a constant local fuel rod power are low. Compressive stresses are created by the pressure differential between the coolant pressure and the rod internal gas pressure. Because of the pre-pressurization with helium, the volume average effective stresses are always less than approximately 10,000 psi at the pressurization level used in this fuel rod design. Stresses due to the temperature gradient are not included in this average effective stress because thermal stresses are, in general, negative at the clad inside diameter and positive at the clad outside diameter and their contribution to the clad volume average stress is small. Furthermore, the thermal stress decreases with time during steady state operation due to stress relaxation. The stress due to pressure differential is highest in the minimum power rod at the beginning-of-life due to low internal gas pressure and the thermal stress is highest in the maximum power rod due to steep

temperature gradient.

Tensile stresses could be created once the clad has come in contact with the pellet. These stresses would be induced by the fuel pellet swelling during irradiation. Fuel swelling can result in small clad strains (< 1 percent) for expected discharge burnups but the associated clad stresses are very low because of clad creep (thermal and irradiation-induced creep). Furthermore, the 1 percent strain criterion is extremely conservative for fuel-swelling driven clad strain because the strain rate associated with solid fission products swelling is very slow.

3. Materials and chemical evaluation

Zircaloy-4 and ZIRLO clad have a high corrosion resistance to the coolant, fuel and fission products. As shown in References [1] and [9], there is considerable PWR operating experience on the capability of Zircaloy-4 and ZIRLO as clad material. Controls on fuel fabrication specify maximum moisture levels to preclude clad hydriding.

Metallographic examination of irradiated commercial fuel rods has shown occurrences of fuel/clad chemical interaction. Reaction layers of < 1 mil in thickness have been observed between fuel and clad at limited points around the circumference. Metallographic data indicates that this interface layer remains very thin even at high burnup. Thus, there is no indication of propagation of the layer and eventual clad penetration.

4. Fretting

Cladding fretting has been experimentally investigated as shown in Reference [9 and 20]. No significant fretting of the cladding is expected during the life of the fuel assembly.

5. Stress Corrosion

Stress corrosion cracking is another postulated phenomenon related to fuel/clad chemical interaction. Out of pile tests have shown that in the presence of high clad tensile stresses, large concentrations of iodine can chemically attack the Zircaloy-4 and ZIRLO tubing and can lead to eventual clad cracking. Extensive post irradation examination has produced no in pile evidence that this mechanism is operative in commercial fuel.

6. Cycling and Fatigue

A comprehensive review of the available strain fatigue models was conducted by Westinghouse as early as 1968. This review include the Langer-O'Donnell model (Ref. [13], the Yao-Munse model and the Manson-Halford model. Upon completion of this review and using the results of the Westinghouse experimental programs, it was concluded that the approach defined by Langer-O'Donnell would be retained and the emperical factors of their correlation bound the results of the Westinghouse testing program.

7. Rod bowing

Reference [12] presents the NRC approved model used for the evaluation of fuel rod bowing. The effects of rod bowing on DNBR are described in Section 4.4.2.1.3.

8. Consequences of power-coolant mismatch

This subject is discussed in Chapter 15.

9. Irradiation Stability of the Cladding

As shown in Reference [1], there is considerable PWR operating experience on the capability of Zircaloy as a cladding material. Extensive experience with irradiated Zircaloy-4 is summarized in Reference [2], and in the Appendices A-E in Reference [9] for ZIRLO.

10. Creep collapse and creepdown

This subject and the associated irradiation stability of cladding have been evaluated using the models described in Reference [6] and [29].

It has been established that the design basis of no clad collapse during planned core life can be satisfied by limiting fuel densification and by having a sufficiently high initial internal rod pressure.

4.2.3.2 Fuel Materials Consideration

Sintered, high density uranium dioxide fuel reacts only slightly with the clad, at core operating temperatures and pressures. In the event of clad defects, the high resistance of uranium dioxide to attack by water protects against fuel deterioration although limited fuel erosion can occur. As has been shown by operating experience and extensive experimental work, the thermal design parameters conservatively account for changes in the thermal performance of the fuel elements due to pellet fracture which may occur during power operation. The consequences of defects in the clad are greatly reduced by the ability of uranium dioxide to retain fission products including those which are gaseous or highly volatile. Observations have shown that fuel pellets can densify under irradiation to a density higher than the manufactured values. Fuel densification and subsequent settling of the fuel pellets can result in local and distributed gaps in the fuel rods. Fuel densification has been minimized by improvements in the fuel manufacturing process and by specifying a nominal 95 percent initial fuel density.

The evaluation of fuel densification effects and their consideration in fuel design are described in References [4], [5] and [10]. The treatment of fuel swelling and fission gas release are described in References [5] and [10].

Waterlogging considerations on fuel behavior are discussed in Section 4.2.3.3.

4.2.3.3 Fuel Rod Performance

In calculating the steady state performance of a nuclear fuel rod, the following interacting factors are considered:

1. Clad creep and elastic deflection.

2. Pellet density changes, thermal expansion, gas release, and thermal properties as a function of temperature and fuel burnup.

3. Internal pressure as a function of fission gas release, rod geometry, and temperature distribution.

These effects are evaluated using a fuel rod design model [5].

The model modifications for time dependent fuel densification are given in References [4] and [5]. With these interacting factors considered, the model determines the fuel rod performance characteristics for a given rod geometry, power history, and axial power shape. In particular, internal gas pressure, fuel and clad temperatures, and clad deflections are calculated. The fuel rod is divided into several axial sections and radially into a number of annular zones. Fuel density changes are calculated separately for each segment. The effects are integrated to obtain the internal rod pressure.

The initial rod internal pressure is selected to delay fuel/clad mechanical interaction and to avoid the potential for flattened rod formation. It is limited, however, by the design criteria for the rod internal pressure, Section 4.2.1.3.

The gap conductance between the pellet surface and the clad inner diameter is calculated as a function of the composition, temperature, and pressure of the gas mixture, and the gap size of contact pressure between clad and pellet. After computing the fuel temperature for each pellet annular zone, the fractional fission gas release is assessed using an empirical model derived from experimental data [5]. The total amount of gas released is based on the average fractional release within each axial and radial zone and the gas generation rate which in turn is a function of burnup. Finally, the gas released is summed over all zones and the pressure is calculated.

The code shows good agreement in fit for a variety of published and proprietary data on fission gas release, fuel temperatures and clad deflections (Refs. [5] and [10]). Included in this spectrum are variations in power, time, fuel density, and geometry.

1. Fuel-cladding mechanical interaction

   One factor in fuel element duty is potential mechanical interaction of fuel and clad. This fuel/clad interaction produces cyclic stresses and strains in the clad, and these in turn consume clad fatigue life. The reduction of fuel/clad interaction is therefore a goal of design. In order to achieve this goal and to enhance the cyclic operational capability of the fuel rod, the technology for using pre-pressurized fuel rods in Westinghouse PWRs has been developed.

   Initially the gap between the fuel and clad is sufficient to prevent hard contact between the two. However, during power operation a gradual compressive creep of the clad onto the fuel pellet occurs due to the external pressure exerted on the rod by the coolant. Clad compressive creep eventually results in the fuel/clad contact. During this period of fuel/clad contact, changes in power level could result in changes in clad stresses and strains. By using pre-pressurized fuel rods to partially offset the effect of the coolant external pressure, the rate of clad creep toward the surface of the fuel is reduced. Fuel rod pre-pressurization delays the time at which fuel/clad interaction and contact occur and hence significantly reduces the number and extent of cyclic stresses and strains

experienced by the clad both before and after fuel/clad contact. These factors result in an increase in the fatigue life margin of the clad and lead to greater clad reliability. If gaps should form in the fuel stacks, clad flattening will be prevented by the rod pre-pressurization so that the flattening time will be greater than the fuel core life.

A two dimensional (r, θ) finite element model has been established to investigate the effects of radial pellet cracks on stress concentrations in the clad. Stress concentration, herein, is defined as the difference between the maximum clad stress in the θ direction and the mean clad stress. The first case has the fuel and clad in mechanical equilibrium and as a result the stress in the clad is close to zero. In subsequent cases the pellet power is increased in steps and the resultant fuel thermal expansion imposes tensile stress in the clad. In addition to uniform clad stresses, stress concentrations develop in the clad adjacent to radial cracks in the pellet. These radial cracks have a tendency to open during a power increase but the frictional forces between fuel and clad oppose the opening of these cracks and result in localized increases in clad stress. As the power is further increased, large tensile stresses exceed the ultimate tensile strength of UO_2 and additional cracks in the fuel are created which limits the magnitude of the stress concentration in the clad.

As part of the standard fuel rod design analysis, the maximum stress concentration evaluated from finite element calculations is added to the volume averaged effective stress in the clad as determined from one dimensional stress/strain calculations. The resultant clad stress is then compared to the temperature dependent Zircaloy/ZIRLO yield stress in order to assure that the stress/strain criteria are satisfied.

Transient Evaluation Method

Pellet thermal expansion due to power increases is considered the only mechanism by which significant stresses and strains can be imposed on the clad. Power increases in commercial reactors can result from fuel shuffling (e.g., Region 3 positioned near the center of the core for Cycle 2 operation after operating near the periphery during Cycle 1), reactor power escalation following extended reduced power operation, and full length control rod movement. In the mechanical design model, lead rods are depleted using best estimate power histories as determined by core physics calculations. During the depletion, the amount of diametral gap closure is evaluated based upon the pellet expansion-cracking model, clad creep model, and fuel swelling model. At various times during depletion, the power is increased locally on the rod to the burnup dependent attainable power density as determined by core physics calculations. The radial, tangential and axial clad stresses resulting from the power increase are combined into a volume average effective clad stress.

2. Irradiation experience

   Westinghouse fuel operational experience is presented in Reference [1].

3. Fuel and cladding temperature

   The methods used for evaluation of fuel rod temperatures are presented in Section 4.4.2.11.

4. Waterlogging

Local cladding deformations typical of water-logging[b] bursts have never been observed in commercial Westinghouse fuel. Experience has shown that the small number of rods which have acquired clad defects, regardless of primary mechanism, remain intact and do not progressively distort or restrict coolant flow. In fact such small defects are normally observed through reductions in coolant activity to be progressively closed upon further operation due to the buildup of zirconium oxide and other substances. Secondary failures which have been observed in defected rods are attributed to hydrogen embrittlement of the cladding. Post-irradiation examinations point to the hydriding failure mechanism rather than a waterlogging mechanism; the secondary failures occur as axial cracks in the cladding and are similar regardless of the primary failure mechanism. Such cracks do not result in flow blockage. Hence, the presence of such fuel, the quantity of which must be maintained below Technical Specification limits, does not in any way exacerbate the effects of any postulated transients.

Zircaloy clad fuel rods which have failed due to waterlogging (Refs. [14] and [15] indicate that very rapid power transients are required for fuel failure. Normal operational transients are limited to about 40 cal/gm-min. (peak rod), while the Spert tests [14] indicate that 120 cal/gm to 150 cal/gm is required to rupture the clad even with very short transients (5.5 msec period).

Local cladding deformations typical of waterlogging bursts have never been observed in commercial Westinghouse fuel. Secondary failures which have been observed in defected rods are attributed to hydrogen embrittlement of the cladding. Post-irradiation examinations point to the hydriding failure mechanism rather than a waterlogging mechanism; the secondary failures occur as axial cracks in the cladding and are similar regardless of the primary failure mechanism. Such cracks, of course, do not result in flow blockage. Therefore, waterlogging is not considered to be a concern and operation with PCI defects poses no problem as long as the coolant activity remains below established levels.

5. Potentially damaging temperature effects during transients

The fuel rod experiences many operational transients (intentional maneuvers) during its residence in the core. A number of thermal effects must be considered when analyzing the fuel rod performance.

The clad can be in contact with the fuel pellet at some time in the fuel lifetime. Clad-pellet interaction occurs if the fuel pellet temperature is increased after the clad is in contact with the pellet. Clad-pellet interaction is discussed in Section 4.2.3.3.

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b. Waterlogging damage of a previously defected fuel rod has occasionally been postulated as a mechanism for subsequent rupture of the cladding. Such damage has been postulated as a consequence of a power increase on a rod after water has entered such a rod through a clad defect of appropriate size. Rupture is postulated upon power increase if the rod internal pressure increase is excessive due to insufficient venting of water to the reactor coolant.

The potential effects of operation with waterlogged fuel are discussed in Section 4.2.3.3 which concluded that waterlogging is not a concern during operational transients.

Clad flattening, as shown in Reference [6], has been observed in some operating power reactors. Thermal expansion (axial) of the fuel rod stack against a flattened section of clad could cause failure of the clad. This is no longer a concern because clad flattening is precluded during the fuel residence in the core (see Section 4.2.3.1).

Potential differential thermal expansion between the fuel rods and the guide thimbles during a transient is considered in the design. Excessive bowing of the fuel rods is precluded because the grid assemblies allow axial movement of the fuel rods relative to the grids. Specifically thermal expansion of the fuel rods is considered in the grid design so that axial loads imposed on the fuel rods during a thermal transient will not result in excessively bowed fuel rods.

6. Fuel element burnout and potential energy release

As discussed in Section 4.4.2.2, the core is protected from DNB over the full range of possible operating conditions. In the extremely unlikely event that DNB should occur, the clad temperature will rise due to the steam blanketing at the rod surface and the consequent degradation in heat transfer. During this time there is a potential for chemical reaction between the cladding and the coolant. However, because of the relatively good film boiling heat transfer following DNB, the energy release resulting from this reaction is insignificant compared to the power produced by the fuel.

7. Coolant flow blockage effects on fuel rods

This evaluation is presented in Section 4.4.4.7.

4.2.3.4 Spacer Grids

The coolant flow channels are established and maintained by the structure composed of grids and guide thimbles. The lateral spacing between fuel rods is provided and controlled by the support dimples of adjacent grid cells. Contact of the fuel rods on the dimples is maintained through the clamping force of the grid springs. Lateral motion of the fuel rods is opposed by the spring force and the internal moments generated between the spring and the support dimples. Grid testing is discussed in References 16 and 20.

As shown in References 16, 20, and 21, grid crushing tests and seismic and LOCA evaluations show that the grids will maintain a geometry that is capable of being cooled under the worst-case accident Condition IV event.

4.2.3.5 Fuel Assembly

4.2.3.5.1 Stresses and Deflections

The fuel assembly component stress levels are limited by the design. For example, stresses in the fuel rod due to thermal expansion and Zircaloy/ZIRLO irradiation growth are limited by the relative motion of the rod as it slips over the grid spring and dimple surfaces. Clearances between the fuel rod ends and nozzles are provided so that Zircaloy/ZIRLO irradiation growth

does not result in rod end interferences. Stresses in the fuel assembly caused by tripping of the rod cluster control assembly have little influence on fatigue because of the small number of events during the life of an assembly. Assembly components and prototype fuel assemblies made from production parts have been subjected to structural tests to verify that the design bases requirements are met.

The fuel assembly design loads for shipping have been established at 4 g's axially and 6 g's laterally and transverse for the Westinghouse optimized fuel. Accelerometers are permanently placed into the shipping cask to monitor and detect fuel assembly accelerations that would exceed the criteria. Past history and experience has indicated that loads which exceed the allowable limits rarely occur. Exceeding the limits requires reinspection of the fuel assembly for damage. Tests on various fuel assembly components such as the grid assembly, sleeves, inserts and structure joints have been performed to assure that the shipping design limits do not result in impairment of fuel assembly function. Seismic analysis of the fuel assembly is presented in Reference [16].

4.2.3.5.2 Dimensional Stability

A prototype fuel assembly has been subjected to column loads in excess of those expected in normal service and faulted conditions [16, 20].

No interference with control rod insertion into thimble tubes will occur during a postulated loss of coolant accident transient due to fuel rod swelling, thermal expansion, or bowing. In the early phase of the transient following the coolant break, the high axial loads, which could be generated by the difference in thermal expansion between fuel clad and thimbles, are relieved by slippage of the fuel rods through the grids. The relatively low drag force restraint on the fuel rods will induce only minor thermal bowing, which is insufficient to close the fuel rod-to-thimble tube gap.

References [16] and [20] shows that the fuel assemblies will maintain a geometry amenable to cooling during a combined seismic and double ended LOCA.

4.2.3.6 Incore Control Components

The components are analyzed for loads corresponding to normal, upset, emergency and faulted conditions. The analysis performed depends on the mode of operation under consideration.

The scope of the analysis requires many different techniques and methods, both static and dynamic.

Some of the loads that are considered on each component where applicable are as follows:

1. Control rod trip (equivalent static load).

2. Differential pressure.

3. Spring preloads.

4. Coolant flow forces (static).

5. Temperature gradients.

6. Differences in thermal expansion

   a. Due to temperature differences.

   b. Due to expansion of different materials.

7. Interference between components.

8. Vibration (mechanically or hydraulically induced).

9. Operational transients.

10. Pump overspeed.

11. Seismic loads (Operating Basis Earthquake and Safe Shutdown Earthquake).

12. Blowdown forces (due to cold or hot leg break).

13. Material swelling and gas generation pressure.

The main objective of the analysis is to satisfy allowable stress limits, to assure an adequate design margin, and to establish deformation limits which are concerned primarily with the functioning of the components. The stress limits are established not only to assure that peak stresses will not reach unacceptable values, but also limit the amplitude of the oscillatory stress component in consideration of fatigue characteristics of the materials. Standard methods of strength of materials are used to establish the stresses and deflections of these components. The dynamic behavior of the reactivity control components has been studied using experimental test data and experience from operating reactors.

Sufficient diametral and end clearances have been provided in the neutron absorber, burnable poison, and source rods to accommodate the relative thermal expansions and material swelling between the enclosed material and the surrounding clad and end plugs. There is no bending or warping induced in the rods although the clearance offered by the guide thimble would permit a postulated warpage to occur if there were no restraint on the rods. Bending, therefore, is not considered in the analysis of the rods.

Experience with incore control components is discussed in Reference [1]. Materials data and evaluations are given in References [2] and [23].

1. Mechanical strength evaluation

   The design of incore component rods provides a sufficient cold void volume within the burnable poison and source rods to limit the internal pressures to a value which satisfies the criteria in Section 4.2.1.6. A gas plenum at the top of source and burnable poison rods provides void volume for the pressure buildup. This is not a concern for the all Ag-In-Cd control rod designs since no gases are generated by this absorber material. The void volume for the helium in the burnable poison rods is also obtained through the use of glass in tubular form which provides a central void along the length of the rods. Helium gas is not released by the Ag-In-Cd neutron absorber rod material. The internal pressure of source rods continues to increase from ambient until end of life. The stress

analysis of reactivity component rods assumes 100 percent gas release to the rod void volume and satisfies the criteria in Section 4.2.1.6.

Based on available data for properties of the borosilicate glass and on nuclear and thermal calculations for the standard burnable poison rods, gross swelling or cracking of the glass tubing is not expected during operation. Some minor creep of the glass at the hot spot on the inner surface of the tube could occur but would continue only until the glass came in contact with the inner liner. The wall thickness of the inner liner is sized to provide adequate support in the event of slumping and to collapse locally before rupture of the exterior cladding if unexpected large volume changes due to swelling or cracking should occur. The top of the inner liner is open to allow communication to the central void by helium which diffuses out of the glass.

The designs of the Wet Annular Burnable Absorber (WABA) provide a sufficient cold void volume to accommodate the internal pressure increase during operation. An annular void volume is provided between the two tubes at the top and along the length of the rods. The stress analysis of the WABA rods assumes a 30% maximum gas release (Reference 19). During normal transient and accident conditions, the void volume limits the internal pressures to values which satisfy the subsection 4.2.1.6.

Analysis on the full length rod cluster control spider indicates the spider is structurally adequate to withstand the various operating loads including the higher loads which occur during the drive mechanism stepping action and rod drop. Experimental verification of the spider structural capability has been completed.

2. Thermal evaluation

The radial and axial temperature profiles have been determined by considering gap conductance, thermal expansion, and neutron or gamma heating of the contained material as well as gamma heating of the clad. The maximum neutron absorber material temperature was found to be less than 850ºF for the Ag-In-Cd which occur axially at only the highest flux region. The maximum standard poison assembly borosilicate glass temperature was calculated to be about 1200ºF and less than 1200ºF for WABA's. This takes place following the initial rise to power. The glass temperature then decreases rapidly for the following reasons: 1) reduction in power generation due to B10 depletion; 2) better gap conductance as the helium produced diffuses to the gap; and 3) external gap reduction (standard poison assembly only) due to boronsilicate glass creep. Rod, guide thimble, and dashpot flow analysis indicates that the flow is sufficient to prevent coolant boiling and maintain clad temperatures at which the clad material has adequate strength to resist coolant operating pressures and rod internal pressures.

3. Irradiation and chemical evaluation

The materials selected are considered to be the best available from the standpoint of resistance to irradiation damage and compatibility to the reactor environment. The materials selected partially dictate the reactor environment (e.g., C1- control in the coolant). The irradiation stability of the absorber material is discussed in Reference [2 and 19]. Irradiation produces no deleterious effects on Ag-In-Cd neutron absorbers.

At high fluences the austenitic materials increase in strength with a corresponding decreased ductility (as measured by tensile tests) but energy absorption (as measured by impact tests) remain quite high. Corrosion of the materials exposed to the coolant is quite low and proper chemical control in the coolant will prevent the occurrence of stress corrosion. All of the austenitic stainless steel base materials used are processed and fabricated by furnace brazing, the procedure used requires that the pieces be rapidly cooled so that the time-at-temperature is minimized. The time that is spent by the control rod spiders in the sensitization range, 800-1500ºF, during fabrication is controlled to preclude sensitization. The 17-4 pH parts are all aged at the highest standard aging temperature of 1100ºF to avoid stress corrosion problems exhibited by aging at lower temperatures.

Based on the preceding considerations, it is judged that the potential for interference with rod cluster control movement due to possible corrosion phenomena is very low. Additional information on irradiation and chemical effects is given in Reference [2].

4. Failure evaluations

Analysis of the full length rod cluster control assemblies show that if the drive mechanism housing ruptures, the rod cluster control assembly will be ejected from the core by the pressure across the drive rod assembly. The ejection is also predicted on the failure of the drive mechanism to retain the drive rod/rod cluster control assembly position.

It should be emphasized that a drive mechanism housing rupture will cause the ejection of only one rod cluster control assembly with the other assemblies remaining in the core. Analysis shows that a pressure drop in excess of 4000 psi must occur across a two-fingered vane to break the vane/spider body joint causing ejection of two neutron absorber rods from the core. Since the greatest pressure drop in the system is only 2250 psi, a pressure drop in excess of 4000 psi is not credible. Thus, the ejection of the neutron absorber rods is not possible.

Ejection of a burnable poison or thimble plug assembly is conceivable based on the postulation that the hold down bar fails and that the base plate and burnable poison rods are severely deformed. In the unlikely event that failure of the hold down bar occurs, the upward displacement of the burnable poison assembly only permits the base plate to contact the upper core plate. Since this displacement is small, the major portion of the borosilicate glass tubing remains positioned within the core. In the case of the thimble plug assembly, the thimble plugs will partially remain in the fuel assembly guide thimbles thus maintaining a majority of the desired flow impedance. Further displacement or complete ejection would necessitate the square base plate and burnable poison rods be forced, thus plastically deformed, to fit up through a smaller diameter hole. It is expected that this condition requires a substantially higher force or pressure drop than that of the hold down bar failure.

The mechanical design of the reactivity control components provides for the protection of the active elements to prevent the loss of control capability and functional failure of critical components. The components have been reviewed for potential failure and consequences of a functional failure of critical parts. The results of the review are summarized below.

Full Length Rod Cluster Control Assembly

a. The basic absorbing materials are sealed from contact with the primary coolant and the fuel assembly and guidance surfaces by a high quality chromium plated stainless steel clad. Potential loss of absorber mass or reduction in reactivity control material due to mechanical or chemical erosion or wear is, therefore, reliably prevented.

b. The silver-indium-cadmium absorber material is relatively inert and would still remain remote from high coolant velocity regions for Conditions I, II and III. For Condition IV situations, refer to Reference 30 for assessment of control rod survivability.

c. The individually clad absorber rods are doubly secured to the retaining spider vane by a threaded joint and a welded lock pin. This joint has been qualified by functional testing and actual service in operating plants.

   It should also be noted that in several instances of control rod jamming caused by foreign particles, the individual rods at the site of the jam have borne the full capacity of the control rod drive mechanism and higher impact loads to dislodge the jam without failure. The conclusion to be drawn from this experience is that this joint is extremely insensitive to potential mechanical damage. A failure of the joint would result in the insertion of the individual rod into the core.

d. The spider finger braze joint by which the individual rods are fastened to the vanes has also experienced the service described above and been subjected to the same jam freeing procedures also without failure. A failure of this joint would also result in insertion of the individual rod into the core.

e. The radial vanes are attached to the spider body by a welded and brazed joint. The joints are designed to a theoretical strength in excess of that of the components joined.

   It is a feature of the design that the guidance of the rod cluster control is accomplished by the inner fingers of these vanes. They are, therefore, the most susceptible to mechanical damage. Since these vanes carry two rods, failure of the vane-to-hub joint does not prevent the free insertion of the rod pair. Neither does such a failure interfere with the continuous free operation of the drive line.

   Failure of the vane-to-hub joint of a single rod vane could potentially result in failure of the separated vane and rod to insert. This could occur only at withdrawal elevations where the spider is above the continuous guidance section of the guide tube (in the upper internals). A rotation of the disconnected vane could cause it to hang on one of the guide cards in the intermediate guide tube. Such an occurrence would be evident from the failure of the rod cluster control to insert below a certain elevation but with free motion above this point.

   This possibility is considered extremely remote because the single rod vanes are subjected to only vertical loads and very light lateral reactions from the rods. The consequences of such a failure are not considered critical since only one drive line

of the reactivity control system would be involved. This condition is readily observed and can be corrected at shutdown.

f. The spider hub being of single unit cylindrical construction is very rugged and of extremely low potential for damage. Rod cluster control assembly spider tests have verified the structural adequacy of the design including the vane-to-hub joint and spring pack load deflection tests (see Section 4.2.4.3). It is difficult to postulate condition to cause failure. Should some unforeseen event cause fracture of the hub above the vanes, the lower portion with the vanes and rods attached would insert by gravity into the core causing a reactivity decrease. The rod could then not be removed by the drive line. Fracture below the vanes cannot be postulated since all loads, including scram impact, are taken above the vane elevation.

g. The guide thimbles of the fuel assemblies provide a clear channel for insertion of the rod cluster control rods. Fuel rod failure due to postulated control rod contact is prevented by providing this physical barrier between the fuel rod and the intended insertion channel. Distortion of the fuel rods by bending cannot apply sufficient force to damage or significantly distort the guide thimble. Fuel rod distortion by swelling, though precluded by design, would be terminated by fracture before contact with the guide thimble occurs. If such were not the case, it would be expected that a force reaction at the point of contact would cause a slight deflection of the guide thimble. The radius of curvature of the deflected shape of the guide thimbles would be sufficiently large to have a negligible influence on rod cluster control insertion.

PWR operating experience has shown that fretting wear has occurred between control rods and guide thimble tubes. A report describing extensive post irradiation examinations of control rod guide thimble tubes in both reactor site and in hot cells [24], along with additional Westinghouse data [25], shows that the guide thimble wear is not a safety concern in Westinghouse facilities.

Burnable Poison Assemblies

The burnable poison assemblies are static temporary reactivity control elements. The axial position is assured by the hold down assembly which bears against the upper core plate. Their lateral position is maintained by the guide thimbles of the fuel assemblies.

The individual rods are shouldered against the underside of the retainer plate and securely fastened at the top by a threaded nut which is then locked in place. The square dimension of the retainer plate is larger than the diameter of the flow holes through the core plate. Failure of the hold down bar or spring pack, therefore, does not result in ejection of the burnable poison rods from the core.

The only incident that could potentially result in ejection of the burnable poison rods is a multiple fracture of the retainer plate. This is not considered credible because of the light loads borne by this component. During normal operation the loads borne by the plate are distributed at the points of attachment. Even a multiple fracture of the retainer plate would result in jamming of the plate

segments against the upper core plate, again preventing ejection. Excessive reactivity increase due to burnable poison ejection is, therefore, prevented.

4.2.4 TESTING AND INSPECTION PLAN

4.2.4.1 Quality Assurance Program

The Quality Assurance Program Plan of the Westinghouse Nuclear Fuel Division, as summarized in References [17] and [22], has been developed to serve the division in planning and monitoring its activities for the design and manufacture of nuclear fuel assemblies and associated components.

This program provides for control over all activities affecting product quality, commencing with design and development and continuing through procurement, materials handling, fabrication, testing and inspection, storage, and transportation. This program also provides for the indoctrination and training of personnel and for the auditing of activities affecting product quality through a formal auditing program.

4.2.4.2 Quality Control

Quality control philosophy is generally based on the following inspections being performed to a 95 percent confidence that at least 95 percent of the product meets specification, unless otherwise noted.

1. Fuel system components and parts

   The characteristics inspected depend upon the component parts and includes dimensional, visual, check audits of test reports, material certification and nondestructive examination such as X-ray and ultrasonic.

   All material used in this core is accepted and released by Quality Control.

2. Pellets

   Inspection is performed for dimensional characteristics such as diameter, density, length and squareness of ends. Additional visual inspections are performed for cracks, chips, missing pellet surfaces and surface conditions according to approved standards.

   Pellet densities are determined and compared to specification requirements. Chemical analyses are taken on a specified sample basis throughout pellet production.

3. Rod inspection

   Fuel rod, control rodlet, burnable poison and source rod inspection consists of the following nondestructive examination techniques and methods, as applicable:

   a. Leak testing

   b. Enclosure welds

c. Dimensional Inspections

d. Plenum dimensions

e. Pellet-to-pellet gaps

f. 100 percent of the fuel rods are active gamma scanned to verify enrichment control prior to acceptance for assembly loading.

g. Traceability

4. Assemblies

Each fuel, control rod, burnable poison and source rod assembly is inspected for drawing and/or specification requirements. Other incore control component inspection and specification requirements are given in Section 4.2.4.3.

5. Other inspections

The following inspections are performed as part of the routine inspection operation:

a. Tool and gage inspection and control. Tool inspection is performed at prescribed intervals on all serialized tools. Complete records are kept of calibration and conditions of tools.

b. Audits are performed of inspection activities and records to assure that prescribed methods are followed and that records are correct and properly maintained.

c. Surveillance inspection where appropriate, and audits of outside contractors are performed to ensure conformance with specified requirements.

6. Process control

To prevent the possibility of mixing enrichments during fuel manufacture and assembly, strict enrichment segregation and other process controls are exercised.

The UO_2 powder is kept in sealed containers which are fully identified by descriptive tagging and preselected color coding or, for the closed system, the material is monitored by a computer data management information system. For the sealed container system, a Westinghouse identification tag completely describing the contents is affixed to the containers before transfer to powder storage. Isotopic content is confirmed by analysis.

Finished pellets are placed on trays and transferred to segregated storage racks within the confines of the pelleting area. Samples from each pellet lot are tested for isotopic content and impurity levels prior to acceptance by Quality Control. Physical barriers prevent mixing of pellets of different nominal densities and enrichments in this storage area. Unused powder and substandard pellets are returned to storage in identified containers.

Pellets are loaded into fuel cladding tubes on isolated production lines. Each production line contains only rods of one type at any one time. A serialized traceability code is placed on each fuel rod which identifies the contract and enrichment. The end plugs are inserted and welded to seal the tube. The fuel tube remains coded, and traceability identified. The traceability code provide a cross reference of the fuel contained in the fuel rods.

At the time of installation into an assembly, a matrix is generated to identify each rod in its position within a given assembly. After the fuel rods are installed, all fuel rods in an assembly are verified to carry the correct identification character describing the fuel enrichment for the core region being fabricated. The top nozzle is inscribed with a permanent identification number providing traceability to the fuel contained in the assembly.

Similar traceability is provided for burnable poison, source rods and control rodlets as required.

4.2.4.3 Incore Control Component Testing and Inspection

Tests and inspections are performed on each reactivity control component to verify the mechanical characteristics. In the case of the full length rod cluster control assembly, prototype testing has been conducted and both manufacturing test/inspections and functional testing at the plant site are performed.

During the component manufacturing phase, the following requirements apply to the reactivity control components to assure the proper functioning during reactor operation:

1. All materials are procured to specifications to attain the desired standard of quality.

2. The spiders are load-tested to insure that the design load requirements are met.

3. All rods are checked for integrity by the methods described in Section 4.2.4.2.3.

4. To assure proper fitup with the fuel assembly, the rod cluster control, burnable poison and source assemblies are installed in the fuel assembly without restriction or binding in the dry condition. Also a straightness of 0.01 in/ft is required on the entire inserted length of each rod assembly.

The full length rod cluster control assemblies are functionally tested, following each refueling outage prior to criticality to demonstrate reliable operation of the assemblies. Each assembly is operated (and tripped) at full flow/hot conditions.

In order to demonstrate continuous free movement of the full length rod cluster control assemblies, and to ensure acceptable core power distributions during operations, partial movement checks are performed on every full length rod cluster control assembly every 31 days during reactor critical operation. In addition, periodic drop tests of the full length rod cluster control assemblies are performed at each refueling shutdown to demonstrate continued ability to meet trip time requirements, to ensure core subcriticality after reactor trip, and to limit potential reactivity insertions from a hypothetical rod cluster control assembly ejection.

If a rod cluster control assembly cannot be moved by its mechanism, adjustments in the boron concentration ensure that adequate shutdown margin would be achieved following a trip. Thus inability to move one rod cluster control assembly can be tolerated. More than one inoperable rod cluster control assembly could be tolerated, but would impose additional demands on the plant operator. Therefore, the number of inoperable rod cluster control assemblies has been limited to one.

4.2.4.4 Tests and Inspections by Others

If any tests and inspections are to be performed on behalf of Westinghouse, the fuel supplier will review and approve the quality control procedures, inspection plans, etc. to be utilized to ensure that they are equivalent to the description provided above and are performed properly to meet all fuel vendor's requirements.

4.2.4.5 In-Service Surveillance

Westinghouse conducts a program to examine detailed aspects of their fuel assembly performance. The Westinghouse program is described in Reference [7]. This document is periodically updated in order to provide recent results of operating experience with Westinghouse fuel and incore control components.

4.2.4.6 Onsite Inspection

Detailed written procedures are used by the station staff for the post shipment inspection of all new fuel and associated components such as control rods, plugs, and inserts. Fuel handling procedures specify the sequence in which handling and inspection takes place.

Loaded fuel containers, when received onsite, are externally inspected to ensure that labels and markings are intact and seals are unbroken. After the containers are opened, the shock indicators attached to the suspended internals are inspected to determine if movement during transit exceeded design limitations.

Following removal of the fuel assembly from the container in accordance with detailed procedures, the fuel assembly polyethylene wrapper is examined for evidence of damage. The polyethylene wrapper is then removed and a visual inspection of the entire bundle is performed.

Control rod assemblies are typically shipped in fuel assemblies and are inspected during fuel receipt operations. The control rod assembly is withdrawn from the fuel assembly to ensure free and unrestricted movement. The exposed section is then visibly inspected for mechanical integrity, replaced in the fuel assembly and stored with the fuel assembly.

4.2.4.7 Post-Irradiation Surveillance

To assure continued fuel integrity, a post-irradiation fuel surveillance program is conducted via visual examination of a sample of post-irradiated fuel assemblies through the use of visual aids, such as binoculars and/or underwater television cameras. During visual inspection, indications of fretting, crud, fuel assembly damage and anomalies are observed. If significant anomalies are encountered that would affect fuel performance, a more detailed examination of those fuel assemblies will be performed.

REFERENCES

1. "Operational Experience with Westinghouse Cores," WCAP-8183, latest revision.

2. Beaumont, M. D., (Ed.), "Properties of Fuel and Core Component Materials," WCAP-9179, Revision 1 (Proprietary), and WCAP-9224 (Non-Proprietary), July 1978 and Appendix A, "Hafnium" and Appendix B (AL_2O_3-B_4C) issued October 1980. Revised pages issued September 1982.

3. Christensen, J. A., Allio, R. J. and Biancheria, A., "Melting Point of Irradiated UO_2" WCAP-6065, February 1965.

4. Hellman, J. M. (Ed.), "Fuel Densification Experimental Results and Model for Reactor Operation," WCAP-8218-P-A (Proprietary) and WCAP-8219-A (Non-Proprietary), March 1975.

5. Foster, J. P., et.al., "Westinghouse Improved Performance Analysis and Design Model (PAD 4.0)," WCAP-15063-P-A, Revision 1, with Errata (Proprietary) and WCAP-15064-NP-A, Revision 1, with Errata (Non-Proprietary), July 2000.

6. George, R. A., et al., "Revised Clad Flattening Model," WCAP- 8377 (Proprietary) and WCAP-8381 (Non-Proprietary), July 1974.

7. Risher, D., et al., "Safety Analysis for the Revised Fuel Rod Internal Pressure Design Basis," WCAP-8963 (Proprietary), November 1976 and WCAP-8964 (Non-Proprietary), August 1977.

8. Cohen, J., "Development and Properties of Silver Base Alloys as Control Rod Materials for Pressurized Water Reactors," WAPD-214, December 1959.

9. Davidson, S. L., Ryan, T. L., "VANTAGE+ Fuel Assembly Reference Core Report," WCAP-12610-P-A and Appendices A-G, April 1995.

10. Weiner, R. Al, et. al., "Improved Fuel Performance Models for Westinghouse Fuel Rod Design and Safety Evaluations," WCAP-10851-P-A (Proprietary) and WCAP-11873-A (Non-Proprietary), August 1988.

11. Demario, E. E., "Hydraulic Flow Test of the 17 x 17 Fuel Assembly," WCAP-8278 (Proprietary) and WCAP-8279 (Non- Proprietary), February 1974.

12. Skartika, J., (Ed.), "Fuel Rod Bowing Evaluation," WCAP-8691, Rev. 1 (Proprietary) and WCAP-8692, Rev. 1 (Non-Proprietary), July 1979.

13. O'Donnell, W. J., and Langer, B. F., "Fatigue Design Basis for Zircaloy Components," Nuclear Science and Engineering, 20, 1-12, 1964.

14. Stephan, L. A., "The Effects of Cladding Material and Heat Treatment on the Response of Waterlogged UO_2 Fuel Rods to Power Bursts," IN-ITR-III, January 1970.

15. Western New York Nuclear Research Center Correspondence with the AEC on February 11 and August 27, 1971, Docket No. 50-57.

16. Gesinski, L., Chiang, D. and Nakazato, S., “Safety Analysis of the 17 x 17 Fuel Assembly for Combined Seismic and Loss of Coolant Accident,” WCAP-8236 (Proprietary) and WCAP-8288 (Non- Proprietary), December 1973.

17. Dollard, W. J., “Nuclear Fuel Division Quality Assurance Program Plan,” WCAP-7800, Revision 4-A, April 1971.

18. Not used.

19. Skaritka, J. et. al., “Westinghouse Wet Annular Burnable Absorber Evaluation Report,” WCAP-10021-P-A, Revision 1 (Proprietary), and WCAP-10377-A, dated October 1983.

20. Beaumont, M. D., et. al., (Ed.), “Verification Testing and Analysis of the 17x17 Optimized Fuel Assembly,” WCAP-9401 (Proprietary) and WCAP-9402, March 1979.

21. Davidson, S. L., Iu, J. A., “Reference Core Report - 17x17 Optimized Fuel Assembly,” WCAP-9500-A, May 1982.

22. Moore, J. S., “Nuclear Fuel Division Quality Assurance Program Plan,” WCAP-7800, Revision 5-A, November 1979.

23. T. M. Anderson (W), Westinghouse letter NS-TMA-1985, to J. F. Stolz (NRC), “Response to Questions on Properties of Fuel and Core Component Materials,” dated November 10, 1978.

24. T. M. Anderson (W), Westinghouse letter NS-TMA-1936, to D. G. Eisenhut (NRC), “Response to NRC Concerning the Integrity of Control Rod Guide Thimble (CRGT) in Westinghouse Facilities,” dated September 12, 1978 (Proprietary).

25. T. M. Anderson (W), Westinghouse letter NS-TMA-1992, to D. G. Eisenhut (NRC), “Response to Additional Information Requested on Westinghouse letter NS-TMA-1936,” dated December 15, 1978 (Proprietary).

26. Not used.

27. Not used.

28. Not used.

29. Kersting, P. J., et. al., “Assessment of Clad Flattening and Densification Power Spike Factor Elimation in Westinghouse Nuclear Fuel,” WCAP-13589-A, March 1995.

30. Letter from PWR Owners Group to Document Control Desk (NRC), “Assessment of Post LOCA Control Rod Survivability, PA-ASC-0313,” OG-07-17, January 9, 2007.

31. “Addendum 1 to WCAP-12488-A, Revision to Design Criteria,” WCAP-12488-A, Addendum 1-A Revision 1, January 2002.

32. Davidson, S. L. (ed.), et al., “Extended Burnup Evaluation of Westinghouse Fuel,” WCAP-10125-P-A, December 1985.

4.3 NUCLEAR DESIGN

4.3.1 DESIGN BASES

This section describes the design bases and functional requirements used in the nuclear design of the fuel and reactivity control system and relates these design bases to the General Design Criteria (GDC) in 10CFR50 Appendix A. Where appropriate, supplemental criteria such as the Final Acceptance Criteria for Emergency Core Cooling Systems are discussed. Before discussing the nuclear design bases, it is appropriate to briefly review the four major categories ascribed to conditions of plant operation.

The full spectrum of plant conditions is divided into four categories, in accordance with the anticipated frequency of occurrence and risk to the public:

1. Condition I - Normal Operation.

2. Condition II - Incidents of Moderate Frequency.

3. Condition III - Infrequent Faults.

4. Condition IV - Limiting Faults.

In general the Condition I occurrences are accommodated with margin between any plant parameter and the value of that parameter which would require either automatic or manual protective action. Condition II incidents are accommodated with, at most, a shutdown of the reactor with the plant capable of returning to operation after corrective action. Fuel damage (fuel damage as used here is defined as penetration of the fission product barrier i.e. the fuel rod clad) is not expected during Condition I and Condition II events. It is not possible, however, to preclude a very small number of rod failures. These are within the capability of the Chemical and Volume Control System (CVCS) and are consistent with the plant design basis.

Condition III incidents shall not cause more than a small fraction of the fuel elements in the reactor to be damaged, although sufficient fuel element damage might occur to preclude immediate resumption of operation. The release of radioactive material due to Condition III incidents should not be sufficient to interrupt or restrict public use of these areas beyond the exclusion radius. Furthermore, a Condition III incident shall not, by itself generate a Condition IV fault or result in a consequential loss of function of the reactor coolant or reactor containment barriers.

Condition IV occurrences are faults that are not expected to occur but are defined as limiting faults which must be designed against. Condition IV faults shall not cause a release of radioactive material that results in exceeding the limits of 10CFR100.

The core design power distribution limits related to fuel integrity are met for Condition I occurrences through conservative design and maintained by the action of the control system. The requirements for Condition II occurrences are met by providing an adequate protection system which monitors reactor parameters. The control and protection systems are described in Chapter 7 and the consequences of Condition II, III and IV occurrences are given in Chapter 15.

4.3.1.1 Fuel Burnup

Basis

The fuel rod design basis is described in Section 4.2. The nuclear design basis is to install sufficient reactivity in the fuel to attain the design region discharge burnup. The above along with the design basis in Section 4.3.1.3 satisfies GDC-10.

Discussion

Fuel burnup is a measure of fuel depletion which represents the integrated energy output of the fuel (MWD/MTU) and is a convenient means for quantifying fuel exposure criteria.

The core design lifetime or design discharge burnup is achieved by installing sufficient initial excess reactivity in each fuel region and by following a fuel replacement program (such as that described in Section 4.3.2) that meets all safety-related criteria in each cycle of operation.

Initial excess reactivity installed in the fuel, although not a design basis, must be sufficient to maintain core criticality at full power operating conditions throughout cycle life with equilibrium xenon, samarium, and other fission products present. The end of design cycle life is defined to occur when the chemical shim concentration is essentially zero.

A limitation on initial installed excess reactivity is not required other than as is quantified in terms of other design bases such as core negative reactivity feedback and shutdown margin discussed below.

4.3.1.2 Negative Reactivity Feedbacks (Reactivity Coefficient)

Basis

The fuel temperature coefficient will be negative and the moderator temperature coefficient of reactivity will be non-positive for full power operating conditions, thereby providing negative reactivity feedback characteristics. The design basis meets GDC-11.

Discussion

When compensation for a rapid increase in reactivity is considered, there are two major effects. These are the resonance absorption effects (Doppler) associated with changing fuel temperature and the spectrum effect resulting from changing moderator density. These basic physics characteristics are often identified by reactivity coefficients. The use of slightly enriched uranium ensures that the Doppler coefficient of reactivity is negative. This coefficient provides the most rapid reactivity compensation. The core is also designed to have an overall negative moderator temperature coefficient of reactivity at most operating conditions so that average coolant temperature or void content provides another, slower compensatory effect. Full power operation is permitted only in a range of overall non-positive moderator temperature coefficient. The non-positive moderator temperature coefficient can be achieved through use of fixed burnable absorber rods, and/or integral fuel burnable absorber (IFBA), and/or control rods by limiting the reactivity held down by soluble boron.

Burnable absorber content (quantity and distribution) is not stated as a design basis other than as it relates to accomplishment of a non-positive moderator temperature coefficient at power operating conditions discussed above.

4.3.1.3 Control of Power Distribution

Basis

The nuclear design basis is that:

1. The fuel will not be operated at greater than 14.51 kW/ft under normal operating conditions, including an allowance of 0.6% for calorimetric uncertainty.

2. With at least a 95 percent confidence level, there at least a 95% probability that, under abnormal conditions including the maximum overpower condition, the fuel peak power will not cause melting as defined in Section 4.4.1.2.

3. With at least a 95 percent confidence level, there is at least a 95% probability that the fuel will not operate with a power distribution that violates the departure from nucleate boiling (DNB) design basis discussed in Section 4.4.1 under Condition I and II events including the maximum overpower condition.

4. Fuel management will be such as to produce rod powers and burnups consistent with the assumptions in the fuel rod mechanical integrity analysis of Section 4.2.

The above basis meets GDC-10.

Discussion

Calculation of extreme power shapes which affect fuel design limits is performed with approved methods. The conditions under which limiting power shapes are assumed to occur are chosen conservatively with regard to any permissible operating state.

Even though there is good agreement between measured peak power calculations and measurements, a nuclear uncertainty margin (see Section 4.3.2.2.7) is applied to calculated peak local power. Such a margin is provided both for the analysis for normal operating states and for anticipated transients.

4.3.1.4 Maximum Controlled Reactivity Insertion Rate

Basis

The maximum reactivity insertion rate due to withdrawal of rod cluster control assemblies at power or by boron dilution is limited. A maximum reactivity change rate for accidental withdrawal of control banks is set such that peak heat generation rate and DNBR do not exceed the maximum allowable at overpower conditions. This satisfies GDC-25.

The maximum reactivity worth of control rods and the maximum rates of reactivity insertion employing control rods are limited so as to preclude rupture of the coolant pressure boundary or

disruption of the core internals to a degree which would impair core cooling capacity due to a rod withdrawal or ejection accident (see Chapter 15).

Following any Condition IV event (rod ejection, steam line break, etc.) the reactor can be brought to the shutdown condition and the core will maintain acceptable heat transfer geometry. This satisfies GDC-28.

Discussion

Reactivity addition associated with an accidental withdrawal of a control bank (or banks) is limited by the maximum rod speed (or travel rate) and by the worth of the bank(s). The maximum control rod speed is 45 inches per minute.

The reactivity change rates are conservatively calculated assuming unfavorable axial power and xenon distributions. The peak xenon burnout rate is significantly lower than the maximum reactivity addition rate for normal operation and for accidental withdrawal of two banks.

4.3.1.5 Shutdown Margins

Basis

Minimum shutdown margin as specified in the Core Operating Limits Report is required at any power operating condition, in the hot standby shutdown condition and in the cold shutdown condition.

In all analyses involving reactor trip, the single, highest worth rod cluster control assembly is postulated to remain untripped in its full-out position (stuck rod criterion). This satisfies GDC-26.

Discussion

Two independent reactivity control systems are provided, namely control rods and soluble boron in the coolant. The control rod system can compensate for the reactivity effects of the fuel and water temperature changes accompanying power level changes over the range from full-load to no-load. In addition, the control rod system provides the minimum shutdown margin under Condition I events and is capable of making the core subcritical rapidly enough to prevent exceeding acceptable fuel damage limits assuming that the highest worth control rod is stuck out upon trip.

The boron system can compensate for all xenon burnout reactivity changes and will maintain the reactor in the cold shutdown. Thus, backup and emergency shutdown provisions are provided by a mechanical and a chemical shim control system which satisfies GDC-26.

Basis

When fuel assemblies are in the pressure vessel and the vessel head is not in place, k_{eff} will be maintained at or below 0.95 with control rods and soluble boron. Further, the fuel will be maintained sufficiently subcritical so that removal of all rod cluster control assemblies will not result in criticality.

Discussion

ANSI Standard Nl8.2 specifies a k_{eff} not to exceed 0.95 in spent fuel storage racks and transfer equipment flooded with pure water and a k_{eff} not to exceed 0.98 in normally dry new fuel storage racks assuming optimum moderation. No criterion is given for the refueling operation; however, a 5 percent margin, which is consistent with spent fuel storage and transfer and the new fuel storage, is adequate for the controlled and continuously monitored operations involved.

The boron concentration required to meet the refueling shutdown criteria is specified in the Technical Specifications. Verification that this shutdown criteria is met, including uncertainties, is achieved using standard design methods. The subcriticality of the core is continuously monitored as described in the Technical Specifications.

4.3.1.6 Stability

Basis

The core will be inherently stable to power oscillations at the fundamental mode. This satisfies GDC-12. Spatial power oscillations within the core with a constant core power output, should they occur, can be reliably and readily detected and suppressed.

Discussion

Oscillations of the total power output of the core, from whatever cause, are readily detected by the loop N-16 power monitors and by the nuclear instrumentation. The core is protected by these systems and a reactor trip would occur if power increased unacceptably, preserving the design margins to fuel design limits. The stability of the turbine/steam generator/core systems and the reactor control system is such that total core power oscillations are not normally possible. The redundancy of the protection circuits ensures an extremely low probability of exceeding design power levels.

The core is designed so that diametral and azimuthal oscillations due to spatial xenon effects are self-damping and no operator action or control action is required to suppress them. The stability to diametral oscillations is so great that this excitation is highly improbable. Convergent azimuthal oscillations can be excited by prohibited motion of individual control rods. Such oscillations are readily observable and alarmed, using the excore long ion chambers. Indications are also continuously available from incore thermocouples and loop power measurements. Moveable incore detectors can be activated to provide more detailed information. In all proposed cores these horizontal plane oscillations are self-damping by virtue of reactivity feedback effects designed into the core.

However, axial xenon spatial power oscillations may occur late in core life. The control banks and excore detectors are provided for control and monitoring of axial power distributions. Assurance that fuel design limits are not exceeded is provided by reactor Overpower N-16 and Overtemperature N-16 trip functions which use the measured axial power imbalance as an input.

4.3.1.7 Anticipated Transients Without Trip

The effects of anticipated transients with failure to trip are not considered in the design bases of the plant. Analysis has shown that the likelihood of such a hypothetical event is negligibly small. Furthermore, analysis of the consequences of a hypothetical failure to trip following anticipated transients has shown that no significant core damage would result, system peak pressures would be limited to acceptable values and no failure of the Reactor Coolant System would result [1]. However, Anticipated Transient Without Scram Mitigation System Actuation Circuitry (AMSAC) has been incorporated into the CPNPP design in accordance with the requirement of 10CFR50.62. For detailed discussion see FSAR Section 7.8.1.

4.3.2 DESCRIPTION

4.3.2.1 Nuclear Design Description

The reactor core consists of a specified number of fuel rods which are held in bundles by spacer grids and top and bottom fittings. The fuel rods are constructed of Zircaloy or ZIRLO cylindrical tubes containing UO_2 fuel pellets. The bundles, known as fuel assemblies, are arranged in a pattern which approximates a right circular cylinder.

Each fuel assembly normally contains a 17 x 17 rod array composed of 264 fuel rods, 24 rod cluster control thimbles and an incore instrumentation thimble. Further details of the fuel assembly are given in Section 4.2.

The core will normally operate approximately one to two years between refuelings, accumulating approximately 14,000 MWD/MTU per year. The exact reloading pattern, initial and final positions of assemblies, number of fresh assemblies and their placement are dependent on the energy requirement for the next cycle and burnup and power histories of the previous cycles.

The core average enrichment is determined by the amount of fissionable material required to provide the desired core lifetime and energy requirements. The physics of the burnout process is such that operation of the reactor depletes the amount of fuel available due to the absorption of neutrons by the U-235 atoms and their subsequent fission. The rate of U-235 depletion is directly proportional to the power level at which the reactor is operated. In addition, the fission process results in the formation of fission products, some of which readily absorb neutrons. These effects, depletion and the buildup of fission products, are partially offset by the buildup of plutonium which occurs due to the non-fission absorption of neutrons in U-238. Typical plutonium buildup is shown in Figure 4.3-2. Therefore, at the beginning of any cycle a reactivity reserve equal to the depletion of the fissionable fuel and the buildup of fission product poisons over the specified cycle life must be “built” into the reactor. This excess reactivity is controlled by removable neutron absorbing material in the form of boron dissolved in the primary coolant, burnable absorber rods, and/or ZrB_2 coated fuel pellets (IFBA).

The concentration of boric acid in the primary coolant is varied to provide control and to compensate for long term reactivity requirements. The concentration of the soluble neutron absorber is varied to compensate for reactivity changes due to fuel burnup, fission product poisoning including xenon and samarium, burnable absorber depletion, and the cold-to-operating moderator temperature change. Using its normal makeup path, the Chemical and Volume Control System (CVCS) is capable of inserting negative reactivity at a rate of approximately

30 pcm/min when the reactor coolant boron concentration is 1000 ppm and approximately 35 pcm/min when the reactor coolant boron concentration is 100 ppm. If the emergency boration path is used, the CVCS is capable of inserting negative reactivity at a rate of approximately 65 pcm/min when the reactor coolant concentration is 1000 ppm and approximately 75 pcm/min when the reactor coolant boron concentration is 100 ppm. The peak burnout rate for xenon is approximately 25 pcm/min (Section 9.3.4 discusses the capability of the CVCS to counteract xenon decay). Rapid transient reactivity requirements and safety shutdown requirements are met with control rods.

As the boron concentration is increased, the moderator temperature coefficient becomes less negative. The use of a soluble poison alone would result in a positive moderator coefficient at beginning-of-life. Therefore, burnable absorber rods are used to reduce the soluble boron concentration sufficiently to ensure that the moderator temperature coefficient is negative for full power operating conditions. During operation the poison content in these rods is depleted, thus adding positive reactivity to offset some of the negative reactivity from fuel depletion and fission product buildup. The depletion rate of the burnable absorber rods is not critical since chemical shim is always available and flexible enough to cover any possible deviations in the expected burnable absorber depletion rate. Figure 4.3-3 is a graph of a typical core depletion with and without burnable absorber rods. Note that even at end-of-life conditions some residual absorber remains in the burnable absorber rods resulting in a net decrease in the first cycle lifetime. In addition to reactivity control, the burnable absorber rods and/or ZrB_2 coated fuel pellets (IFBA) are strategically located to provide a favorable radial power distribution.

Control rods are located for use in the core to provide control for rapid changes in reactivity. The total reactivity worth of the control rods is dependent on the particular absorber material used and the power distribution is affected by the number, worth, and location of the inserted control rods.

4.3.2.2 Power Distributions

Two methods can be employed for performing core power distribution calculations. The Power Distribution Monitoring System (PDMS) generates a continuous measurement of the core power distribution using the methodology documented in Reference 12. The measured core power distribution is used to determine the most limiting core peaking factors, which are used to verify that the reactor is operating within the design limits.

The PDMS requires information on current plant and core conditions in order to determine the core power distribution using the core peaking factor measurement and measurement uncertainty methodology described in Reference 12. The core and plant condition information is used as input to the continuous core power distribution measurement software that continuously and automatically determines the current core peaking factor values. The core power distribution calculation software provides the measured peaking factor values at nominal one-minute intervals to allow confirmation that the core peaking factors are within design limits. In order for the PDMS to accurately determine the peaking factor values, the core power distribution measurement software requires accurate information about the current reactor power level average reactor vessel inlet temperature, control bank positions, the power range detector currents, and the core exit thermocouples.

Data obtained from the movable neutron flux detectors, described in Section 7.7.1, are used to calibrate the PDMS, and may also be used independent of the PDMS to generate a flux map of

the core power distribution. The accuracy of power distribution calculations has been confirmed through approximately one thousand flux maps during some twenty years of operation under conditions similar to those expected. Details of this confirmation are given in Reference 2 and in Section 4.3.2.2.7.

4.3.2.2.1 Definitions

Power distributions are quantified in terms of hot channel factors. These factors are a measure of the peak pellet power within the reactor core and the total energy produced in a coolant channel and are expressed in terms of quantities related to the nuclear or thermal design namely:

Power density is the thermal power produced per unit volume of the core (kW/liter).

Linear power density is the thermal power produced per unit length of active fuel (kW/ft). Since fuel assembly geometry is standardized, this is the unit of power density most commonly used. For all practical purposes, it differs from kW/liter by a constant factor which includes geometry and the fraction of the total thermal power which is generated in the fuel rod.

Average linear power density is the total thermal power produced in the fuel rods divided by the total active fuel length of all rods in the core.

Local heat flux is the heat flux at the surface of the cladding ($Btu\text{-}ft^{-2}\text{-}hr^{-1}$). For nominal rod parameters this differs from linear power density by a constant factor.

Rod power or rod integral power is the length integrated linear power density in one rod (kW).

Average rod power is the total thermal power produced in the fuel rods divided by the number of fuel rods (assuming all rods have equal length).

The hot channel factors used in the discussion of power distributions in this section are defined as follows:

F_Q, Heat Flux Hot Channel Factor, is defined as the maximum local heat flux on the surface of a fuel rod divided by the average fuel rod heat flux, allowing for manufacturing tolerances on fuel pellets and rods.

F^N_Q, Nuclear Heat Flux Hot Channel Factor, is defined as the maximum local fuel rod linear power density divided by the average fuel rod linear power density, assuming nominal fuel pellet and rod parameters.

F^E_Q, Engineering Heat Flux Hot Channel Factor, is the allowance on heat flux required for manufacturing tolerances. The engineering factor allows for local variations in enrichment, pellet density and diameter, surface area of the fuel rod and eccentricity of the gap between pellet and clad. Combined statistically the net effect is a factor of 1.03 to be applied to fuel rod surface heat flux.

$F_{\Delta}{}^{N}{}_{H}$, Nuclear Enthalpy Rise Hot Channel Factor, is defined as the ratio of the integral of linear power along the rod with the highest integrated power to the average rod power.

Manufacturing tolerances, hot channel power distribution and surrounding channel power distributions are treated explicitly in the calculation of the DNBR described in Section 4.4.

It is convenient for the purposes of discussion to define subfactors of F_Q, however, design limits are set in terms of the total peaking factor.

F_Q = Total peaking factor of heat flux hot-channel factor

= Maximum KW/ft / Average KW/ft

without densification effects

$F_Q = F^N_Q \times F^E_Q$

$= F^N_{XY} \times F^N_Z \times F^N_U \times F^E_Q$

where

F^N_Q and F^E_Q are defined above.

F^N_U = factor for conservatism, assumed to be 1.05

F^N_{XY} = ratio of peak power density to average power density in the horizontal plane of peak local power

F^N_Z = ratio of the power per unit core height in the horizontal plane of peak local power to the average value of power per unit core height. If the plane of peak local power coincides with the plane of maximum power per unit core height the F^N_Z is the core average axial peaking factor.

To include the allowances made for densification effects, which are height dependent, the following quantities are defined.

S(Z) = the allowance made for densification effects at height Z in the core (see Section 4.3.2.2.5).

P(Z) = ratio of the power per unit core height in the horizontal plane at height Z to the average value of power per unit core height.

Then

F_Q = Total peaking factor

= Maximum KW/ft / Average KW/ft

Including densification allowance

$$F_Q = \max (F^N_{XY}(Z) \times P(Z) \times S(Z)) \times F^N_{U\,X}\, F^E_Q$$

4.3.2.2.2 Radial Power Distributions

The power shape in horizontal sections of the core at full power is a function of the fuel and burnable absorber loading patterns and the presence or absence of a single bank of full length control rods.

Thus, at any time in the cycle a horizontal section of the core can be characterized as unrodded or with group D control rods. These two situations combined with burnup effects determine the radial power shapes which can exist in the core at full power. The effect on radial power shapes of power level, xenon, samarium and moderator density effects are considered also but these are quite small. The effect of non-uniform flow distribution is negligible. While radial power distributions in various planes of the core are often illustrated, the core radial enthalpy rise distribution as determined by the integral of power up each channel is of greater interest. Figures 4.3-6 through 4.3-11 show typical radial power distributions for one eighth of the core for representative operating conditions. These conditions are: 1) hot full power (HFP) at beginning-of-life (BOL) - unrodded - no xenon, 2) HFP at BOL - unrodded - equilibrium xenon, 3) HFP at BOL - bank D in - equilibrium xenon, 4) HFP near middle-of-life (MOL) - unrodded - equilibrium xenon, 5) HFP at end-of-life (EOL) - unrodded - equilibrium xenon, and 6) HFP at EOL - bank D in - equilibrium xenon.

Since hot channel location varies from time to time, a single reference radial design power distribution is selected for DNB calculations. This reference power distribution, normalized to core average power, is chosen conservatively to concentrate power in one area of the core, minimizing the benefits of flow redistribution.

4.3.2.2.3 Assembly Power Distributions

For the purpose of illustration, typical assembly power distributions from BOL and EOL conditions corresponding to Figures 4.3-7 and 4.3-10 respectively, are given in Figures 4.3-12 and 4.3-13, respectively. The power distributions provided are consistent with the burnable poison rod patterns shown in Figure 4.3-5.

Since the detailed power distribution surrounding the hot channel varies from time to time, a conservatively flat assembly power distribution is assumed in the DNB analysis, described in Section 4.4, with the rod of maximum integrated power artificially raised to the design value of

$F^N\Delta_H$. The nuclear design considers all fuel cycles and all operating conditions to ensure that a flatter assembly power distribution does not occur with limiting values of $F^N\Delta_H$.

4.3.2.2.4 Axial Power Distributions

The shape of the power profile in the axial or vertical direction is largely under the control of the operator either through the manual operation of the full length control rods or automatic motion of full length rods responding to manual operation of the CVCS. Nuclear effects which cause variations in the axial power shape include moderator density, Doppler effect on resonance absorption, spatial xenon, and burnup. Automatically controlled variations in total power output and full length rod motion are also important in determining the axial power shape at any time. Signals are available to the operator from the excore ion chambers which are long ion chambers outside the reactor vessel which contain four active sections running parallel to the axis of the core. Signals from the top two and bottom two sections are averaged to obtain top and bottom signals. The difference between top and bottom signals from each of four multi-section of detectors is displayed on the control panel and called the flux difference, ΔI. Calculations of core average peaking factor for many plants and measurements from operating plants under many operating situations are associated with either ΔI or axial offset in such a way that an upper bound can be placed on the peaking factor. For these correlations axial offset is defined as:

$$\text{axial offset} = \frac{\varphi t - \varphi b}{\varphi t + \varphi b}$$

and φt and φb are the top and bottom detector readings.

Representative axial power shapes for BOL, MOL, and EOL conditions are shown in Figures 4.3-14 through 4.3-16. These figures cover a wide range of axial offset including values not permitted at full power.

The typical radial power distributions shown in Figures 4.3-8 and 4.3-11 involving the insertion of control rods represent a synthesis of power shapes from the rodded and unrodded planes. The applicability of the separability assumption upon which this procedure is based is assured through extensive three-dimensional calculations of possible rodded conditions. As an example, Figure 4.3-17 compares the axial power distribution for several assemblies at different distances from inserted control rods with the core average distribution. The only significant difference from the average occurs in the low power peripheral assemblies, thus confirming the validity of the separability assumption.

4.3.2.2.5 Local Power Peaking

Fuel densification, which has been observed to occur under irradiation in several operating reactors, causes the fuel pellets to shrink both axially and radially. The pellet shrinkage combined with random hangup of fuel pellets results in gaps in the fuel column when the pellets below the hung-up pellet settle in the fuel rod. The gaps vary in length and location in the fuel rod. Because of decreased neutron absorption in the vicinity of the gap, power peaking occurs in the adjacent fuel rods resulting in an increased power peaking factor. A quantitative measure of this local peaking is given by the power spike factor S(Z) where Z is the axial location in the core.

In previous analyses of power peaking factors, it was necessary to apply a penalty on calculated overpower transient F_Q values to allow for interpellet gaps caused by pellet hang-ups and pellet shrinkage due to densification. This penalty is known as the densification spike factor. However, studies have shown (Reference 3) that this penalty can be eliminated for the fuel type present in the Comanche Peak Units 1 and 2 cores.

4.3.2.2.6 Limiting Power Distributions

According to the ANSI classification of plant conditions (see Chapter 15), Condition I occurrences are those which are expected frequently or regularly in the course of power operation, maintenance, or maneuvering of the plant. As such, Condition I occurrences are accommodated with margin between any plant parameter and the value of that parameter which would require either automatic or manual protective action. Inasmuch as Condition I occurrences occur frequently or regularly, they must be considered from the point of view of affecting the consequences of fault conditions (Conditions II, III, and IV). In this regard, analysis of each fault condition described is generally based on a conservative set of initial conditions corresponding to the most adverse set of conditions which can occur during Condition I operation.

The list of steady state and shutdown conditions, permissible deviations and operational transients is given is Section 15.0. Implicit in the definition of normal operation is proper and timely action by the reactor operator. That is, the operator follows recommended operating procedures for maintaining appropriate power distributions and takes any necessary remedial actions when alerted to do so by the plant instrumentation. Thus, as stated above, the worst or limiting power distribution which can occur during normal operation is to be considered as the starting point for analysis of ANSI Conditions II, III, and IV events.

Improper procedural actions or errors by the operator are assumed in the design as occurrences of moderate frequency (ANSI Condition II). Some of the consequences which might result are discussed in Section 15.0. Therefore, the limiting power shapes which result from such Condition II events are those power shapes which deviate from the normal operating condition at the recommended axial offset band, e.g. due to lack of proper action by the operator during a xenon transient following a change in power level brought about by control rod motion. Power shapes which fall in this category are used for determination of the reactor protection system setpoints so as to maintain margin to overpower or DNB limits.

The means for maintaining power distributions within the required hot channel factor limits are described in the Technical Specifications. A complete discussion of power distribution control in Westinghouse PWRs is included in Reference 6. Detailed background information on the following design constraints on local power density in a Westinghouse PWR, the defined operating procedures, and the measures taken to preclude exceeding design limits are presented in the topical report on power distribution control and load following procedures (Reference 13). The following paragraphs summarize these reports and describe the calculations used to establish the upper bound on peaking factors.

The calculations used to establish the upper bound on peaking factors, F_Q and $F_{\Delta}{}^{N}{}_{H}$, include all of the nuclear effects which influence the radial and/or axial power distributions throughout core life for various modes of operation including load follow, reduced power operation, and axial xenon transients.

The core average axial profile, however, can experience significant changes which can occur rapidly as a result of rod motion and load changes and more slowly due to xenon distribution. Since the properties of the nuclear design dictate what axial shapes can occur, boundaries on the limits of interest can be set in terms of the parameters which are readily observed on the plant. Specifically, the nuclear design parameters which are significant to the axial power distribution analysis are:

1. Core power level.

2. Core height.

3. Coolant temperature and flow.

4. Coolant temperature program as a function of reactor power.

5. Fuel cycle lifetimes.

6. Rod bank worths.

7. Rod bank overlaps.

Normal operation of the plant assumes compliance with the following conditions:

1. Control rods in a single bank move together with no individual rod insertion differing by more than 12 steps (indicated) from the bank demand position;

2. Control banks are sequenced with overlapping banks;

3. The control full length bank insertion limits are not violated;

4. Axial power distribution procedures, which are given in terms of flux difference control and control bank position, are observed.

The axial power distribution procedures referred to above are part of the required operating procedures which are followed in normal operation. During normal operation with relaxed axial offset control (RAOC), they require control of the axial flux difference at power levels above 50% RTP within a permissible operating space. The limits on axial flux difference (AFD) are given in the COLR. This minimizes xenon transient effects on the axial power distribution, since the procedures essentially keep the xenon distribution in phase with the power distribution.

Calculations are performed for normal operation of the reactor at various points in the operating cycles. Different operation maneuvers are assumed prior to calculating the effect on the axial power distributions. For a given plant and fuel cycle, a finite number of maneuvers are studied to determine the general behavior of the local power density as a function of core elevation.

These cases represent many possible reactor states in the life of one fuel cycle, and they have been chosen as sufficiently definitive of the cycle by comparison with much more exhaustive studies performed on some 20 or 30 different, but typical, plant and fuel cycle combinations. The cases are described in detail in Reference 7, and they are considered to be necessary and sufficient to generate a local power density limit which, when increased by 5 percent for

conservatism, will not be exceeded with a 95-percent confidence level. Many of the points do not approach the limiting envelope. However, they are part of the time histories which lead to the hundreds of shapes which do define the envelope. They also serve as a check that the reactor studied is typical of those studied more exhaustively.

Thus it is not possible to single out any transient or steady state condition which defines the most limiting case. It is not even possible to separate out a small number which form an adequate analysis. The process of generating a myriad of shapes is essential to the philosophy that leads to the required level of confidence. A maneuver which provides a limiting case for one reactor fuel cycle is not necessarily a limiting case for another reactor or fuel cycle with different control bank worths, enrichments, burnup, coefficients, etc. Each shape depends on the detailed history of operation up to that time and on the manner in which the operator conditioned xenon in the days immediately prior to the time at which the power distribution is calculated.

The calculated points are synthesized from axial calculations combined with radial factors appropriate for rodded and unrodded planes. In these calculations, the effects on the unrodded radial peak of xenon redistribution that occurs following the withdrawal of a control bank (or banks) from a rodded region is obtained from two-dimensional X-Y calculations. A conservative factor is applied on the unrodded radial peak that was obtained from calculations in which xenon distribution was preconditioned by the presence of control rods and then allowed to redistribute for several hours. A detailed discussion of this effect may be found in Reference 7. The calculated values have been increased by a factor of 1.05 for conservatism and a factor of 1.03 for the engineering factor for the overpower (kW/ft) evaluations.

The envelope drawn over the calculated [max (F_Q • Power)] points represents an upper bound envelope on local power density versus elevation in the core. It should be emphasized that this envelope is a conservative representation of the bounding values of local power density. Expected values are considerably smaller and, in fact, less conservative bounding values may be justified with additional analysis or surveillance requirements.

This upper bound envelope is verified for operation within an allowed AFD operating space, per the RAOC methodology outlined in Reference 13. Axial offset control is detailed in the Technical Specifications, with AFD limits specified in the COLR, and followed by relying only upon excore surveillance supplemented by the normal monthly power distribution measurement requirement and by computer-based alarms on deviation from the allowed AFD operating space.

Allowing for fuel densification effects the average linear power is 5.77 kW/ft at 3612 MWt. The conservative upper bound value of normalized local power density, including uncertainty allowances, is 2.50.

To determine reactor protection system setpoints, with respect to power distributions, three categories of events are considered, namely rod control equipment malfunctions, operator errors of commission, and operator errors of omission. In evaluating these three categories of events, the core is assumed to be operating within the four constraints described above.

The first category comprises uncontrolled rod withdrawal (with rods moving in the normal bank sequence) for full length banks (automatic rod withdrawal is no longer available). Also included are motions of the full-length banks below their insertion limits, which could be caused, for example, by uncontrolled dilution or primary coolant cooldown. Power distributions were

calculated throughout these occurrences, assuming short term corrective action. That is, no transient xenon effects were considered to result from the malfunction. The event was assumed to occur from typical normal operating situations, which include normal xenon transients. It was further assumed in determining the power distributions that total core power level would be limited by reactor trip to below 118.5 percent. Since the study is to determine protection limits with respect to power and axial offset, no credit was taken for trip setpoint reduction due to flux difference. The peak power density which can occur in such events, assuming reactor trip at or below 118.5 percent, is less than that required for center-line melt, including uncertainties and densification effects.

The second category assumes that the operator mispositions the full-length rod bank in violation of the insertion limits and creates short-term conditions not included in normal operating conditions. The third category assumes that the operator fails to take action to correct a flux difference violation. The results for FQ are multiplied by a factor to include an allowance for calorimetric error. The peak linear power does not exceed that required for centerline melt. Since the peak kW/ft is below the limit, no flux difference penalties are required for overpower protection. It should be noted that a reactor overpower accident is not assumed to occur coincident with an independent operator error. Additional detailed discussion of these analyses is presented in Reference 7.

Analyses of possible operating power shapes show that the appropriate hot channel factors F_Q and $F^N\Delta_H$ for peak local power density and for DNB analysis at full power are the values addressed in the Technical Specifications and the COLR.

The maximum allowable F_Q can be increased with decreasing power, as shown in the Technical Specifications. Increasing $F^N\Delta_H$ with decreasing power is permitted by the DNB protection set points and allows radial power shape changes with rod insertion to the insertion limits, as described in Section 4.4.4.3. The allowance for increased permitted becomes a design basis criterion which is used for establishing acceptable control rod patterns and control bank sequencing. Like-wise, fuel loading patterns for each cycle are selected with consideration of this design criterion. The worst values of for possible rod configurations occurring in normal operation are used in verifying that this criterion is met. The worst values generally occur when the rods are assumed to be at their insertion limits. Maintenance of relaxed axial offset control establishes rod positions which are at or above the allowed rod insertion limits. As discussed in Section 3.2 of Reference 8, it has been determined that the Technical Specification limits are met, provided the above conditions a and b are observed. These limits are taken as input to the thermal-hydraulic design basis, as described in Section 4.4.4.3.1.

When a situation is possible in normal operation which could result in local power densities in excess of those assumed as the precondition for a subsequent hypothetical accident, but which would not itself cause fuel failure, administrative controls and alarms are provided for returning the core to a safe condition.

4.3.2.2.7 Experimental Verification of Power Distribution Analysis

This subject is discussed in depth in Reference 2. A summary of this report is given below. The measured and calculational comparison is normally followed periodically throughout the cycle lifetime of the reactor as required by Technical Specifications.

In a measurement of the heat flux hot channel factor, F_Q, with the movable detector system described in Sections 7.7.1 and 4.4.6, the following uncertainties have to be considered:

1. Reproducibility of the measured signal.

2. Errors in the calculated relationship between detector current and local flux.

3. Errors in the calculated relationship between detector flux and peak rod power some distance from the measurement thimble.

The appropriate allowance for category 1 above has been quantified by repetitive measurements made with several inter-calibrated detectors by using the common thimble features of the incore detector system. This system allows more than one detector to access any thimble. Errors in category 2 above are quantified to the extent possible, by using the fluxes measured at one thimble location to predict fluxes at another location which is also measured. Local power distribution predictions are verified in critical experiments on arrays of rods with simulated guide thimbles, control rods, burnable poisons, etc. These critical experiments provide quantification of errors of types 2 and 3 above.

Reference 2 describes critical experiments performed at the Westinghouse Reactor Evaluation Center and measurements taken on two Westinghouse plants with incore systems of the same type as used in this plant. The report concludes that the uncertainty associated with F_Q (heat flux) is 4.58 percent at the 95 percent confidence level with only 5 percent of the measurements greater than the inferred value. This is the equivalent of a 1.645 limit on a normal distribution and is the uncertainty to be associated with a full core flux map with movable detectors reduced with a reasonable set of input data incorporating the influence of burnup on the radial power distribution. The uncertainty is usually rounded up to 5 percent.

In comparing measured power distributions (or detector currents) against the calculations for the same situation it is not possible to subtract out the detector reproducibility. Thus a comparison between measured and predicted power distributions has to include some measurement error. Such a comparison is given in Figures 4.3-24 for one of the maps used in Reference 2. Since the first publication of the report, hundreds of maps have been taken on these and other reactors. The results confirm the adequacy of the 5 percent uncertainty allowance on the calculated F_Q.

A similar analysis for the uncertainty in $F^N_{\Delta H}$ (rod integral power) measurements results in an allowance of 3.65 percent at the equivalent of a 1.645 confidence level. For historical reasons an 8 percent uncertainty factor is allowed in the nuclear design calculational basis; that is, the predicted rod integrals at full power must not exceed the design $F^N_{\Delta H}$ less 8 percent. This 8 percent may be reduced in final design to 4 percent to allow a wider range of acceptable axial power distributions in the DNB analysis and still meet the design bases of Section 4.3.1.3.

A recent measurement in the second cycle of a 121 assembly, 12 foot, core is compared with a simplified one dimensional core average axial calculation in Figure 4.3-25. This calculation does not give explicit representation to the fuel grids.

The accumulated data on power distributions in actual operation was basically of three types:

1. Much of the data was obtained in steady state operation at constant power in the normal operating configuration;

2. Data with unusual values of axial offset were obtained as part of the excore detector calibration exercise which was performed every 92 EFPD;

3. Special tests have been performed in load follow and other transient xenon conditions which have yielded useful information on power distributions.

These data are presented in detail in Reference [8]. Figure 4.3-26 contains a summary of measured values of F_Q as a function of axial offset for five plants from that report.

4.3.2.2.8 Testing and Operations Support

A very extensive series of physics tests is performed on the first cores. These tests and the criteria for satisfactory results are described in Chapter 14. Since not all limiting situations can be created at BOL, the main purpose of the tests is to provide a check on the calculational methods used in the predictions for the conditions of the test. Limited tests are performed at the beginning of each reload cycle to verify that the core is loaded as designed and can be safely operated.

4.3.2.2.9 Monitoring Instrumentation

The adequacy of instrument numbers, spatial deployment, required correlations between readings and peaking factors, calibration and errors are described in References 2, 6, and 8. The relevant conclusions are summarized here in Sections 4.3.2.2.7 and 4.4.6.

Reference 12 describes the instrumentation requirements and calibration of the PDMS and the uncertainties applied to the calculated peaking factors.

Provided the limitations given in Section 4.3.2.2.6 on rod insertion and flux difference are observed, the excore detector system provides adequate online monitoring of power distributions. Further details of specific limits on the observed rod positions and flux difference are given in the Technical Specifications together with a discussion of their bases.

Limits for alarms, reactor trip, etc. are given in the Technical Specifications. Descriptions of the systems provided are given in Section 7.7.

4.3.2.3 Reactivity Coefficients

The kinetic characteristics of the reactor core determine the response of the core to changing plant conditions or to operator adjustments made during normal operation, as well as the core response during abnormal or accidental transients. These kinetic characteristics are quantified in reactivity coefficients. The reactivity coefficients reflect the changes in the neutron multiplication due to varying plant conditions such as power, moderator or fuel temperatures, or less significantly due to a change in pressure or void conditions. Since reactivity coefficients change during the life of the core, ranges of coefficients are employed in transient analysis to determine the response of the plant throughout life. The results of such simulations and the reactivity coefficients used are presented in Chapter 15. The reactivity coefficients are calculated

on a corewise basis using diffusion theory methods. The effect of radial and axial power distribution on core average reactivity coefficients is implicit in those calculations and is not significant under normal operating conditions. For example, a skewed xenon distribution which results in changing axial offset by 5 percent changes the moderator and Doppler temperature coefficients by less than 0.01 pcm/°F and 0.03 pcm/°F respectively. An artificially skewed xenon distribution which results in changing the radial $F^N_{\Delta H}$ by 3 percent changes the moderator and Doppler temperature coefficients by less than 0.03 pcm/°F and 0.001 pcm/°F respectively. The spatial effects are accentuated in some transient conditions (for example, in postulated rupture of the main steam line break and rupture of rod cluster control assembly mechanism housing described in Sections 15.1.5 and 15.4.8) and are included in these analyses.

The analytical methods and calculational models used in calculating the reactivity coefficients are given in Section 4.3.3.

Quantitative information for calculated reactivity coefficients, including fuel-Doppler coefficient, moderator coefficients (density, temperature, pressure, void) and power coefficient is given in the following sections.

4.3.2.3.1 Fuel Temperature (Doppler) Coefficient

The fuel temperature (Doppler) coefficient is defined as the change in reactivity per degree change in effective fuel temperature and is primarily a measure of the Doppler broadening of U-238 and Pu-240 resonance absorption peaks. Doppler broadening of other isotopes such as U-236, Np-237, etc. are also considered but their contributions to the Doppler effect is small. Fission products have a negligible effect on the Doppler coefficient. An increase in fuel temperature increases the effective resonance absorption cross sections of the fuel and produces a corresponding reduction in reactivity.

The fuel temperature coefficient is calculated with the standard nuclear design methods described in Section 4.3.3.

A typical Doppler temperature coefficient is shown in Figure 4.3-27 as a function of the effective fuel temperature (at BOL and EOL conditions). The Doppler-only contribution to the power coefficient, defined later, is shown in Figure 4.3-28 as a function of relative core power. The integral of the differential curve on Figure 4.3-28 is the Doppler contribution to the power defect and is shown in Figure 4.3-29 as a function of relative power. The Doppler coefficient becomes more negative as a function of life as the Pu-240 content increases, thus increasing the Pu-240 resonance absorption, but overall becomes less negative since the fuel temperature changes with burnup as described in Section 4.3.3.1. The upper and lower limits of Doppler coefficient used in accident analyses are given in Chapter 15.

4.3.2.3.2 Moderator Coefficients

The moderator coefficients is a measure of the change in reactivity due to a change in specific coolant parameters such as density, temperature, pressure or void. The coefficients so obtained are moderator density, temperature, pressure and void coefficients.

Moderator Density and Temperature Coefficients

The moderator temperature (density) coefficient is defined as the change in reactivity per degree change in the moderator temperature. Generally, the effect of the changes in moderator density as well as the temperature are considered together. A decrease in moderator density means less moderation which results in a negative moderator coefficient. An increase in coolant temperature, keeping the density constant, leads to a hardened neutron spectrum and results in an increase in resonance absorption in U-238, Pu-240 and other isotopes. The hardened spectrum also causes a decrease in the fission to capture ratio in U-235 and Pu-239. Both of these effects make the moderator coefficients more negative. Since water density changes more rapidly with temperature as temperature increases, the moderator temperature coefficient become more negative with increasing temperature.

The soluble boron used in the reactor as a means of reactivity control also has an effect on moderator density coefficient since the soluble boron poison density as well as the water density is decreased when the coolant temperature rises. A decrease in the soluble poison concentration introduces a positive component in the moderator density coefficient.

Thus, if the concentration of soluble poison is large enough, the net value of the coefficient may be positive. With the burnable absorber rods present, however, the initial hot full power boron concentration is sufficiently low that the moderator temperature coefficient is negative at operating temperatures. The effect of control rods is to make the moderator coefficient more negative by reducing the required soluble boron concentration and by increasing the “leakage” of the core.

With burnup, the moderator coefficient becomes more negative primarily as a result of boric acid dilution but also to a significant extent from the effects of the buildup of plutonium and fission products.

The moderator coefficient is calculated for the various plant conditions discussed above with the standard nuclear design methods described in Section 4.3.3. A typical moderator coefficient is shown as a function of core temperature and boron concentration for the unrodded and rodded core in Figures 4.3-30 through 4.3-32. The temperature range covered is from cold (68°F) to about 600°F. The contribution due to Doppler coefficient (because of change in moderator temperature) has been subtracted from these results. Figure 4.3-33 shows a representative hot full power moderator temperature coefficient plotted as a function of cycle lifetime for the just critical boron concentration condition based on the design boron letdown condition.

The moderator coefficients presented here are calculated on a corewide bases, since they are used to describe the core behavior in normal and accident situations when the moderator temperature changes can be considered to affect the entire core.

Moderator Pressure Coefficient

The moderator pressure coefficient relates the change in moderator density, resulting from a reactor coolant pressure change, to the corresponding effect on neutron production. This coefficient is of much less significance in comparison with the moderator temperature coefficient. A change of 50 psi in pressure has approximately the same effect on reactivity as a half-degree change in moderator temperature. This coefficient can be determined from the moderator temperature coefficient by relating change in pressure to the corresponding change in density.

The moderator pressure coefficient is negative over a portion of the moderator temperature range at BOL but is always positive at operating conditions and becomes more positive during life.

Moderator Void Coefficient

The moderator void coefficient relates the change in neutron multiplication to the presence of voids in the moderator. In a PWR this coefficient is not very significant because of the low void content in the coolant. The core void content is less than 1/2 of 1 percent and is due to local or statistical boiling. The void coefficient varies from approximately 50 pcm/percent void at BOL and at low temperatures to approximately -250 pcm/ percent void at EOL and at operating temperatures. The negative void coefficient at operating temperature becomes more negative with fuel burnup.

4.3.2.3.3 Power Coefficient

The combined effect of moderator temperature and fuel temperature change as the core power level changes is called the total power coefficient and is expressed in terms of reactivity change per percent power change. A typical power coefficient at BOL and EOL conditions is given in Figure 4.3-34.

It becomes more negative with burnup reflecting the combined effect of moderator and fuel temperature coefficients with burnup. A typical defect (integral reactivity effect) at BOL and EOL is given in Figure 4.3-35.

4.3.2.3.4 Comparison of Calculated and Experimental Reactivity Coefficients

Section 4.3.3 discusses the comparison of calculated and experimental reactivity coefficients.

Experimental evaluation of the calculated coefficients will be completed during the physics startup tests described in Chapter 14.

4.3.2.3.5 Reactivity Coefficients Used in Transient Analysis

The limiting values of reactivity coefficients are used as design limits in the transient analysis. The exact values of the coefficient used in the analysis depend on whether the transient of interest is examined at the BOL or EOL, whether the most negative or the most positive (least negative) coefficients are appropriate, and whether spatial nonuniformity must be considered in the analysis. Conservative values of coefficients, considering various aspects of analysis are used in the transient analysis. This is described in Chapter 15.

The reactivity coefficients shown in Figures 4.3-27 through 4.3-35 are typical best estimate values. The limiting values are chosen to encompass the best estimate reactivity coefficients calculated for the current cycle. The most positive as well as the most negative values are selected to form the design basis range used in the transient analysis. A direct comparison of the best estimate and design limit values can be misleading since in many instances, the most conservative combination of reactivity coefficients is used in the transient analysis even though the extreme coefficients assumed may not simultaneously occur at the conditions of lifetime, power level, temperature and boron concentration assumed in the analysis. The need for a re-evaluation of any accident in a subsequent cycle is contingent upon whether or not the

coefficients for that cycle fall within the identified range used in the analysis presented in Chapter 15 with due allowance for the calculational uncertainties given in Section 4.3.3. Reactivity components for the shutdown margin calculation are given in Table 4.3-3 for the first cycle and for a hypothetical equilibrium cycle. These latter numbers are provided for information only and their validity in a particular cycle would be an unexpected coincidence.

4.3.2.4 Control Requirements

To ensure the shutdown margin stated in the Core Operating Limits Report under conditions where a cooldown to ambient temperature is required, concentrated soluble boron is added to the coolant.

For all core conditions including refueling, the boron concentration is well below the solubility limit. The rod cluster control assemblies are employed to bring the reactor to the hot shutdown condition. The minimum required shutdown margin is given in the Core Operating Limits Report.

The ability to accomplish the shutdown for hot conditions is demonstrated in Table 4.3-3 by comparing the typical difference between the rod cluster control assembly reactivity available with an allowance for the worst struck rod with that required for control and protection purposes. The shutdown margin includes an allowance of 10 percent for analytic uncertainties (see Section 4.3.2.4.9). The largest reactivity control requirement appears at the EOL when the moderator temperature coefficient reaches its peak negative value as reflected in the larger power defect.

The control rods are required to provide sufficient reactivity to account for the power defect from full power to zero power and to provide the required shutdown margin. Typical Ag-In-Cd absorber design control rod reactivity worths listed in Table 4.3-3 are the most limiting values for ensuring shutdown margin. The reactivity addition resulting from power reduction consists of contributions from Doppler, variable average moderator temperature, flux redistribution, and reduction in void content as discussed below.

4.3.2.4.1 Doppler

The Doppler effect arises from the broadening of U-238 and Pu-240 resonance peaks with an increase in effective pellet temperature. This effect is most noticeable over the range of zero power to full power due to the large pellet temperature increase with power generation.

4.3.2.4.2 Variable Average Moderator Temperature

When the core is shutdown to the hot, zero power condition, the average moderator temperature changes from the equilibrium full load value determined by the steam generator and turbine characteristics (steam pressure, heat transfer, tube fouling, etc.) to the equilibrium no load value, which is based on the steam generator shell side design pressure.

Since the moderator coefficient is negative at most operating conditions, there is typically a reactivity addition with power reduction. The moderator coefficient becomes more negative as the fuel depletes because the boron concentration is reduced. This effect is the major contributor to the increased requirement at EOL.

4.3.2.4.3 Redistribution

During full power operation the coolant density decreases with core height; and this, together with partial insertion of control rods, results in less fuel depletion near the top of the core. Under steady state conditions, the relative power distribution will be slightly asymmetric towards the bottom of the core. On the other hand, at hot zero power conditions, the coolant density is uniform up the core, and there is no flattening due to Doppler. The result will be a flux distribution which at zero power can be skewed toward the top of the core. The reactivity insertion due to the skewed distribution is calculated with an allowance for effects of xenon distribution.

4.3.2.4.4 Void Content

A small void content in the core is due to nucleate boiling at full power. The void collapse coincident with power reduction makes a small reactivity contribution.

4.3.2.4.5 Rod Insertion Allowance

At full power, the control bank is operated within a prescribed band of travel to compensate for additional reactivity changes. Since the insertion limit is set by a rod travel limit, a conservatively high calculation of the inserted worth is made which exceeds the normally inserted reactivity.

4.3.2.4.6 Burnup

Excess reactivity is installed at the beginning of each cycle to provide sufficient reactivity to compensate for fuel depletion and fission products throughout the cycle. This reactivity is controlled by the addition of soluble boron to the coolant and by burnable absorber. Since the excess reactivity for burnup is controlled by soluble boron and/or burnable poison, it is not included in control rod requirements.

4.3.2.4.7 Xenon and Samarium Poisoning

Changes in xenon and samarium concentrations in the core occur at a sufficiently slow rate, even following rapid power level changes, that the resulting reactivity change is controlled by changing the soluble boron concentration.

4.3.2.4.8 pH Effects

Changes in reactivity due to a change in coolant pH, if any, are sufficiently small in magnitude and occur slowly enough to be controlled by the boron system. Further details are provided in Reference 11.

4.3.2.4.9 Experimental Confirmation

Following a normal shutdown, the total core reactivity change during cooldown with a stuck rod has been measured on a 121 assembly, 10 foot high core and 121 assembly, 12 foot high core. In each case, the core was allowed to cooldown until it reached criticality simulating the steamline break accident. For the ten foot core, the total reactivity change associated with the cooldown is overpredicted by about 0.3 percent $\Delta\rho$ with respect to the measured result. This represents an error of about 5 percent in the total reactivity change and is about half the uncertainty allowance for this quantity. For the 12 foot core, the difference between the

measured and predicted reactivity change was an even smaller 0.2 percent $\Delta\rho$. These measurements and others demonstrate the capability of the methods described in Section 4.3.3.

4.3.2.4.10 Control

Core reactivity is controlled by means of a chemical poison dissolved in the coolant, rod cluster control assemblies, and burnable poison rods as described below.

4.3.2.4.11 Chemical Poison

Boron in solution as boric acid is used to control relatively slow reactivity changes associated with:

1. The moderator temperature defect in going from cold shutdown at ambient temperature to the hot operating temperature at zero power,

2. The transient xenon and samarium poisoning, such as that following power changes or changes in rod cluster control position,

3. The excess reactivity required to compensate for the effects of fissile inventory depletion and buildup of long-life fission products.

4. The burnable poison depletion.

4.3.2.4.12 Rod Cluster Control Assemblies

Full length rod cluster control assemblies exclusively are employed in this reactor. The full length rod cluster control assemblies are used for shutdown and control purposes to offset fast reactivity changes associated with:

1. The required shutdown margin in the hot zero power, stuck rod condition,

2. The reactivity compensation as a result of an increase in power above hot zero power (power defect including Doppler, and moderator reactivity changes),

3. Unprogrammed fluctuations in boron concentration, coolant temperature, or xenon concentration (with rods not exceeding the allowable rod insertion limits),

4. Reactivity ramp rates resulting from load changes.

The allowed full length control bank reactivity insertion is limited at full power to maintain shutdown capability. As the power level is reduced, control rod reactivity requirements are also reduced and more rod insertion is allowed. The control bank position is monitored and the operator is notified by an alarm if the limit is approached. The determination of the insertion limit uses conservative xenon distributions and axial power shapes. In addition, the rod cluster control assembly withdrawal pattern determined from these analyses is used in determining power distribution factors and in determining the maximum worth of an inserted rod cluster control assembly ejection accident. For further discussion, refer to the Technical Specifications on rod insertion limits.

Power distribution, rod ejection and rod misalignment analyses are based on the arrangement of the shutdown and control groups of the rod cluster control assemblies shown in Figure 4.3-36. All shutdown rod cluster control assemblies are withdrawn before withdrawal of the control banks is initiated. In going from zero to 100 percent power, control banks A, B, C and D are sequentially withdrawn. The limits of rod positions and further discussion on the basis for rod insertion limits are provided in the Technical Specifications.

4.3.2.4.13 Reactor Coolant Temperature

Reactor coolant (or moderator) temperature control has added flexibility in reactivity control of the Westinghouse PWR. This feature takes advantage of the negative moderator temperature coefficient inherent in a PWR to:

1. Maximize return to power capabilities,

2. Provide ±5 percent power load regulation capabilities without requiring control rod compensation, and

3. Extend the time in cycle life for which daily load follow operations can be accomplished.

Reactor coolant temperature control supplements the dilution capability of the plant by lowering the reactor coolant temperature to supply positive reactivity through the negative moderator coefficient of the reactor. After the transient is over, the system automatically recovers the reactor coolant temperature to the programmed value.

Moderator temperature control of reactivity, like soluble boron control, has the advantage of not significantly affecting the core power distribution. However, unlike boron control, temperature control can be rapid enough to achieve reactor power change rates of 5 percent/minute.

4.3.2.4.14 Burnable Absorber Rods

The burnable absorber rods (discrete and/or integral) provide partial control of the excess reactivity available during the fuel cycle. In doing so, these rods prevent the moderator temperature coefficient from being positive at normal full power operating conditions. They perform this function by reducing the requirement for soluble poison in the moderator at the beginning of the fuel cycle as described previously. For purposes of illustration, a typical burnable absorber rod pattern in the core together with the number of rods per assembly is shown in Figure 4.3-5, while the arrangements within an assembly are displayed in Figure 4.3-4. The boron in the rods is depleted with burnup but at a sufficiently slow rate so that the resulting critical concentration of soluble boron is such that the moderator temperature coefficient remains non-positive for full power operating conditions.

4.3.2.4.15 Peak Xenon Startup

Compensation for the peak xenon buildup is accomplished using the boron control system. Startup from the peak xenon condition is accomplished with a combination of rod motion and boron dilution. The boron dilution may be made at any time, including during the shutdown period, provided the shutdown margin is maintained.

4.3.2.4.16 Load Follow Control and Xenon Control

During load follow maneuvers, power changes are accomplished using control rod motion and dilution or boration by the boron system as required. Control rod motion is limited by the control rod insertion limits on full length rods as provided in the Technical Specifications and discussed in Sections 4.3.2.4.12 and 4.3.2.4.13. The power distribution is maintained within acceptable limits through the location of the full length rod bank. Reactivity changes due to the changing xenon concentration can be controlled by rod motion and/or changes in the soluble boron concentration. Late in cycle life, extended load follow capability is obtained by augmenting the limited boron dilution capability at low soluble boron concentrations by temporary moderator temperature reductions.

Rapid power increases (5 percent/min) from part power load follow operation are accomplished with a combination of rod motion, moderator temperature reduction, and boron dilution. Compensation for the rapid power increase is accomplished initially by a combination of rod withdrawal and moderator temperature reduction. As the slower boron dilution takes affect after the initial rapid power increase, the moderator temperature returns to the programmed value.

4.3.2.4.17 Burnup

Control of the excess reactivity for burnup is accomplished using soluble boron and/or burnable absorber. The boron concentration must be limited during operating conditions to ensure the moderator temperature coefficient is within the Technical Specifications. Sufficient burnable absorber is installed at the beginning of a cycle to give the desired cycle lifetime without exceeding the boron concentration limit.

4.3.2.5 Control Rod Patterns and Reactivity Worth

The full length rod cluster control assemblies are designated by function as the control groups and the shutdown groups. The terms “group” and “bank” are used synonymously throughout this report to describe a particular grouping of control assemblies. The rod cluster assembly pattern is displayed in Figure 4.3-36 which is not expected to change during the life of the plant. The control banks are labeled A, B, C, and D and the shutdown banks are labeled SA, SB, etc., as applicable. Each bank, although operated and controlled as a unit, is comprised of two subgroups. The axial position of the full length rod cluster control assemblies may be controlled manually or automatically. The rod cluster control assemblies are all dropped into the core following actuation of reactor trip signals.

Two criteria have been employed for selection of the control groups. First the total reactivity worth must be adequate to meet the control requirements (i.e., fuel temperature, moderator temperature, redistribution and rod insertion allowance). Typical values for these parameters are listed in Table 4.3-3. Second, in view of the fact that these rods may be partially inserted at power operation, the total power peaking factor should be low enough to ensure that the power capability requirements are met. Analyses indicate that the first requirement can be met either by a single group or by two or more banks whose total worth equals at least the required amount. The axial power shape would be more peaked following movement of a single group of rods worth three to four percent $\Delta\rho$; therefore, four banks (described as A, B, C and D in Figure 4.3-36) each worth approximately one percent $\Delta\rho$ have been selected.

The position of control banks for criticality under any reactor condition is determined by the concentration of boron in the coolant. On an approach to criticality, boron is adjusted to ensure that criticality will be achieved with control rods above the insertion limit set by shutdown and other considerations (see the Technical Specifications). Early in the cycle there may also be a withdrawal limit at low power to maintain the moderator temperature coefficient within limits assumed in the safety analysis.

Allowable deviations due to misaligned control rods are discussed in the Technical Specifications.

A representative calculation for two banks of control rods withdrawn simultaneously (rod withdrawal accident) is given in Figure 4.3-37.

Calculation of control rod reactivity worth versus time following reactor trip involves both control rod velocity and differential reactivity worth. A representative rod position versus time of travel after rod release assumed is given in Figure 4.3-38. For nuclear design purposes, the reactivity worth versus rod position is calculated by a series of steady state calculations at various control rod positions assuming all rods out of the core as the initial position in order to minimize the initial reactivity insertion rate. Also to be conservative, the rod of highest worth is assumed stuck out of the core and the flux distribution (and thus reactivity importance) is assumed to be skewed to the bottom of the core. The result of a typical calculation is shown on Figure 4.3-39.

The shutdown groups provide additional negative reactivity to assure an adequate shutdown margin. Shutdown margin is defined as the amount by which the core would be subcritical at hot shutdown if all rod cluster control assemblies are tripped, but assuming that the highest worth control assembly remains fully withdrawn and no changes in xenon or boron take place. The loss of control rod worth due to the material irradiaiton is negligible since only bank D may be in the core under normal operating conditions.

The values given in Table 4.3-3 show that the available reactivity in withdrawn rod cluster control assemblies provides the design bases minimum shutdown margin allowing for the highest worth cluster to be at its fully withdrawn position. An allowance for the uncertainty in the calculated worth of N-1 rods is made before determination of the shutdown margin.

4.3.2.6 Criticality of the Reactor During Refueling and Criticality of Fuel Assemblies

The basis for maintaining the reactor subcritical during refueling is presented in Section 4.3.1.5 and a discussion of how control requirements are met is given in Sections 4.3.2.4 and 4.3.2.5.

Criticality of fuel assemblies outside the reactor is precluded by adequate design of fuel transfer and fuel storage facilities and by administrative control procedures. This section identifies those criteria important to criticality safety analyses.

New fuel is stored in 21 inch center to center racks in the new fuel storage facility with no water present but which are designed so as to prevent accidental criticality even if unborated water is present. For the flooded condition assuming new fuel of the highest anticipated enrichment in place, the effective multiplication factor does not exceed 0.95. For the normally dry condition the effective multiplication factor does not exceed 0.98 with fuel of the highest anticipated enrichment in place assuming possible sources of moderation such as those that could arise during fire

fighting operations. Credit is taken for the inherent neutron-absorbing effect of materials of construction of the racks.

In the analysis for the new fuel storage facilities, the fuel assemblies are assumed to be in their most reactive condition, namely fresh or undepleted and with no control rods or removable neutron absorbers present. Assemblies can not be closer together than the design separation provided by the storage facility except in special cases such as in fuel shipping containers where analyses are carried out to establish the acceptability of the design. The mechanical integrity of the fuel assembly is assumed.

A mechanical description of the fuel storage racks can be found in Section 9.1. A discussion of the SFP Region I and Region II oversized inspection cells is contained in the Technical Specification Bases 3.7.17.

High Density Region I Racks in Spent Fuel Pool 1 (SFP1) (any storage configuration) and in Spent Fuel Pool 2 (SFP2) (any storage configuration)

For High Density Region I Racks in SFP1/SFP2 (any storage configuration), fuel assembly spacing is such that subcriticality is ensured, $k_{eff} \leq 0.95$ taking credit for soluble boron and $k_{eff} < 1.0$ assuming unborated water.

Verification that the design criteria for fuel storage are met is achieved through the use of the three dimensional KENO5a Monte Carlo code as described in Reference 42.

High Density Region II Racks in SFP1/SFP2

For High Density Region II Racks in SFP1/SFP2, fuel assembly spacing is such that subcriticality is ensured, $k_{eff} \leq 0.95$ taking credit for soluble boron and $k_{eff} < 1.0$ assuming unborated water.

Verification that the design criteria for fuel storage are met is achieved through the use of the taking credit for soluble boron and $k_{eff} < 1.0$ assuming unborated water as described in Reference 42.

Low Density Racks in Containment Refueling Cavity (5 x 5 module, any storage configuration)

For the low density racks in the containment refueling cavity (any storage configuration), fuel assembly spacing is such that subcriticality is ensured ($k_{eff} \leq 0.95$) even if assemblies are immersed in unborated water.

Verification that the design criteria for fuel storage are met is achieved through the use of the three dimensional Monte Carlo code package NITAWL-KENO5a, as described in References 39 and 45.

A boron concentration equal to or greater than the Technical Specification limit assures that a dilution event which will result in a k_{eff} greater than 0.95 is not credible. This is demonstrated by a boron dilution analysis performed for the CPNPP Spent Fuel pools. This conclusion is based on the following: (1) a substantial amount of water is needed in order to dilute the SFP to the design k_{eff} of 0.95, (2) since such a large water volume turnover is required, a SFP dilution event would be readily detected by plant personnel via alarms, flooding in the fuel and auxiliary

buildings or by normal operator rounds through the SFP area, and (3) evaluations indicate that, based on the flow rates of non-borated water normally available to the SFP, taken in conjunction with significant operator errors, and equipment failures, sufficient time is available to detect and respond to a dilution event. In addition, there is significant conservatism built into this evaluation; for example, the cooling of the spent fuel pools can be performed by one train supplying common water to both pools. This cooling configuration would allow credit of the volume of both pools and substantially increase the dilution time estimates presented. However, because the flexibility exists for the cooling system to be totally dedicated to one pool, only one pool volume is considered in this evaluation. As discussed above, calculations have been performed on a 95/95 basis to show that the spent fuel rack k_{eff} remains less than 1.0 with non-borated water in the pool. Thus, even if the SFP were diluted to concentrations approaching zero ppm, the fuel in the racks would remain subcritical.

4.3.2.7 Stability

4.3.2.7.1 Introduction

The stability of the PWR cores against xenon-induced spatial oscillations and the control of such transients are discussed extensively in References 6, 14, 15, and 16. A summary of these reports is given in the following discussion and the design bases are given in Section 4.3.1.6.

In a large reactor core, xenon-induced oscillations can take place with no corresponding change in the total power of the core. The oscillation may be caused by a power shift in the core which occurs rapidly by comparison with the xenon-iodine time constants. Such a power shift occurs in the axial direction when a plant load change is made by control rod motion and results in a change in the moderator density and fuel temperature distributions. Such a power shift could occur in the diametral plane of the core as a result of abnormal control action.

Due to the negative power coefficient of reactivity, PWR cores are inherently stable to oscillations in total power. Protection against total power instabilities is provided by the Control and Protection System as described in Section 7.7. Hence, the discussion on the core stability will be limited here to xenon-induced spatial oscillations.

4.3.2.7.2 Stability Index

Power distributions, either in the axial direction or in the X-Y plane, can undergo oscillations due to perturbations introduced in the equilibrium distributions without changing the total core power. The overtones in the current PWRs, and the stability of the core against xenon-induced oscillations can be determined in terms of the eigenvalues of the first flux overtones. Writing, either in the axial direction or in the X-Y plane, the eigenvalue ς of the first flux harmonic as:

$$\varsigma = b + ic \qquad (4.3\text{-}1)$$

then b is defined as the stability index and T = 2 /c as the oscillation period of the first harmonic. The time-dependence of the first harmonic in the power distribution can now be represented as:

$$(t) = Ae^{t} = ae^{bt}\cos ct \qquad (4.3\text{-}2)$$

where A and a are constants. The stability index can also be obtained approximately by:

$$b = \frac{1}{T} \ln \frac{An + 1}{An} \qquad (4.3\text{-}3)$$

where A_n, A_{n+1} are the successive peak amplitudes of the oscillation and T is the time period between the successive peaks.

4.3.2.7.3 Prediction of the Core Stability

The stability of the core described herein (i.e., with 17 x 17 fuel assemblies) against xenon-induced spatial oscillations is expected to be equal to or better than that of earlier designs. The prediction is based on a comparison of the parameters which are significant in determining the stability of the core against the xenon-induced oscillations, namely: 1) the overall core size is unchanged and spatial power distributions will be similar, 2) the moderator temperature coefficient is expected to be similar to or slightly more negative, and 3) the Doppler coefficient of reactivity is expected to be equal to or slightly more negative at full power.

Analysis of both the axial and X-Y xenon transient tests, discussed in Section 4.3.2.7.5, shows that the calculational model is adequate for the prediction of core stability.

4.3.2.7.4 Stability Measurements

1. Axial measurements

   Two axial xenon transient tests conducted in a PWR with a core height of 12 feet and 121 fuel assemblies is reported in Reference 17, and will be briefly discussed here. The tests were performed at approximately 10 percent and 50 percent of cycle life.

   Both a free-running oscillation test and a controlled test were performed early in cycle life. The second test at mid-cycle consisted of a free-running oscillation test only. In each of the free-running oscillation tests, a perturbation was introduced to the equilibrium power distribution through an impulse motion of the control Bank D and the subsequent oscillation period. In the controlled test conducted early in the cycle, the part length rods were used to follow the oscillations to maintain an axial offset within the prescribed limits. The axial offset of power was obtained from the excore ion chamber readings (which had been calibrated against the incore flux maps) as a function of time for both free-running tests as shown in Figure 4.3-40.

   The total core power was maintained constant during these spatial xenon tests, and the stability index and the oscillation period were obtained from a least-square fit of the axial offset data in the form of Equation (4.3-2). The axial offset of power is the quantity that properly represents the axial stability in the sense that it essentially eliminates any contribution from even order harmonics including the fundamental mode. The conclusions of the tests are:

   a. The core was stable against induced axial xenon transients both at the core average burnups of 1550 MWD/MTU and 7700 MWD/MTU. The measured

stability indices are -0.041 hr^{-1} for the first test (Curve 1 of Figure 4.3-40) and -0.014 hr^{-1} for the second test (Curve 2 of Figure 4.3-40). The corresponding oscillation periods are 32.4 hrs. and 27.2 hrs., respectively.

b. The reactor core becomes less stable as fuel burnup progresses and the axial stability index was essentially zero at 12,000 MWD/MTU.

2. Measurements in the X-Y plane

Two X-Y xenon oscillation tests were performed at a PWR plant with a core height of 12 feet and 157 fuel assemblies. The first test was conducted at a core average burnup of 1540 MWD/MTU and the second at a core average burnup of 12900 MWD/MTU. Both of the X-Y xenon tests show that the core was stable in the X-Y plane at both burnups. The second test shows that the core became more stable as the fuel burnup increased and all Westinghouse PWRs with 121 and 157 assemblies are expected to be stable throughout their burnup cycles.

In each of the two X-Y tests, a perturbation was introduced to the equilibrium power distribution through an impulse motion of one rod cluster control unit located a long the diagonal axis. Following the perturbation, the uncontrolled oscillation was monitored using the moveable detector and thermocouple system and the excore power range detectors. The quadrant tilt difference (QTD) is the quantity that properly represents the diametral oscillation in the X-Y plane of the reactor core in that the differences of the quadrant average powers over two symmetrically opposite quadrants essentially eliminates the contribution to the oscillation from the azimuthal mode. The QTD data were fitted in the form of Equation (4.3-2) through a least-square method. A stability index of -0.076 hr^{-1} with a period of 29.6 hours was obtained from the thermocouple data shown in Figure 4.3-41.

It was observed in the second X-Y xenon test that the PWR core with 157 fuel assemblies had become more stable due to an increased fuel depletion and the stability index was not determined.

4.3.2.7.5 Comparison of Calculations with Measurements

The analysis of the axial xenon transient tests was performed in an axial slab geometry using a flux synthesis technique. The direct simulation of the axial offset data was carried out using the PANDA Code [18]. The analysis of the X-Y xenon transient tests was performed in an X-Y geometry using TORTISE which is an advanced version of TURTLE Code [10]. Both the PANDA and TORTISE codes solve the two-group time-dependent neutron diffusion equation with time-dependent xenon and iodine concentrations. The fuel temperature and moderator density feedback is limited to a steady-state model. All the X-Y calculations were performed in an average enthalpy plane.

The basic nuclear cross-sections used in this study were generated from the unit cell depletion program ARK, which has evolved from the codes LEOPARD [19] and CINDER [20]. The detailed experimental data during the tests including the reactor power level, enthalpy rise and the impulse motion of the control rod assembly, as well as the plant follow burnup data were closely simulated in the study.

The results of the stability calculation for the axial tests are compared with the experimental data in Table 4.3-5. The calculations show conservative results for both of the axial tests with a margin of approximately -0.01 hr^{-1} in the stability index.

An analytical simulation of the first X-Y xenon oscillation test shows a calculated stability index of -0.081 hr^{-1}, in good agreement with the measured value of -0.076 hr^{-1}. As indicated earlier, the second X-Y xenon test showed that the core had become more stable compared to the first test and no evaluation of the stability index was attempted. This increase in the core stability in the X-Y plane due to increased fuel burnup is due mainly to the increased magnitude of the negative moderator temperature coefficient.

Previous studies of the physics of xenon oscillations, including three dimensional analysis, are reported in the series of topical reports, References 14, 15 and 16. A more detailed description of the experimental results and analysis of the axial and X-Y xenon transient tests is presented in Reference 17 and Section 1 of Reference 21.

4.3.2.7.6 Stability Control and Protection

The excore detector system is utilized to provide indications of xenon-induced spatial oscillations. The readings from the excore detectors are available to the operator and also form part of the protection system.

1. Axial power distribution

   For maintenance of proper axial power distributions, the operator is instructed to maintain an axial offset within a prescribed operating band, based on the excore detector readings. Should the axial offset be permitted to move far enough outside this band, the protection limit will be reached and power will be automatically reduced.

   Twelve foot PWR cores become less stable to axial xenon oscillations as fuel burnup progresses. However, free xenon oscillations are not allowed to occur except for special tests. The fuel length control rod banks are sufficient to dampen and control any axial xenon oscillations present. Should the axial offset be inadvertently permitted to move far enough outside the control band due to an axial xenon oscillation, or any other reason, the protection limit on axial offset will be reached and power will be automatically reduced.

2. Radial power distribution

   The core described herein is calculated to be stable against X-Y xenon induced oscillations at all times in life.

   The X-Y stability of large PWRs has been further verified as part of the startup physics test program for cores with 193 fuel assemblies. The measured X-Y stability of the cores with 157 and 193 assemblies was in good agreement with the calculated stability as discussed in Sections 4.3.2.7.4 and 4.3.2.7.5. In the unlikely event that X-Y oscillations occur, back-up actions are possible and would be implemented, if necessary, to increase the natural stability of the core. This is based on the fact that several actions could be

taken to make the moderator temperature coefficient more negative (less positive), which will increase the stability of the core in the X-Y plane.

Provisions for protection against non-symmetric perturbations in the X-Y power distribution that could result from equipment malfunctions are made in the protection system design. This includes control rod drop, rod misalignment and asymmetric loss of coolant flow.

A more detailed discussion of the power distribution control in PWR cores is presented in References 6 and 7.

4.3.2.8 Vessel Irradiation

A brief review of the methods and analyses used in the determination of neutron and gamma-ray flux attenuation between the core and the pressure vessel is given below. A more complete discussion on the pressure vessel irradiation and surveillance program is given in Section 5.3.

The materials that serve to attenuate neutrons originating in the core and gamma rays from both the core and structural components consist of the core baffle, core barrel, neutron pads and associated water annuli all of which are within the region between the core and the pressure vessel.

In general, few group neutron diffusion theory codes are used to determine fission power density distributions within the active core, and the accuracy of these analyses is verified by incore measurements on operating reactors. Region and rodwise power sharing information from the core calculations is then used as source information in two-dimensional S_n transport calculations which compute the flux distributions throughout the reactor.

The neutron flux distribution and spectrum in the various structural components varies significantly from the core to the pressure vessel. Representative values of the neutron flux distribution and spectrum are presented in Table 4.3-6. The values listed are based on time averaged equilibrium cycle reactor core parameters and power distributions; and, thus, are suitable for long term nvt projections and for correlation with radiation damage estimates.

As discussed in Section 5.3, the irradiation surveillance program utilizes actual test samples to verify the accuracy of the calculated fluxes at the vessel.

4.3.3 ANALYTICAL METHODS

Calculations required in nuclear design consist of the following three distinct types which are performed in sequence:

1. Determination of effective fuel temperatures

2. Generation of macroscopic few-group parameters

3. Space-dependent, few-group diffusion calculations

4.3.3.1 Fuel Temperature (Doppler) Calculations

Temperatures vary radially within the fuel rod, depending on heat generation rate in the pellet, the conductivity of the materials in the pellet, gap and cladding, and coolant temperature.

Fuel temperatures for use in most nuclear design Doppler calculations are obtained from a simplified version of the Westinghouse fuel rod design model described in Section 4.2.1.3.1, which considers the effect of radial variation of pellet conductivity, expansion-coefficient and heat generation rate, elastic deflection of the cladding, and a gap conductance which depends on the initial fill gas, the hot open gap dimension, and the fraction of the pellet over which the gap is closed. The fraction of the gap assumed closed represents an empirical adjustment to produce good agreement with observed reactivity data at BOL. Further gap closure occurs with burnup and accounts for the decrease in Doppler defect with burnup which has been observed in operating plants.

Radial power distributions in the pellet as a function of burnup are obtained from LASER (Reference 4) calculations.

The effective U-238 temperature for resonance absorption is obtained from the radial temperature distribution by applying a radially dependent weighting function. The weighting function was determined from REPAD (Reference 5) Monte Carlo calculations of resonance escape probabilities in several steady state and transient temperature distributions. In each case, a flat pellet temperature was determined which produced the same resonance escape probability as the actual distribution. The weighting function was empirically determined from these results. The effective Pu-240 temperature for resonance absorption is determined by a convolution of the radial distribution of Pu-240 number densities from LASER burnup calculations and the radial weighting function. The resulting temperature is burnup dependent, but the difference between U-238 and Pu-240 temperatures, in terms of reactivity effects, is small.

The effective pellet temperature for pellet dimensional change is that value which produces the same outer pellet radius in a virgin pellet as that obtained from the temperature model. The effective cladding temperature for dimensional change is its average value.

The temperature calculational model has been validated by plant Doppler defect data and Doppler coefficient data. Stability index measurements also provide a sensitive measure of the Doppler coefficient near full power (see Section 4.3.2.7). It can be seen that Doppler defect data are typically within 0.2 percent Δp of prediction.

Starting with Unit 2 Cycle 19 and Unit 1 Cycle 22, fuel temperature coefficients are calculated with NEXUS/PARAGON/ANC (References 25 and 26).

4.3.3.2 Macroscopic Group Constants

The PHOENIX-P computer code is a two-dimensional, multi-group, transport based lattice code and capable of providing all necessary data for PWR analysis. Being a dimensional lattice code, PHOENIX-P does not rely on pre-determined spatial/spectral interaction assumptions for a heterogeneous fuel lattice, hence, provides a more accurate multi-group flux solution than versions of LEOPARD/CINDER. The PHOENIX-P computer code is approved by the USNRC as the lattice code for generating macroscopic and microscopic few group cross sections for PWR analysis (Reference 22).

The solution for the detailed spatial flux and energy distribution is divided into two major steps in PHOENIX-P. In the first step, a two-dimensional fine energy group nodal solution is obtained which couples individual subcell regions (pellet, cladding and moderator) as well as surrounding pins. PHOENIX-P uses a method based on the Carlvik's collision probability approach and heterogeneous response fluxes which preserves the heterogeneity of the pin cells and their surroundings. The nodal solution provides accurate and detailed local flux distribution, which is then used to spatially homogenize the pin cells to fewer groups.

The second step in the solution process solves for the angular flux distribution using a standard S4 discrete ordinates calculation. This step is based on the group-collapsed and homogenized cross sections obtained from the first step of the solution. The S4 fluxes are then used to normalize the detailed spatial and energy nodal fluxes. The normalized nodal fluxes are used to compute reaction rates, power distribution and to deplete the fuel and burnable absorbers. A standard B1 calculation is employed to evaluate the fundamental mode critical spectrum and to provide an improved fast diffusion coefficient for the core spatial codes.

The PHOENIX-P code employs a 70 energy group library, which has been derived mainly from ENDF/B-VI files. The PHOENIX-P cross sections library was designed to properly capture integral properties of the multi-group data during group collapse, and enabling proper modeling of important resonance parameters. The library contains all neutronic data necessary for modeling fuel, fission products, cladding and structural data, coolant, and control/burnable absorber materials present in Light Water Reactor cores.

Group constants for burnable absorber cells, guide thimbles, instrument thimbles, control rod cells and other non-fuel cells can be obtained directly from PHOENIX-P without any adjustments such as those required in the cell or 1D lattice codes.

PARAGON has been approved by the NRC as the new generation of Westinghouse lattice code (Reference 25). PARAGON is a replacement for PHOENIX-P and its primary use will be to provide the same types of input data that PHOENIX-P generates for use in three dimensional core simulator codes. This includes macroscopic cross sections, microscopic cross sections for feedback adjustments to the macroscopic cross sections, pin factors for pin power reconstruction calculations, discontinuity factors for a nodal method solution, and other data needed for safety analysis or other downstream applications.

PARAGON is based on collision probability – interface current cell coupling methods. PARAGON provides flexibility in modeling that was not available in PHOENIX-P including exact cell geometry representation instead of cylinderization, multiple rings and regions within the fuel pin and the moderator cell geometry, and variable cell pitch. The solution method permits flexibility in choosing the quality of the calculation through both increasing the number of regions modeled within the cell and the number of angular current directions tracked at the cell interfaces.

The calculation scheme in PARAGON is based on the conventional lattice modules: resonance calculation, flux solution, leakage correction and depletion. The detailed theory of these modules is described in Reference 25. The cross-section resonance calculation module is based on the space dependent Dancoff method (Reference 25); it is a generalization of the PHOENIX-P methodology that permits to subdivide the fuel pin into many rings and therefore generates space dependent self-shielded isotopic cross-sections. The flux solution module uses the interface

current collision probability method and permits a detailed representation of the fuel cells (Reference 25). The other two modules (leakage and depletion) are similar to the ones used in PHOENIX-P.

The current PARAGON cross section library is a 70-group library, based on the ENDF/B basic nuclear data, with the same group structure as the library currently used with PHOENIX-P. The PARAGON qualification library has been improved through the addition of more explicit fission products and fission product chains (Reference 25). PARAGON is however designed to employ any number of energy groups.

The new NEXUS cross-section generation system uses PARAGON as the lattice code (Reference 26).

4.3.3.3 Spatial Few-Group Diffusion Calculations

ANC (Advanced Nodal Code) is a multidimensional nodal analysis program used to predict nuclear reactor core reactivity and assembly and rod distributions for normal and off-normal conditions (Reference 9).

Nodal three-dimensional calculations are carried out to determine the critical boron concentrations and power distributions. The moderator coefficient is evaluated by varying the inlet temperature in the same calculations used for power distribution and reactivity predictions. Validation of reactivity calculations is associated with the validation of the group constants themselves, as discussed in Section 4.3.3.2. Validation of the Doppler calculations is associated with the fuel temperature validation, as discussed in Section 4.3.3.1. Validation of the moderator coefficient calculation is obtained by comparison with plant measurements at HZP conditions.

Axial calculations are used to determine differential control rod worth curves (reactivity versus rod insertion) and axial power shapes during steady state and transient xenon conditions. Group constants are obtained from three-dimensional nodal calculations homogenized by flux volume weighting.

Validation of the spatial codes for calculating power distributions involves the use of incore and excore detectors, and is discussed in Section 4.3.2.2.7.

Beginning with Unit 2 Cycle 19 and Unit 1 Cycle 22, the reference code used for spatial calculations is ANC (including Ref. 27) with NEXUS/PARAGON (Ref. 26) generated cross-section sets.

REFERENCES

1. “Westinghouse Anticipated Transients Without Trip Analysis,” WCAP-8330, August 1974.

2. Langford, F. L. and Nath, R. J., “Evaluation of Nuclear Hot Channel Factor Uncertainties,” WCAP-7308-L (Proprietary) and WCAP-7810 (Non-Proprietary), December 1971.

3. Kersting P.J., et.al., “Assessment of Clad Flattening and Densification Power Spike Factor Elimination in Westinghouse Nuclear Fuel”, WCAP-13589-A (Proprietary), March 1995.

4. C. G. Poncelet, LASER - A Depletion Program for Lattice Calculations Based on MUFT and THERMOS, WCAP-6073, April 1966.

5. J. E. Olhoeft, The Doppler Effect for a Non-Uniform Temperature Distribution in Reactor Fuel Elements, WCAP-2048, July 1962.

6. J. S. Moore, “Power Distribution Control of Westinghouse Pressurized Water Reactors,” WCAP-7811, December 1971.

7. Morita, T., et al., “Power Distribution Control and Load Follow Procedures,” WCAP-8385 (Proprietary) and WCAP-8403 (Non-Proprietary), September 1974.

8. McFarlane, A. F., “Power Peaking Factors,” WCAP-7912-P-A (Proprietary) and WCAP-7912-A (Non-Proprietary), January 1975.

9. Davidson, S. L., ed, et al., "ANC: Westinghouse Advanced Nodal Computer Code," WCAP-10965-P-A, September 1986.

10. Barry, R. F. and Altomare, S., “The TURTLE 24.0 Diffusion Depletion Code,” WCAP-7213-P-A (Proprietary) and WCAP-7758-A (Non-Proprietary), January 1975.

11. Cermak, J. O., et al., “Pressurized Water Reactor pH - Reactivity Effect Final Report,” WCAP-3696-8 (EURAEC-2074), October 1968.

12. WCAP-12472-P-A, “BEACON Core Monitoring and Operations Support System,” August 1994 (including “Addendum 1-A,” January 2000).

13. Miller, R. W., et al., "Relaxation of Constant Axial Offset Control, FQ Surveillance Technical Specification," WCAP-10216-P-A, June 1983.

14. Poncelet, C. G. and Christie, A. M., “Xenon-Induced Spatial Instabilities in Large PWRs,” WCAP-3680-20, (EURAEC-1974) March 1968.

15. Skogen, F. B. and McFarlane, A. F., “Control Procedures for Xenon-Induced X-Y Instabilities in Large PWRs,” WCAP-3680-21, (EURAEC-2111), February 1969.

16. Skogen, F. B. and McFarlane, A. F., “Xenon-Induced Spatial Instabilities in Three Dimensions,” WCAP-3680-22 (EURAEC-2116), September 1969.

17. Lee, J. C., et al., “Axial Xenon Transient Tests at the Rochester Gas and Electric Reactor,” WCAP-7964, June 1971.

18. Barry, R. F. et al., “The PANDA Code,” WCAP-7048-P-A (Proprietary) and WCAP-7757-A (Non-Proprietary), January 1975.

19. Barry, R. F., “LEOPARD - A Spectrum Dependent Non-Spatial Depletion Code for the IBM-7094,” WCAP-3269-26, September 1963.

20. England, T. R., “CINDER - A One-Point Depletion and Fission Product Program,” WAPD-TM-334, August 1962.

21. Eggleston, F. T., “Safety-Related Research and Development for Westinghouse Pressurized Water Reactors, Program Summaries,” WCAP-8768, latest revision.

22. Nguyen, T. Q., et al., "Qualification of the PHOENIX-P/ANC Nuclear Design System for Pressurized Water Reactor Cores," WCAP-11596-P-A, June 1988.

23. Deleted.

24. Nodvik, R. J., et al., “Supplementary Report on Evaluation of Mass Spectrometric and Radiochemical Analyses of Yankee Core I Spent Fuel, Including Isotopes of Elements Thorium Through Curium,” WCAP-6086, August 1969.

25. Ouisloumen, M. et al., “Qualification of the Two-Dimensional Transport Code PARAGON,” WCAP-16045-P-A, Westinghouse, August 2004.

26. Zhang, B. et al., “Qualification of the NEXUS Nuclear Data Methodology,” WCAP-16045-P-A, Addendum 1, Westinghouse, November 2005.

27. Zhang, B., et. al., “Qualification of the New Pin Power Recovery Methodology,” WCAP-10965-P-A, Addendum 2-A, September, 2010.

28. Deleted.

29. Deleted.

30. Deleted.

31. Deleted.

32. Deleted.

33. Deleted.

34. Deleted.

35. Deleted.

36. Deleted.

37. Letter from R. L. Tedesco (NRC) to T. M. Anderson (W), Safety Evaluation of WCAP-9500, “Reference Core Report - 17x17 Optimized Fuel Assembly,” NRC SER dated May 22, 1981.

38. Deleted.

39. Letter from W. J. Cahill, Jr. (TU Electric) to NRC, “License Amendment Request 92-05, Increase in Fuel Enrichment,” TXX-92468, October 16, 1992.

40. Deleted.

41. Deleted.

42. “Comanche Peak Nuclear Power Plant Units 1 and 2 Spent Fuel Pool Criticality Safety Analysis”, WCAP-17728-P, Revision 1, October 2013.

43. Deleted.

44. Deleted.

45. Letter from C. L. Terry (TXU Electric) to NRC, “Submittal of License Amendment Request 94-009, Increase in Fuel Enrichment,” TXX-94048, April 22, 1994.

TABLE 4.3-1
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TABLE 4.3-2A
NUCLEAR DESIGN PARAMETERS
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TABLE 4.3-2B
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TABLE 4.3-3
TYPICAL REACTIVITY PARAMETERS FOR ROD CLUSTER CONTROL ASSEMBLIES

Reactivity Effects, percent $\Delta\rho$		Beginning-of-Life (First Cycle)	End-of-Life (First Cycle)	End-of-Life (Equilibrium Cycle)
1.	Control requirements			
	Fuel temperature, Doppler (%$\Delta\rho$)	1.36	1.12	1.10
	Moderator temperature[(a)] (%$\Delta\rho$)	0.10	1.17	1.34
	Redistribution (%$\Delta\rho$)	0.50	0.85	0.85
	Rod insertion allowance (%$\Delta\rho$)	0.50	0.50	0.50
2.	Total control (%$\Delta\rho$)	2.46	3.64	3.79
3.	Estimated rod cluster control assembly worth (53 rods, Ag-In-Cd or Hf)			
	a. All full length assemblies inserted (%$\Delta\rho$)	7.59	7.68	7.30
	b. All but one (highest worth) assemblies inserted (%$\Delta\rho$)	6.62	6.52	6.20
4.	Estimated rod cluster control assembly credit with 10 percent adjustment to accomodate uncertainties, 3b - 10 percent (%$\Delta\rho$)	5.96	5.87	5.58
5.	Shutdown margin available, 4-2 (%$\Delta\rho$)	3.50	2.23	1.79

a) Includes void effects.

TABLE 4.3-4
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TABLE 4.3-5
AXIAL STABILITY INDEX PRESSURIZED WATER REACTOR CORE WITH A 12 FOOT HEIGHT

Burnup (MWD/MTU)	F_Z	C_B (ppm)	Stability Index (hr^{-1}) Exp	Calc
1550	1.34	1065	-0.041	-0.032
7700	1.27	700	-0.014	-0.006
		Difference:	+0.027	+0.026

TABLE 4.3-6
TYPICAL NEUTRON FLUX LEVELS (n/cm^2-sec) AT FULL POWER

	E>1.0MeV	0.111MeV < E <1.0MeV	0.3eV < E <0.111MeV	<E 0.3eV
Core Center	9.98×10^{13}	1.11×10^{14}	2.17×10^{14}	5.36×10^{13}
Core Outer Radius At Mid-Height	4.24×10^{13}	4.85×10^{13}	9.52×10^{13}	2.21×10^{13}
Core Top, on Axis	2.62×10^{13}	2.13×10^{13}	1.31×10^{14}	4.35×10^{13}
Core Bottom, on Axis	2.70×10^{13}	2.25×10^{13}	1.33×10^{14}	4.74×10^{13}
Pressure Vessel Inner Diameter Azimuthal Peak, Core Mid-Height	2.08×10^{10}	2.83×10^{10}	6.18×10^{10}	1.20×10^{11}

TABLE 4.3-7
DELETED

TABLE 4.3-8
DELETED

TABLE 4.3-9
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TABLE 4.3-10
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TABLE 4.3-11
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4.4 THERMAL AND HYDRAULIC DESIGN

4.4.1 DESIGN BASES

The overall objective of the thermal and hydraulic design of the reactor core is to provide adequate heat transfer which is compatible with the heat generation distribution in the core such that heat removal by the Reactor Coolant System (RCS) or the Emergency Core Cooling System (ECCS) (when applicable) assures that the following performance and safety criteria requirements are met:

1. Fuel damage (defined as penetration of the fission product barrier, i.e, the fuel rod clad) is not expected during normal operation and operational transients (Condition I) or any transient conditions arising from faults of moderate frequency (Condition II). It is not possible, however, to preclude a very small number of rod failures. These will be within the capability of the plant cleanup system and are consistent with the plant design bases.

2. The reactor can be brought to a safe state following a Condition III event with only a small fraction of fuel rods damaged (see above definition) although sufficient fuel damage might occur to preclude resumption of operation without considerable outage time.

3. The reactor can be brought to a safe state and the core can be kept subcritical with acceptable heat transfer geometry following transients arising from Condition IV events.

To satisfy the above requirements, the following design bases have been established for the thermal and hydraulic design of the reactor core.

4.4.1.1 Departure from Nucleate Boiling Design Basis

Basis

There will be at least a 95 percent probability that departure from nucleate boiling (DNB) will not occur on the limiting fuel rods during normal operation and operational transients and any transient conditions arising from faults of moderate frequency (Condition I and II events) at a 95 percent confidence level.

Discussion

The design method employed to meet the DNB design basis for both OFA and VANTAGE 5 fuel assemblies is the Revised Thermal Design Procedure (RTDP), Reference 3. Note, the VANTAGE+ fuel design is often referred to as VANTAGE 5 or OFA to distinguish assemblies containing IFMs from assemblies without IFMs, respectively. With the RTDP methodology, uncertainties in plant operating parameters, nuclear and thermal parameters, fuel fabrication parameters, computer codes and DNB correlation predictions are considered statistically to obtain DNB uncertainty factors. Based on the DNB uncertainty factors, RTDP design limit DNBR values are determined such that there is at least a 95 percent probability at a 95 percent confidence level that DNB will not occur on the most limiting fuel rod during normal operation and operational transients and during transient conditions arising from faults of moderate frequency (Condition I and II events as defined in ANSI N18.2).

Uncertainties in the plant operating parameters (pressurizer pressure, primary coolant temperature, reactor power, reactor coolant system flow, etc.) have been evaluated. Only the random portion of the plant operating parameter uncertainties is included in the statistical combination. Instrumentation bias is treated as a direct DNBR penalty. Since the parameter uncertainties are considered in determining the RTDP design limit DNBR values, the plant safety analyses are performed using input parameters at their nominal values.

The RTDP design limit DNBR for OFA fuel and the WRB-1 correlation is 1.23 for both typical and thimble cells. The RTDP design limit DNBR values are 1.23 and 1.22 for the typical and thimble cells, respectively, for VANTAGE 5 fuel and the WRB-2 correlation. The design limit DNBR values are used as a basis for the technical specifications and for consideration of the applicability of unreviewed safety questions as defined in 10CFR 50.59.

To maintain DNBR margin to offset DNB penalties such as that due to fuel rod bow (see paragraph 4.4.2.1.5), the safety analyses were performed to DNBR limits higher than the design limit DNBR values. The difference between the design limit DNBRs and the safety analysis limit DNBRs results in available DNBR margin. The net DNBR margin, after consideration of all penalties, is available for operating and design flexibility.

The Standard Thermal Design Procedure (STDP) is used for those analyses where RTDP is not applicable. In the STDP method the parameters used in analysis are treated in a conservative way from a DNBR standpoint. The parameter uncertainties are applied directly to the plant safety analyses input values to give the lowest minimum DNBR. The DNBR limit for STDP is the appropriate DNB correlation limit increased by sufficient margin to offset the applicable DNBR penalties.

By preventing DNB, adequate heat transfer is assured between the fuel clad and the reactor coolant, thereby preventing clad damage as a result of inadequate cooling. Maximum fuel rod surface temperature is not a design basis as it will be within a few degrees of coolant temperature during operation in the nucleate boiling region. Limits provided by the nuclear control and protection systems are such that this design basis will be met for transients associated with Condition II events including overpower transients. There is an additional large DNBR margin at rated power operation and during normal operating transients.

4.4.1.2 Fuel Temperature Design Basis

Basis

During modes of operation associated with Condition I and Condition II events, there is at least a 95 percent probability that the peak kW/ft fuel rods will not exceed the UO_2 melting temperature at the 95 percent confidence level. The melting temperature of UO_2 is taken as 5080°F, Reference [1], unirradiated and decreasing 58°F per 10,000 MWD/MTU. By precluding UO_2 melting, the fuel geometry is preserved and possible adverse effects of molten UO_2 on the cladding are eliminated. To preclude center melting and as a basis for overpower protection system setpoints, a calculated centerline fuel temperature of 4700°F has been selected as the overpower limit. This provides sufficient margin for uncertainties in the thermal evaluations as described in Section 4.4.2.6.1.

Discussion

Fuel rod thermal evaluations are performed at rated power, maximum overpower and during transients at various burnups. These analyses assure that this design basis as well as the fuel integrity design bases given in Section 4.2 are met. They also provide input for the evaluation of Condition III and IV events given in Chapter 15.

4.4.1.3 Core Flow Design Basis

Basis

Most of the thermal flow rate will pass through the fuel rod region of the core and be effective for fuel rod cooling. Coolant flow through the thimble tubes as well as the leakage from the core barrel-baffle region into the core are not considered effective for heat removal.

Discussion

Core cooling evaluations are based on the thermal flow rate (minimum flow) entering the reactor vessel. A small fraction of this value is allotted as bypass flow. This includes Rod Cluster Control (RCC) guide thimble cooling flow, head cooling flow, baffle leakage, and leakage to the vessel outlet nozzle. The amount of the bypass flow is a function of the fuel design and is confirmed for each reload.

4.4.1.4 Hydrodynamic Stability Design Basis

Basis

Modes of operation associated with Condition I and II events shall not lead to hydrodynamic instability.

4.4.1.5 Other Considerations

The above design bases together with the fuel clad and fuel assembly design bases given in Section 4.2.1 are sufficiently comprehensive so additional limits are not required.

Fuel rod diametral gap characteristics, moderator-coolant flow velocity and distribution, and moderator void are not inherently limiting. Each of these parameters is incorporated into the thermal and hydraulic models used to ensure the above mentioned design criteria are met. For instance, the fuel rod diametral gap characteristics change with time and the fuel rod integrity is evaluated on that basis. The effect of the moderator flow velocity and distribution and moderator void distribution are included in the core thermal-hydraulic evaluation and thus affect the design bases.

Meeting the fuel clad integrity criteria covers possible effects of clad temperature limitations. As noted in Section 4.2, the fuel rod conditions changed with time. A single clad temperature limit for Condition I or Condition II events is not appropriate since of necessity it would be overly conservative. A clad temperature limit is applied to the loss of coolant accident (Section 15.6.5), control rod ejection accident (Section 15.4.8), and locked rotor accident (Section 15.3.3).

4.4.2 DESCRIPTION OF THERMAL AND HYDRAULIC DESIGN OF THE REACTOR CORE

4.4.2.1 Critical Heat Flux Ratio or Departure from Nucleate Boiling Ratio and Mixing Technology

DNBRs are calculated using approved correlations and the definitions described in the following sections. An approved core thermal-hydraulic computer code is used to determine the flow distribution in the core and the local conditions in the hot channel for use in the DNB correlation. The use of hot channel factors is discussed in Section 4.4.4.3.1 (nuclear hot channel factors) and in Section 4.4.2.1.4 (engineering hot channel factors).

4.4.2.1.1 DNB Technology

The primary DNB correlation used for the analysis of 17 x 17 OFA fuel is the WRB-1 correlation (Reference 83). The WRB-1 correlation was developed based exclusively on the large bank of mixing vane grid rod bundle CHF data (over 1100 points) that Westinghouse has collected. The WRB-1 correlation, based on local fluid conditions, represents the rod bundle data with better accuracy over a wide range of variables than the previous correlation used in design. This correlation accounts directly for both typical and thimble cold wall cell effects, uniform and nonuniform heat flux profiles, and variations in rod heated length and in grid spacing. A correlation limit DNBR of 1.17 is applicable for the WRB-1 correlation.

The applicable range of parameters for the WRB-1 correlation is:

Pressure	$1440 \leq P \leq 2490$ psia
Local mass velocity	$0.9 \leq G_{loc}/10^6 \leq 3.7$ lb/ft^2-hr
Local quality	$-0.2 \leq X_{loc} \leq 0.3$
Heated length, inlet to CHF location	$L_h \leq 14$ ft
Grid spacing	$13 \leq g_{sp} \leq 32$ in.
Equivalent hydraulic diameter	$0.37 \leq d_e \leq 0.60$ in.
Equivalent heated hydraulic diameter	$0.46 \leq d_h \leq 0.59$ in.
Distance from last grid to CHF site	$0 \leq d_g \leq g_{sp}$

The primary DNB correlation used for the analysis of VANTAGE 5 fuel is the WRB-2 correlation (Reference 97). The WRB-2 DNB correlation was developed to take credit for the VANTAGE 5 intermediate flow mixer (IFM) grid design. A limit of 1.17 is applicable for the WRB-2 correlation.

The applicable range of parameters for the WRB-2 correlation is:

Pressure	$1440 \leq P \leq 2490$ psia
Local mass velocity	$0.9 \leq G_{loc}/10^6 \leq 3.7$ lb/ft^2-hr
Local quality	$-0.1 \leq X_{loc} \leq 0.3$
Heated length, inlet to CHF location	$L_h \leq 14$ ft
Grid spacing	$10 \leq g_{sp} \leq 26$ in.
Equivalent hydraulic diameter	$0.33 \leq d_e \leq 0.5101$ in.
Equivalent heated hydraulic diameter	$0.45 \leq d_h \leq 0.66$ in.

The W-3 DNB correlation, References 5 and 6, is used for both fuel types where the primary DNBR correlations are not applicable. The WRB-1 and WRB-2 correlations were developed based on mixing vane data and therefore are only applicable in the heated rod spans above the first mixing vane grid. The W-3 correlation, which does not take credit for mixing vane grids, is used to calculate DNBR values in the heated region below the first mixing vane grid. In addition, the W-3 correlation is applied in the analysis of accident conditions where the system pressure is below the range of the primary correlations. For system pressures in the range of 500 to 1000 psia, the W-3 correlations limit is 1.45, Reference 19. For system pressures greater than 1000 psia, the W-3 correlation limit is 1.30. A cold wall factor, Reference 8, is applied to the W-3 DNB correlation to account for the presence of the unheated thimble surfaces.

4.4.2.1.2 Definition of DNBR

The DNB heat flux ratio (DNBR) as applied to typical cells (flow cells with all walls heated) and thimble cells (flow cells with heated and unheated walls) is defined as:

$$\mathrm{DNBR} = \frac{q''_{\mathrm{DNB,N}}}{q''_{\mathrm{loc}}} \qquad (1)$$

where:

$$q''_{\mathrm{DNB,N}} = \frac{q''_{\mathrm{DNB,EU}}}{F} \qquad (2)$$

$q''_{\mathrm{DNB,EU}}$ = the uniform DNB heat flux as predicted by the WRB-1, WRB-2, or W-3 DNB correlation (typical cell only).

F = the flux shape factor to account for nonuniform axial heat flux distributions [10] with the term “C” modified as in Reference 5.

q''_{loc} = the actual local heat flux.

The DNBR as applied to the W-3 DNB correlation when a cold wall (CW) is present is:

$$\mathrm{DNBR} = \frac{q''_{\mathrm{DNB,N,CW}}}{q''_{\mathrm{loc}}} \qquad (3)$$

where:

$$q''_{\mathrm{DNB,N,CW}} = \frac{q''_{\mathrm{DNB,EU,Dh}} \times \mathrm{CWF}}{F} \qquad (4)$$

where:

$q''_{\mathrm{DNB,EU,Dh}}$ = the uniform DNB heat flux as predicted by the W-3 cold wall DNB correlation (5) when not all flow cell walls are heated (i.e. - thimble CW cell).

$$CWF = \frac{q''_{coldwall}}{q''_{W-3,D_h}} = 1.0 - R_u\left[13.76 - 1.372e^{1.78\chi} - 4.732\left(\frac{G}{10^6}\right)^{-0.0535} - 0.0619\left(\frac{P}{1000}\right)^{0.14} - 8.509D_h^{0.107}\right]$$

$$Ru = 1.0\text{-}De/Dh$$

4.4.2.1.3 Mixing Technology

The rate of heat exchange by mixing between flow channels is proportional to the difference in the local mean fluid enthalpy of the respective channels, the local fluid density, and the flow velocity. The proportionality is expressed by the dimensionless thermal diffusion coefficient (TDC) which is defined as:

$$TDC = \frac{w'}{\rho V a}$$

where:

w' = flow exchange rate per unit length (lbm/ft-s).

ρ = fluid density (lbm/ft^3).

V = fluid velocity (ft/s).

a = lateral flow area between channels per unit length (ft^2/ft).

The application of the TDC in the thermal hydraulic analysis for determining the overall mixing effect or heat exchange rate is presented in Reference 7.

Various mixing tests have been performed at Columbia University.(12) These series of tests, using the "R" mixing vane grid design on 13-, 26-, and 32-in. grid spacing, were conducted in pressurized water loops at Reynolds numbers similar to that of a pressurized water reactor (PWR) core under the following single- and two-phase (subcooled boiling) flow conditions:

- Pressure 1500 to 2400 psia
- Inlet temperature 332 to 642°F
- Mass velocity 1.0 to 3.5 x 10^6 lbm/hr-ft^2
- Reynolds number 1.34 to 7.45 x 10^5
- Bulk outlet quality -52.1 to -13.5 percent

DC was determined by comparing the THINC code predictions with the measured subchannel exit temperatures. TDC is found to be independent of the Reynolds number, mass velocity, pressure, and quality over the ranges tested. The two-phase data (local, subcooled boiling) fell within the scatter of the single-phase data. The effect of two-phase flow on the value of TDC has been demonstrated by Cadek,(12) Rowe and Angle,(13)(14) and Gonzalez-Santalo and Griffith.(15) In the subcooled boiling region, the values of TDC were indistinguishable from the single-phase values. In the quality region, Rowe and Angle show that in the case with rod spacing similar to that in PWR core geometry, the value of TDC increased with quality to a point and then decreased, but never below the single-phase value. Gonzalez-Santalo and Griffith show that the mixing coefficient increased as the void fraction increased. The data from these tests on the R-grid showed that a design TDC value of 0.038 (for 26-in. grid spacing) can be used in determining the effect of coolant mixing in the THINC analysis. A mixing test program similar to the one described above was conducted at Columbia University for the current 17 x 17 geometry and mixing vane grids on 26-in. spacing.(16) The mean value of TDC obtained from these tests was 0.059, and all data were well above the current design value of 0.038.

The inclusion of three intermediate flow mixer grids in the upper span of the VANTAGE 5 fuel assembly results in a grid spacing of approximately 10 inches. Per Reference 97, a TDC value of 0.038 was chosen as a conservatively low value for use in VANTAGE 5 to determine the effect of coolant mixing in the core thermal performance analysis.

4.4.2.1.4 Hot Channel Factors

The total hot channel factors for heat flux and enthalpy rise are defined as the maximum-to-core average ratios of these quantities.

The heat flux hot channel factor considers the local maximum linear heat generation rate at a point (the hot spot), and the enthalpy rise hot channel factor involves the maximum integrated value along a channel (the hot channel).

Each of the total hot channel factors considers a nuclear hot channel factor (see Section 4.4.4.3) describing the neutron power distribution and an engineering hot channel factor, which allows for variations in flow conditions and fabrication tolerances. The engineering hot channel factors are made up of subfactors which account for the influence of the variations of fuel pellet diameter, density, enrichment and eccentricity; fuel rod diameter pitch and bowing inlet flow distribution; flow redistribution; and flow mixing.

Heat Flux Engineering Hot Channel Factor, F^{E}_{Q}

The heat flux engineering hot channel factor is used to evaluate the maximum heat flux. This subfactor is determined by statistically combining the tolerances for the fuel pellet diameter and has a value of 1.03 at the 95-percent probability level with 95-percent probability level with 95-percent confidence, density, enrichment, eccentricity and the fuel rod diameter. As shown using approved methodologies [90] and [92], no DNB penalty need be taken for the relatively low intensity heat flux spikes caused by variations in the above parameters, as well as fuel pellet eccentricity and fuel rod diameter variation.

Enthalpy Rise Engineering Hot Channel Factor,$F^{E}_{\Delta H}$

The effect of variations in flow conditions and fabrication tolerances on the hot channel enthalpy rise is directly considered in the core thermal-hydraulic analysis under any reactor operating condition. The items considered contributing to the enthalpy rise engineering hot channel factor are discussed below:

1. Pellet diameter, density and enrichment:

   Variations i pellet diameter, density, and enrichment are considered statistically in establishing the limit DNBRs (see Section 4.4.1.1) for the Revised Thermal Design Procedure (Reference 101) employed in this application. Variances in these design values are met for 95 percent of the limiting channels at a 95 percent confidence level.

2. Inlet flow maldistribution:

   The consideration of inlet flow maldistribution in core thermal performances is discussed in Section 4.4.4.2.2. A design basis of 5 percent reduction in coolant flow to the hot assembly is used in the core thermal-hydraulic analysis.

3. Flow redistribution:

   The flow redistribution accounts for the reduction in flow in the hot channel resulting from the high flow resistance in the channel due to the local or bulk boiling. The effect of the non-uniform power distribution is inherently considered in the core thermal-hydraulic analysis for every operating condition which is evaluated.

4. Flow mixing:

   The spacer grid design induces additional flow mixing between the various flow channels in a fuel assembly as well as between adjacent assemblies. This mixing reduces the enthalpy rise in the hot channel resulting from local power peaking or unfavorable mechanical tolerances.

4.4.2.1.5 Effects of Rod Bow on DNBR

The phenomenon of fuel rod bowing, as described in Reference 81, must be accounted for in DNBR safety analysis of Condition I and Condition II vents for each plant application. The methodology described in Reference 81 has been accepted by the NRC for use in design applications (Reference 99) and used in the analysis of fuel rod bowing.

The rod bow penalties accounted for in the design safety analysis are based on an assembly average of 24,000 MWD/MTU. Credit is taken for the effect of $F^{N}_{\Delta H}$ burndown due to the decrease in fissionable isotopes and the buildup of fission product inventory (Reference 100).

In the upper spans of the VANTAGE+ fuel assembly, additional restraint is provided with the intermediate flow mixer (IFM) grids such that the grid-to-grid spacing in those spans with IFM grids is approximately 10 inches compared to approximately 20 inches in the other spans. Therefore, no rod bow DNBR penalty is required in the 10 inch spans in the VANTAGE+ safety analyses.

For the safety analysis for the CPNPP units, sufficient DNBR margin is maintained to accommodate DNBR penalties due to rod bow.

4.4.2.2 Linear Heat Generation Rate

The core average and maximum linear heat generation rates are given in Table 4.4-1. The method of determining the maximum linear heat generation rate is given in Section 4.3.

4.4.2.3 Void Fraction Distribution

The void models used in the VIPRE-01 computer code are described in Subsection 4.4.2.6.3.

4.4.2.4 Normalized Core Flow and Enthalpy Rise Distributions

Normalized core flow and enthalpy rise distributions are shown in Figures 4.4-2, 4.4-3, and 4.4-4 which would be typical of the Cycle 1 core design. These general distributions would also be typical of later operating cycles.

4.4.2.5 Core Pressure Drops and Hydraulic Loads

4.4.2.5.1 Core Pressure Drops

The core pressure drop includes the fuel assembly, lower core plate, and upper core plate pressure drops. These pressure drops are based on the best estimate flow for expected plant operating conditions as described in Section 5.1.1. Section 5.1.1 also defines and describes the thermal design flow (minimum flow) which is the basis for reactor core thermal performance and the mechanical design flow (maximum flow) which is used in the mechanical design of the reactor vessel internals and fuel assemblies. Since the best estimate flow is that flow which is most likely to exist in an operating plant, the calculated core pressure drop values are based on this best estimate flow rather than the thermal design flow.

Uncertainties associated with the core and vessel pressure drop values are discussed in Section 4.4.2.8.2.

4.4.2.5.2 Hydraulic Loads

The fuel assembly hold down springs are designed to keep the fuel assemblies in contact with the lower core plate under all Condition I and II events with the exception of the turbine overspeed transient associated with a loss of external load. The hold down springs are designed to tolerate the possibility of an over deflection associated with fuel assembly liftoff for this case and to provide contact between the fuel assembly and the lower core plate following this transient. More adverse flow conditions occur during a loss of coolant accident. These conditions are presented in Section 15.6.5.

Hydraulic loads at normal operating conditions are calculated considering the best estimate flow which is described in Section 5.1, accounting for the best estimate core bypass flow and adding margin. Core hydraulic loads at cold plant startup conditions are based on the cold best estimate flow, but are adjusted to account for the coolant density difference. Conservative core hydraulic loads for a pump overspeed transient, which could possibly create flow rates 18 to 20 percent

greater than the best estimate flow, are evaluated to be approximately twice the fuel assembly weight.

Core hydraulic loads were measured during the prototype assembly tests of the optimized fuel assembly. Reference [86] contains a detailed discussion of the results.

4.4.2.6 Correlation and Physical Data

4.4.2.6.1 Surface Heat Transfer Coefficients

Forced convection heat transfer coefficients are obtained from the familiar Dittus-Boelter correlation (Reference 20), with the properties evaluated at bulk fluid conditions:

$$\frac{hDe}{K} = 0.023\left(\frac{D_e G}{\mu}\right)^{0.8}\left(\frac{C_p \mu}{K}\right)^{0.4} \qquad (6)$$

where:

h = heat transfer coefficient, (Btu/hr-ft^2°F)

De = equivalent diameter, (ft)

K = thermal conductivity, (Btu/hr-ft-°F)

G = mass velocity, (lb_m/hr-ft^2)

μ = dynamic viscosity, (lb_m/ft-hr)

C_P = heat capacity, (Btu/lb_m -°F)

This correlation has been shown to be conservative (Reference 21) for rod bundle geometries with pitch-to-diameter ratios in the range used by PWRs. The onset of nucleate boiling occurs when the clad wall temperature reaches the amount of superheat predicted by Thom's correlation, Reference 22. After this occurrence the outer clad wall temperature is determined by:

$$\Delta T_{sat} = (0.072 \exp(-P/1260))\,(q'')^{0.5} \qquad (7)$$

where:

ΔT_{sat} = wall superheat, $T_w - T_{sat}$, (°F)

q" = wall heat flux, (Btu/hr-ft^2)

P = pressure, (psia)

T_w = outer clad wall temperature, (°F)

T_{sat} = saturation temperature of coolant at P, (°F)

4.4.2.6.2 Total Core and Vessel Pressure Drop

Unrecoverable pressure losses occur as a result of viscous drag (friction) and/or geometry changes (form) in the fluid flow path. The flow field is assumed to be incompressible, turbulent, single- phase water. These assumptions apply to the core and vessel pressure drop calculations for the purpose of establishing the primary loop flow rate. Two-phase considerations are neglected in the vessel pressure drop evaluation because the core average void is negligible. Two-phase flow considerations in the core thermal subchannel analyses are considered and the models are discussed in Subsection 4.4.4.2.3. Core and vessel pressure losses are calculated by equations of the form:

$$P_L = \left(K + \frac{FL}{D_e}\right)\frac{\rho V^2}{2g_c(144)} \qquad (8)$$

Where:

P_L = unrecoverable pressure drop, (lbf/in^2)

ρ = fluid density, (lbm/ft^3)

L = length, (ft)

D^e = equivalent diameter, (ft)

V = fluid velocity, (ft/sec)

g_c = $32.174 \frac{lb_m - ft}{lb_f - sec^2}$

K = form loss coefficient, dimensionless

F = friction loss coefficient, dimensionless

Fluid density is assumed to be constant at the appropriate value for each component in the core and vessel. Because of the complex core and vessel flow geometry, precise analytical values for the form and friction loss coefficients are not available. Therefore, experimental values for these coefficients are obtained from geometrically similar models.

Values are quoted in Table 4.4-1 for unrecoverable pressure loss across the reactor vessel. The results of full-scale tests of core components and fuel assemblies were utilized in developing the core pressure loss characteristic. The pressure drop for the vessel was obtained by combining the core loss with correlation of 1/7th scale model hydraulic test data on a number of vessels

(References 23 and 24) and form loss relationships (Reference 25). Moody curves (Reference 26) were used to obtain the single-phase friction factors.

4.4.2.6.3 Void Fraction Correlation

Empirical correlations are used in VIPRE to model the void fraction in two-phase flow. The subcooled void correlation used to model the non-equilibrium transition from single phase to nucleate boiling is given in Reference 27. The bulk (saturated) void model relates flow quality with void fraction which can account for phase slip.

4.4.2.7 Thermal Effects of Operation Transients

DNB core safety limits are generated as a function of coolant temperature, pressure, core power and axial power imbalance. Steady- state operation within these safety limits insures that the minimum DNBR is not less than the DNBR limit.

The overtemperature N-16 system provides adequate protection against anticipated operational transients that are slow with respect to fluid transport delays in the primary system. For fast transients, e.g., uncontrolled rod bank withdrawal at power incident (Section 15.4.2) specific protection functions are provided as described in Section 7.2 and the use of these protection functions is described in Chapter 15.

4.4.2.8 Uncertainties in Estimates

4.4.2.8.1 Uncertainties in Fuel and Clad Temperatures

As discussed in Section 4.4.2.10, the fuel temperature is a function of crud, oxide, clad, pellet-clad gap, and pellet conductances. Uncertainties in the fuel temperature calculation are essentially of two types: fabrication uncertainties such as variations in the pellet and clad dimensions and the pellet density; and model uncertainties such as variations in the pellet conductivity and the gap conductance. These uncertainties have been quantified by comparison of the thermal model to inpile thermocouple measurements, by out-of pile measurements of the fuel and clad properties, and by measurements of the fuel and clad dimensions during fabrication. The resulting uncertainties are then used in all evaluations involving the fuel temperature. The effect of densification on fuel temperature uncertainties is also considered.

In addition to the temperature uncertainty described above, the measurement uncertainty in determining the local power, and the effect of density and enrichment variations on the local power are considered in establishing the heat flux hot channel factor. These uncertainties are described in Section 4.3.2.2.1.

Reactor trip setpoints, as specified in the Technical Specifications, include allowance for instrument and measurement uncertainties such as calorimetric error, instrument drift and channel reproducibility, temperature measurement uncertainties, noise, and heat capacity variations. Uncertainty in determining the cladding temperature results from uncertainties in the crud and oxide thicknesses. Because of the excellent heat transfer between the surface of the rod and the coolant, the film temperature drop does not appreciably contribute to the uncertainty.

4.4.2.8.2 Uncertainties in Pressure Drops

Core and vessel pressure drop values are based on the best estimate flow, as described in Section 5.1. The uncertainties are based on the uncertainties in both the test results and the analytical extension of these values to the reactor application.

A major use of the core and vessel pressure drop value is to determine the primary system coolant flow rates as discussed in Section 5.1. In addition, as discussed in Section 4.4.5.1, tests on the primary system prior to initial criticality will be made to verify that a conservative primary system coolant flow rate has been used in the design and analyses of the plant.

4.4.2.8.3 Uncertainties Due to Inlet Flow Maldistribution

The effects of uncertainties in the inlet flow maldistribution criteria used in the core thermal analyses are discussed in Section 4.4.4.2.2.

4.4.2.8.4 Uncertainty in DNB Correlations

The uncertainty in the DNB correlations can be written as a statement on the probability of not being in DNB based on the statistics of the DNB data. This is discussed in Section 4.4.4.2.2.

4.4.2.8.5 Uncertainties in DNBR Calculations

The uncertainties in the DNBRs calculated by the core thermal-hydraulic analysis due to uncertainties in the nuclear peaking factors are accounted for by applying conservatively high values of the nuclear peaking factors and including measurement error allowances in the statistical evaluation of the limit DNBR using the Revised Thermal Design Procedure (Reference 101). In addition, conservative values for the engineering hot channel factors and flow mixing coefficients are used. The results of sensitivity studies [18] and [92] show that the minimum DNBR in the hot channel is relatively insensitive to variations in the core-wide radial power distribution (for the same value of $F^N_{\Delta H}$).

Studies have been performed [18] and [92] to determine the sensitivity of the minimum DNBR in the hot channel to the void fraction correlation; the inlet velocity and exit pressure distributions assumed as boundary conditions for the analysis; and the grid pressure loss coefficients. The results of these studies show that the minimum DNBR in the hot channel is relatively insensitive to variations in these parameters. The range of variations considered in these studies covered the range of possible variations in these parameters.

4.4.2.8.6 Uncertainties in Flow Rates

The uncertainties associated with loop flow rates are discussed in Section 5.1. For core thermal performance evaluations, a thermal design loop flow is used which accounts for both prediction and measurement uncertainties. In addition, a small fraction of the thermal design flow is assumed to be ineffective for core heat removal capability because it bypasses the core through the various available vessel flow paths described in Section 4.4.4.2.1.

4.4.2.8.7 Uncertainties in Hydraulic Loads

As discussed in Section 4.4.2.5.2, hydraulic loads on the fuel assembly are evaluated for a pump overspeed transient which could possibly create flow rates 18 to 20 percent greater than the mechanical design flow. The mechanical design flow as stated in Section 5.1 is greater than the best estimate or most likely flow rate value for the actual plant operating condition.

4.4.2.9 Flux Tilt Considerations

Significant quadrant power tilts are not anticipated during normal operation since this phenomenon is caused by some asymmetric perturbation. A dropped or misaligned rod cluster control assembly could cause changes in hot channel factors; however, these events are analyzed separately in Chapter 15. This discussion will be confined to flux tilts caused by conditions such as x-y xenon transients, inlet temperature mismatches, and enrichment variations within tolerances.

In addition to unanticipated quadrant power tilts as described above, other readily explainable asymmetries may be observed during calibration of the excore detector quadrant power tilt alarm. During operation, incore maps are taken at least once per month and, periodically, additional maps are obtained for calibration purposes. Each of these maps is reviewed for deviations from the expected power distributions. Asymmetry in the core, from quadrant to quadrant, is frequently a consequence of the design when assembly and/or components shuffling and rotation requirements do not allow exact symmetry preservation. In each case, the acceptability of an observed asymmetry, planned or otherwise depends solely on meeting the required accident analyses assumptions.

In practice, once acceptability has been established by review of the incore maps, the quadrant power tilt alarms and related instrumentation are adjusted to indicate zero Quadrant Power Tilt Ratio as the final step in the calibration process. This action ensures that the instrumentation is correctly calibrated to alarm in the event an unexplained or unanticipated change occurs in the quadrant to quadrant relationships between calibration intervals. Proper functioning of the quadrant power tilt alarm is significant because no allowances are made in the design for increased hot channel factors due to unexpected developing flux tilts since all likely causes are prevented by design or procedures or specifically analyzed. Finally in the event that unexplained flux tilts do occur, the Technical Specifications provide appropriate corrective actions to ensure continued safe operation of the reactor.

4.4.2.10 Fuel and Cladding Temperatures

Consistent with the thermal-hydraulic design bases in Section 4.4.1, the following discussion pertains mainly to the fuel pellet temperature evaluation. A discussion of fuel clad integrity is presented in Section 4.2.3.1.

The thermal-hydraulic design assures that the maximum fuel temperature is below the melting point of UO_2. The temperature distribution within the fuel pellet is predominantly a function of the local power density and the UO_2 thermal conductivity. However, the computation of radial fuel temperature distributions combines crud, oxide, clad gap and pellet conductances. The factors which influence these conductances, such as gap size (or contact pressure), internal gas pressure, gas composition, pellet density, time dependent fuel densification and radial power

distribution within the pellet, etc., have been combined into a semi-empirical thermal model. This thermal model enables the determination of these factors and their net effects on temperature profiles. The temperature predictions have been compared to inpile fuel temperature measurements with good results.

Fuel rod thermal evaluations (fuel centerline, average and surface temperatures) are determined throughout the fuel rod lifetime with consideration of time dependent densification. To determine the maximum fuel temperature, various burnup rods, including the highest burnup rod, are analyzed over the rod linear power range of interest. The principal factors which are employed in the determination of the fuel temperature are discussed below.

4.4.2.10.1 UO_2 Thermal Conductivity

The thermal conductivity of uranium dioxide was evaluated from data reported from a number of measurements.

At the higher temperatures, thermal conductivity is best obtained by utilizing the integral conductivity to melt which can be determined with more certainty. From an examination of the data, it has been concluded that the best estimate for the value of 2800°C Kdt is 93 watts/cm.

The design curve is in excellent agreement with the recommendation of the IAEA panel (Reference 28).

4.4.2.10.2 Radial Power Distribution in UO_2 Fuel Rods

An accurate description of the radial power distribution as a function of burnup is needed for determining the power level for incipient fuel melting and other important performance parameters such as pellet thermal expansion, fuel swelling and fission gas release rates. Radial power distribution in UO2 fuel rods is determined with the neutron transport code LASER. The LASER code has been validated by comparing the code predictions on radial burnup and isotopic distributions with measured radial microdrill data (References 29 and 30). A “radial power depression factor,” f, is determined using radial power distributions predicted by LASER. The factor, f, enters into the determination of the pellet centerline temperature, TC, relative to the pellet surface temperature, T_S, through the expression:

$$\int_{T_s}^{T_c} K(T)dT = \frac{q'f}{4\pi}$$

where:

K(T) = the thermal conductivity for UO_2 with a uniform density distribution

q' = the linear power generation rate

4.4.2.10.3 Gap Conductance

The temperature drop across the pellet-clad gap is a function of the gap size and the thermal conductivity of the gas in the gap. The gap conductance model is selected such that when

combined with the UO_2 thermal conductivity model, the calculated fuel centerline temperatures reflect the inpile temperature measurements. A more detailed discussion of the gap conductance model is presented in Reference 31.

4.4.2.10.4 Surface Heat Transfer Coefficients

The fuel rod surface heat transfer coefficients during subcooled forced convection and nucleate boiling are presented in Subsection 4.4.2.7.1.

4.4.2.10.5 Fuel Clad Temperatures

The outer surface of the fuel rod at the hot spot operates at a temperature of slightly higher than the bulk fluid temperature at the hot spot for steady state operation at rated power throughout core life due to nucleate boiling.

Initially (beginning-of-life), this temperature is that of the clad metal outer surface.

During operation over the life of the core, the buildup of oxides and crud on the fuel rod surface causes the clad surface temperature to increase. Allowance is made in the fuel center melt evaluation for this temperature rise. Since the thermal-hydraulic design basis limits DNB, adequate heat transfer is provided between the fuel clad and the reactor coolant so that the core thermal output is not limited by considerations of clad temperature.

4.4.2.10.6 Treatment of Peaking Factors

The total heat flux hot channel factor, F_Q, is defined by the ratio of the maximum to core average heat flux. The design value of F_Q for normal operation is 2.42 for the 3458 MWt design and 2.5 for the uprated design providing 3612 MWt. This results in peak linear powers at full-power conditions of 13.36 and 14.41 kW/ft respectively.

The centerline temperature at the peak linear power resulting from overpower transients/ overpower errors must be below the UO_2 melt temperature over the lifetime of the rod, including allowances for uncertainties. The fuel temperature design basis is discussed in Section 4.4.1.2. The centerline temperature at the peak linear power resulting from overpower transients/ overpower errors is below that required to produce melting.

4.4.3 DESCRIPTION OF THE THERMAL AND HYDRAULIC DESIGN OF THE REACTOR COOLANT SYSTEM

4.4.3.1 Plant Configuration Data

Plant configuration data for the thermal-hydraulic and fluid systems external to the core are provided in the appropriate Chapters 5, 6, and 9. Implementation of the Emergency Core Cooling System (ECCS) is discussed in Chapter 15.

4.4.3.2 Operating Restrictions on Pumps

The minimum net positive suction head (NPSH) and minimum seal injection flow rate must be established before operating the reactor coolant pumps. With the minimum labyrinth seal

injection flow rate established, the operator will have to verify that the system pressure satisfies NPSH requirements.

4.4.3.3 Power-Flow Operating Map (BWR)

Not applicable to the CPNPP.

4.4.3.4 Temperature-Power Operating Map

A representative relationship between Reactor Coolant System temperature and power is shown in Figure 4.4-21.

The effects of reduced core flow due to inoperative pumps is discussed in Sections 15.3.1 and 15.3.2. Natural circulation capability of the system is discussed in Section 15.2.6.

4.4.3.5 Load Following Characteristics

The Reactor Coolant System is designed on the basis of steady state operation at full power heat load. The reactor coolant pumps utilize constant speed drives as described in Section 5.4 and the reactor power is controlled to maintain average coolant temperature at a value which is a linear function of load, as described in Section 7.7.

4.4.3.6 Thermal and Hydraulic Characteristics Summary Table

The thermal and hydraulic characteristics are given in Tables 4.1-1 and 4.4-1.

4.4.4 EVALUATION

4.4.4.1 Critical Heat Flux

The critical heat flux correlations utilized in the core thermal analyses is explained in detail in Section 4.4.2.

4.4.4.2 Core Hydraulics

4.4.4.2.1 Flow Paths Considered in Core Pressure Drop and Thermal Design

The following flow paths for core bypass flow are considered:

1. Flow through the spray nozzles into the upper head for head cooling purposes.

2. Flow entering into the RCC guide thimbles.

3. Leakage flow from the vessel inlet nozzle directly to the vessel outlet nozzle through the gap between the vessel and the barrel.

4. Flow introduced between the baffle and the barrel for the purpose of cooling these components and which is not considered available for core cooling.

5. Flow in the gaps between the fuel assemblies on the core periphery and the adjacent baffle wall.

The above contributions are evaluated for each cycle to confirm that the design value of the core bypass flow is met.

4.4.4.2.2 Inlet Flow Distributions

Data have been considered from several 1/7 scale hydraulic reactor model tests, References [23], [24], and [62], in arriving at the core inlet flow maldistribution criteria used in the core thermal-hydraulic analyses.

The effect of the total flow rate on the inlet velocity distribution was studied in the experiments of Reference [23]. As was expected, on the basis of the theoretical analysis, no significant variation could be found in inlet velocity distribution with reduced flow rate.

4.4.4.2.3 Empirical Friction Factor Correlations

Two empirical friction factor correlations are used in the VIPRE-01 computer code (described in Subsection 4.4.4.5.1).

The friction factor in the axial direction, parallel to the fuel rod axis, is evaluated using the correlations described in Reference 27).

The flow in the lateral directions, normal to the fuel rod axis, views the reactor core as a large tube bank. Thus, the lateral friction factor proposed by Idel'chik (Reference 25) is applicable. This correlation is of the form:

$$F_L = A\, Re_L^{-0.2}$$

where:

A is a function of the rod pitch and diameter as given in Reference 25.

Re_L is the lateral Reynolds number based on the rod diameter.

4.4.4.3 Influence of Power Distribution

The core power distribution which is largely established at beginning- of-life by fuel enrichment, loading pattern, and core power level is also a function of variables such as control rod worth and position, and fuel depletion throughout lifetime. Radial power distributions in various planes of the core are often illustrated for general interest; however, the core radial enthalpy rise distribution as determined by the integral of power up each channel is of greater importance for DNB analyses. These radial power distributions, characterized by $F^N_{\Delta H}$ defined in Section 4.3.2.2.1) as well as axial heat flux profiles are discussed in the following two sections.

4.4.4.3.1 Nuclear Enthalpy Rise Hot Channel Factor, $F^N_{\Delta H}$

Given the local power density q' (kW/ft) at a point x, y, z in a core with N fuel rods and height H,

$$F^N_{\Delta H} = \frac{\text{hot rod power}}{\text{average rod power}} = \frac{\text{Max}\int_o^h q'(xo, yo, z)dz}{\frac{1\Sigma}{N_{\text{all rods}}}\int_o^h q'(x, y, z)dz}$$

The way in which $F^N_{\Delta H}$ is used in the DNB calculation is important. The location of minimum DNBR depends on the axial profile and the value of DNBR depends on the enthalpy rise to that point. Basically, the maximum value of the rod integral is used to identify the most likely rod for minimum DNBR. An axial power profile is obtained, which when normalized to the design value of $F^N_{\Delta H}$, recreates the axial heat flux along the limiting rod. The surrounding rods are assumed to have the same axial profile with rod average powers which are typical distributions found in hot assemblies. In this manner, worst case axial profiles can be combined with worst case radial distributions for reference DNB calculations.

It should be noted again that $F^N_{\Delta H}$ is an integral and is used as such in DNB calculations. Local heat fluxes are obtained by using hot channel and adjacent channel explicit power shapes which take into account variations in horizontal power shapes throughout the core. The sensitivity of the core thermal-hydraulic analysis to radial power shapes is discussed in References [18] and [92]. For operation at a fraction of full power, the design $F^N_{\Delta H}$ used is given by:

$$F^N_{\Delta H} = F^{RTD}_{\Delta H}[1 + PF_{\Delta H}(1-P]$$

$F^{RTP}_{\Delta H}$ is the limit at rated thermal power (RTP) specified in the Core Operating Limits Report (COLR).

$PF_{\Delta H}$ is the power factor multiplier for $F^N_{\Delta H}$ specified in the COLR.

P is the fraction of rated thermal power.

The permitted relaxation of $F^N_{\Delta H}$ is included in the DNB protection setpoints and allows radial power shape changes with rod insertion to the insertion limits, (Reference 32) thus allowing greater flexibility in the nuclear design.

4.4.4.3.2 Axial Heat Flux Distributions

As discussed in Section 4.3.2.2, the axial heat flux distribution can vary as a result of rod motion, power change, or due to spatial xenon transients which may occur in the axial direction. Consequently, it is necessary to measure the axial power imbalance by means of the excore nuclear detectors (as discussed in Section 4.3.2.2.7) and protect the core from excessive axial power imbalance.

The Reactor Trip System provides automatic reduction of the trip setpoint in the overtemperature N-16 channels on excessive axial power imbalance; that is, when an extremely large axial offset corresponds to an axial shape which could lead to a DNBR which is less than that calculated for the reference DNB design axial shape.

To determine the penalty to be taken in protection setpoints for extreme values of flux difference, this reference shape is supplemented by other axial shapes skewed to the bottom and top of the core. The initial conditions for the accidents for which DNB protection is required are assumed to be those permissible within the constant axial offset control strategy for the load maneuvers described in Reference [67]. The course of those accidents in which DNB is a concern is analyzed in Chapter 15, assuming that the protection setpoints have been set on the basis of these shapes. In many cases, the axial power distribution in the hot channel changes throughout the course of the accident due to rod motion, coolant temperature and power level changes.

4.4.4.4 Core Thermal Response

A general summary of the steady-state thermal-hydraulic design parameters including thermal output, flow rates, etc., is provided in Table 4.4-1. As stated in Section 4.4.1, the design bases for the thermal hydraulic design of the reactor core are to prevent DNB and to prevent fuel melting for Condition I and II events. The protective systems described in Chapter 7 are designed to meet these bases. The response of the core to Condition II transients is given in Chapter 15.

4.4.4.5 Analytical Techniques

4.4.4.5.1 Core Analysis

The objective of reactor core thermal design is to determine the maximum heat-removal capability in all flow subchannels and to show that the core safety limits are not exceeded using the most conservative power distribution. The thermal design takes into account local variations in dimensions, power generation, flow redistribution, and mixing. VIPRE-01 is a realistic three-dimensional matrix model which has been developed to account for hydraulic and nuclear effects on the enthalpy rise in the core (Reference 27). The behavior of the hot assembly is determined by superimposing the power distribution among the assemblies upon the inlet flow distribution while allowing for flow mixing and flow distribution between assemblies. The local variations in power, fuel rod and pellet fabrication, and mixing within the hottest assembly are superimposed on the average conditions of the hottest assembly in order to determine the conditions in the hot channel.

4.4.4.5.2 Steady State Analysis

The VIPRE-01 computer program and subchannel analysis methodology, as approved by the NRC (Reference 27) is used to determine coolant density, mass velocity, enthalpy, vapor void, static pressure, and DNBR distributions within the reactor core hot subchannel under all expected operating conditions. The VIPRE-01 code is described in detail in Reference 27, including models and correlations used.

4.4.4.5.3 Experimental Verification

Experimental verification of VIPRE-01 is presented in References 33 and 27. The VIPRE-01 analysis methodology is based on a knowledge and understanding of the heat transfer and hydrodynamic behavior of the coolant flow and the mechanical characteristics of the fuel elements. VIPRE-01 analysis provides a realistic evaluation of the core performance and is used in the thermal analyses as described above.

4.4.4.5.4 Transient Analysis

The approved VIPRE-01 methodology (Reference 27) was shown to be conservative for transient thermal-hydraulic analysis.

4.4.4.6 Hydrodynamic and Flow Power Coupled Instability

Boiling flows may be susceptible to thermohydrodynamic instabilities [72]. These instabilities are undesirable in reactors since they may cause a change in thermohydraulic conditions that may lead to a reduction in the DNB heat flux relative to that observed during a steady flow condition or to undesired forced vibrations of core components. Therefore, a thermohydraulic design was developed which states that modes of operation under Condition I and II events shall not lead to thermohydrodynamic instabilities.

Two specific types of flow instabilities are considered for Westinghouse PWR operation. These are the Ledinegg or flow excursion type of static instability and the density wave type of dynamic instability.

A Ledinegg instability involves a sudden change in flow rate from one steady state to another. This instability occurs[72] when the slope of the reactor coolant system pressure drop-flow rate curve $(\partial \Delta P / \partial G_{internal})$ becomes algebraically smaller than the loop supply (pump head) pressure drop-flow rate curve $(\partial \Delta P / \partial G_{external})$. The criterion for stability is thus $(\partial \Delta P / \partial G_{internal} > \partial \Delta P / \partial G_{external})$. The Westinghouse pump head curve has a negative slope $(\partial \Delta P / \partial G_{external} < 0)$, whereas the reactor coolant system pressure drop-flow curve has a positive slope $(\partial \Delta P / \partial G_{internal} > 0)$ over the Condition I and Condition II operational ranges. Thus, the Ledinegg instability will not occur.

The mechanism of density wave oscillations in a heated channel has been described by Lahey and Moody[73]. Briefly, an inlet flow fluctuation produces an enthalpy perturbation. This perturbs the length and the pressure drop of the single-phase region and causes quality or void perturbations in the two-phase regions which travel up the channel with the flow. The quality and length perturbations in the two-phase region create two-phase pressure drop perturbations. However, since the total pressure drop across the core is maintained by characteristics of the fluid system external to the core, the two-phase pressure drop perturbation feeds back to the single-phase region. These resulting perturbations can be either attenuated or self-sustained.

A simple method has been developed by Ishii[74] for parallel closed channel systems to evaluate whether a given condition is stable with respect to the density wave type of dynamic instability. This method had been used to assess the stability of typical Westinghouse reactor

designs[70,71,75,76,77] under Condition I and II operation. The results indicate that a large margin to density wave instability exists, e.g., increases on the order of 200% of rated reactor power would be required for the predicted inception of this type of instability.

The application of the method of Ishii[74] to Westinghouse reactor designs is conservative due to the parallel open channel feature of Westinghouse PWR cores. For such cores, there is little resistance to lateral flow leaving the flow channels of high power density to low power density channels. There is also energy transfer from channels of high power density to lower power density channels. This coupling with cooler channels has led to the opinion that an open channel configuration is more stable than the above closed channel analysis under the same boundary conditions. Flow stability tests[77] have been conducted where the closed channel systems were shown to be less stable than when the same channels were cross connected at several locations. The cross connections were such that the resistance to channel-to-channel crossflow and enthalpy perturbations would be greater than that which would exist in a PWR core which has a relatively low resistance to crossflow.

Flow instabilities which have been observed have occurred almost exclusively in closed channel systems operating at low pressures. Kao, Morgan and Parker[77b] analyzed parallel closed channel stability experiments simulating a reactor core flow. These experiments were conducted at pressures up to 2200 psia. The results showed that for flow and power levels typical of power reactor conditions, no flow oscillations could be induced above 1200 psia.

Additional evidence that flow instabilities do not adversely affect thermal margin is provided by the data from the rod bundle DNB tests. Many Westinghouse rod bundles have been tested over wide ranges of operating conditions with no evidence of premature DNB or of inconsistent data which might be indicative of flow instabilities in the rod bundle.

In summary, it is concluded that thermohydrodynamic instabilities will not occur under Condition I and II modes of operation for W PWR reactor designs. A large power margin, greater than doubling rated power, exists to predicted inception of such instabilities. An analysis has been performed which shows that minor plant to plant differences in Westinghouse reactor designs such as fuel assembly arrays, core power to flow ratios, fuel assembly length, etc., will not result in gross deterioration of the above power margins.

4.4.4.7 Fuel Rod Behavior Effects from Coolant Flow Blockage

Coolant flow blockages can occur within the coolant channels of a fuel assembly or external to the reactor core. The effects of fuel assembly blockage within the assembly on fuel rod behavior are more pronounced than external blockages of the same magnitude. In both cases, the flow blockages cause local reductions in coolant flow. The amount of local flow reduction, where it occurs in the reactor, and how far along the flow stream the reduction persists are considerations which will influence the fuel rod behavior. The effects of coolant flow blockages in terms of maintaining rated core performance are determined both by analytical and experimental methods. The experimental data are usually used to augment analytical tools. Inspection of the DNB correlations (Subsection 4.4.2 and Reference 8) shows that the predicted DNBR is dependent upon the local values of quality and mass velocity.

Thermal-hydraulic codes are capable of predicting the effects of local flow blockages on DNBR within the fuel assembly on a subchannel basis, regardless of where the flow blockage occurs.

In Reference 49, it is shown that for a fuel assembly similar to the Westinghouse design, the flow distribution within the fuel assembly when the inlet nozzle is completely blocked can be accurately predicted. Full recovery of the flow was found to occur about 30 inches downstream of the blockage. With the reference reactor operating at the nominal full power conditions specified in Table 4.4-1, the effects of an increase in enthalpy and decrease in mass velocity in the lower portion of the fuel assembly would not result in the reactor reaching a minimum DNBR below the safety analysis limit. From a review of the open literature, it is concluded that flow blockage in “open lattice cores” similar to the CPNPP cores causes flow perturbations which are local to the blockage. For instance, Ohtsubo [78], et. al., show that the mean bundle velocity is approached asymptotically about four inches downstream from a flow blockage in a single flow cell. Similar results were also found for two and three cells completely blocked. Basmer [79], et. al., tested an open lattice fuel assembly in which 41 percent of the subchannels were completely blocked in the center of the test bundles between spacer grids. Their results show the stagnant zone behind the flow blockage essentially disappears after 1.65 L/De or about 5 inches for their test bundle. They also found that leakage flow through the blockage tended to shorten the stagnant zone or, in essence, the complete recovery length. Thus, local flow blockages within a fuel assembly have little effect on subchannel enthalpy rise. The reduction in local mass velocity is then the main parameter which affects the DNBR. If the plant was operating at full power and nominal steady state conditions as specified in Table 4.4-1, a substantial reduction in local mass velocity would be required to reduce the DNBR close to the DNBR Safety Analysis Limits. The above mass velocity effect on the DNB correlations was based on the assumption of fully developed flow along the full channel length. In reality, a local flow blockage is expected to promote turbulence and thus would likely not affect DNBR at all.

Coolant flow blockages induce local crossflows as well as promote turbulence. Fuel rod behavior is changed under the influence of a sufficiently high crossflow component. Fuel rod vibration could occur, caused by this crossflow component, through vortex shedding or turbulent mechanisms. If the crossflow velocity exceeds the limit established for fluid elastic stability, large amplitude whirling results. The limits for a controlled vibration mechanism are established from studies of vortex shedding and turbulent pressure fluctuations. The crossflow velocity required to exceed fluid elastic stability limits is dependent on the axial location of the blockage and the characterization of the crossflow (jet flow or not). These limits are greater than those for vibratory fuel rod wear. Crossflow velocity above the established limits can lead to mechanical wear of the fuel rods at the grid support locations. Fuel rod wear due to flow induced vibration is considered in the fuel rod fretting evaluation (Section 4.2).

4.4.5 TESTING AND VERIFICATION

4.4.5.1 Tests Prior to Initial Criticality

A reactor coolant flow test is performed following fuel loading but prior to initial criticality. Reactor coolant flow is measured using data obtained from installed elbow tap differential pressure instrumentation. This data allows determination of the coolant flow rates at reactor operating conditions. This test verifies that proper coolant flow rates have been used in the core thermal and hydraulic analysis.

4.4.5.2 Initial Power and Plant Operation

Core power distribution measurements are made at several core power levels (see Chapter 14). These tests are used to insure that conservative peaking factors are used in the core thermal and hydraulic analysis.

4.4.5.3 Component and Fuel Inspections

Inspections performed on the manufactured fuel are delineated in Section 4.2.4. Fabrication measurements critical to thermal and hydraulic analysis are obtained to verify that the engineering hot channel factors employed in the design analyses (Section 4.4.2.1.4) are met.

4.4.6 INSTRUMENTATION REQUIREMENTS

4.4.6.1 Incore Instrumentation

Instrumentation is located in the core so that by correlating movable neutron detector information with fixed thermocouple information, radial, axial, and azimuthal core characteristics may be obtained for all core quadrants.

The incore instrumentation system is comprised of thermocouples, positioned to measure fuel assembly coolant outlet temperatures at preselected positions, and fission chamber detectors, positioned in guide thimbles which run the length of selected fuel assemblies to measure the neutron flux distribution. Figure 4.4-20 shows the number and location of instrumented assemblies in the core.

The core-exit thermocouples provide a backup to the flux monitoring instrumentation for monitoring power distribution. The routine, systematic collection of thermocouple readings by the operator provides a data base. From this data base, abnormally high or abnormally low readings, quadrant temperature tilts, or systematic departures from a prior reference map can be deduced.

The movable incore neutron detector system would be used for more detailed mapping if the thermocouple system indicated an abnormality. These two complementary systems are more useful when taken together than either system alone would be. The incore instrumentation system is described in more detail in Section 7.7.1.9.

The incore instrumentation is provided to obtain data from which fission power density distribution in the core, coolant enthalpy distribution in the core, and fuel burnup distribution may be determined.

4.4.6.2 Overtemperature and Overpower N-16 Instrumentation

The overtemperature N-16 trip protects the core against low DNBR. The overpower N-16 trip protects against excessive power (fuel rod rating protection).

As discussed in Section 7.2.1.1.2, factors included in establishing the overtemperature N-16 and overpower N-16 trip setpoint include the reactor coolant temperature in each loop and the axial distribution of core power through the use of the four section excore neutron detectors.

4.4.6.3 Instrumentation to Limit Maximum Power Output

The output of the three ranges (source, intermediate, and power) of the detectors, with the electronics of the nuclear instruments, is used to limit the maximum power output of the reactor within their respective ranges.

There are six radial locations containing a total of eight neutron flux detectors installed around the reactor in the primary shield, and two proportional counters for the source range installed on opposite “flat” portions of the core containing the primary startup sources at an elevation approximately one quarter of the core height. Two compensated ionization chambers for the intermediate range, located in the same instrument wells and detector assemblies as the source range detectors, are positioned at an elevation corresponding to one half of the core height; also positioned are four, four-section uncompensated ionization chamber assemblies for the power range installed vertically at the four corners of the core and located equidistant from the reactor vessel at all points and, to minimize neutron flux pattern distortions, within one foot of the reactor vessel. Each power range detector provides two signals corresponding to the neutron flux in the upper and in the lower sections of the core. The three ranges of detectors are used as inputs to monitor neutron flux from a completely shutdown condition to 120 percent of full power with the capability of recording overpower excursions up to 200 percent of full power.

The output of the power range channels is used for:

1. The rod speed control function,

2. To alert the operator to an excessive power imbalance between the quadrants,

3. Protect the core against rod ejection accidents, and

4. Protect the core against adverse power distributions.

Details of the neutron detectors and nuclear instrumentation design and the control and trip logic are given in Chapter 7. The trip setpoints are given in the Technical Specifications.

4.4.6.4 Loose Parts Monitoring System

The Loose Parts Monitoring System uses an array of active and passive accelerometers to detect metal to metal impacts indicative of loose parts in the RCS. The accelerometers are permanently installed at selected locations where a loose part would tend to collect or impact. The locations of the accelerometers provide a high degree of reliability in the detection of metal to metal impact in the reactor coolant system. Two accelerometers are located at each of the following locations:

1. Reactor Pressure Vessel - Upper Head Region

2. Reactor Pressure Vessel - Lower Region

3. All Steam Generators - Primary Coolant inlet Region

One accelerometer is located at each of the following locations:

1. All Reactor Coolant Pumps

2. All Steam Generators - Upper Region

The major source of noise is expected to be due to primary flow turbulence, reactor coolant pump vibrations, feedwater and steam flow turbulences. A spectral comparison of the measured local metal-to-metal acoustical resonances and the normal background is performed to minimize the noise-effect on the LPMS. From this comparison, narrow band filters are tuned to eliminate/ minimize the effects of background noise on the LPMS. In addition automatic gain control is used to maintain background noise at fairly constant level to allow easier detection of metal-to-metal impact. (The accelerometers operate on the piezoelectric crystal effect with deformation of the crystal producing a measurable proportional charge. This signal is amplified, filtered, and conditioned to accentuate the frequency band known by measurement to correspond to metal to metal impacts.) The active channels are designed to alarm the presence of unusual noises above normal background noise present in the plant. Baseline data will be taken during the initial startup of the plant to allow determination of the appropriate alarm settings for each channel and to use it later as a comparison for loose part detection and analysis. The system comprises eight active and twelve passive accelerometer channels. The active channels are monitored continuously for detection of loose parts. The passive channels are used for diagnostics (location determination) of a loose part that has been detected. Detection of a loose part by the active channels will activate an audible alarm in the main control room and the data is digitally recorded on a hard disk. The LPMS microprocessor is capable of printing out noise location information. The control room operator responds to the alarm upon referring to his annunciator response manual and no additional operator training program is necessary.

The LPMS vendor has seismically analyzed the LPMS and found it to be functional following an OBE. CPNPP has installed the system as a non safety related system.

The retention of plant records including those of the LPMS is addressed in plant administrative procedures.

REFERENCES

1. Christensen, J. A., Allio, R. J. and Biancheria, A., “Melting Point of Irradiated UO_2,” WCAP-6065, February 1965.

2. Motley, F. E., Wenzel, A. H. and Cadek, F. F., “Critical Heat Flux Testing of 17 x 17 Fuel Assembly Geometry with 22 Inch Grid Spacing,” WCAP-8536 (Proprietary) and WCAP-8537 (Non- Proprietary), May 1975.

3. Friedland, A.J., and Ray, S., "Revised Thermal Design Procedure," WCAP-11397-P-A (Proprietary), April 1989.

4. Tong, L. S., “Boiling Heat Transfer and Two-Phase Flow,” John Wiley & Sons, New York, 1965.

5. Tong, L. S., “Boiling Crisis and Critical Heat Flux,” AEC Critical Review Series, TID-25887, 1972.

6. Tong, L. S., “Critical Heat Fluxes in Rod Bundles,” in “Two- Phase Flow and Heat Transfer in Rod Bundles,” pp. 31-41, American Society of Mechanical Engineers, New York, 1969.

7. Chelemer, H., Weisman, J. and Tong, L. S., “Subchannel Thermal Analysis of Rod Bundle Cores,” WCAP-7015, Revision 1, January 1969.

8. Motley, F. E. and Cadek, F. F., “DNB Test Results for New Mixing Vane Grids (R)”, WCAP-7695-P-A (Proprietary) and WCAP-7958-A (Non-Proprietary), January 1975.

9. Motley, F. E. and Cadek, F. F., “DNB Test Results for R Grid Thimble Cold Wall Cells,” WCAP-7695-Addendum 1-P-A (Proprietary) and WCAP-7958-Addendum 1-A (Non-Proprietary), January 1975.

10. Tong, L. S., “Prediction of Departure from Nucleate Boiling for an Axially Non-Uniform Heat Flux Distribution,” J. Nucl. Energy, Vol. 21, pp. 241-248, (1967).

11. Owen, D. B., “Factors for One-Sided Tolerance Limits and for Variable Sampling Plans,” SCR-607, March 1963.

12. Cadek, F. F., Motley, F. E. and Dominicis, D. P., “Effect of Axial Spacing on Interchannel Thermal Mixing with the R Mixing Vane Grid,” WCAP-7941-P-A (Proprietary) and WCAP-7959-A (Non- Proprietary), January 1975.

13. Rowe, D. S. and Angle, C. W., “Crossflow Mixing Between Parallel Flow Channels During Boiling, Part II Measurements of Flow and Enthalpy in Two Parallel Channels,” BNWL-371, part 2, December 1967.

14. Rowe, D. S. and Angle, C. W., “Crossflow Mixing Between Parallel Flow Channels During Boiling, Part III Effect of Spacers on Mixing Between Two Channels,” BNWL-371, part 3, January 1969.

15. Gonzalez-Santalo, J. M. and Griffith, P., “Two-Phase Flow Mixing in Rod Bundle Subchannels,” ASME Paper 72-WA/NE-19.

16. Motley, F. E., Wenzel, A. H. and Cadek, F. F., “The Effect of 17 x 17 Fuel Assembly Geometry on Interchannel Thermal Mixing,” WCAP-8298-P-A (Proprietary) and WCAP-8299-A (Non-Proprietary), January 1975.

17. Cadek, F. F., “Interchannel Thermal Mixing with Mixing Vane Grids,” WCAP-7667-P-A (Proprietary) and WCAP-7755-A (Non-Proprietary), January 1975.

18. Hochreiter, L. E. and Chelemer, H., “Application of the THINC-IV Program to PWR Design,” WCAP-8054 (Proprietary) and WCAP-8195 (Non-Proprietary), October 1973.

19. Scherder, W.J. (ed.), et al. “Reactor Core Response to Excessive Secondary Steam Releases,” WCAP-9226-P-A, Revision 1, February 1998.

20. Dittus, F. W., and Boelter, L. M. K., "Heat Transfer in Automobile Radiators of the Tubular Type," Calif. Univ. Publication In Eng., 2, No. 13, 443461 (1930).

21. Weisman, J., "Heat Transfer to Water Flowing Parallel to Tube Bundles," Nucl. Sci. Eng., 6, 78-79 (1959).

22. Thom, J. R. S., Walker, W. M., Fallon, T. A. and Reising, G. F. S., "Boiling in Sub-Cooled Water During Flow up Heated Tubes or Annuli," Prc. Instn. Mech. Engrs., 180, Pt. C, 226-46 (1955-66).

23. Hetsroni, G., “Hydraulic Tests of the San Onofre Reactor Model,” WCAP-3269-8, June 1964.

24. Hetsroni, G., “Studies of the Connecticut-Yankee Hydraulic Model,” NYO-3250-2, June 1965.

25. Idel'chik, I. E., "Handbook of Hydraulic Resistance," AEC-TR-6630, 1960.

26. Moody, L. F., "Friction Factors for Pipe Flow," Transaction of the American Society of Mechanical Engineers, 66, 671-684 (1944).

27. WCAP-14565, "VIPRE-01 Modeling and Qualification for Pressurized Water Reactor Non-LOCA Thermal-Hydraulic Safety Analysis," Y. X. Sung, et al., April 1997.

28. International Atomic Energy Agency, "Thermal Conductivity of Uranium Dioxide," Report of the Panel held in Vienna, April 1965, IAEA Technical Reports Series, No. 59, Vienna, The Agency, 1966.

29. Poncelet, C. G., "Burnup Physics of Heterogeneous Reactor Lattices," WCAP-6069, June 1965.

30. Nodvick, R. J., "Saxton Core II Fuel Performance Evaluation," WCAP-3385-56, Part II, "Evaluation of Mass Spectrometric and Radiochemical Analyses or Irradiated Saxton Plutonium Fuel," July 1970.

31. Weiner, R. A., et al., "Improved Fuel Performance Models for Westinghouse Fuel Rod Design-and Safety Evaluation," WCAP-10851-P-A, August 1988.

32. McFarlane, A. F., "Power Peaking Factors," WCAP-7912-P-A (Proprietary) and WCAP-7912-A (Nonproprietary), January 1975.

33. NP-2511-CCM Volumes 1-5, "VIPRE-01: A Thermal-Hydraulic Code for Reactor Cores", Electric Power Research Institute.

34. Deleted

35. Deleted

36. Deleted

37. Deleted

38. Deleted

39. Deleted

40. Deleted

41. Deleted

42. Deleted

43. Deleted

44. Deleted

45. Deleted

46. Deleted

47. Deleted

48. Deleted

49. Hochreiter, L. E., Chelemer, H. and Chu, P. T., “THINC-IV An Improved Program for Thermal-Hydraulic Analysis of Rod Bundle Cores,” WCAP-7956, June 1973.

50. Deleted

51. Deleted

52. Deleted

53. Deleted

54. Deleted

55. Deleted

56. Deleted

57. Deleted

58. Deleted

59. Deleted

60. Deleted

61. Deleted

62. Carter, F. D., “Inlet Orificing of Open PWR Cores,” WCAP-7836, January, 1972.

63. Shefcheck, J., “Application of the THINC Program to PWR Design,” WCAP-7359-L (Proprietary), August 1969 and WCAP-7838 (Non- Proprietary), January 1972.

64. Deleted

65. Deleted

66. Deleted

67. Morita, T., et. al., “Power Distribution Control and Load Follow Procedures,” WCAP-8385 (Proprietary) and WCAP-8403 (Non- Proprietary), September 1974.

68. Deleted

69. Deleted

70. S. Kakac, T. N. Veziroglu, K. Akyuzlu, O. Berkol, “Sustained and Transient Boiling Flow Instabilities in a Cross- Connected Four-Parallel-Channel Upflow System,” Proc. of 5th International Heat Transfer Conference, Tokyo, September 3-7, 1974.

71. H. S. Kao, C. D. Morgan, and W. B. Parker, “Prediction of Flow Oscillation in Reactor Core Channel,” Trans. ANS Vol. 16, 1973, pp. 212-213.

72. J. A. Boure, A. E. Bergles, and L. S. Tong, “Review of Two-Phase Flow Instability,” Nucl. Engr. and Design Vol. 25, (1973), pp. 165-192.

73. R. T. Lahey and F. J. Moody, “The Thermal Hydraulics of a Boiling Water Reactor,” American Nuclear Society, 1977.

74. P. Saha, M. Ishii, and N. Zuber, “An Experimental Investigation of the Thermally Induced Flow Oscillations in Two-Phase Systems,” November 1976, pp. 616-622.

75. Virgil C. Summer FSAR, Docket #50-395.

76. Byron/Braidwood FSAR, Docket #50-456.

77. South Texas FSAR, Docket #50-498.

78. Ohtsubo, A. and Uruwashi, S., “Stagnant Fluid due to Local Flow Blockage,” J. Nucl. Sci. Technol., 9, No. 7, pp. 433-434, (1972).

79. Basmer, P., Kirsh, D. and Schultheiss, G. F., “Investigation of the Flow Pattern in the Recirculation Zone Downstream of Local Coolant Blockages in Pin Bundles,” Atomwirtschaft, 17, No. 8, pp. 416-417, (1972). (In German).

80. Deleted

81. Skaritka, J., (ed.), “Fuel Rod Bow Evaluation,” WCAP-8691, Rev. 1 (Proprietary) and WCAP-8692, Rev. 1 (Non-Proprietary), July 1979.

82. Deleted

83. Motley, F. E., Hill, K. W., Cadek, F. F. and Shefcheck, J., “New Westinghouse Correlation WRB-1 for Predicting Critical Heat Flux in Rod Bundles with Mixing Vane Grids,” WCAP-8762, July 1976 (Proprietary) and WCAP-8763, July 1976 (Non-Proprietary).

84. Letter from J. F. Stolz (NRC) to E. Eicheldinger (Westinghouse); Subject: Staff Evaluation for WCAP-7956, WCAP-8054, WCAP-8567, WCAP-8762, April, 19, 1978.

85. Chelemer, H., Boman, L. H. and Sharp, D. R., “Improved Thermal Design Procedures,” WCAP-8567-P, July 1975 (Proprietary) and WCAP-8568 July 1975 (Non-Proprietary).

86. Davidson, S. L., Motley, F. E., Lee, Y. C., Bogard, T., Bryan, W. J., “Verification Testing and Analyses of the 17 x 17 Optimized Fuel Assembly,” WCAP-9401, March 1979 (Proprietary) and WCAP-9402, March 1979, (Non-Proprietary).

87. Motley, F. E., Cadek, F. F., “Application for Modified Spacer Factor to L. Grid Typical and Cold Wall Cell DNB,” WCAP-7988 (Westinghouse Proprietary) and WCAP-8030-A (Non-Proprietary), October 1972.

88. Motley, F. E., Wenzel, A. H. and Cadek, F. F., “Critical Heat Flux Testing of 17 x 17 Fuel Assembly Geometry with 22 Inch Grid Spacing”, WCAP-8536 (Proprietary), May 1975 and WCAP-8537, May 1975.

89. Hill, K. W., Motley, F. E., Cadek, F. F., Wenzel, A. H., “Effect of 17 x 17 Fuel Assembly Geometry on DNB”, WCAP-8296-P-A (Westinghouse Proprietary) and WCAP-8297-A (Non-Proprietary), February 1975.

90. Hill, K. W., Motely, F. E., and Cadek, F. F., “Effect of Local Heat Flux Spikes on DNB in Non-Uniform Heated Rod Bundles”, WCAP-8174, August 1973 (Proprietary) and WCAP-8202, August 1973 (Non-Proprietary).

91. Deleted

92. Letter from C. Eicheldinger (Westinghouse) to John F. Stolz (NRC); Subject: THINC-IV information, dated November 2, 1977.

93. Deleted

94. Deleted

95. Deleted

96. Deleted

97. WCAP-10444-P-A, “Reference Core Report VANTAGE 5 Fuel Assembly,” September 1985 (Technical Specification 5.6.5b Item 16.

98. Deleted

99. Letter from Thomas, C. O. (NRC) to Rahe, E. P., Jr. (Westinghouse), “Acceptance for Referencing of Licensing Topical Report WCAP-8691(P)/WCAP-8692(NP),” December 29, 1982.

100. Letter from Berlinger, C. (NRC) to Rahe, E. P., Jr (Westinghouse), “Request for Reduction in Fuel Assembly Burnup Limit for Calculation of Maximum Rod Bow Penalty,” June 18, 1986.

101. Friedland, A. J. and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

TABLE 4.4-1
THERMAL AND HYDRAULIC COMPARISON TABLE

(Sheet 1 of 2)

Design Parameters[a]	Pre-Uprate Design	Uprate Design
Reactor core heat output (MW_t)[b]	3458	3612
Reactor core heat output (10^6) BTU/h)[b]	11,799	12,325
Heat generated in fuel (%)	97.4	97.4
System pressure, nominal (psia)	2250	2250
Minimum DNBR at nominal conditions		
Typical flow channel	>2.46	>2.46
Thimble (cold wall) flow channel	>2.34	>2.34
Minimum DNBR for design transients		
Typical flow channel		
WRB-1	1.23	1.23
WRB-2	1.23	1.23
Thimble (cold wall) flow channel		
WRB-1	1.23	1.23
WRB-2	1.22	1.22
DNB correlations[c]	WRB-2, W-3 & WRB-1	WRB-2, W-3 & WRB-1
Coolant conditions		
Vessel minimum measured flow rate (MMF		
gpm	396,400	396,400
Vessel thermal design flow rate (TDF)[c]		
10^6 lbm/h	142.04	142.3
gpm	382,800	382,800
Effective flow-rate for heat transfer		
(based on TDF, excluding bypass)[d]		
10^6 lbm/h	133.80	134.04
gpm	360,598	360,598
Effective flow area for heat transfer (ft^2)	54.13	54.13
Average mass velocity along fuel rods		
(based on TDF, excluding bypass)[d]		
ft/s	14.84	14.84
Average mass velocity		
(based on TDF, excluding bypass)[d]		
10^6 lbm/h-ft^2	2.47	2.48

TABLE 4.4-1
THERMAL AND HYDRAULIC COMPARISON TABLE

(Sheet 2 of 2)

Design Parameters[a]	Pre-Uprate Design	Uprate Design
Coolant temperature		
Nominal inlet (°F)	559.2	558.0
Average rise in vessel (°F)	60.0	62.4
Average rise in core (°F)	63.3	65.8
Average in core (°F)	592.6	592.8
Average in vessel (°F)	589.2	589.2
Heat transfer		
Active heat transfer surface area (ft^2)	57,505	57,505
Average heat flux (BTU/h-ft^2)	199,900	208,802
Maximum heat flux for normal[e][f]operation (BTU/h-ft^2)	483,758	522,005
Average linear power (kW/ft)[f]	5.52	5.77
Peak linear power for normal operation (kW/ft)[e][f]	13.36	14.41
Peak linear power resulting from overpower transients/operator errors, assuming a maximum overpower of 118.5% (kW/ft)	$\leq$22.4	$\leq$22.4
Peak linear power for prevention of centerline melt (kW/ft)	22.4	22.4
Fuel central temperature Peak at peak linear power for prevention of centerline melt (°F)	4,700	4,700
Pressure drop across core[g]	29.1	28.9

a) Unit 1 flow rates are used for comparison. The lower Unit 1 flow rates are more limiting with respect to DNB. Also, these flows are to be used for both units for uprate conditions.

b) The proposed power level of 3612 MW_t has been used for all thermal-hydraulic design analyses.

c) See paragraph 4.4.2.1.1 for the use of the W-3 correlation.

d) Based on thimble plugs inserted.

e) Based on maximum F_Q of 2.42 for current design and 2.50 for uprate design.

f) Based on densified active fuel length.

g) Based on best estimate flow and design bypass flow rates.

TABLE 4.4-1B
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TABLE 4.4-2
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TABLE 4.4-3
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TABLE 4.4-4
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4.5 REACTOR MATERIALS

4.5.1 CONTROL ROD SYSTEM STRUCTURAL MATERIALS

4.5.1.1 Materials Specifications

All parts exposed to reactor coolant are made of metals which resist the corrosive action of the water. Three types of metals are used exclusively: stainless steels, nickel-chrome-iron and cobalt based alloys. In the case of stainless steels, only austenitic and martensitic stainless steels are used, the martensitic stainless steels are not used in the heat treated conditions which cause susceptibility to stress corrosion cracking or accelerated corrosion in the Westinghouse pressurized water reactor (PWR) water chemistry.

1. Pressure vessel

   All pressure containing materials comply with Section III of the American Society of Mechanical Engineers (ASME) Boiler and Pressure Vessel Code, and are fabricated from austenitic (Type F304LN for Unit 1, Type 304 for Unit 2) stainless steel.

2. Coil stack assembly

   The coil housings require a magnetic material. Both low carbon cast steel and ductile iron have been successfully tested for this application. The choice, made on the basis of cost, indicates that ductile iron will be specified on the control rod drive mechanism (CRDM). The finished housings are zinc plated or flame sprayed to provide corrosion resistance.

   Coils are wound on bobbins of molded Dow Corning 302 material, with double glass insulated copper wire. Coils are then vacuum impregnated with silicon varnish. A wrapping of mica sheet is secured to the coil outside diameter. The result is a well insulated coil capable of sustained operation at 200 degrees centigrade.

3. Latch assembly

   Magnetic pole pieces are fabricated from Type 410 stainless steel. All non-magnetic parts, except pins and springs, are fabricated from Type 304 stainless steel. Haynes 25 is used to fabricate link pins. Springs are made from nickel-chrome-iron alloy (Inconel-X). Latch arm tips are clad with Stellite-6 to provide improved wearability. Hard chrome plate and Stellite-6 are used selectively for bearing and wear surfaces.

4. Drive rod assembly

   The drive rod assembly utilizes a Type 410 stainless steel drive rod. The coupling is machined from Type 403 stainless steel. Other parts are Type 304 stainless steel with the exception of the springs which are nickel-chrome-iron alloy and the locking button which is Haynes 25.

4.5.1.2 Fabrication and Processing of Austenitic Stainless Steel Components

The discussions provided in Section 5.2.3 concerning the processes, inspections, and tests on austenitic stainless steel components to assure freedom from increased susceptibility to intergranular corrosion caused by sensitization, and the discussions provided in Section 5.2.3 on the control of welding of austenitic stainless steels, especially control of delta ferrite, are applicable to the austenitic stainless steel pressure housing components of the CRDM.

4.5.1.3 Contamination Protection and Cleaning of Austenitic Stainless Steel

The CRDM's are cleaned prior to delivery in accordance with the guidance of ANSI 45.2.1. Process specifications in packaging and shipment are discussed in Section 5.2.3. Although the procedure at the construction site is not in the Westinghouse Nuclear Steam Supply System scope of supply, Westinghouse personnel do conduct surveillance of these operations to assure that manufacturers and installers adhere to appropriate requirements as discussed in Section 5.2.3.

4.5.2 REACTOR INTERNALS MATERIALS

4.5.2.1 Materials Specifications

All the major material for the reactor internals is Type 304 stainless steel. Parts not fabricated from Type 304 stainless steel include bolts and dowel pins which are fabricated from Type 316 stainless steel and radial support key bolts which are fabricated of Inconel- 750. These materials are listed in Table 5.2-4. There are no other materials used in the reactor internals or core support structures which are not otherwise included in the ASME Code, Section III, Appendix I.

4.5.2.2 Controls on Welding

The discussions provided in Section 5.2.3 are applicable to the welding of reactor internals and core support components.

4.5.2.3 Nondestructive Examination of Wrought Seamless Tubular Products and Fittings

The discussion provided in Appendix 1A(N) verifies conformance of reactor internals and core support structures with Regulatory Guide 1.66.

4.5.2.4 Fabrication and Processing of Austenitic Stainless Steel Components

The discussions provided in Section 5.2.3 and Appendix 1A(N) verify conformance of reactor internals and core support structures with Regulatory Guide 1.44.

Regulatory Guide 1.36 is not applicable to the reactor vessel internals since no insulation material of any kind is used on these structures.

The discussions provided in Section 5.2.3 and Appendix 1A(N) verify conformance of reactor internals and core support structures with Regulatory Guide 1.31.

The discussion provided in Appendix 1A(N) verifies conformance of reactor internals with Regulatory Guide 1.34.

The discussion provided in Appendix 1A(N) verifies conformance of reactor internals and core support structures with Regulatory Guide 1.71.

4.5.2.5 Contamination Protection and Cleaning of Austenitic Stainless Steel

The discussions provided in Section 5.2.3 and Appendix 1A(N) are applicable to the reactor internals and core support structures and verify conformance with ANSI 45 specifications and Regulatory Guide 1.37.

4.6 FUNCTIONAL DESIGN OF REACTIVITY CONTROL SYSTEMS

4.6.1 INFORMATION FOR CONTROL ROD DRIVE SYSTEM (CRDS)

The CRDS is described in Section 3.9N.4.1. Figures 3.9N-5 and 3.9N-6 provide the details of the control rod drive mechanisms, and Figure 4.2-8 provides the layout of the CRDS. No hydraulic system is associated with its functioning. The instrumentation and controls for the Reactor Trip System are described in Section 7.2 and the Reactor Control System is described in Section 7.7.

4.6.2 EVALUATION OF THE CRDS

The CRDS has been analyzed in detail in a failure mode and effects analysis (FMEA) [1]. This study, and the analyses presented in Chapter 15, demonstrates that the CRDS performs its intended safety function, a reactor trip, by putting the reactor in a subcritical condition when a safety system setting is approached, with any assumed credible failure of a single active component. The essential elements of the CRDS (those required to ensure reactor trip) are isolated from non-essential portions of the CRDS (the Rod Control System) as described in Section 7.2.

Despite the extremely low probability of a common mode failure impairing the ability of the Reactor Trip System to perform its safety function, analyses have been performed in accordance with the requirements of WASH-1270. These analyses documented in References [2] and [3] have demonstrated that acceptable safety criterion would not be exceeded even if the CRDS were rendered incapable of functioning during a reactor transient for which their function would normally be expected.

The design of the control rod drive mechanism is such that failure of the control rod drive mechanism cooling system will, in the worst case, result in an individual control rod trip or a full reactor trip (see Section 9.2).

4.6.3 TESTING AND VERIFICATION OF THE CRDS

The CRDS is extensively tested prior to its operation. These tests may be subdivided into five categories, 1) prototype tests of components, 2) prototype CRDS tests, 3) production tests of components following manufacture and prior to installation, 4) onsite preoperational and initial startup tests, and 5) periodic inservice tests. These tests which are described in Sections 3.9N.4.4, 4.2, 14.2, and the Technical Specifications are conducted to verify the operability of the CRDS when called upon to function.

4.6.4 INFORMATION FOR COMBINED PERFORMANCE OF REACTIVITY SYSTEMS

As is indicated in Chapter 15, the only postulated events which assume credit for reactivity control systems other than a reactor trip to render the plant subcritical are the steam line break, feedwater line break, and loss of coolant accident. The reactivity control systems for which credit is taken in these accidents are the reactor trip and the Safety Injection System (SIS). Additional information on the CRDS is presented in Section 3.9N.4 and on the SIS in Section 6.3. Note that no credit is taken for the boration capabilities of the Chemical and Volume Control System (CVCS) as a system in the analysis of transients presented in Chapter 15. Information on the capabilities of the CVCS is provided in Section 9.3.4. The adverse boron dilution possibilities due to the operation of the CVCS are investigated in Section 15.4.6. Prior proper operation of

the CVCS has been presumed as an initial condition to evaluate transients and appropriate Technical Specifications have been prepared to ensure the correct operation or remedial action.

4.6.5 EVALUATION OF COMBINED PERFORMANCE

The evaluation of the steam line break, feedwater line break and loss of coolant accident which presume the combined actuation of the Reactor Trip System to the CRDS and the SIS are presented in Sections 15.1.5, 15.2.8, and 15.6.5. Reactor trip signals and safety injection signals for these events are generated from functionally diverse sensors and actuate diverse means of reactivity control, i.e., control rod insertion and injection of soluble poison.

Non-diverse but redundant types of equipment are only utilized in the processing of the incoming sensor signals into appropriate logic which initiates the protective action. This equipment is described in detail in Section 7.2 and 7.3. In particular, note that protection from equipment failures is provided by redundant equipment and periodic testing. Effects of failures of this equipment have been extensively investigated as reported in Reference [4]. This FMEA verifies that any single failure will not have a deleterious effect upon the Engineering Safety Features Actuation System. Adequacy of the Emergency Core Cooling System and SIS performance under faulted conditions is verified in Section 6.3.

4.6.6 STARTUP REPORTS

A summary report of unit startup and power escalation testing shall be submitted following: (1) receipt of an Operating License, (2) amendment to the license involving a planned increase in power level unless the power increase is based on measurement uncertainty recapture within the existing safety analyses, (3) installation of fuel that has a different design or has been manufactured by a different fuel supplier, and (4) modifications that may have significantly altered the nuclear, thermal or hydraulic performance of the unit.

The initial Startup Report shall address each of the startup tests identified in Chapter 14 of the Final Safety Analysis Report and shall include a description of the measured values of the operating conditions or characteristics obtained during the test program and a comparison of these values with design predictions and specifications. Any corrective actions that were required to obtain satisfactory operation shall also be described. Any additional specific details required in license conditions based on other commitments shall be included in this report. Subsequent Startup Reports shall address startup tests that are necessary to demonstrate the acceptability of changes and/or modifications. Startup Reports shall be submitted within: (1) 90 days following completion of the Startup Test Program, (2) 90 days following resumption or commencement of commercial power operation, or (3) 9 months following initial criticality, whichever is earliest. If the Startup Report does not cover all three events (i.e, initial criticality, completion of Startup Test Program, and resumption or commencement of commercial operation), supplementary reports shall be submitted at least every 3 months until all three events have been completed.

REFERENCES

1. Shopsky, W. E., “Failure Mode and Effects Analysis (FMEA) of the Solid State Full Length Rod Control System,” WCAP-8976, September 1977.

2. “Westinghouse Anticipated Transients Without Trip Analysis,” WCAP-8330, August 1974.

3. Gangloff, W. C. and Loftus, W. D., “An Evaluation of Solid State Logic Reactor Protection in Anticipated Transients,” WCAP-7706-L (Proprietary) and WCAP-7706 (Non-Proprietary), February 1971.

4. Eggleston, F. T., Rawlins, D. H., Petrow, J. R., “Failure Mode and Effects Analysis (FMEA) of the Engineering Safeguard Features Actuation System,” WCAP-8584 (Proprietary) and WCAP-8760 (Non- Proprietary), April 1976.

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5.1 SUMMARY DESCRIPTION

The Reactor Coolant System (RCS) shown in Figure 5.1-1, Sheets 1 and 2 consist of similar heat transfer loops connected in parallel to the reactor pressure vessel. Each loop contains a reactor coolant pump, steam generator and associated piping and valves. In addition, the system includes a pressurizer, a pressurizer relief tank, interconnecting piping and instrumentation necessary for operational control. All the above components are located in the Containment Building.

During operation, the RCS transfers the heat generated in the core to the steam generators where steam is produced to drive the turbine generator. Borated demineralized water is circulated in the RCS at a flow rate and temperature consistent with achieving the reactor core thermal-hydraulic performance. The water also acts as a neutron moderator and reflector, and as a solvent for the neutron absorber used in chemical shim control.

The RCS pressure boundary provides a barrier against the release of radioactivity generated within the reactor, and is designed to ensure a high degree of integrity throughout the life of the plant.

RCS pressure is controlled by the use of the pressurizer where water and steam are maintained in equilibrium by electrical heaters and water sprays. Steam can be formed (by the heaters) or condensed (by the pressurizer spray) to minimize pressure variations due to contraction and expansion of the reactor coolant. Spring loaded safety valves and a power operated relief valve are mounted on the pressurizer and discharge to the pressurizer relief tank, where the steam is condensed and cooled by mixing with water.

Reactor coolant system high point vents are provided on the reactor vessel head and on the pressurizer (see Figure 5.1-1). These vents are designed to vent noncondensible gases which may inhibit core cooling during natural circulation. Venting is directly to open areas of the Containment to assure good mixing with the containment air. The vents are sized to limit the required makeup flow rate to within the capability of the Chemical and Volume Control System.

The extent of the RCS is defined as:

1. The reactor vessel including control rod drive mechanism housings.

2. The reactor coolant side of the steam generators.

3. The reactor coolant pumps up to and including the number 1 seal.

4. A pressurizer attached to one of the reactor coolant loops.

5. The pressurizer relief tank.

6. The safety and relief valves.

7. The interconnecting piping, valves and fittings between the principal components listed above.

8. The piping, fittings and valves leading to connecting auxiliary or support systems up to and including either: 1) a 3/8 inch inside diameter flow restrictor, or 2) the second of two isolation valves either normally closed or capable of automatic or remote manual closure.

Reactor Coolant System Components

Reactor Vessel

The reactor vessel is cylindrical, with a welded hemispherical bottom head and a removable, flanged and gasketed, hemispherical upper head. The vessel contains the core, core supporting structures, control rods and other parts directly associated with the core.

The vessel has inlet and outlet nozzles located in a horizontal plane just below the reactor vessel flange but above the top of the core. Coolant enters the vessel through the inlet nozzles and flows down the core barrel-vessel wall annulus, turns at the bottom and flows up through the core to the outlet nozzles.

Steam Generators

The steam generators are vertical shell and U-tube evaporators with integral moisture separating equipment. The reactor coolant flows through the inverted U-tubes, entering and leaving through the nozzles located in the hemispherical bottom head of the steam generator. Steam is generated on the shell side and flows upward through the moisture separators to the outlet nozzle at the top of the vessel.

Reactor Coolant Pumps

The reactor coolant pumps are identical single-speed centrifugal units driven by air-cooled, three-phase induction motors. The shaft is vertical with the motor mounted above the pumps. A flywheel on the shaft above the motor provides additional inertia to extend pump coastdown. The inlet is at the bottom of the pump; discharge is on the side.

Piping

The reactor coolant loop piping is specified in sizes consistent with system requirements.

The hot leg inside diameter is 29 inches and the inside diameter of the cold leg return line to the reactor vessel is 27-1/2 inches. The piping between the steam generator and the pump suction is increased to 31 inches in inside diameter to reduce pressure drop and improve flow conditions to the pump suction.

Pressurizer

The pressurizer is a vertical, cylindrical vessel with hemispherical top and bottom heads. Electrical heaters are installed through the bottom head of the vessel while the spray nozzle, relief and safety valve connections are located in the top head of the vessel.

Pressurizer Relief Tank

The pressurizer relief tank is a horizontal, cylindrical vessel with hemispherical ends. Steam from the pressurizer safety and relief valves is discharged into the pressurizer relief tank through a sparger pipe under the water level. This condenses and cools the steam by mixing it with water that is near ambient temperature.

Safety and Relief Valves

The pressurizer safety valves are of the totally enclosed pop-type. The valves are spring loaded, self-activated with back-pressure compensation. The power operated relief valves limit system pressure for large power mismatch. They are operated automatically or by remote manual control. Remotely operated valves are provided to isolate the inlet to the power operated relief valves if excessive leakage occurs.

High Point Vent Valves

The high point vent valves are solenoid type and are operated by remote manual control from the Control Room.

Reactor Coolant System Performance Characteristics

Tabulations of important design and performance characteristics of the RCS are provided in Table 5.1-1.

Reactor Coolant Flow

The reactor coolant flow, a major parameter in the design of the system and its components, is established with a detailed design procedure supported by operating plant performance data, by pump model tests and analysis, and by pressure drop tests and analyses of the reactor vessel and fuel assemblies. Though data from all operating plants have indicated that the actual flow has been well above the flow specified for the thermal design of the plant, the following is provided for monitoring for postulated flow reductions:

1. Technical Specification requirements to verify RCS flow and to perform calorimetric power checks;

2. For global flow changes:

   a. Flow meter on each RCS loop

   b. Inconsistencies between RCS temperatures and reactor power;

3. For local flow changes, inconsistencies in the incore flux map and/or the core exit thermocouple temperature indications.

By applying the design procedure described below, it is possible to specify the expected operating flow with reasonable accuracy.

Three reactor coolant flow rates are identified for the various plant design considerations. The definitions of these flows are presented in the following paragraphs. Many of the original plant design parameters were based on these flow rates. Actual flow rates may differ. See Chapter 15 for current values of the thermal design flow rate.

Best Estimate Flow

The best estimate flow is the calculated value predicted to occur for the actual plant operating condition. This flow is based on the best estimate of the reactor vessel, steam generator and piping flow resistance, and on the best estimate of the reactor coolant pump head-flow capacity, with no uncertainties assigned to either the system flow resistance or the pump head. The effects of differences in resistance due to changes in fuel assembly designs or other components are considered when necessary. System pressure drops, based on best estimate flow, are presented in Table 5.1-1. Although the best estimate flow is the value expected to occur in operation, more conservative flow rates are applied in some of the thermal and mechanical designs.

Thermal Design Flow

Thermal design flow is the basis for the reactor core thermal performance, the steam generator thermal performance, and the nominal plant parameters used throughout the design. To provide the required margin, the thermal design flow accounts for the uncertainties in reactor vessel, steam generator and piping flow resistances, reactor coolant pump head, and the methods used to measure flow rate. The thermal design flow is confirmed when the plant is placed in operation. Tabulations of important design and performance characteristics of the RCS as provided in Table 5.1-1 are based on the thermal design flow.

Mechanical Design Flow

Mechanical design flow is the conservatively high flow used in the mechanical design of the reactor vessel internals and fuel assemblies. To assure that a conservatively high flow is specified, the mechanical design flow is based on a reduced system resistance and on increased pump head capability. The mechanical design flow is greater than the best estimate flow.

Pump overspeed, due to a turbine generator overspeed of 20 percent, results in a peak reactor coolant flow of 120 percent of the mechanical design flow. The overspeed condition is applicable only to operating conditions when the reactor and turbine generator are at power.

Flows with One Pump Shutdown

The design procedure for calculation of flows with one pump shutdown is similar to the procedure described above for calculating flows with all pumps operating. For the case where reverse flow exists in the idle loop, the system resistance incorporates the idle loop with a locked rotor pump impeller reverse flow resistance as a flow path in parallel with the reactor vessel internals. The thermal design flow uncertainty includes a conservative application of parallel flow uncertainties (reactor internals high, idle loop low) as well as the usual component, pump and flow measurement uncertainties, thereby resulting in a conservatively low reactor flow rate for the thermal design. The mechanical design flow uncertainty is increased slightly to account for the slightly higher uncertainties at the higher pump flows.

Interrelated Performance and Safety Functions

The interrelated performance and safety functions of the RCS and its major components are listed below:

1. The RCS provides sufficient heat transfer capability to transfer the heat produced during power operation and when the reactor is subcritical, including the initial phase of plant cooldown, to the Steam and Power Conversion System.

2. The system provides sufficient heat transfer capability to transfer the heat produced during the subsequent phase of plant cooldown and cold shutdown to the Residual Heat Removal System.

3. The system heat removal capability under power operation and normal operational transients, including the transition from forced to natural circulation, shall assure no fuel damage within the operating bounds permitted by the Reactor Control and Protection Systems.

4. The RCS provides the water used as the core neutron moderator and reflector and as a solvent for chemical shim control.

5. The system maintains the homogeneity of soluble neutron poison concentration and rate of change of coolant temperature such that uncontrolled reactivity changes do not occur.

6. The reactor vessel is an integral part of the RCS pressure boundary and is capable of accommodating the temperatures and pressures associated with the operational transients. The reactor vessel functions to support the reactor core and control rod drive mechanisms.

7. The pressurizer maintains the system pressure during operation and limits pressure transients. During the reduction or increase of plant load, reactor coolant volume changes are accommodated in the pressurizer via the surge line.

8. The reactor coolant pumps supply the coolant flow necessary to remove heat from the reactor core and transfer it to the steam generators.

9. The steam generators provide high quality steam to the turbine. The tube and tubesheet boundary are designed to prevent or control to acceptable levels the transfer of activity generated within the core to the secondary system.

10. The RCS piping serves as a boundary for containing the coolant under operating temperature and pressure conditions and for limiting leakage (and activity release) to the Containment atmosphere. The RCS piping contains demineralized borated water which is circulated at the flow rate and temperature consistent with achieving the reactor core thermal and hydraulic performance.

5.1.1 SCHEMATIC FLOW DIAGRAM

The RCS is shown schematically in Figure 5.1-2. Included on this figure is a tabulation of principal pressures, temperatures, and the flow rate of the system under normal steady state full

power operating conditions. These parameters are based on the best estimate flow at the pump discharge. RCS volume under the above conditions is presented in Table 5.1-1.

5.1.2 PIPING AND INSTRUMENTATION DIAGRAM

A simplified version of the piping and instrumentation diagram of the RCS is shown on Figure 5.1-1, Sheets 1 and 2. The diagrams show the extent of the systems located within the Containment, and the points of separation between the RCS and the secondary (heat utilization) system. As part of the Reactor Coolant System, the four reactor coolant system wide range pressure sensors (PT403, PT437, PT3616 and PT405) are installed as depicted on Figure 5.1-1 Sheet 1, and are inside Containment as shown. One of these sensors is Veritrak Model 76 (PH2); the others are Rosemount Model 1154SH.

5.1.3 ELEVATION DRAWING

The components of the RCS are surrounded by concrete structures which provide support, radiation shielding and missile protection.

A detailed plant elevation drawing illustrating principal dimensions of the RCS in relation to the surrounding concrete structures is shown in Figure 5.1-3.

RCS shielding permits limited access to the Containment during power operation. The reactor vessel is installed in a thick concrete cavity formed by the primary shield. The entire RCS is enclosed by the secondary shield.

TABLE 5.1-1A (Unit 1)
SYSTEM DESIGN AND OPERATING PARAMETERS

(Sheet 1 of 2)

Plant design life, (years)	40
RCS design pressure, (psig)	2,485
Nominal operating pressure, (psig)	2,235
Total system volume including pressurizer and surge line, (ft^3)	13,400
System liquid volumes, including pressurizer water at maximum guaranteed power, (ft^3)	12,600
System water and steam volume at nominal Tavg of 589.5ºF, (ft^3)	12,135 ± 100
System design temperature, (except Pressurizer) ºF	650
Pressurizer design temperature, ºF	680
Pressurizer spray rate, maximum (gpm)	900
Pressurizer heater capacity, (kW)	1,800
Pressurizer relief tank volume, (ft^3)	1,800

SYSTEM THERMAL AND HYDRAULIC DATA - 3475MWt

	T_{AVG} = 589.2ºF[(a)(b)]	T_{AVG} = 574.2ºF[(a)(b)]
NSSS power (Mwt)	3,475	3,475
Reactor power, (MWt)	3,458	3,458
Thermal design flows, (gpm)		
Active loop	95,700	95,700
Reactor	382,800	382,800
Total reactor flow, (10^6 lb/hr)	142.0	145.2

TABLE 5.1-1A (Unit 1)
SYSTEM DESIGN AND OPERATING PARAMETERS

(Sheet 2 of 2)

Temperatures, (°F)		
Reactor vessel outlet	619.2	605.0
Reactor vessel inlet	559.2	543.5
Steam generator outlet	558.9	543.1
Steam generator steam	547.1	530.9
Feedwater	390/444.6	390/444.6 390/444.6
Steam pressure, (psia)	1021	892
Total steam flow, (10^6 lb/hr)	14.35/15.46	14.27/15.37
Best estimate flows, (gpm)		
Active loop	102,500	102,500
Reactor	410,000	410,000
Mechanical design flows, (gpm)		
Active loop	109,000	109,000
Reactor.	436,000	436,000

SYSTEM PRESSURE DROPS - 3475 MWt[(a)]

	17 x 17 OFA	RFAw/1FMs
Reactor vessel P, (psi)	49.5	51.5
Steam generator P, (psi)	33.7	33.2
Hot leg piping P, psi	1.3	1.3
Pump suction piping P, (psi)	3.4	3.3
Cold leg piping P, (psi)	3.6	3.5
Pump head, (feet)	279	283

a) 0% SGTP

b) Parameters bound both current WEC 17x17 OFA and RFA with IFMs fuel. RFA fuel may be implemented in the future.

TABLE 5.1-1B (Unit 2)
SYSTEM DESIGN AND OPERATING PARAMETERS

(Sheet 1 of 2)

Plant design life, (years)	40
RCS design pressure, (psig)	2,485
Nominal operating pressure, (psig)	2,235
Total system volume including pressurizer and surge line, (ft^3)	12,500
System liquid volumes, including pressurizer water at maximum guaranteed power, (ft^3)	12,000
System water and steam volume at nominal Tavg of 589.5ºF, (ft^3)	12,135 ± 100
System design temperature, (except Pressurizer) ºF	650
Pressurizer design temperature, ºF	680
Pressurizer spray rate, maximum (gpm)	900
Pressurizer heater capacity, (kW)	1,662
Pressurizer relief tank volume, (ft^3)	1,800

SYSTEM THERMAL AND HYDRAULIC DATA

	4 Pumps Running
NSSS power (Mwt)	3,474
Reactor power, (MWt)	3,458
Thermal design flows, (gpm)	
Active loop	95,700
Idle loop	-
Reactor	382,800
Total reactor flow, (10^6 lb/hr)	142.0

TABLE 5.1-1B (Unit 2)
SYSTEM DESIGN AND OPERATING PARAMETERS

(Sheet 2 of 2)

Temperatures, (°F)	
Reactor vessel outlet	618.8
Reactor vessel inlet	559.6
Steam generator outlet	559.3
Steam generator steam	544.6
Feedwater	441.5
Steam pressure, (psia)	995
Total steam flow, (10^6 lb/hr)	15.36
Best estimate flows, (gpm)	
Active loop	100,800
Idle loop	-
Reactor	402,000
Mechanical design flows, (gpm)	
Active loop	109,000
Idle loop	-
Reactor.	424,000

SYSTEM PRESSURE DROPS

Reactor vessel P, (psi)	46.5
Steam generator P, (psi)	38.0
Hot leg piping P, psi	1.3
Pump suction piping P, (psi)	3.3
Cold leg piping P, (psi)	3.3
Pump head, (feet)	287

5.2 INTEGRITY OF REACTOR COOLANT PRESSURE BOUNDARY

Per Regulatory Guide 1.70, Revision 2, this section presents a discussion of the measures employed to provide and maintain the integrity of the reactor coolant pressure boundary (RCPB) for the plant design lifetime. In this context, the RCPB is as defined in Section 50.2 of 10CFR50. In that definition, the RCPB extends to the outermost Containment isolation valve in system piping which penetrates the Containment and is connected to the Reactor Coolant System (RCS). Since other sections of this report already describe the components of these auxiliary fluid systems in detail, the discussions in this section will be limited to the components of the RCS as defined in Section 5.1, unless otherwise noted.

For additional information on the RCS and for components which are part of the RCPB (as defined in 10CFR50.) but are not described in this section, refer to the following sections:

Section 6.3 - For discussions of the RCPB components which are part of Emergency Core Cooling System.

Section 9.3.4 - For discussions of the RCPB components which are part of the Chemical and Volume Control System.

Section 3.9N.1 - For discussions of the design loadings, stress limits, and analyses applied to the RCS and American Society of Mechanical Engineers (ASME) Code Class 1 components.

Section 3.9N.3 - For discussions of the design loadings, stress limits and analyses applied to ASME Code Class 2 and 3 components.

The phrase, RCS, as used in this section is as defined in Section 5.1. When the term RCPB is used in this section, its definition is that of Section 50.2 of 10CFR50.

5.2.1 COMPLIANCE WITH CODES AND CODE CASES

5.2.1.1 Compliance with 10CFR Section 50.55a

RCS components are designed and fabricated in accordance with the rules of 10CFR50., Section 50.55a, “Codes and Standards” except as described below. The actual addenda of the ASME Code applied in the design of each component is listed in Table 5.2-1.

All components located within the reactor coolant pressure boundary (as defined by 10CFR50.2) are Safety Class 1 as required by 10CFR50.55a with the exception of the 3/4” instrument lines and other 3/4” piping connected to the pressurizer above the normal water level and the pressurizer relief and safety valve discharge line. A rupture of one of these lines may result in a rapid depressurization of the reactor coolant system and ECCS actuation on low pressurizer pressure. Relief from the requirements of 10CFR50.55a was authorized by the NRC in accordance with 10CFR50.55a(a)(c)(ii). [8]

5.2.1.2 Applicable Code Cases

Regulatory Guides 1.84 and 1.85 are discussed in Appendix 1A, and Regulatory Guide 1.147. Code Cases N-20-4, 2124-1 and 2143-1 were used in the manufacture of the Unit 1 RSGs.

Prior to the implementation date of Regulatory Guide 1.85, Code Case 1528, which is not included in the Regulatory Guide 1.85 list of Nuclear Regulatory Commission (NRC) endorsed code cases, was used in the manufacture of the Comanche Peak Nuclear Power Plant (CPNPP) steam generators (Unit 2). Westinghouse is conducting a test program and compiling data which demonstrates the adequacy of Code Case 1528 material. Preliminary results of the test program (for review) and a request for approval of the use of Code Case 1528 have been submitted to the NRC via Reference [1].

A list of all Code Cases used on Safety Class 1 components within the Reactor Coolant Pressure Boundary is provided in Table 5.2-7.

5.2.2 OVERPRESSURE PROTECTION

RCS overpressure protection is accomplished by the utilization of safety valves along with the Reactor Protection System and associated equipment. Combinations of these systems provide compliance with the overpressure requirements of the ASME Code, Section III, paragraphs NB-7300 and NC-7300, for pressurized water reactor systems.

Auxiliary or emergency systems connected to the RCS are not utilized for prevention of RCS overpressurization.

5.2.2.1 Design Bases

Overpressure protection is provided for the RCS by the pressurizer safety valves which discharge to the pressurizer relief tank by common header. The transient on which the design requirements are set for the primary system overpressure protection is a complete loss of steam flow to the turbine with credit taken for steam generator safety valve operation and main feedwater flow maintained. However, for the sizing of the pressurizer safety valves, no credit is taken for reactor trip nor the operation of the following:

1. Pressurizer power operated relief valves.

2. Steam line relief valve.

3. Steam Dump System.

4. Reactor Control System.

5. Pressurizer Level Control System.

6. Pressurizer spray valve.

For this transient, the peak RCS and peak steam system pressure must be limited to 110 percent of their respective design values.

Assumptions for the overpressure analysis include: 1) the plant is operating at the power level corresponding to the engineered safeguards design rating, and 2) the RCS average temperature and pressure are at their maximum values. These are the most limiting assumptions with respect to system overpressure.

Overpressure protection for the steam system is provided by steam generator safety valves. The steam system safety valve capacity is based on providing enough relief to remove 105 percent of the engineered safeguards design steam flow. This must be done by limiting the maximum steam system pressure to less than 110 percent of the steam generator shell side design pressure.

Blowdown and heat dissipation systems of the Nuclear Steam Supply System (NSSS) connected to the discharge of these pressure relieving devices are discussed in Section 5.4.11.

Steam generator blowdown systems for the balance of plant are discussed in Section 10.4.8.

Postulated events and transients on which the design requirements of the overpressure protection system are based and discussed in Reference [2].

The criterion for sizing the pressurizer and steam generator safety valves is presented in Reference 2 in Chapter 2. Differences in plant parameters between CPNPP and Reference 2 are shown in Table 5.2- 6. Due to these differences, the transient analysis results will change. Chapter 15 of the CPNPP FSAR presents transient analysis using actual plant information. Values for trip setpoints, time delays and error allowances are also provided in Chapter 15. The results and conclusions presented there indicate that for all postulated transient conditions both primary and secondary system pressures remain within 110% of the respective design values. Therefore, the conclusions presented in Reference 2 are valid for Comanche Peak.

5.2.2.2 Design Evaluation

The relief capacities of the pressurizer and steam generator safety valves are determined from the postulated overpressure transient conditions in conjunction with the action of the Reactor Protection System. An evaluation of the functional design of the system and an analysis of the capability of the system to perform its function is presented in Reference [2]. The report describes in detail the types and number of pressure relief devices employed, relief device description, locations in the systems, reliability history, and the details of the methods used for relief device sizing based on typical worst condition. The description of the analytical model used in the analysis of the overpressure protection system and the basis for its validity is discussed in Reference [3].

A description of the pressurizer safety valves performance characteristics along with the design description of the incidents, assumptions made, method of analysis and conclusions are discussed in Chapter 15.

5.2.2.3 Piping and Instrumentation Diagrams

Overpressure protection for the RCS is provided by pressurizer safety valves shown in Figure 5.1-1, Sheet 2. These discharge to the pressurizer relief tank by a common header.

The main steam supply system safety valves are discussed in Section 10.3 and are shown on Figure 10.3-1, Sheet 1 of 2.

5.2.2.4 Equipment and Component Description

The operation, significant design parameters, number and types of operating cycles, and environmental qualification of the pressurizer safety valves are discussed in Section 5.4.13.

A discussion of the equipment and components of the main steam supply system overpressure protection is discussed in Section 10.3

5.2.2.5 Mounting of Pressure-Relief Devices

Westinghouse provides the architect engineer with installation guidelines and suggested physical layout. This information is transmitted to the architect engineer as part of a systems standard design criteria document. The architect engineer is required by Westinghouse to limit the piping reaction loads on the safety valves to acceptable values.

Westinghouse provides mounting brackets on the pressurizer which can be used to support the pressurizer safety valves. The architect engineer is responsible for the design and mounting of the supports of these valves. They are also responsible for determining reactions on the pressurizer mounting brackets.

Mounting of the components of the main steam supply system overpressure protection is discussed in Section 10.3

5.2.2.6 Applicable Codes and Classification

The requirements of ASME Code, Section III, paragraphs NB-7300 (Over-pressure Protection Report) and NC-7300 (Overpressure Protection Analysis), are followed and complied with for pressurized water reactor systems.

Piping, valves and associated equipment used for overpressure protection are classified in accordance with American Nuclear Society (ANS) N18.2, “Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants.” These safety class designations are delineated on Table 3.2-2 and shown on Figure 5.1-1.

For further information, refer to Section 3.9N.

5.2.2.7 Material Specifications

Refer to Section 5.2.3 for a description of material specifications.

5.2.2.8 Process Instrumentation

Each pressurizer safety valve discharge line incorporates a control board temperature indicator and alarm to notify the operator of steam discharge due to either leakage or actual valve operation. For a further discussion on process instrumentation associated with the system, refer to Chapter 7.

5.2.2.9 System Reliability

The reliability of the pressure relieving devices is discussed in Section 4 of Reference [2].

5.2.2.10 Testing and Inspection

Testing of the overpressure protection system is conducted in the following manner:

1. A test is performed to assure operability of system electronics in accordance with Technical Specification requirements.

2. A test for valve operability is conducted as specified in ASME Code Section XI.

3. Subsequent to system, valve or electronics maintenance, a test on that portion(s) of the system is performed prior to declaring the system operable.

Testing and inspection of the overpressure protection components are discussed in Section 5.4.13.4 and Chapter 14.

5.2.2.11 RCS Pressure Control During Low Temperature Operation

Administrative procedures are developed to aid the operator in controlling RCS pressure during low temperature operation. These include ensuring a path to a RHR suction relief valve is open whenever the RCS is water solid. However, to provide a back-up to the operator and to minimize the frequency of RCS overpressurization, an automatic system is provided to mitigate the pressure excursion to address the allowable pressure, obtained from the Pressure and Temperature Limits Report.

Analyses have shown that either one PORV or one RHR suction relief valve is sufficient to prevent violation of the RCS pressure limits as a result of anticipated mass and heat input transients.

However, redundant protection against such over-pressurization is provided through use of two PORV's or two RHR Suction Relief Valves, or a combination of a PORV and a RHR Suction Relief. The PORVs are automatically armed and activated to mitigate any potential pressure transients during low temperature water solid operation.

5.2.2.11.1 System Operation

Both pressurizer power operated relief valves are supplied with actuation logic to ensure that a redundant and independent RCS pressure control back-up feature is provided for the operator during low temperature operations. This system provides the capability for RCS inventory letdown, thereby maintaining RCS pressure within allowable limits. Refer to Sections 5.4.7, 5.4.10, 5.4.13, 7.7 and 9.3.4 for additional information on RCS pressure and inventory control during other modes of operation.

Each of the two PORV's is supplied with an independent, seismically designed supply of nitrogen which is sized to assure that no operator action is required to terminate the transient for 10 minutes. High pressure nitrogen will be regulated down to the required operating pressure for

the PORV actuators. Relief valves provide protection against over-pressurizing the actuators due to regulator failure.

The basic function of the system logic is to continuously monitor RCS temperature and pressure conditions whenever plant operation is at low temperatures. An auctioneered system temperature will be continuously converted to an allowable pressure and then compared to the actual RCS pressure. This system logic will first annunciate a main control board alarm whenever the measured pressure approaches a pre-determined amount of the allowable pressure thereby indicating a pressure transient is occurring. On a further increase in measured pressure, an actuation signal is transmitted to the power operated relief valves when required to mitigate pressure transient.

5.2.2.11.2 Evaluation of Low Temperature Overpressure Transients

Pressure Transient Analyses

ASME Section III, Appendix G, establishes guidelines and upper limits for RCS pressure, primarily for low temperature conditions (≤350°). The mitigation system discussed in this section satisfies these conditions, as discussed in the following paragraphs.

Transient analyses were performed to determine the maximum pressure for the postulated mass input and heat input events.

The mass input pressure transient which is most likely during the course of normal plant operation would involve letdown isolation with charging pumps delivering an input less than or equal to 120 gpm. However, the mass input analysis was performed assuming letdown isolation with two charging pumps operating in a configuration producing maximum delivery rates. This more unlikely and more severe configuration was chosen to provide additional system flexibility for pressure control.

The maximum setpoint overpressure achieved for two charging pump operation with letdown isolation will vary with valve setpoint pressure. For example, at low RCS temperatures with a setpoint pressure of 505 psig the peak overpressure for PORV discharge is calculated to be 565 psi. At an RCS temperature of 200° and a setpoint pressure of 640 psig, the water discharge peak overpressure would be 700 psig.

The heat input transient analysis is performed over the entire RCS shutdown temperature range. This analysis also assumes an inadvertent reactor coolant pump startup with a 50°F mismatch.

Both heat input and mass input analyses took into account the single failure criteria and therefore, only one power operated relief valve (PORV) was assumed to be available for pressure relief. The above events have been evaluated considering the allowable pressure/ temperature limits established in the Pressure and Temperature Limits Report. The evaluation of the transient results conclude that the allowable limits will not be exceeded and therefore will not constitute an impairment to vessel integrity and plant safety.

In addition, the probability of a loss of a complete DC bus concurrent with a low temperature overpressure event concurrent with a PORV failure was considered. In the CPNPP design, the failure of the Train A DC bus will cause the letdown valve and Train A RCS PORV to fail in the closed position, i.e., not being able to open on demand. However, the Train B PORV is not the

only remaining available means to mitigate on overpressurization event under these conditions. There are several methods by which the operator can be made aware of this scenario, including an alarm when the DC bus fails, an alarm before the pressure relief setpoint is reached, and panel indication of system pressure. The operator has several methods to mitigate the accident, including automatic or manual operation of the Train B PORV, manual operation of an RCS vent, stopping RCS heatup, stopping the centrifugal charging pumps, and increasing RCS reactor coolant pump seal return flow. The probability of an overpressurization event under these conditions is considered to be very small; when the alternative means of mitigation as described above are taken into account along with the very low probability, the CPNPP design is considered adequate.

5.2.2.11.3 Operating Basis Earthquake Evaluation

A fluid systems evaluation has been performed considering the potential for overpressure transients following an Operating Basis Earthquake (OBE). The most likely failure of the cold overpressure mitigation system during an OBE is the loss of plant nitrogen supply system. Therefore, this system employs seismically qualified nitrogen reservoirs which ensure that motive force to the PORVs will be maintained during an OBE.

The PORVs have been designed in accordance with the ASME code to provide the integrity required for the reactor coolant pressure boundary. The PORVs and associated valve operators have been analyzed for accident loads and for loads imposed by seismic events and have been shown to maintain their integrity. As a minimum, this means that pressure boundary joints remain leaktight. Yokes, frames and similar structures will not break, actuators will not freeze or bind, and the structural integrity of the valve internals will not be degraded. Should an OBE occur, the operability of the pressurizer PORVs can be easily checked by closing the block valves and testing the PORVs. If any problems are found, the operator would take additional precautionary measures during low temperature conditions to ensure that an overpressurization does not occur and that pressure excursions are quickly mitigated.

5.2.2.11.4 Administrative Procedures

Although the system described in Section 5.2.2.11.1 is installed to mitigate the pressure excursion to address the allowable pressure limits, administrative procedures are recommended for minimizing the potential for any transient that could actuate the overpressure relief system. The following discussion highlights these procedural controls, listed in hierarchy of their function in preventing RCS cold overpressurization transients.

Of primary importance is the basic method of operation of the plant. Normal plant operating procedures will maximize the use of a pressurizer cushion (steam/nitrogen bubble) during periods of low pressure, low temperature operation. This cushion will dampen the plants response to potential transient generating inputs, providing easier pressure control with slower response rates.

An adequate cushion substantially reduces the severity some potential pressure transients such as reactor coolant pump induced heat input and slows the rate or pressure rise for others. In conjunction with the previously discussed alarms, this provides reasonable assurance that most potential transients can be terminated by operator action before the overpressure relief system actuates.

However, for those modes of operation when water solid operation may still be possible, procedures will minimize the potential for developing an overpressurization transient. The following procedural controls will be effected:

1. Both residual heat removal inlet lines from the reactor coolant loops will not be isolated unless the charging pumps are stopped. This precaution is to assure there is a relief path from the reactor coolant loop to a residual heat removal suction line relief valve when the RCS is at low pressure (less than 500 psi) and is water solid.

2. When the plant is water solid and the reactor coolant pressure is being maintained by the low pressure letdown control valve, letdown flow will bypass the normal letdown orifices, and the valve in the bypass line will be in the full open position. During this mode of operation, all three letdown orifices will also remain open.

3. If all reactor coolant pumps (RCP) have stopped for more than 5 minutes during plant heatup, and the reactor coolant temperature is greater than the charging and seal injection water temperature, RCP restart will not be attempted unless a steam bubble is formed in the pressurizer. This precaution will minimize the pressure transient when the pump is started and the cold water previously injected by the charging pumps is circulated through the warmer reactor coolant components. The steam bubble will accommodate the resultant expansion as the cold water is rapidly warmed.

4. If all reactor coolant pumps are stopped and the RCS is being cooled down by the RHR heat exchangers, a non-uniform temperature distribution may occur in the reactor coolant loops. RCP restart will not be a attempted unless a steam bubble is formed in the pressurizer.

5. During plant cooldown, all steam generators are either connected to the header or isolated. Procedures controls will be established to assure a uniform RCS cooldown when the MSIVs are closed.

6. At least one reactor coolant pump will remain in service until the reactor coolant temperature is reduced to 160°F.

7. RCPs will not be started in a water solid RCS, if the temperature difference between the steam generator and the RCS is greater than 50°F.

These special precautions back-up the normal operational mode of maximizing periods of steam bubble operation so that cold overpressure transient prevention is continued during periods of transitional operations.

The configurations of ECCS testing and alignment will also minimize the potential for developing a cold overpressurization transient. During these limited periods of plant operation, the following procedural controls will be effected:

1. To preclude inadvertent emergency core cooling system (ECCS) actuation during heatup and cooldown, procedures will require blocking the low pressurizer pressure and low compensated steam line pressure, safety injection signal actuation logic below the P-11 interlock setpoint. Block features associated with pressurizer and steam line safety injection signals are discussed in Section 7.3.2.2.6.

2. During further cooldown, closure and power lockout of the accumulator isolation valves will be performed at RCS conditions less than 1000 psig and 350°F, and power lockout of charging pumps and safety injection pumps will be performed at RCS conditions less than 350°F as required by Technical Specifications. This provides additional back-up to step 1.) above.

3. The preferred procedure for periodic ECCS pump performance testing will be to test the pumps during normal power operation or at hot shutdown conditions. This precludes any potential for developing a cold overpressurization transient.

   When cold shutdown testing of the pumps is required, the test will be done by one of the following means, precluding overpressurization potential:

   a. With the reactor vessel head off, or

   b. With the reactor vessel head on and with the necessary ECCS pump discharge valve closure (either a closed isolation valve with power removed from the valve operator, or a manual isolation valve secured in the closed position) to isolate potential ECCS pump input. During such testing, which takes place during RHR system operation, the RHR suction relief valves provide some additional backup relief protection.

4. "S" signal circuitry testing, if done during cold shutdown, will also require RHRS alignment and non-operating ECCS pumps power lockout to preclude developing cold overpressurization transients.

The above procedural considerations covering normal operations with a steam bubble, transitional operations where potentially water solid, and specific testing operations provide in-depth cold overpressure preventions or reductions. These procedures thereby augment the installed automatic overpressure relief system.

5.2.3 REACTOR COOLANT PRESSURE BOUNDARY MATERIALS

5.2.3.1 Material Specifications

Material specifications used for the principal pressure retaining applications in each component comprising the RCPB are listed in Table 5.2-2 for ASME Class 1 primary components and Table 5.2-3 for ASME Class 1 and 2 auxiliary components. Tables 5.2-2 and 5.2-3 for ASME Class 1 and 2 auxiliary components. Tables 5.2-2 and 5.2-3 include the unstabilized austenitic stainless steel material specifications used for the: 1) RCPB, 2) systems required for reactor shutdown, and 3) systems required for emergency core cooling.

The unstabilized austentic stainless steel material for the reactor vessel internals which are required for emergency core cooling for any mode of normal operation or under postulated accident conditions and for core structural load bearing members are listed in Table 5.2-4.

The materials are procured in accordance with the material specification requirements and include the special requirements of the ASME Code, Section III, plus addenda and code cases as are applicable and appropriate to meet Appendix B of 10CFR50.

In some cases, Table 5.2-3 may not be totally inclusive of the material specifications used in the listed applications. However, the listed specifications are representative of those materials utilized. All of the materials used are procured in accordance with ASME Code requirements.

The welding materials used for joining the ferritic base materials of the RCPB, conform to or are equivalent to ASME Material Specifications SFA 5.1, 5.2, 5.5, 5.17, 5.18 and 5.20. They are tested and qualified to the requirements of the ASME Code, Section III. In addition the ferritic materials of the reactor vessel beltline are restricted to the following maximum limits of copper, phosphorous and vanadium to reduce sensitivity to irradiation embrittlement in service:

Element	Base Metal(%)	As Deposited Weld Metal(%)
Copper	0.10 (ladle) 0.12 (check)	0.10
Phosphorous	0.012 (ladle) 0.017 (check)	0.015
Vanadium	0.05 (check)	0.05 (as residual)

The welding materials used for joining the austenitic stainless steel base materials of the RCPB conform to ASME Material Specifications SFA 5.4 and 5.9. They are tested and qualified according to the requirements of the ASME Code, Section III.

The welding materials used for joining nickel-chromium-iron alloy in similar base material combination and in dissimilar ferritic or austenitic base material combination conform to ASME Material Specifications SFA 5.11 and 5.14. They are tested and qualified to the requirements of the ASME Code, Section III.

5.2.3.2 Compatibility With Reactor Coolant

5.2.3.2.1 Chemistry of Reactor Coolant

The RCS chemistry specifications are given in Table 5.2-5.

The RCS water chemistry is selected to minimize corrosion. A periodic analysis of the coolant chemical composition is performed to verify that the reactor coolant quality meets the specifications.

The Chemical and Volume Control System provides a means for adding chemicals to the RCS which control the pH and oxygen content of the coolant. The oxygen content and pH limits for power operations are shown in Table 5.2-5.

The pH control chemical employed is lithium-7 hydroxide. This chemical is chosen for its compatibility with the materials and water chemistry of borated water/stainless steel/zirconium/Inconel systems. In addition, lithium is produced in solution from the neutron irradiation of the dissolved boron in the coolant. The lithium-7 hydroxide is introduced into the RCS via the charging flow. The solution is prepared in the laboratory and poured into the

chemical mixing tank. Reactor makeup water is then used to flush the solution to the suction header of the charging pumps. The concentration of lithium-7 hydroxide in the RCS is maintained in the range specified for pH control. If the concentration exceeds this range, the cation bed demineralizer is employed in the letdown line in series operation with the mixed bed demineralizer. Since the amount of lithium to be removed is small and its buildup can be readily calculated and determined by analysis, the flow through the cation bed demineralizer is not required to be full letdown flow.

Hydrazine is employed as an oxygen scavenging agent during reactor startup from cold condition or during reactor shutdown to a cold condition. The hydrazine solution is introduced into the RCS in the same manner as described above for the pH control agent.

Dissolved hydrogen is employed to control and scavenge oxygen produced due to radiolysis of water in the core region. Sufficient partial pressure of hydrogen is maintained in the volume control tank such that the specified equilibrium concentration of hydrogen is maintained in the reactor coolant. A self-contained pressure control valve maintains a minimum pressure in the vapor space of the volume control tank. This can be adjusted to provide the correct equilibrium hydrogen concentration.

5.2.3.2.2 Compatibility of Construction Materials With Reactor Coolant

All the ferritic low alloy and carbon steels which are used in principal pressure retaining applications are provided with corrosion resistant cladding on all surfaces that are exposed to the reactor coolant. This cladding material has a chemical analysis which is at least equivalent to the corrosion resistance of Types 304 and 316 austenitic stainless steel alloys or nickel-chomium-iron alloy, martensitic stainless steel and precipitation hardened stainless steel. The cladding on ferritic type base materials receives a post- weld heat treatment.

Ferritic low alloy and carbon steel nozzles are safe ended with either stainless steel wrought materials, stainless steel weld metal analysis A-7, or nickel-chromium-iron alloy weld metal F-Number 43. The latter buttering material requires further safe ending with austenitic stainless steel base material after completion of the post-weld heat treatment when the nozzle is larger than a 4 inch nominal inside diameter and/or the wall thickness is greater than 0.531 inches.

All of the austenitic stainless steel and nickel-chromium-iron alloy base materials with primary pressure retaining applications are used in the solution anneal heat treat condition. These heat treatments are as required by the materials specifications.

During subsequent fabrication, these materials are not heated above 800°F other than locally by welding operations. The solution annealed surge line material is subsequently formed by hot bending followed by a resolution annealing heat treatment.

Components with stainless steel sensitized in the manner expected during component fabrication and installation will operate satisfactorily under normal plant chemistry conditions in pressurized water reactor systems because chlorides, fluorides and oxygen are controlled to very low levels.

5.2.3.2.3 Compatibility With External Insulation and Environmental Atmosphere

In general, all of the materials listed in Tables 5.2-2 and 5.2-3 which are used in principal pressure retaining applications and which are subject to elevated temperature during system operation are in contact with thermal insulation that covers their outer surfaces.

The thermal insulation used on the RCPB is either reflective stainless steel type or to be made of compounded materials which yield low leachable chloride and/or fluoride concentrations. The compounded materials in the form of blocks, boards, cloths, tapes, adhesives, cements, etc., are silicated to provide protection of austenitic stainless steels against stress corrosion which may result from accidental wetting of the insulation by spillage, minor leakage or other contamination from the environmental atmosphere. Appendix 1A.(N) includes a discussion which indicates the degree of conformance with Regulatory Guide 1.36, “Nonmetallic Thermal Insulation for Austenitic Stainless Steel.”

In the event of coolant leakage, the ferritic materials will show increased general corrosion rates. Where minor leakage is anticipated from service experience, such as valve packing, pump seals, etc., only materials which are compatible with the coolant are used. These are as shown in Tables 5.2-2 and 5.2-3. Ferritic materials exposed to coolant leakage can be readily observed as part of the inservice visual and/or nondestructive inspection program to assure the integrity of the component for subsequent service.

5.2.3.3 Fabrication and Processing of Ferritic Materials

5.2.3.3.1 Fracture Toughness

The fracture toughness properties of the RCPB components meet the requirements of the ASME Code, Section III, paragraphs NB-2300, NC- 2300 and ND-2300 as appropriate.

Limiting steam generator and pressurizer RT_{NDT} temperatures are guaranteed at 60°F for the base materials and the weldments. These materials will meet the 50 ft-lb absorbed energy and 35 mils lateral expansion requirements of the ASME Code, Section III at 120°F. The actual results of these tests are provided in the ASME material data reports which are supplied for each component and are submitted to Texas Utilities at the time of shipment of the component. Fracture toughness tests for RCPB components are performed in accordance with written procedures, by operators qualified by training and experience.

Calibration of temperature instruments and charpy impact test machines are performed to meet the requirements of the ASME Code, Section III, paragraph NB-2360.

Westinghouse has conducted a test program to determine the fracture toughness of low alloy ferritic materials with specified minimum yield strengths greater than 50,000 pounds per square inch (psi) to demonstrate compliance with Appendix G of the ASME Code, Section III. The results of this program are discussed in reference [7].

SA-508 Class 2a and SA-533 Grade A Class 2 material was used in the fabrication of Comanche Peak Units 1 and 2 pressurizers. The conclusions of Westinghouse topical report WCAP-9292 [7] are applicable to the Comanche Peak Units.

5.2.3.3.2 Control of Welding

All welding is conducted utilizing procedures qualified according to the rules of Sections III and IX of the ASME Code. Control of welding variables, as well as examination and testing, during procedures qualification and production welding is performed in accordance with ASME Code requirements.

Appendix 1A(N) includes discussions which indicate the degree of conformance of the ferritic materials components of the RCPB with Regulatory Guides 1.34, “Control of Electroslage Properties,” 1.43, “Control of Stainless Steel Weld Cladding of Low-Alloy Steel Components,” 1.50, “Control of Preheat Temperature for Welding of Low-Alloy Steel,” 1.66, “Nondestructive Examination of Tubular Products,” and 1.71, “Welder Qualification for Areas of Limited Accessibility.”

5.2.3.4 Fabrication and Processing of Austenitic Stainless Steel

Sections 5.2.3.4.1 through 5.2.3.4.5 address Regulatory Guide 1.44, “Control of the Use of Sensitized Stainless Steel,” and present the methods and controls utilized by Westinghouse to avoid sensitization and prevent intergranular attach of austenitic stainless steel components. Also, Appendix 1A(N) includes a discussion which indicates the degree of conformance with Regulatory Guide 1.44.

5.2.3.4.1 Cleaning and Contamination Protection Procedures

It is required that all austenitic stainless materials used in the fabrication, installation and testing of nuclear steam supply components and systems be handled, protected, stored and cleaned according to recognized and accepted methods which are designed to minimize contamination which could lead to stress corrosion cracking. The rules covering these controls are stipulated in Westinghouse process specifications. As applicable these process specifications supplement the equipment specifications and purchase order requirements of every individual austenitic stainless steel component or system which Westinghouse procures for the CPNPP NSSS, regardless of the ASME Code classification.

The process specifications which define these requirements and which are in compliance with ANSI N-45 committee specifications are as follows:

Process Specification Number	
82560HM	Requirements for Pressure Sensitive Tapes for use on Austenitic Stainless Steels.
83336KA	Requirements for Thermal Insulation Used on Austenitic Stainless Steel Piping and Equipment.
83860LA	Requirements for Marking of Reactor Plant Components and Piping.

84350HA	Site Receiving Inspection and Storage Requirements for Systems, Material and Equipment.
84351NL	Determination of Surface Chloride and Fluoride on Austenitic Stainless Steel Materials.
85310QA	Packaging and Preparing Nuclear Components for Shipment and Storage.
292722	Cleaning and Packaging Requirements of Equipment for Use in the NSSS.
597756	Pressurized Water Reactor Auxiliary Tanks Cleaning Procedures.
597760	Cleanliness Requirements During Storage Construction, Erection and Startup Activities of Nuclear Power Systems.

Appendix 1A(N) includes a discussion which indicates the degree of conformance of the austenitic stainless steel components of the RCPB with Regulatory Guide 1.37, “Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants”.

5.2.3.4.2 Solution Heat Treatment Requirements

The austenitic stainless steels listed in Tables 5.2-2, 5.2-3 and 5.2- 4 are utilized in the final heat treated condition required by the respective ASME Code, Section II materials specification for the particular type or grade of alloy.

5.2.3.4.3 Material Inspection Program

The Westinghouse practice is that austenitic stainless steel materials of product forms with simple shapes need not be corrosion tested provided that the solution heat treatment is followed by water quenching. Simple shapes are defined as all plates, sheets, bars, pipe and tubes, as well as forgings, fittings, and other shaped products which do not have inaccessible cavities or chambers that would preclude rapid cooling when water quenched. When testing is required, the tests are performed in accordance with ASTM-A-262-70, Practices A or E, as amended by Westinghouse Process Specification 84201 MW.

5.2.3.4.4 Prevention of Intergranular Attack of Unstabilized Austenitic Stainless Steels

Unstabilized austenitic stainless steels are subject to intergranular attack (IGA) provided that three conditions are present simultaneously. These are:

1. An aggressive environment, e.g., an acidic aqueous medium containing chlorides or oxygen.

2. A sensitized steel.

3. A high temperature.

If any one of the three conditions described above is not present, intergranular attack will not occur. Since high temperatures cannot be avoided in all components in the NSSS, Westinghouse relies on the elimination of conditions 1 and 2 to prevent intergranular attack on wrought stainless steel components.

The water chemistry in the RCS of a Westinghouse pressurized water reactor is rigorously controlled to prevent the intrusion of aggressive species. In particular, the maximum permissible oxygen and chloride concentrations are 0.10 and 0.15 parts per million (ppm), respectively. Reference [4] describes the precautions taken to prevent the intrusion of chlorides into the system during fabrication, shipping, and storage. The use of hydrogen over pressure precludes the presence of oxygen during operation. The effectiveness of these controls has been demonstrated by both laboratory tests and operating experience. The long time exposure of severely sensitized stainless in early plants to pressurized water reactor coolant environments has not resulted in any sign of intergranular attack. Reference [4] describes the laboratory experimental findings and the Westinghouse operating experience. The additional years of operations since the issuing of Reference [4] have provided further confirmation of the earlier conclusions. Severely sensitized stainless steels do not undergo any intergranular attack in Westinghouse pressurized water reactor coolant environments.

In spite of the fact there never has been any evidence that pressurized water reactor coolant water attacks sensitized stainless steels, Westinghouse considers it good metallurgical practice to avoid the use of sensitized stainless steels in the NSSS components. Accordingly, measures are taken to prohibit the purchase of sensitized stainless steels and to prevent sensitization during component fabrication. Wrought austenitic stainless steel stock used for components that are part of: 1) the RCPB, 2) systems required for reactor shutdown, 3) systems required for emergency core cooling, and 4) reactor vessel internals that are relied upon to permit adequate core cooling for normal operation or under postulated accident conditions is utilized in one of the following conditions:

1. Solution annealed and water quenched.

2. Solution annealed and cooled through the sensitization temperature range within less than approximately 5 minutes.

It is generally accepted that these practices will prevent sensitization. Westinghouse has verified this by performing corrosion tests (ASTM-393) on as-received wrought material. Westinghouse recognizes that the heat affected zones of welded component must, of necessity, be heated into the sensitization temperature range, 800 to 1500°F. However, severe sensitization, i.e., continuous grain boundary precipitates of chromium carbide, with adjacent chromium depletion, can still be avoided by control of welding parameters and welding processes. The heat input[a] and associated cooling rate through the carbide precipitation range are of primary importance. Westinghouse has demonstrated this by corrosion testing a number of weldments.

Of 25 production and qualification weldments tested, representing all major welding processes, and a variety of components, and incorporating base metal thicknesses from 0.10 to 4.0 inches, only portions of two were severely sensitized. Of these, one involved a heat input of 120,000 joules, and other involved a heavy socket weld in relatively thin walled material. In both cases, sensitization was caused primarily by high heat inputs relative to the section thickness. However, in only the socket weld did the sensitized condition exist at the surface, where the material is

exposed to the environment. The component has been redesigned and a material change has been made to eliminate this condition.

Westinghouse controls the heat input in all austenitic pressure boundary weldments by:

1. Prohibiting the use of block welding.

2. Limiting the maximum interpass temperature to 350°F.

3. Exercising approval rights on all welding procedures.

To further assure that these controls are effective in preventing sensitization, Westinghouse will, if necessary, conduct additional intergranular corrosion tests of qualification mock-ups of primary pressure boundary and core internal component welds, including the following:

1. Reactor vessel safe ends.

2. Pressurizer safe ends.

3. Surge line and reactor coolant pump nozzles.

4. Control rod drive mechanisms head adaptors.

5. Control rod drive mechanisms seal welds (Unit 2 only).

6. Control rod extensions.

7. Lower instrumentation penetration tubes.

To summarize, Westinghouse has a four point program designed to prevent intergranular attack of austenitic stainless steel components.

1. Control of primary water chemistry to ensure a benign environment.

2. Utilization of materials in the final heat treated condition and the prohibition of subsequent heat treatments in the 800 and 1500°F temperature range.

3. Control of welding processes and procedures to avoid heat affected zone sensitization.

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a. Heat input is calculated according to the formula:

Where $H = \frac{(E)(I)(60)}{S}$

H = joules/in
E = volts
I = amperes
S = travel speed (in/min)

4. Confirmation that the welding procedures used for the manufacture of components in the primary pressure boundary and of reactor internals do not result in the sensitization of heat affected zones.

Both operating experience and laboratory experiments in primary water have conclusively demonstrated that this program is 100 percent effective in preventing intergranular attack in Westinghouse NSSS's utilizing unstabilized austenitic stainless steel.

5.2.3.4.5 Retesting Unstabilized Austenitic Stainless Steels Exposed to Sensitization Temperatures

It is not normal Westinghouse practice to expose unstabilized austenitic stainless steels to the sensitization range of 800 to 1500°F during fabrication into components. If, during the course of fabrication, the steel is inadvertently exposed to the sensitization temperature range, 800 to 1500°F, the material may be tested in accordance with ASME-A-393 or A-262 as amended by Westinghouse Process Specification 84201 MW to verify that it is not susceptible to intergranular attack, except that testing is not required for:

1. Cast metal or weld metal with a ferite content of 5 percent or more.

2. Material with a carbon content of 0.03 percent or less that is subjected to temperatures in the range of 800 to 1500°F for less than 1 hour.

3. Material exposed to special processing provided the processing is properly controlled to develop a uniform product and provided that adequate documentation exists of service experience and/or test data to demonstrate that the processing will not result in increased susceptibility to intergranular stress corrosion.

If it is not verified that such material is not susceptible to intergranular attack, the material will be resolution annealed and water quenched or rejected.

5.2.3.4.6 Control of Welding

The following paragraphs address Regulatory Guide 1.31, "Control of Stainless Steel Welding," and present the methods used, and the verification of these methods, for austenitic stainless steel welding.

The welding of austenitic stainless steel is controlled to mitigate the occurrence of microfissuring or hot cracking in the weld. Although published data and experience have not confirmed that fissuring is detrimental to the quality of the weld, it is recognized that such fissuring is undesirable in a general sense. Also, it has been well documented in the technical literature that the presence of delta ferrite is one of the mechanisms for reducing the susceptibility of stainless steel welds to hot cracking. However, there is insufficient data to specify a minimum delta ferrite level below which the material will be prone to hot cracking. It is assumed that such a minimum lies somewhere between 0 and 3 percent delta ferrite.

The scope of these controls discussed herein encompasses welding processes used to join stainless steel parts in components designed, fabricated or stamped in accordance with ASME Code, Section III, Class 1 and 2, and core support components. Delta ferrite control is appropriate for the above welding requirements except where no filler metal is used or for other

reasons such control is not applicable. These exceptions include electron beam welding, autogenous gas shielded tungsten arc welding, explosive welding, and welding using fully austenitic welding materials.

The fabrication and installation specifications require welding procedure and welder qualification in accordance with the ASME Code, Section III, and include welding materials that are used for welding qualification testing and for production processing. Specifically, the undiluted weld deposits of the "starting" welding materials are required to contain a minimum of 5 percent delta ferrite[b] as determined by chemical analysis and calculation using the appropriate weld metal constitution diagrams in the ASME Code, Section III. When new welding procedure qualification tests are evaluated for these applications, including repair welding of raw materials, they are performed in accordance with the requirements of Sections III and IX of the ASME Code.

The results of all the destructive and nondestructive tests are reported in the procedure qualification record in addition to the information required by the ASME Code, Section III.

The "starting" welding materials used for fabrication and installation welds of austenitic stainless steel materials and components meet the requirements of the ASME Code, Section III. The austenitic stainless steel welding material conforms to ASME weld metal analysis A-7 (designated A8 in the 1974 Edition of the ASME B&PV Code), Type 308 for all applications except Type 308L weld metal analysis may be substituted for consumable inserts when used for weld root closures. Bare weld filler metal, including consumable inserts, used in inert gas welding processes conform to ASME SFA-5.9, and are procured to contain not less than 5 percent delta ferrite according to the ASME Code, Section III. Weld filler materials used in flux shielded welding processes conform to ASME SFA-5.4 or SFA-5.9 and are procured in a wire-flux combination to be capable of providing not less than 5 percent delta ferrite in the deposit according to the ASME Code, Section III. Welding materials are tested using the welding energy inputs to be employed in production welding.

Combinations of approved heat and lots of "starting" welding materials are used for all welding processes. The welding quality assurance program include identification and control of welding material by lots and heats as appropriate. All of the weld processing is monitored according to approved inspection programs which include review of "starting" materials, qualification records and welding parameters. Welding systems are also subject to quality assurance audit including calibration of gages and instruments: identification of "starting" and completed materials; welder and procedure qualifications; availability and use of approved welding and heat treating procedures; and documentary evidence of compliance with materials, welding parameters and inspection requirements. Fabrication and installation welds are inspected using nondestructive examination methods according to the ASME Code, Section III rules.

To assure the reliability of these controls, Westinghouse has completed a delta ferrite verification program, described in Reference [5] which has been approved as a valid approach to verify the Westinghouse hypothesis and is considered an acceptable alternative for conformance with the NRC interim position on Regulatory Guide 1.31. The NRC acceptance letter and topical report evaluation were received on December 30, 1974. The program results, which do support the hypothesis presented in Reference [5], are summarized in Reference [6].

b. The equivalent ferrite number may be substituted for percent delta ferrite.

Appendix 1A(N) includes discussions which indicate the degree of conformance of the austenitic stainless steel components of the RCPB with Regulatory Guides 1.34, “Control of Electroslag Properties,” 1.66, “Nondestructive Examination of Tubular Products,” and 1.71, “Welder Qualification for Areas of Limited Accessibility.”

5.2.4 INSERVICE INSPECTION AND TESTING OF THE REACTOR COOLANT PRESSURE BOUNDARY

This section describes the Inservice Inspection Program which will ensure the integrity of the reactor coolant pressure boundary. The Inservice Inspection Program for Class 1 systems and component supports is written in accordance with subsections IWB and IWF of Section XI, ASME Boiler & Pressure Vessel Code (Edition and Addenda as required by 10CFR50.55a).

The Inservice and Preservice Inspection Programs shall be in accordance with the guidelines of Subsections IWA and IWB of Section XI, ASME Boiler and Pressure Vessel Code (Edition and Addenda as required by 10CFR50.55a). During the preservice inspection, Class 1 systems were inspected to the 1980 Edition for Unit 1 and to the 1983 Edition with Code Case N-408 for Unit 2 except the reactor pressure vessel was for 1980 Edition. Details of the preservice inspection, including relief requests, are included in the Preservice Inspection Plan for each unit.

5.2.4.1 System Boundary Subject To Inspection

The system boundary subject to inspection includes all Safety Class 1 pressure-retaining components (and their integral attachments) as defined in IWB-1200.

5.2.4.2 Accessibility

The reactor coolant pressure boundary is designed to the extent practical so that provisions for access, as required by Subarticle IWA-1500 of ASME Section XI, can be followed.

The components, the equipment layout, and the support structures of the Reactor Coolant System have been designed to the extent practical to permit access for inspection purposes.

Reactor Vessel

Specific provision to be made for inspection access in the design of the reactor vessel, system layout and other major primary coolant components will be as follows:

1. All reactor internals will be completely removable. The tools and storage space required to permit reactor internals removal for these inspections will be provided.

2. The reactor vessel shell in the core area will be designed with a clean, uncluttered cylindrical inside surface to permit future positioning of test equipment without obstruction.

3. The reactor vessel cladding will be improved in finish by the vessel manufacturer by grinding to the extent necessary to permit meaningful examination of the vessel welds and adjacent base metal in accordance with the Code.

4. The cladding to base metal interface will be ultrasonically examined by the vessel manufacturer to assure satisfactory bonding to allow the volumetric inspection of the vessel welds and base metal from the vessel inside surface.

5. The reactor closure head will be stored in a dry condition on the operating deck during refueling, allowing direct access for inspection.

6. The insulation on the vessel closure and lower heads will be removable, allowing access for nondestructive examination of head welds.

7. All reactor vessel studs, nuts, and washers may be removed to dry storage during refueling, allowing inspection in parallel with refueling operations.

8. Access holes will be provided in the core barrel flange allowing access for the remote visual examination of the clad surface of the vessel without removal of the lower internals assembly.

9. Removable plugs will be provided in the primary shield providing access for the surface and volumetric examination of the primary nozzle O.D. safe-end-welds.

The use of conventional non-destructive, volumetric test techniques can be applied to the inspection of all primary loop components except for the reactor vessel. The reactor vessel presents special problems because of the radiation levels and remote underwater accessibility to this component.

Pressurizer

A manway is provided in the pressurizer top head to allow access for internal inspection.

Steam Generator

Manways will be provided in the steam generator channel head to provide access for internal inspection.

Reactor Coolant Pumps

The external surfaces of the pump casings are accessible for inspection. The internal surface of the pump is available for inspection by removing the pump motor and pump internals.

Piping

The reactor coolant piping, fittings, and attachments to the piping external to the primary shield are accessible for external surface and volumetric inspections in areas where inspection is required.

Insulation

The insulation covering all component and piping welds and adjacent base metal is designed for ease of removal and replacement in areas where external inspection will be planned.

Components' Location Relative To Structure and Adjacent Components

The primary loop compartments are designed to allow personnel entry during refueling operations in order to permit direct inspections access to the external portion of piping and components. Provisions for adequate distance between component areas requiring inspection and adjacent components and structures have been factored into the design and layout of components.

5.2.4.3 Examination Techniques and Procedures

Visual, surface, and volumetric examination techniques and procedures, including any special technique and procedure, are in accordance with the requirements of ASME Section XI.

5.2.4.4 Inspection Intervals

The required examinations and pressure tests will be completed during each 10-year interval of service in accordance with ASME Section XI.

5.2.4.5 Examination Categories and Requirements

The categories and requirements appropriate for each examination area follow the categories and requirements specified in ASME Section XI. An inservice inspection program which discusses the examination categories will be provided.

5.2.4.6 Evaluation of Examination Results

Evaluation of examination results for Class 1 components will be conducted in accordance with ASME Section XI.

Unacceptable indications will be repaired in accordance with the requirements of ASME Section XI.

5.2.4.7 System Leakage Tests

The System Leakage tests for the reactor pressure vessel and reactor coolant pressure boundary will be conducted in accordance with the requirements of ASME Section XI.

5.2.5 DETECTION OF LEAKAGE THROUGH REACTOR COOLANT PRESSURE BOUNDARY

The leakage-detection systems are intended to sense leakage from the reactor coolant and auxiliary systems into the Containment and to provide the means to locate such leakage.

The safety significance of leaks through the reactor coolant pressure boundary (RCPB) can vary widely depending on the source of the leak as well as the leakage rate and duration. Therefore, the detection and monitoring of reactor coolant leakage into the Containment is necessary.

The leakage-detection systems provide information which permits the plant operators to take immediate corrective action should a leak be evaluated as detrimental to the safety of the plant.

Leakage-detection system design objectives are in accordance with the requirements of 10 CFR Part 50, GDC 30, and NRC Regulatory Guide 1.45.

5.2.5.1 Leakage Classification and Limits

RCPB leakage is classified as identified or unidentified and methods for physically separating the leakage into these classifications are provided to supply prompt and quantitative information about the leakage to the plant operators. Definitions of identified and unidentified leakage are as follows:

5.2.5.1.1 Identified Leakage

Identified leakage is reactor coolant leakage into the containment area (i.e. into a closed system or containment atmosphere) that is specifically located, can be detected, collected and to the extent practical, isolated from the containment atmosphere so as not to mask any potentially serious leak should it occur.

Identified leakage is comprised of:

1. Leakage, from reactor head flange leak-offs and valve packing leak-offs, that is captured and conducted to the reactor coolant drain tank (RCDT). FSAR Figure 11.2-2 shows the Reactor Coolant Drain Subsystem. All reactor coolant pressure boundary valves that use the compressed packing method of sealing the working fluid are equipped with piped leak-off connections and the potential leaks are classified as identified. Since the leakage detection system is closed, it is essentially isolated from the Containment atmosphere and cannot mask any potentially serious leakage to the atmosphere from unidentified sources including a flaw in the RCPB.

2. Reactor coolant leakage through steam generators to the secondary system.

3. Leakage through the closed pressurizer safety and relief valves.

4. Leakage of the seal water through the reactor coolant pumps seal number 2 directed to the Reactor Coolant Drain Tank (reactor coolant pump seal leak-off number 2).

5. Leakage of the seal water through the reactor coolant pumps seal number 3 directed to the Containment Sump (reactor coolant pump seal leak-off number 3).

6. Leakage of the reactor coolant to the Component Cooling Water System (CCWS) resulting from leakage from the Reactor Coolant Pumps thermal barrier.

7. Intersystem leakage through Reactor Coolant System Pressure Isolation Valves.

All identified leakage (except primary-to-secondary leakage, intersystem leakage, RC pump seal No. 3 leakage and closed pressurizer safety and relief valves leakage) is collected in Reactor Coolant Drain Tank.

The only circulating pumps located inside the Containment Building that are operational during normal plant operation are the reactor coolant drain tank pumps and the containment floor drain sump pumps (not considering the reactor coolant pumps).

The reactor coolant drain tank pumps are of “canned” type. The configuration of the pump shaft seal system is of a completely closed system with no communication with the containment atmosphere.

The shaft seal leakage of the containment floor drain sump pumps are returned to the floor sump and are not considered as system outleakage.

Based on the above consideration leakage through the pump shaft seals is not entering a closed system or containment atmosphere and as such considered as completely leaktight.

5.2.5.1.2 Unidentified Leakage

Unidentified leakage is all leakage which is not identified leakage or controlled leakage. It is impractical to completely eliminate unidentified leakage, but efforts are made to reduce this leakage to a small background flow rate permitting the leakage detection systems to detect positively and rapidly any small increase in unidentified leakage flow rate.

5.2.5.1.3 Controlled Leakage

Controlled leakage is the seal water flow supplied to the reactor coolant pump seal number 1 (approximately 8-gpm per pump). The reactor coolant pump seal subsystem is discussed in Section 9.3.

5.2.5.1.4 Limits for Reactor Coolant Leakage

Leakage through the RCPB is limited to the following:

1. Identified leakage

    a. 150 gpd through any one steam generator

    b. 10-gpm total leakage (Items 1, 2, 3, 4, 5, 6, and 7 in Section 5.2.5.1.1)

2. Unidentified leakage

    a. 1-gpm limit.

5.2.5.1.5 Maximum Allowable Identified Leakage

As stated in Technical Specifications, Limiting Conditions for Operation, operation of the reactor is permitted with a total primary system identified leakage not exceeding flow rates as specified in Section 5.2.5.1.4.

For a maximum allowable capacity of the reactor coolant makeup system of 120-gpm, the ratio of the maximum allowable identified leakage to the makeup rate is 0.083.

For a normal capacity of the containment water removal system (sump pump) of 80-gpm, the ratio of the maximum allowable identified leakage to this normal capacity is 0.125.

5.2.5.1.6 Maximum Allowable Unidentified Leakage

Selection of 1.0-gpm as the maximum permissible unidentified leakage is based on the leakage behavior of cracks in piping under pressure as described in WCAP-7503 and discussed in the following paragraphs.

The lengths of through-wall cracks that are calculated to leak 0.5-gpm in 2-in. lines, 1-gpm in 3-in. lines and 2 gpm in 4 in. lines and larger are given in WCAP-7503 and are defined as detectable leakage crack lengths. This study also gives curves for detectable leakage crack length versus detectable leak rate for longitudinal and circumferential through-wall cracks.

In addition, this study gives the critical crack length which leads to pipe failure for both longitudinal and circumferential cracks. In all cases, the detectable leakage crack length is less than the critical crack length. The study shows the margins of safety and the number of years for a detectable leakage crack to grow to a critical crack size. If a leak is detected from a detectable leakage crack, there is sufficient time to shut down the plant and make all necessary repairs.

Although the 1-gpm maximum permissible unidentified leakage rate is larger than the 0.5-gpm leakage rate analyzed for cracks in 2-in lines, core cooling analyses have shown that for small breaks, i.e., breaks up to the equivalent of the cross-sectional area of 4-in- diameter lines, no reactor fuel-cladding damage is expected. Thus, the selection of 1-gpm maximum permissible unidentified leakage rate is a conservative choice.

5.2.5.2 Leakage Detection Methods

5.2.5.2.1 Identified Leakage

In general, identified leakage that is conducted to the Reactor Coolant Drain Tank and to the pressurizer relief tank is measured by means of a level change in the respective tank. Leakage into the RCDT may also be measured by a flow totalizer on the discharge of the RCDT level control valve. Identified leakage conducted to the Containment Sumps is measured by means of a flow totalizer in the Containment Sump Pumps discharge header (outside the Containment).

1. Steam Generator primary-to-secondary leakage is detected by the Steam Generator Blowdown Process Sample (SGBPS) Monitor (see Section 11.5.2.7.1), the Steam Generator Leak Rate (SGLRM) Monitors (see Section 11.5.2.6.13), and the Condenser Off-Gas Monitor (see Section 11.5.2.6.5).

2. Reactor head flange leak-off system

   The seal for the reactor vessel head is formed by two concentric o-rings with two intermediate leak-off lines. If leakage past the innermost o-ring occurs, it is detected by a surface mounted temperature detector located on the common leak-off line.

3. Pressurizer safety and relief valves

   Leakage from the closed pressurizer safety and relief valves is detected by a surface mounted temperature detector located on each safety valve discharge pipe on the common discharge header of the pressurizer relief valves.

4. Reactor Coolant Drain Tank

Leakage from valve stem packing leak-offs and from the RCS pump seal leak-offs to the Reactor Coolant Drain Tank is measured by means of a flow totalizer downstream of the level control valve or level change in the RCDT.

Monitoring of total leakage is done by observations of fluctuation of RCDT levels coupled with evaluation of operational occurrences. If a leakage rate that requires action is reached, the leaking component will be identified by entering the containment for inspection and valve packing leaks will be identified by taking piping surface temperature readings of the leak-off piping.

5. Leakage to the Component Cooling Water System

Any leakage of the reactor coolant to the Component Cooling Water System (CCWS) is detected by radiation monitors in the CCWS (see Section 11.5.2.7.6). Also, leakage into the CCWS can be detected as an increase in the level of the CCWS surge tank.

6. Pressure Isolation Valve Leakage

An SI injection header alternate relief path has been provided on Unit 2 (only) which may be used to manually bypass the injection header relief valves. This path includes the SI System Header Pressure Isolation Valve Leakage Monitoring System (Unit 2 only) which may be used to measure the RCS pressure isolation valve leakage. See Section 6.3.5.3.9 for more details.

5.2.5.2.2 Unidentified Leakage

Primary detection of unidentified leakage to the Containment atmosphere is provided by air particulate monitors, containment sump flow monitors and condensate flow rate measuring system. In addition, containment dewpoint, indication of gross leakage and liquid inventory and radioactive gas monitor are other indications available to the operator for determination of unidentified leakage.

In normal operation, the primary monitors show a background level which is indicative of the normal magnitude of unidentified leakage inside the Containment. Variations in airborne reactor coolant corrosion products or condensate flow rate above the normal level signifies an increase in unidentified leakage rates and signal to the plant operators that corrective action may be required. Similarly, increases in Containment sump flow and radioactive gaseous concentration in the containment atmosphere signifies an increase in unidentified leakage.

A description of these primary unidentified leakage monitors is provided below along with a description of other leakage indications available to the plant operator.

1. Containment Air Particulate Monitor

Air particulate monitors take continuous air samples from the containment atmosphere and measure the particulate activity collected on a filter paper system. After passing through an iodine and noble gas monitor downstream of the particulate monitor, the air returns to the Containment. (See Section 11.5.2.6.2)

The sensitivity of the Containment air particulate monitor to an increase in reactor coolant leak rate is dependent upon the magnitude of the normal baseline leakage into the Containment. Sensitivity is greatest where baseline leakage is lowest.

2. Radioactive Gas Monitor

The radioactive gas monitor indicates the presence of containment gaseous activity originating from fuel-cladding defects. It measures the gaseous beta radioactivity by continuously sampling the containment atmosphere. (See Section 11.5.2.6.2) The radioactive gas monitor is less sensitive to an increase in reactor coolant leak rate than the containment particulate monitor.

3. Containment Sump Flow Monitoring

After collection in containment sump 1, containment sump 2, or the reactor cavity sump, leakage is pumped via a common header to floor drain tank 1 or to the waste hold-up tank. In this common discharge header is a flow totalizer/indicator that measures flow and facilitates the recording of this total flow in the control room (see Figures 9.3-5 sh 1 and 2).

The sumps also have several other features as follows:

a. Any one of the 6 pumps in these 3 sumps starting causes the “ANY CONT SUMP PUMP RUN” audible/visual alarm to actuate in the control room.

b. Each of the 3 sumps has a separate high level audible/visual alarm in the control room.

c. Sumps 1 and 2 have an additional level switch arranged with a time delay and the appropriate logic so that an increase by 1 gpm or greater flow into the sump will actuate a “CONT SUMP FILL RATE INCREASE” audible/visual alarm in the control room.

The sump discharge line may be sampled from outside of the Containment to provide additional aid in identifying the leakage source.

4. Specific Humidity Monitors

Specific humidity monitors are sensitive to vapor originating from the reactor coolant, steam, feedwater, and auxiliary systems in the Containment. Therefore, these monitors provide a means of detecting unidentified leakage from both radioactive and non-radioactive sources. Humidity detection is accomplished either by measuring the condensate from the Containment air cooling coils or by monitoring the dewpoint temperature in the Containment. These methods are described as follows:

a. Condensate Flow Rate Measurement

Humidity detection is accomplished by measuring the condensate flow rate from the Containment cooling coils. The containment specific humidity increases proportionately with time and leakage until the dewpoint is reached at the

Containment recirculation unit cooling coils. If the specific humidity increases above this point, the heat removal needed to cool the air-steam mixture to its dewpoint temperature increases above this point.

Therefore, since the cooling coils are designed to remove heat at a constant rate, an increase in specific humidity results in increased condensate flow. The condensate measuring system consists of a vertical standpipe with an internal self-siphoning device which empties the condensate in the standpipe to the sump when the standpipe is nearly full. The condensate measuring system permits measurement of the condensate flow rate from each Containment recirculation unit by means of a derivative unit which measures the rate of change in the standpipe level. Should the leakage inside the Containment increase, the condensate flow also increases, thereby increasing the rate of change of the standpipe level. The rate of level change in the standpipe is continuously recorded on strip chart recorders in the Control Room. An alarm for high rate of level change is provided to warn Control Room personnel of an increase in the condensate flow rate. An alarm is also provided if condensate flow is greater than the amount of flow that the siphon can discharge to the sump. Through accurate measurements of condensate flow, a reliable estimate of the total leakage rate to the Containment can be made. The condensate flow measuring system is shown in Figure 9.3-5.

b. Containment Dewpoint Monitors

The Containment humidity sensing system consists of dewpoint sensors, signal conditioning units, cabling, indicators, and plant process computer inputs, all packaged in a system capable of the continuous, unattended, automatic operation for remote monitoring of the dewpoint of the Containment atmosphere. Dewpoint sensors are strategically located in five representative areas of the Containment and are capable of detecting and reading out a change of 1ºF in dewpoint. The signal conditioning units provide a linear output signal for transmission to a Control Room board-mounted analog indicator.

5. Containment Temperature Monitors

An increase in Containment temperature can indicate a leak of high temperature fluid from the RCPB or other high temperature systems.

A description of these monitors is provided in Section 7.5

6. Containment Pressure Monitors

An increase in Containment pressure can indicate a leak of high temperature fluid from the RCPB or other high temperature systems.

A description of these monitors is provided in Section 7.3.

7. Frequency and Duration of Operation of the Containment Sump Pumps.

Each pump is provided with a running time indicator which indicates in seconds the duration of pump operation. This indicator can be used to estimate gross leakage rates and can act as a backup to the discharge flow monitors.

8. Gross Leakage Indications

Gross leakage in the Containment can be indicated by:

a. Decrease in pressurizer level

b. Increase in the rate of supply of reactor coolant makeup water

c. Containment temperature monitors

d. Containment pressure monitors

e. Containment sump level high alarm

9. Liquid inventory

Reactor coolant volume can be indicative of system leakages. Net level changes in the pressurizer and volume control tank are functions of the system leakage because the Chemical Volume Control System is a closed loop system. Abnormal makeup requirements can be indicative of system leakage.

5.2.5.2.3 Intersystem Leakage Detection Method

Leakage of reactor coolant into secondary and auxiliary systems can occur as a result of equipment defects. As discussed previously, the principal leak path for primary coolant into other systems is through the steam generator tubes into the secondary side of the steam generator. An additional path is through faulty equipment (primarily RHR and SIS components) into the Component Cooling Water System (CCWS).

Leakage from the RCS into connected systems does not present a significant safety issue. During normal operation, the CVCS letdown line and charging line have fluid flow through them from and to the RCS, respectively; therefore, the question of RCS leakage detection is not applicable to these lines. In all other cases where the RCS and the ECCS, RHRS and CVCS are interconnected, the auxiliary systems are isolated from the RCS by at least seat leakage. Where appropriate, these isolation valves comply with the requirements of BTP EICSB 3, “Isolation of Low Pressure Systems from High Pressure Reactor Coolant Systems.”

Because of the isolation valves provided, any leakage into the connected systems would be at very low rates (cc’s/hr). For closed systems with design pressure equal to that of the RCS, the only effect of the leakage would be to pressurize a portion of the piping to RCS operating pressure. For closed systems with design pressures less than that of the RCS, the only effect would be to lift the relief valve provided. For open systems (surge volumes present), the only effect would be a slow increase in the surge volume level.

Leakage into the lower pressure connected systems is detected directly either by lifting of the relief valve or by any increase in the surge volume level. Other leakage detection methods discussed previously in Section 5.2.5.2.

Pressure indication is provided to monitor ECCS piping pressure during normal plant operations. Refer to Section 6.3.5.2.6 for more details.

5.2.5.3 Sensitivity and Response Time of Selected Leakage-Detection Systems

1. Containment Air Particulate Monitor

The Containment air particulate monitor is the most sensitive instrument available for detection of reactor coolant leakage into the Containment. This instrument is capable of detecting particulate activity in concentrations as low as 5×10^{-11} microcuries per cubic centimeter ($\mu Ci/cm^3$) in the Containment air sampled.

Using this concentration, calculations show that the particulate monitor, for a reference nuclide of Cs-137, can conservatively detect a 1.0-gpm increase in unidentified leakage within the Containment in less than one hour after the leak begins. The sensitivity of the containment airborne particulate and gas monitors for detection of 1 gpm primary coolant leakage is dependent of both the primary coolant activity level and the background radiation level in containment which vary with Reactor Power.

Conservative analysis indicates that the maximum pre-existing containment background levels that will not prevent reliable leak detection by the particulate monitor (without spurious alarms) will vary with changes in the primary coolant activity. The relation between failed fuel fraction, primary coolant activity and background levels for any specified condition may be derived from the general sensitivity equation in ANSI 13.10 and the time constants specified for the monitor used in this service. Operating experience has shown the particulate background radiation levels have remained low relative to the expected activity increase from 1gpm leak. Therefore, this monitor is expected to detect a 1gpm leak within one hour.

2. Containment Radioactive Gas Monitor

This system is less sensitive than the Containment air particulate monitor but gives a positive indication of leakage in the event that reactor coolant gaseous activity exists as a result of fuel-cladding defects. A one gallon per minute leakage from the primary coolant pressure boundary in the containment can be detected in less than one hour provided the containment atmosphere activity is below the level that would mask the activity corresponding to the leakage. If this system is one of the two required Leakage Detection System inoperable in MODES 1, 2, 3 and 4 operations may continue up to 30 days provided grab samples of the containment atmosphere are obtained and analyzed at least once per 24 hours.

Analysis also shows that the maximum pre-existing containment radioactive gaseous background levels for which reliable detection is possible will vary directly with the activity levels in the primary coolant. With primary coolant concentrations less than equilibrium levels, such as during startup, the increase in detector count rate due to leakage will be

partially masked by the statistical variation of the minimum detector background count rate, rendering reliable detection of a 1 gpm leak uncertain.

Operating experience has shown elevated gaseous background radiation levels will partially mask the detection of a 1 gpm leak. However, the monitor is capable of qualitatively detecting an RCS to containment atmosphere leak.

In conclusion, reliable leak detection is possible, provided that the equilibrium activity of the containment atmosphere is below the level that would mask the change in activity corresponding to a 1 gpm leak in one hour. Given the above limitations, the intent of the leak detection requirements of Regulatory Guide 1.45 is met in the following manner. The monitors are seismologically qualified as required in Section C of Regulatory Guide 1.45. The minimum sensitivities of the containment air particulate and the radioactive gas monitors are 5×10^{-11} μCi/ml (reference nuclide Cs-137) and 1×10^{-6} μCi/ml (reference nuclide Xe-133), respectively. These are the minimum detectable activities when situated in a 2.5 mR/hr, of 1 Mev Gamma background field, which is the normal maximum anticipated at the location of the monitors. These sensitivities meet or exceed the sensitivities required of these monitors by Section B of Regulatory Guide 1.45.

3. Condensate Measuring System

The measurement of the condensate flow from containment recirculation unit cooling coils gives a sensitive indication of increases in unidentified leakage into the Containment. Condensate flow from approximately 0-gpm to 4.0-gpm can be measured with this system. In the event of very low reactor coolant activity levels, this system provides the most sensitive indication of unidentified leakage.

4. Dewpoint Temperature Monitors

These instruments are sensitive to an increase in dew point of 1°F or greater.

5. Containment Sump Fill Rate Increased By ≥ 1 GPM Alarm

All open floor areas and the equipment room floors are sloped toward floor drains with the drain piping routed to containment sumps 1 or 2. By design, flow to the sumps is unimpeded. This ensures that all liquid leakage will be routed to the sumps.

The containment sumps are equipped with level switches and an alarm timer. A sump inleak increase of ≥ 1 gpm will activate the sump leak detection system by initiating the alarm timer which is set at an interval which allows detection of ≥ 1 gpm leak increase in less than one hour.

To ensure continued reliability of the leakage-detection system, the equipment described in this section complies with Paragraph 4.10 of IEEE 279-1971.

5.2.5.4 Seismic Performance

The airborne particulate and gaseous radioactivity monitoring systems are designed to remain functional when subjected to the SSE. In addition, the Containment temperature and pressure

monitors, which serve as backup, gross leakage indicators, are also part of the post accident monitoring system and are qualified for the SSE.

The Containment sump flow monitor is capable of performing its function following seismic events not requiring plant shutdown.

The remaining systems described herein are considered to perform alarm and backup indication functions only and are not seismically qualified.

5.2.5.5 Leakage Information Available In The Control Room

1. Identified Leakage

Information about the identified leakage described in Subsection 5.2.5.2.1 is provided in the Control Room as follows:

a. Reactor Coolant Drain Tank process status (level, pressure, flow) are available at the liquid waste processing panel. A common trouble alarm is sent to the Control Room.

b. Radiation levels exceeding preset levels in the steam generator blowdown and the condenser off-gas systems are alarmed.

c. The temperature of the reactor head flange leak-off connection is indicated and alarmed.

d. Temperatures of the pressurizer safety and relief valve discharge piping is indicated and alarmed.

e. Component cooling water surge tank level is indicated, recorded and alarmed. Component cooling water radiation monitors are provided with readouts.

f. Primary-to-secondary leak rate, as detected my N-16 levels in the Main Steam Lines, exceeding preset levels is alarmed.

2. Unidentified Leakage

The primary monitors described in Subsection 5.2.5.2.2 are all indicated, recorded and alarmed.

The 1-gpm sump fill rate increase alarm is not recorded; sump discharge flow rate is recorded and totalized, but not alarmed.

5.2.5.6 Sensitivity and Operability Tests

1. Radiation Monitors

The tests proposed to demonstrate operability and relative accuracy of the air particulate and radioactive gas monitors are discussed in Section 11.5 and 12.3.4.

2. Containment Sump Flow Monitoring

   The testing proposed to demonstrate operability of the Containment Sump Flow Monitoring System is discussed in Section 9.3.3.4.

3. Containment Sump Fill Rate Increase Of ≥ 1 GPM Alarm

   During refueling shutdown, water is introduced to each sump to confirm switch settings and alarm annunciation.

5.2.5.7 Differentiation Between Identified and Unidentified Leaks

Any increases above the background level of 1.0-gpm of unidentified leakage and 10-gpm identified leakage (other than primary-to-secondary leakage) are investigated and evaluated by the operator in order to locate the sources of leakage.

Examples of techniques which will be employed by the operator in locating the area of leakage are provided below:

1. Leakage occurring from the reactor vessel head to vessel closure joint is identifiable by an increase in temperature in the leak-off line provided at this joint.

2. Leakage occurring from the main steam supply system, feedwater system, or CCWS is identifiable by an increase in condensate monitor indication without associated increase in background radioactivity. The increased frequency of sump pump operation is also an indication as is the 1-gpm leak increase alarm in the Control Room.

3. Leakage occurring from the RCPB is usually identifiable by a simultaneous increase in condensate and radioactivity monitor indications.

4. Dewpoint temperature plant process computer data can assist operators in locating leakage points because of the various locations of the dew cells in the Containment.

5. Steam generator primary-to-secondary leakage is detected by the Steam Generator Blowdown Process Sample (SGBPS) Monitor (see Section 11.5.2.7.1), the Steam Generator Leak Rate (SGLRM) Monitors (see Section 11.5.2.6.13), and the Condenser Off-Gas Monitor (see 11.5.2.6.5).

   The SGBPS, SGLRM's, and the condenser vacuum pump gas monitors meet the criteria for detection of primary-to-secondary leakage at the limits noted in Section 5.2.5.1.4, Item 1. This is conservatively based on the reactor coolant activities given in Table 11.1-4 and the dilution of the reactor coolant by the entire mass of coolant in the secondary system. The analysis is also based on no condensate cleanup, steam generator moisture carryover, partition of iodine in the steam generator or condenser, or correction for detector efficiency, as well as zero blowdown flow.

   It should be noted that blowdown flowrate, moisture carryover, and condensate cleanup do not effect the condenser vacuum pump gas monitor which is calibrated to specifically detect Xe-133, generally the predominate noble gas in the system. However, when argon

is injected, as described in Section 11.5.2.6.5, Ar-41 will be the predominate gas in the system.

It should also be noted that the steam generator leak rate monitors are on-line monitors located on the Main Steam lines and are calibrated to specifically detect N-16. Therefore, blowdown flowrate, moisture carryover, and condensate cleanup do not affect the steam generator leak rate monitors.

6. Leakage of the reactor coolant to the CCWS is detectable by means of the radiation monitor (see Sections 9.2.2 and 11.5) in the CCWS.

7. Leakage of the reactor coolant outside Containment is detectable by plant vent gas monitors (see Section 11.5) and the airborne radioactivity monitors of the Safeguards Building (see Section 12.3.4).

8. An alternate relief path (Unit 2 only) is available to measure intersystem leakage as described in Section 5.2.5.2.1.

5.2.5.8 Adequacy of the Leakage Detection System

A normal level of 1-gpm or less in unidentified leakage is expected. The leakage-detection systems are capable of detecting leakage as low as 0.1-gpm using the air particulate monitor and as low as 1-gpm using the condensate flow rate and the sump level alarm. The sensitivity is reasonably adequate to detect an increase in unidentified leakage rate. In addition, the capacity of the reactor coolant makeup system and containment water removal are well above the proposed leakage limits provided in the Technical Specifications.

1. Identified Leakage

Identified leakage collected in the Reactor Coolant Drain Tank is measured indirectly. The operator calculates the leakage in gpm based on the level change of the tank or based on total quantity (gallons) of fluid discharged and the known time interval between readings.

Identified leakage flow rate to the steam generators is measured by means of radiation monitors on the Main Steam Lines (SGLRM's), Steam Generator Blowdown System and Condenser Off-Gas System.

2. Unidentified Leakage

The condensate flow from the containment cooling condensate measuring system has direct reading in gpm.

Gross flow from the containment sumps is totalized, and the total flow in gallons and the time interval between sump evacuations is available to the operator. The operator also has the running time for each pump (80-gpm pumps). A rate of level change increase in the sumps of 1 gpm or greater leakage is alarmed to the operators as "The Containment Sump Fill Rate Increase" and requires no readout conversion.

REFERENCES

1. Letter NS-CE-1228, dated October 4, 1976, C. Eicheldinger (Westinghouse) to J. F. Stolz (NRC).

2. Cooper, L., Miselis, V. and Starek, R. M., “Overpressure Protection for Westinghouse Pressurized Water Reactors,” WCAP- 7769, Revision 1, June 1972 (also letter NS-CE-622, dated April 16, 1975, C. Eicheldinger (Westinghouse) to D. B. Vassallo (NRC), additional information on WCAP-7769, Revision 1).

3. Burnett, T. W. T., et al., “LOFTRAN Code Description,” WCAP-7907, June 1972.

4. Golik, M. A., “Sensitized Stainless Steel in Westinghouse PWR Nuclear Steam Supply Systems,” WCAP-7477-L (Proprietary), March 1970 and WCAP-7735 (Non-Proprietary), August 1971.

5. Enrietto, J. F., “Control of Delta Ferrite in Austenitic Stainless Steel Weldments,” WCAP-8324-A, June 1975.

6. Enrietto, J. F., “Delta Ferrite in Production Austenitic Stainless Steel Weldments,” WCAP-8693, January 1976.

7. W. A. Logsdon, J. A. Begley, C. L. Gottshall, “Dynamic Fracture Toughness of ASME SA-508 Class 2a and ASME SA-533 Grade A Class 2 Base and Heat affected Zone Material and Applicable Weld Metals,” WCAP-9292, March 1978.

8. NRC Letter dated April 14, 2003, from Robert A. Gramm to C. L. Terry, “Comanche Peak Nuclear Power Plant, Units 1 and 2 – Re: Relief from the Requirements of the American Society of Mechanical Engineers Boiler and Pressure Vessel Code, Section III, Concerning Pressurizer Upper Level Instrumentation and Other Lines and Associated Components (TAC No.s MB6427 and MB6428).

TABLE 5.2-1
APPLICABLE CODE ADDENDA FOR REACTOR COOLANT SYSTEM COMPONENTS

Reactor vessel	ASME III, 1971 Ed. thru Winter 72
Reactor vessel closure head	ASME III, 1989 Ed. (Unit 1) ASME III, 1971 Ed. thru Summer 72 (Unit 2)
Steam generator	ASME III, 1989 Ed. (Unit 1) ASME III, 1971 Ed. thru Summer 73 (Unit 2)
Pressurizer	ASME III, 1974 Ed.
CRDM housing	ASME III, 1989 Ed. (Unit 1)
Full length	ASME III, 1974 Ed. (Unit 2)
CRDM head adapter	ASME III, 1989 Ed. (Unit 1) ASME III, 1971 Ed. thru Winter 72 (Unit 2)
Reactor coolant pump	ASME III, 1971 Ed. thru Summer 73
Reactor coolant pipe	ASME III, 1974 Ed. thru Summer 75
Surge lines	ASME III, 1974 Ed. thru Winter 75
Valves	
Pressurizer safety	ASME III, 1971 Ed. thru Winter 72
Motor operated	ASME III, 1974 Ed. thru Summer 74
Manual (3" and larger)	ASME III, 1971 Ed. thru Summer 73
Control	ASME III, 1974 Ed. thru Summer 75

TABLE 5.2-2
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1
PRIMARY COMPONENTS

(Sheet 1 of 4)

Reactor Vessel Components	
Shell plates (other than core region)	SA-533 Gr. A, B or C, Class 1 or 2 (vacuum treated) (Unit 1)
Shell and head plates (other than core region)	SA-533, Gr. A, B or C, Class 1 or 2 (vacuum treated) (Unit 2)
Shell plates (core region)	SA-533, Gr. A or B, Class 1 (vacuum treated)
Replacement reactor vessel closure head	SA-508, Class 3 (Unit 1)
Shell, flange and nozzle forgings, nozzle safe ends	SA-508, Class 2 or 3, SA-182, Type F304 or F316
CRDM and/or ECCS appurtenances, upper head	SB-166 or 167 and SA-182, Type F304
Instrumentation tube appurtenances, lower head	SB-166 or 167 and SA-182, Type F304, F304L or F316
Closure studs, nuts, washers	SA-540, Class 3, Gr. B24
Core support pads	SB-166 with carbon less than 0.10%
Monitor tubes and vent pipe	SA-312 or 376, Type 304 or 316 or SB-167
Vessel supports, seal ledge and heat lifting lugs	SA-516, Gr. 70 quenched and tempered or SA-533, Gr. A, B C, Class 1 or 2 (vessel supports may be of weld metal build up of equivalent strength)
Cladding and buttering	Stainless steel weld metal analysis A-7 and Ni-Cr-Fe weld metal F-number 43

TABLE 5.2-2
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1
PRIMARY COMPONENTS

(Sheet 2 of 4)

Steam Generator Components	
Pressure plates	SA-533, Gr. A, B or C, Class 1 or 2 (Unit 2)
Pressure forgings (including nozzles and tubesheet)	SA-508, Class 3a (Unit 1) SA-508, Class 2 or 3 (Unit 2)
Primary Nozzle safe ends	SA-336, Class 316LN (Forging) (Unit 1) Stainless steel weld metal (Unit 2)
Channel heads	SA-508, Class 3a (Unit 1) SA-533, Gr. A, B or C, Class 1 or 2 or SA-216, Gr. WCC (Unit 2)
Tubes	SB-163, Ni-Cr-Fe annealed Alloy 690 thermally treated (Unit 1) Alloy 600 thermally treated (Unit 2)
Cladding and buttering	Stainless steel weld metal analysis A-8 (Unit 1), A-7 (Unit 2) and Ni-Cr-Fe weld metal F-Number 43
Closure hardware	
Studs	SA-193, Gr. B7
Nuts	SA-194, Gr.7
Washers	SA-194, Gr. 7 (Unit 1) ASTM F-436 (Unit 2)
Pressurizer Components	
Pressure plates	SA-533, Gr. A, B or C, Class 1 or 2

TABLE 5.2-2
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1
PRIMARY COMPONENTS

(Sheet 3 of 4)

Pressure forgings	SA-508, Class 2 or 3
Nozzle safe ends	SA-182 or 376, Type 316 or 316L and Ni-Cr-Fe weld metal F- Number 43
Cladding and buttering	Stainless steel weld metal analysis A-7 and Ni-Cr-Fe weld metal F-Number 43
Closure hardware	
Studs	SA-193, Gr. B7
Nuts	SA-194, Gr. 7
Washers	ASTM F-436
Pressurizer safety valve forgings	SA-182, Type F316
Reactor Coolant Pump	
Pressure forgings	SA-182, Type F304, F316, F347 or F348
Pressure casting	SA-351, Gr. CF8, CF8A or CF8M
Tube and pipe	SA-213, 376 or 312, seamless Type 304 or 316
Pressure plates	SA-240, Type 304 or 316
Bar material	SA-479, Type 304 or 316
Closure bolting	SA-193, 320, 540 or 453, Gr. 660
Flywheel	SA-533, Gr. B, Class 1

TABLE 5.2-2
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1
PRIMARY COMPONENTS

(Sheet 4 of 4)

Reactor Coolant Piping	
Reactor coolant pipe	SA-376, Gr. 304N or SA-351, Gr. CF8A centrifugal casting.
Reactor coolant fittings, branch nozzles	SA-351, Gr. CF8A and SA-182, Code Case 1423-2, Gr. 304N
Surge line	SA-376, Gr. 304, 316 or F304N
Auxiliary piping 1/2 through 12 inch and wall schedules 40S through 80S (ahead of second isolation valve)	ANSI B36.19
All other auxiliary piping (ahead of second isolation valve)	ANSI B36.10
Socket weld fittings	ANSI B16.11
Piping flanges	ANSI B16.5
Full Length CRDM	
Pressure housing	SA-336 Type 304LN (Unit 1) SA-182, Gr. F304 or SA-351, Gr. CF8 (Unit 2)
Pressure forgings	SA-182, Gr. F304 or SA-336, Gr. F8
Bar material	SA-479, Type 304
Welding materials	SFA 5.4 and 5.9, Type 308 or 308L

TABLE 5.2-3
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1 AND 2
AUXILIARY COMPONENTS

(Sheet 1 of 2)

Valves	
Bodys	SA-182, Type F316 or SA-351, Gr. CF8 or CF8m
Bonnets	SA-182, Type F316 or SA-351,Gr. CF8 or CF8M or SA-479 T316
Discs	SA-182, Type F316 or SA-564, Gr. 630, cond. 1100°F heat treatment A-567 Grade 1 (Stellite 21)
Stems	SA-182, Type F316 or SA-564, Gr. 630, cond. 1100°F heat treatment (Note 1)
Pressure retaining bolting	SA-453, Gr. 660
Pressure retaining nuts	SA-453, Gr. 660 or SA-194, Gr. 6
Auxiliary Heat Exchangers	
Heads	SA-240, Type 304
Nozzle necks	SA-182, Gr. F304
Tubes	SA-213, Type 304
Tubesheets	SA-182, Gr. F304
Shells	SA-240 and 312, Type 304
Auxiliary Pressure Vessels, Tanks, Filters, etc.	
Shells and heads	SA-240, Type 304 or SA-264 consisting of SA-537, Gr. C11 with stainless steel weld metal analysis A-8 cladding
Flanges and nozzles	SA-182, Gr. F304 and SA-105 or 350, Gr. LF2 and LF3 with stainless steel weld metal analysis A-8 cladding
Piping	SA-312 and 240, Type 304 or 316 seamless
Pipe fittings	SA-403, Type 304 seamless
Closure bolting and nuts	SA-193, Gr. B7 and SA-194, Gr. 2H

TABLE 5.2-3
REACTOR COOLANT PRESSURE BOUNDARY MATERIALS CLASS 1 AND 2
AUXILIARY COMPONENTS

(Sheet 2 of 2)

Auxiliary Pumps	
Pump casing and heads	SA-351, Gr. CF8 or CF8M and SA-182, Gr. F304 or F316
Flanges and nozzles	SA-182, Gr. F304 or F316 and SA-403, Gr. WP316L seamless
Piping	SA-312, Type 304 or 316 seamless
Stuffing or packing box cover	SA-351, Gr. CF8 or CF8M and SA-240, Type 304 or 316
Pipe fittings	SA-403, Gr. WP316L seamless
Closure bolting and nuts	SA-193, Gr. B6, B7 or B8M and SA-194, Gr. 2H or 8M, SA-193, Gr. B6, B7 or B8M, SA-453, Gr. 660, and nuts, SA-194, Gr. 2H, 8M and 6

Note 1: Stems may also be A-582, Type T416 or other suitable materials identified in the Code Data Reports.

TABLE 5.2-4
REACTOR VESSELS INTERNALS, INCLUDING EMERGENCY CORE COOLING SYSTEMS

Forgings	SA-182, Type F304
Plates	SA-240, Type 304
Pipes	SA-312, Type 304 seamless or SA-376, Type 304
Tubes	SA-213, Type 304
Bars	SA-479, Type 304 and 410
Castings	SA-351, Gr. CF8 or CF8A
Bolting	SA-193, Gr. B8M (65 MYS/90MTS) Code Case 1618 inconel-750 SA-461, Gr. 688
Nuts	SA-193, Gr. B8
Locking devices	SA-479, Type 304
Weld buttering	Stainless steel weld metal analysis A-7

TABLE 5.2-5
REACTOR COOLANT WATER CHEMISTRY SPECIFICATIONS

Oxygen, maximum (ppm)	Oxygen concentration of the reactor coolant is maintained at or below 0.10 ppm for plant operation above 250°F.
Chloride, maximum (ppm)	0.15
Fluoride, maximum (ppm)	0.15
Hydrogen (cc(STP)/Kg H_20)	25 to 50
Reactor power level above 1 MWt, excluding decay heat during subcritical operation	
pH control agent (Li^7OH)	Maintain Lithium in relationship with Boron within the required values to support a minimum hot full power operating pH_t range of $\geq$ 6.9 to a maximum target pH_t of 7.4 with a maximum lithium concentration of 6.0 ppm.
Boric acid (ppm B)	Variable from 0 to approximately 2500

TABLE 5.2-6A (Unit 1 - Historical)
PARAMETERS FROM WCAP-7769 WHICH ARE DIFFERENT FOR
COMANCHE PEAK

Engineered safeguards power (Mwt)	3579
Total steam generator safety valve capacity (1bm/hr)	18,191,000
Coolant flow (RCS) (gpm)	377,600
Average core mass velocity (10^6 1bm/hr-ft^2)	2.62
Inlet temperature (°F)	558
Average power density (kw/1)	124.5
Fuel loading (kg/1)	2.7
Full power steam flow per loop (1bm/sec)	1051
Nominal shell-side steam generator water mass per loop (1bm)	104,000
Nominal NSSS power (Mwt)[(a)]	3474

a) Unit 1 was 3427 MWt until completion of 1RF09.

TABLE 5.2-6B (Unit 2)
PARAMETERS FROM WCAP-7769 WHICH ARE DIFFERENT FOR
COMANCHE PEAK

Engineered safeguards power (Mwt)	3579
Total steam generator safety valve capacity (1bm/hr)	18,191,000
Coolant flow (RCS) (gpm)	377,600
Average core mass velocity (10^6 1bm/hr-ft^2)	2.62
Inlet temperature (°F)	558
Average power density (kw/1)	124.5
Fuel loading (kg/1)	2.7
Full power steam flow per loop (1bm/sec)	1051
Nominal shell-side steam generator water mass per loop (1bm)	104,000
Nominal NSSS power (Mwt)	3474

TABLE 5.2-7
CODE CASES USED ON SAFETY CLASS 1 COMPONENTS WITHIN THE REACTOR COOLANT PRESSURE BOUNDARY

(Sheet 1 of 2)

Steam Generator:

Unit 1

N-20-4	-	SB-163 Nickel-Chromium-Iron Tubing (Alloy 600 and 690) and Nickel-Iron Chromium Alloy 800 at a Specified Minimum Yield Strength of 40.0 Ksi and Cold Worked Alloy 800 at Yield Strength of 47.0 Ksi, Section III, Division 2, Class 2
2142-1	-	F Number grouping for Ni-Cr-Fe Classification UNS N06052 Filler Metal
2143-1	-	F Number grouping for Ni-Cr-Fe Classification UNS W86152 Weld Electrode

Unit 2

1528	-	High Strength Steel SA-508, Class 2 and SA-541, Class 2 Forgings, Section III, Class 1 Components
1484	-	SB-163 Nickel-Chromium-Iron Tubing (Alloy 600 and 690) and Nickel-Iron-Chromium Alloy 800 at a Specified Minimum Yield Strength of 40.0 Ksi Section III, Division 2, Class 2
1493-1	-	Post-Weld Heat Treatment Section I, III and VIII, Division 1 and 2
1355	-	Electroslag Welding, Section I, III and VIII, Division 1 and 2

Pressurizer:

1528	-	Same As Above

Piping:

1423-2	-	Wrought Type 304 and 316 with Nitrogen Added, Sections I, III, VIII, Division 1 and 2
N-122	-	January 21, 1985 “Stress Indices for Integral Structural Attachments, Class 1”
N-319	-	July 13, 1984 “Alternate Procedure for Evaluation of Stresses in Butt Welding Elbows in Class 1 Piping, Section III, Division 1”

TABLE 5.2-7
CODE CASES USED ON SAFETY CLASS 1 COMPONENTS WITHIN THE REACTOR COOLANT PRESSURE BOUNDARY

(Sheet 2 of 2)

N-391	-	November 28, 1983 “Procedure for Evaluation of the Design of Hollow Circular Cross Section Welded Attachments on Class 1 Piping”
N-397	-	February 20, 1984 “Alternate Rules to the Spectral Broadening Procedures of N-1226.3 for Class 1, 2 and 3 Piping”
N-411	-	September 17, 1984 “Alternate Damping Values for Seismic Analysis for Class 1, 2 and 3 Piping Systems”

Valves:

1712		Nameplates and Stamping for Section III, Division 1, Class 1, 2, 3 and MC (N-101)
1539		Metal Bellows and Metal Diaphragm Stem Sealed Valves, Section III, Classes 1, 2, and 3 (N-30-1)

Flux Thimble Tubing:

1612	-	Use of Type 308 Stainless Steel Rod and Bar for Section III, Class 1, 2, 3 and CS construction
1649	-	Modified SA 453-GR 660 for Class 1, 2, 3 and CS construction
1553-1	-	Upset Heading and Roll Threading of SA-453 for High Temperature Bolting, Section III, Classes I, 2, 3 and MC

Reactor Vessel Bottom-Mounted Instrumentation:

N-378	-	Seal Table Material

5.3 REACTOR VESSEL

5.3.1 REACTOR VESSEL MATERIALS

5.3.1.1 Material Specifications

Material specifications are in accordance with the American Society of Mechanical Engineers (ASME) Code requirements and are given in Section 5.2.3.

5.3.1.2 Special Processes Used for Manufacturing and Fabrication

1. The vessel is Safety Class 1. Design and fabrication of the reactor vessel is carried out in strict accordance with ASME Code, Section III, Class 1 requirements. The head flanges and nozzles are manufactured as forgings. The cylindrical portion of the vessel is made up of several shells, each consisting of formed plates joined by full penetration longitudinal weld seams. The hemispherical heads are made from dished plates. The reactor vessel parts are joined by welding, using the single or multiple wire submerged arc.

2. The use of severely sensitized stainless steel as a pressure boundary material has been prohibited and has been eliminated by either a select choice of material or by programming the method of assembly.

3. Unit 1

   The surfaces of the guide studs are chrome plated to prevent possible galling of the mated parts.

   Unit 2

   The control rod drive mechanism head adaptor threads and surfaces of the guide studs are chrome plated to prevent possible galling of the mated parts.

4. At all locations in the reactor vessel where stainless steel and Inconel are joined, the final joining beads are Inconel weld metal in order to prevent cracking.

5. Core region shells fabricated of plate material have longitudinal welds which are angularly located away from the peak neutron exposure experienced in the vessel, where possible.

6. Unit 1

   The location of full penetration weld seams in the vessel bottom head are restricted to areas that permit accessibility during inservice inspection. There are no full penetration welds in the upper closure head.

   Unit 2

   The location of full penetration weld seams in the upper closure head and vessel bottom head are restricted to areas that permit accessibility during inservice inspection.

7. The stainless steel clad surfaces are sampled to assure that composition and delta ferrite requirements are met.

8. The procedure qualification for cladding low alloy steel (SA-508, Class 2) requires a special evaluation to assure freedom from underclad cracking.

9. Minimum preheat requirements have been established for pressure boundary welds using low alloy material. The preheat must be maintained either until (at least) an intermediate post weld heat treatment is completed or until the completion of welding. In the latter case, upon completion of welding, a low temperature (400°F minimum) post weld heat treatment is applied for 4 hours, followed by allowing the weldment to cool to ambient temperature. For primary nozzle to shell welds, preheat must be maintained until an intermediate or full post weld heat treatment is completed.

5.3.1.3 Special Methods for Nondestructive Examination

The nondestructive examination of the reactor vessel and its appurtenances is constructed in accordance with the ASME Code, Section III requirements; also numerous examinations are performed in addition to ASME Code, Section III requirements. Nondestructive examination of the vessel is discussed in the following sections and the reactor vessel quality assurance program is given in Table 5.3-1.

5.3.1.3.1 Ultrasonic Examination

1. In addition to the design code straight beam ultrasonic test, angle beam inspection of 100 percent of plate material is performed during fabrication to detect discontinuities that may be undetected by longitudinal wave examination.

2. In addition to the ASME Code, Section III, nondestructive examination, all full penetration ferritic pressure boundary welds and heat affected zones in the reactor vessel are ultrasonically examined during fabrication. This test is performed upon completion of the welding and intermediate heat treatment but prior to the final post weld heat treatment. See Section 5.3.3.7 for additional discussion of this examination.

3. In addition to ASME Code, Section III, nondestructive examination, all full penetration ferritic pressure boundary welds in the reactor vessel are ultrasonically inspected after hydrostatic testing to establish additional assurance that the vessel will pass the ASME Code, Section XI, preservice inspection. See Section 5.3.3.7 for additional discussion of this shop examination.

5.3.1.3.2 Penetrant Examinations

The partial penetration weld for the control rod drive mechanism head adaptors and the bottom instrumentation tubes are inspected by dye penetrant after the root pass in addition to code requirements. Core support block attachment welds are inspected by dye penetrant after the first layer of weld metal and after each 1/2 inch of weld metal. All clad surfaces and other vessel and head internal surfaces are inspected by dye penetrant after the hydrostatic test.

5.3.1.3.3 Magnetic Particle Examination

The magnetic particle examination requirements below are in addition to the magnetic particle examination requirements of Section III of the ASME Code. All magnetic particle examinations of materials and welds shall be performed in accordance with the following:

1. Prior to the final post weld heat treatment - Only by the Prod, Coil or Direct Contact Method.

2. After the final post weld heat treatment - Only by the Yoke Method.

The following surfaces and welds shall be examined by magnetic particle methods. The acceptance standards shall be in accordance with Section III of the ASME Code.

5.3.1.3.3.1 Surface Examinations

1. Magnetic particle examine all exterior vessel and heat surfaces after the hydrostatic test.

2. Magnetic particle examine all exterior closure stud surfaces and all nut surfaces after final machining or rolling. Continuous circular and longitudinal magnetization shall be used.

3. Magnetic particle examine all inside diameter surfaces of carbon and low alloy steel products that have their properties enhanced by accelerated cooling. This inspection to be performed after forming and machining (if performed) and prior to cladding.

5.3.1.3.3.2 Weld Examination

Magnetic particle examination of the weld metal buildup for vessel support welds attaching the closure head lifting lugs and refueling seal ledge to the reactor vessel after the first layer and each 1/2 inch of weld metal is deposited. All pressure boundary welds shall be examined after back chipping or back grinding operations.

5.3.1.4 Special Controls for Ferritic and Austenitic Stainless Steels

Welding of ferritic steels and austenitic stainless steels is discussed in Section 5.2.3. Section 5.2.3 includes discussions which indicate the degree of conformance with Regulatory Guides 1.31, “Control of Stainless Steel Welding,” and 1.44, “Control of the Use of Sensitized Stainless Steel.” Appendix 1A(N) discusses the degree of conformance with Regulatory Guides 1.34, “Control of Electroslag Weld Properties,” 1.43, “Control of Stainless Steel Weld Cladding of Low-Alloy Steel Components,” 1.50, “Control of Preheat Temperature for Welding of Low-Alloy Steels,” 1.71, “Welder Qualification for Areas of Limited Accessibility,” and 1.99, “Effects of Residual Elements on Predicted Radiation Damage to Reactor Vessel Materials.”

5.3.1.5 Fracture Toughness

Assurance of adequate fracture toughness of ferritic materials in the reactor coolant pressure boundary (ASME Code, Section III, Class 1 components) is provided by compliance with the requirements for fracture toughness testing included in NB-2300 to Section III of the ASME Code and Appendix G of 10CFR50.

The fasteners for the steam generators, pressurizers, reactor coolant pumps, and valves designed and fabricated to the ASME Code Summer 1973 Addendum or later (as specified in Table 5.2-1) comply with the requirements of 10CFR50 Appendix G as shown below. The only primary pressure boundary fasteners built to an earlier Code are the pressurizer safety valve fasteners, which were built to the ASME Code 1971 Edition, Winter 1972 Addendum.

		Fastener		Charpy Tests
Code/Regulation	Diameter	ft. lbs.	MLE	Temperature
10CFR50 Appendix G	≤ 1" > 1"	No test required. 45	 25	Lower of preload or lowest service temperature.
ASME Code. 1971 Ed. Summer 1973 Addendum Later	≤ 1" > 1"	No test required. 45	 25	Lower of preload or lowest service temperature
ASME Code, 1971 Ed. Winter 1972 Addendum	≤ 1" > 1"	No test required. No req't	 25	40°F or lower

Since the fasteners for the pressurizer safety valves are less than 1 inch in diameter and since the preload and lowest service temperatures are expected to be greater than 40°F, compliance with the ASME Code ensures compliance with 10CFR50 Appendix G requirements. Fracture toughness properties of the reactor vessel fasteners are shown in Tables 5.3-3A, 5.3-3B, 5.3-4A, 5.3-4B and 5.3-4C.

The initial Charpy V-notch minimum upper shelf fracture energy levels for the reactor vessel beltline (including welds) shall be 75 foot pounds as required per Appendix G of 10CFR50. Materials having a section thickness greater than 10 inches with an upper shelf of less than 75 foot pounds shall be evaluated with regard to effects of chemistry (especially copper content), initial upper shelf energy and fluence to assure that a 50 foot pound shelf energy as required by Appendix G of 10CFR50 is maintained throughout the life of the vessel. The specimens shall be oriented as required by NB-2300 of Section III of the ASME Code.

Fracture toughness data for the Unit 1 and Unit 2 vessels are presented in Table 5.3-2A and B, respectively. Unit 1 and 2 reactor vessel beltline region material composition is given in Tables 5.3-6A and 5.3-6B. Fracture toughness properties at the beltline region for Units 1 and 2 reactor vessels are given in Tables 5.3-7A and 5.3-7B. Reactor vessel beltline weld and base metal charpy plots are given in Figures 5.3-2A through 5.3-4A for Unit 1, and Figures 5.3-2B through 5.3-4B for Unit 2. The same information is presented in tabular form in Tables 5.3-8A through 5.3-14A for Unit 1 and Tables 5.3-8B through 5.3-14B for Unit 2.

The reactor vessel non-beltline weld metal toughness properties are given in Table 5.3-15A for Unit 1 and Table 5.3-15B for Unit 2.

The steam generator and pressurizer base metal fracture toughness data are given in Table 5.3-16A for Unit 1 and Table 5.3-16B for Unit 2. The steam generator and pressurizer weld metal fracture toughness data are provided in Table 5.3-17A for Unit 1 and Table 5.3-17B for Unit 2. Plate locations are shown in Figures 5.3-1A and 5.3-1B.

Analysis had demonstrated that the closure flange regions for Units 1 and 2 are less limiting than the beltline regions [3].

5.3.1.6 Material Surveillance

5.3.1.6.1 Reactor Vessel Material Surveillance Capsules

In the surveillance program, the evaluation of the radiation damage is based on pre-irradiation testing of Charpy V-notch and tensile specimens and post-irradiation testing of Charpy V-notch, tensile and 1/2 T (thickness) compact tension (CT) fracture mechanics test specimens. The program is directed toward evaluation of the effect of radiation on the fracture toughness of reactor vessel steels based on the transition temperature approach and the fracture mechanics approach. The program will conform with ASTM E-185, “Recommended Practice for Surveillance Tests for Nuclear Reactor Vessels,” and 10CFR50, Appendix H. The reports detailing the reactor vessel surveillance programs are listed in Reference section of the Pressure and Temperature Limits Report (PTLR) for Unit 1 and Unit 2 respectively.

The reactor vessel surveillance program uses six specimen capsules. The capsules are located in guide baskets welded to the outside of the neutron shield pads and are positioned directly opposite the center portion of the core. The capsules can be removed when the vessel head is removed and can be replaced when the internals are removed. The six capsules contain reactor vessel steel specimens, oriented both parallel and normal (longitudinal and transverse) to the principal rolling direction of the limiting base material located in the core region of the reactor vessel and associated weld metal and weld heat affected zone metal. The six capsules contain 54 tensile specimens, 360 Charpy V-notch specimens (which include weld metal and weld heat affected zone material), and 72 CT specimens. Archive material sufficient for two additional capsules will be retained.

Dosimeters, including nickel (Ni), copper (Cu), iron (Fe), cobalt- aluminum (Co-Al), cadmium (Cd) shielded Co-A1, Cd shielded neptunium-237 (Np-237), and Cd shielded uranium-238 (U-238), are placed in filler blocks drilled to contain them. The dosimeters permit evaluation of the flux seen by the specimens and the vessel wall. In addition, thermal monitors made of low melting point alloys are included to monitor the maximum temperature of the specimens. The specimens are enclosed in a tight fitting stainless steel sheath to prevent corrosion and ensure good thermal conductivity. The complete capsule is helium leak tested. As part of the surveillance program, a report of the residual elements in weight percent to the nearest 0.01 percent will be made for surveillance material and as deposited weld metal.

Each of the six capsules contains the following specimens:

Material	Number of Charpys	Number of Tensiles	Number of CT's
Limiting base material[a]	15	3	4
Limiting base material[b]	15	3	4
Weld metal[c]	15	3	4
Heat affected zone	15	-	-

a) Specimens oriented in the major rolling or working direction.

b) Specimens oriented normal to the major rolling or working direction.

c) Weld metal to be selected per ASTM-E-185.

The following dosimeters and thermal monitors are included in each of the six capsules:

1. Dosimeters
    a. Fe.
    b. Cu.
    c. Ni.
    d. Co-A1 (0.15 percent Co).
    e. Co-A1 (Cd shielded).
    f. U-238 (Cd shielded).
    g. Np-237 (Cd shielded).
2. Thermal monitors
    a. 97.5 percent palladium (Pd), 2.5 percent silver (Ag)(579°F melting point).
    b. 97.5 percent Pd, 1.75 percent Ag, 0.75 percent tin (Sn)(590°F melting point).

The fast neutron exposure of the specimens occurs at a faster rate than that experienced by the vessel wall, with the specimens being located between the core and the vessel. Since these specimens experience accelerated exposure and are actual samples from the materials used in the vessel, the transition temperature shift measurements are representative of the vessel at a

later time in life. Data from CT fracture toughness specimens are expected to provide additional information for use in determining allowable stresses for irradiated material.

Correlations between the calculations and the measurements of the irradiated samples in the capsules, assuming the same neutron spectrum at the samples and the vessel inner wall, are described in Section 5.3.1.6.1.

They have indicated good agreement. Evaluation of surveillance specimen data will include the effects of specimen caused perturbations in the fast neutron flux and energy distribution. The calculated wall exposure will be verified and adjusted as necessary using data from all capsules removed. The schedule for removal of the capsules for post-irradiation testing is found in Section 2.4 of PTLR for the respective units.

5.3.1.6.1.1 Measurement of Integrated Fast Neutron (E > 1.0 MeV) Flux at the Irradiation Samples

In order to effect a correlation between fast neutron (E > 1.0 MeV) exposure and the radiation induced properties changes observed in the test specimens, a number of fast neutron flux monitors are included as an integral part of the reactor vessel surveillance program. In particular the surveillance capsules contain detectors employing the following reactions.

Fe^{54} (n,P) Mn^{54}

Ni^{58} (n,P) Co^{58}

Cu^{63} (n,α) Co^{60}

Np^{237}(n,f) Cs^{137}

U^{238} (n,f) Cs^{137}

In addition, thermal neutron flux monitors, in the form of bare and Cd shielded Co-A1 wire, are included within the capsules to enable an assessment of the effects of isotopic burnup on the response of the fast neutron detectors.

The use of activation detectors such as those listed above does not yield a direct measure of the energy dependent neutron flux level at the point of interest. Rather, the activation process is a measure of the integrated effect that the time and energy dependent neutron flux has on the target material. An accurate estimate of the average neutron flux level incident on the various detectors may be derived from the activation measurements only if the parameters of the irradiation are well known. In particular, the following variables are of interest:

1. The operating history of the reactor.

2. The energy response of the given detector.

3. The neutron energy spectrum at the detector location.

The procedure for the derivation of the fast neutron flux from the results of the Fe^{54}(n,P) Mn^{54} reaction is described below. The measurement technique for the other dosimeters, which are sensitive to different portions of the neutron energy spectrum, is similar.

The Mn^{54} product of the Fe^{54}(n,P) Mn^{54} reaction has a half life of 314 days and emits gamma rays of 0.84 MeV energy which are easily detected using a NaI scintillator. In irradiated steel samples, chemical separation of the Mn^{54} may be performed to ensure freedom from interfering activities. This separation is simple and very effective, yielding sources of very pure Mn^{54} activity. In some samples, all of the interferences may be corrected for by the gamma spectrometric methods without any chemical separation.

The analysis of the sample requires that two procedures be completed. First, the Mn^{54} disintegration rate per unit mass of sample and the Fe content of the sample must be measured as described above. Second, the neutron energy spectrum at the detector location must be calculated.

For this analysis, the DOT [1], two-dimensional multigroup discrete ordinates transport code is employed to calculate the spectral data at the location of interest. Briefly, the DOT calculations utilize a 21 group energy scheme, an S_8 order of angular quadrature, and a P_1 expansion of the scattering matrix to compute neutron radiation levels within the geometry of interest. The reactor geometry employed here includes a description of the radial regions internal to the primary concrete (core barrel, neutron pad, pressure vessel and water annuli) as well as the surveillance capsule and an appropriate reactor core fuel loading pattern and power distribution. Thus, distortions in the fission spectrum due to the attenuation of the reactor internals are accounted for in the analytical approach.

Having the measured activity, sample weight, and neutron energy spectrum at the location of interest, the calculation of the threshold flux is as follows:

The induced Mn^{54} activity in the Fe flux monitors may be expressed as:

$$D = \frac{No}{A} f_i \int_E \sigma(E)\varphi(E) \sum_{J=1}^{n} F_j(1 - e^{-\lambda\tau}J)e^{-\lambda\tau}d$$

where

D = induced Mn^{54} activity (dps/gm_{Fe})

No = Avogadro's number (atom/gm-atom)

A = atomic weight of Fe (gm/gm-atom)

fi = weight fraction Fe^{54} in the detector

$\sigma(E)$ = energy dependent activation cross section for the Fe^{54} (n,P)Mn^{54} reaction (barns)

$\varphi(E)$ = energy dependent neutron flux at the detector at full reactor power (n/cm^2-sec)

λ = decay constant of Mn^{54} (1/sec)

F_j = fraction of full reactor power during the Jth time interval, τ_J

τ_j = length of the Jth irradiation period (sec)

τd = decay time following the Jth irradiation period (sec).

The parameters F_j, τ_J, and τ_d depend on the operating history of the reactor and the delay between capsule removal and saple counting.

The integral term in the above equation may be replaced by the following relation:

$$\int \sigma(E)\varphi(E) = \bar{\sigma}\bar{\varphi}_{E_{TH}} = \left\{ \frac{\sum_{0}^{\infty} \sigma_s(E)\varphi_s(E)}{\sum_{E_{TH}}^{\infty} \varphi_S \,/\, \varphi} \right\} \bar{\varphi}\, E_{TH}$$

where

$\bar{\sigma}$ = effective spectrum average reaction cross section for neutrons above energy, E_{TH}

$\bar{\varphi}E_{TH}$ = average neutron flux above energy, E_{TH}

$\sigma_s(E)$ = multigroup $Fe^{54}(n,P)Mn^{54}$ reaction cross sections compatible with the DOT energy group structure

$\varphi_s(E)$ = multigroup energy spectra at the detector location obtained from the DOT analysis.

Thus,

$$D = \frac{No}{A} f_i \, \bar{\sigma}\bar{\varphi} E_{TH} \sum_{J=1}^{n} F_J (1 - e^{-\lambda\tau} J) e^{-\lambda\tau} d$$

or, solving for the threshold flux:

$$\bar{\varphi}E_{TH} = \frac{D}{\frac{No}{A} f_i \bar{\sigma} \sum_{J=1}^{n} F_J(1 - e^{-\lambda\tau}J)e^{-\lambda\tau}d}$$

The total fluence above energy E_{TH} is given by:

$$\varphi_{E_{TH}} = \bar{\varphi}_{E_{TH}} \sum_{J=1}^{n} F_j \, \tau_J$$

where $\sum_{J=1}^{n} F_J \, \tau_J$ represents the total effective full power seconds of reactor operation up to the time of capsule removal.

Because of the relatively long half life of Mn^{54}, the fluence may be accurately calculated in this manner for irradiation periods up to about 2 years. Beyond this time, the calculated average flux begins to be weighted toward the later stages of irradiation and some inaccuracies may be introduced. At these longer irradiation times, therefore, more reliance must be placed on Np^{237} and U^{238} fission detectors with their 30 year life product (Cs^{137}).

No burnup correction was made to the measured activities, since burnout of the Mn^{54} product is not significant until the thermal flux level is about 10^{14} n/cm^2-sec.

The error involved in the measurement of the specific activity of the detector after irradiation is estimated to be ± 5 percent.

5.3.1.6.1.2 Calculation of Integrated Fast Neutron (E > 1.0 MeV) Flux at the Irradiation Samples

The energy and spatial distribution of neutron flux within the reactor geometry is obtained from the DOT [1] two-dimensional Sn transport code. The radial and azimuthal distributions are obtained from an R,θ computation wherein the reactor core as well as the water and steel annuli surrounding the core is modeled explicitly. The axial variations are then obtained from an R, Z DOT calculation using the equivalent cyclindrical core concept. The neutron flux at any point in the geometry is then given by:

$$\varphi\ (E, R, \theta, Z) = \varphi\ (E, R, \theta)F(Z)$$

Where φ (E, R, θ) is obtained directly from the R, θ calculation and F(Z) is a normalized function obtained from the R, Z analysis. The core power distributions used in both the R, θ and R,Z computations represent the expected average over the life of the station.

Having the calculated neutron flux distributions within the reactor geometry, the exposure of the capsule as well as the lead factor between the capsule and the vessel may be determined as follows:

The neutron flux at the surveillance capsule is given by:

$$\varphi_c = \varphi\ (E, R_c, \theta_c, Z_c)$$

and the flux at the location of peak exposure on the pressure vessel inner diameter is:

$$\varphi_{\ v-max} = \varphi\ (E, R_v, \theta_{\ v-max}, Z_{v-max})$$

The lead factor then becomes:

$$LF = \frac{\varphi_c}{\varphi_{\ v-max}}$$

Similar expressions may be developed for points within the pressure vessel wall, and, thus, together with the surveillance program dosimetry, serve to correlate the radiation induced damage to test specimens with that of the reactor vessel.

5.3.1.6.2 Ex-Vessel dosimetry

To achieve the goals of the Reactor Cavity Neutron Measurement Program, two types of measurements are made. Comprehensive sensor sets including radiometric monitors (RM) are employed at discrete locations within the reactor cavity to characterize the neutron energy spectrum variations axially and azimuthally over the beltline region of the reactor vessel. In addition, stainless steel gradient chains are used in conjunction with the encapsulated dosimeters to complete the mapping of the neutron environment between the discrete locations chosen for spectrum determinations.

5.3.1.6.2.1 Sensor Sets

The Reactor Cavity Neutron Measurement Program employs advanced sensor sets that are recommended by and are designed to the latest ASTM neutron dosimetry standards. The sensor sets consist of the following encapsulated dosimeters and gradient chains. The Table below lists the neutron reactions that are of interest.

1. Radiometric Monitors (RM) - these include cadmium shielded foils of the following metals: copper, titanium, iron, nickel, and cobalt-aluminum. Cadmium shielded fast fission reactions include ^{238}U and ^{237}Np in vanadium encapsulated oxide detectors. Bare iron and cobalt monitors are also included.

2. Gradient Chains - These stainless steel bead chains connect and support the dosimeter capsules containing the radiometric monitors. These segmented chains provide iron, nickel, and cobalt reactions that are used to complete the determination of the axial and azimuthal gradients. The high purity iron, nickel, and cobalt-aluminum foils contained in

the multiple foil sensor sets provide a direct correlation with the measured reaction rates from these gradient chains. These cross-comparisons permit the use of the gradient measurements to derive neutron flux distributions in the reactor cavity with a high level of confidence.

Table

Material	Reaction of nterest	Neutron Energy Response[a]	Product Half-Life	Dosimeter Capsule Position[b]	Gradient Chain[c]
Copper	$^{63}Cu(n,\alpha)^{60}Co$	6.13-11 MeV	5.271 yr	2-Cd	No
Titanium	$^{48}Ti(n,p)^{46}Sc$	3.86-9.4 MeV	83.81 dy	2-Cd	No
Iron	$^{54}Fe(n,p)^{54}Mn$	2.47-7.8 MeV	312.5 dy	1-B & 2-Cd	Yes
Nickel	$^{57}Ni(n,p)^{58}Co$	2.09-7.6 MeV	70.82 dy	2-Cd	Yes
^{238}U[d]	$^{238}U(n,f)^{137}Cs$	1.51-6.7 MeV	30.17 yr	3-Cd	No
^{237}Np[d]	$^{237}Np(n,f)^{137}Cs$	0.67-5.7 MeV	30.17 yr	3-Cd	No
Cobalt-Al	$^{59}Co(n,y)^{60}Co$	Thermal	5.271 yr	1-B & 2-Cd	Yes

Notes:

a) Energies between which 90% of activity is produced (^{235}U fission spectrum)

b) B denotes bare and Cd denotes cadmium shielded

c) Determined with additional chemical analysis

d) For the fission monitors ^{95}Zr (64.02 dy) and ^{103}Ru (39.27 dy) activities are also reported

5.3.1.6.2.2 Support Hardware and Installation

The sensor sets are located in the annular space between the reactor vessel reflective insulation and the primary biological shield at Comanche Peak Unit 1 and 2. The dosimetry installation consists of small aluminum dosimeter capsules connected to and supported by stainless steel bead chain.

The design of the dosimetry support system and the gradient chains and chain stops ensures correct and repeatable axial and azimuthal placement of the dosimetry relative to well known reactor features.

5.3.1.6.2.3 Supporting Analysis

Plant specific discrete ordinates transport theory calculations are performed to determine the neutron and gamma ray environment within the reactor geometry. The specific methods applied are consistent with those described in WCAP-15557, Qualification of the Westinghouse Pressure Vessel Neutron Fluence Evaluation Methodology, S. L. Anderson, August 2000 and will meet the requirements of Regulatory Guide RG-1.190, Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence, March 2001.

The transport calculations are carried out using the following three-dimensional synthesis technique:

$$\phi(r, \theta, z) = \phi(r, \theta) x \frac{\phi(r, z)}{\phi(r)}$$

where

$\phi(r, \theta, z)$ is the synthesized three-dimensional flux distribution,

$\phi(r, \theta)$ is the transport solution in r, θ geometry,

$\phi(r, z)$ is the two-dimensional solution for a cylindrical reactor model using the actual axial core power distribution, and

$\phi(r)$ is the one-dimensional solution for a cylindrical reactor model using the same source per unit height as that used in the two-dimensional r, θ calculation.

The transport calculations are performed using the DORT discrete ordinates transport theory code (ORNL RSICC CCC-650, August 1996) and the BUGLE-96 cross-section library (ORNL RSICC DLC-185, March 1996). The ENDF/B-VI based BUGLE-96 library is a 67 energy-group (47 neutron, 20 gamma ray) coupled neutron-gamma ray cross-section data set produced specifically for light-water reactor applications. In the transport analysis anisotropic scattering will be treated with a P_5 expansion of the cross-sections and angular discretization will be modeled with an S_{16} angular quadrature. An energy- and space-dependent fixed distributed source based on fuel cycle specific core power, burnup, and enrichment distributions will be used. In addition, system operating temperatures will be treated on a fuel cycle specific basis.

A detailed geometric model of the core, reactor internals, in-vessel surveillance capsules, reactor vessel, reflective insulation, and biological shield configuration is included in the transport calculations in order to ensure an accurate assessment of the relationship between the reactor vessel exposure and the measurements made in the surveillance capsules and in the reactor cavity.

The synthesis procedure combining the $\phi(r, \theta)$, $\phi(r, z)$, and $\phi(r)$ transport solutions into the three-dimensional flux / fluence maps within the reactor geometry is carried out using the Westinghouse developed SYNTHE post-processing code.

Evaluations of the neutron dosimetry sensor sets is done using the current state-of-the-art least squares methodology described in WCAP-15557. Best estimates for key exposure parameters

such as fast neutron flux (E > 1.0 MeV) or iron atom displacement rate (dpa/s) along with their uncertainties are then easily obtained from the adjusted spectrum. In general, the least squares methods, as applied to dosimetry evaluations, act to reconcile the measured sensor reaction rate data, dosimetry reaction cross-sections, and the calculated neutron energy spectrum within their respective uncertainties. For example,

$$R_i \pm \delta_{R_i} = \sum_g (\sigma_{ig} \pm \delta_{\sigma_{ig}})(\phi_g \pm \delta_{\phi_g})$$

relates a set of measured reactions, to a single neutron spectrum, through the multigroup dosimeter reaction cross-section, each with an uncertainty. The primary objective of the least squares evaluation is to produce unbiased estimates of the neutron exposure parameters at the location of the measurement.

The least squares evaluation of the dosimetry sets employs the FERRET code (ORNL RSICC PSR-145, January 1980) to combine the results of the plant specific neutron transport calculations and the sensor reaction rate measurements to determine best estimate values of fast neutron exposure parameters; e.g. neutron flux (E > 1.0 MeV), neutron flux (E > 0.1 MeV), and iron atom displacement rate (dpa/s) along with associated uncertainties.

The application of the least squares methodology requires the following input:

- The calculated neutron energy spectrum and associated uncertainties at the measurement location.
- The measured reaction rates and associated uncertainty for each sensor contained in the multiple foil sensor sets, and
- The energy dependent dosimetry reaction cross-sections and associated uncertainties for each sensor contained in the multiple foil sensor sets.

The calculated neutron energy spectrum is obtained from the results of the plant specific neutron transport calculations described above. The sensor reaction rates is derived from the measured specific activities for each dosimetry capsule analyzed. The dosimetry reaction cross-sections and uncertainties is obtained from the ENDF/B-VI based SNLRML dosimetry cross-section library (ORNL RSICC DLC-178, July 1994).

This approach is consistent with the requirements of Regulatory Guide RG-1.190, Calculational and Dosimetry Methods for Determining Pressure Vessel Neutron Fluence, March 2001.

5.3.1.7 Reactor Vessel Fasteners

The reactor vessel closure studs, nuts, and washers are designed and fabricated in accordance with the requirements of the ASME Code, Section III. The closure studs are fabricated of SA-540, Class 3, Grade B24. The closure stud material meets the fracture toughness requirements of the ASME Code, Section III and 10CFR50, Appendix G. Conventional nuts and washers were originally supplied with the reactor vessel. Hydraulic nuts were procured to be used with the reactor vessel studs, as appropriate.

Compliance with Regulatory Guide 1.65, “Materials and Inspections for Reactor Vessel Closure Studs,” is discussed in Appendix 1A(N). Nondestructive examinations are performed in

accordance with the ASME Code, Section III. Fracture toughness data for Unit 1 and Unit 2 bolting materials are presented in Tables 5.3-3A,B and 5.3-4A,B,C.

Westinghouse refueling procedures require the studs, nuts and washers to be removed from the reactor closure and be placed in storage racks or stud covers installed over studs left in place during preparation for refueling. The storage racks are then removed from the refueling cavity and stored at convenient locations on the Containment operating deck prior to removal of the reactor closure head and refueling cavity flooding. Therefore the reactor closure studs are never exposed to the borated refueling cavity water.

Surface treatments and lubricants applied to the reactor vessel closure stud bolting are evaluated in accordance with the requirements of Regulatory Guide 1.65, Regulatory Position 1.b(3).

If reactor studs were removed then the stud holes in the reactor flange are sealed with special plugs before removing the reactor closure thus preventing leakage of the borated refueling water into the stud holes.

5.3.2 PRESSURE-TEMPERATURE LIMITS

5.3.2.1 Limit Curves

Limitations on pressure and temperature during startup and shutdown are based on controlling the stress in the reactor pressure vessel wall [2]. Rates of temperature change are controlled so the combination of thermally induced stress and stress from internal pressure will not exceed the stress needed to cause a flow, postulated in the vessel wall, to grow. The methods outlined in Appendix G of the ASME Code Section III describe the postulated flaw and the analysis that assures protection against brittle failure.

The beltline portion of the reactor vessel is nearest to the core and is therefore exposed to the most intense radiation. This intense radiation lowers the ability of the material to resist brittle failure and causes a corresponding decrease in the fracture toughness (K_I) of the material. Lower values of fracture toughness are related to higher values of nil-ductility temperature RT_{NDT}. Methods for determining the reference nil-ductility temperature are outlined in Regulatory Guide 1.99, “Radiation Embrittlement of Reactor Vessel Materials”, Revision 2 (See Section 1A(N) for Regulatory Guide 1.99).

The operating curves including pressure-temperature limitations are calculated in accordance with 10CFR50, Appendix G and ASME Code, Section III, Appendix G, requirements and are included in the Pressure Temperature Limits Report (PTLR). Changes in fracture toughness of the beltline plates of forgings, weldments and associated heat affected zones due to radiation damage will be monitored by a surveillance program which conforms with ASTM-E-185 and 10CFR50, Appendix H. Capsules containing specimens of these materials are located adjacent to the inner wall of the reactor pressure vessel, near the core midheight. Capsules are removed at intervals prescribed in Section 5.3.1.6.

Prior to withdrawal of the first capsule, the shift in the Adjusted Reference Temperature (ART) is determined from a calculated fluence and the copper and nickel content of the steel. When the capsule withdrawal sequence begins, the shift in reference temperature is determined by comparing Charpy V-notch results from specimens tested during preirradiation testing with

Charpy V-notch results from specimens in the capsules. Updated values of reference temperature will be used to revise operating curves if RT_{NDT} determined from the surveillance program is greater than the predicted RT_{NDT}.

Temperature limits for preservice hydrotests and inservice leak and hydrotests will be calculated in accordance with 10CFR50, Appendix G and are included in the Technical Specification 3.04.3 and the PTLR with the exception of the preservice limits.

5.3.2.2 Operating Procedures

The transient conditions that are considered in the design of the reactor vessel are presented in Section 3.9N.1.1. These transients are representative of the operating conditions that should prudently be considered to occur during plant operation. The transients selected form a conservative basis for evaluation of the RCS to insure the integrity of the RCS equipment.

Those transients listed as upset condition transients are listed in Table 3.9N-1. None of these transients will result in pressure-temperature changes which exceed the heatup and cooldown limitations as described in Section 5.3.2.1 and in Table 3.9N-1a.

5.3.2.2.1 10CFR50.61 Screening Values

The values of pressurized thermal shock reference temperature (RT_{PTS}) were calculated for each reactor vessel using the methodology required by 10CFR50.61. The Unit 1 reactor vessel has beltline region limiting end of life (EOL) RT_{PTS} value of 100°F. The Unit 2 reactor vessel has a beltline region limiting EOL RT_{PTS} value of 94°F. These values are well below the 10CFR50.61 screening values of 270°F and 300°F. Therefore no corrective actions associated with 10CFR50.61 are expected to be required during the operating life of CPNPP.

5.3.3 REACTOR VESSEL INTEGRITY

5.3.3.1 Design

The reactor vessel is cylindrical with a welded hemispherical bottom head and a removable, bolted, flanged, and gasketed, hemispherical upper head. The reactor vessel flange and head are sealed by two hollow metallic O-rings. Seal leakage is detected by means of two leakoff communications: one between the inner and outer ring and one outside the outer O-ring. The vessel contains the core, core support structures, control rods, and other parts directly associated with the core.

Unit 1

The Unit 1 reactor vessel closure head contains head adapters. These head adaptors are tubular members, attached by partial penetration welds to the underside of the closure head. The upper end of these allow for welding to the control rod drive mechanisms. There are also two penetrations for the Reactor Vessel Level Measuring System (RVLMS) and four penetrations for the Core Exit Thermocouple Nozzle Assemblies (CFTNA).

Unit 2

The Unit 2 reactor vessel closure head contains head adapters. These head adaptors are tubular members, attached by partial penetration welds to the underside of the closure head. The upper end of these adaptors contains asme threads for the assembly of control rod drive mechanisms or instrumentation adaptors. The seal arrangement at the upper end of these adaptors consists of a welded flexible canopy seal. Mechanical canopy seal clamp assemblies (CSCAs) may be used to contain or prevent leaks in the canopy seal. Leakage past the canopy seal or CSCA is not pressure boundary leakage.

Inlet and outlet nozzles are located symmetrically around the vessel. Outlet nozzles are arranged on the vessel to facilitate optimum layout of the RCS equipment. The inlet nozzles are tapered from the coolant loop vessel interfaces to the vessel inside wall to reduce loop pressure drop.

The bottom head of the vessel contains penetration nozzles for connection and entry of the nuclear incore instrumentation. Each nozzle consists of a tubular member made of either an Inconel or an Inconel-stainless steel composite tube. Each tube is attached to the inside of the bottom head by a partial penetration weld.

Internal surfaces of the vessel which are in contact with primary coolant are weld overlay with 0.125 inch minimum of stainless steel or Inconel. The exterior in the reactor vessel is insulated with canned stainless steel reflective sheets. The insulation is a minimum of 3 inches thick and contoured to enclose the top, sides and bottom of the vessel. All the insulation modules are removable but the access to vessel side insulation is limited by the surrounding concrete. The noncrushable nozzle insulation, which is a minimum of 5 inches thick, is removable provided the pressure restrictor plate and associated structures do not inhibit removal.

The reactor vessel is designed and fabricated in accordance with the requirements of the ASME Code, Section III.

Principal design parameters of the reactor vessel are given in Table 5.3-5.

There are no special design features which would prohibit the in-situ annealing of the vessel. If the unlikely need for an annealing operation was required to restore the properties of the vessel material opposite the reactor core because of neutron irradiation damage, a metal temperature greater than 650°F for a period of 168 hours maximum would be applied. Various modes of heating may be used depending on the temperature.

The reactor vessel materials surveillance program is adequate to accommodate the annealing of the reactor vessel. Sufficient specimens are available to evaluate the effects of the annealing treatment.

Cyclic loads are introduced by normal power changes, reactor trip, startup and shutdown operations. These design base cycles are selected for fatigue evaluation and constitute a conservative design envelope for the projected plant life. Vessel analysis results in a usage factor that is less than 1.

The design specifications require analysis to prove that the vessel is in compliance with the fatigue and stress limits of the ASME Code, Section III. The loadings and transients specified for the analysis are based on the most severe conditions expected during service. The heatup and

cooldown rates imposed by Technical Specifications are limited to 100°F per hour. The 100°F per hour limiting rate is reflected in the vessel design specifications.

5.3.3.2 Materials of Construction

The materials used in the fabrication of the reactor vessel are discussed in Section 5.2.3.

5.3.3.3 Fabrication Methods

The fabrication methods used in the construction of the reactor vessel are discussed in Section 5.3.1.2.

5.3.3.4 Inspection Requirements

The nondestructive examinations performed on the reactor vessel are described in Section 5.3.1.3.

5.3.3.5 Shipment and Installation

The reactor vessel is shipped in a horizontal position on a shipping sled with a vessel lifting truss assembly. All vessel openings are sealed to prevent the entrance of moisture and an adequate quantity of desiccant bags are placed inside the vessel. These are usually placed in a wire mesh basket attached to the vessel cover. All carbon steel surfaces are painted with a heat resistant paint before shipment except for the vessel support surfaces and the top surface of the external seal ring.

Unit 1

The Unit 1 replacement closure head is shipped with a protective shipping cover and skid. Two support plates within the shipping cover protect the control rod drive mechanism housings. All head openings are sealed to prevent the entrance of moisture and an adequate quantity of desiccant bags are placed inside the head. In addition, nitrogen purge is maintained to protect the control rod drive mechanism internals. The shipping cover also acts as a lifting frame for handling the replacement closure head.

Unit 2

The Unit 2 closure head is also shipped with a shipping cover and skid. An enclosure attached to the ventilation shroud support ring protects the control rod mechanism housings. All head openings are sealed to prevent the entrance of moisture and an adequate quantity of desiccant bags are placed inside the head. These are placed in a wire mesh basket attached to the head cover. All carbon steel surfaces are painted with heat resistant pain before shipment. A lifting frame is provided for handling the vessel head.

5.3.3.6 Operating Conditions

Operating limitations are presented in Section 5.3.2 and in the Core Operating Limits Report (COLR). The procedures and methods used to ensure the integrity of the reactor vessel under the most severe postulated conditions are described in Section 3.9N.1.4.

5.3.3.7 Inservice Surveillance

The internal surface of the reactor vessel is capable of inspection periodically using visual and/or nondestructive techniques over the accessible areas. During refueling, the vessel cladding is capable of being inspected in certain areas between the closure flange and the primary coolant inlet nozzles, and, if deemed necessary, the core barrel is capable of being removed, making the entire inside vessel surface accessible.

The closure head is accessible for examination during refueling. Optical devices permit a selective inspection of the cladding, control rod drive mechanism nozzles, and the gasket seating surface. For Unit 2, the knuckle transition piece, which is the area of highest stress of the closure head, is accessible on the outer surface for visual inspection, dye penetrant or magnetic particle, and ultrasonic testing. The closure studs can be inspected periodically using visual, magnetic particle and/or ultrasonic techniques.

The full penetration welds in the following areas of the installed irradiated reactor vessel are available for visual and/or nondestructive inspection.

1. Vessel shell - from the inside surface.

2. Primary coolant nozzles - from the inside surface.

3. Closure head - from the inside and outside surfaces. (Note: Unit 1 replacement head is a one-piece forging; therefore, full penetration weld on closure head has been eliminated.) Bottom head - from the outside surface.

4. Closure studs, nuts and washers.

5. Field welds between the reactor vessel, nozzles and the main coolant piping.

6. Vessel flange seal surface.

The design considerations which have been incorporated into the system design to permit the above inspection are as follows:

1. All reactor internals are completely removable. The tools and storage space required to permit these inspections are provided.

2. The closure head is stored dry on the vessel head storage stand during refueling to facilitate direct visual inspection.

3. All reactor vessel studs, nuts and washers can be removed to dry storage during refueling.

4. Removable plugs are provided in the primary shield. The insulation covering the nozzle welds may be removed.

The reactor vessel presents access problems because of the radiation levels and remote underwater accessibility to this component. Because of these limitations on access to the reactor vessel, several steps have been incorporated into the design and manufacturing

procedures in preparation for the periodic nondestructive tests which are required by the ASME inservice inspection code. These are:

1. Shop ultrasonic examinations are performed on all internally clad surfaces to an acceptance and repair standard to assure an adequate cladding bond to allow later ultrasonic testing of the base metal from inside surface. The size of cladding bonding defect allowed is 1/4 inch by 3/4 inch with the greater direction parallel to the weld in the region bounded by 2 T (T = wall thickness) on both sides of each full penetration pressure boundary weld. Unbounded areas exceeding 0.442 square inches (3/4 inch diameter) in all other regions are rejected.

2. The design of the reactor vessel shell is a clean, uncluttered cylindrical surface to permit future positioning of the test equipment without obstruction.

3. The weld deposited clad surface on both sides of the weld to be inspected is specifically prepared to assure meaningful ultrasonic examinations.

4. During fabrication, all full penetration ferritic pressure boundary welds are ultrasonically examined in addition to ASME Code, Section III, requirements. These examinations are performed with manual contact techniques using methods and acceptance criteria of Section XI of the ASME Code, 1974 Edition with addenda up to and including the Summer 1975.

5. After the shop hydrostatic testing, all full penetration ferritic pressure boundary welds are ultrasonically examined in addition to ASME Code, Section III, requirements. These examinations are performed with manual contact techniques using methods and acceptance criteria of Section XI of the ASME Code, 1974 Edition with addenda up to and including the Summer 1975.

The vessel design and construction enables inspection in accordance with the ASME Code, Section XI.

5.3.3.8 Degradation of Control Rod Drive Mechanism Nozzle and Other Vessel Closure Head Penetrations (GL 97-01)

Beginning in 1986, leaks were reported in several Alloy 600 pressurizer instrument nozzles at both domestic and foreign reactors from several different NSSS vendors. The NRC staff identified primary water stress corrosion cracking (PWSCC) as an emerging technical issue to the Commission in 1989, after cracking was noted in Alloy 600 pressurizer heater sleeve penetrations at a domestic PWR facility. The NRC staff reviewed the safety significance of the cracking that occurred, as well as the repair and replacement activities at the affected facilities. The NRC staff determined that the cracking was not of immediate safety significance because the cracks were axial, had a low growth rate, were in a material with an extremely high flaw tolerance (high fracture toughness) and, accordingly, were unlikely to propogate very far. These factors also demonstrated that any cracking would result in detectable leakage and the opportunity to take corrective action before a penetration would fail. Further, with the exception of the leak found at Bugey 3 during hydrostatic testing, the NRC staff is not aware of any failure of any Alloy 600 vessel closure head penetration during plant operation. The NRC staff issued Information Notice (IN) 90-10, “Primary Water Stress Corrosion Cracking (PWSCC) of Inconel 600,” dated February 23, 1990, to inform the nuclear industry of the issue.

Unit 2 (Unit 1 RRVCH replaced Alloy 600 with Alloy 690)

CPNPP, as with most other PWRs, has Alloy 600 CRDM nozzle and other vessel head closure penetrations (VHPs) that extend above the reactor pressure vessel head. The stainless steel housing of the CRDM is screwed and seal-welded onto the top of the nozzle penetration. The weld between the nozzle top and bottom pieces is a dissimilar metal weld, which is also called a bimetallic weld. The nozzles protrude below the vessel head, thus exposing the inside surface of the nozzles to reactor coolant. The areas of interest for potential cracking are the weld between the nozzle and reactor vessel head, and the portion of the nozzle inside the reactor vessel head above the nozzle-to-vessel weld.

On April 1, 1997, the staff issued GL 97-01 to the industry, requesting that addressees provide a description of their plans to inspect the vessel head penetrations (VHPs) at their respective pressurized water reactor (PWR) designed plants. The Westinghouse Owners Group (WOG), in coordination with the efforts of the Nuclear Energy Institute (NEI) and the other PWR Owners Groups, determined that it was appropriate for its members to develop a cooperative integrated inspection program in response to GL 97-01. The WOG submitted two Topical Reports, WCAP-14901, Revision 0, “Background and Methodology for Evaluation of Reactor Vessel Closure Head Penetration Integrity for the Westinghouse Owners Group” and WCAP-14902, Revision 0, “Background Material for Response to NRC Generic Letter 97-01: Reactor Vessel Closure Head Penetration Integrity for the Westinghouse Owners Group” on behalf of the member utilities in the WOG. In these reports, the WOG provided descriptions of the two models, the Electric Power Research Institute (EPRI)/Dominion Engineering crack initiation and growth susceptibility model and the Westinghouse Model, that were being used to rank the VHPs at the participating plants in the owners group. CPNPP endorsed the probabilistic susceptibility model in Westinghouse Topical Report WCAP-14901, Revision 0, as being applicable to the assessment of VHPs at CPNPP, Units 1 and 2 [4].

REFERENCES

1. Soltesz, R. G., et al., “Nuclear Rocket Shielding Methods, Modification, Updating, and Input Data Preparation, Volume 5 - Two-Dimensional Discrete Ordinates Technique,” WANL-PR-(LL)-034, August 1970.

2. Hazelton, W. S., et al., “Basis for Heatup and Cooldown Limit Curves,” WCAP-7924-A, April 1975.

3. Letter TXX-4170, dated May 16, 1984, H.C. Schmidt (TUGCO) to B.J. Youngblood (NRC).

4. WCAP-14901, Revision 0, “Background and Methodology for Evaluation of Reactor Vessel Closure Head Penetration Integrity for the Westinghouse Owners Group”.

TABLE 5.3-1
REACTOR VESSEL QUALITY ASSURANCE PROGRAM

(Sheet 1 of 2)

	RT[a]	UT[b]	PT[c]	MT[d]
Forgings				
Closure Head (Unit 1)		yes		yes
Vessel Flange (Unit 1)		yes		yes
Vessel and head flanges (Unit 2)		yes		yes
Studs and nuts		yes		yes
Head adapters		yes	yes	
Head adapter tube		yes	yes	
Instrumentation tube		yes	yes	
Main nozzles		yes		yes
Nozzle safe ends		yes	yes	
Plates		yes		yes
Weldments				
Main seam	yes	yes		yes
Control rod drive head adapter connection			yes	
Instrumentation tube connection			yes	
Main nozzle	yes	yes		yes
Cladding		yes	yes	
Nozzle safe ends (if forging)	yes	yes	yes	
Nozzle safe ends (if weld deposit)	yes	yes	yes	
Head adapter forging to head adapter tube	yes		yes	
All ferritic welds accessible after hydrotest		yes		yes

TABLE 5.3-1
REACTOR VESSEL QUALITY ASSURANCE PROGRAM

(Sheet 2 of 2)

	RT[a]	UT[b]	PT[c]	MT[d]
All non-ferritic welds accessible after hydrotest		yes	yes	
Seal ledge				yes
Head lift lugs				yes
Core pad welds			yes	

a) RT - Radiographic.

b) UT - Ultrasonic.

c) PT - Dye penetrant.

d) MT - Magnetic particle.

TABLE 5.3-2A
UNIT 1 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

(Sheet 1 of 2)

Component	Material	Code Number	Cu (%)	P (%)	Ni (%)	^{T}NDT(°F)	50 ft-lb 35 Mil. Temperature (°F)	^{RT}NDT (°F)	Average Shelf Energy MWD[a] (ft-lb)	Average Shelf Energy NMWD[b] (ft- b)
Closure Head Forging	A508, C3	-	0.05	-	0.87	-50	10	-50	-	184.0
Vessel Flange	A508, C1.2	R1101-1	-	-	.72	10	70	10	-	97.0
Inlet Nozzle	A508, C1.2	R1105-1	0.09	-	.82	-10	50	-10	-	147.0
Inlet Nozzle	A508, C1.2	R1105-2	0.11	-	.84	-20	40	-20	-	136.5
Inlet Nozzle	A508, C1.2	R1105-3	0.11	-	.81	-10	50	-10	-	134.0
Inlet Nozzle	A508, C1.2	R1105-4	0.09	-	.82	-10	50	-10	-	156.5
Outlet Nozzle	A508, C1.2	R1106-1	-	-	.68	-20	40	-20	-	135.0
Outlet Nozzle	A508, C1.2	R1106-2	-	-	.62	-10	50	-10	-	111.0
Outlet Nozzle	A508, C1.2	R1106-3	-	-	.64	-20	50	-10	-	135.5
Outlet Nozzle	A508, C1.2	R1106-4	-	-	.65	-20	40	-20	-	117.5
Upper Shell	A533B, C1.1	R1104-1	0.07	0.012	.61	-30	100	40	-	83.0
Upper Shell	A533B, C1.1	R1104-2	0.08	0.011	.67	-50	100	40	-	75.0
Upper Shell	A533B, C1.1	R1104-3	0.05	0.010	.60	-20	70	10	-	107.5
Inter. Shell	A533B, C1.1	R1107-1	0.06	0.010	.65	-20	70	10	111.5	93.5
Inter. Shell	A533B, C1.1	R1107-2	0.06	0.010	.64	-20	50	-10	123.5	103.0
Inter. Shell	A533B, C1.1	R1107-3	0.05	0.007	.68	-20	70	10	131.0	88.0
Lower Shell	A533B, C1.1	R1108-1	0.08	0.008	.64	-20	60	0	119.0	85.0
Lower Shell	A533B, C1.1	R1108-2	0.05	0.006	.59	-30	80	20	124.5	78.0

TABLE 5.3-2A
UNIT 1 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

(Sheet 2 of 2)

Component	Material	Code Number	Cu (%)	P (%)	Ni (%)	^{T}NDT(°F)	50 ft-lb 35 Mil. Temperature (°F)	^{RT}NDT (°F)	Average Shelf Energy MWD[a] (ft-lb)	Average Shelf Energy NMWD[b] (ft- b)
Lower Shell	A533B, C1.1	R1108-3	0.07	0.008	.64	-30	60	0	122.0	98.0
Bottom Head Torus	A533B, C1.1	R1112-1	0.13	0.010	.62	-50	50	-10	-	112.0
Bottom Head Dome	A533B, C1.1	R1113-1	0.08	0.010	.60	-50	70	10	-	90.0
Inter. & Lower Shell[c] (Long & Girth Weld Seams)		G1.67	0.04	0.008	.17	-70	-10	-70	-	150.0

a) Major working direction (Longitudinal)

b) Normal to major working direction (Transverse)

c) B4 Weld Wire HT 88112 & Linde 0091 Flux Lot No. 0145

TABLE 5.3-2B
UNIT 2 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

(Sheet 1 of 2)

Component	Material	Code Number	Cu (%)	Ni (%)	P (%)	NDT (°F)	50 ft-lb 35 Mil. Temperature (°F)	^{RT}NDT (°F)	Average Shelf Energy	
									MWD[a] (ft-lb)	NMWD[b] (ft- b)
Closure Head Dome	A533B, C1.1	R3811-1	0.15	.65	0.014	-40	60	0	-	131
Closure Head Torus	A533B, C1.1	R3810-1	0.15	.69	0.011	-30	30	-30	-	143
Closure Head Flange	A508, C1.2	R3802-1	-	.71	0.013	40	<100	40	-	152
Vessel Flange	A508, C1.2	R3801-1	-	.70	0.009	-10	< 50	-10	-	121
Inlet Nozzle	A508, C1.2	R3803-1	-	.84	0.009	-10	< 50	-10	-	138
Inlet Nozzle	A508, C1.2	R3803-2	0.10	.91	0.008	-20	< 40	-20	-	136
Inlet Nozzle	A508, C1.2	R3803-3	-	.91	0.010	-10	< 50	-10	-	146
Inlet Nozzle	A508, C1.2	A3803-4	-	.86	0.009	-20	< 40	-20	-	136
Outlet Nozzle	A508, C1.2	A3805-1	-	.64	0.006	0	< 60	0	-	132
Outlet Nozzle	A508, C1.2	A3805-2	-	.66	0.005	0	< 60	0	-	119
Outlet Nozzle	A508, C1.2	A3805-3	-	.66	0.004	0	< 60	0	-	117
Outlet Nozzle	A508, C1.2	A3805-4	-	.67	0.005	0	< 60	0	-	119
Nozzle Shell	A533B, C1.1	A3806-1	0.05	.61	0.010	-10	100	40	-	76
Nozzle Shell	A533B, C1.1	A3806-2	0.06	.62	0.009	-30	70	10	-	87
Nozzle Shell	A533B, C1.1	A3806-3	0.06	.70	0.007	-30	100	40	-	86
Inter. Shell	A533B, C1.1	A3807-1	0.06	.64	0.006	-20	< 40	-20	133	108
Inter. Shell	A533B, C1.1	A3807-2	0.06	.64	0.007	-20	70	10	122	101
Inter. Shell	A533B, C1.1	A3807-3	0.05	.60	0.007	-20	40	-20	120	105

TABLE 5.3-2B
UNIT 2 REACTOR VESSEL FRACTURE TOUGHNESS PROPERTIES

(Sheet 2 of 2)

Component	Material	Code Number	Cu (%)	Ni (%)	P (%)	NDT (°F)	50 ft-lb 35 Mil. Temperature (°F)	^{RT}NDT (°F)	Average Shelf Energy MWD[a] (ft-lb)	Average Shelf Energy NMWD[b] (ft- b)
Lower Shell	A533B, C1.1	A3816-1	0.05	.59	0.001	-30	30	-30	136	107
Lower Shell	A533B, C1.1	A3816-2	0.03	.65	0.002	-30	60	0	131	106
Lower Shell	A533B, C1.1	A3816-3	0.04	.63	0.008	-40	20	-40	139	108
Bottom Head Torus	A533B, C1.1	A3813-1	0.12	.65	0.009	-60	0	-60	-	123
Bottom Head Dome	A533B, C1.1	A3814-1	0.12	.66	0.009	-70	-10	-70	-	112
Weld Metal[c] (Inter. to Lower Shell Girth Seam)			0.05	.03	0.004	-60	<0	-60	-	96
Weld Metal[d] (Inter. & Lower Shell Long Seams)			0.07	.05	0.005	-50	<10	-50	-	172

a) Major working direction (Longitudinal)

b) Normal to major working direction (Transverse)

c) B4 Weld Wire Ht. 89833 & Linde 124 Flux Lot No. 1061

d) B4 Weld Wire Ht. 89833 & Linde 0091 Flux Lot No. 1054

TABLE 5.3-2C
UNIT 1 REACTOR VESSEL BELTLINE REGION TENSILE PROPERTIES AND MATERIAL HEAT TREATMENT

Component	Material	Test Temp.	0.2% YS (ksi)	UTS (ksi)	Elong. (%)	RA (%)
Inter. Shell	Plate R1107-1	Room	71.3	90.8	25.0	63.8
Inter. Shell	Plate R1107-1	Room	69.7	88.8	25.0	62.9
Inter. Shell	Plate R1107-2	Room	68.7	89.0	25.0	64.5
Inter. Shell	Plate R1107-2	Room	69.1	89.6	25.0	64.5
Inter. Shell	Plate R1107-3	Room	71.3	92.6	25.0	62.6
Inter. Shell	Plate R1107-3	Room	71.9	93.0	25.0	61.3
Inter. Lower	Plate R1108-1	Room	68.3	88.9	25.0	59.5
Inter. Lower	Plate R1108-1	Room	68.8	88.8	25.0	59.9
Inter. Lower	Plate R1108-2	Room	65.7	86.6	24.0	56.6
Inter. Lower	Plate R1108-2	Room	66.0	87.1	24.0	58.4
Inter. Lower	Plate R1108-3	Room	66.1	86.6	26.0	60.7
Inter. Lower	Plate R1108-3	Room	65.3	86.6	26.0	63.8
Inter. to Lower Shell Girth Weld Seam	G1.67[(a)]	Room	77.1	88.5	28.0	71.1

a) B4 Weld Wire Heat 88112 & Linde 0091 Flux Lot 0145

Component	Heat Treatment
Inter. and Lower Shell Plates	1600°F ±25°F - 4 Hours. Water quenched 1225°F ±25°F - 4 Hours 1150°F +50°F - 40 Hours. Furnace Cooled to 600°F
Inter. and Lower Shell Welds	1150°F +50°F - 40 Hours. Furnace Cooled to 600°F

TABLE 5.3-2D
UNIT 2 REACTOR VESSEL BELTLINE REGION TENSILE PROPERTIES AND MATERIAL HEAT TREATMENT

Component	Material	Test Temp.	0.2% YS (ksi)	UTS (ksi)	Elong. (%)	RA (%)
Inter. Shell	Plate R3807-1	Room	66.9	85.8	27.5	64.8
Inter. Shell	Plate R3807-1	Room	66.7	85.6	27.0	64.1
Inter. Shell	Plate R3807-2	Room	67.8	88.5	25.0	63.0
Inter. Shell	Plate R3807-2	Room	67.9	88.3	25.0	62.1
Inter. Shell	Plate R3807-3	Room	68.8	87.9	27.0	62.3
Inter. Shell	Plate R3807-3	Room	68.5	88.0	25.0	63.5
Inter. Lower	Plate R3816-1	Room	67.8	87.6	26.0	66.5
Inter. Lower	Plate R3816-1	Room	67.8	87.4	26.0	66.5
Inter. Lower	Plate R3816-2	Room	67.3	88.1	27.0	62.6
Inter. Lower	Plate R3816-2	Room	67.2	87.6	27.0	62.9
Inter. Lower	Plate R3816-3	Room	68.7	89.8	25.5	62.9
Inter. Lower	Plate R3816-3	Room	69.0	89.1	25.5	64.2
Inter. to Lower Shell Girth Weld Seam	E3.10[a]	Room	70.8	86.4	27.0	65.4
Inter. to Lower Shell Long. Weld Seam	G3.03[b]	Room	67.9	80.6	30.0	71.0

a) B4 Weld Wire Heat 89833 & Linde 124 Flux Lot 1061

b) B4 Weld Wire Heat 89833 & Linde 0091 Flux Lot 1054

Component	Heat Treatment
Inter. and Lower Shell Plates	1600°F ±25°F - 4 Hours. Water quenched 1225°F ±25°F - 4 Hours 1150°F +50°F - 40 Hours. Furnace Cooled to 600°F
Inter. and Lower Shell Welds	1150°F +50°F - 40 Hours. Furnace Cooled to 600°F

TABLE 5.3-3A
UNIT 1 REACTOR PRESSURE VESSEL CLOSURE BOLTING MATERIAL PROPERTIES – STUDS

Heat Number	Material	Bar Number	0.2% Yield Stress (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Brinell Hardness Number	Energy at 10°F (ft-lbs)	Lateral Expansion(mils)
84176	A540, B24	412	148.7	163.5	15.0	49.2	352	46, 47, 46	25, 28, 26
84176	A540, B24	412-1	143.5	158.0	16.5	51.5	341	51, 50, 53	28, 29, 28
84176	A540, B24	419	147.5	161.5	16.0	51.4	341	50, 50, 50	28, 29, 28
84176	A540, B24	419-1	142.5	157.0	16.5	52.8	331	50, 51, 50	31, 33, 30
84176	A540, B24	423	144.0	159.0	16.5	51.9	341	48, 50, 50	26, 29, 30
84176	A540, B24	423-1	147.5	163.0	16.0	52.2	331	50, 52, 51	30, 31, 30
84176	A540, B24	425	150.2	164.0	15.0	50.0	341	45, 46, 45	28, 27, 26
84176	A540, B24	425-1	146.5	161.0	16.0	51.8	341	49, 49, 49	31, 29, 33
84176	A540, B24	428	148.5	163.0	17.0	51.4	341	48, 49, 48	28, 29, 28
84176	A540, B24	428-1	142.5	156.0	16.5	49.0	341	48, 49, 49	29, 31, 29
83833	A540, B24	410	137.7	155.0	17.0	55.2	321	53, 54, 53	34, 34, 32
83833	A540, B24	410-1	148.2	164.0	16.0	51.4	341	51, 50, 50	30, 29, 27
43320	A540, B24	459	137.2	153.0	17.0	53.4	331	48, 47, 48	30, 30, 28
43320	A540, B24	459-1	136.2	151.5	17.0	53.2	331	49, 48, 46	31, 28, 28

TABLE 5.3-3B
UNIT 2 REACTOR VESSEL CLOSURE HEAD BOLTING MATERIAL PROPERTIES – STUDS

(Sheet 1 of 2)

Heat Number	Material	Bar Number	0.2% Yield Stress (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Brinell Hardness Number	Energy at 10°F (ft-lbs)	Lateral Expansion(mils)
85695	SA540, B24	695	138.5	153.0	18.0	56.2	331	54, 54, 52	33, 33, 31
85695	SA540, B24	695-1	140.0	155.0	17.5	55.2	341	50, 52, 52	30, 34, 30
85695	SA540, B24	699	142.5	157.0	17.5	56.0	341	49, 50, 50	30, 31, 32
85695	SA540, B24	699-1	139.5	154.0	18.0	56.2	341	50, 49, 48	31, 29, 30
85695	SA540, B24	700	142.5	157.0	17.5	53.6	331	50, 49, 48	29, 29, 30
85695	SA540, B24	700-1	140.0	155.0	17.5	54.1	331	52, 53, 53	31, 33, 33
85695	SA540, B24	706	141.5	156.0	17.0	55.7	341	51, 51, 52	30, 32, 33
85695	SA540, B24	706-1	140.0	154.0	18.0	56.5	341	52, 53, 53	32, 33, 33
85695	SA540, B24	711	138.5	152.0	17.5	55.7	331	52, 54, 51	34, 34, 31
85695	SA540, B24	711-1	138.2	153.0	18.0	56.2	321	50, 50, 49	32, 31, 30
85695	SA540, B24	713	141.0	155.5	17.0	54.1	331	48, 50, 51	29, 29, 31
85695	SA540, B24	713-1	142.0	157.0	17.5	54.7	341	50, 50, 50	32, 29, 30
85919	SA540, B24	740	142.0	156.0	17.5	51.4	331	55, 54, 55	35, 35, 34
85919	SA540, B24	740-1	142.0	155.0	17.5	52.5	341	54, 53, 55	34, 33, 34
85919	SA540, B24	744	135.2	150.0	17.5	51.9	331	55, 56, 56	34, 36, 35
85919	SA540, B24	744-1	144.7	158.0	16.5	50.6	331	52, 54, 52	32, 33, 31

TABLE 5.3-3B
UNIT 2 REACTOR VESSEL CLOSURE HEAD BOLTING MATERIAL PROPERTIES – STUDS

(Sheet 2 of 2)

Heat Number	Material	Bar Number	0.2% Yield Stress (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Brinell Hardness Number	Energy at 10°F (ft-lbs)	Lateral Expansion(mils)
85919	SA540, B24	748	137.2	152.0	17.0	53.3	341	54, 55, 55	34, 36, 33
85919	SA540, B24	748-1	141.0	154.5	17.0	52.2	341	56, 55, 57	33, 34, 34
85919	SA540, B24	752	142.0	156.0	16.5	50.0	331	56, 56, 55	34, 34, 36
85919	SA540, B24	752-1	140.0	155.0	17.0	53.3	341	55, 53, 54	33, 33, 31
85919	SA540, B24	754	140.0	155.0	16.5	50.0	331	56, 56, 54	36, 37, 33
85919	SA540, B24	754-1	142.2	156.0	17.0	52.5	341	57, 54, 56	35, 34, 38
85919	SA540, B24	760	141.0	155.5	17.5	53.3	341	58, 58, 57	37, 38, 36
85919	SA540, B24	760-1	142.7	157.5	17.0	52.5	341	52, 52, 53	31, 31, 33

TABLE 5.3-4A
UNIT 1 REACTOR PRESSURE VESSEL CLOSURE BOLTING MATERIAL PROPERTIES - NUTS AND WASHERS

Heat Number	Material	Tube Number	0.2% Yield Stress (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Brinell Hardness Number	Energy at 10^{o}F (ft-lbs)	Lateral Expansion(mils)
63182	A540, B24	132	148.0	162.0	17.5	57.3	331	51, 52, 51	31, 32, 30
63182	A540, B24	132-1	148.7	162.0	17.0	54.7	331	49, 48, 49	29, 26, 29
63182	A540, B24	133	147.2	161.0	17.0	55.2	321	52, 50, 51	31, 30, 30
63182	A540, B24	133-1	y149.2	162.5	17.5	54.7	331	51, 51, 49	29, 31, 27
63182	A540, B24	135	147.5	161.0	17.0	53.0	321	49, 49, 51	28, 29, 30
63182	A540, B24	135-1	143.2	157.0	17.5	55.2	321	55, 54, 52	33, 32, 31
63182	A540, B24	137	145.0	159.0	16.5	54.8	331	54, 54, 53	33, 33, 29
63182	A540, B24	137-1	147.0	160.0	17.0	55.7	321	54, 55, 54	34, 36, 33
63182	A540, B24	143	145.0	159.0	18.0	58.1	331	55, 54, 54	33, 32, 32
63182	A540, B24	143-1	147.0	160.0	17.0	57.3	321	54, 50, 52	33, 29, 30
63182	A540, B24	145	145.0	159.0	17.0	56.0	321	54, 54, 55	34, 35, 34
63182	A540, B24	145-1	146.2	159.7	17.0	57.0	331	56, 55, 54	36, 35, 36
63182	A540, B24	148	144.0	157.5	17.5	56.5	331	56, 55, 55	33, 34, 34
63182	A540, B24	148-1	148.5	162.0	17.0	55.6	321	52, 51, 52	33, 28, 30
63182	A540, B24	150	144.7	158.0	17.5	55.7	331	55, 55, 54	33, 30, 31
63182	A540, B24	150-1	145.7	160.0	17.0	56.5	331	53, 50, 52	33, 30, 31

TABLE 5.3-4B
UNIT 2 REACTOR PRESSURE VESSEL HEAD CLOSURE BOLTING MATERIAL PROPERTIES - NUTS AND WASHERS

Heat Number	Material Spec. No.	Bar Number	0.2% Yield Stress (ksi)	Ultimate Tensile Strength (ksi)	Elongation (%)	Reduction in Area (%)	Brinell Hardness Number	Energy at 10°F (ft-lbs)	Lateral Expansion(mils)
85408	SA540, B24	208	147.5	161.0	17.5	54.7	331	54, 54, 52	35, 35, 35
85408	SA540, B24	208-1	149.0	162.0	16.5	51.9	321	50, 50, 49	31, 30, 30
85408	SA540, B24	210	149.0	162.5	17.5	55.5	341	52, 53, 51	33, 34, 30
85408	SA540, B24	210-1	144.7	157.0	18.0	56.2	331	55, 55, 54	36, 36, 33
85408	SA540, B24	214	151.7	165.0	18.0	54.4	321	50, 49, 50	30, 28, 30
85408	SA540, B24	214-1	150.0	164.0	17.5	53.3	321	47, 49, 47	28, 29, 27
85408	SA540, B24	216	147.0	160.0	16.5	54.1	321	50, 50, 51	30, 30, 32
85408	SA540, B24	216-1	150.0	162.5	16.5	53.2	331	52, 52, 53	31, 32, 33
85408	SA540, B24	219	148.0	161.0	17.5	54.4	321	51, 49, 49	30, 28, 28
85408	SA540, B24	219-1	147.0	161.0	17.5	53.0	341	53, 55, 54	30, 34, 33
85408	SA540, B24	222	149.5	163.5	17.0	53.8	341	49, 49, 48	29, 29, 26
85408	SA540, B24	222-1	148.2	162.0	17.5	53.4	341	49, 50, 51	30, 30, 32
85408	SA540, B24	225	148.5	161.0	17.0	53.6	331	52, 50, 52	34, 32, 32
85408	SA540, B24	225-1	150.0	163.0	17.5	54.7	331	51, 50, 50	32, 30, 32

TABLE 5.3-4C
UNIT 1 AND 2 REACTOR PRESSURE VESSEL CLOSURE BOLTING MATERIAL PROPERTIES - HYDRANUTS AND WASHERS

Heat Number	Material	0.2% Yield Stress (ksi) (min, max)	Ultimate Tensile Strength (ksi) (min, max)	Elongation (%) (min, max)	Reduction in Area (%) (min, max)	Brinell Hardness Number (min, max)	Energy at 10°F (ft-lbs) (min, max)	Lateral Expansion(mils) (min, max)
13522	A540, B24	144.75, 151.0	158.0, 165.0	16.0, 17.5	51.7, 54.7	321, 341	47, 54	27, 35
19632	A540, B24	142.0, 150.5	155.0, 162.0	16.0, 17.5	51.9, 55.7	321, 331	47, 52	27, 34
43432	A540, B24	142.5, 150.0	157.0, 164.0	16.5, 18.0	54.6, 57.5	321, 331	49, 55	26, 32
44415	A540, B24	140.0, 151.25	155.0, 164.0	16.5, 18.0	53.6, 57.3	321, 341	49, 56	29, 36
63182	A540, B24	143.25, 149.25	157.0, 162.5	16.5, 18.0	53.0, 58.1	321, 331	48, 56	26, 36
85408	A540, B24	144.75, 151.75	157.0, 165.0	16.5, 18.0	51.9, 56.2	321, 341	47, 55	26, 36

TABLE 5.3-5
REACTOR VESSEL DESIGN PARAMETERS

Design/operating pressure (psig)	2485/2317
Design temperature (°F)	650
Overall height of vessel and closure head, bottom head outside diameter to top of control rod mechanism adapter (ft-in.)	43-6.4 (Unit 1) 43-10 (Unit 2)
Thickness of insulation, minimum (in.)	3
Number of reactor closure head studs	54
Diameter of reactor closure head/studs, minimum shank (in.)	6-13/16
Inside diameter of flange (in.)	167
Outside diameter at shell (in.)	205
Inside diameter at shell (in.)	173
Inlet nozzle inside diameter (in.)	27-1/2
Outlet nozzle inside diameter (in.)	29
Clad thickness, minimum (in.)	1/8
Lower head thickness, minimum (in.)	5-3/8
Vessel belt-line thickness, minimum (in.)	8.63
Closure head thickness (in.)	7 (Unit 1) 6-1/2 (Unit 2)

TABLE 5.3-6A
CHEMICAL COMPOSITION OF UNIT NO. 1 REACTOR VESSEL BELTLINE REGION MATERIAL

Element	Intermediate Shell Plate			Lower Shell Plate			Weld Control No. G1.67[a]
	R1107-1	R1107-2	R1107-3	R1108-1	R1108-2	R1108-3	
C	.22	.22	.25	.22	.21	.22	.16
Mn	1.42	1.33	1.44	1.39	.136	1.40	1.23
P	.010	.010	.007	.008	.006	.008	.008
S	.009	.013	.011	.016	.012	.014	.008
Si	.24	.19	.26	.24	.23	.22	.17
Ni	.65	.64	.68	.64	.59	.64	.17
Cr	.05	.03	.03	.06	.02	.04	.02
Mo	.57	.58	.59	.60	.52	.59	.54
Cu	.06	.06	.05	.08	.05	.07	.04
V	.006	.004	.004	.005	.002	.003	.006
Cb	<.01	<.01	<.01	<.01	<.01	<.01	<.01
Pb	ND	ND	ND	ND	ND	ND	<.001
W	.01	.03	.01	<.01	<.01	<.01	<.01
As	.006	.008	.004	.003	.004	.005	<.001
Sn	.004	.007	.004	.002	.002	.003	.001
Co	.012	.012	.013	.013	.012	.013	.006
N_2	.012	.009	.008	.010	.007	.008	.006
A1	.019	.025	.025	.008	.008	.014	.006
B	<.001	<.001	<.001	<.001	<.001	<.001	<.001
Ti	<.01	<.01	<.01	<.01	<.01	<.01	<.01
Zr	.001	.001	.001	.001	.001	.001	<.001

a) Submerged arc weld - Type B4 Wire Heat No. 88112 and Linde 0091 Flux Lot No. 0145 - used to fabricate all beltline region weld seams.

TABLE 5.3-6B
CHEMICAL COMPOSITION OF UNIT NO. 2 REACTOR VESSEL BELTLINE REGION MATERIAL

	Intermediate Shell Plate			Lower Shell Plate			Weld Control No. G3.03[a]	Weld Control No. E3.10[b]
Element	R3807-1	R3807-2	R3807-3	R3816-1	R3816-2	R3816-3		
C	.21	.22	.22	.23	.23	.22	.16	.088
Mn	1.42	1.40	1.30	1.48	1.48	1.50	1.32	1.33
P	.006	.007	.007	.001	.002	.008	.005	.004
S	.015	.016	.009	.004	.012	.008	.011	.010
Si	.25	.24	.19	.19	.21	.19	.16	.51
Ni	.64	.64	.60	.59	65	.63	.05	.03
Cr	.05	.04	.06	.03	.03	.04	.02	.03
Mo	.60	.59	.58	.49	.50	.52	.54	.54
Cu	.06	.06	.05	.05	.03	.04	.07	.05
V	.002	.003	.002	.003	.003	003	.004	.003
Cb	<.01	<.01	<.01	<.01	<.01	<.01		
Pb	<.001	<.001	<.001	<.001	<.001	<.001		
W	<.01	<.01	<.01	<.01	<.01	<.01		
As	.004	.005	.005	.009	.011	.015		
Sn	.003	.004	.003	.001	.001	.002		
Co	.012	.013	.009	.020	.012	.012		
N_2	.009	.010	.007	.028	.014	.014		
A1	.020	.025	.023	.026	.026	.018		
B	<.001	<.001	<.001	<.001	<.001	<.001		
Ti	<.01	<.01	<.01	<.01	<.01	<.01		
Zr	<.001	<.001	<.001	<.001	<.001	<.001		

a) Submerged arc weld - Type B4 Wire Heat No. 89833 and Linde 0091 Flux Lot No. 1054 - used in seam 101-124 & 101-142A, B, C

b) Submerged arc weld - Type B4 Wire Heat No. 89833 and Linde 124 Flux Lot No. 1061 - used in seam 101-171

TABLE 5.3-7A
UNIT NO. 1 REACTOR VESSEL BELTLINE REGION FRACTURE TOUGHNESS SUMMARY

Plate No. or Welds	Chemistry		Chem. Factor (CF)	Fluence[(a)] Factor (1/4 Thickness)	ΔRT_{NDT}[(c)] (°F)	Initial RT_{NDT} (°F)	Margin[(b)] $2\sqrt{\sigma_I^2 \sigma_\Delta^2}$	Limiting[(d)] ART (°F)
	Cu (Wt %)	Ni (Wt %)						
R1107-1	0.06	0.65	37	0.99	37	10	34	81
R1107-2	0.06	0.64	37	0.99	-	-10	Non Limiting Plate	
R1107-3	0.05	0.68	31	0.99	-	10	Non Limiting Plate	
R1108-1	0.08	0.64	51	0.99	51	0	34	85
R1108-2	0.05	0.59	31	0.99	31	20	31	82
R1108-3	0.07	0.64	44	0.99	-	0	Non Limiting Plate	
Longitudinal	0.04	0.17	40	0.99	-	-70	Non Limiting Plate	
Circumferential	0.04	0.17	40	0.99	-	-70	Non Limiting Plate	

a) Fluence Factor = $f^{(0.28-0.10 \log f)}$
Where: $f = f_{surf}\ e^{-0.24X}$
and: $f_{surf} = 1.59 \times 10^{19} N/Cm^2$, E>1 MeV at 16 EFPY (Westinghouse Design Basis)
X = Distance From Inner (Wetted) Surface (Inches); (Beltline Thickness = 8.63 inches)

b) Error: σ_Δ = 17°F for Base Material and 28°F for Welds (Assume ΔL 0.5 ΔRT_{NDT})
σ_1 = O (Initial RT_{NDT} Values Obtained from “Upper Bound” of Test Results)

c) ΔRT_{NDT} = CF x Fluence Factor

d) Adjusted Reference Temperature (ART) = Initial RT_{NDT} + ΔRT_{NDT} + Margin

TABLE 5.3-7B
UNIT NO. 2 REACTOR VESSEL BELTLINE REGION FRACTURE TOUGHNESS SUMMARY

Plate No. or Welds	Chemistry		Chem. Factor (CF)	Fluence[a] Factor (1/4 Thickness)	ΔRT_{NDT}[c] (°F)	Initial RT_{NDT} (°F)	Margin[b] $2\sqrt{\sigma_I^2 \sigma_\Delta^2}$	Limiting[d] ART (°F)
	Cu (Wt %)	Ni (Wt %)						
R3807-1	0.06	0.64	37	0.99	-	-20	Non Limiting Plate	
R3807-2	0.06	0.64	37	0.99	37	10	34	81
R3807-3	0.05	0.60	31	0.99	-	-20	Non Limiting Plate	
R3816-1	0.05	0.59	31	0.99	-	-30	Non Limiting Plate	
R3816-2	0.03	0.65	20	0.99	-	0	Non Limiting Plate	
R3816-3	0.04	0.63	26	0.99	-	-40	Non Limiting Plate	
Longitudinal	0.07	0.05	38	0.99	-	-50	Non Limiting Weld	
Circumferential	0.05	0.03	29	0.99	-	-60	Non Limiting Weld	

a) Fluence Factor = $f^{(0.28-0.10 \log f)}$
Where: $f = f_{surf}\ e^{-0.24X}$
and: $f_{surf} = 1.59 \times 10^{19} N/Cm^2$, E>1 MeV at 16 EFPY (Westinghouse Design Basis)
X = Distance From Inner (Wetted) Surface (Inches); (Beltline Thickness = 8.63 inches)

b) Error: σ_Δ = 17°F for Base Material and 28°F for Welds (Assume $\Delta \underline{L}$ 0.5 ΔRT_{NDT})
σ_1 = O (Initial RT_{NDT} Values Obtained from “Upper Bound” of Test Results)

c) RT_{NDT} = CF x Fluence Factor

d) Adjusted Reference Temperature (ART) = Initial RT_{NDT} + ΔRT_{NDT} + Margin

TABLE 5.3-8B
UNIT 2 BELTLINE REGION WELD METAL CHARPY V-NOTCH IMPACT DATA

Inter. To Lower Shell Girth Seam Weld Code No. E3.10				Inter. and Lower Shell Long. Seams Weld Code No. G3.03			
Temp (°F)	Energy (Ft-Lb)	Lat. Exp (Mils)	Shear (percent)	Temp. (°F)	Energy (Ft-Lb)	Lat. Exp. (Mils)	Shear (percent)
-80	12	6	0	-80	29	16	5
-80	7	3	0	-80	18	9	0
-80	15	9	0	-80	25	14	5
-40	39	29	20	-40	60	43	30
-40	26	20	10	-40	53	38	25
-40	43	32	20	-40	47	32	25
0	54	39	50	10	105	65	70
0	65	46	50	10	98	62	60
0	55	40	50	10	95	63	60
60	87	61	80	60	120	81	80
60	88	64	80	60	139	82	100
60	86	62	80	60	116	79	80
100	92	74	95	100	152	87	100
100	97	73	95	100	151	85	100
100	94	73	95	100	155	86	100
160	96	76	100	160	168	86	100
160	95	75	100	160	183	87	100
160	96	75	100	160	164	87	100

TABLE 5.3-9A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-1107-1

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	11	0	6
-40	12	0	8
-40	12	0	7
+10	25	10	17
+10	33	15	23
+10	30	15	19
+40	38	20	26
+40	42	20	29
+40	39	20	26
+60	48	20	34
+60	47	25	40
+60	45	20	32
+70	51	25	37
+70	53	25	38
+70	53	25	39
+100	60	30	47
+100	69	40	52
+100	65	40	50
+160	90	95	68
+160	99	95	73
+160	96	95	72
+212	100	100	72
+212	90	100	69
+212	91	100	70

TABLE 5.3-8A
UNIT 1 BELTLINE REGION WELD METAL CHARPY V-NOTCH IMPACT DATA
WELD CODE NO. G1.67

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-104	19	9	0
-104	18	10	0
-104	19	10	0
-80	20	10	5
-80	35	20	10
-80	23	12	0
-40	62	37	30
-40	72	43	40
-40	74	44	40
-10	95	54	60
-10	97	58	60
-10	95	55	60
10	124	70	80
10	119	68	80
10	122	73	80
50	131	80	95
50	125	74	90
50	139	82	100
100	149	85	100
100	152	88	100
100	148	82	100

TABLE 5.3-9B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-3807-1

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	7	0	2
-40	13	0	6
-40	11	0	5
+0	44	20	31
+0	52	25	35
+0	33	15	24
+40	62	30	45
+40	68	30	47
+40	65	30	45
+100	92	60	60
+100	90	60	63
+100	81	40	50
+160	116	100	69
+160	108	100	72
+160	109	100	74
+212	103	100	76
+212	111	100	77
+212	99	100	70

TABLE 5.3-10A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-1107-2

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	9	0	5
-40	11	0	6
-40	15	0	9
+10	31	15	22
+10	38	20	26
+10	38	20	25
+40	50	25	34
+40	41	20	26
+40	54	30	37
+50	65	35	42
+50	60	30	40
+50	52	25	37
+60	70	35	47
+60	51	25	35
+60	60	30	41
+100	72	50	54
+100	69	50	50
+100	77	60	58
+160	95	95	67
+160	98	95	70
+160	92	95	66
+212	105	100	77
+212	100	100	70
+212	104	100	75

TABLE 5.3-10B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-3807-2

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-20	22	5	16
-20	21	0	14
-20	19	0	13
+40	43	20	31
+40	40	20	28
+40	55	25	38
+60	71	40	48
+60	48	20	36
+60	64	30	43
+70	65	50	49
+70	63	40	44
+70	65	40	46
+100	79	70	55
+100	78	70	52
+100	74	60	49
+160	99	95	69
+160	96	90	64
+160	103	95	72
+212	98	100	66
+212	104	100	67
+212	100	100	66

TABLE 5.3-11A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-1107-3

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
+0	22	0	17
+0	19	0	16
+0	16	0	12
+40	42	20	27
+40	34	15	24
+40	39	15	26
+60	40	20	30
+60	54	30	41
+60	51	30	36
+70	53	25	37
+70	51	25	37
+70	53	30	36
+100	65	30	45
+100	70	40	51
+100	65	35	48
+160	92	95	72
+160	86	90	64
+160	90	95	71
+212	87	100	64
+212	92	100	70
+212	85	100	66

TABLE 5.3-11B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL
INTERMEDIATE SHELL PLATE CODE R-3807-3

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	16	0	8
-40	16	0	8
-40	14	0	7
+0	28	10	18
+0	28	10	18
+0	34	15	24
+40	50	25	36
+40	51	25	35
+40	50	25	36
+100	85	80	59
+100	78	80	53
+100	61	50	46
+160	106	100	70
+160	102	100	71
+160	94	100	64
+212	104	100	71
+212	104	100	70
+212	106	100	72

TABLE 5.3-12A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-1108-1

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	16	0	10
-40	18	0	12
-40	16	0	11
+0	32	15	24
+0	28	10	21
+0	31	15	22
+40	43	20	29
+40	35	15	27
+40	37	15	24
+50	53	25	36
+50	46	20	32
+50	45	20	35
+60	52	25	37
+60	61	30	43
+60	60	30	41
+100	60	70	49
+100	74	80	57
+100	66	70	51
+160	90	100	66
+160	76	90	59
+160	86	100	62
+212	83	100	62
+212	81	100	62
+212	86	100	64

TABLE 5.3-12B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-3816-1

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	14	0	7
-40	21	5	12
-40	22	5	14
+0	52	25	36
+0	42	20	28
+0	43	20	28
+30	54	20	37
+30	61	25	42
+30	56	25	40
+60	70	40	47
+60	83	40	51
+60	75	40	49
+100	107	90	66
+100	98	90	67
+100	105	90	70
+160	103	100	70
+160	109	100	73
+160	110	100	71

TABLE 5.3-13A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-1108-2

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	9	0	4
-40	8	0	4
-40	9	0	5
+0	24	10	19
+0	24	10	19
+0	23	10	18
+30	26	15	19
+30	29	15	20
+30	29	15	22
+70	53	30	40
+70	48	25	38
+70	56	30	42
+80	51	30	38
+80	56	30	42
+80	55	30	42
+100	53	30	42
+100	57	30	44
+100	62	40	49
+160	78	95	58
+160	76	95	54
+160	73	95	53
+212	78	100	55
+212	79	100	55
+212	77	100	54

TABLE 5.3-13B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-3816-2

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	18	0	9
-40	15	0	8
-40	20	0	10
+0	31	10	20
+0	23	5	17
+0	28	10	21
+30	42	20	33
+30	48	20	38
+30	38	20	24
+50	46	20	35
+50	54	25	42
+50	55	25	39
+60	73	35	47
+60	59	30	39
+60	66	35	41
+100	85	50	61
+100	88	60	61
+100	86	60	58
+160	98	100	67
+160	107	100	71
+160	104	100	68
+212	107	100	68
+212	106	100	68
+212	105	100	68

TABLE 5.3-14A
UNIT 1 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-1108-3

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-40	7	0	5
-40	15	0	11
-40	10	0	7
+0	27	10	19
+0	29	10	20
+0	34	15	23
+30	43	20	22
+30	38	15	20
+30	53	25	35
+50	48	25	35
+50	47	20	34
+50	63	30	42
+60	50	25	36
+60	63	30	47
+60	61	30	45
+100	82	50	57
+100	72	40	53
+100	70	40	49
+160	93	90	62
+160	91	90	60
+160	96	90	64
+212	102	100	69
+212	97	100	68
+212	96	100	65

TABLE 5.3-14B
UNIT 2 CHARPY V-NOTCH IMPACT DATA FOR REACTOR VESSEL LOWER SHELL PLATE CODE R-3816-3

(Transverse Orientation)

Test Temp. (°F)	Energy (FT-LB)	Shear (percent)	Lateral Expansion (MILS)
-80	7	0	5
-80	11	0	7
-80	13	0	7
-40	32	10	17
-40	25	5	14
-40	22	5	12
+0	39	15	24
+0	38	15	26
+0	44	20	29
+20	58	30	41
+20	52	25	36
+20	59	30	41
+60	77	35	47
+60	89	40	52
+60	84	40	54
+100	102	90	64
+100	94	90	65
+100	94	100	61
+160	109	100	71
+160	105	100	65
+160	103	100	69
+212	105	100	63
+212	108	100	67
+212	111	100	68

TABLE 5.3-15A
UNIT 1 REACTOR VESSEL NON-BELTLINE WELD METAL TOUGHNESS PROPERTIES

Weld Location	Weld Seam No.	Weld Mat'l Code No.	T_{NDT} °F	50 Ft-Lb/35 Mil Temp. °F	RT_{NDT} °F
Vessel Flange to Nozzle Shell	101-121	G 1.71	-50	< 10	-50
Vessel Flange to Nozzle Shell	101-121	G 2.08	-60	< 0	-60
Nozzle Shell Long. Seams	101-122 A,B and C	G 1.59	-50	< 10	-50
Inlet Nozzle to Nozzle Shell	105-121 A and C	G 3.09	-70	< -10	-70
Inlet Nozzle to Nozzle Shell	105-121 B and D	G 2.28	-70	< -10	-70
Inlet Nozzle to Nozzle Shell	105-121 D	G 2.23	-60	< 0	-60
Outlet Nozzle to Nozzle Shell	107-121 A and C	G 2.28	-70	< -10	-70
Outlet Nozzle to Nozzle Shell	107-121 B and D	G 3.09	-70	< -10	-70
Nozzle Shell to Inter. Shell	103-121	G 2.12	-60	< 0	-60
Lower Shell to Bottom Head Torus	101-141	E 2.09	-70	< -10	-70
Bottom Head Torus Long. Seams	101-154 A,B,C and D	A 2.19	-70	< -10	-70
Bottom Head Torus to Bottom Head Dome	102-151	G 3.09	-70	< -10	-70

TABLE 5.3-15B
UNIT 2 REACTOR VESSEL NON-BELTLINE WELD METAL TOUGHNESS PROPERTIES

Weld Location	Weld Seam No.	Weld Mat'l Code No.	T_{NDT} °F	50 Ft-Lb/35 Mil Temp. °F	RT_{NDT} °F
Closure Head Dome to Closure Head Torus	103-101	E 3.08	-80	< -20	-80
Closure Head Torus Long. Seams	101-104 A,B,C and D	A 2.16	-80	< -20	-80
Closure Head Torus to Closure Head Flange	101-101	E 3.08	-80	< 20	-80
Vessel Flange to Nozzle Shell	101-121	E 2.10	-80	< 30	-30
Vessel Flange to Nozzle Shell	101-121	E 2.09	-70	< -10	-70
Vessel Flange to Nozzle Shell	101-121	E 2.17	-50	< 10	-50
Nozzle Shell Long. Seams	101-122 A,B and C	G 3.07	-80	< -20	-80
Inlet Nozzle to Nozzle Shell	105-121 A and C	E 3.08	-80	< -20	-80
Inlet Nozzle to Nozzle Shell	105-121 B and D	E 3.02	-70	< -10	-70
Outlet Nozzle to Nozzle Shell	107-121 A and C	E 3.02	-70	< -10	-70
Outlet Nozzle to Nozzle Shell	107-121 B and D	E 3.08	-80	< -20	-80
Nozzle Shell to Inter. Shell	103-121	E 2.13	-80	< -20	-80
Lower Shell to Bottom Head Torus	101-141	E 4.05	-80	< 0	-60
Bottom Head Torus Long. Seams	101-154 A,B,C and D	A 2.33	-50	< 10	-50
Bottom Head Torus to Bottom Head Dome	102-151	E 4.01	-80	< -10	-70

TABLE 5.3-16A
UNIT 1 STEAM GENERATOR AND PRESSURIZER BASE METAL FRACTURE TOUGHNESS DATA

Component	Component Part	Test Number	2NB Drop Wt °F	T_{NDT} °F	°F for 50 ft lb/35 MLE	RT_{NDT} °F
SG 1 CH2	Channel Head	EM 4636	70	60	< 120	60
	Tube Sheet	EM 4591	70	60	< 120	60
SG 2 CH2	Channel Head	EM 4648	70	60	< 120	60
	Tube Sheet	EM 4592	70	60	< 120	60
SG 3 CH2	Channel Head	EM 4649	70	60	< 120	60
	Tube Sheet	EM 4602	70	60	< 120	60
SG 4 CH2	Channel Head	EM 4650	70	60	< 120	60
	Tube Sheet	EM 4610	70	60	< 120	60
Pressurizer	Lower Head	4902-1	70	60	< 120	60
	Surge Nozzle	4742-1	70	60	< 120	60
	Upper Head	4896-1	70	60	< 120	60
	Manway Nozzle	4708-4	70	60	< 120	60
	Safety Nozzle	4749-12	70	60	< 120	60
	Safety Nozzle	4749-4	70	60	< 120	60
	Safety Nozzle	4749-11	70	60	< 120	60
	Relief Nozzle	4749-16	70	60	< 120	60
	Spray Nozzle	4743-4	70	60	< 120	60
	Manway Cover	5775	70	60	< 120	60
	Shell Barrel	4906	70	60	< 120	60
	Shell Barrel	4905	70	60	< 120	60
	Shell Barrel	4926	70	60	< 120	60
	Shell Barrel	4921	70	60	< 120	60

TABLE 5.3-16B
UNIT 2 STEAM GENERATOR AND PRESSURIZER BASE METAL FRACTURE TOUGHNESS DATA

Component	Component Part	Test Number	2NB Drop Wt °F	T_{NDT} °F	°F for 50 ft lb/35 MLE	RT_{NDT} °F
SG 2211	Channel Head	5012	70	60	< 120	60
	Tube Sheet	4338	70	60	< 120	60
SG 2212	Channel Head	5219	70	60	< 120	60
	Tube Sheet	4337	70	60	< 120	60
SG 2213	Channel Head	5263	70	60	< 120	60
	Tube Sheet	4685	70	60	< 120	60
SG 2214	Channel Head	5013	70	60	< 120	60
	Tube Sheet	4882	70	60	< 120	60
Pressurizer	Lower Head	4985	70	60	< 120	60
	Surge Nozzle	5672	70	60	< 120	60
	Upper Head	5743	70	60	< 120	60
	Manway Nozzle	5653	70	60	< 120	60
	Safety Nozzle	5514-4	70	60	< 120	60
	Safety Nozzle	5418-9	70	60	< 120	60
	Safety Nozzle	5418-11	70	60	< 120	60
	Relief Nozzle	5418-10	70	60	< 120	60
	Spray Nozzle	5697	70	60	< 120	60
	Manway Cover	6467	70	60	< 120	60
	Shell Barrel	5828	70	60	< 120	60
	Shell Barrel	5829	70	60	< 120	60
	Shell Barrel	5839	70	60	< 120	60
	Shell Barrel	5867	70	60	< 120	60

TABLE 5.3-17A
UNIT 1 STEAM GENERATOR AND PRESSURIZER WELD METAL FRACTURE TOUGHNESS DATA

Weld Type	Test Number Electrode	Test Number Flux	2NB Drop Wt °F	T_{NDT} °F	°F for 50 ft lb/35 MLE	RT_{NDT} °F
SMAW	5167	---	20	10	< 70	10
	5199	---	20	10	< 70	10
	5200	---	20	10	< 70	10
	5299	---	20	10	< 70	10
	5300	---	20	10	< 70	10
	5301	---	20	10	< 70	10
	5315	---	20	10	< 70	10
	5318	---	20	10	< 70	10
	5319	---	20	10	< 70	10
	5429	---	20	10	< 70	10
	5431	---	20	10	< 70	10
	5444	---	20	10	< 70	10
	5447	---	20	10	< 70	10
	5450	---	20	10	< 70	10
	5451	---	20	10	< 70	10
	5455	---	20	10	< 70	10
SAW	4110	5330	20	10	< 70	10
	4109	5330	20	10	< 70	10
	4109	5126	20	10	< 70	10
	5164	5511	20	10	< 70	10
	5478	5511	20	10	< 70	10
	5477	6033	20	10	< 70	10

TABLE 5.3-17A
UNIT 1 STEAM GENERATOR AND PRESSURIZER WELD METAL FRACTURE TOUGHNESS DATA

SMAW[(a)]	3071	---	20	10	<70	10
	5310	---	20	10	<70	10
SAW[a]	128734.1	128734.1	20	10	<70	10

a) These values are for the Unit 1 Model Delta 76 Steam Generator. The values for the original installation steam generator have been maintained in this Table for historical information.

TABLE 5.3-17B
COMPANCHE PEAK UNIT 2 STEAM GENERATOR AND PRESSURIZER WELD METAL FRACTURE TOUGHNESS DATA

(Sheet 1 of 2)

Component	Test Number Electrode	Test Number Flux	2NB Drop Wt °F	T_{NDT} °F	°F for 50 ft lb/35 MLE	RT_{NDT} °F
SMAW	5458	---	20	10	< 70	10
	6026	---	20	10	< 70	10
	6027	---	20	10	< 70	10
	6041	---	20	10	< 70	10
	6049	---	20	10	< 70	10
	6164	---	20	10	< 70	10
	6165	---	20	10	< 70	10
	6182	---	20	10	< 70	10
	6193	---	20	10	< 70	10
	6787	---	20	10	< 70	10
	6790	---	20	10	< 70	10
	7386	---	20	10	< 70	10
	7404	---	20	10	< 70	10
	7417	---	20	10	< 70	10
	7643	---	20	10	< 70	10
	7960	---	20	10	< 70	10
	7967	---	20	10	< 70	10
	7969	---	20	10	< 70	10

TABLE 5.3-17B
COMPANCHE PEAK UNIT 2 STEAM GENERATOR AND PRESSURIZER WELD METAL FRACTURE TOUGHNESS DATA

(Sheet 2 of 2)

Component	Test Number Electrode	Test Number Flux	2NB Drop Wt °F	T_{NDT} °F	°F for 50 ft lb/35 MLE	RT_{NDT} °F
	7973	---	20	10	< 70	10
	7978	---	20	10	< 70	10
	8297	---	20	10	< 70	10
	8303	---	20	10	< 70	10
	8308	---	20	10	< 70	10
	7961	---	20	10	< 70	10
	7977	---	20	10	< 70	10
	3415	---	20	10	< 70	10
	4514	---	20	10	< 70	10
SAW	4109	5330	20	10	< 70	10
	4110	5330	20	10	< 70	10
	5478	5511	20	10	< 70	10
	5477	6033	20	10	< 70	10
	6180	6033	20	10	< 70	10
	6181	6185	20	10	< 70	10
	7653	6769	20	10	< 70	10
	6330	6769	20	10	< 70	10
	8301	6769	20	10	< 70	10

5.4 COMPONENT AND SUBSYSTEM DESIGN

5.4.1 REACTOR COOLANT PUMPS

5.4.1.1 Design Bases

The reactor coolant pump ensures an adequate core cooling flow rate and hence sufficient heat transfer, to maintain a departure from nucleate boiling ratio (DNBR) greater than the limit value within the parameters of operation. The required net positive suction head is by conservative pump design always less than that available by system design and operation.

Sufficient pump rotation inertia is provided by a fly wheel, in conjunction with the impeller and motor assembly, to provide adequate flow during coastdown. This flow following an assumed loss of pump power aids in the development of a thermal gradient for initiation of natural circulation while providing the core with adequate cooling.

The reactor coolant pump motor is tested, without mechanical damage, at overspeeds up to and including 125 percent of normal speed. The integrity of the flywheel during a loss of coolant accident (LOCA) is demonstrated in Reference [1] which is undergoing generic review by the NRC staff.

Ongoing steam/water tests being planned jointly by Westinghouse, Framatone, and the French Atomic Energy Commission (CEA) are described in Section 1.5. The ultimate use of the data from these tests will be to develop an empirical two phase flow pump performance model. It is expected that this new model will confirm that the present pump model conservatively predicts performance in all LOCA conditions and thus increases the safety margin available in Emergency Core Cooling System (ECCS) and reactor coolant pump overspeed analyses.

The reactor coolant pump is shown in Figure 5.4-1. The reactor coolant pump design parameters are given in Table 5.4-1.

Code and material requirements are provided in Section 5.2.

5.4.1.2 Design Description

The reactor coolant pump is a vertical, single stage, centrifugal, shaft seal pump designed to pump large volumes of main coolant at high temperatures and pressures.

The pump consists of three areas from bottom to top. They are the hydraulics, the shaft seals, and the motor.

1. The hydraulic section consists of an impeller, diffuser, casing, thermal barrier, heat exchanger, lower radial bearing, main flange, motor stand, and pump shaft.

2. The shaft seal section consists of three devices. They are the number 1 controlled leakage, film riding face seal and the number 2 and 3 rubbing face seals. These seals are contained within the main flange and seal housing.

   The number 1 seal insert houses a separate SHIELD Shutdown Seal (SSDS) component. The SSDS is a passive device actuated by high seal flow temperature resulting from a

loss of seal injection and component cooling water (CCW) cooling to the thermal barrier heat exchanger.

In the event of a loss of seal injection and CCW flow to the thermal barrier heat exchanger, reactor coolant system (RCS) fluid begins to travel along the reactor coolant pump (RCP) shaft and displaces the cooler seal injection water. Once the temperature within the number 1 seal reaches the actuation temperature range of the SSDS, the SSDS will activate to limit leakage from the RCS through the RCP seals. The loss of reactor coolant through the RCP seal package is limited when the SSDS polymer ring activates (clamps down) around the number 1 seal sleeve.

3. The motor section consists of a vertical solid shaft, squirrel cage induction type motor, and oil lubricated double Kingsbury type thrust bearing, two oil lubricated radial bearings, and a flywheel.

Attached to the bottom of the pump shaft is the impeller. The reactor coolant is drawn up through the impeller, discharged through passages in the diffuser, and out through the discharge nozzle in the side of the casing. Above the impeller is a thermal barrier heat exchanger which limits heat transfer between hot system water and seal injection water. Component cooling water is supplied to the thermal barrier heat exchanger.

High pressure seal injection water is introduced through the thermal barrier wall. A portion of this water flows through the radial bearing and the seals; the remainder flows down the shaft through the thermal barrier where it acts as a buffer to prevent system water from entering the radial bearing and seal section of the unit. The thermal barrier heat exchanger provides a means of cooling system water to an acceptable level in the event that seal injection flow is lost. The water lubricated journal type pump bearing, mounted above the thermal barrier heat exchanger, has a self-aligning spherical seat.

The reactor coolant pump motor bearings are of conventional design. The radial bearings are the segmented pad type, and the thrust bearings are tilting pad Kingsbury bearings. All are oil lubricated. The lower radial bearing and the thrust bearings are submerged in oil, and the upper radial bearing is oil fed from a viscosity pump integral with the thrust runner. Component cooling water is supplied to the two oil coolers on the pump motor.

The motor is a water/air cooled, Class B thermalastic epoxy insulated, squirrel cage induction motor. The rotor and stator are of standard construction and are cooled by air. Six resistance temperature detectors are located throughout the stator to sense the winding temperature. The top of the motor consists of a flywheel and an anti-reverse rotation device.

The internal parts of the motor are cooled by air. Integral vanes on each end of the rotor draw air in through cooling slots in the motor frame. This air passes through the motor with particular emphasis on the stator end turns. It is then routed to the external water/air heat exchangers, which are supplied with component cooling water.

Each motor has two such coolers, mounted diametrically opposed to each other. In passing through the coolers the air is cooled to below 122°F so that minimum heat is rejected to the Containment from the motors.

Each of the reactor coolant pumps is equipped with two vibration pickups. The shaft monitoring system is mounted near the coupling hub, in a horizontal plane to pickup radial vibrations of the pump shaft. The frame monitoring system is similar to the shaft monitoring system and is mounted at the top of the motor support. Signals from all the reactor coolant pumps are sent to a Control Room Monitoring Cabinet. The signals may be read on a vibration meter which shows the amplitude of pump shaft and frame vibration.

All parts of the pump in contact with the reactor coolant are austenitic stainless steel except for seals, bearings and special parts.

A removable shaft segment, the spool piece, is located between the motor coupling flange and the pump coupling flange; the spool piece allows removal of the pump seals with the motor in place. The pump shaft, seal housing, thermal barrier, main flange and motor stand can be removed from the casing without disturbing the reactor coolant piping. The flywheel is available for inspection by removing the cover.

5.4.1.3 Design Evaluation

5.4.1.3.1 Pump Performance

The reactor coolant pumps are sized to deliver flow at rates which equal or exceed the required flow rates. Initial Reactor Coolant System (RCS) tests confirm the total delivery capability. Thus, assurance of adequate forced circulation coolant flow is provided prior to initial plant operation.

The performance characteristic, shown in Figure 5.4-2, is common to all of the fixed speed mixed flow pumps, and the “knee” at about 45 percent design flow introduces no operational restrictions, since the pumps operate at full speed.

The Reactor Trip System ensures that pump operation is within the assumptions used for loss of coolant flow analyses, which also assures that adequate core cooling is provided to permit an orderly reduction in power if flow from a reactor coolant pump is lost during operation.

An extensive test program has been conducted for several years to develop the controlled leakage shaft seal for pressurized water reactor applications. Long term tests were conducted on less than full scale prototype seals as well as on full size seals. Operating plants continue to demonstrate the satisfactory performance of the controlled leakage shaft seal pump design.

The support of the stationary member of the number 1 seal (“seal ring”) is such as to allow large deflections, both axial and tilting, while still maintaining its controlled gap relative to the seal runner. Even if all the graphite were removed from the pump bearing, the shaft could not deflect far enough to cause opening of the controlled leakage gap. The “spring-rate” of the hydraulic forces associated with the maintenance of the gap is high enough to ensure that the ring follows the runner under very rapid shaft deflections.

Testing of pumps with the number 1 seal entirely removed (full system pressure on the number 2 seal) shows that relatively small leakage rates would be maintained for long periods of time; even if the number 1 seal fails entirely the number 2 seal would maintain these small leakage rates. The plant operator is warned of number 1 seal damage by the increase in number 1 seal leakoff. Following warning of excessive seal leakage conditions, the plant operator should secure the pumps then close the number 1 seal leakoff valve to the pump, to reduce possible damage.

Gross leakage from the pump does not occur if the proper operator action is taken subsequent to warning of excessive seal leakage conditions.

Seal Injection from the Chemical and Volume Control System (CVCS), to the Reactor Coolant Pumps is supplied with a bypass valve which may be opened if pump bearing temperature or seal water inlet temperature limits are exceeded and RCS pressure is greater than 100 psig. Opening the bypass valve ensures adequate cooling to the lower bearing when seal leakoff flow is insufficient to cool the radial bearing. Continued pump operation with the seal bypass line open, although not recommended, will not result in damage to the number 1 seal, nor will the performance of the number 2 and number 3 seals be affected.

The effect of loss of offsite power on the pump itself is to cause a temporary stoppage in the supply of injection flow to the pump seals and also of the cooling water for seal and bearing cooling. The emergency diesel generators are started automatically due to loss of offsite power so that component cooling flow is automatically restored. Seal water injection flow is subsequently restored by a charging pump automatically restarting on diesel power.

Consequences of the loss of component cooling water flow to RCS pumps are discussed in Section 9.2.2.

An extensive test program has been conducted for several years to develop the SHIELD Shutdown Seal (SSDS), a separate component within the number 1 seal insert. The SHIELD Shutdown Seal (SSDS) test program has demonstrated that the SSDS will remain in place and will not inadvertently actuate as a result of operational vibration, leakoff fluid flow, and plant design conditions. The SHIELD Shutdown Seal (SSDS) test program has demonstrated that the SSDS will not impact the performance of the number 1, number 2 or number 3 seals. SSDS actuation due to a loss of seal injection cooling will limit seal leakoff flow at the number 1 seal insert.

5.4.1.3.2 Coastdown Capability

It is important to reactor operation that the reactor coolant continues to flow for a short time after reactor trip. In order to provide this flow in a station blackout condition, each reactor coolant pump is provided with a flywheel. Thus, the rotating inertia of the pump, motor and flywheel is employed during the coastdown period to continue the reactor coolant flow. The coastdown flow transients are provided in the figures in Section 15.3.

After a loss of seal injection cooling, reference Section 5.4.1.2, the actuation of the SHIELD Shutdown Seal (SSDS) on a reactor coolant pump that is still rotating will not have any measurable effect on the reactor coolant pump coastdown.

The pump/motor system is designed for the Safe Shutdown Earthquake at the site and the integrity of the bearings is described in Section 5.4.1.3.3. Hence, it is concluded that the coastdown capability of the pumps is maintained even under the most adverse case of a blackout coincident with the Safe Shutdown Earthquake. Core flow transients and figures are provided in Sections 15.3.1 and 15.4.4.

5.4.1.3.3 Bearing Integrity

The design requirements for the reactor coolant pump bearings are primarily aimed at ensuring a long life with negligible wear, so as to give accurate alignment and smooth operation over long periods of time. The surface-bearing stresses are held at a very low value, and even under the most severe seismic transients do not begin to approach loads which cannot be adequately carried for short periods of time.

Because there are no established criteria for short time stress related failures in such bearings, it is not possible to make a meaningful quantification of such parameters as margins to failure, safety factors, etc. A qualitative analysis of the bearing design, embodying such considerations, gives assurance of the adequacy of the bearing to operate without failure.

Low oil levels in the lube oil sumps signal an alarm in the Control Room and require shutting down of the pump. Each motor bearing contains embedded temperature detectors, and so initiation of failure, separate from loss of oil, is indicated and alarmed in the Control Room as a high bearing temperature. This, again, requires pump shutdown. If these indications are ignored, and the bearing proceeded to failure, the low melting point of Babbitt metal on the pad surfaces ensures that sudden seizure of the shaft will not occur. In this event the motor continues to operate, as it has sufficient reserve capacity to drive the pump under such conditions. However, the high torque required to drive the pump will require high current which will lead to the motor being shutdown by the electrical protection systems.

5.4.1.3.4 Locked Rotor

It may be hypothesized that the pump impeller might severely rub on a stationary member and then seize. Analysis has shown that under such conditions, assuming instantaneous seizure of the impeller, the pump shaft fails in torsion just below the coupling to the motor, disengaging the flywheel and motor from the shaft. This constitutes a loss of coolant flow in the loop. Following such a postulated seizure, the motor continues to run without any overspeed, and the flywheel maintains its integrity, as it is still supported on a shaft with two bearings. Flow transients are provided in the figures in Section 15.3.3 for the assumed locked rotor.

There are no other credible sources of shaft seizure other than impeller rubs. Any seizure of the pump bearing is precluded by graphite in the bearing. SHIELD Shutdown Seal (SSDS) actuation on a reactor coolant pump that is still rotating will not effect the cooling flow to the reactor core. Any seizure in the seals results in a shearing of the antirotation pin in the seal ring. The motor has adequate power to continue pump operation even after the above occurrences. Indications of pump malfunction in these conditions are initially by high temperature signals from the bearing water temperature detector, and excessive number 1 seal leakoff indications respectively. Following these signals, pump vibration levels are checked. Excessive vibration indicates mechanical trouble and the pump is shutdown for investigation.

5.4.1.3.5 Critical Speed

The reactor coolant pump shaft is designed so that its operating speed is below its first critical speed. This shaft design, even under the most severe postulated transient, gives low values of actual stress.

5.4.1.3.6 Missile Generation

Precautionary measures taken to preclude missile formation from primary coolant pump components assure that the pumps will not produce missiles under any anticipated accident condition. Each component of the primary pump motors has been analyzed for missile generation. Any fragments of the motor rotor would be contained by the heavy stator. The same conclusion applies to the pump impeller because the small fragments that might be ejected would be contained by the heavy casing. Further discussion and analysis of missile generation is contained in Reference [1].

5.4.1.3.7 Pump Cavitation

The minimum net positive suction head required by the reactor coolant pump at running speed is approximately a 192 foot head (approximately 85 pounds per square inch (psi)). Proper operation of the controlled leakage seal is assured by requiring a minimum differential pressure of 200 psi across the seal. The corresponding RCS pressure is approximately 325 psi during a normal fill and vent evolution. During a vacuum fill evolution, the RCS pressure requirement is 250 psid across the seal. At this pressure the net positive suction head requirement is exceeded and the pump can be successfully operated.

5.4.1.3.8 Pump Overspeed Considerations

For turbine trips actuated by either the Reactor Trip System or the Turbine Protection System (with the exception of turbine trips caused by low lube oil pressure, thrust bearing, and local manual trip), the generator and reactor coolant pumps are maintained connected to the external network for 30 seconds to prevent any pump overspeed condition.

An electrical fault requiring immediate trip of the generator (with resulting turbine trip) could result in an overspeed condition. However, the Turbine Control System and the turbine intercept valves limit the overspeed to less than 120 percent. As additional back-up, the Turbine Protection System has redundant and diverse overspeed protection systems, usually set at about 110 percent (of turbine speed). In case a generator trip de-energizes the pump buses, the reactor coolant pump motors will be transferred to offsite power within 6 to 10 cycles. Further discussion of pump overspeed considerations is contained in Reference [1].

5.4.1.3.9 Anti-Reverse Rotation Device

Each of the reactor coolant pumps is provided with an anti-reverse rotation device in the motor. This anti-reverse mechanism consists of five pawls mounted on the outside diameter of the flywheel, a serrated ratchet plate mounted on the motor frame, a spring return for the ratchet plate, and two shock absorbers.

After the motor has slowed and come to a stop, the dropped pawls engage the ratchet plate and, as the motor tends to rotate in the opposite direction, the ratchet plate also rotates until it is stopped by the shock absorbers. The rotor remains in this position until the motor is energized again. When the motor is started, the ratchet plate is returned to its original position by the spring return.

As the motor begins to rotate, the pawls drag over the ratchet plate. When the motor reaches sufficient speed, the pawls are bounced into an elevated position and are held in that position by

friction resulting from centrifugal forces acting upon the pawls. Considerable shop testing and plant experience with the design of these pawls have shown high reliability of operation.

5.4.1.3.10 Shaft Seal Leakage

Leakage along the reactor coolant pump shaft is controlled by three shaft seals arranged in series such that reactor coolant leakage to the Containment is essentially zero. Charging flow is directed to each reactor coolant pump via a seal water injection filter (See Table 9.3-7). It enters the pumps through a connection on the thermal barrier flange and is directed down to a point between the pump shaft bearing and the thermal barrier cooling coils. Here the flow enters the shaft annulus; a portion flows down through the thermal barrier labyrinth, past the cooling coils and into the RCS; the remainder flows up the pump shaft annulus cooling the lower shaft bearing. This flow provides a backpressure on the number 1 seal and a controlled flow through the seal. During normal operation of the pump and seal, most of the flow leaves the pump through the number 1 seal leakoff line. Minor flow passes through the number 2 seal and leakoff line. A back flush injection of 800 cubic centimeters per hour (cc/hr) from a standpipe flows into the number 3 seal between its “double dam” seal area. At this point the flow divides with half flushing through one side of the seal and out the number 2 seal leakoff while the remaining half flushes through the other side and out the number 3 leakoff. This arrangement assures essentially zero leakage of reactor coolant or trapped gases from the pump.

In the event that seal injection and CCW flow to the thermal barrier heat exchanger is lost, reactor coolant begins to travel along the RCP shaft and displace the cooler seal injection water. Once the temperature downstream of the number 1 seal reaches the SSDS activation temperature range, the SSDS activates by the retraction of a thermal actuator, causing the SSDS piston and polymer rings to constrict around the shaft sleeve. SSDS actuation controls shaft seal leakage and limits the loss of reactor coolant through the RCP seal package.

5.4.1.3.11 Seal Discharge Piping

Discharge pressure from the number 1 seal is reduced to that of the volume control tank. Water from each pump number 1 seal is piped to a common manifold, and through the seal water return filter and through the seal water heat exchanger where the temperature is reduced to that of the volume control tank. The number 2 leakoff lines dump number 2 seal leakage to the reactor coolant drain tank; the number 3 leakoff lines discharge number 3 seal leakage to the Containment sump. A description of the RCS pump seal flow is given in Section 9.3.4.1.2.1. Details are shown on Figure 9.3-10 sheet 1.

5.4.1.4 Tests and Inspections

The reactor coolant pumps can be inspected in accordance with the American Society of Mechanical Engineers (ASME) Code, Section XI, for inservice inspection of nuclear reactor coolant systems.

All full penetration pressure boundary welds were prepared with a smooth transition between the weld and parent metal to facilitate Non-destructive examination (NDE) required by the construction Code and for in-service examinations required by ASME Section XI.

The design enables disassembly and removal of the pump internals for usual access to the internal surface of the pump casing.

The reactor coolant pump quality assurance program is given in Table 5.4-2.

5.4.1.5 Pump Flywheels

The integrity of the reactor coolant pump flywheel is assured on the basis of the following design and quality assurance procedures.

5.4.1.5.1 Design Basis

The calculated stresses at operating speed are based on stresses due to centrifugal forces. The stress resulting from the interference fit of the flywheel on the shaft is less than 2000 psi at zero speed, but this stress becomes zero at approximately 600 revolutions per minute (rpm) because of radial expansion of the hub. The primary coolant pumps run at approximately 1190 rpm and may operate briefly at overspeeds up to 109 percent (1295 rpm) during loss of outside load. For conservatism, however, 125 percent of operating speed was selected as the design speed for the primary coolant pumps. The flywheels are given a preoperational test of 125 percent of the maximum synchronous speed of the motor.

5.4.1.5.2 Fabrication and Inspection

The flywheel consists of two thick plates bolted together. The flywheel material is produced by a process that minimizes flaws in the material and improves its fracture toughness properties, such as vacuum degassing, vacuum melting, or electroslag remelting. Each plate is fabricated from SA-533, Grade B, Class 1 steel. Supplier certification reports are available for all plates and demonstrate the acceptability of the flywheel material on the basis of the requirements of Regulatory Guide 1.14.

Flywheel blanks are flame-cut from the SA-533, Grade B, Class 1 plates with at least 1/2 inch of stock left on the outer and bore radii for machining to final dimensions. The finished machined bores, keyways, and drilled holes are subjected to magnetic particle or liquid penetrant examinations. The finished flywheels are subjected to 100 percent volumetric ultrasonic inspection per paragraphs NB-2532.1 and NB-2532.2 of the ASME Code, Section III.

The reactor coolant pump motors are designed such that the flywheel is available by removing the cover to provide access to allow an inservice inspection program in accordance with requirements of Section XI of the ASME code. The inservice inspection program is in accordance with Technical Specification 5.5.7.

5.4.1.5.3 Acceptance Criteria and Compliance with Regulatory Guide 1.14

The reactor coolant pump motor flywheel shall conform to the following material acceptance.

1. The nil-ductility transition temperature (NDTT) of the flywheel material shall be obtained by two drop weight tests (DWT) which will exhibit “no-break” performance at 20°F in accordance with ASTM-E-208. The above drop weight tests demonstrate that the NDTT of the flywheel material is no higher than 10°F.

2. A minimum of three Charpy V-notch impact specimens from each plate shall be tested at ambient (70°F) temperature in accordance with the specification ASTM-E-23. The Charpy V- notch (Cv) energy in both the parallel and normal orientation with respect to the

rolling direction of the flywheel material shall be at least 50 foot pounds at 70°F to demonstrate compliance with Regulatory Guide 1.14. A lower bound K_{Id} reference curve (see Figure 5.4-3) has been constructed from dynamic fracture toughness data generated in SA-533, Grade B, Class 1 steel [2]. All data points are plotted on the temperature scale relative to the NDTT. The construction of the lower bound curve below which no single test point falls, combined with the use of dynamic data when flywheel loading is essentially static, together represent a large degree of conservatism. Reference of this curve to the guaranteed NDTT of +10°F indicates that, at the predicted flywheel operating temperature of 110°F, the minimum fracture toughness is in excess of 10 KSI-in$^{1/2}$. This conforms to the Regulatory Guide 1.14 requirement that the dynamic stress intensity factor must be at least 100 KSI-in$^{1/2}$.

The fracture toughness properties of the Unit 1 and Unit 2 reactor coolant pump flywheels are shown in Table 5.4-19 and Table 5.4-19A, respectively.

Thus, it is concluded that flywheel plate materials are suitable for use and can meet Regulatory Guide 1.14 acceptance criteria on the bases of suppliers certification data.

5.4.2 STEAM GENERATOR

5.4.2A Unit 1 Steam Generators

5.4.2A.1 Steam Generator Materials

5.4.2A.1.1 Selection and Fabrication of Materials

All pressure boundary materials used in the steam generator are selected and fabricated in accordance with the requirements of Section III of the ASME Code. A general discussion of materials specifications is given in Section 5.2.3, with types of materials listed in Tables 5.2-2 and 5.2-3. Fabrication of reactor coolant pressure boundary materials is also discussed in Section 5.2.3, particularly in Sections 5.2.3.3 and 5.2.3.4.

Testing has justified the selection of corrosion-resistant thermally treated Inconel Alloy 690, a nickel-chromium-iron alloy (ASME-SB-163), for the steam generator tubes and divider plate. The interior surfaces of the reactor coolant channel heads and nozzles are clad with austenitic stainless steel. The primary side of the tubesheet is weld clad with Inconel (ASME-SB-163). The tubes are hydraulically expanded into the tubesheet holes after the ends are tack rolled and seal welded. The flush autogenous welds made to the tubesheet cladding are performed in compliance with Sections III and IX of the ASME Code and are thoroughly inspected before each tube is expanded.

Code cases used in material selection are discussed in Section 5.2.1. The extent of conformance with Regulatory Guides 1.84 and 1.85 is also discussed in Section 5.2.1 and Appendix 1A(N).

During manufacture, cleaning is performed on the primary and secondary sides of the steam generator in accordance with written procedures which follow the guidance of Regulatory Guide 1.37, “Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants,” and the ANSI Standard N45.2.1-1980,

"Cleaning of Fluid Systems and Associated Components for Nuclear Power Plants." Onsite cleaning and cleanliness control is in accordance with Westinghouse recommendations given in Westinghouse process specifications, as discussed in Section 5.2.3.4.

The fracture toughness of the materials is discussed in Section 5.2.3.3 and 5.3.1.5. Adequate fracture toughness of ferritic materials in the reactor coolant pressure boundary is provided by compliance with Appendix G of 10CFR50 and with NB-2300 of Section III of the ASME Code. Per the discussion in Section 5.4.2.3, consideration of fracture toughness is only necessary for materials used for Class 1 components.

5.4.2A.1.2 Steam Generator Design Effects on Materials

Several features are employed to control the regions where deposits would tend to accumulate. To avoid extensive crevice areas between the tube and tubesheet, the tubes are hydraulically expanded for the full depth surface of the tubesheet, after their ends are seal welded to the Inconel (ASME-SB-163) cladding on the primary side of the tubesheet. In addition, all tube support plates have trefoil-shaped holes configured to maximize flow adjacent to the tube so as to reduce the potential for chemical concentration in the regions where the tubes pass through the tube support plate.

5.4.2A.1.3 Compatibility of Steam Generator Tubing with Primary and Secondary Coolants

As mentioned in Section 5.4.2A.1.1, corrosion tests, which subjected the steam generator tubing material Inconel-600 (ASME-SB-163) to simulated steam generator water chemistry, have indicated that the loss due to general corrosion over the 40 year plant life is insignificant compared to the tube wall thickness. Testing to investigate the susceptibility of heat exchanger construction materials to stress corrosion in caustic and chloride aqueous solutions has indicated that Inconel-600 has excellent resistance to general corrosion in severe operating water conditions. Many reactor years of successful operation have shown the same low general corrosion rates as indicated by the laboratory tests. Recent operating experience, however, has revealed areas on secondary surfaces where localized corrosion rates were significantly greater than the low general corrosion rates. Both intergranular corrosion and tube wall thinning were experienced in localized areas, although not at the same location or under the same environmental conditions (water chemistry, sludge composition). These localized areas of corrosion posed no threat to the public health and safety but were of concern because of their possible effect on plant availability.

To eliminate these localized areas of corrosion over the long term operation of the unit, it was decided that the use of phosphates for steam generator control would be eliminated. The adoption of the all volatile treatment (AVT) control program will minimize the possibility for recurrence of the tube wall thinning phenomenon related to poor phosphate chemistry control. Successful AVT operation requires maintenance of low concentrations of impurities in the steam generator water, thus reducing the potential for formation of highly concentrated solutions in low flow zones, one of the precursors for a corrosion mechanism. By restriction of the total alkalinity in the steam generator and prohibition of extended operation with free alkalinity, the AVT program will prevent the recurrence of intergranular corrosion in localized areas due to excessive levels of free caustic.

Laboratory testing has shown that the Inconel-600 tubing is compatible with the AVT environment. Isothermal corrosion testing in AVT water has shown that Inconel-600 at

engineering stresses does not suffer intergranular stress corrosion cracking in extended exposures. Model boiler tests being conducted by Westinghouse have shown quite favorable results for AVT to date.

AVT chemistry control has been employed in plant operations successfully for considerable periods. Plants with stainless steel tubes which have demonstrated successful AVT operation include Selni, Sena and Yankee-Rowe. Selni has operated with AVT since 1964, Sena since 1966, and Yankee-Rowe since 1967.

AVT is the water treatment of preference for plants having Inconel-600 steam generator tubing. Among the plants which have operated successfully with AVT is Maine-Yankee (since 1972). The defects encountered in Inconel-600 tubed steam generators treated with AVT water treatment are primarily the result of the use of copper alloy materials in the secondary coolant side and continued operation with in-leakage of condenser cooling water. The defects are not expected to occur or will be minimized at CPNPP since the major components in the CPNPP secondary coolant side, including the condenser and moisture separator reheaters, are composed of non-copper base alloys. In addition, the appropriate corrective actions will be performed should condenser cooling water in-leakage occur.

Operating experience, verified in numerous steam generator inspections, indicates that the tube degradation associated with phosphate water treatment is not occurring where only AVT has been utilized. Adherence to the AVT chemical specifications and close monitoring of the condenser integrity will assure the continued good performance of the steam generator tubing.

Additional extensive operating data has been accumulated since the conversion to AVT chemistry. A comprehensive program of steam generator inspections, including the requirements of Regulatory Guide 1.83, with the exceptions as stated in Appendix 1A(N) has ensured detection and correction of any unanticipated degradation that occurred in the steam generator.

Even though the tube degradation associated with phosphate water treatment did not occur, numerous other modes of degradation mechanisms have occurred in A600MA tubing after several EFPY of operation. The major modes of degradation experienced are primary water stress corrosion cracking (PWSCC), which occurs on the inside diameter of the steam generator tubing or the primary side, and outside diameter stress corrosion cracking (ODSCC) which occurs on the secondary side of the steam generator tubing. For the purposes of this discussion, ODSCC will also include contribution of intergranular attack (IGA).

The degradation of A600MA steam generators has resulted in high operational costs to the industry as a result of increased inspection and maintenance requirements and also through significant losses in availability. Consequently, the large majority of the A600MA Steam Generators has been replaced.[Reference 8] This includes Comanche Peak Unit 1. The replacement material for Comanche Peak Unit 1 is A690TT. This alloy has been shown through extensive laboratory testing and limited operational experience to be superior to A600MA tubing in both primary and secondary water chemistry environments. The results have been extensively evaluated and documented by EPRI in References 9-11, and will be summarized in the following discussion, including assessment of A600TT, which is the tubing material in Comanche Peak Unit 2.

The Unit 2 Model D5 was specifically considered in Reference 2. Improvement factors for the performance of A600TT tubing in the Model D5 Steam Generator versus A600MA tubing were calculated for specific degradation mechanisms as follows:

- The improvement factor for PWSCC = 5
- The improvement factor for ODSC = 4.5
- The improvement factor for the tube support ODSCC = 2.2

The overall predicted degradation for Model D5 (A600TT Tubing) Steam Generators calculations were based on an assumed hot leg temperature of 618 Degree F. Aggregate failures for all degradation modes at an estimated end of life of 51.0 EFPY are predicted to be 6.5% of the tubes. This represents a large improvement compared to A600MA performance.

Reference 9 calculated the predicted degradation for A690TT Tubing Steam Generators also based on an assumed hot leg temperature of 618 Deg. F. The results of this predictions was complemented by the improvement factor for PWSCC that was developed in another EPRI study, MRP-111, Reference 10. MRP-111 did not develop an improvement factor for ODSCC, therefore no comparison can be made.

The specific results of the predictions from Reference 9 and Reference 10 will not covered in detail as Reference 11 provided an extensive update to the improvement factors for A690TT. However, it should be noted from Reference 9 that the results from AVB wear was not specifically modeled due to the improved design and degradation history to date. Therefore the new designs were assumed to be essentially immune to AVB wear. AVB wear can be considered to be evaluated in the miscellaneous category. For the aggregate of the degradation mechanisms, no tubes are predicted to be plugged at the end of life of 51.0 EFPY. The miscellaneous category was predicted to be the leading cause of degradation and is predicted to be 5 tubes per 10,000 Steam Generator tubes at end of life.

Reference 11 provided an updated improvement factor for A690TT utilizing the following factors:

- Material improvement factors derived from additional laboratory testing data
- Overall improvement factors derived from plant experience was introduced into the evaluation, even though the actual experience time is still too limited to determine a improvement factor that is not very conservative
- Environmental weighing factors were considered by the derivation of an average material improvement factor for the environments tested in the laboratory studies
- Internals design improvement factors were considered such as the following:
    - Tube sheet expansion mechanism
    - Tube sheet crevice depth
    - Tube support material and geometry

- Overall improvement factors were calculated for replacement versus original steam generators using the Ginna Steam Generator (SG) replacement case

The updated improvement factors for A690TT versus A600MA for each specific degradation mechanism were calculated as follows:

- Expansion zone circumferential PWSCC = >40
- Expansion zone axial PWSCC = >200
- Denting PWSCC = >800
- Tube sheet ODSCC = >230
- Tube support ODSCC = >60

Utilizing these improvement factors, the time for the first observation of these failures were calculated for the Ginna replacement SG's with A690TT tubing based on the first observation of these failures in the original SG's with A600MA tubing. The results are as follows:

- PWSCC may be experienced in the Ginna replacement SG's after 200 EFPY.
- ODSCC may be experienced in the Ginna replacement SG's after 1450 EFPY.

While these estimates extend well beyond a reasonable extrapolation time for a typical plant life, they provide a great deal of confidence in the fact that replacement design Steam Generators with A690TT will operate well beyond their expected lifetime without experiencing significant degradation.

5.4.2A.1.4 Monitoring of Secondary Side Water Chemistry

The chemistry of the steam generator water and condensate is continuously monitored as described in Section 9.3.2 and 10.4.16 respectively. The conductivity and pH of the secondary side are continuously measured as are dissolved oxygen, sodium and hydrazine content. Addition rates of secondary side chemicals are controlled by the continuous on-line analyzers. Steam generator blowdown is continuous; the rate can be adjusted using the water chemistry as a basis. The steam generator blowdown processing system is described in Section 10.4.8.

The approach to monitoring secondary side water chemistry complies with the approach outlined in Branch Technical Position MTEB 5-3.

5.4.2A.1.5 Cleanup of Secondary Side Materials

Several methods are employed to clean operating steam generators of corrosion causing secondary side deposits. Sludge lancing, a procedure in which a hydraulic jet inserted through an access opening (inspection port) loosens deposits which are removed by means of a suction pump, can be performed when the need is indicated by the results of steam generator tube inspection. Blowdown procedures are performed as deemed necessary by regular water chemistry testing. For the Delta 76 steam generators, the top of the tubesheet suction holes

located along the tubelane facilitates the efficient removal of impurities that have accumulated on the tubesheet.

5.4.2A.2 Steam Generator Inservice Inspection

5.4.2A.2.1 Steam Generator Design Characteristics For Inservice Inspection

The steam generator is designed to permit inservice inspection of Class 1 and 2 components, including individual tubes. The design aspects that provide access for inspection and the proposed inspection program comply with the edition of Section XI of the ASME Code, Division 1, “Rules for Inspection and Testing of Components of Light- Water-Cooled Plants,” required by 10 CFR 50.55a, paragraph g. A number of access openings make it possible to inspect and repair or replace a component according to the techniques specified. These openings include four manways, two of them for inspection and maintenance of the steam dryer. Also the Unit 1 steam generators have four 6.0 inch diameter handholes, four 4.0 inch diameter inspection openings and sixteen 2.5 inch diameter inspection openings.

5.4.2A.2.2 Program For Inservice Inspection Of Steam Generator Tubing

Steam generator tubing will be inspected in accordance with:

1. The recommendations given in Regulatory Guide 1.83, “Inservice Inspection of Steam Generator Tubes,” Revision 1, July 1975, and

2. The requirements of ASME Section XI (Edition and Addenda as required by 10CFR50.55a), Subarticle IWB-2413.

3. Comanche Peak Nuclear Power Plant Technical Specifications Section 5.5.9.

The program consists of the following areas:

Inspection Equipment & Procedures

Eddy current testing equipment will be used to inspect the tubing and shall be sensitive enough to detect imperfections of 20 percent or more through the tube wall.

Baseline Inspection

1. All tubes in the steam generators shall be inspected by eddy current or alternative techniques prior to service to establish a baseline condition of the tubing.

2. If a major change in their secondary water chemistry is made during plant lifetime, a baseline inspection will be conducted before resumption of power operation.

Steam Generator Sample Selection and Inspection

Steam Generator Sample Selection and Inspection shall be consistent with CPNPP TS 5.5.9.

Steam Generator Tube Inspection Techniques and Tubes Left in Service Based on Indication Size (GL 97-05).

Traditionally, eddy-current inspection techniques are relied upon to assess the condition of steam generator tubes. Although the eddy-current method is a proven technique for detecting tube degradation, the ability to depth size indications is possible only for specific modes of degradation. Specifically, tube degradation from inter-granular attack (IGA) and stress corrosion cracking (SCC), major modes of steam generator tube degradation, are difficult to size with eddy-current inspection techniques because of a number of complicating variables. Theoretically, there is a relationship between the depth of penetration of a defect and the eddy-current signal response; in practice, however, the relationship between signal voltage or phase angle and the degradation depth is influenced by many other variables. Oxide deposits, variability of tube material properties and geometry, degradation morphology, human factors, and eddy-current data analysis and acquisition practices are some of the factors that can significantly alter a depth estimation of steam generator tube degradation. The depth of several specific forms of volumetric steam generator tube degradation can be sized with a reasonable degree of accuracy; however, qualifying techniques for sizing of some forms of degradation, e.g., IGA and SCC, are typically problematic.

CPNPP has adopted the guidelines of Energy Institute (NEI) letter 97-06, “Steam Generator Program Guidelines”. NEI 97-06 requires following the inspection guidelines contained in the latest revision of the Electric Power Research Institute (EPRI) Pressurized Water Reactor (PWR) Steam Generator Examination Guidelines.

Appendix H, “Performance Demonstration for Eddy Current Examination,” of the EPRI “PWR Steam Generator Examination Guidelines,” latest revision, provides guidance on the qualification of steam generator tubing examination techniques and equipment used to detect and size indications. Damage mechanisms are divided into the following categories: thinning, pitting, wear, outside diameter inter-granular attack/stress-corrosion cracking (IGA/SCC), primary-side SCC, and impingement damage for qualification.

For qualification purposes, test samples are used to evaluate detection and sizing capabilities. While pulled tube samples are preferred, fabricated samples may be used. If fabricated test samples are used, the samples are verified to produce signals similar to those being observed in the field in terms of signal characteristics, signal amplitude, and signal-to-noise ratio. Samples are examined to determine the actual through wall defect measurements as part of the Appendix H qualification process.

The procedures at CPNPP are developed in accordance with Appendix H which specifies the essential variables for each technique. These essential variables are associated with an individual instrument, probe, cable, or particular equipment configurations. Additionally, certain techniques have undergone testing and review to quantify sizing performance. The sizing data set includes the detection data set for the technique with additional requirements for number and composition of the grading units.

At CPNPP only tubes with indications of wear type degradation found at anti-vibration bars and tube support plates and which are less than the Technical Specification value of 40% are permitted to remain in service. All tubes with crack-like degradation are plugged upon detection.

5.4.2A.3 Design Bases

Steam generator design data are given in Table 5.4-3. Code classifications of the steam generator components are given in Section 3.2. Although the ASME classification for the

secondary side is specified to be Class 2, the current philosophy is to design all pressure retaining parts of the steam generator, and thus, both the primary and secondary pressure boundaries, to satisfy the criteria specified in Section III of the ASME Code for Class 1 components. The design stress limits, transient conditions and combined loading conditions applicable to the steam generator, are discussed in Section 3.9N.1. Estimates of radioactivity levels anticipated in the secondary side of the steam generators during normal operation, and the bases for the estimates, are given in Chapter 11. The accident analysis of a steam generator tube rupture is discussed in Chapter 15.

The internal moisture separation equipment is designed to ensure that moisture carryover does not exceed 0.10 percent of weight under the following conditions:

1. Steady state operation up to 100 percent of full load steam flow, with water at the normal operating level.

2. Loading or unloading at a rate of 5 percent of full power steam flow per minute in the range from 15 to 100 percent of full load steam flow.

3. A step load change of 10 percent of full power in the range from 15 to 100 percent full load steam flow.

The water chemistry on the reactor side is selected to provide the necessary boron content for reactivity control and to minimize corrosion of RCS surfaces. The water chemistry of the steam side and its effectiveness in corrosion control are discussed in Chapter 10. Compatibility of steam generator tubing with both primary and secondary coolants is discussed further in Section 5.4.2A.1.3.

The steam generator is designed to prevent unacceptable damage from mechanical or flow induced vibration. Tube support adequacy is discussed in Section 5.4.2A.5.3. The tubes and tubesheet are analyzed and confirmed to withstand the maximum accident loading condition as it is defined in Section 3.9(N).1. Further consideration is given in Section 5.4.2A.5.4 to the effect of tube wall thinning on accident condition stresses.

5.4.2A.4 Design Description

The steam generator shown in Figure 5.4-4A is a Delta 76, vertical shell and U-tube evaporator with integral moisture separating equipment. On the primary side, the reactor coolant flows through the inverted U-tubes, entering and leaving through nozzles located in the hemispherical bottom head of the steam generator. The head is divided into inlet and outlet chambers by a vertical divider plate extending from the head to the tubesheet. Steam is generated on the shell side, flows upward and exits through the outlet nozzle at the top of the vessel. During normal operation, subcooled feedwater is supplied from the plant feedwater system and enters the steam generator through the main (or sometimes auxiliary) feedwater inlet nozzle located in the upper shell of the steam generator. Inside the steam generator, the feedwater is joined by the recirculating water separated from the steam-water mixture by the moisture separators, producing a subcooled mixture in the upper plenum that is slightly below saturation temperature. The subcooled mixture flow down the downcomer annulus formed by the I.D. of the steam generator shell and the O.D. of the wrapper, then enters the lower tube bundle through an opening between the lower edge of the wrapper and the secondary surface of the tubeplate. Subsequently the water-steam mixture flows upward through the tube bundle and into the steam

drum section, where individual centrifugal moisture separators remove most of the entrained water from the steam. The steam continues to the secondary separators for further moisture removal, increasing its quality to a minimum of 99.9 percent. The moisture separators recirculate the separated water through the annulus between the shell and tube bundle wrapper. The returning flow then combines with the feedwater for another passage through the steam generator. Dry steam exits through the outlet nozzle which is provided with a steam flow restrictor, described in Section 5.4.4.

5.4.2A.5 Design Evaluation

5.4.2A.5.1 Forced Convection

The limiting case for heat transfer capability is the “nominal 100 percent design” case. The steam generator effective heat transfer coefficient is based on the coolant conditions of temperature and flow for this case. The best estimate for the heat transfer coefficient applied in steam generator design calculations and plant parameter selection is approximately 1250 Btu/hr-ft2-oF. The coefficient incorporates a specified fouling factor resistance of 0.00011 hr/ft2-oF/Btu which is the value selected to account for the differences in the measured and calculated heat transfer performance as well as provide design margin. Although margin for tube fouling is available, operating experience to date has not indicated that steam generator performance decreases over a long time period. Adequate tube area is selected to ensure that the full design heat removal rate is achieved.

5.4.2A.5.2 Natural Circulation Flow

The driving head created by the change in coolant density as it is heated in the core and rises to the outlet nozzle initiates convection circulation. This circulation is enhanced by the fact that the steam generators, which provide a heat sink, are at a higher elevation than the reactor core which is the heat source. Thus natural circulation is assured for the removal of decay heat during hot shutdown in the unlikely event of loss of forced circulation.

5.4.2A.5.3 Mechanical and Flow Induced Vibration Under Normal Operation

In the design of Westinghouse steam generators, the potential for tube wall degradation attributable to mechanical or flow induced excitation has been thoroughly evaluated. The evaluation included detailed analyses of the tube support systems for various mechanisms of tube vibration.

The primary cause of tube vibration in heat exchangers is hydrodynamic excitation due to secondary fluid flow on the outside of the tubes. In the range of normal steam generator operating conditions, the affects of primary fluid flow inside the tubes and mechanically induced tube vibration are considered to be negligible.

To evaluate flow induced tube vibration in the tube bundle, detailed analyses have been performed employing data from operating plants, full and partial scale model tests and analytical tube vibration models. Parallel flow analyses indicate that the flow velocities are sufficiently low such that they result in negligible fatigue and vibratory amplitudes. The support system, therefore, is deemed adequate with regard to parallel flow excitation.

To evaluate cross flow at the exit of the downcomer flow to the tube bundle and at the top of the bundle in the U-bend area, Westinghouse performed an experimental research program of cross flow in tube arrays with the specific parameters of the steam generator. Air and water model tests were employed. The results of this research indicate that these regions of the bundle are not subject to the vortex shedding mechanism of tube excitation. Vortex shedding was found not to be a significant mechanism in these two regions for the following reasons:

1. Flow turbulence in the downcomer and tube bundle inlet region inhibit the formation of Von Karman vorticies.

2. Both axial and cross flow velocity components exist on the tubes. The axial flow component disrupts the Von Karman vortices.

This research program was also the basis for evaluation of the fluidelastic mechanism due to cross flow at the tubesheet and at the top of the bundle in the U-bend area. The evaluation showed the adequacy of the tube support arrangement.

Flow turbulence can result in some tube excitation in these regions. This excitation is of little concern, however, since:

1. Maximum stresses in the tubes are below the fatigue endurance limit of the tube material, and

2. Tube support arrangements preclude significant vibratory motion.

In summary, tube vibration has been thoroughly evaluated. Mechanical and primary flow excitation are considered negligible. Secondary flow excitation has been evaluated. From this evaluation, it is concluded that if tube vibration does occur, the magnitude will be limited. Tube fatigue due to the vibration is judged to be negligible. Any tube wear resulting from the tube vibration would be limited and would progress slowly. This allows use of a periodic tube inservice inspection program for detection and follow of any tube wear. This inservice inspection program, in conjunction with tube plugging criteria, provides for safe operation of the steam generators.

5.4.2A.5.4 Allowable Tube Wall Thinning Under Accident Conditions

For tube plugging, the criteria is found in Section 1(A)N.

5.4.2A.6 Quality Assurance

The steam generator quality assurance program is given in Table 5.4-4A.

Radiographic inspection and acceptance standard shall be in accordance with the requirements of Section III of the ASME Code.

Liquid penetrant inspection is performed on weld deposited tubesheet cladding, channel head cladding, and tube-to-tubesheet weldments. Liquid penetrant inspection and acceptance standards are in accordance with the requirements of Section III of the ASME Code.

Magnetic particle inspection is performed on the tubesheet forging, channel head forging, nozzle forgings, and the following weldments:

1. Nozzles to shell.

2. Instrument connection (primary and secondary).

3. Temporary attachments after removal.

4. All accessible pressure containing welds after hydrostatic test.

Magnetic particle inspection and acceptance standard are in accordance with requirements of Section III of the ASME Code.

An ultrasonic test is performed on the tubesheet forging, tubesheet cladding, channel head forging, secondary shell and head forging and nozzle forgings.

The heat transfer tubing is subjected to eddy current and ultrasonic tests.

Hydrostatic tests are performed in accordance with Section III of the ASME Code.

In addition, the heat transfer tubes shall be subjected to a hydrostatic test pressure prior to installation into the vessel which is not less than 1.25 times the primary side design pressure.

5.4.2B Unit 2 Steam Generators

5.4.2B.1 Steam Generator Materials

5.4.2B.1.1 Selection and Fabrication of Materials

All pressure boundary materials used in the steam generator are selected and fabricated in accordance with the requirements of Section III of the ASME Code. A general discussion of materials specifications is given in Section 5.2.3, with types of materials listed in Tables 5.2-2 and 5.2-3. Fabrication of reactor coolant pressure boundary materials is also discussed in Section 5.2.3, particularly in Sections 5.2.3.3 and 5.2.3.4.

Testing has justified the selection of corrosion-resistant Inconel- 600, a nickel-chromium-iron alloy (ASME-SB-163), for the steam generator tubes and divider plate. The interior surfaces of the reactor coolant channel heads and nozzles are clad with austenitic stainless steel. The primary side of the tubesheet is weld clad with Inconel (ASME-SB-163). The tubes are mechanically or explosively expanded into the tubesheet holes after the ends are tack rolled and seal welded. The recessed fusion welds made to the tubesheet cladding are performed in compliance with Sections III and IX of the ASME Code and are thoroughly inspected before each tube is expanded.

Approximately 100 tubes in the preheat section of unit 1 and 2 steam generators have been hydraulically expanded at the “B” and “D” support plate elevations to reduce the tube vibration phenomena experienced in Westinghouse counterflow preheat steam generators. The material properties of the expanded section of the tubes are discussed in Reference 4.

Code cases used in material selection are discussed in Section 5.2.1. The extent of conformance with Regulatory Guides 1.84 and 1.85 is also discussed in Section 5.2.1 and Appendix 1A(N).

During manufacture, cleaning is performed on the primary and secondary sides of the steam generator in accordance with written procedures which follow the guidance of Regulatory Guide 1.37, “Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants,” and the ANSI Standard N45.2.1- 1973, “Cleaning of Fluid Systems and Associated Components for Nuclear Power Plants.” Onsite cleaning and cleanliness control is in accordance with Westinghouse recommendations given in Westinghouse process specifications, as discussed in Section 5.2.3.4.

The fracture toughness of the materials is discussed in Section 5.2.3.3 and 5.3.1.5. Adequate fracture toughness of ferritic materials in the reactor coolant pressure boundary is provided by compliance with Appendix G of 10CFR50 and with NB-2300 of Section III of the ASME Code. Per the discussion in Section 5.4.2.3, consideration of fracture toughness is only necessary for materials used for Class 1 components.

5.4.2B.1.2 Steam Generator Design Effects on Materials

Several features are employed to control the regions where deposits would tend to accumulate. To avoid extensive crevice areas between the tube and tubesheet, the tubes are roller, explosively or hydraulically expanded for the full depth surface of the tubesheet, after their ends are seal welded to the Inconel (ASME-SB-163) cladding on the primary side of the tubesheet. A flow distribution plate located below the preheat section encourages recirculating flow to sweep the tubesheet before turning upward through the tube bundle. This plate also serves to separate the tubesheet from the colder feedwater entering at the preheat section. A separate auxiliary feedwater nozzle provided in the upper shell avoids introducing cold water into the preheat section, and, thus, maximizes the integrity of steam generator materials.

5.4.2B.1.3 Compatibility of Steam Generator Tubing with Primary and Secondary Coolants

As mentioned in Section 5.4.2.B1.1, corrosion tests, which subjected the steam generator tubing material Inconel-600 (ASME-SB-163) to simulated steam generator water chemistry, have indicated that the loss due to general corrosion over the 40 year plant life is insignificant compared to the tube wall thickness. Testing to investigate the susceptibility of heat exchanger construction materials to stress corrosion in caustic and chloride aqueous solutions has indicated that Inconel-600 has excellent resistance to general corrosion in severe operating water conditions. Many reactor years of successful operation have shown the same low general corrosion rates as indicated by the laboratory tests. Recent operating experience, however, has revealed areas on secondary surfaces where localized corrosion rates were significantly greater than the low general corrosion rates. Both intergranular corrosion and tube wall thinning were experienced in localized areas, although not at the same location or under the same environmental conditions (water chemistry, sludge composition). These localized areas of corrosion posed no threat to the public health and safety but were of concern because of their possible effect on plant availability.

To eliminate these localized areas of corrosion over the long term operation of the unit, it was decided that the use of phosphates for steam generator control would be eliminated. The adoption of the all volatile treatment (AVT) control program will minimize the possibility for recurrence of the tube wall thinning phenomenon related to poor phosphate chemistry control. Successful AVT operation requires maintenance of low concentrations of impurities in the steam generator water, thus reducing the potential for formation of highly concentrated solutions in low flow zones, one of the precursors for a corrosion mechanism. By restriction of the total alkalinity

in the steam generator and prohibition of extended operation with free alkalinity, the AVT program will prevent the recurrence of intergranular corrosion in localized areas due to excessive levels of free caustic.

Laboratory testing has shown that the Inconel-600 tubing is compatible with the AVT environment. Isothermal corrosion testing in AVT water has shown that Inconel-600 at engineering stresses does not suffer intergranular stress corrosion cracking in extended exposures. Model boiler tests being conducted by Westinghouse have shown quite favorable results for AVT to date.

AVT chemistry control has been employed in plant operations successfully for considerable periods. Plants with stainless steel tubes which have demonstrated successful AVT operation include Selni, Sena and Yankee-Rowe. Selni has operated with AVT since 1964, Sena since 1966, and Yankee-Rowe since 1967.

AVT is the water treatment of preference for plants having Inconel-600 steam generator tubing. Among the plants which have operated successfully with AVT is Maine-Yankee (since 1972). The defects encountered in Inconel-600 tubed steam generators treated with AVT water treatment are primarily the result of the use of copper alloy materials in the secondary coolant side and continued operation with in-leakage of condenser cooling water. The defects are not expected to occur or will be minimized at CPNPP since the major components in the CPNPP secondary coolant side, including the condenser and moisture separator reheaters, are composed of non-copper base alloys. In addition, the appropriate corrective actions will be performed should condenser cooling water in-leakage occur.

A number of design changes have been incorporated in the Model D5 steam generators for CPNPP Unit 2. These changes have been incorporated to reduce the consequences of adverse secondary side environmental conditions and hence improve overall steam generator reliability.

The tube support plates used in the Model D5 will be ferritic stainless steel which has been shown in laboratory tests to be resistant to corrosion in the AVT environment. When corrosion of ferritic stainless steel does occur, the volume of the corrosion products is equivalent to the volume of the consumed material. The support plates have also been designed with broached tube holes (quatrefoil design) rather than drilled holes. The broached tube hole design promotes high velocity flow along the tube sweeping any impurities away from the support plate location.

Additional measures are incorporated in the Model D5 design to prevent areas of dryout in the steam generator and accumulations of sludge in low velocity areas. Modification to the wrapper have increased water velocities across the tubesheet. A flow distribution baffle is provided which forces the low flow area to the center of the bundle. Increased capacity blowdown pipes have been added to enable continuous blowdown of the steam generators at a high volume. The intakes of these blowdown pipes are located below the center cut-out section of the flow distribution baffle in the low velocity region where sludge may be expected to accumulate. Continuous blowdown provides maximum protection against inleakage of impurities from the condenser.

Thermal treatment of Inconel tubes has been shown to be effective in limiting stress corrosion cracking, especially in the U-bend region and the expanded region at the tubesheet. Tubing used in the Model D5 steam generators (Unit 2) have been thermally treated at the factory to provide additional margin against inner diameter primary water stress corrosion cracking.

Operating experience, verified in numerous steam generator inspections, indicates that the tube degradation associated with phosphate water treatment is not occurring where only AVT has been utilized. Adherence to the AVT chemical specifications and close monitoring of the condenser integrity will assure the continued good performance of the steam generator tubing.

Additional extensive operating data is presently being accumulated with the conversion to AVT chemistry. A comprehensive program of steam generator inspections, including the requirements of Regulatory Guide 1.83, with the exceptions as stated in Appendix 1A(N) will ensure detection and correction of any unanticipated degradation that might occur in the steam generator.

5.4.2B.1.4 Monitoring of Secondary Side Water Chemistry

The chemistry of the steam generator water and condensate is continuously monitored as described in Section 9.3.2 and 10.4.16 respectively. The conductivity and pH of the secondary side are continuously measured as are dissolved oxygen, sodium and hydrazine content. Addition rates of secondary side chemicals are controlled by the continuous on-line analyzers. Steam generator blowdown is continuous; the rate can be adjusted using the water chemistry as a basis. The steam generator blowdown processing system is described in Section 10.4.8.

The approach to monitoring secondary side water chemistry complies with the approach outlined in Branch Technical Position MTEB 5-3.

5.4.2B.1.5 Cleanup of Secondary Side Materials

Several methods are employed to clean operating steam generators of corrosion causing secondary side deposits. Sludge lancing, a procedure in which a hydraulic jet inserted through an access opening (inspection port) loosens deposits which are removed by means of a suction pump, can be performed when the need is indicated by the results of steam generator tube inspection. Blowdown procedures are performed as deemed necessary by regular water chemistry testing. The location of the blowdown piping suction, adjacent to the tubesheet and in a region of relatively low flow velocity, facilitates the efficient removal of impurities that have accumulated on the tubesheet.

5.4.2B.2 Steam Generator Inservice Inspection

5.4.2B.2.1 Steam Generator Design Characteristics For Inservice Inspection

The steam generator is designed to permit inservice inspection of Class 1 and 2 components, including individual tubes. The design aspects that provide access for inspection and the proposed inspection program comply with the edition of Section XI of the ASME Code, Division 1, “Rules for Inspection and Testing of Components of Light- Water-Cooled Plants,” required by 10 CFR 50.55a, paragraph g. A number of access openings make it possible to inspect and repair or replace a component according to the techniques specified. These openings include four manways, two of them for inspection and maintenance of the steam dryer. Also the Unit 2 steam generators have five 6.0 inch diameter handholes and three 2.5 inch diameter inspection openings for additional access through the secondary side pressure boundary.

5.4.2B.2.2 Program For Inservice Inspection Of Steam Generator Tubing

Steam generator tubing will be inspected in accordance with:

1. The recommendations given in Regulatory Guide 1.83, “Inservice Inspection of Steam Generator Tubes,” Revision 1, July 1975, and

2. The requirements of ASME Section XI (Edition and Addenda as required by 10CFR50.55a), Subarticle IWB-2413.

3. Comanche Peak Nuclear Power Plant Technical Specifications Section 5.5.9.

The program consists of the following areas:

Inspection Equipment & Procedures

Eddy current testing equipment will be used to inspect the tubing and shall be sensitive enough to detect imperfections of 20 percent or more through the tube wall.

Baseline Inspection

1. All tubes in the steam generators shall be inspected by eddy current or alternative techniques prior to service to establish a baseline condition of the tubing.

2. If a major change in their secondary water chemistry (e.g., phosphate to volatile treatment) is made during plant lifetime, a baseline inspection will be conducted before resumption of power operation.

Steam Generator Sample Selection and Inspection

Steam Generator Sample Selection and Inspection shall be consistent with CPNPP TS 5.5.9.

Steam Generator Tube Inspection Techniques and Tubes Left in Service Based on Indication Size (GL 97-05)

Traditionally, eddy-current inspection techniques are relied upon to assess the condition of steam generator tubes. Although the eddy-current method is a proven technique for detecting tube degradation, the ability to depth size indications is possible only for specific modes of degradation. Specifically, tube degradation from inter-granular attack (IGA) and stress corrosion cracking (SCC), major modes of steam generator tube degradation, are difficult to size with eddy-current inspection techniques because of a number of complicating variables. Theoretically, there is a relationship between the depth of penetration of a defect and the eddy-current signal response; in practice, however, the relationship between signal voltage or phase angle and the degradation depth is influenced by many other variables. Oxide deposits, variability of tube material properties and geometry, degradation morphology, human factors, and eddy-current data analysis and acquisition practices are some of the factors that can significantly alter a depth estimation of steam generator tube degradation. The depth of several specific forms of volumetric steam generator tube degradation can be sized with a reasonable degree of accuracy; however, qualifying techniques for sizing of some forms of degradation, e.g., IGA and SCC, are typically problematic.

CPNPP has adopted the guidelines of Energy Institute (NEI) letter 97-06, “Steam Generator Program Guidelines”. NEI 97-06 requires following the inspection guidelines contained in the latest revision of the Electric Power Research Institute (EPRI) Pressurized Water Reactor (PWR) Steam Generator Examination Guidelines. Appendix H, “Performance Demonstration for Eddy Current Examination,” of the EPRI “PWR Steam Generator Examination Guidelines,” latest revision, provides guidance on the qualification of steam generator tubing examination techniques and equipment used to detect and size indications. Damage mechanisms are divided into the following categories: thinning, pitting, wear, outside diameter inter-granular attack/stress-corrosion cracking (IGA/SCC), primary-side SCC, and impingement damage for qualification.

For qualification purposes, test samples are used to evaluate detection and sizing capabilities. While pulled tube samples are preferred, fabricated samples may be used. If fabricated test samples are used, the samples are verified to produce signals similar to those being observed in the field in terms of signal characteristics, signal amplitude, and signal-to-noise ratio. Samples are examined to determine the actual through wall defect measurements as part of the Appendix H qualification process.

The procedures at CPNPP are developed in accordance with Appendix H which specifies the essential variables for each technique. These essential variables are associated with an individual instrument, probe, cable, or particular equipment configurations. Additionally, certain techniques have undergone testing and review to quantify sizing performance. The sizing data set includes the detection data set for the technique with additional requirements for number and composition of the grading units.

At CPNPP only tubes with indications of wear type degradation found at anti-vibration bars and tube support plates and which are less than the Technical Specification value of 40% are permitted to remain in service. All tubes with crack-like degradation are plugged upon detection.

5.4.2B.3 Design Bases

Steam generator design data are given in Table 5.4-3. Code classifications of the steam generator components are given in Section 3.2. Although the ASME classification for the secondary side is specified to be Class 2, the current philosophy is to design all pressure retaining parts of the steam generator, and thus, both the primary and secondary pressure boundaries, to satisfy the criteria specified in Section III of the ASME Code for Class 1 components. The design stress limits, transient conditions and combined loading conditions applicable to the steam generator, are discussed in Section 3.9N.1. Estimates of radioactivity levels anticipated in the secondary side of the steam generators during normal operation, and the bases for the estimates, are given in Chapter 11. The accident analysis of a steam generator tube rupture is discussed in Chapter 15.

The internal moisture separation equipment is designed to ensure that moisture carryover does not exceed 0.25 percent of weight under the following conditions:

1. Steady state operation up to 100 percent of full load steam flow, with water at the normal operating level.

2. Loading or unloading at a rate of 5 percent of full power steam flow per minute in the range from 15 to 100 percent of full load steam flow.

3. A step load change of 10 percent of full power in the range from 15 to 100 percent full load steam flow.

The water chemistry on the reactor side is selected to provide the necessary boron content for reactivity control and to minimize corrosion of RCS surfaces. The water chemistry of the steam side and its effectiveness in corrosion control are discussed in Chapter 10. Compatibility of steam generator tubing with both primary and secondary coolants is discussed further in Section 5.4.2B.1.3.

The steam generator is designed to prevent unacceptable damage from mechanical or flow induced vibration. Tube support adequacy is discussed in Section 5.4.2B.5.3. The tubes and tubesheet are analyzed in Reference [3] and confirmed to withstand the maximum accident loading condition as it is defined in Section 3.9N.1. Further consideration is given in Section 5.4.2B.5.4 to the effect of tube wall thinning on accident condition stresses.

The preheat section of the steam generator is arranged to provide the maximum amount of counter flow feasible and, therefore, more efficient heat transfer.

A separate auxiliary feedwater nozzle is provided in the upper shell in order to avoid introducing cold water into the possible hot and empty preheat section. The integrity of the steam generator design is, thus, maximized.

5.4.2B.4 Design Description

The steam generator shown in Figure 5.4-4B is a model D4, vertical shell and U-tube evaporator with integral moisture separating equipment. On the primary side, the reactor coolant flows through the inverted U-tubes, entering and leaving through nozzles located in the hemispherical bottom head of the steam generator. The head is divided into inlet and outlet chambers by a vertical divider plate extending from the head to the tubesheet. Steam is generated on the shell side, flows upward and exits through the outlet nozzle at the top of the vessel. During normal operation, feedwater flows through a flow restrictor, directly into the counter flow preheat section and is heated almost to saturation temperature before entering the boiler section. Subsequently the water-steam mixture flows upward through the tube bundle and into the steam drum section, where individual centrifugal moisture separators remove most of the entrained water from the steam. The steam continues to the secondary separators for further moisture removal, increasing its quality to a minimum of 99.75 percent. The moisture separators recirculate the separated water through the annulus between the shell and tube bundle wrapper via the space formed by the distribution plate, between the tubesheet and the preheat section. The returning flow then combines with the already preheated water-steam mixture for another passage through the steam generator. Dry steam exits through the outlet nozzle which is provided with a steam flow restrictor, described in Section 5.4.4.

5.4.2B.5 Design Evaluation

5.4.2B.5.1 Forced Convection

The limiting case for heat transfer capability is the “nominal 100 percent design” case. The steam generator effective heat transfer coefficient is based on the coolant conditions of temperature and flow for this case. The best estimate for the heat transfer coefficient applied in steam generator design calculations and plant parameter selection is 1301 Btu/hr-ft^2-°F. This coefficient is approximately 5 to 10 percent less than the heat transfer performance experienced at a number of operating plants. The coefficient incorporates a specified fouling factor resistance of 0.00005 hr/ft^2-°F/Btu (0.0001 in the preheat section) which is the value selected to account for the differences in the measured and calculated heat transfer performance as well as provide the margin indicated above. Although margin for tube fouling is available, operating experience to date has not indicated that steam generator performance decreases over a long time period. Adequate tube area is selected to ensure that the full design heat removal rate is achieved.

5.4.2B.5.2 Natural Circulation Flow

The driving head created by the change in coolant density as it is heated in the core and rises to the outlet nozzle initiates convection circulation. This circulation is enhanced by the fact that the steam generators, which provide a heat sink, are at a higher elevation than the reactor core which is the heat source. Thus natural circulation is assured for the removal of decay heat during hot shutdown in the unlikely event of loss of forced circulation.

5.4.2B.5.3 Mechanical and Flow Induced Vibration Under Normal Operation

In the design of Westinghouse steam generators, the potential for tube wall degradation attributable to mechanical or flow induced excitation has been thoroughly evaluated. The evaluation included detailed analyses of the tube support systems for various mechanisms of tube vibration.

The primary cause of tube vibration in heat exchangers is hydrodynamic excitation due to secondary fluid flow on the outside of the tubes. In the range of normal steam generator operating conditions, the affects of primary fluid flow inside the tubes and mechanically induced tube vibration are considered to be negligible.

To evaluate flow induced tube vibration in the preheater region of the tube bundle, Westinghouse undertook an extensive program employing data from operating plants, full and partial scale model tests and analytical tube vibration models. Operating plant data consisted of tube wear data from pulled tube evaluations and eddy current tests, and tube motion data from accelerometers installed inside selected tubes. Model testing generated tube wear data, flow velocity distributions, tube motion parameters and flow induced tube vibration forcing functions. The tube vibration analyses applied the forcing functions to produce tube motion data. The results of this evaluation were consistent with the early operating experience of preheat steam generators.

On the basis of an extensive model test and analysis program, Westinghouse designed, verified and implemented a modification to the steam generator to reduce tube vibratory response to preheater inlet flow excitation. Additionally, the magnitude of the flow forcing function was

reduced through implementation of a preheater flow bypass arrangement in the feedwater system. The verification of the performance of the modifications in reducing tube excitation and response was done with input from a full scale test under simulated conservative flow and tube support conditions.

Fatigue of the tubes in the preheater region which are subject to flow induced excitation is not a concern since the maximum resultant stresses in the tube are below the endurance limit of the material.

For areas of the tube bundle other than the preheater, parallel flow analyses were performed to determine the vibratory deflections. These analyses indicate that the flow velocities are sufficiently low such that they result in negligible fatigue and vibratory amplitudes. The support system, therefore, is deemed adequate with regard to parallel flow excitation.

To evaluate cross flow at the exit of the downcomer flow to the tube bundle and at the top of the bundle in the U-bend area, Westinghouse performed an experimental research program of cross flow in tube arrays with the specific parameters of the steam generator. Air and water model tests were employed. The results of this research indicate that these regions of the bundle are not subject to the vortex shedding mechanism of tube excitation. Vortex shedding was found not to be a significant mechanism in these two regions for the following reasons:

1. Flow turbulence in the downcomer and tube bundle inlet region inhibit the formation of Von Karman vortices.

2. Both axial and cross flow velocity components exist on the tubes. The axial flow component disrupts the Von Karman vortices.

This research program was also the basis for evaluation of the fluidelastic mechanism due to cross flow at the tubesheet. The evaluation showed the adequacy of the tube support arrangement.

Flow turbulence can result in some tube excitation in these regions. This excitation is of little concern, however, since:

1. Maximum stresses in the tubes are at least an order of magnitude below the fatigue endurance limit of the tube material, and

2. Tube support arrangements preclude significant vibratory motion.

In summary, tube vibration has been thoroughly evaluated. Mechanical and primary flow excitation are considered negligible. Secondary flow excitation has been evaluated. From this evaluation, it is concluded that if tube vibration does occur, the magnitude will be limited. Tube fatigue due to the vibration is judged to be negligible. Any tube wear resulting from the tube vibration would be limited and would progress slowly. This allows use of a periodic tube inservice inspection program for detection and follow of any tube wear. This inservice inspection program, in conjunction with tube plugging criteria, provides for safe operation of the steam generators.

5.4.2B.5.4 Allowable Tube Wall Thinning Under Accident Conditions

An evaluation is performed to determine the extent of tube wall thinning that can be tolerated under accident conditions. Under such a postulated design basis accident, vibration is of short enough duration that there is no endurance problem. The results of a study made on “D series” (0.75 inch nominal diameter, 0.043 inch nominal thickness) tubes under accident loading are discussed in Reference [3] and show that a minimum wall thickness of 0.026 inches would have a maximum faulted condition stress (i.e., due to combined LOCA and Safe Shutdown Earthquake loads) that is less than the allowable limit. This thickness is 0.010 inches less than the minimum steam generator tube wall thickness 0.039 reduced to 0.036 inches by the assumed general corrosion and erosion loss of 0.003 inches. However, 0.026 inches is not the limiting minimum wall thickness but rather is a more conservative value. For tube plugging, the criteria is found in Section 1(A)N.

The corrosion rate is based on a conservative weight loss rate for mill annealed Inconel tubing in flowing 650°F primary side reactor coolant fluid. The weight loss, when equated to a thinning rate and projected over a 40 year plant life with appropriate reduction after initial hours, is equivalent to 0.080 mils thinning. The assumed corrosion rate of 3 mils leaves a conservative 2.2 mils for general corrosion thinning on the secondary side. These values, as noted, have been developed for mill annealed material; a comparable set of data for thermally treated Inconel is being generated.

The steam generator tubes, existing originally at their minimum wall thickness and reduced by a very conservative general corrosion loss, still provide quite an adequate safety margin. Thus, it can be concluded that the ability of the steam generator tubes to withstand accident loadings is not affected by a lifetime of general corrosion losses.

5.4.2B.5.5 Steam Generator Denting

Localized steam generator tube diameter reductions were first discovered during the April 1975 steam generator inspection at the Surry Unit No. 2 plant. This discovery was evidenced by eddy current signals, resembling those produced by scanning dents, and by difficulty in passing the standard 0.700 in. diameter eddy current probe through the tubes at the intersections with the support plates. Subsequent to the initial finding, steam generator inspections at other operating plants revealed essentially identical results.

Denting is a term which describes a group of related phenomena resulting from corrosion of carbon steel in the crevices formed between the tubes and the tube support plates or tubesheets. The term “denting” has been applied to the secondary effects which include:

1. Tube diameter reduction

2. Tube support plate hole dilation

3. Tube support plate flow hole distortion, flow slot hourglassing

4. Tube support plate expansion

5. Tube leakage

6. Wrapper distortion

The mechanism which produces the effects cited involves an acid chloride environment in the tube crevices. In sequence, the process appears to occur as follows:

The crevice between the tube and the support plate is blocked as a result of deposition of chemical species present in the bulk water, including phosphate compounds, secondary system corrosion products and minimal tube corrosion products. Once plugged, the annulus provides a site for concentration of various nominally soluble contaminants, such as chlorides, sulfates, etc. Recent studies indicate that in the absence of nonvolatile, alkalizing species, there may exist the potential for production of an acid solution by hydrolysis of such compounds as magnesium chloride, nickel phosphate, cupric chloride, various ferrrous salts, etc. In an acid chloride solution, the corrosion film on the carbon steel is converted from protective in character, to a thick, non-protective oxide of high density which assumes a laminar configuration subject to disruption due to the volume mismatch between the oxide and the base metal. The buildup of the thick oxide in the nominal 14 mil radial gap between the tube and the support plate causes sufficient force to be exerted against the tube to cause plastic deformation locally and correspondingly dilation of the tubehole in the plate. The transmission of these forces can cause distortion of the circulation holes in the plate (both the flow holes between the tubes and the central flow slots between the inlet and outlet halves of the tube bundle). In the most extreme cases as corrosion proceeds and in-plate forces accumulate, the entire plate increases in diameter and the ligaments between the holes in the plate may break. Ovalization of the tubes at the intersections results in high strains, leading to tensile stress on the tube ID and possible leakage by intergranular stress assisted cracking. A similar result may be induced at the apex of the first row (i.e., the smallest radius) U-bend if sufficient distortion of the top support plate flow slots occurs, resulting in leg displacement, ovalization, and high strains. Development to this stage presupposes condenser leakage which results in chloride contamination of the steam generator liquid.

Approximately 100 tubes in the Preheat section of the Unit 1 and Unit 2 steam generators have expanded sections at the “B” and “D” support plate elevations. These tubes have reduced nominal clearances at these support plates. The denting phenomena has been investigated for expanded tubes and it has been found that they are not adversely affected. (See reference 4)

The tube leakage and support plate effects do not pose a safety problem with respect to release of radioactivity or effects on accident calculations, but the frequency of leakage and resultant repair shutdowns does present an economic concern to the operators. The utilization of preventive plugging, therefore, serves to maintain availability and to permit orderly planning for long-term corrective action.

The occurrence of denting has thus far been associated exclusively with plants having a history of chloride contamination due to condenser leakage. Moreover, it has recently been noted that Maine Yankee and Millstone Point 2, non-Westinghouse plants which have used AVT exclusively, have apparently incurred denting also; sea water is used for cooling the condensers at both of these plants.

Research into the causes of denting was initiated shortly after the discovery of the denting condition. Initially, dented tubes were removed for laboratory examination. Subsequently tube

support plate samples containing sections of tubing were also removed for analysis from operating plants.

The initial hard data on the nature of the denting phenomenon were derived from these tube/ support plate samples which revealed the thick oxide buildup, the tube diameter reduction, and chemical makeup of the crevice-filling materials. It was demonstrated that there was only minor corrosive attack on the tube material, approximately 0-2 mil circumferential thinning, and that the crevice contained a thick layer of almost pure magnetite (Fe_30_4); other chemical constituents included Inconel-metal-phosphate corrosion products close to the tube, and general secondary system contaminants between the Fe_30_4 and the phosphate layer. There was evidence of nickel and copper deposits and the oxide was laced with chlorides.

Armed with those general observations, a series of crevice-with- contaminants tests geometries were evaluated; denting was produced first in reverse as “bulging” when a carbon steel plug was inserted into an Inconel tube to form the crevice; later heated crevice assemblies with heat transfer were shown to be effective dent simulators; finally denting in model boilers equipped with plant-type geometrical configurations was demonstrated. While pure, uncontaminated AVT environments have to date been found to be innocuous, it has been shown that the PO_4 to AVT transition was unnecessary to initiate the denting process. Only the presence of acid chloride solutions has been found to be common factor. Nickel chloride, ferrous or cupric chloride solutions have been shown to be corrosive, and have also produced measurable denting. Thus far, test data indicate that phosphates, calcium hydroxide, zinc oxide and borates seem to retard the dent process; morpholine, among the common volatile amines, shows a beneficial effect on the corrosion rate of carbon steel.

Model boiler tests have been used to evaluate the adequacy of the AVT chemistry specifications adopted in 1974. With one significant alteration, the specifications appear to be adequate to preserve tube integrity: the frequency and the length of time above the chloride limit for normal operation (0.15 ppm) must be limited. Westinghouse is working to prepare a uniform specification to be applied to all plants, which will limit the chloride concentration, the number of consecutive days beyond the normal specification, and the number of incidents per year.

As has been discussed, tube denting is a result of support plate corrosion products compressing the tube while dilating the tube hole. Therefore, measures to inhibit denting concentrate on removing the corrosion mechanisms, medium or materials.

The Comanche Peak steam generators are of the preheat design. The water entering the preheater will be of feedwater quality and any impurities contained therein will not be subject to concentration due to the absence of recirculation. This and thermal-hydraulic characteristics minimize the potential for chemical hideout in the preheater. Since sludge may still settle out at the tubesheet elevation outside the preheater blowdown, pipes have been designed to achieve maximum removal of the sludge from the steam generators. Should condenser in-leakage occur, specific continuous corrective actions will be taken to minimize its impact.

Operating experience, verified in numerous steam generator inspections, indicates that the tube degradation associated with phosphate water treatment is not occurring where only AVT has been utilized. Adherence to the AVT chemical specifications and close monitoring of the condenser integrity will assure the continued good performance of the steam generator tubing.

5.4.2B.6 Quality Assurance

The steam generator quality assurance program is given in Table 5.4-4.

Radiographic inspection and acceptance standard shall be in accordance with the requirements of Section III of the ASME Code.

Liquid penetrant inspection is performed on weld deposited tubesheet cladding, channel head cladding, and tube-to-tubesheet weldments. Liquid penetrant inspection and acceptance standards are in accordance with the requirements of Section III of the ASME Code.

Magnetic particle inspection is performed on the tubesheet forging, channel head casting, nozzle forgings, and the following weldments:

1. Nozzles to shell.

2. Support brackets.

3. Instrument connection (primary and secondary)

4. Temporary attachments after removal.

5. All accessible pressure containing welds after hydrostatic test.

Magnetic particle inspection and acceptance standard are in accordance with requirements of Section III of the ASME Code.

An ultrasonic test is performed on the tubesheet forging, tubesheet cladding, secondary shell and head plate and nozzle forging.

The heat transfer tubing is subjected to eddy current and ultrasonic tests.

Hydrostatic tests are performed in accordance with Section III of the ASME Code.

In addition, the heat transfer tubes shall be subjected to a hydrostatic test pressure prior to installation into the vessel which is not less than 1.25 times the primary side design pressure.

5.4.3 REACTOR COOLANT PIPING

5.4.3.1 Design Bases

The RCS piping is designed and fabricated to accommodate the system pressures and temperatures attained under all expected modes of plant operation or anticipated system interactions. Stresses are maintained within the limits of Section III of the ASME Code. Code and material requirements are provided in Section 5.2.

Materials of construction are specified to minimize corrosion/erosion and ensure compatibility with the operating environment.

The piping in the RCS is Safety Class 1 and is designed and fabricated in accordance with the ASME Code, Section III, Class 1 requirements.

Stainless steel pipe conforms to ANSI B36.19 for sizes 1/2 inch through 12 inches and wall thickness Schedules 40S through 80S. Stainless steel pipe outside of the scope of ANSI B36.19 conforms to ANSI B36.10.

The minimum wall thickness of the loop pipe and fittings are not less than that calculated using the ASME Code, Section III, Class 1 formula of paragraph NB-3641.1(3) with an allowable stress value of 17,550 psi. The CPNPP Unit 1 and 2 pressurizer surge line design utilizes schedule 160 piping with the exception of a one foot section of schedule 140 piping in the Unit 1 pressurizer surge line. The minimum pipe bend radius is 5 nominal pipe diameters; ovality does not exceed 6 percent.

All butt welds, branch connection nozzle welds, and boss welds shall be of a full penetration design.

Processing and minimization of sensitization are discussed in Section 5.2.3.

Flanges conform to ANSI B16.5.

Socket weld fittings and socket joints conform to ANSI B16.11.

Inservice inspection is discussed in Section 5.2.4.

For Unit 1, the Mechanical Stress Improvement Process (MSIP) was applied to the Reactor Vessel Nozzles (Hot Leg and Cold Leg) to prevent and mitigate PWSCC at the nozzle to safe-end welds during Refueling Outage 1RF20.

MSIP is applied by the use of large pipe clamps and box presses that will mechanically "squeeze" the pipe near the Dissimilar Metal (DM) welds using hydraulic pressure. The result of this "squeeze" is a small reduction in pipe circumference (Approximately 1 to 2 percent) resulting in insignificant effect on flow and, more importantly, will also result in a residual compressive stress field in the location of the DM weld during all plant conditions. This compressive stress field will mitigate the propagation of any Primary Water Stress Corrosion Cracking.

5.4.3.2 Design Description

Principal design data for the reactor coolant piping are given in Table 5.4-5.

Pipe and fittings are cast, seamless without longitudinal welds and electroslag welds, and comply with the requirements of the ASME Code, Section II (Part A and C), Section III, and Section IX.

The RCS piping is specified in the smallest size consistent with system requirements. This design philosophy results in the reactor inlet and outlet piping diameters given in Table 5.4-5. The line between the steam generator and the pump suction is larger to reduce pressure drop and improve flow conditions to the pump suction.

The reactor coolant piping and fittings which make up the loops are austenitic stainless steel. There will be no electroslag welding on these components. All smaller piping which comprise

part of the RCS such as the pressurizer surge line, spray and relief line, loop drains and connecting lines to other systems are also austenitic stainless steel. The nitrogen supply line for the pressurizer relief tank is carbon steel. All joints and connections are welded, except for the pressurizer code safety valves, where flanged joints are used. A thermal sleeve is installed on the pressurizer surge nozzle located on the bottom of the pressurizer vessel.

All piping connections from auxiliary systems are made above the horizontal centerline of the reactor coolant piping, with the exception of:

1. Residual heat removal pump suction lines, which are 45 degrees down from the horizontal centerline. This enables the water level in the RCS to be lowered in the reactor coolant piping while continuing to operate the Residual Heat Removal System, should this be required for maintenance.

2. Loop drain lines and the connection for temporary level measurement of water in the RCS during refueling and maintenance operation.

3. The differential pressure taps for flow measurement, which are downstream of the steam generators on the first 90 degree elbow.

4. The pressurizer surge line, which is attached at 45 degrees above the horizontal centerline.

5. The 3 inch branch connections to accommodate the 1 1/2 inch cold leg safety injection lines are located on the horizontal centerline.

6. Two of the three taps in each resistance temperature detector hot leg connection.

Penetrations into the coolant flow path are limited to the following:

1. The spray line inlet connections extend into the cold leg piping in the form of a scoop so that the velocity head of the reactor coolant loop flow adds to the spray driving force.

2. The reactor coolant sample system taps protrude into the main stream to obtain a representative sample of the reactor coolant.

3. Hot leg thermowells to accommodate in line narrow range RTDs.

4. The wide range temperature detectors are located in resistance temperature detector wells that extend into both the hot and cold legs of the reactor coolant pipes.

5. Cold leg RTD thermowells are for the fast response T_{cold} measurement.

In the previous design, a hot leg temperature measurement was made in the hot leg RTD manifold. This was combined with a cold leg temperature measurement in the cold leg RTD manifold to provide a ΔT signal which was used as an indication of reactor power in the protection system. The ΔT measurement was also used off-line in conjunction with a secondary system power measurement to perform a calorimetric calculation of primary system flow rate.

Two cold leg RTD thermowells in each loop are installed to replace the cold leg RTD manifold. The cold leg bypass nozzles and crossover leg bypass return nozzles are installed in the main loop piping; however, all of these connections have been plugged and are not used.

In the new configuration as discussed in Section 7.2.1.1.4.2, there are four N-16 detectors mounted externally on each hot leg piping. The two upstream detectors are used to provide indication of reactor power for use in the protection system and can also be used by the transit time flow meter to measure reactor coolant flow. The two down stream detectors are only used to provide an input to the transit time flow meter. There are narrow range hot leg RTDs in Unit 1 only for this new configuration.

The RCS piping includes those sections of piping interconnecting the reactor vessel, steam generator, and reactor coolant pump. It also includes the following:

1. Charging line and alternate charging line from the system isolation valve up to the branch connections on the reactor coolant loop.

2. Letdown line and excess letdown line from the branch connections on the reactor coolant loop to the system isolation valve.

3. Pressurizer spray lines from the reactor coolant cold legs to the spray nozzle on the pressurizer vessel.

4. Residual heat removal lines to or from the reactor coolant loops up to the designated check valve or isolation valve.

5. Safety injection lines from the designated check valve to the reactor coolant loops.

6. Accumulator lines from the designated check valve to the reactor coolant loops.

7. Drain, sample[a], and instrument[a] lines to or from the designated isolation valve to or from the reactor coolant loops.

8. Pressurizer surge line from one reactor coolant loop hot leg to the pressurizer vessel inlet nozzle.

9. Pressurizer spray scoop, sample connection[a] with scoop, reactor coolant temperature element installation boss, and the temperature element well itself.

10. All branch connection nozzles attached to reactor coolant loops.

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a. Lines with a 3/8 inch flow restricting orifice qualify as Safety Class 2; in the event of a break in one of these Safety Class 2 lines, the normal makeup system is capable of providing makeup flow while maintaining pressurizer water level.

11. Pressure relief lines from nozzles on top of the pressurizer vessel up to and through the power operated pressurizer relief valves and pressurizer safety valves.

12. Seal injection water and labyrinth differential pressure lines to or from the reactor coolant pump inside reactor Containment.

13. Auxiliary spray line from the isolation valve to the pressurizer spray line header.

14. Sample lines[a] from pressurizer to the isolation valve.

Details of the materials of construction and codes used in the fabrication of reactor coolant piping and fittings are discussed in Section 5.2.

5.4.3.3 Design Evaluation

Piping load and stress evaluation for normal operating loads, seismic loads, blowdown loads, and combined normal, blowdown and seismic loads is discussed in Section 3.9(N).

5.4.3.3.1 Material Corrosion/Erosion Evaluation

The water chemistry is selected to minimize corrosion. A periodic analysis of the coolant chemical composition is performed to verify that the reactor coolant quality meets the specifications.

The design and construction are in compliance with the ASME Code, Section XI, Subarticle IWA-1500. Pursuant to this, all pressure containing welds out to the second valve that delineates the RCS boundary are available for examination with removable insulation.

Components constructed with stainless steel will operate satisfactorily under normal plant chemistry conditions in pressurized water reactor system, because chlorides, fluorides, and particularly oxygen, are controlled to very low levels (see Section 5.2.3).

Periodic analysis of the coolant chemical composition is performed to monitor the adherence of the system to desired reactor coolant water quality listed in Table 5.2-5. Maintenance of the water quality to minimize corrosion is accomplished using the Chemical and Volume Control System and Sampling System which are described in Chapter 9.

5.4.3.3.2 Sensitized Stainless Steel

Sensitized stainless steel is discussed in Section 5.2.3.

5.4.3.3.3 Contaminant Control

Contamination of stainless steel and Inconel by copper, low melting temperature alloys, mercury and lead is prohibited. Colloidal graphite is the only permissible thread lubricant.

Prior to application of thermal insulation, the austenitic stainless steel surfaces are cleaned and analyzed to a halogen limit of 0.0015 mg Cl/dm^2 and 0.0015 F/dm^2.

5.4.3.4 Tests and Inspections

The RCS piping quality assurance program is given in Table 5.4-6.

Volumetric examination is performed throughout 100 percent of the wall volume of each pipe and fitting in accordance with the applicable requirements of Section III of the ASME Code for all pipe 27-1/2 inches and larger. All unacceptable defects are eliminated in accordance with the requirements of the same section of the code.

A liquid penetrant examination is performed on both the entire outside and inside surfaces of each finished fitting in accordance with the criteria of the ASME Code, Section III. Acceptance standards are in accordance with the applicable requirements of the ASME Code, Section III.

The pressurizer surge line conforms to SA-376, Grade 304, 304N, or 316 with supplementary requirements S2 (transverse tension tests), and S6 (ultrasonic test). The S2 requirement applies to each length of pipe. The S6 requirement applies to 100 percent of the piping wall volume.

The end of pipe sections, branch ends and fittings are machined back to provide a smooth weld transition adjacent to the weld path.

5.4.4 MAIN STEAM LINE FLOW RESTRICTOR

5.4.4.1 Design Basis

The outlet nozzle of the steam generator is provided with a flow restrictor designed to limit steam flow in the unlikely event of a break in the main steam line. A large increase in steam flow will create a backpressure which limits further increase in flow. Several protective advantages are thereby provided: rapid rise in Containment pressure is prevented, the rate of heat removal from the reactor coolant is kept within acceptable limits, thrust forces on the main steam line piping are reduced, and most important, stresses on internal steam generator components, particularly the tubesheet and tubes, are limited. The restrictor is also designed to minimize the unrecovered pressure loss across the restrictor during normal operation.

5.4.4.2 Design Description

The flow restrictor (Figure 5.4-5) consists of seven Inconel (ASME-SB-163) venturi inserts which are inserted into the holes in a integral steam outlet low alloy steel forging. The inserts are arranged with one venturi at the centerline of the outlet nozzle and the other six equally spaced around it. After insertion into the low alloy steel forging holes, the inconel venturi nozzles are welded to the inconel cladding on the inner surface of the forging.

5.4.4.3 Design Evaluation

The flow restrictor design has been sufficiently analyzed to assure its structural adequacy. The equivalent throat diameter of the steam generator outlet is 16 inches, and the resultant pressure drop through the restrictor at 100 percent steam flow is approximately 3.4 psi.

This is based on a design flow rate f 3.78 x 10^6 lb/hr. Materials of construction and manufacturing of the flow restrictor are in accordance with Section III of the ASME Code.

5.4.4.4 Tests and Inspections

Since the restrictor is not a part of the steam system boundary, no tests and inspection beyond those during fabrication, are anticipated.

5.4.5 MAIN STEAM LINE ISOLATION SYSTEM

The Main Steam Line Isolation System is described in Sections 6.2.4 and 10.3.

5.4.6 REACTOR CORE ISOLATION COOLING SYSTEM

This section is not applicable to the Comanche Peak Nuclear Power Plant (CPNPP).

5.4.7 RESIDUAL HEAT REMOVAL SYSTEM

The Residual Heat Removal System (RHRS) transfers heat from the RCS to the Component Cooling Water System to reduce the temperature of the reactor coolant to the cold shutdown temperature at a controlled rate during the second part of normal plant cooldown and maintains this temperature until the plant is started up again.

Parts of the RHRS also serve as parts of the ECCS during the injection and recirculation phases of a LOCA (see Section 6.3).

The RHRS also is used to transfer refueling water between the refueling cavity and the Refueling Water Storage Tank at the beginning and end of the refueling operations.

Nuclear plants employing the same RHRS design as the CPNPP are given in Section 1.3.

5.4.7.1 Design Bases

RHRS design parameters are listed in Table 5.4-7.

For normal plant cooldowns, the RHRS is required to cool the RCS to a temperature adequate for refueling or maintenance in a time that is consistent with refueling or maintenance operations. The RHRS is designed to be placed in operation approximately 4 hours after reactor shutdown when the temperature and pressure of the RCS are approximately 350°F and 425 psig, respectively. Assuming that two heat exchangers and two pumps are in service and that each heat exchanger is supplied with component cooling water at design flow and maximum temperature, the RHRS is designed to reduce the temperature of the reactor coolant from 350°F to 140°F within 28 hours. The time required, under these conditions, to reduce reactor coolant temperature from 350°F to 212°F is less than 4 hours. The heat load handled by the RHRS during the cooldown transient includes residual and decay heat from the core and reactor coolant pump heat. The design heat load is based on the ANSI/ANS-5.1-1979 American National Standard for Decay Heat in Light Water Reactors.

For abnormal plant cooldowns, the RHRS is required to cool the RCS from hot shutdown (350°F) to cold shutdown (200°F) in a reasonable period of time, assuming the most limiting single failure. See Appendix 5A for an evaluation of compliance with Branch Technical Position RSB 5-1.

In the event of a fire, the RHRS is required to cool the RCS from hot shutdown to cold shutdown as described in the Fire Protection Report.

In the event of a design basis accident in one unit, the other unit is required to be capable of an orderly shutdown and cooldown. [GDC-5]

The RHRS is also required to cool the RCS from hot standby to cold shutdown in less than 24 hours starting 12 hours after reactor shutdown assuming only one heat exchanger and pump are in service and the Safe Shutdown Impoundment is conservatively assumed at 102°F. This ensures the RHRS is capable of satisfying action times in the Technical Specifications.

The RHRS is designed to be isolated from the RCS whenever the RCS pressure exceeds the RHRS design pressure. The RHRS is isolated from the RCS on the suction side by two motor operated valves in series on each suction line. Each motor operated valve is interlocked to prevent its opening if RCS pressure is greater than 425 psig. The RHRS is isolated from the RCS on the discharge side by two check valves in each return line. Also provided on the discharge side is a normally open motor operated valve downstream of each RHRS heat exchanger. (These check valves and motor operated valves are not considered part of the RHRS; they are shown as part of the ECCS; see Figure 6.3-1).

Each inlet line to the RHRS is equipped with a pressure relief valve designed to relieve the combined flow of all the charging pumps at the relief valve set pressure. These relief valves also protect the system from inadvertent overpressurization during plant cooldown or startup. Each discharge line from the RHRS to the RCS is equipped with a pressure relief valve designed to relieve the maximum possible backleakage through the valves isolating the RHRS from the RCS.

The RHRS is designed for a single nuclear power unit and is not shared among nuclear power units.

The RHRS is designed to be fully operable from the Control Room for normal operation following the restoration of power to both inlet motor operated valves in each suction line. Power is removed to prevent inadvertent or spurious actuation. Control Room manual operations required of the operator are: opening the suction isolation valves, positioning the flow control valves downstream of the RHRS heat exchangers, and starting the residual heat removal pumps. By nature of its redundant two train design, the RHRS is designed to accept all major component single failures with the only effect being an extension in the required cooldown time. For two low probability electrical system single failures, i.e., failure in the suction isolation valve interlock circuitry, or diesel generator failure in conjunction with loss of offsite power, limited operator action outside the Control Room is required to open the suction isolation valves. Although Westinghouse considers it to be of low probability, spurious operation of a single motor operated valve can be accepted without loss of function as a result of the redundant two train design.

Missile protection, protection against dynamic effects associated with the postulated rupture of piping, and aseismic design are discussed in Sections 3.5, 3.6N and 3.7N respectively.

5.4.7.2 System Design

5.4.7.2.1 Schematic Piping and Instrumentation Diagrams

The RHRS, as shown in Figures 5.4-6 and 5.4-7, consists of two residual heat exchangers, two residual heat removal pumps, and the associated piping, valves, and instrumentation necessary for operational control. The inlet lines to the RHRS are connected to the hot legs of two reactor coolant loops, while the return lines are connected to the cold legs of each of the reactor coolant loops. These return lines are also the ECCS low head injection lines (see Figure 6.3-1).

The RHRS suction lines are isolated from the RCS by two motor operated valves in series and a relief valve, all located inside the Containment. Each discharge line is isolated from the RCS by two check valves located inside the Containment and by a normally open motor operated valve located outside the containment. (The check valves and the Motor operated valve on each discharge line are not part of the RHRS; these valves are shown as part of the ECCS, see Figure 6.3-1).

During RHRS operation, reactor coolant flows from the RCS to the residual heat removal pumps, through the tube side of the residual heat exchangers, and back to the RCS. The heat is transferred to the component cooling water circulating through the shell side of the residual heat exchangers.

Coincident with operation of the RHRS, a portion of the reactor coolant flow may be diverted from downstream of the residual heat exchangers to the Chemical and Volume Control System low pressure letdown line for cleanup and/or pressure control. By regulating the diverted flow rate and the charging flow, the RCS pressure may be controlled. Pressure regulation is necessary to maintain the pressure range dictated by the fracture prevention criteria requirements of the reactor vessel and by the number 1 seal differential pressure and net positive suction head requirements of the reactor coolant pumps.

The RCS cooldown rate is manually controlled by regulating the reactor coolant flow through the tube side of the residual heat exchangers. A line containing a flow control valve bypasses each residual heat exchanger and is used to maintain a constant return flow to the RCS. Instrumentation is provided to monitor system pressure, temperature and total flow.

The RHRS is also used for filling the refueling cavity before refueling. After refueling operations, water is pumped back to the Refueling Water Storage Tank until the water level is brought down to the flange of the reactor vessel. The remainder of the water is removed via a drain connection at the bottom of the refueling canal.

When the RHRS is in operation, the water chemistry is the same as that of the reactor coolant. Provision is made for the Process Sampling System to extract samples from the flow of reactor coolant downstream of the residual heat exchangers. A local sampling point is also provided on each residual heat removal train between the pump and heat exchangers.

The RHRS functions in conjunction with the high head portion of the ECCS to provide injection of borated water from the Refueling Water Storage Tank into the RCS cold legs during the injection phase following a LOCA.

In its capacity as the low head portion of the ECCS, the RHRS provides long term recirculation capability for core cooling following the injection phase of the LOCA. This function is accomplished by aligning the RHRS to take fluid from the Containment sump, cool it by circulation through the residual heat exchangers, and supply it to the core directly as well as via the centrifugal charging pumps and safety injection pumps.

The use of the RHRS as part of the ECCS is more completely described in Section 6.3.

The RHR pumps, in order to perform their ECCS function, are interlocked to start automatically on receipt of a safety injection signal (see Section 6.3).

The RHR suction isolation valves in each inlet line from the RCS are separately interlocked to prevent both their being opened when RCS pressure is greater than 425 psig. This interlock is described in more detail in Sections 5.4.7.2.4 and 7.6.2.

The RHR suction isolation valves are also interlocked to prevent their being opened unless the isolation valves in the following lines are closed:

1. Recirculation lines from the residual heat exchanger outlets to the suctions of the safety injection pumps and centrifugal charging pumps.

2. Residual heat removal pump suction line from the Refueling Water Storage Tank.

3. Residual heat removal pump suction line from the Containment sump.

The motor operated valves in the RHR miniflow bypass lines are interlocked to open when the RHR pump discharge flow is less than 500 gpm and close when the flow exceeds 1000 gpm. This interlock is also described in more detail in Section 5.4.7.2.4. The logic circuitry for the interlock is shown in Figure 5.4-20.

5.4.7.2.2 Equipment and Component Descriptions

The materials used to fabricate RHRS components are in accordance with the applicable code requirements. All parts of components in contact with borated water are fabricated or clad with austenitic stainless steel or equivalent corrosion resistant material. Component parameters are given in Table 5.4-8.

Residual Heat Removal Pumps

Two pumps are installed in the RHRS. Component cooling water is supplied to RHR pump seal coolers under all modes of plant operation. The RHR pumps operate during startup, cooldown, and cold shutdown operations, and during safeguards operations.

The pumps are sized to deliver reactor coolant flow through the residual heat exchangers to meet the plant cooldown requirements. The use of two separate residual heat removal trains assures that cooling capacity is only partially lost should one pump become inoperative.

The residual heat removal pumps are protected from overheating and loss of suction flow by miniflow bypass lines that assure flow to the pump suction. A valve located in each miniflow line is regulated by a signal from the flow transmitters located in each pump discharge header. The

control valves open when the residual pump discharge flow is less than 500 gpm and close when the flow exceeds 1000 gpm.

Each RHR train is provided with control board position indication of the miniflow and suction valves, and a low flow alarm to alert the operator to the inadvertent closure of a suction valve on a running RHR pump. A pressure sensor in each pump discharge header provides a signal for an indicator in the Control Room. A high pressure alarm is also actuated by the pressure sensor.

The two pumps are vertical, centrifugal units with mechanical seals on the shafts. All pump surfaces in contact with reactor coolant are austenitic stainless steel or equivalent corrosion resistant material.

The residual heat removal pumps also function as the low head safety injection pumps in the ECCS (see Section 6.3 for further information and for the residual heat removal pump performance curves).

Residual Heat Exchangers

Two residual heat exchangers are installed in the system. The heat exchanger design is based on heat load and temperature differences between reactor coolant and component cooling water existing 28 hours after reactor shutdown when the temperature difference between the two systems is small.

The installation of two heat exchangers in separate and independent residual heat removal trains assures that the heat removal capacity of the system is only partially lost if one train becomes inoperative.

The residual heat exchangers are of the shell and U-tube type. Reactor coolant circulates through the tubes, while component cooling water circulates through the shell. The tubes are welded to the tubesheet to prevent leakage of reactor coolant.

The residual heat exchangers also function as part of the ECCS (see Section 6.3).

Residual Heat Removal System Valves

Valves that perform a modulating function are equipped with two sets of packings and an intermediate leakoff connection that discharges to the drain header.

Manual and motor operated valves typically have backseats to facilitate repacking. Leakage connections are provided where required by valve size and fluid conditions.

5.4.7.2.3 System Operation

Reactor Startup

Generally, while at cold shutdown condition, decay heat from the reactor core is being removed by the RHRS. The number of pumps and heat exchangers in service depends upon the heat load at the time.

At initiation of the plant startup, the RCS is completely filled, and the pressurizer heaters are energized. The RHRS is operating and is connected to the Chemical and Volume Control System via the low pressure letdown line to control reactor coolant pressure. During this time, the RHRS acts as an alternate letdown path. The manual valves downstream of the residual heat exchangers leading to the letdown line of the Chemical and Volume Control System are opened to permit letdown flow to CVCS. The normal letdown pressure control valve in CVCS is then adjusted to regulate letdown flow and RCS pressure.

After the reactor coolant pumps are started, the residual heat removal pumps may continue to run to assist in controlling the heatup rate. The RHRS pumps are stopped when RCS pressure is increased to approximately 400 psig, but pressure control via the RHRS and the low pressure let down line is continued until the pressurizer steam bubble is formed. Indication of steam bubble formation is provided in the Control Room by the damping out of the RCS pressure fluctuations, and by pressurizer level indication. The RHRS is then isolated from the RCS (prior to reaching 350°F) and the system pressure is controlled by normal letdown and the pressurizer spray and pressurizer heaters.

Power Generation and Hot Standby Operation

During power generation and hot standby operation, the RHRS is not in service but is aligned for operation as part of the ECCS.

Reactor Cooldown

Reactor cooldown is defined as the operation which brings the reactor from no-load temperature and pressure to cold conditions.

The initial phase of reactor cooldown is accomplished by transferring heat from the RCS to the Steam and Power Conversion System through the use of the steam generators.

When the reactor coolant temperature and pressure are reduced to approximately 350°F and 425 psig, approximately 4 hours after reactor shutdown, the second phase of cooldown starts with the RHRS being placed in operation.

Startup of the RHRS includes a warmup period during which time reactor coolant flow through the heat exchangers is limited to minimize thermal shock. The rate of heat removal from the reactor coolant is manually controlled by regulating the coolant flow through the residual heat exchangers. By adjusting the control valves downstream of the residual heat exchangers the mixed mean temperature of the return flows is controlled. Coincident with the manual adjustment, each heat exchanger bypass valve is regulated to give the required total flow.

Provisions exist to isolate the RHR lines and restore the ECCS capability for core cooling should a leak or rupture develop during cooldown (below 400 psig). Any leakage of the RHR System piping would be expected to occur when the System is initially pressurized at 400 psig. The RCS is at this time under manual control by the reactor operator, who is monitoring the pressurizer level and the RCS loop pressure. Any significant leakage from the RHR System would be immediately detected. Upon detection, the operator would isolate the RHR System and identify the location and cause. Since decay heat generation four hours after shutdown is about 1.2% of

full power, the RCS fluid temperature is at about 350^{o}F, and the core stored energy is essentially removed, the operator would have ample time to isolate the RHR loop.

The reactor cooldown rate is limited by RCS equipment cooling rates based on allowable stress limits, as well as the operating temperature limits of the Component Cooling Water System. As the reactor coolant temperature decreases, the reactor coolant flow through the residual heat exchangers is increased by adjusting the control valve in each heat exchanger's tube side outlet line.

As cooldown continues, the pressurizer is filled with water and the RCS is operated in the water solid condition.

At this stage, pressure control is accomplished by regulating the charging flow rate and the rate of letdown from the RHRS to the Chemical and Volume Control System.

After the reactor coolant pressure is reduced and the temperature is 140°F or lower, the RCS may be opened for refueling or maintenance.

The potential for exceeding the allowable cooldown rate has been analyzed. The maximum cooldown rate which can result if both RHR flow control valves and both RHR bypass valves all simultaneously fail in such a manner as to permit maximum flow through the RHR heat exchangers depends on several factors including the RHR flow rate, the component cooling water system flow rates and temperatures, and other heat loads on the component cooling water system. (This simultaneous failure is considered to be a low probability event considering the few hours a year when it could cause any effect).

One of the key factors is the RCS temperature, since the heat removal rate depends on the temperature differential between the RHR (RCS) flow and the component cooling water flow in the RHR heat exchanger. Typically, it is impossible to maintain a cooldown rate even as big as 50°F/hr when the RCS temperature is less than 250°F, even with the maximum flow through the RHR heat exchangers. Even if maximum flow through the RHR heat exchangers was experienced at the instant of initiating RHR operation and no operator actions was taken, it is unlikely that the cooldown would exceed 100°F over the first hour. The maximum possible cooldown rate for the period from 350°F to 250°F would not exceed 200°F/hr.

Calculations have been performed which show that, from a stress standpoint, a cooldown rate greater than 200°F/hr is acceptable for such a hypothetical cooldown from 350° to 250°F even though it is difficult to maintain a cooldown rate of 50°F/hr at or below an RCS temperature of 250°F. Although such a hypothetical cooldown event is acceptable from a stress standpoint, assuming no operator action, it should be noted that the operator can significantly limit the maximum possible cooldown rate by merely stopping one of the RHR pumps. See Table 5.4-17.

Refueling

Both residual heat removal pumps are utilized during refueling to pump borated water from the Refueling Water Storage Tank to the refueling cavity. During this operation, the isolation valves in the inlet lines of the RHRS are closed, and the isolation valves to the Refueling Water Storage Tank are opened.

The reactor vessel head is lifted slightly. The refueling water is then pumped into the reactor vessel through the normal RHRS return lines and into the refueling cavity through the open reactor vessel. The reactor vessel head is gradually raised as the water level in the refueling cavity increases. After the water level reaches the normal refueling level, the inlet isolation valves are opened, the Refueling Water Storage Tank supply valves are closed, and residual heat removal is resumed.

During Mode 6 with the water level ≥ 23 ft above the top of the reactor vessel flange, one RHR loop shall be OPERABLE and in operation as required per TS 3.9.5. In the event of a residual heat removal system malfunction, an alternate cooling path may be used upon loss of the RHR pumping design function (i.e., failure of the RHR Pump or motor/power). If the RHR pumping capability was lost, the discharge path could be realigned to the suction of the Safety Injection Pump which is in alignment with the emergency boration flow path. The RHR minimum flow valves will be closed so that all flow is in the design direction. In lieu of normal injection to the cold legs, flow will be directed to the suction of the Safety Injection pump. Flow will be controlled by HCV-0606/0607 (valves which normally controls RHR flow) or by 8717 (a manual valve). Because the RWST will remain aligned, both NPSH and emergency boration water supply will remain within design. The SI pump discharge will not change and will supply water cooled at the RHR heat exchanger to the cold legs. This method provided the RHR function via a SI pump and remains within the bounding design pressures, temperatures and flows of the RHR/SI systems.

During refueling, the RHRS is maintained in service with the number of pumps and heat exchangers in operation as required by the heat load.

Following refueling, the residual heat removal pumps are used to drain the refueling cavity to the top of the reactor vessel flange by pumping water from the RCS to the Refueling Water Storage Tank.

The residual heat removal pumps may also be used during filling of Safety Injection Accumulators.

5.4.7.2.4 Control

Each inlet line to the RHRS is equipped with a pressure relief valve sized to relieve the combined flow of all the charging pumps at the relief valve set pressure. Sufficient capacity in RHR relief valves is available to also relieve thermal expansion. These relief valves also protect the system from inadvertent overpressurization during plant cooldown or startup. Each valve has a relief flow capacity of 900 gpm at a set pressure of 450 psig. An analysis has been conducted to confirm the capability of the RHRS relief valve to prevent overpressurization in the RHRS. All credible events were examined for their potential to overpressurize the RHRS. These events included normal operating conditions, infrequent transients, and abnormal occurrences. The analysis confirmed that one relief valve has the capability to keep the RHRS maximum pressure within code limits.

Each discharge line from the RHRS to the RCS is equipped with a pressure relief valve to relieve the maximum possible backleakage through the valves separating the RHRS from the RCS. Technical specifications places maximum limits on RCS leakage. Further, the technical specifications require the plant to be shutdown within specified time periods if these values are exceeded. Each valve has a relief flow capacity of 20 gpm at a set pressure of 600 psig; therefore, even if it were conservatively assumed that all RCS leakage was in one RHR train the

relief value still provides significant margin. These relief valves are located in the ECCS (see Figure 6.3-1).

The fluid discharged by the suction side relief valves is collected in the pressurizer relief tank. The fluid discharged by the discharge side relief valves is collected in the recycle holdup tank of the Boron Recycle System.

The design of the RHRS includes two motor operated gate isolation valves in series on each inlet line between the high pressure RCS and the lower pressure RHRS. They are closed during normal operation and are only opened for residual heat removal during a plant cooldown after the RCS pressure is reduced to 425 psig or lower and RCS temperature is reduced to approximately 350°F. During a plant startup the inlet isolation valves are shut after drawing a bubble in the pressurizer and prior to increasing RCS pressure above 600 psig. These isolation valves are provided with a “prevent-open” interlock which is designed to prevent possible exposure of the RHRS to normal RCS operating pressure.

The two inlet isolation valves in each residual heat removal subsystem are separately and independently interlocked with pressure signals to prevent being opened when RCS pressure is greater than RHRS operational pressure.

The interlock signal provided to the isolation valve closest to the RCS is independent and diverse from the interlock signal provided to the isolation valves closest to the RHRS.

Overpressure protection is provided for the piping between each pair of RHR isolation valves via check valves in a bypass line. Should a pressure build-up occur either from water trapped between these normally closed inner and outer isolation valves or from thermal expansion of the water in this section of line due to a rapid heat-up of containment (e.g. following a postulated LOCA or steam break), pressure is relieved through these check valves (See Figure 5.4-6).

The use of two independently powered motor operated valves in each of the two inlet lines, along with two independent pressure interlock signals for each function assures a design which meets applicable single failure criteria. Not only more than one single failure but also different failure mechanisms must be postulated to defeat the function of preventing possible exposure of the RHRS to normal RCS operating pressure. These protective interlock designs, in combination with plant operating procedures, provide diverse means of accomplishing the protective function.

For a discussion of the function of the RHR suction line isolation valve arrangement, see FSAR Section 6.2.4.1.3.5. For further information on the instrumentation and control features, see Section 7.6.2.

The RHR inlet isolation valves are provided with red-green position indicator lights on the main control board.

Isolation of the low pressure RHRS from the high pressure RCS is provided on the discharge side by two check valves in series. These check valves are located in the ECCS and their testing is described in Section 6.3.4.2.

The motor-operated valves in the mini-flow bypass lines between the RHR pump suction and discharge is normally open. These valves are interlocked to flow transmitters in the pump discharge lines to close when the RHR pump discharge flow goes above an upper limit, and

reopen when the pump discharge flow fails below a minimum value. The interlock ensures that the flow through the RHR pump will be sufficient to cool the pump when the pressure in the lines to which the pump discharge flow is directed is greater than pump discharge pressure. The margin between the opening and closing signals prevents valve operator cycling due to flow restrictions. The separation of this interlock is in accordance with IEEE 384-1974. The Train A interlock is electrically independent from the Train B interlock. A single random electrical failure of the interlock in one train will not prevent operation of the interlock in the redundant train. In addition, the interlock is environmentally and seismically qualified to the appropriate criteria.

5.4.7.2.5 Applicable Codes and Classifications

The entire RHRS is designed as Nuclear Safety Class 2. Component codes and classifications are given in Section 3.2.

5.4.7.2.6 System Reliability Considerations

General Design Criterion 34 requires that a system to remove residual heat be provided. The safety function of this system is to transfer fission product decay heat and other residual heat from the core at a rate sufficient to prevent fuel or pressure boundary design limits from being exceeded. Safety grade systems are provided in the plant design, both NSSS scope and balance of plant scope, to perform this safety function. The NSSS scope safety grade systems which perform this function, for all plant conditions except a LOCA, are the RCS and steam generators, which operate in conjunction with the Auxiliary Feedwater System, the steam generator safety valves, and the steam generator power operated relief valves; and the RHRS which operates in conjunction with the Component Cooling Water System and the Service Water System. The balance of plant scope safety grade systems which perform this function, for all plant conditions except LOCA, are the Auxiliary Feedwater System, the steam generator safety valves, and the steam generator power operated relief valves, which operate in conjunction with the RCS and the steam generators; and the component cooling water and service water systems, which operate in conjunction with the RHRS. For LOCA conditions, the safety grade system which performs the function of removing residual heat from the reactor core is the ECCS, which operates in conjunction with the Component Cooling Water System and the Service Water System.

The RHRS is provided with two residual heat removal pumps, and two residual heat removal heat exchangers arranged in two separate, independent flow paths. To assure reliability, each residual heat removal pump is connected to a different vital bus. Each residual heat removal train is isolated from the RCS on the suction side by two motor operated valves in series. Each motor operated valve receives power via a separate motor control center and the two valves in series in each train receive their power from a different vital bus. Each suction isolation valve is also interlocked to prevent exposure of the RHRS to the normal operating pressure of the RCS (see Section 5.4.7.2.4).

RHRS operation for normal conditions and for major failures is accomplished completely from the Control Room following the restoration of power to both inlet motor operated valves in each suction line. Power is removed to prevent inadvertent or spurious actuation. The redundancy in the RHRS design provides the system with the capability to maintain its cooling function even with major single failures, such as failure of a residual heat removal pump, valve, or heat exchanger since the redundant train can be used for continued heat removal. Consequences of the loss of component cooling water flow to RHR pumps are discussed in Section 9.2.2.

Although such major system failures are within the system design basis, there are other less significant failures which can prevent opening of the residual heat removal suction isolation valves from the Control Room. Since these failures are of a minor nature, improbable to occur, and easily corrected outside the Control Room, with ample time to do so, they have been realistically excluded from the engineering design basis.

Such failures are not likely to occur during the limited time period in which they can have any effect (i.e., when opening the suction isolation valves to initiate RHR operation); however, even if they should occur, they have no adverse safety impact and can be readily corrected. In such a situation, the Auxiliary Feedwater System and steam generator power operated relief valves can be used to maintain the plant at 350°F until the RHRS is made available.

One failure of this type is a failure in the interlock circuitry which is designed to prevent exposure of the RHRS to the normal operating pressure of the RCS (see Section 5.4.7.2.4). In the event of such a failure, RHRS operation can be initiated by defeating the failed interlock through corrective action at the solid State Protection System cabinet or at the individual effected motor control centers.

The other type of failure which can prevent opening the residual heat removal suction isolation valves (8701B and 8702A, as shown in Figure 5.4-6) from the Control Room is a failure of normal electrical power train. Such a failure is extremely unlikely to occur during the few minutes out of a year's operating time during which it can have any consequence. If such an unlikely event should occur, several alternatives are available. The most realistic approach would be to obtain restoration of offsite power, which can be expected to occur in less than 1/2 hour. Other alternatives are to restore the emergency diesel generator to operation or to bring in an alternate power source to valves 8701B and 8702A as described below.

Normally, valve 8701B is supplied via a Train A power supply and 8702A via a Train B power supply. In the event one power supply train becomes unavailable, alternate connections are made as follows:

The description applies to valve 8701B, however, a similar procedure applies to 8702A. Two metal enclosed junction boxes (one for Train A and one for Train B) are connected by rigid steel conduit to a common metal enclosed junction box. Power and control cables from Train A and Train B motor control centers are terminated by female connectors in the Train A and Train B junction boxes respectively.

Train A cables from the valve motor and limit switch are terminated by male cable connectors and routed through the common junction box. Sufficient slack in these cables are provided so that the male connectors can be plugged into the appropriate Train A or Train B female connectors as required. Normally, the connectors are mated in the Train A junction box. When required, cables are unplugged and removed from the Train A junction box and replugged in the Train B junction box.

A combination motor starter, with control switch and indicating lights, provided in the Train B motor control center then permits valve operation from the Train B motor control center.

The only impact of either of the above types of failures is some delay in initiating residual heat removal operation, while action is taken to open the residual heat removal suction isolation valves. This delay has no adverse safety impact because the Auxiliary Feedwater System and

steam generator power operated relief valves can be used to maintain the plant at 350° until the RHRS is made available.

A failure mode and effects analysis (FMEA) of the Residual Heat Removal System is provided as Table 5.4-17. Although the FMEA specifically addresses single active mechanical failures in the RHR System, spurious movement of powered components and operator errors by the nature of the system and the FMEA are enveloped in the analysis. Passive failures of non-active components, although not addressed in the FMEA, will affect a single train, due to provisions for isolation of the independent RHR trains.

5.4.7.2.7 Manual Actions

The RHRS is designed to be fully operable from the Control Room for normal operation following the restoration of power to both valves in each of the suction lines. Power is removed to prevent inadvertent or spurious actuation. Control Room manual operations required of the operator are: opening the suction isolation valves, positioning the flow control valves down stream of the RHRS heat exchangers, and starting the residual heat removal pumps.

Manual actions required outside the Control Room under conditions of single failure, are discussed in Section 5.4.7.2.6.

5.4.7.2.8 Leakage Detection Capability for RHR System Leakage

The major portion of the RHRS is contained in the Safeguards Building between elevations 773'-0" and 810'-6". All leakages originating at RHR components will be collected by the floor drain system.

The effects of the leaks will be detected in the Control Room via the floor drain system alarms and area radiation monitoring alarms. Large leaks in the RHRS will be detected by interpretation of RHR flow parameters, area radiation monitoring alarms and high level alarms of floor drain sumps nos. 1 and 3. Small leaks will be alarmed in the Control Room by the area radiation monitors in the Safeguards Building. (See the listing for these monitors in Table 12.3-8.)

By interpretation of process parameters and alarms, the operators will determine the area where the leakage has occurred. Further information will be obtained by visual observation. System design permits complete isolation of the faulted RHR loop with no impact on plant safety.

Depending on the severity of the leak, the operator will make a determination on the proper course of action. After a loop is taken out of service, it can be drained and vented, and prepared for maintenance.

The occurrence of leaks will be alarmed in the Control Room. The Safeguards sumps alarms and area radiation monitoring alarms are not designed in accordance with the requirements of IEEE-279; however, the area radiation monitoring system is designed in accordance with the test and calibration requirements of IEEE-279.

5.4.7.3 Performance Evaluation

The performance of the RHRS in reducing reactor coolant temperature is evaluated through the use of heat balance calculations on the RCS, and the Component Cooling Water System at

stepwise intervals following the initiation of residual heat removal operation. Heat removal through the residual heat removal and component cooling water heat exchangers is calculated at each interval by use of standard water-to-water heat exchanger performance correlations; the resultant fluid temperatures for the residual heat removal and component cooling water systems are calculated and used as input to the next interval's heat balance calculation.

Assumptions utilized in the series of heat balance calculations describing plant residual heat removal cooldown are as follows:

1. Residual heat removal operation is initiated after reactor shutdown as follows:

Normal Plant Cooldown	4 hours
BTP RSB 5-1 Cooldown	9 hours
Fire Safe Shutdown	14 hours
Cooldown in the event of a DBA in the opposite unit	4 hours
Cooldown in accordance with technical specifications	12 hours

2. Residual heat removal operation begins at a reactor coolant temperature of 350°F.

3. Thermal equilibrium is maintained throughout the RCS during the cooldown.

4. Component cooling water temperature during cooldown is limited to a maximum of 122°F.

5. The effect of common (shared) auxiliary heat loads on the CCWS operation are assumed as follows:

Normal Plant Cooldown	The opposite unit can assume the common auxiliary heat loads.
BTP RSB 5-1 Cooldown	The opposite unit can assume the common auxiliary heat loads.
Fire Safe Shutdown	Each unit must be capable of shutting down without system support of the other.
Cooldown in the event of a DBA in the opposite unit	The cooldown unit can assume common auxiliary loads.
Cooldown in accordance with technical specifications	The opposite unit can assume the common auxiliary heat loads.

Cooldown curves calculated using this method are provided for the case of all residual heat removal components operable (Figure 5.4-8) and for the case of a single train residual heat removal cooldown (Figure 5.4-9).

5.4.7.4 Preoperational Testing

Preoperational testing of the RHRS is addressed in Chapter 14.

5.4.8 REACTOR WATER CLEANUP

This section is not applicable to the CPNPP.

5.4.9 MAIN STEAM LINE AND FEEDWATER PIPING

The main steam line is described in Section 10.3 and the feedwater piping is described in Section 10.4.7.

5.4.10 PRESSURIZER

5.4.10.1 Design Bases

The general configuration of the pressurizer is shown in Figure 5.4-10. The design data of the pressurizer are given in Table 5.4-9. Codes and material requirements are provided in Section 5.2.

The pressurizer provides a point in the RCS where liquid and vapor can be maintained in equilibrium under saturated conditions for pressure and control purposes.

5.4.10.1.1 Pressurizer Surge Line

The surge line is sized to limit the pressure drop between the RCS and the safety valves with maximum allowable discharge flow from the safety valves. Overpressure of the RCS does not exceed 110 percent of the design pressure.

The surge line and the thermal sleeves at each end are designed to withstand the thermal stresses resulting from volume surges of relatively hotter or colder water which may occur during operation.

The pressurizer surge line nozzle diameter is given in Table 5.4-9 and the pressurizer surge line dimensions are shown in Figure 5.1-1, Sheet 2.

5.4.10.1.2 Pressurizer

The volume of the pressurizer is equal to, or greater than, the minimum volume of steam, water, or total of the two which satisfies all of the following requirements:

1. The combined saturated water volume and steam expansion volume is sufficient to provide the desired pressure response to system volume changes.

2. The water volume is sufficient to prevent the heaters from being uncovered during a step load increase of 10 percent at full power.

3. The steam volume is large enough to accommodate the surge resulting from 50 percent reduction of full load with automatic reactor control and 40 percent steam dump without the water level reaching the high level reactor trip point.

4. The steam volume is large enough to prevent water relief through the safety valves following a loss of load with the high water level initiating a reactor trip, without reactor control or steam dump.

5. The pressurizer will not empty following reactor trip and turbine trip.

6. The emergency core cooling signal is not activated during reactor trip and turbine trip.

5.4.10.2 Design Description

5.4.10.2.1 Pressurizer Surge Line

The pressurizer surge line connects the pressurizer to one reactor hot leg. The line enables continuous coolant volume pressure adjustments between the RCS and the pressurizer.

5.4.10.2.2 Pressurizer

The pressurizer is a vertical, cylindrical vessel with hemispherical top and bottom heads constructed of carbon steel, with austenitic stainless steel cladding on all surfaces exposed to the reactor coolant. A stainless steel liner or tube may be used in lieu of cladding in some nozzles.

The surge line nozzle and removable electric heaters are installed in the bottom head. The heaters are removable for maintenance or replacement. A thermal sleeve is provided to minimize stresses in the surge line nozzle. A screen at the surge line nozzle and baffles in the lower section of the pressurizer prevent an insurge of cold water from flowing directly to the steam/water interface and assist mixing.

Spray line nozzle, relief and safety valve connections are located in the tip head of the vessel. Spray flow is modulated by automatically controlled air operated valves. The spray valves also can be operated manually by a switch in the Control Room.

A small, continuous spray flow is provided through a manual bypass valve around the power operated spray valves to assure that the pressurizer liquid is homogeneous with the coolant and to prevent excessive cooling of the spray piping.

During an outsurge from the pressurizer, flashing of water to steam and generating of steam by automatic actuation of the heaters keep the pressure above the minimum allowable limit. During an insurge from the RCS, the spray system, which is fed from two cold legs, condenses steam in the vessel to prevent the pressurizer pressure from reaching the setpoint of the power operated relief valves for normal design transients. Heaters are energized on high water level during insurge to heat the subcooled surge water that enters the pressurizer from the reactor coolant loop.

Material specifications are provided in Table 5.2-2 for the pressurizer, pressurizer relief tank, and the surge line. Design transients for the components of the RCS are discussed in

Section 3.9(N).1. Additional details on the pressurizer design cycle analysis are given in Section 5.4.10.3.5.

Pressurizer Instrumentation

Refer to Chapter 7 for details of the instrumentation associated with pressurizer pressure, level, and temperature.

Spray Line Temperatures

Temperatures in the spray lines from two loops are measured and indicated. Alarms from these signals are actuated by low spray water temperature. Alarm conditions indicate insufficient flow in the spray lines.

Safety and Relief Valve Discharge Temperatures

Temperatures in the pressurizer safety and relief valve discharge lines are measured and indicated. An increase in a discharge line temperature is an indication of leakage or relief through the associated valve.

5.4.10.3 Design Evaluation

5.4.10.3.1 System Pressure

Whenever a steam bubble is present within the pressurizer, RCS pressure is maintained by the pressurizer. Analyses indicate that proper control of pressure is maintained for the operating conditions.

A safety limit has been set to ensure that the RCS pressure does not exceed the maximum transient value allowed under the ASME Code, Section III, and thereby assures continued integrity of the RCS components.

Evaluation of plant conditions of operation which follow indicate that this safety limit is not reached.

During startup and shutdown, the rate of temperature change is controlled by the operator. The maximum heatup/cooldown rate of the pressurizer is governed by the Technical Specifications. Heatup rate is controlled by pump energy and by the pressurizer electrical heating capacity. This heatup rate takes into account the continuous spray flow provided to the pressurizer. When the reactor core is shutdown, the heaters are de-energized.

When the pressurizer is filled with water, i.e., during initial system heatup, and near the end of the second phase of plant cooldown, RCS pressure is maintained by the letdown flow rate via the RHRS.

5.4.10.3.2 Pressurizer Performance

The pressurizer has a minimum free internal volume. The normal operating water volume at full load conditions is a percentage of the free internal vessel volume. Under part load conditions, the water volume in the vessel is reduced for proportional reductions in plant load. The various

plant operating transients are analyzed and the design pressure is not exceeded with the pressurizer design parameters as given in Table 5.4-9.

5.4.10.3.3 Pressure Setpoints

The RCS design and operating pressure together with the safety, power relief and pressurizer spray valves setpoints, and the protection system setpoint pressures are listed in Table 5.4-10. The design pressure allows for operating transient pressure changes. The selected design margin considers core thermal lag, coolant transport times and pressure drops, instrumentation and control response characteristics, and system relief valve characteristics.

The pressurizer power-operated relief valve (PORV) setpoint value listed on Table 5.4-10 is that of the PORV automatic opening at power (2/2) logic function. When the pressurizer pressure goes above this value, the PORV will automatically open. The purpose of this logic function is to limit Reactor Coolant System (RCS) pressure when the reactor is at power or at hot standby and to prevent actuation of the pressurizer high pressure Reactor Trip when at power on certain operational transients, such as a load reduction. This 2/2 logic function has two inputs; the first input is from a control channel whose setpoint is effectively 2335 psig, which is the value listed on Table 5.4-10; the second input is an interlock which is derived from an independent pressurizer pressure channel and has a setpoint of 2185 psig such that when the pressurizer pressure is above this value the PORV automatic opening control is armed but will not cause the PORV to open as long as pressurizer pressure is below the setpoint of the control channel. The purpose of the interlock (i.e., the input set at 2185 psig) is to prevent depressurization of the RCS by automatically closing the PORV in the event the control input fails high.

5.4.10.3.4 Pressurizer Spray

Two separate, automatically controlled spray valves with remote manual overrides are used to initiate pressurizer spray. In parallel with each spray valve is a manual throttle valve which permits a small continuous flow through both spray lines to reduce thermal stresses and thermal shock when the spray valves open, and to help maintain uniform water chemistry and temperature in the pressurizer. Temperature sensors with low alarms are provided in each spray line to alert the operator to insufficient bypass flow. The layout of the common spray line piping to the pressurizer forms a water seal which prevents the steam buildup back to the control valves.

The spray rate is selected to prevent the pressurizer pressure from reaching the operating setpoint of the power relief valves during a step reduction in power level of 10 percent of full load.

The pressurizer spray lines and valves are large enough to provide adequate spray using as the driving force the differential pressure between the surge line connection in the hot leg and the spray line connection in the cold leg. The spray line inlet connections extend into the cold leg piping in the form of a scoop so that the velocity head of the reactor coolant loop flow adds to the spray driving force. The spray valves and spray line connections are arranged so that the spray will operate when one reactor coolant pump is not operating. The line may also be used to assist in equalizing the boron concentration between the reactor coolant loops and the pressurizer.

A flow path from the Chemical and Volume Control System to the pressurizer spray line is also provided. This additional facility provides auxiliary spray to the vapor space of the pressurizer during cooldown if the reactor coolant pumps are not operating. The thermal sleeves on the

pressurizer connection and the spray piping are designed to withstand the thermal stresses resulting from the introduction of cold spray water.

5.4.10.3.5 Pressurizer Design Analysis

The occurrences for pressurizer design cycle analysis are defined as follows:

1. The temperature in the pressurizer vessel is always, for design purposes, assumed to equal saturation temperature for the existing RCS pressure increase. In this case the temperature of the steam space will exceed the saturation temperature since an isentropic compression of the steam is assumed.

   The only exception of the above occurs when the pressurizer is filled solid during plant startup and cooldown.

2. The temperature shock on the spray nozzle is assumed to equal the temperature of the nozzle minus the cold leg temperature and the temperature shock on the surge nozzle is assumed to equal the pressurizer water space temperature minus the hot leg temperature.

3. Pressurizer spray is assumed to be initiated instantaneously to its design value as soon as the RCS pressure increases 40 psi above the nominal operating pressure. Spray is assumed to be terminated as soon as the RCS pressure falls 40 psi below normal operating pressure.

4. Unless otherwise noted, pressurizer spray is assumed to be initiated once per occurrence of each transient condition. The pressurizer surge nozzle is also assumed to be subject to one temperature transient per transient condition, unless otherwise noted.

5. At the end of each transient, except the faulted conditions, the RCS is assumed to return to a load condition consistent with the plant heatup transient.

6. Temperature changes occurring as a result of pressurizer spray are assumed to be instantaneous. Temperature changes occurring on the surge nozzle are also assumed to be instantaneous.

7. Whenever spray is initiated in the pressurizer, the pressurizer water level is assumed to be at the no-load level.

5.4.10.4 Test and Inspections

The pressurizer is designed and constructed in accordance with the ASME Code, Section III.

To implement the requirements of the ASME Code, Section XI, the following welds are designed and constructed to present a smooth transition surface between the parent metal and the weld metal.

The path is ground smooth for ultrasonic inspection.

1. Support skirt to the pressurizer lower head.

2. Surge nozzle to the lower head.

3. Nozzles to the safety, relief, and spray lines.

4. Nozzle to safe end attachment welds.

5. All girth and longitudinal full penetration welds.

6. Manway attachment welds.

The liner within the safe end nozzle region extends beyond the weld region to maintain a uniform geometry for ultrasonic inspection.

Peripheral support rings are furnished for the removable insulation modules.

The pressurizer quality assurance program is given in Table 5.4-11.

5.4.11 PRESSURIZER RELIEF DISCHARGE SYSTEM

The Pressurizer Relief Discharge System collects, cools and directs for processing the steam and water discharged from the various safety and relief valves in the Containment. The system consists of the pressurizer relief tank, the safety and relief valve discharge piping, the relief tank spray header and associated piping, and the tank nitrogen supply, the vent to Containment and the drain to the Waste Processing System.

5.4.11.1 Design Bases

Codes and materials of the pressurizer relief tank and associated piping are given in Section 5.2. Design data for the tank are given in Table 5.4-12.

The system design is based on the requirement to absorb a discharge of steam equivalent to 110 percent of the full power pressurizer steam volume. The steam volume requirement is approximately that which would be experienced if the plant were to suffer a complete loss of load accompanied by a turbine trip but without the resulting reactor trip.

The minimum volume of water in the pressurizer relief tank is determined by the energy content of the steam to be condensed and cooled, by the assumed initial temperature of the water, and by the desired final temperature of the water volume. The initial water temperature is assumed to be 120°F, which corresponds to the design maximum expected Containment temperature for normal conditions. Provisions is made to permit cooling the tank should the water temperature rise above 120°F, which allows the contents of the tank to be drained directly to the Waste Processing System without cooling.

The vessel saddle supports and anchor bolt arrangement are designed to withstand the loadings resulting from a combination of nozzle loadings acting simultaneously with the vessel seismic and static loadings.

5.4.11.2 System Description

The piping and instrumentation diagram for the Pressurizer Relief Discharge System is given in Figure 5.1-1, Sheet 2.

The steam and water discharged from the various safety and relief valves inside the Containment is routed to the pressurizer relief tank if the discharged fluid is of reactor grade quality. Table 5.4-13 provides an itemized list of valves discharging to the tank together with references of the corresponding piping and instrumentation diagrams.

The tank normally contains water and a predominantly nitrogen atmosphere. In order to obtain effective condensing and cooling of the discharged steam, the tank is installed horizontally and the steam is discharged through a sparger pipe located near the bottom, under the water level. The sparger holes are designed to insure a resultant steam velocity close to sonic.

The tank is also equipped with an internal spray and a drain which are used to cool the water following a discharge. Cold water is drawn from the Reactor Makeup Water System, or the content of the tank is circulated through the reactor coolant drain tank heat exchanger of the Waste Processing System and back into the spray header.

The nitrogen gas blanket is used to control the atmosphere in the tank and to allow room for the expansion of the original water plus the condensed steam discharge. The tank gas volume is calculated using a final pressure based on an arbitrary design pressure of 100 psig. The design discharges raises the worst case initial conditions to 50 psig, a pressure low enough to prevent fatigue of the rupture discs. Provisions are made to permit the gas in the tank to be periodically analyzed to monitor the concentration of hydrogen and/or oxygen.

The contents of the vessel can be drained to the waste holdup tank in the Waste Processing System or the recycle holdup tank in the Boron Recycle System via the reactor coolant drain tank pumps in the Waste Processing System.

5.4.11.2.1 Pressurizer Relief Tank

The general configuration of the pressurizer relief tank is shown in Figure 5.4-11. The tank is a horizontal, cylindrical vessel with elliptical dished heads. The vessel is constructed of austenitic stainless steel and is overpressure protected in accordance with ASME Code, Section VIII, Division 1, by means of two safety heads with stainless steel rupture discs.

A flanged nozzle is provided on the tank for the pressurizer discharge line connection to the sparger pipe. The tank is also equipped with an internal spray connected to a cold water inlet and with a bottom drain, which are used to cool the tank following a discharge.

5.4.11.3 Safety Evaluation

The Pressurizer Relief Discharge System does not constitute part of the reactor coolant pressure boundary per 10CFR50, Section 50.2, since all of its components are downstream of the RCS safety and relief valves. Thus, General Design Criteria 14 and 15 are not applicable.

Furthermore, complete failure of the auxiliary systems serving the pressurizer relief tank will not impair the capability for safe plant shutdown.

The design of the system piping layout and piping restraints is consistent with Regulatory Guide 1.46. Westinghouse requires the architect engineer to meet Regulatory Guide 1.46 by restraining the safety and relief valve discharge piping so that integrity and operability of the valves are maintained in the event of a rupture.

Regulatory Guide 1.67 is not applicable since the system is not an open discharge system.

The Pressurizer Relief Discharge System is capable of handling the design discharge of steam without exceeding the design pressure and temperature.

The volume of water in the pressurizer relief tank is capable of absorbing the heat from the assumed discharge maintaining the water temperature below 200°F. If a discharge exceeding the design basis should occur, the relief device on the tank would pass the discharge through the tank to the Containment.

The rupture discs on the relief tank have a relief capacity equal to or greater than the combined capacity of the pressurizer safety valves. The tank design pressure is twice the calculated pressure resulting from the design basis safety valve discharge described in Section 5.4.11.1. The tank and rupture discs holders are also designed for full vacuum to prevent tank collapse if the content cools following a discharge without nitrogen being added.

Westinghouse, using piping layouts and conservative assumptions as to conditions in the pressurizer relief tank, calculated the backpressure at the safety valves. These calculations confirmed that the discharge piping from the safety and relief valves to the relief tank is sufficiently large to prevent backpressure at the safety valves from exceeding 20 percent of the setpoint pressure at full flow.

The overpressure transient analysis, requiring Safety Valve actuation, assumes the valves operate at their design maximum relieving capacity (see Table 5.4-16). The backpressure (see Table 5.4-16) is consistent with the valves' balanced bellows operating capability and is low enough not to affect the valve relieving capacity.

5.4.11.4 Instrumentation Requirements

The pressurizer relief tank pressure transmitter provides an indication of pressure relief tank pressure. An alarm is provided to indicate high tank pressure.

The pressurizer relief tank level transmitter supplies a signal for an indicator with high and low level alarms. The high level alarm setpoint for the pressurizer relief tank (PRT) will be less than the maximum PRT level that would prevent the PRT from exceeding 50 psig (stated in Section 5.4.11.2) during the design safety valve discharge to the PRT.

The temperature of the water in the pressurizer relief tank is indicated, and an alarm actuated by high temperature informs the operator that cooling of the tank contents is required.

5.4.11.5 Inspection and Testing Requirements

The system components are subject to nondestructive and hydrostatical testing during construction in accordance with Section VIII, Division 1 of the ASME Code.

During plant operation, periodic visual inspections and preventive maintenance are conducted on the system components according to normal industrial practice.

5.4.12 VALVES

5.4.12.1 Design Bases

As noted in Section 5.2, all valves out to and including the second valve normally closed or capable of automatic or remote closure, larger than 3/4 inch, are ANS Safety Class 1, and ASME III, Code Class 1 valves. For Unit 2, all 3/4 inch or smaller valves in lines connected to the RCS are Class 2 since the interface with the Class 1 piping is provided with suitable orificing for such valves. For Unit 1, all 3/4 inch or smaller valves in lines connected to the RCS are Class 2 since the interface with the Class 1 piping is provided with suitable orificing for such valves except for the lower level instrument taps (valves 1RC-8054A, 1RC-8054B, and 1RC-8054C) which are Class 1 because the valves are upstream of the orifice. Design data for the RCS valves are given in Table 5.4-14.

For a check valve to qualify as part of the RCS it must be located inside the containment system. When the second of two normally open check valves is considered part of the RCS (as defined in Section 5.1), means are provided to periodically assess back-flow leakage of the first valve when closed.

To ensure that the valves will meet the design objectives, the materials of construction minimize corrosion/erosion and ensure compatibility with the environment, leakage is minimized to the extent practicable by design, and Class 1 stresses are maintained within the limits of the ASME Code, Section III.

5.4.12.2 Design Description

All manual and motor operated valves of the RCS which are 3 inches and larger are provided with double packed stuffing boxes and intermediate latern ring leakoff connections. All throttling control valves, regardless of size, provided with double packed stuffing boxes and with stem leakoff connections. In general, RCS leakoff connections are piped to a closed collection system. Leakage to the atmosphere is essentially zero for these valves.

Gate valves at the engineered safety features interface are wedge design and are essentially straight through. The wedges are flex- wedge or solid. All gate valves have backseats. Globe valves are “T” and “Y” style. Check valves are swing type for sizes 2-1/2 inches and larger. All check valves which contain radioactive fluid are stainless steel and do not have body penetrations other than the inlet, outlet and bonnet. The check hinge is serviced through the bonnet.

The accumulatory check valve is designed such that at the required flow, the resulting pressure drop is within the specified limits. All operating parts are contained within the body. The disc has limited rotation to provide a change of seating surface and alignment after each valve opening.

5.4.12.3 Design Evaluation

The design/analysis requirements for Class 1 valves, as discussed in Section 5.2, limit stresses to levels which ensure the structural integrity of the valves. In addition, the testing program

described in Section 3.9N demonstrate the ability of the valves of operate as required during anticipated and postulated plant conditions.

Reactor coolant chemistry parameters are specified in the design specifications to assure the compatibility of valve construction materials with the reactor coolant. To ensure that the reactor coolant continues to meet these parameters, the chemical composition of the coolant will be analyzed periodically.

The above requirements and procedures, coupled with the previously described design features for minimizing leakage, ensure that the valves will perform their intended functions as required during plant operation.

5.4.12.4 Tests and Inspections

All RCS valves are tested in accordance with the requirements of the ASME Code, Section III. The tests and inspections discussed in Section 3.9N are performed to ensure the operability of active valves. In place operational testing is performed on valves, as defined in the ASME OM Code.

There are no full penetration welds within valve body walls. Valves are accessible for disassembly and internal visual inspection to the extent practical. Plant layout configurations determine the degree of inspectability. The valve quality assurance program is given in Table 5.4-15. Inservice inspection is discussed in Section 5.2.4.

5.4.13 SAFETY AND RELIEF VALVES

5.4.13.1 Design Bases

The combined capacity of the pressurizer safety valves is designed to accommodate the maximum surge resulting from complete loss of load.

This objective is met without reactor trip or any operator action by the opening of the steam safety valves when steam pressure reaches the steam side safety setting.

The pressurizer power operated relief valves are designed to limit pressurizer pressure to a value below the fixed high pressure reactor trip setpoint. They are designed to fail to the closed position on a loss of nitrogen supply. The valves are classified as active. Each of the two PORVs is supplied with an independent, seismically designed supply of nitrogen. See Section 5.2.2.11.1.

5.4.13.2 Design Description

The pressurizer safety valves are of the pop type. The valves are spring loaded, open by direct fluid pressure action, and backpressure compensation features.

The pipe connecting the pressurizer nozzles to their respective safety valves are shaped in the form of a loop seal. Condensate resulting from normal heat losses accumulates in the loop. The water prevents any leakage of hydrogen gas or steam through the safety valve seats. If the pressurizer pressure exceeds the set pressure of the safety valves, they start lifting, and the water from the seal discharges during the accumulation period.

The pressurizer power operated relief valves are pneumatic actuated valves which respond to a signal from a pressure sensing system or to manual control. Remotely operated block valves provided to isolate the power operated relief valves if excessive leakage develops.

Temperatures in the pressurizer safety and relief valve discharge lines are measured and indicated. An increase in a discharge line temperature is an indication of leakage or relief through the associated valve. Design parameters for the pressurizer spray control, safety, and power relief valves are given in Table 5.4-16.

5.4.13.3 Design Evaluation

The pressurizer safety valves prevent RCS pressure from exceeding 110 percent of system design pressure, in compliance with the ASME Code.

The Maximum design basis water relief rate through the three safety valves is a total of approximately 17.4 ft^3/sec. The loop seal discharge transient is bounding for all other discharge transient cases including feedwater line break.

The pressurizer safety valves have been analyzed for the FWLB conditions consistent with the work performed for WCAP-11677. The conclusions in WCAP-11677 that the Crosby 6M6 valves can pass slightly subcooled water at least three times without damage applies.

The pressurizer power relief valves prevent actuation of the fixed reactor high pressure trip for all design transients up to and including the design step load decreases with steam dump. The relief valves also limit undesirable opening of the spring loaded safety valves. Note that setpoint studies to date indicate that the pressure rise in a four-loop plant for the design step load decrease of 10 percent from full power is limited to 60 psi. These studies also indicate that the design step load decrease of 10 percent under N-1 loop operation is limited to approximately 50 psi. In both cases, the pressure rise is not sufficient to actuate the power operated relief valves, and thus this design is conservative.

5.4.13.4 Tests and Inspections

All safety and relief valves are subjected to hydrostatic tests, seat leakage tests, operational tests, and inspections as required. For safety valves that are required to function during a faulted condition, additional tests are performed. These tests are described in Section 3.9N. Also, see Section II.D.1 for the RCS relief and safety valve testing program for valve operability under expected flow conditions for design basis transients and accidents.

There are no full penetration welds within the valve body walls. Valves are accessible for disassembly and internal visual inspection.

5.4.14 COMPONENT SUPPORTS

5.4.14.1 Design Bases

Component supports allow virtually unrestrained lateral thermal movement of the loop during plant operation and provide restraint to the loops and components during accident conditions. The loading combinations and design stress limits are discussed in Sections 3.9N.1.1 and

3.9N.1.4.7. Material properties are discussed in Section 5.2.3. Support design is in accordance with the ASME Code, Section III, Subsection NF. The design maintains the integrity of the RCS boundary for normal and accident conditions and satisfies the requirements of the piping code. Results of piping and supports stress evaluation are presented in Section 3.9N.

5.4.14.2 Description

The support structures are welded structural steel sections. Linear type structures (tension and compression struts, columns, and beams) are used in all cases except for the reactor vessel supports, which are plate type structures. Attachments to the supported equipment are nonintegral type that are bolted to or bear against the components. The supports-to-concrete attachments are either embedded anchor bolts or fabricated assemblies.

The supports permit virtually unrestrained thermal growth of the supported systems but restrain vertical, lateral, and rotational movement resulting from seismic and pipe break loadings. This is accomplished using spherical bushings in the columns for vertical support and girders, bumper pedestals, hydraulic snubbers, and tie-rods for lateral support.

Because of manufacturing and construction tolerances, ample adjustment in the support structures must be provided to ensure proper erection alignment and fit-up. This is accomplished by shimming or grouting at the supports-to-concrete interface and by shimming at the supports-to-equipment interface.

The supports for the various components are described in the following paragraphs.

5.4.14.2.1 Reactor Pressure Vessel

Supports for the reactor vessel (Figures 5.4-12, 12A and 12B) are individual, air-cooled rectangular box structures beneath the vessel nozzles bolted to the primary shield wall concrete. Each box structure consists of a horizontal top plate that receives loads from the reactor vessel shoe, a horizontal bottom plate supported by short columns which transfer the loads to the primary shield wall concrete, and connecting vertical plates. The supports are air cooled to maintain the supporting concrete temperature within acceptable levels.

5.4.14.2.2 Steam Generator

As shown in Figure 5.4-13, the steam generator supports will consist of the following elements:

1. Vertical support

   Four individual columns will provide vertical support for each steam generator. These will be bolted at the top to the steam generator and at the bottom to the concrete structure.

   Spherical ball bushing at the top and bottom of each column will allow unrestrained lateral movement of the steam generator during heatup and cooldown. The column base design permits both horizontal and vertical adjustment of the steam generator for erection and adjustment of the system.

2. Lower lateral support

   Lateral support is provided at the generator tubesheet by fabricated steel girders and struts. These are bolted to the compartment walls and include bumpers that bear against the steam generator but permit unrestrained movement of the steam generator during changes in system temperature. Stresses in the beam caused by wall displacements during compartment pressurization are considered in the design.

3. Upper lateral support

   The upper lateral support of the steam generator is provided by a built up ring plate girder at the operating deck. Hydraulic snubbers are used in RCS component supports on the steam generator upper lateral support for Unit 2, and are intended for shock arrest only. Thus, the two-way acting snubbers restrain sudden seismic or blowdown induced motion, but permit the normal thermal movement of the steam generator. Movement perpendicular to the thermal growth direction of the steam generator is prevented by struts. The exact configuration of the upper support and location of snubbers varies depending on the details of the supporting concrete. The snubbers have been eliminated for Unit 1.

   The design specification for snubbers establishes the design, performance, fabrication, inspection, testing, packaging, shipping and installation requirements for the hydraulic shock struts (snubbers) to be used in conjunction with the RCS component supports. It is the intent of the specification to provide ASME Class I nuclear quality equipment and be in compliance with ASME B&PV Code Section III, including Subsection NF. The snubbers are built to the intent of ASME Code but are not stamped.

5.4.14.2.3 Reactor Coolant Pump

Three individual columns, similar to those used for the steam generator, provide the vertical support for each pump. Lateral support for seismic and blowdown loading is provided by three lateral tension tie bars. The pump supports are shown in Figure 5.4-14.

5.4.14.2.4 Pressurizer

The supports for the pressurizer, as shown in Figure 5.4-15, consist of:

1. A steel ring plate between the pressurizer skirt and the supporting concrete slab. The ring serves as a leveling and adjusting member for the pressurizer and may also be used as a template for positioning the concrete anchor bolts.

2. The upper lateral support consists of struts cantilevered off the compartment walls that bear against the “seismic lugs” provided on the pressurizer. The configuration of the lateral struts depends on the location of the concrete walls and piping within the compartment, as well as the orientation of the pressurizer.

5.4.14.2.5 Pipe Restraints

The application of leak-before-break technology to the CPNPP main loop piping has resulted in the exclusion of the main loop piping breaks from the design basis for dynamic effects

(Sections 3.1.1.4 and 3.6B). The reactor coolant loop piping is qualified for LOCA dynamic effects from the loop branch nozzle breaks (breaks 12 and 13, Table 3.6B-2) without the main loop piping restraints (Figures 3.6B-89 to 92). These piping restraints are therefore not required for CPNPP Units 1 and 2. The crossover leg vertical run restraints, crossover leg restraints, and hot leg restraints have been functionally removed from the design by removal of saddle blocks and shims. Although not functionally required, the lateral restraints, illustrated in Figure 5.4-19 and the non-crushable insulation, illustrated in Figure 3.6B-89, are to remain as installed.

5.4.14.3 Evaluation

Detailed evaluation ensures the design adequacy and structural integrity of the reactor coolant loop and the primary equipment supports system. This detailed evaluation is made by comparing the analytical results with established criteria for acceptability. Structural analyses are performed to demonstrate design adequacy for safety and reliability of the plant in case of a large or small seismic disturbance and/or LOCA conditions. Loads which the system is expected to encounter often during its lifetime (thermal, weight, pressure) are applied and stresses are compared to allowable values as described in Section 3.9N.1.4.7.

The Safe Shutdown Earthquake and design basis LOCA resulting in a rapid depressurization of the system are required design conditions for public health and safety. The methods used for the analysis of the Safe Shutdown Earthquake and LOCA conditions are given in Section 3.9N.1.

5.4.14.4 Tests and Inspections

Weld inspection and standards are specified in accordance with Section V of the ASME Code. Welder qualifications and welding procedures are specified in accordance with Section IX of the ASME Code.

Testing of hydraulic snubbers is in accordance with the Technical Requirements Manual (TRM) and the Inservice Inspection Program Plan.

REFERENCES

1. “Reactor Coolant Pump Integrity in LOCA,” WCAP-8163, September 1973.

2. Shabbits, W. O., “Dynamic Fracture Toughness Properties of Heavy Section A533 Grade B Class 1 Steel Plate,” WCAP-7623, December 1970.

3. “Evaluation of Steam Generator Tube, Tubesheet and Divider Plant Under Combined LOCA Plus SSE Conditions,” WCAP-7832, December 1973.

4. Westinghouse Report “Counterflow Preheat Steam Generator Tube Expansion Report,” June 1983.

5. WCAP-11175 (Proprietary), WCAP-11174 (Non-Proprietary), ”Row 1 and Row 2 U-Bend Heat Treatment Licensing Report for Comanche Peak Unit 1”, July 1986.

6. WCAP-11127 (Proprietary), WCAP-11128 (Non-Proprietary), “Shotpeening Licensing Report for Comanche Peak Unit 1”, May 1986.

7. WCAP-13698, Rev. 3, “Laser Welded Sleeves for ¾ Inch Diameter Tube Feed Ring Type and Westinghouse Preheater Steam Generators Generic Sleeving Report.”

8. WCAP-15090, “Specific Application of Laser Welded Sleeves for the Comanche Peak Unit 1 and 2 Steam Generators.”

9. Steam Generator Progress Report: Revision 15, EPRI, Palo Alto, CA: 2000, 1000805.

10. Pressurized Water Reactor Generic Tube Degradation Predictions: U.S. Recirculating Steam Generators with Alloy 600TT and Alloy 690TT Tubing, EPRI, Palo Alto, CA: 2003, 1003589.

11. Materials Reliability Program (MRP), Resistance to Primary Water Stress Corrosion Cracking of Alloys 690, 52, and 152 in Pressurized Water Reactors (MRP-111), EPRI, Palo Alto, CA, U.S. Department of energy, Washington, DC: 2004, 1009801.

12. Alloy 690 Improvement Factor Update: Application of Improvement Factor Data to the Analysis of a Secondary System Chemistry Upset at Ginna. EPRI, Palo Alto, CA and Constellation Energy Group, Inc., Baltimore, MD: 2006, 1013640.

TABLE 5.4-1
REACTOR COOLANT PUMP DESIGN PARAMETERS

(Sheet 1 of 2)

Unit design pressure (psig)	2485
Unit design temperature (°F)	650[a]
Unit overall height (ft)	26.93
Seal water injection (gpm)	8
Seal water return (gpm)	3
Cooling water flow (gpm)	495
Maximum continuous cooling water inlet temperature (°F)	108
Pump	
Capacity (gpm)	99,000
Developed head (ft)	288
NPSH required (ft)	Figure 5.4-2
Suction temperature (°F)	557.8
Pump discharge nozzle, inside diameter (in)	27-1/2
Pump suction nozzle, inside diameter (in)	31
Speed (rpm)	1183
Water volume (ft^3)	78.6[b]
Weight, dry (lbs)	205,330
Motor	
Type	Drip proof, squirrel cage induction, water/air cooled
Power (hp)	7000
Voltage (volts)	6600
Phase	3
Frequency (hz)	60
Insulation class	Class B, thermalastic epoxy insulation

TABLE 5.4-1
REACTOR COOLANT PUMP DESIGN PARAMETERS

(Sheet 2 of 2)

Starting	
Current	3000 amp @ 6600 volts
Input, hot reactor coolant	~489 amp
Input, cold reactor coolant	~669 amp
Pump moment of inertia, maximum ($1b/ft^2$)	
Flywheel	70,000
Motor	22,500
Shaft	520
Impeller	1,980

a) Design temperature of pressure retaining parts of the pump assembly exposed to the reactor coolant and injection water on the high pressure side of the controlled leakage seal shall be that temperature determined for the parts for a primary loop temperature of 650°F.

b) Composed of reactor coolant in the casing and of injection and cooling water in the thermal barrier.

TABLE 5.4-2
REACTOR COOLANT PUMP QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]
Castings	yes		yes	
Forgings				
Main shaft		yes	yes	
Main studs		yes	yes	
Flywheel (rolled plate)		yes		
Weldments				
Circumferential	yes		yes	
Instrument connections			yes	

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.

TABLE 5.4-3
STEAM GENERATOR DESIGN DATA

Design pressure, reactor coolant side (psig)	2485
Design pressure, steam side (psig)	1285
Design temperature, reactor coolant side (°F)	650
Design temperature, steam side (°F)	600
Total heat transfer surface area (ft^2)	76,000 (Unit 1) 48,300 (Unit 2)
Maximum moisture carryover (wt percent)	0.10 (Unit 1) 0.25 (Unit 2)
Overall height (ft-in)	67-9 (Unit 1) 67-8 (Unit 2)
Number of U-tubes	Unit 1 – 5532 Unit 2 – 4570
U-tube nominal diameter (in)	0.750
Tube wall nominal thickness (in)	0.043
Number of manways	4
Inside diameter of manways (in)	16
Number of inspection ports	
2.0" Diameter	4 (Unit 2)
2.5" Diameter	16 (Unit 1) 2 (Unit 2)
4.0” Diameter	4 (Unit 1)
6.0" Diameter	4 (Unit 1) 5 (Unit 2)
Design fouling factor	0.00005 (Unit 2)
Preheat section fouling factor	0.00010 (Unit 2)

TABLE 5.4-4A
STEAM GENERATOR QUALITY ASSURANCE PROGRAM (UNIT 1)

	RT[a]	UT[a]	PT[a]	MT[a]	ET[a]
Tubesheet					
Forging		yes		yes	
Cladding		yes[b]	yes		
Channel head					
Forging		yes		yes	
Cladding			yes		
Secondary shell and head					
Forging		yes			
Tubes		yes			yes
Nozzles (forgings)		yes		yes	
Weldments					
Shell, circumferential	yes			yes	
Cladding (channel head-tubesheet joint cladding restoration)			yes		
Steam and feedwater nozzle to shell	yes			yes	
Support brackets				yes	
Tube to tubesheet			yes		
Instrument connections (primary and secondary)				yes	
Temporary attachments after removal				yes	
After hydrostatic test (all welds – where accessible)				yes	
Primary nozzle safe ends (forgings)		yes	yes		

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.
ET - Eddy current.

b) Flat surfaces only.

TABLE 5.4-4B
STEAM GENERATOR QUALITY ASSURANCE PROGRAM (UNIT 2)

	RT[a]	UT[a]	PT[a]	MT[a]	ET[a]
Tubesheet					
Forging		yes		yes	
Cladding		yes[b]	yes		
Channel head					
Casting	yes			yes	
Cladding			yes		
Secondary shell and head					
Plates		yes			
Tubes		yes			yes
Nozzles (forgings)		yes		yes	
Weldments					
Shell, longitudinal	yes			yes	
Shell, circumferential	yes			yes	
Cladding (channel head-tubesheet joint cladding restoration)			yes		
Steam and feedwater nozzle to shell	yes			yes	
Support brackets				yes	
Tube to tubesheet			yes		
Instrument connections (primary and secondary)				yes	
Temporary attachments after removal				yes	
After hydrostatic test (all welds and complete cast channel head – where accessible)				yes	
Nozzle safe ends (if forgings)	yes		yes		
Nozzle safe ends (if weld deposit)			yes		

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.
ET - Eddy current.

b) Flat surfaces only.

TABLE 5.4-5
REACTOR COOLANT PIPING DESIGN PARAMETERS

Reactor inlet piping, inside diameter (in)	27-1/2
Reactor inlet piping, nominal wall thickness (in)	2.32
Reactor outlet piping, inside diameter (in)	29
Reactor outlet piping, nominal wall thickness (in)	2.45
Coolant pump suction piping, inside diameter (in)	31
Coolant pump suction piping, nominal wall thickness (in)	2.60
Pressurizer surge line piping, nominal pipe size (in)	14
Pressurizer surge line piping, nominal wall thickness (in)	1.406
Reactor coolant loop piping	
Design/operating pressure (psig)	2485/2235
Design temperature (°F)	650
Pressurizer surge line	
Design pressure (psig)	2485
Design temperature (°F)	680
Pressurizer safety valve inlet line	
Design pressure (psig)	2485
Design temperature (°F)	680
Pressurizer (power operated) relief valve inlet line	
Design pressure (psig)	2485
Design temperature (°F)	680
Pressurizer relief tank inlet line	
Design pressure (psig)	700
Design temperature (°F)	600

TABLE 5.4-6
REACTOR COOLANT PIPING QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]
Fittings and pipe (castings)	yes		yes
Fittings and pipe (forgings)		yes	yes
Weldments			
Circumferential	yes		yes
Nozzle to runpipe (except no RT for nozzles less than 6 inches)	yes		yes
Instrument connections			yes
Castings	yes		yes (after finishing)
Forgings		yes	yes (after finishing)

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.

TABLE 5.4-7
DESIGN BASES FOR RESIDUAL HEAT REMOVAL SYSTEM OPERATION

Residual Heat Removal System startup	~4 hours after reactor shutdown
Reactor Coolant System initial pressure (psig)	~425
Reactor Coolant System initial temperature (°F)	~350
Component cooling water maximum temperature (°F)	122
Cooldown time (hours after initiation of Residual Heat Removal System operation)	28
Reactor Coolant System temperature at end of cooldown (°F)	140
Decay heat generation	ANS 5.1 – 1979

TABLE 5.4-8
RESIDUAL HEAT REMOVAL SYSTEM COMPONENT DATA

Residual Heat Removal Pump		
Number	2	
Design pressure (psig)	600	
Design temperature (°F)	400	
Design flow (gpm)	3800	
Design head (ft)	350	
NPSH required at 3800 gpm (ft)	18	
Power (hp)	450	
Residual Heat Exchanger		
Number	2	
Design heat removal capacity (Btu/hr)	39.1×10^6	
Estimated UA (Btu/hr-°F)	2.3×10^6	
	Tube side	Shell side
Design pressure (psig)	600	165
Design temperature (°F)	400	200
Design flow (lb/hr)	1.9×10^6	3.8×10^6
Inlet temperature (°F)	140	105
Outlet temperature (°F)	119.4	115.2
Material	Austenitic stainless steel	Carbon steel
Fluid	Reactor coolant	Component cooling water

TABLE 5.4-9
PRESSURIZER DESIGN DATA

Design pressure (psig)	2485
Design temperature (°F)	680
Surge line nozzle diameter (in)	14
Heatup rate of pressurizer using heaters only (°F/hr)	55
Internal volume (ft^3)	1800

TABLE 5.4-10
REACTOR COOLANT SYSTEM DESIGN PRESSURE SETTINGS

	Psig
Hydrostatic test pressure	3107
Design pressure	2485
Safety valves (begin to open)	2485
High pressure reactor trip	2385
High pressure alarm	2310
Power operated relief valves	2335(a)
Pressurizer spray valves (full open)	2310
Pressurizer spray valves (begin to open)	2260
Proportional heaters (begin to operate)	2250
Operating pressure	2235
Proportional heater (full operation)	2220
Backup heaters on	2210
Low pressure alarm	2210
Pressurizer power operated relief valve interlock	2185(b)
Low pressure reactor trip (typical, but variable)	1885

a) At 2335 psig, a pressure signal initiates actuation (opening) of these valves. Remote manual control is also provided.

b) Below 2185 psig, the PORV is automatically closed to prevent depressurization of the RCS in the event the control input to the PORV (2335 psig) fails high.

TABLE 5.4-11
PRESSURIZER QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]	MT[a]
Heads				
Plates		yes		
Cladding			yes	
Shell				
Plates		yes		
Cladding			yes	
Heaters				
Tubing[b]		yes	yes	
Centering of element	yes			
Nozzle (forgings)		yes	yes[c]	yes[c]
Weldments				
Shell, longitudinal	yes			yes
Shell, circumferential	yes			yes
Cladding			yes	
Nozzle safe end (if forging)	yes		yes	
Instrument connection			yes	
Support skirt, longitudinal seam	yes			yes
Support skirt to lower head		yes		yes
Temporary attachments (after removal)				yes
All external pressure boundary welds after shop hydrostatic test				yes

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.
MT - Magnetic particle.

b) Or a UT and ET.

c) MT or PT.

TABLE 5.4-12
PRESSURIZER RELIEF TANK DESIGN DATA

Design pressure (psig)		100
Rupture disc release pressure (psig)	Nominal: Range:	91 86-100
Design temperature (°F)		340
Total rupture disc relief capacity at 100 psig (lb/hr)		1.6×10^6

TABLE 5.4-13
RELIEF VALVE DISCHARGE TO THE PRESSURIZER RELIEF TANK

Reactor Coolant System		
3	Pressurizer safety valves	Figure 5.1-1
2	Pressurizer power operated relief valves	Figure 5.1-1
Residual Heat Removal System		
2	Residual heat removal pump suction line from the Reactor Coolant System hot legs	Figure 5.4-6
Chemical and Volume Control System		
1	Seal water return line	Figure 9.3-10
1	Letdown line	Figure 9.3-10

TABLE 5.4-14
REACTOR COOLANT SYSTEM VALVE DESIGN PARAMETERS

Design/normal operating pressure (psig)	2485/2235
Preoperational plant hydrotest (psig)	3107
Design temperature (°F)	650

TABLE 5.4-15
REACTOR COOLANT SYSTEM VALVES QUALITY ASSURANCE PROGRAM

	RT[a]	UT[a]	PT[a]
Boundary Valves, Pressurizer Relief and Safety Valves			
Castings (larger than 4 inches)	yes		yes
(2 inches to 4 inches)	yes[b]		yes
Forgings (larger than 4 inches)	(c)	(c)	yes
(2 inches to 4 inches)			yes

a) RT - Radiographic.
UT - Ultrasonic.
PT - Dye penetrant.

b) Weld ends only.

c) Either RT or UT.

TABLE 5.4-16
PRESSURIZER VALVES DESIGN PARAMETER

Pressurizer Spray Control Valves	
Number	2
Design pressure (psig)	2485
Design temperature (°F)	650
Design flow for valves full open, each (gpm)	450
Pressurizer Safety Valves	
Number	3
Saturated steam relieving capacity, ASME rated flow (lb/hr)	420,000
Set pressure (psig)	2485
Design temperature (°F)	650
Transient condition (°F) Superheated steam	680
Backpressure	
Normal (psig)	3 to 5
Expected during discharge (psig)	500
Water relieving capacity at 2575 psig per valve (lb/sec)	178.4
Pressurizer Power Relief Valves	
Number	2
Design pressure (psig)	2485
Design temperature (°F)	650
Saturated steam relieving capacity at 2335 psig per valve (lb/hr)	210,000
Transient condition (°F) Superheated steam	680
Water relieving capacity at 2235 psig per valve (lb/sec)	97.2

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 1 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
1.	Motor operated gate valve 1-8702A (1-8701A analogous)	a.	Fails to open on demand (open manual mode CB switch selection)	a.	Failure blocks reactor coolant flow from hot leg of RC loop #1 through train “A” of RHRS. Fault reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements will be met by reactor coolant flow from hot leg of RC loop #4 flowing through train “B” of RHRS, however, time required to reduce RCS temperature will be extended.	a.	Valve position indication (closed to open position change) at CB; RC loop hot leg pressure indication at CB; RHR train “A” discharge flow indication and low flow alarm at CB; and RHR pump discharge pressure indication at CB.	1. 2.	Valve is electrically interlocked with the containment sump and RWST isolation valves 1-8811A and 1-8812A, with RHR to charging pump suction line isolation valve 1-8804A and with “prevent-open” pressure interlock of RC loop hot leg. The valve can not be opened remotely from the CB if one of the indicated isolation valves is open or if RC loop pressure exceeds 425 psig. If both trains of RHRS are unavailable for plant cooldown due to multiple component failures, the auxiliary feedwater system and S.G. power operated relief valves can be used to perform the safety function of removing residual heat.

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 2 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
2.	Motor operated gate valve 1-8702B (1-8701B analogous)	a.	Same failure modes as those stated for item #1	a.	Failure blocks reactor coolant flow from hot leg of RC loop #4 through train “B” of RHRS. Fault reduces redundancy of RHR cooling trains provided. No effect on safety for system operation. Plant cooldown flow from hot leg of RC loop #1 flowing through train “A” of RHRS. However, time required to reduce RCS temperature will be extended.	a.	Valve position indication (Closed to open position change) at CB; RC loop hot leg pressure indication at CB; RHR train “B” discharge flow inidcation and low flow Alarm at CB; and RHR pump discharge pressure indication at CB.	1.	Valve is electrically interlocked with the containment sump and RWST isolation valves 1-8811B and 1-8812B with RHR to safety injection pump suction line isolation valve 1-8804B and with “prevent-open” pressure interlock of RC loop hot leg. The valve can not be opened remotely from the CB if one of the indicated isolation valves is open or if RC loop pressure exceeds 425 psig.
								2.	Same as for Item #1.
3.	Residual heat removal pump #1, RHR pump #2 analogous)	a.	Fails to deliver working fluid	a.	Failure results in loss of reactor coolant flow from hot leg of RC loop #1 through train “A” of RHRS. Fault reduces redundancy of RHR coolant trains provided. No effect on safety for system operation. Plant cooldown requirements will be met by reactor coolant flow from hot leg of RC loop #4 flowing through train “B” of RHRS, however, time required to reduce RCS temperature will be extended.	a.	Open pump switchgear circuit breaker indication at CB; circuit breaker close position monitor light for group monitoring of component at CB; common breaker trip alarm at CB; RC loop #1 hot leg pressure indication(PI-405) at CB; RHR train A discharge flow indication (FI-618) and low flow alarm at CB; and pump discharge pressure indication (PI-614)	1.	The RHRS shares components with the ECCS. Pumps are tested as part of the ECCS testing program(see Section 6.3.4). Pump failure may also be detected during ECCS testing.

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 3 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
4.	Motor operated globe valve 1-FCV-610 (1-FCV-611 analogous)	a.	Fails to open on demand (open manual mode CB switch selection)	a.	Failure blocks miniflow line to suction of RHR pump “A” during cooldown operation of checking boron concentration level of coolant in train “A” of RHRS. No effect on safety for system operation. Operator may establish miniflow for RHR pump “A” operation by opening of CVCS letdown control valve (1-HCV-128) and manual valve 1-8734A to allow flow to CVCS.	a.	Valve position indication (closed to open position change) at CB.	1.	Valve is automatically controlled to open when pump discharge is less than 500 gpm and close when the discharge exceeds 1000 gpm. The valve protects the pump from dead-heading during ECCS operation. CB switch set to “Auto” position for automatic control of valve positioning.
		b.	Fails to close on demand (“Auto” mode CB switch selection)	b.	Failure allows for a portion of RHR heat exchanger “A” discharge flow to be bypassed to suction of RHR pump “A”. RHRS train “A” is degraded for the regulation of coolant temperature by RHR heat exchanger “A”. No effect on safety for system operation. Cooldown of RCS within established specification cooldown rate may be accomplished through operator action of throttling flow control valve 1-HCV-606 and controlling cooldown with redundant RHRS Train “B”.	b.	Valve position indication (open to closed position change) and at CB. RHRS Train “A” discharge flow indication (FI-618) at CB.		

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 4 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
5.	Air Diaphragm operated butterfly valve 1-FCV-618 (1-FCV-619 analogous)	a.	Fails to open on demand (“Auto” mode CB switch selection)	a.	Failure prevents coolant discharged from RHR pump “A” from bypassing RHR heat exchanger “A” resulting in mixed mean temperature of coolant flow to RCS being low. RHRS train “A” is degraded for the regulation of controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve 1-HCV-606 and controlling cooldown with redundant RHRS train “B”	a.	RHR pump discharge temperature and RHRS train “A” discharge to RCS cold leg temperature recording (TR-612) at CB; and RHRS train “A” discharge to RCS cold leg flow indication (FI-618) at CB.	1.	Valve is designed to fail”closed” and is electrically wired so that electrical solenoid of the air diaphragm operator is energized to open the valve. Valve is normally “closed” to align RHRS for ECCS the valve. Valve operation during plant power operation and load follow.
		a.	Fails to close on demand (“Auto” mode CB switch selection	b.	Failure allows coolant discharged from RHR pump “A” to bypass RHR heat exchanger “A” resulting in mixed mean temperature of coolant flow to RCS being high. RHRS train “A” is degraded for the regulation of controlling temperature of coolant. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished through operator action of throttling flow control valve 1-HCV-606 and controlling cooldown with redundant RHRS train “B”, however, cooldown time will be extended.	b.	Same method of detection as those stated above.		

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 5 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
6.	Air diaphragm operated butterfly valve 1-HCV-606 (1-HCV-607 analogous)	a.	Fails to close on demand for flow reduction	a.	Failure prevents control of coolant discharge flow from RHR heat exchanger “A” resulting in loss of mixed mean temperature coolant flow adjustment to RCS. No effect on safety for system operation. Cooldown of RCS within established specification rate may be accomplished by operator action of controlling cooldown with redundant RHRS train “B”.	a.	Same methods of detection as those stated for item #5. In addition, monitor light (valve not open) for group monitoring of components at CB.	1.	Valve is designed to fail “open”, The valve is normally “open” to align RHRS for ECS operation during plant power operation and load follow.
		b.	Fails to open on demand for increased flow	b.	Same effect on system operation as that stated above for failure mode “Fails to close on demand for flow reduction.”	b.	Same methods as those stated above for failure mode “Fails to close on demand for flow reduction.		
7.	Manual globe valve 1-8734A (1-8734B analogous)	a.	Fails closed	a.	Failure blocks flow from train “A” of RHRS to CVCS letdown heat exchanger. Fault prevents (during the initial phase of plant cooldown) the adjustment of boron concentration level of coolant in lines of RHRS train “A” so that it equals the concentration level in the RCS using the RHR cleanup line to CVCS. No effect on safety for system operation. Operator can balance boron concentration levels by cracking open flow control valve 1-HCV-606 to permit flow to cold leg of loop #1 of RCS in order to balance levels using normal CVCS letdown flow.	a.	CVCS letdown flow indication (FI-132) at CB.	1.	Valve is normally “closed” to align the RHRS for ECCS operation during plant power operation and load follow.

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 6 of 7)

	Component		Failure Mode		[a]Effect on System Operation		[b]Failure Detection Method		Remarks
8.	Air diaphragm operated globe valve 1-HCV-128	a.	Fails to open on demand	a.	Failure blocks flow from train “A” and “B” of RHRS to CVCS letdown heat exchanger. Fault prevents use of of RHR cleanup line to CVCS for balancing boron concentration levels of RHR trains “A” and “B” with RCS during initial cooldown operation and later in plant cooldown for letdown flow. No effect on safety for system operation. Operator can balance boron concentration levels with similar actions, using pertinent flow control valve 1-HCV-606 and 1-HCV-607, as stated above for item #7. Normal CVCS letdown flow can be used for purification if RHRS cleanup line is not available.	a.	Valve position indication (degree of opening) at CB and CVCS letdown flow indication (FI-132) at CB.	1. 2.	Same remark as that stated above for item #7. Valve is the boundary between CVCS and RHRs. It provides a path for RCS letdown pressure control and purification during shutdown conditions.
9.	Motor operated gate valve 1-8812A (1-8812B analogous)	a.	Fails to close on demand	3.	Failure reduces the redundancy of isolation valves provided to flow isolate RHRS train “A” from RWST. No effect on safety for system operation. Check valve 1-8958A in series with MO-valve provides the primary isolation against the bypass of RCS coolant flow from the suction of RHR pump”A” to RWST.	a.	Valve position indication (open to closed position change) at CB and valve (closed) monitor light for group monitoring at CB.	1.	Valve is a component of the ECCS that performs an RHR function dduring plant cooldown. Valve is normally “open” to align the RHRS for ECCS operation during plant power operation and load

List of acronyms and abbreviations

Auto - Automatic

CB - Control Board

CVCS - Chemical and Volume Control System

TABLE 5.4-17
FAILURE MODE AND EFFECTS ANALYSIS - RESIDUAL HEAT REMOVAL SYSTEM ACTIVE COMPONENTS - PLANT COOLDOWN OPERATION

(Sheet 7 of 7)

Component	Failure Mode	[a]Effect on System Operation	[b]Failure Detection Method	Remarks

ECCS - Emergency Core Cooling System

MO - Motor Operated

RC - Reactor Coolant

RCS - Reactor Coolant System

RHR - Residual Heat Removal

RHRS - Residual Heat Removal System

RWST - Refueling Water Storage Tank

SG - Steam Generator

a) See list at end of table for definition of acronym and abbreviations used.

b) As part of plant operation, periodic tests, surveillance inspections and instrument calibrations are made to monitor equipment and performance. Failures may be detected during such monitoring of equipment in addition to detection methods noted.

TABLE 5.4-18
TABLE DELETED

TABLE 5.4-19
UNIT 1 RCP FLYWHEEL DATA

Motor Unit	Melt Number	Charpy V-Notch (ft-lbs)	Lateral Expansion (mils)	Test Temp (°F)	T_{NDT}[(a)] (°F)	RT_{NDT} (°F)	Yield Strength (psi)	Tensile Strength (psi)	Elongation (% in 2")
01	B9692-2	58, 61, 60 (T)	51, 49, 53, (T)	70	10	10	69,300	91,100	24 (T)
		118, 116, 116 (L)	79, 81, 80 (L)	70			76,500	97,500	22 (L)
	B0510-1	80, 78, 72 (T)	73, 71, 68 (T)	70	10	10	70,900	92,500	23 (T)
		178, 180, 180 (L)	97, 98, 96 (L)	70			75,100	94,500	23 (L)
02	B0510-4	81, 82, 79 (T)	74, 70, 72 (T)	70	10	10	72,200	93,500	26 (T)
		150, 154, 151 (L)	95, 92, 96 (L)	70			73,900	96,500	24 (L)
	B0510-1	See data above for Motor Unit 01.							
03	B9692-2	See data above for Motor Unit 01.							
	B0510-1	See data above for Motor Unit 01.							
04	B0510-1	See data above for Motor Unit 01.							
	B0510-4	See data above for Motor Unit 02.							

a) Based on two drop weight test exhibiting no breaks at 20°F

TABLE 5.4-19A
UNIT 2 RCP FLYWHEEL DATA

Motor Unit	Melt Number	Charpy V-Notch (ft-lbs)	Lateral Expansion (mils)	Test Temp (°F)	T_{NDT} (°F)	RT_{NDT}* (°F)	Yield Strength (psi)	Tensile Strength (psi)	Elongation (% in 2")
01	B2934-3	90, 82, 92 (T)	77, 70, 78, (T)	70	10	10	70,100	92,900	20 (T)
		136, 128, 130 (L)	86, 84, 88 (L)	70			63,000	87,000	22 (L)
	D6482-3	110, 108, 112 (T)	81, 76, 80 (T)	70	10	10	70,100	90,500	26 (T)
		120, 118, 116 (L)	80, 82, 78 (L)	70			67,500	90,000	24 (L)
02	B2685-1	96, 98, 92 (T)	77, 76, 72 (T)	70	10	10	81,800	97,500	23 (T)
		166, 170, 168 (L)	94, 96, 92 (L)	70			67,100	91,000	23 (L)
	A5246-3	110, 116, 108 (T)	90, 86, 89 (T)	70	10	10	65,700	89,600	29 (T)
		128, 130, 130 (L)	94, 93, 92 (L)	70			65,800	88,800	27 (L)
03	B2934-3	See data above for Motor Unit 01.							
	D6482-3	See data above for Motor Unit 01.							
04	D6482-3	See data above for Motor Unit 01.							
	A5246-3	See data above for Motor Unit 02.							

5A EVALUATION OF COMPLIANCE WITH NRC BRANCH TECHNICAL POSITION RSB 5-1 ON DESIGN REQUIREMENTS OF THE RESIDUAL HEAT REMOVAL SYSTEM

The following describes the manner by which CPNPP complies with the technical requirements of BTP RSB 5-1. Comanche Peak is a Class 2 plant (as defined by the implementation section of BTP RSB 5-1) and is, thus, subject to the technical requirements of RSB 5-1 only as they apply to Class 2 plants. Only partial compliance with the technical position is required where manual actions or repairs can be demonstrated to be an acceptable alternative to strict compliance. The safe shutdown design basis for Comanche Peak is hot standby. The functional requirements of BTP RSB 5-1, the CPNPP Cold Shutdown Scenario, a Single Failure Analysis, and a discussion on Natural Circulation are provided below.

I. FUNCTIONAL REQUIREMENTS OF BTP RSB 5-1

1. Provyide safety grade steam generator dump valves, operators, air and power supplies which meet the single failure criterion.

   One safety grade steam generator power-operated atmospheric relief valve (PORV) is provided for each of the four steam generators. The PORV control circuitry and the valve operator meets safety grade criteria (Class 1E). A permanent safety grade air supply bottle for each PORV has been added to the CPNPP design. These air reservoirs (safety class 3 and Seismic Category I) will provide motive power to the PORV's until local manual operator control is available in the event that the plant air supply is unavailable. See Section 9.3.1.2 for related information. See Section II.1 (Cold Shutdown Scenario - Residual Heat Removal) and Section III.1 (Single Failure Evaluation - Residual Heat Removal) as provided below.

2. Provide the capability to cooldown to cold shutdown in approximately 36 hours assuming the most limiting single failure and loss of offsite power or show that manual actions inside or outside containment or return to hot standby until the manual actions or maintenance can be performed to correct the failure provides an acceptable alternative.

   The plant can be maintained in a safe hot standby condition while any necessary manual actions are taken. The plant is capable of being cooled via natural circulation and reaching cold shutdown conditions within 36 hours including the time required to perform any manual actions. See Sections II.1 and III.1 as provided below.

3. Provide the capability to depressurize the Reactor Coolant System with only safety grade systems assuming a single failure and loss of offsite power or show that manual actions inside or outside containment or remaining at hot standby until manual actions or repairs are complete provides an acceptable alternative.

   The plant can be maintained in a safe hot standby condition while any required manual actions are taken. See Section II.3 (Cold Shutdown Scenario-depressurization) and Section III.3 (Single Failure Evaluation - Depressurization) as provided below.

4. Provide the capability for borating with only safety grade systems assuming a single failure and loss of offsite power or show that manual actions inside or outside containment or remaining at hot standby until manual actions or repairs are completed provides an acceptable alternative.

   The plant can be maintained in a safe hot standby condition while any required manual actions are taken. See Section II.2 (Cold Shutdown Scenario - Boration and Inventory Control) and Section III.2 (Single Failure Criterial - Boration and Inventory Control) as provided below.

5. Provide the system and component design features necessary for the prototype testing of both the mixing of the added borated water and cooldown under natural circulation conditions with and without a single failure of a steam generator atmospheric dump valve. These tests and analyses will be used to obtain information on cooldown times and the corresponding auxiliary feedwater (AFW) requirements.

   The plant design provides the capability for conducting natural circulation cooldown tests if required. However, the CPNPP test program does not include tests to verify boration or cooldown under natural circulation conditions since a utility is permitted to satisfy the testing requirement of the BTP through a comparison of plant systems of previously tested plants of a similar design. The natural circulation evaluation provided here presents a comparison of CPNPP Units 1 and 2 with Diablo Canyon Unit 1 to show that the natural circulation tests performed at Diablo Canyon are representative of the natural circulation cooldown, depressurization and boron mixing capabilities at CPNPP. The Diablo Canyon test results indicated that natural circulation flow rates were adequate to ensure that core decay heat removal, boron mixing and plant cooldown/depressurization were maintained throughout the test. The response of RCS temperatures indicated stable natural circulation conditions existed throughout the test.

6. Commit to providing specific procedures for cooling down using natural circulation and submit a summary of these procedures.

   Procedures for cooling down using natural circulation have been prepared; they were developed in accordance with applicable criteria of Regulatory Guide 1.33. These procedures are based upon generic Westinghouse Owners' Group Emergency Response Guidelines and contain plant specific information where applicable. A summary of the required actions is provided in Section II and Section III, below.

7. Provide a seismic Category I AFW supply for a least 4 hours at Hot Shutdown plus cooldown to the RHR system cut-in based on the longest time (for only onsite or offsite power and assuming the worst single failure), or show that an adequate alternate seismic Category I source will be available.

   Sufficient emergency feedwater is provided in the Seismic Category I condensate storage tank to permit four hours operation at hot standby plus cooldown to RHRS initiation conditions. In addition, a long term source of auxiliary feedwater is

provided by a connection to the Seismic Category I Service Water System. See Sections II.1 and III.1, as provided below.

8. Provide for collection and containment of RHR pressure relief or show that adequate alternative methods of disposing of discharge are available.

   The RHR relief valves discharge to the pressurizer relief tank, located inside containment.

II. COLD SHUTDOWN SCENARIO

The safe shutdown design basis of CPNPP is hot standby. The cold shutdown capability of the plant has been evaluated in order to demonstrate how the plant can be maintained in a safe hot standby condition while necessary manual actions are taken to achieve cold shutdown conditions following a safe shutdown earthquake, loss of offsite power, and the most limiting single failure. Under such conditions, the plant is capable of achieving cold shutdown conditions (approximately 200°F) in approximately 36 hours. To achieve and maintain cold shutdown, the following key functions must be performed: 1) residual heat removal, 2) boration and inventory control, and 3) depressurization.

1. Residual Heat Removal

   The function of residual removal is performed in two stages in accomplishing the cooldown from hot standby to cold shutdown.

   The first stage is from hot standby to 350°F. During this stage, circulation of the reactor coolant is provided by natural circulation with the reactor core as the heat source and the steam generators as the heat sink. Steam is initially released via the steam generator safety valves to maintain a hot standby. This occurs automatically as a result of turbine and reactor trip. Steam release for cooldown occurs via the steam generator power-operated atmospheric relief valves. As the cooldown proceeds, the operator adjusts these valves to increase the amount of steam dump, to permit a reasonable cooldown rate. Feedwater makeup is provided by the Auxiliary Feedwater System.

   The steam generator safety valves are Seismic Category I spring-loaded valves that can automatically maintain the plant in a safe hot standby for an extended period of time. The steam generator power-operated atmospheric relief valves are also seismically qualified. The control grade plant air supply system that powers these valves can be loaded on the emergency train A bus. Should a seismic event result in the loss of the air supply system, motive power for the PORV's is available from the seismically qualified air supply bottles. The steam generator power operated relief valves may be operated by means of manual handwheel operators when the air supply is exhausted. Should a single failure render one of the PORV's inoperable, the plant could be cooled down to the RHRS initiation temperature via one of the three active loops. Additionally the manual valve upstream of the failed PORV could be closed while the failed valve was repaired or replaced.

The Auxiliary Feedwater System is comprised of two separate Seismic Category I subsystems with sufficient alignment capability and flow capacity to ensure that feedwater can always be provided to all four generators. The first subsystem is comprised of two motor-driven pumps each powered from a different emergency power train. The two pumps and associated piping are normally isolated from each other and each is dedicated to two of the four steam generators. Operator action could be taken to open the normally closed isolation valves enabling either pump to provide feedwater to any of the four steam generators. The second subsystem incorporates a turbine-driven pump which can receive motive steam from either of two steam generators and can supply feedwater to all four steam generators.

The Auxiliary Feedwater System is capable of providing feedwater for an extended period of time. The primary source of feedwater is the Seismic Category I condensate storage tank (see Section 10.4.7.2 for additional details). There are additional sources of backup feedwater which can be manually accessed. Initial backup is provided by the Demineralized and Reactor Makeup Water System. Ultimate backup is from the Seismic Category I service Water System. In the unlikely event that sufficient auxiliary feedwater was not available in the condensate storage tank to complete the cooldown to RHR cut-in conditions, switchover to backup sources of water would be performed by the operator upon receipt of the condensate storage tank low level signal.

The status of each steam generator can be monitored using Class 1E instrumentation located in the Control Room. Separate indication channels for both steam generator pressure and water level are available. Operation of the Auxiliary Feedwater System can be monitored using Class 1E instrumentation located in the control room. This includes indication of the flows into each steam generator, the operating status light for the motor-driven pumps, and condensate storage tank level indication. There is also local indication of the turbine-driven pump suction and discharge pressure.

The second stage of cooldown is from 350°F to cold shutdown. During this stage, the RHRS is brought into operation. Circulation of the reactor coolant is provided by the RHR pumps and the heat exchangers in the RHRS act as the means of heat removal from the RCS. In the RHR heat exchangers, the residual heat is transferred to the Component Cooling Water System which ultimately transfers the heat to the Service Water System.

The RHRS is a fully redundant system. Each RHR subsystem includes a RHR pump and a heat exchanger. Each RHR pump is powered from different emergency power trains and each RHR heat exchanger is cooled by a different Component Cooling Water System loop. The Component Cooling Water and the Service Water Systems are both designed to Seismic Category I. If any component in one of the RHR subsystems were rendered inoperable as the result of a single failure, cooldown of the plant would not be compromised; however, the time for cooldown would be extended. The operation of the RHRS can be monitored using Class IE instrumentation in the Control Room.

There is indication of the pump discharge flow, the pump operating status and the component cooling flow from the discharge of the RHR heat exchangers.

2. Boration and Inventory Control

Boration is accomplished using portions of the Chemical and Volume Control System (CVCS). The boric acid transfer pumps supply four weight percent (7000 ppm) boric acid from the boric acid tanks to the suction of the centrifugal charging pumps which inject into the Reactor Coolant System (RCS) via the normal charging, safety injection auxiliary pressurizer spray, and/or reactor coolant pump seal injection flow paths. Makeup in excess of that required for boration can be provided from the refueling water storage tank (RWST) using paths as described for boration. Two motor-operated valves, different emergency diesels and connected in parallel, transfer the suction of the charging pumps to the RWST.

The boric acid transfer pumps and centrifugal charging pumps are train oriented and can be loaded on the emergency diesels. Should a common valve make both the normal and alternate charging lines unavailable, the reactor coolant pump seal injection flow would be sufficient for boration. The RCS can be borated to the cold shutdown concentration by accommodating the boration flow in the steam space of the pressurizer and in the space made available as the RCS shrinks due to cooling. Should this not be sufficient to accommodate all additions to RCS inventory, an alternative means of letdown would be via the reactor vessel head vent system.

Boration and makeup can be monitored using Class 1E instrumentation in the Control Room. Indications available include boric acid transfer and centrifugal charging pump operating status and boric acid tank and RWST level. Sampling can be done intermittently from several sampling connections in the normal letdown path, if it is available, or from two separate RCS hot legs. In the worst case situation, the amount of boron injected can be calculated by monitoring the inventory in the boric acid tanks.

3. Depressurization

Depressurization is accomplished using portions of the CVCS. Either four weight percent boric acid or refueling water can be used as desired for depressurization with the flow path being from the centrifugal charging pumps to the auxiliary spray valve to the pressurizer.

The centrifugal charging pumps of the CVCS are Seismic Category I, and can be powered by the emergency diesels.

The two means of depressurization of the RCS are the pressurizer sprays and the pressurizer PORVs. Pressurizer spray valves are fail closed valves and require instrument air which is not safety related. Therefore, operator actions may be required to open them locally as allowed by BTP RSB 5-1. The pressurizer PORVs are also fail closed valves; however, they are provided with safety related nitrogen supplys as described in FSAR Section 5.2.2.11. In addition, the PORVs

and the their appurtenances are environmentally qualified (FSAR Section 3.11N and Appendix 3A) for a harsh containment environment. The pressurizer PORVs discharge to the pressurizer relief tank described in Section 5.4.11.

Should the auxiliary spray valve fail to open following a seismic event, every effort would be made to open it. As an alternative, depressurization could be accomplished by discharging RCS inventory from the pressurizer via the pressurizer power-operated relief valves to the pressurizer tank (PRT). The design requirements of the PRT are provided in Section 5.4.11, Pressurizer Relief Discharge System. This operation can be integrated with the cooldown function near the end of the cooldown to 350°F. As RCS inventory is relieved to the PRT, the pressurizer temperature and pressure is reduced, thus, reducing the pressure in the RCS. Make-up is provided as necessary to maintain a minimum level in the pressurizer. RCS pressure and temperature and pressurizer level can be monitored using Class 1E instrumentation in the Control Room.

4. Instrumentation

Class 1E instrumentation is available in the Control Room to monitor the key functions associated with achieving cold shutdown. This instrumentation is discussed in Section 7.5 and includes the following:

a. RCS wide range temperature

b. RCS wide range pressure

c. Pressurizer water level

d. Steam generator water level (per steam generator)

e. Steam line pressure (per steam line)

f. Condensate storage tank level

g. RWST level

h. Boric acid tank level (per boric acid tank)

i. Reactor building pressure

This instrumentation is sufficient to monitor the key functions associated with cold shutdown and to maintain the RCS within the desired pressure, temperature and inventory relationships. Operation of the auxiliary systems that service the RCS can be monitored by the Control Room operator, if desired, via remote communication with an operator in the plant.

5. Maintaining RCS Temperature and Pressure During Cooldown

The plant is maintained in a hot standby condition while the operator evaluates the initial plant conditions and the availability of equipment and systems (including

non-safety grade equipment) that can be used in shutdown. Prior to initiating cooldown, the operator will determine the boration requirements and the method by which the plant can be taken to cold shutdown. In performing the cooldown, the operator integrates the functions of heat removal, boration and makeup, and depressurization, attempting to accomplish these functions without letdown from the RCS. Once the plant is cooled to 350°F and depressurized to 425 psig, RHRS operation is initiated and the RCS is taken to cold shutdown conditions.

Boration, cooldown, and depressurization is accomplished in a series of short steps arranged to keep RCS temperature and pressure and pressurizer level in the desired relationships. However, to demonstrate that boration and depressurization can be done without letdown, a simpler scenario can be used. First, the operator integrates the cooldown and boration functions, taking advantage of the steam space available in the pressurizer and the RCS inventory contraction resulting from the cooldown. Then the operator uses auxiliary spray from the CVCS to depressurize the plant to RHRS initiating conditions. Finally, the RCS is cooled to cold shutdown using the RHRS and continuing to make up with borated water.

The calculation to demonstrate this capability assumes worst case boration requirements based on core end-of-life/peak xenon conditions and the following RCS initial conditions following plant trip:

RCS Temperature	557°F
RCS Pressure	2250 psia
Pressurizer Water Volume	450 ft^3
Pressurizer Steam Volume	1350 ft^3

The cooldown from 557°F to 350°F decreases the volume of water in the RCS by approximately 1620 cubic feet. This assumes that the pressurizer is not cooled and the water level is maintained at the initial condition. Makeup for contraction is supplied by four weight percent boric acid stored in the boric acid tanks. A boric acid tank volume of approximately 1460 cubic feet expands to approximately 1620 cubic feet as it is heated to the RCS temperature at 350°F assuming an initial tank temperature of 70°F. The volume required for boration at 350°F, to maintain the reactor within the technical specification shutdown requirements, is less than the contraction volume available at 350°F.

To calculate if depressurization can be accomplished without letdown and without taking the plant water solid, it is assumed that the pressurizer is at saturated conditions with 450 cubic feet of water, 1350 cubic feet of steam, and the pressurizer metal all at 653°F (2250 psia). It is further assumed that no additional water is removed from the pressurizer by the cooldown contraction. With these assumptions, and including the effect of heat input from the pressurizer metal, spraying in approximately 620 cubic feet of 70°F water produces saturated conditions at 425 psia (450°F) with a water volume of 1250 cubic feet and a steam volume of 550 cubic feet.

Once depressurized to RHR operational pressure, RHRS operation is initiated and cooldown is continued to cold shutdown conditions. The cooldown from 350°F to 200°F further decreases the volume of water in the RCS by approximately 550 cubic feet. This assumes that the pressurizer is not cooled and the pressurizer water level is maintained at the level resulting from depressurization. Makeup for contraction is again supplied by four weight percent boric acid from the boric acid tank volume of approximately 530 cubic feet as it is heated to the RCS temperature of 200°F. The additional volume required for boration at 200°F, approximately 260 cubic feet of four weight percent boric acid which expands to approximately 270 cubic feet, is less than the additional contraction volume available at 200°F. This illustrates the technical specification requirements for cold shutdown conditions are satisfied.

The results of the calculation described above demonstrate that, based on the assumed initial conditions, boration and depressurization can be accomplished without letdown and without taking full credit for the available volume created by the cooldown contraction. Should boration without letdown prove impractical due to any combination of plant conditions or equipment failure, the operator could initiate the alternate methods of boration and or depressurization, i.e., boration could be accommodated via letdown through the reactor vessel head vent line and depressurization could be accomplished by discharging RCS inventory via the pressurizer power-operated relief valves to the PRT.

III. SINGLE FAILURE EVALUATION

1. Residual Heat Removal

A. From Hot Standby to 350°F

*Reactor coolant loops and steam generator - Four reactor coolant loops and steam generators are provided, any one of which can provide natural circulation flow for adequate core cooling. Even with the most limiting single failure (of a steam generator power-operated relief valve), three of the reactor coolant loops and steam generators remain available.

*Steam generator power-operated atmospheric relief valves - Four valves are provided (one per generator), any one of which is sufficient for residual heat removal. In the event of a single failure, three power-operated relief valves remain available.

*Auxiliary feedwater pumps - Two motor driven and one steam driven auxiliary feedwater pumps are provided. In the event of a single failure, adequate pumps remain available to provide sufficient feedwater. A pump can be aligned to provide water to all steam generators.

*Auxiliary feedwater flow control valves - Air-operated, fail open valves are provided. In the event of a single failure of one flow control valve (which affects flow to one steam generator from either a motor-driven pump or the steam-driven pump), auxiliary feedwater can still be provided to all four steam generators from the other pumps.

*Condensate storage tank - Upon depletion of the primary source of auxiliary feedwater in the Seismic Category I condensate storage tank, a backup source of auxiliary feedwater can be supplied from either train of the Seismic Category I Service Water System. (See Section II.1).

B. From 350°F to Cold Shutdown

*RHR pumps 1 and 2 - Two RHR pumps are provided, either one of which can provide adequate circulation of the reactor coolant. Each pump is powered from a different emergency power train. In the event of a single failure, either pump can provide sufficient RHR flow.

*RHR suction isolation valves 8701A and 8702A (to RHR pump 1) and 8701B and 8702B (to RHR pump 2) - The two valves in each RHR subsystem are each powered from different emergency power trains. Failure of either power train can prevent initiation of RHR cooling in the normal manner. In the event of such a failure, the affected valves(s) can be deenergized and opened with its handwheel or can be opened using alternate power via programmed operator action outside of the control room. Any other single failure can be tolerated as it would only affect one of the RHR subsystems and adequate cooling can be provided by the redundant subsystem.

*RHR heat exchanger 1 and 2 - If either heat exchanger is unavailable for any reason, the remaining heat exchanger can provide sufficient heat removal capability.

*RHR flow control valves HCV-606 and HCV-607 - If either of these normally open, fail open valves closes spuriously sufficient RHR cooling can be provided by the unaffected RHR subsystem.

*RHR/Safety Injection System cold leg isolation valves 8809 A and B - If either of these normally open, motor-operated valves, which are powered from different emergency power trains, closes spuriously, sufficient RHR cooling can be provided by the unaffected RHR subsystem. The affected valve can be deenergized and opened with its handwheel.

*Component Cooling Water System - Two redundant subsystems are provided for safety-related loads. Either subsystem can provide sufficient heat removal via one of the RHR heat exchangers.

*Service Water System - Two redundant subsystems are provided for safety-related loads. Either subsystem can provide sufficient heat removal via one of the Component Cooling Water System heat exchangers.

2. Boration and Inventory Control

*Boric Acid Tanks 1 and 2 - Two boric acid tanks are provided. Each tank contains sufficient four weight percent boric acid to borate the RCS for cold shutdown.

*Boric Acid Transfer Pumps 1 and 2 - Each pump is powered from a different emergency power train. In the event of a single failure, either pump can provide sufficient boric acid flow.

*Isolation Valve 8104 - If valve 8104, which is supplied from emergency power and is normally closed, cannot be opened due to power train or operator failure, it can be opened locally with its handwheel. If valve 8104 cannot be opened with its handwheel, an alternate flow path is available via a) air-operated, fail open valve FCV-110A and normally closed manual valve 8439 or b) gravity feed through normally closed manual valves 8507 and 8509.

*Refueling Water Storage Tank Isolation Valves LCV-112D and LCV-112E - Each valve is powered from a different emergency power train; only one of these normally closed motor-operated valves needs to be opened to provide a makeup flow path from the RWST to the centrifugal charging pumps.

*Centrifugal Charging Pumps 1 and 2 - Pumps 1 and 2 are powered from a different emergency power train. In the event of a single failure, any one pump can provide sufficient boratin or makeup flow.

*Flow Control Valve FCV-121 - This normally open valve fails open on loss of air. If FCV-121 closes spuriously, the centrifugal charging pumps can safely operate on their miniflow circuits. Efforts would be made to open it. The charging flow could be aligned to the RCS using the safety injection pathway, via valves 8801 A/B. Alternately, boration can be accomplished by starting the positive displacement charging pump.

*Normal Charging Flow Control Valve HCV-182 - This normally open valve fails open on loss of air or power. If HCV-182 closes spuriously, the charging pumps can operate on their miniflow circuits until operator action can open bypass valve 8403.

*Normal Charging Isolation Valves 8105 and 8106 - If either of these normally open, motor-operated valves, each of which is powered from a different emergency power train, closes spuriously, operator can be used to deenergize the valve operator and reopen the valve with its handwheel.

*Alternate Charging Isolation Valve 8402A - If this normally open motor-operated valve closes spuriously, operator action can be used to deenergize the valve operator and reopen the valve with its handwheel.

*Normal Charging Isolation Valve 8146 - If this normally open valves closes spuriously, alternate charging valve 8147, which fails open, can be used.

*Reactor Coolant Pump Seal Injection Valves 8351 A, B, C and D - If any of these normally open, motor-operated valves closes spuriously, operator action can be used to deenergize the valve operator and reopen the valve with its handwheel.

*Reactor Vessel Head Vent - This path provides an alternative means of letdown to accommodate the addition to RCS inventory during boration, i.e., if it cannot be accommodated by system shrinkage.

3. Depressurization

*Auxiliary Spray Valve - This normally closed valve fails closed on loss of air. If 8145 is struck closed as a result of a single failure, the redundant pressurizer power-operated relief valves can be used to depressurize the RCS by discharging the pressurizer inventory to the pressurizer relief tank.

*Charging Valves 8146 and 8147 - These valves fail open on loss of air decreasing the effectiveness of the auxiliary pressurizer spray. If either is struck open, the redundant pressurizer power-operated relief valves can be used to depressurize the RCS by discharging the pressurizer inventory to the pressurizer relief tank.

*RHR Suction Isolation Valve 8701 A and B and 8702 A and B - The RHR suction isolation valves are qualified for the steam line break environment. Therefore, they are qualified for the less severe environment that would result if, as described in the above A and B, the RCS is depressurized by discharging the pressurizer inventory to the pressurizer relief tank.

4. Instrumentation

Sufficient instrumentation is provided to monitor from the Control Room the key functions associated with cold shutdown. All necessary indications are redundant. Thus, in the event of a single failure, the operator can make comparisons between duplicate information channels or between functionally related channels in order to identify the particular malfunction. Refer to Section 7.5 for applicable details.

IV. NATURAL CIRCULATION

Comanche Peak Unit 1 and Unit 2 and Diablo Canyon Unit 1 have been compared in detail to ascertain any differences between the two plants that could potentially affect natural circulation flow and attendant boron mixing. Because of the similarity between the plants, it was concluded that the natural circulation capabilities would be similar, and, therefore, the results of the natural circulation cooldown tests conducted at Diablo Canyon are representative of the capability at Comanche Peak.

The general configuration of the piping and components in each reactor coolant loop is the same in both Comanche Peak and Diablo Canyon. The elevation head represented by these components and the system piping is similar in both plants.

To compare the natural circulation capabilities of Comanche Peak and Diablo Canyon, the hydraulic resistance coefficients were compared. The hydraulic resistance coefficients applicable to normal flow conditions are as follows:

	Diablo Canyon Unit 1	Comanche Peak	
	($X10^{-10}$ft/(GPM)^2)	($X10^{-10}$ft/(GPM)^2)	
		Unit 1[a]	Unit 2
Reactor Core and Internals	129.0	110.2	116.0
Reactor Nozzles	36.1	27.2	27.2
RCS Piping	20.9	25.1	25.1
Steam Generator	112.0	112.4[b]	114.1
Total Hydraulic Resistance Coefficients	298.0	274.8[c]	282.4

a) Values for Steam Generator and Total Hydraulic Resistance Coefficients have been determined for the Unit 1 Δ76 Replacement Steam Generator (RSG) (feedring design).

b) The Unit 1 RSG values are 94.9 and 112.7 for 0% and 10% SGTP respectively.

c) The Unit 1 RSG values are 257.4 and 275.2 for 0% and 10% SGTP respectively.

The flow ratio of Comanche Peak to Diablo Canyon is proportional to the square root of the inverse of the resistance coefficients.

Flow Ratio:

$$\frac{\text{Comanche Peak Unit 1}}{\text{Diablo Canyon}}\left[\frac{298.0}{274.8}\right]^{1/2} = 1.0414 \quad \text{Unit 1 old SG}$$

$$\frac{\text{Comanche Peak Unit 2}}{\text{Diablo Canyon}}\left[\frac{298.0}{282.4}\right]^{1/2} = 1.0272$$

RSG 0% SGTP

$$\frac{\text{Comanche Peak Unit 1 RSG}}{\text{Diablo Canyon}}\left[\frac{298.0}{257.4}\right]^{1/2} = 1.0760$$

RSG 10% SGTP

$$\frac{\text{Comanche Peak Unit 1 RSG}}{\text{Diablo Canyon}}\left[\frac{298.0}{275.2}\right]^{1/2} = 1.0406$$

Note: The hydraulic resistance and the natural circulation flow thermal driving head for the Comanche Peak Unit 1 RSGs (Δ76 SGs) are comparable to those of the old

Model D4 steam generators. Therefore the RSGs will have no adverse impact on the natural circulation cooling capability of the plant [1].

The general arrangement of the reactor core and internals is the same for Comanche Peak and Diablo Canyon. The slight variation in the hydraulic resistance coefficient is primarily due to the specific design details of the vessel and internals (i.e., flow area, upper/lower support plate designs, thermal design flow, elevations, etc.).

The reactor vessel outlet nozzle configuration for both plants is the same. The radius in curvature between the vessel inlet nozzle and downcomer section of the vessel in the two plants is different. Based on 1/7 scale model testing performed by Westinghouse, the radius on the vessel nozzle/vessel downcomer juncture influences the hydraulic resistance of the flow turning from the nozzle to the downcomer. The Diablo Canyon vessel inlet nozzle radius is significantly smaller than that of Comanche Peak as reflected by the higher coefficient for Diablo Canyon.

The resistance coefficient for the RCS piping for both plants is similar, as reflected in the resistance coefficients previously tabulated.

Steam generator units were also compared to ascertain any variation that could affect natural circulation capability by changing the effective elevation of the heat sink or the hydraulic resistance seen by the primary coolant. The Comanche Peak design utilizes a Model D4 (Unit 1) and D5 (Unit 2) steam generator design while the Diablo Canyon design utilizes a Model 51. The major design differences between the generators are that the Model D4 and D5 incorporate a preheater in the lower tube bundle region and the tube bundle elevation is shorter than that of the Model 51. For the natural circulation conditions present, the presence of the preheater unit has negligible effects on flow conditions because auxiliary feedwater flow does not pass directly through the preheater. The differences of tube bundle elevations has an effect on the natural circulation driving head established by the system. The longer tube bundle in the Model 51 SG for Diablo Canyon Unit 1 would result in approximately a 6.3 ± 2.5% (3.8 to 8.8%) higher driving head when compared to the Model D4 or D5 SGs installed at Comanche Peak. This variance in net driving head is relatively small and should not significantly affect the natural circulation flow rate. The heat transfer area between the two generators is also very similar. Even though the Model D4 and D5 have a shorter tube bundle elevation, they have a greater number of tubes as compared to the Model 51. This increase in the number of tubes offsets the shorter tube bundle length of the D4/D5 steam generators with respect to heat transfer area. Additionally, the RCP impeller designs for each plant were also considered to determine any impact on natural circulation flow. The impeller designs are similar for the Diablo Canyon and Comanche Peak RCPs. However, it is noted that the RCP loss coefficients are slightly different. The effects of the slight differences in the driving heads and RCP loss coefficients are quantified below.

As indicated, the overall hydraulic loss coefficient (excluding RCPs) for Comanche Peak is lower than that of Diablo Canyon. It is expected that the relative effect of the coefficients seen would be the same under natural circulation conditions. Therefore, based on the flow ratio correlation utilizing the total hydraulic loss coefficient for each plant (and excluding the differences in driving heads and RCP loss coefficients), the Comanche Peak units will have a natural circulation flow rate approximately 2 - 4% higher than of that for Diablo Canyon. If the slight differences in the thermal driving head and

RCP flow resistance (as detailed previously) are accounted for, the natural circulation flow rate for Comanche Peak will be approximately 4% lower than that for Diablo Canyon. A Comanche Peak natural circulation flow of this magnitude will be sufficiently similar to that obtained for Diablo Canyon such that the Diablo Canyon natural circulation test results are applicable to Comanche Peak.

The data provided above reflects the flowrate and associated heat removal capability of an individual loop in the plant. The comparison, therefore, does not take into consideration the number of loops available nor the core heat to be removed. An evaluation of the Comanche Peak steam relief and auxiliary feedwater systems has been performed to demonstrate that cooling can be provided via three steam generators following the most limiting single active failure, i.e., the failure of an atmospheric relief valve.

Loop circulation flow is dependent on reactor core decay heat which is a function of time based on core rating and its power operating history. Under natural circulation flow conditions, flow into the upper head area will constitute only a small percentage of the total core natural circulation flow and therefore will not result in an unacceptable reduction in the natural circulation flow required to cool the core.

For typical 4-loop plants there are two potential flow paths by which flow crosses the upper head region boundary in a reactor. These paths are the head cooling spray nozzles, and the guide tubes. The head cooling spray nozzle is a flow path between the downcomer region and the upper head region. The temperature of the flow which enters the head via this path corresponds to the cold leg value (i.e., T_{cold}). Fluid may also be exchanged between the upper plenum region (i.e., the portion of the reactor between the upper support plate) and the upper head region via the guide tubes.

Guide tubes are dispersed in the upper plenum region from the center to the periphery. Because of the nonuniform pressure distribution at the upper core plate elevation and the flow distribution in the upper plenum region, the pressure in the guide tube varies from location to location. These guide tube pressure variations create the potential for flow to either enter or exit the upper head region via the guide tubes.

To ascertain any difference between the upper head cooling capabilities between Diablo Canyon and Comanche Peak a qualitative comparison is made. Since Comanche Peak is a Tcold plant, the upper head region is expected to cool at a rate comparable to or exceeding that of Diablo Canyon. Due primarily to differences in the upper support plate design between the two plants, the upper head region for Comanche Peak is approximately 87% larger than that of Diablo Canyon. The reactor vessel spray nozzles between the downcomer and upper head region have a flow area more than 10 times larger for Comanche Peak (vs. Diablo Canyon) providing the enhanced flow mixing capability which maintains the upper head region at a temperature near the cold leg temperature. Therefore, adequate upper head cooling for Comanche Peak is expected during a natural circulation cooldown scenario.

BORON MIXING

A boron mixing calculation has been performed for Comanche Peak to determine the time necessary, under natural circulation conditions, to achieve an increase in RCS boron concentration similar to that exhibited by Diablo Canyon Unit 1 during the natural circulation boron mixing test (i.e., 300 ppm). The calculation assumes no let down and the normal charging line plus the RCP seals provide the boron injection paths at a total boration rate of 75 gpm. For Comanche Peak, boron will normally be supplied from the 7000 ppm (minimum) boron solution of the boric acid tanks to the suction of the centrifugal charging pumps by the boric acid transfer pumps. Makeup in excess of that needed for boration can be provided from the RWST. The results of the calculation indicate that a time of approximately one (1) hour is needed to achieve a 300 ppm increase to the initial RCS boron concentration. The BAT is adequately sized to provide this quantity of borated water.

Conclusion

Based on the above evaluations, it can be concluded that the results of the natural circulation cooldown tests performed at Diablo Canyon are representative of the natural circulation and boron mixing capability of Comanche Peak. Therefore, the Comanche Peak units satisfy the test requirements of Branch Technical Position RSB 5-1 via this comparison.

References

1. Westinghouse report, WCAP-16469-P, Rev. 1, Comanche Peak Unit 1 Replacement Steam Generator Program NSSS Engineering Report, June 2006 (VL-06-001760).

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6.1N ENGINEERED SAFETY FEATURE MATERIALS

6.1N.1 METALLIC MATERIALS

6.1N.1.1 Materials Selection and Fabrication

Typical materials specifications used for components in the engineered safety features (ESF) are listed in Table 6.1N-1. In some cases, this list of materials may not be totally inclusive. However, the listed specifications are representative of those materials used. Materials utilized are procured in accordance with the materials specification requirements of the American Society of Mechanical Engineers (ASME) Code, Section III, plus applicable and appropriate addenda and code cases.

The welding materials used for joining the ferritic base materials of the ESF conform to or are equivalent to ASME Material Specifications SFA 5.1, 5.2, 5.5, 5.17, 5.18, and 5.20. The welding materials used for joining nickel-chromium-iron alloy in similar base material combination and in dissimilar ferritic or austenitic base material combination conform to ASME Material Specifications SFA 5.11 and 5.14. The welding materials used for joining the austenitic stainless steel base materials conform to ASME Material Specifications SFA 5.4 and 5.9. These materials are tested and qualified to the requirements of the ASME Code, Section III and Section IX rules and are used in procedures which have been qualified to these same rules. The methods utilized to control delta ferrite content in austenitic stainless steel weldments are discussed in Section 5.2.3.

All parts of components in contact with borated water are fabricated of or clad with austenitic stainless steel or equivalent corrosion resistant material. The integrity of the safety-related components of the ESF is maintained during all stages of component manufacture. Austenitic stainless steel is utilized in the final heat treated condition as required by the respective ASME Code, Section II, material specification for the particular type or grade of alloy. Furthermore, it is required that austenitic stainless steel materials used in the ESF components be handled, protected, stored, and cleaned according to recognized and accepted methods which are designed to minimize contaimination which could lead to stress corrosion cracking. The rules covering these controls are stipulated in Westinghouse process specifications, which are discussed in Section 5.2.3. Additional information concerning austenitic stainless steel, including the avoidance of sensitization and the prevention of intergranular attack, can be found in Section 5.2.3. No cold worked austenitic stainless steels having yield strengths greater than 90,000 pounds per square inch (psi) are used for components of the ESF within the Westinghouse standard scope.

Westinghouse supplied ESF components within the Containment that would be exposed to core cooling water and containment sprays in the event of a loss of coolant accident (LOCA) utilize materials listed in Table 6.1N-1. These components are manufactured primarily of stainless steel or other corrosion resistant, high temperature material. The integrity of the materials of construction for ESF equipment when exposed to post design basis accident (DBA) conditions has been evaluated. Post DBA conditions were conservatively represented by test conditions. The test program [1] performed by Westinghouse considered spray and core cooling solutions of the design chemical compositions, as well as the design chemical compositions contaminated with corrosion and deterioration products which may be transferred to the solution during recirculation. The effects of sodium (free caustic), chlorine (chloride), and fluorine (fluoride) on austenitic stainless steels were considered. Based on the results of this investigation, as well as

testing by Oak Ridge National Laboratory (ORNL) and others, the behavior of austenitic stainless steels in the post DBA environment will be acceptable. No cracking is anticipated on any equipment even in the presence of postulated levels of contaminants, provided the core cooling and spray solution pH is maintained at an adequate level. The inhibitive properties of alkalinity (hydroxyl ion) against chloride cracking and the inhibitive characteristic of boric acid on fluoride cracking have been demonstrated. Coatings on exposed surfaces within the Containment are not subject to breakdown under exposure to the spray solution and can withstand the temperature and pressure expected in the event of a LOCA.

Information concerning the degree to which the materials used comply with Regulatory Guides 1.31, 1.37, and 1.44 can be found in Appendix 1A(N).

6.1N.1.2 Composition, Compatibility, and Stability of Containment and Core Spray Coolants

The only Westinghouse supplied vessels used for storing ESF coolants, are the accumulators.

The accumulators are carbon steel clad with austenitic stainless steel. Because of the corrosion resistance of these materials, significant corrosive attack on the storage vessels is not expected.

The accumulators are vessels filled with borated water and pressurized with nitrogen gas. The nominal boron concentration, as boric acid, is provided in Table 6.1B-4. Samples of the solution in the accumulators are taken periodically for checks of boron concentration. Principal design parameters of the accumulators are listed in Table 6.3-1.

The method of establishing containment spray and recirculation sump pH following a LOCA is discussed in Sections 6.2.2 and 6.2.3. Information concerning hydrogen release by the corrosion of containment metals and the control of the hydrogen and combustible gas concentrations within the containment following a LOCA is discussed in Section 6.2.5.

6.1N.2 ORGANIC MATERIALS

Protective coatings for use in the reactor containment have been evaluated as to their suitability in post DBA conditions. Tests have shown that the epoxy and modified phenolic systems are the most desirable of the generic types evaluated for in Containment use. This evaluation considered resistance to high temperature and chemical conditions anticipated following a LOCA, as well as high radiation resistance [2]. Information regarding compliance with Regulatory Guide 1.54 can be found in Appendix 1A(N).

6.1N.3 POSTACCIDENT CHEMISTRY

Refer to Section 6.1B.3.

REFERENCES

1. "Behavior of Austenitic Stainless Steel in Post Hypothetical Loss of Coolant Accident Environment," WCAP-7798-L (Proprietary) and WCAP-7803 (Non-Proprietary), January 1972.

2. "Evaluation of Protective Coatings for use in Reactor Containment," WCAP-7198-L (Proprietary), April 1969 and WCAP-7825 (Non-Proprietary), December 1971.

TABLE 6.1N-1
ENGINEERED SAFETY FEATURE MATERIALS

(Sheet 1 of 2)

Valves	
Bodys	SA-182, Type F316 or SA-351, Gr. CF8 or CF8M
Bonnets	SA-182, Type F316 or SA-351, Gr. CF8 or CF8M
Discs	SA-182, Type F316 or SA-564, Gr. 630 cond. 1100° heat treatment or SA-351, Dr. CF8 or CF8M
	A-567 Grade 1 (Stellite 21)
Pressure retaining bolting	SA-453, Gr. 660
Pressure retaining nuts	SA-453, Gr. 660 or SA-194, Gr. 6
Auxiliary Heat Exchangers	
Heads	SA-240, Type 304
Nozzle necks	SA-182, Gr. F304; SA-312, Type 304 SA-240, Type 304
Tubes	SA-213, Type 304; SA-249, Type 304
Tubesheets	SA-182, Gr. F304; SA-240, Type 304 SA-516, Gr. 70 with Stainless Steel Cladding A-7 Analysis
Shells	SA-240 and SA-312 Type 304
Auxiliary Pressure Vessels, Tanks, Filters, etc.	
Shells and heads	SA-351, Gr. CF8A; SA-240, Type 304; SA-264 Clad Plate of SA-537, Gr. B with SA-240, Type 304 Clad and Stainless Steel Weld Overlay A-8 Analysis
Flanges and nozzles	SA-182, Gr. F304; SA-350, Gr. LF2 with SA-240, Type 304 and Stainless Steel Weld Overlay A-8 Analysis
Piping	SA-312 and SA-240, Type 304 or Type 316 Seamless
Pipe fittings	SA-403, Type 304 Seamless
Closure bolting and nuts	SA-193, Gr. B7 and SA-194, Gr. 2H

TABLE 6.1N-1
ENGINEERED SAFETY FEATURE MATERIALS

(Sheet 2 of 2)

Auxiliary Pumps	
Pump casing and heads	SA-182, Gr. F304
Flanges and nozzles	SA-182, Gr. F304
Stuffing or packing box cover	SA-182, Gr. F304
Closure bolting and nuts	SA-193, Gr. B6 and Gr. B7; SA-453, Gr. 600; and Nuts Gr. 6 and Gr. 7

6.1B ENGINEERED SAFETY FEATURE MATERIALS

6.1B.1 METALLIC MATERIALS

This section contains information concerning the materials used for the major pressure retaining components in the engineered safety features (ESF) not covered by Section 6.1N. The information provided presents acceptable means for the control of welding procedures, control of the use of sensitized stainless steel, and selection of materials with regard to their compatibility with specific fluids used.

The systems affected include the Containment Spray System (CSS) [2], [3], and Containment Isolation System. Tables 6.1B-1 and 6.1B-2 contain major components and material specifications for non-Westinghouse procured ESF systems.

Selection and procedures of the above system components materials are in compliance with the ASME B&PV Code, Section III. The control of delta ferrite in welding will comply with Subsections 6.1B.1.1.3 or 6.1B.1.1.4. Avoidance of sensitization will be in compliance with the recommendations of NRC Regulatory Guide [6] except for position C6. Intergrannular corrosion tests for each welding procedure will not be made because the control of weld heat input will be implemented as described in Subsection 6.1B.1.1.2, 1.e.

6.1B.1.1 Materials Selection and Fabrication

6.1B.1.1.1 Specification for Principal Pressure-Retaining Materials and ESF Construction Materials Exposed to Core Cooling and Containment Spray Water

The internal and external surfaces of the ESF components covered in this section are subjected to three types of core cooling and Containment spray water exposures:

1. Internal exposure only - components located outside the Containment

2. External exposure only - components located inside the Containment and not part of the ESF fluid systems

3. Internal and external exposure - components located inside the Containment and part of the ESF fluid systems

All materials used for the ESF components which are subjected to any type of core cooling or Containment spray water exposure, or both, are chosen to be corrosion resistant and their integrity and performance in other operating plants was satisfactory. A listing of construction materials for the ESF components (other than valves) is given in Table 6.1B-1.

Typical materials specifications used for valves in the engineered safety features (ESF) are listed in Table 6.1B-2. In some cases, this list of materials may not be totally inclusive. However, the listed specifications are representative of those materials used. All materials for pressure retaining components are procured in accordance with the ASME B&PV Code, Section III.

Criteria for pH level of postaccident emergency core cooling and Containment spray water established in Branch Technical Position MTEB 6-1 [12] are implemented in design of the CSS as discussed in Sections 6.2.2 and 6.5.2.

6.1B.1.1.2 Integrity of ESF Components

The following sections provide information pertaining to the integrity of the safety-related components of the ESF during all stages of component manufacture.

1. Avoidance of Sensitization

In order to limit the stress corrosion cracking in the unstabilized austenitic stainless steel materials Types 304 and 316 used for components and piping of safety-related systems, the following provisions are established:

a. Only the materials listed in Tables 6.1B-1 and 6.1B-2 are used.

They have been processed and fabricated using the methods described in this section to minimize local sensitization. In addition, all austenitic stainless steel materials are supplied in the solution-annealed condition as prescribed by the ASME B&PV Code, Section II.

For austenitic stainless steel materials of product forms with complex shapes which have inaccessible cavities or chambers that would preclude rapid cooling when water quenched a specimen from each heat treating procedure and each heat of material is intergranular corrosion tested in accordance with ASTM A 262-70, Practice A or E. This testing is not performed on product forms with simple shapes such as plates, sheets, bars, pipe, and tubes as well as forgings and castings with simple shapes which do not preclude rapid cooling.

b. The cleaning and contamination protection procedures, as described in Part 2 of this section, are applied to all surfaces of the austenitic stainless steel materials prior to any elevated temperature treatment and prior to hydrotests.

c. The material used is not subjected to sensitizing temperatures above 800°F during fabrication except for welding and hot bending. Hot bending is followed by solution annealing.

d. Welding is in compliance with ASME B&PV Code, Sections III and IX, using qualified welding procedures and welders.

e. In order to avoid sensitization in the heat affected zone, arc heat input of any weld bead will be calculated by the following formula and documented in the procedure qualification records:

$$H = \frac{E \times I \times 60}{S}$$

where

H = energy input, Joules/in.
E = welding voltage, V

I = welding current, A
S = travel speed, in./min

Energy input as determined by this formula is less than or equal to the following maximum values for the various welding processes shown:

Welding Process Method	Max. Energy Input (kJ/in.)
Manual shielded tungsten arc (GTAW)	50
Manual shielded metallic arc (SMAW)	120
Semiautomatic gas shielded metal arc (GMAW)	60
Automatic gas-shielded tungsten arc - hot wire (GTAW)	75
Automatic gas-shielded tungsten arc - cold wire[(a)] (GTAW)	45
Automatic submerged arc (SAW)	140
Flux-core welding	100

a) Except for specimens having essentially the same thickness, the weld procedures qualifications specimens for the cold wire GTAW process are corrosion tested on ASTM A 262-70, Practice E, [10] and will be so documented.

Exceptions to the control of weld are heat input are the two (2) Chemical Additive Tanks. The above criteria were not used in the welding of these tanks. Using written procedures, welds were simulated and then tested by an independent laboratory to Practice E of ASTM A-262. The heat affected zones were proven to be free of intergrannular corrosion attack.

As an alternate to the foregoing, the heat input for austenitic stainless steel welding shall be controlled by the following:

1. The welding procedure qualification shall include corrosion tests in accordance with ASTM A 262-70, Practice E as modified by Westinghouse P.S. 84201MW-MX Revision 3 dated November 28, 1973.

2. Interpass temperatures shall be limited to 350°F maximum.

3. Production welding shall be monitored for the essential variables listed in paragraph 4 following.

4. Welding parameters of maximum amperage, voltage and maximum bead width shall be considered essential variables and shall not be exceeded without requalification in accordance with (1) above.

f. Hardsurfacing

All hardsurfacing procedures are submitted for approval and are established to preclude sensitization of stainless steel components.

2. Cleaning and Contamination Protection Procedures

Stainless steel materials used for fabrication, installation, and testing of components that are part of the ESF systems are thoroughly cleaned and descaled.

Pickling and passivating during component fabrication will be used only if deemed necessary to ensure cleanliness. They are done in accordance with the practices recommended in ASTM A 380 [11]. The stainless steel materials are thoroughly cleaned including removal of any potentially harmful markings and coatings (adhesive paper, masking tape, or plastic coating) before any elevated temperature treatment and hydrotests.

When all cleaning solutions, processing compounds, degreasing agents, and other foreign materials are removed, austenitic stainless steel materials are protected against contaminants hazardous to austenitic stainless steel throughout the fabrication, shipment, storage, construction, testing, and operation of components.

Exceptions to this are two (2) Recycle Hold-up Tanks, and two (2) Boric Acid Storage Tanks. These items were not given the protection described during construction. Construction site personnel have cleaned and maintained the cleanness of the equipment mentioned in accordance with ANSI N45.2.1-1973 "Cleaning of Fluid Systems and Associated Components During Construction Phase of Nuclear Power Plants", Class B level.

The cleaning and contamination protection procedures and acceptance criteria are established in the stainless steel component design specifications as required by ASME B&PV Code, Section III, Appendix B.

Plugs and closures made from elastomers, plastics, wood, composite board, and metal of the same or compatible composition as the component are placed over all openings as protective covers to prevent contamination during storage and shipment.

3. Thermal Insulation

Within the Containment Building, stainless steel reflective metallic insulation is used to insulate the bulk of the piping.

The other types of insulation used inside the Containment Building are: anti-sweat insulation and high efficiency thermal insulation.

a. Anti-sweat insulation is used to prevent condensation from forming on cold lines. This insulation is composed of molded or resin bonded fiberglass with an outside vapor barrier.

b. High efficiency thermal insulation is used only on specific piping and is installed underneath u-bar pipe restraints and adjacent to high energy absorbing bumpers to reduce the acceleration gap and minimize the restraint loads. This insulation is composed of fibrous media and very fine heat resistant particulate matter totally encased in stainless steel.

Outside of the Containment, nonmetallic asbestos-free insulation composed of hydrous calcium silicate, anti-sweat or thermal fiberglass are used. All hydrous calcium silicate insulation and finishing cements are composed of calcium silicate mineral fibers and binders. Sodium silicate is added as required to meet fluoride and chloride content restrictions. All calcium silicate insulation is covered by an aluminum or stainless steel jacket. Antisweat or thermal fiberglass insulation, as previously described, is also used outside of the Containment.

A representative sample from each production lot shall undergo a chemical analysis to determine leachable chloride, fluoride, sodium, and silicate ion concentration as described in NRC Regulatory Guide 1.36. The lot shall be accepted if the requirements of NRC Regulatory Guide 1.36 [5] are met. In addition, each type of thermal insulation material shall also be qualified for use by a stress corrosion test to ensure that the insulation composition does not induce stress corrosion. Either of the following methods will be used:

a. ASTM C692-71, Evaluating Stress Corrosion Effect of Wicking-Type Thermal Insulations on Stainless Steel (Dana Test) [13]. The material shall be rejected if more than one of five specimens crack.

b. RDT M12-IT, Test Requirements for Thermal Insulating Materials for Use on Austenitic Stainless Steel, Section 5 (Knolls Atomic Power Laboratory, KAPL Test) [14]. The material shall be rejected if more than one of four specimens cracks.

All insulation materials will be adequately packaged and protected to prevent moisture pickup or contamination by external sources. Insulation will be stored in a clean, dry area.

Before the insulation is applied, all metal surfaces are adequately cleaned to remove unacceptable levels of harmful contaminents. Insulation which is wet or shows signs of having previously been wet is unacceptable for use.

6.1B.1.1.3 Control of Delta Ferrite

This section describes the method of implementing Appendix A, GDC 1 [1] and Appendix B of 10 CFR Part 50 with regard to control of welding of austenitic stainless steel and components and systems important to safety.

The welding materials used for production of weld joints must have a minimum of five percent delta ferrite (including consumable inserts).

The welding procedures and welder qualifications are determined according to ASME B&PV Code, Section IX, Welding Qualifications.

When the new welding procedure qualification tests are evaluated for these applications, including repair welding of materials, the following examinations are performed in addition to the requirements of ASME B&PV Code, Section III:

1. Chemical Analysis

   Actual chemical analysis is compared to the Shaeffler Diagram (American Society for Metals Handbook, Vol. 6, pp. 246-47) [15]. Samples will be taken from the undiluted weld metal.

2. Magne-gage method

   The determination of the delta ferrite content is made on undiluted weld metal using calibrated magnetic measuring devices conforming to the AWS A4.2-74, Standard Procedures for Calibrating Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic Stainless Steel Weld Metal [16].

3. Macroscopic Examination

   A visual examination is performed on the procedure-qualification-tested bend specimens using 5 to 10 times magnification. In addition to the evaluation of open defects, the small fissures or cracks are observed and evaluated on the bases of their length and number per unit of area. This information is reported for the procedure qualification records.

4. Heat Input Control

   The procedure qualification record also includes the nonessential elements which affect the quality of a weld. The segregation and fissuring can be reduced if the temperature gradient, i.e., heat input, is kept low. Consequently, the procedure qualification records contain the calculation of the heat input using the nonessential elements as described in Subsection 6.1B.1.1.3.

5. Interpass Temperature Control

   The maximum interpass temperature in all welding processes is 350°F and is reported in the procedure qualification records.

6. Destructive and Nondestructive Testing

   The results of all the destructive and nondestructive tests are reported in the procedure qualification record in addition to the information required by ASME B&PV Code, Section III.

7. Welding Materials

   The welding materials used for production welds meet the requirements of Section III and Section II, Part C, of the ASME B&PV Code. According to ASME B&PV Code Section III,

bare weld filler metal (including consumable inserts) used in inert gas-welding process conforms to ASME SFA-5.9 and contains not less than five percent delta ferrite in the deposit.

Weld filler metal materials used in flux shielded welding processes conform to ASME SFA-5.4 or SFA-5.9 and are procured in a wire flux composition to be capable of providing not less than five percent delta ferrite in the deposit according to ASME B&PV Code, Section III.

8. Control of Production Welds

The welding procedures are qualified using nondestructive and destructive mechanical tests as specified in ASME B&PV Code, Sections III and IX. Both the essential and the nonessential variables are recorded and certified and are accessible to the authorized inspector.

The procedure qualification for stainless steel welding will record all the essential variables and in addition, the following:

a. Process variables such as slag pool depth and electrode feed rate oscillation

b. Welder performance qualifications, as required in ASME B&PV Code, Section IX, are determined and recorded. Those welders qualified to make austenitic stainless steel welds must follow the established welding procedures in order to ensure the minimum 5 percent of delta ferrite in production welds.

9. Quality Assurance

The welding quality assurance program includes identification and control of welding material by lots and heats as appropriate. All of the weld processing is monitored according to approved inspection programs, which include review of "starting" materials, qualification records, and welding parameters. Welding systems are also subject to quality assurance audit including calibration of gages and instruments, identification of "starting" and completed materials, welder and procedure qualifications, availability and use of approved welding and heat treating procedures, and documentary evidence of compliance with materials, welding parameters, and inspection requirements.

6.1B.1.1.4 Control of Delta Ferrite - Alternate Method

As an alternate to Subsection 6.1B.1.1.3, the control of delta ferrite will follow the recommendations of NRC Regulatory Guide 1.31, Revision 1 [4], or Revision 2 [4a].

An exception to this is the Containment Spray Pumps. In general, Revision 2 of Regulatory Guide 1.31 will apply in the control of delta ferrite. However, where welding was done prior to establishing the acceptability of the welding materials used, and where these materials are no longer available, determination of delta ferrite was made in accordance with position C5 of Regulatory Guide 1.31, Revision 1.

6.1B.1.2 Composition and Compatibility of Containment and Core Spray Coolants

The method of establishing and controlling Containment spray and recirculation sump pH following a loss-of-coolant accident (LOCA) is discussed in Sections 6.2.2 and 6.5.2. The method ensures a pH greater than 8.25 for all modes of operation of the spray system. Use of the alkaline spray enhances iodine removal and reduces the possibility of stress corrosion cracking of austenitic stainless steel. Use of metals, such as aluminum and zinc, which react with the alkaline spray forming hydrogen are carefully controlled. This use conforms to Branch Technical Position MTEB 6-1. Information concerning hydrogen release by the corrosion of Containment metals and other sources and control of hydrogen gas concentrations within the Containment following a LOCA is presented in Section 6.2.5.

Materials used for the Containment and for components therein are chosen for their compatibility with the spray solution (LOCA environment) and for their intended normal service. Materials used in the ESF system are chosen for their compatibility with the spray solution which consists of borated water and 30 percent sodium hydroxide. Details concerning the spray additive subsystem are discussed in Sections 6.2.2 and 6.5.2.

6.1B.2 ORGANIC MATERIALS

A list of organic materials used in the Containment is presented in Table 6.1B-3.

The significant coating systems used inside the containment are as follows:

	Primer	Finish
Steel	Inorganic Zinc	Epoxy
Concrete	Epoxy surfacer	Epoxy

The protective coatings program for ongoing Service Level I inspections, maintenance, repair and modifications is in accordance with the following guidance:

- EPRI Report 1003102, “Guideline on Nuclear Safety-Related Coatings”, Revision 1 (formerly TR-109937) [Ref. 7]

- ASTM D 5144-00, “Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants. [Ref. 8]

A detailed “suitability of application review” of the protective coatings inside containment for Units 1 and 2 has been performed based on the guidance in EPRI Report 1003102 “Guideline on Nuclear Safety-Related Coatings”, Revision 1 (formerly TR-109937) and ASTM D 5144-00, Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants. [Ref. 17]

ASTM D 3911-03, Standard Test Method for Evaluating Coatings Used in Light Water Nuclear Power Plants at Simulated Design Basis Accident (DBA) Conditions is followed in the evaluation of protective coating system test results. Testing of each coating system to a minimum of 3.0 E+08 Rads gamma is adequate for a Service Level I system to be qualified. However, testing of each coating system to a minimum of 2.0 E+08 Rads gamma is also adequate for a Service

Level I system to be qualified as long as the individual coatings have passed radiation tolerance testing equal to, or in excess of 3.0 E+08 Rads.

A specifically structured quality assurance program based on the guidance in EPRI Report 1003102 is applicable to the Service Level I coatings and program as described in Appendix 17A.

The Containment building coating systems are applied over surfaces prepared in accordance with approved procedures and the coating manufacturers' instructions. The primary criteria for the selection of maintenance protective coatings used within the Containment building are:

1. Capability of being easily decontaminated

2. Capability of not reacting chemically with spray solutions

3. Capability of preventing the formation of gaseous or solid waste products

4. Capability of resisting the environmental radiation for the life of the plant

5. Capability of withstanding D.B.A. conditions individually tested as described above in accordance with ASTM D 3911-03 [Ref. 9]

The surveillance program for testing, inspection of documentation of the Service Level I protective coating systems inside the Containment buildings includes:

A - Qualification and training of inspection personnel.

B - Inspection and testing procedures which specify:

1. Operational methods for each inspection and test;
2. Inspection equipment;
3. Frequency of testing and inspection;
4. Acceptance criteria for each inspection and test; and
5. Record keeping to document inspections and tests.

C - Verification of storage and handling of protective coatings.

D - Calibration of measuring and test equipment.

E - Reporting, disposition, and tracking of coating degradation and deficiencies

F - Completion, issuance, and control of documentation.

G - Maintenance and Control of the Coatings Exempt Log.

Charcoal and oil present inside the containment are located within filter housings and mechanical components respectively. Thus, charcoal and oil will not be exposed to the containment spray and will not create debris or methane.

6.1B.3 POSTACCIDENT CHEMISTRY

In the event of an accident, sodium hydroxide and boric acid solutions are present in the Containment sump. Parameter ranges used for calculation of equilibrium sump solution pH following an accident are given in Table 6.5-3. For sources of boron in the sump following an accident, see Table 6.1B-4. The range of equilibrium sump solution pH provided by the Containment Spray System (CSS) and its chemical additive subsystem is described in Section 6.5.2.

Adequate mixing of these solutions is assured by draining all major volumes in the Containment to the Containment sump. The only volume potentially capable of retaining a significant volume of spray solution, thereby preventing mixing of that volume, is the reactor cavity. The solution held up in the reactor cavity has a pH equal to that of the spray solution during injection, and loss of this volume is considered in calculating the minimum Containment sump pH.

Since the pH range of the Containment sump solution is relatively high, corrosion of aluminum components is limited by restricting the use of aluminum inside Containment as discussed in Section 6.2.5.

REFERENCES

1. 10 CFR Part 50, Appendix A, GDC 1, Quality Standards and Records.

2. 10 CFR Part 50, Appendix A, GDC 38, Containment Heat Removal.

3. 10 CFR Part 50, Appendix A, GDC 41, Containment Atmosphere Cleanup.

4. NRC Regulatory Guide 1.31, Control of Stainless Steel Welding, Revision 1, June 1973, U. S. Nuclear Regulatory Commission.

4a. NRC Regulatory Guide 1.31, Control of Ferrite Content in Stainless Steel Weld Metal, Revision 2, May 1977, U.S. Nuclear Regulatory Commission.

5. NRC Regulatory Guide 1.36, Nonmetallic Thermal Insulation for Austenitic Stainless Steel, February 1973, U.S. Nuclear Regulatory Commission.

6. NRC Regulatory Guide 1.44, Control of the Use of Sensitized Stainless Steel, May 1973, U.S. Nuclear Regulatory Commission.

7. EPRI Report 1003102, “Guideline on Nuclear Safety-Related Coatings”, Revision 1 (formerly TR-109937).

8. ASTM D 5144-00, Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants.

9. ASTM D 3911-03, Standard Test Method for Evaluating Coatings Used in Light-Water Nuclear Power Plants at Simulated Design Basis Accident (DBA) Conditions.

10. ASTN A-262-70 Practice E, Copper-Copper Sulfate-Sulfuric Test for Detecting Susceptibility to Intergranular Attack in Stainless Steels, Annual Book of ASTM Standards, Part 3, American Society for Testing and Materials.

11. ASTM A-380, Recommended Practice for Cleaning and Descaling Stainless Steel Parts, Equipment and Systems.

12. Branch Technical Position MTEB 6-1, pH for Emergency Coolant Water, appended to Standard Review Plan 6.1.1.

13. STM C692-71, Evaluating Stress Corrosion Effect of Wicking-Type Thermal Insulations on Stainless Steel.

14. RDT M12-IT, Text Requirements for Thermal Insulating Materials for Use on Austenitic Stainless Steel, Section 5, Knolls Atomic Power Laboratory.

15. American Society for Metals Handbook, Vol. 6, pp. 246-471.

16. AWS A4.2-74, Standard Procedures for Calibration Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic Stainless Steel Weld Metal.

17. ER-ME-124, Evaluation of CPSES Protective Coatings, Revision 0, November 28, 2007.

18. Deleted.

19. Model Testing of the Recirculation Containment Sump, Comanche Peak Units 1 and 2, Western Canada Hydraulic Labratories, November 1981.

TABLE 6.1B-1
ESF COMPONENT MATERIALS SPECIFICATIONS

(Sheet 1 of 2)

Component	Material
1. Containment Spray System	
Heat Exchanger	
Shell	SA-516 Gr. 70
Bonnet	SA-240 TP 304
Tube Support	SA-285 Gr. C FBQ
Baffles	SA-285 Gr. C FBQ
Tubes	SA-249 TP 304
Tubesheets	SA-240 TP 304
Gaskets	Flexi Carb
Nozzles (shell)	SA-106 Gr. B
Nozzles (bonnet)	SA-358 Class I TP 304 or 316
Couplings	SA-182 F 304 or 316
Pump	
Casing	SA-351 CF8M
Impeller	SA-351 CF8M or CF3M
Shaft	SA-182 Gr. 316 L
Shaft Sleeves	SA-182 Gr. 316 L
Radial sleeve bearings	A-48 CL30
Ball thrust bearings	A-295
Mechanical seal	SA-182 F 304/316
Coupling	Forged Alloy Steel
Gasket	Permanite XJ534 or SF3500
Chemical Additive Tank	
Shell	SA-240 TP 304
Heads	SA-240 TP 304
Nozzles	SA-312 TP 304
Flanges	SA-182 TP 304
Refueling Water Storage Tank	
Liner	SA-240 TP 304

TABLE 6.1B-1
ESF COMPONENT MATERIALS SPECIFICATIONS

(Sheet 2 of 2)

Component	Material
Spray Nozzles	SA-351 Gr. CF8 TP 304
Chemical Eductor	
Body	SA-182 Gr. F 304
Motive Nozzle	SA-182 Gr. F 304
Motive Flange	SA-240 TP 304
Suction Nozzle	SA-312 TP 304
Suction Flange	SA-182 Gr. F 304
Discharge Nozzle	SA-182 Gr. F 304
Discharge Flange	SA-182 Gr. F 304
Recirculation Sump Screens	SA-249 TP 316
Piping	SA 312 and SA 358 CL1 TP 304 or TP 316

TABLE 6.1B-2
ESF VALVE MATERIALS SPECIFICATIONS

Pressure Retaining Components

Part	Material
Bodies	SA-351 Grade CF8 or Grade CF8M SA-182 Grade F 316 SA-105 SA-216 Grade WCB or WCC SA-515 Grade 70
Bonnets	SA-351 Grade CF8 or Grade CF8M SA-182 Grade F 316 SA-479 Type 316 SA-182 Grade F 11 SA-105 SA-217 Grade WC6 SA-350 Grade LF1 or LF2
Discs[(a)]	SA-182 Grade F 316 SA-351 Grade CF8M with N: Plating SA-479 Type 416 or Type 316 SA-182 Grade F11 A-567 Grade 1 (Stellite 21)
Closure Bolting and Nuts	SA-453 Grade 660 SA-193 Grade B6 or Grade B7 SA-564 Grade 630 SA-194 Grade 2H SA-194 Grade 8M

a) Diaphragm material for Saunders type valves is EPT Grade M

TABLE 6.1B-3
ORGANIC MATERIALS USED INSIDE CONTAINMENT

Material		Quantity (Per Unit)
Charcoal (filters)		11,544 lb
Plastic (cable insulation and jacket)		approximately 60,200 lb
Coatings	Steel:	333,000 sq. ft.
	Concrete:	285,000 sq. ft.
Oil		approximately 1435 gal.

TABLE 6.1B-4
SOURCES OF BORON IN SUMP FOLLOWING A LOCA

Source	Borated Water (gal)	Boron Concentration (ppm)
RWST	388,677 to 481,919	2400 to 2600
Accumulators	24,476 to 26,388	2300 to 2600
RCS	88,232.7	0 to 1850

6.2 CONTAINMENT SYSTEMS

6.2.1 CONTAINMENT FUNCTION DESIGN

6.2.1.1 Containment Structure

6.2.1.1.1 Design Bases

The Containment Building (Reactor Building) design meets the requirements of General Design Criteria (GDC) 16 and 50 and is in accordance with the following:

1. Postulated LOCAs and secondary system breaks are considered in combination with loss of offsite power and single active failures in determining the maximum pressure and temperature.

2. The Containment Building and associated systems are designed such that the calculated Containment pressure following a LOCA or a secondary system break is below the design pressure.

3. The sources and amounts of mass and energy released to the Containment are conservatively determined as discussed in Subsections 6.2.1.3 and 6.2.1.4.

4. The Containment pressure is reduced to less than 50 percent of the Containment design pressure within 24 hr after the postulated accident. This is consistent with the assumptions used in the offsite dose analyses presented in Chapter 15.

5. The minimum calculated Containment pressure is not less than the pressure used in the analysis of the ECCS performance capability.

   The minimum Containment pressure analysis is discussed in Subsection 6.2.1.5.

6. The maximum calculated external pressure differential resulting from inadvertent actuation of the Containment Spray System (CSS) is below the Containment external design pressure.

7. The Containment internal structures are designed to withstand the differential pressure loadings resulting from pipe breaks within the Containment subcompartments. The details are presented in Subsection 6.2.1.2.

8. Instrumentation capable of operating in the postaccident environment is provided to monitor the Containment pressure and temperature and the sump water level after an accident. The details of the instrumentation provided are given in Section 7.5.

6.2.1.1.2 Design Features

The Containment consists of a steel-lined, reinforced concrete structure which houses the nuclear steam supply system (NSSS) and certain portions of the ESF systems. The general arrangement drawings are included in Section 1.2.

The Containment is designed to limit the leakage of radioactive fission products to the environment in the event of an accident inside the Containment. The structural design of the Containment is presented in Section 3.8.1. The design provisions to protect the Containment structure and the ESF systems from pipe whip and missiles resulting from postulated accidents are discussed in Sections 3.5 and 3.6. Applicable codes and standards are listed in Section 3.8. The Containment Ventilation System is discussed in Section 9.4.

The principal design parameters of the Containment are listed in Table 6.2.1-1.

6.2.1.1.3 Design Evaluation

6.2.1.1.3.1 Spectrum Of Breaks Analyzed

1. LOCA

   The Containment pressure temperature transient response is analyzed for a large area rupture of the RCS at two distinct locations:

   a. Hot leg (between reactor vessel and steam generator)

   b. Pump suction (between steam generator and reactor coolant pump)

   The specific breaks analyzed are

   a. Double-ended pump suction guillotine (DEPSG)

   b. Double-ended hot leg guillotine (DEHLG)

   The DEPSG break is analyzed assuming minimum ECCS operation (minimum ESF) and assuming maximum ECCS operation (one out of two Containment Spray trains operating - Minimum Containment Spray System Operation). The DEHLG is only analyzed through blowdown, so ECCS operation is not applicable.

   The use of identical mass and energy release data (Reference 17) Component Cooling Water (CCW) system flow rates (affecting the Residual Heat Removal (RHR) and the CSS heat exchanger), the CSS actuation delay time, and containment spray header fill time in the analysis of postulated LOCAs for both Units 1 and 2 results in identical containment pressure and temperature responses to the postulated scenarios. Section 6.2.1 refers to Tables and Figures that represent both Units 1 and 2 analysis results.

2. Steam Line Breaks

   The Containment response is presented for a spectrum of main steam line breaks. Four power levels (100.6 percent, 70 percent and 30 percent of 3628 MWt, and hot shutdown) are considered. A full double-ended rupture and a split rupture are presented at each power level.

6.2.1.1.3.2 Results

The calculated peak pressures and temperatures resulting from postulated LOCAs and steam line breaks for Unit 1 are summarized in Tables 6.2.1-2 and 6.2.1-2A, respectively.

The pressure margins and percent pressure margins are also listed.

Conservative values of the initial steam generator water and steam inventories are used.

The results of the Containment pressure temperature analysis (limiting cases) for MSLB are graphically presented in Figures 6.2.1-3 and 6.2.1-4.

Among the steam line breaks analyzed, for Units 1 and 2 the 4.3 ft^2 split rupture at 100.6% power results in the maximum containment temperature while the 4.7 ft^2 split rupture at 30% power results in the maximum containment pressure, respectively.

The containment pressure and temperature analysis for LOCA used for the environmental qualification evaluations are based on a GOTHIC run of the DEPS MINSI case.

The temperature and pressure profiles utilized for EQ evaluations as described in section 3.11B bounds the limiting temperature and pressure transients for postulated LOCA and MSLB events.

6.2.1.1.3.3 Method Of Analysis

The Containment pressure and temperature responses due to a spectrum of the LOCAs and main steam line break accidents are analyzed using the computer program, GOTHIC.

The analytical model for the GOTHIC program is described in Section 6.2.1.1.3.11.

6.2.1.1.3.4 Mass And Energy Data

Mass and energy release analysis for the LOCA is presented in Subsection 6.2.1.3. Energy inventories, mass and energy balances, and release data for the spectrum of breaks analyzed are also included in Subsection 6.2.1.3.

The generic mass and energy release analysis for the steam line breaks is described in Reference [6]. A discussion of the mass and energy releases for the Comanche Peak Nuclear Power Plant (CPNPP) is included in Subsection 6.2.1.4.

6.2.1.1.3.5 Containment Initial Conditions

The initial Containment pressure and temperature, used in the analysis are shown in Table 6.2.1-5. Table 6.2.1-5 also lists the temperatures of the refueling water, component cooling water and service water.

6.2.1.1.3.6 Failure Mode Analysis

1. LOCA

The DEPSG breaks are analyzed assuming minimum effectiveness of the CSS, i.e., only one out of two spray trains is operating. A single failure analysis of the CSS is included in Subsection 6.2.2.

Both maximum and minimum ECCS operation is considered in the analysis of the design basis LOCA: the DEPSG break.

2. Steam Line Breaks

The following safety system failures are considered in the analysis of steam line breaks:

a. Failure of the feedwater line isolation valve to close

b. Failure of steam line stop valve

c. Failure of one Containment spray train

A discussion of the failures considered is given in Subsection 6.2.1.4.

6.2.1.1.3.7 Structural Heat Sinks

The passive structural heat sinks assumed in the analysis are tabulated in Table 6.2.1-6.

An interface thermal conductivity of 0.0161 Btu/hr-ft-F, and volumetric heat capacity of 0.0167 Btu/ft^3-F is used between the steel liner and the concrete for the dome an cylindrical portion of the containment wall.

All heat sinks in Containment are assumed to be painted. Paint thermal conductivities of 0.0875 and 0.0700 Bth/hr-ft-F are used for steel and concrete surfaces, respectively. Also, a paint volumetric heat capacity of 0.1 Btu/ft^3-F is assumed for both steel and concrete structures. The contact between paint layers and heat sink outer surfaces is considered to be adhesive, and accordingly, thermal contact resistance is taken as zero.

The thermophysical properties of the heat sink materials are tabulated in Table 6.2.1-8.

The heat absorption calculations are based on the Diffusion Layer Model (DLM) heat transfer coefficient for the LOCA and for the MSLB analyses.

6.2.1.1.3.8 Accident Chronology

For the design basis LOCA, the significant events and their times of occurrence are shown in Table 6.2.1-9. For the design basis steam line break, the chronology of major events is presented in Table 6.2.1-10.

6.2.1.1.3.9 External Pressure Analysis

The Containment is designed for an external differential pressure of five psi (See Section 3.8.1.3), which is well above the maximum differential pressure resulting from inadvertent operation of the CSS.

The differential pressure resulting from inadvertent CSS actuation is calculated using the following assumptions:

1. The initial containment pressure is assumed to be 14.2 psia.

2. The containment temperature is assumed to be reduced from an initial value of 120°F to the minimum expected containment spray (refueling water) temperature of 40°F.

3. The relative humidity is assumed to be 100 percent.

With the previously mentioned assumptions, the maximum external pressure differential is calculated to be 3.79 psi. This value is below the design value of five psid.

The Containment is capable of withstanding the design external pressure differential; therefore, no special provision or operator action is required to protect the Containment against inadvertent spray actuation.

6.2.1.1.3.10 Boron Injection Tank Removal Analysis

The main steamline ruptures were analyzed based on the removal of the Boron Injection Tank (BIT). See Section 6.2.1.1.3.2 for results.

6.2.1.1.3.11 GOTHIC Computer Program and Containment Analysis Analytical Model

Calculation of the containment response following a postulated LOCA was analyzed by use of the digital computer code GOTHIC. GOTHIC version 7.2a was used for the LOCA containment response analysis. The GOTHIC Technical Manual (Reference 18) provides a description of the governing equations, constitutive models, and solution methods in the solver. The GOTHIC Qualifications Report (Reference 19) provides a comparison of the solver results with both analytical solutions and experimental data.

The GOTHIC containment modeling for CPNPP is consistent with the recent NRC approved Ginna evaluation model (Reference 20). The latest code version is used to take advantage of the diffusion layer model (DLM) heat transfer option. This heat transfer option was approved by the NRC (Reference 20) for use in Ginna containment analyses with the condition that mist be excluded from what was earlier termed as the mist diffusion layer model (MDLM). The GOTHIC containment modeling for CPNPP has followed the conditions of acceptance placed on Ginna. Ginna and CPNPP both have large dry containment designs. The CPNPP containment volume is considerably larger because CPNPP is a 4-loop plant and Ginna is a 2-loop plant. The only other notable difference is that CPNPP has two independent and redundant containment spray systems whereas Ginna has one spray system and service water cooled fan coolers. None of the user-controlled enhancements added to version 7.2a were implemented in the CPNPP containment model.

The containment response for design basis LOCA and steam line break containment integrity is an American Nuclear Society (ANS) Condition IV event, an infrequent fault. The relevant requirements to satisfy NRC acceptance criteria are as follows:

1. GDC-16 and -50: In order to satisfy the requirement of GDC-16 and -50, the peak calculated containment pressure should be less than the containment design pressure of 50 psig.

2. GDC-38: In order to satisfy the requirement of GDC-38, the calculated pressure at 24 hours should be less than 50 percent of the peak calculated value. (This is related to the criteria for containment leakage assumptions as affecting doses at 24 hours.)

6.2.1.2 Containment Subcompartments

6.2.1.2.1 Design Bases

The Containment Building subcompartments are the fully or partially enclosed volumes within the Containment which contain high energy lines. These subcompartments are designed to limit the adverse effects of a postulated high energy pipe rupture within them. Subcompartment multi-node pressurization analyses have been performed to investigate the consequences of such high energy line ruptures and to ensure the adequacy of the subcompartment design, both with respect to the integrity of the structure itself and with respect to possible asymmetric pressure loads on safety related components situated within the affected subcompartments. All postulated pipe break locations and types for the analysis were chosen in accordance with NRC Regulatory Guide 1.46 in order to select a design basis rupture yielding the highest mass and energy release rates consistent with break location criteria. However, in accordance with the 1987 revision to GDC-4 (Refer to Section 3.6B.2), the dynamic effects of RCS main loop piping breaks and RCS branch line breaks 10 inch diameter and larger have been eliminated from consideration. Specifically, subcompartment pressurization analyses were performed for the following subcompartments and design basis pipe ruptures:

1. Postulated Pipe Break Types and Locations

   a. Steam Generator Compartment

      The following breaks are used in evaluating both the compartment pressurization and the RCS component asymmetric pressurization:

      1. Main steam line full guillotine break in steam generator compartment No. 3.

      2. Main steam line full guillotine break in steam generator compartment No. 4.

      3. Feedwater line full guillotine break in steam generator compartment No. 4.

      4. Residual heat removal 6" line full guillotine break in steam generator compartment No. 4.

5. Auxiliary feedwater 6" line full guillotine break in steam generator compartment No. 4.

Note: Except for the main steam line break in steam generator compartment No. 3, the results obtained from steam generator compartment No. 4 are representative of the other compartments.

b. Main Steam Penetration Area

A full guillotine break was assumed. This break determined compartment structural loads only as no components were subjected to differential pressure loading.

c. Feed Water Penetration Area

A full guillotine break was assumed. This break determined compartment structural loads only, as no components were subjected to differential pressure loading.

d. Pressurizer Compartment

A full guillotine spray line break at the nozzle was assumed for evaluating both the compartment structural design and the component supports design.

6.2.1.2.2 Design Features

Figures 1.2-8 and 1.2-11 through 1.2-21 provide detailed plant and evaluation drawings of the containment subcompartments. The internal structures and equipment locations are shown on these drawings. The volume and vent areas for each subcompartment are discussed in Section 6.2.1.2.3.

6.2.1.2.3 Design Evaluation

Containment subcompartment analyses are performed to calculate the differential pressure transient across major equipment and walls that will result from postulated pipe ruptures. The resulting pressure gradients are used to evaluate the loads and moments on major equipment supports. The differential pressures across the subcompartment walls are used as the design basis for the structures. Section 3.8.3 provides the acceptance criteria for the structures.

A model is developed for each subcompartment to predict a conservative pressure response. Each subcompartment is subdivided into a network of control volumes or nodes. Boundaries between control volumes, which represent junctions or vent paths, are located at physical discontinuities where geometric influences are expected to create a pressure differential. A detailed description of each subcompartment model is given later in this section.

The assumed initial conditions for the subcompartment volumes are conservatively chosen to maximize the resultant differential pressure responses. The values selected are as follows:

Maximum temperature (°F)	120.0
Minimum air partial pressure (psia)	14.2
Minimum relative humidity (%)	0.1

The containment subcompartment pressurization analyses are performed with the RELAP4/MOD5 computer program [Reference 16]. The critical flow correlation selected for each vent path is the homogeneous equilibrium model (HEM).

The vent loss coefficient depends on the geometry of the particular vent. The values of the total loss coefficients for both forward and reverse flow directions are simply the sum of the loss coefficients for each portion of the vent path. These total loss coefficients consist of the following; contraction and expansion losses which are determined as a function of the ratio of the upstream and/or downstream cross-sectional area to the cross-sectional area of the vent, form losses which are due to objects in the flow path such as grating. Sensitivity analyses to evaluate the effects of additional steel grating in the steam generator loop rooms as well as in the pressurizer compartments were performed using RELAP4/MOD5 computer program.

Mass and energy release rate data for the pressurizer spray line break are calculated by Westinghouse, as described in Reference [17], using the SATAN-V computer code. A 10 percent safety margin is applied to the release rates for Unit 1 (covers the 9 percent penalty associated with the replacement steam generator program and 4.5% uprate), while a one percent safety margin is applied to the release rates for Unit 2. The methods used by SATAN-V short-term mass and energy release analyses for evaluation of critical flow were reviewed. The results indicated that critical flow through an orifice is higher than critical flow through a pipe, and that SATAN-V, in terms of pressure, break flow and break energy, yields conservative results.

The SATAN-V critical flow calculation is dependent upon the fluid conditions at the break location. For subcooled fluid conditions, a modification of the Zaloudek correlation is applied with a discharge coefficient of 1.0. The parameters which determine subcooled critical flow are the reservoir pressure and the saturation pressure corresponding to reservoir conditions. The Moody correlation, a thermodynamic equilibrium critical flow model also with a discharge coefficient of 1.0, is used for saturated and two-phase fluid conditions. The Moody model is a function of stagnation properties. (See Reference [9] for more detailed description of these break flow models.)

These critical flow models were compared to the subcooled and saturated data presented in Reference [10]. Reference [10] contains representative data for nozzles of varying entrance geometry, length and diameter. In the two-phase region, the Moody model underpredicted some of the applicable data. By applying a multiplier to the SATAN-V correlation, the data fell beneath the curve generated from this modified correlation. Similarly, the Zaloudek model was non-conservative when compared to several subcooled data points. A multiplier on the Zaloudek equation provided a correlation which represented an upper bound to the data.

In order to determine the effect of these modified correlations, the adjusted critical flow model was programmed into the SATAN-V code for all break locations analyzed. Results indicate that for all break locations, with the exception of the reactor cavity hot leg break, the analyses

performed using the adjusted critical flow models compare favorably to the analyses which used the unmodified correlations. The largest variation in break flows was less than 1%. For the reactor cavity hot leg break, the effect of the multiplier on break flow becomes somewhat more pronounced, and the resulting peak release rate increased by approximately 3.5%. Due to the fact that this break is not the most limiting for a reactor cavity break, these results do not justify any change in the data used for subcompartment analyses nor do they require reanalysis of subcompartment pressures. The trend in the mass release transient is reflected in the resulting differential pressure transient; thus, the peak differential pressure will be determined by the dominant reactor cavity break in terms of break flow (the cold leg break). Since a 40% margin was applied to the calculated pressure differentials, the results currently reported are sufficiently conservative under review of the current critical flow data. Of this 40%, 10% is directly attributable to calculated mass and energy releases. Only 1% of the 10% margin on the short-term pressurizer spray line break mass and energy releases remains because 9% of the margin was needed to cover the effects of the replacement steam generator and 4.5% uprate program.

The mass and energy release rates for the main steam, feedwater, and auxiliary feedwater line breaks were calculated based on Westinghouse Safety Analysis Standard 12-1 [Reference 15]. The mass and energy release rates for the residual heat removal line break were developed using the Zaloudek correlation, as developed in Westinghouse report WCAP-8312-A [Reference 7].

The peak loads on the steam generators, reactor coolant pumps, and the pressurizer due to asymmetric pressurization of the surrounding cubicle are provided in Table 6.2.1-94 for each of the postulated pipe ruptures evaluated. The cubicle number and postulated rupture producing the loads are also specified.

These loads are resolved about the steam generator at the elevation of the upper support, the reactor coolant pump at the elevation of the pump feet, and the pressurizer at the elevation of the upper support.

Descriptions of the subcompartment geometric models and results are presented below:

STEAM GENERATOR COMPARTMENTS

The steam generator compartment analysis employs two nodalization models due to geometry differences between compartments. They are steam generator compartment No. 3 model and steam generator compartment No. 4 model which also represents Steam Generator No. 1 and No. 2. Both models employ fine vertical and horizontal subdivisions to maximize the local peak pressure for various breaks. All major flow area changes are included in the models. Thus, no further sensitivity study is performed.

This analysis assumes flow through the various gratings located in the steam generator compartments. Grating blockage caused by insulation debris is considered negligible. Other major vent paths exist which are free of grating. In addition, grating free area (80%) is conservatively input as sixty percent of total grating area.

Conservative design for regions removed from a particular break location is assured by evaluating potential break locations closer to these regions.

The effects of the additional sludge lancing platforms in the steam generator compartment were evaluated. The addition of this grating affects the flow areas and resistance coefficients. The effects of the additional steel grating were determined to be negligible; thus, the original analysis remains valid.

MAIN STEAM AND FEEDWATER PENETRATION AREAS

Nodal boundaries are chosen at the most restrictive locations in order to maximize local peak pressure. A nodalization sensitivity study was not performed, since the node flow areas are very large in comparison to the areas available at the nodal boundaries.

Conservative design is assured for regions removed from a particular break location by considering additional break locations in these downstream volumes.

There are no safety-related components subjected to asymmetric pressurization in the containment penetration area. Maximum wall differential pressures are given in Table 6.2.1-92.

There are no safety-related components subjected to asymmetric pressurization in the containment penetration area. Maximum wall differential pressures are given in Table 6.2.1-78.

The analysis of the pressure versus time provides the basis for design evaluation of these structures.

PRESSURIZER COMPARTMENT

Node boundaries are chosen at the most restrictive flow areas to maximize local peak pressures. The pressurizer compartment spray line break analysis considers all vertical and horizontal flow obstacles.

The effects of the additional steel grating in the pressurizer compartment were analyzed. The effects of the additional steel grating were determined to be negligible; thus, the results of the original RELAP-4 analysis remain valid.

6.2.1.3 Mass and Energy Release Analyses for Postulated Loss-of-Coolant Accidents

The M&E release rates described in this section form the basis of further computations to evaluate the containment following the postulated accident. Discussed in this section are the long-term LOCA M&E releases for the hypothetical double-ended pump suction (DEPS) rupture with minimum safeguards and maximum safeguards and double-ended hot-leg (DEHL) rupture break cases. These LOCA cases are used for the long-term containment integrity analyses in Section 6.2.1.1.3.2.

6.2.1.3.1 Input Parameters and Assumptions

The M&E release analysis is sensitive to the assumed characteristics of various plant systems, in addition to other key modeling assumptions. Where appropriate, bounding inputs are utilized and instrumentation uncertainties are included. For example, the RCS operating temperatures are chosen to bound the highest average coolant temperature range of all operating cases and a temperature uncertainty allowance of +5.9°F is then added. Nominal parameters are used in certain instances. For example, the RCS pressure in this analysis is based on a nominal value of

2,250 psia plus an uncertainty allowance +30.0 psi. All input parameters are chosen consistent with accepted analysis methodology.

Some of the most critical items are the RCS initial conditions, core decay heat, safety injection flow, and primary and secondary metal mass and steam generator (SO) heat release modeling. Specific assumptions concerning each of these items are discussed in the following paragraphs. Tables 6.2.1-56 and 6.2.1-57 present key data assumed in the analysis of CPNPP Units 1 and 2. Only one analysis is performed because Unit 1 with the model Δ76 steam generators bounds Unit 2 with the model D5 pre-heat steam generators.

The core rated power of 3,612 MWt plus the adjustment for the plant specific calorimetric error of 0.6 percent was used in the analysis. As previously noted, RCS operating temperatures were used to bound the highest average coolant temperature range as bounding analysis conditions. The use of higher temperatures is conservative because the initial fluid energy is based on coolant temperatures that are at the maximum levels attained in steady-state operation. Additionally, an allowance to account for instrument error and deadband is reflected in the initial RCS temperatures. The selection of 2,250 psia plus uncertainty as the limiting pressure is considered to affect the blowdown phase results only, since this represents the initial pressure of the RCS. The RCS rapidly depressurizes from this value until the point at which it equilibrates with containment pressure.

The rate at which the RCS blows down is initially more severe at the higher RCS pressure. Additionally, the RCS has a higher fluid density at the higher pressure (assuming a constant temperature) and subsequently has a higher RCS mass available for releases. Thus, 2,250 psia plus uncertainty was selected for the initial pressure as the limiting case for the long-term M&E release calculations.

The selection of the fuel design features for the long-term M&E release calculation is based on the need to conservatively maximize the energy stored in the fuel at the beginning of the postulated accident (that is, to maximize the core stored energy). The core stored energy that was selected to bound the 17x17 fuel product that will be used at CPNPP Units 1 and 2 was 3.75 full-power seconds (FPS). The margins in the core stored energy include a statistical uncertainty to address the thermal fuel model and associated manufacturing uncertainties and the time in the fuel cycle for maximum fuel densification. Thus, the analysis very conservatively accounts for the stored energy in the core.

Margin in RCS volume of 3 percent (which is composed of 1.6 percent allowance for thermal expansion and 1.4 percent allowance for uncertainty) was modeled.

A uniform steam generator tube plugging (SGTP) level of 0 percent was modeled. This assumption maximizes the reactor coolant volume and fluid release by virtue of consideration of the RCS fluid in all SG tubes. During the post-blowdown period, the SGs are active heat sources since significant energy remains in the secondary metal and secondary mass that has the potential to be transferred to the primary side. The 0 percent tube plugging assumption maximizes the heat transfer area and, therefore, the transfer of secondary heat across the SG tubes. Additionally, this assumption reduces the reactor coolant loop resistance, which reduces the pressure drop (i.e., ΔP) upstream of the break for the pump suction breaks and increases break flow. Thus, the analysis conservatively accounts for the level of SGTP.

The secondary-to-primary heat transfer is maximized by assuming conservative heat transfer coefficients. This conservative energy transfer is ensured by maximizing the initial internal energy of the inventory in the SG secondary side. This internal energy is based on full-power operation plus uncertainties.

Regarding safety injection (SI) flow, the M&E release calculation considered configurations/ failures to conservatively bound respective alignments. The cases include a minimum safeguards case that represents a failure of one diesel generator with one centrifugal charging pump (CCP), one SI pump, and one residual heat removal (RH) pump; and a maximum safeguards case with two CCPs, two SI pumps, and two RH pumps. In addition, the containment backpressure is assumed to be equal to the containment design pressure during the early portion of the transient. This assumption was shown in WCAP-I0325-P-A (Reference 17) to be conservative for the generation of M&E releases.

In summary, the following assumptions were employed to ensure that the M&E releases are conservatively calculated, thereby maximizing energy release to containment:

- Maximum expected operating temperature of the RCS (100 percent full-power conditions)
- Allowance for RCS temperature uncertainty (+5.9°F)
- Margin in RCS volume of 3 percent (which is composed of 1.6 percent allowance for thermal expansion and 1.4 percent allowance for uncertainty)
- Core rated power of 3,612 MWt
- Allowance for calorimetric error (+0.6 percent of power)
- Conservative heat transfer coefficients (that is, SG primary/secondary heat transfer, and RCS metal heat transfer)
- Allowance in core stored energy for effect of fuel densification
- A margin in core stored energy (statistical uncertainty to account for manufacturing tolerances)
- An allowance for RCS initial pressure uncertainty (+30.0 psi)
- A maximum containment backpressure equal to design pressure (50.0 psig)
- SGTP leveling (0 percent uniform)
    - Maximizes reactor coolant volume and fluid release
    - Maximizes heat transfer area across the SG tubes
    - Reduces coolant loop resistance, which reduces the ΔP upstream of the break for the pump suction breaks and increases break flow

- Main feedwater addition is modeled. The feedwater control valve closure time is based on the time to reach the SI signal, plus electronic delay and valve stroke time.

- The SG initial fluid mass was calculated at full (100 percent), accounting for level and methodology uncertainties.

Thus, based on the previously discussed conditions and assumptions, an analysis of CPNPP Units1 and 2 was made for the release of M&E from the RCS in the event of a LOCA at 3,612 MWt plus the plant specific power uncertainty of 0.6 percent.

6.2.1.3.1.1 Decay Heat Model

ANS Standard 5.1 (Reference 23) was used in the LOCA M&E release model for CPNPP Units 1 and 2 for the determination of decay heat energy. This standard was balloted by the Nuclear Power Plant Standards Committee (NUPPSCO) in October 1978 and subsequently approved. The official standard (Reference 23) was issued in August 1979. Table 6.2.1-49 lists the decay heat curve used in the CPNPP Units 1 and 2 SPU Program M&E release analysis.

Significant assumptions in the generation of the decay heat curve for use in the LOCA M&E releases analysis include the following:

1. The decay heat sources considered are fission product decay and heavy element decay of U-239 and Np-239.

2. The decay heat power from fissioning isotopes other than U-235 is assumed to be identical to that of U-235.

3. The fission rate is constant over the operating history of maximum power level.

4. The factor accounting for neutron capture in fission products is taken from Table 10 of the ANS Standard 5.1 (Reference 23).

5. The fuel is assumed to be at full power for 10^8 seconds.

6. The number of atoms of U-239 produced per second is assumed to be equal to 70 percent of the fission rate.

7. The total recoverable energy associated with one fission is assumed to be200 MeV/ fission.

8. An uncertainty of two sigma (two times the standard deviation) has been applied to the fission product decay.

Based upon NRC staff review (Safety Evaluation Report (SER) of the March 1979 evaluation model [Reference 17]), use of the ANS Standard-5.1, November 1979 decay heat model was approved for the calculation of M&E releases to the containment following a LOCA.

6.2.1.3.1.2 Application of Single-Failure Criterion

An analysis of the effects of the single-failure criterion has been performed on the M&E release rates for each break analyzed. An inherent assumption in the generation of the M&E release is that offsite power is lost. This results in the actuation of the emergency diesel generators (EDG's), required to power the Safety Injection System (SIS). Actuation of the EDG to provide power delays the star of operation of the SIS, that is essential to mitigate the transient. Since none of the powered safety systems were assumed to be operational during the blowdown phase, the application of single-failure criteria is not an issue for the blowdown period, which is limited by the DEHL break.

Two cases have been analyzed to assess the effects of a single failure. The first case assumes minimum safeguards SI flow based on the postulated single failure of one diesel generator. This results in the loss of one of each ECCS pump type and a loss of some containment heat removal systems. The second case assumes maximum safeguards SI flow based on no postulated failures that would impact the amount of ECCS flow. The failure for this second case would be taken as the loss of one train of containment spray pumps. The analysis of the cases described provides confidence that the effect of credible single failures is bounded.

6.2.1.3.2 Acceptance Criteria for Analyses

A large-break LOCA (LBLOCA) accident is classified as an American Nuclear Society (ANS) Condition IV event, an infrequent fault. To satisfy the NRC acceptance criteria presented in the Standard Review Plan, Section 6.2.1.3 (Reference 24), the relevant requirements are the following:

- 10 CPR 50, Appendix A
- 10 CPR 50, Appendix K, paragraph I.A

To meet these requirements, the following must be addressed:

- Sources of energy
- Break size and location
- Calculation of each phase of the accident
- M&E release data

The bounding M&E releases and related analysis information for CPNPP Units 1 and 2 are presented in Section 6.2.1.3.4. The containment response is provided in Section 6.2.1.1.

6.2.1.3.3 Description of Analyses

The evaluation model used for the long-term LOCA M&E release calculations is the March 1979 model described in WCAP-10325-P-A (Reference 17). The NRC review and approval letters for the evaluation model can be found in References 17, 25, and 27.

This report section presents the limiting long-term LOCA M&E releases generated in support of the CPNPP Units 1 and 2 SPU Program. These M&E releases are then used in the containment integrity analysis presented in Section 6.2.1.1.

6.2.1.3.3.1 LOCA Mass and Energy Release Phases

The containment system receives M&E releases following a postulated rupture in the RCS. These releases continue over a time period, which, for the LOCA M&E analysis, is typically divided into four phases.

Blowdown - the period of time from accident initiation (when the reactor is at steady-state operation) to the time that the RCS and containment reach an equilibrium state.

Refill - the period of time when the lower plenum is being filled by accumulator and Emergency Core Cooling System (BCCS) water. At the end of blowdown, a large amount of water remains in the cold legs, downcomer, and lower plenum. To conservatively consider the refill period for containment M&E releases, it is assumed that this water is instantaneously transferred to the lower plenum along with sufficient accumulator water to completely fill the lower plenum. This allows an uninterrupted release of M&E to containment. Thus, the refill period is conservatively neglected in the M&E release calculation.

Reflood - begins when the water from the lower plenum enters the core and ends when the core is completely quenched.

Post-reflood (FROTH) - describes the period following the reflood phase. For the pump suction break, a two-phase mixture exits the core, passes through the hot legs, and is superheated in the SGs prior to exiting the break as steam. After the broken-loop SG cools, the break flow becomes two phase.

6.2.1.3.3.2 Computer Codes

The WCAP-10325-P-A (Reference 17) M&E release evaluation model comprises M&E release versions of the following codes: SATAN VI, WREFLOOD, FROTH, and EPITOME. These codes were used to calculate the long-term LOCA M&E releases for the CPNPP Units 1 and 2 SPU Program.

SATAN VI calculates blowdown, the first portion of the thermal-hydraulic transient following break initiation, including pressure, enthalpy, density, M&E flow rates, and energy transfer between primary and secondary systems as a function of time.

The WREFLOOD code addresses the portion of the LOCA transient when the core reflooding phase occurs after the primary coolant system has depressurized (blowdown) due to the loss of water through the break and when water supplied by the ECCS refills the reactor vessel and provides cooling to the core. The most important feature of WREFLOOD is the steam/water mixing model (see subsection 6.2.1.3.4.2).

FROTH models the post-reflood portion of the transient. The FROTH code is used for the SG heat addition calculation from the broken and intact-loop SGs.

EPITOME continues the FROTH post-reflood portion of the transient from the time at which the secondary equilibrates to containment design pressure to the end of the transient. It also compiles a summary of data on the entire transient, including formal instantaneous M&E release tables and M&E balance tables with data at critical times.

6.2.1.3.3.3 Break Size and Location

Generic studies (Reference 17) have been performed with respect to the effect of postulated break size on the LOCA M&E releases. The double-ended guillotine break has been found to be limiting due to larger mass flow rates during the blowdown phase of the transient. During the reflood and FROTH phases, the break size has little effect on the releases.

Three distinct locations in the RCS loop can be postulated for a pipe rupture for M&E release purposes:

- Hot leg (between vessel and SG)
- Cold leg (between pump and vessel)
- Pump suction (between SG and pump)

The break locations analyzed for this program are the double-ended pump suction (DEPS) rupture with a total break area of 10.48 ft^2 and the double-ended hot leg (DEHL) rupture with a total break area of 9.17 ft^2. Break M&E releases have been calculated for the blowdown, reflood, and post-reflood phases of the LOCA for the DEPS cases. For the DEHL case, the releases were calculated only for the blowdown. The following information provides a discussion on each break location.

The DEHL rupture has been shown in previous studies to result in the highest blowdown M&E release rates. Although the core flooding rate would be the highest for this break location, the amount of energy released from the SG secondary is minimal because the bulk of the fluid that exits the core vents directly to containment, bypassing the SGs. As a result, the reflood M&E releases are reduced significantly as compared to either the pump suction or cold leg break locations where the core-exit mixture must pass through the SGs before venting through the break. For the hot leg break, generic studies (Reference 17) have confirmed that there is no reflood peak that is, from the end of the blowdown period, the containment pressure would continually decrease. Therefore, only the M&E releases for the hot leg break blowdown phase are calculated and presented in this section of the report.

The cold leg break location has also been found in previous studies to be much less limiting in terms of the overall containment energy releases. The cold leg blowdown is faster than that of the pump suction break, and more mass is released into the containment. However, the core heat transfer is greatly reduced, and this results in a considerably lower energy release into containment. Studies (Reference 17) have determined that the blowdown transient for the cold leg is, in general, less limiting than that for the pump suction break. During reflood, the flooding rate is greatly reduced and the energy release rate into the containment is reduced. Therefore, the cold leg break is bounded by other breaks and no further evaluation is necessary.

The pump suction break combines the effects of the relatively high core flooding rate, as in the hot leg break, and the addition of the stored energy in the SGs. As a result, the pump suction break yields the highest energy flow rates during the post-blowdown period by including all the available RCS energy in calculating the releases to containment. Thus, only the DEHL and DEPS cases are used to analyze long-term LOCA containment integrity.

6.2.1.3.4 Mass and Energy Release Data

6.2.1.3.4.1 Blowdown Mass and Energy Release Data

The SATAN-VI code is used for computing the blowdown transient. The code utilizes the control volume (element) approach with the capability for modeling a large variety of thermal fluid system configurations. The fluid properties are considered uniform and thermodynamic equilibrium is assumed in each element. A point-kinetics model is used with weighted feedback effects. The major feedback effects include moderator density, moderator temperature, and Doppler broadening. A critical flow calculation for subcooled (modified Zaloudek), two-phase (Moody), or superheated break flow is incorporated into the analysis. The methodology for the use of this model is described in WCAP-10325-P-A (Reference 17).

Table 6.2.1-39 presents the calculated M&E release for the blowdown phase of the DEHL break for CPNPP Units 1 and 2. For the hot leg break M&E release tables, break path 1 refers to the M&E exiting from the reactor vessel side of the break. Break path 2 refers to the M&E exiting from the SG side of the break. Table 6.2.1-54 present the M&E balance data for the Units 1 and 2 DEHL case.

Table 6.2.1-3 presents the calculated M&E releases for the blowdown phase of the DEPS breaks. For the pump suction breaks, break path 1 in the M&E release tables refers to the M&E exiting from the SG side of the break. Break path 2 refers to the M&E exiting from the pump side of the break.

6.2.1.3.4.2 Reflood Mass and Energy Release Data

The WREFLOOD code is used for computing the reflood transient. The WREFLOOD code consists of two basic hydraulic models: one for the contents of the reactor vessel and one for the coolant loops. The two models are coupled through the interchange of the boundary conditions applied at the vessel outlet nozzles and at the top of the downcomer. Additional transient phenomena such as pumped SI and accumulators, reactor coolant pump (RCP) performance, and SG releases are included as auxiliary equations that interact with the basic models as required. The WRFLOOD code permits the capability to calculate variations during the core reflooding transient of basic parameters such as core flooding rate, core and downcomer water levels, fluid thermodynamic conditions (pressure, enthalpy, density) throughout the primary system, and mass flow rates through the primary system. The code permits hydraulic modeling of the two flow paths available for discharging steam and entrained water from the core to the break, that is, the path through the broken loop and the path through the unbroken loops.

A complete thermal equilibrium mixing condition for the steam and ECCS injection water during the reflood phase has been assumed for each loop receiving ECCS water. This is consistent with the usage and application of the WCAP-10325-P-A (Reference 17) M&E release evaluation model in recent analyses, for example, D.C. Cook Docket (Reference 25). Even though the WCAP-10325-P-A (Reference 17) model credits steam/water mixing only in the intact loop and

not in the broken loop, the justification, applicability, and NRC approval for using the mixing model in the broken loop has been documented (Reference 25). Moreover, this assumption is supported by test data and is further discussed below.

The model assumes a complete mixing condition (that is, thermal equilibrium) for the steam/water interaction. The complete mixing process, however, is made up of two distinct physical processes. The first is a two-phase interaction with condensation of steam by cold ECCS water. The second is a single-phase mixing of condensate and ECCS water. Since the steam release is the most important influence to the containment pressure transient, the steam condensation part of the mixing process is the only part that must be considered. (Any spillage directly heats only the sump.)

The most applicable steam/water mixing test data have been reviewed for validation of the containment integrity reflood steam/water mixing model. This data was generated in 1/3-scale tests (Reference 26), which are the largest scale data available and thus most clearly simulate the flow regimes and gravitational effects that would occur in a pressurized water reactor (PWR). These tests were designed specifically to study the steam/water interaction for PWR reflood conditions.

A group of 1/3-scale tests discussed in Reference 26 corresponds directly to containment integrity reflood conditions. The injection flow rates for this group cover all phases and mixing conditions calculated during the reflood transient. The data from these tests were reviewed and discussed in detail in WCAP-10325-P-A (Reference 17). For all of these tests, the data clearly indicate the occurrence of very effective mixing with rapid steam condensation. The mixing model used in the containment integrity reflood calculation is, therefore, wholly supported by the 1/3-scale steam/water mixing data.

Additionally, the following justification is also noted. The post-blowdown limiting break for the containment integrity peak pressure analysis is the pump suction double-ended rupture break. For this break, there are two flow paths available in the RCS by which M&E may be released to containment. One is through the outlet of the SG, the other via reverse flow through the RCP. Steam that is not condensed by ECCS injection in the intact RCS loops passes around the downcomer and through the broken loop cold leg and pump in venting to containment. This steam also encounters ECCS injection water as it passes through the broken loop cold leg, upon which complete mixing occurs and a portion of it is condensed. It is the portion of steam that is condensed that is taken credit for in this analysis. This assumption is justified based on the postulated break location, and the actual physical presence of the ECCS injection nozzle. Descriptions of the test and test results are contained in WCAP-10325-P-A (Reference 17) and operating license Amendment No. 126 for D.C. Cook (Reference 25).

Table 6.2.1-3A presents the calculated M&E releases for the reflood phase of the pump suction double-ended rupture with the minimum safeguards (i.e., the diesel failure). Table 6.2.1-41 presents the reflood phase M&E releases for CPNPP Units 1 and 2 for the maximum safeguards cases.

The principal parameters during reflood are given in Table 6.2.1-3B and Table 6.2.1-42.

6.2.1.3.4.3 Post-Reflood Mass and Energy Release Data

The FROTH code (Reference 7) is used for computing the post-reflood transient. The FROTH code calculates the heat release rates resulting from a two-phase mixture present in the SG tubes. The M&E releases that occur during this phase are typically superheated (Reference 27) due to the depressurization and equilibration of the broken loop and intact loop SGs. During this phase of the transient, the RCS has equilibrated with the containment pressure. However, the SGs contain a secondary inventory at an enthalpy that is much higher than the primary side. Therefore, there is a significant amount of reverse heat transfer that occurs. Steam is produced in the core due to core decay heat. For a pump suction break, a two-phase fluid exits the core, flows through the hot legs, and becomes superheated as it passes through the SG. Once the broken loop cools, the break flow becomes two phase. During the FROTH calculation, ECCS injection is addressed for both the injection phase and the recirculation phase. The FROTH code calculation stops when the secondary side equilibrates to the saturation temperature (T_{sat}) at the containment design pressure, after which the EPITOME code completes the SG depressurization.

The methodology for the use of this model is described in WCAP-10325-P-A (Reference 17). The M&E release rates are calculated by FROTH and EPITOME until the time of containment depressurization. After containment depressurization (14.7 psia), the M&E release available to containment is generated directly from core boil-off/decay heat.

Tables 6.2.1-4 and 6.2.1-43 present the two-phase post-reflood M&E release data for the pump suction double-ended break cases.

6.2.1.3.4.4 Steam Generator Equilibration and Depressurization

SG equilibration and depressurization is the process by which secondary-side energy is removed from the SGs in stages. The FROTH computer code calculates the heat removal from the secondary mass until the secondary temperature is the saturation temperature (T_{sat}) at the containment design pressure. After the FROTH calculations, the EPITOME code continues the FROTH calculation for SG cooldown removing SG secondary energy at different rates (that is, first- and second-stage rates). The first-stage rate is applied until the SG reaches T_{sat} at the user-specified intermediate equilibration pressure, when the secondary pressure is assumed to reach the actual containment pressure. Then the second-stage rate is used until the final depressurization, when the secondary reaches the reference temperature of T_{sat} at 14.7 psia, or 212°F. The heat removal of the broken-loop and intact-loop SGs is calculated separately.

During the FROTH calculations, SG heat removal rates are calculated using the secondary-side temperature, primary-side temperature, and a secondary-side heat transfer coefficient determined using a modified McAdam's correlation. SG energy is removed during the FROTH transient until the secondary-side temperature reaches T_{sat} at the containment design pressure. The constant heat removal rate used during the first heat removal stage is based on the final heat removal rate calculated by FROTH. The SG energy available to be released during the first-stage interval is determined by calculating the difference in secondary energy available at the containment design pressure and that at the (lower) user-specified intermediate equilibration pressure, assuming saturated conditions. This energy is then divided by the first-stage energy removal rate, resulting in an intermediate equilibration time. At this time, the rate of energy release drops substantially to the second-stage rate. The second-stage rate is determined as the

fraction of the difference in secondary energy available between the intermediate equilibration and final depressurization at 212°F, and the time difference from the time of the intermediate equilibration to the user-specified time of the final depressurization at 212°F. With the current methodology, all the secondary energy remaining after the intermediate equilibration is conservatively assumed to be released by imposing a mandatory cooldown and subsequent depressurization to atmospheric pressure at 3,600 seconds, that is, 14.7 psia and 212°F (the M&E balance tables have this point labeled as "Available Energy")

6.2.1.3.4.5 Long- Term Mass and Energy Releases

The long-term post-one hour mass and energy releases (boil-off from core at the decay heating rate) are performed through user defined input functions in the GOTHIC code (Reference 18). This method of determining the long-term mass and energy releases is consistent with past long-term mass and energy releases is consistent with past applications of Westinghouse methodology.

6.2.1.3.4.6 Sources of Mass and Energy

The sources of mass and energy considered in the LOCA M&E release analysis are given in Table 6.2.1-54 for the DEHL break for CPNPP Units 1 and 2. The sources of mass for the DEPS break cases for CPNPP Units 1 and 2 are given in Tables 6.2.1-50 and 6.2.1-51. These sources are the RCS, accumulators, and pumped SI.

The energy inventories considered in the DEPS break M&E release analysis for CPNPP Units 1 and 2 are also given in Tables 6.2.1-50 and 6.2.1-51. The energy sources are the following:

- RCS water
- Accumulator water (all inject)
- Pumped SI water
- Decay heat
- Core-stored energy
- RCS metal (includes SG tubes)
- SG metal (includes transition cone, shell, wrapper, and other internals)
- SG secondary energy (includes fluid mass and steam mass)
- Secondary transfer of energy (feedwater into and steam out of the SG secondary)

The analysis used the following energy reference points:

- Available energy :212°F: 14.7 psia (energy available that could be released)
- Total energy content: 32°F; 14.7 psia (total internal energy of the RCS)

The M&E inventories are presented at the following times, as appropriate:

- Time zero (initial conditions)
- End of blowdown time
- End of refill time
- End of reflood time
- Time of broken-loop SG equilibration to pressure setpoint
- Time of intact-loop SG equilibration to pressure setpoint
- Time of full depressurization (3,600 seconds)

The energy release from the metal-water reaction rate is considered as part of the WCAP-10325-P-A (Reference 17) methodology. Based on the way that the energy in the fuel is conservatively released to the vessel fluid, the fuel cladding temperature does not increase to the point where the metal-water reaction is significant. For the LOCA M&E release calculation, the energy created by the metal-water reaction value is small and is not explicitly provided in the energy balance tables. The energy that is determined is part of the M&E releases and is therefore already included in the overall M&E releases for the CPNPP units.

The sequence of events for the CPNPP Units 1 and 2 LOCA design basis accident (DEPS MINSI) is shown in Table 6.2.1-9. The sequence of events tables for the DEHL and DEPS MAXSI break cases can be found in Table 6.2.1-11 and Table 6.2.1-12 respectively.

6.2.1.4 Mass and Energy Release Analysis for Postulated Secondary System Pipe Ruptures

6.2.1.4.1 Plant Power Level

Steam line breaks can be postulated to occur with the plant in any operating condition ranging from hot shutdown to full power. Since steam generator mass decreases with increasing power level, breaks occurring at lower power generally result in a greater total mass release to the plant Containment. However, because of increased energy storage in the primary plant, increased heat transfer in the steam generators, and the additional energy generation in the nuclear fuel, the energy release to the Containment from breaks postulated to occur during power operation may be greater than for breaks occurring with the plant in a hot shutdown condition. Additionally, steam pressure and the dynamic conditions in the steam generators change with increasing power and have significant influence on both the rate of blowdown and the amount of moisture entrained in the fluid leaving the break following a steam line break event.

Because of the opposing effects of changing power level on steam line break mass and energy releases, no single power level can be singled out as a worst case initial condition for a steam line break. Therefore, a spectrum of power levels spanning the operating range (100.6 percent, 70 percent, and 30 percent of 3628 MWt), as well as the hot shutdown condition, has been considered. See Table 6.2.1-2A for the types of steam line breaks and power levels analyzed.

6.2.1.4.2 Break Type, Area, and Location

1. Break Type

There are two possible types of pipe ruptures which must be considered in evaluating steam line breaks.

The first is a split rupture in which a hole opens at some point on the side of the steam pipe or steam header but does not result in a complete severance of the pipe. A single, distinct break area is fed uniformly by all steam generators until steam line isolation occurs. The blowdown from the individual steam generator is not independent since fluid coupling exists among all steam lines.

The second break type is the double-ended guillotine rupture in which the steam pipe is completely severed and the ends of the break displace from each other. The effective break area is limited by the cross-sectional area of the flow restrictor that is integral to the steam generator. The reverse break area is the cross-sectional area of the pipe.

2. Break Area

The breaks analyzed include two break types (full double-ended and split rupture) at all four initial power levels as follows:

a. A full double-ended pipe rupture downstream of the steam line flow restrictor. For this case, the actual break area equals the cross-sectional area of the steam line, but the blowdown from the steam generator with the broken line is controlled by the flow restrictor throat area (1.4 ft^2). The reverse flow from the intact steam generators is controlled by the cross-sectional area of the pipe.

b. A split break that represents the largest break which will neither generate a steam line isolation signal from the primary protection equipment nor result in moisture entrainment. Steam and feedwater line isolation signals will be generated by high containment pressure signals for these cases.

3. Break Location

Break location affects steam line mass and energy releases by virtue of the pressure losses which would occur in the length of piping between the steam generator and the break. The effect of the pressure loss is to reduce the effective break area seen by the steam generator.

Although this would reduce the rate of depressurization, it would not significantly change the total release of energy to the Containment. Therefore, piping loss effects have been conservatively ignored in all blowdown results.

6.2.1.4.3 Main Feedwater Addition Prior to Feedwater Line Isolation

All of the double-ended ruptures generate main steam and feedwater isolation signals very quickly following the break. Isolation of these lines is assumed to be complete following a time

delay sufficiently long to conservatively allow for the instrument response time and signal processing delay and valve closing time (7 sec for feedwater line isolation and 7 sec for main steam line isolation).

For the split ruptures, feedwater line isolation signals (on the Safety Injection Actuation Signal) and main steam line isolation signals result from high Containment pressure protective trips. The setpoints for feedwater line and steam line isolation signals are assumed to be 5 psig and 8 psig, respectively. The isolation is assumed to be complete after a time delay sufficiently long to conservatively allow for the isolation as described above.

Prior to complete isolation, the depressurization of the steam generator results in significant amounts of feedwater being added to the broken loop steam generator through the main feedwater system. The quantity of feedwater added is conservatively evaluated using the following assumptions:

1. The two main feedwater pumps are conservatively operating at 5740 rpm (maximum turbine speed).

2. Prior to receipt of a valve isolation signal, all main feedwater control valves maintain the position that exists at "valves wide open" power operation, i.e., the valve feeding the faulted steam generator is fully open.

3. The main feedwater control valves in the intact loops maintain their initial flow until a valve isolation signal is received.

4. Immediate closure of the feedwater isolation valves and control valves in the intact loops upon receipt of the isolation signal. The control valves operate as a backup for the isolation valves. Control valves were originally procured to the requirements of ASME Code, Section III, Class 3; however, they are not required to be safety grade.

5. No flow reduction through the feedwater isolation valve and control valve in the broken loop prior to complete closure (i.e., no credit is taken for decrease in flow during valve closure or pump trip and coastdown).

6. For double-ended ruptures, the faulted loop main feedwater flow is modeled as a function of steam generator pressure, while the intact loops are kept constant at an initial flow. For split ruptures, the analysis conservatively models the main feedwater flow equal to the steam flow.

These assumptions were used along with the Feedwater System hydraulic resistances and pump performance curves to determine the amount of feedwater added to the steam generator with the broken loop over the range of steam generator pressures projected prior to feedwater isolation for each power level evaluated.

Based on the above assumptions (specifically, assumption number 3), the check valve in the feedwater line is not considered in a closed position. The analysis assumptions and results are consistent with those presented in Reference [6].

6.2.1.4.4 Auxiliary Feedwater System Design

Generally within the first minute following a steam line break, the Auxiliary Feedwater System is initiated on any one of several protection system signals. Addition of auxiliary feedwater to the steam generators increases the secondary mass available for release to the Containment as well as increasing the heat transferred to the secondary fluid.

The effects on steam generator mass are maximized in the calculation by assuming full auxiliary feedwater flow to the faulted steam generator starting at the time the ESF setpoint is reached and continuing until manually stopped by the plant operator. The time for manual action is assumed to be 600 sec after the break. The information available to the operator to manually isolate the auxiliary feedwater line to the faulted steam generator is described in Section 10.4.9.

The Auxiliary Feedwater System is equipped with flow restrictors which would limit the flow to a maximum of 1380 gpm to the steam generator with the broken line.

The mass and energy release data presented are conservatively based on an auxiliary feedwater addition of 1380 gpm to the steam generator with the broken line from the time of the protection system signal until 600 seconds after the break initiation.

6.2.1.4.5 Fluid Stored in the Feedwater Piping Prior to Isolation

The blowdown data were determined assuming a large value of 547 ft^3 of unisolated volume in the feedline supplying the steam generator with the broken line.

6.2.1.4.6 Fluid Stored in the Steam Piping Prior to Isolation

All the steam in the steam lines up to the turbine stop valve is assumed to be released to the Containment following the break. The volume of the initial steam released to the containment is 10,000 ft^3.

6.2.1.4.7 Availability of Offsite Power

Loss of offsite power following a steam line rupture would result in tripping of the reactor coolant pumps and the steam-driven main feedwater pumps, and a possible delay of auxiliary feedwater initiation due to emergency or diesel generator starting delays. Each of these occurrences aids in mitigating the effects of the steam line break releases by either reducing the fluid inventory available to feed the blowdown or reducing the energy transferred from the primary coolant system to the steam generators. Thus, blowdowns occurring in conjunction with a loss of station power are less severe than cases where offsite power is available; these cases are not presented.

6.2.1.4.8 Safety System Failures

1. Failure of Main Feedwater Line Isolation Valve

   One feedwater isolation valve and one feedwater control valve are located in series in each feedwater line. In this analysis, the assumed valve closure time is the design maximum main feedwater isolation time of 7 seconds.

Failure of one valve following a steam line break would increase the unisolable feedwater line by the volume between the two valves. This effect is included in the mass and energy release data presented.

No further consideration need to be given to this type of failure.

2. Failure of the Auxiliary Feedwater Pump Runout Protection

Limitation of maximum auxiliary feedwater flow to the broken loop steam generator is achieved through passive flow restrictors. Failure of passive flow restrictors is not considered credible.

3. Failure of Main Feedwater Pump Trip

No credit is taken for feedwater pump trip and coastdown in calculating main feedwater addition prior to feedwater line isolation. Therefore, this failure has no effect on the results presented.

4. Failure of a Steam Line Stop Valve

Failure of a main steam line stop valve increases the volume of steam piping which empties into the Containment.

The effects of this failure on calculated Containment pressures and temperatures were compared with the effects of the failure of one Containment spray train. With respect to the maximum Containment pressure, calculations showed that the adverse effect of a steam line stop valve failure was more than that of one Containment spray train failure for the full power cases and less than that of one containment spray train failure for the 70 percent, 30 percent and hot zero power cases. With respect to the maximum Containment temperature, the adverse effect of a steam line stop valve failure was greater than that of one containment spray train failure.

5. Failure of One Containment Spray Train

The failure of one of the two redundant Containment spray trains following a main steam line break results in reduced ability to remove energy from the Containment atmosphere.

A time delay following a Containment pressure signal is necessary to incorporate delays for the instrumentation signal response, diesel generator start, pump load sequencing, pump acceleration, Containment isolation valve opening and system fill time.

The greater timing of containment spray setpoints and delays are modeled:

- 74.3 seconds after the containment reaches the Containment S signal setpoint (5 psig), or

- 52.5 seconds after the containment reaches the Containment P signal setpoint (20 psig)

Full flow conditions are achieved for both after the delay time is assumed.

6.2.1.5 Minimum Containment Pressure Analysis for Performance Capability Studies of Emergency Core Cooling System

The Containment backpressure is calculated using the methods and assumptions discussed in Section 15.6.5.3. Input parameters including the Containment initial conditions, net free Containment volume, passive heat sink materials, thicknesses, and surface areas and starting time and number of Containment cooling systems used in the analysis are described in the following paragraphs.

6.2.1.5.1 Mass and Energy Release Data

The mathematical models which calculate the mass and energy releases to the Containment are described in Reference 9. These methods are similar to the models used to perform the original analysis of the large break LOCA to meet the requirements of Appendix K. Since the requirements of Appendix K of 10 CFR Part 50 are very specific in regard to the modeling of the RCS during blowdown and the models used are in conformance with Appendix K, no alterations to those models have been made in regard to the mass and energy releases. A break spectrum analysis is performed that considers various break sizes, break locations, and Moody discharge coefficients for the DECLGs which do affect the mass and energy released to the Containment. This effect is considered for each case analyzed.

6.2.1.5.2 Initial Containment Internal Conditions

The initial values used in the analysis are provided in Section 15.6.5.

The Containment initial conditions are representatively low values anticipated during normal full power operation. In addition, the RWST temperature and outside air temperature assume minimum operational values.

6.2.1.5.3 Containment Volume

The net free volume is calculated by subtracting the volume of internal structures and equipment from the gross volume of the Containment.

Only the following are considered in the determination of the volume of internal structures and equipment:

1. Volume of internal concrete walls and floors

2. Volume of the major components of the RCS

The volumes of internal steel structures and supports, cranes, tanks, and other miscellaneous equipment are neglected.

This results in a conservatively high Containment net free volume of 3.063×10^6 ft^3.

6.2.1.5.4 Active Heat Sinks

The CSS operates to remove heat from the Containment.

CSS data has been determined as follows:

1. The maximum flow rate is based on maximum water level in the Refueling Water Storage Tank (RWST) and minimum Containment pressure necessary for Containment spray actuation. Both of these conditions concurrently will produce minimum Containment spray pump RDH during injection and consequent maximum flow.

   This is conservative because the actual RWST level will be falling and the actual Containment pressure will be higher.

2. The minimum time for post LOCA initiation of Containment Spray System operation with a loss of offsite power is based on the time required for diesel generator starting and loading of the spray pumps plus the minimum time in which the spray pumps can fill the spray headers up to the first spray nozzle. This is conservative since the spray pump flow will actually be limited during pump startup to a lower rate to protect the pumps from a runout trip.

The sump temperature was not used in the analysis because the maximum peak cladding temperature occurs prior to initiation of the recirculation phase for CSS.

6.2.1.5.5 Steam Water Mixing

Water spillage rates from the broken loop accumulator are determined as part of the core reflooding calculation and are included in the Containment calculation model.

6.2.1.5.6 Passive Heat Sinks

The surface areas and thicknesses of internal concrete walls and floors were calculated using detailed structural drawings.

The surface areas were increased to include an estimated error in the calculations.

The concrete floor thicknesses were increased by maximum thickness of floor finish.

The surface areas and thicknesses of mechanical equipment were calculated using detailed manufacturer's drawings. The surface areas and thickness of piping and HVAC ducts were calculated using detailed mechanical drawings of actual layout.

Surface areas and thicknesses were calculated to be conservatively high.

The resistance to heat absorption caused by layers of protective coating on heat sinks is conservatively neglected. Similarly, any potential airgap between the steel liner and concrete is also ignored.

The thermophysical properties of steel and concrete are in conformance with Reference 4.

6.2.1.5.7 Heat Transfer to Passive Heat Sinks

The condensing heat transfer coefficients used for heat transfer to the steel Containment structures are included in the Containment calculation model.

6.2.1.5.8 Other Parameters

No other parameters have a substantial effect on the minimum Containment pressure analysis.

6.2.1.6 Testing and Inspection

Preoperational and periodic inservice tests are conducted to ensure the functional capability of the Containment and associated systems. The tests are discussed in the following Sections:

Section	Test
5.4.7	Residual heat removal
6.2.2	Containment spray and sump tests
6.2.6	Containment leakage test including integrated leakage rate test, penetration leakage rate test, and isolation value leakage rate test

6.2.1.7 Instrumentation Requirements

The reactor coolant pressure boundary (RCPB) leakage during normal plant operation is detected by Containment air particulate monitors, radioactive gas monitors, Containment sump flow monitors, Containment dewpoint monitors, Containment cooling coil condensate measurement and other means as discussed in Section 5.2.5. If the level of leakage is excessive, Control Room alarms are sounded, and ventilation air ducts are automatically or manually closed in order to isolate the Containment atmosphere from the environment.

In addition, instrumentation which monitors process parameters is used to actuate ESF systems. These parameters are pressurizer pressure, Containment pressure, steam line pressure, steam line differential pressures, steam flows, and RCS average temperature. When the values of these parameters fall outside of predetermined setpoints, the actuating signals are generated. The design details and instrumentation logic are presented in Sections 7.2 and 7.3.

The postaccident monitoring is discussed in Section 7.5.

REFERENCES

1. Hargroves, D. W., Metcalfe, L. J., "CONTEMPT-LT/028 A Computer Program for Predicting Containment Pressure-Temperature Response to Loss-of-Coolant Accident", NUREG/CR-02555, March 1979.

2. McAdams, W. H., 1956. Heat Transmission, Third Edition.

3. Slaughterback, D. C., 1970. A Review of Heat Transfer Coefficients for Condensing Steam in a Containment Building Following a Loss-of-Coolant Accident. Interim Task Report, Subtask 4.2.2.1, Idaho Nuclear Corp.

4. NRC Branch Technical Position CSB6-1, Minimum Containment Pressure Model for PWR ECCS Performance Evaluation, NUREG 75/087, 11/75.

5. Wheat, L.L., Wagner, R.J., Niederauer, G.F., and Obenchain, C.F., CONTEMPT LT: A computer program for predicting Containment Pressure Temperature Response to a Loss-of-Coolant Accident, ANCR 1219, Aerojet Nuclear Co., June 1975.

6. Land, RE., "Mass and Energy Releases Following a Steam Line Rupture," WCAP-8822 (Proprietary) and WCAP-8860 (Nonproprietary), September 1976; Osborne, M.P., and Love, D.S., "Supplement 1 - Calculations of Steam Superheat in Mass/Energy Releases Following a Steam Line Rupture," WCAP-8822-S1-P-A Proprietary) and WCAP-8860-S1-A (Nonproprietary), September 1986; Butler, J.C., and Linn, P.A., "Supplement 2 - Impact of Steam Superheat in Mass/Energy Releases Following a Steam Line Rupture for Dry and Subatmospheric Containment Designs," WCAP-8822-S2-P-A (Proprietary) and WCAP-8860-S2-A Nonproprietary), September 1986.

7. Shepard, R.M., Massie, H.W., Mark, R.H., and Docherty, P.J., Westinghouse Mass and Energy Release Data For Containment Design, WCAP-8264-P-A, June 1975 (Proprietary) and WCAP-8312-A, Revision 2, August 1975 (Nonproprietary).

8. Bordelon, F.M., Massie, H.W., and Zordan, T.A., Westinghouse Emergency Core Cooling System Evaluation Model - Summary, WCAP- 8339, June 1974.

9. USAEC, Division of Reactor Licensing 1970. Safety Evaluation Report for Virginia Electric and Power Company. North Anna Power Station Units 1 and 2. Docket 50-338 and 50-339.

10. USAEC, Division of Reactor Licensing 1972b. Safety Evaluation Report for Virginia Electric Power Company, Surry Power Station Units 1 and 2. Docket 50-280 and 50-281.

11. USAEC, Division of Reactor Licensing 1972c. Safety Evaluation Report for Maine Yankee Atomic Power Station. Docket 50-309.

12. USNRC Office of Nuclear Reactor Regulations; NUREG 1057; Safety Evaluation Report Related to the Operations of Beaver Valley Power Station, Unit 2, Docket No. 50-412; Duquesne Light Company, et. al.; October 1985; Section 6.2.1.1, pp. 6-4 to 6-6.

13. USAEC, Directorate of Licensing 1974c. Safety Evaluation Report for the Dequesne Light Company, Toledo Edison Company, Pennsylvania Power Company, Beaver Valley Power Station Unit 1. Docket 50-334.

14. USNRC Office of Nuclear Reactor Regulation; NUREG 1031, Safety Evaluation Report Related to the Operation of Millstone Nuclear Power Station, Unit No. 3; Docket No. 50-423; Northeast Nuclear Energy Company; Section 6.2.1.1, pp. 6-4 to 6-6.

15. Not Used.

16. Aerojet Nuclear Company (ANC) 1976. RELAP4/MOD 5: A Computer Program for Transient Thermal Hydraulic Analysis of Nuclear Reactors and Related Systems. User's Manual Vol. I-III, Report ANCR-NUREG-1335.

17. Westinghouse LOCA Mass and Energy Release Model for Containment Design March 1979 Version, WCAP-10325-P-A, May 1983 (Proprietary) and WCAP-10326-A, May 1983 (Non-proprietary).

18. NAI 8907-06, Rev. 16, “GOTHIC Containment Analysis Package Technical Manual,” Version 7.2a, January 2006.

19. NAI-8907-09, Rev 9, “GOTHIC Containment Analysis Package Qualification Report,” Version 7.2a, January 2006.

20. Docket No. 50-244, Safety Evaluation by the Office of Nuclear Reactor Regulation Related to Amendment No. 97 to Renewed Facility Operating License No. DPR-18 R E. Ginna Nuclear Power Plant Inc., R E. Ginna Nuclear Power Plant, Nuclear Regulatory Commission, July 11, 2006.

21. USNRC Office of Nuclear Reactor Regulation, NUREG-0138, "Staff Discussion of Fifteen Technical Issues Listed in Attachment to November 3, 1976 Memorandum from Director, NRR to NRC Staff", November 1976.

22. NAI 8907-02, Rev. 17, "GOTHIC Containment Analysis Package User Manual," Version 7.2a, January 2006.

23. ANSI/ANS-5.1 1979, "American National Standard for Decay Heat Power in Light Water Reactors," August 29, 1979.

24. U.S Nuclear Regulatory Commission, Office of Nuclear Reactor Regulation, NUREG-0800, Standard Review Plan for the Review of Safety Analysis Reports for Nuclear Power Plants, LWR Edition, Section 6.2.1, "Containment Function Design," Revision 2, July 1981.

25. Docket No. 50-315, Amendment No. 126, Facility Operating License No. DPR-58 (TAC No. 71062), for D. C. Cook Nuclear Plant Unit 1, June 9, 1989.

26. WCAP-8423, EPRI 294-2, Mixing of Emergency Core Cooling Water with Steam; 1/3-Scale Test and Summary, Final Report, June 1975.

27. Letter from Herbert N. Berkow, Director (NRC) to James A. Gresham (Westinghouse), "Acceptance of Clarifications of Topical Report WCAP-10325-P-A, 'Westinghouse LOCA Mass and Energy Release Model for Containment Design - March 1979 Version' (TAC No. MC7980)," October 18, 2005.

28. Westinghouse Letter WPT-16922, “Evaluation of Short Term Pressurizer Spray Line Break Mass and Energy Releases,” April 9, 2007.

29. Westinghouse Letter WPT-16890, “License Report Section NSSS Parameters,” February 16, 2007.

6.2.2 CONTAINMENT HEAT REMOVAL SYSTEMS

The Containment Spray System (CSS) is designed to remove heat from the Containment environment following a LOCA, a main steam line break accident, or a feedwater line break accident. Major components of the CSS are the RWST, Containment spray pumps, Containment spray heat exchangers, spray headers, spray nozzles, and Containment recirculation sumps.

Each unit of the CPNPP is equipped with two redundant Containment spray trains, each designed to provide emergency Containment heat removal in the event of a LOCA. This system, in conjunction with the ECCS, removes postaccident thermal energy from the Containment environment, thereby reducing the Containment pressure and temperature.

6.2.2.1 Design Bases

The CSS is a nuclear-safety-related system and is classified as seismic Category I. The system is designed in accordance with NRC Regulatory Guides 1.1, 1.26, 1.29, and 1.82 and in accordance with GDC 38, 39, 40, 41, 50, and 56 of Appendix A of 10 CFR Part 50.

The CSS operation is divided into two phases, initially operating in the injection phase and then the recirculation phase. Two redundant and physically separated spray trains are provided for the CSS. Each train draws spray water from a common RWST during the injection phase and from separate recirculation sumps during the recirculation phase. The failure of any single active component during the injection or recirculation phase or failure of a single passive component during the recirculation phase does not prevent safe operation of the system.

Each train is designed to deliver a minimum of 5800 gpm of cold water (≤120°F) to the spray nozzles during the injection phase of system operation. This is sufficient to maintain the Containment pressure and temperature below Containment design values. The postaccident Containment pressure is reduced to half its value in 24 hours. The sources and amount of energy taken into consideration are described in Subsection 6.2.1.

Each Containment spray train is designed with a heat removal capacity during the recirculation phase sufficient to maintain the pressure and temperature in the Containment below design values for a long period of time until the decay heat generated by the reactor fuel is reduced to the level at which the heat can be removed from the Containment by other systems or by natural heat transfer.

Each Containment spray train is provided with a chemical additive subsystem which is designed to remove fission products from the Containment atmosphere. The chemical additive subsystem is described in Section 6.5.

To ensure proper performance during accident conditions, the CSS equipment and components are designed for the environmental conditions discussed in Section 3.11B. Additional equipment design criteria are given in Table 6.2.2-1.

The CSS has been designed in accordance with the requirements of the ASME B&PV Code, Section III, and NRC Regulatory Guide 1.29 to ensure the capability of the system to withstand the Safe Shutdown Earthquake (SSE) without loss of its safety function. Dynamic effects resulting from the fluid transient of the CSS operation have been evaluated at CPNPP.

Pipe whip restraints are located on adjacent high-energy lines to prevent a broken pipe from whipping and damaging components unacceptably in the CSS as described in Sections 3.5 and 3.6.

The system is designed to allow periodic determination of proper functioning to demonstrate system readiness. Routine testing is performed periodically to verify the operability of system components.

6.2.2.2 System Design

The CSS, which is shown on Figure 6.2.2-1, is an ESF specifically provided to mitigate the consequences of a LOCA, a steam line break, or a feedwater line break inside the Containment. The CSS has the dual function of removing heat and fission products, especially radioactive iodine, from the postaccident Containment atmosphere. Acting in conjunction with the ECCS, which maintains cooling of the reactor core after a LOCA, the CSS ensures that a LOCA does not jeopardize Containment functional design. To reduce transient pressure and temperature and to remove the fission products in the Containment following an accident, water is sprayed into the Containment atmosphere through a large number of spray nozzles mounted on ring headers which are located in the Containment dome and on headers in lower areas of the Containment. Because of the large surface area of the solution drops sprayed into the Containment atmosphere, large amounts of heat and elemental iodine are absorbed into the spray drops as they fall through the Containment atmosphere to the sump. The spray water flow path is as follows: water flows from the spray nozzles, falls through the Containment atmosphere, washes the Containment walls, washes down the steam generators, reactor coolant pumps and piping, and floors, and thereby reaches the Containment sump suction lines.

The CSS has two modes of operation, the injection phase and the recirculation phase.

Initiation of the injection phase begins after a LOCA, when the Containment spray pumps start on an S signal. The Containment spray header discharge valves open on a P signal. A detailed description of the generation of S and P signals is provided in Section 7.3, Engineered Safety Features Actuation System. Set points for the ESF Actuation System are given in unit Technical Specifications. These set points have been established to provide sufficient time for initiation of CSS operation and performance of its safety function.

The pumps take suction from the RWST and discharge the solution inside the Containment. To enhance iodine removal and ensure retention of the iodine and to adjust the pH value of the spray solution to inhibit chloride stress corrosion of stainless steel components, a chemical additive subsystem is provided to inject sodium hydroxide into the spray water during this phase (Section 6.5.2.). The sequence of events which occur prior to initiation of Containment spray, as well as the timing involved, is summarized in Table 6.2.1-9.

After ECCS Switchover (Section 6.3.2.8) is complete, the suction of the Containment spray pumps is shifted by the operator to the Containment recirculation sumps where injection water, reactor coolant, and accumulator spillage have collected. This water is sprayed into the Containment again after having given up its heat to the component cooling water via the Containment spray heat exchangers.

Makeup to the RWST is obtained from the Reactor Makeup Water Storage Tank (RMWST) by manual operation using the reactor makeup water transfer pumps and the CVCS blender to maintain required boron concentration in the RWST.

Using the demineralized water transfer pumps, makeup to the spray headers inside the Containment is obtained from the demineralized water storage tank by manual operation. The makeup connection is provided above the operating floor at elevation 905 ft 9 inches. A level switch is provided to monitor the level in the spray header riser during plant operation.

Makeup to the containment spray pump suctions piping located downstream of the containment isolation valves is also obtained from the demineralized water storage tank by manual operation using the demineralized water transfer pumps, a local drain valve and hose connection.

Redundancy requirements of the CSS are satisfied by providing duplicate components of 100 percent capacity. The total heat removal function can be accomplished with the use of two pumps, one heat exchanger, and one train of nozzles. Each train is supplied by one suction line from a Containment spray recirculation sump. With the exception of the RWST and the chemical additive subsystem, both Containment spray trains are completely separated for the heat removal function. For more information on the CSS design parameters and the Containment design parameters, refer to Tables 6.2.2-2 and 6.2.2-3, respectively.

6.2.2.2.1 Component Description

Additional equipment design criteria is given in Table 6.2.2-1. A detailed description of environmental qualification verifications performed on CSS components is given in Section 3.11. For a detailed description of the chemical additive subsystem, spray headers, and nozzles, refer to Section 6.5.2. Table 17A-1 (System 7) lists the safety class, applicable standard or code, code class, seismic category and quality assurance for the CSS components. The CSS components are represented in FSAR Figure 6.2.2-1.

1. Containment Spray Heat Exchangers

   Two Containment spray heat exchangers are provided for each unit. The heat exchangers are of the vertical shell and U-tube type. They are designed in accordance with the requirements of the ASME B&PV Code, Section III, and in accordance with TEMA Class R for those items not covered by the ASME B&PV Code Section III. The tube side of each heat exchanger is classified ASME B&PV Code Class 2, and the shell side is ASME B&PV Code Class 3. The heat exchangers are designed to seismic Category I requirements.

2. Containment Spray Pumps

   Four Containment spray pumps are provided for each unit and are of the horizontal, double-suction, split-case, centrifugal type. The pumps are designed to the requirements of the ASME B&PV Code, Section III, Class 2, and are designed to seismic Category I requirements.

   Each pump is capable of delivering 3000 gpm with a total dynamic head (TDH) of 585 ft, which includes the sum of the Containment design pressure (50 psig), the nozzles elevation head, the nozzle pressure drop (40 psi), the line losses, and a 5-percent

margin. The pump design temperature and pressure are 300°F and 325 psig, respectively.

Each pump is driven by a three-phase, 60-Hz, 6600-V, 700-hp motor.

3. Valve Isolation Tanks

The Containment recirculation sump isolation valves are enclosed in valve isolation tanks. These tanks are not part of the Containment barrier. Therefore, the tanks are not tested to Containment design conditions.

4. Refueling Water Storage Tank

One RWST is provided for each unit. This tank, located outdoors, serves as a source of emergency borated cooling water during injection and as the source of refueling water during refueling operations. The tank consists of a concrete structure lined with stainless steel plate and is designed to seismic Category I requirements. It is designed to withstand tornado loadings and tornado generated missiles. Freeze protection is provided by operating a Containment spray pump in a recirculation mode through the test line, which returns to the tank, to maintain the tank temperature above 40°F. If necessary additional heat can be provided, by operating a residual heat removal (RHR) pump in addition to the Containment spray pump.

5. Spray Nozzles and Ring Headers

The nozzles are of one-piece construction, with a 3/8-in. diameter orifice and produce a hollow cone spray pattern.

The spray nozzles are installed on ring headers in the Containment dome and spray headers at lower elevations in the Containment. The nozzle arrangement is designed to provide maximum coverage inside the Containment. Refer to Section 6.5.2 for a detailed description of the spray header location and spray coverage. The spray header piping is designed in accordance with the ASME B&PV Code, Section III, Class 2, and the requirements of seismic Category I. The spray nozzles are designed in accordance with manufacturer's standards which include applicable requirements of the ASME B&PV Code, Section III, Class 2.

6. Piping and Valves

The safety related piping and valves of the CSS are designed in accordance with the ASME B&PV Code, Section III, and seismic Category I requirements. They have the following characteristics:

a. The pump discharge piping and valves for the CSS are designed to 325 psig and 300°F. The pump suction lines from the recirculation sumps and the RWST are designed to 70 psig and 300°F.

b. The motor-operated isolation valves in the recirculation lines are totally enclosed in valve isolation tanks. The recirculation lines that run from the sumps to the Containment spray pumps are protected by concentric guard piping up to the

valve isolation tanks. These tanks are used to control leakage and are not tested at Containment design conditions.

c. All piping and valves used for the CSS are stainless steel.

7. Containment Spray Recirculation Sumps

The design of the Containment spray recirculation sumps satisfies the requirements of NRC Regulatory Guide 1.82, Sumps for Emergency Core Cooling and Containment Spray Systems as described in Appendix 1A(B). Two sumps are provided, one for each safety train. The sumps are physically separated and are located at the lowest elevation of the Containment Building exclusive of the reactor vessel cavity. Sump covers are provided to protect the sumps against falling debris. Stainless steel strainers are provided to preclude clogging of the recirculation lines and any of the system's components. The strainers have nominal 0.095 inch holes in perforated plate. The size of the opening ensures that the 3/8-in. diameter spray nozzle orifices and the grid assemblies in the reactor core do not clog. It is required that suction piping to the containment recirculation pumps be arranged such that vortexing does not occur (i.e., no vertical pipe with downward flow or horizontal pipe with inadequate submergence). To prevent the possibility of vortex formation, the suction is fitted with a conical opening and a grating cage (Figure 6.2.2-3). The sump strainer design provides natural vortex suppression in addition to the suction inlet design. The arrangement of the recirculation sumps, including design provisions for the prevention of vortex formation in the recirculation sump piping of RHR and containment spray lines, is shown in Figure 6.2.2-3 and 6.2.2-3A.

ECCS fluid collecting in the reactor refueling cavity is returned to the containment sump through the drain lines provided in the design of the refueling cavity. To prevent blockage in the main refueling cavity, debris strainers are provided for the 4 inch drain and debris screens are provided for the two 6 inch drains. The part plan and elevation of the reactor refueling cavity at elevation 831'-6" are shown in Figure 6.2.2-4 and Figure 6.2.2-5. The refueling cavity consists of an upper internals storage stand area (floor elevation 831'-0"), a central portion (floor elevation 834'-0 1/2") and the lower internals storage stand area (floor elevation 823'-0 1/2"). The upper internals storage stand area and the lower internal storage stand area communicate hydraulically through a 4 inch equalization line. This line is imbedded and not subject to mechanical damage. Following an accident the ECCS fluid reaching the upper internals storage stand area will drain to the lower internals storage stand area from where it will be drained to the 808'-0" elevation through the drain lines as shown on Figure 6.2.2-5. The fluids that reach the fuel transfer upender area will be drained to the 808'-0" elevation through a separate drain line. To prevent blockage in the upender area of refueling cavity, a debris strainer is provided for the 4 inch drain.

6.2.2.2.2 Electrical Requirements

Each train is electrically supplied from a separate and independent Class 1E bus which is capable of supplying the minimum safety-related loads required following a LOCA or a loss of offsite power (blackout), or both. Each bus can be powered from two independent offsite power sources or by one onsite power source (diesel generator) assigned to the bus. Loads which are

required for safe shutdown of the unit are supplied power from a Class 1E power supply. Other loads are supplied power from a non-Class 1E power supply.

In the event of a LOCA or loss of all offsite power (blackout), or both, Containment spray pumps and their associated motor operated valves are automatically sequenced onto their respective emergency buses as follows:

Component	Start Time After SIAS (Sec)
Motor-operated valves	10(1)
Containment spray pumps	25

1) For motor operated valves, this is the time at which the respective motor control centers are re-energized. The valves are capable of operation, but will not operate unless a "P" signal is received.

Motor-operated valves stop automatically when valve action is completed. Containment spray pumps must be manually stopped.

6.2.2.3 Design Evaluation

6.2.2.3.1 Containment Spray Nozzles

Spray Engineering Company's model 1713A nozzle has been found acceptable in experiments with Containment spray iodine removal systems. This type of nozzle is discussed in more detail in Section 6.5.2.

Clogging of nozzles is precluded by providing a 0.115-in. mesh screen over the recirculation sumps. Since the nozzle orifice is 3/8-in.diameter, and the screen does not pass particles larger than 0.115-in. nominal size, nozzle clogging is prevented.

The CSS is designed to maintain a 40 psi differential pressure across each nozzle. All spray nozzles on one train are permanently connected through the system with two Containment spray pumps and one heat exchanger.

The total flow of one train is distributed so that each single spray nozzle passes 15.2 gpm nominal flow, and on the basis of this information, the drop size spectrum emitted from the nozzle has been established. The criteria used concerning drop size is as follows:

1. The data concerning the drop size distribution are taken from testing performed by Spray Engineering Company's laboratory on the spray nozzle model 1713A [1].

2. The droplet size and droplet distribution were recorded by high speed photographic methods, using a droplet measurement chamber. This method is considered the most accurate known, and among the 65 nozzles which have been tested, variance from nozzle to nozzle was insignificant.

3. Since the method of measurement uses high speed photographic techniques which involve stop motion, the data are considered as a spatial distribution, whereas a temporal distribution would be the true distribution after taking into consideration the effect of velocity.

4. A cross section through the spray cone was analyzed. For this measurement, a chamber was constructed to house the photographic equipment making it possible to penetrate the spray cone. The photographic equipment was mounted on a traversing rack, which could be traversed outward from the spray axis.

5. Each quadrant of the cone was divided into eight zones, permitting the collection of a total of 32 sample points. From a quantity of 325 nozzles, 65 nozzles were randomly selected for testing.

In the high temperature postaccident Containment atmosphere an increase in drop size can occur as a result of steam condensing on the spray drops. Through the use of a mass balance approach, Pasedag and Gallagher [2] have shown that the change in drop size distribution can be modeled by a step increase to the equilibrium size immediately after the drops emerge from the nozzle.

Plan and elevation drawings are shown on Figure 6.5-4.

The Containment coverage is described in more detail in Section 6.5.2.

6.2.2.3.2 Heat Removal Capability

The CSS limits the effects of postblowdown energy addition so that the effects are within the design capabilities of the Containment during the injection phase following a LOCA.

Containment pressure-temperature analyses detailed in Section 6.2.1 assume that Containment cooling capability is reduced to one spray train. This is the minimum equipment available considering the single failure criterion in the emergency power or the CSS.

The heat removal capability of the CSS is dependent on the ability of spray droplets to condense steam from the high-temperature postaccident Containment environment. A spray droplet, which enters the Containment after a LOCA encounters a saturated steam-laden atmosphere. Since the vapor pressure of the droplet is much lower than the partial pressure of steam in the Containment air, steam diffuses to the drop surface, where condensation occurs. This mass diffusion of steam is a vehicle for energy transfer to the droplet, which causes the droplet temperature, the resultant vapor pressure, and the size of the droplet to increase until thermal equilibrium is reached with the partial pressure of steam. When this occurs, condensation ceases, and no further heat removal is possible by that droplet.

Calculations of heat and mass transfer of a droplet are based on the use of heat transfer and mass transfer coefficients, which are derived from the Nusselt number for heat transfer (Nu_t) and the Nusselt number for mass transfer (Nu_m). These may be calculated from the following equations of Ranz and Marshall:

$$Nu_t = 2 + 0.6\ (Re)^{1/2}\ (Pr)^{1/3}$$

$$Nu_m = 2 + 0.6\ (Re)^{1/2}\ (Sc)^{1/3}$$

where

Re = Reynold's number
Pr = Prandtl's number
Sc = Schmidt's number

Analysis shows that a droplet with the mean droplet diameter reaches equilibrium even before it falls half the available spray fall height, which is conservatively assumed to be the distance from the lowest ring header to the operating deck.

Calculations performed based on the methodology established by Parsly [3] [4] show that the overall thermal efficiency of spray droplets in cooling the Containment atmosphere is greater than 0.99. However, a spray thermal efficiency lower than 0.99 at full spray flow was used in the analysis.

6.2.2.3.3 Recirculation Sump Design

In response to Generic Letter 2004-02 [Ref. 6], the emergency sump design was modified to replace the flat screen based design with a complex strainer based design with an effective factor of 20 increase in surface area. An analysis of the susceptibility of the ECCS and CSS recirculation functions for Units 1 and 2 was performed. This analysis provides plant specific evaluations of upstream effects, debris generation, and debris transport to the ECCS and CSS recirculation sump. The head loss associated with debris accumulation, and its associated effect on available net positive suction head were demonstrated by testing. The structural capability of the sump strainers under debris loadings was also evaluated. The downstream effects of debris that passes through the screens on components in the recirculation flow path such as pumps, valves, orifices, spray nozzles, and core components were also evaluated. The testing and analyses provide the basis to show compliance with the applicable regulatory requirements including 10CFR50.46; 10 CFR 50 Appendix A, General Design Criteria 35, 38 and 41; and 10CFR100.

The NRC has approved the methodology for meeting Generic Letter 2004-02 using the guidance of Nuclear Energy Institute (NEI) document titled "*Pressurized-Water Reactor (PWR) Sump Performance Methodology*," dated May 28, 2004 as approved and supplemented by the NRC in an SER dated December 6, 2004. The sump performance methodology and the associated NRC SER have been issued collectively as NEI Report NEI 04-07, "Pressurized Water Reactor Sump Performance Evaluation Methodology," Revision 0, dated December 2004. [REF. 7]

The methodology used employs plant specific refinements, as allowed by the NRC SER. Additional data and methodology from ongoing research on specific issues such as downstream effects, chemical effects, and coatings were also used to the extent possible. The methodology was supplemented with plant specific design and licensing basis information and contractor specific proprietary information and data as appropriate with the current state of knowledge. Exceptions and/or interpretations being taken to the guidance given in NEI 04-07 as modified by the SER are described in the responses to the Generic Letter.

All of the thermal insulation inside the Containment on both piping and equipment is of the reflective (mirror) type, composed of stainless steel. All antisweat insulation inside containment is low to medium density fiberglass jacketed in stainless steel. Vapor barrier materials are not permitted.

All of the high efficiency thermal insulation is composed of fibrous media and very fine heat resistant particulate matter, totally encased in stainless steel.

Thermal insulation is attached by use of quick release fasteners, buckles, straps, or bands. All reflective insulation, with the exception of the reactor coolant pipe insulation inside the primary shield concrete, is designed to remain in place during an SSE. Sample typical panels of insulation are seismically tested to confirm the design. A series of pressurization tests are also performed to ensure that the insulation maintains its structural integrity under postaccident pressures as well as Containment structural acceptance test and leakage rate test pressures. A thermal transient test is performed on sample insulation panels to ensure that the insulation maintains its structural integrity during postaccident temperature transients. This test consists of heating the sample panel to 650°F and quenching with cold water.

The reflective insulation assemblies are specified to withstand seismic forces resulting from acceleration of 3g in both horizontal directions and 3g in the vertical direction caused by the SSE. As a result of the Steam Generator Replacement Project, the accelerations for the Unit 1 steam generators, the new main feedwater piping and the new auxiliary feedwater piping are location specific. The insulation structural mounting frames and panel attachments to the mounting frames are designed to maintain their structural integrity during the SSE.

In order to verify that the insulation meets the required seismic criteria, the insulation supplier has tested a typical assembly on a generic basis. The tests consisted on an initial sinusoidal input frequency between 3 and 100 Hz to determine the resonant frequency condition followed by an endurance test at the lowest resonant frequency. The insulation assembly was subjected to 10g's in both the horizontal and vertical directions with following visual inspections. No damage or distortion to the structure was observed.

It is also anticipated that only that portion of insulation which is covering the actual break area and any other insulation subject to jet impingement effects are dislodged during a LOCA. Since the thermal insulation in the vicinity of postulated Reactor Coolant System (RCS) or main steam line breaks is of an all-metal reflective type or for the main steam and feedwater piping underneath the pipe whip restraints, high efficiency thermal insulation totally encased in stainless steel, the major portion of dislodged insulation sinks rapidly and remains in the area of the break. As demonstrated by testing, any pieces of reflective insulation which are carried by high-velocity streams of water out of the steam generator compartment tend to settle in the low-velocity area approaching the recirculation sumps. Analysis and testing of potential debris sources has shown that the primary debris of concern for sump performance is the combination of fibrous debris, particulate, and chemical precipitate. Fibers from fiberglass antisweat insulation located on cooling and chilled water lines and from latent debris are capable of transporting to the strainer surface. The covers for lead shielding blankets also contain fiberglass which could be within the zone of influence of a LOCA. High efficiency thermal insulation (Min-K) is made of both fibrous and particulate materials. Particulate of concern includes latent debris and coating debris. The chemical precipitates of concern result from the interaction of containment spray with aluminum. Debris generation analyses have conservatively determined bounding quantities of these and other materials that could be generated by a loss of coolant accident or a secondary line break.

The radiation protection doors from the steam generator compartments located at Elevation 808 are designed with bars in lieu of wire mesh to ensure that blockage of water and debris from a LOCA or a secondary line break does not occur. The door to the incore instrumentation guide tube room and the floor barrier around the tubes are similarly designed to ensure blockage does not occur for a LOCA or secondary line break. This design, in conjunction with the debris interceptor around the emergency sumps, ensures that transport of small debris to the large inactive sump under the reactor vessel is maximized. Debris which is transported to the inactive sump cannot affect sump performance during recirculation. Debris transport analysis has conservatively determined bounding quantities of the materials identified in the debris generation analyses that could be transported to the vicinity of the recirculation sumps. In addition to particulate and fiber, latent debris was assumed to include labels, tape, and other miscellaneous materials which could be present in containment. The results of the debris generation and debris transport analyses are combined to determine the design basis debris load for strainer qualification testing. See Section 6.2.2.3.4.

Containment sump strainer design is in accordance with the intent of Regulatory Guide 1.82. See Appendix 1A(B) for a discussion of conformance to RG 1.82.

Each of the two Containment recirculation sump strainers has a total surface are of approximately 3,947 square feet and consists of 36 modules, each with seven vertical (stacked) disks made from perforated plate. The sump strainer design ensures that the strainer are capable of withstanding the force of full debris loading, hydrodynamic loads and inertial effects of water in the containment recirculation pool, in conjunction with all design basis design conditions, including seismic, without collapse or structural damage. The sump strainer design is robust enough so that the disks also perform the same function as a trash racks. Large floating debris would not be able to block significant flow or surface area of the strainer. Trash racks are provided on the inboard side of the strainers to protect them from incidental damage during outages. Because the strainers perform the trash rack function, the trash racks are non-safety related, seismic Category II and are not required to meet RG 1.82.

A one foot tall solid debris interceptor is provided around each of the strainers. The safety function of this interceptor is to stop tumbling debris such as reflective insulation, coatings, labels, and other miscellaneous debris. This design feature will significantly reduce the debris which could reach the strainer in a design basis event. The largest quantity of debris generated by a LOCA jet would be reflective insulation which transports by tumbling along the floor. Testing has shown that reflective metal insulation debris will not transport to the strainers and that this debris is beneficial in that it would capture, and/or impede the transport of, fibrous debris. However, no credit for the beneficial aspects of RMI was taken in the analyses or testing.

Containment inspection procedures are in place to ensure the containment recirculation sumps are cleared of items that could potentially block the sumps. In addition, plant operating procedures provide information to the operators concerning indicators, probable cause, and actions to be taken in the event of low flow due to sump blockage or vortexing.

The refueling cavity is equipped with two (2) 4 inch drains located at el. 823'-0.5" and 819'-6 3/4" respectively, both discharging to el. 808'-6". These drains are covered by debris strainers making their clogging unlikely. Two redundant 6 inch drains are also provided in the refueling cavity at elevation 823’-0.5”. Each is covered with a speciality designed debris screen to prevent blockage by large insulation debris. One of these 6” drains is sufficient to drain the main area of the refueling cavity. The 4 inch drain and debris strainer in the main area of the refueling cavity

provides a diverse drain path which is capable of assuring minimal water holdup. The upender area of the refueling cavity, with the drain at el. 823', is connected to the main area of the refueling cavity by a gate opening at approximately el. 834'. Therefore, the 6 inch drains provide a diverse drain path which is capable of assuring limited water holdup.

There are no sand plug materials or structures inside the Containment designed to be displaced by accident pressures to provide vent areas. Consequently, this is not a source of debris.

The coating systems specified for use inside the Containment are qualified for postaccident Containment environment. See Section 6.1B.2 for additional details.

Any concrete particles or miscellaneous steel such as ductwork stairs or grating which may be displaced by differential pressure or jet impingement tend to sink in the low-velocity area approaching the screens.

All temporary materials used during refueling or maintenance outages, such as paper, plastic sheeting, or temporary wooden scaffolding, are removed from the Containment prior to operation.

6.2.2.3.4 Net Positive Suction Head

Sufficient net positive suction head (NPSH) is available to the Containment spray pumps for both the injection and recirculation modes of operation.

During the injection phase, the NPSH is calculated using the atmospheric pressure in the RWST, the static head between the suction sparger and the pump elevation, the piping losses, and the vapor pressure of water at 120°F.

During the recirculation phase, adequate NPSH for the Containment spray pump is ensured by the design of the CSS in accordance with NRC Regulatory Guide 1.1. It is assumed that the Containment ambient pressure is equal to the vapor pressure of the sump liquid.

The CPNPP design assumes that a water volume equal to the volume of the reactor incore instrumentation pit (between elevations 783'-7 and 808'-0) is trapped immediately following an accident and not available for recirculation. This trapped volume of water was also deducted from total volume when the available NPSH for containment spray pumps was calculated.

The available NPSH for the containment spray pumps during the recirculation phase was determined as follows:

NPSH = Static Head - Piping Friction Losses - Entrance Losses - CSHL

Where: CSHL is the Clean Strainer Head Loss.

The static head calculation considers only the difference between the elevation of the Containment Spray pumps' impeller centerline and the minimum calculated containment water level during the containment spray system recirculation phase.

Figure 6.2.2-2 shows the relationship between the available NPSH and the pump flow during the injection and recirculation phases and shows the required NPSH. Design parameters for the pumps are shown in Table 6.2.2-1.

The NPSH margin is calculated based on a clean strainer and minimum containment water levels during containment spray recirculation. The design basis debris head loss is determined by prototypical testing of a full size trainer with the design basis debris load as described in Section 6.2.2.3.3 scaled to the test configuration. This testing has shown that significant NPSH margin remains after the design basis debris head loss is subtracted from the clean strainer NPSH margin.

6.2.2.3.5 Single Failure Analysis

A failure analysis of all active components of the heat removal systems during the injection phase and all active and passive components during the recirculation phase shows that the failure of any single component does not prevent performance of the design function. This analysis is summarized in Table 6.2.2-5.

6.2.2.4 Tests and Inspections

6.2.2.4.1 Inspections

The CSS is designed to permit periodic determination of proper functioning to demonstrate system readiness.

The Containment spray pumps and heat exchangers are located outside the Containment to permit access for periodic inspection and testing during normal plant operation. The pressure-containing systems are inspected for leaks from pump seals, valve packings, flange joints, and safety valves during system testing. A full flow test line is provided for the pumps to allow for periodic testing, of each train during which water is recirculated to the RWST.

The spray nozzles are not to be tested insitu with water. However, a CSS full-flow ring header test was performed at Zion Nuclear Power Plants - Units 1 and 2 (Docket Nos. 50-295 and 50-304). This test demonstrated that the flow which passed through the system was as designed, that there was no discernible movement in the ring headers when subjected to spray flow forces, and that the Containment coverage was as predicted.

The CSS is subject to periodic inservice inspection by volumetric, surface and visual examination techniques in accordance with Section 6.6. Also, control equipment is checked to verify that the CSS is in a state of readiness. Performance curves and NPSH requirements have been supplied by shop tests for new pumps and expected performance curves based on generic and field tests when new impellers were provided in place of pumps, to confirm pump design characteristics. The associated NPSH set of these curves is shown on Figure 6.2.2-2.

Performance data for spray nozzles are provided in Section 6.5.2.2.1 to verify maximum droplet mean diameter. These data are the result of manufacturer's shop tests.

6.2.2.4.2 Preoperational Testing

Outside the Containment, piping welds are subjected to 100-percent radiographic inspection and hydrotesting, while inside the Containment, the spray header welds are subjected to 100-percent radiographic inspection. These tests demonstrate that the system is adequate to meet the design pressure and temperature conditions.

Air under pressure is passed through test connections to check that the spray nozzles and ring headers are free of obstructions and are not clogged.

The Containment spray signal is used to trigger the actuation of valves and pumps to demonstrate the proper operation of remotely operated valves.

Each Containment spray pump runs at minimum flow, and the flow is directed through the recirculation path back to the RWST. Then the pump runs at full flow through the test line connected to the RWST. During this time, the closing of the miniflow control valves is checked. This test is performed to verify the operation of the Containment spray pumps.

6.2.2.4.3 Operational Testing

Routine testing is performed periodically to verify the operability of active CSS components.

1. Testing of valves and pumps is performed by shutting the manual valve on the Containment spray line inside the Containment, shutting the manual valve on the chemical additive supply line, and triggering a dummy actuation signal. All automatic valves and the pumps are checked for proper operation.

2. Each pump is run at full flow and the flow is directed back to the RWST.

3. The following provisions are made for periodic testing of the spray nozzles:

   a. Compressed air test connections

   b. Special facility permitting access to all ring headers

   Operators will confirm that all nozzles are free and unobstructed.

6.2.2.5 Instrumentation Requirements

CSS operation is automatically initiated by signals generated by the ESF Actuation System as described in Section 7.3. The minimum flow recirculation lines are opened when flow is less than 25%. The recirculation lines are automatically closed when 25 percent of the spray pump design flow is achieved or the Containment spray isolation valve for that pump is open or associated containment sump suction valve is opened. Flow instruments are provided in each pump discharge line. Four RWST level channels are provided and used for ECCS and CSS operation. (Refer to Section 6.3.2 for use with the ECCS.)

Level instruments provide remote indication, and high, low, low-low and empty level alarms.

A low-low [two out of four channels] level signal is provided for automatic initiation of ECCS changeover from injection mode to recirculation mode when a minimum ECCS injection volume of approximately 300,000 gallons from the minimum usable volume of approximately 440,300 gallons have been drawn by the ECCS and the CSS.

A empty level alarm is provided to annunciate the necessity to manually stop any ECCS pump still taking suction from the RWST. This alarm is initiated when approximately 30,700 gallons remain to provide pump protection above the centerline of the sparger.

Each Containment spray header riser is provided with a pressure switch to alarm in the Control Room if the level in the riser falls below the minimum level required.

Two water level instruments are provided for each Containment as shown in Figure 6.2.2-1 and 7.1-3. Both are indicated in the Control Room.

The Containment spray pump discharge pressures and flows are indicated and recorded in the Control Room.

Safety-related display instrumentation is provided as described in Table 7.5-7A for the following CSS parameters:

RWST levels

Containment spray pump flow

Containment water level

REFERENCES

1. Spray Engineering Company, Spray Analysis on Sprayco Model 1713A Nozzle by Spray Engineering Company, January 1973.

2. Pasedag, W.F., and Gallagher, J.L., Drop Size Distribution and Spray Effectiveness, WNES, Pittsburgh, Nuclear Technology, vol. 10, April 1971.

3. Parsly, L.F., Spray Tests are the Nuclear Safety Pilot Plant, in Nuclear Safety Program Annual Progress Report for Period Ending December 31, 1970, ORNL-4647, p. 82, Oak Ridge National Laboratory, 1971.

4. Parsly, L.F., Spray Program at the Nuclear Safety Pilot Plant, Nuclear Technology, vol. 10, p.472, 1971.

5. NUREG-0797, SSER No. 9 Related to the Operation of CPSES Units 1 and 2, Appendix "L", dated March 1985.

6. NRC Generic Letter 2004-02, “Potential Impact of Debris Blockage on Emergency Recirculation During Design Basis Accidents at Pressurized-water Reactors”

7. NEI-04-07, “Pressurized Water Reactor Sump Performance Evaluation Methodology,” Revision 0, dated December 2004.

8. TXX-05162 dated September 1, 2005, Response to Requested Information Part 2 of NRC Generic Letter 2004-2, “Potential Impact of Debris Blockage on Emergency Recirculation During Design Basis Accidents at Pressurized-water Reactors”

9. TXX-08033 dated February 29, 2008, Supplement to Response to NRC Generic Letter (GL) 2004-02, “Potential Impact of Debris Blockage on Emergency Recirculation During Design Basis Accidents at Pressurized-water Reactors”

6.2.3 SECONDARY CONTAINMENT FUNCTIONAL DESIGN

The CPNPP does not utilize a secondary Containment.

6.2.4 CONTAINMENT ISOLATION SYSTEM

The design objective of the Containment Isolation System is to allow the normal or emergency passage of fluids through the Containment boundary while preserving the ability of the boundary to limit the escape of fission products from postulated accidents.

The Containment Isolation System is designed to monitor the development of gross leakages in the Containment and to limit radioactive emission from the Containment during normal operation and in the event of loss-of-coolant accident (LOCA). All Containment Building piping penetrations are considered potential paths for the escape of radioactivity and are equipped with isolation barriers located as close to the Containment as practical. All piping penetrating the Containment and all Containment isolation barriers are rated to withstand pressures consistent with the Containment design and to operate in normal as well as postulated accident environments.

6.2.4.1 Design Bases

6.2.4.1.1 Governing Conditions

Containment isolation is mandatory in the event of a LOCA. Containment isolation is not required but could occur as a result of a main steam or feedwater line rupture inside the Containment. The system isolates the Containment to prevent or limit the escape of fission products that may result from postulated accidents. Chapter 7 describes containment isolation and other ESF actuation signals which isolate the steam generators to prevent excessive cooldown of the Reactor Coolant System (RCS) or overpressurization of the Containment.

The Containment leakage test program is described in Subsection 6.2.6. RCS leak detection is described in Section 5.2.5. Gross leakage inside the Containment is detected by the Containment air particulate monitor, radioactive gas monitor, and specific humidity monitor. Further indications of such leakage are an increased frequency of operation of the Containment sump pumps and an increase in demand for reactor coolant makeup water. Each of the above actuates Control Room alarms. In addition, the Containment air radiation monitors are provided to actuate Containment ventilation isolation valves upon sensing high levels of radiation. Refer to Table 11.5-1; the Containment isolation actuation logic diagram is presented in Figure 7.2-1. Leakage outside Containment is detected by accident monitoring instrumentation as described in Section 7.5.

6.2.4.1.2 Isolation Criteria - Fluid Systems Penetrating the Containment

In the event of a LOCA the Containment Isolation System is designed to minimize the leakage of radioactive materials through fluid lines penetrating the Containment to a low enough rate to prevent the radioactivity level from exceeding the boundary doses specified in 10 CFR Part 100.

The requirements of 10 CFR Part 50, Appendix A, General Design Criteria (GDC) 54, 55, 56, and 57, are followed with respect to the piping systems and to the number and location of independent isolation valves provided for fluid lines penetrating the Containment. Section 3.1, Conformance with NRC General Design Criteria, discusses compliance with the GDC.

All Containment piping penetrations are located in radiation controlled areas of the Auxiliary, Fuel and Safeguards buildings which are monitored for Containment leakage after a LOCA as described in Section 7.5. This is consistent with GDC-54 requirements for leak detection.

The three types of fluid lines that penetrate the Containment wall and require isolation valves are as follows:

1. Type A - Lines which form part of the reactor coolant pressure boundary (RCPB); Section 5.4.3.2 of ANSI N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, is followed to define the limits of the RCPB.

2. Type B - Lines which connect directly with the Containment atmosphere.

3. Type C - Lines which are part of a closed system, i.e., lines that are neither part of the RCPB nor connected to the Containment atmosphere

GDC 55 is applied to Type A penetrations; except as described in Section 6.2.4.1.3, below, these penetrations are provided with one of the following isolation schemes:

1. One locked-closed isolation valve inside and one locked-closed isolation valve outside the Containment

2. One automatic isolation valve inside and one locked-closed isolation valve outside the Containment

3. One locked-closed isolation valve inside and one automatic isolation valve outside the Containment; a simple check valve is not used as the automatic isolation valve outside the Containment.

4. One automatic isolation valve inside and one automatic isolation valve outside the Containment; a simple check valve is not used as the automatic isolation valve outside the Containment.

GDC 56 is applied to Type B penetrations; these penetrations are provided with isolation schemes identical to those set forth for Type A penetrations as well as the following additional isolation schemes:

1. One blind flange inside Containment and one locked-closed isolation valve outside the Containment.

2. One blind flange inside Containment and one blind flange outside Containment.

Penetrations which are required to be closed during accident conditions are defined as GDC-55 and GDC-56 lines provided with containment automatic isolation valves which receive a containment isolation signal (Phase A, Phase B, or Containment Ventilation Isolation) or locked closed valves, blind flanges, or deactivated automatic valves secured in their position in accordance with Technical Specifications.

GDC 57 is applied to Type C penetrations; these penetrations are provided with at least one Containment isolation valve which is automatic, locked-closed, or capable of remote manual operation. These valves are located outside the Containment and as close to it as practicable; a simple check valve is not used as the automatic isolation valve.

6.2.4.1.3 Special Containment Isolation Provisions

1. Remote Manual Operation of Isolation Valves in Safeguard Lines

Valves in lines which are required to perform a safeguard function following an accident must be opened or must remain open for the Safeguard System to operate. Automatic closing of these valves would defeat their intended purpose; therefore, these valves are designed to be operated manually from the Control Room. Lines which fall into this category include low head safety injection lines (Figure 6.2.4-1, Valve Arrangement 8), high head safety injection lines (Figure 6.2.4-1, Valve Arrangement 9, 33 and 46), Containment spray lines (Figure 6.2.4-1, Valve Arrangement 25), hydrogen purge lines (Figure 6.2.4-1, Valve Arrangement 20 and 21), Auxiliary Feedwater Lines (Figure 6.2.4-1, Valve Arrangement 36) and Main Steam Supply to the Auxiliary Feedwater Pump Turbine (Figure 6.2.4-1, Valve Arrangement 17).

Accident monitoring instrumentation described in Section 7.5 is provided to detect leakage from lines associated with these valves.

Pressure monitoring instrumentation described in Section 6.3.5.2.6 is provided to monitor pressure within high head safety injection lines associated with these valves.

2. Containment Sump Recirculation Lines

The Containment sump recirculation lines, which supply suction to the low-head safety injection (RHR) pumps, and the Containment spray pumps (Figure 6.2.4-1, Valve Arrangement 2) are each provided with a single remote-manual gate valve outside the Containment. This valve is enclosed in a valve isolation tank (Figure 6.2.2-3) and the piping from the sump to the valve is enclosed in a concentric guard pipe.

The guard pipe and valve isolation tank are not considered part of the barrier between Containment and external environment and are not tested at Containment design conditions. The reason for this is that these moderate energy lines are designed to meet the requirements of Branch Technical Position MEB 3-1 (SRP 3.6.2) with stress levels less than 0.4 ($1.2S_h + S_a$). The penetrations are designed and fabricated per ASME Section III CL2 and MC (Article NE1000) Summer '76.

The valves and isolation tanks are located in the safeguard building which are served by the ESF emergency filtration system. System arrangement also provides for valve leakage control. Valve stem leakage is piped and routed via the valve stem leakoff connections to the safeguards building sumps and then to the Floor Drain Tank No. 1. Valve bonnet leakage will be collected at the bottom of the tanks which are equipped with normally closed drain connections. These are also piped and routed via the safeguard building sumps to Floor Drain Tank No. 1. Each tank is also provided with level switches and alarms which will alert the operator in the event of significant liquid accumulation.

The isolation valves are equipped with tapered stems above the gate to provide a seating surface in the backseat position to provide additional leakage control while the valve is in an open position.

The isolation valves are also equipped with relief valves and piping from the bonnet chamber to the outlet nozzle. The relief valves prevent pressure build-up in the bonnet chamber which could result in pressure locking of the valves.

3. Relief Valves Inside Containment

The RHR suction line (Figure 6.2.4-1, Valve Arrangement 11) and component cooling water supply line to the excess letdown and reactor coolant drain tank heat exchanger are all equipped with relief valves inside the reactor Containment and form part of the Containment boundary. Containment atmosphere can impinge only on the discharge side of these valves which are held closed by spring pressure. When these valves operate, discharge is into the Containment.

Relief valve set pressure are verified by periodic bench tests. Isolation valves outside the Containment, with the exception of the RHR suction line discussed in Subsection 6.2.4.1.3, Item 5 which are inside Containment, act as redundant Containment barriers.

4. Seal Injection Lines

The seal injection lines (Figure 6.2.4-1, Valve Arrangement 15) allow flow from the charging pumps to the main seals on the reactor coolant pumps in order to provide cooling to the seals and shafts. Due to the sensitive nature of the seals, seal injection flow should be maintained when the Reactor Coolant System is not completely depressurized or when the Reactor Coolant System inventory is moving through the seal package. The charging pumps can also provide high pressure inflow following an accident. Therefore, no automatic trip valves are required for these lines; instead, a remote manual isolation valve is furnished which can be closed when the charging pumps have completed their safeguard function.

5. Residual Heat Removal System Suction Lines

Each line from the RCS hot legs to the RHR pump suction lines contains two remote-manual, motor-operated valves which are closed during normal plant operation. These RHR isolation valves are interlocked and controlled as described in Section 7.6.2 as shown in Figure 6.2.4-1, Valve Arrangement 11. The valves are interlocked so that they cannot be opened when the RCS pressure is greater than the design pressure of the

RHR System. The valves closer to the RCS are interlocked with one pressure transmitter while the valves closer to containment are interlocked with a separate transmitter. The valve which is located closer to the RCS inside the missile barrier is not considered a containment isolation valve. The second valve defines the limit of the RCPB. This valve also provides the Containment isolation barrier inside the Containment, and is considered to be locked closed. Both these valves are located inside the Containment.

The RHR hot legs suction lines are connected to the safety injection (low head) Containment recirculation lines, which are filled with sump fluid after a LOCA. At least one of these recirculation lines is in operation during the safety injection recirculation phase. It is unnecessary to provide an isolation valve outside the Containment in the RHR hot legs suction lines, since the single failure criterion is satisfied as follows:

If a leak occurs in the recirculation system outside the Containment, the sump valve is closed to prevent loss of sump water and the closed valve in the RHR suction line prevents any Containment atmosphere from entering the system outside the Containment.

If a leak occurs in the line upstream of the isolation valve (towards the reactor coolant system) the leakage will be stopped by the closed isolation valve.

If a leak occurs in the short length of pipe between the valve inside the Containment and the Containment, any Containment atmosphere will get only as far as the fluid-filled system. Since this system is filled with sump water, no gas can escape to the outside. The fluid in the RHR suction line drops to approximately the level of fluid in the sump and any Containment atmosphere which does leak into the line is contained in this length of closed piping. In order to minimize the length of pipe between the Containment wall and the first RHR isolation valve in each RHR suction line, this valve is located within 10 linear pipe feet from the Containment wall.

Another closed valve in the line would do nothing except somewhat decrease the length of pipe outside the Containment which would be exposed to Containment atmosphere following a leak. It is possible that a valve in this section of pipe would increase the probability of leakage of gas through the stem packing and cannot be considered as tight as a clean length of pipe. No single failure of any active or passive component anywhere in the present system can cause any release of Containment atmosphere to the outside. Any additional valves would complicate normal RHR operation and are unnecessary for Containment isolation.

This arrangement meets GDC 55 on the "other defined basis" criteria in that system reliability is enhanced by a single valve and there is at least a single mechanical barrier after a single active failure.

6. Leak Rate Test and Maintenance Connections (Figure 6.2.4-1, Valve Arrangements 31 and 38)

The leak rate test and maintenance connections are each equipped with a bolted blind flange with locked closed valves outside the Containment, or a bolted blind flange both inside and outside the Containment. The arrangement provided satisfies the functional requirement of GDC 56 by providing redundant isolation barriers.

7. Turbine Driven Auxiliary Feedwater (TDAFW) Pump Steam Supply and MSIV Bypass Valves (Figure 6.2.4-1, Valve Arrangement 17)

   The warm-up bypass valves around the steam supply valves in the TDAFW pump steam supply lines (loop 1 & 4 only, see Figure 6.2.4-1 Sheet 5 of 10) are normally locked closed. The bypass warm-up valves are not required during an emergency cold start of the TDAFW pump.

   Similarly, the manual bypass valves around the Main Steam Isolation Valves (MSIVs) are normally locked closed, and may be opened during plant startup, per Technical Specification Bases requirements, to warm up the system piping downstream of the MSIVs.

8. Local Vent, Drain and Test Connection Valves

   To ensure that Containment integrity is maintained, local vent, drain, and test connection valves within the penetration boundary are locked closed and are under administrative control. See Section 6.2.6 for leak testing requirements.

   The penetration boundary is defined as that portion of a GDC 55, 56 or 57 line:

   a. Between containment isolation valves for normal two containment isolation valve GDC 55 or 56 lines (i.e., one valve inside and one valve outside containment).

   b. Between the containment and the containment isolation valve outside containment for GDC 57 lines.

   c. Between the containment isolation valve and the containment for single containment isolation valve GDC 55 or GD 56 lines (i.e., those penetrations for which special containment isolation provisions have been provided in FSAR Section 6.2.4.1.3).

9. Thermal Relief Valves (Figure 6.2.4-1, Valve Arrangements 1, 4, 5, 6, 22, 25, 26, 27, 28, 29, 30).

   Thermal relief valves are provided for those penetrations which may contain stagnant fluid between closed penetration isolation valves and may be subject to external heat sources after the fluid is isolated. The valves relieve potential pressure increases in the fluid due to the heating process.

   Relief valves function as Containment isolation valves, and the set pressures are greater than 1.5 times the Containment design pressure.

10. Airlock equalization valves (Figure 6.2.4-1, Valve Arrangements 41, 42 and 45)

    The Unit 1 personnel hydraulically operated airlock equalization valves in valve arrangement 41 are mechanically interlocked, the Unit 1 & 2 emergency airlock equalization valves in valve arrangement 42 are mechanically interlocked, and the Unit 2 personnel hydraulically operated airlock equalization valves in valve arrangement 45 are electrically interlocked, so that only one at a time can be open. These valves are the

normal equalization valves and are controlled by the airlock door operating mechanisms so that they are closed except when opening the associated door. This ensures one containment boundary is closed at all times when containment integrity is required.

Power to the hydraulic pump for these hydraulically operated valves is tripped by a safety injection signal to ensure that no failure in the non-class 1E control circuits or a spurious signal can cause the valves to open coincident with a LOCA. This satisfies the requirements of 10CFR50.49 and BTP ICSB-18.

The personnel airlock hydraulic units are supplying hydraulic power for containment airlock operation fed from non-safety related Train C power via MCC 1, 2EB1-2. In order to preclude post accident environmental conditions inside containment from causing spurious opening of the airlock doors/equalization valves, the power supply is tripped via an SI signal from both Trains. The Train A trip is accomplished by tripping the Class 1E feeder breaker to MCC 1, 2EB1-2 the Train A 480V switchgear 1, 2EB1. The Train B trip is accomplished by tripping Class 1E Train B circuit breaker CP1, CP2-BSDSEB-01 in the feeder circuit of the hydraulic unit fed from MCC 1, 2EB1-2.

The feeder circuit cables inside the personnel airlock feeder circuit control panel CP1, CP2-BSCPEB-01 are trained to preclude a short between the breaker line and load side cables. The non-Class 1E cables used in the hydraulic pump circuit cannot fail to override the safety function of opening the breaker because the cables can only short, open or fault to ground. In all such cases, power to the hydraulic pumps is terminated thus supporting the safety function. Similarly, non-Class 1E motor failure (i.e. short circuit, open circuit and short to ground) removes the motive power from the hydraulic pump thereby supporting the safety function. Therefore, tripping the power to the hydraulic pumps following an accident resulting in a safety injection signal meets the redundancy requirements of 1EEE 279.

The Unit 1 and Unit 2 personnel airlock manual equalization valves in valve arrangements 41 and 45, respectively, perform the same functions as those above except they are normally locked closed and are provided in the event the normal equalization valves are not available to relieve differential pressure across a door. They are administratively controlled to ensure one containment boundary is intact at all times when containment integrity is required.

This arrangement meets GDC-56 since redundant isolation barriers are provided and the isolation function is maintained assuming any single active or passive failure.

11. Airlock hydraulic system valves (Figure 6.2.4-1, Valve Arrangements 40 and 44).

The Unit 1 hydraulic system valves in valve arrangement 40 are locked closed except during ingress and egress as allowed by Technical Specifications. The valves must be opened for normal operation of the personnel airlock operating mechanism and interlocks. The inboard portion of the hydraulic system meets the requirements of NUREG-0800, Section 6.2.4, II.6, paragraph o, for closed systems inside Containment.

This arrangement meets GDC-57 since redundant isolation barriers are provided and the isolation function is maintained assuming any single active or passive failure.

The Unit 2 hydraulic system valves in valve arrangement 44 are locked closed. These valves are only opened under administrative controls when it becomes necessary to use the hand pump to unlatch the doors.

This arrangement meets GDC-56 since redundant isolation barriers are provided and isolation function is maintained assuming any single active or passive failure.

12. Airlock Instrumentation (pressure gauges)

The pressure gauges on the airlocks are classified as Nuclear Safety Related, Seismic Category I because they form part of the containment isolation boundary and isolation valves may not be accessible post-accident.

6.2.4.1.4 Isolation Criteria - Fluid Instrument Lines Penetrating the Containment

Instrument lines penetrating the Containment are designed in accordance with NRC Regulatory Guide 1.11.

There are four instrument lines which penetrate the Containment that are required to remain functional following a LOCA or steam line break. These lines sense the pressure of the Containment atmosphere and transmit this pressure to instruments outside the Containment.

Signals derived from these instruments initiate the safety injection, Containment isolation and Containment spray. Therefore, it is essential that these lines remain open following an accident. Isolation is provided by means of sealed bellows connected to a fluid-filled tube. This tubing, along with the transmitter and bellows, is conservatively designed and subject to strict quality control and regular inspections to assure integrity. The arrangement consists of a double isolation barrier. If the instrument line breaks outside the Containment, leakage of the Containment atmosphere is prevented by virtue of the sealed bellows. If the instrument line breaks inside the Containment, leakage is prevented by a leaktight diaphragm installed in the pressure instrument which is designed to withstand the full Containment design pressure.

6.2.4.1.5 Design Requirements for Containment Isolation Barriers

Quality standards and seismic design classification of the Containment Isolation barriers follow the recommendations of NRC Regulatory Guides 1.26, Rev. 3 and 1.29, Rev. 2, respectively. The Containment Isolation barriers are designed to seismic Category I standards, as well as temperature, pressure, humidity, and radiation conditions which exceed those expected in the event of a design basis accident (DBA).

Analyses are performed to demonstrate the integrity of the isolation valves and the connecting piping under the application of dynamic forces which would result from inadvertent closure of a valve during operating conditions (e.g., closure of a steam line isolation valve under full steaming rate). These analyses examine the pressure transients which develop in such situations. The forces and stresses resulting from these pressure transients are calculated, and displacements and support reactions are determined. As a result of these calculations, the location of support restraints is established and assurance is provided that the isolation valve pipe stresses are within allowable limits.

The equipment is designed to assure the operability of Containment Isolation System components following a DBA and during normal plant operating conditions. A detailed description of seismic qualification testing and analysis performed on safety-related mechanical equipment to assure operability during and after a postulated earthquake is given in Section 3.9.2.2. Section 3.11 discusses the environmental conditions considered in the design of the Containment Isolation System. That section also includes a discussion of the tests and analyses conducted to assure the adequacy of component performance under the specified environmental conditions.

The closed systems used as isolation barriers inside and outside the Containment satisfy the following design requirements:

1. The systems do not communicate with either the RCS or the Containment atmosphere.

2. The systems are protected against missiles and pipe whip.

3. The systems are designated seismic Category I.

4. The systems are classified ANS Safety Class 2.

5. The systems are designed to withstand temperature at least equal to the Containment design temperature.

6. The systems are designed to withstand the external pressure from the Containment structural acceptance test.

7. The systems are designed to withstand the loss-of-coolant accident transient and environmental conditions.

The valve types used for Containment isolation services are designs which have minimum leakage; essentially no leakage is anticipated through the Containment isolation valves when in the closed position.

The design pressure of piping and components within the isolation boundaries are equal to, or greater than, the design pressure of the Containment structure. Process piping, valves, and Containment penetrations are designed, constructed, and installed in accordance with the requirements of ASME B&PV Code, Section III, Code Class MC and Class 2. The Process Sampling System tubing, fittings, valves, and supports are designed, constructed, and installed in accordance with the requirements as described in FSAR Table 17A-1, Items 8 and 32. The Unit 1 Personnel Airlock Hydraulic System is designed in accordance with the requirements as described in FSAR Table 17A-1, Item 8.

6.2.4.2 System Design

A schematic diagram of the various isolation valving applications, indicating the locations with respect to the Containment barriers of all isolation valves and fluid systems penetrating the Containment wall, is shown in Figure 6.2.4-1.

Tables 6.2.4-1, 2 and 3 summarize the types of isolation valves and isolation schemes provided for the Containment Isolation System. For completeness and use with Section 6.2.6, these

Tables and Tables 6.2.4-4 and 6.2.4-6 also include all Containment mechanical penetrations and have corresponding item numbers for all five Tables for ease of cross reference. See Section 6.2.6 for Containment Leakage Testing.

These tables contain the following information:

1. Containment penetration number
2. Applicable GDC or NRC Regulatory Guide
3. System name
4. Fluid contained
5. Line size
6. Engineered safety feature system (yes or no)
7. Arrangement type
8. Isolation valve number
9. Location of valve (inside or outside the Containment)
10. Type of leakage test (A, B or C)
11. Length of pipe from Containment to outermost isolation valve
12. Valve type and operator
13. Primary mode of valve actuation
14. Secondary mode of valve actuation
15. Normal valve position
16. Shutdown valve position
17. Postaccident valve position
18. Power failure valve position
19. Containment isolation signals
20. Valve closure time
21. Power source

The Containment Isolation System is automatically actuated by signals developed by the engineered safety features actuation system described in Section 7.3.1.1.4. (See Tables 7.3-1, 7.3-2, and 7.3-3, and Figure 7.2-1).

The Containment Isolation System is designed in accordance with 10 CFR Part 50, Appendix A, GDC 54, 55, 56, 57 requirements discussed in Subsection 6.2.4.1.2.

Instrument lines penetrating the Containment are designed in accordance with NRC Regulatory Guide 1.11 as described in Subsection 6.2.4.1.4.

A detailed discussion of missile protection can be found in Section 3.5. The design features and measures against jet forces and pipe whip are described in Section 3.6, and the seismic design bases for the safety-related systems are provided in Section 3.7.

In case of loss of control voltage to the associated solenoid valve, or in case of loss of air, the air-operated Containment isolation valves move to the position of greatest safety. Motor-operated Containment isolation valves fail in the as is position. Emergency power from the standby diesel generators is provided to the AC valves to ensure system operation in the event of a loss of offsite power.

All Containment isolation valves which receive signals to close from Containment isolation phase A, Containment isolation phase B, steamline isolation or feedwater isolation have valve closure times as fast as practical, consistent with the type of valves and valve operators, with consideration given to water hammer effects. Those lines which provide a direct connection from the Containment atmosphere to the environment are equipped with isolation valves having closing times of five seconds or less or are locked closed. Valve closure time for each valve is shown in Table 6.2.4-3. To ensure operability, the Containment Isolation System is designed to meet the single failure criteria with no loss of function.

The possibility of debris becoming entrained in escaping fluid and preventing tight closure of an isolation valve is of concern only for penetrations which are open to the Containment atmosphere during power operation. The Containment pressure relief line is the only penetration which falls in this category. The following provisions are made to ensure that debris does not become entrained in escaping fluid and prevent tight closing of the isolation valves in this line:

1. Pipe provided for this function is separate from other duct systems inside of the Containment and is qualified to seismic Category I by analysis; no registers or other potential sources of debris are used in this ductwork.

2. The pipe is routed to a clear area inside Containment as close to the penetration as possible; the clear area is free of potential sources of debris such as piping insulation, and so forth.

3. A debris screen covered by a flow restriction orifice plate is provided as described in Section 9.4A.

4. The isolation valves are fast closing valves which close in less than three seconds.

The redundancy requirement is satisfied by having two isolation barriers in series, one on each side of Type A and Type B penetrations. Reliability is assured by conducting periodic tests to

check the operability of the isolation valves, actuators, and controls. Furthermore, a fail-safe feature is incorporated into air- operated and solenoid-operated isolation valve design, so that in the event of actuating power loss, the valve assumes the position that ensures safety.

The power operated isolation valves may be operated manually from the Control Room to provide a secondary means of actuation. To confirm performance capabilities, i.e., closing time and valve status (open or closed), indicator lights are checked and observed during the periodic testing.

The primary and secondary modes of actuation are shown in Table 6.2.4-2.

As shown in the Figure 6.2.4-1, penetrations which require leakage testing are provided with appropriate test connections.

6.2.4.3 Design Evaluation

The design of the Containment Isolation System meets the requirements for system integrity, response, operation, and reliability. Isolation valve and piping design and their location ensure Containment integrity for any postulated single failure.

Regular functional testing of the Containment Isolation System during shutdown periods assures operability of all isolation valves. Leakage rate testing during the same periods ensures that the leakage through isolation valves and piping penetrations does not exceed values commensurate with offsite radiation doses under accident conditions, given in the Technical Specification. The use of double isolation barriers ensures that no single failure of any active or passive component renders the Containment Isolation System either partially or wholly inoperable. Open or closed isolation valve status during normal plant operation is regularly checked and controlled, particularly with regard to manually-operated isolation valves. In addition, automatic isolation valves, whether actuated remote-manually or by isolation signals, are designed to assume a fail safe position. These are tabulated in Tables 6.2.4-1, 2 and 3.

No system is provided for continuous leaktightness monitoring. However, the use of double isolation barriers, periodic testing, administrative control of manual isolation valves, and surveillance of automatic isolation valves ensure that the Containment Isolation System performs its intended function. Radiation monitors, valve isolation tank level instrumentation, and Safeguards Building sump instrumentation provide post-accident leak detection in accordance with GDC-54.

All system paths penetrating the Containment wall are evaluated on the basis of their function at the time of an accident. Because some system piping penetrating the Containment performs more than one function, each system path is evaluated. Each system path has a unique Containment isolation valving arrangement as identified in Table 6.2.4-1. The system paths have been classified either as essential or non-essential.

1. Essential System Paths

   A system is designated as essential if:

   a. it is required to mitigate an accident (as ESF systems), or,

   b. unavailability could increase the magnitude of the accident.

2. Non-Essential System Paths

   A system is designated non-essential if it is not classified as essential. These systems are automatically isolated from the Containment during Containment isolation, or their lines are closed during normal operation and kept closed during an accident.

The purpose of the above selection is to identify essential systems which allow an emergency path of fluids through the Containment boundary and ensure integrity of the Containment boundary during an accident.

Table 6.2.4-6 lists all penetrations comprising the Containment isolation system together with fluid system path classifications, normal operating function and post-accident function of the Containment isolation barriers (i.e., Containment isolation valves) assuming a "P" signal has occurred.

6.2.4.4 Tests and Inspections

A rigorous program of tests and inspections is performed in accordance with 10 CFR Part 50, Appendix J, Option B, to ensure Containment Isolation System pressure integrity, leakage rate, and reliability of operation. Subsection 6.2.6 gives a detailed description. These tests ensure that the leakages from the Containment Isolation System are held within allowable Appendix J, Option B leakage rate limits. Furthermore, these tests verify the operability of the Containment isolation valving. Detailed test procedures utilized and results of tests performed are provided following completion of the tests.

Table 6.2.4-2 list the isolation valves provided for each penetration and indicates the direction in which the isolation valves will be tested. The testing arrangement for containment isolation valves will be controlled via administrative procedures and as described in FSAR Section 6.2.4 and 6.2.6.

All containment penetrations that utilize expansion bellows can be tested at P_a. Table 6.2.4-2 lists all the containment penetrations which are part of fluid system process and instrumentation piping and shows the type of local leakage testing for each penetration. Typical electrical penetration assembly is shown in Figure 3.8-8. List of all electrical penetration assemblies is shown on Figure 8.3-16. Testing of electrical penetration assemblies is described in Section 6.2.6.

Type C tests are performed on Containment isolation valves as indicated in Table 6.2.4-2. Specific exemptions from Type C testing are provided as footnotes to the table. In general these justifications include but are not limited to:

- closed systems meeting the requirements of NUREG-0800 Section 6.2.4, II.6 paragraph e and o.

- valves in systems which are inservice post accident at a pressure in excess of containment design pressure.

- valves in systems which are water filled for a period of 30 days following an accident.

Figure 6.2.4-1 shows the arrangement of test connections and test vent which permit the isolation valves to be leak tested. During testing the test vent (TV) connection is open and the pressure is applied through the test connection (TC). Test gas is applied at each test connection to establish a test volume in the piping so that the valve is exposed to gas at Containment calculated peak internal pressure. Equipment is laid out in order to minimize the size of the test volume. When necessary, test vents are supplied to ensure that the side of the valve opposite to the test gas is at ambient pressure during the test. Valves are tested in the direction of leakage from the Containment, i.e., from the center of the Containment outwards, with the following exceptions:

1. In valve arrangements 18, 19, 20, and 21 of Figure 6.2.4-1, the butterfly valves inside the Containment are tested towards the center of the Containment. Butterfly valve disc leakage is the same in either direction due to the symmetrical design of the valve.

2. In valve arrangement 22 of Figure 6.2.4-1, the diaphragm valve inside Containment will be tested toward the center of the Containment. Diaphragm valve leakage is the same in either direction due to the symmetrical design of the valve.

3. In valve arrangements 41 and 42 of Figure 6.2.4-1, the ball valves on the inboard side of the personnel and emergency airlocks are tested toward the center of the Containment. Ball valve leakage is the same in either direction due to the symmetrical design of the valve.

4. In valve arrangement 45, the manual spring closed valves are tested as part of the barrel test and the two valves on the containment side are tested in the direction away from the reactor due to the valves being unsymmetrical. Under DBA conditions, all of these manual spring closed valves are oriented in the direction which results in increasing seating force (i.e. DBA pressure loads the discharge side). Therefore, leak testing as part of the barrel test is conservative.

Containment isolation valve leakage rates are evaluated by methods discussed in Section 6.2.6.3.

Environmental qualification tests performed on the Containment Isolation System components are discussed in Section 3.11.

6.2.5 COMBUSTIBLE GAS CONTROL IN CONTAINMENT

Following a DBA, hydrogen gas may be generated inside the Containment by reactions such as zirconium metal with water, corrosion of materials of construction, exposure of the organic cable materials to radiation and radiolysis of aqueous solution in the core and sump. In addition a small amount of methane is generated by the irradiation of the cables as discussed, in

Section 6.1B.2. The following section is presented to describe the design of the Combustible Gas Control System.

6.2.5.1 Design Bases

6.2.5.1.1 Generation, Accumulation, and Mixing of Combustible Gases

1. A combustible mixture can be formed when hydrogen gas concentration in the Containment atmosphere is greater than four volume percent (v/o).

2. A volume of hydrogen is generated by radiolysis in the core and the sump and is released in the compartment where the LOCA occurred and in the Containment sump where the coolant is collected.

   All subcompartments are provided with vents at the top and drains at the bottom. The vents provide for the release, caused by buoyancy, of any hydrogen generated within or beneath the subcompartment. The drains prevent the accumulation of water within a subcompartment, thus preventing substantial generation of hydrogen by radiolysis within that subcompartment.

   Arrangement of the subcompartments with bottom and top openings creates a stack effect. In addition to the driving forces generated by diffusion rate, the natural ventilation going through the subcompartments provides mixing and avoids hydrogen stratification. Therefore, the flow caused by the stack effect yields a hydrogen concentration within a subcompartment that does not substantially differ from the bulk Containment conditions.

3. Although operation of the containment spray effectively prevents hydrogen stratification, neither the containment spray nor the recirculation fans are required to ensure adequate mixing. Use of containment spray during post LOCA conditions would enhance natural circulation by causing a temperature gradient in addition to the driving force of falling drops.

6.2.5.1.2 Electric Hydrogen Recombiners

The Electric Hydrogen Recombiners for Unit 1 and for Unit 2 have been abandoned in place.

6.2.5.1.3 Hydrogen Purge System

The following design descriptions apply to the Hydrogen Purge System:

The Hydrogen Purge System functions to provide controlled purging of the containment atmosphere to aid in cleanup in accordance with GDC 60.

As required by GDC 41, when the system operates, it is capable of maintaining the hydrogen concentration in the Containment below the lower flammability limit following an accident.

The manually actuated Hydrogen Purge System has a process capacity of 700 cfm.

The system is required to be capable of operating with a Containment pressure range of 0 to 5.8 psig and temperature range of 50 to 160°F.

Protection is provided to preclude damage by missiles. All materials are selected to be compatible with accident and normal operating environments.

6.2.5.1.4 Containment Hydrogen Monitoring System

The Containment Hydrogen Monitoring System monitors the hydrogen partial pressure in several well-ventilated areas of the Containment Building in order to obtain typical values for hydrogen gas concentration.

The plant has two hydrogen monitoring systems. Each monitoring system consists of four (4) sensor modules and one (1) microprocessor analyzer. Of the four (4) sensor modules in each system, two (2) are located in each Containment. The microprocessor analyzer is thus shared by Units 1 and 2. The system can be operational within 90 minutes after an accident and is designed for continuous duty during normal plant operation. The hydrogen gas analyzers alarm at 3 v/o (wet) hydrogen.

The sensor modules and microprocessors are qualified to function under Seismic Category I requirements and post accident conditions as described in CPNPP FSAR Appendix 3A.

6.2.5.2 System Design

The primary means of reducing hydrogen concentration in the Containment following a LOCA is by the use of a Hydrogen Purge System which is available for use to aid in cleanup by providing controlled purging of the Containment atmosphere. A Containment Hydrogen Monitoring System is provided to sample the Containment atmosphere in various locations to determine the hydrogen concentration.

6.2.5.2.1 Electric Hydrogen Recombiners

Abandoned in place.

6.2.5.2.2 Hydrogen Purge System

The Hydrogen Purge System shown in Section 9.4 on Figure 9.4-6 is common to both units. This system is not used during normal operation but is capable of operating intermittently or continuously after an accident. The Hydrogen Purge system consists of two 700 cfm blowers for supply, inlet and outlet ductwork, and piping, isolation valves, a flow control valve, two atmospheric cleanup systems and two exhaust fans. The blowers are capable of transporting 700 cfm of the fresh, filtered air to the Containment. Air is drawn from either Containment as required, passed through a filterplenum (particulate, iodine adsorbers, HEPA filters) and discharged through the plant discharge duct. A demister and heater are used to maintain the humidity entering the filters below 70 percent. Two trains are provided (one train is required to operate), each capable of controlling the design airflow of 700 cfm when the containment is less than 5.8 psig.

The Hydrogen Purge System is manually operated and is isolated from the Containment by normally closed valves. Mixing of the Containment atmosphere is by natural convection.

Any radioactivity discharged is measured by the plant vent stack monitoring system.

The expected efficiencies of the filters are in accordance with NRC Regulatory Guide 1.140.

6.2.5.2.3 Containment Hydrogen Monitoring System

The hydrogen concentration in each Containment is monitored by four (4) sensors located on four (4) different elevations of the containment. Two (2) sensors from each Containment are coupled to one of the two hydrogen analyzer microprocessors located in the control room. Each microprocessor is supplied from a safety-related uninterrupted power supply train. Thus, two independent analysis trains, each monitoring two points inside each Containment, are provided for measurement.

The analyzers continuously monitor the hydrogen content of the Containment atmosphere during normal plant operation and will be operational within 90 minutes following a LOCA. This monitoring system does not rely on the hydrogen recombiner installation or operation.

The analyzer system meets with the following requirements:

Sensitivity	0.1 percent hydrogen by volume
Accuracy	±2.0 percent of full scale
Range	0-10 percent hydrogen by volume
Calibration	Fully automatic sequencing for feeding known gaseous mixtures to the sensor modules and adjustment

The sensor modules are of the in-Containment measurement type using an electrochemical sensor for specific measurement of hydrogen partial pressure.

Each sensor module consists of the following major components mounted on an integral rack: hydrogen sensor, calibration mechanism, calibration gas bottles, solenoid valves (calibration gas isolation), RTD temperature transducer and an electronics interface terminal box. One absolute pressure transducer is provided with each pair of sensor modules. This transducer is mounted on one of the sensor modules.

The analyzer microprocessor modules accept, process and condition the sensor output signal. The microprocessor has a digital display for the following:

- Hydrogen volume percent (wet)
- Hydrogen volume percent (dry)
- Hydrogen partial pressure
- Temperature
- Pressure

The control room operators are able to select any display for instantaneous readout. The microprocessors also have two buffered 0- 10 volt dc output signals for remote analog display of hydrogen volume percent (wet) on the Main Control Board.

The alarms from the microprocessor modules are from solid state relays and indicate the following conditions: high hydrogen concentration, power failure and system error.

The Containment Hydrogen Monitoring System is designated as IEEE Class 1E and qualified per requirements of IEEE 323-1974.

6.2.5.3 Design Evaluation

6.2.5.3.1 Hydrogen Generation

Based on the revision to 10CFR50.44 effective October 16, 2003, the calculation of hydrogen generation following LOCA is no longer needed.

6.2.5.3.2 Hydrogen Mixing

As described in Subsection 6.2.5.1.1, all subcompartments are provided with vents to aid in hydrogen mixing and to avoid high concentration pockets of hydrogen. These vents cause a stack effect which maintains the subcompartments at virtually the same hydrogen concentration as the remainder of the Containment.

This stack effect is governed by the following formula:

$$Q = 7.2A\ ht$$

where

Q	=	air flow (ft^3/min)
A	=	area of bottom or top openings, whichever is smaller (ft^2)
h	=	height from bottom to top openings (ft)
t	=	temperature difference between the subcompartment and the bulk Containment atmosphere (F)
7.2	=	constant of proportionality, for conditions not favorable

At a minimum temperature gradient of 1°F, the number of air changes per hour is always in the range of 2 to 3.

With regard to mixing of post-LOCA hydrogen, the following aspects have been considered:

1. Mixing in the bulk Containment above operating floor at elevation of 905 ft 9 in.

   Experimental results from spray experiments conducted at Oak Ridge National Laboratories have substantiated the adequacy of the sprays to ensure mixing in the bulk

Containment volume. These results apply to the Region A described in Section 6.5, Figure 6.5-2. While spray operation is a sufficient condition to ensure mixing, it is not a necessary condition. The stack effect described above would also be effective in this region under LOCA conditions, as would a thermal/buoyancy plume formed from the effluent of the break.

2. Subcompartments are enclosed between floor elevations 808 ft 0 in., 832 ft 6 in., 860 ft 0 in., and 905 ft 9 in.

These regions are described in Section 6.5 and shown on Figure 6.5-2, where they are referenced as Region B, C, and D. To avoid accumulation of hydrogen between floors, each of them is provided with openings to permit mixing with the bulk Containment.

Between the Region D and the other regions, the total opening is approximately 900 ft^2. Although the hottest subcompartment is the steam generator subcompartment, a temperature difference between the upper Containment and the lower floors exists as a result of the cold water sprayed in the dome which induces natural convection. This natural convection ensures a general mixing of hydrogen in the Containment, after spray operation. If sprays are not in operation, the mixture in Region D will be entrained into the hotter steam generator compartment, or into the openings on the floor, as a result of the stack effect. A mixture descending along the liner will be colder as a result of condensation on the liner. This natural circulation loop ensures mixing even if sprays are not operating. Alternatively, conditions also can induce the formation of a thermal/ buoyancy plume which will entrain steam and gases as it rises. The plume hydrogen concentration is near the average ambient concentration at the upper containment elevations.

These three regions are partially sprayed by nozzles (see Section 6.5, Table 6.5-5) when operating containment spray during the injection and recirculation phases. In addition, the driving force of falling drops enhances air circulation.

3. Subcompartments Where LOCA Occurs

These subcompartments are the steam generator subcompartments, the connecting pipe tunnel to reactor vessel cavity, and the pressurizer subcompartment.

A large part of the hydrogen generated in the Containment is released in the subcompartment where the break occurs, as a result of radiolysis in the core.

a. Steam Generator Subcompartments

Each steam generator subcompartment is fully open at the top and provided with two main openings in the bottom (for communication with another steam generator subcompartment and personnel access). This arrangement has the following effects:

1. Coverage of the steam generator subcompartment with spray, ensuring a mixing within the subcompartment

2. Release of hydrogen through the top as a result of its low atomic weight

3. Mixing of the contents of the steam generator subcompartment with the contents of the bulk Containment through natural convection effects. A sensible energy is introduced during the long-term with the core recirculation flow in the subcompartment and provides the driving forces for mixing

b. Pressurizer Subcompartment

Hydrogen entering this subcompartment is vented through a 36-ft^2 top vent. This opening is in the vertical wall underneath the slab. Hydrogen is released above elevation 905 ft 9 in. in Region A, which is the bulk Containment volume (see Section 6.5, Figure 6.5-2).

4. Subcompartments Where There Is No LOCA

a. Pressurizer Relief Tank Subcompartment

This subcompartment has a bottom drain opening of 100 ft^2. Hydrogen released within the subcompartment and hydrogen entering the subcompartment is vented through the top vents.

b. Cubicles at Elevation 808 ft 0 in.

These cubicles are for excess letdown heat exchanger, reactor coolant drain tank and pumps, and reactor coolant drain tank heat exchanger. The Containment Building arrangement is such that the sump water will enter these cubicles and will generate hydrogen which could accumulate in the upper parts.

Consequently, all cubicles located at elevation 808 ft 0 in. are provided with a vent at their highest points to avoid local high level hydrogen concentration.

c. Reactor Cavity

Part of the fluid spilled out of the break and part of the water volume sprayed fill the reactor cavity. Hydrogen generated in the cavity by radiolysis is released through the gap around the reactor vessel and through the relief openings located in each floor (elevations 808 ft 0 in., 832 ft 6 in., and 849 ft 0 in.).

6.2.5.3.3 Electric Hydrogen Recombiners

The Electric Hydrogen Recombiners have been abandoned in place.

6.2.5.3.4 Hydrogen Purge System

The Hydrogen Purge System is capable of continually or intermittently processing a minimum of 700 cfm.

If a supply blower or exhaust fan fails, redundant fans will be able to supply or exhaust air by changing the valve and damper arrangement. Air supply and exhaust lines are arranged so as not to be rendered inoperative by accumulation of water in the line from Containment spray, condensation, or flooding.

All equipment is leaktight, and the filter housings are designed to facilitate replacement without undue exposure of personnel to radioactive sources.

The Hydrogen Purge exhaust air filtration units meet the requirements of NRC Regulatory Guide 1.140 as discussed in Appendix 1A(B).

The supply and exhaust lines are routed through different Containment penetrations. Each fan is connected to an emergency standby diesel generator bus. (Section 8.3)

The Containment isolation valves, the piping inside the Containment, and the piping between the isolation valves are ANS Safety Class 2. The exhaust equipment beyond the outboard Containment isolation valves is non-nuclear safety, seismic category II.

6.2.5.3.5 Containment Hydrogen Monitoring System

The Containment Hydrogen Monitoring System is capable of determining the hydrogen concentration at four elevations in the Containment. Four sensor modules and two microprocessors analyzers are provided to ensure that sufficient redundancy is available.

6.2.5.4 Tests and Inspections

Test programs for preoperational testing and periodic test are implemented.

1. Deleted.

2. Hydrogen Purge System

   Component qualification tests demonstrate the characteristics of materials incorporated into components (e.g., efficiency of charcoal filter).

   Component acceptance tests demonstrate the capability of the components incorporated. Fans are tested by the manufacturers to determine that their characteristic curves are within design limits.

   A post-installation test is performed to demonstrate system compliance with design requirements. During this test, fans are testing in accordance with the standards of the Air Moving and Conditioning Association (AMCA), and filters are tested in accordance with NRC Regulatory Guide 1.140 (See Appendix 1A(B)). All ductwork/piping of the Hydrogen Purge Air cleanup system are quantitatively leak tested from the outboard containment isolation valves.

   After installation the Hydrogen Purge System can be tested. The system is normally idle. Periodic tests are performed on major components to demonstrate their ability to function.

3. Containment Hydrogen Monitoring System

The Containment Hydrogen Monitoring System is calibrated when installed and periodically recalibrated in accordance with manufacturer's instructions.

The sensors will be automatically recalibrated using known calibration gases containing two (2) and six (6) percent hydrogen in high purity nitrogen. The calibration cycle will be automatically initiated at regular intervals by the microprocessor system, although manual initiation is also possible.

The Containment Hydrogen Monitoring System will be field tested in accordance with Regulatory Guide 1.68, "Preoperational and Initial Startup Test Programs for Water-Cooled Power Reactors."

6.2.5.5 Instrumentation Requirements

The hydrogen monitoring system is used in determining Containment hydrogen concentration. This measurement can be taken from any of four sensor locations within the Containment. Instrumentation is provided to both monitor the hydrogen concentration in the Containment and to monitor the Hydrogen Purge System operation. Two hydrogen indicators are provided, one mounted on the Main Control Board and the second one on the microprocessor analyzer.

The hydrogen purge supply blowers and exhaust fans are manually started from the Control Room. A humidity control heater located in the filter is interlocked with the fan to come on when the fan is started and to shut off when the fan is stopped. A thermistor is provided on the discharge side of the iodine adsorber to provide a high temperature signal to the Fire Protection Systems panel. Differential pressure switches and/or indicating switches are provided to monitor the differential pressure across the fans and exhaust filtration units. A low alarm is annunciated from the fan switch and a high alarm from the filter bank.

6.2.5.6 Materials

The materials of construction for the electric hydrogen recombiners are selected for their compatibility with the post-LOCA environment.

The major structural components are manufactured from 300-Series stainless steel. Incoloy-800 is used for the heater sheaths and Inconel-600 for other parts such as the heat duct, which operates at high temperature.

There are no radiolytic or pyrolytic decomposition products from these materials. Materials of construction for Containment Hydrogen Purge System components are listed in Table 6.2.5-6.

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 41, Containment Atmosphere Cleanup.

2. 10 CFR Part 50, Appendix A, General Design Criterion 42, Inspection of Containment Atmosphere Cleanup Systems.

3. 10 CFR Part 50, Appendix A, General Design Criterion 43, Testing of Containment Atmosphere Cleanup Systems.

4. Deleted.

5. Deleted.

6. NRC Regulatory Guide 1.22, Periodic Testing of Protection System Actuation Functions, February 1972, U.S. Nuclear Regulatory Commission.

7. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam-, and Radioactive-Waste-Containing Components of Nuclear Power Plants, Rev. 3, February 1976, U.S. Nuclear Regulatory Commission.

8. NRC Regulatory Guide 1.29, Seismic Design Classification, Rev. 2, February 1976, U.S. Nuclear Regulatory Commission.

9. ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973.

10. Branch Technical Position APCSB 9-2, Residual Decay Energy for Light Water Reactors for Long-Term Cooling.

11. Branch Technical Position CSB 6-2, Control of Combustible Gas Concentrations In Containment Following a Loss of Coolant Accident, Nov. 24, 1975.

12. J. F. Wilson, Electric Hydrogen Recombiner for PWR Containments, WCAP-7709 (Proprietary) and WCAP-7820, Supplements 1, 2, 3, 4 and 5 (Nonproprietary).

13. TID-14844, Calculation of Distance Factors for Power and Test Reactor Sites, March 23, 1962.

14. ANSI N101.1, Efficiency Testing of Air-Cleaning Systems Containing Devices for Removal of Particles, 1972.

15. Letter from D. B. Vassallo (NRC) to C. Eicheldinger (Westinghouse) dated May 1, 1975.

16. Letter from J. Stolz (NRC) to T. M. Anderson (Westinghouse) dated June 22, 1978.

6.2.5A HYDROGEN PRODUCTION AND ACCUMULATION

This section has been deleted.

6.2.6 CONTAINMENT LEAKAGE TESTING

The Containment Building, Containment penetrations, and Containment isolation barriers are designed to permit periodic leakage rate testing as required by GDC 52, 53, and 54 of Title 10, Code of Federal Regulations, Part 50, Appendix A [2] [3] [4], in accordance with the requirements of 10 CFR Part 50, Appendix J, Option B [1]. Containment leakage tests are performed to verify that Containment leakage is maintained below the acceptable limits stated in the Technical Specifications.

The leakage testing program consists of the following types of leakage tests:

1. Type A Tests

   Type A tests are those tests, as defined in 10 CFR Part 50, Appendix J, Option B, to measure the containment system overall integrated leakage rate, conducted after the Containment System has been completed, and is ready for operation, and at a periodic interval based on the historical performance of the overall Containment System.

2. Type B Tests

   Type B tests are those tests, as defined by 10 CFR Part 50 Appendix J, Option B, which are performed to measure leakage rates across pressure retaining or leakage limiting boundaries other than valves.

3. Type C Tests

   Type C tests are those tests, as defined by 10 CFR Part 50, Option B, which are performed to measure Containment isolation valve leakage rates.

6.2.6.1 Containment Integrated Leakage Rate Test (Type A Test)

The maximum allowable Containment integrated leakage rate for the Comanche Peak Nuclear Power Plant (CPNPP) is 0.10 weight percent per day at a pressure of P_a equal to the calculated peak Containment internal pressure as specified in the Technical Specifications. After completion of construction of the Containment and the installation of all mechanical, fluid, electrical, and instrumentation systems penetrating the Containment pressure boundary, a Containment structural acceptance test was performed as described in Section 3.8.1. A detailed description of the preparation and inspection of the Containment structures, which is performed prior to the Containment structural acceptance test, is given in Section 3.8.1. For Unit 1 initial Type A leakage rate test was performed at a reduced pressure P_t equal to 0.5 P_a following completion of the structural acceptance test. The reduced pressure leakage-rate test was initiated when the Containment pressure had been reduced to P_t. A peak pressure test was then performed at the calculated peak internal pressure P_a. Leakage-rate tests are performed periodically during the operating life of the plant in accordance with 10CFR50, Appendix J, Option B.

Containment integrated leakage rate tests are performed in accordance with the requirements of 10 CFR Part 50 Appendix J, Option B, ANSI/ANS-56.8-1994 “Containment System Leakage Testing Requirements,” NEI 94-01 “Industry Guideline for Implementing Performance - Based Option of 10CFR Part 50 Appendix J”, and also allowed performance of reduced duration periodic tests using the Bechtel Topical Report BN-TOP-1 criteria and methodology.

Containment System Leakage Testing Requirements. For periodic integrated tests, minimum pathway local leakage rate test corrections will be used, where applicable, for isolated penetrations.

Prior to performance of Type A leakage-rate tests, the Containment isolation valves are closed by normal operation without any adjustment. All piping assemblies are equipped with low point

drains and high point vents. This will allow exposing the piping systems as applicable, including the portions which are part of the Containment isolation system, to Containment atmosphere during Containment integrated leakage rate (type A) test. Portions of some systems which penetrate the Containment and are located inside the Containment or are directly connected to the penetration outside the Containment are drained and vented during the leakage-rate test. The portion inside the Containment is vented to the Containment atmosphere and the portion outside is vented to the environment to ensure that the full Containment test air pressure is imposed on the isolation valves. The portions of the RCS constituting the reactor coolant pressure boundary (RCPB) need not be drained or vented since no portion of the RCPB penetrates the Containment. The RHR system lines and those safety injection system lines which will be filled with water and required to operate during an accident are not drained and vented during the leakage-rate test. Table 6.2.4-4 lists penetrations which are filled with water during a Containment leakage rate (Type A) test and therefore not exposed to Containment atmosphere. These fluid lines are part of systems that during post-accident recovery are operating at a pressure significantly higher than Containment pressure, or are required for safe and efficient conduct of the test, or are lines designed in accordance with NRC Regulatory Guide 1.11. Penetrations with isolation valve arrangements conforming to GDC 57 (Section 6.2.4.1.2) per Table 6.2.4-1 are not necessarily drained or vented during the Type A test and are not listed in Table 6.2.4-4. In addition, for planning and scheduling purposes or ALARA considerations, penetrations which are Type B or C test either within the previous 24 calendar months prior to the ILRT or during the ILRT outage, need not be vented or drained during the Type A test and are not necessarily listed on Table 6.2.4-4.

The Containment is pressurized to the test pressure P_a and air mass allowed to stabilize for a minimum of four hours. During this stabilization period as well as during the test, the Containment Air Recirculation and Cooling System may be operated, if required, to circulate the Containment atmosphere and provide a homogeneous mixture. Containment cooling may be provided to maintain a constant temperature. If possible, the Containment integrated leakage-rate tests are performed during periods of relatively stable weather conditions.

Upon commencement of the leakage-rate test following the period of stabilization, the following parameters are measured and recorded for the duration of the test:

1. Containment absolute pressure

2. Containment dry bulb temperature

3. Containment dew point temperature or relative humidity (to determine vapor pressure of moisture in Containment atmosphere)

Using the Mass Point method of ANSI/ANS-56.8-1994 a periodic test extends for at least 8 hours. Reduced duration periodic tests may also be performed using the Bechtel Topical Report BN-TOP-1 criteria and methodology. The temperature measurements are taken from a sufficient number of separate locations in the Containment to permit determination of a mean representative temperature. Leakage-rates are calculated using previously mentioned data and the methods from ANSI/ANS 56.8-1994 and/or BN-TOP-1. The Containment pressure is corrected for the partial pressure of water vapor.

Upon completion of the leakage-rate test the results of the test are validated by a verification test in accordance with the method outlined in ANSI N56.8-1994 and/or BN-TOP-1. This verification essentially consists of imposing a calibrated leakage on the Containment vessel in addition to existing leakages. Readings of Containment parameters are resumed and a new leakage-rate is calculated to be compared with the originally calculated leakage rate.

Acceptance criteria for the Containment integrated leakage-rate tests are as follows:

1. Preoperational Tests

   Peak pressure test leakage rate L_{am} is acceptable if it is less than 75 percent of L_a and not greater that L_d as previously defined by 10 CFR Part 50, Appendix J.

2. Periodic Tests

   The “As Left” containment integrated leakage rate is acceptable if the leakage at the 95% UCL plus all applicable additions and corrections is less than or equal to 0.75 La as specified in the Technical Specifications.

   The “As Found” containment integrated leakage rate is acceptable if the leakage at the 95% UCL plus all applicable additions, corrections, and leakage improvements (from repairs/adjustments performed prior to the ILRT), is less than or equal to 1.0 L_a as specified in the Technical Specifications. For preoperational/prestart test program tests, the sum of the post repaired minimum path local leakage rate valves was added to the upper confidence limit (UCL) per ANSI 56.8-1987.

6.2.6.2 Containment Penetration Leakage Rate Test (Type B Test)

Tables 6.2.4-1, 6.2.4-2 and 6.2.4-3 list all Containment mechanical penetrations as well as the type of penetration and leakage testing method. Figure 8.3-16 lists all Containment electrical penetrations (which are Type B tested). The Containment pressure sensing instrument penetrations are not subject to Type B testing. Each Containment pressure sensing instrument constitutes a closed, liquid-filled system sealed on the inside of the Containment by a diaphragm and on the outside by the pressure transmitter; both are designed to withstand Containment pressure and temperature and postaccident environment. The sealed pressure sensing system is designed to withstand the SSE and is protected from postulated missiles as well as pipe whip and jet impingement effects resulting from postulated pipe ruptures.

The leakage-rate tests performed on the various types of penetrations are described in the following paragraphs:

1. The equipment hatch, personnel airlock, emergency airlock, and fuel transfer tube are provided with covers sealed by gaskets designed so that the space between the gasket sealing surfaces can be pressurized. The seal is tested by either pressurized air or nitrogen P_a and measuring the leakage rate.

   The hydraulic system which constitutes the personnel airlock operating mechanism (See Section 3.8.1), contains some non-metallic seals and component necessary for it to perform its design function. These materials have been environmentally qualified for

post-LOCA conditions. In normal use during the opening and closing of the airlock door, these seals and components are normally pressurized to between 100 psig to 500 psig. Any leakage or failure would be readily visible at the point of Containment ingress and egress. In addition, the Unit 1 hydraulic system boundary is included in the Type A and airlock barrel tests. Therefore, no additional testing is required.

2. The fuel transfer tube penetration consists of a sleeve embedded in the Containment wall and welded to the liner through which the transfer tube passes. The sleeve is sealed to the transfer tube by two bellows expansion joints, one on each side of the penetration. Figure 6.2.6-1 shows the fuel transfer tube leak test arrangement.

   A test connection is provided so that the space between the transfer tube and the sleeve with connecting bellows can be pressurized to Pa in order to measure the leakage rate of the bellows or attachment welds.

3. The Containment recirculation sump penetrations consist of sleeves embedded in the Containment mat with the process pipe seal welded to the sleeve by a seal ring inside the Containment. The sleeve is welded to the Containment liner. Each isolation valve outside the Containment is enclosed within a valve isolation tank which is sealed to the sleeve by a 24 inch guard pipe and to the process pipe downstream of the isolation valve by a bellows expansion joint (see Figure 6.2.2-3). The bellows, guard pipe and isolation tank assembly do not require testing as discussed in Section 6.2.4.1.3.

   All other mechanical penetrations do not incorporate any expansion joints or resilient seals. They consist of either a pipe embedded in the Containment wall concrete and welded to the Containment liner or a sleeve embedded and welded to the liner with the process pipe passing through the sleeve and sealed by a flued head welded to the sleeve. These penetrations are tested by a type C test performed on the isolation valves as described in Subsection 6.2.6.3.

4. Maintenance and Containment Leakage Rate Test

   Pressurization/pressure sensing penetrations consist of pipes or sleeves embedded in the Containment wall concrete, welded to the Containment liner and blind flanged shut at each end. A test connection is provided so that the space between the end flanges can be pressurized to P_a in order to measure the leakage rate.

5. Electrical penetrations are tested as follows:

   The design of each electrical penetration assembly includes two static seals at pressure retaining boundary interfaces. The pressure retaining integrity of these seals is monitored by a pressure gauge located outside of the Containment. To determine leak rate of the penetration pressure system, the assembly is pressurized with dry nitrogen to a pressure equal to P_a.

   The results of Containment penetration leakage-rate tests are acceptable if the combined leakage rates of all penetrations and valves subject to type B and C tests are less than or equal to 60 percent of L_a as defined by Technical Specifications.

Local test connections, vent and drain valves (TVDs) within the penetration boundary are subject to the same leak rate testing as the other containment isolation valves in the associated penetration, including all applicable leak testing exceptions (see FSAR Table 6.2.4-2 including notes). However, TVDs in the penetration boundary that are one inch or less in size, consist of a double barrier (such as capped or isolated by blind flange) and under administrative controls are not required to be leak rate tested.

6.2.6.3 Containment Isolation Valve Leakage Rate Test (Type C Test)

All Containment isolation valves are listed on Tables 6.2.4-1, 2 and 3, Part B, with applicability of Type C testing indicated for each valve or set of valves. Each valve or set of valves subject to Type C testing will be tested by pressurizing with air or nitrogen to maintain a differential pressure of at least Pa across the valve(s). Test connections and test vents are provided in the piping to facilitate Type C testing. The results of Containment isolation valve leakage rate testing are acceptable if the combined leakage rates of all penetrations and valves subject to Type B and C testing are less than or equal to 60 percent of L_a as defined by Technical Specifications.

Local test connections, vent and drain valves (TVDs) within the penetration boundary are subject to the same leak rate testing as the other containment isolation valves in the associated penetration, including all applicable leak testing exceptions (see FSAR Table 6.2.4-2 including notes). However, TVDs in the penetration boundary that are one inch or less in size, consist of a double barrier (such as capped or isolated by blind flange), and under administrative controls are not required to be leak rate tested.

6.2.6.4 Scheduling and Reporting of Periodic Tests

1. Containment integrated leakage-rate test (Type A) are performed at least once per 48 months with possible extension to 10 yr. intervals based upon acceptable performance history as determined in accordance with NEI 94-01 and Reg. Guide 1.163, as modified by the exceptions in Technical Specification 5.5.16.a. Type A tests are only conducted while the plant is in the shutdown condition.

2. Containment penetration leakage-rate tests (Type B) are performed at 24 month intervals with possible extension up to 10 yr. intervals as determined in accordance with NEI 94-01. The personnel airlock and emergency airlock door seals are tested after opening and overall leak rate tested at 30 month intervals.

3. Containment isolation valve leakage-rate test (Type C) are performed at 24 month intervals with possible extension to 5 yr. intervals as determined in accordance with NEI 94-01 and Reg. Guide 1.163.

4. The preoperational test results are documented in accordance with the requirements of 10 CFR 50, Appendix J, Section V and periodic test results are documented in accordance with the requirements of 10 CFR Part 50, Appendix J, Option B, Section IV.

6.2.6.5 Special Testing Requirements

Any major modification or replacement of a component which is part of the primary reactor Containment performed after the preoperational leakage-rate test will be followed by a Type A,

Type B, or Type C test as applicable. Testing may be deferred to the next regularly scheduled Type A test for the following repairs or modifications:

- Welds of attachments to the surface of steel pressure-retaining boundary;
- Repair cavities, the depth that does not penetrate required design steel wall by more than 10%; or
- Welds attaching to steel pressure-retaining boundary penetrations, where the nominal diameter of the welds or penetrations does not exceed one inch.

All other requirements for regularly scheduled leakage-rate tests apply to the previously mentioned special tests.

REFERENCES

1. 10 CFR Part 50, Appendix J, Primary Reactor Containment Leakage Testing for Water-Cooled Power Reactors, Option B, Performance Based Requirements.
2. 10 CFR Part 50, Appendix A, General Design Criterion 52, Capability for Containment Leakage Rate Testing.
3. 10 CFR Part 50, Appendix A, General Design Criterion 53, Provisions for Containment Testing and Inspection.
4. 10 CFR Part 50, Appendix A, General Design Criterion 54, Piping Systems Penetrating Containment.
5. ANSI N45.4-1972, Leakage-Rate Testing of Containment Structures for Nuclear Reactors, 1972.
6. ANSI/ANS 56.8-1994, Containment System Leakage Testing Requirements.
7. ANSI/ANS 56.8-1987, Containment System Leakage Testing Requirements, 1987.
8. Bechtel Topical Report BN-TOP-1, Testing Criteria for Integrated Leakage Rate Testing of Primary Containment Structures for Nuclear Power Plants, Rev. 1, November 1, 1972.
9. WCAP-8204, Westinghouse Mass and Energy Release Data for Containment Design.
10. NEDO-13418, Critical Flow of Saturated and Subcooled Water at High Pressure, Sozzi, G. L. and Sutherland, W. A.
11. Regulatory Guide 1.163, “Performance-Based Containment Leak-Test Program”, September 1995.
12. NEI 94-01, Rev. 0, Nuclear Energy Institute “Industry Guideline for Implementing Performance-Based Option of 10 CFR Part 50, Appendix J”, July 26, 1995.

TABLE 6.2.1-1

THIS TABLE HAS BEEN DELETED

TABLE 6.2.1-2
CONTAINMENT PEAK PRESSURE AND TEMPERATURE ANALYSIS FOR LOCA

BREAK	INITIAL CONTAINMENT CONDITIONS P/T/RH[a] (PSIA/°F/%)	SINGLE FAILURE	CONTAINMENT PEAK PRESSURE (PSIG)	TIME FOR PEAK PRESSURE (SEC)	CONTAINMENT PEAK TEMPERATURE (°F)	TIME FOR PEAK TEMPERATURE (SEC)	PRESSURE MARGIN (psi)	PRESSURE MARGIN AS A PERCENT OF PEAK PRESS (%)
DEPSG	16.2/120/15	1 CS TRAIN	40.2	74.6	258.6	24.0	9.8	24.4
DEPSG[b]	16.2/120/15	1 ESF TRAIN	45.2	629.7	266.0	629.6	4.8	10.6
DEPSG	16.2/120/15	NONE - BLOWDOWN ONLY	41.1	22.0	261.0	21.5	8.9	21.6

a) RH = Relative Humidity

b) The limiting LOCA scenarios with respect to peak injection phase pressure and/or temperature, highest pressure or temperature at 24 hours, and highest recirculation phase temperature are reanalyzed to evaluate modifications to analysis input models.

c) The temperature and pressure profiles utilized for EQ evaluations bound the limiting temperature and pressure transients for postulated LOCA and MSLB events.

Notes:

1) CS - Containment Spray

2) ESF - Engineered Safety Features (ECCS & CSS)

TABLE 6.2.1-2A
CONTAINMENT PEAK PRESSURE AND TEMPERATURE FOR A SPECTRUM OF STEAM LINE BREAKS
(Sheet 1 of 2)

BREAK DESCRIPTION	POWER LEVEL	ASSUMED FAILURE	INITIAL CONTAINMENT CONDITIONS P/T/RH (PSIA/°F/%)	PEAK PRESSURE (PSIG)	TIME AFTER BREAK TO PEAK PRESSURE (SEC)	PRESSURE MARGIN (PSI)	PRESSURE MARGIN AS A PERCENT OF PEAK PRESS (%)	PEAK TEMPERATURE (°F)	TIME AFTER BREAK TO PEAK TEMPERATURE (SEC)
FULL DE	100.6%	MSIV	16.2/120/15	33.0	200.6	17.0	51.5	316.9	50.5
FULL DE	70%	MSIV	16.2/120/15	33.9	240.2	16.1	47.5	313.8	50.1
FULL DE	30%	MSIV	16.2/120/15	34.7	360.2	15.3	44.1	312.0	50.1
FULL DE	0%	MSIV	16.2/120/15	36.1	260.2	13.9	38.5	305.9	60.1
3 FT^2 SPLIT RUPTURE	100.6%	MSIV	16.2/120/15	34.5	200.2	15.5	44.9	324.9	40.1
5 FT^2 SPLIT RUPTURE	70%	MSIV	16.2/120/15	35.6	240.2	14.4	40.4	324.3	40.1
7 FT^2 SPLIT RUPTURE	30%	MSIV	16.2/120/15	36.9	380.2	13.1	35.5	324.1	30.1
7 FT^2 SPLIT RUPTURE	0%	MSIV	16.2/120/15	37.7	280.2	12.3	32.6	318.2	30.1
FULL DE	100.6%	1 CS TRAIN	16.2/120/15	32.7	620.2	17.3	52.9	307.1	80.2
FULL DE	70%	1 CS TRAIN	16.2/120/15	34.3	620.2	15.7	45.8	304.7	80.1
FULL DE	30%	1 CS TRAIN	16.2/120/15	36.6	620.2	13.4	36.6	303.2	90.2
FULL DE	0%	1 CS TRAIN	16.2/120/15	36.4	260.2	13.6	37.4	299.5	100.2
4.3 FT^2 DE SPLIT RUPTURE	100.6%	1 CS TRAIN	16.2/120/15	34.2	620.2	15.8	46.2	312.9	70.1
4.5 FT^2 DE SPLIT RUPTURE	70%	1 CS TRAIN	16.2/120/15	35.9	620.2	14.1	39.3	311.6	60.1

TABLE 6.2.1-2A
CONTAINMENT PEAK PRESSURE AND TEMPERATURE FOR A SPECTRUM OF STEAM LINE BREAKS
(Sheet 2 of 2)

BREAK DESCRIPTION	POWER LEVEL	ASSUMED FAILURE	INITIAL CONTAINMENT CONDITIONS P/T/RH (PSIA/°F/%)	PEAK PRESSURE (PSIG)	TIME AFTER BREAK TO PEAK PRESSURE (SEC)	PRESSURE MARGIN (PSI)	PRESSURE MARGIN AS A PERCENT OF PEAK PRESS (%)	PEAK TEMPERATURE (°F)	TIME AFTER BREAK TO PEAK TEMPERATURE (SEC)
4.7 FT^2 DE SPLIT RUPTURE	30%	1 CS TRAIN	16.2/120/15	39.0	620.2	11.0	28.2	311.1	50.2
4.7 FT^2 DE SPLIT RUPTURE	0%	1 CS TRAIN	16.2/120/15	38.3	280.2	11.7	30.5	305.3	50.1

a) The temperature and pressure profiles utilized for EQ evaluations bounds the limiting temperature and pressure transients for postualated LOCA and MSLB events.

TABLE 6.2.1-2B
THIS TABLE HAS BEEN DELETED

TABLE 6.2.1-2C
THIS TABLE HAS BEEN DELETED

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 1 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.00000	0.0	0.0	0.0	0.0
0.00109	89631.0	50303.7	43101.2	24133.7
0.004	43153.4	24164.6	42333.3	23699.9
0.006	43135.8	24155.7	41599.8	23285.4
0.009	43116.1	24145.5	40856.0	22865.0
0.011	43097.5	24135.9	40108.4	22442.4
0.014	43076.1	24124.8	39178.4	21916.8
0.016	43058.8	24115.9	38344.4	21445.4
0.019	43040.5	24106.5	37383.3	20902.2
0.021	43022.0	24097.1	36306.7	20293.8
0.024	43004.2	24088.0	35172.9	19653.1
0.026	42986.1	24078.8	33921.1	18946.0
0.029	42969.5	24070.4	32682.6	18246.9
0.031	42952.2	24061.8	31319.8	17478.4
0.034	42936.4	24054.0	30029.2	16751.0
0.036	42921.8	24046.7	28809.9	16064.8
0.039	42906.5	24039.2	27539.5	15351.1
0.041	42890.7	24031.6	26283.7	14647.2
0.044	42876.7	24024.9	25293.9	14093.6
0.046	42864.9	24019.3	24543.4	13675.1
0.049	42851.7	24013.1	23805.7	13265.1

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 2 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.051	42837.5	24006.5	23127.0	12889.7
0.054	42827.5	24002.0	22721.0	12666.4
0.056	42814.3	23996.1	22275.6	12422.9
0.059	42805.3	23992.2	22027.1	12288.0
0.061	42795.3	23988.0	21795.9	12163.5
0.064	42786.5	23984.5	21631.3	12075.8
0.066	42779.4	23981.9	21522.0	12018.4
0.069	42773.7	23980.4	21450.6	11982.1
0.071	42770.9	23980.5	21414.7	11964.9
0.074	42771.3	23982.4	21405.7	11962.3
0.076	42774.5	23986.1	21417.9	11971.3
0.079	42778.4	23989.8	21439.8	11985.0
0.081	42784.7	23995.4	21480.5	12009.3
0.084	42791.0	24000.6	21523.7	12034.6
0.087	42800.6	24008.4	21595.2	12075.8
0.089	42809.2	24015.2	21663.1	12114.6
0.091	42817.8	24021.8	21733.0	12154.2
0.094	42828.6	24030.3	21819.4	12203.2
0.096	42837.1	24036.7	21886.5	12241.1
0.099	42850.5	24046.7	21986.3	12297.4
0.101	42863.1	24056.0	22070.0	12344.7

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 3 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.126	43019.4	24167.3	22862.3	12792.5
0.151	43221.4	24308.7	23621.2	13220.4
0.176	43476.7	24484.0	24161.5	13524.6
0.20	48419.9	27309.4	24508.3	13720.4
0.23	48409.3	27341.2	24689.3	13824.7
0.25	48257.2	27295.8	24768.4	13872.7
0.28	48100.7	27248.7	24812.5	13900.5
0.30	48038.0	27258.0	24752.9	13870.3
0.33	47973.9	27268.0	24624.9	13802.2
0.35	47932.9	27291.1	24454.8	13710.6
0.38	47892.0	27319.3	24246.6	13597.8
0.40	47818.5	27330.0	24036.2	13483.0
0.43	47926.4	27445.4	23786.3	13345.7
0.45	48023.3	27557.7	23509.2	13193.1
0.48	48118.8	27672.0	23238.3	13043.9
0.50	48196.0	27777.4	22997.7	12911.2
0.70	46604.5	27354.8	21223.0	11921.9
0.90	46836.2	27954.2	20165.7	11331.9
1.10	45532.7	27575.9	19583.3	11007.0
1.30	43508.8	26704.9	19417.5	10915.1
1.50	41618.5	25893.4	19307.9	10853.4

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 4 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
1.70	39619.5	25013.9	19250.8	10820.7
1.90	37364.4	23997.8	19209.1	10797.0
2.10	34901.2	22850.2	19070.6	10718.7
2.30	32406.6	21633.6	18820.8	10578.3
2.50	30049.4	20432.1	18181.4	10217.6
2.50	30035.9	20425.1	18177.2	10215.3
2.50	30032.0	20423.5	18176.0	10214.6
2.75	25157.4	17454.5	17611.4	9899.9
3.00	21748.6	15405.9	17056.2	9590.1
3.25	19367.2	13890.6	16566.4	9318.1
3.50	17655.5	12775.5	16139.6	9081.7
3.75	16354.3	11919.8	15793.5	8890.8
4.00	15380.1	11268.7	15475.8	8715.7
4.25	14697.8	10802.3	15185.6	8555.8
4.50	14183.9	10434.0	14910.7	8404.4
4.75	13815.4	10152.0	14638.5	8253.7
5.00	13547.5	9928.9	16258.5	9174.0
5.25	13338.0	9737.7	15995.5	9025.6
5.50	13238.1	9620.2	15666.9	8842.0
5.75	13205.8	9548.0	15567.2	8788.6
6.00	13219.3	9509.8	15402.0	8698.4

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 5 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
6.25	13261.2	9494.4	15215.5	8596.1
6.50	13318.5	9491.5	15051.4	8506.2
6.75	13422.7	9519.4	14942.9	8447.2
7.00	13569.0	9565.3	14761.8	8345.4
7.25	13771.4	9639.7	14570.3	8237.2
7.50	14029.2	9744.4	14408.3	8145.4
7.75	14540.6	10015.5	14230.7	8044.4
8.00	14183.1	9753.9	14214.1	8035.5
8.25	13259.4	9497.5	13893.6	7851.4
8.50	12130.8	9000.4	13714.0	7749.9
8.75	12031.0	8924.8	13519.7	7640.4
9.00	12153.1	8938.9	13278.0	7502.9
9.25	12197.7	8897.5	13055.8	7376.8
9.50	12276.8	8887.4	12840.2	7254.3
9.75	12303.3	8830.0	12607.1	7121.4
10.00	12243.0	8722.3	12410.4	7009.7
10.25	12164.8	8612.3	12200.1	6889.6
10.50	11954.3	8418.6	12011.2	6782.0
10.75	11640.9	8176.7	11845.5	6687.4
10.75	11638.4	8174.8	11844.0	6686.6
10.75	11637.1	8173.8	11843.3	6686.1

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 6 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
10.75	11636.0	8173.1	11842.6	6685.8
11.00	11346.6	7962.3	11671.8	6588.1
11.25	11015.1	7731.1	11525.7	6504.6
11.50	10704.2	7529.3	11378.2	6420.2
11.75	10454.7	7372.4	11224.8	6332.4
12.00	10179.3	7196.2	11094.8	6258.3
12.25	9937.4	7048.2	10946.5	6173.3
12.50	9706.6	6903.5	10817.1	6099.4
12.75	9463.3	6752.5	10686.6	6024.6
13.00	9251.2	6624.8	10554.5	5948.9
13.25	9027.6	6488.0	10435.9	5881.0
13.50	8822.0	6365.9	10308.2	5807.9
13.75	8615.9	6243.4	10194.3	5742.8
14.00	8437.3	6140.2	10069.5	5671.7
14.25	8255.8	6031.4	9953.5	5606.0
14.50	8081.9	5926.7	9820.7	5531.0
14.75	7890.6	5810.1	9676.2	5450.0
15.00	7683.1	5683.4	9530.6	5369.3
15.25	7472.9	5549.3	9381.8	5287.4
15.50	7291.8	5425.0	9248.2	5214.4
15.75	7147.1	5316.9	9115.8	5142.3

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 7 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
16.00	7021.7	5221.3	8988.4	5074.3
16.25	6894.2	5131.0	8847.0	4999.4
16.50	6746.2	5037.9	8688.1	4915.9
16.75	6555.8	4930.9	8466.3	4797.4
17.00	6358.2	4841.5	8208.3	4658.4
17.25	6141.4	4756.6	8034.6	4569.2
17.50	5893.9	4658.7	7713.5	4394.7
17.75	5643.3	4561.4	7403.7	4229.5
18.00	5396.3	4473.6	7133.1	4090.3
18.25	5224.4	4444.4	6717.2	3960.5
18.50	5111.6	4559.8	6273.8	3784.1
18.75	4467.7	4503.2	5790.5	3505.0
19.00	3647.1	4222.8	5328.8	3231.1
19.25	3130.0	3830.2	4830.5	2946.8
19.50	2714.0	3356.9	4325.2	2673.1
19.75	2386.1	2966.7	3861.3	2419.2
20.00	2104.9	2626.9	3449.5	2187.1
20.25	1856.6	2324.6	3079.2	1973.1
20.50	1637.1	2055.1	2751.9	1783.7
20.75	1445.9	1819.5	2448.4	1599.3
21.00	1279.4	1613.3	2178.5	1414.0

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 8 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
21.25	1135.3	1434.1	1940.7	1231.8
21.50	1002.2	1268.4	1729.4	1071.4
21.75	876.5	1111.0	1547.6	957.5
22.00	765.6	971.4	1382.2	861.4
22.25	671.5	852.9	1232.3	769.0
22.50	578.7	735.6	1085.7	749.3
22.75	494.4	629.0	917.6	701.3
23.00	423.4	539.0	793.1	653.9
23.25	385.6	491.2	703.6	653.4
23.50	358.3	456.6	569.3	613.8
23.75	220.5	280.8	488.8	530.1
24.00	165.7	211.6	319.3	328.5
24.25	122.1	156.2	220.5	181.1
24.50	129.2	165.4	188.1	131.6
24.75	161.3	206.3	224.1	138.8
25.00	196.9	251.7	331.6	182.2
25.25	200.5	256.3	478.9	253.8
25.50	234.3	299.4	484.6	330.9
25.75	217.1	277.6	284.7	305.4
26.00	221.2	282.8	298.4	340.8
26.25	221.1	282.8	322.3	341.9

Table 6.2.1-3
BLOWDOWN MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION
(Sheet 9 of 9)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
26.50	261.3	334.2	146.7	177.5
26.75	296.6	379.1	323.5	297.9
27.00	288.1	368.4	105.2	104.9
27.25	290.9	372.2	145.7	181.9
27.50	363.1	464.6	133.8	174.3
27.75	450.9	576.6	266.9	271.4
28.00	404.7	517.3	557.1	353.6
28.25	255.5	327.0	476.9	267.4
28.50	111.4	142.8	105.3	77.4
28.75	0.0	0.0	0.0	0.0

a) M&E exiting from the steam generator side of the break.

b) M&E exiting from the broken loop reactor coolant pump side of the break.

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 1 of 8)

	Break Path No. 1[(a)]		Break Path No. 2[(b)]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
28.8	0.0	0.0	0.0	0.0
29.2	0.0	0.0	0.0	0.0
29.4	0.0	0.0	0.0	0.0
29.5	0.0	0.0	0.0	0.0
29.6	0.0	0.0	0.0	0.0
29.7	0.0	0.0	0.0	0.0
29.8	0.0	0.0	0.0	0.0
29.9	88.3	104.1	0.0	0.0
30.0	36.0	42.4	0.0	0.0
30.1	32.4	38.2	0.0	0.0
30.2	40.0	47.1	0.0	0.0
30.3	48.8	57.5	0.0	0.0
30.4	54.5	64.2	0.0	0.0
30.5	60.1	70.9	0.0	0.0
30.6	65.4	77.1	0.0	0.0
30.7	70.4	83.0	0.0	0.0
30.8	75.1	88.6	0.0	0.0
30.9	79.7	93.9	0.0	0.0
31.0	84.0	99.1	0.0	0.0
31.1	88.2	104.0	0.0	0.0
31.2	92.2	108.8	0.0	0.0

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 2 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
31.3	96.1	113.4	0.0	0.0
31.4	100.7	118.8	155.2	13.7
31.5	104.7	123.5	155.2	13.7
31.6	108.5	127.9	155.2	13.7
31.7	112.2	132.3	155.2	13.7
31.8	115.7	136.5	155.2	13.7
32.8	147.4	173.9	155.2	13.7
33.8	173.9	205.3	155.2	13.7
34.9	589.4	700.0	5256.0	750.4
35.0	597.5	709.7	5310.2	765.3
35.9	604.6	718.3	5357.4	785.5
36.9	597.2	709.5	5300.3	779.8
37.9	588.6	699.1	5230.8	771.9
38.9	579.5	688.3	5157.2	763.2
39.9	570.5	677.5	5082.6	754.2
40.9	561.6	666.8	5008.6	745.2
41.9	552.9	656.4	4935.8	736.3
42.9	544.5	646.3	4864.7	727.5
43.9	536.4	636.6	4795.5	719.0
44.0	535.6	635.6	4788.7	718.2
44.9	528.5	627.1	4728.3	710.7

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 3 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
45.9	520.9	618.1	4662.9	702.6
46.9	430.0	509.9	3547.0	633.9
47.9	412.7	488.8	3653.1	588.1
48.9	557.1	661.3	366.1	282.9
49.9	571.0	678.0	371.6	291.1
50.9	552.4	655.8	363.7	280.6
51.9	533.1	632.7	355.7	269.7
52.9	515.9	612.0	348.5	260.1
53.9	499.6	592.6	341.8	251.1
54.9	484.2	574.1	335.4	242.5
55.9	469.5	556.5	329.3	234.5
56.0	468.0	554.8	328.7	233.7
56.9	455.5	539.8	323.6	226.9
57.9	442.2	523.9	318.1	219.6
58.9	429.4	508.7	313.0	212.8
59.9	417.3	494.3	308.0	206.3
60.9	405.7	480.4	303.3	200.1
61.9	394.6	467.2	298.9	194.2
62.9	384.1	454.6	294.6	188.7
63.9	373.9	442.6	290.6	183.4
64.9	364.3	431.1	286.7	178.3

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 4 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
65.9	355.0	420.1	283.1	173.6
66.9	346.2	409.5	279.6	169.0
67.9	337.7	399.5	276.2	164.7
68.9	329.7	389.9	273.1	160.6
69.9	322.0	380.7	270.1	156.7
70.9	314.6	372.0	267.2	152.9
71.9	307.5	363.6	264.5	149.4
72.1	306.2	362.0	263.9	148.7
72.9	300.8	355.6	261.9	146.1
73.9	294.4	348.0	259.4	142.9
74.9	288.3	340.7	257.1	139.9
75.9	282.4	333.8	254.8	137.0
76.9	276.9	327.2	252.7	134.3
77.9	271.6	320.9	250.7	131.7
78.9	266.5	314.9	248.8	129.2
79.9	261.7	309.2	247.0	126.9
80.9	257.1	303.8	245.3	124.7
81.9	252.8	298.6	243.6	122.6
82.9	248.6	293.7	242.1	120.7
83.9	244.7	289.0	240.6	118.8
84.9	241.0	284.6	239.3	117.0

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 5 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
85.9	237.4	280.4	238.0	115.4
86.9	234.0	276.4	236.7	113.8
87.9	230.9	272.6	235.6	112.3
88.9	227.8	269.1	234.5	110.9
89.9	225.0	265.7	233.4	109.6
91.9	219.7	259.5	231.5	107.2
93.9	215.0	253.9	229.8	105.0
94.2	214.4	253.1	229.6	104.7
95.9	210.9	249.0	228.3	103.1
97.9	207.2	244.6	227.0	101.4
99.9	203.9	240.8	225.8	100.0
101.9	201.1	237.4	224.8	98.7
103.9	198.5	234.4	223.9	97.5
105.9	196.3	231.7	223.1	96.5
107.9	194.4	229.5	222.4	95.7
109.9	192.7	227.5	221.8	94.9
111.9	191.3	225.8	221.3	94.3
113.9	190.1	224.4	220.9	93.7
115.9	189.0	223.1	220.5	93.2
117.9	188.2	222.1	220.2	92.8
119.9	187.5	221.3	219.9	92.5

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 6 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
120.8	187.2	220.9	219.8	92.4
121.9	186.9	220.6	219.7	92.2
123.9	186.4	220.0	219.5	92.0
125.9	186.1	219.6	219.4	91.8
127.9	185.8	219.3	219.3	91.7
129.9	185.6	219.1	219.2	91.5
131.9	185.5	219.0	219.1	91.5
133.9	185.5	218.9	219.1	91.4
135.9	185.5	218.9	219.1	91.4
137.9	185.6	219.0	219.1	91.4
139.9	185.7	219.2	219.1	91.4
141.9	185.8	219.3	219.1	91.4
143.9	186.0	219.5	219.1	91.5
145.9	186.2	219.8	219.2	91.5
147.9	186.4	220.1	219.2	91.6
149.3	186.6	220.3	219.2	91.6
149.9	186.7	220.4	219.3	91.7
151.9	186.9	220.7	219.3	91.7
153.9	187.2	221.0	219.4	91.8
155.9	187.5	221.3	219.5	91.9
157.9	187.8	221.6	219.5	92.0

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 7 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
159.9	188.1	222.0	219.6	92.1
161.9	188.4	222.3	219.7	92.2
163.9	188.7	222.7	219.7	92.3
165.9	189.0	223.1	219.8	92.4
167.9	189.3	223.5	219.9	92.5
169.9	189.6	223.8	220.0	92.6
171.9	190.0	224.2	220.1	92.7
173.9	190.3	224.6	220.2	92.8
175.9	190.6	225.0	220.2	92.9
177.9	190.9	225.4	220.3	93.0
178.7	191.1	225.5	220.4	93.0
179.9	191.3	225.8	220.4	93.1
181.9	191.6	226.2	220.5	93.2
183.9	191.9	226.6	220.6	93.3
185.9	192.3	226.9	220.7	93.4
187.9	192.6	227.3	220.8	93.5
189.9	192.9	227.7	220.9	93.7
191.9	193.3	228.1	220.9	93.8
193.9	193.6	228.5	221.0	93.9
195.9	193.9	228.9	221.1	94.0
197.9	194.3	229.3	221.2	94.1

Table 6.2.1-3A
REFLOOD MASS AND ENERGY RELEASE RATES
FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 8 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	**Flow (lb_m/sec)**	**Energy (10^3 Btu/sec)**	**Flow (lb_m/sec)**	**Energy (10^3 Btu/sec)**
199.9	194.6	229.7	221.3	94.2
201.9	194.9	230.1	221.4	94.3
203.9	195.2	230.4	221.5	94.4
205.9	195.5	230.8	221.6	94.5
207.9	195.8	231.2	221.6	94.6
209.0	196.0	231.4	221.7	94.7

a) M&E exiting from the steam generator side of the break.

b) M&E exiting from the broken loop reactor coolant pump side of the break.

TABLE 6.2.1-3B
PRINCIPAL PARAMETERS DURING REFLOOD FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 1 of 2)

	Flooding						Total	Injection Accumulator	Spill	
Time (s)	Temp (°F)	Rate (in/s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction		(lbm/s)		Enthalpy
28.8	240.4	0.000	0.000	0.00	0.00	0.250	0.0	0.0	0.0	0.00
29.5	236.9	22.688	0.000	0.65	1.60	0.000	8010.6	8010.6	0.0	89.82
29.7	234.9	25.137	0.000	1.05	1.52	0.000	7962.5	7962.5	0.0	89.82
30.9	233.2	2.724	0.313	1.50	4.86	0.345	7660.2	7660.2	0.0	89.82
31.3	233.1	2.665	0.373	1.56	6.15	0.357	7575.7	7575.7	0.0	89.82
31.4	233.0	2.674	0.386	1.57	6.48	0.360	8175.6	7554.8	0.0	89.68
31.5	233.0	2.672	0.399	1.59	6.83	0.363	8155.1	7534.2	0.0	89.68
35.0	231.3	5.471	0.645	2.01	16.11	0.627	6780.8	6240.6	0.0	89.68
35.9	230.5	5.250	0.678	2.14	16.12	0.625	6602.3	6065.4	0.0	89.67
38.9	228.7	4.741	0.724	2.50	16.12	0.622	6229.7	5686.7	0.0	89.66
44.0	226.8	4.304	0.747	3.01	16.12	0.615	5733.9	5180.2	0.0	89.64
47.9	226.1	3.544	0.752	3.34	16.12	0.572	4391.7	3810.1	0.0	89.58
48.9	226.0	4.371	0.754	3.41	15.91	0.632	549.3	0.0	0.0	88.00
49.9	225.9	4.405	0.755	3.50	15.54	0.633	543.1	0.0	0.0	88.00
56.0	226.6	3.658	0.756	4.00	13.68	0.624	566.4	0.0	0.0	88.00
63.9	229.6	2.996	0.754	4.54	12.10	0.610	584.8	0.0	0.0	88.00
72.1	234.2	2.529	0.751	5.00	11.14	0.595	596.1	0.0	0.0	88.00
82.9	240.6	2.136	0.748	5.52	10.57	0.574	604.3	0.0	0.0	88.00
94.2	246.2	1.904	0.746	6.00	10.48	0.557	608.5	0.0	0.0	88.00

TABLE 6.2.1-3B
PRINCIPAL PARAMETERS DURING REFLOOD FOR THE DESIGN BASIS ACCIDENT - DEPSG
(MINIMUM SAFETY INJECTION)
(Sheet 2 of 2)

	Flooding						Total	Injection Accumulator	Spill	
Time (s)	Temp (°F)	Rate (in/s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction		(lbm/s)		Enthalpy
107.9	251.9	1.766	0.746	6.53	10.73	0.545	610.8	0.0	0.0	88.00
120.8	256.6	1.711	0.748	7.00	11.15	0.540	611.6	0.0	0.0	88.00
135.9	261.5	1.691	0.752	7.54	11.71	0.539	611.9	0.0	0.0	88.00
143.9	263.9	1.689	0.754	7.82	12.02	0.540	611.8	0.0	0.0	88.00
149.3	265.4	1.690	0.755	8.00	12.23	0.540	611.8	0.0	0.0	88.00
163.9	269.2	1.693	0.759	8.50	12.80	0.542	611.7	0.0	0.0	88.00
178.7	272.8	1.699	0.763	9.00	13.37	0.544	611.5	0.0	0.0	88.00
193.9	276.0	1.705	0.768	9.51	13.95	0.546	611.3	0.0	0.0	88.00
209.0	279.0	1.711	0.772	10.00	14.51	0.548	611.1	0.0	0.0	88.00

TABLE 6.2.1-4

POST-REFLOOD MASS AND ENERGY RELEASE RATES FOR THE DESIGN BASIS ACCIDENT - DEPSG (MINIMUM SAFETY INJECTION)

(Sheet 1 of 3)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
209.10	262.9	328.2	357.8	140.7
214.10	261.5	326.5	359.2	140.9
219.10	261.2	326.1	359.5	140.7
224.10	260.9	325.7	359.8	140.6
229.10	260.6	325.3	360.1	140.5
234.10	260.2	324.8	360.5	140.4
239.10	259.8	324.3	361.0	140.3
244.10	259.3	323.7	361.4	140.2
249.10	258.8	323.1	361.9	140.1
254.10	259.3	323.8	361.4	139.8
259.10	258.8	323.1	362.0	139.7
264.10	258.1	322.3	362.6	139.7
269.10	257.5	321.4	363.2	139.6
274.10	257.8	321.8	362.9	139.3
279.10	257.0	320.9	363.7	139.3
284.10	256.2	319.8	364.5	139.3
289.10	256.3	320.0	364.4	139.1
294.10	255.4	318.8	365.3	139.1
299.10	255.4	318.8	365.3	138.9
304.10	255.3	318.7	365.4	138.7
309.10	254.2	317.3	366.6	138.8
314.10	253.9	317.0	366.8	138.7
319.10	253.6	316.6	367.2	138.5
324.10	253.1	316.0	367.6	138.4
329.10	252.6	315.3	368.1	138.4
334.10	251.9	314.5	368.8	138.3

TABLE 6.2.1-4

POST-REFLOOD MASS AND ENERGY RELEASE RATES FOR THE DESIGN BASIS ACCIDENT - DEPSG (MINIMUM SAFETY INJECTION) (Sheet 2 of 3)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
339.10	252.1	314.7	368.7	138.1
344.10	251.2	313.6	369.5	138.1
349.10	251.0	313.4	369.7	137.9
354.10	250.7	312.9	370.0	137.8
359.10	250.2	312.3	370.6	137.7
364.10	249.5	311.4	371.3	137.7
369.10	249.3	311.3	371.4	137.5
374.10	249.0	310.8	371.7	137.3
379.10	248.3	310.0	372.4	137.3
384.10	248.1	309.8	372.6	137.1
389.10	247.5	309.1	373.2	137.1
394.10	246.6	307.8	374.2	137.1
399.10	246.5	307.7	374.2	136.9
404.10	245.9	307.0	374.8	136.8
409.10	245.3	306.2	375.4	136.8
414.10	245.0	305.9	375.7	136.6
419.10	244.8	305.6	376.0	136.5
424.10	244.1	304.7	376.7	136.5
429.10	243.8	304.4	376.9	136.3
434.10	243.2	303.7	377.5	136.3
439.10	242.9	303.3	377.8	136.1
444.10	242.3	302.5	378.4	136.1
449.10	241.7	301.8	379.0	136.0
615.00	241.7	301.8	379.0	136.0
615.10	99.4	123.1	521.3	173.1
619.10	99.3	122.9	521.5	172.9

TABLE 6.2.1-4

POST-REFLOOD MASS AND ENERGY RELEASE RATES FOR THE DESIGN BASIS ACCIDENT - DEPSG (MINIMUM SAFETY INJECTION) (Sheet 3 of 3)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
1631.40	99.3	122.9	521.5	172.9
1631.50	80.6	99.6	391.4	181.6
1687.04	80.6	99.6	391.4	181.6
1687.14	78.8	90.7	393.2	97.3
2000.00	75.2	86.6	396.8	97.9
2000.10	75.2	86.6	396.8	101.2
2500.00	72.1	83.0	399.9	101.8
2500.10	72.1	83.0	399.9	99.4
3000.00	69.0	79.4	403.0	100.0
3000.10	69.0	79.4	403.0	97.6
3500.00	65.8	75.7	406.2	98.2
3500.10	65.8	75.7	406.2	97.3
3600.00	65.2	75.0	406.8	97.4

a) M&E exiting from the steam generator side of the break.

b) M&E exiting from the broken loop reactor coolant pump side of the break.

TABLE 6.2.1-4A

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TABLE 6.2.1-4B
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TABLE 6.2.1-5
CONTAINMENT INITIAL CONDITIONS

Reactor Coolant System (100.6 percent ESDR)	3634
Average coolant temperature, F[(a)]	595.1
Mass of reactor coolant, lbm	572.12×10^3
Energy of reactor coolant, Btu	340.8×10^6
Containment	
Pressure, psig	-0.5 to 1.5
Inside temperature, F	120
Component cooling water temperature, F	105 (LOCA only)
Service water temperature, F	102 (LOCA only)
Refueling water temperature, F	120
Accumulators	
Total Volume, ft^3	3392.4
Cover gas pressure, psia	728
Temperature, F	120

a) RCS Coolant temperatures include uncertainty of +5.9 °F.

TABLE 6.2.1-6

STRUCTURAL HEAT SINKS
(Sheet 1 of 2)

No.	Description	Material	Area (ft^2)	Thickness (inches)	Initial Temp (°F)
1	Containment dome hemisphere	Paint on steel	28,628	0.006996	120
		Steel		0.500400	
		Linear air gap		0.124800	
		Concrete		29.499600	
		Paint on concrete		0.027996	
2	Misc. steel (filters, coils, hangers)	Paint on steel	23,819	0.007	120
		Stee		0.263	
3	.Misc. steel (piping, shielding, beams)	Paint on steel	4,586	0.006996	120
		Steel		0.160980	
4	Slabs and walls, 2-sided	Paint on concrete	172,676	0.027996	120
		Concrete		12.000000	
5	Ventilation ducts	Paint on steel	40,257	0.006996	120
		Steel		0.060000	
6	Misc. steel (platforms, ladders, ...)	Paint on steel	6,334	0.006996	120
		Steel		0.049992	
7	Cylindrical section of containment wall	Paint on steel	81,642	0.006996	120
		Steel		0.375600	
		Liner air gap		0.124800	
		Concrete		53.624400	
		Paint on concrete		0.027996	
8	Misc. steel (crane sheel, beams, ...)	Paint on steel	1,866	0.006996	120
		Steel		0.750000	
9	Misc. steel (wheels, girders, ...)	Paint on steel	528	0.006996	120
		Steel		4.002000	

TABLE 6.2.1-6

STRUCTURAL HEAT SINKS
(Sheet 2 of 2)

No.	Description	Material	Area (ft^2)	Thickness (inches)	Initial Temp (°F)
10	Foundation mat	Paint on concrete	11,395	0.027996	120
		Concrete		30.00000	
		Liner air gap		0.124800	
		Steel		0.240000	
		Liner air gap		0.124800	
		Concrete		143.964000	
11	Misc. steel (boxes, pumps, ...)	Paint on steel	23,359	0.006996	120
		Steel		0.150000	
12	Misc. steel (piping, rims, crane, ...)	Paint on steel	21,684	0.006996	120
		Steel		0.279600	
13	Misc. steel (trolley girders, ...)	Paint on steel	1,528	0.006996	120
		Steel		2.000400	
14	Girders	Paint on steel	16,631	0.006996	120
		Steel		2.250000	
15	Trolley girders	Paint on steel	22,987	0.006996	120
		Steel		0.096000	

TABLE 6.2.1-7
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TABLE 6.2.1-8
THERMOPHYSICAL PROPERTIES OF STRUCTURAL HEAT SINK MATERIALS

Material	Thermal Conductivity (Btu/hr-ft-°F)	Volumetric Heat Capacity (Btu/hr-ft^3-°F)	Specific Heat (Btu/lbm-°F)	Density (lbm/ft^3)
1) Steel	26	53.9	0.11	490
2) Concrete	0.8	23.2	0.16	145
3) Paint on steel	0.0875	0.1	0.1	1
4) Paint on concrete	0.07	0.1	0.1	1
5) Liner air gap	0.0161	0.0167	0.20875	0.08

TABLE 6.2.1-9

CHRONOLOGY OF EVENTS FOR LOCA DESIGN BASIS ACCIDENT DOUBLE-ENDED PUMP SUCTION GUILLOTINE, MINIMUM SAFETY INJECTION
(Sheet 1 of 2)

Time (sec)	Event Description
0.0	Break Occurs and Loss of Offsite Power is Assumed
0.68	Containment Spray Actuation Pressure Setpoint (19.7 psia; Analysis Value) Reached
3.42	Compensated Pressurizer Pressure for Reactor Trip (1859.7 psia) Reached and Turbine Trip Occurs
4.34	Low-Pressurizer Pressure SI Setpoint (1,715 psia) Reached - Feedwater Isolation Signal
11.35	Feedwater Isolation Valves Closed
15.2	Broken Loop Accumulator Begins Injecting Water
15.4	Intact Loop Accumulator Begins Injecting Water
28.75	End of Blowdown Phase
28.75	Accumulator Mass Adjustment for Refill Period
31.34	Pumped Safety Injection Begins (Includes 27 second Diesel Delay)
46.76	Broken Loop Accumulator Water Injection Ends
48.01	Intact Loop Accumulator Water Injection Ends
74.98	Containment Spray Pump Injection from RWST begins
209.01	End of Reflood for Minimum Safeguards Case
615.10	M&E Release Assumption: Broken Loop SG Equilibration at Containment Pressure of 54.7 psia
629.6	Containment Peak Vapor Temperature Occurs (266.0°F)
629.7	Containment Peak Pressure Occurs (59.9 psia)
1631.5	Switchover to Cold Leg Recirculation Begins
1687.04	M&E Release Assumption: Intact Loop SG Equilibration at Containment Pressure of 44.7 psia

TABLE 6.2.1-9

CHRONOLOGY OF EVENTS FOR LOCA DESIGN BASIS ACCIDENT DOUBLE-ENDED PUMP SUCTION GUILLOTINE, MINIMUM SAFETY INJECTION
(Sheet 2 of 2)

Time (sec)	Event Description
2953.6	Containment Spray Injection Ends, Recirculation Begins
3600.0	Transient Modeling Terminated

TABLE 6.2.1-10
CHRONOLOGY OF EVENTS FOR MAIN STEAM LINE BREAK WHICH RESULTS IN THE PEAK PRESSURE 4.7 FT^2 SPLIT RUPTURE AT 30 PERCENT POWER

INITIAL CONTAINMENT PRESSURE = 1.5 psig

Event	Time (sec)
Break occurs	0.0
Feedwater isolation valves close	11.0
Main steam isolation valves close	13.5
Maximum temperature of 311.1°F is reached	50.2
Initiation of containment spray	76.7[(a)]
Auxiliary feedwater is manually isolated	600.0
Peak pressure of 39.0 psig is reached	620.2
Dryout Time	800.0
End of analysis	86400.0

a) See Section 6.2.1.4.8, Item 5.

TABLE 6.2.1-11

DOUBLE-ENDED HOT LEG BREAK SEQUENCE OF EVENTS

Time (sec)	Event Description
0.0	Break Occurs and Loss of Offsite Power is Assumed
3.04	Compensated Pressurizer Pressure for Reactor Trip (1859.7 psia) Reached and Turbine Trip Occurs
3.95	Low-Pressurizer Pressure SI Setpoint (1,715 psia) Reached - Feedwater Isolation Signal
10.95	Feedwater Isolation Valves Closed
12.4	Broken Loop Accumulator Begins Injecting Water
12.5	Intact Loop Accumulator Begins Injecting Water
22.0	Containment Peak Vapor Temperature Occurs (260.8°F) Containment Peak Pressure Occurs (55.8 psia)
22.5	End of Blowdown Phase
22.5	Transient Modeling Terminated

TABLE 6.2.1-12

DOUBLE-ENDED PUMP SUCTION BREAK SEQUENCE OF EVENTS (MAXIMUM SAFEGUARDS)

(Sheet 1 of 2)

Time (sec)	Event Description
0.0	Break Occurs and Loss of Offsite Power is Assumed
0.68	Containment Spray Actuation Pressure Setpoint (19.7 psia; Analysis Value) Reached
3.42	Compensated Pressurizer Pressure for Reactor Trip (1859.7 psia) Reached and Turbine Trip Occurs
4.34	Low-Pressurizer Pressure SI Setpoint (1,715 psia) Reached - Feedwater Isolation Signal
11.35	Feedwater Isolation Valves Closed
15.2	Broken Loop Accumulator Begins Injecting Water
15.4	Intact Loop Accumulator Begins Injecting Water
24.0	Containment Peak Vapor Temperature Occurs (258.6°F)
28.75	End of Blowdown Phase
28.75	Accumulator Mass Adjustment for Refill Period
31.34	Pumped Safety Injection Begins (Includes 27 second Diesel Delay)
47.08	Broken Loop Accumulator Water Injection Ends
48.28	Intact Loop Accumulator Water Injection Ends
74.6	Containment Peak Pressure Occurs (54.94 psia)
74.98	Containment Spray Pump Injection from RWST begins
194.23	End of Reflood for Maximum Safeguards Case
1092.93	M&E Release Assumption: Broken Loop SG Equilibration at Containment Pressure of 44.7 psia
1222.00	Switchover to Cold Leg Recirculation Begins
1748.70	M&E Release Assumption: Intact Loop SG Equilibration at Containment Pressure of 34.7 psia

TABLE 6.2.1-12

DOUBLE-ENDED PUMP SUCTION BREAK SEQUENCE OF EVENTS (MAXIMUM SAFEGUARDS) (Sheet 2 of 2)

Time (sec)	Event Description
2577.0	Containment Spray Injection Ends, Recirculation Begins
3600.0	Transient Modeling Terminated

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TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 1 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.00000	0.0	0.0	0.0	0.0
0.00110	47523.0	30918.9	47521.3	30916.5
0.004	46192.6	30049.2	45336.1	29476.0
0.006	44863.9	29190.5	42909.4	27876.1
0.009	43947.9	28605.4	40934.5	26574.4
0.011	43180.8	28127.5	38761.1	25142.0
0.014	42622.7	27783.8	36744.2	23813.9
0.016	42152.3	27499.8	34770.2	22517.2
0.019	41844.8	27327.8	32872.6	21273.6
0.021	41703.7	27258.9	31470.8	20357.7
0.024	41615.6	27227.3	30001.8	19402.6
0.026	41633.4	27262.7	28945.9	18720.5
0.029	41737.1	27354.0	28149.5	18210.6
0.031	41912.7	27492.2	27468.8	17779.6
0.034	42132.3	27657.3	27128.6	17570.5
0.036	42387.7	27843.0	26962.3	17473.7
0.039	42686.9	28056.6	26911.3	17451.9
0.042	42981.6	28264.3	26950.0	17485.8
0.044	43256.4	28455.2	27030.5	17544.0
0.046	43539.4	28651.2	27139.1	17619.9

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 2 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.049	43794.6	28827.2	27250.7	17696.1
0.051	44051.7	29003.3	27368.8	17775.2
0.054	44271.4	29153.4	27470.0	17842.3
0.056	44500.7	29309.2	27575.4	17911.4
0.059	44677.7	29429.1	27657.7	17965.0
0.061	44855.5	29549.5	27742.9	18020.1
0.064	44975.9	29630.8	27806.4	18061.1
0.066	45087.1	29705.8	27875.8	18106.0
0.069	45172.8	29763.5	27944.9	18150.7
0.072	45241.5	29809.7	28025.4	18203.0
0.074	45273.6	29831.2	28098.4	18250.4
0.076	45279.5	29835.7	28165.8	18294.1
0.079	45256.1	29820.2	28245.0	18345.5
0.081	45209.2	29789.4	28381.6	18433.9
0.084	45128.8	29736.6	28440.1	18471.2
0.086	45030.2	29672.0	28481.7	18497.6
0.089	44905.6	29590.4	28510.1	18515.3
0.091	44737.2	29480.3	28523.6	18523.1
0.094	44526.3	29342.8	28516.0	18516.9
0.097	44315.0	29205.9	28489.4	18498.4

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 3 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.099	44128.6	29084.5	28455.1	18475.0
0.101	43873.0	28918.4	28394.1	18433.8
0.125	40966.0	27036.1	27292.2	17702.3
0.151	38233.6	25278.4	26134.1	16929.8
0.175	36872.1	24411.3	25432.9	16447.1
0.20	36702.9	24314.1	24852.9	16031.2
0.23	36940.3	24474.4	24086.3	15494.9
0.25	37048.9	24548.0	23358.0	14985.7
0.28	36911.8	24461.8	22929.4	14666.8
0.30	36638.0	24286.3	22621.2	14423.2
0.33	36337.9	24093.9	22299.2	14168.3
0.35	36097.0	23940.3	21916.7	13878.4
0.38	35916.5	23826.7	21652.9	13664.0
0.40	35778.2	23741.7	21476.0	13506.3
0.43	35655.9	23668.0	21294.3	13346.4
0.45	35527.2	23591.0	21096.1	13177.5
0.48	35392.3	23510.3	20906.6	13015.9
0.50	35270.7	23438.7	20764.8	12888.1
0.53	35160.3	23376.1	20624.5	12760.3
0.70	34384.2	22952.1	19887.8	12071.7

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 4 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
0.90	32921.9	22122.7	19316.2	11532.3
1.10	31694.0	21474.0	18949.6	11176.7
1.30	30739.6	21000.8	18716.1	10937.8
1.50	29600.3	20317.0	18659.2	10821.4
1.70	28905.7	19915.7	18704.3	10776.7
1.90	27884.4	19275.7	18754.2	10745.3
2.10	26877.5	18634.4	18791.6	10717.1
2.25	26308.4	18291.3	18808.6	10696.5
2.50	25173.2	17553.6	18765.9	10634.2
2.75	24233.9	16932.0	18659.9	10550.1
3.00	23435.6	16386.9	18489.4	10441.2
3.25	22755.1	15895.1	18256.0	10306.6
3.50	22198.5	15470.1	17970.9	10150.3
3.75	21757.9	15110.5	17621.6	9963.3
4.00	21423.0	14815.0	17199.4	9738.9
4.25	21156.5	14563.7	16745.3	9499.1
4.50	20970.4	14367.8	16282.2	9255.2
4.75	20878.4	14246.0	15822.1	9012.9
5.00	20838.5	14166.2	15424.1	8805.4
5.25	20880.9	14112.5	15008.4	8585.3

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 5 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
5.50	20974.2	14088.2	14582.1	8357.6
5.75	21079.1	14065.9	14186.5	8146.9
6.00	21223.4	14063.4	13821.6	7953.3
6.25	21421.7	14089.3	13458.0	7758.8
6.50	21677.0	14143.7	13089.7	7560.6
6.75	22007.5	14236.0	12753.8	7380.5
7.00	15914.7	12013.0	12427.3	7204.9
7.25	16910.2	11767.7	12112.3	7035.0
7.50	17076.0	11809.9	11812.9	6873.6
7.75	17200.6	11791.3	11544.3	6729.1
8.00	17394.3	11835.8	11262.4	6575.4
8.25	17561.8	11859.6	10997.3	6431.1
8.50	17696.3	11861.2	10730.9	6285.7
8.75	17828.1	11897.5	10472.6	6144.9
9.00	17946.5	11883.5	10213.0	6003.6
9.25	17762.9	11709.3	9958.6	5865.6
9.50	17995.9	11764.0	9704.8	5728.4
9.75	18229.8	11823.1	9454.4	5593.8
10.00	18530.8	11924.7	9207.8	5461.6
10.25	19318.3	12307.9	8963.6	5331.1

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 6 of 8)

	Break Path No. 1[(a)]		Break Path No. 2[(b)]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
10.50	19832.2	12597.9	8718.0	5200.5
10.75	20594.3	13040.7	8473.1	5070.7
11.00	24262.6	15327.9	8219.7	4937.1
11.25	22973.7	14428.7	7965.0	4804.2
11.50	22507.8	14068.1	7680.3	4655.0
11.75	22221.3	13850.7	7390.9	4507.1
12.00	22182.4	13810.9	7095.8	4359.6
12.25	21856.9	13615.7	6802.2	4215.7
12.50	21522.4	13382.1	6518.7	4079.5
12.75	21237.6	13187.8	6244.2	3949.5
13.00	20854.6	12956.7	5979.6	3824.7
13.25	20305.2	12649.0	5714.2	3698.4
13.50	8629.1	6101.4	5422.7	3557.9
13.75	9851.7	6819.7	5131.7	3424.1
14.00	10043.3	6883.5	4833.3	3294.8
14.25	10030.5	6876.3	4552.7	3175.2
14.50	9970.6	6887.2	4300.7	3058.3
14.75	9740.2	6818.6	4115.3	2953.1
15.00	9490.3	6769.0	4010.2	2859.8
15.25	9168.7	6683.0	3979.5	2788.5

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 7 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
15.50	8798.6	6567.7	4003.6	2745.8
15.75	7623.5	6409.9	4043.2	2715.7
16.00	6573.5	6228.3	4073.7	2690.1
16.25	5766.6	5841.7	4087.4	2665.7
16.50	5082.1	5179.5	4067.6	2634.2
16.75	4652.1	4720.2	4017.5	2598.2
17.00	4393.3	4438.6	3930.5	2553.4
17.25	4126.5	4181.2	3802.5	2499.4
17.50	3876.4	3956.9	3645.9	2444.3
17.75	3696.8	3778.3	3458.0	2384.0
18.00	3508.3	3591.2	3249.1	2321.6
18.25	3270.3	3372.0	3027.8	2255.6
18.50	3086.2	3154.0	2789.2	2177.9
18.75	2835.2	2941.5	2548.2	2103.9
19.00	2612.3	2763.2	2298.5	2033.1
19.25	2407.9	2603.7	2020.7	1965.9
19.50	2230.1	2458.3	1720.8	1889.8
19.75	2067.4	2322.5	1446.4	1746.2
20.00	1918.8	2188.8	1221.1	1505.3
20.25	1798.0	2062.7	1074.6	1332.0

TABLE 6.2.1-39

DOUBLE ENDED HOT LEG GUILLOTINE (DEHLG) BLOWDOWN MASS AND ENERGY RELEASE RATE

(Sheet 8 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
20.50	1323.2	1617.9	987.3	1226.8
20.75	1020.5	1260.6	828.9	1030.9
21.00	818.1	1018.1	559.7	698.5
21.25	676.2	844.4	426.3	535.7
21.50	504.8	628.8	341.8	429.8
21.75	402.3	502.0	296.2	373.5
22.00	252.9	313.4	256.7	324.3
22.25	64.0	78.5	81.2	103.0
22.50	0.0	0.0	0.0	0.0

a) M&E exiting from the reactor vessel side of the break.

b) M&E exiting from the steam generator side of the break.

TABLE 6.2.1-40

THIS TABLE HAS BEEN DELETED

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 1 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
28.8	0.0	0.0	0.0	0.0
29.2	0.0	0.0	0.0	0.0
29.4	0.0	0.0	0.0	0.0
29.5	0.0	0.0	0.0	0.0
29.6	0.0	0.0	0.0	0.0
29.7	0.0	0.0	0.0	0.0
29.8	0.0	0.0	0.0	0.0
29.9	88.3	104.1	0.0	0.0
30.0	36.0	42.4	0.0	0.0
30.1	32.4	38.2	0.0	0.0
30.2	40.0	47.1	0.0	0.0
30.3	48.8	57.5	0.0	0.0
30.4	54.5	64.2	0.0	0.0
30.5	60.1	70.9	0.0	0.0
30.6	65.4	77.1	0.0	0.0
30.7	70.4	83.0	0.0	0.0
30.8	75.1	88.6	0.0	0.0
30.9	79.7	93.9	0.0	0.0
31.0	84.0	99.1	0.0	0.0
31.1	88.2	104.0	0.0	0.0

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 2 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
31.2	92.2	108.8	0.0	0.0
31.3	96.1	113.4	0.0	0.0
31.4	103.1	121.7	304.9	26.8
31.5	107.3	126.6	304.9	26.8
31.6	111.3	131.2	304.9	26.8
31.7	115.1	135.7	304.9	26.8
31.8	118.8	140.1	304.9	26.8
32.8	151.7	179.0	304.9	26.8
33.8	179.3	211.6	304.9	26.8
34.9	644.1	765.8	5708.4	814.5
35.9	646.2	768.3	5721.5	825.2
36.9	638.7	759.3	5664.8	819.2
37.9	630.1	749.0	5597.2	811.3
38.5	624.8	742.6	5554.6	806.2
38.9	621.2	738.3	5525.8	802.7
39.9	612.2	727.5	5453.5	793.9
40.9	603.4	716.9	5381.8	785.0
41.9	594.8	706.6	5311.4	776.3
42.9	586.5	696.6	5242.5	767.7
43.3	583.2	692.7	5215.5	764.3

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 3 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
43.9	578.4	687.0	5175.5	759.3
44.9	570.6	677.6	5110.3	751.1
45.9	563.1	668.6	5047.0	743.1
46.9	555.8	659.9	4985.6	735.4
47.9	457.0	541.7	4091.8	634.0
48.9	390.9	462.8	517.5	195.5
49.0	390.8	462.6	517.8	195.5
49.9	389.4	461.0	521.1	195.6
50.9	387.8	459.1	525.0	195.7
51.9	386.2	457.2	528.8	195.8
52.9	384.6	455.2	532.5	195.9
53.9	382.9	453.3	536.3	196.0
54.9	381.2	451.2	540.0	196.1
55.9	379.5	449.2	543.7	196.2
56.3	378.8	448.4	545.2	196.2
56.9	377.7	447.1	547.5	196.3
57.9	376.0	445.0	551.2	196.4
58.9	374.2	442.9	554.9	196.5
59.9	372.4	440.7	558.7	196.6
60.9	370.5	438.5	562.4	196.8

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 4 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
61.9	368.6	436.3	566.2	196.9
62.9	366.7	434.0	570.1	197.0
63.9	364.8	431.7	573.9	197.2
64.9	362.8	429.3	577.8	197.3
65.9	360.8	426.9	581.8	197.5
66.9	358.8	424.5	585.8	197.7
67.9	356.7	422.0	589.8	197.9
68.9	354.6	419.5	593.9	198.1
69.9	352.4	416.9	598.0	198.3
70.9	350.2	414.3	602.2	198.6
71.9	348.0	411.7	606.5	198.8
72.9	345.7	408.9	610.9	199.1
73.9	343.3	406.2	615.3	199.4
74.9	341.0	403.3	619.8	199.7
75.9	338.5	400.4	624.3	200.0
76.9	336.0	397.4	629.0	200.3
77.9	333.5	394.4	633.7	200.7
78.9	330.8	391.3	638.5	201.1
79.9	328.1	388.1	643.5	201.5
80.9	325.4	384.8	648.5	201.9

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 5 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
81.9	322.5	381.4	653.6	202.3
82.9	319.6	377.9	658.9	202.8
83.9	316.6	374.3	664.3	203.3
84.9	313.5	370.6	669.7	203.8
85.9	310.3	366.8	675.4	204.3
86.9	307.0	362.9	681.1	204.9
87.9	303.6	358.9	687.0	205.5
88.9	300.1	354.8	693.0	206.1
89.5	297.9	352.2	696.7	206.5
89.9	296.5	350.5	699.2	206.8
91.9	288.9	341.5	712.1	208.3
93.9	280.8	331.9	725.7	209.9
95.9	272.1	321.6	740.2	211.7
97.9	262.7	310.4	755.6	213.7
99.9	252.6	298.4	772.1	216.0
101.9	241.4	285.2	790.0	218.5
103.9	229.1	270.6	809.5	221.5
105.9	215.4	254.3	831.0	224.9
107.9	199.7	235.7	855.2	229.0
109.9	196.8	232.4	860.4	228.5

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 6 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
110.6	196.5	231.9	861.2	228.3
111.9	195.7	231.1	862.7	228.0
113.9	194.7	229.8	865.0	227.4
115.9	193.6	228.5	867.2	226.9
117.9	192.5	227.2	869.5	226.3
119.9	191.4	225.9	871.7	225.8
121.9	190.3	224.7	873.9	225.3
123.9	189.3	223.4	876.1	224.8
125.9	188.2	222.1	878.3	224.2
127.9	187.1	220.9	880.5	223.7
129.9	186.1	219.6	882.7	223.2
131.9	185.0	218.4	884.9	222.7
133.9	183.9	217.1	887.0	222.2
135.7	183.0	216.0	889.0	221.7
135.9	182.9	215.9	889.2	221.7
137.9	181.8	214.6	891.3	221.2
139.9	180.8	213.3	893.5	220.7
141.9	179.7	212.1	895.6	220.2
143.9	178.6	210.8	897.8	219.7
145.9	177.6	209.6	899.9	219.2

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 7 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
147.9	176.5	208.3	902.0	218.7
149.9	175.4	207.0	904.2	218.2
151.9	174.3	205.8	906.3	217.7
153.9	173.3	204.5	908.4	217.2
155.9	172.2	203.2	910.6	216.7
157.9	171.1	201.9	912.7	216.2
159.9	170.0	200.6	914.9	215.8
161.9	168.9	199.3	917.0	215.3
163.3	168.1	198.4	918.5	214.9
163.9	167.8	198.0	919.2	214.8
165.9	166.7	196.7	921.3	214.3
167.9	165.5	195.3	923.4	213.9
169.9	164.4	194.0	925.6	213.4
171.9	163.3	192.7	927.7	212.9
173.9	162.2	191.4	929.9	212.5
175.9	161.0	190.0	932.0	212.0
177.9	159.9	188.7	934.2	211.5
179.9	158.8	187.3	936.3	211.1
181.9	157.6	186.0	938.5	210.6
183.9	156.5	184.6	940.7	210.2

TABLE 6.2.1-41

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - REFLOOD MASS AND ENERGY

(Sheet 8 of 8)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
185.9	155.3	183.3	942.8	209.7
187.9	154.2	181.9	945.0	209.3
189.9	153.0	180.5	947.1	208.9
191.9	151.8	179.1	949.3	208.4
193.9	150.7	177.8	951.4	208.0
194.2	150.5	177.6	951.8	207.9

a) M&E exiting from the steam generator side of the break.

b) M&E exiting from the broken loop reactor coolant pump side of the break.

TABLE 6.2.1-42
DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - PRINCIPAL PARAMETERS DURING REFLOOD
(Sheet 1 of 2)

	Flooding						Total	Injection Accumulator	Spill	
Time (s)	Temp (°F)	Rate (in/s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction		(lb_m/s)		Enthalpy (Btu/lb_m)
28.8	240.4	0.000	0.000	0.00	0.00	0.250	0.0	0.0	0.0	0.00
29.5	236.9	22.688	0.000	0.65	1.60	0.000	8010.6	8010.6	0.0	89.82
29.7	234.9	25.137	0.000	1.05	1.52	0.000	7962.5	7962.5	0.0	89.82
30.9	233.2	2.724	0.313	1.50	4.86	0.345	7660.2	7660.2	0.0	89.82
31.3	233.1	2.665	0.373	1.56	6.15	0.357	7575.7	7575.7	0.0	89.82
34.9	231.1	5.774	0.648	2.02	16.12	0.635	7201.9	6137.7	0.0	89.55
35.9	230.2	5.480	0.683	2.17	16.12	0.635	7024.0	5962.6	0.0	89.54
38.5	228.5	5.025	0.723	2.50	16.12	0.634	6703.5	5632.0	0.0	89.53
43.3	226.4	4.590	0.747	3.00	16.12	0.628	6236.6	5145.4	0.0	89.50
48.9	225.2	3.419	0.753	3.50	16.12	0.561	1165.5	0.0	0.0	88.00
49.0	225.2	3.417	0.753	3.50	16.12	0.561	1165.5	0.0	0.0	88.00
56.3	226.3	3.304	0.757	4.00	16.12	0.559	1168.7	0.0	0.0	88.00
63.9	229.5	3.187	0.759	4.50	16.12	0.557	1172.5	0.0	0.0	88.00
71.9	234.3	3.056	0.761	5.00	16.12	0.552	1177.1	0.0	0.0	88.00
80.9	240.5	2.892	0.763	5.53	16.12	0.544	1183.1	0.0	0.0	88.00
89.5	245.9	2.708	0.764	6.01	16.12	0.531	1190.1	0.0	0.0	88.00
99.9	251.5	2.427	0.765	6.53	16.12	0.502	1200.8	0.0	0.0	88.00
110.6	256.2	2.103	0.763	7.00	16.12	0.453	1212.4	0.0	0.0	88.00
123.9	261.2	2.022	0.765	7.54	16.12	0.452	1212.8	0.0	0.0	88.00

TABLE 6.2.1-42
DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - PRINCIPAL PARAMETERS DURING REFLOOD
(Sheet 2 of 2)

	Flooding						Total	Injection Accumulator	Spill	
Time (s)	Temp ($^{\circ}$F)	Rate (in/s)	Carryover Fraction	Core Height (ft)	Downcomer Height (ft)	Flow Fraction		(lb_m/s)		Enthalpy (Btu/lb_m)
135.7	265.1	1.953	0.766	8.00	16.12	0.452	1213.1	0.0	0.0	88.00
149.9	269.2	1.871	0.768	8.53	16.12	0.451	1213.5	0.0	0.0	88.00
163.3	272.7	1.794	0.770	9.00	16.12	0.449	1214.0	0.0	0.0	88.00
179.9	276.4	1.699	0.772	9.55	16.12	0.447	1214.5	0.0	0.0	88.00
194.2	279.2	1.618	0.774	10.00	16.12	0.444	1215.0	0.0	0.0	88.00

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 1 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
194.3	167.8	209.5	1052.0	226.2
199.3	167.0	208.5	1052.8	226.1
204.3	167.6	209.3	1052.2	225.7
209.3	167.1	208.7	1052.7	225.6
214.3	166.6	208.0	1053.3	225.5
219.3	167.2	208.8	1052.6	225.1
224.3	166.7	208.2	1053.1	225.0
229.3	166.2	207.5	1053.7	224.9
234.3	166.8	208.3	1053.1	224.5
239.3	166.2	207.6	1053.6	224.4
244.3	166.8	208.4	1053.0	224.0
249.3	166.3	207.7	1053.5	223.9
254.3	165.8	207.1	1054.0	223.8
259.3	166.3	207.8	1053.5	223.4
264.3	165.8	207.1	1054.0	223.4
269.3	166.4	207.8	1053.4	223.0
274.3	165.8	207.1	1054.0	222.9
279.3	166.4	207.8	1053.5	222.5
284.3	165.8	207.1	1054.0	222.4
289.3	165.3	206.4	1054.6	222.3

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 2 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
294.3	165.8	207.1	1054.0	221.9
299.3	165.2	206.4	1054.6	221.8
304.3	165.7	207.0	1054.1	221.4
309.3	165.1	206.3	1054.7	221.3
314.3	165.6	206.9	1054.2	221.0
319.3	165.1	206.2	1054.8	220.9
324.3	165.5	206.7	1054.3	220.5
329.3	164.9	206.0	1054.9	220.4
334.3	165.4	206.6	1054.4	220.1
339.3	164.8	205.8	1055.0	220.0
344.3	165.2	206.4	1054.6	219.6
349.3	164.6	205.6	1055.2	219.5
354.3	165.0	206.1	1054.8	219.2
359.3	164.4	205.3	1055.4	219.1
364.3	164.8	205.8	1055.0	218.7
369.3	164.2	205.0	1055.7	218.6
374.3	164.5	205.5	1055.3	218.3
379.3	163.9	204.7	1055.9	218.2
384.3	164.2	205.1	1055.6	217.9
389.3	163.6	204.3	1056.2	217.8

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 3 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
394.3	163.9	204.7	1055.9	217.4
399.3	163.2	203.9	1056.6	217.4
404.3	163.7	204.4	1056.1	217.0
409.3	163.2	203.8	1056.6	216.9
414.3	163.6	204.3	1056.2	216.5
419.3	163.1	203.7	1056.8	216.4
424.3	163.5	204.2	1056.3	216.1
429.3	163.9	204.7	1055.9	215.7
434.3	163.3	204.0	1056.5	215.6
439.3	163.7	204.4	1056.1	215.3
444.3	163.1	203.7	1056.7	215.2
449.3	163.4	204.1	1056.4	214.8
454.3	162.8	203.4	1057.0	214.8
459.3	163.2	203.8	1056.7	214.4
464.3	163.5	204.2	1056.4	214.1
469.3	162.8	203.4	1057.0	214.0
474.3	163.1	203.7	1056.7	213.7
479.3	162.4	202.9	1057.4	213.6
484.3	162.7	203.2	1057.1	213.3
489.3	162.9	203.5	1056.9	213.0

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 4 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
494.3	163.1	203.7	1056.7	212.7
499.3	162.4	202.8	1057.4	212.6
504.3	162.6	203.0	1057.2	212.3
509.3	162.7	203.2	1057.1	212.0
514.3	162.8	203.4	1057.0	211.7
519.3	162.1	202.4	1057.8	211.7
524.3	162.1	202.5	1057.7	211.4
529.3	162.2	202.6	1057.6	211.1
534.3	162.2	202.6	1057.6	210.8
539.3	162.2	202.6	1057.6	210.6
544.3	162.2	202.6	1057.6	210.3
549.3	162.1	202.5	1057.7	210.1
554.3	162.0	202.4	1057.8	209.8
559.3	161.9	202.2	1057.9	209.6
564.3	161.7	202.0	1058.1	209.4
569.3	161.6	201.8	1058.3	209.2
574.3	162.1	202.5	1057.7	208.8
579.3	161.8	202.1	1058.0	208.6
584.3	161.5	201.7	1058.3	214.0
589.3	161.9	202.2	1057.9	213.6

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 5 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
594.3	161.5	201.7	1058.3	213.4
599.3	161.7	202.0	1058.1	213.0
604.3	161.3	201.4	1058.6	212.9
609.3	161.4	201.6	1058.4	212.6
614.3	161.5	201.8	1058.3	212.2
619.3	161.5	201.8	1058.3	211.9
624.3	161.5	201.7	1058.3	211.7
629.3	161.3	201.4	1058.5	211.4
634.3	161.0	201.1	1058.8	211.2
639.3	161.2	201.4	1058.6	210.9
644.3	160.7	200.7	1059.1	210.7
649.3	160.7	200.7	1059.2	210.4
654.3	161.0	201.1	1058.8	210.1
659.3	160.5	200.5	1059.3	209.9
664.3	160.4	200.3	1059.4	209.6
669.3	160.5	200.5	1059.3	209.3
674.3	160.2	200.1	1059.6	209.1
679.3	160.5	200.5	1059.3	208.7
684.3	160.1	200.0	1059.7	208.5
689.3	160.4	200.3	1059.4	208.2

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 6 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)	Flow (lb_m/sec)	Energy (10^3 Btu/sec)
694.3	160.3	200.2	1059.5	207.9
699.3	160.1	199.9	1059.8	207.6
704.3	160.1	200.0	1059.7	207.3
709.3	159.8	199.6	1060.0	207.1
714.3	159.7	199.4	1060.1	206.8
1092.8	159.7	199.4	1060.1	206.8
1092.9	88.2	109.2	1131.6	220.1
1094.3	88.2	109.1	1131.6	219.9
1219.3	86.2	106.6	1133.6	220.0
1222.0	86.1	106.6	1151.6	310.3
1500.0	86.1	106.6	1151.6	310.3
1500.1	81.7	101.0	1156.0	320.7
1748.7	81.7	101.0	1156.0	320.7
1748.8	78.0	89.7	1159.7	222.4
2000.0	75.1	86.4	1162.6	222.9
2000.1	75.1	86.4	1162.6	220.4
2500.0	72.0	82.8	1165.7	221.0
2500.1	72.0	82.8	1165.7	217.3
3000.0	68.8	79.2	1168.9	217.8
3000.1	68.8	79.2	1168.9	215.4

TABLE 6.2.1-43

DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MAXIMUM ECCS - POST REFLOOD MASS AND ENERGY RELEASE RATE

(Sheet 7 of 7)

	Break Path No. 1[a]		Break Path No. 2[b]	
Time (sec)	**Flow (lb_m/sec)**	**Energy (10^3 Btu/sec)**	**Flow (lb_m/sec)**	**Energy (10^3 Btu/sec)**
3500.0	65.7	75.6	1172.0	215.9
3500.1	65.7	75.6	1172.0	212.2
3600.0	65.1	74.9	1172.6	212.3

a) M&E exiting from the steam generator side of the break.

b) M&E exiting from the broken loop reactor coolant pump side of the break.

TABLE 6.2.1-44

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TABLE 6.2.1-45

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TABLE 6.2.1-46

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TABLE 6.2.1-47

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TABLE 6.2.1-48

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TABLE 6.2.1-49
(Sheet 1 of 2)
LOCA M&E RELEASE ANALYSIS CORE DECAY HEAT FRACTION

Time (sec)	Decay Heat Generation Rate (Btu/Btu)
10	0.053876
15	0.050401
20	0.048018
40	0.042401
60	0.039244
80	0.037065
100	0.035466
150	0.032724
200	0.030936
400	0.027078
600	0.024931
800	0.023389
1,000	0.022156
1,500	0.019921
2,000	0.018315
4,000	0.014781
6,000	0.013040
8,000	0.012000
10,000	0.011262
15,000	0.010097
20,000	0.009350
40,000	0.007778
60,000	0.006958
80,000	0.006424

TABLE 6.2.1-49
(Sheet 2 of 2)
LOCA M&E RELEASE ANALYSIS CORE DECAY HEAT FRACTION

Time (sec)	Decay Heat Generation Rate (Btu/Btu)
100,000	0.006021
150,000	0.005323
200,000	0.004847
400,000	0.003770
600,000	0.003201
800,000	0.002834
1,000,000	0.002580
2,000,000	0.001909
4,000,000	0.001355
6,000,000	0.001091
8,000,000	0.000927
10,000,000	0.000808

TABLE 6.2.1-50
MASS AND ENERGY BALANCE DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) WITH MAXIMUM ECCS
(Sheet 1 of 2)

TIME (SEC)		0.00	28.75	28.75	194.23	1092.93	1748.70	3600.00
		MASS (10^3 LBM)						
INITIAL	IN RCS AND ACCUMULATOR	781.03	781.03	781.03	781.03	781.03	781.03	781.03
ADDED MASS	PUMPED INJECTION	0.00	0.00	0.00	194.00	1290.16	2099.49	4390.85
	TOTAL ADDED	0.00	0.00	0.00	194.00	1290.16	2099.49	4390.85
***** TOTAL AVAILABLE *****		**781.03**	**781.03**	**781.03**	**975.03**	**2071.19**	**2880.52**	**5171.88**
DISTRIBUTION	REACTOR COOLANT	571.17	107.07	107.14	169.57	169.57	169.57	169.57
	ACCUMULATOR	209.86	115.42	115.35	0.00	0.00	0.00	0.00
	TOTAL CONTENTS	781.03	222.49	222.49	169.57	169.57	169.57	169.57
EFFLUENT	BREAK FLOW	0.00	558.52	558.52	794.14	1890.30	2699.66	4991.01
	ECCS SPILL	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	558.52	558.52	794.14	1890.30	2699.66	4991.01
***** TOTAL ACCOUNTABLE *****		**781.03**	**781.03**	**781.01**	**963.71**	**2059.87**	**2869.22**	**5160.58**

TABLE 6.2.1-50
MASS AND ENERGY BALANCE DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) WITH MAXIMUM ECCS
(Sheet 2 of 2)

TIME (SEC)		0.00	28.75	28.75	194.23	1092.93	1748.70	3600.00
		MASS (10^3 LBM)						
INITIAL ENERGY	IN RCS, ACCUMULATORS AND STEAM GENERATORS	994.92	994.92	994.92	994.92	994.92	994.92	994.92
ADDED ENERGY	PUMPED INJECTION	0.00	0.00	0.00	17.07	113.53	236.31	627.39
	DECAY HEAT	0.00	8.78	8.78	29.17	107.20	153.17	261.57
	HEAT FROM SECONDARY	0.00	5.04	5.04	5.04	5.04	5.04	5.04
	TOTAL ADDED	0.00	13.82	13.82	51.28	225.78	394.52	894.00
***** TOTAL AVAILABLE *****		**994.92**	**1008.74**	**1008.74**	**1046.20**	**1220.70**	**1389.45**	**1888.92**
DISTRIBUTION	REACTOR COOLANT	340.87	24.28	24.28	45.20	45.20	45.20	45.20
	ACCUMULATOR	18.85	10.37	10.36	0.00	0.00	0.00	0.00
	CORE STORED	24.20	9.61	9.61	4.91	4.47	4.17	3.33
	PRIMARY METAL	173.99	163.51	163.51	143.69	89.72	71.26	56.70
	SECONDARY METAL	121.28	118.33	118.33	107.69	71.95	53.03	42.14
	STEAM GENERATOR	315.72	325.86	325.86	292.45	187.31	135.83	107.29
	TOTAL CONTENTS	994.92	651.96	651.96	593.93	398.65	309.50	254.66
EFFLUENT	BREAK FLOW	0.00	356.20	356.20	449.18	818.96	1084.51	1639.81
	ECCS SPILL	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	356.20	356.20	449.18	818.96	1084.51	1639.81
***** TOTAL ACCOUNTALBE *****		**994.92**	**1008.16**	**1008.16**	**1043.12**	**1217.61**	**1394.01**	**1894.47**

TABLE 6.2.1-50A

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TABLE 6.2.1-51
MASS AND ENERGY BALANCE DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MINIMUM ECCS
(Sheet 1 of 2)

TIME (SEC)		0.00	28.75	28.75	209.01	615.10	1687.04	3600.00
		MASS (10^3 LBM)						
INITIAL	IN RCS AND ACCUMULATOR	781.03	781.03	781.03	781.03	781.03	781.03	781.03
ADDED MASS	PUMPED INJECTION	0.00	0.00	0.00	106.70	358.72	1011.56	1918.74
	TOTAL ADDED	0.00	0.00	0.00	106.70	358.72	1011.56	1918.74
***** TOTAL AVAILABLE *****		**781.03**	**781.03**	**781.03**	**887.73**	**1139.75**	**1792.59**	**2699.77**
DISTRIBUTION	REACTOR COOLANT	571.17	107.07	107.14	166.40	166.40	166.40	166.40
	ACCUMULATOR	209.86	115.42	115.35	0.00	0.00	0.00	0.00
	TOTAL CONTENTS	781.03	222.49	222.49	166.40	166.40	166.40	166.40
EFFLUENT	BREAK FLOW	0.00	558.52	558.52	710.02	962.03	1619.13	2522.04
	ECCS SPILL	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	558.52	558.52	710.02	962.03	1619.13	2522.04
***** TOTAL ACCOUNTABLE *****		**781.03**	**781.01**	**781.01**	**876.41**	**1128.42**	**1785.52**	**2688.44**

TABLE 6.2.1-51
MASS AND ENERGY BALANCE DOUBLE ENDED PUMP SUCTION GUILLOTINE (DEPSG) MINIMUM ECCS
(Sheet 2 of 2)

TIME (SEC)		0.00	28.75	28.75	209.01	615.10	1687.04	3600.00
		MASS (10^3 LBM)						
INITIAL ENERGY	IN RCS, ACCUMULATORS AND STEAM GENERATORS	994.92	994.92	994.92	994.92	994.92	994.92	994.92
ADDED ENERGY	PUMPED INJECTION	0.00	0.00	0.00	9.39	31.57	92.40	249.01
	DECAY HEAT	0.00	8.78	8.78	30.74	69.06	149.10	261.70
	HEAT FROM SECONDARY	0.00	5.04	5.04	5.04	5.04	5.04	5.04
	TOTAL ADDED	0.00	13.82	13.82	45.17	105.67	246.54	515.75
***** TOTAL AVAILABLE *****		**994.92**	**1008.74**	**1008.74**	**1040.10**	**1100.59**	**1241.46**	**1510.67**
DISTRIBUTION	REACTOR COOLANT	340.87	24.28	24.28	44.32	44.32	44.32	44.32
	ACCUMULATOR	18.85	10.37	10.36	0.00	0.00	0.00	0.00
	CORE STORED	24.20	9.61	9.61	4.91	4.71	4.21	3.33
	PRIMARY METAL	173.99	163.51	163.51	144.62	105.42	75.47	56.66
	SECONDARY METAL	121.28	118.33	118.33	108.65	88.19	57.56	42.10
	STEAM GENERATOR	315.72	325.86	325.86	295.69	235.12	148.06	107.19
	TOTAL CONTENTS	994.92	651.96	651.96	598.18	477.76	329.61	253.59
EFFLUENT	BREAK FLOW	0.00	356.20	356.20	438.83	619.75	936.67	1283.53
	ECCS SPILL	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	356.20	356.20	438.83	619.75	936.67	1283.53
***** TOTAL ACCOUNTALBE *****		**994.92**	**1008.16**	**1008.16**	**1037.01**	**1097.51**	**1266.29**	**1537.12**

TABLE 6.2.1-52

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TABLE 6.2.1-54
MASS AND ENERGY BALANCE DOUBLE-ENDED HOT LEG GUILLOTINE
(DEHLG)

(Sheet 1 of 2)

TIME (SECONDS)		0.00	22.50	22.50
MASS (THOUSAND LBM)				
INITIAL	IN RCS AND ACC	781.03	781.03	781.03
ADDED MASS	PUMPED INJECTION	0.00	0.00	0.00
	TOTAL ADDED	0.00	0.00	0.00
*** TOTAL AVAILABLE ***		781.03	781.03	781.03
DISTRIBUTION	REACTOR COOLANT	571.17	80.60	89.31
	ACCUMULATOR	209.86	152.34	143.63
	TOTAL CONTENTS	781.03	232.94	232.94
EFFLUENT	BREAK FLOW	0.00	548.07	548.07
	ECCS SPILL	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	548.07	548.07
*** TOTAL ACCOUNTABLE ***		781.03	781.01	781.01

TABLE 6.2.1-54
MASS AND ENERGY BALANCE DOUBLE-ENDED HOT LEG GUILLOTINE (DEHLG)

(Sheet 2 of 2)

TIME (SECONDS)		0.00	22.50	22.50
ENERGY (MILLION BTU)				
INITIAL ENERGY IN RCS, ACC, S, GEN		994.92	994.92	994.92
ADDED ENERGY	PUMPED INJECTION	0.00	0.00	0.00
	DECAY HEAT	0.00	8.16	8.16
	HEAT FROM SECONDARY	0.00	5.33	5.33
	TOTAL ADDED	0.00	13.49	13.49
*** TOTAL AVAILABLE ***		994.92	1008.41	1008.41
DISTRIBUTION	REACTOR COOLANT	340.87	22.40	23.18
	ACCUMULATOR	18.85	13.68	12.90
	CORE STORED	24.20	9.52	9.52
	PRIMARY METAL	173.99	162.92	162.92
	SECONDARY METAL	121.28	117.17	117.17
	STEAM GENERATOR	315.72	322.12	322.12
	TOTAL CONTENTS	994.92	647.81	647.81
EFFLUENT	BREAK FLOW	0.00	360.12	360.12
	ECCS SPILL	0.00	0.00	0.00
	TOTAL EFFLUENT	0.00	360.12	360.12
*** TOTAL ACCOUNTABLE ***		994.92	1007.93	1007.93

TABLE 6.2.1-55

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TABLE 6.2.1-56
LOCA MASS AND ENERGY RELEASES BASES FOR ANALYSIS

Plant model	4 loop, 12-ft core
Core power (license application), MWt	3612
Engineered safeguards design rating, MWt	3628
Nominal inlet temperature, F	558.0
Nominal outlet temperature, F	620.4
Steam pressure, psia	1032
Rod array	17 x 17
Total accumulator mass, lbm (minimum)	202367
Accumulator temperature, F	120
Assumed containment design pressure, psia	64.7
Pumped ECCS Injection (assumed for froth)	
Minimum	See Table 6.2.1-57
Maximum	See Table 6.2.1-57

TABLE 6.2.1-57

LOCA MASS AND ENERGY RELEASES ECCS SAFETY INJECTION FLOW VS. BACK PRESSURE

Pressure (psia)	Minimum Flow Rate (ft^3/sec)	Maximum Flow Rate (ft^3/sec)
14.7	11.8	22.3
34.7	11.1	21.3
54.7	10.4	20.3
74.7	9.7	19.2
94.7	9	18.1
114.7	8.3	16.8

TABLE 6.2.1-58

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TABLE 6.2.1-59

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TABLE 6.2.1-60

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TABLE 6.2.1-77

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TABLE 6.2.1-78
MAXIMUM PRESSURES - FEEDWATER LINE BREAK AT CONTAINMENT PENETRATION

VOL. NO.	PRESSURE (PSIA)	DIFFERENTIAL PRESSURE (PSI)	TIME (sec)
1	17.409 (19.0842)	2.709 (0.1712)	0.075 (10.0)
2	17.3436 (19.0429)	2.6436 (0.1299)	0.076 (10.0)
3	17.3479 (19.0327)	2.6479 (0.1197)	0.076 (10.0)
4	22.4316 (19.998)	7.7316 (1.085)	0.067 (10.0)
5	18.913	-----	10.0

NOTE:

Values for end of run (t = 10.0 sec) are given in parentheses.

TABLE 6.2.1-79
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TABLE 6.2.1-92
MAIN STEAM LINE BREAK AT CONTAINMENT PENETRATION MAXIMUM DIFFERENTIAL PRESSURES

VOLUME NO.	BREAK VOLUME NO.	WALL DIFFERENTIAL PRESSURE (PSI.)	TIME TO REACH PEAK (SECONDS)
1	1	16.32	0.027
2	2	11.72	0.023

TABLE 6.2.1-93

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TABLE 6.2.1-94
PEAK LOADS ON PRIMARY COMPONENTS DUE TO ASYMMETRIC PRESSURIZATION

(Sheet 1 of 2)

COMPONENT	LOAD (Maximum/Minimum)				
Rupture (Cubicle)	F_x	F_y	F_z	M_x	M_z
STEAM GENERATOR					
Main Steam (4)	36.8	22.6	86.9	17107	10746
	-51.4	-126.4	-20.9	-9327	-4889
Main Steam (3)	57.6	0.0	125.9	11774	23749
	-118.6	-172.9	-102.5	-12425	-5042
Feedwater (4)	118.9	79.7	20.0	33812	30728
	-64.5	-2.5	-125.0	-2082	-5478
RHR (4)	34.3	42.7	52.8	9736	10427
	-66.1	-7.0	-37.4	-17217	-21855
Aux. Feed (4)	87.2	15.0	132.6	29217	9953
	-37.4	-11.9	-114.9	-22017	-19243
RC Pump					
Main Steam (4)	5.9	12.7	5.3	642	774
	-5.0	-21.0	-7.6	-984	-823
Main Steam (3)	8.4	12.5	12.2	999	791
	-6.4	-13.8	-11.1	-1144	-894
Feedwater (4)	104.4	5.4	61.5	4113	1436
	-6.1	-7.9	-3.8	-864	-7051

TABLE 6.2.1-94
PEAK LOADS ON PRIMARY COMPONENTS DUE TO ASYMMETRIC PRESSURIZATION

(Sheet 2 of 2)

COMPONENT	LOAD (Maximum/Minimum)				
Rupture (Cubicle)	F_x	F_y	F_z	M_x	M_z
RHR (4)	16.0	10.4	10.1	534	743
	-16.9	-1.2	-9.6	-435	-784
Aux. Feed (4)	3.9	6.1	5.5	495	373
	-3.4	-3.9	-5.9	-432	-339
Pressurizer					
Press. Spray	40.0	1.0	43.2	8107	6287
	-29.5	-91.2	-36.8	-9035	-5508

Notes:

(1) Units are kips and inch-kips

(2) M_y is always zero

(3) Loads due to a RHR Line break are multiplied by 1.0058 for Unit 2

TABLE 6.2.1-95

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TABLE 6.2.1-96

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TABLE 6.2.1-97

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TABLE 6.2.1-98

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TABLE 6.2.1-99

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TABLE 6.2.1-100

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TABLE 6.2.1-101

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TABLE 6.2.1-102

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TABLE 6.2.1-103

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TABLE 6.2.1-104

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TABLE 6.2.1-105
THIS TABLE HAS BEEN DELETED

TABLE 6.2.2-1
CONTAINMENT SPRAY SYSTEM COMPONENT DESIGN PARAMETERS

(Sheet 1 of 3)

1. Containment Spray Pump

Quantity	4
Type	Horizontal centrifugal
Design pressure, psig	325
Design temperature, F	300
Design flow rate, gpm	3000
Total design head, ft	585
NPSH required, ft	13

2. Containment Spray Nozzle

Quantity 761/753 Unit 1/Unit 2	
Type	Spraco 1713A
Design flow per nozzle at 40 psi, gpm	15.2 ± 5 percent

3. Refueling Water Storage Tank

Quantity, per unit	1
Usable volume, gal	506,000
Total content, gal	525,000
Boric acid concentration, ppm	2400-2600
Design pressure, ft	50

TABLE 6.2.2-1
CONTAINMENT SPRAY SYSTEM COMPONENT DESIGN PARAMETERS

(Sheet 2 of 3)

Design temperature, F	120
Operating pressure, ft	41 (Hydraulic head)
4. Containment Spray Heat Exchanger	
Quantity	2
Type	Shell and U tube
Overall heat transfer	
(Btu/hr ft^2 F) coefficient	580.0
Flow, gpm	
Shell side	6080
Injection	5800
Recirculation	7200
Inlet Temperature, F	
Shell side, maximum	135
Tube side, maximum	243
Design Pressure, psig	
Shell	165
Tube	325
Design Temperature, F	
Shell	200
Tube	300

TABLE 6.2.2-1
CONTAINMENT SPRAY SYSTEM COMPONENT DESIGN PARAMETERS

(Sheet 3 of 3)

5. Piping and Valves	
Spray Discharge Lines	
Design pressure, psig	325
Design temperature, F	300
Spray Suction Lines	
Design pressure, psig	70
Design temperature, F	300
Chemical Supply Line	
Design pressure, psig	20
Design temperature, F	150
Eductor Piping	
Design pressure, psig	325
Design temperature, F	300
6. Valve Isolation Tank	
Quantity, per unit	2
Type	Vertical
Design pressure, psig	50
Design temperature, F	280

TABLE 6.2.2-2
CONTAINMENT SPRAY SYSTEM DESIGN PARAMETERS

Number of spray trains	2
Flow rate per spray train, gpm (minimum)	
Injection	5800
Recirculation	7200
Number of headers per spray train	7
Spray initiation - time after LOCA (sec.)	Note 1

Note 1:

See Section 6.2.1.4.8, Item 5

TABLE 6.2.2-3
CONTAINMENT DESIGN PARAMETERS

SEE TABLE 6.2.1-1

TABLE 6.2.2-4
CONTAINMENT SPRAY SYSTEM MATERIALS[a]

Component	Quantity	Material
Chemical additive tank	1	SA-240 Type 304
Recirculation sump strainers	2	ASTM 300 Series stainless steel
Spray Pump	4	
Casing		SA-351 CF 8M
Shaft		SA-182 Grade 316L
Impeller		SA-351 CF 8M or CF 3M
Chemical eductor	4	SA-182 Grade F304
Heat exchanger	2	
Tubes		SA-249 Type 304
Shell		SA-516 Grade 70
Valves, 2 1/2 in. or larger	40	SA-351 CF 8
Piping		SA-312 or SA-358
RWST	1	TP 304 or 316
Liner		SA-240 Type 304L
Structural wall		Reinforced concrete
Stop valves, chemical additive tank	4	SA-351 CF 8

a) All pressure boundary materials conform to ASME B&PV Code, Section III, Class 2 or 3.

TABLE 6.2.2-5
SINGLE FAILURE ANALYSIS - CONTAINMENT SPRAY SYSTEM

Component	Malfunction	Comments and Consequences
1. Spray nozzles	Clogged	Large number of nozzles precludes clogging of a significant number.
2. Pumps	Fail to start	Four 50-percent pumps are provided (two on each train). Operation of two on one train is required.
3. Valves		
Automatically operated Containment spray pump isolation valves[a]	Fail to open	Two valves are provided. Operation of one is required.
Containment sump recirculation isolation valves (operated from Control Room)	Fail to open	Two lines are in parallel, one to each pair of spray pumps. Operation of one valve in one line is required.
Miniflow recirculation valves	Fail to close	Operator must ensure valve is closed prior to switchover.
4. Heat Exchanger	Clogging of tubes or loss of component cooling water	Two heat exchangers are provided. One must remain operable.
5. Recirculation sump	Clogged	Two sumps are incorporated into system. One must remain unclogged and operable.

a) Open on coincidence of two out of four high-3 containment pressure signals

TABLE 6.2.2-6
THIS TABLE HAS BEEN DELETED.

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 1 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
1	MI-1	MS	Main Steam From Steam Generator #1	32	57	17	Sat. Steam	No	10.3-1
2	MI-1	MS	MSIV Bypass From Steam Generator #1	4	57	17	Sat. Steam	No	10.3-1
3	MI-1	MS	Drain From Main Steamline #1	2	57	17	Sat. Water	No	10.3-1
4	MI-1	MS	Main Steam to Aux. F.P.T. From Steam Line #1	4	57	17	Sat. Steam	Yes	10.3-1
4a	MI-1	MS	TDAFW Pump Bypass Warm-up Valve	1	57	17	Sat. Steam	No	10.3-1
5	MI-1	MS	Main Steam from Steam Generator #1	8	57	17	Sat. Steam	No	10.3-1
5a	MI-1	MS	N_2 Supply to Steam Generator #1	3/4	57	17	Sat. Water	No	10.3-1
5b	MI-1	MS	Main Steam Safety Valves for Steam Generator #1	6	57	17	Sat. Steam	No	10.3-1
6	MI-2	MS	Main Steam From Steam Generator #2	32	57	17	Sat. Steam	No	10.3-1
7	MI-2	MS	MSIV Bypass From Steam Generator #2	4	57	17	Sat. Steam	No	10.3-1
8	MI-2	MS	Drain From Main Steamline #2	2	57	17	Sat. Water	No	10.3-1
9	MI-2	MS	Main Steam From Steam Generator #2	8	57	17	Sat. Steam	No	10.3-1
9a	MI-2	MS	N_2 Supply to Steam Generator #2	3/4	57	17	Sat. Water	No	10.3-1
9b	MI-2	MS	Main Steam Safety Valves From Steam Generator #2	6	57	17	Sat. Steam	No	10.3.1
10	MI-3	MS	Main Steam From Steam Generator #3	32	57	17	Sat. Steam	No	10.3-1

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 2 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
11	MI-3	MS	MSIV Bypass From Steam Generator #3	4	57	17	Sat. Steam	No	10.3-1
12	MI-3	MS	Drain From Main Steamline #3	2	57	17	Sat. Water	No	10.3-1
13	MI-3	MS	Main Steam From Steam Generator #3	8	57	17	Sat. Steam	No	10.3-1
13a	MI-3	MS	N_2 Supply to Steam Generator #3	3/4	57	17	Sat. Water	No	10.3-1
13b	MI-3	MS	Main Steam Safety Valves From Steam Generator #3	6	57	17	Sat. Steam	No	10.3-1
14	MI-4	MS	Main Steam From Steam Generator #4	32	57	17	Sat. Steam	No	10.3-1
15	MI-4	MS	MSIV Bypass From Steam Generator #4	4	57	17	Sat. Steam	No	10.3-1
16	MI-4	MS	Drain From Main Steam Line #4	2	57	17	Sat. Water	No	10.3-1
17	MI-4	MS	Main Steam to Aux. F.P.T. From Steam line #4	4	57	17	Sat. Steam	Yes	10.3-1
17a	MI-4	MS	TDAFW Pump Bypass Warm-up Valve	1	57	17	Sat. Steam	No	10.3-1
18	MI-4	MS	Main Steam From Steam Generator #4	8	57	17	Sat. Steam	No	10.3-1
18a	MI-4	MS	N_2 Supply to Steam Generator #4	3/4	57	17	Sat. Water	No	10.3-1
18b	MI-4	MS	Main Steam Safety Valves From Steam Generator #4	6	57	17	Sat. Steam	No	10.3-1
19	MI-5	FW	Feedwater to Steam Generator #1	18	57	16	Water	No	10.4-9
20	MI-5	FW	Feedwater Sample (FW to Steam Generator #1)	3/4	57	16	Water	No	10.4-9
20a	MV-18	FW	Auxiliary Feedwater to Steam Generator #1	4	57	36	Water	Yes	10.4-11

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 3 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
20b	MI-5	FW	N_2 Supply to Steam Generator #1	3/4	57	16	Water	No	10.4-9
20c	MV-18	FW	Feedwater Preheater Bypass Line to S.G. #1	6	57	36	Water	No	10.4-9
20d	MI-5	FW	Feedwater Bypass Line S.G.#1	3	57	16	Water	No	10.4-9
21	MI-6	FW	Feedwater to Steam Generator #2	18	57	16	Water	No	10.4-9
22	MI-6	FW	Feedwater Sample (FW to Steam Generator #2)	3/4	57	16	Water	No	10.4-9
22a	MV-19	FW	Auxiliary Feedwater to Steam Generator #2	4	57	36	Water	Yes	10.4-11
22b	MI-6	FW	N_2 Supply to Steam Generator #2	3/4	57	16	Water	No	10.4-9
22c	MV-19	FW	Feedwater Preheater Bypass Line to S.G. #2	6	57	36	Water	No	10.4-9
22d	MI-6	FW	Feedwater Bypass Line S.G. #2	3	57	16	Water	No	10.4-9
23	MI-7	FW	Feedwater to Steam Generator #3	18	57	16	Water	No	10.4-9
24a	MV-20	FW	Auxiliary Feedwater to Steam Generator #3	4	57	36	Water	Yes	10.4-11
24b	MI-7	FW	N_2 Supply to Steam Generator #3	3/4	57	16	Water	No	10.4-9
24c	MV-20	FW	Feedwater Preheater Bypass Line to S.G. #3	6	57	36	Water	No	10.4-9
24d	MI-7	FW	Feedwater Bypass Line S.G. #3	3	57	16	Water	No	10.4-9
25	MI-8	FW	Feedwater to Steam Generator #4	18	57	16	Water	No	10.4-9
26a	MV-17	FW	Auxiliary Feedwater to Steam Generator #4	4	57	36	Water	Yes	10.4-11
26b	MI-8	FW	N_2 Supply to Steam Generator #4	3/4	57	16	Water	No	10.4-9
26c	MV-17	FW	Feedwater Preheater Bypass Line to S.G. #4	6	57	36	Water	No	10.4-9

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 4 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
26d	MI-8	FW	Feedwater Bypass Line S.G.#4	3	57	16	Water	No	10.4-9
27	MI-9	MS	Blowdown From Steam Generator #3	3	57	35	Sat. Water	No	10.3-1
28	MI-10	MS	Blowdown From Steam Generator #2	3	57	35	Sat. Water	No	10.3-1
29	MI-11	MS	Blowdown From Steam Generator #1	3	57	35	Sat. Water	No	10.3-1
30	MI-12	MS	Blowdown From Steam Generator #4	3	57	35	Sat. Water	No	10.3-1
31	MI-13	-	Spare	12	50	-	-	-	-
32	MII-1	CS	Letdown Line to Letdown Heat Exchanger	3	55	37	Water	No	9.3-10
33	MII-2	RH	R.H.R From Hot Leg Loop #4	12	55	11	Water	No	5.4-6
34	MII-3	RH	R.H.R From Hot Leg Loop #1	12	55	11	Water	No	5.4-6
35	MII-4	SI	R.H.R to Cold Leg Loops #1 and #2	10	55	8	Water	Yes	6.3-1
36	MII-5	SI	R.H.R To Cold Leg Loops #3 and #4	10	55	8	Water	Yes	6.3-1
37	MII-6	-	Spare	12	50	-	-	-	-
38	MII-7	-	Spare	24	50	-	-	-	-
39	MII-8	-	Spare	24	50	-	-	-	-
40a	MII-9(A)	-	Maintenance Penetration	2	56	31	Air	No	9.4-6
40b	MII-9(B)	-	Maintenance Penetration	1 1/2	56	31	Air	No	9.4-6
40c	MII-9(C)	-	Maintenance Penetration	2	56	31	Air	No	9.4-6
41	MIII-1	RC	Reactor Make Up Water to Pressurizer Relief Tank & R.C. Pump Stand Pipe	3	56	4	Water	No	5.1-1

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 5 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
41a	MIII-1	RC	Penetration Thermal Relief	3/4	56	4	Water	No	5.1-1
42	MIII-2	SI	S.I. To Cold Leg Loops #1, #2, #3, & #4	3	55	9	Water	Yes	6.3-1
43	MIII-3	SI	S.I. to R.C. System Hot Leg Loops #2 & #3	4	55	8	Water	Yes	6.3-1
44	MIII-4	SI	S.I. To R.C. System Hot Leg Loops #1 & #4	4	55	8	Water	Yes	6.3.1 (Sh. 3)
45	MIII-5	SI	S.I. To R.C. System Cold Leg Loops #1, #2, #3 & #4	4	55	33	Water	Yes	6.3-1
46	MIII-6	CS	Charging Line to Regenerative Heat Exchanger	3	55	25	Water	No	9.3-10
47	MIII-7	CS	Seal Injection to R.C. Pump (Loop #1)	2	55	15	Water	No	9.3-10
48	MIII-8	CS	Seal Injection to R.C. Pump (Loop #2)	2	55	15	Water	No	9.3-10
49	MIII-9	CS	Seal Injection to R.C. Pump (Loop #3)	2	55	15	Water	No	9.3-10
50	MIII-10	CS	Seal Injection to R.C. Pump (Loop #4)	2	55	15	Water	No	9.3-10
51	MIII-11	CS	Seal Water Return And Excess Letdown	2	55	24	Water	No	9.3-10
52	MIII-12	WP	R.C.D.T Heat Exchanger To Waste Hold Up Tank	3	56	27	Water	No	11.2-2
52a	MIII-12	WP	Penetration Thermal Relief	3/4	56	27	Water	No	11.2-2
53	MIII-13	-	Spare	10	50	-	-	-	-
54	MIII-14	CT	Containment Spray To Spray Header (TR. B)	16	56	25	Water	Yes	6.2.2-1

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 6 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
55	MIII-15	CT	Containment Spray to Spray Header (TR. A)	16	56	25	Water	Yes	6.2.2-1
56	MIII-16	SF	Refueling Water Purification to Refueling Cavity	4	56	14	Air[(a)]	No	9.1-13
57	MIII-17	LT	Containment Leak Rate Test	10	56	31	Air	No	9.4-6
58	MIII-18	VA	Hydrogen Purge Supply	12	56	20	Air	No	9.4-6
59	MIII-19	VA	Hydrogen Purge Exhaust	12	56	21	Air	No	9.4-6
60	MIII-20	DD	Demineralized Water Supply	3	56	5	Water	No	9.2-5
60a	MIII-20	DD	Penetration Thermal Relief	3/4	56	5	Water	No	9.2-5
61	MIII-21	VD	Containment Sump Pump Discharge	4	56	22	Water	No	9.3-5
61a	MIII-21	VD	Penetration Thermal Relief	3/4	56	22	Water	No	9.3-5
62	MIII-22	CI	Instrument Air To Containment	3	56	7	Air	No	9.3-1
63	MIII-23	SI	R.H.R To Hot Leg Loops #2& #3	10	55	8	Water	Yes	6.3-1 (Sh. 3)
64	MIII-24	-	Spare	12	50	-	-	-	-
65	MIII-25	-	Spare	12	50	-	-	-	-
66	MIII-26	-	Spare	12	50	-	-	-	-
67	MIII-27	SF	Refueling Cavity To Refueling Water Purification Pump	4	56	14	Air[(a)]	No	9.1-13
68	MIII-28	-	Spare	12	50	-	-	-	-
69	MIII-29	-	Spare	12	50	-	-	-	-
70	MIII-30	LT	Containment Leak Rate Test Pressurization	12	56	31	Air	No	9.4-6

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 7 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
71	MIII-31	SF	Refueling Cavity Skimmer Pump Discharge (System no longer used)	3	56	14	Air	No	9.1-13
72	MIII-32	-	Spare	12	50	-	-	-	-
73	MIV-1(a)	MS	Sample From Steam Generator #1	3/4	57	35	Sat. Water	No	10.3-1
74	MIV-1(b)	PS	R.C. Sample From Hot Legs	3/8	55	29	Sat. Water	No	9.3-4
74a	MIV-1(b)	PS	Penetration Thermal Relief	3/4	55	29	Sat. Water	No	9.3-4
75	MIV-1(c)	-	Spare	2	50	-	-	-	-
76	MIV-2(a)	MS	Sample From Steam Generator #2	3/4	57	35	Sat. Water	No	10.3-1
77	MIV-2(b)	PS	Pressurizer Liquid Space Sample	3/4	55	1	Sat. Water	No	9.3-4
77a	MIV-2(b)	PS	Penetration Thermal Relief	3/4	55	1	Sat. Water	No	9.3-4
78	MIV-2(c)	PS	Pressurizer Steam Space Sample	3/4	55	1	Steam/Water	No	9.3-4
78a	MIV-2(c)	PS	Penetration Thermal Relief	3/4	55	1	Steam/Water	No	9.3-4
79	MIV-3(a)	MS	Sample From Steam Generator #3	3/4	57	35	Sat. Water	No	10.3-1
80	MIV-3(b)	PS	Sample From Accumulators	3/4	55	30	Water	No	9.3-4
80a	MIV-3(b)	PS	Penetration Thermal Relief	3/4	55	30	Water	No	9.3-4
81	MIV-3(c)	WP	RC Pass Sample Discharge to RCDT	3/4	55	43	Water	No	11.2-2
81a	MIV-3(c)	WP	Penetration Thermal Relief	3/4	55	26	Water	No	11.2-2
82	MIV-4(a)	MS	Sample From Steam Generator #4	3/4	57	35	Sat. Water	No	10.3-1
83	MIV-4(b)	SI	Accumulator Test & Fill	3/4	55	6	Water	No	6.3-1
83a	MIV-4(b)	SI	Penetration Thermal Relief	3/4	55	6	Water	No	6.3-1
84	MIV-4(c)	PS	Containment Air PASS Return	1	56	34	Air	No	9.4-6
85	MIV-5(a)	-	Spare	1	50	-	-	-	-
86	MIV-5(b)	-	Spare	2	50	-	-	-	-

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 8 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
87	MIV-5(c)	-	Spare	2	50	-	-	-	-
88	MIV-6(a)	-	Spare	1	50	-	-	-	-
89	MIV-6(b)	-	Spare	1	50	-	-	-	-
90	MIV-6(c)	-	Spare	2	50	-	-	-	-
91	MIV-7(a)	-	Spare	1	50	-	-	-	-
92	MIV-7(b)	ESFAS	Containment Press. Sensing PT-937/PT-938	1/2	1.11	10	Hydraulic Fluid	Yes	9.4-6
93	MIV-7(c)	-	Spare	2	50	-	-	-	-
94	MIV-8(a)	RM	Radiation Monitoring Sample	1	56	3	Air	No	9.4-6
95	MIV-8(b)	ESFAS	Containment Press. Sensing PT-936/PT-939	1/2	1.11	10	Hydraulic Fluid	Yes	9.4-6
96	MIV-8(c)	-	Spare	N/A	50	-	-	-	-
97	MIV-9(a)	PS	Containment Air PASS Inlet	1	56	3	Air	No	9.4-6
98	MIV-9(b)	ESFAS	Containment Press. Sensing PT-935	1/2	1.11	10	Hydraulic Fluid	Yes	9.4-6
99	MIV-9(c)	-	Spare	2	50	-	-	-	-
100	MIV-10(a)	PS	Containment Air PASS Inlet	1	56	3	Air	No	9.4-6
101	MIV-10(b)	ESFAS	Containment Pressure Sensing PT-934	1/2	1.11	10	Hydraulic Fluid	Yes	9.4-6
102	MIV-10(c)	RM	Radiation Monitoring Sample Return	1	56	34	Air	No	9.4-6
103	MIV-11(a)	-	Spare	1	50	-	-	-	-
104	MIV-11(b)	SI	N_2 Supply To Accumulators	1	56	7	Nitrogen	No	6.3-1
105	MIV-11(c)	WP	H_2 Supply to R.C. Drain Tank	3/4	56	13	Hydrogen	No	11.2-2
106	MIV-12(a)	-	Spare	2	50	-	-	-	-

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 9 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
107	MIV-12(b)	-	Spare	2	50	-	-	-	-
108	MIV-12(c)	-	Spare	2	50	-	-	-	-
109	MV-1	VA	Containment Purge Air Supply	48	56	18	Air	No	9.4-6
110	MV-2	VA	Containment Purge Air Exhaust	48	56	19	Air	No	9.4-6
111	MV-3	CC	CCW Supply To Excess Letdown & R.C. Drain Tank Heat Exchanger	4	57	32	Water	No	9.2-3
112	MV-4	CC	CCW Return From Excess Letdown & R.C. Drain Tank Heat Exchanger	4	57	12	Water	No	9.2-3
113	MV-5	CA	Service Air to Containment	3	56	7	Air	No	9.3-1
114	MV-6	CC	Containment CCW Drain Tank Pumps Discharge	2	56	26	Water	No	9.2-3
114a	MV-6	CC	Penetration Thermal Relief	3/4	56	26	Water	No	9.2-3
115	MV-7	LT	Containment Leak Rate Test Pressure Sensing	8	56	38	Air	No	9.4-6
116	MV-8	RC	Nitrogen Supply to PRT	1	56	13	Nitrogen	No	5.1-1
117	MV-9	CC	CCW Return From R.C.P.'s Motors	8	56	24	Water	No	9.2-3
118	MV-10	CC	CCW Supply to R.C.P.'s Motors	10	56	25	Water	No	9.2-3
119	MV-11	CC	CCW Return From R.C.P.'s Thermal Barrier	4	56	24	Water	No	9.2-3
120	MV-12	CH	Chilled Water Supply to Containment Coolers	6	56	25	Water	No	9.4-11
120a	MV-12	CH	Penetration Thermal Relief	3/4	56	25	Water	No	9.4-11
121	MV-13	CH	Chilled Water Return From Containment Coolers	6	56	28	Water	No	9.4-11
121a	MV-13	CH	Penetration Thermal Relief	3/4	56	28	Water	No	9.4-11
122	MV-14	VA	Containment Pressure Relief	18	56	19	Air	No	9.4-6

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 10 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
123	MV-15	-	Maintenance Penetration	1	56	31	Air	No	9.4-6
124	MV-16	FP	Fire Protection System Isolation	4	56	39	Air[b]	No	9.5-48
125	MS-1	SI	Containment Recirculation	16	56	2	Water	Yes	6.3-1
			Sump To R.H.R. Pump (TR. A) Thermal Relief	3/4	56	2	Water	No	
126	MS-2	SI	Containment Recirculation	16	56	2	Water	Yes	6.3-1
			Sump to R.H.R. Pump (TR. B) Thermal Relief	3/4	56	2	Water	No	
127	MS-3	CT	Containment Recirculation	16	56	2	Water	Yes	6.2.2-1
			Sump to Spray Pumps (TR. A) Thermal Relief	3/4	56	2	Water	No	
128	MS-4	CT	Containment Recirculation	16	56	2	Water	Yes	6.2.2-1
			Sump To Spray Pumps (TR. B) Thermal Relief	3/4	56	2	Water	No	
129	MS-5	SF	Fuel Transfer Tube	20	50	23	Air[a]	No	9.1-13 6.2.6-1
130	Unit 1								
	Personnel Airlock Hydraulics	BS/CB	Airlock Hydraulic System	3/4	57	40	Hydraulic	No	3.8-22
131	Unit 1								
	Personnel Airlock	BS/CB	Hydraulically Operated Airlock Equalization	3	56	41	Air	No	3.8-22
131a	Unit 1								
	Personnel Airlock	BS/CB	Manual Airlock Equalization	3/4	56	41	Air	No	3.8-22

TABLE 6.2.4-1
CONTAINMENT ISOLATION VALVING APPLICATION

(Sheet 11 of 11)

Item	Penetration Number	System	Line or Service	Line Size (Inches)	NRC General Design Criterion or Reg. Guide Met	Isolation Valving Arrangement (Fig. 6.2.4-1)	Fluid Contained	Engineered Safeguard Feature	FSAR Figure Number
132	Emergency Airlock	BS/CB	Manual Airlock Equalization	2	56	42	Air	No	3.8-23
133	Unit 2								
	Personnel Airlock Hydraulics	BS/CB	Airlock Hydraulic System	3/4	56	44	Hydraulic	No	3.8-23
134	Unit 2								
	Personnel Airlock	BS/CB	Hydraulically Operated Airlock Equalization	3	56	45	Air	No	3.8-22
134a	Unit 2 Personnel Airlock	BS/CB	Manual Airlock Equalization	3/4	56	45	Air	No	3.8-23

a) Fluid: Air During Normal Operation, Water During Refueling - kThe valve (SF-0001) associated with this penetration (ITEM 129) is Open during Refueling. Valve SF-0001 is a non-safety related valve that has no direct Containment Isolation function during MODES 1-4. This valve is Normally Closed in MODES 1-4. The valve may be opened in MODES 1-4 except when the Fuel Transfer Canal is flooded to maintain the fuel Transfer Flange as Leak Rate Tested. Opening the valve allows for equipment testing and to support preparations for Refueling Outages.

b) This penetration is drained and purged with nitrogen prior to plant operation.

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 1 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
1	HV-2333A	Outside	Note 1	N/A	40'/38'-1"	Y - Globe/Hydr. N_2 Actuator	Auto close	Remote Manual
2	HV-2333B	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
3	HV-2409	Outside	Note 1	N/A	-	Globe/Air	Auto close	Remote Manual
4	HV-2452-2	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
	MS-101 (Note 16)	Outside	Note 1	N/A	-	Gate/Gear	Local Manual	N/A
4a	MS-711	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
5	PV-2325	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
5a	MS-390	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
5b	MS-021	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-022	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-023	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-024	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-025	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
6	HV-2334A	Outside	Note 1	N/A	38'/35'-9"	Y - Globe/Hydr. N_2 Actuator	Auto close	Remote Manual
7	HV-2334B	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
8	HV-2410	Outside	Note 1	N/A	-	Globe/Air	Auto close	Remote Manual
9	PV-2326	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
9a	MS-387	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
9b	MS-058	Outside	Note 1	N/A	-	Safety Relief	Self Actuated	N/A
	MS-059	Outside	Note 1	N/A	-	Safety Relief	Self Actuated	N/A
	MS-060	Outside	Note 1	N/A	-	Safety Relief	Self Actuated	N/A
	MS-061	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-062	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
10	HV-2335A	Outside	Note 1	N/A	40'/38'-5"	Y - Globe/Hydr. N_2 Actuator	Auto close	Remote Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 2 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
11	HV-2335B	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
12	HV-2411	Outside	Note 1	N/A	-	Globe/Air	Auto close	Remote Manual
13	PV-2327	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
13a	MS-384	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
13b	MS-093	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-094	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-095	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-096	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-097	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
14	HV-2336A	Outside	Note 1	N/A	38'/35'-9"	Y - Globe/Hydr. N_2 Actuator	Auto close	Remote Manual
15	HV-2336B	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
16	HV-2412	Outside	Note 1	N/A	-	Globe/Air	Auto close	Remote Manual
17	HV-2452-1	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
	MS-128 (Note 16)	Outside	Note 1	N/A	-	Gate/Gear	Local Manual	N/A
17a	MS-712	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
18	PV-2328	Outside	Note 1	N/A	-	Globe/Air	Remote Manual	N/A
18a	MS-393	Outside	Note 1	N/A	-	Globe/Manual	Local	N/A
18b	MS-129	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-130	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-131	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-132	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
	MS-133	Outside	Note 1	N/A	-	Safety Relief	Self-Actuated	N/A
19	HV-2134	Outside	Note 1	N/A	10'/12'-10"	Gate/Hydr. N_2 Actuator	Auto close	Remote Manual
20	FW-0116	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
20a	HV-2491A	Outside	Note 1	N/A	50'/48'-9"	Gate/Motor	Remote Manual	Local Manual
	HV-2491B	Outside	Note 1	N/A	50'/51'-9"	Gate/Motor	Remote Manual	Local Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 3 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
20b	FW-106	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
20c	2-FV-2193	Outside	Note 1	N/A	30'-9"	Globe/Air	Auto close	Remote Manual
20d	HV-2185	Outside	Note 1	N/A	12'-7"/11'-3"	Globe/Air	Auto close	Remote Manual
21	HV-2135	Outside	Note 1	N/A	10'/11'-8"	Gate/Hydr. N_2 Actuator	Auto close	Remote Manual
22	FW-0113	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
22a	HV-2492A	Outside	Note 1	N/A	50'/51'-8"	Gate/Motor	Remote Manual	Local Manual
	HV-2492B	Outside	Note 1	N/A	50'/54'-7"	Gate/Motor	Remote Manual	Local Manual
22b	FW-104	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
22c	2-FV-2194	Outside	Note 1	N/A	30'-9"	Globe/Air	Auto close	Remote Manual
22d	HV-2186	Outside	Note 1	N/A	12'-8"/11'-8"	Globe/Air	Auto close	Remote Manual
23	HV-2136	Outside	Note 1	N/A	10'/15'-4"	Gate/Hydr. N_2 Actuator	Auto close	Remote Manual
24a	HV-2493A	Outside	Note 1	N/A	50'/50'-5"	Gate/Motor	Remote Manual	Local Manual
	HV-2493B	Outside	Note 1	N/A	50'/53'-5"	Gate/Motor	Remote Manual	Local Manual
24b	FW-102	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
24c	2-FV-2195	Outside	Note 1	N/A	33'-0"	Globe/Air	Auto close	Remote Manual
24d	HV-2187	Outside	Note 1	N/A	12'-8"/11'-9"	Globe/Air	Auto close	Remote Manual
25	HV-2137	Outside	Note 1	N/A	10'/15'-3"	Gate/Hydr. N_2 Actuator	Auto close	Remote Manual
26	-	-	-	-	-	-	-	-
26a	HV-2494A	Outside	Note 1	N/A	50'/50'-4"	Gate/Motor	Remote Manual	Local Manual
	HV-2494B	Outside	Note 1	N/A	50'/53'8"	Gate/Motor	Remote Manual	Local Manual
26b	FW-108	Outside	Note 1	N/A	-	Globe/Manual	Local Manual	N/A
26c	2-FV-2196	Outside	Note 1	N/A	33'-1"	Globe/Air	Auto close	Remote Manual
26d	HV-2188	Outside	Note 1	N/A	12'-8"/11'-9"	Globe/Air	Auto close	Remote Manual
27	HV-2399	Outside	Note 1	N/A	17'/16'-6"	Globe/Air	Auto close	Remote Manual
28	HV-2398	Outside	Note 1	N/A	17'/16'-2"	Globe/Air	Auto close	Remote Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 4 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
29	HV-2397	Outside	Note 1	N/A	17'/16'-0"	Globe/Air	Auto close	Remote Manual
30	HV-2400	Outside	Note 1	N/A	17'/16'-8"	Globe/Air	Auto close	Remote Manual
31	-	-	A	N/A	-	-	-	-
32	8152	Outside	C	YES	7'/7'-7"	Globe/Air	Auto close	Remote Manual
	8160	Inside	C	YES	-	Globe/Air	Auto close	Remote Manual
33	8701B	Inside	Note 17	N/A	N/A	Gate/Motor	Remote Manual	Remote Manual
	8708B	Inside	Note 17	N/A	N/A	Relief	Self-Actuated	N/A
34	8701A	Inside	Note 17	N/A	N/A	Gate/Motor	Remote Manual	Local Manual
	8708A	Inside	Note 17	N/A	N/A	Relief	Self-Actuated	N/A
35	8809A	Outside	Note 3	N/A	8'/9'-11"	Gate/Motor	Remote Manual	Local Manual
	8818A	Inside	Note 11	N/A	-	Check	Self-Actuated	N/A
	8818B	Inside	Note 11	N/A	-	Check	Self-Actuated	N/A
	8890A	Inside	Note 11	N/A	-	Globe/Air	Auto close	Remote Manual
36	8809B	Outside	Note 3	N/A	3'/5'-9"	Gate/Motor	Remote Manual	Local Manual
	8818C	Inside	Note 11	N/A	-	Check	Self-Actuated	N/A
	8818D	Inside	Note 11	N/A	-	Check	Self-Actuated	N/A
	8890B	Inside	Note 11	N/A	-	Globe/Air	Auto close	Remote Manual
37	-	-	A	N/A	-	-	-	-
38	-	-	A	N/A	-	-	-	-
39	-	-	A	N/A	-	-	-	-
40a	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
40b	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
40c	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
41	8046	Inside	C	YES	-	Check	Self-Actuated	N/A
	8047	Outside	C	YES	16'/18'-1"	Diaphragm/Air	Auto close	Remote Manual
41a	RC-036	Outside	C	YES	-	Relief	Self-Actuated	N/A

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 5 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
42	8815	Inside	Note 2	N/A	-	Check	Self-Actuated	N/A
	8801A	Outside	Note 7	N/A	13'/16'-1"	Gate/Motor	Remote Manual	N/A
	8801B	Outside	Note 7	N/A	13'/16'-1"	Gate/Motor	Remote Manual	N/A
	8843	Inside	Note 2	N/A	-	Globe/Air	Auto close	Remote Manual
43	8802A	Outside	Note 12	N/A	11'/13'-0"	Gate/Motor	Remote Manual	Local Manual
	SI-8905B	Inside	Note 12	N/A	-	Check	Self-Actuated	N/A
	SI-8905C	Inside	Note 12	N/A	-	Check	Self-Actuated	N/A
	8881	Inside	Note 12	N/A	-	Globe/Air	Auto close	Remote Manual
44	8802B	Outside	Note 12	N/A	12'/13'-9"	Gate/Motor	Remote Manual	Local Manual
	SI-8905A	Inside	Note 12	N/A	-	Check	Self-Actuated	N/A
	SI-8905D	Inside	Note 12	N/A	-	Check	Self-Actuated	N/A
	8824	Inside	Note 12	N/A	-	Globe/Air	Auto close	Remote Manual
45	8835	Outside	Note 13	N/A	16'/19'-1"	Gate/Motor	Remote Manual	Local Manual
	SI-8819A	Inside	Note 13	N/A	-	Check	Self-Actuated	N/A
	SI-8819B	Inside	Note 13	N/A	-	Check	Self-Actuated	N/A
	SI-8819C	Inside	Note 13	N/A	-	Check	Self-Actuated	N/A
	SI-8819D	Inside	Note 13	N/A	-	Check	Self-Actuated	N/A
	8823	Inside	Note 13	N/A	-	Globe/Air	Auto close	Remote Manual
46	8105	Outside	C	YES	3'/10'-7"	Gate/Motor	Auto close	Remote Manual
	8381	Inside	C	YES	-	Check	Self-Actuated	N/A
47	CS-8368A	Inside	Note 4	N/A	-	Check	Self-Actuated	N/A
	8351A	Outside	Note 4	N/A	6'/15'-7"	Globe/Motor	Remote Manual	Local Manual
48	CS-8368B	Inside	Note 4	N/A	-	Check	Self-Actuated	N/A
	8351B	Outside	Note 4	N/A	7'/9'-9"	Globe/Motor	Remote Manual	Local Manual
49	CS-8368C	Inside	Note 4	N/A	-	Check	Self-Actuated	N/A
	8351C	Outside	Note 4	N/A	2'/11'-11"	Globe/Motor	Remote Manual	Local Manual
50	CS-8368D	Inside	Note 4	N/A	-	Check	Self-Actuated	N/A
	8351D	Outside	Note 4	N/A	10'/10'-3"	Globe/Motor	Remote Manual	Local Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 6 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
51	8100	Outside	C	YES	7'/7'-1"	Globe/Motor	Auto close	Remote Manual
	8112	Inside	C	YES	-	Globe/Motor	Auto close	Remote Manual
	CS-8180	Inside	C	YES	-	Check	Self-Actuated	N/A
52	7136	Inside	C	YES	-	Diaph./Air	Auto close	Remote Manual
	LCV-1003	Outside	C	YES	30'/23'-11"	Globe/Air	Auto close	Remote Manual
	7135	Outside	C	YES	30'/24'-2"	Diaph./Manual	Local Manual	N/A
52a	WP-7176	Outside	C	YES	-	Relief	Self-Actuated	N/A
53	-	-	A	N/A	-	-	-	-
54	HV-4777	Outside	Note 3	N/A	15'/16'-7"	Gate/Motor	Remote Manual	N/A
	CT-145	Inside	Note 3	N/A	-	Check	Self-Actuated	N/A
55	HV-4776	Outside	Note 3	N/A	9'/16'-9"	Gate/Motor	Remote Manual	N/A
	CT-142	Inside	Note 3	N/A	-	Check	Self-Actuated	N/A
56	SF-011	Outside	C	YES	11'/18'-6"	Diaphragm/Man.	Local Manual	N/A
	SF-012	Inside	C	YES	-	Diaphragm/Man.	Local Manual	N/A
57	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
58	HV-5542	Outside	C	YES	5'-6"/1'-5"	Butterfly/Motor	Auto close	Remote Manual
	HV-5543	Inside	C	NO	-	Butterfly/Motor	Auto close	Remote Manual
	HV-5563	Inside	C	NO	-	Butterfly/Motor	Auto close	Remote Manual
59	HV-5540	Outside	C	YES	5'-6"/5'-10"	Butterfly/Motor	Auto close	Remote Manual
	HV-5541	Inside	C	NO	-	Butterfly/Motor	Auto close	Remote Manual
	HV-5562	Inside	C	NO	-	Butterfly/Motor	Auto close	Remote Manual
60	HV-5365	Outside	C	YES	12'/14'-8"	Globe/Air	Auto close	Remote Manual
	HV-5366	Inside	C	YES	-	Globe/Air	Auto close	Remote Manual
60a	DD-430	Outside	C	YES	-	Relief	Self-Actuated	N/A
61	HV-5157	Outside	C	YES	14'/16'-2"	Diaphragm/Air	Auto close	Remote Manual
	HV-5158	Inside	C	NO	-	Diaphragm/Air	Auto close	Remote Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 7 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
61a	1VD-907 (2VD-0896)	Outside	C	YES	-	Relief	Self-Actuated	N/A
62	HV-3487	Outside	C	YES	14'/17'-3"	Globe/Air	Auto close	Remote Manual
	CI-030	Inside	C	YES	-	Check	Self-Actuated	N/A
63	8840	Outside	Note 3	N/A	26'/27'-8"	Gate/Motor	Remote Manual	Local Manual
	8825	Inside	Note 14	N/A	-	Globe/Air	Auto close	Remote Manual
	8841A	Inside	Note 14	N/A	-	Check	Self-Actuated	N/A
	8841B	Inside	Note 14	N/A	-	Check	Self-Actuated	N/A
64	-	-	A	N/A	-	-	-	-
65	-	-	A	N/A	-	-	-	-
66	-	-	A	N/A	-	-	-	-
67	SF-021	Inside	C	YES	-	Diaphragm/Man.	Local Manual	N/A
	SF-022	Outside	C	YES	11'/8'-2"	Diaphragm/Man.	Local Manual	N/A
68	-	-	A	-	-	-	-	-
69	-	-	A	-	-	-	-	-
70	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
71	1SF-053 (2SF-055)	Inside	C	YES	-	Diaphragm/Man.	Local Manual	N/A
	1SF-054 (2SF-056)	Outside	C	YES	7'/4'-10"	Diaphragm/Man.	Local Manual	N/A
72	-	-	A	N/A	-	-	-	-
73	HV-2405	Outside	Note 1	N/A	5'-3"/11'-8"	Globe/Air	Auto close	Remote Manual
74	HV-4170	Outside	C	YES	6'-0"/8'-0"	Angle/Air	Auto close	Remote Manual
	HV-4168	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
	HV-4169	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
74a	PS-503	Outside	C	YES	-	Relief	Self-Actuated	N/A
75	-	-	A	N/A	-	-	-	-
76	HV-2406	Outside	Note 1	N/A	5'-6"/12'-11"	Globe/Air	Auto close	Remote Manual

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 8 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
77	HV-4167	Outside	C	YES	3'-8"/8'-1"	Angle/Air	Auto close	Remote Manual
	HV-4166	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
77a	PS-501	Outside	C	YES	-	Relief	Self-Actuated	N/A
78	HV-4176	Outside	C	YES	4'-8"/11'-10"	Angle/Air	Auto close	Remote Manual
	HV-4165	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
78a	PS-502	Outside	C	YES	-	Relief	Self-Actuated	N/A
79	HV-2407	Outside	Note 1	N/A	4'-9"/7'-6"	Globe/Air	Auto close	Remote Manual
80	HV-4175	Outside	C	YES	5'-8"/10'-11"	Angle/Air	Auto close	Remote Manual
	HV-4171	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
	HV-4172	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
	HV-4173	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
	HV-4174	Inside	C	YES	-	Angle/Air	Auto close	Remote Manual
80a	PS-500	Outside	C	YES	-	Relief	Self-Actuated	N/A
81	HV-7311	Outside	C	YES	15'/17'-3"	Globe/Air	Auto close	Remote Manual
	HV-7312	Inside	C	YES	-	Globe/Air	Auto close	Remote Manual
81a	WP-7177	Outside	C	YES	-	Relief	Self-Actuated	N/A
82	HV-2408	Outside	Note 1	N/A	6'-3"/12'-6"	Globe/Air	Auto close	Remote Manual
83	8871	Inside	C	YES	-	Globe/Air	Auto close	Remote Manual
	8888	Outside	C	YES	21'/22'-7"	Globe/Air	Auto close	Remote Manual
	8964	Outside	C	YES	15'/19'-0"	Globe/Air	Auto close	Remote Manual
	SI-8961	Outside	Note 5	N/A	15'/16'-6"	Globe/Manual	Manual	N/A
83a	1SI-8972 (2SI-8983)	Outside	C	YES	-	Relief	Self-Actuated	N/A
84	HV-5556	Outside	C	YES	5'-9"/5'-9"	Globe/Solenoid	Auto close	Remote Manual
	HV-5557	Inside	C	YES	-	Globe/Solenoid	Auto close	Remote Manual
85	-	-	A	N/A	-	-	-	-
86	-	-	A	N/A	-	-	-	-
87	-	-	A	N/A	-	-	-	-

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 9 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
88	-	-	A	N/A	-	-	-	-
89	-	-	A	N/A	-	-	-	-
90	-	-	A	N/A	-	-	-	-
91	-	-	A	N/A	-	-	-	-
92	None	-	A	N/A	-	N/A	-	-
93	-	-	A	N/A	-	-	-	-
94	HV-5544	Outside	C	YES	4'-0"/3'-2"	Globe/Solenoid	Auto close	Remote Manual
	HV-5545	Inside	C	YES	-	Globe/Solenoid	Auto close	Remote Manual
95	None	-	A	N/A	-	N/A	-	-
96	-	-	A	N/A	-	-	-	-
97	HV-5558	Outside	C	YES	5'-0"/3'-8"	Globe/Solenoid	Auto close	Remote Manual
	HV-5559	Inside	C	YES	-	Globe/Solenoid	Auto close	Remote Manual
98	None	-	A	N/A	-	N/A	-	-
99	-	-	A	N/A	-	-	-	-
100	HV-5560	Outside	C	YES	4'-0"/4'-9"	Globe/Solenoid	Auto close	Remote Manual
	HV-5561	Inside	C	YES	-	Globe/Solenoid	Auto close	Remote Manual
101	None	-	A	N/A	-	N/A	-	-
102	HV-5546	Outside	C	YES	2'0"/2'-11"	Globe/Solenoid	Auto close	Remote Manual
	HV-5547	Inside	C	YES	-	Globe/Solenoid	Auto close	Remote Manual
103	-	-	A	N/A	-	-	-	-
104	8880	Outside	C	YES	28'/28'-2"	Globe/Air	Auto close	Remote Manual
	SI-8968	Inside	C	YES	-	Check	Self-Actuated	N/A
105	7126	Inside	C	YES	-	Diaphragm/Air	Auto close	Remote Manual
	7150	Outside	C	YES	19'/22'-11"	Diaphragm/Air	Auto close	Remote Manual
106	-	-	A	N/A	-	-	-	-
107	-	-	A	N/A	-	-	-	-
108	-	-	A	N/A	-	-	-	-

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 10 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
109	HV-5536	Outside	C	YES	1'-0"/1'-7"	Butterfly/Air	Auto close	Remote Manual
	HV-5537	Inside	C	NO	-	Butterfly/Air	Auto close	Remote Manual
110	HV-5538	Outside	C	YES	2'/1'-11"	Butterfly/Air	Auto close	Remote Manual
	HV-5539	Inside	C	NO	-	Butterfly/Air	Auto close	Remote Manual
111	HV-4710	Outside	Note 1	N/A	12'/16'-8"	Globe/Air	Auto close	Remote Manual
112	HV-4711	Outside	Note 1	N/A	14'/16'-1"	Globe/Air	Auto close	Remote Manual
113	CA-016	Inside	C	YES	-	Check	Self-Actuated	N/A
	HV-3486	Outside	C	N/A	11'/11'-1"	Globe/Air	Auto close	Remote Manual
114	HV-4725	Inside	C	YES	-	Globe/Air	Auto close	Remote Manual
	HV-4726	Outside	C	YES	11'/11'-1"	Globe/Air	Auto close	Remote Manual
114a	1CC-1067 (2CC-1090)	Outside	C	YES	-	Relief	Self-Actuated	N/A
115	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
116	8027	Outside	C	YES	25'/4'-1"	Diaphragm/Air	Auto close	Remote Manual
	8026	Inside	C	YES	-	Diaphragm/Air	Auto close	Remote Manual
117	HV-4708	Outside	C	YES	8'/14'-7"	Gate/Motor	Auto close	Remote Manual
	HV-4701	Inside	C	YES	-	Gate/Motor	Auto close	Remote Manual
	CC-629	Inside	C	YES	-	Check	Self-Actuated	N/A
118	HV-4700	Outside	C	YES	18'/12'-7"	Gate/Motor	Auto close	Remote Manual
	CC-713	Inside	C	YES	-	Check	Self-Actuated	N/A
119	HV-4709	Outside	C	YES	18'/19'-1"	Gate/Motor	Auto close	Remote Manual
	HV-4696	Inside	C	YES	-	Gate/Motor	Auto close	Remote Manual
	CC-831	Inside	C	YES	-	Check	Self-Actuated	N/A
120	HV-6084	Outside	C	YES	16'/16'-1"	Gate/Motor	Auto close	Remote Manual
	CH-024	Inside	C	YES	-	Check	Self-Actuated	N/A
120a	1CH-271 (2CH-0281)	Outside	C	YES	-	Relief	Self-Actuated	N/A

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 11 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
121	HV-6082	Outside	C	YES	15'/16'-8"	Gate/Motor	Auto close	Remote Manual
	HV-6083	Inside	C	YES	-	Gate/Motor	Auto close	Remote Manual
121a	1CH-272 (2CH-0282)	Outside	C	YES	-	Relief	Self-Actuated	N/A
122	HV-5548	Outside	C (Note 15)	YES	2'/1'-7"	Butterfly/Air	Auto close	Remote Manual
	HV-5549	Inside	C (Note 15)	NO	-	Butterfly/Air	Auto close	Remote Manual
123	N/A	N/A	B	N/A	N/A	N/A	N/A	N/A
124	HV-4075B	Outside	C	YES	-	Gate/Motor	Auto close	Remote Manual
	HV-4075C	Inside	C	YES	-	Gate/Motor	Auto close	Remote Manual
125	8811A	Outside	Note 9	N/A	6'/7'-5"	Gate/Motor	Remote Manual	Local Manual
	SI-0182	Outside	Note 9	N/A	-	Relief	Self-Actuated	N/A
126	8811B	Outside	Note 9	N/A	6'/6'9"	Gate/Motor	Remote Manual	Local Manual
	SI-0183	Outside	Note 9	N/A	-	Relief	Self-Actuated	N/A
127	HV-4782	Outside	Note 9	N/A	6'/6'-9"	Gate/Motor	Remote Manual	Local Manual
	CT-0309	Outside	Note 9	N/A	-	Relief	Self-Actuated	N/A
128	HV-4783	Outside	Note 9	N/A	6'/7'-5"	Gate/Motor	Remote Manual	Local Manual
	CT-0310	Outside	Note 9	N/A	-	Relief	Self-Actuated	N/A
129	Bellows	N/A	B	N/A	N/A	N/A	N/A	N/A
	Flange	N/A	B	N/A	N/A	N/A	N/A	N/A
130	1BS-0016	Outside	Note 1	N/A	1	Globe/Manual	Local Manual	N/A
	1BS-0017	Outside	Note 1	N/A	1	Globe/Manual	Local Manual	N/A
131	1BS-0030	Inside	C	No	1	Ball/Hydraulic	Local Hydraulic	N/A
	1BS-0025	Inside	C	Yes	1	Ball/Hydraulic	Local Hydraulic	N/A

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 12 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary
131a	1BS-0056	Inside	C	No	1	Ball/Manual	Local Manual	N/A
	1BS-0044	Inside	C	No	1	Ball/Manual	Local Manual	N/A
	1BS-0029	Inside	C	Yes	1	Ball/Manual	Local Manual	N/A
	1BS-0015	Outside	C	Yes	1	Ball/Manual	Local Manual	N/A
132	BS-0202	Inside	C	No	1	Ball/Manual	Local Manual	N/A
	BS-0203	Inside	C	Yes	1	Ball/Manual	Local Manual	N/A
133	2BS-0016	Inside	C	No	1	Globe/Manual	Local Manual	N/A
	2BS-0017	Inside	C	No	1	Globe/Manual	Local Manual	N/A
	2BS-0039	Inside	C	Yes	1	Globe/Manual	Local Manual	N/A
	2BS-0040	Inside	C	Yes	1	Globe/Manual	Local Manual	N/A
134	2BS-0030	Inside	C	No	1	Ball/Hydraulic	Local Hydraulic	N/A
	2BS-0025	Outside	C	Yes	1	Ball/Hydraulic	Local Hydraulic	N/A
134a	2BS-0056	Inside	C	No	1	Relief/Spring Closed	Local Manual	N/A
	2BS-0044	Inside	C	No	1	Relief/Spring Closed	Local Manual	N/A
	2BS-0029	Inside	C	Yes	1	Relief/Spring Closed	Local Manual	N/A
	2BS-0015	Outside	C	Yes	1	Relief/Spring Closed	Local Manual	N/A

Notes:

1. These are closed systems which meet the requirements of NUREG-0800, Section 6.2.4, II.6, paragraph o. These valves are therefore not required to be tested.
2. These valves inside containment are part of closed systems outside containment which are in service post-accident at a pressure in excess of containment design pressure and satisfy single active failure criteria. These valves are therefore not required to be tested.
3. These are closed systems outside containment which are in service post-accident and have a water filled loop seal on the containment side of the valves for a period greater than 30 days following the accident. These valves are either open or are closed providing a third barrier to containment leakage. A water seal if maintained both inside and outside containment. These valves are therefore not required to be leakrate tested.
4. These ESF valves are normally open and remain open during post-accident conditions. Post-accident they are continually pressurized in excess of containment pressure from an ESF source which meets the single active failure criteria. These valves are therefore not to be tested.
5. This valve is not required to be leakage tested per 10 CFR 50 Appendix J since the downstream pressure indicator sensing element is dead-ended, and is hydrostatically tested to 1.5 times the design pressure of the pipe, thus providing a leaktight barrier similar to vent and drain connections having capped ends and under administrative controls.

TABLE 6.2.4-2
CONTAINMENT ISOLATION VALVING APPLICATION (Note 8)

(Sheet 13 of 13)

Item	Isolation Valve No (Note 6)	Location in Relation to Containment	Type of Leakage Rate Test	Direction of Test (Note 10)	Length of Pipe U1/U2 to Outermost Isolation Valve (ft)	Valve Type/Operator	Method of Actuation Primary	Secondary

6. Unit 1 and Unit 2 Tag Numbers are generally the same except for the prefix or as otherwise noted.
7. These are parallel ESF valves that are normally closed, but are designed to open during post-accident conditions. Failure of one valve to open will not prevent system pressurization on both sides of both valves in excess of containment pressure. These valves are therefore not required to be tested.
8. Table does not list local vent, drain and test connections as they are a special class of containment isolation valve per FSAR Section 6.2.4.1.3. These valves are locked closed and capped to meet containment isolation criteria if located within the pressure boundary. These valves do not require Type C testing.
9. These penetrations are normally closed and upon initial ESF actuation remain closed with a water seal from the RWST. During the ESF recirculation phases, the RHR and Containment Spray suction side isolation valves are open and supplying water to their respective ESF pumps. Water in the containment sump provides a seal between the containment atmosphere (post-accident) and these valves.
10. "YES" signifies that the isolation valve test pressure is applied in the same direction as the pressure existing when the valve is required to perform its containment isolation function.
    "NO" signifies that the isolation valve test pressure is not applied in the same direction as the pressure existing when the valve is required to perform its safety function.
11. This penetration is an engineered safety feature system supplying RHR pump flow (valves opened) to the cold legs of the RCS during cold leg injection and cold leg recirculation modes of operation. During hot leg recirculation this penetration is not in service (valves closed) but is pressurized by the residual heat removal pumps to a pressure in excess of 1.1 times the containment design pressure. The outside containment motor operated valve and the closed system outside containment provide two boundries in addition to this valve. These valves are therefore not required to be Type C tested.
12. This penetration is an Engineered Safety Feature System supplying SI pump flow (valves opened) to the hot legs of the RCS during hot leg recirculation mode of operation. During cold leg injection and cold leg recirculation this penetration is not in service (valves closed) but is pressurized by the safety injection pumps to a pressure in excess of 1.1 times the containment design pressure. This ensures that leakage path for containment atmosphere does not exist during a LOCA. Therefore, these valves are not required to be Type C tested.
13. This penetration is an Engineered Safety Feature System supplying SI pump flow (valves opened) to the cold legs of the RCS during cold leg injection and cold leg recirculation modes of operation. During hot leg recirculation this penetration is not in service (valves closed) but is pressurized by the safety injection pumps to a pressure in excess of containment design pressure. This ensures that a leakage path for containment atmosphere does not exist during a LOCA. Therefore, these valves are not required to be Type C tested.
14. This penetration is an Engineered Safety Feature System supplying RHR pump flow (valves opened) to the hot legs of the RCS during hot leg recirculation mode of operation. During cold leg injection this penetration is not in service (valve closed) but is pressurized by the residual heat removal pumps to a pressure in excess of containment design pressure. During cold leg recirculation, this penetration is not in service but is isolated by a minimum of two additional valves outside containment. These valves are therefore not required to be Type C tested.
15. Due to the piping arrangement, these valves cannot be tested individually; as such the leak test will result in a combined leak rate for both isolation valves under test conditions.
16. These valves function as local manual isolation for penetrations MI-1, MI-4 following exhaustion of air accumulators for HV-2452-1, -2 (see FSAR Section 9.3.1.2 discussion regarding accumulators for main steam supply to AFW pump turbine).
17. These valves do not require Appendix J Type C leak rate testing. An effective fluid seal on these penetrations is provided by the suction sources to the residual heat removal pumps during and following an accident. In addition, these containment isolation valves are non-automatic, are not required to operate post-accident and are located inside containment. See Section 6.2.4.1.3, items 3 and 5 for details.

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 1 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
1	-	Opened	Opened/Closed	Closed	Closed (Note 2)	5	A/B	Closes on Steam Line Isolation Signal
2	-	Closed/opened	Closed	Closed	N/A	N/A	-	Main Steam Isolation Bypass Valve (Note 1)
3	-	Opened	Open/Closed	Closed	Closed	5	A/B	Closes on Steam Line Isolation Signal
4	-	Closed	Closed	Opened/Closed	Opened	N/A	B	(Note 5)
4a	-	Closed	Opened/Closed	Closed	N/A	N/A	-	TDAFW Pump Bypass Warm-up Valve
5	-	Closed	Closed	Opened/Closed	Closed	N/A	A	
5a	-	Closed	Closed	Closed	N/A	N/A	-	
5b	-	Closed	Closed	Closed	N/A	N/A	-	
6	-	Opened	Open/Closed	Closed	Closed(Note 2)	5	A/B	Closes on Steam Line Isolation Signal
7	-	Closed/Opened	Closed	Closed	N/A	N/A	-	Main Steam Isolation Bypass Valve (Note 1)
8	-	Opened	Opened/Closed	Closed	Closed	5	A/B	Closed on Steam Line Isolation Signal
9	-	Closed	Closed	Opened/Closed	Closed	N/A	B	
9a	-	Closed	Closed	Closed	N/A	N/A	-	
9b	-	Closed	Closed	Closed	N/A	N/A	-	
10	-	Opened	Opened/Closed	Closed	Closed(Note 2)	5	A/B	Closed on Steam Line Isolation Signal
11	-	Closed/Opened	Closed	Closed	N/A	N/A	-	Main Steam Isolation Bypass Valve (Note 1)
12		Opened	Open/Closed	Closed	Closed	5	A/B	Closed on Steam Line Isolation Signal
13	-	Closed	Closed	Opened/Closed	Closed	N/A	A	
13a	-	Closed	Closed	Closed	N/A	N/A	-	

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 2 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
13b	-	Closed	Closed	Closed	N/A	N/A	-	
14	-	Opened	Open/Closed	Closed	Closed(Note 2)	5	A/B	Closed on Steam Line Isolation Signal
15		Closed/Opened	Closed	Closed	N/A	N/A	-	Main Steam Isolation Bypass Valve (Note 1)
16	-	Opened	Opened/Closed	Closed	Closed	5	A/B	Closed on Steam Line Isolation Signal
17	-	Closed	Closed	Opened/Closed	Opened	N/A	A	(Note 5)
17a	-	Closed	Opened/Closed	Closed	N/A	N/A	-	TDAFW Pump Bypass Warm-up Valve
18	-	Closed	Closed	Opened/Closed	Closed	N/A	B	
18a	-	Closed	Closed	Closed	N/A	N/A	-	
18b	-	Closed	Closed	Closed	N/A	N/A	-	
19	-	Opened	Closed	Closed	FAI	5	A/B	Closed on Feedwater Isolation Signal
20		Closed	Closed	Closed	Closed	N/A	B	Remote Manual Isolation satisfies CDC-57
20a	-	Opened	Opened	Opened/Closed	FAI	N/A	A	
-	Opened	Opened	Opened/Closed	FAI	N/A	B		
20b	-	Closed	Closed/Opened	Closed	N/A	N/A	-	
20c	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closes on Feedwater Isolation Signal (Unit 2 only, valve deleted Unit 1)
20d	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal
21	-	Opened	Closed	Closed	FAI	5	A/B	Closed on Feedwater Isolation Signal
22		Closed	Closed	Closed	Closed	N/A	B	Remote Manual Isolation satisfies CDC-57

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 3 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
22a	-	Opened	Opened	Opened/Closed	FAI	N/A	A	
-	Opened	Opened	Opened/Closed	FAI	N/A	B		
22b	-	Closed	Closed/Opened	Closed	N/A	N/A	-	
22c	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal (Unit 2 only, valve deleted Unit 1)
22d	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal
23	-	Opened	Closed	Closed	FAI	5	A/B	Closed on Feedwater Isolation Signal
24a	-	Opened	Opened	Opened/Closed	FAI	N/A	A	
-	Opened	Opened	Opened/Closed	FAI	N/A	B		
24b	-	Closed	Closed/Opened	Closed	N/A	N/A	-	
24c	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal (Unit 2 only, valve deleted Unit 1)
24d	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal
25	-	Opened	Closed	Closed	FAI	5	A/B	Closed on Feedwater Isolation Signal
26a	-	Opened	Opened	Opened/Closed	FAI	N/A	A	
-	Opened	Opened	Opened/Closed	FAI	N/A	B		
26b	-	Closed	Closed/Opened	Closed	N/A	N/A	-	
26c	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal (Unit 2 only, valve deleted Unit 1)
26d	-	Closed	Opened/Closed	Closed	Closed	5	A/B	Closed on Feedwater Isolation Signal
27	Phase A	Opened	Closed	Closed	Closed	5	A/B	Remote Manual Isolation Satisfies CDC-57

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 4 of 12)

Item	Containment Isolation Signal	Valve Position Normal	Valve Position Shutdown	Valve Position Post-Accident	Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
28	Phase A	Opened	Closed	Closed	Closed	5	A/B	Remote Manual Isolation Satisfies CDC-57
29	Phase A	Opened	Closed	Closed	Closed	5	A/B	Remote Manual Isolation Satisfies CDC-57
30	Phase A	Opened	Closed	Closed	Closed	5	A/B	Remote Manual Isolation Satisfies CDC-57
31	-	-	-	-	-	-	-	
32	Phase A	Opened	Closed	Closed	Closed	10	B	
	Phase A	Opened	Closed	Closed	Closed	10	A	
33	-	Closed/Opened	Opened	Closed	FAI	N/A	B	Relief valve is
		Closed	Closed	Closed	N/A	N/A	-	closed in backflow direction at all times
34	-	Closed/Opened	Opened	Closed	FAI	N/A	A	Relief valve is
		Closed	Closed	Closed	N/A	N/A	-	closed in backflow direction at all times
35	-	Opened	Opened	Opened/Closed	FAI	N/A	A	
	-	Closed	Closed	Opened/Closed	-	-	-	Check Valve
	Phase A	Closed	Closed	Closed	Closed	15	A	Air operated valve on test line
36	-	Opened	Opened	Opened	FAI	-	B	
	Closed	Closed	Opened/Closed	-	-	-		Check Valve
	Phase A	Closed	Closed	Closed	Closed	15	B	Air operated valve on test line
37	-	-	-	-	-	-	-	
38	-	-	-	-	-	-	-	
39	-	-	-	-	-	-	-	
40(a)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Flanged
40(b)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Flanged
40(c)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Flanged
41	-	Closed	Closed	Closed	-	-	-	Check Valve
	Phase A	Closed	Closed	Closed	Closed	10	B	
41a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 5 of 12)

Item	Containment Isolation Signal	Valve Position: Normal	Valve Position: Shutdown	Valve Position: Post-Accident	Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
42	-	Closed	Closed	Open	-	-	-	Check Valve
	-	Closed	Closed	Opened	FAI	N/A	A	
	-	Closed	Closed	Opened	FAI	N/A	B	
	Phase A	Closed	Closed	Closed	Closed	10	B	Air operated valve on test line
43	-	Closed	Closed	Closed/Opened	FAI	N/A	A	
	-	Closed	Closed	Closed/Opened	-	-	-	Check Valve
	Phase A	Closed	Closed	Closed	Closed	10	A	Air operated valve on test line
44	-	Closed	Closed	Closed/Opened	FAI	N/A	B	
	-	Closed	Closed	Closed/Opened	-	-	-	Check Valve
	Phase A	Closed	Closed	Closed	Closed	10	A	Air operated valve on test line
45	-	Opened	Opened	Opened	FAI	N/A	B	
	-	Closed	Closed	Opened/Closed	-	-	-	Check Valve
	-	Closed	Closed	Opened/Closed	-	-	-	Check Valve
	-	Closed	Closed	Opened/Closed	-	-	-	Check Valve
	-	Closed	Closed	Opened/Closed	-	-	-	Check Valve
	Phase A	Closed	Closed	Closed	Closed	10	A	Air operated valve on test line
46	-	Opened	Closed	Closed	FAI	10	B	Closed on Safety Injection
		Opened	Closed	Closed	-	-	-	Check Valve
47	-	Opened	Opened/Closed	Opened	-	-	-	Check Valve
	-	Opened	Opened/Closed	Open	FAI	N/A	B	
48	-	Opened	Open/Closed	Open	-	-	-	Check Valve
	-	Opened	Opened/Closed	Open	FAI	N/A	B	

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 6 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
49	-	Opened	Opened/Closed	Opened	-	-	-	Check Valve
	-	Opened	Opened/Closed	Open	FAI	N/A	B	
50	-	Opened	Opened/Closed	Opened	-	-	-	Check Valve
	-	Opened	Opened/Closed	Open	FAI	N/A	B	
51	Phase A	Opened	Opened/Closed	Closed	FAI	10	B (8100)	
	Phase A	Opened	Opened/Closed	Closed	FAI	10	A (8112)	
	-	Closed	Closed	Open/Closed	-	-	-	Thermal Relief Check Valve
52	Phase A	Opened	Opened	Closed	Closed	10	B	
	Phase A	Opened	Opened	Closed	Closed	10	A	
	-	Closed	Closed	Closed	N/A	N/A	-	Manual Valve
52a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
53	-	-	-	-	-	-	-	
54	-	Closed	Closed	Closed/Opened	FAI	N/A	B	
	-	Closed	Closed	Closed/Open	-	-	-	Check Valve
55	-	Closed	Closed	Closed/Opened	FAI	-	A	
	-	Closed	Closed	Closed/Open	-	-	-	Check Valve
56	-	Closed	Closed	Closed	N/A	N/A	-	
	-	Closed	Opened/Closed	Closed	N/A	N/A	-	
57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
58	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	B (HV-5542)	(Note 3)
	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	A (HV-5543)	(Note 3)
	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	B (HV-5563)	(Note 3)

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 7 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
59	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	B (HV-5540)	(Note 3)
	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	A (HV-5541)	(Note 3)
	Containment Vent. Isolation	Closed	Closed	Closed/Open	FAI	N/A	B (HV-5562)	(Note 3)
60	Phase A	Closed	Opened	Closed	Closed	5	B	Provides automatic fire protection water
	Phase A	Closed	Opened	Closed	Closed	5	A	
60a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
61	Phase A	Open	Closed	Closed	Closed	5	B	
	Phase A	Open	Closed	Closed	Closed	5	A	
61a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
62	Phase A	Open	Open	Closed/Open	Closed	5	B	
	-	Open	Open	Closed/Open	-		-	Check Valve
63	-	Closed	Closed	Closed/Opened	FAI	A/N	B	
	Phase A	Closed	Closed	Closed	Closed	15	A	Air operated valve on test line
	-	Closed	Closed	Closed/Opened	-	-	-	Check Valve
	-	Closed	Closed	Closed/Opened	-	-	-	Check Valve
64	-	-	-	-	-	-	-	
65	-	-	-	-	-	-	-	
66	-	-	-	-	-	-	-	
67	-	Closed	Closed	Closed	N/A	N/A	-	
	-	Closed	Closed	Closed	N/A	N/A	-	
68	-	-	-	-	-	-	-	
69	-	-	-	-	-	-	-	
70	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
71	-	Closed	Closed	Closed	N/A	N/A	-	
	-	Closed	Closed	Closed	N/A	N/A	-	

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 8 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
72	-	-	-	-	-	-	-	
73	Phase A	Opened	Closed	Closed	Closed	5	B	Remote Manual Isolation Satisfies CDC-57
74	Phase A	Closed	Closed	Closed/Open	Closed	5	B	
	Phase A	Closed	Closed	Closed/Open	Closed	5	A	
	Phase A	Closed	Closed	Closed/Open	Closed	5	A	
74a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
75	-	-	-	-	-	-		
76	Phase A	Opened	Closed	Closed	Closed	5	B	Remote Manual Isolation Satisfies CDC-57
77	Phase A	Closed	Closed	Closed	Closed	5	B	
	Phase A	Closed	Closed	Closed	Closed	5	A	
77a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
78	Phase A	Closed	Closed	Closed	Closed	5	B	
	Phase A	Closed	Closed	Closed	Closed	5	A	
78a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
79	Phase A	Opened	Closed	Closed	Closed	5	B	Remote Manual Isolation Satisfies CDC-57
80	Phase A	Closed	Closed	Closed	Closed	5	B	
	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	A	
80a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
81	Phase A	Closed	Closed	Closed	Closed	5	B	
	Phase A	Closed	Closed	Closed	Closed	5	A	
81a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
82	Phase A	Opened	Closed	Closed	Closed	5	B	Remote Manual Satisfies Isolation CDC-57

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 9 of 12)

Item	Containment Isolation Signal	Valve Position			Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
		Normal	Shutdown	Post-Accident				
83	Phase A	Closed	Closed	Closed	Closed	10	A	
	Phase A	Closed	Closed	Closed	Closed	10	B	
	Phase A	Closed	Closed	Closed	Closed	10	B	
83a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
84	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	B	
85	-	-	-	-	-	-	-	
86	-	-	-	-	-	-	-	
87	-	-	-	-	-	-	-	
88	-	-	-	-	-	-	-	
89	-	-	-	-	-	-	-	
90	-	-	-	-	-	-	-	
91	-	-	-	-	-	-	-	
92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Bellows Seal
93	-	-	-	-	-	-	-	
94	Phase A	Opened	Opened	Closed	Closed	5	B	
	Phase A	Opened	Opened	Closed	Closed	5	A	
95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Bellows Seal
96	-	-	-	-	-	-	-	
97	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	B	
98	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Bellows Seal
99	-	-	-	-	-	-	-	
100	Phase A	Closed	Closed	Closed	Closed	5	A	
	Phase A	Closed	Closed	Closed	Closed	5	B	
101	-	N/A	N/A	N/A	N/A	-	-	
102	Phase A	Opened	Opened	Closed	Closed	5	B	
	Phase A	Opened	Opened	Closed	Closed	5	A	
103	-	-	-	-	-	-	-	

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 10 of 12)

Item	Containment Isolation Signal	Valve Position Normal	Valve Position Shutdown	Valve Position Post-Accident	Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
104	Phase A	Opened	Opened	Closed	Closed	10	B	
	-	Opened	Opened	Closed	-	-	-	Check Valve
105	Phase A	Closed	Closed	Closed	Closed	10	A	
	Phase A	Closed	Closed	Closed	Closed	10	B	
106	-	-	-	-	-	-	-	
107	-	-	-	-	-	-	-	
108	-	-	-	-	-	-	-	
109	Containment Vent Isolation	Closed	Opened	Closed	Closed	N/A	B	(Note 3)
	Containment Vent Isolation	Closed	Opened	Closed	Closed	N/A	A	(Note 3)
110	Containment Vent Isolation	Closed	Opened	Closed	Closed	N/A	B	(Note 3)
	Containment Vent Isolation	Closed	Opened	Closed	Closed	N/A	A	(Note 3)
111	Phase A	Opened	Opened/Closed	Closed	Closed	5	B	Remote Manual Isolation Satisfies CDC-57
112	Phase A	Opened	Opened/Closed	Closed	Closed	5	B	Remote Manual Isolation Satisfies CDC-57
113	-	Closed	Opened	Closed	-	-	-	Check Valve
	Phase A	Closed	Opened	Closed	Closed	5	B	
114	Phase A	Opened	Opened	Closed	Closed	10	A	
	Phase A	Opened	Opened	Closed	Closed	10	B	
114a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
115	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Flanged
116	Phase A	Closed	Closed	Closed	Closed	10	B	
	Phase A	Closed	Closed	Closed	Closed	10	A	
117	Phase B	Opened	Opened	Closed	FAI	30	B (HV-4708)	
	Phase B	Opened	Opened	Closed	FAI	30	A (HV-4701)	
	-	Closed	Closed	Closed/Opened	-	-	-	Thermal Relief Check Valve

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 11 of 12)

Item	Containment Isolation Signal	Valve Position: Normal	Valve Position: Shutdown	Valve Position: Post-Accident	Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
118	Phase B	Opened	Opened/	Closed	FAI	30	B	
	-	Opened	Opened	Closed	-	-	-	Check Valve
119	Phase B	Opened	Opened	Closed	FAI	15	B (HV-4709)	
	Phase B	Opened	Opened	Closed	FAI	15	A	
	-	Closed	Closed	Closed/Open	-	-	-	Thermal Relief Check Valve
120	Phase A	Opened	Opened	Closed	FAI	15	B	
	-	Opened	Opened	Closed	-	-	-	Check Valve
120a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
121	Phase A	Opened	Opened	Closed	FAI	15	B (HV-6082)	
	Phase A	Opened	Opened	Closed	FAI	15	A (HV-6083)	
121a	-	Closed	Closed	Open/Closed	N/A	N/A	-	Thermal Relief Valve
122	Containment Vent Isolation	Closed/Opened	Closed	Closed	Closed	3 (Note 4)	B	Intermittently opened in normal
	Containment Vent Isolation	Closed/Opened	Closed	Closed	Closed	3 (Note 4)	A	operation to function as part of the containment pressure relief system.
123	N/A	N/A	N/A	N/A	N/A	N/A	N/A	Flanged
124	Phase A	Closed	Closed	Closed	FAI	10	B(4075B)	
	Phase A	Closed	Closed	Closed	FAI	10	A(4075C)	
125	-	Closed	Closed	Closed/Opened	FAI	N/A	A	
	-	Closed	Closed	Closed	N/A	N/A	-	Thermal Relief
126	-	Closed	Closed	Closed/Opened	FAI	N/A	B	
	-	Closed	Closed	Closed	N/A	N/A	-	Thermal Relief
127	-	Closed	Closed	Closed/Opened	FAI	N/A	A	
	-	Closed	Closed	Closed	N/A	N/A	-	Thermal Relief
128	-	Closed	Closed	Closed/Opened	FAI	N/A	B	
	-	Closed	Closed	Closed	N/A	N/A	-	Thermal Relief
129	N/A	N/A	N/A	N/A	N/A	N/A	-	

TABLE 6.2.4-3
CONTAINMENT ISOLATION VALVING APPLICATION (Note 1)

(Sheet 12 of 12)

Item	Containment Isolation Signal	Valve Position Normal	Shutdown	Post-Accident	Valve Power Failure	Valve Closure Time (Sec.)	Power Source	Remarks
130	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
131	-	Closed	N/A	Closed	FAI	N/A	N/A	
	-	Closed	N/A	Closed	FAI	N/A	N/A	
131a	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
132	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
133	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
134	-	Closed	N/A	Closed	FAI	N/A	N/A	
	-	Closed	N/A	Closed	FAI	N/A	N/A	
134a	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	
	-	Closed	N/A	Closed	N/A	N/A	N/A	

Notes:

1 - MSIV Bypass valves are locked closed except during startup when one may be opened to warm up main steam piping.

2 - Valve fails closed on loss of hydraulic pressure. Either pilot solenoid must be energized to cause valve close when hydraulic pressure is available.

3 - Sealed closed in Modes 1-4.

4 - Technical Requirements Table 2.1.1, “Containment Isolation Valves,” requires a maximum isolation time of 5 seconds for valves HV-5548 and HV-5549 with a footnote stating that the 5 second time includes the instrumentation delays of the containment ventilation isolation signal from Pressurizer Pressure Low.”

5 - Manual isolation valves upstream of TDAFWP steam supply valves are normally open. Closed for steam line isolation concurrent with loss of air to the Steam Supply Valve

TABLE 6.2.4-4
PENETRATIONS THAT ARE NOT DRAINED AND VENTED DURING
CONTAINMENT INTEGRATED LEAKAGE RATE (TYPE A) TEST**

Item	Penetration No.	System
33	MII-2	RHR
34	MII-3	RHR
35	MII-4	SI
36	MII-5	SI
42	MIII-2	SI
43	MIII-3	SI
44	MIII-4	SI
45	MIII-5	SI
47*	MIII-7	CS
48*	MIII-8	CS
49*	MIII-9	CS
50*	MIII-10	CS
54	MIII-14	CT
55	MIII-15	CT
63	MIII-23	SI
92	MIV-7 (b)	VA
95	MIV-8 (b)	VA
98	MIV-9 (b)	VA
101	MIV-10 (b)	VA
120*	MV-12	CH
120a*	MV-13	CH
121*	MV-13	CH
121a	MV-13	CH
125	MS-1	SI
126	MS-2	SI
127	MS-3	CT
128	MS-4	CT

* Items may be water filled during Type A testing but are not necessarily water filled post-accident.

** Type C penetrations, conforming to GDC-57, are not necessarily drained or vented during the Type A test. See Table 6.2.4-1, Items 1 to 30, 73, 76, 79, 82, 111, 112, 130

In addition other penetrations not shown on this table may not be vented and drained as discussed in Section 6.2.6.1.

TABLE 6.2.4-5
TABLE 6.2.4-5 HAS BEEN DELETED.

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 1 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
1	MI-1	MS	Main Steam From Steam Generator #1	non-essential	none
2	MI-1	MS	MSIV Bypass From Steam Generator #1	non-essential	none
3	MI-1	MS	Drain From Main Steamline #1	non-essential	none
4	MI-1	MS	None	essential	Main Steam to Aux. F.P. Turbine From Steamline #1
4a	MI-1	MS	None	non-essential	none
5	MI-1	MS	Main Steam From Steam Generator #1	essential	Atmospheric Relief
5a	MI-1	MS	N_2 Supply to Steam Generator #1	non-essential	none
5b	MI-1	MS	Main Steam From Steam Generator #1	essential	Main Steam Pressure Relief
6	MI-2	MS	Main Steam From Steam Generator #2	non-essential	none
7	MI-2	MS	MSIV Bypass From Steam Generator #2	non-essential	none
8	MI-2	MS	Drain From Steam Line #2	non-essential	none
9	MI-2	MS	Main Steam From Steam Generator #2	essential	Atmospheric Relief
9a	MI-2	MS	N_2 Supply to Steam Generator #2	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 2 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
9b	MI-2	MS	Main Steam From Steam Generator #2	essential	Main Steam Pressure Relief
10	MI-3	MS	Main Steam From Steam Generator #3	non-essential	none
11	MI-3	MS	MSIV Bypass From Steam Generator #3	non-essential	none
12	MI-3	MS	Drain From Steam Line #3	non-essential	none
13	MI-3	MS	Main Steam From Steam Generator #3	essential	Atmospheric Relief
13a	MI-3	MS	N_2 Supply to Steam Generator #3	non-essential	none
13b	MI-3	MS	Main Steam from Steam Generator #3	essential	Atmospheric Relief
14	MI-4	MS	Main Steam From Steam Generator #4	non-essential	none
15	MI-4	MS	MSIV Bypass From Steam Generator #4	non-essential	none
16	MI-4	MS	Drain From Steam Line #4	non-essential	none
17	MI-4	MS	None	essential	Main Steam to Aux. F.P. Turbine from Steam Line #4
17a	MI-4	MS	None	non-essential	none
18	MI-4	MS	Main Steam From Steam Generator #4	essential	Atmospheric Relief
18a	MI-4	MS	N_2 Supply to Steam Generator #4	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 3 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
18b	MI-4	MS	Main Steam From Steam Generator #4	essential	Main Steam Pressure Relief
19	MI-5	FW	Feedwater to Steam Generator #1	non-essential	none
20	MI-5	FW	Feedwater Sample (FW to Steam Generator #1)	non-essential	none
20a	MV-18	FW/AFW	Auxiliary Feedwater to Steam Generator #1	essential	Auxiliary Feedwater to Steam Generator
20b	MI-5	FW	N_2 Supply to Steam Generator #1	non-essential	none
20c	MV-18	FW	Feedwater Preheater Bypass to Steam Generator #1 (Unit 2 only)	non-essential	none
20d	MI-5	FW	Feedwater Bypass to Steam Generator #1	non-essential	none
21	MI-6	FW	Feedwater to Steam Generator #2	non-essential	none
22	MI-6	FW	Secondary Sample (FW to Steam Generator #2)	non-essential	none
22a	MV-19	FW/AFW	Auxiliary Feedwater to Steam Generator #2	essential	Auxiliary Feedwater to Steam Generator
22b	MI-6	FW	N_2 Supply to Steam Generator #2	non-essential	none
22c	MV-19	FW	Feedwater Preheater Bypass to Steam Generator #2 (Unit 2 only)	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 4 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
22d	MI-6	FW	Feedwater Bypass to Steam Generator #2	non-essential	none
23	MI-7	FW	Feedwater to Steam Generator #3	non-essential	none
24a	MV-20	FW/AFW	Auxiliary Feedwater to Steam Generator #3	essential	Auxiliary Feedwater to Steam Generator
24b	MI-7	FW	N_2 Supply to Steam Generator #3	non-essential	none
24c	MV-20	FW	Feedwater Preheater Bypass to Steam Generator #3 (Unit 2 only)	non-essential	none
24d	MI-7	FW	Feedwater Bypass to Steam Generator #3	non-essential	none
25	MI-8	FW	Feedwater to Steam Generator #4	non-essential	none
26a	MV-17	FW/AFW	Auxiliary Feedwater to Steam Generator #4	essential	Auxiliary Feedwater to Steam Generator
26b	MI-8	FW	N_2 Supply to Steam Generator #4	non-essential	none
26c	MV-17	FW	Feedwater Preheater Bypass to Steam Generator #4 (Unit 2 only)	non-essential	none
26d	MI-8	FW	Feedwater Bypass to Steam Generator #4	non-essential	none
27	MI-9	MS	Blowdown From Steam Generator #3	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 5 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
28	MI-10	MS	Blowdown From Steam Generator #2	non-essential	none
29	MI-11	MS	Blowdown From Steam Generator #1	non-essential	none
30	MI-12	MS	Blowdown From Steam Generator #4	non-essential	none
31	MI-13	-	Spare	-	-
32	MII-1	CS	Letdown Line to Letdown Heat Exchanger	non-essential	none
33	MII-2	RH	R.H.R. From Hot Leg Loop #4	non-essential	none
34	MII-3	RH	R.H.R. From Hot Leg Loop #1	non-essential	none
35	MII-4	SI	None	essential	Injection of cooling water into the cold leg loops #1 and #2
36	MII-5	SI	None	essential	Injection of cooling water into the cold leg loops #3 and #4
37	MII-6	-	Spare	-	-
38	MII-7	-	Spare	-	-
39	MII-8	-	Spare	-	-
40a	MII-9A	-	Maintenance Penetration	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 6 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
40b	MII-9B	-	Maintenance Penetration	non-essential	none
40c	MII-9C	-	Maintenance Penetration	non-essential	none
41	MIII-1	RC	Reactor Make Up Water To Pressure Relief Tank & R.C. Pump Stand Pipe	non-essential	none
41a	MIII-1	RC	None	non-essential	none
42	MIII-2	SI	None	essential	Safety Injection to Cold Leg Loops #1, #2, #3, & #4
43	MIII-3	SI	None	essential	S.I. To R.C. System Hot Leg Loops #2 & #3
44	MIII-4	SI	None	essential	S.I. To R.C. System Hot Leg Loops #1 & #4
45	MIII-5	SI	None	essential	S.I. To R.C. System Cold Leg Loops #1, #2, #3, & #4
46	MIII-6	CS	Charging Line Regenerative Heat Exchanger	non-essential	none
47	MIII-7	CS	Seal Injection to R.C. Pump (Loop #1)	Essential (Note 2)	Seal Water Injection

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 7 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
48	MIII-8	CS	Seal Injection to R.C. Pump (Loop #2)	Essential (Note 2)	Seal Water Injection
49	MIII-9	CS	Seal Injection to R.C. Pump (Loop #3)	Essential (Note 2)	Seal Water Injection
50	MIII-10	CS	Seal Injection to R.C. Pump (Loop #4)	Essential (Note 2)	Seal Water Injection
51	MIII-11	CS	Seal Water Return to R.C. Excess Letdown	non-essential	none
52	MIII-12	WP	R.C.D.T. Heat Exchanger To Waste Hold Up Tank	non-essential	
52a	MIII-12	WP	None	non-essential	none
53	MIII-13	-	Spare	-	-
54	MIII-14	CT	None	essential	Containment Spray To Spray Header (Train B)
55	MIII-15	CT	None	essential	Containment Spray To Spray Header (Train A)
56	MIII-16	SF	Refueling Water Purification to Refueling Cavity	non-essential	none
57	MIII-17	LT	None	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 8 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
58	MIII-18	VA	None	non-essential	none
59	MIII-19	VA	None	non-essential	none
60	MIII-20	DD	Demineralized Water Supply	non-essential	none
60a	MIII-20	DD	None	non-essential	none
61	MIII-21	VD	Containment Sump Pump Discharge	non-essential	none
61a	MIII-21	VD	None	non-essential	none
62	MIII-22	CI	Instrument Air to Containment	non-essential	Post-Accident Sampling
63	MIII-23 #2 & #3	SI	R.H.R. To Hot Leg Loops	essential	Core Cooling
64	MIII-24	-	Spare	-	-
65	MIII-25	-	Spare	-	-
66	MIII-26	-	Spare	-	-
67	MIII-27	SF	Refueling Cavity to Refueling Water Purification Pump	non-essential	none
68	MIII-28	-	Spare	-	-

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 9 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
69	MIII-29	-	Spare	-	-
70	MIII-30	LT	none	non-essential	none
71	MIII-31	SF	Refueling Cavity Skimmer Pump Discharge (System no longer used)	non-essential	none
72	MIII-32	-	Spare	-	-
73	MIV-1(a)	MS	Sample From Steam Generator #1	non-essential	none
74	MIV-1(b)	PS	R.C. Sample From Hot Legs	non-essential	Reactor Coolant Sampling
74a	MIV-1(b)	PS	None	non-essential	None
75	MIV-1(c)	-	Spare	-	-
76	MIV-2(a)	MS	Sample From Steam Generator #2	non-essential	None
77	MIV-2(b)	PS	Pressurizer Liquid Space Sample	non-essential	None
77a	MIV-2(b)	PS	None	non-essential	None
78	MIV-2(c)	PS	Pressurizer Steam Space Sample	non-essential	None
78a	MIV-2(c)	PS	None	non-essential	None
79	MIV-3(a)	MS	Sample From Steam Generator #3	non-essential	None

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 10 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
80	MIV-3(b)	PS	Sample From Accumulators	non-essential	None
80a	MIV-3(b)	PS	None	non-essential	None
81	MIV-3(c)	WP	None	non-essential	None
81a	MIV-3(c)	WP	None	non-essential	None
82	MIV-4(a)	MS	Sample From Steam Generator #4	non-essential	None
83	MIV-4(b)	SI	Accumulator Test & Fill	non-essential	None
83a	MIV-4(b)	SI	None	non-essential	None
84	MIV-4(c)	PS	None	non-essential	None
85	MIV-5(a)	-	Spare	-	-
86	MIV-5(b)	-	Spare	-	-
87	MIV-5(c)	-	Spare	-	-
88	MIV-6(a)	-	Spare	-	-
89	MIV-6(b)	-	Spare	-	-
90	MIV-6(c)	-	Spare	-	-

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 11 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
91	MIV-7(a)	-	Spare	-	-
92	MIV-7(b)	ESFAS	Containment Pressure Sensing PT-934	essential measurement	Containment pressure
93	MIV-7(c)	-	Spare	-	-
94	MIV-8(a)	RM	Radiation Monitoring Sample	non-essential	None
95	MIV-8(b)	ESFAS	Containment Pressure Sensing PT-935	essential measurement	Containment pressure
96	MIV-8(c)	-	Spare	-	-
97	MIV-9(a)	PS	None	non-essential	None
98	MIV-9(b)	ESFAS	Containment Pressure Sensing PT-936	essential measurement	Containment Pressure
99	MIV-9(c)	-	Spare	-	-
100	MIV-10(a)	PS	None	non-essential	None
101	MIV-10(b)	ESFAS	Containment Pressure Sensing PT-937	essential measurement	Containment Pressure
102	MIV-10(c)	RM	Radiation Monitoring Sample return	non-essential	None

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 12 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
103	MIV-11(a)	-	Spare	-	-
104	MIV-11(b)	SI	N^2 Supply To Accumulators	non-essential	None
105	MIV-11(c)	WP	H^2 Supply to RC Drain Tank	non-essential	None
106	MIV-12(a)	-	Spare	-	-
107	MIV-12(b)	-	Spare	-	-
108	MIV-12(c)	-	Spare	-	-
109	MV-1	VA	Containment Purge Air Supply	non-essential	none
110	MV-2	VA	Containment Purge Air Exhaust	non-essential	none
111	MV-3	CC	C.C.W. Supply to Excess Letdown & RC Drain Tank Heat Exchanger	non-essential	none
112	MV-4	CC	C.C.W. Return from Excess Letdown & RC Drain Tank Heat Exchanger	non-essential	none
113	MV-5	CA	Service Air to Containment	non-essential	none
114	MV-6	CC	Containment CCW Drain Tank Pumps Discharge	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 13 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
114a	MV-6	CC	None	non-essential	none
115	MV-7	LT	Containment Leak Rate Test Pressure Sensing	non-essential	none
116	MV-8	RC	N_2 Supply to Pressurizer Relief Tank	non-essential	none
117	MV-9	CC	CCW Return From RCP Motors	essential (Note 1)	Cooling of RCP motors
118	MV-10	CC	CCW Supply to RCP Motors and Thermal Barrier	essential (Note 1)	Cooling of RCP motors and Thermal Barriers
119	MV-11	CC	CCW Return From RCP Thermal Barrier	essential (Note 1)	Cooling of RCP Thermal Barrier
120	MV-12	CH	Chilled Water Supply To Containment Coolers	non-essential	none
120a	MV-12	CH	None	non-essential	none
121	MV-13	CH	Chilled Water Return From Containment Coolers	non-essential	none
121a	MV-13	CH	None	non-essential	none
122	MV-14	VA	Containment Pressure Relief	non-essential	none
123	MV-15	-	Maintenance Penetration	non-essential	none
124	MV-16	FP	Fire Protection System Isolation	non-essential	none

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 14 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
125	MS-1	SI	None	essential	Containment Recirculation Sump To R.H.R. Pump (Train A)
126	MS-2	SI	None	essential	Containment Recirculation Sump To R.H.R. Pump (Train B)
127	MS-3	CT	None	essential	Containment Recirculation Sump To Spray Pumps (Train A)
128	MS-4	CT	None	essential	Containment Recirculation Sump To Spray Pumps (Train B)
129	MS-5	-	Fuel Transfer Tube	non-essential	none
130	Unit 1 Personnel Airlock Hydraulics	BS/CB	Door Operating Mechanism	Non-essential	None
131	Unit 1 Personnel Airlock	BS/CB	Equalization of Door Differential Pressure	Non-essential	None
131a	Unit 1 Personnel Airlock	BS/CB	Equalization of Door Differential Pressure	Non-essential	None

TABLE 6.2.4-6
CLASSIFICATION OF SYSTEMS PATHS PENETRATION CONTAINMENT WALL

(Sheet 15 of 15)

Item	Penetration Number	System	Normal Operating Function	Classification	Post-Accident Function
132	Emergency Airlock	BS/CB	Equalization of Door Differential Pressure	Non-essential	None
133	Unit 2 Personnel Airlock Hydraulics	BS/CB	Door Operating Mechanism	Non-essential	None
134	Unit 2 Personnel Airlock	BS/CB	Equalization of Door Differential Pressure	Non-essential	None
134a	Unit 2 Personnel Airlock	BS/CB	Equalization of Door Differential Pressure	Non-essential	None

Notes:

1 - Essential for those accident conditions where the "P" has not been generated. The system is classified as non-essential after "P" signal actuation.

2 - See FSAR Section 6.2.4.1.4 paragraph 4.

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TABLE 6.2.5-5
FAILURE MODE AND EFFECTS ANALYSIS

Component of System	Malfunction	Comments and Consequences
Containment Spray System	Fails to operate and provide mixing of containment atmosphere	Two redundant containment spray trains are provided. Operation of one train is adequate for mixing of hydrogen in the post- LOCA containment atmosphere.

TABLE 6.2.5-6
HYDROGEN PURGE SYSTEM COMPONENT MATERIALS SPECIFICATION

COMPONENT	MATERIAL
Exhaust Housing	A36CS
Prefilters	Fiberglass
HEPA Filters	Fiberglass
Charcoal Adsorbers	Perforated Stainless Steel

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6.3 EMERGENCY CORE COOLING SYSTEM

6.3.1 DESIGN BASES

The Emergency Core Cooling System (ECCS) is designed to cool the reactor core and provide shutdown capability following initiation of the following accident conditions:

1. Pipe breaks in the Reactor Coolant System (RCS) which cause a discharge larger than that which can be made up by the normal makeup system, up to and including the instantaneous circumferential rupture of the largest pipe in the RCS.

2. Rod cluster control assembly ejection accident.

3. Pipe breaks in the steam system, up to and including the instantaneous circumferential rupture of the largest pipe in the steam system.

4. A steam generator tube failure.

The primary function of the ECCS is to remove the stored and fission product decay heat from the reactor core during accident conditions.

The ECCS is designed to tolerate a single active failure (short term) or single active or passive failure (long term). See Section 6.3.2.5 for details. It can meet its minimum required performance level with onsite or offsite electrical power.

The ECCS consists of the centrifugal charging, safety injection and residual heat removal pumps, accumulators, residual heat removal heat exchangers, Refueling Water Storage Tank along with the associated piping, valves, instrumentation and other related equipment.

The design bases for selecting the functional requirements of the ECCS are derived from Appendix K limits for fuel cladding temperature, etc., following any of the above accidents as delineated in 10CFR50.46. The subsystem functional parameters are selected to integrate so that the Appendix K requirements are met over the range of anticipated accidents and single failure assumptions.

Reliability of the ECCS has been considered in selection of the functional requirements, selections of the particular components and location of components and connected piping. The Equipment Specification for the SI/RHR pumps employed at Comanche Peak require them to be capable of performing their long-term cooling function for one year. Environmental qualification durations for ECCS equipment are described in Sections 3.11N and 3.11B. The same type of pump has been used extensively in other operating plants. Their function during recurrent normal cooldowns in such plants as Zion, D.C. Cook, and Trojan has successfully demonstrated their performance capability. Periodic inspections have further confirmed their long-term operability. Nevertheless, design provisions were included that would allow maintenance on either pump if it were necessary during long-term operation.

Redundant components are provided where the loss of one component would impair reliability. Valves are provided in series where isolation is desired and in parallel when flow paths are to be established for ECCS performance. Redundant sources of the ECCS actuation signal are available so that the proper and timely operation of the ECCS will not be inhibited. Sufficient

instrumentation is available so that a failure of an instrument will not impair readiness of the system. The active components of the ECCS are powered from separate buses which are energized from offsite power supplies.

In addition, redundant sources of auxiliary onsite power are available through the use of the emergency diesel generators to assure adequate power for all ECCS requirements. Each diesel is capable of driving all pumps, valves and necessary instruments associated with one train of the ECCS. Other auxiliary systems required to support operation of the ECCS include Component Cooling Water (CCW) for cooling of ECCS pumps and RHR heat exchanger (Section 9.2.2.2) and ESF ventilation for cooling ECCS motors (Section 9.4.5.1).

Spurious movement of a motor operated valve due to an electrical fault in the motor actuation circuitry, coincident with a loss of coolant accident (LOCA), has been analyzed and found to be a very low probability event. However, to comply with the NRC's present position on this issue, the applicant has committed to compliance with BTP- EICSB-18. Compliance is accomplished by providing a control board control power cut-off switch for each valve whose spurious movement could result in degraded ECCS performance. The motor operated valves on the ECCS that require power lockout in order to meet BTP ICSB-18 are identified in Table 6.3-3. The details of the power lockout design is described in Sections 6.3.2.2.12 and 7.6.4. Also see Figure 7.6-3 and elementary diagrams E1-0062, sheets 07, 08, 13, 16 thru 21, 26, 68 and 69 (FSAR Section 1.7). The applicant, nevertheless, reserves the right to retract this commitment in light of WCAP-8966 [See Section 1.6].

Administrative procedures will ensure that motor-operated ECCS valves are properly aligned and have power locked out to prevent spurious operation.

Westinghouse has prepared a Topical Report, WCAP-8966, "Evaluation of Mispositioned ECCS Valves," which substantiates and quantifies the Westinghouse position that spurious movement of a motor operated valve due to an electrical fault coincident with a LOCA is a very low probability event. The bases and assumptions along with the results of this analysis are described in WCAP-8966. Other component failure probabilities (e.g., from maintenance) were also considered in this analysis to justify spurious movement as a very low probability event.

The elevated temperature of the sump solution during recirculation is well within the design temperature of all ECCS components. In addition, consideration has been given to the potential for corrosion of various types of metals exposed to the fluid conditions prevalent immediately after the accident or during long term recirculation operations.

Environmental testing of ECCS equipment inside the Containment, which is required to operate following a LOCA, is discussed in Section 3.11N.

6.3.2 SYSTEM DESIGN

The ECCS components are designed such that a minimum of three accumulators, one charging pump, one safety injection pump, and one residual heat removal pump together with their associated valves and piping will assure adequate core cooling in the event of a design basis LOCA. The redundant onsite emergency diesels assure adequate emergency power to all electrically operated components in the event that a loss of offsite power occurs simultaneously with a LOCA, even assuming a single failure in the Emergency Power System such as the failure of one diesel to start.

All ECCS equipment, with the exception of the ECCS high head cold leg injection throttle valves, has been designed to perform its design function for at least one year without any periodic maintenance. The ECCS high head cold leg injection throttle valves have a service life of six months without any periodic maintenance, however, this service life exceeds the design function requirement for the high head injection flow path. Environmental qualification durations for ECCS equipment are described in Sections 3.11N and 3.11B.

6.3.2.1 Schematic Piping and Instrumentation Diagrams

Flow diagrams of the ECCS are shown in Figures 6.3-1 and 6.3-2. Figure 6.3-2 is provided for illustrative purposes and is based on enveloping conservative layout assumptions. Pertinent design and operating parameters for the components of the ECCS are given in Table 6.3-1. Information in this table has been derived from the actual CPNPP layout. The codes and standards to which the individual components of the ECCS are designed are listed in Table 17A-1.

The component interlocks used in different modes of system operation are listed below.

1. The safety injection signal is interlocked with the following components and initiates the indicated action:

   a. Centrifugal charging pumps start on "S" signal.

   b. Refueling Water Storage Tank suction valves to charging pumps open on "S" signal.

   c. High head safety injection isolation valves (parallel) open on "S" signal.

   d. Normal charging path valves close on "S" signal.

   e. Charging pump miniflow valves close on "S" signal. Alternate miniflow isolation valves (8511A and 8511B) open on “S” signal.

   f. Safety injection pumps start on "S" signal.

   g. The residual heat removal pumps start on "S" signal.

   h. Accumulator isolation valves receive a confirmatory "S" signal; however, these valves are open and have power removed above 1000 psig in accordance with Technical Specifications.

   i. Volume control tank outlet isolation valves close on "S" signal coincident with RWST suction valves full open.

   j. Refueling Water Storage Tank discharge isolation valves to the Spent Fuel Pool Cooling and Cleanup System close on "S" signal.

2. Switchover from the injection mode to recirculation involves the following interlocks.

   a. The suction valves from the sump open when a “S” signal is present and two-out-of-four low level transmitters indicate a low level in the Refueling Water Storage Tank.

   b. The safety injection pump and charging pump recirculation suction isolation valves can be opened provided that the safety injection pump miniflow lines have been isolated.

6.3.2.2 Equipment and Component Descriptions

The component design and operating conditions are specified as the most severe conditions to which each respective component is exposed during either normal plant operation, or during operation of the ECCS. For each component, these conditions are considered in relation to the code to which it is designed. By designing the components in accordance with applicable codes, and with due consideration for the design and operating conditions, the fundamental assurance of structural integrity of the ECCS components is maintained. Components of the ECCS are designed to withstand the appropriate seismic loadings in accordance with their safety class as given in Table 17A-1.

The major mechanical components of the ECCS follow and the specific component parameters are listed in Table 6.3-1.

6.3.2.2.1 Accumulators

The accumulators are pressure vessels partially filled with borated water and pressurized with nitrogen gas. During normal operation each accumulator is isolated from the RCS by two check valves in series. Should the RCS pressure fall below the accumulator pressure, the check valves open and borated water is forced into the RCS. One accumulator is attached to each of the cold legs of the RCS. Mechanical operation of the swing-disc check valves is the only action required to open the injection path from the accumulators to the core via the cold leg.

Connections are provided for remotely adjusting the level and boron concentration of the borated water in each accumulator during normal plant operation as required. Accumulator water level may be adjusted either by draining to the recycle holdup tank or by pumping borated water from the Refueling Water Storage Tank to the accumulator. Samples of the solution in the accumulators are taken periodically for checks of boron concentration.

Accumulator pressure is provided by a supply of nitrogen gas, and can be adjusted as required during normal plant operation; however, the accumulators are normally isolated from this nitrogen supply. Gas relief valves on the accumulators protect them from pressures in excess of design pressure.

The accumulators are located within the Containment but outside of the secondary shield wall which protects them from missiles.

Accumulator gas pressure is monitored by indicators and alarms. The operator can take action as required to maintain plant operation within the requirements of the Technical Specification covering accumulator operability. The accumulator borated water volume and nitrogen

cover-pressure will be provided as operability limits in the Technical Specifications prior to issuance of the operating license. Refer to Table 15.6-5 for the input SIS accumulator parameters utilized in the LOCA accident analysis.

6.3.2.2.2 Boron Injection Tank

The boron injection tank and all associated equipment has been deleted from the CPNPP design.

6.3.2.2.3 Boron Injection Surge Tank

The boron injection surge tank has been deleted from the CPNPP design.

6.3.2.2.4 Residual Heat Removal Pumps

In the event of a LOCA the residual heat removal pumps are started automatically on receipt of an "S" signal. The residual heat removal pumps deliver water to the RCS from the Refueling Water Storage Tank during the injection phase and from the containment sump during the recirculation phase. Each residual heat removal pump is a single stage, vertical position centrifugal pump.

A minimum flow bypass line is provided for the pumps to recirculate and return the pump discharge fluid to the pump suction should these pumps be started with their normal flow paths blocked. Once flow is established to the RCS, the bypass line is automatically closed. This line prevents deadheading of the pumps and permits pump testing during normal operation.

During the recirculation phase, the centrifugal charging pumps and the safety injection pumps are aligned to the discharge of the RHR pumps. If the RWST level continues to drop below the empty setpoint, the RHR pump discharge pressure will eventually close the check valve leading back to the RWST before air can be drawn through a postulated open motor-operated valve (MOV).

Each RHR pump contains a dedicated MOV in the suction line to the RWST. Each MOV is independently powered from a separate electrical train. The physical plant arrangement is such that, if one MOV fails to close and the sump isolation valve has opened, it is not possible for air to be drawn into an RHR pump through the open MOV. In addition, the redundant RHR train would not be affected by the single failure and would provide adequate plant protection.

In addition, the Emergency Operating Procedures includes the following caution:

If the RWST empty level alarm is received at any time subsequent to the accident, immediately stop any pumps still taking suction from the RWST, then complete the switchover and restart any pump which was stopped, beginning with the RHR pumps.

The residual heat removal pumps are discussed further in Section 5.4.7, and pump performance curve is given in Figure 6.3-3.

Related motor-operated valves are discussed further in Section 6.3.2.2.12, and manual actions required of the operator in the changeover from the injection to the recirculation modes is discussed in Section 6.3.2.8.

6.3.2.2.5 Centrifugal Charging Pumps

In the event of an accident, the charging pumps are started automatically on receipt of an "S" signal and are automatically aligned to take suction from the Refueling Water Storage Tank during injection. During recirculation, suction is provided from the residual heat removal pump discharge.

These pumps deliver flow to the RCS at the prevailing RCS pressure. Each centrifugal charging pump is a multistage diffuser design, barrel-type casing with vertical suction and discharge nozzles. To prevent water hammer, ECCS lines associated with the centrifugal charging pumps are maintained full of water since this piping is used for charging and makeup functions of the RCS during all phases of plant operation (refer to Section 9.3.4).

A minimum flow bypass line is provided on each pump discharge to recirculate flow to the pump suction after cooling via the seal water heat exchanger during normal plant operation. The minimum flow bypass line contains two valves in series which close on receipt of the safety injection "S" signal. This signal also opens the charging pump/Refueling Water Storage Tank suction valves and closes the valves to isolate the normal charging line and volume control tank to align the high head portion of the ECCS for injection. A charging pump suction line high point vent is provided to vent non-condensable gases to the VCT during normal operations. Redundant isolation valves in the line are interlocked with the VCT suction isolation valves to open and close on the same signals. An alternate minimum flow line is provided on each pump discharge line to recirculate flow to the RWST as described below. The charging pumps may be tested during power operation via the minimum flow bypass line or through normal charging flow path. A pump performance curve is given in Figure 6.3-4.

The two normally closed alternate miniflow isolation valves (8511A and 8511B) are required to open automatically in the presence of an “S” signal to provide a recirculation path to protect the centrifugal charging pumps from being deadheaded by high RCS pressure during accident conditions such as feed line breaks. When the valve strokes open, each centrifugal charging pump is connected to its associated alternate miniflow relief valve that discharges sufficient pump flow back to the RWST to prevent pump damage. Each normally closed valve is powered from the same train as its associated charging pump. A normally open isolation valve powered from the opposite train is provided in each line for redundant isolation. These valves are required to be closed manually prior to taking suction from the containment sump to prevent the alternate miniflow system from discharging contaminated sump water into the RWST through a leaking or stuck-open miniflow relief valve.

6.3.2.2.6 Safety Injection Pumps

In the event of an accident the safety injection pumps are started automatically on receipt of an "S" signal.

These pumps deliver water to the RCS from the Refueling Water Storage Tank during the injection phase and from the containment sump via the residual heat removal pumps during the recirculation phase. Each intermediate head safety injection pump is driven directly by an induction motor.

A minimum flow bypass line is provided on each pump discharge to recirculate flow to the Refueling Water Storage Tank in the event that the pumps are started with the normal flow paths

blocked. This line may also be used for testing during normal plant operation. Two parallel valves in series with a third (SI Pump Mini-Flow Valve) downstream of a common header, are provided in this line. These valves are manually closed from the Control Room as part of the ECCS realignment from the injection to the recirculation mode. A pump performance curve is shown in Figure 6.3-5.

After the SI Pump Mini-Flow Valve is brought to the required open or closed position by the "OPEN-CLOSE" control switch, the power is removed by a key operated power cut off switch. The key is removable in the "OFF" position only. Both these switches are located on the control board in the same module. Placing the key operated switch in the "OFF" position disconnects both the hot and neutral legs of the control power supply for the Motor Operated Valve starter coils, not just the control circuitry on the main control board. This switch however does not remove 480V power directly from the valve operator. To do so would require bringing 480V power to the main control board where the present circuit voltages are limited to nominal 120V AC and 125V DC and would violate the intent of Regulatory Guide 1.75, paragraph 12 (Rev. 1, January 1975).

6.3.2.2.7 Boron Injection Recirculation Pumps

These pumps have been deleted from the CPNPP design.

6.3.2.2.8 Residual Heat Exchangers

The residual heat exchangers are conventional shell and U-tube type units. During normal cooldown operation, the residual heat removal pumps recirculate reactor coolant through the tube side while component cooling water flows through the shell side. During emergency core cooling recirculation operation, water from the containment sump flows through the tube side. The tubes are seal welded to the tubesheet.

A further discussion of the residual heat exchangers is provided in Section 5.4.7.

6.3.2.2.9 Valves

Design parameters for all types of valves used in the ECCS are given in Table 6.3-1.

Design features employed to minimize valve leakage include:

1. Where possible, packless valves are used.

2. Other valves which are normally open, except check valves and those which perform a control function, are provided with backseats to limit stem leakage.

3. Normally closed globe valves are installed with recirculation fluid pressure under the seat to prevent stem leakage of recirculated (radioactive) water.

4. Relief valves are enclosed; i.e., they are provided with a closed bonnet.

6.3.2.2.9.1 Motor Operated Valves

The seating design of all motor operated gate valves is of the crane flexible wedge design.

This design releases the mechanical holding force during the first increment of travel so that the motor operator works only against the frictional component of the hydraulic unbalance on the disc and the packing box friction. The disc is guided throughout the full disc travel to prevent chattering and to provide ease of gate movement. The seating surfaces are hard faced to prevent galling and to reduce wear.

Where a gasket is employed for the body-to-bonnet joint, it is either a fully trapped, controlled compression, spiral wound gasket with provisions for seal welding, or it is of the pressure seal design with provisions for seal welding. The valve stuffing boxes are designed with a lantern ring leakoff connection with a minimum of a full set of packing below the lantern ring and a minimum of 1/2 of a set of packing above the lantern ring. A full set of packing is defined as a depth of packing equal to 1-1/2 times the stem diameter. The following exceptions to these general guidelines exist: Motor operated valves two inches and under in size do not have a leak connection, but have a minimum of a full set of stem packing. Also, motor operated RHR pump minimum flow bypass 3-inch globe valves FCV-0610 and FCV-0611are supplied without “provisions for seal welding” (weld preparation of the bonnet flanges).

The motor operator incorporates a "hammer blow" feature that allows the motor to impact the discs away from the backseat upon opening or closing. This "hammer blow" feature not only impacts the disc but allows the motor to attain its operational speed prior to impact; thus, providing a more positive opening and closing mechanism. Valves which must function against system pressure are designed such that they function with a pressure drop equal to full system pressure across the valve disc. This feature is an accepted commercial design, not limited to application within the nuclear industry. Considerable valve operating experience, and Westinghouse's valve qualification and operability programs, have demonstrated the reliability of the valve design feature. Surveillance procedures incorporated in Technical Specifications for ECCS valves ensure continued valve operability, including the "hammer blow" feature.

6.3.2.2.9.2 Manual Globes, Gates and Check Valves

Gate valves employ a wedge design and are straight through. The wedge is either split or solid. All gate valves have backseats and outside screws and yokes.

Globe valves, "T" and "Y" style are full ported with outside screw and yoke construction.

Check valves are spring loaded, lift piston types or nozzle check valve types for sizes 2 inches and smaller. Check valves are swing type for size 2-1/2 inches to 4 inches and tilting disc type for size 4 inches and larger. Stainless steel check valves have no penetration welds other than the inlet, outlet and bonnet. The check hinge is serviced through the bonnet.

The stem packing and gasket of the stainless steel manual globe and gate valves are similar to those described above for motor operated valves. Carbon steel manual valves are employed to pass nonradioactive fluids only and therefore do not contain the double packing and seal weld provisions.

6.3.2.2.9.3 Accumulator Check Valves (Swing-disc)

The accumulator check valve is designed with a low pressure drop configuration with all operating parts contained within the body.

Design considerations and analyses which assure that leakage across the check valves located in each accumulator injection line will not impair accumulator availability are as follows:

1. During normal operation the check valves are in the closed position with a total differential pressure across the discs of both check valves of approximately 1650 pounds per square inch (psi). Since the valves remain in this position except for testing or when called upon to open following an accident, and are therefore not subject to the abuse of flowing operation or impact loads caused by sudden flow reversal and seating, they do not experience significant wear of the moving parts, and are expected to function with minimal backleakage. This backleakage can be checked via the test connection as described in Section 6.3.4.

2. Periodically, the check valves are tested for leakage as described in Section 6.3.4.2. This test confirms the seating of the disc and whether or not there has been an increase in the leakage since the last test.

3. The experience derived from the check valves employed in the emergency injection systems indicate that the system is reliable and workable; check valve leakage has not been a problem. This is substantiated by the satisfactory experience obtained from operation of Robert Emmett Ginna (Docket No. 50-244) and subsequent plants where the usage of check valves is identical to the CPNPP.

4. The accumulators can accept some inleakage from the RCS without affecting availability. The availability and operability of the SIS accumulator tanks will be unaffected as long as the accumulator in-leakage does not violate the accumulator pressure/volume technical specification limits. Additionally, accumulator in-leakage is considered when determining RCS identified leakage, the total of which is limited by the technical specifications.

6.3.2.2.9.4 Relief Valves

Relief valves are installed in various sections of the ECCS to protect lines which have a lower design pressure than the RCS. The valve stem and spring adjustment assembly are isolated from the system fluids by a bellows seal between the valve disc and spindle. The closed bonnet provides an additional barrier for enclosure of the relief valves. Table 6.3-2 lists the systems relief valves with their capacities and setpoints.

6.3.2.2.9.5 Butterfly Valves

Each main residual heat removal line has an air operated butterfly valve which is normally open and is designed to fail in the open position. The actuator is arranged such that air pressure on the diaphragm overcomes the spring force, causing the linkage to move the butterfly to the closed position. Upon loss of air pressure, the spring returns the butterfly to the open position. These valves are left in the full open position during normal operation to maximize flow from this system to the RCS during the injection mode of the ECCS operation. These valves are used during normal Residual Heat Removal System (RHRS) operation to control cooldown flow rate.

Each residual heat removal heat exchanger bypass line has an air operated butterfly valve which is normally closed and is designed to fail closed. These valves are used during normal cooldown to avoid thermal shock to the residual heat exchanger.

6.3.2.2.10 Net Positive Suction Head

Available and required net positive suction head for ECCS charging and safety injection pumps are shown in Table 6.3-1. This Table used the actual CPNPP Unit 1 station piping layout to determine the NPSH available at runout conditions. Conservative assumptions have been included (e.g., RWST at minimum level and all other safeguard pumps at maximum runout flow) to minimize the NPSH available.

The safety intent of Regulatory Guide 1.1 is met by the design of the ECCS such that adequate net positive suction head is provided to system pumps. In addition to considering the static head and suction line pressure drop, the calculation of available net positive suction head in the recirculation mode assumes that the vapor pressure of the liquid in the sump is equal to the Containment ambient pressure. This assures that the actual available net positive suction head is always greater than the calculated net positive suction head.

The calculation of available net positive suction head is as follows:

$$(\text{Net positive suction head})_{actual} = (h)_{ambient\ pressure}$$

$$- (h)_{vapor\ pressure} + (h)_{static\ head} - (h)_{loss}$$

$$(\text{Net positive suction head})_{calc} = (h)_{static\ head} - (h)_{loss}$$

The most limiting ECCS condition for available NPSH is with the charging pumps aligned to the RWST, minimum RWST level, and maximum runout for both charging pumps.

ECCS pump specifications include a specified maximum required NPSH which the pump is required to meet. Adequate net positive suction head is shown to be available for all pumps as follows:

1. Residual heat removal pumps

   The net positive suction head of the residual heat removal pumps is evaluated for normal plant cooldown operation, and for both the injection and recirculation modes of operation for the design basis accident. Recirculation operation gives the limiting net positive suction head requirement, and the net positive suction head available is determined from the Containment sump level relative to the pump elevation, the clean strainer head loss, and the pressure drop in the suction piping from the sump to the pumps. The net positive suction head evaluation is based on one residual heat removal pump delivering to two Reactor Coolant System loops and both safety injection and both charging pump suctions.

   The NPSH margin is calculated based on a clean strainer and minimum containment water levels during full sump recirculation. The design basis debris head loss is determined by prototypical testing of a full size strainer with the design basis debris load as described in Section 6.2.2.3.3 scaled to the test configuration. This testing has shown that significant NPSH margin remains after the design basis debris head loss is subtracted from the clean strainer NPSH margin.

2. Safety injection and centrifugal charging pumps

The net positive suction head for safety injection pumps and the centrifugal charging pumps is evaluated for both the injection and recirculation modes of operation for the design basis accident. The end of injection mode of operation gives the limiting net positive suction head available. The net positive suction head available is determined from the elevation head and vapor pressure of the water in the refueling water storage tank, which is at atmospheric pressure, and the pressure drop in the suction piping from the tank to the pumps. When a pre-determined low refueling water storage tank level is reached the safety injection and charging pumps are manually aligned to take suction from the residual heat removal pump discharge headers. The net positive suction head requirements of these pumps are therefore satisfied by the discharge head of the residual heat removal pumps.

6.3.2.2.11 Accumulator Motor Operated Valve Controls

As part of the plant shutdown administrative procedures, the operator is required to close these valves. This prevents a loss of accumulator water inventory to the RCS and is done when the accumulators are no longer required operable by the Technical Specifications. The redundant pressure and level alarms on each accumulator would remind the operator to close these valves, if any were inadvertently left open. Control power is disconnected to these valves after closure by locking open the valve breakers.

During plant startup, the operator is instructed via procedures to energize and open these valves and remove power when the accumulators are required operable by the Technical Specifications. Monitor lights in conjunction with an audible alarm will alert the operator should any of these valves be left inadvertently closed once the RCS pressure increases beyond the safety injection unblock setpoint.

The accumulator isolation valves are not required to move during power operation or in a post accident situation. For a discussion of limiting conditions for operation and surveillance requirements of these valves, refer to Section 3.5.1 of the Technical Specifications.

For further discussions of the instrumentation associated with these valves refer to Sections 6.3.5, 7.3.1 and 7.6.4.

6.3.2.2.12 Motor Operated Valves and Controls

Remotely operated valves for the injection mode which are under manual control (i.e., valves which normally are in their ready position and do not require a safety injection signal) have their positions indicated on a common portion of the control board. If a component is out of its proper position, its monitor light will indicate this on the control panel. At any time during operation when one of these valves is not in the ready position for injection, this condition is shown visually on the board. Table 6.3-3 is a listing of motor operated isolation valves in the ECCS showing interlocks, automatic features and position indications.

The ECCS delivery lag times are given in Chapter 15. The accumulator injection time varies as the size of the assumed break varies since the RCS pressure drop will vary proportionately to the break size.

Spurious movement of a motor operated valve due to an electrical fault in the motor actuation circuitry, coincident with a LOCA, has been analyzed and found to be a very low probability event [WCAP-8966]. However, to comply with the NRC's present position on this issue, the applicant has committed to compliance with BTP-EICSB-18. Compliance is accomplished by providing a control board control power cut-off switch for each valve whose spurious movement could result in degraded ECCS performance. All valves that require removal of power shall be listed in the Technical Specifications per BTP-EICSB-18.

The position in which those valves are located (open or closed) and locked into position are those which ensure ECCS operability. The consequences of a LOCA under those conditions are therefore as described in Chapter 15.

Administrative procedures will require control room power lockout during normal operation with the motor operated valves in their correct position. With these design provisions any concern over damage to both safety injection pumps due to spurious valve movement is eliminated.

Further, subsequent to an accident, power can easily be reinstituted and the valve repositioned as required during the switchover from the injection to recirculation mode. Valve 8806 is in series with check valve 8926 which closes when valves 8804A and 8804B are opened to supply the safety injection pumps from the RHR. This isolates the RWST. Thus, a single failure of valve 8806 to close will not result in a loss of core cooling or a path that permits the release of radioactive material from the containment to the environment.

Administrative controls, procedures, and checklists are employed to assure that manual valves are always restored to their correct position and that lockable valves are locked in their correct position during operation (e.g.) SI-047 and SI-048 are locked open).

The following manually operated valves which could degrade ECCS operation if moved to the incorrect position are locked in their correct position during operation.

8822 A, B, C & D
8810 A, B, C & D

8717

8816 A, B, C & D
SI-047
SI-048

Since these valves are not motor-operated, they do not fall under the requirements of RSB BTP 6-1, and are not subject to single active failure.

The above design and administrative provisions are adequate to ensures ECCS valves are maintained in the correct position.

Periodic visual inspection and operability testing of the motor operated valves in the ECCS insures that there is no potential for impairment of valve operability due to boric acid crystallization which could result from valve stem leakage.

6.3.2.3 Applicable Codes and Classifications

Applicable industry codes and classifications for ECCS components are discussed in Section 3.2.

6.3.2.4 Materials Specifications and Compatibility

Materials employed for components of the ECCS are given in Table 6.3-4. Materials are selected to meet the applicable material requirements of the codes in Table 3.2-2 and the following additional requirements:

1. All parts of components in contact with borated water are fabricated of or clad with austenitic stainless steel or equivalent corrosion resistant material.

2. All parts of components in contact (internal) with sump solution during recirculation are fabricated of austenitic stainless steel or equivalent corrosion resistant material.

3. Valve seating surfaces are hard faced with Stellite Number 6 or manufactured from materials suitable to prevent galling and to reduce wear.

4. Valve stem materials are selected for their corrosion resistance, high tensile properties, and resistance to surface scoring by the packing.

6.3.2.5 System Reliability

Reliability of the ECCS is considered in all aspects of the system from initial design to periodic testing of the components during plant operation.

The ECCS is a two train, fully redundant standby safeguard feature. The system has been designed and proven by analysis to withstand any single credible active failure during injection, or active or passive failure during recirculation and maintain the performance objectives desired in Section 6.3.1. This capability is demonstrated by the failure mode and effects analysis (FMEA) presented in Table 6.3-10. Operator error although not specifically discussed, is inherently addressed in the FMEA and the compliance with BTP-EICSB-18. Failure of components connected to the ECCS are addressed in either the FMEA for the ECCS (e.g., LCV 112 B&C or MOV 8105) or in other sections of the FSAR (e.g., see Section 9.2.1 and 9.2.2 for failure analysis of the station service water system and the component cooling water system, respectively). Two trains of pumps, heat exchangers, and flow paths are provided for redundancy as only one train is required to satisfy the performance requirements. The initiating signals for the ECCS are derived from independent sources as measured from process (e.g., low pressurizer pressure) or environmental variables (e.g., Containment pressure). Redundant as well as functionally independent variables are measured to initiate the safeguards signals. Each train is physically separated and protected where necessary so that a single event cannot initiate a common failure. Power sources for the ECCS are divided into two independent trains supplied from the emergency buses from offsite power. Sufficient diesel generating capacity is maintained onsite to provide required power to each train. The diesel generators and their auxiliary systems are completely independent and each supplies power to one of the two ECCS trains.

The reliability program extends to the procurement of the ECCS components such that only designs which have been proven by testing and/or past use in similar or more severe

applications are acceptable for use. The quality assurance program as described in Chapter 17 assures receipt of components only after manufacture and test to the applicable codes and standards.

The preoperational testing program assures that the systems, as designed and constructed, will meet the functional requirements as calculated in design.

To prevent the occurrence of water hammer in the ECCS lines, the lines are maintained full of water at all times. Refer to Section 6.3.3.7 for a discussion of ECCS piping between containment penetrations and the RWST, and Section 6.3.2.2.5 for a discussion of ECCS lines associated with the centrifugal charging pumps. The water solid condition of the remaining ECCS piping is assured each time the operability of the system is tested. Test procedures ensure that at the conclusion of the test, the system is left in the water solid condition. The ECCS is designed with the ability for on-line testing of most components so the availability and operational status can be readily determined.

In addition to the above, the integrity of the ECCS is assured through examination of critical components during the routine inservice inspection.

1. Active Failure Criteria

   The ECCS is designed to accept a single failure following the incident without loss of its protective function. The system design will tolerate the failure of any single active component in the ECCS itself or in the necessary associated service systems at any time during the period of required system operations following the incident.

   A single active failure analysis is presented in Table 6.3-5 and demonstrates that the ECCS can sustain the failure of any single active component in either the short or long term and still meet the level of performance for core cooling. For ECCS performance evaluation for a loss of coolant accident (LOCA), the single failure assumed is the loss of a low head SI (RHR) pump for a large break LOCA and the loss of an SI train for a small break LOCA. See Section 15.6.5.3.2 for a discussion of the most limiting single failure assumed.

   Since the operation of the active components of the ECCS following a steam line rupture is identical to that following a LOCA, the same analysis is applicable and the ECCS can sustain the failure of any single active component and still meet the level of performance for the addition of shutdown reactivity.

2. Passive Failure Criteria

   The following philosophy provides for necessary redundancy in component and system arrangement to meet the intent of the General Design Criteria on single failure as it specifically applies to failure of passive components in the ECCS. Thus, for the long term, (i.e., after initial switchover from cold leg recirculation to hot leg recirculation) the system design is based on accepting either a passive or an active failure.

A single passive failure analysis is presented in Table 6.3-6. It demonstrates that the ECCS can sustain a single passive failure during the long term phase and still retain an intact flow path to the core to supply sufficient flow to maintain the core covered and effect the removal of decay

heat. The procedure followed to establish the alternate flow path also isolates the component which failed. Figure 6.3-2 is a simplified illustration of the ECCS. The notes provided with Figure 6.3-2 contain information relative to the operation of the ECCS in its various modes. The modes of operation illustrated are full operation of all ECCS components, cold leg recirculation with only residual heat removal pump number 2 operating, and hot leg recirculation with residual heat removal pump number 1 operating. These are representative of the operation of the ECCS during accident conditions.

6.3.2.5.1 Redundancy of Flow Paths and Components for Long Term Emergency Core Cooling

In design of the ECCS, the following criteria was utilized:

1. During the long term cooling period following a LOCA, the emergency core cooling flow paths shall be separable into two subsystems, either of which can provide minimum core cooling functions and return spilled water from the floor of the Containment back to the RCS.

2. Either of the two subsystems can be isolated and removed from service in the event of a leak outside the Containment.

3. Adequate redundancy of check valves is provided to tolerate failure of a check valve during the long term as a passive component.

4. Should one of these two subsystems be isolated in this long term period, the other subsystem remains operable.

5. Provisions are also made in the design to detect leakage from components outside the Containment, to collect this leakage and to provide for maintenance of the affected equipment. Specific provisions are available to maintain major components (e.g., RHR pump and heat exchanger, SI pumps, charging pumps) as well as isolation valves, high point vents and drains during recovery from a LOCA. These provisions assist in the reduction of radiation exposure levels by providing a means for isolating, draining and flushing components which may require maintenance and connected piping. Exposure limits are also reduced by the building layout; it provides shielding between individual components and adequate ventilation of equipment cubicles. Isolation valves are provided with back seats to facilitate stempacking replacement and to limit maintenance exposure.

Thus, for the long term emergency core cooling function, adequate core cooling capacity exists with one flow path removed from service.

6.3.2.5.2 Subsequent Leakage from Components in Safeguards Systems

With respect to piping and mechanical equipment outside the Containment, considering the provisions for visual inspection and leak detection, leaks will be detected before they propagate to major proportions. A review of the equipment in the system indicates that the largest sudden leak potential would be the sudden failure of a pump shaft seal. Evaluation of the shaft leak rate assuming the complete failure of the primary seal, with only the presence of a seal retention ring

(or throttle bushing) for the RHR pump design, showed a leak rate significantly less than 50 gallons per minute (gpm).

Assuming a design basis passive failure of 50 gpm occurs, the water level in the sump will rise at a rate of approximately five inches per minute.

The plant operator receives warning by Control Room alarms, as follows:

1. The first floor drain sump pump will start when the water level reaches the Hi-1 setpoint, less than two minutes after the design basis passive failure occurs. The first level alarm (Hi-1) alerts the Control Room operator that a Safeguard Buildings sump pump has been started.

2. The second floor drain sump pump will start when the sump water level reaches the Hi-2 setpoint. Assuming the first pump failed to start, the second level alarm (Hi-2) alerts the Control Room operator that a second Safeguards Building sump pump has been started.

3. Assuming at least one of the two pumps start, the sump water level will cycle at least once every 5 minutes. This alarm cycle alerts the Control Room operator that a major leak may have occurred and an operator is dispatched to identify the affected train.

After the faulted ECCS train is identified, it can be isolated from the Control Room by operator action. This is done by stopping the corresponding pumps and isolating the respective RHR or containment spray suction lines with appropriate remote operated valves.

Piping leaks, valve packing leaks, or flange gasket leaks have been of a nature to buildup slowly with time and are considered less severe than the pump seal failure.

Larger leaks in the ECCS are prevented by the following:

1. The piping is classified in accordance with ANS Safety Class 2 and receives the ASME Class 2 quality assurance program associated with this safety class.

2. The piping, equipment and supports are designed to ANS Safety Class 2 seismic classification permitting no loss of function for the safe shutdown earthquake.

3. The system piping is located within a controlled area on the plant site.

4. The piping system receives periodic pressure tests and is accessible for periodic visual inspection.

5. The piping is austenitic stainless steel which, due to its ductility, can withstand severe distortion without failure.

Based on this review, the design of the Auxiliary Building and related equipment is based upon handling of leaks up to a maximum of 50 gpm. Means are also provided to detect and isolate such leaks in the emergency core cooling flow path within 30 minutes. See Section 9.3.3.5 for additional details on leak detection.

6.3.2.5.3 Lag Times

Lag times for initiation and operation of the ECCS is limited by pump startup time and consequential loading sequence of these motors onto the safeguard buses. Most valves are normally in the position conducive to safety; therefore valve opening time is not considered for these valves. In the case of an accident coincident with a blackout, a 10 second delay is assumed for diesel startup; then pumps are loaded according to the sequencer. The charging pumps and all valves will be applied to the buses in 10 seconds, the safety injection pumps will start in 15 seconds and the residual heat removal pumps in 20 seconds. These times refer to the maximum delay after receipt of an "S" signal assuming loss of offsite power. If there is no loss of offsite power, the same pump starting sequence is followed after a one second sequencer delay with the first valve load being started upon receipt of the "S" signal. See Section 8.3.1.1.5.3.

6.3.2.5.4 Potential Boron Precipitation

Boron precipitation in the reactor vessel is prevented by a backflush of cooling water through the core to reduce boiloff and resulting concentration of boric acid in the water remaining in the reactor vessel.

Three flow paths are available for hot leg recirculation of sump water. Each safety injection pump can discharge to two hot legs with suction taken from the residual heat removal pump discharge. In addition the residual heat removal pumps can discharge through the common cross connect line and inject water through two hot legs. Loss of one pump or one flow path will not prevent hot leg recirculation since two redundant flow paths are available for use.

6.3.2.5.5 Submerged Valve Motors

An evaluation of the potential for the submersion of ECCS valve motors concludes that all motors are above the local maximum post-accident water level.

The RHR suction isolation valves (Numbers 1-8701A, 1-8701B, 1-8702A and 1-8702B) are located inside the Containment at approximately elevation 817 ft. This elevation is above the local maximum post-accident water level.

6.3.2.6 Protection Provisions

The provisions taken to protect the system from damage that might result from dynamic effects are discussed in Section 3.6N. The provisions taken to protect the system from missiles are discussed in Section 3.5. The provisions to protect the system from seismic damage are discussed in Sections 3.7N, 3.9N and 3.10N. Thermal stresses on the RCS are discussed in Section 5.2.

6.3.2.7 Provisions for Performance Testing

Test lines are provided for performance testing of the ECCS as well as individual components. These test lines and instrumentation are shown in Figure 6.3-1. All pumps have miniflow lines which may be used for testing operability. Additional information on testing can be found in Section 6.3.4.2.

6.3.2.8 Manual Actions

No manual actions are required of the operator for proper operation of the ECCS during the injection mode of operation. Only limited manual actions are required by the operator to realign the system for the cold leg recirculation mode of operation, and, after approximately 3 hours, the hot leg recirculation mode of operation. These actions are delineated in Table 6.3-7.

The changeover from the injection mode to recirculation mode is initiated automatically and completed manually by operator action from the Control Room. The ECCS switchover from safety injection to cold leg recirculation is initiated automatically upon receipt of the RWST switchover initiation (RWST low-low level) signal, and is completed via timely operator action at the main control board. Protection logic is provided to automatically open the two Safety Injection System (SIS) recirculation sump isolation valves (8811 A&B), when two of four Refueling Water Storage Tank (RWST) level channels indicate an RWST level less than a low-low level setpoint, in conjunction with the engineered safeguards actuation signal ("S" signal). This automatic action aligns the suction of the two Residual Heat Removal (RHR) pumps to the containment recirculation sump to ensure continued availability of a suction source. It should be noted that the RHR pumps continue to operate during this changeover from injection mode to recirculation mode.

The two charging pumps and the two safety injection pumps would continue to take suction from the RWST, following the above automatic action, until manual operator action is taken to align these pumps in series with the residual heat removal pumps.

The RWST low-low level protection logic consists of four level channels with each level channel assigned to a separate process control protection set. Four RWST level transmitters provide level signals to corresponding normally deenergized level channel bistables. Each level channel bistable would be energized on receipt of an RWST low-low level signal.

A two out of four coincident logic is utilized in both protection cabinets A and B to ensure a trip signal in the event that two of the four level channel bistables are energized. This trip signal, in conjunction with the "S" signal, provides the actuation signal to automatically open the corresponding containment sump isolation valves.

The low-low RWST level signal is also alarmed to inform the operator to initiate the manual action required to realign the charging and safety injection pumps for the recirculation mode. The manual switchover sequence that must be performed by the operator is delineated in Table 6.3-7.

The switchover procedure is designed to minimize the time required to align the ECCS pumps to the containment sump, utilizing switchover steps in which the operator simultaneously switches both trains of the ECCS from injection to recirculation. As the switchover actions are completed, RWST outflow is reduced. Manual actions 1 thru 6 of Table 6.3-7 must be performed following switchover initiation prior to loss of the RWST transfer allowance to ensure that all ECCS pumps are protected with suction flow available from the containment recirculation sump.

Following the automatic and manual switchover sequence, the two residual heat removal pumps would take suction from the containment recirculation sump and deliver borated water directly to the RCS cold legs. A portion of the number 1 residual heat removal pump discharge flow would be used to provide suction to the two charging pumps which would also deliver directly to the RCS cold legs. A portion of the discharge flow from the number 2 residual heat removal pump

would be used to provide suction to the two safety injection pumps which would also deliver directly to the RCS cold legs. As part of the manual switchover procedure (Table 6.3-7, Action 6), the suctions of the safety injection and charging pumps are cross connected so that one residual heat removal pump can deliver flow to the RCS and both safety injection and charging pumps, in the event of the failure of the second residual heat removal pump.

The minimum delay between the RWST "low" level signal and the "low-low" level signal, with all systems operating (based on maximum allowable Technical Specification instrument errors in each of the four RWST level channels involved is greater than 10 minutes. Operator action to initiate ECCS switchover is not required until 30 seconds later (following automatic opening of the recirculation sump isolation valves (8811A & B).

Thus, operator action to initiate ECCS switchover is not required prior to 10 minutes after event initiation.

The RWST switchover is then performed by transferring the ECCS to cold leg recirculation followed by transferring containment spray to recirculation.

The time available for switchover is dependent on the flowrate out of the RWST as the switchover manual actions are performed. As valves are repositioned, the flowrate out of the RWST is reduced in magnitude. In order to analyze the shortest time available for switchover, the following conservative bases were established.

1. The minimum ECCS transfer allowance available between Low-Low and Empty is greater than 117,760 gallons.

2. Containment and RCS pressures for large break conditions are conservatively assumed to be 0 psig; thus, no credit is taken for the reduction in RWST outflow that will result with higher containment and RCS pressures following a large break. The same assumption is made for the small break conditions, except that RCS pressure is assumed to be greater than RHR pump shutoff head resulting in no RHR pump flow to the RCS for small break conditions.

3. Flow out of the RWST during switchover includes allowances for both pumped flow to the RCS and containment and the possibility of gravity flow (backflow) to the containment sump based on the 0 psig containment pressure assumption and ECCS operating conditions. Specific flowrate allowances are addressed in Table 6.3-11.

4. Flowrate out of the RWST for the worst single failure condition is determined assuming one of the RWST isolation valves (8812A or B) fails to close on demand. This single failure maximizes RWST outflow during switchover. Flowrates out of the RWST assume no operator corrective action to mitigate the single failure (i.e., stop the affected RHR pump and close the appropriate sump isolation valves).

5. Containment flood level above RWST Low-Low plus instrument setpoint uncertainty is assumed constant throughout ECCS switchover to minimize sump flow.

6. Containment flood level above RWST 6% indication is conservatively assumed constant throughout containment spray switchover to minimize sump flow.

7. Flowrate out of the RWST for the worst containment spray single failure condition is determined assuming one of the sump isolation valves (HV-4782 or HV-4783) fails to open on demand. This single failure is bounded by the single failure in the ECCS. Flowrate out of the RWST assumes the operator will stop the affected containment spray pump.

8. The operator initiates the opening of the sump to containment spray pump isolation valves (HV-4782 and HV-4783) at 6% after receipt of the Empty Alarm (at 9%). There is sufficient volume between the 6% indication minus instrument uncertainty and the minimum level for pump suction for the tank isolation valves to fully close.

Sufficient NPSH is available at the time of assumed changeover of the ESF pumps; the single RHR pump (due to the single failure assumption) and the four containment spray pumps continue to draw suction from the RWST. The RHR pump minimum NPSH requirement is below the elevation of the RWST discharge nozzle; therefore, the operator has adequate time to shut down the RHR pump associated with the failed valve. The NPSH requirement for the containment spray pumps is also below the elevation of the RWST discharge nozzle. A monitor light indication is provided to the operator during the recirculation mode when the valve is not in its proper position.

Based on the above criteria, the calculated flowrates out of the RWST as a function of switchover manual action are itemized in Table 6.3-11 for a large break with a single failure, which constitutes the condition where RWST outflow is greatest. Table 6.3-11 also identifies the operator action time assumed, as well as the change in RWST volume, per switchover step. Analyzing the flowrate out of the RWST for the large LOCA with the worst single failure indicates that less than 117,760 gallons are consumed prior to the completion of RWST switchover manual actions. This volume is less than the ECCS transfer allowance which ensures that the switchover steps necessary to protect all ECCS pumps can be accomplished before the transfer allowance is depleted.

See Section 7.5 for process information available to the operator in the control room following an accident. For further information on RHR pumps, and related operator action, see Section 6.3.2.2.4.

Although startup and shutdown are transient events and accidents are not to be considered coincidentally, the following protection is afforded the plant for a secondary side pipe rupture. Safety injection actuation on low pressurizer pressure and low steam line pressure may be manually blocked when NSSS pressure falls below P-11. Specific blocking features are addressed in Section 7.3.2.2.6. Between P-11 and 1000 psig, all safety equipment in the ECCS is aligned for safety injection with the exception of the disarmed pressurizer and steam line pressure safety injection signals. At this time, the operator is monitoring the pressurizer pressure and water level and RCS temperature per the plant cooldown procedure. Also, Technical Specifications impose minimum temperature requirements as a function of pressure on the operator to avoid exceeding NDT limitations. The operator, as a matter of course, has available the pressurizer pressure and water level and RCS temperature measurements on the control board strip chart recorders.

For large LOCAs, sufficient mass and energy is released to the containment to automatically actuate safety injection when the containment high pressure setpoint is reached. At this time, the operator is alerted of the occurrence of a LOCA by the following safety-related indications:

1. Loss of pressurizer level,

2. Rapid decrease of RCS pressure,

3. Containment pressure increase.

In addition, the following indications are normally available to the operator at the control board:

1. Radiation alarms inside containment,

2. Sump water level increasing,

3. Accumulator water levels and pressure decreasing,

4. ECCS valve and pump position and status light in ECCS energized indication. Annunciators will light as safeguards equipment becomes energized,

5. Flow from ECCS pumps is indicated on control board.

For small LOCA's, approximately less than two inches in diameter, where the containment high pressure setpoint may not be reached, the operator observes the safety-related indications plus the first two normally available indications. In addition, there is a charging flow/letdown mismatch which provides the operator with another indication of leakage from the RCS. Since the operator is observing the pressurizer level and is getting additional indications that a LOCA has occurred, the operator immediately initiates manual safety injection. As noted in WCAP-8356, ECCS Plant Sensitivity Studies, the time to uncover the core following a small break is relatively long (e.g., greater than 10 minutes for a 2-inch break). The operator, therefore, has sufficient time to manually initiate safety injection.

At less than 1000 psig, the operator closes and locks out the safety injection accumulator isolation valves. At less than 350 F, the operator renders the high head safety injection pumps inoperable in accordance with Technical Specifications. At this time, at least one low head safety injection pump and at least one charging pump are available from either automatic or manual safety injection actuation. At less than 350 F, the operator aligns the RHR System. The valves in the line from the RWST are closed.

The significance of these actions on the mitigation of a LOCA when power is locked out to the isolation valves is that between 1000 psig and reaching 350 F, a portion of the ECCS can be actuated automatically (containment hi-1 signal) or manually by the operator. The equipment that can be energized are two low head safety injection and two high head charging pumps, two high head safety injection pumps and, subsequent to the operator reinstituting power to the accumulator isolation valves.

Below 350 F, the system is in the RHR cooling mode. The operator realigns the RHR system per plant startup procedure; places all safeguards systems valves in the required positions for ECCS operation; and then manually actuates the individual ECCS components.

Comparing plant cooldown and heatup, the limiting case for a LOCA is during cooldown rather than heatup because the core decay heat generation is higher. The ECCS analysis presented in Chapter 15 conforms to the acceptance criteria of 10CFR50.46 (initiation of the LOCA is at

100.6% of full licensed power rating for Unit 1 and Unit 2 with corresponding RCS conditions), and is more limiting than a LOCA during shutdown, since:

1. A LOCA initiated during shutdown has reduced decay heat generation; the reactor would have been at zero power for an extended period of time.

2. The core stored energy during shutdown is reduced due to the RCS isothermal condition at a reduced temperature.

3. The energy content of the RCS is lower.

Furthermore, the probability of an occurrence of a LOCA during this time along with the critical flaw size needed to rupture the RCS piping at reduced pressure is considered to be incredible. (Note: Unit 1 LOCA initiation is at 102% of full licensed power until completion of 1RF09).

Emergency operating procedures are written such that the operator is permitted to manually reset the safety injection signal after two minutes following protection system actuation. This enables the operator to shut down a safeguards pump or to change the position of a valve receiving an SIS signal, as necessary. In the unlikely event that a blackout occurs following an SIS reset, ECCS valve positions remain unchanged; the operator only needs to depress the manual safety injection switches to restart safety injection. Manual resets are discussed in greater detail in Section 7.3.2.2.6.

6.3.3 PERFORMANCE EVALUATION

Accidents which require ECCS operation are the following:

1. Inadvertent opening of a steam generator relief or safety valve (see Section 15.1.4).

2. Loss of reactor coolant from small ruptured pipes or from cracks in large pipes which actuates the ECCS (see Section 15.6.5).

3. Major reactor coolant system pipe ruptures (LOCA) (see Section 15.6.5).

4. Steam system piping failure (see Section 15.1.5).

5. Steam generator tube failure (see Section 15.6.3).

The ECCS is actuated from any of the following:

1. Two out of four low pressurizer pressure signals.

2. Two out of three high-1 containment pressure signals.

3. Two out of three low steam line pressure signals in any one loop.

4. Manual initiation.

6.3.3.1 Inadvertent Opening of a Steam Generator Relief or Safety Valve

The adequacy of the ECCS is demonstrated through compliance with the relevant event acceptance criteria, as described in Section 15.1.4.

6.3.3.2 Loss of Reactor Coolant from Small Ruptured Pipes or from Cracks in Large Pipes Which Actuate the Emergency Core Cooling System

The adequacy of the ECCS is demonstrated through compliance with the relevant event acceptance criteria, as described in Section 15.6.5.

6.3.3.3 Major Reactor Coolant System Pipe Ruptures (Loss of Coolant Accident)

The adequacy of the ECCS is demonstrated through compliance with the relevant event acceptance criteria, as described in Section 15.6.5.

6.3.3.4 Steam System Piping Failure

The adequacy of the ECCS is demonstrated through compliance with the relevant event acceptance criteria, as described in Section 15.1.5.

6.3.3.5 Steam Generator Tube Failure

The adequacy of the ECCS is demonstrated through compliance with the relevant event criteria, as described in Section 15.6.3.

6.3.3.6 Existing Criteria Used to Judge the Adequacy of the ECCS

The relevant acceptance criteria for the each event that requires the operation of the ECCS are described in Chapter 15.

6.3.3.7 Use of Dual Function Components

The ECCS contains components which have no other operating function as well as components which are shared with other systems. Components in each category are as follows:

1. Components of the ECCS which perform no other function are:

   a. One accumulator for each loop which discharges borated water into its respective cold leg of the reactor coolant loop piping.

   b. Two safety injection pumps, which supply borated water for core cooling to the RCS. (May be used during check valve testing also and filling of Safety Injection Accumulators.)

   c. Associated piping, valves and instrumentation.

2. Components which also have a normal operating function are as follows:

   a. The residual heat removal pumps and the residual heat exchangers: These components are normally used during the latter stages of normal reactor cooldown and when the reactor is held at cold shutdown for core decay heat removal. However, during all other plant operating periods, they are aligned to perform the low head injection function. (May also be used during filling of Safety Injection Accumulators.)

   b. The centrifugal charging pumps: These pumps are normally aligned for charging service. As a part of the Chemical and Volume Control System, the normal operation of these pumps is discussed in Chapter 9.

   c. The Refueling Water Storage Tank: This tank is used to fill the refueling cavity for refueling operations. However, during other plant operations it provides an alternate suction to the charging pumps which is automatically aligned on the receipt of a safety injection signal, a low-low level in the VCT, or a boron dilution mitigation signal. During Power Operations (Mode 1), Startup (Mode 2), Hot Standby (Mode 3), and Hot Shutdown (Mode 4), the tank is aligned to provide suction to the safety injection pumps, the containment spray pumps, and the residual heat removal pumps when the latter are not aligned in the RHR cooling mode.

      To prevent the occurrence of water hammer, ECCS piping between containment penetrations and the RWST is kept full of water by gravity in MODES 1 through 4. The highest point in the piping is at El. 825'-6" which is below the minimum RWST level of 850'-5". All lines are equipped with high point vents which ensures complete filling and absence of air pockets.

      Figure 6.3-7 shows the location of the refueling water storage tank vents. Each vent line alone provides the required venting capacity for normal plant operation. In the event of a LOCA, both RWST vents are required to provide the required venting capacity. Tornadoes are not postulated coincident with LOCAs; however, the vents are conservatively designed to break away if struck by a tornado missile to ensure the RWST availability for all design basis events. The vents are located above the tank overflow level, and their blocking by freezing water is not considered feasible. The vents are classified as non-nuclear safety related.

      Blocking of the vents by ice buildups on the tank roof is prevented by the self-draining capability of the roof structure.

An evaluation of all components required for operation of the ECCS demonstrates that either:

1. The component is not shared with other systems, or

2. If the component is shared with other systems, it is either aligned during normal plant operation to perform its accident function or if not aligned to its accident function, two valves in parallel are provided to align the system for injection, and two valves in series are provided to isolate portions of the system not utilized for injection. These valves are automatically actuated by the safety injection signal.

Table 6.3-8 indicates the alignment of components during normal operation, and the realignment required to perform the accident function.

In all cases of component operation, safety injection has the priority usage such that an "S" signal will override all other signals and start or align systems for injection.

6.3.3.8 Limits on System Parameters

The analyses show that the design basis performance characteristic of the ECCS is adequate to meet the requirements for core cooling following a LOCA with the minimum engineered safety feature equipment operating. In order to ensure this capability in the event of the simultaneous failure to operate any single active component, Technical Specifications are established for reactor operation.

Normal operating status of ECCS components is given in Table 6.3-9.

The ECCS components are available whenever the coolant energy is high and the reactor is critical. During low temperature physics tests there is a negligible amount of stored energy in the coolant and low decay heat; therefore, an accident comparable in severity to accidents occurring at operating conditions is not possible and ECCS components are not required.

The principal system parameters and the number of components which may be out of operation in test, quantities and concentrations of coolant available, and allowable time in a degraded status are illustrated in the Technical Specifications. If efforts to repair the faulty component are not successful, the plant is placed into a lower operational status, i.e., hot standby to hot shutdown, hot shutdown to cold shutdown, etc.

6.3.4 TESTS AND INSPECTIONS

6.3.4.1 ECCS Performance Tests

6.3.4.1.1 Preoperational Test Program at Ambient Conditions

Preliminary operational testing of the ECCS can be conducted during the hot functional testing of the RCS following flushing and hydrostatic testing, with the system cold and the reactor vessel head removed. Provision should be made for excess water to drain into the refueling canal. The ECCS must be aligned for normal power operation. Simultaneously, the safety injection block switch is reset and the breakers on the lines supplying offsite power are tripped manually so that operation of the emergency diesels is tested in conjunction with the SIS. This test should provide information including the following facets:

1. Satisfactory safety injection signal generation and transmission.

2. Proper operation of the emergency diesel generators, including sequential load pickup.

3. Valve operating times.

4. Pump starting times.

5. Pump delivery rates at runout conditions (one point on the operating curve).

6.3.4.1.2 Components

1. Pumps

Separate flow tests of the pumps in the ECCS are conducted during the operational startup testing (with the reactor vessel head off) to check capability for sustained operation. The centrifugal charging, safety injection, and residual heat removal pumps will discharge into the reactor vessel through the injection lines, the overflow from the reactor vessel passing into the refueling canal. Each pump will be tested separately with water drawn from the Refueling Water Storage Tank. In the case of the RHR pumps, water may optionally be drawn from the Reactor Coolant System hot legs. Data will be taken to determine pump head and flow at this time. Pumps will then be run on miniflow circuits and data taken to determine a second point on the head flow characteristic curve.

2. Accumulators

Each accumulator is filled with water from the Refueling Water Storage Tank and pressurized with the motor operated valve on the discharge line closed. Then the valve is opened and the accumulator allowed to discharge into the reactor vessel as part of the operational startup testing with the reactor cold and the vessel head off.

Conformance with Regulatory Guide 1.79 is discussed in Appendix 1A(B).

6.3.4.2 Reliability Tests and Inspections

Routine periodic testing of the ECCS components and all necessary support systems at power is planned. Valves which operate after a LOCA are operated through a complete cycle, and pumps are operated individually in this test. If such testing indicates a need for corrective maintenance, the redundancy of equipment in these systems permits such maintenance to be performed without shutting down or reducing load under certain conditions. These conditions include considerations such as the period within which the component should be restored to service and the capability of the remaining equipment to provide the minimum required level of performance during such a period.

The operation of the remote stop valve and the check valve in each accumulator tank discharge line may be tested by opening the remote test line valves just downstream of the stop valve and check valve, respectively. Flow through the test line can be observed on instruments, and the opening and closing of the discharge line stop valve can be sensed on this instrumentation.

The ECCS is designed with the capability to determine leakage from the RCS to the ECCS through any of the series check valves which are located in each accumulator, residual heat removal pump, safety injection pump, and charging pump cold leg or hot leg injection line. Tests are performed to verify that each series check valve can independently sustain a differential

pressure across its disc and also to verify that the valve is in its closed position. The following check valves are provided with leakage detection capability:

8818 A thru D
SI-8819 A thru D
8948 A thru D
8956 A thru D
8949 A thru D
SI-8905 A thru D
SI-8900 A thru D
8841 A and B
8815

Where series pairs of checks valves form the high pressure to low pressure isolation barrier between the RCS and the SIS piping outside the Containment (8818 A thru D, 8819 A thru D, 8948 A thru D & 8956 A thru D), periodic testing of these check valves must be performed to provide assurance that certain postulated failure modes will not result in a loss of coolant from the low pressure system outside Containment with a simultaneous loss of safety injection pumping capacity. The SIS test line subsystem provides the capability for determination of the integrity of the pressure boundary formed by series check valves. As noted above, the tests performed verify that each of the series check valves can independently sustain differential pressure across its disc, and also verify that the valve is in its closed position. The required periodic tests are to be performed after each refueling just prior to plant startup, after the RCS has been pressurized.

Lines in which the series check valves are to be tested include the safety injection pump injection lines, the residual heat removal pump injection lines, and the SIS accumulator discharge lines.

To implement the periodic component testing requirements, Technical Specifications have been established. See the Technical Specifications for the selection of test frequency, acceptability of testing, and measured parameters for the testing described in this Section. During periodic system testing, a visual inspection of pump seals, valve packings, flanged connections, and relief valves is made to detect leakage. Leakage must be within Technical Specification limits for RCS leakage.

Design measures have been taken to assure that the following testing can be performed:

1. Active components may be tested periodically for operability (e.g., pumps, certain valves, etc.).

2. An integrated system actuation test[(a)] can be performed when the plant is cooled down and the RHRS is in operation. The ECCS will be arranged so that no flow will be introduced into the RCS for this test.

3. An initial flow test of the full operational sequence can be performed.

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a. Details of the testing of the sensors and logic circuits associated with the generation of a safety injection signal together with the application of this signal to the operation of each active component are given in Section 7.2.

The design features which assure this test capability are specifically:

1. Power sources are provided to permit individual actuation of each active component of the ECCS.

2. The safety injection pumps can be tested periodically during plant operation using the minimum flow recirculation lines provided.

3. The residual heat removal pumps are used every time the RHRS is put into operation. They can also be tested periodically when the plant is at power using the miniflow recirculation lines.

4. The centrifugal charging pumps are either normally in use for charging service or can be tested periodically.

5. Remote operated valves can be exercised during routine plant maintenance.

6. Level and pressure instrumentation is provided for each accumulator tank, for continuous monitoring of these parameters during plant operation.

7. Flow from each accumulator tank can be directed at any time through a test line to determine check valve leakage and to demonstrate operation of the accumulator motor operated valves.

8. A flow indicator is provided in the safety injection pump header, and in the residual heat removal pump headers. Pressure instrumentation is also provided in these lines.

9. An integrated system test can be performed when the plant is cooled down and the RHRS is in operation. This test does not introduce flow into the RCS but does demonstrate the operation of the valves, pump circuit breakers, and automatic circuitry including diesel starting and the automatic loading of ECCS components of the diesels (by simultaneously simulating a loss of offsite power to the vital electrical buses).

Inservice inspection provides further confirmation that no significant deterioration is occurring in the ECCS fluid boundary. A description of the inservice inspection program is provided in the Unit 1 and Unit 2 Inservice Inspection Program Plans. ECCS components and systems are designed to meet the intent of the ASME Code, Section XI for inservice inspection.

6.3.5 INSTRUMENTATION REQUIREMENTS

Instrumentation and associated analog and logic channels employed for initiation of ECCS operation is discussed in Section 7.3. This section describes the instrumentation employed for monitoring ECCS components during normal plant operation and also ECCS post accident operation. All alarms are annunciated in the Control Room.

6.3.5.1 Temperature Indication

6.3.5.1.1 Boron Injection Tank Temperature

Deleted from the CPNPP design; refer to section 6.3.2.2.2.

6.3.5.1.2 Residual Heat Exchanger Outlet Temperature

The fluid temperature at the outlet of each residual heat exchanger is recorded in the Control Room.

6.3.5.1.3 Boron Injection Surge Tank Temperature

Deleted from the CPNPP design; refer to Section 6.3.2.2.3.

6.3.5.2 Pressure Indication

6.3.5.2.1 Boron Injection Tank Pressure

Deleted from the CPNPP design; refer to Section 6.3.2.2.2

6.3.5.2.2 Safety Injection Pump Discharge Pressure

Safety injection pump discharge pressure is indicated in the Control Room.

6.3.5.2.3 Accumulator Pressure

Duplicate pressure channels are installed on each accumulator. Pressure indication in the Control Room and high and low pressure alarms are provided by each channel.

6.3.5.2.4 Test Line Pressure

A local pressure indicator used to check for proper seating of the accumulator check valves between the injection lines and the RCS is installed on the leakage test line.

6.3.5.2.5 Residual Heat Removal Pump Discharge Pressure

Residual heat removal discharge pressure for each pump is indicated in the Control Room. A high pressure alarm is actuated by each channel.

6.3.5.2.6 Pressure Monitoring

In response to Generic Letter 2008-01 [Ref. 1], pressure transmitters are provided to monitor ECCS piping that may be affected by potential gas sources. Monitoring of ECCS piping while the systems ar in normal stand-by status provided to ensure that the systems are in compliance with regulations and the operating license. The monitoring and TS Surveillance Requirements are designed to ensure that the ECCS systems are water solid, where required, and provide the basis to show compliance with the applicable regulatory requirements including 10 CFR50 Appendix A, General Design Criteria 1, 34, 10 CFR 50.36(c)(3).

Pressure indication is provided within the RHRS Pump Discharge Header, the SI Pump Discharge Header, the SI Hot Leg Injection Headers, the CCP Injection Header to Cold Leg Injection Header, and the SI Test Line Header to monitor ECCS piping pressure while the systems are in normal stand-by status. Pressure indication is retrieved from the Plant Computer System.

6.3.5.3 Flow Indication

6.3.5.3.1 Boric Acid Recirculation Flow

Deleted from the CPNPP design.

6.3.5.3.2 Charging Pump Injection Flow

Injection flow through the reactor cold leg is indicated in the Control Room.

6.3.5.3.3 Safety Injection Pump Header Flow

Flow through the safety injection pump header is indicated in the Control Room.

6.3.5.3.4 Residual Heat Removal Pump Injection Flow

Flow through each residual heat removal injection and recirculation header leading to the reactor cold or hot legs is indicated in the Control Room.

6.3.5.3.5 Test Line Flow

Local indication of the leakage test line flow is provided to check for proper seating of the accumulator check valves between the injection lines and the RCS.

6.3.5.3.6 Residual Heat Removal Return Line Flow

The return flow of reactor coolant from the residual heat removal loop during normal plant cooldown is recorded in the Control Room.

6.3.5.3.7 Safety Injection Pump Minimum Flow

A flow indicator is installed in the safety injection pump minimum flow line.

6.3.5.3.8 Residual Heat Removal Pump Minimum Flow

A flowmeter installed in each residual heat removal pump discharge header provides control for the valve located in the pump minimum flow line.

6.3.5.3.9 Pressure Isolation Valve Leakage Monitoring System

Unit 2 is provided with a leak measurement line from valve 2SI-8975 equipped with flow measurement instruments to measure intersystem leakage through RCS pressure isolation valves into the SI injection header. This alternate relief path may be used to route this leakage to the recycle holdup tank in lieu of via the SI injection header relief valves. The line is designed to limit the SI system loss to less than or equal to 0.2 gpm when the isolation valve is open even in the event of an SSE. When using the line unattended in this mode, this design leakage will also be tracked as part of the leakage limit for primary coolant sources outside the containment structure as discussed in TMI Section III.D.1.1 of the FSAR to ensure the 1 gpm limit is not exceeded in the event of an accident.

6.3.5.4 Level Indication

6.3.5.4.1 Refueling Water Storage Tank Level

Four water level indicator channels, which indicate in the Control Room, are provided for the Refueling Water Storage Tank. Each channel is provided with a high, low, low-low and empty level alarm.

The high level alarm is provided to protect against possible overflow of the Refueling Water Storage Tank.

The low level alarm is provided to assure that a sufficient volume (473,731 gallons) of water is always available in the Refueling Water Storage Tank (RWST) in conformance with the Technical Specifications. Receipt of a low level alarm, concurrent with receipt of an "S" signal, also alerts the operator that: 1) ECCS cold leg injection is in progress, and 2) that manual action will be required to effect ECCS switchover to the recirculation mode when the RWST low-low level alarm is received.

The low-low level alarm automatically initiates ECCS switchover (by opening the recirculation sump isolation valves). Concurrently, the RWST low-low level alarm alerts the operator of the automatic initiation of ECCS switchover and that manual actions (as delineated in Table 6.3-7) must be performed to realign the ECCS from the injection mode to the recirculation mode following an accident.

The empty alarm indicates that the usable volume of the Refueling Water Storage Tank is nearing depletion. Upon receipt of the empty alarm, the operator stops any ECCS pump still taking suction from the RWST. After performing any pump protection required for the ECCS pumps, the operator would complete any required actions to isolate the containment spray pumps from the RWST or ensure pump protection from loss of suction. The empty alarm ensures sufficient volume is available to stop any pump still taking suction from the RWST. Each channel also provides level indication in the Control Room.

6.3.5.4.2 Accumulator Water Level

Duplicate water level channels are provided for each accumulator. Both channels provide indication in the Control Room and actuate high and low water level alarms.

6.3.5.4.3 Boron Injection Surge Tank Level

Deleted from the CPNPP design.

6.3.5.5 Valve Position Indication

Valve positions which are indicated on the control board are done so by a "normal off" system; i.e., should the valve not be in its proper position, a bright white light will be lit and thus give a highly visible indication to the operator.

The accumulator motor operated valves are provided with red (open) and green (closed) position indicating lights located at the control switch for each valve. These lights are powered by valve control power and actuated by valve motor operator limit switches. These position indicators

function when control power is removed by a key lock switch at the Main Control Board in accordance with BTP ICSB-18. However, these position indicators do not function when power is removed at the Motor Control Center.

A monitor light that is on when the valve is not fully open is provided in an array of monitor lights that are all off when their respective valves are in proper position enabling safeguards operation. This light is energized from a separate monitor light supply and actuated by a valve motor operated limit switch.

To meet the intent of BTP ICSB-18, an alarm annunciator point is activated by both a valve motor operator limit switch and by a valve position limit switch activated by stem travel whenever an accumulator valve is not fully open for any reason with the system at pressure (the pressure at which the safety injection block is unblocked is approximately 1900 psig). A separate annunciator point is used for each accumulator valve. This alarm will be recycled at approximately 1 hour intervals to remind the operator of the improper valve lineup.

All ECCS motor operated and air operated valves have position indication in the control room by monitor lights and/or red/green position indication lights (see Table 6.3-3). The position in which the valves are located (open or closed) and locked into position are those which ensure ECCS operability.

The position of manual valves 1SI-047 and 1-8717 is indicated and alarmed by means of the SSII (Safety System Inoperable Indication - see FSAR Section 7.1.2.6). 1SI-047 is on the line from the RWST to the ECCS pump suction. If 1SI-047 is not open, an SSII module light will flash and a horn will sound. 1-8717 is on the return line from the RHRS pump discharge to the RWST. If 1-8717 is not closed, an SSII module light will flash and a horn will sound.

REFERENCES

1. NRC Generic Letter 2008-01, “Managing Gas Accumulation in Emergency Core Cooling, Decay Heat Removal, and Containment Spray Systems”, dated January 11, 2008.

TABLE 6.3-1
EMERGENCY CORE COOLING SYSTEM COMPONENT PARAMETERS

(Sheet 1 of 3)

Component	Parameter
Accumulators	
Number	4
Design pressure (psig)	700
Design temperature (°F)	300
Operating temperature (°F)	70-150
Maximum operating pressure (psig)	693
Minimum operating pressure (psig)	603
Total volume (ft^3)	1350 each
Minimum water volume (gal)	6119 each
Volume nitrogen gas (ft^3)	500
Boric acid concentration, nominal (ppm)	2400
Boric acid concentration, minimum (ppm)	2300
Relief valve setpoint (psig)	700
Centrifugal Charging Pumps	
Number	2
Design pressure (psig)	2800
Design temperature (°F)	300
Design flow rate[(a)] (gpm)	150
Design head (ft)	5800
Maximum flow rate (gpm)	550
Head at maximum flow rate (ft)	1400
Discharge head at shutoff (ft)	6000
Motor rating[(b)] (php)	600

TABLE 6.3-1
EMERGENCY CORE COOLING SYSTEM COMPONENT PARAMETERS

(Sheet 2 of 3)

Component	Parameter
Required NPSH (ECCS) maximum flow rate (ft)	22
Available NPSH (ft)	26.7
Safety Injection Pumps	
Number	2
Design pressure (psig)	1860
Design temperature (°F)	200
Design flow rate (gpm)	425
Design head (ft)	2680
Maximum flow rate (gpm)	650
Head at maximum flow rate (ft)	1650
Discharge head (ft)	3600
Motor rating[(b)] (bhp)	450
Required NPSH (ft)	25
Available NPSH (ft)	>34
Residual Heat Removal Pumps	
(See Section 5.4.7 for design parameters)	
Residual Heat Exchangers	
(See Section 5.4.7 for design parameters)	
Valves	

Motor operated valves	Valve Size (in.)	Max.open.or closing time
8814A & B	1.5	10 sec.

TABLE 6.3-1
EMERGENCY CORE COOLING SYSTEM COMPONENT PARAMETERS

(Sheet 3 of 3)

Component		Parameter
8110, 8111, 8813	2	10 sec.
8511 A & B, 8512 A & B	2	10 sec.
FCV-610, FCV-611	3	15 sec.
8105, 8106	3	10 sec.
8801 A & B	4	10 sec.
8802 A & B, 8835	4	15 sec.
LCV-112 B & C	4	10 sec.
8821 A & B	4	15 sec.
8807 A & B, 8923 A & B, 8924	6	15 sec.
8804 A & B, 8806, LCV-112 D & E	8	15 sec.
8716 A & B	10	15 sec.
8809 A & B, 8840	10	20 sec.
8811 A & B, 8812 A & B	14	20 sec.
8701 A & B, 8702 A & B	12	120 sec.
8808 A - D	10	20 sec.

a) Includes miniflow.

b) 1.15 service factor not included.

TABLE 6.3-2
EMERGENCY CORE COOLING SYSTEM RELIEF VALVE DATA

Description	Fluid Discharge	Fluid Inlet Temperature (Normal)	Set Pressure (Psig)	Back Pressure Constant	Psig Buildup	Capacity
Nitrogen supply to accumulators	Nitrogen	120	750 Tag # (1-8857) 750 Tag # (2-8857)	0	0	1500 scfm
Safety injection pump discharge	Water	100	1860	3	50	20 gpm
Residual heat removal pump safety injection line	Water	120	600	3	50	20 gpm
Safety injection pumps suction header	Water	120	220	3	50	20 gpm
Accumulator to Containment	Water or nitrogen gas	120	700	0	0	1500 scfm

TABLE 6.3-3
MOTOR OPERATED ISOLATION VALVES IN THE EMERGENCY CORE COOLING SYSTEM

(Sheet 1 of 3)

Location Indication	Valve Identification	Interlocks	Automatic Features	Position Indication	Alarms	Monitor Light Box
Accumulator	8808 A,B,C,D	Power lockout Cannot be closed with "S" signal or when manually locked out	Open on "S" signal, or RCS pressure greater than unblock	MCB	Yes-out of position	Yes
Safety injection pump suction from RWST	8806	Power lockout	None	MCB	Yes-out of position	Yes
RHR suction from RWST	8812 A&B	Cannot be opened unless sump valve closed	None	MCB	Yes-out of position	Yes
Safety injection pump suction isolation valves	8923 A&B	None	None	MCB	No	Yes
RHR discharge to safety injection/charging pump	8804 A&B	Cannot be opened unless safety injection pump miniflow isolated and RHR suction from RCS isolated	None	MCB	No	Yes
Safety injection hot leg injection	8802 A&B	Power lockout	None	MCB	No	Yes
RHR hot leg injection	8840	Power lockout	None	MCB	No	Yes
Containment sump isolation valve	8811 A & B	Cannot be opened in normal operation unless RHR suction valves from RWST and one of two RCS suction series valves are closed.	Opens on RWST Lo-Lo with "S" signal	MCB	No	Yes
CVCS suction from RWST	LCV-112 D&E	None	Opens on "S" signal, or VCT Lo-Lo level	MCB	No	Yes

TABLE 6.3-3
MOTOR OPERATED ISOLATION VALVES IN THE EMERGENCY CORE COOLING SYSTEM

(Sheet 2 of 3)

Location Indication	Valve Identification	Interlocks	Automatic Features	Position Indication	Alarms	Monitor Light Box
CVCS normal suction	LCV-112 B& C	None	With CVCS suction from RWST open, closes on "S" signal, or VCT Lo-Lo level	MCB	No	Yes
Safety injection pump to cold leg	8835	Power Lockout	None	MCB	No	Yes
CVCS normal discharge	8105 8106	None	Closes on "S" signal	MCB	No	Yes
High head safety injection isolation valves	8801 A&B	None	Opens on "S" Signal	MCB	No	Yes
Charging and safety injection pump header from RHR	8807 A&B 8924	None None	None None	MCB MCB	No None	Yes Yes
RHR to RCS cold legs	8809 A&B	Power lockout	None	MCB	No	Yes
Safety injection pump miniflow	8813	Power lockout. Cannot be opened unless RHR discharge to SI/CHG pump valves closed.	None	MCB	No	Yes
	8814 A&B	Cannot be opened unless RHR discharge to SI/CHG pump valves closed.				
RHR cross connect	8716 A&B	None	None	MCB	No	Yes
Safety injection pump cross connect	8821 A&B	None	None	MCB	No	Yes
Charging pump miniflow isolation valves	8110 8111	None	Closes on "S" signal	MCB	No	Yes

TABLE 6.3-3
MOTOR OPERATED ISOLATION VALVES IN THE EMERGENCY CORE COOLING SYSTEM

(Sheet 3 of 3)

Location Indication	Valve Identification	Interlocks	Automatic Features	Position Indication	Alarms	Monitor Light Box
RHR hot leg suction isolation valves	8701 A, B 8702 A, B	Cannot be opened unless there is no RCS High-1 pressure and Containment sump isolation valve, RHR suction from RWST valve and RHR discharge to safety injection/charging pump suction valves are closed.		MCB	Yes-out of position	No
Charging pump alternate miniflow isolation valves	8511 A, B	Cannot be opened manually from MCB unless the CVCS suction from the VCT valve is closed and the RHR discharge to safety injection/charging pump suction valves are closed.	Opens on "S" signal	MCB	No	Yes
Charging pump alternate miniflow isolation valve	8512 A, B	Cannot be opened unless the RHR discharge to safety injection/charging pump suction valves are closed.	None	MCB	No	Yes
RHR pump recirculation	FCV 610, 611	None	Opens when flow from the RHR pump is low. Closes when flow from the RHR pump is High 10 sec after the pump has started.	MCB	No	No

Power lockout - Control board control power cut-off switch to prevent spurious movement. Valve remains "as is" when control power is cut-off.

Note 1 - Monitor and hand switch

MCB-Main Control Board

TABLE 6.3-4
MATERIALS EMPLOYED FOR EMERGENCY CORE COOLING SYSTEM COMPONENTS

(Sheet 1 of 2)

Component	Material
Accumulators	Carbon steel, clad with austenitic stainless steel
Pumps	
Centrifugal charging	Austenitic stainless steel
Safety injection	Austenitic stainless steel
Residual heat removal	Austenitic stainless steel
Residual heat exchangers	
Shell	Carbon steel
Shell end cap	Carbon steel
Tubes	Austenitic stainless steel
Channel	Austenitic stainless steel
Channel cover	Austenitic stainless steel
Tubesheet	Austenitic stainless steel
Valves	
Motor operated valves containing radioactive fluids	
Pressure containing parts	Austenitic stainless steel or equivalent
Body-to-bonnet	ASME SA-453 grade 660 (NUT-ASME 193, B6, Type 410) (All other parts - SA-194)
Seating surfaces	Stellite No. 6 or other material suitable for the application
Stems	Austenitic stainless steel or 17-4 pH stainless

TABLE 6.3-4
MATERIALS EMPLOYED FOR EMERGENCY CORE COOLING SYSTEM COMPONENTS

(Sheet 2 of 2)

Component	Material
Motor operated valves containing nonradioactive, boron-free fluids	
Body, bonnet and flange	Carbon steel
Stems	Corrosion resistance steel
Diaphragm valves	Austenitic stainless steel
Accumulator check valves	
Parts contacting borated water	Austenitic stainless steel
Relief valves	
Stainless steel bodies	Stainless steel
Carbon steel bodies	Carbon steel
All nozzles, discs, spindles and guides	Austenitic stainless steel
Bonnets for stainless steel valves without a balancing bellows	Stainless steel or plated carbon steel
All other bonnets	Carbon steel
Piping	
All piping in contact with borated water	Austenitic stainless steel

TABLE 6.3-5
SINGLE ACTIVE FAILURE ANALYSIS FOR EMERGENCY CORE COOLING SYSTEM COMPONENTS

(Sheet 1 of 3)

Component	Malfunction	Comments
Short Term Phase		
1. Pumps		
a. Centrifugal charging	Fails to start	Two provided, evaluation based on operation of one.
b. Safety injection	Fails to start	Two provided, evaluation based on operation of one.
c. Residual heat removal	Fails to start	Two provided, evaluation based on operation of one.
2. Automatically operated valves		
a. High head safety injection isolation valve	Fails to open	Two parallel lines; one valve in either line required to open.
b. Residual heat removal pumps suction line to containment sump	Fails to open	Two parallel lines; only one valve in either line required to open.
c. Centrigugal charging pumps		
1) Suction line from refueling water storage tank	Fails to open	Two parallel valves; only one valve required to open.

TABLE 6.3-5
SINGLE ACTIVE FAILURE ANALYSIS FOR EMERGENCY CORE COOLING SYSTEM COMPONENTS

(Sheet 2 of 3)

Component			Malfunction	Comments
	2)	Discharge line to the normal charging pathvalve	Fails to close	Two valves in series; only one required to close.
	3)	Miniflow bypass line	Fails to close	Two valves in series; only one valve required to close.
	4)	Suction from volume control tank	Fails to close	Two valves in series; only one valve required to close.
	5)	Suction line high point vent	Fails to close	Two valves in series; only one valve required to close.
Long Term Phase				
1.	Valves operated manually from the control room			
	a.	Residual heat removal pumps suction line from refueling water storage tank	Fails to close	Check valve in series with one gate valve; operation of only one valve required.
	b.	Safety injection pump suction line from refueling water storage tank	Fails to close	Check valve in series with one gate valve; operation of only one valve required.
	c.	Centrifugal charging pump suction line from refueling water storage tank	Fails to close	Check valve in series with two parallel gate valves; operation of either check valve or both of the gate valves required.

TABLE 6.3-5
SINGLE ACTIVE FAILURE ANALYSIS FOR EMERGENCY CORE COOLING SYSTEM COMPONENTS

(Sheet 3 of 3)

Component		Malfunction	Comments
d.	High head pump suction line at discharge of residual heat exchanger	Fails to open	Separate and independent high head injection paths to safety injection pumps and charging pumps taken suction from discharge of residual heat exchangers; operation of only one valve required.
e	Residual heat removal cross connect line	Fails to close	Two valves in series; operation of one required.
f.	Safety injection pump miniflow lines	Fails to close	Two parallel valves provided in series with a third; operation of either both parallel valves or series valve required.
g.	Safety injection/charging cross connect line in suction header	Fails to open	Two parallel valves provided; operation of either one required.
h.	Safety injection/residual heat removal hot leg isolation valves	Fails to open	Three flow paths available; adequate flow to core is assured by any two.
i.	Safety injection/residual heat removal cold leg isolation valves	Fails to close	Redundant valves provided with suitable arrangements to preclude pump runout.

TABLE 6.3-6
EMERGENCY CORE COOLING SYSTEM RECIRCULATION PIPING PASSIVE FAILURE ANALYSIS LONG TERM PHASE

Flow Path	Indication of Loss of Flow Path	Alternate Flow Path
Low Head Recirculation		
From containment sump to the low head injection header via the residual heat removal pumps and the residual heat exchangers	Accumulation of water in a residual heat removal pump compartment or the Safeguards Building sump	Via the independent, identical low head flow path utilizing the second residual heat exchanger and residual heat removal pump
High Head Recirculation		
From containment sump to the high head injection header via the residual heat removal pump, residual heat exchanger and high head injection pumps	Accumulation of water in a residual heat removal pump compartment or the Safeguards Building sump or safety injection or charging pump compartments	From containment sump to the high head injection headers via alternate residual heat removal pump, residual heat exchanger, safety injection or charging pump

TABLE 6.3-7
SEQUENCE OF SWITCHOVER OPERATIONS (BASED ON NO SINGLE FAILURES)

(Sheet 1 of 4)

SWITCHOVER FROM INJECTION TO COLD LEG RECIRCULATION

The following manual operator actions are required to perform the switchover operation from the ECCS cold leg injection mode to the cold leg recirculation mode. Following event initiation and upon receipt of the RWST low level alarm, the operator is in the process of anticipating the RWST low-low level switchover initiation alarm. The operator monitors the RWST level and containment recirculation sump in anticipation of switchover. During the cold leg injection mode and prior to the receipt of the RWST Low-Low level alarm, the operator is to:

(a) verify that all ECCS pumps are operating and are delivering flow to the RCS cold legs, and

(b) monitor the RWST and the containment recirculation sump levels.

Upon receipt of the RWST low-low level signal, the operator is required to reset SI and to verify that the component cooling water isolation valves for the heat exchangers are open. The emergency procedure includes cautions and notes, as required. The operator is then required to verify the sump isolation valves are open and to perform the manual actions listed below in an orderly and timely manner, and in the proper sequence. Manual actions 1 thru 6 function to align the suction of the residual heat removal pumps to the containment sump and to align the suction of the charging & safety injection pumps to the discharge of the residual heat removal pumps, thereby assuring an available suction source for all ECCS pumps. The remaining operator actions serve only to provide redundant isolation of the RWST from the recirculation fluid.

SWITCHOVER STEPS (Note 1)

The RWST low-low level signal automatically initiates opening of the containment sump isolation valves (8811 A&B). Upon receipt of the RWST low-low level signal the operator is to immediately enter the emergency procedure for transfer to cold leg recirc and perform the following actions:

STEP 1: When each sump isolation valve (8811 A&B) has reached the full open position, take immediate action to close the corresponding RWST to residual heat removal pump suction isolation valve (8812 A&B).

STEP 2: Close the three Safety Injection Pump miniflow valves (8813, 8814A, and 8814B).

STEP 3: Close the two valves in the crossover line downstream of the residual heat removal heat exchangers (8716 A&B).

STEP 4: Close the CCP miniflow to RWST valves (8511A, 8511B, 8512A, 8512B).

TABLE 6.3-7
SEQUENCE OF SWITCHOVER OPERATIONS (BASED ON NO SINGLE FAILURES)

(Sheet 2 of 4)

STEP 5: Open the two parallel valves in the common suction lines between the charging pump suction and the safety injection pump suction (8807 A&B).

STEP 6: Open each valve from each residual heat removal pump discharge line to the charging pump suction and to the safety injection pump suction (8804 A&B respectively).

All ECCS pumps are now aligned with suction flow from the containment sump. See Table 6.3-11 for additional information on these switchover steps.

STEP 7: Verify proper operation and alignment of all ECCS components. Complete the following manual actions to provide redundant isolation of the RWST from the recirculation fluid.

STEP 8: Close the two parallel valves in the line from the RWST to the charging pump suction (LCV-112 D&E).

STEP 9: Restore power to and close the valve in the common line from the RWST to both safety injection pumps (8806).

The ECCS is now aligned for cold leg recirculation as follows:

a. Both residual heat removal pumps are delivering from the containment sump directly to the RCS cold legs and are also delivering to the suction of the safety injection and charging pumps.

b. Both safety injection and charging pumps are delivering to the RCS cold legs.

Notes (1) The operator actions for switchover from injection to cold leg recirculation are not to be interrupted until all of the steps in the switchover are completed; however, if the RWST EMPTY level alarm is received any time during the switchover, immediately stop any ECCS pumps still taking suction from the RWST, then complete the switchover and restart any pump which was stopped, starting with the residual heat removal pump.

The RWST level indication then informs the operator that sufficient water has been injected into containment and to initiate switchover of the containment spray system.

STEP 10: Switchover all four (4) containment spray pumps.

TABLE 6.3-7
SEQUENCE OF SWITCHOVER OPERATIONS (BASED ON NO SINGLE FAILURES)

(Sheet 3 of 4)

SWITCHOVER FROM COLD LEG-RECIRCULATION TO HOT LEG RECIRCULATION

At approximately 3 hours after the accident, hot leg recirculation shall be initiated. The following manual operator actions are required to perform the switchover operation from the cold leg recirculation mode to the hot leg recirculation mode. These steps are the general steps required to complete the switchover for both trains of ECCS. Plant procedures may direct completion of steps by individual train as necessary based on component availability.)

SWITCHOVER STEPS

STEP 1: Close the residual heat removal pump discharge cold leg header isolation valves (8809 A/B).

STEP 2: Open the residual heat removal pump discharge crossover isolation valves (8716 A/B).

[a]STEP 3: Open the residual heat removal pump discharge hot leg header isolation valve (8840).

STEP 4: Stop safety injection pump No. 1.

STEP 5: Close the corresponding safety injection pump discharge crossover header isolation valve (8821A).

STEP 6: Open the corresponding safety injection pump discharge hot leg header isolation valve (8802A).

[a]STEP 7: Restart safety injection pump No. 1.

STEP 8: Stop safety injection pump No. 2.

STEP 9: Close the corresponding safety injection pump discharge crossover isolation valve (8821B).

STEP 10: Open the corresponding safety injection pump discharge hot leg header isolation valve (8802B).

[a]STEP 11: Restart safety injection pump No. 2.

STEP 12: Close the safety injection pump discharge cold leg header isolation valve (8835).

STEP 13: Open Valve 8821A or 8821B. No preference, however, only one valve may be opened.

TABLE 6.3-7
SEQUENCE OF SWITCHOVER OPERATIONS (BASED ON NO SINGLE FAILURES)

(Sheet 4 of 4)

The ECCS is now aligned for hot leg recirculation as follows:

a. Both residual heat removal pumps are delivering from the containment sump directly to the RCS hot legs and are also delivering to the suction of the safety injection and charging pumps.

b. Both safety injection pumps are delivering to the RCS hot legs. Hot leg flow will dilute the reactor vessel boron concentration by passing relatively dilute boron solution from the hot leg through the vessel to the cold leg break location.

c. Both charging pumps are delivering to the RCS cold legs. High head charging flow will continue to be provided to the RCS cold legs to preclude boron concentration buildup in the vessel for breaks in the hot leg.

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a) Note:

If the alignment of a residual heat removal or safety injection pump for hot leg recirculation is not successful, the affected pump is returned to delivering cold leg recirculation flow, plant staff is consulted, and applicable steps not yet performed are taken to align remaining pump(s) for hot leg recirculation.

TABLE 6.3-8
EMERGENCY CORE COOLING SYSTEM SHARED FUNCTIONS EVALUATION

Component	Normal Operating Arrangement	Accident Arrangement
Refueling water storage tank	Lined up to suction of centrifugal charging, safety injection, residual heat removal, and containment spray pumps	Lined up to section of centrifugal charging, safety injection, residual heat removal, and containment spray pumps
Centrifugal charging pumps	Lined up for charging service	Lined up to cold leg injection header. Valves for realignment meet single failure criteria
Residual heat removal pumps	Lined up to cold legs of reactor coolant piping	Lined up to cold legs of reactor coolant piping
Residual heat removal (RHR) heat exchangers	Lined up to cold legs of reactor coolant piping	Lined up to cold legs of reactor coolant piping

TABLE 6.3-9
NORMAL OPERATING STATUS OF EMERGENCY CORE COOLING SYSTEM
COMPONENTS FOR CORE COOLING

Number of safety injection pumps operable	2
Number of charging pumps operable	2
Number of residual heat removal pumps operable	2
Number of residual heat exchangers operable	2
Refueling water storage tank required water volume (gal) minimum	473,731
Boron concentration in refueling water storage tanks, minimum (ppm)	2,400
Boron concentration in accumulator, minimum (ppm)	2,300
Number of accumulators	4
Minimum accumulator pressure (psig)	603
Minimum accumulator water volume (gal)	6119
System valves, interlocks, and piping required for the above components which are operable	All

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 1 of 13)

Component	Failure Mode	ECCS Operation Phase	[(a)]Effect on System Operation	Failure Detection Method	Remarks
1. Motor operated gate valve 1-LCV-112B (1-LCV-112C analogous)	Fails to close on demand	Injection - cold legs of RC loops	Failure reduces redundancy of providing VCT discharge isolation. No effect on safety for system operation; valve (1-LCV-112C) provides backup tank discharge isolation.	Valve position indication (open to closed position change) at CB. Valve close position monitor light for group monitoring of components at CB. Periodic tests and inspection in compliance with surveillance requirements of plant's technical specification.	Valve is electrically interlocked to close on actuation by a SI "S" signal or a "low-low level" VCT signal providing isolation valve 1-LCV-112D (1-LCV-112E analogs.) is at a full open position.
2. Motor operated gate valve 1-LCV-112D (1-LCV-112E analogous)	Fails to open on demand	Injection - cold legs of RC loops	Failure reduces redundancy of providing fluid flow from RWST to section of HHSI/CH pumps. No effect on safety for system operation. Alternate isolation valve (1-LCV-112E) opens to provide backup flow path to suction of HHSI/CH pumps.	Same methods of detection as that stated for item #1 except open position monitor light for group monitoring of components and closed to open position change indication at CB.	Valve is electrically interlocked to open upon actuation by a SI "S" signal, or by a "Low-low level" VCT signal.
3. Centrifugal charging pump #1, 1-APCH (pump #2 analogous)	Fails to deliver working fluid.	Injection and recirculation cold legs of RC loops.	Failure reduces redundancy of providing emergency coolant to the RCS at prevailing incident RCS pressure. Fluid flow from HHSI/CH pump #1 w/be lost. Minimum flow requirements at prevailing high RCS pressures will be met by HHSI/CH pump #2 delivery via RCS cold legs.	HHSI/CH pump discharge header flow (FI-917) at CB. Open pump switch gear circuit breaker indication on CB. Circuit breaker close position monitor light for group monitoring of components at CB. Common breaker trip alarm at CB. Periodic tests in compliance with surveillance requirements of plant's technical specification.	One HHSI/CH pump is used for normal charging of RCS during plant operation. Pump circuit breaker aligned to close on actuation by a SI "S" signal.

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 2 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
4. Motor operated globe valve 1-8110 (1-8111 analogous)	Fails to close on demand.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing isolation of HHSI/CH pump miniflow line. No effect on safety for system operation. Alternate isolation valve (1-8111) in miniflow line provides backup isolation.	Same method of detection at that stated for item #1. In addition, close position alarm for group monitoring of components at CB.	Valve aligned to close upon actuation by a SI “S” signal.
5. Motor operated globe valve 1-8105 (1-8106 analogous)	Fails to close on demand.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing isolation of HHSI/CH pump discharge to normal charging line of CVCS. No effect on safety for system operation. Alternate isolation valve (1-8106) provides backup normal CVCS charging line isolation.	Same method of detection as that stated for item #1.	Valve aligned to close upon actuation by a SI "S" signal.
6. Motor operated gate valve 1-8801A (1-8801B analogous)	Fails to open on demand.	Injection - cold legs of RC loops.	Failure reduces redundancy of fluid flow paths from HHSI/CH pumps to the RCS. No effect on safety for system operation. Alternate isolation valve (1-8801B) opens to provide backup flow path from HHSI/CH pumps to the RCS.	Same method of detection as that stated for item #2. In addition, open position alarm for group monitoring of components at CB.	Valve aligned to open upon actuation by a SI "S" signal.

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 3 of 13)

Component	Failure Mode	ECCS Operation Phase	[(a)]Effect on System Operation	Failure Detection Method	Remarks
7. Motor operated globe valve 1-FCV-610 (1-FCV-611 analogous)	a. Fails to close on demand.	Injection - cold legs of RC loops.	a. Failure reduces working fluid delivered to RCS from RHR pump #1. Minimum flow requirements for LHSI will be met by LHSI/ RHR pump #2 delivering working fluid to RCS.	a. Valve position indication (open to closed position change) at CB. RHR pump return line to cold legs flow indication (FI-618) at CB. Periodic test and inspection in compliance with surveillance requirements of plant's technical specification.	Valve is regulated by signal from flow transmitter located in pump discharge header. The control valve opens when the RHR pump discharge flow is less than 500 gpm and closes when the flow exceeds 1000 gpm.
	b. Fails closed.	Injection - cold legs of RC loops.	b. Failure results in an insufficient fluid flow through LHSI/RHR pump #1 for a small LOCA or steam line break resulting in possible pump damage. If pump becomes inoperative minimum flow requirements for LHSI will be met by LHSI/RHR pump #2 delivering working fluid to RCS.	b. Same as that stated above for failure mode "fails open" except closed to open position change indication at CB.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 4 of 13)

Component	Failure Mode	ECCS Operation Phase	[(a)]Effect on System Operation	Failure Detection Method	Remarks
8. Residual heat removal pump #1, 1ARPH (Pump #2 analogous)	Fails to deliver working fluid.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing emergency coolant to the RCS from the RWST at low RCS pressure (195 psig). Fluid flow from LHSI/RHR #1 will be lost. Minimum flow requirements for LHSI will be met by LHSI/RHR pump #2 delivering working fluid.	RHR pump return line to cold legs flow indication (FI-618) and low flow alarm at MCB. RHR pump discharge pressure (PI-614) at CB. Open pump switchgear circuit breaker indication at CB. Circuit breaker close position monitor light for group monitoring of components at CB. Common breaker trip alarm at CB. Periodic tests in compliance with surveillance requirements of plant's technical specification.	The RHR pump is sized to deliver reactor coolant through the RHR heat exchanger to meet plant cooldown requirements & is used during plant cooldown & startup operations. The pump circuit brkr. is aligned to close on actuation by a SI "S" signal.
9. Safety injection pump #1, 1APSI (Pump #2 analogous)	Fails to deliver working fluid.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing emergency coolant to the RCS from the RWST at high RCS pressure(1520 psi). Fluid flow from HHSI/SI pump #1 will be lost. Minimum flow requirements for HHSI will be met by HHSI/SI pump #2 delivering working fluid.	SI pumps discharge pressure (PI-919) at CB. SI pump discharge flow (FI-918) at CB. Open pump switchgear circuit breaker indication at CB. Circuit breaker close position monitor light for group monitoring of components at CB. Common breaker trip alarm at CB. Periodic tests in compliance with surveillance requirements of plant's technical specification.	Pump circuit breaker aligned to close on actuation by a SI "S" signal.

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 5 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
10. Motor operated gate valve 1-8811A (1-8811B analogous)	Fails to open on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing fluid from the Containment Sump to the RCS during recirculation. LHSI/RHR pump #1 will not provide recirculation flow. Minimum LHSI flow requirements will be met through opening of isolation valve 1-8811B and recirculation of fluid by LHSI/RHR pump #2.	Same method of detection as that stated for item #7. In addition failure may be detected through monitoring of RHR pump return line to cold legs flow indication (FI-618) and RHR pump discharge pressure (PI-614) at CB.	Valve is actuated to open by SI "S" signal in coincidence with two-out-of-four "Low-Low Level" RWST signals. Valve is electrically interlocked from manually being opened by isolation valves 1-8812A, 1-8701A, & 1-8702A.
11. Motor operated gate valve 1-8812A (1-8812B analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing flow isolation of Containment Sump from RWST. No effect on safety for system operation. Alternate check isolation valve (1-8958A) provides backup isolation.	Same method of detection as that stated for item #4.	Valve is electrically interlocked with isolation valve 1-8811A & may not be opened unless valve 1-8811A is closed.
12. Motor operated gate valve 1-8716A (1-8716B analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing LHSI/RHR pump train separation for recirculation of fluid to cold legs of RCS. No effect on safety for system operation. Alternate isolation valve (1-8716B) provides backup isolation for LHSI/ RHR pump train separation.	Same method of detection as that stated for item #4.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 6 of 13)

Component	Failure Mode	ECCS Operation Phase	[(a)]Effect on System Operation	Failure Detection Method	Remarks
13. Motor operated globe valve 1-8813	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing isolation of HHSI/SI pump's miniflow line isolation from RWST. No effect on safety for system operation. Alternate isolation valve (1-8814A and 1-8814B) in each pump's miniflow line provide back up isolation.	Same method of detection as that stated for time #4.	Valve is electrically interlocked with isolation valves 1-8804A & 1-8804B & may not be opened unless these valves are closed.
14. Motor operated globe valve 1-8814A (1-8814B analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing isolation of HHSI/SI pump #1 miniflow isolation from RWST. No effect on safety for system operation. Alternate isolation valve (1-8813) in main miniflow line provides backup isolation.	Same method of detection as that stated for item #4.	Same remark as that stated for item #16.
15. Motor operated gate valve 1-8804A.	Fails to open on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing NPSH to suction of HHSI/CH pumps from LHSI/RHR pumps. No effect on safety for system operation Minimum NPSH to HHSI/CH pump suction will be met by flow from LHSI/RHR pump #2 via cross-tie line and opening of isolation valve 1-8807A or 1-8807B and isolation valve 1-8804B.	Same method of detection as that stated for item #7.	Valve is electrically interlocked with isolation valves 1-8814A, 1-8814B, 1-8813, 1-8701A, 1-8702A, 1-8511A, 1-8511B, 1-8512A and 1-8512B. Valve cannot be opened unless 1-8813 or 1-8814A&B are closed, 1-8701A or 1-8702A is closed, 1-8511A or1-8512B is closed and 1-8511B or 1-8512A is closed.

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 7 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
16. Motor operated gate valve 1-8804B	Fails to open on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing NPSH to suction of HHSI/SI pumps from LHSI/RHR pumps. No effect on safety for sytem operation. Minimum NPSH to HHSI/SI pump suction will be met by flow from LHSI/RHR pump #1 via cross-tie line and opening of isolation valve 1-8807A or 1-8807B and isolation valve 1-8804A.	Same method of detection as that stated for item #7.	Same remark as that stated for item #15 except interlocked with valves 1-8701B and 1-8702B.
17. Motor operated gate valve 1-8807-A (1-8807B analogous)	Fails to open on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing fluid flow through cross-tie between suction of HHSI/CH pumps and HHSI/IS pumps. No effect on safety for system operaation. Alternate isolation valve (1-8807B) opens to provide backup flow path through cross-tie line.	Same method of detection as stated for item #7.	
18. Motor operated gate valve 1-8806	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing flow isolation of HHSI/SI pump suction from RWST. No effect on safety for system operation. Alternate check isolation valve (1-8926) provides backup isolation.	Same method of detection as that stated for item #4.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 8 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
19. Motor operated gate valve 1-LCV-112D (1-LCV-112E analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing flow isolation of suction of HHSI/CH pumps from RWST. No effect on safety for system operation. Alternate check isolation valve (1-8546) provides backup isolation.	Same method of detection as that stated previously for failure of item during injection phase of ECCS operation.	
20. Residual heat removal pump #1, 1APRH (pump #2 analogous)	Fails to deliver working fluid.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump. Fluid flow from LHSI/RHR pump #1 will be lost. Minimum recirculation flow requirements for LHSI flow will be met by LHSI/RHR pump #2 delivering working fluid.	Same method of detection as that stated previously for failure of item during injection phase of ECCS operation.	
21. Safety injection pump #1, 1APSI (pump #2 analogous)	Fails to deliver working fluid.	Recirculation - cold or hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump to cold legs of RC loops via RHR and SI pumps. Fluid flow from HHSI/SI pump #1 will be lost. Minimum recirculation flow requirements for HHSI flow will be met by HHSI/SI pump #2 delivering working fluid.	Same method of detection as that stated previously for failure of item during injection phase of ECCS operation.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 9 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
22. Motor operated gate valve 1-8809A	Fails to close on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump to hot legs of RC loops. Fluid flow from LHSI/RHR pump #1 will continue to flow to cold legs of RC loops. Minimum recirculation flow requirements to hot legs of RC loops will be met by LHSI/RHR pump #2 recirculating fluid to RC hot legs via HHSI/SI pumps.	Same method of detection as that stated for item #4.	
23. Motor operated gate valve 1-8716A (1-8716B analogous)	Fails to open on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump to the hot legs of RC loops. Fluid flow from LHSI/RHR pump #1 will be lost. Minimum recirculation flow requirements to hot legs of RC loops will be met by LHSI/RHR pump #2 recirculating fluid to RC hot legs via HHSI/SI pumps.	Same method of detection as that stated previously for failure of item during recirculation into cold legs of RC loops ECCS operational phase. In addition, RHR pump discharge pressure (PI-614) at CB.	
24. Motor operated gate valve 1-8840	Fails to open on demand.	Recirculation - hot legs of RC loops.	Same effect on system operation as that stated for item #26.	Same method of detection as that stated for item #7 In addition, RHR pump discharge pressure (PI-614) at CB.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 10 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
25. Motor operated gate valve 1-8809B.	Fails to close on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump to hot legs of RC loops. Fluid flow from LHSI/RHR pump #2 will continue to flow to cold legs of RC loops. Minimum recirculation flow requirements to hot legs of RC loops will be met by LSHI/RHR pump #1 recirculating fluid to RC hot legs.	Same method of detection as that stated for item #4.	
26. Motor operated gate valve 1-8821A (1-8821B analogous)	Fails to close on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing flow isolation of HHSI/SI pump flow to cold legs of RC loops. No effect on safety for system operation. Alternate isolation valve (1-8835) provides backup isolation against flow to cold legs of RC loops.	Same method of detection as that stated for item #4.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 11 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
27. Motor operated gate valve 1-8802A (1-8802B analogous)	Fails to open on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the hot legs of RCS from the Containment Sump via HHSI/SI pumps. Minimum recirculation flow requirements to hot legs of RC loops will be met by LHSI/RHR pump #1 recirculating fluid from Containment Sump to hot legs of RC loops and HHSI pump #2 recirculating fluid to hot legs 1 and 4 of RC loops through the opening of isolation valve 1-8802B.	Same method of detection as that stated for item #7. In addition, SI pump discharge pressure (PI-919) and flow (FI-918) at CB.	
28. Motor operated gate valve 1-8835	Fails to close on demand.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing flow isolation of HHSI/SI pump flow to cold legs of RC loops. No effect on safety for system operation. Alternate isolation valves(1-8821A and 1-8821B) in cross-tie line between HHSI/SI pumps provides backup isolation agains flow to cold legs of RC loops.	Same method of detection as that stated for item #4.	

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 12 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
29. Residual heat removal pump #1, 1APRH (Pump #2 analogous)	Fails to deliver working fluid.	Recirculation - hot legs of RC loops.	Failure reduces redundancy of providing recirculation of coolant to the RCS from the Containment Sump to the hot legs of RC loops. Fluid flow from LHSI/RHR pump #1 will be lost. Minimum flow requirements to hot legs of RC loop will be met by LHSI/RHR pump #2 recirculating fluid to RC hot legs via HHSI/SI pumps.	Same method of detection as that stated previously for failure of item during injection phase of ECCS operation.	
30. Motor operated globe valve 1-8511A (1-8511B analogous)	Fails to open on demand.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing miniflow to Train A CCP. No effect on safety for system operation.	Valve position indication(open to closed position change) at CB. Valve open position monitor light for group monitoring of components at CB. Periodic tests and inspection in compliance with surveillance requirements of plant's technical specification.	Valve is aligned to open upon actuation by a SI "S" signal. Alternate miniflow valve 1-8511B provided for Train B CCP.
31. Motor operated globe valve 1-8511A (1-8511B analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of isolation of Train A CCP alternate miniflow line to RWST. No effect on safety for system operation.	Valve position indication (open to closed position change) at CB.	Alternate miniflow isolation valve 1-8512B provided for redundant isolation of Train A CCP mini flow to the RWST.
32. Motor operated globe valve 1-8512B (1-8512A analogous)	Fails to close on demand.	Recirculation - cold legs of RC loops.	Failure reduces redundancy of isolation of Train A CCP alternate miniflow line to RWST. No effect on safety for system operation.	Valve position indication (open to closed position change) at CB.	Alternate miniflow isolation valve 1-8511A provided for redundant isolation of Train A CCP miniflow to the RWST.

TABLE 6.3-10
FAILURE MODE AND EFFECTS ANALYSIS – EMERGENCY CORE COOLING SYSTEM – ACTIVE COMPONENTS

(Sheet 13 of 13)

Component	Failure Mode	ECCS Operation Phase	[a]Effect on System Operation	Failure Detection Method	Remarks
33. Air operated ball valve HV-8220 (HV-8221 analogous)	Fails to close on demand.	Injection - cold legs of RC loops.	Failure reduces redundancy of providing VCT isolation. No effect on safety for system operation; valve (HV-8221) provides backup high point isolation.	Valve position indication (open to closed position change) at CB. Valve close position monitor light for group monotoring of components at CB. Periodic tests and inspection in compliance with surveillance requirements of plant's technical specifications.	Valve is electrically interlocked with LCV-112B (LCV-112C analogous) to close on actuation by SI "S" signal. A "low-low level" VCT signal providing isolation valve LCV-112D (LCV-112E analogous) is at a full open position.

List of abbreviations and acronyms

LOCA - Loss of Coolant Accident

RWST - Refueling Water Storage Tank

BIR - Boron Injection Recirculation

CB - Control Board

SI - Safety Injection

NPSH - Net Positive Suction Head

VCT - Volume Control Tank

CH - Charging

RC - Reactor Coolant

HHSI- High Head Safety Injection

RCS - Reactor Coolant System

LHSI- Low Head Safety Injection

RHR - Residual Heat Removal

a) See list at end of Table for definition of acronyms and abbreviations used.

TABLE 6.3-11
RWST OUTFLOW LARGE BREAK - WORST SINGLE FAILURE[9]

Step [1]	Table 6.3-7 Step	Action	Time Req. Per Step (sec) [3] [5] [10]	RWST Outflow Per Step (gpm) [2] [6] [7]	Cumulative Change in RWST Vol. (gal)
1	-	Reset SI	30	24,694	12,347
2	-	Verify CCW flow from RHR heat exchangers	30	21,596	23,145
3a [4]	-	Verify 8811A&B open (emergency sump to RHRPs)	30	21,245	33,767
3b	1	Close 8812A&B (RWST to RHRPs)	50	20,888	51,174
3c	2	Close 8814A&B, 8813 (SIP miniflow)	40	19,290	64,034
3d	3	Close 8716A&B (RHRP crosstie)	45	18,999	78,282
3e	4	Close 8511A&B, 8512A&B (CCP Alt Miniflow)	40	18,609	90,688
3f	5	Open 8807A&B (SI to Charging Suction Header Crosstie)	45	18,254	104,379
3g [8]	6	Open 8804A&B (RHRP to SIP Suction)	45	17,842	117,760
End of ECCS Switchover Volume			-	-	-
Start of Spray Switchover Volume					
4	7	At 6%, Open HV-4782/HV-4783, Close HV-4758/HV-4759	70	Note 11	12,464

Notes:

(1) Emergency operating procedure steps for transfer to cold leg recirculation. See Table 6.3-7 switchover steps 1 through 6 for a description of the steps 3b through 3g (manual switchover actions for ECCS).

(2) It is assumed that the RHR, SI, charging and containment spray pumps operate at conservative flowrates. Pumped flowrates are assumed to be constant during each step of switchover and the specified number of pumps of each type are operating:

RHR pump	=	2 pumps operating
CCHG pump	=	2 pumps operating
SI pump	=	2 pumps operating
CS pump	=	4 pumps operating

(3) ECCS Valve operating times are maximum operating times from Table 6.3-1.

(4) An allowance of time for valves 8811A/B to automatically open occurs prior to completion of step 1.

(5) Time required to complete the required action for ECCS Switchover includes a conservative 30 seconds for operator response time for each manual procedure.

(6) The flowrate in this column is assumed to occur during the entire time interval for its respective step. This is conservative since valve repositioning may reduce the flowrate during the time interval.

(7) Flow out of the RWST during switchover includes allowances for both pumped flow to the RCS and containment and for backflow to the containment sump.

(8) Following the completion of this step all ECCS pumps are aligned with suction flow from the containment sump with the exception of one RHR pump due to the single failure. The containment spray pumps continue to take suction from the RWST until the RWST level indication is less than or equal to 24% and the operator initiates switchover of the Containment Spray System.

(9) Based on a Large Break LOCA in conjunction with a single failure of one of the RWST to RHR pump isolation valves (e.g., 8812A or 8812B fails to close on demand).

(10) Containment Spray valve operating times are maximum operating times of 20 seconds for HV-4782 and HV-4783 (sump isolation valves) and 30 seconds for HV-4758 and HV-4759 (RWST isolation valves).

(11) Step 4 assumes the operator is anticipating spray switchover after receipt of the empty alarm and can begin the step in 10 seconds. The minimum tank inventory for spray switchover based on 6% indicated level is 13,450 gallons.

6.4 HABITABILITY SYSTEMS

6.4.1 DESIGN BASES

6.4.1.1 Control Room Envelope

The Control Room pressurized envelope, as defined in Section 6.4.2.1, includes the Control Room and all areas adjacent to the Control Room on elevation 831' 6" of the Electrical and Control Building containing plant information and equipment that may be needed during an emergency including kitchen, sanitary facilities, and computer rooms.

Control Room design is based upon the safe occupation of the Control Room envelope during normal operation and for a period of not less than 30 days after a loss-of-coolant accident (LOCA). Habitability systems ensure that the personnel occupying the Control Room during these times will not be exposed to radiation doses exceeding 5 rem whole body gamma dose, 50 rem thyroid dose, and 50 rem beta skin dose. The allowable unprotected beta skin dose may be increased to 75 rem when special protective clothing and eye protection is used. The Control Room is designed in accordance with NRC General Design Criterion (GDC) 19 [1]. The Control Room envelope contains adequate medical supplies and the necessary kitchen and sanitary facilities to sustain plant personnel for a period of 30 days following a DBA. The necessary food and water for five plant personnel for five days will be permanently stored in the Control Room.

6.4.1.2 Radiation and Toxic Gas Protection

Control Room shielding is designed to limit the dose from external sources to a level compatible with the dose criteria given in Subsection 6.4.1.1 based on the inventories given in Table 6.4-1.

The Control Room HVAC system is designed to maintain a positive pressure with respect to the environs during normal and emergency modes of operation.

Airborne radioiodine is limited to levels compatible with the dose criteria given in Subsection 6.4.1.1, based on the radioiodine activities given in Table 6.5-6, and a Containment leak rate of 0.1- percent for the first 24 hr following an accident and one-half this value for the balance of the accident. Refer to Subsection 15.6.5.4 for an analysis of the inhalation dose to the Control Room operators. In the event of a toxic gas release, the control room may be manually isolated from the outside environment by placing the control room ventilation system in the isolation mode. For CPNPP, the probability of simultaneous occurrence of a toxic gas release and radiological release caused by a loss of coolant accident (LOCA) is assumed to be extremely low. Therefore, the event of concurrent releases is not considered in the design basis. See Section 2.2.3 for a discussion of toxic gas releases and analyses.

Airborne radioactive material in the Control Room atmosphere is controlled after an accident by the emergency recirculation filtration units and emergency pressurization filtration units. These atmosphere cleanup units are used in the event of a release of airborne radioactive material. Inlet bubble tight dampers are used in the event of a toxic gas release or fire to provide inlet isolation of either or both trains of outside air into the control room envelope. The Control Room Ventilation System is described in detail in Section 9.4. The limitations of the Control Room environment following a LOCA are listed in Table 6.4-3.

6.4.1.3 Respiratory, Eye, and Skin Protection for Emergencies

Portable self-contained breathing apparatus and protective clothing are provided in the Control Room envelope for use by the plant personnel required to leave this controlled zone during the emergency recirculation mode of operation. There will be an adequate supply of air to sustain the five-man emergency team for a six-hr period. At least one portable self-contained breathing apparatus will be provided for each member of the emergency team. Replenishment capability for the breathing apparatus is also located offsite.

Further air supplies will be replenished as needed by the emergency organization.

Breathable air bottles will be replenished by using the services of one or several commercial air suppliers in the Dallas/Fort Worth metroplex area. Empty bottles will be transported to the commercial air suppliers for replenishment and full bottles will be returned to CPNPP by commercial and/or company vehicle. Once at the site the full air bottles will be distributed to the control room by the members of the Emergency Organization.

6.4.1.4 Habitability System Operation During Emergencies

A detailed description of the Control Room Air-Conditioning System emergency modes of operation is presented in Section 9.4.

6.4.1.5 Emergency Monitors and Control Equipment

Radiation monitors used to switch the Control Room Air-Conditioning System into the emergency recirculation mode are located in the Control Room outside-air intakes. The outside-air intake monitors, located at opposite sides of the Control Building, are used to sample makeup and pressurization air flows introduced into the Control Room envelope. Ionizing smoke detectors are provided in the control room air intakes to provide alarms and indication to the operator of the presence of smoke. Upon detection of smoke in the control room, the operator may manually initiate the isolation mode of the control room ventilation system.

6.4.1.6 Fire Protection Criteria

The Fire Protection System is designed to safeguard equipment and personnel. Combustible materials are excluded as far as practical from the Control Room to lessen the possibility of a fire. The fire stops serve a dual function. Fire stops are incorporated on all cables entering the Control Room to prevent the entry of a fire originating outside the Control Room. They also form a leak boundary which limits exfiltration of air from the Control Room envelope.

Because any fire in the control panels would be very limited, due to the amount of combustible materials present, Control Room evacuation is not considered a necessity; however, remote shutdown capability is available as described in Section 7.4. Codes and guides used in the design of the Fire Protection System are given in Subsection 9.5.1.

6.4.2 SYSTEM DESIGN

6.4.2.1 Definition of Control Room Envelope

The Control Room pressurized envelope consists of the following areas where continuous or frequent operator or technical support personnel occupancy may be required during emergency operation:

Space	Elevation
East Control Room	830'-0"
West Control Room	830'-0"
Console and Control Room Unit 1	830'-0"
Console and Control Room Unit 2	830'-0"
Instrument Room Unit 1	830'-0"
Instrument Room Unit 2	830'-0"
Computer Room Unit 1	830'-0"
Computer Room Unit 2	830'-0"
Shift Manager's Office	830'-0"
Production Supervisor's Office	830'-0"
Corridor	830'-0"
Men's Toilet	830'-0"
Women's Toilet	830'-0"
Kitchen and Janitor Closet	830'-0"
Charts and Supplies Storage Room	830'-0"
Technical Support Center (Office and Corridor)	840'-6"
Offices (2)	840'-6"
Electrical Equipment Corridors (2)	840'-6"

The Control Room envelope also includes the Control Room Air Conditioning System (CRACS) mechanical equipment rooms, Trains A and B, located in the Control Building above the Control Room complex at elevation 854 ft 4 in. These rooms are pressurized and may require infrequent access by a Control Room operator during an emergency condition. The components located in the CRACS mechanical equipment rooms are described in detail in Section 9.4.

6.4.2.2 Ventilation System Design

The Control Room Air-Conditioning and Ventilation System pressurizes the control room with filtered outside air during post-LOCA operation. The Post-LOCA system is designed to control the level of airborne contamination in the Control Room atmosphere and to control the temperature and humidity for personnel safety and comfort. The flow diagrams of the system are shown on Figure 9.4-1. These diagrams include equipment, dampers, instrumentation, and flow paths for normal and emergency operation. Redundant atmosphere cleanup units (emergency filtration units) are used to remove particulate matter and other contaminants from the Control Room air. The design of emergency filtration units is in accordance with NRC Regulatory Guide 1.52 [3] except as noted in Appendix 1A(B). Each filtering unit consists of a particulate, HEPA, iodine adsorber, and HEPA filters, and a booster fan to draw the air through the unit. See Section 9.4 for design parameters and capacities of the filters and related equipment. See Figure 6.4-1 for filtration unit drawing.

The system operation is discussed in Section 9.4. The performance objectives of the Control Room Air Conditioning and Ventilation System and the associated design basis necessary to ensure habitability during and after a LOCA are given in Table 6.4-3.

Redundant emergency pressurization units are used to pressurize the Control Room envelope during emergency recirculation. Redundant inlet bubble tight dampers are installed at each control room ventilation inlet to minimize filtered inleakage into the control room during accident conditions or toxic gas releases.

The emergency pressurization units supply outside air filtered through a particulate, HEPA, iodine adsorber, and HEPA filters to the supply header of the emergency filtration units. A booster fan, demister, and heater are used to circulate the air and maintain the humidity of the incoming outside air below 70 percent. See Figure 6.4-2 for pressurization unit drawing. Seismic classifications for the system are listed by component in Appendix 17A. Layout drawings of the equipment for this area, including doors, corridors, stairwells, and shielding walls, are shown on Figures 1.2-33 and 1.2-34. A detail of the Control Room air inlet is shown on Figure 6.4-3. A description of the instrumentation and controls for the Control Room Air- Conditioning System is given in Section 9.4. Radiation detectors are provided to control ventilation system operation. The redundant train oriented radiation detectors monitor the Control Room ventilation outside-air intakes. A complete description of the emergency filtration and pressurization filter trains is given in Section 6.5.

The Control Room air-conditioning and filtration units, fans, dampers, ductwork, and associated equipment essential to the operation of the Control Room HVAC and Filtration System are located within a missile- protected structure. The Control Room HVAC and filtration air intakes and exhausts are also protected against the damaging effects of a tornado-generated missile.

6.4.2.3 Leaktightness

During the Emergency Recirculation mode, the Control Room envelope is maintained at an overpressure of 0.125-in. water gauge (wg) to prevent infiltration of unfiltered and unmonitored air from the adjacent areas. Any differential greater than 0.125-in. wg causes difficulties in opening and closing doors leading to the area served by the Control Room Air-Conditioning System. During normal operating mode, the Control Room is maintained at a nominal positive pressure of 0.125 in wg with respect to the outdoor environment to minimize dust infiltration. During this mode of operation, other areas in the Control Room envelope are maintained at a slightly positive pressure. This overpressure is maintained by modulating exhaust dampers during normal plant operations and by emergency pressurization units during accident conditions. The boundaries of the Control Room envelope consist of concrete walls and floors which exhibit low leakage characteristics.

To minimize this leakage all joints and penetrations are sealed; all doors are gasketed. All doors are designed to swing inward except the missile resistant door which is airtight and opens outward. In addition, the inlet bubble tight dampers provide assurance of minimal filtered inleakage from the standby train of control room ventilation. The maximum flow rate of the pressurization unit for emergency operation is 800 ft^3/min which is sufficient to pressurize the Control Room envelope to 0.125-in. wg as indicated in Table 6.4-4. However, leakage is expected to be less. Periodic testing of the Control Room envelope is performed to verify this value and to ensure that adequate pressurization is maintained. (See Subsection 6.4.5.) For an

analysis of the dose received by Control Room occupants in the unlikely event of a LOCA, see Subsection 15.6.5.4.

The infiltration rate when the control room is isolated will be much less than the exfiltration rate when pressurized to 0.125 inch water gauge since infiltration is due to wind loadings and much less than half the leakage paths in Table 6.4-4 are exposed to wind loadings.

6.4.2.4 Interaction With Other Zones and Pressure-Containing Equipment

The Control Room envelope is isolated and maintained pressurized during an accident involving the release of radioactive gases in surrounding zones. The Control Room Air-Conditioning System is operated in the emergency recirculation mode, with outside filtered air used to maintain Control Room pressurization. The standby train of control room ventilation is isolated via the use of redundant inlet bubble tight dampers. Doors are designed to open against positive pressure to ensure closure at all times. The use of fire extinguishers located in the Control Room envelope will not yield a hazardous concentration of toxic gas. All piping not connected or related to Control Room equipment is routed outside the pressurized boundary. In the unlikely event of a large release of toxic gas to the outside ambient or surrounding zones, the operator may line up the control room ventilation system in the isolation mode by closing the respective inlet bubble tight dampers. In addition, portable self-contained breathing apparatus are readily available for use by the plant operators.

6.4.2.5 Shielding Design

The shielding design for the Control Room is based on the requirements specified in 10 CFR Part 50, Appendix A, GDC 19. The Control Room is designed to provide radiation protection for personnel occupancy under accident conditions so that no individual will receive exposures in excess of 5-rem whole-body gamma dose, 50-rem thyroid dose and 75-rem unprotected beta skin dose (with special protective clothing and eye protection). To achieve this goal, the shielding design of the Control Room considers airborne contaminants within the Control Room and other DBA sources of radiation. Specifically, these other sources are fission products released to the reactor Containment atmosphere, airborne radioactive contaminants surrounding the Control Room, and sources of radiation caused by potentially contaminated equipment in the vicinity of the Control Room. They are considered to be dominant sources of radiation, and they are among the principal parameters for the shielding design of the Control Room.

Shield thicknesses of structural concrete provided for the Control Room are shown on Figure 12.3-14. The 2 ft-0 in. structural shielding walls surrounding the Control Room, combined with the roof and floor slabs above the Control Room, provide more radiation protection for personnel in the Control Room. In addition to shield thicknesses, distances that separate dominant radiation sources from the Control Room are included on the scaled layout and arrangement drawings of the facility in Section 1.2. Radioactive decay for each isotope of the DBA source is taken into consideration in the analysis of the dose to Control Room occupants shown in Subsection 15.6.5.4. A layout drawing of the Control Room and associated structures is presented on Figure 1.2-33.

DBA sources of radiation surrounding the Control Room and shielding related considerations are presented in Section 12.2, Section 12.3, and Subsection 15.6.5.4. A plan view drawing of the Control Room and associated structures identifying distances and shield thicknesses is shown in Figure 12.3-14.

6.4.3 SYSTEM OPERATIONAL PROCEDURES

The following modes of operation characterize the Control Room Air-Conditioning System:

1. Normal operation

2. Emergency recirculation

3. Emergency ventilation

4. Isolation (emergency recirculation without pressurization)

The Control Room Air-Conditioning System is automatically switched to the emergency recirculation mode upon receipt of signals as outlined in Section 9.4. In addition, the system may be manually switched to the emergency ventilation mode of operation by the operator from the Control Room ventilation panels. This feature enables the removal of traces of smoke that remain in the atmosphere even after the Control Room has been exhausted. The smoke exhaustion from the Control Room and the sequence of automatic events for switching modes are described in Section 9.4.

The Control Room Air-Conditioning System is switched automatically when going from normal operation to the emergency recirculation mode of operation. The sequence of events for switching to emergency recirculation is described in Section 9.4.1.2. The emergency recirculation mode is considered the optimum emergency mode. During this mode of operation, the Control Room envelope is pressurized with filtered outside air to ensure a safe environment for the operators under shutdown conditions. Switching to the emergency ventilation mode can only be done manually, and only from the emergency recirculation mode.

The Control Room Air Conditioning System can only be manually switched to the isolation mode.

The Control Room Air-Conditioning System components are laid out to allow access by Control Room personnel for manual operation of the equipment, particularly dampers. The manual operation of the equipment would be abnormal and it is unlikely that it would have to be utilized.

6.4.4 DESIGN EVALUATIONS

6.4.4.1 Radiological Protection

An evaluation of radiological exposures to plant operators in the event of a DBA is discussed in Subsection 15.6.5.4.

6.4.4.2 Toxic Gas Protection

A hazards analysis for each toxic material was performed as recommended in NRC Regulatory Guide 1.78 [4] and is presented in Section 2.2. The habitability of the Control Room envelope was evaluated to determine if a site-related accident involving a release of hazardous chemicals exceeds the toxicity limits as specified in NRC Regulatory Guide 1.78. For additional information pertaining to this evaluation, see Section 2.2.3.

The Computer Rooms for Unit 1 and Unit 2 and the Technical Support Center, which are located inside the Control Room pressure boundary, employ ten non-seismic non-safety related supplementary cooling units. These areas do not contain safety related equipment and are not needed for continuous occupancy. An analysis based on Reference 10 has been performed to demonstrate that refrigerant concentrations in these areas due to the release of the total refrigerant inventory associated with these units after a seismic event (DBE) will be within the limits specified in ANSI/ASHRAE 15-78 [10].

6.4.4.3 Evaluation of Heating, Ventilation, Air-Conditioning, and Filtration System

The HVAC and Filtration System readiness is ensured by the periodic testing program described in Section 6.4.5. Safe operation is ensured by having redundant equipment for the Control Room HVAC and Filtration System. A complete safety evaluation is given in Section 9.4.

6.4.5 TESTING AND INSPECTION

Preoperational tests are conducted on the Control Room HVAC and Filtration System to ensure that all equipment satisfies the design criteria during all modes of operation. Tests are also performed, as described in Section 9.4 and consistent with the Control Room Envelope Habitability Program in the Technical Specifications, to ensure overall system performance. The leakage tests will be conducted by closing all the access points to the Control Room.

Control Room pressure will be established by controlling the outside air intake flow of the emergency pressurization units until the design pressure is achieved. Should the outside makeup airflow through the emergency pressurization unit exceed the maximum allowable flow of approximately 800 scfm, a survey shall be conducted to locate points of excessive leakage and attempt to seal them. Tests shall be repeated as often as necessary until the above criteria are established.

Control Room pressure is measured by the permanently installed differential pressure transmitters.

The result of the Control Room leak test is considered acceptable if the emergency pressurization airflow does not exceed 800 scfm with the Control Room envelope being maintained at 0.125-in. wg.

Planned leakage tests will be performed to verify that adequate pressurization of the Control Room envelope is maintained during Emergency Recirculation mode. The Control Room envelope will be maintained at 0.125-in. wg positive pressure relative to the outside atmosphere with a maximum makeup airflow of approximately 800 scfm.

In-place testing of air cleaning components will be performed in accordance with test methods and acceptance criteria described in ANSI N510 [7].

Control Room equipment will also be tested in accordance with the methods described in ANSI N509 [6].

Testing requirements, acceptability and frequencies for ESF components are established in the Technical Specifications. The Control Room Emergency Filtration System and the Control Room

Envelope are tested as part of the Control Room Envelope Habitability Program as established in the Technical Specifications.

6.4.6 INSTRUMENTATION REQUIREMENTS

Sufficient indications are provided in the Control Room for the operator to monitor HVAC system performance. Annunciators indicate HVAC system or component malfunctions. See Sections 7.3 and 9.4 for more detailed discussions of the instrumentation.

Fire protection and alarm devices are annunciated in the Control Room as described in Subsection 9.5.1.

Ionization and radiation sensors detect smoke and unsafe levels of radiation in the incoming air. The system automatically switches to emergency recirculation if a radiation detector fails. Subsequent to automatic initiation of the emergency recirculation mode, the operator can regain manual control over the system from the Control Room ventilation panels. A Control Room vertical panel mounted selector switch allows the operator to select normal operation or emergency ventilation modes as required for the operation conditions.

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 19, Control Room.

2. NRC Regulatory Guide 1.4, Assumptions Used for Evaluating the Potential Radiological Consequences of a Loss of Coolant Accident for Pressurized Water Reactors. Revision 2, June 1974, United States Nuclear Regulatory Commission.

3. NRC Regulatory Guide 1.52, Design, Testing, and Maintenance Criteria for Engineered-Safety-Feature Atmosphere Cleanup System Air Filtration and Adsorption Units of Light-Water-Cooled Nuclear Power Plants, Revision 1, July 1976, United States Nuclear Regulatory Commission.

4. NRC Regulatory Guide 1.78, Assumptions for Evaluating the Habitability of a Nuclear Power Plant Control Room During a Postulated Hazardous Chemical Release, June 1974, United States Nuclear Regulatory Commission.

5. Deleted.

6. ANSI N509, Nuclear Power Plant Air Cleaning Units and Components.

7. ANSI N510, Field Testing of Nuclear Air Cleaning Systems.

8. Leakage Characteristics of Openings for Reactor Housing Components, NAA-SR-MEMO-5137, Atomics International, Div. of North American Aviation, Inc., June 20, 1960.

9. Deleted.

10. ANSI/ASHRAE 15-78, Safety Code for Mechanical Refrigeration.

TABLE 6.4-1
NOBLE GAS AND HALOGEN INVENTORIES RELEASED TO CONTAINMENT
AS THE RESULT OF A MAXIMUM CREDIBLE ACCIDENT

Noble Gases		Halogens	
Curies Released From Core		Curies Released From Core	
Nuclide	(100% Core Inventory)	Nuclide	(50% Core Inventory)[a]
Kr-83m	1.3×10^7		
Kr-85m	3.0×10^7		
Kr-85	7.3×10^5		
Kr-87	5.4×10^7		
Kr-88	7.7×10^7		
		I-131	5.4×10^7
		I-132	7.9×10^7
Xe-131m	7.7×10^5	I-133	1.1×10^8
Xe-133m	3.2×10^7	I-134	1.2×10^8
Xe-133	2.2×10^8	I-135	1.0×10^8
Xe-135m	4.4×10^7		
Xe-135	4.6×10^7		
Xe-138	1.8×10^8		

a) Per R.G. 1.195 half of the iodines in the core are assumed to be available for release to the containment atmosphere immediately.

TABLE 6.4-2
THIS TABLE HAS BEEN DELETED.

TABLE 6.4-3
LIMITATIONS OF CONTROL ROOM ENVIRONMENT

Ambient pressure, in. wg	+0.125[a]
Ambient temperature, FDB	75 ±5
Ambient relative humidity, percent	35 to 50
Noxious substances	Not Applicable
Maximum Radiation doses to operators:	
Whole Body Gamma Dose	5 rem
Beta Skin Dose	75 rem[b]
Thyroid Inhalation Dose	50 rem[c]

a) Except during detection of smoke or toxic gas, at which time the operator may manually initiate control room isolation mode and maintain the control room at atmospheric pressure.

b) The 75 rem limit for beta skin dose is allowed due to the use of protective clothing and eye protection. The dose limit is 50 rem if credit is not taken for protective clothing.

c) Regulatory Guide 1.195 Control Room Thyroid limit.

TABLE 6.4-4
POTENTIAL LEAK PATHS AND THEIR APPROPRIATE LEAKAGE CHARACTERISTICS[a]

Components	Description of Components	Leak Rate (ft^3/min)
Pipe penetrations	For all pipe penetration in the Control Room	Negligible
Cable penetrations	For all cable penetration in the Control Room	Negligible
Doors (assume hollow doors with metal interlocking and gasket-seal weather strip)	Total of twelve doors (Ref. Figures 1.2-33, 34, and 36)	282 (total)
Dampers	CPX-VADPMU-05, CPX-VADPMU-06	9 (total)
	CPX-VADPOU-27, CPX-VADPOU-28	
	All others	Negligible
Redundant inlet bubble tight dampers	Failure of one damper	Negligible
Tornado Blowout Panels	Ten (10) panels 24 (total)	24
Leakage Due to Ingress Egress	All normal ingress and egress to stairwell and interior rooms which serve as the equivalent of vestibules.	10
Ductwork	Uncontrolled Access Area & Office & Service Area Supply & Return ductwork inside the Control Room Pressure boundary	2
Unidentified Leakage	Fan margin	464
TOTAL		800

a) The criteria used to establish leakages is based on a pressure differential of 1/4 in.wg.as specified in NRC Regulatory Guide 1.78[4].

Note:

1. The Control Room is pressurized to 1/8 in. wg. in the Emergency Recirculation mode of operation. The leakrate will be ≤800 ft^3/min at 1/8 in. wg. positive pressure.

2. Total in-leakage is confirmed to be consistent with the Control Room Habitability Program as established in the Technical Specifications.

6.5 FISSION PRODUCT REMOVAL AND CONTROL SYSTEM

Following the postulated DBA, ESF fission-product removal and control systems are required to perform a safety-related function as described in this section.

1. GDC 19 of Appendix A to 10 CFR Part 50 and 10 CFR 100 require adequate radiation protection during access and occupancy of the Control Room and control of radioactive releases to the environment. The secondary atmosphere cleanup systems satisfying GDC 19 and 10 CFR 100 are described in Subsection 6.5.1.

   Since the Containment ventilation system does not perform safeguard functions, i.e., its operation is not required in the event of a postulated LOCA, a description of this system is not provided in this section. The details of the Containment ventilation system are provided in Section 9.4.

   Since the non-ESF units of the Primary Plant Ventilation System do not perform a nuclear safety function, a description of these units is provided in Section 9.4.

2. GDC 41, 42, and 43 of Appendix A to 10 CFR Part 50 require that Containment atmosphere cleanup systems be provided as necessary to reduce the amount of radioactive material released to the environment following a postulated DBA and that these systems be designed to permit appropriate periodic inspection and testing to ensure their integrity, capability, and operability.

   For this purpose, the CSS and the Containment spray chemical additive subsystem are included as ESF in the design of CPNPP and are described in Subsection 6.5.2.

6.5.1 ENGINEERED SAFETY FEATURE (ESF) FILTER SYSTEMS

The atmosphere cleanup systems summarized below are classified as ESF filter systems. Table 6.5-7 indicates which filters perform safety- related functions following a DBA.

	System	Function		Number of Filtration Units	Redundancy
1.	Control room ventilation system	a.	Emergency pressurization	2	100 percent
		b.	Emergency filtration	2	100 percent
2.	Primary plant ventilation ESF exhaust units		Maintenance of Auxiliary, Fuel Handling and Safeguards buildings at a negative pressure during LOCA	4	100 percent

6.5.1.1 Design Bases

As required in GDC 19 of Appendix A to 10 CFR Part 50 and 10 CFR Part 100, the ESF exhaust units are provided to reduce the amount of radioactive material released to the environment following a DBA. As required in GDC 19 of Appendix A to 10 CFR Part 50, the Control Room filtration units are provided to ensure a safe environment and to permit access and occupancy of the Control Room during and after a DBA. The ESF exhaust units are designed to maintain the Auxiliary, Fuel Handling and Safeguards Building at a slightly negative pressure during a LOCA. See Section 9.4.5 for system mode of operation of the ESF ventilation system. Each Control Room emergency pressurization unit is designed to pressurize the Control Room envelope to 0.125-in. wg overpressure based on the combined leak rates shown in Table 6.4-4.

Each emergency filtration unit is designed to handle approximately 16 percent of the required airflow supplied within the Control Room envelope. This percentage is based on an air change rate. The emergency recirculation airflow rate allows the Control Room volume to be filtered approximately once every hour. Control Room area ventilation system operation is explained in Section 9.4.1.

HEPA filters and iodine adsorbers used in these filter trains comply with the construction and efficiency requirements of NRC Regulatory Guide 1.52 (See Appendix 1A(B)). Efficiencies of the HEPA filters and iodine adsorbers are in accordance with Table 9.4-4.

6.5.1.2 System Design

For each ESF atmosphere cleanup system, the design features and fission product removal capability are in compliance with the six positions detailed in NRC Regulatory Guide 1.52 (See Appendix 1A(B)) as shown in Table 6.5-1. Each design item to which exception is taken is justified. Figure 6.5-1 illustrates a typical layout for the ESF atmosphere cleanup units.

6.5.1.3 Design Evaluation

The ESF atmosphere cleanup systems conform to the criteria established in the regulatory positions of NRC Regulatory Guide 1.52 (See Appendix 1A(B)) as shown in Table 6.5-1. Efficiencies of HEPA filters in new condition and data for the iodine adsorbers are presented in Table 9.4-4. Filters are designed to seismic Category I requirements. Redundancy of equipment and power supplies enables the systems to sustain a single, active failure without loss of function during LOCA, loss of offsite power, and normal plant operation.

The safety related ductwork was leak tested per ANSI-N509-1980. The maximum allowable leakage rates are listed in Table 4-3 of this standard. This ductwork is designated as Seismic Category I.

6.5.1.4 Tests and Inspections

Inspection and testing of the ESF atmosphere cleanup units are consistent with the inspection and test requirements outlined in NRC Regulatory Guide 1.52 (See Appendix 1A(B)). These requirements are described in Section 9.4.1.4. The criteria for keeping occupational radiation exposures as low as reasonably achievable (ALARA) during replacement of filters and adsorbers are described in Section 12.3.3.5. Equipment is both factory and in-place tested as per NRC

Regulatory Guide 1.52 (See Appendix 1A(B)). The maintenance procedures are in accordance with position 4.

6.5.1.5 Instrumentation Requirements

Each ESF atmosphere cleanup unit is provided with instrumentation as described in this section.

A pressure indicator connected across the filter train is provided to monitor the overall resistance of each unit. Abnormal differential pressure alarms in the Control Room alert the operator. Local pressure indicators are provided to monitor the resistance of each individual filter bank, while high differential pressure indicates clogged or dirty filters. Each adsorber bed has a temperature-monitoring system which actuates a high temperature alarm in the Control Room in accordance with the temperatures listed in Table 6.5-1. The actions that are subsequently taken are also described in this table.

A pressure differential switch is located across each fan with a Control Room alarm for low fan differential pressure.

It is not necessary to record or monitor filter flow rates and pressure drops from the Control Room because this action is not essential for the ESF Filtration Units to perform their safety related function. Operability assurance of the Filtration Units is maintained in accordance with the Technical Specifications. For those units provided for postulated DBA conditions, alarm annunciators in the Control Room are utilized to monitor design limits (e.g., flow, pressure drops, temperature, etc.) as well as system malfunction (e.g., damper, fan). This design conforms to the intent of NRC Regulatory Guide 1.52 (See Appendix 1A(B)) System Design Criteria, C.2.g. Design details and the system logic are shown on the drawings listed in Section 1.7.

6.5.1.6 Materials

Each ESF filter housing is of all steel construction and welded and it is also brazed or bolted, or a combination of both, in accordance with the design requirements of ORNL-NSIC-65, Design Construction and Testing of High Efficiency Air Filtration Systems for Nuclear Applications. Specific information relating to commercial name and chemical composition of the materials used is provided in Table 9.4-4. HEPA filters and prefilters are fabricated of glass fiber and adsorbers are fabricated of activated charcoal. Ductwork is of sheet metal (galvanized steel) construction. Redundant ESF atmosphere cleanup units are provided for the system outlined in Subsection 6.5.1.1 and are physically separated as shown on Figure 9.4-9 to ensure that any radiolytic or pyrolytic decomposition of the materials used in or on a particular filter system does not interfere with the safe operation of that system or any other ESF system.

6.5.2 CONTAINMENT SPRAY SYSTEMS

This section describes the chemical additive subsystem of the CSS and the spray header locations and volumetric spray coverage. A detailed description of the CSS and its energy removal function can be found in Section 6.2.2.

Containment atmosphere purification and cleanup is accomplished by the CSS. A chemical additive subsystem is provided to inject sodium hydroxide into the spray water. The sodium hydroxide maintains the pH of the spray water at an acceptable value for fission product removal from the post LOCA Containment atmosphere.

6.5.2.1 Design Bases

Following a LOCA, the release of radioactive iodine isotopes (Table 6.5-6) in the gaseous state presents a hazardous condition which is alleviated by the CSS.

The CSS and its chemical additive subsystems are used to remove post-LOCA fission products, principally elemental iodine, from the containment atmosphere. To achieve a fission-product-removal capability which ensures that offsite doses are well below the limits of 10 CFR Part 100, sodium hydroxide solution is added to the borated spray water via the chemical additive subsystem. Component sizing, layout, and the spray solution chemistry are governed by the following principal criteria:

1. Volumetric Containment Coverage with one out of two redundant spray trains in operation must be maximum.

2. The minimum spray fall height above the operating floor is equal to the elevation of the lowest ring header, which is 115 ft 9 in. The average fall height is 126 feet.

3. The system must deliver a minimum flow rate of 5800 gpm to one train of spray nozzles, despite a single failure in the CSS.

4. The system must be capable of permanent removal of iodine from the Containment atmosphere during a LOCA by absorption into the spray droplets and retention in the Containment sump.

5. The Containment spray chemical additive subsystem components are to be designed to seismic Category I requirements.

6. The radioactive iodine isotopes available for release following a LOCA are tabulated in Table 6.5-6.

7. The iodine removal coefficient is calculated in Subsection 6.5.2.3.

6.5.2.2 System Design (for Fission Product Removal)

The CSS has the dual function of removing heat as well as fission-product iodine from the containment atmosphere. Equipment descriptions and principal design parameters for those system components required for the heat removal function are delineated in Section 6.2.2.

The figures mentioned in Subsections 6.5.2.2.2, and 6.5.2.2.4 are based on Unit 1 drawings, but the representation of CSS and nozzle arrangements shown in the figures can be applied to Unit 2 because of its similar design. Specific differences in the total number of containment spray nozzles between Unit 1 and Unit 2 are discussed in Subsection 6.5.2.2.3.

6.5.2.2.1 Modes of Operation

The chemical additive subsystem of the CSS begins operation when the chemical additive tank isolation valves open and sodium hydroxide from the tanks is educted into the borated water. These isolation valves are auto-opened simultaneously with the opening of the containment spray valves. See Section 6.2.2 for details of the modes of operation of the CSS.

The sodium hydroxide is added to the borated water during the injection phase and may continue during the recirculation phase. Vacuum breakers are provided to prevent a negative pressure during chemical injection.

The chemical additive subsystem is isolated automatically to prevent air ingestion when chemical addition motor operated tank isolation valves auto-close on chemical additive tank low level.

An equilibrium sump pH condition exists when both the chemical addition tank isolation valves have closed and the CSS is in the recirculation mode. The spray pH at this point is identical to the equilibrium sump pH.

6.5.2.2.2 Major Components

The mechanical components described in this section are either additions to the CSS or part of the system which has the dual function of heat removal and fission product removal.

The Containment spray chemical additive subsystem is composed of one chemical additive tank, four chemical eductors, connecting piping, and valves. (The Containment spray pumps supply kinetic energy to the motive fluid in the eductors.) The flow diagram of the CSS, including the chemical additive subsystem, is presented on Figure 6.2.2-1. See Table 6.5-2 for component design parameters.

1. Chemical Additive Tank

   The chemical additive tank is sized to hold a sufficient quantity of 28 to 30 weight percent sodium hydroxide solution to maintain equilibrium sump water alkalinity within a design pH range of 8.25 to 10.5. The solution has a crystallization temperature of 35°F.

   The equilibrium sump water is above 8.25. (See Table 6.5-3.) An inert blanket of nitrogen gas is used to cover the sodium hydroxide solution.

2. Containment Spray Chemical Eductors

   The 28 to 30 weight percent sodium hydroxide solution is added to the spray water mainstream by means of liquid jet eductors. A portion of the Containment spray pump discharge is recirculated through the eductor, where the sodium hydroxide is drawn from the chemical additive tank and discharged into the pump suction. The eductors and piping are designed to have a pressure drop compatible with the Containment spray pump suction line in order not to affect the pump NPSH.

3. Piping and Valves

   See Section 6.2.2.

6.5.2.2.3 Containment Spray Headers and Nozzles

The CSS, including the spray headers, is divided into two redundant trains. Each train has headers and nozzles in four regions (A, B, C, and D) of the Containment. For information on the volumes covered by the Containment spray, see Subsection 6.5.2.2.4.

Region A consists of four ring headers per train located within 2 ft of the Containment dome and a minimum of 115 ft 9 in. above the operating floor. Each train in region A contains 274 nozzles for Unit 2 and a minimum 272 nozzles for Unit 1.

Ring 4 is located at a radius of 61 ft 8 in. from the centerline of the Containment with the upper ring (train A) elevation of 1022 ft 10 in. and a lower ring (train B) elevation of 1021 ft 6 in. This ring has 120 (119 for Unit 1) equally spaced nozzles per header, 60 (59 for Unit 1) of which face vertically downward. Of the remaining 60 nozzles, 20 point upward and toward the center of the Containment at a 45 degree angle, with the remaining 40 pointing horizontally toward the center of the Containment.

Ring 3 is located at a radius of 51 ft 4 in. from the Containment centerline with a upper ring (train A) elevation at 1041 ft 3 in. and a lower ring (train B) elevation of 1039 ft 11 in. This ring consists of 60 equally spaced nozzles per header, 40 of which are pointed horizontally toward the center of the Containment. The remaining 20 nozzles are pointed upward and toward the center of the Containment at a 45 degree angle. Ring 2 is located at a radius of 30 ft 3 in. from the Containment centerline with an upper ring (train A) elevation of 1058 ft 8 3/8 in. and a lower ring (train B) elevation of 1057 ft 4 3/8 in. This ring contains 48 (47 for Train A of Unit 1) equally spaced nozzles per header, 32 (31 for Train A of Unit 1) of which are pointed downward. The remaining 16 point upward and toward the center of the Containment at a 45 degree angle.

Ring 1 consists of 46 nozzles and is located at a radius of 13 ft 3 in. from the Containment centerline with an upper ring (train A) elevation of 1065 ft 2 in. and a lower ring (train B) elevation of 1063 ft 10 in. This ring contains 46 nozzles per header with 24 pointing downward and away from the center of the containment at a 45 degree angle, 18 pointing downward, and 4 pointing upward and toward the center of the Containment at a 45 degree angle.

Regions B, C, and D are large open areas below the operating floor of the Containment where if not for Containment spray, airborne fission products could collect during an accident. For additional information on spray volumes, see Subsection 6.5.2.2.4.

The following discussion of containment spray nozzles for Regions B, C and D reveals minor differences between Unit 1 and Unit 2. For Unit 2, a total of eleven spray nozzles have been omitted from Regions B, C and D due to construction material requirements. For the purposes of containment spray iodine removal and thermal effectiveness in containment analysis, these nozzles are considered to be completely obstructed. There is no significant difference between the effective (unobstructed) number of spray nozzles in Unit 1 and Unit 2 as reflected in Table 6.5-5, and there is no significant difference between the analysis results for Unit 1 and Unit 2 due to the containment spray nozzles.

On Unit 1, the two spray headers in region B contain 67 spray nozzles per header. The header on Train A has 67 spray nozzles with 29 pointing downward, 15 pointing downward at 45° below the horizontal, 18 pointing downward at 30° from vertical, 4 pointing downward at 20° from vertical and 1 pointing downward at 40° from vertical.

The header on Train B has 67 spray nozzles with 32 pointing downward, 16 pointing downward 45 degrees from the horizontal, and 19 pointing downward 60 degrees from vertical. On Unit 2, the two spray headers in region B contain 64 spray nozzles on Train A and 67 spray nozzles on Train B. The header on Train A has 30 spray nozzles pointing downward, 15 pointing downward 45 degrees below the horizontal, and 19 pointing downward 30 degrees above the horizontal.

The header on Train B has 67 nozzles with 31 pointing downward, 16 pointing downward at 45 degrees below the horizontal, 17 pointing downward at 30 degrees above the horizontal, 1 pointing downward at 60 degrees above the horizontal, 1 pointing downward at 30 degrees below the horizontal and 1 pointing downward at 36 degrees below the horizontal. For both Unit 1 and Unit 2, these headers are located at an approximate elevation of 900 ft 0 in. and spray an open volume between the secondary shield walls and the Containment liner between elevation 905 ft 9 in. and 860 ft 0 in.

On Unit 1, the two headers in region C contain 14 nozzles per header, all pointing downward and away from the center of the Containment at a 45 degree angle; except two nozzles which are at 52 degrees. On Unit 2, the two spray headers in region C contain 14 spray nozzles on Train A and 13 spray nozzles on Train B, all pointing downward and away from the center of the Containment. The header on Train A has 9 spray nozzles pointing downward 45 degrees from the horizontal, 1 spray nozzle pointing downward 49 degrees from the horizontal, 1 spray nozzle pointing downward 50 degrees from the horizontal, 2 spray nozzles pointing downward 51 degrees from the horizontal, and 1 spray nozzle pointing downward 53 degrees from the horizontal. The header on Train B has 13 spray nozzles pointing downward 45 degrees from the horizontal. For both Unit 1 and Unit 2, these headers are located at an approximate elevation of 855 ft 7 in. and spray an open volume between the secondary shield walls and the Containment liner between elevations 860 ft 0 in. and 832 ft 6 in.

On Unit 1, the two headers in region D contain 27 nozzles per train with four pointing downward and away from the center of the Containment at a 45 degree angle and the remainder pointing downward. On Unit 2, the two spray headers in region D contain 23 spray nozzles on Train A and 24 spray nozzles on Train B, all pointing downward and away from the center of the Containment. The header on Train A has 20 spray nozzles pointing downward and 3 spray nozzles pointing downward 45 degrees from the horizontal. The header on Train B has 21 spray nozzles pointing downward and 3 spray nozzles pointing downward 45 degrees from the horizontal. For both Unit 1 and Unit 2, these headers are located at an approximate elevation of 826 ft 0 in. and spray a space between the secondary shield walls and the Containment liner located between elevations 832 ft 6 in. and 808 ft 0 in.

Spray Engineering Company's Spraco-1713A nozzle has been found acceptable in experiments with Containment spray iodine removal systems. The nozzles are of one piece construction, with a 3/8 in. diameter orifice that produces a hollow cone spray pattern. The nozzle arrangement is designed to provide a maximum coverage inside the Containment. The spray header piping is designed in accordance with the ASME B&PV Code, Section III, Class 2, and the requirements of seismic Category I. The spray nozzles are designed in accordance with manufacturer's standards, which include applicable requirements of ASME B&PV Code, Section III, Class 2. For information on other equipment in the CSS, see Section 6.2.2. For information on drop size, see Section 6.2.2.3.1.

6.5.2.2.4 Containment Coverage

Figure 6.5-2 is a schematic of the Containment that shows the locations of the spray nozzles and sprayed regions, labeled A, B, C, and D.

The calculation of the spray removal coefficient is presented in Subsection 6.5.2.3

The spray system characteristics for each region inside the Containment are described in Table 6.5-5.

1. Bulk Containment Volume

   Region A is covered by 274 (minimum 272 for Unit 1) nozzles per train (Figures 6.5-2 and 6.5-4, Sheets 1 and 2). The nozzle orientation and spacing is such that the volume covered is maximized. The spray density throughout the Containment is as uniform as possible. The design of the lowest ring headers located in the Containment dome is such that the following result:

   a. The minimum fall height is 117 ft 1 in. and 115 ft 9 in. for the lowest ring header of trains A and B, respectively, considering the operating floor at 905 ft 9 in.

   b. The area covered on the operating floor by a nozzle spraying vertically downward is approximately tangent to the Containment wall to minimize spraying of the walls and also to avoid a large unsprayed annulus.

      The ring headers are located less than two ft from the Containment liner to facilitate the piping support design. Each Containment spray train is provided with nozzles spraying horizontally, vertically downward, upward at 45 degrees, and downward at 45 degrees.

      The spray nozzle layout at elevation 905 ft 9 in. is shown on Figures 6.5-2 and 6.5-4, Sheet 3. By using nozzles spraying upward at 45 degrees, the zone between the ring headers and the Containment liner in the dome is covered by spray.

2. Volume Underneath Operating Floor

   Regions B, C, and D (see Figure 6.5-4) (Sheets 3, 4 and 5) are partially covered by spray. The sprayed volume for each region is described in Table 6.5-5.

3. Containment Free Volumes

   The total Containment free volume is 3.031×10^6 ft^3. This volume consists of the following as shown on Figure 6.5-2:

   a. Region A

      The total volume of region A is 2.309×10^6 ft^3 of which 1.665×10^6 ft^3 is covered by spray (72.1 percent), includes the following volumes:

      1. Above operating floor 905 ft 9 in.

      2. Above refueling cavity between 905 ft 9 in. and 860 ft 0 in.

      3. Refueling cavity

4. Fuel storage area

5. Reactor vessel head storage area

6. Steam generator compartments

b. Region B

This volume is 0.168 x 10^6 ft^3, of which 0.035 x 10^6 ft^3 (20.9 percent) are covered by spray. The volume is located between 905 ft 9 in. and 860 ft 0 in. (between the secondary shield wall and the Containment wall).

c. Region C

This volume is 0.073 x 10^6 ft^3, of which 0.0026 x 10^6 ft^3 (3.6 percent) are covered by spray. The volume is located between 860 ft 0 in. and 832 ft 6 in. (between the secondary shield wall and the Containment wall).

d. Region D

This volume is 0.125 x 10^6 ft^3, of which 0.0034 x 10^6 ft^3 (2.7 percent) are covered by spray. The volume is located between 832 ft 6 in. and 808 ft 0 in. (between the secondary shield wall and the Containment wall).

e. Region E

All sub-volumes included in the Containment total free volume and not calculated to be part of Regions A-D are combined together in Region E as "unsprayed volume". These sub-volumes are separate from each other, but linked to Regions A-D by flow paths which permit varying degrees of convective mixing. The sub-volumes which are less open to convective mixing, such as the reactor cavity, are separated from the Containment liner (and potential leakage paths) by the six inch radial gap which is open to convective mixing. Region E has a total volume of 0.356 x 10^6 ft^3 and includes the following compartments:

1. Cavity beneath reactor vessel

2. Elevator

3. Rod position indication room (860 ft 0 in.)

4. Pressurizer relief tank compartment

5. Pressurizer compartment

6. In-core instrumentation room (849 ft 0 in.)

7. Seal table room (832 ft 6 in.)

8. Neutron flux detector operating room (808 ft 0 in.)

9. Heat exchanger compartments

10. Miscellaneous passageways

The effective spray volume consists of the sprayed volumes in regions A, B, C, and D.

The total sprayed volume from all sprayed regions is 1.706×10^6 ft^3. This volume represents 56.3 percent of the Containment free volume.

Regions A, B, C and D are linked to each other and to the various sub-regions which comprise region E via numerous air flowpaths (e.g. six inch radial gap between the concrete floors and the inner wall of the containment building, gated doors, floor grating, etc.). These flowpaths are sufficient to allow an assumption that convective mixing occurs between the sprayed and unsprayed volumes at a rate of two turnovers per hour.

6.5.2.3 Design Evaluation

1. Iodine Absorption

The Containment spray chemical additive subsystem incorporates provisions to enhance absorption of radioactive iodine from the Containment atmosphere by the Containment spray droplets. A favorable alkalinity level is established by the addition of sodium hydroxide solution to the borated water. This alkalinity increases the solubility of iodine to the extent that the rate of iodine absorption is mainly limited by the mass transfer rate through the gas film surrounding the spray water droplets. Furthermore, the alkalinity level is also favorable in preventing the dissolved iodine from escaping from the Containment sump water.

Radioiodine, in its various forms, is the fission product of primary concern in the evaluation of a LOCA. The major benefit of the Containment spray is its capacity to absorb elemental iodine from the Containment atmosphere. To enhance the iodine-absorption capacity of the spray, the spray solution is adjusted to an alkaline pH which promotes iodine hydrolysis to nonvolatile forms.

According to the known behavior of elemental iodine in highly dilute solutions, the hydrolysis reaction

$$I_2 + OH \rightarrow HIO + I$$

is nearly completed at pH>8. The iodine form is highly soluble, and HIO readily undergoes additional reactions to form iodate.

The overall reaction is

$$3I_2 + 3H_2O \rightarrow 5HI + HIO_3$$

2. Solution pH

Following a design basis accident, the spray pH may be as low as 4.5 or as high as 12.5 for a short duration under bounding single failure scenarios. The sump and therefore spray attains an equilibrium pH value ranging between 8.25 to 9.2. The parameter variations used to calculate the equilibrium pH values are given in Table 6.5-3. However, long term sump pH valves used for iodine retention are not impacted by this short term out of range pH.

To determine the Containment sump solution pH, all sources of borated water such as the RCS accumulators, the RWST, and system piping are considered. The mixing of the Containment spray solution and the spilled water is considered complete in the Containment sumps.

3. Containment Spray Iodine Removal Coefficient

The spray effectiveness evaluation was based on the experimental summaries and modeling presented in WASH 1329, A Review of Mathematical Models for Predicting Spray Removal of Fission Products in Reactor Containment Vessels [1], as recommended in Standard Review Plan 6.5.2.

The removal of radioiodine is considered to be a first order with respect to iodine concentration and is mathematically described as follows:

$$\frac{dC}{dt} = -\lambda C \qquad (6.5\text{-}1)$$

which integrates to

$$C = C_o e^{-\lambda t} \qquad (6.5\text{-}2)$$

where

C = iodine concentration in the containment atmosphere

λ = iodine removal coefficient

C_o = initial iodine concentration in the containment atmosphere

t = time after the spray operation

Pursuant to WASH 1329, the removal coefficient of elemental iodine by sprays with a sodium hydroxide composition has been evaluated.

The elemental iodine removal coefficient is related to the spray parameter as follows:

$$\lambda = \frac{6VdhF}{VdU} \qquad (6.5\text{-}3)$$

where

V_d = overall deposition velocity

h = spray drop fall height

F = spray flow rate

V = Containment sprayed volume

U = spray drop terminal velocity

d = drop diameter

The deposition velocity V is evaluated from

$$\frac{1}{Vd} = \frac{1}{Kg} + \frac{1}{HK_L} \qquad (6.5\text{-}4)$$

where

V_d = overall deposition velocity

k_g = gas film mass transfer coefficient

k_L = liquid film mass transfer coefficient

H = iodine partition coefficient

The liquid film mass transfer coefficient is predicted by:

$$k_L = \frac{2\pi D_L}{3d} \qquad (6.5\text{-}5)$$

where

D_L = diffusivity of iodine in water

d = drop diameter

The gas film mass transfer coefficient is predicted by:

$$Kg = \frac{D}{d}(2.0 + 0.6R_e^{0.5}S_c^{0.33}) \qquad (6.5\text{-}6)$$

where

D = diffusivity of iodine in gas phase

R_e = Reynolds number

S_c = Schmidt number

The mathematical model described in Equations (6.5-1) through (6.5-6) is evaluated numerically using input parameters which are chosen to yield conservative estimates of washout rates. The washout rate predicted from Equation (6.5-3) is reduced by numerical factors which account for physical factors not directly included in the model. The parameters, related factors, and assumptions are described in the following paragraphs:

a. The removal mechanisms are assumed to be first order behavior which required the removal process to be independent of time and iodine concentration. Physical properties such as density, viscosity, diffusivity, as well as the partition coefficient H, must be independent of iodine concentration at the very low iodine concentration involved in Containment sprays. However, the iodine partition coefficient is not. Conservatively, a small value of 5000 is chosen for H as recommended in WASH 1329 [1]. To be independent of time, there must be no approach to an equilibrium between the iodine concentration in the containment atmosphere and in the spray solution. This assumption is conservative because of the assumption that the elemental iodine removal effectiveness of the spray system will cease after a maximum decontamination factor for 100 for the first two hr in the containment atmosphere has been achieved.

b. The radioiodine is conservatively assumed to be instantaneously released to the Containment atmosphere at the time of the accident.

c. The gas space in the sprayed portion of each region is perfectly mixed so that the drops are exposed to the same iodine concentration through its entire fall height.

d. Drops are assumed to fall vertically at their terminal velocity.

e. The drop fall height assumed for each region is the minimum fall height for the region.

f. Steam condensation effects are neglected. Based on an analysis contained in WASH 1329, steam condensation on the drops results in a negligible increase in washout. For the Spraco 1713A Nozzle, the effective drop diameter was conservatively assumed to be 1500 microns.

g. Impingement of the spray drops on the Containment wall will lower the washout by a maximum of ten percent, and any buildup of fine drops has a negligible effect on washout.

h. Calculation is done at a conservative temperature of 100°C.

i. D, R_e, S_c, and U are calculated using the methodology on ORNL-TM-1911, Removal Elemental Iodine from Steam - Air Atmosphere by Reactive Spray [2], and D_L is computed using the methodology of Reference [3].

The iodine removal coefficient is evaluated in accordance with the CSS characteristics delineated in Table 6.5-5.

The spray washout model for particulate iodine is represented in equation form as follows:

$$\lambda\rho = \frac{3hEF}{2dV} \qquad (6.5\text{-}7)$$

where

λp = particulate iodine removal coefficient

h = drop fall height

E = total collection efficiency for a single drop

F = spray flow rate

d = drop diameter

V = containment spray volume

This model embodies assumptions equivalent to most assumptions applied to the washout of elemental iodine. The drop collection efficiency is modeled using a value of 10/m (this value is reduced to 1.0/m after the airborne activity is reduced by a factor of 50). The particulate iodine removal coefficient is listed in Table 6.5-5.

For each region calculated, elemental iodine removal coefficients are above 10 hr^{-1}, respectively. The removal coefficients used in LOCA offsite dose calculation (see Section 15.6.5.3) have been limited to a maximum value of 10 hr^{-1} for elemental iodine [1]. The most conservative particulate iodine removal coefficient 11.4 hr^{-1} is also taken into account on the LOCA offsite dose calculation until the inventory is reduced by a factor of 50, after which a value of 1.14 hr^{-1} is used.

4. Single Failure Analysis

The containment spray chemical additive subsystem is designed to function reliably in the event of a LOCA. Redundancy requirements are met by providing two electrically

separated motor-operated chemical additive tank stop valves in parallel such that, in case of single active failure, there still remains a full 100-percent capacity to reduce the elemental iodine concentration inside the Containment. In addition, an air-operated normally open valve is provided in series with each of the motor-operated valves to ensure that the line may be closed when the chemical additive tank is empty and thus prevent air from being entrained in the suction of the containment spray pumps. There is no credit for the air-operated valve (passive, fails open) in the pH analysis. A single failure analysis is presented in Table 6.5-4 for the chemical additive subsystem.

Reproductibility of subsystem design parameters is ensured by using performance tested chemical eductors and spray nozzles, as well as constant speed containment spray pumps. Furthermore, the only moving parts in the system are those associated with the pumps, valves, and motors. System component parameters are verified by testing.

6.5.2.4 Tests and Inspections

1. Preoperational Tests

Component testing is performed on the motor-operated stop valves at the outlet of the chemical additive tank because these are the only active components of the Containment spray chemical additive subsystem.

Testing is performed on the Containment spray chemical additive subsystem in conjunction with the testing of the CSS as described in Section 6.2.2.4. A test signal simulating the Containment spray actuation signal is artificially initiated to demonstrate the operation of the spray system up to the isolation valves on the pump discharge lines. The isolation valves are closed for the test and are checked separately. The functional test of the onsite power system described in Chapter 8 demonstrates proper transfer to the emergency diesel generator power source in the event of a loss of offsite power. The chemical eductors are tested in the manufacturer's shop with a fluid comparable to the chemical additive solution. Preoperational testing is conducted using water as a test fluid. The Chemical Additive Tank (CAT) is aligned to the eductors and acceptable flow is demonstrated for each eductor over the range of the CAT levels.

For evaluation of test data, equivalent chemical solution and water flows are calculated accounting for physical differences (e.g., density, viscosity)

2. Inservice Surveillance

Inservice surveillance of the chemical eductors and chemical additive tank and associated valves is performed in a manner similar to preoperational testing as discussed above, except as described below.

Routine testing of the chemical additive system components is performed periodically, during shutdown and refueling operations.

The test line from the RWST to the chemical additive discharge line, upstream from the chemical additive flow measuring element, is opened, along with the full flow test line from the discharge of the Containment spray pumps which runs back to the RWST. The Containment spray pumps are started from the Control Room. Borated water flows from

the RWST through the spray additive flowmeters into the suction of the eductors. The fluid then flows through the Containment spray pump test line.

A pressure gauge located upstream from the eductor and flow throttling in the test line from the RWST are used to simulate the suction head from the chemical additive tank.

The chemical additive tank isolation valves are operated periodically for testing, during shutdown and refueling operations. The contents of the tank are periodically sampled to determine that the required concentration is maintained.

During these tests, the equipment is visually inspected for leaks. Leaking seals, packing, or flanges are corrected to eliminate the leak. Valves are operated and inspected after any maintenance to ensure proper operation. The inservice surveillance testing of the onsite power system discussed in Section 8.3 demonstrates proper transfer to the emergency diesel generator power source in the event of a loss of offsite power.

6.5.2.5 Instrumentation Requirements

The logic for initiation of the CSS operation is presented in Sections 7.2 and 7.3 and on Figure 7.2-1. The following describes the instrumentation which is employed for monitoring the chemical additive subsystem during normal plant operation and during postaccident operation. All alarms are annunciated in the Control Room.

1. Containment Spray Chemical Additive Tank Pressure

   A locally mounted indicator on the nitrogen line provides the means to continuously monitor chemical additive tank pressure. High and low pressure are alarmed in the Control Room from a pressure switch on the nitrogen line.

2. Chemical Additive Flow and Pressure

   A flow element is located in the discharge line from the chemical additive tank. Readout is local to provide flow indication during flow testing described in 6.5.2.4, above.

   A low flow alarm is provided in the Control Room. Pressure indication is provided in each chemical eductor suction line.

3. Chemical Additive Tank Level

   Two level switches automatically close the motor operated tank isolation valves when low level is reached. If one isolation valve does not operate and tanks continue to empty, a second air operated valve closes at a lower level to isolate the tank. A level transmitter provides indication and low level alarm in the Control Room alerting the operator to check the tank isolation valves. As noted in the single failure analysis, above, the motor operated valves perform the safety function.

6.5.2.6 Materials

The materials used in the CSS components are listed in Table 6.2-4.

6.5.3 FISSION PRODUCT CONTROL SYSTEMS

6.5.3.1 Primary Containment

The primary Containment is equipped with a Containment Spray System. The Containment Spray System is designed primarily to remove heat generated within the primary containment following a DBA. The Containment Spray System, which also serves as a fission product control system, is described in detail in Section 6.5.2.

The Containment Spray System is the only primary containment fission product control system designed to function after a DBA.

6.5.3.2 Secondary Containments

There are no secondary containments installed at the CPNPP. The primary containment is the sole means relied upon to contain radioactive material following an accident.

6.5.4 ICE CONDENSER AS A FISSION PRODUCT CLEANUP SYSTEM

Ice condensers are not installed at the CPNPP.

REFERENCES

1. Postma, A. K., and Pasedag, W. F., WASH 1329, A Review of Mathematical Models for Predicting Spray Removal of Fission Products in Reactor Containment Vessels, June 1974.

2. Parshy, Jr., L. F., ORNL-TM-1911, Removal of Elemental Iodine from Steam - Air Atmosphere by Reactive Sprays, October 1967.

3. Griffiths, V., UKAEA Health and Safety Branch Report AHSE (S)-R- 45, The Removal of Iodine from the Atmosphere by Sprays, 1963.

TABLE 6.5-1
ANALYSIS OF ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEMS WITH RESPECT TO EACH POSITION OF NRC REGULATORY GUIDE 1.52, Rev. 1

(Sheet 1 of 5)

Criteria	Control Room HVAC System 1. Emergency Pressurization Atmospheric Cleanup System [a] 2. Emergency Filtration Atmospheric Cleanup System [a]	Primary Plant Exhaust Atmoshperic Cleanup System[a] (ESF) [b]
Regulatory Position 1: Environmental Design Criteria		
a. Each ESF atmoshpere cleanup system is based on conditions resulting from DBA.	Designed to limit dose resulting from DBA within limits of GDC 19	System is designed to contain radioactive particulates and radioactive iodine resulting from a fuel accident or ESF component leakage.
b. System design is based on 30-day integrated radiation dose.	Yes; filters are designed for a 30-day integrated dose.	Yes
c. Adsorber design is based on iodine concentration.	Yes	Yes
d. Compatibility of atmosphere cleanup system with other ESFs	Yes	Yes
e. Components of system designed for both the lowest and highest outdoor temperatures	Yes	Yes
Regulatory Position 2: System Design Criteria		
a. Redundancy of atmosphere cleanup system if designed for mitigation of accident doses	Two filter trains, heater, mist eliminators, particulate, HEPA filters, fan, and housing are the components included in the design of each unit, with redundancy in active and passive components. Heater and mist eliminator are not included on emergency filtration units.	Two filter trains, heater mist eliminator, HEPA and carbon filters, fan, and the components included in the design of each unit, with redundancy in active and passive components.
b. Physical separation of redundant atmosphere cleanup systems	Yes	Yes
c. Atmosphere cleanup system designated seismic Category I to prevent release of fission products	Yes	Yes
d. Atmosphere cleanup system pressure surge protection	NA	NA

TABLE 6.5-1
ANALYSIS OF ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEMS WITH RESPECT TO EACH POSITION OF NRC REGULATORY GUIDE 1.52, Rev. 1

(Sheet 2 of 5)

Criteria		Control Room HVAC System 1. Emergency Pressurization Atmospheric Cleanup System [a] 2. Emergency Filtration Atmospheric Cleanup System [a]	Primary Plant Exhaust Atmoshperic Cleanup System[a] (ESF) [b]
e.	Atmoshpere cleanup system construction materials; effective performance if exposed to radiation	Yes	Yes
f.	Maximum and (required) volumetric airflow rate per atmosphere cleanup system train, respectively**	Filtration units: 8000 ft^3/min (8000 ft^3/min) Pressurization units: 1000 ft^3/min (800 ft^3/min)	15,000 ft^3/min (15,000 ft^3/min)
g.	Atmosphere cleanup system instrumentation provided	Yes[c][d]	Yes[c][d]
h.	Electrical distribution and power supply conforming to IEEE Standards	Yes	Yes
i.	Radiation protection for workers in order to perform maintenance	Removal of individual components and ensuring that exposure to operating personnel is ALARA.	Removal of individual components and ensuring that exposure to operating personnel is ALARA.
j.	Minimization of meteorological effects on outdoor air intakes	Yes	Yes
k.	Atmosphere cleanup system housing and ductwork maximum total leak rate limitations	In accordance with ANSI N-509	In accordance with ANSI N-509.

TABLE 6.5-1
ANALYSIS OF ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEMS WITH RESPECT TO EACH POSITION OF NRC REGULATORY GUIDE 1.52, Rev. 1

(Sheet 3 of 5)

Criteria		Control Room HVAC System 1. Emergency Pressurization Atmospheric Cleanup System [a] 2. Emergency Filtration Atmospheric Cleanup System [a]	Primary Plant Exhaust Atmoshperic Cleanup System[a] (ESF) [b]
Regulatory Position 3: Component Design Criteria and Qualification Testing			
a.	Demister performance and qualification requirements	Demisters[e] are provided for the pressurization units.	A loss of fuel[e] pool cooling may result in relative humidity to 100 percent. Demister is necessary.
b.	Effective operating condition on adsorption units	Yes; pressurization units equipped with heaters. Emergency filtration unit relative humidity controlled by Control Room air conditioning system.	Yes; units are equipped with heaters.
c.	Requirements of prefilter materials subjected to radiation	Materials used in prefilters withstand the radiation levels and environmental conditions prevalent during the postulated DBA, based on the use of Cambridge Model 3 CP-90-24-24-12 or equal.	Materials used in prefilters withstand the radiation levels and evironmental conditions prevalent during the posulated DBA, based on the use of Cambridge Model 3 CP-90-24-24-12 or equal.
d.	HEPA filter requirements and standards performance	Yes; based on the use of Flanders 1500-cfm filter model number 7083-NL or equal.	Yes; based on the use of Flanders 1500-cfm filter model number 7083-NL or equal.
e.	Filter and adsorber mounting frames (design and construction)	In accordance with ANSI N-509	In accordance with ANSI N-509
f.	Filter and adsorber bank arrangement recommendations	In accordance with ANSI N-509[d]	In accordance with ANSI N-509[d]
g.	System filter housings design and construction	In accordance with ANSI N-509[f]	In accordance with ANSI N-509[f]
h.	Water drain recommendations	In accordance with ORNL-NSIC-65[g]	In accordance with ORNL-NSCI-65[g][h]
i.	Removal of gaseous iodine by absorber (carbon) material	Carbon efficiency in accordance with Table 5-1, N509-1976	Carbon efficiency in accordance with Table 5-1, N509-1976
j.	Pleated-bed absorbent canister design recommendations	Yes	Yes

TABLE 6.5-1
ANALYSIS OF ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEMS WITH RESPECT TO EACH POSITION OF NRC REGULATORY GUIDE 1.52, Rev. 1

(Sheet 4 of 5)

Criteria		Control Room HVAC System 1. Emergency Pressurization Atmospheric Cleanup System [a] 2. Emergency Filtration Atmospheric Cleanup System [a]	Primary Plant Exhaust Atmoshperic Cleanup System[a] (ESF) [b]
k.	Fire prevention in adsorber	Ignition >600°F; desorption >300°F; isolation of the affected unit prior to desorption, water spray to inhibit adsorber fires	Ignition >600°F; desorption >300°F; isolation of the affected unit prior to desorption, water spray to inhibit adsorber fires
l.	System fans provided with sufficient capacity and pressure	Yes	Yes
m.	Atmosphere cleanup system fan or blower operates under the environmental conditions postulated.	Yes	Yes
n.	Ductwork designed in accordance with recommendations	Yes; meets requirements of ANSI N-509[d]	Yes; meets requirements of ANSI N-509[d]
o.	System contains a minimum of ledges, protrusions, crevices, and similar items which impede personnel or create a hazard.	Yes	Yes
Regulatory Position 4: Maintenance			
a.	Personnel safety and ready removal of the elements	Yes	Yes
b.	Easy access of components	Yes	Yes
c.	Definite mounting frame separation distances	Yes[i]	Yes[i]
d.	Permanent test probes with external connections	Test Ports are provided externally[d]	Test Ports are provided externally[d]
e.	Periodic operation of standby atmosphere cleanup system	Yes; cyclic use of components	Yes; cyclic use of components

TABLE 6.5-1
ANALYSIS OF ENGINEERED SAFETY FEATURE ATMOSPHERE CLEANUP SYSTEMS WITH RESPECT TO EACH POSITION OF NRC REGULATORY GUIDE 1.52, Rev. 1

(Sheet 5 of 5)

Criteria	Control Room HVAC System 1. Emergency Pressurization Atmospheric Cleanup System [a] 2. Emergency Filtration Atmospheric Cleanup System [a]	Primary Plant Exhaust Atmoshperic Cleanup System[a] (ESF) [b]
f. Atmosphere cleanup system components installed after active construction	Yes	Yes
Regulatory Position 5: In-Place Testing Criteria[j]		
a. In-place testing of atmosphere cleanup system	Acceptance tests and periodic tests during plant operation in accordance with ANSI-N510, 1980	Acceptance tests and periodic tests during operation in accordance with ANSI-N510 1980
b. Testing the airflow distribution to the filters (HFPA and adsorbers)	Acceptance tests and periodic tests during plant operation in accordance with ANSI-N510, 1980. For the control room pressurization unit, this is not necessary since there is only one HEPA filter.	Acceptance tests and periodic tests during operation in accordance with ANSI-N510, 1980
c. The in-place testing of HEPA filters conforms to ANSI standards or is replaced.	Acceptance tests and periodic tests during plant operation in accordance with ANSI-N510, 1980	Acceptance tests and periodic tests during plant operation in accordance with ANSI-N510, 1980
d. The adsorbers are leak-tested.	Acceptance tests and periodic tests during plant operation in accordance with ANSI-N510, 1980	Acceptance tests and periodic tests during operation in accordance with ANSI-N510, 1980
Regulatory Position 6: Laboratory Testing Criteria for Activated Carbon[f][j]		
a. If the activated carbon adsorber meets the regulatory requirements, the adsorber section is assigned the decontamination efficiencies. If not, the carbon is not used in ESF adsorbers.	Testing of samples to determine efficiencies (in accordance with ASTM D3803-1989)	Testing of samples to determine efficiencies (in accordance with ASTM D3803-1989)
b. The efficiency of the activated carbon adsorber section is determined by laboratory testing of representative samples of the activated carbon exposed simultaneously to the same service conditions as the adsorber section.	Yes; see Subsection 6.5.1.4. The methyl iodide penetration allowable for the control room pressurization and emergency filtration units is 0.5% by using safety factor 2 and 99% efficiency of the filter units.	Yes; see Subsection 6.5.1.4. The methyl iodide penetration allowable for the primary plant ventilation filtration units is 2.5% by using safety factor 2 and 95% efficiency of the filter units.

a) Components of Atmospheric Cleanup Systems are as described in Regulatory Guide 1.52.

b) Features of the modular exhaust filtration system used as follows: These units are connected in parallel. Four units of the modular system are considered as ESF units.

c) It is not necessary to record or monitor filter flow rates and pressure drops from the Control Room because this action is not essential for the ESF Filtration Units to perform their safety related function. Operability assurance of the Filtration Units is maintained in accordance with the Technical Specifications. For those units provided for postulated DBA conditions, alarm annunciators in the Control Room are utilized to monitor design limits (e.g., flow, pressure drops, temperature, etc.) as well as system malfunction (e.g., damper, fan).

d) See Appendix 1A(B) for additional information.

e) Demisters are designed in accordance with the recommendations of Mine Safety Analysis Research 71-45 and meet Underwriters; Laboratories, Inc., Class I requirements.

f) In lieu of the standards and their revisions recommended by Regulatory Guide 1.52, ANSI N509-1980 has been used. In-place testing of the HEPA filter banks and adsorber will be required following any significant painting, fire or chemical release. This design conforms to the intent of NRC Regulatory Guide 1.52, Revision 2 (03/78). See Appendix 1A(B).

g) Check valves have been utilized on some drains in lieu of seals or traps. See Appendix 1A(B).

h) The drains have not been provided with seals, traps or check valves in the demister compartments. Any minimal airflow through the demister drains will be filtered prior to discharge through the stack. See Appendix 1A(B).

i) There are cases where the minimum spacing requirements are not met. See Appendix 1A(B) for the discussion on Regulatory Guide 1.52.

j) In lieu of the standards and their revisions recommended Regulatory Guide 1.52, ANSI N510-1980 has been used. In-place testing of the HEPA filter banks and adsorber will be required following any significant painting, fire or chemical release. This design conforms to the intent of NRC Regulatory Guide 1.52, Revision 2 (03/78). See Appendix 1A(B).

TABLE 6.5-2
CONTAINMENT SPRAY CHEMICAL ADDITIVE SUBSYSTEM COMPONENT
DESIGN PARAMETERS

(Sheet 1 of 2)

1. Chemical Additive Tank	
Quantity	One
Volume (empty), gal	5500
Volume of contents, gal	4500
Contents	Sodium hydroxide
Concentration, % by weight	30
Design pressure, psig	15
Design temperature, F	200
2. Chemical Eductors	
Quantity	Four
Motive Fluid	Borated water
Flow rate, gpm	100
Concentration, ppm	2400
Design pressure, psig	325
Design temperature, F	300
Suction Fluid	Sodium hydroxide solution
Flow rate, gpm	45
Concentration, % by weight	30
Design pressure	Hydraulic head
Design temperature	ambient
Specific gravity	1.34
Viscosity, cP	10

TABLE 6.5-2
CONTAINMENT SPRAY CHEMICAL ADDITIVE SUBSYSTEM COMPONENT
DESIGN PARAMETERS

(Sheet 2 of 2)

3. Piping and Valves	
Chemical Supply Piping	
Design Pressure, psig	150
Design temperature	200
Eductor Piping	
Design pressure, psig	325
Design temperature, F	300

TABLE 6.5-3
PARAMETER RANGES USED FOR CALCULATION OF EQUILIBRIUM SUMP SOLUTION pH

	Minimum pH 8.25	Maximum pH 9.7
RWST Volume (gal)	506000 (Approx.)	316000 (Approx.)
Boron Concentration in RWST (ppm)	2600	2400
Reactor Coolant System Liquid Volume (gal)	88233	0
Boron Concentration in RCS (ppm)	1850	N/A
Safety Injection Accumulator Volume (gal)	26388	24476
Volume of Fluid in Chemical Additive Tank (CAT) (gal)	4900	5314
NaOH Concentration in Chemical Additive Tank (WT%)	28	30
Single active failure postulated	None	Failure of CAT isolation valve to close at Low Level

TABLE 6.5-4
SINGLE FAILURE ANALYSIS CONTAINMENT SPRAY CHEMICAL ADDITIVE SUBSYSTEM

Component	Malfunction	Remarks
Motor-operated chemical additive tank stop valve	Valve fails to open	Two valves are provided; operation of one is required.
	Valve fails to close	Two valves are provided; operation of one is required.
Chemical eductor	Eductor orifice is clogged.	Four eductors are provided (one pair for each train); one pair is required to operate.

TABLE 6.5-5
CONTAINMENT SPRAY SYSTEM CHARACTERISTICS

	Regions					
	A	B	C	D	E	Total
Total volume, ft^3	2,309,461	167,645	72,686	124,910	356,333	3,031,035
Sprayed containment volumes, ft^3	1,664,901	35,040	2,640	3,364	0.0	1,705,945
Unsprayed containment volumes ft^3	644,560	132,605	70,046	121,546	356,333	1,325,090
Spray drops fall height, ft	126[a]	38	23	20	-	-
Number of nozzles per train [b]	274 (272-Unit 1 train A) (273-Unit 1 train B)	67 [36][c] (64-Unit 2 train A) (67-Unit 2 train B)	14 [5][c] (14-Unit 2 train A) (13-Unit 2 train B)	27 [10][c] (23-Unit 2 train A) (24-Unit 2 train B)	-	382 (380-Unit 1 train A) (381-Unit 1 train B) (375-Unit 2 train A) (378-Unit 2 train B)
Elemental iodine removal coefficient, hr^{-1}[d]	60.4	106.9	119.4	162.9		
Particulate iodine removal coefficient, hr^{-1}	11.4	38.7	63.2	76.3	-	-

a) Average value

b) 15.2 gpm/nozzle

c) Unobstructed spray nozzle (train A)

d) Maximum X_e allowed to be used in analysis is 10 hr^{-1}.

TABLE 6.5-6
RADIOACTIVE IODINE ISOTOPES AVAILABLE FOR RELEASE TO OUTSIDE ATMOSPHERE FROM THE CONTAINMENT FOLLOWING A LOCA

Isotope	Activity[a] (Ci) ($x10^7$)
I-131	2.7
I-132	4.0
I-133	5.5
I-134	6.1
I-135	5.2

a) Based on NRC Regulatory Guide 1.195 assumptions, half of the iodine activity released from the core to the containment is available for release to the outside atmosphere.

TABLE 6.5-7
ESF FILTRATION UNITS EMPLOYED DURING DESIGN BASIS ACCIDENTS

	Design Basis Accident	Filtration Units Employed	Reference Section
1.	Loss-of-coolant accident	Control Room Emergency pressurization units	15.6.5.4(4)(b)
		Emergency filtration units	15.6.5.4(4)(b)
		Primary plant exhaust unit	15.6.5.4(2)(f)
2.	Waste gas system failure	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.7.1
3.	Steam generator tube rupture	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.6.3
4.	Fuel handling accident	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.7.4
5.	Control rod ejection accident	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.4.8
6.	CVCS letdown line break	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.6.2
7.	Main Steam line	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.1.5
8.	Liquid waste tank failure	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.7.2
9.	Locked Rotor	Control Room Emergency Pressurization Unit and Emergency Filtration Units	15.3.3

6.6 INSERVICE INSPECTION OF ASME CODE CLASS 2 & 3 COMPONENTS

6.6.1 COMPONENTS SUBJECT TO EXAMINATION

Fluid system components for CPNPP that are important to nuclear safety have been classified by Vistra OpCo as Safety Class 1, 2, and 3 and correspond to the ASME Boiler and Pressure Vessel Code, Section III, Classes 1, 2, and 3. The inservice inspection (ISI) program for Class 1 components is contained in Section 5.2.4. Class 2 components and systems are examined in accordance with Table IWC-2500-1, of ASME Section XI. Class 3 components and systems are examined in accordance with Table IWD-2500-1 of ASME Section XI.

During the preservice inspection, Class 1, 2 and 3 systems were inspected to the 1980 Edition for Unit 1 and to the 1983 Edition with Code Case N-408 for Unit 2. Details of the preservice inspection, including relief requests, are included in the Preservice Inspection Plan for each unit.

The Inservice Inspection Program for Class 1, 2, and 3 systems and component supports is written to Subsections IWB, IWC, IWD, and IWF of ASME Section XI (Edition and Addenda as required by 10CFR 50.55a).

6.6.2 ACCESSIBILITY

Class 2 and 3 components and systems have been designed and arranged for accessibility to the extent practical in order to conduct the ISI program in accordance with Articles IWC-2000 and IWD-2000 at the frequency specified by Subarticles IWC-2400 and IWD-2400.

6.6.3 EXAMINATION TECHNIQUES AND PROCEDURES

Class 2 systems and components are inservice inspected by volumetric, surface and visual examination techniques to meet the requirements of Article IWC-2000, of ASME Section XI. Ultrasonic Techniques, are generally employed where volumetric examination is required, and either liquid penetrant or magnetic particle techniques are employed where surface examination is required. Visual examinations are conducted in accordance with the requirements of Subarticle IWA-2210, of ASME Section XI.

Class 3 systems and components are visually examined during system inservice tests, component functional tests, or system pressure tests to detect evidence of component leakage, structural distress of corrosion, in accordance with the requirements of Subarticle IWD-2600, of ASME Section XI. Visual inspection of integral attachments is performed in accordance with the requirements of Table IWD-2500-1, of ASME Section XI.

6.6.4 INSPECTIONS INTERVALS

An inspection schedule for Class 2 and 3 system components will be developed in accordance with guidelines of Subarticles IWC-2400 and IWD-2400, and IWF-2400, of ASME Section XI.

6.6.5 EXAMINATION CATEGORIES AND REQUIREMENTS

The examination categories and requirements will be in accordance with the guidelines of Table IWC-2500-1 to the extent practicable. The examination requirements for Class 3

components will be in accordance with the guidelines of Subarticle IWD-2600 to the extent practicable.

An inservice inspection program, which will discuss the examination categories, will be submitted within nine (9) months of the date of the operating license, in compliance with the requirements of 10 CFR Part 50 Section 50.55a, paragraph (g).

6.6.6 EVALUATION OF EXAMINATION RESULTS

Articles IWC-3000 and IWD-3000, concerning evaluation of examination results for class 2 and 3 components, have not yet been prepared. After publication, Articles IWC-3000 and IWD-3000 will be reviewed and incorporated in this section if found applicable.

Presently, Class 2 and 3 components will be evaluated in accordance with Article IWB-3000.

Components with unacceptable indications will be repaired or replaced in accordance with the guidelines of Articles IWC-4000 and IWD-4000, except if the guidelines of IWC-4000 & IWD-4000 are inappropriate for the components, then the guidelines of Article IWA-4000 will apply.

The above evaluation and repair programs conform to ASME Section XI.

6.6.7 SYSTEM PRESSURE TEST

Class 2 systems subject to system pressure tests will be tested in accordance with Articles IWA-5000 and IWC-5000 and Table IWC-2500-1, of ASME Section XI.

Class 3 systems subject to system pressure tests will be tested in accordance with the requirements of Articles IWA-5000, IWD-5000, and Table IWD-2500-1, of ASME Section XI.

6.6.8 AUGMENTED INSERVICE INSPECTION

Main Steam and main feedwater piping located in the containment penetration areas and designated "break exclusion piping" is subject to augmented inservice inspection in accordance with paragraph B.2.d of APCSB 3-1. 100 percent of all circumferential and longitudinal welds on this piping, except as may be exempted by ASME Code Section XI paragraph IWC-1220, is volumetrically inspected each 10 year inspection interval.

6.7 MAIN STEAM LINE ISOLATION VALVE LEAKAGE CONTROL SYSTEM

This section is not applicable to Comanche Peak Seam Electric Station.

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7.1 INTRODUCTION

This chapter presents the various plant instrumentation and control systems by relating the functional performance requirements, design bases, system descriptions, design evaluations, and tests and inspections for each. The information provided in this chapter emphasizes those instruments and associated equipment which constitute the protection system as defined in the Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971, "IEEE Standard: Criteria for Protection Systems for Nuclear Power Generating Stations."

The primary purpose of the instrumentation and control systems is to provide automatic protection and exercise proper control against unsafe and improper reactor operation during steady state and transient power operations (American Nuclear Society (ANS) Conditions I, II and III) and to provide initiating signals to mitigate the consequences of faulted conditions (ANS Condition IV). ANS conditions are discussed in Chapter 15. Consequently, the information presented in this chapter emphasizes those instrumentation and control systems which are central to assuring that the reactor can be operated to produce power in a manner that ensures no undue risk to the health and safety of the public.

These systems meet the applicable criteria and codes, such as General Design Criteria and IEEE Standards, concerned with the safe generation of nuclear power. Table 7.1-1 lists applicable criteria and directs the reader to the sections where conformance is discussed. Tables 7.1-2.2 through 7.1-2.7 provide a matrix which identifies the systems to which the applicable criteria of Table 7.1-1 is applied.

Definitions

Terminology used in this chapter is based on the definitions given in IEEE Standard 279-1971. In addition, the following definitions apply:

1. Degree of redundancy

   The difference between the number of channels monitoring a variable and the number of channels which when tripped will cause an automatic system trip.

2. Minimum degree of redundancy

   The degree of redundancy below which operation is prohibited, or otherwise restricted by the Technical Specifications.

3. Cold shutdown condition

   When the reactor is subcritical by at least 1 percent k/k; percent rated thermal power, excluding decay heat, is zero; and T_{avg} is $\leq$ 200°F.

4. Hot shutdown condition

   When the reactor is subcritical by at least 1 percent k/k; percent rated thermal power, excluding decay heat, is zero; and 350°F > T_{avg} > 200°F.

5. Hot standby condition

When the reactor is subcritical by at least 1 percent k/k; percent rated thermal power, excluding decay heat, is zero; and $T_{avg} \geq 350°F$.

6. Phase A Containment Isolation

Closure of all nonessential process lines which penetrate the Containment, initiated by the safety injection signal. Phase A Containment isolation is on an “S” signal.

7. Phase B Containment Isolation

Closure of remaining process lines, initiated by Containment hi-3 pressure signal (process lines do not include engineered safety features lines). Phase B Containment isolation is on a “P” signal.

8. System response times

a. Reactor Trip System Response Time

The time delays are defined as the time required for a signal to reach the reactor trip breakers from the time the setpoint is reached following a step change in the variable being monitored from 5 percent below (or above) to 5 percent above (or below) the trip setpoint.

b. Engineered Safety Features Actuation System Response Time

The interval required for the engineered safety features sequence to be initiated subsequent to the point in time that the appropriate variable(s) exceed setpoints. The response time includes sensor/process (analog) and logic (digital) delay.

9. Reproducibility

This definition is taken from Scientific Apparatus Manufactures Association (SAMA) Standard PMC-20.2-1973, Process Measurement and Control Terminology: “the closeness of agreement among repeated measurements of the output for the same value of input, under normal operating conditions over a period of time, approaching from both directions.” It includes drift due to environmental effects, hysteresis, long term drift, and repeatability. Long term drift (aging of components, etc.) is not an important factor in accuracy requirements since, in general, the drift is not significant with respect to the time elapsed between testing. Therefore, long term drift may be eliminated from this definition. Reproducibility, in most cases, is a part of the definition of accuracy (see item 10).

10. Accuracy

This definition is derived from SAMA Standard PMC-20.1-1973, Process Measurement and Control Terminology. An accuracy statement for a device falls under note 2 of the SAMA definition of accuracy, which means reference accuracy or the accuracy of that device at reference operating conditions: “Reference accuracy includes conformity,

hysteresis and repeatability." To adequately define the accuracy of a system, the term reproducibility is useful as it covers normal operating conditions. The following terms, "trip accuracy" and "indicated accuracy" etc., will then include conformity and reproducibility under normal operating conditions. Where the final result does not have to conform to an actual process variable but is related to another value established by testing, conformity may be eliminated, and the term reproducibility may be substituted for accuracy.

11. Normal Operating Conditions

For this document, these conditions cover all normal process temperature and pressure changes. Also included are ambient temperature changes around the transmitter and racks. This document does not include any accuracies under "post-accident" conditions.

12. Readout Devices

For consistency the final device of a complete channel is considered a readout device. This includes indicators, recorders, isolators (nonadjustable), and controllers.

13. Channel Accuracy

This definition includes accuracy of primary element, transmitter and rack modules. It does not include readout devices or rack environmental effects, but does include process and environmental effects on field mounted hardware. Rack environmental effects are included in the next two definitions to avoid duplication due to dual inputs.

14. Indicated and/or Recorded Accuracy

This definition includes channel accuracy, accuracy of readout devices and rack environmental effects.

15. Trip Accuracy

This definition includes comparator accuracy, channel accuracy, each input, and rack environmental effects. This is the tolerance expressed in process terms (or percent of span) within which the complete channel must perform its intended trip function. This includes all instrument errors but no process effects such as streaming. The term "actuation accuracy" may be used where the word "trip" might cause confusion (for example, when starting pumps and other equipment).

16. Control Accuracy

This definition includes channel accuracy, accuracy of readout devices (isolator and controller), and rack environmental effects. Where an isolator separates control and protection signals, the isolator accuracy is added to the channel accuracy to determine control accuracy, but credit is taken for tuning beyond this point; i.e., the accuracy of these modules (excluding controllers) is included in the original channel accuracy. It is simply defined as the accuracy of the control signal in percent of the span of that signal. This will then include gain changes where the control span is different from the span of the measured variable. Where controllers are involved, the control span is the input span

of the controller. No error is included for the time in which the system is in a nonsteady state condition.

7.1.1 IDENTIFICATION OF SAFETY-RELATED SYSTEMS

7.1.1.1 Safety-Related Systems

The Nuclear Steam Supply System (NSSS) and the balance of plant (BOP) instrumentation discussed in Chapter 7 that is required to function to achieve the system responses assumed in the safety evaluations, and those needed to shutdown the plant safely are given in this section.

Refer to Figure 7.1-3 for location layout drawings of Class 1E instrumentation. These figures pertain to location layout drawing requirements as discussed in Sections 7.2, 7.3 and 7.5.

7.1.1.1.1 Reactor Trip System

The Reactor Trip System (RTS) is a functionally defined system described in Section 7.2. The RTS responsibility falls primarily under the NSSS supplier, with the exceptions of underfrequency, and undervoltage, and turbine trip signal which are under BOP scope (See Figure 7.2-1). The equipment which provides the trip functions is identified and discussed in Section 7.2. Design bases for the RTS are given in Section 7.1.2.1. Figure 7.1-1 is a block diagram of this system.

7.1.1.1.2 Engineered Safety Features Actuation System

The Engineered Safety Features Actuation System (ESFAS) is a functionally defined system described in Section 7.3. The equipment which provides the actuation functions is identified and discussed in Section 7.3. Design bases for the ESFAS are given in Section 7.1.2.1.

The ESF and ESF Support Systems requiring actuation are as follows:

ESF Systems:

1. Emergency Core Cooling System (Section 6.3)

   a. Safety Injection System

   b. Residual Heat Removal System (partial)

   c. Chemical and Volume Control System (partial)

   Designed by NSSS vendor.

2. Containment Spray System (Section 6.2.2)

   a. Containment Spray Chemical Additive Subsystem (Section 6.5.2)

   Designed under the Architect-Engineer's (A-E) specifications.

3. Containment Isolation System (Section 6.2.4)

   Containment isolation valves for the following systems are furnished by the NSSS vendor.

   a. Chemical and Volume Control System.

   b. Residual Heat Removal System.

   c. Safety Injection System.

   d. Waste Processing System.

   The remaining Containment isolation valves are built to the specifications of the A-E.

4. Deleted.

5. Control Room Air Conditioning System (Sections 6.4 and 9.4).

   Designed and built to the specifications of the A-E.

6. Auxiliary Feedwater System (Section 10.4.9).

   Designed and built to the specifications of the A-E.

7. ESF Filter Systems (Section 6.5)

   a. Control Room Air Conditioning System (partial)

   b. Primary Ventilation System (partial)

   Designed and built to the specification of A-E.

ESF Support System:

1. Component Cooling Water System (Section 9.2.2)

   Designed and built to the specifications of the A-E.

2. Service Water System (Section 9.2.1)

   Designed and built to the specifications of the A-E.

3. Onsite Power Supply System

   a. Diesel Generator Sets (Section 8.3)

   b. Diesel Generator F.O. Storage and Transfer System (Section 9.5.4)

   c. Diesel Generator Cooling Water System (Section 9.5.5)

d. Diesel Generator Starting System (Section 9.5.6)

e. Diesel Generator Lube Oil System (Section 9.5.7)

f. Diesel Generator Combustion Air Intake and Exhaust System (Section 9.5.8)

g. Diesel Generator Ventilation System (Section 9.4C)

h. D.C. System (Section 8.3.2)

Designed and built to the specifications of the A-E.

4. ESF Ventilation System (Section 9.4.5)

Designed and built to the specification of the A-E.

5. Safety Chilled Water System (Section 9.4F)

Designed and built to the specifications of the A-E.

6. Service Water Intake Structure Ventilation System (Section 9.4B)

Designed and built to the specifications of the A-E.

7. UPS Ventilation System (Section 9.4C.8)

Designed and built to the specifications of the A-E.

7.1.1.1.3 Class 1E 118V AC Uninterruptible Power System

Design bases for the Class 1E 118V AC Uninterruptible Power System (UPS) are given in Section 7.1.2.1. Further description of this system is provided in Section 7.6.1. The UPS for the reactor protection system is described in the above mentioned sections. For the Class 1E Static Uninterruptible Power System for the Balance of Plant furnished safety-related instrumentation refer to Section 8.3.1.

7.1.1.2 Safety-Related Display Instrumentation

Display instrumentation provides the operator with information to enable him to monitor the results of engineered safety features actions following a Condition II, III, or IV event. Section 7.5 (Table 7.5-7A, B & C) provides information required to maintain the plant in a hot shutdown condition, or to proceed to cold shutdown.

7.1.1.3 Instrumentation and Control System Designers

All systems discussed in Chapter 7 have definitive functional requirements developed on the basis of the Westinghouse NSSS design. Figure 7.2-1 defines scope interface. Regardless of the supplier, the functional requirements necessary to assure plant safety and proper control are clearly delineated.

7.1.1.4 Plant Comparison

System functions for all NSSS systems discussed in Chapter 7 are similar to those of the William B. McGuire Nuclear Station (Docket No. 50-369 and 50-370). A plant comparison table is provided in Section 1.3.

7.1.2 IDENTIFICATION OF SAFETY CRITERIA

Section 7.1.2.1 gives design bases for the systems given in Section 7.1.1.1. Design bases for nonsafety-related systems are provided in the sections which describe the systems. Conservative considerations for instrument errors are included in the accident analyses presented in Chapter 15. Functional requirements, developed on the basis of the results of the accident analyses, which have utilized conservative assumptions and parameters, are used in designing these systems and a preoperational testing program verifies the adequacy of the design. Accuracies are given in Table 7.2-3, and Section 7.3.1.2.6.

The documents listed in Table 7.1-1 were considered in the design of the systems given in Section 7.1.1. In general, the scope of these documents is given in the document itself. This determines the systems or parts of systems to which the document is applicable. A discussion of compliance with each document for systems in its scope is provided in the referenced sections given in Table 7.1-1 for each criterion. A matrix which identifies the systems to which the applicable criteria of Table 7.1-1 is applied is given in Tables 7.1-2.2 through 7.1-2.7. Because some documents were issued after design and testing had been completed, the equipment documentation may not meet the format requirements of some standards. Justification for any exceptions taken to each document for systems in its scope is provided in the referenced sections.

7.1.2.1 Design Bases

7.1.2.1.1 Reactor Trip System

The RTS acts to limit the consequences of Condition II events (faults of moderate frequency), such as loss of feedwater flow, by, at most, a shutdown of the reactor and turbine, with the plant capable of returning to operation after corrective action. The RTS features impose a limiting boundary region to plant operation which ensures that the reactor safety limits are not exceeded during Condition II events and that these events can be accommodated without developing into more severe conditions. Reactor trip setpoints will be provided in the Technical Specifications.

The design requirements for the RTS are derived by analyses of plant operating and fault conditions where automatic rapid control rod insertion is necessary in order to prevent or limit core or reactor coolant boundary damage. The design bases addressed in IEEE Standard 279-1971 are discussed in Section 7.2.1. The design limits specified by Westinghouse for the RTS are:

1. Minimum departure from nucleate boiling ratio (DNBR) shall not be less than the limit value as a result of any anticipated transient or malfunction (Condition II events).

2. Power density shall not exceed the rated linear power density for Condition II events.

3. The stress limit of the Reactor Coolant System for the various conditions shall be as specified in Sections 5.2, 5.3 and 5.4.

4. Release of radioactive material shall not be sufficient to interrupt or restrict public use of those areas beyond the exclusion radius as a result of any Condition III event.

5. For any Condition IV event, release of radioactive material shall not result in an undue risk to public health and safety.

7.1.2.1.2 Engineered Safety Features Actuation System

The ESFAS acts to limit the consequences of Condition III events (infrequent faults such as primary coolant spillage from a small rupture which exceeds normal charging system makeup and requires actuation of the Safety Injection System). The ESFAS acts to mitigate Condition IV events (limiting faults, which include the potential for significant release of radioactive material).

The design bases for the ESFAS are derived from the design bases given in Chapter 6 for the engineered safety features. Design bases requirements of IEEE Standard 279-1971 are addressed in Section 7.3.1.2. General design requirements are given below.

1. Automatic actuation requirements

   The primary requirement of the ESFAS is to receive input signals (information) from the various on-going processes within the reactor plant and Containment and automatically provide, as output, timely and effective signals to actuate the various components and subsystems comprising the Engineered Safety Features System.

2. Manual actuation requirements

   The ESFAS must have provisions in the Control Room for manually initiating the functions of the Engineered Safety Features System.

7.1.2.1.3 Class 1E 118V AC Uninterruptible Power System

The UPS provides continuous, reliable, regulated single phase alternating current (AC) power to all Reactor Protection System instrumentation and control equipment required for plant safety. Details of this system are provided in Section 7.6. The design bases are given below:

1. The inverter shall have the capacity and regulation required for the AC output for proper operation of the equipment supplied.

2. Redundant loads shall be assigned to different distribution panels which are supplied from different inverters.

3. Auxiliary devices that are required to operate dependent equipment shall be supplied from the same distribution panel to prevent the loss of electric power in one protection set from causing the loss of equipment in another protection set. No single failure shall cause a loss of power supply to more than one distribution panel.

4. Each of the distribution panels shall have access only to its respective inverter supply and a standby power supply.

5. The system shall comply with IEEE Standard 308-1974 Section 5.4.

7.1.2.1.4 Class 1E Onsite AC Power System

Design bases and system description for the Emergency Power Supply System is provided in Section 8.3.

7.1.2.1.5 Interlocks

Interlocks are discussed in Sections 7.2, 7.3, 7.6 and 7.7. The protection (P) interlocks are given in Tables 7.2-2 and 7.3-3. The safety analyses demonstrate that even under conservative critical conditions for either postulated or hypothetical accidents, the protective systems ensure that the NSSS will be put into and maintained in a safe state following an ANS Condition II, III or IV event commensurate with applicable Technical Specifications and pertinent ANS Criteria. Therefore the protective systems have been designed to meet IEEE Standard 279-1971 and are entirely redundant and separate, including all permissives and blocks. All blocks of a protective function are automatically cleared whenever the protective function would be required to function in accordance with General Design Criteria 20, 21 and 22 and Sections 4.11, 4.12 and 4.13 of IEEE Standard 279-1971. Control interlocks (C) are identified on Table 7.7-1. Because control interlocks are not safety-related, they have not been specifically designed to meet the requirements of IEEE 279 protection system standards.

7.1.2.1.6 Bypasses

Bypasses are designed to meet the requirements of IEEE Standard 279-1971, Sections 4.11, 4.12, 4.13 and 4.14. A discussion of bypasses provided is given in Sections 7.2 and 7.3.

The capability of bypass testing is provided for the 7300 Process Protection System Reactor Trip and Engineered Safety Features Actuation functions and the Nuclear Instrumentation System Reactor Trip functions.

The Bypass Test Instrumentation which allows testing in a bypassed condition instead of a tripped condition conforms to applicable regulatory criteria including IEEE 279-1971 and Regulatory Guide 1.47 as well as prior regulatory guidance concerning test in bypass. Additional information concerning test in bypass can be found in WCAP-14096.

7.1.2.1.7 Equipment Protection

The criteria for equipment protection are given in Chapter 3. Equipment related to safe operation of the plant is designed, constructed and installed to protect it from damage. This is accomplished by working to accepted standards and criteria aimed at providing reliable instrumentation which is available under varying conditions. As an example, equipment is seismically qualified in accordance with the discussion in Chapter 3. Independence and separation is achieved, as required by IEEE Standard 279-1971, IEEE Standard 384-1974 and Regulatory Guide 1.75, either by barriers, physical separation or demonstration test. This serves to protect against complete destruction of a system by fires, missiles or other natural hazards.

7.1.2.1.8 Diversity

Functional diversity has been designed into the system. Functional diversity is discussed in Reference [1]. The extent of diverse system variables has been evaluated for a wide variety of postulated accidents. Generally, two or more diverse protection functions would automatically terminate an accident before unacceptable consequences could occur.

For example, there are automatic reactor trips based upon neutron flux measurements, reactor coolant loop flow measurements, reactor coolant loop hot leg N-16 measurements pressurizer pressure and level measurements, and reactor coolant pump underfrequency and undervoltage measurements, as well as manually, and by initiation of a safety injection signal.

Regarding the ESFAS for a loss of coolant accident, a safety injection signal can be obtained manually or by automatic initiation from two diverse parameter measurements.

1. Low pressurizer pressure.

2. High Containment pressure (hi-1).

For a steam line break accident, safety injection signal actuation is provided by:

1. Low compensated steamline pressure.

2. For a steam line break inside Containment, high Containment pressure (hi-1) provides an additional parameter for generation of the signal.

All of the above sets of signals are redundant and physically separated and meet the requirements of IEEE Standard 279-1971.

7.1.2.1.9 Bistable Trip Setpoints

The setpoint methodology is basically the square root of the sum of the squares (SRSS) of the statistically independent parameters. Dependent parameters are arithmetically summed prior to systematic combination with other terms. The total combination of error terms is identified as the channel statistical allowance (CSA).

Three values applicable to reactor trip and engineered safety features actuation are specified:

1. Safety Analysis limit

2. Nominal Safety System Setting

3. Limiting Safety System Setting

The safety analysis limit is the value assumed in the accident analysis and is the least conservative value.

The nominal safety system setting is the technical specification “Trip Setpoint” and is determined by subtracting the channel’s “total allowance” (Total Allowance = CSA + Margin) from the safety analysis limit.

The limiting safety system setting is the technical specification “allowable value” and a setpoint exceeding this value indicates that a channel may be inoperable. The allowable value is determined by either adding the arithmetic sum of the error components encountered during periodic surveillances (including drift) to the nominal safety system setting or by subtracting the sum of margin plus a statistical combination of channel error terms except those encountered during periodic surveillances from the safety analysis limit. The most conservative result is used as the allowable value.

The trip setpoint is determined by factors other than the most accurate portion of the instrument’s range. The safety analysis limit setpoint is determined only by the accident analysis. As described above, allowance is then made for process uncertainties, instrument error, instrument drift, and calibration uncertainty to obtain the nominal setpoint value which is actually set into the equipment. The only requirement on the instrument’s accuracy value is that over the instrument span, the error must always be less than or equal to the error value allowed in the accident analysis. The instrument does not need to be the most accurate at the setpoint value as long as it meets the minimum accuracy requirement. The accident analysis accounts for the expected errors at the actual setpoint.

Range selection for the instrumentation covers the expected range of the process variable being monitored consistent with its application. The design of the reactor protection and engineered safety features systems is such that the bistable trip setpoints do not require process transmitters to operate within 5 percent of the high and low end of their calibrated span or range. Functional requirements established for every channel in the reactor protection and engineered safety features systems stipulate the maximum allowable errors on accuracy, linearity, and reproducibility. The protection channels have the capability for, and are tested to ascertain that the characteristics throughout the entire span in all aspects are acceptable and meets functional requirement specifications. As a result, no protection channel operates normally within 5 percent of the limits of its specified span.

In this regard, it should be noted that the specific functional requirements for response time, setpoint, and operating span is finalized based on the plant specific safety studies. Emphasis is placed on establishing adequate performance requirements under both normal and faulted conditions. This includes consideration of process transmitters margins such that even under a highly improbable situation of full power operation at the limits of the operating map (as defined by the high and low pressure reactor trip, N-16 Overpower and Overtemperature trip lines (DNB protection) and the steam generator safety valve pressure setpoint) that adequate instrument response is available to ensure plant safety.

7.1.2.1.10 Engineered Safety Features Motor Specifications

The minimum voltage for the residual heat removal pump motor and Engineered Safety Features Auxiliary System pump motors rated 6600 volt and 460 volt is 80 percent of rated voltage at the motor terminals to start and accelerate the driven equipment. The motors are capable of accelerating the driven equipment from rest to operating speed within 5 seconds.

The minimum margin of motor torque over the pump full load torque (as defined by the pump speed/torque curve) will be sufficient to accelerate all the driven equipment within 5 seconds and with 80 percent of rated voltage at the motor terminals from standstill to operating speed.

Verification of the engineered safety features pump motors capability to operate within design temperature ratings, including the NEMA Test Specification MG1-20.43 (“number of starts”), is based on the design tests of the prototype motor that are performed at the manufacturer’s test facilities, rather than by means of initial or periodic tests in the field.

For conditions where the motor stalls or fails to start, the influence of these conditions is best monitored from the current versus time characteristics, and equipment protection for this is provided by the circuit breaker trip function.

7.1.2.2 Independence of Redundant Safety-Related Systems

The safety-related systems in Section 7.1.1.1 are designed to meet the independence and separation requirements of Criterion 22 of the 1971 General Design Criteria and Section 4.6 of IEEE Standard 279-1971.

The electrical power supply, instrumentation, and control wiring for redundant circuits of a nuclear plant have physical separation to preserve the redundancy and to ensure that no single credible event will prevent operation of the associated function due to electrical conductor damage. Critical circuits and functions include power, control and analog instrumentation associated with the operation of the RTS or ESFAS. Credible events shall include, but not be limited to, the effects of short circuits, pipe rupture, missiles, fire, etc. and are considered in the basic plant design. Control board details are given in Section 7.7.1.12. In the control board, separation of redundant circuits is maintained as described in Section 8.3.1.4.

7.1.2.2.1 General

The criteria for the installation and routing of cables is discussed in Section 8.3.1.4.

The physical separation criteria for redundant safety-related system sensors, sensing lines, wireways, cables, and components on racks within Westinghouse NSSS scope meet recommendations [2] contained in Regulatory Guide 1.75 with the following comments:

1. The Westinghouse design of the protection system does not rely on over-current devices to prevent malfunctions in one circuit from causing unacceptable influences on the functioning of the protection system [2]. The protection system uses redundant instrumentation channels and actuation trains and incorporates physical and electrical separation to prevent faults in one channel from degrading any other protection channel.

   Within a protection system, however, the Westinghouse design does rely on over-current devices as isolation devices to prevent malfunctions in Non-Class 1E portions of a circuit from causing unacceptable influences on the Class 1E function of the circuit [4].

   For discussion on control grade instrumentation power circuits powered from protection grade power supplies, see paragraph 8.3.1.2.1, item 7 “Compliance with NRC Regulatory Guide 1.75 [5] and IEEE 384 [6]”.

2. Separation recommendations for redundant instrumentation racks are not the same as those given in Regulatory Position C.16 of Regulatory Guide 1.75, Revision 1, for the control boards because of different functional requirements. Main control boards contain redundant circuits which are required to be physically separated from each other.

However, since there are no redundant circuits which share a single compartment of an NSSS protection instrumentation rack, and since these redundant protection instrumentation racks are physically separated from each other, the physical separation requirements specified for the main control board do not apply.

However, redundant, isolated control signal cables leaving the protection racks are brought into close proximity elsewhere in the plant, such as the control board. It could be postulated that electrical faults, or interference, at these locations might be propagated into all redundant racks and degrade protection circuits because of the close proximity of protection and control wiring within each rack. Regulatory Guide 1.75 (Regulatory Position C.4) and IEEE Standard 384-1974 (Section 4.5(3)) provide the option to demonstrate that the absence of physical separation could not significantly reduce the availability of Class 1E circuits.

The Nuclear Instrumentation System and Solid State Protection System were included in the "Westinghouse Protection System Noise Tests" report submitted and accepted by the Nuclear Regulatory Commission (NRC) in support of the Diablo Canyon application (Docket No. 50-275 and 50-323). The tests on the Process Control System - 7300 Series are reported in Reference [2], the conclusions having been accepted by the NRC.

The protection and surveillance upgrade package cabinets that house the protection equipment and components including isolators are 7300 series process control equipment similar to that reported in Reference [2].

Provisions are made to provide assurance that maximum credible fault voltages and conditions which could be postulated in the Comanche Peak Nuclear Power Plant (CPNPP), as a result of BOP cable routing design, will not exceed those used in the tests.

These Westinghouse tests demonstrated that protection systems performance would not be degraded even if subjected to abnormal electrical conditions which far exceed those which can be reasonably postulated.

The Auxiliary Relay Racks, Safeguards Test Cabinets, and Reactor Trip Switchgear utilize Class 1E relays to provide isolation between redundant Class 1E circuits and between Class 1E/non Class 1E circuits. Refer to FSAR Sec. 8.3.1.2.1.7 item b(1). This equipment implements similar wiring practices and layout as implemented for the 7300 series process control system cabinets supplied by Westinghouse. The 7300 series process control system noise test (WCAP-8892A) demonstrated that a fault will not propagate wire to wire even if the wires are in close proximity (i.e., touching). Separate auxiliary relay racks are utilized for redundant trains further ensuring integrity of divisional circuits. For the Reactor Trip Switchgear, the additional concern of an accidental short between the coil and contact terminals of the cross train interlock relays was evaluated with the conclusion that the ability of the breakers to perform their trip function is not impacted. Therefore, internal separation of the wiring internal to these cabinets is not required.

3. The physical separation criteria for instrument cabinets (including the protection and surveillance upgrade package cabinets) within Westinghouse NSSS scope meet the recommendations contained in Section 5.7 of IEEE Standard 384-1974.

4. The BOP Analog Instrumentation System utilizes redundant safety trains and incorporates physical and electrical separation to prevent faults in one safety train from degrading any other safety train. Each BOP Analog Rack cabinet contains only single safety related train wiring and devices and therefore the separation requirement (between trains) is not applicable in this case.

   The train orientation and wiring techniques used in the 7300 series process control system has been demonstrated by tests to meet regulatory requirements per Regulatory Guide 1.75 (Ref. WCAP-8892-A, "Westinghouse 7300 Series Process System Noise Tests," June 1977). In the BOP Analog Instrumentation System, Westinghouse utilized similar wiring techniques. Westinghouse certificate of qualification, CQ-W9525, establishes an auditable link between hardware furnished under the BOP Analog Instrumentation System and the 7300 Series Process Control Equipment. Additionally it has been demonstrated that close proximity of safety train wiring to non-safety train wiring within individual BOP Analog Instrumentation cabinets will not degrade the safety functions of Class 1E circuits should faults occur in the non-safety train circuits.

   Therefore, the physical and electrical separation criteria for the BOP Analog Instrumentation System supplied by Westinghouse meet the recommendations contained in Regulatory Guide 1.75.

7.1.2.2.2 Specific Systems

Independence is maintained throughout the system, extending from the sensor through to the devices actuating the protective function. Physical separation is used to achieve separation of redundant transmitters. Separation of wiring is achieved using separate wireways, cable trays, conduit runs and Containment penetrations for each redundant protection channel set. Redundant analog equipment is separated by locating modules in different protection rack sets. Each redundant channel set is energized from a separate AC power feed.

There are four separate process analog sets. Separation of redundant analog channels begins at the process sensors and is maintained in the field wiring, Containment penetrations and analog protection cabinets to the redundant trains in the logic racks. Redundant analog channels are separated by locating modules in different protection cabinets. Since all equipment within any cabinet is associated with a single protection set, there is no requirement for separation of wiring and components within the cabinet.

Two three-bay cabinets housing the protection and surveillance upgrade functions involving the N-16 Power Measuring System are included in process instrumentation and control system. Each of the two three-bay cabinets contains equipment for two protection sets with appropriate physical and electrical separation and provisions for top and/or bottom entry of the cables. One of the three-bay cabinets contains the surveillance instrumentation (in its center bay which has been abandoned in place) with the appropriate separation provided. Contained in each protection set is instrumentation to perform the N-16 power calculation utilizing a T_{cold} input from process protection and N-16 detector signals from two hot leg detectors. In addition, the four section neutron power range detector signals are brought into the protection racks, summed (to form Q_{UPPER} and Q_{LOWER}), and sent to Nuclear Instrumentation System (NIS) rack (see Figure 7.1-2).

In the Nuclear Instrumentation System, Process Systems, and the Solid State Protection System input cabinets where redundant channel instrumentation is physically adjacent, there are no wireways, or cable penetrations which would permit, for example, a fire resulting from electrical failure in one channel to propagate into redundant channels in the logic racks. Redundant analog channels are separated by locating modules in different cabinets. Since all equipment within any cabinet is associated with a single protection set, there is no requirement for separation of wiring and components within the protection cabinet.

Two reactor trip breakers are actuated by two separate logic matrices which interrupt power to the control rod drive mechanisms. The breaker main contacts are connected in series with the power supply so that opening either breaker interrupts power to all full length control rod drive mechanisms, permitting the rods to free fall into the core.

1. Reactor Trip System

    a. Separate routing shall be maintained for the four basic RTS channel sets analog sensing signals, bistable output signals and power supplies for such systems. The separation of these four channel sets shall be maintained from sensors to instrument cabinets to logic system input cabinets.

    b. Separate routing of the redundant reactor trip signals from the redundant logic system cabinets shall be maintained, and in addition, they shall be separated (by spatial separation or by provision of barriers or by separate cable trays or wireways) from the four analog channel sets.

2. Engineered Safety Features Actuation System

    a. Separate routing shall be maintained for the four basic sets of ESFAS analog sensing signals, bistable output signals and power supplies for such systems. The separation of these four channel sets shall be maintained from sensors to instrument cabinets to logic system input cabinets.

    b. Separate routing of the engineered safety features actuation signals from the redundant logic system cabinets shall be maintained. In addition, they shall be separated by spatial separation or by provisions of barriers or by separate cable trays or wireways from the four analog channel sets.

    c. Separate routing of control and power circuits associated with the operation of engineered safety features equipment is required to retain redundancies provided in the system design and power supplies.

3. Class 1E Static Uninterruptible Power System

    The separation criteria presented also apply to the power supplies for the load centers and buses distributing power to redundant components and to the control of these power supplies.

The RTS and ESFAS analog and digital circuits may be routed in the same wireways provided circuits have the same power supply and channel set identified (I, II, III or IV).

7.1.2.2.3 Fire Protection

For the electrical equipment it has been specified that noncombustible or fire retardant materials be used. Specification requirements ensure that materials will not be used which may ignite or explode from an electrical spark, flame, or from overheating, or will independently support combustion. These reviews also have included assurance of conservative current carrying capacities of all instrument cabinet wiring, which precludes electrical fires resulting from excessive overcurrent (I^2R) losses. For example, wiring used for instrument cabinet construction for the NSSS has teflon or tefzel insulation and is adequately sized based on current carrying capacities set forth by the National Electric Code. Braided sheathed material is noncombustible. The cables in power trays are sized using derating factors listed in IPCEA Publications P-46-426 and P-54-440.

For early warning protection against propagation of electrical fires, smoke or other high sensitivity detectors are provided for fire detection and alarm in remote wireways or other unattended areas where large concentrations of cables are installed. Refer to Section 8.3.3.

Details of the Fire Protection System are provided in Section 9.5.1.

7.1.2.3 Physical Identification of Safety-Related Equipment

There are four separate protection sets identifiable with process equipment associated with the RTS and ESFAS. A protection set may be comprised of more than a single process equipment cabinet. The color coding of each process equipment rack nameplate coincides with the color code established for the protection set of which it is a part. Redundant channels are separated by locating them in different equipment cabinets. Separation of redundant channels begins at the process sensors and is maintained in the field wiring, Containment penetrations and equipment cabinets to the redundant trains in the logic racks. The Solid State Protection System input cabinets are divided into four isolated compartments, each serving one of the four redundant input channels. Horizontal 1/8 inch thick solid steel barriers, coated with fire retardant paint, separate the compartments. Four, 1/8 inch thick solid steel, wireways coated with fire retardant paint enter the input cabinets vertically, even in its own quadrant. The wireway for a particular compartment is open only into that compartment so that flame could not propagate to affect other channels. At the logic racks the protection set color coding for redundant channels is clearly maintained until the channel loses its identity in the redundant logic trains. The color coded nameplates described below provide identification of equipment associated with protective functions and their channel set association:

Protection Set	Color Coding
I	RED with WHITE lettering
II	WHITE with BLACK lettering
III	BLUE with WHITE lettering
IV	YELLOW with BLACK lettering

All non-cabinet mounted protective equipment and components are provided with an identification tag or nameplate. Small electrical components such as relays have nameplates on

the enclosure which houses them. All cables are numbered with identification tags. In congested areas, such as under or over the control boards, instrument racks, etc., cable trays and conduits containing redundant circuits shall be identified using permanent markings. The purpose of such markings is to facilitate cable routing identification for future modification or additions. Positive permanent identification of cables and/or conductors shall be made at all terminal points. There are also identification nameplates on the input panels of the Solid State Logic Protection System.

7.1.2.4 Conformance to Criteria

A listing of applicable criteria and the sections where conformance is discussed is given in Table 7.1-1. A matrix of the applicability of this criteria to instrumentation and control systems is provided in Tables 7.1-2.2 through 7.1-2.7.

7.1.2.5 Conformance to Regulatory Guide 1.22

Periodic testing of the RTS and ESFAS, as described in Sections 7.2.2 and 7.3.2, complies with Regulatory Guide 1.22, “Periodic Testing of Protection System Actuation Functions.”

Where the ability of a system to respond to a bona fide accident signal is intentionally bypassed from the safeguards test cabinet for the purpose of performing a test during reactor operation, each bypass condition is automatically indicated to the reactor operator in the main Control Room by a separate annunciator for the train in test. Test circuitry does not allow two trains of the SSPS to be tested at the same time so that extension of the bypass condition to the redundant system is prevented. Administrative and procedural control are used to prevent testing of more than one protection set of the analog circuitry simultaneously. For those devices without additional test circuits that will be manually disabled to prevent operation, the above indication will be manually initiated prior to disabling.

The actuation logic for the RTS and ESFAS is tested as described in Sections 7.2 and 7.3. As recommended by Regulatory Guide 1.22, where actuated equipment is not tested during reactor operation, it has been determined that:

1. There is no practicable system design that would permit operation of the equipment without adversely affecting the safety or operability of the plant;

2. The probability that the protection system will fail to initiate the operation of the equipment is, and can be maintained, acceptably low without testing the equipment during reactor operation; and

3. The equipment can routinely be tested when the reactor is shutdown.

The list of equipment that cannot be tested at full power so as not to damage equipment or upset plant operation is:

1. Manual actuation switches.

2. Reactor coolant pump breakers.

3. Turbine trip equipment from reactor trip.

4. Main steam line isolation valves (close).

5. Main feedwater isolation valves (close).

6. Main feedwater control valves (close).

7. Reactor coolant pump component cooling water supply and return isolation valves (close).

8. Reactor coolant pump seal water return isolation valves (close).

9. Unit 2 Feedwater splitflow bypass valves (close).

10. Feedwater isolation bypass Valves (close).

11. Unit 2 Feedwater preheater bypass valves (close).

12. Feedwater control bypass valves (close).

13. Chilled water supply and return header to chilled water recirculation pump isolation valves (close).

14. Instrument air to containment isolation valve (close).

15. Ventilation chiller CCW supply and discharge control valves (close).

16. Intentionally blank.

17. Generator trip on turbine trip.

18. Residual Heat Removal Pump containment sump suction isolation valve (close).

19. Normal charging containment isolation valves (close).

20. Normal letdown containment isolation valves (close).

21. RWST - charging pump suction isolation valves (close).

22. VCT outlet isolation valves (close).

23. High head safety injection isolation valves (open).

24. Selected valves from Inservice Testing Plan.

The justifications for not testing the above 24 items at full power are discussed below.

1. Manual actuation switches

These would cause initiation of their protection system function at power causing plant upset and/or reactor trip. It should be noted that the reactor trip function that is derived from the automatic safety injection signal is tested at power as follows:

a. The analog signals, from which the automatic safety injection signal is derived, is tested at power in the same manner as the other analog signals and as described in Section 7.2.2.2.3.(10). The processing of these signals in the Solid State Protection System wherein their channel orientation converts to a logic train orientation is tested at power by the built-in semiautomatic test provisions of the Solid State Protection System. The reactor trip breakers are tested at power as discussed in Section 7.2.2.2.3.(10).

2. Tripping of reactor coolant pump breakers

No credit is taken in the accident analyses for a reactor coolant pump breaker opening causing a direct reactor trip. Since testing them at power would cause plant upset, the reactor coolant pump breakers do not need to be tested at power.

3. Turbine

The generation of a turbine trip on demand from the RTS is not fully testable at power. Testing of the Turbine Trip Block (hydraulic 2 out of 3 logic) is not practical at power, since partial or complete exercising will cause a turbine trip. The Turbine Trip Block solenoid valves and associated hydraulic pistons are individually tested at power. In addition, the Turbine Trip Black is functionally tested during each startup.

4. Closing the main steam line isolation valves

Main steam isolation valves are full stroke tested during refueling outages in accordance with the IST Plan. Testing of the main steam isolation valves to closure at power is not practical. As the plant power is increased, the core average temperature is programmed to increase. If the valves are closed under these elevated temperature conditions, the steam pressure transient would unnecessarily operate the steam generator relief valves and possibly the steam generator safety valves. The steam pressure transient produced would cause shrinkage in the steam generator level, which would cause the reactor to trip on low-low steam generator water level.

The valves are capable of being periodically partial stroke tested. Each valve can be actuated at slow speed and is limited to approximately 10 percent of full travel.

Based on the above identified problems incurred with full stroke periodic testing of the main steam line isolation valves at power and since 1) no practical system design will permit operation of the valves without adversely affecting the safety or operability of the plant, 2) the probability that the Protection System will fail to initiate the actuated equipment is acceptably low due to test up to final actuation, and 3) these valves will be tested during refueling outages in accordance with the IST plan, the proposed resolution meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

5. Closing the main feedwater isolation valves

The feedwater isolation valves are tested during refueling outages in accordance with the IST Plan. Periodic testing of the closure of these isolation valves at full power would induce steam generator water level transients and oscillations which would lead to a reactor trip. The operational status of these isolation valves is periodically verified in order to provide assurance of valve/valve actuator operability. The valves are capable of partial stroke tests limited to approximately 10 percent of valve stroke.

Since, 1) no practical system design will permit operation of these valves without adversely affecting the safety or operability of the plant, 2) the probability that the Protection System will fail to initiate the activated equipment is acceptably low due to testing up to final actuation and 3) these valves will be tested during refueling outages in accordance with the IST Plan, the proposed resolution meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

6. Closing the main feedwater control valves

These valves are routinely tested during refueling outages. To close them at power would adversely affect the operability of the plant. The verification of operability of feedwater control valves at power is assured by confirmation of proper operation of the steam generator water level system. The actual actuation function of the solenoids, which provides the closing function, is periodically tested at power as discussed in Section 7.3.2.2.5. The operability of the slave relay which actuates the solenoid, which is the actuating device, is verified during this test. Although the actual closing of these control valves is blocked when the slave relay is tested, all functions are tested to assure that no electrical malfunctions have occurred which could defeat the protective function. It is noted that the solenoids work on the de-energize-to-actuate principle, so that the feedwater control valves will fail close upon either the loss of electrical power to the solenoids or loss of air pressure.

Based on the above, the testing of the isolating function of feedwater control valves meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

7. Reactor coolant pump component cooling water supply and return isolation valves (close)

Component cooling water supply and return Containment isolation valves and component cooling water Non-Safeguard Loop isolation valves are routinely tested during refueling outages. Testing of these valves while the reactor coolant pumps are operating introduces an unnecessary risk of costly damage to all the reactor coolant pumps. Loss of component cooling water to these pumps is of economic consideration only, as the reactor coolant pumps are not required to perform any safety-related function.

The reactor coolant pumps will not seize due to complete loss of component cooling. Information from the pump manufacturer indicates that the bearing babbitt would eventually break down but not so rapidly as to overcome the inertia of the flywheel. If the pumps are not stopped within 3 to 10 minutes after component cooling water is isolated, pump damage could be incurred.

Additional Containment penetrations and Containment isolation valves would introduce additional unnecessary potential pathways for radioactive leakage following a postulated accident. Also, since the component cooling water flow rates and temperatures are about

equal during both plant power operation and plant refueling, periodic tests of these valves during a refueling outage would duplicate accident conditions. Additionally, possibility of failure of Containment isolation is remote because an additional failure of the low pressure fluid system in addition to failure of both isolation valves would have to occur to open a path through the Containment.

The Non-Safeguard Loop isolation valves provide reactor coolant pump cooling in series with the containment isolation valves and the considerations above are also applicable to them.

Based on the above described potential reactor coolant pump damage incurred with periodic testing of the subject component cooling water isolation valves at power, the duplication of at-power operating conditions during refueling outages, and since 1) no practical system design will permit operation of these valves without adversely affecting the safety or operability of the plant, 2) the probability that the Protection System will fail to initiate the activated equipment is acceptably low due to testing up to final actuation, and 3) these valves will be routinely tested during refueling outages when the reactor coolant pumps are not operating, the proposed resolution meets the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22.

8. Reactor coolant pump seal water return isolation valves (close)

Seal return line isolation valves are routinely tested during refueling outages. Closure of these valves during operation would cause the safety valve to lift, with the possibility of valve chatter. Valve chatter would damage this relief valve. Testing of these valves at power would cause equipment damage. Therefore, these valves will be tested during scheduled refueling outages. As above, additional Containment penetrations and Containment isolation valves introduce additional unnecessary potential pathways for radioactive release following a postulated accident. Thus, the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

9. Feedwater split flow bypass valves (close) (Unit 2 only)

Feedwater split flow bypass valves are routinely tested during refueling outages. Closure of these valves during operation would divert all feedwater flow to the preheater region of the steam generator which may cause flow induced tube failures in the preheater region of the steam generator, and may induce excessive thermal transients in the nozzle and the connecting piping when flow is transferred to the auxiliary feedwater nozzle from the main feedwater line during unloading. As above, closure of these valves may damage other equipment. Thus, the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

10. Feedwater isolation bypass valves (close)

These valves are routinely tested during refueling outages. Slave relay tests are accomplished as described in Section 7.3.2.2.5. To test closure on a Feedwater isolation signal at power would require that the valves be manually isolated and that the interlocks with the associated Unit 2 FWIV be defeated on all four Unit 2 FIBVs at the same time. Therefore, the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

11. Feedwater preheater bypass valves (close) (Unit 2 only)

These valves are routinely tested during refueling outages. Slave relay tests are accomplished as described in Section 7.3.2.2.5. To test closure on a Feedwater isolation signal at power would require that the valves be manually isolated and that the interlocks with the associated FWIV be defeated on all four FPBVs at the same time. Therefore, the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met. Control circuits allow the valves to be stroked individually using the associated hand switch; however, stroking the valves in this manner requires that they first be manually isolated.

12. Feedwater control bypass valves (close)

These valves are routinely tested during refueling outages. To stroke them at power would adversely affect normal operation of the plant. These valves are required to be open during startup and low loads operation prior to opening the feedwater control valves. Therefore the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

13. Closing chilled water supply and return header to chilled water recirculation pump isolation valves.

These valves are routinely tested during refueling outages. To close them at power would adversely affect the operability of neutron detector well cooling units. Therefore the guidelines of Regulatory Position D.4 of regulatory guide 1.22 are met.

14. Closing instrument air to containment isolation valve.

This valve is routinely tested during refueling outages. To close it at power would adversely affect the operability of pneumatic equipment inside the containment. Therefore the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

15. Closing ventilation chiller CCW supply and discharge control valves.

These valves are routinely tested during refueling outages. To close them at power would adversely affect the operability of the ventilation chiller and letdown chiller. Therefore the guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

16. Intentionally blank.

17. Generator trip on turbine trip

Generator trip cannot be tested at power because to test it, turbine trip has to be initiated. This circuit is routinely tested during refueling outages.

18. RHR containment sump suction isolation valves

Valves 1-8811A and 1-8811B, RHR containment recirculation sump isolation valves, cannot be safely exercised with the plant at power. The stroke test of these valves requires isolating the RHR pumps from the RWST to prevent gravity draining the RWST to the containment sump.

The absence of a check valve in the RHR suction line potentially could push water into the containment sump which would then require removal. The RHR pump suction header would then have to be partially drained to prevent water back flow to the sump rendering the pumps inoperable. An additional concern involves the ability to adequately vent and fill the system after testing. The additional risks encountered and the amount of time to perform testing do not justify the additional assurance gained by quarterly testing. These valves are routinely tested during refueling outages.

19. Normal charging isolation valves

Valves 1-8105 and 1-8106, normal charging isolation valve, cannot be exercised during power operation since this would interrupt normal charging flow which could result in loss of pressurizer level control. This valve will be full-stroke exercised during cold shutdown.

20. Normal letdown isolation valves

Valves 1-8152 and 1-8160, CVCS letdown containment isolation valves, cannot be exercised during power operation since this would result in a loss of normal letdown flow, which would stop the preheating of the charging water going through the regenerative heat exchanger. This lack of preheating would cause an unnecessary thermal shock to the charging line penetration. These valves will be exercised during refueling outages.

21. RWST charging pump suction isolation valves

Valves 1-LCV-112D and E, charging pump suction from RWST isolation valves, cannot be exercised during power operation since this would result in defeating the chemical balance attained by normal use of the CVCS. The subsequent reactivity transient could result in an uncontrolled plant shutdown. These valves will be exercised during refueling outages.

22. VCT outlet isolation valve

Valves 1-LCV-112B and C, volume control tank outlet isolation valves, cannot be exercised during plant operation since this would cause a loss of pressurizer level control, RCP seal injection, and letdown regenerative heat exchanger cooling. These valves will be exercised during refueling outages in accordance with the IST Plan.

23. High head safety injection isolation valves

Valves 1-8801A and B, high head safety injection isolation valves, cannot be opened during plant operation because opening the valves results in unnecessary thermal transients on the reactor coolant system cold legs and an uncontrolled level increase in the pressurizer. In addition, the unit shutdown is required to leak check the reactor coolant system pressure isolation check valves located downstream of the valves. These valves will be exercised during refueling outages in accordance with the IST Plan.

24. Selected valves from Inservice Testing Plan

FSAR Section 3.9B.6.2 states that the inservice testing of valves shall be in accordance with the Inservice Testing Plan. Inservice Testing Plan identifies which valves cannot be

tested at full power so as not to damage equipment or upset plant operation (as allowed by the ASME Code). These valves are identified by a footnote and a justification as to why the valve cannot be tested at full power. The guidelines of Regulatory Position D.4 of Regulatory Guide 1.22 are met.

7.1.2.6 Conformance to Regulatory Guide 1.47

The Bypassed and Inoperable Status Indication System (called Safety System Inoperable Indication (SSII) on CPNPP) conforms to the requirements of Regulatory Guide 1.47. The system is designed as follows:

1. The SSII is located in the Control Room and consists of one control board display for each safety train. Within each display are bypass lights for each of the systems listed in item 6.

   The system for each train consists of a lamp box, a horn and an electronics logic chassis. The lamp box is mounted on the front of the control board. The electronics logic chassis and the horn are mounted inside the control board.

   Each lamp box consists of twenty backlighted pushbuttons and “acknowledge,” “reset” and two “test” pushbuttons. Each backlighted pushbutton is capable of red and amber displays and there are separate test pushbuttons for the red and amber lamps.

   The inputs to each backlighted pushbutton are divided into two categories, primary inputs which give a red display and secondary inputs which give an amber display. The primary inputs are those inputs to a given system, from components within the given system itself, that will cause the given system to be bypassed or inoperative. The secondary inputs are those inputs from other (supporting) systems or other primary displays to the given system that will cause the given system to be bypassed or inoperative.

   The system functions like an annuniciator system in that activation of a field contact, or momentary actuation of a backlighted pushbutton causes the light in that pushbutton to flash and the horn to come on. Pressing the acknowledge button, silences the horn and the light becomes steady. Upon return to normal, the light stays on until “reset” button is pressed.

   For electrical schematic drawings, see drawings 2323-E1-0071, sheets 46 through 65, 67 and 68.

2. The SSII does not perform functions essential to mitigate the consequences of an accident, nor do administrative procedures require operator manual actions based solely on SSII displays. SSII is implemented as a non-safety grade system consistent with R.G. 1.47.

3. Electrical separation is based on analysis provided in Section 8.3. The interface with safety systems and components is such that no degradation of safety systems will occur because of an SSII failure.

4. The SSII display is automatically initiated for those inoperable conditions reasonable expected to occur more frequently than once per year when the affected system is normally required to be operable.

5. A means for manually initiating the SSII light exists for those maintenance or bypass activities not automatically initiated. The manual initiation consists of pressing the backlighted pushbutton used for display. This can only be cleared by again actuating the pushbutton.

6. An audible alarm is sounded when any bypass is automatically initiated.

   The list of systems included on each SSII display, along with a reference to the description of the system is as follows:

   a. Residual Heat Removal, (Section 5.4.7)

   b. Safety Injection, (Section 6.3)

   c. Containment Spray, (Section 6.5.2)

   d. Onsite power, diesel (Section 8.3)

   e. Preferred offsite power, (Section 8.3)

   f. Alternate offsite power, (Section 8.3)

   g. 480 VAC, (Section 8.3)

   h. Station Service Water, (Section 9.2.1)

   i. Auxiliary Feedwater, (Section 10.4.9)

   j. Component Cooling Water, (Section 9.2.2)

   k. Control Room HVAC, (Section 9.4.1)

   l. 125 VDC, (Section 8.3.2)

   m. 118 VAC, (Section 8.3.1)

   n. Safety Chilled Water System, (Section 9.4F)

   o. Primary Plant ESF ventilation exhaust system, (Section 9.4)

   A logic diagram showing typical implementation of this system is shown on Figure 7.1-4 for the Containment Spray System.

The design was verified as part of the Preoperational Test Program.

7.1.2.7 Conformance to Regulatory Guide 1.53 and IEEE Standard 379-1972

1. NSSS Systems:

The principles described in IEEE Standard 379-1972 were used in the design of the Westinghouse Protection System. The system complies with the intent of this standard and the additional guidance of Regulatory Guide 1.53 although the formal analyses gone beyond the required analyses and has performed a fault tree analysis, Reference [1]. The referenced topical report provides details of the analyses of the protection systems previously made to show conformance with the single failure criterion set forth in Section 4.2 of IEEE Standard 279-1971. The interpretation of single failure criterion provided by IEEE Standard 379-1972 does not indicate substantial differences with the Westinghouse interpretation of the criterion except in the methods used to confirm design reliability. Established design criteria in conjunction with sound engineering practices form the bases for the Westinghouse protection systems. The RTS and ESFAS are each redundant safety systems. The required periodic testing of these systems will disclose any failures or loss of redundancy which could have occurred in the interval between tests, thus ensuring the availability of these systems.

2. BOP Systems:

Listed below are the BOP portions of the plant safety system, and the auxiliary systems required for supprt of the safety systems:

System		Described in Section
ESF System:		
a.	Containment Spray	6.2.2, 6.5.2
b.	Containment Isolation	6.2.4, 10.3 and 10.4.7
c.	Control Room Air Conditioning	6.4.2.2 and 9.4
d.	Auxiliary Feedwater	10.4.9
e.	ESF Filters	6.5
ESF Support System:		
a.	Component Cooling Water	9.2.10
b.	Station Service Water	9.2.1
c.	Onsite Power Supply	8.3, 9.5 and 9.4C
d.	ESF Ventilation	9.4.5
e.	Safety Chilled Water	9.4F
f.	Service Water Intake Structure Ventilation	9.4B
g.	UPS Ventilation System	9.4C.8

The instrumentation and controls, and associated power supplies, for the listed systems are designed to meet the principle of redundancy and the requirements for electrical independence and physical separation of redundant trains. The redundant components, their locations, and arrangements are protected from common mode failures resulting from missiles, pipe whip, fire, harsh environment etc. For each of the listed systems, a failure mode and effects analysis has been determined to ensure compliance with the single failure criterion. Required periodic testing, as described in Section 7.1.2.5, will detect potential failures or loss of redundance which may have occurred between test intervals. Also, anomalous indication and alarms during normal plant operation provide additional means for detection of failures.

Thus the BOP safety systems and auxiliary systems required for support of safety systems meet the requirements of Regulatory Guide 1.53 and IEEE STD 379-1972.

7.1.2.8 Conformance to Regulatory Guide 1.63

Refer to Section 8.3.1.2.

7.1.2.9 Conformance to IEEE Standard 317-1972

Refer to Section 8.3.1.2.

7.1.2.10 Conformance to IEEE Standard 336-1971

Refer to Section 8.3.1.2.

7.1.2.11 Conformance to IEEE Standard 338-1971

The periodic testing of the RTS and ESFAS conforms to the requirements of IEEE Standard 338-1971 with the following comments:

1. The surveillance requirements of the Technical Specifications for Protection System ensure that the system functional operability is maintained comparable to the original design standards. Periodic tests at frequent intervals verify this capability for the system, excluding sensors.

   Overall protection systems response times are verified by test procedures detailed by Vistra OpCo. Sensors are adequately demonstrated for the design by vendor testing, insitu tests in operating plants with appropriately similar design, or by suitable type testing. The Nuclear Instrumentation System detectors are excluded since they exhibit response time characteristics such that delays attributable to them are negligible in the overall channel response time required for safety.

   Sensors will be periodically tested to ensure their response times are within the minimum performance requirements of the design basis. This may be done by either (1) in place on site or off site test measurements or (2) utilizing the replacement sensors with certified response times.

   For pressure and differential pressure transmitters, in place on site response time measurements will involve application of step inputs about each setpoint for which

response time determination is required. This is consistent with original Westinghouse specifications as well as with qualification testing.

The steps' initial value will be 5% of span, in the reset direction from the setpoint of interest. The steps' final value will be 5% of span, in the tripped direction, from the setpoint of interest. The response time will be that time required for the transmitter's output to transition from its initial value at the time of initiation of the step input to the value corresponding to the setpoint of interest. For each transmitter, these steps are applied about all associated setpoints for which a response time determination is required.

For those transmitters only requiring response time determination about a single setpoint, an additional step is applied as a linearity check. This additional step will utilize the same ±5% of span for initial and final values as described above. However, these values will be applied about 50% of span versus an actual setpoint.

An alternative to the step-input Sensor Response Time Testing (SRTT) determination method for pressure and differential pressure devices is the White Noise Analysis technique. This involves measuring the natural fluctuations existing at the transmitters' output during normal operation. The signal is then mathematically analyzed and a response time is extracted. The resultant White Noise response time is then correlated to the step-input response time. Subsequent White Noise tests are compared to the initial White Noise response to determine if degradation of time response has occurred.

For resistance temperature detectors (RTDs), in place/on site response time testing will be conducted using the Loop Current Step Response (LCSR) method described in EPRI report NP-834. Westinghouse design specifications placed on their RTD suppliers reference a plunge type test for determination of RTD response time. However, a study conducted by Analysis and Measurement Services (AMS) for CPNPP (Report AMS-TU8402RO), compared plunge testing with LCSR testing for the type RTDs and thermal wells used at CPNPP. The report concludes that the agreement between the two test methods demonstrates the LCSR testability of the CPNPP RTDs.

Technical Specifications require response time verification on at least 18 month intervals. Each verification shall include at least one logic train such that both logic trains are verified at least once per 36 months and one channel per function such that all channels are verified at least once every N times 18 months, where N is the total number of redundant channels in a specific protective function.

Verification of the overall response time at the specified time intervals provides assurance that the protective and engineered safety features action function associated with each channel is completed within the time limit assumed in the accident analyses.

2. The reliability goals specified in Section 4.2 of IEEE Standard 338-1971 and adequacy of time intervals are described in Reference 1.

3. The periodic time interval discussed in Section 4.3 of IEEE Standard 338-1971, and specified in the Technical Specifications, is conservatively selected to assure that equipment associated with protection functions has not drifted beyond its minimum performance requirements. If any protection channel appears to be marginal or requires

more frequent adjustments due to plant condition changes, the time interval will be decreased to accommodate the situation until the marginal performance is resolved.

4. The test interval discussed in Section 5.2 of IEEE Standard 338-1971 is developed primarily on past operating experience and modified if necessary to assure that system and subsystem protection is reliably provided. Analytic methods for determining reliability are not used to determine test interval.

5. Testing of the underfrequency relay for Reactor Coolant Pumps, described in Section 7.2.2.2.3.10, conforms to the requirements of IEEE Standards 279-1971 and 338-1971 with the following comment.

   Testing is accomplished by introducing and varying a substitute input to the sensor of the same nature as the introduced variable. The inoperable condition is administratively controlled and is also indicated as soon as the relay setpoint is verified.

Based on the scope definition given in IEEE Standard 338-1971, no other instrumentation described in Chapter 7 is required to comply with this standard.

REFERENCES

1. Gangloff, W. C. and Loftus, W. D., “An Evaluation of Solid State Logic Reactor Protection in Anticipated Transients,” WCAP-7706-L (Proprietary) and WCAP-7706 (Non-Proprietary), February 1973.

2. Siroky, R. M. and Marasco, F. W., “7300 Series Process Control System Noise Tests,” WCAP-8892-A, April 1977.

3. Morgan, C. E., “Elimination of Periodic Protection Channel Response Time Tests,” WCAP-14036-P-A, Revision 1, October 6, 1998.

4. Letter NS-CE-604, dated March 31, 1975, C. Eicheldinger (Westinghouse) to the Secretary of the Nuclear Regulatory Commission.

5. NRC Regulatory Guide 1.75, Physical Independence of Electric Systems, Revision 1, January 1975, U.S. Nuclear Regulatory Commission.

6. IEEE 384-1974, Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits.

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 1 of 8)

	Criteria	Title	Conformance Discussed In:
1.	General Design Criteria (GDC), Appendix A to 10CFR50		
	GDC 1	Quality Standards and Records	3.1, 7.2.2.2
	GDC 2	Design Bases for Protection Against Natural Phenomena	3.1, 7.2.1.1.11
	GDC 3	Fire Protection	3.1, 9.5
	GDC 4	Environmental and Missile Design Bases	3.1, 7.2.2.2
	GDC 5	Sharing of Structures, Systems, and Components	3.1
	GDC 10	Reactor Design	3.1, 7.2.2.2
	GDC 12	Suppression of Reactor Power Oscillations	3.1
	GDC 13	Instrumentation and Control	3.1, 7.3.1, 7.3.2
	GDC 15	Reactor Coolant System Design	3.1, 7.2.2.2
	GDC 19	Control Room	3.1, 6.4, 9.4
	GDC 20	Protection System Functions	3.1, 7.1.2.1.5, 7.2.2.2, 7.3.1, 7.3.2

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 2 of 8)

Criteria	Title	Conformance Discussed In:
GDC 21	Protection System Reliability and Testability	3.1, 7.1.2.1.5, 7.2.2.2, 7.3.2
GDC 22	Protection System Independence	3.1, 7.1.2.1.5, 7.1.2.2, 7.2.2.2, 7.3.1, 7.3.2
GDC 23	Protection System Failure Modes	3.1, 7.2.2.2, 7.3.1, 7.3.2
GDC 24	Separation of Protection and Control Systems	3.1, 7.2.2.2, 7.3.1, 7.3.2
GDC 25	Protection System Requirements for Reactivity Control Malfunctions	3.1, 7.3.2
GDC 26	Reactivity Control System Redundancy and Capability	3.1, 7.7
GDC 27	Combined Reactivity Control Systems Capability	3.1, 7.3.1, 7.3.2
GDC 28	Reactivity Limits	3.1, 7.3.1, 7.3.2
GDC 29	Protection Against Anticipated Operational Occurrences	3.1, 7.2.2.2
GDC 30	Quality of Reactor Coolant Pressure Boundary	3.1, 5.2.5, 7.6.2
GDC 33	Reactor Coolant Makeup	3.1, 6.3, 9.3
GDC 34	Residual Heat Removal	3.1, 5.4.7

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 3 of 8)

Criteria	Title	Conformance Discussed In:
GDC 35	Emergency Core Cooling	3.1, 7.3.1, 7.3.2
GDC 37	Testing of Emergency Core Cooling System	3.1, 7.3.1, 7.3.2
GDC 38	Containment Heat Removal	3.1, 6.2.2
GDC 40	Testing of Containment Heat Removal System	3.1, 6.2.2
GDC 41	Containment Atmosphere Cleanup	3.1, 6.2.2, 6.5.2
GDC 43	Testing of Containment Atmosphere Cleanup System	3.1, 6.5.2
GDC 44	Cooling Water	3.1, 9.2.2
GDC 46	Testing of Cooling Water System	3.1, 9.2.2
GDC 50	Containment Design Basis	3.1
GDC 54	Piping Systems Penetrating Containment	3.1, 6.2.4
GDC 55	Reactor Coolant Pressure Boundary Penetrating Containment	3.1, 6.2.4
GDC 56	Primary Containment Isolation	3.1, 7.3.1, 6.2.4
GDC 57	Closed Systems Isolation Valves	3.1, 6.2.4

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 4 of 8)

Criteria		Title	Conformance Discussed In:
	GDC 63	Monitoring Fuel & Waste Storage	3.1, 11.5, 12.3.4
	GDC 64	Monitoring Radioactivity Releases	3.1, 11.5, 12.3.4 12.3.5
2.	Institute of Electrical and Electronics Engineers (IEEE) Standards:		
	IEEE Std 279-1971 (ANSI N42.7-1972)	Criteria for Protection Systems for Nuclear Power Generating Stations	7.1, 7.2, 7.3, 7.4 7.5, 7.6, 7.7
	IEEE Std 308-1974	Criteria for Class 1E Electric Systems for Nuclear Power Generating Stations	7.6.1, 7.1.1.1.3, 8.3.1
	IEEE Std 317-1976 (ANSI N45.3-1973)	Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations	8.3.1.2
	IEEE Std 323-1974	Qualifying Class 1E Equipment for Nuclear Power Generating Stations	3.11B, 3.11N
	IEEE Std 336-1971 (ANSI N45.2.4-1972)	Installation, Inspection and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations	8.3.1.2

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 5 of 8)

Criteria		Title	Conformance Discussed In:
	IEEE Std 338-1971 (ANSI N41.3)	Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection Systems	7.1.2.11
	IEEE Std 344-1971 (ANSI N41.7)	Trial-Use Guide for Seismic Qualification of Class I Electric Equipment for Nuclear Power Generating Stations	3.10B, 3.10N
	IEEE Std 379-1972 (ANSI N41.2)	Trial-Use Guide for the Application of the Single Failure Criterion to Nuclear Power Generating Station Protection Systems	7.1.2.7
	IEEE Std 384-1974 (ANSI N41.14)	Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits	7.1.2.2.1
3.	Regulatory Guides (RG)		
	RG 1.6	Independence Between Redundant Standby (Onsite) Power Sources and Between Their Distribution Systems	1A(B), 8.3
	RG 1.7	Control of Combustible Gas Concentrations in Containment Following a Loss of Coolant Accident	1A(B), 6.2.5
	RG 1.11	Instrument Lines Penetrating Primary Reactor Containment	1A(B), 1A(N), 7.3.1.1.2

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 6 of 8)

Criteria	Title	Conformance Discussed In:
RG 1.12	Instrumentation for Earthquakes	1A(B)
RG 1.22	Periodic Testing of Protection System Actuation Functions	1A(N), 1A(B), 7.1.2.5, 7.3.2.2.5
RG 1.29	Seismic Design Classification	1A(B), 1A(N)
RG 1.30	Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment	1A(B), 1A(N), Chapter 17
RG 1.32	Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants	1A(B), 8.3
RG 1.40	Qualifications Tests of Continuous- Duty Motors Installed Inside the Containment of Water-Cooled Nuclear Power Plants	1A(B), 1A(N)
RG 1.45	Reactor Coolant Pressure Boundary Leakage Detection Systems	1A(B), 5.2.5
RG 1.47	Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems	1A(B), 7.1.2.6
RG 1.53	Application of the Single Failure Criterion to Nuclear Power Plant Protection Systems	1A(N), 7.1.2.7, 1A(B)

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 7 of 8)

Criteria	Title	Conformance Discussed In:
RG 1.62	Manual Initiation of Protection Actions	1A(N), 7.3.2.2.7
RG 1.63	Electric Penetration Assemblies in Containment Structures for Light Water-Cooled Nuclear Power Plants	1A(B), 8.3.1.2
RG 1.67	Installation of Overpressure Protection Devices	1A(B), 3.9N.3.3, 3.9B.3.3
RG 1.68	Initial Test Programs for Water-Cooled Reactor Power Plants	1A(B), Chapter 14
RG 1.70	Standard Format and Content of Safety Analysis Reports for Nuclear Power Plants	1A(B), Chapter 7
RG 1.73	Qualification Test of Electric Operators Installed Inside the Containment of Nuclear Power Plants	1A(B), 1A(N), 3.10B, 3.10N, 3.11B, 3.11N
RG 1.75	Physical Independence of Electric Systems	1A(B), 1A(N), 7.1.2.2.1, 8.3
RG 1.78	Assumptions for Evaluating the Habitability of a Nuclear Power Power Plant Control Room During a Postulated Hazardous Chemical Release	1A(B), 2.2, 6.4
RG 1.80	Preoperational Testing of Instrument Air Systems	1A(B), Chapter 14

TABLE 7.1-1
LISTING OF APPLICABLE CRITERIA

(Sheet 8 of 8)

Criteria	Title	Conformance Discussed In:
RG 1.89	Qualification of Class 1E Equipment for Nuclear Power Plants	1A(B), 1A(N), 3.11B, 3.11N
RG 1.95	Protection of Nuclear Power Plant Control Room Operators against an Accidental Chlorine Release	1A(B)
RG 1.97	Instrumentation for Light-Water-Cooled Nuclear Power Plants to Assess Plant Conditions During & Following an Accident	1A(N), 1A(B), 7.5
RG 1.100	Seismic Qualification of Electric Equipment for Nuclear Power Plants	1A(N), 1A(B), 3.10B, 3.10N
RG 1.105	Instrument Spans and Setpoints	1A(N), 1A(B)
RG 1.106	Thermal Overload Protection for Electric Motors on Motor-Operated Valves	1A(B), 8.3
RG 1.118	Periodic Testing of Electric Power & Protection Systems	1A(N), 1A(B), 7.1.2.11
RG 1.120	Fire Protection Guidelines for Power Plants	1A(N), 1A(B), 9.5

TABLE 7.1-2.2
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR REACTOR TRIP SYSTEM (1,7)

(Sheet 1 of 4)

	RTS & NSSS Inputs	Turbine Trip-Reactor Trip	RC Pump Under Voltage-Under Frequency Trips
GDC 1	X	X	X
GDC 2	X	X	X
GDC 3	X	X	X
GDC 4	X	X	X
GDC 5	X	X	X
GDC 10	X	X	X
GDC 12	X	-	-
GDC 13	X	X	X
GDC 15	X	X	X
GDC 19	X	X	X
GDC 20	X	X	X
GDC 21	X	X	X
GDC 22	X	X	X
GDC 23	X	X	X
GDC 24	X	X	X
GDC 25	X	-	-
GDC 26	X	-	-
GDC 27	X	-	-
GDC 28	X	-	-
GDC 29	X	X	X
GDC 30	-	-	-
GDC 33	-	-	-

TABLE 7.1-2.2
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR REACTOR TRIP SYSTEM (1,7)

(Sheet 2 of 4)

	RTS & NSSS Inputs	Turbine Trip-Reactor Trip	RC Pump Under Voltage-Under Frequency Trips
GDC 34	-	-	-
GDC 35	-	-	-
GDC 37	-	-	-
GDC 38	-	-	-
GDC 40	-	-	-
GDC 41	-	-	-
GDC 43	-	-	-
GDC 44	-	-	-
GDC 46	-	-	-
GDC 50	-	-	-
GDC 54	-	-	-
GDC 55	-	-	-
GDC 56	-	-	-
GDC 57	-	-	-
GDC 63	-	-	-
GDC 64	-	-	-
RG 1.6	-	-	-
RG 1.7	-	-	-
RG 1.11	-	-	-
RG 1.12(2)			
RG 1.22	X	X	X
RG 1.29	X	X	X

TABLE 7.1-2.2
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR REACTOR TRIP SYSTEM (1,7)

(Sheet 3 of 4)

	RTS & NSSS Inputs	Turbine Trip-Reactor Trip	RC Pump Under Voltage-Under Frequency Trips
RG 1.30	X	X	X
RG 1.32	-	-	-
RG 1.45	-	-	-
RG 1.47	X	X	X
RG 1.53	X	X	X
RG 1.62	X	-	-
RG 1.63(3)			
RG 1.67(4)			
RG 1.68(5)			
RG 1.70	X	X	X
RG 1.75	X	X	X
RG 1.78	-	-	-
RG 1.80(5)			
RG 1.89	X	X	X
RG 1.95	-	-	-
RG 1.97	-	-	-
RG 1.100	X	X	X
RG 1.105	X	X	X
RG 1.118	X	X	X
RG 1.120(6)			
IEEE STD 279	X	X	X

TABLE 7.1-2.2
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR REACTOR TRIP SYSTEM (1,7)

(Sheet 4 of 4)

	RTS & NSSS Inputs	Turbine Trip-Reactor Trip	RC Pump Under Voltage-Under Frequency Trips
IEEE STD 308	X	-	-
IEEE STD 317(3)			
IEEE STD 323	X	X	X
IEEE STD 336	X	X	X
IEEE STD 338	X	X	X
IEEE STD 344	X	X	X
IEEE STD 379	X	X	X
IEEE STD 384	X	X	X

Notes:

1. The scope of applicability is instrumentation and control and NSSS electrical power supply only.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3 and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section 9.5.1 and Appendix 1A(B).
7. X Signifies compliance as discussed or qualified in the FSAR sections referenced in Table 7.1-1.

   - Not applicable.

TABLE 7.1-2 3
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR ENGINEERED SAFETY FEATURES SYSTEM (1,7)

(Sheet 1 of 5)

	ESF								ESF SUPPORT						
	EMERGENCY				CONTROL ROOM				COMP. STATION			SAFETY	SERVICE WATER	ONSITE	
	ESFAS	CORE COOLING	CONT. SPRAY	CONT. ISOLATION	AUX LIARY FEEDWATER	AIR CONDITION NG	COMBUSTIBLE GAS CONTROL	ESF FILTER	COOLING WATER	SERVICE WATER	ESF VENT.	CHILLED WATER	NTAKE STRUC. HVAC	POWER SUPPLY	UPS VENT.
GDC 1	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 2	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 5	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 10	X	X	-	-	X	-	-	-	-	-	-	-	-	-	-
GDC 12	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 13	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 15	X	X	-	-	X	-	-	-	-	-	-	-	-		-
GDC 19	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
GDC 20	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 21	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 22	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 23	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 24	X	X	X	X	X	X	X	X	X	X	X	X	X	X	-
GDC 25	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 26	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

TABLE 7.1-2 3
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR ENGINEERED SAFETY FEATURES SYSTEM (1,7)

(Sheet 2 of 5)

	ESF								ESF SUPPORT						
	EMERGENCY				CONTROL ROOM				COMP. STATION			SAFETY	SERVICE WATER	ONSITE	
	ESFAS	CORE COOLING	CONT. SPRAY	CONT. ISOLATION	AUX LIARY FEEDWATER	AIR CONDITION NG	COMBUSTIBLE GAS CONTROL	ESF FILTER	COOLING WATER	SERVICE WATER	ESF VENT.	CHILLED WATER	NTAKE STRUC. HVAC	POWER SUPPLY	UPS VENT.
GDC 27	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 29	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 30	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 33	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 34	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 35	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 37	-	X	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 38	-	-	X	-	-	-	-	-	-	-	-	-	-	-	-
GDC 40	-	-	X	-	-	-	-	-	-	-	-	-	-	-	-
GDC 41	-	-	X	-	-	-	X	-	-	-	-	-	-	-	-
GDC 43	-	-	X	-	-	-	X	-	-	-	-	-	-	-	-
GDC 44	-	-	-	-	-	X	-	-	X	X	-	X	-	-	-
GDC 46	-	-	-	-	-	X	-	-	X	X	-	X	-	-	-
GDC 50	-	-	X	-	-	-	-	-	-	-	-	-	-	-	-
GDC 54	-	X	X	X	X	-	X	-	X	-	-	-	-	-	-
GDC 55	-	-	-	X	-	-	-	-	-	-	-	-	-	-	-

TABLE 7.1-2 3
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR ENGINEERED SAFETY FEATURES SYSTEM (1,7)

(Sheet 3 of 5)

	ESF								ESF SUPPORT						
	EMERGENCY				CONTROL ROOM				COMP. STATION			SAFETY	SERVICE WATER	ONSITE	
	ESFAS	CORE COOLING	CONT. SPRAY	CONT. ISOLATION	AUX LIARY FEEDWATER	AIR CONDITION NG	COMBUSTIBLE GAS CONTROL	ESF FILTER	COOLING WATER	SERVICE WATER	ESF VENT.	CHILLED WATER	NTAKE STRUC. HVAC	POWER SUPPLY	UPS VENT.
GDC 56	-	-	X	X	-	-	X	-	X	-	-	-	-	-	-
GDC 57	-	-	-	X	X	-	-	-	X	-	-	-	-	-	-
GDC 63	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GDC 64	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.6	-	-	-	-	-	-	-	-	-	-	-	-	-	X	-
RG 1.7	-	-	-	-	-	-	X	-	-	-	-	-	-	-	-
RG 1.11	X	-	-	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.12[2]															
RG 1.22	X	X	X	X	X	X	-	X	X	X	X	X	-	X	X
RG 1.29	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.30	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.32	-	-	-	-	-	-	-	-	-	-	-	-	-	X	-
RG 1.45	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.47	X	X	X	X	X	X	-	X	X	X	-	X	-	X	X
RG 1.53	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.62	X	-	-	-	-	-	-	-	-	-	-	-	-	X	-

TABLE 7.1-2 3
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR ENGINEERED SAFETY FEATURES SYSTEM (1,7)

(Sheet 4 of 5)

	ESF								ESF SUPPORT						
	EMERGENCY				CONTROL ROOM				COMP. STATION			SAFETY	SERVICE WATER	ONSITE	
	ESFAS	CORE COOLING	CONT. SPRAY	CONT. ISOLATION	AUX LIARY FEEDWATER	AIR CONDITION NG	COMBUSTIBLE GAS CONTROL	ESF FILTER	COOLING WATER	SERVICE WATER	ESF VENT.	CHILLED WATER	NTAKE STRUC. HVAC	POWER SUPPLY	UPS VENT.
RG 1.63[3]															
RG 1.67[4]															
RG 1.68[5]															
RG 1.70	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.73	-	X	-	X	-	-	X	-	X	-	-	-	-	-	-
RG 1.75	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.78	-	-	-	-	-	X	-	-	-	-	-	-	-	-	-
RG 1.80[5]															
RG 1.89	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.95	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.97	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.100	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.105	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
RG 1.118	X	-	-	-	-	-	-	-	-	-	-	X	-	X	-
RG 1.120[6]															
IEEE - 279	X	-	-	-	-	-	-	-	-	-	-	-	-	X	-
IEEE - 308	X	-	-	-	-	-	-	-	-	-	-	-	-	X	-

TABLE 7.1-2 3
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR ENGINEERED SAFETY FEATURES SYSTEM (1,7)

(Sheet 5 of 5)

	ESF								ESF SUPPORT						
	EMERGENCY				CONTROL ROOM				COMP. STATION			SAFETY	SERVICE WATER	ONSITE	
	ESFAS	CORE COOLING	CONT. SPRAY	CONT. ISOLATION	AUX LIARY FEEDWATER	AIR CONDITION NG	COMBUSTIBLE GAS CONTROL	ESF FILTER	COOLING WATER	SERVICE WATER	ESF VENT.	CHILLED WATER	NTAKE STRUC. HVAC	POWER SUPPLY	UPS VENT.
IEEE - 317[3]															
IEEE - 323	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IEEE - 336	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IEEE - 338	X	X	X	X	X	X	-	X	X	X	X	X	-	X	X
IEEE - 344	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IEEE - 379	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
IEEE - 384	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Notes:

1. The scope of applicability is instrumentation and control and NSSS electrical power supply only.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3. and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section 9.5.1 and Appendix 1A(B).
7. X Signifies compliance as discussed or qualified in the FSAR sections referenced in Table 7.1-1.

   - Not applicable.

TABLE 7.1-2.4
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SYSTEMS REQUIRED FOR SAFE SHUTDOWN (1,8)

(Sheet 1 of 5)

	HOT STANDBY											
	REQUIRED			SUPPORT								
	AUXILIARY FEEDWATER	CVCS BORON ADDITION	ATMOS. STEAM RELIEF (7)	COMP. COOLING WATER	STATION SERVICE WATER	CONTROL ROOM AIR COND.	ON SITE POWER SUPPLY	SAFETY CHILLED WATER	UPS VENT	ESF VENT	REMOTE SHUTDOWN	COLD SHUTDOWN
GDC 1	X	X	X	X	X	X	X	X	X	X	X	-
GDC 2	X	X	X	X	X	X	X	X	X	X	X	-
GDC 3	X	X	X	X	X	X	X	X	X	X	X	-
GDC 4	X	X	X	X	X	X	X	X	X	X	X	-
GDC 5	X	X	X	X	X	X	X	X	X	X	X	-
GDC 10	X	X	X	-	-	-	-	-	-	-	-	-
GDC 12	-	-	-	-	-	-	-	-	-	-	-	-
GDC 13	X	X	X	X	X	X	X	X	X	X	X	X
GDC 15	X	-	X	-	-	-	-	-	-	-	-	-
GDC 19	X	X	X	X	X	X	X	X	X	X	X	X
GDC 20	-	-	-	-	-	-	-	-	-	-	-	-
GDC 21	-	-	-	-	-	-	-	-	-	-	-	-
GDC 22	-	-	-	-	-	-	-	-	-	-	-	-
GDC 23	-	-	-	-	-	-	-	-	-	-	-	-
GDC 24	X	X	-	X	X	X	X	X	-	-	X	X
GDC 25	-	-	-	-	-	-	-	-	-	-	-	-
GDC 26	-	X	-	-	-	-	-	-	-	-	-	-
GDC 27	-	X	-	-	-	-	-	-	-	-	-	-

TABLE 7.1-2.4
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SYSTEMS REQUIRED FOR SAFE SHUTDOWN (1,8)

(Sheet 2 of 5)

	HOT STANDBY											
	REQUIRED			SUPPORT								
	AUXILIARY FEEDWATER	CVCS BORON ADDITION	ATMOS. STEAM RELIEF (7)	COMP. COOLING WATER	STATION SERVICE WATER	CONTROL ROOM AIR COND.	ON SITE POWER SUPPLY	SAFETY CHILLED WATER	UPS VENT	ESF VENT	REMOTE SHUTDOWN	COLD SHUTDOWN
GDC 28	-	X	-	-	-	-	-	-	-	-	-	-
GDC 29	-	X	-	-	-	-	-	-	-	-	X	X
GDC 30	-	-	-	-	-	-	-	-	-	-	-	-
GDC 33	-	X	-	-	-	-	-	-	-	-	-	-
GDC 34	-	-	-	-	-	-	-	-	-	-	X	X
GDC 35	-	-	-	-	-	-	-	-	-	-	-	-
GDC 37	-	-	-	-	-	-	-	-	-	-	-	-
GDC 38	-	-	-	-	-	-	-	-	-	-	-	-
GDC 40	-	-	-	-	-	-	-	-	-	-	-	-
GDC 41	-	-	-	-	-	-	-	-	-	-	-	-
GDC 43	-	-	-	-	-	-	-	-	-	-	-	-
GDC 44	-	-	-	X	X	X	-	X	-	-	-	-
GDC 46	-	-	-	X	X	X	-	X	-	-	-	-
GDC 50	-	-	-	-	-	-	-	-	-	-	-	-
GDC 54	X	X	-	X	-	-	-	-	-	-	-	-
GDC 55	-	X	-	-	-	-	-	-	-	-	-	-
GDC 56	-	-	-	X	-	-	-	-	-	-	-	-
GDC 57	-	-	-	X	-	-	-	-	-	-	-	-

TABLE 7.1-2.4
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SYSTEMS REQUIRED FOR SAFE SHUTDOWN (1,8)

(Sheet 3 of 5)

	HOT STANDBY											
	REQUIRED			SUPPORT								
	AUXILIARY FEEDWATER	CVCS BORON ADDITION	ATMOS. STEAM RELIEF (7)	COMP. COOLING WATER	STATION SERVICE WATER	CONTROL ROOM AIR COND.	ON SITE POWER SUPPLY	SAFETY CHILLED WATER	UPS VENT	ESF VENT	REMOTE SHUTDOWN	COLD SHUTDOWN
GDC 63	-	-	-	-	-	-	-	-	-	-	-	-
GDC 64	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.6	-	-	-	-	-	-	X	-	-	-	-	-
RG 1.7	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.11	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.12(2)												
RG 1.22	X	X	-	X	X	X	X	X	X	X	-	-
RG 1.29	X	X	-	X	X	X	X	X	X	X	X	-
RG 1.30	X	X	-	X	X	X	X	X	X	X	X	-
RG 1.32	-	-	-	-	-	-	X	-	-	-	-	-
RG 1.45	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.47	X	X	-	X	X	X	X	X	X	X	X	X
RG 1.53	X	-	-	X	X	X	X	X	X	X	X	-
RG 1.62	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.63(3)												
RG 1.67(4)												
RG 1.68(5)												
RG 1.70	X	X	-	X	X	X	X	X	X	X	X	X

TABLE 7.1-2.4
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SYSTEMS REQUIRED FOR SAFE SHUTDOWN (1,8)

(Sheet 4 of 5)

	HOT STANDBY											
	REQUIRED			SUPPORT								
	AUXILIARY FEEDWATER	CVCS BORON ADDITION	ATMOS. STEAM RELIEF (7)	COMP. COOLING WATER	STATION SERVICE WATER	CONTROL ROOM AIR COND.	ON SITE POWER SUPPLY	SAFETY CHILLED WATER	UPS VENT	ESF VENT	REMOTE SHUTDOWN	COLD SHUTDOWN
RG 1.73	-	-	-	X	-	-	-	-	-	-	-	-
RG 1.75	X	X	-	X	X	X	X	X	X	X	X	-
RG 1.78	-	-	-	-	-	X	-	-	-	-	-	-
RG 1.80(5)												
RG 1.89	X	X	-	X	X	X	X	X	X	X	-	-
RG 1.95	-	-	-	-	-	X	-	-	-	-	-	-
RG 1.97(9)	-	-	-	-	-	-	-	-	-	-	-	-
RG 1.100	X	X	-	X	X	X	X	X	X	X	-	-
RG 1.105	X	X	-	X	X	X	X	X	X	X	-	-
RG 1.118	-	-	-	-	-	-	X	-	-	-	-	-
RG 1.120(6)												
IEEE STD 279	-	-	-	-	-	-	-	-	-	-	X	-
IEEE STD 308	-	-	-	-	-	-	X	-	-	-	-	-
IEEE STD 317(3)												
IEEE STD 323	X	X	-	X	X	X	X	X	X	X	-	-
IEEE STD 336	X	X	-	X	X	X	X	X	X	X	X	-
IEEE STD 338	X	-	-	X	X	X	X	X	X	X	-	-
IEEE STD 344	X	X	-	X	X	X	X	X	X	X	-	-

TABLE 7.1-2.4
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SYSTEMS REQUIRED FOR SAFE SHUTDOWN (1,8)

(Sheet 5 of 5)

	HOT STANDBY											
	REQUIRED			SUPPORT								
	AUXILIARY FEEDWATER	CVCS BORON ADDITION	ATMOS. STEAM RELIEF (7)	COMP. COOLING WATER	STATION SERVICE WATER	CONTROL ROOM AIR COND.	ON SITE POWER SUPPLY	SAFETY CHILLED WATER	UPS VENT	ESF VENT	REMOTE SHUTDOWN	COLD SHUTDOWN
IEEE STD 379	X	-	-	X	X	X	X	X	X	X	X	-
IEEE STD 384	X	X	-	X	X	X	X	X	X	X	X	-

Notes:

1. The scope of applicability is instrumentation & control and electrical power supply only.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3. and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section 9.5.1 and Appendix 1A(B).
7. Atmospheric steam relief is provided by self-actuated code safety valves, which have no associated 1 & C or electrical power supplies. GDC's indicated apply to these safety valves.
8. X Signifies compliance as discussed or qualified in the FSAR Sections referenced in Table 7.1-1.

   - Not applicable.
9. See Table 7.1-2.3.

TABLE 7.1-2.5
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SAFETY RELATED DISPLAY INSTRUMENTS (1, 8)

(Sheet 1 of 4)

	ACCIDENT MONITORS
GDC 1	X
GDC 2	X
GDC 3	X
GDC 4	X
GDC 5	X
GDC 10	-
GDC 12	-
GDC 13	X
GDC 15	-
GDC 19	X
GDC 20	-
GDC 21	-
GDC 22	-
GDC 23	-
GDC 24	X
GDC 25	-
GDC 26	-
GDC 27	-
GDC 28	-
GDC 29	-
GDC 30	-
GDC 33	-
GDC 34	-

TABLE 7.1-2.5
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SAFETY RELATED DISPLAY INSTRUMENTS (1, 8)

(Sheet 2 of 4)

	ACCIDENT MONITORS
GDC 35	-
GDC 37	-
GDC 38	-
GDC 40	-
GDC 41	-
GDC 43	-
GDC 44	-
GDC 46	-
GDC 50	-
GDC 54	-
GDC 55	-
GDC 56	-
GDC 57	-
GDC 63	-
GDC 64	X
RG 1.6	-
RG 1.7	-
RG 1.11	X
RG 1.12(2)	-
RG 1.22	-
RG 1.29	X

TABLE 7.1-2.5
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SAFETY RELATED DISPLAY INSTRUMENTS (1, 8)

(Sheet 3 of 4)

	ACCIDENT MONITORS
RG 1.30	X
RG 1.32	-
RG 1.45	-
RG 1.47	-
RG 1.53	X
RG 1.62	-
RG 1.63 (3)	
RG 1.67 (4)	
RG 1.68 (5)	
RG 1.70	X
RG 1.75	X
RG 1.78	-
RG 1.80 (5)	
RG 1.89	X
RG 1.95	-
RG 1.97	X
RG 1.100	X
RG 1.105	-
RG 1.116	-
RG 1.120 (6)	

TABLE 7.1-2.5
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR SAFETY RELATED DISPLAY INSTRUMENTS (1, 8)

(Sheet 4 of 4)

	ACCIDENT MONITORS
IEEE STD 279	-
IEEE STD 308	
IEEE STD 317 (3)	
IEEE STD 323	X
IEEE STD 336	X
IEEE STD 338	-
IEEE STD 344	X
IEEE STD 379	X
IEEE STD 384	X

Notes:

1. The scope of applicability is instrumentation.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3. and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section9.5.1 and Appendix 1A(B).
7. Refer to FSAR Section 7.1.
8. X Signifies compliance as discussed or qualified in the FSAR Section referenced in Table 7.1-1.

   - Not applicable.

TABLE 7.1-2.6
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY (1,7)

(Sheet 1 of 5)

	I & C POWER SUPPLY(3)	RHR ISOLATION VALVES	REFUELING INTERLOCKS(8)	ACCUMULATOR MO. VALVES	SWITCHOVER INJECTION TO RECIRCULATION	PROCESS & EFF RADIATION MONITORS	RCPB LEAK DETECTION	INTERLOCKS RCS PRESSURE CONTROL	FIRE PROTECTION(6)
GDC 1		X	-	X	X	X	X	-	-
GDC 2		X	-	X	X	X	X	X	-
GDC 3		X	-	X	X	X	X	X	-
GDC 4		X	-	X	X	X	X	X	-
GDC 5		X	-	X	X	X	X	X	-
GDC 10		X	-	-	-	-	-	-	-
GDC 12		-	-	-	-	-	-	-	-
GDC 13		X	-	X	X	X	X	X	-
GDC 15		X	-	-	-	-	-	X	-
GDC 19		X	-	X	X	X	X	X	-
GDC 20		-	-	-	-	-	-	-	-
GDC 21		-	-	-	-	-	-	-	-
GDC 22		-	-	-	-	-	-	-	-
GDC 23		-	-	-	-	-	-	-	-
GDC 24		X	-	X	X	X	X	X	-
GDC 25	-	-	-	-	-	-	-	-	-
GDC 26		-	-	-	-	-	-	-	-
GDC 27		-	-	-	-	-	-	-	-

TABLE 7.1-2.6
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY (1,7)

(Sheet 2 of 5)

	I & C POWER SUPPLY(3)	RHR ISOLATION VALVES	REFUELING INTERLOCKS(8)	ACCUMULATOR MO. VALVES	SWITCHOVER INJECTION TO RECIRCULATION	PROCESS & EFF RADIATION MONITORS	RCPB LEAK DETECTION	INTERLOCKS RCS PRESSURE CONTROL	FIRE PROTECTION(6)
GDC 28		-	-	-	-	-	-	-	-
GDC 29		-	-	-	-	-	-	-	-
GDC 30		-	-	-	-	-	X	X	-
GDC 33		-	-	-	-	-	-	-	-
GDC 34		X	-	-	-	-	-	-	-
GDC 35		X	-	X	X	-	-	-	-
GDC 37	-	-	-	X	X	-	-	-	-
GDC 38	-	-	-	-	X	-	-	-	-
GDC 40	-	-	-	-	X	-	-	-	-
GDC 41		-	-	-	-	-	-	-	-
GDC 43		-	-	-	-	-	-	-	-
GDC 44		-	-	-	-	-	-	-	-
GDC 46		-	-	-	-	-	-	-	-
GDC 50		-	-	-	-	-	-	-	-
GDC 54		-	-	-	-	-	-	-	-
GDC 55		X	-	-	-	-	-	-	-
GDC 56		-	-	-	-	-	-	-	-

TABLE 7.1-2.6
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY (1,7)

(Sheet 3 of 5)

	I & C POWER SUPPLY(3)	RHR ISOLATION VALVES	REFUELING INTERLOCKS(8)	ACCUMULATOR MO. VALVES	SWITCHOVER INJECTION TO RECIRCULATION	PROCESS & EFF RADIATION MONITORS	RCPB LEAK DETECTION	INTERLOCKS RCS PRESSURE CONTROL	FIRE PROTECTION(6)
GDC 57		-	-	-	-	-	-	-	-
GDC 63		-	-	-	-	X	-	-	-
GDC 64		-	-	-	-	X	-	-	-
RG 1.6		-	-	-	-	-	-	-	-
RG 1.7		-	-	-	-	-	-	-	-
RG 1.11		-	-	-	-	-	-	-	-
RG 1.12(2)									
RG 1.22		-	-	-	X	-	-	-	-
RG 1.29		X	-	X	X	X	X	-	-
RG 1.30		X	-	X	X	X	X	-	-
RG 1.32		-	-	-	-	-	-	-	-
RG 1.45		-	-	-	-	X	X	-	-
RG 1.47		-	-	-	X	-	-	-	-
RG 1.53		X	-	X	-	X	-	-	-
RG 1.62		-	-	-	-	-	-	-	-
RG 1.63(3)									

TABLE 7.1-2.6
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY (1,7)

(Sheet 4 of 5)

	I & C POWER SUPPLY(3)	RHR ISOLATION VALVES	REFUELING INTERLOCKS(8)	ACCUMULATOR MO. VALVES	SWITCHOVER INJECTION TO RECIRCULATION	PROCESS & EFF RADIATION MONITORS	RCPB LEAK DETECTION	INTERLOCKS RCS PRESSURE CONTROL	FIRE PROTECTION(6)
RG 1.67(4)									
RG 1.68(5)									
RG 1.70		X	-	X	X	X	X	X	-
RG 1.73		-	-	X	-	-	-	-	-
RG 1.75		X	-	X	X	-	-	-	-
RG 1.78		-	-	-	-	-	-	-	-
RG 1.80(5)						78			
RG 1.89		X	-	X	X	X	-	-	-
RG 1.95		-	-	-	-	-	-	-	-
RG 1.97		-	-	X	X	X	-	-	-
RG 1.100		X	-	X	X	X	-	-	-
RG 1.105		X	-	X	X	X	-	-	-
RG 1.118		-	-	-	-	-	-	-	-
RG 1.120(6)						78			
IEEE STD 279		-	-	-	-	-	-	-	-
IEEE STD 308		-	-	-	-	-	-	-	-

TABLE 7.1-2.6
SAFETY RELATED INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY (1,7)

(Sheet 5 of 5)

	I & C POWER SUPPLY(3)	RHR ISOLATION VALVES	REFUELING INTERLOCKS(8)	ACCUMULATOR MO. VALVES	SWITCHOVER INJECTION TO RECIRCULATION	PROCESS & EFF RADIATION MONITORS	RCPB LEAK DETECTION	INTERLOCKS RCS PRESSURE CONTROL	FIRE PROTECTION(6)
IEEE STD 317(3)									
IEEE STD 323		X	-	X	X	X	-	-	-
IEEE STD 336		X	-	X	X	X	X	-	-
IEEE STD 338		-	-	-	X	-	-	-	-
IEEE STD 344		X	-	X	X	X	-	-	-
IEEE STD 379		X	-	X	X	X	-	-	-
IEEE STD 384		X	-	X	X	X	-	-	-

Notes:

1. The scope of applicability is instrumentation & control.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3. and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section 9.5.1 and Appendix 1A(B).
7. X Signifies compliance as discussed or qualified in the FSAR sections referenced in Table 7.1-1. - Not applicable.
8. Refer to FSAR Sections 7.6.3 and 9.1.4.

TABLE 7.1-2.7
INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR CONTROL SYSTEMS NOT REQUIRED FOR SAFETY (1,7)

(Sheet 1 of 4)

	NSSS CONTROL SYSTEMS	BOP CONTROL SYSTEMS
GDC 1	-	-
GDC 2	-	-
GDC 3	-	-
GDC 4	-	-
GDC 5	-	-
GDC 10	-	-
GDC 12	-	-
GDC 13	X	X
GDC 15	-	-
GDC 19	X	X
GDC 20	-	-
GDC 21	-	-
GDC 22	-	-
GDC 23	-	-
GDC 24	X	X
GDC 25	-	-
GDC 26	X	-
GDC 27	-	-
GDC 28	-	-
GDC 29	-	-
GDC 30	-	-
GDC 33	-	-

TABLE 7.1-2.7
INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR CONTROL SYSTEMS NOT REQUIRED FOR SAFETY (1,7)

(Sheet 2 of 4)

	NSSS CONTROL SYSTEMS	BOP CONTROL SYSTEMS
GDC 34	-	-
GDC 35	-	-
GDC 37	-	-
GDC 38	-	-
GDC 40	-	-
GDC 41	-	-
GDC 43	-	-
GDC 44	-	-
GDC 46	-	-
GDC 50	-	-
GDC 54	-	-
GDC 55	-	-
GDC 56	-	-
GDC 57	-	-
GDC 63	-	-
GDC 64	-	-
RG 1.6	-	-
RG 1.7	-	-
RG 1.11	-	-
RG 1.12(2)		
RG 1.22	-	-

TABLE 7.1-2.7
INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR CONTROL SYSTEMS NOT REQUIRED FOR SAFETY (1,7)

(Sheet 3 of 4)

	NSSS CONTROL SYSTEMS	BOP CONTROL SYSTEMS
RG 1.29	-	-
RG 1.30	-	-
RG 1.32	-	-
RG 1.45	-	-
RG 1.47	-	-
RG 1.53	-	-
RG 1.62	-	-
RG 1.63(3)		
RG 1.67(4H4)		
RG 1.68(5)		
RG 1.70	X	X
RG 1.75	-	-
RG 1.78	-	-
RG 1.80(5)		
RG 1.89	-	-
RG 1.95	-	-
RG 1.97	-	-
RG 1.100	-	-
RG 1.105	-	-
RG 1.118	-	-
RG 1.120(6)		

TABLE 7.1-2.7
INSTRUMENTATION & CONTROL SYSTEMS/CODES, STANDARDS & GUIDES/APPLICABILITY MATRIX FOR CONTROL SYSTEMS NOT REQUIRED FOR SAFETY (1,7)

(Sheet 4 of 4)

	NSSS CONTROL SYSTEMS	BOP CONTROL SYSTEMS
IEEE STD 279[8]	X	-
IEEE STD 308	-	-
IEEE STD 317[3]		
IEEE STD 323	-	-
IEEE STD 336	-	
IEEE STD 338	-	-
IEEE STD 344	-	-
IEEE STD 379	-	-
IEEE STD 384	-	-

Notes:

1. The scope of applicability is instrumentation & control.
2. Refer to FSAR Section 3.7B.4 and Appendix 1A(B).
3. Refer to FSAR Section 8.3.1.2.1 and Appendix 1A(B).
4. Refer to FSAR Section 3.9B.3.3. and Appendix 1A(B).
5. Refer to FSAR Section 14.2 and Appendix 1A(B).
6. Refer to FSAR Section 9.5.1 and Appendix 1A(B).
7. X Signifies compliance as discussed or qualified in the FSAR Sections referenced in Table 7.1-1.

   - Not applicable.
8. Compliance with Paragraph 4.7 of IEEE Std. 279, Control and Protection System Interaction.

TABLE 7.1-3
EICSB BRANCH TECHNICAL POSITIONS

(Sheet 1 of 2)

Criteria	Title	Information Related to Conformance Discussed In:
BTP EICSB-1	Backfitting of the Protection and Emergency Power Systems of Nuclear Reactors	Not Applicable
BTP EICSB-3	Isolation of Low Pressure Systems From the High Pressure Reactor Coolant System	5.4.7.2.4, 7.6.2
BTP EICSB-4 (PSB)	Requirements on Motor-Operated Valves in the ECCS Accumulator Lines	6.3.1, 6.3.2, 7.6
BTP EICSB-5	Scram Breaker Test Requirements - Technical Specifications	7.2.2.2.3.10
BTP EICSB-9	Definition and Use of “Channel - Calibration” - Technical Specifications	7.2.2.2.3.10
BTP EICSB-10	Electrical and Mechanical Equipment Seismic Qualification Program	See RG 1.100, 3.10(N), 3.10(B), 1A(N), 1A(B)
BTP EICSB-12	Protection System Trip Point Changes for Operation with Reactor Coolant Pumps Out of Service	7.2.2.2.1,
BTP EICSB-13	Design Criteria for Auxiliary Feedwater Systems	10.4.9, Table 10.4-9
BTP EICSB-14	Spurious Withdrawals of Single Control Rods in Pressurized Water Reactors	3.1.3.1, 3.1.3.6, Chapter 15, 7.7.2.2
BTP EICSB-15(PSB)	Reactor Coolant Pump Breaker Qualifications	7.2.1.1.2.4, 8.3.1.1.2, 15.3, 5.4, 15.2, 7.1.2.5
BTP EICSB-16	Control Element Assembly (CEA) Interlocks in Combustion Engineering Reactors	Not Applicable

TABLE 7.1-3
EICSB BRANCH TECHNICAL POSITIONS

(Sheet 2 of 2)

Criteria	Title	Information Related to Conformance Discussed In:
BTP EICSB-18	Application of the Single Failure Criteria to Manually-Controlled Electrically-Operated Valves	6.2.4.1.3, 6.3.1, 6.3.2.2.12, 6.3.2.5, 6.3.5.5, 7.6.4
BTP EICSB-19	Acceptability of Design Criteria for Hydrogen Mixing and Drywell Vacuum Relief Systems	Not Applicable
BTP EICSB-20	Design of Instrumentation and Controls Provided to Accomplish Changeover from Injection to Recirculation Mode	7.6.5, 6.3.2.8, Table 6.3-7
BTP EICSB-21	Guidance for Application of Regulatory Guide 1.47	7.1.2.6
BTP EICSB-22	Guidance for Application of Regulatory Guide 1.22	7.1.2.5
BTP EICSB-23	Qualification of Safety-Related Display Instrumentation for Post-Accident Condition Monitoring and Safe Shutdown	7.5
BTP EICSB-24	Testing of Reactor Trip System and Engineered Safety Feature Actuation System Sensor Response Times	7.1.2.11
BTP EICSB-25	Guidance for the Interpretation of General Design Criterion 37 for Testing the Operability of the Emergency Core Cooling System as a Whole	3.1, 6.3.4
BTP EICSB-26	Requirements for Reactor Protection System Anticipatory Trips	7.2.1.1.2.(6)
BTP EICSB-27	Design Criteria for Thermal Overload Protection for Motors of Motor-Operated Valves	8.3.1.2

7.2 REACTOR AND TRIP SYSTEM

7.2.1 DESCRIPTION

7.2.1.1 System Description

The Reactor Trip System (RTS) automatically keeps the reactor operating within a safe region by shutting down the reactor whenever the limits of the region are approached. The safe operating region is defined by several considerations such as mechanical/hydraulic limitations on equipment, and heat transfer phenomena. Therefore the RTS keeps surveillance on process variables which are directly related to equipment mechanical limitations, such as pressure, pressurizer water level (to prevent water discharge through safety valves, and uncovering heaters) and also on variables which directly affect the heat transfer capability of the reactor (e.g., flow and reactor coolant temperatures). Still other parameters utilized in the RTS are calculated from various process variables. In any event, whenever a direct process or calculated variable exceeds a setpoint the reactor will be shutdown in order to protect against either gross damage to fuel cladding or loss of system integrity which could lead to release of radioactive fission products into the Containment.

The following systems make up the RTS (see References [1], [2], [3], and [5] for additional background information).

1. Process Instrumentation and Control System.

2. Nuclear Instrumentation System.

3. Solid State Logic Protection System.

4. Reactor trip switchgear.

5. Manual actuation circuit.

The RTS consists of sensors which, when connected with analog circuitry consisting of two to four redundant channels monitor various plant parameters, and digital circuitry, consisting of two redundant logic trains, which receives inputs from the analog protection channels to complete the logic necessary to automatically open the reactor trip breakers.

Each of the two trains, A and B, is capable of opening a separate and independent reactor trip breaker, RTA and RTB, respectively. The two trip breakers in series connect three phase alternating current (AC) power from the rod drive motor generator sets to the rod drive power cabinets, as shown on Figure 7.2-1, Sheet 2. During plant power operation, a direct current (DC) undervoltage coil on each reactor trip breaker holds a trip plunger against its spring, allowing the power to be available at the rod control power supply cabinets. For reactor trip, a loss of DC voltage to the undervoltage coil releases the trip plunger and trips open the breaker. When either of the trip breakers opens, power is interrupted to the rod drive power supply, and the control rods fall, by gravity, into the core. The rods cannot be withdrawn until the trip breakers are manually reset. The trip breakers cannot be reset until the abnormal condition which initiated the trip is corrected. Bypass breakers BYA and BYB are provided to permit testing of the trip breakers, as discussed in Section 7.2.2.2.3.

7.2.1.1.1 Functional Performance Requirements

The RTS automatically initiates reactor trip:

1. Whenever necessary to prevent fuel damage for an anticipated operational transient (Condition II).

2. To limit core damage for infrequent faults (Condition III),

3. So that the energy generated in the core is compatible with the design provisions to protect the reactor coolant pressure boundary for limiting fault conditions (Condition IV).

The RTS initiates a turbine trip signal whenever reactor trip is initiated to prevent the reactivity insertion that would otherwise result from excessive reactor system cooldown to avoid unnecessary actuation of the Engineered Safety Features Actuation System.

The RTS provides for manual initiation of reactor trip by operator action.

7.2.1.1.2 Reactor Trips

The various reactor trip circuits automatically open the reactor trip breakers whenever a condition monitored by the RTS reaches a preset level. To ensure a reliable system, high quality design, components, manufacturing, quality control and testing is used. In addition to redundant channels and trains, the design approach provides a RTS which monitors numerous system variables, therefore providing protection system functional diversity. The extent of this diversity has been evaluated for a wide variety of postulated accidents.

Table 7.2-1 provides a list of reactor trips which are described below.

1. Nuclear overpower trips

   The specific trip functions generated are as follows:

   a. Power range high neutron flux trip

      The power range high neutron flux trip circuit trips the reactor when two out of the four power range channels exceed the trip setpoint.

      There are two bistables, each with its own trip setting used for a high and a low range trip setting. The high trip setting provides protection during normal power operation and is always active. The low trip setting, which provides protection during startup, can be manually bypassed when two out of the four power range channels read above approximately 10 percent power (P-10). Three out of the four channels below 10 percent automatically reinstate the trip function. Refer to Table 7.2-2 for a listing of all protection system interlocks.

   b. Intermediate range high neutron flux trip

      The intermediate range high neutron flux trip circuit trips the reactor when one out of the two intermediate range channels exceed the trip setpoint. This trip, which

provides protection during reactor startup, can be manually blocked if two out of four power range channels are above approximately 10 percent power (P-10). Three out of the four power range channels below this value automatically reinstate the intermediate range high neutron flux trip. The intermediate range channels (including detectors) are separate from the power range channels. The intermediate range channels can be in-bypassed at the nuclear instrumentation racks to permit channel testing during plant shutdown or prior to startup. This bypass action is annunciated on the control board.

c. Source range high neutron flux trip

The source range high neutron flux trip circuit trips the reactor when one out of the two source range channels exceeds the trip setpoint. This trip, which provides protection during reactor startup and plant shutdown, can be manually bypassed when one out of the two intermediate range channels reads above the P-6 setpoint value and is automatically reinstated when both intermediate range channels decrease below the P-6 setpoint value. This trip is also automatically bypassed by two out of four logic from the power range protection interlock (P-10). This trip function can also be reinstated below P-10 by an administrative action requiring manual actuation of two control board mounted switches. Each switch will reinstate the trip function in one out of the two protection logic trains. The source range trip point is set between the P-6 setpoint (source range cutoff power level) and the maximum source range power level. The channels can be individually bypassed at the nuclear instrumentation racks to permit channel testing during plant shutdown or prior to startup. This bypass action is annunciated on the control board.

d. Power range high positive neutron flux rate trip

This circuit trips the reactor when a sudden abnormal increase in nuclear power occurs in two out of four power range channels. This trip provides protection against uncontrolled RCCA bank withdrawal and rod ejection accidents of low worth from midpower and is always active.

Figure 7.2-1, Sheet 3, shows the logic for all of the nuclear overpower and positive rate trips.

2. Core thermal overpower trips

The N-16 system installed at CPNPP provides protection equivalent to that of over-power ΔT protection assumed in WCAP-9227 (Reference 5 of 15.1).

In addition to the specific trip functions generated, as discussed in the following subsections, a spectrum of steam line break accidents were analyzed to determine any part of the spectrum in which protection from thermal overpower trips (overpower or overtemperature N-16) might be required. Based on this analysis, it was determined that thermal overpower trips would be required to function for a range of small to intermediate size steam line breaks. Calculation of the containment pressure and temperature transients associated with this range of steam line breaks shows that the maximum containment temperature for which the thermal overpower trips will be required is 200°F.

If the containment temperature exceeds 200°F, a high containment pressure safety injection and reactor trip signal will be actuated, and the thermal overpower instrumentation will not be required. Thus, 200°F has been established as the maximum qualification temperature required for the N-16 detectors which feed into the thermal overpower trips.

The basis for the equations for the Overtemperature and Overpower setpoints can be understood by examining Figure 15.0-1 in Chapter 15. The Overpower and Overtemperature N-16 protection limits are established such that these two trips, in conjunction with the High and Low Pressurizer Pressure reactor trips and the steam generator safety valve limits, will provide protection against exceeding the core thermal limits. These trips are equivalent to the Overpower and Overtemperature ΔT trips, using N-16 as a power measurement rather than ΔT, and using T_{avg} as a combination of N-16 and T_{cold} rather than a combination of T_{hot} and T_{cold} as in the $\Delta T/T_{avg}$ design.

The specific trip functions generated are as follows:

a. Overtemperature N-16 trip

This trip protects the core against low DNBR and trips the reactor on coincidence as listed in Table 7.2-1 with one set of temperature measurements per loop. The setpoint for this trip is continuously calculated by analog circuitry for each loop by solving the following equation:

$$Q_{setpoint} = K_1 - K_2 \frac{(1+\tau_1 S)}{(1+\tau_2 S)} (T_{cold} - T°cold) + K_3(P - 2235) - f_1(\Delta\varphi)$$

Where:

T_{cold}	=	Measured value of Cold Leg Temperature (°F)
T°cold	=	indicated T_{cold} at rated thermal power
P	=	pressurizer pressure (psig)
K_1	=	preset bias
K_2	=	preset gain which compensates for the effect of temperature on the DNB limits
K_3	=	preset gain which compensates for the effect of pressure on the DNB limits
τ_1, τ_2	=	preset constants which compensate for piping and instrument time delay

S	=	laplace transform operator (seconds^{-1})
$f_1(\Delta\phi)$	=	a function of the neutron flux difference between average of the upper two and average of the lower two calibrated ion chambers readings (refer to Figure 7.2-2)

A separate ion chamber unit supplies the flux signal for each Overtemperature N-16 trip channel.

The basis for the Overtemperature N-16 is to prevent exceeding the DNB limits. As shown in Figure 15.0-1, for a given flow rate and power distribution, these limits are a function of T_{cold} and pressure; hence the Overtemperature N-16 setpoint equation is a function of T_{cold} and pressure. Dynamic compensation terms are included to account for the response time of the T_{cold} measurement. The $f_1(\Delta\phi)$ term is used to reduce the trip setpoint to account for the effects of axial power distributions more severe than the design values illustrated in Figure 15.0-1. Again, this is the same basis and same form of setpoint equations as the Overtemperature ΔT trip.

Increases in $f_1(\Delta\phi)$ beyond a pre-defined deadband result in a decrease in trip setpoint. Refer to Figure 7.2-2.

The required one pressurizer pressure parameter per loop is obtained from separate sensors connected to three pressure taps at the top of the pressurizer. Four pressurizer pressure signals are obtained from the three taps by connecting one of the taps to two pressure transmitters. Refer to Section 7.2.2.3.3 for an analysis of this arrangement.

Figure 7.2-1, Sheet 5, shows the logic for Overtemperature N-16 trip function.

The actual core power, which is used for comparison to the trip setpoint described above, is measured with an N-16 power meter as described in reference 5. There is one power meter on each loop. Figure 7.7-18 shows the complete functional diagram of the Comanche Peak N-16 system, including all compensation terms.

Some minor differences exist between the Comanche Peak N-16 system and that described in WCAP-9190 as described below. The functional diagram given in WCAP-9190 applies to a digital system while that given for Comanche Peak (Figure 7.7-18) applies to an analog system. The Comanche Peak system contains a temperature compensation term not contained in WCAP-9190. Finally, the N-16 system in WCAP-9190 contains a background radiation compensation term which is not in the N-16 system given herein. This compensation is only required if significant fuel failures exists, as discussed in WCAP-9190. If fuel failures should occur during operation, the N-16 power meter will read high which means the trip point will be reached earlier than required. Thus, the plant is always protected.

b. Overpower N-16 trip

This trip protects against excessive power (fuel rod rating protection) and trips the reactor on coincidence as listed in Table 7.2-1, with one set of temperature measurements per loop. The setpoint for each channel is continuously calculated using the following equation:

$$Q \text{ setpoint} = K_4 - f_2(\Delta\phi)$$

Where

$f_2(\Delta\phi)$ = a function of the neutron flux difference between the average of the upper two and average of the lower two calibrated ion chamber current readings. However, $f_2(\Delta\phi)=0$.

K_4 = A preset bias

The source of flux information is identical to that of the Overtemperature N-16 trip and the resultant Q setpoint is compared to the same Q. Figure 7.2-1, Sheet 5, shows the logic for this trip function.

The basis for the Overpower N-16 trip is to prevent the core thermal power from exceeding 118% of rated power consistent with the design basis discussed in Sections 4.4.2.11.6 and 4.3.2.2.6. This basis is the same as for the Overpower ΔT setpoint although the resulting equation is somewhat simpler due to the direct measurement of power. (For the ΔT equation, additional terms are required to account for the change in the heat capacity of water with temperature and for the relatively longer measurement time response associated with the T_{avg} and ΔT measurements.) The $f_2(\Delta\phi)$ term was retained, identical to the Overpower ΔT setpoint, to allow the capability for dealing with severe axial power distributions.

3. Reactor Coolant System pressurizer pressure and water level trips

The specific trip functions generated are as follows:

a. Pressurizer low pressure trip

The purpose of this trip is to protect against low pressure which could lead to DNB. The parameter being sensed is reactor coolant pressure as measured in the pressurizer. Above P-7 the reactor is tripped when the pressurizer pressure measurements (compensated for rate of change) fall below preset limits. This trip is blocked below P-7 to permit startup. The trip logic and interlocks are given in Table 7.2-1.

The trip logic is shown on Figure 7.2-1, Sheet 6.

b. Pressurizer high pressure trip

The purpose of this trip is to protect the Reactor Coolant System against system overpressure.

The same sensors and transmitters used for the pressurizer low pressure trip are used for the high pressure trip except that separate bistables are used for trip. These bistables trip when uncompensated pressurizer pressure signals exceed preset limits on coincidence as listed in Table 7.2-1. There are no interlocks or permissives associated with this trip function.

The logic for this trip is shown on Figure 7.2-1, Sheet 6.

c. Pressurizer high water level trip

This trip is provided as a back-up to the high pressurizer pressure trip and serves to prevent water relief through the pressurizer safety valves. This trip is blocked below P-7 to permit startup. The coincidence logic and interlocks of pressurizer high water level signals are given in Table 7.2-1.

The trip logic for this function is shown on Figure 7.2-1, Sheet 6.

4. Reactor Coolant System low flow trips

These trips protect the core from DNB in the event of a loss of coolant flow situation. Figure 7.2-1, Sheet 5, shows the logic for these trips. The means of sensing the loss of coolant flow are as follows:

a. Low reactor coolant flow

Reactor coolant flow is sensed by three differential pressure transmitters which are connected to a network of four elbow taps in each coolant loop. Each transmitter provides a signal to a low flow bistable. A trip signal from two out of three bistables indicates a low flow in the loop.

The coincidence logic and interlocks are given in Table 7.2-1.

b. Reactor coolant pump undervoltage trip (BOP Scope)

This trip is required in order to protect against low flow which can result from loss of voltage to more than one reactor coolant pump motor (e.g., from plant blackout or reactor coolant pump breakers opening).

There is one undervoltage sensing relay connected for each pump at the motor side of each reactor coolant pump breaker. These relays provide an output signal when the pump voltage goes below approximately 70 percent of rated voltage. Signals from these relays are time delayed to prevent spurious trips caused by short term voltage perturbations. The coincidence logic and interlocks are given in Table 7.2-1.

c. Reactor coolant pump underfrequency trip (BOP Scope)

This trip protects against low flow resulting from pump underfrequency, for example a major power grid frequency disturbance. The function of this trip is to trip the reactor for an underfrequency condition greater than approximately 2.4 hertz (Hz)/ second. The setpoint of the underfrequency relays is adjustable between 54 and 59 Hz.

There is one underfrequency sensing relay for each reactor coolant pump motor. Signals from relays for any two of the pump motors (time delayed up to approximately 0.1 second to prevent spurious trips caused by short term frequency perturbations) will trip the reactor if the power level is above P-7.

Undervoltage, underfrequency and associated time delay relays for the four channels are housed in a protection relay rack. The protection relay rack is divided into four separate compartments, similar to the Solid State Protection System input cabinet described in Section 7.1.2.3 to maintain channel independence. Potential transformers from which the input signals for undervoltage and underfrequency relays are derived and the protection relay rack are classified as Class 1E and are located in a Seismic Category I structure.

5. Steam generator trip/Low-low steam generator water level trip

The specific trip function generated is the low-low steam generator water level trip. This trip protects the reactor from loss of heat sink in the event of a loss of feedwater to the steam generators. This trip is actuated on two out of four low-low water level signals occurring in any steam generator.

The logic is shown on Figure 7.2-1, Sheet 7.

6. Reactor trip on a turbine trip (anticipatory) (BOP Scope)

The reactor trip on a turbine trip is actuated by two out of three logic from emergency trip fluid pressure signals or by all closed signals from the turbine steam stop valves. A turbine trip causes a direct reactor trip above P-9. The reactor trip on turbine trip provides additional protection and conservatism beyond that required for the health and safety of the public. This trip is included as part of good engineering practice and prudent design. Though the trip is non-safety related, calibration and maintenance of the associated hardware is subject to an operational QA program. No credit is taken in any of the safety analyses (Chapter 15) for this trip.

The turbine provides anticipatory trips to the Reactor Protection System from contacts which change position when the turbine stop valves close or when the turbine emergency trip fluid pressure goes below its setpoint.

The turbine trip fluid pressure switches and turbine stop valve closed limit switches are procured as Class 1E. Seismic qualification is not required as described herein. Wiring from these switches in the turbine building to the reactor protection system cabinets is routed in accordance with the same separation criteria, applicable to protection

channels I, II, III and IV and is in rigid steel conduits dedicated exclusively for these signals. These conduits are shown on plant physical layout drawings.

The reactor trip system interfaces with the turbine trip system as shown on Figure 7.2-1, sheet 16. The turbine trip fluid is dumped by non-Class 1E turbine trip solenoid valves. The reactor trip system is separated and isolated from the non-Class 1E main turbine trip solenoids by Class 1E isolation devices, typically described in Section 8.3.1.2.1 subparagraph b.1.

One of the design bases considered in the Protection System is the possibility of an earthquake. With respect to these contacts, their functioning is unrelated to a seismic event in that they are anticipatory to other diverse parameters which cause reactor trip. The contacts are shut during plant operation and open to cause reactor trip when the turbine is tripped. No power is provided to the Protection System from the contacts; they merely serve to interrupt power to cause reactor trip. This design functions in a de-energize-to-trip fashion to cause a plant trip if power is interrupted in the trip circuitry. This ensures that the Protection System will in no way be degraded by this anticipatory trip because seismic design considerations do not form part of the design bases for anticipatory trip sensors. (The Reactor Protection System cabinets which receive the inputs from the anticipatory trip sensors are, of course, seismically qualified as discussed in Section 3.10N.) The anticipatory trips thus meet the Institute of Electrical and Electronic Engineers (IEEE) Standard 279-1971, including redundancy, separation, single failure, etc. Seismic qualification of the contacts sensors is not required.

The logic for this trip is shown on Figure 7.2-1, Sheet 16.

7. Safety injection signal actuation trip

A reactor trip occurs when the Safety Injection System is actuated. The means of actuating the Safety Injection System are described in Section 7.3. This trip protects the core against any event which would actuate safety injection.

Figure 7.2-1, Sheet 8, shows the logic for this trip.

8. Manual Trip

The manual trip consists of two switches with two outputs on each switch. One output is used to actuate the train A trip breaker, the other output actuates the train B trip breaker. Operation of either switch de-energizes the undervoltage coils in both breakers through both logic trains. At the same time the breaker shunt coils in both breakers are energized. Figure 7.2-3 shows the switch arrangement and Figure 7.2-1 Sheet 3 shows the manual trip logic. There are no interlocks which can block this trip.

The bases for this design are in Regulatory Guide 1.62 and single failure criteria of IEEE-279 and IEEE-379. It provides that either switch will initiate the required action at the system level (both trains). Failure of one switch will not prevent system actuation to open both reactor trip breakers.

In order to maintain separation between wiring associated with different trains, redundant safety train wiring is not terminated on a single device. Backup manual actuation

switches link the separate trains by mechanical means to provide greater reliability of operator action for the manual reactor trip (as well as Engineered Safety Features actuations). The linked switches are themselves redundant so that operation of either set of linked switches will actuate safety trains “A” and “B” simultaneously.

7.2.1.1.3 Reactor Trip System Interlocks

1. Power escalation permissivies

The overpower protection provided by the out of core nuclear instrumentation consists of three discrete, but overlapping, ranges. Continuation of startup operation or power increase requires a permissive signal from the higher range instrumentation channels before the lower range level trips can be manually blocked by the operator.

A one out of two intermediate range permissive signal (P-6) is required prior to source range trip blocking and detector high voltage cutoff. Source range trips are automatically reactivated and high voltage restored when both intermediate range channels are below the permissive (P-6) setpoint. There are two manual reset switches for administratively reactivating the source range level trip and detector high voltage when between the permissive P-6 and P-10 set points, if required. Source range level trip block and high voltage cutoff are always maintained when above the permissive P-10 setpoint.

The intermediate range level trip and power range (low setpoint) trip can only be blocked after satisfactory operation and permissive information are obtained from two out of four power range channels. Four individual blocking switches are provided so that the low range power range trip and intermediate range trip can be independently blocked (one switch for each train). These trips are automatically reactivated when any three out of four power range channels are below the permissive (P-10) setpoint, thus ensuring automatic activation to more restrictive trip protection.

The development of permissives P-6 and P-10 is shown on Figure 7.2-1, Sheet 4. All of the permissives are digital; they are derived from analog signals in the nuclear power range and intermediate range channels.

Refer to Table 7.2-2 for the list of Protection System interlocks.

2. Blocks of reactor trips at low power

Interlock P-7 blocks a reactor trip at low power (below approximately 10 percent of full power) on a low reactor coolant flow in more than one loop, reactor coolant pump undervoltage, reactor coolant pump underfrequency, pressurizer low pressure or pressurizer high water level. Refer to Figure 7.2-1, Sheets 5 and 6 for permissive applications. The low power signal is derived from three out of four power range neutron flux signals below the setpoint in coincidence with two out of two turbine impulse chamber pressure signals below the setpoint (low plant load). See Figure 7.2-1, Sheets 4 and 16, for the derivation of P-7.

The P-8 interlock blocks a reactor trip when the plant is below approximately 48 percent of full power, on a low reactor coolant flow in any one loop. The block action (absence of the interlock signal) occurs when three out of four neutron flux power range signals are

below the setpoint. Thus, below the P-8 setpoint, the reactor will be allowed to operate with one inactive loop and trip will not occur until two loops are indicating low flow. Refer to Figure 7.2-1, Sheet 4, for derivation of P-8, and Sheet 5 for applicable logic.

The P-9 interlock blocks a reactor trip following a turbine trip when the plant is below approximately 50% of full power. The plant is designed for 50% load rejection capability (40% steam dump and 10% control rods insertion). The absence of the P-9 interlock (three out of four power range flux signals below the setpoint) blocks the immediate reactor trip following a turbine trip. Refer to Figure 7.2-1, Sh. 4 for derivation of P-9 and Sh. 16 for applicable logic.

Refer to Table 7.2-2 for the list of Protection System blocks.

7.2.1.1.4 Coolant Temperature, N-16 and Power Range Neutron Detector Sensor Arrangement

The bypass line T_{hot} and T_{cold} measurements are, as a result of incorporation of the Comanche Peak Protection and Surveillance Upgrade Package, being replaced with an N-16 power monitor and an in-line T_{cold} measurement. In addition, the present two-section power range neutron detectors are being replaced with four-section detectors (used for both protection and control functions).

7.2.1.1.4.1 In-Line Fast Response Thermowell Installation of Cold Leg Temperature Measurement

The fast response in-line T_{cold} measurement is provided by an RTD in a thin wall thermowell installed in the reactor coolant cold leg piping. The thin-wall thermowell is designed to provide improved time response for a T_{cold} measurement which uses a thermowell installation. The system includes two cold leg RTD's per loop, one active RTD and one installed spare, both wired to the process protection cabinet. Utilization of thermowells allows replacement of the RTD's without violating the reactor coolant boundary.

The cold leg bypass nozzles, and crossover leg bypass return nozzles were installed in the Comanche Peak reactor coolant system piping; however, all of these connections have been plugged and are not used.

7.2.1.1.4.2 N-16 Sensor Arrangement

The N-16 power monitor (see Reference 5) measures the thermal power of the NSSS by detecting the amount of N-16 present in the coolant system. The N-16 population (an isotope of the nitrogen generated by neutron activation of oxygen contained in the water) which is present in the primary coolant is directly proportional to the fission rate in the core. Decay of the N-16 isotope produces high-energy gamma rays which penetrate the wall of the high-pressure piping. As a result, the N-16 concentration in the primary coolant can be monitored by measuring the gamma radiation outside of the primary coolant piping.

The N-16 gamma radiation is detected by ion chambers located on the hot leg piping of each coolant loop. Detectors are mounted on opposite sides of the hot leg so that asymmetries due to geometry effects can be compensated for the measured N-16.

A total of four N-16 gamma detectors and mounting assemblies are provided for each loop and will be strapped on the RC piping. All four are mounted on the hot leg outside the biological shield and as close to the reactor vessel as possible.

An N-16 Transit Time Flow Meter (TTFM) is also included in the upgrade package. The removal of the RTD bypass lines requires that an alternative method of flow verification measurement be provided during plant startup and periodically (once per cycle) throughout plant life. The TTFM is portable and not connected to the N-16 detectors during normal operation; therefore, it is normally independent from the protection equipment. Interface by plug-in connectors is made only at the time of flow calibration. The N-16 TTFM performance data will be collected during the first cycle of CPNPP Unit 1 for review and evaluation. This data will be submitted to the NRC to further support the uncertainty analysis of the N-16 TTFM.

The N-16 Flow Measurement System allows the use of the two N-16 power monitor detectors, in conjunction with two additional N-16 detectors, to verify the flow in each of the reactor coolant loops to an improved flow calibration accuracy. Thus four N-16 detectors per loop are used for flow measurement, with two of the four per loop also being used for power measurements.

Since the N-16 detectors are externally mounted on the reactor coolant system piping, no loop penetrations are required. This system requires no flow calibration and introduces no pressure drop. Therefore, no primary system changes are needed. The system requires only the availability of containment penetrations to connect the detector output to an electronics rack in the control room.

To check the flow in any loop, a portable N-16 TTFM electronic rack would be positioned at the appropriate N-16 power monitor channel and connected to the gamma detectors by means of an interface panel located in an upgrade protection cabinet. Prior to connection of the TTFM, all of the bistables associated with that N-16 power monitor channel would be put in the tripped or bypass mode (as is done for test purposes) to preclude any control (TTFM) and protection (N-16 power) interaction. The TTFM utilizes the detectors of the N-16 power monitor.

The Transit Time Flow Meter (TTFM) accurately measures the reactor coolant primary loop flow by sensing the radioactive signature of the N-16 isotope. Cross-correlation techniques are used to determine the mean transit time of the N-16 signature from an upstream to a downstream detector. By knowing the distance between the detectors, the measured transit time, and the piping internal diameter, the volumetric flow in each coolant loop can be determined. This direct conversion of N-16 activity to RCS flow is accomplished with an accuracy of $\pm$ 1.3 percent. For additional conservatism, the Technical Specifications state an accuracy value of $\pm$ 1.8 percent.

7.2.1.1.4.3 Power Range Neutron Detector Assembly Installation

Four-section power range neutron detector assemblies (four sets) are provided to replace the two-section detectors and mount in the existing detector assembly.

Each of the four-section power range detector assemblies is supplied with a junction box (mounted outside the biological shield) for distribution of the common high voltage to each of the four sections. A similar junction box is supplied for each set of four N-16 detectors (mounted in the hot leg area).

7.2.1.1.5 Pressurizer Water Level Reference Leg Arrangement

The design of the pressurizer water level instrumentation employs the usual tank level arrangement using differential pressure between an upper and a lower tap on a column of water. A reference leg connected to the upper tap is kept full of water by condensation of steam at the top of the leg.

7.2.1.1.6 Analog System

The analog system consists of two instrumentation systems: the Process Instrumentation System and the Nuclear Instrumentation System.

Process instrumentation includes those devices (and their interconnection into systems) which measure temperature, pressure, fluid flow, fluid level as in tanks or vessels, and occasionally physiochemical parameters such as fluid conductivity or chemical concentration. Process instrumentation specifically excludes nuclear and radiation measurements. The process instrumentation includes the process measuring devices, power supplies, indicators, recorders, alarm actuating devices, controllers, signal conditioning devices, etc., which are necessary for day-to-day operation of the Nuclear Steam Supply System as well as for monitoring the plant and providing initiation of protective functions upon approach to unsafe plant conditions.

The primary function of nuclear instrumentation is to protect the reactor by monitoring the neutron flux and generating appropriate trips and alarms for various phases of reactor operating and shutdown conditions. It also provides a secondary control function and indicates reactor status during startup and power operation. The Nuclear Instrumentation System uses information from three separate types of instrumentation channels to provide three discrete protection levels. Each range of instrumentation (source, intermediate, and power) provides the necessary overpower reactor trip protection required during operation in that range. The overlap of instrument ranges provides reliable continuous protection beginning with source level through the intermediate and low power level. As the reactor power increases, the overpower protection level is increased by administrative procedures after satisfactory higher range instrumentation operation is obtained. Automatic reset to more restrictive trip protection is provided when reducing power.

Various types of neutron detectors, with appropriate solid state electronic circuitry, are used to monitor the leakage neutron flux from a completely shutdown condition to 120 percent of full power. The power range channels are capable of recording overpower excursions up to 200 percent of full power. The neutron flux covers a wide range between these extremes. Therefore, monitoring with several ranges of instrumentation is necessary.

The lowest range (“source” range) covers six decades of leakage neutron flux. The lowest observed count rate depends on the strength of the neutron sources in the core and the core multiplication associated with the shutdown reactivity. This is generally greater than two counts per seconds. The next range (“intermediate” range) covers eight decades. Detectors and instrumentation are chosen to provide overlap between the higher portion of the source range and the lower portion of the intermediate range. The highest range of instrumentation (“power” range) covers approximately two decades of the total instrumentation range. This is a linear range that overlaps with the higher portion of the intermediate range.

The system described above provides control room indication and recording of signals proportional to reactor neutron flux during core loading, shutdown, startup and power operation, as well as during subsequent refueling. Startup rate indication for the source and intermediate range channels is provided at the control board. Reactor trip, rod stop, control and alarm signals are transmitted to the Reactor Control and Protection System for automatic plant control. Equipment failures and test status information are annunciated in the control room.

Refer to References [1] and [2] for additional background information on the process and nuclear instrumentation.

7.2.1.1.7 Solid State Logic Protection System

The Solid State Logic Protection System takes binary inputs (voltage/no voltage) from the process and nuclear instrument channels corresponding to conditions (normal/abnormal) of plant parameters. The system combines these signals in the required logic combination and generates a trip signal (no voltage) to the undervoltage coils of the reactor trip circuit breakers when the necessary combination of signals occur. The system also provides annunciator, status light and computer input signals which indicate the condition of bistable input signals, partial trip and full trip functions and the status of the various blocking, permissive and actuation functions. Refer to Reference [3] for additional background information.

In addition the system includes means for semiautomatic testing of the logic circuits.

Problems have been experienced by other operating nuclear power plants in the testing of the solid state protection system. The problems identified include failure of the automatic test equipment to test the integrated circuit ZIC (a NAND gate in the scram breaker undervoltage card), an isolation problem in the general warning circuit, and the use of diodes with insufficient voltage rating. Though these problems appear to be different, they all relate to the same concern; in this case, an undetectable failure in the General Warning Alarm reactor trip circuits in the Solid State Protection System. The consequences of the failure were reported by the individual operating plant, but independent reporters have produced different descriptions of the same thing. Westinghouse assessed the incident, reported the generic impact to the NRC, and submitted for approval circuit modifications to resolve the deficiency. NRC Inspection and Enforcements witnessed verification tests which Westinghouse conducted after receipt of NRC acceptance (see Reference [6]). These changes have been implemented in the System for CPNPP, and are confirmed and documented in accordance with established Westinghouse PWR OA procedures.

The NRC has also indentified another potential problem, which has been found at other operating plants, and that is the modification of previously qualified components during production by their vendors which upon modification could not satisfy the original functional-requirements. Such a modification was found to have occurred at CPNPP with Westinghouse-supplied AR relays with latch assembly used in protection system equipment. This particular modification has been completed prior to shipments of the equipment to the site. Confirmation and documentation of the modification is in accordance with established Westinghouse PWR QA procedures. Changes to the equipment resolve a tolerance problem discovered and reported to the NRC by Westinghouse. The modified relay was requalified for the nuclear application.

In addition, the NRC noted that at some operating plants BFD-31 relays were not qualified for the equalizing voltage. The I & C protection systems for CPNPP do not employ BFD relays. Refer to Section 7.2.2.2.3 for further discussion of solid state logic testing.

7.2.1.1.8 Isolation Amplifiers

In certain applications, Westinghouse considers it advantageous to employ control signals derived from individual protection channels through isolation amplifiers contained in the protection channel, as permitted by IEEE Standard 279-1971.

In all of these cases, analog signals derived from protection channels for nonprotective functions are obtained through isolation amplifiers located in the analog protection racks. By definition, nonprotective functions include those signals used for control, remote process indication, and computer monitoring.

7.2.1.1.9 Energy Supply and Environmental Variations

The energy supply for the RTS, including the voltage and frequency variations, is described in Section 7.6 and Chapter 8. The environmental variations, throughout which the system will perform, is given in Sections 3.11N and 8.3.

The reactor protection system is designed in a manner to prevent the load dispatch system from interfering with the safety actions. This is accomplished by means of not interconnecting the Vistra OpCo load dispatch system to the CPNPP reactor protection system.

Also the turbine loading is dependent on the reactor operator action on the turbine control system.

7.2.1.1.10 Setpoints

The setpoints that require trip action are given in the Technical Specifications. A detailed discussion on setpoints is found in Section 7.1.2.1.9.

7.2.1.1.11 Seismic Design

The seismic design considerations for the RTS are given in Section 3.10N. This design meets the requirements of Criterion 2 of the 1971 General Design Criteria (GDC).

7.2.1.2 Design Basis Information

The information given below presents the design basis information requested by Section 3 of IEEE Standard 279-1971. Functional logic diagrams are presented in Figure 7.2-1.

7.2.1.2.1 Generating Station Conditions

The following are the generating station conditions requiring reactor trip.

1. DNBR approaching the safety analysis limit applicable to the specific DNB correlation being used.

2. Power density (kilowatts per foot) approaching rated value for Condition II events (see Section 4.2 for fuel design limits).

3. Reactor Coolant System overpressure creating stresses approaching the limits specified in Section 5.2.

7.2.1.2.2 Generating Station Variables

The following are the variables required to be monitored in order to provide reactor trips (see Table 7.2-1). Except for the Reactor Coolant Pump Underfrequency/Undervoltage sensors, refer to Figure 7.1-3 for the location of the sensors which provide these trip inputs. Refer to explanatory note 7 on Sheet 1 of Figure 7.1-3 for the location of the and Reactor Coolant Pump Underfrequency/Undervoltage sensors.

1. Neutron flux.

2. Reactor coolant temperature.

3. Reactor Coolant System pressure (pressurizer pressure).

4. Pressurizer water level.

5. Reactor coolant flow.

6. Reactor coolant pump operational status (voltage and frequency).

7. Steam generator water level.

8. Turbine-generator operational status (trip fluid pressure and stop valve position).

9. N-16 gamma flux.

7.2.1.2.3 Spatially Dependent Variables

The following variable is spatially dependent:

1. The $_{Tavg}$ input for which the measurement of N-16 is used is the only indirect measurement which has spatial dependence because the N-16 power measurement has radial and azimuthal spatial dependence. As discussed in Section 7.2.1.1.4.2. the N-16 gamma radiation is detected by ion chambers located on the hot leg piping of each coolant loop. Detectors are mounted on the opposite sides of the hot leg so that asymmetries due to geometrical effects can be averaged out for the measured N-16. The signals from the two detectors are summed in the N-16 power module, to form one loop's uncompensated N-16 signal.

7.2.1.2.4 Limits, Margins and Setpoints

The parameter values that will require reactor trip are given in Section 15.0 and the Technical Specifications. Chapter 15 proves that the setpoints used in the Technical Specifications are conservative.

The setpoints for the various functions in the RTS have been analytically determined such that the operational limits so prescribed will prevent fuel rod clad damage and loss of integrity of the Reactor Coolant System as a result of any American Nuclear Society (ANS) Condition II event (anticipated malfunction). As such, during any ANS Condition II event, the RTS limits the following parameters to:

1. Minimum DNBR ≥ applicable limit value.

2. Maximum system pressure = 2750 pounds per square inch absolute (psia).

3. Fuel rod maximum linear power for determination of protection setpoints = 18.0 kilowatts per foot (kw/ft)

The accident analyses described in Chapter 15 demonstrate that the functional requirements as specified for the RTS are adequate to meet the above considerations, even assuming, for conservatism, adverse combinations of instrument errors. A discussion of the safety limits associated with the reactor core and Reactor Coolant System, plus the limiting safety system setpoints, are presented in the Technical Specifications.

7.2.1.2.5 Abnormal Events

The malfunctions, accidents or other unusual events which could physically damage RTS components or could cause environmental changes and the protection criteria followed are:

1. Earthquakes (see Sections 3.7N and 3.7B).

2. Fire (see Section 9.5.1).

3. Deleted.

4. Missiles (see Section 3.5).

5. Flood (see Section 3.4).

6. Wind and tornadoes (see Section 3.3).

7. Loss of coolant accidents (see Section 6.2)

8. Steam line breaks (see Section 6.2)

9. Loss of ventilation (see Section 9.4)

The RTS fulfills the requirements of IEEE Standard 279-1971 to provide automatic protection and to provide initiating signals to mitigate the consequences of faulted conditions.

7.2.1.2.6 Minimum Performance Requirements

1. RTS response times

   RTS response time is defined in Section 7.1. The maximum allowable time delays tabulated in Table 7.2-3 represent functional design values as opposed to acceptance criteria for response time testing. The Technical Specification response time requirements ensure that reactor protection system functional operability is maintained relative to these reference values. Refer to Section 7.1.2.11 for a discussion of periodic response time verification capabilities.

2. Reactor trip accuracies

   Accuracy is defined in Section 7.1. Reactor trip accuracies are tabulated in Table 7.2-3. An additional discussion on accuracy is found in Section 7.1.2.1.9.

3. Protection System ranges

   The Protection System ranges are tabulated in Table 7.2-3. Range selection for the instrumentation covers the expected range of the process variable being monitored during power operation. Limiting setpoints are at least 5 percent from the end of the instrument span.

7.2.1.3 Final Systems Drawings

Functional block diagrams, electrical elementaries and other drawings required to assure electrical separation and perform a safety review are provided in the Safety-Related Drawing Package (see Section 1.7).

7.2.2 ANALYSES

7.2.2.1 Failure Mode and Effects Analyses

An analysis of the RTS has been performed. Results of this study and a fault tree analysis are presented in Reference [4].

7.2.2.2 Evaluation of Design Limits

While most setpoints used in the Reactor Protection System are fixed, there are variable setpoints, most notably the Overtemperature N-16 setpoint. All setpoints in the RTS have been selected on the basis of engineering design or safety studies. The capability of the RTS to prevent loss of integrity of the fuel cladding and/or Reactor Coolant System pressure boundary during Condition II and III transients is demonstrated in Chapter 15. These accident analyses are carried out using those setpoints determined from results of the engineering design studies. Setpoint limits are presented in the Technical Specifications. A discussion of the intent for each of the various reactor trips and the accident analyses (where appropriate) which utilizes this trip is presented below. It should be noted that the selected trip setpoints all provide for margin before protection action is actually required to allow for uncertainties and instrument errors. The design meets the requirements of Criteria 10 and 20 of the 1971 GDC.

7.2.2.2.1 Trip Setpoint Discussion

The DNBR existing at any point in the core for a given core design can be determined as a function of the core inlet temperature, power output, operating pressure and flow. Consequently, core safety limits in terms of a DNBR equal to a unit specific or applicable DNBR limit value for the hot channel can be developed as a function of core power, T_{cold} and pressure for a specified flow as illustrated by the solid lines in Figure 15.0-1. Also shown as a dashed line in Figure 15.0-1 is a line equivalent to 118 percent of power representing the overpower (kW/ft) limit on the fuel. The dashed lines indicate the maximum permissible setpoint (N-16) as a function of T_{cold} and pressure for the overtemperature and overpower reactor trip. Actual setpoint constants in the equation representing the dashed lines are as given in the Technical Specifications. These values are conservative to allow for instrument errors. The design meets the requirements of Criteria 10, 15, 20 and 29 of the 1971 GDC.

DNBR is not a directly measurable quantity; however, the process variables that determine DNBR are sensed and evaluated. Small isolated changes in various process variables may not individually result in violation of a core safety limit, whereas the combined variations, over sufficient time, may cause the overpower or overtemperature safety limit to be exceeded. The design concept of the RTS takes cognizance of this situation by providing reactor trips associated with individual process variables in addition to the overpower/overtemperature safety limit trips. Process variable trips prevent reactor operation whenever a change in the monitored value is such that a core or system safety limit is in danger of being exceeded should operation continue. Basically, the high pressure, low pressure and Overpower/Overtemperature N-16 trips provide sufficient protection for slow transients as opposed to such trips as low flow or high flux which will trip the reactor for rapid changes in flow or flux, respectively, that would result in fuel damage before actuation of the slower responding N-16 trips could be effected.

Therefore, the RTS has been designed to provide protection for fuel cladding and Reactor Coolant System pressure boundary integrity where: 1) a rapid change in a single variable or factor could potentially result in exceeding a core or a system safety limit, and 2) a slow change in one or more variables will have an integrated effect which will cause safety limits to be exceeded. Overall, the RTS offers diverse and comprehensive protection against fuel cladding failure and/or loss of Reactor Coolant System integrity for Condition II and III accidents. This is demonstrated by Table 7.2-4 which lists the various trips of the RTS, the corresponding Technical Specification on safety limits and safety system settings and the appropriate accident discussed in the safety analyses in which the trip could be utilized.

It should be noted that the RTS automatically provides core protection during nonstandard operating configurations, i.e., operation with a loop out of service. Although operating with a loop out of service over an extended time is considered to be an unlikely event, no Protection System setpoints need to be reset. This is because the nominal value of the power (P-8) interlock setpoint restricts the power such that a DNBR less than the limit value will not be realized during any Condition II transients occurring during this mode of operation. This restricted power is considerably below the boundary of permissible values as defined by the core safety limits for operation with a loop out of service. Thus the P-8 interlock acts essentially as a high nuclear power reactor trip when operating with one loop not in service. The Technical Specifications do not allow continued operation in Modes 1 or 2 with less than four reactor coolant loops in service.

The design meets the requirements of Criterion 21 of the 1971 GDC.

Preoperational testing is performed on RTS components and systems to determine equipment readiness for startup. This testing serves as a further evaluation of the system design.

Analyses of the results of Condition I, II, III and IV events including considerations of instrumentation installed to mitigate their consequences are presented in Chapter 15. The instrumentation installed to mitigate the consequences of load rejection and turbine trip is given in Section 7.4.

7.2.2.2.2 Reactor Coolant Flow Measurement

The elbow taps used on each loop in the primary coolant system are instrument devices that indicate the status of the reactor coolant flow. The basic function of this device is to provide information as to whether or not a reduction in flow has occurred. The correlation between flow and elbow tap signal is given by the following equations:

$$P = k(w)^2 \text{ and } P_o = k(w_o)^2$$

where P_o is the pressure differential at the reference flow w_o, P is the pressure differential at the corresponding flow w, and k is a constant of proportionality. The full flow reference point is established during plant startup following each refueling outage. The low flow trip point is then established by extrapolating along the correlation curve. The expected absolute accuracy of the channel is within ± 10 percent of full flow and field results have shown the repeatability of the trip point to be within ± 1 percent.

7.2.2.2.3 Evaluation of Compliance to Applicable Codes and Standards

The RTS meets the criteria of the GDC as indicated. The RTS meets the requirements of Section 4 of IEEE Standard 279-1971, as indicated below.

1. General functional requirement

   The Protection System automatically initiates appropriate protective action whenever a condition monitored by the system reaches a preset level. Functional performance requirements are given in Section 7.2.1.1.1. Section 7.2.1.2.4 presents a discussion of limits, margins and levels; Section 7.2.1.2.5 discusses unusual (abnormal) events; and Section 7.2.1.2.6 presents minimum performance requirements.

2. Single failure criterion

   The Protection System is designed to provide two, three, or four instrumentation channels for each protective function and two logic train circuits. These redundant channels and trains are electrically isolated and physically separated. Thus, any single failure within a channel or train will not prevent protective action at the system level when required. Loss of input power, the most likely mode of failure, to a channel or logic train will result in a signal calling for a trip. This design meets the requirements of Criterion 23 of the 1971 GDC.

To prevent the occurrence of common mode failures, such additional measures as functional diversity, physical separation, and testing as well as administrative control during design, production, installation and operation are employed, as discussed in Reference [4]. The design meets the requirements of Criteria 21 and 22 of the 1971 GDC.

3. Quality of components and modules

For a discussion on the quality of the components and modules used in the RTS, see References [1], [2], [3], and [5]. The quality assurance applied conforms to Criterion 1 of the 1971 GDC.

4. Equipment qualification

For a discussion of the type tests made to verify the performance requirements, refer to Section 3.11N and 3.11B. The test results demonstrate that the design meets the requirements of Criterion 4 of the 1971 GDC.

5. Channel integrity

Protection System channels required to operate in accident conditions maintain necessary functional capability under extremes of conditions relating to environment, energy supply, malfunctions, and accidents. The energy supply for the RTS is described in Sections 7.6 and 8.3.1.1.13. The environmental variations throughout which the system will perform is given in Section 3.11N.

6. Independence

Channel independence is carried throughout the system, extending from the sensor through to the devices actuating the protective function. Physical separation is used to achieve separation of redundant transmitters. Separation of wiring is achieved using separate wireways, cable trays, conduit runs and Containment penetrations for each redundant channel. Redundant analog equipment is separated by locating modules in different protection cabinets. Each redundant protection channel set is energized from a separate AC power feed. This design meets the requirements of Criterion 21 of the 1971 GDC.

Two reactor trip breakers are actuated by two separate logic matrices which interrupt power to the control rod drive mechanisms. The breaker main contacts are connected in series with the power supply so that opening either breaker interrupts power to all full length control rod drive mechanisms, permitting the rods to free fall into the core. Refer to Figure 7.1-1.

The design philosophy is to make maximum use of a wide variety of measurements. The Protection System continuously monitors numerous diverse system variables. Generally, two or more diverse protection functions would terminate an accident before intolerable consequences could occur. This design meets the requirements of Criterion 22 of the 1971 GDC.

7. Control and Protection System interaction

The Protection System is designed to be independent of the Control System. In certain applications the control signals and other nonprotective functions are derived from individual protective channels through isolation amplifiers. The isolation amplifiers are classified as part of the Protection System and are located in the analog protective racks. Nonprotective functions include those signals used for control, remote process indication, and computer monitoring. The isolation amplifiers are designed such that a short circuit, open circuit, or the application of credible fault voltages from within the cabinets on the isolated output portion of the circuit (i.e., the nonprotective side of the circuit) will not affect the input (protective) side of the circuit. The signals obtained through the isolation amplifiers are never returned to the protective racks. This design meets the requirements of Criterion 24 of the 1971 GDC and Section 4.7 of IEEE Standard 279-1971.

The results of applying various malfunction conditions on the output portion of the isolation amplifiers show that no significant disturbance to the isolation amplifier input signal occurred.

8. Derivation of system inputs

To the extent feasible and practical, Protection System inputs are derived from signals which are direct measures of the desired variables. Variables monitored for the various reactor trips are listed in Section 7.2.1.2.2.

9. Capability for sensor checks

The operational availability of each system input sensor during reactor operation is accomplished by cross checking between channels that bear a known relationship to each other and that have readouts available. Channel checks are discussed in the Technical Specifications Bases.

10. Capability for testing

The RTS is capable of being tested during power operation. Where only parts of the system are tested at any one time, the testing sequence provides the necessary overlap between the parts to assure complete system operation. The testing capabilities are in conformance with Regulatory Guide 1.22 as discussed in Section 7.1.2.5.

The Protection System is designed to permit periodic testing of the analog channel portion of the RTS during reactor power operation without initiating a protective action unless a trip condition actually exists. This is because of the ability to test the analog system in bypass and the coincidence logic required for reactor trip. These tests may be performed at any plant power from cold shutdown to full power. Before starting any of these tests with the plant at power, all redundant reactor trip channels associated with the function to be tested must be in the normal (untripped) mode in order to avoid spurious trips. Setpoints are referenced in the CPNPP technical specifications and/or the Instrument and Control Equipment List.

Analog Channel Tests

Analog channel testing is performed at the analog instrumentation rack set by individually introducing dummy input signals into the instrumentation channels and observing the tripping of the appropriate output bistables. Process analog output to the logic circuitry may be bypassed during individual channel test by a test switch which, when thrown, maintains the associated logic input. The bypass capability is designed so that credible failures (e.g., relays) will not result in a function being automatically placed in a bypassed condition. Both a local status light and a main control room annunciator are provided to indicate a bypassed condition. Interruption of the bistable output to the logic circuitry for any cause (test, maintenance purposes, or removed from service) will cause that portion of the logic to be actuated (partial trip) accompanied by a partial trip alarm and channel status light actuation in the Control Room. Each channel contains those switches, test points, etc., necessary to test the channel. Refer to References [1] and [2] for additional background information.

The following periodic tests of the analog channels of the protection circuits are performed:

a. T_{avg} and N-16 protection channel testing.

b. Pressurizer pressure protection channel testing.

c. Pressurizer water level protection channel testing.

d. Steam generator water level protection channel testing.

e. Reactor coolant low flow protection channels.

f. Impulse chamber pressure channel testing.

g. Steam pressure protection channels.

h. Containment pressure.

Nuclear Instrumentation Channel Tests

The power range channels of the Nuclear Instrumentation System may be tested by superimposing a test signal on the actual detector signal being received by the channel at the time of testing. These channels which produce a rod stop, permissive, or a reactor trip on one-out-of-four, two-out-of-four, or three-out-of-four logic are provided with a bypass function to prevent initiation of an undesired action from that system function during the period that one channel is in test. To permit testing in the bypass condition, a test panel is provided on each of the four NIS protection sets. Use of administrative controls will ensure that not more than one channel will be bypassed at a time. The bypass capability is designed so that credible failures (e.g., relays) will not result in a function being automatically placed in a bypassed condition. Annunciators in the main control room and bypass status lights on the bypass test panels are provided to indicate the bypassed condition.

To test a power range channel, a “TEST-OPERATE” switch is provided to require deliberate operator action and operation of which will initiate the “CHANNEL TEST” annunciator in the Control Room. Bistable operation may be tested by increasing the test signal to its trip setpoint and verifying bistable relay operation by control board annunciator and trip status lights.

If a channel is tested by superimposing a test signal on the actual detector signal, it should be noted that a valid trip signal would cause the channel under test to trip at a lower actual reactor power level. A reactor trip would occur when a second bistable trips.

A Nuclear Instrumentation System channel which can cause a reactor trip through one out of two protection logic (source or intermediate range) is provided with a bypass function which prevents the initiation of a reactor trip from that particular channel during the short period that it is undergoing test. These bypasses are annunciated in the Control Room.

The following represents the technical concept (not a circuit analysis) for testing flux rate trips while the plant is at power.

a. During normal operation at steady power level, the percent full power meter reading at the Nuclear Instrumentation System cabinets is recorded. The reading is the result of summing the outputs from detectors A and B in the circuits (A + B = % power).

b. A test signal, B' is added to the detector B signal such that:

   A + B + B' = X%

   where: X is a power level greater than 100% which is specified by station procedure. Additionally, a test signal A' is added to the detector A signal such that:

   A + A' + B + B' = Y% and

   Y - X = A' = the positive/negative rate trip percentage.

c. The test signal A' is added/removed by operation of the Operation Selector switch to effect a rapid positive/negative change in power level equal to A'. This is seen by the rate sensing circuits. The positive rate trip is confirmed in accordance with Technical Specifications.

The following periodic tests of the Nuclear Instrumentation System are performed (listed also are the modes during which that testing is permitted):

a. Source Range:	Analog Channel Operational Test - Any Mode[(a)] Channel Calibration - Any Mode[(a)] Analog Channel Calibration, High Flux at Shutdown - Any Mode[(a)] Signal Conditioning (Channel Calibration) - Any Mode[(a)]
b. Intermediate Range:	Analog Channel Operational Test - Mode 1 > 10% or, in modes 3, 4, 5, or 6. Neutron detector plateau - > 10% power
c. Power Range:	Analog Channel Operational Test - Any Mode Channel Calibration - Any Mode Detector Plateau - > 10% power

a) Not permitted with core alterations in progress

Any deviations noted during the performance of these tests are investigated and corrected in accordance with the established calibration and trouble shooting procedures provided in the plant technical manual for the Nuclear Instrumentation System. Control and protection trip settings are indicated in the CPNPP Technical Specifications and/or the Instrument and Controls Equipment List.

For additional background information on the Nuclear Instrumentation System, refer to Reference [2].

Undervoltage and Underfrequency Channel Tests

The test procedure for the undervoltage and underfrequency channels is as follows. Individual undervoltage channels are tested by a test switch which, when operated, de-energizes the undervoltage relay, operates the associated time delay relay, and in turn will actuate the corresponding portion of the logic (partial trip) accompanied by a partial trip alarm and channel status light actuation in the Control Room. Individual underfrequency channels are tested in the same way, except that the activation of an underfrequency relay requires that the test switch, when operated, connects the underfrequency relay to an external variable frequency test source.

Undervoltage, underfrequency and time delay relays are mounted in drawout cases to permit periodic testing of setpoint and accuracy of these relays.

Solid State Logic Testing

The reactor logic trains of the RTS are designed to be capable of complete testing at power. After the individual channel testing is complete, the logic matrices are tested from

the train A and train B logic rack test panels. This step provides overlap between the channel and logic portions of the test program. During this test, all of the logic inputs are actuated automatically in all combinations of trip and non-trip logic. Trip logic is not maintained sufficiently long enough to permit opening of the reactor trip breakers. The reactor trip undervoltage coils are “pulsed” in order to check continuity. During logic testing of one train, the other train can initiate any required protective functions. Annunciation is provided in the Control Room to indicate when a train is in test (train output bypassed) and when a reactor trip breaker is bypassed. Logic testing can be performed in less than 30 minutes.

A direct reactor trip resulting from undervoltage or underfrequency on the reactor coolant pump buses is provided as discussed in Section 7.2.1 and shown on Figure 7.2-1. The logic for these trips is capable of being tested during power operation. When parts of the trip are being tested, the sequence is such that an overlap is provided between parts so that a complete logic test is provided. Undervoltage and underfrequency sensors are tested as described above. Thus complete testing of the equipment discussed above is possible.

This design complies with the testing requirements of IEEE Standard 279-1971 and IEEE Standard 338-1971 discussed in Section 7.1.2.11.

The permissive and block interlocks associated with the RTS and Engineered Safety Features Actuation System are given on Tables 7.2-2 and 7.3-3 and designated protection or “P” interlocks. As a part of the Protection System, these interlocks are designed to meet the testing requirements of IEEE Standard 279-1971 and IEEE Standard 338-1971.

Testing of all Protection System interlocks is provided by the logic testing and semiautomatic testing capabilities of the Solid State Protection System. In the Solid State Protection System the undervoltage coils (reactor trip) and master relays (engineered safeguards actuation) are pulsed for all combinations of trip or actuation logic with and without the interlock signals. For example, reactor trip on low flow (two out of four loops showing two out of three low flow) is tested to verify operability of the trip above P-7 and non-trip below P-7 (see Figure 7.2-1, Sheet 5). Interlock testing may be performed at power.

Testing of the logic trains of the RTS includes a check of the input relays and a logic matrix check.

a. Check of input relays

Each Process Instrumentation System and Nuclear Instrumentation System channel bistable is placed in a trip mode causing one input relay in train A and one in train B to de-energize. A contact of each relay is connected to a universal logic printed circuit card. This card performs both the reactor trip and monitoring functions. Each reactor trip input relay contact causes a status lamp and an annunciator on the control board to operate. Either the train A or train B input relay operation will light the status lamp and annunciator.

Each train contains a multiplexing test switch. At the start of input relay testing, this switch (in either train) is placed in the A + B position. The A + B position alternately

allows information to be transmitted from the two trains to the control board. A steady status lamp and annunciator indicates that input relays in both trains have been de-energized. A flashing lamp means that the input relays in the two trains did not both de-energize. Contact inputs to the logic protection system such as reactor coolant pump bus underfrequency relays operate input relays which are tested by operating the remote contacts as described above and using the same type of indications as those provided for bistable input relays.

Actuation of the input relays provides the overlap between the testing of the logic protection system and the testing of those systems supplying the inputs to the logic protection system. Test indications are status lamps and annunciators on the control board. Inputs to the logic protection system are checked one channel at a time, leaving the other channels in service. For example, a function that trips the reactor when two out of four channels trip becomes a one out of three trip when one channel is placed in the trip mode. Both trains of the logic protection system remain in service during this portion of the test.

b. Check of logic matrices

Logic matrices are checked one train at a time. Input relays are not operated during this portion of the test. Reactor trips from the train being tested are inhibited with the use of the input error inhibit switch on the semiautomatic test panel in the train. At the completion of the logic matrix tests, one bistable in each protection set of process instrumentation or nuclear instrumentation is tripped to check closure of the input error inhibit switch contacts.

The logic test scheme uses pulse techniques to check the coincidence logic. All possible trip and non-trip combinations are checked. Pulses from the tester are applied to the inputs of the universal logic card at the same terminals that connect to the input relay contacts. Thus there is an overlap between the input relay check and the logic matrix check. Pulses are fed back from the reactor trip breaker undervoltage coil to the tester. The pulses are of such short duration that the reactor trip breaker undervoltage coil armature cannot respond mechanically.

Test indications that are provided are an annunciator in the Control Room indicating that reactor trips from the train have been blocked and that the train is being tested, and green and red lamps on the semiautomatic tester to indicate a good or bad logic matrix test. Protection capability provided during this portion of the test is from the train not being tested.

The testing capability meets the requirements of Criterion 21 of the 1971 GDC.

Testing of Reactor Trip Breakers

Normally, reactor trip breakers 52/RTA and 52/RTB are in service, and bypass breakers 52/BYA and 52/BYB are withdrawn (out of service). In testing the protection logic, pulse techniques are used to avoid tripping the reactor trip breakers. The following procedure describes the method used for testing the trip breakers:

a. With bypass breaker 52/BYA in the test position, manually close and trip it to verify its operation.

b. Rack in and close 52/BYA. Manually trip 52/RTA through a protection system logic matrix or by a Reactor Trip Test pushbutton via a keylocked test selector switch.

c. Reset 52/RTA.

d. Trip and rack out 52/BYA.

e. Repeat above steps to test trip breaker 52/RTB using bypass breaker 52/BYB.

Auxiliary contacts of the bypass breakers are connected into the alarm system of their respective trains such that if either train is placed in test while the bypass breaker of the other train is closed, both reactor trip breakers and both bypass breakers will automatically trip.

Auxiliary contacts of the bypass breakers are also connected in such a way that if an attempt is made to close the bypass breaker in one train while the bypass breaker of the other train is already closed, both bypass breakers will automatically trip.

The two bypass breakers train A and train B alarm systems operate separate annunciators in the Control Room. Bypassing of a protection train with either the bypass breaker or with the test switches will result in audible and visual indications.

The complete RTS is normally required to be in service. However, to permit on-line testing of the various protection channels or to permit continued operation in the event of a subsystem instrumentation channel failure, Technical Specification 3.3.1, defining the required number of operable channels, has been formulated. This Technical Specification also defines the required restriction to operation in the event that the channel operability requirements cannot be met.

11. Channel bypass or removal from operation

The Protection System is designed to permit periodic testing of the analog channel portion of the RTS during reactor power operation without initiating a protective action unless a trip condition actually exists. This is because of the ability to test the analog system in bypass and the coincidence logic required for reactor trip. Additional information is given in Section 7.2.2.2.3, item 10.

12. Operating bypasses

Where operating requirements necessitate automatic or manual bypass of a protective function, the design is such that the bypass is removed automatically whenever permissive conditions are not met. Devices used to achieve automatic removal of the bypass of a protective function are considered part of the Protective System and are designed in accordance with the criteria of this section. Indication is provided in the control room if some part of the system has been administratively bypassed or taken out of service.

13. Indication of bypasses

Channels associated with protection logic are provided with bypass functions which are annunciated in the control room, with the exception of undervoltage and underfrequency protection circuits.

Underfrequency and undervoltage relay testing is under administrative controls.

Each bypass breaker is separately annunciated in the Control Room by visible and audible alarms.

See discussion under item 10 above.

14. Access to means for bypassing

The design provides for administrative control of access to the means for manually bypassing channels or protective functions.

15. Multiple setpoints

For monitoring neutron flux, multiple setpoints are used. When a more restrictive trip setting becomes necessary to provide adequate protection for a particular mode of operation or set of operating conditions, the Protective System circuits are designed to provide positive means or administrative control to assure that the more restrictive trip setpoint is used. The devices used to prevent improper use of less restrictive trip settings are considered part of the Protective System and are designed in accordance with the criteria of this section.

16. Completion of protective action

The Protection System is so designed that, once initiated, a protective action goes to completion. Return to normal operation requires action by the operator.

17. Manual initiation

Switches are provided on the control board for manual initiation of protective action. Failure in the automatic system does not prevent the manual actuation of the protective functions. Manual actuation relies on the operation of a minimum of equipment.

18. Access

The design provides for administrative control of access to all setpoint adjustments, module calibration adjustments, and test points.

19. Identification of protective actions

Protective channel identification is discussed in Section 7.1.2.3. Indication is discussed in Item 20 below.

20. Information read out

The Protective System provides the operator with complete information pertinent to system status and safety. All transmitted signals (flow, pressure, temperature, etc.) which can cause a reactor trip will be either indicated or recorded for every channel, including all neutron flux power range currents (sum of two top detectors, sum of two bottom detectors, and difference of the two summed detector signals).

Any reactor trip will actuate an alarm and an annunciator. Such protective actions are indicated and identified down to the channel level.

Alarms and annunciators are also used to alert the operator of deviations from normal operating conditions so that he may take appropriate corrective action to avoid a reactor trip. Actuation of any rod stop or trip of any reactor trip channel will actuate an alarm.

21. System repair

The system is designed to facilitate the recognition, location, replacement, and repair of malfunctioning components or modules. Refer to the discussion in Item 10 above.

7.2.2.3 Specific Control and Protection Interactions

7.2.2.3.1 Neutron Flux

Four power range neutron flux channels are provided for overpower protection. An isolated auctioneered high signal is derived by auctioneering of the four channels for automatic rod control. If any channel fails in such a way as to produce a low output, that channel is incapable of proper overpower protection but will not cause control rod movement because of the auctioneer. Two out of four overpower trip logic will ensure an overpower trip if needed even with an independent failure in another channel.

In addition, channel deviation signals in the control system will give an alarm if any neutron flux channel deviates significantly from the average of the flux signals. Also, the control system will respond only to rapid changes in indicated neutron flux; slow changes or drifts are compensated by the temperature control signals. Finally, an over-power signal from any nuclear power range channel will block manual and automatic rod withdrawal. The setpoint for this rod stop is below the reactor trip setpoint.

7.2.2.3.2 Coolant Temperature

The accuracy of the in-line fast response thermowell-installed resistance temperature detector temperature measurements is demonstrated during plant startup tests by comparing temperature measurement from all resistance temperature detectors with one another as well as with the temperature measurements obtained from the wide range resistance temperature detector located in the hot leg and cold leg piping of each loop. The comparisons are done with the Reactor Coolant System in an isothermal condition. The linearity of the N-16 power measurements as a function of plant power is also checked during plant startup tests. The absolute value of N-16 versus plant power is not important, per se, as far as reactor protection is concerned. The RTS setpoints are based upon percentages of the indicated N-16 at nominal full power rather than on absolute values of ΔT. This is done to account for loop differences which

are inherent. Therefore the percent N-16 power scheme is relative, not absolute, and therefore provides better protective action without the expense of accuracy. For this reason, the linearity of the N-16 power signals as a function of power is of importance rather than the absolute values of N-16 power.

Reactor control is based upon signals derived from Protection System channels after isolation by isolation amplifiers such that no feedback effect can perturb the protection channels.

One of the inputs to the Rod Control System, which controls RCCA motion, is the average of the average temperatures (T-avg) for the four loops. Should one T-avg indication fail high, the average T-avg indication will be greater than the reference temperature, resulting in a demand for inward rod motion. With no operator intervention, the resulting mismatch between reactor power and turbine power will likely result in a reactor trip on low pressurizer pressure. If one T-avg indication was to fail low, the average T-avg indication would be less than the reference temperature, resulting in a demand for outward rod motion. The resulting T-err signal could be sufficient to drive rods out at maximum speed. Outward rod motion dies off as the following occurs:

1. The lead-lag characteristics of the T-avg circuit causes the circuit output to drop.

2. Auctioneered reactor power rises due to rod withdrawal.

Whether or not a high flux reactor trip occurs depends upon:

1. Initial bank D position.

2. The magnitude of the negative moderator temperature coefficient (MTC).

3. Magnitude of the doppler only power coefficient (DOPC).

As reactor power and temperature increase, the following can occur: C-2, C-3, and C-4 rod blocks: C-3 and C-4 turbine runbacks.

The scenario is bounded by the analyses of the control rod withdrawal events presented to FSAR Section 15.4. As needed operator intervention will consist of placing the rod control and pressurizer level control systems in manual until the failed channel is defeated. Once defeated, the averaging unit will use one average temperature input twice when developing the average T-avg.

In addition, channel deviation signals in the control system will give an alarm if any temperature channel deviates significantly from the average value. Automatic rod withdrawal blocks and turbine runback (power demand reduction) will also occur if any two out of four Overtemperature or Overpower N-16 channels indicate an adverse condition.

7.2.2.3.3 Pressurizer Pressure

The pressurizer pressure protection channel signals are used for high and low pressure protection and as inputs to the Overtemperature N-16 trip protection function. Isolated output signals from these channels are used for pressure control. These are used to control pressurizer

spray and heaters and power operated relief valves. Pressurizer pressure is sensed by fast response pressure transmitters.

A spurious high pressure signal from one channel can cause decreasing pressure by actuation of either spray or relief valves. Additional redundancy is provided in the low pressurizer pressure reactor trip and in the logic for safety injection to ensure low pressure protection.

Overpressure protection is based upon the positive surge of the reactor coolant produced as a result of turbine trip under full load, assuming the core continues to produce full power. The self-actuated safety valves are sized on the basis of steam flow from the pressurizer to accommodate this surge at a setpoint of 2500 psia and an accumulation of 3 percent. Note that no credit is taken for the relief capability provided by the power operated relief valves during this surge. However, the Power Operated Relief Valves (PORVs) provide operational control that limits the RCS pressure increase following transients. Therefore, the PORV actuation instrumentation is required to have a channel calibration every 18 months.

In addition, operation of any one of the power operated relief valves can maintain pressure below the high pressure trip point for most transients. The rate of pressure rise achievable with heaters is slow, and ample time and pressure alarms are available to alert the operator of the need for appropriate action.

Redundancy is not compromised by having a shared tap (see Section 7.2.1.1.2, item 2.a) since the logic for this trip is two out of four. If the shared tap is plugged, the affected channels will remain static. If the impulse line bursts, the indicated pressure will drop to zero. In either case the fault is easily detectable, and the protective function remains operable.

7.2.2.3.4 Pressurizer Water Level

The pressurizer has three differential pressure transmitters. Each transmitter employs an open column reference leg with a condensing pot to ensure that the reference leg maintains a constant level. The operation of the pressurizer water level measurement system is identical to that employed for measuring steam generator water level. See Section 7.2.2.3.5.

The pressurizer water level is used for the following safety functions: reactor trip on high water level and post-accident monitoring. Pressurizer water level provides no trip function following an accident which results in an adverse environment inside containment.

Isolated signals from the three pressurizer water level channels are used for pressurizer water level control. A failure in the level control system could fill or empty the pressurizer at a slow rate (on the order of half an hour or more). The high water level trip setpoint provides sufficient margin such that the undesirable condition of discharging liquid coolant through the safety valves is avoided. Even at full power conditions, which would produce the worst thermal expansion rates, a failure of the water level control would not lead to any liquid discharge through the safety valves. This is due to the automatic high pressurizer pressure reactor trip actuating at a pressure sufficiently below the safety valve setpoint.

A graph depicting level measurement error due to system pressure changes is shown in Figure 7.2-6. A plot similar to this is made available to the operators to ensure they are aware of potential level measurement error.

During required manual safety functions such as maintaining charging flow control within specified limits, the operator follows the applicable Emergency Operating Procedures (EOPs). This assures the operator of remaining within specified limits (accounting for adverse environmental errors). For example, the operator is instructed by the EOPs to re-establish operation of the pressurizer heaters only after verification of sufficient pressurizer heaters. Also, the operator is instructed to restore normal pressurizer level control after the containment temperatures are low enough to assure proper operation of the level control system. These actions ensure that, accounting for the post-accident level indication errors, the pressurizer heater will not be uncovered and the pressurizer will not become water solid.

7.2.2.3.5 Steam Generator Water Level

The steam generator narrow range and wide range water level detection system is located within containment and used to initiate safety actions. The steam generator narrow range water level detection system consists of four differential pressure measurement channels per steam generator. Each channel measures differential pressure between an upper and lower tap using an open column reference leg, a condensing pot to ensure that the reference leg maintains a constant level, and a differential pressure (level) transmitter.

The steam generator narrow range water level is used for the following safety functions:

1. Turbine trip and feedwater isolation on high-high steam generator water level,

2. Reactor trip on low-low steam generator water level,

3. Auxiliary feedwater initiation on low-low steam generator water level,

4. Post-accident monitoring.

Each steam generator has one wide range differential pressure transmitter. Each transmitter employs an open column reference leg with condensing pot to ensure that the reference leg maintains a constant level. When the steam generator level is high, the differential pressure between the vessel and the reference leg is smallest, as the steam generator level drops, the differential pressure increases. The steam generator wide range water level also has a post-accident monitoring function.

Transmitters are purchased such that the upper and lower range limits for the transmitters are greater than the upper and lower range values, i.e., the calibrated instrument range is not the design limit for the device.

The basic function of the reactor protection circuits associated with low-low steam generator water level is to preserve the steam generator heat sink for removal of long-term residual heat. Should a complete loss of feedwater occur, the reactor would be tripped on low-low steam generator water level. In addition redundant auxiliary feedwater pumps are provided to supply feedwater in order to maintain residual heat removal after trip. This reactor trip acts before the steam generators are dry. This reduces the required capacity, increases the time interval before auxiliary feedwater pumps are required, and minimizes the thermal transient on the Reactor Coolant System and steam generators. Therefore, a low-low steam generator water level reactor trip circuit is provided for each steam generator to ensure that sufficient initial thermal capacity is available in the steam generator at the start of the transient. Two-out-of-four low-low

steam generator water level trip logic ensures a reactor trip if needed even with an independent failure in another channel used for control and when degraded by an additional second postulated random failure.

A spurious low signal from the feedwater flow channel being used for control would cause an increase in feedwater flow. The mismatch between steam flow and feedwater flow produced by the spurious signal would actuate alarms to alert the operator of the situation in time for manual correction. If the condition continues, a two-out-of-three high-high steam generator water level signal in any loop, independent of the indicated feedwater flow, will cause feedwater isolation and trip the turbine. The high-high steam generator water level trip is an equipment protective trip preventing excessive moisture carryover which could damage the turbine blading.

The level control system for the steam generator uses one of the three steam generator level signals or a fourth signal, which is not used in the high-high steam generator logic. The fourth level signal shall always be used at CPNPP for steam generator level control, except when the signal is not operable or for testing. When this fourth level signal is used, the failure of a single level signal will not cause both a steam generator level logic to two-out-of-two. When the other level signal which is available for steam generator level control is selected and steam generator control is in automatic, the high-high steam generator isolation signal for the channel selected for steam generator level control will be placed in the tripped condition within the time limits specified in the Technical Spefications and shall remain in the tripped condition whenever steam generator level control is in automatic until the fourth level signal is operable and selected for control. This shall ensure that the requirements of IEEE-279 Section 4.7.3 and 10 CFR Part 50.55a(h) are adequately met.

In addition, the three element feedwater controller incorporates reset action on the level error signal, such that with expected controller settings a rapid increase or decrease in the flow signal would cause only a small change in level before the controller would compensate for the level error. A slow change in the feedwater signal would have no effect at all. A spurious low or high steam flow signal would have the same effect as high or low feedwater signal, discussed above.

A spurious high steam generator water level signal from the protection channel used for control will tend to close the feedwater valve. A spurious low steam generator water level signal will tend to open the feedwater valve. Before a reactor trip would occur, two-out-of-four channels in a loop would have to indicate a low-low water level. Any slow drift in the water level signal will permit the operator to respond to the level alarms and take corrective action.

High energy line breaks inside containment can result in the heating of level measurement reference legs. Increased reference leg water column temperature results in a decrease of the water column density with a consequent apparent increase in the indicated water level, i.e., apparent level exceeding actual level. The following formula can be used to calculate the magnitude of this bias:

$$E = \frac{H_L}{H} \frac{(P_L, cal - P_L)}{(P_f, cal - P_g, cal)}$$

Where:

E	=	Level error due to reference leg heatup, as a fraction of level span,
H	=	Level span = vertical distance between pressure taps - 233 in., nominal
H_L	=	Height of reference leg - 235 in., nominal
P_L, cal	=	Water density at containment temperature and system pressure for which the level indication system was calibrated. At CPNPP, the containment temperature is 120°F and the steam generator pressure is 1053 psia (70% power),
P_L	=	Water density in reference leg at the time of interest,
(P_f, cal-P_g, cal)	=	Difference between saturated water density and dry saturated steam density at the system pressure for which the level indication system was calibrated, 3.g., 1053 psia for CPNPP steam generators.

This procedure is based on the assumption that the tubing from the upper and lower taps, below the elevation of the lower tap, have the same temperatures at all times. Figure 7.2-4 shows the steam generator level bias as a function of reference leg temperature for CPNPP conditions.

In addition to the above reference leg density change under subcooled conditions, boiling could conceivably occur in the reference leg following depressurization of any steam generator with high containment temperature. This combination of conditions could only occur following a steam line or feedline rupture inside containment. If such boiling were to occur, it could cause a major bias in the indicated level for a short time period, in the extreme case indicating 100 percent level when the vessel is actually empty. However, containment analyses performed by Westinghouse indicate that such boiling would not occur.

A bias in indicated water level may also be introduced by changes in pressurizer or steam generator pressure, due to changes in the density of the saturated water and steam within those vessels. While prediction of the effects of rapid depressurization requires complex calculations for each specific case, the bias which would exist at low power under quiescent conditions can be calculated directly, using the following formula:

$$E = \frac{H_L}{H}\frac{(P_g - P_g, cal)}{(P_f, cal - P_g, cal)} + \frac{L}{H}\frac{(P_f - P_g)}{(P_f, cal - P_g, ccal)} - \frac{L}{H}$$

Where:

E	=	Level error due to density changes in both the vessel and the reference leg, as a fraction of level span,
L	=	True water level in the vessel, above the lower level tap,

P_f = Saturated water density at the pressure of interest,

P_g = Dry saturated steam density at the pressure of interest.

Note: Other symbols have the same meaning as in the previous formula.

For example, Figure 7.2-5 shows the true water level as a function of steam generator pressure and indicated level using the following CPNPP calibration conditions: containment temperature = 120°F, steam generator pressure = 1053 psia, and the reference leg at 120°F.

The only high energy line rupture within containment for which the steam generator water level provides the primary trip function is a secondary high energy line rupture from an initial high power condition. Based on its analyses and investigation, Westinghouse asserts that functions which have 95% probability setpoints (when including uncertainties using the approved Westinghouse setpoint methodology) within 5% of the top or bottom of the instrument range will not respond any differently than any other protection function. Because large steam generator pressure changes (Section 7.2.2.3.4) are not expected before trip, only the reference leg heatup effects need be considered, and not the effects of system pressure changes.

The basis for determination of the low-low setpoint is the loss of normal feedwater and feedline break events. The setpoints were determined by considering the level used in each of the analyses for each unit (0% of span in both cases). For each unit, the setpoint was determined by considering the following errors for feedline break:

1. Normal error, e.g., normal channel accuracy,

2. Post-accident effects on transmitter due to temperature,

3. Reference leg effects (post-accident heatup).

For loss of normal feedwater, only normal errors were considered since it does not result in an adverse environment.

At CPNPP the reference leg effects have been addressed in the determination of the low-low setpoints. Considering a steam line break containment temperature of 345°F, which bounds conditions for the feedline break, the error due to reference leg heatup will be approximately 14% span.

High-high steam generator water level trip is not required for accident situations that could cause significant errors in level indication. The setpoint of this trip will remain unchanged.

Steam generator wide range level does not provide any automatic trip functions.

A graph depicting the level measurement error due to steam generator reference leg heatup is shown in Figure 7.2-4. Error due to system pressure changes are shown in Figure 7.2-5. A plot similar to the latter figure is made available to the operators to ensure they are aware of the potential level measurement errors.

In addition, a remote possibility exists that the fluid in the open reference legs may flash to steam in the depressurized steam generators following a secondary high energy line rupture. To alert the operator to the possibility of erroneous indications, Westinghouse recommends that a Caution be inserted in all plant emergency instructions for indicated steam generator water level, which states that the operator should not rely upon steam generator water level indications in any depressurized steam generators following a high energy line rupture inside containment due to the possibility of reference leg boiling. The Westinghouse reference Emergency Operating Instructions take the post-accident indicated water level errors into account in the specification of the minimum levels required for safety injection terminations. Similar cautions are included in CPNPP emergency procedures.

7.2.2.4 Additional Postulated Accidents

Loss of plant instrument air or loss of component cooling water is discussed in Section 7.3.2. Load rejection and turbine trip are discussed in further detail in Section 7.7.

The control interlocks, called rod stops, that are provided to prevent abnormal power conditions which could result from excessive control rod withdrawal are discussed in Section 7.7.1.4 and listed on Table 7.7-1. Excessively high power operation (which is prevented by blocking of automatic rod withdrawal), if allowed to continue, might lead to a safety limit (as given in the Technical Specifications) being reached. Before such a limit is reached, protection will be available from the RTS. At the power levels of the rod block setpoints, safety limits have not been reached; and therefore these rod withdrawal stops do not come under the scope of safety-related systems, and are considered as control systems.

7.2.3 TESTS AND INSPECTIONS

The RTS meets the testing requirements of IEEE Standard 338-1971, as discussed in Section 7.1.2.11. The test-ability of the system is discussed in Section 7.2.2.2.3. The initial test intervals are specified in the Technical Specifications. Written test procedures and documentation, conforming to the requirements of IEEE Standard 338-1971, will be available for audit by responsible personnel. Periodic testing complies with Regulatory Guide 1.22 as discussed in Sections 7.1.2.5 and 7.2.2.2.3.

REFERENCES

1. Reid, J.B., “Process Instrumentation for Westinghouse Nuclear Steam Supply Systems,” WCAP-7913, January 1973.

2. Lipchak, J. B., “Nuclear Instrumentation System,” WCAP-8255, January 1974.

3. Katz, D. N., “Solid State Logic Protection System Description,” WCAP-7488-L (Proprietary), March 1971 and WCAP-7672 (Non- Proprietary), May 1971.

4. Gangloff, W. C. and Loftus, W. D., “An Evaluation of Solid State Logic Reactor Protection In Anticipated Transients,” WCAP-7706-L (Proprietary) and WCAP-7706 (Non-Proprietary), February 1971.

5. Graham, K. F., “N-16 Power Measuring System,” WCAP 9190 (Proprietary) and WCAP 9191 (Non-Proprietary), December, 1977.

6. Letter from R. Heineman (NRC) to C. Eicheldinger (Westinghouse) dated June 21, 1976 re: documentation of NRC acceptance of Solid State Protection System circuit modifications.

TABLE 7.2-1
LIST OF REACTOR TRIPS

(Sheet 1 of 2)

	Reactor Trip	Coincidence Logic	Interlocks	Comments
1.	Power range high neutron flux	2/4	Manual block of low setting permitted by P-10	High and low setting; manual block and automatic reset of low setting by P-10
2.	Intermediate range high neutron flux	1/2	Manual block permitted by P-10	Manual block and automatic reset
3.	Source range high neutron flux	1/2	Manual block permitted by P-6, interlocked with P-10	Manual block and automatic reset; automatic block above P-10
4.	Power range high positive neutron flux rate	2/4	No interlocks	-
5.	Not Used			
6.	Overtemperature N-16	2/4	No interlocks	-
7.	Overpower N-16	2/4	No interlocks	-
8.	Pressurizer low presure	2/4	Interlocked with P-7	Blocked below P-7
9.	Pressurizer high pressure	2/4	No interlocks	-
10.	Pressurizer high water level	2/3	Interlocked with P-7	Blocked below P-7

TABLE 7.2-1
LIST OF REACTOR TRIPS

(Sheet 2 of 2)

	Reactor Trip	Coincidence Logic	Interlocks	Comments
11.	Low reactor coolant flow	2/3 in any loop	Interlocked with P-7 and P-8	Low flow in one loop will cause a reactor trip when above P-8 and a low flow in two loops will cause a reactor trip when above P-7; blocked below P-7
12.	Reactor coolant pump undervoltage	2/4	Interlocked with P-7	Low voltage on all buses permitted below P-7.
13.	Reactor coolant pump underfrequency	2/4	Interlocked with P-7	Underfrequency on 2 motors will trip all reactor coolant pump breakers and cause reactor trip; reactor trip blocked below P-7
14.	Low-low steam generator water level	2/4 in any loop	No interlocks	-
15.	Safety injection signal	Coincident with actuation of safety injection	No interlocks	(See Section 7.3 for engineered safety features actuation conditions)
16.	Turbine (anticipatory) trip			
	a. Low trip fluid pressure	2/3	Interlocked with P-9	Blocked below P-9
	b. Turbine stop valve close	4/4	Interlocked with P-9	Blocked below P-9
17.	Manual	1/2	No interlocks	-

TABLE 7.2-2
PROTECTION SYSTEM INTERLOCKS

(Sheet 1 of 2)

Designation	Derivation	Function
	I Power Escalation Permissives	
P-6	Presence of P-6: 1/2 neutron flux (intermediate range) above setpoint	Allows manual block of source range reactor trip
	Absence of P-6: 2/2 neutron flux (intermediate range) below setpoint	Defeats the block of source range reactor trip
P-10	Presence of P-10: 2/4 neutron flux (power range) above setpoint	Allows manual block of power range (low setpoint) reactor trip
		Allows manual block of intermediate range reactor trip and intermediate range rod stops (C-1)
		Blocks source range reactor trip (back-up for P-6)
	Absence of P-10: 3/4 neutron flux (power range) below setpoint	Defeats the block of power range (low setpoint) reactor trip
		Defeats the block of intermediate range reactor trip and intermediate range rod stops (C-1)
		Input to P-7

TABLE 7.2-2
PROTECTION SYSTEM INTERLOCKS

(Sheet 2 of 2)

Designation	Derivation	Function
	II Blocks of Reactor Trips	
P-7	Absence of P-7: 3/4 neutron flux (power range) below set point (from P-10) and 2/2 turbine impulse chamber pressure below setpoint (from P-13)	Blocks reactor trip on: Low reactor coolant flow in more than one loop, undervoltage, underfrequency, pressurizer low pressure, and pressurizer high level
P-8	Absence of P-8: 3/4 neutron flux (power range) below setpoint	Blocks reactor trip on low reactor coolant flow in a single loop
P-9	Absence of P-9: 3/4 neutron flux (power range) below setpoint	Blocks reactor trip on turbine trip below approximately 50% power
P-13	2/2 turbine impulse chamber pressure below setpoint	Input to P-7

TABLE 7.2-3
REACTOR TRIP SYSTEM INSTRUMENTATION[a]

(Sheet 1 of 2)

	Reactor Trip Signal	Range	Trip Accuracy	Time Response(sec)
1.	Power range high neutron flux	1 to 120% full power	± 1% of full power	0.2
2.	Intermediate range high neutron flux	8 decades of neutron flux overlapping source range by 2 decades	± 5% of full scale ± 1% of full scale from 10% to 150% full power[b]	0.2
3.	Source range high neutron flux	6 decades of neutron flux (1 to 10^6 counts/ sec)	± 5% of full scale	0.2
4.	Power range high positive neutron flux rate	+15% of full power	± 5%[b]	0.2
5.	Not Used			
6.	Overtemperature N-16	N-16 0 to 150% power T_C 510 to 630°F P_{PRZR} 1700 to 2500 psig $F_{(\Delta\phi)}$ -50 to +50	± 4.0% span	N-16 1.0 T_C 6.0 P_{PRZR} 1.0
7.	Overpower N-16	N-16 0 to 150% power T_C 510 to 630°F	± 4.0% span	N-16 1.0 T_C 6.0
8.	Pressurizer low pressure	1700 to 2500 psig	± 18 psi (compensated signal)	0.6
9.	Pressurizer high pressure	1700 to 2500 psig	± 18 psi (non-compensated signal)	0.6

TABLE 7.2-3
REACTOR TRIP SYSTEM INSTRUMENTATION[a]

(Sheet 2 of 2)

	Reactor Trip Signal	Range	Trip Accuracy	Time Response(sec)
10.	Pressurizer high water level	Entire cylindrical portion of pressurizer (distance between taps)	2.25% of full range $\Delta\rho$ between taps at design temperature and pressure	1.2
11.	Low reactor coolant flow	$\Delta\rho$ equivalent of 0 to 120% of rated flow	2.75% $\Delta\rho$ span	0.3
12.	Reactor coolant pump undervoltage	0 to 100% rated voltage	±1%	0.7
13.	Reactor coolant pump underfrequency	50 to 65 Hz	±0.1 Hz	0.3
14.	Low-low steam generator water level	total distance between narrow range steam generator level taps	2.25% of $\Delta\rho$ span over pressure range of 700 to 1200 psig	1.2
15.	Turbine trip			0.3
	a. Stop valve position	N/A	±5% stroke (at closed position) of valve travel	
	b. Trip fluid pressure	0 to 200 psig	±10% of full scale	

a) Values represents functional requirements only

b) Reproducibility (see definitions in Section 7.1)

Accuracy of steam flow signal is ±3 percent of maximum calculated flow over the pressure range of 700 to 1200 psig.

TABLE 7.2-4
REACTOR TRIP CORRELATION

(Sheet 1 of 5)

	Trip[a]	Accident[b]	Technical Specification[c]
1.	Power range high neutron flux trip (low setpoint)	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal From a Subcritical or Low Power Startup Condition (15.4.1) Feedwater System Malfunctions that Result in a Decrease in Feedwater Temperature (15.1.1) Spectrum of Rod Cluster Control Assembly Ejection Accidents (15.4.8)	3.3.1, Table 3.3.1-1 #2b
2.	Power range high neutron flux trip (high setpoint)	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal From a Subcritical or Low Power Startup Condition (15.4.1) Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (15.4.2) Startup of an Inactive Reactor Coolant Pump at an Incorrect Temperature (15.4.4) Feedwater System Malfunctions that Result in a Decrease in Feedwater Temperature (15.1.1) Excessive Increase in Secondary Steam Flow (15.1.3) Inadvertent Opening of a Steam Generator Relief or Safety Valve (15.1.4) Spectrum of Steam System Piping Failures Inside and Outside of Containment in a PWR (15.1.5) Spectrum of Rod Cluster Control Assembly Ejection Accidents (15.4.8)	3.3.1, Table 3.3.1-1 #2a
3.	Intermediate range high neutron flux trip (15.4.1)	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal From a Subcritical or Low Power Startup Condition	See Note d[d], 3.3.1, Table 3.3.1-1 #4

TABLE 7.2-4
REACTOR TRIP CORRELATION

(Sheet 2 of 5)

	Trip[a]	Accident[b]	Technical Specification[c]
4.	Source range high neutron flux trip	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal From a Subcritical or Low Power Startup Condition (15.4.1)	See Note d,[d] 3.3.1, Table 3.3.1-1 #5
5.	Power range high positive neutron flux rate trip	Uncontrolled RCCA Bank Withdrawal (15.4.2) Spectrum of Rod Cluster Control Assembly Ejection Accidents (15.4.8)	3.3.1, Table 3.3.1-1 #3
6.	Not Used		
7.	Overtemperature N-16 trip	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (15.4.2) Chemical and Volume Control System Malfunction that Results in a Decrease in the Boron Concentration in the Reactor Coolant (15.4.6) Loss of External Electrical Load (15.2.2) Turbine Trip (15.2.3) Feedwater System Malfunctions that Result in a Decrease in Feedwater Temperature (15.1.1) Excessive Increase in Secondary Steam Flow (15.1.3) Inadvertent Opening of a Pressurizer Safety or Relief Valve (15.6.1) Inadvertent Opening of a Steam Generator Relief or Safety Valve (15.1.4) Loss of Coolant Accidents Resulting from the Spectrum of Postulated Piping Breaks within the Reactor Coolant Pressure Boundary (15.6.5)	3.3.1, Table 3.3.1-1 #6 Note 1

TABLE 7.2-4
REACTOR TRIP CORRELATION

(Sheet 3 of 5)

	Trip[a]	Accident[b]	Technical Specification[c]
8.	Overpower N-16 Trip	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (15.4.2)	3.3.1, Table 3.3.1-1 #7
		Feedwater System Malfunctions that Result in a Decrease in Feedwater Temperature (15.1.1)	
		Excessive Increase in Secondary Steam Flow (15.1.3)	
		Inadvertent Opening of a Steam Generator Relief or Safety Valve (15.1.4)	
		Spectrum of Steam System Piping Failures Inside and Outside of Containment in a PWR (15.1.5)	
9.	Pressurizer low pressure trip	Inadvertent Opening of a Pressurizer Safety or Relief Valve (15.6.1)	3.3.1, Table 3.3.1-1 #8a
		Loss of Coolant Accidents Resulting from the Spectrum of Postulated Piping Breaks within the Reactor Coolant Pressure Boundary (15.6.5)	
		Steam Generator Tube Failure (15.6.3)	
		Rod Cluster Control Assembly Misalignment (15.4.3)	
10.	Pressurizer high pressure trip	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (15.4.2)	3.3.1, Table 3.3.1-1 #8b
		Loss of External Electrical Load (15.2.2)	
		Turbine Trip (15.2.3)	
11.	Pressurizer high water level trip	Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (15.4.2)	3.3.1, Table 3.3.1-1 #9
		Loss of External Electrical Load (15.2.2)	
		Turbine Trip (15.2.3)	

TABLE 7.2-4
REACTOR TRIP CORRELATION

(Sheet 4 of 5)

	Trip[a]	Accident[b]	Technical Specification[c]
12.	Low reactor coolant flow	Partial Loss of Forced Reactor Coolant Flow (15.3.1)	3.3.1, Table 3.3.1-1 #10
		Loss of Non-Emergency AC Power To the Station Auxiliaries (15.2.6)	
		Complete Loss of Forced Reactor Coolant Flow (15.3.2)	
		Reactor Coolant Pump Shaft Seizure (Locked Rotor) (15.3.3)	
13.	Reactor coolant pump undervoltage trip	Complete Loss of Forced Reactor Coolant Flow (15.3.1)	3.3.1, Table 3.3.1-1 #12
14.	Reactor coolant pump underfrequency trip	Complete Loss of Forced Reactor Coolant Flow (15.3.1)	3.3.1, Table 3.3.1-1 #13
15.	Low-low steam generator water level trip	Loss of Normal Feedwater Flow (15.2.7)	3.3.1, Table 3.3.1-1 #14
		Loss of Non-Emergency AC Power to the Station Auxiliaries (15.2.6)	
		Feedwater System Pipe Break (15.2.8)	
16.	Reactor trip on turbine trip	Loss of External Electrical Load (15.2.2)	See Note d,[d] 3.3.1, Table 3.3.1-1 #16
		Turbine Trip (15.2.3)	
		Loss of Non-Emergency AC Power to the Station Auxiliaries (15.2.6)	See Note d[d]

TABLE 7.2-4
REACTOR TRIP CORRELATION

(Sheet 5 of 5)

	Trip[a]	Accident[b]	Technical Specification[c]
17.	Safety injection signal actuation trip	Inadvertent Opening of a Steam Generator Relief or Safety Valve (15.1.4)	See Note e[e] 3.3.1 Table 3.3.1-1 #17
		Steam System Piping Failure (15.1.5)	
18.	Manual trip	Available for all Accidents (Chapter 15)	See Note d 3.3.1 Table 3.3.1-1 #1

a) Trips are listed in order of discussion in Section 7.2.

b) References refer to accident analyses presented in Chapter 15.

c) References refer to Technical Specifications.

d) A Technical Specification is not required because this trip is not assumed to function in the accident.

e) Accident assumes that the reactor is tripped at end-of-life which is the worst initial condition for this case.

7.3 ENGINEERED SAFETY FEATURES SYSTEMS

In addition to the requirements for a reactor trip for anticipated abnormal transients, the facility is provided with adequate instrumentation and controls to sense accident situations and initiate the operation of necessary engineered safety features. The occurrence of a limiting fault, such as a loss of coolant accident or a steam line break, requires a reactor trip plus actuation of one or more of the engineered safety features in order to prevent or mitigate damage to the core and Reactor Coolant System components, and ensure containment integrity.

In order to accomplish these design objectives, timely initiating signals are to be supplied by the sensors, transmitters and logic components making up the various instrumentation channels of the Engineered Safety Features Actuation System (ESFAS). The details on the ESFAS signals and design are provided in Figure 7.3-4. The equipment arrangement is shown in Reference [2].

The protective functions pertaining to steamline break protection signal (an ESFAS Signal) are the Steamline isolation actuation and safety injection actuation signals generated when exceeding the setpoints of the following parameters.

1. Low pressurizer pressure - Initiates SI

2. High Containment Pressure - Hi-1 (initiates SI)

   Hi-2 initiates main steam line isolation - MSLI

3. Low Steamline Pressure - Initiates SI and MSLI

4. High Steam Pressure Rate - Initiates MSLI (only during normal cooldown and heat-up)

7.3.1 DESCRIPTION

The ESFAS uses selected plant parameters, determines whether or not predetermined safety limits are being exceeded and, if they are, combines the signals into logic matrices sensitive to combinations indicative of primary or secondary system boundary ruptures (Condition III or IV events). Once the required logic combination is completed, the system sends actuation signals to the appropriate engineered safety features components. The ESFAS meets the requirements of General Design Criteria (GDC) 13 and 20.

7.3.1.1 System Description

The ESFAS is a functionally defined system described in this section. The equipment which provides the actuation functions identified in Section 7.3.1.1.1 is listed below and discussed in this section. (For additional background information refer to References [1], [2], and [3].)

1. Process Instrumentation and Control System (Reference [1]).

2. Solid State Logic Protection System (Reference [2]).

3. Engineered Safety Features Test Cabinet (Reference [3]).

4. Manual Actuation Circuits.

The ESFAS consists of two discrete portions of circuitry: 1) an analog portion consisting of three to four redundant channels per parameter or variable to monitor various plant parameters such as the Reactor Coolant System and steam system pressures, temperatures and flows and containment pressures; and 2) a digital portion consisting of two redundant logic trains which receive inputs from the analog protection channels and perform the logic needed to actuate the engineered safety features. Each digital train is capable of actuating the engineered safety features equipment required. The intent is that any single failure within the ESFAS shall not prevent system action when required.

The redundant concept is applied to both the analog and logic portions of the system. Separation of redundant analog channels begins at the process sensors and is maintained in the field wiring, with containment vessel penetrations and analog protection racks terminating at the redundant safeguards logic racks. The design meets the requirements of Criteria 20, 21, 22, 23 and 24 of the 1971 GDC.

The variables are sensed by the analog circuitry as discussed in Reference [1] and in Section 7.2. The inputs from the analog channels 6, 7, and 8. Tables 7.3-1 and 7.3-2 give additional information pertaining to logic and function.

The interlocks associated with the ESFAS are outlined in Table 7.3-3. These interlocks satisfy the functional requirements discussed in Section 7.1.2.

Manual actuation from the control board of containment isolation Phase A is provided by operation of either one of the redundant momentary containment isolation Phase A controls. Each control consists of two back-up linked actuation switches. The separate trains are thereby linked by mechanical means in a fashion similar to that shown in Figure 7.2-3. Also on the control board is manual actuation of safety injection by one of the redundant controls and a manual activation of containment isolation Phase B by either of the two sets of controls.

Manual controls are also provided to switch from the injection to the recirculation phase after a loss of coolant accident.

7.3.1.1.1 Function Initiation

The specific functions which rely on the ESFAS for initiation are:

1. A reactor trip, provided one has not already been generated by the Reactor Trip System.

2. Cold leg injection isolation valves which are opened for injection of borated water by safety injection pumps into the cold legs of the Reactor Coolant System.

3. Charging pumps, safety injection pumps, residual heat removal pumps and associated valving which provide emergency makeup water to the cold legs of the Reactor Coolant System following a loss of coolant accident.

4. Containment spray pumps startup.

5. Those pumps which serve as part of the heat sink for containment cooling (e.g., service water and component cooling water pumps).

6. Motor driven auxiliary feedwater pumps, Turbine Driven Auxiliary Feedwater Pumps (open Steam Supply Valve)

7. Phase A containment isolation, whose function is to prevent fission product release. (Isolation of all lines not essential to reactor protection).

8. Steam line isolation to prevent the continuous, uncontrolled blowdown of more than one steam generator and thereby uncontrolled Reactor Coolant System cooldown.

9. Main feedwater line isolation as required to prevent or mitigate the effect of excessive cooldown.

10. Start the emergency diesels to assure back-up supply of power to emergency and supporting systems components.

11. Initiate the Control Room ventilation emergency recirculation to meet Control Room occupancy requirements following a loss of coolant accident.

12. Containment spray actuation which performs the following functions:

    a. Confirms spray pump startup (see item 4 above).

    b. Initiates containment spray by opening the spray valves to reduce containment pressure and temperature following a loss of coolant or steam line break accident inside of containment.

    c. Initiates Phase B containment isolation which isolates the containment following a loss of reactor coolant accident, or a steam or feedwater line break within containment to limit radioactive releases. (Phase B isolation together with Phase A isolation results in isolation of all but safety injection and containment spray lines penetrating the containment.)

7.3.1.1.2 Analog Circuitry

The process analog sensors and racks for the ESFAS are covered in Reference [1]. Discussed in this report are the parameters to be measured including pressures, flows, tank and vessel water levels, and temperatures as well as the measurement and signal transmission considerations. These latter considerations include the transmitters, orifices and flow elements, resistance temperature detectors, as well as automatic calculations, signal conditioning and location and mounting of the devices.

The sensors monitoring the primary system are located as shown on the piping flow diagrams in Chapter 5. The secondary system sensor locations are shown on the steam system flow diagrams given in Section 10.3.

Containment pressure is sensed by four physically separated differential pressure transmitters mounted by strong supports outside of the containment, (which are connected to containment atmosphere by a filled and sealed hydraulic transmission system). The distance from penetration to transmitter is kept to a minimum, and separation is maintained. This arrangement,

together with the pressure sensors external to the containment, forms a double barrier and conforms to GDC 56 and Regulatory Guide 1.11.

7.3.1.1.3 Digital Circuitry

The engineered safety features logic racks are discussed in detail in Reference [2]. The description includes the considerations and provisions for physical and electrical separation as well as details of the circuitry. Reference [2] also covers certain aspects of on-line test provisions, provisions for test points, considerations for the instrument power source, considerations for accomplishing physical separation. The outputs from the analog channels are combined into actuation logic as shown on Sheets 5, 6, 7, 8, and 14 of Figure 7.2-1.

To facilitate engineered safety features actuation testing, four cabinets (two per train) are provided which enable operation, to the maximum practical extent, of safety features loads on a group-by-group basis until actuation of all devices has been checked. Final actuation testing is discussed in detail in Section 7.3.2.

7.3.1.1.4 BOP Furnished Engineered Safety Features Systems

The BOP furnished engineered safety features systems (ESFS) are:

1. Containment Spray System
2. Containment Isolation System
3. Item Deleted.
4. Control Room Air-Conditioning System
5. Auxiliary Feedwater System
6. ESF Filter System

The BOP furnished engineered safety features support systems are:

7. Component Cooling Water System
8. Station Service Water System
9. Onsite Power Supply System
10. ESF Ventilation System
11. Safety Chilled Water System
12. Service Water Intake Structure Ventilation System
13. UPS Ventilation System (see Section 9.4C.8)

The ability to manually initiate operation of the systems described above is provided in accordance with IEEE-279-1971, Section 4.17.

The monitor light displays, located on the Control Board in the vicinity of the ESF system controls, provide an overview of the ESF functions. See Section 7.5.3.8.

In order to correspond with the NSSS criteria requirements to validate the assumptions and bases made in Chapter 15, Accident Analysis, instrumentation and control interface between the systems listed above and the NSSS is as follows:

The Engineered Safety Features Actuation System (ESFAS) is furnished by the NSSS vendor. It senses selected plant parameters and determines if predetermined safety limits are being exceeded or not. If they are, the ESFAS provides the contact closure outputs necessary to initiate those ESFS, including 1, 2, 4, 5, 7, 8, and 9 listed above, whose function best serves the requirements of the accident.

The ESFAS sensors, logic and initiating outputs to the various systems are per Figure 7.2-1. ESFAS sensors and other Class 1E instruments required for interlocks described below are located on Figure 7.1-3.

1. Containment Spray System

The CSS functions include containment cooling as well as fission product cleanup following a LOCA. The cooling function is described in Section 6.2.2 while the fission product cleanup function is in Section 6.5. The flow diagram is shown in Figure 6.2.2-1.

a. Initiating circuits

The Containment Spray System is initiated by either of the following signals:

1. Spray actuation signal (P signal)

2. Manual start from Main Control Board

The Containment Spray pumps are started by any of the following signals:

1. Spray actuation signal (P signal)

2. Safety injection sequence (S signal)

3. Manual start from Main Control Board

b. Logic

See Instrumentation & Control Diagrams listed under “Containment Spray System” in Tables 1.7-1 and 1.7-2.

c. Bypass

The Containment Spray System “bypass” is indicated on the Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and illustrated typically in Figure 7.1-4.

d. Interlocks

The Containment Spray pumps cannot be started manually when the safety injection sequencing or blackout sequencing is in progress.

e. Redundancy

The Containment Spray System consists of two separate and independent full capacity safety-related loops, each capable of fulfilling the design requirements. Each train of the Containment Spray System is spatially and electrically separated.

f. Diversity

The Containment Spray System is actuated by such diverse signals as:

1. Spray actuation signal

2. Manual command.

g. Actuated devices

For a list of all devices actuated by the Containment Spray System see Table 7.3-5.

h. Supporting systems

1. Class 1E Electric Power (See Section 8.3)

2. Containment spray pump room fan coolers start upon a containment spray pump start (See Section 9.4.5)

3. Component Cooling Water System supplies cooling water to the containment spray pump seal coolers and to the containment spray heat exchangers (See Section 9.2.2)

4. Station Service Water System supplies cooling water to the containment spray pump bearings (See Section 9.2.1)

i. Design basis

See Section 6.2.2.1 for design basis information.

j. Electrical schematic drawings

See Table 1.7-1 for Unit 1 and 1.7-2 for Unit 2 schematics (electrical) associated with “Containment Spray.”

k. Portion of system not required for safety

Except for those instruments required for actuating Containment Spray System components listed in Table 7.3-5, or the safety related instrumentation described in Section 7.5 (Tables 7.5-7A, B & C), the instruments and monitoring equipment are not required for safety. Local indicators, plant computer points, plant annunciator and bypass monitoring systems are not required for safety.

l. Control of system operation

The spray additive tank discharge valves, initially opened by the spray actuation signal, are closed by actuation of level switches when the tank is emptied. The Containment Spray System is designed with on-off controls. Once the system is actuated, the pumps operate with constant flow.

m. Monitoring of system operation

The Containment Spray System is provided with Post Accident Monitors as described in Section 7.5. In addition, local indication is provided for spray-additive tank level and pressure, spray-additive flow and Containment Spray pump discharge pressure. The spray additive tank is provided with level indication, low-low level alarm, and a high/low pressure alarm in the Control Room. Additional Control Room alarms are provided for valve isolation tank high level, refueling water storage tank low temperature and spray additive low flow.

Each power operated valve is provided with position indicating lights in the Control Room. ESF monitor lights are provided for pumps and valves.

n. Sequencing

The sequencing of containment spray pumps upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

2. Containment Isolation System

Containment isolation is carried out in two phases: Phase A isolation closes all nonessential process lines penetrating the Containment, and Phase B isolation closes remaining process lines, except for those lines required for ESFAS. Containment Ventilation Isolation is initiated by Phase A isolation or a high containment radioactivity signal. The overriding of one type of safety actuation signal does not cause the blocking of any other type of safety actuation signal for those valves that have no function other than containment isolation. The Containment Isolation System is discussed in further detail in Section 6.2.4 and 7.3.1.1.5.

a. Initiating circuits

The initiating circuits for all Containment Isolation phases are shown on Figure 7.2-1, Sheet 8, and listed on Table 7.3-2, Item 1.

Phase A Isolation is initiated by the Safety Injection signal, as well as by manual actuation of Phase A isolation at the Control Board. Table 7.3-1, Item 1 lists the functions required to generate the Safety Injection signal.

Containment Ventilation Isolation is initiated by Phase A Isolation described above as well as by the Containment Spray Actuation manual controls on the Control Board, or high containment radioactivity (particulate or gas detectors). The instrumentation systems which initiate Containment Ventilation Isolation and are part of the ESFAS are qualified to Class 1E criteria.

Containment Ventilation Isolation occurs with an ESFAS “T” signal derived from three diverse variables. In addition, a non-safeguard, non-redundant, non-Class 1E radioactivity channel has also been added to the Containment Ventilation Isolation Logic. See Figure 7.2-1, Sheet 8, and Table 7.1-2, Safety Related Instrumentation and Control Systems/Codes, Standards and Guides/ Applicability Matrix.

The high radiation signal and a “T” signal, which provide input to Containment Ventilation Isolation, share a retentive-memory-with-actuation-block gate; however, the high radiation signal is a pulsed signal. If the gate was set because of a high radiation signal, resetting the gate after the pulse would not invoke the actuation-block feature; thus, making it receptive to a subsequent ESFAS “T” signal. See Figure 7.2-1, Sheet 8. This part of the CPNPP design was verified as part of the Preoperational Test Program.

Phase B Isolation is initiated by a “P” signal, generated by hi-3 containment pressure in 2 out of 4 channels, or by manual controls on the Control Board for Containment Spray actuation.

The discussion in 7.3.2.2.7 is referred to for the manual actuation switches described above. The manual Containment Isolation controls require simultaneous operation of two Control Board switches, as shown on Figure 7.2-1, Sheet 8.

b. Logic

The Containment Isolation logic is shown on Figure 7.2-1, Sheet 8. The design of the control systems for automatic containment isolation valves is such that resetting the isolation signal will not result in automatic reopening of the containment isolation valves. To reopen each containment valve requires deliberate operator action. In addition, See Tables 6.2.4-1, 2 and 3 for a complete list of Containment Isolation valves and their isolation signals.

c. Bypass

Some valves actuated by the Containment Ventilation Isolation signal or the Containment Isolation System can be manually operated from the Control Room, one at a time, by holding their control switches in the open or closed position, as applicable. The control switches for the isolation valves or dampers are of the spring-return-to-automatic type, so that as soon as the operator releases the switch, the component will again be returned to automatic control and the jurisdiction of the Containment Isolation System or the Containment Ventilation Isolation signal. This control switch feature provides adequate administrative control to ensure that the overriding of one type of safety actuation signal does not cause the blocking of any other type.

This manual override does not constitute a bypass of a protective function, as defined by IEEE-279, in that the channel is not removed from service. The component level manual override does not interfere with the protective function. This momentary manual override feature is not indicated on the Safety System Inoperable Indication System because it is apparent to the Control Room operator who is, in fact, performing the override function. (See Section 7.3.2.2.8).

Note that if the operator wanted to regain control over all components tripped by either the Containment Isolation or Containment Ventilation Isolation signals, he would do so by resetting the system-level actuation-block latch on ESFAS. See Figure 7.2-1, Sheet 8.

Also note that overriding or resetting an ESFAS signal does not cause any equipment to change position. Equipment must be subsequently reset at a lower level (e.g., at the individual component controls, by resetting the sequencer) before equipment will reposition. This was verified as part of the Preoperational Test Program.

d. Interlocks

Refer to discussion in Section 1A(B) regarding thermal overload interlocks for motor operated valves. There are no other interlocks associated with the Containment Isolation System.

e. Redundancy

The Containment Isolation redundancy requirements are satisfied by having two isolation barriers in series for Type A, Type B, and Type C penetrations (see Section 6.2.4.1.2). When two power operated isolation valves are in series, the power supply to the valves comes from separate and redundant buses. Failure of any one of two safety-related buses will not prevent Containment Isolation System from fulfilling its design requirements.

f. Diversity

There is some diversity associated with the generation of Phase A isolation signal, depending on which postulated event is assumed. Refer to discussion in Section 7.3.1.1.5, Item 2.

There is no diversity in the generation of Phase B isolation signal.

g. Actuated devices

The isolation valves relating to containment isolation are the actuated devices and are listed in Tables 6.2.4-1, 2, and 3.

h. Supporting systems

The Containment Isolation System is powered by two separate and independent Class 1E load centers, (See Section 8.3).

i. Design basis

Refer to Section 6.2.4.1 for design information.

j. Electrical schematic drawings

See Tables 6.2.4-1, 2 and 3 for a list of Containment Isolation valves and Tables 1.7-1 and 1.7-2 for electrical schematic drawings.

k. Portion of system not required for safety

Plant annunciator and plant computer are not required for safety.

l. Control of system operation

The Containment Isolation System is completely digital in nature, with on-off controls only. The manual system level control of the isolation function is provided by the Phase A, Phase B (Spray Actuation) and steamline isolation control board switches shown on Figure 7.2-1, Sheet 8.

In addition to the system level manual controls described above, each power-operated valve or damper is provided with a component-level control switch, with open, automatic and close positions. These switches spring return to the automatic position, through which the automatic containment isolation functions are achieved.

m. Monitoring of System Operation

Each component actuated by the Containment Isolation System has closed/not closed position indicating lights in the Control Room with the exception of specific containment isolation valves identified in Tables 7.5-7A and 7.5-7C. Containment

isolation valves whose method of actuation is “local-manual” have their stem position under direct observation when being operated.

In addition, a redundant backup is provided by monitor light displays. The monitor light displays provide a convenient overview of the isolation function, since all equipment actuated by a given isolation signal is grouped together.

n. Sequencing

This item is not applicable to the Containment Isolation System.

3. Item Deleted.

4. Control Room Air-Conditioning System

The Control Room Air-Conditioning System is discussed in Section 9.4-1. The system flow diagrams is shown in Figure 9.4-1.

a. Initiating Circuits

The Control Room Air-Conditioning system is started by any of the following signals:

1. Safety Injection Sequence (from either Unit 1 or Unit 2)

2. Blackout Sequence (from either Unit 1 or Unit 2)

3. Manual Start

b. Logic

See instrumentation and control diagrams listed under “Control Room Air-Conditioning” in Table 1.7-1.

c. Bypass

Bypass is indicated on Safety System Inoperable Indicator described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

Control Room Air-Conditioning units cannot be started manually while Safety Injection Sequence or Blackout Sequence is in progress.

e. Redundancy

Separate system actuation switches and circuitry is provided for redundant system components. Physical and electrical separation is provided as discussed in Section 8.3.

f. Diversity

The Control Room Air-Conditioning System is started by such diverse signals as:

1. Safety Injection Sequence (from either Unit 1 or Unit 2)

2. Blackout Sequence (from either Unit 1 or Unit 2)

3. Manual Command

Control Room Emergency recirculation mode is started by such diverse signals as:

1. Safety Injection Signal (from either Unit 1 or Unit 2)

2. Blackout Sequence (from either Unit 1 or Unit 2)

3. High Radiation Concentration In The Intake Duct

4. Manual Command From Control Room

Control Room Isolation Mode is started by:

1. Manual Action

Control Room Emergency Ventilation Mode is started manually from the Control Room only when Emergency recirculation signal is present.

g. Actuated Devices

1. For a list of devices actuated within the Control Room Air-Conditioning System, by the Safety Injection signal refer to Table 7.3-4.

2. For devices actuated directly by the Control Room Air- Conditioning System see associated instrumentation and control diagrams listed in Table 1.7-1.

h. Supporting Systems

The systems required for proper air-conditioning operation are:

1. Class 1E Electric Power (See Section 8.3)

2. Component Cooling Water (See Section 9.2.2)

i. Design Basis

For information concerning basis of design refer to Section 9.4.1.1.

j. Electrical Schematic Drawings

See Table 1.7-1 for schematics (electrical) associated with “Control Room and Office Facility HVAC.”

k. Portion of System Not Required For Safety

Except for those instruments required for actuating the Control Room Air Conditioning System and components listed in Table 7.3-4, local indicators, the plant annunciator, the plant computer and the bypass indication system are not required for safety.

l. Control of System

There are four modes of operation: Normal Operation, Emergency Ventilation, Isolation and Emergency Recirculation. For each mode of operation, the final status of fans and dampers is indicated in Figure 9.4-1.

For the Normal Operating Mode, starting of the fans and actuation of dampers are carried out manually by the operator from the Control Room. Design ambient conditions of the Control Room are maintained by the air-conditioning units. Slightly positive Control Room pressure is maintained by throttling the exhaust dampers.

The Control Room Air-Conditioning system will be automatically switched to Emergency Recirculation Mode on receipt of any of the following signals:

1. Control Room Intake Duct Radiation Hi

2. Safety Injection (from either Unit 1 or Unit 2)

3. Blackout Sequence (from either Unit 1 or Unit 2)

Emergency Recirculation can also be manually initiated.

m. Monitoring of System Operation

System component status indicating lights, system failure alarms, Control Room to outside environs differential pressure indicator, and Control Room area radiation monitors are provided in the Control Room to enable the operator to evaluate the system performance.

n. Sequencing

The sequencing of the Control Room Air-Conditioning upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

5. Auxiliary Feedwater System

The Auxiliary Feedwater System is outlined in Subsection 10.4.9. The system flow diagram is shown in Figure 10.4-11.

a. Initiating Circuits

The motor-driven pumps are started by any of the following signals:

1. Safety Injection Sequence
2. Blackout Sequence
3. Trip of both main feedwater pumps (pumps 1-A and 1-B)
4. Steam generator low-low level (two of four from any steam generator)
5. Manual start (from Control Room or Hot Shutdown Panel)
6. AMSAC

The turbine-driven pump is started by any of the following signals:

1. Steam generators low-low level (two of four from any two steam generators)
2. Blackout (loss of voltage) signal
3. Manual start (from Control Room or Hot Shutdown Panel)
4. AMSAC

The auto start signals that starts any of the three auxiliary feedwater pumps closes the blowdown and sample line isolation valves for all the steam generators, and closes the condensate makeup and reject line isolation valves and the Unit 2 feedwater split flow bypass valves.

b. Logic

See instrumentation and control diagrams listed under “Auxiliary Feedwater” in Tables 1.7-1 and 1.7-2.

c. Bypass

Bypass is indicated on Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and generally illustrated in Figure 7.1-4.

d. Interlocks

1. The operator lockout (OL) or automatic lockout (AL) feature of the sequencer can, on the appearance of a safety injection or blackout signal, annul either manual or autostart of the AFW System motor-driven pumps in order to properly sequence them on the diesel. For local override control at the Hot Shutdown Panel refer to Section 7.4.1.3.1.

2. Automatic starts of the Auxiliary Feedwater System also close the steam generator blowdown isolation and sample line isolation valves, and close the condensate makeup and reject line isolation valves and the Unit 2 feedwater splitflow bypass valves.

e. Redundancy

Steam admission valves and controls for turbine drive are redundant. (See instrumentation and control drawings listed in Section 1.7)

Motor-driven pumps controls and instrumentation are redundant.

Steam Generator level measurement is quadruplicated. (See Figure 7.2-1, Sheet 7)

Auxiliary feedwater flow to each steam generator is redundant (See Figure 7.3-4)

f. Diversity

The AFW System is started by such diverse signals as:

1. Trip of Both Main FW Pumps

2. Steam Generator Low-Low Level

3. Safety Injection Sequence

4. Blackout Sequence

5. Manual Command at Either the Control Room or Hot Shutdown Panel (See Figure 7.2-1, Sheet 15).

6. AMSAC

g. Actuated Devices

Actuated devices are listed in Table 7.3-4.

h. Supporting Systems

The following support systems are required by the AFW System:

1. Class 1E Electric Power (See Section 8.3)

2. Engineered Safety Features Ventilation System (See Section 9.4.5 and Figure 9.4-4)

3. Safety Chilled Water System (See Section 9.4F).

i. Design Basis

See Section 10.4.9.1 for the design bases of the AFW System.

j. Electric Schematic Drawings

See Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2 schematics (electrical) associated with “Auxiliary Feedwater.”

k. Portion of System Not Required For Safety

Except for those instruments required for actuating Auxiliary Feedwater System components listed in Table 7.3- 4, or the safety-related instrumentation described in Section 7.5 (Tables 7.5-7A, B & C), the instruments and monitoring equipment are not required for safety. Local indicators, plant computer points, plant annunciator and bypass monitoring systems are not required for safety.

l. Control of System Operation

The AFW System is used for startup when there is no steam to drive the feedwater pumps, for shutdown until the hot-leg temperature is down to 350°F, at which time the RHR System goes into operation, or following the receipt of any of the automatic signals described above.

Pneumatically operated flow control valves are provided downstream of the motor and steam turbine driven AFW pumps to maintain water level in the steam generators.

The pneumatically operated flow control valves downstream of the motor driven pumps are automatically tripped to full open position on auto start of the pump. After the initial lock out the valves are manually controlled from the Control Room.

The normally open pneumatically operated flow control valves downstream of the steam turbine driven pump are manually controlled from the control room.

All other valves in the system with the exception of the steam admission valves to the auxiliary feed pump turbine are normally in the ready position to permit auxiliary feedwater flow.

The pumps deliver full flow upon receipt of the actuation signal. See section 10.4.9 for additional details.

The steam admission valves, normally closed, will open upon receipt of the actuation signal.

Protection against excessive flow to containment and pump runout in the event of a drop in the steam generator pressure is provided by the flow orifices located on the pump discharge lines.

m. Monitoring of System Operation

The AFW System is provided with display instrumentation as follows:

1. Steam Generator Level (Accident Monitors, Type A, B and D variables, see Tables 7.5-2, 3, 5)

2. Auxiliary Feedwater Flow To Each Steam Generator (Accident Monitors, Type A, B, D variables, Tables 7.5-2, 3, 5).

3. Pump Flow

4. Pump Discharge Pressure

5. Pump Suction Pressure

6. Condensate Storage Tank Level (Accident Monitors, Type A, D variables, see Tables 7.5-2, 5)

n. Sequencing

The sequencing of motor-driven AFW pumps upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

6. ESF Filter System

The ESF Filter System is described in Section 6.5.1. Figure 6.5-1 illustrates a typical layout for the ESF filter units.

a. Initiating Circuit

No automatic initiating circuit is provided. ESF filters are passive components. Inlet dampers, exhaust fans and filter assembly heaters for primary plant ESF exhaust filters are started either manually or by a Safety Injection Actuation Signal from either Unit 1 or Unit 2. Inlet dampers, exhaust fans and filter assembly heaters for the Control Room Emergency Pressurization and Emergency Filtration units are started either manually or by Control Room Emergency Recirculation signal. Heaters are not included in Control Room Emergency filtration units.

b. Logic

See instrumentation and control diagram listed under “Control Room Air-Conditioning”, and “Ventilation Equipment Primary Plant” listed in Table 1.7-1.

c. Bypass

There are no bypasses associated with this system.

System level SSII for Control Room Emergency Pressurization and Emergency Filtration Units is “Control Room HVAC” and System level SSII for ESF exhaust filtration units is “Primary Plant ESF Ventilation Exhaust”.

d. Interlocks

There are no interlocks associated with this system.

e. Redundancy

Redundant ESF filters are provided which are physically and electrically separated from one another.

f. Diversity

Control Room Emergency Pressurization and Emergency Filtration Units inlet dampers, exhaust fans and associated filter assembly heaters are started by such diverse signals as:

1. Control Room Emergency Recirculation Signal

2. Manual Command at Control Room.

g. Actuated Devices

1. ESF Filter Inlet Dampers

2. ESF Filter Exhaust Fans

3. ESF Filter Assembly Heaters

h. Supporting System

ESF Filter System is supported by Class 1E electric power (See Section 8.3).

i. Design Basis

See Section 6.5.1.1 for the design basis of ESF Filter System.

j. Electric Schematic Drawings

See Table 1.7-1 for schematics (electrical) associated with “Control Room and Office Facility HVAC” and “Primary Plant Ventilation.”

k. Portions of System Not Required For Safety

Except for those instruments required to actuate the ESF Filter System components listed in Item g above, or the safety-related instrumentation described in Section 7.5 (Tables 7.5-7A, B & C), the instruments and monitoring equipment are not required for safety. Local indicators, plant computer points, plant annunciator and bypass monitoring systems are not required for safety.

l. Control of System Operation

All ESF Filter System components necessary for operation are controlled from the Control Room. Temperature detection components provide Hi and Hi-Hi temperature information in the fire protection panels located in the same building as the corresponding filter units. These fire protection panels provide Hi and Hi-Hi temperature alarm and trouble alarm information to the Control Room.

m. Monitoring of System Operation

Each ESF filter and exhaust fan are supplied with differential pressure alarms to alert the Control Room operator of a fan malfunction or a clogged filter. Local pressure indicators are provided to monitor the resistance of each individual filter bank. Each adsorber bed has a temperature monitoring system which indicates bed temperature and actuates a high temperature alarm in the Control Room. System component status indicating lights are provided in the Control Room.

n. Sequencing

The sequencing of the primary plant ESF filter units upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

7. Component Cooling Water System

The Component Cooling Water (CCW) System is discussed in Section 9.2.2. The System flow diagram is shown in Figure 9.2-3.

a. Initiating Circuits

The Component Cooling Water pumps are started by any of the following signals:

1. Safety Injection Sequence

2. Blackout Sequence

3. Service Water Pump Auto Start On Low Pressure

4. Low Pressure In The Discharge Header Of Redundant Component Cooling Water Pump.

5. Manual Start (From Control Room or Hot Shutdown Panel)

b. Logic

See instrumentation and control diagram listed under “Component Cooling Water” in Tables 1.7-1 and 1.7-2.

c. Bypass

Bypass is indicated on Safety System Inoperable Indicator described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

The Component Cooling Water pumps cannot be started either manually or automatically when the Safety Injection Sequencing or Blackout Sequencing is in progress. For local override control at the Hot Shutdown Panel, refer to Section 7.4.1.3.1.

e. Redundancy

Separate switches and actuation circuitry are provided for redundant components which are physically and electrically separated from one another.

f. Diversity

The Component Cooling System is started by such diverse signals as:

1. Safety Injection Sequence

2. Blackout Sequence

3. Service Water Pump Autostart On Low Pressure

4. Low Pressure In The Discharge Header of Redundant Component Cooling Water Pump.

5. Manual Command At Either The Control Room or Hot Shutdown Panel.

g. Actuated Devices

Actuated devices are listed in Table 7.3-4, (for all equipment actuated by “s” signal) and Table 7.3-5 (for all equipment actuated by “p” signal).

h. Supporting Systems

The following support systems are required by the Component Cooling Water System:

1. Station Service Water System (See Section 9.2.1)

2. Class 1E Electric Power (See Section 8.3)

3. Engineered Safety Features Ventilation System (See Section 9.4.5 and Figure 9.4-2, Sheet 1).

i. Design Basis

See Section 9.2.2.2.2 for the design basis of the Component Cooling Water System.

j. Electric Schematic Drawings

See Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2 schematics (electrical) associated with “Component Cooling Water.”

k. Portion of System Not Required For Safety

Except for those instruments required to actuate the Component Cooling Water System components listed in Table 7.3-5 or the safety related instrumentation described in Section 7.5 (Tables 7.5-7A, B & C), the instruments and monitoring equipment are not required for safety. Local indicators, the plant annunciator and the plant computer are also not required for safety.

l. Control of System Operation

The CCW System is designed as an on/off system during normal operation. Pump recirculation valves are provided to maintain required minimum pump flow under abnormal conditions.

The CCW pumps are automatically started by low pressure in the pump discharge header.

The CCW surge tank level is used to detect system leakage. The control system for each half of each partitioned tank consists of the following:

1. Level Indication - Local and Control Room

2. Level Recording

3. Hi-Hi/Lo Alarms

4. Automatic Make-up From Reactor Make-Up Water on Lo-Lo Level

5. Automatic Make-Up “Initiated” Alarm

6. Automatic Make-Up Termination On Hi-Level

7. Manual Make-up From Demineralized Water.

8. Automatic train separation on Empty Level

m. Monitoring of System Operation

The CCW System is provided with monitoring instrumentation as described in Table 7.5-7A and B. In addition, each component is supplied with local temperature indication. Sufficient local flow indication is provided to allow flow balancing of the system. Each flow indicator has a low flow alarm contact to alert the operator of a system malfunction. Flow indication is provided in the Control Room for Containment Spray and RHR heat exchangers.

Each power operated valve is supplied with a control switch and position indicating lights in the Control Room.

Radiation monitors are provided with readout in the Control Room for each of the three loops in each subsystem.

n. Sequencing

The sequencing of CCW pumps upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

8. Station Service Water

The Station Service Water (SSW) System is discussed in Section 9.2.1. The System Flow diagram is shown in Figure 9.2-1.

a. Initiating Circuits

The Station Service Water System is started by any of the following signals:

1. Safety Injection Sequence

2. Blackout Sequence

3. Component Cooling Water Pump Start Signal

4. Low Pressure Signal of Train B Supply Header Starts Train A Service Water Pump and Vice Versa.

5. Manual Command Either at Control Board or at Hot Shutdown Panel.

b. Logic

See instrumentation and control diagrams listed under “Station Service Water System” in Tables 1.7-1 and 1.7-2.

c. Bypass

Station Service Water System Bypass is indicated on the Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

The Station Service Water pumps cannot be started either manually or automatically when the Safety Injection Sequencing or Blackout Sequencing is in progress. For local override control at the Hot Shutdown Panel, refer to Section 7.4.1.3.1.

e. Redundancy

Separate switches and actuation circuitry are provided for redundant components which are physically and electrically separated from one another.

f. Diversity

The Service Water System may be controlled from the following locations:

1. Each Service Water pump may be controlled by:

   (a) A Control Switch on the Main Control Board

   (b) A Local Control Switch on the Hot Shutdown Panel

   (c) Blackout Sequence

   (d) Safety Injection Sequence

   (e) Component Cooling Water Pump Start Signal

In addition to above, low pressure signal of Train B supply header starts Train A Service Water pump and vice versa.

g. Actuated Devices

For devices actuated directly by the “S” signal see Table 7.3-4.

For devices actuated directly by the Station Service Water System, refer to Station Service Water instrumentation and control drawings listed in Tables 1.7-1 and 1.7-2.

h. Support Systems

The Service Water System is supported by:

1. Class 1E Electric Power (See Section 8.3)

2. Service Water Intake Structure Ventilation Exhaust System (See Section 9.4B and Item 6 below)

i. Design Basis

For information concerning basis of design see Section 9.2.1.1.

j. Electrical Schematic Drawings

See Table 1.7-1 for Unit 1 and 1.7-2 for Unit 2 schematics (electrical) associated "Station Service Water."

k. Portion of System Not Required for Safety

Except for those instruments required for actuating Service Water System components listed in Table 7.3-4, or the safety-related instrumentation described in Section 7.5 (Tables 7.5-7A, B & C), the instruments and monitoring equipment are not required for safety. Local indicators, plant computer points, plant annunciator and bypass monitoring systems, are not required for safety.

l. Control of System Components

The SSW System is designed to operate with a constant system pressure during normal operation.

m. Monitoring of System Operation

The SSW System is provided with monitoring instrumentation as described in Section 7.5. In addition, each component is provided with local temperature indication. Sufficient local flow indication is provided to allow flow balancing of the system.

Each power operated valve is supplied with a control switch and position indicating lights in the Control Room.

Radiation monitors are provided with readout in the Control Room for each of the return headers of the CCW system.

n. Sequencing

The sequencing of SSW pumps upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

9. Onsite Power Supply System

The Onsite AC Power Supply System is discussed in Section 8.3. Two independent diesel generators are provided for each unit. The starting and loading diagrams for the diesel generators are shown in Figure 8.3-3.

The station DC systems provide power for the plant instrumentation and control under all modes of plant operation. For details see Section 8.3.2.

a. Initiating Circuits

AC - Starting of diesel generator sets, automatic stripping and loading of buses are discussed in Section 8.3.1.1.5.

DC - The DC power supplies are designed to be continuously available during all modes of plant operation. Thus, no initiating circuits are required.

b. Logic

AC - See Figure 8.3-3.

DC - See Figure 8.3-14.

c. Bypass

AC - Bypass is indicated on Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

DC - The system is indicated on Safety System Inoperable Indicator (SSII) as described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

AC - See Section 8.3.1.1.11, item 2.

DC - Normally, one battery charger is operating and the other is kept as a spare. The circuit breakers connecting these battery chargers to the DC distribution bus are mechanically interlocked such that only one battery charger remains connected to the bus at any time. For details see Section 8.3.2.1, Item 2b.

e. Redundancy

AC - Separate control switches and actuation circuitry are provided for starting each emergency diesel generator and actuating emergency bus breakers. The Standby Power System consists of redundant diesel generator sets, each having redundant air starting systems and fuel transfer systems. DC power from a redundant battery system furnishes power for the circuitry. The DC system is described in Section 8.3.2.

DC - Two independent and redundant 125V DC systems are provided for each unit. Each system consists of two batteries, each having one main distribution bus with molded case circuits breakers and fusible switches, two static battery chargers (one spare) and local distribution panels (see Figures 8.3-14 and 8.3-15B).

f. Diversity

AC - The diesel generators are actuated by such diverse signals as:

1. Safety Injection Signal

2. Undervoltage On Its Respective Emergency Bus

3. Manual Command.

DC - A spare battery charger is provided for each DC system battery.

g. Actuated Devices

AC - Actuated devices are diesel generator sets which are listed in Table 7.3-4.

DC - No devices require actuation in this system.

h. Supporting Systems

AC - Protection systems, diesel generator fuel oil storage and transfer system, cooling and heating system, Lube Oil System, Combustion Air Intake and Exhaust System and Ventilation System for the Standby Power System are discussed in Sections 8.3.1.1.11, 9.5.4, 9.5.5, 9.5.7, 9.5.8, and 9.4C, respectively.

DC - Battery chargers and ventilation systems are described in Section 8.3.2.1, Items 2b and 3.

i. Design Basis

AC - See Section 8.3.1 for the design basis of Onsite Power Supply System.

DC - See Section 8.3.2.

j. Electric Schematic Drawings

AC - See Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2 schematics.

DC - See Figure 8.3-14

k. Portion of System Not Required For Safety

AC - Local indicators and annunciator system are not required for safety.

DC - Local indicators and annunciator system are not required for safety.

l. Control of System Operation

AC - Once the system is actuated, the diesel generator voltage and frequency are automatically controlled.

Each diesel generator set has its own speed control system and voltage regulator. Manual controls are not necessary for proper system functioning. Manual backup for voltage and frequency controls are provided locally and in the Control Room. Control switches are also provided locally and in the Control Room for manually starting the diesel generators and operating the generator breakers. Once the diesel generator is started in emergency mode, shutdown can only be accomplished by operator's manual action, except under generator differential protection operation, and overspeed protection.

DC - Except for adjusting battery charger voltage for periodic equalizing charge, no other manual or automatic controls are required. For details see Section 8.3.2.1.

m. Monitoring of System Operation

AC - Control Room indication, alarm and status instrumentation are provided to enable the operator to evaluate system performance and detect malfunctions. Diesel generator current, voltage, and frequency are indicated. Alarms are provided to indicate diesel generator malfunction or trip.

Instrumentation and control systems are provided as discussed in Section 8.3.1.1.7.

DC - Battery float and discharge current and distribution bus voltage are monitored at the switchboard and in the Control Room. For details, see Section 8.3.2.2, Item 1.

n. Sequencing

This is not applicable to the Onsite Power System.

10. ESF Ventilation System

The ESF Ventilation System is discussed in Section 9.4.5. The system flow diagram is shown in Figure 9.4-4.

a. Initiating Circuit

Emergency fan coil units of safety-related pump rooms are started either manually or by the start signal from the equipments they serve. Emergency fan coil units for electrical area are started manually or automatically. Diesel oil day tank area ventilation fans are started either manually or by the start signal from the diesel generator.

b. Logic

See instrumentation and control diagram listed under “Ventilation Safeguards and Electrical Area” listed in Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2.

c. Bypass

Bypass is indicated on Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

There are no interlocks in this system.

e. Redundancy

Redundant fan coil units and redundant diesel oil day tank area ventilation fans are powered from redundant electric power trains.

f. Diversity

Fan coil units for safety-related pump rooms and diesel oil day tank area ventilation fans are started by such diverse signals as:

1. Start Signal From the Equipment They Serve

2. Manual Command Either at Control Room or Local Switch Station.

Fan coil units for electrical area are started by such diverse signals as:

1. Manual Command either at Control Room or locally

2. Safety Injection Sequence

3. Blackout Sequence

g. Actuated Devices

1. Emergency Fan Coil Units for Safety-Related Pump Rooms

2. Emergency Fan Coil Units for Electrical Area

3. Diesel Oil Day Tank Area Ventilation Fans

h. Supporting System

ESF Ventilation System is supported by Class 1E electric power (See Section 8.3) and the Safety Chilled Water System (See Section 9.4F).

i. Design Basis

See Section 9.4.5.1 for the design basis of ESF Ventilation System.

j. Electrical Schematic Drawings

See Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2 schematic (electrical) associated with “Safeguard and Electrical Area Ventilation.”

k. Portion of System Not Required For Safety

Remote and local indications, Safeguard Building electrical area exhaust and supply fans, Safeguard Buildings main steam and feedwater exhaust and supply fans, room supply and exhaust dampers, fan heaters and the plant annunciator are not required for safety.

l. Control of System Operation

The ESF Ventilation System components necessary for operation are controlled from the Control Room. In addition, the emergency fan coil units of safety-related pump rooms can be controlled by local control. Each set of room supply dampers and exhaust valves is provided with a control switch and position indicating lights in the Control Room.

m. Monitoring of System Operation

Each safety-related pump room is provided with local temperature indicator and a Hi/Lo temperature alarm in the Control Room. Electrical area is also provided with temperature indicators mounted near the fan coolers and Hi-Hi temperature alarms in the Control Room. Transfer of controls of fan coil units from remote to local actuates an alarm in the Control Room. System component status indicating lights are provided in the Control Room.

n. Sequencing

The sequencing of the emergency fan coil units upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

11. Safety Chilled Water System

The Safety Chilled Water System is described in Section 9.4F. The System flow diagram is shown in Figure 9.4-12.

a. Initiating Circuit

The Safety Chilled Water recirculation pumps are started by any of the following signals:

1. Safety Injection Sequence

2. Blackout Sequence

3. Manual Start From Control Room

4. Start of CCW Pump

5. Manual Start From HSP For “A” Train Pump

b. Logic

See instrumentation and control diagrams listed under “Ventilation Safety Chilled Water” in Tables 1.7-1 and 1.7-2.

c. Bypass

Bypass is indicated on Safety System Inoperable Indicator described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4.

d. Interlocks

Chilled water recirculation pumps cannot be started manually while Safety Injection Sequence or Blackout Sequence is in progress. For local override control at the Hot Shutdown Panel, refer to Section 7.4.1.3.1.

e. Redundancy

Separate switches and actuation circuitry are provided for redundant components which are physically and electrically separated from one another.

f. Diversity

The Safety Chilled Water Recirculation pumps are started by such diverse signals as:

1. Safety Injection Sequence

2. Blackout Sequence

3. Manual Command at the Control Room.

4. Start of CCW Pump

5. Manual Start From HSP For “A” Train Pump

g. Actuated Devices

1. Safety Chilled Water Recirculation Pumps

2. Water Chillers

h. Supporting Systems

The following supported systems are required by the Safety Chilled Water System:

1. Class 1E Electric Power (See Section 8.3)

2. Component Cooling Water (See Section 9.2.2).

i. Design Basis

See Section 9.4F.1 for the design basis of the Safety Chilled Water System.

j. Electric Schematic Drawings

See Table 1.7-1 for Unit 1 and Table 1.7-2 for Unit 2 schematic (electrical) associated with “Ventilation Chilled Water and Safety Chilled Water.”

k. Portion of System Not Required For Safety

The following portions of the Safety Chilled Water System are not required for safety:

1. Remote and Local Indicator

2. Plant Annunciators

3. Plant Computer

4. Inlet valves to surge tank from reactor make-up storage tank and demineralized water storage tank.

l. Control of System Operation

The Safety Chilled Water System is designed as an on/off system during normal operation.

Surge tank level is used to monitor the system for leakage. The control system for each half of the partitioned surge tank consists of the following:

1. Level indication - local

2. Hi-Hi/Lo-Lo alarm

3. Automatic make-up from demineralized or reactor makeup water on lo level

4. Automatic make-up termination on Hi level

5. Manual make-up from demineralized or reactor make-up water.

Each power operated valve is supplied with a control switch and position indicating lights in the Control Room.

m. Monitoring of System Operation

The Safety Chilled Water System is provided with display instrumentation and monitoring as follows:

1. Pump discharge pressure indication - local

2. Low pump discharge pressure alarm in the Control Room

3. Evaporator package discharge temperature local indications, discharge flow indication and Lo-flow alarm in the Control Room.

4. Surge tank level indication - local

5. Emergency fan coil unit temperature - local

6. Status lights in the Control Room for Safety Chilled Water recirculation pumps and all power operated valves.

n. Sequencing

The sequencing of safety chilled water pumps upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

12. Service Water Intake Structure Ventilation

The Service Water Intake Structure Ventilation is described in Section 9.4B and the flow diagram is shown in Fig. 9.4-7.

a. Initiating circuit

Service Water pump area exhaust system is started manually by the intake structure operator or automatically by locally mounted temperature switches.

b. Logic

See instrumentation and control diagrams listed under “Ventilation Service Water Intake Structure and Misc. Building” in Table 1.7-1.

c. Bypass

There are no bypasses associated with this system.

d. Interlocks

There are no interlocks in this system.

e. Redundancy

Separate switches and actuation circuitry are provided for redundant exhaust fans which are physically and electrically separated from one another.

f. Diversity

Service Water Intake Structure area fans are started manually by the intake structure operator or automatically by locally mounted temperature switches. Status of the fans are indicated both locally and remotely on the Control Board.

g. Actuated devices

Service Water Intake Structure area fans

h. Supporting system

Service Water Intake Structure Ventilation is supported by Class 1E electric power (See Section 8.3).

i. Design basis

See Section 9.4B.1 for the design basis of Service Water Intake Structure Ventilation System.

j. Electric schematic diagram

See Table 1.7-1 for schematic (electrical) associated with “Service Water.”

k. Portion of system not required for safety

Local indication and alarms are not required for safety.

l. Control of system operation

The HVAC System for the Service Water Intake Structure is designed to maintain the ambient temperature within the structure as indicated in Section 9.4B.1. A

local start, stop, spring-return-to-auto switch is available for exhaust fan control. In addition, six non-safety-related, thermostatically controlled, unit heaters are provided for winter service. Generally, exhaust fan operation is not required during the winter.

m. Monitoring of system operation

Exhaust fan “Start”, “Stop”, and “Trip” is indicated in the Control Room for each fan. In addition, fan “Trip” is alarmed in the Control Room.

A “High” and “Lo-Lo” temperature alarm is available in the Control Room derived from local temperature indicating switches. In addition, a combined unit heater trip alarm is available in the Control Room.

n. Sequencing

This Load is not sequenced automatically. The fans start automatically by locally mounted temperature switches. See Table 8.3-1.

13. UPS Ventilation System

The UPS Ventilation System is discussed in Section 9.4C.8. The System Flow diagram is shown in Figure 9.4-15.

a. Initiating circuits

The UPS Ventilation System is started by any of the following signals:

1. Safety Injection Sequence (from either Unit 1 or Unit 2)

2. Blackout Sequence (from either Unit 1 or Unit 2)

3. Temperature Switches located in the UPS and Distribution Room

4. Manual command at Control Board.

b. Logic

See instrumentation and control diagrams listed under “UPS Ventilation System” in Table 1.7-1.

c. Bypass

There are no bypasses associated with this system.

d. Interlocks

The UPS Ventilation A/C units and fans cannot be started manually from the Control Panel when the Safety Injection Sequencing or Blackout Sequencing is in progress.

e. Redundancy

Separate switches and actuation circuitry are provided for redundant components which are physically and electrically separated from one another.

f. Diversity

A/C units and fans are started by such diverse signals as:

1. Safety Injection Sequence (from either Unit 1 or Unit 2)

2. Blackout Sequence (from either Unit 1 or Unit 2)

3. Temperature switches located in the Unit 1 and Unit 2 UPS and distribution rooms.

4. Manual command from Control Room or local control switch.

g. Actuated devices

For devices actuated directly by the “S” signal see Table 7.3-4.

For devices actuated directly by the UPS Ventilation System, refer to UPS Ventilation System instrumentation and control drawings listed in Table 1.7-1 and 1.7-2.

h. Support Systems

The UPS Ventilation System is supported by:

1. Class 1E electric power (See Section 8.3)

2. Component Cooling Water System (See Section 9.2.2)

i. Design basis

For information concerning basis of design see Section 9.4C.8.

j. Electrical schematic drawings

See Table 1.7-1, “BOP Misc. Systems” for the schematics (electrical).

k. Portion of system not required for safety

Except for those instruments required for actuating UPS Ventilation System components listed in Table 7.3-4, the instruments and monitoring equipment are not required for safety. Local indicators, plant annunciator and bypass monitoring systems, are not required for safety.

l. Control of system components

The UPS Ventilation system is designed to remove heat rejected by both Unit 1 and Unit 2 UPS inverters and chargers to maintain ambient temperature below design limits. During normal operation, one train is in service with the opposite train in standby.

m. Monitoring of system operation

UPS Ventilation A/C Unit and fan status monitoring lights, A/C fan and the booster return fan differential pressure alarms, and UPS and distribution room high temperature alarm are provided for the monitoring of system operation.

n. Sequencing

The sequencing of the UPS Ventilation A/C units and fans upon loss of offsite power concurrent with a LOCA is shown in Table 8.3-1.

7.3.1.1.5 Final Actuation Circuitry

The outputs of the Solid State Logic Protection System (the slave relays) are energized to actuate, as are most final actuators and actuated devices. These devices are listed as follows:

1. Emergency Core Cooling. Refer to Section 6.3 and Figure 7.2-1 for flow diagrams and additional information.

2. Containment isolation Phase A is by a “T” signal which isolates all nonessential process lines on receipt of the safety injection signal, as well as by manual actuation of Phase A containment isolation at the control board. The process parameters which are relied on for deriving the safety injection signal are listed under Item 1 of Table 7.3-1 and their performance requirements are discussed in Section 7.3.1.2.6. For a given design condition, there is an initial containment isolation function derived from a primary generating station variable. For example, for a primary system accident (see Section 7.3.1.2.2), containment isolation is initially derived from low pressurizer pressure. However, another containment isolation function may be derived from high-1 containment pressure during the course of a postulated accident and depending on what postulated event is assumed. Criteria to provide diversity is not a requirement, but as indicated in the forgoing, some may be available for Phase A containment isolation.

   Containment isolation Phase B is by a “P” signal which isolates the remaining process lines (which do not include safety injection lines) on receipt of two out of four Hi-3 containment pressure signal. For further information, refer to Section 6.2.4.

3. Service water pump and valve actuators (see Section 7.3.1.1.4 and 9.2.1)

4. Auxiliary feedwater pumps start (see Sections 7.3.1.1.4 and 10.4.9)

5. Diesel start (see Sections 7.3.1.1.4.6 and Chapter 8)

6. Feedwater isolation (see Section 10.4.7.3 and Fig. 7.2-1)

7. Control Room Ventilation isolation valve and damper actuators (see Section 7.3.1.1.4.7, Section 9.4.1.1, and Table 9.4-10)

8. Steamline isolation valve actuators (see Section 10.3 and Fig. 7.2-1)

9. Containment spray pump and valve actuators (see Sections 7.3.1.1.4 and 6.2.2)

Power interruption to the engineered safety features system in conjunction with a LOCA or other postulated accident is a highly improbable event. To satisfy GDC 35 requirements, accident analyses assume a loss of offsite power coincident with certain postulated events such as a LOCA, steam line break, and feedline break (see Chapter 15). It is also assumed that a single failure occurs which causes the loss of one of the two onsite emergency diesel generators. In the unlikely event that a loss of offsite power were to occur following the onset of a LOCA or other accident, the consequences could be mitigated by administrative procedures. If an accident is assumed to occur coincident with loss of offsite power, the engineered safety features loads must be sequenced onto the diesel generators to prevent overloading them. This sequence is discussed in Chapter 8. The design meets the requirements of Criterion 35 of the 1971 GDC.

7.3.1.2 Design Basis Information

The functional diagrams presented in Figure 7.2-1, Sheets 5, 6, 7 and 8 provide a graphic outline of the functional logic associated with requirements for the ESF Systems. Additional requirements for the Engineered Safety Features System are given in Chapter 6. Given below is the design bases information required in IEEE Standard 279- 1971 Reference [4].

7.3.1.2.1 Generating Station Conditions

The following is a summary of those generating station conditions requiring protective action:

1. Primary System

    a. Rupture in small pipes or cracks in large pipes

    b. Rupture of a reactor coolant pipe (loss of coolant accident)

    c. Steam generator tube rupture

2. Secondary System

    a. Minor secondary system pipe breaks resulting in steam release rates equivalent to flow through a single dump, relief or safety valve

    b. Rupture of a major steam pipe.

7.3.1.2.2 Generating Station Variables

The following list summarizes the generating station variables required to be monitored for the automatic initiation of safety injection during each accident identified in the preceding section. Post-accident monitoring requirements are given in Table 7.5-1.

1. Primary System accidents

    a. Pressurizer pressure

    b. Containment pressure (not required for steam generator tube rupture)

2. Secondary System accidents

    a. Pressurizer pressure

    b. Steam line pressures and steam line pressure rates

    c. Containment pressure

7.3.1.2.3 Spatially Dependent Variables

The ESFAS has no inputs derived from direct measurements of the desired variables that have spatial dependence.

7.3.1.2.4 Limits, Margins and Levels

Prudent operational limits, available margins and setpoints before onset of unsafe conditions requiring protective action are discussed in Chapter 15 and Technical Specifications.

7.3.1.2.5 Abnormal Events

The malfunctions, accidents, or other unusual events which could physically damage Protection System components or could cause environmental changes are as follows:

1. Earthquakes (see Sections 3.7N and 3.7B).

2. Fire (see Section 9.5.1).

3. Deleted.

4. Missiles (see Section 3.5).

5. Flood (see Section 3.4).

6. Wind and tornadoes (see Section 3.3)

7. Loss of Coolant Accidents (see Section 6.2)

8. Steam line breaks (see Section 6.2)

9. Loss of ventilation (see Section 9.4)

7.3.1.2.6 Minimum Performance Requirements

Minimum performance requirements are as follows:

1. System response times

   The ESFAS response time is defined as the interval required for the engineered safety features sequence to be initiated subsequent to the point in time that the appropriate variable(s) exceed setpoints. The response time includes sensor/process (analog) and logic (digital) delay plus, the time delay associated with tripping open the reactor trip breakers and control and latching mechanisms, although the engineered safety features actuation signal occurs before or simultaneously with engineered safety features sequence initiation (see Figure 7.2- 1, Sheet 8). Therefore, the response times to initiating engineered safety features presented herein are conservative. The values listed herein are maximum allowable times consistent with the safety analyses and are systematically verified during plant preoperational startup tests. These maximum delay times thus include all compensation and therefore require that any such network be aligned and operating during verification.

   The ESFAS is always capable of having response time tests performed using the same methods as those tests performed during the preoperational test program or following significant component changes. Refer to Section 7.1.2.11 for a discussion of periodic response time verification capabilities.

   Maximum allowable time delays in generating the Engineered Safeguards actuation signal are justified through the accident analyses and specified in the Technical Requirements Manual.

2. System accuracies

   Accuracies required for generating the required actuation signals for loss of coolant protection are:

a.	Pressurizer pressure	±2.25 percent of span

   Accuracies required in generating the required actuation signals for a steam line break protection are given:

a.	Steam line pressure	±2.25 percent of span
b.	Steam line pressure rate	±5 psi/sec
c.	Containment pressure signal	±1.75 percent of full scale

3. Ranges of sensed variables to be accommodated until conclusion of protective action is assured.

Ranges required in generating the required actuation signals for loss of coolant protection are given:

a.	Pressurizer pressure	1700 to 2500 psig
b.	Containment pressure	0 to 60 psig

Ranges required in generating the required actuation signals for steam line break protection are given:

a.	Steam line pressure	0 to 1300 psig

7.3.1.3 Final System Drawings

The schematic diagrams for the systems discussed in this section are listed in Section 1.7 and are submitted in support of this application.

Figure 7.1-3 depicts the location layout of Engineered Safety Features instrumentation for systems discussed in this section.

7.3.2 ANALYSIS

7.3.2.1 Failure Mode and Effects Analyses

Failure mode and effects analyses have been performed on Engineered Safety Features Systems (ESFS) equipment within the scope of Westinghouse. The results verify that these systems meet protection single failure criteria as required by IEEE Standard 279-1971.

The failure mode and effects analysis (FMEA) which was performed on Engineered Safety Features (ESF) equipment within the scope of Westinghouse was for a typical Westinghouse ESF actuation system (ESFAS). See Reference [5].

Appendix B and C of revision 1 to WCAP-8584 provides interface criteria for the failure modes and effects analysis. This criteria is included in the Westinghouse generic interface criteria which has been provided to CPNPP and has been incorporated into the BOP design.

The analysis has application to the Comanche Peak Nuclear Power Plant (CPNPP) ESFAS as well as generic application to all Westinghouse ESFAS of the CPNPP vintage for the following reasons:

1. The CPNPP ESFAS equipment is designed to equivalent safety design criteria.

2. The actuation of the CPNPP Engineered Safety Features Systems is functionally the same as the systems studies in this analyses.

3. The FMEA was of an ESFAS that employed the Solid State Protection System (SSPS) and the 7300 series Process Control System (PCS).

4. The FMEA has been performed down to the replaceable component level, such as transmitters, relays, modules, and cards which are the same family of components as those applied to the CPNPP.

The conclusion is that the analysis (1) qualitatively demonstrates the reliability of the CPNPP ESFAS to perform is intended function and (2) shows that the ESFAS does comply with the single failure criterion, because no single failure was found which could prevent the ESFAS from generating the proper actuation signal on demand for an engineered safety feature. Random single failures are either in a safe direction or a redundant channel or train ensures the necessary actuation capability.

The basis of an FMEA is principally that single failures are detectable, identifiable, and random. They are not systematic (common mode). The systematic failure considerations applied to equipment hardware, as well as actuation functions, are addressed elsewhere in the FSAR, such as:

1. Seismic qualification of Seismic Category I instrumentation and electrical equipment (Section 3.10). This conforms to Section 4.7.4.2 of IEEE 279-1971.

2. Environmental design of mechanical and electrical equipment (Section 3.11). This conforms to Section 4.7.4.2 of IEEE 279-1971.

3. The Nuclear Instrumentation System, the Solid State Protection System, and the 7300 Series Process Control System noise tests (See Section 7.1.2.2.1 and Reference 2 in Section 7.2).

4. Manual initiation of protective actions (See Section 7.3.2.2.7).

For the engineered safety features and support equipment, failure mode and effects analyses are as follows:

1. Emergency Core Cooling System, Section 6.3 and Table 6.3-10.

2. Containment Spray System, Section 6.2.2 and Table 6.2.5-5.

3. Containment Isolation System, Section 6.2.4 - Failure mode and effects analysis is not applicable; Refer to Table 6.2.4-1 for Valve Arrangement and General Design Criteria applicability.

4. Combustible Gas Control Systems, Section 6.2.5 and Table 6.2.5- 5.

5. Control Room Air-Conditioning System, Section 9.4 and Table 9.4-8.

6. Auxiliary Feedwater System, Section 10.4.9, Table 10.4-9.

7. ESF Filter System, Section 6.5, Tables 9.4-8 and 9.4-9.

8. Component Cooling Water System, Section 9.2.2 and Table 9.2-5.

9. Service Water System, Section 9.2.1 and Table 9.2-1.

10. On Site Power Supply System, Sections 8.3.1, 9.5.4, 9.5.5 and Tables 8.3-3, 9.5-11, 9.5-12.

11. ESF Ventilation System, Section 9.4.5 discusses failure modes.

12. Safety Chilled Water, Section 9.4F discusses system redundancy and single failure.

13. Service Water Intake Structure Ventilation Section 9.4B discusses system redundancy and single failure.

7.3.2.2 Compliance With Standards and Design Criteria

Discussion of GDC are provided in various sections of Chapter 7 where a particular GDC is applicable. Applicable GDC include Criteria 13, 20, 21, 22, 23, 24, 25, 35, 37, 40, 43, and 46. Compliance with certain IEEE Standards is presented in Sections 7.1.2.7, 7.1.2.9, 7.1.2.10, and 7.1.2.11. Compliance with Regulatory Guide 1.22 is discussed in Section 7.1.2.5. The discussion given below shows that the ESFAS complies with IEEE Standard 279-1971, Reference [4]. For the list of references to the discussions of conformance to applicable criteria, see Table 7.1-1.

7.3.2.2.1 Single Failure Criteria

The discussion presented in Section 7.2.2.2.3 is applicable to the ESFAS, with the following exception.

In the engineered safety features, a loss of instrument power will call for actuation of engineered safety features equipment controlled by the specific bistable that lost power (containment spray and RWST Low-Low excepted). The actuated equipment must have power to comply. The power supply for the protection systems is discussed in Section 7.6 and in Chapter 8. For containment spray, the final bistables are energized to trip to avoid spurious actuation. In addition, manual containment spray requires a simultaneous actuation of two manual controls. This is considered acceptable because spray actuation on hi-3 containment pressure signal provides automatic initiation of the system via protection channels meeting the criteria in Reference [3]. Moreover, two sets (two switches per set) of containment spray manual initiation switches are provided to meet the requirements of IEEE Standard 279-1971. Also, it is possible for all engineered safety features equipment (valves, pumps, etc.) to be individually manually actuated from the control board. Hence, a third mode of containment spray initiation is available. The design meets the requirements of Criteria 21 and 23 of the 1971 GDC.

7.3.2.2.2 Equipment Qualification

Equipment qualifications are discussed in Sections 3.10B, 3.10N, 3.11B and 3.11N.

7.3.2.2.3 Channel Independence

The discussion presented in Section 7.2.2.2.3 is applicable. The engineered safety features slave relay outputs from the solid state logic protection cabinets are redundant, and the actuations associated with each train are energized up to and including the final actuators by the separate alternating current (AC) power supplies which power the logic trains.

7.3.2.2.4 Control and Protection System Interaction

The discussions presented in Section 7.2.2.2.3 are applicable.

7.3.2.2.5 Capability for Sensor Checks and Equipment Test and Calibration

The discussions of system testability in Section 7.2.2.2.3 are applicable to the sensors, analog circuitry, and logic trains of the ESFAS.

The following discussions cover those areas in which the testing provisions differ from those for the Reactor Trip System.

Testing of Engineered Safety Features Systems

The engineered safety features systems are tested to provide assurance that the systems will operate as designed and will be available to function properly in the unlikely event of an accident. The testing program meets the requirements of Criteria 21, 37, 40 and 43 of the 1971 GDC and Regulatory Guide 1.22 as discussed in Section 7.1.2.5. The tests described in Section 7.3.2.2.3 and further discussed in Section 6.3.4 meet the requirements on testing of the Emergency Core Cooling System as stated in GDC 37 except for the operation of those components that will cause an actual safety injection. The test, as described, demonstrates the performance of the full operational sequence that brings the system into operation, the transfer between normal and emergency power sources and the operation of associated cooling water systems. The safety injection and residual heat removal pumps are started and operated and their performance verified in a separate test discussed in Section 6.3.4. When the pump tests are considered in conjunction with the Emergency Core Cooling System test, the requirements of GDC 37 on testing of the Emergency Core Cooling System are met as closely as possible without causing an actual safety injection.

Testing as described in Sections 6.3.4, 7.2.2.2.3 and 7.3.2.2.3 provides complete periodic testability during reactor operation of all logic and components associated with the Emergency Core Cooling System. This design meets the requirements of Regulatory Guide 1.22 as discussed in the above sections. The program is as follows:

1. Prior to initial plant operations, Engineered Safety Features System tests will be conducted.

2. Subsequent to initial startup, Engineered Safety Features System tests will be conducted during each regularly scheduled refueling outage.

3. During on-line operation of the reactor, all of the engineered safety features analog and logic circuitry will be fully tested. In addition, essentially all of the engineered safety features final actuators will be fully tested. The remaining few final actuators whose operation is not compatible with continued on-line plant operation will be checked by means of continuity testing.

4. During normal operation, the operability of testable final actuation devices of the engineered safety features systems will be tested by manual initiation from the Control Room.

Performance Test Acceptability Standard for the “S” (Safety Injection Signal) and for the “P” (the Automatic Demand Signal for Containment Spray Actuation) Actuation Signals Generation

During reactor operation the basis for ESFAS acceptability will be the successful completion of the overlapping tests performed on the initiating system and the ESFAS, see Figure 7.3-2. Checks of process indications verify operability of the sensors. Analog checks and tests verify the operability of the analog circuitry from the input of these circuits through the channel bistable output. Channel calibrations verify the operability of the channel bistable output through to and including the logic input relays except for the input relays associated with the containment spray function which are tested during the solid state logic testing. Solid state logic testing also checks the digital signal path from the logic input relay contacts through the logic matrices and master relays and perform continuity tests on the coils of the output slave relays; final actuator testing operates the output slave relays and verifies operability of those devices which require safeguards actuation and which can be tested without causing plant upset. A continuity check is performed on the actuation circuitry of the untestable devices. Operation of the Final devices is confirmed by control board indication or visual observation that the appropriate pump breakers close and automatic valves have completed their travel.

The basis for acceptability for the engineered safety features interlocks will be control board indication of proper receipt of the signal upon introducing the required input at the appropriate setpoint.

Frequency of Performance of Engineered Safety Features Actuation Tests

During reactor operation, complete system testing (excluding sensors or those devices whose operation would cause plant upset) is performed periodically as specified in the Technical Specifications. Testing, including the sensors, is also performed during scheduled plant shutdown for refueling.

Engineered Safety Features Actuation Test Description

The following sections describe the testing circuitry and procedures for the on-line portion of the testing program. The guidelines used in developing the circuitry and procedures are:

1. The test procedures must not involve the potential for damage to any plant equipment.

2. The test procedures must minimize the potential for accidental tripping.

3. The provisions for on-line testing must minimize complication of engineered safety features actuation circuits so that their reliability is not degraded.

Description of Initiation Circuitry

Several systems comprise the total Engineered Safety Features System, the majority of which may be initiated by different process conditions and be reset independently of each other.

The remaining functions (listed in Section 7.3.1.1.1) are initiated by a common signal (safety injection) which in turn may be generated by different process conditions.

In addition, operation of all other vital auxiliary support systems, such as auxiliary feedwater, component cooling and service water, is initiated by the safety injection signal.

Each function is actuated by a logic circuit which is duplicated for each of the two redundant trains of engineered safety features initiation circuits.

The output of each of the initiation circuits consists of a master relay which drives slave relays for contact multiplication as required. The logic, master, and slave relays are mounted in the solid state logic protection cabinets designated train A, and train B, respectively, for the redundant counterparts. The master and slave relay circuits operate various pump and fan circuit breakers or starters, motor operated valve contactors, solenoid operated valves, emergency generator starting, etc.

Analog Testing

Analog testing is identical to that used for reactor trip circuitry and is described in Section 7.2.2.2.3.

Solid State Logic Testing

Except for containment spray channels, solid state logic testing is the same as that discussed in Section 7.2.2.2.3. During logic testing of one train, the other train can initiate the required engineered safety features function. For additional details, refer to Reference [2].

Actuator Testing

At this point, testing of the initiation circuits through operation of the master relay and its contacts to the coils of the slave relays has been accomplished. The ESFAS logic slave relays in the Solid State Protection System output cabinets are subjected to coil continuity tests by the output relay tester in the Solid State Protection System cabinets. Slave relays (K601, K602, etc.) do not operate because of reduced voltage applied to their coils by the mode selector switch (TEST/ OPERATE). A multiple position master relay selector switch chooses different master relays and corresponding slave relays to which the coil continuity is applied. The master relay selector switch is returned to “OFF” before the mode selector switch is placed back in the “OPERATE” mode. However, failure to do so will not result in defeat of the protective function. The ESFAS slave relays are activated during testing by the on-line test cabinet, so that overlap testing is maintained.

The ESFAS final actuation device or actuated equipment testing shall be performed from the engineered safeguards test cabinets. These cabinets are located near the Solid State Logic Protection System equipment. There is one set of test cabinets provided for each of the two protection trains A and B. Each set of cabinets contains individual test switches necessary to actuate the slave relays. To prevent accidental actuation, test switches are of the type that must be rotated and then depressed to operate the slave relays. Assignments of contacts of the slave relays for actuation of various final devices or actuators has been made such that groups of devices or actuated equipment, can be operated individually during plant operation without causing plant upset or equipment damage. In the unlikely event that a safety injection signal is initiated during the test of the final device that is actuated by this test, the device will already be in its safeguards position. Redundant devices in the opposite train will be functional for those components that will be made inoperable due to establishing plant conditions to support testing.

During this last procedure, close communication between the main Control Room operator and the operator at the test panel is required. Prior to the energizing of a slave relay, the operator in the main Control Room assures or establishes plant conditions that will permit operation of the equipment to be actuated by the relay. Appropriate LCOs will be entered for those devices rendered inoperable due to test set-up requirements. After the tester has energized the slave relay, the main Control Room operator observes that all equipment has operated as indicated by appropriate indicating lamps, monitor lamps and annunciators on the control board and records all operations. For the test of slave relays associated with the turbine trip from Reactor Trip or Engineered Safety Features System actuation, the Operator will observe on the OM computer workstation screen that all relay-activated equipment has operated as indicated. After observing all the results, the Operator then resets all devices and prepares for operation of the next slave relay actuated equipment.

The following methodology will be used for those final actuation device circuits that cannot be operated by an individual slave relay or actuation of which will cause plant upset/equipment damage and for which no additional block testing circuitry is provided. The end device will be declared inoperable and then disabled from operating by removing fuses/opening breakers, etc. The slave relay will be energized as described above. Proper slave relay state change will then be verified through local control circuit continuity measurements. Restoration of the slave will be as above followed by restoration of equipment to the status required per Technical Specifications.

Automatic actuation circuitry of all engineered safety features devices actuated by ESFAS initiation circuits, with the exceptions noted in Section 7.1.2.5 under a discussion of Regulatory Guide 1.22, are tested by means of the procedures outlined above.

Actuator Blocking and Continuity Test Circuits

This section describes methodology used for those final actuation devices that cannot be designed to be actuated during plant operation (discussed in Section 7.1.2.5) and that have been assigned to slave relays for which additional test circuitry has been provided to individually block actuation of a final device upon operation of the associated slave relay during testing. Operation of these slave relays, including contact operations, and continuity of the electrical circuits associated with the final devices control are checked in lieu of actual operation. The circuits provide for monitoring of the slave relay contacts, the devices control circuit cabling, control voltage and the devices actuation solenoids. Interlocking prevents blocking the output from more than one output relay in a protection train at a time. Interlocking between trains is also provided to prevent continuity testing simultaneously in both trains; therefore, the redundant device associated with the protection train not under test will be available in the event the protection action is required. If an accident occurs during testing, the automatic actuation circuitry will override testing as noted above. An exception to this is that if the accident occurs while testing a slave relay whose output must be blocked, the final actuation devices associated with this slave relay will not be overridden; however, the redundant devices in the other train would be operational and would perform the required safety function. Actuation devices to be blocked are identified in Section 7.1.2.5.

The continuity test/indication circuits for these components with blocking relay circuits are verified by proving lights on the safeguards test racks, or by indication on OM computer workstations in the control room.

The typical schemes for blocking operation of selected protection function actuator circuits are shown in Figure 7.3-3 as details A and B. The schemes operate as explained below and are duplicated for each safeguards train. The method for blocking operation of turbine trip (from Reactor Trip or Engineered Safety Features System actuation) is an exception to these schemes. This method is also explained below.

Detail A shows the circuit for contact closure for protection function actuation. Under normal plant operation, and equipment not under test, the test lamps “DS*” for the various circuits will be energized. Typical circuit path will be through the normally closed test relay contact “K8*” and through test lamp connections 1 to 3. Coil “X1” will be capable of being energized for protection function actuation upon closure of solid state logic output relay contacts “K*”. Coil “X1” is typical for a breaker closing auxiliary coil, motor starter master coil, coil of a solenoid valve, auxiliary relay, etc. When the contacts “K8*” are opened to block energizing of coil “X1”, the white lamp is de-energized, and the slave relay “K*” may be energized to perform continuity testing. To verify operability of the blocking relay in both blocking and restoring normal service, open the blocking relay contact in series with lamp connections - the test lamp should be de-energized; close the blocking relay contact in series with the lamp connections - the test lamp should now be energized, which verifies that the circuit is now in its normal, i.e., operable condition.

Detail B shows the circuit for contact opening for protection function actuation. Under normal plant operation, and equipment not under test, the white test lamps “DS*” for the various circuits will be energized, and green test lamp “DS*” will be de-energized. Typical circuit path for white lamp “DS*” will be through the normally closed solid state logic output relay contact “K*” and through test lamp connections 1 to 3. Coils “Y1” and “Y2” will be capable of being de- energized for protection function actuation upon opening of solid state logic output relay contacts “K*.” Coil “Y2” is typical for a solenoid valve coil, auxiliary relay, etc. When the contacts “K8*” are closed to block de-energizing of coils “Y1” and “Y2”, the green test lamp is energized and the slave relay “K*” may be energized to verify operation (opening of its contacts). To verify operability of the blocking relay in both blocking and restoring normal service, close the blocking relay contact to the green lamp - the green test lamp should now be energized also; open this blocking relay contact - the green test lamp should be de- energized, which verifies that the circuit is now in its normal, i.e., operable position.

Time Required for Testing

It is estimated that analog testing can be performed at a rate of several channels per hour. Logic testing of both trains A and B can be performed in less than 30 minutes. Testing of actuated components (including those which can only be partially tested) will be a function of Control Room operator availability. It is expected to require several shifts to accomplish these tests. During this procedure automatic actuation circuitry will override testing, except for those few devices associated with a single slave relay whose outputs must be blocked (and then only while blocked) and for those devices rendered inoperable due to test set-up requirements. It is anticipated that testing associated with one of these slave relays could take several minutes. During this time the redundant devices in the other trains would be functional.

Summary of On-Line Testing Capabilities

The procedures described provide capability for checking completely from the process signal to the logic cabinets and from there to the individual pump and fan circuit breakers or starters, valve contactors, pilot solenoid valves, etc., including all field cabling actually used in the circuitry

called upon to operate for an accident condition. For those few devices whose operation could adversely affect plant or equipment operation, the same procedure provides for checking from the process signal to the logic rack. Actuation of the final actuation device circuitry is checked through continuity testing.

The procedures require testing at various locations.

1. Analog testing and verification of bistable setpoint are accomplished at process analog racks. Verification of bistable relay operation is done at the main Control Room status lights.

2. Logic testing through operation of the master relays and low voltage application to slave relays is done at the logic rack test panel.

3. Testing of pumps, fans and valves is done at a test panel located in the vicinity of the logic racks in combination with the Control Room operator.

4. Continuity testing for those circuits with additional block testing circuits is done at the same test panel mentioned in item 3 above.

5. Control circuit continuity testing of contacts associated with those untestable slave relays without additional block testing circuitry will be performed locally.

For the case of turbine trip (from Reactor Trip or Engineered Safety Features System actuation), the trip and blocking relay contacts input directly to the Turbine Generator Digital Protection System. Test lamps (as described for Details A and B above) do not exist for these circuits. During normal plant operation with no equipment test in progress, trip and block indications are absent on the Turbine Protection System operator workstation. When the blocking relay is actuated, the block is indicated on the operator workstation, and the turbine trip output signal is blocked in the system software. When the trip relay is actuated, the trip is indicated on the operator workstation, confirming operation of the trip relay contacts. The trip and blocking relays are then restored. The associated operator workstation indications clear to indicate that the relays have been satisfactorily returned to the normal operating condition. As long as the block signal is present, annunciation is provided on the operator workstation to ensure that the block is removed after testing is complete.

Testing During Shutdown

Emergency Core Cooling System tests will be performed periodically in accordance with the Technical Specifications with the Reactor Coolant System isolated from the Emergency Core Cooling System by closing the appropriate valves. A test safety injection signal will then be applied to initiate operation of active components (pumps and valves) of the Emergency Core Cooling System. This is in compliance with Criterion 37 of the 1971 GDC.

Containment Spray System tests will be performed at each major fuel reloading. The tests will be performed with the isolation valves in the spray supply lines at the containment and spray additive tank blocked closed and are initiated by tripping the normal actuation instrumentation.

Periodic Maintenance Inspections

The maintenance procedures which follow will be accomplished per applicable plant procedures. The frequency will depend on the operating conditions and requirements of the reactor power plant. If any degradation of equipment operation is noted, either mechanically or electrically, remedial action is taken to repair, replace, or readjust the equipment. Optimum operating performance must be achieved at all times.

Maintenance procedures include the following:

1. Check cleanliness of all exterior and interior surfaces.

2. Check all fuses for corrosion.

3. Inspect for loose or broken control knobs and burned out indicator lamps.

4. Inspect for moisture and condition of cables and wiring.

5. Mechanically check all connectors and terminal boards for looseness, poor connection, or corrosion.

6. Inspect the components of each assembly for signs of overheating or component deterioration.

7. Perform complete system operating check.

The balance of the requirements listed in Reference [4] (Sections 4.11 through 4.22) are discussed in Section 7.2.2.2.3. Section 4.20 of Reference [4] receives special attention in Section 7.5.

7.3.2.2.6 Manual Resets and Blocking Features

Manual reset and block functions for the ESFAS are shown on Figure 7.2-1, Sheet 8. The purpose for each is as follows:

1. Safety injection system level reset is required after an accident to terminate safety injection and allow switchover to recirculation in the case of operator actions following a LOCA, to permit the operator to take over manual control of a tripped reactor, and, to complete the shutdown process following a transient or accident that initiates safety injection, as well as to recover from a false signal.

2. Containment spray actuation reset is needed to allow for recovery from all accidents that initiate containment spray, as well as to recover from a false signal.

3. Containment isolation phase A reset is needed to allow for defeat of containment isolation phase A during recovery from an accident, as well as to recover from a false signal.

4. Containment isolation phase B reset is needed to allow for defeat of containment isolation phase B during recovery from an accident that initiated phase B isolation, as well as to recover from a false signal.

5. Containment ventilation isolation reset is needed to allow for defeat of containment ventilation isolation during recovery from an accident that initiated it. In recovering from a false signal, the false signal is expected to be cleared prior to reset.

6. Control Room intake duct isolation and emergency filter actuation reset is needed to permit the operator to bring the Control Room air conditioning system into its normal operational mode, as well as to recover from a false signal. The functional logic for this reset function is shown in instrumentation and control diagrams listed under “Control Room Air Conditioning” in Table 1.7-1.

7. Main steam line isolation reset is a retentive memory circuit only, and does not include the maintained-actuation-signal-block function. This manual reset will allow opening the steam line stop valves following a transient should the transient automatically generate main steam line isolation actuation after the automatic actuation signal has been eliminated for reasons that include the use of the steam dump through the non-faulted steam generator if offsite power is available.

The manual reset feature associated with containment spray actuation, and the manual block features associated with pressurizer and steam line safety injection signals, are discussed in further detail in the following paragraphs.

The manual reset feature associated with containment spray actuation is provided in the standard design of the Westinghouse Solid State Protection System design for two basic purposes. First, the feature permits the operator to start an interruption procedure of automatic containment spray in event of false initiation of an actuate signal. Second, although spray system performance is automatic, the reset feature enables the operator to start a manual takeover of the system to handle unexpected events which can be better dealt with by operator appraisal of changing conditions following an accident.

It is most important to note that manual control of the spray system does not occur, once actuation has begun, by just resetting the associated logic devices alone. Components will seal in (latch) so that removal of the actuate signal, in itself, will neither cancel or prevent completion of protective action or provide the operator with manual override of the automatic system by this single action. In order to take complete control of the system to interrupt its automatic performance, the operator must deliberately unlatch relays which have “sealed in” the initial actuate signals in the associated motor control center, in addition to tripping the pump motor circuit breakers, if stopping the pumps is desirable or necessary.

The manual reset feature associated with containment spray, therefore, does not perform a bypass function. It is merely the first of several manual operations required to take control from the automatic system or interrupt its completion should such an action be considered necessary.

In event that the operator anticipates system actuation and erroneously concludes that it is undesirable or unnecessary and imposes a standing reset condition in one train (by operating and holding the corresponding reset switch at the time the initiate signal is transmitted) the other train will automatically carry the protective action to completion. In the event that the reset condition is imposed simultaneously in both trains at the time the initiate signals are generated, the automatic sequential completion of system action is interrupted and control has been taken by the operator. Manual takeover will be maintained, even though the reset switches are

released, if the original initiate signal exists. Should the initiate signal then clear and return again, automatic system actuation will repeat.

Note also that any time delays imposed on the system action are to be applied after the initiating signals are latched. Delay of actuate signals for fluid systems line-up, load sequencing, etc., do not provide the operator time to interrupt automatic completion, with manual reset alone, as would be the case if time delay was imposed prior to sealing of the initial actuate signal.

The manual block features associated with pressurizer and steam line safety injection signals provide the operator with the means to block initiation of safety injection during plant startup and shutdown.

Safety injection actuation on low pressurizer pressure and low steam line pressure may be manually blocked when NSSS pressure falls below P-11. If a steam line rupture occurs while both of these safety injection actuation signals are blocked, steam line isolation will occur on high negative steam pressure rate.

Prior to blocking safety injection, the shutdown margin at hot standby, as listed in Technical Specifications, will be increased by the amount necessary to makeup for the shutdown margin to be lost via cooldown and to maintain the Technical Specification shutdown margin for the applicable mode of operation. Thus, before cooldown is initiated, the boron concentration will be at least as great as the maximum required boron concentration during cooldown. This increased margin ensures that the steam line rupture return to power from hot zero power (see Section 6.3.3.3) is higher than the case where safety injection actuation is manually blocked on low steam line pressure and low pressurizer pressure. Therefore, no further protection is required.

During a plant cooldown, the operator is instructed to manually block the automatic safety injection signals when pressure falls below P-11. This action disarms the safety injection signals from the pressurizer pressure transmitters along with the steam line pressure transmitters. The other safety injection signal, containment high pressure, is armed and will actuate safety injection if the setpoint is exceeded. Manual safety injection actuation is also available.

During the post-LOCA recovery, or recovery from any event where safety injection has been actuated, the reactor operator is instructed to reset Safety Injection and, as conditions allow, secure various ECCS components. If plant conditions degrade such that ECCS actuation is again required, the reactor operator is instructed to individually restart any required ECCS components.

These block features meet the requirements of Section 4.12 of IEEE Standard 279-1971 in that automatic removal of the block occurs when plant conditions require the protection system to be functional.

7.3.2.2.7 Manual Initiation of Protective Actions (Regulatory Guide 1.62)

There are four individual main steam stop valve momentary control switches (one per loop) mounted on the control board. Each switch when actuated will isolate one of the main steam lines. In addition, there will be two system level switches. Each switch will actuate all four main steam line isolation valves of the system level. Manual initiation of switchover to recirculation is in compliance with Section 4.17 of IEEE Standard 279-1971 with the following comment.

Manual initiation of either one of two redundant safety injection actuation main control board mounted switches provides for actuation of the components required for reactor protection and mitigation of adverse consequences of the postulated accident, including delayed actuation of sequenced started emergency electrical loads as well as components providing switchover from the safety injection mode to the cold leg recirculation mode following a loss of primary coolant accident. Therefore, once safety injection is initiated, those components of the Emergency Core Cooling System (see Section 6.3) which are automatically realigned as part of the semiautomatic switchover go to completion on low-low refueling water storage tank (RWST) water level without any manual action. Manual operation of other components or manual verification of proper position as part of emergency procedures is not precluded nor otherwise in conflict with the above described compliance to Section 4.17 of IEEE Standard 279- 1971 of the semiautomatic switchover circuits.

No exception to the requirements of IEEE Standard 279-1971 has been taken in the manual initiation circuit of safety injection. Although Section 4.17 of IEEE Standard 279-1971 requires that a single failure within common portions of the protective system shall not defeat the protective action by manual or automatic means, the standard does not specifically preclude the sharing of initiated circuitry logic between automatic and manual functions. It is true that the manual safety injection initiation functions associated with one actuation train (e.g., train A) shares portions of the automatic initiation circuitry logic of the same logic train; however, a single failure in shared functions does not defeat the protective action of the redundant actuation train (e.g., train B). A single failure in shared functions does not defeat the protective action of the safety function. It is further noted that the sharing of the logic by manual and automatic initiation is consistent with the system level action requirements of the IEEE Standard 279-1971, Section 4.17, and the minimization of complexity.

Manual actuation of main steam line isolation (all valves), containment isolation (Phase A), and containment spray actuation conforms to the same criteria herein described for the manual safety injection manual actuation functions.

7.3.2.2.8 Component Control Switches

The control switches for ESF final actuators, e.g., pumps, valves and dampers which receive an ESF actuation signal are either the spring-return-to-automatic type or are maintained in one manual (i.e., non-automatic) position. The safe position of the ESF final actuator will be obtained in response to an ESF actuation signal or the maintained position of the control switch will place the actuator in the safe position required by the ESF signal. This design assures the completion of the protective function once it has been initiated, regardless of the position of the actuator control switch.

When operating conditions necessitate, the operator can manually override the automatic operation of individual components. The manual overrides are accomplished in the following manner:

1. Valves and Dampers - The control switch must be held by the operator in the alternate position for the duration of the period that the alternate action is required. The automatic action is restored as soon as the operator releases the control switch.

2. Pumps - For those pumps that receive an auto start signal from the ESF sequencer, the operator can stop the pump by holding the control switch in the stop position. After the

sequencer has cycled, the pump will not restart when the operator releases the control switch. This override will be continuously indicated in the control room and both the override and the indication will be automatically removed whenever permissive conditions for the bypass are not met.

The alternate operating mode is provided with permissives or operator lockouts that prevents the operator from intervening in the automatic operation of the final actuator when the ESF sequencer is in operation.

This manual override meets the intent of IEEE Std 279 requirements for operating bypasses. The channel is not removed from service during the period that the operator is manually operating the component and the operator has status indicating lamps to inform him of the status of the final actuator.

7.3.2.3 Further Considerations

In addition to the considerations given above, a loss of instrument air or loss of component cooling water to vital equipment has been considered. Neither the loss of instrument air nor the loss of cooling water (assuming no other accident conditions) can cause safety limits as given in the Technical Specifications to be exceeded. Likewise, loss of either one of the two will not adversely affect the core or the Reactor Coolant System nor will it prevent an orderly shutdown if this is necessary. Furthermore, all pneumatically operated valves and controls will assume a preferred operating position upon loss of instrument air or safety related air accumulators are provided (see Section 9.3.1). It is also noted that, for conservatism during the accident analysis (see Chapter 15), credit is not taken for the instrument air systems nor for any control system benefit.

The design does not provide any circuitry which will directly trip the reactor coolant pumps on a loss of component cooling water. Normally, indication in the Control Room is provided whenever component cooling water is lost. The reactor coolant pumps can run about 10 minutes after a loss of component cooling water. This provides adequate time for the operator to correct the problem or trip the plant if necessary.

A loss of Component Cooling Water (CCW) flow to the Reactor Coolant Pumps (RCP) is alarmed in the Control Room. The alarms are not expected to be operable during or after a LOCA or SSE condition. The alarm system that is actuated on CCW loss of flow to the RCP consists of:

1. Flow detection bistables or switches.

2. Annunciator electronics, lamp cabinets, audible horns.

Although the alarm system is not designed to remain operational with single failure in either “a” or “b” above, item “b” does contain certain features for increased reliability as follows:

1. Redundant input power sources, 125 volts DC and 120V AC with auto transfer feature.

2. Display windows in Lamp Cabinets with dual lamps.

3. Multiple audible horns

Furthermore, the alarm contacts which are normally closed, open to alarm. This provides a self monitoring feature which detects an open circuit condition, such as caused by an open wire.

The alarm system, as described above is classified as non safety related, and is not qualified to IEEE-323-1974 and 344-1975.

The alarms for loss of CCW to the RCP upper bearing lube-oil cooler or to the lower bearing lube-oil cooler are originated by in-line rotameters. The time delay between the loss of CCW flow and the initiation of the alarm, is between 4 to 7 seconds time lag is due mainly to the time it takes the float to change positions. The time lag for the remaining instrument loop is in the order of milli-secs. The alarms for loss of CCW to the RCP Motor Air Cooler or to the RCP thermal barrier are originated by differential pressure transmitter, square-root-extractor and bistable. The time delay is, in this case, less than a second and is the combined time of the transmitter to sense the change of flow and the bistable relay to operate. The time lag for the remaining instruments of the loop is in the order of the milli-seconds.

7.3.2.4 Summary

The effectiveness of the ESFAS is evaluated in Chapter 15, based on the ability of the system to contain the effects of Condition III and IV events, including loss of coolant and steam break accidents. The ESFAS parameters are based upon the component performance specifications which are given by the manufacturer or verified by test for each component. Appropriate factors to account for uncertainties in the data are factored into the constants characterizing the system.

The ESFAS must detect Condition III and IV events and generate signals which actuate the engineered safety features. The system must sense the accident condition and generate the signal actuating the protection function reliably and within a time determined by and consistent with the accident analyses in Chapter 15.

Much longer times are associated with the actuation of the mechanical and fluid system equipment associated with engineered safety features. This includes the time required for switching, bringing pumps and other equipment to speed and the time required for them to take load.

Operating procedures require that the complete ESFAS normally be operable. However, redundancy of system components is such that the system operability assumed for the safety analyses can still be met with certain instrumentation channels out of service. Channels that are out of service are to be placed in the tripped mode or bypass mode in the case of containment spray.

7.3.2.4.1 Loss of Coolant Protection

By analysis of loss of coolant accident and in system tests it has been verified that except for very small coolant system breaks which can be protected against by the charging pumps followed by an orderly shutdown, the effects of various loss of coolant accidents are reliably detected by the low pressurizer pressure and water level signal; the Emergency Core Cooling System is actuated in time to prevent or limit core damage.

For large coolant system breaks the passive accumulators inject first, because of the rapid pressure drop. This protects the reactor during the unavoidable delay associated with actuating the active Emergency Core Cooling System phase.

High containment pressure also actuates the Emergency Core Cooling System. Therefore, emergency core cooling actuation can be brought about by sensing this other direct consequence of a primary system break; that is the ESFAS detects the leakage of the coolant into the containment. The generation time of the actuation signal of about 1.5 seconds, after detection of the consequences of the accident, is adequate.

Containment spray will provide additional emergency cooling of the containment and also limit fission product release upon sensing elevated containment pressure (hi-3) to mitigate the effects of a loss of coolant accident.

The delay time between detection of the accident condition and the generation of the actuation signal for these systems is assumed to be about 1.0 second, well within the capability of the Protection System equipment. However, this time is short compared to that required for startup of the fluid systems.

The analyses in Chapter 15 show that the diverse methods of detecting the accident condition and the time for generation of the signals by the protection systems are adequate to provide reliable and timely protection against the effects of loss of coolant.

7.3.2.4.2 Steam Line Break Protection

The Emergency Core Cooling System is also actuated in order to protect against a steam line break. About 2.0 seconds elapses between sensing low steam line pressure and generation of the actuation signal. Analysis of steam line break accidents assuming this delay for signal generation shows that the Emergency Core Cooling System is actuated for a steam line break in time to limit or prevent further core damage for steam line break cases. There is a reactor trip but the core reactivity is further reduced by the highly borated water injected by the Emergency Core Cooling System.

Additional protection against the effects of steam line break is provided by feedwater isolation which occurs upon actuation of the Emergency Core Cooling System. Feedwater line isolation is initiated in order to prevent excessive cooldown of the reactor vessel and thus protect the Reactor Coolant System boundary.

Additional protection against a steam line break accident is provided by closure of all steam line isolation valves in order to prevent uncontrolled blowdown of all steam generators. The generation of the Protection System signal (about 2.0 seconds) is again short compared to the time to trip the main steam isolation valves which are designed to close in less than 5 seconds after receipt of an isolation signal.

In addition to actuation of the engineered safety features, the effect of a steam line break accident may also generate a signal resulting in a reactor trip on overpower or following Emergency Core Cooling System actuation. However, the core reactivity is further reduced by the highly borated water injected by the Emergency Core Cooling System.

The analyses in Chapter 15 of the steam line break accidents and an evaluation of the Protection System instrumentation and channel design shows that the ESFAS is effective in preventing or mitigating the effects of a steam line break accident.

REFERENCES

1. Reid, J. B., “Process Instrumentation for Westinghouse Nuclear Steam Supply System (4 Loop Plant using WCID 7300 Series Process Instrumentation),” WCAP-7913, March 1973.

2. Katz, D. N., “Solid State Logic Protection System Description,” WCAP-7488-L (Proprietary), March 1971 and WCAP-7672 (Non- Proprietary), May 1971.

3. Swogger, J. W., “Testing of Engineered Safety Features Actuation System,” WCAP-7705, Revision 2, January 1976.

4. The Institute of Electrical and Electronics Engineers, Inc., “IEEE Standard: Criteria for Protection System for Nuclear Power Generating Stations,” IEEE Standard 279-1971.

5. Eggleston, F. T., Rawlins, D. H., Petrow J. R., “Failure Mode and Effects Analysis (FMEA) of the Engineering Safeguard Features Actuation System,” WCAP-8584, (Proprietary), Revision 1, February 1980 and WCAP-8760 (Non-Proprietary), Revision 1, February 1980.

6. Morgan, C. E., “Elimination of Periodic Protection Channel Response Time Tests,” WCAP-14036-P-A, Revision 1, October 6, 1998.

TABLE 7.3-1
INSTRUMENTATION OPERATING CONDITIONS FOR ENGINEERED SAFETY FEATURES

No	Functional Unit	No. of Channels	No of Channels to Trip
1.	Safety injection		
	a. Manual	2	1
	b. High containment pressure (hi-1)	3	2
	c. Low compensated steam line pressure[(a)]	12 (3/steam line)	2 in any steam line
	d. Pressurizer low pressure[(a)]	4	2
2.	Containment spray		
	a. Manual[(b)]	4	2
	b. Containment pressure (hi-3)	4	2

a) Permissible bypass if reactor coolant pressure less than 2000 psig.

b) Manual actuation of containment spray is accomplished by actuating either of two sets (two switches per set). Both switches in a set must be actuated to obtain a manually initiated spray signal. The sets are wired to meet separation and single failure requirements of IEEE Standard 279-1971. Simultaneous operation of two switches is desirable to prevent inadvertent spray actuation.

TABLE 7.3-2
INSTRUMENT OPERATING CONDITIONS FOR ISOLATION FUNCTIONS

No.	Functional Unit	No. of Channels	No. of Channels to Trip
1.	Containment isolation		
a.	Automatic safety injection (Phase A)	See item 1 (b) through (d) of Table 7.3-1	
b.	Containment pressure (Phase B)	See item 2 (b) of Table 7.3-1	
c.	Manual		
1)	Phase A	2	1
2)	Phase B	See item 2 (a) of Table 7.3-1	
2.	Steam line isolation		
a.	Low steam line pressure[a]	12 (3/steam line)	2 in any one steam line
b.	Containment pressure (hi-2)	3	2
c.	High steam pressure rate	12 (3/steam line)	2 in any one
d.	Manual	2[b]	1[b]
3.	Feedwater line isolation		
a.	Safety injection	See item 1 of Table 7.3-1	
b.	Steam generator high-high level 2/3 on any steam generator	12 (3/Steam generator)	2 in any one steam generator
c.	Low T_{avg}(Average Temperature) Coincident with Reactor Trip	See Figure 7.2-1 Sh. 2, 5 & 13	See Figure 7.2-1 Sh. 2,5 & 13

a) Permissible by pass if reactor coolant pressure less than p-11 setpoint.

b) Additionally there is a switch for each loop that will actuate the mainsteam isolation valve for its respective loop.

TABLE 7.3-3
INTERLOCKS FOR ENGINEERED SAFETY FEATURES ACTUATION SYSTEM

Designation	Input	Function Performed
P-4	Reactor trip	Actuates turbine trip
		Closes main and bypass feedwater valves on Low T_{avg} below setpoint.
		Prevents opening of main and bypass feedwater valves which were closed by injection or high-high steam generator water level
		Allows manual block of the automatic reactuation of safety injection
	Reactor not tripped	Defeats the block preventing automatic reactuation of safety injection
P-11	2/3 pressurizer pressure below setpoint	Allows manual block of safety injection actuation on low pressurizer pressure signal
		Allows manual block of safety injection actuation on low compensated steam line pressure signal
	2/3 pressurizer pressure above setpoint	Defeats manual block of safety injection actuation
P-12	2/4 Lo-Lo T_{avg} below setpoint	Blocks steam dump.
	3/4 Lo-Lo T_{avg} above setpoint	Defeats the manual bypass of steam dump block
P-14	2/3 steam generator water level above setpoint on any steam generator	Closes all feedwater control valves and isolation valves
		Trips all main feedwater pumps which closes the pumps discharge valves
		Actuates turbine trip

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 1 of 20)

Equipment Identification[a]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-FCV-510	SG 1 Main Feedwater Flow Control Valve	A/B	Close	M1-2203-05
1-FCV-520	SG 2 Main Feedwater Flow Control Valve	A/B	Close	M1-2203-05
1-FCV-530	SG 3 Main Feedwater Flow Control Valve	A/B	Close	M1-2203-05
1-FCV-540	SG 4 Main Feedwater Flow Control Valve	A/B	Close	M1-2203-05
1-LV-2162	SG 1 Main Feedwater Bypass Control Valve	A/B	Close	M1-2203-05
1-LV-2163	SG 2 Main Feedwater Bypass Control Valve	A/B	Close	M1-2203-05
1-LV-2164	SG 3 Main Feedwater Bypass Control Valve	A/B	Close	M1-2203-05
1-LV-2165	SG 4 Main Feedwater Bypass Control Valve	A/B	Close	M1-2203-05
1-HV-2134	SG #1 Feedwater Isolation Valve	A/B	Close	M1-2203-04
1-HV-2135	SG #2 Feedwater Isolation Valve	A/B	Close	M1-2203-04
1-HV-2136	SG #3 Feedwater Isolation Valve	A/B	Close	M1-2203-04
1-HV-2137	SG #4 Feedwater Isolation Valve	A/B	Close	M1-2203-04
1-LCV-1003	RC Drain Tank Pressure & Level Control Valve	A	Close	M1-2264-02

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 2 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1HV-2185	FWIV Bypass Valve	A/B	Close	M1-2203-07
1HV-2186	FWIV Bypass Valve	A/B	Close	M1-2203-07A
1HV-2187	FWIV Bypass Valve	A/B	Close	M1-2203-07A
1HV-2188	FWIV Bypass Valve	A/B	Close	M1-2203-07A
SJ14S031	Trip Fluid Solenoid Valve Channel 1	A/B	Trip	E1-0022-C
SJ14S032	Trip Fluid Solenoid Valve Channel 2	A/B	Trip	E1-0022-C
SJ14S033	Trip Fluid Solenoid Valve Channel 3	A/B	Trip	E1-0022-C
1-HV-3486	Service Air HDR to Containment Isolation Valve	B	Close	M1-2216-03
1-HV-3487	Instrument Air HDR to Containment Isolation Valve	B	Close	M1-2216-03
1-HV-4075B	Fire Prot. Wtr. to Standpipes Isolation Valve	B	Close	M1-2225-01B
1-HV-4075C	Fire Prot. Wtr. to Standpipes Isolation Valve	A	Close	M1-2225-01B
1-HV-4165	Pressurizer Stm. Space - Prim. Samp. Sys. Valve	A	Close	M1-2228-01
1-HV-4166	Pressurizer Stm. Space to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-01

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 3 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-4167	Pressurizer Liq. Space to Prim. Samp. Sys Control Valve.	B	Close	M1-2228-01A
1-HV-4168	RC Loop 1 HOT LEG to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-03
1-HV-4169	RC Loop 4 HOT LEG to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-03
1-HV-4170	RC Loops 1 & 4 HOT LEG to Prim. Samp. Sys Control Valve.	B	Close	M1-2228-04
1-HV-4171	Accumulator 1 Liq. Space to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-01
1-HV-4172	Accumulator 2 Liq. Space to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-01
1-HV-4173	Accumulator 3 Liq. Space to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-01
1-HV-4174	Accumulator 4 Liq. Space to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-01
1-HV-4175	Accumulator 1 through 4 Liq. Space to Prim. Samp. Sys. Control Valve	B	Close	M1-2228-01A
1-HV-4176	Pressurizer Stm. Space to Prim. Samp. Sys. Control Valve	B	Close	M1-2228-01A
1-HV-4178	RHR Loop to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-02
1-HV-4179	RHR Loop to Prim. Samp. Sys. Control Valve	B	Close	M1-2228-02
1-HV-4182	Containment Sump to Prim. Samp. Sys. Control Valve	A	Close	M1-2228-05
1-HV-5157	Containment Sump Drain Valve	B	Close	M1-2238-02

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 4 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-5158	Containment Sump Drain Valve	A	Close	M1-2238-02
1-HV-5544	Containment Atmos. to Rad. Monitor Sys. Control Valve	B	Close	M1-2301-06
1-HV-5545	Containment Atmos. to Rad. Monitor Sys. Control Valve	A	Close	M1-2301-06 (M2-2301-04)
1-HV-5546	Containment Rad. Monitor Sys. Exh. Control Valve	B	Close	M1-2301-06 (M2-2301-04)
1-HV-5547	Containment Rad. Monitor Sys. Exh. Control Valve	A	Close	M1-2301-06 (M2-2301-04)
1-7126	RC Drain Tank Vent Valve	A	Close	M1-2264-06
1-7136	RC Drain Tank Disch. Pump Suct. Isolation Valve	B	Close	M1-2264-07
1-7150	RC Drain Tank Vent Isolation Valve	B	Close	M1-2264-09
1-HV-7311	PASS to RC Drain Tank Control Valve	B	Close	M1-2264-10
1-HV-7312	PASS to RC Drain Tank Control Valve	A	Close	M1-2264-10
1-8026	Pressurizer Rel. Tk. Vent to WPS Isolation Valve	A	Close	M1-2251-08
1-8027	Pressurizer Rel. Tk. Vent to WPS Isolation Valve	B	Close	M1-2251-08
1-8047	Rctr. Makeup Wtr. to Pressurizer Rel. Tank Isolation Valve	B	Close	M1-2251-09

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 5 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-LCV-0459	Letdown Isolation Valve	A	None	M1-2253-08
1-LCV-0460	Letdown Isolation Valve	A	None	M1-2253-08
1-8100	RCP Seal Wtr. Isolation Valve	B	Close	M1-2253-09
1-8112	RCP Seal Wtr. Isolation Valve	A	Close	M1-2253-09
1-8152	Letdown Line Isolation Valve	B	Close	M1-2253-14
1-8160	Letdown Line Isolation Valve	A	Close	M1-2253-15
1-HV-8220	Charging Pump Suction High Point Vent Valve	A	Close	M1-2255-07
1-HV-8221	Charging Pump Suction High Point Vent Valve	B	Close	M1-2255-07
1-8888	Accumulator Fill Line Control Valve	B	Close	M1-2263-11
1-8871	Test Line Header Control Valve	A	Close	M1-2262-05
1-8823	Safety Injection Pump Test Line Control Valve	A	Close	M1-2263-08
1-8824	Safety Injection Pump Test Line Control Valve	A	Close	M1-2263-08
1-8825	Safety Injection Pump Test Line Control Valve	A	Close	M1-2263-08
1-8843	Charging Pump Test Line Isolation Valve	B	Close	M1-2261-08

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 6 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-8880	Accumulator N_2 Supply Isolation Valve	B	Close	M1-2262-07
1-8881	Safety Injection Pump Hot Leg Test Line Control Valve	A	Close	M1-2263-10
1-8890A	RHR Pump Test Line Control Valve	A	Close	M1-2263-11
1-8890B	RHR Pump Test Line Control Valve	B	Close	M1-2263-11
1-8964	Test Line Header Control Valve	B	Close	M1-2262-08
1-HV-5365	Demineralized Water, Containment Isolation Valve	B	Close	M1-2242-01
1-HV-5366	Demineralized Water, Containment Isolation Valve	A	Close	M1-2242-01
1-HV-4710	CCW System, Containment Isolation Valve	B	Close	M1-2231-05
1-HV-4711	CCW System, Containment Isolation Valve	B	Close	M1-2231-05
1-HV-4725	CCW System, Containment Isolation Valve	A	Close	M1-2231-07
1-HV-4726	CCW System, Containment Isolation Valve	B	Close	M1-2231-07
1-HV-5536	Containment Purge Air Supply Isolation Valve	B	None	M1-2301-04 (M2-2301-02)
1-HV-5537	Containment Purge Air Supply Isolation Valve	A	None	M1-2301-04 (M2-2301-02)

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 7 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-5538	Containment Purge Air Exh. Isolation Valve	B	None	M1-2301-04 (M2-2301-02)
1-HV-5539	Containment Purge Air Exh. Isolation Valve	A	None	M1-2301-04 (M2-2301-02)
1-HV-5540	Containment Hydrogen Controlled Purge Exh. Isolation Valve	B	Close	M1-2301-05 (M2-2301-3)
1-HV-5541	Containment Hydrogen Controlled Purge Exh. Isolation Valve	A	Close	M1-2301-05 (M2-2301-03)
1-HV-5542	Containment Hydrogen Controlled Purge Supply Isolation Valve	B	Close	M1-2301-05 (M2-2301-03)
1-HV-5543	Containment Hydrogen Controlled Purge Supply Isolation Valve	A	Close	M1-2301-05 (M2-2301-03)
1-HV-5548	Containment Purge Relief Disch. Isolation Valve	B	Close	M1-2301-07 (M2-2301-05)
1-HV-5549	Containment Purge Relief Disch. Isolation Valve	A	Close	M1-2301-07 (M2-2301-05)
1-HV-5556	Containment Atmos. Samp. Ret. to Containment Isolation Valve	A	Close	M1-2301-08 (M2-2301-06A)

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 8 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-5557	Containment Atmos. Samp. Ret. to Containment Isolation Valve	B	Close	M1-2301-08 (M2-2301-06)
1-HV-5559	Containment Atmos. Samp. from Containment Isolation Valve	B	Close	M1-2301-08 (M2-2301-06)
1-HV-5561	Containment Atmos. Samp. from Containment Isolation Valve	B	Close	M1-2301-08 (M2-2301-06)
1-HV-5562	Exh. Header to Hydrogen Purge Fans Control Valve	B	Close	M1-2301-09 (M2-2301-07)
1-HV-5563	Exh. Header to Hydrogen Purge Fans Control Valve	B	Close	M1-2301-09 (M2-2301-07)
1-HV-6082	Chilled Wtr. Ret. Hdr. to Circulation Pump Control Valve	B	Close	M1-2307-05
1-HV-6083	Chilled Wtr. Ret. Hdr. to Circulation Pump Control Valve	A	Close	M1-2307-05
1-HV-6084	Chilled Wtr. Supply Hdr. to Containment Unit Coolers Control Valve	B	Close	M1-2307-06
1-HV-5558	Containment Atmos. Samp. from Containment Isolation Valve	A	Close	M1-2301-08A (M2-2301-06A)
1-HV-5560	Containment Atmos. Samp. from Containment Isolation Valve	A	Close	M1-2301-08A (M2-2301-06A)

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 9 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
CP1-AFAPMD-01	Motor Driven Aux. FW Pump	A	Start	2323-M1-2206-01
CP1-AFAPMD-02	Motor Driven Aux. FW Pump	B	Start	M1-2206-01
CP1-DDAPRM-01	Reactor Make-Up Water Pump	A	Stop	M1-2241-03
CPX-DDAPRM-01	Reactor Make-Up Water Pump	B	Stop	M1-2241-03
CP1-CHCICE-05	HVAC Water Chiller	A	Start	E1-0033 Sh. 29
CP1-CHCICE-06	HVAC Water Chiller	B	Start	E1-0033 Sh. 31
CPX-VAACUP-01	UPS and Distr. Room A/C Unit	A	Start	M1-2313-01
CPX-VAACUP-02	UPS and Distr. Room A/C Unit	B	Start	M1-2313-01
CP1-VAAUSE-15	Electrical Area Fan	B	Start	M1-2302-08
CP1-VAAUSE-16	Electrical Area Fan	B	Start	M1-2302-08
CP1-VAAUSE-17	Electrical Area Fan	A	Start	M1-2302-08
CP1-VAAUSE-18	Electrical Area Fan	A	Start	M1-2302-08
CPX-VAACCR-01	Control Room A/C Unit	A	Start	M1-2304-05
CPX-VAACCR-02	Control Room A/C Unit	A	Start	M1-2304-05

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 10 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
CPX-VAACCR-03	Control Room A/C Unit	B	Start	M1-2304-05
CPX-VAACCR-04	Control Room A/C Unit	B	Start	M1-2304-05
CPX-SFAPSF-01	Spent Fuel Pool Pump	A	Stop	M1-2235-01
CPX-SFAPSF-02	Spent Fuel Pool Pump	B	Stop	M1-2235-01
TBX-RHAPRH-01	Residual Heat Removal Pump No. 1	A	Start	M1-2260-01
TBX-RHAPRH-02	Residual Heat Removal Pump No. 2	B	Start	M1-2260-01
TBX-SIAPSI-01	Safety Injection Pump No. 1	A	Start	M1-2263-01
TBX-SIAPSI-02	Safety Injection Pump No. 2	B	Start	M1-2263-01
TBX-CSAPCH-01	Charging Pump No. 1	A	Start	M1-2255-01
TBX-CSAPCH-02	Charging Pump No. 2	B	Start	M1-2255-01
1-LCV-112B	Volume Control Tank Isolation Valve	A	Close	M1-2255-07
1-LCV-112C	Volume Control Tank Isolation Valve	B	Close	M1-2255-07
1-LCV-112D	SIS Refueling Water to Charging Pump Header, Isolation Valve	A	Open	M1-2255-08
1-LCV-112E	SIS Refueling Water to Charging Pump Header, Isolation Valve	B	Open	M1-2255-08

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 11 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-8105	Charging Pump Discharge to RCS Isolation Valve	B	Close	M1-2255-18
1-8106	Charging Pump Discharge to RCS Isolation Valve	A	Close	M1-2255-18
1-8110	Charging Pump Miniflow Isolation Valve	A	Close	M1-2255-19
1-8111	Charging Pump Miniflow Isolation Valve	B	Close	M1-2255-19
1-8800A	RWST Isolation Valve to SFPCS Pump	A	Close	M1-2261-05
1-8800B	RWST Isolation Valve to SFPCS Pump	B	Close	M1-2261-05
1-8801A	High Head Safety Injection Valve	A	Open	M1-2261-05
1-8801B	High Head Safety Injection Valve	B	Open	M1-2261-05
1-8808A	SIS Accumulator #1 Isolation Valve	A	Open	M1-2262-04
1-8808B	SIS Accumulator #2 Isolation Valve	B	Open	M1-2262-04
1-8808C	SIS Accumulator #3 Isolation Valve	A	Open	M1-2262-04
1-8808D	SIS Accumulator #4 Isolation Valve	B	Open	M1-2262-04
1-8811A	Sump Control Valve to RHR Pump-01	A	Open	M1-2263-06
1-8811B	Sump Control Valve to RHR Pump-02	B	Open	M1-2263-06

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 12 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-8202A	Suction Stabilizer Vent Isolation Valve	A	Close	M1-2255-20
1-8202B	Suction Stabilizer Vent Isolation Valve	B	Close	M1-2255-20
1-8210A	Nitrogen Supply to Suction Stabilizer Tank	A	Close	M1-2255-20
1-8210B	Nitrogen Supply to Suction Stabilizer Tank	B	Close	M1-2255-20
1-A	Feedwater Pump Turbine	A/B	Trip	M1-2203-02
1-B	Feedwater Pump Turbine	A/B	Trip	M1-2203-02
1-HV-2484	Condensate Makeup Tank to Condensate System Isolation Valve	A	Close	M1-2206-09
1-HV-2397	SG #1 Blowdown Isolation Valve	A/B	Close	M1-2202-06
1-HV-2397A	SG #1 Blowdown Isolation Valve	B	Close	M1-2202-06A
1-HV-2398	SG #2 Blowdown Isolation Valve	A/B	Close	M1-2202-06
1-HV-2398A	SG #2 Blowdown Isolation Valve	B	Close	M1-2202-06A
1-HV-2399	SG #3 Blowdown Isolation Valve	A/B	Close	M1-2202-06
1-HV-2399A	SG #3 Blowdown Isolation Valve	B	Close	M1-2202-06A
1-HV-2400	SG #4 Blowdown Isolation Valve	A/B	Close	M1-2202-06

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 13 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-2400A	SG #4 Blowdown Isolation Valve	B	Close	M1-2202-06A
1-HV-2401A	SG #1 Drum Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2401B	SG #1 Blowdowm Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2402A	SG #2 Drum Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2402B	SG #2 Blowdown Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2403A	SG #3 Drum Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2403B	SG #3 Blowdown Sample Isolation Valve	A	Close	M1-2202-06B
1-PV-2453A	MDAFW To SG1 Flow Control Valve	A	Trip to Open with 10 sec Operator Lockout	M1-2206-03
1-PV-2453B	MDAFW to SG2 Flow Control Valve	A	Trip to Open with 10 sec. Operator Lockout	M1-2206-03
1-PV-2454A	MDAFW to SG3 Flow Control Valve	B	Trip to Open with 10 sec Operator Lockout	M1-2206-03A
1-PV-2454B	MDAFW to SG4 Flow Control Valve	B	Trip to Open with 10 sec. Operator Lockout	M1-2206-03A
1-HV-2404A	SG #4 Drum Sample Isolation Valve	A	Close	M1-2202-06B

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 14 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-2404B	SG #4 Blowdown Sample Isolation Valve	A	Close	M1-2202-06B
1-HV-2485	Condensate Make-up Tank to Condensate System Isolation Valve	B	Close	M1-2206-09
1-HV-2405	SG #1 Sample Isolation Valve	B	Close	M1-2202-06C
1-HV-2406	SG #2 Sample Isolation Valve	B	Close	M1-2202-06C
1-HV-2407	SG #3 Sample Isolation Valve	B	Close	M1-2202-06C
1-HV-2408	SG #4 Sample Isolation Valve	B	Close	M1-2202-06C
CP1-CTAPCS-01	Containment Spray Pump	A	Start	M1-2232-03
CP1-CTAPCS-03	Containment Spray Pump	A	Start	M1-2232-03
CP1-CTAPCS-02	Containment Spray Pump	B	Start	M1-2232-03
CP1-CTAPCS-04	Containment Spray Pump	B	Start	M1-2232-03
1-HV-5640A	Motor Driven Aux. FWP Room Supply Damper (Unit 1 Only)	BB	None	M1-2302-05
1-HV-5640B	Motor Driven Aux. FWP Room Exhaust Valve (Unit 1 Only)	BB	None	M1-2302-05
1-HV-5641A	Motor Driven Aux. FWP Room Supply Damper (Unit 1 Only)	AA	None	M1-2302-05
1-HV-5641B	Motor Driven Aux. FWP Room Exhaust Valve (Unit 1 Only)	AA	None	M1-2302-05

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 15 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
1-HV-5658A	RHR Pump Room Supply Damper	AA	None	M1-2302-05
1-HV-5658B	RHR Pump Room Exhaust Valve	AA	None	M1-2302-05
1-HV-5659A	Containment Spray Pump Room Supply Damper	AA	None	M1-2302-05
1-HV-5659B	Containment Spray Pump Room Exhaust Valve	AA	None	M1-2302-05
1-HV-5660A	Safety Injection Pump Room Supply Damper	AA	None	M1-2302-05
1-HV-5660B	Safety Injection Pump Room Exhaust Valve	AA	None	M1-2302-05
1-HV-5663A	Safety Injection Pump Room Supply Damper	BB	None	M1-2302-05
1-HV-5663B	Safety Injection Pump Room Exhaust Valve	BB	None	M1-2302-05
1-HV-5664A	Containment Spray Room Supply Damper	BB	None	M1-2302-05
1-HV-5664B	Containment Spray Room Exhaust Valve	BB	None	M1-2302-05
1-HV-5665A	RHR Pump Room Supply Damper	BB	None	M1-2302-05
1-HV-5665B	RHR Pump Room Exhaust Valve	BB	None	M1-2302-05
X-HV-5747A	Charging Pump Room Supply Damper	BB	None	M1-2303-05A
X-HV-5747B	Charging Pump Room Exhaust Valve	BB	None	M1-2303-05A

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 16 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
X-HV-5762B	Charging Pump Room Exhaust Valve	AA	None	M1-2303-05A
X-HV-5764B	Component Cooling Water Pump Room Exhaust Valve	AA	None	M1-2303-05A
X-HV-5765A	Component Cooling Water Pump Room Supply Damper	BB	None	M1-2303-05A
X-HV-5765B	Component Cooling Water Pump Room Exhaust Valve	BB	None	M1-2303-05A
X-HV-5766A	Component Cooling Water Pump Room Supply Damper	BB	None	M1-2303-05A
X-HV-5766B	Component Cooling Water Pump Room Exhaust Valve	BB	None	M1-2303-05A
X-HV-5774B	Charging Pump Room Exhaust Valve	AA	None	M1-2303-05A
X-HV-5779A	Charging Pump Room Supply Damper	BB	None	M1-2303-05A
X-HV-5779B	Charging Pump Room Exhaust Valve	BB	None	M1-2303-05A
CPX-VAFNCB-07	Primary Plant Vent ESF Exhaust Fan	A	Start	M1-2309-01
CPX-VAFNCB-08	Primary Plant Vent ESF Exhaust Fan	B	Start	M1-2309-01
CPX-VAFNCB-21	Primary Plant Vent ESF Exhaust Fan	A	Start	M1-2309-01
CPX-VAFNCB-22	Primary Plant Vent ESF Exhaust Fan	B	Start	M1-2309-01
CP1-VAFNID-07	Battery Room Exhaust Fan	A	Start	M1-2305-04

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 17 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
CP1-VAFNID-08	Battery Room Exhaust Fan	A	Start	M1-2305-04
CP1-VAFNID-09	Battery Room Exhaust Fan	B	Start	M1-2305-04
CP1-VAFNID-10	Battery Room Exhaust Fan	B	Start	M1-2305-04
1-FV-4536	Component Cooling Water Loop 1 Recirc. Control Valve	A	Close	M1-2229-04
1-FV-4537	Component Cooling Water Loop 2 Recirc. Control Valve	B	Close	M1-2229-04
1-HV-4572	RHR Heat Exchanger Discharge Control Valve	A	Part-Open	M1-2229-06
1-HV-4573	RHR Heat Exchanger Discharge Control Valve	B	Part-Open	M1-2229-06
1-HV-4631A	CCW Non-Safeguard Loop to Primary Sampling System Valve	A	Close	M1-2230-01
1-HV-4631B	Primary Sample Sys. to CCW Return HDR Valve	A	Close	M1-2230-01
1-FV-4650A	Vent Chillers CCW Supply Control Valve	B	Close	M1-2230-02
1-FV-4650B	Vent Chillers CCW Disch. Control Valve	B	Close	M1-2230-02
CP1-CHAPCP-05	Chilled Water Recirc. Pump-05	A	Start	M1-2311-01
CP1-CHAPCP-06	Chilled Water Recirc. Pump-06	B	Start	M1-2311-01
CP1-CCAPCC-01	Component Cooling Water Pump-01	A	Start	M1-2229-03

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 18 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
CP1-CCAPCC-02	Component Cooling Water Pump-02	B	Start	M1-2229-03
CP1-MEDGEE-01	Diesel Generator -01	A	Start	E1-0067-95
CP1-MEDGEE-02	Diesel Generator -02	B	Start	E1-0067-95
CP1-SWAPSW-01	Station Service Water Pump-01	A	Start	M1-2233-01
CP1-SWAPSW-02	Station Service Water Pump-02	B	Start	M1-2233-01
X-HV-5825B	Control Room Make-Up Supply Damper	A	Shut	M1-2304-01B
X-HV-5828B	Control Room Make-Up Supply Damper	B	Shut	M1-2304-01B
X-HV-5837	Control Room Intake Damper	A	Open	M1-2304-03
X-HV-5838	Control Room Intake Damper	B	Open	M1-2304-03
X-HV-5825A	Make-Up Units Inlet Damper	A	Close	M1-2304-01
X-HV-5828A	Make-Up Units Inlet Damper	B	Close	M1-2304-01
X-HV-5831	Emergency Pressurization Units Inlet Damper	A	Open	M1-2304-02
X-HV-5834	Emergency Pressurization Units Inlet Damper	B	Open	M1-2304-02
X-HV-5839	Emergency Filtration Units Damper	A	Open	M1-2304-03

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 19 of 20)

Equipment Identification[(a)]	Description	Train	ESFAS Function	SIS Signal Drawing Number
X-HV-5840	Emergency Filtration Units Damper	B	Open	M1-2304-03
X-PV-5855	Control Room Exh. Fan Inlet Control Damper	A	Close	M1-2304-06
X-PV-5856	Control Room Exh. Fan Inlet Control Damper	B	Close	M1-2304-06
X-HV-5857	Kitchen & Toilet Exh. Fan Inlet Control Damper	A	Close	M1-2304-07
X-HV-5858	Kitchen & Toilet Exh. Fan Inlet Control Damper	B	Close	M1-2304-07
CPX-VAFNAV-37	Control Room Air Makeup Fan	A	Stop	M1-2304-01
CPX-VAFNAV-38	Control Room Air Makeup Fan	B	Stop	M1-2304-01
CPX-VAFNCB-05	Control Room Pressurization Fan	A	Start	M1-2304-02
CPX-VAFNCB-06	Control Room Pressurization Fan	B	Start	M1-2304-02
CPX-VAFNCB-23	Emergency Filtration Fan	A	Start	M1-2304-04
CPX-VAFNCB-24	Emergency Filtration Fan	B	Start	M1-2304-04
CPX-VAFNID-01	Control Room Main Exhaust Fan	A	Stop	M1-2304-06
CPX-VAFNID-02	Control Room Main Exhaust Fan	B	Stop	M1-2304-06
CPX-VAFNID-03	Control Room Complex Kitchen and Toilet Exhaust Fan	A	Stop	M1-2304-07

TABLE 7.3-4
SAFETY INJECTION ACTUATED EQUIPMENT LIST

(Sheet 20 of 20)

Equipment Identification[a]	Description	Train	ESFAS Function	SIS Signal Drawing Number
CPX-VAFNID-04	Control Room Complex Kitchen and Toilet Exhaust Fan	B	Stop	M1-2304-07
X-HV-5883	Control Room Vent Control Damper	A/B	Partial-Open	M1-2304-13
1-8511A	Charging Pump Alternate Mini-flow Isolation Valve	A	Open	M1-2255-21
1-8511B	Charging Pump Alternate Mini-flow Isolation Valve	B	Open	M1-2255-21
--	Containment Personnel Airlock Hydraulic Unit	A/B	Trip	M1-2245-01 (Note 1)

a) This table lists only Unit 1 and Common safety related equipment actuated directly by a Safety Injection signal. Unless otherwise noted, Unit 2 equipment identification and drawing numbers are the same except for unit prefix. For any subsidiary equipment, in turn started by this equipment, refer to applicable drawings.

Notes:

1: The Train A trip is accomplished by tripping the Class 1E feeder breaker to MCC 1EB1-2 at the Train A 480V switchgear 1EB1. The Train B trip is accomplished by tripping Class 1E Train B circuit breaker CP1-BSDSEB-01 in the feeder circuit of the hydraulic unit fed from MCC 1EB1-2.

TABLE 7.3-5
ESFAS ACTUATED COMPONENTS OF THE CONTAINMENT SPRAY SYSTEMS AND ITS SUPPORTING SYSTEMS

(Sheet 1 of 3)

Equipment Identification[(a)]	Description	ESF Train	ESFAS Signal Function	Drawing Number
CP1-CTAPCS-01	Containment Spray Pump No. 1-1	A	Start	M1-2232-03
CP1-CTAPCS-02	Containment Spray Pump No. 1-2	B	Start	M1-2232-03
CP1-CTAPCS-03	Containment Spray Pump No. 1-3	A	Start	M1-2232-03
CP1-CTAPCS-04	Containment Spray Pump No. 1-4	B	Start	M1-2232-03
1-HV-4776	Containment Spray Heat Exchanger-01 Outlet Valve	A	Open	M1-2232-04
1-HV-4777	Containment Spray Heat Exchanger-02 Outlet Valve	B	Open	M1-2232-04
1-LV-4754	Chemical Additive Tank Discharge Valve	A	Open	M1-2232-01
1-LV-4755	Chemical Additive Tank Discharge Valve	B	Open	M1-2232-01
1-HV-4572	RHR Heat Exchanger-01 Outlet Control Valve	A	Part-Open	M1-2229-06
1-HV-4573	RHR Heat Exchanger-02 Outlet Control Valve	B	Part-Open	M1-2229-06
1-HV-4574	Containment Spray Heat Exchanger Discharge Control Valve	A	Part-Open	M1-2229-06
1-HV-4575	Containment Spray Heat Exchanger Discharge Control Valve	B	Part-Open	M1-2229-06

TABLE 7.3-5
ESFAS ACTUATED COMPONENTS OF THE CONTAINMENT SPRAY SYSTEMS AND ITS SUPPORTING SYSTEMS

(Sheet 2 of 3)

Equipment Identification[a]	Description	ESF Train	ESFAS Signal Function	Drawing Number
1-HV-4512	CCW Non-Safeguards Loop Return Header Control Valve	A	Close	M1-2229-02
1-HV-4513	CCW Non-Safeguards Loop Return Header Control Valve	B	Close	M1-2229-02
1-HV-4514	CCW Non-Safeguards Loop Return Header Control Valve	A	Close	M1-2229-04B
1-HV-4515	CCW Non-Safeguards Loop Return Header Control Valve	B	Close	M1-2229-02
1-HV-4524	CCW Heat Exchanger to Non-Safeguards Loop Isolation Valve	A	Close	M1-2229-04B
1-HV-4525	CCW Heat Exchanger to Non-Safeguards Loop Isolation Valve	B	Close	M1-2229-02
1-HV-4526	CCW Heat Exchanger to Non-Safeguards Loop Isolation Valve	A	Close	M1-2229-04B
1-HV-4527	CCW Heat Exchanger to Non-Safeguards Loop Isolation Valve	B	Close	M1-2229-02
1-HV-4696	CCW System Containment Isolation Valve	A	Close	M1-2231-02
1-HV-4699	CCW System Isolation Valve	A	Close	M1-2231-03
1-HV-4700	CCW System Containment Isolation Valve	B	Close	M1-2231-03

TABLE 7.3-5
ESFAS ACTUATED COMPONENTS OF THE CONTAINMENT SPRAY SYSTEMS AND ITS SUPPORTING SYSTEMS

(Sheet 3 of 3)

Equipment Identification[a]	Description	ESF Train	ESFAS Signal Function	Drawing Number
1-HV-4701	CCW System Containment Isolation Valve	A	Close	M1-2231-03
1-HV-4708	CCW System Containment Isolation Valve	B	Close	M1-2231-05
1-HV-4709	CCW System Containment Isolation Valve	B	Close	M1-2231-05

a) This table lists only Unit 1 safety related equipment actuated directly by a Spray Actuation or Containment Isolation Phase B signal. Unit 2 equipment identification and drawing numbers are the same except for unit prefix. For any subsidiary equipment in turn started by this equipment, refer to applicable drawings.

TABLE 7.3-6
STEAMLINE ISOLATION ACTUATED EQUIPMENT LIST

Equipment Identification[a]	Description	ESF Train	Function	ESFAS Signal Drawing Number
1-HV-2333A	Steam Generator #1 MSIV	A/B	Close	M1-2202-02
1-HV-2334A	Steam Generator #2 MSIV	A/B	Close	M1-2202-02
1-HV-2335A	Steam Generator #3 MSIV	A/B	Close	M1-2202-02
1-HV-2336A	Steam Generator #4 MSIV	A/B	Close	M1-2202-02
1-HV-2409	Main Steam Line #1 Before MSIV Drip Pot Isolation Valve	A/B	Close	M1-2202-07
1-HV-2410	Main Steam Line #2 Before MSIV Drip Pot Isolation Valve	A/B	Close	M1-2202-07
1-HV-2411	Main Steam Line #3 Before MSIV Drip Pot Isolation Valve	A/B	Close	M1-2202-07
1-HV-2412	Main Steam Line #4 Before MSIV Drip Pot Isolation Valve	A/B	Close	M1-2202-07

a) This table lists only Unit 1 safety related equipment actuated directly by a Steam Line Isolation signal. Unit 2 equipment identification and drawing numbers are the same except for unit prefix.

7.4 SYSTEMS REQUIRED FOR SAFE SHUTDOWN

The systems required for safe shutdown are controlled and monitored by instrumentation channels associated with those systems in both the Nuclear Steam Supply System (NSSS) and balance of plant (BOP) systems. These systems are normally aligned to serve a variety of operational functions, including startup and shutdown, as well as protective functions. There are no uniquely identified safe shutdown systems per se. However, prescribed procedures for securing and maintaining the plant in a safe condition can be instituted by appropriate alignment of selected systems.

Two kinds of shutdown conditions are addressed in this section: hot standby and cold shutdown. Hot standby is a stable condition of the reactor achieved shortly after a programmed or emergency shutdown of the plant and is the safe shutdown design basis for CPNPP. Cold shutdown is a stable condition of the plant achieved after the residual heat removal process has brought the primary coolant temperature below 200°F. In either case, the reactivity control systems maintain a subcritical condition of the core. The plant technical specifications explicitly define both hot standby and cold shutdown conditions.

The instrumentation and control functions required to be aligned for maintaining safe shutdown of the reactor that are discussed in this section are the minimum number under nonaccident conditions. These functions permit the necessary operations that will:

1. Prevent the reactor from achieving criticality in violation of the technical specifications.

2. Provide an adequate heat sink so that design and safety limits are not exceeded.

The designation of systems that can be used for safe shutdown depends on identifying those systems which provide the following capabilities:

1. Reactivity control - reactor trip and boration

2. Decay heat removal - auxiliary feedwater (AFW) supply and residual heat removal (RHR) [The RHRS is required to achieve and maintain cold shutdown.]

7.4.1 DESCRIPTION

In the event of a unit shutdown, the unit will be brought to and maintained at a safe shutdown condition from the main Control Room or the Hot Shutdown Panel (see Section 7.4.1.3).

The portions of the Reactor Trip System required to achieve the shutdown condition are described in Section 7.2. The minimum systems, support systems, component controls, and monitoring indicators required under nonaccident conditions to maintain hot standby are tabulated and discussed in Section 7.4.1.1 and those required to maintain cold shutdown are tabulated and discussed in Section 7.4.1.2. Shutdown from outside the Control Room is discussed in Section 7.4.1.3.

7.4.1.1 Hot Standby

Hot standby is maintained by providing decay heat removal via at least one steam generator. The following systems, support systems, and monitoring indicators are required for hot standby:

1. Essential Systems

   a. Auxiliary Feedwater System (AFS)

   b. Steam Generator safety valves

   c. Chemical and Volume Control System (CVCS), boron addition portion.

2. Support Systems

   a. Station Service Water System (Sections 9.2.1 and 7.3.1.1.4)

   b. Component Cooling Water System (Sections 9.2.2 and 7.3.1.1.4)

   c. Onsite power system (Sections 8.3 and 7.3.1.1.4), including diesel generators (Section 9.5)

   d. Control Room ventilation system (Sections 9.4.1 and 7.3.1.1.4)

   e. ESF Ventilation System (Sections 9.4.5 and 7.3.1.1.4)

   f. Safety Chilled Water System (Sections 9.4F and 7.3.1.1.4)

   g. UPS Ventilation System (Sections 9.4C.8)

3. Monitoring Indicators

   a. Steam Generators (SG's)

      1. Water level for each SG

      2. Pressure for each SG

   b. Reactor Coolant System

      1. Pressurizer water level

      2. Pressurizer pressure

   c. Condensate storage tank level

7.4.1.1.1 Auxiliary Feedwater System

At hot standby, decay heat is removed by supplying auxiliary feedwater to the steam generators. Heat is dumped to the condenser (if offsite power is available), to the atmosphere via the steam generator power-operated relief valves, or to the atmosphere via the steam generator safety valves.

For a description of the Auxiliary Feedwater System refer to Section 10.4.9.

For a description of the Auxiliary Feedwater System control refer to Section 7.3.1.1.4, Item 5. A discussion of instrumentation is found in Section 10.4.9.5.

The system flow diagram is shown on Figure 10.4-11.

The system failure mode analysis is contained in Table 10.4-9 and the pertinent illustration is Figure 10.4-12.

7.4.1.1.2 Steam Generator Atmospheric Relief Valves and Safety Valves

For normal plant cooldown, the Steam Dump System, described in Section 10.4.4, is used to control the cooldown rate by directing steam from the steam generators to the condenser. The steam generator atmospheric relief valves enable heat removal from the steam generators to the atmosphere when the condenser is not in service. These valves, described in Section 10.3.2.2, will allow gradual cooldown of the RCS to the point where the RHRS can be employed to continue plant cooldown (less than 425 psig and 350°F).

The main steam system is protected against overpressurization by the ASME B&PV Code-certified safety valves described in Section 10.3.2.1.

In the event of loss of the Steam Dump System and loss of the steam generator atmospheric relief valves, the safety valves will be used to dump steam to the atmosphere, thereby maintaining the plant at hot standby.

1. Initiating Circuits

   A pressure transmitter and controller is provided for each steam generator atmospheric relief valve. Steam pressure developed in the SG is limited by the controller set point selected. Either automatic or manual control can be selected on the controller located in the Control Room.

   These valves are also equipped with handwheels to enable local manual operation. For operation outside the Control Room, access to the valve's control circuitry is provided and manual operation of each valve is accomplished with a local controller at the Hot Shutdown Panel.

   The initiation device for the safety valve is an intrinsic part of the valve.

2. Logic

   See instrumentation and control diagrams (ICD's) listed under "Main Steam Reheat and Steam Dump System" in Tables 1.7-1 and 1.7-2 for steam generator atmospheric relief valve control.

   The safety valves are spring-loaded valves that open on reaching the pressure set point.

3. Bypass

   There are no electrical bypasses on safety valves nor on steam generator atmospheric relief valves.

The steam generator atmospheric relief valves have manually operated isolation valves upstream of the relief valves for relief valve maintenance. Administrative procedures will assure the proper position of these isolation valves.

4. Interlocks

There are none.

5. Redundancy

The safety valves and steam generator atmospheric relief valves are redundant in that two of four steam generators are required for safe shutdown.

6. Diversity

The normally power-operated atmospheric relief valves can also be operated diversely and manually via the valve handwheel.

7. Actuated Devices

None, other than the valves themselves.

8. Supporting Systems

The following supporting systems are required for operation of steam generator atmospheric relief valves.

a. Class 1E electric power (see Section 8.3)

b. Compressed Air System (see Section 9.3.1 and Figure 9.3-1)

9. Design Basis Information

a. See Table 10.3-2 for the design basis of the main steam safety valves.

b. See Table 10.3-3 for the design basis of the steam generator atmospheric relief valves.

10. Electric Schematic Drawings

See Table 1.7-1 for Unit 1 and common and Table 1.7-2 for Unit 2 schematics (electrical) associated with “Main Steam Reheat & Steam Dump System.”

11. Portions of System Not Required for Safety

The steam generator atmospheric relief valves and pressure transmitter output indication displayed on the control board are not required for safety.

7.4.1.1.3 Chemical and Volume Control System, Boron Addition Portion

The functions of the CVCS discussed in Section 9.3.4 are those associated with normal operation. Safe and economical operation design parameters (see Table 9.3-6) dictate the design of the CVCS for normal operation. For the safety evaluation of the overall CVCS operation refer to Section 9.3.4.1.3 and Table 9.3-9 for the failure modes and effects analysis (FMEA). If safe shutdown operation alone is to be considered, then the primary functions of the CVCS are that it provide a means, along with the proper operator action, to maintain adequate shutdown margin by sufficiently borating the core and to maintain pressurizer level. These functions are those required to maintain adequate Reactor Coolant System (RCS) inventory. The instrumentation, control and electrical features associated with maintaining adequate RCS inventory are those applied to the following groups of equipment:

1. Boric acid transfer pumps

2. Centrifugal charging pumps

3. Letdown orifice isolation valves

Further considerations, as would apply to one unit, are as follows:

1. Initiating Circuits

   Initiation is by automatic or manual commands. For discussion of automatic commands of the CVCS proper, refer to Sections 9.3.4 and 9.3.4.1.2.3, and for a discussion of the pressurizer water level control portion, see Section 7.7. Maintenance of hot standby and cold shutdown is discussed in 9.3.4.1.2.6. Manual control may be taken either in the Control Room or outside the Control Room at the Hot Shutdown Panel. For the instrumentation application, refer to Section 9.3.4.1.5.

2. Logic

   At the Shutdown Transfer Panel (Train A) and Hot Shutdown Panel (HSP) (Train B), operator action may transfer control of the boric acid transfer pumps, centrifugal charging pumps and letdown orifice isolation valves from the Control Room to the Hot Shutdown panel. At this location, start-stop operation of the pumps and open-close operation of the valves in question can then be undertaken. If the redundant normal suction paths to centrifugal charging pumps that establish boration are not available, normally closed motor operated valve 1-8104 (refer to Figure 9.3-10) can be opened either remotely in the Control Room or by handwheel at the local station. Manual or motor operated valves may also be opened to allow the centrifugal charging pumps to take suction on either the Boric Acid Storage tank or the Refueling Water Storage Tank, respectively.

3. Bypasses

   Control from the Control Room of the above groups of equipment is manually bypassed at the Shutdown Transfer Panel (Train A) and at the Hot Shutdown Panel (Train B) when control is taken over at the Hot Shutdown Panel.

4. Interlocks

With control of those above groups of equipment at the Hot Shutdown Panel, there are no interlocks imposed on their operation.

5. Redundancy

There are two centrifugal charging pumps, two boric acid transfer pumps, and three letdown orifice isolation valves.

6. Diversity

Diversity is not required for the boric acid transfer pumps for the charging pumps can take a direct suction from the storage tanks. In addition to the centrifugal charging pumps, there is a positive displacement charging pump. There are other valves in the letdown line, which, in addition to letdown orifice isolation valves, can also be used to isolate the letdown line.

7. Actuated Devices

The motor control centers for the centrifugal charging pumps and boric acid transfer pumps and the solenoid valves for the letdown orifice isolation valves are the actuation devices of the CVCS required for safe shutdown.

8. Supporting Systems

The centrifugal charging pumps, the boric acid transfer pumps, and the letdown orifice isolation valve solenoid valves are powered from Class 1E buses. Additionally, the Safety Chilled Water System supplies chilled water to the charging pump fan coil unit (see Section 9.4F) and the Station Service Water System supplies cooling water to the centrifugal charging pump lube oil coolers (see Section 9.2.1).

9. Design Basis Information

The CVCS is not specifically designed to be a protection system. Section 3, “Design Bases, “ of IEEE Standard 279 does not specifically apply to non-protection systems such as the CVCS. Portions of the CVCS, which are shared with the Safety Injection System, come under IEEE Standard 279 design bases as discussed in Section 7.3.

10. Drawings

For the CVCS flow diagram, refer to Figure 9.3-10. Also refer to the flow diagrams in Section 6.3 for shared functions of the Safety Injection System, as well as the electrical drawing list in Section 1.7.

11. Portion of System Not Required for Safety

Refer to item 9 above

7.4.1.2 Cold Shutdown

Cold shutdown is achieved following hot standby and includes passing through hot shutdown. Four key functions are required to achieve and maintain cold shutdown. They are circulation of the reactor coolant, removal of residual heat, boration, and depressurization. The required means for performing these functions are described below.

1. Circulation of the reactor coolant is provided first by natural circulation that is effected by the reactor core providing the heat source and the steam generators the heat sink, and then by the residual heat removal pumps.

2. Removal of residual heat is accomplished first via the Auxiliary Feedwater System and then via the residual heat removal heat exchanger. Hot standby is maintained by releasing steam via the safety grade steam generator safety valves. Cooldown to 350°F is accomplished by releasing steam via operation of the steam generator power-operated relief valves. Then cooldown to cold shutdown conditions is achieved with the Residual Heat Removal System. A sufficient seismic Category I supply of deaerated auxiliary feedwater to permit four hours operation at hot standby followed by cooldown to Residual Heat Removal System initiation conditions is provided by the condensate storage tank.

   A backup seismic Category I source for the Auxiliary Feedwater System is the Station Service Water System. This system provides an assured long-term supply of auxiliary feedwater.

3. Boration is accomplished using portions of the Chemical and Volume Control System. Boric acid from the boric acid tanks is supplied to the suction of the centrifugal charging pumps by the boric acid transfer pumps or the charging pumps may take a direct suction on the tanks. The centrifugal charging pumps inject the boric acid into the reactor coolant system via the safety injection flow paths or the normal charging and reactor coolant pump seal injection flow paths. As an alternate source, or for additional boration, the refueling water storage tank is available.

4. Depressurization is accomplished using portions of the Chemical and Volume Control System. Either boric acid from the boric acid tanks or refueling water from the refueling water storage tank is used as desired for depressurization with the flow path being via the centrifugal charging pumps and auxiliary spray valve to the pressurizer.

In addition to the systems required for hot standby delineated in Section 7.4.1.1, the following systems, support systems, and monitoring indicators are required to achieve and maintain cold shutdown.

1. Essential Systems

   a. Residual Heat Removal System (RHRS)

   b. Pressurizer pressure control during RCS depressurization

2. Support Systems - No additional support systems required other than those listed in Section 7.4.1.1, Number 2.

3. Monitoring Indicators

   a. RCS wide range temperature

   b. Boric acid tank level (per boric acid tank)

Additionally, the safety injection signal trip circuit must be defeated and the accumulator isolation valves closed during the later stages of plant cooldown.

7.4.1.2.1 Residual Heat Removal System (RHRS)

The Residual Heat Removal System is designed to remove residual heat from the core and reduce the temperature of the Reactor Coolant System (RCS) during plant cooldown to cold shutdown conditions. The system is permitted to be manually initiated when the reactor coolant temperature and pressure are reduced to 350°F and 400 psig or below. A detailed discussion of the RHRS is provided in Section 5.4.7. RHRS operation for reactor cooldown is discussed in Section 5.4.7.2.3. Manual actions required of the operator, both inside and outside Control Room, is discussed in Section 5.4.7.2.7. Further considerations are as follows.

1. Initiating Circuits

   Manual control of the RHRS is from the Control Room (or from the Hot Shutdown Panel for Train A). For manual actions outside the Control Room, refer to Section 5.4.7.2.7.

2. Logic

   For the logic diagram for control of the RHRS inlet isolation valves, see Figure 7.6-2.

3. Bypasses

   RHRS Bypass is indicated on the Safety System Inoperable Indicator (SSII) described in Section 7.1.2.6 and typically illustrated in Figure 7.1-4. Isolation of the RHRS cannot be manually bypassed unless permissive conditions have been automatically met as described in Section 7.6.2.

4. Interlocks

   For discussion of the RHRS inlet isolation valves interlocks, refer to Section 7.6. Section 5.4.7.2.7 discusses application of the single failure criteria.

5. Redundancy

   Two independent residual heat removal pumps are provided, either of which can provide the necessary heat removal capacity for the RCS for the safe shutdown condition. Each RHR pump is powered from a separate Class 1E bus.

6. Diversity

For opening permissives of the RHR inlet isolation valves, diverse RCS pressure transmitters, made by different manufacturers, are used. There is no other diversity in the RHRS control.

7. Actuated Devices

Electrical control equipment for the RHRS pumps, the mini-flow control, and the RHRS heat exchanger are the principal actuated devices.

8. Supporting Systems

The RHRS is powered from Class 1E buses. The RHRS heat exchangers and pumps are cooled with component cooling water (see Section 9.2.2).

9. Design Basis Information

The RHRS is not specifically designed as a protection system. Section 3, "Design Basis," of IEEE Standard 279 does not specifically apply to non-protection systems like the RHRS. Portions of the Safety Injection System which are shared with the RHRS came under IEEE Standard 279 design basis as discussed in Section 7.3.

10. Drawings

For the RHRS flow diagram, refer to Figures 5.4-6 and 5.4-7. For the RHRS inlet isolation valve interlocks, refer to Figure 7.6-2. Refer to the Section 1.7 list for applicable electrical drawings.

11. Portion of System Not Required for Safety

Refer to item 9 above.

7.4.1.2.2 Pressurizer Pressure Control

Pressurizer heater control is provided to maintain the RCS at operating pressure following a reactor trip to prevent excessive cooling of the pressurizer fluid and subsequent depressurization of the system. Normal operation is automatic via the proportional and backup heaters, as described in Section 7.7. If, for any reason, the normal pressure-regulating system is not available, the operator will control pressure with the backup heater groups in the pressurizer by manual ON-OFF control switches. Air operated valve 1-8145 in the auxiliary spray flow path, with manual control in the Control Room, provides spray for depressurizing when normal pressurizer spray control is not available. Means for depressurizing the RCS are also available from the pressurizer power-operated relief valves, with manual control in the Control Room, as well as from the normal pressurizer pressure spray control, which is active when the pressurizer pressure control system is operating. This pressurizer pressure control system, which also automatically controls the pressurizer heaters, as described in Section 7.7, is non-safety related. The pressurizer safety valves, which prevent the RCS from exceeding its design pressure, are safety-related but are self- activated and do not require instrumentation, control, and electrical circuitry for operation.

1. Initiating Circuits

   In normal automatic operation, the pressurizer pressure control system is controlled by pressurizer pressure transmitters. In addition, the backup heater groups are provided with direct manual control by the operator.

2. Logic

   See Figures 7.2-1 (sheets 11 and 12) and 7.7-4.

3. Bypass

   The control circuit provides no bypass of the manual backup heater control function other than maintenance provisions.

4. Interlocks

   All pressurizer heater groups are connected with a low-level switch to cut off power to the heaters on pressurizer low-low water level, thereby preventing heater damage. This interlock is not used when manual control of the pressurizer backup heaters is transferred to the Hot Shutdown Panel.

5. Redundancy

   Two sources of power for backup heater groups are provided, either of which can provide the necessary energy input to the pressurizer fluid to develop the requisite pressure for the safe shutdown condition. Either of two power-operated relief valves will perform the depressurization function.

6. Diversity

   No diversity of control of power supply is provided between the backup heater groups. Diversity of depressurization is provided by power-operated relief valves, safety valves, spray valves, and auxiliary spray valves.

7. Actuated Devices

   The electrical control equipment for the pressurizer backup heater groups, power-operated relief valves, safety valves, spray valves, and auxiliary spray valves are the actuated devices.

8. Supporting Systems

   Provisions have been included to feed the pressurizer backup heaters from independent Class 1E power supplies. The pressurizer power-operated relief valves and the pressurizer auxiliary spray valve utilize air from the Instrument Air System and control power from Class 1E DC buses.

9. Design Basis Information

   Section 3, “Design Basis,” of IEEE Standard 279 does not apply because the pressurizer pressure control system, except for the safety valves, is not designed as a safety system. The design basis is derived from operational objectives. The safety valves are self actuated.

10. Drawings

    For piping and instrumentation flow diagrams for the pressurizer pressure control system, refer to Sections 5.4.10 and 9.8.4.1. Figure 7.2-1, sheets 11 and 12, illustrate the logic.

11. Portions of System Not Required for Safety

    Refer to item 9 above.

7.4.1.3 Shutdown From Outside the Control Room

A common Control Room is provided that contains all controls and instrumentation necessary for the operation of Units 1 and 2 reactors, turbine-generators, and auxiliary and emergency systems under normal or accident conditions. Sufficient radiation shielding, containment integrity, missile protection and habitability provisions are provided to permit access, exit, and continued occupancy of the control room for the duration of accident conditions such that aggregate radiation exposure to personnel would be below that specified by 10CFR50, Appendix A, General Design Criterion 19.

Considering the detailed station design provisions to ensure continuous Control Room access, it is unlikely that the necessity could arise for evacuation of the Control Room. Nevertheless, provisions have been made to maintain the reactors in a safe hot standby condition if access to the Control Room is lost. Hot standby is maintained as described in Section 7.4.1.3.2. Furthermore, cold shutdown can be achieved from outside the Control Room through the use of suitable procedures as described in Section 7.4.1.3.3.

7.4.1.3.1 Design Criteria

In designing for safe reactor shutdown in the event of Control Room evacuation, the following design criteria are applied.

1. Access back into the Control Room will generally be achieved prior to the initiation of cold shutdown except for a fire that destroys the Control Room or either Cable Spreading Room, or the Control Room HVAC Mechanical Equipment Room. The capability for bringing the reactors to cold shutdown conditions exist outside the Control Room through the use of suitable procedures and secondary controls as described in Section 7.4.1.3.3.

2. Except for a fire in the alternate shutdown areas (Control Room, Cable Spreading Rooms, or Control Room HVAC Mechanical Equipment Room), Control Room evacuation is initiated for an undefined cause (for example, control room environment not habitable) and the event which causes Control Room evacuation does not degrade Control Room equipment.

3. The design basis for Control Room evacuation does not consider a concurrent Condition II, III, or IV event, nor a single failure. However, loss of offsite power has been considered.

4. For five specific plant fires, shutdown capability is provided as follows:

Fire Location	Loss	Shutdown From
Unit 1 or Unit 2 Cable Spreading Room (CSR)	Both Trains	Unit 1 and Unit 2 Hot Shutdown Panel using Train A, Shutdown Transfer Panel transfer and local controls.
Control Room (CR)	Both Trains	Unit 1 and Unit 2 Hot Shutdown Panel using Train A, Shutdown Transfer Panel transfer and local controls.
Control Room HVAC Mechanical Equipment Room	Both Trains	Unit 1 and Unit 2 Hot Shutdown Panel using Train A, Shutdown Transfer Panel transfer and local controls.
Unit 1 or Unit 2 Hot Shutdown Panel (HSP)	Train B	Control Room using Train A
Shutdown Transfer Panel (STP)	Train A	Control Room using Train B

For any other postulated plant fire, shutdown is accomplished from the Control Room utilizing surviving equipment.

5. Except for a fire in the alternate shutdown areas, loss of safety system redundancy does not occur as a result of the event requiring control room evacuation and all equipment in the control room and all automatic controls continue to function. To prevent any adverse impact on controls at the HSP or local control stations when Control Room is evacuated due to a fire, the surveillance test program validates the opening of isolation contacts for transfer switches used to transfer controls in the event of a control room fire.

6. The respective unit's Hot Shutdown Panel (HSP) and the Shutdown Transfer Panel (STP) including Class 1E equipment mounted on them, are designed to withstand an SSE with no loss of Class 1E function. The essential local control stations are also designed to withstand an SSE with no loss of essential functions.

7. The Hot Shutdown Panels are normally unattended and are surrounded by a locked enclosure. Opening the Hot Shutdown Panel's cabinet doors will initiate an alarm in the Control Room. The Shutdown Transfer Panels are also normally unattended and access to them is restricted via normally locked doors. Opening the Shutdown Transfer Panel's cabinet doors will initiate an alarm in the Control Room.

8. The Hot Shutdown Panels, located in the switchgear area of each Safeguards Building, at elevation 831'-6", the Shutdown Transfer Panels, located one floor below at elevation 810'-6", and other local controls are easily accessible to Control Room operators through controlled access areas.

9. Electrical separation for the Hot Shutdown Panels and the Shutdown Transfer Panels follows the same criteria as corresponding Control Room equipment. Loss of control or indication for one train for any reason will not affect its redundant counterpart.

10. Controls on the Hot Shutdown Panels (HSP) are provided with a transfer switch (at the Shutdown Transfer Panel for Train A and Hot Shutdown Panel for Train B) that transfers control of equipment from the Control Room to the Hot Shutdown Panel. Placing the Train A or Train B transfer switch in the HSP position will electrically isolate all Control Room and Cable Spreading Room equipment from the rest of the circuitry with the exception of some Train B equipment not required for alternate shutdown. Placing either the Train A or Train B switch in the HSP position will provide audible and visible indication in the Control Room and will turn off status lights on the respective main control board. Interlocks which are generated in the Control Room or Cable Spreading Rooms, such as the lockout signals from the sequencer, are also isolated when control is transferred to the Hot Shutdown Panels.

11. Each control circuit consists of cables that 1) connect the transfer switches to control switches in the Control Room, 2) connect the transfer switches to the HSP-mounted control switches, 3) connect the transfer switches to the pertinent motor control center. Each of these cables is inherently separated from its redundant counterpart. Loss of a control circuit in the Control Room/Cable Spreading Room areas does not mean loss of function since control circuits are available at the HSP.

12. To prevent a single event due to a fire (e.g., short-circuit) from affecting both the Control Room controls and the HSP controls of required alternate shutdown systems, control-circuit fuses are either located in equipment accessible in such an event (e.g., 6.9 KV and 480V switchgear) or separate fuses, located in separate fire zones, are provided for both the Control Room control circuit and the HSP control circuit.

The controls and monitoring indicators on the Hot Shutdown Panels provided for hot standby are listed in Table 7.4-1. Instrumentation and controls on the Hot Shutdown Panels provided for cold shutdown are listed in Table 7.4-2. Those controls and indicators provided for Alternate Shutdown (see section 7.4.1.3.4) are identified on each table. Some controls and indicators on these tables are only provided for operating convenience. Switches provided on the Shutdown Transfer Panels are listed in Table 7.4-3.

7.4.1.3.2 Hot Standby From Outside The Control Room

Should the Control Room become uninhabitable, the reactors will be manually tripped, the neutron level and control rod position will be verified before evacuation takes place. Also, the reactors can be tripped locally at the respective reactor trip switchgear which is in close proximity to the associated Hot Shutdown Panel.

Sufficient controls are provided outside the Control Room on the Seismic Category I Hot Shutdown Panels (See Table 7.4-1) and other locations to:

- Achieve prompt hot standby of both reactors

- Maintain both units in a safe condition during hot standby.

7.4.1.3.3 Cold Shutdown From Outside the Control Room

Cold shutdown can be achieved from outside the Control Room through the use of suitable procedures and by virtue of local control of the systems listed in Section 7.4.1.2. The design bases for the achievement and maintenance of cold shutdown are as listed in Section 7.4.1.3.1. In addition, certain local manipulations of controls and initiating devices, as described below, are required.

For cold shutdown from outside the Control Room due to a Control Room or Cable Spreading Room fire, see Section 7.4.1.3.4. The basic procedure to established cold shutdown from the hot standby condition, for reasons other than fire in the Control Room or Cable Spreading Room and assuming the Control Room is uninhabitable, is as follows.

1. Transfer control from the Control Room to the Hot Shutdown Panels, as required, using the transfer switches at the Shutdown Transfer Panels for Train A and the transfer switches at the Hot Shutdown Panels for Train B.

2. Borate to the cold shutdown boron concentration using boric acid transfer pumps, if available, and charging pumps.

   Location: Hot Shutdown Panels.

   Available Indications: Reactor Coolant System (RCS) boron concentration determined by sampling.

3. Cooldown the RCS by use of the steam generator power-operated atmospheric relief valves.

   Location: Hot Shutdown Panels and local patch panels or manual handwheel operation.

   Available Indications: RCS temperature, pressurizer pressure and level, steam generator level, auxiliary feedwater flow and condensate storage tank level.

4. Depressurize the RCS by throttling the pressurizer spray valve while maintaining pressurizer level.

   Location: Hot Shutdown Panels

   Available Indications: Pressurizer pressure and level, charging and letdown flow.

5. At 1900 psig in the RCS, block the low pressurizer pressure safety injection (SI) and the low steamline pressure SI signals.

   Location: Cable spreading rooms

   Available Indications: Pressurizer pressure indication on the Hot Shutdown Panels.

6. At less than 1000 psig in the RCS, close the accumulator discharge isolation valves. Rack out the safety injection pump and containment spray pump breakers at approximately 350°F and 200°F respectively.

   Location: Switchgear and motor control centers.

   Available Indications: Pressurizer pressure indication on the Hot Shutdown Panels.

7. At less than 350°F and RCS pressure less than RHRS operational pressure, align the RCS for cooldown with the Residual Heat Removal System.

   Location: Switchgear and Hot Shutdown Panels.

   Available Indications: RCS pressure indication and RCS temperature indication on the Hot Shutdown Panel.

7.4.1.3.4 Alternate Shutdown System

The Hot Shutdown Panels are designed to enable control to a hot standby condition if, for unspecified but non-catastrophic reasons, the Control Room (CR) had to be evacuated. The circuits required for shutdown on the Hot Shutdown Panels have been redesigned to comply with the CPNPP fire protection requirements. The present design enables control via Train A equipment even if a complete loss of the Control Room or Cable Spreading Room is postulated due to fire. Thus, the Hot Shutdown Panels, Shutdown Transfer Panels and specific portions of the plant systems noted in this section function as the Alternate Shutdown System for CPNPP for a fire in the Control Room, either Cable Spreading Room, or Control Room HVAC Mechanical Equipment Room (alternate shutdown areas).

The Hot Shutdown Panels and Shutdown Transfer Panels are each located in different fire areas which are, in turn, different from the fire areas associated with the alternate shutdown areas.

Since the postulated complete loss of either Cable Spreading Room, Control Room or Control Room HVAC Mechanical Equipment Room would also cause the loss of both Train A and Train B shutdown control and associated monitoring, alternate shutdown capability was provided to survive this highly improbable catastrophe by:

1. Providing transfer of required Train A circuits on the Shutdown Transfer Panels,

2. Providing required Train A fire safe shutdown control circuits and process monitoring circuits independent of the alternate shutdown areas, and

3. Providing monitoring circuits for required Train B fire safe shutdown components which are independent of the alternate shutdown areas equipment and cabling.

If the postulated fire occurs in the alternate shutdown areas, the transfer of Train A will take place at the Shutdown Transfer Panels and the control of the required Train A shutdown equipment is established at the Hot Shutdown Panels. For this scenario, Train A has been chosen for alternate shutdown control with certain Train B and non-safety related indications independent of the Control Room and Cable Spreading Rooms. The shutdown procedures are similar to those as described in Section 7.4.1.3.2 and 7.4.1.3.3. For cold shutdown, however, step 6 of Section 7.4.1.3.3 is carried out immediately.

To summarize, loss of the Control Room, either Cable Spreading Room or Control Room HVAC Mechanical Equipment Room will require Train A transfer from the Shutdown Transfer Panels and Train A control at the Hot Shutdown Panels. Loss of a Hot Shutdown Panel will require Train A control in the Control Room. Loss of a Shutdown Transfer Panel will require Train B control in the Control Room.

7.4.2 ANALYSIS

Hot standby is a stable plant condition that is automatically reached following a reactor trip from power. Additionally, the plant design features permit the achievement of cold shutdown as described in Section 7.4.1.2. In the unlikely event that access to the Control Room is restricted, the plant can be safely kept at hot standby by the use of the monitoring indicators and controls listed in Section 7.4.1.3 until the Control Room can be reentered. The administrative controls section of the technical specifications are immediately violated by any forced evacuation of the control room. Furthermore, cold shutdown conditions can be achieved from outside the Control Room through the use of suitable procedures as described in Sections 7.4.1.3.3 and 7.4.1.3.4 and by virtue of local control of the equipment listed in Section 7.4.1.2, in conjunction with the instrumentation and controls provided on the Hot Shutdown Panels and the Shutdown Transfer Panels (Tables 7.4-1, 7.4-2 and 7.4-3).

The discussions below demonstrate conformance to applicable General Design Criteria (GDC), Regulatory Guides, and IEEE Standard 279-1971. See also Table 7.1-2 for criteria applicable to all systems, Section 7.3, and other sections as referred to in the text.

7.4.2.1 General Analysis

1. Conformance to NRC General Design Criteria

   a. GDC 19

      The Hot Shutdown Panels and essential local control stations, discussed in Section 7.4.1.3, provide adequate controls and indications at locations outside the Control Room to maintain the reactors and the reactor coolant systems in the safe shutdown condition and to ensure decay heat removal in the event the Control Room must be evacuated.

b. GDC 34

Essential controls are provided outside the control room on the Hot Shutdown Panels to ensure adequate decay heat removal from the Reactor Coolant Systems in the event the main Control Room must be evacuated.

2. Conformance to NRC Regulatory Guides

a. RG 1.29

The Hot Shutdown Panels, Shutdown Transfer Panels, and essential local control stations are designed to withstand the effects of a Safe Shutdown Earthquake (SSE) without loss of Class 1E function or physical damage. The Hot Shutdown Panels, Shutdown Transfer Panels, and essential local control stations are classified Seismic Category I components.

b. RG 1.75

Proper separation in accordance with Regulatory Guide 1.75 and IEEE Standard 384 is maintained within the Hot Shutdown Panels for Train A, Train B and non-Class 1E Equipment and wiring.

3. Conformance to IEEE Standard 279-1971

The Hot Shutdown Panel and Shutdown Transfer Panel, including essential controls and indications, and the essential local control stations are designed to conform to applicable portions of IEEE Standard 279-1971. The control circuits at the Hot Shutdown Panel and essential local control stations are designed such that a single failure will not prevent proper protective action (maintaining safe hot standby) during postulated events. This is accomplished by providing independent Class 1E power for the fully redundant controls for the systems required for safe shutdown. The single failure criteria does not apply to postulated fires.

To prevent interaction between the redundant systems, the control channels are wired independently and separated with no electrical connections between redundant control systems. Non-Class 1E control circuits and non-Class 1E monitor circuits are electrically isolated from Class 1E controls and indications to prevent jeopardizing the reliability of the systems required for safe shutdown.

4. Demonstration of the ability of the plant to achieve safe shutdown using the minimum set of equipment designated for fire safe shutdown is provided in the fire safe shutdown analysis. Supporting analysis for equipment selection and event assumptions are provided in such documents as the WCAP-11331 “CPSES Thermal Hydraulic Analysis of Fire Safe Shutdown Scenario” dated 10/30/86.

7.4.2.2 Analysis for Shutdown From Outside the Control Room

The discussion found in Section 7.4.2.1 is applicable. The additional guides, criteria, and standards listed in Table 7.1-1 apply only to the Class 1E instrumentation and controls required for safe shutdown from outside the control room.

Also see the criteria applicability matrix, Table 7.1-2, for further information.

7.4.2.3 Consideration of Selected Plant Contingencies

7.4.2.3.1 Loss of Instrument Air Systems

Since electric-powered instrumentation is supplied from the Class 1E Power System, loss of the Instrument Air System will not degrade instrumentation required for safe shutdown. An analysis of the effect of loss of plant instrument air is provided in Section 9.3.1 and Table 9.3-3.

7.4.2.3.2 Loss of Cooling Water to Vital Equipment

Cooling water for safety-related systems is supplied by the Safety Chilled Water System (Section 9.4F), the Component Cooling Water System (Section 9.2.2), and the Station Service Water System (Section 9.2.1). Each of these systems is redundant; therefore the loss of any one cooling loop or instrumentation associated with that loop will not degrade the safety-related equipment serviced by the system. Also see the analyses presented in Sections 9.4F, 9.2.1 and 9.2.2 and Tables 9.2-1 and 9.2-5.

7.4.2.3.3 Plant Load Rejection, Turbine Trip, and Loss of Offsite Power

In the event of loss of offsite power associated with plant load rejection or turbine trip, power for safe shutdown is provided by the onsite Class 1E power systems. The description of the power systems is presented in Section 8.3. The standby diesel generators will provide power for the operation of safety-related equipment. The station batteries will provide DC power for the operation of control and instrumentation required to actuate and control essential components.

7.4.2.3.4 Loss of Class 1E and Non-Class 1E Bus During Operation

A review of the Class 1E and Non-Class 1E busses supplying power to safety and non-safety-related instrumentation and control systems which could affect the ability to achieve cold shutdown was performed. See Section 8.3.1.2. Upon the loss of inverter output voltage (<95% of rated), an alarm is provided to alert the operator about the loss of power to the bus. A review of I&C loads concluded that a loss of power of any one instrumentation and control bus will not result in an inability to achieve cold shutdown.

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 1 of 7)

Identification[c]	Function	
LI-501A	(d)	Steam generator 1 - level
LI-502A	(d)	Steam generator 2 - level
LI-503A		Steam generator 3 - level
LI-504A		Steam generator 4 - level
PI-514B	(d)	Steam generator 1 - pressure
PI-524B	(d)	Steam generator 2 - pressure
PI-534B		Steam generator 3 - pressure
PI-544B		Steam generator 4 - pressure
LI-459B	(d)	Pressurizer - level
LI-460B		Pressurizer - level
PI-455B	(d)	Pressurizer - pressure
LI-2478B	(d)	Condensate storage tank - level
LI-2479B		Condensate storage tank - level
HS-2450C	(d)	Motor-driven AFW pump 01 - control switch
HS-2451C		Motor-driven AFW pump 02 - control switch
HS-2451B		Motor-driven AFW pump 02 - control transfer switch
1/1-APCH1L	(d)	Charging pump 1 - control switch
1/1-APCH2L		Charging pump 2 - control switch
43/1-APCH2L		Charging pump 2 - control transfer switch
1/1-APBA1L		Boric acid transfer pump 1 - control switch
1/1-APBA2L		Boric acid transfer pump 2 - control switch

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 2 of 7)

Identification[c]	Function	
43/1-APBA2L		Boric acid transfer pump 2 - control transfer switch
HS-4250C	[d]	SSW pump 01 - control switch
HS-4251C		SSW pump 2 - control switch
HS-4251B		SSW pump 2 - control transfer switch
HS-4518C	[d]	CCW pump 01 - control switch
HS-4519C		CCW pump 2 - control switch
HS-4519B		CCW pump 2 - control transfer switch
43/1-456FT	[d]	Pressurizer PORV control transfer switch
1/1-456FL	[d]	Pressurizer PORV control switch
HS-2456FL	[d]	Motor-driven AFWP-1/recirculation valve control switch
HS-4286FL	[d]	SSW pump-1 discharge valve control switch
HS-4393FL	[d]	Diesel generator-1 Cooler Service Water return valve control switch
HS-4699FL		RCP/Thermal barrier cooler CCW supply isolation valve control switch
HS-4701FL		RCP motor air and lube oil coolers CCW return header isolation valve control switch
HS-4514FL	[d]	CCW heat exchanger to non-safety loop-1 isolation valve control switch
HS-4524FL	[d]	Non-safety loop return to CCW header isolation valve control switch
HS-4526FL		CCW to non-safety loop isolation valve control switch

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 3 of 7)

Identification[c]	Function	
1/1-8106FL	(d)	Charging pumps to Reactor Coolant System isolation valve control switch
1/1-8801AF	(d)	Charging pumps to Reactor Coolant System SIS isolation valve control switch
43/1-8153FT	(d)	Reactor Coolant system excess letdown valve control transfer switch
1/1-8153FL	(d)	Reactor Coolant system excess letdown valve control switch
1/1-8110FL	(d)	Charging pump miniflow isolation valve control switch
1/1-8701AF	(d)	RHR loop-1 inlet isolation valve control switch
1/1-8701BF	(d)(e)	RHR Loop-2 inlet isolation valve control switch
HS-2333FL	(d)	Main Steam loop-1 isolation valve control switch
HS-2334FL	(d)	Main Steam loop-2 isolation valve control switch
HS-2335FL	(d)	Main Steam loop-3 isolation valve control switch
HS-2336FL	(d)	Main Steam loop-4 isolation valve control switch
1/1-455AFL	(d)	Pressurizer power relief valve control switch
HS-6700FL	(d)	Safety Chilled water recirculation pump 5 control switch
CS-BT-1EA1L	(d)	Tie breaker BT-1EA1 control switch
CS-T1EB1L	(d)	Bus 1EA1 transformer T1EB1 feeder breaker control switch
CS-1EB1-1L	(d)	Incoming breaker 1EB1-1 control switch
CS-T1EB3L	(d)	Bus 1EA1 transformer T1EB3 feeder breaker control switch

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 4 of 7)

Identification[c]	Function	
CS-1EB3-1L	(d)	Incoming breaker 1EB3-1 control switch
CS-BT-1EB13L	(d)	Tie breaker BT-1EB13 control switch
TR-410F/420F/413F/423F	(d)	RCS Wide Range Temp. Cold/Hot Legs (Loops 1 & 2)
TR-430F/440F/433F/443F		RCS Wide Range Temp. Cold/Hot Legs (Loops 3 & 4)
NI-50A-3	(d)	Source Range Neutron Flux Monitoring
NI-50B-3		Source Range Neutron Flux Monitoring
1/1-8104L		Emergency Boration Valve - Control Switch
43/1-8104L		Emergency Boration Valve – Control Transfer Switch
CS-1EG1-L	(d)	Diesel Generator Breaker 1EG1 Control Switch
FK-121A	(d)	Charging Flow Control Valve (Pressurizer Level Control)
1/1-8149AL	(d)	Letdown Orifice Isolation (45 gpm) Control Switch
1/1-8149BL	(d)	Letdown Orifice Isolation (75 gpm) Control Switch
1/1-8149CL	(d)	Letdown Orifice Isolation (75 gpm) Control Switch
1/1-PCPR1L		Pressurizer Heater Backup Group A - Control Switch
1/1-PCPR2L		Pressurizer Heater Backup Group B - Control Switch
43/1-PCPR2L		Pressurizer Heater Backup Group B - control transfer switch
SK-2452B[f]		Turbine-driven AFW pump - speed controller

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 5 of 7)

Identification[c]	Function	
SI-2452B		Turbine-driven AFW pump - speed indication
HS-2452C	(d)	Turbine-driven AFW pump Main Steam line No. 4 supply valve - control switch
HS-2452D		Turbine-driven AFW pump Main steam line No. 1 supply valve - control transfer switch
HS-2452E		Turbine-driven AFW pump Main Steam line No. 1 supply valve - control switch
FK-2459B[f]		Turbine-driven AFW pump to steam generator 1-valve controller
FK-2453C[f]		Motor-driven AFW pump 01 to steam generator 1-valve controller
FK-2460B[f]		Turbine-driven AFW pump to steam generator 2-valve controller
FK-2453D[f]		Motor-driven AFW pump 01 to steam generator 2-valve controller
FK-2461B[f]		Turbine-driven AFW pump to steam generator 3-valve controller
FK-2454C[f]		Motor-driven AFW pump 02 to steam generator 3-valve controller
FK-2462B[f]		Turbine-driven AFW pump to steam generator 4-valve controller
FK-2454D[f]		Motor-driven AFW pump 02 to steam generator 4-valve controller
FI-121B		Charging Pump to CVCS Charging and RCP Seals - Flow
FI-2463D		AFW to steam generator 1 - flow
FI-2463B		AFW to steam generator 1 - flow
FI-2464D		AFW to steam generator 2 - flow

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 6 of 7)

Identification[c]	Function
FI-2464B	AFW to steam generator 2 - flow
FI-2465D	AFW to steam generator 3 - flow
FI-2465B	AFW to steam generator 3 - flow
FI-2466D	AFW to steam generator 4 - flow
FI-2466B	AFW to steam generator 4 - flow
PI-2477B	Turbine-driven AFW pump - suction pressure
PI-2455B	Turbine-driven AFW pump - discharge pressure
PI-2475B	Motor-driven AFW pump 01 - suction pressure
PI-2453B	Motor-driven AFW pump 01 - discharge pressure
PI-2476B	Motor-driven AFW pump 02 - suction pressure
PI-2454B	Motor-driven AFW pump 02 - discharge pressure
MLB-63	Train B Valve Positions
LCV-0112B [d]	Motor-driven Charging Pump - Suction Pressure
LCV-0112D [d]	Motor-driven Charging Pump - Suction Pressure

TABLE 7.4-1
INSTRUMENTATION AND CONTROL LOCATED ON HOT SHUTDOWN PANEL
PROVIDED FOR HOT STANDBY[a][b]

(Sheet 7 of 7)

Identification[c]	Function

a) These instruments and controls are also provided for cold shutdown.

b) All controls in this table are provided with transfer switch

c) Where Unit 1 equipment has been identified (i.e., 1/1-APCH1L, 43/1-APCH2L, 1EA1, BT-1EA1, CS-1EG1-L) Unit 2 Identification will be similar except for prefix (i.e., 1/2..., 43/2..., CS-2..., 2EA1, BT-2...)

d) Available to accomplish the desired action for alternate shutdown during fire.

e) Valve is closed and breaker opened during normal operation to prevent fire induced spurious operation.

f) Hand/Auto station with built-in transfer device to block control room signal and initiate control room local override alarm

SSW - Station Service Water System
CCW - Component Cooling Water System
AFW - Auxiliary Feedwater System

TABLE 7.4-2
OTHER INSTRUMENTATION AND CONTROLS LOCATED ON HOT SHUTDOWN PANEL PROVIDED FOR COLD SHUTDOWN[a]

(Sheet 1 of 3)

Identification[b]	Function
FI-183B	Emergency Borate Flow
FI-132B	Letdown - Flow
1/1-TCV-129L	Letdown divert valve TCV-129 - control switch
HS-5405C	Containment recirculation fan 01 - control switch
HS-5409B	Containment recirculation fan 02 - control switch
HS-5409C	Containment recirculation fan 02 - control transfer switch
HS-5413C	Containment recirculation fan 03 - control switch
HS-5417C	Containment recirculation fan 04 - control switch
HS-5417B	Containment recirculation fan 04 - control transfer switch
FI-4258B	SSW Train A - Flow
FI-4259B	SSW Train B - Flow
PI-4252B	SSW Pump 01 - discharge pressure
PI-4253B	SSW Pump 02 - discharge pressure
V-1EA1-L	6900-V bus 1EA1 - voltage
F-1EA1-L	6900-V bus 1EA1 - frequency
A-1EA1-1L	6900-V bus 1EA1, preferred offsite source - amperes
A-1EG1-L	6900-V bus 1EA1, onsite source - amperes
A-1EA1-2L	6900-V bus 1EA1, alternate offsite source - amperes
CS-1EA1-1L	Incoming breaker 1EA1-1 - control switch
CS-1EA1-2L	Incoming breaker 1EA1-2 - control switch
V-1EA2-L	6900-V bus 1EA2 - voltage
F-1EA2-L	6900-V bus 1EA2, frequency
A-1EA2-2L	6900-V bus 1EA2, alternate offsite source - amperes
A-1EG2-L	6900-V bus 1EA2, onsite source - amperes

TABLE 7.4-2
OTHER INSTRUMENTATION AND CONTROLS LOCATED ON HOT SHUTDOWN PANEL PROVIDED FOR COLD SHUTDOWN[a]

(Sheet 2 of 3)

Identification[b]	Function
A-1EA2-1L	6900-V bus 1EA2, preferred offsite source - amperes
CS-1EA2-2L	Incoming breaker 1EA2-2 - control switch
CS-1EG2-L	Diesel Generator 2 breaker 1EG2 - control switch
CS-1EA2-1L	Incoming breaker 1EA2-1 - control switch
43-1EA2-2	Breaker 1EA2-2 control transfer switch
43-1EG2	Diesel Generator 2 breaker 1EG2 control transfer switch
43-1EA2-1	Breaker 1EA2-1 control transfer switch
ZL-2476B	Motor-driven AFWP-2 suction pressure status light
ZL-2454C	Motor-driven AFWP-2 discharge to SG-3 control valve position
ZL-2454D	Motor-driven AFWP-2 discharge to SG-4 control valve position
ZL-2459B	Turbine-driven AFWP discharge to SG-1 control valve position
ZL-2460B	Turbine-driven AFWP discharge to SG-2 control valve position
ZL-2453C	Motor-driven AFWP-1 discharge to SG-1 control valve position
ZL-2475B	Motor-driven AFWP-1 suction pressure status light
ZL-2453D	Motor-driven AFWP-1 discharge to SG-2 control valve position
ZL-2461B	Turbine-driven AFWP discharge to SG-3 control valve position
ZL-2462B	Turbine-driven AFWP discharge to SG-4 control vavle position
ZL-455CF	Pressurizer spray valve position

TABLE 7.4-2
OTHER INSTRUMENTATION AND CONTROLS LOCATED ON HOT SHUTDOWN PANEL PROVIDED FOR COLD SHUTDOWN[a]

(Sheet 3 of 3)

Identification[b]		Function
ZL-455BF		Pressurizer spray valve position
ZL-PCPX1F		RCP-1 running
ZL-PCPX2F		RCP-2 running
ZL-PCPX3F		RCP-3 running
ZL-PCPX4F		RCP-4 running
HS-5180F		Steam generator heat exchanger outlet pressure control switch
HC-2325		SG-1 atmospheric relief valve control
HC-2326		SG-2 atmospheric relief valve control
HC-2327		SG-3 atmospheric relief valve control
HC-2328		SG-4 atmospheric relief valve control
HC-455C		RC Loop 4 pressurizer spray valve control
HC-606A	[c]	RHR Heat Exchanger Flow Control
HC-618	[c]	RHR Heat Exchanger Bypass Flow Control
1/1-APRH1F	[c]	RHR pump 1 control switch

a) The items in Table 7.4-1 are also provided for cold shutdown

b) Where Unit 1 equipment has been identified (i.e., 1/1-TCV-129L, 1EA1, V-1EA1-L) Unit 2 identification will be similar except for prefix (i.e., 1/2..., V-2..., 2EA1).

c) Available to accomplish the desired action for alternate shutdown during fire

SSW - Station Service Water System
CCW - Component Cooling Water System
AFW - Auxiliary Feedwater System

TABLE 7.4-3
TRANSFER SWITCHES LOCATED ON SHUTDOWN TRANSFER PANEL (STP)

(Sheet 1 of 3)

Identification[a]		Function
HS-2450B	(b)	Motor-driven AFWP-1 remote local
HS-2453AF	(b)	Motor-driven AFWP-1 SG-1 flow control valve remote local
HS-2453BF	(b)	Motor-driven AFWP-1 to SG-2 control valve remote local
HS-2456FT	(b)	Motor-driven AFWP-1 recirculation valve remote local
43/1-121-FT	(b)	Charging flow control remote local
HS-4250B	(b)	SSW pump-1 remote local
HS-4286FT	(b)	SSW pump-1 discharge valve remote local
HS-4393FT	(b)	Diesel generator-1 SSW return valve remote local
HS-4699FT		RCP/Thermal Barrier Cooler CCW supply Isolation valve remote local
HS-4701FT		RCP cooler CCW return isolation remote local
HS-4518B	(b)	CCW pump-1 remote local
HS-4514FT	(b)	Safeguard Loop CCW supply isolation remote local
HS-4524FT	(b)	Non-Safeguard Loop CCW Return valve remote local
HS-4526FT	(b)	Non-Safeguard Loop CCW Supply Valve remote control
HS-5405B		Containment recirculation fan 01 remote local
43/1-APCH1L	(b)	Centrifugal charging pump-1 remote local
43/1-APBA1L		Boric acid transfer pump-1 remote local
HS-5413B		Containment recirculation fan 03 remote local

TABLE 7.4-3
TRANSFER SWITCHES LOCATED ON SHUTDOWN TRANSFER PANEL (STP)

(Sheet 2 of 3)

Identification[a]	Function	
43/1-8106FT	(b)	Charging pumps to Reactor Coolant System isolation valve remote local
43/1-8801AF	(b)	Charging pumps to Reactor Coolant System SIS isolation valve remote local
43/1-8149AL	(b)	CVCS letdown orifice isolation valve remote local
43/1-8149BL	(b)	CVCS letdown orifice isolation valve remote local
43/1-8149CL	(b)	CVCS letdown orifice isolation valve remote local
43/1-8110FT	(b)	Charging pump miniflow isolation valve remote local
43/1-APRH1F	(b)	RHR pump-1 remote local
43/1-8701AF	(b)	RHRP 1 Hot Leg Recirculation Isolation
43/1-8701BF	(b)(c)	RHRP 2 Hot Leg Recirculation Isolation
HS-2452B	(b)	Turbine-driven AFWP steam supply line 4 valve remote local
43/1-TCV-129		Letdown to demineralizer or volume control tank remote local
HS-2333FT	(b)	Main Steam loop 1 isolation valve remote local
HS-2334FT	(b)	Main Steam loop 2 isolation valve remote local
HS-2335FT	(b)	Main Steam loop 3 isolation valve remote local
HS-2336FT	(b)	Main Steam loop 4 isolation valve remote local
43/1-455AFT	(b)	Pressurizer power relief valve remote local
43/1-PCPR1L		Pressurizer heater backup group-A remote local

TABLE 7.4-3
TRANSFER SWITCHES LOCATED ON SHUTDOWN TRANSFER PANEL (STP)

(Sheet 3 of 3)

Identification[a]	Function	
HS-6710B	(b)	Safety Chill Water Chiller Bypass Lockout
HS-67OOFT	(b)	Chilled water recirculation pump-5 remote local
43/1EA1-1	(b)	Incoming breaker 1EA1-1 remote local
43/1EA1-2	(b)	Incoming breaker 1EA1-2 remote local
43/1EG1	(b)	Diesel generator breaker 1EG1 remote local
43/BT-1EA1	(b)	Tie breaker BT-1EA1 remote local
43/T1EB1	(b)	Bus 1EA1 transfer T1EB1 feeder breaker remote local
43/1EB1-1	(b)	Incoming breaker 1EB1-1 remote local
43/T1EB3	(b)	Bus 1EA1 transfer 1EB3 feeder breaker remote local
43/1EB3-1	(b)	Incoming breaker 1EB3-1 remote local
43/BT-1EB13		Tie breaker BT-1EB13 remote local
LCV-0112B	(b)	Motor-driven Charging Pump - Suction Pressure
LCV-0112D	(b)	Motor-driven Charging Pump - Suction Pressure

a) Where Unit 1 equipment has been identified (i.e., 43/1-121-/ft, 1EA1) Unit 2 Identification will be similar except for prefix (i.e., 43/2..., 2EA1-1, BT-2...).

b) Available to accomplish the desired action for alternate shutdown during fire.

c) Valve is closed and breaker opened during normal operation to prevent fire induced spurious operation.

SSW - Station Service Water System
CCW - Component Cooling Water System
AFW - Auxiliary Feedwater System

7.5 INFORMATION SYSTEMS IMPORTANT TO SAFETY

7.5.1 DESCRIPTION OF INFORMATION SYSTEMS

The plant safety analyses and evaluations define the design basis accident event scenarios for which pre-planned operator actions are required. Accident monitoring instrumentation is necessary to permit the operator to take required actions to address these analyzed situations. However, instrumentation is also necessary for unforeseen situations (i.e., to ensure that, should plant conditions evolve differently than predicted by the safety analyses, the control room staff has sufficient information to monitor the course of the event). Additional instrumentation is also needed to indicate to the operator whether the integrity of the in-core fuel clad, the reactor coolant system pressure boundary, or the reactor containment has degraded beyond the prescribed limits defined as a result of the plant safety analyses and other evaluations.

Five classifications of variables have been identified to provide this instrumentation. These five classification supercede the “PAMS” classification previously required by U.S. NRC Regulatory Guide 1.97, Revision 1. Operator manual actions identified in the emergency response procedures, associated with design basis accident events are pre-planned. Those variables that provide information needed by the operator to perform these manual actions are designated Type A. The basis for selecting Type A variables is given in Section 7.5.1.2.1.

Those variables needed to assess that the plant critical safety functions are being accomplished or maintained, as identified in the plant safety analysis and other evaluations, are designated Type B.

Variables used to monitor for the gross breach or the potential for gross breach of the in-core fuel clad, the reactor coolant system pressure boundary, or the primary reactor containment, are designated Type C. Variables used to monitor the potential breach of containment have an arbitrarily determined extended range. The extended range is chosen to minimize the probability of instrument saturation even if conditions exceed those predicted by the safety analysis. The response characteristics of Type C information display channels allows the control room staff to detect conditions indicative of gross failure of any of the three fission product barriers or the potential for gross failure of these barriers. Although variables selected to fulfill Type C functions may rapidly approach the values that indicate an actual gross failure, it is the final steady-state value reached that is important. Therefore, a high degree of accuracy and a rapid response time are not necessary for Type C information display channels.

Those variables needed to assess the operation of individual safety systems and other systems important to safety are designated Type D.

The variables that are required for use in determining the magnitude of the postulated releases and continually assessing any such releases of radioactive materials are designated Type E.

The five classifications are not mutually exclusive, in that, a given variable (or instrument) may be included in one or more types. When a variable is included in one or more of the five classifications, the equipment monitoring this variable will meet the highest category as noted in Tables 7.5-7A, B & C.

Three categories of design and qualification criteria have been identified. The differentiation is made in order that an importance of information hierarchy can be recognized in specifying

accident monitoring instrumentation. Category 1 instrumentation has the highest pedigree and should be utilized for information which is essential to the control room operating staff in order for them to determine if the plant safety functions are being performed. Category 2 and Category 3 instruments are of lesser importance in determining the state of the plant and do not require the same level of operational assurance.

Category 1 displays located on the Main Control Boards will be uniquely identified.

The primary differences between category requirements are in qualification, application of single failure, power supply, and display requirements. Category 1 requires seismic and environmental qualification, the application of a single failure criteria, utilization of emergency standby power, and a continuous display. Category 2 requires qualification commensurate with the required function but does not require the single failure criteria, emergency standby power, or a continuous display. Category 2 requires, in effect, a rigorous performance verification for a single instrument channel. Category 3 does not require qualification, single failure criteria, emergency standby power, or a continuous display.

7.5.1.1 Definitions

7.5.1.1.1 Design Basis Accident Events

Those events, any one of which could occur during the lifetime of a particular unit and those events not expected to occur but postulated because their consequences would include the potential for release of significant amounts of radioactive gaseous, liquid, or particulate material to the environment are Design Basis Accident Events. Excluded are those events (defined as “normal” and “anticipated operational occurrences” in 10 CFR 50) expected to occur more frequently than once during the lifetime of a particular unit. The limiting accidents that are used to determine instrument functions are: 1) LOCA, 2) Steamline Break, 3) Feedwater Line Break, and 4) Steam Generator Tube Rupture.

7.5.1.1.2 Safe Shutdown

The safe shutdown condition is completely defined in CPNPP/FSAR Section 7.4.

7.5.1.1.3 Deleted

7.5.1.1.4 Critical Safety Functions

Those safety functions that are essential to prevent a direct and immediate threat to the health and safety of the public. These are the accomplishing or maintaining of:

1. Subcriticality (reactivity control)

2. Reactor coolant system integrity

3. Reactor coolant system inventory

4. Reactor core cooling

5. Heat sink

6. Containment integrity

7.5.1.1.5 Immediately Accessible Information

Information that is visually available to the control room operator immediately after he has made the decision that the information is needed.

7.5.1.1.6 Primary Information

Information that is essential for the direct accomplishment of the pre-planned manual actions necessary to bring the plant into a safe condition in the event of a Design Basis Accident Event; it does not include those variables that are associated with contingency actions.

7.5.1.1.7 Contingency Actions

Those manual actions that address conditions beyond the Design Basis Accident Events.

7.5.1.1.8 Key Variables

Those variables which provide the most direct measure of the information required.

7.5.1.1.9 Preferred Backup Variables

Those backup variables that provide the most direct information for the key variable being monitored.

7.5.1.1.10 Backup Variable

That information, made up of additional variables, that provides supplemental and/or confirmatory information to the Control Room Operating Staff.

7.5.1.1.11 Diverse Variable

Where failure of an accident monitoring channel results in information ambiguity that can lead the operator to defeat or fail to accomplish a required safety function, a second variable shall be identified to allow the operator to deduce the actual conditions of the plant. This second variable is called a diverse variable.

7.5.1.2 Variable Types

The accident monitoring variables and information display channels are those that are required to enable the Control Room Operating Staff to perform the functions defined by Types A, B, C, D, and E below.

7.5.1.2.1 Type A

Those variables that provide the primary information required to permit the Control Room Operating Staff to:

1. Perform the diagnosis specified in the applicable CPNPP Emergency Response Guidelines (ERG),

2. Take the specified pre-planned manually controlled actions, for which no automatic control is provided, that are required for safety systems to accomplish their safety function in order to recover from the Design Basis Accident Event, and

3. Reach and maintain a safe shutdown condition.

Consistent with the definition of Type A variables in Regulatory Guide 1.97 Revision 2, the verification of the actuation of safety systems has been excluded from the definition of Type A. The variables which provide this verification are included in the definition of Type D.

Variables in Type A are restricted to pre-planned actions for Design Basis Accident Events. Contingency actions and additional variables which might be utilized will be in Types B, C, D, and E.

7.5.1.2.2 Type B

Those variables that provide to the Control Room operating staff information to assess the process of accomplishing or maintaining critical safety functions, i.e., reactivity control, reactor coolant system integrity, RCS inventory, reactor core cooling, heat sink, and containment integrity.

7.5.1.2.3 Type C

Those variables that provide to the Control Room Operating Staff information to monitor (1) the extent to which variables, which indicate the potential for causing a gross breach of a fission product barrier, have exceeded the design basis values and (2) that the in-core fuel cladding, the reactor coolant system pressure boundary or the primary reactor containment may have been subject to gross breach. These variables include those required to initiate the early phases of the emergency plan.

Type C variables used to monitor the potential for breach of a fission product barrier have an arbitrarily-determined extended range.

7.5.1.2.4 Type D

Those variables that provide to the control room operating staff sufficient information to monitor the performance of:

1. plant safety systems employed for mitigating the consequences of an accident and subsequent plant recovery to attain a safe shutdown condition. These include verification of the automatic actuation of safety systems, and

2. other systems normally employed for attaining a safe shutdown condition.

7.5.1.2.5 Type E

Those variables to be monitored as required for use in determining accessibility for required post accident actions, estimating the potential magnitude of the release of radioactive materials, and continually assessing such releases.

7.5.1.3 Variable Categories

The qualification requirements of the Type A, B, C, D and E accident monitoring instrumentation are subdivided into three categories (1, 2, 3). Descriptions of the three categories are given below. Table 7.5-1 briefly summarizes the design and qualification requirements of the three designated categories.

7.5.1.3.1 Category 1

7.5.1.3.1.1 Selection Criteria for Category 1

The selection criteria for Category 1 variables have been subdivided according to the variable type. For Type A, those key variables used for diagnosis or providing information for preplanned operator action have been designated Category 1. For Type B, those key variables which are used for monitoring the process of accomplishing or maintaining critical safety functions have been designated Category 1. For Type C, those key variables which are used for monitoring the potential for breach of a fission product barrier have been designated Category 1.

7.5.1.3.1.2 Qualification Criteria for Category 1

The instrumentation is environmentally and seismically qualified in accordance with Sections 3.11 and 3.10, respectively, of the FSAR. Instrumentation shall continue to read within the required accuracy following but not necessarily during a seismic event. At least one redundant instrumentation channel for each Category 1 variable is qualified from sensor to a direct-indicating meter or recording device. For the balance of instrumentation channels, qualification applies up to and includes the channel isolation device. (Refer to Section 7.5.1.3.4 in regard to extended range instrumentation qualification).

7.5.1.3.1.3 Design Criteria for Category 1

1. No single failure within either the accident-monitoring instrumentation, its auxiliary supporting features, or its power sources, concurrent with the failures that are a condition of or result from a specific accident, will prevent the operator from being presented the required information. Where failure of one accident-monitoring channel results in information ambiguity (e.g., the redundant displays disagree), additional information is provided to allow the operator to deduce the actual conditions in the plant. This may be accomplished by providing additional independent channels of information of the same variable (addition of an identical channel), or by providing independent channels which monitor different variables which bear known relationships to the multiple channels (addition of a diverse channel(s)). Redundant or diverse channels are electrically independent and physically separated from each other to the extent practicable with two train separation, and from equipment not classified important to safety in accordance with 10 CFR 50 Appendix A, Criteria 22 and 24, and derived regulatory guides consistent with the licensing basis for CPNPP. See FSAR Sections 1A(N) and 1A(B).

When diversity is employed in lieu of redundancy, detailed procedures will be established to detect and resolve any ambiguity that may exist. These procedures recognizes such factors as electrical independence and physical separation of the channels employed.

2. The instrumentation is energized from station emergency standby power sources, battery backed where momentary interruption is not tolerable.

3. The out-of-service interval is based on normal Technical Specification requirements for the system it serves where applicable or where specified by other requirements.

4. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the required interval between testing is less than the normal time interval between generating station shutdowns, a capability for testing during power operation is provided.

5. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

6. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

7. The monitoring instrumentation design minimizes the development of conditions that would cause meters, annunciators, recorders, alarms, etc., to give anomalous indications potentially confusing to the operator.

8. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

9. To the extent practicable, monitoring instrumentation inputs will be from sensors that directly measure the desired variables.

10. Periodic checking, testing, calibration, and calibration verification is in accordance with the CPNPP Technical Specification as implemented by CPNPP procedures. When no CPNPP Technical Specifications are applicable, CPNPP plant specific procedures are implemented.

7.5.1.3.1.4 Information Processing and Display Interface Criteria for Category 1

The interface criteria specified here provide requirements to be implemented in the processing and displaying of the information.

1. The operator has immediate access to the information from redundant or diverse channels in units familiar to the operator (i.e., for temperature readings degrees should be used, not volts). Where two or more instruments are needed to cover a particular range, overlapping of instrument spans are provided.

2. A historical record of a minimum of one instrumentation channel for each process variable is maintained. The Plant Computer System has a long-term archival system with disk storage which can archive at least one fuel cycle of data from the process variables. In

addition graphs of critical parameters during any immediately preceding thirty minute period will be available.

7.5.1.3.2 Category 2

7.5.1.3.2.1 Selection Criteria for Category 2

The selection criteria for Category 2 variables are subdivided according to the variable type. For Types B and C those variables which provide preferred backup information are designated Category 2. For types D, and E, (except the Containment Radiation Level (High Range) which is designated category 1) key variables have been included under Category 2.

7.5.1.3.2.2 Qualification Criteria for Category 2

Category 2 instrumentation is environmentally qualified when it is subjected to adverse environments caused by the DBA during the time it must serve its intended function. Category 2 instrumentation is seismically qualified from the sensor up to and including the channel isolation device when the instrument is a part of a safety related system and its failure could degrade the system (e.g., the channel is Class 1E).

7.5.1.3.2.3 Design Criteria for Category 2

1. The instrumentation is energized from a highly reliable on-site power source, not necessarily the emergency standby power, which is battery backed where momentary interruption is not tolerable.

2. The out-of-service interval is based on normal Technical Specification requirements for the system it serves where applicable or where specified by other requirements.

3. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the required interval between testing is less than the normal time interval between generating station shutdowns, a capability for testing during power operation is provided.

4. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

5. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

6. The monitoring instrumentation design minimizes the development of conditions that would cause meters, annunciators, recorders, alarms, etc., to give anomalous indications potentially confusing to the operator.

7. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

8. To the extent practicable, monitoring instrumentation inputs are from sensors that directly measure the desired variables.

9. Periodic checking, testing, calibration, and calibration verification is in accordance with CPNPP plant specific procedures.

7.5.1.3.2.4 Information Processing and Display, Interface Criteria for Category 2

The instrumentation signal is, as a minimum, processed for display on demand. See Section 7.5.1.3.1.4 Item 2.

7.5.1.3.3 Category 3

7.5.1.3.3.1 Selection Criteria for Category 3

For all types, those variables which provide backup information are designated Category 3.

7.5.1.3.3.2 Qualification Criteria for Category 3

The instrumentation is high quality commercial grade, selected to withstand its' service environment.

7.5.1.3.3.3 Design Criteria for Category 3

1. Servicing, testing, and calibration programs are specified to maintain the capability of the monitoring instrumentation. For those instruments where the required interval between testing is less than the normal time interval between generating station shutdowns, a capability for testing during power operation is provided.

2. Whenever means for removing channels from service are included in the design, the design facilitates administrative control of the access to such removal means.

3. The design facilitates administrative control of the access to all setpoint adjustments, module calibration adjustments, and test points.

4. The monitoring instrumentation design minimizes the development of conditions that would cause meters, annunciators, recorders, alarms, etc., to give anomalous indications potentially confusing to the operator.

5. The instrumentation is designed to facilitate the recognition, location, replacement, repair, or adjustment of malfunctioning components or modules.

6. To the extent practicable, monitoring instrumentation inputs are from sensors that directly measure the desired variables.

7. The instrumentation signal is, as a minimum, processed for display on demand. See Section 7.5.1.3.1.4 Item 2.

7.5.1.3.3.4 Extended Range Instrumentation Qualification Criteria

The qualification environment for extended range information display channel components is based on the Design Basis Accident Events, except the assumed maximum value of the monitored variable shall be the value equal to the specified maximum range for the variable. The

decay for this variable is considered proportional to the decay for this variable associated with the Design Basis Accident Events. No additional qualification margin needs to be added to the extended range variable. All environmental envelopes except that pertaining to the variable measured by the information display channel will be those associated with the Design Basis Accident Events. The environmental qualification requirement for extended range equipment does not account for steady-state elevated levels that may occur in other environmental parameters associated with the extended range variable. For example, a sensor measuring containment pressure must be qualified for the measured process variable range (i.e., 3 times design pressure for concrete containments), but the corresponding ambient temperature is not mechanistically linked to that pressure. Rather, the ambient temperature value is the bounding value for Design Basis Accident Events analyzed in Chapter 15 of the FSAR. The extended range requirement is to ensure that the equipment will continue to provide information if conditions degrade beyond those postulated in the safety analysis. Since extended variable ranges are non-mechanistically determined, as in the example of containment pressure and ambient temperature, extension of associated parameter levels is not justifiable and is therefore not required.

7.5.2 DESCRIPTION OF VARIABLES

7.5.2.1 Type A Variables

Type A variables are defined in Section 7.5.1.2.1. They are the variables which provide primary information required to permit the Control Room operating staff to:

1. Perform the diagnosis specified in the applicable CPNPP ERG's;

2. Take specified pre-planned manually controlled actions for which no automatic control is provided, that are required for safety systems to accomplish their safety function to recover from the Design Basis Accident Event (verification of actuation of safety systems is excluded from Type A variables and is included as Type D);

3. Reach and maintain a safe shutdown condition.

Key Type A variables have been designated Category 1. These are the variables which provide the most direct measure of the information required. The KEY Type A variables are:

1. RCS Wide Range Pressure

2. Wide Range Hot Leg Reactor Coolant Temperature (T(Hot))

3. Wide Range Cold Leg Reactor Coolant Temperature (T(Cold))

4. RCS Subcooling (Saturation Margin)

5. Narrow Range Steam Generator Water Level (NR)

6. Pressurizer Water Level

7. Containment Pressure - Intermediate range

8. Main Steam Line Pressure (S/G Pressure)

9. Refueling Water Storage Tank (RWST) Level

10. Deleted

11. Condensate Storage Tank (CST) Level

12. Auxiliary Feedwater (AFW) Flow to each S/G

13. Core Exit Temperature

All Type A Variables have been designated Category 1.

A summary of Type A Variables is provided in Table 7.5-2.

7.5.2.2 Type B Variables

Type B variables are defined in Section 7.5.1.2.2. They are the variables that provide to the Control Room Operating Staff information to assess the process of accomplishing or maintaining critical safety functions, i.e.:

1. Subcriticality (Reactivity Control)

2. Reactor Coolant System integrity

3. Reactor Coolant System Inventory

4. Reactor Core Cooling

5. Heat Sink

6. Containment integrity

Variables which provide the most direct indication (i.e., Key variable) to assess each of the critical safety functions have been designated Category 1. Preferred backup variables have been designated Category 2. All other backup variables are Category 3. These are listed in Table 7.5-3.

7.5.2.3 Type C Variables

Type C variables are defined in Section 7.5.1.2.3. Basically, they are the variables that provide to the Control Room Operating Staff information to monitor the potential for breach or actual gross breach of:

1. In-core fuel clad;

2. Reactor Coolant System Boundary;

3. Containment Boundary.

(Variables associated with monitoring of radiological releases from the plant are included in Type E).

Those Type C key variables which provide the most direct measure of the POTENTIAL for breach of one of the 3 fission product boundaries have been designated Category 1. Variables which indicate actual breach have been designated as preferred backup information and are designated Category 2. All other backup variables are designated Category 3.

Table 7.5-4 summarizes the selection of Type C variables.

7.5.2.4 Type D Variables

Type D variables are defined in Section 7.5.1.2.4. They are those variables that provide sufficient information to the Control Room Operating Staff to monitor the performance of:

1. Plant safety systems employed for mitigating the consequences of an accident and subsequent plant recovery to attain a safe shutdown condition, including verification of the automatic actuation of safety systems; and

2. Other systems normally employed for attaining a safe shutdown condition.

Type D key variables are designated Category 2. Backup variables are designated Category 3.

The following systems or major components have been identified as requiring Type D variables to be monitored:

1. Reactor Coolant Pump (to assure RCS pressure boundary integrity by maintaining seal integrity)

2. Pressurizer level and pressure control to assess status of RCS following return to normal pressure and level control under certain post-accident conditions

3. Chemical and Volume Control System (CVCS) (employed for attaining a safe shutdown under certain post-accident conditions)

4. Main Steam and Feedwater Systems (employed for restoring/maintaining a secondary heat sink under post-accident conditions)

5. Emergency Core Cooling System (ECCS)

6. Auxiliary Feedwater

7. Containment Systems

8. Component Cooling Water

9. Service Water

10. Heating, Ventilation, Air Conditioning (HVAC) if required for ESF operation

11. Safety Chilled Water

12. Vital Electrical Power Supplies

13. Residual Heat Removal

14. Radioactive Waste System

Table 7.5-5 summarizes the selection of Type D variables.

7.5.2.5 Type E Variables

Type E variables are defined in Section 7.5.1.2.5. They are those variables that provide the Control Room operating staff with information to:

1. Monitor the habitability in the control room,

2. Monitor the plant areas where access may be required to service equipment necessary to monitor or mitigate the consequences of an accident and the post accident radiation environment warrants continuous monitoring,

3. Estimate the magnitude of the radiation source available for potential release,

4. Estimate the magnitude of release of radioactive materials through identified pathways, and

5. Estimate the magnitude of unplanned releases of radioactive material from unidentified pathways.

Containment Radiation Level has been designated Category 1. All other Key Type E variables have been designated Category 2. Backup Type E variables have been designated Category 3.

Table 7.5-6 summarizes the selection of Type E variables.

7.5.3 ANALYSIS OF INFORMATION SYSTEMS IMPORTANT TO SAFETY

7.5.3.1 Compliance With General Design Criteria 2

All Category 1 instrumentation and that Category 2 instrumentation which is part of a safety related system and whose failure could degrade the system are seismically qualified as noted in Tables 7.5-7A, B & C. See CPNPP FSAR Section 3.10.

7.5.3.2 Compliance With General Design Criteria 4

All Category 1 instrumentation and that Category 2 instrumentation which is subjected to adverse environments when it must perform its intended function are environmentally qualified as noted in Tables 7.5-7A, B & C. See CPNPP FSAR Section 3.11.

7.5.3.3 Compliance With General Design Criteria 13

The instrumentation identified in CPNPP FSAR Tables 7.5-7A, B & C monitors the proper variables over adequate ranges to assure adequate safety.

7.5.3.4 Compliance With General Design Criteria 19

The instrumentation provided allows for safe operation of the plant from the control room and allows the maintenance of a safe condition under design basis accident conditions. Adequate instrumentation is also provided outside the control room to allow for safe shutdown and cold shutdown (see CPNPP FSAR Section 7.4).

7.5.3.5 Compliance With U.S. NRC Regulatory Guide 1.47

The bypassed and inoperable status system (Safety System Inoperable Indication (SSII) are described in Section 7.1.2.6 under a discussion of conformance to U.S. NRC Regulatory Guide 1.47. The SSII is in accordance with the Regulatory Guide as described in the referenced section. Also, see Figure 7.1-4 for typical implementation of SSII for an ESF System.

7.5.3.6 Compliance With U.S. NRC Regulatory Guide 1.97

As described throughout CPNPP FSAR Section 7.5, CPNPP meets the intent of U.S. NRC Regulatory Guide 1.97.

Table 7.5-7A provides information on instrument range, environmental/seismic qualification, quality assurance, redundance and sensor location and the power supply for the variables listed in Table 2 to the Regulatory Guide 1.97, Rev. 2. Table 7.5-7B provides similar information pertaining to those variables included in the CPNPP accident monitoring design which are not listed in Table 2 to Regulatory Guide 1.97, Rev. 2.

Table 7.5-7C pertains to the notes to Tables 7.5-7A and 7.5-7B. Table 7.5-7D lists specific deviations which did not meet or exceed the guidance of the Regulatory Guide as identified in Table 7.5-7A. Table 7.5-7E includes the supporting justifications and alternatives that correspond to each of the deviations listed and reference numbers in Table 7.5-7D. Table 7.5-7F includes general deviations from the Regulatory Guide and provides the defferences between the specific wording of Section C in the Regulatory GUide and CPNPP design along with any justifications.

7.5.3.7 Compliance With U.S. NRC Regulatory Guide 1.105

The accuracies and ranges of the instruments selected to monitor the variables specified are consistent with the assumptions used in the accident analysis. See FSAR Section 1A(N) and 1A(B).

7.5.3.8 Other Information Systems and Human Factors Evaluations

Other information systems such as the safety parameter display system and the ERF (emergency response facility) data display systems (see FSAR Section III.A.1.2) are integrated with the CPNPP instrumentation described in this section. In order to provide the operator adequate information to prevent and cope with accidents, these displays have been included in human factors engineering reviews as discussed in CPNPP FSAR Section I.D.1.

Monitor light displays consist of two independent trains of status monitoring equipment, one for each safety train. The displays, located on the Control Board in the vicinity of the ESF system controls, are light box arrays which monitor the status of ESF support system valves, dampers, pumps, and fans. The displays provide visual indication that the components are not in their correct position, in the case of valves and dampers, or in their correct operational mode in the case of pumps and fans, following receipt of a safety actuation signal. The monitor light displays provide a convenient overview of the ESF functions and main control board indication for some of the accident monitoring instrumentation. These light displays are qualified seismic Class 1E. See Sections 7.5, 7.5.3.6, Tables 7.5-7A, B, C, D, E, and F.

TABLE 7.5-1
SUMMARY OF MINIMUM DESIGN QUALIFICATION, AND INTERFACE REQUIREMENTS

Qualification	Category 1	Category 2	Category 3
Enviromental	Yes	As appropriate	No
Seismic	Yes	As appropriate	No
Design			
Single Failure	Yes	No	No
Power Supply	Emergency Standby	Highly Reliable (On-Site)	Non-1E
Interface			
Indication	Immediately Accessible	On Demand	On Demand
Recording	ERF Computer	No	No

TABLE 7.5-2
SUMMARY OF TYPE A VARIABLES

Variable	Variable Function	Type/ Category
RCS Pressure (WR)	Key	A1
T(HOT) (WR)	Key	A1
T(COLD) (WR)	Key	A1
RCS Subcooling Water	Key	A1
Steam Generator Water Level (NR)	Key	A1
Pressurizer Level	Key	A1
Containment Pressure (IR)	Key	A1
Main Steam Line Pressure (S/G Pressure)	Key	A1
RWST Level	Key	A1
CST Level	Key	A1
Auxiliary Feedwater Flow to each S/G	Key	A1
Core Exit Temperature	Key	A1

WR - Wide Range NR - Narrow Range IR - Intermediate Range

TABLE 7.5-3
SUMMARY OF TYPE B VARIABLES

(Sheet 1 of 2)

Function Monitored	Variable	Variable Function	Type/ Category
Subcriticality	Neutron Flux	KeyB1	
Reactivity	Wide Range T(Hot)	Backup (P)	B2
Control	Wide Range T(Cold)	Backup (P)	B2
	Control Rod Position	Backup	B3
Reactor	RCS Pressure (WR)	Key	B1
Coolant	Wide Range T (Hot)	Key	B1
System	Wide Range T (Cold)	Key	B1
Integrity	Core Exit Temperature	Key	B1
	Containment Radiation Level	Backup (P)	B2
	Condenser Off-gas Radiation	Backup (P)	B2
	S/G Blowdown Radiation Level	Backup (P)	B2
	Containment Water Level	Backup (P)	B2
Reactor	Pressurizer Level	Key	B1
Coolant	Reactor Vessel Water	Key	B1
System	Level (RVLIS)		
Inventory	Containment Water Level	Backup (P)	B2
Reactor	Core Exit Temperature	Key	B1
Core	RCS Subcooling	Key	B1
Cooling	Reactor Vessel Water	Key	B1
	Level (RVLIS)		
	Wide Range T (Hot)	Backup (P)	B2
	Wide Range T (Cold)	Backup (P)	B2
	RCS Pressure (WR)	Backup (P)	B2

TABLE 7.5-3
SUMMARY OF TYPE B VARIABLES

(Sheet 2 of 2)

Function Monitored	Variable	Variable Function	Type/ Category
Heat Sink	Steam Generator Level (NR)	Key	B1
	Steam Generator Level (WR)	Key	B1
	Auxiliary Feedwater Flow to each S/G	Key	B1
	Main Steamline Pressure (S/G Pressure)	Key	B1
Containment Integrity	Containment Pressure (IR)	Key	B1
	Containment Water Level	Key	B1
	Containment Hydrogen Concentration	Backup	B3

Backup (P) - Preferred backup

IR - Intermediate Range

TABLE 7.5-4
SUMMARY OF TYPE C VARIABLES

Function Monitored	Variable	Variable Function	Type/ Category
Fuel Cladding	Core Exit Temperature	Key	C1
	Reactor Vessel Water Level (RVLIS)	Backup (P)	C2
RCS Pressure Boundary	RCS Pressure (WR)	Key	C1
	Containment Pressure (IR)	Backup (P)	C2
	Containment Water Level	Backup (P)	C2
	Containment Radiation Level (HR)	Backup (P)	C2
	S/G Blowdown Radiation Level	Backup (P)	C2
	Condenser Off-gas Radiation	Backup (P)	C2
Containment Boundary	Containment Pressure (WR)	Key	C1
	Containment Hydrogen Concentration	Backup	C3
	Plant Vent Effluent radioactivity and flow	Backup (P)	C2
	RCS Pressure (WR)	Backup (P)	C2
	Containment Isolation Valve Status[(a)]	Backup (P)	C2
	Containment Pressure (IR)	Backup (P)	C2
	Area Radiation Level Adjacent Containment	Backup (P)	C3

a) Excludes local manual, check, relief and safety valves.

IR - Intermediate Range
WR - Wide Range
HR - High Range

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 1 of 6)

System/Component	Variable	Variable Function	Type/ Category
Reactor Coolant Pump	CCW to RCP Valve Status	Key	D2
	RCP Seal Injection Isolation Valve Status	Key	D2
	Motor Current	Backup	D3
	Seal Injection Flow	Backup	D3
Pressurizer Level and Pressure Control	PORV Valve Status	Key	D2
	Pressurizer PORV Block Valve Status	Key	D2
	Safety Valve Status	Key	D2
	Pressurizer Level	Key	D2
	RCS Pressure (WR)	Key	D2
	Heater Current	Key	D2
	Pressurizer Relief Tank Level	Backup	D3
	Pressurizer Relief Tank Temperature	Backup	D3
	Pressurizer Relief Tank Pressure	Backup	D3
Chemical and Volume Control System (CVCS)	Charging system makeup flow	Key	D2
	Letdown flow	Key	D2
	VCT level	Key	D2
	CVCS valve status	Key	D2
	Boric Acid Tank Level	Key	D2

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 2 of 6)

System/Component	Variable	Variable Function	Type/ Category
Main Steam and Feedwater Systems	S/G PORV status	Key	D2
	MSIV and bypass Valve status	Key	D2
	S/G safety valve status	Key	D2
	S/G blowdown isolation valve status	Key	D2
	S/G Sampling isolation valve status	Key	D2
	AFW Pump Turbine Main Steam Header Isolation Valve status	Key	D2
	CST water level	Key	D2
	Main Steamline pressure (S/G Pressure)	Key	D2
	Auxiliary Feedwater flow to each S/G	Key	D2
	S/G water level (WR)	Key	D2
	S/G water level (NR)	Key	D2
	MFW control and bypass valve status	Key	D2
	MFW isolation and bypass valve status	Key	D2
	MFW flow	Backup	D3
	Turbine Stop Valve status	Backup	D3
	Unit 2 Feedwater Split-flow bypass valves status	Key	D2

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 3 of 6)

System/Component	Variable	Variable Function	Type/ Category
Emergency Core Cooling System (ECCS)	RWST Level	Key	D2
	Safety Injection Pump flow	Key	D2
	Safety Injection Pump Status	Key	D2
	RHR flow	Key	D2
	RHR Pump Status	Key	D2
	Centrifugal Charging Pump injection flow	Key	D2
	Centrifugal Charging Pump Status	Key	D2
	Containment Water Level	Key	D2
	ECCS valve status	Key	D2
	SI accumulator isolation valve status	Key	D2
	SI accumulator tank pressure	Key	D2
	SI accumulator tank level	Backup	D3
	Control Rod Position	Backup	D3
Auxiliary Feedwater	Flow to each S/G	Key	D2
	Pump Status	Key	D2
	Valve status	Key	D2
	CST level	Key	D2
	AFW Pump Turbine Main Steam Header Isolation Valve Status	Key	D2

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 4 of 6)

System/Component	Variable	Variable Function	Type/ Category
Containment Systems	Containment spray flow	Key	D2
	Containment Spray Pump status	Key	D2
	Containment water level	Key	D2
	Spray valve status	Key	D2
	Containment atmosphere temperature	Key	D2
	Containment Pressure (IR)	Key	D2
	Containment Spray Additive Tank Outlet Valve Status	Key	D2
	Containment Pressure (WR)	Key	D2
	Containment Spray Additive Tank Level	Key	D2
Component Cooling Water System (CCW)	Header Pressure	Key	D2
	Header Temperature	Key	D2
	Surge tank level	Key	D2
	CCW flow	Key	D2
	CCW Pump Status	Key	D2
	Valve status	Key	D2
Service Water System	Header Flow	Key	D2
	SSW Pump Status	Key	D2

IR - Intermediate Range

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 5 of 6)

System/Component	Variable	Variable Function	Type/ Category
ESF Ventilation	CR A/C Units	Key	D2
	CR Vent Damper Position	Key	D2
	CR Emergency Filtration Fans	Key	D2
	CR Emergency Pressurization Fans	Key	D2
	Primary Plant Exhaust Fan Cooler	Key	D2
	Electrical Area Fan Cooler	Key	D2
	Diesel Generator Fuel Oil Day Tank Area Vent Fan	Key	D2
	Diesel Generator Area Vent Fan	Key	D2
	SSW Intake Structure Exhaust Fan	Key	D2
	Battery Room Exhaust Fan	Key	D2
	RHR Room Fan Cooler	Key	D2
	Motor Driven AFW Pump Room Fan Cooler	Key	D2
	CCW Pump Room Fan Cooler	Key	D2
	SI Pump Room Fan Cooler	Key	D2
	Containment Spray Pump Room Fan Cooler	Key	D2
	Centrifugal Charging Pump Room Fan Cooler	Key	D2
	UPS Ventilation	Key	D2

TABLE 7.5-5
TYPE D - VARIABLES

(Sheet 6 of 6)

System/Component	Variable	Variable Function	Type/ Category
Safety Chilled Water System	Flow	Key	D2
Electric Power	Vital bus(s) voltage	Key	D2
RHR	Heat Exchanger discharge temperature	Key	D2
	Flow	Key	D2
	RHR Pump Status	Key	D2
	Valve status	Key	D2
	RCS pressure (WR)	Key	D2
	Heat Exchanger Bypass Valve Status	Backup	D3
	RHR Heat Exchanger Inlet Temperature	Backup	D3
Radioactive Waste System	High-level Radioactive Liquid Tank Level	Backup	D3
	Radioactive Gas Holdup Tank pressure	Backup	D3

TABLE 7.5-6
TYPE E - VARIABLES

Function	Variables	Variable Function	Type/ Category
Estimating Potential Release	Containment Radiation level	Key	E1
Area Radiation	Control Room Radiation	Backup	E3
	Area Radiation in locations or Areas where Access may be required[a]	Backup	E3
Assess Radioactive Releases	Plant Vent Effluent Radioactivity and flow	Key	E2
	Main Steamline Radiation	Key	E2
	Steam Generator PORV status	Key	E2
	Steam Generator Safety Valve Status	Key	E2
	Concentration of Radioactive Material discharged from liquid pathways[b]	Key	E2
	Environs Radiation	Backup	E3
	Meterological Parameters (Wind speed, Wind direction, and Atmospheric stability)	Backup	E3
	Plant Vent Effluent Particulate and Halogens	Backup	E3

a) RHR Pump Room, Sampling Room, Plant Vent Stack Sample Area, Hot Lab Area

b) Liquid Waste Effluent, Turbine Building Drains

TABLE 7.5-7
IS DELETED

PERTINENT INFORMATION IS IN TABLE 7.5-7A THRU 7.5-7C

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 1 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display: CR Displays	Location of Display: TSC/EOF Location (25)
NEUTRON FLUX	B1 (B1) NE-50A	2 PER UNIT Figure 7.1-3 NE-50B	YES %FP(WR) $0.1\text{-}10^5$ cps (SR)	$10^{-8}\text{-}2\text{x}10^2$	EQ, SQ, QA	1E	ERFCS INDICATION	YES
CONTROL ROD POSITION	B3, D3 (B3)	1 PER CONTROL ROD (53 RODS)	N/A AT VESSEL	0 to 228 steps	NONE	NON 1E	LED DISPLAY 1 PER ROD, PCS	YES
RCS SOLUABLE BORON CONCENTRATION	NONE (B3)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
T(HOT) RCS (WR)	A1, B1, B2 (B1)	1 PER LOOP TE-413A TE-423A TE-433A TE-443A	YES FIGURE 7.1-3	0-700°F	EQ, SQ, QA	1E	ERFCS, RECORDER (ALL) INDICATION (LOOPS 1 & 2)	YES
T(COLD) RCS (WR)	A1, B1, B2 (B1, B3)	1 PER LOOP TE-413B TE-423B TE-433B TE-443B	YES FIGURE 7.1-3	0-700°F	EQ, SQ, QA	1E	ERFCS, RECORDER (ALL) INDICATION (LOOPS 1 & 2)	YES
RCS PRESSURE (WR)	A1, B1, B2, C1, D2, C2 (B1, C1)	3 PER UNIT PT-403 PT-3616 PT-437	YES FIGURE 7.1-3	0-3000 psig	EQ, SQ, QA	1E	ERFCS, RECORDER INDICATION	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 2 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
CORE EXIT TEMPERATURE	A1, B1, C1 (C1, B3)	25 PER TRAIN TE-001 to TE-050 (27)	YES FIGURE 7.1-3	0-2300°F	EQ, SQ, QA	1E	ERFCS INDICATION (1 PER TRAIN FOR HIGHEST TEMP.)	YES
REACTOR VESSEL WATER LEVEL	B1, C2	2 PROBES (28) PER UNIT TE-3613A TE-3613B	YES IN VESSEL FIGURE 7.1-3	UPPER CORE PLATE TO TOP OF REACTOR VESSEL	EQ, SQ, QA	1E	ERFCS, LED DISPLAY	YES
RCS SUBCOOLING	A1, B1 (B2)	2 PER UNIT TY-3611 TY-3612	YES FIGURE 7.1-3	-300 to 300°F	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CONTAINMENT WATER LEVEL (NOTE 21)	B1 B2, C2, D2 (NR-B2, C2; WR-B1, C1)	2 PER UNIT LT-4779 LT-4781	YES FIGURE 7.1-3	808'-3" to 817'-6"	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CONTAINMENT PRESSURE (IR)	A1, B1 C2, D2 (B1, C1)	4 per unit PT-934 TO 937	YES FIGURE 7.1-3	-5 to 60 psig	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CONTAINMENT ISOLATION VALVE STATUS	C2 (B1)	1 PER VALVE (NOTE 14)	N/A AT VALVES TABLE 6.2.4-1 AND 2	CLOSED/NOT CLOSED	EQ, SQ, QA	1E	ERFCS, INDICATION (NOTE 24)	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 3 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display: CR Displays	Location of Display: TSC/EOF Location (25)
RADIATION LEVEL IN PRIMARY COOLANT	NONE (C1)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ACCIDENT SAMPLING	NONE (C3 PRIMARY COOLANT) (E3 PRIMARY COOLANT AND SUMP) (E3 CONTAINMENT AIR)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
CONTAINMENT RAD LEVEL (HR)	B2, C2, E1 (E1, C3)	2 PER UNIT RE-6290A&B	YES FIGURE 7.1-3	1-10^{7}R/hr.	EQ, SQ, QA	1E	ERFCS, RAD MONITOR CONSOLE	YES
CONDENSER OFF-GAS RADIATION	B2, C2 (C3)	1 PER UNIT RE-2959	N/A FIGURE 7.1-3	10^{-5} TO 10^{-1} uCi/cc	EQ, QA (NOTE 2)	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
CONTAINMENT HYDROGEN CONCENTRATION	B3, C3 (C1)	4 PER UNIT AE-5506A THRU D	YES FIGURE 7.1-3	0-10%	EQ, SQ, QA	1E	ERFCS, INDICATION (1 PER TRAIN)	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 4 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
CONTAINMENT PRESSURE (WR)	C1, D2 (C1)	2 PER UNIT PT-938 PT-939	YES FIGURE 7.1-3	0-150 psig	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
PLANT VENT EFFLUENT RADIOACTIVITY AND FLOW	C2, E2 (C1, E2)	1 PER VENT STACK X-RE-5570 A & B	N/A FIGURE 7.1-3	10^{-6} - 10^{5} uCi/cc 0-140,000 cfm	EQ, QA (NOTE 3)	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
AREA RAD. LEVELS ADJACENT CONTAINMENT	C3 (C2)	11 PER UNIT RE-6259 A&B RE-6291 A&B RE-6292 THRU RE-6297	N/A FIGURE 7.1-3	10^{-1} - 10^{4} R/hr	NONE	NON 1E	ERFCS, RAD. MONITOR CONSOLE	YES
		RE-5637 3 (SHARED)		10^{-4}-10^{o} uCi/cc				
		XRE-6273 XRE-6275		10^{-1}-10^{4} mR/hr				
		XRE-5702		10^{-4}-10^{o} uCi/cc				
EFFLUENT RADIOACTIVITY AND FLOW NOBLE GASES-AREAS ADJACENT CONTAINMENT	C2, E2 (C2, E2)	1 PER VENT STACK X-RE-5570 A & B	N/A FIGURE 7.1-3	10^{-6} - 10^{5} uCi/cc 0-140,000 cfm	EQ, QA (NOTE 3)	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 5 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
RHR HEAT EXCHANGER DISCHARGE TEMPERATURE	D2 (D2)	1 PER TRAIN TE-604 TE-605	N/A FIGURE 7.1-3	50-400°F	EQ, QA (NOTE 20)	NON 1E	ERFCS, RECORDER	YES
SI ACCUMULATOR TANK LEVEL	D3 (D2)	2 PER TANK LT-950 TO 957	N/A FIGURE 7.1-3	0-100%	NONE (NOTES 5 AND 6)	NON 1E	ERFCS, INDICATION	YES
SI ACCUMULATOR TANK PRESSURE	D2 (D2)	2 PER TANK PT-960 TO 967	N/A FIGURE 7.1-3	0-700 psig	EQ, QA (NOTE 5 AND 20)	NON 1E	ERFCS, INDICATION	YES
SI ACCUMULATOR ISOLATION VALVE STATUS	D2 (D2)	1 PER VALVE 8808A, B, C, AND D	N/A AT VALVE	OPEN/NOT OPEN	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
BORIC ACID CHARGING FLOW (NOTE 10)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
CENTRIFUGAL CHARGING PUMP INJECTION FLOW (NOTE 10)	D2 (D2)	1 PER UNIT FT-917	N/A FIGURE 7.1-3	0-1000 gpm	EQ, QA (NOTES 5 AND 20)	NON 1E	ERFCS, INDICATION	YES
SAFETY INJECTION PUMP FLOW (HPI SYSTEM) (NOTE 10)	D2 (D2)	1 PER PUMP FT-918 FT-922	N/A FIGURE 7.1-3	0-800 gpm	EQ, QA (NOTES 5 AND 20)	NON 1E	ERFCS, INDICATION	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 6 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
RHR FLOW (LPI SYSTEM) (NOTE 10)	D2 (D2)	1 PER TRAIN FT-618 FT-619	N/A FIGURE 7.1-3	0-5500 gpm	EQ, QA (NOTE 5 AND 20)	NON 1E	ERFCS, INDICATION	YES
RWST LEVEL	A1, D2 (D2)	4 PER UNIT LT-930 TO 933	YES FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
REACTOR COOLANT PUMP STATUS (MOTOR CURRENT)	D3 (D3)	1 PER MOTOR 1PCPX1,2PCPX1 1PCPX2,2PCPX2 1PCPX3,2PCPX3 1PCPX4,2PCPX4	N/A AT SWITCHGEAR	0-800 amp	NONE	NON 1E	ERFCS, INDICATION	YES
PRESSURIZER PORV STATUS	D2 (D2)	1 PER VALVE PCV-455A PCV-456	N/A AT VALVE	CLOSED/NOT CLOSED OPEN/NOT OPEN	EQ, SQ, QA	1E	ERFCS, INDICATION (NOTE 15)	YES
PRESSURIZER SAFETY VALVE STATUS	D2 (D2)	1 PER VALVE 8010A 8010B 8010C	N/A AT VALVE	CLOSED/NOT CLOSED	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
PRESSURIZER LEVEL	A1, B1, D2 (D1)	3 PER UNIT LT-459 LT-460 LT-461	YES FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	ERFCS, RECORDER, INDICATION	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 7 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
PRESSURIZER HEATER CURRENT	D2 (D2)	1 PER HEATER BANK PCPR PCPR1 PCPR2 PCPR3	N/A AT SWITCHGEAR	0-1000A	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
PRESSURIZER RELIEF TANK LEVEL (QUENCH TANK)	D3 (D3)	1 PER UNIT LT-470	N/A FIGURE 7.1-3	0-100%	NONE	NON 1E	ERFCS, INDICATION	YES
PRESSURIZER RELIEF TANK TEMPERATURE (QUENCH TANK)	D3 (D3)	1 PER UNIT TE-468	N/A FIGURE 7.1-3	50-350°F	NONE	NON 1E	ERFCS, INDICATION	YES
PRESSURIZER RELIEF TANK PRESSURE (QUENCH TANK)	D3 (D3)	1 PER UNIT PT-469	N/A FIGURE 7.1-3	0-100 psig	NONE	NON 1E	ERFCS, INDICATION	YES
STEAM GENERATOR WATER LEVEL (WR)	B1, D2 (D1)	1 PER STEAM GENERATOR LT-501 LT-502 LT-503 LT-504	N/A FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	ERFCS, RECORDER, INDICATION	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 8 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
STEAM GENERATOR WATER LEVEL (NR)	A1, B1, D2 (NONE-PLANT SPECIFIC)	4 PER STEAM GENERATOR LT-517 TO 519 LT-527 TO 529 LT-537 TO 539 LT-547 TO 549 LT-551 TO 554	YES FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
MAIN STEAMLINE PRESSURE (S/G PRESSURE)	A1, B1, D2 (D2)	3 PER LOOP PT-514 TO 516 PT-524 TO 526 PT-534 TO 536 PT-544 TO 546	YES FIGURE 7.1-3	0-1300 psig	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
STEAM GENERATOR SAFETY VALVE STATUS	D2, E2 (D2, E2)	1 PER VALVE PV-2444A TO E PV-2445A TO E PV-2446A TO E PV-2447A TO E	N/A AT VALVE	CLOSED/NOT CLOSED	EQ, QA (NOTES 6 and 20)	NON 1E	ERFCS	YES
STEAM GENERATOR PORV STATUS	D2, E2 (D2, E2)	1 PER VALVE PV-2325 THRU PV-2328	N/A AT VALVE	CLOSED/NOT CLOSED OPEN/NOT OPEN	EQ, SQ, QA	1E	ERFCS, INDICATION (NOTE 15)	YES
MAIN FEEDWATER FLOW	D3 (D3)	1 PER STEAM GENERATOR FT-511 FT-521 FT-531 FT-541	N/A FIGURE 7.1-3	0 TO $5X10^6$ lb/hr	NONE	NON 1E	ERFCS, INDICATION, RECORDER	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 9 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
AUXILIARY FEEDWATER FLOW	A1, B1, D2 (D2)	2 PER STEAM GENERATOR FT-2463 A&B FT-2464 A&B FT-2465 A&B FT-2466 A&B	YES FIGURE 7.1-3	0-550 gpm	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CST WATER LEVEL	A1, D2 (D1)	2 PER UNIT LT-2478 LT-2479	YES FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CONTAINMENT SPRAY FLOW	D2 (D2)	1 PER PUMP FT-4772-1&2 FT-4773-1&2	N/A FIGURE 7.1-3	0-4000 gpm	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
HEAT REMOVAL BY CONTAINMENT FAN HEAT REMOVAL SYSTEM (NOTE 7)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
CONTAINMENT ATMOSPHERE TEMPERATURE	D2 (D2)	4 PER UNIT TE-5400 TO TE-5403	N/A FIGURE 7.1-3	0 TO 360°F	EQ, SQ, QA	1E	ERFCS, INDICATION FOR AVERAGE TEMPERATURE	YES
CONTAINMENT SUMP WATER TEMPERATURE (NOTE 8)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

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Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
CVCS CHARGING SYSTEM MAKEUP FLOW	D2 (D2)	1 PER UNIT FT-121	N/A FIGURE 7.1-3	0-200 gpm	EQ, QA (NOTE 5 AND 16)	NON 1E	ERFCS, INDICATION	YES
CVCS LETDOWN FLOW	D2 (D2)	1 PER UNIT FT-132	N/A FIGURE 7.1-3	0-200 gpm	EQ, QA (NOTE 5 AND 16)	NON 1E	ERFCS, INDICATION	YES
VOLUME CONTROL TANK LEVEL	D2 (D2)	1 PER UNIT LT-112	N/A FIGURE 7.1-3	0-100%	EQ, QA (NOTE 5 AND 16)	NON 1E	ERFCS, INDICATION	YES
CCW HEADER TEMPERATURE	D2 (D2)	1 PER HEADER TE-4530 TE-4534	N/A FIGURE 7.1-3	30-150°F	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
CCW FLOW	D2 (D2)	1 PER TRAIN FT-4536A FT-4537A	N/A FIGURE 7.1-3	0-20,000 gpm	EQ, SQ, QA	1E	ERFCS, INDICATION RECORDER	YES
SERVICE WATER HEADER FLOW	D2 (D2)	1 PER TRAIN FT-4258 FT-4259	N/A FIGURE 7.1-3	0-20,000 gpm	EQ, SQ, QA	1E	ERFCS, INDICATION, RECORDER	YES
HIGH LEVEL RADIOACTIVE LIQUID TANK LEVEL (NOTE 9)	D3 (D3)	1 PER TANK LT-1003 X-LT-1001	N/A FIGURE 7.1-3	0-100%	NONE	NON 1E	NO	NO

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 11 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display: CR Displays	Location of Display: TSC/EOF Location (25)
RADIOACTIVE GAS HOLDUP TANK PRESSURE (NOTE 9)	D3 (D3)	1 PER TANK X-PT-1036 TO 1039 X-PT-1052 TO 1057	N/A FIGURE 7.1-3	0-150 psig	NONE	NON 1E	NO	N0
CR VENT DAMPER POSITION (EMERGENCY VENTILATION DAMPER POSITION) (NOTE 17)	D2 (D2)	1 PER DAMPER X-HV-5826, 5829, 5825 A&B 5828 A&B 5857, 5858, 5883, X-PV-5855, X-PV-5856	N/A AT DAMPER	CLOSED/NOT CLOSED	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
		X-HV-5831 5834, 5837 5838, 5839 5840, 5847 5848, 5851 5853		OPEN/NOT OPEN				
VITAL BUS VOLTAGE AVAILABILITY	D2 (D2)	1 PER BUS 1EA1, 1EA2 1ED1, 1ED2 1ED3, 1ED4	N/A AT SWITCHGEAR	0-9000 VAC 100-150 VDC	EQ, SQ, QA	1E	ERFCS, INDICATION	YES
		1EB1, 1EB2 1EB3, 1EB4 2EA1, 2EA2		0-600 VAC			ERFCS	

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

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Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
		2ED1, 2ED2 2ED3, 2ED4 2EB1, 2EB2 2EB3, 2EB4 1 PER PANEL 1PC1, 1PC2 1PC3, 1PC4 1EC1, 1EC3 1EC2, 1EC4 1EC5, 1EC6 2PC1, 2PC2 2PC3, 2PC4 2EC1, 2EC3 2EC2, 2EC4 2EC5, 2EC6		0-150 VAC				
CR AREA RADIATION	E3 (E2)	2 PER PLANT XRE-6281 XRE-6282	N/A FIGURE 7.1-3	10^{-1} - 10^{4} mR/hr	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
RHR PUMP ROOM AREA RADIATION	E3 (E2)	1 PER ROOM RE-6260A RE-6260B	N/A FIGURE 7.1-3	10^{-1} - 10^{4} R/hr (NOTE 22)	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
PASS ROOM AREA RADIATION	E3 (E2)	1 PER ROOM RE-6261	N/A FIGURE 7.1-3	10^{-1} - 10^{4} mR/hr (NOTE 22)	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
PLANT VENT	E3	2 PER SITE	N/A	10^{-1} - 10^{4}	NONE	NON 1E	ERFCS,	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 13 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
STACK SAMPLE AREA RADIATION	(E2)	1-RE-6259 2-RE-6259	FIGURE 7.1-3	mR/hr (NOTE 22)			RAD MONITOR CONSOLE	
HOT LAB AREA RADIATION	E3 (E2)	1 PER SITE X-RE-6283	N/A FIGURE 7.1-3	10^{-1} - 10^{4} mR/hr (NOTE 22)	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
COMMON PLANT VENT INCLUDING CONTAINMENT PURGE, AUXILIARY BUILDING AND OTHER AREAS NOBLE GASES, VENT FLOW, PARTICULATES, AND HALOGENS	C2, E2 (C2, E2)	1 PER VENT STACK X-RE-5570 A&B	N/A FIGURE 7.1-3	10^{-6} - 10^{5} uCi/cc 0-140,000 cfm	EQ, QA (NOTE 3)	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
AIRBORNE RADIO HALOGENS AND PARTICULATES	E3 (E3)	GRAB SAMPLE	N/A	10^{-9} - 10^{-3} uCi/cc	NONE	NON 1E	N/A	N/A
REACTOR SHIELD BUILDING ANNULUS	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PLANT VENT EFFLUENT PARTICULATE AND HALOGEN	E3 (E3)	GRAB SAMPLE	10^{-3} - 10^{-2} uCi/cc	NONE	NON 1E	N/A	N/A	

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

(Sheet 14 of 15)

Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display CR Displays	Location of Display TSC/EOF Location (25)
MAIN STEAMLINE RADIATION	E2 (E2)	1 PER STEAMLINE RE-2325 RE-2326 RE-2327 RE-2328	N/A FIGURE 7.1-3	10^{-1} - 10^{3} uCi/cc	EQ, QA (NOTE 2)	NON 1E	ERFCS, RAD MONITOR CONSOLE	YES
RADIATION EXPOSURE METERS	N/A (NOTE 23)	N/A	N/A	N/A	N/A	N/A	N/A	N/A
ENVIRONS RADIATION	E3 (E3)	AS REQUIRED (PORTABLE)	N/A	10^{-3} - 10^{4} R/HR FOR PHOTONS 10^{-3} - 10^{4} RADS/HR FOR BETA AND LOWER ENERGY PHOTONS	NONE	SELF CONTAINED BATTERY	NO	NO
WIND DIRECTION	E3 (E3)	3 PER SITE X-ZE-4115 X-ZE-4116 X-ZE-4126	N/A METEOROLOGICAL TOWERS	0-540 DEGREES	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE RECORDER	YES
WIND SPEED	E3 (E3)	3 PER SITE X-SE-4117 X-SE-4118 X-SE-4128	N/A METEOROLOGICAL TOWERS	0-100 mph	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE, RECORDER	YES

TABLE 7.5-7A
INSTRUMENT SUMMARY DATA FOR VARIABLES IN TABLE 2 OF REG. GUIDE 1.97, REV 2

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Variable	Type/ Category CPNPP (RG 1.97)	Quantity Tag Numbers	Redundance and Sensor Location (11)	Instrument Range (1)	QA and Qualification (19)	Power Supply	Location of Display: CR Displays	Location of Display: TSC/EOF Location (25)
ATMOSPHERIC STABILITY	E3 (E3)	2 PER SITE X-TY-4119 X-TY-4120	N/A METEOROLOGICAL PRIMARY TOWER	-5 TO +15°F	NONE	NON 1E	ERFCS, RAD MONITOR CONSOLE, RECORDER	YES
POST ACCIDENT SAMPLING SYSTEM	NONE (C3 PRIMARY COOLANT) (E3 PRIMARY COOLANT AND SUMP) (E3 CONTAINMENT AIR)	N/A	N/A	N/A	N/A	N/A	N/A	N/A

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
STEAM GENERATOR BLOWDOWN RAD.	B2, C2	1 per unit RE-4200		N/A FIGURE 7.1-3	10^{-5}-5x10^{-2} uCi/cc	EQ, QA (NOTE 2)	NON 1E	RAD MONITOR CONSOLE
RHR HEAT EXCHANGER BYPASS VALVE	D3	1 PER VALVE FCV-618 FCV-619		N/A AT VALVE	CLOSED/ NOT CLOSED	NONE	NON 1E	INDICATION
RHR HEAT EXCHANGER INLET TEMPERATURE	D3	1 PER TRAIN TE-612 TE-613		N/A FIGURE 7.1-3	50-400°F	NONE	NON 1E	ERFCS RECORDER
RHR VALVE	D2	1 PER VALVE		N/A	OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION
STATUS		8701-A&B 8702-A&B FCV-610 FCV-611		AT VALVE	CLOSED/ NOT CLOSED			
BORIC ACID TANK LEVEL	D2	2 PER TANK (4 SHARED) X-LT-102 X-LT-104 X-LT-105 X-LT-106		N/A FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	INDICATION
ECCS VALVE STATUS	D2	1 PER VALVE LCV-112B&C 8105 8801A&B 8812A&B 8811A&B		N/A AT VALVE	*VARIOUS (NOTE 13)	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
		8809A&B 8808 A,B,C,D 8804A&B 8106 8716A&B 8807A&B 8924 8840 8802A&B 8821A&B 8835 8814A&B 8813						
	D2	8923A&B 8806 8220 8221 8110 8111 8511A&B 8512A&B 8202A&B 8210 A&B LCV-112D LCV-112E 8701A&B 8702A&B						
CCW TO RCP VALVE STATUS	D2	1 PER VALVE HV-4699, 4512 HV-4700, 4513 HV-4709,4515 HV-4708, 4524		N/A	OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
		HV-4701, 4525 HV-4696, 4526 HV-4514, 4527						
RCP SEAL WATER INJECTION FLOW	D3	1 PER RCP FT-142 FT-143 FT-144 FT-145		N/A FIGURE 7.1-3	0-20 gpm	NONE (NOTE 6)	NON 1E	INDICATION
S/G BLOWDOWN ISOLATION VALVE STATUS	D2	1 PER VALVE HV-2397 -2397A HV-2398 -2398A HV-2399 -2399A HV-2400 -2400A		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION
MSIV & BYPASS VALVE STATUS	D2	1 PER VALVE HV-2333 A&B HV-2334 A&B HV-2335 A&B HV-2336 A&B		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
AFW PUMP TURBINE MAIN STEAM HEADER ISOLATION VALVE STATUS	D2	1 PER VALVE HV-2452-1 HV-2452-2		N/A AT VALVE	CLOSED/ NOT CLOSED OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION
MAIN FEEDWATER CONTROL AND BYPASS VALVE STATUS	D2	1 PER VALVE FCV-510 520, 530 540, LCV-2162 LCV-2163 LCV-2164 LCV-2165		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, QA	1E	INDICATION
MAIN FEEDWATER ISOLATION AND BYPASS VALVE STATUS	D2	1 PER VALVE HV-2134 to 2137 2-FV-2193 to 2196 HV-2185 to 2188		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION
FEEDWATER SPLIT-FLOW BYPASS VALVES STATUS	D2	1 per valve 2-FV-2181 TO 2184		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION
TURBINE STOP VALVE STATUS	D3	1 PER VALVE UV-2413 TO 2416 UV-2428 TO 2431		N/A AT VALVE	CLOSED/ NOT CLOSED	NONE	NON 1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
AUXILIARY FEEDWATER VALVE STATUS	D2	1 PER VALVE PV-2453A,B PV-2454A,B HV-2459 to HV-2462 HV-2491A,B HV-2492A,B HV-2493A,B HV-2494A,B FV-2456 FV-2457		N/A AT VALVE	CLOSED/ NOT CLOSED OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION
CONTAINMENT SPRAY VALVE STATUS	D2	1 PER VALVE HV-4777, HV-4773-1&2 HV-4759 HV-4776 FV-4772-1&2 HV-4758 HV-4782 HV-4783		N/A AT VALVE	*VARIOUS (NOTE 13)	EQ, SQ, QA	1E	INDICATION
CCW HEADER PRESSURE	D2	1 PER HEADER PT-4520 PT-4521		N/A FIGURE 7.1-3	0-200 psig	EQ, SQ, QA	1E	INDICATION
CCW SURGE TANK LEVEL	D2	1 PER TRAIN LT-4500 LT-4501		N/A FIGURE 7.1-3	0-100%	EQ, SQ, QA	1E	INDICATION RECORDER
CVCS VALVE STATUS	D2	1 PER VALVE 8100, 8112 8160, 8152 8146, 8147		N/A AT VALVE	*VARIOUS (NOTE 13)	EQ, SQ, QA NOTE 26	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
		LCV-459, LCV-460, 8145, 8149A,B,C 8153, 8154 LCV-112B&C 8104, 8105 8106						
CCW VALVE STATUS	D2	1 PER VALVE HV-4513 HV-4512 HV-4572 HV-4573 HV-4574 HV-4575 FV-4537 FV-4536 HV-4514 HV-4515 HV-4524 THRU 4527		N/A AT VALVE	*VARIOUS (NOTE 13)	EQ, SQ, QA	1E	INDICATION
CR A/C UNITS	D2	4 (SHARED) CPX-VAACCR 01 THRU 04		N/A FIGURE 9.4-1	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CR EMERGENCY FILTRATION FANS	D2	2 (SHARED) CPX-VAFNCB-23 CPX-VAFNCB-24		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CR EMERGENCY PRESSURIZATION FAN	D2	2 (SHARED) CPX-VAFNCB-05 CPX-VAFNCB-06		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
PRIMARY PLANT	D2	4 (SHARED) CPX-VAFNCB -07, 08, 21, 22		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	EXHAUST FAN
ELECTRICAL AREA FAN COOLER	D2	4 PER UNIT CP1/2-VAAUSE- -15,16,17,18		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
DIESEL GENERATOR FUEL OIL DAY TANK AREA VENT FAN	D2	2 PER UNIT CP1/2-VAFNCB -04, 05		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
DIESEL GENERATOR AREA VENT FAN	D2	8 PER UNIT CP1/2-VAFNAV-25 26, 27, 28 29, 30, 31 32		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
SSW INTAKE STRUCTURE EXHAUST FAN	D2	8 (SHARED) CPX-VAFNWV -02, 03, 04, 05, 06, 07, 08, 09		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
BATTERY ROOM EXHAUST FAN	D2	4 PER UNIT CP1/2-VAFNID -07, 08 09, 10		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
RHR ROOM FAN COOLER	D2	2 PER UNIT CP1/2-VAAUSE -01, 02		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
MOTOR DRIVEN AFW PUMP ROOM FAN	D2	2 PER UNIT CP1/2-VAAUSE -07, 08		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CCW PUMP ROOM FAN COOLER	D2	2 PER UNIT CP1/2-VAAUSE -09, 10		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
SI PUMP ROOM FAN COOLER	D2	2 PER UNIT CP1/2-VAAUSE -05, 06		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CONTAINMENT SPRAY PUMP ROOM FAN COOLER	D2	4 PER UNIT CP1/2-VAAUSE -11, 12, 13, 14		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CENTRIFUGAL CHARGING PUMP ROOM FAN COOLER	D2	2 PER UNIT CP1/2-VAAUSE -03, 04		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
UPS VENTILATION	D2	4 (SHARED) CPX-VAACUP -01, 02, and CPX-VAFNAV 42, 43		N/A AT MCC	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
SAFETY CHILLED WATER FLOW	D2	1 PER TRAIN FT-6708 FT-6709		N/A FIGURE 7.1-3	0-400 gpm	EQ, QA (NOTE 4 and 6)	NON 1E	INDICATION
LIQUID WASTE EFFLUENT RAD.	E2	1 (SHARED) X-RE-5253		N/A FIGURE 7.1-3	10^{-5} - $5x10^{-2}$ uCi/cc	EQ, QA (NOTE 3)	NON 1E	RAD MONITOR CONSOLE

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
TURBINE BUILDING DRAINS RADIATION	E2	1 PER UNIT RE-5100		N/A FIGURE 7.1-3	10^{-5} - $5x10^{-2}$ uCi/cc	EQ, QA (NOTE 3)	NON 1E	RAD MONITOR CONSOLE
RCP SEAL INJECTION ISOLATION VALVE STATUS	D2	1 PER VALVE 8351A-D		N/A AT VALVE	OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION
PRESSURIZER PORV BLOCK VALVE STATUS	D2	1 PER VALVE 8000 A&B		N/A AT VALVE	CLOSED/ NOT CLOSED OPEN/NOT OPEN	EQ, SQ, QA	1E	INDICATION
STEAM GENERATOR SAMPLING ISOLATION VALVE STATUS	D2	1 PER VALVE HV-2405, 2406, 2407 2408, 2401 A&B 2402 A&B 2403 A&B 2404 A&B		N/A AT VALVE	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION
LIQUID WASTE EFFLUENT RAD	E2	1(SHARED) X-RE-5251A		N/A FIGURE 7.1-3	10^{-5}-5X10^{-2} uCi/cc	EQ, QA (NOTE 3)	NON IE	RAD MONITOR CONSOLE
CENTRIFUGAL CHARGING PUMPS (CCP) STATUS	D2	1 PER PUMP TBX/TCX-CSAPCH -01, 02		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
SAFETY INJECTION PUMP (SIP) STATUS	D2	1 PER PUMP TBX/TCX-SIAPSI -01, 02		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

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VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
RESIDUAL HEAT REMOVAL PUMP (RHR) STATUS	D2	1 PER PUMP TBX/TCX-RHAPRH -01, 02		N/A AT SWICHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
RHR PUMP SUCTION PRESSURE	02	1 PER PUMP	PT-601A PT-602A	N/A AT PUMP SUCTION	0-60 PSIG	EQ, QA NOTES 6 AND 20	N1E	ERFCS
CONTAINMENT SPRAY PUMP (CSP) STATUS	D2	1 PER PUMP CP1/2-CTAPCS -01, 02, 03, 04		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
CONTAINMENT SPRAY PUMP SUCTION PRESSURE	D2	1 PER TRAIN	PT-4806A PT-4807A	N/A AT PUMP SUCTION	0-60 PSIG	EQ, QA NOTES 6 AND 20	N1E	ERFCS
CONTAINMENT SPRAY ADDITIVE TANK LEVEL	D2	2 PER TANK LT-4752 LT-4753		N/A AT TANK	0-100%	EQ, SQ, QA	1E	INDICATION
CONTAINMENT SPRAY ADDITIVE TANK OUTLET VALVE	D2	1 PER VALVE LV-4752 THRU LV-4755		N/A AT VALVE OPEN/NOT OPEN	CLOSED/ NOT CLOSED	EQ, SQ, QA	1E	INDICATION
COMPONENT COOLING WATER PUMP STATUS	D2	1 PER PUMP CP1/2-CCAPCC -01, 02		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION
AUXILIARY FEEDWATER PUMP STATUS	D2	1 PER PUMP CP1/2-AFAPMD 01 & 02		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7B
INSTRUMENT SUMMARY DATA FOR ACCIDENT MONITORING VARIABLES FOR CPNPP NOT IN TABLE 2 OF REG. GUIDE 1.97, REV.2

(Sheet 11 of 11)

VARIABLE	CPNPP TYPE/ CATEGORY	QUANTITY TAG NUMBERS	TAG NUMBERS	REDUNDANCE AND SENSOR LOCATION (11)	INSTRUMENT RANGE (1)	QA AND QUALIFICATION(19)	POWER SUPPLY	LOCATION OF DISPLAY (18) CR DISPLAY
STATION SERVICE WATER PUMP STATUS	D2	1 PER PUMP CP1/2-SWAPSW -01, 02		N/A AT SWITCHGEAR	RUNNING/ NOT RUNNING	EQ, SQ, QA	1E	INDICATION

TABLE 7.5-7C
NOTES AND ABBREVIATIONS FOR TABLES 7.5-7A AND 7.5-7B

(Sheet 1 of 4)

1. Instrumentation ranges may vary from Regulatory Guide 1.97 Revision 2. However, CPNPP instrument ranges envelope the requirements of Reg. Guide 1.97 Rev. 2 unless otherwise specified in Table 7.5-7D.

2. Main Steamline Radiation, Condenser Off-Gas Radiation, and Steam Generator Blowdown Radiation are only required for steam generator tube rupture detection. These Non-Class 1E Monitors are in mild environments for the duration of this event and therefore do not require environmental qualification.

3. These radiation monitors are located in a mild environment for the duration of the events for which they are required.

4. These instruments are located in a mild environment for the duration of all design basis accident events. They are included in the Qualification Maintenance Program as described in Appendix 3A.

5. Instruments are seismically qualified for pressure boundary integrity.

6. Instruments are seismically mounted.

7. CPNPP does not take credit for this system during an accident.

8. Containment air temperature and RHR heat exchanger temperature provide equivalent information for Containment Sump Water Temperature.

9. These instruments provide indication at the local waste processing panels and annunciation from the panels is provided in the control room to alarm when there is a potential for over filling the tanks.

10. Boric acid charging flow at CPNPP is not an accident monitoring variable because it is not used for Safety Injection. Safety Injection Flow is provided by Centrifugal Charging Pump Flow, Safety Injection Flow, and RHR Flow.

11. Redundancy is listed as N/A for all Category 2 and 3 instrument channels since it is not required (although redundancy may, in fact, exist). A response of “Yes” means that separate safety- related trains are used and the redundant channels, including sensors, have the required separation and independence.

12. The QA and Qualification information applies to the Class 1E isolation valves only.

13. “*Various” for Range means that different positions are monitored for different valves including open/not open and/or closed/not closed and/or intermediate positions.

14. All valves listed in FSAR Table 6.2.4-2 except local manual, check, relief and safety valves.

TABLE 7.5-7C
NOTES AND ABBREVIATIONS FOR TABLES 7.5-7A AND 7.5-7B

(Sheet 2 of 4)

15. ERFCS displays the valve position requirements (e.g. Closed/Not Closed) of Reg. 1.97, Rev. 2 Table 2.

16. These parameters are D2 variables (Non 1E Devices) as they monitor other systems normally employed for safe shutdown but are not required during or following an accident (per section 7.5.1.1.1). When required to function, these instruments are in a mild environment and therefore do not require environmental qualification.

17. Only CR Ventilation dampers are listed on Table 7.5-7A because we understand that the intent of the variable “Emergency Ventilation Dampers” in Table 2 of Regulatory Guide 1.97, Rev. 2, was limited to the dampers required to realign the Control Room Ventilation System for its post-accident emergency ventilation modes. At CPNPP, we have included additional ESF ventilation parameters as accident monitoring variables (See Table 7.5-7B).

18. Table 7.5-7B does not identify which variables are displayed in the Control Room, on the ERF Console, in the TSC or in the EOF as these displays are not required by Reg. Guide 1.97, Rev. 2, Table 2.

19. EQ: Environmental Qualification, as described in Section 3.11B and Appendix 3A, is applicable to Category 1 and 2 instruments located in a harsh environment when they are required to perform their function. Category 1 and 2 instruments located in a mild environment when they are required to perform their function are also included in the Qualification Maintenance Program as described in Appendix 3A.
SQ: Seismic Qualification is applicable to Class 1E, seismic Category I instruments.
QA: Appendix B QA for Class 1E or Augmented Quality (FSAR Table 17A-2, Note C) for Non-1E.

20. QA is provided as necessary to support Environmental Qualification.

21. The containment water level covers the entire range of expected water level (from prior to the start of switchover from injection to recirculation to the maximum containment flood level) in the containment for post-accident conditions. The water level in the sump is assured by the RWST Low-Low level setpoint for the start of switchover. Operator action based on containment water level is not required. Therefore at CPNPP Containment Water Level (NR) is not considered as required for accident monitoring.

22. Since portable instrumentation is primarily used, the fixed radiation monitors perform as backup instrumentation. Therefore Category 3 is appropriate for this variable.

23. Radiation monitoring following an accident is covered by the CPNPP Emergency Plan; Radiation Exposure meters are not required at CPNPP.

24. Position indication is not required in the ERF computer for remote manual valves.

TABLE 7.5-7C
NOTES AND ABBREVIATIONS FOR TABLES 7.5-7A AND 7.5-7B

(Sheet 3 of 4)

25. See response to the NRC Action Plan developed as a result of the TMI-2 accident, Section III.A.1.2

26. These Parameters are D2 variables (1E Devices) as they monitor other systems normally employed for safe shutdown but are not required during or following an accident (Section 7.5.1.1.1). When required to function these instruments are in a mild environment. Environmental qualification is in accordance with Section 3.11B.

27. Technical Specifications require two core exit thermocouples operable per quadrant per train. In the event one or more thermocouple(s) is (are) found inoperable, and as long as the requirements of the Technical Specifications are met, leads of the inoperable thermocouple(s) may be temporarily lifted until the inoperable thermocouple(s) is(are) repaired or replaced.

28. Technical Specifications requires four or more reactor vessel water level sensors operable per channel, one or more in the upper section and three or more in the lower section. In the event, one or more sensor(s) is(are) found inoperable, and as long as the requirements of the Technical Specifications are met, leads of the inoperable sensor(s) may be temporarily lifted until the inoperable sensor(s) is(are) repaired or replaced.

Gen.	-	generator	CVCS	-	Chemical and Volume Control System
WR	-	wide range	N/A	-	not applicable
NR	-	narrow range	MSIV	-	main steam isolation valve
RCS	-	Reactor Coolant System	ECCS	-	Emergency Core Cooling System
Aux.	-	auxiliary	SI	-	Safety Injection
RWST	-	Refueling Water Storage Tank	CR	-	Control Room
CST	-	Condensate Storage Tank	A/C	-	air conditioning
Rad.	-	radiation	Vent	-	ventilation
RHR	-	Residual Heat Removal			
SR	-	source range	EQ	-	Environmentally Qualified As Stated in CPNPP FSAR 3.11

TABLE 7.5-7C
NOTES AND ABBREVIATIONS FOR TABLES 7.5-7A AND 7.5-7B

(Sheet 4 of 4)

CCW	-	Component Cooling Water	SQ	-	Seismically Qualified as stated in CPNPP FSAR 3.10
RCP	-	Reactor Coolant Pump			
Mon.	-	Monitor	AFW	-	Auxiliary feedwater
			QA	-	Quality Assurance Complies As Stated in CPNPP FSAR Chapter 17
ERFCS	-	ERF Console in Control Room			
TSC	-	ERF Console in the TSC	Rad Mon	-	Rad Mon Console in Control Room
EOF	-	ERF Console in the EOF			
Sh.	-	Sheet			

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 1 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
RCS Soluable Boron Concentration	Variable	Required	Not Required	(10)
T(HOT) RCS (WR)	Instrument Range	50°F to 750°F	0-700°F	(1)
T(COLD) RCS (WR)	Instrument Range	50°F to 750°F	0-700°F	(1)
Reactor Vessel Water Level	Instrument Range	Bottom of core to top of Vessel	Upper core plate to top of Reactor Vessel	(1)
Containment Water Level	Instrument Range	Narrow range (sump) Wide range (bottom of containment to 600,000 gal. level)	808'-3”-817'6"	(1)
	Additional Information	Required	Not available	(18)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 2 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
Containment Isolation Valve Status	Category	1	2	(3)
	Redundancy/Diversity	Required	Not available	(7)
Radiation Level in Circulating Primary Coolant	Category	1	Not Required	(4)
Containment Pressure (WR)	Instrument Range	10 psia to 3x design pressure	0-150 psig	(1)
	Additional information	Required	Not available	(22)
Area Radiation Levels Adjacent Containment	Instrument Range	10^{-1} R/hr to 10^{4} R/hr	10^{-4} to 10^{o} uCi/cc (RE-5637 & X-RE-5702)	(16)
			10^{-1} to 10^{4}mR/hr (XRE-6273 & 6275)	(17)
	Category	2	3	(19)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 3 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
RHR Heat Exchanger Discharge Temperature	Instrument Range	32°F to 350°F	50-400°F	(1)
SI Accumulator Tank Level	Category	2	3	(5)
	Instrument Range	10% to 90%	0-100%	(1)
	QA and Qualification	Required	Not available	(7)
SI Accumulator Tank Pressure	Instrument Range	0-750 psig	0-700 psig	(1)
Boric Acid Charging Flow	Variable	Required	Not designated as an accident monitoring variable	(9)
Pressurizer Level	Instrument Range	Bottom to top	0-100%	(1)
Pressurizer Relief Tank Temperature	Instrument Range	50°F to 750°F	50-350°F	(1)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 4 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
Main Steamline Pressure	Instrument Range	From atmospheric to 20% above the lowest safety valve setting	0-1300 psig	(1)
Heat Removal by Containment Fan Heat Removal System	Variable	Required	Not available	(11)
Containment Atmospheric Temperature	Instrument Range	40°F to 400°F	0-360°F	(1)
Containment Sump Water Temperature	Variable	Required	Not available	(12)
High-Level Radioactive Liquid Tank Level	Display	Control Room	Local,Waste processing panels, and CR alarm	(13)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 5 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
Radioactive Gas Holdup Tank Pressure	Instrument Range	0 to 150% design pressure	0-150 psig	(1)
	Display	Control Room	Local, Waste processing panels and CR alarm	(13)
Main Steamline Radiation	Vent Flow Rate	Required	Release calculated	(14)
Reactor Shield Building Annulus Effluent	Variable	Required (if in design)	Not available (noy in design)	Not in CPNPP design
Condenser Off-gas Radiation	Instrument Range	10^{-6} to 10^{-2} uCi/cc	10^{-5} to 10^{-1} uCi/cc	(1)
Radiation Exposure Meters	Variable	Required	Not available	(15)
Post Accident Sampling System	Variable	Required	Not Required	(10)
Atmospheric Stability	Instrument Range	-5°C to 10°C	-5 to +15°F	(1)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 6 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
Containment Radiation	Additional Information	Required	Not available	(20)
CST Water Level	Additional Information	Required	Not Available	(21)
CR Area Radiation	Range Category	10^{-1}R/hr to 10^{4}R/hr Category 2	10^{-1}mR/hr to 10^{4}mR/hr Category 3	(1) (6)
PASS Room Area Radiation	Range Category	10^{-1}R/hr to 10^{4}R/hr Category 2	10^{-1}mR/hr to 10^{4}mR/hr Category 3	(1) (6)
Plant Vent Stack Area Radiation	Range Category	10^{-1}R/hr to 10^{4}R/hr Category 2	10^{-1}mR/hr to 10^{4}mR/hr Category 3	(1) (6)
Hot Lab Area Radiation	Range Category	10^{-1}R/hr to 10^{4}R/hr Category 2	10^{-1}mR/hr to 10^{4}mR/hr Category 3	(1) (6)
RHR Pump Room Area Radiation	Category	Category 2	Category 3	(6)

TABLE 7.5-7D
SPECIFIC DEVIATIONS FROM THE GUIDANCE IN REG. GUIDE 1.97, REV. 2

(Sheet 7 of 7)

Variable	Item	R.G. 1.97 Rev. 2	CPNPP	Table 7.5-7E Reference (Justification)
High Level Radioactive Liquid Tank Level	Instrument Range	Top to Bottom	0-100%	(1)
Plant and Environs Radioactivity	Instrument Range	Multichannel gamma-ray spectrometer	Hot Lab NOSF	(23)
CCW Header Temperature	Range	32°F-200°F	30°F-150°F	(2)
Hydrogen Monitors	Category	Category 1	Category 3	(24)

TABLE 7.5-7E
REFERENCE FOR TABLE 7.5-7D

(Sheet 1 of 3)

Supporting justification of alternatives for deviations from the guidance in Reg. Guide 1.97, Rev. 2

(1) These instrument ranges deviate from the specific ranges listed in Table 2 of Reg. Guide 1.97, Rev. 2, but have been found acceptable for CPNPP's specific requirements based on the finding that the CPNPP specified ranges cover the anticipated ranges for normal operation, anticipated operational occurrences and accident conditions.

(2) The highest temperature in the component cooling water header is 136°F. CPNPP selected an upper range of 150°F since it provides a better scale reading on the control board indicator.

(3) Table 2 to Reg. Guide 1.97, Rev. 2, designates the variable, Containment Isolation Valve Position, as category 1 and as a type B variable under Maintaining Containment Integrity. In the CPNPP plant specific analysis (see the CPNPP FSAR, Section 7.5) the key type B variables for Containment Environment are Containment Pressure (NR), Containment Radiation Level, and Containment Water Level. Containment Isolation Valve Status was considered a type C backup variable for monitoring Containment Boundary Integrity. Therefore, Containment Isolation Valve Status is category 2 for CPNPP. Each penetration into the containment has two isolation boundaries.

(4) Table 2 to Reg. Guide 1.97, Rev. 2, designates the variable, Radiation Level in Primary Coolant, as category 1 and as a type C variable under Fuel Cladding. In the CPNPP plant specific analysis (see the CPNPP FSAR, Section 7.5) the type C key variable for monitoring In-Core Fuel Clad integrity is Core Exit Temperature with the primary backup variable Reactor Vessel Water Level Indicating System (RVLIS). Radiation Level in Primary Coolant is not required for CPNPP.

(5) Table 2 to Reg. Guide 1.97, Rev. 2, designates the variable, Accumulator Tank Level, as category 2 and as a type D variable under Safety Injection Systems. In the CPNPP plant specific analysis (see the CPNPP FSAR, Section 7.5) the type D key variables for monitoring ECCS are RWST Level, Safety Injection Pump Flow, RHR Pump Flow, Centrifugal Charging Pump Injection Flow, Containment Water Level, ECCS Valve Status, SI Accumulator Isolation Valve Status, and SI Accumulator Tank Pressure. The backup variable is SI Accumulator Tank Level. Therefore, SI Accumulator Tank Level is category 3 for CPNPP.

(6) Table 2 to Reg. Guide 1.97, Rev. 2, designates the variable, Radiation Exposure Rate, as category 2 and as a type E variable under Area Radiation. Since portable instrumentation is primarily use for this function the fixed radiation monitors perform as backup instrumentation. Therefore, category 3 is appropriate for this variable.

(7) Not required based on the category assigned by the CPNPP plant specific analysis (Refer to Section 7.5).

TABLE 7.5-7E
REFERENCE FOR TABLE 7.5-7D

(Sheet 2 of 3)

(8) NOT USED

(9) Table 2 to Reg. Guide 1.97, Rev. 2, lists the variable, Boric Acid Charging Flow, as a Type D variable under Safety Injection Systems along with the other variables such as Flow in HPI System and Flow in LPI System. The equipment for these three variables at CPNPP are Centrifugal Charging Pump Flow, Safety Injection Pump Flow and RHR Pump Flow. These three variables are included in the CPNPP Accident Monitoring design. For the CPNPP design, Boric Acid Charging Flow is not a measure of Safety Injection and therefore is not an Accident Monitoring variable.

(10) Post Accident Sampling is no longer required for CPNPP in accordance with License Amendment 91.

(11) The CPNPP design does not include a Fan Heat Removal System, inside containment, that is designed to operate following a design basis accident.

(12) The variable, Containment Sump Water Temperature, is not monitored at CPNPP and was not considered a required variable per the CPNPP plant specific accident monitoring analysis (see the CPNPP FSAR, Section 7.5). In addition, it should be noted that diverse information can be obtained by monitoring Containment Atmospheric Temperature, RHR Heat Exchanger Inlet Temperature and RHR Heat Exchanger Discharge Temperature.

(13) The only immediate concern of the Control Room operator following an accident with respect to the level in the High Level Radioactive Liquid Tank and the pressure in the Radioactive Gas Holdup Tank is inadvertently over-filling these tanks. This is prevented by providing an alarm in the Control Room when there is a potential for over-filling one of the tanks.

(14) Table 2 to Reg. Guide 1.97, Rev. 2, includes Vent Flow Rate as a variable to be measured for all identified release points. CPNPP complies with this requirement except for the Steam Generator Relief and Safety Valve discharge path. For this path, Main Steamline Radiation is measured and the volume released is calculated based on the duration for which the valves are open and the physical parameters involved. This is the best way to measure such an intermittent, high energy release.

(15) Radiation Exposure Meters are not required at CPNPP. Radiation monitoring following an accident is covered by the CPNPP Emergency Plan. Issuance of R.G. 1.97, Rev. 3 superseded the errata and deleted this requirement.

(16) An equivalent measure of containment breach is indicated in these areas by means of process (ventilation exhaust) monitors with a range of 10^{-4} to 10^{0} $_{u}$Ci/cc.

TABLE 7.5-7E
REFERENCE FOR TABLE 7.5-7D

(Sheet 3 of 3)

(17) The Spent Fuel Pool area monitors function as normal area monitors in the fuel building. The range of these monitors (10^{-1} to 10^{4}mR/hr) is suitable for monitoring areas adjacent to the containment building.

(18) Containment Water level transmitters were intended to be multipoint sensors which would prevent an ambiguous failure from occurring. The original indicators were never replaced, however, and the original analog indicators rely on an additive logic to compute the water level. This results in the potential for a failed indicator to cause a false reading of ± 1 foot. The containment water level indicators are for information only and will not adversely affect the ability to safety mitigate an accident.

(19) Table 2 to Reg. Guide 1.97 Rev. 2, designates the variable, Area Radiation Levels Adjacent Containment, as Category 2 and as a type C variable under Containment. In the CPNPP plant specific analysis (see the CPNPP FSAR, Section 7.5) the key type C variables for Containment Boundary is Containment Pressure (WR). Adjacent Building Radiation is considered a backup variable for monitoring the Containment Boundary. Therefore this variable is Category 3.

(20) Containment Radiation has redundant channels but additional information in the form of a diverse channel does not exist. Ambiguities can be resolved using portable instruments outside containment and prompt communications to the control room and this will not lead operators to defeat or fail to accomplish a required safety function.

(21) CST Water Level has redundant channels but additional information in the form of another identical channel or a diverse channel does not exist as a qualified display in the control room. Local, such as indication gauge level and pump suction pressure indications are available to resolve ambiguities. Sufficient time is available to resolve ambiguities before the operator must act on CST Water Level Information.

(22) Containment Pressure Wide Range has qualified redundant channels but additional information is not available to resolve ambiguities over the full range of the instruments. The additional information provided by the Containment Pressure Narrow Range is sufficient.

(23) CPNPP field monitoring teams are dispatched, as needed, to collect samples. These samples are transported to the Nuclear Operations Support Facility or the plant Hot Lab for analysis. At either location gamma spectroscophy equipment is used to analyze the samples. Instrumentation provided for this capability meets the intent of Regulatory Guide 1.97, Revision 2.

(24) The Hydrogen Monitors are capable of diagnosing beyond design-basis accidents. The Hydrogen Monitors were changed from Category 1 to Category 3 based on the revision to 10CFR50.44 on October 16, 2003 and License Amendment 117.

TABLE 7.5-7F
GENERAL DEVIATIONS FROM REGULATORY GUIDE 1.97, REV. 2

(Sheet 1 of 2)

Topic	Guidance	CPNPP Design and Justification
Environmental Qualification (EQ)	Regulatory Guide 1.89 and NUREG-0588 (Positions C1.3.1a & C1.3.2a)	EQ at CPNPP is discussed in the CPNPP FSAR, Sections 3.11 and Appendix 3A.
Quality Assurance (QA)	Specific list of regulatory guides(Positions C1.3.1e & C1.3.2d)	The CPNPP Quality Assurance program and the applicable regulatory guides are addressed in the CPNPP FSAR and in particular in Chapter 17.
Operability	1EEE Standard 279 for category 1 (or Tech Specs) (Position C1.3.1d)	CPNPP Operability of Accident Monitoring instrumentation shall be in compliance with the CPNPP Tech Specs.
EQ for category 2 variables	EQ is required for all category 2 variables. (Position C1.3.2a)	Category 2 instrumentation is environmentally qualified when it is subjected to adverse environments caused by the DBA during the time it must serve its intended function. This position is compatible with the EQ rule, 10 CFR 50.49.
Isolation devices	The transmission of signals from Category1 or 2 instrumentation for other than accident-monitoring or systems operation and effluent monitoring should be through isolation devices that are part of the accident monitoring channel and meet the provisions of this Reg. Guide. (Position C1.4a)	The CPNPP design meets this requirement when the power source for the channel is class 1E. The use of Class 1E isolation devices is only justified for Class 1E circuits.

TABLE 7.5-7F
GENERAL DEVIATIONS FROM REGULATORY GUIDE 1.97, REV. 2

(Sheet 2 of 2)

Topic	Guidance	CPNPP Design and Justification
Instrument identification	Type A, B & C category 1 & 2 should be specifically identified (Position C1.4b)	Category 1 and 2, Types A, B and C instrumentation is uniquely identified with the exception of the steam generator blowdown: Condenser-Offgas; and Plant Vent Stack radiation monitors and Containment Isolation Valve (CIV) Status Indication. The radiation monitors are part of a computer based system in which the information is displayed on demand on a LCD. The CIV status indication is provided by control switches and/or monitor light boxes located on the main control board.
Checking, testing, calibration & calibration verification	Comply with Reg. Guide 1.118 (Position C1.4h)	Compliance with Reg. Guide 1.118 is as discussed in Section 1A(B) of the CPNPP FSAR. The surveillance requirements for accident monitoring channels will comply with the CPNPP Tech. Specs. as implemented by CPNPP test procedures (see FSAR Section 13.5.2.2.5).

7.6 ALL OTHER INSTRUMENTATION SYSTEMS REQUIRED FOR SAFETY

7.6.1 INSTRUMENTATION AND CONTROL POWER SUPPLY SYSTEM

7.6.1.1 Description

The following is a description of the Instrumentation and Control Power Supply System (also referred to as the Class 1E 118V AC Uninterruptible Power System, see Section 7.1.1):

1. Refer to Figure 8.3-15 for a single line diagram of Class 1E, 118V AC Instrumentation and Control Power Supply System.

2. There are four Class 1E inverters per train, two for the reactor protection system and the other two for the balance of plant systems. Each inverter is connected independently to one Class 1E distribution panel. In addition to the above, there is one spare installed inverter per train. The spare inverter can be aligned to substitute any of the four inverters in that train. Procedural controls and interlocks ensure that the spare inverter can feed the loads of only one inverter at a time and the power source of the spare inverter is the same as of the substituted inverter. This is accomplished using the selector switches provided on the inverters.

3. The inverters provide a source of 118V, 60 hertz (Hz) grounded uninterruptible power for the operation of the Reactor Protection System instrumentation and other 118 VAC system vital loads. To assure continued operation of the 118V Class 1E AC System, the incoming power to the inverters is provided by the Class 1E DC switchboards. The Class 1E DC switchboards are normally powered from the 480V AC system via the battery chargers and in case of loss of battery chargers or station blackout, the station batteries feed the DC switchboards and the inverters. Each inverter is provided with a 120V AC bypass source. In case of UPS failure or fault condition the transfer of load from the inverter to the bypass source takes place automatically.

4. Each of the above distribution panels is also provided with a 120V AC backup source. Mechanically interlocked, transfer circuit breakers are provided in the distribution panels to prevent paralleling the inverters with the backup source.

5. For the power supply system for the balance of plant furnished nonsafety-related instrumentation and additional details of Class 1E UPS system, refer to Section 8.3.1.

7.6.1.2 Analysis

There are two Class 1E batteries and four battery chargers per train. One battery charger and a spare are provided for each battery. Each battery is attached to a bus serving one Reactor Protection System inverter and one BOP System inverter. Channel I & Channel III RPS inverters are powered by Train A and Channel II & Channel IV RPS inverters are powered by Train B. In addition, there is one spare installed inverter per train that can be manually aligned to feed the loads of any inverter in that train. Procedural controls and interlocks ensure that the spare inverter can feed the loads of only one inverter at a time and the power source of the spare inverter is the same as of the substituted inverter.

Since one BOP inverter and one RPS inverter is connected to one DC bus in each train, therefore the loss of a single DC bus can only affect one of the four RPS inverters and one of the four BOP system inverters. Each inverter is independently connected to its respective distribution panel, so the loss of an inverter cannot affect more than its associated distribution panel. In case of loss of an inverter the spare inverter can be manually aligned to feed the loads of the out of service inverter.

In addition, two independent sources (one independent source per train) of backup 120V AC power are provided to the inverter panels. Four of the eight distribution panels are connected to each source. Each distribution panel can receive power from the 120V AC backup source under operator control. The mechanically interlocked, transfer circuit breakers are provided in the distribution panels to prevent paralleling the inverter with the backup source.

Therefore no single failure in the Instrumentation and Control Power Supply System or its associated power supplies can cause a loss of power to more than one of the redundant channel loads and one of vital bus loads.

The inverters are designed to maintain their outputs within acceptable limits. The loss of DC input is alarmed in the control room, as is the loss of an inverter output. Inverter DC input and AC output breakers have local controls only. There are no inverter breaker controls on the control board as no manual action is required for the switch over to the DC battery source if there were loss of 480V AC power sources to the battery charger. Upon loss of battery chargers the inverters are automatically powered from the station batteries.

Physical separation and provisions to protect against fire are discussed in Chapter 8.

Based on the scope definitions presented in References [1] through [3], the criteria which are applicable to the Instrumentation and Control Power Supply System is the Institute of Electrical and Electronic Engineers (IEEE) Standard 308-1974. The design is in compliance with IEEE Standard 308-1974 and Regulatory Guide 1.32. Availability of this system is continuously indicated by the operational status of the systems it serves (see Figure 8.3-15) and is verified by periodic testing performed on the served systems. The inverters have been seismically qualified as discussed in Section 3.10B.

7.6.1.3 Control System Failures

The following evaluation consists of postulating failures which affect the major NSSS control systems and demonstrating that for each failure, the resulting event is within the bounds of existing accident analyses. The events which are considered are:

1. Loss of any single instrument

2. Break of any single instrument line

3. Loss of power to all system powered by a single power supply system (i.e. single inverter)

The analysis is conducted for all five major NSSS control systems:

1. Reactor control system

2. Steam dump system

3. Pressurizer pressure control system

4. Pressurizer level control system

5. Feedwater control system

The initial conditions for the analysis are assumed to be anywhere within the full operating power range of the plant (i.e. 0-100%) where applicable.

The results of the analysis indicate that, for any of the postulated events considered in a) through c) above, the condition II accident analyses given in Chapter 15 of the Comanche Peak FSAR are bounding.

7.6.1.3.1 Loss of Any Single Instrument

TABLE 7.6-1 Loss of Any Single Instrument, is a sensor-by-sensor evaluation of the effect on the control systems itemized above caused by a sensor failing either high or low. The particular sensor considered is given, along with the number of channels which exist, the failed channel, the control systems impacted by the sensor, the effects on the control systems for failures in both directions, and the bounding FSAR accident. Where no control action occurs or where control action is in a safe direction, no bounding accident is given.

The table clearly shows that for any single instrument failure, either high or low, the condition II events itemized in the FSAR Chapter 15 are bounding.

7.6.1.3.2 Loss of Power to an Inverter, Control Group, or Protection Set

The Comanche Peak NSSS instrument power supply consists of four instrument distribution panels 1PC1, 1PC2, 1PC3 and 1PC4 receiving power through four inverters (for convenience called inverters I through IV). Each instrument distribution panel powers a single control group and a single protection set (panel 1PC1, powered by inverter I, distributes power to control group I and protection set I; panel 1PC2, powered by inverter II, distributes power to control group 2 and protection set II, etc.). In addition, each control group cabinet receives a backup power from a non-safety instrument distribution panel. Therefore, a loss of a single inverter can cause only the loss of one protection set; the backup power supply, through auctioneering will continue to supply power to the control-grade cabinets. However, this backup power is not available for control grade functions requiring 118VAC power. Tables 7.6-2 through 7.6-5, Loss of Power to Protection Sets I through IV, respectively, analyze the effects on the control systems caused by the loss of power to a single inverter. Identical arrangement applies to Unit 2. In the tables, the control systems affected, the sensors affected, the failure direction, the effect on the control systems, and the bounding FSAR accident are given. Where no control action occurs or where control action is in a safe direction, no bounding accident is given.

All of the above described inverters, control groups, and protection sets are unique to the NSSS. In addition, there are four Class 1E and four non-Class 1E static uninterruptible power supplies for the Balance of Plant (BOP) control system and sensor power. In the absence of inverter output of these power supplies, the static switch automatically disconnects the inverter from the load and connects the load to an alternate power supply. This design ensures uninterruptible power supply for the BOP safety and non-safety-related control systems and sensors.

Besides the loss of power to a protection set, there is the chance of having an electrical fault on one of the control system circuit cards. The control systems are designed so that each card is used in only one control system. A circuit card failure cannot directly impact more than one control system. A failure on a control card would cause the controller to generate either an “off” or a “full on” output, depending on the type of failure. This result would be similar to having a fault in a sensor feeding the control system. Therefore, the failure of or loss of power in any control system circuit card would be bounded by the Loss of Any Single Instrument analysis described in Table 7.6-1.

The tables show that for a loss of power to any protection set, the Condition II events analyzed in the FSAR Chapter 15 are bounding.

7.6.1.3.3 Loss of Common Instrument Lines

Table 7.6-6, Loss of Common Instrument Lines, considers the scenario whereby an instrument line which supplies more than one signal ruptures, causing faulty sensor readings.

Two sets of sensors are located in common lines:

1. Loop steam flow (control groups 1 through 4 for steam generators 1 through 4, respectively) and narrow range steam generator level (protection set II, any steam generator)

2. Pressurizer level (protection sets, I, II, or III) and pressurizer pressure (protection sets I, II, III, or IV)

Not shown on the tables since they are not part of the control system but are used just for protection are the loop flow transmitters. There are three flow transmitters in each loop, with each transmitter having a common high pressure tap but separate and unique low pressure taps. Therefore, a break at the high pressure flow transmitter tap would result in disabling all three flow transmitters in one loop, resulting in a low flow reading for all three transmitters. This would result in a reactor trip if the plant is above the P-8 setpoint, or an annunciation if it is below P-8.

The only malfunctions mode explicitly analyzed was a break in the common instrument line at the tap. Another possibility is to have a complete blockage in the sensor tap, causing the sensor to read a constant (before blockage) value. However, this last failure mode is not analyzed since it is really not a credible event. There is no anticipated agent available that would cause a tap blockage. The Reactor Coolant System piping and fittings, and the instrument impulse line tubing are all stainless steel, so no products of corrosion are expected. Also, the water chemistry is of high quality, which along with high temperature operation, precludes the presence of solids in the water and assures the maintenance of the solubility of chemicals in the water. Therefore, the hypothesis of the presence of a complete blockage of the sensor tap is not sufficiently credible to warrant its consideration as a design basis.

In the extremely unlikely event that a complete instrument line blockage were to occur, the condition is detectable because the reading would become static (no variations over time). In an unblocked channel, a reading would always vary somewhat due to noise (i.e. flow induced noise in flow channels) or slight controller action (i.e. cycling operation of spray and heaters in pressurizer). By a comparison of the static channel to the redundant unblocked channels, the operator would be informed that a blockage in one channel has occurred.

7.6.2 RESIDUAL HEAT REMOVAL ISOLATION VALVES

7.6.2.1 Description

The Residual Heat Removal System (RHRS) isolation valves are normally closed and are only opened for residual heat removal after system pressure is reduced to approximately 425 pounds per square inch gage (psig) and system temperature has been reduced to approximately 350°F. They are the same type of valve and motor operator as those used for accumulator isolation, but they differ in their controls and in their indications in the following respect.

The residual heat removal valves are provided with red (open) and green (closed) position indicating lights located at the control switch for each valve. These lights are powered by valve control power and actuated by valve motor operator limit switches.

There are two motor operated valves in series in each of the two RHR pump suction lines from the Reactor Coolant System (RCS) hot legs. The two valves nearest the RCS (8702A, B) are designated as the inner isolation valves, while the two valves nearest the RHR pumps (8701A, B) are designated as the outer isolation valves. The interlock features provided for the outer isolation valves, shown on Figure 7.6-2, Sheet 1 are identical to those provided for the inner isolation valves, shown on Figure 7.6-2, Sheet 2.

Different types of pressure transmitters are used in the inner and outer isolation valve interlocks to provide the required diversity. One is a Veritrak Model 76 (PH2) while the other is a Rosemount Model 1154 SH.

Each valve is interlocked so that it cannot be opened unless the RCS pressure is below approximately 425 psig. This interlock prevents the valve from being opened when the RCS pressure plus the RHR pump pressure would be above the RHRS design pressure.

7.6.2.2 Analysis

Based on the scope definitions presented in References [2] and [3], these criteria do not apply to the residual heat removal isolation valve interlocks; however, in order to meet NRC requirements and because of the possible severity of the consequences of loss of function, the requirements of IEEE Standard 279-1971 will be applied with the following comments.

1. For the purpose of applying IEEE Standard 279-1971, to this circuit, Protection System is defined as the two valves in series in each line and all components of their interlocking circuits.

2. IEEE Standard 279-1971, Section 4.10

The above mentioned pressure interlock signals and logic will be tested online to the maximum extent possible without adversely affecting safety. This test will include the analog signal through to the train signal which activates the slave relay (the slave relay provides the final output signal to the valve control circuit). This is done in the best interests of safety since an actual actuation to permit opening the valve could potentially leave only one remaining valve to isolate the low pressure RHRS from the RCS.

3. IEEE Standard 279-1971, Section 4.15

This requirement does not apply, as the setpoints are independent of mode of operation and are not changed.

Environmental qualification of the valves and wiring are discussed in Section 3.11N.

7.6.3 REFUELING INTERLOCKS

7.6.3.1 Instrumentation Installed to Prevent Refueling Accidents

7.6.3.1.1 Initiating Circuits

The Fuel Handling System operates under manual local control. The refueling interlocks operation is completely automatic and stops any action by the operator that could be detrimental to the system.

7.6.3.1.2 Logic

Refer to Section 9.1.4.3.1 for a description of the refueling interlock logic.

7.6.3.1.3 Interlock Bypasses

1. Fuel Transfer System

There are three interlock bypasses in the Fuel Transfer System.

a. Transfer Car Permissive Switch Interlock Bypass - The refueling canal upender (lifting arm) is interlocked with the manipulator crane (refueling machine). The upender cannot be operated unless the manipulator crane gripper tube (mast) is in the fully retracted position or the crane is over the core. The bypass control for this interlock is located on the Fuel Transfer System reactor side control console. The upender on the Fuel Building side is similarly interlocked with the fuel handling bridge crane with bypass control located on the Fuel Transfer System Fuel Building side control console.

b. Transfer Car Lifting Arm Interlock Bypass - The transfer car is interlocked with both upenders such that both upenders must be in the down position before the transfer car can be moved. The bypass for this interlock is located on the Fuel Transfer System Reactor and Fuel Building side control consoles.

c. Lifting Arm Transfer Car Position Interlock Bypass - The conveyor interlock allows upender operation only when the transfer car is at either end of its travel. The bypass for this interlock is located on both control consoles.

2. Manipulator Crane (Refueling Machine)

There are six interlock bypasses on the manipulator crane. The control switches for these bypasses are located on the manipulator crane control console.

a. Overload Bypass - The hoist master switch up circuit is interlocked with a load cell contact which opens when the load on the hoist is greater than 250 pounds above the indicated suspended weight for wet conditions. This load cell contact can be bypassed by the overload bypass switch.

b. Hoist Interlock Bypass - The hoist-gripper position interlock consists of two separate circuits that work in tandem such that one circuit must be permissive for the hoist to operate. This criteria can be eliminated by the hoist interlock bypass switch.

c. Bridge Left Interlock Bypass and

d. Bridge Right Interlock Bypass - These interlock bypasses allow the bridge to operate beyond its travel limit switches.

e. Trolley Interlock Bypass - This interlock bypass allows the trolley to operate beyond it travel limit switches.

f. Gripper Up Disengaged Bypass - Providing a fuel assembly is not attached to the gripper, bridge and trolley movement is allowed without the gripper tube being in the maximum up position by bringing it up approximately 12 inches into the outer mast. The gripper up disengaged bypass allows the gripper tube to be raised to the full up position and allows bridge and trolley movement only if the gripper tube is in the maximum up position.

7.6.3.1.4 Interlocks, Redundancy

In the analysis below, the refueling interlocks have only been analyzed for failure in the permissive mode. The consequences of interlock failure in the interlocked mode are the same for all; the failed interlock would prevent normal operation of the equipment until repaired. There is no case where failure in the interlocked mode can result in a hazardous situation.

The manipulator crane electrical interlocks are as listed in Section 9.1.4.3.1

1. Travel Limit Switches on Bridge and Trolley

Failure of travel limit switches in the permissive mode could result in collision of the mast with the canal walls, vessel guide studs or the upper internals in their stored position. The fuel assembly would be protected from damage by the outer mast. The mast is a 16 inch diameter, 3/4 inch wall pipe that completely encloses the fuel. Within the mast, the fuel is

restrained by guide bars at all four corners that limit the lateral movement of the fuel to 1/4 inch maximum and supports it along its length.

2. Bridge, Trolley and Hoist Drive Mutual Interlocks

   These interlocks are redundant and satisfy the single failure criterion.

3. Bridge Trolley Drive - Gripper Tube Up

   This interlock is redundant and satisfies the single failure criterion.

4. Gripper Interlock

   This interlock is redundant and satisfies the single failure criterion.

5. Excessive Suspended Weight

   This interlock is redundant and satisfies the single failure criterion.

6. Hoist-Gripper Position Interlock

   The hoist-gripper position interlock consists of two separate circuits that work in tandem such that one circuit must be permissive for the hoist to operate. The interlock has a self monitoring circuit such that if one or both interlocking circuits fail in the permissive mode an audible and visual alarm on the console is actuated. The interlock, therefore, is not redundant but does satisfy the single failure criterion since both an interlocking circuit and the monitoring circuit must fail to cause a hazardous condition.

7. Bridge - Trolley Drive Interlock

   This interlock is part of the travel limit switch circuits described in (a).

8. Bridge and Trolley Hold-down devices are not properly defined as interlocks. They are utilized in the seismic restraint of the bridge and trolley.

The fuel transfer system interlocks are as listed in subsection 9.1.4.3.1

1. Transfer Car - Transverse Interlock

   This interlock satisfies the single failure criterion. The interlock is primarily designed to protect the equipment from overload and possible damage.

   Assuming the interlock fails in the permissive mode, there are two accidents to be considered. First, the lifting frame in the transfer canal is up or partially up and the car is driven towards the position. The car passes under the frame and is stopped by a mechanical stop at the upending position. The second case assumes that the fuel container is in the upright position in the Containment and the conveyor car permissive switch circuit fails in the permissive condition. The car is mechanically held in the frame and remains in the upright position. Movement is permitted only when the fuel container is in the full down position.

2. Lifting Arm - Transfer Car Position Interlock

   This interlock is redundant and satisfies the single failure criterion.

3. Transfer Car - Valve Open Interlock

   This interlock does not satisfy the single failure criteria. If the interlock fails in the permissive mode, the car can be started with the valve not fully open. The conveyor car contacts the valve gate stopping the conveyor car movement. As soon as the valve is opened fully, the conveyor car is freed.

4. Lifting Arm - Manipulator Crane Interlock

   This interlock is redundant and satisfies the single failure criteria.

5. Lifting Arm - Fuel Handling Bridge Crane (Fuel Handling Machine) Interlock

   This interlock satisfies the single failure criteria.

7.6.3.1.5 Actuated Devices

The only device actuated on an interlock failure is the gripper interlock failure alarm horn located on each of the crane control consoles. This alarm occurs when the gripper engaged and gripper disengaged limit switches are both closed or both open.

7.6.3.1.6 Design Bases Information

1. Design Bases

   The design bases for the portions of the Fuel Handling System important to safety is to provide the necessary interlocks, controls, and alarms to prevent fuel assembly damage during refueling operations. Refer to Section 9.1.4.3 for a description of the provisions made to ensure safe handling of the fuel assemblies.

2. Conformance to IEEE 279-1971

   These interlocks are not specifically designed to meet IEEE 279-1971 since the scope of IEEE-279, which is for the Station Protection System, does not apply to the Fuel Handling System.

   The Station Protection System, as explained in the IEEE-279 scope section, includes signals, such as those which actuate Reactor Trip and engineered safety features, which are all those that protect the plant during normal power operation. It should be noted that the Fuel Handling System is not operational during normal plant operation and is therefore not categorized as a Nuclear Power Generating Station Protection System.

7.6.3.2 Analysis

7.6.3.2.1 NRC General Design Criteria

Implementation of the requirements of NRC General Design Criteria is described in Section 3.1.

7.6.3.2.2 Single Failure Criterion

See Section 7.6.3.1.4 entitled “Interlocks, Redundancy” above.

7.6.3.2.3 Conformance to Branch Technical Positions ISCB 3, 4, and 20

Information related to conformance with Branch Technical Positions may be found by reference in Table 7.1-3.

7.6.3.3 Instrumentation Installed to Mitigate the Consequences of Refueling Accidents

7.6.3.3.1 Description

The instrumentation installed to mitigate the consequences of refueling accidents is included in the Area Radiation Monitoring System (ARMS). A detailed discussion of the ARMS is contained in Section 12.3.

7.6.3.3.2 Analysis

The analyses of effects and consequences of the postulated fuel handling accidents in Section 15.7.4 do not take any credit for the instrumentation installed to mitigate the consequences of these accidents. With no credit taken for this instrumentation, the calculated doses from these accidents are well below the limits defined in 10 CFR 100.

7.6.4 ACCUMULATOR MOTOR OPERATED VALVES

The design of the interconnecting of these signals to the accumulator isolation valve meets the criteria established in Nuclear Regulatory Commission (NRC) Branch Technical Position EICSB-4 on this matter. However, some provisions of the BTP are no longer applicable as noted below.

1. Automatic opening of the accumulator valves is not required. Technical Specifications require the isolation valves to be open with power removed above 1000 psig.

2. Visual indication in the control room of the open or closed status of the valve. (Refer to Section 6.3.5.5)

3. Bypassed and inoperable status indication in accordance to regulatory Guide 1.47 is not required because the valves are not allowed to be closed when required operable by technical specifications. (Refer to Section 7.1.2.6)

4. Utilization of a safety injection signal to automatically remove (override) any bypass features that are provided is not required because the Technical Specifications require the isolation valves to be open with power removed above 1000 psig.

The valves are not required to meet the above BTP requirements; however, the current control circuit for these valves (functional block diagram of valves 8808A, B, C, and D) is still as shown on Figure 7.6-3. The valves and control circuits are further discussed in Sections 6.3.2.2.11 and 6.3.5.

In order to comply with NRC Branch Technical Position EICSB-18, power is removed from the accumulator discharge valves when the valves are required open by Technical Specifications.

The four main control board position switches for these valves provide a “spring return to auto” from the open position and a “maintain position” from the closed position.

The “maintain closed” position is not used in MODES 1-3 above 1000 psig nor is it used for meeting fire protection requirements during shutdown.

During plant shutdown, the accumulator valves are in a closed position. To prevent an inadvertent opening of these valves due to a fire during that period the accumulator valve breakers are opened or removed. Administrative control is again required to ensure that these valve breakers are closed during the pre-startup procedures.

These normally open motor operated valves have alarms, indicating a malpositioning (with regard to their Emergency Core Cooling System function during the injection phase). The alarms sound in the control room.

An alarm will sound for either accumulator isolation valve under the following conditions when the RCS pressure is above the “SI unblocking pressure.”:

1. Valve motor operator limit switch indicates valve not open.

2. Valve stem limit switch indicates valve not open. The alarms on this switch will report itself at given intervals.

7.6.5 SWITCHOVER FROM INJECTION TO RECIRCULATION

The details of achieving cold leg recirculation following safety injection are given in Section 6.3.2.8 and on Table 6.3-7. Figure 7.6-4 shows the logic which will be used to open automatically the sump valves.

The logic functions derived from the refueling water storage tank level sensors are depicted in Figure 7.6-4, Sheet 1.

As noted in Section 6.3.2.8, protection logic is provided to open automatically the two Safety Injection System (SIS) recirculation sump isolation valves (8811A and 8811B) when two of four (2/4) Refueling Water Storage Tank (RWST) level signals are less than the Lo-Lo level setpoint in conjunction with the initiation of the engineered safety features actuation signal (“S” signal). The 2/4 Lo-Lo RWST level is the trip signal shown on Fig. 7.6-4, Sheet 1.

This automatic action would align the two residual heat removal pumps to take suction from the containment sump and to deliver directly to the RCS. The automatic action, which bypasses the blocks preventing operator opening valves during normal operation, is shown on Figure 7.6-4, Sheet 2, as a coincidence logic function of the Trip signal, the main control board switch in the

auto position (which is designed to be spring return to this position), and the presence of the "S" signal. The design provides for the retention of the "S" signal, should one be generated, that allows for this automatic changeover. The "S" signal is retained by sealing it in (i.e., it is latched). The retention of this signal is necessary since emergency procedures are written to permit the operator (by means of the manual (S.I.) reset and block signal, shown on Fig. 7.2-1, Sheet 8) to reset the "S" signal at a time significantly in advance of the RWST Lo-Lo water level signal generation that initiates the changeover. However, if this has not occurred prior to Lo-Lo, it is performed as part of ECCS transfer. A manual reset pushbutton is provided on the main control board to permit the operator to remove the actuation signal in the event the corresponding sump isolation valve must be closed and retained in a closed position following a LOCA, such as, for maintenance purposes.

The Trip Signal logic consists of four RWST water level transmitters, each of which provides a level signal to one of the four RWST level channel bistables. The RWST level channel bistables are:

1. normally de-energized

2. de-energized on loss of power

3. energized on Lo-Lo setpoint

Each level channel bistable is assigned to a separate instrumentation and control power supply. A trip signal is provided from both Train A and Train B Solid State Protection System (SSPS) cabinets to the corresponding sump isolation valve's logic, should two of the four water level channel bistables receive a RWST level signal lower than the Lo-Lo level setpoint following the generation of an S signal.

On-line test provisions are designed such that the test operator cannot open the sump isolation valves unless the corresponding RWST/RHR suction isolation valves are closed and either the corresponding RHR inner or outer isolation valves are closed. The test button (TB) signal is, therefore, blocked from simulating the 2/4 RWST low water level signal until the above is accomplished.

The recirculation sump isolation valves are also interlocked such that they must be closed before the following valves can be opened:

1. RWST/RHR pump suction isolation valves, 8812A and 8812B.

2. RHR inner or outer isolation valves 8701A, 8701B, 8702A, and 8702B.

A lo-lo, two-out-of-four, RWST level signal is provided for automatic change-over from injection mode to recirculation mode, as described in Table 6.3-7. RWST depletion due to injection and changeover to recirculation mode is discussed in Section 6.3.

An RWST empty level alarm is provided to the operator to communicate the need to initiate pump protection by stopping any ECCS pumps still taking suction. The RWST "empty" level alarm has a ±2.3 percent uncertainty band. This is the maximum loop uncertainty including process measurement accuracy.

The operator has the following status indication information available in the control room to guide manual switch over of the CSS from the RWST to the containment sumps:

RWST lo-lo level alarm (any of 4-channels)

RWST empty alarm (any of 4-channels)

RWST level indicator (See Section 7.5)

See Sections 6.2.2 and 6.5 for the design bases, system design, design evaluation, tests and inspection, and instrumentation requirements of the CSS.

7.6.6 PROCESS AND EFFLUENT RADIOLOGICAL MONITORS

Seismic design, redundancy and emergency power for process and effluent radiological monitors are discussed in Section 11.5.

7.6.7 REACTOR COOLANT PRESSURE BOUNDARY LEAKAGE DETECTION SYSTEMS

7.6.7.1 Description

The leakage detection systems will be intended to sense radioactive and nonradioactive leakage from the reactor coolant and auxiliary loops.

Means will be available to locate the leakage, and if necessary, corrective actions will be taken to ensure that total leakage is below levels consistent with safe operation of the plant.

7.6.7.1.1 Leakage Detection Methods

The methods for detecting leakage to the containment are by containment air particulate monitors, radioactive gas monitors, specific humidity monitors and sump level control and flow monitors. Refer to the description of these methods in Section 5.2.5.

7.6.7.1.2 Instrumentation and Controls

1. Containment air particulate monitors

   Indication, recording and high radiation alarm are provided in the control room.

2. Radioactive gas monitors

   Indication, recording and high radiation alarm are provided in the control room.

3. Specific humidity monitors

   High level and high rate of level increase alarms are provided for the condensate measuring systems.

   Indications and plant process computer inputs for the dew point temperature monitoring system are provided in the control room.

4. Sump level control and flow monitors

   The containment sump pumps are automatically controlled by mechanical alternators on a pre-determined increasing level. One sump pump will start and continue to run until a pre-determined decreasing low level. The second sump pump will start if, with one pump operation, the sump level continues to increase. These conditions were alarmed in the control room. Pumps status indication is provided in the control room. Pump discharge flow is recorded in the control room.

7.6.7.2 Analysis

The Reactor Coolant Pressure Boundary Leakage Detection System is formed by four diverse systems. The containment air particulate monitor is qualified to function through the Safe Shutdown Earthquake.

The system is in accordance with 10CFR50, General Design Criterion 30, Regulatory Guide 1.45 as discussed in Section 1A(B) and IEEE Standard 279-1971, Section 4.l0.

7.6.8 INTERLOCKS FOR RCS PRESSURE CONTROL DURING LOW TEMPERATURE OPERATION

The basic function of the RCS pressure control during low temperature operation is discussed in Section 5.2.2. As noted in Section 5.2.2, this pressure control includes automatic actuation logic for two pressurizer power operated relief valves (PORV's). The function of this actuation logic is to continuously monitor RCS temperature and pressure conditions, with the actuation logic only unblocked when plant operation is at a temperature below the Reference Nil Ductility Temperature (RNDT). The monitored system temperature signals are processed to generate the reference pressure limit program which is compared to the actual monitored RCS pressure. This comparison will provide an actuation signal to an actuation device which will cause the PORV to automatically open if necessary to prevent pressure conditions from exceeding allowable limits. See Figure 7.6-5 for the block diagram showing the interlocks for RCS pressure control during low temperature operation. As shown on this figure, the generating station variables required for this interlock are channelized as follows:

1. Protection Set I

   a. Wide Range RCS Temperature from Hot Legs 1 & 2

   b. Wide Range RCS Temperature from Cold Legs 3 & 4

   c. Wide Range RCS System Pressure (PT 405)

2. Protection Set II

   a. Wide Range RCS Temperature from Cold Legs 1 & 2

   b. Wide Range RCS Temperature from Hot Legs 3 & 4

3. Protection Set IV

a. Wide Range RCS System Pressure (PT 403)

The wide range temperature signals, as inputs to the Protection Sets I and II, continuously monitor RCS temperature conditions whenever plant operation is at a temperature below the RNDT. In Protection Set I, the existing RCS cold and hot leg wide range temperature channels will supply continuous analog input through an isolator to two auctioneering devices, which are located in the Control Group No 1.

The lowest reading as selected by one auctioneer is input to a function generator which calculates the reference pressure limit program considering the plant's allowable pressure and temperature limits. Also available from Protection Set I is the wide range RCS pressure signal which is sent through an isolation device to Control Group 1. The reference pressure from the function generator is compared to the actual RCS pressure monitored by the wide range pressure channel. The error signal derived from the difference between the reference pressure and the actual measured pressure will first annunciate a main board alarm whenever the actual measured pressure approaches, within a predetermined amount, the reference pressure. On a further increase in measured pressure, the error signal will generate an annunciated actuation signal. The actuation signal available from Auxiliary Relay Rack "A" will control PORV "A" whenever a temperature-dependent permissive signal from Process Control Group 4 is present. The two auctioneer devices mentioned above select the lowest temperature. One low temperature is used as a permissive; the other, for use in the reference pressure limit program. The temperature-dependent permissive to the PORV's actuation device effectively disarms (blocks) the actuation signal at temperatures greater than the range of concern. This will prevent unnecessary system actuation when at normal RCS operating conditions as a result of a failure in the process sensors.

The monitored generating station variables that generate the actuation signal for the "B" PORV are processed in similar manner. In the case of PORV "B", the reference temperature is generated in Process Control Group 4 from the lowest auctioneered wide range cold or hot leg temperature; the auctioneering device deriving its input from the RCS wide range temperature in Protection Set II and the actual measured pressure signal is available from Protection Set IV. Therefore, the generating station variables used for PORV "B" are derived from a Protection Set that is independent of the sets from which generating station variables used for PORV A are derived. The error signal derivation itself used for the actuation signals is available from the control group.

Upon receipt of the actuation signal, the actuation device will automatically cause the PORV to open. Upon sufficient RCS inventory letdown, the operating RCS pressure will decrease, clearing the actuation signal. Removal of this signal causes the PORV to close.

7.6.8.1 Analysis of Interlock

Many criteria presented in IEEE-279-1971 and IEEE-338-1971 standards do not apply to the interlocks for RCS pressure control during low temperature operation, because the interlocks do not perform a protective function but rather provide automatic pressure control at low temperatures as a backup to the operator. However, although IEEE-279 criteria do not apply, some advantages of the dependability and benefits of an IEEE-279 design have accrued by including selected elements as noted above in the protection sets and by organizing the control

of two (either of which can accomplish the RCS pressure function) into dual channels wherever practical. Either of the two PORV's can accomplish the RCS pressure control function.

The design of the low temperature interlocks for RCS pressure control is such that pertinent features include:

1. No credible failure at the output of the protection set racks, after the output leaves the racks to interface with the interlocks, will prevent the associated protection system channel from performing its protective function because such outputs that leave the racks go through an isolation device as shown in Figure 7.6-5.

2. Testing capability for elements of the interlocks within (not external to) the Protection System is consistent with the testing principles and methods discussed in Section 7.2.2.2.3.10. It should be noted that there is an annunciator which provides an alarm when there is low auctioneered RCS temperature (below RNDT) coincident with a closed position of the motor operated (MOV) pressurizer relief isolation valve. This MOV is in the same fluid path as the PORV, with a separate MOV used and alarmed associated with the second PORV.

3. A loss of offsite power will not defeat the provisions for an electrical power source for the interlocks because these provisions are through onsite power which is described in Section 8.3.

7.6.9 MONITORING COMBUSTIBLE GAS IN THE CONTAINMENT

The primary means of measuring the concentration of combustible gases in the Containment following a severe accident is by the Containment Hydrogen Monitoring.

Section 6.2.5 provides a description of the Containment Combustible Gas Control .

7.6.10 FIRE DETECTION SYSTEM

The fire detection system along with the fire-extinguishing system are part of the Fire Protection System discussed in Section 9.5.1.

7.6.11 INSTRUMENTATION FOR MITIGATING CONSEQUENCES OF INADVERTENT BORON DILUTION

7.6.11.1 Description

Instrumentation is provided to mitigate the consequences of inadvertent addition of unborated, primary grade water into the Reactor Coolant System.

Figure 7.6-6 is a simplified system block diagram showing the flux doubling detection system (2 ϕ meter) and the protection system output for isolation valve actuation.

In the event of a boron dilution transient, the Nuclear Instrumentation Source Range in conjunction with the 2 ϕ meter will detect a doubling of the neutron flux. This information is sent to the Solid State Protection System. An alarm is sounded at the time for plant operators to indicate that flux doubling has occurred.

7.6.11.2 Analysis

The analysis of effects and consequences of inadvertent boron dilution transients is covered in Section 15.4.6.

7.6.11.3 Qualification

Qualification of the instrumentation is discussed in Sections 3.10 and 3.11.

7.6.12 MITIGATION OF ENVIRONMENTAL EFFECTS OF PIPE BREAKS OUTSIDE CONTAINMENT

An environmental analysis of the compartments outside containment as a result of pipe breaks was performed for CPNPP Units 1 & 2, as described in Section 3.6. The results showed that with detection and mitigation added for three systems, Auxiliary Steam (SA), Steam Generator Blowdown (SGB) and Chemical Volume and Control (CVCS), minimal environmental effects would be imposed on the safety-related and non-safety related equipment and structures required for safe shutdown.

From the analysis, the environmental conditions of the equipment required for shutdown was identified. Alternate shutdown paths were evaluated when essential equipment was not qualified for the harsh environmental conditions. Where alternate shutdown paths could not be found, mitigation of the pipe break as described below was used to insure an adequate environment for equipment required for safe shutdown.

7.6.12.1 Auxiliary Steam System Line Break Mitigation

Two (2) redundant Auxiliary Steam pressure switches are provided for each of two (2) Auxiliary Steam header lines where the break is postulated to occur. These pressure switches (X-PS-3226A, 3226B, 3227A, 3227B) are located remote from the environment resulting from the postulated Auxiliary Steam line break. These switches, shown on Figure 10.4-16 (Sh.3), are Non-Class 1E and Non-Seismic Category I. See Section 10.4.13 for a description of the Auxiliary Steam System. The isolation valves, shown on Figure 10.4-16 (Sh.2), are also Non-Class 1E and Non-Seismic Category I.

Instrument air (non-Class 1E, non-seismic category I) and offsite power are essential for auxiliary steam system line break mitigation.

Tabulated below are the major sources of steam to the Auxiliary Steam System and their isolation valves.

		Isolation Valves
	Steam Source (Located Upstream of Pipe Breaks)	
1.	Main Steam (Unit 1 or 2) or Aux. Boiler	2 in Series
2.	Extraction Steam Unit 1	2 in Series
3.	Extraction Steam Unit 2	2 in Series

Rapid depressurization occurs as a result of a line break at any point in the two (2) Auxiliary Steam headers. The mitigation scheme is such that any one of the four (4) pressure switches is designed to automatically close all of the Auxiliary Steam source isolation valves, except the Turbine Gland Steam Header, thereby preventing a turbine trip. In addition, the Auxiliary Steam System alarm and the High Energy Line Break (HELB) common trouble alarm are activated at the main control board.

A two (2) position switch ("Engage-Disengage") is provided at the Auxiliary Steam panel section of the main control board. This switch enables the operator to engage or disengage the mitigation circuitry, thus bypassing the isolation signal when the Auxiliary Steam System is being started up and auxiliary steam pressure is low. Placing the switch in the "Disengage" position activates the system alarm and also the HELB common trouble alarm, both located at the main control board. The alarm signal in this system is transmitted to both the Unit 1 and Unit 2 control rooms. The switch is normally placed in the "Engage" position.

7.6.12.2 Steam Generator Blowdown (SGB) System Line Break

One (1) flow indicating switch (FIS) is provided for each of the four (4) SGB lines. Each SGB line is provided with two isolation valves in series, between the SG and the postulated break. Extremely high flow measurement on three (3) FIS occurs as a result of a line break downstream or upstream of the FIS in any of the four (4) SGB lines. Any one of the four (4) FIS is designed to automatically close all of the SGB isolation valves. In addition, the SGB system alarm and the HELB common trouble alarm are activated at the main control board. These switches (FIS-5175, 5176, 5177 and 5178), shown on Figure 10.4-10 (Sh.1), are Non-Class 1E and Non-Seismic Category I.

See Section 10.4.8 for a description of the Steam Generator Blowdown System. The isolation valves, shown on Figure 10.3-1(Sh.1) are Nuclear Safety Related, Seismic Category I.

7.6.12.3 Chemical Volume and Control System (CVCS) Line Break

For a CVCS line break, mitigation is accomplished by detection and 10 minute operator action to isolate the system.

Rapid depressurization occurs as a result of a CVCS line break. Two pressure switches have been provided for the CVCS system to detect low system pressure. Each low pressure signal activates an alarm at the main control board and the operator manually isolates the system. The pressure switches are located remote from the environment resulting from a CVCS line break.

REFERENCES

1. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Criteria for Class 1E Electric Systems for Nuclear Power Generating Stations," IEEE Standard 308-1974.

2. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Standard: Criteria for Protection Systems for Nuclear Power Generating Stations," IEEE Standard 279-1971.

3. The Institute of Electrical and Electronic Engineers, Inc., "IEEE Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection System," IEEE Standard 338-1971.

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 1 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Feedpump Discharge Pressure	1 per plant	_____	Feedwater Control	Lo	FW pump speed increases if in auto mode. (FW control valves close due to increased flow if in auto mode.)	If FW pump in manual - no event. If FW pump and FCV in auto – new steady state w/higher pump speed and decr. FCV lift. If FW pump in auto and FCV is manual - bounding event is Excessive FW Flow (FSAR 15.1.2)
				Hi	FW pump speed decreases if in auto mode. (FW control valves open due to decreased flow if in auto mode).	If FW pump in manual - no event. Other modes - result in a decreased FW flow over time, hence bounding event is loss of Normal FW Flow (FSAR 15.2.7).
Steam Header Pressure	1 per plant	_____	Feedwater Control Steam Dump (TAVG Mode)	Lo	FW pump speed decreases if in auto mode. (FW control valves open due to decreased flow if in auto mode).	If FW pump in manual - no event. Other modes - result in a decreased FW flow over time, hence bounding event is loss of Normal FW Flow (FSAR 15.2.7).

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 2 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	FW pump speed increases if in auto mode. (FW control valves close due to decreased flow if in auto mode).	If FW pump in manual - no event. If FW pump and FCV in auto – new steady state w/higher pump speed and decr. FCV lift. If FW pump in auto and FCV in manual - bounding event is Excessive FW Flow (FSAR 15.1.2).
Steam Header Pressure	1 per plant	_____	Feedwater Control Steam Dump (Pressure Mode)	Lo	FW pump speed decreases if in auto mode. (FW control valves open due to decreased flow if in auto mode).	If FW pump in manual - no event. Other modes - result in a decreased FW flow over time, hence bounding event is loss of Normal FW Flow (FSAR 15.2.7).

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 3 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	FW pump speed increases if in auto mode. (FW control valve close due to decreased flow if in auto mode).Dump valves open (Steam dump blocked on Lo-Lo TAVG (P-12).	Steam dump in pressure mode at hot standby or very low power only. Hence, dump valves will open for only a very short time till lo-lo TAVG (P-12) is reached. If FW pump speed is in manual or FW pump and FCV in auto, then this event is bounded by Excessive Increase in Secondary Steam Flow (FSAR 15.1.3). If FW pump in auto and FCV in manual, get increase in FW flow causing excessive cooling. Bounding event is Excessive FW Flow (FSAR 15.1.2).
Loop Steam Flow	2 per loop	1 selected for control	Feedwater Control	Lo	FW pump speed decreases if in auto mode. FW valves close if in auto mode.	If FW pump and FCV in manual – no event. Other modes result in decreased FW flow, bounding event is Loss of Normal FW Flow (FSAR 15.2.7).

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 4 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	FW pump speed increases if in auto mode. FW valves open if in auto mode.	If FW pump and FCV in manual – no event. Other modes - result in Increased FW flow, bounding event is Excessive FW Flow (FSAR 15.1.2).
Loop FW Flow	2 per loop	1 selected for Control	Feedwater Control	Lo	FW valve opens if in auto mode	If FCV in manual - no event. If FCV in auto, results is Excessive FW Flow (FSAR 15.1.2).
				Hi	FW valve closes if in auto mode	If FCV in manual - no event. If FCV is auto, result is decreased FW flow. Bounding event is Loss of Normal FW Flow (FSAR 15.2.7).
Narrow Range Level	4 per Steam Generator (two available for control)	1 selected for control I or II	Feedwater Control	Lo	FW valve opens if in auto mode	If FCV in manual - no event. If FCV in auto, result is Excessive FW Flow (FSAR 15.1.2).

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 5 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	FW valves closes if in auto mode	If FCV in manual - no event. If FCV is auto, result is decreased FW flow. Bounding event is Loss of Normal FW Flow (FSAR 15.2.7).
Pressurizer Level (Control)	3 per unit	I or III	Prz. Level Control	Lo	Charging flow increases. Heaters turn off (except for local control). Letdown isolated (VCT empties, charging pumps take suction from RWST.)	Bounding event is Increased Reactor Coolant Inventory (FSAR 15.5.2).
				Hi	Charging flow decreases Backup heaters on (Later, letdown isolation from interlock channel, heaters blocked from interlock channel.)	While heaters are on, no net depressurization of RCS. After heaters are blocked, decreased charging flow acts to depressurize RCS. Depressurization event is therefore bounded by Inadvertent Opening of a Pressurizer Safety or Relief Valve (FSAR 15.6.1)

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 6 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Pressurizer Level (Interlock)	3 per unit	II or II	Prz. Level Control	Lo	Letdown isolated. Prz. Heaters blocked (except for local control). (Charging flow reduced to maintain level).	Steady-state reached at slightly high level. No event.
				Hi	No control action, get Hi level annunciation.	Not applicable
Pressurizer Pressure	4 per unit	I	Prz. Pressure Control (Pos. 1 or 2)[(a)]	Lo	Turn on Backup Heaters. PORV 455A blocked from opening. PORV 456 opens if required, closes when pressure falls below dead band. Spray remains off.	Heaters being on causes increase in Prz. pressure to PORV 456 actuation. No event.
				Hi	PORV 455A opens, closes when pressure falls below deadband. Spray turned on.	Result is bounded by Inadvertent Opening of a Prz. Safety or Relief Valve (FSAR 15.6.1).
			(Pos. 3)[(a)]		Channel not connected	Not applicable

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 7 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Pressurizer Pressure	4 per unit	II	Prz. Pressure Control (Pos. 2 or 3)[a]	Lo	No control action. PORV 456 blocked from opening. PORV 455A Opens if required, closes when pressure falls below deadband.	Not applicable
				Hi	PORV 456A Opens, closes when pressure falls below deadband Spray turned on.	Result is bounded by Inadvertent Opening of a Prz. Safety or Relief Valve (FSAR 15.6.1)
			(Pos. 1)[a]		Channel not connected	Not applicable
Pressurizer Pressure	4 per unit	III	Prz. Pressure Control (Pos. 3)[a]	Lo	Turn on Backup Heaters. PORV 455A and 456 blocked from opening Spray remains off.	Heaters being on causes increase in Prz. pressure, possibly to safety valve actuation and reactor trip. Bounding event is Loss of Normal FW Flow (FSAR 15.1.2).
				Hi	PORV 455A Opens, closed on low pressure interlock. Spray turned on. PORV 456 unblocked.	Result is bounded by Inadvertent Opening of a Prz. Safety or Relief Valve (FSAR 15.6.1)

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 8 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
			(Pos. 1 or 2)[(a)]	Lo	Block PORV 456 from opening; no control action.	Not applicable
				Hi	Unblock PORV 456; no control action.	Not applicable
Pressurizer Pressure	4 per unit	IV	Prz. Pressure Control (Pos. 1)[(a)]	Lo	Block PORV 456 & 455A from opening; no control action.	Not applicable
				Hi	PORV 455A unblocked. PORV 456 opens, closes when pressure falls below deadband.	Result is bounded by Inadvertent Opening of a Prz. Safety or Relief Valve (FSAR 15.6.1).
			(Pos. 2 or 3)[(a)]	Lo	Block PORV 455A from opening; no control action	Not applicable
				Hi	Unblock PORV 455A; no control action	Not applicable

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 9 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
TAVG	one per loop	Any TAVG	Steam Dump Reactor Control Prz. Level Control	Lo	• Stop turbine loading, defeat remote dispatching (C-16 Annunciation occurs). • Outward rod motion (non-conservative direction), reactor trip on N16 over temperature trip. If C2, C3, C4 block rods so power does not significantly increase, reactor trip on High Flux trip, whether or not a high flux reactor trip occurs depends upon initial bank D position; the magnitude of the negative moderator temperature coefficient (MTC), magnitude of the doppler-only power coefficient (DOPC).	Bounded by analysis of control rod withdrawal events in FSAR Section 15.4.

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 10 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
					• Charging flow decreases to minimum causing Pressurizer level decrease to a minimum level.	
		Auct. Lo	Turbine Loading Dispatching	Hi	• Rods in (safe direction) at max speec results in lo pressure reactor trip or over temperature N16 reactor trip. • Charging flow increases until full power Prz, level is reached (if at reduced power). If reactor trips steam dump enabled and dump valves open until steam dump stops when Lo-lo TAVG is reached.	No event unless reactor trips, then dump valves open and bounding event is Excessive Increase in Secondary Steam Flow (FSAR 15.1.3)
Steamline Pressure	1 per loop for control different from those used for protection	Control Channel	Steam Dump	Lo	No control action	Not applicable

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 11 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	S. GEN. relief valve opens.	Bounding event is Inadvertent Opening of a Steam Generator Relief or Safety Valve (FSAR 15.1.4)
Intermediate Range Flux	2 per plant	I or II	Reactor Control	Lo	No control action.	Not applicable
				Hi	Reactor trips if below P-10 interlock.	Not applicable
Turbine Impulse Chamber Pressure (Control)	2 per turbine	I (Pos. 1)[(a)] or II (Pos. 2)[(a)]	Steam Dump (TAVG Mode) Reactor Control FW Control	Lo	Rods in (safe direction), auto rods withdrawal blocked (C-5). (If reactor trip occurs, steam dump unblocked and dump valves modulate until no load TAVG is reached). No effect on FW control since have constant S.G. level program.	Not applicable

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 12 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
				Hi	Rods out until blocked by Hi flux, overpower, or overtemperature, rod stop, or until programmed TREF limit is reached. (If reactor trip occurs, steam dump unblocked and dump valves open until no load TAVG is reached). No effect of FW control since have constant S.G. level program.	Result is bounded by Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (FSAR 15.4.2)
Turbine Impulse Chamber Pressure (Control)	2 per turbine	I (Pos. 1)[(a)] or II (Pos. 2)[(a)]	Steam Dump (Pressure Mode) Reactor Control FW Control	Lo	Rods in, (safe direction) auto rod withdrawal blocked (C-5). No effect on FW control since have constant S.G. level program.	Not applicable
				Hi	Rods out until blocked by Hi flux, overpower, or overtemperature rod stop. Dump valves open to keep steam header pressure at or below setpoint. No effect on FW control.	Result is bounded by Uncontrolled Rod Cluster Control Assembly Bank Withdrawal at Power (FSAR 15.4.2)

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 13 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Turbine Impulse Chamber Pressure (Interlock)	2 per turbine	II	Steam Dump (TAVG Mode)	Lo	Unblock steam dump	Not applicable
				Hi	Block steam dump.	Not applicable
Turbine Impulse Chamber Pressure (Interlock)	2 per turbine	I II	Steam Dump (Pressure Mode)	Lo or Hi	No control Action.	Not applicable
Power Range Flux	4 per unit	Any	Reactor Control FW Control	Lo	No control action (auctioneered Hi)	Not applicable
				Hi	Auto and manual rod withdrawal blocked (C-2), rods in (in safe direction). No effect on FW bypass valve due to nuclear feedforward gain set to zero. (If reactor trip occurs, dump valves open until no load TAVG is reached).	No event.

TABLE 7.6-1
LOSS OF ANY SINGLE INSTRUMENT

(Sheet 14 of 14)

SENSOR	NUMBER OF CHANNELS	FAILED CHANNEL	SYSTEM	ASSUMED FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Condenser Available	2 per condenser	Any	Steam Dump	Lo	No control action - steam dump unblocked, condenser available.	Not applicable
				Hi	No control action - steam dump blocked, condenser unavailable.	Not applicable
Steam Flow Pressure Compensator	2 per loop	Control Channel	Steam Flow	Lo	Identical to Loop Steam Flow channel failing low. See analysis above.	See above
				Hi	Identical to Loop Steam Flow channel failing high. See analysis above.	See above

a) Signals for pressurizer and turbine impulse chamber pressure can be obtained from different channels. Selection of desired channels is done by manual switches in the control room. Resulting accident due to failed instrument is dependent on switch positions.

TABLE 7.6-2
LOSS OF POWER TO PROTECTION SET I

CONTROL SYSTEMS AFFECTED	SIGNALS AFFECTED	FAILURE DIRECTION	ITEMIZED EFFECTS	BOUNDING EVENT
Steam Dump	Turbine Pressure (Control or Interlock)	Low	Steam dump demanded but blocked from interlock. If reactor trips, steam dump valves modulate.	
Reactor Control	Power Range Flux (Control)	Low	If Turbine pressure on control channel, rods in (safe direction), power decreases and stop turbine loading/defeat remote dispatching. Otherwise, no control action.	Bounding event is Excessive FW Flow (FSAR 15.1.2). Hi-Hi SG level closes FW control and isolation valves, trips FW turbine, and main turbine. Increased charging flow and pressurizer transients have little effect in comparison.
	Turbine Pressure (Control)	Low		
	TAVG (Loop 1)	Low		
FW Control	Narrow Range Level SG No. 1 and 4	Low	If affected level signal used for control, FCV opens in loops 1 and 4. If affected steam flow pressure compensation used, control logic would cause FW flow in loops 2 and 3 to decrease with subsequent return to normal condition. Steam pressure compensation signal effects in loops 1 and 4 negligible in comparison to level signal effects.	
	Steam Flow Pressure Compensation (All Loops)	Low		
Pressurizer	Pzr. Level (Control and Interlock)	Low	Letdown isolated, heaters blocked if affected level signal used for control. Charging flow increases.	
Pressurizer Pressure	Pzr. Pressure	Low	PORV 455A stays closed. (PORV 456 available if required.) If affected pressure signal used for control spray off.	

TABLE 7.6-3
LOSS OF POWER TO PROTECTION SET II

CONTROL SYSTEMS AFFECTED	SIGNALS AFFECTED	FAILURE DIRECTION	ITEMIZED EFFECTS	BOUNDING EVENT
Steam Dump	Turbine Pressure (Control or Interlock)	Low	No control action, and steam dump blocked. If reactor trips main steam relief valves available.	Bounding event is Excessive FW Flow (FSAR 15.1.2). Hi-Hi SG level closes FW control and isolation valves, trips FW turbine, and main turbine.
Reactor Control	Power Range Flux (Control)	Low	If Turbine pressure on control channel, rods in (safe direction), power decreases and stop turbine loading/defeat remote dispatching. Otherwise, no control action.	
	Turbine Pressure (Control)	Low		
	TAVG (Loop 1)	Low		
FW Control	Narrow Range Level SG No. 2and 3	Low	If affected level signal used for control, FCV opens in loops 2 and 3 If affected steam flow pressure compensation used, control logic would cause FW flow in loops 1 and 4 to decrease with subsequent return to normal condition. Steam pressure compensation signal effects in loops 2 and 3 negligible in comparison to level signal effects.	
	Steam Flow Pressure Compensation (All Loops)	Low		
Pressurizer	Pzr. Level (Interlock)	Low	If affected level signal used for interlock, heaters off, letdown isolated, charging flow unaffected. Otherwise, channel not connected, no control action.	
Pressurizer Pressure	Pzr. Pressure	Low	PORV 456 stays closed. (PORV 455A available if required.) Otherwise, channel not connected, no control action.	

TABLE 7.6-4
LOSS OF POWER TO PROTECTION SET III

CONTROL SYSTEMS AFFECTED	SIGNALS AFFECTED	FAILURE DIRECTION	ITEMIZED EFFECTS	BOUNDING EVENT
Steam Dump	None	---	No signals affected, no control action	With htrs on, and spray off, high pressurizer pressure trip is actuated. Event is bounded by loss of normal feedwater (FSAR 15.2.7). Bounding events due to charging flow increase or letdown isolated are slowly developing; alarms in the control room would allow the operator ample time to restore power.
Reactor Control	Power Range Flux (Control)	Low	Stop turbine loading, defeat remote dispatching	
	TAVG (Loop 3)	Low		
FW Control	None	---	No signals affected, no control action	
Pressurizer Level	Pzr. Level (Control or Interlock)	Low	If affected level signal used for control, charging flow increases, letdown isolated, heaters blocked. If used for interlock, heaters blocked and letdown isolated. Otherwise, channel not connected, no control action.	
Pressurizer Pressure	Pzr. Pressure (Interlock and Control)	Low	PORV 455A and 456 stay closed. If affected pressure signal used for control, backup heaters on (if allowed by level signal, see above) and spray off. Otherwise, spray available if required.	

TABLE 7.6-5
LOSS OF POWER TO PROTECTION SET IV

CONTROL SYSTEMS AFFECTED	SIGNALS AFFECTED	FAILURE DIRECTION	ITEMIZED EFFECTS	BOUNDING EVENT
Steam Dump	None	---	No signals affected, no control action	
Reactor Control	Power Range Flux (Control)	Low	Stop turbine loading, defeat remote dispatching	
	TAVG (Loop 4)	Low		
FW Control	None	---	No signals affected, no control action	No event is initiated due to loss of power, therefore bounding event is not applicable.
Pressurizer Level	None	---	No signals affected, no control action	
Pressurizer Pressure	Pzr. Pressure (Control and Interlock)	Low	PORV 455A and 456 stay closed.	

TABLE 7.6-6
LOSS OF COMMON INSTRUMENT LINES (ASSUMED BREAK IN LINE)

SENSORS	FAILED CHANNELS	SYSTEM	FAILURE DIRECTION	EFFECT	BOUNDING EVENT
Loop Steam Flow and Narrow Range Level	II	Feedwater Control	Lo Hi	FW valve closes in affected S.G., pump speed decreases	Bounding event is Loss of Normal FW Flow (FSAR 15.1.2)
Pressurizer Level (Control) and Pressurizer Pressure	I (Level and Pressure)	Pzr. Level Control Pzr. Pressure Control	Hi Lo	PORV 455A stays closed. Spray unavailable. Charging flow decreases (Control). Backup heaters on (Control). (On low level, letdown isolated and heaters blocked from interlock channel).	Bounding event is inadvertent opening of a Prz. Safety or Relief Valve (FSAR 15.6.1)
Pressurizer Level (Interlock) and Pressurizer Pressure	III (Level and Pressure)	Pzr. Level Control Pzr. Pressure Control	Hi Lo	Scenario dependant upon selector switch position, bracketing effects are: PORV 455A and 456 stay closed, spray unavailable, backup htrs on, charging flow decrease, on low level letdown isolated, htrs blocked.	Bounding event is Loss of Normal FW Flow (FSAR 15.1.2)
Pressurizer Level (control or Interlock) and Pressurizer Pressure	II (Level) II and IV (Pressure)	Pzr. Pressure Control	Hi	PORV 455A and 456 stay closed. Spray, charging flow, letdown isolation, and htr controls unaffected.	No event

7.7 CONTROL SYSTEMS NOT REQUIRED FOR SAFETY

The general design objectives of the plant control systems are:

1. To establish and maintain power equilibrium between primary and secondary system during steady state unit operation.

2. To constrain operational transients so as to preclude unit trip and re-establish steady state unit operation.

3. To provide the reactor operator with monitoring instrumentation that indicates all required input and output control parameters of the systems and provides the operator the capability of assuming manual control of the system.

7.7.1 DESCRIPTION

The plant control systems described in this section perform the following functions:

1. Reactor Control System

    a. Enables the nuclear plant to accept a step load increase or decrease of 10 percent and a ramp increase or decrease of 5 percent per minute within the load range of 15 percent to 100 percent without reactor trip, steam dump, or pressurizer relief actuation, subject to possible xenon limitations.

    b. Maintains reactor coolant average temperature within prescribed limits by creating the bank demand signals for moving groups of full length rod cluster control assemblies during normal operation and operational transients. The reactor coolant average temperature control also supplies a signal to pressurizer water level control, and steam dump control.

2. Rod Control System

    a. Provides for reactor power modulation by manual or automatic control of full length control rod banks in a pre-selected sequence and for manual operation of individual banks.

    b. Systems for monitoring and indicating

        1. Provide alarms to alert the operator if the required core reactivity shutdown margin is not available due to excessive control rod insertion.

        2. Display control rod position.

        3. Provide alarms to alert the operator in the event of control rod deviation exceeding a preset limit.

3. Plant Control System interlocks

   a. Prevent further withdrawal of the control banks when signal limits are approached that predict the approach of a departure from nucleate boiling ratio (DNBR) limit or kilowatt per feet (kW/ft) limit.

   b. Inhibit automatic turbine load change as required by the Nuclear Steam Supply System (NSSS).

4. Pressurizer pressure control

   a. Maintains or restores the pressurizer pressure to the design pressure +30 pounds per square inch (psi) (which is well within reactor trip and relief and safety valve actuation setpoint limits) following normal operational transients that induce pressure changes by control (manual or automatic) of heaters and spray in the pressurizer.

      Provides steam relief by controlling the power relief valves.

5. Pressurizer water level control

   a. Establishes, maintains, and restores pressurizer water level within specified limits as a function of the average coolant temperature. Changes in level are caused by coolant density changes induced by loading, operational, and unloading transients. Level changes are produced by means of charging flow control (manual or automatic) as well as by manual selection of letdown orifices. Maintaining coolant level in the pressurizer within prescribed limits by actuating the charging and letdown system thus provides control of the reactor coolant water inventory.

6. Steam generator water level control

   a. Establishes and maintains the steam generator water level to within pre-determined physical limits during normal operating transients.

   b. Restores the steam generator water level to within predetermined limits at unit trip conditions. Regulates the feedwater flow rate such that under operational transients the heat sink for the Reactor Coolant System (RCS) is maintained.

7. Steam dump control

   a. Permits the nuclear plant to accept a sudden loss of load without incurring reactor trip. Steam is dumped to the condenser and/or the atmosphere as necessary to accommodate excess power generation in the reactor during turbine load reduction transients.

   b. Ensures that stored energy and residual heat are removed following a reactor trip to bring the plant to equilibrium no-load conditions without actuation of the steam generator safety valves.

c. Maintains the plant at no-load conditions and permits a manually controlled cooldown of the plant.

8. Incore instrumentation

Provides information on the neutron flux distribution and on the core outlet temperatures at selected core locations.

7.7.1.1 Reactor Control System

The Reactor Control System enables the nuclear plant to follow load changes automatically including the acceptance of step load increase or decreases of 10 percent and ramp increases or decreases of 5 percent per minute within the load range of 15 percent to 100 percent without reactor trip, steam dump, or pressure relief (subject to possible xenon limitations). The system is also capable of restoring coolant average temperature to within the programmed temperature deadband following a change in load. Manual control rod operation may be performed at any time.

The Reactor Control System controls the reactor coolant average temperature by regulation of control rod bank position. The reactor coolant loop average temperatures are determined from N-16 power and cold leg measurements in each reactor coolant loop. There is an average coolant temperature (T_{avg}) computed for each loop, as shown on Figure 7.7-18:

$$T_{avg} = T_{cold} + K \times Q_{N-16}$$

The error between the programmed reference temperature (based on turbine first stage pressure) and the average of the T_{avg} measured temperatures (which is processed through a lead-lag compensation unit) from each of the reactor coolant loops constitutes the primary control signal as shown in general on Figure 7.7-1 and in more detail on the functional diagrams shown in Figure 7.2-1, Sheet 9. The system is capable of restoring coolant average temperature to the programmed value following a change in load. The programmed coolant temperature increases linearly with turbine load from zero power to the full power condition. The T_{avg} also supplies a signal to pressurizer level control and steam dump control.

The temperature and N-16 channels needed to derive the temperature input signals for the Reactor Control System are fed from protection channels via isolation amplifiers.

An additional control input signal is derived from the reactor power versus turbine load mismatch signal. This additional control input signal improves system performance by enhancing response and reducing transient peaks.

The core axial power distribution is controlled during load follow maneuvers by changing (a manual operator action) the boron concentration in the RCS. The control board displays (see Section 7.7.1.3.1) indicate the need for an adjustment in the axial power distribution. Adding boron to the reactor coolant will reduce T_{avg} and cause the rods (through the Rod Control System) to move toward the top of the core. This action will reduce power peaks in the bottom of the core. Likewise, removing boron from the reactor coolant will move the rods further into the core to control power peaks in the top of the core.

7.7.1.2 Rod Control System

The full length rod control system receives rod speed and direction signals from the T_{avg} control system. The rod speed demand signal varies over the corresponding range of 5 to 45 inches per minute (8 to 72 steps/minute) depending on the magnitude of the input signal. Manual control is provided to move a control bank in or out at a prescribed fixed speed.

When the turbine load reaches approximately 15 percent of rated load, the operator may select the “AUTOMATIC” mode, and rod motion is then controlled by the reactor control systems. A permissive interlock C-5 (see Table 7.7-1) derived from measurements of turbine first stage pressure prevents automatic control when the turbine load is below 15 percent. In the “AUTOMATIC” mode, the rods are again withdrawn (or inserted) in a pre-determined programmed sequence by the automatic programming with the control interlocks (see Table 7.7-1).

The shutdown banks are always in the fully withdrawn position during normal operation, and are moved to this position at a constant speed by manual control prior to criticality. A reactor trip signal causes them to fall by gravity into the core. There are five shutdown banks.

The control banks are the only rods that can be manipulated under automatic control. Each control bank is divided into two groups to obtain smaller incremental reactivity changes per step. All rod control cluster assemblies in a group are electrically paralleled to move simultaneously. There is individual position indication for each rod cluster control assembly.

Power to rod drive mechanisms is supplied by two motor generator sets operating from two separate 480 volt, three phase buses. Each generator is the synchronous type and is driven by a 200 horsepower (hp) induction motor. The alternating current (AC) power is distributed to the rod control power cabinets through the two series connected reactor trip breakers.

The variable speed rod drive programmer affords the ability to insert small amounts of reactivity at low speed to accomplish fine control of reactor coolant average temperature about a small temperature deadband, as well as furnishing control at high speed. A summary of the rod cluster control assembly sequencing characteristics is given below.

1. Two groups within the same bank are stepped such that the relative position of the groups will not differ by more than one step.

2. The control banks are programmed such that withdrawal of the banks is sequenced in the following order: control bank A, control bank B, control bank C, and control bank D. The programmed insertion sequence is the opposite of the withdrawal sequence; i.e., the last control bank withdrawn (bank D) is the first control bank inserted.

3. The control bank withdrawals are programmed such that when the first bank reaches a preset position, the second bank begins to move out simultaneously with the first bank which continues to move toward its fully withdrawn position. When the second bank reaches a preset position, the third bank begins to move out, and so on. This withdrawal sequence continues until the unit reaches the desired power level. The control bank insertion sequence is the opposite.

4. Overlap between successive control banks is adjustable between 0 to 50 percent (0 and 115 steps), with an accuracy of ±1 step.

5. Rod speeds for either the shutdown banks or manual operation of the control banks are capable of being controlled between a minimum of 6 steps per minute and a maximum of 68 steps per minute.

7.7.1.3 Plant Control Signals for Monitoring and Indicating

7.7.1.3.1 Monitoring Functions Provided by the Nuclear Instrumentation System (NIS)

The power range channels are important because of their use in monitoring power distribution in the core within specified safe limits. They are used to measure power level, axial flux imbalance, and radial flux imbalance. These channels are capable of recording overpower excursions up to 200 percent of full power. Suitable alarms are derived from these signals as described below.

Basic power range signals are:

1. Total current from a power range detector (four such signals from a power range detector assembly, i.e., one detector assembly per core quadrant); each detector set consists of 4 power range detectors.

2. Current from the sum of the upper two detectors of each power range detector assembly (four such signals).

3. Current from the sum of the lower two detectors, of each power range detector assembly (four such signals).

Derived from these basic signals are the following (including standard signal processing for calibration).

4. Indicated nuclear power (four such).

5. Indicated axial flux difference ($\Delta\phi$), derived from the sum of the upper two detectors in an assembly minus the sum of the lower two detectors in an assembly (four such).

Alarm functions derived are as follows:

6. Deviation (maximum minus minimum of four) in indicated nuclear power.

7. Upper radial tilt (maximum to average of four) on average of the upper two detector sets' currents.

8. Lower radial tilt (maximum to average of four) on average of the lower two detector sets' currents.

Provision is made to continuously record, on strip charts on the control board, the 8 groups of the ion chamber signals, i.e., average of the upper and average of the lower currents for each detector assembly. Nuclear power and axial unbalance is selectable for recording as well. Indicators are provided on the control board for nuclear power and for axial flux difference.

The axial flux difference deviation alarms are derived from the plant process computer which determines the 1 minute averages of the excore detector outputs to monitor $\Delta\phi$ in the reactor core and alerts the operator where $\Delta\phi$ alarm conditions exist. When power level is 50 percent or greater, an alarm message is output immediately upon determining a delta flux exceeding the Technical Specifications (this alarm requires 2 of the 4 AFD channels to exceed the power dependent limits). The signals from the four section excore detectors are summed and calibrated to a power measurement made using a secondary system calorimetric. The signals from the upper two sections and lower two sections are combined to give an upper flux and lower flux signal as shown in Figure 7.1-2. These upper and lower signals must be calibrated to agree with axial offset measurements derived from flux maps made using the movable incore system or power distribution monitoring system. The NIS provides both slope and zero-offset adjustments to achieve this calibration.

Additional background information on the Nuclear Instrumentation System can be found in Reference [1].

7.7.1.3.2 Rod Position Monitoring of Full Length Rods

Two separate systems are provided to display control rod position as described below:

1. Digital Rod Position Indication System

   The Digital Rod Position Indication System (DRPIS) measures the actual position of each full length rod using a detector which consists of discrete coils mounted concentrically with the rod drive pressure housing. The coils are located axially along the pressure housing and magnetically sense the entry and presence of the rod drive shaft through its centerline. For each detector, the coils are interlaced into two data channels, and are connected to the Containment electronics (Data A and B) by separate multi-conductor cables. By employing two separate channels of information, the DRPIS can continue to function (at reduced accuracy) when one channel fails. Multiplexing is then used to transmit the digital position signals from the Containment electronics to the control board display unit.

   The control board display unit contains a column of light- emitting-diodes (LED) for each rod. At any given time, the one LED illuminated in each column shows the position for that particular rod. Since shutdown rods are always fully withdrawn with the plant at power, their position is displayed in 6 step increments with an accuracy of ±4 steps only from rod bottom to 18 steps and from 210 steps to 228 steps. All intermediate positions of the rod are represented by a single “transition” LED. Each rod of the control banks has its position displayed in 6 step increments with an accuracy of ±4 steps throughout its range of travel.

   Included in the system is a rod at bottom signal for each rod that operates an alarm. Also a Control Room annunciator is actuated for any misaligned rod at bottom.

2. Demand Position System

   The Demand Position System counts pulses generated in the rod drive control system to provide a digital readout of the demanded bank position.

The demand position and digital rod position indication systems are separate systems, but safety criteria were not involved in the separation, which was a result only of operational requirements. Operating procedures require the reactor operator to compare the demand and indicated (actual) readings from the DRPIS so as to verify operation of the Rod Control System.

7.7.1.3.3 Control Bank Rod Insertion Monitoring

When the reactor is critical, the normal indication of reactivity status in the core is the position of the control bank in relation to reactor power and coolant average temperature. These parameters are used to calculate insertion limits for the control banks. Two alarms are provided for each control bank.

1. The “low” alarm alerts the operator of an approach to the rod insertion limits requiring boron addition by following normal procedures with the Chemical and Volume Control System.

2. The “low-low” alarm alerts the operator to take action to ensure shutdown margin is within limits or restored within limits per Technical Specification.

The purpose of the control bank rod insertion monitor is to give warning to the operator of excessive rod insertion. The insertion limit maintains sufficient core reactivity shutdown margin following reactor trip and provides a limit on the maximum inserted rod worth in the unlikely event of a hypothetical rod ejection, and limits rod insertion such that acceptable nuclear peaking factors are maintained. Since the amount of shutdown reactivity required for the design shutdown margin following a reactor trip increases with increasing power, the allowable rod insertion limits must be decreased (the rods must be withdrawn further) with increasing power. The rod insertion monitor uses the N-16 power parameters for each control rod bank as follows:

$$Z_{LL} = A(\text{N-16 power})_{auct} + C$$

where

Z_{LL}	=	maximum permissible insertion limit for affected control bank
$(\text{N-16})_{auct}$	=	highest N-16 of all loops
A,C	=	constant chosen to maintain $Z_{LL} \geq$ actual limit based on physics calculations

The control rod bank demand position (Z) is compared to Z_{LL} as follows:

1. If $Z - Z_{LL} \leq D$ a low alarm is actuated.

2. If $Z - Z_{LL} \leq E$ a low-low alarm is actuated.

Since the highest value of N-16 power is chosen by auctioneering, a conservatively high representation of power is used in the insertion limit calculation.

Actuation of the low alarm alerts the operator of an approach to a reduced shutdown reactivity situation. Administrative procedures require the operator to add boron through the Chemical and Volume Control System. Actuation of the low-low alarm requires the operator to verify shutdown margin within limits or to take actions to restore the shutdown margin in accordance with Technical Specification. The value for "D" is chosen such that the low alarm would normally be actuated before the insertion limit is reached. This allows the operator to follow normal boration procedures. Figure 7.7-2 shows a block diagram representation of the control rod bank insertion monitor. The monitor is shown in more detail on the functional diagrams shown in Figure 7.2-1, Sheet 9. In addition to the rod insertion monitor for the control banks, the plant computer, which monitors individual rod positions, provides an alarm that is associated with the rod deviation alarm discussed in Section 7.7.1.3.4 is provided to warn the operator if any shutdown rod cluster control assembly leaves the fully withdrawn position.

Rod insertion limits are established by:

1. Establishing the allowed rod reactivity insertion at full power consistent with the purposes given above.

2. Establishing the differential reactivity worth of the control rods when moved in normal sequence.

3. Establishing the change in reactivity with power level by relating power level to rod position.

4. Linearizing the resultant limit curve. All key nuclear parameters in this procedure are measured as part of the initial and periodic physics testing program.

Any unexpected change in the position of the control bank under automatic control, or a change in coolant temperature under manual control, provides a direct and immediate indication of a change in the reactivity status of the reactor. In addition, samples are taken periodically of coolant boron concentration. Variations in concentration during core life provide an additional check on the reactivity status of the reactor, including core depletion.

7.7.1.3.4 Rod Deviation Alarm

A rod deviation function is performed as part of the DRPIS where an alarm is generated if a preset limit is exceeded as a result of a comparison of any control rod against the other rods in a bank. The deviation alarm of a shutdown rod is based on a preset insertion limit being exceeded.

The demanded and measured rod position signals are also monitored by the plant computer which provides a visual display and an audible alarm whenever an individual rod position signal deviates from the other rods in the bank by a preset limit. The alarm can be set with appropriate allowance for instrument error and within sufficiently narrow limits to preclude exceeding core design hot channel factors.

Figure 7.7-3 is a block diagram of the rod deviation comparator and alarm system implemented by the plant computer. Additionally, the DRPIS contains rod deviation circuitry that detects and alarms the following conditions:

1. When any two rods within the same control bank are misaligned by a preset distance (≥ 12 steps), and

2. When any shutdown rod is below the full-out position by a preset distance (18 steps).

7.7.1.3.5 Rod Bottom Alarm

A rod bottom signal for the full length rods in the digital rod position system is used to operate a control relay, which generates the “ANY ROD AT BOTTOM” AND/OR “≥ 2 ROD AT BOTTOM” alarm.

7.7.1.4 Plant Control System Interlocks

The listing of the plant control system interlocks, along with the description of their derivations and functions, is presented in Table 7.7-1. It is noted that the designation numbers for these interlocks are preceded by “C”. The development of these logic functions is shown in the functional diagrams (Figure 7.2-1, Sheets 9 through 16).

7.7.1.4.1 Rod Stops

Rod stops are provided to prevent abnormal power conditions which could result from excessive control rod withdrawal initiated by either a control system malfunction or operator violation of administrative procedures.

Rod stops are the C-1, C-2, C-3, C-4, C-5, and C-11 control interlocks identified in Table 7.7-1. The C-3 rod stop derived from Overtemperature N-16 and the C-4 rod stop, derived from Overpower N-16 are also used for turbine runback, which is discussed below.

7.7.1.4.2 Automatic Turbine Load Runback

Automatic turbine load runback is initiated by an approach to an overpower or overtemperature condition. This will prevent high power operation that might lead to an undesirable condition, which, if reached, will be protected by reactor trip.

Turbine load reference reduction is initiated by either an Overtemperature or Overpower N-16 signal. Two out of four coincidence logic is used.

A rod stop and turbine runback are initiated when:

$$Q_{N\text{-}16} > Q_{rod\ stop}$$

for both the overtemperature and the overpower condition.

For either condition in general:

$$Q_{rod\ stop} = Q_{setpoint} - C$$

where

C = a Constant

where $Q_{setpoint}$ refers to the Overtemperature N-16 reactor trip value and the Overpower N-16 reactor trip value for the two conditions.

The turbine runback is continued until N-16 power is equal to or less than $Q_{rod\ stop}$.

This function serves to maintain an essentially constant margin to trip.

In addition to the fast turbine runback initiated by Overtemperature or Overpower N-16 parameters, an automatic turbine load runback is initiated by the following condensate/feedwater system malfunctions:

1. Main feedwater pump trip.

2. Condensate pump trip.

3. Heater drain pump trip.

The above runbacks are within the capability of automatic reactor load following. However, a condensate pump trip may result in a plant trip due to loss of feedwater or low levels in the steam generators.

7.7.1.4.3 Turbine Loading Stop

An interlock (C-16) is provided to limit turbine loading during a rapid return to power transient when a reduction in reactor coolant temperature is used to increase reactor power (through the negative moderator coefficient). This interlock limits the drop in coolant temperature to exceed cooldown accident limits and preserves satisfactory steam generator operating conditions. Subsequent automatic turbine loading can begin after the interlock has been cleared by an increase in coolant temperature which is accomplished by reducing the boron concentration in the coolant.

7.7.1.5 Pressurizer Pressure Control

The RCS pressure is controlled by using either the heaters (in the water region) or the spray (in the steam region) of the pressurizer plus steam relief for large transients. The electrical immersion heaters are located near the bottom of the pressurizer. A portion of the heater group is proportionally controlled to correct small pressure variations. These variations are due to heat losses, including heat losses due to a small continuous spray. The remaining (back-up) heaters are turned on when the pressurizer pressure controlled signal demands approximately 100 percent proportional heater power.

The spray nozzles are located on the top of the pressurizer. Spray is initiated when the pressure controller spray demand signal is above a given setpoint. The spray rate increases proportionally with increasing spray demand signal until it reaches a maximum value.

Steam condensed by the spray reduces the pressurizer pressure. A small continuous spray is normally maintained to reduce thermal stresses and thermal shock and to help maintain uniform water chemistry and temperature in the pressurizer.

Power relief valves limit system pressure for large positive pressure transients. In the event of a large load reduction, not exceeding the design plant load rejection capability, the pressurizer power operated relief valves might be actuated for the most adverse conditions, e.g., Beginning of Life conditions. The relief capacity of the power operated relief valves is sized large enough to limit the system pressure to prevent actuation of high pressure reactor trip for the above condition.

A block diagram of the Pressurizer Pressure Control System is shown on Figure 7.7-4.

7.7.1.6 Pressurizer Water Level Control

The pressurizer operates by maintaining a steam cushion over the reactor coolant. As the density of the reactor coolant adjusts to the various temperatures, the steam water interface moves to absorb the variations with relatively small pressure disturbances.

The water inventory in the RCS is maintained by the Chemical and Volume Control System. During normal plant operation, the charging flow varies to produce the flow demanded by the pressurizer water level controller. The pressurizer water level is programmed as a function of coolant average temperature, with the average temperature being used. The pressurizer water level decreases as the load is reduced from full load. This is a result of coolant contraction following programmed coolant temperature reduction from full power to low power. The programmed level is designed to match as nearly as possible the level changes resulting from the coolant temperature changes.

To control pressurizer water level during startup and shutdown operations, the charging flow is manually regulated from the main Control Room.

A block diagram of the Pressurizer Water Level Control System is shown on Figure 7.7-5.

7.7.1.7 Steam Generator Water Level Control

Each steam generator is equipped with a three element feedwater flow controller which maintains a programmed water level which is a function of turbine load. The three element feedwater controller regulates the feedwater valve by continuously comparing the feedwater flow signal, the water level signal, the programmed level and the pressure compensated steam flow signal. In addition, the main turbine feedwater pump speed is varied to maintain a programmed pressure differential between the steam header and the feed pump discharge header. The speed controller continuously compares the actual ΔP with a programmed ΔP_{ref} which is a linear function of steam flow. Continued delivery of feedwater to the steam generators is required as a sink for the heat stored and generated in the reactor following a reactor trip and turbine trip. An override signal closes the feedwater valves when the average coolant temperature is below a given temperature and the reactor has tripped. Manual override of the feedwater control system is available at all times.

Block diagrams of the Steam Generator Water Level Control System and the Main Feedwater Pump Speed Control System are shown in Figures 7.7-6 and 7.7-7.

7.7.1.8 Steam Dump Control

The Steam Dump System in conjunction with the Rod Control System is designed to accept a 50 percent loss of net load without tripping the reactor.

The automatic Steam Dump System is able to accommodate this abnormal load rejection and to reduce the effects of the transient imposed upon the RCS. By bypassing main steam directly to the condenser an artificial load is thereby maintained on the primary system. The Rod Control System can then reduce the reactor temperature to a new equilibrium value without causing overtemperature and/or overpressure conditions. The steam dump steam flow capacity is greater than 40 percent of full load steam flow at full load steam pressure.

If the difference between the reference T_{avg} (T_{ref}) based on turbine first stage pressure and the lead-lag compensated average T_{avg} exceeds a pre-determined amount, and the interlock mentioned below is satisfied, a demand signal will actuate the steam dump to maintain the RCS temperature within control range until a new equilibrium condition is reached.

To prevent actuation of steam dump on small load perturbations, an independent load rejection sensing circuit is provided. This circuit senses the rate of decrease in the turbine load as detected by the turbine first stage pressure. It is provided to unblock the dump valves when the rate of load rejection exceeds a preset value corresponding to a 10 percent step load decrease or a sustained ramp load decrease of 5 percent/minute.

A block diagram of the Steam Dump Control System is shown on Figure 7.7-8.

7.7.1.8.1 Load Rejection Steam Dump Controller

This circuit prevents large increase in reactor coolant temperature following a large, sudden load decrease. The error signal is a difference between the lead-lag compensated average T_{avg} and the reference T_{avg} based on turbine first stage pressure.

The T_{avg} signal is the same as that used in the RCS. The lead-lag compensation for the T_{avg} signal is to compensate for lags in the plant thermal response and in valve positioning. Following a sudden load decrease, T_{ref} is immediately decreased and T_{avg} tends to increase, thus generating an immediate demand signal for steam dump. Since control rods are available, in this situation steam dump terminates as the error comes within the maneuvering capability of the control rods.

7.7.1.8.2 Plant Trip Steam Dump Controller

Following a reactor trip, as monitored by the reactor trip signal (P-4), the load rejection steam dump controller is defeated and the turbine trip steam dump controller becomes active. Since control rods are not available in this situation, the demand signal is the error signal between the lead-lag compensated average T_{avg} and the no-load reference T_{avg}. When the error signal exceeds a pre-determined setpoint the dump valves are tripped open in a prescribed sequence.

As the error signal reduces in magnitude (indicating that the RCS T_{avg} is being reduced toward the reference no-load value) the dump valves are modulated by the plant trip controller to regulate the rate of removal of decay heat and thus gradually establish the equilibrium hot shutdown condition.

7.7.1.8.3 Steam Header Pressure Controller

Residual heat removal is maintained by the steam generator pressure controller (manually selected) which controls the amount of steam flow to the condensers. This controller operates a portion of the same steam dump valves to the condensers which are used during the initial transient following turbine reactor trip on load rejection.

7.7.1.9 Incore Instrumentation

The Incore Instrumentation System consists of chromel-alumel thermocouples at fixed core outlet positions and movable miniature neutron detectors which can be positioned at the center of selected fuel assemblies, anywhere along the length of the fuel assembly vertical axis. The basic system for insertion of these detectors is shown in Figure 7.7-9.

7.7.1.9.1 Thermocouples

Chromel-alumel thermocouples are inserted into guide tubes that penetrate the reactor vessel head through seal assemblies, and terminate at the exit flow end of the fuel assemblies. The Unit 2 thermocouples are provided with two primary seals, a conoseal/grafoil seal and swage type seal from conduit to head. The Unit 1 thermocouples are provided with a grafoil seal and swage type seal from conduit to head. Thermocouple readings are monitored by the plant process/ERF computers. Information from the incore instrumentation is available even if the computers are not in service.

7.7.1.9.2 Movable Neutron Flux Detector Drive System

Miniature fission chamber detectors can be remotely positioned in retractable guide thimbles to provide flux mapping of the core. The stainless steel detector shell is brazed to the leading end of helical wrap drive cable and to stainless steel sheathed coaxial cable. The retractable thimbles, into which the miniature detectors are driven, are pushed into the reactor core through conduits which extend from the bottom of the reactor vessel down through the concrete shield area and then up to a thimble seal table. Their distribution over the core is nearly uniform with about the same number of thimbles located in each quadrant.

The thimbles are closed at the leading ends, are dry inside, and serve as the pressure barrier between the reactor water pressure and the atmosphere. Mechanical seals between the retractable thimbles and the conduits are provided at the seal table. During reactor operation, the retractable thimbles are stationary. They are extracted downward from the core during refueling to avoid interference within the core. A space above the seal table is provided for the retraction operation.

The drive system for the insertion of the miniature detectors consists basically of drive assemblies, five path rotary transfer assemblies, and 10 path rotary transfer assemblies, as shown in Figure 7.7-9. The drive system pushes hollow helical wrap drive cables into the core

with the miniature detectors attached to the leading ends of the cables and small diameter sheathed coaxial cables threaded through the hollow centers back to the ends of the drive cables. Each drive assembly consists of a gear motor which pushes a helical wrap drive cable and a detector through a selective thimble path by means of a special drive box and includes a storage device that accommodates the total drive cable length.

Manual isolation valves (one for each thimble) are provided for closing the thimbles. When closed, the valves form a 2500 pounds per square inch gage (psig) barrier. The manual isolation valves are not designed to isolate a thimble while a detector/drive cable is inserted into the thimble. The detector/drive cable must be retracted to a position above the isolation valve prior to closing the valve.

A small leak would probably not prevent access to the isolation valves and thus a leaking thimble could be isolated during a hot shutdown. A large leak might require cold shutdown for access to the isolation valve.

7.7.1.9.3 Control and Readout Description

The Control and Readout System provides means for inserting the miniature neutron detectors into the reactor core and withdrawing the detectors while plotting neutron flux versus detector position. The control system is located in the Control Room. Limit switches in each transfer device provide feedback of path selection operation. Each gear box drives an encoder for position feedback. One five path operation selector is provided for each drive unit to insert the detector in one of five functional modes of operation. One 10 path operation selector is also provided for each drive unit that is then used to route a detector into any one of up to 10 selectable paths. A common path is provided to permit cross calibration of the detectors.

The Control Room contains the necessary equipment for control, position indication, and flux recording for each detector.

A “flux-mapping” consists, briefly, of selecting (by panel switches) flux thimbles in given fuel assemblies at various core quadrant locations. The detectors are driven to the top of the core and stopped automatically. An x-y plot (position versus flux level) is initiated with the slow withdrawal of the detectors through the core from top to a point below the bottom. In a similar manner other core locations are selected and plotted. Each detector provides axial flux distribution data along the center of a fuel assembly.

Various radial positions of detectors are then compared to obtain a flux map for a region of the core.

The number and location of these thimbles have been chosen to permit measurement of local to average peaking factors to an accuracy of ±5 percent (95 percent confidence). Measured nuclear peaking factors will be increased by 5 percent to allow for this accuracy. If the measured power peaking is larger than acceptable, reduced power capability will be indicated.

Operating plant experience has demonstrated the adequacy of the incore instrumentation in meeting the design bases stated.

7.7.1.9.4 Power Distribution Monitoring System

An alternative to the Movable Incore Detector System for developing full core flux maps while above 25% power, is the BEACON Power Distribution Monitoring System (Reference 2). This system obtains data from the plant computer and processes the data into a 3-dimensional core model. The inputs to this system are core exit thermocouples, control rod bank position, Tcold, Reactor power level, and the nuclear instrumentation Power Range section signals. Per the Technical Requirements Manual, there is a minimum number of each type of input necessary to consider the system operable.

7.7.1.10 Boron Concentration Measurement System

The Boron Concentration Measurement System is abandoned in place.

7.7.1.11 Balance of Plant Systems

The following balance of plant control systems not required for safety are completely independent of any protective functions and as such can not impair the ability of the protection systems to function.

1. Condensate System (see Section 10.4.7).
2. Condensate Cleanup System (see Section 10.4.6).
3. Service Air System (see Section 9.3.1).
4. Instrument Air System (see Section 9.3.1).
5. Secondary Plant Sampling (see Section 10.4.16).
6. Water Treatment System (see Section 9.2.3).
7. H_2 and N_2 Supply System (see Section 10.4.15).
8. Potable and Sanitary Water System (see Section 9.2.4).
9. Heater Drains Systems (see Section 10.4.11).
10. Circulating Water Systems (see Section 10.4.5).
11. Turbine Plant Cooling Water Systems (see Section 10.4.12).
12. Turbine Building Area Ventilation System (see Section 9.4.4).
13. Condenser Evacuation System (see Section 10.4.2).
14. Turbine Gland Sealing System (see Section 10.4.3).
15. Steam Dump System (see Section 10.4.4).

16. Fire Protection System (see Section 9.5.1).

17. Turbine Generator (see Section 10.2).

18. Extraction Steam System (see Section 10.4.10).

19. Feedwater System (upstream of isolation valve) (see Section 10.4.7).

20. Steam Generator Blowdown System (see Section 10.4.8).

21. Auxiliary Steam System (see Section 10.4.13).

22. Main Steam System (downstream from first moment restraint after the isolation valve) (see Section 10.3.3).

23. Turbine Oil Purification System (see Section 10.4.14).

The following balance of plant control systems are not required for safety but have an interface with the protective system. The protective system function is assured for the following reasons:

1. Turbine Control System (see Section 7.2).

2. Plant Computer System

   The Plant Computer System provides monitoring of various processes to the Control Room. The Plant Computer provides secondary support functions to enhance operation efficiency and traceability of operational events.

   The Plant Computer System has also integrated the Emergency Response Facility (ERF) computer functions. ERF computer performs continuous monitoring and evaluation of selected process parameters for display in the Control Room, TSC, and EOF. Included with the ERF functions are the Safety Parameter Display System (SPDS) and the Emergency Response Data System (ERDS). The SPDS provides displays of plant parameters from which the safety status of operation may be assessed in the emergency facilities. The ERDS processes and transmits to the NRC timely and accurate data on a limited set of parameters whose values indicate the condition of the plant following a declaration of an alert or higher emergency classification. Further description of the ERF computer can be found in FSAR TMI Response sections I.D.2, III.A.1.2, and III.A.2.

   The plant computer will monitor the status of the protection system. The inputs to the computer will be designed in accordance with the isolation criteria of the Institute of Electrical and Electronics Engineers (IEEE) Standard 279-1971.

   The terms “plant computer”, “Plant Computer System”, “PCS”, “process computer”, “plant process computer”, “Emergency Response Facility Computer System”, “ERFCS”, “PCS workstation”, “Satellite Display System”, “SDS”, “TSC Data Display”, “EOF Data Display”, “Safety Parameter Display System”, “SPDS Computer System”, “SPDS display”, “SPDS”, and combinations of these terms refer to functions that are performed by digital computers and their associated data acquisition systems (DAS), analog-to-digital

conversion equipment, alpha-numeric and/or graphical video display consoles, printers, keyboards, mass storage devices, telecommunications devices, and other peripheral. Except where specifically described, the use of these separate terms does not imply that separate, dedicated computer systems or equipment exist for the function associated with a particular term. The specific equipment used to accomplish particular functions is described in the associated Design Basis Documents (DBDs) which provide the engineering bases for the plant installed equipment.

3. Main feedwater pump control (see Section 7.7.1.7).

   The tripping of the main feedwater pump interacts with the protection system by starting the motor driven auxiliary feedwater pumps. The circuit is designed in accordance with IEEE Standard 279-1971 with the exception that the sensors and associated wiring being located by necessity in non-Seismic Category I structures will not be seismically qualified.

4. Blowdown and sample controls

   The blowdown and sample isolation valves from the secondary side at the steam generator are closed on the start of the auxiliary feedwater pumps from the protection system. These actuation signals are designed in accordance with IEEE Standard 279-1971. Following a start of the Auxiliary Feedwater System, the sample system isolation valves may be reopened from the sample room by manually blocking the closing signal. This capability permits identification, and subsequent isolation, of a steam generator responsible for fission product transfer from the primary to the secondary system. Similarly, the blowdown system isolation valves may be reopened from the Control Room by manually blocking the closing signal, thereby enabling blowdown to the processing system.

7.7.1.12 Control Room Operating Console

The Control Room is the central operating location in a nuclear power station. It is from the Control Room that the plant operators control the NSSS, the turbine-generator systems, the auxiliary systems and the emergency systems. The Control Room operating console layout is a functional layout, based on operators' ease in relating the control board devices to the plant equipment and being able to determine, at a glance, the status of the equipment.

Figure 7.7-14A shows the arrangement of the Control Room panel. Figure 7.7-14 shows the functional layout of the control console. The chemical and volume control, nuclear instrumentation and rod control systems, systems that require almost continuous supervision during plant normal operation, are in the center of the "U" shaped consoles. Next to those systems are the pressurizer and main steam control systems. Far from the center are the auxiliary equipment that requires little supervision or will go in service automatically under emergency or accident conditions.

Due to the functional layout of the Control Room console, it is not possible to locate redundant controls in separate panels. Separation of redundant, safety-related controls is achieved by either a fire retardant material barrier or by maintaining a minimum 6 inch air separation. The Control Room horseshoe areas encompass two-tiered operating platforms that support operation of Plant Computer, Radiation Monitoring and Communication equipment.

The control switches and associated lights are furnished in enclosed metallic boxes (modules), which provide for physical protection of the switch, lights and internal wiring and for separation requirements between safety-related switches of redundant trains.

Redundant circuits which serve the same protective function enter the control board through separate apertures and terminate in separate terminal blocks.

Intra-panel wiring separation is achieved by providing 6 inch air separation between redundant train wiring or by separate enclosed metal wireways for redundant trains.

7.7.2 ANALYSIS

The plant control systems are designed to assure high reliability in any anticipated operational occurrences. Equipment used in these systems is designed and constructed with a high level of reliability.

Proper positioning of the control rods is monitored in the Control Room by bank arrangements of the individual position columns for each rod cluster control assembly. A rod deviation alarm alerts the operator of a deviation of one rod cluster control assembly from the other rod in that bank position. There are also insertion limit monitors with visual and audible annunciation. A rod bottom alarm signal is provided to the Control Room for each full length rod cluster control assembly. Four excore long ion chambers also detect asymmetrical flux distribution indicative of rod misalignment.

Overall reactivity control is achieved by the combination of soluble boron and rod cluster control assemblies. Long term regulation of core reactivity is accomplished by adjusting the concentration of boric acid in the reactor coolant. Short term reactivity control for power changes is accomplished by the Rod Control System which automatically moves rod cluster control assemblies. This system uses input signals including neutron flux, coolant temperature, and turbine load.

The axial core power distribution is controlled by moving the control rods through changes in RCS boron concentration. Adding boron causes the rods to move out thereby reducing the amount of power in the bottom of the core, this allows power to redistribute toward the top of the core. Reducing the boron concentration causes the rods to move into the core thereby reducing the power in the top of the core, the result redistributes power towards the bottom of the core.

The plant control systems will prevent an undesirable condition in the operation of the plant that, if reached, will be protected by reactor trip. The description and analysis of this protection is covered in Section 7.2. Worst case failure modes of the plant control systems are postulated in the analysis of off-design operational transients and accidents covered in Chapter 15 such as the following:

1. Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low power startup condition.

2. Uncontrolled rod cluster control assembly bank withdrawal at power.

3. Rod cluster control assembly misalignment.

4. Loss of external electrical load.

5. Turbine trip.

6. Loss of all non-emergency AC power to the station auxiliaries.

7. Feedwater system malfunctions that result in a decrease in feedwater temperature.

8. Excessive increase in secondary steam flow.

9. Inadvertent opening of a steam generator relief or safety valve.

These analyses show that a reactor trip setpoint is reached in time to protect the health and safety of the public under those postulated incidents and that the resulting coolant temperatures produce a DNBR well above the limit value. Thus, there is no cladding damage and no release of fission products to the RCS under the assumption of these postulated worst case failure modes of the Plant Control System.

7.7.2.1 Separation of Protection and Control System

In some cases, it is advantageous to employ control signals derived from individual protection channels through isolation amplifiers contained in the protection channel. As such, a failure in the control circuitry does not adversely affect the protection channel. Test results have shown that a short circuit or the application (credible fault voltage from within the cabinets) of 118 volt AC or 140 volt DC on the isolated output portion of the circuit (nonprotection side of the circuit) will not affect the input (protection) side of the circuit.

Where a single random failure can cause a control system action that results in a generating station condition requiring protective action and can also prevent proper action of a protection system channel designed to protect against the condition, the remaining redundant protection channels are capable of providing the protective action even when degraded by a second random failure. This meets the applicable requirements of Section 4.7 of IEEE Standard 279-1971.

The pressurizer pressure channels needed to derive the control signals are electrically isolated from control.

7.7.2.2 Response Considerations of Reactivity

Reactor shutdown with control rods is completely independent of the control functions since the trip breakers interrupt power to the full length rod drive mechanisms regardless of existing control signals. The design is such that the system can withstand accidental withdrawal of control groups or unplanned dilution of soluble boron without exceeding acceptable fuel design limits. The design meets the requirements of the 1971 General Design Criterion 25.

No single electrical or mechanical failure in the Rod Control System could cause the accidental withdrawal of a single rod cluster control assembly from the partially inserted bank at full power operation. The operator could deliberately withdraw a single rod cluster control assembly in the control bank; this feature is necessary in order to retrieve a rod, should one be accidentally dropped. In the extremely unlikely event of simultaneous electrical failures which could result in

single rod cluster control assembly withdrawal, rod deviation would be displayed on the plant annunciator, and the individual rod position readouts would indicate the relative positions of the rods in the bank. Withdrawal of a single rod cluster control assembly by operator action, whether deliberate or by a combination of errors, would result in activation of the same alarm and the same visual indications.

Each control bank and shutdown bank A and B is divided into two groups (group 1 and group 2) and shutdown banks C, D, and E have one group (group 1). Each group can have up to four mechanisms each. The rods comprising a group operate in parallel through multiplexing thyristors. The two groups in a bank move sequentially such that the first group is always within one step of the second group in the bank. The group 1 and group 2 power circuits are installed in different cabinets as shown in Figure 7.7-15, which also shows that one group is always within one step (5/8 inch) of the other group. A definite schedule of actuation or deactuation of the stationary gripper, moveable gripper, and lift coils of a mechanism is required to withdraw the rod cluster control assembly attached to the mechanism. Since the four stationary grippers, moveable grippers, and lift coils associated with the rod cluster control assemblies of a rod group are driven in parallel, any single failure which could cause rod withdrawal would affect a minimum of one group of rod cluster control assemblies. Mechanical failures are in the direction of insertion, or immobility.

Figure 7.7-16 is provided for a discussion of design features that assure that no single electrical failure could cause the accidental withdrawal of a single rod cluster control assembly from the partially inserted bank at full power operation.

The Figure 7.7-16 shows the typical parallel connections on the lift, movable and stationary coils for a group of rods. Since single failures in the stationary or movable circuits will result in dropping or preventing rod (or rods) motion, the discussion of single failure will be addressed to the lift coil circuits. 1) Due to the method of wiring the transformers which turn on the lift coil multiplex thyristors, three of the four thyristors in a rod group could remain turned off when required to turn on, if for example the gate signal lead failed open at point X_1. Upon “up” demand, one rod in group 1 and 4 rods in group 2 would withdraw. A second failure at point X_2 in the group 2 circuit is required to withdraw one rod cluster control assembly; 2) Timing circuit failures will affect the four mechanisms of a group or the eight mechanisms of the bank and will not cause a single rod withdrawal; 3) More than two simultaneous component failures are required (other than the open wire failures) to allow withdrawal of a single rod.

The identified multiple failure involving the least number of components consists of open circuit failure of the proper two out of 16 wires connected to the gate of the lift coil thyristors. The probability of open wire (or terminal) failure is 0.016×10^{-6} per hour by MIL-HDB-217A. These wire failures would have to be accompanied by failure, or disregard, of the indications mentioned above. The probability of this occurrence is therefore too low to have any significance.

Concerning the human element, to erroneously withdraw a single rod cluster control assembly, the operator would have to improperly set the bank selector switch, the lift coil disconnect switches, and the “in-hold-out” lever. In addition, the three indications would have to be disregarded or ineffective. Such series of errors would require a complete lack of understanding and administrative control. A probability number cannot be assigned to a series of errors such as these.

The DRPIS provides direct visual displays of each control rod assembly position. The plant computer alarms for deviation of rods from their banks. In addition a rod insertion limit monitor provides an audible and visual alarm to warn the operator of an approach to an abnormal condition due to dilution. The low-low insertion limit alarm alerts the operator to follow emergency boration procedures. The facility reactivity control systems are such that acceptable fuel damage limits will not be exceeded even in the event of a single malfunction of either system.

An important feature of the Control Rod System is that insertion is provided by gravity fall of the rods.

In all analyses involving reactor trip, the single, highest worth rod cluster control assembly is postulated to remain untripped in its full out position.

One means of detecting a stuck control rod assembly is available from the actual rod position information displayed on the control board. The control board position readouts, one for each full length rod, give the plant operator the actual position of the rod in steps. The indications are grouped by banks (e.g., control bank A, control bank B, etc.) to indicate to the operator the deviation of one rod with respect to other rods in a bank. This serves as a means to identify rod deviation.

The plant computer monitors the actual position of all rods. Should a rod be misaligned from the other rods in that bank by more than 12 steps, the rod deviation alarm is actuated.

Misaligned rod cluster control assemblies are also detected and alarmed in the Control Room via the flux tilt monitoring system which is independent of the plant computer.

Isolated signals derived from the Nuclear Instrumentation System are compared with one another to determine if a preset amount of deviation of average power level has occurred. Should such a deviation occur, the comparator output will operate a bistable unit to actuate a control board annunciator. This alarm will alert the operator to a power imbalance caused by a misaligned rod. By use of individual rod position readouts, the operator can determine the deviating control rod and take corrective action. The design of the plant control systems meets the requirements of the 1971 General Design Criterion 23.

Refer to Section 4.3 for additional information on response considerations due to reactivity.

7.7.2.3 Step Load Changes Without Steam Dump

The Rod Control System restores equilibrium conditions, without a trip, following a plus or minus 10 percent step change in load demand, over the 15 to 100 percent power range for automatic control. Steam dump is blocked for load decrease less than or equal to 10 percent.

The Rod Control System minimizes the reactor coolant average temperature deviation during the transient within a given value and restores average temperature to the programmed setpoint. Excessive pressurizer pressure variations are prevented by using spray and heaters and power relief valves in the pressurizer.

The control system must limit nuclear power overshoot to acceptable values following a 10 percent increase in load to 100 percent.

7.7.2.4 Loading and Unloading

Ramp loading and unloading of 5 percent per minute can be accepted over the 15 to 100 percent power range under automatic control without tripping the plant. The function of the control system is to maintain the coolant average temperature as a function of turbine-generator load.

The coolant average temperature increases during an unloading transient and causes a continuous insurge to the pressurizer as a result of coolant expansion. The sprays limit the resulting pressure increase. Conversely, as the coolant average temperature is decreasing during a loading transient, there is a continuous outsurge from the pressurizer resulting from coolant contraction. The pressurizer heaters limit the resulting system pressure decrease. The pressurizer water level is programmed such that the water level is above the setpoint for heater cut out during the loading and unloading transients. The primary concern during loading is to limit the overshoot in nuclear power and to provide sufficient margin in the Overtemperature N-16 setpoint.

The automatic load controls are designed to adjust the unit generation to match load requirements within the limits of the unit capability and licensed rating.

During rapid loading transients, a drop in reactor coolant temperature is sometimes used to increase core power. This mode of operation is applied when the control rods are not inserted deeply enough into the core to supply all the reactivity requirements of the rapid load increase (the boron control system is relatively ineffective for rapid power changes). The reduction in temperature is initiated by continued turbine loading past the point where the control rods are completely withdrawn from the core. The temperature drop is recovered and nominal conditions restored by a boron dilution operation.

Excessive drops in coolant temperature are prevented by interlock C-16. This interlock circuit monitors the auctioneered low coolant temperature indications and the programmed reference temperature which is a function of turbine first stage pressure and causes a turbine loading stop when the temperature difference reaches the setpoint.

The core axial power distribution is controlled during the reduced temperature return to power by placing the control rods in the manual mode when the operating limits are approached. Placing the rods in manual will stop further changes in axial flux imbalance due to rod motion and it will also initiate the required drop in coolant temperature. Normally power distribution control is not required during a rapid power increase and the rods will proceed, under the automatic rod control system, to the top of the core. The bite position is reestablished at the end of the transient by decreasing the coolant boron concentration.

7.7.2.5 Load Rejection Furnished by the Steam Dump System

When a load rejection occurs, if the difference between the required temperature setpoint of the RCS and the actual average temperature exceeds a predetermined amount, a signal will actuate the steam dump to maintain the RCS temperature within control range until a new equilibrium condition is reached.

The reactor power is reduced at a rate consistent with the capability of the Rod Control System. Reduction of the reactor power is automatic. The steam dump flow reduction is as fast as rod cluster control assemblies are capable of inserting negative reactivity.

The Rod Control System can then reduce the reactor temperature to a new equilibrium value without causing overtemperature and/or overpressure conditions. The design steam dump steam flow capacity is 40 percent of full load steam flow at full load steam pressure.

The steam dump flow reduces proportionally as the control rods act to reduce the average coolant temperature. The artificial load is therefore removed as the coolant average temperature is restored to its programmed equilibrium value.

The dump valves are modulated by the reactor coolant average temperature signal. The required number of steam dump valves can be tripped quickly to stroke full open or modulate, depending upon the magnitude of the temperature error signal resulting from loss of load.

7.7.2.6 Turbine-Generator Trip with Reactor Trip

Whenever the turbine-generator unit trips at an operating power level above 50 percent power, the reactor also trips. The unit is operated with a programmed average temperature as a function of load, with the full load average temperature significantly greater than the equivalent saturation pressure of the steam generator safety valve setpoint. The thermal capacity of the RCS is greater than that of the secondary system, and because the full load average temperature is greater than the no-load temperature, a heat sink is required to remove heat stored in the reactor coolant to prevent actuation of steam generator safety valves for a trip from full power. This heat sink is provided by the combination of controlled release of steam to the condenser and by makeup of feedwater to the steam generators.

The Steam Dump System is controlled from the reactor coolant average temperature signal whose setpoint values are programmed as a function of turbine load. Actuation of the steam dump is rapid to prevent actuation of the steam generator safety valves. With the dump valves open, the average coolant temperature starts to reduce quickly to the no-load setpoint. A direct feedback of temperature acts to proportionally close the valves to minimize the total amount of steam which is bypassed.

Following the turbine trip, the feedwater flow is cut off when the average coolant temperature decreases below a given temperature or when the steam generator water level reaches a given high level.

Additional feedwater makeup is then controlled manually to restore and maintain steam generator water level while assuring that the reactor coolant temperature is at the desired value. Residual heat removal is maintained by the steam header pressure controller (manually selected) which controls the amount of steam flow to the condensers. This controller operates a portion of the same steam dump valves to the condensers which are used during the initial transient following turbine and reactor trip.

The pressurizer pressure and level fall rapidly during the transient because of coolant contraction. The pressurizer water level is programmed so that the level following the turbine and reactor trip is above the heaters. However, if the heaters become uncovered following the trip, the Chemical and Volume Control System will provide full charging flow to restore water level in the pressurizer. Heaters are then turned on to restore pressurizer pressure to normal.

The steam dump and feedwater control systems are designed to prevent the average coolant temperature from falling below the programmed no load temperature following the trip to ensure adequate reactivity shutdown margin.

REFERENCES

1. Lipchak, J. B. and Stokes, R. A., “Nuclear Instrumentation System,” WCAP-8255, January 1974.

2. WCAP-12472-P-A, “BEACON Core Monitoring and Operations Support System,” August 1994.

TABLE 7.7-1
PLANT CONTROL SYSTEM INTERLOCKS

(Sheet 1 of 2)

Designation	Derivation	Function
C-1	1/2 neutron flux (intermediate range) above setpoint	Blocks automatic and manual control rod withdrawal
C-2	1/4 neutron flux (power range) above setpoint	Blocks automatic and manual control rod withdrawal
C-3	2/4 Overtemperature N-16 above setpoint	Blocks automatic and manual control rod withdrawal
		Actuates turbine runback via load reference
		Defeats remote load dispatching (if remote load dispatching is used)
C-4	2/4 Overpower N-16 above setpoint	Blocks automatic and manual control rod withdrawal
		Actuates turbine runback via load reference
		Defeats remote load dispatching (if load dispatching is used)
C-5	1/1 turbine impulse chamber pressure below setpoint	Defeats remote load dispatching (if load dispatching is used)
		Blocks automatic control rod withdrawal
C-7	1/1 time derivative (absolute value) of turbine impulse chamber pressure (decrease only) above setpoint	Makes steam dump valves available for either tripping or modulation
C-8	Turbine trip, 2/3 turbine emergency trip fluid pressure below setpoint	Input signal to non-NSSS turbine/ generator trip logic
	or	
	4/4 turbine valves closed	

TABLE 7.7-1
PLANT CONTROL SYSTEM INTERLOCKS

(Sheet 2 of 2)

Designation	Derivation	Function
C-9	Any condenser pressure above setpoint, or 3/4 circulation water pump breakers open	Blocks steam dump to condenser
C-11	1/1 bank D control rod position above setpoint	Blocks automatic rod withdrawal
C-16	Reduced limit in coolant temperature above normal setpoint	Stops automatic turbine loading until condition clears
C-20	2/2 Turbine Impulse Chamber Pressure	Arms AMSAC; below setpoint blocks AMSAC (See FSAR Section 7.8)

TABLE 7.7-2
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7.8 ATWS MITIGATION SYSTEM ACTUATION CIRCUITRY (AMSAC)

7.8.1 DESCRIPTION

7.8.1.1 System Description

The ATWS (Anticipated Transient Without Scram) Mitigation System Actuation Circuitry (AMSAC) provides a backup to the Reactor Trip System (RTS) and ESF Actuation System (ESFAS) for initiating turbine trip and auxiliary feedwater flow in the event an anticipated transient results (e.g., the complete loss of main feedwater). The AMSAC is independent of and diverse from the Reactor Trip System and the ESF Actuation System with the exception of the analog steam generator level and turbine first stage pressure inputs, and the final actuation devices. It is a highly-reliable, microprocessor-based, non-safety related circuitry.

The AMSAC continuously monitors level in the steam generators, which is an anticipatory indication of a loss of heat sink. AMSAC initiates certain functions when the level drops below a predetermined setpoint for a preselected time and for three of the four steam generator levels. These initiated functions are turbine trip, auxiliary feedwater initiation, and steam generator blowdown and sample lines isolation.

The AMSAC is designed to be highly reliable, resistant to inadvertent actuation, and easily maintained. Reliability is assured through the use of internal redundancy and continual self-testing within the system. Inadvertent actuations are minimized through the use of internal redundancy and majority voting at the output stage of the system. The time delay on low steam generator level and the coincidence logic also minimize inadvertent actuations.

The AMSAC automatically performs its actuations when the plant is above a preselected power level which is determined using turbine impulse chamber pressure. This signal remains armed sufficiently long after the pressure drops below the setpoint to ensure that its function will be performed in the event of a turbine trip.

7.8.1.2 Equipment Description

The AMSAC equipment is located in a seismically qualified cabinet.

The design of the AMSAC is based on the industry standard Intel multibus format, which permits the use of various readily available and widely used microprocessor cards on a common data bus for various functions.

The AMSAC consists of the following:

1. Steam Generator (SG) Level

   SG level is measured with four existing differential pressure-type level transmitters, for each of the main steam generators.

2. Turbine First State Pressure

   Turbine First Stage Pressure is measured with two existing pressure transmitters located in the steam supply line near the turbine.

3. System Hardware

The system hardware consists of two primary systems: the Actuation Logic System (ALS) and the Test/Maintenance System (T/MS).

Actuation Logic System

The ALS monitors the analog inputs, performs the required functional logic, provides actuation outputs to trip the turbine and initiate auxiliary feedwater flow, and provides status information to the Test/Maintenance System. The ALS consists of three groups of input/output (I/O) modules, three actuation logic processors (ALPs), two majority voting modules, and two output relay panels. The I/O modules provide signal conditioning, isolation, and test features for interfacing between the ALS and T/MS. Conditioned signals are sent to three identical ALPs for analog-to-digital conversion, setpoint comparison, and coincidence logic performance. Each of the ALPs perform identical logic calculations using the same inputs and derive component actuation demands, which are then sent to the majority voting modules. The majority voting modules perform a two-out-of-three vote on the ALP demand signals. These modules drive the relays providing outputs to the existing turbine trip and auxiliary feedwater initiation circuits. A simplified block diagram of the AMSAC ALS architecture is presented in Figure 7.8-1.

Test/Maintenance System

The Test/Maintenance System provides the AMSAC with automated and manual testing as well as a maintenance mode. Automated testing is the continuously performed self-check done by the system during normal operation. ALS status is monitored by the T/MS and sent to the plant computer and the main control board. Manual testing of the system by the maintenance staff can be performed on-line to provide assurance that the ALS system is fully operational. The maintenance mode permits the maintenance staff, under administrative control, to modify channel setpoints, channel status and timer values, and initiate channel calibration.

The T/MS consists of a test/maintenance processor, a digital-to- analog conversion board, a memory board, expansion boards, a self-health board, digital output modules, a test/maintenance panel, and a portable terminal/printer.

4. Equipment Actuation

The output relay panels provide component actuation signals through the isolation relays. These signals drive the final actuation circuitry for auxiliary feedwater initiation and turbine trip. Existing actuation devices of the component are used.

7.8.1.3 Functional Performance Requirements

The AMSAC automatically initiates auxiliary feedwater, trips the turbine, and isolates steam generator blowdown and sampling lines. Analyses have shown that the most limiting ATWS event is a loss of feedwater event without a reactor trip. Therefore, AMSAC performs the following mitigative actuations:

1. Ensures a secondary heat sink following an anticipated transient (ANS Condition II) without a reactor trip.

2. Limits core damage following an anticipated transient without a reactor trip.

3. Ensures that the energy generated in the core is compatible with the design limits to protect the reactor coolant pressure boundary by maintaining the reactor coolant pressure to within ASME Stress Level C.

7.8.1.4 AMSAC Interlocks

A single interlock, designated as C-20 (See Table 7.7-1), is provided to allow for the automatic arming and blocking of the AMSAC. The system is blocked at sufficiently low reactor power levels when the actions taken by the AMSAC following an ATWS need not be automatically initiated. Turbine impulse chamber pressure in a two-out-of-two logic scheme is used for this permissive. Turbine impulse chamber pressure above the setpoint will automatically defeat any block, i.e., will arm the AMSAC. Dropping below this setpoint will automatically block the AMSAC. Removal of the C-20 permissive is automatically delayed for a predetermined time. The operating status of the AMSAC is displayed on the main control board.

7.8.1.5 Steam Generator Level Sensor Arrangement

SG level is determined by a differential pressure transmitter, measuring the level drop in the steam generator. These SG level signals are used as inputs to the AMSAC and are isolated signals from the Process Protection Cabinets.

7.8.1.6 Turbine Impulse Chamber Pressure Arrangement

Turbine impulse chamber pressure is determined by a pressure transmitter, measuring the pressure rise in the turbine. These pressure signals are used as input into AMSAC and are isolated signals from the Process Protection Cabinets.

7.8.1.7 Trip System

The differential pressure that is measured in the steam generator is used by the AMSAC to determine trip demand. Signal conditioning is performed on the transmitter output and used by each of the ALPs to derive a component actuation demand. If three of the four steam generators have a low level at a power level greater than the C-20 permissive, then a trip demand signal is generated. This signal drives output relays for performing the necessary mitigative actions.

7.8.1.8 Isolation Devices

AMSAC is independent of the Reactor Trip (RTS) and Engineered Safety Features Actuation Systems (ESFAS). The AMSAC inputs for measuring turbine impulse chamber pressure and narrow range steam generator water level are derived from existing transmitters and channels within the Process Protection System. Connections to these channels are made downstream of Class 1E isolation devices which are located within the Process Protection Cabinets. These isolation devices ensure that the existing protection system continues to meet all applicable safety criteria. Buffering of the AMSAC outputs from the safety related final actuation device circuits is achieved through Class 1E isolation relays located within the AMSAC cabinet. A

credible fault occurring in the nonsafety-related AMSAC will not propagate through and degrade the RTS and ESFAS.

7.8.1.9 AMSAC Diversity from the Reactor Trip and Engineered Safety Features Actuation System

The AMSAC utilizes equipment diverse from the RTS and ESFAS to prevent common mode failures that might affect the AMSAC and the RTS or ESFAS. The AMSAC is a digital, microprocessor-based system with the exception of the analog SG level and turbine impulse pressure transmitter inputs. The reactor trip system utilizes an analog based protection system. Also where similiar components are utilized for the same function in both AMSAC and the reactor trip system, the components used in AMSAC are provided from a different manufacturer.

Common mode failure of identical components in the analog portion of the Reactor Protection System (RPS) that could result in the inability of the system to generate a reactor trip signal, will not impact the ability of the digital AMSAC to generate its required mitigative functions. Similarly, a postulated common mode failure affecting analog components in ESFAS, which could affect its ability to initiate auxiliary feedwater, will not impact the ability of the digital based AMSAC to automatically initiate auxiliary feedwater.

7.8.1.10 Power Supply

The AMSAC power supply is a non-Class 1E Uninterruptable AC Power Supply (UPS) source, and battery backed. The AMSAC is an energize-to-actuate system capable of performing its mitigative functions with a loss of offsite power.

7.8.1.11 Environmental Variations

AMSAC equipment is not designed as safety-related equipment with the exception of the output isolation relays. Therefore, AMSAC is not fully required to be qualified as safety related. The AMSAC equipment is located in a controlled environment such that variations in the ambient conditions are minimized. No AMSAC equipment is located inside containment. The transmitters (steam generator level and turbine impulse chamber pressure) that supply the input into AMSAC are located inside containment and the turbine building, respectively.

7.8.1.12 Setpoints

The AMSAC makes use of two setpoints in the coincidence logic to determine if mitigative functions are required. Water level in each steam generator is sensed to determine if a loss of secondary heat sink is imminent. The low level setpoint is selected in such a manner that a true lowering of the level will be detected by the system. The normal small variations in steam generator level will not result in a spurious AMSAC signal.

The C-20 permissive setpoint is selected in order to be consistent with ATWS investigations showing that the mitigative actions performed by the AMSAC need not be automatically actuated below a certain power level. The maximum allowable value of the C-20 permissive setpoint is defined by these investigations.

To avoid inadvertent AMSAC actuation on the complete loss of main feedwater, AMSAC actuation is delayed by a preselected time. This will ensure the Reactor Trip System will provide the first trip signal.

To ensure that the AMSAC remains armed sufficiently long to permit its function in the event of a turbine trip, the C-20 permissive is maintained for a preset time delay after the turbine impulse chamber pressure drops below the setpoint.

The AMSAC setpoints and setpoint modifications are administratively controlled.

7.8.2 ANALYSIS

7.8.2.1 Safety Classification/Safety-Related Interface

The AMSAC is not a safety-related system and therefore need not meet the requirements of IEEE 279-1971. The AMSAC has been implemented such that the Reactor Trip System and the ESF Actuation System continue to meet all applicable safety-related criteria. The AMSAC is independent of the RTS and ESFAS. The isolation provided, between the RTS and the AMSAC and between the ESFAS and the AMSAC, by the isolator modules and the isolation relays ensures that the applicable safety-related criteria are met for the RTS and the ESFAS.

7.8.2.2 Redundancy

Since AMSAC is a backup non-safety related system to the redundant RPS, redundancy is not required. To ensure high system reliability, portions of the AMSAC have been implemented as internally redundant, such that a single failure of an input channel or ALP will neither actuate not prevent actuation of the AMSAC.

7.8.2.3 Diversity from the Existing Trip System

Diverse equipment has been selected in order that common mode failures affecting both the RTS and the AMSAC or both the ESFAS and the AMSAC will not render these systems inoperable simultaneously. A more detailed discussion of the diversity, between the RTS and the AMSAC and between the ESFAS and the AMSAC, is presented in Section 7.8.1.9.

7.8.2.4 Electrical Independence

The AMSAC is electrically independent of the RTS and ESFAS from the sensor output up to but not including the final actuation devices. Isolation devices are provided to isolate the non-safety AMSAC circuitry from the safety-related input sensors and actuation circuits of the auxiliary feedwater system.

7.8.2.5 Physical Separation from the RTS and ESFAS

AMSAC is physically separated from the existing protection system hardware. The AMSAC outputs are provided from separate relay panels within the cabinets. The three trains (non-1E Train C, 1E Train A, and 1E Train B) are separated within the AMSAC cabinet by a combination of metal barriers, conduit and distance.

7.8.2.6 Environmental Qualification

Equipment related to the AMSAC is qualified to operate under conditions resulting from anticipated operational occurrences for the respective equipment location. The AMSAC equipment, with the exception of the isolation devices, located outside containment in a mild environment follows the same design standard that currently exists for non-class 1E equipment.

7.8.2.7 Seismic Qualification

It is required that only the isolation devices comply with seismic qualification. The AMSAC output isolation device which is located within the AMSAC cabinet, is qualified in accordance with the requirements of IEEE Standard 344-1975, “IEEE Standard for Seismic Qualification of Class 1E Electrical Equipment for Nuclear Power Generating Stations”.

7.8.2.8 Test, Maintenance, and Surveillance Quality Assurance

The quality assurance requirements for testing, maintenance, and surveillance of the AMSAC are, at a minimum, consistent with plant procedures for quality related equipment. Specific quality assurance requirements are defined by the procurement specification for inspections, examinations, storage, shipment, and testing of the AMSAC.

Acceptance testing and program validation of the AMSAC was completed prior to shipment. Functional testing of the system was conducted as part of the preoperational test program. Periodic testing will be performed both automatically through the use of the system automatic self-checking capability, and manually via the AMSAC test/maintenance panel in accordance with plant procedures.

7.8.2.9 Testability at Power

The AMSAC is testable at power. This testing is done via the system test/maintenance panel. The capability of the AMSAC to perform its mitigative actuations is bypassed at a system level while in the test mode. Total system testing is performed as a set of three sequential, partial, overlapping tests. The first of the tests checks the analog input portions of the AMSAC in order to verify accuracy. Each of the analog input modules is checked separately. The second test checks each of the ALPs to verify that the appropriate coincidence logic is sent to the majority voter. Each ALP is tested separately. The last test exercises the majority voter and the integrity of the associated output relays. The majority voter and associated output relays are tested by exercising all possible input combinations to the majority voter. The integrity of each of the output relays is checked by confirming continuity of the relay coils without operating the relays. The capability to individually operate the output relays confirm the integrity of the associated field wiring. Operation of the corresponding isolation relays and final actuation devices at plant shutdown is provided.

7.8.2.10 Inadvertent Actuation

The AMSAC is designed such that the frequency of inadvertent actuations is minimized. This high reliability is ensured through use of three redundant ALPs and a majority voting module. A single failure in any of these modules will not result in a spurious AMSAC actuation. In addition, the three-out-of-four low steam generator level coincidence logic and time delay minimize the potential for inadvertent actuations.

7.8.2.11 Maintenance Bypasses

The AMSAC is blocked at the system level during maintenance, repair, calibration or testing. While the system is blocked, the bypass conditions is continuously indicated on the main control board.

7.8.2.12 Operating Bypasses

The AMSAC has been designed to allow for operational bypasses with the inclusion of the C-20 permissive. Above the C-20 setpoint, the AMSAC is automatically unblocked (i.e., armed); below the setpoint, the system is automatically blocked. The operating status of the AMSAC is continuously indicated on the main control board via an annunciator window.

7.8.2.13 Indication of Bypasses

Whenever the mitigative capabilities of the AMSAC are bypassed or deliberately rendered inoperable, this condition is continuously indicated on the main control board. In addition to the operating bypass, any manual maintenance bypass is indicated via the AMSAC general warning that is sent to the main control board.

7.8.2.14 Means for Bypassing

A permanently installed system bypass selector switch is provided to bypass the system. This is a two-position selector switch with “NORMAL” and “BYPASS” positions. At no time is it necessary to use any temporary means, such as installing jumpers or pulling fuses, to bypass the system.

7.8.2.15 Completion of Mitigative Actions Once Initiated

The AMSAC mitigative actions are initiated when the coincidence logic is satisfied and the time delay requirements are met. If the flow in the feedwater lines is re-initiated before the timer expires and the SG water level increases to above the low low setpoint, then the coincidence logic will no longer be satisfied and the actuation signal disappears. If the coincidence logic conditions are maintained for the duration of the time delay, then the mitigative actions are initiated. The auxiliary feedwater initiation signal is latched in at the component actuating devices and the turbine trip is latched at the turbine electro-hydraulic control system. Deliberate operator action is then necessary to terminate auxiliary feedwater flow, clear the turbine trip signal using the main control board turbine trip reset switch, and proceed with the reopening of the turbine stop valves.

7.8.2.16 Manual Initiation

Manual initiation of the AMSAC is not provided. The capability to initiate the AMSAC mitigative functions manually, i.e., initiate auxiliary feedwater, trip the turbine, and isolate steam generator blowdown and sampling lines, exists at the main control board.

7.8.2.17 Information Readout

The AMSAC has been designed such that the operating and maintenance staffs have accurate, complete and timely information pertinent to the status of the AMSAC. A system level general

warning alarm is indicated in the control room. Diagnostic capability exists from the test/maintenance panel to determine the cause of any unanticipated inoperability or deviation.

7.8.3 COMPLIANCE WITH STANDARDS AND DESIGN CRITERIA

The AMSAC meets the applicable requirements of Part 50.62 of Title 10 of the Code of Federal Regulations and the quality assurance requirements of NRC Generic Letter 85-06.

8.0 ELECTRIC POWER

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8.1 INTRODUCTION

8.1.1 UTILITY GRID DESCRIPTION

The Texas Reliability Entity (TRE) is one of 8 regional reliability councils in the North American Electric Reliability Corporation (NERC). As a NERC member, the primary responsibility of the TRE is to facilitate reliable power grid operations in the Electric Reliability Council of Texas (ERCOT) region. ERCOT is the Independent System Operator (ISO) of the ERCOT Transmission Grid (Grid).

ERCOT is located entirely within Texas; consequently, the Public Utility Commission of Texas (PUCT) is ERCOT's principal regulatory authority. A Board of Directors, made up of members from each of ERCOT's electricity market groups, together with independent members, governs ERCOT.

ERCOT oversees the operation of the Grid and manages the activities related to the electric industry. ERCOT is the regional security coordinator for the ERCOT Grid and is responsible for operating the ERCOT Grid and the ERCOT Control Area in compliance with good utility practice, NERC standards, ERCOT Operating Guides and ERCOT Protocols, and individual Transmission/Distribution Service Provider (TDSP) limitations. Transmission Operator (TO) System Operations Center works with ERCOT and carries out dispatch instructions directly or on behalf of ERCOT.

TO is the transmission owner and transmission planner for CPNPP area. TO's transmission system is an integrated part of ERCOT system and TO maintains it's transmission system in conformance with the ERCOT Operating Guides and ERCOT Protocols.

The TO transmission systems consist of 345-kV lines for bulk supply and includes 138-kV and 69-kV lines to transmit power to load-serving substations. Composition of generation sources connected to the TO transmission system include fossil fuel plants (lignite, gas/oil,) and the CPNPP nuclear plant (interconnected).

The CPNPP output is connected to the 345-kV transmission system via the TO's CPNPP 345-kV switchyard. The startup and shutdown power for the units are derived from the 138-kV and 345-kV switchyards as described in Section 8.2.

8.1.2 ONSITE ELECTRIC SYSTEM DESCRIPTION

The onsite electric system includes power supplies, distribution equipment, and instrumentation and control to supply power to the unit auxiliary loads (normal and safety-related) during startup, normal operation, and normal and emergency shutdown. Connection of the generator outputs to the 345-kV switchyard is via isolated-phase bus (generator main leads), step-up transformers, and transmission lines.

Power to the unit 6900-V auxiliary bus systems is furnished through either the unit auxiliary, station service or startup transformers.

Normally, the non-safety-related auxiliaries are supplied by the main generators through the unit auxiliary transformers. These transformers are connected to the main generator leads, between

the generator and main transformers, by means of isolated-phase bus. Safety-related auxiliaries are normally supplied by the preferred offsite power systems.

Two separate and physically independent startup transformers provide startup, preferred and alternate shutdown power to the safety-related auxiliaries of the units on an immediate basis. One transformer is connected to the 345-kV switchyard while the second transformer is connected to the 138-kV switchyard; these transformers are connected to the safety-related 6900-V auxiliary bus systems and, as such, provide two independent means of supplying the safety-related equipment from the offsite power system without relying on the main generator.

Two station service transformers provide power to the non-safety-related auxiliaries. These transformers are connected to the 345-kV switchyard. One transformer is connected to the non-safety-related 6900-V auxiliary buses of one unit while the second transformer is connected to the non-safety-related 6900-V buses of the other unit.

In addition, the 25-kV Plant Support Power Loop, fed from the 138-kV switchyard, supplies power to non-safety-related equipment. The 25-kV Plant Support Power Loop also supplies alternate power (through manual transfer switches) to certain essential equipment in plant Modes 5 and 6 during safety-related bus outages.

Upon loss of all Offsite AC power, station standby power sources, consisting of four diesel generators (two per unit) are provided to satisfy the loading requirements of the AC safety-related loads. System redundancy precludes loss of all onsite power as a result of any single failure.

The DC safety-related loads of each unit are supplied by four independent Class 1E 125-VDC battery systems, which are divided into two redundant trains. The DC loads that are non-safety-related receive power through independent 125/250-VDC, 125-VDC, and 24/48-VDC battery systems. Switchyard DC load requirements are supplied by independent systems located in each switchyard. The AC and DC safety-related loads of each unit are divided into redundant load groups, each energized from an independent emergency power supply. There are no interconnections between redundant load groups.

8.1.3 SAFETY-RELATED LOADS

The safety-related loads that require electric power to perform their safety functions are identified in Table 8.1-1. This table includes the safety load, safety functions performed, and type of electric power (AC or DC, or both).

8.1.4 DESIGN CRITERIA

The design bases, criteria, NRC regulatory guides, standards, and other documents that are implemented in the design of the safety-related systems (as discussed in Sections 8.2 and 8.3) are listed as follows:

8.1.4.1 United States Code of Federal Regulations Title 10, Energy (10 CFR Part 50)

10 CFR Part 50, Appendix A, General Design Criteria for Nuclear Power Plants (2/4/72)

8.1.4.2 Nuclear Regulatory Commission

1. NRC Regulatory Guide 1.6, Independence Between Redundant Standby (Onsite) Power Sources and Between Their Distribution Systems (3/10/71)

2. NRC Regulatory Guide 1.9, Selection of Diesel Generator Set Capacity for Standby Power Supplies (3/10/71)

3. NRC Regulatory Guide 1.22, Periodic Testing of Protection System Actuation Functions (2/17/72)

4. NRC Regulatory Guide 1.29, Seismic Design Classification (Revision 2, 2/76)

5. NRC Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment (8/11/72)

6. NRC Regulatory Guide 1.32, Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants (Revision 2, 2/77)

7. NRC Regulatory Guide 1.41, Preoperational Testing of Redundant On-Site Electric Power Systems to Verify Proper Load Group Assignments (3/16/73)

8. NRC Regulatory Guide 1.47, Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems (5/73)

9. NRC Regulatory Guide 1.53, Application of the Single Failure Criterion to Nuclear Power Plant Protection Systems (6/73)

10. NRC Regulatory Guide 1.62, Manual Initiation of Protective Actions (10/73)

11. NRC Regulatory Guide 1.63, Electric Penetration Assemblies in Containment Structures for Water-Cooled Nuclear Power Plants (Rev. 2, July 1978)

12. NRC Regulatory Guide 1.73, Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants (1/74)

13. NRC Regulatory Guide 1.75, Physical Independence of Electric Systems (Revision 1, 1/75)

14. NRC Regulatory Guide 1.81, Shared Emergency and Shutdown Electric Systems for Multi-Unit Nuclear Power Plants (Rev. 1, 1/75)

15. NRC Regulatory Guide 1.89, Qualification of Class 1E Equipment for Nuclear Power Plants (11/74)

16. NRC Regulatory Guide 1.93, Availability of Electric Power Sources (12/74)

17. NRC Regulatory Guide 1.106, Thermal Overload Protection for Electric Motors on Motor Operated Valves (Revision 1, March 1977)

18. NRC Regulatory Guide 1.108, Periodic Testing of Diesel Generator Units used as Onsite Electric Power Systems at Nuclear Power Plants (Revision 1, August 1977)

19. NRC Regulatory Guide 1.118, Periodic Testing of Electric Power and Protection Systems (June 1976)

20. NRC Regulatory Guide 1.129, Maintenance, Testing and Replacement of Large Lead Storage Batteries for Nuclear Power Plants (Revision 1, February 1978)

21. NRC Regulatory Guide 1.131, Qualification Tests of Electric Cables, Field Splices, and Connections for Light-Water-Cooled Nuclear Power Plants. 8/77 (for comments)

8.1.4.3 Institute of Electrical and Electronics Engineers (IEEE)

1. IEEE 279-1971, Criteria for Protection Systems for Nuclear Power Generating Stations (Revision 1)

2. IEEE 308-1974, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Stations

3. IEEE 317-1976, Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations

4. IEEE 323-1974, Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations

5. IEEE 334-1974, Standard for Type Tests of Continuous Duty Class 1E Motors for Nuclear Power Generating Stations

6. IEEE 336-1971 (ANSI N45.2.4), Installation, Inspection, and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations

7. IEEE 338-1971, Trial-Use Criteria for Periodic Testing of Nuclear Power Generating Station Protection Systems

8. IEEE 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations

9. IEEE 352-1972, Trial-Use Guide General Principles for Reliability Analysis of Nuclear Power Generating Station Protection Systems

10. IEEE 379-1972, Guide for the Application of the Single Failure Criterion to Nuclear Power Generating Station Protection Systems

11. IEEE 382-1972 (ANSI N41.6), Guide for Type Test of Class 1 Electric Valve Operators for Nuclear Power Generating Stations

12. IEEE 383-1974, Standard for Type Test of Class 1E Electric Cables, Field Splices, and Connections for Nuclear Power Generating Stations

13. IEEE 384-1974, Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits

14. IEEE 387-1977, Criteria for Diesel Generator Units Applied as Standby Power Supplies for Nuclear Power Generating Stations

15. IEEE 420-1973, Trial-Use Guide for Class 1E Control Switchboards for Nuclear Power Generating Stations

16. IEEE 422-1973, Guide for the Design and Installation of Cable Systems in Power Generating Stations (Draft 3)

17. IEEE-450-1995, IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead-Acid Batteries for Stationary Applications

18. IEEE 494-1974, Standard Method for Identification of Documents Related to Class 1E Equipment and Systems for Nuclear Power Generating Stations

8.1.4.4 Insulated Cable Engineers Association (ICEA)

1. IPCEA P-46-426 (IEEE S-135), Power Cable Ampacities, Volume 1, Copper Conductors, and Volume 2, Aluminum Conductors, 1962

2. ICEA P-54-440 (NEMA WC-51-1975), Ampacities, Cables in Open-Top Cable Trays, Rev. 2, August, 1979

8.1.4.5 American National Standards Institute (ANSI)

1. ANSI C37, Power Switchgear

2. ANSI C57, Transformers, Regulators, and Reactors

8.1.4.6 National Electrical Manufacturers Association (NEMA)

1. NEMA SG 3-1971, Low Voltage Power Circuit Breakers (9/71)

2. NEMA ICS, Industrial Controls and Systems With Revision 6

3. NEMA SG 4-1968, A-C High-Voltage Circuit Breakers

4. NEMA SG 5-1971, Power Switchgear Assemblies (3/71)

5. NEMA SG 6-1966, Power Switching Equipment

6. NEMA TR1-1971, Transformers, Regulators, and Reactors

7. NEMA MG 1-1972, Motors and Generators

8. NEMA VE 1-1971, Cable Tray Systems

9. NEMA AB 1-1975, Molded Case Circuit Breakers

10. NEMA FU 1-1972, Low-Voltage Cartridge Fuses

11. NEMA PB 1-1971, Panelboards With Revision 1

12. NEMA PB 2-1972, Dead-Front Distribution Switchboards With Revision 1

8.1.4.7 National Fire Protection Association (NFPA)

No. 70-1971, National Electrical Code

8.1.4.8 Underwriters' Laboratories, Inc. (UL)

1. UL-50, Electrical Cabinets and Boxes (1975)

2. UL-67, Electric Panelboards (Revision, 10/75)

3. UL-891, Dead-Front Electrical Switchboards (1975)

8.1.4.9 Illuminating Engineering Society (IES)

IES Lighting Handbook, Application Volume, 1981

8.1.5 COMPLIANCE WITH NRC REGULATORY GUIDES AND IEEE STANDARDS

The extent to which the recommendations of the NRC regulatory guides and IEEE standards are complied with is described in the following sections:

8.1.5.1 NRC Regulatory Guides

For description of compliance to the Regulatory Guides, see Appendix 1A(B) and 1A(N).

8.1.5.2 IEEE Standards

1. IEEE 338-1971, Trial-Use Criteria for Periodic Testing of Nuclear Power Generating Station Protection Systems

   Periodic testing of protection systems conforms to the requirements of this standard.

2. IEEE 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations

   All seismic Category I Class 1E equipment is seismically qualified in accordance with this standard.

   For details, see Sections 3.10B and 3.10N.

3. IEEE 387-1977, Criteria for Diesel Generator Units Applied as Standby Power Supplies for Nuclear Generating Stations

The design criteria and the qualification and testing requirements of the diesel generator units conform to the requirements of this standard.

4. IEEE 420-1973, Trial-Use Guide for Class 1E Control Switchboards for Nuclear Power Generating Station.

   All Class 1E control boards are designed in accordance with IEEE 420-1973 requirements with following clarification:

   Lens and Buttons for approximately forty (40) safety system inoperable indication lights and for sixteen (16) non-safety valve control devices are constructed of “LEXAN” (poly carbonate) and “CELON” respectively. These indicating lights/buttons are grouped and enclosed in a metal housing to provide separation between them and other equipment on the main control board. The housings are flush mounted on the front of the main control board. Cable connectors are provided on the back plate of the housing for external wiring. Internal fire (unlikely event) in the housing will not significantly degrade the integrity of the main control board.

   Wire splices are used in limited applications on field cables that terminate in certain Class 1E panels, cabinets or racks. The normal design is to terminate field cables without the use of wire splices. The wire splices are only used where additional length is required for the field wire and it was not judged reasonable to pull a new field cable. The use of such wire splices has been minimized.

   The wire splices are butt splices. The crimping technique, device and materials used for the splices are identical to those used for the terminal lugs in that panel. The wire splices are only allowed on low power applications such as control cables. The inspection procedure verifies the operability of the spliced circuit by completion of a continuity check. The wire splices used are qualified for anticipated service conditions, and the splices are staggered or separated within the panel so that they are not adjacent to each other or pressing against one another in the same wire bundle. Since previously accepted crimping methods and materials are used, the splices are limited to low power circuits and to field cables that already terminate in the panel, and the required wire separation and wire bundle support is maintained, the wire splices are not expected to significantly alter the heat load in the panel, the probability of a fire or the operability of any equipment or cables in that panel.

5. IEEE 494-1974, Standard Method for Identification of Documents Related to Class 1E Equipment for Nuclear Power Generating Stations.

   All station design documents and drawings containing Class 1E equipment and systems, in whole or in part, are identified with the term “Nuclear Safety Related” in accordance with this standard or, as an alternative, with the equivalent term “CLASS I”. Vendor supplied documents and drawings are used in conjunction with station design documents and/or drawings which are properly identified as to their safety classification; therefore, vendor supplied documents and drawings are not necessarily identified as “nuclear safety related” in accordance with IEEE 494-1974.

TABLE 8.1-1
SAFETY LOADS AND FUNCTION

(Sheet 1 of 2)

Safety Load	Function	Power
Safety injection pumps	Provide emergency core cooling	AC
Charging pumps	Provide emergency core cooling	AC
Residual heat removal (RHR) pumps	Provide emergency core cooling and reactor heat removal during refueling operations	AC
Containment spray pumps	Provide cooling spray in containment following a loss-of-coolant accident (LOCA)	AC
Service water pumps	Provide cooling water for Component Cooling Water System (CCWS) heat exchangers, and emergency diesel generators	AC
Component cooling water pumps	Provide cooling water to safety-related equipment	AC
Auxiliary feedwater pumps	Provide adequate water to steam generators in the event of a unit trip coupled with a loss of offsite power	AC
Spent-fuel pool cooling and cleanup pumps	Cool spent fuel pool water	AC
Control Room emergency	Maintain safe environmental conditions for operating personnel air cooling units and limit ambient air temperature in safety-related compartments heating, ventilating, and air conditioning.	AC
Heating, ventilating,and air-conditioning (HVAC) water chiller	Provide cooling fluid for emergency air cooling units	AC
Motor-operated valves small motors, fans and heaters, associated with Safety-related equipment	Insure coordinated operation of safety-related systems	AC

TABLE 8.1-1
SAFETY LOADS AND FUNCTION

(Sheet 2 of 2)

Safety Load	Function	Power
Reactor Protection System and Engineered Safety Features (ESF) Actuation System	Provide safe plant shutdown	AC and DC
Plant instrumentation	Provide safe reactor operation	AC
Instrument buses	Provide power to instrumentation and control equipment	AC
Shutdown control and instrumentation	Provide control to shutdown plant from outside of Control Room.	AC and DC
Instrument bus inverters	Provide power to instrument buses	DC

8.2 OFFSITE POWER SYSTEM

8.2.1 DESCRIPTION

TO's transmission system serves as the main outlet and source of offsite power for the CPNPP. Connection of the station outputs to the system is achieved via 345-kV overhead lines to the 345-kV switchyard. Separate connections to the 138-kV switchyard and the 345-kV switchyard provide independent and reliable offsite power sources to the Class 1E systems of each unit.

Because the 345-kV system forms the backbone of the TO transmission system, it provides a highly reliable source of continuous power for plant shutdown. Another reliable source is the 138-kV network.

The TDSP's high-voltage (HV) switchyards at CPNPP consist of 345-kV switching facilities and 138-kV switching facilities and are an integral part of the TO transmission system. The network interconnections to CPNPP switchyards are made through seven 345-kV and two 138-kV transmission lines to other switching stations within the TO transmission system as shown on Figure 8.2-4. There are no interconnections between the 138-kV switchyard and the 345-kV switchyard at the CPNPP site. The 138-kV switchyard is physically and electrically independent of the 345-kV switchyard.

Essentially, the 345-kV and the 138-kV switchyards each consist of a two-bus arrangement as shown in Figure 8.2-1. Power can be supplied to each switchyard from any of their respective transmission circuits.

The CPNPP switchyards are located approximately 600 ft due west of the Turbine Building. Figure 8.2-1 shows the physical orientation and separation of the high-voltage switchyards, main, station service and startup transformers, and transmission lines routing from the transformers, and transmission lines routing from the transformers to the switchyards. The CPNPP HV switchyards' configurations are shown on Figure 8.2-4.

Two three-phase, half-size, step-up transformers are provided for each unit to raise the 22-kV generator voltage to 345-kV prior to transmission via overhead lines to the CPNPP 345-kV switchyard. Each CPNPP unit output line is connected to both 345-kV buses through two breakers which function as generator circuit breakers. The units are synchronized to the system across the generator circuit breakers. In the event of a unit trip, these breakers isolate the associated generator from the system. If required, the generators can be isolated individually by removing the bus links in the main isolated-phase bus, permitting energization of the non-safety-related 6900-V auxiliary bus system by closing the generator circuit breakers and backfeeding through the main and unit auxiliary transformers.

The CPNPP offsite power source line for start up transformer XST1 is connected to both 138-kV buses thru two breakers which function as a bus tie. The offsite power source line for start up transformer XST2 is connected to both 345-kV buses thru two breakers which function as a bus tie.

Startup transformer XST1 and alternate startup transformer XST1A are connected to a common overhead line from the 138-kV switchyard. Each transformer is provided with a 138-kV motor-operated air switch such that each transformer can be energized independent of the other transformer.

Startup transformer XST2, alternate startup transformer XST2A and station service transformer 1ST are connected to a common overhead line from the 345-kV switchyard. Each transformer is provided with a 345-kV motor-operated air switch such that each transformer can be energized independent of the other transformer.

Alternate startup transformers XST1A and XST2A are provided as shown in Figure 8.2-1. XST1A is functionally redundant to XST1 and either can be in service to provide power to the safety buses for Units 1 and 2 from the 138 kV switchyard. Likewise, XST2A is functionally redundant to XST2 and either can be utilized to provided power to the safety buses for Units 1 and 2 from the 345 kV switchyard.

Each of the four startup transformers has been equipped with a non-safety-related monitoring system for detection of a potential open-phase condition. The OPC monitoring system is not required for compliance with Technical Specification requirements for offsite power sources.

Station service transformer 2ST is connected to the 345-kV switchyard east and west bus via circuit breakers and overhead line.

The 138-kV and 345-kV circuit breakers are provided with an energy storage mechanism that allows the operation of the individual circuit breaker without having an external source of power. The circuit breakers have two separate avenues of relay protection termed primary and secondary or backup to provide a high degree of operational reliability.

The 125-VDC supply, for each of the switchyard relays, is segregated into two completely independent systems, one for the primary relays and the other for the backup relays. These two systems are independent of both the station DC systems and the DC systems of the other switchyard.

The 345-kV switchyard TO circuit breakers may be operated from either the 345-kV Switchyard Control Building or remotely from Oncor's System Operations Center. Control of CPNPP 345-kV switchyard circuit breakers and motor-operated air switches is administered from the plant Control Room except for the motor-operated air switch for transformer XST2A, which has local motor control. The 138-kV switchyard circuit breakers may be operated from either the 138-kV Switchyard Relay House or remotely from TO's System Operations Center. The 138-kV motor-operated air switch (DXST1) may be operated from the Control Room. The 125-VDC control power for circuit breakers and motor-operated air switches in the 345-kV system is independent of the 125-VDC control power for circuit breakers and motor-operated air switches in the 138-kV system.

Layout of the 138 kV and 345 kV switchyards are shown in Figure 8.2-1 and Figure 8.2-4.

The substations that are connected to the CPNPP switchyards (as shown on Figure 8.2-4) are as follows:

- DeCordova (138-kV)
- Stephenville (138-kV)
- Mitchell Bend (345-kV)

- Wolf Hollow (345-kV)
- Everman (345-kV)
- Johnson Switch (345-kV)
- Comanche Switch (345-kV)
- Parker No. 1 (345-kV)
- Parker No. 2 (345-kV)

The layout of transmission lines from TO's CPNPP switchyards to other switching stations, in the vicinity of CPNPP switchyards is shown in Figure 8.2-12.

The following combinations of lines form double-circuits routed on common transmission towers:

345-kV Parker No. 1 and 345-kV Parker No. 2

345-kV Comanche Switch and 138-kV Stephenville

345-kV Everman and 345-kV Johnson Switch

345-kV Mitchell Bend and 345-kV Wolf Hollow

The 138-kV line to DeCordova is supported by independent transmission towers and shares the same right-of-way as the Wolf Hollow and Mitchell Bend 345-kV circuits (Figure 8.2-12). Routing of this circuit was selected to avoid crossings with the 345-kV overhead lines, thereby maintaining a high degree of circuit reliability and availability (Figures 8.2-1 and 8.2-12).

The 138-kV line to Stephenville passes beneath the 345-kV lines to Mitchell Bend, Wolf Hollow, Everman, and Johnson Switch. The minimum overhead clearance between the Stephenville line and the 345-kV lines is 20 ft.

In addition to the transmission line circuit breakers in the 138-kV switchyard, there are two Plant Support Power System transformers. These fully redundant transformers provide a reliable source of power to plant support loads.

Two physically independent and redundant sources of offsite power are available on an immediate basis for the safe shutdown of either unit. The preferred source to Unit 1 is the 345-kV offsite supply from the 345-kV switchyard and the startup transformer, XST2; the preferred source to Unit 2 is the 138-kV offsite supply from the 138-kV switchyard and the startup transformer, XST1. Each of the startup transformers (XST1 and XST2) normally energizes its related Class 1E buses; i.e., XST1 normally energizes Unit 2 Class 1E buses and XST2 normally energizes Unit 1 Class 1E buses. This eliminates the need for automatic transfer of safety-related loads in the event of unit trips. In the event one startup transformer (e.g., XST1, a preferred source) becomes unavailable to its normally fed class 1E buses, power is made available from the other startup transformer (e.g., XST2, an alternate source) by an automatic transfer scheme, described in Section 8.3.1.1.5. This limits plant transients due to loss of a preferred source to only one unit. Capability is provided to test the transfer scheme, during

power operation, by manually causing loss of voltage to the undervoltage relays which results in tripping the appropriate startup transformer breaker in the 6.9-kV safety related switchgear. Periodic testing of the automatic transfer scheme is performed as specified in the Technical Specifications.

Additionally, two station service transformers are provided for the non-safety-related auxiliaries, one for each unit. The source for these non-safety-related buses is the 345-kV offsite power supply from the switchyard and transformers 1ST (Unit 1) and 2ST (Unit 2).

The two startup transformers which feed the emergency buses (XST1 and XST2) are connected to the safety-related 6900-V auxiliary bus systems by separate, metal-enclosed, ventilated, louvered top cover, cable bus ducts and short sections of ventilated, solid top cover, cable tray. Cable spacing is maintained within each bus duct and cable tray. Figure 8.2-11 shows the layout and connections of the cable bus ducts/cable trays.

The preferred and alternate power source bus ducts and cable trays for each unit, where routed in the same area, are in areas which are free of hazards from rotating equipment.

For fires that affect both preferred and alternate power source bus ducts, the CPNPP fire protection program has provided adequate protection to ensure safe shutdown can be achieved. For other single external events, the bus duct and cable tray construction, separation and independence provided between the preferred and alternate power source bus ducts assure that simultaneous failure of both bus ducts/cable trays will not occur. For example, separate independent circuits are provided for the bus duct connections to the onsite distribution systems. The possibility of a single external fire near the west wall of the Auxiliary Building causing damage to bus ducts A2, B1 and B2, is minimized by the inaccessible location of the subject ducts (see Figure 8.2-11), and the following features:

1. There are no fixed external ignition sources located in this area;

2. Inaccessibility to the area restricts the introduction of transient combustible materials;

3. Thermal type fire detectors and a fixed wet pipe sprinkler system are provided in this area (refer to the CPNPP Fire Protection Report);

4. Bus duct enclosures consist of totally enclosed ventilated metal type construction;

5. Bus duct cables meet the flame test requirements of IEEE 383-1974; and,

6. Bus ducts and cable trays are separated from the rooms below by a concrete fire barrier roof.

The Fire Detection System and the protection of the preferred offsite power system is described in Section 9.5.1.

8.2.1.1 Design Basis

ERCOT requires the TO to uphold Generators (CPNPP) Federal (NRC) licensing requirements. TO has implemented CPNPP specific guidelines in their Transmission Planning and CPNPP Switchyards Working Procedures to assure adequacy of CPNPP offsite power sources.

The ERCOT requirements to uphold CPNPP licensing requirements and the implementation of the CPNPP specific requirements by TO enhances the reliability and availability of CPNPP offsite sources. The offsite power systems are capable of providing reliable sources of power to the Class 1E systems of each unit in compliance with NRC General Design Criterion (GDC) 17, NRC Regulatory Guide 1.32 [3], and IEEE 308 [6]. Design of the offsite power systems for CPNPP exceeds the minimum requirements cited in these regulations.

8.2.1.2 Compliance with Standards

8.2.1.2.1 Compliance with General Design Criteria 17

Two independent power sources are available on an immediate basis following a DBA to ensure operation of the vital safety functions. Because the Class 1E buses of each unit are normally energized from the preferred offsite power source (see Section 8.3), switching is not required to make this source available. The second offsite source (alternate) is available within seconds. These two sources of power exceed the minimum requirements of GDC 17 which stipulate that one circuit shall be available within seconds.

Physical separation and independence of the transformers and transmission lines minimize the likelihood of their simultaneous failure. A layout presenting the arrangement and separation of the transformer and transmission lines is shown on Figure 8.2-1. The electrical outputs from Units 1 and 2 are connected to separate positions located at opposite ends of the switchyard. These unit output lines and the 345-kV offsite power source to the Class 1E systems are arranged with adequate separation to preclude failure of either output line which would create a failure on the 345-kV offsite power circuit. There are no transmission line crossings from the switchyard to the plant.

Disruption of both offsite power sources resulting from multiple equipment failures caused by tornadoes, ice storms, thunderstorms, or large missiles is considered highly improbable.

The overhead line design meets the requirements of the National Electrical Safety Code [8] for heavy loading district, Grade B construction, and is based on a lightning performance of less than one outage per 100 miles per year. The transmission lines have two overhead ground wires for protection from lightning. Design of switchyard components is in accordance with IEEE, ANSI, and NEMA standards.

8.2.1.2.2 Compliance with General Design Criteria 18

The offsite power system circuitry is designed to permit periodic testing of operability and functional performance of power supplies, relays, and switches.

Under conditions as close to design as practical, testing the operational sequence that brings the system into operation and transfer of power between the preferred and standby power sources is performed during the refueling shutdown.

8.2.1.2.3 Compliance with IEEE 308 and NRC Regulatory Guide 1.32

The offsite power systems described previously exceed the minimum requirements of IEEE 308, which stipulates that each safety load group has access to a preferred power supply consisting of one or more power sources. Furthermore, the design conforms to the preferred design outlined

in NRC Regulatory Guide 1.32, i.e., inclusion of two immediate access circuits from the transmission network.

Considerations for multi-unit stations given in IEEE 308 permit sharing of the preferred power supply between units as long as sufficient capacity is provided to carry the Engineered Safety Features (ESF) for a Design Basis Accident (DBA) on one unit and on those systems required for a concurrent safe shutdown on the second unit. The preferred offsite power systems for CPNPP are supplied through two startup transformers common to both units. These transformers have more than ample capacity to operate the ESF loads for DBAs on both units simultaneously, although the design criteria requires consideration of a DBA on one unit only.

8.2.1.2.4 Compliance with NRC Regulatory Guide 1.93

As described in the Technical Specifications, power operation is initiated and continued without restriction only when the Limiting Conditions for Operation (LCO) are met. If the LCO is not met, power operation is restricted, as explained in the Technical Specification.

8.2.1.2.5 Compliance with IEEE 336 (7) and NRC Regulatory Guide 1.30 (2)

The Quality Assurance Program for the Class 1E portions of the CPNPP preferred power system is based on the requirements of IEEE 336 and Regulatory Guide 1.30. For details see Chapter 17 and Appendix 1A(B).

8.2.1.2.6 Compliance with NRC Regulatory Guide 1.47

The surveillance of the off-site preferred power system operability status is based on the requirements of Regulatory Guide 1.47 augmented by Branch Technical Position ICSB 21 as described herein:

A system level indication is provided to indicate if a preferred power source is unavailable. This indication for Train A Unit 1 is activated on:

1. Loss of preferred source voltage

2. Breaker 1EA1-1 control switch in the pull to lock position

3. Operator manual action

Train B is similar to Train A. A duplicate scheme is provided for Unit 2.

8.2.2 ANALYSIS

Offsite power sources are not obtained from a common switchyard. The 138-kV switchyard is physically separated from the 345-kV switchyard. There is no interconnection between the CPNPP 345-kV and 138-kV switchyards. (See Figure 8.2-1). The control supply of the 138-kV circuit breakers is independent of the control supply of the 345-kV circuit breakers. The source of DC power provided for the 138-kV switchyard is separate from the source of DC power for the 345-kV switchyard. See description in 8.2.1. The offsite power source lines from 138-kV and 345-kV switchyards to startup transformers XST1 and XST2 do not cross each other or any other transmission line from the switchyards to the plant. The power from 138-kV and 345-kV

switchyard buses, through startup transformers XST1 and XST2, to Safety related buses constitute the offsite power sources for CPNPP.

The 345-kV transmission lines (Parker No. 1 and Parker No. 2 double-circuit line, Everman and Johnson Switch double-circuit line, Comanche Switch line, and the Wolf Hollow and Mitchell Bend double-circuit line), which connect to the CPNPP 345-kV switchyard, are also physically separated from one another. The transmission lines, which emanate from the CPNPP switchyards, do not cross each other, with the exception of the Stephenville 138-kV line, which crosses underneath the Wolf Hollow and Mitchell Bend double circuit and Everman and Johnson Switch double-circuit 345-kV lines. Failure of a tower carrying the Comanche Switch and Stephenville lines would not impact the continued availability of either offsite source. Similarly, failure of a tower carrying the Wolf Hollow and Mitchell Bend double-circuit or Everman and Johnson Switch double-circuit lines would not impact the continued availability of either offsite source. The failure of a tower carrying the Parker No. 1 and Parker No. 2 double-circuit line would not impact the continued availability of either offsite source. The failure of a tower carrying the 138-kV Decordova line would not impact the continued availability of 345-kV offsite source; however, it may impact the immediate availability of 138-kV offsite source. The physical separation of switchyards, primary and backup protection systems, and the stable transmission grid system minimizes the probability of simultaneous failures of offsite power sources.

The bulk transmission system of the ERCOT and the TO transmission system are designed to withstand the loss of the largest power plant and to retain the integrity of the remaining bulk transmission system.

The full load capacity of CPNPP at the time of Unit 1 installation represented two to three percent of the ERCOT-estimated peak load and, with the addition of the second unit, was equivalent to approximately five to six percent of the estimated peak. Actual disturbances on the ERCOT system have occurred where large amounts of capacity were lost, one as high as 10 percent, with no integrity degradation of the transmission system observed.

Studies confirm that loss of the CPNPP plant when 100-percent loaded will not impair the integrity of the bulk transmission system for conditions representative of those projected at the time of installation of Unit 1 or both Units 1 and 2.

The stability studies demonstrate both the effect on the transmission system when one or both of the CPNPP units are lost and when the plant auxiliaries are transferred to the standby source. It is evident from the studies that loss of one or both of the nuclear units will not cause the loss of auxiliary power to the station. In addition, the system remains stable for all disturbances near CPNPP which are cleared by primary or backup relaying.

Simultaneous loss of either unit and the most critical Generator does not affect the capability of either offsite source to furnish shutdown power on an uninterrupted basis.

Simultaneous loss of either unit and the most critical transmission line does not adversely affect the capability of the system to furnish shutdown power on an uninterrupted basis. The steady-state and transient stability analyses are described in Reference [9].

The normal operating voltages at CPNPP for the 345-kV and 138-kV (nominal) offsite power grid are approximately 354-kV and 142-kV. These normal operating voltages are subject to periodic

review and adjustment based on the requirements of the TO system which may change during the operating life of CPNPP.

The maximum voltages at CPNPP are 361-kV and 144-kV. These maximum grid voltages are based on equipment ratings (circuit breakers, transformers, etc.) and are controlled by generator excitation (throughout the grid), and switching of shunt reactors and capacitors. The minimum voltage for the respective grids at CPNPP has been calculated to be 340-kV and 135-kV for normal and credible contingency conditions respective to CPNPP.

On the basis of these voltage ranges of 340-kV to 361-kV and 135-kV to 144-kV for the offsite power sources, class 1E bus voltage ranges are determined to be within the out-of- tolerance voltage values of Table 8.3-1C. All class 1E equipment can operate continuously at these voltages for all modes of plant operation to perform their safety function.

Grid voltages lower than those calculated could occur for situations involving contingencies in the grid system. Periodically, such situations are studied. These studies are conducted in accordance with the ERCOT Planning Criteria and the TO Transmission Planning Procedures. See Section 8.3.1.1.5 for discussion of the automatic transfers which occur for condition of undervoltage.

The nominal frequency on the ERCOT system is 60 ± .03 hz. On occasion when large amounts of generation are lost, the frequency will drop to 59.6 hz and recover within a few seconds to 59.8 hz and within a few minutes to normal, 60.0 hz. The deviations in frequency are well within the operating ranges of the Class 1E equipment.

Studies have been and are continually being made where catastrophic disturbances are postulated to test the ERCOT system against events beyond the design criteria. The lowest frequency observed on these studies has been 57.5 hz, for which the system recovered with no cascading shutdowns.

The frequency decay rates observed on the ERCOT system during daily operation are usually less than 1.5 hz/sec. Location or origin of disturbance should have little effect on frequency decay rate. The type of system disturbance postulated in the above studies resulted in a maximum calculated frequency decay rate of 2.4 hz/sec.

One inconceivable situation has been studied to determine a maximum frequency decay rate at Comanche Peak which resulted in a calculated decay rate of 4.37 hz/sec. This situation envisioned the total Dallas/Fort Worth metro area load being dumped on the Comanche Peak unit instantaneously and all other sources of generation removed.

In order to satisfy offsite power requirements, the TO should maintain 345kV grid system voltage at CPNPP switchyard between the voltage range of 340kV to 361kV and 138-kV grid system voltage at CPNPP switchyard between the voltage range of 135-kV to 144-kV.

Grid configuration and reliable operation is maintained through tested and proven operating procedures and guidelines. TO is a member of ERCOT which continually plans (coordinates) the capacity on line (to serve load), the amount of responsive reserve, the maintenance of units, the maintenance of grid components, and the scheduled transfer of energy between members. Many entities in ERCOT have their own control centers where the systems are monitored. Parameters monitored in these control centers are: system frequency, grid voltages (at many

points on system), generation (all units), transmission line flows, and reserve capacity. Through use of these tools and adherence to operating guides, experience has proven grid integrity over a long period of time.

The responsive reserve on the ERCOT system is coordinated by the ERCOT. ERCOT maintains a minimum responsive reserve of 2300 MW. The amount of responsive reserve is subject to periodic review and adjustment based on the requirements of ERCOT, which may change during the operating life of CPNPP. This reserve is shared among all operating plants and loads acting as resources (interruptible loads) throughout the system and not concentrated in specific areas or any particular plant. No restrictions are placed on specific responsive reserve in relation to Comanche Peak.

The operators at the Comanche Peak plant have available to them in the control room:

1. Grid frequency,

2. 345-kV and 138-kV Comanche Peak bus voltages, and

3. Constant communication (dedicated circuits) with TO System Operation Center.

Loss of the Comanche Peak unit or largest system load should not adversely affect shutdown power because of logical reasons:

1. Loss of further generation during lower than normal frequency would result in automatic shedding of load which acts to restore frequency,

2. When the system loses load during low frequency or low voltage conditions, the system automatically moves toward a more nearly normal condition, and

3. If the system voltage drops below the minimum calculated values described above and causes the voltages at the plant switchgear buses to drop below the degraded voltage protection settings, the plant would be isolated from the offsite power sources and the standby diesels would automatically start to ensure a source of shutdown power.

REFERENCES

1. 10 CFR Part 50, General Design Criteria for Nuclear Power Plant Construction Permits, U.S. Nuclear Regulatory Commission (2/4/72).

2. NRC Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment, August 11, 1972, U. S. Nuclear Regulatory Commission.

3. NRC Regulatory Guide 1.32, Criteria for Safety-Related Electric Power Systems for Nuclear Power Plants (Revision 2, 2/77).

4. NRC Regulatory Guide 1.47, Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems, May 1973, U.S. Nuclear Regulatory Commission.

5. NRC Regulatory Guide 1.93, Availability of Electric Power Sources, December 1974, U.S. Nuclear Regulatory Commission.

6. IEEE 308-1974, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Stations.

7. IEEE 336-1971 (ANSI N45.2.4-1972), Installation, Inspection, and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations.

8. National Bureau of Standards of U.S. Department of Commerce, National Electrical Safety Code, Handbook 81 (ANSI C2.2-1960) 6th ed.

9. 1980-1982 Texas Utilities Company System Comanche Peak Stability Study.

10. M. M. Merrian and P. J. VandeWalle, 1976, The Effect of Grid Frequence Decay Transients on Pressurized Water Reactors, IEEE Transactions on Power Apparatus and Systems, Vol. PAS-95, No. 1, p. 269.

11. 1980/81-1982/83 Winter Peak TUCS Supplemental Comanche Peak Report, April 9, 1974.

TABLE 8.2-1
THIS TABLE IS DELETED.

8.3 ONSITE POWER SYSTEMS

8.3.1 AC POWER SYSTEMS

The onsite AC Power Systems consist of various auxiliary electrical systems designed to provide reliable electrical power to Class 1E and non-Class 1E station loads. (See Figure 8.3-1.) The AC Essential Lighting System connected to Class 1E power buses is described in Section 9.5.3.

Redundancy of Class 1E onsite AC Power Systems ensures safe reactor shutdown during a Safe Shutdown Earthquake (SSE) or DBA coincident with any single failure within the standby AC Power System or the 118-V uninterruptible AC Power System.

The standby AC Power System ensures safe plant shutdown when the preferred and alternate offsite power sources are not available. The 118-V uninterruptible AC Power supply feeds power to reactor protection instrumentation and control systems and to other Class 1E components and systems essential to safe reactor operation.

8.3.1.1 Description

8.3.1.1.1 System Structure (Network)

Any one of the following systems can furnish power to the Class 1E onsite AC power distribution systems:

1. Preferred power system (offsite power sources, 345-kV source for Unit 1 and 138-kV source for Unit 2)

2. Alternate power system (alternate offsite power source, 138-kV source for Unit 1 and 345-kV source for Unit 2)

3. Standby power system (diesel generators)

   CPNPP has also provided a set of non-safety related Alternate Power Diesel Generators (APDGs) for each unit with the capability to connect to a Class 1E train at a time to provide defense-in-depth for safe shutdown of a unit during outages or during extended duration of an inoperable offsite circuit on occurrence of concurrent loss of offsite power and failure of EDGs. The APDGs may provide 3450 kVA to provide long term cooling of each unit. See Subsection 8.3.1.2.1.7.h for a discussion of the sizing of APDG cables.

Additionally, the system structure during unit startup is described in item 4.

These systems perform as described in the following paragraphs:

1. Preferred Power System

   The preferred power sources supply power to the Class 1E buses during plant startup, normal operation, emergency shutdown, and upon a unit trip. These sources originate as part of the 138-kV and 345-kV offsite power systems (discussed in Section 8.2) and supply power to the 6900-V Class 1E auxiliary bus systems through startup transformers; as shown on Figure 8.3-1.

Each startup transformer is rated 35/46.7/58.3 MVA, three phase, 60 Hz; each has two secondary windings, x and y, each rated 17.5/23.3/29.2 MVA, three phase, 60 Hz. The secondary windings are connected to the 6900-V Class 1E buses as described below and in Subsection 8.3.1.1.2.

Either transformer XST1 or XST1A is connected to the 138-kV Switchyard via an overhead transmission line as shown on Figure 8.2-1. The secondary connections at 6900-V are made as follows: the x-winding is connected to Class 1E buses 1EA1 and 1EA2 of Unit 1, and the y-winding is connected to Class 1E buses 2EA1 and 2EA2 of Unit 2. Startup transformer XST1 or XST1A serves as the preferred power source to Unit 2 Class 1E buses 2EA1 and 2EA2 and as the alternate power source to Unit 1 Class 1E buses 1EA1 and 1EA2 as shown on Figure 8.3-1.

The second offsite power source is provided through 345-kV transformer XST2 or XST2A. The transformer primary is connected to the 345-kV Switchyard, located approximately 600 ft away, via an overhead transmission line. This connection is shown on Figure 8.2-1. The switchyard offers a highly reliable source of power because it can be supplied by seven separate transmission lines from various switching stations on the TO system in conjunction with the two CPNPP unit outputs (discussed in Section 8.2).

The secondary connections at 6900-V are made as follows: the x-winding is connected to Unit 2 Class 1E buses 2EA1 and 2EA2, and the y-winding is connected to Class 1E buses 1EA1 and 1EA2 of Unit 1. Startup transformer XST2 or XST2A serves as the preferred power source to Unit 1 Class 1E buses 1EA1 and 1EA2 and as the alternate power source to Unit 2 Class 1E buses 2EA1 and 2EA2 as shown on Figure 8.3-1.

2. Alternate Power System

As described previously, either startup transformer XST1 or XST1A serves as an alternate source of offsite power to Unit 1 Class 1E buses 1EA1 and 1EA2 as shown on Figure 8.3-1.

Either startup transformer XST2 or XST2A serves as an alternate source of offsite power to the Unit 2 Class 1E buses 2EA1 and 2EA2 as shown on Figure 8.3-1.

Each startup transformer has the capacity to carry the required Class 1E loads of both units during all modes of plant operation. Transformer XST1(2) and XST1(2)A are fully redundant and neither is ‘preferred’ to the other.

In determining the capacity of 138-kV startup transformer XST1, the following abnormal conditions were considered to occur simultaneously, and represent the worst case loading condition.

a. Preferred (345-kV) off site power supply to Unit 1 safeguards buses is unavailable and these buses are powered from the 138-kV source.

b. Unit 1 trips at full power.

c. DBA occurs on Unit 2.

The above conditions result in the in-service 138-kV startup transformer carrying the required safeguards load of Unit 1 and Unit 2.

The above conditions apply equally to the 345-kV startup transformer when considering loss of the in-service 138-kV startup transformer, a DBA on Unit 1 and a Unit 2 trip from full power.

Under DBA conditions, the safety loads are always started in sequence via the safeguards sequencer as discussed in Section 8.3.1.1.5.

Therefore, the Class 1E buses of each unit can be supplied by two independent and reliable immediate-access offsite power sources. Sharing of these offsite power sources between the two units has no effect on the station electrical system reliability because each transformer is capable of supplying the required safety-related loads of both units if it becomes necessary to safely shut down both units simultaneously.

3. Standby Power System (Diesel Generator)

The standby AC Power System is an independent, onsite, automatically starting system designed to furnish reliable and adequate power for Class 1E loads to ensure safe plant shutdown and standby when preferred and alternate power sources are not available. Electrical equipment associated with starting of the diesel generators are enclosed in cabinets and appropriate procedures are in place to ensure the proper degree of cleanliness is maintained on a plant wide basis commensurate with the design and operating requirements of the equipment, in order to minimize diesel generator failure due to accumulation of dust or other particles on these equipment. Four independent diesel generator sets, two per unit, are provided. The manufacturer/model number for the diesel engine is Delaval/DSRV-16-4; the manufacturer/model number for the generator is EP Portec/L-11091.

Each generator is driven by a single prime mover and is capable of sequentially starting and supplying the minimum power requirements for a DBA or blackout in one unit. The four diesel generators are electrically and physically independent. They are located above the water level of the probable maximum flood (PMF) level (789.7 ft). Each diesel generator and its associated equipment is located in a separate room with walls designed to protect the diesel generators and associated equipment against an SSE, tornadoes, missiles, and fire. See Figures 1.2-11 and 1.2-17.

4. Unit Startup

Startup of Unit 1 is accomplished using non-Class 1E buses 1A1, 1A2, 1A3, and 1A4 which are supplied from 345-kV station service transformer 1ST and Class 1E buses 1EA1 and 1EA2 from 345-kV startup transformer XST2 or XST2A. Following the main generator synchronization to the system, the aforementioned non-Class 1E buses are transferred live to unit auxiliary transformer 1UT.

For startup of Unit 2, non-Class 1E buses 2A1, 2A2, 2A3, and 2A4 are supplied from 345-kV station service transformer 2ST, and Class 1E buses 2EA1 and 2EA2 from 138-kV startup transformer XST1 or XST1A. Following the main generator

synchronization to the system, the non-Class 1E buses are transferred live to unit auxiliary transformer 2UT.

Bus XA1 can be powered from either 345-kV station service transformer 1ST or 2ST.

5. Plant Support Power System

The 25 KV Plant Support Power Loop, feeding certain non-safety-related equipment, is supplied from two Plant Support Power System transformers located in the 138 KV switchyard. The system can also provide power, via manual transfer switches (inside key-locked enclosures), to certain safety-related equipment during bus outages in plant Modes 5 and 6.

8.3.1.1.2 Busing Arrangement, Interconnections, and Load Assignment

Two independent and redundant 6900-V Class 1E buses are provided for each unit, each capable of supplying the required safety-related loads to safely shut down the unit following a DBA. Each Class 1E bus can be fed from two independent offsite power sources or the diesel generator assigned to the bus. Arrangement of buses is shown on Figure 8.3-1; assignment of emergency loads is indicated in Tables 8.3-8 and 8.3-9. Redundant safety-related loads are divided between Trains A and B so that loss of either train does not impair fulfillment of the minimum shutdown safety requirements. There are no manual or automatic connections between Class 1E buses and loads of redundant trains. Electrical separation of redundant trains is maintained through all voltage levels, including DC and instrumentation.

FSAR figures are based on Unit 1 and Common drawings and bus configurations. Unit 2 uses the same design criteria as Unit 1 and no Unit 2 specific FSAR figures are submitted. Unit 2 electric equipment have the same tag numbers as the Unit 1 electrical equipment shown on the figures in FSAR Section 8.3 except for the unit designator which is “2” versus “1”. Minor differences in regard to load bus assignment, cable numbers, compartment numbers and references exist between Unit 1 and Unit 2 are not reflected in these figures.

Safety-related (Class 1E) 6900-V buses 1EA1 and 1EA2 for Unit 1 and 2EA1 and 2EA2 for Unit 2 are fed directly from dedicated startup transformers XST1 or XST1A and XST2 or XST2A. There are no interconnections between safety-related and non-safety-related 6900-V buses.

The reactor coolant pump motors are connected to the 6900-V non-Class 1E buses located in the non-seismic category area, as shown on Figures 1.2-25 and 1.2-30 and as discussed in Section 3.2. The reactor coolant pump motors are not Class 1E.

8.3.1.1.3 Redundant Bus Separation

All Class 1E buses are arranged in such a way that train A buses are electrically and physically isolated from train B buses to satisfy the single failure criteria. Physical locations of these buses are shown on Figures 1.2-11 and 1.2-13.

The redundant 6900-V and 480-V Class 1E switchgear assemblies are located in separate rooms in the seismic Category I Safeguards Buildings. Each room contains only electrical equipment, therefore minimizing exposure to mechanical, water, or fire damage caused by

equipment such as steam lines, waterlines, pumps, and motors. Separation between train A and train B switchgear is accomplished by placing the equipment on two different floor elevations.

8.3.1.1.4 Equipment Capacities

All switchgear is adequately sized and coordinated to permit safe and reliable operation under normal and short-circuit conditions.

Interrupting ratings of Class 1E switchgear, motor control centers (MCCs), and distributions boards are as follows:

Equipment	Class	Interrupting Current
6.9-kV switchgear	7.2 kV	40,000 A at 7.2 kV
480-V switchgear	600 V	50,000 A at 480 V
480-V motor control centers	600 V	25,000 A at 480 V Exception: 24,000A at 480V for MCC XEB1-1, XEB1-2, XEB2-1, XEB2-2
208/120-VAC, 120-VAC, and 118-VAC distribution panels	277 V	10,000 A at 208 V
125-VDC Class 1E switchboards	250 V	20,000 A at 125 V
125-VDC Class 1E distribution panels	250 V	10,000 A at 125 V

The diesel generators are sized so that each set is capable of carrying the required load for one unit in the unlikely case of a DBA or blackout. Based on the demands given in Tables 8.3-1 and 8.3-2, the continuous output rating of each diesel generator is 7000 kW at 0.8 power factor, 6900 V, three phase, 60 Hz, and has a normal operating speed of 450 rpm. Each diesel generator has a two hr rating of 7700 kW. The diesel generator excitation characteristic curve is provided in Figure 8.3-19.

The design and continuous rating is consistent with NRC Regulatory Guide 1.9 and IEEE 387-1977 [32].

The capacities of individual loads are determined on the bases of nameplate rating, pump pressure and flow conditions, rated BHP, or pump runout conditions. Basis of selection is noted in Tables 8.3-1 and 8.3-2.

8.3.1.1.5 Automatic Transfers, Loading and Load Shedding

8.3.1.1.5.1 Non-Class 1E Bus - Automatic Transfer

When Unit 1 trips, the power source for Bus 1A1 is automatically transferred from main generator 1G to transformer 1ST.

The logic, and typical electric schematics for Bus 1A1 are shown in Figure 10.2-1. Depending on the system conditions, this automatic transfer will be either “fast” or “slow” as described below:

Energizing of relays 86-1/1G, 86-2/1G, 94-1B/1G and 94-2B/1G results in tripping of the main generator. Upon operation of any of these relays, the normally closed incoming breaker 1A1-1 receives a trip signal.

An “early 52-b” contact of this breaker will send a signal to automatically close the offsite source incoming breaker 1A1-2 by:

1. Fast transfer, provided the

    a. Offsite source voltage is available (this voltage is monitored by undervoltage relays 27/1ST set to drop out at approximately 85% of nominal system voltage of 6900 volts) and,

    b. Offsite source and the bus 1A1 voltages are not out of phase by more than 40 degrees (as seen by the synchronism check relay 25/1A1).

    c. This transfer is completed in the minimum time possible (maximum of 10 cycles), however, it is disabled after 15 cycles from initiation to eliminate the possibility of accidental closure in case of any contact malfunction in the close circuit.

A successful fast transfer assures maintenance of voltage on bus 1A1 without interruption.

2. Slow transfer, if the requirements of a. are not satisfied within approximately 15 cycles after the closure of the “early 52-b” contact of the normally closed incoming breaker, provided:

    a. Voltage on the bus (to be transferred) has decreased to approximately 30% of the nominal voltage of 6900 volts,

    b. Offsite source voltage is available as described in 1.a) and

    c. All motor feeder breakers on bus 1A1 have tripped as evidenced by closure of their “52-b” contacts. Tripping of these motor feeder breakers is accomplished by bus undervoltage relays set at approximately 30% of the nominal voltage of 6900 volts via timers set at approximately 0.5 seconds.

The main generator 1G tripping is delayed by 30 seconds subsequent to a reactor trip (see FSAR Figure 7.2-1 Sheet 16 and Figure 10.2.1). This ensures full coolant flow for 30 seconds after a reactor trip before any bus transfer is made.

The automatic transfers of the other non-Class 1E buses (1A2, 1A3, 1A4, 2A1, 2A2, 2A3, and 2A4) are the same as discussed for 1A1 above.

8.3.1.1.5.2 Class 1E Buses-Automatic Transfer

1. Undervoltage Sensors to detect loss of offsite power at the safety buses.

   The sensors that detect loss of preferred and alternate offsite sources are located in 6.9 kV safety related switchgear buses 1EA1 and 1EA2 for Unit 1 and 2EA1 and 2EA2 for Unit 2. The scheme consists of instantaneous undervoltage relays which initiate tripping of the source breakers via time delay pick up relays.

   Loss of voltage detector relay setting is selected to preclude operation during momentary low voltage conditions that prevail during large motor starting. The selected setting is to provide a motor terminal voltage of ≥75% of motor rated voltage and it is lower than the lowest momentary voltage fluctuation for all plant load conditions. Settings for the time delay pickup relay is set long enough to preclude their operation during a three phase fault when system voltage could drop below the undervoltage relay setting, for a few cycles, until the fault is removed by opening the appropriate circuit breakers.

2. Undervoltage protection for Degraded Grid Voltages

   Degraded grid undervoltage protection is applied at the 6.9KV and 480V safety related buses as shown in Figure 8.3-18. The Degraded bus undervoltage relays are set to provide a motor terminal voltage of ≥90% of motor rated voltage for motors fed from the bus. These relays via time delays will trip the 6.9KV bus source breakers to assure that the bus loads exposure to degraded voltage is limited to 60 seconds. In the event of an ‘s’ signal, after the degraded condition has been confirmed that it is not due to a motor start, the Preferred source breaker is tripped instantly and the Alternate source breaker is tripped after a time delay, if alternate source does not restore the bus voltage.

   At the 480V bus, one set of Low Grid undervoltage relays, set at voltages higher than the degraded undervoltage relay settings, via time delays will initiate an alarm to indicate a degrading grid. In the event of an ‘s’ signal, after the low grid condition has been confirmed to persist, the 6.9KV bus preferred source breaker is tripped instantly and the alternate source breaker is tripped after a time delay if the alternate source does not restore the bus voltage.

3. Undervoltage protection for 6.9KV Dead Bus (6.9KV Class 1E bus loss of voltage)

   These relays monitor the loss of bus voltage and are set to assure that the bus voltage has decreased to a low enough value to be energized from another source and not result in equipment damage. After a time delay to allow bolted three-phase faults to be cleared by over current protection at the bus incoming breaker, the relay will trip all the 6.9KV motors, provide an alarm indicating loss of bus voltage, and permit the alternate source breaker to be closed to the 6.9KV bus. The relay also starts the D/G after a time delay to allow for the alternate source breaker to be closed.

   Degraded voltage conditions are annunciated by the operation of voltage relays on the high side of the startup transformers which detect/alarm small voltage variations in the

offsite power system grid voltage when outside the range stated in Section 8.2. The second level undervoltage relay set points are selected above the minimum out of tolerance values for the 6.9KV and 480 V safety related buses (switchgears and load centers) provided in Table 8.3-1C in order to allow continued operation of equipment without any damage. The degraded grid undervoltage protection relay settings assure a minimum of 90% of motor rated voltage at the motor terminals during steady state conditions. These relay settings, in conjunction with the settings of loss of power relays, assure that, on their actuation, the motors are not exposed to a sustained voltage of less than 75% of their rating and exposure to a voltage between 75% to 90% of motor rating is limited to a maximum of 60 seconds to preclude any motor damage. Motor damage on bus reenergization is precluded by the dead bus relay setting which assures that the bus residual voltage has decreased to a low enough value, before allowing closure of the alternate source breaker, such that on bus reenergization the motors are not exposed to a voltage of more than 1.33 V/Hz. Motor damage on bus reenergization by Diesel Generator is precluded by providing a time delay for bus voltage decay which assures that the bus residual voltage has decreased to a low enough value, before allowing closure of the diesel generator breaker, such than on bus reenergization the motors are not exposed to a voltage of more than 1.33 V/Hz. Automatic tripping of the source breaker to the safety bus by the second level undervoltage relay prevents loss of equipment function due to sustained severe undervoltages. Undervoltage and time delay relay set points are compatible with the equipment capabilities. Low Grid undervoltage relays provide an early alarm for a degrading grid. The safety related equipment can operate continuously at the low grid voltages, however, if the low grid condition persists the safety buses are isolated from the degrading grid on receipt of an SIAS, as a precaution only, to prevent safety related equipment exposure to degrading voltage.

Overvoltage/undervoltage relays coincidence logic are identified in Figure 8.3-18. The undervoltage relays coincidence logic preclude spurious tripping of offsite power source.

4. Automatic Transfer

When the source breakers for the preferred offsite power source are tripped, the Class 1E buses are transferred to the alternate power source by a slow transfer in a manner similar to that described for the non-Class 1E buses. For class 1E busses, the motor trip signals are concurrently initiated with alternate source breaker closing permissive signal.

The alternate power source of Unit 1 safety-related buses 1EA1 and 1EA2 is startup transformer XST1 or XST1A and the alternate power source of Unit 2 safety-related buses 2EA1 and 2EA2 is startup transformer XST2 or XST2A. The preferred and alternate power source startup transformers (XST1 or XST1A and XST2 or XST2A) are designed to feed only safety related buses. This eliminates the possibility of overloads on a startup transformer during any mode of plant operation. In the event that the preferred offsite source is lost, the Class 1E buses would automatically transfer to the alternate source. The transfer capabilities of the source feeders to the emergency busses were tested in the Startup Program as described in Table 14.2-2, AC Power Distribution Test Summary. Operation of the unit could continue for a limited period of time based upon the constraints defined in the plant Technical Specifications.

Low voltage (after an appropriate time delay) on the Class 1E bus shall start the respective emergency diesel generators for that bus should the alternate source not

return power to the Class 1E buses. The time allowed for a DG start signal, after detection of loss of preferred or alternate offsite source, assures that the DG will start in time to meet the accident mitigation requirements for accidents requiring DG LOOP start. If both the preferred and alternate power sources are not available, the emergency diesel generator will power the bus after it has reached rated voltage and frequency provided the time delay, for bus voltage to decay to an acceptable level, has elapsed and there is no bus fault. The bus will be powered by the emergency diesel generator within 10 seconds after the diesel receives a starting signal.

8.3.1.1.5.3 Sequencer Loading

1. Sequencer

For each unit, two independent and redundant solid-state safeguards sequencer (SSSS) cabinets are provided for sequential loading of the safeguard buses, one for Train A and one for Train B. Each cabinet houses two sequencers, one for the safety injection mode sequencing (SIS) and one for loss of offsite power (blackout) only mode sequencing (BOS). Both Sequencer cabinets are located in the main control room (Elevation 830'-0") of the Electrical and Control Buildings (see Figure 1.2-33). Power to the Solid-State Safeguards Sequencers (SSSS) is supplied from uninterruptible 118-volt ac panels. Each Sequencer cabinet will provide the logic for loading one 6.9 kV emergency bus in a pre-established time sequence in the event of loss and subsequent restoration of bus voltage and/or in the event of a LOCA.

Each sequencer is basically a group of output electromechanical relays operated by solid state logic circuits and timers. The separate and independent timer circuits are initiated for each loading step by solid state logic circuits. The SSSS accepts and combines safety injection signals from the solid-state protection cabinet and undervoltage (UV) input signals from the safeguards bus, and provides output control signals to the safety related equipment required for the safe shutdown of the plant. The output signals are shown in the functional logic diagram Figure 8.3-4. Mode decision logic responds to the status of the inputs by initiating the proper mode of operation.

The sequencer control logics each contain a time delay circuit to provide reset and restart should an undervoltage or safety injection occur during manual test or when the automatic test is operating. This circuit could delay the sequencer start up to one second during the normal auto-testing mode.

During normal operation, the sequencers do not function except for testing. Operation is required upon receipt of a safety injection signal, or loss and subsequent restoration of the 6.9 kV safeguards bus voltage, and for brief periods during periodic testing. Solid-state digital circuits are used for the logic and timing functions; sequencer equipment compatibility with existing systems is accomplished through input isolation buffer circuits and output relays.

Prior to seismic testing, sequencer equipment was aged in accordance with IEEE 323-1974. The System was tested as a completely assembled, operating, deliverable unit, not on a component basis. Contact chatter and time delay of all relays in the Sequencer System were monitored during the seismic tests, and the relays were maintained in their normal state (open or closed), during the tests. In the specific case of

the sequencer, the chatter monitors were capable of detecting discontinuities as small as 0.2 milliseconds, and none were detected. (The generally accepted industry criterion for relays is no more than 2 milliseconds.) The System tested fully energized, and all devices were monitored for full operability and functionality before, during and after all qualification tests. In addition, the relays were cycled from energized to deenergized or deenergized to energized state where appropriate relative to their function within the System. The Sequencer System met all performance requirements without any failures or anomalies; all electrical circuits functioned properly during the tests. See Section 3.10B for testing and documentation requirements.

Reliability of each sequencer is assured by continuous testing from the input diode matrix through the logics and timers to the coils of the output relays. If a fault is detected it is alarmed in the control room. A review of the sequencer circuitry reveals that the sequencer design is relatively direct and simple, not subject to sneak circuits and is highly reliable.

Each SSSS is also designed to combine manual testing with an overlapping automatic test to verify SSSS equipment operation from contact inputs to contact outputs. The manual test facilities include input test switches, output test switches, output blocking switches, local annunciator lamps and digital display. The manual test provides the capability for exercising all of the SSSS logic functions and verifying the time delay setting of each sequencer step output. Additionally, isolated contact outputs are provided for remote annunciation of system blocked for test, operator and auto lockouts operated, measurement of the time delay of each sequencer step output and input status.

The automatic test provides the capability for continuous surveillance of the SSSS operation from the logic input signals through the logic and counter states and up to and including continuity of the output buffer and relay coils. The automatic test does not interfere with system requirements or cause any output relay actuation during normal system operation. The automatic test monitors the SSSS, and upon occurrence of any improper response, displays the step number of the failed test and energizes a fault relay for remote annunciation. The automatic test may be conveniently interrupted at any time by means of the control panel AUTO TEST ON/RESET switch. The automatic test consists of four distinct parts: input diode matrix, test input generator, fault logic, and buffer logic. The automatic test sequence is initiated when the AUTO TEST ON/RESET switch is positioned to ON. This action releases the automatic test, resets and allows the test input generator and the input diode matrix to generate the test inputs necessary for testing. When testing the sequencer circuit, or other time dependent circuit, an automatic test frequency is substituted for the circuits normal time base clock frequency to maintain the circuits time response compatible with the operating rate of the AUTO TEST. During this time, the fault logic monitors appropriate outputs to verify SSSS operation. If the fault logic detects an improper response, a fault condition is initiated. All system accuracy and function requirements are maintained with automatic test implementation. Detailed descriptions of the test circuitry are contained in controlled copies of vendor manuals and CPNPP procedures, which are maintained at the plant site.

2. Loading

In the event of safety injection actuation, the following sequence of operation is initiated.

a. The diesel generator sets receive starting signals and the sequencer receives a signal to arm the Safety Injection Sequencer (SIS).

b. All non-Class 1E loads connected to Class 1E 6.9KV and 480V buses are tripped except those which are isolated in accordance with Section 8.3.1.2.1.7a.3.

c. Power source for the Class 1E buses is established as discussed above. Diesel generator breaker will close, if required, after the diesel generator rated voltage and frequency have been established provided the time delay, for bus voltage to decay to an acceptable level after both preferred and alternate source breakers are open, has elapsed and there is no bus fault. SIS is started on restoration of bus voltage. Large loads required during a DBA are started in sequence by the safety injection sequencer. Small loads (less than 20 hp) are generally started in accordance with their respective circuit logic. See Table 8.3-1 for details.

This mode functions to recognize a one-out-of-six condition of the safety injection input signals from the solid-state protection system cabinet without a UV condition. A single safety injection signal, without a UV condition, initiates the Safety Injection Sequencer (SIS) timing cycle. The SIS provides 10 independent, time-separated, sequential loading signals to equipment loads listed in Table 8.3-1. A two-out-of-six (or more) condition causes SIS circuits to function in a like manner. Additionally, the two-out-of-six condition causes the SIS to generate SIS auto-lockout and SIS operator lockout signals. The last SIS step, in addition to providing an output loading signal, also provides a reset signal to SIS operator lockout circuits. The SIS sequential loading signals can be reset manually; this also resets the SIS auto-lockout signals.

On loss of both preferred and alternate offsite sources, the incoming breakers on the source buses are tripped or prevented from closing if open.

If both offsite sources are lost (blackout), but no “S” signal is present, the following sequence of operation is initiated:

a. Undervoltage signal arms the Blackout Sequencer (BOS) and starts the diesel generator sets.

b. Undervoltage shed the required loads from the Class 1E buses. The load shedding system consists of shedding of loads at three levels.

   1. Loads powered from Class 1E 6.9kV switchgear are shed by undervoltage relays when the bus voltage decays to approximately 29.3 percent of the rated voltage.

   2. Loads powered from Class 1E 480V AC switchgear are shed by undervoltage relays when the bus voltage decays to approximately 70 percent of the rated voltage.

   3. Loads powered from Class 1E 480V AC MCC are shed automatically on loss of power be deenergization of circuit holding coils which are powered from the same source as the MCC.

c. Power source for the Class 1E buses is established as discussed above. Diesel generator breaker will close, if required, after the diesel generator rated voltage and frequency have been established provided the time delay, for bus voltage to decay to an acceptable level after both preferred and alternate source breakers are open, has elapsed and there is no bus fault. The BOS is started on restoration of bus voltage. Large loads required during a Loss of Offsite Power (blackout) mode are started in sequence by the blackout sequencer (BOS). Small loads (less than 20 hp) are generally started in accordance with their respective circuit logic. See Table 8.3-2 for details.

This mode functions to recognize a two-out-of four or three-out-of four condition of the UV input signal from a 6.9 kV safeguards bus with no safety injection signal present. A two-out-of-four UV condition (true indication of loss of bus voltage) places the Blackout Sequencer (BOS) circuits in a standby condition. When the UV input status returns to a one-out-of-four or less condition (true indication of acceptable bus voltage), the BOS circuits begin their timing cycle and provide 10 independent, time-separated, sequential loading signals to the blackout loads listed in Table 8.3-2. Output relays are shown in Figure 8.3-4.

A three-out-of-four (or more) condition of UV input signals (as shown in Figure 8.3-4) causes the BOS circuits to function as previously described. Additionally, it also causes the BOS to generate BOS auto-lockout and BOS operator lockout signals.

The last BOS step, in addition to providing an output loading signal, also provides a reset signal to the BOS operator lockout circuits. The BOS sequential loading signals can be reset manually; this also resets the BOS auto-lockout signals.

In the event of safety injection, plus blackout, the SSSS functions to recognize the simultaneous or sequential occurrence of UV and safety injection signals. In this mode, the time-step output signals of both sequencers are deenergized and remain deenergized until the status of the UV input returns to normal. When the UV condition returns to normal, the SIS proceeds through its operation as described for safety injection alone. See Figure 8.3-4.

The load shedding function is retained after diesel generators are connected to their respective buses. If load shedding function bypass feature were to be adopted, any malfunction on the bypass system may prevent the required load shedding prior to diesels connected to their buses.

8.3.1.1.6 Safety-Related Power System Equipment Identification

Safety-related power system electrical equipment is uniquely numbered so that identification as safety equipment is evident. The numbering sequence used for Class 1E equipment is different from the one used for non-Class 1E equipment. The equipment numbering system is discussed in Subsection 8.3.1.3. In addition, color-coded nameplates conspicuously identify the major equipment as Class 1E. Plant personnel can determine the train or channel that the equipment is associated with by this color.

Color delineation of the trains and reactor protection instrumentation channels is as follows:

1. Engineered Safety Features (ESF) Systems

Train A	orange
Train B	green

2. Reactor Protection System and ESF Systems at Channel Level

Channel I	red
Channel II	white
Channel III	blue
Channel IV	yellow

These color designations are consistent throughout the plant to allow for immediate identification. Cable and cable tray identifications are discussed in Subsections 8.3.1.1.3 and 8.3.1.4. These standards do not apply to the identification of control panels.

8.3.1.1.7 System Instrumentation and Control

Bus voltage and frequency monitors include the voltmeters for 6900 volts bus 1EA1, 1EA2, 2EA1 and 2EA2, for transformers XST1/XST1A and XST2/XST2A High-Voltage Sides as well as the frequency recorder for 345KV buses East or West and frequency meter for 6900 volts bus 1EA1, 1EA2, 2EA1 and 2EA2.

Remote instrumentation for the 6900-V Class 1E switchgear consists of ammeters in the Control Room and on the hot shutdown panel for the preferred power source and the alternate power source. The remote instrumentation for the standby power source (diesel generator unit) consists of ammeter, voltmeter, frequency meter, wattmeter, and varmeter in the Control Room and an ammeter on the hot shutdown panel. Various annunciators and bus voltmeters and frequency meters are located in the Control Room. Instrumentation for each 480-V Class 1E switchgear consists of ammeters on the 6900-V switchgear breaker for that switchgear and a bus voltmeter located on the 480-V switchgear. This is shown on Figures 8.3-6 and 8.3-8.

The diesel generator breaker control and synchronizing switches are located in the Control Room for corrective action by the operator. Selector switches located on the shutdown transfer panel and on the hot shutdown panel, and diesel generator breaker control switches located on the hot shutdown panel, are provided for corrective action by the operator in the unlikely event the Control Room becomes inaccessible.

All control switches for 6900-V Class 1E incoming supply breakers, tie breakers, and feeder breakers for Class 1E loads are located on the main control board in the Control Room. In addition, the control switches for Train “A” 6900-V Class 1E incoming breakers, tie breaker, and selected feeder breakers are located on the hot shutdown panel. All 6900-V Class 1E circuit breakers have local test switches at the respective switchgear which are operable only when the breaker is in the “test” (disengaged) position.

Similarly, all control switches for 480-V Class 1E incoming supply breakers, tie breakers, and selected feeder breakers are located on the main control board in the Control Room. In addition, the control switches for Train "A" 480-V Class 1E incoming supply breakers, tie breaker, and selected feeder breakers are located on the hot shutdown panel. All 480-V Class 1E circuit breakers have local test switches at the respective switchgear which are operable only when the breaker is in the "test" (disengaged) position.

Control required for safe shutdown in the event of Control Room uninhabitability is discussed in Section 7.4.1.3.

The described control and instrumentation is used in testing the diesel generator and in monitoring 6900-V and 480-V Class 1E switchgear during normal and Loss of Offsite Power conditions.

Control power for Class 1E equipment is provided as follows:

Class 1E Equipment	Class 1E Control Power Source
6900-V breaker and associated protective relaying	125-VDC system
480-V load center breakers and relaying	125-VDC system
480-V motor control center starters	120-VAC derived through control transformers
Control relays and panels and instrument racks	125-VDC and 118-V uninterruptible AC

8.3.1.1.8 System Testing During Power Operation

Onsite and offsite power sources are not paralleled except for testing of the diesel generators. During the testing mode, only one onsite source is paralleled to the offsite source at any one time. When a diesel generator is not under test, the standby power source is separated from the preferred power source. CPNPP design does not provide for automatic paralleling of these sources by the sequencer. Consequently, no single failure in the sequencer will either compromise separation of redundant portions of these two power sources, or render both preferred power sources and both redundant onsite power sources unavailable.

The transfer of power from preferred to alternate to standby power sources and for load shedding prior to sequential loading are discussed in Subsection 8.3.1.1.5.

The testability of safeguard actuation systems is discussed in Section 7.3.2.

8.3.1.1.9 Sharing of Equipment between Two Units

Nuclear-safety-related loads associated with each unit are powered exclusively from Class 1E systems of that particular unit. However, nuclear-safety-related loads common to both units are powered from Class 1E MCCs and distribution panels which have supplies from each unit.

These dual unit supplies are interlocked to preclude supplying power to one MCC or distribution panel from both units simultaneously. Incoming feeders to train A MCCs or distribution panels common to both units are supplied only from train A power systems of both units (a similar arrangement exists for train B equipment and incoming feeders). This ensures the proper train separation between the equipment common to both units. Train separation is further discussed in Subsection 8.3.1.1.3.

8.3.1.1.10 Basis of Loading

Power requirements for each safety load (Class 1E) are based on motor nameplate rating, pump pressure and flow, or on pump runout conditions. The basis of selection is noted in Tables 8.3-1 and 8.3-2. Continuous duty safety-related (Class 1E) motors have a nameplate service factor of 1.15, except for some totally enclosed Class 1E motors. These motors fall into two categories: first, motors rated 15 percent in excess of the load design brake horsepower (bhp), with a unity nameplate service factor, and NEMA Class-H insulation system, with temperature rise in accordance with NEMA Class-B requirements; and second, totally enclosed air-over fan motors which have an air-over service factor of not less than 115 percent at design conditions and have a nameplate service factor of 1.0.

8.3.1.1.11 Onsite Emergency Power Sources (Diesel Generators)

The standby power system structure is discussed in Subsection 8.3.1. The Diesel Generator System is shown on the main and auxiliary one-line diagrams (Figures 8.3-1 and 8.3-6).

1. Starting Circuits

   Each diesel generator set automatically starts in emergency mode whenever any of the following conditions occur:

   a. Undervoltage on its respective emergency bus

   b. Safety injection actuation signal (SIAS)

   The means of actuating the Safety Injection System are described in Section 7.3.

   The emergency buses are normally fed from the preferred offsite power source, but upon loss of the preferred power supply, automatic bus transfer to the alternate power source occurs after the undervoltage relays on the Class 1E buses have shed the required loads on the Class 1E buses. The diesel generators are not started if the alternate source returns power to the class 1E buses. However, if the alternate source is not available, the diesel generator starts and the breaker will automatically close when the diesel generator has reached rated voltage and frequency. Sequential application of the loads will occur to avoid excessive loading of the diesel generator. For details, see Subsection 8.3.1.1.5.

Automatic starting and loading logic for diesel generator 1EG1 is shown on Figure 8.3-3. This logic is typical for diesel generators 1EG2, 2EG1, and 2EG2.

Each diesel generator is capable of attaining rated voltage and frequency and is ready to accept load within 10 sec after receiving a starting signal. Generator reactances and the characteristics of the static exciter and voltage regulator are coordinated to provide satisfactory starting and acceleration of sequenced loads. The exciter regulator system design ensures rapid voltage recovery when starting large motor loads. Voltage drops between sequencing steps do not exceed the limits established in NRC Regulatory Guide 1.9 [3], with the following comment: the voltage may dip below 75 percent of nominal voltage when the diesel generator breaker closes and energizes the two 2000/2666-kVA, 6.9-kV/480-V unit substation transformers supplied from each diesel generator. This dip is due to transformer magnetizing inrush current which exists for two to three cycles. The diesel generators are designed for recovering to 80 percent of nominal voltage within 10 cycles for this transient. Therefore, the dip may delay motor starting a maximum of 10 cycles. However, since all Class 1E motors have acceleration times within 5 seconds and safe stall times are considerably longer than 10 cycles this potential starting delay has no significant effect on the motors' availability or performance.

Each EDG uses a digital excitation/voltage regulator. This digital exciter/voltage regulator is a Safety Related Siemens THYRIPART generator field excitation unit. This THYRIPART excitation unit utilizes a digital control system consisting of a 32-bit technology module and communications module. This system controls the output of a chopper controller circuit to achieve $\pm$0.5 percent Diesel Generator voltage regulation, and has capability of both manual as well as automatic control.

The THYRIPART excitation units have three modes of operation, two automatic and one manual. The normal operable mode of operation is the AVR mode, in which all exciter regulation functions occur automatically. Under unusual circumstances, a bumpless transfer is manually initiated to the manual (ECR) mode with a trouble indication, in which the field current can be manually adjusted directly from the control room via RAISE and LOWER commands. An open-loop control mode is available in which the entire control of the exciter is conducted manually.

If Diesel Generator voltage goes outside the Technical Specifications limits, the digital control system is automatically disconnected, while EDG voltage remains sufficient for performance of its required loads. The controller then operates in isochronous mode with the magnetics portion of the system maintaining the Diesel Generator voltage within the Technical Specification limits.

Starting of the diesel engines is accomplished by a Compressed Air System consisting of independent and redundant air compressors, receivers, and solenoid valves. Each of the redundant receivers is sized for sufficient compressed air storage for five starts. Section 9.5.6 presents a more detailed description of the starting air system.

Fast starting and load acceptance are enhanced by maintaining engine temperature by heating and forced circulation of cooling water and lube oil. In addition, the units are located in heated and ventilated rooms.

The manual starting of the diesel generators from the Control Room is incorporated into the design to permit periodic testing of the units.

Each EDG has a digital governor system. The digital governor system allows slow starting of the EDG. Slow starting is better for the EDG. It reduces wear by avoiding the full fuel rack starts. Soft start capability is a recommendation of NUREG 1366 and 1431.

Each diesel generator has a manual start/stop switch in the Control Room and on the local panel.

2. Protection Systems

The diesel generator protection systems initiate automatic and immediate protective actions to prevent or limit equipment damage and allow restoration of the equipment upon correction of the trouble. Automatic tripping of the diesel generators occurs for any of the following reasons, when the diesel generator is started in normal mode.

a. Lube oil low pressure

b. Engine overspeed

c. Crankcase high pressure

d. Generator differential

e. Generator reverse power

f. Jacket water high temperature

g. Generator outboard bearing high temperature

h. Generator loss of excitation

i. Lube oil high temperature

j. Generator negative sequence

k. Generator voltage restrained time overcurrent

l. Generator stator ground voltage

m. Generator field ground

n. Engine high vibration

o. Turbo high vibration

p. Engine bearing high temperature

q. Left or right bank turbo low oil pressure

r. Generator overexitation

s. Generator neutral ground overcurrent

t. Generator underfrequency

u. With the S600 switch (EDG AVR Trip/Reset Switch) in Auto, any of the following signals will trip the EDG:

1. Loss of excitation system 24VDC supply

2. Loss of DC supply to Masterdrive

3. Loss of DC supply to DC field transducer

4. Loss of DC supply to internal binary inputs

5. Masterdrive T-400 trip

6. Masterdrive CUVC trip

7. AVR trip

With the exception of generator differential and engine overspeed protection, the preceding trips are bypassed when the diesel generators auto start in emergency mode because of a Safety Injection Actuation Signal (SIAS), or associated bus undervoltage or when the diesel generators are manually started in emergency mode. The diesel generator differential relays are General Electric type IJD with a high degree of field-proven reliability.

A diesel generator trouble alarm in the Control Room is provided to alert the operators of any of the abnormal conditions listed previously. Appropriate action can then be taken by the operator as required.

Overcurrent relay protection is provided on each load feeder. Motor feeders connected to the 6900-V switchgear buses are protected by phase and ground overcurrent relays; 480-V feeders are protected by phase overcurrent relays only. See Figures 8.3-5 through 8. If the feeder breaker fails to clear, backup relaying trips the incoming circuit breaker. Smaller motors are connected to MCCs containing combination starters, where short-circuit protection is provided by circuit breakers and overload protection by thermal relays.

Class 1E motor-operated valve motor starters are provided with thermal overload relays connected to alarm only. The valve motors are protected against sustained fault conditions by circuit breakers. The valve motors may be protected by fused disconnect switches if full load currents are less than 1.3 amps.

The unit auxiliary and startup transformers are protected by differential relays and overcurrent backup relaying. Faults within the differential relay zone of the unit auxiliary transformer cause the respective unit to trip and open the unit output 345-kV breaker in

the switchyard. The non-Class 1E 6900-V buses then fast transfer to the offsite power source. A fault within the differential zone around either startup transformer opens the appropriate high-voltage breaker in the appropriate switchyard and respective 6900-V breakers and causes the Class 1E 6.9-kV buses to transfer to the alternate power source with no loss of unit operation.

All relay settings are coordinated to isolate a fault condition without interfering with the effective operation of the rest of the system. Settings allow primary relaying to function before the backup relays.

Interlocks are provided to ensure safe and proper operation of the electrical systems. Typical applications are as follows:

a. To lock out all possible sources of power to a bus if a fault exists on the bus

b. To prevent closing of the diesel generator breakers if voltage is present on the bus with the control switch in the automatic mode

c. To prevent closing of the diesel generator breaker until the diesel generator achieves rated voltage and frequency

d. To prevent the two incoming breakers on the Class 1E buses from operating simultaneously in the closed position

e. To prevent paralleling both battery chargers associated with each 125-VDC system

f. To prevent out-of-phase bus transfers

3. Testability and Maintenance

Circuit design provisions incorporate test capability to periodically monitor the operational capability of the safety-related Class 1E systems during power operation. Initially, all safety-related equipment is tested during the startup testing phase.

Diesel generator sets are tested after final assembly and preliminary startup. Site acceptance tests are given to each diesel generator set to demonstrate its required capabilities.

The following tests are administered to certify the adequacy of the units for the intended service:

a. Starting tests

b. Load acceptance tests

c. Rated load tests

d. Design load tests

e. Load rejection tests

f. Electrical tests

g. Subsystem tests

The objectives and requirements of the preceding tests are established in IEEE 387-1977 [32].

Periodic testing of the diesel generators will be performed as specified in the Technical Specifications to verify their continued capability and availability to perform their design function after commercial operation of the plant. Typical tests consist of availability and operational tests as outlined in IEEE 387-1977.

The CPNPP preventive maintenance program is established to prevent failures. Should failures occur, however, the program will function to identify root cause of any malfunctions and to perform the required repairs or component replacement. On completion of the repairs/maintenance, final equipment check is performed prior to starting of any tests to assure a good start. On satisfactory completion of the post-maintenance testing, control of equipment is transferred to the control room operator. These are accomplished by maintenance department procedure.

4. Diesel Generator Fuel Oil Storage and Transfer System

For description of fuel oil storage and transfer system, see Section 9.5.4.

5. Cooling and Heating Systems

a. For description of diesel generator cooling and heating system, see Section 9.5.5.

b. For description of diesel generator starting system, see Section 9.5.6.

c. For description of diesel generator lubricating system, see Section 9.5.7.

6. Instrumentation and Control Systems

The diesel generator starting mode selector switch has remote/local/maintenance positions and is normally in the remote mode of operation position. The only time it is in the local position is for local manual starting in the event of Control Room evacuation and nonroutine testing. If the switch is not returned to the remote position from the local or maintenance position, an alarm in the Control Room persists until the switch is returned to the remote mode. Other manual controls of the auxiliary system of the diesel generator are also alarmed if failure to return to their auto start position would inhibit automatic operation of the diesel generators.

Normal and emergency diesel generator switches (start/stop) are provided in the Control Room and at the local panel.

Control power is obtained from redundant 125-VDC systems. Train A loads of Unit 1 and Unit 2 receive power from batteries BT1ED1 (Unit 1) and BT2ED1 (Unit 2), respectively.

Train B loads of Unit 1 and Unit 2 are fed from batteries BT1ED2 (Unit 1) and BT2ED2 (Unit 2), respectively. For details see Subsection 8.3.2.

The source of control power to each switchgear is indicated below:

Unit No.	Battery Source	Classification (Safety Train)	Switchgear 6.9kV	No. 480V
1	BT1ED1	A	1EA1	1EB1,1EB3
	BT1ED2	B	1EA2	1EB2,1EB4
	BT1D2	Non-safety	1A1[(a)],1A3[(a)]	1B1, 1B3
	BT1D4	Non-safety	1A2[(a)],1A4[(a)]	1B2, 1B4
2	BT2ED1	A	2EA1	2EB1,2EB3
	BT2ED2	B	2EA2	2EB2,2EB4
	BT2D2	Non-safety	2A1[(a)], 2A3[(a)]	2B1, 2B3
	BT2D4	Non-safety	2A2[(a)], 2A4[(a)]	2B2, 2B4
1 & 2	BT1D2[(b)] and BT2D2	Non-safety	XA1	

a) control power for Unit 1 and 2 RCP breakers is supplied from batteries BT1D5 and BT2D5 respectively

b) via transfer switch

The DC power is required by each diesel generator for controls, alarms, protective relays, air-starting solenoid valves, and generator field flashing from the station batteries.

The following instrumentation is provided to monitor the operability of each diesel generator in the Control Room:

a. Voltmeter

b. Ammeter

c. Frequency meter

d. Varmeter

e. Wattmeter

Instrumentation and alarms provided for diesel generator cooling, starting, lubricating, and ventilation systems are discussed in Sections 9.5.5, 9.5.6, and 9.5.7, respectively. Locations of the previously mentioned instruments in Items a through e are shown on Figure 8.3-6.

Local indication or alarm devices for the standby power system are provided as follows:

a. Bus voltage and frequency

b. Diesel generator status

c. Protective relaying operational alarms

Status indication of safety-related switchgear breakers and MCC breakers is also provided.

7. Prototype Qualification Program

The test program to be implemented for qualification of diesel generator sets is briefly outlined as follows:

a. One full load test is performed on the diesel generator to demonstrate the start and load capability of these units at the two hour rating.

b. Prior to initial criticality of Unit 1, at least 300 valid start and load tests are performed to demonstrate a 0.99 reliability factor with a 50-percent confidence level. This includes all valid tests performed offsite. [270 valid start and load tests are performed from cold ambient conditions with loading to at least 50 percent of the continuous rating within the required 10-sec interval and continued operation until temperature equilibrium is attained. Cold ambient condition is defined as the ambient condition of the diesel generator during normal plant operation, i.e., the diesel generator not running and the water jacket and lube oil heated. 30 valid start and load tests are performed from design hot equilibrium temperature conditions (normal operating temperature equilibrium conditions) with loading to at least 50% of continuous rating within the required 10 second interval.] The onsite testing is to be performed using the diesel generator auxiliary systems and the actual connected loads.

   A failure rate in excess of one per hundred in a total of 300 start and load tests requires further testing, as well as a review of system design adequacy.

   At least five on-site start and load tests are required to be performed following installation of each diesel generator unit.

c. The diesel generator supplier has performed in his facility one full load test and two margin tests to demonstrate capability in excess of the design requirements on one CPNPP unit.

The seismic testing and analysis of the diesel generator set is performed per IEEE 344-1975 [26].

8.3.1.1.12 Class 1E Equipment Design Criteria

1. All safety-related continuous duty Class 1E motors are sized with a nameplate service factor of 1.15, except for some totally- enclosed Class 1E motors. For service factor of totally-enclosed Class 1E motors see Subsection 8.3.1.1.10.

2. All safety-related Class 1E motors are designed to accelerate their driven loads with 80 percent of the motor rated voltage available at the terminals except for Class 1E Motor Operated Valve (MOV) motors. For Class 1E MOV motors, the available voltages at motor terminals are evaluated for adequacy of MOVs to perform their functions.

3. Motor starting torque is selected based on load speed torque curves and combined wk^2 of motor and driven load for the above terminal voltage and acceleration time (less than five seconds) requirements. In accordance with the loading sequence established in Table 8.3-1, motor acceleration time is specified to achieve motor rated speed in less than five seconds. Motor driven loads not requiring sequencing such as fans have acceleration times in excess of five seconds.

   In addition some fan motors that are part of the sequencing cycle also have acceleration times in excess of five seconds, which will not adversely affect the safety related system performance.

4. The minimum motor torque margin over pump torque through acceleration period is 20 percent for the BOP motors. For Class 1E pump motors furnished by Westinghouse as part of NSSS, the minimum margin of motor torque over the pump full load torque (as defined by the pump speed/torque curve) is sufficient to accelerate the driven equipment within five sec and with 80 percent of rated voltage at the terminals from standstill to operating speed.

5. Motor insulation is selected on the basis of ambient temperature and expected temperature rise based on worst-case loading conditions. In general, the motors have Class B insulation.

6. All 6.6-kV motor stator windings are provided with two temperature detectors embedded in each phase. In case a temperature detector fails, the temperature detectors are abandoned in place and may be replaced when the motors are rewound.

7. Interrupting capacities of switchgear, load centers, motor control centers, and distribution panels are selected on the basis of short-circuit calculations. Transformer impedances are selected to permit starting of the largest motor on the bus without exceeding a maximum voltage drop of 20 percent and at the same time remaining within the interrupting and momentary capabilities of the breakers.

8. Electric circuit protection is provided to prevent damage to the equipment, maintain operational continuity, and reduce the safety hazard to plant personnel. Fast-acting relays are used to respond to overload undervoltage or faults on feeders or buses to initiate corrective actions and isolate the affected equipment.

9. System grounding and equipment grounding are provided to reduce the safety hazard to plant personnel.

   The 6900-V system is grounded through a low-resistance and designed to trip on ground faults to protect the 6900-V equipment. The diesel generator units are high resistance grounded to allow the generator to continue to supply power without tripping on a single ground fault condition. The 480-V system is high-resistance grounded with a ground detection scheme. This system does not trip on ground fault condition. For each transformer, a voltage relay device is connected across the secondary of a potential transformer (PT) and the primary of the PT is connected across the grounding resistor. If a ground fault occurs, one of these relays will pick up; a ground fault condition in the 480-V system or in the Diesel Generator System is then annunciated in the Control Room.

   The grounding grid of the plant is designed to achieve a ground resistance of less than 0.5 ohm. To achieve this, a ground grid of bare copper cables is buried approximately 2.5 ft below grade level. A cable size of 500 thousand circular mils (MCM) is used for the main loop and interconnections and No. 4/0 AWG cable size for equipment ground connections. These criteria are in accordance with IEEE Guide No. 80, 1971 [40].

8.3.1.1.13 118-V Uninterruptible AC Power

The 118V AC single phase, 60HZ, grounded uninterruptible power is supplied to critical instrumentation and control circuits. 118V AC Class 1E, UPS is described in Section 7.6.1.1. Each unit has two non Class 1E uninterruptible power supplies (UPS) for BOP, non Class 1E, instrument and control systems. In addition each unit has two uninterruptible power supplies (UPS) for the plant computer and other plant loads. Each UPS consists of an inverter, automatic static switch and a manual bypass switch.

For details see Figures 8.3-15, 8.3-15B and 8.3-15C. Each Class 1E inverter receives incoming power supply from the class 1E DC switchboard. The DC switchboard is normally fed from the battery charger. The battery charger receives its incoming power from a class 1E motor control center that has access to the two offsite sources and one onsite source.

In case of loss of the battery charger, the station batteries will feed the DC switchboard and there will be no interruption of power to the inverter loads.

In the absence of inverter output, the static switch disconnects the inverter from the load and connects the load to the alternate supply line from the bypass (nonregulating) transformer within 1/4 cycle. The manual bypass switch operation is make before break.

Each bypass transformer receives incoming power supply from a Class 1E 480-V motor control center. The inverter will automatically resynchronize with respect to frequency and phase with the bypass source following a loss and restoration of this reference source. Under normal operating conditions, the static transfer switch connects the inverter to the load. However, a manual bypass transfer switch, provided for equipment maintenance, can be used to bypass the static switch and supply power to the load directly from the bypass transformer. Operation of the bypass switch to any position will not cause power interruption to the load. Bypassing of the static switch or inverter is annunciated locally and in the Control Room. In the case of inverter

supplying power to the plant computer and other non-Class 1E instruments, the DC power input is provided from the non-Class 1E 125/250 VDC system.

The AC instrument buses have two incoming circuit breakers to power the loads from either the UPS or directly from the bypass transformers. Common AC instrumentation buses have two incoming circuit breakers to power the loads from either Unit 1 AC instrumentation bus or Unit 2 instrumentation bus. These manually operated circuit breakers are mechanically interlocked to prevent paralleling of both sources.

The mechanical interlock between the bypass source and the inverter source breakers of panels 1EC1, 1EC2, 1EC5, 1EC6, 2EC1, 2EC2, 2EC5, 2EC6, 1PC1, 1PC2, 1PC3 and 1PC4 may be removed during inverter maintenance. When the interlock is removed, appropriate procedural controls are exercised to prevent paralleling the inverter output with the bypass source.

Class 1E buses feeding common systems loads shared by both units receives power from buses having an incoming manual transfer switch that can select power from either unit. Transfer switch design is such that power can not be supplied from both units simultaneously. Train separation is maintained by supplying these shared buses from the same train of both units.

On loss of offsite power and subsequent loss of battery charger output, 118 VAC and 125 VDC buses feeding common loads will be automatically fed by the battery. If the battery charger output can not be restored, by restoration of offsite source or by EDG, during battery duty cycle, the common 118 VAC panels shall be manually transferred to other unit and the common 125 VDC panels shall be automatically transferred to the other unit.

A spare inverter is provided in each train of 118 VAC system. The spare inverter can be manually aligned to substitute any of the four inverters in that train. Procedural controls and interlocks ensure that the spare inverter can feed the loads of only one inverter at a time and the power source of the spare inverter is the same as that of the substituted inverter.

The AC output voltage provided by the inverters remains regulated within the following limits: voltages of 120V +/-2%, Frequency of 60 HZ. +/-0.5%. The inverters are provided with a synchronizing circuit to synchronize its output sine wave with the 120V AC bypass source.

1. Tests and Inspection

   Prior to placing the Class 1E UPS systems in operation, the system components are tested to ensure their proper operation. The inverters are checked for output voltage and frequency, and transfer between normal and bypass sources, while operating on either the normal or bypass supplies. Panel-mounted instruments monitoring the inverter are calibrated and annunciator and static switch operation checked. During plant power operations, the UPS systems are periodically tested and inspected to ensure their continued capabilities to perform their operations.

   The inverter can be removed from service for inspection and test by manually transferring to the bypass power source. The surveillance instrumentation provides continuous monitoring of the system.

8.3.1.1.14 Physical Arrangement of Class 1E Equipment

Physical locations of Class 1E equipment are selected to minimize vulnerability to physical damage. The degree of separation takes into account the potential hazards in a particular area. Separation is achieved by locating equipment and circuits in separate rooms, maintaining distance, or by use of barriers. The potential hazard of non-safety-related equipment failure on safety-related redundant equipment is considered in choice of equipment location or protection.

Class 1E switchgear and equipment located below the probable maximum flood level of the reservoir (elevation 789.7 ft) are protected as described in Section 3.4. Wherever practicable, electrical equipment is located away from mechanical piping in order to minimize damaging effects of pipe ruptures. All floor levels above 810 ft have adequate drainage provisions to preclude accumulation of water if a pipe rupture occurs. Plant levels 810 ft and below are provided with adequate sumps to limit water accumulation.

Location and separation of major Class 1E equipment are shown in Figures 1.2-11 through 1.2-19 and 1.2-32 and 1.2-33. Separation of major electrical distribution equipment is as follows:

1. Unit auxiliary, station service and startup transformers are located outdoors, physically separated from each other as shown on Figure 8.2-1. For fire protection features, refer to Section 9.5.1.

   Lightning arrestors are used to protect the main transformers, station service transformers, unit auxiliary transformers and startup transformers against lightning. Although these transformers are not classified as Class 1E equipment, careful attention is given to the design of these facilities to ensure reliable and continuous operation.

2. Separate rooms in the seismic Category I Safeguards Buildings provide adequate separation for the redundant Class 1E 6900-V switchgear buses and 480-V load centers. These rooms are located above the PMF level. Piping containing fluids is excluded from these rooms. Rooms containing train A equipment are separated from train B equipment by floor elevations with concrete floor slabs having a minimum thickness of 12 inches. Any failure, whether electrical or physical, in one room will not have any effect on the redundant equipment in the second room. The 6900-V Class 1E switchgear rooms are located adjacent to the diesel generator buildings to minimize the length of the cable runs between the switchgear and associated diesel generator and reduce the possibility of mechanical damage to these connections. Additional protection is provided to the generator leads by running the cables entirely in conduits.

3. The 480-V MCCs are located in areas of electrical load concentration. Redundant Class 1E MCCs are located within seismic Category I structures and separated by location in separate rooms, on different floor elevations, or physically spaced a minimum of 20 ft apart to preclude a single failure from defeating the operation of both trains.

4. Two seismic Category I Diesel Generator Buildings, which are part of the Safeguards Buildings, are provided as shown in Figures 1.2-11 and 1.2-17. One building contains diesel generators 1EG1 and 1EG2, installed as part of Unit 1 construction; the second building contains 2EG1 and 2EG2, installed as part of the Unit 2 construction. Separation between redundant diesel generators within each building is accomplished by locating each diesel generator in a separate room partitioned by a concrete wall designed to

withstand an SSE, fire, or missiles. Each diesel generator and its associated starting equipment and auxiliaries are located in the same room. This arrangement results in a complete system independent and isolated from the redundant diesel generator and its systems. Each diesel generator is provided with independent room ventilation air intake and independent air discharge and engine inlet and exhaust ducts. Any credible single failure does not immobilize both diesel generator systems (see Section 9.4.8.1).

For diesel generator combustion, air intake, and exhaust systems, see Section 9.5.8.

The fire protection provisions for the diesel generators are provided in Section 9.5.1.

5. Containment electric penetration assembly arrangement design provides access for connecting cables, leak testing during operation, and space for removal and replacement. Separation of penetrations meets single-failure criteria.

   Physical separation between redundant circuits is achieved by establishing three independent penetration areas on different floor elevations. Location and separation of penetrations are shown on Figure 8.3-16. Separate penetrations are provided for medium voltage and low voltage power, control, and instrumentation (includes fiber optic circuits) functions. The maximum possible separation between Class 1E electrical penetrations of redundant trains and any large piping penetration is provided to prevent damage from steam line or waterline rupture.

   Redundant penetrations are located at different floor elevations, where floor slabs are a minimum of 12 in. thick. The minimum centerline separation between any two nonredundant electrical penetrations is 2 ft 2 in. Protection Channels I and III penetrations are located at elevation 842 ft 6 inches. Protection Channels II and IV penetrations are located at elevation 870 ft 0 inches. Minimum separation (clear air space) between any two channel penetrations is 3 ft horizontal and 5 ft vertical. The design and qualification tests are in accordance with IEEE 317-1976 [21] and ASME Boiler and Pressure Vessel Code [45].

8.3.1.2 Analysis

The onsite AC, DC, and 118-V uninterruptible AC electric systems are in conformance with General Design Criteria (GDC), NRC Regulatory Guides, IEEE standards, and other applicable criteria as listed in Section 8.1.4 and as described briefly in Section 8.1.5.

A review of the Class 1E and Non-Class 1E busses supplying power to safety and non-safety-related instrumentation and control systems which could affect the ability to achieve cold shutdown was performed. All power supplies are train-related with the exception of the charging pump discharge flow valve, which will fail open upon loss of power. The loss of power to this valve will not affect the ability to achieve cold shutdown, since the valve fails open. A review of the Emergency Operating Procedures found:

1. Alarm and system operating procedures specify the actions to be taken for loss of a Class 1E or non-Class 1E power supply for instrumentation and control. Actions to restore the supply to normal are in the system operating procedures.

2. Abnormal Conditions Procedures (listed in Table 13.5-3) specify alternate instrumentation and/or controls which can be used by the operator upon loss of power to a particular bus.

8.3.1.2.1 Compliance

1. Compliance With GDC 17 [1]

The safety-related systems are designed with sufficient capacity, independence, and redundancy (as described in Subsections 8.3.1 and 8.3.2) to ensure performance of their safety functions assuming a single failure. The offsite electrical power system also provides independence and redundancy (as discussed in Section 8.2) to ensure an available source of power to the safety-related loads.

Upon loss of the preferred power source to any 6.9 kV Class 1E bus, the alternate power source is automatically connected to the bus and the diesel generator starts should the alternate source not return power to the Class 1E buses. Loss of both offsite power sources to any 6.9 kV Class 1E bus, although highly unlikely, results in the diesel generator providing power to the Class 1E bus.

As discussed in Subsection 8.3.1, two independent diesel generators and their distribution systems are provided for each unit to supply power to the redundant onsite AC Power System. Each diesel generator and its distribution system is designed and installed to provide a reliable source of redundant onsite-generated (standby) AC power and is capable of supplying the Class 1E loads connected to the Class 1E bus which it serves.

As discussed in Subsection 8.3.2, four independent Class 1E 125-V batteries and their distribution systems are provided for each unit to supply power to the redundant DC systems. Each Class 1E battery and its distribution system is designed and installed to provide a reliable source of redundant onsite DC power. Each Class 1E battery is capable of supplying power for four hours to the Class 1E loads connected to the Class 1E bus which it serves.

Redundant parts within the AC and DC systems are physically and electrically independent to the extent that a single event or single electrical fault can not cause a loss of power to both Class 1E load groups.

2. Compliance With GDC 18 [1]

The electric power systems are designed to permit inspection and testing of all Class 1E systems. Periodic testing is performed on a scheduled basis to demonstrate the operability and continuity of all safety-related systems and components. The testing capability provided for the diesel generators and Class 1E batteries is described in Subsections 8.3.1.1 and 8.3.2.1, respectively. Testing capability for solid-state safeguards sequencers (SSSS) is discussed in Section 8.3.1.1.5.3. Plant design also provides testing capability of other Class 1E equipment as required by IEEE 308 [20].

3. Compliance With NRC Regulatory Guide 1.6 [2]

The CPNPP design is in compliance with the provisions of NRC Regulatory Guide 1.6.

The electrically powered safety loads, both AC and DC, are separated into two redundant and completely independent load groups for each unit. There are no automatic or manual ties between redundant load groups.

No single failure can prevent operation of the minimum number of required safety loads and loss of any one group will not prevent the minimum safety functions from being performed. Each Class 1E AC bus has access to two offsite power sources and an onsite standby power source. There are no automatic or manual ties between redundant buses.

Two diesel generators are provided for each unit. Each diesel generator is connected exclusively to its associated 6.9-kV Class 1E bus, which ensures independence in the onsite standby power sources.

Each Class 1E DC bus can be energized either by a battery or by one of two battery chargers (one spare) or combination of battery and battery charger. There are no automatic or manual ties between Class 1E redundant DC load groups. Arrangement of the AC and DC systems is described in Subsections 8.3.1 and 8.3.2, respectively.

Because there are no bus ties between redundant load groups, interlocks are not required.

4. Compliance With NRC Regulatory Guide 1.9 [3]

The rating of the diesel generators is based on the maximum continuous load demand. This rating exceeds the sum of the conservatively rated loads. Motor loads are based on nameplate rating, pump runout conditions, or flow pressure conditions. 6600-V motor efficiency is based on design data. Low-voltage motor efficiency is assumed to be 80 percent.

During preoperational testing, the maximum continuous load demand is verified by tests.

Each diesel generator set is capable of starting and accelerating to rated speed all Class 1E loads in the required sequence.

Sequencing of large loads at 5-sec intervals ensures that large motors have reached rated speed and that voltage and frequency have stabilized before the succeeding loads are applied. The voltage may dip below 75 percent of nominal voltage when the diesel generator breaker closes and energizes the two 2000/2666 kVA, 6.9 kV/480-V unit substation transformers supplied from each diesel generator. This dip is due to magnetizing inrush current which exists for two to three cycles. The diesel generators are designed to recover to 80 percent of nominal voltage within 10 cycles for this transient. The effect on the first load group would, therefore, be a maximum possible delay of 12 to 13 cycle after closure of the diesel generator breaker. However, the objective of first load group and subsequent load groups is not affected. During recovery from transients caused by step load increases or resulting from the disconnection of the largest single load, the speed of the diesel generator set should not exceed the nominal speed plus 75 percent of the difference between nominal speed and the overspeed trip setpoint or 115 percent of nominal, whichever is lower. The voltage is restored to within 10 percent of nominal; and the frequency is restored to within two percent of nominal in less than

40 percent of each load sequence time interval. The diesel generator supplier has successfully performed these tests in his facility on one CPNPP diesel generator set.

The prototype qualification test program of

a. Start and load capability at full load, and

b. 300 valid start and load tests

on the diesel generator are discussed in Section 8.3.1.1.11.

5. Compliance With Regulatory Guide 1.32 [7]

The offsite power system includes the preferred design stated in NRC Regulatory Guide 1.32: namely, two immediate access circuits from the transmission network are available to the emergency (Class 1E) bus systems.

Each battery charger is sized to handle the combined steady-state loads while recharging the battery from the design minimum charge state to the fully charged state under all modes of plant operation.

6. Compliance With NRC Regulatory Guide 1.63 [12]

The electric penetration assembly design complies with the intent of NRC Regulatory Guide 1.63.

The propagation of light thru fiber optic cable in the Electrical Penetration Fiber Optic modules does not generate heat. Therefore, the Electrical Penetration seals and Electrical Penetration concrete interface are not impacted by Fiber Optic Circuits. As such, the circuit protection requirements of NRC Regulatory Guide 1.63 are not applicable to Fiber Optic Circuits of Electrical Penetrations.

In reference to Regulatory Position C.1 of NRC Regulatory Guide 1.63, the electric penetration assembly design, for electrical circuits, is capable of withstanding, without loss of mechanical integrity, the maximum current versus time conditions permitted by backup protective devices. The adequacy of penetration protective devices to protect the penetrations is established by detailed calculations which demonstrate that the fault current-versus time conditions for which the penetrations are designed and qualified will not be exceeded.

Circuits using fiber optic cables are not required to have overcurrent protection. The penetration assembly modules for fiber optic cables will only contain fiber optic circuits and, therefore will not impact the fault current vs. time conditions of the penetrations.

The electrical distribution system design incorporates backup protective devices for all power circuits. Control circuits have also been provided with backup protective devices.

Fuses or fusible links within the penetration assembly are not incorporated in the design because of the physical limitations of the standard penetration designs available.

The circuit design complies with IEEE 279 [19] in that the protective devices are redundant. Circuit independence and physical separation are limited in that both protective devices are in series in the same circuit as described in the following paragraphs. IEEE-279 is not applicable to individual penetration circuit conductors since this standard applies to protective and actuation signals that are part of the multichannel systems required to perform safety functions.

Incorporation of online testability, bypassing, manual initiation, and other requirements of IEEE 279 [19] for every electric penetration assembly circuit is extremely difficult to implement and is not required. The application of backup protection devices and periodic testing of primary and backup devices is sufficient to ensure that the penetration assembly integrity is not violated if one device fails to operate.

Protective devices are tested by the manufacturer in accordance with appropriate NEMA and ANSI standards to assure the devices are qualified to perform their protective functions. In addition, periodic testing verifies the ability of protective devices to interrupt during abnormal operating conditions.

The primary and backup protective device schemes for all power circuits are described below. Also described below are the protection schemes for control circuits. Justification is provided where the control level penetration assembly integrity is maintained without primary and/or backup protective devices.

a. 6900-V Circuits

   Reactor coolant pumps are the only 6900-V loads inside the Containment. The primary and backup protection of these circuits is provided by means of a reactor coolant pump breaker and bus incoming supply breakers, respectively.

b. 480-V Switchgear Motor Feeder Circuits

   The primary protection of these circuits is provided by means of feeder breakers. The backup protection of these circuits are provided by means of an incoming bus supply breaker or a bus tie-breaker.

c. 480 Volt Motor Control Centers (MCC) Outgoing Feeders -

   1. For all starter circuits feeding motors and for all feeder circuits, primary protection is generally provided with a circuit breaker and/or starter with a thermal overload relay in series with a circuit breaker as backup protection or two circuit breakers in series for primary and backup protection are provided in each circuit.

   2. For starter circuits feeding motors having full load currents less than 1.3A, primary and backup protection may be provided by means of a fused disconnect switch and a circuit breaker respectively.

d. Control Circuits of 480-V MCC Outgoing Feeders

Two fuses are used in series in the ungrounded leg of each 120V AC control circuit, one as a primary and the other as a backup device.

e. 125-VDC Power and Control Circuits

1. 125V DC For S.O.V.'s -

The primary and backup protection of these feeder circuits is provided by means of fuses and a circuit breaker respectively or two pairs of fuses.

2. 125V DC Miscellaneous Control Circuits -

The primary and back-up protection of these circuits are provided by means of fuses and a circuit breaker respectively or by two pairs of fuses.

f. Rod Control System Lift and Gripper Coil Circuits -

The primary and backup protection for the Lift and Gripper Coil circuits is provided by fuses.

g. Pressurizer Heater Circuits -

1. Control Group -

The primary and backup protection of these circuits is provided by two feeder breakers in series.

2. Backup Groups -

The primary and backup protection of these circuits is provided by the branch circuit breaker and the panel incoming breaker.

h. Motor Space Heaters

1. Heaters of motors fed from 480V SWGR -

Primary and backup protection are provided by fuses and circuit breakers, respectively.

2. Heaters of motors fed from 480V MCC's -

Fuses are used as primary and backup protection devices.

i. A. C. Lighting Panels Feeders and Polar Crane Feeder-

Primary and backup protection are provided by separate circuit breakers.

j. Liquid Waste Process Panel -

Primary and backup protection are provided by means of a fuse and a circuit breaker respectively.

k. RCP Motor Differential Protective CT Penetration Circuits-

During the normal operation of the RCP motor, there is no current in the CT secondary circuits which penetrate the containment. The primary and backup protection of these circuits are provided by means of a RCP motor feeder breaker and bus incoming supply breakers, respectively.

l. Personnel and Emergency Air Locks -

Primary and backup protection are provided by means of fuse and circuit breaker respectively.

m. 118V AC Control Circuits From Miscellaneous Control Cabinets -

Primary and backup protection to the control circuits originating from NSSS Relay Racks, BOP Relay Racks and PASS Containment Isolation Valve Control Panels are provided by means of a fuse and a circuit breaker respectively or two identical fuses in series.

n. Instrument Distribution Panelboard -

The primary and backup protection for the control power feed to the Instrument Distribution Panelboards inside containment are provided by means of two fuses in series.

o. Monitor and Status Light Circuits -

Primary and backup protection are provided by means of two fuses in series.

p. Annunciator Circuits -

Primary and backup protection of these circuits are provided by means of two fuses in the supply unit of the annunciator cabinet.

q. Solid State Isolation Cabinets -

The maximum available short circuit current at the penetration conductors of these circuits is less than the continuous current rating of the conductors. Therefore, the penetration conductors are self protected and primary and backup protection is not required.

r. Fire Protection/Detection System -

Primary and backup protection is provided by fuses.

s. Fuel Transfer System -

115V, AC control circuit conductors of this system penetrate containment. These circuits, including penetration conductors are protected against overcurrent by two fuses in series (primary and backup protection).

t. Emergency Evacuation System Warning Lights -

The primary and backup protection of these circuits is provided by means of a fuse and a circuit breaker, respectively.

u. In-Core Flux Mapping System -

The primary and backup protection for this system is provided by means of two fuses in series.

v. Plant Communication System -

The primary and backup protection for this system is provided by means of a fuse and circuit breaker, respectively.

w. Rod Position Indication Cabinets -

Power circuits from 120V distribution panels. Breakers are used as primary and backup protection devices.

7. Compliance With NRC Regulatory Guide 1.75 [15] and IEEE 384 [31]

The CPNPP design complies with the intent of NRC Regulatory Guide 1.75 and IEEE 384 (Refer to Appendix 1A(B)). Physical separation of redundant safety-related equipment and wiring is achieved by location in separate rooms or by providing barriers. Isolation devices are provided to preclude interaction between Class 1E and associated circuits and non-Class 1E circuits, as described in the following paragraphs.

Electrical isolation methods are used as required in power, control and instrumentation circuits to maintain the independence of redundant circuits and equipment such that protective functions required during and following any design basis event is accomplished. Different types of isolation devices are used for power, control and instrumentation circuits.

Isolation devices meet the criteria and performance requirements specified in IEEE 279-1971, Revision 1, and are qualified in accordance with IEEE 323-1974 and 344-1975.

Associated circuits shall comply with one of the following:

a. They shall be uniquely identified as such and shall remain with, or be separated the same as, those Class 1E circuits with which they are associated.

b. They shall be in accordance with (1) above from the Class 1E equipment to and including an isolation device. Beyond the isolation device a circuit is not subject to the requirements of IEEE Std. 384-1974, provided it does not again become associated with a Class 1E system.

c. They shall be analyzed or tested to demonstrate that Class 1E circuits are not degraded below an acceptable level.

The following paragraphs describe the various conditions:

a. Power Circuits

The following types of devices are used in the CPNPP design for isolation of power circuits:

1. Circuit breaker coordinated with the upstream protective device and tripped by a safety injection signal.

2. Circuit protective device of a starter coordinated with the upstream protective device and starter contactor opened by a safety injection signal.

3. Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective device, and the circuit breakers periodically tested.

The non-Class 1E loads connected to Class 1E power buses are isolated with an isolation device as described above. These power circuit loads are identified in Table 8.3-11 and in the following figures:

Figure No.
8.3-6 (Sheets 1&2)
8.3-8 thru 12

The above figures identify the circuits with non-Class 1E loads listed in Table 8.3-11 which remain associated after the isolation device. Table 8.3-10 identifies the non-safety loads fed by associated cables, after isolation, where separation from internal non-Class 1E circuits within the equipment is not required. The cables that remain associated after the isolation device meet all the requirements imposed on Class 1E cables and are routed with the train associated with their respective Class 1E power source. The Class 1E isolation device combined with Class 1E circuit protection device, as well as the qualification and routing of these cables, assures that any fault in the connected non-Class 1E load will not degrade the Class 1E power source or Class 1E cables routed with the associated cables.

b. Control and Instrumentation Circuits

The isolation devices mentioned below are classified as Class 1E for control and instrumentation circuits. Separation between the Class 1E and non-Class 1E wiring is maintained within the cabinets in which these devices are mounted. Maximum wiring separation at the isolation device is limited by the physical design of the device.

1. Qualified auxiliary relays provide isolation between Class 1E, redundant Class 1E, and non-Class 1E circuits for both coil to contact and contact to contact isolation. Electrically, Class 1E circuits are equal or superior to non-Class 1E circuits. Physically, Class 1E circuits are subject to more stringent analysis than non-Class 1E circuits. Therefore, devices qualified for non-Class 1E-to-Class 1E isolation applications provide more than adequate train-to-train isolation.

2. Phototransistor coupled pairs (light emitting diode and transistor) provide isolation of Class 1E contacts to non-Class 1E monitoring devices.

3. A SIAS signal provides isolation between Class 1E and non-Class 1E circuits by relay actuation, which opens the circuit.

4. Current Transformers provide isolation between their Class 1E primary circuit and non-Class 1E secondary circuits. This application is based on testing (Reference 46).

5. Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective device, and the circuit breakers periodically tested, provide isolation between Class 1E and non-Class 1E circuits.

6. Within the NSSS protection system cabinets, a fuse or breaker, coordinated with the upstream protective device, is an isolation device to prevent malfunctions in Non-Class 1E portions of a circuit from causing unacceptable influences on the Class 1E function of the circuit (Reference 48). This exception to the isolation devices described in the preceding paragraph (5), is applicable only to the NSSS circuit design by Westinghouse. See Section 7.1.2.2.1.

7. Frequency Transducers, provide isolation between Class 1E circuit and non-Class 1E Frequency Indicators.

Cables that remain associated after the isolation device meet all the requirements imposed on Class 1E cables and are routed with the train associated with their respective Class 1E system. The circuit protection and Class 1E isolation device, as well as the qualification and routing of these cables, assures that any fault in the connected non-Class 1E device will not degrade the Class 1E system including the Class 1E cables routed with the associated cables.

Safety-related equipment, exposed raceways, and cables are identified by distinct color markers so that the plant personnel can distinguish, without resorting to any reference material, between the various redundant Class 1E systems and between redundant Class 1E systems and non-Class 1E circuits.

The physical separation and identification of circuits are described in detail in Subsections 8.3.1.4 and 8.3.1.3, respectively.

c. NSSS Instrumentation Power Circuits

Instrumentation power for Unit 1 NSSS is provided from distribution panels 1PC1, 1PC2, 1PC3 and 1PC4. See Figure 8.3-15. These panels power safety loads as well as non-safety loads.

Each non-safety circuit powered from these panels will have a non-safety circuit breaker or fuse connected in series with the panel circuit breaker.

Protection channel wiring, safety-train wiring and non-safety train wiring within panels 1PC1, 1PC2, 1PC3 and 1PC4 will be in different wire bundles. These bundles will be separated to the maximum extent practicable.

The same criteria described above is also applicable to Unit 2 NSSS instrumentation circuits and their respective distribution panels.

d. Lighting System

The non-Class 1E security lighting circuits and the non-Class 1E AC lighting inside the containment building are isolated from their Class 1E power source with two separate Class 1E breakers connected in series. These breakers are coordinated with their supply breakers and will be tested periodically to ensure that coordination is maintained.

The non-Class 1E AC essential lighting circuits are isolated from Class 1E power sources with two separate Class 1E breakers (i.e., the 480V MCC supply breaker and the main breaker within the Class 1E lighting distribution panel) connected in series. These breakers are coordinated with their supply breaker and will be tested periodically to ensure that coordination is maintained.

The non-Class 1E AC essential lighting circuits use interconnecting cable (i.e., from the lighting distribution panel feeder breaker to the lighting load) routed in conduit. The routing of the circuits in conduit ensures the physical and electrical independence from Class 1E circuits beyond the second isolation breaker.

The non-Class 1E DC emergency lighting circuits connected to dedicated batteries are routed in conduit. The routing of the circuits in conduit ensures physical and electrical independence from Class 1E circuits.

The lighting circuits routed in conduit meet the separation criteria of Section 8.3.1.4.

e. Diesel Generator Neutral Ground System

The non-Class 1E diesel generator neutral grounding transformer is connected to the neutral of the Class 1E diesel generator. The diesel generators have 7200-120/240 volt resistance grounding transformers on their neutrals which limit the ground fault current in the associated 6.9 kV system to a low level. This enables the diesel generators to supply power to safety loads during an emergency with a ground fault on the 6.9 kV system. A voltage relay is connected across the secondary of the neutral grounding transformers.

An analysis has been performed which demonstrates that a fault on the non-Class 1E portion of the circuit will not cause an unacceptable influence on the Class 1E system. If a ground fault occurs in the 6.9 kV safeguards electrical distribution system, while it is being powered by a diesel generator, the associated voltage relay will pick up and activate an alarm on the local diesel generator control panel. In addition, a “Diesel Generator Trouble” alarm will be activated in the Control Room, which will indicate that an alarm condition exists at the local panel. If a safety injection signal does not exist concurrently with the actuation of the voltage relay, a diesel generator trip will occur when a ground fault is detected. If a safety injection signal does exist prior to the actuation of the voltage relay, the diesel generator will not trip and the alarm will notify the operator of trouble.

In addition, the interconnecting cable is routed within the diesel generator room. The cable is routed in dedicated raceway and is inspected to Class 1E requirements.

f. Safety System Inoperable Indication (SSII) Panel Isolation and Separation

The Safety System Inoperable Indication (SSII) panels have input signal circuits originating from dry contacts in Class 1E field devices. The SSII panel indication performs no safety-related function nor is operator manual action required based solely on SSII displays. The equipment is non-Class 1E and is fed from a non-Class 1E power source. The logic between the non-Class 1E power circuits and the input circuits from the Class 1E sources is not electrically isolated based on the following analysis.

The output of the SSII power pack is at 115-VDC which is protected by a 0.5 amp fuse. The circuit to the individual field contact is current limited at the logic card to a value of 0.767 mA by means of a series resistor. In addition, logic panel input power is protected by two 3 amp fuses in series. Field contacts at Class 1E devices are rated for 125-VDC and 0.5 amp or higher, which is more than adequate to handle the current and, as such, a fault at the SSII logic panel will not affect the Class 1E field contacts or devices.

The multiconductor cable between the SSII logic panel located in the control room and the termination cabinet in the cable spreading room meet the requirements of Regulatory Guide 1.75, Revision 1, Regulatory Position C.4 except for Class 1E environmental qualification. The cable materials meet IEEE Standard 383-1974 for the attributes of flame and radiation resistance. The cable is installed in a mild environment. One cable is used per ESF train. The insulation is rated at 600V.

Any individual conductor is sized at a minimum #18 AWG while carrying a maximum 0.767 mA (current is limited at the SSII logic card as noted above). The system is protected by a 0.5 amp fuse at the 115-VDC SSII power supply. Based on the above analysis, a fault at the SSII logic panel will not impact the availability of adjacent Class 1E cables with which these associated cables are routed.

g. Non Class 1E Indicating Lights

The emergency diesel generator engine control panel has non Class 1E indicating lights which are connected to a Class 1E power supply and are not provided with isolation devices.

The above indicating lights are located in a mild environment, seismically installed and all cabling is made to Class 1E requirements. Since the lights are seismically installed they will cause no damage to other Class 1E devices in the Control Panel during a seismic event. Hence the only possible failure mode of the lights is open circuiting of the lamp filament which will result in isolation of the light from the Class 1E power supply and loss of indication. Since the lights are for non Class 1E loads (e.g., DG Air Compressor, Fuel Oil Booster Pump, Jacket Water Keep Warm Pump, Pre-lube Pump, Aux Lube Oil Pump, and Aux Jacket Water Pump), opening of the lamp filament and loss of indication is not a concern and it will not degrade the Class 1E power supply.

The above design assures that non Class 1E indicating lights, wired to Class 1E requirements, and seismically installed in Class 1E equipment with no isolation devices will not degrade the Class 1E power supply.

h. Associated cables from non-Class 1E alternate power transfer switch to Class 1E 6.9kV switchgear

CPNPP Class 1E 6.9kV switchgear can be connected to a non-Class 1E alternate power diesel generator (APDG) through a non-Class 1E alternate power transfer switch. With the exception of a brief test window during setup of the APDGs during modes 5 and 6 and in the unlikely event of loss of all AC power, the above cables will remain de-energized during all modes of plant operation and the 6.9kV circuit breakers located in the Class 1E switchgear remain racked out and their locking mechanisms in engaged positions thus isolating the Class 1E 6.9kV bus from the APDG power feed. In view of the above it is concluded that any cable fault (open circuit, short circuit or ground) in the out door section of the above associated cables is of no consequence and it will not affect the integrity of the Class 1E circuits with which these cables are associated.

In Modes 5 and 6 when the APDG may be feeding a train of Class 1E 6.9kV switchgear and during brief period of test window when APDG is being aligned to a train in outage, a cable fault in the out door section of the associated cables will expose the associated portion inside the building to a fault current contributed by the motor loads on the 6.9kV switchgear. The associated cable rating and size are adequate to withstand the magnitude and duration of the fault current provided by the motor loads and as such have no adverse effect on the integrity of the Class 1E circuits with which these cables are associated. Because these

cables inside the building are inherently protected and will adequately withstand the exposure due to fault outside the building, Class 1E protection devices to isolate the fault to protect Class 1E circuit with which these cables are associated, are not required.

In view of the above analysis it is concluded that the non-seismic supports and routing of these associated cables outdoor, do not degrade the Class 1E circuit with which these cables are associated. Additionally, Class 1E devices are not required to protect these cables because these cables inside the seismic building are inherently protected from faults caused outside the building.

i. Level Switch Electronics Boxes (X-LY-4849A-1, X-LY-4849A-2, X-LY-4849B-1 and X-LY-4849B-2)

The Class 1E Electronics Boxes (X-LY-4849A-1, X-LY-4849A-2, X-LY-4849B-1 and X-LY-4849B-2) are connected to the Non-Class 1E annunciator in the Spent Fuel Pool Panel CPX-EIPRLV-06. At the Electronics Box, the Class 1E Cables from the level switches and 120V AC power supply cable enter the box from the top where as the Non-Class 1E cables from the annunciator enter the box from the bottom. The cables from the Class 1E Electronics Boxes to the plant computer are Non-Class 1E. The conductors originate at a Class 1E I/O board and is routed through an isolation device to the terminal block within the Electronics Box. To ensure the isolation device is protected from hot shorts, each conductor is independently fused between the isolation device and the terminal block. Two 1/4 amp, 250 VAC Class 1E fuses on the “+” and “-” of the isolator output are used for protection. These fuses will open under an abnormal faulted circuit condition to prevent damage to the isolator. The circuit is considered to be Non-Class 1E after the Class 1E fuses. The conductors are routed away from any Class 1E device inside the Electronics Boxes, and are landed below the annunciator circuits on the terminal block. However, inside the Electronics Box the Class 1E and Non-Class 1E Cables both terminate on the same terminal block. The terminal block is heavy duty, barrier type, rated 600 Volts and 75 Amps AC, with breakdown voltage 13,000 V RMS line to line. The line to line spacing between the terminals is 0.66 inches. Non-Class 1E Cables are rated at 600 V AC and are fire retardant.

The annunciator circuit rated voltage is 12 V DC and the maximum fault current that the annunciator power supply can furnish is limited to 1.3 mA. As the cables are routed in dedicated conduits the probability of these cables being imposed with any other voltage due to external cable faults is precluded. Inside the Spent Fuel Pool Panel the devices are seismically installed and cable terminations made per the requirements of ES-100. There are Non-Class 1E Cables that carry 120 V AC and are terminated in the vicinity of above annunciator cables terminations. Since the cables are trained to terminate on particular terminals, it is not credible that the 120V AC carrying cable terminals will get loose and create a hot short at the annunciator cable terminals that are several terminals away.

A 250 ohm resistor, connected in parallel to the output of the isolation device and mounted on the terminal block, is being used to provide a 1-5 VDC output signal to the plant computer. Train “C” conduit is used to route the cable from the

Electronics Boxes to a level 4 tray and is then run to the Safeguards building in this tray. The level 4 tray system, which has level 3 trays running above it, is supported as Seismic Cat. II, at minimum. Therefore, the entire tray system will be substantially rigid in case of a seismic event. If the tray(s), running above the tray system carrying the output cable, were to collapse, the highest voltage level which could possibly come in contact with the Non-Class 1E cable would not be greater than 120 VAC. Since it has been determined that the isolation device, in conjunction with the fuses, could withstand a transient voltage exceeding 120 VAC, there is no reason to suspect that an accident of any magnitude could induce voltage which would cause a credible failure of any kind. Therefore, an accident will not create a credible failure to a Class 1E device within the Electronics Box. Dedication testing has provided data to show, with the inclusion of the fuses to the output of the isolation device, the isolation device can withstand open circuits, shorts across the output terminals and application of abnormal circuit voltages to the isolator output without damage to the Class 1E input circuit. Therefore, these cables do not pose any new failure modes.

The devices inside the Electronics Boxes are seismically installed, terminations made per the requirements of ES-100, the Non-Class 1E terminals are located below the Class 1E terminals and the wires are trained to terminate on particular terminals. In view of the above it is not credible that Non-Class 1E terminals will get loose and cause a hot short at the Class 1E terminals above them.

The Non-Class 1E Cables that originate from the Spent Fuel Panel CPX-EIPRLV-06 annunciator are wired to the dry contacts of Class 1E relays installed in the Electronics Box. The relay's contact is rated two Amps at 115 V AC, which is more than adequate to handle the maximum fault current of 1.3 mA DC. The Class 1E relays (American Zettler Relay Type AZ2428-C56-40L) used for annunciation purposes do not do any other function. Since the annunciator circuit maximum fault current is limited to 1.3 mA and the circuit voltage cannot be more than 12V DC any faulted condition in the Non-Class 1E cables will have no adverse impact either on the Class 1E Cables, circuits or on the relay contacts.

In view of the above discussion it is concluded that the Non-Class 1E Cables connected to Class 1E Electronics Boxes are acceptable as it will not degrade the operation of Class 1E Electronics Boxes. Hence, the Electronics Boxes (X-LY-4849A-1, X-LY-4849A-2, X-LY-4849B-1 and X-LY-4849B-2) are exempt from separation distances between the Class 1E Cables/Terminations and the Non-Class 1E Cables/Terminations. Also no Electrical Isolation is required between the Electronics Boxes and cable from the annunciators in the Spent Fuel Panel (CPX-EIPRLV-06).

j. Non Class 1E Emergency Diesel Generator Stator RTDs

The Emergency Diesel Generator stator RTDs do not perform a safety related function, and therefore are classified as non-class 1E. The diesel generator vendor has evaluated the various failure modes and determined that the failure of these RTDs have no effect on the safety function of the generator.

RTDs are passive devices. The only credible failure modes for the RTDs are an open circuit or a short circuit. These failure modes can cause false temperature indication. However, the temperature indications derived from these RTDs do not have a safety function. Consequently, any incorrect temperature indications derived will not have an adverse effect on safety.

Unit 1 and Unit 2 Emergency Diesel Generator RTDs are input through a digital multifunction protective device (MPR-1) that generates a common alarm locally as well as a trouble alarm to the control room. This MPR-1 device output alarm functions are non-safety.

k. Associated control cables interfacing with the non-Class 1E control cables at the Instrument Air Compressor Termination Cabinet CP1-CICACO-01A for Unit 1 and at the Instrument Air Compressor Control Panel CP2-CICACO-01B for Unit 2. These associated cables are considered adequately isolated and are exempt from separation requirements where they interface with non-Class 1E circuits at the termination cabinet CP1-CICACO-01A or at the control panel CP2-CICACO-01B. This configuration has been evaluated and found to be acceptable on the following basis:

   1. The power for these control circuits is tripped on a SIS. Therefore, during any event where a SIS is present, this circuit configuration can have no adverse effect because the entire circuit is de-energized and a failure of the Non-1E components could not affect the 1E components.

   2. Worse case credible potential cable faults have been evaluated. Characteristics of the circuit design including double fuses at the control power transformer for Unit 1 and a fuse/breaker arrangement for Unit 2, ensure that no cable damage results from even the worse case faults and no propagation to nearby Class 1E circuits is credible.

l. Personnel Airlock Hydraulic Unit Power Feeder Control Panel CP1, CP2-BSCPEB-01.

   The following analysis demonstrates that non-Class 1E Train C 480V power cables connected to the line and load side of breaker CP1, CP2-BSDSEB-01 located in personnel airlock power feeder control panel CP1, CP2-BSCPEB-01 will not cause an unacceptable influence on Class 1E Train B wiring/cabling powering Class 1E Train B shunt trip coil of the breaker due to lack of physical separation.

   The non-Class 1E cables connected to the breaker are 600V Class 1E cables used in a non-1E application and therefore meet IEEE 383 requirements. Shunt trip Class 1E cables are also 600V IEEE 383 rated. These cables are trained and secured within panel CP1, CP2-BSCPEB-01 so as to maximize the degree of spatial separation between them. Breaker CP1, CP2-BSDSEB-01 and panel CP1, CP2-BSCPEB-01 are Class 1E Train B and installed and supported Seismic Category I. All cable terminations to the breaker, non-Class 1E line and load side cables and Class 1E shunt trip cables are terminated to meet the

Class 1E cable termination requirements. This precludes any seismically induced failure of the breaker, panel mounting or cable terminations.

The non-Class 1E cables are fire retardant, adequately designed (voltage level and load service) and are protected by at least two breakers in series; therefore, under any cable fault condition, the non-Class 1E cables will be protected. The fault will not propagate or effect the Class 1E Train B cable. Given the maximized spacing between wiring/cabling, Seismic Category I mounting of the panel, breaker and cable terminations, it is not credible that the Class 1E Train B cables and non-Class 1E Train C cables will short.

Based on the above analysis, the internal spatial separation, seismic panel mounting, Class 1E termination, circuit protection and fire retardant cable assures that any fault in the connected non-Class 1E cabling will not degrade the Class 1E system.

m. Unit 2 Turbine Driven Auxiliary Feedwater Pump Control Panel (CP2-EIPRLV-48)

CP2-EIPRLV-48 is the Unit 2 Turbine Driven Auxiliary Feedwater Pump (TDAFWP) Control panel. The panel does not serve a safety related function and contains one non-safety related cable along with several Train A associated cables. The non-safety related cable is part of a circuit that provides an annunciation interlock between the lube oil trouble alarm and the tachometer. Power to the Associated cables is tripped upon receipt of an SI signal. The maximum credible fault current in the non-safety related cable is less than the current carrying capacity of the cable and will not affect any of the Associated cables in its vicinity (even touching).

Therefore, the separation between non-safety and Associated Train A cables in CP2-EIPRLV-48 is not required.

n. Non-Class 1E SSW Traveling Screens Differential Level Transmitters fed by a Class 1E Power Supply

The Class 1E Power Supply (X-LQ-4288A, X-LQ-4288B, X-LQ-4289A and X-LQ-4289B) is protected by an internal (1 Amp) fuse. A current regulator limits the current to 45 mA.

Three resistors (internally mounted in the power supply) are connected in series with a 4 to 20 mA level transmitter (X-LT-4288A, X-LT-4288B, X-LT-4289A and X-LT-4289B). The transmitter can be represented as a variable resistor in the loop. A resistor (392 Ω) is used to compensate the resistance of the loop, keeping the current between 4 and 20 mA. The voltage across another resistor (250 Ω) is used to verify that the loop is working correctly. Another resistor (30.1 Ω) is used to provide input voltage to an Operational Amplifier. The Operational Amplifier output is processed to provide an indication or an alarm signal.

If an open circuit occurs, due to a failure of the level transmitter, no current will flow in the loop. The voltage across the 30.1 Ω resistor will not be normal and an indication or an alarm may initiate. If a short circuit occurs, due to a malfunction of

the level transmitter, the current flowing in the loop will be greater than 20 mA. The voltage across the 30.1 Ω resistor will not be normal and an indication or an alarm may initiate. A fault to ground of the transmitter will have the same effect as a short circuit. These indications and alarms are not a safety-related function. Therefore, the malfunction of the level transmitters is not a safety concern.

A loop voltage and a short circuit current test on the power supply card, required by the vendor to verify the operability of the card, is performed by connecting a 100 Ω 1 W resistor which simulates a short of the transmitter, the 392 Ω and 250 Ω resistor. The test required by the vendor simulates a more severe case than the failure of the transmitter only. As such, the power supply will not be challenged by the failure of the transmitter only and the failures in the transmitter will not adversely affect the Class 1E power source. Therefore, Non-Class 1E Travel Screen Differential Level Transmitters fed by a Class 1E Power Supply is not a safety concern.

o. Electrical Isolation/Separation of Class 1E Partial Discharge Monitor Bus Couplers for Unit 1 and Unit 2 Station Service Water Pump and Component Cooling Water Pump Motors and Emergency Diesel Generators.

The Bus Coupler consists of a Class-1E, 15 kV rated, non-shielded jumper cable, Class-1E epoxy mica capacitor (EMC), and Non-Class 1E low voltage, low energy coaxial cable for each motor/generator phase connection and a common Non-Class 1E termination box. The termination box is used to connect Non-Class 1E diagnostic equipment, one phase at a time, when partial discharge monitoring is performed.

The 15 kV jumper is spliced to the motor feeder cable. The other end of this jumper connects to the high voltage side of the EMC. The coaxial cable connects to the low voltage side of the EMC and the other end of this coaxial cable terminates at a BNC connector in the termination box.

The EMC is an 80 pico-farad capacitor that has an impediance of 33 meg-ohms at 60 hertz. Thus the EMC essentially acts as an open circuit at the normal operating frequency of 60 hertz. Only the partial discharge pulses (on the order of nano seconds duration and 100-500 mili-volts) are passed through this capacitor. The diagnostic equipment employs 120 VAC (60 hertz) plant power. Any short on the low voltage side would be of insufficient magnitude to damage the 6.6 kV rated windings of the motor or generator. An open circuit on the low voltage side will not affect the performance of the Station Service Water Pump Motors, Component Cooling Water Pump Motors, and Emergency Diesel Generators, since there is no voltage applied to the EMC under this condition. Also, the capacitor essentially acts as an on circuit at this frequency. Therefore, electrical separation is not required.

8. Compliance With NRC Regulatory Guide 1.81 [16]

The CPNPP design is in compliance with the provisions of Regulatory Guide 1.81 with an exception to Regulatory Position C1 as described below.

Safety-related loads shared between both units are powered from common MCCs, 120-VAC panels, 118-VAC panels, and 125-VDC panels as described in Subsections 8.3.1.1.9, 8.3.1.1.13 and 8.3.2.1. Indication of source of power associated with the common electrical equipment mentioned above is provided on a common panel located in the Control Room and accessible to both unit operators.

A single failure at the system level will not affect the capability to automatically supply minimum ESF loads in any one unit and safely shutdown the other unit assuming a loss of off-site power because the redundancy of common buses is maintained the same as redundancy for Unit 1 and Unit 2 buses.

On-site power capacity to energize sufficient seismic Category I equipment to attain a safe and orderly cold shutdown of both the units, assuming the loss of off-site power and most severe design basis event and a single failure in the on-site electrical system, is not compromised as a result of common buses because each unit system is designed to have sufficient capacity to feed common bus loads in addition to the unit specific loads.

The CPNPP design is controlled such that only common loads are fed from common buses except for some DC / 118-VAC common panels which feed Unit 1 loads also and the normal source of power for the panels is Unit 1. This does not affect the capability of any unit to feed these loads adequately. Because the unit specific loads of only one unit are fed from a common panel and the common panel normal power source is the same unit, therefore, under normal operation, the interaction between the units for maintenance and test operation will be no different than what is required for a common panel. The time when such common panel is aligned to the unit other than the one whose specific loads it feeds will be limited, therefore, any additional interaction needed for maintenance and test activities will be limited also.

9. Compliance With NRC Regulatory Guide 1.93 [18]

CPNPP power operation procedure is in compliance with NRC Regulatory Guide 1.93 as described in technical specifications. The power operation procedure is initiated and continued without restriction only when the limiting conditions for operation (LCO) are met. If the LCO are not met, the power operation will be restricted in accordance with the technical specification.

10. Compliance With IEEE 308-1974 [20]

Class 1E electrical equipment and power sources are designed to satisfy the functional requirements under conditions produced by the DBA listed in IEEE 308. The capacity of each onsite Class 1E power source is sufficient to operate all required Class 1E loads during and after the DBA.

Each Class 1E distribution system is capable of transmitting sufficient energy to start and operate all required loads in that system. A failure of any onsite Class 1E power source does not jeopardize the capability of the redundant onsite Class 1E power source to start and run the required shutdown systems.

Separation, redundancy, and independence of components eliminate the probability of a common failure mode. Class 1E equipment is located in seismic Category I structures

except as noted in Section 8.3.1.4 item 1 and qualified in accordance with IEEE 344-1975 [26]. Seismic design of electrical equipment is discussed in Section 3.10. Surveillance of Class 1E systems indicates readiness to perform their intended safety functions. Availability and operability of these systems are monitored by means of periodic testing.

All aspects of the electrical station design comply with IEEE 308-1974 with the exception of test intervals for the battery performance discharge test. (See Subsection 8.3.2.)

11. Compliance with IEEE 336 [24] and NRC Regulatory Guide 1.30 [6]

    Quality Assurance Program for the CPNPP onsite Class 1E AC power system is based on the requirements of IEEE 336 and Regulatory Guide 1.30. For details see Chapter 17 and Appendix 1A(B).

12. Compliance with NRC Regulatory Guide 1.47 [9]

    The surveillance of each on-site Class 1E AC power system operability status is based on the requirements of Regulatory Guide 1.47, augmented by Branch Technical Position ICSB 21.

    A system level safety system inoperable indication (SSII) is provided for each train to indicate if the on-site power source is unavailable.

    The Diesel Generator Power Window (DG pwr) on the SSII panel is activated by those conditions that render the D-G inoperable for auto start. The same SSII window is also activated if the following conditions exist.

    a. Diesel generator remote-local-maintenance switch in local or maintenance position

    b. 6.9kV generator breaker control switch in the lockout position

    c. Operator manual action

    d. Service water system inoperable

    e. Loss of 125 volts DC

    f. Diesel generator disabled

Although the conditions listed above may not be electrically interlocked with the D-G, it is recognized that they may render the D-G inoperable or, are otherwise important enough to advise the operator of it's existence.

The condition that renders the D-G incapable of responding to an automatic emergency start signal are:

1. 125V DC not available

2. Overspeed trip not reset

3. Differential lock-out relay not reset

4. Remote-Local-Maintenance Switch in local or maintenance mode

5. Starting air pressure low

8.3.1.2.2 Analysis of Uninterruptible Power Systems

Class 1E 118-V uninterruptible AC Power Systems, which provide power to the Class 1E instrumentation and control circuits, are designed to the same criteria as those for the onsite Class 1E power system.

Distribution panels and equipment they feed as a minimum meet the requirements of GDC 17 [1] and 18 [1], NRC Regulatory Guides 1.6 [2], and IEEE 308 [20], 344-1975 [26] and 384-1974 [31].

8.3.1.2.3 Failure Mode Analysis

Verification that the safety-related auxiliary AC and DC systems satisfy the single-failure criteria is demonstrated by the failure mode analysis given in Tables 8.3-3 and 8.3-7. Component failure and the effects of the failure are noted. Item numbers corresponding to those in the tables appear on Figures 8.3-1 and 8.3-13.

8.3.1.2.4 Class 1E Equipment in a Potentially Harsh Environment

Wherever possible, electrical equipment is located to avoid or minimize the effects of potentially harsh environments during all modes of plant operation. All Class 1E equipment is specified to perform its intended function under the maximum expected environmental conditions at the equipment location.

For details, see Sections 3.11N and 3.11B and Appendix 3A. Appendix 3A delineates postulated environmental extremes for Class 1E equipment located in a potentially harsh environment.

Class 1E equipment located in a potentially harsh environment is designed, fabricated, and qualified in accordance with the requirements of IEEE 323 [22] and applicable IEEE standards for particular equipment (e.g., IEEE 382 [29] for valve motor operators, IEEE 383 [30] for cables, and IEEE 317 [21] and ASME Boiler and Pressure Vessel Code [45] for electric penetrations).

8.3.1.3 Physical Identification of Class 1E Power Systems Equipment

The identification method by which onsite power system equipment can be distinguished as redundant Class 1E systems, associated Class 1E circuits, and non-Class 1E systems is described below:

1. Equipment Tag No.

   Electrical equipment has its own tagging scheme developed by equipment type. Many equipment types follow the tagging scheme for mechanical equipment with a modification of the eighth and ninth character. For equipment using a modified mechanical equipment tagging scheme, the eighth character is generally either “E” or “N”. “E” designates

equipment which is Class 1E and “N” designates equipment which is non safety-related. The ninth character typically indicates bus voltage.

Motors supplied with pumps are identified by the same tag numbers as the pumps, usually with the suffix “M” appended

2. Cables and Raceway Tag Nos.

All Class 1E system cables and the seismic Category I raceway system are identified by nine alphanumeric character tag numbers as follows:

a. Raceway System

The fourth character of a cable tray identification number identifies whether or not the given raceway contains safety-related cables. The fourth character is obtained from the first letter of the applicable train or reactor protection channel color code as described below. For non-safety related raceways, the fourth character is K. The second character indicates the Unit (i.e., Unit 1 = (1); Unit 2 = (2); common areas = (0)).

b. Cables

The second character in the cable number identifies the cable color code. In addition, the first character of each cable number indicates whether the cable is safety train or channel oriented (E), associated train (A), or non-safety related (N). The third character indicates the Unit (i.e., Unit 1 = (1); Unit 2 = (2); common areas (0)). All conduits/cables not given an identification number in the above manner are termed “Unscheduled”. All unscheduled conduits/cables are categorized as non-Class 1E. These include Heat Tracing, Cathodic Protection, Neoweld, communication, Security, Fire Protection, Fire Detection, and branch circuit lighting conduits originating at Lighting Distribution Panels.

3. Color Coding (Identification) System for Equipment, Raceways, Conduits, and Cables (excluding control panels)

In addition to the tag numbers, Class 1E equipment, raceways, and cables in raceways are identified by the following color coding system:

a. ESF System

Train A	Orange (O with a slash)
Train B	Green (G)

b. Reactor Protection System and ESF Systems at Channel Level

Channel I	Red	(R)
Channel II	White	(W)
Channel III	Blue	(B)
Channel IV	Yellow	(Y)

c. Associated Circuits

Associated Train A	Orange with white stripes
(Train AA)	(0 with a slash)
Associated Train B	Green with white stripes
(Train BB)	(G)

d. Non-Class 1E Circuits

Train C	Black	(K)

e. Non-Class 1E Cable Raceways and Equipment — Natural raceway color or natural equipment color

f. Electrical Equipment

Electrical equipment is also identified by a color coded label. The label has an outer and inner border. The outer border identified the unit as follows: Unit 1 (Blue), Unit 2 (Yellow), Common (White with black border), No Unit (White). The inner border identifies the train/channel by color. The inner border colors and corresponding train or channel use the same coloring scheme noted above except as follows:

Associated Train A is orange, Associated Train B is green, Channel II is white with a black border. Equipment with both Train A and Train B has a border that is half orange and half green

4. Method of Equipment, Raceway System, and Cable Color Coding (Excluding Control Panels)

a. Equipment

The preceding color coding is applied to the Class 1E equipment on an area which is readily visible (e.g., close to the nameplate or identification tag number).

b. Raceways

Exposed raceways containing Class 1E cables are marked by the color codes described previously in a distinct permanent manner at intervals not to exceed 15 ft and at points of entry to and exit from enclosed areas. These raceways are marked prior to the installation of their cables.

c. Cables

In general, all Class 1E cables and associated cables are jacket color-coded throughout their entire length. Cable jackets requiring field color coding are color coded at intervals not to exceed every 5 feet for exposed cable runs (cable not in raceway, i.e., cable not in cable tray, conduit or pull box) and at each end. One hour fire rated cable has an overall stainless steel sheath. The stainless steel sheath will be field color-coded similar to that of cable jackets. Cable jackets that require field color coding after installation (due to reclassification of cables from associated Class 1E to Class 1E, or vice versa, after their initial pull) will be field color coded as follows:

1. Where entering and exiting equipment, raceway and inside junction/pull boxes.

2. All exposed portions of the cables will be worked at intervals not to exceed five feet.

3. Portions of installed cables in conduit or trays will not be field color coded.

d. Non-Class 1E Equipment, Raceways, and Cables

Non-Class 1E equipment, raceways, and cables in raceways are not marked by color code, but are left in their natural color. For AC Essential Lighting conduits which have been declassified as non-Class 1E, the originally installed color markers are not required to be removed since the conduit number provides identification. In general all non-Class 1E cables have a black outer jacket. However, non-Class 1E cables whose natural color is not black will be field color coded black at intervals not to exceed every 5 feet for exposed cable runs (cable not in raceway, i.e. cable not in cable tray, conduit or pull box) and at each end.

8.3.1.4 Independence of Redundant Systems

The criteria which have been used to establish the minimum requirements for preserving the independence of redundant Class 1E systems are stated in IEEE 308 [20] and 384 [31] and NRC Regulatory Guides 1.6 [2] and 1.75 [15]. Class 1E equipment and circuits are clearly identified on documents and drawings in accordance with IEEE 494-1974 [36] as discussed in Section 8.1.5.2 (Item 5). The electrical cable system for Class 1E systems is described in the following subsections

Physical arrangement and separation between redundant Class 1E systems are based on the potential hazards in the particular area.

Separation of equipment is achieved by one of the following means:

1. Distance

2. Separate rooms

3. Barriers

Possible hostile effects of non-safety-related equipment failure on safety-related equipment are also considered in determining adequate separation of components.

1. Equipment Arrangement

   Each diesel generator including its associated auxiliaries is located in a separate room. The electrical switchgear of train A is separated from that of train B by locating them on different elevations.

   A separate and independently ventilated room is provided for the 125V Class 1E batteries of each train in each unit. Battery chargers, distribution switchboards, and SUPS of one train are separated from those of other trains by locating them in separate rooms. In general plant areas, Class 1E MCCs and distribution panels of one train are separated from those of the other train by a distance of at least 20 ft, or where it is not possible to maintain a 20 ft distance, barriers are provided to maintain proper separation. The Control Room and cable spreading area do not contain high-energy equipment such as switchgear, MCC's, transformers, rotating equipment, high-energy piping, or other potential sources of missiles.

   The following equipment are located in the Control Building Mezzanine Area (EL-840'-0"):

   a. Regulating transformers TXEC3 and TXEC4 (in Room # 148D).

   b. Air conditioner unit CP1-VAACTC-01 (in Room # 148D).

   c. Air conditioner unit CPX-VAACTC-01 (in Room # 148).

   d. Air conditioner unit CP2-VAACTC-01 (in Room # 148C).

   The above rooms in the Control Building Mezzanine Area are separated from the Control Room by a seismic Category II partial height gypsum wall and concrete floor. The ratings of these transformers and air conditioners do not categorize them as high energy equipment.

   All Class 1E equipment is located in seismic Category I Structures except for the following devices, which by necessity, are located in the non-seismic Turbine Building:

   a. Main Turbine, stop-valve limit switches

   b. Main Turbine, trip, fluid-pressure switches

c. Main Turbine impulse pressure transmitters PT505 and PT506, associated with bistables PB505A and PB506A respectively.

d. Feed-pump turbine, trip, fluid-pressure switches

For devices A, B and C, see Figure 7.2-1 sheet 16. For devices D, see Figure 7.2-1 sheet 15.

2. Criteria and Basis for Installation of Cables for Class 1E Systems

Separation between Class 1E cables is provided to preserve the redundancy and independence of redundant electrical circuits. The criteria for cable installation are derived from the basis stated previously and are in compliance with the intent of NRC Regulatory Guide 1.75 [15] and IEEE 384 [31]. Compliance with NRC Regulatory Guide 1.75 [15] is discussed in Subsection 8.3.1.2.1., Appendix 1A(B) and Appendix 8A.

3. Cable Routing

Cables are routed in separate raceway systems according to voltage and function as follows:

a. 6.9-kV power

b. Low-voltage power (less than 600-VAC) and 125-VDC power

c. AC and DC control (may contain 125-VDC or 120-VAC power cables provided the current they normally carry does not exceed the maximum ampacity of control conductors in the raceway system).

d. Instrumentation cables

For Class 1E circuits, analog or other low level type signal conductors (potentials less than 100 volts) are not routed in cable trays containing power or control cables (potentials greater than 100 volts). Analog or other low level type signal conductors for non-Class 1E circuits are not routed in cable trays with conductors at AC or DC potentials exceeding 120V AC (RMS) and 140V DC.

In a vertical stack of trays, except for short distances at vertical to horizontal transitions, the highest voltage level is on top, with lower trays descending in order of voltage level, control trays and finally instrumentation trays at the lowest level.

Cables are installed in ladder-type trays, totally-enclosed trays, conduits, and underground ductbanks. In general, the cable tray system conforms to NEMA VE 1-1971 [39], Cable Tray System. Trays requiring seismically qualified supports are analyzed and supports are located as required.

Non-Class 1E instrumentation cables not installed in raceways, are run inside CPNPP Unit 1 Reactor Building for computer inputs from temperature and thermal movement sensors. Non-Class 1E instrumentation cables not installed in raceways, are also run inside CPNPP Unit 1 and 2 Reactor Buildings for data acquisition from Reactor Coolant

Pump Motor Stator temperature sensors. All of these cables carry low level signals and they are qualified to the IEEE 383 standard as well as installed to seismic Category II requirements and to meet separation requirements of Regulatory Guide 1.75.

When Class 1E cables are routed between adjacent seismic Category I Buildings, (leave one building and enter another building) one building wall is provided with embedded rigid metal conduit sleeves and the other building wall has a relatively large opening depending on the number and size of embedded sleeves. Class 1E cables are routed between seismic Category I building walls by either of the following methods:

a. When cable tray is installed on the side with wall openings, the tray will either protrude into the opening so that cables will be routed through the embedded sleeves onto cable tray, or embedded sleeves will be extended through the wall opening with short lengths of conduit enabling cables to be routed onto cable tray running along the wall. At the opposite end of the embedded sleeves, continuous conduit or cable tray will be installed.

b. When continuous conduit is run between two such adjacent buildings, a flexible connection is achieved by using liquid tight flexible metal conduit. The flexible conduit is installed on the side with the wall opening.

Class IE cables are routed between non-adjacent seismic Category I buildings via Category I duct banks. The duct banks consist of rigid metal conduits encased in concrete resting on bedding and backfill materials. Duct bank rigid conduit ends and building rigid conduit ends are connected by means of flexible metal conduits. Both ends of the flexible conduits are provided with liquid tight and threaded connectors. Similar connections are provided at the manholes.

Cable slack is provided for the above conditions to prevent possible damage to Class 1E cables under the maximum possible movement of adjacent seismic Category I Building walls or wall and duct bank relevant to each other. Similarly, cable slack is provided at both the inboard and outboard ends of the Reactor Building’s electric penetration assemblies.

4. Cable Separation

The cable and raceway separation criteria are based on preservation of independence of redundant systems. Cables of redundant Class 1E systems are separated from each other as well as from cables of non-Class 1E systems. Cables of redundant Class 1E circuits are run in separate cable trays, conduits, ducts, and penetrations.

Lesser separation distances than those given in Regulatory Guide 1.75, Revision 1, are being used at CPNPP in several locations between Class 1E wiring and non-Class 1E Area Radiation Monitoring detector wiring and Public Address System speaker wiring based on the following:

a. Area Radiation Monitoring detectors utilize Geiger-Muller Tubes and Ionization Chambers. These devices require very low currents but at high voltage levels. Since the power supplies need to supply 1 milliamp or less, they are designed to provide 2 watts under normal operation conditions and have a design limit of

5 watts. The power supply is not capable of igniting a shorted detector cable because of its “fold over regulation” characteristic which turns off even this low current flow. Therefore any damage will be limited to internal damage and will not be propagated to nearby Class 1E circuits.

b. The Public Address System speaker wire is a similar case with both low current and low voltage requirements. (The National Electric Code recognizes such low power circuits under article 725’s Class II wiring). The speaker amplifier is designed to provide a maximum of 12 watts at 14 VDC, which is not capable of igniting the circuit under a fault condition. Therefore any damage in speaker wire shall be limited to internal damage and will not be propagated to nearby Class 1E circuits.

Heliax and Radiax radio antenna coaxial cables are used in the non-Class 1E Radio Communication System. The energy carried by antenna cables is not sufficient to cause internally generated fire in these cables and, therefore, separation between these cables and Class 1E cables is not required.

Thermistor is similar in nature to mineral insulated cable and is a thermal detector used in Fire Detection circuits. The energy carried by Thermistor is not sufficient to cause internally generated fires, and therefore separation between Thermistor and Class 1E cables is not required.

Fiber optic cables used in non-Class 1E monitoring circuits carry no electrical energy by themselves and, therefore, are not required to maintain physical separation from Class 1E circuits.

The raceways of one train are separated from those of the other train by locating them in separate structures or on opposite sides of large rooms or spaces. Where this is not possible, separation is maintained as described below or by providing barriers. The Class 1E cables are routed such that any single failure in one train system does not cause a failure in another train system. The separation of associated circuit cables is maintained on a train basis in the same manner and degree as the Class 1E circuit cables with which they are associated.

Where cables are exposed to such potential hazards as pipe whip, flammable material, and missiles, separation requirements are evaluated on a case-by-case basis to ensure an acceptable level of redundant circuit independence.

5. Minimum Separation Requirement

In plant areas which are free from potential hazards such as missiles, external fires, and pipe whip, the minimum separation between redundant cable trays is three feet between trays separated horizontally and five feet between trays separated vertically.

For interaction involving instrumentation cables only, in the non hazard areas (Cable spreading area and the Control Room) and limited hazard areas (Plant areas which are free from potential hazards such as missiles, external fires, and pipe whip), the minimum separation between redundant instrumentation cables and between instrumentation cables and redundant instrumentation trays is one inch horizontally and three inches

vertically. These minimum separation requirements for instrumentation cables are based on IEEE 384-1992 as endorsed by RG 1.75 Rev. 3.

In the cable spreading area and the Control Room, the minimum separation between redundant cable trays is one foot between trays separated horizontally and three feet between trays separated vertically.

In general plant areas and in the cable spreading area, these separation criteria are also applied to cables which are not located in raceway and not inside equipment as if the cable were in an open ladder-type tray.

Where plant arrangement precludes maintaining the above minimum separation criteria, the separation requirements described below are applicable. For cables in large power circuits (6.9KV circuits or 480V circuits using #4/0 or larger power cables), separation from cables in redundant large power circuits is provided by a minimum of 1 inch and two barriers. However, if the cable for the large power circuit is in conduit, and

- the conduit is non-safety related, the minimum separation required in any direction between redundant 6.9KV and large 480V power cable trays or cables in air is one inch, or

- the conduit is safety-related and it is below the cable tray side rail or below the bottom of the cable, the minimum separation required between redundant 6.9KV and large 480V power cable trays or cables in air is one inch.

Separation of cables in large power circuits from all other redundant power and control cables is provided by a minimum of 1 inch and one barrier.

A single barrier is provided and 1" separation or the separation described below is maintained for all other power and control cables or raceways. The minimum separation required between redundant conduit and redundant cable tray is 1". Similarly, the minimum separation required between redundant conduit and redundant cables is 1". A minimum separation of 1" is required if one or both of the conduits are power circuits. However, a minimum of 1/8" is required between a Class 1E conduit and a non-Class 1E instrumentation or control conduit. A minimum separation of 1/8" is required for redundant conduits when both are control or instrumentation.

The above minimum separation criteria are in accordance with Regulatory Guide 1.75, Revision 1, [15] and IEEE-384-1974 [31] and have been demonstrated by testing or analysis specifically for CPNPP or have been demonstrated by testing performed by other utilities where the cables are of the same or similar construction as used at CPNPP. Accordingly, the test results and conclusions are applicable to CPNPP installations. The CPNPP electrical separation criteria generally require greater separation distances than demonstrated in the test configuration.

All Nuclear Instrumentation System (NIS) cables are routed in conduit according to their channel assignment. To preclude introduction of noise in the NIS circuits, a minimum separation of 6 feet is maintained between NIS conduits and raceway containing 6.9 kV circuits. Also, a minimum separation of 2 feet is maintained from NIS conduits running parallel to raceways containing electrical noise sources such as low voltage power and

rod control cables. Any deviations are justified on a case by case basis to ensure that unacceptable noise is not induced onto the NIS signal cables.

The minimum separation distance between redundant Class 1E and between Class 1E and non-Class 1E equipment and circuits internal to control equipment is six inches. In this case, the wire and cables are flame-retardant with self-extinguishing and nonpropagating characteristics. Other components such as terminal blocks, wire troughs, wire cleats, raceways, cable ties, glastic barriers, and so forth are manufactured from self-extinguishing material.

Separation within the Class 1E inverters listed in Table 8.3-10 between Class 1E train related input cables and the Class 1E channel related output cables is not required since these cables are integrally associated with each other. Procedural controls and interlocks ensure that only one output circuit cable from the spare inverter is energized at a time. The remaining three output cables from the spare inverter remain deenergized and disconnected at both ends. Therefore, the separation between the output cables of the spare inverter internal or external to the spare inverter is not required. However, separation from all other circuits is maintained as required. For the Class 1E Nuclear Instrumentation System (NIS) neutron flux preamplifiers, signal processors, and isolation expansion assemblies listed in Table 8.3-10, separation between Class 1E train related cables and Class 1E channel related cables is not required inside each cabinet, since their circuits are integrally associated with each other.

Separation between associated circuits and non-Class 1E circuits in fire panels CPX-EIPRLV-29, CPX-EIPRLV-29A, and CPX-EIPRLV-30 (listed in Table 8.3-10) is not required based on analysis. Separation between trains and channels within the Westinghouse-supplied panel, racks and cabinets listed in Table 8.3-10 is not required based on testing or analysis. These circuits carry low energy fire detection control signals and the postulated failure/fault in the associated circuits inside the fire protection local panel does not degrade the Class 1E contacts or degrade the Class 1E function of the fire detection panel based on the following:

a. The associated circuits link normally open contacts of Class 1E auxiliary relays with the non-Class 1E alarm relays. The alarm relays and the alarm functions for CPNPP are non-safety. Degradation of the associated circuits due to their proximity with the 120-V AC black circuits can occur due to imposition of 120-V AC on the associated circuit wires due to a hot short, or a line-to-ground fault. Since contacts of the Class 1E relays in the local fire detection panels are normally open and are required to close only during a fire in the respective fire detection zone, no current could flow through the same under a non-fire condition.

   Occurrence of an electrical fault coincident with an external fire in two different plant areas is not credible because the 120-V cables (1) are rated for 600-V insulation, i.e., no dielectric failure hazard; (2) carry low current for their size (less than 5 Amps), i.e., no thermal hazard; and (3) run internal to the panel, i.e., no mechanical hazard.

b. The intra-panel non-Class 1E circuits are 24-28-V DC circuits energized by a low capacity 120-V AC to 24-V DC transformer-rectifier system. The internal wires are 600-V grade insulated wires which are used for low energy level circuits. The

low-level power supply and the printed electronic circuit cards have a very small capability to feed a dead short (a maximum current flow of three amps will open circuit on the circuit cards). This magnitude of fault current is considered too small to jeopardize the associated circuits and the Class 1E equipment associated with them.

Minimum physical separation between Class 1E cables and non Class 1E differential relay (87/ST1) and (87/ST2) protection cables are not maintained per IEEE 384 inside the 6.9kV Class 1E switchgears. The installation is acceptable as analyzed below:

a. Open Circuit - The Class 1E CTs are qualified isolation devices between Class 1E primary (bus) and secondary output. These CTs are used as isolation devices between Class 1E bus and on Class 1E differential relay cables. The cables from CT lead to Class 1E terminal block inside the 6.9kV switchgear are Class 1E and the cables from the Class 1E terminal block to the differential relay terminal are non Class 1E. However, these cables are 600V rated, procured as Class 1E and are qualified to IEEE 838 fire retardant requirement. Any discontinuity of CTs secondary side circuit will cause the CT open circuit. 600V rated cables tested at 2500V for 5 minutes will adequately withstand the momentary CT open circuit high voltage.

b. Short Circuit - The differential relay cables, are adequately sized to carry CTs secondary current corresponding to 3-φ bus fault condition. Since the cables were tested for 2500V for 5 minutes the installation will withstand CT secondary voltage during 3-φ bus fault condition.

c. Hot Short - The output contacts of differential relays are connected to the lockout relay energization circuit which is protected by 20A fuses. The differential relays are mounted in Protective Relay Board (PRB) located inside the Control Room. The PRB is mounted as seismic Cat. II. The differential relay mounting details are similar to equivalent Class 1E relays. Therefore, the “hot short” between the CT lead wires connected to the differential relay coil and relay output contact is not credible and no further analysis has been done.

d. The cable size is such that it will function adequately and will not fail due to circuit voltage and current conditions to which it might be exposed due to open circuit and short circuit condition. The cable will maintain its integrity and will not adversely affect the Class 1E cables in the vicinity of these cables inside the cubicles. Therefore, these non-1E cables used in differential relay protection for startup transformers are not required to meet the minimum separation requirement of IEEE =-384 inside the 6.9kV Class 1E switchgear.

Separation within other equipment listed in Table 8.3-10 is not required since circuits are isolated from Class 1E buses by isolation devices.

Minimum separation for control and instrumentation cables or raceways inside equipment is 1". Conduit to conduit minimum separation is 0". Cables #10 AWG and larger feeding power to control equipment from distribution panels and all power cables inside power equipment maintain 6" separation or are enclosed or separated by a barrier.

Control and instrumentation cables entering control equipment through BISCO fire sealant maintain a minimum separation of 1" and cables #10 AWG and larger feeding power to control equipment from distribution panels maintain 6" separation.

The above separation criteria has been demonstrated by testing and analysis (refer to References 41 and 42) to meet or exceed Regulatory Guide 1.75 [15] and IEEE-384 [31].

For the purpose of electrical cable separation, acceptable barriers include rigid metal conduit, electrical metallic tubing (EMT), flexible metallic conduit, cable tray covers (both solid and ventilated types), cable bus enclosures, equipment and device enclosures, enclosed metal wireways inside equipment, a wrap of woven silicon dioxide, and one or three-hour fire rated materials (Thermo-Lag and one hour fire rated cable).

A wrap of woven silicon dioxide, and one hour fire rated cable are considered equivalent to conduit with respect to protection from electrical failures. Thermolag and one hour fire rated cable shall only be used as a Regulatory Guide 1.75 electrical separation barrier when installed to satisfy the requirements described in FSAR Section 9.5.1.2.

When a cable, conduit or tray is protected by one or three-hour fire rated Thermo-lag installed to satisfy the requirements as described in FSAR section 9.5.1.2; there is no separation requirement from the Thermo-lag protected cable, conduit or tray, to the redundant cable, conduit or tray outside the Thermo-lag enclosure as analyzed below:

a. External Cable Fault: The Thermo-lag installation tests have demonstrated the ability to protect the internal cables from damage that would affect their functionality when subjected to an external one or three-hour duration standard time-temperature curve fire. This is a much more intense and longer lasting fire exposure than the one that could be generated from the short duration of heat exposure from the external cable fault.

b. Internal Cable Fault: The Thermo-lag installation also provides adequate protection to the outside cable from fault generated fire inside the enclosure for the following reasons:

   1. The fire resistance property of the material is same regardless of the fire protection.

   2. The burning characteristics of both the IEEE 383 cable and the Thermo-lag are such that any smoldering or burning that may occur during the fault condition will self-extinguish once the cable fault is isolated by breaker/ fuse. In addition tightness of the enclosure will deprive the fire of oxygen necessary to support a significant, long term combustion in the enclosure.

   3. The exterior banding, tie wires and other structural features of the enclosure maintain the enclosure integrity during the external fire test. These exterior banding, tie wires and other structural features will also maintain the structural integrity during the limited heat exposure induced by an internal cable fault.

c. In view of the above, it is concluded that there is no separation requirement from the Thermo-lag protected cable, conduit or tray to the redundant cable, conduit or tray outside the Thermo-lag enclosure.

Metal Clad (MC) cables include copper sheathed (CS) cable, Aluminum sheathed (ALS) cable and Galvanized Steel Sheathed cable (GS). MC cable conductor size is limited to #10 AWG and below with a maximum of four (4) conductors and will be used only in non-class IE, 120 Vac/125Vdc applications. GS cable shall not be limited to 4 conductors for non-Class 1E instrumentation applications.

CS cable is constructed of continuous corrugated 16 mil thick copper tube with no outer jacket and 600V XHHW, 90°C insulation. CS cable will be used only inside the containment and only in the lighting system.

ALS cable is constructed of continuous corrugated 25 mil. thick seamless aluminum tube with an outer thermosetting chlorosulphonated polyethylene jacket and 600V XHHW, 90° C insulation ALS cable will only be used outside of the containment building in the lighting, fire protection, heat tracing and communication systems.

Galvanized steel sheathed cable (GS) is constructed of continuous interlocked 25 mil thick galvanized steel armor with or without an outer jacket. The internal cable is 600V, 90°C, FR-XLPE with an overall jacket. GS cable without an outer jacket is acceptable for use inside containment.

Testing performed by other utilities has demonstrated the adequacy of the above materials to be used as enclosed raceway and barriers for Regulatory Guide 1.75 [15] separation purposes.

a. Flexible Conduit: Tests documented in Reference 43 for Niagara Mohawk Power Corporation's Nine Mile Point Nuclear Station Unit 2 demonstrated the adequacy of BOA stainless steel flexible conduit and Anaconda steel flexible conduit as an enclosed raceway. Anaconda, BOA and similar quality flexible conduit are used at CPNPP. The power cable was manufactured by Okonite. The control cable was manufactured by Rockbestos. The cables are of the same manufacture and similar construction to those used at CPNPP. Accordingly, the test results and conclusions are applicable to CPNPP installations.

b. Ventilated Tray Covers and Cable Bus Enclosures: Tests documented in Reference 44 for Duquesne Light Company's Beaver Valley Power Station Unit 2 (Configuration No. 4) demonstrated the equivalency of a fluted, ventilated tray cover to a solid tray cover for the purpose of physical separation. The test results were also used to justify considering cable bus enclosures the same as enclosed raceway. Both the power cable tested and that used at CPNPP are of the same manufacture and similar construction. Cable tray, tray covers and cable bus enclosures used at CPNPP are of similar design and construction as those tested/ analyzed for Beaver Valley 2 and accordingly, the test results and conclusions are applicable to CPNPP installations.

c. Protective Wrap: Tests documented in Reference 44 for Duquesne Light Company's Beaver Valley Unit 2 (Configurations 1 and 2) concluded that cables

covered with Siltemp protective wrap and 3M No. 69 glass tape combined with a 3/8 in. air space provides adequate protection regardless of whether the faulted cable is within the wrap or outside of the wrap. The results justify considering the protective wrap equivalent to a conduit with respect to protection from electrical failures. The power and control cables tested are of the same manufacture and similar construction to those used at CPNPP. Accordingly, these test results and conclusions can be applied to the CPNPP installation.

Ampacity tests included in the same report demonstrated that no additional derating, beyond design ampacities, was required when a cable was enclosed in protective wrap. Design ampacities for cables used at CPNPP are similarly derated to those tested (from ICEA table values), and need not be additionally derated.

d. Metal Clad (MC) Cable: MC cable includes Aluminum sheathed (ALS) cable, Copper sheathed (CS) cables and Galvanized steel sheathed cable (GS). Tests documented in Reference 43 for Niagara Mohawk Power Corporation's Nine Mile Point Nuclear Station Unit 2 and Reference 41 for CPNPP demonstrated the adequacy of BOA stainless steel flexible conduit and Anaconda steel flexible conduit as an enclosed raceway.

The ALS cable is enclosed in a 25 mil. thick corrugated seamless aluminum tube which provides mechanical integrity comparable to the 6 mil. thick BOA stainless steel flexible conduit tested in Reference 41. The seamless aluminum tube will contain any cable ignition within the tube. In addition, ALS cable has the annulus between the cable and the tube tightly packed with flame retardant filler materials, which will inhibit flame propagation due to the absence of free-air surrounding the cables. As ALS cable conductor size is limited to #10 AWG with a maximum number of 4 conductors, the cross sectional area of conductors used in the cable is much smaller than the 2/0 fault cable tested in the Anaconda steel flexible conduit in Reference 43. Therefore, the fault currents would be lower than that tested.

ALS cable has thermosetting insulation which is similar to the Anaconda steel flexible conduit tested in Reference 43. The jacketing materials used in ALS cable are also flame retardant.

The conductor material, insulation and filler materials used in CS and GS cable are the same as those used in ALS cable. The CS cable and GS cable construction will exhibit the same flame retardancy characteristics as discussed above for ALS cable. In addition, CS cable is enclosed in a 16 mil thick corrugated copper tube which provides mechanical integrity comparable to ALS cable. The 25 mil thick interlocking Galvanized Steel sheath also provides mechanical integrity comparable to the ALS cable. The GS cable size for #10 AWG and smaller (maximum 4 conductors) is much smaller than the 2/0 fault cable tested in reference 43, therefore the fault currents would be lower than that tested. (GS cable is not limited to 4 conductors for non-1E instrumentation applications). Based on Wyle Laboratories Test #53575 [47], CS cable may be considered as cable in enclosed raceway and one inch separation provides adequate protection to prevent degradation of redundant circuits.

Only MC cable meeting the flame test requirements of IEEE-383 [30] will be used.

Based on the above testing and analysis, MC cable is equivalent to cable inside conduit for electrical separation.

e. One hour fire rated cable:

One hour fire rated cable is used to satisfy the fire safe shutdown requirements as described in FSAR Section 9.5.1.2. The cable is used in power and control circuits outside containment where the total radiation dose is less than or equal to 50 MRADS gamma, and in cable sizes of #8 AWG and smaller. The cable is constructed of a continuously welded corrugated 12 mil thick stainless steel sheath with high temperature nickel-clad copper conductors, glass braid cable jacket and silicone rubber insulation. The cable is Class 1E qualified per IEEE 323-1974 and IEEE 383-1974 for flame retardancy (unaged cables only). See FSAR Section 1A(B), Regulatory Guide 1.131. One hour fire rated cable meets the requirements of ASTM E-119-1971 for a fire resistance rating of one hour. This cable is considered equivalent to cable in conduit for the purposes of electrical separation.

Gaps of 3/8 in. or less at cable tray side rail joints, unfilled square holes 7/16 in. or less on tray side rails and 1/4 in. diameter Ty-rap holes on tray bottom are not considered significant enough to degrade the tray surface as a barrier. These gaps are comparable to openings on ventilated tray covers which have been tested and found acceptable as effective barriers to meet separation requirements per Regulatory Guide 1.75 [15].

f. Stainless Steel Clad Mineral Insulated Cable:

Stainless steel clad mineral insulated cable is used in the Reactor Vessel Level Indication System in the Containment Building. This cable is Class 1E qualified to IEEE 323-1974 and to IEEE 383-1974 for flame retardancy. This cable is constructed with a minimum of 15 mil thick stainless steel cladding which is comparable to the 6 mil thick stainless steel conduit tested in Reference 41. This cable is considered equivalent to cable inside conduit for purposes of electrical separation.

Separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits in the Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A and 02A, listed in Table 8.3-10, is not required based on the following analysis. These panels are re-classified as Multi-Train cabinets. Both the AC and DC circuits for the Diesel Generator Engine Control Panels are supplied by a Class 1E power supply. The non-Class 1E circuits in the control panels are electrically isolated by an approved design having two circuit breakers in series for electrical isolation to assure that the Class 1E power source is not adversely affected by a fault in the non-Class 1E circuit.

Internal Train Separation Exemption:

Per Regulatory Guide 1.75, non-Class 1E circuits shall be physically separated from Class 1E circuits and non-Class 1E circuits shall be physically separated

from associated Class 1E circuits by the minimum separation requirements. All cables installed in the Diesel Generator Engine Control Panels are flame retardant and meet IEEE-383 flame test requirement. Any non-Class 1E cable fault, short circuit, short to ground, or over load condition will be isolated by an isolation breaker and will not cause the cable temperature to exceed normal operating temperature. The non-Class 1E cables inside the control panel will not be degraded or will not reach the temperature that can affect the Class 1E or associated Class 1E circuits. Open non-Class 1E circuit, have no impact on Class 1E or associated Class 1E circuits. The non-Class 1E cables when maintained within their rated temperature, have no adverse impact on adjacent Class 1E or associated Class 1E cables with no physical separation because the Class 1E and associated Class 1E cables also have the same temperature ratings.

Therefore, the adequacy of the safety-related circuits to perform their safety function is maintained with no separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits inside the Diesel Generator Engine Control Panels. Any faults in the non-Class 1E circuits will not compromise the safety function of the Class 1E or associated Class 1E circuits internal to the Diesel Generator Engine Control Panels. Therefore, there is no need for internal electrical separation requirements within Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A and 02A.

6. Cable Spreading Area and Control Room

The cable spreading area is the space below the Control Room where instrumentation and control cables converge prior to entering the control, termination, or instrument panels. This area does not contain high-energy equipment such as switchgear, transformers, rotating equipment, or potential sources of missiles or pipe whip. Flammable material is not stored or installed in this area. Cable constructions are qualified in accordance with IEEE 383 [30]. The circuits in this area are limited to control and instrument functions and those power supply circuits serving the Control Room. Power circuits are not routed in this area, except power cables serving instrument and Control Room distribution panels. These power cables are encased in concrete or run in rigid steel conduits from the point where they enter this area, except for the 120 VAC power circuits, that may be installed using MC cables underneath the Control Room raised flooring.

In this area, a minimum separation of one ft horizontal and three ft vertical is preferred between redundant trays. Where raceway arrangements preclude maintaining the minimum separation distance, the redundant circuits are run in enclosed raceways or barriers are provided between redundant circuits.

Where it is impractical to meet the above requirements, for cables and raceways which are limited to instrumentation and control, the minimum separation distances are as listed in paragraph 8.3.1.4, Item 5.

7. Electric Penetration Area

Except for six penetrations, all individual penetrations are classified according to function and are restricted to exclusive use for power, control, or instrumentation. These six penetrations are exclusively used for six different motor operated valves for both power

and control. In addition, penetrations used for the NIS cables are not used for any other purpose.

There are three electric penetration areas, one on each floor elevation. Class 1E penetrations are located on two different floor elevations, one train on each floor. The third (middle) floor area contains the four channels of the NIS and two channels of the RPS system. The remaining two channels of the Reactor Protection System (RPS) are located on the floor where train B penetrations are located. The minimum center line separation between redundant NIS penetrations is six feet. The minimum center line separation between penetrations of redundant channels of RPS, between RPS and NIS, and between these channels and any other electrical penetrations, is 5 feet. The minimum centerline separation between any two (2) non-Class 1E penetrations or any two (2) same train penetrations is approximately 2-1/2 feet. The minimum centerline separation between any Class 1E penetration and non-Class 1E penetration is 3 feet, except for Non-Class 1E fiber optic penetration modules which are installed in Class 1E penetrations.

8. Hostile Environments

Routing of cables for Class 1E systems through an area where there is potential for accumulation of meaningful quantities of oil or other combustible material is avoided. Where such routing is unavoidable, only one system of redundant cables is allowed in any such area, and the cables are protected by installing them in conduits or solid bottom trays with solid covers. In areas containing potential missiles, physical arrangement, protective barriers, or pipe restraints preclude loss of redundant systems.

9. Sharing of Cable Trays

Non-Class 1E cables are separated from Class 1E cables and from associated cables. Non-Class 1E cables, when they share the same raceway with Class 1E cables or by virtue of power supply connection to Class 1E buses, are associated cables and are so designated. As described in preceding sections, these associated cables are separated from non-Class 1E and from Class 1E channels and redundant train cables with which they are not associated. This separation is maintained throughout the length of the cable (or circuit) until it passes through an isolation device.

Lesser separations are being used in several locations between Class 1E wiring and non-Class 1E Area Radiation Monitoring detector wiring and Public Address System speaker wiring based on analysis. See paragraph 8.3.1.4, Item 4.

10. Spacing of Wiring and Components in Control Boards, Panels, and Relay Racks

Separation of redundant devices and components is accomplished by mounting them on physically separated control boards. Where this mounting is not feasible from a plant operational point of view, and where operational design dictates that redundant equipment be in close proximity, separation is achieved by a six inch minimum airspace or by a fire-retardant barrier. Where this is not possible, an analysis will be provided to show adequacy of separation. Redundant circuits that serve the same protective or control function enter the control boards or panels through separated aperatures and terminate on separate terminal blocks or device connectors.

Where it is impractical to meet the above requirements, for cables and raceways which are limited to instrumentation and control, the minimum separation distances are as listed in paragraph 8.3.1.4, Item 5.

Fiber optic cables used in non-Class 1E monitoring circuits carry no electrical energy by themselves and therefore are not required to maintain physical separation from Class 1E circuits.

11. Cable and Raceway Identification, Fire Protection, and Fire Detection

Raceways and Class 1E cables are identified with unique tag numbers and a color coding system as discussed in detail in Subsection 8.3.1.3. Tag numbers are cross-referenced with cable schedules. Tag numbers and color coding of cables and raceways permit plant personnel to identify Class 1E cables by train and channel without referring to cable schedules.

Cables are installed with consideration given to the cable ampacities. The criteria for cable derating, cable tray fill, and cable fire protection and detection are discussed in Subsections 8.3.3 and 9.5.1.

8.3.1.5 Vital Supporting Systems

The vital supporting systems required for the proper operation of the Class IE load and systems include:

1. Air conditioning, heating, cooling and ventilation systems. (For design bases and description of air conditioning, heating, cooling and ventilation systems see FSAR Section 9.4.)

2. Station Service Water System

(For design bases and description of station service water system, see FSAR Section 9.2.1.)

3. Component Cooling Water System

(For design bases and description of component cooling system, see FSAR Section 9.2.2.)

8.3.2 DC POWER SYSTEMS

8.3.2.1 Description

The station DC systems supply power to the plant instrumentation and control under all modes of plant operation. In addition, upon loss of AC power, the DC systems also provide power for emergency lighting and auxiliary turbine-generator motors. The DC Emergency Lighting System is described in Section 9.5.3.

The DC systems for each unit consist of one 125-V, one 125/250-V, and one 24/48-V non-Class 1E battery systems, and two independent and redundant Class 1E 125-V battery systems.

The Class 1E 125-V battery systems supply power to Class 1E loads without interruption during normal operations or DBA conditions. The Class 1E l25-V, seismic Category I battery systems conform to IEEE 308 [20], and NRC Regulatory Guides 1.32 [7], and 1.6 [2].

Each redundant Class 1E 125-V system consists of two independent batteries each having, one main distribution bus with molded case circuit breakers, fusible switches, two static battery chargers (one spare), and local distribution panels (See Figures 8.3-14 and 8.3-14A). Redundancy of components precludes loss of both systems as a result of a single failure. For Unit 1, batteries BT1ED1 and BT1ED3 feed all train A load requirements, while batteries BT1ED2 and BT1ED4 supply train B load requirements. For Unit 2, batteries BT2ED1 and BT2ED3 feed all train A load requirements, while batteries BT2ED2 and BT2ED4 supply train B load requirements. There are no bus ties or sharing of power supplies between redundant trains. Class 1E equipment associated with systems shared by both units receives power from panel boards having an incoming automatic transfer switch which can select power from either unit. Transfer switch design is such that power cannot be supplied from both units simultaneously. Train separation is maintained by supplying these shared panelboards from the same train of both units. Independence and separation are maintained throughout the AC supply circuitry as well as in the DC distribution network.

Power is supplied to non-Class 1E loads from the 125-VDC, 125/250-VDC, and 24/48-VDC systems. Each of these systems is completely independent, both of the Class 1E 125-VDC systems and of each other.

The 125-VDC system is primarily a part of uninterruptible power supply (UPS) system to Plant computer and peripherals. The system consists of a 125-V battery, two battery chargers (one spare), and a main distribution bus with molded case circuit breakers (see Figure 8.3-15C). The battery system supplies two 15 kVA inverters (see Section 8.3.1.1.13) which provide the required regulated 118VAC voltage for distribution. The non-Class 1E 125/250-VDC system consists of two 125-V batteries, three 125-V battery chargers (one spare), a main distribution bus with molded case circuit breakers, and fusible switches (see Figure 8.3-14A). The battery system supplies two 10 kVA inverters (see Section 8.3.1.1.13) which provide the required regulated 118 VAC voltage for distribution. The loads on the 125/250-VDC system essentially consist of the following:

1. Turbine-generator emergency bearing oil pump (250-V)

2. Turbine-generator emergency hydrogen seal oil pump (250-V)

3. Feedwater turbine emergency oil pumps (250-V)

4. Control power for non-class 1E switchgear (125-V)

5. Non-Class 1E lighting and distribution panels (125-V)

6. Non-class 1E 10KVA inverters (125-V)

The 24/48-VDC system for each unit is provided for turbine-generator control and instrumentation. The system consists of two 24-V batteries, three battery chargers (one spare), and a main distribution bus. The system in each unit is completely independent of the Class 1E 125-VDC systems and 125/250-V systems described previously.

Implementation of the criteria set forth in the design of the Class 1E 125-VDC systems is described in the following paragraphs.

1. Separation

   The Class 1E 125-V batteries of each redundant train, are located in a separate, seismic Category I room of the Electrical Building at elevation 792 ft. In addition to providing protection against the SSE, the walls of these rooms act as fire barriers to maintain the integrity of the redundant systems. The battery chargers and distribution boards associated with a particular battery are located in a room adjacent to the battery room and are of similar construction. Separation of batteries from associated equipment eliminates possible harmful effects of any corrosive fumes emanating from the batteries, thereby maintaining a high degree of system reliability and availability. Electrical separation is also maintained to ensure that a single failure in one train does not cause a failure in the redundant train. There is no sharing between redundant Class 1E trains of equipment such as batteries, battery chargers, or distribution panels.

2. Capacity

   a. Batteries

      Each Class 1E 125-VDC system has the capacity to continuously supply all the connected normal running load while maintaining its respective battery in a fully charged condition. Each battery is capable of carrying the essential load continuously for a period of four hours in the event of a total loss of onsite and offsite AC power. The battery capacity for BT1ED1 and BT1ED2, at an eight-hr discharge rate with a final voltage of 1.75-V/cell, is 1944 Ah. The battery capacity for BT1ED3 and BT1ED4, at an eight-hr discharge rate with a final voltage of 1.75- V/cell, is 1200 Ah. Battery capacities for the Unit 2 Class 1E 125-V batteries are similar

   b. Battery chargers

      Two 100-percent-capacity static battery charger units are provided for each Class 1E 125-V battery system. Normally, one charger is operating, and the other is kept as a spare. Two circuit breakers connecting these battery chargers to the DC distribution bus are mechanically interlocked such that only one charger remains connected to the bus at any time. Each battery charger has a “Float” and “Equalize” mode. A front panel mounted switch is provided to select either mode of operation. The float and equalize voltages may be independently set at 132 volts ±5% and 140 volts ±5% respectively. The equalize voltage is set not to exceed 140-V. Output voltage is regulated within ±0.5 percent of setpoint from no load to full load. Backup protection is incorporated by a overvoltage relay mounted on the dc switchboard which annunciates an overvoltage condition in the control room. All connected loads are specified and designed for operation with a

maximum input voltage of 140-VDC. Loads are protected from overvoltage by presetting the maximum equalize voltage not to exceed 140-VDC.

Each battery charger is sized to recharge the battery from the design minimum charge state to the fully charged state within 24 hrs while supplying the steady state loads under all modes of plant operation. Input power to these chargers is obtained through independent 480-V, three-phase, AC supply from Class 1E MCCs. Protection is incorporated in the battery chargers to preclude the AC supply source from becoming a load on the battery as a result of power feedback upon loss of AC input power.

3. Ventilation

All battery rooms are ventilated to prevent the accumulation of gases produced during charging operations. Each Class 1E 125-V battery room is provided with redundant ventilation fans and a separate exhaust duct is provided for each room.

In addition, the battery cells are provided with explosion- resistant vent caps that prevent the ignition of gases within the cell from an ignition source outside the cell.

4. Loading

The Class 1E 125-V battery load requirements of Units 1 and 2 are described in Table 8.3-4.

5. Redundancy

Two independent and redundant Class 1E, 125-VDC systems are provided for each unit. Electrical and physical separation as previously described precludes the loss of both systems as a result of a single failure.

6. Testing and Inspection

The preoperational testing of the Class 1E 125-VDC systems is done in compliance with NRC Regulatory Guides 1.41 [8] and 1.68 [13].

Periodic inspection and testing of the DC systems are performed to monitor the condition of the equipment to ensure reliable operation. Visual inspections, liquid level, specific gravity, and cell voltage checks, and performance discharge tests are performed at regular intervals on each battery. All maintenance and testing procedures and criteria for replacement are in accordance with IEEE 450-1995 [35], and Regulatory Guide 1.129 [18D]. Visual checks and performance tests are also scheduled for the battery chargers.

The periodic tests of all Class 1E DC equipment are performed to satisfy the requirements of GDC 18 and 21 [1].

7. Identification of Safety Loads

The safety loads connected to the Class 1E 125-VDC systems are identified in Figures 8.3-14 and 8.3-15B. In addition, the table 8.3-4 indicates the maximum length of

operating time required for each battery total load upon loss of AC power. The method of distinguishing between Class 1E and non-Class 1E loads is defined in Subsection 8.3.1.3.

8.3.2.2 Analysis

The design of the Class 1E 125-VDC systems is in accordance with the requirements of GDC-17 [1], GDC-18 [1], NRC Regulatory Guides 1.6 [2], 1.32 [7], 1.75 [15] and IEEE 308 [20]. Compliance with these criteria is described in Subsection 8.3.1.2.1. The seismic requirements are specified in Section 3.10B.

Redundant power supplies and equipment satisfy GDC 17 for a single failure. A failure mode analysis is presented in Table 8.3-7. Quality assurance, cable routing, separation, and equipment identification are covered in Subsections 8.3.1.2 and 8.3.1.4.

1. Surveillance and Monitoring

   Each battery charger is equipped with a DC voltmeter, a DC ammeter, AC failure relays, and low battery voltage relays. Malfunction of the chargers annunciates in the Control Room. A system level inoperable status indication is provided for the Class 1E 125-VDC system in accordance with NRC Regulatory Guide 1.47 [9].

   Battery float and discharge current and distribution bus voltage are monitored at the switchboard and in the Control Room. Ground detection, undervoltage relays, and overvoltage relays are provided on the main distribution buses; off-normal conditions are annunciated in the Control Room.

   The overall system design including function requirements, redundancy, capacity, and availability is in conformance with IEEE 308 [20] criteria for Class 1E systems with the exception of battery performance discharge test intervals which is in accordance with IEEE 450-1995.

2. Physical Identification and Separation

   The physical identification of the Class 1E DC power systems is combined with the identification of Class 1E AC Power Systems and is described in Subsection 8.3.1.3. The physical separation of the Class 1E DC systems is maintained in accordance with the intent of NRC Regulatory Guide 1.75 [15].

3. Independence of Redundant Class 1E 125-VDC Systems

   Redundancy of power sources and distribution equipment is provided in the DC system. This redundancy extends from the station batteries and battery chargers through distribution panels, cabling, and switchgear. Each redundant DC system and its associated distribution equipment can independently provide the required DC power for safe shutdown of the plant.

   Each redundant Class 1E 125-V battery system is located in a separate seismic Category I battery room. The Class 1E battery chargers and main distribution buses associated with each Class 1E 125-V battery system are located in a separate seismic

Category I room. The local distribution panels, feeders, and control and instrumentation cables raceway system associated with each Class 1E battery system are separated as described in the previous section. Separation of train A and train B battery systems is maintained up through the Class 1E 6900-VAC buses and is consistent with the train designations of such equipment.

The quality assurance program discussed in Chapter 17 ensures compliance with established criteria in IEEE 336-1971 [24] and NRC Regulatory Guide 1.30 [6]. The Class 1E 125-VDC equipment and circuits are identified on documents and drawing in accordance with the requirements of IEEE 494-1974 [36] as discussed in Section 8.1.5.2 (Item 5).

8.3.3 FIRE PROTECTION FOR CABLE SYSTEMS

The Fire Protection System for cables is a part of the integrated system of the fire detection and protection for the entire plant. Cable fire prevention measures such as power cable derating, restricted percentage of cable tray fill, and the installation of nonpropagating and self-extinguishing type cable, insulation, and jacket material are used. These measures are described in detail in the following sections. In general, ionization and thermal type fire detectors are located in areas of heavy cable concentration to detect a cable fire. These fire detectors form a part of the Fire Detection System installed throughout the plant. The system provides early warning at local panels and in the Control Room to alert the operator and subsequently the plant fire brigade to take immediate action to extinguish a fire.

Portable fire extinguishers and hose stations located throughout the plant are available to extinguish a cable fire.

The fire suppression systems and specific equipment used to extinguish a cable fire are described in detail in Section 9.5.1 for each area containing cables.

8.3.3.1 Cable Derating and Cable Raceway Fill

Cable ampacities are established based on the following:

1. IPCEA P-46-426 (IEEE S-135), Power Cable Ampacities, Volume 1, Copper Conductors, and Volume 2, Aluminum Conductors, 1962.

2. ICEA P-54-440 (NEMA WC-51-1975), Ampacities, Cables in Open-Top Cable Trays, Rev. 2, August, 1979.

3. National Electrical Code (NEC), 1975

4. Manufacturers Standards

Cables are derated in accordance with IPCEA P-46-426 [37], ICEA P-54- 440 [38], the National Electrical Code (1975), and manufacturers recommendations. The following factors are considered in determining the cable sizes:

1. Normal and emergency load currents

2. Short-circuit heating capacity for 6.9-kV loads and 480-V loads fed from the unit substations

3. Voltage regulation

4. Load factor of 100 percent in ductbanks

5. Grouping derating

6. Load diversity

7. Derating for fire stops and tray covers

8. Ambient temperature

The cable type and cable insulation are selected to minimize the rate of deterioration during the design life of the plant as a result of temperature, humidity, and radiation. Environmental Type Tests are performed on cables that are required to function during and following a LOCA.

Cable tray fill criteria generally limit the summation of the cross-sectional areas of control/ instrumentation cables and power cables to a maximum of 40 and 30 percent, respectively, of the usable cross section of the tray. However, these percentages may be exceeded provided the following conditions are satisfied:

1. Cables do not extend above the side rails of the cable tray except as allowed below.

2. For power cables - the thermal rating of the cable is not exceeded.

3. Cable tray support design is adequate.

Cable tray fill for power cables, when sized for maintained spacing, is limited to the requirements of IPCEA P-46-426.

Cable trays may have cables extending above the top of the cable tray siderails. Cables may extend above the top of cable tray siderails if the situation does not result in an electrical or mechanical separation problem and the cables are not routed near potential hazards which may cause physical damage to the cables.

During construction of CPNPP Unit 1, a one time review was made at the request of the NRC to identify where cables extend above the top of Class 1E cable tray siderails and exceed the percent fill limit and can not be corrected by reasonable means. Four such cases were identified at that time. Vistra OpCo recognizes the need to minimize the number of cases where cables extend above the top of Class 1E cable tray siderails and exceed the percent fill limit and has implemented measures to control future installations such that additional instances will only be implemented subsequent to evaluation by Vistra OpCo.

Conduit fill for power, control and instrumentation cables is limited to the requirements delineated below:

Number of contained cables	1	2	3 or more
Percent Fill Limit	53	31	40

However, these percentages may be exceeded provided the following conditions are satisfied:

1. Cable pulling tension is not exceeded.

2. For power cables - the thermal rating of the cable is not exceeded.

3. For fire seal application the limits of conduit fill for the qualified seal are not exceeded.

4. Conduit support design is adequate.

8.3.3.2 Fire Detection and Protection Devices

As stated previously, fire detection equipment and protection equipment in the area where cables are installed is described in detail in Section 9.5.1.

8.3.3.3 Fire Barriers and Tray Supports

The fire barriers, where required between redundant trays, are described in detail in Section 9.5.1. The separation between redundant trays is described in Subsection 8.3.1.4.

8.3.3.4 Fire Stops

Fire stops are provided at penetrations in walls and floors designated as fire barriers as described in Section 9.5.1.

REFERENCES

1. 10 CFR Part 50, General Design Criteria For Nuclear Power Plants, February 4, 1972, U.S. Nuclear Regulatory Commission.

2. NRC Regulatory Guide 1.6, Independence Between Redundant Standby (Onsite) Power Sources and Between Their Distribution Systems, March 10, 1971, U.S. Nuclear Regulatory Commission.

3. NRC Regulatory Guide 1.9, Selection of Diesel Generator Set Capacity For Standby Power Supplies, March 10, 1971, U.S. Nuclear Regulatory Commission.

4. NRC Regulatory Guide 1.22, Periodic Testing of Protection System Actuation Functions, February 17, 1972, U.S. Nuclear Regulatory Commission.

5. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976, U.S. Nuclear Regulatory Commission.

6. NRC Regulatory Guide 1.30, Quality Assurance Requirements for the Installation, Inspection, and Testing of Instrumentation and Electric Equipment, August 11, 1972, U.S. Nuclear Regulatory Commission.

7. NRC Regulatory Guide 1.32, Use of IEEE Std. 308-1974, Criteria For Class 1E Electric Systems For Nuclear Power Generating Stations, Revision 2, February 1977, U.S. Nuclear Regulatory Commission.

8. NRC Regulatory Guide 1.41, Preoperational Testing of Redundant On-site Electric Power Systems to Verify Proper Load Group Assignments, March 16, 1973, U.S. Nuclear Regulatory Commission.

9. NRC Regulatory Guide 1.47, Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems, May 1973, U.S. Nuclear Regulatory Commission.

10. NRC Regulatory Guide 1.53, Application of Single-Failure Criterion to Nuclear Power Plant Protection Systems, June 1973, U.S. Nuclear Regulatory Commission.

11. NRC Regulatory Guide 1.62, Manual Initiation of Protective Actions, October 1973, U.S. Nuclear Regulatory Commission.

12. NRC Regulatory Guide 1.63, Electric Penetration Assemblies in Containment Structures for Water-Cooled Nuclear Power Plants, Revision 2, July 1978, U.S. Nuclear Regulatory Commission.

13. NRC Regulatory Guide 1.68, Preoperational Initial Startup Test Programs for Water-Cooled Power Reactors, Revision 2, August 1978, U.S. Nuclear Regulatory Commission.

14. NRC Regulatory Guide 1.73, Qualification Tests of Electric Valve Operators Installed Inside the Containment of Nuclear Power Plants, January 1974, U.S. Nuclear Regulatory Commission.

15. NRC Regulatory Guide 1.75, Physical Independence of Electric Systems, Revision 1, January 1975, U.S. Nuclear Regulatory Commission.

16. NRC Regulatory Guide 1.81, Shared Emergency and Shutdown Electric Systems for Multi-Unit Nuclear Power Plants, Revision 1, January 1975, U.S. Nuclear Regulatory Commission.

17. NRC Regulatory Guide 1.89, Qualification of Class 1E Equipment or Nuclear Power Plants, November 1974, U.S. Nuclear Regulatory Commission.

18. NRC Regulatory Guide 1.93, Availability of Electric Power Sources, December 1974, U.S. Nuclear Regulatory Commission.

18A. NRC Regulatory Guide 1.106, Thermal Overload Protection for Electric Motors on Motor Valves, Revision 1, March 1977, U.S. Nuclear Regulatory Commission.

18B. NRC Regulatory Guide 1.108, Periodic Testing of Diesel Generator Units used as Onsite Electric Power Systems at Nuclear Power Plants, Revision 1, August 1977, U.S. Nuclear Regulatory Commission.

18C. NRC Regulatory Guide 1.118, Periodic Testing of Electric Power and Protection Systems, June 1976, U.S. Nuclear Regulatory Commission.

18D. NRC Regulatory Guide 1.129, Maintenance, Testing, and Replacement of Large Lead Storage Batteries for Nuclear Power Plants, Revision 1, February 1978, U.S. Nuclear Regulatory Commission.

18E. NRC Regulatory Guide 1.131, Qualification Tests of Electric Cables, Field Splices, and Connections for Light-Water-Cooled Nuclear Power Plants, 8/77 (for comments), U.S. Nuclear Regulatory Commission.

19. IEEE 279-1971, Revision 1, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Stations.

20. IEEE 308-1974, Standard Criteria for Class 1E Power Systems for Nuclear Power Generating Systems.

21. IEEE 317-1976, Electric Penetration Assemblies in Containment Structures for Nuclear Power Generating Stations.

22. IEEE 323-1974, Standard for Qualifying Class 1E Equipment for Nuclear Power Generating Stations.

23. IEEE 334-1974, Standard for Type Tests of Continuous-Duty Class 1E Motors for Nuclear Power Generating Stations.

24. IEEE 336-1971 (ANSI N45.2.4-1972), Installation, Inspection, and Testing Requirements for Instrumentation and Electric Equipment During the Construction of Nuclear Power Generating Stations.

25. IEEE 338-1971, Trial-Use Criteria for the Periodic Testing of Nuclear Power Generating Station Protection Systems.

26. IEEE 344-1975, Recommended Practices for Seismic Qualification of Class 1E Equipment for Nuclear Power Generating Stations.

27. IEEE 352-1972, Guide for General Principles of Reliability Analysis of Nuclear Power Generating Station Protection Systems.

28. IEEE 379-1972, Guide for the Application of the Single- Failure Criterion to Nuclear Power Generating Station Protection Systems.

29. IEEE 382-1972 (ANSI N41.6-1972), Guide for Type Test of Class 1 Electric Valve Operators for Nuclear Power Generating Stations.

30. IEEE 383-1974, Standard for Type Test of Class 1E Electric Cables, Field Splices, and Connections for Nuclear Power Generating Stations.

31. IEEE 384-1974, Trial-Use Standard Criteria for Separation of Class 1E Equipment and Circuits.

32. IEEE 387-1977, Criteria for Diesel-Generator Units Applied as Standby Power Supplies for Nuclear Power Generation Stations.

33. IEEE 420-1973, Trial-Use Guide for Class 1E Control Switchboards for Nuclear Power Generating Stations.

34. IEEE 422-1973, (draft 3), Guide for the Design and Installation of Cable Systems in Power Generating Stations.

35. IEEE 450-1995, IEEE Recommended Practice for Maintenance, Testing, and Replacement of Vented Lead Acid Batteries for Stationary Applications.

36. IEEE 494-1974, Standard Method for Identification of Documents Related to Class 1E Equipment for Nuclear Power Generating Stations.

37. IPCEA P-46-426 (IEEE S-135), Power Cable Ampacities, Volume 1, Copper Conductors, and Volume 2, Aluminum Conductors, 1962.

38. ICEA P-54-440 (NEMA WC-51-1975), Ampacities, Cables in Open-Top Cable Trays, Rev. 2, August, 1979.

39. NEMA VE 1-1971, Cable Tray Systems, National Electrical Manufacturers Association.

40. IEEE 80-1961 (Reaff. 1971), Guide for Safety in AC Substation Grounding.

41. Wyle Laboratories Test Report No. 48037-02, Electrical Raceway Separation Verification Testing for the Texas Utilities Generating Company for use in the Comanche Peak Steam Electric Station Units 1 and 2, February 6, 1986.

42. Wyle Laboratories Test Report No. 48422-1, Cable Separation Test Program for the Texas Utilities Generating Company Comanche Peak Steam Electric Station Units 1 and 2, August 14, 1986.

43. Wyle Laboratories Test Report No. 47906-02, Test Report on Electrical Separation Verification Testing for the Stone & Webster Engineering Corporation for Use In Niagra Mohawk Power Corporation Nine Mile Point Nuclear Station - Unit 2, November 22, 1985, Configuration No.5.

44. Wyle Laboratories Test Report No. 17666-02, Test Report on Electrical Separation Verification Testing for the Stone & Webster Engineering Corporation for Use In Duquesne Light Company’s Beaver Valley Power Station - Unit 2, April 19, 1985, Configurations 1, 2, 3, 4 and 6.

45. ASME Boiler and Pressure Vessel Code, Sections II, III, V, and IX.

46. Brown Boveri Electric, Inc., Test Report K-82089-K1, Test Date May 27, 1982.

47. Wyle Laboratories Test Report No. 53575, Test Report on Separation Verification Testing for Bechtel Energy Corporation for Houston Lighting and Power's South Texas Project, Configuration #1, Test #2.

48. Letter NS-CE-604, dated March 31, 1975, C. Eicheldinger (Westinghouse) to the Secretary of the Nuclear Regulatory Commission.

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 1 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
CENTRIFUGAL CHARGING PUMP						
TBX-CSAPCH-02 (TBX-CSAPCH-01)	600	YES	10	1	(a)(25)	(4)
MOTOR-OPERATED VALVES						
VARIOUS (VARIOUS)	---	NO	10 (14)	AS REQ'D	(3)	(6)
DIESEL OIL DAY TANK AREA FAN						
CP1-VAFNCB-05 (CP1-VAFNCB-04)	1.5	NO	10 (14)	1	(d)	(4)
DIESEL GENERATOR ROOM FANS						
CP1-VAFNAV-29, 30, 31, 32 (CP1-VAFNAV-25, 26, 27, 28)	40	NO	10 (14)	4	(d)	(4)
DIESEL GENERATOR FUEL OIL TRANSFER PUMP						
CP1-DOAPFT-03, 04 (CP1-DOAPFT-01, 02)	3	NO	10 (10)	2	(d)	(6)
SAFEGUARDS BUILDING FLOOR DRAIN SUMP PUMP						
CP1-WPAPSS-01, 02 (CP1-WPAPSS-03, 04)	5	NO	10 (10)	2	(b)	(6)
CONTROL ROOM PRESSURIZATION FAN (2)						
CPX-VAFNCB-06 (CPX-VAFNCB-05)	5	NO	10 (14) (15)	1	(d)	(5)
CONTROL ROOM EMERGENCY PRESSURIZATION HEATER (2)						
CPX-VAFUPK-22 (CPX-VAFUPK-21)	10 KW	NO	10 (10) (14) (15)	1	(d)	(5)

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 2 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
CR. RADIATION MONITOR SAMPLE PUMP TRANSFORMER (2)						
CPX-RMTRET-04 (03)	2.070	NO	10 (14)	1	(d)	(4)
UPS ROOM FAN COIL UNIT						
CP1-VAAUPR-02 (CP1-VAAUPR-01)	5	NO	10 (14)	1	(d)	(4)
CONTROL ROOM EMERGENCY FILTRATION FAN						
CPX-VAFNCB-24 (2) (CPX-VAFNCB-23)	30	NO	10 (14) (15)	1	(b)	(5)
CENTRIFUGAL CHARGING PUMP ROOM FAN COOLER						
CP1-VAAUSE-04 (CP1-VAAUSE-03)	3	NO	10 (10)	1	(d)	(4)
BATTERY ROOM EXHAUST FANS						
CP1-VAFNID-09, 10 (CP1-VAFNID-07, 08)	1.5	NO	10 (14)	2	(d)	(4)
LIGHTING TRANSFORMERS						
LIGHTING - TRAIN B (LIGHTING - TRAIN A)	450 KVA	NO	10 (14)	---	(d)	(4)
BATTERY CHARGERS						
CP1-EPBCED-02, 04, 06, 08 (CP1-EPBCED-01, 03, 05, 07)	300 ADC	NO	10 (14)	2	(d)	(4)

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 3 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
UPS BYPASS TRANSFORMER						
CP1-ECTRET-02 LOAD INCLUDES X-RE-5895B (CP1-ECTRET-01) LOAD INCLUDES X-RE-5896A	25 KVA	NO	10 (14)	1	(d)	(4)
AUXILIARY BUILDING REGULATING TRANSFORMER						
CPX-EPTRNT-43 (2) (CPX-EPTRNT-42)	45 KVA	NO	10 (14)	1	(d)	(4)
REGULATING TRANSFORMER - ALT. SHUTDOWN						
---NONE--- (CP1-EPTRNT-28)	--- 5 KVA	--- NO	--- 10 (14)	--- 1 (TRAIN A ONLY)	--- (d)	--- (4)
SAFEGUARDS BUILDING TRANSFORMER						
CP1-EPTRET-06 (CP1-EPTRET-05)	45 KVA	NO	10 (14)	1	(d)	(4)
SERVICE WATER INTAKE STRUCTURE VENTILATION EXHAUST FAN (24)						
CPX-VAFNWV-08, 09 (CPX-VAFNWV-06, 07)	3	NO	10 (10)	2	(d)	(5)
REGULATING TRANSFORMER						
CPX-EPTRNT-45 (CPX-EPTRNT-44)	30 KVA	NO	10 (14)	1	(d)	(4)
HIGH PRESSURE CHEMICAL FEED ROOM EXHAUST FAN						
CP1-VAFNCB-09 (NONE)	3 ---	NO ---	10 (10) ---	1 --- (TRAIN B ONLY)	(d) ---	(5) ---

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 4 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
HIGH PRESSURE CHEMICAL FEED ROOM SUPPLY FAN						
CP1-VAFNCB-08	3	NO	10 (10)	1	(d)	(5)
(NONE)	---	---	---	--- (TRAIN B ONLY)	---	---
BATTERY ROOM 117 AND 124 UNIT HEATER						
CP1-VAHEUH-02 (CP1-VAHEUH-03)	25 KW	NO	10 (10)	1	(d)	(5)
BORIC ACID TRANSFER PUMP						
TBX-CASPBA-02 (TBX-CASPBA-01)	20.78	NO	10 (10)	1	(b)	(4)
SAFETY INJECTION PUMP						
TBX-SIAPSI-02 (TBX-SIAPSI-01)	450	YES	15	1	(a) (25)	(4)
CHILLED WATER RECIRCULATING PUMP						
CP1-CHAPCP-06 (CP1-CHAPCP-05)	25	YES	15	1	(d)	(4)
SAFETY INJECTION PUMP ROOM FAN						
CP1-VAAUSE-06 (CP1-VAAUSE-05)	3	NO	15 (10)	1	(d)	(4)
RESIDUAL HEAT REMOVAL PUMP						
TBX-RHAPRH-02 (TBX-RHAPRH-01)	450	YES	20	1	(a) (25)	(4)
RESIDUAL HEAT REMOVAL PUMP ROOM FAN						
CP1-VAAUSE-02 (CP1-VAAUSE-01)	3	NO	20 (10)	1	(d)	(4)

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 5 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
CONTAINMENT SPRAY PUMP						
CP1-CTAPCS-02, 04 (CP1-CTAPCS-01, 03)	700	YES	25	2	(c) (25)	(4)
CONTAINMENT SPRAY PUMP ROOM FAN						
CP1-VAAUSE-12, 14 (CP1-VAAUSE-11, 13)	3	NO	25 (10)	2	(d)	(4)
COMPONENT COOLING WATER PUMP						
CP1-CCAPCC-02 (CP1-CCAPCC-01)	1000	YES	30	1	(c) (25)	(4)
COMPONENT COOLING WATER PUMP ROOM FAN COOLER						
CP1-VAAUSE-10 (CP1-VAAUSE-09)	3	NO	30 (10)	1	(d)	(4)
STATION SERVICE WATER PUMP						
CP1-SWAPSW-02 (CP1-SWAPSW-01)	900	YES	35	1	(c) (25)	(4)
AUXILIARY FEEDWATER PUMP						
CP1-AFAPMD-02 (CP1-AFAPMD-01)	700	YES	40	1	(c) (25)	(4)
MOTOR DRIVEN AFW PUMP ROOM FAN						
CP1-VAAUSE-08 (CP1-VAAUSE-07)	3	NO	40 (10)	1	(d)	(4)
PRIMARY PLANT VENT EXHAUST FAN (2)						
CPX-VAFNCB-08, -22 (CPX-VAFNCB-07, -21)	60	YES	40	2	(d)	(5)

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 6 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
PRIMARY PLANT ESF HEATER (2)						
CPX-VAFUPK-02, -16 (CPX-VAFUPK-01, -15)	100 KW	NO	40 (10)	2	(d)	(5)
HVAC CENTRIFUGAL WATER CHILLER						
CP1-CHCICE-06 (CP1-CHCICE-05)	154	YES	75	1	(d)	(5)
UPS AND DISTRIBUTION ROOM A/C UNIT (2)						
CPX-VAACUP-02 (CPX-VAACUP-01)	65.7 KVA	YES	90	1	(d)	(5)
UPS AND DISTRIBUTION ROOM BOOSTER RETURN FAN (2)						
CPX-VAFNAV-43 (CPX-VAFNAV-42)	20	NO	90 (10)	1	(b)	(5)
CONTROL ROOM A/C UNIT (2)						
CPX-VAACCR-03, 04 (CPX-VAACCR-01, 02)	232.6 KVA	YES	90	2 (18)	(d)	(5)
CONTROL ROOM A/C HEATER (2)						
CPX-VAACCR-03, 04 (CPX-VAACCR-01, 02)	30 KW	NO	90 (10)	2 (18)	(d)	(5)
ELECTRICAL AREA FAN						
CP1-VAAUSE-15, 16 (CP1-VAAUSE-17, 18)	5	YES	90	2	(d)	(4)
SPENT FUEL POOL HX AND PUMP ROOM FAN COOLER (2)						
CPX-VAAUSE-02 (CPX-VAAUSE-01)	3	NO	120 (9)	1	(d)	(4)

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 7 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP
SPENT FUEL POOL COOLING WATER PUMP (2)						
CPX-SFAPSF-02 (CPX-SFAPSF-01)	250	NO	120 (9)	1	(b)	(4)
REACTOR MAKE-UP WATER PUMP (2)						
CPX-DDAPRM-01 (CP1-DDAPRM-01)	40	NO	120 (11)	1	(b)	(4)
DG AIR COMPRESSOR						
CP1-MECAED-03, 04 (CP1-MECAED-01, 02)	20	NO	120 (10) (16)	1	(d)	(5)
CONTROL ROOM MAKE-UP SUPPLY FAN (2)						
CPX-VAFNAV-38 (CPX-VAFNAV-37)	2	NO	AS REQ'D (11) (16)	1	(b)	(5)
CONTROL ROOM EXHAUST FAN (2)						
CPX-VAFNID-02 (CPX-VAFNID-01)	5	NO	AS REQ'D (11) (16)	1	(d)	(5)
CONTROL ROOM KIT/TOILET EXHAUST FAN (2)						
CPX-VAFNID-04 (CPX-VAFNID-03)	1.5	NO	AS REQ'D (11) (16)	1	(d)	(5)

SUMMARY:

Total demand KW for loss of coolant accident coincident with loss of offsite power is less than 6300 KW. (Unit 1 Train A or B, Unit 2 Train A or B).

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 8 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP

GENERAL NOTES:

* Where “Pump (Fan)” components are listed, it is understood that the pump (fan) motor is the actual electrical load component.

(A) Train B shown, components powered by Train ‘A’ are shown in parenthesis.

(B) Letters indicated in the “Basis for “KW” column denote the following:

- (a) Pump Runout Condition
- (b) Rated BHP
- (c) Based on Actual Demand
- (d) Nameplate Rating
- (e) Since the MOVs operate intermittently only, this load is not reflected in the running and continuous loading of the D-G. However, all MOVs that may start during diesel generator sequencing and have an auto function to close or open are conservatively assumed to start at the first load step. In addition, 10% of the combined starting load of these MOVs is assumed to occur at each subsequent load step.

(C) Auto sequencer signal: “YES” indicates a direct signal from the sequencer contacts to close component feeder breaker for the step noted in the “START TIME AFTER SIAS” column. “NO” indicates that there is no direct sequencer signal and that the start time is clarified by a note in the “START TIME AFTER SIAS” column.

SPECIFIC NOTES:

1. Maximum closing time to close D-G circuit breaker is 10 seconds after the receipt of start signal including D-G to come up to rated speed and voltage. The “10 seconds” delay indicated in the table is coincident with the D-G circuit breaker closing and step 1 of the sequencer. Step 1 of the sequencer can be delayed by one second when loads are energized from an offsite power source. There is no time delay interposed for non-sequenced loads (Note 14) when the loads are energized from an offsite power source. See Section 8.3.1.1.5.3.
2. Equipment is shared between two units.
3. Not used.
4. Manually stopped.
5. Stops automatically with assigned diesel or pump, temperature, pressure, level, etc.
6. Motor stops automatically when valve action is completed, or receives signal to stop (e.g., sump pump stops on low water level).

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 9 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP

7. For other DBAs which are mitigated by the SIAS such as a Main Steam Line Break accident, automatic loading sequence is same as Loss-of-Coolant Accident condition. See DBD-ME-011, Diesel Generator Sets, Attachment 3 for other DBAs where loss of offsite power is mitigated by the BOS.
8. For Feedwater Linebreak DBAs, the EDG may be initially started by LOOP and loaded by BOS. (The loss of generation, caused by turbine trip as a result of FWLB, is assumed to initiate a loss of offsite power.) After the initiation of the accident a Safety Injection Actuation Signal (SIAS) is generated based on plant parameter conditions. This (SIAS) initiates the SI Sequencer (SIS). All loads not sequenced by BOS which are required for FWLB are sequenced by SIS. Some non-Class 1E loads are isolated by the SIAS. Thus after the initiation of SIAS only loads also required to be sequenced by SIS remain loaded on the DG. The SIS will send signals to all sequenced loads, the loads already running will keep running and the loads not running will be started. This loading by SIS will result in the final EDG load to be bounded by that represented in the LOCA loading table. The EDG loading calculations are required to evaluate the acceptability of EDG capability to start SIS step loads not running during BOS.
9. As required following diesel loading sequence. Indicated times are estimates and do not restrict plant operation based upon diesel generator loading.
10. Starts automatically with assigned load or upon temperature, pressure, level switch signal, etc.
11. Manually started as required.
12. Not used.
13. Not used.
14. Load is automatically energized as the voltage is restored to the ESF Buses. This load therefore, is not considered a sequenced load.
15. The CREFS is started in the event of a LOCA. If a loss of offsite power occurs, the components will load when power is restored to the bus.
16. Times shown are minimum start times. The SIAS must be reset prior to component start.
17. Not Used.
18. The Control Room HVAC (CRAC) A/C Units and A/C Heaters are mutually exclusive loads. The larger load (CRAC Unit) is considered in EDG loading.
19. Thru 23. Not Used.
24. Components CPX-VAFNWV-08 & 09 are fed from Unit 1, Train B. Components CPX-VAFNWV-06 & 07 are fed from Unit 1, Train A. Components CPX-VAFNWV-04 & 05 are fed from Unit 2, Train B. Components CPX-VAFNWV-02 & 03 are fed from Unit 2, Train A.
25. Calculated pump runout condition or actual load demand.

TABLE 8.3-1
LOADING REQUIREMENTS FOR LOSS-OF-COOLANT ACCIDENT COINCIDENT WITH LOSS OF OFFSITE POWER
(Sheet 10 of 10)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER SIAS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD (B)	TIME TO STOP

Abbreviations:

Component cooling water pump (CCWP); safety injection system (SIS); residual heat removal (RHR); heating, ventilating and air conditioning (HVAC); balance of plant (BOP); Nuclear Safety Related (NSR); air conditioning (A/C).

TABLE 8.3-1A
ALL SHEETS HAVE BEEN DELETED

TABLE 8.3-1B
ALL SHEETS HAVE BEEN DELETED

TABLE 8.3-1C
OUT OF TOLERANCE VOLTAGE VALUES (CLASS 1E 6.9KV SWITCHGEAR AND 480 VOLT LOAD CENTERS)

BUS	NOMINAL VOLTAGE	MINIMUM ALLOWABLE VOLTAGE	MAXIMUM ALLOWABLE VOLTAGE
6.9KV Switchgear 1EA1, 2EA1, 1EA2, 2EA2	6900	5965	7260
480V Switchgear 1EB1, 2EB1, 1EB3, 2EB3, 1EB2, 2EB2, 1EB4, 2EB4	480	435	510

TABLE 8.3-1D
THIS TABLE HAS BEEN DELETED

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 1 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
CENTRIFUGAL CHARGING PUMP						
TBX-CSAPCH-02 (TBX-CSAPCH-01)	600	YES	10	1	(a)(25)	(4)
MOTOR-OPERATED VALVES						
VARIOUS (VARIOUS)	---	NO	10 (14)	AS REQ'D	(3)	(6)
DIESEL OIL DAY TANK AREA FAN						
CP1-VAFNCB-05 (CP1-VAFNCB-04)	1.5	NO	10 (14)	1	(d)	(4)
DIESEL GENERATOR ROOM FANS						
CP1-VAFNAV-29, 30, 31, 32 (CP1-VAFNAV-25, 26, 27, 28)	40	NO	10 (14)	4	(d)	(4)
DIESEL GENERATOR FUEL OIL TRANSFER PUMP						
CP1-DOAPFT-03, 04 (CP1-DOAPFT-01, 02)	3	NO	10 (10)	2	(d)	(5)
SAFEGUARDS BUILDING FLOOR DRAIN SUMP PUMP						
CP1-WPAPSS-01, 02 (CP1-WPAPSS-03, 04)	5	NO	10 (10)	2	(b)	(6)
CHILLED WATER RECIRCULATION PUMP						
CP1-CHAPCP-06 (CP1-CHAPCP-05)	25	NO	10 (14)	1	(d)	(4)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 2 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
CONTROL ROOM PRESSURIZATION FAN (2)						
CPX-VAFNCB-06 (CPX-VAFNCB-05)	5	NO	10 (14)	1	(d)	(5)
CONTROL ROOM EMERGENCY PRESSURIZATION HEATER (2)						
CPX-VAFUPK-22 (CPX-VAFUPK-21)	10 KW	NO	10 (10)	1	(d)	(5)
CONTROL ROOM EMERGENCY FILTRATION FAN (2)						
CPX-VAFNCB-24 (CPX-VAFNCB-23)	30	NO	10 (14)	1	(b)	(5)
FUEL OIL BOOSTER PUMP						
CP1-DOAPBP-02 (CP1-DOAPBP-01)	3	NO	10 (10)	1	(d)	(4)
UPS ROOM FAN COIL UNIT						
CP1-VAAUPR-02 (CP1-VAAUPR-01)	5	NO	10 (14)	1	(d)	(4)
FUEL OIL DRIP RETURN PUMP						
CP1-DOAPDW-02 (CP1-DOAPDW-01)	1	NO	10 (10)	1	(d)	(4)
CENTRIFUGAL CHARGING PUMP ROOM FAN COOLER						
CP1-VAAUSE-04 (CP1-VAAUSE-03)	3	NO	10 (10)	1	(d)	(4)
BATTERY ROOM EXHAUST FANS						
CP1-VAFNID-09, 10, 12 (CP1-VAFNID-07, 08, 11)	1.5	NO	10 (14)	3	(d)	(4)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 3 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
LIGHTING TRANSFORMERS						
LIGHTING - TRAIN B (LIGHTING - TRAIN A)	495 KVA	NO	10 (14)	---	(d)	(4)
BATTERY CHARGERS						
CP1-EPBCED-02, 04, 06, 08 (CP1-EPBCED-01, 03, 05, 07)	300 ADC	NO	10 (14)	2	(d)	(4)
BATTERY CHARGER						
CP1-EPBCND-02, 04 (NONE)	200 ADC ---	NO ---	10 (14) ---	1 (TRAIN B --- ONLY)	(d) ---	(4) ---
CONTROL ROOM RADIATION MONITOR SAMPLE PUMP TRANSFORMER (2)						
CPX-RMTRET-04(03)	1.5	NO	10 (14)	1	(d)	(4)
VENTILATION STACK RADIATION MONITOR SYSTEM SAMPLE PUMP (2)						
X-RE-5567B (X-RE-5567A)	1.5	NO	10 (14)	1	(d)	(4)
CONTAINMENT AIR RADIATION MONITOR SAMPLE PUMP						
1-RE-5502-03-66 (NONE)	1.5 ---	NO ---	10 (14) ---	1 (TRAIN B --- ONLY)	(d) ---	(4) ---
SUPS BYPASS TRANSFORMER						
CP1-ECTRET-02 LOAD INCLUDES X-RE-5895B (CP1-ECTRET-01) LOAD INCLUDES X-RE-5896A	25 KVA	NO	10 (14)	1	(d)	(4)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 4 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
CONTROL BUILDING BYPASS TRANSFORMER						
NONE	---	---	---	---	---	---
(CP1-EPTRMT-12)	10 KVA	NO	10 (14)	1 (TRAIN A ONLY)	(d)	(4)
AUXILIARY BUILDING REGULATING TRANSFORMER						
CPX-EPTRNT-43	45 KVA	NO	10 (14)	1	(d)	(4)
(CPX-EPTRNT-42)						
REGULATING TRANSFORMER - ALT. SHUTDOWN						
NONE	---	---	---	---	---	---
(CP1-EPTRNT-28)	5 KVA	NO	10 (14)	1 (TRAIN A ONLY)	(d)	(4)
SAFEGUARDS BUILDING TRANSFORMER						
CP1-EPTRET-06	45 KVA	NO	10 (14)	1	(d)	(4)
(CP1-EPTRET-05)						
REGULATING TRANSFORMER						
CPX-EPTRNT-45	30 KVA	NO	10 (14)	1	(d)	(4)
(CPX-EPTRNT-44)						
HIGH PRESSURE CHEMICAL FEED ROOM EXHAUST FAN						
CP1-VAFNCB-09	3	NO	10 (10)	1 (TRAIN B	(d)	(5)
(NONE)	---	---	---	--- ONLY)	---	---
HIGH PRESSURE CHEMICAL FEED ROOM SUPPLY FAN						
CP1-VAFNCB-08	3	NO	10 (10)	1 (TRAIN B	(d)	(5)
(NONE)	---	---	---	--- ONLY)	---	---

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 5 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
BATTERY ROOM 117 AND 124 UNIT HEATER						
CP1-VAHEUH-02 (CP1-VAHEUH-03)	25 KW	NO	10 (10)	1	(d)	(5)
BORIC ACID TRANSFER PUMP						
TBX-CASPBA-02 (TBX-CASPBA-01)	20.78	NO	10 (10)	1	(b)	(4)
ROOM ADJACENT TO REACTOR MAKEUP WATER STORAGE TANK HEATER						
CP1-VAEHUH-28 (CP1-VAEHUH-25)	10 KW	NO	10 (14)	1	(d)	(5)
ROOM ADJACENT TO COND. WATER STORAGE TANK HEATER						
CP1-VAEHUH-29 (CP1-VAEHUH-26)	10 KW	NO	10 (14)	1	(d)	(5)
ROOM ADJACENT TO REFUELING WATER STORAGE TANK HEATER						
CP1-VAEHUH-30 (CP1-VAEHUH-27)	7.5 KW	NO	10 (14)	1	(d)	(5)
AIR DRYER CONTROL PANEL FDR						
CP1-CIDYIA-02 (CP1-CIDYIA-01)	18.3 KW	NO	10 (14)	1	(d)	(4)
SERVICE WATER PUMPHOUSE VENTILATION (24) EXHAUST FAN						
CPX-VAFNWV-08, 09 (CPX-VAFNWV-06, 07)	3	NO	10 (10)	2	(d)	(5)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 6 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
COMPONENT COOLING WATER PUMP FAN COOLER						
CP1-VAAUSE-10 (CP1-VAAUSE-09)	3	NO	30 (10)	1	(d)	(4)
COMPONENT COOLING WATER PUMP						
CP1-CCAPCC-02 (CP1-CCAPCC-01)	1000	YES	30	1	(c) (25)	(4)
CRDM VENT FAN						
CP1-VAFNCB-02 (CP1-VAFNCB-01)	125	YES	30	1	(b)	(4)
STATION SERVICE WATER PUMP						
CP1-SWAPSW-02 (CP1-SWAPSW-01)	900	YES	35	1	(c) (25)	(4)
AUXILIARY FEEDWATER PUMP						
CP1-AFAPMD-02 (CP1-AFAPMD-01)	700	YES	40	1	(c) (22)(25)	(4)
MOTOR DRIVEN AFW PP ROOM FAN						
CP1-VAAUSE-08 (CP1-VAAUSE-07)	3	NO	40 (10)	1	(d)	(4)
CONTAINMENT RECIRCULATION FAN						
CP1-VAFNAV-02, 04 (CP1-VAFNAV-01, 03)	125	YES	50	2	(b)	(4)
NEUTRON DETECTOR WELL FAN						
CP1-VAFNAV-10 (CP1-VAFNAV-09)	40	YES	50	1	(b)	(4)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 7 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
VENT CHILLED WATER PUMP						
CPX-CHAPCP-02, 04 (CPX-CHAPCP-01, 03)	125	YES	60	2	(d)	(4)
AUXILIARY BUILDING VENT EQUIPMENT ROOM EXHAUST FAN						
CPX-VAFNAV-28 (CPX-VAFNAV-27)	40	YES	60	1	(b)	(4)
HVAC CENTRIFUGAL WATER CHILLER						
CP1-CHCICE-06 (CP1-CHCICE-05)	154	YES	75	1	(d)	(4)
HVAC CENTRIFUGAL WATER CHILLER (2), (2A)						
CPX-CHCICE-02, (CPX-CHCICE-01)	670	YES	85 (20)	1	(c)	(4)
HVAC CENTRIFUGAL WATER CHILLER OIL PUMP (2)						
CPX-CHCICE-02B, 04B (CPX-CHCICE-01B, 03B)	0.25	NO	85 (10)(20)	2	(d)	(5)
UPS AND DISTRIBUTION ROOM A/C UNIT (2)						
CPX-VAACUP-02 (CPX-VAACUP-01)	65.7 KVA	YES	90	1	(d)	(4)
UPS AND DISTRIBUTION ROOM BOOSTER RETURN FAN (2)						
CPX-VAFNAV-43 (CPX-VAFNAV-42)	20	NO	90 (10)	1	(b)	(5)
CONTROL ROOM A/C HEATER (2)						
CPX-VAACCR-03, 04 (CPX-VAACCR-01, 02)	30 KW	NO	90 (10)	2 (18)	(d)	(5)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 8 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
CONTROL ROOM A/C UNIT (2)						
CPX-VAACCR-03, 04 (CPX-VAACCR-01, 02)	232.6 KVA	YES	90	2 (18)	(d)	(5)
ELECTRICAL AREA FAN						
CP1-VAAUSE-15, 16 (CP1-VAAUSE-17, 18)	5	YES	90	2	(d)	(4)
INSTRUMENT AIR COMPRESSOR						
CP1-CICACO-02 (CP1-CICACO-01)	200	NO	90 (10)	1	(d)	(4)
DG AIR COMPRESSOR						
CP1-MECAED-03, 04 (CP1-MECAED-01, 02)	20	NO	120 (10)	1	(d)	(4)
AUXILIARY LUBE OIL PP						
CP1-MEAPLO-04 (CP1-MEAPLO-02)	60	NO	120 (11)	1	(23)	(23)
AUXILIARY JACKET WTR PP						
CP1-MEAPJW-04 (CP1-MEAPJW-02)	75	NO	120 (11)(23)	1	(d)	(4)
PRIMARY PLANT VENT EXHAUST FAN						
CPX-VAFNCB-10,12,14,16,18,20 (CPX-VAFNCB-09,11,13,15,17,19)	60	NO	120 (11)	2 (MAX)	(d)	(5)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 9 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
POSITIVE DISPLACEMENT CHARGING PUMP (16)						
NONE (TBX-CSAPPD-01)	---- 200	---- NO	---- 120 (11)	---- 1 (TRAIN A ONLY)	---- (16)	---- (4)
POSITIVE DISPLACEMENT CHARGING PUMP FAN COOLER						
NONE (CP1-VAFNCB-03)	---- 1.5	---- NO	---- 120(11)(17)	---- 1 (TRAIN A ONLY)	---- (16)	---- (4)
REACTOR MAKE-UP WATER PUMP						
CPX-DDAPRM-01 (CP1-DDAPRM-01)	40	NO	120 (11)	1	(b)	(5)
SPENT FUEL POOL COOLING WATER PUMP (2)						
CPX-SFAPSF-02 (CPX-SFAPSF-01)	250	NO	30 MIN (9)	1	(b)	(4)
SPENT FUEL POOL HX AND PUMP ROOM FAN COOLER (2)						
CPX-VAAUSE-02 (CPX-VAAUSE-01)	3	NO	30 MIN (9)	1	(d)	(4)
REGULATING TRANSFORMER FOR PRESSURIZER HEATERS (19)						
CP1-EPTRNT-06, 08 (CP1-EPTRNT-05, 07)	485 KW	NO	60 MIN (9)	1	(c)	(4)
RESIDUAL HEAT REMOVAL PUMP						
TBX-RHAPRH-02 (TBX-RHAPRH-01)	45O	NO	4 HOURS (9)	1	(a) (25)	(4)

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 10 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP
RESIDUAL HEAT REMOVAL PUMP ROOM FAN						
CP1-VAAUSE-02 (CP1-VAAUSE-01)	3	NO	4 HOURS (9)	1	(d)	(4)
CONTROL ROOM MAKE-UP SUPPLY FAN (2)						
CPX-VAFNAV-38 (CPX-VAFNAV-37)	2	NO	AS REQ'D (11)	1	(b)	(5)
CONTROL ROOM EXHAUST FAN (2)						
CPX-VAFNID-02 (CPX-VAFNID-01)	5	NO	AS REQ'D (11)	1	(d)	(5)
CONTROL ROOM KIT/TOILET EXHAUST FAN (2)						
CPX-VAFNID-04 (CPX-VAFNID-03)	1.5	NO	AS REQ'D (11)	1	(d)	(5)
SERVICE WATER SCREEN WASH PUMP						
CPX-SWAPTS-02 (CPX-SWAPTS-01)	15	NO	AS REQ'D (11)	1	(d)	(5)
SERVICE WATER TRAVELING SCREEN						
CPX-SWTSTS-02 (CPX-SWTSTS-01)	1.5	NO	AS REQ'D (10)	1	(d)	(5)

SUMMARY:

Total demand KW for loss of offsite power is less than 6300 KW (Unit 1 Train A or B, Unit 2 Train A or B).

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 11 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP

GENERAL NOTES:

* Where “Pump (Fan)” components are listed, it is understood that the pump (fan) motor is the actual electrical load component.

(A) Components powered by Train ‘A’ are shown in parenthesis.

(B) Letters indicated in the “Basis for “KW” column denote the following:

(a) Pump Runout Condition

(b) Rated BHP

(c) Based on Actual Demand

(d) Nameplate Rating

(e) Since the MOVs operate intermittently only, this load is not reflected in the running and continuous loading of the D-G. However, all MOVs that may start during diesel generator sequencing and have an auto function to close or open are conservatively assumed to start at the first load step. In addition 10% of the combined starting load of these MOVs is assumed to occur at each subsequent load step.

(C) Auto sequence signal: “YES” indicated a direct signal from the sequencer contacts to close component feeder breaker for the step noted in the “START TIME AFTER BOS” column. “NO” indicates that there is no direct sequencer signal and that the start time is specified by a note in the “START TIME AFTER BOS”. The component may not start due to status prior to BOS operation. For example, a Ventilation Chiller in standby prior to BOS operation will not start but the feeder breaker will be closed by the BOS signal.

(D) Shared components prefixed with CPX.

SPECIFIC NOTES:

1. Maximum closing time to close D-G circuit breaker is 10 seconds after the receipt of start signal including D-G to come up to rated speed and voltage. The “10 seconds” delay indicated in the table is coincident with the D-G circuit breaker closing and step 1 of the sequencer. There is no time delay interposed for non-sequenced loads (Note 14) when the loads are energized from an offsite power source. See Section 8.3.

2. Equipment is shared between two units.

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 12 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP

3. The HVAC centrifugal water chiller are powered from the 6.9KV buses: CPX-CHCICE-01 from Unit 1 Train A, CPX-CHCICE-02 from Unit 1 Train B, CPX-CHCICE-03 from Unit 2 Train A, CPX-CHCICE-04 from Unit 2 Train B.
4. Manually stopped.
5. Stops automatically with assigned diesel or pump, temperature, pressure, level, etc.
6. Motor stops automatically when valve action is completed, or receives signal to stop (e.g., sump pump stops on low water level).
7. Not Used.
8. For Feedwater Linebreak DBAs, the EDG may be initially started by LOOP and loaded by BOS. (The loss of generation, caused by turbine trip as a result of FWLB, is assumed to initiate a loss of offsite power.) After the initiation of the accident a Safety Injection Actuation Signal (SIAS) is generated based on plant parameter conditions. This (SIAS) initiates the SI Sequencer. All loads not sequenced by BOS which are required for FWLB are sequenced by SIS. Some non-Class 1E loads are isolated by the SIAS. Thus after the initiation of SIAS only loads also required to be sequenced by SIS remain loaded on the DG. The SIS will send signals to all sequenced loads, the loads already running will keep running and the loads not running will be started. This loading by SIS will result in the final EDG load to be bounded by that represented in the LOCA loading table. The EDG loading calculations are required to evaluate the acceptability of EDG capability to start SIS step loads not running during BOS.
9. As required following diesel loading sequence. Indicated times are estimates and do not restrict plant operation based upon diesel generator loading.
10. Starts automatically with assigned load or upon temperature, pressure, level switch signal, etc.
11. Manually started as required.
12. Not used.
13. Not used.
14. Load is automatically energized as the voltage is restored to the ESF Buses. This load therefore, is not considered a sequenced load.
15. Deleted
16. Operates in lieu of the centrifugal charging pump (CCP). The CCP is included in the total continuous load.
17. Operates in association with positive displacement charging pump.

TABLE 8.3-2
EMERGENCY ELECTRICAL LOADING REQUIREMENTS FOR LOSS OF OFFSITE POWER (BLACKOUT) CONDITIONS
(Sheet 13 of 13)

EQUIPMENT DESCRIPTION & TAG NUMBERS* (A)	NAMEPLATE HP (KW/KVA/A IF INDICATED)	AUTO SEQUENCER SIGNAL (C)	START TIME AFTER BOS (SEC) (1) (8)	NUMBER REQUIRED PER TRAIN	BASIS FOR KW REQD.(B)	TIME TO STOP

18. The Control Room HVAC (CRAC) A/C units and A/C heaters are mutually exclusive loads. The larger load (CRAC unit) is considered in the EDG loading.

19. Four groups of pressurizer heaters are installed. Groups A and B, are 485 KW and Group C and D are 416 KW; any group can be energized manually as required after the loss of power.

20. Start signal is from 85 second step. There may be additional delay due to equipment interlocks.

21. Not used.

22. The basis for the Auxiliary Feedwater Pump required BHP is documented by calculation. During the Loss of Offsite Power event, the BHP requirement for a single pump is not mopre than 750 HP. Actual pump demand decreases as the plant cools down and flow to the steam generators is manually controlled by the control room operator to maintain SG level.

23. Auxiliary Lube Oil Pump is only required if Shaft Driven Pump fails. Mechanical failure of not more than one pump is postulated. To be conservative, Jacket Water Pump (75 HP) loading has been included; hence, no additional loading is required due to Auxiliary Lube Oil Pump (60 HP) failure.

24. Components CPX-VAFNWV-08 & 09 ar fed from Unit 1, Train A. Components CPX-VAFNWV-06 & 07 are fed from Unit 1, Train B. Components CPX-VAFNWV-04 & 05 are fed from Unit 2, Train A. Components CPX-VAFNWV-02 & 03 are fed from Unit 2, Train B.

25. Calculated pump runout condition or actual load demand.

Abbreviations:

Component cooling water pump (CCWP); blackout signal (BOS); residual heat removal (RHR); heating, ventilating, and air conditioning (HVAC); balance of plant (BOP); Nuclear Safety Related (NSR); air conditioning (A/C).

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 1 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
1	138-kV offsite power source	supplies power to startup transformer XST1 (XST1A)	fails to provide electrical power to XST1 (XST1A)	transmission line fails; transmission line circuit breakers open.	loss of electrical power to transformer XST1 (XST1A)	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available.
2	345-kV offsite power source	supplies power to startup transformer XST2 (XST2A)	fails to provide electrical power to XST2 (XST2A)	transmission line fails; transmission line circuit breakers open.	loss of electrical power to transformer XST2 (XST2A)	protective relays; annunciation in control room	None: offsite power source (Item 1) or diesel generators are available.
3	Startup transformer XST1 (XST1A)	supplies POPS to Unit 2 safety buses through tap box #8 and CPX-EPTSST-04Y and AOPS to Unit 1 safety buses through CPX-EPTSST-04X and CPX-EPTSST-05X	fails to deliver power	internal fault; component failure; overheating; fire	loss of POPS to buses 2EA1 and 2EA2 and loss of AOPS for buses 1EA1 and 1EA2	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available. Can also realign to startup transformer XST1A (XST1)
3A	15-kV cable	supplies power from XST1 (XST1A) to manual transfer panels CPX-EPTSST-04X and CPX-EPTSST-05X	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of AOPS to buses 1EA1 and 1EA2 and loss of POPS to buses 2EA1 and 2EA2. For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: Offsite power source (Item 2) or diesel generators are available. Can also realign to startup transformer XST1A (XST1)

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 2 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
3B	15-kV cable (Note 5)	supplies power from XST1 (XST1A) to tap box #8 and manual transfer panel CPX-EPTSST-04Y	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of POPS to buses 2EA1 and 2EA2 and loss of AOPS to buses 1EA1 and 1EA2. For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available. Can also realign to startup transformer XST1A (XST1)
3C	15-kV manual transfer panels CPX-EPTSST-04X and CPX-EPTSST-05X.	supplies AOPS from XST1 (XST1A) to buses 1EA1 and 1EA2	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of AOPS to buses 1EA1 and 1EA2, For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available.
3D	tap box #8 and 15-kV manual transfer panel CPX-EPTSST-04Y.	supplies POPS from XST1 (XST1A) to buses 2EA1 and 2EA2	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of POPS to buses 2EA1 and 2EA2, For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available.

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 3 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
3E	15-kV cable	supplies AOPS from manual transfer panel CPX-EPTSST-04X (CPX-EPTSST-05X) to bus 1EA1 (1EA2)	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of AOPS to bus 1EA1 (1EA2) and POPS to bus 2EA1 (2EA2). For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available.
3F	15-kV cable	supplies POPS from tap box #8 and manual transfer panel CPX-EPTSST-04Y to bus 2EA1 (2EA2)	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of POPS to bus 2EA1 (2EA2) and AOPS for bus 1EA1 (1EA2). For short circuit: Loss of POPS to buses 2EA1, 2EA2, and loss of AOPS for buses 1EA1 and 1EA2.	protective relays; annunciation in control room	None: offsite power source (Item 2) or diesel generators are available.
4	startup transformer XST2 (XST2A)	supplies POPS to Unit 1 safety buses through CPX-EPTSST-02Y and CPX-EPTSST-03Y; and AOPS to Unit 2 safety buses through CPX-EPTSST-02X and CPX-EPTSST-03X	fails to deliver power	internal fault; component failure; overheating; fire	loss of POPS to buses 1EA1, and 1EA2 and loss of AOPS for buses 2EA1 and 2EA2	protective relays; annunciation in control room	None: offsite power source (Item 1) or diesel generators are available. Can also realign to startup transformer XTS2A (XST2).

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 4 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
4A	15-KV cable	supplies power from XST2 (XST2A) to transfer panel CPX-EPTSST-02Y and CPX-EPTSST-03Y	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of POPS to buses 1EA1 and 1EA2. For short circuit: Loss of POPS to buses 1EA1, 1EA2 and loss of AOPS for buses 2EA1 and 2EA2.	Protective relays; annunciation in control room	None: offsite power source (Item 1) or diesel generators are available. Can also realign to startup transformer XTS2A (XST2).
4B	15-KV cable	supplies power from XST2 (XST2A) to transfer panel CPX-EPTSST-02X and CPX-EPTSST-03X	short circuit; open circuit	mechanical damage; fire	For open circuit: Loss of AOPS to buses 2EA1 and 2EA2. For short circuit: Loss of POPS to buses 1EA1, 1EA2 and loss of AOPS for buses 2EA1 and 2EA2.	protection relays; annunciation in control room	None: offsite power source (Item 1) or diesel generators are available. Can also realign to startup transformer XTS2A (XST2).
4C	15-KV Transfer Panel CPX-EPTSST-02X and CPX-EPTSST-03X (Note 6)	Supplies AOPS from XST2 (XST2A) to buses 2EA1 and 2EA2	Short circuit; open circuit	Mechanical damage; fire	For open circuit: Loss of AOPS to buses 2EA1 and 2EA2. For short circuit: Loss of POPS to buses 1EA1, 1EA2, and loss of AOPS for buses 2EA1 and 2EA2.	Protective relays; annunciation in the control room	None; offsite power sources (Item 1) or diesel generators are available.

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 5 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
4D	15-KV Transfer Panel CPX-EPTSST-02Y and CPX-EPTSST-03Y (Note 7)	Supplies POPS from XST2 (XST2A) to buses 1EA1 and 1EA2	Short circuit; open circuit	Mechanical damage; fire	For open circuit: Loss of POPS to buses 1EA1 and 1EA2. For short circuit; Loss of POPS to buses 1EA1, 1EA2 and loss of AOPS for buses 2EA1 and 2EA2.	Protective relays; annunciation in the contol room	None; offsite power source (Item 1) or diesel generators are available.
4E	15-KV Cable (Note 8)	Supplies POPS from transfer panel CPX-EPTSST-02Y and CPX-EPTSST-03Y to buses 1EA1 and 1EA2	Short circuit; open circuit	Mechanical damage; fire	For open circuit: Loss of POPS to buses 1EA1 and 1EA2. For short circuit: Loss of POPS to buses 1EA1, 1EA2 and loss of AOPS for buses 2EA1 and 2EA2.	Protective relays; annunciation in the control room	None; offsite power source (Item 1) or diesel generators are available
4F	15-KV Cable (Note 9)	Supplies AOPS from transfer panel CPX-EPTSST-02X and CPX-EPTSST-03X to buses 2EA1 and 2EA2	Short circuit; open circuit	Mechanical damage; fire	For open circuit: Loss of AOPS to buses 2EA1 and 2EA2. For short circuit: Loss of POPS to buses 1EA1, 1EA2 and loss of AOPS for buses 2EA1 and 2EA2.	Protective relays; annunciation in the control room	None; offsite power source (Item 1) or diesel generators are available.

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 6 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
5	6.9-kV circuit breaker 1EA1-2	protects bus 1EA1	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Failure to open could cause loss of bus 1EA1 & XFMR XST1 (XST1A) 2. Fails to close: power cannot be supplied from XST1 (XST1A)	1. Annunciation 2. Periodic testing for operational readiness	None: 1. Redundant equipment is provided on bus 1EA2. 2. Power can be supplied from offsite power source (Item 2) or diesel generator 1EG1.
6	6.9-kV circuit breaker 1EA1-1	protects bus 1EA1	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Failure to open could cause loss of bus 1EA1 & XFMR XST2 (XST2A) 2. Fails to close: power cannot be supplied from power source XST2 (XST2A)	1. Annunciation 2. Periodic testing for operational readiness	None: 1. Redundant equipment is provided on bus 1EA2. 2. Power can be supplied from offsite power source (Item 1) or diesel generator 1EG1.
7	6.9-kV circuit breaker 1EA2-2	protects bus 1EA2	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Failure to open could cause loss of bus 1EA2 & XFMR XST1 (XST1A) 2. Fails to close: power cannot be supplied from power source XST1 (XST1A)	1. Annunciation 2. Periodic testing for operational readiness	None: 1. Redundant equipment is provided on bus 1EA1. 2. Power can be supplied from offsite power source (Item 2) or diesel generator 1EG2.

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 7 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
8	6.9-kV circuit breaker 1EA2-1	protects bus 1EA2	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Failure to open could cause loss of bus 1EA2 & XFMR XST2 (XST2A) 2. Fails to close: power cannot be supplied from power source XST2 (XST2A)	1. Annunciation 2. Periodic testing for operational readiness	None: 1. Redundant equipment is provided on bus 1EA1. 2. Power can be supplied from offsite power source (Item 1) or diesel generator 1EG2.
9	Bus 1EA1	distributes electrical power to Train A loads	fails to deliver power	short circuit; overload; fire; loss of ventilation	loss of Train A safety load group	1. protective relays 2. Annunciation in Control Room	None: Redundant Train B from bus 1EA2 available
10	Bus 1EA2	distributes electrical power to Train B loads	fails to deliver power	short circuit overload; fire; loss of ventilation	loss of Train B safety load group	1. protective relays 2. Annunciation in Control Room	None: redundant Train A from from bus 1EA1 available
11	6.9-kV circuit breaker	protects Unit Substation transformer T1EB1	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Fails to open: back up breaker opens and loss of 1EA1 Bus 2. Fails to close: loss of power to 1EB1	1. protective relays 2. Annunciation redundant load group in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
11A	8-kV cable	connects circuit breaker and transformer T1EB1	short circuit; open circuit	Mechanical damage; fire	loss of power to bus 1EB1	1. protective relays 2. Annunication in Control Room	None: load supplied by redundant load group provided from bus 1EA2

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 8 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
12	6.9-kV circuit breaker	protects unit substation transformer T1EB3	fails to open; fails to close	mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Fails to open: back up breaker opens and loss of 1EA1 Bus 2. Fails to close: loss of power to 1EB3	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
12A	8-kV cable	connects circuit breaker and transformer T1EB3	short circuit; open circuit	mechanical damage; fire	loss of power to bus 1EB3	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
13	6.9-kV circuit breaker	Protects Unit substation transformer T1EB2	fails to open; fails to close	Mechanical failure; stuck contacts; relay failure; control power fails; fire; loss of ventilation	1. Fails to open: back up breaker opens and loss of 1EA2 bus 2. Fails to close: loss of power to 1EB2	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
13A	8-kV cable	connects circuit breaker and transformer T1EB2	short circuit; open circuit	mechanical damage; fire	loss of power to bus 1EB2	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
14	6.9-kV circuit breaker	protects Unit substation transformer T1EB4	fails to open; fails to close	mechanical failure; stuck contacts relay failure; control power fails; fire; loss of ventilation	1. Fails to open: back up breaker opens and loss of 1EA2 bus 2. Fails to close: loss of power to 1EB4	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
14A	8-kV cable	connects circuit breaker and transformer T1EB4	short circuit; open circuit	mechanical damage; fire	loss of power to bus 1EB4	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 9 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
15	6.9-kV circuit breaker	diesel generator 1EG1 supply breaker	fails to close; fails to open	control failure; mechanical failure; relay failure; stuck contacts; fire; loss of ventilation	1. Fails to close; 1EG1 cannot supply power to bus 1EA1. 2. Fails to open: possible damage to bus 1EA1 and diesel generator 1EG1	1. periodic testing 2. Annunciation in Control Room	None: redundant loads supplied from bus 1EA2 by diesel generator 1EG2
15A	8-kV cable	connects 1EG1 to bus 1EA1 circuit breaker (Item 15)	short circuit; open circuit	mechanical damage; fire	does not transmit power from 1EG1 to bus 1EA1	1. periodic testing 2. Annunciation in Control Room	None: redundant loads supplied from bus 1EA2 by diesel generator 1EG2
15B	Diesel generator 1EG1	provides standby power for Train A loads	delivers no power to bus 1EA1	does not start; component failure; fire; loss of heating or ventilation, or both; spurious relay trip; loss of Control Power	does not transmit power from 1EG1 to bus 1EA1	1. relays and annunciation in control room 2. Periodic testing	None: redundant loads supplied from bus 1EA2 by diesel generator 1EG2
15C	15kV jumper cable	connects 1EG1 to PD bus coupler	short circuit	mechanical damage; fire	does not transmit power from 1EG1 to bus 1EA1	1. periodic testing 2. annunciation in control room	None: redundant loads supplied from bus 1EA2 by diesel generator 1EG2
15D	6.9kV bus coupler	provides isolation between 1EG1 and non-safety PD signal	short circuit	mechanical damage; fire	does not transmit power from 1EG1 to bus 1EA1	1. periodic testing 2. annunciation in control room	Non: redundant loads supplied from bus 1EA2 by diesel generator 1EG2

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 10 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
16	6.9-kV circuit breaker	diesel generator 1EG2 supply breaker	fails to close; fails to open	control failure; mechanical failure; relay failure; stuck contacts; fire; loss of ventilation	1. Fails to close: 1EG2 cannot supply power to bus 1EA2. 2. Fails to open: possible damage to bus 1EA2 and diesel generator 1EG2	1. periodic testing 2. Annunciation in Control Room	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1
16A	8-kV cable	connects 1EG2 to bus 1EA2 circuit breaker (Item 16)	short circuit; open circuit	mechanical damage; fire	does not transmit power from 1EG2 to bus 1EA2	1. periodic testing 2. Annunciation in Control Room	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1
16B	Diesel generator 1EG2	provides standby power for Train B loads	delivers no power to bus 1EA2	does not start; component failure; fire; loss of heating or ventilation, or both; spurious relay trip; loss of Control Power	does not transmit power from 1EG2 to bus 1EA2	1. relays and annunciation in control room 2. Periodic Testing	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1
16C	15kV jumper cable	connects 1EG2 to PD discharge bus coupler	short circuit	mechanical damage; fire	does not transmit power from 1EG2 to bus 1EA2	1. periodic testing 2. annunciation in control room	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1
16D	6.9 kV bus coupler	provides isolation between 1EG2 and non-safety PD discharge signal	short circuit	mechanical damage; fire	does not transmit power from 1EG2 to bus 1EA2	1. periodic testing 2. annunciation in control room	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 11 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
17	Deleted						
17A	Deleted						
17B	Deleted						
17C	Deleted						
17D	Deleted						
18	6.9-kV circuit breaker BT-1EA1	isolates 6.9-kV bus from 480 V load centers for diesel generator testing, only during plant shutdown	fails to close; fails to open	control failure; mechanical failure; relay failures; stuck contacts; fire; loss of ventilation	1. Fails to close: 1EG1 cannot supply power to bus 1EA1. 2. Fails to open; Diesel generator cannot be sequentially loaded without interrupting power to 480 V loads.	1. protective relays; annunciation in control room 2. Periodic Testing	None: redundant loads supplied from bus 1EA2 by diesel generator 1EG2
19	6.9-kV circuit breaker BT-1EA2	isolates 6.9-kV bus from 480 V load centers for diesel generator testing, only during plant shutdown	fails to close; fails to open	control failure; mechanical failure; relay failure; stuck contacts; fire; loss of ventilation	1. Fails to close: 1EG2 cannot supply power to bus 1EA2. 2. Fails to open: Diesel generator cannot be sequentially loaded without interrupting power to 480 V loads.	1. Protective relays; annunciation in control room 2. Periodic Testing	None: redundant loads supplied from bus 1EA1 by diesel generator 1EG1

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 12 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
20	Unit substation transformer T1EB1	supplies power at 480 V to bus 1EB1	fails to transform power	mechanical damage; short circuit; over-heating; fire; loss of ventilation; loss of Control Power	loss of power on bus 1EB1	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
20A	Transformer T1EB1 throat connection	connects transformer to bus circuit breaker	open circuit; short circuit;	mechanical failure; fire	loss of power on bus 1EB1	1. Protective relays 2. Undervoltage 3. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
21	Unit substation transformer T1EB3	supplies power at 480 V to bus 1EB3	fails to transform power	mechanical damage; short circuit; over-heating; fire; loss of ventilation; loss of Control Power	loss of power on bus 1EB3	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA2
21A	Transformer T1EB3 throat connection	connects transformer to bus circuit breaker	open circuit; short circuit	mechanical failure; fire	loss of power on bus 1EB3	1. Protective relays 2. Undervoltage 3. Annunciation in Control Room	None: load supplied by redundant load group provided from bus 1EA2
22	Unit substation transformer T1EB2	supplies power to 480-V bus 1EB2	fails to transform power	mechanical damage; short circuit; over heating; fire; loss of ventilation; loss of Control Power	loss of power on bus 1EB2	1. protective relays 2. Annunciation in Control Room	None: load supplied by redundant load group provided from bus 1EA1

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 13 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
22A	Transformer T1EB2 throat connection	connects transformer to bus circuit breaker	open circuits; short circuit	mechanical failure; fire	loss of power on bus 1EB2	1. Protective relays 2. Undervoltage 3. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
23	Unit substation transformer T1EB4	supplies power to 480-V bus 1EB4	fails to transform power	mechanical damage; short circuit; overheating; fire; loss of ventilation; loss of Control Power	loss of power on bus 1EB4	1. protective relays 2. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
23A	Transformer T1EB4 throat connection	Connects transformer to bus circuit breaker	open circuit; short circuit	mechanical failure; fire	loss of power on bus 1EB4	1. Protective relays 2. Undervoltage 3. Annunciation in Control Room	None: Load supplied by redundant load group provided from bus 1EA1
24	480-V circuit breaker 1EB1-1	protects bus 1EB1	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens and loss of bus 1EB1 2. Failure to close: bus not energized	annunciation; periodic testing	None: Redundant load supplied by Train B
25	480-V circuit breaker 1EB3-1	protects bus 1EB3	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens and loss of bus 1EB3 2. Failure to close: bus not energized	annunciation; periodic testing	None: Redundant loads supplied by Train B

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 14 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
26	480-V circuit breaker 1EB2-1	protects bus 1EB2	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens and loss of bus 1EB2 2. Failure to close: bus not energized	annunciation; periodic testing	None: Redundant loads supplied by Train A
27	480-V circuit breaker 1EB4-1	protects bus 1EB4	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens and loss of bus 1EB4 2. Failure to close: bus not energized	annunciation; periodic Testing	None: Redundant loads supplied by train A
28	480-V Bus 1EB1	distributes 480-V power to Train A loads	fails to deliver power	mechanical failure; short circuit; overload; fire; loss of ventilation	loss of power to 480-V loads	1. protective relay operation 2. Annunciation	None: Redundant loads supplied by Train B
29	480-V Bus 1EB3	distributes 480-V power to Train A loads	fails to deliver power	mechanical failure; short circuit; overload; fire; loss of ventilation	loss of power to 480-V loads	1. protective relays operation 2. Annunciation	None: Redundant loads supplied by Train B
30	480-V Bus 1EB2	distributes 480-V power to Train B loads	fails to deliver power	mechanical failure; short circuit; overload; fire; loss of ventilation	loss of power to 480-V loads	1. protective relay operations 2. Annunciation	None: Redundant loads supplied by Train A

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 15 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
31	480-V Bus 1EB4	distributes 480-V power to Train A loads	fails to deliver power	mechanical failure; short circuit; overload; fire; ventilation	loss of power to 480-V loads	1. protective relay operation 2. Annunciation	None: Redundant loads supplied by Train A
32	480-V tie breaker BT-1EB13	provides transfer to alternate power supply	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Fails to open: backup breaker opens and lose buses 1EB1 and 1EB3 2. Fails to close: cannot transfer to alternate supply	annunciation; periodic testing	None: Redundant loads supplied by Train B
33	480-V tie breaker BT-1EB24	provides transfer to alternate power supply	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Fails to open: Backup breaker opens and lose buses EB2 and 1EB4 2. Fails to close: cannot transfer to alternate supply	annunciation; periodic testing	None: Redundant loads supplied by Train A
34	480-V circuit breaker	protects unit 1 MCCS supplied by Train A SWGRS.	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens. 2. Failure to close: MCC not energized	annunciation; periodic testing	None: Redundant equipment supplied from other MCCs
35	600-V cable	transmits power to MCCs (Item 34)	open circuit; short circuit	mechanical damage; fire	loss of power to MCC	Annunciation	None: redundant equipment supplied from other MCCs

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 16 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
36	480-V circuit breaker	protects common MCCs supplied by Train A SWGRS	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens. 2. Failure to close: MCC not energized	annunciation; periodic testing	None: Redundant equipment supplied from other MCCs
37	600-V cable	transmits power to MCCs (Item 36)	open circuit; short circuit	mechanical damage; fire	loss of power to MCC	Annunciation	None: Redundant equipment supplied from other MCCs
38	480-V circuit breaker	protects unit 1 MCCs supplied by Train B SWGRS	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts; fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens. 2. Failure to close: MCC not energized	annunciation; periodic testing	None: Redundant equipment supplied from other MCCs
39	600-V cable	transmits power to MCCs (Item 38)	open circuit; short circuit	mechanical damage; fire	loss of power to MCC	Annunciation	None: Redundant equipment supplied from other MCCs
40	480-V circuit breaker	protects common MCCs supplied by Train B switchgears	fails to open; fails to close	mechanical failure; relay failure; control power fail; stuck contacts: fire; loss of ventilation	1. Failure to open could damage bus. Backup breaker opens. 2. Failure to close: MCC not energized	annunciation; periodic testing	None: Redundant equipment supplied from other MCCs
41	600-V cable	transmits power to MCCs (Item 40)	open circuit; short circuit	Mechanical damage; fire	loss of power to MCC	Annunciation in Control Room	None: Redundant equipment supplied from other MCCs
42	480-V automatic transfer SW. for Common MCC's, Train A	automatically transfers power from unit 1 or 2 to common MCCs Train A	no undervoltage transfer	stuck contacts; short circuit.	power not available from one of the units	annunciation; in control room	None: redundant common MCCs supplied from opposite train.

TABLE 8.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY AC POWER SYSTEM

(Sheet 17 of 17)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	Failure Detection	Effects on System
43	480-V automatic transfer SW. for Common MCC's Train B	automatically transfers power from unit 1 or 2 to common MCCs Train B	no undervoltage transfer	stuck contacts; short circuit.	power not available from one of the units	annunciation; in control room	None: redundant common MCCs supplied from opposite train.
44	Unit 1 Train A MCCs supplied by Train A switchgears	distributes 480-V power	fails to deliver power	mechanical failure; short circuit; over-load; fire; loss of ventilation	loss of MCC	Annunciation in control room	None: Item redundant equipment on unit 1 MCCs supplied by Train B switchgears
45	Common Train A MCCs supplied by Train A unit 1 or 2 switchgears	distributes 480-V power	fails to deliver power	mechanical failure; short circuit; over-load; fire; loss of ventilation	loss of MCC	Annunciation in control room	None: redundant equipment on common MCCs supplied by Train B switchgears
46	Unit 1 Train B MCCs supplied by Train B switchgears	distributes 480-V power	fails to deliver power	mechanical failure; short circuit; over-load; fire; loss of ventilation	loss of MCC	Annunciation in control room	None: redundant equipment on unit 1 MCCs supplied by Train A switchgears.
47	Common Train B MCCs supplied by Train B unit 1 or 2 switchgears	distributes 480-V power	fails to deliver power	mechanical failure; short circuit; over-load; fire; loss of ventilation	loss of MCC	Annunciation in Control Room	None: redundant equipment on common MCCs supplied by Train A switchgears.

Notes:

1. The items listed in this Table are shown on the plant one-line diagram, Figure 8.3-1.

2. The analysis listed is for Unit 1; Unit 2 is similar, except for equipment identification numbers.

3. Analysis of the dc system is given in Table 8.3-7.

4. Flooding considerations for the previously mentioned equipment are discussed in Section 8.3.1.1.14.

5. The cables from XST1 Y-winding to tap box #8 and manual transfer switches CPX-EPTSST-04Y are routed through tap box #3 and tap box #6.

6. The removable link boxes CPX-EPTSST-02X and CPX-EPTSST-03X and Eaton bus bar connection of these boxes constitute the manual transfer switch for the X-winding of XST2 and XST2A.

7. The removable link boxes CPX-EPTSST-02Y and CPX-EPTSST-03Y and Eaton bus bar connection of these boxes constitute the manual transfer switch for the Y-winding of XST2 and XST2A.

8. The cables from the Y-winding manual transfer switch to Unit 1 safety buses are routed through tap box #2 and tap box #5.

9. The cables from the X-winding manual transfer switch to Unit 2 safety buses are routed through tap box #1 and tap box #7.

Abbreviations:

1. POPS - Preferred Offsite Power Source

2. AOPS - Alternate Offsite Power Source

3. PD - Partial Discharge

TABLE 8.3-4
125-Vdc CLASS 1E BATTERY LOAD REQUIREMENTS

Battery	Amperes Per Time Interval After Loss of AC Power		
	0 to 1 Min.	1 to 239 Min.	239 to 240 Min.
BT1ED1, (BT1ED2)	482.28 (449.28)	210 (213)	310 (313)
BT2ED1, (BT2ED2)	481.12 (458.12)	223 (224)	323 (324)
		0 to 240 Minutes	
BT1ED3, (BT1ED4)		144 (156)	
BT2ED3, (BT2ED4)		144 (156)	

Notes:

1. During the battery duty cycle the batteries maintain a voltage of 105V or greater that will provide adequate voltage for operation of all required loads.

TABLE 8.3-4A
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TABLE 8.3-4B
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TABLE 8.3-4C
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TABLE 8.3-5
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TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 1 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
1	Batteries BT1ED1 and BT1ED3	supplies Train A dc power	fails to provide power	short circuit; cell or component failure	loss of power supply to Train A buses	annunciation; periodic testing	None: Power supplied by battery chargers.
1A	Batteries BT1ED2 and BT1ED4	supplies Train B dc power	fails to provide power	short circuit; cell or component failure	loss of power supply to Train B buses	annunciation; periodic testing	None: Power supplied by battery chargers
2	Battery chargers Train A	supplies rectified dc power from 480-Vac system Train A	fails to provide power	component failure; short circuit; fire	momentary loss of charging to Train A batteries	annunciation; periodic testing	None: Power supplied by station battery or redundant charger.
2A	Battery chargers Train B	supplies rectified dc power from480-Vac system Train B	fails to provide power	component failure; short circuit; fire	momentary loss of charging to Train B batteries	annunciation; periodic testing	None: Power supplied by station battery or redundant charger.
3	600-V cable	transmits dc power to Train A buses	open circuit; short circuit	mechanical damage; fire	loss of charging to Train A Batteries	annunciation; periodic testing	None: Power supplied by station battery.
3A	600-V cable	transmits dc power to Train B buses	open circuit; short circuit	mechanical damage; fire	loss of charging to Train B Batteries	annunciation; periodic testing	None: Power supplied by station battery.
4	Fusible Switch (Station Battery)	protects Train A buses and batteries	Blows at less than rated current	material defect	battery cannot supply power to bus	annunciation in control room	None: Charger will feed Train A bus.
4A	Fusible Switch (Station Battery)	protects Train B buses and batteries	Blows at less than rated current	material defect	battery cannot supply power to bus	annunciation in control room	None: Charger will feed Train B bus.

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 2 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
5	Circuit Breaker (Battery Charger)	protects Train A bus and charger	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open: backup breaker opens and could damage charger; fails to close: power not available from battery charger	annunciation in control room	None: Redundant charger or station battery will be connected to Train A bus
5A	Circuit Breaker (Battery Charger)	protects Train B bus and charger	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open: backup breaker opens and could damage charger; fails to close: power not available from battery charger	annunciation in control room	None: Redundant charger or station battery will be connected to Train B bus
6	Bus 1ED1 and 1ED3	distributes Train A dc power	fails to deliver power	short circuit or overload; fire	loss of Train A load group	annunciation in control room	None: redundant loads are supplied by Train B buses
6A	Bus 1ED2 and 1ED4	distributes Train B dc power	fails to deliver power	short circuit or overload; fire	loss of Train B load group	annunciation in control room	None: redundant loads are supplied by Train A buses
7	Circuit breaker	protects Train A inverter	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open; could damage inverter, backup breaker opens and loss of DC Bus 1ED1; fails to close: loss of dc power to inverter	annunciation in control room	None: Redundant loads are supplied by Train B inverter

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 3 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
7A	Circuit breaker	protects Train B inverter	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open; could damage inverter, backup breaker opens and loss of DC Bus 1ED2; fails to close: loss of dc power to inverter	annunciation in control room	None: Redundant loads are supplied by Train A inverter
8	Circuit breaker	protects channel inverter supplied from Train A (IV1PC1 & IV1PC3)	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open; could damage inverter, backup breaker opens and loss of DC Bus 1ED1; fails to close: loss of dc power to inverter	annunciation in control room	None: Redundant loads are supplied by inverters powered from Train B (IV1PC2 & IV1PC4)
8A	Circuit breaker	protects channel inverter supplied from Train B (IV1PC2 & IV1PC4)	fails to open; fails to close	mechanical failure; stuck contacts; fire	fails to open; could damage inverter, backup breaker opens and loss of DC Bus 1ED2; fails to close: loss of dc power to inverter	annunciation in control room	None: Redundant loads are supplied by inverters powered from Train A (IV1PC1 & IV1PC3)
9	Fusible switch	protects Train A distribution panel	blows at less than rated current	material defect	loss of power to panel	annunciation in control room	None: redundant safety-related equipment supplied by redundant panel of Train B
9A	Fusible switch	protects Train B distribution panel	blows at less than rated current	material defect	loss of power to panel	annunciation in control room	None: redundant safety-related equipment supplied by redundant panel of Train A

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 4 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
10	600-V cable	transmits dc power to Train A inverter	open circuit short circuit	mechanical damage; fire	loss of dc power to inverter	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train B
10A	600-V cable	transmits dc power to Train B inverter	open circuit short circuit	mechanical damage; fire	loss of dc power to inverter	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train A
11	600-V cable	transmits dc power to channel inverter from Train A	open circuit short circuit	mechanical damage; fire	loss of dc power to inverter	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train B
11A	600-V cable	transmits dc power to channel inverter from Train B	open circuit short circuit	mechanical damage; fire	loss of dc power to inverter	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train A
12	600-V cable	transmits dc power to Train A distribution panel	open circuit short circuit	mechanical damage; fire	loss of dc power to distribution panel	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train B
12A	600-V cable	transmits dc power to Train B distribution panel	open circuit short circuit	mechanical damage; fire	loss of dc power to distribution panel	annunciation; periodic testing	None: redundant safety-related equipment supplied by redundant panel of Train A
13	Inverter Train A	supplies 118-Vac, single phase power	loss of ac output from inverter	Component failure, fire	loss of power to instrument bus from inverter	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train B instrument bus 2. Alternate 120-V, single-phase supply is available

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 5 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
13A	Inverter Train B	supplies 118-Vac, single phase power	loss of ac output from inverter	component failure, fire	loss of power to instrument bus from inverter	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train A instrument bus 2. Alternate 120-V, single-phase supply is available
14	Inverter IV1PC1 and IV1PC3	supplies 118-Vac, single phase power	loss of ac output from inverter	component failure, fire	loss of power to instrument bus from inverter	annunciation; periodic testing	None: 1. Redundant equipment supplied on instrument buses 1PC2 and 1PC4 2. Alternate 120-V, single phase supply is available
14A	Inverter IV1PC2 and IV1PC4	supplies 118-Vac, single phase power	loss of ac output from inverter	component failure, fire	loss of power to instrument bus from inverter	annunciation periodic testing;	None: 1. Redundant equipment supplied on instrument buses 1PC1 and 1PC3 2. Alternate 120-V, single phase supply is available
15	DELETED						
15A	DELETED						
16	DELETED						
16A	DELETED						

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 6 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
17	600-V cable	supplies 120-V, single phase power to Train A bus (alternate supply)	open circuit; short circuit	mechanical damage; fire	alternate supply does not furnish power to bus	annunciation; periodic testing	None: inverter supply is available
17A	600-V cable	supplies 120-V, single phase power to Train B bus (alternate supply)	open circuit; short circuit	mechanical damage; fire	alternate supply does not furnish power to bus	annunciation; periodic testing	None: inverter supply is available
18	600-V cable	transmits inverter power to Train A instrument bus	open circuit; short circuit	mechanical damage; fire	momentary loss of power to bus; manual transfer to alternate supply	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train B instrument buses 2. Alternate 120-V, single phase supply is available
18A	600-V cable	transmits inverter power to Train B instrument bus	open circuit; short circuit	mechanical damage; fire	momentary loss of power to bus; manual transfer to alternate supply	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train A instrument buses 2. Alternate 120-V, single phase supply is available
19	600-V cable	transmits inverter power to buses 1PC1 and 1PC3	open circuit; short circuit	mechanical damage; fire	momentary loss of power to bus; manual transfer to alternate supply	annunciation; periodic testing	None: 1. Redundant equipment supplied on instrument buses 1PC2 and 1PC4 2. Alternate 120-V, single phase supply is available

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 7 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
19A	600-V cable	transmits inverter power to buses 1PC2 and 1PC4	open circuit; short circuit	mechanical damage; fire	momentary loss of power to bus; manual transfer to alternate supply	annunciation; periodic testing	None: 1. Redundant equipment supplied on instrument buses 1PC1 and 1PC3 2. Alternate 120-V, single phase supply is available
20	Circuit breaker	supplies power from inverter to Train A BOP instrument bus	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by 120-V alternate source	annunciation; periodic testing	None: redundant equipment to Train B buses; inverter supply available
20A	Circuit breaker	supplies power from inverter to Train B BOP instrument bus	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by 120-V alternate source	annunciation; periodic testing	None: redundant equipment to Train A buses; inverter supply available
21	Circuit breaker	supplies power from inverter to Train A BOP instrument bus	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by Train A inverter	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train B instrument buses 2. Alternate 120-V, single phase supply is available

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 8 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
21A	Circuit breaker	supplies power from inverter to Train B BOP instrument bus	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by Train B inverter	annunciation; periodic testing	None: 1. Redundant equipment supplied on Train A instrument buses 2. Alternate 120-V, single phase supply is available
22	Circuit breaker	protects instrument buses 1PC1 and 1PC3	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by 120-V alternate source	annunciation; periodic testing	None: redundant equipment on buses 1PC2 and 1PC4; inverter supply available
22A	Circuit breaker	protects instrument buses 1PC2 and 1PC4	fails to open fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by 120-V alternate source	annunciation; periodic testing	None: redundant equipment on buses 1PC1 and 1PC3; inverter supply available
23	Circuit breaker	protects instrument buses 1PC1 and 1PC3	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by IV1PC1 and IV1PC3	annunciation; periodic testing	None: 1. Redundant equipment supplied on instrument buses inverters 1PC2 and 1PC4 2. Alternate 120-V, single phase supply

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 9 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
23A	Circuit breaker	protects instrument buses 1PC2 and 1PC4	fails to open; fails to close	mechanical failure; stuck contacts, fire	fails to open: could damage bus; fails to close: power cannot be supplied by inverters IV1PC2 and IV1PC4	annunciation; periodic testing	None: 1. Redundant equipment supplied on instrument bus 1PC1 and 1PC3 2. Alternate 120-V, single phase supply is available
24	BOP instrument bus Train A	supplies ac power to BOP instrumentation	fails to deliver power	short circuit or overload; fire	loss of power to instrument loads	annunciation; periodic testing	None: redundant BOP loads provided on Train B buses
24A	BOP instrument bus Train B	supplies ac power to BOP instrumentation	fails to deliver power	short circuit or overload; fire	loss of power to instrument loads	annunciation; periodic testing	None: redundant BOP loads provided on Train A buses
25	Instrumentation buses 1PC1 and 1PC3	distributes instrumentation power to systems	fails to deliver power	short circuit or overload; fire	loss of one load group of instrumentation	annunciation; periodic testing	None: redundant loads provided by buses 1PC2 and 1PC4
25A	Instrumentation buses 1PC2 and 1PC4	distributes instrumentation power to systems	fails to deliver power	short circuit or overload; fire	loss of one load group of instrumentation	annunciation; periodic testing	None: redundant loads provided by buses 1PC1 and 1PC3
26	Train A distribution panel	distributes power to auxiliaries	fails to deliver power	short circuit or overload; fire	loss of panel	annunciation; periodic testing	None: redundant loads provided by Train B buses
26A	Train B distribution panel	distributes power to auxiliaries	fails to deliver power	short circuit or overload; fire	loss of panel	annunciation; periodic testing;	None: redundant loads provided by Train A buses

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 10 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
27	120-V bypass transformer Train A	supplies 120-V, single phase power to instrument buses (alternate supply) and BOP inverter's static switch	open circuit; short circuit	mechanical damage; short circuit; fire	alternate supply does not furnish power to bus	annunciation; periodic testing	None: inverter supplies are available
27A	120-V, bypass transformer Train B	supplies 120-V, single phase power to instrument buses (alternate supply) and BOP inverter's static switch	open circuit; short circuit	mechanical damage; short circuit; fire	alternate supply does not furnish power to bus	annunciation; periodic testing	None: inverter supplies are available
28	600-V cable	supplies Unit 1 dc power to Train A common distribution panel	open circuit; short circuit	mechanical damage; fire	loss of Unit 1 power to panel	annunciation; periodic testing	None: 1. Unit 2 power is available 2. Redundant loads are supplied by Train B common panel
28A	600-V cable	supplies Unit 1 dc power to Train B common distribution panel	open circuit; short circuit	mechanical damage; fire	loss of Unit 1 power to panel	annunciation; periodic testing	None: 1. Unit 2 power is available 2. Redundant loads are supplied by Train A common panel

TABLE 8.3-7
FAILURE MODE AND EFFECT ANALYSIS FOR AUXILIARY DC POWER SYSTEM

(Sheet 11 of 11)

Item	Description	Function	Failure Mode	Causes of Failure	Effects of Failure	How Failure Is Detected	Effects on System
29	600-V cable	supplies Unit 2 dc power to Train A common distribution panel	open circuit; short circuit	mechanical damage; fire	loss of Unit 2 power to panel	annunciation; periodic testing	None: 1. Unit 1 power is available 2. Redundant loads are supplied by Train B common panel
29A	600-V cable	supplies Unit 2 dc power to Train B common distribution panel	open circuit; short circuit	mechanical damage; fire	loss of Unit 2 power to panel	annunciation; periodic testing	None: 1. Unit 1 power is available 2. Redundant loads are supplied by Train A common panel
30	Automatic transfer switch	automatically transfers power from Unit 1 or 2 to common Train A panel	no undervoltage transfer	stuck contacts; short circuit, fire	power not available from one of the units	annunciation; periodic testing	None: redundant loads supplied by Train B common panel
30A	Automatic transfer switch	automatically transfer power from Unit 1 or 2 to common Train B panel	no undervoltage transfer	stuck contacts; short circuit, fire	power not available from one of the units	annunciation; periodic testing	None: redundant loads supplied by Train A common panel

Notes:

1. The items listed in this table are shown on the one-line diagram Figure 8.3-13.

2. Analysis listed is for Unit 1: Unit 2 similar except for equipment identification numbers

TABLE 8.3-8
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TABLE 8.3-9
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TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 1 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
1-NY-0050A-1	NIS NEUTRON FLUX PREAMPLIFIER ASSEMBLY (1E)	ORANGE, BLUE	1
1-NY-0050A-2	NIS NEUTRON FLUX SIGNAL PROCESSOR (1E)	ORANGE, BLUE, BLACK	1
1-NY-0050A-3	NIS NEUTRON FLUX ISOLATOR EXPANSION ASSEMBLY (1E)	ORANGE, BLUE, BLACK	1
1-NY-0050B-1	NIS NEUTRON FLUX PREAMPLIFIER ASSEMBLY (1E)	GREEN, YELLOW	1
1-NY-0050B-2	NIS NEUTRON FLUX SIGNAL PROCESSOR (1E)	GREEN, YELLOW, BLACK	1
1-NY-0050B-3	NIS NEUTRON FLUX ISOLATOR EXPANSION ASSEMBLY (1E)	GREEN, YELLOW, BLACK	1
1-SSII-1	SAFETY SYSTEM INOPERABLE INDICATION PANEL 1	ASSOCIATED ORANGE TO BLACK	5
1-SSII-2	SAFETY SYSTEM INOPERABLE INDICATION PANEL 2	ASSOCIATED GREEN TO BLACK	5
2-NY-0050A-1	NIS NEUTRON FLUX PREAMPLIFIER ASSEMBLY (1E)	ORANGE, BLUE	1
2-NY-0050A-2	NIS NEUTRON FLUX SIGNAL PROCESSOR (1E)	ORANGE, BLUE, BLACK	1
2-NY-0050A-3	NIS NEUTRON FLUX ISOLATOR EXPANSION ASSEMBLY (1E)	ORANGE, BLUE, BLACK	1
2-NY-0050B-1	NIS NEUTRON FLUX PREAMPLIFIER ASSEMBLY (1E)	GREEN, YELLOW	1
2-NY-0050B-2	NIS NEUTRON FLUX SIGNAL PROCESSOR (1E)	GREEN, YELLOW, BLACK	1
2-NY-0050B-3	NIS NEUTRON FLUX ISOLATOR EXPANSION ASSEMBLY (1E)	GREEN, YELLOW, BLACK	1
2-SSII-1	SAFETY SYSTEM INOPERABLE INDICATION PANEL 1	ASSOCIATED ORANGE TO BLACK	5
2-SSII-2	SAFETY SYSTEM INOPERABLE INDICATION PANEL 2	ASSOCIATED GREEN TO BLACK	5
CP1-ECDPPC-01 (1PC1)	INSTRUMENT POWER PANEL	ORANGE, RED, BLACK	8
CP1-ECDPPC-02 (1PC2)	INSTRUMENT POWER PANEL	GREEN, WHITE, BLACK	8
CP1-ECDPPC-03 (1PC3)	INSTRUMENT POWER PANEL	ORANGE, BLUE, BLACK	8
CP1-ECDPPC-04 (1PC4)	INSTRUMENT POWER PANEL	GREEN, YELLOW, BLACK	8
CP1-EIPRC1-01 (1-CI-01)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, BLUE	6
CP1-IEPRCI-02 (1-CI-02)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, BLACK	6
CP1-ELDPEC-01(ESB1)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 2 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CP1-ELDPEC-02(ESB2)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP1-EPDSNA-01	ALTERNATE POWER TRANSFER SWITCH	ASSOCIATED ORANGE TO ASSOCIATED GREEN ASSOCIATED ORANGE OR GREEN TO BLACK	13
CP1-EPTRNT-12 (T1C3)	BYPASS TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	2
CP1-EC1VEC-05	SPARE INVERTER (1E)	ORANGE AND RED OR ORANGE AND BLUE	16
CP1-EC1VEC-06	SPARE INVERTER (1E)	GREEN AND WHITE OR GREEN AND YELLOW	16
CP1-CICACO-01A	INSTRUMENT AIR COMPRESSOR TERMINATION CABINET	ASSOCIATED ORANGE TO BLACK	2
CP1-ELDPEC-03(ESB3)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP1-ELDPEC-04(ESB4)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP1-ELDPEC-05(ESB5)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP1-ELDPEC-06(ESB6)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP1-ELDPEC-07(ESB7)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CP1-ELDPEC-08(ESB8)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CP1-ELDPEC-09(ESB9)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP1-ELDPEC-10(ESB10)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP1-ELDPEC-11(ESB11)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP1-ELDPEC-12(ESB12)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP1-ELDPEC-13(ESB13)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CP1-ELDPEC-14(ESB14)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CP1-ELTRNT-18(XF-SC1-3)	LIGHTING TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	3
CP1-ELTRNT-19(XF-SC2-4)	LIGHTING TRANSFORMER (NON 1E)	ASSOCIATED GREEN TO BLACK	3
CP1-EPBCND-02 (BC1D2)	BATTERY CHARGER (NON 1E)	ASSOCIATED GREEN TO BLACK	2
CP1-EPBCND-04 (BC1D4)	BATTERY CHARGER (NON 1E)	ASSOCIATED GREEN TO BLACK	2

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 3 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CP1-EPTRNT-28 (T1EC3-3)	ISOLATION TRANSFORMER (NON 1E) ASSOCIATED	ASSOCIATED ORANGE TO BLACK	3
CP1-ECTRET-03,04,05 & 06	PT CABINET (1E)	--	12
CP2-ECTRET-03,04,05 & 06	PT CABINET (1E)	--	12
CP2-ECDPPC-01(2PC1)	INSTRUMENT POWER PANEL	ORANGE, RED, BLACK	8
CP2-CICACO-01B	INSTRUMENT AIR COMPRESSOR CONTROL PANEL	ASSOCIATED ORANGE TO BLACK	2
CP2-ECDPPC-02(2PC2)	INSTRUMENT POWER PANEL	GREEN, WHITE, BLACK	8
CP2-ECDPPC-03(2PC3)	INSTRUMENT POWER PANEL	ORANGE, BLUE, BLACK	8
CP2-ECDPPC-04(2PC4)	INSTRUMENT POWER PANEL	GREEN, YELLOW, BLACK	8
CP2-EIPRCI-01 (2-CI-01)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, BLACK	6
CP2-EIPRCI-02 (2-CI-02)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, BLACK	6
CP2-ELDPEC-01(2ESB1)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CP2-ELDPEC-02(2ESB2)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP2-ELDPEC-03(2ESB3)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP2-EPDSNA-01	ALTERNATE POWER TRANSFER SWITCH	ASSOCIATED ORANGE TO ASSOCIATED GREEN ASSOCIATED ORANGE OR GREEN TO BLACK	13
CP2-EIPRLV-48	TURBINE DRIVEN AFW PUMP INSTRUMENT RACK	ASSOCIATED ORANGE TO BLACK	5
CP2-ELDPEC-04(2ESB4)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP2-ELDPEC-05(2ESB5)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP2-ELDPEC-06(2ESB6)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP2-ELDPEC-07(2ESB7)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CP2-ELDPEC-08(2ESB8)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CP2-ELDPEC-09(2ESB9)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP2-ELDPEC-10(2ESB10)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 4 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CP2-ELDPEC-11(2ESBD1)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CP2-ELDPEC-12(2ESBD2)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CP2-ELDPEC-13(2ECB3)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CP2-ELDPEC-14(2ECB4)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CP2-ELTRNT-18 (XF-2SC1-3)	LIGHTING TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	3
CP2-ELTRNT-19 (XF-2SC2-4)	LIGHTING TRANSFORMER (NON 1E)	ASSOCIATED GREEN TO BLACK	3
CP2-EPBCND-02 (BC2D2)	BATTERY CHARGER (NON 1E)	ASSOCIATED GREEN TO BLACK	2
CP2-EPBCND-04 (BC2D4)	BATTERY CHARGER (NON 1E)	ASSOCIATED GREEN TO BLACK	2
CP2-EPTRNT-28 (T2EC3-3)	ISOLATION TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	3
CPX-EIPRCI-06 (X-CI-06)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, ASSOCIATED ORANGE, BLACK	6
CPX-EIPRCI-07 (X-CI-07)	BOP ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, ASSOCIATED GREEN, BLACK	6
CPX-EIPRLV-29 (X-LV-29)	LOCAL FIRE DET. PANEL (NON 1E)	ASSOCIATED GREEN TO BLACK	4
CPX-EIPRLV-29A (X-LV-29A)	LOCAL FIRE DET. PANEL (NON 1E)	ASSOCIATED ORANGE TO BLACK	4
CPX-EIPRLV-30 (X-LV-30)	LOCAL FIRE DET. PANEL (NON 1E)	ASSOCIATED ORANGE & ASSOCIATED GREEN TO BLACK	4
CPX-ELDPEC-03 (ECB1)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CPX-ELDPEC-04 (ECB2)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CPX-ELDPEC-05 (ECB5)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CPX-ELDPEC-06 (ECB6)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CPX-ELDPEC-07 (EAB1)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-08 (EAB2)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-ELDPEC-09 (EAB3)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-10 (EAB4)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 5 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CPX-ELDPEC-11 (EAB5)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-12 (EAB6)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-ELDPEC-13 (EAB7)	LTG DISTR PNLBD (1E)	ORANGE AND ASSOCIATED ORANGE TO BLACK	3
CPX-ELDPEC-14 (EAB8)	LTG DISTR PNLBD (1E)	GREEN AND ASSOCIATED GREEN TO BLACK	3
CPX-ELDPEC-15 (EAB9)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-16 (EAB10)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-ELDPEC-17 (EAB11)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-18 (EAB12)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-ELDPEC-21 (EABD1)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-22 (EABD2)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-ELDPEC-23 (ECBD3)	LTG DISTR PNLBD (1E)	ORANGE TO BLACK	3
CPX-ELDPEC-24 (ECBD4)	LTG DISTR PNLBD (1E)	GREEN TO BLACK	3
CPX-EPMCEB-05 (XEB3-3)	MOTOR CONTROL CENTER (NON 1E)	ASSOCIATED ORANGE TO BLACK	2
CPX-EPMCEB-06 (XEB4-3)	MOTOR CONTROL CENTER (NON 1E)	ASSOCIATED GREEN TO BLACK	2
CPX-EPTRNT-42 (TXEC1)	REGULATING TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	3
CPX-EPTRNT-43 (TXEC2)	REGULATING TRANSFORMER (NON 1E)	ASSOCIATED GREEN TO BLACK	3
CPX-EPTRNT-44 (TXEC3)	REGULATING TRANSFORMER (NON 1E)	ASSOCIATED ORANGE TO BLACK	3
CPX-EPTRNT-45 (TXEC4)	REGULATING TRANSFORMER (NON 1E)	ASSOCIATED GREEN TO BLACK	3
TBX-ASELIV-01 (IV1C1)	INVERTER (NON 1E)	ASSOCIATED ORANGE TO BLACK	2
TBX-ESELAR-01 (1-AR-01)	NSSS AUXILIARY RELAY CABINETS	ORANGE, GREEN, ASSOC. ORANGE, ASSOC. GREEN, BLACK	9
TBX-ESELIV-01 (IV1PC1)	INVERTER (1E)	ORANGE TO RED	1
TBX-ESELIV-02 (IV1PC2)	INVERTER (1E)	GREEN TO WHITE	1

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 6 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
TBX-ESELIV-03 (IV1PC3)	INVERTER (1E)	ORANGE TO BLUE	1
TBX-ESELIV-04 (IV1PC4)	INVERTER (1E)	GREEN TO YELLOW	1
TBX-ESELSP-01 (1-SP-01)	REACTOR SOLID STATE PROTECTION PANEL	ORANGE, GREEN, ASSOCIATED ORANGE, ASSOCIATED GREEN, RED, WHITE, BLUE, YELLOW, BLACK	6, 7
TBX-ESELTC-01 (1-ETC-01)	REACTOR SAFEGUARDS TEST CABINET	ORANGE, GREEN, ASSOCIATED ORANGE, ASSOCIATED GREEN	6
TBX-ESPDTS-01 (1-TS-01)	REACTOR TRIP SWITCHGEAR CABINETS	ORANGE, GREEN, BLACK	6
TBX-NIELCA-01 (1-LCA-01)	NUCLEAR INSTRUMENTATION RACKS	RED, WHITE, BLUE, YELLOW, BLACK	6
TBX-XIELRK-01 (1-RK-01)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, RED, BLACK	6
TBX-XIELRK-02 (1-RK-02)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, WHITE, BLACK	6
TBX-XIELRK-03 (1-RK-03)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, BLUE, BLACK	6
TBX-XIELRK-04 (1-RK-04)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, YELLOW, BLACK	6
TBX-XIELSS-50IX (1-SS-50IX)	UPGRADE SURVEILLANCE AND PROTECTION PANEL	ORANGE, RED, BLUE, BLACK	6
TBX-XIELSS-50X (1-SS-50X)	UPGRADE SURVEILLANCE AND PROTECTION PANEL	GREEN, WHITE, YELLOW, BLACK	6
TCX-ESELAR-01 (2-AR-01)	NSSS AUXILIARY RELAY CABINETS	ORANGE, GREEN, ASSOC. ORANGE, ASSOC. GREEN, BLACK	9
TCX-ESELIV-01 (IV2PC1)	INVERTER (1E)	ORANGE TO RED	1
TCX-ESELIV-02 (IV2PC2)	INVERTER (1E)	GREEN TO WHITE	1
TCX-ESELIV-03 (IV2PC3)	INVERTER (1E)	ORANGE TO BLUE	1
TCX-ESELIV-04 (IV2PC4)	INVERTER (1E)	GREEN TO YELLOW	1
TCX-ESELSP-01 (2-SP-01)	REACTOR SOLID STATE PROTECTION PANEL	ORANGE, GREEN, ASSOCIATED ORANGE, ASSOCIATED GREEN, RED, WHITE, BLUE, YELLOW, BLACK	6, 7
TCX-ESELTC-01 (2-ETC-01)	REACTOR SAFEGUARDS TEST CABINET	ORANGE, GREEN, ASSOCIATED ORANGE, ASSOCIATED GREEN	6
TCX-ESPDTS-01 (2-TS-01)	REACTOR TRIP SWITCHGEAR CABINETS	ORANGE, GREEN, BLACK	6

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 7 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
TCX-NIELCA-01 (2-LCA-01)	NUCLEAR INSTRUMENTATION RACKS	RED, WHITE, BLUE, YELLOW, BLACK	6
TCX-XIELRK-01 (2-RK-01)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, RED, BLACK	6
TCX-XIELRK-02 (2-RK-02)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, WHITE, BLACK	6
TCX-XIELRK-03 (2-RK-03)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	ORANGE, BLUE, BLACK	6
TCX-XIELRK-04 (2-RK-04)	NSSS ANALOG PROCESS INSTRUMENTATION RACKS	GREEN, YELLOW, BLACK	6
TCX-XIELSS-50IX (2-SS-50IX)	UPGRADE SURVEILLANCE AND PROTECTION PANEL	ORANGE, RED, BLUE, BLACK	6
TCX-XIELSS-50X (2-SS-50X)	UPGRADE SURVEILLANCE AND PROTECTION PANEL	GREEN, WHITE, YELLOW, BLACK	6
CPX-EPTSEB-01	MANUAL TRANSFER SWITCH	GREEN TO BLACK	14
CPX-EPTSEB-02	MANUAL TRANSFER SWITCH	ASSOCIATED GREEN TO BLACK	11
CPX-EPTSEB-03	MANUAL TRANSFER SWITCH	ASSOCIATED ORANGE TO BLACK	11
CPX-EPTSEB-04	MANUAL TRANSFER SWTICH	ORANGE TO BLACK	11
CP1-EPTSEB-01	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP1-EPTSEB-03	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP1-EPTSEB-06	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPTSEB-07	MANUAL TRANSFER SWITCH	ASSOCIATED GREEN TO BLACK	11
CP1-EPTSEB-08	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPTSEB-09	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPTSEB-11	MANUAL TRANSFER SWITCH	ASSOCIATED GREEN TO BLACK	11
CP1-EPTSEB-12	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPTSEB-14	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPTSEB-16	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP1-EPTSEB-17	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP1-EPTSEB-18	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 8 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CP2-EPTSEB-01	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP2-EPTSEB-02	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP2-EPTSEB-03	MANUAL TRANSFER SWITCH	ASSOCIATED GREEN TO BLACK	11
CP2-EPTSEB-04	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP2-EPTSEB-06	MANUAL TRANSFER SWTICH	ORANGE TO BLACK	11
CP2-EPTSEB-07	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP2-EPTSEB-08	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP2-EPTSEB-09	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP2-EPTSEB-10	MANUAL TRANSFER SWITCH	ORANGE TO BLACK	11
CP2-EPTSEB-11	MANUAL TRANSFER SWITCH	ASSOCIATED ORANGE TO BLACK	11
CP2-EPTSEB-12	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP2-EPTSEB-13	MANUAL TRANSFER SWITCH	GREEN TO BLACK	11
CP1-EPSWEA-01	6.9kV SAFEGUARD BUS 1EA1	ORANGE, BLACK	10
CP1-EPSWEA-02	6.9kV SAFEGUARD BUS 1EA2	GREEN, BLACK	10
CP2-EPSWEA-01	6.9kV SAFEGUARD BUS 2EA1	ORANGE, BLACK	10
CP2-EPSWEA-02	6.9kV SAFEGUARD BUS 2EA2	GREEN, BLACK	10
X-LY-4849A-1	ELECTRONICS BOX FOR LEVEL SWITCH X-LS-4849A-1	ORANGE, BLACK	14
X-LY-4849A-2	ELECTRONICS BOX FOR LEVEL SWITCH X-LS-4849A-2	ORANGE, BLACK	14
X-LY-4849B-1	ELECTRONICS BOX FOR LEVEL SWITCH X-LS-4849B-1	GREEN, BLACK	14
X-LY-4849B-2	ELECTRONICS BOX FOR LEVEL SWITCH X-LS-4849B-2	GREEN, BLACK	14
1-TE-3419-2A	1EG1 GEN. STATOR RTD	-	15
1-TE-3419-2B	1EG1 GEN. STATOR RTD	-	15
1-TE-3419-2C	1EG1 GEN. STATOR RTD	-	15

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 9 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
1-TE-3419-2D	1EG1 GEN. STATOR RTD	-	15
1-TE-3419-2E	1EG1 GEN. STATOR RTD	-	15
1-TE-3419-2F	1EG1 GEN. STATOR RTD	-	15
1-TE-3420-2A	1EG2 GEN. STATOR RTD	-	15
1-TE-3420-2B	1EG2 GEN. STATOR RTD	-	15
1-TE-3420-2C	1EG2 GEN. STATOR RTD	-	15
1-TE-3420-2D	1EG2 GEN. STATOR RTD	-	15
1-TE-3420-2E	1EG2 GEN. STATOR RTD	-	15
1-TE-3420-2F	1EG2 GEN. STATOR RTD	-	15
CP1-BSCPEB-01	PERSONNEL AIRLOCK HVD FDR CONT. PNL	GREEN, BLACK	17
CP2-BSCPEB-01	PERSONNEL AIRLOCK HVD FDR CONT. PNL	GREEN, BLACK	17
2-TE-3419-2A	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3419-2B	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3419-2C	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3419-2D	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3419-2E	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3419-2F	2EG1 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2A	2EG2 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2B	2EG2 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2C	2EG2 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2D	2EG2 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2E	2EG2 GEN. STATOR RTD	GREEN, BLACK	15
2-TE-3420-2F	2EG2 GEN. STATOR RTD	GREEN, BLACK	15

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 10 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES
CP1-MEDGEE-01A	DIESEL GENERATOR 1-01 DIESEL ENGINE CONTROL PANEL		18
CP1-MEDGEE-02A	DIESEL GENERATOR 1-02 DIESEL ENGINE CONTROL PANEL		18
CP2-MEDGEE-01A	DIESEL GENERATOR 2-01 DIESEL ENGINE CONTROL PANEL		18
CP2-MEDGEE-02A	DIESEL GENERATOR 2-02 DIESEL ENGINE CONTROL PANEL		18

1. Separation is not required since they are integrally associated with each other.
2. Separation is not required since the associated circuit is integrally related to the non-Class 1E circuit and isolated by a Class 1E breaker tripped by an “S” signal.
3. Separation is not required since the Class 1E or the associated circuit is integrally related to the non-Class 1E circuit and isolated by two Class 1E breakers connected in series and coordinated with an upstream supply breaker (these three breakers are upstream of the portion of the circuit which does not require separation).
4. Separation between the identified associated circuits and black circuits is not required as these circuits carry low energy control signals.
5. Separation between the identified associated circuits and black circuits is not required as these circuits carry low energy instrumentation signals.
6. Equipment supplied by Westinghouse has been tested and is exempt from internal separation requirements (Reference FSAR Section 7.1.2.2).
7. Exemption from internal separation requirements does not apply to the field cables entering the input section of this cabinet.
8. Wiring of different trains/channels will be in different wire bundles, and will be separated to the maximum extent practicable (Reference FSAR Para. 8.3.1.2.1.7.C and SSER #1 Section 8.4.4).
9. Equipment supplied by Westinghouse has been analyzed and is exempt from internal separation requirements (Reference FSAR Section 7.1.2.2).
10. Equipment supplied by ITE has been analyzed and is exempt from internal separation requirements (Ref. FSAR Section 8.3.1.4.5).
11. Separation is not required since the manual transfer switches can only be aligned to Plant Support Power during Plant Modes 5 and 6 under procedural control. These procedures prevent 1E and non-class 1E power from being present at the transfer switch simultaneously.
12. Under all plant conditions connecting of non-class 1E cables from RCP circuits to class 1E PT input terminals does not degrade the PT ability to perform its function. The cables are fire retardant and routed in dedicated conduits which will preclude external non-class 1E cables-related fire damage to the PT cabinet. Hence in view of this, the PT cabinets are exempt from physical separation and also isolation is not required between non-class 1E cables and PT input fuse terminals.
13. Separation is not required because these associated circuits cannot be energized in modes 1 through 4 and never (in any mode) be energized at the same time.
14. Equipment supplied by FCI has been analyzed and is exempt from internal separation requirements (Reference FSAR Section 8.3)
15. Analysis indicates that connection of non-class 1E generator stator RTDs to 1E cables does not degrade the 1E circuit.

TABLE 8.3-10
ELECTRICAL EQUIPMENT NOT REQUIRING INTERNAL CABLE SEPARATION

(Sheet 11 of 11)

EQUIPMENT TAG NO.	DESCRIPTION	DIFFERENT COLOR-CODED CABLE WITHIN	NOTES

16. The separation between the output circuits of the spare inverters is accomplished by the procedural controls and interlocks. The procedural controls and the interlocks. The procedural controls and the interlocks ensure that only one output circuit cable from the spare inverter may be energized at a time. The remaining three output cables remain deenergized and disconnected at both ends thus eliminating the need for separation. Unit 2 spare inverters are not listed here as their input and output cables have the same classifications.

17. This equipment has been analyzed and is exempt from internal minimum separation distance between Class 1E and non-Class 1E cables. Component trips power to the personnel airlock hydraulic unit on SI signal. Power and control cabling are trained and secured to maximize spatial separation.

18. Equipment supplied by Delaval Engine & Compressor has been analyzed and is exempt from internal separation requirement. (Reference FSAR Section 8.3.1.4.5.)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 1 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CPX-CHCICE-03	HVAC CENTRIFUGAL WATER CHILLER #13	CP2-EPSWEA-01	14	(1)
CPX-CHCICE-01	HVAC CENTRIFUGAL WATER CHILLER #11	CP1-EPSWEA-01	4	(1)
CPX-CHCICE-04	HVAC CENTRIFUGAL WATER CHILLER #14	CP2-EPSWEA-02	11	(1)
CPX-CHCICE-02	HVAC CENTRIFUGAL WATER CHILLER #12	CP1-EPSWEA-02	7	(1)
CPX-EPMCNB-01	AUX BLDG NON SAFEGUARD MCC XEB1-3	CP1-EPSWEB-01	5B	(2)
CP1-EPMCNB-03	AUX, SG & FUEL BLDGS NON-SAFEGUARDS MCC 1EB1-3	CP1-EPSWEB-01	5D	(2)
CP1-EPMCNB-01	CONTAINMENT NON-SAFEGUARDS MCC 1EB1-2	CP1-EPSWEB-01	5C	(2)
CP1-VAFNAV-01	CONTAINMENT RECIRC FAN	CP1-EPSWEB-01	3B	(1)
TBX-CSAPPD-01	POSITIVE DISPLACEMENT CHARGING PUMP	CP1-EPSWEB-01	2B	(1)
CP1-EPTRNT-07	ISOL. TRANSFORMER FOR PRESSURIZER HTR CTRL GROUP-C	CP1-EPSWEB-01	6D	(1)
CP1-EPTRNT-05	ISOL. TRANSFORMER FOR PRESSURIZER HTR BK-UP GRP-A	CP1-EPSWEB-03	11B	(1)
CP1-VAFNAV-03	CONTAINMENT RECIRC FAN	CP1-EPSWEB-03	9B	(1)
CP1-VAFNCB-01	CRDM VENT FAN	CP1-EPSWEB-03	8B	(1)
CPX-EPMCNB-03	FUEL BLDG NON-SAFEGUARD MCC XEB3-1	CP1-EPSWEB-03	11C	(2)
CPX-FPAPFP-01	FIRE PUMP (FIRE BRIGADE TRAINING/EMERG FILL PUMP)	CP1-EPSWEB-03	9C	(7)
CP1-EPMCNB-04	AUX, SG & FUEL BLDGS NON-SAFEGUARD MCC 1EB2-3	CP1-EPSWEB-02	5D	(2)
CP1-EPMCNB-02	CONTAINMENT NON-SAFEGUARD MCC IEB2-2	CP1-EPSWEB-02	5C	(2)
CP1-VAFNAV-02	CONTAINMENT RECIRC FAN	CP1-EPSWEB-02	3B	(1)
CP1-EPDSNA-01	ALTERNATE POWER TRANSFER SWITCH	ALTERNATE POWER SOURCE		(10)
CP1-EPTRNT-06	ISOL. TRANSFORMER FOR PRESSURIZER HTR BK-UP GRP-B	CP1-EPSWEB-02	5B	(1)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 2 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CP1-EPTRNT-08	ISOL. TRANSFORMER FOR PRESSURIZER HTR BK-UP GRP-D	CP1-EPSWEB-04	11B	(1)
CP1-VAFNAV-04	CONTAINMENT RECIRC FAN	CP1-EPSWEB-04	9B	(1)
CP1-VAFNCB-02	CRDM VENT FAN	CP1-EPSWEB-04	8B	(1)
CPX-EMPCNB-04	FUEL BLDG NON-SAFEGUARD MCC XEB4-1	CP1-EPSWEB-04	11C	(2)
CP1-ELTRNT-26	LTG TRANSFORMER XFS1 POWER SUPPLY	CP1-EPMCEB-01	9E	(5)
CP1-EIPRLV-48	TURBINE DRIVEN AFW PUMP CONTROL PNL	CP1-EPMCEB-01	2KL	(1)
CP1-VAFNCB-03	POSITIVE DISPLACEMENT CHARGING PUMP ROOM FAN	CP1-EPMCEB-03	4F	(1)
CP1-CICACO-01	INSTRUMENT AIR COMPRESSOR 1-01	CP1-EPSWEB-03	11D	(1)
CP1-VAFNID-11	BATTERY ROOM 1-3 (C) EXHAUST FAN	CP1-EPMCEB-03	3C	(1)
MOV 1-8109	POSITIVE DISPLACEMENT PUMP BYPASS VALVE	CP1-EPMCEB-03	10J	(1)
CP1-CIDYIA-01	AIR DRYER CONTROL PANEL FDR	CP1-EPMCEB-03	2BR	(1)
CP1-EPTRNT-12 (T1C3)	BYPASS TRANSFORMER	CP1-EPMCEB-03	1CR	(1)
1-REK-5502/03/66	RADIATION MONITOR	CP1-EPMCEB-02	11D	(1)
CP1-ELTRNT-27	LTG TRANSFORMER XFS2 POWER SUPPLY	CP1-EPMCEB-02	10D	(5)
CP1-EPBCND-02	BATTERY CHARGER BC1D2 FOR BATTERY BT1D2	CP1-EPMCEB-02	1FR	(1)
CP1-VAFNID-12	BATTERY ROOM 1-3(C) EXHAUST FAN 12	CP1-EPMCEB-04	3C	(1)
CP1-EPBCND-04	BATTERY CHARGER BC1D4 FOR BATTERY BT1D4	CP1-EPMCEB-04	10DR	(1)
CP1-CICACO-02	INSTRUMENT AIR COMPRESSOR CONTROL PANEL	CP1-EPSWEB-04	11D	(1)
CP1-CIDYIA-02	INSTRUMENT AIR DRYER CONTROL PANEL	CP1-EPMCEB-04	12FL	(1)
CP1-VAFNAV-09	NEUTRON DETECTOR WELL FAN 09	CP1-EPMCEB-05	8M	(1)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 3 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CP1-ELTRNT-18	CONTAINMENT LTG TRANSFORMER/LTG PNLS SC1 & SC3	CP1-EPMCEB-05	7M	(5)
MOV 1-HV-6074	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-05	5G	(5)
MOV 1-HV-6076	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-05	4G	(5)
MOV 1-HV-6078	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-05	4M	(5)
CP1-EPTRNT-28	ISOL. TRANFORMER TIEC3-3	CP1-EPMCEB-05	1BR	(5)
CP1-ELTRNT-30	LTG TRANSFORMER XFS3	CP1-EPMCEB-06	4B	(5)
CP1-ELTRNT-31	LTG TRANSFORMER, PANEL S4	CP1-EPMCEB-07	1J	(5)
CP1-VAFNAV-10	NEUTRON DETECTOR WELL FAN 10	CP1-EPMCEB-06	7M	(1)
CP1-ELTRNT-19	CONTAINMENT LTG TRANSFORMER/LTG PNLS SC2 & SC4	CP1-EPMCEB-06	6M	(5)
MOV 1-HV-6075	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-06	4G	(5)
MOV 1-HV-6077	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-06	3G	(5)
MOV 1-HV-6079	VENT CONTROL COOL UNIT COOLER/CHILLED WATER	CP1-EPMCEB-06	3M	(5)
CP1-MEDGEE-02K	AUXILIARY JACKET WATER PUMP	CP1-EPMCEB-10	1M	(1)
CP1-MEDGEE-02J	AUXILIARY LUBE OIL PUMP	CP1-EPMCEB-10	3M	(1)
CP1-MEDGEE-02P	JACKET WATER HEATER	CP1-EPMCEB-10	4M	(1)
CP1-MEDGEE-02V	LUBE OIL HEATER	CP1-EPMCEB-10	2M	(1)
CP1-MECAED-03	DG AIR COMPRESSOR #3	CP1-EPMCEB-10	1G	(1)
CP1-MECAED-04	DG AIR COMPRESSOR #4	CP1-EPMCEB-10	2G	(1)
CP1-MEDGEE-02M	PRELUBE PUMP	CP1-EPMCEB-10	6M	(1)
CP1-MEDGEE-02L	JACKET WATER KEEP WARM PUMP	CP1-EPMCEB-10	7J	(1)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 4 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CP1-MEDGEE-02N	FUEL OIL BOOSTER PUMP	CP1-EPMCEB-10	7M	(1)
CP1-MEDGEE-02G	FUEL OIL DRIP RETURN PUMP	CP1-EPMCEB-10	8M	(1)
CP1-VAEHUH-30	ROOM ADJACENT TO REFUEL WATER STORAGE TANK HTR	CP1-EPMCEB-10	2BR	(1)
CP1-VAEHUH-28	ROOM ADJACENT TO REACTOR MK-UP WTR STORAGE TANK HTR	CP1-EPMCEB-10	4BL	(1)
CP1-VAEHUH-29	ROOM ADJACENT TO COND. WATER STORAGE TANK HTR	CP1-EPMCEB-10	4BR	(1)
CPX-CHAPCP-03	VENT CHILLED WATER PUMP 03	CPX-EPMCEB-01	3M	(1)
CPX-VAFNCB-09	PRIMARY PLANT VENT EXHAUST FAN 09	CPX-EPMCEB-01	4M	(7)
CPX-VAFNCB-11	PRIMARY PLANT VENT EXHAUST FAN 11	CPX-EPMCEB-01	5G	(7)
CPX-VAFNCB-13	PRIMARY PLANT VENT EXHAUST FAN 13	CPX-EPMCEB-01	5M	(7)
CPX-VAFNCB-03	CONTAINMENT HYDROGEN PURGE SUPPLY FAN 03	CPX-EPMCEB-01	3E	(1)
CPX-CHAPCP-04	VENT CHILLED WATER PUMP 4	CPX-EPMCEB-02	3M	(1)
CPX-VAFNCB-10	PRIMARY PLANT VENT EXHAUST FAN 10	CPX-EPMCEB-02	4M	(7)
CPX-VAFNCB-12	PRIMARY PLANT VENT EXHAUST FAN 12	CPX-EPMCEB-02	5G	(7)
CPX-VAFNCB-14	PRIMARY PLANT VENT EXHAUST FAN 14	CPX-EPMCEB-02	5M	(7)
CPX-VAFNCB-04	CONTAINMENT HYDROGEN PURGE SUPPLY FAN 04	CPX-EPMCEB-02	3E	(1)
CPX-EPTRNT-43	AUX BLDG REGULATING TRANSFORMER TXEC2	CPX-EPMCEB-02	5BR	(5)
CPX-CHAPCP-01	VENT CHILLED WATER PUMP 1	CPX-EPMCEB-07	4M	(1)
CPX-VAFNAV-27	AUXILIARY BLDG VENT EQUIP ROOM EXHAUST FAN 27	CPX-EPMCEB-07	3M	(1)
CPX-EPTRNT-44	REGULATING TRANSFORMER TXEC3	CPX-EPMCEB-07	1BR	(5)
CPX-EPTRNT-42	REGULATING TRANSFORMER TXEC1	CPX-EPMCEB-03	7DR	(5)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 5 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CPX-CHAPCP-02	VENT CHILLED WATER PUMP 2	CPX-EPMCEB-08	3M	(1)
CPX-VAFNAV-28	AUXILIARY BLDG VENT EQUIP ROOM EXHAUST FAN 28	CPX-EPMCEB-08	4M	(1)
X-REK-5567A	VENT STACK “1” RADIATION MONITOR SYS SAMPLE PUMP	CPX-EPMCEB-03	6C	(1)
CPX-VAFNCB-15	PRIMARY PLANT VENT EXHAUST FAN 15	CPX-EPMCEB-03	4G	(7)
CPX-VAFNCB-17	PRIMARY PLANT VENT EXHAUST FAN 17	CPX-EPMCEB-03	4M	(7)
CPX-VAFNCB-19	PRIMARY PLANT VENT EXHAUST FAN 19	CPX-EPMCEB-03	5G	(7)
CPX-CHCICE-01	HVAC CENTRIFUGAL WATER CHILLER OIL PUMP	CPX-EPMCEB-03	5BL	(5)
CPX-CHCICE-03	HVAC CENTRIFUGAL WATER CHILLER OIL PUMP	CPX-EPMCEB-07	5BL	(5)
X-REK-5567B	VENT STACK “2” RADIATION MONITOR SYS SAMPLE PUMP	CPX-EPMCEB-04	6G	(1)
CPX-VAFNCB-16	PRIMARY PLANT VENT EXHAUST FAN 16	CPX-EPMCEB-04	4G	(7)
CPX-VAFNCB-18	PRIMARY PLANT VENT EXHAUST FAN 18	CPX-EPMCEB-04	4M	(7)
CPX-VAFNCB-20	PRIMARY PLANT VENT EXHAUST FAN 20	CPX-EPMCEB-04	5G	(7)
CPX-EPTRNT-45	REGULATING TRANSFORMER TXEC4	CPX-EPMCEB-04	7C	(5)
CPX-CHCICE-02	HVAC CENTRIFUGAL WATER CHILLER OIL PUMP	CPX-EPMCEB-04	5BL	(5)
CPX-CHCICE-04	HVAC CENTRIFUGAL WATER CHILLER OIL PUMP	CPX-EPMCEB-08	2BR	(5)
CP1-MEDGEE-01M	PRELUBE PUMP	CP1-EPMCEB-09	6M	(1)
CP1-MEDGEE-01L	JACKET WATER KEEP WARM PUMP	CP1-EPMCEB-09	7J	(1)
CP1-MEDGEE-O1N	FUEL OIL BOOSTER PUMP	CP1-EPMCEB-09	7M	(1)
CP1-MEDGEE-01G	FUEL OIL DRIP RETURN PUMP	CP1-EPMCEB-09	8M	(1)
CP1-VAEHUH-27	RM ADJ TO REFUEL WATER STORAGE TANK HTR	CP1-EPMCEB-09	2BR	(1)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 6 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CP1-VAEHUH-25	RM ADJ TO REACTOR MK-UP WATER STORAGE TANK HTR	CP1-EPMCEB-09	4BL	(1)
CP1-VAEHUH-26	RM ADJ TO COND. WATER STORAGE TANK HTR	CP1-EPMCEB-09	4BR	(1)
CP1-MEDGEE-01K	AUXILIARY JACKET WATER PUMP	CP1-EPMCEB-09	1M	(1)
CP1-MEDGEE-01J	AUXILIARY LUBE OIL PUMP	CP1-EPMCEB-09	3M	(1)
CP1-MEDGEE-01P	JACKET WATER HEATER	CP1-EPMCEB-09	2M	(1)
CP1-MEDGEE-01V	LUBE OIL HEATER	CP1-EPMCEB-09	4M	(1)
CP1-MECAED-01	DG AIR COMPRESSOR #1	CP1-EPMCEB-09	1G	(1)
CP1-MECAED-02	DG AIR COMPRESSOR #2	CP1-EPMCEB-09	2G	(1)
CPX-EPMCEB-05	MOTOR CONTROL CENTER XEB3-3	CP2-EPMCEB-07	2M	(4)
CPX-EPMCEB-05	MOTOR CONTROL CENTER XEB3-3	CP1-EPMCEB-07	2E	(4)
CPX-SWTSTS-01	SERVICE WATER TRAVELING SCREEN	CPX-EPMCEB-05	3C	(4)
CP1-SWAPTS-01	SERVICE WATER SCREEN WASH PUMP	CPX-EPMCEB-05	3F	(4)
CPX-ELTRNT-36	LIGHTING TRANSFORMER (LTG PNL SWP1)	CPX-EPMCEB-05	3HL	(4)
WRCPT2	WELDING RECEPTACLE	CPX-EPMCEB-05	1L	(3)
CPX-MESCSW-01	SERVICE WATER INTAKE STRUCTURE CRANE	CPX-EPMCEB-05	1FL	(3)
CPX-VAEHUH-08	SERVICE WATER PUMPHOUSE UNIT HEATER 08	CPX-EPMCEB-05	1FR	(3)
CPX-MEMHCH-12	SERVICE WATER TRAVELING SCREEN HOIST	CPX-EPMCEB-05	1BL	(3)
CPX-VAEHUH-06	SERVICE WATER PUMPHOUSE UNIT HEATER 06	CPX-EPMCEB-05	1BR	(3)
CPX-VAEHUH-09	SERVICE WATER PUMPHOUSE UNIT HEATER 09	CPX-EPMCEB-05	1DL	(3)
CPX-RUDMSW-01	ROLL UP DOOR	CPX-EPMCEB-05	1DR	(3)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 7 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CPX-ELTRNT-37	LIGHTING TRANSFORMER (LTG PNL SWP2)	CPX-EPMCEB-06	2HL	(4)
CPX-VAFNAV-41	DIESEL FIRE PUMP ROOM VENTILATION EXH FAN 41	CPX-EPMCEB-06	5F	(3)
CPX-SWEHSG-01	SCREEN STOP GATE HOIST (2HP)	CPX-EPMCEB-06	5MR	(3)
WRCPT1	WELDING RECEPTACLE (60 AMPS)	CPX-EPMCEB-06	4L	(3)
CPX-ELTRNT-02	SWIS CHLOR. BLDG DIST TRANSFORMER (LTG PNL SWN1&SWCB1)	CPX-EPMCEB-06	4BL	(3)
CPX-VAEHUH-04	SERVICE WATER PUMPHOUSE UNIT HEATER 04	CPX-EPMCEB-06	4BR	(3)
CPX-VAEHUH-05	SERVICE WATER PUMPHOUSE UNIT HEATER 05	CPX-EPMCEB-06	4DL	(3)
CPX-ECCPRT-05	SWIS CATHODIC PROTECTION RECTIFIER #5	CPX-EPMCEB-06	4DR	(3)
CPX-EPTRNT-21	HEAT TRACING TRANSFORMER TXHT-2 (HT DIST PNL XHT-2)	CPX-EPMCEB-06	4FL	(3)
CPX-VAEHUH-07	SERVICE WATER PUMPHOUSE UNIT HEATER 07	CPX-EPMCEB-06	4FR	(3)
CPX-EPMCEB-06	MOTOR CONTROL CENTER XEB4-3	CP2-EPMCEB-08	2M	(4)
CPX-EPMCEB-06	MOTOR CONTROL CENTER XEB4-3	CP1-EPMCEB-08	2E	(4)
CPX-SWTSTS-02	SERVICE WATER TRAVELING SCREEN 02	CPX-EPMCEB-06	2C	(4)
CPX-SWAPTS-02	SERVICE WATER SCREEN WASH PUMP 02	CPX-EPMCEB-06	2F	(4)
CPX-FPAPFP-03	JOCKEY FIRE PUMP (Abandoned in place)	CPX-EPMCEB-06	3C	(4)
CP1-ELTRET-01	XFMR FOR LTG DIST PNL ESB1 & ESB3	CP1-EPMCEB-01	1DL	(5),(6)
CP1-ELTRET-02	XFMR FOR LTG DIST PNL ESB5, ESB7, ESB9 & ESBD-1	CP1-EPMCEB-01	1DR	(5),(6)
CP1-ELTRET-03	XFMR FOR LTG DIST PNL ESB2 & ESB4	CP1-EPMCEB-02	1BL	(5),(6)
CP1-ELTRET-04	XFMR FOR LTG DIST PNL ESB6, ESB8, ESB10 & ESBD-2	CP1-EPMCEB-02	1BR	(5),(6)
CPX-ELTRET-09	XFMR FOR LTG DIST PNL EAB5, EAB7 & EABD-3	CPX-EPMCEB-01	4BL	(5),(6)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 8 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
1EB1-1/8M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-01	8M	(1)
1EB3-1/9M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-03	9M	(1)
1EB2-1/10M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-02	10M	(1)
1EB4-1/11M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-04	11M	(1)
1EB3-2/7C/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1- EPMCEB-05	7C	(1)
1EB4-2/6C/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-06	6C	(1)
1EB4-4/8J/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-10	8J	(1)
XEB1-2/6M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-01	6M	(1)
XEB2-2/6M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-02	6M	(1)
XEB1-1/2M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-07	2M	(1)
XEB2-1/5M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-08	5M	(1)
XEB3-2/2F/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-03	2F	(1)
XEB4-2/2F/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CPX-EPMCEB-04	2F	(1)
1EB3-3/3M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-07	3M	(1)
1EB3-4/8J/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-09	8J	(1)
1EB4-3/3M/TR	MCC AND MOTOR SPACE HEATER SUPPLY	CP1-EPMCEB-08	3M	(1)
CPX-ELTRET-06	XFMR FOR LTG DIST PNL ECB2 & EAB10	CPX-EPMCEB-02	6C	(5),(6)
CPX-ELTRET-10	XFMR FOR LTG DIST PNL EABD4, EAB6 & EAB8	CPX-EPMCEB-02	4BL	(5),(6)
CP1-ELTRET-05	XFMR FOR LTG DIST PNL ECB5 & ECB3	CP1-EPMCEB-07	5BL	(5),(6)
CP2-ELTRET-05	XFMR FOR LTG DIST PNL 2ECB3	CPX-EPMCEB-07	5BR	(5),(6)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 9 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CPX-ELTRET-05	XFMR FOR LTG DIST PNL EAB9 & ECB1	CPX-EPMCEB-07	1BL	(5),(6)
CP2-ELTRET-06	XFMR FOR LTG DIST PNL 2ECB4	CPX-EPMCEB-08	2BL	(5),(6)
CP1-ELTRET-06	XFMR FOR LTG DIST PNL ECB4 & ECB6	CPX-EPMCEB-08	2BR	(5),(6)
CPX-ELTRET-07	XFMR FOR LTG DIST PNL EAB1, EAB3, EAB11 & EABD1	CPX-EPMCEB-03	4BL	(5),(6)
CPX-ELTRET-08	XFMR FOR LTG DIST PNL EAB2, EAB4, EAB12 & EABD2	CPX-EPMCEB-04	6A	(5),(6)
CP1-MEDGEE-01H	GENERATOR SPACE HEATER	CP1-EPMCEB-09	2BL	(8)
CP1-MEDGEE-02H	GENERATOR SPACE HEATER	CP1-EPMCEB-10	2BL	(8)
CPX-VAFNCB-01	Hydrogen Purge System (HPS) Exhaust Fan 01	CPX-EPMCEB-01	6J	(1)
CPX-VAFNCB-02	Hydrogen Purge System (HPS) Exhaust Fan 02	CPX-EPMCEB-02	6J	(1)
CPX-VAFUPK-19	HPS Filtration Unit Heaters 19	CPX-EPMCEB-01	2BR	(1)
CPX-VAFUPK-20	HPS Filtration Unit Heaters 20	CPX-EPMCEB-02	2BR	(1)
CPX-VAFULV-19	Fire Protection Panel 19	CPX-ECDPEC-03	9	(5)
CPX-VAFULV-20	Fire Protection Panel 20	CPX-ECDPEC-04	9	(5)
X-HV-5526	HPS Motorized Valve	CPX-EPMCEB-03	2C	(5)
X-HV-5579	HPS Motorized Valve	CPX-EPMCEB-03	6F	(5)
X-HV-5529	HPS Motorized Valve	CPX-EPMCEB-04	2C	(5)
X-HV-5580	HPS Motorized Valve	CPX-EPMCEB-04	6D	(5)
CP2-CICACO-01	INSTRUMENT AIR COMPRESSOR 2-01	CP2-EPSWEB-03	10C	(1)
CP2-CICACO-02	INSTRUMENT AIR COMPRESSOR 2-02	CP2-EPSWEB-04	10D	(1)
CP2-CIDYIA-01	INSTRUMENT AIR DRYER 2-01	CP2-EPMCEB-03	2BR	(1)

TABLE 8.3-11
NON-CLASS 1E EQUIPMENT CONNECTED TO SAFETY RELATED POWER CIRCUITS

(Sheet 10 of 10)

EQUIPMENT ID NO.	DESCRIPTION	POWER SOURCE ID NO.		NOTE
CP2-CIDYIA-02	INSTRUMENT AIR DRYER 2-02	CP2-EPMCEB-06	9FL	(1)

NOTES:

(1) In accordance with Regulatory Guide 1.75, January 1975, Position C.1, Automatically Tripped on SIAS (accident signal). Reconnection requires operator action(s) after resetting SIAS.

(2) Breaker trips on SIAS, requires operator action to reset, and connecting cable is black and routed separately. In addition, these loads are also tripped on LOOP.

(3) This portion of the non-Class 1E MCC is tripped on SIAS or Blackout (Loss of Offsite Power) signal, and cable is in dedicated raceway.

(4) Same as No. 1, except MCC's are tripped from either Unit 1 or 2 independent power supply, thus MCC's, associated and non-Class 1E loads are isolated by SIAS signal. During normal operation MCC's are powered by Unit 2, and Unit 1 power is locked out.

(5) Non-Class 1E loads fed from Class 1E supplies are protected by two separate Class 1E breakers, or Class 1E breaker and Class 1E fuse or two separate Class 1E fuses connected in series. These breakers/fuses are coordinated with their supply breakers, and breakers will be tested and calibrated periodically to ensure coordination.

(6) The Class 1E transformers and lighting distribution panels feed non-Class 1E loads.

(7) Breaker trips on SIAS (accident signal), reconnection requires operator action to reset, and connecting cable is associated. In addition, this load is also tripped on LOOP.

(8) Breaker trips on SIAS, requires operator action to reset, and connecting cable is black and routed separately. These loads are also tripped by diesel generator breaker auxiliary contacts when the breaker closes to power the Class 1E bus. Therefore the loads will not be on the bus during a safety injection or loss of offsite power.

(9) Equipment listed, is for Unit 1; Unit 2 is similar, except for equipment identification numbers.

(10) Transfer switch is isolated from Class 1E bus by administratively controlled normally open circuit breakers. There are no automatic connections to the 1E bus. It is only connected to either bus during modes 5 and 6, and only if the plant experiences a loss of offsite power coincident with failure of the Class 1E Emergency Diesel Generators

8A ANALYSIS TO JUSTIFY CABLE SPLICES IN RACEWAYS

8A.1 PURPOSE

The purpose of this analysis is to show that the limited use of splices in raceways, incorporated in the CPNPP design, does not degrade the Class 1E circuits and does not pose any undue hazard of a fire. This analysis is developed to satisfy the requirements of NRC Regulatory Guide 1.75, Rev. 1, Regulatory Position C.9.

8A.2 SCOPE

This analysis covers all cable splices in raceways utilized in the CPNPP plant design. The term "raceways" shall include open or enclosed trays and rigid or flexible steel conduits and condulets. This analysis does not discuss splices inside an equipment enclosure, splices in junction/terminal boxes furnished as an integral part of an equipment, or splices in junction/terminal boxes in raceways, since the purpose of such boxes is the termination of cables. This analysis does not discuss splices in yard non-Class I (non-seismic) manholes and handholes since these manholes/handholes do not contain and are not in the vicinity of Class 1E cables and equipment. Splices in Class I (seismic) manholes containing Class 1E cables are not discussed since one of the purposes of manholes is to accommodate splices and individual manholes do not contain cables of opposite trains.

Although this analysis covers cable splices in raceways, the description provided also applies to other Class 1E splices performed at CPNPP.

In CPNPP cable splices which are made in raceways can be categorized in the following three groups:

1. Group 1 - Field routed power, control and instrumentation cables which are connected to pigtail cables from Electric Penetration Assembly (EPA) and Core Exit Thermocouple (CET) Integral Reference Junction (IRJ), Control Rod Drive Mechanism (CRDM) disconnect panel connectors and Electric Conductor Seal Assemblies (ECSAs) pigtail cables by means of in-line splices located in trays.

2. Group 2 - Pigtail cables from local mounted devices (LMDs - solenoid valves, limit switches, level switches, etc.) which are connected to field routed control and instrument grade cables by means of parallel or in-line butt splices located in flexible conduit or condulets.

3. Group 3 - Pigtail cables from local mounted devices which are connected to the pigtail conductors of the Electric Conductor Seal Assembly (ECSA) by means of in-line butt splices located in the flexible conduit of the ECSA.

8A.3 REGULATORY POSITION REQUIREMENT

The CPNPP FSAR commits to IEEE Std. 384-1974 and NRC Regulatory Guide 1.75, Revision 1 dated January 1975. Section 5.1.1.3 of IEEE Standard 384, with the Regulatory Guide 1.75, Regulatory Position C.9 supplement, reads as follows:

"5.1.1.3 The minimum separation distances specified in Section 5.1.3 and 5.1.4 are based on open ventilated cable trays of either the ladder or trough type as defined in NEMA VE1-1971, Cable Tray Systems. Where these distances are used to provide adequate physical separation:

(1) Cables and raceways involved shall be flame retardant

(2) The design basis shall be that the cable trays will not be filled above the side rails

(3) Hazards shall be limited to failures or faults internal to the electric equipment or cables

(4) Cable splices in raceways should be prohibited.

If lesser separation distances are used they shall be established as in Section 5.1.1.2."

In plant areas where hazards are limited to failures or faults internal to the electric equipment or cables, CPNPP design permits the minimum separation distances specified in Sections 5.1.3 and 5.1.4 of the IEEE Standard 384 to achieve independence between redundant trains. Use of these specified separation distances in plant design therefore requires compliance with all of the above conditions unless justified by analysis.

Per requirement 3 above, hazards are limited to failures or faults internal to the electric equipment or cables. For this analysis then, the only hazard of concern would be an electrically generated fire inside the raceway.

Regulatory Position C.9 prohibits splices in raceways even though separation distance is adequate to prevent a fire in the raceways of one train from affecting cables in a redundant train. However, it also states under the "Basis" of this position (paragraph C.9) that "Splices are not, by themselves, unacceptable. If they exist, the resulting design should be justified by analyses." Use of splices in raceways must therefore be justified. For this analysis, it will be shown that (1) the minimum separation distances utilized in the plant design is adequate to maintain independence between redundant trains for a postulated fire originating at a cable splice and (2) cable splices in raceways do not pose any undue hazard of initiating a fire.

8A.4 DETERMINATION OF ACCEPTABILITY OF CABLE SPLICES IN RACEWAYS

8A.4.1 INDEPENDENCE OF REDUNDANT TRAINS

Minimum separation distances utilized in CPNPP design to achieve independence between redundant trains meet the requirements of Sections 5.1.3 and 5.1.4 of IEEE Standard 384. Per the requirements of IEEE Standard 384, as modified by Regulatory Guide 1.75, these minimum separation distances are adequate provided (1) cable and raceway materials involved are flame retardant, (2) cable trays are not filled above their side rails, (3) hazards are limited to fire originating in the raceway and (4) there should be no cable splices in raceways.

Since the design may not comply with the fourth requirement, the question would be whether the specified minimum separation distances are adequate for providing independence between redundant trains in case of a fire electrically generated at a cable splice.

Where there are no splices, the specified separation distances per the Standard, or as allowed by tests or analysis, are adequate to satisfy this objective for the case of an electrically generated cable fire. If the fire generated at a cable splice is no worse than a cable fire, the specified minimum separation distances would be adequate. We will show that the degree of potential damage from a cable splice fire is no more than the degree of potential damage from a fire in a cable without splices.

All cable splices within the scope of this analysis are made of essentially two types of material: one is the metallic connector part for conducting current and the second is the non-metallic insulation part. The metallic part is non-combustible like the copper conductor in a cable. The insulation materials applied at the splice are of a flame retardant type similar to the insulation and jacket material of a cable.

The insulation and jacket materials of CPNPP cables in raceways consist of EPR, XLPE, CPE, ETFE and CSPE. The materials, in their constructed configurations, are flame retardant and are self extinguishing. This has been shown by documented tests performed by the cable vendors in order to qualify their cables per Section 2.5 test requirements of IEEE Standard 383-1974. The oxygen index of these cable insulating materials ranges from 22 to 34. The higher the oxygen index, the quicker the fire extinguishes. Uninsulated splices employ Raychem heat-shrink type insulating material or Okonite tape. Insulated splices use AMP PIDG and Pre-Insulated Environmental Splices (PIES) connectors with PVF2 (Kynar) insulation material. The materials involved in the Raychem splices consist of WCSF for sleeves, "-52" for molded parts and S-1119 used as a sealant and adhesive. WCSF has passed the IEEE Standard 383 flame test. The -52 material is a cross-linked Polyolefin similar to WCSF and has the same oxygen index of 28. Therefore a similar performance to WCSF is expected. S-1119 adhesive has also passed the flame test with WCSF. In the installed configuration, the major portion of the S-1119 material does not have access to air. Access to air is esential for a burning process to continue. The oxygen index of Kynar is 44. The oxygen index of the remaining non-metallic splice products ranges from 28-44, which is higher than the minimum oxygen index of the cable insulation and jacket materials. The Okonite jacketing and insulation tapes have successfully passed the IEEE Standard 383 flame test in the installed configuration. Therefore, it can be seen that all cable splices, as installed, are flame retardant, non-propagating and self-extinguishing to a degree equal to or greater than the cable insulating material itself.

Based on the above, it is concluded that the degree of potential damage due to a fire at a cable splice is no more than that from a fire in a cable. Therefore, use of the minimum required separation distances specified in Sections 5.1.3 and 5.1.4 of IEEE Standard 384-1974 is justified for the splices utilized in the CPNPP design.

8A.4.2 ASSESSMENT OF FIRE HAZARD DUE TO CABLE SPLICES IN RACEWAYS

8A.4.2.1 Approach

In Section 4.1 of this analysis we have shown that the consequences of a postulated fire originating at a cable splice are acceptable. In this Section we will show that the likelihood of such a postulated fire in a splice is no greater than that in a cable itself.

A fire can originate in the insulating materials of a cable splice due to (1) excessive heating of the internal current-carrying metallic parts and (2) breakdown of the dielectric property of the insulating material. In order to assess the probability of such a fire, we need to evaluate the

control of splices which have been used in raceways and those attributes which can cause such heating and dielectric breakdown.

The purpose of this analysis is not to prove that a cable splice would not generate a fire under any circumstances. The attributes which have the potential of starting a fire in a cable (e.g., failure of a circuit protective device to interrupt a fault current) may also start a fire at a cable splice. What will be shown is that introduction of a good quality cable splice (installed using approved procedures by trained personnel using approved materials) in a continuous cable run does not create a weak link; that is, it is at least as reliable as the continuous cable run. We will therefore analyze all the attributes of a splice which are essential to ensure that consistently good quality splices are provided in the CPNPP design and construction.

8A.4.2.2 Review of Attributes and Their Effects

1. Attribute: Limited and controlled use of splices.

   Discussion: The electrical erection specification, used for construction of CPNPP, requires that cables be installed without splices unless splices are specifically called for on the drawings, cable and raceway schedule, or when approved by the Engineer. All splices in CPNPP are made per the electrical drawings. These drawings show typical details for each group of splices. The details identify the materials to be used to make field splices. Per procedures, the Engineer's approval of design changes is recorded via Design Change Authorizations. These procedures ensure application of quality control for site performed splices.

2. Attribute: Quality assurance and quality control.

   Discussion: All materials involved in the splice connections are procured under the quality assurance program in compliance with the applicable ANSI N45.2 series of standards. Splices are made only by trained personnel as per the installation procedures. This procedure ensures adherence to splice manufacturer's recommended methods for installation and all construction drawings and specifications.

   Use of the proper compression tool is assured since all AMP tools are matched by design to specific connectors. The tools are serialized and periodically checked for calibration by the on- site calibration laboratory. The tools are of the ratchet type and each crimp is brought to a full compression before the tool can be released. This ensures compliance with the manufacturer's crimping requirement and excludes any possibility of an under or over crimp condition. The tools are logged in and out of a tool room each day and are checked with a "go/no go" gauge at the end of each day. The craftsman records on the tool card the use made of the tool on each occasion. Tools that do not pass the "go/no go" test are taken out of service and the connectors installed with them that day are removed.

   All Class 1E splices are inspected in accordance with quality control procedures and each splice inspection is documented by an inspection report. QC inspects all physical attributes that make up the splice, e.g., selection of proper connector and crimping tool, proper stripping of cable insulation and jacket, proper insertion of conductor into the connector barrel, adequacy of crimp, etc. If a bolted connection is being made, QC verifies correct bolting materials and witnesses torquing. For splices which are insulated

at site by installation of heat shrink tubing, QC verifies proper cleaning of cable, proper selection of heat shrink material and proper installation. All these steps ensure that the splices are made per manufacturers' installation procedures and thereby produce qualified high quality splices. The QC inspection reports and the cable connection sign-off cards are maintained as a permanent QA record. Non-Class 1E splices are installed under craft supervision in accordance with approved procedures, using attributes similar to the Class 1E splice attributes.

The above shows that adequate quality assurance and control measures are implemented to ensure a good quality splice.

3. Attribute: Effect of temperature on splice materials.

Discussion: All CPNPP splices utilize either insulated or uninsulated type connectors. All splice connectors are made of fine grade high conductivity copper and have tin plating to resist corrosion. The connectors are crimped on the cable conductors by using calibrated compression tools which give positive indication of completion of compression. The compression tooling has features to prevent under or over crimping and provides dot or wire size coding for quality control verification.

The AMP PIDG and PIES connectors meet the requirements of MIL-T-7928. Their use is restricted to #10 AWG size cable and smaller. All other wire connectors utilized in CPNPP are UL 486A Listed and all splicing wire connectors are UL 486C Listed. As part of the UL Listing requirements, the connectors must pass the static heating test. The maximum acceptable temperature rise when tested with UL specified currents is 50 deg. C. Table A below shows the UL specified test current and maximum design limit currents utilized in CPNPP design for different conductor sizes.

TABLE A

Conductor Size	UL Static Heating Specified Test Current-Amp UL 486A	UL 486C	CPS ES Maximum Design Limit Current-Amp
# 12 AWG	35	35	25
# 10 AWG	50	50	33
# 8 AWG	70	70	46
# 6 AWG	95	95	62
# 4 AWG	125	-	81
# 2 AWG	170	-	116
# 2/0AWG	265	-	182
# 4/0AWG	360	-	249
350 MCM	505	-	405

500 MCM	620	-	509
750 MCM	785	-	658

From Table A above, it can be seen that the UL specified test currents which can produce a maximum 50 deg. C connector temperature rise are 119 to 154 percent higher than the maximum CPNPP design limit currents. The IEEE 323 qualification test for AMP preinsulated splice connectors has shown a maximum temperature rise of 37.7 deg. C during the post-radiation and LOCA static heating test with UL specified test currents. Therefore the connector temperature rise for CPNPP will be much less than 40 deg. C, which is the design limit temperature rise for CPNPP cables.

All insulating materials applied at the splices (i.e., Raychem WCSF and "-52", AMP Kynar materials, and Okonite tapes) have a demonstrated 40 year qualified life at 90 deg. C operating temperature, which is the minimum rated insulation temperature of CPNPP cables.

Therefore, it is concluded that the conducting material of the splices, which operate at a lower temperature than the conductor, are not hot spots which can degrade the insulating materials.

4. Attribute: Aging of splice insulating materials.

Discussion: The insulation materials utilized for uninsulated connectors are Raychem WCSF or "-52" material. The AMP preinsulated splice connectors have PVF2 Kynar insulation. All splices are made using material manufacturers' procedures.

The manufacturers have qualified the splice configurations (recommended in their procedures) per requirements of IEEE Standards 323 and 383. The splices were thermally aged to simulate a 40 year installed life using Arrhenius methodology. The DBE testings included vibration and LOCA exposure. The qualification tests have demonstrated that all splices in raceways are capable of performing their required functions throughout their 40 year qualified life and during Design Basis Events of seismic, radiation exposure, LOCA and post-LOCA conditions, as applicable (PIES are used only in mild environment areas). Therefore, aging of the splices will not create a weak link in the cable run.

Selected Class 1E splices not in raceways, for high voltage applications outside containment, may have a shorter qualified life and will be replaced prior to expiration of their qualified life.

5. Attribute: Mechanical integrity.

Discussion: Splicing wire connectors (in-line butt type or parallel connectors) or wire connectors (bolted back to back solid tongue (with hole(s)) type lugs) are used for making splice connections. The connectors are crimped on the cable conductors by using manufacturer's certified compression tools. Bolted joints are torqued to the values specified in the installation procedures. The splicing wire connections are capable of withstanding the UL 486C pull-out test requirements and the wire connections (bolted

tongue connections) are capable of withstanding the UL 486A pull-out test requirements. The AMP PIDG and PIES connectors meet MIL-T-7928 pull out test requirements which are more severe than UL 486A and UL 486C. Additionally, splices at terminations can only be made after the cables are pulled. Thus, pulling stresses which have potential to degrade the splices are eliminated. Hence, mechanical integrity of CPNPP splices will not be degraded during or after installation.

8A.5 SUMMARY AND CONCLUSION

It has been shown that the splice materials, tools, qualification, drawings, training and installation procedures used at CPNPP are designed to ensure that all splices made will be of high quality. Limited use of splices in raceways in the CPNPP design does not compromise independence of redundant Class 1E trains nor significantly increase the likelihood of fire. As such, splices, as applied, do not pose any undue hazard nor do they introduce weak links in the cable run and they can be expected to function at least as good as the continuous cable in which they are installed. Therefore, the CPNPP design meets the intent of the Regulatory Guide 1.75, Regulatory Position C.9.

8B STATION BLACKOUT

To assess the compliance of Comanche Peak Units 1 and 2 with the Station Blackout rule (10CFR50.63) an evaluation [Ref.2] following the guidance provided by RG 1.155 [Ref.1] was performed. This evaluation determined that both units are capable of coping with a station blackout (SBO) for 4 hours as AC Independent plants and that no modifications were required.

Subsequent SBO evaluations and other clarifications of sections of Attachment 1 to Ref.2 are as follows:

1. Section A.3.3: With the exception of two train EDG outages, an EDG on the Non-SBO unit will energize on the loss of offsite power and provide power to common equipment in support of the SBO unit; however, as this is a normal electrical line up, it does not constitute the “Alternate AC” approach to coping with SBO. In the event that a single EDG is not available in the Non-SBO Unit, such as during a two train EDG outage, no power is required to support the SBO Unit, since no common equipment is required in support of the SBO Unit.

2. Sections A.4.2 and D.3.12: Although they are assumed to function normally, no credit for SBO coping is required for the non-safety related turbine stop valves, feedwater control valves or feedwater control bypass valves. The safety related main steam and feedwater isolation valves do not require AC power to close.

3. Section A.4.2: Steam generator blowdown and sampling valves close on a loss of air or power.

4. Section A.4.3: The pressurizer PORVs should not cycle during an SBO. They are environmentally qualified to not fail open in an adverse environment.

5. Section A.10: Containment isolation valves on the secondary side of the steam generators are excluded from those of concern by Ref. 3. Closure of these valves for containment isolation during SBO coping is not required.

6. Section E.3: Air operated valves that are needed for SBO can be initially operated remotely with air stored in accumulators and locally manually when air is exhausted. The air accumulators for the Atmospheric Relief Valves (ARVs) on the steam generators are sized for Steam Generator Tube Rupture Mitigation as described in Section 9.3.1.2 and are not required to provide air for the entire SBO duration.

7. Section F.b: NUMARC 87-00 [Ref. 3] considers it necessary for plants, such as CPNPP, to further analyze control room air conditioning if it is a common control room with partial operational capability remaining during the SBO scenario. In the event that a single EDG is not available in the Non-SBO Unit, such as during two train EDG outages, the control room environment is analyzed without control room air conditioning. It is noted that Unit outages that remove two trains of SSW or two trains of CCW would put the Control Room air conditioning units in the same configuration as a two train EDG outage.

8. Section F.c: Modifications to the UPS equipment and ventilation systems have been made which enhance SBO coping. In the event that a single EDG is not available in the Non-SBO Unit, such as during two train EDG outages, credit is not taken for these

modifications to the ventilation systems. It is noted that Unit outages that remove two trains of SSW or two trains of CCW or two trains of Safety Chilled Water would put the UPS cooling units in the same configuration as a two train EDG outage. Equipment qualification limitations have been increased from 120°F by replacement with inverters which have a 131°F limit. Ventilation modifications are described in References 4 and 5 and in FSAR Section 9.4C.8

The 4-hour coping duration was determined by approved methods based on the redundancy and reliability of onsite emergency AC power sources, the expected frequency of loss of offsite power, and the probable time needed to restore offsite power.

The reactor coolant system with associated support systems were analyzed and determined to be capable of maintaining the appropriate core cooling and containment integrity during a 4-hour postulated SBO event.

The analysis ensured that the applicable emergency response guidelines facilitate proper monitoring of plant parameters and mitigation of the SBO effects.

Because CPNPP Units 1 and 2 have certain common cooling systems, the evaluation showed that no equipment modifications were required for a blacked out unit to cope with a Station Blackout event. In the event that a single EDG is not available in the Non-SBO Unit, such as during two train EDG outages, no credit is taken for these common cooling systems for the SBO Unit.

SBO coping is not directly covered by Technical Specification (TS) requirements; however, SBO coping equipment are covered by various Technical Specification requirements which are consistent with the coping analysis assumptions when the units are in MODES 1-6. In the event that a single EDG is not available in the Non-SBO Unit, such as during two train EDG outages, credit is not taken for a Non-SBO Unit's EDG. This is because it has been demonstrated that the SBO unit can cope without the common systems powered by the Non-SBO Unit's EDG. Unit outages that remove two trains of SSW or two trains of CCW or two trains of Safety Chilled Water would put the UPS cooling units in the same configuration as a two train EDG outage. In addition, Unit outages that remove two trains of SSW or two trains of CCW would put the Control Room air conditioning units in the same configuration as a two train EDG outage.

REFERENCES

1. Regulatory Guide 1.155, Station Blackout (August 1988)

2. TXX-92447, dated October 1, 1992

3. NUMARC 87-00, Guideline and Technical Basis for NUMARC Initiatives Addressing Station Blackout at Light Water Reactors, Revision 1, August 1991

4. TXX-96405, dated July 10, 1996

5. TXX-96475, dated October 1, 1996

6. TXX-19057, dated May 16, 2019

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9.1 FUEL STORAGE AND HANDLING

9.1.1 NEW FUEL STORAGE

9.1.1.1 Design Bases

New fuel is stored in racks (Figure 9.1-1) composed of individual vertical cells fastened together in any number to form a module which can be firmly bolted to anchors in the floor of the new fuel storage pit. The new fuel storage racks are designed to include storage for two thirds core at a center-to-center spacing of 21 inches. If the new fuel assemblies are stored dry, this spacing provides a minimum separation between adjacent fuel assemblies of 12 in., which is sufficient to maintain a subcritical array as described in Section 4.3.2.6. All surfaces that come into contact with the fuel assemblies are made of annealed austenitic stainless steel.

The racks are designed to withstand normal operating loads as well as Safe Shutdown Earthquake (SSE) and Operating Basis Earthquake (OBE) seismic loads meeting ANS Safety Class 3 [12] and ASME B&PV Code, Section III, Appendix XVII requirements. The new fuel racks are designed to withstand a maximum uplift force of 5000 lb.

9.1.1.2 Facilities Description

Both units of the CPNPP are serviced by a common Fuel Building which houses facilities for the storage and transfer of new and spent fuel. The Fuel Building is a controlled leakage building designed to seismic Category I requirements. For a description of the structural design considerations, see Section 3.8. The ventilation system is discussed in Section 9.4.4. The locations of the fuel storage areas within the station complex are shown on plan and elevation drawings; see Section 1.2 and Figures 1.2-38 through 1.2-40. The fuel storage and handling facilities are built in accordance with NRC Regulatory Guide 1.13.

New fuel assemblies are delivered to the site in United States Department of Transportation (DOT) approved containers. The containers are brought into the new fuel receiving area by the Fuel Building crane. Here a container is opened and the assemblies are unloaded and inspected.

Once the inspection is completed, the new fuel assembly is inserted in the new fuel storage rack (see Figure 9.1-1). The protective cover on each fuel assembly must be removed from the fuel assembly or must be open at the bottom so that water will not collect in the protective cover.

New fuel assemblies and control rods are stored in a reinforced concrete pit located in the Fuel Building. The pit, an integral part of the Fuel Building, is provided for temporary dry storage and is equipped with storage racks of sufficient capacity for approximately one-third core for each unit (total 132 fuel assemblies).

All surfaces that come into contact with fuel assemblies are made of austenitic stainless steel, thus precluding significant materials compatibility problems.

For the structural design considerations, including the loading criteria (loading and load combinations) for the Fuel Building, see Section 3.8.

The probability of a dropped mass damaging a new fuel assembly is very remote, for the following reasons:

1. New fuel racks located in the new fuel pit area are protected from dropped objects by a protective steel cover.

2. Administrative controls or interlocks, or both are used to prevent the handling of loads heavier than a fuel assembly and the associated handling tools over the new fuel storage area.

3. Safe handling features of the new fuel assembly handling tool are discussed in Subsection 9.1.4.2.3.

In preparation for refueling, the individual fuel assemblies are transported from the new fuel storage racks to the new fuel elevator using the fuel handling bridge crane or the Fuel Building overhead crane equipped with the new fuel handling tool. When an assembly has been lowered by the elevator, the fuel handling bridge crane equipped with the spent fuel handling tool can be used to place it either in the spent fuel pool for interim storage or in the Fuel Transfer System fuel basket for immediate transport into the Containment. For additional information on the fuel handling system, see Subsection 9.1.4.2.

The manipulation of new fuel assemblies will be performed by personnel trained in proper fuel handling techniques and, in addition, will use fuel handling procedures which contain provisions to assure that damage to fuel assemblies during movement is prevented.

Details of the seismic design and testing of the new fuel storage area are presented in Section 3.7B.

For general arrangement of new fuel storage facilities, see Section 1.2 and Figures 1.2-38, 1.2-39, and 1.2-40.

9.1.1.3 Safety Evaluation

The design of normally dry new fuel storage racks is such that the effective multiplication factor (keff) does not exceed the criteria described in Section 4.3.2.6. Consideration is given to the inherent neutron absorbing effect of the materials of construction. The detailed criticality safety evaluation is discussed in Section 4.3.2.6.

The design of the fuel storage rack assembly is such that it is impossible to insert the new fuel assemblies in other than prescribed locations, thereby preventing any possibility of accidental criticality.

The fuel storage racks are designed to withstand shipping, handling, and normal operating loads (dead loads of fuel assemblies), as well as SSE loads; these racks meet ANS Safety Class 3 requirements. The fuel storage racks are also designed to meet the seismic Category I requirements of NRC Regulatory Guide 1.29, as discussed in Section 1A(B).

The potential for inadvertent criticality has been evaluated for accidents that could result in alteration of the geometric spacing of fuel assemblies. For the normally dry conditions of the new fuel storage racks, fuel with an enrichment of 5 w/o U235 or less cannot be made critical, even

considering any damage or deformation that could occur as a result of dropped heavy loads or crane uplift forces on the racks. Further, it is not necessary to assume two unlikely, independent, concurrent events to ensure protection against a criticality accident. Thus, wet moderated conditions need not be considered coincident with accidental deformation or damage to the normally dry new fuel storage racks. It is concluded that such accidents would not results in advertent criticality.

Shielding requirements are discussed in Subsection 9.1.4.3.4.

Design of this storage facility is in accordance with NRC Regulatory Guide 1.13, Revision 1, December 1975, ensuring a safe condition under normal and postulated accident conditions.

9.1.2 SPENT FUEL STORAGE IN THE SPENT FUEL POOLS AND CONTAINMENTS

9.1.2.1 Design Bases

High density spent fuel storage is provided in the two common pools, Spent Fuel Pool 1 (SFP1) and Spent Fuel Pool 2 (SFP2), and low density spent fuel storage is provided inside each reactor containment. The storage in each spent fuel pool is comprised in two regions: Region I high density storage racks and Region II high density storage racks. The criticality design basis for Region I and II racks is provided in Section 4.3.2.6. Region I racks are primarily designed for refueling operations and the Region II racks are designed for long term storage. The design bases for the Region I and Region II racks is contained in Reference 18. The low density storage inside containment is designed for temporary storage in support of refueling operations.

The racks maintain a separation between spent fuel assemblies sufficient to maintain subcriticality as described in Section 3.1.6.3 and Section 4.3.2.6. All surfaces that come into contact with fuel assemblies are made of annealed austenitic stainless steel which is resistant to corrosion during normal and emergency water quality conditions.

Spent fuel storage racks are designed to withstand shipping, handling, normal operating loads (dead loads of fuel assemblies), as well as SSE loads; these racks meet ANS Safety Class 3 and ASME B&PV Code, Section III, Appendix XVII requirements. The spent fuel storage racks are also designed to meet the seismic Category I requirements of Reg. Guide 1.29, Revision 2, February 1976.

The spent fuel storage racks have adequate energy absorption capabilities to withstand the impact of a dropped spent fuel assembly from the maximum lift height of the fuel handling bridge crane. Cranes capable of carrying loads heavier than a spent fuel assembly are prevented by interlocks or administrative controls, or both, from traveling over the spent fuel storage areas when fuel is stored in them.

The spent fuel storage racks can withstand an uplift force equal to the uplift force of the spent fuel pool bridge hoist.

Shielding requirements are discussed in Subsection 9.1.4.3.4.

9.1.2.2 Facilities Description

Two pools are provided for CPNPP spent fuel storage. Spent fuel assemblies and irradiated control rods are stored underwater in racks after transfer from the reactor. The fuel assemblies and control rods are held vertically in the racks located on the floor of the spent fuel storage pools. The two reinforced concrete pools are stainless- steel lined and are an integral part of the Fuel Building. For the structural design considerations of the Fuel Building, including the loading criteria, see Section 3.8. The spent fuel racks are designed to accommodate an SSE, shipping, and handling loads, and the dead load of the spent fuel assemblies.

The spent fuel assemblies in SFP1 are stored in high density Region I and Region II racks with a usable capacity of 1,684 storage cells. The spent fuel assemblies in SFP2 are stored in high density Region I and Region II racks with a usable capacity of 1,689 storage cells. This provides a total storage space for the two pools of 3,373 fuel assemblies.

The high density Region I rack (Figure 9.1-2B) is composed of vertical cells fastened together in a checkerboard arrangement to produce a matrix structure. The cells are welded to a baseplate and to one another to form an integral structure without the use of a supporting grid structure. The high density Region I racks in Spent Fuel Pool 1 (SFP1) and Spent Fuel Pool 2 (SFP2) are free standing and self supporting. The center to center spacing between cells within a Region I rack is a nominal 10.65 inch by a nominal 11.05 inch. The Region I racks use a flux trap design and have neutron absorbing “Boral” panels between adjacent storage cells to provide neutron attenuation.

The high density Region II rack (Figure 9.1-2A) is composed of vertical cells fastened together in a checkerboard arrangement to produce a matrix structure. The cells are welded to a baseplate and to one another to form an integral structure without the use of a supporting grid structure. The high density Region II racks in SFP1 and SFP2 are free standing and self supporting. The center to center spacing between cells within a Region II rack is a nominal 9.0 inches. The Region II racks do not use a flux trap design and have no special neutron absorbing material.

SFP1 contains two (2) 10 x 8 Region I rack modules, one (1) 9 x 8 Region I rack module, six (6) 12 x 14 Region II rack modules, and three (3) 11 x 14 Region II rack modules (twelve racks total). Some of the Region I cells have been modified at the top to prevent interference with the spent fuel pool swing gate. Some of the Region II cells in SFP1 have been modified to allow for fuel inspection.

SFP2 contains two (2) 10 x 8 Region I rack modules, one (1) 9 x 8 Region I rack module, six (6) 12 x 14 Region II rack modules, and three (3) 11 x 14 Region II rack modules (twelve racks total). Some of the Region I cells in SFP2 have been modified to allow for fuel inspection. Some of the Region I cells in SFP2 have been modified at the top to prevent interference with the spent fuel pool swing gate.

The containment refueling cavity of each unit has additional interim storage space for one (1) low density 5 x 5 rack module. Each low density rack is composed of individual vertical cells fastened together within a grid structure to form a module (Figure 9.1-2). The low density rack in the containment refueling cavity is firmly bolted to the floor of the cavity. The center to center spacing between cells within the containment refueling cavity storage rack is a nominal 16.0 inches.

See Section 9.1.2.1 for description of the design basis. See Section 3.1.6.3 and Section 4.3.2.6 for a more detailed discussion of prevention of criticality and storage patterns.

Each spent fuel pool is designed to safely store the irradiated fuel assemblies. A separate pit is provided as a loading area for the spent fuel shipping cask. The refueling cavities, spent fuel pools, and cask pit are connected with a common transfer canal. Each connection between the transfer canal and the spent fuel pools can be closed by using gates (see Section 9.1.4.2.3 and Figure 1.2-39).

The spent fuel pools, transfer canal, and cask pit are lined with stainless steel plate.

The reactor is refueled using equipment that handles the spent fuel assemblies underwater from the time they leave the reactor vessel until they are placed in a cask for shipment from the site.

The Fuel Building fuel handling bridge crane, provided for spent fuel handling, has a wheel mounted walkway which spans spent fuel pools, the transfer canal, and cask pit. The bridge carries an electric monorail hoist on an overhead structure which is provided with an antiderailing device and is designed to withstand an SSE. The fuel assemblies are moved within a spent fuel pool by means of a long-handled tool suspended from the hoist. For general arrangement of spent fuel storage facilities, see Section 1.2, Figures 1.2-12, 1.2-13, 1.2-15, 1.2-18, 1.2-19, 1.2-38, 1.2-39, and 1.2-40.

The manipulation of spent fuel assemblies will be performed by personnel trained in proper fuel handling techniques and, in addition, will use fuel handling procedures which contain provisions to assure that damage to fuel assemblies during movement is prevented.

Once the fuel is stored in the spent fuel pool, the Spent Fuel Pool Cooling and Cleanup System ensures continuous cooling. (See Subsection 9.1.3.3.) There are no drains or permanently connected systems or other features that can cause a loss of coolant that would uncover fuel.

Normal makeup water, to compensate evaporation losses, is supplied from the reactor makeup water system or the demineralized water supply system. In the case of a failure or malfunction of the demineralized water supply, the safety-related (seismic Category I, Safety Class 3, and redundant) Reactor Makeup Water System supplies reactor coolant purity water to the spent fuel pools. For a detailed discussion, see Section 9.2.3. Water level monitoring equipment is discussed in Subsection 9.1.3.

To limit the dose rate at the surface of the pools to 2.5 mR/hr a minimum water shielding depth of 10 ft is provided above a fuel assembly. The design low-water level provides a positive margin to the minimum water shielding requirement with a fuel element located above the spent fuel storage racks during fuel handling operations. The maximum height to which the fuel elements can be lifted is limited by the design of the hoist and the spent fuel handling tool controls. For a detailed description of shielding design, see Section 12.1. Consideration of criticality safety analysis is discussed in Section 3.1.6.3, Section 4.3.2.6 and Subsection 9.1.2.3.

Details of seismic design and testing are presented in Section 3.7B.

Sealed bearings or other measures, such as protective pans, are used to prevent the lubricant of the cranes from contaminating the spent fuel pools. The crane control and power systems are

capable of permitting continuous operation at minimum speed or frequent jogging without detrimental effects on any circuit or component.

Either spent fuel pool can be used for storage of fuel assemblies from both reactors as there are no adverse implications of sharing. In fact, sharing between the two pools permits greater flexibility.

The fuel storage facilities are designed in accordance with NRC Regulatory Guide 1.13.

When fuel assembly decay heat has reached an acceptable level, the fuel assembly can be removed from the spent fuel pool and loaded into a spent fuel shipping cask.

The following design features of the Fuel Building Overhead Crane are provided in order to prevent a cask from dropping:

1. The crane is designed to the requirements of seismic Category I. As such it can retain the maximum design load during a SSE and remain in place under all postulated seismic loadings.

2. To preclude any swinging or pendulum action of the block upon failure of one system, each wire rope system is reeved to both sides of the bottom block and upper block system.

The Fuel Building Overhead Crane is prevented by interlocks from moving over the new fuel pit during cask handling operations. The maximum lifting height for a loaded spent fuel cask is less than 30 feet. Mechanical antiderailing devices which prevent crane from being dislodged from the rail due to horizontal and vertical motion during an earthquake are provided on the Fuel Building Overhead Crane and designed to withstand an SSE. The concrete floors can withstand a fully loaded cask drop from the maximum lifting height of 29.25 feet.

A more detailed description of the Fuel Building Overhead Crane is provided in Section 9.1.4.

9.1.2.3 Safety Evaluation

Design of this storage facility in accordance with NRC Regulatory Guide 1.13, Revision 1, December 1975, ensures a safe condition under normal and postulated accident conditions. The spent fuel storage racks are designed to prevent criticality by the means of physical separation of the fuel assemblies, administrative control of storage configurations in the racks (based on initial enrichment and minimum burnup requirements), special neutron absorbing material, and crediting of soluble boron (for both normal and accident conditions). Placement of fuel in the spent fuel pool outside of a rack module, such as in the gap between the rack module and the pool wall, is administratively controlled. Consideration of criticality safety analysis is discussed in Section 3.1.6.3 and Section 4.3.2.6.

The Spent Fuel Pool Cooling and Cleanup System is discussed in Subsection 9.1.3.

All surfaces that come into contact with fuel assemblies are made of materials that are resistant to corrosion during normal and emergency water quality conditions.

9.1.3 SPENT FUEL POOL COOLING AND CLEANUP SYSTEM

9.1.3.1 Design Bases

The Spent Fuel Pool Cooling and Cleanup System, a common system for both units, is designed in compliance with Title 10, Code of Federal Regulations, Part 50 Appendix A, General Design Criteria (GDC) 1, 2, 3, 4, 5, 44, 45, 46, 56, 61 and 63 [1], [2], [3], [4], [5], [6], [7] to perform the following principal functions:

1. To remove heat generated by stored spent fuel elements from the station's spent fuel pools

2. To maintain the clarity and purity of water in the spent fuel pools, the transfer canal, the wet cask pit, the RWST, and the refueling cavities

The calculations for the amount of thermal energy to be removed by the spent fuel pool cooling system are in accordance with BTP ASB 9-2, "Residual Decay Energy for Light Water Reactors for Long-Term Cooling" (Rev. 2).

Two cooling loops are provided, each capable of simultaneously servicing both of the station spent fuel pools. Two cleanup loops are also provided [14]. System design parameters are presented in Table 9.1-1.

The water depth above the top of the fuel assemblies as well as the removal of fission products and other contaminants by the system's purification loop limits the dose rate at the surface of the pools to 2.5 mr/hr.

9.1.3.1.1 Spent Fuel Pool Cooling

The Spent Fuel Pool Cooling and Cleanup System is designed to limit the temperature of the spent fuel pools in the following cases:

1. Maximum Design Condition

   The maximum design condition bounds the maximum normal heat loads which occur during refueling outages (RFOs). Temperature limits are in accordance with the ACI Code and ANSI N210.

   The spent fuel pool bulk water temperatures are maintained at less than 150^{o}F for normal operation based on decay heat generation from a normal full core offload at 125 hours after shutdown, plus decay heat from the opposite unit's last refueling discharge plus decay heat from fuel assemblies from a maximum number of previous refuelings in both pools. At least 193 spaces in the spent fuel pools are assumed to remain available to accept one full core in accordance with ANSI N18.2 [15].

   In actual practice, the 193 spaces assumed to remain available for one full core offload may be used for fuel assembly storage, with the exception of 32 fuel storage locations which must remain available for dry cask offloading in accordance with the requirements of the Cask Technical specifications. Based on the conservative assumptions in the

design conditions and the insignificant decay heat from additional assemblies in these in these spaces, the analyses remain valid even when these spaces are used for assembly storage.

Back to back refueling outages are assumed for conservatism. The first outage is assumed to be complete prior to the start of the second. The start of the second outage is conservatively assumed to begin 4 months after the first. A normal full core offload is conservatively assumed to start at 75 hours after the reactor is subcritical and complete at 125 hours. Off-load rates do not affect the analyses. Refueling discharges (which remain in the pools after refueling) are assumed to be up to 96 fuel assemblies for 18 month fuel cycles. The outages are assumed to begin and end during normal refueling periods (September 15th through May).

The SSI conditions are assumed to be representative of one unit operation at full power and one unit shutdown during normal refueling periods.

The normal design SFP HX outlet temperature is 140^{o}F to protect the resins in the cleanup system.

2. Maximum Summer Design Conditions

The maximum summer design condition bounds the maximum normal heat loads which occur during normal power operation of both units. Temperature limits are in accordance with the ACI Code and ANSI N210.

The spent fuel pool water temperatures are maintained at less than 150°F for normal operation based on decay heat from the most recent refueling discharge at the end of the outage plus decay heat from the opposite unit's previous refueling discharge plus decay heat from a maximum number of refuelings in both pools. At least 193 spaces in the spent fuel pool are assumed to remain available to accept one full core in accordance with ANSI N18.2 [15]. In actual practice, the 193 spaces assumed to remain available for one full core offload may be used for fuel assembly storage, with the exception of 32 fuel storage locations which must remain available for dry cask offloading in accordance with the requirements of the Cask Technical Specifications. Based on the conservative assumptions in the design conditions and the insignificant decay heat from additional assemblies in these spaces, the analyses remain valid even when these spaces are used for assembly storage.

The SSI temperature is assumed to be normal maximum (102°F).

The normal design spent fuel pool heat exchanger outlet temperature is 140°F to protect the resins in the cleanup system.

3. Abnormal Maximum Design Conditions

The abnormal maximum design condition bounds the abnormal heat load from an emergency core offload (ECO) from either unit immediately after back to back refuelings of both units.

The spent fuel pool water temperatures are maintained at less than 212°F for two loop operation based on an emergency core offload 125 hours after shutdown, plus the most recent refueling discharge 31 days prior to shutdown, plus the opposite unit's previous refueling discharge 5 months prior to shutdown, plus decay heat from a maximum number of previous refuelings in both pools.

Two spent fuel cooling loops are assumed to be available if required to meet temperature limits for this case; a single active failure need not be considered for an emergency core offload. Also, no other coincident events are assumed.

The SSI temperature is assumed to be normal maximum (102°F).

For fuel assembly loading in the spent fuel pools versus time, see Table 9.1-4.

One train operation is not normal during maximum design conditions. For the maximum normal heat load with normal cooling systems in operation, and assuming a single active failure, the design maximum pool temperature is 200°F; however, the design spent fuel pool heat exchanger outlet temperature is 140°F to protect the resins in the cleanup system. The level in the pools is maintained by makeup from the Reactor Makeup Water System which also meets the single active failure criterion.

Spent Fuel Pool Cooling to one or both pools could be lost temporarily due to an upset, emergency or faulted plant condition. There is sufficient time to restore forced spent fuel pool cooling prior to boiling. The spent fuel pool cooling system is designed to maintain water temperatures less than 212°F for one loop operation during and after plant upset, emergency, and faulted conditions coincident with maximum design or maximum summer design conditions.

The spent fuel pool water temperature in the above cases is based on the corresponding component cooling water temperature at the inlet to the spent fuel pool heat exchanger. The maximum component cooling water supply temperature is 122°F during normal cooldown with Residual Heat Removal System operation. This condition coincident with maximum spent fuel pool heat loads is considered an unlikely event which is expected to result in a small temperature increase for a short period of time during the transient.

Table 9.1-4 provides the number of spent fuel assemblies in spent fuel pools by refueling outage. Table 9.1-1 provides the decay heat parameters for the three design conditions above corresponding to the spent fuel storage in Table 9.1-4.

9.1.3.1.2 Water Purification

Two purification loops are provided for removal of fission products and other contaminates by means of filtration and ion exchange. Each purification loop is capable of purifying flow from either the spent fuel pool cooling pumps or the refueling water purification pumps. The use of two loops ensures maintenance of acceptable activity and purity levels in the spent fuel pools in the event of failure of one loop. Each purification loop limits the activity of fission and corrosion products in the spent fuel water to a maximum of 5×10^{-9} Ci/cm^3, exclusive of tritium. Purification is sufficient to permit unrestricted access to the spent fuel storage area.

The optical clarity of the spent fuel pool water surface is maintained by use of the skimmer, strainer, and skimmer filter of the system. The purification loops similarly provide for cleanup of water in the Refueling Water Storage Tank, refueling cavities, transfer canal, and the cask loading pits.

Evaporation and gaseous activity released to the atmosphere from the spent fuel pools are controlled by an air sweep system which provides a high-velocity air curtain across the pools (see Section 9.4.2).

9.1.3.2 System Description

The Spent Fuel Pool Cooling and Cleanup System consists of two cooling loops, two purification loops, and one surface skimmer loop. The system flow diagram is shown on Figure 9.1-13. Each cooling loop includes a pump, heat exchanger, and associated piping, valving, and instrumentation. Heat is transferred via the spent fuel pool heat exchanger to the Component Cooling Water System (CCWS). Depending on heat loads the system can be lined up with each train of equipment cooling one or both pools. If a heat exchanger is out of service (ie. CCW flow isolated) the affected train can be used for mixing by lining up to both pools. When heat loads are low a single train can be used to cool both pools.

Except during abnormal design conditions, one spent fuel pool cooling water pump and one spent fuel heat exchanger are capable of cooling both pools in the event that one train is out of service. Spent fuel pool cooling pump 01 (Train A) is protected from low suction pressure by the Train A level instruments in Pool 1. Spent fuel pool cooling pump 02 (Train B) is protected from low suction pressure by the Train B level instruments in Pool 2. Both spent fuel cooling pumps 01 and 02 (Train A and Train B) are protected with identical devices. These devices have 2 out of 2 logic which protects the pumps from erroneous trips. Redundant instruments require an instrument failure or LO-LO level pump trip signal from each device to trip the pumps. Pump 1 protection is provided by level instrumentation located in pool 1, therefore it is preferred that an operating pump be aligned to its associated pool (pump 1 to pool 1 and pump 2 to pool 2). When the pools are cross connected the pool levels will remain approximately equal and the pump protection provided by the LO-LO pool level pump trip will protect the cross connected pump. The suction lines, protected by spent fuel pool suction screens, are located approximately four ft below the normal spent fuel pool water level. The return lines terminate approximately six ft above the fuel assemblies, which prevents siphoning below this point in the event of a pipe break. To further ensure that siphoning does not occur, each return line contains an antisiphon hole approximately six in. below the low water level which corresponds to 12 in. below the normal water level.

Wide range spent fuel pool instrumentation is installed for monitoring SFP level in compliance with Order EA-12-051. The SFP level instrumentation is used as part of the mitigation strategies implemented for NRC Order EA-12-049. See Sections 1.2.5.1 and 1.2.5.2 for additional information.

During heat removal operations, a portion of the spent fuel pool water may be diverted through a demineralizer and filter in either of the purification loops to maintain spent fuel pool water clarity and purity. Transfer canal water may also be circulated through a purification loop by opening either of the two spent fuel pool gates and opening the valves in the cooling loop discharge lines to the transfer canal. In addition, the water in the transfer canal or the cask pits may be purified

by aligning the cask pit and transfer canal drain pump to take suction from the pit or canal and to discharge through a purification loop and back to the same pit or canal.

To allow maintenance of the fuel transfer equipment, the transfer canal is drained by the cask pit and transfer canal drain pump. The transfer canal water is pumped through the purification loop and discharged into the recycle holdup tank, which is part of the Boron Recycle System (BRS). After maintenance, an auxiliary discharge is provided in the BRS to return water to the refueling transfer canal using the recycle evaporator feed pumps.

The cask pits are drained in a similar manner. The cask pit and transfer canal drain pump impels the water through the purification loop and into the recycle holdup tank and returns water to the pits by way of the recycle evaporator feed pumps.

The demineralizer and filter are isolated manually from the heat removal portion of the system. The purification equipment can thus be used to maintain refueling water purity while spent fuel pool heat removal operations proceed simultaneously. Connections are provided so that the refueling water may be pumped from either the Refueling Water Storage Tanks or the refueling cavities through a filter and demineralizer and discharged back to either the refueling cavities or the Refueling Water Storage Tanks. Purification flow is obtained by way of the refueling water purification pump.

The valve arrangement of the purification loops is such that either loop may be used to maintain refueling water purity while the heat removal portion of the system is isolated manually. It is also possible to simultaneously use one purification loop for spent fuel pools and one purification loop for refueling water.

To further assist in maintaining water clarity in the spent fuel pools, the water surface is cleaned by a skimmer loop. Water is removed from the surface by the skimmers, pumped through a strainer and filter, and then returned to the pool at remote locations from the skimmers. Temporary equipment is used as required to assist in refueling cavity cleanup. The refueling cavity skimmer system is no longer used.

The spent fuel pools are filled with water of approximately the same (or greater) boron concentration than the RWSTs. The RWST Technical Specification requirements are verified prior to entering the applicable MODE of operation should intermixing of the borated water occur between the RWST and the SFP during outages.

The reactor makeup system shown in Figure 9.2-5, as described in Section 9.2.3, is a seismic Category I system. Ventilation requirements are discussed in Section 9.4.2. Makeup water to compensate for evaporation losses is also available from the demineralized water supply or reactor makeup water system (see Section 9.2.3).

A secondary function of the spent fuel pool cooling and cleanup system is to form a small portion of a BR cross tie flowpath between the BRS and the SI systems. This Y-shaped flow path allows the efficient transfer of RHUT water to either RWST. See Figure 9.1-13 and Section 9.3.4.2 for a more detailed description of this cross tie.

9.1.3.2.1 Component Description

Codes and safety classifications for Spent Fuel Pool Cooling and Cleanup System components are given in Table 9.1-2.

All process lines shown on Figure 9.1-13 and identified as nuclear safety class are classified Seismic Category I. The boundary between the Seismic Category I piping and non-Seismic Category I piping coincides with the boundary between safety class piping and non-safety class piping. This separation appears on Figure 9.1-13 as safety class 3 to piping class 5 (NSS) transition except on vent, drain and test lines which are NNS downstream of the root valve.

All piping in contact with spent fuel pool water is made of stainless steel. The piping is welded except where flanged connections are used to facilitate maintenance.

For instrumentation applications, see Subsection 9.1.3.5.

9.1.3.3 Safety Evaluation

Spent fuel pool water is cooled by two redundant cooling loops, each of which contains a pump, heat exchanger, piping, valves, and instrumentation. In the event of a failure of spent fuel pool cooling pump or heat exchanger, the other loop ensures the continuity of effective cooling.

In case of spent fuel stored in both pools or a closely spaced refueling of both reactors, the two cooling loops may be used. In the event of a failure of one loop, the second loop ensures a minimum cooling and limits the water temperature to the cleanup system to less than 140°F to protect the demineralizer resins.

To detect leakage through the spent fuel pool liner welds, a channel is provided in back of the welds to form a leak chase. Concrete troughs are formed under the welds in the floor plate. Sections of welds which are leaking can be determined by observing which leak chase the water is coming from before the leak chases merge into a common drain header. Any leaks detected will be evaluated.

Furthermore, as indicated in Table 9.1-1, the pool capabilities are sufficiently large so that an extended cooling outage is required before pool temperatures reach 212°F. Thus the system can be shut down safely for reasonable time periods for maintenance or replacement of malfunctioning components. The effect of the evaporation rate from the pools on humidity are described in Section 9.4.2.

The suction lines inside the spent fuel pools are positioned to take suction four ft below the normal water level in order to minimize vortexing and the possibility of floating debris entering the system. Return lines from the spent fuel pool heat exchangers are located so that cooled water is discharged downward approximately six ft above the fuel assemblies. This ensures adequate dispersion of the cooled water around the stored spent fuel assemblies. The suction and return lines are located on opposite sides of the pools to prevent channeling and to obtain maximum circulation.

To protect against loss of water from the spent fuel pools, the spent fuel pool cooling pump suction lines penetrate the pool wall and terminate approximately four ft below the normal water level and the return lines terminate six ft above the fuel assemblies. The return lines contain

antisiphon holes. This arrangement precludes gravity draining of the pools in the event of a pipe break and ensures that sufficient shielding is maintained.

There are no drain lines connected to the pool. Appropriate redundancy, including a seismic Category I source, is provided for makeup water to the pools. Draining of either pool below the design water level is not considered credible. The rate of makeup water is greater than the rate of water loss. The radiological evaluation of the cleanup system is presented in Chapters 11 and 12.

9.1.3.4 Inspection and Testing Requirements

The active components of this system are in either continuous or intermittent use during normal plant operation. Periodic visual inspections and preventive maintenance are conducted as necessary. All components are accessible for periodic inspection except one section of each cooling pump suction line and one section of the cooling water return line. These sections, of all-welded construction, are embedded in concrete in the vicinity of the spent fuel pool and cask storage area.

To ensure that proper operational conditions exist for the spent fuel pool, periodic chemical analyses and operational surveillance shall be performed when this system is in use. Chemical analyses will be performed weekly for determining concentrations of chloride, fluoride and boron. Radioactivity levels and pH will be determined, as a minimum, on a weekly frequency. The chemical limits used in the monitoring of the spent fuel pool are, as follows:

Chlorides	0.15 ppm (maximum)
Fluorides	0.15 ppm (maximum)
PH	Variable
Boron Concentration	2400 ppm (minimum)
Radioactivity Levels	Activity levels shall be maintained as low as reasonably achievable (ALARA)

The bases for these limits are to minimize the potential for corrosion attack, to ensure the proper reactivity control and to maintain the radioactivity levels as low as reasonably achievable (ALARA).

For information on the sampling and monitoring of the spent fuel pool demineralizers and filters, see Section 12.2.1.2.2.

9.1.3.5 Instrument Requirements

The instrumentation for the Spent Fuel Pool Cooling and Cleanup System is discussed in the following paragraphs.

1. Temperature

Local temperature indicators are provided at the spent fuel panel for spent fuel pools and the refueling cavities and also provided at the outlet of the spent fuel pool heat exchanger.

Annunciation is given in the spent fuel pool panel and a common trouble alarm in the control room when the normal temperature is exceeded.

2. Pressure

Pressure gauges are provided at the discharge of the pumps used in this system. Local differential pressure indicators are connected across the spent fuel pool filter, spent fuel pool skimmer filter, spent fuel pool demineralizer and resin trap, with an alarm on the spent fuel pool panel and a common trouble alarm in the control room.

3. Flow

Local indicators are provided to indicate the flows through the purification loop and in the spent fuel pool return lines. Low flow in the pool return lines is also alarmed in the spent fuel pool panel and a common trouble alarm in the control room.

4. Level

The spent fuel pool water levels are measured to give alarms in the local panel and a common alarm in the Control Room for high and low water level. A LO-LO water level trip is provided for cooling pump protection. The level instrumentation is designed to provide 2 out of 2 logic for cooling pump protection on LO-LO water level. Continuous level indication is provided to the Control Room via the plant computer.

See FSAR Section 1.2.5.2 for a description of the wide range level instrumentation for beyond design basis events.

5. Radiation

Area radiation monitors are located in the fuel pool area. See Section 12.3 for a description of the area radiation monitors.

9.1.4 FUEL HANDLING SYSTEM

9.1.4.1 Design Bases

The fuel handling system (FHS) consists of equipment and structures used for conducting the refueling operation in a safe manner; this system conforms to GDC 61 and 62 of 10 CFR Part 50, Appendix A.

The following design bases apply to the FHS:

1. Fuel handling devices have provisions to avoid dropping or jamming of fuel assemblies during transfer operation.

2. Handling equipment has provisions to avoid dropping of fuel handling devices during the fuel transfer operation.

3. Handling equipment used to raise and lower spent fuel has a limited maximum lift height so that the minimum required depth of water shielding is maintained.

4. The Fuel Transfer System (FTS), where it penetrates the Containment, has provisions to preserve the integrity of the Containment pressure boundary.

5. Criticality during fuel handling operation is prevented by the geometrically safe configuration of the fuel handling equipment.

6. In the event of an SSE, handling equipment cannot fail in such a manner as to damage seismic Category I equipment.

7. The inertial loads imparted to the fuel assemblies or core components during handling operations are less than potential damage-causing loads.

8. Physical safety features are provided for personnel who operate handling equipment.

9.1.4.2 System Description

The FHS consists of the equipment needed for the refueling operation on the reactor core. Basically, this equipment is comprised of fuel assemblies, core component and reactor component hoisting equipment, handling equipment, and a FTS. The structures associated with the fuel handling equipment are the refueling cavity, including the fuel storage area in the Containment, and the refueling canal, cask pit, and spent fuel pools in the Fuel Building.

In addition, heavy loads associated with fuel handling are lifted by the Containment Polar Crane and by the Fuel Building Overhead Crane. The Containment Polar Crane is used to remove the reactor vessel head and internals for refueling. The Fuel Building Overhead Crane is used to lift new and spent fuel casks.

The elevation and arrangement drawings of the fuel handling facilities are on Figures 1.2-12, 1.2-13, 1.2-18, 1.2-19, 1.2-38, 1.2-39, and 1.2-40.

9.1.4.2.1 Fuel Handling Description

New fuel assemblies received for initial refueling are removed one at a time from the shipping container and moved to the new fuel assembly inspection area. After completion of inspection, the acceptable new fuel assemblies are stored in the new fuel storage racks or are lowered by the new fuel elevator and stored in the spent fuel racks in the spent fuel pools.

The fuel handling equipment is designed to handle the spent fuel assemblies underwater from the time they leave the reactor vessel until they are placed in a container for shipment from the site. Underwater transfer of spent fuel assemblies provides an effective, economic, and transparent radiation shield, as well as a reliable cooling medium for removal of decay heat. The boric acid concentration in the water is sufficient to preclude criticality.

The refueling cavity in Containment is divided by a refueling gate into two areas: the reactor vessel area and Containment fuel transfer area. These areas are flooded only during plant shutdown for refueling. The entire refueling cavity is kept full of water whenever spent fuel is being moved or stored in it. The refueling gate is used to isolate either area of the refueling cavity for drain down if required. The associated fuel handling structures in the Fuel Building are divided by refueling gates into three areas: the fuel transfer canal, the cask pit, and the spent fuel pools. These areas are kept full of water whenever spent fuel is being moved or stored in them. The Containment fuel transfer area and the fuel transfer canal are connected by a fuel transfer tube which is fitted with a blind flange on the Containment end and a gate valve on the Fuel Building end. The blind flange is in place except during refueling to ensure Containment integrity. Fuel is carried through the tube on an underwater transfer car.

Fuel is moved between the reactor vessel and the Containment fuel transfer area by the refueling machine. A control rod cluster changing fixture is provided for transferring control elements from one fuel assembly to another. Control elements may also be changed in the Fuel Building. The FTS is used to move fuel assemblies between the Containment Building and the Fuel Building.

In the Fuel Building, fuel assemblies are moved about by the fuel handling bridge crane and the Fuel Building overhead crane. Fuel movements are limited such that no more than two fuel assemblies are outside of an approved shipping container, fuel inspection station, storage rack or fuel transfer tube at any one time and a minimum 12-inch edge-to-edge distance is maintained between such assemblies. For underwater fuel handling operations, a minimum 12-inch edge-to-edge distance is maintained between any assembly that is outside of an approved shipping container, storage rack, or fuel transfer tube and any other assembly; including those in shipping containers, storage racks, or fuel transfer tubes. These spacing requirements, in combination with fuel assembly design and enrichment characteristics, assure that Keff remains below 0.95 for dry (unmoderated) as well as flooded (unborated water) conditions.

A discussion of the spent fuel pool Region I and Region II oversized inspection cells is contained in the Technical Specification Bases 3.7.17.

Decay heat, generated by the spent fuel assemblies in the spent fuel pools, is removed by the Spent Fuel Pool Cooling and Cleanup System. After a sufficient decay period, the spent fuel assemblies are removed from the fuel racks and loaded into a spent fuel shipping cask for removal from the site.

9.1.4.2.2 Refueling Procedure

The refueling operation follows a detailed procedure which provides safe and efficient refueling. Prior to initiating refueling operation, the RCS is borated and cooled down to refueling shutdown conditions as specified in the Technical Specifications. Criticality protection for refueling operations is specified in the Technical Specifications. The following significant points are ensured by the refueling procedure:

1. The refueling water and the reactor coolant contain sufficient boron so that with the negative reactivity of control rods, the core can be maintained shutdown during the refueling operation. It is also sufficient to maintain the core subcritical in the unlikely event, that all of the rod cluster control assemblies (RCCA) were removed from the core.

2. The water level in the refueling cavity is high enough to keep the radiation levels within acceptable limits when the fuel assemblies are being removed from the core.

The refueling operation is divided into four major phases: preparation, reactor disassembly, fuel handling, and reactor assembly. A general description of a typical refueling operation through the phases is given as follows:

1. Phase I Preparation

   The reactor is shut down and borated to Refueling shutdown condition as specified in the Technical Specifications. The coolant level in the reactor vessel is lowered to a point slightly below the vessel flange and the fuel transfer equipment and refueling machine are checked for proper operation.

2. Phase II Reactor Disassembly

   The external air duct is removed from the CRDM Ventilation System's Air Handling Units and plenum (Unit 1 only). Cables are disconnected to allow removal of the roll-away missile shield (Unit 2 only) and the vessel head. Air ducts (Unit 2 only) and insulation are removed from the vessel head. The refueling cavity is then prepared for flooding by checking the underwater lights, tools, and FTS, closing the refueling canal drain lines, and removing the blind flange from the fuel transfer tube. With the refueling cavity prepared for flooding, selected portions may be filled in preparation for vessel head removal. With the water level providing a safe shielding depth (see Subsection 9.1.4.3.4), the vessel head is taken to its storage pedestal. The remainder of the refueling cavity is flooded to the depth specified in the Technical Specifications. The control rod drive shafts are disconnected and, with the upper internals, are removed from the vessel. The fuel assemblies and RCCA are now free from obstructions and the core is ready for refueling.

3. Phase III Fuel Handling

   The refueling sequence is started with the refueling machine. The positions of partially spent assemblies are changed, and new assemblies are added to the core.

   The refueling machine is positioned over a selected fuel assembly and the fuel assembly is lifted to a predetermined height sufficient to clear the reactor vessel. The refueling machine is moved to align the fuel assembly with the fuel transfer system and the fuel assembly is placed into the fuel assembly container of the transfer car. The container is pivoted to the horizontal position by the lifting arm and is moved through the fuel transfer tube to the Fuel Building by the transfer car. The fuel assembly container is pivoted to the vertical position. The fuel assembly is unloaded by the fuel handling bridge crane using the spent fuel handling tool and placed in its designated storage location.

   Fuel and insert components can be shuffled in either Containment or Fuel Building using the tools designated for components.

   Any partially spent fuel not designated for removal from the core is shuffled to its desirable new location for the next cycle as the reactor is reloaded. Fuel is reloaded into the reactor by reversing the steps necessary for fuel removal.

4. Phase IV Reactor Assembly

Reactor assembly, following refueling, is essentially achieved by reversing the operations given in Phase II, Reactor Disassembly.

9.1.4.2.3 Component Description

1. Refueling Machine

The refueling machine (Figure 9.1-3) is a rectilinear bridge and trolley system with a vertical mast extending down into the refueling water. The bridge spans the refueling cavity and runs on rails set into the edge of the refueling cavity. The bridge and trolley motions are used to position the vertical mast over a fuel assembly in the core. A long tube with a pneumatic gripper on the end is lowered down out of the mast to grip the fuel assembly and the tube is long enough so that the upper end is still contained in the mast when the gripper end contacts the fuel. Mounted on the trolley, a winch raises the gripper tube and fuel assembly up into the mast tube. The fuel is transported while inside the mast tube to its new position.

The refueling machine uses a programmable logic controller and a variable frequency control system that gives stepless variable speeds from zero to full speed.

All controls for the refueling machine are mounted on a console in the trolley. The bridge and trolley are positioned by use of indexing plates. A video camera and television monitor provides a picture for the operator of the north-south indexing plate. The drives for the bridge, trolley, and winch are variable speeds and include a separate inching control on the winch. The maximum speeds for the bridge, trolley and main hoist are 60, 26 and 26 ft/min respectively. The auxiliary monorail hoist on the refueling machine has a two-step magnetic controller to give hoisting speeds of approximately 7 and 20 ft/min.

Electrical interlocks and limit switches on the bridge and trolley drives prevent damage to the fuel assemblies. The winch is also provided with limit switches to prevent a fuel assembly from being raised above a safe shielding depth should the limit switch fail. In an emergency, the bride, trolley, and winch can be operated manually using a handwheel on the motor shaft.

The refueling machine is provided with a television system which permits viewing of fuel assembly positions and fuel movements.

Additionally, the refueling machine is provided with a fuel sipping system. This “In-Mast Sipping System” is a set of hardware that provides a means of performing on-line, quantitative leak testing of fuel assemblies in the refueling mast during normal fuel handling operations. Included in the sipping system are: mechanical equipment connected to the refueling machine mast, a control module, a pump and valve module, and a detection and recording module.

2. Fuel Handling Bridge Crane

The Fuel Handling Bridge Crane (FHBC) (Figure 9.1-4), is a wheel-mounted walkway spanning the spent fuel pools, which carries two electric trolley hoist on an overhead structure. This crane is used for handling fuel assemblies within the spent fuel pools, refueling canal, and cask pits by means of a long-handled tool suspended from the hoist. A load monitoring device is integral to the crane for monitoring fuel assembly loads. The hoist travel and tool length are designed to limit the maximum lift of a fuel assembly to a safe shielding depth.

Although the FHBC has two trolley-hoist assemblies, redundant and diverse interlocks are provided to ensure the FHBC will not have the ability to suspend more than one load (e.g. fuel assembly) at a time. No bypass functions are provided for these interlocks.

The fuel handling bridge and trolley have single speed motor controlled by a variable frequency controller. The programmed maximum speed of the bridge and trolley is 40 feet per minute. The hoist has a single speed motor controlled by a variable frequency controller. The hoist speeds are limited to 3 feet per minute in the slow zones and 21 feet per minute outside slow zones.

For loads equal to or greater than the combined weight of a fuel assembly and the handling tool, the maximum hoist speed is automatically limited to 21 fpm. For loads less than the combined weight of a fuel assembly and the handling tool, the maximum hoist speed is automatically allowed to operate at the higher ultra speed of 31.5 fpm.

To ensure safe handling of the fuel, the FHBC allows bridge movement only if the hoist is at zero speed. The bridge can only operate at microspeed (3 feet per minute) if the hoist is not at full up position when loaded with a Fuel Assembly and the Fuel Assembly Handling Tool. Hoist operation is not permitted while either the bridge or the trolley is moving. Trolley operation is not permitted while the hoist is moving. The trolley can only operate at microspeed if the hoist is not at the full up position when loaded with a Fuel Assembly and the Fuel Assembly Handling Tool. Bridge and trolley motion can only be combined if the hoist is completely up or if the suspended weight is less than the combined weight of a Fuel Assembly and the Fuel Assembly Handling Tool (i.e. equal to 1200 pounds or less). If this is not the case, only the first activated movement (bridge or trolley) is active.

Additional automatic features include:

During fuel movement, when the weight of a fuel assembly is detected, bridge and trolley motion is not possible while hoisting or lowing unless the fuel assembly is indicated to be clear of the rack or upender.

The fuel transfer cart upender cannot be raised unless these functions are satisfied:

Bridge axis is clear of the upender.

Trolley axis is clear of the upender.

Hoist with fuel is fully raised.

Hoist is above the lower slow zone without fuel.

Trolley movement is limited in travel at each end of the girder. The trolley automatically slows then stops as the trolley approaches the end stops. This is an added feature which prevents the trolley from impacting the end stops at full speed.

Bridge speed is limited in travel at each end of the runway. The bridge automatically slows then stops the bridge movement as it approaches the end stops.

The FHBC control station is automatically moved by a motor assembly so that it remains with the hoist-trolley it is currently controlling in order to assure that the fuel handler remains with the fuel assembly being moved with the crane. A switch to allow the fuel handler to override this feature is provided.

In addition to these automatic features, all fuel handling operations are carried out using strict administrative controls.

3. Containment Fuel Handling Bridge Crane

The Containment fuel handling bridge crane is a wheel-mounted walkway spanning the Containment upender area, which carries an electric trolley hoist on an overhead structure. The Containment fuel handling bridge crane is used exclusively for handling fuel assemblies within the Containment by means of a long-handled tool suspended from the hoist. The hoist travel and tool length are designed to limit the maximum lift of a fuel assembly to a safe shielding depth.

4. New Fuel Elevator

The new fuel elevator (Figure 9.1-5) consists of a box-shaped elevator assembly with its top end open and sized to house one fuel assembly. It is used to lower a new fuel assembly to the bottom of the cask pit refueling equipment area where it is transported to the storage racks by the fuel handling bridge crane.

It is also used to raise the fuel assembly for manipulation during fuel reconstitution. As a precaution, additional restraints are established during this mode to prevent the elevator from raising the fuel within 10 feet of fuel pool water level.

5. Fuel Transfer System

The FTS (Figure 9.1-6) includes an underwater, electric motor driven transfer car that runs on tracks extending from the refueling canal, through the transfer tube, and into the Containment fuel storage area; a hydraulically actuated lifting arm is at each end of the transfer tube. The fuel container in the Containment fuel storage area receives a fuel assembly in the vertical position from the refueling machine. The fuel assembly is then lowered to a horizontal position for passage through the transfer tube. After passing through the tube, the fuel assembly is raised to a vertical position for removal by a tool suspended from a hoist mounted on a fuel handling bridge crane in the Fuel Building.

The fuel handling bridge crane then moves to a storage loading position and places the spent fuel assembly in the spent fuel storage racks.

During reactor operation, the transfer car is stored in the fuel storage area. A blind flange is bolted on the Containment end of the transfer tube to seal the reactor containment. The terminus of the tube outside the Containment is closed by a gate valve.

6. Rod Cluster Control Changing Fixture

The rod cluster control changing fixture is supplied for periodic rod cluster control element inspections and for transfer of rod cluster control elements from one fuel assembly to another if this is ever required (Figure 9.1-7). The major subassemblies which comprise the changing fixture are the frame and track structure, the carriage, the guide tube, the gripper, and the drive mechanism. The carriage is a movable container supported by the frame and track structure. The tracks provide a guide for the four flanged carriage wheels and allows horizontal movement of the carriage during changing operation.

In order to locate each of the three carriage compartments directly below the guide tube, the positioning stops are provided on both the carriage and frame. Two of these compartments are designed to hold individual fuel assemblies; the third is made to support a single rod cluster control element. Situated above the carriage and mounted on the refueling canal wall is the guide tube. It provides for the guidance and proper orientation of the gripper and rod cluster control element as they are being raised and lowered. The gripper is a pneumatically actuated mechanism responsible for engaging the rod cluster control element. It has two flexure fingers which can be inserted into the top of the rod cluster control element when air pressure is applied to the gripper piston. Normally the fingers are locked in a radially extended position. Mounted on the operating deck is the drive mechanism assembly, which is made up of the manual carriage drive mechanism, the revolving stop operating handle, the pneumatic selector valve for actuating the gripper piston, and the electric hoist for elevation control of the gripper.

7. Spent Fuel Assembly Handling Tool

The spent fuel assembly handling tool (Figure 9.1-8) is used to handle new and spent fuel assemblies in the fuel storage areas. It is a manually actuated tool, suspended from the fuel handling bridge cranes, which uses four cam actuated latching fingers to grip the underside of the fuel assembly top nozzle. The operating handle to actuate the fingers is located at the top of the tool. When the fingers are latched, a pin is inserted into the operating handle which prevents the fingers from being accidently unlatched during fuel handling operations.

8. New Fuel Assembly Handling Tool

The new fuel assembly handling tool (Figure 9.1-9) is used to lift and transfer fuel assemblies from the new fuel shipping containers to the new fuel storage racks or new fuel elevator. A manually actuated tool suspended from the Fuel Building overhead crane or fuel handling bridge crane uses four cam-actuated latching fingers to grip the underside of the fuel assembly top nozzle. The operating handles which actuate the fingers are located on the side of the tool. When the fingers are latched, the safety screw is turned in to prevent the accidental unlatching of the fingers.

9. Reactor Vessel Head Lifting Device

The reactor vessel head lifting device consists of a welded and bolted structural steel frame with suitable rigging to enable the crane operator to lift the head and store it during refueling operations. Attached to the head lifting device are the monorail and hoists for the reactor vessel stud tensioners.

10. Reactor Internals Lifting Device

The reactor internals lifting device (Figure 9.1-10) is a structural frame suspended from the polar crane. The frame is lowered onto the guide tube support plate of the internals and is mechanically connected to the support plate by three breech-lock-type connectors. Bushings on the frame engage guide studs in the vessel flange to provide guidance during removal and replacement of the internals package.

11. Reactor Vessel Stud Tensioner

The stud tensioners (Figure 9.1-11) are used to secure the head closure joint at every refueling. The stud tensioner is a hydraulically operated device that uses oil as the working fluid. The device permits preloading and unloading of the reactor vessel closure studs at cold shutdown conditions. Stud tensioners minimize the time required for the tensioning or unloading operation. The studs are tensioned to their operational load in an approved sequence to prevent high stresses in the flange region and unequal loadings in the studs. Relief valves on each tensioner prevent overtensioning of the studs because of excessive pressure.

12. Containment Fuel Storage Rack

This rack is located in the Containment fuel transfer area to facilitate fuel assembly movements. This rack may be used for interim storage of new or spent fuel. Spent fuel or irradiated components (i.e., RCCAs, BPRAs, Thimble Plugs, Secondary Sources, and Reactor Vessel Irradiated Specimens) are not stored in this rack unless the entire refueling cavity is full.

13. Not used.

14. Containment Polar Crane

The Containment Polar Crane is a single trolley traveling bridge crane rotating on a single rail circular track and provided with cab and radio control stations. See Figure 9.1-14.

The crane is used for construction lifts and for refueling and maintenance operations with occasional lifts at the maximum rated crane lifting capabilities. See Section 9.1.4.3.2 for the Polar Crane seismic considerations. See Section 3.8.3.4.10 for the applicable industry standard.

A telescopic jib crane is also provided inside Containment for refueling and maintenance. It is supported by a structural steel support mounted on east-west divider wall between S.G. compartments at elevation 905’-9”. Both the Polar Crane and Telescopic Jib Crane operating at the same time constitutes Multi-crane Operation which requires special

administrative controls to reduce the likelihood of unplanned crane interactions. See Section 3.8.3.1.2 for a description of the administrative controls and design features for the Polar Crane and the Telescopic Jib Crane.

During normal operation of the plant the crane is set in the pre-established parked position (see Figure 1.2-8). Prior to entering outages, crane operation is required to perform certain inspections, functional checks and preventative maintenance (PM) activities. The polar crane may be energized during MODES 1-4. The crane will remain essentially in the park position (FSAR Figure 1.2-8) during MODES 1 through 4 unless maintenance is being performed on it. The Containment Access Rotating Platform is allowed to be used for maintenance on (including modifications to) the Polar Crane in Modes 1 through 4. During Polar Crane maintenance, the polar crane may be operated through its full 360 degree travel range. The trolley is to remain as close to the liner plate/ crane rail as possible. The hook is not to be moved outside the park position. The hook should not be placed in a location to impact any safety related SSC. No load will be carried by the polar crane during MODES 1-4. A compensatory measure (a dedicated person in direct communication with the control room, at the power supply disconnect outside containment) is established to prevent inadvertent operation of crane due to High Energy Line Break in containment.

The main components of the Containment Polar Crane are:

a. Bridge with welded box-section-type girders, cab footwalk and trolley rails (see Figure 9.1-14, Sheet 1 of 3).

b. Bridge end sills and trucks.

c. Crane trolley consisting of:

   Frame

   Mainhoist machinery (175 tons/475 tons)

   Auxiliary hoisting machinery (20 tons)

   Upper sheave assembly

   Bottom block

   Upper equalizer bar

   Main and auxiliary drums

   Main and auxiliary ropes

   (see Figure 9.1-14, Sheet 2 of 3)

d. Electrical equipment

e. Control system equipment

15. Fuel Building Overhead Crane

The Fuel Building Overhead Crane is the primary means of transporting nuclear fuel in and out of the fuel handling area of the Fuel Building. Its range includes the spent fuel cask loading area, the new fuel storage pit, the cask handling area, the new fuel receiving area and the railroad loading and unloading area. (see Figures 1.2-9 and 1.2-40). The crane is designed as a traveling bridge crane with a single trolley and is provided with a cab.

Reference [17] provides information regarding the design of the Fuel Building Overhead Crane.

16. Refueling Gates

The refueling gates consist of lift type gates and hinged gates. Lift gates are used for prohibiting a transfer of water across the gate from one section of the refueling cavity or canal to the other sections of the cavity or canal. Hinged gates are used for prohibiting the exit of water from the spent fuel pool into an empty refueling canal. The gates are designed to withstand forces resulting from the simultaneous combination of seismic forces and the seismically induced hydrodynamic fluid pressure and maintain their leakproof integrity.

The refueling gates are provided with inflatible seals to maintain a watertight barrier under all conditions. The seals are designed to be passive devices and do not require a continuous pneumatic supply. A minimum of one seal suitably isolated from its NNS pneumatic supply is required to perform a gate's Nuclear Safety functions. See Section 9.3.1 for a description of the Compressed Air Systems.

During spent fuel transfer in the transfer canal adjacent to a pool, the spent fuel pool involved in the transfer is normally open and full to its normal operating level. If an adjacent pool (not involved in the transfer) is below its normal operating level and is isolated by its gate from the canal, the level of water in the pool is maintained such that any failure of the gate would not result in loss of spent fuel cooling or in excessive radiation levels from a fuel assembly in transit.

The single-failure-proof Fuel Building Overhead Crane is used to move and maintain the lift gate in the Fuel Building. The single-failure-proof Wonderhoist is used to move the hinged (swing) gates from their installed positions to and from the Wet cask Pit for maintenance. Interlocks are provided to prevent the Wonderhoist from moving over spent fuel stored in the spent fuel pools [See Appendix 1A(B), Regulatory Guides 1.13 and 1.104].

17. Rod Cluster Control Change Tools

Either of two Rod Control Cluster Change Tools are used for removing or replacing an RCC in a fuel assembly. One tool is fully mechanical, the other is mechanical with an electro-pneumatic winch. Tool selection is at the discretion of the refueling crew.

The fully mechanical tool uses telescoping concentric tubes that are manually locked or unlocked by the operator in the fully extended (RCC removed from fuel) or collapsed

(RCC inserted into fuel) positions. Keyed slots in the tubes assure proper operation by only allowing the repositioning of control levers at specific elevations of the tool. The gripper is latched or unlatched via a control lever that spreads or relaxes the gripper jaw.

The mechanical tool with electro-pneumatic winch consists of an electrically-powered winch and controls for gripper actuation and winch operation. Internal limit switches and an overload switch protect the RCC from inadvertent unlatching during RCC movement, and prevent over-travel in either direction of winch movement. The gripper is latched or unlatched via an air-operated cylinder and slug that either spread (RCC latched) or relax (RCC unlatched) the gripper jaw.

The bottom of each tool is equipped with locating pins to properly orient the tool with respect to the fuel assembly top nozzle.

18. Portable Underwater Lights

Portable underwater lighting adequate for safe handling of irradiated fuel assemblies is provided. These lights are used as needed in the refueling cavity, transfer canals, and spent fuel pools.

9.1.4.2.4 Applicable Design Codes

The design codes and standards used for the FHS are given in Subsection 9.1.4.3 and in Table 17A-1.

The Containment Polar Crane and the Fuel Building Overhead Crane are designed to seismic Category I requirements (as defined in Section 3.2 and listed in Table 17A-1) as discussed in Section 9.1.4.3.2.

As stated in Section 3.8.3.2.3, the criteria of CMAA Specification No. 70 are also used for the design, testing and maintenance of both cranes.

9.1.4.3 Safety Evaluation

9.1.4.3.1 Safe Handling

Design criteria for the FHS are as follows:

1. The primary design requirement of the equipment is reliability. A conservative design approach is used for all load bearing parts. Where possible, components are used that have a proven record of reliable service. Throughout the design, consideration is given to the fact that the equipment spends long idle periods stored in a high humidity and temperature atmosphere (up to 122°F and 100% humidity).

2. Except as otherwise specified, the refueling machine is designed and constructed in accordance with Crane Manufacturers Association of America, Inc. (CMAA) Specification 70.

The fuel handling bridge crane designed and constructed in accordance with ASME NUM-1, Type B for Hoisting equipment and ASME NOG-1. Type II for Trolley and rest of the crane.

3. Design load for the refueling machine and Fuel Building fuel handling bridge crane shall be normal dead and live loads plus maximum hoist load.

4. The allowable stresses for the refueling machine structures supporting the weight of the fuel are as specified in Subarticle XVII-2200 of the ASME B&PV Code, Section III, Appendix XVII (1974).

   Hoisting components loaded in simple tension have an allowable stress of 0.20 ultimate stress maximum.

5. All components critical to the operation of the refueling equipment used over the reactor-vessel or located so that parts can fall into the reactor are assembled with the fasteners positively restrained from loosening under vibration. This is applicable to the following equipment: refueling machine, and the control rod drive shaft unlatching tool (critical fasteners).

   Spring- or tooth-type lockwashers are not considered as positive restraint. Fasteners on purchased components that cannot be wired or tackwelded are coated with an acceptable locking compound or, alternatively, are positively restrained by another acceptable method.

Industrial codes and standards used in the design of the fuel handling equipment are as follows:

1. The refueling machine conforms to applicable sections of CMAA Specification 70.

   The fuel handling bridge crane conforms to applicable sections of ASME NUM-1, Type B for Hoisting equipment and ASME NOG-1. Type II for Trolley and the rest of the crane.

2. The refueling machine's structural components conforms to the ASME B&PV Code, Section III, appendix XVII, Subarticle XVII-2200 (1974).

3. Electrical equipment conforms to the applicable standards and requirements of National Electrical Code, NFPA 70, and NEMA Standards MGI and ICS for design, installation, and manufacturing.

4. Materials conform to the specifications of ASTM standards.

5. Safety standards include the OSHA standards, 29 CFR Parts 1910 and 1926, including load testing requirements, the requirements and ANSI N18.2, NRC Regulatory Guide 1.29, and GDC 61 and 62.

The refueling machine design includes the following provision to ensure safe handling of fuel assemblies:

1. Electrical Interlocks

a. Bridge, Trolley, and Hoist Drive Mutual Interlocks

Bridge, trolley, and winch drives are mutually interlocked, using redundant interlocks to prevent simultaneous operation of any two drives; therefore, they can withstand a single failure.

b. Bridge and Trolley Drive, Gripper Tube Up

Bridge and trolley drive operation is prevented except when the gripper tube up position switches are actuated. The interlock is redundant and can withstand a single failure.

c. Gripper Interlock

An interlock is supplied which prevents the opening of a solenoid valve in the air line to the gripper except when zero suspended weight is indicated by a force gauge. As backup protection for this interlock, the mechanical weight-actuated lock in the gripper prevents operation of the gripper under load even if air pressure is applied to the operating cylinder. This interlock is redundant and can withstand a single failure.

d. Excessive Suspended Weight

Two redundant excessive-suspended-weight switches open the hoist drive circuit in the up direction when the loading is in excess of 110 percent of a fuel assembly weight. The interlock is redundant and can withstand a single failure.

e. Hoist-Gripper Position Interlock

An interlock in the hoist drive circuit in the up direction permits the hoist to be operated only when either the open or closed indicating switch on the gripper is actuated. The hoist-gripper position interlock consists of two separate circuits that work in parallel so that one circuit must be closed for the hoist to operate. If one or both interlocking circuits fail in the closed position, an audible and visual alarm on the console is actuated. The interlock, therefore, is not redundant but can withstand a single failure since both an interlocking circuit and the monitoring circuit has to fail to cause a hazardous condition.

2. Bridge and Trolley Holddown Devices

Both refueling machine bridge and trolley are horizontally restrained on the rails by two pairs of guide rollers, one pair at each wheel location on one truck only. The rollers are attached to the bridge truck and contact the vertical faces on either side of the rail to prevent horizontal movement. Vertical restraint is accomplished by antirotation bars located at each of the four wheels for both the bridge and trolley. The antirotation bars are bolted to the trucks and, for the bridge restraints, extended under the rail flange; while, for the trolley restraints, they extend beneath the top flange of the bridge grider which supports the trolley rail. Both horizontal and vertical restraints are designed to adequately withstand the forces and overturning moments resulting from the SSE.

3. Design Load

The design load for structural components supporting the fuel assembly is the dead weight plus 4800 lb (three times the fuel assembly weight).

4. Main Hoist Braking System

The main hoists are equipped with two independent braking systems. A solenoid release-spring-set electric brake is mounted on the motor shaft. This brake operates in the normal manner to release, upon application of current to the motor, and set when current is interrupted. The second brake is a mechanically actuated load brake internal to the hoist gearbox that sets if the load starts to overhaul the hoist. It is necessary to apply torque from the motor to raise or lower the load. In raising, this motor cams the brake open; in lowering, the motor slips the brake allowing the load to lower. This brake actuates upon loss of torque from the motor for any reason and is not dependent on any electrical circuits. On the main hoist the motor brake is rated at 350-percent operating load and the mechanical brake at 300 percent.

5. Fuel Assembly Support System

The main hoist system is supplied with redundant paths of load support such that failure of any one components does not result in free fall of the fuel assembly. Two wire ropes are anchored to the winch drum and carried over independent sheaves to a load equalizing mechanism on the top of the gripper tube. In addition, supports for the sheaves and equalizing mechanism are backed up by passive restraints to pick up the load in the vent of failure of this primary support. Each cable system is designed to support 13,750 lb or 27,500 lb acting together.

The working load of fuel assembly plus gripper is approximately 2500 lb.

The gripper itself has four fingers that grip the fuel and support the fuel assembly weight.

The gripper mechanism contains a spring-actuated mechanical lock which prevents the gripper from opening unless the gripper is under a compressive load.

The following safety features are provided for in the FTS.

1. Transfer Car Permissive Switch

The transfer car controls are located in both Fuel and Reactor Building and the conditions of the lifting arm as well as transfer car are displayed on both consoles during normal operating condition of the consoles. The transfer car permissive interlock allows either operator to move the car in either direction as required. The operator however can only operate the lifting arm on side that the operator is located on.

Transfer car operation is possible only when both lifting arms are in the down position as indicated by the limit switches. The load system and frame up limit switch are the backups for the lifting arm interlock. Assuming the fuel container is in upright position in the Reactor Building, and the lifting arm interlock circuit fails in the permissive condition,

neither operator can operate the car because of the PLC interlock and mechanical interlock. The interlock, therefore can withstand a single failure.

2. Lifting Arm - Transfer Car Position

   Two redundant interlocks allow lifting arm operation only when the transfer car is at the respective end of its travel and therefore can withstand a single failure.

   Of the two redundant interlocks which allow lifting arm operation only when the transfer car is at the end of its travel, one interlock is a position limit switch in the control circuit. The backup interlock is a mechanical latch device on the lifting arm that is opened by the car moving into position.

3. Transfer Car - Valve Open

   The interlock on the transfer tube valve permits transfer car operation only when the transfer tube valve position switch indicates that the valve is fully open. However, if the interlock fails in the permissive mode, the car can be started with the valve not fully open. The transfer car contacts the valve, stopping the transfer car movement. The car would stall without damage to the equipment or fuel. As soon as the valve is opened fully, the transfer car is freed.

4. Transfer Car - Lifting Arm

   The transfer car lifting arm interlock is primarily designed to protect the equipment from overload and possible damage if an attempt is made to move the car while the fuel container is in the vertical position. This interlock is redundant and can withstand a single failure. The basic interlock is a position limit switch in the control circuit. The backup interlock is a mechanical latch device that is opened by the weight of the fuel container when it is in the horizontal position

5. Lifting Arm - Refueling Machine

   The containment side lifting arm is interlocked with the refueling machine. Whenever the transfer car is located in the refueling canal, the lifting arm cannot be operated unless the refueling machine mast is in the fully retracted position or the refueling machine is over the core.

6. Lifting Arm - Fuel Building Fuel Handling Bridge Crane

   The Fuel Building side lifting arm is interlocked with the fuel handling bridge crane. The lifting arm cannot be operated unless the fuel handling bridge crane is not over the lifting arm area.

The Fuel Building fuel handling bridge crane includes the following safety features:

1. The fuel handling bridge crane controls are interlocked to prevent simultaneous operation of bridge drive and hoist.

2. Bridge drive operation is prevented except when the hoist is in the full up position.

3. An overload protection device is included on the hoist to limit the uplift force which could be applied to the fuel storage racks. The protection device limits the hoist load to the rated 2-ton-hoist capacity of 4000 lb and can withstand a single failure.

4. The design load on the hoist is the weight of one fuel assembly (1600 lb), weight of one failed fuel container (1000 lb), and the weight of the tool which gives it a total weight of approximately 3000 lb.

5. Restraining bars are provided on each truck to prevent the bridge from overturning.

All fuel handling tools and equipment handled over an open reactor vessel are designed to prevent inadvertent decoupling from machine hooks; i.e., lifting rigs are pinned to the machine hook, and safety latches are provided on hooks supporting tools.

Tools required for handling internal reactor components are designed with fail-safe features that prevent disengagement of the component in the event of operating mechanism malfunction. These safety features apply to the following tools:

1. Control rod Drive Shaft Unlatching Tool

   The air cylinders actuating the gripper mechanism are equipped with backup springs which close the gripper in the event of loss of air to the cylinder.

   Air-operated valves are equipped with safety locking rings to prevent inadvertent actuation.

2. Spent Fuel Handling Tool

   When the fingers are latched a pin is inserted into the operating handle and prevents inadvertent actuation. The tool weighs approximately 385 lb and is preoperationally tested at 125 percent the weight of one fuel assembly (1600 lb).

3. New Fuel Assembly Handling Tool

   When the fingers are latched a safety screw is screwed in, preventing inadvertent actuations. The tool weighs approximately 100 lb and is preoperationally tested 125 percent the weight of one fuel assembly (1600 lb).

The means have been developed to assure the safe handling of heavy loads at CPNPP. The consequences of dropping heavy loads were determined through a detailed analysis (Reference [17]), and the implementation of the CPNPP Control of Heavy Loads Program resulted in a significant reduction in the probability of a load drop occurrence.

For load handling systems where a potential load drop could affect the safe shutdown of the plant or result in unacceptable fuel damage, preventative measures have been developed and implemented as described in Reference [17]. Specifically, cranes which handle heavy loads in the area of spent fuel are designed with inherent safety features which prevent damage to fuel in the event of load drop. These inherent safety features are:

1. Containment Polar Crane

The Containment Polar Crane is used during the plant construction phase for lifts up to 475 tons (for handling the reactor vessel and steam generators) prior to its intended normal service.

The use of the crane during the construction phase does not imply any nuclear safety related condition.

During refueling or maintenance operations, the Containment Polar Crane handles a maximum critical load of 175 tons. The heaviest load expected to be lifted is the reactor vessel head assembly.

Due to the seismic design of the Containment Polar Crane, its structural integrity is sustained and its wheels are not dislodged from the track during an SSE.

The Containment Polar Cranes' main hoists have been equipped with single-failure-proof features. A detailed analysis of the features of the Containment Polar Crane has been made against the guidelines of NUREG-0554 "Single-Failure-Proof Cranes for Nuclear Power Plants." A summary of this analysis is provided in Reference [17]. The special safety features incorporated into the design of the main hoisting system of the Containment Polar Crane precludes a load drop accident by preventing a load drop in the event of a single failure in the hoisting or braking systems (Reference [17]). In addition, CPNPP design employs an integral steel missile shield (Unit 1 only) and a roll-away steel missile shield (Unit 2 only) over the reactor instead of a concrete-type shield. Therefore, no concrete structures are lifted in the vicinity of, or over, spent fuel.

2. Fuel Building Overhead Crane

The Fuel Building Overhead Crane cannot travel over the spent fuel pools. In addition, a spent fuel shipping cask, if dropped, cannot roll into the spent fuel pools.

However, to avoid placing undue restrictions on routine crane operations (including handling of spent fuel shipping casks) in the Fuel Building, the Fuel Building Overhead Crane's main hoist was equipped for spent fuel dry cask operations.

During refueling and dry cask storage operations, the heaviest loads handled by the Fuel Building Overhead Crane will be the spent fuel shipping cask and the loaded HI-TRAC transfer cask. The HI-TRAC transfer cask, with or without spent fuel, is considered a critical load. Heavy loads over an MPC with spent fuel, such as the MPC lid, HI-TRAC lid, Mating Device and HI-STORM lid, are considered critical loads. In addition, the transfer of an MPC with spent fuel between the HI-TRAC and the HI-STORM storage cask is a critical load.

The special safety features incorporated into the design of the main hoisting system of the Fuel Building Overhead Crane precludes a load drop accident by preventing a load drop in the event of a single failure in the hoisting or braking systems. Information regarding the Fuel Building Overhead Crane's compliance with the single-failure-proof provisions of Regulatory Guide 1.104 "Single-Failure-Proof Overhead Crane Handling Systems for Nuclear Power Plants" (Draft 3, Revision 1, October 1978) is provided in Reference [17].

The detailed evaluation of heavy load handling systems (Reference [17], safe load paths and areas, single-failure-proof cranes and hoists, and administrative control support the safe operation of CPNPP.

9.1.4.3.2 Seismic Considerations

The safety classification and seismic categories for all fuel handling and storage equipment are listed under Systems 33, 34, and 35 in Table 17A-1. These safety classes and seismic categories provide criteria for the seismic design of the various components. Safety Class 1 and Safety Class 2 equipment are designed to withstand the forces of the OBE and SSE. For normal conditions plus OBE loadings, the resulting stresses are limited to allowable working stresses as defined in the ASME B&PV Code, Section III, Appendix XVII, Subarticle XVII-2200 for normal and upset conditions. For normal conditions plus SSE loadings, the stresses are limited to within the allowable values given by ASME B&PV Code, Section III, Appendix XVII, Subarticle XVII-2110 for critical parts of the equipment which are required to maintain the capability of the equipment to perform its safety function. Permanent deformation is allowed for the loading combination which includes the SSE to the extent that there is no loss of safety function.

The Safety Class 3 fuel handling and storage equipment satisfies the Safety Class 1 and Safety Class 2 criteria for the SSE. Consideration is given to the OBE only insofar as failure of the Safety Class 3 equipment might adversely affect Safety class 1 or Safety Class 2 equipment.

The containment polar crane and fuel building overhead crane are listed in FSAR Table 17A-1 as seismic Category I. The safety related functional requirements are the retention of lifted loads by the main hoist during and after seismic events. The cranes are not otherwise required to remain functional after a seismic event. The loading combinations for the fuel building overhead crane include OBE and SSE with lifted load. Because of the extremely low probability of SSE in conjunction with a critical (heavy load, i.e., > 2150 lbs.) lifted load for the polar crane, the loading combinations include OBE with lifted load and SSE without lifted load.

The refueling gates are designated seismic category I in accordance with Reg. Guide 1.29 and Reg. Guide 1.13.

For non-nuclear-safety related equipment, design for the SSE is considered if failure might adversely affect a Safety Class 1, Safety Class 2, or Safety Class 3 component. Design for the OBE is considered if failure of the non-nuclear-safety component might adversely affect a Safety Class 1 or Safety Class 2 component.

Seismic design is not required for portable underwater lights. The use of lights containing mercury within the containment or fuel building are subject to precautions and limitations that assure nuclear safety is not compromised.

9.1.4.3.3 Containment Pressure Boundary Integrity

The fuel transfer tube which connects the refueling canal (inside the reactor containment) and the fuel storage area (outside the containment) is closed on the refueling canal side by a blind flange at all times except during refueling operations. Two seals are located around the periphery of the blind flange with leak-check provisions between them.

9.1.4.3.4 Radiation Shielding

During all phases of spent fuel transfer, the gamma dose rate at the surface of the water is 2.5 mrem/hr or less. This is accomplished by maintaining a minimum of 10 ft of water above the top of the fuel assembly during all handling operations.

The two fuel handling devices used to lift spent fuel assemblies are the refueling machine and the fuel handling bridge crane. The refueling machine contains positive stops which prevents the top of a fuel assembly from being raised to within a minimum of 10 ft of the normal water level in the refueling cavity.

The hoist on the fuel handling bridge crane and the containment fuel storage area crane moves spent fuel assemblies with a long-handled tool. Hoist travel and tool length likewise limit the maximum lift of a fuel assembly to within a minimum of 10 ft of the normal water level in the fuel storage area.

9.1.4.4 Inspection and Testing Requirements

The test and inspection requirements for the equipment in the FHS are as follows:

1. Fuel Handling Bridge Crane, Refueling Machine, RCC Changing Fixture, and New Fuel Elevator

   The minimum acceptable tests at the shopsite include the following:

   a. Hoist and cable are load-tested at 125 percent of the rated load.

   b. The equipment is assembled and checked for proper functional and running operation.

   The following maintenance and checkout tests are recommended to be performed prior to using the equipment:

   a. Visually inspect for loose or foreign parts. Keep free of dirt and grease.

   b. Lubricate wheels and exposed gears with proper lubricant.

   c. Inspect hoist cables for worn or broken strands.

   d. Visually inspect all limit switches and limit switch actuators for any sign of damaged or broken parts.

   e. Check the equipment for proper functional and running operation.

2. Head Lifting Rig and Internals Lifting Rig

   The minimum acceptable tests at the shopsite include the following:

   a. The rigs are load-tested to 125 percent of the rated load.

b. The rigs are assembled to ensure proper component fit-up.

The following maintenance and checkout tests are recommended to be performed prior to using the tools:

a. Visually inspect for loose or foreign parts or damaged surfaces.

b. Visually inspect all engagement surfaces and lubricate with proper lubricant.

c. On the internals lifting rig, check for the proper functioning of the engagement and protective rig operators.

3. New Fuel Assembly Handling Tool and Spent Fuel Assembly Handling Tool

The minimum acceptable tests at the shopsite include the following:

a. The tools are load-tested to 125 percent of the rated load.

b. The tools are assembled and checked for the proper functional operation.

The following maintenance and checkout tests are recommended to be performed prior to using the tools:

a. Visually inspect the tools for dirt, for loose hardware, and for any signs of damage such as nicks and burns.

b. Check the tools for proper functional operation.

4. Fuel Transfer System

The minimum acceptable tests at the shopsite include the following:

a. The system is assembled and checked for proper functional and running operation.

b. The fuel container is held in a 45-degree position for an unpowered condition.

The following maintenance and checkout tests are recommended to be performed prior to using the tools:

a. Visually inspect for loose or foreign parts. Keep free of dirt and grease.

b. Lubricate wheels and exposed gears with proper lubricant.

c. Visually inspect all limit switches and limit switch actuators for any sign of damaged or broken parts.

d. Check the system for proper functional and running operation.

5. Reactor Vessel Stud Tensioner

The minimum acceptable test at the shopsite is that the tensioner be assembled and checked for proper functional and running operation.

The following maintenance and checkout tests are recommended to be performed prior to using the equipment.

a. Visually inspect for loose or foreign parts.

b. Inspect hydraulic lines for wear or damage.

c. Check the hydraulic unit for proper pressurization and if any leaks occur at operating pressure.

6. Portable Underwater Lights

A visual inspection of lamp modules containing mercury is required prior to use.

9.1.4.5 Instrumentation Requirements

The control systems for the refueling and fuel handling machines and FTS are discussed in Subsection 9.1.4.2.3, Component Description. A discussion of additional electrical controls, such as the interlocks and the main hoist breaking system for the FHS, are discussed in Subsection 9.1.4.3.1, Safe Handling.

9.1.5 SPENT FUEL DRY CASK STORAGE

The CPNPP has established an Indpendent Spent Fuel Storage Installation (ISFSI) on an approximately 1.75 acre site located about 2900 feet east of the Unit 1 Containment Building and along the souther edge of the CPNPP site peninsula located within the Squaw Creek Reservoir.

The Spent Fuel Dry Cask Stoarge opertions at CPNPP will be conducted under a general license in accordance with Subpart K of 10 CFR Part 72. The general license issued by 10 CFR 72.210, “General license issued,” authorizes a 10 CFR Part 50 nuclear power plant license to store spent fuel at an onsite ISFSI. Subpart K of 10 CFR Part 72 also includes 10 CFR 72.212, “Conditions of general license issued under 10 CFR 72.210,” which requires the use of a dry cask storage system that is pre-approved by the Nuclear Regulatory Commission, as evidenced by its listing in 10 CFR 72.214.

The ISFSI site boundary has a security fence to establish a separate protected aea for the ISFSI. A nuisance fence is included on the perimeter of the outside isolation zone to minimize intrusion detection alarms.

The CPNPP ISFSI uses the Holtec HI-STORM 100S Version B vertical cask storage overpack and the Holtec MPC-32 multi-purpose canister (MPC), as described in the HI-STORM 100 Cask System FSAR [19] and approved by the Nuclear Regulatory Commission via the HI-STORM Certificate of Compliance No. 1014 [20]. The ISFSI concrete pad has a capacity for 84 vertical spent fuel storage casks.

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 2, Design Bases for Protection Against Natural Phenomena.

2. 10 CFR Part 50, Appendix A, General Design Criterion 3, Fire Protection.

3. 10 CFR Part 50, Appendix A, General Design Criterion 4, Environmental and Missile Design Bases.

4. 10 CFR Part 50, Appendix A, General Design Criterion 5, Sharing of Structures, Systems, and Components.

5. 10 CFR Part 50, Appendix A, General Design Criterion 44, Cooling Water.

6. 10 CFR Part 50, Appendix A, General Design Criterion 45, Inspection of Cooling Water System.

7. 10 CFR Part 50, Appendix A, General Design Criterion 46, Testing of Cooling Water System.

8. 10 CFR Part 50, Appendix A, General Design Criterion 61, Fuel Storage and Handling and Radioactivity Control.

9. 10 CFR Part 50, Appendix A, General Design Criterion 62, Prevention of Criticality in Fuel Storage and Handling.

10. 10 CFR Part 50, Appendix A, General Design Criterion 63, Monitoring Fuel Waste and Storage.

11. NRC Regulatory Guide 1.13, Fuel Storage Facility Design Basis, March 1971.

12. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam-, and Radioactive-Waste-Containing Components of Nuclear Power Plants, Revision 3, February 1976.

13. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976.

14. Branch Technical Position APCSB 3-1, Protection Against Postulated Piping Failure in Fluid Systems Outside Containment.

15. ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973.

16. DELETED

17. Design Basis Document, DBD-ME-006, Control of Heavy Loads at Nuclear Plants (Incorporation by Reference), (Ref. 17 Originally “CPSES Final Response to NUREG-0612”, June 1983, as submitted by TXX-3659 dated June 8, 1983.)

18. License Amendment Request 00-05, Revision to Technical Specification Spent Fuel Assembly Storage Racks and Fuel Storage Capacity, Docket Nos. 50-445 and 50-446

a. TXU Electric letter logged TXX-00144, from C. L. Terry to USNRC dated October 4, 2000

b. TXU Electric letter logged TXX-01052, from C. L. Terry to USNRC dated March 21, 2001

c. TXU Electric letter logged TXX-01074, from C. L. Terry to USNRC dated April 30, 2001

d. TXU Electric letter logged TXX-01102, from C. L. Terry to USNRC dated June 18, 2001

e. TXU Electric letter logged TXX-01115, from C. L. Terry to USNRC dated June 27, 2001

f. TXU Electric letter logged TXX-01118, from C. L. Terry to USNRC dated July 18, 2001

g. TXU Electric letter logged TXX-01183, from C. L. Terry to USNRC dated January 22, 2002.

19. Holtec International Final Safety Analysis Report for the HI-STORM 100 Cask System, Revision No. 9, USNRC Docket No. 72-1014, Holtec Report No.: HI-2002444, February 13, 2010.

20. USNRC Certificate of Compliance No. 1014, Docket No. 72-1014, Amendment No. 7, for the HI-STORM 100 Cask System, December 28, 2009.

21. NUREG/CR-6407 Classification of Transportatin Packaging and Dry Spent Fuel Storage System Components according to Importance to Safety.

TABLE 9.1-1
SPENT FUEL POOL COOLING AND CLEANUP SYSTEM DECAY HEAT PARAMETERS (NOTE 1)

PARAMETER (NOTE 3)	MAX. DESIGN		MAX. SUMMER DESIGN		ABNORMAL MAX. DESIGN	
	MAX POOL	MIN POOL	MAX POOL	MIN POOL	MAX POOL	MIN POOL
Decay Heat Produced (x 10^6 BTU/hr) (Note 3)	52.0	9.09	23.28	6.08	62.92	6.6
SSI Temperature (°F)	94	-	102	-	102	-
Maximum SFP Temp (°F)	<150	<150	<150	<150	<212	<212
Time to boiling (hrs) (Note 3)	>3	-	>3	-	N/A	N/A

NOTES:

1. See Section 9.1.3.1.1 for the design conditions in this table.
2. Deleted.
3. Decay heat loads in “Max Pool” maximize temperatures and minimize time to boiling.
4. Assuming cooling is temporarily lost, the time to boiling is evaluated for the temperature rise from 150°F to 212°F.

TABLE 9.1-2
SPENT FUEL POOL COOLING AND CLEANUP SYSTEM CODE AND SAFETY CLASS REQUIREMENTS

Component	Safety Class	Code
Spent fuel pool cooling water pump	3	ASME III, Class 3
Refueling water purification pump	NNS	Mfrs. standard
Spent fuel pool skimmer pump	NNS	Mfrs. Standard
Spent fuel pool heat exchanger	3	ASME III, Class 3
Spent fuel pool demineralizer	3	ASME III, Class 3
Spent fuel pool filter	NNS	ASME VIII
Spent fuel pool skimmer filter	NNS	ASME VIII
Spent fuel pool suction screens	3	Mfrs. standard
Spent fuel pool skimmer	NNS	Mfrs. Standard
Spent fuel pool skimmer strainer	NNS	Mfrs. standard
Spent fuel pool cooling system pressure reduction orifice	3	ASME III, Class 3
Purification loop resin trap	3	ASME III, Class 3
Cask pit and transfer canal drain pump	NNS	Mfrs. standard
Piping and valves (nuclear)	3	ASME III, Class 3
Piping and valves (nuclear)	2	ASME III, Class 2
Piping and valve (non-nuclear)	NNS	ANSI B31.1
Refueling cavity purification pressure reduction orifice	NNS	Mfrs. standard

TABLE 9.1-3
TABLE DELETED

TABLE 9.1-4
NUMBER OF FUEL ASSEMBLIES IN SPENT FUEL POOLS BY REFUELING OUTAGE

(Sheet 1 of 2)

Outage Year	U1 Discharged Assemblies	U2 Discharged Assemblies	Total Discharged Assemblies Following Refueling	Remaining Storage Capability
1991	56	0	56	500
1992	61	0	118	440
1993	88	0	206	350
1994	0	83	289	267
1995	97	0	386	170
1996	85	96	567	715[(a)]
1997	0	77	644	638
1998	89	0	733	549
1999	89	89	911	371
2000	0	93	1004	799[(b)]
2001	93	0	1097	675
2002	89	90	1276	1706[(c)]
2003	0	81	1357	1598
2004	89	0	1446	1479
2005	85	101	1632	1231
2006	0	88	1720	1114
2007	89	0	1809	995
2008	93	89	1991	752
2009	0	93	2084	628
2010	93	0	2177	504
2011	92	92	2361	259

TABLE 9.1-4
NUMBER OF FUEL ASSEMBLIES IN SPENT FUEL POOLS BY REFUELING OUTAGE

(Sheet 2 of 2)

Outage Year	U1 Discharged Assemblies	U2 Discharged Assemblies	Total Discharged Assemblies Following Refueling	Remaining Storage Capability
2012	0	89	2162	524[(d)]

a) Use of Region II racks in SFP2 in a high density (2/4) storage configuration. (capacity of 1291).

b) Full Use of Region II racks in SFP2 in a high density (4/4) storage configuration (capacity of 2026).

c) Fuel Use of Region I and Region II racks in SFP1 and SFP2 (capacity of 3373).

d) Implementation of Dry Cask Storage at CPNPP to maintain SFP Storage capability.

9.2 WATER SYSTEMS

This section provides a discussion of each of the auxiliary water systems associated with the plant. These systems are: 1) Station Service Water System (SSWS); 2) Component Cooling Water System (CCWS); 3) Demineralized Water Makeup System; 4) Potable and Sanitary Water System; 5) Ultimate Heat Sink; 6) Condensate Storage Facilities.

9.2.1 STATION SERVICE WATER SYSTEM

9.2.1.1 Design Bases

The SSWS removes heat from the CCWS heat exchangers and from the emergency diesel generators, and supplies cooling water to the safety injection, centrifugal charging pump lube oil coolers and the containment spray pump bearing oil coolers. In conjunction with the CCWS (Section 9.2.2), the SSWS supplies cooling water to meet the plant cooling requirements during normal operation, shutdown, and during or after a postulated loss-of-coolant accident (LOCA) of either unit [5]. The required cooling water is taken from the safe shutdown impoundment (SSI), which is the ultimate heat sink and is designed in accordance with the guidelines in NRC Regulatory Guide 1.27 [11]. The SSWS also acts as a backup water supply for the Auxiliary Feedwater System if the Condensate Storage Tank is depleted.

The SSI contains a water supply for a minimum of 30 days of reactor decay heat removal, without outside makeup. The SSWS is designed to properly operate with water in the SSI at the lowest level during this period of time. (Sections 2.4.11.5 and 2.4.11.6)

The SSWS is designed to seismic Category I requirements [12].

The maximum calculated SSWS supply temperature during a two unit-two train cool down is 117.8°F. The maximum calculated SSWS supply temperature is 116.9°F during LOCA conditions. See Section 9.2.5 for additional details. One system for each unit is provided as described in Section 9.2.1.2.

The SSWS is designed on the basis of the following:

1. Flow from one of the two redundant trains is continuously delivered to one of the two emergency diesel generators of each unit. During a loss of all SSWS, Fire Protection water may be aligned to the jacket water heat exchangers for emergency diesel generators.

2. Flow is continuously delivered to one of two lube oil coolers per unit for the safety injection and centrifugal charging pump and two of four containment spray pump bearing oil coolers as shown on Figure 9.2-1 (Sheet 3 of 3). During a loss of all SSWS, Fire Protection water may be aligned to the lube oil coolers for safety injection pumps and centrifugal charging pumps.

3. Normal cooldown of one unit is accomplished with two SSWS pumps delivering water to two CCWS heat exchangers to cool the reactor coolant system (RCS) from 350°F to 140°F. The SSWS outlet temperature is maintained below 133°F.

4. An orderly shutdown can be achieved with one SSWS pump and one CCWS heat exchanger, although at a slower cooldown rate than outlined in Item 3 above.

5. One SSWS pump in conjunction with one CCWS heat exchanger provides one unit with enough cooling capacity for safe recovery after a LOCA (see Section 9.2.2.1).

The SSWS is vital to plant safety and is provided with redundant components so that no single failure denies cooling to equipment required for safe shutdown.

Equipment necessary for shutdown is supplied with emergency diesel generator power if normal and offsite power sources fail. In this event, at least one SSWS pump per unit is operative within 60 sec from the beginning of blackout (including emergency diesel generators starting time, sequencing delays, and pump starting time).

Chloride contamination of the CCWS is detrimental to the physical integrity of various stainless steel components supplied with component cooling water, such as reactor coolant pump heat exchangers, letdown heat exchangers, and so forth. In order to prevent such contamination through a tube leak in a CCWS heat exchanger, the component cooling water is at a higher pressure than the station service water.

9.2.1.2 System Description

9.2.1.2.1 General Description

Figure 9.2-1 (Sheets 1, 2 and 3) shows a schematic flow diagram of the SSWS. The major components of the system are described in Section 9.2.1.7.

The SSWS consists of two separate and independent full-capacity, safety-related trains. Cross connections between the two trains add operational flexibility to the SSWS.

The safety-related trains are redundant in that the components supplied by one train are sufficient to perform the minimum required safety functions. Two full capacity SSWS pumps and two full capacity supply and return headers are provided for each unit. Both pumps normally operate. This provides a continuous cooling water supply to the two redundant safety-related trains.

Controls are provided to allow remote operation from the Control Room or from the hot shutdown panel. A cross connection between Unit 1 and Unit 2 SSWS is provided.

The SSWS pump motors have couplers installed for connection of partial discharge monitoring equipment. The installation of diagnostic equipment is used for long term reliability monitoring and is not required for system operation.

During low water levels in the Squaw Creek Reservoir, the SSI is maintained at or above el. 769 ft, 6 in. by the safe shutdown dam, connecting canal. (Section 9.2.5)

A bleed line is provided from the Circulating Water System, at the intake structure, to the Service Water Intake Structure pump suction pit. See Section 9.2.5.2 for additional details.

The SSWS has a separate system which injects sodium hypochlorite and sodium bromide to control organic fouling. The effect of some fouling on heat transfer surfaces is considered in the design of the heat exchangers in the SSWS. A combined fouling factor of 0.0020 hr-ft^2-°F/BTU has been used on both the shell and tube side of the component cooling water heat exchanger for the evaluation of containment heat removal systems and CCW safety functions.

Bounding analyses are performed using the bounding design heat loads combined CCW heat exchanger fouling factor of 0.0020 hr-ft-°F/BTU and the maximum normal service water intake temperature of 102°F with no tube plugging. When service water intake temperatures are less than maximum, the fouling may exceed 0.0020 without any loss of quality or functional capability. The tube plugging margin is accounted for in the periodic fouling monitoring described below.

The quality of the SSI water is very similar to that of Squaw Creek Reservoir water. Stability and saturation index calculations have shown the reservoir water to be corrosive. In addition, to control the Asian clam population in the service water system and to help prevent the establishment of corrosion mechanisms within equipment, a non-oxidizing liquid biocide injection system is provided.

A chemical addition system is used to control corrosion and fouling in the Service Water System to protect the carbon steel pipe. A coordinated chemical treatment program with phosphate, organic phosphate, and copolymar is used in addition to the existing biocide treatments for this added corrosion protection.

Scale-forming tendencies are reduced as a result of the following:

1. Low water temperatures in the system (less than 110°F with a maximum of 133°F at the CCWS heat exchangers outlet for short periods during initial unit cooldown)

2. High water velocities are used in the system piping design.

Periodic monitoring determines when scaling is affecting design tube cleanliness factors. Safety-related systems serviced by SSWS are provided with redundant components so that if scaling affects tube cleanliness factors units are taken out of service during shutdown periods and cleaned.

This CCW heat exchanger fouling monitoring program (Section 9.2.1.8, item 5) ensures there is no loss of quality or functional capability between cleanings. In addition, more restrictive limits on system line ups, on SSI temperature and/or on CCW heat exchanger fouling are imposed on the operating (non-outage) unit as necessary during abnormal condition such as maintenance during outages.

9.2.1.2.2 Normal Operation

Normal operation of one unit requires the operation of both pumps associated with Train A and Train B. Operation of both SSWS trains continuously is desirable for corrosion control.

To insure reliability, these two pumps are connected to two separate emergency diesel buses.

9.2.1.2.3 Normal Unit Shutdown

For normal cooldown of one unit, two trains are used. The two trains provide the CCWS heat exchangers with the cooling capacity required to meet the design cooldown rate.

9.2.1.2.4 Operation During and After a LOCA

Train A or Train B provides enough cooling capacity for operation of the engineered safeguard features (ESF) required during the injection and recirculation phases of operation.

Normally during either phase, two trains are operating and both trains are aligned to cool down the unit under a LOCA.

During both the injection phase and the recirculation phase, only one train is required for safe recovery from the accident. However, both trains operate for more efficient energy removal from the Containment.

9.2.1.3 Leakage Detection and Control

All headers and major equipment are furnished with flow indication, which is used to detect abnormalities in the system flow rates. Abnormal flow conditions are annunciated in the Control Room. Abnormal leakage from systems located in plant buildings results in high sump levels, which are detected by level switches and also annunciated in the Control Room. A description of the means of detection of leakage through the CCWS heat exchangers is given in Sections 9.2.2.3 and 9.2.2.4. Upon detection of SSWS leakage, the leaking component can be identified by successive isolation of components until the leaking component is identified.

9.2.1.4 Safety Evaluation

The SSWS is designated ANSI N18.2 Safety Class 3 and designed to the requirements of seismic Category I [12], [15]. The SSWS pumps and related valves, operators, and instrumentation are located in the SSWS pumphouse which is a Category I structure, and is designed to withstand tornado effects. Piping between the SSWS pumphouse and the Auxiliary Building and from the Auxiliary Building to the SSI is buried underground with an amount of earth cover sufficient to protect the piping from tornado missiles. Other components are located inside the Auxiliary Building, which is a Category I structure and is designed against tornadoes. Pump motors, valve operators, and controls in the pumphouse are located above the highest postulated water level in the SSI. Valve operators and controls inside the Auxiliary Building also are located above the highest water level that might occur due to equipment failures within the building.

The SSW intake trash racks and traveling screens and a small portion of discharge piping are not protected from Tornado missiles. See Section 3.5.1.4 for details.

All of the previously mentioned conditions make the system capable of withstanding adverse environmental conditions, such as postulated earthquakes, tornadoes, and tornado missiles [1],[2]. (For details on seismic Category I structures, see Sections 3.2.1, 3.3, and 3.5.)

The SSWS is a moderate energy piping system. For a discussion of postulated pipe ruptures in this system, refer to Section 3.6 [14].

Pumps and pump motors inside the pumphouse are physically separated from each other by walls, as shown on Figures 1.2-45 and 1.2-46. These walls are designed to preclude coincident damage to redundant equipment in the event of a postulated pipe rupture, equipment failure, or missile generation.

Flooding is not considered possible because the floor of the pump compartments is at el. 796 ft, which is above the probable maximum water level of 791.3 ft. In the event of a pipe rupture within the pump compartment, the water drains back to the SSI through large holes in the compartment floor.

The single-intake bay for all pumps, downstream of the intake channels and traveling screens, ensures flow to all pumps in the service water pumphouse.

The performance of all essential equipment can be monitored from the Control Room.

It is anticipated that no radioactive material will leak into the SSWS. The provisions for monitoring the SSWS for radioactive contamination are described in Section 11.5.

Environmental design conditions for seismic Category I self-actuated valves are described in Section 3.11.

9.2.1.5 Safety Implication Related to Sharing

There is no sharing of any safety-related component or safety-related function between the two units. The SSWSs include two redundant trains per unit [3]. Each train is separated from each other by two normally closed valves. GDC-5 is satisfied by a single locked closed valve in the cross-connection provided between units. The intake bay is provided with two traveling screens and associated screen wash equipment. There is no safety implication of this arrangement.

In the event of a loss of all essential cooling water on an operating unit, a cross-connect may be opened to provide backup cooling capability. One service water pump is capable of providing essential cooling to both units for this event by manually aligning and flow balancing. Therefore, there is no adverse safety implications of this back-up function (Ref. 17 and 18).

The cross-connects may also be opened under appropriate administrative controls for periodic flushing in accordance with Reference 19.

9.2.1.6 Single-Failure Analysis

A single-failure analysis is presented in Table 9.2-1.

This analysis shows that the failure or malfunction of any single active or passive component does not result in the loss of the minimum required safeguards functions [5], [11].

9.2.1.7 Components

The SSWS components safety classification is summarized in Section 3.2.2 and is indicated on the system's flow diagram on Figure 9.2-1. The major components in the SSWS are described as follows:

1. The four SSWS pumps are of the vertical centrifugal wet pit type, rated at 17,000 gpm at 140 ft of head.

2. The SSWS valves are either 316 stainless steel, carbon steel, nickel-plated carbon steel, or aluminum-bronze.

3. SSWS piping for sizes up to and including 3 1/2 in. is made of 304 or 316 stainless steel.

   Four inch diameter or larger piping is either carbon steel or 316 stainless steel.

   All joints and connections are welded, except at certain components where flanged connections are used to facilitate maintenance.

9.2.1.8 Testing and Inspection Requirements

Components in operation are interchanged periodically to enable testing and inspection [6],[7]. The standby components are inspected completely to find and correct incipient malfunctions. The inspection includes the following:

1. Pumps and Drive Motors

   Each pump is started and operated for sufficient time to insure its proper operation. Maintenance is provided as required.

2. Safety-Related Manual Valves

   At least once quarterly, Station Service Water System cross-connect valves XSW-0006, XSW-0007, XSW-0008, XSW-0028 and XSW-0029 will be stroked through their full range to insure that they are in operating condition.

3. Electrically Operated Safety-Related Valves

   All electrically operated safety-related valve control circuits are designed with the ability for on-line testing so that the availability and operational status can be readily determined.

4. Instrument and annunciator operation is checked periodically for proper operation and accuracy.

5. Heat Exchangers

   Safety-related heat exchangers are monitored periodically to establish effective capacity.

9.2.1.9 Instrumentation Requirements

A detailed description of instrumentation applications is given in Section 7.3.

9.2.1.10 Service Water Intake Structure Crane

9.2.1.10.1 Description

The Service Water Intake Structure (SWIS) Crane is located in the SWIS building (See Figure 1.2-46) and is used to install and maintain four service water pumps, three fire pumps (two of which are abandoned in place) and associated piping and equipment. The SWIS crane is an overhead traveling crane with an underhung 7.5 ton capacity hoist (see Figure 9.2-14). The main components of the SWIS crane are:

1. Runway beams with stops

2. Motor driven bridge and hoist

3. Runway conductors and collectors

4. Limit and overload switches

5. Pendant push button control station.

9.2.1.10.2 Design Criteria

The SWIS crane is designed to seismic category II (as defined in Appendix 17A and listed in Table 17A-1) and classified as a non-nuclear safety component.

9.2.1.10.3 Safety Evaluation

The SWIS crane is required to handle occasional non-critical loads and operate during normal operation of the plant including scheduled maintenance. If the Safe Shutdown Earthquake occurs, the load-bearing components, such as girders, wheels and runways are conservatively designed to remain in place with the wheels being prevented from leaving the tracks.

The separation and redundancy of the Station Service Water System ensures that, should an accidental load drop disable one train, at least two trains are still available for safe shutdown of both units.

9.2.2 COMPONENT COOLING WATER SYSTEM

9.2.2.1 Design Bases

The Component Cooling Water System (CCWS) provides an intermediate barrier between radioactive or potentially radioactive heat sources and the Station Service Water System (SSWS). It is designed to remove rejected heat from various plant components in a manner which precludes direct leakage of radioactive fluids into the Safe Shutdown Impoundment (SSI).

Leakage of the station service water into the component cooling water is detrimental to the physical integrity of various stainless steel components, such as reactor coolant pumps, heat exchangers, letdown heat exchangers and so forth, because of possible chloride-induced stress corrosion. To prevent such an occurrence, the CCWS is maintained at a higher pressure than the SSWS.

The CCWS is designed to supply cooling water for the components which are part of the Reactor Coolant System (RCS), Emergency Core Cooling System (ECCS), ESF systems, CVCS, Spent Fuel Pool Cooling and Cleanup System, Waste Processing System (WPS), ventilation system, and the Instrument Air System. The CCWS is required to operate according to the modes of operation as described in Table 9.2-4.

The CCWS is normally required to be operating during all phases of plant operation including startup, power operation, shutdown, refueling, and the injection and recirculation phases following a loss-of-coolant accident (LOCA). However, during outages, it may be necessary to take both safeguards trains and/or the non-safeguards loop of CCW (see Section 9.2.2.2) out of service for maintenance or to perform other maintenance which makes two trains of CCW unavailable (e.g. maintenance on support system such as SSW, Safety Chilled Water, and EDGs). In these cases, the common (shared) components are cooled by the operating unit when required. In this abnormal alignment, more restrictive limits may be imposed as described in Section 9.2.2.2.1.

The design cooldown rate, based on reducing the temperature of the reactor coolant from 350°F to 140°F will be achieved using two CCWS pumps and two CCWS heat exchangers. A slower but acceptable cooldown rate may be accomplished using only one CCWS pump and one CCWS heat exchanger. Under both conditions the CCWS supplies cooling water to its RHR heat exchanger at a maximum temperature of 122°F during its early part of cooldown. The supply water temperature will gradually return to 108°F towards the end of cooldown.

Because the system is required to perform its safety function during the short-term and long-term plant accident conditions, the safety related passive components as well as the active components are designed to meet the single failure criteria [12]. An analysis of postulated cracks in moderate energy systems is in Section 3.6. All ANSI Class 2 and 3 components of the CCWS are designed to meet seismic Category I requirements. Equipment that is necessary for shutdown and equipment that is required to mitigate the effects of an accident is supplied with emergency diesel power, should normal and offsite power sources fail.

The CCWS is designed in accordance with the following criteria, regulatory guides, and codes:

1. Title 10, Code of Federal Regulations, Part 50, Appendix A, General Design Criteria for Nuclear Power Plants GDC 2 [1], 4 [2], 5 [3], 44 [5], 45 [6], 46 [7], 56 [8], and 57 [9]

2. NRC Regulatory Guides 1.26 [10], 1.29 [12], and 1.48 [13]

3. Codes

    a. ASME B&PV Code, Section III, Nuclear Power Plant Components

    b. ASME B&PV Code, Section XI, Inservice Inspection

    c. ANSI B31.1.0 Power Piping

    d. TEMA, Standard of Tubular Exchanger Manufacturers Association

    e. ASME OM Code

4. ANSI N18.2-1973 [15]

9.2.2.2 System Description

One system per unit is provided and each consists of three subsystems. Each system contains a Train A safeguards loop, a Train B safeguards loop, and a non-safeguards loop. The design allows common components, such as control room A/C condensers, to be supplied by either Unit 1 or Unit 2 CCW or a combination of the two. The two systems operate independently but can be cross-connected if needed.

Figure 9.2-3 presents the flow diagram of the Unit 1 system.

Each system consists of two 100-percent-capacity CCWS pumps and heat exchangers, with one pump and one heat exchanger operating and one pump and one heat exchanger on standby. Each system consists of two separate, redundant and independent full-capacity safeguards loops to service the engineered safeguards components and a nonredundant nonsafeguards loop with ANSI safety Class 3 and non-safety-class portions.

The CCW pump motors have couplers installed for connection of partial discharge monitoring equipment. The installation of diagnostic equipment is used for long term reliability monitoring and is not required for system operation.

During unit power operation, components in the same safeguards loop can be inoperable at any one time, provided that the other safeguards loop remains operative.

The equipment comprising the previously described loops is listed in the following paragraphs.

1. Each of the two safeguards loops consists of cooling to the following:

   a. Two Containment spray pump seal coolers

   b. One RHR pump seal cooler

   c. One RHR heat exchanger

   d. One Containment spray heat exchanger

   e. One Chilled Water System condenser (nuclear)

   f. Two Control Room air-conditioning condensers (common)

   g. One UPS air-conditioning condenser (common)

   In addition, train-A of the safeguards loop supplies cooling to the reactor coolant post accident sampling system sample cooler.

2. The nonsafeguards loop consists of cooling to the following:

   a. Two catalytic recombiner heat exchangers (common)

b. Two waste gas compressor seal water coolers (common)

c. One positive displacement charging pump hydraulic coupling oil cooler and lube oil cooler

d. One letdown heat exchanger (CVCS letdown cooling)

e. One seal water heat exchanger (CVCS reactor coolant pumps seals)

f. Two spent fuel pool heat exchangers (common)

g. One boron recycle evaporator package consisting of one distillate cooler, one evaporator condenser, and one vent condenser (common)

h. One waste evaporator package consisting of one distillate cooler, one evaporator condenser, and one vent condenser (common)

i. One floor drain evaporator package consisting of one distillate cooler, one evaporator condenser, and one vent condenser (common)

j. Four reactor coolant pump packages, each consisting of one upper bearing lube oil cooler, two motor air coolers, one lower bearing lube oil cooler, and one thermal barrier cooler (located inside Containment)

k. One excess letdown heat exchanger and one reactor coolant drain tank heat exchanger (CVCS equipment located inside Containment)

l. Four ventilation chillers (common)

m. One letdown chiller package condenser

n. Two rotary instrument air compressor packages, each consisting of one intercooler, one aftercooler and one oil cooler.

o. Deleted

p. Process sample coolers

Items c to k inclusive above are ANSI Safety Class 3 components; items a, b and l to p inclusive above are non-safety-class-components. Common components are designed to be cooled by either Unit 1 or 2 CCW with the demand being shared about equally between the two.

The two safeguards loops per unit are redundant in that the components supplied with cooling water by either of the safeguards loops can perform the minimum required safeguards functions.

Two branches from the nonsafeguards loop penetrate the Containment; one supplies cooling water to the reactor coolant pumps, and the other supplies the excess letdown heat exchanger and reactor coolant drain tank heat exchanger. A third branch supplies the spent fuel pool heat exchangers.

The Unit 1 and 2 CCW are normally separated from each other by at least one closed valve.

Component cooling water flow to the non-safeguards loop is terminated by means of redundant, motor-operated valves upon initiation of containment spray and phase B containment isolation.

The safeguards loops are isolated either by locked-closed valves or by redundant, motor-operated valves actuated by a P or phase B isolation signal, or upon a surge tank Empty Level signal.

A partitioned surge tank for each unit vented to the atmosphere is provided with a separate chamber and separate piping for each safeguards loop to accommodate expansion, contraction, and supply makeup water from a redundant Safety Class 3 source. The CCWS surge tank is located at an elevation which assures adequate net positive suction head (NPSH) to the CCWS pumps, and is the highest point in the system. The surge tank partition has one opening at the top, and another above the “Empty Level” set point to equalize the levels of both chambers. The partition is designed to maintain its integrity with one side of the surge tank empty.

The CCWS is initially filled with demineralized water to which a suitable corrosion inhibitor is added. Water chemistry is described in Table 9.2-3. A pot-type, manual, chemical addition system is provided for makeup during system operation.

To preclude discharging corrosion-inhibited component cooling and chilled water to the environment, one drain tank for each unit is provided in the Safeguards Building to collect component cooling water from piping and equipment being drained and from the discharge of most of the CCW thermal relief valves. Drainage flows to the drain tank by gravity. Two 100-percent-capacity CCWS drain pumps are provided per unit. Discharge lines from some CCW thermal relief valves and overflow lines from the CCWS surge tank are routed to the floor drains.

One CCWS drain pump from each unit is used to pump the drains to the truck discharge connection, or to the Wastewater Management System (see Section 9.2.8), depending on the quality and quantity of the water collected.

One Containment CCWS drain tank for each unit is provided to collect component cooling water from components in the Containment. Two 100- percent-capacity Containment CCWS drain pumps are provided per unit. One pump for each unit is used to pump the drains back to the main drain tank in the Safeguards Building. One Containment penetration is provided for the drain’s line.

The CCWS water chemistry (see Table 9.2-3) is controlled via a filter/demineralizer/chemical addition skid (one per Unit) and by the addition of chemicals via the chemical addition tank. Samples are periodically taken from sample taps to verify that water chemistry is as required. If dilution is required, water is bled off to the drain tank through one of the existing drain connections. Noninhibited water is added through the CCWS surge tank.

9.2.2.2.1 Modes of Operation

1. Normal Operation

During normal operation, each system has one CCWS pump and one CCWS heat exchanger in operation. A second CCWS pump along with its heat exchanger is on standby (redundant) and aligned to replace the pump and heat exchanger in service. If the CCWS pump in operation trips, the redundant CCWS pump automatically starts and is operative within 60 sec.

To ensure reliability, these two pumps are connected to two separate, emergency diesel buses. Each safeguards loop is supplied with emergency power from a separate diesel generator. Component cooling water is normally supplied through the operating heat exchanger to both the safeguards loops and to the nonsafeguards loops, including one spent fuel pool heat exchanger and two ventilation condensers. Component cooling water is not supplied to RHR and Containment spray heat exchangers during normal operation.

Manual Valves CC-0107, CC-0109, CC-0157, and CC-0158 have modified discs which have been drilled to serve as flow restriction orifices during MODES 1 and 2. These valves are required to be closed to provide acceptable CCW flow balancing for Design Basis Accidents.

2. Normal Unit Shutdown

The cooldown phase requiring use of the RHR System starts approximately four hrs after reactor shutdown. At this time two CCWS heat exchangers and two CCWS pumps are aligned to all loops of the subsystem serving the unit being cooled down. Component cooling water is supplied to both RHR heat exchangers.

Manual Valves CC-0109 and CC-0157, which provide a flow limiting function in MODES 1, 2, and 3, may be opened in MODE 3 at or below 400°F as needed to support RHR cooldown in MODE 4, 5 and 6. Manual Valves CC-0107 and CC-0158, which provide a flow limiting function in MODES 1 and 2, may be opin MODE 4, 5, and 6.

3. Operation During and After LOCA

Component cooling water is continuously supplied to both safeguards loops so that any of the redundant safeguards pumps can be immediately started. Following Containment spray initiation, the nonsafeguards loops are isolated and do not receive component cooling water.

In this case, equipment cooled by the nonsafeguards loop is cooled by the nonaccident unit. If cooling from the non-accident unit is not available (e.g. due to maintenance), cooling to the nonsafeguards loop from the accident unit may be restored when the heat load has decreased sufficiently to accept the additional heat and remain within temperature limits for the safeguards loops.

The safeguards loops are isolated from each other so that each loop is supplied by a separate CCWS heat exchanger. Component cooling water is supplied to the RHR heat exchanger and the Containment spray heat exchangers of both loops, as well as to the Control Room air-conditioning condensers and the Chilled Water System condenser, although one loop is sufficient to maintain the Containment temperature and pressure within the design limits and to supply sufficient cooling for condensers. A single failure of

any component does not prevent the system from supplying water to one safeguards loop, as described in Table 9.2-5.

Manual Valves CC-0107, CC-0109, CC-0157, and CC-0158, which provide a flow limiting function in the short term after a Design Basis Accident (e.g. LOCA), may be opened to accelerate cooldown after accident heat loads have sufficiently decayed if desired and if the valve locations are accessible.

4. Failure-Mode Analysis

A single-failure analysis is presented in Table 9.2-5 and shows that the failure or malfunction of any single active or passive component does not prevent fulfillment of the safeguards functions.

9.2.2.2.2 Equipment Design Bases

The following are the design bases for the major components of the CCWS. A more detailed equipment listing and description showing design, performance, and material data is found in Table 9.2-2.

1. CCWS Heat Exchangers

Two 100-percent-capacity, single-pass shell and straight-tube heat exchangers are provided for heat transfer between the CCWS and the SSWS. Service water, which has a greater tendency to foul heat transfer surfaces than component cooling water, flows through the straight tubes to facilitate cleaning. As stated under design bases, since service water can contain chloride which is detrimental to stainless steel components, the component cooling water pressure in the heat exchanger is higher than the service water pressure to prevent service water in-leakage.

The heat exchangers have straight tubes and fixed tubesheets, with tubes rolled into the tubesheets and with channels and removable channel covers (TEMA Type CEN). They are designated ANSI Safety Class 3, seismic Category I. Both tubes and shell sides of the heat exchangers are designed to the requirements of the ASME B&PV Code, Section III, Code Class 3, and to TEMA Class R standards for items not specifically covered by the ASME B&PV Code.

2. CCWS Pumps

Two 100-percent-capacity pumps are provided. The pumps are horizontal centrifugal, double suction split case.

They are designated as ANSI Safety Class 3, seismic Category I and are designed to the requirements of the ASME B&PV Code, Section III, Code Class 3.

3. CCWS Surge Tank

The CCWS surge tank accommodates surges resulting from component cooling water thermal expansion and contraction and collects any water which may leak into the system from components which are being cooled. In the event that these surges are extreme,

alarms will be triggered that require manual action by the operator. However, this will not jeopardize the safety function of the CCWS system. The surge tank also contains a supply of water to provide makeup as required until a leaking cooling line can be isolated. The CCWS surge tank is the highest point of the system to facilitate proper filling and venting of the CCWS system and to ensure adequate net positive suction head (NPSH) for the pumps during normal operation and accident conditions.

Two horizontal cylindrical surge tanks are provided (one per unit). Each surge tank has a vertical partition normal to centerline to provide two compartments, one compartment for each CCWS pump. The NPSH for the CCWS pumps are summarized in Table 9.2-2

The tank components are designated as ANSI Safety Class 3, seismic Category I, and are designed to the requirements of the ASME B&PV Code, Section III, Code Class 3.

4. Piping and Valves

The principal branches of the CCWS and their respective pipe categories are summarized in Table 9.2-2. Valves of the CCWS are made of carbon steel. Relief valves are made of either carbon steel or stainless steel. Thermal relief valves are secondary equipment overpressure devices, positioned near or on the component to relieve the volumetric expansion which occurs if the cooling water lines to the component are isolated and the water temperature rises. The thermal relief valve set pressures are less than or equal to 110% of the system or component design pressure.

The CPNPP technical specifications include a surveillance requirement to verify that each automatic valve serving safety related equipment actuates to its correct position on all applicable test signal. Periodic test procedures will be written to satisfy these surveillance requirements.

9.2.2.3 Safety Evaluation

Each component cooling water system is comprised of two full-capacity safeguards loops and one nonsafeguards loop. During emergency operation, the nonsafeguards loop is isolated from the safeguards loops by two automatic valves. The two safeguards loops are separated from each other either by redundant, remotely operated valves or by locked-closed valves.

The nonsafeguards loop penetrates the Containment to supply the reactor coolant pumps, excess letdown heat exchanger, and reactor coolant drain tank heat exchanger. Containment isolation valves are supplied at each penetration in accordance with the requirements of GDC 56 and 57. (See Section 6.2.4.)

The check valves supplied on the CCWS supply lines inside the Containment prevent reverse flow in case of pipe rupture.

The performance of all essential equipment can be monitored from the Control Room. Low flow, high temperature, and high radiation level, which are indicative of system malfunction, are annunciated in the Control Room. System pressure indication is provided in the Control Room.

Leakage from any system being cooled into the CCWS can be detected as an increase in the level of the CCWS surge tank or as an increase in system radiation level when the system being cooled is contaminated. Details of the radiation monitoring equipment are given in Section 11.5.

Leakage from the CCWS to the SSWS or to the atmosphere is detected as a decrease in the CCWS surge tank level.

The ability to detect leakages to or from the CCWS permits the operator to take appropriate action before degradation of both boundaries between normally radioactive fluid and the SSWS can occur.

The partition in the surge tank below its equalizing opening provides separate surge volumes for each safeguards loop. If one of them develops a leak and is taken out of service, the operation of the other loop is unaffected. If a leak develops in the nonsafeguards loop, it can be isolated from the safeguards loops and their operation left unaffected. The partition is designed to maintain its integrity with one side of the surge tank empty.

In the unlikely event that leakage from the system exceeds the makeup rate such that the level in the surge tank drops faster than the operator can respond, each safeguards loop and the non-safeguards loop are automatically isolated from one another to prevent loss of inventory from both safeguards loops.

All major components of the system, piping, and its appurtenances are located inside the seismic Category I structure. Each CCWS pump is located in an individual, compartment and at an elevation above the highest water level that can occur as a result of equipment failures within the building.

Manual Valves CC-0107, CC-0109, CC-0157, and CC-0158 have modified discs which have been drilled to serve as flow restriction orifices. These valves are closed to provide acceptable CCW flow balancing for Design Basis Accidents to limit heat addition to CCW. These valves are not required to be opened to mitigate a Design Basis Accident (e.g. LOCA); however, they may be opened to accelerate cooldown after accident heat loads have sufficiently decayed if desired and if the valve locations are accessible. However, CC-0109 and CC-0157 are classified as Active Valves (see Table 3.9B-10) for an open safety function because they may be required to be opened to satisfy the functional requirements for cooldown of BTP RSB 5-1 (see Appendix 5A).

The non-safety radiation monitors and their sample piping on the safeguards loops are seismically supported to ensure their failure does not adversely affect the CCWS.

The previously mentioned conditions and the design of the CCWS as seismic Category I make the system capable of withstanding adverse environmental occurrences such as postulated earthquakes, tornadoes, and tornado missiles (for details on seismic Category I structures, see Sections 3.2, 3.3, and 3.5).

The compartment walls around each pump preclude coincident damage to redundant equipment in the event of a postulated pipe rupture or an equipment failure with consequent pipe whip, jet impingement, and missile generation.

CCW containment isolation valves in the return line from the RCPs thermal barrier close automatically in the event of a RCP thermal barrier tube break. CCW flow from the thermal barrier of all four RCPs is stopped automatically whenever outlet flow or temperature reaches a high value, indicative of a thermal barrier break. Failure of the nonsafeguards loop (loss of CCW) has the following effect upon the reactor coolant pumps:

1. The chemical and volume control system continues to provide seal injection water to the RCPs; the seal injection flow is sufficient to prevent damage to the seals with a loss of thermal barrier cooling.

2. The motor winding temperature will increase as a result of the loss of CCW to the motor air cooler.

3. Lube oil temperature will increase, with a corresponding rise in bearing metal temperature, as a result of the loss of CCW to the motor bearing oil coolers. Testing, performed by Westinghouse, has shown that the manufacturer's recommended maximum bearing operating temperature will be reached in approximately 10 minutes. Therefore, the RCPs will incur no damage with a CCW flow interruption of 10 minutes.

4. No circuitry is provided to directly trip the RCPs on loss of component cooling water.

The following component cooling water instrumentation is provided:

1. Cooling water flow and temperature for all components associated with each RCP are indicated in the Control Room.

    a. RCP motor bearing high temperature alarms for each RCP are provided on the plant computer.

    b. Temperature indicator and alarm for the CCWS return line from each RCP are located on the main control board.

    c. CCWS flow indicator and alarm for each RCP oil cooler are located on the main control board.

    d. CCWS isolation valve monitor lights, indicating valve closure, are also located on the main control board.

2. Low cooling water flows are annunciated in the Control Room.

    Westinghouse has conducted a human engineering analysis of a loss of CCW to the RCPs, considering (1) the instrumentation provided, (2) the psychological stress induced on the average trained operator, and (3) the response time required of the operator to trip the reactor and stop the RCPs. It was found that the stress is much less than that induced by a LOCA (which would cause a response time delay of one minute), and that the operator actions required are not complicated and are a direct logical result of the event symptoms, as alarmed and indicated. In view of these factors, 10 minutes is a conservative and appropriate operator response time for this event during normal operation. In addition, since testing performed by Westinghouse has demonstrated that the RCPs are capable of operation for a 10-minute CCW flow interruption to the oil coolers without damage to the pumps, a period of ten minutes is adequate for the operator to take corrective actions or to trip the pumps, if necessary.

    During startup, cooldown and cold shutdown, two CCW pumps are normally used to supply all CCW requirements for one unit, including two RHR pumps. Failure of one of the CCW pumps or a single incorrect valve closure or a single moderate energy line crack

would cause a temporary reduction of CCW flow to individual components while the CCWS is realigned. As discussed above, all low flow alarms and control for realignment are contained in the Control Room, and the time period for operator response would be no more than 10 minutes.

Depending on the type of failure, the reduced flow rate would be no less than 50% of the normal flow rate. The RHR pumps would continue to operate as required under the circumstances. At no time would there be a cessation of CCW flow to either of the RHR pumps except in the case of a pipe crack in certain lines supplying an RHR pump. The realignment in this case would cause complete loss of CCW flow to one of the RHR pumps; the other RHR pump would continue to be supplied with CCW and cooldown would be accomplished using the single RHR pump. Cold shutdown may be achieved using only one CCW pump, if the other pump is not available.

Cold shutdown may be maintained using only one operating CCW pump with the other pump on standby. Failure of the operating pump would result in the automatic start of the standby pump within 30 seconds. Loss of CCW flow to the RHR pumps for this period of time can be tolerated.

During safeguards operation, the CCWS is train aligned. At this time, one CCW pump supplies one safeguards train, including one RHR pump. The other CCW pump supplies the other safeguards train, including the other RHR pump. Loss of any part of one safeguards train can be tolerated.

9.2.2.3.1 Safety Implications Related to Sharing

The CCWS includes two redundant CCWS pumps, two redundant CCWS heat exchangers, two redundant safeguards loops, and one nonsafeguards loop. The interconnections between Unit 1 safeguards loops and Unit 2 safeguards loops are blocked by at least one locked-closed isolation valve. Component cooling water for control room air-conditioning and UPS air-conditioning may be provided by either unit as these are shared systems (see FSAR Section 9.4.1 and 9.4C.8, respectively). There is no sharing of any other safeguards component or safeguards function between the two units.

9.2.2.3.2 Leakage Detection and Control

Leakage from any component being cooled into the CCWS can be detected as an increase in the level of the CCWS surge tank or as an increase in CCWS radiation level when the component being cooled is contaminated. Leakage from the CCWS to the SSWS or to the atmosphere is detectable by a decrease in CCWS surge tank level.

Detection of leakage in the CCWS surge tank is followed by the manual operation of either the feed system from demineralized water or the feed system from reactor makeup water, or the alarmed automatic actuation of the emergency feed system from reactor makeup water. Each CCWS surge tank level is recorded in the Control Room to facilitate leakage detection.

The CCWS surge tank contains water to provide a continuous component cooling water supply until a leaking CCWS loop can be taken out of service, with flow transferred to the other CCWS loop. The partition in the surge tank provides two compartments for separate surge volumes for each safeguards redundant loop. If one loop develops a leak and is taken out of service, the

operation of the other loop is unaffected. A leaking safeguards redundant loop can be identified by isolating each loop of the system with the water level below the equalizing opening, and observing the water level in the related surge tank compartment volume. If the level stops decreasing when the safeguards loop is isolated, the source of leakage must be, by elimination of sources, in the nonsafeguards loop. In this case, if outside the Containment, the leaking component can be identified by visual inspection. A leaking component inside the Containment is identified by abnormal flow indication from that component in the Control Room.

9.2.2.4 Tests and Inspection

Prior to installation in the CCWS, each component is inspected and cleaned.

Preoperational testing consists of calibrating the instruments, testing the automatic controls for actuation of the proper set points, and checking the operability and limits of alarm functions.

Periodic chemical examination of the component cooling water is made for pH, chloride content, and corrosion inhibitor content; manual adjustment is made if required. Sufficient instruments are provided to monitor system performance. Periodic visual inspection and preventive maintenance can be conducted as necessary without interruption of the CCWS operation.

9.2.2.5 Instrumentation and Control Requirements

9.2.2.5.1 General System Monitoring

Each power-operated valve is supplied with a control switch and a position indicating light on the control board in the Control Room. Radiation monitors are provided with readout in the Control Room.

Separate switches and actuation circuitry are provided for redundant components which are physically and electrically separated from one another.

9.2.2.5.2 Instrumentation

1. Pressure

   Pressure indication for each CCWS pump discharge is provided both locally and in the Control Room. Pressure indication is provided locally for CCWS pump suction.

2. Temperature

   a. CCWS heat exchanger outlet temperature is indicated in the Control Room. A high outlet-temperature alarm is also provided in the Control Room.

   b. Containment spray heat exchanger and RHR heat exchanger outlet temperatures are indicated in the Control Room.

   c. Local temperature indication is provided for all components, except for the process sample coolers and the components inside the Containment. The process sample coolers have temperature test wells, and the components inside the Containment have indication in the Control Room.

3. Flow

   a. Flow indication is provided in the Control Room for component cooling water recirculation, component cooling water main flow to the loops, RHR heat exchangers, and Containment spray heat exchangers, in addition to all components inside the Containment (reactor coolant pumps accessories, excess letdown heat exchanger, and reactor coolant drain tank heat exchanger). The component cooling water flow is also recorded. Sufficient flow indication at all CCWS components is provided for flow balancing the system.

   b. Low flow alarms are provided for all CCWS components except for the process sample coolers, the control room air- conditioning condensers, the safety chilled water system condensers, and the ventilation chillers. The ventilation chillers have high-flow alarms.

4. Level

   CCWS surge tank level indication is provided for each compartment locally and in the Control Room. It is also alarmed and recorded. CCWS drain tanks level indication is provided locally and is alarmed.

9.2.2.5.3 Control

1. CCWS Pumps and Remote-Operated Valves

   a. All pumps and remote-operated valves are operated from the Control Room.

   b. All remote-operated valves are supplied with open-close position indicating lights in the Control Room. Both lights are on when the valve is in the intermediate position.

   c. CCWS pumps for both trains A & B have control switches located on the hot shutdown panel (HSP). In addition, CCWS Pump Train A transfer switch (CR-HSP) is located on the shutdown transfer panel, while CCWS pump train B transfer switch is located on the hot shutdown panel. Local control position alarms are provided in the Control Room.

   d. If a low CCWS pressure signal is received, the redundant CCWS pump and its associated SSWS pump automatically start. The CCWS pump and the SSWS pumps are train associated pumps.

   e. If a SSWS pump automatic start on low SSWS pressure occurs, the corresponding CCWS pump is automatically started to ensure correct alignment of the safeguard train.

   f. During blackout conditions, CCW pumps are automatically loaded onto the emergency diesel generator within 30 seconds of blackout.

2. Makeup

Makeup to the CCWS Surge tank is normally obtained from the Demineralized Water Storage Tank (DWST) or the Reactor Makeup Water Storage Tank (RMWST) by remote manual operation of valves for either surge tank compartment. The operator is informed of the need for makeup by a low-level alarm for each compartment. The demineralized water pumps are used to transfer makeup water from the demineralized water system. If the alarm signal does not reach the operator because of equipment failure or operator inattention, makeup is automatically supplied by the emergency makeup system from the Reactor Makeup Water Storage Tank (RMWST). A low-low-level signal opens valves for either surge tank compartment and actuates a low-low-level alarm in the Control Room. The low-low-level actuated valves close automatically when the water in the surge tank reaches the high level.

Makeup can also be supplied by manual operation of level control valves, should the Instrument Air System become unavailable.

3. CCWS Pump Recirculation

If the CCWS pump flow falls below the required minimum, the CCWS recirculation valve opens automatically. Once the component cooling water flow increases above the required value, the valve closes automatically. The recirculation valve is also automatically closed upon receipt of an “S” signal.

4. Safeguards Chilled Water Condenser/Nuclear

The cooling of the chilled water is based on the condenser refrigerant pressure. A pressure controller on the refrigerant side of the condenser adjusts the CCW regulating valve on the condenser outlet piping. This valve is air operated with accumulator tank backup. The accumulator tank is sized to supply air for a minimum of 30 minutes after an instrument air header low pressure condition. The valve is handwheel equipped for manual operation by a control room operator when instrument air is exhausted and CCW temperature is below 70 degrees F.

5. Evaporator Flow Control

Component cooling water to the evaporator condensers of the various evaporator packages is modulated within the evaporator package to meet demand.

6. Ventilation Chillers and Letdown Chillers

Flow to the ventilation chillers is stopped automatically when it reaches an abnormally high value indicative of leak or system malfunction. In addition, the CCWS inlet and outlet valves to the ventilation chiller and letdown chiller condensers are closed automatically on an “S” signal.

The cooling of the ventilation chiller is based on the condenser refrigerant pressure. A pressure indicating controller (PIC) on the refrigerant side of the condenser adjusts the CCW regulating valve (200-2000 gpm) to the condenser and inversely the CCW bypass valve (1800 - 0 gpm) for a total flow around/across the condenser of 2000 gpm.

7. Thermal Barrier Cooler

RCP thermal barrier CCW outlet flow and temperature instrumentation provides Class 1E, redundant and diverse signals for Class 1E indication in the main control room and automatic closure of CCW containment isolation valves upon sensing RCP thermal barrier tube break. A Train A flow transmitter in the return from each thermal barrier (FT-4678, FT-4682, FT-4686, or FT-4690) will automatically close Train A Containment Isolation Valve HV-4709 on high flow. A Train B temperature element in the return from each thermal barrier (TE-4691, TE-4692, TE-4693, or TE-4694) will automatically close Train B Containment Isolation Valve HV-4696 on high temperature (See Figure 9.2-3).

In addition, flow from the thermal barrier of any RCP is stopped automatically with control grade valves (TV-4691, TV-4692, TV-4693, or TV-4694) whenever the outlet temperature (TE-4691, TE-4692, TE-4693, or TE-4694) reaches an abnormally high value indicative of a thermal barrier break. These control valves provide an additional level of redundancy and diversity of protection from an intersystem LOCA. Control grade alarms are provided for outlet high flow and temperature (See Figure 9.2-3).

8. Drain Tank Pumps

CCWS drain tank pumps are interlocked to stop on low drain tank level.

9. S Signal

When an S signal occurs, the following operations take place:

a. Both CCWS pumps are started.

b. The Containment CCWS drain tank pumps stop.

c. The pump recirculation valves close.

d. The RHR heat exchanger CCWS isolation valves open fully then partially close to a preset throttle position.

e. CCWS Containment phase “A” isolation valves close as described in Section 6.2.4.

f. The ventilation chiller condenser’s and letdown chiller condenser’s isolation valves close.

g. The Primary Sampling System CCW isolation valves close as listed in Table 7.3-4.

10. P Signal

When a P signal occurs the following operations take place.

a. The RHR heat exchanger CCWS isolation valves open fully then partially close.

b. The Containment spray heat exchanger CCWS isolation valves open fully then partially close to a preset throttle position.

c. CCWS containment Phase “B” isolation valves close as described in Section 6.2.4.

d. The safeguards loop isolation valves close.

e. The non-safeguards loop isolation valves close.

9.2.2.5.4 Radiation Monitoring

Radiation monitors are provided on the return headers in each loop of the CCWS. Any of these monitors can trigger a high-activity-level alarm signal in the Control Room. A temperature switch is provided to trip the monitor sample pump if the temperature exceeds 120°F to protect the detector.

A radiation monitor is also provided in the common discharge line of the CCW Drain Tanks, Auxiliary Building Sumps 3 and 11, and the Diesel Generator sumps. A high radiation signal or radiation monitor failure diverts the discharge from the LVW System to the Cocurrent Waste System.

9.2.3 DEMINERALIZED AND REACTOR MAKEUP WATER SYSTEM

9.2.3.1 Design Bases

The design of the Demineralized and Reactor Makeup Water System is in accordance with the following criteria: 10CFR50, Appendix-A, General Design Criteria (GDC) 1, 2, 3, 4, 5, 52, 54, and 56; NRC Regulatory Guide 1.26 and 1.29; and Branch Technical Position APCSB 3-1. The related electrical systems are designed to comply with the requirements of regulatory guides, standards and other documents as described in Section 8.1.4. The related Containment Penetrations are designed to comply with the requirements described in Section 6.2.4. The Demineralized and Reactor Makeup Water System is a single system designed to provide an adequate supply of deaerated demineralized water of reactor coolant purity to other systems as makeup and to provide other plant demineralized water requirements for both units of the CPNPP.

The Demineralized Water Storage Tank (DWST) receives water from the Water Treatment System. The demineralized water chemistry is in accordance with the EPRI PWR Primary Water Chemistry Guidelines (including Westinghouse supplements). This water normally serves as a source of demineralized water for initial filling of the Reactor Coolant Systems (RCSs) including the pressurizer relief tanks, the boric acid tanks, the Refueling Water Storage Tanks, the CCWSs, the Turbine Plant Cooling Water (TPCW) Systems, and the spent fuel pools. Demineralized water is also transferred to condenser hot wells, Condensate Storage Tanks, and Reactor Makeup Water Storage Tanks (RMWSTs) as required by their respective levels. The demineralized water is deaerated in a vacuum degasifier in the water treatment plant to remove dissolved oxygen before it is transferred to the Condensate Storage Tanks and the RMWSTs. The RMWST has a nitrogen sparger installed inside the tank. Nitrogen from the plant bulk low pressure nitrogen gas system is injected into the tank to reduce and maintain low dissolved oxygen levels.

Demineralized water is also provided via a transfer pump to the containment pre-access filtration units charcoal deluge system, and to the fire protection standpipe for the containment hose stations. For a detailed description of the Fire Protection System refer to Section 9.5.1.

Demineralized water is also required to perform turbine generator primary flow rate testing during refueling outages. The source of the demineralized water is provided at the inlet line to the turbine plant cooling water head tank.

Demineralized water is supplied to some eyewash stations and showers which are located upstream of any connections to radioactive portions of the system (e.g. Reactor Makeup Water) or connections to radioactive components (e.g. Refueling Water Storage Tanks) such that they are adequately protected from inadvertent radiation contamination. Either aerated or deaerated water may be used for this purpose. Deaerated water becomes aerated upon discharge from the spray nozzles and is acceptable for this use. Water downstream of the Reactor Makeup Water Storage Tank is not used for eyewash stations or showers.

The RMWST serves as a source of reactor makeup water and may contain trace amounts of tritium if it has been recycled from the RCS. Demineralized deaerated water is provided to the RCS and the auxiliary equipment, where the presence of tritium is not objectionable. These include supply to evaporators, gas strippers, pumps, demineralizer tanks, and pipelines for cleaning and flushing operations, and to the RCS as a diluent. The reactor makeup water chemistry is in accordance with the EPRI PWR Primary Water Chemistry Guidelines (including Westinghouse supplements).

9.2.3.2 System Description

The Water Treatment System (which includes surface water pretreatment) is used to produce demineralized water, through three new trains with a total capacity of 420 gpm, which is fed directly to the DWST. Figure 9.2-4 shows the paths used for treating lake water or well water in the Water Treatment System. The mixed-bed effluent flows directly to the DWST.

Sulfuric acid and sodium hydroxide are used to regenerate the Water Treatment System demineralizers. The spent regenerants are pumped to the Wastewater Management System for treatment prior to discharge, as discussed in Section 9.2.8.2.

Water from the DWST is used as a source for initial filling of the RMWST and Reactor Makeup Water System.

The RMWST, with makeup from the DWST if required, accommodates the makeup requirements resulting from a cold shutdown, followed by startup from cold conditions. A minimum quantity of water must be available during normal plant operation to achieve and maintain a safe, cold shutdown.

The RMWST is also required to provide a seismic category I supply of makeup water for the Chemical and Volume Control System, the Safety Chilled Water System, the Component Cooling Water System, and the Spent Fuel Pool Cooling System.

In the event of a loss of water from the spent fuel pools to the minimum suction elevation, there is normally sufficient volume in each RMWST to provide makeup to restore and maintain levels until cooling is restored.

As a minimum, one reactor makeup water pump per unit is normally operating.

If the system load demand is low, the pump recirculation insures that sufficient flow of tank water passes through the outdoor piping to prevent icing during severe outdoor temperatures.

The Demineralized and Reactor Makeup Water System is shown schematically in Figure 9.2-5.

The original system design intent was for Comanche Peak to be a “no release” plant, thereby recycling and reusing all effluent streams. For ALARA reasons, this philosophy has changed and the intent now is to prevent any tritiated or otherwise contaminated effluent from reaching the Reactor Makeup Water Storage Tanks. For this reason the isolation valves between the WP Waste Evaporator Condensate Pumps and the DD/RMW system are presently administratively controlled in a closed position. Also, the flowpath from the recycle evaporator package to the RMWST has been removed. In any case, previously designed processing of effluents from the waste evaporator is possible by simply opening these valves and aligning other valves as appropriate.

9.2.3.3 Safety Evaluation

The Demineralized and Reactor Makeup Water System has sufficient storage capacity to supply makeup water to the system should the Water Treatment System be out of service for short periods of time. Two 50% capacity pumps and two trains of ion exchangers are provided to ensure the reliability of the Water Treatment System.

The Water Treatment System does not contain, treat, or produce any radioactive material in its operation. Therefore, any waste produced by this system can be deposited directly into the Low Volume Waste Treatment Facility.

The RMWST, reactor makeup water pumps, associated valves and piping, and all connections to the boric acid blenders, the safety chilled water surge tanks, the component cooling water surge tanks, and the spent fuel pools are designated seismic Category I and Safety Class 3. Containment penetrations are described in Section 6.2.4. Components of the Demineralized and Reactor Makeup Water System are compatible with all chemicals used throughout the system. Stainless steel piping is used throughout the system as a deterrent against corrosion in the transfer of corrosive fluids.

The RMWST is designed to withstand the effects of natural phenomena such as safe shutdown earthquake (SSE), probable maximum flood (PMF), and tornado missile. All components located in the Containment, Safeguards, Auxiliary, Fuel and Electrical, and Control buildings are located inside seismic Category I structures, which are also designed to withstand tornado effects. These seismic Category I structures are capable of withstanding adverse environmental conditions such as postulated earthquakes, tornadoes, and tornado missiles. (For details on seismic Category I structures, see Section 3.2.1, 3.3, and 3.5.)

Any failure of non-safety-class-design equipment associated with the Demineralized and Reactor Makeup Water System will not cause any failure of safety-related systems or components.

Flow restriction orifices are provided in each Unit’s Reactor Makeup Water supply line to the Chemical and Volume Control System. These orifices are designed to limit the rate at which unborated water can be added to the RCS. For further information see section 15.4.6.

9.2.3.4 Testing and Inspection Requirements

The equipment in the Demineralized and Reactor Makeup Water System is initially inspected and tested to insure system integrity and completeness. The inspection includes the following:

1. Pumps and Motor Drives

   Each pump is started and runs for sufficient time to insure its proper operation. Maintenance is provided as required.

2. Valves

   Each valve is operated through its complete range to ensure that it is in normal operating condition.

3. Safety Class Isolation Valves

   An electrical signal is transmitted to ensure that isolation valving systems (safety-related) are operating properly.

4. Instruments and Annunciators

   Operation is checked periodically for proper operation and accuracy.

9.2.3.5 Instrumentation Requirements

Control for the Demineralized and Reactor Makeup Water System is monitored from the main control panel or from local panels: the potable and demineralized water panel (PDP) or the water treatment panel (WTP), which are located adjacent to each other.

The Demineralized and Reactor Makeup Water System is controlled from a local PDP. System operation is designed to be fully automatic, but with provision for manual control. The final water quality of the Water Treatment System is continuously monitored for conductivity. High conductivity is alarmed locally. Deviations from specified water quality trip the demineralizers to prevent poor quality water from entering the DWST. Regeneration is manually initiated and then proceeds automatically or manually until completion.

Level indicators are provided in both the local panel and the Control Room for the DWST (PDP), RMWST (PDP), and vacuum deaerator (WTP) to provide level indications. The vacuum deaerator provides local level indication only. For the RMWST, temperature measurement is also provided with an indication on the main control panel. Vacuum deaerator pressure is also monitored and indicated locally. Pump discharge pressure is monitored and indicated locally for all pumps.

Flow measurements are provided for the main demineralized water supply line and the discharge line of the deaerated water transfer pumps. Signals from flow transmitters are indicated on the local panel with total flow indication on the main control panel.

9.2.4 POTABLE AND SANITARY WATER SYSTEM

9.2.4.1 Design Bases

The station potable and sanitary water system is designed to provide the following:

1. Water for toilets, sinks, showers, and drinking purposes in all permanent personnel areas of the plant site, as required.

2. Water for emergency eyewash and showers, as required.

3. Water to fire protection hoses for various on-site buildings.

4. Water to fill and to provide normal makeup to the Fire Protection Storage Tanks.

9.2.4.2 System Description

The distribution system that provides potable water to the plant and associated support structures and buildings for both Unit 1 and 2 is supplied by the Somervell County Water District (SCWD) public water system.

The potable water is chlorinated to conform with the Texas Department of Health regulations.

Domestic wastes generated on-site are transferred to and treated in, the 90,000-gpd domestic waste treatment facility. Domestic waste treatment effluent is treated with high intensity ultraviolet light for disinfection and odor reduction and is discharged to the Squaw Creek Reservoir in accordance with regulatory agency permitted effluent limits. The treatment facility sludge is dewatered and bagged for appropriate disposal. (See Figure 9.2-7)

9.2.4.3 Safety Evaluation

The potable and sanitary water system is designed without interconnection with, and is physically separated from, any radioactive sources, thus precluding the possibility of radioactive contamination. It is completely divorced from the laundry and hot shower portion of the Liquid Waste Processing System described in Section 11.2. Wastes produced by the potable and sanitary water system contain no radioactive materials and can therefore be safely treated in the domestic waste treatment facility.

The maximum radioactive release from the plant is below the limits established by 10 CFR Part 20, Appendix B. Contamination of ground water for normal operation is not considered significant; the concentration is well within 10 CFR Part 20, Appendix B limits. For an evaluation of the effects of accidental releases of liquid radioactive material, see Sections 2.4.12 and 2.4.13.3. Monitoring and safeguard requirements are discussed in Section 2.4.13.4.

Because the system is common to both units and is independent of their operation, a shutdown of either or both units does not affect the supply of potable water. Additionally, potable water can be trucked to the site and distributed in portable containers.

In case of water contamination (radiological or otherwise) or an accident where the piping is forced out of service, potable water can be trucked to the site and distributed in portable containers.

9.2.4.4 Tests and Inspection

1. Sampling of Potable Water

   Water samples are taken at various locations throughout the Potable Water System. Samples are collected and tested in accordance with regulatory agency standards.

9.2.5 ULTIMATE HEAT SINK

9.2.5.1 Design Bases

The function of the ultimate heat sink is to dissipate heat rejected from the Station Service Water System during postaccident shutdown and normal cooldown conditions. It has been sized to provide adequate cooling capacity for the CPNPP as required by NRC Regulatory Guide 1.27, Rev. 2.

The ultimate heat sink complex of the CPNPP consists of the Safe Shutdown Impoundment (SSI) (the body of cooling water) and the SSI dam, as shown in Section 2.4 and on Figure 2.4-1.

There are two possible conditions of operation for the SSI: 1) the simultaneous shutdown and cooldown of both units or 2) the shutdown and cooldown of one unit concurrent with the dissipation of post-design-basis-accident (DBA) rejected heat from the other unit. As required by NRC Regulatory Guide 1.27, the SSI has been sized for the most severe of these two alternatives, which, in the case of the CPNPP, is the first condition. In addition, it is assumed that the Squaw Creek Reservoir (SCR) is lost coincident with the DBA and that no makeup water is added to the SSI for the 30-day period of evaluation following the DBA. Therefore, all of the 30-day postaccident cooling requirements are satisfied by the amount of water contained in the SSI with the initial water level at elevation 770 ft 0 in. followed by loss to elevation 769 ft 6 in., which is the limit of entrainment set by the invert of the equalization channel between the SSI and the SCR. (See Section 2.4.)

The ultimate heat sink design is in accordance with the following GDC and NRC Regulatory Guides:

1. GDC 2 [1], 4 [2], 5 [3], 44 [5], 45 [6], and 46 [7].

2. NRC Regulatory Guides 1.27 [11] and 1.29 [12].

Specifically, the ultimate heat sink complex can withstand the effects of the SSE, tornado, or flooding effects of the probable maximum flood (PMF) without loss of safety function.

9.2.5.2 System Description

The SSI is an enclosed body of water formed from a cove of the SCR and is retained by a seismic Category I dam, designed and constructed to withstand the most severe postulated

natural phenomena, as described in Chapter 2. The SSI dam separates the SSI from the SCR. A complete description of the SSI dam is given in Section 2.4.

An equalization channel allows each body of water to adjust to a common level above the minimum water level. In the event the SCR dam fails, this channel limits the low water level in the SSI to 769 ft 6 in., at which point the volume of water contained is approximately 284 acre-feet, allowing for 40 years of sedimentation. During normal operation, the water level is at elevation 775 ft with a volume of 467 acre-feet contained. A non-intrusive, floating raft system has been placed near the south end of the equalization channel. The raft system has been designed to allow water to freely pass under the raft for all water levels without blocking flow during severe natural phenomena such as a postulated seismic or PMF event.

A bleed line is provided from the Circulating Water System upstream of the main condensers, at the CWS intake structure, to the Service Water Intake Structure pump suction pit. The circulating water bleed-off was designed to ensure continuous makeup to the SSI and provide constant outflow (blowdown) to prevent excessive dissolved solids concentration in the SSI. The bleed line is not in operation due to concerns about the piping's internal liner. Make-up is not required because of the equalization channel described above. Dissolved solids are reduced by normal rainfall and runoff.

9.2.5.3 Safety Evaluation

The heat rejection capabilities of the SSI are a function of meteorological conditions, the volume and surface area of the SSI and the location of structures. All three of these features can be represented by using a numerical, three-dimensional, time-varying hydrodynamic and transport model to simulate the performance of the SSI.

An analysis of 39 years of offsite meteorological data showed that August 31, 1990 was the most severe period for ultimate heat sink performance, both for the 24 hour transient analysis and for peak SSI intake temperatures. These data are shown in Table 2.3-7B. Similarly, the period June 25, 1980 to July 25, 1980 was shown to be the most severe period of evaporative water loss. The meteorological data for this period is shown in Table 2.3-7C. See Section 2.3.1.2.10 for details.

The hydrodynamic and transport model numerically solves the three-dimensional momentum, mass and constituent balance equations. The model requires a finite difference representation of the SSI bathymetry, locations and dimensions of the discharge and intake structures and of the equalization channel connecting the SSI with SCR. These data were generated from topographic and design drawings.

Heat rates of the cooling water discharged onto the SSI are based on two safety trains in operation per unit, and are developed from the nuclear steam supply system (NSSS) manufacturer's functional requirements, design criteria for residual decay heat removal, and from balance-of-plant (BOP) heat load requirements. In generating the spent fuel pool heat load, it was assumed that the DBA occurs in one unit during normal two unit operation during summer conditions. The spent fuel pool heat loads for summer conditions are described in Section 9.1.3.1.

The decay heat rate, however, is based on BTP ASB 9-2. This results in a high heat input which reflects a conservative spent fuel pool heat loading in conjunction with a conservative containment spray and RHR heat loading.

1. Compliance with NRC Regulatory Guide 1.27

   The intent of NRC Regulatory Guide 1.27 is met by SSI. The thermal ability of the SSI to act as an ultimate heat sink for 30 days is covered in Item 3, Thermal Performance Evaluation. As a single source ultimate heat sink serving two units, the SSI is able to withstand the effects of natural phenomena such as earthquakes, tornadoes, hurricanes, and floods, taken individually, without loss of capability to perform its safety functions. The natural phenomena and their magnitude are selected in accordance with their probability of occurrence, and designs are based upon the most severe of the natural phenomena recorded for the site, with an appropriate margin to account for uncertainties in historical data. Such phenomena and design criteria are discussed in Sections 2.4, 3.3, and 3.4.

2. Hydraulic Performance

   The SSI is supplied with a bleed flow from the Circulating Water System that may be operated to supply a continuous flow of water. This flow is directed into the service water pump intake structure and can be used if required as a blowdown flow from the SSI to the SCR through the equalization channel. If, as a result of an earthquake, the SCR dam fails, the equalization channel invert maintains the water level in the SSI at elevation 769 ft 6 in. A floating raft system in the equalization channel is non-intrusive relative to flow through the channel and does not adversely affect the performance of the function of the SSI. Surface area and volume in the SSI as a function of elevation is discussed in Section 2.4.

   Hydraulic short circuiting is prevented by the physical separation of the intake structure and discharge piping outfall and the orientation of the discharge. The intake and discharge points are over 1800 ft apart, and the exit velocity of the discharge water carries it upstream initially, away from the area of the intake structure, allowing extra time for the transfer of heat to the atmosphere.

   The service water system effluent flows through an open channel type discharge canal prior to entering the SSI. Cooling occurs throughout the approximately 1300 ft. length of the discharge canal before the service water effluent mixes with the SSI bulk fluid. The discharge canal outlet enters the SSI in a direction away from the service water intake structure. Mixing and dispersion of the effluent occurs when the SSI fluid reverses direction in order to pass to vicinity of the intake structure. The minimum shore line distance between the discharge canal and intake structure is 1500 ft. Further cooling of the mixed fluid occurs in route. Because the intake is submerged and the discharge is on the surface, there is vertical as well as horizontal separation. See Figure 1.2-46 for the intake structure.

3. Thermal Performance Evaluation

   To simulate the operation of the SSI preceding and following LOCA, the time varying meteorological data must be aligned with the heat load data such that the peak SSI intake

temperature due to the imposed heat load occurs at the same time that the natural temperature peak occurs. An additional consideration is that the simulation be run for a sufficiently long time so that the SSI temperature reaches stationary state with respect to the pre-LOCA heat load. Both these issues were addressed with sensitivity simulations. To determine the initialization time, steady meteorological data were used with a steady heat load to determine that the SSI temperatures reached steady values after 37 days. To determine the start of LOCA relative the single worst day for atmospheric cooling, the time-varying heat loads were run with steady meteorological data. These tests showed that the time to peak intake temperature due to time varying heat loads and plant pumping was 10 days. To further refine this number, additional sensitivity tests were made in which the time to peak was varying under both time-varying meteorological and heat load conditions. The resulting time to peak was found to be seven days.

The LOCA intake temperature reaches a maximum on the evening of the seventh day after a LOCA at less than 117°F (two train).

A 24-hour transient analysis for a two train ESF LOCA case was conducted with the model to determine how quickly intake temperatures would rise immediately after a LOCA. The lag between discharge and intake temperature rises was determined to be more than 5 hours.

Component cooling water temperature is a function of heat load on the system and the service water temperature. Component cooling water supply temperature in the DBA unit peaks at approximately 135°F, upon initiation of containment spray recirculation, with service water temperature at 102°F. The maximum component cooling water temperature of the shutdown unit, with only one heat exchanger operable, is limited to 122°F. This temperature occurs upon initiation of RHR, several hours after the DBA occurred in the other unit. The preceding maxima of 135°F and 122°F ensure that the system will perform satisfactorily in mitigating the event in the DBA unit concurrent with orderly cooldown of the shutdown unit.

During the postulated 100-year drought conditions and after 40 years of sedimentation, the SSI is determined to have 284-acre feet of water. Although the design basis (RG1.27) requirement is for a 30-day cooling supply, the consumption analysis was based on a 39-day post-accident shutdown period. The maximum consumption of SSI water during the 39-day postaccident shutdown cooldown period amounts to approximately 92.4 acre-feet, resulting in a decrease in surface elevation of approximately 4 ft., allowing adequate margin for post-30-day operation without exceeding the service water pump submergence requirements. Refer to Sections 2.4.11.5 and 2.4.11.6 for further discussion of the heat sink dependability requirements.

A loss of offsite power for both units was evaluated using the same methodology as for LOCA described above. The intake temperature for this scenario reaches a maximum on the evening of the seventh day at 117.8°F. The maximum water consumption at 30 days amounts to approximately 72.3 acre-feet, resulting in a surface elevation of 766 ft. 7 in. The maximum water consumption at 39 days is approximately 92.4 acre-feet, resulting in a surface elevation of 765 ft. 8 in.

Consumption of SSI water by the Auxiliary Feedwater System for supply to the steam generators, if a beyond the design basis failure of one Condensate Storage Tank is

assumed, amounts to only approximately 0.67 acre-feet, based on a 60 gal of water per thermal MW rating of the steam supply system.

The time variation of decay power, based on the ANS 5.1 fission product curve, is corrected for a finite operating time and includes the ANS uncertainty factors. The related curve has been used to develop the values of decay heat rates shown in Table 9.2-13.

The heat rejection capabilities of the SSI are a function of the volume/surface area relationship of this body of water. The SSI is sufficiently sized to accept plant-rejected heat under the most severe conditions specified in NRC Regulatory Guide 1.27, Rev. 2.

9.2.6 CONDENSATE STORAGE FACILITIES

9.2.6.1 Design Bases

The condensate storage facilities (Figure 10.4-11) provide makeup and surge capacity for secondary system inventory changes caused by different operational conditions, thermal effects, and the draining and recharging of any part of the system. It also provides sufficient water storage (see Section 10.4.7.2 for additional details) for emergency decay heat removal by the Auxiliary Feedwater System in the event the Condensate and Feedwater System is inoperable. This amount of feed quality water is required to hold one unit for four hours at hot standby and then cool down the RCS at 50°F per hour for five hours, at which point the RHR System can take over [5].

The facilities are designed to seismic Category I, nuclear Safety Class 3 requirements [10],[12],[15]. The design conditions and criteria are those defined in Chapter 3. In addition, these facilities meet the requirements of the applicable codes listed in Section 3.8.4.

9.2.6.2 System Description

The condensate storage facility consists of one reinforced concrete, stainless steel-lined tank per unit, located in the plant yard adjacent to the respective diesel generator building (see Table 10.4-8 for additional details). The tank contains deaerated water of feedwater quality. A floating diaphragm arrangement prevents air leakage into the tank. The tank's stainless steel liner will not be attacked by deaerated demineralized water. The tank contains water which is used as storage for the Condensate System (see Section 10.4.7.2 for additional details). The reserve auxiliary feedwater (see Section 10.4.7.2 for additional details) cannot be drained by the non-nuclear-safety-related systems because of the elevation of the outlet nozzles. The tank receives makeup water from the Demineralized Water System.

Inadvertent drainage of the tank is prevented by normally closed double valves. In addition, automatically operated valves isolate all other uses of the Condensate Storage Tank when auxiliary feed operation is required, thereby maximizing the volume of water available for safety-related purposes. Refer to Section 10.4.9 and Figure 10.4-11 for the Condensate Storage Tank design parameters.

9.2.6.3 Safety Evaluation

The Condensate Storage Tank is an ANS Safety Class 3 structure. Since this facility is designed to seismic Category I requirements (Section 3.7), failure is not postulated, and there are no

structural and resultant environmental effects. Adequate ability of the tank to withstand earthquakes is ensured by designing the tank in accordance with the requirements for seismic Category I structures. The tank is also designed to withstand tornadoes and missile impact (Sections 3.3 and 3.5) [1],[2],[14]. Adequate protection against corrosion is ensured by a corrosion-resistant, ASTM A 240, chromium nickel stainless steel type 304L, 3/16-in.-thick liner.

Figure 10.4-11 shows the location of the CST vents. Each vent line alone provides the required venting capacity for normal operation and design basis accidents. The vents are located above the tank overflow level and their blocking by freezing water is not credible. The CST vents are classified as non-Nuclear Safety related and, in accordance with ANSI N18.2a, are designed to break away if struck by a tornado missile.

It is not anticipated that any significant radioactive material is present in the system.

Piping for the Condensate Storage Tank is located in the pipe tunnel and tank room. These areas are open to the Safeguards Building and are provided with thermostatically controlled area heaters. Freezing or icing under the most severe outdoor temperatures is not anticipated.

Environmental design conditions for seismic Category I self-actuated valves are described in Section 3.11.

9.2.6.4 Tests and Inspections

The Condensate Storage Tanks are tested during the preoperational test program for both the Condensate System and the Auxiliary Feedwater System. Periodic visual inspections are performed to ensure integrity of the tank [6],[7].

9.2.6.5 Instrumentation Requirements

Filling of the Condensate Storage Tank may be performed manually by the operator aligning demineralized water, or automatically by associated level instrumentation when the demineralized water source is continuously aligned. In the automatic makeup mode, Condensate Storage Tank makeup is supplied automatically whenever the tank level is below set point level. The demineralized water supply valve is opened to supply makeup until a high tank level is reached and the makeup valve is then closed. Makeup water can be supplied manually from a control board switch. Tank level is indicated locally and remotely in the Control Room. High-high, Low, and low-low tank level alarms are provided [4]. Redundant level transmitters are used. One level transmitter is used for makeup control and indications; the other transmitter is used for level monitoring and alarms.

The Condensate Storage Tank is isolated from its non-safety-related users by automatically closing motor-operated valves when the motor-driven or turbine-driven auxiliary feedwater pumps start. The condensate make-up and reject line is isolated by automatically closing motor operated valves when the condensate storage tank reaches a HI-HI level. Valves can be opened manually with control board switches with three positions, close, auto, and open, with spring return to auto.

9.2.7 SURFACE WATER PRE-TREATMENT SYSTEM

9.2.7.1 Design Bases

Surface Water Pre-Treatment System is designed to mitigate the possible consequences of ground water pumping at CPNPP by providing all water required for plant make-up.

9.2.7.2 System Description

Figure 9.2-4A shows a schematic flow diagram of Surface Water Pre-Treatment System. This treatment facility takes raw water from Squaw Creek Reservoir at the circulating water intake structure. Water is supplied from supply pumps installed at the intake structure. Removal of suspended matter and chlorination is accomplished by a pre-treatment (micro filtration) system. A reverse osmosis system reduces the dissolved solids. The micro filtration units produce a maximum of 800 gpm of filtered water. Storage tanks for filtered water and reverse osmosis product water provide system reliability and operation flexibility.

The supply pumps supply approximately 240 gpm to 800 gpm to the micro filtration system. The pre-treatment facility provides for combined filtration, collection, and discharge of suspended matter. A chemical feed system is provided for the addition of chemicals to aid in chlorination, which is automatically controlled to adjust varying flow rates. Feed concentration of chemicals is adjusted periodically to meet the changing inlet water conditions.

Flow from the micro filtration system supplies the Filtered Water Storage Tank to control changes in flow rates.

9.2.7.3 Safety Evaluation

The Surface Water Pre-Treatment System is designed without interconnection with, and is physically separated from, any radioactive sources, thus precluding the possibility of radioactive contamination. It is completely divorced from the laundry and hot shower portions of the Liquid Waste Processing System described in Section 11.2. Wastes produced by the blowdown of the water treatment equipment contains no radioactive materials and can therefore be pumped to the Wastewater Management System, for treatment prior to discharge.

Because the system is common to both units and is independent of their operation, a shutdown of either or both units does not affect the supply of water.

9.2.7.4 Inspection and Testing Requirements

As the components of this system are in either continuous or intermittent use during normal plant operation, no additional periodic tests are required, although periodic visual inspections and preventive maintenance are conducted as necessary. All components are accessible for periodic inspection.

9.2.7.5 Instrumentation Requirements

The Surface Water Pre-Treatment System is designed for semiautomatic operation. The system is provided with audio and visual annunciation to alert the operator of abnormal conditions.

9.2.7.6 Water Treatment System

The Water Treatment System will normally produce 420 gpm of processed water for plant operational use. The primary equipment required to support this production rate includes two micro filtration units, Multi Media Filters (normally all three Multi Media Filters are in service), First Pass Reverse Osmosis Banks, the Forced Draft Decarbonator, Second Pass Reverse Osmosis Banks, the Vacuum Degasifier and Mixed Bed Demineralizers.

A flow control valve to the micro filtration units modulates flow into the unit. The chemical feeds provided for the micro filtration units include Sodium Hypochlorite, Caustic and Acid. Each chemical has at least one storage tank and a metering pump.

The micro filtration units are a membrane based filtration system which removes suspended solids and achieves water quality requirements in a one-step filtration process.

The Filtered Water Storage Tank receives the water, once it has been filtered. The Filtered Water Forwarding Pumps transfer water from the Filtered Water Storage Tank to the Multi Media Filters. Filtered water also serves as the backwash media for the Multi Media Filters.

Water from the Multi Media Filters is further processed with bisulfite, Cartridge Filters, and antiscalant injection. The bisulfite, sulfuric acid and antiscalant have at least one storage tank and two metering pumps each. There are three Cartridge Filter units.

Water is then transferred via the First Pass Reverse Osmosis Feed Pumps to the three First Pass Reverse Osmosis Banks. The First Pass Reverse Osmosis Banks are designed to reduce the total dissolved solid content of the water.

From the First Pass R.O. product, water is sent through a Forced Draft Decarbonator, where carbon dioxide concentration in the water is reduced.

After decarbonation, the Second Pass Reverse Osmosis Feed Pumps will transfer the processed, decarbonated water to the Second Pass Reverse Osmosis Feed Banks. The Second Pass R.O. Feed Banks will further reduce the effluents total dissolved solids content. Following this treatment, water is transferred to the R.O. Product Water Storage Tank.

R.O. Product Water Forwarding Pumps transfer water from the R.O. Product Water Storage Tank to the Vacuum Degasifier. The Vacuum Degasifier will reduce the dissolved oxygen content of the effluent. Effluent from the Vacuum Degasifier then enters the Mixed Bed Demineralizer/Polisher Units.

The Mixed Bed Units provide the final demineralizing/polishing treatment of the water which is then transferred to the Demineralized Water Storage Tank for use in the plant.

9.2.8 WASTE MANAGEMENT SYSTEM

9.2.8.1 Design Bases

The Waste Management System (WMS) provides wastes collection, retention, conventional pollutant treatment, and discharge of normally non-radioactive wastewaters in accordance with the regulatory requirements of the Texas Commission on Environmental Quality (TCEQ) and the

Texas Pollutant Discharge Elimination System (TPDES) permits. The WMS is shown on Figure 9.2-15 and 9.2-16.

The WMS collects wastewaters for processing in either the Low Volume Waste (LVW) or Co-Current Waste (COW) treatment facilities. The WMS also provides for low level radioactive contaminated Condensate Polisher resins to be transferred to the LVW treatment facilities for retention and subsequent proper disposal in accordance with regulatory requirements.

9.2.8.2 System Description

9.2.8.2.1 Low Volume Waste Treatment Facilities

The Low Volume Waste (LVW) treatment facilities provide collection, treatment and discharge of normally non-radioactive wastewaters. The LVW facilities also provide for settling and retention for normally non-radioactive Condensate Polisher resins and the SWPS clarifier sludge. As shown in Figures 9.2-15 and 9.2-16, wastes from several sources are normally processed through the LVW treatment facilities. These sources are:

1. Turbine Building sumps

2. Auxiliary Building Sump #11

3. Diesel Generator Room sump (2 per unit)

4. Component Cooling Water Drain Tanks (1 per Unit)

5. Surface Water Pretreatment System backwash

6. Surface Water Pretreatment System Reverse Osmosis reject and rinse water

7. Condensate Polisher Backwash Waste Cycle wastes

8. Chemical sump wastes from DWMS regeneration and leakage

9. Condensate Polishing Unit Phase Separators Solids

10. Surface Water Pretreatment System (SWPS) Sludge blowdown

The LVW Settling Ponds consist of two 1,750,000 gallon capacity lined settling ponds and a 6,700,000 gallon capacity emergency settling pond. The 1,750,000 gallon settling ponds are sized for average water flow of approximately 455,000 gallons per day to be turned over in accordance with the ODCM limits and waste water discharge permit (batch or continuous flow). These ponds are lined with a double synthetic liner with a leachate collection system.

The non-radioactive low volume wastes from secondary support systems include equipment, floor, laboratory, and sample drains; water treatment wastes from demineralizer regeneration, reverse osmosis (RO) systems operation, condensate polisher system and other miscellaneous water treatment blowdown and backwash operations; periodic drainage and flushing of various system components: and intermittent boiler blowdown from the plant's auxiliary boiler. These low volume wastes and the previously monitored wastes discharged from the Metal Cleaning Waste

Outfall are routed to a Low Volume Waste Management System (WMS) and discharged via the Low Volume Waste Outfall. The WMS consists of the following components: surge basin, API oil separator, clarifier blowdown and condensate polisher decant basins, three separate but interconnected low volume waste retention ponds with double synthetic liners and leachate collection systems, and a synthetic lined concrete equalization basin. The complete WMS provides for oil and grease removal, suspended solids reduction through settling or further treatment, if necessary, and pH control through addition of acid or caustic, if required, to meet TCEQ permit limitations.

The treated non-radioactive low volume waste are normally commingled with the wastes which are potentially low level radioactive and are discharged via the Low Volume Waste Outfall to the condenser cooling water; the commingled waste stream is sampled prior to mixing with cooling water for compliance with the appropriate effluent limitations.

The LVW retention pond effluents are normally within Texas Commission on Environmental Quality (TCEQ) and TPDES permit limits for discharge without additional treatment. Monitoring to verify compliance with these permit limits is performed.

All LVW discharges flow through the Discharge Flume for flow rate monitoring.

A Metal Cleaning Waste (MCW) Pond has been dedicated for the storage and treatment of metal cleaning wastes. Metal cleaning wastes are generated by the chemical cleaning of plant equipment and are evaluated prior to their collection and retention in the pond. Unsuitable wastes area treated and/or disposed by alternate methods; e.g., off-site contract disposal.

Wastewater from the pond, as well as treatable metal cleaning wastes which might be collected in temporary facilities, is treated for metals and pH adjustment prior to discharge via the Metal Cleaning Waste Outfall to the Low Volume Waste retention ponds associated with the Low Volume Waste Outfall. The treated wastes so discharged will be commingled as a previously monitored effluent with the non-radioactive low volume wastes prior to discharge via the Low Volume Waste Outfall.

Chemical cleaning wastes or other metal cleaning wastes can be trucked to the 1,000,000 gallon MCW Pond. The wastes from this pond can be treated for release through the Metal Cleaning Waste Outfall or retained to allow evaporation or further processing. The MCW Pond is sized to accommodate six (6) volumes of the four (4) steam generators (per unit) and associated piping. This pond is lined with a double synthetic liner with a leachate collection system.

Several components of the old evaporation pond wastewater treatment system have been abandoned in place. These components include but are not limited to: batch neutralization tank, LVW oil/water separator skid, soda ash mix tank, sulfuric acid tanks, LVW transfer pumps, clear well transfer pumps and other associated electrical and mechanical equipment. These components have been deleted from the FSAR figures and will be removed from the plant and salvaged.

9.2.8.2.2 Co-Current Waste Treatment Facilities

The Co-Current Waste (COW) treatment facilities provide collection, batch hold-up, conventional pollutant treatment and discharge of low level radioactive waste in accordance with environmental and NRC regulatory requirements. Except for powdex resins, potentially

contaminated low level radioactive waste normally transferred to the LVW treatment facilities are transferred to the COW treatment facilities when radioactive contamination is detected in excess of the limits specified in the Radiological Effluent Controls Program required by Technical Specifications. As shown in Figure 9.2-15, potentially low level radioactive waste collected in the following areas can be diverted to the COW treatment facilities:

1. Turbine Building sumps

2. Auxiliary Building Sump #11

3. Diesel Generator Room sump (2 per unit)

4. Component Cooling Water Drain Tank (1 per unit)

5. Condensate Polisher Backwash Waste Cycle wastes

6. Auxiliary Building Sump #3

Diverted wastes are collected in the Waste Water Hold-up Tanks to facilitate batching. The batched wastes are thoroughly mixed by mechanical mixing and/or placing the hold-up tank contents in a recirculation mode. While recirculating, the batched wastes can be sampled and analyzed to quantify low level radioactive contents.

The environmental regulatory requirements for pH, oil and grease, and total suspended solids in the discharged wastes are specified in Texas Commission on Environmental Quality (TCEQ)-TPDES permits.

Batched volumes requiring pH adjustment are manually neutralized. Following pH adjustment the batched wastes may be discharged through the environmental Low Volume Waste Outfall. If required, the wastes are treated for emulsified oil removal and/or suspended solids reduction prior to discharge through the Low Volume Waste Outfall. Alternately, the wastes may also be drained to Turbine Building Sump No. 4 and discharged to the Low Volume Waste retention pond if radioactivity concentrations are below levels specified in the Radiological Effluent Controls Program required by the Technical Specifications.

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 2, Design Bases for Protection Against Natural Phenomena.

2. 10 CFR Part 50, Appendix A, General Design Criterion 4, Environmental and Missile Design Bases.

3. 10 CFR Part 50, Appendix A, General Design Criterion 5, Sharing of Structures, Systems, and Components.

4. 10 CFR Part 50, Appendix A, General Design Criterion 19, Control Room.

5. 10 CFR Part 50, Appendix A, General Design Criterion 44, Cooling Water.

6. 10 CFR Part 50, Appendix A, General Design Criterion 45, Inspection of Cooling Water System.

7. 10 CFR Part 50, Appendix A, General Design Criterion 46, Testing of Cooling Water System.

8. 10 CFR Part 50, Appendix A, General Design Criterion 56, Primary Containment Isolation.

9. 10 CFR Part 50, Appendix A, General Design Criterion 57, Closed System Isolation Valves.

10. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam-, and Radioactive- Waste-Containing Components of Nuclear Power Plants, Revision 3, February 1976, U.S. Nuclear Regulatory Commission.

11. NRC Regulatory Guide 1.27, Ultimate Heat Sink for Nuclear Power Plants, Revision 1, March 1974, U.S. Nuclear Regulatory Commission.

12. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976, U.S. Nuclear Regulatory Commission.

13. NRC Regulatory Guide 1.48, Design Limits and Loading Combinations for Seismic Category I Fluid System Components, May 1973, U.S. Nuclear Regulatory Commission.

14. Branch Technical Position APCSB 3-1, Protection Against Postulated Piping Failure in Fluid Systems Outside the Containment.

15. ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973.

16. American Nuclear Society Draft Standard ANS 5.1, Decay Energy Release Rates Following Shutdown of Uranium Fueled Thermal Reactors, October 1973.

17. Generic Letter 91-13, “Request for Information Related to the Resolution of Generic Issue 130, Essential Service Water System Failures at Multi-Unit Sites, Pursuant to 10 CFR 50.54(f),” dated September 19, 1991.

18. TXX-92410, License Amendment Request 92-002, Combined Unit 1 and 2 Technical Specifications, from W. J. Cahill, Jr., to U. S. Nuclear Regulatory Commission, dated August 31, 1992.

19. Generic Letter 89-13, “Service Water Problems Affecting Safety Related Equipment”, dated July 18, 1989, and Supplement 1, dated April 4, 1990.

TABLE 9.2-1
SINGLE-FAILURE ANALYSIS OF STATION SERVICE WATER SYSTEM

Component	Malfunction	Effect on System Safeguard Performance	Comments
Service water pump	stops pumping	No effect	Two 100-percent-capacity pumps are provided; one is required.
SSW pump discharge isolation valve	a. Fails to close b. Unwanted closure c. Fails to open	a. No effect b. No effect c. No effect	a. Pump shutdown and check valve prevent flow in either direction. b. Redundant train is provided. c. Redundant train is provided
Emergency diesel generator isolation valve	a. Fails to open b. Fails to close	a. No effect b. No effect	a. Two 100-percent-capacity trains are provided; one is required. b. Not required to close to accomplish a safety function.
Auxiliary feedwater isolation valve	a. Fails to open b. Fails to close	a. No effect b. No effect	a. Two 100-percent-capacity trains are provided; one is required. b. Required minimum flow and sufficient NPSH required are available to AF pumps.
Emergency diesel generator	Fails to start	No effect	Two 100-percent-capacity diesel generators are provided per unit.
Component cooling heat exchanger	Loss of flow	No effect	Two 100-percent-capacity diesel generators are provided per unit.

TABLE 9.2-2
COMPONENT COOLING WATER SYSTEM EQUIPMENT CHARACTERISTICS

(Sheet 1 of 2)

	Shell Side	Tube Side
1. Heat Exchanger		
Design flow, gpm	14,700	14,000
Design pressure drop, psi	12.4	5.3
Design pressure, psig	165	150
Design temperature, °F	225	225
Temperatures, °F		
Outlet	105.0	108.4
Inlet	114.5	98.4
Design heat transfer rate	70×10^6 BTU/hr.	
Tubes Material	ASME SB-111 Type 706	
2. Pumps		
Type	Centrifugal, horizontal	
	Design	Runout
Capacity gpm	14,700	16,400
Total Dynamic Head, ft	226	210
NPSH, ft		
Minimum required	30	36
Available (minimum)	60	60
Design pressure, psig	165	
Temperature, °F		
Normal	115	
Normal (max.)	180	
Design	225	
Material	Carbon steel	

TABLE 9.2-2
COMPONENT COOLING WATER SYSTEM EQUIPMENT CHARACTERISTICS

(Sheet 2 of 2)

3. Surge Tanks	
Outside diameter, ft	6.5
Length, ft	22
Volume, gal	4600
Design pressure, psig	10
Design temperature, °F	200
Material	Carbon steel

4. Piping

All CCWS piping is carbon steel material, ASME SA 106, Grade B, with welded joints and connections except at components where flanged connections are used to facilitate maintenance.

TABLE 9.2-3
COMPONENT COOLING WATER SYSTEM WATER CHEMISTRY

1. Acceptable Corrosion Inhibitors

   Any of the following corrosion inhibitors can be used:

   a. Sodium Molybdate

   b. Tolyltriazole

   c. Hydrazine

   d. Calgon Corporation’s H-300 (Gluteraldehyde)

   e. Sodium Nitrite

   f. Borax

   g. Chromates

   h. Sodium Hydroxide

2. Chloride, ppb <150

3. Fluoride, ppb <150

TABLE 9.2-4
COMPONENT COOLING WATER SYSTEM IN RELATION WITH CODES AND STANDARDS

Plant Conditions	Condition I Normal Operation	Condition II Incidents of Moderate Frequency	Condition III Infrequent Incidents	Condition IV Limiting Faults
NRC Regulatory Guide 1.48 plant conditions	Normal	Upset	Emergency	Faulted
ASME Section III Component Conditions				
Pumps, valves and vessels	Normal	Normal	Normal	Normal
Piping	Normal	Upset	Emergency	Emergency
System Operation	Startup; standby; power range; refueling; cooldown; 4 hrs; cooldown, 20 hrs	Loss of outside power; reactor-turbine mismatch	Small ruptures	LOCA

TABLE 9.2-5
SINGLE FAILURE CRITERIA ANALYSIS OF COMPONENT COOLING WATER SYSTEM

(Sheet 1 of 4)

Component			Malfunction		Effect on Safeguards Systems	Comments
1.	Component cooling water pump		Stops pumps or fails to start		No effect	Redundant train is provided.
2.	Component cooling heat exchanger		Tube or shell rupture		No effect	Redundant train and automatic isolation is provided.
3.	Remotely Operated Stop Valves					
	a.	Containment isolation	1)	Fails to close	No effect	Backup provided by redundant valve or by closed system inside containment.
			2)	Unwanted closure	No effect	Closure of nonsafeguards valves does not affect safe shutdown.
	b.	Containment spray and RHR heat exchanger isolation	1)	Fails to open	No effect	Valve has a manual opeator; heat exchanger redundancy is provided.
			2)	Unwanted closure	No effect	Valve has a manual operator; heat exchanger redundancy is provided.
	c.	Non-safety loop isolation	1)	Fails to close	No effect	Valve redundancy is provided.
			2)	Unwanted closure	No effect	Valve has a manual operator.

TABLE 9.2-5
SINGLE FAILURE CRITERIA ANALYSIS OF COMPONENT COOLING WATER SYSTEM

(Sheet 2 of 4)

Component			Malfunction		Effect on Safeguards Systems	Comments
	d.	Safety loops	1)	Fails to close	No effect	Valve redundancy is provided.
			2)	Unwanted closure	No effect	Loop redundancy is provided
4.	Safety-related headers		Pipe rupture		No effect	Redundant headers and automatic isolation are provided.
5.	Emergency diesel generators		Fails to start		No effect	Redundant diesel generator is provided.
6.	Motor Control					
	a.	CCW Pump Motor	Fails to start due to:			
			1)	Power loss	No effect	Redundant and independent train provided for backup pump on second loop. ESFAS circuitry activates both pumps' motors.
			2)	Failure in control	No effect	
	b.	Safety class and non-safety class loop isolation valves.	Fail to close:			Redundant and independent train provided for backup valve. ESFAS activtes all loop isolation valves in both trains.
			1)	Power loss	No effect	
			2)	Failure in control circuitry	No effect	

TABLE 9.2-5
SINGLE FAILURE CRITERIA ANALYSIS OF COMPONENT COOLING WATER SYSTEM

(Sheet 3 of 4)

Component		Malfunction		Effect on Safeguards Systems	Comments
c.	Containment spray and RHR heat exchanger Isolation Valves	Fail to open			Redundant and independent train provided for isolation valve on backup safety class loop.
		1)	Power loss	No effect	
		2)	Failure in control circuitry	No effect	
d.	CCW system phase B Containment Isolation Valves	Fail to close			Redundant and independent train provided for backup valve. ESFAS activates valves on both trains.
		1)	Power loss	No effect	
		2)	Failure in control circuitry	No effect	
7.	Air Operated Valves				
a.	Phase “A” CCW system Containment Isolation Valves	1)	Loss of Instrument Air	No effect	Valves fail in safe position on loss of air.
		2)	Power loss	No effect	Valves fail in safe position on loss of power
b.	Safety Injection CCW system ventilation condenser isolation	1)	Loss of Instrument Air	No effect	Valves fail in safe position on loss of air.
		2)	Power loss	No effect	Valves fail in safe position on loss of power.
		3)	Fail to close (1 valve)	No effect	Two valves are provided to stop flow to the condensers.

TABLE 9.2-5
SINGLE FAILURE CRITERIA ANALYSIS OF COMPONENT COOLING WATER SYSTEM

(Sheet 4 of 4)

Component		Malfunction		Effect on Safeguards Systems	Comments
c.	Surge Tank Makeup Water valves	1)	Loss of air	No effect	Reactor makeup water valve fails in safe position.
		2)	Loss of power	No effect	Fill valves are provided with handwheels.
		3)	Reactor Makeup Valve fails to open	No effect	Same as loss of air. Valve will be forced open by operator.

TABLE 9.2-6
NOT USED

TABLE 9.2-7
NOT USED

TABLE 9.2-8
WELL WATER ANALYSIS

Constituent	Concentration (mg/1)
Calcium, as Ca	17
Magnesium, as Mg	8
Sodium, as Na	141
Carbonate, as CO3	0
Bicarbonate, as HCO3	378
Sulfate, SO^4	39
Chloride, as C1	22
Silica, as SiO2	12
Total ion, as Fe	0.05
Manganese, as Mn	0.02
Total dissolved solids	615
Phenolphthalein Alkalinity, as CaCO3	0
Total alkalinity, as CaCO3	310
Turbidity, units	1
Color, units	5
PH	8.55
Total Hardness, as CaCO3	77
Free carbon dioxide, as Co2	16

TABLE 9.2-9
DELETED

TABLE 9.2-10
DELETED

TABLE 9.2-11
THIS TABLE HAS BEEN DELETED.

TABLE 9.2-12
--DELETED--

TABLE 9.2-13
DECAY HEAT RATES[a]

TIME AFTER LOCA (SEC)	TOTAL HEAT RATE 10^6 BTU/HR
1×10^{-1}	1047.0
1×10^{0}	955.4
2×10^{0}	919.8
4×10^{0}	842.0
6×10^{0}	804.5
8×10^{0}	795.0
1×10^{1}	767.5
2×10^{1}	695.0
4×10^{1}	621.5
6×10^{1}	585.8
8×10^{1}	549.4
1×10^{2}	514.8
2×10^{2}	432.1
4×10^{2}	371.4
6×10^{2}	339.0
8×10^{2}	306.5
1×10^{3}	285.8
2×10^{3}	224.2
4×10^{3}	179.9
6×10^{3}	161.5
8×10^{3}	143.4
1×10^{4}	132.7
2×10^{4}	108.6
4×10^{4}	89.4
6×10^{4}	81.7
8×10^{4}	74.0
1×10^{5}	69.0
2×10^{5}	55.0
4×10^{5}	43.0
6×10^{5}	37.7
8×10^{5}	32.4
1×10^{6}	29.3
2×10^{6}	21.6
2.6×10^{6}	19.3

a) Total decay heat found by using the rated power level of 3458 Mwt

TABLE 9.2-14
THIS TABLE HAS BEEN DELETED.

9.3 PROCESS AUXILIARY

9.3.1 COMPRESSED AIR SYSTEMS

The Compressed Air System (CAS) consists of two separate systems, Instrument Air System (CI) and Service Air System (CA). The Instrument Air System is designed to provide a reliable supply of clean, dry, oil-free air of suitable quality and pressure for pneumatic instruments and controls and pneumatically operated valves for normal plant operation. The Service Air System is designed to provide the necessary compressed air for pneumatic tools and general plant usage. The CA, via a filtering skid, also provides suitable air to inflate the refueling lift gate seals.

9.3.1.1 Design Bases

The CAS serves no safety function because it is not required to achieve safe shutdown or to mitigate the consequences of a DBA. Therefore, the CAS is not a nuclear-safety-class-designed system.

However, the lines penetrating into containment are designed in accordance with NRC GDC 2, 4, 5, and 56 [1], [2], [3], [4], NRC Regulatory Guides 1.26 and 1.29 [5], [6], and applicable Branch Technical Positions [7], [8], as described in detail in Section 6.2.4, and ANSI N18.2 [9].

Where it is desired to maintain control of selected air-operated valves and dampers after safe shutdown, a safety related air accumulator and a double set of check valves are provided. For the purposes of design basis, functional, and testing requirements, this accumulator and its check valves are considered an extension of the component which it serves.

The air accumulators, the air piping (between the first check valve and the pneumatic control valve) and the associated valves are designated as ANS Safety Class 3 and designed in accordance with the requirements of ASME B&PV Code Section III, Class 3. In addition, they are designed to meet the requirements of seismic Category I in compliance with NRC Regulatory Guide 1.29. (See Tables 3.2-2 and 17A-1)

1. System Design

   The design of the CAS is based on an instrument air normal requirement per unit of 450 scfm (600 scfm with breathing air in service) with a transient maximum of 500 scfm (650 scfm with breathing air in service). The design of the CAS is also based on the service air maximum requirement of 1300 acfm for both units.

   The compressed air supplied has the following design bases:

   a. Instrument and service air regulates 80 to 120 psig nominal pressure

   b. Maximum instrument air dewpoint which meets the requirements of ANSI ISA S7.0.01 – 1996.

   c. Instrument air supplied to the system at the outlet of the dryer will be oil free and cleaned to remove particulate using one micron filters which are 98% or greater efficient.

d. Quantity of instrument air sufficient to provide for all pneumatic controls and valve and damper operators expected to operate simultaneously

e. Instrument air which is oil free of Grade “D” quality for use as breathing air

f. Quantity of service air sufficient for all equipment expected to operate simultaneously plus an allowance for the use of maintenance tools

g. Failure of the service air distribution system not to cause a loss of instrument air

h. Air systems piping is in accordance with ANSI B31.1, piping between Containment isolation valves is per ASME Section III, Code Class 2.

i. Quality of instrument air supplied to individual components will be consistent with manufacturer’s recommendations for air quality.

System operation is not required to initiate operation of engineered safeguards equipment because all air-operated valves and dampers are designed to fail-safe upon loss of control air. Air accumulators are provided for safety-related air-operated equipment where control is desired after safe shutdown.

System operation is not required to maintain the refueling gate seals (Figure 9.3-1) as described in section 9.1.4.2.3, item 16, because one bladder is always isolated from the instrument system (i.e., is passive) when the gate is closed.

2. Equipment Design

a. Instrument Air System

The equipment design parameters are given in Table 9.3-1. System components are located in the Electrical Building and Turbine Building. The main components have the following design characteristics:

1. Instrument Air Compressors

Six instrument air compressors are provided for both Units. Each compressor is a two stage, heavy duty, oil free, water cooled rotary air compressor. Each Lead train with it’s spare or back-up unit will supply dry, oil free air at 100 PSIG nominal pressure to meet the maximum instrument air requirement of each Unit. For both Units 1 and 2, the designation of which compressor is the lead and which is the backup is manually selectable.

Each instrument air compressor is provided with a dry filter-silencer in the intake line. The unit air compressors are cooled by the respective unit’s non-safety loop of the CCWS. The two common air compressors are cooled by the Turbine Plant Cooling Water (TPCW) system. Each of the Unit 2 and common air compressors have a dedicated trim cooler to provide supplemental cooling for the compressors. Non-safety chilled water is routed to the trim coolers to maintain CCW and TPCW cooling

temperatures within operating limits. The cooling water supply piping for the Unit 1 compressors in the Electrical Controls Building (ECB) is in series with the corresponding Unit 1 compressor in the TB allowing a single trim cooler to cool the water supply. The compressors located in the Turbine Building have an additional non-safety chilled water ventilation cooler that ducts cooled air into the compressor air inlet, ensuring intake parameters are within operational limits. The cooling water supply for the common compressor in the ECB is in parallel with the corresponding common compressor in the TB.

2. Instrument Air Aftercoolers

   The aftercoolers for the six rotary air compressors are an integral part of the compressor packages. Each aftercooler is provided with a moisture separator and a moisture trap for automatic removal of the separated water.

3. Instrument Air Receivers

   Six instrument air receivers are provided, one for each rotary air compressor.

4. Prefilters and Afterfilters

   Each air dryer package is provided with two prefilters and two afterfilters to remove moisture aerosol and solid particles that may be carried by the air.

5. Air Dryers

   Six air dryers of the heater type are provided, one for each rotary air compressor. The dryers are of the dual tower internally heated regenerative, desiccant type. The heating elements aid in desiccant regeneration. Each dryer has two towers charged with desiccant beads which adsorb a definite quantity of moisture vapor from the instrument air. During normal operation, one tower provides the drying or adsorption action while the other is being regenerated. Through a system of actuators, linkages, interconnecting piping, and valves, a continuous automatic alternating or cycling of the two towers is possible. Each air dryer is a completely packaged, self-contained unit, which is capable of drying the full instrument airflow requirements of one air compressor to -40°F dewpoint at a line pressure of 100 psig.

6. Piping

   Instrument air piping materials are as follows:

   (a) From the compressor area to the instrument air distribution header: stainless steel ASTM A312 Type 304 or 316

(b) From the distribution header up to the Containment penetration: copper tubing ASTM B88

(c) Between the Containment isolation valves: stainless steel SA312 Type TP304 or 316

(d) Piping inside the Containment and the instrument air yard piping: stainless steel ASTM A312 Type 304 or 316

(e) Tubing inside Containment, stainless steel ASTM A213, Type 316

b. Service Air System

The equipment design parameters are given in Table 9.3-2.

The common compressor is located west of the unit 1 Turbine Building. The main components have the following characteristics:

1. Service Air Compressor

One service air compressor is shared by both Units 1 and 2. It is a single stage, oil injected, air cooled rotary screw, 1300 ACFM air compressor. The compressor is sized to supply air at 100 psig normal pressure to meet the maximum service air requirements of both units.

2. Service Air Receivers

One service air receiver is provided for both units. The receiver is cylindrical and horizontal and is provided with an automatic moisture trap.

3. Piping

The service air piping material is carbon steel ASTM 106 Grade B. The piping code, when applicable, is ANSI B31.1, except for the Containment penetration which has an ANS Safety Class 2 classification and is designed to the requirements of the ASME B&PV Code, Section III, Code Class 2 as discussed in Section 6.2.4. Piping associated with accumulators (between the first check valve and the pneumatic control valve) is designated ANS Safety Class 3 and designed to meet ASME B&PV Code, Section III, Code Class 3.

9.3.1.2 System Description

The CAS is shown schematically on Figure 9.3-1 and Figure 9.3-2.

1. Instrument Air System

During normal system operation, instrument air to each unit is supplied by a train of equipment consisting of a lead instrument air compressor package, air receiver, and air dryer package. Also, a backup train automatically comes “on line” if the air header

pressure decreases. The lead trains for Unit 1 and 2 are manually selectable regarding which is the lead or backup train. Two complete common standby train packages are available (normally one common train is aligned to each unit) and automatically come “on line” if the unit compressors cannot maintain header pressure. The common compressors can be manually selected as the lead or backup compressor for either unit.

Before being delivered to the distribution network, the instrument air is filtered, dried, and filtered again. For this purpose, each air compressor is provided with a dedicated air dryer skid consisting of two 100% capacity prefilters, two 100% capacity air dryer towers and two 100% after filters.

The instrument air lines which penetrate the Containment are provided with two isolation valves. In addition, a flow-limiting orifice is provided for each penetration.

A pressure control valve and local supply station are provided for the refueling gates in the Fuel Building

Air-operated valves throughout the plant are arranged for safe failure in the absence of air.

Where it is desired to maintain control of selected air-operated valves and dampers after safe shutdown, an accumulator and a double set of check valves are provided.

The air piping between the first of the two check valves and the control valve is designed to seismic Category I requirements. The accumulators, in addition to meeting seismic Category I criteria, are designed to the requirements of the ASME B&PV Code, Section III, Code Class 3. The check valves are spring loaded and have resilient seats to ensure bubble-tight shutoff over the range of differential pressures to 120 psi.

Accumulators are provided for the auxiliary feedwater control valves in the individual feed lines and the motor-driven pumps recirculation valves which share accumulators with the flow control valves, the steam admission valves in lines supplying the auxiliary feedwater turbine, each Steam Generator PORV, and the CCW return pressure controller for the safety chillers. The accumulators at each auxiliary feedwater control valve are sized on the basis of allowing the operator remote manual control to isolate a faulted steam generator for a period of 30 minutes after loss of air. The basis for the recirculation valves is one valve cycle, then maintain the valve in the closed position for the remainder of the 30 minute period. See Section 10.4.9.2 for more details. The accumulators at the steam admission valves in lines supplying the auxiliary feedwater turbine are sized to have sufficient air capacity to drive the valve closed and maintain it closed for 7 hours post-accident plus an additional one-half hour for the AFWPT steam supply lines to then be locally-manually isolated by operator action, including any applicable air leakage criteria. The Steam Generator PORV accumulators are sized for Steam Generator Tube Rupture mitigation. They also provide motive power for hot standby and cooldown (See Section 5A) as long as the supply lasts followed by manual operator control to complete cooldown to hot shutdown condition. Each Steam Generator PORV accumulator can provide for 15 positionings over a four hour period after loss of instrument air. The CCW pressure control valves for the safety chillers have accumulators which are sized for 30 minutes of post accident operation.

Accumulators are provided for the control room air inlet control dampers. The dampers also have manual controls. This conservative sizing criteria, used to select tank volume for fabrication (to permit 10 operations of the dampers for a period of 30 days following the loss of the normal air supply system), ensures that the minimum system requirements are met. For the control room air inlet control dampers, the minimum system requirement is met if the air accumulator can hold the damper open on loss of instrument air until it is restored, the need for pressurization of the control room is past, or the operator can lock the damper open (not less than 2 hours after loss of instrument air). An evaluation of post-accident access to these dampers is provided in the Response to the NRC Action Plan Developed as a Result of the TMI-2 Accident, Section II.B.2.4.

2. Service Air System

During normal operation, service air to each unit is supplied by a service air compressor and a service air receiver. The service air is then delivered to the distribution system. The service air line which penetrates the Containment is provided with two isolation valves. In addition, a flow limiting orifice is provided for each penetration.

A pressure control valve and local supply station is provided for the refueling gate in each Containment Building.

9.3.1.3 Safety Evaluation

1. Instrument Air System

The Instrument Air System is not required for the safe shutdown of the plant, however, the Instrument Air System supplies air to the components important to safety. Therefore, all control and actuation devices associated with safeguards are electrically operated or, if pneumatic, have a fail-safe position. Certain valves are provided with local individual air accumulators. Accumulators are designed and sized to provide the required air quantity for a short period of time after loss of the main air system.

Although the Instrument Air System does not have a safeguard function, it is necessary for the operation of the plant, and the availability of air greatly facilitates the plant recovery following an emergency. For this reason, the unit compressors which are automatically tripped by a BOS will be automatically loaded after 90 seconds. An S signal will automatically trip the air compressors and dryers. The system is designed for high reliability, covering possible component failures with adequate backups and redundancies.

Generally, the accumulators are sized to ensure a minimum of one operating cycle of the valve (i.e., open - close - open) or to maintain the valve in the appropriate position until manual action can be provided (i.e., 30 min, but dependent on valve accessibility).

The valves and dampers which have accumulators to enable a safety function, are the auxiliary feedwater flow control valves and the motor-driven pumps recirculation valves which share accumulators with the flow control valves, the steam admission valves in lines supplying the auxiliary feedwater turbine, the Steam Generator PORV’s, and the Control Room air dampers. The valves and dampers fail in a safe position on loss of air.

2. Service Air System

The system is not nuclear-safety-related.

3. Failure Mode Analysis

The CAS is used to inflate the pneumatic seals on the refueling gates. The CAS and nitrogen bottles back-up are disconnected from at least one seal whenever a gate is in Seismic Category I service. Therefore, no failure or malfunction in the NNS CAS can affect the safety related functions of the refueling gates.

A failure mode and effects analysis for air-operated valves is given in Table 9.3-3. All valves within this table are safety-related in that they are components of safety-related systems and meet the applicable requirements of the ASME B&PV Code, Section III, which insures the pressure-retaining capability of the valve body. In addition, the valve actuator mechanisms are qualified for the service conditions (including seismic qualification for active valves) and are capable of controlled opening or closing against specified maximum pressure differentials and within specified time limits where required.

As indicated in Table 9.3-3 loss of instrument air does not jeopardize plant safety nor is operator action required to place the plant in a safe condition.

The results of the failure mode analysis demonstrate that a failure of the non-nuclear-safety-related CAS has no effect on plant safety because the valves fail to the specified positions by spring force alone. Valves the operator may wish to control beyond the specified failure position are equipped with air accumulators, as previously described, and listed in Table 9.3-3. The types of failure being analyzed are as follows:

a. Power Failures

The motor drivers for the six instrument air compressors are connected to six separate electrical buses. Four of the six air compressors are fed from class 1E buses to assure availability after an off-site electrical power failure. The remaining two machines (common spares) are fed from non-class 1E buses.

In the exceptional case where all electrical power is interrupted, the air compressors stop operating. Air-operated valves throughout the plant are arranged for safe failure in the absence of air. In this case, the valves are positioned to preserve the safety of plant and personnel.

b. Dryer Failures

The duplex dryer towers are arranged so that one regenerates while the other is in service. The two dryer towers interchange automatically on a time basis and an alarm at the main control room is provided to indicate a high differential pressure in any filter unit or failure in the microprocessing unit. In addition if a dryer for one of the lead rotary air compressors fails or malfunctions, the unit back-up train can be brought on line.

c. Filter Plug-Up

Normally, plant operators clean filters on a regular basis and switch them into service before plugging can occur. In the exceptional case that a filter does plug, a differential pressure alarm warns the operator, who manually shifts the airstream to the second filter.

d. Air Header Ruptures

A separate header is provided for each CAS and rupture of one header leaves the other headers unaffected. As with a power failure discussed in Subsection 9.3.1.3, Item 3.a, if a rupture of the instrument air header occurs, the plant can be shut down in a safe manner. Air headers are routed to prevent affecting safety-related equipment in the event of a rupture.

e. Equipment Sharing

Both units share a service air compressor. Only the common spare instrument air compressors can be aligned to either unit. Failure of the common service air compressor will not impact a safe shutdown or mitigate the consequences of a DBA.

9.3.1.4 Tests and Inspections

The systems are inspected and cleaned prior to service. Instruments are calibrated during testing, and automatic controls are tested for actuation at the proper set points. Alarm functions are checked for operability and limits during plant operational testing. The system is operated and initially tested with regard to flow paths, flow capacity, and mechanical operability.

9.3.1.5 Instrumentation Requirements

The instrument air compressors are cycled to maintain the set desired pressures. Both Unit 1 and 2 lead compressors are cycled on/off based on a discharge air pressure switch integral to the packages. The backup air compressors are cycled on/off based on an abnormally low pressure in either of their designated air receivers. It is anticipated that during normal operation the lead air compressor will run continuously with the backups cycling on/off as needed. The pressure at the air header is indicated locally and in the Control Room.

9.3.2 PROCESS SAMPLING SYSTEM

9.3.2.1 Design Bases

A separate sampling system is furnished for each unit. Each system is designed to provide liquid and gas samples at controlled temperatures and pressures for laboratory analysis. The samples are drawn from the points designated in Table 9.3-4. To ensure representative samples, samples are taken from turbulent flow zones, i.e., downstream of pump discharges or in tank recirculation lines. Pumped recirculation of tank contents and sample withdrawal from recirculation lines avoids sampling from low points or potential sediment traps. These samples are routed to a central location where the components of the Process Sampling System are located. The fluid properties at the designated sample points are critical to proper functioning of the plant and require frequent testing, as detailed in Technical Specifications.

Local sampling is performed where batch processing is required, i.e., Liquid Waste Processing System (LWPS), Gaseous Waste Processing System (GWPS), Boron Recycle System (BRS), and for plant effluent streams. In addition, local samples can be taken from the boric acid tanks, boric acid batching tank, Refueling Water Storage Tank, and the chemical additive tank. Refer to applicable Flow Diagrams. In general, the local sampling points require less frequent testing and are therefore not routed to the Process Sampling System. Local and effluent sampling is discussed in Section 11.5.4.

The Process Sampling System (PSS) may be used to obtain samples post-accident. During a loss-of-coolant accident (LOCA), PSS sample lines are isolated on both sides of the Containment boundary. See Section 6.2.4 and II.B.3 for PSS sample line isolation details.

9.3.2.2 System Description

9.3.2.2.1 System Operation

Samples from the points designated in Table 9.3-4 are conveyed to the sampling room in stainless steel tubing. The line sizes are determined to meet analyzer and grab sample flow requirements. The distance between the sample point and Process Sampling System has been kept to a minimum to reduce possible plateout and precipitation.

Samples enter the sample conditioning panel which houses the sample coolers, pressure-reducing and pressure-regulating valves, and local pressure and temperature indicators. The requirements for conditioning of a sample are based on the design temperature and pressure at the point of sampling. After conditioning, sample temperature and pressure are less than 120°F and 5 to 80 psig max., respectively.

All the sample lines are provided with grab sample valves or sample vessels with quick disconnect fittings, as shown on the Primary Sampling System piping and instrumentation diagram (P&ID), Flow Diagram M1(2)-0228.

Automatic online analyzers for cation conductivity, high and low pH specific conductivity, and sodium ion concentration are provided for the steam generator blowdown samples. An inline radiation monitor is provided for the steam generator blowdown samples to indicate steam generator tube leakage.

The samples are segregated so that primary and secondary fluids are not mixed. The routing of each sample line is shown on the Primary Sampling System P&ID, Flow Diagram M1(2)-0228.

Automatic online analyzer provided for the CVCS Letdown Upstream of Purification Demineralizer sample to measure RCS dissolved hydrogen and oxygen.

9.3.2.2.2 Components Description

1. Samples Coolers

   The samples coolers are of tube-in-shell design with pressure relief valves on the shells. The samples flow through the tube side and component cooling water flows through the shells. The sample cooler tubes are of stainless steel construction and the shells are carbon steel.

2. Pressure-Reducing and Pressure-Regulating Equipment

The pressure-reducing and pressure-regulating equipment consists of a variable capillary device or a velocity-controlled pressure-reducing valve in series with a pressure-regulating valve to maintain sample stream pressure at 5 to 80 psig max. downstream of the regulating valve. Relief valves are provided on each sample line as an additional safety feature. The relief valves discharge to the waste holdup tank or to an appropriate drain system as shown on Flow Diagram M1(2)-0228.

3. Sample Vessels

Sample vessels or bombs are used for all gas sampling and the reactor coolant, pressurizer, and RHR System liquid samples. The vessels are of stainless steel construction designed to withstand the reactor coolant design pressure and temperature.

4. Sample Sink

Each sample sink is located in a hooded enclosure equipped with an exhaust ventilator ducted to the Containment vent header. The sinks and enclosures are of stainless steel construction and have raised edges to minimize liquid spillage.

9.3.2.2.3 Manual Sampling Techniques

1. Liquid Sampling

Liquid sampling is accomplished by purging the sampling lines. This allows the sample to flow for a sufficient time to ensure that representative samples are obtained.

2. Sample Vessel

Sample vessels, used to collect gaseous samples and certain liquid samples, are used by first adequately purging the sample vessels and then closing inlet and outlet vessel valves. The sample vessel is then removed from its connection lines after the sample flow has been stopped.

3. Sample Line Purging

Sample line purging is a function of the sample line diameter and distance to the point of sampling. Sample line purge flow rates are recommended not to be less than those values as identified in Table 9.3-4.

9.3.2.2.4 Codes and Standards

The equipment in the sampling is designed, furnished, and installed to meet applicable codes and standards. NRC Regulatory Guides 1.21, 1.26, and 1.29 and ANSI N13.1-1969 were applied to the design of the Process Sampling System as follows:

1. NRC Regulatory Guide 1.21

   Monitoring of effluents is not a function of the Process Sampling System. Effluent sampling is discussed in Section 11.5.

2. NRC Regulatory Guide 1.26

   All valves, piping, and tubing originating within the Containment are designed to meet the requirements of ASME B&PV Code, Section III, Class 2. All valves and tubing downstream of the outside Containment isolation are non-nuclear-safety class.

3. NRC Regulatory Guide 1.29

   All valves, piping and tubing located inside the Containment and outside the Containment up to the second isolation valve (the safety related portion of the system) are designed to the requirements of the seismic Category I. All sample lines (including valves) downstream of the second isolation valve (outside the containment) leading to the sampling panels (the non-safety related portion of the system) are designated as Class 5.

   A Class 1E panel is provided to operate the sample line Containment isolation valves from the hallway outside the sample room.

4. ANSI N13.1

   Monitoring of airborne radioactive materials inside the plant and in effluents is not a function of the Process Sampling System. Section 12.3.4 discusses this subject.

9.3.2.3 Safety Evaluation

A separate sampling system is furnished for each unit so that there is no sharing of equipment between units.

The operation of the sampling system is not necessary for safe plant shutdown after an accident. Therefore, in the event of an accident, all sample lines which pass through the Containment are automatically isolated by fail-closed air-operated valves on either side of the Containment. The Containment Isolation System is described in Section 6.2.4.

Similarly, no alternate paths to the Process Sampling Systems during accident conditions are required because the system cannot cause loss of function of active components whose operation is necessary for safe shutdown of the plant.

The routing of high-temperature and high-pressure sample lines outside the Containment is not considered hazardous because of their limited flow, intermittent use, and nonessential nature. Also, the location of sample temperature-reduction and pressure-reduction equipment outside the Containment is considered beneficial from a components maintenance standpoint. Passive flow restrictions are not provided because of intermittent use and the small size (3/8 in.) of the sample lines. The use of hot leg sample line taps with a 0.234" bore provides flow restriction to limit reactor coolant loss from a rupture of a sample line in compliance with the intent of Standard Review Plan Section 9.3.2, Acceptance Criterion 2.g. For further clarification, see FSAR Section 5.4.3.2 and Figure 5.1-1.

The reactor coolant loop samples are passed through 3/8 inch tubing; configured to act as delay coils located inside the Containment to permit the decay of short-lived isotopes. Samples lines are shielded, as required, to protect plant personnel from radiation in the course of their normal duties. Radioactivity releases resulting from break of a reactor coolant sample line are discussed in Section 15.6.2.

9.3.2.4 Testing and Inspection

Each component is inspected and cleaned prior to installation. Instruments and analyzers are calibrated during testing. Automatic controls are tested for actuation at the proper set point. The system is operated and tested initially with regard to flow paths, flow capacity, thermal capacity, and mechanical operability.

9.3.2.5 Instrumentation Application

Local temperature and pressure indicators are provided on all high-temperature and high-pressure sample lines. These local instruments enable the operator to determine if the sample coolers and pressure-regulating equipment are functioning properly. Local flow indicators are used to manually adjust the flow of samples to analyzers and to the closed sample vessels as shown on the Flow Diagram M1(2)-0228.

Radioactivity in the steam generator blowdown fluid is measured by a single inline radiation monitor which sees a combined sample from the four steam generators. If the monitor alarms, the blowdown and blowdown sample outboard Containment isolation valves and inboard process isolation valves close. These valves are reopened from the Control Room and the samples can pass through the monitor one at a time by manually operating local valves to ascertain which steam generator is leaking.

Cation conductivity, pH specific conductivity, and sodium ion concentration are continuously monitored for each steam generator. These parameters guide the selection of chemical quantities to be fed to the condensate and the regulation of steam generator blowdown flow rate. (See Sections 10.3.5 and 10.4.8, respectively.) By using the monitors in conjunction with proper chemical treatment and sufficient blowdown flow, the steam generator water chemistry can be kept within prescribed limits, as detailed in Section 10.3.5. High conductivity, high pH and high sodium levels are alarmed on a local annunciator, which has a common alarm relay to the Control Room.

Specific conductivity is used to monitor the performance of the blowdown cleanup demineralizers.

An electrical panel for operating the Containment isolation valves on the sample lines is provided in the hallway outside the sample room. By first aligning valves on the sample conditioning panel and the sample hood, the operator can then open the appropriate Containment isolation valves to draw a sample. Isolation valves are of the fail-closed type and close automatically on Containment isolation signal.

9.3.3 EQUIPMENT AND FLOOR DRAINAGE SYSTEM

9.3.3.1 Design Bases

Liquid wastes, valve and pump leakoffs, tank overflows, and tank drains are collected by the equipment and floor drainage system. Drains are separated according to their activity and quality as follows [10]:

1. Aerated tritiated drains of reactor coolant quality are collected in a waste holdup tank for treatment, recycling or disposal (Drain channel A subsystem).

2. Tritiated and nontritiated drains from the Containment sumps, Auxiliary Building sumps, Auxiliary Building and Safeguards Building floor drains, and laboratory drains, as outlined in Section 11.2, are collected in the floor drain tank for treatment, recycle, or disposal (Drain channel B subsystem).

3. Laundry and hot shower wastes and decontamination wastes are collected in the laundry and hot shower tank for treatment, recycle, or disposal (Drain channel C subsystem).

4. Tritiated aerated radioactive leakage outside the Containment from valve and pump seal leakoffs, recycle holdup tank overflows, and other equipment is routed through the equipment drain header to the waste holdup tank for treatment, recycle, or disposal.

5. Tanks containing potentially radioactive fluids have high-level alarms or level indicators as shown in Table 9.3-5. Therefore, the operator is alerted prior to an overflow condition. Any overflows are collected as shown on Table 9.3-5.

6. Provisions are made for routing relief valve discharges from potentially radioactive fluid tanks to the Liquid Waste Processing System. These discharges are collected as shown on Table 9.3-5.

   Monitoring and sampling of potentially radioactive fluids is described in Section 11.5.

   The drainage system used to carry clean water is completely separate from the drainage system used to carry potentially radioactively contaminated water to prevent the accidental release of radioactive material to the environment. Equipment drains are also provided in some areas in addition to floor drains.

   Turbine Building floor drains, which normally handle clean water but can potentially carry contaminated water caused by equipment failures or accident conditions, are provided with radiation monitors which detect radioactive contamination before discharging the drains to the environment.

   Sumps, tanks, and pumps are located at lower levels to facilitate transfer to the Waste Processing System (WPS). For radiological considerations of the WPS under normal operating conditions and postulated spills and accidents, see Chapters 11, 12, and 15.

   The essential portions of this system are those drains serving areas with safety-related equipment. The floor drainage system serving areas with equipment required for the safe shutdown of the plant is designed to handle postulated equipment gross leakages.

The floor drainage system serving areas with safety-related equipment and components is segregated into independent systems where a postulated flooding of one area does not impair the safety of the redundant equipment.

9.3.3.2 System Description

Segregated drain headers are provided for each level of the Auxiliary Building and Containment, as described in Section 11.2.

Collection and transfer of liquids from the lower levels of buildings are facilitated by sump pumps, sumps, and drain collection tanks. Spare, full-capacity sump pumps are provided at each transfer point in the drainage system.

Backwater valves are provided, as necessary, to prevent water from backing up through a drain line and possibly impairing the function of safety-related equipment. Ball Float devices, which allow water to drain from an area but prevent steam propagation or unfiltered air inleakage to other interconnecting areas, are provided as necessary for the protection of essential systems and components due to postulated piping failures per FSAR Section 3.6B and for control room habitability during a DBA. For the equipment and floor drain system flow diagrams, see Figures 9.3-5, 9.3-6, 9.3-7, 9.3-8, and 9.3-9.

9.3.3.2.1 Containment Building Floor Drains

Leakage from all Containment Building floors is directed to one of the Containment sumps. The reactor cavity is also provided with a sump at the lowest point in the Containment Building. The sump pumps from all three sumps are typically aligned with its respective floor drain tank. These sumps can be alternately aligned to the waste holdup tank.

An increase in the leakage rate inside the Containment can be detected by the following:

1. An increase in airborne contamination

2. An increase in condensate from the Containment cooling units

3. An increase in humidity in the Containment air

4. An increase in the frequency and/or run time of sump pumps in containment.

For more information about Containment leakage detection within the Containment, see Section 5.2.5.

The Containment Building drains do not function during an accident and are isolated by two fail-closed, air-operated Containment isolation valves, one inside and one outside the Containment. The Containment Building drain piping is non-safety class piping except for the Containment penetration piping, and both Containment isolation valves which are ANSI N18.2 Safety Class 2 [5], [9].

9.3.3.2.2 Safeguards Building Floor Drains

Engineered safeguard pumps are located on the lower elevation of the Safeguards Building. The pumps are divided into two redundant trains with each located in an area segregated from its backup train by a watertight wall, preventing a total flooding of one area from interfering with the operation of the pumps located in the other area.

Each area at or below 785 ft 6 in. is provided with a sump, independent from the sump serving the other area. These areas have floor drain headers discharging into the closed Safeguards Building sumps. Drains from areas above 785 ft 6 in. are routed directly to the floor drain tank or to one of the sumps. The Diesel Generator Room sump pumps are normally aligned for discharge into the Wastewater Management System, for treatment prior to discharge.

CCW Tank Room Sump No. 3, which collects drainage from the component cooling water drain tank, Containment spray and RHR pumps, is aligned to discharge into the Wastewater Management System. CCW Tank Room Sump No. 3 is only fed by drains which collect fluid from normally non-radioactive sources.

Safeguards Building floor drain sumps 1 and 2 (for each unit) are normally aligned for discharge into the floor drain tank and have an alternate line to the waste holdup tank. The floor drains from the floor drain tank (no. 1 for Unit 1 and no. 2 for Unit 2) and floor drain tank room utilize a locked closed valve to prevent recirculation of the tank contents back into Sump No. 1 in the event of failure of the tank or piping within the tank cubicle.

Piping and pumps required to prevent flooding of safety-related equipment are ANSI N18.2 Safety Class 3 [5], [9]. All other equipment is non-nuclear-safety class.

9.3.3.2.3 Auxiliary Building Floor Drains (ABFD)

Leakage from the Auxiliary Building is collected in various sumps or to gravity feed drains to the floor drain tanks. Drainage from the Battery Room, Secondary Sampling Room, Chemical Storage Room, Air Compressor Room, Mechanical Equipment Room, and Cable Room is routed to floor drain Sump 11. The ABFD Sump 11 is normally aligned to discharge to the Wastewater Management System. ABFD Sumps 3 and 10 receive drainage from the component cooling water heat exchangers and safety related chillers, respectively. ABFD Sump 3 additionally receives drainage from the discharge of the Service Water Tunnel Sumps 1 and 2. ABFD Sump 10 is normally aligned to discharge to the Unit 1 Component Cooling Water Drain Tank, with an alternate connection to the Unit 2 Component Cooling Water Drain Tank. ABFD Sump 3 pump is normally aligned to discharge directly to the LVW System. ABFD Sumps 4, 5, and 9 receive drainage from the southern half of the Auxiliary Building below elevation 792 ft 0 in. ABFD Sumps 4, 5, and 9 are aligned to discharge to floor drain tank No. 1. The northern half of the Auxiliary Building is drained into various sumps. ABFD Sumps 6 and 8 take drainage from the waste evaporator condensate tank and recycle evaporator feed pump areas, respectively. In addition, ABFD Sump 8 takes drains from a portion of Unit 2 turbine building elevation 810 ft. 6 in. ABFD Sumps 6 & 8 are aligned to discharge into the waste holdup tanks. ABFD Sumps 1, 2, 7, and 12 collect the remainder of the drainage from the northern half of the Auxiliary Building. ABFD Sumps 1, 2, 7, & 12 are typically aligned to discharge to floor drain tank No. 1. ABFD Sump 2 has an alternate connection to Floor Drain Tank #3.

Discharges from ABFD Sumps 1, 7, and 12 have an alternate connection to floor drain tank 2.

Boron recycle tanks and waste holdup tanks are located within watertight compartments to prevent leakage into other areas if a tank break occurs. Floor drains from these areas have a locked-closed, manually operated gate valve located outside each compartment.

9.3.3.2.4 Turbine Building Floor Drains

In the Turbine Building, the bulk of the drainage is collected in the two Turbine Building sumps. The Turbine Building sumps are normally aligned to eventually discharge to the Low Volume Waste (LVW) Pond of the Wastewater Management System (WMS), for conventional treatment prior to discharge. However, in the event of radioactive contamination in the sump discharge, in excess of limits established in the Radiological Effluent Controls Program, release to the LVW Pond will be automatically diverted to the Co-Current Waste (COW) Treatment System, for conventional and radiological treatment. A leak of radioactive material into Turbine Building sump water is detected and alarmed in the Control Room. The radiation level of Turbine Building sump water can be monitored from the Control Room. For information on radioactive leakage, see Section 11.2.

Other means of drainage in the Turbine Building are as follows:

1. Drains in the hot and cold labs drain by gravity to the floor drain tank for Unit 1. The sinks in the chemistry instrument lab drain by gravity to the floor drain tank for Unit 1.

2. Drains in the personnel decontamination and chemical instrument lab drain by gravity to the laundry and hot shower drain tank.

3. Leakoffs for various secondary components drain to the atmospheric drain tank. The tank is drained to the main condenser spray headers or main condenser gravity drains. Overflow from the tank is drained to Turbine Building sump No. 2.

   All drainage piping and equipment in the Turbine Building are non-nuclear-safety-related and are not required to function after an accident or for safe reactor shutdown.

9.3.3.2.5 Fuel Building Floor Drains

All drainage from the Fuel Building, with the exception of the Service Water Tunnel, is directed into one of two sumps. Most drainage from the Fuel Building is directed to Fuel Building sump No. 1, which is located at elevation 800 ft 2 in. Fuel Building Sumps 1 and 2 are aligned to discharge to floor drain tank No. 1, which is located in the Safeguards Building at elevation 773 ft 0 in.

Drainage from the Fuel Building Service Water Tunnel is directed into two sumps. Service Water Tunnel Sumps 1 and 2, which are located at elevation 785 ft 6 in, are aligned to discharge to the ABFD Sump 3.

9.3.3.3 Safety Evaluation

Each compartment that houses safety-related components is designed to accommodate a design basis flood resulting from a 50 gpm leak for 30 minutes in any one compartment. The highest resulting flood level is 28 inches above the floor for the centrifugal charging pump rooms. All components important to safety are installed above the level resulting from the postulated

flood. Thus, the maximum flood level will not affect operation of the safety-related electrical equipment in the room.

Also, since none of the backwater valves are seismically qualified, it is assumed that more than one valve will fail as a result of a safe shutdown earthquake. If communication between rooms via the floor drain system is also assumed, the postulated flood (backflow) will be distributed over a wider area (multiple compartments) with a resultant flood level lower than if the flood level was confined to one compartment only and, as a result, be in acceptable levels in individual compartments. Therefore, the safety level of the plant will not be affected.

Although some backwater valves are credited to operate for the mitigation of postulated piping failures per FSAR Section 3.6B, a seismic event is not assumed in the analysis of these piping failures.

Compartments containing engineered safeguards pumps, recycle holdup tanks, and waste holdup tanks are designed to seismic Category I requirements [6].

Except for specific LOCA conditions, all portions of the Equipment and Floor Drainage System in the Safeguards Building that are at 773' and are safety related are operative at all times. Continued operation of the Containment portion of the system during a LOCA is not required and is prevented by automatic closure of the Containment drain piping isolation valves.

Main drain headers are at least four in. in diameter. Header size is sufficient to minimize total flooding of the piping and to ensure that the drain lines are not pressurized and ruptured [7], [8].

Floor drains and sump pumps in the lowest elevation of Safeguards Building are designed to accommodate a leakage of 50 gpm without flooding of adjacent areas.

Each sump is equipped with two pumps, either of which can handle the design leakage rate. This leakage is postulated to occur as a result of a gross flange gasket failure or a severely damaged pump seal.

The compartments housing engineered safeguards pumps In the Safeguards Building are designed to contain a leakage rate of 50 gpm for 30 min without causing flooding of adjacent areas, assuming all floor drains from this area are clogged. This design protects safety-related equipment outside the pump compartment from flooding due to leakage.

The Auxiliary Building floor drainage system is designed to accommodate normal expected leakage without localized flooding.

If an accident occurs, i.e., a rupture of a tank, a crack in large piping, and so forth, total flooding of a floor is prevented by the following:

1. Operator action to isolate the affected system

2. An arrangement of floor drains ensuring that an increased number of floor drains are used as the flooding expands

The combination of these two actions ensures that deep flooding of a building floor does not occur.

If a leak becomes large enough or is undetected for such a time that a floor is flooded, the water will begin to escape to the Auxiliary Building lowest elevation through stairways and so forth. However, with the exception of the component cooling water heat exchangers and the emergency chillers, no safety-related equipment is located on this floor.

A partial outside flooding of these heat exchangers does not impair their operability or integrity. The emergency chillers are located In a separate compartment under the Control Room, where they are not subjected to leakage from equipment at higher elevations.

The boron recycle and waste holdup tanks are located in separate watertight compartments. Each compartment is drained to the Auxiliary Building sumps, with locked-closed valves in each line outside the compartments. As a result of the administrative control placed on the valve, the possibility of leaving the valve inadvertently open is not considered credible.

Overflow or drainage of a tank is detected by the tank high-low level instrumentation. Except as indicated on Table 9.3-5, all potentially radioactive overflows are routed to receiving tanks within the liquid waste disposal system, thereby ensuring that sufficient holdup and monitoring capabilities are available prior to the ultimate disposal to the environment or recycling, as applicable [10].

9.3.3.4 Testing and Inspection Requirements

Leaktightness, flow capacity, and flow paths are tested prior to the initial operation of the system. Pumps and level controls are adjusted for maintenance of proper sump levels.

9.3.3.5 Instrumentation Requirements

Each safeguards area sump collecting the floor drainage from one Containment spray pump compartment, one safety injection pump compartment, one RHR pump compartment, and so forth has high-1, high-2, and high-high (high-1 < high-2 < high-high) level alarms in the Control Room to indicate that water is entering the sump. Upon receiving the high-1 signal, one sump pump begins to operate. If the level continues to increase to the high-2 level, the second pump starts. The operational status and running time of the sump pumps are indicated on the local panel. A high-high level alarm alerts the operator if the sump level continues to increase with both sump pumps in operation. Based on this information and the sump pumps design flow, the operator can approximate the leakage inflow to each sump.

The Auxiliary Building sump and the diesel generator floor drain sump have a high-water alarm, which alerts the Control Room operator. The operational status and running time of these pumps are indicated on the local panel.

The Containment sump pump discharge lines have a flow totalizer to indicate total flow as part of the overall Containment leakage control.

All sump pumps operate automatically; i.e., the start of the pumps is controlled by the sump levels.

As shown In Table 9.3-5, storage tanks which contain potentially radioactive fluids are provided with a high-water level alarm and level indication.

Refer to Section 9.2.3 for a discussion of the RMWST, Section 9.2.6 for the Condensate Storage Tank, and Section 11.2 and Table 11.2-1 for the liquid waste processing tanks.

9.3.4 CHEMICAL AND VOLUME CONTROL SYSTEM (INCLUDING BORON RECYCLE SYSTEM)

9.3.4.1 Chemical and Volume Control System

The Chemical and Volume Control System (CVCS), shown in Figures 9.3-10 and 9.3-11 is designed to provide the following services to the Reactor Coolant System (RCS):

1. Maintenance of programmed water level in the pressurizer, i.e., maintain required water inventory in the RCS.

2. Maintenance of seal-water injection flow to the reactor coolant pumps.

3. Control of reactor coolant water chemistry conditions, activity level, soluble chemical neutron absorber concentration and makeup.

4. Emergency core cooling (part of the system is shared with the Emergency Core Cooling System).

5. Provide means for filling, draining and pressure testing of the RCS.

9.3.4.1.1 Design Bases

Quantitative design bases are given in Table 9.3-6 with qualitative descriptions given below.

9.3.4.1.1.1 Reactivity Control

The CVCS regulates the concentration of chemical neutron absorber (boron) in the reactor coolant to control reactivity changes resulting from the change in reactor coolant temperature between cold shutdown and hot full-power operation, burnup of fuel and burnable poisons, buildup of fission products in the fuel, and xenon transients.

Reactor Makeup Control

1. The CVCS is capable of borating the RCS through either one of two flow paths and from either one of two boric acid sources.

2. The amount of boric acid stored in the CVCS always exceeds that amount required to borate the RCS to cold shutdown concentration assuming that the control assembly with the highest reactivity worth is stuck in its fully withdrawn position. This amount of boric acid also exceeds the amount required to bring the reactor to hot shutdown and to compensate for subsequent xenon decay.

Boron Thermal Regeneration

The CVCS was designed to control the changes in reactor coolant boron concentration to compensate for the xenon transients during load follow operations without adding makeup for

either boration or dilution. To accomplish this, the boron thermal regeneration process was designed to allow load follow operations as required by the design load cycle. Load follow operation is currently not utilized at Comanche Peak.

Because load follow operation is currently not utilized, there is no need for chilled water to be provided to the Letdown Chiller Heat Exchanger. Therefore, the BTRS chiller and associated components in the chilled water loop are normally removed from service and drained so as not to provide chilled water to the shell side of the Letdown Chiller Heat Exchanger. Also because load follow operation is currently not utilized, the heating source for the letdown reheat heat exchanger, CVCS water flow through the tube side of the heat exchanger, is normally isolated and the tube side of the heat exchanger is drained. There is no near-term plan to utilize these components for load follow operation. If the BTRS chiller and associated components are ever returned to service, flow to and from the thermal regeneration demineralizers and piping vibration will be verified prior to the return to service.

The BTRS is currently used to reduce reactor coolant boron concentration at the end of the core cycle. Additionally, the BTRS demineralizers may be loaded with various types of resin in order to provide supplemental or enhanced cleanup of CVCS letdown flow.

When the BTRS is placed in service, BTRS water taken from the RCS may flow through the tube side of the Letdown Chiller Heat Exchanger, as well as through the shell side of the Letdown Reheat Heat Exchanger. There is the possibility for leakage of water containing radioactivity from the tube side into the normally drained shell side of the Letdown Chiller Heat Exchanger, or from the shell side into the normally drained tube side of the Letdown Reheat Heat Exchanger.

Leakage into the shell side of the Letdown Chiller Heat Exchanger passes through to the Chiller Surge Tank. The high level alarm of the Chiller Surge Tank provides in-leakage detection when the BTRS is in service. Leakage water entering the tube side of the Letdown Reheat Heat Exchanger passes through open drain valves into floor drains which direct the water to Floor Drain Tanks 1 and 2, both of which have high level alarms. Tank overflow protection is further addressed in Table 9.3-5, item 9 of Section 11.2.2.4.2, and Section 11.2.2.5.2.

9.3.4.1.1.2 Regulation of Reactor Coolant Inventory

The CVCS maintains the coolant inventory in the RCS within the allowable pressurizer level range for all normal modes of operation including startup from cold shutdown, full power operation and plant cooldown. This system also has sufficient makeup capacity to maintain the minimum required inventory in the event of minor RCS leaks (see the Technical Specifications for a discussion of maximum allowable RCS leakage).

9.3.4.1.1.3 Reactor Coolant Purification

The CVCS is capable of removing fission and activation products, in ionic form or as particulates, from the reactor coolant in order to provide access to those process lines carrying reactor coolant during operation and to reduce activity releases due to leaks.

9.3.4.1.1.4 Chemical Additions for Corrosion Control

The CVCS provides a means for adding chemicals to the RCS which control the pH of the coolant during initial startup and subsequent operation, scavenge oxygen from the coolant during

startup, and counteract the production of oxygen in the reactor coolant due to radiolysis of water in the core region.

The CVCS is capable of maintaining the oxygen content and pH of the reactor coolant within limits specified in Table 5.2-5.

9.3.4.1.1.5 Seal Water Injection

The CVCS is able to continuously supply filtered water to each reactor coolant pump seal, as required by the reactor coolant pump design.

9.3.4.1.1.6 Hydrostatic Testing of the Reactor Coolant System

The CVCS is capable of supplying water at the maximum test pressure specified to verify the integrity of the RCS. The hydrostatic test is performed prior to initial operation and as part of the periodic RCS inspection program.

9.3.4.1.1.7 Emergency Core Cooling

The centrifugal charging pumps in the CVCS also serve as the high-head safety injection pumps in the Emergency Core Cooling System. Other than the centrifugal charging pumps and associated piping and valves, the CVCS is not required to function during a loss of coolant accident (LOCA). During a LOCA, the CVCS is isolated except for the centrifugal charging pumps and the piping in the safety injection path.

9.3.4.1.2 System Description

The CVCS is shown in Figure 9.3-10 (piping and instrumentation diagram) with system design parameters listed in Table 9.3-6. The codes and standards to which the individual components of the CVCS are designed are listed in Section 3.2. The CVCS consists of several subsystems: the Charging, Letdown and Seal Water System; the Reactor Coolant Purification and Chemistry Control System; the Reactor Makeup Control System; and the Boron Thermal Regeneration System.

9.3.4.1.2.1 Charging, Letdown and Seal Water System

The charging and letdown functions of the CVCS are employed to maintain a programmed water level in the RCS pressurizer, thus maintaining proper reactor coolant inventory during all phases of plant operation. This is achieved by means of continuous feed and bleed process during which the feed rate is automatically controlled based on pressurizer water level. The bleed rate can be chosen to suit various plant operational requirements by selecting the proper combination of letdown orifices in the letdown flow path.

Reactor coolant is discharged to the CVCS from a reactor coolant loop cold leg; it then flows through the shell side of the regenerative heat exchanger where its temperature is reduced by heat transfer to the charging flow passing through the tubes. The coolant then experiences a large pressure reduction as it passes through the letdown orifice(s) and flows through the tube side of the letdown heat exchanger where its temperature is further reduced. Downstream of the letdown heat exchanger a second pressure reduction occurs. This second pressure reduction is

performed by the low pressure letdown valve, the function of which is to maintain upstream pressure thus preventing flashing downstream of the letdown orifices.

The coolant then flows through one of the mixed bed demineralizers. The flow may then pass through the cation bed demineralizer which is used intermittently when additional purification of the reactor coolant is required. Two mixed bed demineralizers may be used in parallel during shutdown for maximum purification.

Alternatively, the flow may be directed such that the mixed bed demineralizers are bypassed. Flow could then be directed to the cation bed demineralizer and/or to the BTRS demineralizer(s) for purification process.

From a point upstream of the Boron Thermal Regeneration System or from a point upstream of the reactor coolant filters, a small sample flow may be diverted from the letdown stream to the Boron Concentration Measurement System. The Boron Concentration Measurement System is abandoned in place.

During reactor coolant boration and dilution operations, especially during load follow, the letdown flow leaving the demineralizers may be directed to the Boron Thermal Regeneration System. The coolant then flows through the reactor coolant filter and into the volume control tank through a spray nozzle in the top of the tank. Hydrogen (from the Gaseous Waste Processing System) is continuously supplied to the volume control tank where it mixes with fission gasses which are stripped from the reactor coolant into the tank gas space. The contaminated hydrogen is vented back to the Gaseous Waste Processing System. The partial pressure of hydrogen in the volume control tank determines the concentration of hydrogen dissolved in the reactor coolant for control of oxygen produced by radiolysis of water in the core.

Three pumps (one positive displacement pump and two centrifugal charging pumps) are provided to take suction from the volume control tank and return the cooled, purified reactor coolant to the RCS. Normal charging flow is handled by one of the three charging pumps. This charging flow splits into two paths. The bulk of the charging flow is pumped back to the RCS through the tube side of the regenerative heat exchanger. The letdown flow in the shell side of the regenerative heat exchanger raises the charging flow to a temperature approaching the reactor coolant temperature. The flow is then injected into a cold leg of the RCS. Two charging paths are provided from a point downstream of the regenerative heat exchanger. A flow path is also provided from the regenerative heat exchanger outlet to the pressurizer spray line. An air operated valve in the spray line is employed to provide auxiliary spray to the vapor space of the pressurizer during plant cooldown. This provides a means of cooling the pressurizer near the end of plant cooldown, when the reactor coolant pumps, which normally provide the driving head for the pressurizer spray, are not operating.

A portion of the charging flow is directed to the reactor coolant pumps (nominally 8 gpm per pump) through a seal water injection filter. It is directed down to a point between the pump shaft bearing and the thermal barrier cooling coil. Here the flow splits and a portion (nominally 5 gpm per pump) enters the RCS through the labyrinth seals and thermal barrier. The remainder of the flow is directed up the pump shaft, cooling the lower bearing, and to the number 1 seal leakoff. The number 1 seal leakoff flow discharges to a common manifold, exits from the Containment, and then passes through the seal water return filter and the seal water heat exchanger to the suction side of the charging pumps, or by alternate path to the volume control tank. A very small portion of the seal flow leaks through to the number 2 seal. A number 3 seal provides a final

barrier to leakage of reactor coolant to the Containment atmosphere. The number 2 leakoff flow is discharged to the reactor coolant drain tank in the Liquid Waste Processing System.

This seal injection flow is assured because there are 3 charging pumps, which individually are capable of providing the normal charging line flow plus the nominal seal injection flow. A number 3 seal provides a barrier to leakage of reactor coolant to the Containment Atmosphere. The number 3 seal injection is supply by Reactor Makeup Water to a stand pipe. Part of the water flows downward, joins with the leakoff from the No. 2 seal and exits through the No. 2 seal leakoff. The remaining water exits through the No. 3 seal leakoff which is discharged to the Containment Sump.

In the unlikely event of a loss of seal injection flow to the RCPs, the Component Cooling Water System (CCWS) continues to provide flow to the thermal barrier heat exchangers. Under these conditions, the thermal barrier heat exchangers, functioning in its backup capacity, cools the reactor coolant before it enters the pump radial bearing and the shaft seal area. The loss of seal injection flow may result in an increase in the number 1 seal leak rate, a temperature increase in the pump bearing area, and a temperature increase in the seal area. If seal injection flow is lost and thermal barrier heat exchanger cooling is maintained, it is recommended that pump operation be continued no longer than 24 hours, provided the previously identified parameters remain within allowable limits. If these parameters reach the maximum allowable values, the pump should be stopped.

The operator will be alerted to a loss of seal injection flow to the RCPs by the low flow alarm (and indication) located on the main control board. If the loss of seal injection flow results in an increase in number 1 seal leakage, the operator will be alerted of a high number 1 seal leak rate by the high flow alarm (and flow recording) located on the main control board. The pump bearing area temperature and seal area temperature indications are monitored by the computer; the operator will be alerted of resultant high temperature in either area of the pump by CCW return high temperature alarms annunciated on the main control board and high bearing temperature alarms on the plant computer.

An extended loss of all seal cooling, both seal water injection and component cooling water cooling of the thermal barrier, will result in an increase in seal leakoff water temperature as the cooler water in the pump shaft cavity is replaced by hot reactor coolant water. When the seal leakoff water temperature reaches the activation temperature of the reactor coolant pump SHIELD Shutdown Seal (SSDS), it will activate to limit reactor coolant pump seal leakoff. However, if the RCP is running, the SSDS will fail and leakage will not be limited.

The excess letdown path is provided as an alternate letdown path from the RCS in the event that the normal letdown path is inoperable. Reactor coolant can be discharged from a cold leg to flow through the tube side of the excess letdown heat exchanger where it is cooled by component cooling water. Downstream of the heat exchanger a remote-manual control valve controls the letdown flow. The flow normally joins the number 1 seal discharge manifold and passes through the seal water return filter and heat exchanger to the suction side of the charging pumps. The excess letdown flow can also be directed to the reactor coolant drain tank or directly into the volume control tank via a spray nozzle. When the normal letdown line is not available, the normal purification path is also not in operation. Therefore, this alternate condition would allow continued power operation for a limited period of time, dependent on RCS chemistry and activity.

During plant heatup, the excess letdown flow path is placed in service and left in service to avoid an excessive thermal transient, which would occur if the system operation were initiated with a large differential temperature near the end of heatup. The excess letdown flow path is used to provide additional letdown capability during the final stages of plant heatup. This path removes some of the excess reactor coolant due to expansion of the system as a result of the RCS temperature increase.

Surges in RCS inventory due to load changes are accommodated for the most part in the pressurizer. The volume control tank provides surge capacity for reactor coolant expansion not accommodated by the pressurizer. The Volume Control Tank water level is controlled via a three way valve which directs letdown/makeup flow to the Volume Control Tank ("VCT" position) or diverts it to the Boron Recycle System via the Recycle Hold-up Tank ("RHT" position). In addition, the three-way valve controller has an "Auto" position which controls the Volume Control Tank water level automatically by diverting all or a portion of the letdown flow to the Recycle Hold-up Tank when the Volume Control Tank exceeds its normal operating water level. The three-way valve is normally maintained in the "VCT" position with the control room operator manually controlling the Volume Control Tank water level. If the Volume Control Tank water level exceeds a preset maximum, an alarm is actuated and the letdown flow is automatically diverted to the Boron Recycle System. Additional alarms have been added to; 1) alert the control room operators when the three-way valve controller is not in the "VCT" position, and 2) provide early indication of an increasing Volume Control Tank water level ("high" water level alarm). These alarms were added to aid the control room operators detection of a potential inadvertent boron dilution event in Modes 3, 4 and 5. See Section 15.4.6 for additional information.

Low level in the volume control tank initiates makeup from the Reactor Makeup Control System. If the Reactor Makeup Control System does not supply sufficient makeup to keep the volume control tank level from falling to a lower level, a low alarm is actuated. Manual action may correct the situation or, if the level continues to decrease, a low-low level signal from both level channels causes the suction of the charging pumps to be transferred to the refueling water storage tank.

The reciprocating charging pump is also used to perform hydrostatic tests which verify the integrity and leak-tightness of the RCS. The pump can pressurize the RCS to the maximum designated test pressure. The hydrostatic test is performed prior to initial operation and is part of the periodic RCS inservice inspection program.

9.3.4.1.2.2 Reactor Coolant Purification and Chemistry Control System

Reactor coolant water chemistry specifications are given in Table 5.2-5.

pH Control

The pH control chemical employed is lithium hydroxide. This chemical is chosen for its compatibility with the materials and water chemistry of borated water/stainless steel/zirconium/ inconel systems. In addition, lithium-7 is produced in the core region due to irradiation of the dissolved boron in the coolant.

The concentration of lithium-7 is maintained as required to support the hot full power pH_t range as specified in Table 5.2-5. If the concentration exceeds this range, as it may during the early stages of a core cycle, the cation bed demineralizer is employed in the letdown line in series

operation with a mixed bed demineralizer. Since the amount of lithium to be removed is small and its buildup can be readily calculated, the flow through the cation bed demineralizer is not required to be full letdown flow. If the concentration of lithium-7 is below the specified limits, lithium hydroxide can be introduced into the RCS via the charging flow. The solution is prepared in the laboratory and poured into the chemical mixing tank. Reactor makeup water is then used to flush the solution to the suction manifold of the charging pumps.

Oxygen Control

Hydrazine is employed as an oxygen scavenging agent during reactor startup from cold condition or during reactor shutdown to a cold condition. Hydrazine addition is required when hydrogen is not effective or limited driving force is present to combine oxygen and hydrogen. The hydrazine solution is introduced into the RCS in the same manner as described above for the pH control agent.

Dissolved hydrogen is employed to control and scavenge oxygen produced due to radiolysis of water in the core region. Sufficient partial pressure of hydrogen is maintained in the volume control tank such that the specified equilibrium concentration of hydrogen is maintained in the reactor coolant. A pressure control maintains a minimum pressure in the vapor space of the volume control tank. This valve can be adjusted to provide the correct equilibrium hydrogen concentration (25 to 50 cc hydrogen at STP per kilogram of water). Hydrogen is supplied from the hydrogen bulk gas storage system.

Reactor Coolant Purification

Mixed bed demineralizers are provided in the letdown line to provide cleanup of the letdown flow. The demineralizers remove ionic corrosion products and certain fission products. During normal power operation one demineralizer is in continuous service and can be supplemented intermittently by the cation bed demineralizer, if necessary, for additional purification. The cation resin removes principally cesium and lithium isotopes from the purification flow. The second mixed bed demineralizer serves as a standby unit for use if the operating demineralizer becomes exhausted during operation.

A bypass line and isolation valve also provide additional flexibility for cleanup of letdown flow. The line and valve allow letdown flow to be aligned such that the mixed bed demineralizers are bypassed and flow is directly to the cation bed demineralizer for cleanup. Letdown can also be aligned to flow to the BTRS demineralizer(s) for cleanup, either directly or in series with either a mixed bed or the cation bed demineralizer.

In Mode 3 below 500 degrees F, during RCS cleanup for outages, purification flow can be increased by opening all three letdown orifices.

A further cleanup feature is provided for use during Residual Heat Removal System operation. A RHR letdown flow control valve admits flow into the CVCS downstream of the containment isolation valves. Both demineralizers can be placed in service to allow for a more rapid cleanup.

Filters are provided at various locations to ensure filtration of particulate and resin fines and to protect the seals on the reactor coolant pumps.

Fission gases are removed from the reactor coolant by purging of the volume control tank to the Gaseous Waste Processing System.

9.3.4.1.2.3 Reactor Makeup Control System

The soluble neutron absorber (boric acid) concentration is controlled by the Reactor Makeup Control System. The Boron Thermal Regeneration System, if properly configured, may also be used. The Reactor Makeup Control System is also used to maintain proper reactor coolant inventory. In addition, for emergency boration and makeup, the capability exists to provide refueling water or 4 weight percent boric acid directly to the suction of the charging pump.

The Reactor Makeup Control System provides a manually pre-selected makeup composition to the charging pump suction header or to the volume control tank. The makeup control functions are those of maintaining desired operating fluid inventory in the volume control tank and adjusting reactor coolant boron concentration for reactivity control. Reactor makeup water and boric acid solution (4 weight percent) are blended together at the reactor coolant boron concentration for use as makeup to maintain volume control tank inventory or they can be used separately to change the reactor coolant boron concentration.

The boric acid is stored in two boric acid tanks. Two boric acid transfer pumps are provided with one pump normally required to provide boric acid to the suction header of the charging pumps, and the second pump in reserve. They are both aligned to take suction from separate boric acid tanks. On a demand signal by the reactor makeup controller, the pump starts and delivers boric acid to the suction header of the charging pumps. The pump can also be used to recirculate the boric acid tank fluid.

All portions of the CVCS which normally contain concentrated boric acid solution (4 weight percent boric acid) are required to be located with a heated area in order to maintain solution temperature at ≥65°F. If a portion of the system which normally contains concentrated boric acid solution is not located in a heated area, it must be provided with some other means (e.g., heat tracing) to maintain solution temperature at ≥65°F. Heat tracing is utilized inside the plant on 4% boric acid solution lines where potential freezing poses a recrystalization problem.

The reactor makeup water pumps take suction from the reactor make-up water storage tank and are employed for various makeup and flushing operations throughout the systems. There is one pump for each unit plus a common pump for both units. One of these pumps normally operates continuously and provides flow to the suction header of the charging pumps or the volume control tank through the letdown line and spray nozzle on demand.

During reactor operation, changes are made in the reactor coolant boron concentration for the following conditions:

1. Reactor startup - boron concentration must be decreased from shutdown concentration to achieve criticality.

2. Load follow - boron concentration must be either increased or decreased to compensate for the xenon transient following a change in load.

3. Fuel burnup - boron concentration must be decreased to compensate for fuel burnup and the buildup of fission products in the fuel.

4. Cold shutdown - boron concentration must be increased to the cold shutdown concentration.

5. Burnable Poiso depletion - dependent on the core design and time in core life, boron concentration may need to be increased to compensate for the depletion of burnable poisons in the fuel.

The Boron Thermal Regeneration System may be used to control boron concentration to compensate for xenon transients during load follow operations. Boron thermal regeneration can also be used in conjunction with dilution operations of the Reactor Makeup Control System to reduce the amount of effluent to be processed by the Liquid Waste System or the Boron Recycle System.

The Reactor Makeup Control System can be set up for the following modes of operation:

1. Automatic makeup

   The “automatic makeup” mode of operation of the Reactor Makeup Control System provides blended boric acid solution, preset to match the boron concentration in the RCS. Automatic makeup compensates for minor leakage of reactor coolant without causing significant changes in the reactor coolant boron concentration.

   Under normal plant operating conditions, the mode selector switch is set in the “automatic makeup” position. This switch position establishes a preset control signal to the total makeup flow controller and establishes positions for the makeup stop valves for automatic makeup. The boric acid flow controller is set to blend to the same concentration of borated water as contained in the RCS. A preset low level signal from the volume control tank level controller causes the automatic makeup control action to start a boric acid transfer pump, open the makeup stop valve to the charging pump suction, and position the boric acid flow control valve and the reactor makeup water flow control valve. The flow controllers then blend the makeup stream according to the preset concentration. Makeup addition to the charging pump suction header causes the water level in the volume control tank to rise. At a preset high level point, the makeup is stopped. This operation may be terminated manually at any time.

   If the automatic makeup fails or is not aligned for operation and the tank level continues to decrease, a low level alarm is actuated. Manual action may correct the situation or, if the level continues to decrease, a low-low level signal opens the stop valves in the refueling water supply line to the charging pumps, and closes the stop valves in the volume control tank outlet line.

2. Dilution

   The “dilute” mode of operation permits the addition of a preselected quantity of reactor makeup water at a preselected flow rate to the RCS. The operator sets the mode selector switch to “dilute,” the total makeup flow controller setpoint to the desired flow rate, the total makeup batch integrator to the desired quantity and initiates system start. This opens the reactor makeup water flow control valve and opens the makeup stop valve to the volume control tank inlet. Excessive rise of the volume control tank water level is prevented by manual or automatic actuation of a three-way diversion valve which routes

the reactor coolant letdown flow to the Boron Recycle System. When the preset quantity of water has been added, the batch integrator causes makeup to stop. The operation may be terminated manually at any time.

Dilution can also be accomplished by operating the Boron Thermal Regeneration System in the boron storage mode.

3. Alternate dilution

The “alternate dilute” mode of operation is similar to the dilute mode except a portion of the dilution water flows directly to the charging pump suction and a portion flows into the volume control tank via the spray nozzle and then flows to the charging pump suction. This decreases the delay in diluting the RCS caused by directing dilution water to the volume control tank.

4. Boration

The “borate” mode of operation permits the addition of a preselected quantity of concentrated boric acid solution at a preselected flow rate to the RCS. The operator sets the mode selection switch to “borate,” the concentrated boric acid flow controller setpoint to the desired flow rate, the concentrated boric acid batch integrator to the desired quantity, and initiates system start. This opens the makeup stop valve to the charging pumps suction, positions the boric acid flow control valve, and starts the selected boric acid transfer pump, which delivers a 4 weight percent boric acid solution to the charging pumps suction header. The total quantity added in most cases is so small that it has only a minor effect on the volume control tank level. When the preset quantity of concentrated boric acid solution is added, the batch integrator causes makeup to stop. The operation may be terminated manually at any time.

Boration can also be accomplished by operating the Boron Thermal Regeneration System in the boron release mode. See Section 9.3.4.1.1.1.

5. Manual

The “manual” mode of operation permits the addition of a preselected quantity and blend of boric acid solution to the charging pump suction and volume control tank inlet, to the refueling water storage tank, to the recycle holdup tanks in the Boron Recycle System, or to some other location via a temporary connection. While in the manual mode of operation, automatic makeup to the RCS is precluded. The discharge flow path must be prepared by opening manual valves in the desired path.

The operator sets the mode selector switch to “manual,” the boric acid and total makeup flow controllers to the desired flow rates, the boric acid and total makeup batch integrators to the desired quantities, and actuates the makeup start switch.

The start switch actuates the boric acid flow control valve and the reactor makeup water flow control valve and starts the boric acid transfer pump.

When the preset quantities of boric acid and reactor makeup water have been added, the batch integrators cause makeup to stop. This operation may be stopped manually by actuating the makeup stop switch.

If either batch integrator is satisfied before the other has recorded its required total, the pump and valve associated with the integrator which has been satisfied will terminate flow. The flow controlled by the other integrator will continue until that integrator is satisfied. In the manual mode, the boric acid flow is terminated first to prevent piping systems from remaining filled with 4 weight percent boric acid solution.

The quantities of boric acid and reactor makeup water injected are totalized by the batch counters and the flow rates are recorded on strip recorders. Deviation alarms sound for both boric acid and reactor makeup water if flow rates deviate from setpoints.

9.3.4.1.2.4 Boron Thermal Regeneration System

Comanche Peak currently does not utilize load follow operation. Some of the design capabilities discussed below are therefore not currently utilized. See Section 9.3.4.1.1.1.

Downstream of the mixed bed demineralizers, the letdown flow can be diverted to the Boron Thermal Regeneration System where part or all of the letdown flow can be treated when boron concentration changes are desired.

Additionally, the BTRS demineralizer(s) can be utilized to provide for supplemental or enhanced cleanup of letdown flow. After processing, the flow is returned to a point upstream of the reactor coolant filter.

Storage and release of boron during load follow operation is determined by the temperature of fluid entering the thermal regeneration demineralizers. A chiller unit and a group of heat exchangers are employed to provide the desired fluid temperatures at the demineralizer inlets for either storage or release operation of the system.

The flow path through the Boron Thermal Regeneration System is different for the boron storage and the boron release operations. During boron storage, the letdown stream enters the moderating heat exchanger and from there it passes through the letdown chiller heat exchanger. These two heat exchangers cool the letdown stream prior to its entering the demineralizers. The letdown reheat heat exchanger is valved out on the tube side and performs no function during boron storage operations. The temperature of the letdown stream at the point of entry to the demineralizers is controlled automatically by the temperature control valve which controls the shell side flow to the letdown chiller heat exchanger. After passing through the demineralizers, the letdown enters the moderating heat exchanger shell side, where it is heated by the incoming letdown stream before going to the volume control tank.

Therefore, for boron storage, a decrease in the boric acid concentration in the reactor coolant is accomplished by sending the letdown flow at relatively low temperatures to the thermal regeneration demineralizers. The resin, which was depleted of boron at high temperature during a prior boron release operation, is now capable of storing boron from the low temperature letdown stream. Reactor coolant with a decreased concentration of boric acid leaves the demineralizers and is directed to the RCS via the charging system.

During the boron release operation, the letdown stream enters the moderating heat exchanger tube side, bypasses the letdown chiller heat exchanger, and passes through the shell side of the letdown reheat heat exchanger. The moderating and letdown reheat heat exchangers heat the letdown stream prior to its entering the resin beds. The temperature of the letdown at the point of entry to the demineralizers is controlled automatically by the temperature control valve which controls the flow rate on the tube side of the letdown reheat heat exchanger. After passing through the demineralizers, the letdown stream enters the shell side of the moderating heat exchanger, passes through the tube side of the letdown chiller heat exchanger and then goes to the volume control tank. The temperature of the letdown stream entering the volume control tank is controlled automatically by adjusting the shell side flow rate on the letdown chiller heat exchanger. Thus, for boron release, an increase in the boric acid concentration in the reactor coolant is accomplished by sending the letdown flow at relatively high temperatures to the thermal regeneration demineralizers. The water flowing through the demineralizers now releases boron which was stored by the resin at low temperature during a previous boron storage operation. The boron enriched reactor coolant is returned to the RCS via the charging system.

Although the Boron Thermal Regeneration System is primarily designed to compensate for xenon transients occurring during load follow, it can also be used to handle boron swings far in excess of the design capacity of the demineralizers. If used for startup dilution, for example, the resin beds are first saturated, then washed off to the Boron Recycle System, then again saturated and washed off. This operation continues until the desired dilution in the RCS is obtained.

A thermal regeneration demineralizer can also be used as a deborating demineralizer, which would be used to dilute the RCS down to very low boron concentrations towards the end of a core cycle. To make such a bed effective, the effluent concentration from the bed must be kept very low, close to zero ppm boron. This low effluent concentration can be achieved by using fresh resin. Use of fresh resin can be coupled with the normal replacement cycle of the resin; one resin bed being replaced during each core cycle.

As an additional function, the thermal regeneration (BTRS) demineralizer(s) may be loaded with various types of resin in order to provide supplemental or enhanced cleanup of letdown flow. This may be accomplished by aligning letdown such that flow will be directly to the BTRS demineralizer(s) or by placing these demineralizer(s) in series with either a mixed bed or the cation bed demineralizer.

9.3.4.1.2.5 Components Description

A summary of principal component design parameters is given in Table 9.3-7, and safety classifications and design codes are given in Section 3.2.

All CVCS piping that handles radioactive liquid is austenitic stainless steel. All piping joints and connections are welded, except where flanged connections are required to facilitate equipment removal for maintenance and hydrostatic testing.

Charging Pumps

Three charging pumps are supplied to inject coolant into the RCS. Two of the pumps are of the single speed, horizontal, centrifugal type and the third is a positive displacement (reciprocating) pump equipped with variable speed drive. All parts in contact with the reactor coolant are

fabricated of austenitic stainless steel or other material of adequate corrosion resistance. The centrifugal pump seals and the reciprocating pump stuffing box are provided with leakoffs to collect the leakage before it can leak to the atmosphere. The reciprocating pump design prevents lubricating oil from contaminating the charging flow. There is a minimum flow recirculation line to protect the centrifugal charging pumps from a closed discharge valve condition.

Charging flow rate is determined from a pressurizer level signal. The means of flow control for the reciprocating pump is by variation of pump speed. The reciprocating charging pump is also used to hydrotest the RCS. When operating a centrifugal charging pump, the flow paths remain the same but charging flow control is accomplished by a modulating valve on the discharge side of the centrifugal pumps. The centrifugal charging pumps also serve as high head safety injection pumps in the Emergency Core Cooling System. A description of the charging pump function upon receipt of a safety injection signal is given in Section 6.3.2.2.5.

Boric Acid Transfer Pumps

Two canned motor pumps are supplied per unit. One pump is normally required to supply boric acid to the suction header of the charging pumps while the second serves as a standby. Manual or automatic initiation of the Reactor Coolant Makeup System will start the one pump to provide normal makeup of boric acid solution to the suction header of the charging pumps. Miniflow from this pump flows back to the associated boric acid tank and helps maintain thermal equilibrium. The recirculation of the boric acid tanks using the boric acid transfer pump allows the location of a sample point at the boric acid transfer pump discharge. For a remote sampling station, purging of sample lines ensures a representative sample will be taken. The standby pump is also aligned and can be used intermittently to circulate boric acid solution through the other tank to maintain thermal equilibrium in this part of the system. Emergency boration, supplying 4 weight percent boric acid solution directly to the suction of the charging pumps, can be accomplished by manually starting either or both pumps. The transfer pumps may also function to transfer boric acid solution from the batching tank to the boric acid tanks. All parts in contact with the solution are of austenitic stainless steel.

Chiller Pumps

Two centrifugal pumps circulate the water through the chilled water loop in the Boron Thermal Regeneration System. One pump is normally operated, with the second serving as a standby.

Because load follow operation is currently not utilized, the Chiller Pumps are normally removed from service, the chiller loops having been drained. See Section 9.3.4.1.1.1.

Regenerative Heat Exchanger

The regenerative heat exchanger is designed to recover heat from the letdown flow by reheating the charging flow, which reduces thermal effects on the charging penetrations into the reactor coolant loop piping.

The letdown stream flows through the shell of the regenerative heat exchanger and the charging stream flow through the tubes. The unit is constructed of austenitic stainless steel, and is of all welded construction.

The temperatures of both outlet streams from the heat exchanger are monitored with indication given in the Control Room. A high temperature alarm is actuated on the main control board if the temperature of the letdown stream exceeds desired limits.

Letdown Heat Exchanger

The letdown heat exchanger cools the letdown stream to the operating temperature of the downstream demineralizers. Reactor coolant flows through the tube side of the exchanger while component cooling water flows through the shell side. All surfaces in contact with the reactor coolant are austenitic stainless steel, and the shell is carbon steel.

The low pressure letdown valve, located downstream of the heat exchanger, maintains the pressure of the letdown flow upstream of the heat exchanger in a range sufficiently high to prevent two phase flow. Pressure indication and high pressure alarm are provided on the main control board.

The letdown temperature controller indicates and controls the temperature of the letdown flow exiting from the letdown heat exchanger. A temperature sensor, which is part of the CVCS, provides input to the controller in the Component Cooling Water System. The exit temperature of the letdown stream is thus controlled by regulating the component cooling water flow through the letdown heat exchanger. Temperature indication is provided on the main control board. If the outlet temperature from the heat exchanger is excessive, a high temperature alarm is actuated and a temperature controlled valve diverts the letdown directly to the volume control tank.

The outlet temperature from the shell side of the heat exchanger is allowed to vary over an acceptable range compatible with the equipment design parameters and required performance of the heat exchanger in reducing letdown stream temperature.

Excess Letdown Heat Exchanger

The excess letdown heat exchanger cools reactor coolant letdown flow. This letdown flow rate is equivalent to the portion of the nominal seal injection flow which flows into the RCS through the reactor coolant pump labyrinth seals.

The excess letdown heat exchanger can be employed either when normal letdown is temporarily out of service to maintain the reactor in operation or it can be used to supplement maximum letdown during the final stages of heatup. The letdown flows through the tube side of the unit and component cooling water is circulated through the shell. All surfaces in contact with reactor coolant are austenitic stainless steel and the shell is carbon steel. All tube joints are welded.

A temperature detector measures the temperature of the excess letdown flow downstream of the excess letdown heat exchanger. Temperature indication and high temperature alarm are provided on the main control board.

A pressure sensor indicates the pressure of the excess letdown flow downstream of the excess letdown heat exchanger and excess letdown control valve. Pressure indication is provided on the main control board.

Seal Water Heat Exchanger

The seal water heat exchanger is designed to cool fluid from three sources: reactor coolant pump number 1 seal leakage, reactor coolant discharged from the excess letdown heat exchanger, and miniflow from a centrifugal charging pump. Reactor coolant flows through the tube side of the heat exchanger and component cooling water is circulated through the tubes. The design flow rate through the tube side is equal to the sum of the nominal excess letdown flow, maximum design reactor coolant pump seal leakage, and miniflow from one centrifugal charging pump. The unit is designed to cool the above flow to the temperature normally maintained in the volume control tank. All surfaces in contact with reactor coolant are austenitic stainless steel and the shell is carbon steel.

Moderating Heat Exchanger

The moderating heat exchanger operates as a regenerative heat exchanger between incoming and outgoing streams to and from the thermal regeneration demineralizers.

The incoming letdown flow enters the tube side of the moderating heat exchanger. The shell side fluid, which comes directly from the thermal regeneration demineralizers, enters at low temperature during boron storage and high temperature during boron release.

Letdown Chiller Heat Exchanger

During the boron storage operation, the process stream enters the tube side of the letdown chiller heat exchanger after leaving the tube side of the moderating heat exchanger. The letdown chiller heat exchanger cools the process stream to allow the thermal regeneration demineralizers to remove boron from the coolant. The desired cooling capacity is adjusted by controlling the chilled water flow rate passed through the shell side of the heat exchanger.

The letdown chiller heat exchanger is also used during the boron release operation to cool the liquid leaving the thermal regeneration demineralizers to ensure that its temperature does not exceed that of normal letdown to the volume control tank.

Because load follow operation is currently not utilized, the Chiller Pumps are normally removed from service, the chiller loops having been drained. See Section 9.3.4.1.1.1.

Letdown Reheat Heat Exchanger

The letdown reheat heat exchanger is used only during boron release operations and it is then used to heat the process stream. Water used for heating is diverted from the letdown line upstream of the letdown heat exchanger, passed through the tube side of the letdown reheat heat exchanger and then returned to the letdown stream upstream of the letdown heat exchanger.

Because load follow operation is currently not utilized, the Letdown Reheat Heat Exchanger tube side is normally isolated and drained. See Section 9.3.4.1.1.1.

Volume Control Tank

The volume control tank provides surge capacity for part of the reactor coolant expansion volume not accommodated by the pressurizer. When the level in the tank reaches a preset maximum level setpoint, the remainder of the expansion volume is accommodated by diversion of the letdown stream to the Boron Recycle System. The tank also provides a means for introducing hydrogen into the coolant to maintain the required equilibrium concentration of 25 to 50 cc hydrogen (at STP) per kilogram of water and is used for degassing the reactor coolant. It also serves as a head tank for the charging pumps.

A spray nozzle located inside the tank on the letdown line provides liquid to gas contact between the incoming fluid and the hydrogen atmosphere in the tank.

Hydrogen (from the hydrogen bulk gas storage system) is supplied to the volume control tank while a remotely operated vent valve, discharging to the Gaseous Waste Processing System, permits continuous removal of gaseous fission products which are stripped from the reactor coolant and collected in this tank. Relief protection, gas space sampling, and nitrogen purge connections are also provided. The tank can also accept the seal water return flow from the reactor coolant pumps although this flow normally goes directly to the suction of the charging pumps.

Argon gas is intermittently injected to the Volume Control Tank to increase radioactivity in the RCS. The Argon gas is injected through a port located in the Process Sampling System and is equipped with over-pressure protection. The increase in radioactivity allows for early detection of low volume, primary to secondary leakage through the steam generators. The Argon concentration should be increased based on the radiological needs for detection and shall remain within the specified design parameters and technical specifications.

Volume control tank pressure is monitored with indication given in the Control Room. An Alarm is actuated in the Control Room for high and low pressure conditions. The volume control tank pressure control valve is automatically closed by the low pressure signal.

Two level channels govern the water inventory in the volume control tank.

If the volume control tank level rises above the normal operating range and the three-way valve is in the “Auto” position, one level channel provides an analog signal to the proportional controller which modulates the three-way valve downstream of the reactor coolant filter to maintain the volume control tank level within the normal operating band. The three-way valve can split letdown flow so that a portion goes to the Boron Recycle System and a portion to the volume control tank. The controller would operate in this fashion during a dilution operation when reactor makeup water is being fed to the volume control tank from the Reactor Makeup Control System.

If the modulating function of the channel fails and the volume control tank level continues to rise, a level alarm will alert the operator to the malfunction. Letdown flow will then be diverted.

During normal power operation, a low level in the volume control tank initiates auto makeup which injects a pre-selected blend of boric acid solution and reactor makeup water into the charging pump suction header. When the volume control tank level is restored to normal, auto makeup stops.

If the automatic makeup fails or is not aligned for operation and the tank level continues to decrease, a low level alarm is actuated.

Manual action may correct the situation or, if the level continues to decrease, a low-low signal from both level channels opens the stop valves in the refueling water supply line, and closes the stop valves in the volume control tank outlet line.

Boric Acid Tanks

The combined boric acid tank capacity is sized to store sufficient boric acid solution for refueling plus enough for a cold shutdown from full power operation immediately following refueling with the most reactive control rod not inserted. The two boric acid tanks are shared between Units 1 and 2. Each tank has a nitrogen sparger installed inside the tank. Nitrogen from the plant bulk low pressure nitrogen gas system is injected into the boric acid solution to reduce and maintain low dissolved oxygen levels within the fluid.

The concentration of boric acid solution in storage is maintained between 4 and 4.4 percent by weight. Periodic manual sampling and corrective action, if necessary, assure that these limits are maintained. As a consequence, measured amounts of boric acid solution can be delivered to the reactor coolant to control the boron concentration.

A temperature sensor provides temperature measurement of each tank's contents. Temperature indication as well as high and low temperature alarms are provided on the main control board.

Two level detectors indicate the level in each boric acid tank. Level indication is provided on the Unit 1 Main Control Board, with high, low, low-low and empty level alarms provided on both the Unit 1 and Unit 2 Main Control Boards. The high alarm indicates that the tank may soon overflow. The low alarm warns the operator to start makeup to the tank. The low-low alarm is set to indicate the minimum level of boric acid in the tank to ensure sufficient boric acid is available for a cold shutdown with one stuck rod. The empty level alarm is set to give warning of loss of pump suction.

Batching Tank

The batching tank is used for mixing a makeup supply of boric acid solution for transfer to the boric acid tanks. One batching tank is shared between Units 1 and 2.

A local sampling point is provided for verifying the solution concentration prior to transferring it out of the tank. The tank is provided with an agitator to improve mixing during batching operations (and thereby provide a representative sample), and a steam jacket for heating the boric acid solution.

Chemical Mixing Tank

The primary use of the chemical mixing tank is in the preparation of caustic solutions for pH control and hydrazine solution for oxygen scavenging.

Chiller Surge Tank

The chiller surge tank handles the thermal expansion and contraction of the water in the chiller loop. The surge volume in the tank also acts as a thermal buffer for the chiller. The fluid level in the tank is monitored with level indication and high and low level alarms provided on the main control board. Due to current operating philosophy, the low level alarm is no longer in service.

Because load follow operation is currently not utilized, the Chiller Surge Tank is normally drained. A high level alarm is required to be in service when the BTRS is in service. The high level alarm provides an alert if water leaks from the shell side to the tube side of the Letdown Chiller Heat Exchanger and fills the Chiller Surge Tank. A low level alarm is required when the BTRS chiller loop is filled and used to provide chilled water to the Letdown Chiller Heat Exchanger. However, the chilled water loop is normally drained because load follow operation is currently not utilized. See Section 9.3.4.1.1.1.

Discharge Dampener Tank

The discharge dampener is a spherical pressure vessel which uses centrifugal flow patterns to dampen pressure pulsations at the discharge of the positive displacement pump.

Suction Stabilizer Tank

The suction stabilizer is a tank designed to reduce fluid pulsations and acceleration head loss. A cover gas provides pulsation absorption capacity. To prevent excessive gas from accumulating, a vent path to the VCT is provided. To make up for any gas absorbed by the water, a gas makeup system is used. When the stabilizer water level increases, a level signal opens a valve to supply gas. The Operator then momentarily opens the two down stream isolation valves. When the level drops to a predetermined level, the supply valve closes. A hi-hi alarm on the main control board is also provided to warn that the stabilizer is approaching a water solid condition. The gas makeup source will be pre-aligned for consistency with the VCT cover gas. A pressure regulator is located in the gas makeup line to restrict flow to prevent perturbations of the stabilizer water level during gas addition. The vent and gas makeup paths are automatically isolated on an “S” signal.

Mixed Bed Demineralizers

Two flushable mixed bed demineralizers assist in maintaining reactor coolant purity. A lithium-form cation resin and hydroxyl-form anion resin may be charged into the demineralizers. The anion resin is converted to the borate form in operation. Both types of resin remove fission and corrosion products. The resin bed is designed to reduce the concentration of ionic isotopes in the purification stream, except for cesium, yttrium and molybdenum, by a minimum factor of 10.

Each demineralizer is capable of removing ionic impurities for one core cycle with one percent of the rated core thermal power being generated by defective fuel rods. If the mixed bed demineralizers are in use, one demineralizer is normally in service with the other in standby.

A temperature sensor monitors the temperature of the letdown flow down stream of the letdown heat exchanger and if the letdown temperature exceeds the maximum allowable resin operating temperature (approximately 140°F), a three-way valve is automatically actuated to bypass the

flow around the demineralizers. Temperature indication and high alarm are provided on the main control board. The air operated three-way valve failure mode directs flow to the volume control tank.

Cation Bed Demineralizers

A flushable demineralizer with cation resin in the hydrogen form may be used intermittently to control the concentration of lithium-7 which builds up in the coolant from the boron-10 (n,α) lithium-7 reaction. The demineralizer also is capable of maintaining the cesium-137 concentration in the coolant below 1.0 μCi/cc with 1 percent defective fuel. The resin bed is designed to reduce the concentration of ionic isotopes, particularly cesium, yttrium, and molybdenum by a minimum factor of 10.

The demineralizer is capable of removing ionic impurities for one core cycle with 1 percent of the rated core thermal power being generated by defective fuel rods.

Thermal Regeneration Demineralizers

The function of the thermal regeneration demineralizers is to store the total amount of boron that must be removed from the RCS to accomplish the required dilution during a load cycle in order to compensate for xenon buildup resulting from a decreased power level. Furthermore, the demineralizers must be able to release the previously stored boron to accomplish the required boration of the reactor coolant during the load cycle in order to compensate for a decrease in xenon concentration resulting from an increased power level.

As an additional function, the thermal regeneration (BTRS) demineralizer(s) may be loaded with various types of resin in order to provide supplemental or enhanced cleanup of letdown flow. This may be accomplished by aligning letdown such that flow will be directly to the BTRS demineralizer(s) or by placing these demineralizer(s) in series with either a mixed bed or the cation bed demineralizer.

The thermally reversible ion storage capacity of the resin applies only to borate ions. The capacity of the resin to store other ions is not thermally reversible. Thus, during boration, when borate ions are released by the resin, there is no corresponding release of the ionic fission and corrosion products stored on the resin.

The thermal regeneration demineralizer resin capacity is directly proportional to the solution boron concentration and inversely proportional to the temperature. Further, the differences in capacity as a function of both boron concentration and temperature are reversible. For the 50 to 140°F temperature cycle this reversible capacity varies from the beginning of a core cycle to the end of core life by a factor of about 2.

The demineralizers are of the type that can accept flow in either direction. The flow direction during boron storage is therefore always opposite to that during release. This provides much faster response when the beds are switched from storage to release and vice versa, than would be the case if the demineralizers could accept flow in only one direction.

Temperature instrumentation is provided upstream of the thermal regeneration demineralizers to control the temperature of the process flow. During boron storage operations, it controls the flow through the shell side of the letdown chiller heat exchanger to maintain the process flow at 50°F

as it enters the demineralizers. During boron release operations, it controls the flow through the tube side of the letdown reheat heat exchanger to maintain the process flow at 140°F as it enters the demineralizers. Temperature indication and a high temperature alarm are provided on the main control board.

An additional temperature instrument is provided to protect the demineralizer resins from a high temperature condition. On reaching the high temperature setpoint, an alarm is sounded on the main control board and the letdown flow is diverted to the volume control tank from a point upstream of the mixed bed demineralizers.

Failure of the temperature controls resulting in hot water flow to the demineralizers would result in a release of boron stored on the resin with a resulting increase in reactor coolant boron concentration and increased margin for shutdown. If the temperature of the resin rises significantly above 140°F, the number of ion storage will gradually decrease, thus reducing the capability of the resin to remove boron from the process stream. Degradation of ion removal capability will occur for temperatures of approximately 160°F and above. The extent of the degradation and rate at which it will occur depend upon the temperature experienced by the resin and the length of time that the resin experiences this elevated temperature.

Failure of the temperature control system resulting in cold water flow to the demineralizers would result in storage of boron on the resin and reduction of the reactor coolant boron concentration. The amount of reduction in reactor coolant boron concentration is limited by the capacity of the resin to remove boron from the water. As the boron concentration is reduced, the control rods would be driven into the core to maintain power level. If the rods were to reach the rod insertion limit setpoint, an alarm would be actuated informing the operator that the shutdown margin should be verified within limits or to take actions to restore the shutdown margin in accordance with Technical Specifications.

Reactor Coolant Filter

The reactor coolant filter is located in the letdown line upstream of the volume control tank. The filter collects resin fines and particulates from the letdown stream. The nominal flow capacity of the filter is greater than the maximum purification flow rate.

Two local pressure indicators are provided to show the pressures upstream and downstream of the reactor coolant filter and thus provide filter differential pressure.

Seal Water Injection Filters

Two seal water injection filters are located in parallel in a common line to the reactor coolant pump seals; they collect particulate matter that could be harmful to the seal faces. Each filter is sized to accept flow in excess of the normal seal water flow requirements.

A differential pressure indicator monitors the pressure drop across each seal water injection filter and gives local indication with high differential pressure alarm on the main control board.

Seal Water Return Filter

This filter collects particulates from the reactor coolant pump seal water return and from the excess letdown flow. The filter is designed to pass the sum of the excess letdown flow and the maximum design leakage from all reactor coolant pumps.

Two local pressure indicators are provided to show the pressures upstream and downstream of the filter and thus provide differential pressure across the filter.

Boric Acid Filter

The boric acid filter collects particulates from the boric acid solution being pumped from the boric acid tanks by the boric acid transfer pumps. The filter is designed to pass the design flow of two boric acid transfer pumps operating simultaneously.

Local pressure indicators indicate the pressure upstream and downstream of the boric acid filter and thus provide filter differential pressure.

Letdown Orifices

Three letdown orifices are provided to reduce the letdown pressure from reactor conditions and to control the flow of reactor coolant leaving the RCS. The orifices are placed into or out of service by remote operation of their respective isolation valves. One orifice is designed for normal letdown flow with the other two serving as standby. One or both of the standby orifices may be used in parallel with the normally operating orifice for either flow control when the RCS pressure is less than normal or greater letdown flow during maximum purification or heatup. Each orifice consists of an assembly which provides for permanent pressure loss without recovery, and is made of austenitic stainless steel or other adequate corrosion resistant material.

A flow monitor provides indication in the Control Room of the letdown flow rate, and a high alarm to indicate unusually high flow.

A low pressure letdown controller located downstream of the letdown heat exchanger controls the pressure upstream of the letdown heat exchanger to prevent flashing of the letdown liquid. Pressure indication and high pressure alarm are provided on the main control board.

Number 1 Seal Bypass Orifice

An orifice in each reactor coolant pump number 1 seal bypass line is only in service during startup or shutdown when the RCS pressure is low. The bypass flow is necessary to ensure adequate flow for cooling of the pump's lower radial bearing and to limit the temperature rise of the water cooling the number 1 seal. The orifice is constructed of austenitic stainless steel and designed to pass adequate flow for the differential pressure existing at the lowest allowable RCS pressure for reactor coolant pump operation.

Chiller

Because load follow operation is currently not utilized, the Chiller is normally removed from service, the chiller loops having been drained. See Section 9.3.4.1.1.1.

The purpose of the chiller is two fold:

1. To cooldown the process stream during storage of boron on the resin.

2. To maintain an outlet temperature from the Boron Thermal Regeneration System at or below 115°F during release of boron.

Valves

Where pressure and temperature conditions permit, diaphragm type valves are used to essentially eliminate leakage to the atmosphere. All packed valves which are larger than 2 inches and which are designated for radioactive services are provided with a stuffing box and lantern leakoff connections. All control (modulating) and three-way valves are either provided with stuffing box and leakoff connections or are totally enclosed. Leakage to the atmosphere is essentially zero for these valves. Basic material of construction is stainless steel for all valves which handle radioactive liquid or boric acid solutions.

Relief valves are provided for lines and components that might be pressurized above design pressure by improper operation or component malfunction.

1. Charging line downstream of regenerative heat exchanger

   If the charging side of the regenerative heat exchanger is isolated while the hot letdown flow continues at its maximum rate, the volumetric expansion of coolant on the charging side of the heat exchanger is relieved to the RCS through a spring loaded check valve.

2. Letdown line downstream of letdown orifices

   The pressure relief valve downstream of the letdown orifices protects the low pressure piping and the letdown heat exchanger from overpressure when the low pressure piping is isolated. The capacity of the relief valve exceeds the maximum flow rate through all letdown orifices. The valve set pressure is equal to the design pressure of the letdown heat exchanger tube side.

3. Letdown line downstream of low pressure letdown valve

   The pressure relief valve downstream of the low pressure letdown valve protects the low pressure piping and equipment from overpressure when this section of the system is isolated. The overpressure may result from leakage through the low pressure letdown valve. The capacity of the relief valve exceeds the maximum flow rate through all letdown orifices. The valve set pressure is equal to the design pressure of the demineralizers.

4. Volume control tank

   The relief valve on the volume control tank permits the tank to be designed for a lower pressure than the upstream equipment. This valve has a capacity greater than the summation of the following items: maximum letdown, normal seal water return, excess letdown and nominal flow from one reactor makeup water pump. The valve set pressure equals the design pressure of the volume control tank.

5. Charging pump suction

   For Unit 1:

   A relief valve on the charging pump suction header relieves pressure that may build up if the suction line isolation valves are closed or if the system is overpressurized. The valve set pressure is equal to the design pressure of the associated piping and equipment.

   For Unit 2:

   A relief valve on the charging pump suction header and relief valves installed on each Unit 2 charging pump suction line relieve pressure that may build up if the suction line isolation valves are closed or if the system is overpressurized. The valve set pressures are equal to the design pressure of the associated piping and equipment.

6. Seal water return line (inside Containment)

   This relief valve is designed to relieve overpressurization in the seal water return piping inside the Containment if the motor operated isolation valve is closed. The valve is designed to relieve the total leakoff flow from the number 1 seals of the reactor coolant pumps plus the design excess letdown flow. The valve is set to relieve at the design pressure of the piping.

7. Seal water return line (charging pumps bypass flow)

   This relief valve protects the seal water heat exchanger and its associated piping from overpressurization. If either of the isolation valves for the heat exchanger are closed and if the bypass line is closed, the piping would be overpressurized by the miniflow from the centrifugal charging pumps. The valve is sized to handle the miniflow from the centrifugal charging pumps. The valve is set to relieve at the design pressure of the heat exchanger.

8. Positive displacement pump discharge

   The pressure relief valve on the positive displacement pump discharge line relieves the rated pumping capacity if the pump is started with the discharge isolation valve closed. The set pressure of the valve is equal to the design pressure of the pump discharge piping.

9. Letdown reheat heat exchanger

   The relief valve is located on the piping leading from the shell side of the heat exchanger. If the shell side were isolated while flow was maintained in the tube side, overpressurization could occur. The valve is set to relieve at the design pressure of the heat exchanger shell side.

   Because load follow operation is currently not utilized, the Letdown Reheat Heat Exchanger tube side is normally isolated and drained. Water that leaks into the drained tube side of the heat exchanger, whether leaking from the shell side into the tube side or leaking past the seats of the isolation valves, is drained into floor drains appropriate for water which potentially contains radioactivity. See Section 9.3.4.1.1.1.

10. Letdown chiller heat exchanger

    The relief valve is located on the piping leading from the shell side of the heat exchanger. If the shell side were isolated while flow was maintained in the tube side, overpressurization could occur. The valve is set to relieve at the design pressure of the heat exchanger shell side.

    Because load follow operation is currently not utilized, the Letdown Chiller Heat Exchanger shell side is normally drained. Water that leaks into the drained shell side of the heat exchanger from the tube side would be directed to the Chiller Surge Tank which has a high level alarm. See Section 9.3.4.1.1.1.

11. Steam line to batching tank

    The relief valve on the steam line to the batching tank protects the low pressure piping and batching tank heating jacket from overpressure when the condensate return line is isolated. The capacity of the relief valve equals the maximum expected steam inlet flow. The set pressure equals the design pressure of the heating jacket.

Piping

All CVCS piping that handles radioactive liquid is austenitic stainless steel. All piping joints and connections are welded, except where flanged connections are required to facilitate equipment removal of maintenance and hydrostatic testing.

9.3.4.1.2.6 System Operation

The following are typical operating methods for this system.

Reactor Startup

Reactor startup is defined as the operations which bring the reactor from cold shutdown to normal operating temperature and pressure.

It is assumed that:

1. Normal residual heat removal is in progress.

2. RCS boron concentration is at the cold shutdown concentration.

3. Reactor Makeup Control System is set to provide makeup at the cold shutdown concentration.

4. RCS is either water solid or drained to minimum level for the purpose of refueling or maintenance. If the RCS is water solid, system pressure is maintained by operation of a charging pump and controlled by the low pressure letdown valve in the letdown line (letdown is achieved via the RHRS).

5. The charging and letdown lines of the CVCS are filled with coolant at the cold shutdown boron concentration. The letdown orifice isolation valve are closed.

If the RCS requires filling and venting:

Two methods are available to fill the RCS. One method is to makeup to the RCS through CVCS via charging and letdown. The Reactor Coolant Pumps are sequentially operated and the system vented until all air is removed. The other method is by establishing a vacuum on the RCS and filling the system with makeup from CVCS with the charging pump suctions aligned preferably to the RWST.

1. One charging pump is started, which provides blended flow from the Reactor Makeup Control System at the cold shutdown boron concentration.

2. The vents on the head of the reactor vessel and pressurizer are opened.

3. The RCS is filled and the vents closed.

The system pressure is raised by using the charging pump and controlled by the low pressure letdown valve. When the system pressure is adequate for operation of the reactor coolant pumps, seal water flow to the pumps, is established and the pumps are operated and vented sequentially until all gases are cleared from the system. Final venting takes place at the pressurizer.

Vacuum Fill:

Filling the Reactor Coolant System is started by establishing a vacuum on the system while controlling the RCS level with adjustment of CVCS pressure control. Once a vacuum has been established the charging pump suctions are aligned and charging flow adjusted. The reduced inventory instrumentation and pressurizer instrumentation are used to monitor the RCS level. The rate of filling the RCS is reduced as the Pressurizer begins to indicate a level increase. When the Pressurizer indicates sufficient level is present, the connections to the RCS are filled. After the connections are filled, the RCS fill is essentially completed, so vacuum is broken. Normal charging is established and the RCS is ready to continue being filled to a solid condition or to establish a pressurizer bubble.

After the filling and venting operations are completed, charging and letdown flows are established. All pressurizer heaters are energized and the reactor coolant pumps are employed to heat up the system. At this point, steam formation in the pressurizer is accomplished by manual control of the charging flow and automatic pressure control of the letdown flow. When the pressurizer water level reaches the no-load programmed setpoint, the pressurizer level control is shifted to control the charging flow to maintain programmed level. The RHRS is then isolated from the RCS and the normal letdown path is established. The pressurizer heaters are now used to increase RCS pressure.

The reactor coolant boron concentration is now reduced either by operating the Reactor Makeup Control System in the “dilute” mode or by operating the Boron Thermal Regeneration System in the boron storage mode and, when the resin beds are saturated, washing off the beds to the Boron Recycle System. The reactor coolant boron concentration is corrected to the point where the control rods may be withdrawn and criticality achieved. Nuclear heatup may then proceed with corresponding manual adjustment of the reactor coolant boron concentration to balance the temperature coefficient effects and maintain the control rods within their operating range. During

heatup, the appropriate combination of letdown orifices is used to provide necessary letdown flow.

Prior to or during the heating process, the CVCS is employed to obtain the correct chemical properties in the RCS. The Reactor Makeup Control System is operated on a continuing basis to ensure correct control rod position. Chemicals are added through the chemical mixing tank as required to control reactor coolant chemistry such as pH and dissolved oxygen content. Hydrogen overpressure is established in the volume control tank to assure the appropriate hydrogen concentration in the reactor coolant.

Power Generation and Hot Standby Operation

1. Base Load

   At a constant power level, the rates of charging and letdown are dictated by the requirements for seal water to the reactor coolant pumps and the normal purification of the RCS. One charging pump is employed and charging flow is controlled automatically from pressurizer level. The only adjustments in boron concentration necessary are those to compensate for core burnup. These adjustments are made at infrequent intervals to maintain the control groups within their allowable limits. Rapid variations in power demand are accommodated automatically by control rod movement. If variations in power level occur, and the new power level is sustained for long periods, some adjustment in boron concentration may be necessary to maintain the control groups within their maneuvering band.

   During normal operation, normal letdown flow is maintained and demineralizer(s) are utilized to provide reactor coolant purification. Reactor coolant samples are taken periodically to check boron concentration, water quality, pH and activity level. The charging flow to the RCS is controlled automatically by the pressurizer level control signal through the discharge header flow control valve or the positive displacement pump speed controller.

2. Load Follow

   Comanche Peak currently does not utilize load follow operation. See Section 9.3.4.1.1.1.

   A power reduction will initially cause a xenon buildup followed by xenon decay to a new, lower equilibrium value. The reverse occurs if the power level increases; initially, the xenon level decreases and then it increases to a new and higher equilibrium value associated with the amount of the power level change.

   The Boron Thermal Regeneration System can be used to vary the reactor coolant boron concentration to compensate for xenon transients occurring when reactor power level is changed. The Reactor Makeup Control System may also be used to vary the boron concentration in the reactor coolant.

   The original design of the BTRS is to control the changes in the reactor coolant concentration to compensate for xenon transients during load follow operation without adding makeup for either boration or dilution. This is not currently utilized at Comanche Peak. The system is used to reduce reactor coolant boron concentration at the end of a

core cycle using BTRS demineralizers loaded with anion resins. As a result, the BTRS chiller and associated components are normally removed from service, and the heating source for the reheat letdown heat exchanger is normally isolated. However, they are available to be used as desired.

The most important intelligence available to the plant operator, enabling him to determine whether dilution or boration of the RCS is necessary, is the position of the control rods. For example, if the control rods are below their desired position, the operator must borate the reactor coolant to bring the rods outward. If, on the other hand, the control rods are above their desired position, the operator must dilute the reactor coolant to bring the rods inward.

During periods of plant loading, the reactor coolant expands as its temperature rises. The pressurizer absorbs this expansion as the level controller raises the level setpoint to the increased level associated with the new power level. The excess coolant due to RCS expansion is letdown and stored in the volume control tank. During this period, the flow through the letdown orifice remains constant and the charging flow is reduced by the pressurizer level control signal, resulting in an increased temperature at the regenerative heat exchanger outlet. The temperature controller downstream from the letdown heat exchanger increases the component cooling water flow to maintain the desired letdown temperature.

During periods of plant unloading, the charging flow is increased to make up for the coolant contraction not accommodated by the programmed reduction in pressurizer level.

3. Hot Standby

If required, for periods of maintenance, or following spurious reactor trips, the reactor can be held subcritical, but with the capability to return to full power within the period of time it takes to withdraw control rods. During this hot shutdown period, temperature is maintained at no-load T_{avg} by initially dumping steam to remove core residual heat, or at later stages, by running reactor coolant pumps to maintain system temperature.

Following shutdown, xenon buildup occurs and increases the degree of shutdown; i.e., initially, with initial xenon concentrations and all control rods inserted, the core is maintained at a minimum of 1 percent $\Delta k/k$ subcritical. The effect of xenon buildup is to increase this value to a maximum of about 4 percent $\Delta k/k$ at about 8 hours following shutdown from equilibrium full power conditions. If hot shutdown is maintained past this point, xenon decay results in a decrease in degree of shutdown. Since the value of the initial xenon concentration is about 3 percent $\Delta k/k$ (assuming that an equilibrium concentration had been reached during operation), boration of the reactor coolant is necessary to counteract the xenon decay and maintain shutdown.

If a rapid recovery is required, dilution of the system may be performed to counteract this xenon buildup. However, after the xenon concentration reaches a peak, boration must be performed to maintain the reactor subcritical as the xenon decays out.

4. Cold Shutdown

Cold shutdown is the operation which takes the reactor from hot shutdown conditions to cold shutdown conditions (reactor is subcritical by at least 1 percent $\Delta k/k$ and $T_{avg} \leq 200°F$).

Before initiating a cold shutdown, the RCS hydrogen concentration is lowered by reducing the volume control tank overpressure, by replacing the volume control tank hydrogen atmosphere with nitrogen, and by continuous purging to the Gaseous Waste Processing System.

Before cooldown and depressurization of the reactor plant is initiated, the reactor coolant boron concentration is increased to the cold shutdown value. After the boration is completed and reactor coolant samples verify that the concentration is correct, the operator resets the Reactor Makeup Control System for leakage makeup and system contraction at the shutdown reactor coolant boron concentration.

Contraction of the coolant during cooldown of the RCS results in actuation of the pressurizer level control to maintain normal pressurizer water level. The charging flow is increased, relative to letdown flow, and results in a decreasing volume control tank level. The volume control tank level controller automatically initiates makeup to maintain the inventory.

After the RHRS is placed in service and the reactor coolant pumps are shutdown, further cooling of the pressurizer liquid is accomplished by charging through the auxiliary spray line. Coincident with plant cooldown, a portion of the reactor coolant flow is diverted from the RHRS to the CVCS for cleanup. Demineralization of ionic radioactive impurities and stripping of fission gases reduce the reactor coolant activity level sufficiently to permit personnel access for refueling or maintenance operations.

9.3.4.1.3 Safety Evaluation

The classification of structures, components and systems is presented in Section 3.2. A further discussion on seismic design categories is given in Section 3.7N. Conformance with NRC General Design Criteria for the plant systems, components and structures are important to safety as presented in Section 3.1. Also Appendix 1A(N) provides a discussion on applicable Regulatory Guides.

9.3.4.1.3.1 Reactivity Control

Any time that the plant is at power, the quantity of boric acid retained and ready for injection always exceeds that quantity required for the normal cold shutdown assuming that the control assembly of greatest worth is in its fully withdrawn position. This quantity always exceeds the quantity of boric acid required to bring the reactor to hot shutdown and to compensate for subsequent xenon decay. An adequate quantity of boric acid is also available in the refueling water storage tank to achieve cold shutdown.

When the reactor is subcritical (i.e., during cold shutdown, hot shutdown, hot standby) the source range nuclear instrumentation continuously monitors the neutron flux to detect any indication of an inadvertent boron dilution transient. Upon the detection of a doubling of the neutron flux during any of the aforementioned modes of operation, an alarm is sounded to alert the operator.

Other alarms in the CVCS system are also available to alert the operator of a potential inadvertent boron dilution transient. (See also Section 15.4.6 for a discussion of the inadvertent boron dilution transient).

The rate of boration, with a single boric acid transfer pump operating, is sufficient to take the reactor from full power operation to 1 percent shutdown in the hot condition, with no rods inserted, in less than 90 minutes. In less than 90 additional minutes, enough boric acid can be injected to compensate for xenon decay, although xenon decay below the equilibrium operating level will not begin until approximately 25 hours after shutdown. Additional boric acid is employed if it is desired to bring the reactor to cold shutdown conditions.

Two separate and independent flow paths are available for reactor coolant boration; i.e., the charging line and the reactor coolant pump seal injection line. A single failure does not result in the inability to borate the RCS.

As described in Section 9.3.4.1.2.1, activation of the reactor coolant pump SHIELD Shutdown Seal with the RCPs running will not result in a reduction in RCP seal leakoff.

If the normal charging line is not available, charging to the RCS is continued via reactor coolant pump seal injection at the rate of approximately 5 gpm per pump. At the charging rate of 20 gpm (5 gpm per reactor coolant pump), approximately 6.0 hours are required to add enough boric acid solution to counteract xenon decay, although xenon decay below the full power equilibrium operating level will not begin until approximately 25 hours after the reactor is shutdown.

A third boration path can be provided, if the normal charging line or reactor coolant pump seal injection paths are not available. The reactor operator can cool down and depressurize the reactor coolant system to a pressure less than the shutoff head of the safety injection pumps without allowing the core to return to a critical condition. Borated water from the Refueling Water Storage Tank could then be delivered to the reactor coolant system via the safety injection pumps.

As backup to the normal boric acid supply, the operator can align the refueling water storage tank outlet to the suction of the charging pumps.

Since inoperability of a single component does not impair ability to meet boron injection requirements, plant operating procedures allow components to be temporarily out of service for repairs. However, with an inoperable component, the ability to tolerate additional component failure is limited. Therefore, operating procedures require immediate action to affect repairs of an inoperable component, restrict permissible repair time, and require demonstration of the operability of the redundant component.

9.3.4.1.3.2 Reactor Coolant Purification

The CVCS is capable of reducing the concentration of ionic isotopes in the purification stream as required in the design basis. This is accomplished during normal operation by passing the letdown flow through one of the mixed bed demineralizers which removes ionic isotopes, except those of cesium, molybdenum and yttrium, with a minimum decontamination factor of 10. Through occasional use of the cation bed demineralizer the concentration of cesium can be maintained below 1.0 c/cc, assuming 1 percent of the rated core thermal power is being produced by fuel with defective cladding. The cation bed demineralizer is capable of passing a

maximum purification letdown flow of 170 gpm, though only a portion of this capacity is normally utilized. Each mixed bed demineralizer is also capable of processing more than the maximum normal purification letdown flow rate. If the operating mixed bed demineralizer's resin has become exhausted, the second demineralizer can be placed in service. During RCS cleanup for outages, the mixed bed demineralizers may be placed in parallel service to allow increased flow and provide a more rapid cleanup. Each demineralizer is capable of being operated for one core cycle with 1 percent defective fuel.

A bypass line and isolation valve also provide additional flexibility for cleanup letdown flow. The line and valve allow letdown flow to be aligned such that the mixed bed demineralizers are bypassed and flow is directly to the cation bed demineralizer for cleanup. Letdown can also be aligned to flow to the BTRS demineralizer(s) for cleanup, either directly or in series with either a mixed bed or the cation bed demineralizer.

A further cleanup feature is provided for use during Residual Heat Removal operations. A RHR letdown flow control valve admits flow into the CVCS downstream of the containment isolation valves.

The maximum temperature that will be allowed for the letdown stream demineralizers is approximately 140°F. If the temperature of the letdown stream approaches this level, the flow will be automatically diverted so as to bypass the demineralizers. If the letdown is not diverted, the only consequence would be a decrease in ion removal capability. Ion removal capability starts to decrease when the temperature of the resin goes above approximately 160°F for anion resin or above approximately 250°F for cation resin. The resins do not lose their exchange capability immediately. Ion exchange still takes place (at a faster rate) when temperature is increased. However, with increasing temperature, the resin loses some of its ion exchange sites along with the ions that are held at the lost sites. The ions lost from the sites may be reexchanged farther down the bed. The number of sites lost is a function of the temperature reached in the bed and of the time the bed remains at the high temperature. Capability for ion exchange will not be lost until a significant portion of the exchange sites are lost from the resin.

There would be no safety problem associated with overheating of the demineralizer resins. The only effect on reactor operating conditions would be the possibility of an increase in the reactor coolant activity level. If the activity level in the reactor coolant were to exceed the limit given in the Technical Specifications, reactor operation would be restricted as required by the Technical Specifications.

9.3.4.1.3.3 Seal Water Injection

Flow to the reactor coolant pump seals is assured by the fact that there are three charging pumps, anyone of which is capable of supplying the normal charging line flow plus the nominal seal water flow.

9.3.4.1.3.4 Hydrostatic Testing of the Reactor Coolant System

The positive displacement pump can pressurize the RCS to its maximum specified hydrostatic test pressure. The pump is capable of producing a hydrostatic test pressure greater than that required.

9.3.4.1.3.5 Leakage Provisions

CVCS components, valves, and piping which see radioactive service are designed to limit leakage to the atmosphere. The following are preventive means which are provided to limit radioactive leakage to the environment.

1. Where pressure and temperature conditions permit, diaphragm type valves are used to essentially eliminate leakage to the atmosphere.

2. All packed valves which are larger than 2 inches and which are designated for radioactive service are provided with a stuffing box and lattern leakoff connections.

3. All control (modulating) and three-way valves are either provided with stuffing box and leakoff connections or are totally enclosed.

4. Welding of all piping joints and connections except where flanged connections are provided to facilitate maintenance and hydrostatic testing.

The volume control tank provides an inferential measurement of leakage from the CVCS as well as the RCS. The amount of leakage can be inferred from the amount of makeup added by the Reactor Makeup Control System.

During normal operation, the hydrogen and fission gases in the volume control tank are purged to the Waste Processing System, to limit the release of radioactive gases through leakage by maintaining the radioactive gas level in the reactor coolant several times lower than the equilibrium level. Also, provided are demineralizers which maintain reactor coolant purity, thus reducing the radioactivity level of the RCS water.

9.3.4.1.3.6 Ability to Meet the Safeguards Function

A failure analysis of the portion of the CVCS which is safety-related (used as part of the Emergency Core Cooling System) is included as part of the Emergency Core Cooling System failure analysis presented in Tables 6.3-5 and 6.3-6.

9.3.4.1.3.7 Heat Tracing

Heat tracing requirements for boric acid solutions depends mainly on the solution concentration. For this plant the concentration of boric acid ranges from 10 ppm to 4 weight percent boric acid. Electrical heat tracing is not required on any CVCS components which contain 4 weight percent boric acid, providing these components are located in a room maintained at 65°F or higher. Redundant temperature alarms are provided to assure room temperature does not go below 65°F. Refer to Section 9.3.4.1.2 for more information.

9.3.4.1.3.8 Abnormal Operation

The CVCS is capable of making up for an RCS leak of approximately 130 gpm using one centrifugal charging pump and still maintaining seal injection flow to the reactor coolant pumps. This also allows for a minimum RCS cooldown contraction. This is accomplished with the letdown isolated.

9.3.4.1.3.9 Failure Mode and Effects Analysis

A failure mode and effects analysis (FMEA) of the Chemical and Volume Control System is provided as Table 9.3-9.

9.3.4.1.4 Tests and Inspections

As part of plant operation, periodic tests, surveillance inspections and instrument calibrations are made to monitor equipment condition and performance. Most components are in use regularly; therefore, assurance of the availability and performance of the systems and equipment is provided by Control Room and/or local indication.

Refer to Chapter 14 for further information.

9.3.4.1.5 Instrumentation Application

Process control instrumentation is provided to acquire data concerning key parameters about the CVCS. The location of the instrumentation is shown on Figure 9.3-10.

The instrumentation furnishes input signals for monitoring and/or alarming purposes. Indications and/or alarms are provided for the following parameters:

1. Temperature.

2. Pressure.

3. Flow.

4. Water level.

The instrumentation also supplies input signals for control purposes. Some specific control functions are:

1. Letdown flow is diverted to the volume control tank upon high temperature indication upstream of the mixed bed demineralizers.

2. Pressure upstream of the letdown heat exchanger is controlled to prevent flashing of the letdown liquid.

3. Charging flow rate is controlled during charging pump operation.

4. Water level is controlled in the volume control tank.

5. Temperature of the boric acid solution in the batching tank is maintained.

6. Reactor makeup is controlled.

7. Temperature of letdown flow to the Boron Thermal Regeneration System is controlled.

8. Temperature of the chilled water flow to the letdown chiller heat exchanger is controlled.

9. Temperature of letdown flow return from the boron thermal regeneration demineralizers is controlled.

9.3.4.2 Boron Recycle System

The Boron Recycle System (BRS) receives and recycles reactor coolant effluent for possible reuse of the boric acid and makeup water. The system decontaminates the effluent by means of demineralization and gas stripping, and uses evaporation to separate and recover the boric acid and makeup water.

9.3.4.2.1 Design Bases

9.3.4.2.1.1 Collection Requirements

The BRS collects and processes effluent which may be reused as makeup to the RCS. For the most part, this effluent is the deaerated, tritiated, borated, and radioactive water from the letdown and process drains.

The BRS was designed to collect, via the letdown line in the CVCS, the excess reactor coolant that results from the following plant operations during one annual core cycle. Because it is operated in batches, the effluent from longer core cycles can be processed. The inputs used to design the system are:

1. Dilution for core burnup from approximately 1200 ppm boron at the beginning of a core cycle to approximately 10 ppm near the end of the core cycle.

2. Hot shutdowns and startups. Four hot shutdowns are assumed to take place during an annual core cycle.

3. Cold shutdowns and startups. Three cold shutdowns are assumed to take place during an annual core cycle.

4. Refueling shutdown and startup.

The BRS may also collect water from the following sources:

1. Reactor coolant drain tank (Liquid Waste Processing System) - collects leakoff type drains from equipment inside the Containment.

2. Volume control tank and charging pump suction pressure reliefs (CVCS) and safety injection pumps pressure reliefs (Emergency Core Cooling System).

3. Boric acid blender (CVCS) - provides storage of boric acid if a boric acid tank must be emptied for maintenance. The boric acid solution is stored in a recycle holdup tank after first being diluted with reactor makeup water by the blender to prevent precipitation of the boric acid in the unheated recycle holdup tank.

4. Accumulators (Safety Injection System) - collects effluent resulting from leak testing of accumulator check valves.

5. Liquid Waste Processing System - provides capability for using the recycle evaporator as a waste evaporator and vice versa.

6. Spent fuel pool pumps (Spent Fuel Pool Cooling and Cleanup System) - provides a means of storing the fuel transfer canal water in case maintenance is required on the transfer equipment.

7. Valve leakoffs and equipment drains.

9.3.4.2.1.2 Capacity Requirement

The BRS is designed to process the total volume of water collected during a core cycle as well as short term surges. The design surge is that produced by a cold shutdown and subsequent startup during the latter part of a core cycle or by a refueling shutdown and startup.

9.3.4.2.1.3 Purification Requirement

The water collected by the BRS contains dissolved gases, boric acid, and suspended solids. Based on reactor operations with 1 percent of the rated core thermal power being generated by fuel elements with defective cladding, the BRS is designed to provide sufficient cleanup of the water to satisfy the chemistry requirements of the recycled reactor makeup water and 4 weight percent boric acid solution.

The maximum radioactivity concentration buildup in the BRS components is based on operation of the reactor at its engineered safeguards design rating with defective fuel rods generating 1 percent of the core thermal power. For each component, the shielding design considers the maximum buildup on an isotopic basis including only those isotopes which are present in significant amounts. Filtration, demineralization, and evaporation are the means by which the activity concentrations may be controlled.

9.3.4.2.2 System Description

The BRS is shown in Figure 9.3-11 (piping and instrumentation diagram). The codes and standards to which the individual components of the BRS are designed are listed in Section 3.2. When water is directed to the BRS, the flow passes first through the recycle evaporator feed demineralizers and filters and then into the recycle holdup tanks. The recycle evaporator feed pumps can be used to transfer liquid from one recycle holdup tank to the other if desired. When sufficient water is accumulated to warrant evaporator operation, the recycle evaporator feed pumps take suction from the selected recycle holdup tank. The fluid then flows through the recycle evaporator. Here, dissolved gases (i.e., hydrogen, fission gases and other gases) are removed in the stripping column before the liquid enters the evaporator shell. These gases are directed to the evaporator vent condenser and then to the plant ventilation system. An alternate flow path to the Gaseous Waste Processing System is available.

During evaporator operation, distillate from the evaporator flows to the Waste Monitor Tanks for discharge. Also located in this flow path, are the recycle evaporator condensate demineralizer and the recycle evaporator condensate filter.

The evaporator concentrates the boric acid solution until approximately 4 weight percent solution is obtained. The accumulated batch is normally transferred directly to the boric acid tanks in the

CVCS through the recycle evaporator concentrates filter. Before transferring the boric acid from the evaporator to the boric acid tanks, it is analyzed, and, if it does not meet the required chemical standards, it can be diverted back to the recycle holdup tanks for reprocessing or to the Liquid Waste Processing System for disposal.

Connections are provided so that, if necessary, the recycle evaporator can be used as a waste evaporator (and vice versa).

All portions of the BRS which contain concentrated boric acid solution are located within a heated area in order to maintain solution temperature at ≥65°F. This is 10°F above the solubility limit for the nominal 4 weight percent boric acid solution. If a portion of the system which normally contains concentrated boric acid solution is not located in a heated area, it must be provided with some other means (e.g., heat tracing) to maintain solution temperature ≥65°F.

The BRS can also provide a direct source of RHUT water to fill the RWSTs. A BR cross tie downstream of the Recycle Evaporator Feed Pumps provides a flowpath through SF piping, Safety Injection valves 1 & 2-8800A & B, and on to the RWSTs through the ECCS suction headers. This flowpath provides a quicker transfer of water than the flowpath through process equipment and the Spent Fuel Pools/Transfer Canal. Before transferring the RHUT water, it is sampled and analyzed to ensure it meets specified chemical standards. This will prevent the RHUT water from diluting the RWSTs' boron concentration or contaminate the RWSTs. See Figures 9.1-13 and 9.3-11.

9.3.4.2.2.1 Component Descriptions

A summary of principal component data is given in Table 9.3-8 and the code requirements are given in Section 3.2.

Recycle Evaporator Feed Pumps

Two centrifugal, canned pumps supply feed to the recycle evaporator package from the recycle holdup tanks. The pumps can also be used to recirculate water from the recycle holdup tanks through the recycle evaporator feed demineralizers for cleanup if desired. An auxiliary discharge connection is provided to return water to the transfer canal from the recycle holdup tanks, if those tanks were used for storage of transfer canal water during refueling equipment maintenance. An auxiliary discharge connection is provided to supply water to the suction of the charging pumps (CVCS) for refilling the RCS after loop or system drain. An auxiliary discharge connection is provided to allow processing of the recycle holdup tank water to the Filter Demineralizer System.

The Recycle Evap Feed Pumps can also be used to transfer RHUT water directly to the RWSTs.

Recycle Holdup Tanks

Two recycle holdup tanks provide storage of radioactive fluid which is discharged from the RCS during startup, shutdown, load changes and boron dilution. The tanks are constructed of austenitic stainless steel.

Each tank has a diaphragm which prevents air from dissolving in the water and prevents the hydrogen and fission gases in the water from mixing with the air. The volume in the tank above

the diaphragm is continuously ventilated with building supply air, and the volume of gas below the diaphragm is intermittently vented to the Waste Processing System (Gas).

In addition to the collection of effluent as described above, the recycle holdup tanks provide the following functions:

1. Serve as a head tank for the recycle evaporator feed pumps.

2. Provide holdup for a RCS drain to the centerline of the reactor vessel nozzles, including the pressurizer and steam generators.

3. Provide storage for transfer canal water during refueling equipment maintenance.

4. Collect discharge from relief valves in borated water systems outside of containment.

5. Provide storage of water for the RWSTs.

Recycle Evaporator Reagent Tank

This tank provides a means of adding chemicals to the evaporator for process treatment.

Recycle Evaporator Feed Demineralizers

Two flushable, mixed bed demineralizers remove fission products from the fluid directed to the recycle holdup tanks. The demineralizers also provide a means of cleaning the recycle holdup tank contents via recirculation.

Recycle Evaporator Condensate Demineralizer

A flushable demineralizer is provided as a polishing demineralizer for distillate from the recycle evaporator. Although the bed may become saturated with boron at the normally low concentration (<10 ppm) leaving the evaporator, it will still remove boron if the concentration increases because of an evaporator upset. The demineralizer also provides a means of cleanup of the Rector Makeup Water Storage Tank contents.

Recycle Evaporator Feed Filter

This filter collects resin fines and particulates from the fluid entering the recycle holdup tanks.

Recycle Evaporator Condensate Filter

This filter collects resin fines and particulate from the boric acid evaporator condensate stream.

Recycle Evaporator Concentrates Filter

This filter removes particulates from the evaporator concentrate as it leaves the evaporator.

Recycle Evaporator

The recycle evaporator package processes dilute boric acid and produces distillate and approximately 4 weight percent boric acid stripped of hydrogen, radioactive gases, and other dissolved gases.

A boric acid solution is fed from the recycle holdup tanks to the evaporator by the recycle evaporator feed pumps. The feed first passes through a heat exchanger where auxiliary steam raises its temperature. The feed then passes into the top of the stripping column. Gases are stripped off as the feed passes over the packing in the tower in contra flow to stripping steam from the evaporator. After stripping, the feed is introduced into the evaporator as makeup. The vapors leaving the boiling pool are stripped of entrained liquid and volatile boron carryover. Pure vapors are then condensed in the condenser section and pumped from the system. When the desired concentration is reached in the boiling pool, the concentrates are pumped from the system.

Radioactive gases, hydrogen, and other non-condensables are discharged from the system into the plant ventilation system.

The recycle and waste evaporators are interconnected so that they can serve as standbys for each other under abnormal conditions.

Recycle Holdup Tank Vent Ejector

The ejector is designed to pull gases from under the diaphragm in a recycle holdup tank and deliver them to the Gaseous Waste Processing System. Nitrogen, provided by the standby waste gas compressor, provides the motive force.

9.3.4.2.2.2 System Operation

The BRS is manually operated with the exception of a few automatic protection functions. These automatic functions protect the recycle evaporator feed demineralizers from a high inlet temperature and a high differential pressure, prevent a high vacuum from being drawn on the recycle holdup tank diaphragm, protect the recycle evaporator feed pumps from low net positive suction head, and prevent high activity recycle evaporator condensate from being sent to the Reactor Makeup Water Storage Tank. The BRS has sufficient instrumentation readouts and alarms to provide the operator information to assure proper system operation.

Evaporation

Water is accumulated in the recycle holdup tank until sufficient quantity exists to warrant an evaporator startup. Prior to startup of the evaporator, the contents of the recycle holdup tank are analyzed and, if necessary, are recirculated through the recycle evaporator feed demineralizers and filter. The flow can be discharged back to the recycle holdup tank or to the evaporator. The evaporator is then operated to produce a batch of 4 weight percent boric acid.

During the operation of the evaporator, condensate may be routed to a Waste Monitor Tank or may be continuously sent to the Recycle Holdup Tank via the recycle evaporator condensate demineralizer.

After a batch of boric acid is concentrated to 4 weight percent, it is analyzed to ensure that it is within specifications for reuse. If it meets the specifications, it is pumped to the boric acid tanks. If it does not, it can be returned to the recycle holdup tank via the recycle evaporator feed demineralizers for reevaporation or, if desired, the concentrated boric acid can be sent to the Liquid Waste Processing System for disposal.

Storage of Water for the RWSTs

The BRS can be used to store water for the RWSTs. A BR cross tie can transfer RHUT water through a small portion of the SF system, through the 1 & 2-8800A & B valves, and on to either RWST. The Recycle Evap Feed Pumps provide the pumping force for this transfer which allows the bypassing of BRS process equipment and the SF pools/Transfer Canal. This results in a more efficient transfer of the water. The water in the RHUTs must be sampled to meet appropriate chemistry requirements prior to the transfer. See Figures 9.1-13 and 9.3-11.

Recycle Holdup Tank Venting

Because hydrogen is dissolved in the reactor coolant at approximately 1 atmosphere overpressure, a portion of the hydrogen along with fission gases will come out of solution in the recycle holdup tank under the diaphragm. The hydrogen and fission gases are vented to the Gaseous Waste Processing System as required. The total integrated flow from the letdown line and the reactor coolant drain tank to the recycle holdup tanks is monitored. An alarm indicates when a sufficient amount of water has passed to the recycle holdup tanks to require venting of the accumulated gases. A recycle holdup tank will be vented after a loop drain or a fuel storage area drain as per the plant operating instructions.

When venting of either recycle holdup tank is required, the following steps are observed:

1. All inlets to the recycle holdup tank are closed.

2. The recycle holdup tank is emptied of water to a low level by either processing with the evaporator or transferring to the other recycle holdup tank.

3. The standby waste gas compressor is lined up to the recycle holdup tank vent ejector. Normally, the standby compressor will feed the other waste gas compressor which is lined up to a catalytic recombiner and a high activity gas decay tank. However, in the event of a recycle holdup tank diaphragm leak or after a RCS loop drain or spent fuel pool drain, a shutdown gas decay tank is used instead of a high activity gas decay tank. This prevents accumulation of air (i.e., nitrogen) in the high activity gas decay tanks.

4. The standby gas compressor is started up and the vent from the holdup tank is opened. The vent flow is throttled to approximately 1 scfm. At this time, a sample of the vent gases can be taken to check the composition.

5. When the gases have been vented from the recycle holdup tank, the pressure in the vent line decreases, which automatically trips the recycle holdup tank vent isolation valve closed.

6. After the vent isolation valve closes, the manual vent valve is closed, the gas compressor is shut down, and the recycle holdup tank inlets and outlets are lined up for normal use.

Maintenance Drains

When large amounts of water must be drained from the RCS or the fuel storage area (or fuel transfer canal) to the BRS, a recycle holdup tank is drained of water and vented to the Gaseous Waste Processing System.

The water can then be stored in this tank until maintenance is completed and, after checking the chemistry, returned. After returning the water, the recycle holdup tank is again vented to the Gaseous Waste Processing System.

Reactor Makeup Water Cleanup

If the reactor makeup water requires purification, it can be recirculated through the recycle evaporator condensate demineralizer until its chemistry is within specifications. If further processing is necessary, water from the Reactor Makeup Water Storage Tank can be directed through the recycle evaporator condensate demineralizer and into the recycle holdup tanks for re-evaporation.

Waste Processing with the Recycle Evaporator

Connections are provided so that the recycle evaporator can be used as a waste evaporator (and vice versa). These consist of a feed connection from the waste holdup tank, a condensate connection to the waste condensate tank, a concentrate connection to the Plant Effluent Tank Room, and an evaporator vent connection to the plant vent. Thus, the recycle evaporator can be used to perform the function of the waste evaporator except that, since heat tracing is not provided for the recycle evaporator, the boric acid would be concentrated to no more than 4 weight percent rather than to 12 weight percent, as is permitted for the waste evaporator.

After using the recycle evaporator to process water from the Liquid Waste Processing System, it is thoroughly rinsed out. During initial recycle processing, the condensate is directed to the waste condensate tank for analysis prior to recycling to the reactor makeup water storage tank. Depending upon the purity of the evaporator bottoms, the concentrated boric acid can be recycled to the boric acid tanks, returned to the recycle holdup tanks for reprocessing, or transferred to the Liquid Waste Processing System for disposal.

9.3.4.2.3 Safety Evaluation

Malfunctions in the BRS do not affect the safety of station operations. The BRS is designed to tolerate equipment faults with criterial functions being met by the use of the two pieces of equipment so that the failure of one will, at most, reduce the capacity of the BRS but not completely shut it down. Because of the large surge capacity of the BRS, the non-availability of the recycle evaporator can be tolerated for periods of time. Also, backup is provided by the waste evaporator.

9.3.4.2.4 Tests and Inspections

The BRS is in intermittent use throughout normal reactor operation. Periodic visual inspection and preventive maintenance are conducted using normal industry practice. Refer to Chapter 14 for further information.

9.3.4.2.5 Instrumentation Application

The instrumentation available for the BRS is discussed below. Alarms are provided as noted. There is also a common alarm on the main control board which indicates any alarms on the BRS panel.

9.3.4.2.5.1 Temperature

Temperature at Inlet to the Recycle Evaporator Feed Demineralizer

Instrumentation is provided to measure the temperature of the inlet flow to the recycle evaporator feed demineralizers and to control a three-way bypass valve. If the inlet temperature becomes too high, the instrumentation aligns the valve to bypass the demineralizers. Local temperature indication and a high temperature alarm on the BRS panel are provided by this instrumentation.

9.3.4.2.5.2 Pressure

Pressure Differential Across the Recycle Evaporator Feed Demineralizers

Instrumentation is provided to measure the pressure differential across the recycle evaporator feed demineralizers and to control the same three-way valve as discussed above (but independently of the temperature control). If the pressure drop through the demineralizers is too high, this instrumentation aligns the valve to divert flow directly to the recycle evaporator feed filters. Local pressure differential indication and a high alarm on the BRS panel are provided by this instrumentation.

Pressure at Inlet and Outlet of Filters

Instrumentation is provided to measure the pressure differential across each recycle evaporator feed filter, the recycle evaporator concentrates filter, and the recycle evaporator condensate filter. Local indication of the pressure in each inlet and outlet line is provided.

Pressure at Discharge of Recycle Evaporator Feed Pump

Instrumentation is provided to measure and give local indication of the discharge pressure of each recycle evaporator feed pump.

Pressure in Vent Line from the Recycle Holdup Tanks

Instrumentation is provided to measure the pressure in the recycle holdup tank vent line and to control a shutoff valve in the vent line. This instrumentation is used during holdup tank venting operations. When the pressure in this line becomes too low, the valve will be automatically closed to protect the holdup tank diaphragm from an excessive differential pressure across it. Local pressure indication and low pressure alarm on the BRS panel are provided.

9.3.4.2.5.3 Flow

Boron Recycle System Flow Totalizer

Instrumentation is provided to monitor the total integrated flow received by the BRS from the letdown line (CVCS) and the reactor coolant drain tank (Waste Processing System). Indication of integrated flow and high alarm are given on the BRS panel. Actuation of the high alarm indicates that the integrated flow has reached a value at which the volume of gases (hydrogen and fission gases) which have come out of solution should be vented from the recycle holdup tanks.

Flow in Vent Line from Recycle Holdup Tanks

Instrumentation is provided which gives local indication of the recycle holdup tank vent purge flow.

Flow in Feed Line to the Recycle Evaporator

Instrumentation is provided which gives local indication of recycle evaporator feed flow.

9.3.4.2.5.4 Level

Level in the Recycle Holdup Tanks

Instrumentation is provided to give an indication of the water level of each recycle holdup tank. Both high level and low level alarms are provided by this instrumentation at the BRS panel. The recycle evaporator feed pumps are stopped on the holdup tank low level signal.

9.3.4.2.5.5 Deleted

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 2, Design Bases for Protection Against Natural Phenomena.

2. 10 CFR Part 50, Appendix A, General Design Criterion 4, Environmental and Missile Design Bases.

3. 10 CFR Part 50, Appendix A, General Design Criterion 5, Sharing of Structures, System, and Components.

4. 10 CFR Part 50, Appendix A, General Design Criterion 56, Primary Containment Isolation.

5. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam-, and Radioactive-Waste- Containing Components of Nuclear Power Plants, U.S. Nuclear Regulatory Commission.

6. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 1, U.S. Nuclear Regulatory Commission.

7. Branch Technical Position APCSB 3-1, Protection Against Postulated Failures in Fluid Systems Outside Containment.

8. Branch Technical Position MEB 3-1, Postulated Break and Leakage Locations in Fluid System Piping Outside Containment.

9. ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973.

10. NRC Regulatory Guide 8.8, Information Relevant to Maintaining Occupational Radiation Exposure As Low As Reasonably Achievable (Nuclear Power), Revision 1, Sept. 1975, U. S. Nuclear Regulatory Commission

TABLE 9.3-1
INSTRUMENT AIR SYSTEM COMPONENT CHARACTERISTICS

(Sheet 1 of 2)

A. DELETED

B. DELETED

C. ROTARY AIR COMPRESSOR PACKAGES AND ASSOCIATED EQUIPMENT

1. Compressor Packages

Number of Units Required	Four rotary screw and two rotary lobe, two stage, oil-free, water cooled Compressors
Safety Class	NNS
Operating Mode	Lead/Backup
Capacity at Design Inlet Conditions (14.5 psia, 122°F, 100% Relative Humidity)	689 ACFM
Capacity at Standard Inlet Conditions (14.7 psia, 60°F, 36% Relative Humidity).	650 SCFM
Motor, hp	200
Compressor Speed, rpm	3580
Intercooler, Aftercooler, & Oil Cooler Exit Cooling Water Temperature based on 95°F Inlet, °F	125 122 (Common and Unit 2)
Total Cooling Water Requirement, gpm	30 (Unit 1) 40 (Common and Unit 2)

2. Instrument Air Receivers

Number of Receivers Required	3 Vertical Cylindrical Tank and 3 Horizontal Cylindrical Tank
Operating Mode	1 Receiver Dedicated to each Air Compressor Package
Design Pressure, psig	125 165 (Common X-01)
Design Temperature, °F	150 (Vertical) 250 (Horizontal) 450 (Horizontal X-01)

TABLE 9.3-1
INSTRUMENT AIR SYSTEM COMPONENT CHARACTERISTICS

(Sheet 2 of 2)

	Volume, ft^3	200 (Vertical) 222 (Horizontal) 204 (Horizontal X-01)
3.	Instrument Air Dryer Packages	
	Number of Dryers Required	6 Dual Tower, Internal Heater, Regenerative Type
	Safety Class	NNS
	Operating Mode	1 Dryer Package Dedicated to each Air Compressor Package
	Capacity, SCFM	1000

Notes:

Aftercoolers are integral to each air compressor package.

Pre & Afterfilters are integral to each dryer package.

Pre & Afterfilters are integral to each dryer.

TABLE 9.3-2
SERVICE AIR SYSTEM COMPONENT CHARACTERISTICS

1. Service Air Compressor Package

Number of units required	1, Single stage, oil injected air cooled, rotary screw compressor
Safety class	NNS
Operating mode	1, 100-percent-capacity unit
Capacity, ft^3/min (29.07 in.Hg abs, 104^oF, and 70 percent relative humidity)	1300, full load
Motor size, hp	250
Compressor, bhp	270
Volts/Hz/phase	460/60/3

2. Service Air Receiver

Number of units required cylindrical	1, horizontal, receiver
Safety class	NNS
Operating mode	1, 100-percent-capacity unit
Maximum operating pressure, psig	107
Ambient temperature, ^oF	125

TABLE 9.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AIR-OPERATED SAFETY-RELATED VALVES

(Sheet 1 of 4)

Valve Designation	System	Failure Mode Note 1	Effect On Plant Safety	Remarks
Power-operated relief Valve	MS	Closed	No effect	Accumulators are provided for SGTR mitigation or to permit remote manual control for up to 4 hours to cooldown from hot standby at 50°F hour, followed by manual operator control if required to cooldown RHRS conditions.
Main Steam Isolation	MS	None	No effect	The MSIVs fail closed on loss of valves hydraulic fluid (See Section 10.3.2.3). The hydraulic pumps are air driven.
Hydrazine and ammonia hydroxide feed isolation valves	CFS	Closed	No effect	
Motor-driven AF pumps recirculation valve	AF	Open	No effect	Accumulators provided to permit automatic control for a 30 minute period. Manual values may be used to throttle or isolate recirculation flow locally as required.
Motor-driven AF pumps discharge valves	AF	Open	No effect	Required AF flow ensured to the steam generators. Manual operators provided at valve locations. Accumulators provided to permit remote manual isolation control for a 30-min period.
Turbine-driven AF pumps discharge valves	AF	Open	No effect	Required AF flow ensured to the steam generators. Manual operators provided at valve locations. Accumulators provided to permit remote manual isolation for a 30-min period.
Component cooling water pumps recirculation valves	CCWS	Closed	No effect	Maximum flow to equipment requiring cooling

TABLE 9.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AIR-OPERATED SAFETY-RELATED VALVES

(Sheet 2 of 4)

Valve Designation	System	Failure Mode Note 1	Effect On Plant Safety	Remarks
Demineralized water supply to CCWS surge tank valve	CCWS	Open	No effect	
Reactor makeup water to CCWS surge tank valve	CCWS	Open	No effect	Maximum makeup water supply ensured to the surge tank. Valves have manual operators.
Safety Chiller Condenser CCWS Regulating Valves	CCWS	Open	No effect	Valves are backed up by Accumulator air supply for a minimum of 30 minutes. They are also equipped with handwheels for manual action.
Pressurizer spray valves	RCS	Closed	No effect	
RHR heat exchanger discharge valves	RHR	Open	No effect	Maximum flow ensured through the RHR heat exchangers.
Containment isolation	Various	Closed	No effect	Valves
Charging pump suction high point vent	CS	Closed	No effect	Fails closed on loss of power, generation of a VCT Lo-Lo water level, generation of a source range neutron flux doubling signal and the generation of a safety injection “S” signal.
CCWS surge tank level control valve	CCWS	Closed	No effect	Prevent flooding of CCWS surge tank. Handwheels provided for manual control.
Process Sampling System Control and Isolation valves	PSS	Closed	No effect	

TABLE 9.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AIR-OPERATED SAFETY-RELATED VALVES

(Sheet 3 of 4)

Valve Designation	System	Failure Mode Note 1	Effect On Plant Safety	Remarks
MS to AF pump turbine isolation valve	MS	Open	No effect	Maximum steam flow to AF pump turbine. Accumulators provided to permit remote manual isolation for a seven hour period.
Blowdown Control valves	MS	Open	No effect	Valves upstream of the isolation valves
Blowdown isolation valves	MS	Closed	No effect	Valves of the Containment Isolation System.
Steam Generator sample control valves	MS	Closed	No effect	Used to select upper or lower sample
Steam generator sample isolation valves	MS	Closed	No effect	Valves of the Containment Isolation System.
Main steam drain pot isolation valve	MS	Closed	No effect	Valves of the Containment Isolation System.
Demineralizer makeup to condensate storage tank isolation valve	AF	Closed	No effect	Makeup not required during an accident
Letdown HX outlet CCWS return control valve	CCWS	Open	No effect	Valves of the CCWS Non-safeguard Loop
Spent fuel pool HX CCWS return control valve	CCWS	Open	No effect	Maximum cooling to spent fuel pool HX

TABLE 9.3-3
FAILURE MODE AND EFFECT ANALYSIS FOR AIR-OPERATED SAFETY-RELATED VALVES

(Sheet 4 of 4)

Valve Designation	System	Failure Mode Note 1	Effect On Plant Safety	Remarks
Ventilation chillers CCWS supply and discharge control valves	CCWS	Closed	No effect	Valves of the CCWS Non-safeguard Loop
Excess letdown CCWS return isolation valves	CCWS	Closed	No effect	Valves of the CCWS Non-safeguard Loop
CCWS supply to excess Letdown and RCS drain tank	CCWS	Closed	No effect	Valves of the CCWS Non-safeguard Loop
Chemical additive tank	CSS	Open	No effect	Maximum amount of chemicals added during injection phase of control valve discharge safety injection
Reactor makeup water to safety chilled water surge tank isolation valves	CWS	Open	No effect	
Demineralized water to safety chilled water surge tank isolation valves	CWS	Closed	No effect	
Safety chilled water surge tank level control valves	CWS	Closed	No effect	Prevents flooding of surge tank. Has manual bypass

NOTES:

1: Failure Mode on loss of air supply

AFS - Auxiliary Feedwater System CCWS - Component Cooling Water System CFS - Chemical Feed System CSS - Containment Spray System CWS - Chilled Water System HX - Heat Exchanger MSSS - Main Steam Supply System PSS - Primary Sampling System SSWS - Station Service Water System

TABLE 9.3-4
PROCESS SAMPLING SYSTEM PARAMETERS

(Sheet 1 of 3)

Sample Point Description	Sampling Conditions		Method of Sampling[a]			Sample Flow (gpm)		Parameter Measured[b]	Process Fluid	Sample Purpose	Application
	Design Temp. (F)	Design Pressure (Psig)	Grab Sample	Sample Vessel	Continuous Analyzers	Maximum (Purge)	Minimum (Purge/Grab Sample)				
SIS accumulator tanks	300[c]	700	X	-	-	1.0	0.264/0.15	B Cl	Reactor coolant	Detect deviation from specified boron and halide concentrations	Guide operation to prevent chloride stress corrosion
RCS pressurizer steam	650	2485	X	X	-	1.0	0.6/0.15	-	Saturated steam without boric gas	Detect accumulation of gross fission gas activity before shutdown	Schedule venting for controlled disposal
RCS pressurizer liquid space	650	2485	X	X	-	1.0	0.75/0.15	B SC pH C1	Reactor coolant	Detect deviation from specified values for conductivity, pH, total alkali, halide	Insure effective corrosion control prevent chloride, stress corrosion
RCS hot legs	650	2485	X	X	-	0.85	0.75/0.15	B SC pH C1	Reactor coolant	Detect deviations in reactor coolant chemistry and detect accumulation of fission product activity	Insure effective corrosion control maintain sufficient, hydrogen control, and guide system operation.
RHR (RHR pump bypass)	400	600	X	X	-	1.0	0.75/0.15	-	-	-	-
Upstream from thermal regeneration demineralizer	250[c]	300	X	-	-	1.0	-	B	-	Detect boron concentration	Monitor and evaluate demineralizer effectiveness

TABLE 9.3-4
PROCESS SAMPLING SYSTEM PARAMETERS

(Sheet 2 of 3)

Sample Point Description	Sampling Conditions		Method of Sampling[a]			Sample Flow (gpm)		Parameter Measured[b]	Process Fluid	Sample Purpose	Application
	Design Temp. (F)	Design Pressure (Psig)	Grab Sample	Sample Vessel	Continuous Analyzers	Maximum (Purge)	Minimum (Purge/Grab Sample)				
Downstream of thermal regeneration demineralizer	250[c]	300	X	-	-	1.0	-	B	-	Detect boron concentration	Monitor and evaluate demineralizer effectiveness
Volume control tank gas phase	250[c]	300	X	-	-	-	-	-	Hydrogen cover gas	Detect accumulation of gross fission production gas activity	Guide to venting and purging to control gaseous activity in reactor coolant
CVCS letdown line upstream from purification demineralizer	250[c]	300	X	-	-	1.0	-	Various	Reactor coolant	Reactor coolant chemistry	Backup to reactor coolant hot-leg
	250[c]	300	-	-	X	Varies	Varies	H_2/O_2	Reactor coolant	Reactor coolant chemistry	Backup to reactor coolant hot-leg
CVCS letdown line downstream of purification demineralizer	250[c]	300	X	-	-	1.0	-	C1	Reactor coolant	Provide reactor coolant activity reduction	Monitor decontamination factor
Steam generator blowdown (steam generator sample taps)	600	1300	X	-	X	1.0	0.4/0.15	CC SC Na R pH	Secondary system (steam generator)	Detect deviation from specified chemical technical specification	Guide to regulation of blowdown rate, and steam generator primary-to-secondary side leakage

TABLE 9.3-4
PROCESS SAMPLING SYSTEM PARAMETERS

(Sheet 3 of 3)

Sample Point Description	Sampling Conditions		Method of Sampling[a]			Sample Flow (gpm)		Parameter Measured[b]	Process Fluid	Sample Purpose	Application
	Design Temp. (F)	Design Pressure (Psig)	Grab Sample	Sample Vessel	Continuous Analyzers	Maximum (Purge)	Minimum (Purge/Grab Sample)				
Spent fuel pool demineralizers (only Unit 1)	150	150	X	-	-	1.0	0.75/0.15	B	Spent fuel pool fluid	Detect deviation from water chemistry specification	Determine resin exhaustion and performance
Downstream blowdown cation demineralizers	130	300	X	-	X	1.0	- /0.05	SC	Processed blowdown	Detect operational problems in blowdown processing system	Determine resin exhaustion and performance
Downstream blowdown mixed-bed demineralizers	130	300	X	-	X	1.0	- /0.05	SC	Processed blowdown	Detect operational problems in the blowdown cleanup system	Determine resin exhaustion and performance

a) X denotes that this method of sampling is used.

b) B = boric acid/boron measurement
CC = cation conductivity measurement
SC = specific conductivity measurement
pH = pH measurement
R = online radioactivity measurement
Na = sodium concentration measurement
Cl = chloride concentration measurement
H_2 = hydrogen
O_2 = oxygen

c) Operating temperature >120°F.

TABLE 9.3-5
POTENTIALLY RADIOACTIVE LIQUID STORAGE TANKS

(Sheet 1 of 4)

Tank	System	Number Supplied (Two Units)	Capacity	Location and Elevation	Leak Detection and/ or Overflow Detection	Overflow or Relief Valve Discharge Receiving Point	Remarks
Pressurizer relief tank	RCS	2	1800 ft^3	Containment, 822 ft 9 in.	Level indication (CR) high and low alarms (CR)	Via Containment sump to FDT (Alternate - WHT)	Relief through rupture discs
Chiller surge tank fluid	CVCS	2	500 gal	Auxiliary Building, 852 ft 6 in.	Level indication (CR) High alarm	To component cooling water drain tank	Possibility of radio-active entering through a letdown chiller heat exchanger tube leak
Volume control tank	CVCS	2	400 ft^3	Safeguards Building, 831 ft 6 in.	Level indication (CR) high-high, high, low and low-low alarms (CR)	Relief to RHT	_____
Accumulator	SIS	8	1350 ft^3	Containment, 832 ft 6 in. and 842 ft 0 in.	Level indication (CR) high and low alarms (CR)	_____	Assumes RC backflow through check valves
Floor drain tank	WPS	3	a. 10,000 gal b. 10,000 gal c. 30,000 gal	a. Safeguards Building Unit No. 1 b. Safeguards Building Unit No. 2 c. Auxiliary Building	Level indication (LWPP) high and low alarms (CR, LWPP)	Overflow to floor	Tank located within water-tight compartment; NC valve in floor drain line for FDT's 1 & 2 FDT-3 Floods to FDT #1 via FD Sump #4
Laundry and hot shower tank	WPS	1	10,000 gal	Auxiliary Building, 790 ft 6 in.	Level indication (LWPP) high and low alarms (CR, LWPP)	Overflows FD Sump #7 to FDT	_____

TABLE 9.3-5
POTENTIALLY RADIOACTIVE LIQUID STORAGE TANKS

(Sheet 2 of 4)

Tank	System	Number Supplied (Two Units)	Capacity	Location and Elevation	Leak Detection and/or Overflow Detection	Overflow or Relief Valve Discharge Receiving Point	Remarks
Reactor coolant drain tank	WPS	2	350 gal	Containment, 808 ft 0 in.	Level indication (LWPP) high and low alarms (CR, LWPP)	Relief to WHT or FDT via the containment sump	_____
Spent resin storage tank	WPS	1	4100 gal	Auxiliary Building, 810 ft 6 in.	Level indication (LWPP, DP) high and low alarms (CR, LWPP)	Relief to FDT via FD Sump #4	_____
Waste evaporator condensate tank	WPS	1	5000 gal	Auxiliary Building, 790 ft 6 in.	Level indication (LWPP) high and low alarms (CR, LWPP)	Overflows to WHT via FD Sump #6	_____
Waste holdup tank	WPS	1	10,000 gal	Auxiliary Building, 790 ft 6 in.	Level indication (LWPP) high-high, high, and low alarms (CR, LWPP)	Overflows on floor	Refer to Subsection 9.3.3.2.3
Waste monitor tank	WPS	2	5000 gal	Auxiliary Building, 790 ft 6 in.	Level indication (LWPP) high and low alarms (CR, LWPP)	Overflows to FDT via FD Sump #5	_____
Component cooling water surge tank	CCWS	2	4600 gal	Auxiliary Building 873 ft 6 in.	Level indication (CR) high-high, high low-low, and low alarms (CR)	Relief to component cooling water drain tank	Assumes a primary component leakage
Component cooling water drain tank	CCWS	2	2300 gal	Safeguards Building, 773 ft 0 in.	Local level indication High alarm (CR)	Overflows to FDT	Assumes a primary component leakage

TABLE 9.3-5
POTENTIALLY RADIOACTIVE LIQUID STORAGE TANKS

(Sheet 3 of 4)

Tank	System	Number Supplied (Two Units)	Capacity	Location and Elevation	Leak Detection and/or Overflow Detection	Overflow or Relief Valve Discharge Receiving Point	Remarks
Reactor makeup water storage tank	RMWS	2	112,000 gal	Outdoor	Level indication (CR, PDP) low alarm (CR, PDP) High alarm (PDP)	Overflows to WHT	Assumes an accidental makeup of radioactive recycled water
Refueling water storage tank	CSS	2	112,000 gal	Outdoor	Level indication (CR, PDP) high, low, and low-low alarms (CR)	Overflows to FDT	_____
Recycle holdup tank	BRS	2	112,000 gal	Auxiliary Building, 790 ft 6 in.	Level indication (BRP) high and low alarms (CR, LWPP)	Overflows to WHT	_____
Steam generator blowdown spent resin storage tank	steam generator blowdown and cleanup system	1	3700 gal	Auxiliary Building, 790 ft 6 in.	Local level indication (DP, SGBP) high and low alarms (CR, SGBP)	Relief Valve discharge to FDT	Assumes a steam generator tube leak
Chemical drain tank	WPS	1	600 gal	Auxiliary Building, 790 ft 6 in.	Level indication (LWPP) high and low alarms (CR, LWPP)	Overflows to FDT via Sump #7	_____
Laundry holdup and monitor tanks	WPS	2	5000 gal	Auxiliary Building 790 ft 6 in.	Level indication high and low alarms (LWPP, CR)	Overflow to FDT via Sump #7	_____
Condensate storage tank	AFW	2	500,000 gal	Outside	Level indication (CR) high-high, low, and low-low alarms (CR)	Overflow to WHT	_____
Waste conditioning tank	WPS	1	2000 gal	Fuel Building 841 ft 0 in.	Level indication high and low alarms (LWPP, CR)	_____	_____

TABLE 9.3-5
POTENTIALLY RADIOACTIVE LIQUID STORAGE TANKS

(Sheet 4 of 4)

Tank	System	Number Supplied (Two Units)	Capacity	Location and Elevation	Leak Detection and/ or Overflow Detection	Overflow or Relief Valve Discharge Receiving Point	Remarks
Hot phase separator	CPS	1	10,000 gal	Fuel Building 860 ft 0 in.	Level indication high and low alarms	Overflow to back-wash recovery system	_____
Plant Effluent Holdup and Monitoring Tanks	WPS	2	30,000 gal	Yard Inside Protected Area 810’	Level indication high, high-high, low, and low-low	Overflow to Fuel Bldg. Sump 2	_____

NOTE:

SIS	-	Safety Injection System
WPS	-	Waste Processing System
CSS	-	Containment Spray System
BRS	-	Boron Recycle System
CPS	-	Condensate Polishing System
RMWS	-	Reactor Makeup Water System
CR	-	Control Room
LWPP	-	liquid waste processing panel
FDT	-	floor drain tank
WHT	-	waste holdup tank
RHT	-	recycle holdup tank
RTP	-	resin transfer panel
DP	-	drumming panel
BRP	-	boron recycle panel
NC	-	normally closed
PDP	-	potable and demineralized water panel

TABLE 9.3-6
CHEMICAL AND VOLUME CONTROL SYSTEM DESIGN PARAMETERS

General	
Seal Water supply flow rate, for four reactor coolant pumps, nominal (gpm)	32
Seal water return flow rate, for four reactor coolant pumps, nominal (gpm)	12
Letdown flow	
Normal (gpm)	75
Maximum (gpm)[(a)]	120
Charging flow (excludes seal water)	
Normal (gpm)	55
Maximum (gpm)	100
Temperature of letdown reactor coolant entering system (°F)	560
Temperature of charging flow directed to Reactor Coolant System (°F)	518
Temperature of effluent directed to Boron Recycle System (°F)	115
Centrifugal charging pump bypass flow, each (gpm)	60
Amount of 4 weight percent boric acid solution required to meet cold shutdown requirements shortly after full power operation (gallons)	17,800
Maximum pressurization required for hydrostatic testing of Reactor Coolant System (psig)	3,107
Temperature	Ambient

a) Letdown flow maximum refers to the letdown configuration involving flow through both a 45 gpm and 75 gpm letdown orifice. The letdown flow maximum is 140 gpm.

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 1 of 14)

Positive Displacement Pump	
Number	1
Design pressure (psig)	3200
Design temperature (°F)	250
Design flow (gpm)	98
Design head (ft)	5800
Material	Austenitic Stainless Steel
Maximum operating pressure, for Reactor Coolant System hydrotest purposes (psig)	3125
Centrifugal Charging Pumps	
Number	2
Design pressure (psig)	2800
Design temperature (°F)	300
Design flow (gpm)	150
Design head (ft)	5800
Material	Austenitic Stainless Steel
Boric Acid Transfer Pump	
Number	2
Design pressure (psig)	150
Design temperature (°F)	250

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 2 of 14)

Design flow (gpm)	75
Design head (ft)	235
Material	Austenitic Stainless Steel
Chiller Pumps	
Number	2
Design pressure (psig)	200
Design temperature (°F)	200
Design flow (gpm)	400
Design head (ft)	150
Material	Carbon Steel
Regenerative Heat Exchanger	
Number	1
Heat transfer rate at design conditions (Btu/hr)	11.0×10^6
Shell Side	
Design pressure (psig)	2485
Design temperature (°F)	650
Fluid	Borated Reactor Coolant
Material	Austenitic Stainless Steel

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 3 of 14)

Tube Side	
Design pressure (psig)	3100
Design temperature (°F)	650
Fluid	Borated Reactor Coolant
Material	Austenitic Stainless Steel
Shell Side (Letdown)	
Flow (lb/hr)	37,300
Inlet temperature (°F)	560
Outlet temperature (°F)	290
Tube Side (Charging)	
Flow (lb/hr)	27,400
Inlet temperature (°F)	130
Outlet temperature (°F)	518
Letdown Heat Exchanger	
Number	1
Heat transfer rate at design conditions (Btu/hr)	16.1×10^6

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 4 of 14)

Shell Side	
Design pressure (psig)	165
Design temperature (°F)	250
Fluid	Component Cooling Water
Material	Carbon Steel
Tube Side	
Design pressure (psig)	600
Design temperature (°F)	400
Fluid	Borated Reactor Coolant
Material	Austenitic Stainless Steel

Shell Side	Heatup	Normal
Flow (lb/hr)	498,000	170,000
Inlet temperature (°F)	105	105
Outlet temperature (°F)	137	143
Tube Side (Letdown)		
Flow (lb/hr)	59,600	37,300
Inlet temperature (°F)	380	290
Outlet temperature (°F)	115	115

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 5 of 14)

Excess Letdown Heat Exchanger	
Number	1
Heat transfer rate at design conditions (Btu/hr)	5.2×10^6

	Shell Side	Tube Side
Design pressure (psig)	165	2485
Design temperature (°F)	250	650
Design flow (lb/hr)	129,000	12,410
Inlet temperature (°F)	105	560
Outlet temperature (°F)	145	165
Fluid	Component Cooling Water	Borated Reactor Coolant
Material	Carbon Steel	Austenitic Stainless Steel

Seal Water Heat Exchanger	
Number	1
Heat transfer rate at design conditions (Btu/hr)	2.4×10^6

	Shell Side	Tube Side
Design pressure (psig)	165	150
Design temperature (°F)	250	250
Design flow (lb/hr)	186,000	42,200

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 6 of 14)

Inlet temperature (°F)	105	172
Outlet temperature (°F)	118	115
Fluid	Component Cooling Water	Borated Reactor Coolant
Material	Carbon Steel	Austenitic Stainless Steel
<u>Moderating Heat Exchanger</u>		
Number		1
Heat transfer rate at design conditions (Btu/hr)	300	2.53×10^6
Design pressure (psig)	300	300
Design temperature (°F)	200	200
Design flow (lb/hr)	59,600	59,600
Design inlet temperature, boron storage mode (°F)	92.4	72.6
Design outlet temperature, boron storage mode (°F)	140	115
Outlet temperature, boron release mode (°F)	123.7	131.3
Material	Austenitic Stainless Steel	Austenitic Stainless Steel

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 7 of 14)

Letdown Chiller Heat Exchanger

Number	1
Heat transfer rate at design conditions, boron storage mode (Btu/hr)	1.65×10^6

	Shell Side	Tube Side
Design pressure (psig)	150	300
Design temperature (°F)	200	200
Design flow, boron storage mode (lb/hr)	175,000	59,600
Design inlet temperature, boron storage mode (°F)	39	72.6
Design outlet temperature, boron storage mode (°F)	48.4	45
Flow, boron release mode (lb/hr)	175,000	59,600
Inlet temperature, boron release mode (°F)	90	123.7
Outlet temperature, boron release mode (°F)	99.4	96.1
Material	Carbon Steel	Austenitic Stainless Steel

Letdown Reheat Heat Exchanger

Number	1
Heat transfer rate at design conditions (Btu/hr)	1.49×10^6

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 8 of 14)

	Shell Side	Tube Side
Design pressure (psig)	300	600
Design temperature (°F)	200	400
Design flow (lb/hr)	59,600	44,700
Inlet temperature (°F)	115	280
Outlet temperature (°F)	140	246.7
Material	Austenitic Stainless Steel	Austenitic Stainless Steel
Volume Control Tank		
Number		1
Volume (ft^3)		400
Design pressure (psig)		75
Design temperature (°F)		250
Material		Austenitic Stainless Steel
Boric Acid Tanks		
Number		2 shared
Capacity (gal)		46,000
Design pressure (psig)		Atm.
Design temperature (°F)		200
Material		Austenitic Stainless Steel

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 9 of 14)

Batching Tank	
Number	1
Capacity (gal)	800
Design pressure (psig)	Atmospheric
Design temperature (°F)	300
Material	Austenitic Stainless Steel
Chemical Mixing Tank	
Number	1
Capacity (gal)	5
Design pressure (psig)	150
Design temperature (°F)	200
Material	Austenitic Stainless Steel
Chiller Surge Tank	
Number	1
Volume (gal)	500
Design pressure (psig)	Atmospheric
Design temperature (°F)	200
Material	Carbon Steel

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 10 of 14)

Discharge Dampener Tank	
Number	1
Volume, ft^3	2.4
Design pressure, psig Internal	2735
Operating pressure, psig Internal	2350
Design temperature, °F	200
Operating temperature, °F	130
Material	Austenitic Stainless Steel
Suction Stabilizer Tank	
Number	1
Volume, ft^3	2
Design pressure, psig Internal	240
Operating pressure, psig Internal	25
Design temperature, °F	200
Operating temperature, °F	115
Material	Austenitic Stainless Steel

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 11 of 14)

Mixed Bed Demineralizers	
Number	1
Design pressure (psig)	300
Design temperature (°F)	250
Design flow (gpm) (a)	120
Resin volume, each (ft^3)	30
Material	Austenitic Stainless Steel

(a) Maximum allowed flow is 170 gpm

Cation Bed Demineralizers	
Number	1
Design pressure (psig)	300
Design temperature (°F)	250
Design flow (gpm) (a)	120
Resin volume (ft^3)	30
Material	Austenitic Stainless Steel

(a) Maximum allowed flow is 170 gpm

Thermal Regeneration Demineralizers	
Number	5

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 12 of 14)

Design pressure (psig)	300
Design temperature (°F)	250
Design flow (gpm)	250
Resin volume (ft^3)	74.3 (Thermal regeneration mode)
Material	Austenitic Stainless Steel

Boric Acid Filter

Number	1 per unit
Design Pressure (psig)	300
Design temperature (°F)	250
Design Flow (gpm)	250
Micron Rating (M)	0.1 to 40 absolute (100% retention)
Material, vessel	Austenitic Stainless Steel

Letdown Orifice	45 gpm	75 gpm
Number	1	2
Design flow (lb/hr)	22,400	37,300
Differential pressure at design flow (psig)	1700	1700
Design pressure (psig)	2,485	2,485
Design temperature (°F)	650	650

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 13 of 14)

Material	Austenitic Stainless Steel	Austenitic Stainless Steel
<u>No. 1 Seal Bypass Orifices</u>		
Number	1/Loop	
Design flow (gpm)	1	
Differential pressure at design flow (psid)	300	
Design pressure (psig)	2485	
<u>Reactor Coolant Filter</u>		
Number	1	
Design pressure (psig)	300	
Design temperature (°F)	250	
Design flow (gpm)	250	
Micron Rating (M)	0.05 to 40 absolute (100% retention)	
Material, vessel	Austenitic Stainless Steel	
<u>Seal Water Injection Filters</u>		
Number	2	
Design pressure (psig)	3100	
Design temperature (°F)	250	
Design flow (gpm)	80	

TABLE 9.3-7
CHEMICAL AND VOLUME CONTROL SYSTEM PRINCIPAL COMPONENT
DATA SUMMARY

(Sheet 14 of 14)

Micron Rating (M)	0.45 to 23 absolute (100% retention)	
Material, Vessel	Austenitic Stainless Steel	
Seal Water Return Filter		
Number	1	
Design pressure (psig)	300	
Design Temperature (°F)	250	
Design flow (gpm)	250	
Micron Rating (M)	0.1 to 40 absolute (100% retention)	
Material, vessel	Austenitic Stainless Steel	
Design temperature (°F)		250
Material		Austenitic Stainless Steel
Chiller		
Number		1
Capacity (Btu/hr)		1.66×10^6
Design flow (gpm)		352
Inlet temperature (°F)		48.3
Outlet temperature (°F)		39

TABLE 9.3-8
BORON RECYCLE SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 1 of 4)

Recycle Evaporator Feed Pumps	
Number	2
Design pressure (psig)	150
Design temperature (°F)	250
Design flow (gpm)	30/100
Design head (ft)	320/250
Material	Stainless Steel
Recycle Holdup Tanks	
Number	2 shared
Capacity	112,000
Design pressure (psig)	Atm.
Design temperature (°F)	200
Material	Stainless Steel
Recycle Evaporator Reagent Tank	
Number	1
Capacity (gal)	5
Design pressure (psig)	150
Design temperature (°F)	200
Material	Stainless Steel

TABLE 9.3-8
BORON RECYCLE SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 2 of 4)

Recycle Evaporator Feed Demineralizers	
Number	2
Design pressure (psig)	300
Design temperature (°F)	250
Design flow (gpm)	120
Resin volume (ft^3)	30
Material	Stainless Steel
Recycle Evaporator Condensate Demineralizer	
Number	1
Design pressure (psig)	300
Design temperature (°F)	250
Design Flow (gpm)	120
Resin volume (ft^3)	30
Material	Stainless Steel
Recycle Evaporator Feed Filter	
Number	1/unit
Design pressure (psig)	300
Design temperature (°F)	250
Design flow (gpm)	250
Micron rating (μM)	1 to 20 absolute (100% retention)
Material, vessel	Stainless steel

TABLE 9.3-8
BORON RECYCLE SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 3 of 4)

Recycle Evaporator Condensate Filter	
Number	1 shared
Design pressure (psig)	200
Design temperature (°F)	250
Design flow (gpm)	35
Micron rating (μM)	2 to 40 absolute (100% retention)
Material, vessel	Stainless steel
Recycle Evaporator Concentrates Filter	
Number	1
Design pressure (psig)	200
Design temperature (°F)	250
Design flow (gpm)	35
Micron rating (μM)	2 to 40 absolute (100% retention)
Material, vessel	Stainless steel
Recycle Evaporator Package	
Number	1 shared
Design flow (gpm)	15
Concentration of concentrate boric acid (wt percent)	4
Concentration of condensate	<10 ppm boron as H_3BO_3
Material	Stainless Steel

TABLE 9.3-8
BORON RECYCLE SYSTEM PRINCIPAL COMPONENT DATA SUMMARY

(Sheet 4 of 4)

Recycle Holdup Tank Vent Ejector	
Number	1 shared
Design pressure (psig)	150
Design temperature (°F)	200
Suction flow (scfm)	1
Motive flow (scfm)	40
Material	Carbon Steel

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 1 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
1. Air diaphragm operated globe valve 1-LCV-459 (1-LCV-460 analogous)	a. Fails open	a. Charging and Volume Control - letdown flow.	a. Failure reduces redundancy of providing letdown flow isolation to protect PRZ heaters from uncovering at low water level in PRZ. No effect on system operation. Alternate isolation valve (1-LCV-460) provides backup letdown flow isolation	a. Valve position indication (open to closed position change) at CB.	a. Valve is designed to fail “closed” and is electrically wired so that the electrical solenoid of the air diaphragm operator is energized to open the valve. Solenoid is de-energized to close the valve upon the generation of a SI ”S” signal or a low level PRZ control signal. The valve is electrically interlocked with the level letdown orifice isolation valves and may not be opened manually from the CB if any of these valves are at an open position.
	b. Fails Closed	b. Charging and Volume Control - letdown flow	b. Failure blocks normal letdown flow to VCT. Minimum letdown flow requirements for boration of RCS to hot standby concentration level may be met by establishing letdown flow through alternate excess letdown flow path If the alternate letdown flow path to VCT is not available due to common mode failure (loss of instrument air supply) affecting the opening operation of isolation valves in each flow path, the plant operator can borate the RCS to a hot standby concentration level without letdown flow by taking advantage of the steam space available in the PRZ.	b. Valve position indication (closed to open position change) at CB; letdown flow temperature indication (TI-127) at CV; letdown flow pressure indication (PI-131) at CB; and VCT level indication (LI-112A) and low level alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 2 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
2. Air Diaphragm operated globe valve 1--8149A 1-8149B and 1-8149C analogous	a. Fails Open	a. Charging and Volume Control - letdown flow	a. Failure prevents isolation of normal letdown flow through regenerative heat exchanger when bringing the reactor to a cold shutdown condition after the RHRS is placed into operation. No effect on hot standby operation. Containment isolation (1-8152) may be remotely closed from the CB to isolate letdown flow through the heat exchanger.	a. Valve position indication (open to closed position change) at CB.	1. Valve is of similar design as that stated for item #1. Solenoid is de-energized to close the valve upon the generation of a low level PRZ signal or closing of letdown isolation valves (1-459 and 1-460) upstream of the regenerative heat exchanger.
	b. Fails closed	a. Charging and Volume Control - letdown flow.	b. Failure blocks normal letdown flow to VCT. Minimum letdown flow requirements for boration of RCS to hot standby concentration level may be met by opening letdown orifice isolation valves 1-LCV-8149B and 1-LCV-8849C. If common mode failure (loss of instrument air) prevents opening of these valves and also prevents establishing alternate flow through path, plant operator can borate the RCS to a hot standby concentration level without letdown flow by taking advantage of steam space available in PRZ.	b. Same methods of detection as those stated for item #1, failure made “Fails closed”	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 3 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
3. Air diaphragm operated globe valve 1-8152 (1-8160 analogous)	a. Fails closed	a. Charging and Volume Control - letdown flow	a. Same effect on system operation as that stated for item #1, failure mode “Fails closed”.	a. Same methods of detection as those stated for item #1, failure mode “fails closed”. In addition, close position group monitoring light at CB.	1. Valve is of similar design as that stated for item #1. Solenoid is de-energized to close the valve upon the generation of an ESF “T” signal.
	b. Fails open.	b. Charging and Volume Control - letdown flow	b. Failure has no effect on CVCS operation during normal plant operation and load follow. However, under accident conditions requiring containment isolation, failure reduces the redundancy of providing isolation of normal letdown line.	b. Valve position indication (open to closed position change) at CB.	
4. Air diaphragm operated globe valve 1-TCV-381B	a. Fails open	a. Boron Concentration Control - boron thermal regeneration (boration)	a. Failure inhibits use of BTRS for load follow operation (boration) due to low temperature of letdown flow entering BTRS demineralizers. Alternate boration of reactor coolant for load follow is possible using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. Letdown heat exchanger tube discharge flow (FI-132) and pressure (PI-131) indications at CB and BTR demineralizer inlet flow temperature indication (TI-381) at CB if BTRS is in operation.	1. Valve is designed to fail “open” and is electrically wired so that the electrical solenoid of the air diaphragm operator is energized to close the valve.
	b. Fails closed	b. Boron Concentration Control - boron thermal regeneration (boration)	b. Failure inhibits use of BTRS for load follow operation (boration) due to loss of temperature control of letdown flow entering BTRS demineralizers.	b. Same method of detection as those stated for item #1, failure mode “Fails closed” except no “closed to open position change” indication at CB.	2. BTRS operation is not required in operations of CVCS systems used to bring the reactor to hot standby condition.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 4 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
			Failure also blocks normal letdown flow to VCT when BTRS is not being used for load follow. Minimum letdown flow requirements for boration of RCS to hot standby concentration level may be met as stated for effect on system operation for item #1, failure mode “fails closed”.		
5. Air diaphragm operated globe valve 1-PCV-131	a. Fails open.	a. Charging and Volume Control - letdown flow	a. Failure prevents control of pressure to prevent flashing of letdown flow in letdown heat exchanger and also allows high pressure fluid to demineralizers. Relief valve (1-8119) open in demineralizer line to release pressure to VCT and valve (1-TCV-129) changes position to divert flow to VCT. Boration of RCS to hot standby concentration level is poss ble with valve failing open.	a. Letdown heat exchanger tube discharge flow indication (FI-132) and high flow alarm at CB; temperature indication (TI-130) and high temperature alarm at CB; and pressure indication (PI-131) at CB.	1. Same remark as stated for item #4, in regards to valve design. 2. As a design transient the letdown heat exchanger is designed for complete loss of charging flow.
	b. Fails Closed.	b. Charging and Volume Control -letdown flow.	b. Same effect on system operation as that for item #1, failure mode “fails closed”.	b. Letdown heat exchanger discharge flow indication (FI-132), and pressure indication (PI-131) and high pressure alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 5 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
6. Air diaphragm operated globe three way valve 1-TCV-129	a. Fails open for flow only to VCT.	a. Charging and Volume Control - letdown flow.	a. Letdown flow bypassed from flowing to demineralizers and BTRS. Failure prevents ionic purification of letdown flow and inhibits operation of BTRS. Boration of RCS to hot standby concentration level is possible with valve failing open for flow only to VCT.	a. Valve position indication (VC Tank) at CB and RCS activity level when sampling letdown flow.	1. Electrical solenoid of air diaphragm operator is electrically wired so that solenoid is energized to open valve for flow to the demineralizers. Valve opens for flow to VCT on “High Letdown Temperature” or on “High Letdown Reheat HX Outlet Temperature”.
	b. Fails open for flow only to demineralizer.	b. Charging and Volume Control - letdown flow.	b. Continuous letdown to demineralizers and BTRS. Failure prevents automatic isolation of demineralizers and BTRS under fault condition of high letdown flow temperatures. These systems may be manually isolated using local valves Boration of RCS to hot standby concentration level is possible with valve failing open for flow only to demineralizer.	b. Valve position indication (Demin.) at CB. If BTRS is in operation, BTR demineralizer return flow indiction (FI-385) indicating flow during an alarm condition of high letdown reheat. Heat exchanger outlet temperature or high letdown temperature.	2. Technical Specifications provide a limit on RCS activity.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 6 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
7. Air diaphragm operated diaphragm valve 1-7054	a. Fails closed.	a. Boron Concentration Control - boron thermal regeneration or storage	a. Failure inhibits use of BTRS for load follow operation (Boration or dilution) due to flow isolation of the BTRS. Alternate boration or dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	a. Valve position indication (closed to open position change) at CB: BTRS operation indication (borate or dilute) at CB and BTR demineralizer return flow indication (FI-385) and inlet flow temperature indication (TI-381) at CB.	1. Valve is designed to fail “closed” and is electrically wired so that the electric solenoid of air diaphragm operator is energized to the open valve. 2. BTRS not required to bring reactor to hot standby condition.
8. Air diaphragm operated diaphragm valve 1-7002A	a. Fails closed.	a. Boron Concentration Control - boron storage	a. Failure inhibits use of BTRS for load follow operation (dilution) due to follow isolation of letdown chiller heat exchanger. Alternate dilution of reactor coolant for load follow may be accomplished using RMCV of CVCS. No effect on operations to bring reactor to hot standby condition.	a. BTRS operation indication (dilute) at CB; letdown reheat heat exchanger outlet temperature (TI-381) at CB; and RCS boron level when sampling letdown flow.	1. Same remarks as those stated above for item #7.
	b. Fails open	b. Boron Concentration Control - boron thermal regeneration	b. Failure inhibits use of BTRS for load follow operation (boration) due to flow through letdown chiller heat exchanger. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	b. BTRS operation indication (boration) at CB: BTRS return flow temperature indication (TI-386) at CB; BTR return flow indication (FI-385) at CB; and RCS boron level when sampling letdown flow.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 7 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
9. Air diaphragm operated diaphragm valve 1-7002B	a. Fails closed.	a. Boron Concentration Control boron storage	a. Same effect on system operation as that stated above for item #8, failure mode “Fails closed”.	a. Same methods of detection as those stated above for item #8 failure mode “Fails closed”.	1. Same remark as those stated above for item #7.
	b. Fails open.	b. Boron Concentration Control - boron thermal regeneration	b. Failure inhibits use of BTRS for load follow operation (boration) due to bypass of letdown flow from letdown reheat heat exchanger. Alternate boration of reactor coolant may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	b. Same methods of detection as those stated above for item #8, failure mode “Fails open”.	
10. Relief Valve 1-8117	a. Fails open.	a. Charging and Volume Control - letdown flow	a. Letdown flow is relieved to pressurizer relief tank. Failure inh bits use of demineralizers for reactor coolant purification and use of BTRS. Normal letdown line can be isolated and minimum letdown flow requirements for hot standby may be met by establishing letdown flow through alternate excess letdown flow path.	a. High temperature relief line indication (TI-125) and alarm at CB and VCT level indication (LI-112A) and low level alarm at CB.	1. Radioactive fluid contained.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 8 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
11. Relief Valve 1-1889	a. Fails open.	a. Charging and Volume Control - letdown flow.	a. Letdown flow is relieved to VCT. Failure inhibits use of demineralizers for reactor coolant purification and use of BTRS. Normal letdown line can be isolated and minimum letdown flow requirement for hot standby may be met by establishing flow through alternate excess letdown flow path.	a. RCS activity level when sampling letdown flow. When BTRS is operating, low BTR demineralizer return flow indication (FI-385) at CB.	1. Radioactive fluid contained.
12. Air diaphragm operated diaphragm valve 1-8245	a. Fails closed.	a. Boron Concentration Control - boron thermal regeneration or storage	a. Purification of reactor coolant using only mixed bed demineralizers cannot be performed. Failure also blocks normal letdown flow. Boration of RCS to hot standby concentration level may be met as stated for effect on system operation for item #1, failure mode “Fails closed”.	a. BTRS operation indication (off) at CB and RCS activity level when sampling letdown flow. Valve position indication (closed to open position change) at CB.	1. Same remark as that stated for item #4.
	b. Fails open.	b. Boron Concentration Control - boron thermal regeneration or storage.	b. Failure inhibits use of BTRS for load follow operation (boration or dilution) due to bypass of letdown flow from BTRS. Alternate boration or dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	b. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (borate or dilute) at CB and low BTR demineralizer return flow indication (FI-385) at CB. Valve position indication (open to closed position change) at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 9 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
13 Air diaphragm operated diaphragm valve 1-7056 (1-7045 analogous)	a. Fails closed.	a. Boron Concentration Control - boron storage.	a. Failure inhibits use of BTRS for load follow operation (dilution) due to flow isolation of BTR demineralizers. Alternate dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (dilute) at CB and low BTR demineralizer return flow indication (FI-385) at CB.	1. Same remarks as those stated for item #7.
	b. Fails open.	b. Boron Concentration Control - boron thermal regeneration.	b. Failure inhibits use of BTRS for load follow operation (boration) due to flow bypass of BTR demineralizers. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	b. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (borate) at CB.	
14. Air diaphragm operated diaphragm valve 1-7057	a. Fails open.	a. Boron Concentration Control - boron storage	a. Failure inhibits use of BTRS for load follow operation (dilution) due to flow bypass of BTR Demineralizers. Alternate dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (dilute) at CB.	1. Same remark as that stated for item #4.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 10 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails closed.	b. Boron Concentration Control - boron thermal regeneration.	b. Failure inhibits use of BTRS for load follow operation (boration) due to flow isolation of BTR demineralizers. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (borate) at CB and low BTR demineralizer return flow indication (FI-385) at CB.	
15. Air diaphragm operated diaphragm valve 1-7040	a. Fails open.	b. Boron Concentration Control - boron thermal regeneration.	a. Same effect on system operation as that stated for item #14, failure mode “Fails open”.	a. Same methods of detection as those stated for item #14, failure mode “Fails open”.	
	b. Fails closed.	b. Boron Concentration Control - boron thermal regeneration.	b. Failure inhibits use of BTRS for load follow operation (boration) due to blockage of return letdown flow from letdown chiller heat exchanger. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	b. Same methods of detection as those stated above for item #14, failure mode, “Fails closed”.	1. Same remark as that stated for item #4.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 11 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
16. Air diaphragm operated diaphragm valve 1-7041	a. Fails open.	a. Boron Concentration Control - boron storage	a. Failure inhibits use of BTRS for load follow operation (dilution) due to flow bypass of letdown chiller heat exchanger. Alternate dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (dilute) at CB and letdown reheat heat exchanger outlet temperature indication (TI-381) at CB.	1. Same remark as that stated for item #4.
	b. Fails closed.	b. Boron Concentration Control - boron thermal regeneration.	b. Failure inhibits use of BTRS for load follow operation (boration) due to flow isolation of letdown reheat heat exchanger and BTR demineralizers. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	b. Same methods of detection as those stated for item #14, failure mode “Fails closed”.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 12 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
17. Air diaphragm operated diaphragm valve 1-7022	a. Fails closed.	a. Boron Concentration Control - boron storage.	a. Failure inhibits use of BTRS for load follow operation (dilution) due to flow blockage of return letdown flow from moderating heat exchanger. Alternate dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (dilute) at CB and low BTR demineralizer return flow indication (FI-385) at CB.	1. Same remarks as those stated for item #7.
	b. Fails open.	b. Boron Concentration Control - boron thermal regeneration.	b. Failure inhibits use of BTRS for load follow operation (boration) due to bypass of flow from letdown chiller heat exchanger of return letdown flow. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	b. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operate indication (borate) at CB and BTRS return flow temperature indication (TI-386) and high temperature alarm at CB.	
18. Air diaphragm operated butterfly valve 1-TCV-386	a. Fails closed.	a. Boron Concentration Control - boron thermal regeneration and storage.	a. Failure inhibits use of BTRS for load follow operation (boration and dilution) due to flow blockage of chiller flow through letdown chiller heat exchanger. Alternate boration and dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operate indication (borate or dilute) at CB; BTRS return flow temperature indication (TI-386) and high temperature alarm at CB; and chiller surge tank temperature indication (TI-379) at CB.	1. Valve is designed to fail “closed”. 2. BTRS not used to bring the reactor to hot standby condition.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 13 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
19. Air diaphragm operated butterfly valve 1-FCV-375.	a. Fails open.	a. Boron Concentration Control - boron thermal regeneration and storage	a. Failure inhibits use of BTRS for load follow operation (boration and dilution) due to flow bypass of chiller flow from letdown chiller heat exchanger. Alternate boration and dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS return flow temperature indication (TI-386) and high temperature alarm at CB and chiller surge tank temperature indication (TI-379) at CB.	1. Valve is designed to fail “open”. 2. BTRS not used to bring the reactor to hot standby condition.
20. Chiller Unit AHCU	a. Fails to cool liquid	a. Boron Concentration Control - boron thermal regeneration and storage.	a. Failure inhibits use of BTRS for load follow operation (boration and dilution) due to loss of cooling capability of let-down chiller heat exchanger. Alternate boration and dilution of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operations to bring reactor to hot standby condition.	a. Same methods of detection as those stated above for item #19. In addition, BTRS operation indication (borate or dilute) at CB.	1. BTRS not used to bring the reactor to hot standby condition.
21. Chiller Pump #1 APCI-1 (Chiller Pump #2 analogous)	a. Fails to deliver working fluid.	a. Boron Concentration Control - boron thermal regeneration and storage.	a. No effect on BTRS system operation. Redundant chiller pump #2 provides necessary delivery of working fluid for chiller unit operation. BTRS not required in operations to bring reactor to hot standby condition.	a. BTRS operation indication (borate or dilute) at CB and local pump discharge flow pressure indication (PI-377A).	1. Both chiller pumps operate simultaneously during BTRS system operation.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 14 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
22. Air diaphragm operated globe valve 1-TCV-381A	a. Fails closed.	a. Boron Concentration Control - boron thermal regeneration.	a. Failure inhibits use of BTRS for load follow operation (boration) due to flow isolation of shell side of letdown reheat heat exchanger. Alternate boration of reactor coolant for load follow may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	a. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (borate) at CB and letdown reheat heat exchanger outlet temperature indication (TI-381) at CB.	1. Same remarks as those stated for item #7.
	b. Fails open	b. Boron Concentration Control - boron storage.	b. Failure inhibits use of BTRS for load follow operation (dilution) due to passage of CVCS letdown flow through tube side of letdown reheat heat exchanger. Alternate dilution of reactor coolant may be accomplished using RMCS of CVCS. No effect on operation to bring reactor to hot standby condition.	b. RCS boron level when sampling letdown flow. If BTRS is operating, BTRS operation indication (dilute) at CB and letdown reheat heat exchanger outlet temperature indication (TI-381) at CB.	
23. Air diaphragm operated globe valve 1-8153 (1-8154 analogous)	a. Fails closed.	a. Charging and Volume Control - letdown flow.	a. Failure inhibits use of the excess letdown fluid system of the CVCS as an alternate system that may be used for letdown flow control during normal plant operation and inh bits use of the excess letdown system to control water level in the pressurizer of the RCS during final stage of plant startup due to flow blockage.	a. Valve position indication (closed to open position change) at CB and excess letdown heat exchanger outlet pressure indication (PI-124) and temperature indication (TI-122) at CB.	1. Same remark as that stated for item #7 in regards to valve design. 2. If normal letdown and excess letdown flow is not available for hot standby operation, plant operator can borate RCS to hot standby concentration using steam space available in PRZ.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 15 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails open.	b. Charging and Volume Control - letdown flow.	b. Failure reduces redundancy of providing excess letdown flow isolation during normal plant operation and for plant startup. No effect on system operation. Alternate isolation valve (1-1854) closes to provide backup flow isolation of excess letdown line.	b. Valve position indication (open to closed position change) at CB.	
24. Air diaphragm operated globe valve 1-HCV-123	a. Fails closed.	a. Charging and Volume Control - letdown flow.	a. Same effect on system operation as stated for item #23, failure mode “Fails closed”.	a. Same methods of detection as those stated for item #23, failure mode “Fails closed” except for valve position indication at CB.	1. Same remarks as those stated above for item #23.
	b. Fails open.	b. Charging and Volume Control - letdown flow.	b. Failure prevents manual adjustment at CB of RCS system pressure downstream of excess letdown heat exchanger to a low pressure consistent with No. 1 seal leakoff back-pressure requirements. When using excess letdown system failure leads to a decrease in seal water pump shaft flow for cooling pump bearings.	b. Excess letdown heat exchanger outlet pressure indication (PI-124) at CB, and seal water return flow recording (FR-157) and low flow alarm at CB.	
25. Air diaphragm operated diaphgram valve 1-LCV-181 (1-LCV-178, 1-LCV-179, and 1-LCV-180 analogous)	a. Fails closed.	a. Charging and Volume Control - seal water flow.	a. No automatic makeup of seal water to seal standpipe that services No. 3 seal of RC pump #1. No effect on operation to bring the plant to hot standby condition.	a. Valve position indication (closed to open position change) and low standpipe level alarm at CB.	1. Same remark as that stated for item #7 in regards to valve design.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 16 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails open.	b. Charging and Volume Control - seal water flow.	b. Overfill of seal water standpipe and dumping of reactor makeup water to containment sump during automatic makeup of water for No. 3 seal of RC pump #1. No effect on operations to bring reactor to hot standby condition.	b. Valve position indication (open to closed position change) and high standpipe level alarm at CB.	2. Low level standpipe alarm conservatively set to allow additional time for RC pump operation without a complete loss of seal water being injected to No. 3 seal after sounding of alarm.
26. Relief valve 1-8121 2CS-8000 (Unit 2)	a. Fails open.	a. Charging and Volume Control - seal water flow.	a. RC pump seal water return flow and excess letdown flow bypassed to PRZ relief tank of RCS. Failure inhibits use of the excess letdown fluid system of the CVCS as an alternate system that may be used for letdown flow control during normal plant operation and inhibits use of excess letdown system to control water level in the PRZ of the RCS during final stage of a plant startup.	a. Decrease in VCT level causing RMCS of CVCS to operate	1. The capacity of the relief valve equals maximum flow from four RC pump seals plus excess letdown flow. 2. Radioactive fluid contained. 3. Same as remark #2 noted for item #23.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 17 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
27. Motor operated globe valve 1-8112 (1-8100 analogous)	a. Fails open.	a. Charging and Volume Control - seal water flow and excess let-down flow.	a. Failure has no effect on CVCS operation during normal plant operation and load follow. However, under accident conditions requiring containment isolation failure reduces redundancy of providing isolation of seal water flow and excess letdown flow.	a. Valve position indication (open to closed position change) at CB.	1. Valve is normally at a full open position and motor operator is energized to close the valve upon the generation of an ESF "T" signal. 2. If normal letdown and excess letdown flow is not available for hot standby operation, plant operator can borate RCS to hot standby concentration using steam space available in PRZ.
	b. Fails closed.	b. Charging and Volume Control - seal water flow and excess let-down flow.	b. RC pump seal water return flow and excess letdown flow blocked. Failure inh bits use of the excess letdown fluid system of the CVCS as an alternate system that may be used for letdown flow control during normal plant operation and degrade cooling capability of seal water in cooling RC pump bearings.	b. Valve position indication (closed to open position change) at CB; group monitoring light and alarm at CB; and seal water return flow recording (FR-157) and low seal water return flow alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 18 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
28. Motor operated gate valve 1-8105 (1-8106 analogous)	a. Fails open.	a. Charging and Volume Control - charging flow.	a. Failure has no effect on CVCS operation during normal plant operation and load follow. However, under accident condition requiring isolation of charging line, failure reduces redundancy of providing isolation of normal charging flow.	a. Valve position indication (open to closed position change) at CB.	1. Valve is normally at a full open position and motor operator is energized to close the valve upon the generation of a Safety Injection “S” signal.
	b. Fails closed.	b. Charging and Volume Control - charging flow.	b. Failure inhibits use of normal charging line to RCS for boration, dilution, and coolant makeup operations. Seal water injection path remains available for boration of RCS to a hot standby concentration level and makeup of coolant during operations to bring the reactor to hot standby condition.	b. Valve position indication (closed to open position change) and group monitoring light (valve closed) at CB; letdown temperature indication (TI-127) and high temperature alarm at CB; charging flow temperature indication (TI-126) at CB; seal water flow pressure indication (PI-120A) at CB; VCT level indication (LI-112A and LI-185) and high level alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 19 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
29. Air diaphragm operated globe valve 1-HCV-182	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure prevents manual adjustment at CB of seal water flow through the control of back pressure in charging header resulting in a reduction of flow to RC pump seals leading to a reduction in flow to RCS via labyrinth seals and pump shaft flow for cooling pump bearing. Boration of RCS to a hot standby concentration level and makeup of coolant during operations to bring reactor to hot standby condition is still possible through normal charging low path.	a. Seal water flow pressure indication (PI-120A) at CB; seal water return recording (FR-157); and low seal water return flow alarm at CB.	1. Same remark as that stated for item #4 in regards to design of valve.
	b. Fails closed.	b. Charging and Volume Control - charging flow.	b. Same effect on system operation as that stated for item #28, failure mode “Fails closed”.	b. Same method of detection as those stated above for item #28, failure mode “Failure closed”.	
30. Motor operated globe valve 1-8110 (1-8111 analogous)	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure has no effect on CVCS operation during normal plant operation and load follow. However, under accident condition requiring isolation of centrifugal charging pump miniflow line, failure reduces redundancy of providing isolation of mini-flow to suction of pumps via seal water heat exchanger.	a. Valve position indication (open to closed position change) at CB.	1. Same remark as that stated for item #28.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

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Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails closed.	b. Charging and Volume Control - charging flow and seal water flow.	b. Failure blocks miniflow to suction of centrifugal charging pumps via seal water heat exchanger. Normal charging flow and seal water flow prevents deadheading of pumps when used. Boration of RCS to a hot standby concentration level and makeup of coolant during operations to bring reactor to hot standby condition is still possible.	b. Valve position indication (closed to open position change) at CB; group monitoring light (valve closed) and alarm at CB: and charging and seal water flow indication (FI-121A) and high flow alarm at CB.	
31. Air diaphragm operated globe valve 1-8146	a. Fails open	a. Charging and Volume Control - charging flow.	a. Failure has no effect on CVCS operation during normal plant operation. Valve is used during cold shutdown operation to isolate normal charging line when using the auxiliary spray during the cooldown of the pressurizer. Cold shutdown of reactor is still poss ble, however, time for cooling down RPZ will be extended.	a. Valve position indication (open to closed position change) at CB.	1. Same remark as that stated for item #4 in regards to design of valve.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 21 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails closed.	b. Charging and Volume Control - charging flow.	b. Failure blocks normal charging flow to the RCS. No effect on CVCS operations during normal plant operation, load follow or hot standby operation. Plant operation can maintain charging flow by establishing flow through alternate charging path by opening of isolation valve (1-8147)	a. Valve position indication (closed to open position change) at CB; charging flow indication (TI-126) at CB; regenerative heat exchanger shell side exit temperature indication (TI-127) and high temperature alarm at CB; and charging and seal water flow indication (FI-121A) and low flow alarm at CB.	
32. Air diaphragm operated globe valve 1-8147	a. Fails closed.	a. Charging and Volume Control - charging flow.	a. Failure reduces redundancy of charging flow paths to RCS. No effect on CVCS operations during normal plant operation, load follow, or hot standby operation. Normal charging flow path remains available for charging flow.	a. Valve position indication (closed to open position change) at CB.	1. Same remark as that stated for item #4 in regards to design of valve.
	b. Fails open.	b. Charging and Volume Control - charging flow.	b. Same effect on system operation and shutdown as that stated above for item #31, failure mode “fails open” if alternate charging line is in use.	b. Valve position indication (open to closed position change) at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 22 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
33. Air diaphragm operated globe valve 1-8145	a. Fails open.	a. Charging and Volume Control - charging flow.	a. Failure results in inadvertent operation of auxiliary spray that results in a reduction of PRZ pressure during normal plant operation and load follow. PRZ heaters operate to maintain required PRZ pressure Boration of RCS to a hot standby concentration level and makeup of coolant during operation to bring reactor to hot standby condition is still possible.	a. Valve position indication (open to closed position change) at CB and PRZ pressure recording (PR-455) and low pressure alarm at CB.	1. Same remark as that stated for item #7 in regards to design valve.
	b. Fails closed	b. Charging and Volume Control - charging flow	b. Failure has no effect on CVCS operation during normal plant operation, load follow and hot standby operation. Valve is used during cold shutdown operation to activate spray cooling down the pressurizer after operation of RHRS.	b. Valve position indication (closed to open position change) at CB.	
34. Relief Valve 1-8123	a. Fails open.	a. Charging and Volume Control - charging flow.	a. Failure results in a portion of a seal water return flow and centrifugal charging pump miniflow being bypassed to VCT. Boration of RCS to a hot standby concentration level makeup of coolant during operations to bring reactor to hot standby condition is still possible.	a. Local pressure indication (PI-118 and PI-119) in discharge line of centrifugal charging pumps.	1. Radioactive fluid contained.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 23 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
35. Relief Valve 1-8118	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure results in a portion of charging flow and seal water from constant displacement pump being bypassed to VCT. No effect on normal plant operation, load follow or bringing reactor to hot standby condition. Constant displacement pump may be taken out of service and an alternate centrifugal charging pump used for delivery of charging and seal water flow.	a. Local pressure indication (PI-117) in discharge line of constant displacement pump.	1. Constant displacement pump may be flow isolation by closing manual gate valves in discharge and suction lines of pump. 2. Radioactive fluid contained.
36. Air diaphragm operated globe valve -1FCV-121	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure reduces redundancy of providing charging and seal water flow to RCS. No effect on normal plant operation, load follow, or bringing reactor to hot standby condition. Constant displacement pump normally used for delivery of charging and seal water flow to RCS. Check valves (1-8481A and 8481B) provide isolation of displacement flow to discharge of centrifugal pump if valve fails “open” during operation of constant displacement pump.	a. Charging and seal water flow indication (FI-121A) and high flow alarm at CB, and PRZ level recording (LR-459) and high level alarm at CB.	1. Same remark as that stated for item #4 in regards to design of valve. 2. Methods of detection apply when a centrifugal charging pump is in operation.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 24 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails closed.	b. Charging and Volume Control - charging flow and seal water flow.	b. Failure reduces redundancy of providing charging and seal water flow to RCS. No effect on system operation during normal plant operation, load follow, or bringing reactor to hot standby condition. Constant displacement pump normally used for delivery of charging and seal water flow to RCS. Valve failing closed under an accident condition requiring flow delivery by centrifugal charging inh bits flow from the pumps.	b. Charging and seal water flow indication (FI-121A) and low flow alarm at CB, and PRZ level recording (LR-459) and low level alarm at CB.	
37. Check valve 1-8497	a. Fails open.	a. Charging and Volume Control - charging flow and and seal water flow.	a. Failure reduces redundancy of providing charging and seal water to RCS. Discharge of constant displacement pump remains open to “back-flow” when a centrifugal charging pump is placed into operation. No effect on normal plant operation, load follow, or bringing reactor to hot standby conditon; constant displacement pump normally used for delivery of charging and seal water flow.	a. Charging and seal water flow indication (FI-121A) and low flow alarm at CB, and PRZ level recording (LR-459) and low level alarm at CB.	1. Constant displacement pump may be isolated by the closing of the manual valves in pump’s suction and discharge lines. 2. Methods of detection apply when centrifugal charging pump #1 is in operation.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 25 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
38. Check valve 1-8481A (1-8481B analogous)	a. Fails open	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure reduces redundancy of providing charging and seal water flow to RCS. Discharge of centrifugal charging pump #1 is open to "back-flow" when centrifugal charging pump #2 is placed into operation after failure of centrifugal charging pump #1 to deliver charging and seal water flow. No effect on normal plant operation, load follow, or bringing reactor to hot standby condition.	a. Same methods of detection as those stated for item #37.	1. Centrifugal charging pump #1 may be isolated by the closing of manual valves in pump's suction and discharge lines.
39. Positive displacement pump APPD	a. Fails to deliver working fluid	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure reduces redundancy of providing charging and seal water flow to RCS. No effect on normal plant operation, load follow, or bringing reactor to hot standby condition. Centrifugal charging pump (#1 or #2) may be placed into operation for delivery of charging and seal water flow.	a. Pump circuit breaker position indication (open) at CB; common pump breaker trip alarm at CB; charging and seal water flow indication (FI-121A) and low flow alarm at CB; and PRZ level recording (LR-459) and low level alarm at CB.	1. Pump stroke is regulated to control amount of charging flow delivered to the PRZ.
40. Centrifugal charging pump #1 APCH-1 (Pump #2 analogous)	a. Fails to deliver working fluid.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure reduces redundancy of providing charging and seal water flow to RCS from the centrifugal charging pumps. No effect on normal plant operation, load follow, or bringing reactor to hot standby condition.	a. Same methods of detection as those stated above for item #39 when centrifugal charging pump #1 is in operation	1. Flow rate for a centrifugal charging pump is controlled by a modulating valve (1-FCV-121) in discharge header for the centrifugal charging pumps.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

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Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
41. Air diaphragm operated globe valve 1-8156.	a. Fails closed.	a. Chemical Control, Purification and Makeup - oxygen control.	a. Failure blocks hydrogen flow to VCT and leads to loss of venting of VCT (vent valve (1-PCV-115 closes on low VCT pressure) resulting in loss of gas stripping of fission products from RCS coolant. No effect on operation to bring the reactor to hot standby condition.	a. VCT pressure indication (PI-115A) and low pressure alarm at CB. Periodic sampling of gas mixture in VCT.	1. Valve is designed to fail “closed”. 2. Plant’s technical specification set limits on RCS activity level.
42. Relief valve 1-8120	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure allows VCT liquid to be relieved to BRS recycle holdup tank resulting in a loss of VCT liquid makeup coolant available for charging and seal water flow during normal plant operation, load follow, and bringing the reactor to a hot standby condition. VCT isolation valves (1-LCV-112B and 1-LCV-112C) close on low-low tank level signal causing the suction of charging pumps to be transferred to the RWST for an alternate supply of borated coolant.	a. Decrease in VCT level causing RMCS to operate; VCT level indications (LI-112A and LI-185) and low level alarms (low and low-low) at CB; and BRS recycle holdup tank level increase.	1. Radioactive fluid contained.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 27 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
43. Motor operated gate valve 1-LCV-112B (1-LCV-112C analogous)	a. Fails open.	a. Charging and Volume Control - charging flow and seal water flow.	a. Failure has no effect on CVCS operation during normal plant operation, load follow, and bringing reactor to a hot standby condition. However, under accident conditions requiring isolation of VCT, failure reduces redundancy of providing isolation for discharge line of VCT.	a. Valve position indication (open to closed position change) at CB.	1. During normal plant operation and load follow valve is at a full open position and the motor operator is energized to close the valve upon the generation of a VCT low-low water level signal or upon the generation of a Safety Injection “S” signal, providing isolation valve 1-LCV-112D (1-LDV-112E analogous) is at a full open position.
	b. Fails closed.	b. Charging and Volume Control - charging flow and seal water flow.	b. Failure blocks fluid flow from VCT during normal plant operation, load follow and when bringing the reactor to a hot standby condition. Alternate supply of borated coolant from the TWST to suction of charging pumps can be established from the CB by the operator through the opening of RWST isolation valves (1-LCV-112D and 1-LCV-112E).	b. Valve position indication (closed to open position change) at CB; group monitoring light (valve closed) at CB; charging and seal water flow indication (FI-121A) and low flow alarm at CB; and RPZ level recording (LR-459) and low level alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 28 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
44. Air diaphragm operated diaphragm valve 1-PCV-115	a. Fails closed.	a. Chemical Control, Purification and Makeup - oxygen control.	a. Failure blocks venting of VCT gas mixture to gas waste processing system (hydrogen recombiners) for stripping of fission products from RCS coolant during normal plant operation and load follow. No effect on operations to bring the reactor to hot standby condition.	a. Valve position indication (closed to open position change) at CB and VCT pressure indication (PI-115A) at CB. Periodic sampling of gas mixture in VCT.	1. Same remark as that stated for item #7 in regards to valve design. 2. Same remark as that stated for item #41 in regards to RCS activity.
45. Air diaphragm operated diaphragm valve 1-FCV-110B	a. Fails closed.	a. Boron Concentration Control - reactor makeup control - boration, auto make-up, and alternate dilution.	a. Failure blocks fluid flow from reactor makeup control system for automatic boric acid addition and reactor water makeup during normal plant operation and load follow. Failure also reduces redundancy of fluid flow paths for dilution of RC coolant by reactor makeup water and blocks fluid flow for boration of the RC coolant when bringing the reactor to a hot standby condition. Boration (at B A tank boron concentration level) of RCS coolant to bring the reactor to hot standby condition is possible by opening of alternate BA tank isolation valve (1-8104) at CB.	a. Valve position indication (closed to open position change) at CB; total makeup flow deviation alarm at CB; and VCT level indications (LI-112A and LI-185) and low level alarms (low and low low) at CB.	1. Same remark as that stated for item #7 in regards to valve design.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 29 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails open	b. Boron Concentration Control - reactor makeup control - boration, auto make-up, and alternate dilution	b. Failure allows for alternate dilute mode type operation for system operation of normal dilution of RCS coolant. No effect on CVCS operation during normal plant operation and load follow, and when bringing the reactor to a hot standby condition.	b. Valve position indication (open to closed change) at CB.	
46. Air diaphragm operated diaphragm valve 1-FCV-111B	a. Fails closed.	a. Boron Concentration Control - reactor makeup control - dilution and alternate dilution.	a. Failure blocks fluid flow from RMCS for dilution of RCS coolant during normal plant operation and load follow. No effect on CVCS operation. Operator can dilute RCS coolant by establishing “alternate dilute” mode of system operation. Dilution of RCS coolant not required when bringing the reactor to a hot standby condition.	a. Same methods of detection as those stated above item #45, failure mode “Fails closed”.	1. Same remark as that stated for item #7 in regards to valve design.
	b. Fails open.	b. Boron Concentration Control - reactor makeup control - dilution and alternate dilution.	b. Failure allows for alternate dilute mode type operation for system operation of boration and auto makeup of RCS cool-ant. No effect on CVCS operation during normal plant operation and load follow and when bringing the reactor to a hot standby condition.	b. Valve position indication (open to closed position change) at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 30 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
47. Relief Valve 1-8124.	a. Fails open.	a. Charging and Volume Control - charging and seal water flow.	a. Failure allow for a portion of flow to suction header of charging pumps to be relieved to BRS recycle holdup tank. Boration of RCS coolant to bring reactor to hot standby condition is still possible.	a. Decrease in VCT level causing RMCS to operate; VCT level indications (KI-112A and LI-185) and low level alarms (low and low-low) at CB; and BRS recycle holdup tank level increase.	1. Radioactive fluid contained.
48. Air diaphragm operated globe valve 1-FCV-110A	a. Fails open.	a. Boron Concentration Control - reactor makeup control - boration and auto makeup	a. Failure prevents the addition of a pre-selected quantity of concentrated boric acid solution at a pre-selected flow rate to the RCS coolant during normal plant operation, load follow and when bringing the reactor to a hot standby condition. Boration to bring the reactor to a hot standby condition is possible, however, flow rate of solution from BA tanks can not be automatically controlled.	a. Valve position indication (open to closed position change) at CB; and boric acid flow recording (FR-110) and flow deviation alarm at CB.	1. Same remark as that stated for item #4 in regards to valve design.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

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Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
	b. Fails closed.	b. Boron Concentration Control - reactor makeup control - boration, and auto makeup	b. Failure blocks fluid flow of boric acid solution from BA tanks during normal plant operation, load follow, and when bringing the reactor to a hot standby condition. Boration (at BA tank boron concentration level) of RCS coolant to bring the reactor to hot standby condition is possible by opening of alternate BA tank isolation valve (1-8104) at CB.	b. Valve position indication (closed to open position change) at CB; and boric acid flow recording (FR-110) and flow deviation alarm at CB.	
49. Air diaphragm operated globe valve 1-FCV-111A	a. Fails closed.	a. Boron Concentration Control - reactor makeup control - dilute, alternate dilute and auto makeup	a. Failure blocks fluid flow of water from reactor makeup control system during normal plant operation and load follow. No effect on system operation when bringing the reactor to a hot standby condition.	a. Valve position indication (closed to open position change) at CB; VCT level indications (LI-112A and LI-183) and low level alarms (low and low-low) at CB; and makeup water flow recording (FR-110) and flow deviation alarm at CB.	1. Same remark as that stated from item #7 in regards to valve design.
	b. Fails open.	b. Boron Concentration Control - reactor makeup control - dilute, alternate dilute and auto makeup.	b. Failure prevents the addition of a pre-selected quantity of water makeup at a pre-selected flow rate to the RCS coolant during normal plant operation and load follow. No effect on system operation when bringing the reactor to a hot standby condition.	b. Valve position indication (open to closed position change) at CB and makeup water flow recording (FR-110) and flow deviation alarm at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 32 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
50. Motor operated globe valve 1-8104	a. Fails closed.	a. Boron Concentration Control - reactor makeup control - boration and auto makeup	a. Failure reduces redundancy of flow paths for supplying boric acid solution from BA tanks to RCS via charging pumps. No effect on CVCS operation during normal plant operation, load follow, or hot standby operation. Normal flow path via RMCS remains available for boration of RCS coolant.	a. Valve position indication (closed to open position change) at CB and flow indication (FI-183A0) at CB.	1. Valve is at a closed position during normal plant RMCS operation. 2. If both flow paths from BA tanks are blocked due to failure of isolation valves (1-FCV-110A and 1-8104), borated water from RWST is available by opening isolation valve 1-LCV-112D or 1-LCV-112E.
	b. Fails open.	b. Boron Concentration Control - reactor makeup control - boration and auto makeup	b. Failure prevents the addition of a pre-selected quantity of concentrated boric acid solution at a pre-selected flow rate to the RCS coolant during normal plant operation, load follow and when bringing the reactor to a hot standby condition. Boration to bring the reactor to a hot standby condition is possible, however, flow rate of solution from BA tanks can not be automatically controlled.	b. Valve position indication (open to closed position change) at CB and flow indication (FI-183A) at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 33 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
51. Boric acid transfer pump #1 APBA-1 (BA transfer pump #2 analogous)	a. Fails to deliver working fluid.	a. Boron Concentration Control - reactor makeup control - boration and auto makeup.	a. No effect on CVCS system operation during normal plant operation, load follow or bringing reactor to hot standby condition. Redundant BA transfer pump #2 provides necessary delivery of working fluid for CVCS system operation.	a. Pump motor start relay position indication (open) at CB and local pump discharge pressure indication (PI-113).	
52. Solenoid operated valve 1-8210A (1-8210B analogous)	Fails open		Failure has no effect on PDP operation during normal plant operation and safe shutdown. However, under accident conditions requiring PDP suction stabilizer isolation, the failure reduces the redundancy of providing isolation of the gas makeup supply to the PDP suction stabilizer. Alternate isolation valve (1-8210B) provides backup isolation capability.	Valve position indication (open to closed position change). In addition, close position group monitoring light at CB.	The valve is designed to fail “closed” and is electrically wired so that the electrical solenoid is energized to open the valve. The solenoid is de-energized to close the valve upon generation of an ESF “S” signal.
53. Solenoid operated valve 1-802A (1-8202B analogous)	Fails open		Failure has no effect on PDP operation during normal plant operation and safe shutdown. However, under accident conditions requiring PDP suction stabilizer isolation, the failure reduces the redundancy of providing isolation of the gas vent line from the PDP suction stabilizer to the VCT. Alternate isolation (1-8202B) provides backup isolation capability.	Same methods of detection as those stated for item #52, failure mode “Fails Open”.	Same remark as stated for item #52 in regard to valve design.

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 34 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks
54. Air operated ball valve HV-8220 (HV-8221 analogous)	Fails open	a. Charging Pump suction high point vent.	Failure has no effect on CVCS operation during normal plant operation, load follow, and bringing reactor to a hot standby condition. However, under accident conditions requiring isolation of VCT, failure reduces the redundancy of providing isolation for vent line to the VCT.	Valve position indication (open to closed position change) at CB.	1. During normal plant operation and load follow, valve is at the full open position and the solenoid operator is de-energized to close the valve upon the generation of a VCT low-low water level signal or upon the generation of a safety injection “S” signal, providing isolation valve 1-LCV-112D (1-LCV-112E analogous) is at full open position.
	Fails closed	b. Charging Pump suction high point vent.	Failure blocks high point vent to VCT during normal plant operation, load follow, and when bringing the reactor to a hot standby condition.	Valve position indication (closed to open position change) at CB; group monitoring light (valve closed) at CB.	2. Valve solenoid operator is also de-energized to close the valve upon the generation of a source range neutron flux doubling signal, providing isolation valve LCV-112D (1-HV-8221 analogous) is at full open position.
55. Motor operated gate valve HV-8402A.	Fails closed.	Charging and Volume Control-charging flow.	Failure inhibits use of normal charging line to RCS for boration, dilution, and coolant makeup operations. Seal water injection path remains available for boration of RCS to a hot standby concentration level and makeup of coolant during operations to bring the reactor to hot standby condition.	Valve position indication (closed to open position change) and group monitoring light (valve closed) at CB.	

TABLE 9.3-9
FAILURE MODE AND EFFECTS ANALYSIS CHEMICAL AND VOLUME CONTROL SYSTEM ACTIVE COMPONENTS – NORMAL PLANT OPERATION AND LOAD FOLLOW

(Sheet 35 of 35)

Component	Failure Mode	CVCS Operation Function	Effect on System Operation and Shutdown[a]	Failure Detection Method[b]	Remarks

Note:

Unit 1 and Unit 2 Tag Numbers are generally the same except for the prefix or as otherwise noted

List of Acronyms and abbreviations

BRS	-	Boron Recycle System
BTR‘	-	Boron Thermal Regeneration
BTRS	-	Boron Thermal Regeneration System
CB	-	Control Board
CVCS	-	Chemical and Volume Control System
Demin.	-	Demineralizer
PRZ	-	Pressurizer
RC	-	Reactor Coolant
RCS	-	Reactor Coolant System
RHRS	-	Residual Heat Removal System
RWST	-	Refueling Water Storage Tank
RMCS	-	Reactor Makeup Control System
VCT	-	Volume Control Tank

a) See list at end of table for definition of acronyms and abbreviations used.

b) As part of plant operation, periodic tests, surveillance inspections and instrument cal brations are made to monitor equipment and performance. Failures may be detected during such monitoring of equipment in addition to detection methods noted.

9.4 AIR CONDITIONING, HEATING, COOLING, AND VENTILATION SYSTEMS

9.4.1 CONTROL ROOM AREA VENTILATION SYSTEM

9.4.1.1 Design Bases

The Control Room HVAC and filtration systems are designed to maintain suitable and safe ambient conditions for operating personnel and equipment during all modes of operation including post-DBA conditions, in the following areas of the Control Building.

Areas on floor elevation 830 ft 0 in.:

East Control Room

West Control Room

Console and Control Room Unit 1

Console and Control Room Unit 2

Instrument Room Unit 1

Instrument Room Unit 2

Computer Room Unit 1

Computer Room Unit 2

Shift Manager’s Office

Production Supervisor’s Office

Corridor

Men’s Toilet

Woman’s Toilet

Kitchen and Janitor Closet

Charts and Supplies Storage Room

Areas on floor elevation 840 ft 6 in.:

Technical Support Center (Office and Corridor)

Offices (2)

Electrical Equipment Corridor

Areas on floor elevation 854 ft 4 in.:

Control Room Air Conditioning System mechanical equipment rooms, trains A and B.

The Control Room, located on elevation 830 ft 0 in, is maintained at 75°F (±5°F) and 35-50 percent relative humidity. The Control Room HVAC and filtration equipment rooms are maintained between 40°F and 104°F. Miscellaneous areas on elevations 830 ft and 840 ft are also maintained between 40°F and 104°F. Other system design parameters are presented in Tables 9.4-1 and 9.4-2.

Design Conditions-Indoors in Table 9.4-2 were based on personnel comfort or equipment design temperature limitation. Heating, ventilating and air-conditioning equipment were sized to provide sufficient heating and cooling during maximum and minimum outdoor conditions as indicated in Table 9.4-1. Temperatures and wind speed identified in Table 9.4-1 were selected on the bases of the meteorological study presented in Section 2.3. More specifically the maximum temperatures are based on Table 2.3-15. The extreme temperatures in Table 9.4-1 were not reached during the period of record indicated in Table 2.3-15. The duration of such extreme temperatures is considered short (one to two hours).

As described in the following paragraphs, the system is provided with sufficient redundancy in equipment and power supplies to enable the system to sustain a single failure of an active component without loss of function.

1. The system is equipped with four modular air-conditioning units. Each unit is rated at 50 percent of the Control Room HVAC and filtration systems capacity. Each pair of air-conditioning units is powered from an independent Class 1E bus and is physically separated by a concrete wall in the Control Room HVAC and filtration mechanical equipment room.

2. The Control Room makeup air supply fans, Control Room exhaust fans, Control Room complex kitchen and toilet exhaust fans, emergency pressurization air supply filtration units and fans, and emergency filtration units and fans are 100-percent redundant.

3. The system is equipped with ten non-safety related local air-conditioning units that provide supplementary cooling to areas which do not contain safety related equipment and are not needed for continued occupancy.

4. The redundant fans and filtration units are powered from independent Class 1E buses. (See Section 8.3.)

5. Redundant outside air intake dampers CPX-VADPOU-14 and CPX-VADPOU-15 (Figure 9.4-1) are provided to supply air during normal operation and to allow air for pressurization during emergency conditions.

6. All dampers are set to fail in the safe position or are provided with separate, bottled air supplies for emergency operation. (See Section 9.3.1)

7. All exhaust fans are equipped with isolation dampers CPX-VADPMU- 05,06 and CPX-VADPOU-27,28 (Figure 9.4-1).

8. Dampers CPX-VADPOU-41 and CPX-VADPOU-42 are provided for the isolation of the emergency filtration units.

9. The condensers for each pair of air-conditioning units have an independent cooling water source.

10. All control valves and dampers are equipped with manual operators at accessible locations to facilitate their operation in the event of power or instrument air failures, or both.

11. Redundant inlet bubble tight dampers are provided at each control room inlet to minimize unfiltered inleakage when the system is in the emergency mode of operation and isolation during toxic gas release or fires.

System components and ductwork are of seismic Category I and ANS Safety Class 3 design to ensure system availability for safe shutdown of the reactor following DBAs.

Failure modes for isolation valves and dampers are set so that their failure does not render the system inoperable.

Radiation protection (see Sections 11.5 and 12.3.3) is provided to permit access and occupancy of the Control Room during normal plant operation and following DBAs without personnel receiving radiation exposures in excess of five rem whole body, or its equivalent, to any part of the body for the duration of the accident, which is in accordance with GDC 19 of 10 CFR Part 50, Appendix A.

Surveillance of activity levels at the fresh air intake ducts to the Control Room is provided by seismically qualified process radiation monitors which are shown on Figure 9.4-1. (See Section 11.5 for design criteria.) The activity level at each intake duct is monitored and indicated in the Control Room; high levels are annunciated on the radiation monitoring system display/operator cabinet.

Continuous surveillance of radiation dose levels in the Control Room is also provided by redundant area radiation monitors which are shown on Figure 9.4-1. (See Section 12.3.4 for design criteria. Radiation dose level is continuously monitored, and indicated in the Control Room and annunciated on the radiation monitoring CRT display console.

9.4.1.2 System Description

The Control Room design shown in Section 1.2, Figure 1.2-33, provides for a single Control Room serving both units. The Control Room HVAC system is designed to serve this area as well as the computer rooms and offices located at the same elevation and at elevation 840 ft 6 in. The Control Room HVAC system is designed to remove all heat generated by the equipment, computers, lighting, personnel, and so forth and to provide a safe operating condition for personnel.

Two modular air-conditioning units provide the cool air required, with two units of the same size on standby to address the single failure criterion. Each modular air-conditioning unit contains a roughing filter, cooling coil, heating element, humidity control equipment, fan, and associated instrumentation controls.

The cooling coils are the direct-expansion, refrigerant type and suitable for refrigerant R-12. The coils of each modular air-conditioning unit are connected to separate and independent compressors and water-cooled condensers.

During normal operation, the air is recirculated through the air-conditioning unit with about six percent outdoor air (3000 ft^3/min fresh air) added to provide for oxygen depletion, odor control, pressure boundary leakage, and provide makeup for the kitchen and toilet and Control Room Exhaust Fans. Makeup air is supplied by one of two redundant fans.

Two redundant emergency filtration units are provided for use following DBAs. Each unit comprises a roughing filter, two high-efficiency particulate absorption (HEPA) filters, and iodine adsorbers and booster fans.

Exhaust air from the toilets, and kitchen areas is discharged directly to the atmosphere through one of the two 100-percent-capacity toilet exhaust fans.

The Control Room is pressurized with respect to the environment to prevent infiltration of unfiltered and unmonitored air and to account for leakages, as specified in Section 6.4. The overpressure is considered sufficient to prevent infiltration because the Control Room building structure is completely airtight, with few penetrations, and designed to withstand the tornado-generated missiles described in Section 3.3.2. This overpressure is maintained by modulating exhaust damper CPX-VADPMU-05 or CPX-VADPMU-06 during normal operation and emergency ventilation. Damper CPX-VADPOU-22 or CPX- VADPOU-23 can be used to reduce the control room pressure down to .125 overpressure for easier access to the control room during emergency recirculation. (See Section 6.4.)

Overpressure during the emergency recirculation mode is accomplished by introducing up to 800 ft^3/min of filtered air into the Control Room. This air is supplied through a 100-percent-redundant air-cleanup unit consisting of a fan, a mist eliminator, a heater, a roughing filter, two HEPA filters, and an iodine adsorber.

Section 6.4.2 and Table 6.4-4 describe the potential leak paths and leakage characteristics.

The orientation of the Control Room doors (with the exception of the missile resistant door) to swing inward is necessary to reduce the exfiltration rate from the Control Room and to prevent a possible depressurization of the Control Room caused by a door being left ajar. Provision of a positive pressure against an inward-opening, gasketed door results in a significant decrease in leakage over an outward-opening door. Consequently, a reduced emergency pressurization flow rate is required during possible post-LOCA operation. The lower outside air intake rate results in a Control Room operator dose well below the allowable limits of GDC 19, 10 CFR Part 50 Appendix A. For an analysis of the dose received by the Control Room occupants in the unlikely event of a LOCA, see Section 15.6.5.4

Operation of the Control Room HVAC system comprises the following modes:

1. Normal operation

2. Emergency recirculation

3. Emergency ventilation

4. Isolation

The four modes of operation are shown on Figure 9.4-1.

During normal operation, one makeup air supply and one main exhaust fan are operating, and their associated dampers are open. The emergency filtration system is isolated, and its bypass damper is open. One kitchen and toilet exhaust fan also operates.

The Control Room HVAC system automatically switches to emergency recirculation when any one of the conditions outlined in Subsection 9.4.1.3 are detected. The procedure is as follows:

1. The makeup air supply fan and all exhaust fans stop.

2. All exhaust dampers close.

3. The isolation and bypass dampers of the emergency filtration units are positioned such that approximately 16 percent (8000 ft^3/min) of the recirculated air flow is directed through each of the filters and booster fans.

4. Both emergency filtration unit fans start.

5. Both emergency pressurization supply unit fans start. The dampers are positioned such that the air is directed through the pressurization air cleanup units to the emergency filtration unit.

In cases 4 and 5 above, the operator, within one (1) hour, stops one pressurization and one filtration unit fan. Redundant inlet isolation dampers are closed to minimize filtered inleakage through the stopped train.

The operation of the emergency ventilation mode is required to replenish the oxygen content in the Control Room atmosphere during the emergency recirculation mode. For brief intervals the makeup fan is operated, supplying filtered outside air (3800 ft^3/min) to the Control Room complex. The main exhaust fan and the kitchen and toilet exhaust fan are also operated simultaneously.

Subsequent to the automatic initiation of the emergency recirculation mode, the operator can regain manual control over the system from the Control Room ventilation panels.

The operator may take manual action to transfer the control room ventilation system from the normal mode to the isolation mode following control room annunciation of smoke at either of the two outside air intakes (see Section 6.4). The return to the normal operation mode also requires manual action

Each Control Room air-conditioning unit is an independent, seismic Category I package unit with a component cooling water cooled condenser. Electric reheat coils are provided for the the production supervisors office, and the main console areas and are powered by the non-safety-related electrical bus.

9.4.1.3 Safety Evaluation

The Control Room atmosphere cleanup units (integrated into the HVAC system design) conform to the criteria established in NRC Regulatory Guide 1.52, Reference [10], except as shown in Appendix 1A(B) and Section 6.5, Table 6.5-1.

The Control Room HVAC system is provided with instrumentation and controls which continuously monitor system performance. In addition, area radiation and fire detection monitors are provided to ensure safe operating conditions for equipment and personnel during all modes of operation.

An ionization detector is located in each fresh air intake to detect smoke in the incoming air. If smoke is detected, the respective inlet and outlet Control Room air duct dampers may be closed and air recirculated in the Control Room. Switching to the normal or emergency ventilation modes of operation is accomplished manually by the operator by means of a Control Room vertical panel-mounted selector switch.

Any contaminants that have entered the Control Room prior to full closure of the dampers are removed by the emergency filtration unit. Removal of heavy concentrations of contaminants caused by a fire in the Control Room is accomplished by portable smoke ejectors.

Audible and visual alarms are provided in the Control Room to alert the operator in the event of system malfunction or unsafe conditions.

The Control Room air-conditioning system is designed to automatically switch to the emergency recirculation mode of operation described in Subsection 9.4.1.2 should the offsite power fail (for operator convenience only). The system also automatically switches to this mode of operation upon receiving a Control Room ventilation high-radiation signal or a safety injection signal from either Unit 1 or Unit 2. The radiation signal originates from radiation monitors which sample the Control Room intake air vents. Radiation detectors of this type are discussed in Section 11.5.

The iodine absorbers are manufactured from impregnated activated carbon and are used for radioiodine compound removal.

The iodine adsorbers construction and efficiencies comply with NRC Regulatory Guide 1.52 (See Appendix 1A(B)). Additional data concerning iodine adsorbers is presented in Table 9.4-6.

Anticipated efficiencies of filters are in accordance with Table 9.4-4.

An analysis of postaccident dose levels in the Control Room is presented in Section 15.6.5.4.

Sufficient redundancy in equipment and power supplies enables the system to sustain a single failure without total loss of function. The emergency pressurization and recirculation filter trains, plus supply and exhaust fans and the associated dampers, are completely redundant. Four parallel air-conditioning units are used to provide a 100 percent (two air conditioners) standby feature.

Redundant, bubble tight inlet isolation dampers are provided to minimize filtered inleakage via a non-running train.

The probability of an electrical fire in the Control Room is low because of the low-voltage and flame-retardant cables used. Fire dampers automatically isolate the affected areas to prevent spreading of the fire and shut off the oxygen supply. The charcoal adsorbers are also provided with a fire protection water spray system. Actuation of the system also activates an alarm in the Control Room. A failure mode and effects analysis is presented in Table 9.4-8.

Alarms in the Control Room alert the operator to any system malfunction so that he can manually actuate the necessary standby units. The maximum operational temperature limit for Control Room instrumentation is 120°F. This temperature is not reached during the short periods of system malfunction prior to the actuation of the standby units.

The Control Room HVAC equipment is located in a seismic Category I structure above the Control Room at elevation 854 feet 4 inches, as shown in Section 1.2, Figure 1.2-6. The Control Room HVAC system is seismic Category I and ANS Safety Class 3. Outside air enters through intake louvers located in the north and south walls of this structure. Concrete walls and slabs, behind the primary wall, are located so as to prevent postulated missiles from entering further into the building. Refer to Figure 6.4-3. Air flow into the building is provided through an opening in one of the tornado missile enclosure walls; the opening is so located that no postulated missile can pass through into the building, thereby protecting the outside air intake ducts and mechanical equipment from tornado-related missiles described in Section 3.3.2. Any damage to an intake louver will not impair the functioning of the air intake.

All safety-related Control Room HVAC equipment required for operation during a loss of offsite power or following a LOCA is powered from the redundant Class 1E buses.

9.4.1.4 Inspection and Testing Requirements

Shop inspection and testing are performed for all equipment, including heating and cooling coils and controls.

The system is initially tested and adjusted for proper flow paths, flow capacities, heating and cooling capacities, mechanical operability, and filter efficiency.

Fans are rated and tested in accordance with the standards of the Air Moving and Conditioning Association (AMCA).

The HEPA filters and iodine adsorbers are tested periodically during plant operation, in accordance with NRC Regulatory Guide 1.52, (see Appendix 1A(B)). Filter units are arranged to facilitate cell replacement.

Heating and cooling coils are tested in accordance with manufacturer's standards. Ductwork and filter are tested in accordance with ANSI N510 and industry standards.

Redundant standby equipment is operated on a cyclic basis to ensure the availability of the equipment.

The inlet bubble tight dampers are shop tested in accordance with ASME CODE AG-1 and then tested periodically as part of the ventilation system, to ensure minimum filtered inleakage into the control room atmosphere when expected to perform their isolation function.

9.4.2 SPENT FUEL POOL AREA VENTILATION SYSTEM

The spent fuel pool area ventilation system is incorporated in the Fuel Building ventilation system and the Primary Plant Ventilation System.

9.4.2.1 Design Bases

The Fuel Building ventilation system is designed to maintain suitable ambient conditions for personnel and equipment during normal plant operations and scheduled shutdowns. Ambient temperature throughout the building is normally maintained as shown in Table 9.4-2. During emergency conditions (LOCA with a loss-of-offsite power) the ambient temperature in the spent fuel pool heat exchanger and pump rooms shall be maintained below 122°F, though the temperature may rise to 129°F for a short duration. System design parameters are shown in Table 9.4-1.

In addition, a slight negative pressure is maintained during normal operation and during a fuel handling accident to prevent the outflow of unfiltered, contaminated air to the environment. Operating the primary plant ventilation exhaust filter trains minimizes the release of radioactive particulate effluents and radioiodine to the environment.

The dissipation of heat from the spent fuel pool cooling pumps during loss of offsite power is accomplished by using emergency fan coil units, which maintain the safety-related pump rooms below the maximum ambient temperature allowed by the equipment design. The emergency fan coil units are supplied with chilled water from the safety-related chilled water system. (See Figure 9.4-12.) The HVAC equipment is designed to operate in and to maintain the required ambient conditions. The supply and exhaust ventilation units are a part of the modular arrangement for the primary plant ventilation system as shown in Figure 9.4-9 and are described in Subsection 9.4.3.

Wherever possible in the Fuel Building, airflow is directed from areas of lower potential radioactivity to areas of higher potential radioactivity.

Tritium concentration buildup and dispersion throughout the Fuel Building is reduced by a local network of exhaust ducts at the spent fuel pool, as shown schematically on Figure 9.4-2. The spent fuel pool area ventilation system is designed to reduce operator dose as a result of the evaporation of tritiated water. Fresh air is supplied over the spent fuel pool and exhausted by exhaust ducts located near the pool surface. This air is passed through a demister and booster fan before entering the primary plant ventilation exhaust system.

Sufficient air changes are provided to maintain the concentration of airborne radioisotopes throughout the area during fuel handling operation below the concentration levels specified in Appendix B to 10 CFR Part 20[a]. (See Section 12.2.2.)

The modular exhaust filtration unit arrangement provides sufficient redundancy in equipment and power supplies for the system to sustain a single active component failure without loss of function. Each emergency fan coil unit is interconnected with the same Class 1E bus as the

a. Design assessment made with the provisions of 10 CFR 20.1-20.601.

equipment it serves. Instrumentation and controls which incorporate visual as well as audible alarms are provided enabling the operator to continuously monitor system performance and manually switch to the standby fan coil unit when required.

The exhaust air is passed through iodine adsorber beds and discharged through the plant ventilation discharge vent, where it is monitored for activity. An in-line detector monitors ventilation air at the outlet of the Fuel Building exhaust header. Data concerning iodine adsorbers are presented in Tables 9.4-4 and 9.4-6. Data concerning prefilters and HEPA filters are presented in Table 9.4-4. The exhaust air ductwork for the fuel pool area is embedded in the walls of the fuel pool, with the bottom of the exhaust registers at approximately six in. above the water surface. Air delivered by the primary plant supply units (supply air outlets located approximately 30 ft) above the pool surface is drawn downward over the surface through the exhaust registers. This arrangement inundates the occupied areas with contaminant-free air, thus minimizing personnel exposure to contaminated air.

Mist eliminators (demisters) are provided in the spent fuel pool exhaust to remove moisture in the exhaust air that exists as a result of evaporation of high-temperature fuel pool water. The entrained water is routed by a drain system to a Fuel Building sump. In addition, care is taken not to impair the visibility for underwater observation by using the following:

1. Maintaining a low sweep velocity over the spent fuel pool surface to prevent ripple propagation.

2. Supplying sufficient air at a temperature calculated to avoid saturation of the air at the spent fuel pool surface.

9.4.2.2 System Description

The Fuel Building ventilation system is shown schematically on Figure 9.4-2. The primary plant air supply system delivers filtered and tempered outside air to each floor of the Fuel Building through a duct distribution system. Each air supply unit consists of a roughing filter, heating and cooling coils, and a 100-percent-capacity fan.

The arrangement of filters and HVAC equipment is in accordance with the requirements of NRC Regulatory Guide 8.8, Revision 2, to achieve ALARA radiation levels. (See Figure 9.4-9.) Air supplied to the spent fuel pool area is heated via an electric heater to prevent fogging at the pool surface. The cooling water supply to the PPV supply unit cooling coils is maintained by the plant ventilation chilled water system only during normal operation. Chilled water from the plant ventilation chilled water system is shown in Figure 9.4-11.

The fuel handling area ventilation system is designed to maintain a slight negative pressure with respect to the environs during normal operation, LOCA or loss of offsite power.

The exhaust system ductwork branches to all areas within the fuel handling area. Exhaust air is discharged to the atmosphere through the plant ventilation discharge vent after passing through the primary plant modular exhaust filtration units. Each modular non-ESF exhaust filtration unit consists of a roughing and two HEPA filters, an iodine adsorber, and a fan. Mist eliminators are provided in the pool exhaust to remove entrained moisture particles from the exhaust air stream. Trapped moisture from these mist eliminators is returned to the spent fuel pool sump via a system of drains.

The Fuel Building exhaust is routed through the Primary Plant Ventilation exhaust system. (See Figures 9.4-2 and 9.4-9).

Emergency fan coil units are provided for the spent fuel pool pump rooms. The fan coil units maintain suitable operating temperatures in the pump rooms during all modes of operation. The cooling medium for these fan coil units is the nuclear safety chilled water system.

A room mounted air handling unit is provided for Storage Room 250A. The air handling unit maintains adequate cooling for non-safety related equipment which is located within the room. The cooling medium for the air handling unit is chilled water which is provided to the cooling coil by a yard mounted air cooled chiller.

9.4.2.3 Safety Evaluation

The fuel handling area ventilation system conforms to the regulatory position established in NRC Regulatory Guide 1.13.

The system complies with the environmental and components design criteria, the qualification testing provisions, and the intent of the system design criteria promulgated in NRC Regulatory Guide 1.140. Even if the system fission-product removal features failed to function properly, the exclusion area boundary doses resulting from a postulated fuel handling accident would still be well below the exposure guidelines of 10 CFR Part 100. (See Section 15.7.4.3 for an analysis of a postulated fuel handling accident.)

During an abnormal condition, as the spent fuel pool water temperature increases from the normal operating temperature the evaporation rate from the pool increases. See Table 9.4-7. For this purpose, mist eliminators are provided in the exhaust from the spent fuel pool areas. The charcoal and the HEPA filters are protected from free water particles by these demisters and from high humidity by the mixing of the Fuel Building exhaust air with the Auxiliary and Safeguards buildings exhaust air.

The Fuel Building is completely separate from the Containment, and its function is totally independent of the reactor operation; it is not required for safe plant shutdown in the event of a severe natural phenomenon or LOCA. Therefore, during a LOCA the normal Fuel Building (i.e., primary plant) ventilation system is shut down, but it remains in operation after a fuel handling accident.

Alarms are provided to alert operators of equipment malfunction and to facilitate the startup of standby units. Radiation levels are continuously monitored. (See Sections 11.5 and 12.3.4).

During fuel handling operations, the system is sized to maintain the concentrations of airborne radioisotopes below the concentration levels specified in Appendix B to 10 CFR Part 20[a]. (See Section 12.2.2.) For a discussion of a fuel handling accident, see Section 15.7.4.3.

The primary plant ventilation system is described in Section 9.4.3.

The Fuel Building air exhaust ductwork is ANS Safety Class 3, seismic Category I, and NNS, seismic Category II. The emergency fan coil units, which are located in the safety related pump rooms, are seismic Category I and ANS Safety Class 3. The ventilation failure modes are

presented in Table 9.4-9. The spent fuel pool exhaust fans are non- nuclear safety related and non-seismic. Operation of these fans is not required to limit Fuel Building exfiltration during normal operation. The Operator verifies the exhaust fans automatically trip following a safety injection signal and manually trips the fans as required.

9.4.2.4 Inspection and Testing Requirements

Shop inspection and testing are performed for all equipment, including heating and cooling coils and controls.

The system is initially tested and adjusted for proper flow paths, flow capacities, heating and cooling capacities, mechanical operability, and filter efficiency.

Fans are rated and tested in accordance with the standards of the Air Moving and Conditioning Association (AMCA).

Heating and cooling coils are tested in accordance with manufacturer's standards. Ductwork and filter are tested in accordance with ANSI N510 and industry standards.

Redundant standby equipment is operated on a cyclic basis to ensure the availability of the equipment.

9.4.3 AUXILIARY BUILDING AND RADWASTE AREA VENTILATION SYSTEM

9.4.3.1 Design Bases

Auxiliary Building (controlled access area) and radwaste area ventilation system is designed to maintain suitable and safe ambient conditions for operating equipment and personnel during normal plant operation.

During normal plant operation and during and after a LOCA, a slightly negative pressure is maintained. The dissipation of heat during abnormal conditions is accomplished by using emergency fans or fan coil units which maintain the safety-related equipment rooms that need cooling below the maximum permissible ambient temperature. The HVAC equipment is designed to operate in and to maintain the required ambient conditions. The supply and exhaust ventilation filtration units are a part of the modular arrangement for the primary plant (controlled access area) ventilation system as shown on Figures 9.4-9 and 9.4-2.

Ambient temperatures throughout the building are normally maintained as shown in Table 9.4-2. During loss of offsite power and LOCA, safety related equipment rooms equipped with emergency fans or fan coil units are kept below 122°F though the temperature may exceed 122°F for a short duration. However, the maximum indoor temperature during loss of offsite power and LOCA is expected to be kept below 131°F. Other system design conditions are presented in Tables 9.4-1 and 9.4-2.

The ductwork layout is arranged so that in areas where airborne radioactivity may be present, airflow is directed from areas of lower potential radioactivity toward areas of higher potential radioactivity. The waste gas evaporator and waste gas compressor areas are provided with direct exhaust duct connections to the exhaust filter units, while the supply air originates from the surrounding areas of lesser possible radioactivity.

The gas decay tanks are located in separate, closed compartments; flexible hose duct connections are provided for attachment to a manifold in the exhaust systems when it is necessary to ventilate these compartments. The compartments are not ventilated unless personnel access is required. The compartments can then be loaded manually and individually onto the ventilation system.

The Recycle Holdup Tanks (see Section 9.3.4) are also located in separate compartments. Should the flexible diaphragms in the tanks rupture, the hydrogen gas released would be contained in the rooms until the ventilation system has removed it or diffusion has reduced it below the lower explosive limit. All penetrations are sealed, except between compartments, and electrical equipment is explosion proof above the elevation 830 feet in the compartments.

The airflow quantities are adequate to ensure that the concentration of radioisotopes in all areas, including the radwaste area, is below the concentration levels specified in Appendix B to 10 CFR Part 20[a].

Emergency fan coil units are provided for cooling the centrifugal charging pump and component cooling water pump rooms. The units are sized to remove the heat dissipated by the pumps and are powered from the same electrical safety-related buses as the components they serve. The cooling units are supplied with chilled water from the plant safety chilled water system (as shown on Figure 9.4-12) during normal operation, during a LOCA, and after a loss of offsite power.

Each emergency fan coil unit is interconnected so that it starts with the equipment it serves. The system is provided with sufficient instrumentation to enable the operator to continuously monitor the system performance and manually switch to the standby units when required.

Differential pressure measurement equipment is provided to ensure that a slightly negative pressure with respect to the environs is maintained during normal operation and during and after a LOCA in the building, preventing the outward leakage of unfiltered contaminated air to the environment.

Exhaust air is passed through HEPA filters and iodine adsorber beds contained in the modular exhaust filtration units of the Primary Plant Ventilation System. Its activity is monitored prior to discharge to the atmosphere. The estimated annual radioactivity released from this source is discussed in Section 11.3.

9.4.3.2 System Description

The Auxiliary Building (controlled access area) and radwaste area ventilation system is shown schematically on Figure 9.4-2. The primary plant air supply system delivers filtered and tempered outside air to each floor of the Auxiliary Building through a duct distribution system. A minimum of three air supply units of the Primary Plant Ventilation System are normally useed for ventilation of the controlled access areas of the Auxiliary, Fuel and Safeguards Buildings.

Each air supply unit consists of a roughing filter, electric heating and cooling coil, and a fan. The arrangement of filters and HVAC equipment is in accordance with the requirements of NRC Regulatory Guide 8.8, Revision 2, to achieve ALARA radiation levels. Chilled water is supplied to the supply air handling units by the plant ventilation chilled water system, as shown by Figure 9.4-11. This system provides chilled water only during normal operation. The Auxiliary

Building ventilation system is designed to maintain a slightly negative pressure with respect to the environs during normal operation and during and after a LOCA. The exhaust system ductwork branches to all areas within the Auxiliary Building. Wherever possible, airflow is directed from areas of lower potential radioactivity to areas of higher potential radioactivity. Unit heaters are provided in specific areas for operator comfort. Exhaust air is discharged to the atmosphere through the plant ventilation discharge duct after filtration through the modular exhaust filtration units. The Auxiliary Building ventilation duct is monitored by an in-line radiation monitor at the outlet of the ventilation duct header. Each modular non-ESF exhaust filtration unit consists of a roughing and two HEPA filters, an iodine adsorber, and a fan.

Fan coil units are provided for the charging and Component Cooling Water pump rooms in the Auxiliary Building. The fan coil units maintain pump room temperature at a suitable level during all modes of pump operation. All emergency fan coil units are seismic Category I and ANS Safety Class 3 and use chilled water supplied by the nuclear safety chilled water system as a cooling medium. The positive displacement charging pump fan coil unit is non-safety-related and is supplied cooling water from the plant ventilation chilled water system.

The Fuel Building, Auxiliary Building, Safeguards buildings, and Containment purge supply and exhaust units are all part of a modular ventilation system. This system consists of twelve identical non-ESF exhaust ventilation units, four ESF exhaust units, and eight air supply units. Each of the non-ESF exhaust ventilation units consists of a roughing filter, HEPA filter, iodine adsorber, HEPA filter, and fan. Each of the eight air supply units consists of a roll filter, heating and cooling coils, and a fan. The total capacity provided is more than required for normal operation; thus adequate redundancy is provided to prevent a single supply or exhaust fan failure from affecting system operation. In addition, supply fans CPX-VAFNAV-17 through CPX-VAFNAV-22 are interlocked with exhaust fans CPX-VAFNCB-09 through CPX-VAFNCB-20 to prevent a possible pressurization of the plant in the event of insufficient exhaust fan capacity. Supply fans CPX-VAFNAV-23 and 24 are not interlocked with any of the exhaust fans.

During a Loss of Offsite Power, the non-ESF exhaust fans trip automatically causing the interlocked supply fans to also trip. The operator manually trips any remaining operating supply fans and energizes one (or two) Non-ESF exhaust fans to ensure monitoring of effluents (see Section 11.5.2.6.3).

During a LOCA, in either Unit 1 or Unit 2, all supply and exhaust fans trip automatically on the safety injection signal while the ESF exhaust fans start automatically. This trip is redundant and meets the single active failure criteria. This will ensure the primary plant ventilation system maintains a slight negative pressure in the negative pressure envelope during all modes of operation.

The supply and exhaust requirements for each building is shown on Figure 9.4-9. Interconnections between the primary plant modular ventilation system and all other buildings are shown on Figure 9.4-13.

Chilled water for the cooling coils is provided by the plant ventilation chilled water system. (See Section 9.4E). Heating coils are electric. The exhaust system is seismic Category I up to the fan discharge. The air supply system is non-nuclear safety seismic Category II except for the fan discharge gravity dampers that shall be safety Class 3 and seismic Category I.

9.4.3.3 Safety Evaluation

The reliability and safety of the Auxiliary Building ventilation system is ensured by the following features:

1. Instrumentation and controls which incorporate audible and visual alarms in the Control Room facilitate continuous monitoring of performance and alert the operator in the event of system malfunction.

2. Standby supply units can be remotely actuated from the Control Room.

3. Deleted

4. The Auxiliary Building exhaust system is of ANS Safety Class 3 and seismic Category I, and NNS, seismic Category II design.

5. Failure modes for dampers are set so that they do not render the system inoperable.

6. Adequate cooling of safeguard equipment is ensured by the operation of the emergency fan coil units.

7. Exhaust air is passed through iodine adsorber filter beds prior to its discharge. It is also monitored by an in-line radiation monitor at the ventilation duct outlet.

8. The safety Class 3 and seismic Category I gravity dampers at the fan discharge of the primary plant ventilation system and the Auxiliary Building Ventilation Equipment Room and the redundant trips of the supply and exhaust fans as discussed in Section 9.4.3.2 will insure the negative pressure envelope boundary.

The Auxiliary Building and radwaste area ventilation exhaust is located in the primary plant ventilation equipment room at elevations 873 ft 6 in. and 886 ft 6 in. of the Auxiliary Building, as shown in Section 1.2, Figure 1.2-35. The Auxiliary Building is a seismic Category I structure, and the air inlet and duct leading to the Auxiliary Building ventilation system are designed to appropriate seismic requirements and to withstand the tornado loads and tornado-related missile conditions described in Section 3.3.2. The intake louvers are facing west, toward the Turbine Building.

9.4.3.4 Inspection and Testing Requirements

1. See Subsection 9.4.2.4.

2. The HEPA and iodine adsorbers of the non-ESF units are tested periodically during plant operation in accordance with NRC Reg. Guide 1.140 (See appendix 1A(B)). Filter units are arranged to facilitate filter replacement.

9.4.4 TURBINE BUILDING AREA VENTILATION SYSTEM

9.4.4.1 Design Bases

The Turbine Building is provided with outside air as required for ventilation and heat removal. (See Figure 9.4-3 and Tables 9.4-1 and 9.4-2.)

The Turbine Building ventilation system is designed to provide sufficient heat to ensure freeze protection of equipment during the winter months and to provide a sufficient airflow quantity in order to remove the heat generated by equipment located in the Turbine Building. The Turbine Building ventilation system is not required to operate following a DBA or loss-of-offsite-power condition.

9.4.4.2 System Description

Each Turbine Building ventilation system consists of vane axial supply fans, roof-mounted exhaust fans with gravity dampers, and propeller-type wall ventilators used for exhaust. The Turbine Building ventilation system also consists of centrifugal supply fans and its ductwork for ventilating the heater drain pump and condensate pump areas. Mushroom type caps are used on some of the roof exhausters which prevent a direct upblast of air. In addition, there are related ductwork, dampers, and intake louvers, which complete the ventilation system. The HVAC system is designed in accordance with the requirements of NRC Regulatory Guide 8.8, Revision 2, to achieve ALARA radiation levels.

The Turbine Building has exhaust roof ventilators installed on a removable steel plate over the Main Feed Pump area to provide additional exhaust capacity. The air is drawn from the louvers on the west end of the building and from areas below.

The supply fans and ductwork are used to supply outdoor air to the areas below elevation 830 ft. including the condensate pump area and Unit 2 heater drain pump area.

The ventilation air from the lower elevations stratifies through grating to the grade elevation above 803 ft 0 in. (the mezzanine level of the Turbine Building). Outdoor air enters the mezzanine level of the Turbine Building through multiple combination louver-dampers mounted on the west wall of the Turbine Building. The outdoor air which enters the mezzanine level and the air that rises from the lowest elevation of the Turbine Building is exhausted through wall exhausters and roof-mounted exhaust fans.

Locations of the exhaust vents are shown on Figures 1.2-3, 1.2-5, 1.2-24 and 1.2-29. Discharge temperature is dependent on ambient temperature conditions.

Electric unit heaters are provided as a source of auxiliary heat required during the normal modes of plant operation or during maintenance outages. The electric unit heaters are located around the perimeter of the mezzanine level.

The Turbine Building lube oil reservoir room is provided with a wall-mounted intake louver and exhaust fan to adequately ventilate this area. Ambient air from the Turbine Building is drawn through the intake louver and exhausted to the outside by a wall ventilator.

The turbine oil room at elevation 778 ft 0 in. is provided with a wall-mounted intake louver and an exhaust system including a wall-mounted fan and associated ductwork. Ambient air is drawn from the basement elevation of the Turbine Building through a wall-mounted intake louver by a wall-mounted exhaust fan. The exhaust fan is connected to an exhaust duct which discharges the room air to the atmosphere.

The thyristor voltage regulator room is provided with an air-cooled, wall-mounted, air-conditioning unit and backup A/C unit which recirculates and cools the ambient air of this solid state electronics equipment room. The Digital Control Room is provided with an air-cooled, air-conditioning unit and a back-up air-conditioning unit which recirculates and cools the ambient air of this solid state electronic equipment room. A wall-mounted intake louver and a wall-mounted exhaust fan are provided to ensure adequate ventilation of the machinery room. Air is drawn from the basement elevation of the Turbine Building through the intake louver by the wall-mounted exhaust fan. The exhaust fan discharges the ambient air back into the basement elevation of the Turbine Building.

All enclosed stairwells in the Turbine Building are pressurized by roof-mounted supply fans during normal operation to prevent smoke infiltration in the event of a fire.

9.4.4.3 Safety Evaluation

The reliability of the Turbine Building ventilation system is ensured by the following features:

1. The use of multiple roof exhaust fans with excess capacity and power supplies enables the system to sustain a single failure of any single fan without adversely affecting the overall ventilation of the Turbine Building.

2. Instrumentation and controls incorporating audible and visual annunciation facilitate continuous monitoring of the system performance and alert the operator in the event of a system malfunction.

3. There is no treatment of the Turbine Building exhaust air because the discharge from the condenser vacuum pumps is routed to the plant vent stack via the controlled access modular exhaust units and therefore these turbine building exhausts are not release points. (See Section 10.4.2.2.)

No radiation detectors monitor the Turbine Building exhaust per se, but the vacuum pump discharge is monitored. (See Section 10.4.2.3.) The Turbine Building ventilation system is non-seismic Category I and non-nuclear safety.

9.4.4.4 Inspection and Testing Requirements

1. Shop inspection and testing are performed for all equipment, including heating and cooling coils and controls.

2. The system is initially tested and adjusted for proper flow paths, flow capacities, heating and cooling capacities, mechanical operability, and filter efficiency.

3. Fans are rated and tested in accordance with the standards of the Air Moving and Conditioning Association (AMCA).

4. Ductwork is tested in accordance with industry standards.

9.4.5 ENGINEERED SAFETY FEATURES VENTILATION SYSTEM

The engineered safety features (ESF) ventilation system is incorporated as parts of the building ventilation systems (Building exhaust and emergency fan coil units).

9.4.5.1 Design Bases

The Safeguards Building ventilation system is designed to maintain suitable and safe ambient conditions for operating equipment and personnel during normal plant operation and maintains portions of the building under a slightly negative pressure with respect to the environment during all modes of operation.

In addition, the operation of the ESF exhaust filtration unit maintains a slightly negative pressure in portions of the building with respect to the environment following a LOCA and loss of offsite power, thus reducing the radioactive effluent released to the environment. Each of the four ESF exhaust filtration units consists of a demister, heater, HEPA filter, iodine adsorber, HEPA filter and fan, and complies with the environmental and components design criteria, the qualification testing provisions, and the intent of systems design criteria promulgated in NRC Regulatory Guide 1.52 as discussed in Section 1A(B). The removal of heat from ESF pump rooms during these periods is accomplished by using emergency fan coil units which maintain the rooms below the maximum environmental qualification temperature (122°F) though the temperature may rise to 129°F for a short duration. Ventilation system equipment is designed to operate in and to maintain the required ambient conditions.

The supply and exhaust ventilation units are a part of the modular arrangement for primary plant (controlled access area) ventilation systems, as shown on Figure 9.4-9 and as described in Section 9.4.3.

Ambient temperatures throughout the building during normal operation are presented in Table 9.4-2.

ESF exhaust fans are automatically energized from their Class 1E bus following a LOCA in either Unit 1 or Unit 2; thus a slightly negative pressure is provided within the Auxiliary, Fuel Handling and portions of the Safeguards Buildings. During a loss of offsite power at least one or more non-ESF exhaust fan(s) may be manually energized as required after 120 seconds (See Table 8.3-2) to achieve and maintain the slight negative pressure.

The system is provided with sufficient redundancy in equipment and power supplies to enable it to sustain a single active component failure without loss of function. Fire dampers in the exhaust duct are seismically qualified to remain open during an SSE. The redundant modular ESF exhaust fans are powered from redundant electric power trains.

To remove the large quantities of heat which are rejected from the motor driven ESF pumps following a DBA, each of the pump compartments is equipped with emergency fan coil units. Each emergency fan coil unit is powered from the same Class 1E bus as the equipment it serves. Also, each unit is connected in such a way as to start with the equipment it serves. The auxiliary cooling units are supplied with chilled water from the safety-related chilled water system shown on Figure 9.4-12 and described in Subsection 9.4F.

The system is provided with sufficient instrumentation to enable the operator to continuously monitor the system performance and manually switch to the standby units when required.

Exhaust air from the Safeguards Building is monitored by an in-line radiation monitor to assess its activity level and is passed through iodine adsorber beds prior to discharge to the atmosphere.

Portions of the exhaust system and all the emergency fan coil units are seismic Category I and ANS Safety Class 3 design. The main air supply system is non-nuclear safety class and seismic Category II. Also see Subsection 9.4.3 for ventilation of ESF equipment.

9.4.5.2 System Description

The Safeguards Building ventilation system is shown schematically on Figure 9.4-4. The primary plant air supply system delivers filtered and tempered outside air to each floor of the safety feature area through a duct distribution system. Each supply unit consists of a roughing filter, heating and cooling coils, and fan. The arrangement of filters and HVAC equipment is in accordance with the requirements of NRC Regulatory Guide 8.8, Revision 2, to achieve ALARA radiation levels. Chilled water is supplied to the cooling coils from the plant ventilation chilled water system as shown by Figure 9.4-11. This system provides chilled water only during normal operation.

Each of the four ESF exhaust filtration units consists of a demister, heater, HEPA filter, iodine adsorber, HEPA filter and fan, and complies with the environmental and components design criteria, the qualification testing provisions, and the intent of systems design criteria promulgated in NRC Regulatory Guide 1.52 as discussed in Section 1A(B).

The supply and exhaust system ductwork branches to all areas within the safety feature area. Wherever possible, airflow is directed from areas of lower potential radioactivity to areas of higher potential radioactivity. Unit heaters are provided in specific areas for operator comfort and to maintain ambient temperature above the lower limits for some process piping.

Emergency fan coil units, each of which comprises a water cooling coil and fan section, are provided for all compartments which contain motor driven ESF pumps and the electrical areas. The cooling medium for these fan coil units is chilled water, which is supplied by the nuclear safety chilled water system.

The emergency fan coil units are operated in conjunction with the equipment that they cool.

9.4.5.3 Safety Evaluation

The reliability and safety of the ESF area ventilation system is ensured by the following features:

1. The modular arrangement as shown on Figure 9.4-9 (with excess capacity and redundant power supplies) enables the system to sustain a loss of a single fan without loss of function. Standby fans can be remotely actuated from the Control Room. Each motor driven ESF pump room has a 100-percent-capacity cooling unit to remove the heat dissipated by the equipment.

2. Emergency fan coil units are supplied with chilled water from the safety chilled water system. Each chilled water system train and the auxiliary cooling units it serves are powered from the same train Class 1E bus.

3. Instrumentation and controls which incorporate audible and visual alarms in the Control Room facilitate continuous monitoring or system performance and alert the operator if the system malfunctions.

4. Failure modes for isolation valves and dampers are set so that their failure does not render the system inoperable. The appropriate safeguards building exhaust dampers (Figure 9.4-4 Sh. C) are abandoned in place in the OPEN position.

5. The cooling units, and safety features chilled water system are of seismic Category I and ANS Safety Class 3 design. The exhaust system is of ANS Safety Class 3, seismic Category I, and NNS, seismic Category II, design. The supply system components are designed to appropriate seismic criteria, where necessary, to negate the possibility of these components interfering with the operation of safety-related components.

6. Exhaust air from portions of the Safeguards Building is passed through iodine adsorber beds prior to its discharge. Data concerning the ESF and non-ESF iodine adsorbers are presented in Table 9.4-6.

7. In case of a DBA and subsequent operation of the ECCS, the emergency fan coil units operate on a recirculation basis and provide cooling of the ECCS pumps.

8. Fire dampers in the exhaust duct which are Seismic Category II are qualified to remain open during SSE.

9. Exhaust air from portions of the Safeguards Building is monitored by an in-line radiation monitor to assess its activity level and is passed through iodine adsorber beds prior to discharge to the atmosphere.

9.4.5.4 Inspection and Testing Requirements

1. See Subsection 9.4.2.4.

2. The HEPA and iodine adsorbers of the ESF Units are tested periodically during plant operation in accordance with NRC R.G 1.52 (See appendix 1A(B)). Filter Units are arranged to facilitate filter replacement.

REFERENCES

1. 10 CFR Part 50, Appendix A, GDC 2, Design Basis for Protection Against Natural Phenomena.

2. 10 CFR Part 50, Appendix A, GDC 4, Environmental and Missile Design Bases.

3. 10 CFR Part 50, Appendix A, GDC 5, Sharing of Structures, Systems, and Components.

4. 10 CFR Part 50, Appendix A, GDC 19, Control Room.

5. 10 CFR Part 50, Appendix A, GDC 60, Control of Releases of Radioactive Materials to the Environment.

6. 10 CFR Part 50, Appendix A, GDC 64, Monitoring Radioactivity Releases.

7. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water, Steam, and Radioactive Waste Containing Components of Nuclear Power Plants, Revision 3, February 1976, U.S. Nuclear Regulatory Commission.

8. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976, U.S. Nuclear Regulatory Commission.

9. NRC Regulatory Guide 1.13, Fuel Storage Facility Design Basis, Revision 1, December 1975, U.S. Nuclear Regulatory Commission.

10. NRC Regulatory Guide 1.52, Design, Testing, and Maintenance Criteria for Atmosphere Cleanup System Air Filtration and Adsorption Units of Light Water Cooled Nuclear Power Plants, Revision 1, July 1976, U.S. Nuclear Regulatory Commission.

11. NRC Regulatory Guide 1.78, Assumptions for Evaluating the Habitability of a Nuclear Power Plant Control Room During a Postulated Hazardous Chemical Release, June 1974, U.S. Nuclear Regulatory Commission.

12. NRC Regulatory Guide 1.95, Protection of Nuclear Power Plant Control Room Operators Against an Accidental Chlorine Release, Revision 1, January 1977, U.S. Nuclear Regulatory Commission.

13. ANSI N510 1975, Testing of Nuclear Air Cleaning Systems.

14. ANSI N509 1976, Nuclear Power Plant Air Cleaning Units and Components.

15. International Report NAA-SR-100, Leakage Characteristics of Openings for Reactor Housing Components.

16. American Air Filter Topical Report AAF-TR-7101, Design and Testing of Fan Cooler Filter Systems for Nuclear Applications, February 1972.

17. Test Code for Air Moving Devices, Air Moving and Conditioning Association (AMCA) 210 67.

18. ANSI N101.1 1972, Efficiency Testing of Air Cleaning Systems Containing Devices for Removal of Particles.

19. American Society of Heating, Refrigeration, and Air Conditioning Engineers, Guide and Data Handbooks Vol. I-IV.

TABLE 9.4-1
DESIGN CONDITIONS - OUTDOORS

Summer Design Temperature, °F	102 DB/80 WB
Extreme Summer Temperature, °F[(a)]	113 DB/79 WB
Winter Design Temperature, °F	20
Extreme Winter Temperature, °F	4
Annual Average Temperature, °F	66 DB
Daily Average Temperature Range, °F	20 DB
Latitude	32
Elevation, Ft.	810
Average Wind Velocity, mph (Approximately)	15

a) This value was utilized to evaluate the impact of extreme outdoor weather condition on safety related HVAC System. This is not a design condition.

TABLE 9.4-2
DESIGN CONDITIONS - INDOOR

(Sheet 1 of 4)

Building or Area		Normal Plant Conditions		
		Maximum DB(°F)	Minimum DB(°F)[(a)]	Relative Humidity (%)
Auxiliary Bldg. (AB)				
1.	Elevator Machine Room	122	40	[(b)]
2.	Heat Exchanger Area (el. 790'-6")	122	40	[(b)]
3.	Valve & Piping Area (el. 790'-6")	122	40	[(b)]
4.	Operating Valve Rooms (el. 822')	122	40	[(b)]
5.	Auxiliary Steam Drain Tank Equipment Room (el. 790'-6")	122	40	[(b)]
6.	Valve Rooms (el. 810'-6" and 831'-6")	122	40	[(b)]
7.	Operating Valve Room (el. 862'-6")	122	40	[(b)]
8.	ESF Pump Rooms	122	40	[(b)]
9.	Boric Acid Storage Tank Rooms	104	65	[(b)]
10.	All other areas	104	40	[(b)]
Electrical and Control Bldg. (ECB)				
1.	Control Room	80	70	35-50
2.	Mechanical Equipment Rooms	104	40	[(b)]
3.	UPS & Distribution Rooms	104	40	[(b)]
4.	Uncontrolled Access Area	104	40	[(b)]
5.	Battery Rooms	104	70	[(b)]
6.	Chiller Equipment Areas (el. 778')	122	40	[(b)]
7.	All other Areas	104	40	[(b)]

TABLE 9.4-2
DESIGN CONDITIONS - INDOOR

(Sheet 2 of 4)

Building or Area	Normal Plant Conditions		
	Maximum DB(°F)	Minimum DB(°F)[a]	Relative Humidity (%)
Fuel Handling Building			
1. Stairs (el. 810'-6" & 841')	122	40	[b]
2. Spent Fuel Pool Cooling Heat Exchanger & Pump Rooms	122	40	[b]
3. New Fuel Storage Pit	122	40	[b]
4. All other Areas	104	40	[b]
Reactor Containment Bldg. (RCB)			
1. Outside Missile Barrier	120	50	[b]
2. Inside Missile Barrier	120	50	[b]
3. CRDM Shroud (Air Temperature)	163 (CRDM Air Handling Unit inlet) (Unit 1) 163 (outlet) (Unit 2) 120 (inlet)	50 50	[b] [b]
4. CRDM Platform	140	50	[b]
5. Detector Well and Reactor Cavity	135[c]	50	[b]
6. Reactor Coolant Pipe Penetrations	200	50	[b]
Safeguards Bldg. (SGB)			
1. MS & FW Piping Area	104	40	[b]
2. Valve Isolation Tank Rooms (el.790'-6")	122	40	[b]

TABLE 9.4-2
DESIGN CONDITIONS - INDOOR

(Sheet 3 of 4)

Building or Area	Normal Plant Conditions		
	Maximum DB(°F)	Minimum DB(°F)[a]	Relative Humidity (%)
3. Valve Rooms (el. 790'-6")	122	40	[b]
4. RHR/Cont. Heat Exchanger Rooms (el. 790'-6")	122	40	[b]
5. ESF Pump Rooms	122	40	[b]
6. All Other Areas	104	40	[b]
Diesel Generator Bldg. (DGB)			
1. Diesel Generator Rooms	122	40[d]	[b]
2. Day Tank	122	40[d]	[b]
3. Equipment Rooms	130	40[d]	[b]
Service Water Intake Structure (SWIS)			
1. SWIS - All Areas	127	40	[b]
Turbine Bldg. (TB)			
1. Switchgear Area	104	40	[b]
2. ERF-Computer Battery Room	104	40	[b]
3. CAS Room, CPU & UPS Rooms	75	68	35-50
4. Battery Rooms & HVAC Equipment Rooms	104	40	[b]
5. Office and Service Area A/C	80	68	[b]
6. Laboratories	80	68	35-50
7. Alternate RCA Access-Unit 2	80	68	[b]
8. Tool Room	80	65	[b]
9. All Other Areas	122	40	[b]

TABLE 9.4-2
DESIGN CONDITIONS - INDOOR

(Sheet 4 of 4)

Building or Area		Normal Plant Conditions		
		Maximum DB(°F)	Minimum DB(°F)[a]	Relative Humidity (%)
Miscellaneous Building (MB)				
1.	Alternate Access Point (AAP) Bldg.	80	68	(b)
2.	Circulating Water Intake Structure Chlorination Bldg.	115	40	(b)
3.	SWIS - Chlorination Building	115	40	(b)
4.	Switchyard Relay House	80	68	(b)
5.	Security Office	80	68	(b)
6.	Battery Rooms on Turbine Building Deck	87	67	(b)

a) The winter minimum design temperature in non-air conditioned areas, including Diesel Generator Area, Electrical Area and Service Water Intake Structure of 40oF. Except for the Diesel Generator Area, this temperature is maintained by the HVAC systems without considering equipment heat released during startup, normal plant operation, hot standby and refueling operation, except for the Electrical Area Ventilation System (see Section 9.4C.3). There is no heating from the ventilation systems available during loss of offsite power. The heat released by safety-related equipment is sufficient to keep the temperature of the room above the freezing point. The Boric Acid Storage Tanks contain concentrated boric acid solution. To preclude recrystalization, these tanks are required to be located in a room maintained at 65°F or higher.

b) Uncontrolled

c) Occasional temperature spikes of 175°F DB may occur. The reactor cavity exhaust temperature is normally 15 degrees higher (e.g. 150°F and 190°F respectively).

d) Temperatures in the Diesel Generator Area are maintained with the help of the Diesel keep warm system.

TABLE 9.4-3
EQUIPMENT LINE-UP FOR CONTROL ROOM AIR-CONDITIONING SYSTEM
MODES OF OPERATION

Information shown on Figure 9.4-1.

TABLE 9.4-4
FILTER DESIGN PARAMETERS

(Sheet 1 of 2)

Prefilters	
Atmospheric Dust Spot Efficiency Rating per ASHRAE[a] Standard 52	80-95%
Initial (clean) resistance (pressure drop) at rated air flow	0.65 in. W.G. (Max)
Final (dirty) resistance (pressure drop) at rated air flow	1 to 2 in. W.G.
Rated Air Flow	1300 CFM/CELL at 0.4 in. W.G.
HEPA Filters	
Initial (clean) resistance (pressure drop) at rated air flow	1.0 in. W.G.
Final (dirty) resistance (pressure drop) at rated air flow	3.5 in. W.G.
Final (dirty) resistance (pressure drop)at rated air flow for Containment Preaccess Filtration Units	2.0 in. W.G.
Rated Air Flow	1500 CFM/CELL at 1.2 in. W.G.
Filtration Efficiency: Penetration through complete filter (medium, frame and gasket) when operated at rated capacity and tested with thermally generated DOP[b] of uniform 0.3 micron droplet size, in accordance with Edgewood Arsenal Manual.	a) less than 0.05% (For Control Room HVAC Filtration and Recirculation Units) b) less than 1.0% (For all other Filtration Units)
Iodine Adsorber	
Material	Impregnated coconut charcoal (8 x 16 mesh granules)

TABLE 9.4-4
FILTER DESIGN PARAMETERS

(Sheet 2 of 2)

Efficiency (Assigned)	a) Engineered Safety Feature (ESF) (R.G. 1.52) Primary Plant Filtration Units 95% Control Room HVAC Filtration and Recirculation Units 99% b) Non-ESF (Normal Ventilation Air Filtration and Adsorption Units) (R.G. 1.140) Primary Plant Filtration Units 90% Hydrogen Purge Filtration Units 90% Containment Preaccess Filtration Units 90%
Leakage acceptance test requirement	R-11 (or approved alternate)
Penetration, % allowable for in place tests.	0.05% (For Control Room HVAC Units), 1.00% (For all other Filtration Units)
Average Residence Time	0.25 sec per 2" of adsorbent bed
Maximum Permissible Loading	2.5 mg of total Iodine (radioactive plus stable) per gram of activated carbon

a) American Society of Heating Refrigerating and Air-Conditioning Engineers

b) Dioctyl Phthalate

TABLE 9.4-5
CONTAINMENT AIR RECIRCULATION UNIT DATA

(Sheet 1 of 2)

Number installed	four
Number required to operate	three
Type of coil	Chilled water
Fan capacity, ft^3/min per cooling unit	65,000
Conditions	
Containment temperature, °F (maximum)	120
Total pressure, psig	Atmospheric
Cooling medium	Ventilation chilled water system

Neutron Detector Well Unit Data

Number installed	two
Number required to operate	one
Type of coil	Chilled water
Fan capacity, ft^3/min per cooling unit	13,100
Temperature, °F (average)	150 (reactor cavity exhaust)
Cooling medium	Ventilation chilled water system

Control Rod Shroud Ventilation Unit Data

Number installed	two
Number required to operate	one
Type of Coil	Chilled Water
Fan capacity, ft^3/min per fan unit	Approximately 60,000

TABLE 9.4-5
CONTAINMENT AIR RECIRCULATION UNIT DATA

(Sheet 2 of 2)

Temperature, °F (outlet air average)	163 maximum
Cooling Medium	Ventilation Chilled Water System

Containment Preaccess Filtration Subsystem

Number installed	two
Number required to operate	two
Fan capacity, ft^3/min per filtration unit	Approximately 15,000
Temperature, °F (air average)	120 maximum

Normal Containment Purge Supply and Exhaust Subsystem

Number installed (ppv supply/exhaust units)	one/two
Number required to operate	one/two
Fan capacity, f^3/min per fan unit	30,000/15,000
Temperature, °F (air average)	60

TABLE 9.4-6
IODINE ADSORBERS

Filter	Bed Depth (in. thick)	Residence Time (sec)	Temperature (°F)	Inlet Humidity (Percent)
Preaccess filtration, two 15,000 ft^3/min units	4	0.50	120 (Max.)	(a)
Controlled access air cleanup, twelve 15,000 ft^3/min units	4	0.50	120 (Max.)	(a)
Controlled access ESF air cleanup, four 15,000 ft^3/min units	4	0.50	155 (Max.)	70 (Max.)
Hydrogen purge, two 1000 ft^3/min units (one standby)	4	0.50	160 (Max.)	70 (Max.)
Control room emergency pressurization, two 800 ft^3/min units (one standby)	4	0.50	120 (Max.)	70 (Max.)
Control Room emergency filtration, two 8000 ft^3/min units (one standby)	4	0.50	86 (Max.)	(b)

a) Humidity not controlled.

b) Controlled by Control Room A/C units

TABLE 9.4-7
EFFECT OF FUEL POOL EVAPORATION AS A FUNCTION OF WATER TEMPERATURE

Air Temperature (R)	Air Velocity (ft/min)	Pool Water Tempurature (F)	Air Relative Humidity (Percent)
115	200	120	58
115	200	130	68
115	200	140	80
115	200	150	97
115	200	160	100
115	200	170	100
115	200	180	100

TABLE 9.4-8
FAILURE MODE AND EFFECT ANALYSIS OF CONTROL ROOM HVAC SYSTEM

(Sheet 1 of 3)

Component	Malfunction	Effect on System Operation	Comments
1. Control Room air conditioning units	a. Unit stops. b. Unit fails to start.	No effect	Sufficient redundancy provided for 100-percent standby feature.
2. Control Room makeup air supply fan units	a. Fan stops. b. Fan fails to start	No effect	Two 100-percent-capacity makeup air supply fan units are provided, one operating and one standby.
3. Control Room exhaust fan units	a. Fan stops. b. Fan fails to start.	No effect	Two 100-percent-capacity exhaust fan units are provided, one one operating and one standby.
4. Control Room kitchen and toilet exhaust fan units	a. Fan stops. b. Fan fails to start.	No effect	Two 100-percent-capacity kitchen and toilet exhaust fan units are provided, one operating and one standby.
5. Control Room emergency pressurization air supply fan units	a. Fan stops. b. Fan fails to start	No effect	Two 100-percent capacity emergency pressurization air supply fan units are provided, one operating and one standby.
6. Control Room emergency filtration system	Filter train is clogged or inoperative.	No effect	Two 100-percent-capacity emergency filtration systems are provided, one operating and one standby.

TABLE 9.4-8
FAILURE MODE AND EFFECT ANALYSIS OF CONTROL ROOM HVAC SYSTEM

(Sheet 2 of 3)

Component		Malfunction	Effect on System Operation	Comments
7.	Control Room emergency pressurization filtration system	Filter train is clogged or inoperative.	No effect	Two 100-percent-capacity emergency pressurization filtration systems are provided, one operating and one standby.
8.	Control Room HVAC system isolation and modulating dampers	Fails to operate properly	No effect	Failure position of dampers will not impede operation of the components listed above.
9.	Radiation detectors	Fails to operate	No effect	Control Room HVAC system automatically switches to emergency recirculation if any one of the detectors fail to operate.
10.	Control Room emergency pressurization air-conditioning system	Fire isolated.	No effect	Effected air-conditioning system is Two 100-percent-capacity filtration systems are provided, one operating and one standby.
11.	Control Room emergency filtration air-conditioning system	Fire isolated.	No effect	Effected air-conditioning system is Two 100-percent-capacity filtration systems are provided, one operating and one standby.

TABLE 9.4-8
FAILURE MODE AND EFFECT ANALYSIS OF CONTROL ROOM HVAC SYSTEM

(Sheet 3 of 3)

Component		Malfunction	Effect on System Operation	Comments
12.	Electric control	Power failure	No effect	Redundant controls, powered from redundant Class 1E power sources, are provided.
13.	HVAC dampers	Air failure	No effect	Air accumulators are provided on dampers CPX-VADPOU-14, 15, all others fail in the position required to operate in Emergency Recirculation Mode.
14.	Fire in an area of the Control Room complex[a]		No effect	Fire dampers are provided which isolate the affected area, preventing the spread of fire.

a) Low probability of electrical equipment fire.
Control Room complex is provided with fire protection equipment (portable chemical extinguishers).

TABLE 9.4-9
FAILURE MODE AND EFFECT ANALYSIS OF FUEL HANDLING AREA HVAC SYSTEM

(Sheet 1 of 2)

Component	Malfunction	Effect on System Operation	Comments
1. Fuel handling area modular supply units	a. Fan fails to start. b. Fan stops. c. Cooling and heating coils fail to operate properly.	No effect	The modular supply arrangement provides sufficient redundancy in order to maintain perferred air flow patterns.
2. Primary plant exhaust fans	a. Fan fails to start b. Fan stops. c. Filter train clogged.	No effect	The modular exhaust arrangement provides sufficient redundancy in order to maintain preferred air flow patterns.
3. Primary plant exhaust	Loss of fuel pool cooling	No effect	Increased humidity will not affect PPV exhaust unit charcoal because of mixing of exhaust air from safeguard and auxiliary buildings.
4. Primary plant ventilation system	Loss of offsite power	No effect	Emergency fan coil units are provided for Spent Fuel Pool Cooling Pump rooms. Concrete heat sinks are adequate for heat dissipation until ventilation is restored.

TABLE 9.4-9
FAILURE MODE AND EFFECT ANALYSIS OF FUEL HANDLING AREA HVAC SYSTEM

(Sheet 2 of 2)

Component	Malfunction	Effect on System Operation	Comments
5. Spent fuel pool area fans	a. Fan fails to start. b. Fan stops.	No effect	Two 50-percent-capacity fuel pool fans are provided. Equipment not required for safe shutdown or fuel handling accident.
6. Spent fuel pool area electric heaters	Fails to operate	No effect	The rate of evaporation from fuel pool will decrease.
7. Spent fuel pool demister	Decrease in efficiency	No effect	During fuel handling this air is exhausted through the Primary Plant Ventilation exhaust system.

TABLE 9.4-10
FAILURE MODE AND EFFECT ANALYSIS OF SAFETY RELATED DAMPERS IN HVAC SYSTEMS

(Sheet 1 of 3)

BUILDING	FUNCTION & TYPE OF DAMPERS	MALFUNCTIONS	FAILURE MODE	EFFECT ON SYSTEM OPERATION	COMMENTS
1. Fuel, Safeguards and Auxiliary Buildings	a. Room isolation dampers	1. Fail to operate properly 2. Instrument air fails 3. Loss of offsite power.	Fail Closed (F.C.)	No effect	Failure mode of damper is such that it will not impede the operation of the system. Where required, the dampers are locked open.
	b. Not used				
	c. Non-ESF filtration systems isolation dampers: 1. Opposed blade type 2. Gravity type	1. Fail to operate properly 2. Instrument air fails. 3. Loss of offsite power.	See Fig. 9.4-9	No effect	Units are provided each with its own set of dampers. The opposed blade dampers are operated manually.
	d. ESF Filtration System isolation dampers: 1. Opposed blade type 2. Gravity type	1. Fail to operate properly. 2. Instrument air failure. 3. Loss of offsite power.	F.O. for ESF filter units	No effect	Failure mode of damper is such that it will not impede the system operation. Units are provided each with its own set of dampers. The opposed blade dampers can be operated manually if required.
2. Control Room	a. Outside air inlet bubble tight type dampers CPX-VADPOU 14 & 15	1. Fail to operate properly. 2. Instrument air fails.	F.C.	No effect	Failure mode of damper will not impede the system operation. Two-100% redundant outside air intakes each with its own damper are provided. Air accumulators are provided.

TABLE 9.4-10
FAILURE MODE AND EFFECT ANALYSIS OF SAFETY RELATED DAMPERS IN HVAC SYSTEMS

(Sheet 2 of 3)

BUILDING	FUNCTION & TYPE OF DAMPERS	MALFUNCTIONS	FAILURE MODE	EFFECT ON SYSTEM OPERATION	COMMENTS
	b. Control Room exhaust and toilet exhaust fans isolation dampers: 1. Opposed blade type 2. Gravity type CPX-VADPMU-05, 06 CPX-VADPOU-27, 28 CPX-VADPGU-28, 29, 30, 31	1. Fail to operate properly. 2. Power failure to motor operated dampers. 3. Gravity dampers not operable. 4. Loss of offsite power.	Failure Mode Fail As Is (FAI)	No effect	Failure mode of dampers will not impede the system operation. Two-100% capacity fan units each unit with its own set of dampers are provided. Motorized dampers are powered from emergency power supply.
	c. Emergency filtration and emergency pressurization filtration unit and fans isolation 1. Opposed Blade type CPX-VADPOU-41 & 42 2. Gravity type CPX-VADPGU-01, 02, 03, 04 3 Bubble Tight type CPX-VADPOU-22 & 23	1. Fail to operate properly 2. Power failure to motor operated dampers. 3. Gravity dampers not operable. 4. Loss of offsite power.	CPX-VADPOU--41, 42 F.O. CPX-VADPOU-22, 23-FAI	No effect	Motored dampers are powered from emergency power supply. Failure mode of dampers will not impede the system operation. Two-100% capacity fan and filtration units with its own set of dampers.
	d. Air conditioning units isolation dampers: 1. Opposed blade type 2. Gravity type CPX-VADPOU-10, 11, 12, 13 CPX-VADPOU-5, 6, 17, 18	1. Fail to operate properly. 2. Instrument air fails. 3. Gravity dampers not operable. 4. Loss of offsite power.	F.O.	No effect	Four-50% capacity air conditioning units with its own set of dampers are provided. Two units are operating and two units are on standby. Failure mode of dampers will not impede the operation of the system.

TABLE 9.4-10
FAILURE MODE AND EFFECT ANALYSIS OF SAFETY RELATED DAMPERS IN HVAC SYSTEMS

(Sheet 3 of 3)

BUILDING	FUNCTION & TYPE OF DAMPERS	MALFUNCTIONS	FAILURE MODE	EFFECT ON SYSTEM OPERATION	COMMENTS
	e. Makeup supply air fans isolation dampers opposed blade type CPX-VADPOU-16 & 19	1. Bubble Tight type CPX-VADPOU-16 & 19 2. Power failure to motor operated dampers. 3. Instrument air 4. Loss of offsite power.		No effect	Two-100% capacity fan units its own damper set, one unit operating and one on standby. Failure mode of dampers will not impede the operation on standby. Failure mode of dampers will not impede the operation of the system. Power to motor operated dampers is supplied from emergency power. Inlet bubble tight dampers provide redundant isolation.
	f. Damper in recirculation lines opposed blade type CPX-VADPOU-25	1. Fail to operate properly 2. Instrument air fails 3. Loss of offsite power.	FC	No effect	Failure mode of damper is such that it will not impede the operation of the system. It is desired to fail closed and that is the safe mode of failure during accident conditions. Damper can be operated manually if required.

APPENDIX 9.4A - CONTAINMENT VENTILATION SYSTEMS

Separate ventilation systems are furnished for the Containments of Units 1 and 2. The Containment ventilation systems, as shown on Figure 9.4-5, consist of the Containment air recirculation and cooling Control Rod Drive Mechanism (CRDM) ventilation, neutron detector well cooling, reactor coolant pipe penetration cooling, preaccess filtration, Containment pressure relief, and Containment purging systems. Specific data for the various systems are presented in Table 9.4-5.

9.4A.1 DESIGN BASES

9.4A.1.1 Containment Air Recirculation and Cooling System

The environment within containment is controlled by the containment recirculation and cooling system during startup, power operation and hot standby.

The Containment air recirculation and cooling system maintains Containment ambient temperature at or below 120°F during normal plant operation. The Containment air recirculation and cooling system together with the CRDM ventilation system is designed to remove all heat generated by equipment within the Containment, with the exception of the reactor coolant pump motors, which are cooled separately by the CCWS.

The system operates during a loss of offsite power and provides mixing of fresh air during refueling and shutdown, but it is not operative following a DBA. Postaccident cooling is provided by the Containment Spray System (CSS). (See Section 6.2.2.)

9.4A.1.2 Control Rod Drive Mechanism Ventilation System

The CRDM ventilation system maintains temperatures within the CRDM shrouds in accordance with NSSS requirements. The CRDM cooling coils supplement the general containment cooling provided by the Containment Air Recirculation and cooling system. Specific data concerning this system are presented in Table 9.4-5.

9.4A.1.3 Neutron Detector Well Cooling System

Neutron detector well cooling units prevent neutron detectors from exceeding their temperature limitations. (See Table 9.4-5.) These units also provide cooling for reactor shield wall concrete and nozzle supports. Two 100-percent capacity cooling units are provided for redundancy, and cooling water is provided by the plant ventilation chilled water system.

9.4A.1.4 Containment Preaccess Filtration

The need for safe, periodic, or emergency access to the Containment necessitates the use of Containment preaccess filtration. This filtration system provides air circulation and filtering throughout all Containment areas. Containment preaccess filtration equipment reduces the concentration of fission product particulate activity in the containment atmosphere prior to personnel entering the Containment. Iodine adsorbers are designed to reduce the airborne halogen activity. The design is based on estimates of reactor coolant leakage supplied by the NSSS manufacturers. The reduced level of airborne particulate by HEPA filters permits access

to the Containment for brief time intervals, without exceeding the limitation of 100 mrem/week. The containment preaccess filtration system minimizes the need for containment purging.

9.4A.1.5 Containment Purge Supply and Exhaust System

Containment purge supply and exhaust equipment satisfies the prerequisites for safe, prolonged access to the Containment following shutdowns.

The system is utilized during shutdown only; it is not used during startup, power or hot standby operations. Since the system will not be used during power operations, its operation is not considered in the ECCS backpressure computation presented in Section 15.6.5.

Tempered fresh air is supplied and circulated throughout the Containment and ventilation equipment is designed to maintain Containment temperature above 50°F during the winter season. (The supply and exhaust filtration units are a part of the modular arrangement for the primary plant ventilation system as shown on Figure 9.4-9.) Exhaust air is passed through an exhaust filtration unit which consists of a prefilter, two HEPA filters, and an iodine adsorber and discharged into the atmosphere through the plant ventilation vent. The rate of release ensures that the offsite concentration of radioactive material is within the limits of 10 CFR Part 20. A discussion of the estimated annual radioactivity discharges from purging is presented in Section 11.1. Specific data on the various systems are shown in Table 9.4-5.

9.4A.1.6 Containment Pressure Relief System

The pressure relief system is used intermittently during startup, power or hot standby operations. The pressure relief system is the only system that can provide an open path from the containment atmosphere to the environs during startup, power or hot standby conditions, and is designed in accordance with Branch Technical Position CSB 6-4. A 3-inch restrictive flow orifice plate is provided so that the containment pressure relief system may be open continuously.

The Containment pressure relief system is designed to relieve excess Containment pressures caused by temperature transients or air leakage from pneumatic actuators during normal operation. The Containment pressure relief system is connected to the Containment purge exhaust duct outside Containment, downstream of both system Containment isolation valves and upstream from the primary plant exhaust filters. This connection ensures that any air from the Containment is filtered prior to release to the atmosphere, via the plant ventilation vent.

The Containment pressure relief system is designed to reduce the Containment pressure from a maximum of 1.5 psig to atmospheric pressure.

9.4A.1.7 Reactor Coolant Pipe Penetration Cooling System

The reactor coolant pipe penetration cooling in primary shield wall system is designed to dissipate heat conducted to the supports by the reactor coolant lines utilizing forced convection to prevent dehydration of concrete.

9.4A.2 SYSTEM DESCRIPTION

9.4A.2.1 Containment Recirculation and Cooling System

During normal operation, air in the Containment is recirculated and maintained at or below 120°F. Four cooling units are provided, each sized for 33 1/3 percent (65,000 scfm) of the normal duty cooling load. Standby units are manually actuated on receipt of a failure alarm in the Control Room. Fans are provided with a connection to the Class 1E buses to preclude loss of cooling caused by loss of offsite power. The cooling medium is water provided from the plant ventilation chilled water supply during normal operation and loss of offsite power, because the plant ventilation chillers are also supplied from the Class 1E buses during loss of offsite power conditions. Instruments are provided to continuously monitor air temperature, humidity, and pressure within the Containment during all phases of reactor operation.

The Containment recirculation units together with the CRDM ventilation system remove the entire Containment heat load during normal operation and loss of offsite power conditions including the CRDM heat load. The neutron detector well heat load is removed by a separate set of coils.

The cooled air is distributed throughout the Containment by the supply ductwork with no return ductwork provided. Instead, the warm air rises through various openings in the floors and returns to the suction side of the fan coil units.

During Cold Shutdown and Refueling (or when the core is offloaded) the Containment Recirculation Cooling System may be used to cool containment. Removable silencers are provided in Unit 2 to reduce noise levels during this mode of operation. (See FSAR Table 17A-1).

9.4A.2.2 Control Rod Drive Mechanism Ventilation System

Containment air is drawn through the CRDM shrouds and through the CRDM cooling coils and discharged to the containment atmosphere. Two 100-percent capacity fans are provided. The standby unit, isolated by dampers in the duct, is manually actuated on receipt of a failure alarm in the Control Room. In the event normal power supplies are interrupted and the reactor is maintained at hot standby, the CRDM fans are operated from the emergency power supply. The cooling medium to the CRDM cooling coils is provided by the plant ventilation chilled water system. The CRDM ventilation system does not operate in the event of a DBA.

9.4A.2.3 Neutron Detector Well Cooling System

The neutron detector well and nozzle support cooling system is a closed loop system. The air is cooled, and directed up around the reactor from the underside of the vessel. Part of the air cools the neutron detectors, and part of the air is directed to the nozzle supports. The air is returned to the fan via piping embedded in the concrete. Two 100-percent capacity cooling units are provided with each consisting of a fan, two cooling coils, and inlet and outlet dampers. Both units are connected to redundant emergency diesel generator buses to preclude loss of cooling in case of a loss of offsite power. A temperature monitoring device continuously monitors air temperature in the return air ducts of the detector well system.

9.4A.2.4 Containment Preaccess Filtration

Preaccess filtration equipment can be operated prior to reactor shutdown for Containment access. Air inside the Containment is recirculated by two 50-percent capacity filtration units. Each filtration unit includes a fan, a roughing filter, two HEPA filters, and one iodine adsorber. Both units are required to operate for the design iodine removal capacity. The equipment is designed to remove fission product iodine gas as well as radioactive particles to permit prompt access to the Containment. Ductwork connected to the suction side of the preaccess filtration unit extends to all levels of the Containment to ensure that the containment atmosphere is uniformly decontaminated. Air is discharged at the suction side of the Containment recirculation system, which supplies the air through ductwork and guarantees proper mixing. The minimum total airflow with clean filters is 30,000 ft3/min, with an iodine adsorber having an assigned radioiodine removal efficiency in compliance with Regulatory Guide 1.140 (See Appendix 1A(B)). Filters are also designed in accordance with the criteria of Regulatory Guide 1.140. Charcoal adsorbers are provided with a fire protection water deluge system. The system operation is described in section 9.5.1.4.2.2.b.2. Data concerning preaccess filtration equipment are presented in Tables 9.4-6, 9.4-4, and 9.4-5.

9.4A.2.5 Containment Purge Supply and Exhaust System

The purge supply and exhaust units are part of the primary plant ventilation systems modular arrangement (see Figure 9.4-9). Description of each system is as follows:

1. Containment Purge Supply

   Fresh, outside air is passed through a filter and heating and cooling coils and discharged near the recirculation supply fans in the Containment. Thermostatically controlled electric heating coils are provided in the purge supply duct to maintain minimum temperatures for personnel comfort during shutdown.

   One 30,000 scfm filter heating and cooling coil plenum and a fan are required. The containment purge supply isolation valves are designed to close automatically on detection of high radiation levels by the containment air radiation monitor or on a Phase A isolation signal as discussed in sections 6.2.4 and 7.3.

2. Containment Purge Exhaust

   Exhaust air is drawn through a prefilter, two HEPA filters, and an iodine adsorber and exhaust fan, and released to the atmosphere via the plant ventilation vent.

   Two filter plenums with associated fans are required to maintain design purge rate. The air is monitored before discharge to the environment, to limit the concentration of contaminants as required by 10 CFR Part 20. The containment purge exhaust isolation valves are designed to close automatically on detection of high radiation levels by the Containment air radiation monitor or on a phase A isolation signal as discussed in Sections 6.2.4 and 7.3.

9.4A.2.6 Containment Pressure Relief

The Containment pressure relief system is manually operated from the Control Room. A high Containment pressure alarm, located in the Control Room, alerts the operators to pressures exceeding normal operating limits. The alarm has a set point dictated by the maximum pressure permitted within the Containment during normal operation.

The pressure relief line, as shown on Figure 9.4-6, is connected to the Containment purge exhaust system; the containment pressure relief system requires only one penetration. The exhaust air is decontaminated by the primary plant modular filtering trains prior to being discharged to the plant ventilation discharge duct.

The only potential leakage path from the containment building in existence during power operation is the 18 in. in diameter pressure relief line, and it will only open to the atmosphere if it has been manually actuated in response to compartment pressurization. Since it is anticipated that the line will be closed most of the time, it is not included in the ECCS backpressure analysis presented in Section 15.6.5. Moreover, even if the pressure relief line were open, it would have an insignificant effect on the pressure calculated during the core reflood transient because fast-closing isolation valves are installed.

Both pressure relief Containment isolation valves automatically close if high radiation is detected by the containment particulate-iodine-gaseous monitor, on a phase A isolation signal, or on containment ventilation isolation, as discussed in Sections 6.2.4 and 7.3. The isolation valves are 18 in. in diameter and will close within three seconds after receiving the signal.

The valve and actuator design bases include consideration or buildup of Containment pressure for the LOCA break spectrum and relief line flows as a function of time up to and during valve closure.

A debris screen is provided inboard of the inside containment isolation valve. The screen is designed to seismic category II criteria, and is placed in the flow path of the containment pressure relief system for protection of the isolation valves from debris.

Additional requirements for the design of the system isolation valves are presented in Section 6.2.4. Leak testing of containment pressure relief isolation valves is specifically addressed in Table 6.2.4-2.

9.4A.2.7 Reactor Coolant Pipe Penetration Cooling System

The Reactor Coolant Pipe Penetration Cooling System supplies high velocity cooled air to each of the reactor coolant hot cold leg pipe penetration supports. This cool air flow removes heat conducted to the supports by the pipe to keep the pipe tunnel concrete temperature below 200°F. The system consists of four fans (2000 scfm each), fan outlet isolation dampers, and ductwork. The system is divided into two subsystems: each subsystem having two 100-percent capacity fans supplying the four penetrations associated with two reactor coolant loops. During normal plant operation, two fans are operative, i.e., one in each subsystem. The fans are the centrifugal, direct drive type, each equipped with differential pressure indicating switches, three position remote manual hand switches located on the main control board in the Control Room, and parallel blade, counterweight operated gravity dampers at each outlet to prevent free wheeling of

the standby fan. Standby fans are automatically actuated on receipt of a fan trip signal in the Control Room.

The Containment recirculation system supplies tempered air to the immediate area surrounding the fan inlets. Air is drawn in by the operating fan and supplied through ductwork directly to each pipe penetration. The warm air is forced out by the supply air and rises up, thus being cooled and recirculated by the Containment recirculation system. The ductwork is arranged with balancing dampers to ensure that equal amounts of air are supplied to the four supports it connects.

The Reactor Coolant Pipe Penetration Cooling System is not required to operate following a loss of offsite power or a DBA.

9.4A.3 SAFETY EVALUATION

The reliability and safety of the Containment ventilation systems is ensured by the following features:

1. Redundancy in equipment and power supplies with the exception of the preaccess filtration system enables the systems to sustain a single active component failure without total loss of function.

2. Instrumentation and controls which incorporate audible and visual alarms in the Control Room facilitate continuous monitoring of the system performances and alert the operator for system malfunctions.

3. Standby units can be remotely actuated from the Control Room manually.

4. Failure modes for equipment shutoff valves and dampers are set so that their failure does not render the system inoperable.

5. The systems inside the Containment are seismic Category II.

6. Penetrations are ANS Safety Class 2 and seismic Category I.

7. Exhaust air is passed through iodine adsorber beds prior to its discharge.

9.4A.4 INSPECTION AND TESTING REQUIREMENTS

1. Shop inspection and testing are performed for all equipment, including heating and cooling coils and controls.

2. The system is initially tested and adjusted for proper flow paths, flow capacities, heating and cooling capacities, mechanical operability, and filter efficiency.

3. Fans are rated and tested in accordance with the standards of the Air Moving and Conditions Association (AMCA).

4. Heating and cooling coils are tested in accordance with the manufacturer’s standards. Ductwork is tested in accordance with industry standards.

5. Redundant standby equipment is operated on a cyclic basis to ensure the availability of the equipment.

6. The HEPA and charcoal adsorber of the containment filtration units are tested in accordance with NRC Reg. Guide 1.140 [See appendix 1A(B) under NRC Reg. 1.140], except for the following:

   The containment preaccess filtration units are in place leak tested after the initial installation and after any major modification or repair. In place leak testing after HEPA or carbon replacement and the periodic in place leak tests for containment preaccess filtration units is not performed, as long as a) periodic visual inspections and pressure drop determinations across HEPA and charcoal filters are performed during scheduled shutdowns and b) charcoal adsorbers are sampled and laboratory tests are made to confirm performance at scheduled refueling shutdowns.

7. The containment purge and pressure relief system isolation valves' testing and inspection requirements are discussed in Section 6.2.4.

8. The pressure relief containment isolation valves are tested to determine the availability of the isolation function and the leak rate during reactor operation.

APPENDIX 9.4B - SERVICE WATER INTAKE STRUCTURE VENTILATION SYSTEM

9.4B.1 DESIGN BASES

The heating and ventilation system is designed to maintain the ambient temperatures within the Service Water Intake Structure as indicated in Table 9.4-2 for normal operation. During emergency modes of operation, the temperature may rise to 132°F.

9.4B.2 SYSTEM DESCRIPTION

The Service Water Intake Structure heating and ventilating system consists of the following:

1. Service water pump area exhaust system

2. Intake structure heating system

3. Diesel-driven fire pump room exhaust system

The Service Water Intake Structure ventilation system is shown schematically on Figure 9.4-7.

The service water pump area exhaust system consists of eight 50-percent-capacity, propeller-type fans, which are mounted on a missile-protected exhaust plenum located in the overhead of the draft damper. The fans are started automatically by locally mounted temperature switches or manually by the intake structure operator.

A high temperature alarm in the Control Room is provided.

Outside air is drawn through grated openings with outside air intake screens in the floor of the intake structure located near the base of the pump motors. This air picks up heat dissipated by the service water pump motors. Then the stratified air is discharged by exhaust fans to the outdoors through a missile-protected opening in the side of the intake structure.

The diesel-driven fire pump room exhaust system consists of one 100- percent-capacity fan (rated at 8000 scfm), a gravity and fire damper, and associated exhaust ductwork. Air is exhausted from the diesel-driven fire pump room and adjacent electric fire pump area into the ventilation plenum located in the overhead of the intake structure. Outside air is drawn through grated openings in the floor of the intake structure by the exhaust system and passes over the equipment to pick up the heat generated by the fire protection components. This opening provides a means for the ventilation and is also used to isolate the pump room in the event of a fire.

Six thermostatically controlled unit heaters, rated at 7.5 kW each, are provided to maintain the ambient temperature conditions within the Service Water Intake Structure to satisfy the winter design conditions. The primary reason for the use of unit heaters in the intake structure is to provide adequate freeze protection. The unit heaters are located around the perimeter of the intake structure and contain a sufficient degree of design margin to provide a reasonable working environment. The unit heaters are classified non-safety-related and are seismically supported and positioned so as not to damage any safety-related equipment housed within the intake structure.

The service water pump area exhaust system is not required to operate during the winter months. Intermittent operation of exhaust fan(s) to remove odors or purge the intake structure is left up to the discretion of the intake structure equipment operator(s).

9.4B.3 SAFETY EVALUATION

The reliability and safety of the Service Water Intake Structure ventilation system is ensured by the following features:

1. Redundancy in equipment and power supplies enables the system to sustain a single active component failure without loss of function.

2. Instrumentation and controls which incorporate audible and visual alarms in the Control Room facilitate continuous monitoring of system performances and alert the operator to system malfunctions.

3. The system is of seismic Category I and ANS Safety Class 3 design.

The Service Water Intake Structure is a seismic Category I structure; the HVAC equipment is designed to seismic Category I requirements and to withstand the tornado loads and tornado-related missile conditions described in Section 3.3.2.

9.4B.4 INSPECTION AND TESTING REQUIREMENTS

See Subsection 9.4C.1.1.

APPENDIX 9.4C - MISCELLANEOUS VENTILATION SYSTEMS

9.4C.1 DIESEL GENERATOR BUILDING VENTILATION SYSTEM

The Diesel Generator Building ventilation system has two functions: removal of excessive heat and provide heating during the winter months.

The design indoor conditions during normal plant conditions with Diesel Generator not in operation can be found in table 9.4-2. The temperature ranges during normal (while the Diesel Generators are being tested) and emergency conditions with Diesel Generator in operation are as follows:

Diesel Generator Rooms	40°F - 122°F
Day Tank Rooms	40°F - 122°F
Equipment Rooms	40°F - 130°F

Each diesel generator compartment is furnished with a ventilation system sufficient to remove heat from the area. The system consists of a set of intake and exhaust louvers and exhaust fans and is seismic Category I and ANS Safety Class 3. No redundancy is provided because the diesels are already redundant. Figure 9.4-4 shows the system schematically.

The Diesel Generator Building Ventilation System (DGBVS) consists of the following:

1. Diesel Generator Rooms Ventilation Sub-System (DGRVS)

2. Day Tank Rooms Ventilation Sub-System (DTRVS)

3. Heating System (Unit Heaters)

The DGRVS consists of (2) two independent trains, one (1) train for each Diesel Generator Area (DGA) (Train A and Train B). Each train consists of (4) four 25 percent capacity vaneaxial exhaust fans, dampers, louvers, ductwork and instrumentation necessary for the operation and control of the system.

The DTRVS consists of (2) two independent trains, one train for each DGA. Each train consists of one (1) 100 percent capacity centrifugal exhaust fan, dampers, louvers, ductwork and instrumentation necessary for the operation and control of the system.

Two (2) electric unit heaters are provided for each DGA (Train A and Train B). Instrumentation and controls are provided for each electric unit heater.

The diesel generator compartment ventilation system air exhausts are located in the roof of the respective compartments, and intake louvers are on the south and north walls of Unit 1 and Unit 2, respectively. The ventilation systems are seismic Category I and designed to withstand the tornado loads and tornado-related missiles described in Section 3.3.2.

The Unit Heaters are non safety but are required to meet the Seismic Category II requirements, so that in the event of an earthquake, the functioning of any Seismic Category I system or component is not adversely affected. The heaters are not required to be powered from class 1E power source because these are not required to operate during emergency condition.

Based on outside air temperature readings, selected diesel generator exhaust fans may be disabled to ensure design temperatures are maintained during winter months. In addition, certain intake louvers will be closed during the approximate time period the diesel generator exhaust fans are disabled.

To be certain that the ventilation fans are operating when the diesels are operating, the required number of fans in each diesel generator compartment start automatically on receipt of diesel generator start signal. During normal plant operation, the plant ventilation is operated manually.

Monitoring of the conditions in the diesel rooms is accomplished with the following Control Room instrumentation:

1. High diesel room temperature alarm

2. Ventilation fan motor tripped alarm

Since the diesel generator building ventilation (DGBV) fans are scheduled for annual maintenance, manual initiation of the diesel generator inoperable indication is provided, in accordance with R. G. 1.47.

Fans are powered from the same safety-related electrical bus as the diesel that is being ventilated.

The reliability and safety of the DGBVs is ensured by the following features:

1. Redundancy in equipment and power supplies enables the system to satisfy a single active component failure without loss of functions.

2. Instrumentation and controls which incorporate audible and visual alarms in the Control Room facilitate continuous monitoring of system performances and alert the operator to system malfunctions.

9.4C.1.1 Testing and Inspection

Shop inspection and testing are performed for equipment, including heating and controls.

The system is initially tested and adjusted for proper flow paths, flow capacities, heating and ventilating capacities, and mechanical operability.

Fan are rated and tested in accordance with the standards of the Air Moving and Conditioning Association (AMCA).

Heating coils are tested in accordance with manufacturer’s standards. Ductwork is tested in accordance with industry standards.

9.4C.2 MAIN STEAM AND FEEDWATER PIPING AREA VENTILATION SYSTEM

The main steam and feedwater piping area ventilation system is designed to provide adequate ventilation, to provide sufficient heat in order to ensure freeze protection of equipment during the winter months, and to remove the heat dissipated by the main steam and feedwater piping and any additional loads such as lighting or motor operators located in this area. This system is designed to maintain the ambient temperatures of the main steam and feedwater areas at between 40°F and 104°F during all normal modes of plant operation. This system is not required to function following a DBA or following a loss-of-offsite power. The Operator verifies the normal ventilation fans automatically trip following a safety injection signal and manually trips the fans if required.

Safety related equipment, located adjacent to the main steam and feedwater piping area are protected against the effects of main steam and feedwater pipe breaks by the use of isolation dampers located where the supply and exhaust ducts penetrate the main steam and feedwater piping area. These isolation dampers are Safety Class 3 and seismic Category I.

The main steam and feedwater piping area ventilation supply system consists of two 50-percent-capacity vane axial fans connected in parallel and located downstream of a supply housing. The supply housing consists of a roll-type roughing filter, cooling coil, and electric heating coil. The supply air is provided fromt he Electrical Area Ventilation system as described in Section 9.4C.3.

Gravity dampers are located downstream of each fan.

Air is provided to the main steam and feedwater ventilation supply system via a tempered air intake from the Electrical Area Ventilation System as described in Section 9.4C.3. The normal electrical area supply exceeds the capacity of the main steam and feedwater area system. Therefore, excess air is normally exhausted through the original fresh air intake. Because air that is drawn into the main steam and feedwater ventilation supply is tempered i.e., warmer in winter, cooler in summer than outside air, the performance of the main steam and feedwater ventilation system is enhanced. Outside air is drawn in, if required, by the supply fans through a louvered, roof-mounted penthouse. The air is filtered, tempered (heated or cooled), and then passed through the vane axial fans and supplied to the main steam and feedwater areas through a duct distribution system.

Each of the roll filters consists of a glass fiber media and casing with a motor operated media advance. The cooling coil is supplied with chilled water from the non-nuclear-safety-related plant ventilation chilled water system. The fans are direct-drive, vane axial type with totally enclosed air over motors. The vane axial fans are provided with adjustable pitch blades to facilitate balancing of the system. The electric heating coil located within the supply housing is the flange type.

The exhaust system consists of two 50-percent roof-mounted exhaust fans and associated ductwork. Air from the area is exhausted and passes through pneumatically actuated dampers, the exhaust fans, and gravity dampers, and is then discharged to atmosphere. The exhaust fans are the direct-drive, propeller type, complete with gravity dampers and a weatherproof housing. An in-duct radiation monitor is provided as described in Section 11.5.

The main steam and feedwater piping area ventilation system is shown schematically on Figure 9.4-4.

9.4C.3 ELECTRICAL AREA (SAFEGUARDS) VENTILATION SYSTEM

The electrical area (safeguards) ventilation system is designed to provide adequate ventilation in order to provide sufficient heat to ensure freeze protection of equipment during the winter months and to remove the heat dissipated by the electrical switchgear and associated electrical equipment (including lighting) contained within this area. The design indoor conditions during normal plant conditions can be found in Table 9.4-2.

The system is designed to maintain the ambient temperature between 40°F and 104°F during all normal modes of plant operation. This system is not required to function following a DBA. The Operator verifies the normal ventilation fans automatically trip following a safety injection signal and manually trips the fans if required. In addition, emergency fan coil units are provided to maintain the safeguards switchgear train A and train B areas below the maximum ambient temperature of 122°F in the event of loss-of-offsite power or a LOCA although the temperature may rise up to 129°F for a short period of time during a LOCA. The ductwork associated with the emergency fan coil units is interconnected with electrical area ventilation system ductwork. This ductwork, as well as the emergency fan coil unit, is classified Safety Class 3 in accordance with ANSI N18.2 and seismic Category I. The coils on the fan coil units are supplied by the safety chilled water system.

The tempered supply air is distributed via a common stairwell, open doors and through a ductwork system to all three electrical area elevations of the Safeguards Building. At elevations 810 ft 6 in. and 852 ft 6 in., emergency fan coil units are interconnected with the supply and exhaust ductwork and isolated from ductwork by Safety Class 3 gravity dampers. The emergency fan coil units are not normally required to operate. The electrical area ventilation exhaust system consists of two 50-percent capacity fans, gravity dampers, and associated ductwork. Air can be exhausted from each of the three electrical area elevations by the roof-mounted exhaust fans and discharged to the atmosphere. The normal exhaust for the Electrical Area Ventilation System is the Main Steam and Feedwater Area Ventilation Supply and its original fresh air intake as described in 9.4C.2, above. Because the system would not be balanced and excessive negative pressure could result during transients which stop the supply fans, the normal electrical area roof mounted ventilation exhaust may not be used when the Main Steam and Fedwater Area Ventilation System is running. One electrical area roof mounted exhaust fan may be run if only one Main Steam And Feedwater Area Ventilation supply fan is running and two Electrical Area Ventilation System supply fans are running.

The electrical area ventilation supply and exhaust system components are similar to the components as described in Subsection 9.4C.2, except for the rated air flow capacity.

The doorway between the electrical area and the equipment room is open to provide tempered air to the room to maintain minimum temperature and provide an exhaust path to the main steam and feedwater area ventilation supply as described above.

The electrical area ventilation system is shown schematically on Figure 9.4-4.

9.4C.4 CONTROL BUILDING UNCONTROLLED ACCESS AREA HEATING, VENTILATION, AND AIR-CONDITIONING SYSTEM

The uncontrolled access area HVAC system is designed to provide adequate ventilation, to provide sufficient heat in order to ensure freeze protection of equipment during the winter months and to remove the heat dissipated by equipment located in cable spreading rooms, battery rooms, and miscellaneous areas. The system maintains the ambient temperature between 40°F min. and 104°F max. during all normal modes of plant operation. It provides sufficient ventilation to the battery rooms to ensure a minimum number of air changes per hour in order to keep the hydrogen concentration of the rooms below the lower flammability limit, i.e., a hydrogen air mixture of 2.0 percent by volume. The battery room exhaust units and associated ductwork are designated ANS Safety Class 3 and seismic Category I and operate in the event of a DBA. Individual room heaters are supplied to maintain the battery rooms at 70°F min.

The supply system consists of two 50-percent-capacity vane axial fans connected in parallel and located downstream of a supply unit. The supply unit consists of a sheet metal housing which contains a roll filter, an electric heater, and a cooling coil which is supplied water from the non-nuclear-safety-related plant ventilation chilled water system. Outside air is drawn through louvered openings and an isolation damper, and passes through the supply unit, vane axial fans, and gravity dampers. The supply air is distributed throughout the Control Building and uncontrolled access areas, which contains cable spreading areas, battery and charger rooms, air compressor areas, and portions of a heating and ventilation equipment room.

The exhaust and recirculation system either returns air to the suction side of the supply unit or exhausts the air directly to atmosphere. This is accomplished through a system of exhaust ductwork located throughout the Control Building and uncontrolled access areas. The exhaust and recirculation system consists of two 50-percent-capacity vane axial fans and gravity dampers, both of which are required to direct the return airflow back to the supply unit or to the atmosphere.

A separate ventilation exhaust system is provided for each of the battery and charger rooms. Each battery room ventilation exhaust system consists of two 100-percent-capacity centrifugal fans, and each battery charger room ventilation exhaust system consists of two 100-percent-capacity axial fans; all the fans have pneumatically actuated isolation dampers located on the suction side of the fans and associated ductwork. Supply air is drawn into the battery and charger rooms from adjacent corridors by one of the two exhaust fans. The exhaust fans are arranged in parallel. The battery and charger room exhaust is discharged directly to the environs through a missile-protected roof exhaust vent.

If the supply air handling unit fails to deliver air due to loss of power or any other reason, make up air for the battery rooms is provided by dampers that open the HVAC equipment room to the atmosphere. The suction created by the battery room exhaust fans draws outside air via the HVAC equipment room and supply ductwork into the battery rooms.

Standby battery room and battery charger room exhaust fans are automatically actuated on receipt of an operating fan differential pressure trip signal. If, after a time delay, the standby fan fails to start, this condition is alarmed in the Control Room.

Electric unit heaters are located in each of the six battery rooms (three per unit). Each unit heater is an integral unit consisting of a propeller-type fan, fin tubular electric heating element,

casing, and thermostat. The battery room ventilation exhaust system is ANS Safety Class 3 and seismic Category I. The unit heaters in the safety related battery rooms are Class 1E.

The uncontrolled access area ventilation system is shown schematically on Figure 9.4-8.

9.4C.5 ELECTRICAL SWITCHGEAR AREA (TURBINE BUILDING) VENTILATION SYSTEM

The electrical switchgear area (Turbine Building) ventilation system requirements are to remove heat generated by the electrical switchgear equipment and to maintain the area ambient temperature between 104°F max. and 40°F min. during all normal mode of plant operation. This system is not required to operate in the event of a DBA.

The switchgear area ventilation requirement during summer operation is 30,000 scfm. The two 50-percent-capacity switchgear area vane axial supply fans, each rated at 15,000 scfm, are required to operate. The two 50-percent-capacity roof-mounted exhaust fans, each rated at 15,000 scfm, are also required during normal operating conditions. Winter operation requires only one vane axial supply fan (15,000 scfm) and one roof-mounted exhaust fan to provide heated ventilation air to the Turbine Building switchgear areas. The electric heating coil automatically operates, depending on the heating demand. The electrical switchgear area is shown schematically on Figure 9.4-3.

9.4C.6 OFFICE AND SERVICE AREA HVAC SYSTEM

The office and service area HVAC system is designed to provide a comfortable environment for operating personnel by maintaining the temperature as indicated in Table 9.4-2 for laboratories or other areas that contain calibration or test equipment. The office and service area HVAC system also prevents odors in the toilets, and locker rooms from diffusing to other occupied areas. Potentially radioactive rooms are exhausted through the plant atmospheric cleanup trains prior to releasing the effluents to the atmosphere. This system is designed to function during all normal modes of plant operation, but it is not required to operate in the event of a DBA or following a loss-of-offsite power. Running of the system affects the performance of the PPVS which exhausts air from a portion of areas served by this system. Redundant and diverse, single active failure proof trips of the Office and Service Area Fans are provided to ensure adequate ESF filtration.

The office and service area HVAC supply system consists of two 50-percent capacity air-handling units each supplying air through a duct distribution system to all the rooms of the office and service area. The air handling units are located in a heating and ventilation equipment room within the control building and consist of a gravity damper, electric heating coil, chilled water cooling coil, and centrifugal fan. Makeup air enters the system through a missile protected inlet. This makeup air is mixed with the return flow from the office and service area rooms and passes through roughing filter. The tempered air is then supplied to the rooms which constitute the office and service area.

The office and service area HVAC system also provides ventilation to the Alternate Access area for Unit 2 personnel to get into the Radiation Control area (RCA). This area is supplemented by the Primary Plant Ventilation system and two (2) fan coil units with auxiliary electric heating. These fan coil units utilize chilled water supplied by the Non-Safety Chilled Water system.

The Alternate Access area tool room is served by a split system air conditioner and electric duct heating coil to maintain the tool room at design condition. The tool room is maintained at a slightly negative pressure by exhausting 400 cfm through the Primary Plant ventiliation exhaust system.

The office and service area exhaust system consists of the following:

1. Toilet exhaust system

2. Filtered recirculation system

The office and service area toilet exhaust system consists of two 50-percent-capacity in-line exhaust fans, gravity dampers, and associated ductwork. The fans draw air from toilet areas, locker room, shower area, and janitor's closet and discharge it to the atmosphere.

The office and service area recirculation system consists of two 50-percent-capacity in-line fans, gravity dampers, and associated ductwork. The fans return air from all areas of the office and service area, except for the toilet and potentially radioactive areas. The fans discharge the return air back to the suction side of the air-conditioning units.

The office and service area is shown schematically on Figure 9.4-14.

9.4C.7 SECURITY OFFICE HVAC SYSTEM

The security office HVAC system is designed to provide a comfortable environment for the security personnel by maintaining the temperature between 68ºF and 80ºF during all normal modes of plant operation. The security office HVAC system also prevents odors in the toilet area from escaping to other occupied areas by exhausting this area directly to the atmosphere. The security office HVAC system is not required to operate in the event of a DBA or following a loss-of-offsite power. The security office HVAC system consists of an air-conditioning unit, roof-mounted toilet exhaust fan, and associated ductwork and accessories, see Figure 9.4-7.

9.4C.8 UNINTERRUPTABLE POWER SUPPLY (UPS) AND DISTRIBUTION ROOMS AIR CONDITIONING SYSTEMS

The UPS and Distribution Rooms Air Conditioning (A/C) System equipment is located at elevation 778'-0" of the Electrical and Control Building. The Fan Coil Units are located at elevation 792'-0" of the Electrical and Control Building. The system's primary function is to provide cooling for Class 1E electrical equipment in the UPS and Distribution Rooms, Trains A & B, Units 1 and 2, (elevation 792'-0") and the mechanical equipment room (elevation 778'-0"). The Fan Coil Units are designed to operate during all modes of plant operation. The air-conditioners are normally on stand-by. However, if a fan coil unit is unavailable, either one of the UPS air-conditioners (Train A or Train B), should be operating.

The Fan Coil Unit portion of the Air Conditioning System is comprised of one 100 percent capacity self-contained Fan Coil Unit. Each unit cools its respective UPS Room. The Heat Sink for the cooling coil is the Safety Chilled Water (SCW) System.

The other portion of the A/C system is comprised of two 100-percent capacity self-contained air-conditioning units located in adjacent rooms that are physically separated by a dividing fire

wall. The redundant A/C units are powered from independent Class 1E buses. The UPS and Distribution Room A/C system is ANS Safety Class 3 and Seismic Category I.

Each A/C unit consists of replaceable-type filters, refrigerant type (R-12) cooling coil, fan, compressor, condenser and all necessary instrumentation and controls. The heat sink on the condenser side of the units is supplied by the Component Cooling Water System (CCW).

The UPS and Distribution Room Air Conditioning system is shown schematically on Figure No. 9.4-15.

9.4C.9 BATTERY AND CHARGING ROOMS AIR CONDITIONING SYSTEM

The Battery and Charging Rooms Air Conditioning System equipment is located on elevation 830'-0" of the Turbine Building deck. The system's primary function is to provide cooling for the battery and charging rooms located on the same deck. The air-conditioning equipment is non-safety, non-seismic and will operate during normal plant operating conditions.

The air conditioning system is comprised of one vertical self-contained packaged unit. It is located in the charging room along with its thermostat. The air conditioning unit consists of replaceable type filters, refrigerant type R-22 cooling coil, heating coil, fan, compressor, condenser and condenser fan, and all necessary instrumentation and controls. The heat sink for the condenser side of the unit is supplied by outside air.

An exhaust fan with an explosion proof motor will provide a continuous exhaust for the battery room while the air conditioning system is operating.

If there is a malfunction of the air conditioning system and/or a level of hydrogen exceeding 2% is detected in the battery rooms, a local alarm will sound and the appropriate annunciator(s) will notify the operator in the control room.

The Battery and Charging Rooms Air Conditioning System is shown schematically on Figure No. 9.4-3.

9.4C.10 HIGH PRESSURE CHEMICAL FEED ROOM VENTILATION SYSTEM (HPCFVS)

The High Pressure Chemical Feed Room Ventilation System is designed to provide adequate ventilation to Room 100 to maintain the equipment below its design qualified temperature during Loss of Offsite Power (LOOP) and Loss of Coolant Accident (LOCA).

The HPCFVS consists of Class 1E power supply, safety related and seismically supported supply and exhaust fans. The supply fan is provided with two in-line gravity dampers upstream of the fan and ductwork to distribute outside air to the room. The exhaust fan is provided with two in-line gravity dampers downstream of the fan to exhaust air directly from the room to the outside.

The HPCFVS is required to operate when train "B" class 1E power source is utilized, and the temperature inside the room increases as a result of heat dissipated from the Motor Control Center (MCC) located in this room. The HPCFVS is to be started and stopped automatically by a temperature indicating switch (TIS) which senses and indicates the room temperature.

During all modes of plant operation, the HPCFVS will be in a standby mode. Room 100 will be maintained at a slightly negative pressure. However, when the HPCFVS is energized, the supply air to Room 100 is greater than the exhaust air to ensure the room is maintained at a positive pressure. The excess air will be exhausted through the primary plant exhaust system ductwork. The exhaust fan will trip if the supply fan fails to run.

INSPECTION AND TESTING REQUIREMENTS OF THE HPCFVS

SHOP TESTING:

Shop inspection and testing are performed for all equipment, including controls.

Each fan motor with fan blades attached, shall be statically and dynamically balanced prior to assembly.

After assembly, the fan shall be vibration tested at the fan rated speed. Results of the balancing test shall be certified.

FIELD TESTING:

The system is initially tested and adjusted for proper flow paths, flow capacities and mechanical operability to demonstrate field performance after the equipment has been installed.

Ductwork is tested in accordance with industry standards.

Fans are rated and tested in accordance with the standards of the Air Moving and Conditioning Association (AMCA).

Standby equipment is operated on a cyclic basis to ensure the availability of the equipment.

APPENDIX 9.4D - PLANT VENTILATION DISCHARGE VENT

9.4D.1 DESIGN BASES

The plant vent stacks are designed to aid in the dispersion of gaseous effluents exhausted by the primary plant ventilation system, including discharges of radioactive gas from the gas decay tanks (see Section 9.4.3 and 11.3), during normal operation.

9.4D.2 SYSTEM DESCRIPTION

The plant vent stacks are the release points of the primary plant ventilation system (see Figure 9.4-9). The stacks are located adjacent to the northwest side of the Unit 1 Containment and the southwest side of the Unit 2 Containment.

The design parameters for the plant vent stacks are the following:

1. Height of Release

   The height of the plant vents are approximately 196 ft above ground level. The release points are approximately 64 ft below the top of the Containment and at least 84 ft 6 in. above other adjacent structures.

2. Discharge Temperature

   As shown in Table 9.4-2, the minimum indoor design condition is 40°F. The temperature of the outside air is site related; therefore, the temperature difference between the gaseous effluent discharge and the outside air is also site related. The discharge temperature is dependent on ambient temperature conditions and mode of plant operation. The minimum discharge temperature is 40°F . The maximum is a function of the mode of plant operation. Information concerning outdoor design conditions is presented in Table 9.4-1. Design conditions (indoor) are tabulated in Table 9.4-2. See Figures 1.2-1 and 1.2-2.

3. Effluent Discharge Quantity

   The maximum quantity of gaseous effluent exhausted via each plant vent stack is shown on Figure 9.4-9.

   The exhaust flow from the ESF unit(s), during design basis events, will be discharged via the two vent stacks.

4. Plant Vent Stack Size and Shape

   The plant ventilation discharge vent is cylindrical in shape with a 66 inch diameter.

9.4D.3 SAFETY EVALUATION

The plant vents stacks are not designed as safety-related and no credit has been taken for the height of the stacks above plant grade in the offsite dose calculations. The vent stacks are

designed as seismic Category II. The system design is such that failure of the vent stack does not obstruct the release of gaseous effluent at the roof level and does not affect the functional integrity of the safety-related exhaust system.

APPENDIX 9.4E - PLANT VENTILATION CHILLED WATER SYSTEM

9.4E.1 DESIGN BASES

The plant ventilation chilled water system is designed to remove heat rejected by equipment and to maintain ambient temperature below design limits within the areas it serves.

This system provides chilled water to the following non-safety related cooling and ventilation systems of both units.

- Primary Plant Ventilation System

- Auxiliary Building HVAC Equipment Room System

- Safeguards Building Electrical Area Ventilation System

- Safeguards Building Main Steam and Feedwater Penetration HVAC System

- Electrical and Control Building Uncontrolled Access Area HVAC System

- Containment Air Recirculation System

- Containment Neutron Detector Well System

- Containment CRDM System

- Positive Displacement Charging Pump Fan Coil Unit

- Turbine Building Switchgear Area HVAC System

- Secondary Plant Sampling Package

- Turbine Building Office and Room Area Coolers

- Instrument Air System Coolers

The plant ventilation chilled water system is not required to mitigate the consequences of an accident or to maintain the plant in a safe shutdown condition. During a loss of offsite power, part of the chilled water system operates to provide chilled water to the containment cooling units and the positive displacement charging pump room fan coil units. It also provides chilled water to the secondary plant sampling package. All the other systems and equipment listed above are required only during normal plant conditions. The containment isolation valves are motor operated and are powered from the diesel generators.

The system is shown on Figure 9.4-11.

Chilled water is distributed by the recirculation pumps to the cooling coils of the systems listed above. The chillers are designed in accordance with the ASME B&PV Code, Section VIII.

The heat sink for the plant ventilation chilled water system are the Component Cooling Water System described in Section 9.2.2 and the Circulating Water System described in Section 10.4.5.

Except for the containment penetrations and containment isolation valves, the plant ventilation chilled water system is non-nuclear-safety and non-seismic. Therefore should a seismic event cause a loss-of-offsite power, the operation of the plant ventilation chilled water system would not be required. The Containment penetrations and the isolation valves are ANS Safety Class 2 and seismic Category I.

9.4E.2 SYSTEM DESCRIPTION

The plant ventilation chilled water system consists of two distinct subsystems as described below:

1. The first subsystem is shown on Figure 9.4-11 and consists of six chillers, four 50-percent capacity chilled water recirculating pumps, three 50-percent capacity chilled water booster pumps, a chilled water surge tank, chilled water cooling coils, and associated piping, valves and instrumentation. Four chillers are located in the Auxiliary Building and two chillers are located in the Unit 1 Turbine Building. The chillers located in the Auxiliary Building are of the centrifugal type with hermetically sealed, electric-motor-driven compressors. The chillers located in the Turbine Building are of the centrifugal type with open electric motor driven compressors. The pumps are electric-motor-driven, single stage, horizontally split, centrifugal type. The evaporators of the six chillers are connected in series. The four chilled water recirculating pumps are connected in parallel, as are the three chilled water booster pumps. This subsystem is operated to support simultaneous operation of, mainly, the containments of Unit No. 1 and Unit No. 2, which includes normal operation, reactor startup, normal reactor shutdown and reactor hot standby during a loss of offsite power.

   a. During reactor startup, normal operation, and normal reactor shutdown, operators will run a sufficient number of chillers to maintain ambient area temperatures within required limits.

   Additionally two out of four chilled water recirculating pumps and two out of three chilled water booster pumps are required to operate. The chilled water flow rate through the series Auxiliary Building chillers is 2400 GPM, and the chilled water flow rate through the series chillers in the Unit 1 Turbine Building is approximately 2600 GPM.

   The equipment served by this subsystem is as follows:

   - Three out of four Unit 1 Containment air recirculation and cooling units
   - One out of two Unit 1 Containment neutron detector well cooling units
   - Unit 1 Containment CRDM cooling unit
   - Three out of four Unit No. 2 Containment air recirculation and cooling units
   - One out of two Unit 2 Containment neutron detector well cooling units

- Unit 2 Containment CRDM cooling unit
- Unit 1 Positive Displacement Charging Pump Room fan coil unit
- Unit 2 Positive Displacement Charging Pump Room fan coil unit
- Secondary plant sampling package
- Unit 1 Turbine Building Switchgear Area supply unit
- Unit 2 Turbine Building Switchgear Area supply unit
- Turbine Building Office and Room Area fan coil units
- Instrument Air System Coolers

The component cooling water from the nonsafeguard loops of the CCWS for both Units is used as the cooling medium for the four chillers located in the Auxiliary Building. The arrangement of the CCWS piping allows the condensers in a specific pair of these chillers to be connected in series. Two chillers are cooled by Unit 1 non-safeguard Component Cooling Water loop; and the other two chillers are cooled by Unit 2 non-safeguard Component Cooling Water loop. The Component Cooling Water flow rate is 2000 GPM per two series condensers in a pair, or a total of 4000 GPM for the three operating chillers.

The two chiller condensers in the Unit 1 Turbine Building are connected in parallel and are supplied with circulating water as the cooling medium. Each condenser receives 2000 GPM of circulating water i.e., a total of 4000 GPM for the two operating chillers.

b. During a loss of offsite power, this subsystem serves Unit 1 and Unit 2 containment cooling units and Unit 1 and Unit 2 positive displacement charging pump room fan coil units. It will also serve the secondary plant sampling package. Only the four chillers located in the Auxiliary Building and two associated chilled water recirculating pumps are loaded on the diesels by the blackout sequencer. The two Unit 1 Turbine Building chillers and their associated chilled water booster pumps are isolated. The chilled water flow rate through the four operating series chillers is 2400 GPM. The CCW flow rate through each operating pair of series chillers is 2000 GPM i.e., a total of 4000 GPM for both operating pairs. The four chillers and the two chilled water recirculating pumps which are selected for operation are powered from the Standby Power System during a loss of offsite power, as described in Section 8.3.

Remote manually actuated air operated valves (Tag nos: X-HV-6056A and X-HV-6057A) are utilized to switch this system from the normal to post-loss of offsite power operation, by isolating the Unit 1 Turbine Building Chillers and associated chilled water booster pumps and vice-versa.

2. The second plant ventilation chilled water subsystem is shown on Figure 9.4-11 and consists of three chillers, each sized for 33.33 percent of the total cooling capacity, four 33.33-percent capacity chilled water circulating pumps, a chilled water surge tank

(which is shared with the first subsystem), chilled water cooling coils, and associated piping, valves and instrumentation. The chillers are centrifugal type, with direct open, electric-motor driven compressors. The chilled water circulating pumps are electric-motor-driven, single stage, horizontally split centrifugal type. The evaporators of the chillers are connected in parallel to common chilled water supply and return headers. The four chilled water circulating pumps are also connected in parallel. This subsystem is operated during normal operation, reactor startup and normal reactor shutdown only. It does not operate during a loss of offsite power. During operation at maximum design load, all three chillers are required to operate to generate a combined cooling capacity of 1980 tons. Additionally, three out of four chilled water circulating pumps are required to operate, with the fourth pump remaining on standby. The chilled water flow rate through the three parallel chillers is 650 GPM per chiller i.e., a total flow rate of 1950 GPM.

The equipment served by this subsystem is as follows:

- Six out of eight Primary Plant Ventilation System (PPVS) supply units
- The Auxiliary Building HVAC Equipment Room Supply Unit
- Unit 1 Safeguards Building Electrical Area supply unit
- Unit 1 Safeguards Building Main Steam and Feedwater Penetration Area supply unit.
- Unit 2 Safeguards Building Electrical Area supply unit
- Unit 2 Safeguards Building Main Steam and Feedwater Penetration Area supply unit.
- Units 1 and 2 Electrical and Control Building Uncontrolled Access Area supply unit.
- Instrument Air System Coolers

The condensers of the three chillers are connected in parallel and are supplied with circulating water as the cooling medium. Each condenser receives 2250 gpm of circulating water i.e., a total of 6750 gpm.

The surge tank is provided to accommodate expansion and contraction within the system and to permit monitoring of the system for leakage.

The makeup water to the tank is supplied from the Demineralized Water System.

Except for cooling units located inside the Containment, Instrument Air System and those for the Turbine Building Office and Room Area, all cooling coils are provided with a thermostatically controlled valve at the outlet of the coil. The thermostatically controlled valve for the Turbine Building Switchgear Area cooling coils is located at the inlet of the coils. The outlet air temperature from the coil controls the valve.

The coils for the cooling units located inside the Containment, except for the CRDM cooling coils, are provided with an electric-motor-operated-on-off valve which is interlocked with the respective cooling unit fan. Flow through the CRDM coils is not affected when the fans, which can be controlled from the Control Room, are stopped.

Vent and drain connections are provided on piping and equipment where necessary to facilitate testing and maintenance operations. All major components are provided with upstream and downstream isolation valves to facilitate maintenance operation.

The system is not required to function following a DBA. During a DBA, the chilled water supply and return Containment penetrations isolation valves are closed, and the system will shut down.

The chilled water is circulated through the system by the chilled water recirculating pumps and circulating pumps.

Although the two plant ventilation chilled water subsystems described above, operate independently of each other, they are interconnected with piping and with proper valving so as to permit, selectively and through administrative decision, the switching of chilled water supply to various cooling equipment, from the first subsystem to the second subsystem, under off-normal conditions (e.g., failure of a chiller, pump, a cooling unit, etc.).

The operating pumps take suction from their respective chilled water return headers and discharge it to the evaporator sections of the associated operating chillers. The cooled chilled water, after rejecting heat to the refrigerant in the chillers, is discharged into the respective chilled water supply main. The chilled water from the supply main flows through the cooling coils of the various cooling units associated with the subsystems. The warm chilled water from the cooling coils flows via the chilled water piping back to the respective chilled water return header, which feeds the suction of the chilled water recirculating pumps, thus completing the closed system.

Instrumentation and controls incorporating audible and visual annuciation facilitate continuous monitoring of the system performance and alert the operator in the event of a system malfunction. To maintain constant chilled water flow, a valve located in a recirculation loop is modulated by a signal from a flow element in the first subsystem and a pressure differential controller in the second subsystem. The Containment isolation valves are motor operated and are powered from the diesel generators.

9.4E.3 SAFETY EVALUATION

The reliability of the plant ventilation chilled water system is ensured by the following features:

The Containment penetrations and isolation valves are ANS Safety Class 2 and seismic Category I.

APPENDIX 9.4F - SAFETY CHILLED WATER SYSTEM

9.4F.1 DESIGN BASES

The safety chilled water system is designed to remove heat rejected by ESF pump motors, Uninterruptable Power Supply (UPS) Equipment and electrical switchgear and to maintain ambient temperature below design limits within the rooms it serves. The chilled water from the system is supplied to the safety-related cooling units. The system is also used during normal operations to aid in heat removal. The system is shown on Figure 9.4-12.

The heat sink for the safety chilled water system is the safety-related Component Cooling Water System described in Section 9.2.2.

The safety chilled water system is ANS Safety Class 3 and seismic Category I. The system is required to operate during all modes of operation.

The chillers are designed in accordance with ASME B&PV Code Sections III & VIII. The piping is designed in accordance with ASME B&PV Code Section III.

Each unit is provided with two 100-percent-capacity chilled water systems each of which are powered from independent Class 1E buses. Thus, the system will sustain a single active component failure without loss of function during all modes of plant operation.

9.4F.2 SYSTEM DESCRIPTION

The safety chilled water system is shown on Figure 9.4-12.

The system for each unit consists of two-100-percent-capacity hermetic centrifugal chillers, two-100-percent-capacity chilled water recirculation pumps, a chilled water surge tank, chilled water fan coil units, and associated piping, valves, and instrumentation. Each chiller is rated for 101 tons of refrigeration at design conditions. Each pump is capable of circulating chilled water at 300 gpm design flow. The required component cooling water flow is 300 gpm at design conditions.

The chilled water by the system is supplied to the cooling units as follows:

1. Component cooling water pump emergency fan coil units

2. Charging pump emergency fan coil unit

3. Spent fuel pool heat exchanger and pump emergency fan coil unit (These fan coil units are common for both units, and the piping arrangement will allow chilled water to be supplied from either the Unit 1 or the Unit 2 chilled water system.)

4. Safety injection pump emergency fan coil unit

5. Containment spray pump emergency fan coil unit

6. RHR emergency fan coil unit

7. Auxiliary Feedwater pump emergency fan coil unit

8. Electric area emergency fan coil units

9. Uninterruptable Power Supply (UPS) Room Fan Coil Units

The chillers are of the centrifugal type with hermetically sealed, electric-motor-driven compressors. The component cooling water from the respective trains of the safeguards loop of the CCWS is used as a cooling media for the condensers.

The pumps are electrical-driven, single-stage, horizontal-split, and centrifugal type.

The surge tank is provided to accommodate expansion and contraction within the system and to permit monitoring of the system for leakage. The makeup water to the tank is supplied from either the Demineralized Water System or the Reactor Makeup Water System, with consideration given to water chemistry for the Safety Chilled Water System corrosion inhibitor. The partition in the surge tank provides separate surge volumes for each safety train. A leak in one train will not affect the other train.

Vent and drain connections are provided on piping and equipment where necessary to facilitate testing and maintenance operations. All major components are provided with upstream and downstream isolation valves to facilitate maintenance operation.

The system is required to operate during post DBA conditions. The system is powered from Class 1E safety buses. During loss-of-offsite power, the power is provided from diesel generators.

Chilled water is circulated through each of the two closed-loop safety trains. The recirculation pump takes suction from the chilled water return line and the chilled water surge tank which connects into the return line. The recirculation pump discharges into the evaporator of the chiller. The then chilled water from the evaporator enters into the supply header and passes through the fan coil units connected in parallel. The return chilled water from the fan coil units enters into the chilled water line, thus completing the closed system.

Except for the electrical area and Uninterruptable Power Supply (UPS) Room fan coil units, all fan coil units are interlocked with respective equipment to start. The fan coil units for the electrical area start on a Safety Injection or Blackout Sequence Signal. The chilled water recirculation pumps start on a Safety Injection, Blackout Sequence Signal or the start of a CCW Pump.

The instrumentation and controls for the safety chilled water system are provided for automatic and remote operation of the system. The operation and supervision of the safety chilled water system following a Control Room evacuation can be accomplished by using the system components local controls (for fan coils), the Remote Shutdown Panel, and the Switch Transfer Panel for Train “A” pump and chiller.

9.4F.3 SAFETY EVALUATION

The safety chilled water system is ANS Safety Class 3 and seismic Category I.

The reliability of the chilled water system is ensured by the following features:

1. The use of two 100-percent-capacity systems, one for each train, and the power supply from redundant Class 1E buses to each electrically operated equipment; therefore, the system meets the requirement of a single failure without loss of function.

2. Instrumentation and controls are available, providing audible and visual annunciation. This permits continuous monitoring of system performance.

3. The safety chilled water system components are located in a seismic Category I structure.

9.4F.4 INSPECTION AND TESTING REQUIREMENTS

See Subsection 9.4.2.4.

9.5 OTHER AUXILIARY SYSTEMS

9.5.1 FIRE PROTECTION SYSTEM

9.5.1.1 General

This section is a description of the Fire Protection Program of the CPNPP units 1 and 2. The evaluation of fire hazards is included in the CPNPP Fire Protection Report (FPR) which follows the format of the U.S. Nuclear Regulatory Commissions's "Supplementary Guidance on Information Needed for Fire Protection Program Evaluation" and the supplementary criteria in their September 30, 1976, letter.

The overall Fire Protection Program was developed utilizing the defense in depth concept. This concept is a combination of:

1. Preventing fires from starting

2. Quickly detecting and suppressing fires that do occur to limit the extent of damage

3. Designing plant safety systems so that a fire that becomes fully established and burns for a considerable time, in spite of the fire protection systems provided, will not prevent essential plant safety functions from being performed.

The FPR quantifies potential fire hazards throughout the plant in terms of combustible heat release loading. The Fire Protection and Detection Systems are designed based on this heat release loading and on the nature of the transient and in situ combustible material in the area. A summary of this information is presented in tabular form in the FPR.

9.5.1.2 Method of Analysis

9.5.1.2.1 Definitions

Several terms with their definitions as they relate to the Fire Protection Program for CPNPP are presented below. Unless the terms are noted below, the definitions are as stated in Section I of Branch Technical Position APCSB 9.5-1 Reference [2].

1. Fire area

   The fire area is that section of a building or the plant that is separated from other areas of the plant by fire barriers with openings and penetrations protected by seals or closures having a fire resistance rating equal to the rating assigned to the barrier. The fire areas extend through more than one elevation where plant design requirements and low amounts of combustible material in a specific area allow. These areas are designated on FPR Figures.

2. Fire Barriers

   Fire barriers are those components of construction (walls, floors, or protective coverings) that are rated by approved laboratories or are constructed in accordance with the

requirements stated by authorities having jurisdiction in hours of resistance to fire and used to prevent spread of fire.

3. Fire Zone

The fire zone is a subdivision or portion of a fire area that is designated on the FPR Figures.

4. Fire Duration

Fire duration is the approximate time expressed in minutes that the tabulated combustible material will burn. The duration is based on the heat release that will produce an exposure equivalent to the standard time-temperature curve (ASTM E-119).

5. Fire Break

The fire break is a physical barrier that prevents fire propagation, that is, the spreading of a fire from one component to another or the direct exposure of a component to the heat and flames of a fire, or both.

6. Design Basis Fire

Design basis fire is a fire that is postulated to occur in a fire area or fire zone assuming no manual, automatic, or other firefighting action has been initiated. The combustibles in the area are totally consumed and the fire burns at a rate modeling the standard time-temperature curve (ASTM E-119).

7. Enclosed

The term “enclosed” means being surrounded by a case which will prevent a person from accidentally contacting live electrical parts. It also applies to flammable liquids which are contained or encased in fire-resistant materials or buildings and to barriers which may or may not be fire rated that surround or encompass fire areas or fire zones.

8. Dry

The term “dry” indicates that the connecting piping between a deluge valve and the nozzles of a water system is not normally pressurized with water.

9. Wet

The term “wet” indicates that the connecting piping between the main loop and a hose station isolation valve or water nozzle is normally pressurized with water.

10. Radiation Zone

Radiation zone is the classification of an area based on the expected dose equivalent rate (mrem/hr) within that area. See Section 12.3 for a detailed description.

11. Fire Safe Shutdown Essential System or Component

    An essential system or component is defined as a system or component which is required to be operational to safely shutdown the plant in the event of a fire.

12. Maximum Permissible Fire Loading

    The Maximum Permissible Fire Loading (MPFL) is the maximum fire loading (BTU/sq ft) which can be expected to be contained within a fire area by the fire area boundaries without compromising safe shutdown capability.

13. Fire Hazards Analysis Evaluation

    A Fire Hazard Analysis Evaluation is an assessment of the impact of a single fire hazard on redundant components or systems used to provide fire safe shutdown functions for the plant. A Fire Hazards Analysis Evaluation is performed by a Fire Protection Engineer and, if required a Systems Engineer. The purpose of a Fire Hazards Analysis Evaluation is to demonstrate compliance with BTP APCSB 9.5-1 Appendix A based on the following considerations:

    - potential transient and in situ combustible hazards are considered.

    - protection provided is commensurate with the hazards.

    - the consequences of a fire on the plant's ability to safely shutdown are considered.

    - The Fire Hazards Analysis Evaluation is written, organized and maintained to facilitate review by a person who is not involved in the evaluation.

    - The conclusions of the FHA Evaluations are summarized in the applicable sections of the Fire Protection Report.

9.5.1.2.2 Assumptions

The FHA Evaluation is based on the following assumptions:

1. Generally, the minimum fire barrier rating is three hours except for the barriers enclosing the stairwells and elevator shafts, which are rated at two hours, the cable tray/conduit fire barriers which are rated at 1 or 3-hour, one hour fire rated cable, and other special cases where a rating of less than three hours is adequate.

2. When it is determined that a fire involving a fire safe shutdown component or system will not affect its redundant counterpart, the redundant system is assumed to operate without failures.

3. The Maximum Permissible Fire Loading for a fire zone assumes a fire burning in the area which follows the characteristics of the standard time-temperature curve, or as noted in the FPR, Reference [19].

4. A fire involving a combustible loading, up to the Maximum Permissible Fire Loading for the fire zone, will be contained within the fire area by the passive and active/fire protection features (i.e. fire wall and sprinklers, etc.). Furthermore, it is assumed that if any of these passive or active fire protection features is inoperable and the compensatory actions required by the Fire Protection Report have been implemented then an equivalent level of protection is provided.

9.5.1.2.3 Methodology

In order to evaluate potential fire hazards, provide adequate fire protection, ensure isolation of fire safe shutdown systems from these hazards, and prevent the release of radioactive material to the environment, the following method of design and analysis has been formulated and implemented for the entire plant:

1. The plant is divided into separate fire areas using plant walls and floors as barriers. Due consideration as-shown below is given to the separation of redundant fire safe shutdown components from each other, from non-fire safe shutdown components and from major concentrations of combustible materials. Considerations were also given to other area characteristics such as electrical cable routing into and through the area, the ductwork supplying and exhausting the area, access and egress routes for the area, and vent area for depressurization during a tornado.

2. For each fire area/fire zone, the heat of combustion for each in-situ combustible is calculated. The calculated heat of combustion for all in-situ combustibles is divided by the floor area to determine the combustible loading (BTU/sq ft) for the fire area/fire zone. In addition, the approximate fire duration (minutes) is determined based on the ASTM E-119 standard time-temperature curve, (except as noted in the FPR Section II). The combination of transient combustibles and the in-situ combustibles will not exceed the Maximum Permissible Fire Loading without implementation of compensatory measures.

3. The fire safe shutdown essential equipment in each area is tabulated.

4. Once the fire area and combustible material information is tabulated, fire protection equipment is located throughout the plant based on the severity and configuration of the fire hazards, the calculated heat release of each fire area and the plant equipment and components located in the fire area.

5. Fire detectors are located in all areas of the plant where there is a significant combustible loading and in all areas containing equipment required for safe shutdown except as described in Section 9.5.1.6.1.

6. Hose stations are installed in all safety related buildings of the plant such that an effective hose stream can reach any location in a safety related building except as described in Section 9.5.1.6.1.

7. Portable extinguishers are located in all safety related buildings in accordance with NFPA 10 requirements.

8. Fixed automatic water suppression systems will generally be installed in safety related plant areas where any of the following conditions exist:

   a. A high fire hazard exists

   b. Redundant safe shutdown equipment or cabling outside the Containment Building is located in the same fire area and is not separated by a three hour fire barrier.

   c. There is a congestion of cabling.

   In areas where condition (a) and in areas where condition (b) described above exists, the type of protection that will be provided as a minimum will be a sprinkler system providing coverage adequate for the hazard in the area unless justification for deviations are provided per reference [19] and as described in 9.5.1.6.1. The water spray design density will be based on Section 9.5.1.6.1-E.3.c.

   Where the condition described in (c) exists, based on Section 9.5.1.D.3.c, sprinkler systems will be provided for cabling to augment other fire protection features in the area.

9. Where redundant fire safe shutdown equipment cabling is located in the same fire area and is not separated by a three hour fire barrier or a horizontal distance of 20 feet with negligible intervening combustibles or fire hazard, one train of this cabling, if not one hour fire rated cable, will be enclosed by a one-hour fire barrier (or radiant energy shield inside containment) unless an alternate shutdown path is utilized or justification for deviations are provided per reference [19] except as described in Section 9.5.1.6.1.

10. The Cable Spreading Room contains equipment and cables belonging to both safety trains. The following fire protection systems will be provided:

    a. Hose stations for manual fire fighting

    b. Fixed Halon primary suppression system

    c. Manual pre-action sprinkler system

    d. Automatic fire detection system

    e. An alternate shutdown system

11. The plant will be capable of being safely shutdown in the event any of the fires postulated in the Fire Protection Report occurs. Alternate shutdown systems and procedures have been developed using shutdown paths available to the operator which are either free from fire damage or otherwise controllable in spite of such fire damage.

9.5.1.3 Fire Hazard Analysis Evaluation

See Reference [19], Fire Protection Report

See Reference [50] regarding Independent Spent Fuel Storage Installation (ISFSI) Haul Path and ISFSI Pad

9.5.1.4 Fire Protection System Description

9.5.1.4.1 General

The Fire Protection System detects, alarms, and extinguishes fires. It is comprised of two subsystems: Fire Detection and Fire Suppression.

The Fire Detection System is a plant-wide system designed to detect fires in the plant, alert the Control Room operators, and alert the plant fire brigade of the fire and its location.

The Fire-Suppression System is designed to extinguish any fire postulated to occur in the Fire Protection Report. It is comprised of a water supply system, fixed water sprinkler and spray systems, Halon systems, fire hose stations, and portable extinguishers.

9.5.1.4.2 System Design Parameters

1. Fire Detection System

   The Fire Detection System consists of the following components:

   a. Fire Detectors

      1. Ionization smoke detectors

      2. Thermal heat detectors

      3. Ultraviolet detectors

      4. Thermistor line detectors

   b. Fire Detection Local Control Panels

      These panels provide local indication of the status of the protected area. Indication provided is annunication of alarms and system trouble status. These panels also provide automatic initiation of fire suppression where applicable.

   c. Fire Detection Main Control Panel

      This panel is located in the control room. Any fire alarm that is detected in the plant will alarm on this panel in the control room. Trouble circuits of each local panel are also monitored.

The fire detectors are strategically located throughout the plant to detect, annunciate, and indicate in the Control Room, the location of a fire.

The power supplies for the Plant Fire Detection system meets the requirements of NFPA 72D Section 2220. The Plant Fire Detection system can be provided power from any one of the following eight (8) sources: two (2) main generators, two (2) offsite power supplies, and four (4) standby diesel generators (plant emergency power supply).

The fire detection system is electrically supervised for a wiring break in the detection and alarm circuits. Loss of supervision causes an audible and visual trouble indication on the main fire detection control panel, located in the Control Room, in accordance with NFPA 72D requirements. Thermistor-line fire detection systems are supervised for a break or short circuit of the sensing element. Ground fault supervision is provided except as noted in Section 9.5.1.6.1-E.1.

Ionization detectors are of the two-chamber-type design. The first chamber is a reference chamber to compensate for sensitivity changes caused by temperature, barometric pressure, and humidity variations. The second chamber is a sensing chamber open to the outside elements through a protective screen which permits combustion products to enter, while preventing insects and foreign matter from entering and causing false alarms.

Thermal detectors are of the fixed-temperature, rate compensation types or continuous strip thermistor line type.

Ultraviolet detectors respond directly to the presence of flame by sensing the ultraviolet radiation emanating from the flame.

2. Fire Suppression Systems

a. Water Supply Systems

The water supply system was designed using NFPA Codes and BTP 9.5-1 Appendix A. The water supply network and the arrangement of the water extinguishing systems are shown on Figures 9.5-43 through 9.5-48 and Figures 9.5-61 and 9.5-62. The water extinguishing systems are designed to operate with the shortest portion of the Fire Protection yard-loop out of service. The water storage capacity is based on supplying water to the largest fixed extinguishing system and the manual hose stream requirements of Appendix A.

Three 50 percent pumps (one electric motor-driven, two diesel engine-driven; each rated at 2000 gpm at a Total Dynamic Head of 370 ft) are provided for protection of both units. In addition, a jockey pump (rated at 60 gpm at a Total Dynamic Head of 300 ft) maintains the required water pressure throughout the system at all times.

The fire pumps are located in a separate fire pump house structure and take suction from two dedicated fire water storage tanks. The fire pumps are single stage horizontal splitcase centrifugal pumps and Underwriters' Laboratory, Inc. listed for fire protection service.

A 12-in. diameter piping network encircles the entire plant. Post indicator gate valves isolate sections of the yard fire-loop and each branch connection extending off the loop. These valves have indicators that show valve position.

Hydrants isolated by individual auxiliary gate valves (curb box valves), are provided adjacent to the yard loop at approximately 250 ft intervals. Each hydrant is equipped with at least two 2-1/2 in. hose outlets. A 2-1/2 in. outside independent hose gate valve is connected to each outlet to provide individual

control of the hoses. A hose house is provided adjacent to each hydrant. Each 2-1/2 in. hose house is equipped with the following accessories:

1. One fire axe
2. One crowbar
3. Two hydrant wrenches
4. Six coupling spanners
5. Two hose and ladder straps
6. Two 2-1/2 in. hose washers (spares)
7. One emergency light
8. Three hundred feet of 2-1/2 in. firehose
9. Two adjustable straight stream-fog type nozzle and play pipe combination

All hose connections and nozzles are provided with NH type threaded connections that are compatible with those of the local fire department.

b. Fixed Water Suppression Systems

Automatic and manual suppression systems typically consist of a hydraulically balanced piping network, an isolation valve supervised as described in Section 9.5.1.6.1-E.3.6, fusible-link, quartzoid bulb or directional solid-cone spray nozzles, a water flow alarm, and a local and remote (control room) audible and visual alarm. The systems are designed and balanced to provide water spray densities as described in Section 9.5.1.6.1-E.3.c. The fixed water suppression systems are divided into six categories. Each category is as follows:

1. Automatic wet-pipe sprinkler systems

   Automatic wet-pipe sprinkler systems are provided for general area coverage and to protect electrical cable trays.

2. Automatic Water Spray Systems

   When a fire is detected its location is annunciated on the local fire detection control panel and in the control room. The Fire Detection Local Control Panel, transmits a signal to open the proper deluge valve. A water-flow alarm sounds locally and in the control room indicating water flowing through the piping network. The deluge valves operate automatically as described above, or can be manually operated locally. Once actuated, deluge valves can only be reset manually.

Water spray systems are provided for the following:

- Turbine Building Hazards (i.e. Lube Oil, Feedwater Turbines, Hydrogen Seal Oil)

Water spray systems are also provided for the following atmospheric cleanup units' (ACUs) charcoal adsorber beds:

- Four Containment preaccess units

Each charcoal adsorber bed is equipped with thermistor strip type heat detectors.

Upon detection of an abnormally high temperature in any one of the charcoal adsorber beds, a high temperature signal is generated locally and in the control room. This signal also initiates demineralized water flow to the respective unit. Should demineralized water be unavailable, the operator is alerted in the control room and he can manually route fire protection water through the demineralized water pipe supplying these water spray systems.

If the temperature of the charcoal adsorber bed continues to rise, the detection system will generate a high-high temperature signal which will automatically open the deluge type valve on the ACU and initiate water spray on its charcoal bed. When the temperature of the charcoal adsorber bed drops below the high-high set point, the deluge valve will close automatically. Deluge type valves for ACUs can only be operated automatically.

The water flow requirements for each unit are based on the manufacturer's suggested flow rate to extinguish a charcoal-adsorber fire.

3. Manual Water Spray Systems

Manually operated water spray systems protect each turbine generator main shaft bearing at locations where the main shaft is exposed to the environment. These systems are provided with an approved deluge valve, a piping network, a flow switch and directional solid-cone spray nozzles. Isolation valves are provided for each exposed section of the shaft to preclude spraying unaffected areas. Manual operation as well as annunciation of a fire alarm condition via the plant fire detection system is provided.

Water spray systems are also provided for the following atmospheric cleanup units' charcoal adsorber beds:

(a) Sixteen primary plant exhaust ventilation units.

(b) Two hydrogen purge exhaust units.

(c) Two Control Room emergency filtration units.

(d) Two Control Room emergency pressurization units.

Each charcoal adsorber bed is equipped with thermistor strip type heat detectors.

Upon detection of an abnormally high temperature in any one of the charcoal adsorber beds, a high temperature signal is generated locally and in the control room. The fire suppression system headers to the ACUs are normally dry. Manual operation of a deluge charging valve charges the headers with water. If the temperature of the charcoal bed continues to rise, the detection system will generate a high-high temperature signal which will automatically open the deluge type valve on the ACU and initiate water spray on its charcoal bed. When the temperature of the charcoal bed drops below the high-high set point, the deluge valve will close automatically. Deluge type valves on the units can only be operated automatically.

4. Manual Preaction Sprinkler Systems

A manual preaction sprinkler system consists of a normally dry piping network, automatic sprinkler heads, latching normally-closed deluge valve, water flow alarm (which transmits an alarm to the Control Room via the local fire detection panel indicating water flow in the piping), and an approved isolation valve complete with tamper switch.

Manual preaction sprinkler systems are provided in areas requiring general area and electrical cable tray protection but where the potential for inadvertent water discharge is to be eliminated.

Plant areas, protected by manual preaction sprinkler systems, include the following:

- Cable Spread Room systems are provided for backup suppression capability for the Halon Systems.

- Switchgear and Hot Shutdown Panel room systems are provided for specific hazards.

Upon detection of fire, the deluge valve is manually opened through operation of an electric pull station allowing water flow into the piping network. Water will only discharge into the protected area through the individual automatic sprinkler heads which witness their rated temperature.

5. Manual Deluge Sprinkler System

A manual deluge sprinkler system protects the Fuel Building loading bay and consists of a normally dry piping network, open sprinkler heads,

deluge valve, water flow alarm and an approved isolation valve with tamper switch.

Upon detection of a fire, the deluge valve is manually operated through operation of an electric pull station allowing water flow into the piping network which will discharge through all sprinkler heads protecting the area.

6. Automatic Preaction Sprinkler Systems

Automatic preaction sprinkler systems protect the diesel fuel oil day tank rooms and consist of a normally dry piping network, automatic sprinkler heads, approved deluge valve, water flow alarm (which transmits an alarm to the control room via the local fire detection panel indicating water flow in the piping), and an approved isolation valve complete with tamper switch.

When a fire is detected by the detection system, the local fire detection control panel transmits a signal to open the system deluge valve. Water will only discharge into the protected area through the individual automatic sprinkler heads which witness their rated temperature. The deluge valves operate automatically as described above, or can be manually operated locally. Once actuated, deluge valves can only be reset manually.

c. Halon Extinguishing Systems

Automatic, total flooding, fixed, Halon extinguishing systems, actuated by ionization detectors, are provided for the cable spreading rooms. Manual actuated, total flooding, Halon extinguishing systems, are provided for the plant computer rooms.

Halon concentrations for each area are in accordance with NFPA suggested concentration except as described in Section 9.5.1.6.1. Each system is provided with two charges of Halon.

d. Fire Hose Stations

Fire hose stations are located strategically throughout the plant for manual fire fighting operations.

The fire hose stations located on the operating deck of the Turbine Building are equipped with 100 ft of 2-1/2 in. hose using NFPA 14 as a guideline for Class 1 service except as noted in Reference [19].

All other fire hose stations throughout the plant are equipped with 100 ft of 1-1/2 in. hose and a nozzle compatible with the type of fire hazard in the area. The respective standpipes which supply water to these hose stations are sized and located throughout the plant using NFPA 14 for class II service as a guideline except as noted in 9.5.1.6.1 and Reference [19].

Fire hose stations are also provided to supply makeup water to the spent fuel pools if normal sources are unavailable. Fire hose stations (EL. 860', see Figure 9.5-47 Sheet 1) in the area of the spent fuel pools are capable of supplying makeup in excess of the boil-off rate. Since the fire protection makeup is from the SSI, the supply of water is essentially unlimited.

The Hose stations for the Containment Building's Control Room and the cable spreading rooms are dry pipe with manual charging required.

The hose stations inside the containment are fed from the Demineralized Water System via a transfer pump. This pump is initiated by hand pull stations inside the containment. Should demineralized water not be available or the system malfunctions, additional hand pull stations are provided inside the containment which allows for the normal fire protection water to be used for fire fighting purposes. The use of demineralized water will minimize the possiblity of introduction of chlorides inside the containment.

e. Portable Fire Extinguishers

Portable fire extinguishers are provided for fire suppression throughout the plant. The quantity and type of extinguishers located in each fire area are based on the type, quantity, and specific hazard conditions in the respective fire areas. All extinguishers are in accordance with the guidelines of NFPA pamphlet No. 10.

9.5.1.4.3 System Description

Fixed water suppression systems, fixed Halon systems, hose stations, and portable extinguishers are used as the primary and secondary means of fire suppression. As shown on Figures 9.5-43, 9.5-61 and 9.5-62 water is supplied to the water suppression systems and the hose station standpipes from two atmospheric storage tanks via an underground piping distribution system and three 50 percent-capacity fire pumps, one electric motor driven and two diesel engine driven.

When pressure in the main loop drops below a set pressure point, the jockey pump starts automatically; when the system is repressurized, the jockey pump stops automatically. If the jockey pump cannot maintain the system pressure, the lead fire pump (the electric motor-driven fire pump) automatically starts. The diesel engine-driven fire pumps sequentially start automatically if the system pressure cannot be maintained by the other pumps for a preset period of time. The three fire pumps can only be shut down manually at the fire pump house location. The three fire pumps are connected to an approved flow meter to facilitate periodic testing of any fire pump. Pressure switches for the three pumps are located between the pump check valve and pump isolation valve to prevent starting an isolated pump. A siamese fire department connection is provided for emergency fill of the system by a fire truck or a portable auxiliary pump. This fill is used as a backup to the pumps. As required by NFPA No. 24, a check valve and a ball drip valve are provided at the connection of the siamese to the main loop. The siamese connection is located adjacent to the Service Water Intake Structure.

As shown in Figures 9.5-44 and 9.5-44A, the Turbine Buildings have an internal loop which supplies the standpipes. This internal loop has multiple connections to the underground loop in Unit 1 and Unit 2. Crosstie lines are provided between the Unit 1 and Unit 2 Turbine Buildings to

facilitate isolation of sections of either loop. Valves are provided in accordance with NFPA 14 to isolate the system. The water spray systems and automatic sprinkler systems are connected to the outside loop via isolation valves located in the fire protection valve rooms in the basement of each Turbine Building. The valve rooms are accessible from inside and outside the Turbine Buildings, as required, to control the water flow to the suppression systems.

As shown on Figure 9.5-46 Sheet 2, the water suppression systems protecting the diesel generators are independently supplied from the main yard loop. Actuation of the wet-pipe sprinkler system protecting one of the diesel generators will not affect the operation of the other diesel generator. Each diesel generator compartment is provided with a watertight door to prevent flooding of the adjacent areas.

The fire pump house structure is divided into five rooms with three hour rated fire barriers. The structure is protected by an automatic wet-pipe sprinkler system. Water flow and valve tamper alarms are provided at the pump house location and in the Control Room. Each room in the fire pump house is provided with detection which annunciates locally and in the Control Room.

The automatic wet pipe, manual preaction, automatic preaction, manual deluge, hose stand pipe and water spray systems are supplied by the respective safety related building interior supply loop.

Yard post indicator valves are located in supply lines to permanent plant auxiliary buildings in accordance with NFPA 24 to shut off the water supply to these buildings.

The Fire Protection System piping that is located throughout the Auxiliary, Electrical and Controls, and the Safeguards Buildings is routed via a sectionalized loop located on the basement elevation: 778'-0" in the Electrical and Controls Building and 790'-6" in the Auxiliary and Safeguards Buildings. The loop which is located in the corridor remote from the safeguards equipment is supplied via two lines running through the instrument and service air compressor area and a third line running through the corridor adjacent to the laundry holdup area. The loop and supply lines are normally wet (i.e., pressurized with water) and the branch connections off the loop to the hose stations and water spray systems are normally dry and only become pressurized during a fire condition. The Fire Protection System piping in these buildings is designed in accordance with the requirements of Reference [43] to assure the system pressure boundary integrity.

Based on the system design requirements and the physical layout of the Fire Protection System supply lines and main loop, the worst assumed crack in the wet piping would result in spraying a motor control center, which is designed per NEMA requirements and provided with drip shields and thus not effect cold shutdown of the plant. Also, the loop is provided with sectionalizing valves as noted previously in this section to isolate portions in event of a crack. A crack is not considered in the dry portion of the System.

Each Cable Spreading Room Halon system consists of a detection system, storage cylinders, manifold and header assembly, control valves, piping, nozzles, and local control panels. The Halon is released automatically after receipt of a fire signal from cross zoned ionization detectors located in the respective area. Each system incorporates a time delay which provides a warning for personnel evacuation of the area. The reserve charge of Halon is provided for automatic protection of the areas during the time the main cylinders are being refilled following a discharge. Halon can also be released manually.

Each Unit Computer Room Halon system is manually actuated by control room personnel and each system consists of storage cylinders, manifold and header assembly, control valves, piping, and nozzles. An ionization detection system provides control room notification of a fire.

A Fire Detection System is provided throughout the plant. When a fire is detected, it is annunciated and indicated by zone on the fire detection panel in the Control Room. An alarm-indicating lamp illuminates the base of the ionization detector showing the actuated detector. The majority of detectors are placed overhead in the monitored areas. Detectors serve a dual purpose: 1) they sound an alarm via the Control Room main fire detection panel and 2) where applicable, they actuate the automatic suppression systems.

9.5.1.4.4 Administrative Controls

The administrative controls related to fire protection at CPNPP are contained in the CPNPP Fire Protection Report and augmented by station procedures as required for effective implementation. These include the limiting conditions for operation required for the fire suppression systems and fire detection instrumentation, the corresponding compensatory measures, and the required surveillance test requirements to assure operability of the systems which were previously contained in the technical specifications. The licensee may make changes to the approved fire protection program without prior approval of the commission only if those changes would not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

9.5.1.5 Plant Fire Protection Design Requirements

9.5.1.5.1 General Plant Arrangement

The various buildings of CPNPP are divided into a series of fire areas. The primary consideration in this division was the separation of fire safe shutdown systems and components from their redundant counterparts and the isolation and separation of fire hazards from fire safe shutdown systems. Consideration was also given to the isolating of combustibles not located in, or exposed to, areas containing fire safe shutdown components and to provide access and egress routes to fire areas for plant personnel and the fire brigade.

9.5.1.5.2 Structural Construction Elements

All structural construction elements are composed of noncombustible materials. Structural walls, floors, and ceilings consist of poured, reinforced concrete, concrete block or structural steel framing with precast concrete panels or metal siding. Where these assemblies are designated as fire barriers, (i.e., 1 hour, 2 hour or 3 hour fire ratings) the construction is in accordance with a Underwriters' Laboratory listed design, a Uniform Building Code design, a specific fire test by a nationally recognized laboratory or as described in Section 9.5.1.6.1.

9.5.1.5.3 Interior Construction Elements

1. Building Partition Characteristics

   Interior walls and partitions which are non-load bearing and designated as fire barriers are constructed of reinforced concrete, concrete block or noncombustible steel studs and gypsum dry wall. Interior non-load bearing walls which act as fire barriers are constructed in accordance with a Underwriters' Laboratory listed design, a uniform building code

design, a specific fire test by a nationally recognized laboratory, or as described in Section 9.5.1.6.1.

Radiation shielding is accomplished using reinforced concrete poured in place, removable concrete block barriers, lead wool, lead brick, leaded glass and high density silicone materials. The lead wool, lead glass and lead brick are not used when the barrier is designated as both shielding and fire barrier. The thickness of the concrete varies as required to meet the specified shielding, but in all cases is greater than the required thickness of a three-hour rated fire barrier.

Where wood construction is used in the Turbine Building all of the wood used is treated with flame retardant.

Interior finishes such as gypsum plaster, ceramic tile, and acoustical ceiling materials are noncombustible. The acoustical tiles are mineral fiber board with a flame spread rating of less than 25 in accordance with ASTM E-84, Surface Burning Characteristics of Building Materials. Containment building protective coating systems have a flame spread rating in accordance with ANSI N101.2-1972. Fire retardant paints and coatings conform to Factory Mutual requirements. When practical, all other non-fire retardant paints and coatings have flame spread and smoke density ratings less than or equal to 25 when tested in accordance with ASTM-84. Vinyl asbestos floor tiles, located in the Control Room, various corridors, and in the office areas in the Turbine Building, have a flame propagation index of less than four. The flame propagation index is in accordance with UL 992, Test Method for Measuring the Flame Propagating Characteristics of Flooring and Floor Covering Materials. Carpeting installed in the Control Room is discussed in Section 9.5.1.6.2.

Steel checker plate hatch covers and removeable concrete hatches are provided in floor openings required for equipment removal. In floors designated as fire barriers, the steel checker plate hatch covers are coated with an approved fire-resistant coating. Protection provided by steel hatch covers has been demonstrated through analysis, in lieu of providing a tested configuration, as described in 9.5.1.6.2. Concrete hatches are constructed such that the designated fire rating of the barrier is maintained.

The reflective piping insulation is composed of stainless steel sheets and foil and the thermal piping insulation is composed of hydrous calcium silicate. Both are 100-percent inorganic and will not burn or support combustion. Anti sweat piping insulation is composed of fiberglass and along with its finishing cement has been tested by UL to the requirements of ASTM E-84 with a flame spread of 25, a fuel contribution of 25, and a smoke development of 50.

2. Penetration Seals

Penetrations in designated fire barriers are sealed with an approved fire stop material except as noted in Subsection 9.5.1.6.2. The penetration seals have fire resistance ratings that meet or exceed the rating designated for the barrier. The majority of the penetrations are sealed with approved silicone materials tested in accordance with the requirements of ASTM E 119 and, in the case of electrical seals, tested in accordance with IEEE 634.

3. Fire Door Assemblies

Door openings in designated fire barriers are provided with approved labeled fire door assemblies except as noted in Subsection 9.5.1.6.2.

9.5.1.5.4 Ventilation System Characteristics

1. Fire Dampers

All ductwork that penetrates a designated fire barrier of two hours or greater is equipped with an approved damper with a rating at least equivalent to that designated for the barrier. Most fire dampers are equipped with heat-responsive elements which automatically release the fire damper blade when the air temperature in the ductwork exceeds the predetermined element operating temperature. Where appropriate, fire dampers are equipped with electro-thermal links. Fire dampers are normally open, but they close during a fire condition. Where applicable, fire dampers located in ductwork are seismically qualified to ensure that the dampers will not close during a seismic event (see Section 9.4.5).

2. Smoke Removal

Smoke will be removed by portable smoke ejectors. Directions for approaching the fire area, placing of the smoke ejectors and routing of the ejector trunk for each fire area will be provided to the fire brigade. This information will be based on the smoke removal study.

The radiation level of the smoke in the subject fire area will be measured. If it is within the allowable limits as outlined in 10 CFR-20, smoke ejection can proceed. However, if the radiation is above the acceptable limits one of the below procedures must be followed:

a. The radiation level will be allowed to decay until it is below the acceptable limit, the smoke then will be released, directly to the atmosphere.

b. The radioactive smoke will be passed through the atmospheric clean-up units, before being released to the atmosphere.

The primary method of smoke removal is via the use of electrically powered portable smoke ejectors. In addition, a sufficient number of gasoline engine powered portable smoke ejectors are provided as a back-up.

Smoke control and heat venting in the Turbine Building will be accomplished by power venting of the mezzanine level using the exhaust fans. Venting of the basement level will be accomplished by natural draft through grating and equipment hatches and openings in the mezzanine floor.

9.5.1.5.5 Electrical Cable and Cable Tray Design - Characteristics

Generally, electrical cables are flame-retardent and nonpropagating in nature and conform to the criterion of IEEE 383- 1974. They will not support combustion in the absence of a sustained ignition source. The cable construction will allow wetting down without structural damage or

electrical faulting. All cable trays, conduits, and their supports are constructed of noncombustible materials.

Outside the Containment buildings, where cable trays containing cabling related to both redundant trains of equipment required to bring the plant to a hot standby condition, and where both trains are located in the same fire area, and are not separated by a negligible combustible horizontal distance of greater than or equal to 20 feet, and are not comprised of one hour fire rated cable, one train of cabling will be protected by at least a one or three-hour rated fire barrier. Where this situation exists, automatic sprinklers are arranged to provide coverage adequate for the hazards in the area or if automatic sprinklers have not been provided in these areas, cabling may be protected within a 3-hour fire barrier. Sprinklers are also provided for cabling where there is a congestion of cable trays see Section 9.5.1.6.1d. Fire stops are provided within the cable trays whenever the cables penetrate walls or floors designated as fire barriers. Fire stops are not provided at intermediate points in vertical or horizontal cable runs, except in long vertical runs. In such instances, fire stops are located at intervals equivalent to floor spacings. It is a general installation practice that vertical tray runs are provided with solid, sheet steel covers for a minimum distance of 4 feet above the floor where necessary for physical protection of the cable. Fire stops are not provided in cable trays inside the Containment Buildings. Conduit fire stops are provided when the conduit penetrates a designated fire barrier and is not run continuously through the fire area (as described in Section 9.5.1.6.1.D.3.d).

9.5.1.5.6 Transformers

All interior transformers are of the air-cooled dry type and do not contain any insulating oil. The main, unit auxiliary, station service and startup transformers are oil-cooled and are located outdoors adjacent to the Turbine Buildings. The main transformers are separated from each other, as well as from the Turbine Buildings, by a blank three-hr rated fire wall. The unit auxiliary transformer and startup transformers XST1, XST1A, and XST2 are separated from the Turbine Buildings by a three-hr rated fire wall. Penetrations in this wall within 50 ft from each side of the center line of the transformer are protected in order to maintain the fire-resistant integrity of the wall. Additional walls are provided extending out from the Turbine Building wall to protect the ventilation openings located in the exterior Turbine Building wall. Station service transformers 1ST and 2ST and alternate transformer XST2A are separated from the Turbine Building by a distance greater than 50 feet. Alternate startup transformer XST1A is separated from adjacent structures by a three-hour rated fire wall.

9.5.1.5.7 Flammable Liquid and Gas Storage

1. Flammable Liquid Storage

   All significant amounts of flammable liquids are stored in separate fire areas that are isolated from the adjacent plant areas by three-hr fire rated barriers. As a minimum, fire detectors are provided in each area, and dependent upon the hazard, a fixed fire extinguishment system is provided. In all instances, such areas do not present a potential hazard for equipment located in the adjacent areas.

2. Flammable Gas Storage

   Bulk storage of all flammable explosive gases is located outside the primary, secondary and turbine plant buildings. The storage facility is an open structure located outdoors in

the yard adjacent to the security fence. An explosion or fire in this area will not affect any of the primary plant buildings.

9.5.1.6 Conclusions

9.5.1.6.1 Comparison with Appendix A of Branch Technical Position APCSB 9.5-1 of Standard Review Plan 9.5.1

As requested by the NRC in their September 30, 1976, letter, the following is a comparison of the CPNPP fire protection program with the guidelines in Appendix A to the above branch technical position.

CPNPP Fire Protection Program

The CPNPP Fire Protection Program is established to ensure that a fire will not prevent safe plant shutdown and will not endanger the health and safety of the public. Fire protection at CPNPP is accomplished using a defense-in-depth approach to include fire detection and extinguishing systems and equipment, administrative controls and procedures, and trained personnel. CPNPP is committed to meeting the requirements contained in the Fire Protection Report.

APCSB 9.5-1 Appendix A

A. Overall Requirements of the Fire Protection Program

A.1 Personnel

Responsibility for the overall fire protection program should be assigned to a designated person in the upper level of management. This person should retain ultimate responsibility even though formulation and assurance of program implementation is delegated. Such delegation of authority should be to staff personnel prepared by training and experience in fire protection and nuclear plant safety to provide a balanced approach in directing the fire protection programs for nuclear power plants.

CPNPP Fire Protection Program

The CPNPP Fire Protection Program meets the requirements of the D. B. Vassallo letter of August 1977 entitled “Nuclear Plant Fire Protection Functional Responsibilities, Administrative Controls and Quality Assurance,” as described in Section 13.3B.

APCSB 9.5-1 Appendix A

A.1 The fire protection staff should be responsible for:

(a) coordination of building layout and systems design with fire area requirements, including consideration of potential hazards associated with postulated design basis fires,

(b) design and maintenance of fire detection, suppression, and extinguishing systems,

(c) fire prevention activities,

(d) training and manual fire-fighting activities of plant personnel and the fire brigade.

CPNPP Fire Protection Program

See Section 13.3B.1.

APCSB 9.5-1 Appendix A

A.1 Subsequently, the FSAR should discuss the training and the updating provisions such as fire drills provided for maintaining the competence of the station fire-fighting and operating crew, including personnel responsible for maintaining and inspecting the fire protection equipment.

CPNPP Fire Protection Program

See Section 13.3B.1.

APCSB 9.5-1 Appendix A

A.1 The qualification requirements for the fire protection engineer or consultant who will assist in the design and selection of equipment, inspect and test the completed physical aspects of the system, develop the fire protection program, and assist in the fire-fighting training for the operating plant should be stated.

CPNPP Fire Protection Program

The personnel who are responsible for the design and selection of equipment, inspection and testing of the completed physical aspects of the system, and development of the fire protection program are Members of the Society of Fire Protection Engineers or equivalent and are knowledgeable in fire protection and detection systems design as well as in the requirements of nuclear plant safety.

APCSB 9.5-1 Appendix A

A.2 Design Bases

The overall fire protection program should be based upon evaluation of potential fire hazards throughout the plant and the effect of postulated design basis fires relative to maintaining ability to perform safety shutdown functions and minimize radioactive releases to the environment.

CPNPP Fire Protection Program

The overall Fire Protection Program is based on the evaluation of potential fire hazards. See Subsections 9.5.1.2.2 and 9.5.1.2.3 for further discussion.

APCSB 9.5-1 Appendix A

A.3 Backup

Total reliance should not be placed on a single automatic fire suppression system. Appropriate backup fire suppression capability should be provided.

CPNPP Fire Protection Program

Appropriate secondary fire suppression capability is provided throughout the plant by fixed water suppression systems, fire hose stations, or portable extinguishers.

APCSB 9.5-1 Appendix A

A.4 Single Failure Criterion

A single failure in the fire suppression system should not impair both the primary and backup fire suppression capability. For example, redundant fire water pumps with independent power supplies and controls should be provided. Postulated fires or fire protection system failures need not be considered concurrent with other plant accidents or the most severe natural phenomena.

CPNPP Fire Protection Program

The fixed fire suppression systems are designed so that a single failure cannot impair both it and the secondary fire suppression system for the particular area. System design parameters and descriptions are stated in Subsection 9.5.1.4.

APCSB 9.5-1 Appendix A

A.4 The effects of lightning strikes should be included in the overall plant fire protection program.

CPNPP Fire Protection Plan

CPNPP is provided with a plantwide lightning protection system. Fire suppression capability is provided such that any fire started by lightning which would affect the plant can be extinguished.

APCSB 9.5-1 Appendix A

A.5 Fire Suppression Systems

Failure or inadvertent operation of the fire suppression system should not incapacitate safety related systems or components. Fire suppression systems that are pressurized during normal plant operation should meet the guidelines specified in APCSB Branch Technical Position 3-1, "Protection Against Postulated Piping Failures in Fluid Systems Outside Containment."

CPNPP Fire Protection Program

Failure or inadvertent operation of the fire suppression system will not incapacitate both paths of systems or components required for safe shutdown. Redundant trains or components that are susceptible to damage from water spray are either physically separated or protection is provided from the effects of water spray. Fire suppression systems located in the Safeguards, Electrical and Control, Auxiliary and Fuel Handling buildings that are pressurized during normal plant operation are in accordance with the specified guidelines. See FSAR Section 3.6B.

APCSB 9.5-1 Appendix A

A.6 Fuel Storage Areas

The fire protection program (plans, personnel and equipment) for buildings storing new reactor fuel and for adjacent fire zones which could affect the fuel storage zone should be fully operational before fuel is received at the site.

CPNPP Fire Protection Program

The Fire Protection Program for the Fuel Building is in effect and the suppression systems operational.

APCSB 9.5-1 Appendix A

A.7 Fuel Loading

The fire protection program for an entire reactor unit should be fully operational prior to initial fuel loading in that reactor unit.

CPNPP Fire Protection Program

The Fire Protection Program for Unit 1 will be in effect and the suppression systems operational prior to initial fuel loading. The same constraint applies to Unit 2.

APCSB 9.5-1 Appendix A

A.8 Multiple Reactor Sites

On multiple-reactor sites where there are operating reactors and construction of remaining units is being completed, the fire protection program should provide continuing evaluation and include additional fire barriers, fire protection capability, and administrative controls necessary to protect the operating units from construction fire hazards. The superintendent of the operating plant should have the lead responsibility for site fire protection.

CPNPP Fire Protection Program

Separation of Unit 1 from Unit 2 is accomplished by the use of equivalent three hour rated fire walls in areas where fire walls are required and by security barriers where fire walls are not required. The Manager, Plant Operations has the lead responsibility for site fire protection.

APCSB 9.5-1 Appendix A

A.9 Simultaneous Fires

Simultaneous fires in more than one reactor need not be postulated, where separation requirements are met. A fire involving more than one reactor unit need not be postulated except for facilities shared between units.

CPNPP Fire Protection Program

This criterion has been incorporated in evaluating the Fire Protection Program for CPNPP

APCSB 9.5-1 Appendix A

B. Administrative Procedures, Controls and Fire Brigade

B.1 Administrative procedures consistent with the need for maintaining the performance of the fire protection system and personnel in nuclear power plants should be provided.

CPNPP Fire Protection Program

Administrative procedures and controls will be utilized to ensure the reliable performance of the Fire Protection System and personnel.

APCSB 9.5-1 Appendix A

B.2 Effective administrative measures should be implemented to prohibit bulk storage of combustible materials inside or adjacent to safety-related buildings or systems during operation or maintenance periods.

CPNPP Fire Protection Program

Effective administrative controls will be utilized to minimize the amount of combustibles that may be exposed in safety-related areas during operation or maintenance period. These controls will govern the following:

1. Proper storage and handling of flammable gases and liquids, HEPA and charcoal filters, dry, unused ion exchange resins and other combustible supplies.

2. Transient fire loads during maintenance and modifications such as flammable liquids, woods, plastic equipment and other combustibles. This control will require an in-plant review of work activities to identify transient fire loads. The supervisor or foreman responsible for supervising the work activity will specify any required additional fire suppression.

3. Waste, debris, scrap, and oil spills resulting from a work activity in a safety-related area will be minimized while work is in progress and removed upon completion of the activity.

4. Periodic inspection for accumulation of combustibles.

APCSB 9.5-1 Appendix A

B.3 Normal and abnormal conditions or other anticipated operations such as modifications (e.g., breaking fire stops, impairment of fire detection and suppression systems) and refueling activities should be reviewed by appropriate levels of management and appropriate special actions and procedures such as fire watches or temporary fire barriers implemented to assure adequate fire protection and reactor safety. In particular:

CPNPP Fire Protection Program

Effective administrative controls will be utilized to protect safety-related equipment from fire damage or loss resulting from work involving ignition sources such as welding, cutting, grinding, or open flame work. Administrative controls will prohibit the use of open flame or combustion smoke for leak testing and other ignition sources in certain areas. Administrative controls will be established to ensure that the following pre-cautions are taken:

APCSB 9.5-1 Appendix A

B.3 (a) Work involving ignition sources such as welding and flame cutting should be done under closely controlled conditions. Procedures governing such work should be reviewed and approved by persons trained and experienced in fire protection. Persons performing and directly assisting in such work should be trained and equipped to prevent and combat fires. If this is not possible, a person qualified in fire protection should directly monitor the work and function as a fire watch.

CPNPP Fire Protection Program

1. Welding, cutting, grinding, or open flame work will be authorized by the responsible foreman or supervisor through a work permit. The responsible foreman or supervisor will have received sufficient firefighting and fire prevention instruction on the use of fire

extinguishers and methods to combat Class A, B and C fires. A fire watch will be established if required by the work permit.

APCSB 9.5-1 Appendix A

B.3 (b) Leak testing, and similar procedures such as air flow determination, should use one of the commercially available aerosol techniques. Open flames or combustion generated smoke should not be permitted.

CPNPP Fire Protection Program

2. Open flames or combustion generated smoke will not be used for leak testing.

APCSB 9.5-1 Appendix A

B.3 (c) Use of combustion material, e.g., HEPA and charcoal filters, dry ion exchange resins or other combustible supplies, in safety related areas should be controlled. Use of wood inside buildings containing safety-related systems or equipment should be permitted only when suitable non-combustible substitutes are not available. If wood must be used, only fire retardent treated wood (scaffolding, lay down blocks) should be permitted. Such materials should be allowed into safety related areas only when they are to be used immediately. Their possible and probable use should be considered in the fire hazard analysis to determine the adequacy of the installed fire protection systems.

CPNPP Fire Protection Program

3. The CPNPP Fire Protection Program uses administrative procedures to control combustibles within safety related areas of the plant.

APCSB 9.5-1 Appendix A

B.4 Nuclear power plants are frequently located in remote areas, at some distance from public fire departments. Also, first response fire departments are often volunteer. Public fire department response should be considered in the overall fire protection program. However, the plant should be designed to be self-sufficient with respect to fire fighting activities and rely on the public response only for supplemental or backup capability.

CPNPP Fire Protection Program

The CPNPP Fire Protection Program is designed for plant self-sufficiency for both fire fighting activities and equipment. Public fire response is regarded as supplemental capability. The public fire department will be called to the site after assessment of the fire condition by the responsible

fire brigade personnel. Fire protection procedures and equipment are compatible with those of the public fire department

APCSB 9.5-1 Appendix A

B.5 through B.7

CPNPP Fire Protection Program

See Section 13.4

APCSB 9.5-1 Appendix A

C. Quality Assurance Program

Quality assurance (QA) programs of applicants and contractors should be developed and implemented to assure that the requirements for design, procurement, installation, and testing and administrative controls for the fire protection program for safety related areas as defined in this Branch Position are satisfied. The program should be under the management control of the QA organization. The QA program criteria that apply to the fire protection program should include the following:

CPNPP Fire Protection Program

Vistra OpCo has developed a comprehensive Quality Assurance program (QA program) as described in Section 17.0. Vistra OpCo has delegated responsibility for the coordination of design, construction and operation of CPNPP to the Nuclear Generation organization. The structures, systems and components covered by the Quality Assurance program are listed in Table 17A-1. The provisions of the QA program apply to all activities which affect the safety-related functions of those systems and components. The preparation, review, and approval of Fire Protection Program procedures will be handled in the same manner as other safety-related procedures.

APCSB 9.5-1 Appendix A

C.1 Design Control and Procurement Document Control

Measures should be established to assure that all design-related guidelines of the Branch Technical Position are included in design and procurement documents and that deviations therefrom are controlled.

CPNPP Fire Protection Program

The fire protection system designer and/or supplier will assure, through a program of design reviews and reviews of procurement documents that the guidelines of the Branch Technical Position are reflected in these documents. Deviation from the requirements of these documents

will be controlled consistent with the original requirements. The applicant shall verify the adequacy of these activities by audit and surveillance.

APCSB 9.5-1 Appendix A

C.2 Instructions, Procedures and Drawings

Inspections, tests, administrative controls, fire drills and training that govern the fire protection program should be prescribed by documented instructions, procedures or drawings and should be accomplished in accordance with these documents.

CPNPP Fire Protection Program

The fire protection program will provide for the availability and use of documented instructions, procedures, or drawings to perform inspections, tests, administrative controls, fire drills and personnel training related to the system. The availability and use of these will be verified through audit and surveillance.

APCSB 9.5-1 Appendix A

C.3 Control of Purchased Material, Equipment and Services

Measures should be established to assure that purchased material, equipment and services conform to the procurement documents.

CPNPP Fire Protection Program

Measures will be established to assure that materials and services related to the fire protection service conform to the procurement documents. These measures will provide for a program of receiving inspection and supplemented by subsequent tests. Procurement activities, whether performed by the applicant or a Fire Protection System supplier, will be covered by the applicant's audit and surveillance program or Underwriters Laboratory surveillance programs.

APCSB 9.5-1 Appendix A

C.4 Inspection

A program for independent inspection of activities affecting fire protection should be established and executed by, or for, the organization performing the activity to verify conformance with documented installation drawings and test procedures for accomplishing the activities.

CPNPP Fire Protection Program

The applicant will, through a program of audit or surveillance, assure that involved organizations, performing activities related to fire protection, establish and execute an internal program of independent inspection. The independent inspection function will verify conformance.

APCSB 9.5-1 Appendix A

C.5 Test and Test Control

A test program should be established and implemented to assure that testing is performed and verified by inspection and audit to demonstrate conformance with design and system readiness requirements. The tests should be performed in accordance with written test procedures; test results should be properly evaluated and acted on.

CPNPP Fire Protection Program

The applicant will verify by a program of audits and surveillance that required inspections are defined, that the tests demonstrate by appropriate acceptance criteria, conformance with design and system readiness requirements. The applicant will verify that tests are performed in accordance with written test procedures, that tests results are properly evaluated, and that observed deficiencies are corrected.

APCSB 9.5-1 Appendix A

C.6 Inspection, Test and Operating Status

Measures should be established to provide for the identification of items that have satisfactorily passed required tests and inspections.

CPNPP Fire Protection Program

The applicant will verify through, audit and surveillance, that tests and inspections are documented as required by specified documentation requirements to provide adequate identification of items that have satisfactorily passed those required tests and inspections.

APCSB 9.5-1 Appendix A

C.7 Non-Conforming Items

Measures should be established to control items that do not conform to specified requirements to prevent inadvertent use or installation.

CPNPP Fire Protection Program

Those items that do not conform to specified requirements will be identified and controlled until proper disposition is made to prevent inadvertent use or installation.

APCSB 9.5-1 Appendix A

C.8 Corrective Action

Measures should be established to assure that conditions adverse to fire protection, such as failures, malfunctions, deficiencies, deviations, defective components, uncontrolled combustible material and non-conformances are promptly identified, reported and corrected.

CPNPP Fire Protection Program

The applicant will assure through a program of testing and inspection that conditions adverse to fire protections, such as failures, malfunctions, deficiencies, deviations, defective components, uncontrolled combustible material and nonconformance are properly identified, reported and corrected.

APCSB 9.5-1 Appendix A

C.9 Records

Records should be prepared and maintained to furnish evidence that the criteria enumerated above are being met for activities affecting the fire protection program.

CPNPP Fire Protection Program

The applicant will assure, through procurement and programmatic requirements, as well as audit and surveillance, that adequate documentation will be prepared and maintained to serve as evidence that the fire protection program is in conformance with the above requirements.

APCSB 9.5-1 Appendix A

C.10 Audits

Audits should be conducted and documented to verify compliance with the fire protection program including design and procurement documents; instructions; procedures and drawings; and inspection and test activities.

CPNPP Fire Protection Program

Audits will be conducted and documented to serve as evidence that the applicant has assured that activities including design, procurement, instructions, procedures, inspections and tests are in compliance with the fire protection program.

APCSB 9.5-1 Appendix A

D. General Guideline for Plant Protection

D.1 Building Design

D.1.a Plant Layouts should be arranged to:

(1) Isolate safety-related systems from unacceptable hazards, and

CPNPP Fire Protection Program

(1) All buildings of the plant are divided into fire areas. The criteria used to develop this arrangement are discussed in Subsection 9.5.1.2.2, 9.5.1.2.3 and 9.5.1.5.1.

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(2) Separate redundant safety related systems from each other so that both are not subject to damage from a single fire hazard.

(2) Alternatives:

(a) Redundant safety-related systems that are subject to damage from a single fire hazard should be protected by a combination of fire retardant coatings and fire detection and suppression systems, or

(b) a separate system to perform the safety function should be provided.

CPNPP Fire Protection Program

(2) (a): Where redundant fire safe shutdown systems, required to bring the plant to a hot standby condition, are located within the same fire area and are subject to damage from a single fire hazard a Fire Hazards Analysis Evaluation demonstrates and documents compliance to that recommended in the guideline by protecting the function with one of the following:

For systems located both inside and outside the Containment Building, see FPR Section II (4.5), Reference [19].

(b) Where a redundant system required to bring the plant to a cold shutdown condition is subject to damage from a single fire hazard, the following will be provided:

1) Fire detection system

2) procedure to repair at least one train of the damaged system within 72 hours.

APCSB 9.5-1 Appendix A

D.1.b In order to accomplish 1.(a) above, safety related systems and fire hazards should be identified throughout the plant. Therefore, a detailed fire hazards analysis should be made. The fire hazards analysis should be reviewed and updated as necessary.

CPNPP Fire Protection Program

The CPNPP Fire Protection Program is based on detailed fire hazard evaluations which satisfy this guideline.

APCSB 9.5-1 Appendix A

D.1.c For multiple reactor sites, cable spreading rooms should not be shared between reactors. Each cable spreading room should be separated from other areas of the plant by barriers (walls and floors) having a minimum fire resistance of three hours. Cabling for redundant safety divisions should be separated by walls having three hour fire barriers.

Alternative guidance for constructed plants is shown in Section F.3, “Cable Spreading Room”.

CPNPP Fire Protection Program

Two cable spreading rooms are included in the design of CPNPP, one for each unit. These rooms are separated by three hour rated fire barriers except as noted in Reference [19]. See Section F.3 for the design description.

APCSB 9.5-1 Appendix A

D.1.d Interior wall and structural components, thermal insulation materials and radiation shielding materials and sound-proofing should be non-combustible. Interior finishes should be non-combustible or listed by a nationally recognized testing laboratory, such as Factory Mutual or Underwriters’ Laboratory, Inc. for flame spread, smoke and fuel contribution of 25 or less in its use configuration (ASTM E-84 Test “Surface Burning Characteristics of Building Materials”).

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in section 9.5.1.6.2. Specific criteria of the structural and interior construction materials are described in Subsection 9.5.1.5.1, 9.5.1.5.2, and 9.5.1.5.3.

APCSB 9.5-1 Appendix A

D.1.e Metal deck roof construction should be non-combustible (see the building materials directory of the Underwriters' Laboratory, Inc.) or listed as Class I by Factory Mutual System Approval Guide.

CPNPP Fire Protection Program

Metal roof deck construction is not used at CPNPP for power block or other safety-related buildings.

APCSB 9.5-1 Appendix A

D.1.f Suspended ceilings and their supports should be of non-combustible construction. Concealed spaces should be devoid of combustibles.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. The use of combustible material is minimized. Also see Guideline D.3.j. See Subsection 9.5.1.5.3.1 for specific criteria.

APCSB 9.5-1 Appendix A

D.1.g High voltage - high amperage transformers installed inside buildings containing safety related systems should be of the dry type or insulated and cooled with non-combustible liquid.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. See Subsection 9.5.1.5.6 for specific criteria.

APCSB 9.5-1 Appendix A

D.1.h Buildings containing safety related systems, having openings in exterior walls closer than 50 feet to flammable oil filled transformers should be protected from the effects of a fire by:

(i) closing of the opening to have fire resistance equal to three hours,

(ii) constructing a three hour fire barrier between the transformers and the wall openings; or

(iii) closing the opening and providing the capability to maintain a water curtain in case of a fire.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in reference [19]. See Subsection 9.5.1.5.6 for specific criteria.

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D.1.i Floor drains, sized to remove expected fire fighting water flow should be provided in those areas where fixed water fire suppression systems are installed. Drains should also be provided in other areas where hand hose lines may be used if such fire fighting water could cause unacceptable damage to equipment in the area. Equipment should be installed on pedestals, or curbs should be provided as required to contain water and direct it to floor drains. (See NFPA 92M, "Waterproofing and Draining of Floors.") Drains in areas containing combustible liquids should have provisions for preventing the spread of the fire throughout the drain system. Water drainage from areas which may contain radioactivity should be sampled and analyzed before discharge to the environment.

In operating plants or plants under construction, if accumulation of water from the operation of new fire suppression systems does not create unacceptable consequences, drains need not be installed.

CPNPP Fire Protection Program

Floor drains are provided in areas protected by fixed water suppression systems. Drainage of water from areas that are not serviced by the floor drain systems is accomplished by using both the floor drains in the corridors adjacent to the respective fire areas and by manual means after a fire. The switchgear and electrical penetration areas, and the Control Room are not provided with floor drains. If a water sprinkler system discharges or if firefighting with hoses is necessary, doors leading into these areas will be opened and the water will run out into adjacent corridor floor drains. In those areas where electrical cabinets are installed, drainage paths are sufficient to prevent excessive water levels. The battery rooms or their adjacent corridors are provided with floor drains. The cable spreading rooms are provided with floor drains. Drainage of areas where combustible liquids are stored in tanks are designed to prevent the spread of a fire through the drain system. The only safety related areas to which this applies is the DG day tank rooms

which do not have floor drains. See Section 9.5.4 for additional details. Drainage from areas that may contain radioactive material is monitored prior to discharge outside the plant.

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D.1.j Floors, walls and ceilings enclosing separate fire areas should have a minimum fire rating of three hours. Penetrations in these fire barriers, including conduits and piping, should be sealed or closed to provide a fire resistance rating at least equal to that of the fire barrier itself. Door openings should be protected with equivalent rated doors, frames and hardware that have been tested and approved by a nationally recognized laboratory. Such doors should be normally closed and locked or alarmed with alarm and annunciation in the control room. Penetrations for ventilation system should be protected by a standard “fire door damper” where required. (Refer to NFPA 80, “Fire Doors with Windows”). The fire hazard in each area should be evaluated to determine barrier requirements. If barrier fire resistance cannot be made adequate, fire detection and suppression should be provided, such as:

(i) water curtain in case of fire,

(ii) flame retardant coatings,

(iii) additional fire barriers.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in Section 9.5.1.6.2 and reference [19]. The barriers, doors, and penetrations are discussed in the applicable sections of Subsection 9.5.1.5. Fire doors located in fire barriers are normally closed or are automatically closed by listed closure devices and released by solenoids, thermal-links, or electro-thermal links. Where doors have been modified for security purposes, the modifications have been accepted by United Laboratory Inc.

Locks and alarms are provided on doors in accordance with the plant fire protection and security program. The fire barrier resistance rating is generally based on the amount of combustible material in the area.

APCSB 9.5-1 Appendix A

D.2 Control of Combustibles

D.2.a Safety-related systems should be isolated or separated from combustible materials. When this is not possible because of the nature of the safety system or the combustible material, special protection should be provided to prevent a fire from defeating the safety system function. Such protection may involve a combination of automatic fire suppression, and construction capable of withstanding and containing a fire that consumes all combustibles present. Examples of such combustible materials that may not be separable from the remainder of its system are:

(1) Emergency diesel generator fuel oil day tanks

(2) Turbine-generator oil and hydraulic control fluid systems

(3) Reactor coolant pump lube oil system.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Separation and protection of required fire safe shutdown systems are discussed for each fire area in the Fire Protection Report.

APCSB 9.5-1 Appendix A

D.2.b Bulk gas storage (either compressed or cryogenic), should not be permitted inside structures housing safety related equipment. Storage of flammable gas such as hydrogen, should be located outdoors or in separate detached buildings so that a fire or explosion will not adversely affect any safety related systems or equipment.

Care should be taken to locate high pressure gas storage containers with the long axis parallel to building walls. This will minimize the possibility of wall penetration in the event of a container failure. Use of compressed gases (especially flammable and fuel gases) inside buildings should be controlled. (Refer to NFPA 6, “Industrial Fire Loss Prevention.”)

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in Section 9.5.1.6.2. Bulk gas storage is discussed in Section 9.5.1.5.7.

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D.2.c The use of plastic materials should be minimized. In particular, halogenated plastics such as polyvinyl chloride (PVC) and neoprene should be used only when substitute non-combustible materials are not available. All plastic materials, including flame and fire retardant materials, will burn with an intensity and BTU production in a range similar to that of ordinary hydrocarbons. When burning, they produce heavy smoke that obscures visibility and can plug air filters, especially charcoal and HEPA. The halogenated plastics also release free chlorine and hydrogen chloride when burning which are toxic to humans and corrosive to equipment.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Plastic material is used when required for radiation resistance reasons. Halongenated plastics such as polyvinyl chloride

(PVC) are not used as an exposed cable insulation or jacketing material at CPNPP except as discussed in Subsection D.3.f.

APCSB 9.5-1 Appendix A

D.2.d Storage of flammable liquids should, as a minimum, comply with the requirements of NFPA 30, “Flammable and Combustible Liquids Code.”

CPNPP Fire Protection Program

The recommendations of NFPA 30 for storage of flammable and combustible liquids were used as guidance in the development of CPNPP plant procedures.

APCSB 9.5-1 Appendix A

D.3 Electric Cable Construction, Cable Trays and Cable Penetrations

D.3.a Only non-combustible materials should be used for cable tray construction.

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. All cable tray construction materials are noncombustible.

APCSB 9.5-1 Appendix A

D.3.b See Section E.3 for fire protection guidelines for cable trays outside the cable spreading room.

CPNPP Fire Protection Program

Criteria are discussed and referenced in Subsection 9.5.1.5.5 and guideline D.3.f.

APCSB 9.5-1 Appendix A

D.3.c Automatic water sprinkler systems should be provided for cable trays outside the cable spreading room. Cables should be designed to allow wetting down with deluge water without electrical faulting. Manual hose stations and portable hand extinguishers should be provided as backup. Safety related equipment in the vicinity of such cable trays, that does not itself require water fire protection, but is subject to unacceptable damage from sprinkler water discharge, should be protected from sprinkler system operation or malfunction.

CPNPP Fire Protection Program

Automatic and manual pre-action water sprinklers are provided for cable trays outside containment and cable spreading rooms where a congestion of cable trays exist (i.e., 4 or more stacked cable trays). Manual hose stations are the primary suppression for cable trays where the following conditions are met:

1. Low cable tray congestion (cable trays are not stacked greater than three high).

2. Cable trays that contain safe shutdown cabling are separated from their redundant trains by three hour rated fire barriers.

3. Smoke detectors are provided in the area of the cables.

Cables are designed to allow wetting down without electrical faulting. Manual hose stations and portable extinguishers are provided as backup. Essential equipment that could be damaged by a sprinkler system discharge is protected. Also see Guideline A.5.

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D.3.d Cable and cable tray penetration of fire barriers (vertical and horizontal) should be sealed to give protection at least equivalent to that fire barrier. The design of fire barriers for horizontal and vertical cable trays should, as a minimum, meet the requirements of ASTM E-119, “Fire Test of Building Construction and Materials,” including the hose stream test.

Alternate criteria: Where installed penetration seals are deficient with respect to fire resistance, these seals may be protected by covering both sides with an approved fire retardant material. The adequacy of using such material should be demonstrated by suitable testing.

CPNPP Fire Protection Program

For conduits which are greater than four (4) inches nominal size, internal seals are installed either at the barrier or on both sides of the barrier at the first opening in the direction of the barrier. These internal seals have a fire rating equal to or greater than that of the fire barrier rating.

For conduits which are less than or equal to four (4) inches nominal size, and automatic suppression and (manually actuated suppression for switchgear and adjacent rooms in Unit 2) detection are provided on both sides of the barrier, internal seals are installed in the barrier; gas and smoke seals are installed at the first opening on both sides of the barrier; or internal seals are installed at the first opening on either side of the barrier with a fire rating equivalent to that of the barrier, unless individually evaluated and documented in Section 9.5.1.6.2 or Reference [19]. For conduits which are less than or equal to four (4) inches nominal size, and automatic suppression (manually actuated suppression for switchgear and adjacent rooms in Unit 2) and detection are not provided on both sides of the barrier, internal seals are installed at the barrier

with a fire rating equivalent to that of the barrier, or gas and smoke seals are installed at the first opening on both sides of the barrier, except as described in 9.5.1.6.2 or Reference [19].

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D.3.e Fire breaks should be provided as deemed necessary by the fire hazards analysis. Flame or flame retardant coatings may be used as a fire break for grouped electrical cables to limit spread of fire in cable ventings. (Possible cable derating owing to use of such coating materials must be considered during design.)

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. Fire breaks are provided as described in Subsection 9.5.1.5.5.

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D.3.f Electric cable constructions should as a minimum pass the current IEEE No. 383 flame test. (This does not imply that cables passing this test will not require additional fire protection).

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with this guideline except as noted in section 9.5.1.6.2. Electrical cable construction is described in Section 9.5.1.5.5.

APCSB 9.5-1 Appendix A

D.3.g To the extent practical, cable construction that does not give off corrosive gases while burning should be used.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guidelines to the extent practical with present day cable insulation and jacket material.

APCSB 9.5-1 Appendix A

D.3.h Cable trays, raceways, conduit, trenches, or culverts should be used only for cables. Miscellaneous storage should not be permitted, nor should piping for flammable or combustible liquids or gases be installed in these areas.

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline.

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D.3.i The design cable tunnels, culverts and spreading rooms should provide for automatic or manual smoke venting as required to facilitate manual fire fighting capability.

CPNPP Fire Protection Program

The Fire Protection Program provides for manual smoke venting to enable manual fire fighting. See Subsection 9.5.1.5.4 for criteria on smoke venting.

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D.3.j Cables in the control room should be kept to the minimum necessary for operation of the control room. All cables entering the control room should terminate there. Cables should not be installed in floor trenches or culverts in the control room.

CPNPP Fire Protection Program

The cables in the Control Room are the minimum necessary for operation of the plant. There are no cables routed in floor trenches in the Control Room. There is, however, a small amount of cabling enclosed in steel conduit, routed above the suspended ceiling in the Control Room. Fire detection is provided for this concealed area.

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D.4 Ventilation

D.4.a The products of combustion that need to be removed from a specific fire area should be evaluated to determine how they will be controlled. Smoke and corrosive gases should generally be automatically discharged directly outside to a safe location. Smoke and gases containing radioactive materials should be monitored in the fire area to determine if release to the environment is within the permissible limits of the Offsite Dose Calculation Manual.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. See Section 9.5.1.5.4 for further information.

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D.4.b Any ventilation system designed to exhaust smoke or corrosive gases should be evaluated to ensure that inadvertent operation or single failures will not violate the controlled areas of the plant design. This requirement includes containment functions for protection of the public and maintaining habitability for operations personnel.

CPNPP Fire Protection Program

Ventilation of smoke or corrosive gases, resulting from a fire will be accomplished manually as required, and subsequent to monitoring for radioactivity and evaluating the products of combustion.

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D.4.c The power supply and controls for mechanical ventilation systems should be run outside the fire area served by the system.

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. The power supplies for smoke ejectors are described in Subsection 9.5.1.5.4.

APCSB 9.5-1 Appendix A

D.4.d Fire suppression systems should be installed to protect charcoal filters in accordance with Regulatory Guide 1.52, “Design Testing and Maintenance Criteria for Atmospheric Cleanup Air Filtration.”

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. All HVAC charcoal absorber beds are provided with integral water spray systems. These systems are described in Subsection 9.5.1.4.2.c.

APCSB 9.5-1 Appendix A

D.4.e The fresh air supply intakes to areas containing safety related equipment or systems should be located remote from the exhaust air outlets and smoke vents of other fire areas to minimize the possibility of contaminating the intake air with the products of combustion.

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. The use of combustible materials is minimized. In general, ventilation air intakes are located on the exterior walls and the outlets are located on the roof.

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D.4.f Stairwells should be designed to minimize smoke infiltration during a fire. Staircases should serve as escape routes and access routes for fire fighting. Fire exit routes should be clearly marked. Stairwells, elevators and chutes should be enclosed in masonry towers with minimum fire rating of three hours and automatic fire doors at least equal to the enclosure construction, at each opening into the building. Elevators should not be used during fire emergencies.

CPNPP Fire Protection Program

The Fire Protection Program complies with this guidance except as noted in section 9.5.1.6.2. Walls enclosing stairwells are constructed of concrete and gypsum dry walls and are provided with Class B labeled automatic fire doors. All enclosed stairwells are pressurized to minimize smoke infiltration.

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D.4.g Smoke and heat vents may be useful in specific areas such as cable spreading rooms and diesel fuel oil storage areas and switchgear rooms. When natural-convection ventilation is used, a minimum ratio of 1 sq. foot of venting area per 200 sq. feet of floor area should be provided. If forced-convection ventilation is used, 300 CFM should be provided for every 200 sq. feet of floor area. See NFPA No. 204 for additional guidance of smoke control.

CPNPP Fire Protection Program

Smoke and heat venting will be accomplished manually as required. See Section 9.5.1.5.4 for discussion.

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D.4.h Self-contained breathing apparatus, using full face positive pressure masks, approved by NIOSH (National Institute for Occupational Safety and Health - approval formerly given by the U.S. Bureau of Mines) should be provided for fire brigade, damage control and control room personnel. Control room personnel may be furnished breathing air by a manifold system piped from a storage reservoir if practical. Service of operating life should be a minimum of one-half hour for the self-contained units.

CPNPP Fire Protection Program

Self-contained breathing apparatus are provided for fire brigade, damage control, and control room personnel. Service life for the self contained units exceed one-half hour.

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D.4.h At least two extra air bottles should be located onsite for each self-contained breathing unit. In addition, an onsite 6-hour supply of reserve air should be provided and arranged to permit quick and complete replenishment of exhausted supply air bottles as they are returned. If compressors are used as a source of breathing air, only units approved for breathing air should be used. Special care must be taken to locate the compressor in areas free of dust and contaminants.

CPNPP Fire Protection Program

The Fire Protection Program complies with this guideline.

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D.4.i Where total flooding gas extinguishing systems are used, area intake and exhaust ventilation dampers should close upon initiation of gas flow to maintain necessary gas concentration. (See NFPA 12, “Carbon Dioxide Systems”, and 12A, “Halon 1301 Systems.”)

CPNPP Fire Protection Program

The Fire Protection Program complies with the guideline. See Subsection 9.5.1.5.4 for description of fire dampers.

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D.5 Lighting and Communication

Lighting and two way voice communication are vital to safe shutdown and emergency response in the event of fire. Suitable fixed and portable emergency lighting and communication devices should be provided to satisfy the following requirements:

(a) Fixed emergency lighting should consist of sealed beam units with individual 8-hour minimum battery power supplies.

(b) Suitable sealed beam battery powered portable hand lights should be provided for emergency use.

CPNPP Fire Protection Program

Areas containing fire safe shutdown equipment required to achieve hot standby, and primary interior egress and access routes between these areas, are provided with DC Emergency Lighting supplied by 8 hour sealed beam or fluorescent lamp battery power pack units (except in

the Control Room). DC Emergency Lighting in the Control Room is supplied power from the non-class 1E dedicated 8-hour batteries. (See FSAR Section 9.5.3.2.1 and 9.5.1.6.2). Supplemental lighting is provided from battery-powered hand held portable lights.

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D.5 (c) Fixed emergency communication should use voice powered head sets at pre-selected stations.

CPNPP Fire Protection Program

The Fire Protection Program provides intra plant portable radio with page-party/public address system backup for use in emergency conditions instead of voice powered head sets. For additional description of the communication systems, see Subsection 9.5.2.2.

APCSB 9.5-1 Appendix A

D.5 (d) Fixed repeater installed to permit use of portable radio communication units should be protected from exposure to fire damage.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with this guideline by providing radio to radio "talkaround" and plant page party/public address system capability in the event of fire damage to the repeater.

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E. Fire Detection and Suppression

E.1 Fire Detection

E.1.a Fire detection systems should as a minimum comply with NFPA 72D, "Standard for the Installation, Maintenance and Use of Proprietary Protective Signaling Systems."

CPNPP Fire Protection Program

The Fire Protection Program complies with this guideline set forth by the NRC Fire detection system design parameters are described in Subsection 9.5.1.4.2. Location and placement of detectors are in accordance with the guidelines of NFPA 72E except where special conditions did not permit. In such cases a Fire Protection Engineer located the detector based on engineering judgement as permitted by NFPA 72E or provided justification per reference [19].

APCSB 9.5-1 Appendix A

E.1.b Fire detection system should give audible and visual alarm and annuncciation in the control room. Local audible alarms should also sound at the location of the fire.

CPNPP Fire Protection Program

Fire detection systems give audible and visual alarms and annunciation in the Control Room. Local audible alarms sound at the local detector control panel. Fire protection procedures provide for the use of the Public Address System (Gaitronics) for audible annunciation of the fire alarm location throughout the plant upon receipt of an alarm in the Control Room except in the case of a fire in the Control Room complex. A separate alarm is provided for audible annunciation of a fire in the Control Room complex. The Gaitronics will be maintained operational at all times with a minimum reasonable outage duration for maintenance or repair.

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E.1.c Fire alarms should be distinctive and unique. They should not be capable of being confused with any other plant system alarms.

CPNPP Fire Protection Program

Fire alarms are distinctive and unique and will not be confused with other plant alarms.

APCSB 9.5-1 Appendix A

E.1.d Fire detection and actuation systems should be connected to the plant emergency power supply.

CPNPP Fire Protection Program

Fire detection and actuation systems are connected to plant emergency power supplies in accordance with NFPA 72D Section 2220.

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E.2 Fire Protection Water Supply Systems

E.2.a An underground yard fire main loop should be installed to furnish anticipated fire water requirements. NFPA 24 - “Standard for Outside Protection” - gives necessary guidance for such installation. It references other design codes and standards developed by such organizations as the American National Standards Institute (ANSI) and the American Water Works Association (AWWA). Lined steel or cast iron pipe should be used to reduce internal tuberculation. Such tuberculation deposits in an unlined pipe over a period of years can significantly reduce water flow through the combination of increased friction and reduced pipe diameter. Means for treating and flushing the systems should be provided. Approved visually indicating sectional control valves, such as Post Indicator Valves, should be provided to isolate portions of the main for maintenance or repair without shutting off the entire system.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. The system is described in Subsection 9.5.1.4.

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E.2.a The fire main system piping should be separate from service or sanitary water system piping.

CPNPP Fire Protection Program

CPNPP is in compliance with the guideline

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E.2.b A common yard fire main loop may serve multi-unit nuclear power plant sites, if cross-connected between units. Section control valves should permit maintaining independence of the individual loop around each unit. For such installations, common water supplies may also be utilized. The water supply should be sized for the largest single expected flow. For multiple reactor sites with widely separated plants (approaching 1 mile or more), separate yard fire main loops should be used.

CPNPP Fire Protection Program

A common yard fire main loop is provided for the entire CPNPP site. Sectional control valves are provided for isolation of sections and to ensure at least one flow path to the individual branch lines provided for various buildings with any one section isolated. This arrangement eliminates the requirement for cross connection between units. See Subsection 9.5.1.4 for a description of the sizing of the common water supply.

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E.2.c If pumps are required to meet system pressure or flow requirements, a sufficient number of pumps should be provided so that 100 percent capacity will be available with one pump inactive (e.g., three 50 percent pumps or two 100 percent pumps). The connection to the yard fire main loop from each fire pump should be widely separated, preferably located on opposite sides of the plant. Each pump should have its own driver with independent power supplies and control. At least one pump (if not powered from the emergency diesels) should be driven by non-electrical means, preferably diesel engine. Pumps and drivers should be located in rooms separated from the remaining pumps and equipment by a minimum three hour fire wall. Alarms indicating pump running, driver availability, or failure to start should be provided in the control room.

Details of the fire pump installation should as a minimum conform to NFPA 20, “Standard for the Installation of Centrifugal Fire Pumps.”

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Three 50 percent capacity pumps are provided. Two of the three pumps are diesel engine-driven pumps and are located in separate fire compartments in the pump house. The other fire pump is an electric motor-driven pump and is located in a separate fire compartment in the pump house. An independent connection to the loop is provided for each pump. See Subsection 9.5.1.4 for further details concerning the pumps.

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E.2.d Two separate reliable water supplies should be provided. If tanks are used, two 100 percent (minimum of 300,000 gallons each) system capacity tanks should be installed. They should be so interconnected that pumps can take suction from either or both. However, a leak in one tank or its piping should not cause both tanks to drain. The main plant fire water supply capacity should be capable of refilling either tank in a minimum of eight hours.

Common tanks are permitted for fire and sanitary or service water storage. When this is done, however, minimum fire water storage requirements should be dedicated by means of a vertical standpipe for other water services.

CPNPP Fire Protection Program

Two 100 percent capacity atmospheric fire water storage tanks are provided, each with a nominal capacity of 524,500 gallons. The tanks are interconnected to facilitate suction from either or both tanks. Refill capability with a separate pump, which takes suction from the safe shutdown impoundment (SSI), is provided to allow either tank to be refilled within 8 hours after using its contents to extinguish a fire.

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E.2.e The fire water supply (total capacity and flow rate) should be calculated on the basis of the largest expected flow rate for a period of two hours, but not less than 300,000 gallons. This flow rate should be based (conservatively) on 1,000 gpm for manual hose streams plus the greater of:

(1) all sprinkler heads opened and flowing in the largest designed fire area; or

(2) the largest open head deluge system(s) operating.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in section 9.5.1.6.2. The fire water supply capacity and flow rate is described in Subsection 9.5.1.4.2

APCSB 9.5-1 Appendix A

E.2.f Lakes or fresh water ponds of sufficient size may qualify as the sole source of water for fire protections, but requires at least two intakes to the pump supply. When a common water supply is permitted for fire protection and the ultimate heat sink, the following conditions should also be satisfied:

(1) The additional fire protection water requirements are designed into the total storage capacity; and

(2) Failure of the fire protection system should not degrade the function of ultimate heat sink.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. CPNPP employs two storage tanks dedicated to fire protection use for water supply requirements.

APCSB 9.5-1 Appendix A

E.2.g Outside manual hose installation should be sufficient to reach any location with an effective hose stream. To accomplish this, hydrants should be installed approximately every 250 feet on the yard main system. The lateral to each hydrant from the yard main should be controlled by a visually indicating or key operated (curb) valve. A hose house, equipped with hose and combination nozzle, and other auxiliary equipment recommended in NFPA 24, “Outside Protection,” should be provided as needed but at least every 1,000 feet.

Threads compatible with those used by local fire departments should be provided on all hydrants, hose couplings and standpipe risers.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Auxiliary gate valves (curb box valves) are provided adjacent to each hydrant in accordance with ANI criteria. A description of the outside hose station layout, auxiliary equipment contained in each cabinet and the type of threads on all fire fighting equipment is provided in Subsection 9.5.1.4.

APCSB 9.5-1 Appendix A

E.3 Water Sprinklers and Hose Standpipe Systems

E.3.a Each Automatic sprinkler system and manual hose station standpipe should have an independent connection to the plant underground water main. Headers fed from each end are permitted inside buildings to supply multiple sprinkler and standpipe systems. When provided, such headers are considered an extension of the yard main system. The header arrangement should be such that no single failure can impair both the primary and backup fire protection systems.

Each sprinkler and standpipe system should be equipped with OS&Y (outside screw and yoke) gate valve, or other approved shut off valve, and water flow alarm. Safety related equipment that does not itself require sprinkler water fire protection, but is subject to unacceptable damage if wetted by sprinkler water discharge should be protected by water shields or baffles.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in Section 9.5.1.6.2. The connection of each automatic sprinkler system and hose station is described in Subsection 9.5.1.4.

Also see guideline D.3.c.

APCSB 9.5-1 Appendix A

E.3.b All valves in the fire water systems should be electrically supervised. The electrical supervision signal should indicate in the control room and other appropriate command locations in the plant (See NFPA 26, “Supervision of Valves”).

Alternate Criteria: When electrical supervision of fire protection valves is not practicable, an adequate management supervision program should be provided. Such a program should include locking valves open with strict key control; tamper proof seals; and periodic, visual check of all valves.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the criteria. All isolation or sectional control valves are electrically supervised, or locked in the appropriate position. Appropriate procedures will be developed for key control and for periodic visual inspection of valves.

APCSB 9.5-1 Appendix A

E.3.c Automatic sprinkler systems should as a minimum conform to requirements of appropriate standards such as NFPA 13, “Standard for the Installation of Sprinkler Systems”, and NFPA 15, “Standard for Water Spray Fixed Systems.”

CPNPP Fire Protection Program

Automatic sprinkler systems comply with requirements of NFPA 13 and NFPA 15. Water spray systems for the charcoal filters are designed to the requirements of Regulatory Guide 1.52.

Design densities for cable tray suppression systems located in congested cable areas meet the applicable NFPA Standards (Reference 24). Specific differences with the applicable NFPA standards are identified and justified in Reference [19].

APCSB 9.5-1 Appendix A

E.3.d Interior manual hose installation should be able to reach any location with at least one effective hose stream. To accomplish this, standpipes with hose connections, equipped with a maximum of 75 feet of 1-1/2 inch woven jacket lined fire hose and suitable nozzles should be provided in all buildings, including containment, on all floors and should be spaced at not more than 100 foot intervals. Individual standpipes should be of at least 4-inch diameter for multiple hose connections and 2-1/2-inch diameter for single hose connections. These systems should follow the requirements of NFPA 14 for sizing, spacing and pipe support requirements (NELPIA).

Hose stations should be located outside entrances to normally unoccupied areas and inside normally occupied areas. Standpipes serving hose stations in areas housing safety related equipment should have shut off valves and pressure reducing devices (if applicable) outside the area.

CPNPP Fire Protection Program

NFPA 14 was used as guidance for installation of Class II service interior manual hose stations. Each hose station is equipped with 100 feet of 1-1/2 inch woven jacket lined fire hose and a nozzle compatible with the type of fire postulated. The spacing of the hose stations ensures that at least one effective hose stream can reach any location in safety-related areas of the plant except where identified and justified in the Fire Protection Report, Reference [19]. NFPA 14 was used as guidance for sizing of Class II type standpipes and hose systems. For a further description of the interior hose stations see Subsection 9.5.1.4.

APCSB 9.5-1 Appendix A

E.3.e The proper type of hose nozzles to be supplied to each area should be based on the fire hazard analysis. The usual combination spray/straight-stream nozzle may cause unacceptable mechanical damage (for example, the delicate electronic equipment in the control room) and be unsuitable. Electrically safe nozzles should be provided at locations where electrical equipment or cabling is located.

CPNPP Fire Protection Program

FOG type nozzles are provided in hose cabinets inside plant buildings. Outside hose houses are provided with combination nozzles.

APCSB 9.5-1 Appendix A

E.3.f Certain fires such as those involving flammable liquids respond well to foam suppression. Consideration should be given to use of any of the available foams for such specialized protection application. These include the more common chemical and mechanical low expansion foams, high expansion foam and the relatively new Aqueous Film Forming Foam (AFFF).

CPNPP Fire Protection Program

No fixed foam systems are used at the plant but manual foam-based fire fighting equipment and training is provided.

APCSB 9.5-1 Appendix A

E.4 Halon Suppression Systems

The use of Halon fire extinguishing agents should as a minimum comply with the requirements of NFPA 12A and 12B, “Halogenated Fire Extinguishing Agent Systems - Halon 1301 and Halon 1211.” Only UL or FM approved agents should be used.

In addition to the guidelines of NFPA 12A and 12B, preventative maintenance and testing of the systems, including check weighing of the Halon cylinders should be done at least quarterly.

Particular consideration should also be given to:

(a) minimum required Halon concentration and soak time;

(b) toxicity of Halon;

(c) toxicity and corrosive characteristics of thermal decomposition products of Halon.

CPNPP Fire Protection Program

The CPNPP Fire Protection Program uses NFPA 12A and 12B, “Halogenated Fire Extinguishing Agent Systems - Halon 1301 and 1211,” as a guideline for the halon suppression system design and installation. Design parameters and a system description are stated in Subsection 9.5.1.4.2. Administrative controls for Halon systems are in accordance with Reference [19].

APCSB 9.5-1 Appendix A

E.5 Carbon Dioxide Suppression Systems

The use of carbon dioxide extinguishing systems should as a minimum comply with the requirements of NFPA 12, "Carbon Dioxide Extinguishing Systems."

Particular consideration should also be given to:

(a) minimum required CO_2 concentration and soak time;

(b) toxicity of CO_2;

(c) possibility of secondary thermal shock (cooling) damage;

(d) offsetting requirements for venting during CO_2 injection to prevent overpressurization versus sealing to prevent loss of agent;

(e) design requirements from overpressurization; and

(f) possibility and probability of CO_2 systems being out of service because of personnel safety consideration. CO_2 systems are disarmed whenever people are present in an area so protected. Areas entered frequently (even though duration time for any visit is short) have often been found with CO_2 systems shut off.

CPNPP Fire Protection Program

The Fire Protection Program does not use fixed carbon dioxide suppression systems.

APCSB 9.5-1 Appendix A

E.6 Portable Extinguishers

Fire extinguishers should be provided in accordance with guidelines of NFPA 10 and 10A, "Portable Fire Extinguishers Installation, Maintenance and Use." Dry chemical extinguishers should be installed with due consideration given to cleanup problems after use and possible adverse effects on equipment installed in the area.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. See Subsection 9.5.1.4.2 for description of portable fire extinguishers.

APCSB 9.5-1 Appendix A

F. Guidelines for Specifics Plant Areas

F.1 Primary and Secondary Containment

F.1.a Normal Operation

Fire protection requirements for the primary and secondary containment areas should be provided on the basis of specific identified hazards. For example:

- Lubricating oil or hydraulic fluid system for the primary coolant pumps
- Cable tray arrangements and cable penetrations
- Charcoal filters

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with this guideline except as noted in Section 9.5.1.6.2. The Fire Protection Report identifies specific hazards inside containment. Fire detection and suppression are provided accordingly.

APCSB 9.5-1 Appendix A

F.1.a Fire suppression systems should be provided based on the fire hazards analysis.

Fixed fire suppression capability should be provided for hazards that could jeopardize safe plant shutdown. Automatic sprinklers are preferred. An acceptable alternate is automatic gas (Halon or CO_2) for hazards identified as requiring fixed suppression protection.

An enclosure may be required to confine the agent if a gas system is used. Such enclosures should not adversely affect safe shutdown, or other operating equipment in containment.

CPNPP Fire Protection Program

Automatic water spray systems are provided for the carbon absorber beds of the pre-access filter units located inside the Containment. See Section 9.5.1.4.2 for description of these systems. The reactor coolant pumps are equipped with an oil collection system. This system is designed to collect and drain, to a safe place, any oil which may be discharged from the reactor coolant pump lubrication system.

APCSB 9.5-1 Appendix A

F.1.a Operation of the fire protection systems should not compromise integrity of the containment or the other safety related systems. Fire protection activities in the containment areas should function in conjunction with total containment requirements such as control of contaminated liquid and gaseous release and ventilation.

CPNPP Fire Protection Program

Operation of the fire protection system does not compromise the integrity of the Containment.

APCSB 9.5-1 Appendix A

F.1.a Fire detection systems should alarm and annunciate in the control room. The type of detection used and the location of the detectors should be most suitable to the particular type of fire that could be expected from the identified hazard. A primary containment general area fire detection capability should be provided as backup for the above described hazard detection. To accomplish this, suitable smoke detection (e.g., visual obscuration, light scattering and particle counting) should be installed in the air recirculation system ahead of any filters.

Automatic fire suppression capability need not be provided in the primary containment atmospheres that are inserted during normal operation. However, special fire protection requirements during refueling and maintenance operations should be satisfied as provided below.

CPNPP Fire Protection Program

Fire detection is provided throughout the Containment in accordance with the results of Fire Hazard Analysis Evaluations, which are summarized in the Fire Protection Report. Alarms are provided in the Control Room to annunciate a fire condition via the plant public address (PA) system.

APCSB 9.5-1 Appendix A

F.1.b Refueling and Maintenance

Refueling and maintenance operations in containment may introduce additional hazards such as contamination control materials, decontamination supplies, wood planking, temporary wiring, welding and flame cutting (with portable compressed fuel gas supply). Possible fires would not necessarily be in the vicinity of fixed detection and suppression systems.

Management procedures and controls necessary to assure adequate fire protection are discussed in Section 3a.

In addition, manual fire fighting capability should be permanently installed in containment. Standpipes with hose stations, and portable fire extinguishers, should be installed at strategic locations throughout containment for any required manual fire fighting operations.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Standpipes and hose stations are located on each elevation in the Containment Buildings such that an effective hose stream can reach any location, except as identified and justified in Reference [19]. Portable extinguishers are provided for each Containment Building. To reduce radiation related deterioration, portable extinguishers and hoses may be removed from the containment during

normal operation. Whenever the containment is to be occupied for longer than 48 hours, the portable extinguishers and hoses are returned to their designated locations in containment.

APCSB 9.5-1 Appendix A

F.1.b. Adequate self-contained breathing apparatus should be provided near the containment entrances for fire fighting and damage control personnel. These units should be independent of any breathing apparatus or air supply systems provided for general plant activities.

CPNPP Fire Protection Program

Adequate, self-contained breathing apparatus are provided at the fire brigade staging area located in the turbine building. By procedure the fire brigade gathers all required equipment from this location for any fire in the plant including containment fires. This location is sufficiently near the containment to provide for prompt action. Additional air cylinders are located at the containment entrance.

APCSB 9.5-1 Appendix A

F.2 Control Room

(a) The Control Room is essential to safe reactor operation. It must be protected against disabling fire damage and should be separated from other areas of the plant by floors, walls and roofs having minimum fire resistance ratings of three hours.

(b) Exposure fire involving combustibles in the general room area.

CPNPP Fire Protection Program

The Fire Protection Program is in general compliance with the guideline, except as noted in FSAR Section 9.5.1.6.2. The Control Room is separated from other areas of the plant by three hour rated fire barriers, except as noted in Reference [19].

APCSB 9.5-1 Appendix A

F.2 Control Room cabinets and consoles are subject to damage from two distinct fire hazards:

(a) Fire originating within a cabinet or console; and

(b) Exposure fire involving combustibles in the general room area.

Manual fire fighting capability should be provided for both hazards.

CPNPP Fire Protection Program

Manual fire fighting capability employing portable water and Halon extinguishers and hose stations are provided inside the Control Room.

APCSB 9.5-1 Appendix A

F.2 Hose stations and portable water and Halon extinguishers should be located in the control room to eliminate the need for operators to leave the control room. An additional hose piping shut off valve and pressure reducing device should be installed outside the control room.

Hose stations adjacent to the control room with portable extinguishers in the control room are acceptable.

Nozzles that are compatible with the hazards and equipment in the control room should be provided for the manual hose station. The nozzles chosen should satisfy actual fire fighting needs, satisfy electrical safety and minimize physical damage to electrical equipment from hose stream impingement.

Fire detection in the control room cabinets, and consoles should be provided by smoke and heat detectors in each fire area. Alarm and annunciation should be provided in the control room. Fire alarms in other parts of the plant should also be alarmed and annunciated in the control room.

CPNPP Fire Protection Program

Manual fire fighting capability employing portable water and Halon extinguishers, as well as hose stations, is provided in the Control Room.

Fire detection is provided in each console in the Control Room except as noted in Section 9.5.1.6.2.

The Fire Detection Main Control Panel is located in the Control Room. Any fire alarm actuated anywhere in the plant is indicated audibly and visually on this panel.

APCSB 9.5-1 Appendix A

F.2 Breathing apparatus for control room operators should be readily available. Control room floors, ceiling, supporting structures, and walls, including penetrations and doors, should be designed to a minimum fire rating of three hours. All penetration seals should be air tight.

CPNPP Fire Protection Program

Breathing apparatus are located in the Control Room

APCSB 9.5.1 Appendix A

F.2 The control room ventilation intake should be provided with smoke detection capability to automatically alarm locally and isolate the control room ventilation system to protect operators by preventing smoke from entering the control room.

CPNPP Fire Protection Program

The Control Room air intake is provided with smoke detection which alarms in the Control Room so that the ventilation system can be manually placed in the isolation mode upon detection of smoke in the intake duct.

APCSB 9.5-1 Appendix A

F.2 Manually operated venting of the control room should be available so that operators have the option of venting for visibility.

CPNPP Fire Protection Program

The ventilation systems can be manually operated as required for smoke venting. Portable equipment can be used for supplementary venting.

APCSB 9.5-1 Appendix A

F.2 Cables should not be located in concealed floor and ceiling spaces. All cables that enter the control room should terminate in the control room. That is, no cabling should be simply routed through the control room from one area to another.

CPNPP Fire Protection Program

All cables entering the Control Room terminate in the Control Room. There is a small amount of cabling which is enclosed in steel conduit, except telephone cable between junction boxes and handset stations, and a small amount of metal clad cable routed above the Control Room suspended ceiling for which detection is provided.

There is a small amount of metal clad cable or cabling enclosed in steel conduit routed under the raised platform of the Control Room center console. Detection is not provided for this under platform location.

APCSB 9.5-1 Appendix A

F.3 Cable Spreading Room

F.3.a The preferred acceptance methods are:

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline.

APCSB 9.5-1 Appendix A

F.3.a.1 Automatic water system such as closed head sprinklers, open head deluge, or open directional spray nozzles. Deluge and open spray systems should have provisions for manual operation at a remote station; however, there should also be provisions to preclude inadvertent operation. Location of sprinkler heads or spray nozzles should consider cable tray sizing and arrangements to assure adequate water coverage. Cables should be designed to allow wetting down with deluge water without electrical faulting. Open head deluge and open directional spray systems should be zoned so that a single failure will not deprive the entire area of automatic fire suppression capability. The use of foam is acceptable, provided it is of a type capable of being delivered by a sprinkler or deluge system, such as an Aqueous Film Forming Foam (AFFF).

CPNPP Fire Protection Program

Primary fire suppression for each Cable Spreading Room is provided by a total-flooding Halon 1301 system. Class “A” detection loop circuitry is used to actuate the Halon suppression system. A manual pre-action sprinkler system is provided as backup to the halon system.

Cables are designed to allow wetting without electrical faulting.

APCSB 9.5-1 Appendix A

F.3.a.2 Manual hoses and portable extinguishers should be provided as backup.

CPNPP Fire Protection Program

Two hose stations and portable extinguishers are provided for each cable spreading area. See Section 9.5.1.4.2.

APCSB 9.5-1 Appendix A

F.3.a.3 Each cable spreading room of each unit should have divisional cable separation, and be separated from the other and the rest of the plant by a minimum three hour rated fire wall (Refer to NFPA 251 or ASTM E-119 for fire test resistance rating).

CPNPP Fire Protection Program

The unit 1 and 2 cable spreading rooms are separated from each other as well as from the adjacent plant areas by three hour rated fire walls except as noted in Reference [19].

APCSB 9.5-1 Appendix A

F.3.a.4 At least two remote and separate entrances are provided to the room for access by fire brigade personnel; and

CPNPP Fire Protection Program

Two remote and separate entrances from two different fire areas are provided for each cable spreading room.

APCSB 9.5-1 Appendix A

F.3.a.5 Aisle separation provided between tray stacks should be at least three feet wide and eight feet high.

CPNPP Fire Protection Program

Most cabling, except conduit and cable trays run adjacent to the walls, is run to provide a minimum height of eight feet between the lowest tray and the floor. This arrangement allows more free area and is more practical for manual firefighting than aisle separation.

APCSB 9.5-1 Appendix A

F.3.b For cable spreading rooms that do not provide divisional cable separation of F.3.a.3, in addition to meeting F.3.a.1, .2, .4 and .5 above, the following should also be provided:

F.3.b.1 Divisional cable separation should meet the guidelines of Regulatory Guide 1.75, “Physical Independence of Electric Systems.”

CPNPP Fire Protection Program

See Section 8.3.1.4.

APCSB 9.5-1 Appendix A

F.3.b.2 All cabling should be covered with a suitable fire retardant coating.

CPNPP Fire Protection Program

See Section 9.5.1.6.2. Also, see Guideline F.3.b.4.

APCSB 9.5-1 Appendix A

F.3.b.3 As an alternate to F.3.a.1 above, automatically initiated gas systems (Halon or CO_2) may be used for primary fire suppression, provided a fixed water system is used as a backup.

CPNPP Fire Protection Program

See Section F.3.a.1.

APCSB 9.5-1 Appendix A

F.3.b.4 Plants that cannot meet the guideline of Regulatory Guide 1.75, in addition to meeting F.3.a.1, .2, .4 and .5 above, an auxiliary shutdown system with all cabling independent of the cable spreading room should be provided.

CPNPP Fire Protection Program

The plant is capable of being safely shutdown in the event a design basis fire occurs in the cable spreading room. Alternate shutdown systems and procedures are provided using shutdown paths which are independent of the cable spreading room. See Section 7.4 for more discussion.

APCSB 9.5-1 Appendix A

F.4 Plant Computer Room

Safety-related computers should be separated from other areas of the plant by barriers having a minimum three hour fire resistant rating. Automatic fire detection should be provided to alarm and annunciate in the control room and alarm locally. Manual hose stations and portable water and halon fire extinguishers should be provided.

CPNPP Fire Protection Program

The plant computers are not safety-related.

APCSB 9.5-1 Appendix A

F.5 Switchgear Rooms

Switchgear rooms should be separated from the remainder of the plant by minimum three hour rated fire barriers to the extent practicable. Automatic fire detection should alarm and annunciate in the control room and alarm locally. Fire hose stations and portable extinguishers should be readily available.

Acceptable protection for cables that pass through the switchgear room is automatic water or gas agent suppression. Such automatic suppression must consider preventing unacceptable damage to electrical equipment and possible necessary containment of agent following discharge.

CPNPP Fire Protection Program

There are two safety related switchgear rooms in each unit: Train A and Train B. Each switchgear room is in a separate fire area. Manual pre-action water sprinklers are provided for general area protection and/or for direct water spray on cable trays wherever there is a concentration of cable trays. Manual hose stations and portable extinguishers are also provided in these areas.

A fire detection system which alarms at the local detection panel and in the control room is provided for both safety-related switchgear areas. Electrical equipment which could be damaged by direct water spray was considered. See Guideline D.3.c.

APCSB 9.5-1 Appendix A

F.6 Remote Safety Related Panels

The general area housing remote safety related panels should be provided with automatic fire detectors that alarm locally and alarm and annunciate in the control room. Combustible materials should be controlled and limited to those required for operation. Portable extinguishers and manual hose stations should be provided.

CPNPP Fire Protection Program

The general area, housing safety related equipment, is provided with automatic detectors that alarm at the local control panel and annunciate in the Control Room as determined in the Fire Protection Report. Administrative procedures have been developed to control combustible materials. Portable extinguishers and manual hose stations are provided.

APCSB 9.5-1 Appendix A

F.7 Station Battery Rooms

Battery rooms should be protected against fire explosions. Battery rooms should be separated from each other and other areas of the plant by barriers having a minimum fire rating of three hours inclusive of all penetrations and openings. (See NFPA 69, “Standard on Explosion Prevention Systems.”) Ventilation systems in the battery rooms should be capable of maintaining the hydrogen concentration well below 2 percent volume hydrogen concentration.

Standpipe and hose and portable extinguishers should be provided.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. Safety related battery rooms are separated from each other and other areas of the plant by three hour fire barriers. The ventilation system is capable of maintaining the hydrogen concentration well below two percent by volume.

Hose stations and portable extinguishers are located in the corridors serving the battery rooms.

APCSB 9.5-1 Appendix A

F.8 Turbine Lubrication and Control Oil Storage and Use Areas

A blank fire wall having a minimum resistance rating of three hours should separate all areas containing safety related systems and equipment from the turbine oil system.

CPNPP Fire Protection Program

Walls separating safety related areas from the Turbine Building have a minimum fire rating of three hours. All penetrations in these walls also have three hour ratings, except for bus duct penetrations which are discussed in D.1.j of Section 9.5.1.6.2.

APCSB 9.5-1 Appendix A

F.9 Diesel Generator Areas

Diesel generators should be separated from each other and other areas of the plant by fire barriers having a minimum fire resistance rating of three hours.

Automatic fire suppression such as AFFF foam, or sprinklers should be installed to combat any diesel generator or lubricating oil fires. Automatic fire detection should be provided to alarm and annunciate in the control room and alarm locally. Drainage for fire fighting water and means for local manual venting of smoke should be provided.

Day tanks with total capacity up to 1100 gallons are permitted in the diesel generator area under the following conditions:

a. The day tank is located in a separate enclosure, with a minimum fire resistance rating of three hours, including doors or penetrations. These enclosures should be capable of containing the entire contents of the day tanks. The enclosure should be ventilated to avoid accumulation of oil fumes.

b. The enclosure should be protected by automatic fire suppression systems such as AFFF or sprinklers.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted in Section 9.5.1.6.2. The day tanks are located in separate fire areas located above their respective diesel generator. The day tank area is protected by an automatic preaction sprinkler system actuated by Class “B” detector loop circuitry.

APCSB 9.5-1 Appendix A

F.10 Diesel Fuel Oil Storage Areas

Diesel fuel oil tanks with a capacity greater than 1100 gallons should not be located inside the buildings containing safety related equipment. They should be located at least 50 feet from any building containing safety related equipment, or if located within 50 feet, they should be housed in a separate building with construction having a minimum fire resistance rating of three hours. Buried tanks are considered as meeting the three hour fire resistance requirements. See NFPA 30, “Flammable and Combustible Liquids Code,” for additional guidance.

When located in a separate building, the tank should be protected by an automatic fire suppression system such as AFFF or sprinklers.

Tanks, unless buried, should not be located directly above or below safety related systems or equipment regardless of the fire rating of separating floors or ceilings.

CPNPP Fire Protection Program

The diesel fuel oil storage tanks are located outside and adjacent to the Safeguards Buildings. These tanks are buried. The alternate guideline criteria are not applicable to the CPNPP design criteria.

APCSB 9.5-1 Appendix A

F.11 Safety Related Pumps

Pump houses and rooms housing safety related pumps or other safety related equipment should be separated from other areas of the plant by fire barriers having at least three hour ratings. These rooms should be protected by automatic sprinkler protection unless a fire hazards analysis can demonstrate that a fire will not endanger other safety related equipment required for safe plant shutdown. Early warning fire detection should be installed with alarm and annunciation locally and in the control room. Local hose stations and portable extinguishers should also be provided.

Equipment pedestals or curbs and drains should be provided to remove and direct water away from safety related equipment.

Provisions should be made for manual control of the ventilation system to facilitate smoke removal if required for manual fire fighting operation.

CPNPP Fire Protection Program

The Fire protection Program provides protection in accordance with the guideline for fire safe shutdown components, for required separation criteria see D.1.a of Section 9.5.1.6.1.

Fire areas containing required fire safe shutdown pumps are separated from other fire areas of the plant by three hour fire rated barriers.

The Fire Protection Report demonstrates that an adequate level of fire protection is provided to separate redundant equipment used for safe plant shutdown in the event of a fire. Automatic detection is provided in these areas except where a Fire Hazards Analysis Evaluation has concluded that an equivalent level of protection is provided.

APCSB 9.5-1 Appendix A

F.12 New Fuel Area

Hand portable extinguishers should be located within this area. Also, local hose stations should be located outside but within hose reach of this area. Automatic fire detection should alarm and annunciate in the control room and alarm locally. Combustibles should be limited to a minimum in the new fuel area. The storage area should be provided with a drainage system to preclude accumulation of water.

The storage configuration of new fuel should always be so maintained as to preclude criticality for any water density that might occur during fire water application.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. A fire detection system which alarms locally and in the Control Room is provided above the new fuel storage vault. Hose stations and portable extinguishers are provided adjacent to this area. Also see Section 9.1.1.1.

APCSB 9.5-1 Appendix A

F.13 Spent Fuel Pool Area

Protection for the spent fuel pool area should be provided by local hose stations and portable extinguishers. Automatic fire detection should be provided to alarm and annunciate in the control room and to alarm locally.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline.

APCSB 9.5-1 Appendix A

F.14 Radwaste Building

The radwaste building should be separated from other areas of the plant by fire barriers having at least three hour ratings. Automatic sprinklers should be used in all areas where combustible materials are located. Automatic fire detection should be provided to annunciate and alarm in the control room and alarm locally. During a fire, the ventilation systems in these areas should be capable of being isolated. Water should drain to liquid radwaste building sumps.

Acceptable alternative fire protection is automatic fire detection to alarm and annunciate in the control room, in addition to manual hose stations and portable extinguishers consisting of hand held and large wheeled units.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the alternate criteria of the guideline except as noted in Section 9.5.1.6.2. The CPNPP design does not have a Radwaste Building. The radwaste areas are located in the Auxiliary and Fuel Buildings. These areas are protected by automatic fire detectors, hand held portable extinguishers, and manual hose stations.

APCSB 9.5-1 Appendix A

F.15 Decontamination Areas

The decontamination areas should be protected by automatic sprinklers if flammable liquids are stored. Automatic fire detection should be provided to annunciate and alarm in the control room and alarm locally. The ventilation system should be capable of being isolated. Local hose stations and hand portable extinguishers should be provided as backup to the sprinkler system.

CPNPP Fire Protection Program

The decontamination areas are provided with automatic fire detectors, manual hose stations and portable extinguishers. Quantities of flammable liquids stored in these areas will not exceed the amounts used in the determination of the combustible loading.

APCSB 9.5-1 Appendix A

F.16 Safety Related Water Tanks

Storage tanks that supply water for safe shutdown should be protected from the effects of fire. Local hose stations and portable extinguishers should be provided. Portable extinguishers should be located in nearby hose houses. Combustible materials should not be stored next to outdoor tanks. A minimum of 50 feet of separation should be provided between outdoor tanks and combustible materials where feasible.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline.

APCSB 9.5-1 Appendix A

F.17 Cooling Towers

CPNPP Fire Protection Program

The CPNPP design does not incorporate cooling towers.

APCSB 9.5-1 Appendix A

F.18 Miscellaneous Areas

Miscellaneous areas such as records storage areas, shops, warehouses, and auxiliary boiler rooms should be so located that a fire or effects of a fire, including smoke, will not adversely affect any safety related systems or equipment. Fuel oil tanks for auxiliary boilers should be buried or provided with dikes to contain the entire tank contents.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline. All miscellaneous areas are separated from areas containing safety-related equipment. The CPNPP design includes a fuel oil fired auxiliary boiler including associated above ground fuel oil tank which is diked to contain the entire content of the tank.

APCSB 9.5-1 Appendix A

G. Special Protection Guidelines

G.1 Welding and Cutting, Acetylene-Oxygen Fuel Gas Systems

This equipment is used in various areas throughout the plant. Storage locations should be chosen to permit fire protection by automatic sprinkler systems. Local hose stations and portable equipment should be provided as backup. The requirements of NFPA 51 and 51B are applicable to these hazards. A permit system should be required to utilize this equipment.

CPNPP Fire Protection Program

Storage of this equipment is located outside of areas containing safety-related equipment. The use of such equipment is addressed in Section B of this comparison. Additional portable

extinguishers are provided in the areas whenever such equipment is brought into and used in the plant.

APCSB 9.5-1 Appendix A

G.2 Storage Areas for Dry Ion Exchange Resins

Dry ion exchange resins should not be stored near essential safety related systems. Dry unused resins should be protected by automatic wet pipe sprinkler installations. Detection by smoke and heat detectors should alarm and annunciate in the control room and alarm locally. Local hose stations and portable extinguishers should provide backup for these areas. Storage areas of dry resin should have curbs and drains. (Refer to NFPA 92M, “Water-proofing and Draining of Floors.”)

CPNPP Fire Protection Program

The Fire Protection Program complies with this guideline except as noted in section 9.5.1.6.2. Dry ion exchange resins are not stored in or adjacent to areas containing safety-related systems. Storage areas for dry ion exchange resins are protected by fire detectors, portable extinguishers, and manual hose stations.

APCSB 9.5-1 Appendix A

G.3 Hazardous Chemicals

Hazardous chemicals should be stored and protected in accordance with the recommendations of NFPA 49, “Hazardous Chemicals Data.” Chemical storage areas should be well ventilated and protected against flooding conditions since some chemicals may react with water to produce ignition.

CPNPP Fire Protection Program

The Fire Protection Program is in compliance with the guideline except as noted below. Hazardous chemicals are stored in tanks approved for such service and in areas not containing safety-related components and systems, with the exception of the chemical additive tanks in the Unit 1 and Unit 2 Safeguards Buildings. The chemical additive tanks are located in partially enclosed alcoves in fire areas SB4 and 2SB4. These tanks are not open to atmosphere and are inerted with nitrogen. The locations of the tanks are not considered hazardous.

APCSB 9.5-1 Appendix A

G.4 Materials Containing Radioactivity

Materials that collect and contain radioactivity such as spent ion exchange resins, charcoal filters, and HEPA filters should be stored in closed metal tanks or containers that are located in areas free from ignition sources or combustibles. These materials should be protected from exposure to fires in adjacent areas as well. Consideration should be given to requirements for removal of isotopic decay heat from entrained radioactive materials.

CPNPP Fire Protection Program

The fire protection program is in compliance with the guideline

9.5.1.6.2 Justification for Items of Noncompliance to Appendix A to Branch Technical Position APCSB 9.5-1

The following statements are justification for items of noncompliance to Appendix A to Branch Technical Position APCSB 9.5-1 of Standard Review Plan 9.5.1, Revision 1, as stated in the applicable items of Subsection 9.5.1.6.1. Any additional deviations have been included in Appendix C of Reference [19].

Guideline D.1.d

Guideline D.1.d limits the flame spread, smoke and fuel contribution to a maximum of 25 for interior wall and structural components, thermal insulation materials and sound proofing.

Pipe and tank insulating materials as well as finishing cement, etc. are used in various areas of the plant. This insulation is rated as Flame Spread-25, Fuel Contribution-25, and Smoke Development-50.

Carpet, installed in the Control Room envelope complies with Class I interior floor finish requirements of NFPA 101, 1991 Edition. The Control Room envelope is described in Section 6.4.2.

Thermal insulation for ducts have ASTM E 84 rating of 25, 50, 50.

Thermal insulation for chiller unit heat exchangers and piping have ASTM E 84 rating less than flame spread 25, fuel contribution 30 and smoke developed 150.

Guideline D.1.j

Guideline D.1.j address 3-hour rated floors, walls and ceilings separating fire areas. The following justifications are provided where installations are shown to be adequate through analysis in lieu of providing a tested configuration.

1. Floors, walls and ceilings

   Stair tower walls are constructed of two hour rated of design. Justification is provided in subsection 9.5.1.6.2, guideline D.4.f.

   Removable concrete block walls are not fire tested design. Justification is provided in reference [19]. This justification applies to all applications.

   Protection provided by metal hatch covers installed in three (3) hour rated floors has been demonstrated through analysis in lieu of providing a tested configuration. The combustible loading below the hatches is less than 15 minutes with automatic suppression and detection above and below the hatches. The hatches are coated with a layer of fire resistive material to provide a three (3) hour structural steel resistance. Based on the combustible loading, automatic suppression and detection, and fire

resistive coating, a one hour fire could occur without breaching the fire barrier through the metal hatches.

2. Penetration Seals

Containment electrical seals are not a fire tested configuration. Justification is provided in reference [19].

Containment mechanical seals are not a fire tested configuration. Justification is provided in reference [19].

Protection provided by the penetration seals installed in bus duct penetrations installed in three (3) hour rated barriers has been demonstrated through analysis in lieu of providing a tested configuration. The penetration seal design is similar to one currently used in the plant which has a three (3) hour fire rating. The seal maintains the thickness and continuity of the barrier. The barrier's purpose is unchanged by the bus duct penetration. The fire protection features in the vicinity of bus duct penetrations are adequate for the hazards of the area. Based on the fire protection features and a review by a Fire Protection Engineer, bus duct penetrations are expected to survive a fire severity of three (3) hours without breaching the barrier.

3. Non-Rated Fire Doors

Missile resistant doors are not fire tested assemblies. Justification is provided per reference [19].

Watertight doors are not fire tested assemblies. Justification is provided per reference [19] for redundant safe shutdown related separation barriers. The justifications are also applicable for doors in fire barriers that do not separate redundant safe shutdown systems.

Bullet resistant and penetration resistant doors are not fire rated assemblies. The door assemblies are of a construction similar to units tested and listed by Underwriters' Laboratory subsequent to procurement and installation of the CPNPP assemblies except for the Cable Spreading Room (BR/PR) door. This door is justified per reference [19].

Containment Air-Locks for personnel and emergency escape use are not fire tested assemblies. Justification is provided per reference [19].

Protection provided by the tornado vent/fire dampers installed in fire rated barriers with frames mounted outside the concrete walls on steel angles has been demonstrated through analysis in lieu of providing a tested configurations. The support frames of the assemblies are protected with an approved coating to yield a fire resistance equal to that of the barrier.

Fire damper support frames are more substantial than those used in standard sleeve installations. UL 555 gives the acceptance criteria which specifies that a damper assembly must remain in the opening during the fire, and during hose stream application and that no through openings be created. Based on the substantial support frames, the high probability of the dampers remaining in the opening, and the UL test acceptance

criteria, the dampers are expected to provide a tortuous path for fire propagation and meet the conditions of acceptance in a fire test.

Fire door frames are mounted in a frame of steel angles. These angles are then coated with Thermo-lag fire proofing material in accordance with U.L. Design No. X-611. These fire door assemblies are not expected to compromise the integrity of their host 3 hour fire barriers when exposed to a postulated fire.

Guideline D.2.b

Guideline D.2.b addresses bulk gas storage and tank orientation with relations to building walls.

The CPNPP bulk gas storage tanks are located 350 feet from the nearest safety-related building (Electrical and Control Building). The special separation between the tanks and nearest primary plant building is well in excess of the NFPA requirements and therefore the tank orientation is considered acceptable.

Guideline D.3.d:

Protection provided by the penetration seals installed in flexible conduit penetrations installed in fire rated barriers which separate buildings has been demonstrated through analysis in lieu of providing a tested configuration. Flexible conduits are sealed on both sides of a barrier, which is similar to a tested configuration. The combustible loading is low in the areas of flexible conduit penetrations, and fire protection features adequate for the hazards in the area have been provided. Based on the similarity of the configurations to tested configurations, detection, automatic suppression, and manual fire fighting capability, any fire zone which has a flexible conduit could have a fire severity of 3 hours without breaching the barrier through any of the flexible conduit penetrations.

Guideline D.3.f

Guideline D.3.f requires electric cable construction to meet as a minimum the current IEEE-383 Flame Test.

Fire test standards IEEE 1202, UL 1581, UL 1685, ICES T-29-520, CSA FT-4, 1EC60332-3, UL 910, and UL 1666 have been evaluated as equal to or better than the IEEE 383-1974 test for evaluating fire retardant characteristics of electrical cables. Therefore, these standards are considered acceptable alternatives to IEEE 383-1974 for non-class 1E cables.

Additionally, non-class 1E cables are not required to meet IEEE-383-1974 flame retardancy requirements if they are totally contained within conduit, non-class 1E equipment, or flame retardant covering. However, isolated and limited instances exist where exposed cabling has been utilized which does not meet IEEE-383 flame test acceptance criteria nor that associated with the fire test standards listed above. These instances are described and technically justified below on a case-by-case basis.

A small portion of low capacitance non-IEEE-383 cable is installed in the Control Room cable spreading room and computer room for Unit 1. This cable is associated with the ERF and Data Acquisition computer systems. Justification is provided in item 5a of reference [19].

Another small amount of non-IEEE-383 cable is used in association with the radiation monitoring, communication (telephone, radio interconnector and radio antenna), security systems, and fuel handling bridge crane. A small portion of cable associated with the telephone and radio interconnection systems, cable associated with the Unit 2 Secondary Sampling System Oxygen Analyzer. These cables are all routed in conduit (except for short flexible connectors to the oxygen analyzer detectors, the interior radio antenna and telephone cable between junction boxes and handset stations, and are designed for low power service. They do not present a fire hazard in the areas where they are installed.

Another small amount of non-IEEE 383 cable is used in association with the Loose Parts Monitoring System. The cable is constructed of a conductor encased in a stainless steel sleeve. The assembly is then covered with an essentially noncombustible fiberglass wrapping. The instrument cables are installed inside containment and do not present an appreciable fire hazard in the areas where they are located.

A small amount of non-IEEE 383 cable and electrical tape (jacketing and insulation) is installed in the connection boxes for the Unit 1 and Unit 2 Station Service Water Pump Motors, Component Cooling Water Pump Motors, and Emergency Diesel Generators. The cables and tape are part of the Partial Discharge Monitoring System Bus Couplers. These couplers are used to facilitate real time connection of partial discharge monitoring diagnostic equipment. The cables and tape are fully contained inside the motor/generator connection box and do not represent a fire hazard in the areas where they are installed.

Guideline D.4.f

Guideline D.4.f addresses the fire rating of elevator towers and stairwells outside containment. Barriers enclosing the elevatorshafts are rated at two hours with 1 1/2 hour UL labeled fire door assemblies at openings to the elevator shaft. The elevators are not used during fire emergencies. Stairwells used for egress routes have a two-hour fire resistance rated walls with 1-1/2 hour UL labeled fire door assemblies at all openings into the stairwell. Based on the intended use of the stairwells, it is determined that the two-hour rating is adequate. The presence of a stairwell or elevator tower in a fire area will not degrade the 3-hour fire rating of that area's fire barrier. This is based on the tower creating two 2-hour rated fire barriers in series between fire areas joined by the common tower.

Guideline D.5.a

Guideline D.5.a addresses fixed emergency lighting requirements in the event of a fire.

The control room is provided with AC Essential Lighting and DC Emergency Lighting. The AC Essential Lighting is powered from the onsite Standby Diesel Generators. The DC Emergency Lighting is powered from the dedicated non-Class 1E batteries which are sized to supply DC power requirements for a minimum of 8-hours. Fires in areas outside of the control room and cable spreading room will not preclude the availability of DC Emergency Lighting for the control room.

Guideline E.2.e

Guideline E.2.e requires that the water supply be sized assuming all sprinkler heads in the largest designed fire area operating, plus a 1000 GPM hose station allowance. This guideline is

overconservative. The CPNPP water supply is sized following the requirements of NFPA Code 13 with 1000 GPM hose stations allowance.

Guideline E.3.a

Guideline E.3.a requires water flow alarms for standpipe systems. This guideline is not justified because standpipe systems are manual systems requiring plant personnel to be aware of the fire condition prior to the operation of hose stations.

Guideline F.1.a

Guideline F.1.a addresses fixed fire protection for hazards that could jeopardize safe plant shutdown due to a fire inside the Containment Building. An analysis was performed to demonstrate that sufficient equipment is available in at least one shutdown path to safely shutdown CPNPP in the event of an exposure fire in the Containment Building. Radiant Energy Shields were added to resolve interactions as a result of this analysis, and justifications for other interactions present were provided in References [19].

Guideline F.2

Guideline F.2 requires the installation of smoke detectors in the control room cabinets. CPNPP has provided detectors inside control room consoles and equivalent detection for all other control room cabinets.

1. The area immediately adjacent to the cabinets is monitored by ionization type smoke detectors.

2. The Alternate Shutdown System provides safe shutdown of the plant independently of the control room.

3. The area is continuously manned, thus ensuring prompt fire detection.

4. The cabinets were constructed with non-flammable materials containing only low power circuits.

Guideline F.3.b.2

Guideline F.3.b.2 addresses installation of fire retardant coatings on cables in the cable spreading rooms of redundant divisions that are not separated by fire barriers rated at 3 hours fire resistance.

In lieu of installation of 3 hour fire barriers between cable divisions or coating all cables in the area, alternate safe shutdown capability independent of the cable spreading rooms is provided.

Guideline F.9

Guideline F.9 addresses the separation criteria, fire protection criteria, and capability of the diesel generator fuel oil day tanks. The day tanks for each of the four diesel generators are separated and protected in accordance with the criteria of the subject guideline.

The 2160 gallon capacity of the day tank is based on the diesel engine operating characteristic and the operation time. As previously stated, the tank is located in a separate enclosure provided with an automatic preaction sprinkler system. Based on the above and the operation requirements, the existing day tank arrangement is acceptable.

Guideline F.14

Guideline F.14 addresses the use of large wheeled extinguishers. CPNPP has provided hand held extinguishers per NFPA 10 to protect against the potential hazard created by the Radwaste Areas.

Guideline G.2

Guideline G.2 addresses automatic fire protection for areas storing dry ion exchange resins. Storage of dry ion exchange resins is located in the Turbine Buildings remote from areas encompassing safety-related equipment and from any potential ignition sources and exposure to fire hazards. The storage arrangement and the type of resins and quantity stored do not present a fire hazard. Based on the above, protection of the area by manual hose stations and portable extinguishers is adequate.

9.5.2 COMMUNICATION SYSTEMS

9.5.2.1 Design Bases

A comprehensive communications system is provided to ensure reliable intraplant communications, plant to offsite telephone and carrier communications, and offsite emergency communications capabilities with public safety agencies. Effective communication between personnel during plant startup, operation, shutdown, refueling, and maintenance activities is provided by the use of Intraplant (IP) telephone, plant radio, sound-powered telephone, Gai-Tronics page/party, plant-to-offsite two-way radio, wireless intercom system or plant-to-offsite emergency telephone systems. These diverse means of communication are independent to prevent the loss of all systems as a result of a single failure.

An emergency alarm system is installed which provides a unique alarm signal to ensure personnel evacuation.

9.5.2.2 System Description

The following systems comprise the intraplant and plant-to-offsite communication systems for both units.

9.5.2.2.1 Gai-Tronics Page/party System

The Gai-Tronics Page/party System provides party lines (two or more persons), plant wide paging system and plant wide alarms signaling. The page-party line loud speakers are supplied from a source which is available upon Loss of Offsite Power.

The system layout permits communication between the Control Room and all plant areas and buildings of the two units. The system also permits two-way communication between two or more locations. Speakers and microphone handsets are installed at locations vital to the

operation of the plant and the safety of personnel. The voice paging channel output is audible over the expected noise levels under both normal and accident conditions.

Three separate paging zones, consisting of the Control Room Zone, Administrative Buildings Zone, and Main Plant Zone are provided to minimize paging interference from the remaining zones. Manually operated selector switches are provided in the Control Room and other key locations to allow plant wide paging.

Four separate party lines are provided to permit communication between handsets only, thereby making the page channel available to others. All four party lines are available at all handset stations, except those in elevators where only one party line is available. Selection of a desired channel is achieved by a multiposition switch provided as a part of the handset station. Both the page channel and the party line channels may be used simultaneously without interference.

A page-party line (with only one party line) handset station is installed in plant elevators to permit communication in emergency situation.

9.5.2.2.2 Intraplant Telephone System

An independent touchtone telephone system, the IP telephone system is provided for uninterrupted private communication between the following areas: the Control Room, Fuel Building, health/physics and instrument shop areas, hot shutdown panel area, hot shop, Guard House, reactor operating platform areas, intake structures, Maintenance Building offices, and Administration Building offices and work areas. The IP telephone system interfaces with the Intraplant Communication System through an isolating device to ensure that a single failure in either one of these two systems does not affect safe and reliable operation of the other system. Power is supplied to the IP telephone system from the non-Class 1E bus. When the IP telephone system's normal AC power supply is lost, predetermined telephone stations remain operable which derive their power from the public telephone system.

The IP telephone system is connected to the public telephone system by trunk lines as described in Subsection 9.5.2.2.5.

9.5.2.2.3 Intraplant Sound-Powered Telephone System

This system consists of one subsystem per unit as follows:

Subsystem One: Maintenance Loops - Consists of a two channel hard-wired communication link between the control room area and critical plant areas.

The headset jack stations are conveniently located on panels in the Control Room and in critical areas.

Communication can be established between the Control Room and any local panel or between two local panels by suitably plugging the headsets into jack stations which are mounted either in the panel or nearby. This system provides standby communication capability and does not depend on external sources of power other than the human voice.

9.5.2.2.4 Plant Radio Transmitter Receiver System

The plant radio transmitter receiver system provides communication for the fire brigade and the plant operators by the use of two separate repeater systems. Each repeater system receives low power signals from radio units on one frequency and transmits at a higher power level allowing extended radio coverage.

“Satellite Receivers” are strategically placed in areas of the plant to ensure clear reception of low power portable transmitters from at least one of the receivers inside the plant area. All receivers are tuned to the same frequency so that, as the portable radio is moved from place to place throughout the plant, another receiver can pick up the signal. All receiver signals are simultaneously voted by a “comparator” unit to select the receiver with the best quality signal. This signal is then sent to the repeater system for retransmission to all monitoring radio units.

Also used to improve reception is “radiax” transmission cable. The radiax transmission line, which is located throughout the plant, leaks a controlled amount of RF (radio frequency) energy over its full length allowing radio communication to be maintained for areas in which radio attenuation of RF signals, due to extensive metal and concrete building construction, makes normal propagation from a single antenna unsuitable.

9.5.2.2.5 Public Telephone System

The public telephone system is interconnected to the Intraplant (IP) Telephone System by trunk lines. This permits access to the public telephone system from the Control Room, health/physics and instrument shop areas, hot shutdown panel, Guard House, Hot Shop, Maintenance Building offices, Administration Building offices and work areas, reactor operating platform area, and intake structures.

9.5.2.2.6 Plant-to-Offsite Two-Way Radio Transmitter-Receiver System

A VHF two-way radio transmitter-receiver system is provided for emergency communication between the plant and offsite public safety agencies.

The system description is provided in the security plan.

9.5.2.2.7 Plant-To-Offsite Emergency Telephone Systems

The emergency telephone system consists of several dedicated leased telephone lines.

1. Emergency Notification System (ENS) - Dedicated telephone line between CPNPP and the NRC Incident Response Center in Bethesda, Maryland to inform the NRC of an emergency.

2. State and County Notification System - Dedicated telephone lines between CPNPP and the designated state office as specified in the Emergency Plan and the local emergency operations centers in Glen Rose and Granbury to inform state and local officials of an emergency.

3. System Dispatch Center Line - A dedicated telephone line from CPNPP to the offsite system dispatching center.

4. Direct Offsite Telephone Lines - Direct telephone lines to offsite that are independent of the IP Telephone System in the event the IP Telephone System fails.

9.5.2.2.8 Wireless Intercom System

The wireless intercom system provides communications for the fuel handling area and reactor operating floor during refueling operations in modes 5 and 6. Communications to the control room from the fuel handling area and reactor operating floor is provided by Gai-Tronics or Plant Radios.

9.5.2.2.9 Emergency Evacuation Alarm System

The evacuation alarm is generated by a solid state multi-tone generator capable of producing five distinctive tones which can be heard over all plant paging zones via the Intraplant Communication System except overhead and wallmounted speakers in the Control Room. One of the distinctive tones, which satisfies the NRC Regulatory Guide 8.5 [3] requirements, is designated for the evacuation alarm signal. The evacuation alarm system includes rotating beam lights as a visual indication for an alarm condition in those high background noise areas where the evacuation alarm is inaudible.

The evacuation alarm system, including the multi-tone generator, is powered by a source available upon loss of offsite power to ensure personnel evacuation in case of an emergency. The alarm is initiated by the Control Room operator in the event of a site evacuation emergency.

9.5.2.2.10 Evaluation

The following evaluation is intended to establish the adequacy and redundancy of the plant communication system design:

1. Intraplant Systems

   Each intraplant system, i.e., Gai-Tronics page/party system, IP telephone system, wireless intercom system, sound-powered telephone system, and plant radio transmitter-receiver system, is designed to provide the required intraplant communications during and after accident conditions as well as for plant operation and maintenance purposes. Failure of any one of the above systems does not result in a failure of any other system. The power supply for the IP telephone system is provided from the non-Class 1E bus. Upon loss of total AC power to the IP telephone system, predetermined telephones remain operable deriving power from the public telephone system.

   The power supply for the intraplant communication system (Gai-Tronics page/party) is provided from a source available upon Loss of Offsite Power. The Gai-Tronics page/party system handsets and speakers are strategically located to cover critical areas. Each area of the plant is served by a separate circuit to confine a system outage to the area served by the faulty circuit. The Gai-Tronics page/party communication system is employed as the primary backup to the plant radio system for use by plant operators during hot shutdown.

The IP telephone and Gai-Tronics page/party systems are connected through an amplifier device which acts as an isolating device. This ensures that a single failure in either of these two systems does not result in a failure of the other system.

The sound-powered telephone system is independent of all external power sources and its headset jack stations are conveniently located throughout the plant. This system can be employed as a backup to the intraplant communication system in critical equipment areas of the plant. In addition, two channels of components and cabling exist in the sound-powered telephone system to prevent failure of both systems because of a single component failure. Reliable service can be expected because of the ruggedness of these components and their independence from any external power source.

Each intraplant radio transmitter-receiver system is provided with DC battery backup. This system is employed as the primary communication system by plant operators and the fire brigade in case of emergency.

Power to the wireless intercom system Master Controller is provided by plant support power and each headset has an individual battery supply. Upon loss of power, backup communications is provided by the Intraplant Radios.

One of the following communication systems is available between the Hot Shutdown Panel and other remote panels required for fire the safe shutdown of the plant:

a. Intraplant Radio Transmitters-Receivers

b. Gai-Tronics Page/Party handset stations and speakers.

2. Plant-to-Offsite Systems

There are three independent plant-to-offsite communication systems available for the use of Control Room operators. The availability of these systems during and after the accident condition is enhanced by the fact that each enters the plant via different means.

The public telephone lines (trunk lines) are connected to the plant IP telephone system. This extends the use of public telephone lines throughout the plant.

The plant-to-offsite two-way radio communication system can be used as a backup to the public telephone system. This provides communication between the plant and public safety agencies. The power to this system is provided from a source available upon loss of offsite power. Also, there is an inherent redundancy in this system since it has two separate independent onsite base stations.

The dedicated telephone line to the system dispatcher provides further redundancy by establishing communication outside the plant area during the plant emergency and the loss of the other two plant-to-offsite communication systems.

9.5.2.3 Inspection and Testing Requirements

All communication systems are inspected and tested (and adjusted if required) at the completion of the installation to ensure proper coverage and audibility under the maximum plant noise levels during the various operating condition, including the accident condition.

Since the communication systems are used on a daily basis, periodic testing is not required.

Periodic testing of the emergency evacuation alarm signal is performed as outlined in ANSI N2.3-1967 [4].

9.5.3 LIGHTING SYSTEMS

9.5.3.1 Normal Lighting System

Normal lighting for the plant is supplied by a grounded 208/120-V, three-phase, four-wire distribution system. Dry-type transformers, which feed lighting panels, are rated at 30 kVA or 45 kVA, 480- 208/120-V, three phase, 60 Hz, delta-wye, and are connected to motor control centers throughout the plant. The transformers and lighting panels which they feed are conveniently located plantwide to permit efficient distribution of the lighting load. Some of the motor control centers are common to both units. Exterior, nonsecurity lighting uses the 208/120-V system described above as well as a 480-V system fed from lighting panels.

The lighting system design is equal to or exceeds the recommendations contained in the Illuminating Engineering Society (IES) Lighting Handbook [17].

9.5.3.2 AC Essential and DC Emergency Lighting Systems

9.5.3.2.1 System Description

AC Essential and/or DC Emergency Lighting systems are provided in those locations where safety-related functions are performed or where personnel safety is involved. They are designed to provide adequate illumination for the safe shutdown of the plant and the evacuation of personnel. These areas include the following:

1. Control Room

2. Diesel generator room

3. Remote shutdown panel locations (which include the Hot Shutdown Panel and the Shutdown Transfer Panel)

4. Areas required for control of safety-related equipment to accomplish alternate shutdown from outside the control room

5. Primary interior access routes to and from the preceding areas

6. Primary exits

The primary plant AC Essential Lighting is provided by AC systems having connections to the onsite standby diesel generators. During all operating conditions or a LOCA and in the event of the loss of all offsite AC power, onsite standby diesel generator power is available to the AC Essential Lighting System. The illumination level provided by the AC Essential Lighting System is as follows:

1. For the Control Room, in the control board area, there is full illumination, which conforms to the requirements of NUREG-0700 [42].

2. For other primary plant areas requiring AC essential lighting, the illumination level ranges from full illumination to approximately one-fifth to one-tenth of full illumination, which conforms to the requirements of Illumination Engineering Society (IES) Handbook [17].

3. Emergency illumination levels provided by AC essential lighting for personnel safety, evacuation, and operation of safe shutdown work stations will meet or exceed the requirements as described under DC Emergency Lighting System in this section.

Outside of the Containment and Fuel Building, AC Essential Lighting is provided in primary plant areas required for safe shutdown and in major interior access/egress routes between these areas. Inside the Containment, AC lighting is provided for emergency egress. This AC lighting is powered from the Class 1E 480V system, which is powered by the standby diesel generators during a loss of offsite power. AC Essential Lighting is not provided in the Fuel Building, Service Water Intake Structure or Turbine Building.

The non-Class 1E DC Emergency Lighting System consists of lights connected to the dedicated batteries or individual battery packs. DC Emergency Lighting is provided in all areas needed for the operation of fire safe shutdown equipment necessary to achieve hot standby and in the primary interior access/egress routes between these areas. This lighting is provided by 8-hour rated battery packs except in the control room where the lighting is provided by dedicated 8-hour batteries. Battery pack lights are flourescent or sealed beam type.

Emergency Lighting is also provided for evacuation paths in the plant.

The DC Emergency Lights in the Control Room are normally deenergized. The contactor in the DC Emergency Lighting panels is normally held open by a feed from the AC Lighting System. DC Emergency Lights are activated by the loss of power to the AC Lighting Systems.

Individual battery packs are normally under a float charge from the AC lighting power supplies for their respective areas. If an AC lighting power supply is lost, the respective DC lights are activated from their individual battery packs by their respective relays.

The lighting (AC and DC) in the Control Room, the primary plant (non-containment) essential control areas and the primary interior access routes between them is arranged in a staggered pattern and alternately fed from the redundant trains and/or from self contained battery packs energized from either Train A, Train B or Normal AC sources. As an alternate, where the engineered safety features (ESF) equipment is energized from a particular train, AC essential and DC emergency lighting from the same train is provided in that area and in the primary access route to it. These practices ensure adequate lighting in these areas under any possible electrical single-failure condition.

The DC Emergency Lighting System provides the following illuminance levels:

1. 0.5 footcandles (fc) average maintained, horizontally, along the center line of fire protection access/egress routes at floor level, with a maximum/minimum uniformity ratio less than 40:1;

2. 0.5 fc minimum maintained at the center point of a slight hazard within the fire protection access/egress route, with 3 fc average maintained along the hazard. Slight hazards are defined, for safety lighting purposes only, as abrupt changes in direction, intersections of two or more corridors or access/egress routes, changes of floor level, at each door, along stairs, and at weather-stripping door sills;

3. 2 fc minimum maintained at the center point of a high hazard within the fire protection access/egress route. High hazards are defined, for safety lighting purposes only, as permanent stationary protrusions (such as piping, cable trays, panelboards, and water-stopping door sills) into a 2ft-6in wide by 6ft-6in high access/egress route envelope. Hazards are to be determined by a field walkdown. Door sill lighting is in horizontal fc; all other hazard lighting is in vertical fc;

4. 10 fc minimum average maintained within the task seeing area of the following work stations:

   Main Control Boards in the Main Control Room

   Hot Shutdown Panels

   Shutdown Transfer Panels

5. The emergency DC lighting illuminance levels at certain additional task areas where specific event-related tasks may be required to be performed for fire protection purposes will be as follows:

   a. 1 fc average maintained within the task seeing area of valves, individual control switches, and other local panel boards required to be illuminated for fire safe shutdown.

   b. 2 fc average maintained within the task seeing area of trip circuits required for fire safe shutdown.

   c. In addition, supplementary lighting will be provided from battery-powered hand held portable lanterns.

9.5.3.2.2 Design Criteria

The non-Class 1E AC Essential Lighting Systems are connected to Class 1E power system buses through Class 1E circuit breakers in 480 volt Motor Control Centers, Class 1E 480 volt-120/208 volt step-down transformers, and Class 1E lighting distribution panels. The interconnection of the non-Class 1E AC essential lighting circuits with the Class 1E power system is discussed in paragraph 7 of Section 8.3.1.2.1 and Appendix 1A(B) of Section 1.0. The power cables from the 480 volt Motor Control Center to the primary side of the 480 volt

step-down transformer and from the secondary side of the step-down transformer to the lighting distribution panel are Class 1E. The cable raceway system for these power cables is designated Seismic Category I. The feeder cables from the lighting distribution panels to the lighting fixture outlet boxes are non-Class 1E.

The AC essential lighting fixtures, fixture primary supports, lamps and bulbs are classified non-nuclear safety grade (NNS).

The DC emergency lighting fixtures, lamps, fixture primary supports, fluorescent and sealed beam lamps and supports, individual sealed beam battery packs and supports are non-nuclear safety grade (NNS) components and material.

Restraints are placed on lighting system fixtures where required in areas of the primary plant to ensure the integrity of safety related equipment during a seismic event.

In the areas in which the use of mercury is restricted, regular or tungsten halogen incandescent lamps are used for the Normal, AC Essential, and DC Emergency Lighting systems. The lighting for security areas is discussed in the security plans.

9.5.3.2.3 Failure Analysis

The AC Essential and DC Emergency Lighting systems are independent of each other. Any electrical failure in one system does not cause the other system to be inoperable. Furthermore, in critical control areas the lighting is either redundant (i.e., overlapping from adjacent areas) or energized from the same train as the equipment in that area and/or provided by self contained battery packs energized from either Train A, Train B or Normal AC sources. The integrated design of these systems provides adequate emergency station lighting in all areas required for control of safety-related equipment and in the major interior access routes to and from these areas.

9.5.3.2.4 Inspections and Testing

The Normal, AC Essential, and DC Emergency Lighting systems are tested and inspected after their installation is complete.

The AC essential lighting is used on a day-to-day basis to provide a part of the ordinary operational lighting. Therefore, periodic tests are not required. Prompt replacement of burned out lamps assures that the AC essential lighting system is capable of performing at light levels well above the minimum requirements.

The DC emergency lights connected to the dedicated batteries can be tested periodically by tripping the circuits fed from the AC lighting system, thereby closing the feeder circuits to the DC emergency lights. For battery packs, the test circuit contained in each individual unit can be used for test purposes. Battery pack batteries are replaced on a periodic basis.

9.5.3.3 Underwater Lighting

High Pressure Sodium underwater lights are used in the spent fuel pools, transfer canals and wet cask loading pit in the Fuel Building. These lights are subject to inspections and precautions and limitations to control the free mercury contained in the bulbs. These controls minimize the

likelihood of the loss of the mercury and requires an evaluation if mercury is lost and not subsequently recovered.

9.5.4 DIESEL GENERATOR FUEL-OIL STORAGE AND TRANSFER SYSTEM

9.5.4.1 Design Bases

The Diesel Generator Fuel-Oil Storage and Transfer System design is in accordance with the following criteria:

1. 10 CFR Part 50, Appendix A, Criterion 2, Design Bases for Protection Against Natural Phenomena

2. 10 CFR Part 50, Appendix A, Criterion 4, Environmental and Missile Design Bases

3. 10 CFR Part 50, Appendix A, Criterion 5, Sharing of Systems, Structures, and Components

4. NRC Regulatory Guide 1.29, Seismic Design Classification

5. NRC Regulatory Guide 1.137, Fuel-Oil Systems for Standby Diesel Generators (1/1978). The CPNPP design conforms to the requirements of this guide with exceptions as described in Section 1A(B).

The criteria for protection against pipe break outside the Containment conforms to the guidelines contained in Branch Technical Positions APCSB 3-1 and MEB 3-1 as described in Section 3.6. The system is designed to meet the single failure criterion. To further its continued availability, the system is designed to withstand the effects of the worst anticipated environmental phenomenon and to meet seismic Category I requirements.

The Diesel Generator Fuel Oil Storage and Transfer System, shown on Figure 9.5-52, contains Nuclear Safety Related portions designed to IEEE 387 and seismic Category I requirements. These portions are designated by a special symbol and accompanying note on the drawing. They have no other significance than this. The nuclear safety class 3 (ANS safety class 3) portion of the system is also designed to the seismic category I requirements (see Table 3.3-2 for safety classification and Section 3.2.1.2 for seismic category of mechanical systems and components). Symbols used for identification of the ANS nuclear safety class portion from that of the ANS non-nuclear safety class are shown on Figure 3.2-1.

The system is designed in accordance with applicable codes as stated in Section 3.2. The system is designed to supply a reliable source of fuel oil for the four emergency diesel generator sets for a period of not less than seven days. The seven day fuel capacity is based on the following:

1. Continuous operation at rated load, as stated in Section 8.3

2. Diesel fuel lower heating value

3. Fuel consumption based on test data

9.5.4.2 System Description

9.5.4.2.1 General

The four diesel generator sets are supplied by four separate, identical, fuel-oil system trains, as shown on Figure 9.5-52. Each train consists of an underground storage tank, two transfer pumps, day tank, piping, instrumentation and controls necessary for reliable operation. Connections are provided for filling and venting the storage tanks, as shown on Figure 9.5-51. Provisions are made for draining and sampling of fuel, and removing condensate from the storage tanks. Each day and storage tank is provided with instrumentation to monitor and alarm the low & high levels of fuel oil. Hi and lo levels are annunciated on the engine control panel and alarm as “Diesel Engine Trouble” in the Control Room.

Two fuel-oil transfer pumps are located in each diesel generator room, as shown on Figure 9.5-52. Power for these pumps is supplied from their respective emergency diesel generator buses, as described in Section 8.3.1. Each pump takes suction from its associated storage tank and discharges to its associated day tank. The general arrangement of the diesel generator area is shown on Figure 9.5-53.

Fuel consumption may be monitored as a function of depth of fuel in the fuel oil day tank. The flow rate of the fuel for the emergency diesel generator fuel oil system may be measured indirectly by level instrumentation.

The diesel fuel oil storage tank is equipped with a vent line with flame arrestor, a level transmitter, a fill line with shut-off valve (as described below) and the capability for use of a stick gauge to measure the fuel level.

The storage tank is filled by truck through a connection located inside a watertight valve box or via an alternate fill method which utilizes a special diffuser tool to fill through the Sample Grab Point quick-disconnect fitting connection, which is also in a watertight box. This prevents dirt and water from mixing with the fuel-oil. In an effort to minimize the creation of turbulence which will result in the stirring of sediment while the Diesel Fuel Oil Storage Tank is being filled, the fill nozzle is equipped with a fill line which is run to within 2 feet of the bottom of the tank, the end of which is capped and the last two (2) feet of which is perforated with approximately sixteen 1" diameter holes. When the alternate fill method is used, the special diffuser tool is also capped at the bottom and perforated with holes. The function of this arrangement of the fill line is to prevent impingement of the new fuel oil on the bottom and to disperse the new fuel oil throughout the fuel oil that is in the tank. A dirt and water collector is provided in order to limit the build-up of sediment and water in the storage tanks. These collectors will be cleaned on an as needed basis.

The growth of algae is not considered a design basis for the diesel generator fuel oil storage and transfer system but may occur at CPNPP. For this reason, a biocide is added to new fuel in order to deter the growth of algae in the fuel. Diesel fuel oil is sampled, tested and maintained in accordance with CPNPP Technical Specifications.

The tank is assumed to contain a low amount of diesel fuel in it during possible flood conditions to provide the most conservative design case. The tank is provided with hold down straps embedded in a concrete foundation. The mass of the concrete foundation counteracts the buoyancy effects and protects against the possibility of tank floatation.

A strainer is located downstream of each redundant fuel oil transfer pump. In addition, duplex strainers are located downstream of the fuel oil day tank on the engine skid. These strainers are provided to catch any suspended sediment.

A 3 to 5 micron duplex filter is located on the engine skid upstream of the fuel supply header as a final measure to collect suspended sediment.

The day tanks supply an immediate source of fuel-oil to the diesel generator sets. Fill, vent, drain connections and a return line to the storage tank for overflow are provided for each tank. In addition, provisions are made for direct filling of the day tanks from tank trucks. Each day tank is provided with level instrumentation which controls fuel-oil transfer pump operation, provides level indication on the engine control panel and initiates low- and high-level alarms in the diesel generator room, which alarm as “diesel engine trouble” in the Control Room.

If new fuel oil does not meet the diesel generator manufacturer’s requirements for absolute specific gravity at 60/60^{o}F of $\geq$ 0.8299 or for API gravity at 60^{o}F of $\leq$ 39^{o}, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer’s recommendations.

The fuel oil transfer and storage system utilizes pressure indications on the discharge lines of the fuel-oil transfer pumps and differential pressure alarms on the strainers, located in the discharge lines (see Figure 9.5-52 sheets 1 and 2). Temperature indication is not provided on the fuel-oil transfer system since the maximum anticipated temperature differential for the underground tanks and piping, is not expected to exceed the diesel generator manufacturers recommendations. The instrumentation described above is utilized during all modes of operation.

9.5.4.2.2 Equipment Design Bases

The following are the equipment design bases for the major components of the Diesel Generator Fuel-Oil Storage and Transfer System:

1. Fuel-Oil Storage Tank

   The combined capacity of the fuel-oil storage and day tanks for each diesel generator set is designed for a seven-day fuel supply at rated load (see Section 9.5.4.1 for basis) plus approximate 15 percent margin to permit periodic testing.

2. Fuel-Oil Day Tank

   The fuel-oil day tank is designed to supply enough fuel for a minimum of 1 hour continuous operation at rated load plus a minimum margin of 10 percent as required by ANSI N195.

   The fuel-oil day tank is located in the diesel generator room and is enclosed by three-hr fire walls. This tank is elevated above the diesel generator to maintain a positive pressure at the suction of the engine fuel pumps.

3. Fuel-Oil Transfer Pump

Each fuel-oil transfer pump has a capacity equal to 125 percent of the engine's full load fuel requirements. Either fuel oil transfer pump satisfies Emergency Diesel Generator operability for TS 3.8.1, TS 3.8.2, and TS 3.8.3.

4. Diesel Generator Sets

Two identical diesel generator sets of equal capacity, characteristics, and interchangeable components are provided for each unit. The diesel generator sets are qualified in accordance with IEEE Standard 387-1977 [16].

The engine is diesel oil, four-cycle, 450 rpm, turbocharged, water-to-water cooled with station service water. The generator is a synchronous machine type, self-ventilated, air-cooled unit with a static exciter, a voltage rating of 6900, and a power factor of 0.8; it is directly connected to its driver.

There is no AC power required to start the diesel generators. The DC power required to start the diesel generators is supplied from Class 1E 125-VDC buses described in Section 8.3.2.

5. Fuel-Oil

The fuel is diesel fuel oil No. 2 in accordance with ASTM D 975-1981 [44] and Technical Specification requirements.

If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60^{o}F of $\geq$ 0.8299 or for API gravity at 60^{o}F of $\leq$ 39^{o}, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations.

6. Piping and Valves

The Diesel Generator Fuel-Oil Storage and Transfer System piping & valves are fabricated from either carbon or stainless steel.

9.5.4.3 Safety Evaluation

Safety classification of the system components meets the intent of ANSI N18,2a [15] as endorsed by RG1.137 and ANSI N195. Safety Class 3 applies to components of the Diesel Generator Fuel-Oil Storage and Transfer System as shown on Figure 9.5-52.

Each diesel generator set is provided with a usable storage capacity for seven days of operation (see Section 9.5.4.1 for basis), in accordance with Regulatory Guide 1.137. Approximate 15 percent margin in storage tank capacity is also provided to preclude the necessity of refilling the tanks after routine testing operations.

To ensure the continued integrity of the system, the system components are designed to seismic Category I requirements. See Section 3.7 for details. In addition, the system is protected from tornado-generated missiles. Underground components are protected from corrosion by a coating and the Diesel Fuel Oil Storage Tank by a cathodic protection system.

The components in the system are designed to the requirements of the ASME B&PV Code, Section III, Class 3. However, when an ASME Class 3 design component is commercially unavailable, the component is proven to be of equivalent quality.

“Equivalent quality”, of a component is interpreted to mean an item designed for normal commercial use upgraded to ASME B&PV Code Section III requirements through seismic design, testing, qualification and documentation.

Component supports are designed to ASME B&PV Code, Section III Class 3. Fabrication is as a minimum in accordance with AISC-1970.

All such components and component supports are listed in Table 17A-1.

Plant layout precludes routing of any high energy lines in the Diesel Generator rooms containing the following systems:

The diesel generator fuel oil system, the diesel generator cooling water system, the diesel generator starting air system, the diesel generator lubrication system and the diesel generator combustion and exhaust system.

Except for fire protection piping, all the piping in each diesel generator compartment is associated with a diesel generator of the same train affiliation. The diesel generator compartments are equipped with watertight doors to protect the electrical equipment area and the flooding resulting from a failure of a moderate energy line that is part of the diesel generator or fire protection systems will be limited to the respective diesel generator and not affect the reliability of the redundant diesel generator system. Each diesel generator compartment is connected to the pipe tunnel shown on Figures 1.2-11 and 1.2-17. The penetrations are sealed with fire barriers which also provide a ventilation system boundary. These penetration seals are not required to be watertight since any flooding would be detected by the safeguards building sumps and isolated before it could affect any other equipment in the safeguards building. Therefore, the requirements of BTP APCSB 3-1 as described in Section 3.6B are met. There is no credit taken for the normal drains for the diesel generator shown on Figure 9.3-6. The floor drains for each compartment drain to its own sump; so, backflooding is precluded.

To minimize the chances of fire in the Diesel Fuel Oil System, specific precautions were taken. All fuel oil lines are routed such that they are remote from lines of elevated temperature which are in the same room. Each Fuel Oil Day tank is enclosed in its own cubicle which is rated for three hours of fire separation and is protected by an automatic preaction sprinkler system. The redundant Diesel Generators are separated from each other by three hour rated fire barriers. Fire fighting actions taken for one diesel generator will not affect the redundant diesel generator.

There are no sources of open flame in these areas.

Failures caused by icing or freezing of the outdoor equipment in the Fuel Transfer System are not considered possible because of the low freezing point of diesel fuel.

A fuel-oil transfer pump is started when the day tank level reaches two thirds capacity, which is the low-level set point, and run until the day tank is filled to high level. In the event of pump failure, the two-thirds minimum capacity allows the operator to take corrective action before the diesel generator set runs out of fuel.

The time interval between the low level alarm and when the day tank becomes empty is three hours. The range of malfunctions and necessary remedial actions are considered in Table 9.5-11. Considerable flexibility is provided for resupplying the storage and day tanks. Fuel-oil is normally brought in by tank truck. However, if circumstances require, railroad tank cars can be brought in on the plant railroad spur. CPNPP is located approximately 90 miles southwest of the Dallas - Ft. Worth area. Dallas - Ft. Worth is a major commercial area which has distributors of diesel fuel that represent the majority of the major oil companies. A listing of the cities capable of supplying diesel fuel with their distance in miles is provided below:

City	Area	Distance from Site In Miles
Ft. Worth, Texas	North Texas	50
Dallas, Texas	North Texas	90
Houston, Texas	Texas Gulf Coast	275
Beaumont, Texas	Texas Gulf Coast	300
San Antonio, Texas	South Central Texas	200
Oklahoma City, Oklahoma	Central Oklahoma	250
Abilene, Texas	West Texas	150
San Angelo, Texas	West Texas	175

The failure analysis of the Diesel Generator Fuel Oil Storage and Transfer System is provided in Table 9.5-11.

A failure of the drip waste return pump has no effect on reliable operation of the diesel generator and, consequently, no impact on safe shutdown of the plant. Thus, the component is not required to be active.

9.5.4.4 Inspection and Testing Requirements

Prior to plant initial operation, the diesel generators are installed and thoroughly tested to demonstrate their ability to perform as designed. During these tests all associated equipment are operated. During routine plant operations, the diesel generators are periodically test run at calculated maximum load to ensure their continued capability. At the same time, associated systems are thoroughly inspected and tested where possible. This test run of the diesel generator assures that the fuel-oil is being supplied and is of sufficient quality to allow the diesel generators to perform adequately.

In addition, a comprehensive sampling and testing program is performed on the diesel fuel oil for the emergency generators to assure the quality of fuel oil. The methodology used in sampling diesel fuel follows that outlined in Reference ASTM D4057-1981 [45]. Fuel oil not meeting the requirements of Reference ASTM D2276-1978 [48] is replaced or corrected in a short period of time (about a week).

If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60°F of ≥ 0.8299 or for API gravity at 60°F of ≤ 39°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations.

All sampling and testing of diesel generator fuel oil system is performed by the following programs:

1. Technical Specifications in accordance with the Surveillance Frequency Control Program.

2. Technical Specifications in accordance with the Diesel Fuel Oil Testing Program.

3. Technical Requirements Manual in accordance with the Preventative Maintenance Program.

A cathodic protection surveillance is also conducted as follows:

1. At intervals not exceeding 12 months, tests are conducted on each underground cathodic protection system to determine whether the protection is adequate.

2. The test leads required for cathodic protection are maintained in such a condition that electrical measurements can be obtained to ensure the system is adequately protected.

3. At intervals not exceeding two months, each of the cathodic protection rectifiers are inspected.

4. Records of each inspection and test are maintained over the life of the facility to assist in the evaluation of the extent of degradation of the corrosion protection systems.

9.5.5 DIESEL GENERATOR COOLING WATER SYSTEM

9.5.5.1 Design Bases

The Diesel Generator Cooling Water System is designed to allow the diesel generator sets to be rapidly loaded and to operate continuously at their maximum ratings. The various components are sized to remove the maximum heat produced by the diesel generator sets using approximately 117°F service water as a cooling medium. Essential system components are designed to seismic Category I requirements and to withstand the worst anticipated environmental phenomenon as described in Section 3.3, 3.4, and 3.5. This system is designed to meet the requirements of 10 CFR Part 50, GDC 2, 4, 5, 44, 45, and 46, NRC Regulatory Guides 1.29, IEEE 387-1977, and is in accordance with Branch Technical Positions APCSB 3-1 and MEB 3-1 [5 through 14, and 16]. Refer to Section 8.3.1, AC Power System (Onsite), for design information and analyses of standby power systems (diesel generators).

9.5.5.2 System Description

Each Diesel Generator Cooling Water System consists of two subsystems: a closed cooling water loop (jacket water) for the cylinder jackets, cylinder heads, turbochargers, air intercoolers, and lube oil cooler; and an external cooling water loop (service water) which cools the jacket

water heat exchanger. (See Figure 9.5-54.) The components for the internal cooling system are tabulated in Table 9.5-12. For design parameters of the jacket water cooler see Table 9.5-18. Operation of the diesel generator for 60 sec without cooling water has no effect on diesel generator performance. If a loss of power occurs, the service water pumps restart within 35 sec after diesel generator power is made available, well within the limits of the Diesel Generator Cooling Water System requirements.

When the diesel generator sets are not operating, a thermostatically controlled electric standby heater and motor-driven keep-warm circulating water pump are provided to reduce the effects of fast start thermal transients on long term maintenance requirements.

The diesel generator sets are capable of operating at rated speed for seven (7) days with no load across the terminals and are capable of assuming design sequenced and continuous loading or any part thereof, during this period. The diesel generator sets have no minimum loading requirements.

The Diesel Generator Cooling Water System is designed as a closed system with a vented standpipe at the highpoint.

The engine driven jacket water pumps required NPSH is 15 feet. The stand pipe provides a static head range of 33 to 44 feet absolute. The stand pipe is also equipped with a low level alarm which is set approximately 2.5 feet below the water level during normal operation.

Assuming that all the water between the alarm set level down to the minimum water level is available for system make-up, then there is potentially 310 gallons of water available to replace a leakage up to 1.5 gallons per hour for seven (7) days of continuous operation. The seven day period is based on pre-licensing Q&R Q040.81 (May 1979) and post-licensing Q40.21 (August 1993).

The Jacket Water Cooling System is supplied with the following instrumentation and alarms for local monitoring of operational conditions of the cooling system and provide indications of any problem that develops within the system:

1. Low-level jacket water standpipe

2. Low-temperature jacket water in

3. Low-temperature jacket water out

4. High-temperature jacket water in

5. High-temperature jacket water out

6. High-high temperature jacket water out trip; the jacket water trips are blocked during safety injection actuation.

7. Low-jacket water pressure

8. Jacket water pressure local indication

9. On-off temperature control for the jacket water heater

Jacket water temperature is maintained constant by the use of an automatic three-way valve which is thermostatically operated. Vents at high points in the system ensure that no air is trapped in the system.

Leakage in the Jacket Water Cooling System can result in a loss of coolant inventory. This will cause a high jacket water temperature alarm followed by a possible trip of running the EDG in NORMAL mode.

Long-term corrosion and organic fouling in the diesel generator cooling water system is minimized by the use of demineralized water mixed with $NaNO_2$.

To prevent degradation of system cooling performance and of the materials utilized, the jacket water heat exchanger is periodically cleaned and flushed during the planned maintenance.

9.5.5.3 Safety Evaluation

Safety classification of the system meets the intent of ANSI N18.2 and N18.2a [15] as shown in Figure 9.5-54.

The system provides an adequate and reliable source of cooling water. The Jacket Water Cooling System and associated service water header for each diesel generator set are totally independent of the others.

The components in the system that are mounted on the engine are constructed in accordance with manufacturers standards to meet performance requirements as stated in IEEE 387, and those which are mounted off the engine are designed in accordance with the ASME B&PV Code, Section III, Class 3. The off engine piping supports are designed in accordance with ASME B&PV Code, Section III, Class 3 and as a minimum fabricated to American Institute of Steel Construction (AISC).

A general diesel trouble alarm which includes pressure, temperature, and coolant system level alarms (which are annunciated separately on the local diesel generator panel) is provided in the Control Room for the cooling water system. In the unlikely event of a failure in one diesel cooling system, the backup diesel generator is available for immediate emergency use. The single failure criteria analysis of the cooling water system is shown in Table 9.5-12.

The system is housed in a seismic Category I structure. Failure of nonseismic components in the area does not prevent the operation of this system.

9.5.5.4 Inspection and Testing Requirements

Testing and inspecting is performed as described in Subsection 9.5.4.4.

9.5.6 DIESEL GENERATOR STARTING SYSTEM

9.5.6.1 Design Bases

The Diesel Generator Starting System is an air-powered system designed to start the diesel generator set. Redundant starting air systems are provided for each diesel generator set. Starting signals are listed in Section 8.3.1.1. The starting system is designed to meet the requirements of 10 CFR Part 50, Appendix A, GDC 2, 4, and 5, NRC Regulatory Guide 1.29, IEEE-387-1977, and the Branch Technical Positions MEB 3-1, APCSB 3-1, and EICSB-17 [5], [6], [7], [12], [13], [14], [16], [18].

9.5.6.2 System Description

The diesel engines are started by injection of compressed air into the cylinders. The redundant starting systems for each diesel generator are shown on Figure 9.5-55. Each diesel generator has two 100- percent-capacity air systems. Each system includes an air compressor, air receiver, an air dryer, and two solenoid starting valves (redundant solenoids for each bank). In-line air dryers are desiccant-type having activated alumina as the drying medium. The air dryers are designed to maintain a dew point of at least 10ºF less than the lowest expected ambient temperature. Either starting air system satisfies Emergency Diesel Generator operability for TS 3.8.1, TS 3.8.2, and TS 3.8.3.

Each receiver is sized to contain enough air for five starts. The air pressure normally available for starting is between 220 and 250 psig. The air receiver relief valve is set at 275 psig. The diesel generator will not start in EMERGENCY mode if the air pressure drops to 150 psig. Low pressure alarms are actuated if the pressure falls to 210 psig. Two air supply lines are provided, each having redundant solenoid starting valves. One valve passes enough air to start the diesel engine, thus ensuring a sufficient supply of starting air if one valve fails to operate. Each receiver also has a valve to permit periodic blowdown of accumulated moisture and foreign material.

Instrumentation is provided to monitor the operation of the system. On low receiver pressure, redundant pressure elements actuated an alarm in the diesel generator room and a trouble alarm in the Control Room.

There are no differential pressure switches on the filters. Air flows through the lines only when a diesel start is called for. At all other times, there is no flow and, consequently, no differential pressure. Normal periodic maintenance will assure that air will flow unimpeded when needed.

The starting air system is supplied with local pressure indication for each air receiver.

9.5.6.3 Safety Evaluation

Safety classification of the system meets the intent of ANSI N18.2 and N18-2a, Safety Class 3 [15] as shown on Figure 9.5-55.

The starting system provides a reliable source of starting power for the diesel generator. The system is designed to seismic Category I requirements and the worst anticipated environmental and climatic phenomena are considered in the design as discussed in Section 3.11B. To further ensure system reliability, the system components which are located off the diesel generator set are designed to the requirements of the ASME B&PV Code, Section III, Class 3. Exceptions are

the compressors, air dryers and the piping up to the class break which are non-safety related. These components are designed to manufacturer's standards and are pressure tested pneumatically.

To minimize fouling of the starting air valves, or filters with contaminants, automatic condensate drains are provided with the compressors and manual blowdown capability is provided on the receivers to collect any liquid carryover. Inline air dryers remove excess moisture. Strainers are located downstream of the receivers to catch rust.

The lines from the air receiver to the engine are sloped such that any contaminates present in the lines will collect at a low point prior to connection to the engine.

Each starting air system is provided with two redundant starting air admission valves to preclude non-operability of a single starting air system in the event of failure of one of the admission valves.

Periodic running of the diesel engine, and blowdown of the drip-legs and low points will minimize the build-up of contaminants in the starting air system.

The single-failure analysis presented in Table 9.5-13 demonstrates the ability of the system to perform its design function while subjected to a single failure.

9.5.6.4 Inspection and Testing Requirements

Testing and inspecting are performed as discussed in Subsection 9.5.4.4.

9.5.7 DIESEL GENERATOR LUBE OIL SYSTEM

9.5.7.1 Design Bases

Each diesel generator lube oil system is designed to provide adequate engine lubrication under all operating conditions, including immediate full-load operation after starting. The diesel generator lubrication is designed to meet the requirements of 10 CFR Part 50 [5], [6], [7], NRC Regulatory Guide 1.29 [12], and Branch Technical Positions APCSB 3-1[13], MEB 3-1[14], IEEE 387-1977[16] and EICSB-17[18].

9.5.7.2 System Description

The diesel generator lube oil system is depicted by Figure 9.5-56. The diesel generator operates as a dry sump system and has one main lube oil pump that is engine driven. The main pump is a positive displacement pump and draws oil from the sump through a course strainer. The lube oil travels through the pump, through the cooler, through the filter, and then through the strainer to the engine. Full-flow filtering is available. This ensures that the oil keeps its required quality during cooling. The lube oil system is supplied with the following instrumentation and alarms on the Diesel Room Control Panel to provide operating information:

1. Lube oil tank, low level alarm

2. Low temperature, lube oil in alarm

3. High temperature, lube oil in alarm

4. Low temperature, lube oil out

5. High temperature, lube oil trip

6. High temperature, lube oil out

7. High differential pressure, lube oil filter

8. High differential pressure, lube oil strainer

9. Low pressure turbo oil, right bank

10. Low pressure turbo oil, left bank

11. Low pressure turbo oil, trip

12. Low pressure engine oil alarm

13. Low pressure engine oil trip

14. High temperature bearings - trip

15. High pressure crankcase - trip

16. An on-off temperature control for the lube oil heater

17. Local pressure and temperature, indication

Alarms will be set based on the engine manufacturers allowable operating conditions. When an alarm sounds on the Diesel Room Control Panel a “DIESEL TROUBLE” signal will also sound in the control room. Personnel will be dispatched to the Diesel Generator Room to ascertain the specific problem and take required action. Alarm response procedures will be available to assist the operator in determining what action should be taken.

The lubrication oil system is cooled by jacket water. The complete system is designed and supplied by the Diesel Generator Manufacturer. The design of the lubrication oil cooler is solely based on parameters established by the Diesel Generator Manufacturer. For design parameters of the lube oil cooler, see Table 9.5-17.

The Lube Oil System is a closed system and during normal operation, there is no way for deliterious materials to enter the system. Refilling of the Lube Oil System is accomplished through a normally capped two inch connection. Administrative controls are established as required to govern the refilling operation.

The required oil quality as specified by the engine manufacturer is maintained by automatically filtering and straining the oil as it is circulated in the engine. Administrative measures taken to maintain the required quality of the oil will be detailed in the preventative maintenance program which deals with lubrication oil quality. The lube oil will be changed, or sampled to determine if

changing is required, at regular intervals based on recommendations from the manufacturer and the lubrication supplier. This will be performed as part of the preventive maintenance program using approved procedures.

The engine crankcase is protected from overpressurization by the following design features. The system is equipped with alarms and trips initiated by high oil temperature and/or pressure, which shut down the diesel generator if operating limits as specified by the engine manufacturer are exceeded. The lubrication oil system trips are blocked by safety injection actuation. The crankcase is equipped with centrifugal blowers for crankcase ventilation during diesel generator operation. The crankcase is also equipped with blowout panels to protect the unit from damage. The lubrication oil sump, the lubrication oil filter and the lubrication oil strainer are all vented.

Periodic monitoring of the level instrumentation associated with the Lubrication Oil Sump Tank may indicate an uncontrolled loss of lube oil. Also an increase in lube oil temperature or a decrease in oil pressure, or both, may be attributed to oil leakage. Any leakage discovered during routine inspections (or, investigating an occurrence described above,) will be corrected in accordance with operating and maintenance procedures.

CPNPP does not have the requirement of a timed prelube before manually starting the diesel engines.

When the engine is not operating, a motor-driven prelube pump automatically starts and draws oil from the sump, passes it through a strainer and a filter, and then into the engine lubricating system. This recirculation of lube oil continues until the engine receives another signal. The oil is heated by an electrical immersion heater in the sump. This ensures continuous prelubrication of the engine and standby heating of the oil.

The heated oil, in this keep warm mode, is circulated throughout the engine block but does not feed the turbocharger's lubrication header. Check valves in the lube oil system insure that oil is kept near these assemblies so that lubrication can be provided by the engine-driven lube oil pump within one-half of one engine revolution on startup.

9.5.7.3 Safety Evaluation

Safety classification of the system meets the intent of ANSI Standard N18.2 and N18.2a, Safety Class 3 [15] as shown on Figure 9.5-56.

The diesel generator lube oil system is designed to seismic Category I requirements.

The ANS Safety Class 3 components in the system are designed to the requirements of the ASME B&PV Code, Section III, Class 3. Component supports are designed in accordance with ASME B&PV Code, Section III, Class 3 but fabricated as a minimum to AISC-1970. When a component is commercially unavailable as ASME B&PV Code, Section III, Class 3 design, the component is proven of equivalent quality.

"Equivalent quality", of a component is interpreted to mean an item designed for normal commercial use upgraded to ASME B&PV Code Section III requirements through seismic design, testing, qualification and documentation.

All such components are listed in Table 17A-1.

Low oil pressure to the main header is alarmed by a pressure switch. Oil pressure is indicated by a pressure gauge locally. Another pressure switch trips the engine on low lube oil pressure (if no safety injection actuation signal is present).

The volume of lubrication oil in the sump is sufficient to ensure continued operation under emergency conditions. A redundant diesel generator unit has been provided to meet the single-failure criterion, should a failure of the diesel generator lube oil system necessitate the stoppage of a diesel generator unit.

Both the high lube oil temperature and the low lube oil pressure monitoring systems have separate alarm and trip switches, since a failure of these components could result in an engine malfunction being undetected. Automatic shutdown of diesel generators is discussed in Section 8.3.1. A failure mode and effects analysis is given in Table 9.5-14.

The diesel generator lube oil system is housed in a seismic Category I structure. Failure of non-seismic components outside this structure does not affect this system. This structure also protects the system from the worst anticipated environmental phenomena as described in Sections 2.3 and 2.4.

9.5.7.4 Inspection and Testing Requirements

Testing and inspection is described in Section 9.5.4.4.

9.5.8 DIESEL GENERATOR COMBUSTION AIR INTAKE AND EXHAUST SYSTEM

9.5.8.1 Design Bases

The Diesel Generator Combustion Air Intake and Exhaust System is designed to provide the following:

1. Adequate combustion air for each emergency diesel generator under all operating conditions

2. Adequate capability to direct the engine exhaust to the outside air under all operating conditions

The system is designed to meet the single failure criterion. In order to further ensure its continued availability, the system is designed to withstand the effects of the worst anticipated environmental phenomena and seismic activity. The system is designed in accordance with 10 CFR Part 50, Appendix A, GDC 2, 4, and 5, NRC Regulatory Guide 1.29, IEEE 387-1977, and applicable codes listed in Apendix 17A [5], [6], [7], [12], [16].

9.5.8.2 System Description

The Diesel Generator Combustion Air Intake and Exhaust System is shown on Figure 9.5-57. It consists of the Air Intake System, turbocharger, turbocharger air intercooler, and the Exhaust Gas System. Each engine has its own independent Diesel Generator Combustion Air Intake and Exhaust System.

The Diesel Generator Combustion Air Intake System consists of a filter, silencer, flexible connection, adapter for connection to turbocharger intake, air intake duct connection, and interconnecting piping. The filter is capable of being cleaned during operation, and is located inside the Diesel Generator Building. The volume of intake air is sufficient to ensure continued operation under emergency conditions. The silencer is of the in-line horizontal type. The flexible connection is provided to relieve the turbocharger of any possible weight or strain due to piping weight, expansion, or contraction.

The turbocharger consists of an air compressor and a gas turbine coupled on the same shaft.

The turbocharger air intercooler transfers heat produced by the compression of combustion air by the turbocharger to the Jacket Water Cooling System.

The Diesel Generator Combustion Exhaust Gas System consists of an outlet adapter for the turbocharger gas turbine outlet, a flexible connection to ensure expansion protection and vibration isolation, a muffler to reduce the exhaust gas sound level, an exhaust duct connection, interconnecting piping, and a relief valve.

The Diesel Generator Combustion Air Intake and Exhaust System is provided with the following instrumentation:

1. A combustion air pressure indicator on engine panel for right bank and left bank of engine intake manifold

2. A digital electronic indicator for cylinder exhaust gas temperatures on engine panel; a selector switch for selecting temperature readout is also mounted on this panel.

The diesel generator combustion air intake is provided with intake air filters to protect the engine from dirt or other contaminants. The missile-protected intake, as shown on Figures 9.5-58 and 9.5-59, is located 36 ft above ground level to minimize the intake of ground dust or debris. The missile-protected inlet protects the Diesel Generator Combustion Air Intake System from external missiles and prevents the entrance of rain or snow. The combustion air intake and the exhaust are located apart so that dilution or contamination of the intake air by exhaust products does not affect operation of the diesel generators. The engine exhaust is directed upward at a point approximately 35 ft higher than the intake and 55 ft horizontal from the intake, which is directed downward. The intake is located at a sufficient distance (approximately 750 ft from the closest diesel generator) from the onsite bulk gas storage to preclude any contamination of the intake air due to an accidental onsite gas release.

The exhaust pipe downstream of the exhaust silencer contains a drain which prevents water from collecting. A full-flow relief valve located upstream from the exhaust silencer permits the engine to continue operating if the portion of the Diesel Generator Combustion Exhaust Gas System exposed on the roof is crushed by a tornado generated missile. If the normal flow path for the exhaust is blocked, this valve will discharge the exhaust gases directly to the atmosphere. There is no piping downstream of the valve.

9.5.8.3 Safety Evaluation

The Diesel Generator Combustion Air Intake and Exhaust System components, excluding interconnecting piping off the engine, are designed in accordance with manufacturers standards

to perform in accordance with IEEE Standard 387 [16]. The interconnecting piping is constructed to Safety Class 3 requirements in accordance with ANSI N18.2 [15] with the exception of the flexible connectors which are Safety Class 3 but non-ASME and that piping on the roof of the Diesel Generator Building and the relief valve which are classified as NNS. The Diesel Generator Combustion Air Intake and Exhaust System components are designed to seismic Category I or II requirements as applicable, with the exception of the relief valve and the flexible connectors located on the foof. [See Table 17A-1]. See Section 3.7 for details. The Diesel Generator Combustion Air Intake and Exhaust System components are not located near any high- or moderate-energy piping systems other than these serving the same respective diesel generator. This system meets the requirements of NRC Branch Technical Positions APCSB 3-1 and MEB 3-1 [14]. The air intakes for the Diesel Generator are protected from tornado generated missiles by a labyrinth shielding arrangement as shown on Figure 9.5-59. Air enters the building from below, thus making a direct missile entrance impossible.

The exhaust piping, to a point downstream of the relief valve, is protected by a shielding structure on the roof, see Figure 9.5-58. The piping downstream of the relief valve exists this protective structure and connects to the exhaust silencer.

A redundant diesel generator unit is provided to meet the single failure criterion if a failure of the Diesel Generator Combustion Air Intake and Exhaust System necessitates the stoppage of a diesel generator unit. The exhaust silencer is located outside the seismic Category I structure and therefore not protected from external missiles. If piping and components located outside are damaged to such an extent that exhaust gases are blocked, a relief valve is provided which provides an alternate discharge path for the exhaust gases. An opening in this protective structure allows exhaust gases from the open relief valve to exit. This opening is located in such a way as to prevent a direct missile from striking the exhaust. Thus, exhaust silencer damage that blocks the exhaust causes the relief valve to open so that the system continues to operate. A failure mode and effects analysis is given in Table 9.5-15.

In addition, the two 6 inch vent pipes from the diesel engine lube oil sump tank and crankcase to the atmosphere are also partially located outside the seismic Category 1 structure. However, the two lines are protected from external missiles by a missile protection enclosure and are seismically analyzed to maintain their functional and structural integrity.

A potential fire in a diesel generator room, together with a failure of the fire protection system is considered not to significantly degrade the quality of the combustion air for the redundant diesel generator due to the following rationale:

1. Taking into account the type of potential fire hazard present, the volume of the diesel generator room and the location of the room ventilation inlet and exhaust louvers, the majority of the smoke and hot gases will rise and flow out through the room ventilation exhaust louvers.

2. The postulated fire must burn unattended for a substantial period of time before causing pressurization of the diesel generator room and subsequently forcing smoke and hot gases out through the room ventilation inlet louvers located at elevation 813'-6" (see Figures 9.5-58 and 9.5-59 for the location and proximity of these louvers).

3. The normal air intake for each diesel generator is approximately 30 percent greater than the air intake required for combustion at 100 percent load. This 30 percent higher flow

capacity means that a large quantity of smoke and extraneous gas would be required to cause significant dilution of the redundant diesel combustion air before effecting the diesel generator performance.

Each diesel generator area is protected by a fixed deluge water spray system. The use of this type of system eliminates the concerns of degrading the performance of possible loss of a diesel generator due to pollution of the combustion air by the fire extinguishing medium.

9.5.8.4 Inspection and Testing Requirements

Testing and inspection are described in Subsection 9.5.4.4.

REFERENCES:

1. National Fire Protection Association Handbook, 14th Edition 1/76.

2. Appendix “A” to Branch Technical Position APCSB 9.5-1, Guidelines for Fire Protection for Nuclear Power Plants, docketed prior to July 1, 1976, dated August 23, 1976.

3. NRC Regulatory Guide 8.5, Immediate Evacuation Signal, February 16, 1973, U.S. Nuclear Regulatory Commission.

4. ANSI N2.3-1967, Immediate Evacuation Signal For Use In Industrial Installations Where Radiation Exposure May Occur.

5. 10 CFR Part 50, Appendix A, General Design Criterion 2, Design Bases for Protection Against Natural Phenomena.

6. 10 CFR Part 50, Appendix A, General Design Criterion 4, Environmental and Missile Design Bases.

7. 10 CFR Part 50, Appendix A, General Design Criterion 5, Sharing of Structures, Systems, and Components.

8. 10 CFR Part 50, Appendix A, General Design Criterion 44, Cooling Water System.

9. 10 CFR Part 50, Appendix A, General Design Criterion 45, Inspection of Cooling Water System.

10. 10 CFR Part 50, Appendix A, General Design Criterion 46, Testing of Cooling Water System.

11. Deleted

12. NRC Regulatory Guide 1.29, Seismic Design Classification Revision 2, 1976, U. S. Nuclear Regulatory Commission.

13. Branch Technical Position APCSB 3-1, Protection Against Postulated Piping Failures in Fluid Systems Outside Containment, attached to Standard Review Plan 3.6.1.

14. Branch Technical Position MEB 3-1, Postulated Break and Leakage Locations in Fluid System Piping Outside Containment, attached to Standard Review Plan 3.6.2.

15. ANSI N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants, 1973. [Including ANSI N18.2a-1975 Revision and addenda]

16. IEEE 387-1977, Criteria for Diesel Generator Units Applied as Standby Power Supplies for Nuclear Power Generating Stations.

17. Illuminating Engineering Society (IES) Lighting Handbook, Application Volume, 1981.

18. Branch Technical Position EICSB-17, Diesel Generator Protective Trip Circuit Bypasses, attached to Standard Review Plan Appendix 7-A.

19. Comanche Peak Nuclear Power Plant Unit 1 “Fire Protection Report”

20. NFPA 10-1981, “Portable Fire Extinguishers”

21. NFPA 12A-1980, “Halon 1301 Fire Extinguishing Systems”

22. NFPA 13-1978 and 1985, “Automatic Sprinkler Systems”

23. NFPA 14-1978, “Standpipe and Hose Systems”

24. NFPA 15-1979, “Water Spray Fixed Systems”

25. NFPA 20-1983, “Centrifugal Fire Pumps”

26. NFPA 24-1973, “Outside Protection”

27. NFPA 49-1975, “Hazardous Chemical Data”

28. NFPA 72D-1975, “Proprietary Protection and Signaling System”

29. NFPA 72E-1978, “Fire Detection Systems”

30. NFPA 80-1977, “Fire Doors and Windows”

31. NFPA 6-1974, “Industrial Fire Loss Prevention”

32. NFPA 26-1976, “Supervision of Valves”

33. NFPA 30-1984, “Flammable Combustible Liquid Code”

34. NFPA 69-1973, “Explosion Prevention Systems”

35. IEEE 383-1974, “IEEE Standard for Type Test of IE Electric Cables, Field Splices, and Connections”

36. IEEE 634-1978, “IEEE Standard Cable Penetration Fire Stop Qualification Test”

37. UL Subject 992, “Test Method for Measuring the Flame Propagating Characteristics of Floor and Floor Covering Materials”

38. ASTM E 84-1976, “Surface Burning Characteristics of Building Materials”

39. ASTM E119-1976, “Fire Test of Building Construction and Materials”

40. NEL-PIA guideline titled “Basic F.P. for Nuclear Power Plants” (1976)

41. Underwriters’ Laboratory (U.L.) “Building Materials Directory” effective edition at time of design-procurement

42. NUREG-0700, “Guidelines for Control Room Design Reviews, September, 1981”.

43. ANSI B31.1, Power Piping.

44. ASTM D975-1981, “Standard Specification for Diesel Fuel Oils”

45. ASTM D4057-1981, “Standard Practice for Manual Sampling of Petroleum and Petroleum Products”

46. ASTM D1552-1979, “Standard Test Method for Sulfur in Petroleum Products (High-Temperature Method)”

47. ASTM D2662-1982, “Standard Test Method for Sulfur in Petroleum Products (X-Ray Spectrographic Method)”

48. ASTM D2276-1978, American National Standard Test Method for Particulate Contaminant in Aviation Turbine Fuels

49. ANSI N195-1976, “Fuel Oil Systems for Standby Diesel Generators”

50. Comanche Peak Nuclear Power Plant 10CFR72.212 Evaluation Report

TABLE 9.5-1
IS NOT USED

TABLE 9.5-2
IS NOT USED

TABLE 9.5-3
IS NOT USED

TABLE 9.5-4
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TABLE 9.5-5
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TABLE 9.5-9
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TABLE 9.5-10
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TABLE 9.5-11
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR FUEL OIL STORAGE AND TRANSFER SYSTEM

Item Described	Function	Failure Mode	Cause of Failure	Effects on Subsystem	Method of Failure Detection	Effects on System
Diesel 1EG1 oil storage tank	Stores 7-day supply of fuel (approx. 15%)	Leaks	Crack, corrosion	Loss of part of 7-day supply	Level indicator, observation	None: 1EG2 available
Transfer pump 1a	Pump fuel to day tank	No output	a. Motor fails	a. None; use pump 1B	a. Level alarm	a. None; redundancy
			b. Pump fails	b. None; use pump 1B	b. Level alarm	b. None; redundancy
			c. Loss of power	c. Cannot pump oil	c. Level alarm	c. None; 1EG2 available
Transfer pump 1b	Same as pump 1a	Same as pump 1a	Same as pump 1a	Same as pump 1a	Same as pump 1a	None: redundancy none: 1EG2 available
Transfer line	Pipe fuel to day tank	Rupture	Crack, corrosion	Only 3 hr oil available [a]	Level alarm	None: 1EG2 available
Day tank	Store 3-hr fuel supply diesel	Rupture	Crack, corrosion	Loss of fuel supply	Level alarm	None 1EG2 available

a) Temporary hose or truck fill can be used, if break is upstream of check valve.

Note:

Diesel generator, Unit 1, No. 1 = 1EG1
Diesel generator, Unit 1, No. 2 = 1EG2
Diesel generator, Unit 2, No. 1 = 2EG1
Diesel generator, Unit 2, No. 2 = 2EG2

Similar failure modes on 1EG2, 2EG1, and 2EG2

TABLE 9.5-12
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR COOLING WATER SYSTEM

Item Described	Function	Failure Mode	Cause of Failure	Effects on Subsystem	Method of Failure Detection	Effect on System
Engine-driven jacket water pump	Pumps jacket water	No output	Mechanical ailure	Cannot pumpwater	High water temperature	None; 1EG2 is available
Jacket water heat exchangers	Heat sink for diesel service and jacket water systems	a. Leaks	Cracks, corrosion	Temperature rises	High water temperature	None; 1EG2 is available.
		b. No flow	Blockage	Temperature rises	High water temperature	None; 1EG2 is available
Jacket water piping and valves	Transports water	a. No flow	Blockage	Cannot transfer water	High water temperature	None; 1EG2 is available.
		b. Leakage	Cracks and corrosion	Temperature rises	High Water temperature	None; 1EG2 is available
Jacket water keep-warm pump	Circulates water for standby readiness	a. No output	Motor fails, pump fails, loss of power	Cannot transport water	Low water temperature	None; 1EG1 is available.
Standby electric heater	Heats water for standby readiness	No heat	Loss of power	Fails to heat water	Low water temperature	None; 1EG1 is available.

Note:

Diesel Generator - Unit 1, No. 1 = 1EG1
Diesel Generator - Unit 1, No. 2 = 1EG2
Diesel Generator - Unit 2, No. 1 = 2EG1
Diesel Generator - Unit 2, No. 2 = 2EG2

Similar failure modes for 1EG2, 2EG1, and 2EG2

TABLE 9.5-13
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR STARTING SYSTEM

Item Described	Function	Failure Mode	Cause of Failure	Effects on Subsystem	Method of Failure Detection	Effects on System
Air lines	Transports air	No flow	Blockage, cracks, corrosion	Subsystem not available	Low tank pressure	None: redundant start air subsystem is available for 1EG1.
Manifold	Delivers air	No flow	Blockage	Subsystem not available	As above pressure	None: redundant start air subsystem is available for 1EG1.
Compressor and Air Dryer	Refills air storage tanks	No air being supplied	Loss of power.	No refilling of air tanks	Low tank pressure	None: redundant compressor sub-system is available. For 1EG1.
Compressor	Refills starting air storage tank	Loss of air	Seismic or other pressure boundary failure	None (Safety Class 3 isolation provided)	Low tank pressure	None: (Safety Class 3 isolation provided)
Tank	Stores air for five attempted starts	Loss of air	Leaks, ruptures	No starting air	Low tank pressure	None: second tank is available. to provide air to 1EG1.

Note:

Diesel Generator, Unit 1 - No. 1 - 1EG1
Diesel Generator, Unit 1 - No. 2 - 1EG2
Diesel Generator, Unit 2 - No. 1 - 2EG1
Diesel Generator, Unit 2 - No. 2 - 2EG2

Similiar failure modes for 1EG2, 2EG1, and 2EG2

TABLE 9.5-14
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR LUBE OIL SYSTEM

Item Described	Function	Failure Mode	Cause of Failure	Method of Failure Detection	Effects on System
Engine-driven lube oil pump	Pumps lube oil	Pump failure	Mechanical Failure	Low oil pressure	None: 1EG2 is available
Lube oil piping and valves lube oil	Transports	1. No flow	1. Blockage	1. Low oil level	None: 1EG2 is available.
		2. Leakage	2. Cracks or corrosion	2. Low oil pressure	
Lube oil heat exchanger	Cools oil	1. Leaks oil or water	1. Cracks or corrosion	1. Low oil level	None: 1EG2 is available.
		2. No heat transfer	2. Blockage	2. High oil temperature	
Lube oil heater	warms oil	Fails to heat	Loss of power	Low oil temperature	None: 1EG1 is available.
Pre-lube pump	Circulates oil for standby readiness	Pump failure	Motor fails; pump fails; loss of power	Low oil temperature	None: 1EG1 is available.
Strainers and filters	Cleans oil	Clogged element	Particulate matter in oil	High pressure drop	None: 1EG2 is available.

NOTE:

Diesel Generator Unit 1, No. 1 = 1EG1
Diesel Generator Unit 1, No. 2 = 1EG2
Diesel Generator Unit 2, No. 1 = 2EG1
Diesel Generator Unit 2, No. 2 = 2EG2

There are similiar failure modes for 1EG2, 2EG1, and 2EG2.

TABLE 9.5-15
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR (COMBUSTION AIR INTAKE AND EXHAUST SYSTEM)

(Sheet 1 of 2)

Item Described	Function	Failure Mode	Cause of Failure	Effects on Subsystem	Method of Failure Detection	Effects on System
Intake air pipes and flexible connectors	Pipe air to the engine	a. Rupture	Crack, corrosion	Loss of outdoor intake air	Visual	None: 1EG1 is still available
		b. Low flow	Blockage	Loss of adequate air	High exhaust gas temperature	None: 1EG2 available
Intake air filters	Cleans intake air	Low flow	Blockage	Loss of adequate clean intake air	High exhaust gas temperature	None: 1EG2 available
Air intake silencer	Reduce intake air sound level	a. Rupture	Crack, corrosion	Loss of outdoor intake air; excessive noise	Excessive noise	None: 1EG1 is still available
		b. Low flow	Blockage	Loss of adequate intake air	High exhaust gas temperature	None: 1EG2 is still available
Turbocharger	Provide combustion air	No air being supplied	Loss of compressor or turbine	No flow	Stopping of engine, high exhaust gas temperature	None: 1EG2 available
Turbocharger air intercooler	Cools intake air	a. Reduced heat	Fouling	Loss of adequately cooled air	Loss of engine output	None: 1EG2 available
		b. Tube rupture	Cracks or corrosion	Induction of water	Engine stops	None: 1EG2 available

TABLE 9.5-15
FAILURE MODE AND EFFECTS ANALYSIS OF DIESEL GENERATOR (COMBUSTION AIR INTAKE AND EXHAUST SYSTEM)

(Sheet 2 of 2)

Item Described	Function	Failure Mode	Cause of Failure	Effects on Subsystem	Method of Failure Detection	Effects on System
Exhaust gas pipes and flexible connection	Pipe exhaust gas to the outside of plant	Rupture	Crack, corrosion	Exhaust gas inside the Engine Room	Excessive noise	None: 1EG1 is still available
Exhaust silencer and outside piping	Reduce exhaust air sound level	a. Rupture	Crack, corrosion	Excessive noise	Excessive noise	None: 1EG1 is still available
		b. Low flow	Blockage	Exhaust relief valve opens to provide alternate flow path	Excessive Noise	None: Relief valve opens

NOTE:

Diesel Generator - Unit 1, No. 1 = 1EG1
Diesel Generator - Unit 1, No. 2 = 1EG2
Diesel Generator - Unit 2, No. 1 = 2EG1
Diesel Generator - Unit 2, No. 2 = 2EG2

Similar Failure Modes For 1EG2, 2EG1, and 2EG2

TABLE 9.5-16
THIS TABLE HAS BEEN DELETED

TABLE 9.5-17
LUBE OIL COOLER DESIGN PARAMETERS

Design heat removal rate, Btu/hr	3.38×10^6
Required heat removal rate, Btu/hr	3.22×10^6
Flow rate (tube side), gpm	900
lb/hr	438,300
Flow rate (shell side), gpm	500
lb/hr	225,153

	Shell	Tube	Diff.
Inlet temperature	185.0	147.2	37.8
Outlet temperature	156.4	154.7	1.7
Temperature differential	28.6 (drop)	7.5 (rise)	

Design margin in heat removal rate:

The design heat removal rate of 3.38×10^6 Btu/hr includes an excess of 0.16×10^6 Btu/hr over the required heat removal rate.

TABLE 9.5-18
JACKET WATER COOLER DESIGN PARAMETERS

The diesel generator is rated for 7,000 Kw. At this condition the Jacket Water Cooler Parameters are:

Required heat removal rate, Btu/hr	16.29 x 106
Flow rate (shell side), gpm	1,250
Flow rate (tube side), gpm	1,500
Shell Inlet Temperature, deg. F	<175
Shell Outlet Temperature, deg. F	147
Tube Inlet Temperature, deg. F	117
Tube Outlet Temperature, deg. F	139
Tube side fouling, hr-ft^2-F/BTU	0.001
Shell side fouling, hr-ft^2-F/BTU	0.001

The Jacket Water Cooler is capable of removing the heat load generated by the diesel generator when operating at the bounding loading requirements of 6,300 Kw (FSAR Tables 8.3-1 and 8.3-2) with up to 85 tubes plugged.

10.0 STEAM AND POWER CONVERSION SYSTEM

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10.1 SUMMARY DESCRIPTION

The steam and power conversion system for each of the two CPNPP units includes one steam turbine generator and all of the auxiliary equipment and support systems necessary for its operation. Each turbine generator unit is independent of the other, as there are no shared components. The following description applies to each of the units.

The turbine is a tandem-compound, four-flow, 1800-rpm machine installed on the Turbine Building operating deck (elevation 830 ft 0 in.) and is supported by the turbine pedestal.

Steam is supplied to the two-flow, high-pressure turbine from four steam generators, which in turn are supplied with heat from the primary loops of the Pressurized Water Reactor (PWR).]

The high-pressure turbine exhaust is directed through two moisture separator/reheaters (MSR) operating in parallel. The steam is dried and reheated in the MSRs and then passed through the two two-flow low-pressure turbines.

Exhaust steam is condensed in a two-shell, single-pressure, single-pass, surface-type condenser. Condensate storage capacity in the condenser hot wells is equivalent to approximately five minutes operation at the maximum steam flow rating. The Condensate and Feedwater System returns feedwater to the steam generators through six stages of extraction feedwater heating.

Heat balances reflecting the original design at an NSSS rated thermal power of 3628 MWt (maximum guaranteed rating) are shown on Figures 10.1-1 and 10.1-2, respectively. The rating of the electric generator is 1,410,000 kVA, 60 Hz, 0.90 power factor.

The description contained within this chapter is a general description of the original design of the steam and power conversion system. The actual operational parameters (fluid flow rates, temperatures, pressures, etc.) may differ from those reflected throughout Chapter 10 due to as-built conditions, steam generator blowdown, and physical and operational changes made to improve net heat rate and maximize electrical power generation at the licensed reactor thermal power.

Uprates in the reactor licensed thermal power of either unit may also be implemented (See Section 1.1.4). An evaluation of the available margin within the steam and power conversion systems is performed as part of any formal unit uprate to determine the extent that the steam and power conversion system's original design margin can support operation at such power levels. Any modifications to those systems are reflected herein consistent with the level of design detail contained in Chapter 10.

The following system diagrams are shown in the FSAR Flow Diagram Volumes:

Flow Diagram	
M1-0202	Main Steam, Reheat & Steam Dump System
M1-0204	Main Condenser Piping Diagram

M1-0211	Main Condenser Evacuation System (Condenser Vacuum)
M1-0223	Turbine Gland Sealing System
M1-0210	Circulating Water System
M1-0240	Circulating Water Chlorination System
M1-0244	Condensate Cleanup System (Condensate Polishing)
M1-0204	Condensate System
M1-0203	Steam Generator Feedwater System
M1-0239	Steam Generator Blowdown Cleanup System
M1-0206	Auxiliary Feedwater System
M1-0205	Extraction Steam System
M1-0207 thru 0209	Heater Drain System
M1-0212	Turbine Plant Cooling Water System
M1-0213	Auxiliary Steam System
M1-0214	Turbine Oil Purification System
M1-0243	Plant Gas Supply (Nitrogen and Hydrogen)
M1-0222	Secondary Plant Sampling System
M1-0228	Process Sampling System
M1-0228	Post Accident Sampling System

The following system diagrams are shown in Section 10.4:

Figure	
10.4-1	Main Condenser Piping Diagram – Plan
10.4-2	Main Condenser Piping Diagram - Elevation
10.4-12	Flow Diagram - Auxiliary Feedwater Failure Mode Analysis
10.4-22	Main Feedwater Loop Seal Arrangement
10.4-23	Unit 1 Integrally Grooved Tube Sheet Pressurization System

Circulating water for the condensers is provided from the Squaw Creek Reservoir (SCR), where heat is primarily rejected to the atmosphere by surface evaporation. Makeup water for the reservoir is furnished from Lake Granbury.

The safety-related features of the steam and power conversion system include portions of the main steam supply system, upstream from and including the first moment restraint beyond the main steam stop valves used for Containment isolation. Included in these portions are the main steam safety valves, atmospheric steam dump power relief valves, main steam line condensate removal drain pots up to and including the pneumatically operated isolation valves, sections of the Steam Generator Blowdown and Process Sampling System up to and including pneumatically operated isolation valves, which form part of the Containment isolation scheme, and the air vents on the safety class portion of the main steam lines. The steam lines supplying the auxiliary feedwater pump turbine driven are also safety-related. The safety-related portions of the Steam Generator Feedwater System include those sections downstream of and including check valves just upstream of the feedwater isolation valves and those portions of the chemical feed, nitrogen supply, and process sampling systems which form part of the Containment pressure retaining boundary. The Auxiliary Feedwater System is also safety-related.

Instrumentation, controls, and alarms are provided to monitor system functions, as described in detail in subsequent sections.

TABLE 10.1-1
MAJOR STEAM AND POWER CONVERSION EQUIPMENT SUMMARY

Component	Maximum Guaranteed Rating Unit 1	Unit 2
Steam Generator[a]		
Steam flow rate (total 4 steam generators), lb/hr	16.26×10^6	16.26×10^6
Feedwater inlet temperature, °F	450.3	450.3
Steam outlet temperature, °F	545.1	543.2
Steam outlet pressure, psia	1005	989
Steam generator reactor coolant inlet temperature, °F	620.4	620.4
Steam generator reactor coolant outlet temperature, °F	558	558
Turbine Generator[b]		
Steam to turbine generator, lb/hr	14,774,975	14,715,715
Throttle inlet pressure, psia	980	948.94
Throttle enthalpy, Btu/lb	1192.3	1192.1
Percent moisture	.21	.4
Power output, kW	1,263,444	1,253,440
Power factor	0.90	0.90
Hydrogen pressure, psig	75	75

a) Reference: WCAP-16840

b) Reference: Unit 1 - WB-10909-1 (Figure 10.1-1)
Unit 2 - WB-10939-1 (Figure 10.1-2)

10.2 TURBINE GENERATOR

10.2.1 DESIGN BASES

The turbine generator unit is a tandem-compound, four-flow, single reheat-type, direct-connected unit, supplied by Allis-Chalmers Power Systems Inc./Siemens Power Corporation. The turbine generator is intended to be base loaded. The turbine receives steam from four steam generators (thermal energy) and converts the thermal energy to mechanical energy through rotation of the turbine shaft. The turbine, in turn, is directly connected to an electric generator, which produces electrical energy upon rotation against an excited field. The turbine generator original design basis is to produce 1,160,706 kWe, when operating, with 15,140,016 lb/hr of saturated steam at 975 psia, 541.5 F, 0.38-percent moisture at the throttle, 3.5 in. Hg absolute at the exhaust. The turbine consists of one high-pressure element and two low-pressure elements. The maximum original guaranteed design basis heat balance condition is as shown on Figure 10.1-1 of Section 10.1.

The Unit 1 turbine generator is designed to produce 1263.4 MWe when operating at 14,774,975 lb/hr saturated steam at 980 psia and 1192.3 Btu/lb with 0.21% moisture at the throttle and 1.414 in Hg absolute at the turbine exhaust. The Unit 2 turbine generator is designed to produce 1253.4 MWe when operating at 14,715,715 lb/hr saturated steam at 948.9 psia and 1192.1 Btu/lb with 0.40% moisture at the throttle and 1.414 in Hg absolute at the turbine exhaust.

The turbine generator includes an electrohydraulic control system which is capable of adjustment to step load changes as required.

The steam turbine has the following loading capabilities:

1. The capability to accept step changes in load of 10 percent rated power and ramp changes of 5 percent rated power per min in the range from 40- to 100-percent full load

2. The capability to accept 50-percent load reduction from rated power without reactor trip. This capability is based on the capability of secondary plant systems to accept 40-percent steam dump and to maintain adequate feedwater supply to the steam generators. The reactor accepts 10 percent of the load rejection during this mode of operation.

3. The capability to accept complete load rejection from the maximum rated power level with reactor and turbine trip

4. The capability to follow generator-demanded load changes automatically when these load changes occur within the automatic control range of 15- to 100-percent full power

The turbine generator and all its appurtenances generally meet the applicable requirements of the following codes, standards, and legislation:

1. American Insurance Association (A.I.A.)

2. American National Standards Institute (ANSI)

3. American Society of Mechanical Engineers (ASME)

4. American Society for Testing and Materials (ASTM)

5. American Welding Society (AWS)

6. Heat Exchange Institute (HEI)

7. Hydraulic Institute (HI)

8. Institute of Electrical and Electronics Engineers (IEEE)

9. National Electrical Safety Code (NESC) ANSI C2

10. National Electrical Manufacturers Association (NEMA)

11. National Fire Protection Association (NFPA)

12. Occupational Safety and Health Act (OSHA)

13. Steel Structures Painting Council (SSPC)

14. Tubular Exchanger Manufacturers Association (TEMA)

Main steam and feedwater piping outside the Containment from the end of the moment restraint in the Safeguards Building to the Turbine Building is designed in accordance with ANSI B31.1. Main steam piping between the moment restraint in the Safeguards Building and the Containment wall and the piping located within the Containment is designed in accordance with Section III of the ASME B&PV Code. Feedwater piping between the moment restraint in the Safeguards Building and the Containment wall and the piping located within the Containment is designed in accordance with Section III of the ASME B&PV Code. Seismic classification conforms to Section 3.2, and the seismic design requirements are defined in Section 3.7.

The turbine-generator unit is not designed for operation under the stresses that could be imposed by the half safe shutdown earthquake (1/2 SSE) or the safe shutdown earthquake (SSE). However, the turbine generator is designed to function under the thermal stresses that could be imposed because of upset conditions, emergency conditions, and faulted conditions as defined in Section 2 of ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plant, August 1973.

10.2.2 DESCRIPTION

10.2.2.1 Turbine

The turbine is a multicasing, tandem compound, four-flow, reaction type, 1800-rpm unit with 46-in. last-stage blades. No radiation shielding is required.

The turbine consists of one double-flow, high-pressure element in tandem with two double-flow, low-pressure elements. Two horizontal cylindrical external combined moisture separator/reheaters (MSRs) are included; one is installed on the operating deck on each side of the turbine at elevation 830 ft 0 in. Exhaust steam from the high-pressure turbine element enters the MSR shells where moisture is removed, and high-pressure main steam passing through

tubes in the reheater heats the lower-pressure steam prior to its entering the low-pressure turbines. Each MSR normally processes half of the steam flow.

The MSRs are pressure vessels containing heat exchanger tube bundles and moisture-separating elements. The single stage reheater consists of U-tube type tube bundle with a fixed tube sheet of carbon steel clad with Inconel and integrally finned tubes of ferritic stainless steel. The moisture separators are of the Chevron type, and are constructed of stainless steel for durability in the wet steam environment.

The MSRs are designed for high reliability with external tube header, welded tube-to-tube sheet joints, ample drainage capability, and provisions for supports with thermal flexibility. Safety relief valves and heating steam supply valves are also supplied.

Steam from the steam generators is supplied to the high-pressure turbine through four pipes, each containing a stop valve in series with a control valve combined into a single compact casing. The valve units are located above the operating deck (elevation 830 ft 0 in.), close to the high-pressure turbine, with two units on either side of the turbine. This location minimizes the entrapped steam volume on valve closure, thus limiting potential overspeed following a load rejection. The above-floor arrangement and turbine rollaway cover simplify valve inspection and maintenance.

The stop valve is opened by control fluid pressure and closes rapidly by spring force when tripped. The control valve is positioned by a hydraulic servomotor but also closes rapidly by spring force after loss of control fluid pressure.

Upstream from each stop and control valve is a permanently installed steam strainer body which includes removable steam strainer.

Steam from the MSRs enters the low-pressure turbine through low-pressure stop and control valves located at each low-pressure turbine inlet. Each of these valves has its own hydraulic actuator. The low-pressure stop valves are operated by the trip system. Low-pressure control valves are operated by the control system. These valves limit overspeed and maintain unit stability on loss of load.

Provisions have been included to allow for automatic testing of the high-pressure and low-pressure stop and control valves while the turbine is in operation.

The turbine generator is equipped with all instrumentation and control devices required to operate the unit. Included is an electrohydraulic-type speed and load control system designed to control speed and load of the unit under the load changes indicated in Subsection 10.2.1. The system has a main electrohydraulic governor to control speed during normal operation with load reject logic to limit top speed of the unit to a value below the emergency overspeed trip setting. With the exception of the hardware overspeed system sharing three speed probes with the governor system, the overspeed systems are completely independent of the governor control system and initiate a trip at an overspeed of approximately 110 percent to immediately close all main steam stop and control valves and the low-pressure turbine stop and control valves. A trip of the turbine generator will initiate a signal which will activate the reactor trip system when greater than 50% reactor power (P9 interlock).

10.2.2.2 Lubrication System

Bearings are lubricated during operation by oil from the main shaft-driven oil pump located in the front pedestal of the turbine. Three half-capacity oil coolers are provided, two active and the other standby, with mechanically interlocked four-way transfer valves on the inlet and outlet to ensure safe onload changeover from one cooler to the other. Dual filters with a changeover valve are provided in the oil pipe which supplies the combined journal-thrust bearing and the shaft-oil lifting pumps.

The bearing oil-supply piping is guarded against fire hazard resulting from leaking by enclosure in steel ducts. Access is provided for inspection of the piping. The oil drain piping is separate from the supply piping.

Three half-capacity auxiliary oil pumps, an emergency bearing oil pump, and a shaft-oil lifting pump are provided for startup, shutdown, and turning-gear operation.

10.2.2.3 Turning Gear

The turbine generator is equipped with an oil-hydraulic turning gear consisting of an impulse turbine wheel solidly connected to the turbine shaft and segmental nozzle boxes mounted within the front bearing pedestal. Oil from the auxiliary oil pump header is supplied to the turning gear through two motor-operated valves and is discharged through the nozzles, thus impinging on the wheel and causing the turbine-generator shaft to turn. In the case of startup from a stationary condition, the initial friction of the shaft in the journal bearings is overcome by the shaft-oil lifting pump which is fed from a high-pressure oil pump into the bottom of the journal bearings.

The oil-hydraulic turning gear turns the turbine generator at a speed in excess of 80 rpm. The relatively high speed ensures that good, stable lubrication films exist in the journal bearings while they are on turning gear. Also, the speed helps the shaft and moving blades cool down evenly after shutdown and circulates the air or steam in the turbine casings to minimize any temperature differences between upper and lower stationary parts, thus preventing thermal distortion of either rotors or casings. Because there are no gear teeth to be engaged or disengaged, the turning gear is not susceptible to damage. Furthermore, the turning gear requires neither mechanical nor electrical interlocking with the shaft-oil lifting system, which eliminates all requirements that the unit coast down to a stationary condition before engaging the turning gear.

10.2.2.4 Steam Sealing System

The steam glands which seal the high-pressure turbine casing-to-rotor clearances are of the axial-flow labyrinth type. Each gland consists of a large number of radial sealing strips of 10 mil thickness at the seal tip that are caulked into grooves in the stationary gland rings. Each stationary ring is composed of six segments. These are spring-mounted in horizontally split gland ring holders. The shaft glands of the low-pressure turbines use similar sealing strips caulked into spring-mounted segmental gland rings; however, the low-pressure shaft is smooth where it passes through the glands to accommodate the relatively large axial expansions of the low-pressure rotors. Each shaft seal has two annular chambers. One is connected to the sealing steam header, and the other is exhausting steam mixed, with a small amount of air leakage to a seal steam condenser.

In the normal operating range, the sealing steam header is supplied from the high-pressure turbine gland leakoffs and crossaround system.

For startup and at low loads, an auxiliary source such as main steam is required. The seal steam header pressure is regulated at slightly above atmospheric pressure by a supply valve and a leak valve, both operated by an pneumatically controlled system that includes a pressure transducer. The leak valve discharges to the lowest pressure feedwater heater. A smaller amount of air and sealing steam are continuously drawn off the outermost section of each gland and discharged to the seal steam condenser, which is maintained at slightly below atmospheric pressure by a motor-operated exhauster. The control system includes provisions for manual operation from the Control Room, with an automatic balancing device providing for switchover from automatic to manual without a sealing pressure transient.

The glands are designed to avoid local overheating of the shaft if abnormal conditions result in a rub at any seal.

In the event of rubbing of either the moving or the stationary sealing strips, the spring-mounted segments yield against the pressure of the springs associated with each segment. If this radial movement is not sufficient to relieve the rub, the thin strips are worn down with only a slight amount of heat, thus avoiding shaft damage from excessive local overheating. If necessary, glands can be easily repaired. This is done by removing the worn strips and caulking new ones into the same grooves.

In the case of the high-pressure turbine, access is gained to the outer gland rings by removing the horizontally split gland ring holders from the ends of the turbine outer casing. If the glands have been damaged, the sealing strips in the ring segments are renewed without having to open the high-pressure turbine casing. The shaft glands of the low-pressure turbines are enclosed within horizontally split gland ring holders attached rigidly to the adjacent bearing pedestals; consequently, they can be inspected or repaired without disturbing the low-pressure turbine outer casings.

Caulked-in radial sealing strips (similar to those used in the shaft glands) also minimize steam leakage over the shrouds of moving and stationary blade rows. Generally, three strips are used to seal each row of shrouded blades. The sealing effect is improved in the moving rows by using stepped shrouds with two different sealing-strip diameters. Each continuous shroud has an integral elevated or depressed cylindrical surface, and strips of different heights (corresponding to the shroud steps) are caulked into circumferential grooves opposite the shroud.

10.2.2.5 Steam Extraction Connections

Turbine steam extraction connections are provided for six stages of feedwater heating and for the MSRs as discussed in Section 10.4.10.

10.2.2.6 Generator

The generator is rated at approximately 1,410,000 kVA, is three-phase, and has a 60 Hz frequency at 0.90 power factor. The generator stator windings and bushings and generator rotor windings are water cooled, and the core and other components are hydrogen cooled at a maximum pressure of 75 psig. The generator is integrated and connected with its own excitation

system. Generator rating, temperature rise, and insulation class are in accordance with ANSI C50 and NEMA MG 1, as applicable.

10.2.2.6.1 Hydrogen Storage and Supply

Hydrogen for cooling the generator is stored outdoors, about 300 feet west of the Turbine Building. (See Figure 1.2-1). The piping arrangement of this facility is shown on Figure 10.4-19, Flow Diagram Plant Gas Supply System-Hydrogen Supply, and described in Section 10.4.15.

Gas supplied from the storage tanks passes through a pressure-regulating station located at the tanks and is distributed through the Turbine Building at approximately 150 psig. Secondary pressure regulators control the gas supply to the individual generators at a maximum pressure of 65 psig. The remote location of the outdoor bulk storage tanks minimizes the hazards of fire or explosive concentrations of hydrogen. In the event of a break in the supply line, the faulted line is automatically isolated by the excess flow control valve.

In the event of a gross supply line rupture, the flow rate across the excess flow valve would go critical, causing an immediate closure. Reaction times are a function of the hydrogen flow rate from the line rupture - typical reaction times are less than one (1) second.

The excess flow control valve in the hydrogen supply system operates on the following principle. Hydrogen flow through the excess flow valve creates a pressure drop proportional to the square of the flow rate. A spring holds a seat plug off the seat. When the pressure drop across this seat plug reaches the set point of the valve, the force against the upstream side of the seat plug overcomes the spring tension and forces the seat plug against the seat, forming a bubble tight seal. Differential pressure holds the valve closed until a manual bypass valve is opened and the pressure equalized across the valve.

10.2.2.6.2 Fire Prevention and Protection

1. Fire Prevention

   Following manufacturing, the empty generator stator frame with attached shields and terminal box is subjected to a hydraulic test to ensure that it will be capable of withstanding maximum hydrogen explosion pressures.

   The use of hydrogen as a coolant in the generator requires special safety equipment to ensure that hazardous operating conditions will not occur. As a precaution against the explosion hazard the air inside the generator is neither directly replaced with hydrogen during generator filling nor the hydrogen replaced directly with air during the emptying procedure. In both cases the generator is scavenged with an inert gas, argon.

   A meter system serves for measuring the H_2 content of the cooling gas in the generator as well as the composition of gas mixture (argon/air and H_2/argon) during filling and emptying of the generator. A second mechanical purity meter system is also supplied.

   Particular precautions are taken with respect to a H_2 leak in the generator bearing compartment. A special vapor exhauster creates a slight vacuum in the bearing compartment. Any hydrogen collecting in the bearing compartment is drawn off by the

exhauster and vented to atmosphere. If the exhauster fails, a second exhauster is automatically started.

To prevent the hydrogen which enters the bearing compartment from escaping via the oil drain pipes, the drain oil is admitted into the turbine oil tank via the Hydrogen Degassing Storage Tank and a loop seal. This loop seal is permanently filled with oil to prevent the escape of gas.

The seal oil drained from the seal oil tank passes into the Hydrogen Degassing Storage Tank to which the vapor exhauster is also connected. The exhauster creates a slight vacuum in the Hydrogen Degassing Storage Tank so that oil saturated with hydrogen is degassed.

2. Fire Protection

For discussion on fire protection, see Section 9.5.1.

10.2.2.7 Automatic Controls

As described in Subsection 10.2.2.1, steam from the steam generators is supplied to the high-pressure turbine through four pipes, each containing a stop valve in series with a control valve combined into a single compact casing. The stop valve is opened by control fluid pressure, and, when tripped, closes rapidly by spring force. The control valve is positioned by a hydraulic servomotor and also closes rapidly by spring force after loss of control fluid pressure.

Steam from the MSR units enters the low-pressure turbines through low-pressure stop and control valves at each low-pressure turbine inlet. Each of the valves has its own hydraulic actuator which is operated by the trip and control system, respectively. The valves operate as low-pressure stop and control valves and help limit overspeed and maintain unit stability on loss of load.

The standard turbine control system is a redundant electrohydraulic control (EHC) system with load rejection logic to mitigate an overspeed event. After a full load rejection the unit will experience a speed transient up to a maximum of 110 percent of rated speed, then it will settle down to a steady-state speed corresponding to 60.3 Hz at idling (no-load) operation.

10.2.2.7.1 Speed and Load Control With EHC

The EHC contains a speed control loop and a load control loop. The speed control is used primarily to bring the turbine up to synchronous speed. However, the speed control always remains in operation over the entire load range. The load control is switched on automatically when a generator breaker is closed.

The speed control compares the actual turbine speed with a reference setting and uses the difference as a control signal. The actual speed is derived from an electrical speed measuring system that contains speed probes mounted in the front bearing pedestal and directed to a toothed wheel. The toothed wheel is mounted on the turbine shaft within the front high-pressure bearing housing. As the toothed wheel turns, a signal of varying frequency is induced in the probes proportional to the turbine speed. The digital signals are converted to analog signals in the EHC cabinet. There are three speed signal channels in this system. If any channel fails, it is

automatically switched out, and an alarm is given; however, operation continues with the remaining channels. The load control compares the actual generator load with a reference load setting and delivers the difference as a load-demand error signal. The actual generator load is obtained from three load transducer channels using generator voltage and current inputs.

During load pickup, the automatic transfer from speed to load control at synchronization/generator output breaker closure is smooth because a balance control is incorporated into the system.

If load is rejected to less than approximately 15 percent, the speed/load control signal is overridden, initiaing fast closure of all control valves. Without adjustment of the reference speed setting, the system can control a full-load rejection to no-load with a residual speed deviation of about 0.75 percent. It can control a load rejection to auxiliary load with less than 0.4 percent residual speed deviation.

The EHC system contains internal self-monitoring equipment for critical functions.

The value of the EHC speed/load signal positions the turbine control valves through two redundant valve lift controllers, amplifiers, proportional valves and followup piston banks. Adjustments are provided to ensure the proper control of high pressure and low pressure valve operation.

10.2.2.7.2 Hydraulic Control Fluid System

The hydraulic portion of the control system uses fire-resistant pure phosphate ester fluid at a maximum pressure of 455 psig. This moderate pressure level helps minimize the possibility of leaks, and the fluid reduces the possibility of a fire even if a leak should occur.

The system includes a control fluid tank with three half-capacity pumps, two active and one standby. The standby pump is automatically activated if the fluid pressure falls below a preset level. Two full-sized fluid coolers are provided, one active and one standby, with mechanically interlocked three-way valves for safe onload transfer to the standby cooler, if necessary.

The fluid is maintained clean by continuously circulating a portion of the tank volume through filters arranged in series. A separate small motor-driven pump provides this circulation. All fluid returning to the tank passes through sieve filters located at the return connection inside the tank. Dual filters are also provided at the control signal inlet to each control valve actuator.

10.2.2.7.3 Overspeed Protection

The important function of overspeed protection is provided by both the redundant EHC control system and the redundant overspeed protection systems.

The EHC system serves as the primary device for mitigation of an overspeed event. Load rejection logic built into the system monitors the generator output. If the system senses a negative gradient over a certain load range, the protection logic will override the control logic and quickly close both the HP and LP control valves for a preset time.

If for some reason the EHC control system was unable to prevent an overspeed event, two independent and diverse electronic overspeed protection systems will trip the turbine at 110

percent of rated speed (1980 rpm). The overspeed trip systems consist of two independent and diverse trip circuits called hardware (HW) and software (SW) overspeed trip systems. Either system will independently trip the turbine and each is configured for 2 of 3 trip logic.

The hardware overspeed trip system consists of a primary hardware trip circuit and a backup software trip circuit sharing three speed probes. The three speed probes are connected to three individual speed measurement cards that compare the speed signal to preset setpoints. If the setpoints are exceeded (<10 rpm and >1980 rpm), a trip signal is sent from each speed measurement card to the hardware and software trip circuits.

The HW trip signals are then processed by three circuit cards with board mounted relays. The three cards work together as one unit to check for a 2-of-3 trip coincidence. The three circuit cards bypass the fail-safe turbine trip system and de-energize the three relays on the output of each of the three channels of the turbine trip system, and consequently de-energize the turbine trip block. As a backup, the HW trip signals for the speed measurement cards are also processed by three channels of the fail-safe turbine trip system, and de-energize the output to the turbine trip block.

Independent of the hardware overspeed system a separate software overspeed trip system is provided serving as an independent and diverse overspeed trip. The software overspeed trip is independent and utilizes separate speed probes, speed measurement cards, fail-safe software-based trip system and is installed in a separate cabinet with independent power supplies.

As with the hardware system, three speed probes feed three individual speed measurement cards that compare the speed signal to preset setpoints. If the setpoints are exceeded (<10 rpm and >1980 rpm), a trip signal is sent from each speed measurement card to the software trip circuits. The trip signals are then processed by the three channels of fail-safe software-based turbine trip systems. Each channel is configured with 2-of-3 trip logic. The outputs of the trip systems will de-energize which will trip the turbine trip block.

Rapid closure of all turbine stop and control valves will result from either the HW or SW overspeed trips. Only one of the system is necessary to trip the turbine. Since the HW and SW overspeed trip systems are independent and actuation of either system will cause a turbine trip, only one of the two systems is necessary to trip the turbine. Since the HW and SW overspeed trip systems are independent and actuation of either system will cause a turbine trip, only one of the two electrical overspeed trip systems is required to be operable.

Proper operation of the HW and SW overspeed trip systems are checked online daily with an automatic test initiated by the protection system. The turbine trip block test is performed per the Preventative Maintenance Program. The HW overspeed trip relays and their initiating relays cannot be tested at power. These relays will be tested each outage to ensure operability.

10.2.2.7.4 Turbine Stress Evaluator

The turbine stress evaluator continuously monitors differential temperatures in critical areas of the turbine. It automatically computes and provides operating limits for safe speed and load changes.

The turbine stress evaluator uses special dual thermocouples which measure the temperature at the wall center and at the inner wall surface of the first main steam stop valve. The temperature

of the high-pressure turbine admission inner wall surface is measured to evaluate the high-pressure rotor thermal stresses. From these measured temperatures it calculates actual thermal stress conditions and compares them with allowable conditions considering the prevailing speed or load. A selection circuit determines which of the critical areas has the smaller margin between actual and permissible temperature differentials. The magnitude of the limiting margins, which are converted into temperature and load allowances, is displayed on Operating and Monitoring (OM) screens at computer workstations in the Control Room.

The limiting values calculated by the turbine stress evaluator and displayed on its indicator are used as input signals to the speed load reference limiters of the EHC equipment, thus closing the loop between the turbine stress evaluator and the turbine control system. Consequently, speed or load changes are only possible within the allowable ranges, as determined and indicated by the turbine stress evaluator.

Temperature values, speed, load, and temperature and load allowances are displayed on the OM screens. If a malfunction occurs, it is still possible to operate the turbine generator with the turbine stress evaluator completely out of service. This is done by using data on the OM screens and criteria provided in the operating instructions.

10.2.2.7.5 Turbine Stop and Control Valves

A trip signal from any designated source causes closure of the stop valves in each of the four high-pressure steam admission lines, the control valve in each high-pressure steam admission line, the low-pressure turbine valves (of which there are two per low-pressure steam admission line), the motorized stop and power-assisted check valve combination in steam extraction lines 1, 2, 3, and 4. Steam extraction line 3 branches off from the cold reheat line. Extraction lines 5 and 6, located at the low-pressure end, are not valved, since they are inaccessible inside the condenser neck and contribute little to overspeed potential because they are low-pressure lines.

The high-pressure stop valves are simple two-position valves which are held open against spring force by control fluid pressure. Loss of this pressure causes rapid closure.

The control valves are also opened by fluid pressure and closed by spring force. They close rapidly on trip command in a manner similar to the stop valves. The stop valves are designed to be leaktight when closed.

Both the stop and control valves have stems, sealing rings, and other sliding surfaces which are designed and manufactured for reliable operation. The stem seals are of the labyrinth type with radial clearance between the stem and sealing rings. This radial clearance is provided as insurance against the possibility of valve sticking caused by galling or buildup of material deposits on sliding surfaces.

The low-pressure stop and control valves in each low-pressure turbine steam admission are of butterfly-type design. Each valve has its own actuator. The low-pressure stop valves are two position valves which close rapidly on receipt of a trip command. The low-pressure control valves are equipped with actuators similar to the high-pressure control valves for position control in response to a control signal.

The high pressure (HP) stop and control valves have closing times of less than 500 millisecond and the butterfly-type low pressure (LP) stop and control valves less than 1500 millisecond.

Four HP control valves are arranged one in each of four HP admission pipe to the Turbine. An independent two position (open-close) HP stop valve is located immediately upstream of the HP control valve. The four HP stop valves and four HP control valves are in series forming a one-out-of-two system in each of the four admission lines. No interconnection between the four separate HP stop valves and control valves is installed for throttle controlled operation with full-arc admission of the HP turbine.

Each admission pipe to an LP turbine is equipped with a butterfly-type stop valve and a butterfly-type control valve in series forming one-out-of-two system in each of the admission pipe. No interconnection between these admission valves is installed.

Overspeed is limited by rapid closure of the turbine control or stop valves whenever turbine power exceeds generator output, as would exist immediately after a load reduction. The normal control system is designed to limit transient overspeed to less than 110% of rated speed after a full load rejection by closing both the HP and LP control valves. The control system consists of a redundant electrohydraulic control (EHC) system and built-in load rejection logic, as described in section 10.2.2.7.1.

In the unlikely event that the EHC control system should fail, the HW and SW overspeed systems (described in 10.2.2.7.3) will act to limit the overspeed to less than 120% of rated speed. The two independent and diverse electronic trips meet the intent of SRP 10.2 Part III, for redundance and independence. An analysis of the electronic overspeed systems is included in an attachment to the “Probability of Turbine Missiles” provided in reference 2. Either the HW or SW overspeed system will de-energize the three trip solenoids on the turbine trip block, draining turbine trip fluid and resulting in rapid closure of all turbine stop and control valves. Only one of the two systems is needed to trip the turbine, with each system utilizing 2-of-3 trip logic.

The turbine overspeed protection system thus consists of three separate and independent sub-systems, any one of which is capable of limiting the overspeed to within acceptable limits.

Therefore, the failure of any one turbine stop or control valve will have no effect on the overspeed protection system because the valves are arranged in series.

The extraction line isolation valves are located close to the turbine to minimize the volume of trapped steam.

The turbine trip signal provides the closing signal for the controlled extraction steam power assisted non-return check valves in the extraction steam piping to heater Nos. 1, 2, 3, and 4. This protects the unit against excessive overspeed due to reverse flow of steam from the extraction system through the turbine. The turbine trip signal will vent the air from the actuator cylinder of these non-return check valves and the spring loaded piston will supply the energy to close the valve within one (1) second to prevent overspeed of the turbine.

Heaters 5 and 6 are located in the condenser neck with the channel end extended out of the condenser. Non-return check valves for these heaters are not provided because: 1) Anti-flash baffles are provided to restrict the reverse flow from these heaters to a sufficiently low flow so

that it cannot adversely affect turbine overspeed, and 2) These lines do not contain enough available stored energy to overspeed the turbine.

10.2.2.7.6 Turbine Valve Testing

Turbine valve testing will be performed on all turbine valves on periodic intervals consistent with the Preventative Maintenance Program. Testing will require each turbine valve to be cycled to demonstrate free operation as the valves close and reopen. Testing will be conducted by an operator verifying valve operation by direct observation.

10.2.2.8 Other Equipment and Features

In addition to the electronic overspeed trips, the turbine is equipped with a manual trip pushbutton at the front of the turbine, a remote pushbutton, two low-vacuum trip systems, an MSR high level trip device, a trip for thrust bearing failure, and a low lube oil pressure trip device. Additional protective features include atmospheric relief diaphragms in the low-pressure turbine casings, a water spray system to avoid excessive exhaust temperature, and alarms for high lube oil temperature, low condenser vacuum, and low control fluid pressure.

A turbine control system supervises the turbine speed and control valve position, bearing and shaft vibration, shaft and casing expansions, shaft eccentricity, and turbine and bearing metal temperatures.

Operating and Monitoring (OM) computer workstations are supplied in the Control Room. These workstations provide indication and alarm as well as computer screen-based controls of the EHC, Seal Steam Control (SSC), Turbine Stress Evaluator (TSE), Generator Temperature Control (GTC), Leakage Water Return Control (LRC), and (MSR) Heating Steam Control Systems, and an automatic turbine tester (ATT).

Thermal insulating material is furnished, and sound-absorbing metal appearance lagging is supplied for enclosing the high-pressure turbine, including the main stop and control valves.

10.2.2.9 Turbine Protective Devices and Alarms

Turbine protective devices and Control Room annunciations are provided as follows:

1. Thrust bearing failure detectors for trip and alarm

2. Atmospheric relief diaphragm mounted in each low-pressure turbine outer casing

3. Water spray system to avoid excessive exhaust temperature

4. Extraction line check valves to protect the turbine from overspeed as a result of reverse flow after a load rejection turbine trip; these check valves also minimize the possibility of water induction because of high water levels in the heaters.

5. Alarm for high lubricating oil temperature

6. Low lubricating oil pressure trip and alarm

7. Alarm for high low-pressure turbine casing temperature

8. High vibration alarm

9. Low-pressure turbine exhaust low vacuum trip and alarm

10. MSR high water level trip and alarm

11. TSE failure

12. EHC failure

13. SSC failure

14. High Moisture in Generator Gas

10.2.2.10 Turbine Trips

Turbine protective trips cause closing of all the turbine admission valves and initiate closure signals to the swing-type turbine extraction check valves. After the closing of turbine stop valves or release of control fluid pressure, a signal is generated for initiating the reactor trip (if greater than 50% reactor power), the steam dump control, and the electrical generator trip. The reactor trip signals are initiated by instrumentation which provides input to the reactor protection system.

Turbine trip is initiated by any of the following:

1. Reactor trip

2. Steam generator high-high level or safety injection

3. Overspeed (two channels)

4. Generator trip

5. Condenser low vacuum (one mechanical and two redundant electrical)

6. Manual turbine generator trip (from Control Room)

7. Manual turbine generator trip (at turbine)

8. Thrust bearing failure

9. Low bearing oil pressure

10. Moisture separator high level (each MSR)

11. Loss of hydraulic control fluid pressure

Electrical generator trip is initiated by the following:

1. All of the turbine trip signals, including loss of lube oil pressure and thrust-bearing failure

2. Generator protection signals, including signals from the main and unit auxiliary transformers and the 345 kV Switchyard breakers.

3. Manual generator trip from Control Room

4. Electronic generator protection (EGP), including signals from primary water flows, primary water temperature, and primary water tank level.

5. Rotor ground protection

6. Pilot exciter short

After an automatic turbine trip, the generator breaker trip is delayed to furnish uninterrupted power to the reactor coolant pump motors for at least 30 sec without relying on the success of a bus transfer, provided the generator conditions permit this. Likewise, following a manual turbine trip, or a turbine trip due to certain turbine faults or certain generator protection signals, the generator trip is delayed approximately 11.5 seconds. Generator trip (after a turbine trip) is also conditional upon detection of reverse power, except for certain generator faults, to minimize the probability and the degree of overspeed after a turbine trip. The trip logic is shown functionally on Figure 7.2-1, Sheet 16 and Figure 10.2-1.

For a narrative and schematics, which describe in detail the sequence of events in a turbine trip, see Reference 3.

For response times the following is provided. These times assume normal control mode of operation which is electrohydraulic control (EHC).

Approx. Time Action (Milliseconds)	
0	Load rejection
10	Load rejection is sensed by the electrical control system.
20	Output signal of the valve lift controller starts to decrease.
40	Piston of the electr. Hydr. Converter which equals actual lift value starts to move into (valve) closed position.
50	Secondary fluid pressure for HP control valves starts to decrease
80	Secondary fluid pressure for LP control valves starts to decrease.
150	HP control valves start to close.
250	LP Control valves start to close.
300	HP control valves closed.

1250 LP control valves closed.

10.2.2.11 Turbine Supervisory Instrumentation

Turbine supervisory instrumentation is provided for the monitoring of the following conditions:

1. Shaft eccentricity
2. Bearing and shaft vibration
3. Shaft and casing expansions
4. Control valve position
5. Turbine speed
6. Turbine metal temperature
7. Bearing metal temperature
8. Generator hydrogen gas and stator cooling water temperature
9. Exhaust hood temperature
10. Condenser vacuum
11. Stator winding temperature
12. Turbine stress evaluator indicator
13. Temperature load allowance turbine stress evaluator

10.2.2.12 Safety Considerations

As described in Section 7.2.1.1.2 the turbine generator hydraulic pressure switches and the stop valve limit switches are part of the anticipatory reactor-trip-on-turbine-trip channels which provide turbine tripped signals to the Reactor Protections System.

Failure in a high or moderate energy piping or failure of the connection from the low pressure turbine to condenser could damage components of the anticipatory reactor trip channels because no protection is provided for the above mentioned switches. The safety analysis does not take any credit for the anticipatory reactor trip as described in Section 15.2. Consequently, there are no safety implications if the switches are damaged as a result of a pipe break.

10.2.3 TURBINE DISK INTEGRITY

This section provides information demonstrating the integrity of the turbine disks and rotor, as failure of these components can result in turbine high-energy missiles as discussed in Section 3.5.1.3. As described in Reference [10], for the purposes of missile formation the most

hazardous missiles are generated from the low pressure turbine. The potential missile energy from the high pressure turbine is less than that from the low pressure turbine because of its much smaller potential missile mass and thicker turbine casing. Therefore, the high pressure turbine potential missiles are bounded by the low pressure turbine potential missiles.

10.2.3.1 Materials Selection

Each two-flow, low-pressure turbine rotor is made from a stepped shaft with a total of eight shrunk-on blade disks arranged in symmetrical groups of four. The material for disks 1, 2, 3, and 4 has the German standard designation of 26 NiCrMoV 145, which is 3.5-percent nickel-alloy steel similar to ASTM A471.

The percent nominal chemical compositions are as follows:

Item	26 NiCrMov 145
Nickel (Ni)	3.5
Chromium (Cr)	1.5
Carbon (C)	0.26
Manganese (Mn)	<0.40
Vanadium (V)	0.15 (max.)
Molybdenum (Mo)	0.4

The mechanical properties (maximum value for tensile strength, minimum for all other values) at 68°F (20°C) are as follows:

Item	Disc 1	Disc 2	Disc 3	Disc 4
Tensile Strength, Ksi (Mpa) 0.2 percent offset	<147 (<1010)	<141 (<970)	<154 (<1060)	<154 (<1060)
Yield Strength, Ksi (Mpa)	113-123 (780-850)	109-119 (750-820)	119-129 (820-890)	119-129 (820-890)
Elongation (L/d=5), Percent	>15	>15	>15	>15
Reduction of Area, Percent	>50	>50	>50	>50
mpact Strength, ft-lb (J) (Average of three Charpy V-Notch Specimens)	>100 (>130)	>100 (>130)	>100 (>130)	>100 (>130)

Item	Disc 1	Disc 2	Disc 3	Disc 4
FATT, °F (°C)	<-110 (<-80)	<-110 (<-80)	<-110 (<-80)	<-110 (<-80)

The compressive residual stress level of the heat-treated disk forgins, as measured by the ring core process, shall not be lower than 14,504 psi (100 Mpa) and higher than 36,260 psi (250 Mpa) in the middle of the hub area; shall not be lower than 29,007 psi (200 Mpa) and higher than 58,015 psi (400 Mpa) in the area of the rim of the hub; shall not be lower than 14,504 psi (100 Mpa) and higher than 36,260 psi (250 Mpa) in the rim area. Residual tensile stresses are not permitted.

The maximum disk stress at the shrink fit is required to remain less than 50 - 60% of the yield strength of the material at nominal conditions. Disk failure in this speed range can only occur if serious material defects exist or if a major design or manufacturing error is made.

The disk forgings are produced from vacuum-degassed alloy steel and are heat treated for an optimum combination of high fracture-toughness throughout the disk volume and for calibrated compressive residual stresses at the hub bore surface.

10.2.3.2 Fracture Toughness

The safety analysis of each disk design is based on the principles of linear elastic fracture mechanics (LEFM). The methodology used is described in Reference 10.

10.2.3.3 High-Temperature Properties

As described in Subsection 10.2.3, the high-pressure rotor cannot generate missiles; consequently, no discussion of the stress-rupture properties need be presented.

10.2.3.4 Turbine Disk Design

The turbine generator is designed and tested for safe operation up to 120 percent of rated speed, including operating speed and all maximum overspeed excursions that can occur during normal operation of the unit. In this speed range, rotor failure can only occur as a result of material defect or a major error in the design or manufacture of the rotor.

10.2.3.5 Preservice Inspection

Each low-pressure disk is subjected to comprehensive design, manufacture, and quality assurance to ensure its reliability throughout the life of the turbine-generator unit as discussed in Reference [2].

The low-pressure disk forgings are produced from vacuum-degassed alloy steel and are heat treated for an optimum combination of high fracture-toughness (throughout the disk volume) and calibrated compressive residual stresses at the hub bore surface. Each disk forging is examined by ultrasonic testing as follows:

1. In the rough as-forged condition before heat treatment

2. Premachined with contours prior to heat treatment for mechanical properties

3. After heat treatment for mechanical properties

Ultrasonic testing equipment and techniques are in accordance with DIN Standards 54 120 and 54 122 and can detect and measure flaws as small as 0.04 in. (1 mm) in equivalent diameter.

The evaluation of ultrasonic inspection is based upon reported indications as follows:

1. Isolated single indications of an equivalent defect size of 0.2 in. (5 mm), in accordance with The Distant Gain Size diagram, and larger

2. All isolated indications causing a decrease of more than 10 percent of the back reflection

3. All indications of the linear type of the area type, as well as clustered indications, regardless of the size of the single indications in the cluster area

4. All indications of defects located within a 2-in. (50 mm) zone surrounding the axial center bore

Material samples are taken from each low-pressure turbine disk forging near the hub bore surface for the determination of the NDTT as a fracture-toughness criterion and of the 0.2-percent offset yield strength as a strength criterion. The results of these mechanical tests, in combination with the ultrasonic testing and residual stress measurement results, are decisive for the acceptance of the forging.

Before machining out the hub, the following test results of the fully heat-treated forging are obtained and scrutinized:

1. Results of ultrasonic test, covering 100 percent of the disk volume, including documentation of all indications required

2. Actual chemical composition of the forging material at specified locations

3. Tensile and drop-weight (NDTT) test results at a specified location

After machining to the dimensions of the order drawing, the bore of the forging is magnetic-particle-tested. In addition, residual stresses are measured by the ring core method described in Reference [2].

By careful control of heat treatment, desirable residual stress characteristics can be built into the disks. These characteristics are verified by measurements at six specified points, before and after machining, for the first of a batch of similar disk forgings. Subsequent disks of the same batch are checked at only one point.

The low-pressure turbine rotor disk and couplings are shrunk onto the shaft with the shaft in the horizontal position and with rotation of the rotor during the process. This procedure provides the following benefits:

1. Defined heat transfer from disk to rotor, even during the initial stage of the shrinking process when the clearance is still greater than zero, because the disk always rests with its weight on the shaft

2. Exact positioning of the disk by the axial compressing device during the initial stage of the shrinking process, when the clearance is still greater than zero and turning gear is in operation

3. Easy correction of rotor runout by stopping the turning gear for a calculable period of time during the initial stage of the shrinking process with the shaft in the correct position

4. Axial and radial runout checks during the entire shrinking process with the shaft in operating position

Before delivery, each completed, bladed, low-pressure rotor is balanced and subjected to an overspeed test at 125 percent of rated speed for 2 min, at a minimum temperature of 59°F. All other turbine and generator rotors are also subjected to a 120-percent overspeed test.

10.2.3.6 Inspection

The inspection program for the turbine generator consists of the following:

At least one high-pressure stop and control valve is dismantled to allow visual and surface examinations of the valve shaft, disc and stem. A visual inspection of the accessible portions of the valve shaft and disc for at least one low-pressure stop valve and one low-pressure control valve are inspected from within the associated piping. If an inspection uncovers unacceptable flaws such as cracks or excessive corrosion, further investigation of valves of that type shall be made per the Preventative Maintenance Program. Inspection of one valve of each type is performed per the Preventative Maintenance Program. All valves are inspected within inspection interval specified in the Preventative Maintenance Program.

The electronic overspeed systems are continually tested on line automatically or can be initiated manually.

The high-pressure stop and control valves and the low-pressure turbine valves are exercised per the Preventative Maintenance Program using the automatic turbine tester or manual test.

The turbine includes an inspection program to provide assurance that disk flaws (which could lead to brittle failures at design speeds) are detected. This inspection includes disassembly of the turbine at approximately 12-yr intervals during plant shutdowns. The program includes visual and surface examinations and the latest volumetric examinations as required. All normally inaccessible parts are inspected, including couplings, coupling bolts, shafts, low-pressure blading, low-pressure disks, and high-pressure rotors.

To preclude the possibility of a failure of a low pressure turbine due to cracking in the turbine disk, an ultrasonic inspection of the disk bore and keyway areas [4] is performed for each low pressure turbine at intervals not to exceed 100,000 operating hours [10].

Inplace visual examination of the turbine assembly at accessible locations is conducted during refueling shutdowns at intervals not exceeding three years.

10.2.4 EVALUATION

The turbine generator and its related steam-handling system is designed in accordance with design criteria based on latest experience with large light water reactor turbines and erected in a manner to eliminate leakage at all joints.

It is possible that activity in the secondary side of the steam generator will occur. This activity is a function of the tube leakage from the primary side of the steam generator.

The reactor coolant circulates from the reactor core, where it removes heat from the fuel elements, to the steam generators, and back to the core. In the steam generators, heat from the pressurized reactor coolant is transferred across metal tube walls to the secondary coolant to generate steam. The steam passes through the turbine(s), is condensed, and returns to the steam generators.

A small percentage of the fuel elements can exhibit small pinholes or cracks over the lifetime of the station, allowing the diffusion of radioactive fission products from the fuel into the reactor coolant. This is discussed in Section 11.1. The reactor coolant is continuously purified in the Chemical and Volume Control System (CVCS). The filters and demineralizers remove a large portion of the iodine, other fission products, and radioactive corrosion products.

A steam generator tube leak coincident with leakage from failed fuel rod is the only condition that could result in fission products entering the secondary coolant system. If a leak exists, some radioactive reactor coolant is transferred to the secondary system. In addition to the steam generator leak rate, the secondary coolant system activity is affected by moisture carryover, blowdown rates, and partition factors in the steam generator and condenser.

The activity in the reactor coolant is a function of factors such as fuel defect level, system volumes, purification flow rates, and CVCS removal factors.

Once a determination of these variables is made, the secondary side activity is calculable as a function of this leak rate. The SGBS (described in Section 10.4.8) and water chemistry (described in Section 10.3.5) limit this activity by acting in conjunction to control the chemical composition of the steam generator secondary-side water. This control is accomplished by a combination of varied blowdown rates and chemical feed, which result in reduced admission of iodine gas into the turbine generator.

The hypothetical turbine overspeed failure is described in reference 2 and 3 Section 2. The reference 3 report expands on the function of the turbine generator speed control system based on diagrams and drawings. The report also defines the response of the various speed control systems during various possible failure modes. Valve closing and test intervals are defined.

The results of a failure mode and the effect analysis for each overspeed protection system is provided in reference 2.

In the event of a high or moderate energy piping failure, it could be possible that the turbine could lose the electrical speed control system. Assuming this event were to occur with a control fluid piping failure, a failure of the electrical speed control system would have no impact on turbine speed because the loss of control fluid pressure would result in closure of the turbine stop and control valves. The turbine will be protected from overspeed by two independent and diverse

channels, trip redundant, failsafe electronic overspeed systems. Any signal failure from any two speed channels or power failure to any two trip solenoids due to piping failure will also trip the turbine.

The system design includes pneumatically operated positive closing non-return check valves to provide protection against the possibilities of overspeed of the turbine. The valves are automatically closed by a turbine trip within one second from the initiation of the reverse flow. The valves are air piston operated and close by venting air.

Heaters 5 and 6 (see Figure 10.4-13) are not provided with positive closing check valves since the extraction steam lines do not contain enough available energy to contribute overspeed of the turbine.

Refer to FSAR Section 10.4.10 for a description of the function of these check valves.

The activity concentrations in the area of the turbine generator are ascertained to be low enough to classify the area as Zone I, as defined in Section 12.1.1. No shielding or controlled access is required.

Further discussion of releases to the Turbine Building and to the environment is given in Section 11.1.

REFERENCES

1. Allis-Chalmer's Power Systems, Inc., Engineering Report No. ER-503, Turbine Missile Analysis for 1800 rpm Nuclear Steam Turbine-Generators with 44-inch Last Stage Blades, July 1975.

2. Engineering Report No. ER-504, Probability of Turbine Missiles from 1800 r/min Nuclear Steam Turbine-Generators with 44-inch Last Stage Blades October 1975.

3. Allis-Chalmer's Power Systems, Inc., Engineering Report No. ER-601, Speed Control of 1800 rpm Steam Turbine - Generators for Light Water Reactor Applications, May, 1976.

4. Utility-Power Corporation Engineering Report ER-8401, "Ultrasonic Disk Inspection", August, 1984.

5. Utility-Power Corporation Engineering Report ER-8402, "Probability of Disk Cracking Due To Stress Corrosion", August, 1984.

6. Utility-Power Corporation Engineering Report ER-8102, Revision 1, "Critical Crack Sizes of Comanche Peak LP Turbine Disks", November, 1981.

7. Deleted.

8. Deleted.

9. Deleted.

10. CT-27331, Revision 8, “Missile Probability Analysis Methodology for Luminant Generation Company LLC, Comanche Peak Units 1 and 2 with Siemens Retrofit Turbines”, August 23, 2016.

11. TP-03143, “Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC,” July 31, 2003.

10.3 MAIN STEAM SUPPLY SYSTEM

10.3.1 DESIGN BASES

The main steam supply system is considered part of the Main Steam, Reheat, and Steam Dump System. It conveys steam from the outlet of the steam generators to the various system components throughout the Turbine Building. The steam is primarily used for driving the main turbine and for heating service in the MSRs. In addition, it is used for various auxiliary services such as the following:

1. Steam generator feed pump turbine

2. Auxiliary feed pump turbine

3. Steam dump system

4. Turbine shaft seal system

5. Plant process steam system

Steam from the outlet of the four steam generators (a total of 15,140,016 lb/hr, at 1000 psia and 0.25 percent moisture for the original plant design conditions) is routed in the main steam piping to the turbine. The piping is sized for a pressure drop of less than 25 psi. The design pressure-temperature rating of the main steam piping is 1200 psig at 650°F. The full-load nominal original design steam flow from the reheaters to the feedwater pump turbines is approximately 242,472 lb/hr at 152.1 psia. The steam supply from the main steam lines to the auxiliary feedwater pump turbine is in the range of 60,297 to 9050 lb/hr at 1236 to 100 psia steam generator pressures, respectively.

The main steam supply system is used for plant cooldown. This is achieved by progressively lowering the pressure of the steam generators; the decay heat and the sensible heat are removed by the generation of steam. When steam generator pressure has been reduced to 100 psia, the Residual Heat Removal (RHR) System is placed in operation.

The main steam supply system is designed to meet all applicable requirements of 10 CFR Part 50. This system is designed in accordance with NRC Regulatory Guides 1.26 and 1.29.

The main steam safety valves are rated to pass 105 percent of the engineered safeguard design (ESD) steam flow at a pressure not exceeding 110 percent of the main steam system design pressure (ESD flow equals 104.5 percent of the original guaranteed 3425 MWt). These safety valves prevent overpressurization of the system as specified in the ASME B&PV Code.

The steam generator atmospheric relief valves pass at least 10 percent of the steam flow used for the plant design at no-load steam pressure (1107 psia).

The main steam isolation valves (MSIVs) are designed to stop flow from either direction within five sec after receipt of signal to close to prevent uncontrolled steam release from more than one steam generator.

The steam dump system design bases are given in Section 10.4.4.

The main steam lines from the steam generator, out through the Containment, and up to the first moment restraint beyond the MSIV, are designed in accordance with the ASME B&PV Code, Section III, Code Class 2. Beyond this moment restraint, the main steam lines to the main turbine and the lines to the feedwater pump turbines, the condensers (steam dump), and the reheaters are non-nuclear-safety-related and are designed in accordance with ANSI B31.1, Power Piping.

The lines to the auxiliary feedwater pump turbine are designed in accordance with the ASME B&PV Code, Section III, Code Class 3.

The MSIVs, integral bypass valves, and bypass piping are of Code Class 1 design (The applicant has optionally upgraded this equipment to Code Class 1.) The safety valves, and atmospheric relief valves are of Code Class 2 design. The energy storing components for the MSIV operators are not within the scope of ASME B&PV Code, section III, subarticles NA-1120 & NA-1130. However, these components for the operators will comply with ASME B&PV code section VIII.

The piping is insulated, and where exposed to the outdoor environment, it is suitably protected from the weather.

The steam generator shell and lines which emanate from the steam generator shell side are barriers against release of containment atmosphere during and after a LOCA.

Steam is conveyed from the steam generators to the main turbine by four steam lines. Upstream from the MSIVs, each line is provided with five spring-loaded safety valves and one atmospheric relief valve. It is essential that the heat load be evenly shared between the four loops of the Nuclear Steam Supply System (NSSS). To achieve this, the four main steam lines are interconnected by a pressure equalizing header downstream of the steam generator isolation valves.

The sizes and routing of the main steam piping are such that under design flow, a pressure drop of approximately 22 psi is obtained between the steam generator nozzle and the turbine stop valve inlet with a steam velocity of approximately 115 ft/sec. Each steam line uses a drain system to remove accumulated condensate from the line.

To ensure steam supply to the auxiliary feedwater pump turbine (even in the steam generator isolation), two separate steam supply lines are provided from two main steam lines.

A direct connection is provided downstream of the isolation valves for the startup of the steam generator feedwater pump turbines.

Table 10.3-1 shows the design bases of the main steam piping.

The environmental design bases are given in Section 3.11.

The inservice inspection requirements are given in Section 6.6.

The inservice inspection program for the turbine generator is described in Section 10.2.3.5.

10.3.2 DESCRIPTION

The main steam supply system and interconnected piping are shown schematically on Figure 10.3-1. The nuclear safety class is applied to the main steam lines from the steam generator, up to and including the first moment restraint beyond the MSIVs located outside the Containment. The nuclear safety class is also applied to the blowdown and process sampling lines from the steam generator, up to and including the pneumatically operated isolation valves. (See Figure 10.3-1, sheet 1 of 2.) The non-nuclear-safety class portion includes the remaining portion downstream of the moment restraint. (See Figure 10.3-1, sheet 2 of 2.)

Each main steam line has a number of branch-off lines located downstream of the main steam isolation valve. All these branch-off lines are provided with steam shut-off valves as listed in Table 10.3-11.

10.3.2.1 Safety Valves

The main steam line from each steam generator is provided with five ASME B&PV code-certified, spring-loaded, double-outlet-designed safety valves (see Figure 10.3-1) to protect against overpressurization of the main steam supply system. These valves are designed to pass a total rated relieving capacity sufficient to prevent a pressure rise greater than 10 percent above system design pressure under any anticipated pressure transients.

The safety valves are installed outside the Containment, downstream of the power-operated relief valves, and upstream from the MSIVs; they discharge to the atmosphere through 18-in. dual-outlet-designed umbrella-type vent stacks.

To avoid lifting during pressure transients, set pressures for safety valves are as high as possible within the requirements of the codes. To prevent chattering during operation of the safety valves, the individual valves in each steam generator bank are set at a different pressure.

The lowest safety valve setting is 1185 psig, which is the design pressure of the main steam supply system. The highest safety valve setting is 1235 psig, which is 105 percent of the lowest safety valve set pressure minus the 10-psi piping pressure loss between the steam generator nozzle and the safety valve.

The resultant total relieving capacity of the safety valves exceeds the maximum steam generator rated flow. With a 3-percent pressure accumulation when the safety valves are operating, the maximum pressure while relieving is 103 percent of the highest safety valve set pressure (1235 x 1.03 + 15 = 1287.5 psia). This prevents the pressure from exceeding a 110-percent system design pressure (Article NC 7000 of ASME B&PV Code, Section III).

The actual capacity of any one safety valve does not exceed the flow rate specified by the steam generator manufacturer at the design pressure of the main steam supply system; this limits excessive core reactivity insertion.

Table 10.3-2 shows the design bases of the main steam safety valves.

Mounting of the exhaust annulus piping surrounding the safety valve exhaust line allows for vertical and horizontal displacement of the main steam piping caused by thermal expansion. The

arrangement of the steam discharge line allows for piping growth to prevent contact with the safety valve body.

Adequate provisions are made in the steam piping for the installation and support of the safety valves, with consideration being given to static and dynamic loads when operating and when subject to seismic shock.

10.3.2.2 Atmospheric Relief Valves

An atmospheric relief valve (ARV) is mounted adjacent to the five spring-loaded safety valves in the steam outlet piping from each steam generator. The atmospheric relief valve is of the modulating type, air operated, and fail closed, with a maximum full stroke time of 20 seconds in the automatic mode. The valve is set to open at approximately 1125 psig, 60 psi below the lowest safety valve set point. The relief valves discharge to the atmosphere and are used to avoid, wherever possible, the opening of the lowest set safety valves.

The atmospheric relief valves provide means for removal of heat from the NSSS to the atmosphere during periods when the condenser is not in service. They are sized to remove the reactor decay heat generated following a reactor shutdown. The atmospheric valves modulate open to relieve excessive pressure in the main steam lines. The pressure control set point can be adjusted with a control board hand auto station. These valves can be manually opened or closed at the valve by using the valve handwheel, or they can be remotely opened or closed by using the control board hand auto station. In the manual control mode from the main control room or hot shutdown panel, the ARVs are capable of opening/closing maximum full stroke within 40 seconds. The ARVs are required to have the capability to be operated remotely from the control room following a safe shutdown earthquake coincident with the loss of offsite power.

In Unit 1 the Westinghouse steam generators model D-4 have been replaced with a newer model. Two new solenoid valves are installed for each ARV. These solenoid valves are supplied from the opposite train power supply. These solenoid valves are operated from a key locked hand switch in the main control room to open the ARVs.

During a Unit 1 steam generator fault, if normal DC power is lost to the Train A or B buses, the ARVs can still be opened if required from the opposite train power to cool and depressurize the reactor and terminate primary to secondary leak.

Since the CPNPP Instrument Air System is not safety related, safety-related air accumulators are provided to operate the ARV's. The valves fail closed on loss of air or electric signal. Valve positions, open or closed, are indicated with control board lights.

Failure of the relief valves to open causes the system pressure to rise to the set point of the first safety valve, which would then open, preventing further pressurization of the system. The atmospheric relief valves do not provide main steam supply system overpressure protection. This overpressure protection is provided entirely by the safety valve system described previously.

The capacity of each Atmospheric Relief Valve is required to be consistent with the following requirements:

1. Sufficiently large to allow the plant to be cooled from no-load temperature to the RHR cut-in temperature of 350° F prior to the time that the condensate storage tank is exhausted.

2. Less than the maximum allowable relief capacity of a main steam safety valve.

3. Sufficiently large to allow the plant to be cooled and depressurized so as to terminate the primary-to-secondary break flow following a steam generator tube rupture event prior to the time the affected steam generator completely fills with liquid.

4. Sufficiently small such that the calculated radiological offsite dose consequences of a steam generator tube rupture are within the guidelines of 10CFR100.

The valves discharge to the atmosphere and are designed to operate over the steam pressure range of 100 to 1300 psia. Each valve inlet pipe is provided with one manual isolation valve for maintenance.

Table 10.3-3 shows the design bases of the atmospheric relief valves.

10.3.2.3 Main Steam Isolation Valves

10.3.2.3.1 General

Each main steam line is provided with a quick-acting isolation valve, and is designed to stop flow from either direction after a steam line break (five sec after receiving the closing signal) to prevent uncontrolled steam release from more than one steam generator. The valves are installed outside the Containment, downstream of the safety valves, and are provided with a manual 4-in. bypass valve for warming the system and equalizing the pressure across the isolation valve. The bypass valve is locked closed during power operation. The MSIVs can be opened manually by the operator in the Control Room without opening the bypass valve.

Each MSIV is provided with a two-train module, three-position control switch mounted on the main control board. The switch has an electrical two-train module so that valves can be closed even if one train fails. The three switch positions are close, auto, and open, with spring return to auto position. Each MSIV also has a two-train module test switch to enable a valve to be closed to a 10 percent-closed position when tested. In addition to these control board mounted switches, there is a trip switch for each of the two trains, either of which can be used to trip all four MSIVs simultaneously. Trip switch positions are close, auto, and reset, with spring return to auto.

The MSIV's are automatically closed on high-high containment pressure or steamline break protection logic (as indicated by high steam pressure rate or low steamline pressure). High steam pressure rate is only effective when steamline SI is manually blocked during startup and cooldown, and low steamline pressure is only effective when the block is removed (see Figure 7.2-1 sheet 7). The MSIV's are closed by operation of the MSIV valve actuators. The actuator is, in effect, a hydraulic cylinder coupled directly to a nitrogen-accumulator. The accumulator is designed as a chamber concentric to the hydraulic cylinder, and it stores the energy required for closing the MSIV in the form of compressed nitrogen gas. Because the accumulator is an integral part of the cylinder, the loss of any external manifolding or system elements will not prevent the actuator from closing the valve. A hydraulic control system which

maintains hydraulic fluid below the valve actuator piston is utilized to regulate valve closure velocity. Extension of the actuator to close the MSIV is accomplished by redundant 1E electric signals which operate two solenoid valves in the hydraulic control system portion of the actuator. These valves permit the hydraulic fluid below the actuator piston to flow into a hydraulic reservoir at a controlled rate as the compressed nitrogen extends the actuator to close the MSIV.

Each component of the hydraulic control system whose presence or function is required to effect the fail-safe extension of the actuator is redundant with a second component capable of performing the required function regardless of the state of operation or failure of the other. Two hydraulic control system manifolds are provided, each of which is capable of providing valve closure capability independently of the other. The MSIV fails closed on a loss of hydraulic fluid.

The initiation and control of main steam isolation is redundant and electrically and physically separated. There is no single failure in the initiation and control portions of the system that will prevent a “main steam isolation” signal from arriving at its destination.

A main steam isolation signal will close the MSIVs.

A main steam isolation signal will also isolate the flow from the drain pots upstream of the MSIVs associated with each steam line. These valves have two train inputs so that valves can be closed even if one train fails and also fail closed on either loss at electric signal or air failure. (See Section 10.3.2.7).

Following a main steam isolation signal, the main turbine will trip on either redundant reactor trip or turbine trip signals, and the closed turbine stop valves will serve as a backup to the MSIVs. Thus, in the event of a failure of a MSIV to close, main steam will still be isolated. There will be an insignificant loss of steam through the drain pots downstream of the MSIVs and an additional loss as the steam dumps to the condenser open to relieve a steam pressure spike after a turbine trip, but the steam generator will not blow down even with a MSIV that failed to close. (See Section 15.1.5)

(For a description of initiation logic, see Figure 7.2-1 Sheet 8 and 7.3.2.4.2).

Each MSIV can be tested with its own control board switch (two-train module, two-position test, or normal with position maintained). If the valve switch is in the test position and the valve does not reach 10-percent closure after a time delay, an alarm is actuated in the Control Room. When the valve is tested, it closes slowly by energizing a test solenoid in the hydraulic circuit as well as energizing the trip-close solenoid. An automatic MSIV trip-close signal overrides the test signal and closes the valve quickly.

Alarms are actuated when MSIVs have low hydraulic oil pressure or low actuator gas pressure. Each valve has position-indication lights on the main control board for open, closed, and test positions. There are monitor lights which light on valve-closed position.

The automatically operated MSIVs serve only a safety function and are not required for power operation. They are required to limit uncontrolled flow of steam from the steam generators in the event of a break in the piping system. These valves operate under the following situations:

1. Break in the Steam Line from One Steam Generator Inside the Containment Building

If the break is within the Containment, steam is discharged into the Containment. The other steam generators act to feed steam through the interconnecting header into the broken line and then into the Containment. A steam line break results in a significant pressure rise in the Containment so that reverse flow protection is necessary to prevent discharge of more than one steam generator. According to calculations, reverse flow must be interrupted to limit the Containment pressure rise to an amount below design pressure. To achieve this, the automatic isolating valves close within five sec from receipt of the initiating signal. Closure of these valves allows for a single failure of an active component.

2. Break in the Steam Line Outside Containment Building and Upstream from the Isolation Valve

In this case, Containment Building pressurization is not a concern. However, the uncontrolled blowdown of more than one steam generator must be prevented. The 5-sec valve closure time established previously satisfies the requirements for this situation.

3. Break in the Steam Line or Header Downstream of the Isolation Valve

The closure time established previously meets the requirements for this situation.

4. Steam Generator Tube Rupture

In this case, a fast-acting valve closure is not required. The isolation valves limit primary coolant leakage during shutdown by isolating the damaged steam generator after the primary system pressure is reduced below the steam generator shell-side design pressure.

Opening the MSIVs in Mode 5 following an outage and maintaining them open throughout Mode 4 and into Mode 3, to warm up the steamlines up to the turbine stop valves is acceptable. The MSIV Bypass valves may be used as described in Section 10.3.2.4, below, if the MSIVs are closed during heatup.

10.3.2.3.2 MSIV Design Requirements

1. Analysis of Accident Conditions

The MSIVs are designed to withstand the conditions created by a large steam line break on either side of them. The valves are designed to withstand the effects of high mass flow rate, moisture carryover, and high fluid velocity. These valves have a controlled speed of closure. The surge pressure caused by the dynamic effect of valve closure does not affect the pressure boundary, in accordance with the calculation of ASME B&PV Code, Section III. The design and construction of the MSIVs ensure positive closure against flow in either direction under the following conditions:

a. Pipe Break - High Moisture Content (Outside the Containment)

Flow conditions are based on the relief area downstream of the MSIV that results in a maximum mass flow rate and a maximum moisture content (approaching 96 percent).

The following tabulated mass flow rate is a function of the flow restrictor throat diameters and the steam generator no-load pressure (reservoir pressure at initiation of transient):

Flow Conditions

Steam generator outlet no-load pressure, psia	1107
Static pressure at valve minimum flow area, atmospheric pressure downstream (preceding pressure rise resulting from valve closing), psi	1090
Mass flow rate through valve preceding valve closure, lb/sec	10,700

b. Pipe Break - Low Moisture Content (Outside the Containment)

Flow conditions are based on a relief area downstream of the MSIV that results in a mass flow rate approximately 300 to 400 percent of the maximum flow rate at normal operating conditions, i.e., low moisture content and high velocity.

The mass flow rate tabulated in this paragraph is a function of the flow restrictor throat diameters and the steam generator no-load pressure (reservoir pressure at initiation of transient). However, the mass flow rate to the relief opening during the first milliseconds (<250 milliseconds) of the transient (i.e., the steam contained between the flow restrictor and valve) is 2400 lb/sec per ft^2.

When this limited quantity of steam has been expelled, the flow rate decreases to the following values as a result of limitations by the flow restrictors:

Flow Conditions

Steam generator outlet no load pressure, psia	1107
Static pressure at valve minimum flow area, atmospheric pressure downstream (preceding pressure rise as a result of valve closing), psia	1090
Mass flow rate through valve preceding valve closure, lb/sec	3350

c. Pipe Break - Reverse Flow, Low Moisture (Inside the Containment)

Initial flow through the relief opening from the piping system consists of dry steam (assumed 100-percent quality) contained in the piping system downstream of the flow restrictors.

Flow conditions are based on a large relief opening area upstream from the MSIV which results in a large reverse steam flow from the interconnected main steam lines. This reverse steam flow results in a mass flow rate approximately 10 times

the maximum flow rate at normal operating conditions, low moisture content, and greatly increased velocity.

Flow Conditions

Steam generator outlet no-load pressure, psia	1107
Static pressure at valve minimum flow area, atmospheric pressure downstream (preceding pressure rise resulting from valve closing), psia	620
Mass flow rate through valve preceding valve closure, lb/sec	11,000

d. Pipe Break - Reverse Flow, High Moisture (Inside the Containment)

After the dry steam has been exhausted, low-quality steam (4-percent steam quality) is discharged by the unfaulted steam generator(s), through its (their) associated flow restrictors and interconnected main steam piping, to the relief opening. This flow results in maximum mass flow rate and maximum moisture content; velocity is approximately three times the normal operating conditions.

Flow Conditions

Steam generator outlet no-load pressure, psia	1107
Static pressure at valve minimum flow area, atmospheric pressure downstream (preceding pressure rise resulting from valve closing), psia	680
Mass flow rate through valve preceding valve closure, lb/sec	32,100

2. Testing

The MSIVs are designed and shop tested in accordance with ASME B&PV Code, Section III.

a. Shell Hydrostatic Testing

Each valve shell is hydrostatically pressure-tested at 2250 psi in accordance with ASME B&PV Code, Section III, table NB3531-9 (for a 600 lb welded end valve).

b. Disc Hydrostatic Test

Each valve disc is hydrostatically pressure-tested at 1500 psi in accordance with ASME B&PV Code, Section III, Para NB3531.2(c). Leakage under disc hydrostatic testing shall not exceed three cc/hr/in. of valve seat diameter.

c. Valve Operator Test

There is zero leakage from all accessible static seals and joints of the system. The test is conducted when the valve has reached the full extreme in the open and closed direction.

d. Closing Rate Test

The complete valve assembly is tested to ensure that the closing time is less than five sec.

3. Leakage

The valve disc and seat materials are such that valve wear does not increase the leakage rate after a minimum of 500 cycles under normal operating conditions.

4. Design Bases

Table 10.3-4 shows the design bases of the MSIVs.

10.3.2.4 Main Steam Isolation Bypass Valves

The MSIVs are provided with 4-in. bypass valves which are normally closed. If the bypass valves were open they would tend to negate the protection provided by the MSIVs. Therefore, all four bypass valves are locked closed during power operation. During startup, hot standby and hot shutdown one MSIV bypass valve may be opened provided the other three bypass valves are locked closed and their associated MSIVs are closed. Table 10.3-5 shows the design bases for the main steam isolation bypass valves.

10.3.2.5 Flow Restrictors

Each steam generator is provided with flow restrictors which are located inside the steam generator outlet nozzle. These restrictors (several venturis arranged in a bundle) limit the steam flow rate in the event of a steam line rupture. These restrictors also minimize the thrust force effects on the steam generator and piping system.

The design basis, description, and test and inspections are included in Section 5.4.4.

10.3.2.6 Auxiliary Feedwater Pump Turbine Steam Supply

A steam supply line is provided upstream from the isolation valve (see Figure 10.3-1), located in the steam outlet line from two steam generators, to supply motive steam to the turbine drive of the auxiliary feedwater pump. This line ensures a source of steam to the turbine-driven auxiliary feedwater pump when steam generators are isolated and are producing steam from reactor

decay heat. Each line is provided with a check valve for isolation in the event of a main steam line break.

Each line is also provided with a pneumatically operated, fail-open, steam supply valve of Safety Class 2 design at its junction with the main steam line. Safety-related air accumulator tanks are provided to permit closing these valves, to satisfy containment isolation or to isolate a depressurized steam generator. A locked-closed manual valve is provided in a warm-up bypass line around each of these steam supply valves. The bypass line can be used to pre-warm the steam supply lines. The remainder of the line is constructed to Safety Class 3 requirements.

10.3.2.7 Main Steam Line Drainage

Each main steam line upstream from the MSIVs is provided with a drain pot and drain piping and valves.

The drain lines which run from the drain pot up to and including the first pneumatically operated drain valves are Safety Class 2. These valves prevent steam generator blowdown by automatic closure upon receipt of a MSIV close signal.

Each valve can be manually opened or closed from a control board switch, which is a two-train module switch. Valves fail closed on electric signal failure or air failure. Indicating lights for valve positions and a monitor light for valve closed position are located on the control board.

10.3.3 EVALUATION

The portions of main steam lines from the steam generators, out through the Containment, and up to and including the first moment restraint beyond the MSIVs, are Safety Class 2.

The safety considerations related to a main steam line break are discussed in Section 3.6. Protective measures and criteria for separation of all portions of steam lines within and outside the Containment are described in Section 3.6.5.

The seismic design is discussed in Sections 3.2 and 3.7.

The attachment of the main steam piping to the steam generators takes into account the movement of, and the allowable forces and moments on, the nozzles, as specified by the steam generator manufacturer for all operating conditions.

The environmental design bases and inservice inspection requirements are given in Subsection 10.3.1.

The valve reaction forces on the piping and the sequential blowing effect of the steam safety valves and atmospheric relief valves, located in the main steam piping, are evaluated for the design.

The interfacing of the main steam lines with the main turbine stop/control valves is determined by taking into account the movement of, and the allowable forces and moments on, the nozzles, as specified by the turbine generator manufacturer.

For the loading combinations and design stress limits relating to the safety-class main steam piping, see Section 3.9.2.

Steam is conducted from each steam generator in a separate line through the Containment Building each line is anchored at the Containment Wall. The lines have the flexibility to absorb the thermal expansion. The main steam lines run from the Containment to the Turbine Building, passing through the highest level of the Safeguards Building. Each main steam line (with its associated isolation, safety, and relief valves) is located in a compartment (connected with accesses) which provides isolation from the rest of the building and separation from the other main steam lines.

10.3.4 INSPECTION AND TESTING REQUIREMENTS

Before placing the system into service, foreign material and oxides are removed from the piping. During cleaning, entry of any fluid into the steam generators is prevented. The main steam lines are hydrostatically tested to confirm leaktightness. The testing of Class 1 and 2 components conforms to the requirements of Articles NB 6000 and NC 6000, respectively, of the ASME B&PV Code, Section III.

The hydrostatic test pressures for the valve body (discharge side) of the Safety Class 2 flanged safety valves are 1.5 times the safety valve outlet discharge pressure. Pipeline expansion and movement from the cold condition to the hot normal operating condition is checked by measuring movements from field bench marks, such as steel columns or pipe supports, as specified on design isometric piping drawings indicating calculated movements along the x, y, and z axes.

10.3.4.1 Safety Valves

The main steam safety valves are individually tested during preoperational tests. Pressure gauges mounted in the steam piping indicate the actual values of opening and closing pressures of the valves. These values are compared to the design values.

10.3.4.2 Atmospheric Relief Valves

The set point of the main steam relief valves is adjusted during preoperational tests. The valves have the capability of being stroked at any time, either manually by handwheel or remotely with manual-auto station, with their isolation valves closed.

10.3.4.3 Main Steam Isolation Valves

Periodic in-plant tests are conducted to demonstrate the capability of the MSIVs to respond to a test close signal and close within the specified time. The valve design provides for regular in-service testing of partial valve stroke. A provision is made for all solenoid-operated valves that respond to a trip signal which is exercised without interrupting availability of the trip mechanism. The preoperational and periodic testing of the MSIVs is discussed in Section 14.2.12.1.

10.3.5 WATER CHEMISTRY

A Secondary Chemistry Program will be developed for CPNPP. The objectives of the program will be to:

1. Minimize corrosion of system materials.

2. Limit the accumulation of sludge in the steam generators.

3. Minimize scale formation on the heat transfer surfaces.

4. Minimize the potential for formation of free caustic.

5. Maintain the dissolved oxygen level to optimum levels.

6. Monitor for primary to secondary leaks.

7. Maintain Steam Generator tube integrity.

8. Minimize Turbine Deposits due to carry over and volatility from the Steam Generator.

These objectives will be achieved by careful chemistry control which will include a comprehensive sampling and analysis of the secondary plant systems.

The chemistry program will entail an all volatile treatment using amines for pH adjustment and oxygen control.

The control of solids and non-volatile radioactivity in the secondary side is detailed in Section 10.4.6, Condensate Cleanup System, and Section 10.4.8, Steam Generator Blowdown System. A description of the continuous monitors is obtained from Table 10.4-20, while flow diagrams trace the location of the sampling points of the secondary system and Section 9.3.2 describes the process sampling system.

To assure the implementation and proper performance of the Secondary Chemistry Program, appropriate procedures have been developed. These procedures are contained in the Chemistry/Radiochemistry Manual and are reviewed and approved as specified by plant procedures. Procedures for this program include:

1. Identification of a sampling schedule for the critical parameters and of control points for these parameters.

2. Identification of the procedures used to measure the value of the critical parameters.

3. Identification of process sampling points.

4. A description for the recording and management of data.

5. A description of actions to be taken for off-control point chemistry conditions.

6. Identification of the authority responsible for the interpretation of the data and the sequence and timing of administrative events required to initiate the corrective action.

The CPNPP secondary water chemistry program developed for plant startup was based on the specification of the NSSS vendor (Reference 1), EPRI Guidelines (Reference 2), and Branch Technical Position MTEB 5-3, Revision 1 (Reference 3).

Due to the complexity of the corrosion phenomena, the understanding and control of it continues to advance along with the state-of-the-art of the methods used to measure its critical parameters. It therefore is the policy of CPNPP to make changes in its Secondary Chemistry Program that reflect additional understanding of the corrosion phenomena and the changes in the state-of-the-art of its control and critical parameter measurements. The secondary chemistry water quality will meet the specifications of the EPRI guidelines, except where CPNPP evaluations justify exceptions. Changes to the EPRI guidelines will be evaluated for applicability at CPNPP.

In the event of a steam generator tube leak, additional amine can be fed to the steam generator to counter the effect of the borated primary coolant, thus allowing proper adjustment of the pH. Operation in the basic pH range causes the disproportionation of iodine in solution to iodate and iodide. This causes a decrease in the ratio of iodine in the gaseous phase to that in solution. Steam generator blowdown is not flashed but cooled in a heat exchanger and is then demineralized, as described in Section 10.4.8. In the condenser, as in the steam generator, operation at basic pH decreases the gaseous phase iodine. Thus, only minimal amounts of iodine are expelled through the condenser vacuum pumps.

Leakage detection and monitoring of water chemistry is accomplished by the process sampling system which is described in Section 9.3.2.

Iodine partition coefficients based on the expected pH levels are 0.01 in the steam generator and 0.15 in the condenser.

10.3.6 MAIN STEAM AND FEEDWATER SYSTEM MATERIALS

The typical material specifications used in the Containment pressure boundary components are listed in Tables 10.3-6 through 10.3-9. In some cases, this list of materials may not be totally inclusive. However, the listed specifications are representative of those materials used.

10.3.6.1 Fracture Toughness

All pressure-retaining ferritic materials in the Containment pressure boundary except the feedwater isolation valves are impact-resistance tested in accordance with NC 2300 and NB 2300 of the ASME B&PV Code, Section III, mentioned in Section 5.2.4. Supplemental impact testing and fracture analysis, in conjunction with external heating of the feedwater isolation valves, are utilized to demonstrate the acceptability of the originally purchased feedwater isolation valves in lieu of impact resistance testing in accordance with the ASME B&PV Code, Division 1, Section III, Subsection NC-2300. Replacement feedwater isolation valves or their pressure retaining components, such as the valve bonnet, will be impact tested in accordance with the ASME B&PV Code, Division 1, Section III, Subsection NC-2300.

The main steam and feedwater piping are Charpy-type-impact tested in accordance with the method specified in ASME SA 370.

10.3.6.2 Materials Selection and Fabrication

The material specification used for pressure-retaining component parts in the Containment isolation boundary are in conformance with Appendix I of the ASME B&PV Code, Section III.

Where austenitic steel is used, the requirements of NRC Regulatory Guide 1.44 are followed. The insulation used with this material is in compliance with NRC Regulatory Guide 1.36. The welding procedures for this material are in compliance with NRC Regulatory Guide 1.31 to the extent specified in Section 6.1.1.1. Further discussion of fabrication, testing, and welding for austenitic steel materials is given in Section 6.1.

The cleaning and handling of all safety-related austenitic components are given in Section 6.1.

All ferritic steel components are thoroughly cleaned, descaled, and coated in accordance with the Applicant's specifications. These specifications meet or exceed the requirements of ANSI N45.2.1-1973 and NRC Regulatory Guide 1.37.

Preheat temperatures for welding low-alloy steel are in accordance with NRC Regulatory Guide 1.50. Welding is performed in accordance with NRC Regulatory Guide 1.71, where applicable.

REFERENCES

1. Westinghouse Guidelines For Secondary Water Chemistry (SGT-5.1.1 - 4468), dated February 1985.

2. EPRI PWR Secondary Water Chemistry Guidelines, Revision 2, date November 1988.

3. Standard Review Plan, Section 5.4.2.1, Branch Technical Position MTEB 5-3, "Monitoring of Secondary Side Water Chemistry in PWR Steam Generators," Revision 1.

TABLE 10.3-1
MAIN STEAM PIPE DESIGN PARAMETERS

	Nuclear-Safety Class 2 Portion[a]	Non-Nuclear-Safety Portion[a]
Outside diameter, in.	32	34
Inside diameter, in.	29.50	31.50
Minimum wall thickness, in.	1.25	1.25
Pipe material	SA 155, Class 1, Grade KCF-70	A 155, Class 1, Grade KC-70
Design pressure, psig	1200	1200
Design temperature, °F	650	650

a) For definition, see Subsection 10.3.1.

TABLE 10.3-2
MAIN STEAM SAFETY VALVES

Quantity per unit	20
Quantity per line	5
Safety class	ASME B&PV Code, Section III, Class 2, seismic Category I
Design pressure, psia	1250
Design temperature, °F	600
Accumulation	3% of set pressure
Blowdown	≤12% of set pressure
Orifice size, in^2	16

Valve Settings and Rated Flow

Set Pressure (psia)	Rated Flow For Each Valve (lb/hr)
1200	893,160
1210	900,607
1220	908,055
1230	915,502
1250	930,397

Connections	Inlet	Dual Outlet
Type	Flanged	Flanged
Size, in.	6	8 by 8
ANSI rating, lb	1500	300

TABLE 10.3-3
ATMOSPHERIC RELIEF VALVES

Quantity per unit	4	
Safety class	ASME B&PV Code, Section III Class 2, seismic Category I	
Design pressure, psia	1300	
Design temperature, °F	700	
Steam flow per valve, lb/hr		
maximum allowable	968,400 @ 1200 psia	
minimum allowable	62,150 @ 100 psia	
Normal operating pressure range, psia	965 to 1107	
Maximum operating range, psia	100 to 1300	
Normal operating temperature, °F	540 to 557	
Connections	Inlet	Outlet
Type	Butt weld	Butt weld
Size, in.	8	8
ANSI rating, lb	900	900
Actuator	Pneumatic	

TABLE 10.3-4
MAIN STEAM ISOLATION VALVES

Quantity per unit	4
Safety class	ASME B&PV Code, Section III, Class 1, seismic Category I
Design pressure, psia	1200
Design temperature, °F	600
Steam flow rate, lb/hr	4.0×10^6
Design pressure drop, psi	2.1
Normal operating pressure, psia	965 to 1107
Normal operating temperature, °F	540 to 557
Size (inlet by outlet), in.	32 by 34
Actuator, hemisphere cylinder assembly including hemisphere cylinder and transition ring	Stored energy (ASME B&PV Code, Section VIII)
Closure time, sec	5

TABLE 10.3-5
MAIN STEAM ISOLATION BYPASS VALVES

Quantity per unit	4
Safety class	ASME B&PV Code, Section III, Class 1, seismic Category I
Design pressure, psia	1200
Design temperature, °F	600
Steam flow rate, lb/hr (approximate)	60,000
Normal operating pressure range, psia	250 to 970
Normal operating temperature range, °F	401 to 541
Pressure drop	Critical
Actuator	Manual (handwheel)
Valve size, in.	4

TABLE 10.3-6
MATERIALS OF MAIN STEAM AND FEEDWATER VALVES AND PIPING

(Sheet 1 of 2)

Pneumatic/Hydraulic and Manual Gate and Check Valves	
Bodies	SA 105
Bonnets	SA 105
Discs	SA 182, F316
Closure bolting and nuts	SA 564, Type 630 and SA 194, Grade 8M
Stem	SA 564, Type 630
Air-Operated Valves	
Bodies	SA 352, Grade LCB or SA 351, Grade CF8M
Bonnets	SA 352, Grade LCB or SA 351, Grade CF8M
Discs	SA 479, Type 316L
Closure bolting and nuts	SA 193, Grade B7 and SA 194, Grade 7 or SA 453, Grade 660 and SA 194, Grade 6
Stem	SA 479, Type 316L
Miscellaneous Valves (2 in. and Smaller)	
Bodies	SA 105, or SA 182, F316
Bonnets	SA 105, or SA 182, F316
Discs	SA 479, Type 410 or SA 182, F316 or A 567, Grade 1 (Stellite No. 21)
Piping	
34 in.	A 155, Class 1, Grade KC 70
26 in. to 32 in.	SA 155 Class 1, Grade KCF 70
24 in. and smaller	SA 333, Grade 6

TABLE 10.3-6
MATERIALS OF MAIN STEAM AND FEEDWATER VALVES AND PIPING

(Sheet 2 of 2)

Fittings	
2 in. and smaller	SA 350, Grade LF 2
3 in. to 24 in.	SA 420, Grade WPL-6[(a)]
26 in. to 32 in.	SA 234, WPPW
34 in.	A 234, WPCW
Flanges	
4 in. to 8 in.	SA 350, Grade LF 2
34 in.	A 105

a) Due to the Unit 1 RSG installation the 18” x 16” reducing elbow material at the RSG nozzle inlet is now SA-508, Grade 2, Class 1.

TABLE 10.3-7
MATERIALS OF MAIN STEAM SAFETY VALVES

Valve body	SA 105
Inlet nozzle	SA 182, F316
Disc insert	SA 182, F316
Guide and guide ring	ACI CF8M nicology (solution annealed)
Studs (inlet flange)	SA 193, Grade B7
Nuts (inlet flange)	SA 194, Grade 2H

TABLE 10.3-8
MATERIALS OF MAIN STEAM RELIEF VALVES

Valve body	SA 216, Grade WCB
Disc insert	SA 479, Type 316
Bonnet	SA 216, Grade WCB
Bonnet studs or bolts	SA 193, Grade B7
Bonnet nuts	SA 194, Grade B7

TABLE 10.3-9
MATERIALS OF MAIN STEAM ISOLATION VALVES

Valve body	SA 216, Grade WCC
Bonnet	SA 105
Bonnet studs	SA 540, Grade B23CL4
Bonnet nuts	SA 194, Grade 7
Gland studs	A 193, Grade B7
Gland nuts	A 194, Grade 2
Disc	SA 182, Grade F11

TABLE 10.3-10
THIS TABLE HAS BEEN DELETED.

TABLE 10.3-11
BRANCH OFF STEAM FLOW PATH BETWEEN MAIN STEAM ISOLATION VALVE AND TURBINE STOP VALVE

SYSTEM IDENTIFICATION	MAXIMUM STEAM FLOW, LB/HR	TYPE OF SHUT-OFF VALVE	SIZE OF VALVE, INCH	QUALITY OF VALVE	DESIGN CODE	CLOSURE TIME OF VALVE	ACTUATION MECHANISM	POWER SOURCE	MOTIVE REMARKS
Secondary Sampling	150	Globe	3/8	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 5
Main Steam Line	N/A	Globe	2	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Condensate Removal	13,000	Globe	2	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 3
Main Steam Strainer	N/A	Globe	2	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Condensate Removal	13,000	Globe	2	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 3
Steam Dump Header	N/A	Globe	2	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Condensate Removal	13,000	Globe	2	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 3
Steam Dump Valve	1,109,000	Globe	8	Commercial	ANSI B16.5	5 sec.	Air Diaphragm	Air	Notes 1, 3
Steam Dump Valve	N/A	Globe	1	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Line Condensate Removal	6,000	Globe	1	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 3
Feedpump Turbine	N/A	Globe	2	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Steam Supply Condensate Removal	13,000	Globe	2	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 3
Auxiliary Steam Supply	10,000	Gate	3	Commercial	ANSI B16.5	15 sec.	Motor	Elect. (Not safe-guards)	Notes 1, 4, 7
Auxiliary Steam Line	N/A	Globe	2	Commercial	Mfg. Standard	N. A.	Manual	N. A.	Notes 1, 2, 5
Condensate Removal		Globe	2	Commercial	Mfg. Standard	N. A.	Air Diaphragm	Air	Notes 1, 6

NOTES:
1. Will not affect reactor shutdown.
2. Steam traps provided downstream of shut-off valves.
3. Manual closure from Control Room, normally closed, fail closed.
4. Flow is intermittent.
5. Manual closure, normally open.
6. Manual closure from the control room, normally closed, fail open.
7. Normally open, closure from the control room.

10.4 OTHER FEATURES OF STEAM AND POWER CONVERSION SYSTEM

10.4.1 MAIN AND AUXILIARY CONDENSERS

Each CPNPP unit is equipped with one main condenser and two auxiliary condensers. The main condensers are described in Subsection 10.4.1.1 and the auxiliary condensers are described in Subsection 10.4.1.2.

10.4.1.1 Main Condensers

The twin-shell main condenser furnishes the heat sink for the main turbine exhaust steam (as discussed in Subsection 10.4.5) and for the steam dump system (as discussed in Subsection 10.4.4). The hot well also provides condensate storage for plant operations. The exhaust steam side of the main condenser is shown on Figures 10.4-8 and 10.4-9 and described in Subsection 10.4.7.

10.4.1.1.1 Design Bases

Design is based on the following conditions (nominal) at full load of the main turbine:

Conditions	Value
Exhaust steam, lb/hr	8.44×10^6 (approx.)
Condenser duty, Btu/hr	8.06×10^9 (approx.)
Condenser pressure at turbine exhaust, in. Hg abs.	3.55
Circulating water data Flow, gpm	1.03×10^6
Number of passes per shell	One
Inlet temperature, °F	95
Outlet temperature, °F	110.8

The condensers function effectively at all loads below maximum guaranteed turbine load. The main condenser is also designed to perform the following functions:

1. To condense steam released to the condenser by the steam dump system (Subsection 10.4.4) during unit startup, normal cooldown, and emergency cooldown

2. To condense turbine exhaust steam and dump steam under larger load reductions than those defined in Section 10.2.1

3. To deaerate the condensate

4. To accept drains from the feedwater heaters and other miscellaneous sources (see Subsection 10.4.11, Figure 10.4-14).

The condenser hot well provides approximately 13,800 ft^3 of condensate storage capacity, which is equivalent to approximately five-minutes operation at maximum load. Automatic rejection of hot well inventory to a volume equivalent to about 4 minutes operation at maximum load is provided. Hot well inventory is typically rejected to a volume equivalent to 3 minutes operation at maximum load during normal operation. Hot well volume equivalent to two minutes operation at maximum load is sufficient to achieve feedwater isolation and auxiliary feedwater initiation.

Following a 50-percent load rejection, the condenser accepts at least 40 percent of the main steam flow through the steam dump system. This condition is accommodated without increasing the condenser back pressure to the turbine trip set point or exceeding the allowable turbine exhaust temperature.

10.4.1.1.2 System Description

The surface condenser is of the deaerating, single pass, fixed-bottom, twin-shell, single-pressure, horizontal surface, divided water box, floor-supported type, and is sized to condense exhaust steam from the main turbine under full load conditions.

It is connected to the exhaust openings of each low-pressure turbine by an expansion joint. An equalizing line between the shells limit turbine exhaust temperature differences.

Erosion protection is provided by installation of baffles and spray headers to prevent direct impingement of steam and/or flashing mixtures which enter the shell at various locations. Condenser materials were specifically chosen to resist the effects of erosion. The titanium used in the condenser tubes has good erosion and steam impingement resistance.

The condenser is cooled by the Circulating Water System (described in Section 10.4.5), which removes the heat rejected to the condenser.

Layout drawings (Figures 10.4-1 and 10.4-2) show the connections to the turbine exhaust and the Circulating Water System. Valves are provided in the Circulating Water System to permit either half of each condenser shell to be removed from service.

Condensate from the auxiliary (steam generator feedwater pump turbine) condensers is drained to the main condenser hot well.

10.4.1.1.3 Safety Evaluation

1. Radioactive Contaminants

   The radioactivity in the main condensers is a function of the percentage of defective fuel cladding, the escape rate coefficients, the primary-to-secondary leak rate, the secondary system flow rate, the blowdown rate, and the steam generator and condenser partition factors. Steam generator tube leakage effects on the secondary system are discussed in Section 10.2.4. The effects of short term exposure of low radioactivity levels on the Condensate Cleanup System are discussed in Subsection 10.4.6.

   Steam generator blowdown processing of radioactive primary-to secondary system leakage is discussed in Subsection 10.4.8.

Radioactive contaminants from steam generator primary-to-secondary coolant leakage reach the main condenser shell through the steam path (i.e., main steam lines-turbine condenser and steam dump lines-condenser) and the liquid path (steam generator blowdown processing system).

The assumptions (partition factors, decontamination factors, and leak rates) for calculating the expected radioactivity concentrations in the Steam and Power Conversion System generally and in the steam generators, turbine and condenser specifically are discussed and tabulated in Section 11.1.

2. Vacuum Loss

There is no direct influence of the condenser operation on the Reactor Coolant System (RCS). Partial loss of condenser vacuum is annunciated in the Control Room. Should vacuum continue to decay, the turbine automatically trips and the steam dump valves to the condenser open automatically if partial vacuum can be maintained.

Following a turbine trip, the turbine trip steam dump controller becomes active. A description of the turbine trip steam dump controller is presented in Section 7.7.1. If the operation of the dump system is blocked by low condenser vacuum, heat removal from the RCS is provided by the main steam safety valves.

3. Air Leakage

The condensers are designed to minimize air leakage. Welded construction is used for the condenser shell and, wherever practicable, for condenser shell connections and penetrations.

Equipment and piping connected to the condenser shell are also designed to minimize air leakage to the condensers through the use of water seals, pressure seals, and leaktight valves. The maximum expected air leakage into the main condensers is anticipated to be 200 lb/hr during normal operation (see Subsection 10.4.2).

4. Safety-Related Equipment

The condensers are not safety-related and there is no safety related equipment in the area of the condenser which can be flooded as a result of a failure of the condenser.

5. Hydrogen Buildup

The potential for hydrogen buildup in the condenser is negligible since there are no potential sources of hydrogen in direct contact with the condenser. Any small amounts of hydrogen originating in the RCS and reaching the condenser through steam generator tube leakage are removed by the condenser evacuation system. Experience from similar plants shows that the concentration of hydrogen is well below any hazardous value.

6. Control and Detection of Condenser Leakage

The main condenser tube bundles and tube sheet assemblies are constructed of titanium and titanium clad carbon steel respectively. The main condenser tubes are rolled into and

welded to the tubesheets. This design is virtually leak tight, but should a tube-to-tubesheet leak develop, it will be collected in salinity troughs.

Tube leaks are detected by condensate conductivity meters and by installed sodium analyzers at the condenser hotwell. In the event of any leaks into the condenser, the full flow condensate polisher can reduce the dissolved and suspended solid contaminants introduced through the condenser leakage. The cation conductivity measurement on the inlet and outlet headers of the Condensate Cleanup System, as well as at the effluent of each vessel, monitor condensate quality and individual vessel performance. Section 10.4.16, Table 10.4-20, Secondary Plant Sampling System Measured Parameters, and Section 10.4.6.5, para. 3, Conductivity Instrumentation for the Condensate Cleanup System, indicate the measures taken for detecting condenser cooling water leakage into the condensate stream.

10.4.1.1.4 Inspection and Testing Requirements

Each condenser shell receives a field hydrostatic test prior to initial operation which consists of filling the condenser shell with water and, with the resulting static head maintained, inspecting all welds and surfaces for visible leakage, or excessive deflection, or both.

Each condenser water box receives a field hydrostatic test, with all joints and external surfaces being inspected.

After the completion of the tubing installation in the condenser, all tube joints are leak tested.

10.4.1.1.5 Instrumentation Requirements

1. Hot Well

   Each condenser shell is provided with local and remote hot well level and pressure indication. The remote indication is by means of indicators and alarms in the Control Room. The condensate level in the condenser hot well is maintained within proper limits by automatic controls which provide for transfer of condensate to and from the condensate storage tank as needed to satisfy the requirements of the steam system. Condensate temperature is measured in the suction lines of the condensate pumps.

2. Exhaust Hood

   Excessive temperature in the turbine exhaust hood causes a thermostatically controlled spray valve to open, thereby preventing further increases in temperature. The valve closes again once a limit for shutting off the spray is met. These limits include settings for minimum turbine speed, maximum exhaust hood temperature, and minimum steam pressure at low-pressure casing inlet. Rupture diaphragms are provided to prevent excessive overpressure of the main condenser shells. There is no safety-related instrumentation in this section.

3. Water Box

   Water box pressure and temperature measurements are provided.

4. Radioactivity

A radiation monitor is provided on the condenser vacuum pumps discharge line to detect the presence of noncondensible radioactive gases in the system and initiates an alarm upon increasing radiation levels (see Section 9.3.2 and 11.5). A radiation monitor on the steam generator process sample line header detects leakage across the tubes in the steam generator (liquid phase) and provides backup information to the condenser off-gas monitor. Turbine drains effluent is monitored for radioactivity prior to release to the low volume waste pond. Indication and alarm is initiated in the Control Room upon detection of radioactivity in excess of established limits. (See Section 11.5.)

10.4.1.2 Auxiliary Condensers

10.4.1.2.1 Design Bases

The auxiliary condensers provide the heat sink for the steam generator feedwater pump turbine exhaust steam. One auxiliary condenser is provided for each steam generator feedwater pump turbine. The condensate flows into the main condenser hot well.

10.4.1.2.2 System Description

Each auxiliary condenser is of the single-pass, nondivided water box, floor-supported type, and is sized to condense exhaust steam from the steam generator feedwater pump turbine under full-load conditions. Condensate from the auxiliary condenser hot wells is drained to the main condenser hot well. Each condenser is connected to the exhaust openings of the turbine by an expansion joint. The auxiliary condenser is cooled by the Circulating Water System (described in Subsection 10.4.5) which removes the heat rejected to the condenser.

An atmospheric relief valve is provided for each condenser which discharges to the atmosphere through a 20 in. vent connection. No provision is made for the monitoring of radioactivity in the effluent from these relief valves since expected potential radioactive concentrations and flow rates provide effluents well below established limits.

10.4.1.2.3 Safety Evaluation

Safety evaluation considerations for the auxiliary condensers are the same as those for the main condensers as described in Subsection 10.4.1.1.3.

10.4.1.2.4 Inspection and Testing Requirements

See Subsection 10.4.1.1.4.

10.4.1.2.5 Instrumentation Requirements

Each condenser shell is provided with a pressure transmitter for remote indication in the Control Room. The condenser hot well is provided with local level indication. The condensate level in the hot well is monitored by means of high water level alarm in the Control Room. A high turbine exhaust hood temperature alarm is provided in the Control Room.

Steam generator feed pump turbine trip is activated on loss of auxiliary condenser vacuum, with condenser back pressure reaching or exceeding a set point of between 12 and 13 in. Hg abs. A high back pressure alarm setting is in the range of 8.5 to 9.5 in. Hg abs.

Water box pressure and temperature measurements are provided.

There is no safety-related instrumentation in this section.

10.4.2 CONDENSER EVACUATION SYSTEM

The Condenser Evacuation System, as shown on Figure 10.4-3, removes air and noncondensible gases from the main and auxiliary condensers, and it removes trapped air and primes those parts of the Circulating Water System which are above lake level.

This system is considered non-nuclear-safety-related in accordance with ANSI N18.2 and is designated non-seismic Category I.

10.4.2.1 Design Bases

The Condenser Evacuation System is designed for initial evacuation of the main condenser shells and the two auxiliary condenser shells at startup, and for removal of noncondensible gases during normal plant operation.

A separate priming portion of this system primes the Circulating Water System by removing air from the (a) high points in the circulating system piping, (b) main and auxiliary condenser water boxes, and (c) the turbine plant cooling water heat exchanger and the condenser vacuum pump heat exchanger water boxes.

As an alternate means of initiating turbine trips, the system incorporates a vacuum breaker arrangement for the main and auxiliary condenser shells. A vacuum breaker is also provided to protect the circulating water system from system transients.

The system is designed to prevent uncontrolled release of radioactive material to the environment, in accordance with 10 CFR Part 50, Appendix I, and General Design Criteria (GDC) 60 and 64. A radiation monitor is used to provide detection of radioactivity.

The capacity of each vacuum pump is as follows:

Hogging capacity, acfm dry air, at 10 in. Hg abs. is 2300. Design holding capacity, scfm, at 1.0 in. Hg abs. is 25.

10.4.2.2 System Description

10.4.2.2.1 System Operation

For each unit of CPNPP, three 100-percent capacity, motor-driven, two-stage, rotary-type condenser exhausting vacuum (CEV) pumps are furnished for the Condenser Evacuation System. These pumps provide hogging and holding functions for the main and auxiliary condenser shells.

The first function of the CEV pumps is the initial evacuation of the main and auxiliary condenser shells prior to the availability of the steam during plant startup. This operation, called hogging, uses the three CEV pumps operating in parallel. The pump suction lines are connected via a common suction header with the main and auxiliary condenser shells. When started, the three CEV pumps provide for the evacuation of air from the condenser shells at a rate of 2300 acfm dry air per pump at approximately 10 in. Hg absolute. The hogging process brings the condenser shell pressures down to 10 in. Hg in approximately one hour.

At approximately 10 in. Hg abs., the CEV pumps automatically switch to the holding mode. If for some reason the condenser shell pressures increase beyond this setting during the holding operation, the CEV pumps automatically revert to the hogging operation.

During the hogging operation, the volume of air handled by the first stage is greater than the inlet capacity of the second stage of each CEV pump. The CEV pumps use an internal arrangement in which the exhaust bypasses the second stage and is separated and finally discharged. At approximately 10 in. Hg abs., the pumps automatically switch from the hogging to the holding mode, and the second stage handles the entire first-stage discharge. The internal bypass arrangement, now closed, allows the two-stage combination to pull the system down to operating pressure. The second-stage exhaust is then separated and discharged.

During the holding sequence, the number of CEV pumps in operation can be reduced. At any load during the winter, with the circulating water inlet temperature at 40°F, two CEV pumps are required to maintain the condenser vacuum at approximately one in. Hg absolute. One CEV pump is sufficient to maintain a condenser pressure of 3.5 in. Hg abs. at any load with 95°F inlet water.

Seal water makeup for the CEV seals is provided by the demineralized and reactor makeup water system. The circulating water system cools the CEV pump seal water in the CEV pump heat exchangers. The cooled seal water is then recirculated through the CEV pumps. Water vapor pulled from the condenser shell which comes into contact with the CEV pump seal water is condensed. Water from the CEV pump seals drains continuously by gravity to Turbine Building Local Floor Drain, which eventually is disposed of by the liquid waste processing system.

A vacuum breaker system is provided for the main and auxiliary condenser shells. This system consists of piping and motor-operated valves which, when actuated, allow air to enter the condensers at a rate in excess of the vacuum pump capability to remove air. This results in high back-pressure in the condenser and initiates a turbine trip, thereby providing another method of turbine trip, should, for any reason, the normal method become inoperative.

For each unit of CPNPP, two 100 percent capacity, motor driven, single stage rotary-type condenser water box vacuum priming pumps provide for priming of the Circulating Water System, and the water boxes of the Main Condenser, the Auxiliary Condensers, the Turbine Plant Cooling Water System heat exchanger, and the seal water heat exchanger of each CEV pump.

The priming operation removes air from the Circulating Water System and several condenser and heat exchanger waterboxes. The partial vacuum causes them to fill with water from Squaw Creek reservoir. During normal operation, the condenser water box vacuum priming pumps continue to remove accumulated air at high points in the Circulating Water System and the several water boxes.

Priming valves and standpipes are provided to prevent carryover of the potentially corrosive circulating water to the condenser water box vacuum priming pumps.

Condenser water box vacuum priming pump seal water is discharged to the Turbine Building Local Floor Drain. Seal water for the condenser water box vacuum priming pumps is provided by the Demineralized and Reactor Makeup Water System.

A vacuum breaker is provided on the main condenser outlet water boxes. The vacuum breaker is designed to protect the tube side components from transients following a simultaneous trip of all four circulating water pumps. This system is comprised of piping and an air operated valve which will open on loss of all four circulating water pumps.

In order to minimize air inleakage at the main condenser shell and water box vacuum breakers, and the auxiliary condenser relief valves and vacuum breakers, these components are sealed by water from the Condensate System.

Safe shutdown of the unit does not rely on the availability of the Condenser Evacuation System. Hence, there is no provision to provide power to this system from the standby diesel generators.

The loads in this system are supplied with power from a non-class 1E power supply.

10.4.2.2.2 Equipment Design Criteria

Design criteria for equipment are listed in Table 10.4-1.

10.4.2.3 Safety Evaluation

The safety evaluation of this system is given in Subsection 10.4.1.1.3, in conjunction with the evaluation of the condensers. The radiological considerations discussed therein result in the maximum anticipated radioactive Containment discharge rates for this system given in Table 11.2-10.

The off-gas CEV pump discharge is potentially radioactive; therefore, if steam generator tube leakage occurs coincident with fuel cladding failures, the CEV pump discharge gases are cascaded into a common discharge line which is directed to the heating, ventilating, and air-conditioning (HVAC) system for controlled release to the atmosphere through an exhaust unit, which consists of roughing, HEPA, and charcoal filters and the plant vent.

A radiation monitor is provided on the condenser vacuum pumps discharge line to continuously measure the radiation level of the condenser off-gases discharged to the atmosphere with indication given in the Control Room. A sample pump diverts a small percentage of the discharge flow through the monitor and back to the discharge line.

Radioactive elements present in the secondary system which could ultimately reach the condenser off-gas can also be detected by the steam generator liquid sample monitor (Section 11.5).

The presence of radiation above a set point is annunciated in the Control Room where, upon receiving this alarm, a manual sample of the off-gas is drawn to determine if the condensers are the source of the activity.

10.4.2.4 Tests and Inspections

All tests and inspections of equipment in the Condenser Evacuation System are performed in accordance with applicable codes.

10.4.2.5 Instrumentation Requirements

Following a turbine trip, the turbine trip steam dump controller becomes active. A description of the turbine trip steam dump controller is presented in Section 7.7.1. If the operation of the dump system is blocked by low condenser vacuum, heat removal from the RCS is provided by the main steam system safety valves.

The Condenser Evacuation System is provided with the following instrumentation to monitor system performance:

1. The suction header of the CEV pumps contains a pressure switch to alarm on low vacuum and a pressure gauge for local indication.

2. The suction header of the water box priming pumps contains a pressure gauge for local indication.

3. The standpipe that prevents water carryover into the condenser water box vacuum priming pump suction header contains a pressure switch that on low vacuum primary header pressure provides input to a main condenser priming trouble alarm.

4. The main condenser vacuum priming trouble alarm is also activated if the main condenser outlet waterbox vacuum breaker air operated valve is not fully closed.

The CEV pumps are controlled from the Control Room. Indicating lights in the Control Room advise operating personnel of the pump or pumps in operation and of the position of the pump suction valves. Motor trips for the CEV pumps are annunciated in the Control Room.

Motor-operated vacuum breaker valves are provided with Control Room mounted control switches and indicating lights.

A local pressure gauge is provided for monitoring the pressure of the CEV Pump seal water makeup supply from the Demineralized and Reactor Makeup Water System.

The condenser water box vacuum priming pumps are controlled from a locally mounted control panel. Indicating lights on the panel advise operating personnel of the pump(s) in operation.

10.4.3 TURBINE GLAND SEALING SYSTEM

The Turbine Gland Sealing System, shown on Figure 10.4-4, is designed to prevent an inflow of air into the main and auxiliary condensers via the shafts of the main turbine and the feedwater pump turbine drivers, respectively. The system is also designed to prevent an outflow of steam from the high-pressure and low-pressure turbine shafts. System components are located in the Turbine Building.

The system is designed as non-nuclear-safety-related in accordance with ANSI N18.2 and is classified as non-seismic Category I [3], [4]. The system is designed with provisions to monitor and control releases of gaseous radioactive material in accordance with GDC 60 and 64 of 10 CFR Part 50 [1], [2].

10.4.3.1 Design Bases

1. Main Turbine Seals

The main turbine is self-sealing at loads above approximately 40 percent. The seal steam supply header has a design flow rate of 14,370 lb/hr at 14.83 psia. At loads below approximately 40 percent, auxiliary steam from the Auxiliary Steam System is supplied at 6832 lb/hr, 164.7 psia, and 368°F.

Design conditions for dumping excess steam to feedwater heaters 6A and 6B are 13,154 lb/hr at 4.43 psia and 3.4-percent moisture. The gland steam condenser and exhauster are designed to maintain one to two in. water vacuum in the seal steam condenser header while condensing 1670 lb/hr of steam and passing 1050 lb/hr of air through the plant radiation monitor and exhaust plenum. The gland steam condenser tubes are designed for 1140 gpm of condensate at a design temperature and pressure of 125°F and 600 psig, respectively.

2. Feedwater Pump Turbine Seals

The seal steam supply header has a normal flow rate of 230 lb/hr at 2 to 4 psig.

At startup, 1150 lb/hr of auxiliary steam from the Auxiliary Steam System is supplied at 2 to 4 psig. During an emergency, seal steam is supplied from the main steam supply system.

The auxiliary gland steam condenser and exhauster are designed to maintain 7- to 10-in. water vacuum in the seal steam condenser header while condensing 780 lb/hr of steam and passing 170 lb/hr of air through the plant radiation monitor and exhaust plenum.

10.4.3.2 System Description

1. Normal Operation Conditions

a. Main Turbine Steam Seals

Both high-pressure and low-pressure turbines have the same physical seal arrangement. At every point where the turbine shaft penetrates the turbine casing, there are two annular chambers separated by labyrinth seals from each other, the atmosphere, and the interior of the turbine.

The high-pressure turbine seals provide seal steam for the low-pressure turbines and prevent the loss of steam to the atmosphere. The high-pressure steam leaks through the labyrinth seals and is bled off from the inner annular chamber to supply steam to the low-pressure turbine seals at 14.83 psia (slightly above atmospheric pressure). The steam that is not bled off is drawn into the outer

annular chamber along with air from the atmosphere. The outer annular chamber is kept at a slight vacuum (1- to 2-in. water vacuum) by the main turbine gland steam condenser.

The low-pressure turbine seals prevent air from entering the turbine and steam from escaping to the atmosphere. Steam (14.83-psia) from the high pressure turbine enters the inner annular chamber and flows both inward and outward along the shaft. The steam that flows outward joins air flowing inward at the outer annular chamber, where it is drawn into the main turbine gland steam condenser.

The main turbine gland steam condenser is kept at a vacuum by the main turbine gland steam condenser exhauster during all modes of turbine gland steam system operation. Air and steam from the outer annular chambers of all main turbines and from the main steam stop and control valve leakoffs is drawn into the main turbine gland steam condenser, where the steam is condensed by cool condensate and drained through a loop seal to the atmospheric drain tank.

Air, noncondensable gases, and some water vapor are ejected by the main turbine gland steam condenser exhauster. The water drains to the atmospheric drain tank through a loop seal, and the air and noncondensable gases are vented to the primary plant ventilation exhaust system, which consists of 16 exhaust filtration units. See Section 9.4 for a description of the exhaust filtration units. Furthermore, the exhaust is monitored for radiation prior to release to the atmosphere.

During startup, shutdown (condenser at partial vacuum), and when turbine load falls below approximately 40 percent, high-pressure turbine seal pressure is insufficient to provide seal steam for the low-pressure turbines. Under these conditions, all main turbine seals operate in the same manner as the low-pressure turbine seals during normal operation. Steam (14.83 psia) is supplied by the Auxiliary Steam System through a pressure-regulating steam supply valve to the inner annular chamber of both the high-pressure and low-pressure turbines. A warmup line is provided upstream from the supply valve. Any overpressure of the seal steam header is relieved through the 10 in. leakoff valve to feedwater heaters 6A and 6B. The feedwater heater tube bundles are protected from thermal stress by an impingement plate and are suitable for this steam temperature.

Piping that may contain condensate during normal operation collect water in drain pots which discharge to the main condenser. There is also a drain pot on the seal steam header which drains to the Turbine Building Drain System through a motor operated valve.

b. Feedwater Pump Turbine Steam Seals

Both high-pressure and low-pressure seals of the feedwater pump turbine have the same seal arrangement as those for the main turbine.

Steam which is slightly above atmospheric pressure (2 to 4 psig) is piped from the high-pressure inner annular chamber to the low-pressure inner annular chamber to prevent air inleakage at the low-pressure end of the turbine which is at

condenser vacuum. The outer annular chambers at both ends of the turbine are held at a slight vacuum (7- to 10- in. water vacuum). Therefore, at the high-pressure end, most of the steam is bled off to seal the low-pressure end, and the rest of the steam plus air is drawn into the auxiliary gland steam condenser. At the low-pressure end, seal steam is supplied at 2-4 psig to prevent the ingress of air, and steam plus air is drawn into the auxiliary gland steam condenser to prevent the escape of steam to the atmosphere.

The pressure at the high-pressure end is insufficient to seal the low-pressure seals. Auxiliary steam from the Auxiliary Steam System is provided at the inner annular chambers through the steam seal supply valve.

During all modes of plant operation, the auxiliary gland steam condenser applies a vacuum to the outer annular chambers of the feedwater pump turbine seals. Vacuum is maintained by the auxiliary gland steam condenser exhauster which dumps air and noncondensable gases to the Primary Plant Ventilation Exhaust System and drains water to the atmospheric drain tank through a loop seal. Steam is condensed by cool condensate, and the condensed steam drains to the atmospheric drain tank through a loop seal to maintain condenser vacuum.

Piping that may contain condensate during normal operation collects water in drain pots which discharge to the main condenser.

2. Abnormal Conditions

a. Main Turbine Steam Seal System

Upon high water level due to the tube break in the main turbine gland steam condenser, a level switch actuates an alarm to indicate the condition. If the water backs up further, it enters an emergency overflow connection, which provides emergency drains to the atmospheric drain tank. No valves are required in the emergency overflow connection because of a loop seal which maintains vacuum during normal operation. The loop seal is initially filled from the condensate system.

Upon loss of the main turbine gland steam condenser exhauster, the system is capable of continued operation for a limited time by venting steam collected from the outer annular chambers to the atmosphere through the emergency bypass to the atmosphere. In this mode of operation, air does not leak into the low-pressure turbines, but steam leaks out of all turbine seals to the Turbine Building.

In the event of primary to secondary leakage in the steam generators, there will be very low releases of radioactivity when the main turbine gland steam condenser is being bypassed.

b. Feedwater Pump Turbine Steam Seal System

In the event of excessive packing wear or blowout, the packing leakage increases drastically; this causes the steam seal supply valve to modulate to maintain sealing pressure of 2-4 psig.

Unit 1 FWP turbine gland steam pressure is controlled by throttling a fail open, supply valve that delivers steam from the auxiliary steam system to the FWP turbine gland seals in conjunction with a fail closed, dump valve that delivers excess gland steam to the associated auxiliary condenser. A single controller operates both valves to maintain FWP turbine gland steam pressure at 2 to 4 psig.

Unit 2 FWP turbine gland steam pressure is controlled by throttling a fail open, supply valve that delivers steam from the auxiliary steam system to the FWP turbine gland seals at 2 to 4 psig. The dump valve to remove excess gland steam, is abandoned in place and is closed and manually isolated. Excess steam is removed by a relief valve set at 20 psig should the supply valve fail open.

Main unit trip reduces the pump turbine load to the point where the steam seal supply valve opens to maintain the seal header pressure at the set point value.

3. Electrical System

Safe shutdown of the unit does not rely on the availability of the Turbine Gland Sealing System. Therefore, there is no provision to provide power to this system from the standby diesel generators.

The loads in this system are supplied power from the non-Class 1E power supply.

10.4.3.3 Safety Evaluation

The amount of noncondensable gases released by the condensed steam is negligible (as shown in Table 11.1-1), and there is no need for an activity monitor at the gland seal condenser exhauster. However, air and noncondensable gases are exhausted to the primary plant exhaust plenum, which is monitored and consists of a series of filtration units.

For more details, refer to Section 11.3.

10.4.3.4 Tests and Inspection

Normal preoperational tests are performed in accordance with applicable codes. Operability of the gland seal system is verified by the maintenance of condenser vacuum and visual inspection of the equipment is performed before and during operation.

10.4.3.5 Instrumentation Requirements

1. Main Turbine Steam Seals

Gland steam header pressure is maintained at a reference value by an automatic control system supplied by the turbine-generator manufacturer. The seal steam supply valve and seal steam leakoff valve are automatically modulated to maintain header pressure.

The valves can also be controlled manually from Operating and Monitoring (OM) screens at computer workstations in the Control Room.

The seal steam condenser exhauster fan is started and stopped manually with a Control Room switch. Water accumulation in the seal steam header is alarmed in the Control Room by a level switch. The operator then manually opens the motor-operated valve.

Seal steam header pressure indication is provided locally and remotely on the OM screens.

The following system abnormalities are alarmed in the Control Room:

a. Seal steam condenser level high

b. Seal steam control valve failure

c. High drip pot levels

d. Seal steam exhaust fan trip

e. Drain seal steam header level high

Forward-reverse pitot tubes are installed in all seal steam supply lines (both main turbines and feed pump turbines). These can be used to measure abnormal gland wear. Temperature test wells are also installed in these lines. A forward reverse pitot tube and a test temperature well are located in the leakoff steam line to the extraction steam system.

2. Feedwater Pump Turbine Steam Seals

Gland steam header pressure is maintained at a 2 to 4 psig by modulation of the steam seal supply valve. A split range controller sets the air pressure at the control valve so that sufficient steam is admitted from the auxiliary Steam System (if needed).

Seal steam header pressure indication is provided locally and remotely in the Control Room.

The following system abnormalities are alarmed in the Control Room:

a. High auxiliary gland steam condenser level

b. Low seal steam supply header pressure

c. High differential pressure across the control valve air supply filter

d. High drip pot level

10.4.4 STEAM DUMP SYSTEM

10.4.4.1 Design Bases

To enable the NSSS to follow turbine load reductions which may exceed a 10-percent step or five percent per minute ramp (Section 10.2.1), the capability of creating an artificial steam load is

incorporated in the Steam and Power Conversion System. This load is created by dumping steam directly to the main condensers, thus bypassing the turbine.

Each unit of the CPNPP is provided with a steam dump system rated at greater than 40 percent of the steam flow at the ESF rating nominally reflected on the Valves Wide Open Heat Balance, Figure 10.1-2. This flow capacity permits the turbine to take a 50-percent load reduction without reactor trip. This system also allows a turbine and reactor trip to occur from full load without lifting the steam generator safety valves.

During normal reactor plant cooldown, the steam dump system is used to control cooldown rate by bleeding steam from the steam generators to the condensers.

Table 10.4-2 shows the design bases of the steam dump system.

The steam dump lines are designed in accordance with ANSI B31.1, Power Piping Code.

10.4.4.2 System Description

10.4.4.2.1 General

The steam dump system is shown on Figure 10.3-1 (Sheets 1 and 2).

Steam is discharged to the condensers through 12 air-operated, partial capacity, automatically controlled steam dump valves, which are installed in the steam dump lines located in the Main Steam Supply System between the steam generator isolation valves and turbine stop and control valves.

The steam dump valves are arranged in parallel so that, when combined, they permit the desired bypass flow to pass. This arrangement limits the steam bypassed to the condenser should a valve open accidentally or stick open, thereby minimizing the potential hazard of an uncontrolled cooldown rate of the primary system. This arrangement also permits the steam dump flow to be evenly shared by the turbine condensers, thus preventing uneven turbine exhaust back pressures.

The steam dump system is designed to bypass steam around the main turbine to the condenser to accomplish the following:

1. To permit a generator load rejection of up to 50 percent from maximum guaranteed power level without a reactor trip

2. To prevent lifting of the main steam safety and power-operated relief valves following a generator load rejection of up to 50 percent from the maximum guaranteed capability power level

3. To discharge to the condenser the steam generated during plant cooldown at a rate that limits thermal transients to allowable values; steam generator depressurization would continue in this mode down to a level in which the plant cooldown process is transferred to the RHR System

4. To maintain steam system header pressure at required pressure during startup, hot standby, and reactor physics testing periods

10.4.4.2.2 Steam Dump Valves

Twelve steam dump valves are provided for each unit, with each valve having a capacity of greater then 530,000 lb/hr. They are designed to fail closed on loss of either air or control signal and are split into two groups to ensure even distribution of heat into each condenser. These valves are arranged for automatic operation and for remote manual control from the control room. The dump valves are provided with local and remote open/closed position indicators. The dump valves also are capable of fast actuation or of operating on throttling service.

The steam dump valve actuation characteristics are as follows:

1. The valves are capable of going from full closed to required flow within three sec after receiving a trip/open signal. This includes the time required to actuate the solenoid valves associated with each dump valve.

2. The valves are capable of going from full open to full closed in five sec after de-energization of the solenoid valves.

3. The valves are capable of being modulated with a maximum full stroke time of 20 seconds to achieve required flow.

4. To ensure adequate dump control at a hot shutdown condition or during cooldown, the steam dump system is designed to control the total dump flows down to 80,000 lb/hr over the steam pressure in the range of 100 to 1300 psia.

5. The dump valves can be actuated either by the reactor coolant average temperature control logic or by the steam system header pressure depending on a selection made by the operator in the control room.

10.4.4.2.3 System Operation

During operating transients for which the plant is designed, the steam dump system is automatically regulated in the average reactor coolant temperature (Tavg) control mode to maintain the programmed Tavg.

A programmed reactor coolant reference temperature (Tavg) corresponds to each turbine load. During load variations when the reactor and turbine outputs are unbalanced, it deviates from the actual Tavg of the primary coolant. The magnitude and the rate of this deviation, which depends on the transients, provides a signal that selects a new control rod pattern and activates the adequate number and mode of operation for the dump valves.

During a load reduction, the valves are modulated by temperature deviation through a load rejection controller. When the mode selection switch is in the steam pressure, the valves are modulated to maintain the steam header pressure setpoint.

On large step load reductions (above 10 percent) or reactor trip, the steam dump valves open rapidly, in three seconds or less. During the three second period, while the turbine valves are

closing and dump valves are opening, there is a temperature rise in the NSSS and an approximately 100-psi pressure rise in the steam generators. In the initial part of a large step load transient, all dump valves go fully open. Then the valves are modulated closed in sequence to obtain a design load change in the reactor of five percent per minute. The valves are fully closed when the reactor power matches the turbine power.

Reactor trip transfers steam dump control from the load rejection controller to the plant trip controller. The reactor trip signals are redundant.

During hot shutdown conditions, the steam dump system can be operated in the steam generator pressure control mode. Under these conditions the steam dump system is set to maintain a selected pressure in the steam generators.

When the transient results in a unit trip, the operator transfers dump control to the pressure mode and the valves are regulated by a main steam pressure signal to maintain no-load pressure. Lower pressures can be maintained automatically by manual adjustment of the pressure setpoint by the operator in the control room. This remote manual operation is performed during startup, cooldown, and reactor physics test periods.

During primary plant cooldown, the steam dump system is operated in the steam generator pressure control mode. The pressure set point is manually reduced to achieve the required cooldown rate.

All dump valves fail closed on loss of control and they are prevented from opening on loss of condenser vacuum. During a loss of condenser vacuum, excess steam pressure is relieved to the atmosphere through the power relief valves or the safety valves, or both.

Valves are blocked or prevented from opening on low-low Tavg by redundant signals which are used to block valve opening. Three cooldown valves may be opened by the control board switches when there is a low-low Tavg, but the other nine dump valves stay blocked.

An adequate drainage system is provided upstream from each dump valve. The dump lines are normally stagnant and therefore produce condensate continuously. This condensate is automatically removed to permit proper system operation.

10.4.4.3 Safety Evaluation

The steam dump system is not essential to safe operation of the plant. It is required, however, to give the plant flexibility of operation.

Because the dump valves are subjected to modulating control, each of them is provided with isolation valves to permit maintenance. The flow capacity of each valve is selected to prevent excessive cooldown should the valve fail in the open position.

When all the valves are out of service, the steam generator safety valves provide the relieving capacity required to maintain the steam system within the design limits.

No effects of pipe breaks are considered since all piping is either located in the Turbine Building or in other areas where the effect of pipe breaks would not jeopardize the safe shutdown of the plant.

The steam dump system blowdown connections are located on the main condenser steam inlet sufficiently high above the condenser tubes in order to prevent direct impingement of the horizontally discharged high velocity steam on the condenser tubes or other condenser components; thereby precluding their failures from the steam dump system blowdown (See Figure 10.4-21).

Failure of the turbine bypass (steam dump) valves to open on demand does not affect operation of the turbine-generator unit, except that such a failure may cause a reactor trip (and subsequent turbine-generator trip) during primary to secondary plant power mismatches.

Failure of the turbine bypass (steam dump) valves to close will also lead to a turbine-generator trip following the reactor trip generated by low steam line pressure.

10.4.4.4 Tests and Inspections

Each dump valve is periodically tested. The test can be performed during power operation by shutting the associated isolation valve and locally checking for performance and timing. The dump valves are also operated during initial startup and during shutdown.

The steam dump lines are hydrostatically tested to confirm leaktightness.

Exterior visual inspection is used to check the condition of welds.

The steam dump valves were required to meet the following requirements during initial shop testing:

1. Hydrostatic tests in accordance with MSS SP-61

2. Seat leakage tests; 0.01 percent of maximum valve capacity with water at shutoff pressure

3. Operator leak tests; no visible leakage

4. Hysteresis and stroke speed tests at 25, 50, and 75 percent for upward and downward travel

10.4.4.5 Instrumentation Requirements

Indicating lights are provided in the Control Room for each dump valve to indicate when the valve is fully open or fully closed. They are also provided with local position indicators.

The magnitude of the modulating dump valve signal is indicated on the control board.

Control board lights are provided to indicate the status of permissive and interlock circuits.

Detailed descriptions of the steam dump instrumentation and controls are provided in Section 7.7.1.8. The logic diagram is shown on Sheet 10, Figure 7.2-1.

The low-low average temperature interlock (P-12) for steam dump block is the only safety-related instrumentation in this system and is designed in accordance with IEEE 279-1971.

10.4.5 CIRCULATING WATER SYSTEM

10.4.5.1 Design Bases

The Circulating Water System supplies approximately 1,100,000 gpm of cooling water to each unit. This flow is sufficient to remove the heat from the main condenser, the two auxiliary condensers, the turbine plant cooling water heat exchanger, the three condenser exhausting vacuum pump heat exchangers, and two Unit 1 or three Unit 2 non-safety ventilation chillers. The total heat removed amounts to approximately 8.8×10^9 Btu/hr, of which about 8.4×10^9 Btu/hr is removed from the main condenser. The Circulating Water System is supplied by the Squaw Creek Reservoir (SCR), which provides water at a design temperature of 95°F. The expected discharge temperature is an approximately 15°F temperature rise above the inlet temperature of SCR. The system is designed to operate with the water in the SCR at its lowest elevation of 770 feet.

10.4.5.2 System Description

Location of the circulating water discharge structure is shown on Figure 1.2-1. The Circulating Water System itself is shown on Figure 10.4-5. Water from the SCR flows to the eight circulating water pumps (both units) through heavy, steel bar trash racks and 12 traveling screens. The trash racks remove any heavy debris from the intake water while the traveling screens remove smaller debris which can also be present. The screens receive periodic cleaning, which is initiated automatically by means of a timer. When the screens become so clogged as to require cleaning prior to or during the periodic cleaning, a high differential pressure is indicated and backwashing is automatically initiated. If the differential pressure increases beyond the set point for automatic backwashing, an alarm is annunciated in the Control Room.

The backwash water is filtered and returned to the reservoir. For maintenance purposes, each screen well is provided with stop logs to allow dewatering of any individual screen well.

For each unit, the water from six screen wells flows to a common suction pit. Four motor-driven, vertical, centrifugal, mixed flow, circulating water pumps take suction from this pit. Each pump has 25 percent of the maximum capacity required for operation of one unit. This system is duplicated for operation of the second unit.

Both inlet and outlet connecting lines to the water boxes of the main and auxiliary condensers and the turbine plant cooling water heat exchanger are equipped with butterfly valves and expansion joints. Respective to each unit, one line branches off the intake tunnel and supplies water, which services the auxiliary condensers, the turbine plant cooling water heat exchanger, the condenser exhausting vacuum pump heat exchangers, and the non-safety ventilation chillers. There is a bleed line, which branches off the intake manifold of each unit, that can supply makeup water to the Safe Shutdown Impoundment (SSI).

The anticipated water quality of SCR is given in Table 10.4-3. Based on the anticipated water quality of the SCR, calculation of both stability and saturation indices indicate that the water tends to form scale and is corrosive.

Materials selected for use in this system are those that best withstand any long-term corrosion, erosion, pitting, or stress cracking when used with this water. The circulating water system is

composed of stainless steel, plastic, and carbon steel piping. Piping which is stainless steel, plastic, or carbon steel less than or equal to 2" in diameter is unlined. Carbon steel piping 2½" in diameter and larger is epoxy-lined. The main condenser water boxes are epoxy-lined. Titanium tubes are used throughout the condenser tube bundle assemblies in both the air removal and condensing sections. In addition, titanium clad carbon steel is utilized on the water box side of all tube sheets to minimize galvanic attack.

10.4.5.3 Safety Evaluation

The circulating water pumps are designed to remain operable with the water level at its lowest anticipated elevation. The pump motors and valve motor operators are supported so that no electrical parts are immersed in water at the highest anticipated elevation of the reservoir. See Section 2.4.11.6.

The Circulating Water System is not required for emergency cooldown or for operation of the engineered safeguard systems; instead, the Station Service Water System and the related SSI fulfill these functions.

The Circulating Water System is not required for cooling during shutdown because Reactor Coolant System (RCS) cooldown can be accomplished by dumping steam to the atmosphere through the power-operated main steam relief valves.

The results of a failure and its effects on safety-related systems will not adversely affect plant safety under any operating condition. Failure of the main condenser circulating water expansion joint can cause rapid flooding of the Turbine Building sump. Consequent flooding of safety-related equipment is prevented by tripping the Circulation Water Pumps before the flooding overflows the Turbine Building pit (elevation 778 feet, see Figures 1.2-26 and 27). A two out of three level trip logic will be used to automatically trip the Circulating Water Pumps for the affected unit.

The level trip has the following features to assure reliability:

1. The power supply for the trip circuit is an uninterruptible power supply 10kva inverter IV1C2 or inverter IV2C2 (Figure 8.3-13).

2. An alarm in the control room will be caused by a high water level trip condition or test bypass switch in the bypass position.

3. The level switches are the same as those qualified for 1E service. The qualification data is listed in CPNPP/EQR Table 5-1 (Purchase Spec MS-620).

4. The cabling from the level switches is in conduit and routed to protect it from jet impingement.

On shutdown of a circulating water pump, water hammer is avoided by ensuring that the pumps coast down as the pump isolation valves close. These valves are 90-inch-diameter butterfly valves which close in approximately 90 seconds. Condenser inlet and outlet water box isolation valves are also provided. On simultaneous trip of all four circulating water pumps, with no power available for closing the pump isolation valves, water hammer is avoided by a vacuum breaker on the outlet condenser water boxes. On startup, water hammer is avoided by the vacuum

priming of the Circulating Water System. This insures that air which can cause large pressure transients is removed. The valve is then opened, and the pump is started when the valve reaches 35 degrees open. The condenser tubes and water boxes are designed for a pressure of 40 psig, which is well above the maximum discharge pressure of the circulating water pumps.

The environmental considerations associated with disposal of waste heat from the CPNPP require dispersion of the warm condenser water in the SCR in order to provide protection of the aquatic life in the reservoir waters.

For each unit, the circulating water is returned to the SCR via a tunnel discharging into an open structure. This discharge structure greatly reduces the velocity of the circulating water from the pipeline to the end of the structure where the water flows into the reservoir. The discharge velocity is approximately 9.8 ft/sec. The low discharge velocity encourages stratification of the heated circulating water. This in turn promotes dissipation of the rejected heat by evaporation and heat transfer to the atmosphere. This mechanism involves the minimum amount of reservoir water in the heat dissipation process.

Corrosion effects are expected to be minimal because of the anticipated water quality and the material selection.

The circulating water is shock treated with a solutions of sodium hypochlorite and sodium bromide to reduce organic fouling.

10.4.5.4 Inspection and Testing Requirements

All active components of the system are accessible for inspection during station operation.

Model tests of performance were conducted in accordance with the Hydraulic Institute standards.

10.4.5.5 Instrumentation Requirements

A high differential pressure alarm is provided in the Control Room, across each condenser waterbox, to inform the operators of a water flow restriction. Local pressure, level, and temperature indicators at the waterbox inlet and outlet are provided. Plant process computer inputs are provided in the Control Room to record inlet and outlet water temperatures.

The circulating water pumps are individually equipped with motor-operated isolation valves. The main and auxiliary condenser waterboxes and TPCW heat exchangers are equipped with gear-operated isolation valves which enable the equipment to be isolated.

The circulating water pump discharge valves are interlocked to open and close on pump start/ stop signals.

There is no safety-related instrumentation in this section.

10.4.5.6 Circulating Water Chemical Treatment Subsystem

10.4.5.6.1 System Description

Circulating water at CPNPP is drawn from the SCR and returned without concentrating dissolved solids or adding chemicals for pH and corrosion control. A solution of sodium hypochlorite and sodium bromide is added to the water to control organic and biological growth in the cooling system.

Figure 10.4-6 shows the major components and flow path for the circulating water chemical treatment subsystem. Sodium hypochlorite and sodium bromide are drawn from storage tanks by 100% capacity pumps and distributed to the circulating water at the intake bays.

10.4.6 CONDENSATE CLEANUP SYSTEM

The Condensate Cleanup System (CCS) of the CPNPP is provided for the removal of both dissolved and suspended solid contaminants in the condensate. The water used for backwashing may be discharged or reclaimed. The system components are located in the Turbine Building at elevations 803 ft and 778 ft, except for the hot phase separator, which is located in the Fuel Building at elevation 860 ft. One CCS is provided for each unit, with the exception of the hot phase separator which serves both units.

This system is not nuclear-safety-related. However, the CCS can, at times, be exposed to low levels of radioactivity caused by leakage of the primary coolant (in the steam generators) to the secondary coolant (condensate). Many radionuclides behave as dissolved solids and thus are adsorbed by the ion exchange resin. The system components are therefore designed to be resistant to the effects of short term exposure to low radioactivity levels. Seismic design criteria are not considered.

10.4.6.1 Design Bases

1. Flow Rate

Total system	100 percent of the condensate flow equal to 21,000 gpm at 150°F
Each vessel	5250 gpm (with four vessels operating in parallel and one vessel on standby)

2. Impurity Levels

The effluent water quality meets the secondary feedwater chemistry specifications of the EPRI guidelines, except where CPNPP evaluations justify exceptions.

3. Codes

The design codes applied to the CCS are as follows:

General	OSHA

Pressure vessels	ASME B&PV Code, Section VIII
Atmospheric tanks	API 620
Piping	ANSI B31.1
Materials and testing	Applicable ASTM and ASME standards
Electrical and control	Applicable IEEE and ISA standards

10.4.6.2 System Description

1. General

The CCS process flow diagram and equipment layout is shown on Figures 10.4-7 and 1.2-23 respectively.

The CCS normally uses powdered ion exchange resin coated on special filter elements to simultaneously filter and demineralize the condensate.

The main components of the CCS are polisher vessel skids, precoat tank and pump skids, overlay tank and pump skid, backwash pumps and tanks, spent resin tank and pump skid, conductivity sampling equipment, and a Programmable logic controller (PLC) panel and instrumentation.

The system has the following four subsystems:

a. Main cleanup vessels

b. Backwash and precoating facilities

c. Backwash recovery/waste facilities

d. Resin disposal facilities

2. Normal Operations

The condensate flows through the inlet header to the online demineralizer vessels. A flow metering and balancing system regulates the condensate flow through each vessel. The condensate enters each filter vessel in service from the outside to the inside of several hundred filter elements. The powdered resin coating is located on the outer surface of the filter elements. The processed condensate then flows to the system outlet header.

As suspended and dissolved matter are picked up by the resin, the pressure drop across the vessel increases. At a preset pressure drop, a high differential pressure alarm sounds notifying the operator to remove the vessel from service.

The conductivity and other selected chemistry parameters of each vessel outlet is monitored. A vessel high-conductivity alarm is available to notify the operator to remove

the vessel from service. Normally the vessels are changed out based on steam generator chemistry, vessel outlet conductivity and sodium, vessel differential pressure, and/or run time. Also, the sodium ion concentration of a single selectable vessel outlet is monitored.

During low flow or off-stream (standby) conditions, a holding pump recirculates condensate at a nominal flow required to maintain the resin coating on the elements of that vessel. The pump is automatically controlled by a low flow alarm signal.

3. Backwash and Precoating

An automatic system is provided to conduct the backwash and precoat operations of the off-stream vessel in the following sequence: the vessel is backwashed, vented, rewashed, filled, precoated, and placed in a standby condition or put into service. The backwash pump takes suction from the demineralized water storage tank and discharges through the vessel to the vessel backwash drain line.

The backwash air blower supplies air under pressure to the backwash header. The water air mixture flows from the inside to the outside of the filter elements, scouring the resin coating from the elements' outer surface. The backwash effluent is directed to the drain and finally to the A tank of the recovery system.

During this process, the vent valve remains open, allowing air to escape from the vessel. The vessel vent is also directed to the A tank. The backwashing process is repeated again; this is known as the rewash step. The rewash effluent is directed to the B tanks. The rewash step, which may be repeated, is of slightly longer duration than the backwash step. The backwash water from the cycle is normally discharged. The pump and the air blower are then stopped, and the vessel is drained. Then refilled with demineralized water.

The next stage of operation in the automatic cycle is precoating. The precoat media slurry, normally powdered resin, is pumped by the precoat pump through the precoat inlet header to the vessel elements where the resin is deposited on the elements and following the precoat cycle, is kept in position by the flow from the holding pump. The precoat water is routed back to the precoat mixing tank and the automatic cycle is terminated by putting the regenerated vessel in a standby condition.

4. Backwash Recovery

The backwash recovery system, working in conjunction with the exhausted vessel, may be used in incidents of severe water shortage to recover approximately 95 percent of the backwash water and produces a settled spent resin sludge of about 20 percent by weight.

Prior to backwash and precoating of the exhausted vessel, the backwash recovery process is initiated. The operator initiates the sequence to open the appropriate valves and to start the backwash recovery pump which draws water from the A tank supernatant and both B tanks.

Recirculation is established from the backwash recovery pump through the exhausted Powdex vessel back to the A or B tank. This recirculation serves to check the recovered backwash water conductivity.

Subject to an acceptable recovered water quality, the flow is diverted to the condenser deaeration section until the contents of the backwash recovery A or B tanks have been depleted, as signaled by closure of a low-level switch contact.

In the event of an unacceptable conductivity test, or other applicable chemistry parameters, for the recovered backwash water, a small quantity of powdered resin is mixed in the overlay tank, to be applied to the exhausted precoat in the vessel, for the purpose of increasing the ion exchange capacity of the exhausted vessel. The overlay step terminates upon closure of the low-level switch on the overlay tank, and the regular recovery process proceeds. If the total suspended solids count is higher than parameters allow then a temporary filtration system may be used to filter the total suspended solids out. This situation would occur in the event of a primary to secondary reactor coolant leak.

5. Resin Disposal

The A tank is used to hold the resin laden portion of the backwash stream and other streams liable to contain resin. The A tank receives 95 percent or more of the spent resin precoat. The conical bottom and the variable speed scraper of the A tank helps settling and thickening of the resin slurry. The settled resin slurry is then pumped by the spent Powdex pump to the phase separator tank. The two B tanks are used for holding the resin free portion of the backwash until it is discharged or recovered.

The phase separator tank is used as a final settling and decanting tank of the A tank resin slurry. The phase separator resin slurry underflow is pumped by the phase separator pump to the Low Volume Waste Treatment Facilities while the liquid portion decants or overflows to the A tank. Requirements for sampling the resins for radioactivity prior to transfer to the LVW Treatment Facilities and a limit on the total inventory of radioactive material contained in powdex resins allowed in the pond are established in the Radiological Effluent Controls Program required by the Technical Specifications.

When activity in the resin slurry exceeds the Radiological Effluent Controls limits, the resins are transferred to a disposal container for processing as radioactive solid waste.

10.4.6.3 Equipment Design

The design criteria for the Holding Pumps and the Polisher Vessels is a design pressure of 600, psig. The design pressure for the Backwash Pump, Precoat Pump, Backwash Recovery Pump, Spent Powdex Pump, and Phase Separator Pump is 150 psig. The Precoat Tank, Overlay Tank, Backwash Recovery A Tank, Backwash Recovery B Tank, Phase Separator and Hot Phase Separator are designed for Atmospheric Pressure. The Safety Class for all the previously referenced equipment is NNS.

10.4.6.4 Safety Evaluation

Since the CCS is not required for maintenance of plant safety, the system can be bypassed for short periods of time (provided there is no condenser leak, or chemistry upset) without significantly affecting plant operations. The system permits limited plant operation during condenser inleakage (depending on the magnitude of the leak) by reducing contaminants in the condensate to minimize the impact on steam generator chemistry.

The effluent water quality meets the secondary water chemistry specifications of the EPRI secondary water chemistry guidelines, except where CPNPP evaluations justify exceptions.

It is estimated that the contribution of impurities from the secondary side of the RCS is negligible because of the use of the CCS, SGBS, all volatile secondary-side chemistry specifications, and infrequent occurrence of pressure differentials for secondary to primary leakage.

Continuous conductivity and differential pressure monitoring ensure the proper functioning of the CCS.

10.4.6.5 Instrumentation and Control

1. Flow Instrumentation

   A flow control system is provided to equalize the flow rate of condensate to each onstream vessel. Vessel flow rate is regulated by a proportional signal from the flow control system to the vessel outlet valves. To continually maintain the precoat on the filter elements, holding pumps (one for each vessel) are activated when flow to a vessel decreases below a preset minimum. Flow instrumentation also includes rate-of-flow indicators for backwash water and air.

2. Pressure Instrumentation

   Differential pressure across each vessel and the entire system is continuously monitored. High differential pressure across a filter vessel causes an alarm to sound, indicating to the operator that the service run of that vessel should be terminated. By using a differential-pressure-indicating controller to automatically open the system bypass valve on increasing system differential pressure, a pressure drop of $\leq$40 psi is constantly maintained.

3. Conductivity Instrumentation

   Cation conductivity is monitored at the system inlet, system outlet, and at each vessel effluent outlet. High conductivity at any analysis point sounds an alarm to notify the operator to either remove a vessel from service or to look elsewhere in the condensate system for possible trouble (i.e., condenser leakage).

   The conductivity rack also contains a stainless steel sample sink for manual sampling.

4. Control Panel

Two redundant state-of-the art local Programmable Logic Controller (PLC) panels, each with an Operator Interface (CRT based) Unit, are provided for each CCS. The two PLCs are cross linked to serve as a back-up for each other. They are mounted independently of the system skids.

The PLCs operate all CCS equipment automatically and precisely, cycle after cycle. Precoat and Backwash cycles are controlled automatically. Display of equipment status, indications of process variables, and alarm status is provided on the CRTs. They also provide Data Acquisition, Trending and Report Generation.

A common trouble alarm is provided in the Control Room.

5. Sodium Instrumentation

Sodium ion concentration is monitored locally at each vessel effluent. A single sodium ion analyzer is provided for this purpose. Each individual vessel effluent may be selected for sodium ion analysis.

10.4.6.6 Tests and Inspections

The polisher vessels and piping are hydrostatically tested in the assembly shop at 1.5 times the system design pressure. The control system is also tested in the shop to ensure proper valve and pump operation sequences. After installation, the system is again hydrostatically tested.

The filter elements are inspected both when installed and at periodic intervals for physical defects which could cause excessive pressure drops or loss of powdered resin. The expected life of the filter element is five to eight years.

In the event the backwash recovery is used, conductivity, millipore, and/or other applicable tests are conducted on the recovered backwash water to determine whether it should be diverted to the condenser deaeration section or be subjected to further processing.

10.4.6.7 Other Features

The CCS design takes into consideration portions of the recommendations of NRC Regulatory Guide 1.56, positions 1 through 4, to the extent of their applicability to PWRs with recirculating-type steam generators. Position 5 is not applicable for this system.

The CCS is located in the Turbine Building. Figure 1.2-23 of the CCS and the Turbine Building shows that no nuclear-safety-related component or system is located close to the high-or moderate-energy pipes of the CCS. The design of the CCS meets Branch Technical Positions APCSB 3-1 and MEB 3-1, as detailed in Section 3.6.

10.4.7 CONDENSATE AND FEEDWATER SYSTEMS

The Condensate and Feedwater Systems return the condensate from the turbine condenser hot wells through the regenerative feed heating cycle to the steam generators while maintaining the water inventories throughout the cycle.

10.4.7.1 Design Bases

The Condensate and Feedwater Systems are designed to provide approximately 16.32×10^6 lb/hr of feedwater at 446.7°F to the steam generators during steady-state operation at the 100% NSSS power (3628 MWt) with 66°F Circulating Water Inlet Temperature.

The condensate portion of the system is designed to supply approximately 9.976×10^6 lb/hr to the suction side of the steam generator feedwater pumps during steady-state operation at maximum guaranteed turbine load. In addition, the Condensate System in conjuction with the Heater Drain System can supply 96 percent of the full-load feedwater flow (15.667×10^6 lb/hr) to the steam generator feedwater pumps during load-drop transients.

The feedwater portion of the system is designed to supply the feedwater required for various loads at steady-state operation and to maintain this flow, as required, during the steam dump conditions following a large load reduction. The system is designed to maintain uniform feedwater flow to all steam generators under all conditions and to maintain proper steam generator water levels automatically during steady-state and transient conditions.

The Condensate Storage Tank is designed to store supply water for the Condensate and Feedwater Systems and the Auxiliary Feedwater System (see Section 9.2.6).

Condensate is transferred from the auxiliary condensers to the main condenser shell A hot well by gravity. The design parameters for the major components of the Condensate and Feedwater Systems are given in Table 10.4-5.

The only-safety related portion of the Condensate and Feedwater Systems is the feedwater piping portion between the moment restraint upstream of the Feedwater isolation valve and the steam generator feedwater nozzle. This portion is designed to seismic Category I requirements and is designed to withstand the worst anticipated environmental phenomena taken individually. The Condensate Storage Tank is designed to seismic Category I requirements and designated Safety Class 3. It is considered part of the Auxiliary Feedwater System.

The Condensate and Feedwater System components are designed and constructed in accordance with the following applicable regulations, codes, and standards:

1. Code of Federal Regulations, 10 CFR Part 50

2. Branch Technical Positions APCSB 3-1 and MEB 3-1 [6] [7]

3. ASME B&PV Code

Section III	Rules for Construction of Nuclear Power Plant Components, Division I, subsection NC, Class 2 Components
Section XI	Rules for Inservice Inspection of Nuclear Power Plant Components
Section VIII	Pressure Vessels (Division 1)

4. NRC Regulatory Guides

1.26	Quality Group Classification and Standards for Water, Steam & Radioactive-Waste containing components of Nuclear Power Plants. [3]
1.29	Seismic Design Classification [5]
1.32	"Criteria for Safety Related Electric Power Systems for Nuclear Power Plants" [22]
1.47	"Bypassed & Inoperable Status Indication for Nuclear Power Plant Safety Systems" [23]
1.53	"Application of the Single Failure Criterion to Nuclear Power Plant Protection Systems" [24]
1.75	"Physical Independence of Electric Systems" [25]

5. American National Standards Institute (ANSI)

B31.1	Power Piping
N18.2	Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants [4]
N271	Containment Isolation Provisions for Fluid Systems

Additional requirements for the portion of the Feedwater System which is within the Containment boundary are given in Section 10.3.6.

10.4.7.2 Systems Description

The Condensate and Feedwater System flow diagrams are shown on Figure 10.4-8, and 10.4-9, respectively.

1. Condensate System

Two motor-driven, constant-speed, vertical, canned-type condensate pumps are supplied, each designed for approximately 65 percent of the total condensate flow. These pumps withdraw condensate from the two main condenser hot wells via a common discharge arrangement which cross connects the two condenser shells, minimizing level differences between the two hot wells.

Both pumps are vented to main condenser shell B to prevent air binding in the pumps. Seal and priming water are supplied to the condensate pumps from the Condensate Storage Tank. The condensate pumps discharge into a common 30-in. header, which carries the flow to the Condensate Polishing System full-flow condensate filter-demineralizer vessels. The Condensate Polishing System is described in Section 10.4.6.

The total condensate flow rate is measured in the 30-in. line from the condensate pump discharge header to the condensate filter demineralizers. This measurement is made to ensure that the flow rate does not fall below the minimum flow requirements of the condensate pumps (2900 gpm per pump). Should the demands of the Feedwater System be low, the recirculation line valve upstream from the drain coolers opens, allowing the additional flow to return to the condenser.

At the outlet of the filter demineralizers, the flow divides to provide cooling water to both the main and auxiliary gland steam condensers under normal- and minimum-recirculation condensate flow conditions. These lines rejoin the balance of the condensate flow in the 30-in. header downstream of the main and auxiliary gland steam condensers. The flow is then divided into two approximately equal streams, each passing in succession through the tube side of the drain cooler and two stages of regenerative heating provided by feedwater heaters Nos. 5 and 6. A cross connection between the inlets of heaters 4A and 4B promotes equalization of temperature. The flow remains separated as it passes through the next two stages of feedwater heating provided by feedwater heaters Nos. 3 and 4.

The condensate flow is then distributed equally to the suction side of the feedwater pumps after passing through a flow measuring device, located in each pump suction line, used to control feedwater pump recirculation. Specific pairs of feed water heaters in each train are provided with normally closed bypass valves which allow for the removal of these heaters from service for maintenance or repair.

An emergency low-pressure heater bypass is included in a line branching from the condensate pump discharge header upstream from the Condensate Polishing System. The line discharges downstream of the flow element located in the heater drain pump discharge line. The flow is then directed to the steam generator feedwater pump suction lines. Under transient conditions, such as a 40-percent steam dump, fluid in the heater drain tanks may flash, causing a low water level and tripping the heater drain pumps resulting in the loss of their feed flow to the feedwater pump suctions. The loss of this flow (6,346,011 lb/hr at full load) would rapidly lead to a loss of pressure at the feedwater pump suctions. This in turn would result in the loss of feedwater flow, and then a turbine trip. To avoid a turbine trip, the bypass is sized so that 15,667,000 lb/hr (96 percent of full flow) is available to the feedwater pumps for supplying steam generator requirements during a load drop transient. The bypass valve opens fully within 2.5 seconds of receipt of the initiating signal.

The exhaust steam from the steam generator feedwater pump turbine drivers is condensed in the two small auxiliary condensers. The condensate from the auxiliary condenser hot wells normally drains under gravity to the main condenser hot well.

Condensate is obtained from the Condensate Storage Tank upon receipt of a low-level signal from the main condenser shell B hot well. Excess condensate is returned to the Condensate Storage Tank through the condensate pump reject control station.

The 150-gpm auxiliary boiler feedwater pumps obtain suction from the condensate makeup and rejection line.

Demineralized water from the Demineralized and Reactor Makeup Water System, which serves both units of CPNPP, can be introduced into the main condenser hot wells via the Condensate Storage Tank.

All normally open Condensate System Condensate Storage Tank connections are set 23 ft 9 in. above the bottom of the tank to prevent automatic drawdown below this level, thus ensuring that a storage of approximately 282,000 gal of water remains in the tank for possible emergency plant shutdown which exceeds the Technical Specification volume requirements. This includes 12,900 gallons of residual capacity which is below the critical depth and is therefore not considered available.

The design basis for the minimum storage capacity is as follows:

a. a volume of water required to maintain the reactor at hot standby for 2 hours followed by a cooldown to 350°F at a rate of 50°F/h for 5 hours with allowance for feedwater line break spillage.

b. volume of water required to maintain the reactor at hot standby for 4 hours followed by a cooldown to 350°F at a rate of 50°F/h for 5 hours.

All heaters are provided with tube side-safety valves to provide thermal relief by guarding against possible overpressurization caused by heating of water trapped between closed isolation valves.

2. Feedwater System

The Feedwater System is of the closed-cycle type and receives water from the Condensate System and the Heater Drains Systems (specifically, drains from heaters 1, 2, and 3, and MSR Separator drain tanks). The feedwater is transported through the final two stages of feedwater heating to the steam generators.

During power operation each steam generator feedwater pump takes suction from the Condensate System and discharges through a common header to the high-pressure feedwater heaters. One steam generator feedwater pump is required to operate during power operation of up to 50% of the rated power. From 50-percent power to full power, both steam generator feedwater pumps are required, with each pump providing 50-percent of the required flow. Prior to aligning feedwater pumps to the steam generators, water is supplied by the Auxiliary Feedwater System.

Each steam generator feedwater pump is fitted with a minimum recirculation control system which protects the pumps from damage at low loads by ensuring a minimum flow.

Leakages through the pump are detected by monitoring the temperature of the seal injection system drains.

The dual admission feedwater pump turbine drivers operate with steam from two sources. During low-load conditions, high pressure steam is supplied to the turbines from the Main Steam Supply System steam dump header. During normal operation, low-pressure steam is supplied from the MSR in the Main Steam Supply System. Gland steam is provided to the turbines from the turbine gland steam seal supply system.

The flow from the two steam generator feedwater pumps combines at the pump discharge, then divides into two streams for the final two stages of regenerative feedwater heating (heaters Nos. 1 and 2). The two heater trains and the common bypass join downstream of the high-pressure heaters to form a single common header for temperature equalization. From this common header, an individual feedwater line supplies each steam generator.

When originally constructed both Unit 1 and Unit 2 incorporated a Feedwater Bypass System on each main feedwater line to a steam generator. The function of the Feedwater Bypass System is to minimize the potential occurrence of a water hammer in the steam generators, and to mitigate the flow induced tube vibration in the steam generators.

With the installation of Delta 76 steam generators on Unit 1, the Feedwater Bypass System is no longer required for the new feedring steam generators and has been removed from Unit 1 with the exception of the small bypass line around the main feedwater isolation valves. (Note: The Unit 1 feedwater system is operated in a different manner after RSG installation. The FIV remain open during startup and a small flow through the valve and not the FIBV is used for purging purposes.) Unit 2 still retains the Feedwater Bypass System in its entirety.

The Unit 2 Feedwater Bypass System associated with an individual Main Feedwater line consists of three lines with associated instrumentation and controls as follows:

The Unit 2 feedwater preheater bypass line connects each main feedwater line, just upstream of the main isolation valve, to the auxiliary feedwater nozzle in the upper portion (above normal water level) of each steam generator. Each bypass line has its own containment isolation valve (Feedwater Preheater Bypass Valve) which is air operated. The objective of this bypass line is to minimize the potential occurrence of pressure transients and to prevent feedwater from entering the main feedwater nozzle at startup and during certain other operating conditions.

The Unit 2 feedwater split flow bypass line connects the main feed line to the feedwater preheater bypass line inside containment to provide a continuous feedwater split flow during normal plant operating conditions in order to minimize thermal transients in the nozzle and connecting piping when flow is transferred to the auxiliary FW nozzle from the main FW line and to minimize the flow induced tube vibration in the steam generators. It contains an Annubar flow element and an air operated butterfly valve.

In addition, a flow restricting orifice has been installed in each Unit 2 main feedwater line just downstream of the bypass line connection to facilitate the required flow split.

In addition, a small bypass line around the feedwater isolation valve is provided to purge cold feedwater from the main feedwater line between the isolation valve and the steam generator feedwater nozzle. This line incorporates a restricting flow orifice and an air operated globe type shutoff valve which also serves as a containment isolation valve.

Connections from the Condensate System are provided both upstream and downstream of the heaters to permit flushing of the heaters. The upstream connection can also be used to fill the system and the steam generators and to provide feedwater directly from the condensate pumps during the early stages of startup.

Full flow flushing of the Condensate and Feedwater Systems is provided to permit piping cleanup before plant start-up. Full flow flushing is accomplished through a 24" diameter pipe routed from the outlet of the HP feedwater heaters to the condenser. Bypass lines with isolation valves are provided around each steam generator feedwater pump for use during flushing operations.

Condensate and feedwater chemistry are controlled as described in Section 10.3.5. This system uses all volatile chemical treatment in conjunction with a Condensate Polishing System, which is described in Section 10.4.6.

Chemical feed to Unit 2 steam generators is introduced at a point upstream of the auxiliary feedwater supply connection to the Feedwater System for adjustments to the water chemistry during steam generator layup periods only.

3. Electrical Systems

Safe shutdown of the plant does not rely upon the availability of either the Condensate System or the Feedwater System. However, the portion of the Feedwater System from the moment restraint upstream from the feedwater isolation valve to the steam generator feedwater nozzle is nuclear-safety-related and automatic isolation of these lines is required during emergency conditions. The split flow bypass line (Unit 2 only) also requires isolation during emergency conditions. To ensure isolation of these lines, the feedwater isolation valves, feedwater isolation bypass valves, feedwater split flow bypass valves (Unit 2 only), and steam generator preheater bypass valves (Unit 2 only) are all tripped closed by separate, redundant control circuits and solenoids and are powered from two Class 1E-125 VDC power systems, maintaining the proper train separation. The independence and redundance of these two Class 1E-125 VDC power systems is described in Section 8.3.2.

In addition, the feedwater control valves and feedwater control bypass valves are also tripped closed by separate, redundant control circuits and solenoids and are powered from two class 1E 125 VDC power systems, maintaining the proper train separation.

The rest of the valves and equipment in the condensate and feedwater systems are powered from non-Class 1E systems.

10.4.7.3 Safety Evaluation

The requirements of 10 CFR Part 50, GDC 57, for Containment isolation are satisfied by one stop valve on each feedwater line outside the Containment (see Section 6.2.4.). With loss of flow in the normal direction, the check valve upstream of the stop valve closes and is held closed by back pressure. In addition, the stop valve can be closed and secured by remote operation.

The Feedwater System from the steam generators, back to and including the moment restraint upstream of the feedwater isolation valve, is designated as Safety Class 2 and is designed to the requirements of seismic Category I systems (see Section 3.2.1). For an analysis of the effects of a break in the Safety Class portion of the Feedwater System, see Section 10.4.9. The remaining portion of the Feedwater System piping inside the Safeguards Building is designated non-nuclear safety seismic Category II. The portion of the system in the turbine building and the Condensate

System is non-seismic except for the Condensate Storage Tank which is designated Safety Class 3, seismic Category I.

The Condensate and Feedwater Systems are designed to the requirements of the codes listed in Subsection 10.4.7.1. The potential for pipe rupture caused by internal pressure and temperature is as discussed in Section 3.6. Short lengths of pipe are installed above the turbine operating floor and could be fractured by a heavy object carried by the gantry crane or by a turbine-generated missile; but such an occurrence is extremely unlikely. A rupture of the condensate piping anywhere in the Turbine Building does not cause failure of safety-related equipment as a result of flooding, as no such equipment is installed therein.

In the Safeguards Building the feedwater pipes are, for most of their length, enclosed in separate, reinforced concrete ducts fitted with individual drains. Elsewhere, floor drains are designed to collect any discharge from the pipe, preventing damage to safety-related equipment resulting from flooding.

Any failure in the non-safety-class portion of the Condensate and Feedwater Systems has no effect on the safety of the reactor, which can be shutdown in an orderly manner. (See Section 3.6 for a further discussion of postulated pipe rupture.) A source of feedwater supply to the steam generators is required for decay heat removal from the reactor following a unit shutdown. In the event that the Condensate and Feedwater Systems are not available, the Auxiliary Feedwater System (see Section 10.4.9) provides the required emergency supply of feedwater.

Condensate available for emergency purposes is stored in the Condensate Storage Tank (see Section 9.2.6), which is a seismic Category I structure (see Section 3.2.1).

Although unlikely, a small amount of radioactivity may be present in the Condensate and Feedwater Systems in the event of a steam generator tube leak. Water from a pipe leak or break in the Condensate and Feedwater Systems is collected by the Equipment and Floor Drainage System. These drains are monitored for radioactive releases and are handled as described in Section 9.3.3.

The Unit 1 Delta 76 RSGs incorporate a feedring design in lieu of a preheater design and this feedring design feature prevents or mitigates the possibility of a steam generator water hammer event (CP1-RSG-05-184).

Significant flow instabilities due to steam void collapse (i.e., feedline waterhammer) are not expected to occur in the Unit 2 main feedwater system during normal operating transients due to the geometry of the system. The pre-requisities for such events have been shown to be (Refs. 18 and 19); uncoverage of the main feedwater inlet nozzle to the steam generator, thus allowing steam to enter the feedwater line, in conjunction with injection of cold feedwater. The design of the steam generator employed on Unit 2 has the main feedwater inlet nozzle located close to the tube sheet (see Figure 5.4-4). Accident analysis indicates that uncoverage of the nozzle is limited to a highly improbable sequence of events associated with some faulted condition transients. The results of a comprehensive test program have been used to demonstrate (Ref. 19) that the worst case waterhammer under such conditions will not result in a loss of primary circuit pressure boundary integrity under associated faulted conditions. Nevertheless, additional means of preventing the conditions under which waterhammer may occur have been implemented. Loop seals located at the feedwater inlet nozzles to the Unit 2

steam generators minimize the volume of feedwater piping that could self drain and fill with steam (see Figure 10.4-22). Furthermore, system design modifications have been implemented on Unit 2 which preclude the introduction of cold feedwater through the main feedwater inlet nozzle. These modifications, described in Section 10.4.9, consist of a feedwater bypass system to a separate auxiliary feedwater inlet nozzle (Figure 5.4-4).

Tests have shown (Ref 19) that pressure transients, due to steam void collapse, can occur in the steam generator and main feedwater piping if feedwater below 250°F is supplied through the main feedwater nozzle concurrent with low steam generator water level (at or below the level of the main feedwater nozzle in the preheater region) or low steam generator pressure. Although analysis of these test results has demonstrated (Ref 19) that the maximum pressure transients produce stresses below the allowable limits, a feedwater bypass system has been incorporated on Unit 2 to minimize the possibility of steam generator preheater and feed water piping pressure transients. The feedwater bypass system consists of a connection between the auxiliary feedwater nozzle and the main feedwater line upstream of the feedwater isolation valve on each steam generator. This bypass line contains a feedwater preheater bypass valve. The feedwater bypass system also include a feedwater isolation bypass valve.

To minimize pressure transient potential, it is necessary to prevent the introduction of cold water to the Unit 2 steam generator through the main feedwater nozzle at any time when significant void may be present. Therefore, total feedwater flow is not aligned to the normal feedwater nozzle during a startup until the feedwater line temperature is above a set limit. Conversely, feedwater flow is diverted to the auxiliary feedwater nozzle from the main feedwater nozzle during a shutdown.

The feedwater split flow bypass line inside the containment, which connects the main feedwater line and the feedwater bypass line, is designed to minimize the thermal transients in the steam generator nozzles and to preclude the flow induced tube vibration in the pre-heater section of the steam generator (see reference 21) by maintaining a feedwater flow split during feedwater injection through the main feedwater line. Main feedwater flow and feedwater split flow bypass flow are monitored during power operations to ensure an acceptable flow rate relationship exists between the two flow paths. A high flow alarm is available to identify when the normal main feedwater flow is exceeded. A vendor assessment of the potential for pre-heater region tube wear due to associated tube flow induced vibrations has been performed; and guidance is provided to establish a flow-time limit criterion in the event of a flow excursion. In addition, selected steam generator tubes in the preheater section have been expanded at two support plate locations to minimize vibration (reference 20). To achieve the required flow split, each split flow bypass line has been modified to minimize hydraulic resistance by incorporating an air-operated butterfly valve and an Annubar flow element to monitor and modulate the flow rate.

In addition, a flow restricting orifice has been installed in each main feedwater line just downstream of the bypass line connection to facilitate the required flow split.

10.4.7.4 Tests and Inspection

Each feedwater heater, drain cooler, pump, and valve receives a shop hydrostatic test. Prior to initial operation, the completed Condensate and Feedwater System is to receive a field hydrostatic test and inspection. These tests are performed according to the applicable codes listed in Subsection 10.4.7.1. Periodic tests and inspections of the system are to be performed in conjunction with scheduled maintenance outages.

Periodic in-plant tests are conducted to demonstrate the ability of the feedwater isolation valves to respond to a test close signal. The valves must close within the specified time. The valves are designed and constructed with provision for periodic inservice testing of partial valve stroke. Provisions are made in the trip mechanism for all solenoid-operated valves to be exercised without interrupting availability of the trip mechanism.

10.4.7.5 Instrumentation Requirements

Condenser hot well level is maintained by adding water on low level by opening makeup control valves from the Condensate Storage Tank and rejecting water to the Condensate Storage Tank on high condenser hot well level. The makeup control valves are modulated open by split-range signals from a level controller in the control loop with a level transmitter on condenser B hot well. The condensate reject valve is opened and closed by level switches on condenser shell A hot well. Tripping of both condensate pumps will close the condensate reject valve.

Makeup water can be manually added by a control board hand auto station. Water can be rejected manually by an open close switch on the control board.

A condensate gland steam condenser bypass valve is used to prevent excessive flow through the gland steam condenser in Unit 2. An orifice in the main condensate line is sized to provide design flow to the aux and gland steam condensers (Unit 1).

Condensate pumps are tripped by a condenser hot well low level. Failure of motor-operated discharge valves to open to a minimum position will prevent pump start. Minimum flow protection is provided by an on-off recirculation control valve. If one condensate pump trips with the plant above 50 percent load, the result is an automatic runback of the main turbine. However, automatic turbine runback initiated by a condensate pump trip is not sufficient to preclude the unit from tripping.

For Unit 1, the emergency low pressure heater bypass valve opens in less than two and a half seconds after receiving two out of three coincident feedwater pump low suction pressure signals. For Unit 2, the emergency low pressure heater bypass valve opens in less than two and a half seconds after receiving potential two out of three coincident feedwater pump low suction pressure signals. Possible water damage to the turbine caused by high water levels in heater Nos. 5A, 5B, 6A, and 6B, located in the condenser neck, is prevented by means of motor-operated isolation and condensate bypass valves.

Minimum flow protection for heater drain pumps is provided by a modulating recirculation control valve; this valve closes on pump trip.

The feedwater pump turbine control maintains proper pressure differentials across the feedwater control valve by using the input signals of the steam generator header pressure and the set point differential pressure obtained as a function of the sum of all four of the steam loop flows.

In case of loss of feedwater control to the feedwater pump turbine or excessive noise, turbine speed signal protection is provided by an automatic signal shift to a speed signal memory which retains the last functioning signal. Loss of signal is alarmed, and the feedwater pump turbine speed automatically transfers to the manual mode of control (also alarmed).

The feedwater pump turbine interlocks include turbine trip on safety injection signal, turbine trip on high-high steam generator level, and turbine trip on low feed pump suction pressure. Trip of both turbines starts the motor-driven auxiliary feedwater pumps. Trip of a single turbine causes a runback of the main turbine. The turbine trip pressure switches, which are used to start the auxiliary feedwater pumps, are redundant Class 1E switches. Feedwater control and bypass valves are tripped closed by the same signals that trip close the feedwater isolation valves. Redundant, Class 1E, “close” solenoids and associated circuitry are provided for each valve assuring the transmission and receipt of a “close” signal at each valve via at least one of two trains in spite of any single failure in any one train.

The feedwater control valves are regulated by the Westinghouse control system. The Westinghouse system is a three-element control which uses signals from the steam generator water level, the steam flow, and the feedwater flow. Below 25 percent load, the feedwater control bypass valve maintains the steam generator water level by using a control from the steam generator water level. Transfer from the feedwater control bypass valve to feedwater control valve is manually initiated.

Feedwater isolation valves are tripped closed upon receipt of the following safety-related instrumentation signals: a steam generator high-high level (two out of three high-high level signals from any steam generator), a safety injection signal, or a low average temperature with reactor trip. When the plant is operating these valves can be tested for partial closure of the valve. Feedwater isolation bypass valves, feedwater control valves, feedwater control bypass valves and feedwater preheater bypass valves (Unit 2 only) are all tripped closed by separate, redundant control circuits and solenoids.

Based on section 10.4.7.3 Unit 1 does not require any instrumentation to mitigate feedline water hammer.

The Unit 1 feedwater system is operated in a different manner after RSG installation. The FIV remains open during startup and a small flow through the valve and not the FIBV is used for purging purposes.

Feed pump discharge valves are opened by feed pump turbine start signals and are closed by feed pump turbine trip signals. The trip signal overrides the start signal.

Process sampling valves are automatically closed with a Containment isolation phase A, train B signal from the Westinghouse system. Sampling means are provided to monitor the quality of the feedwater. Temperature measurements, provided for each stage of heating, include the temperature into and out of each feedwater heater for the water side of the system and temperature out for the steam side. Steam side temperature measurements into each feedwater heater are provided for extraction lines that may contain dry steam.

Pressure measurements for wet steam lines will suffice to give the saturated temperature in the line. Steam-pressure measurements are provided at each feedwater heater.

In order to preclude water hammer in the steam and feedwater systems, the following interlocks and controls are provided. These water hammer interlocks are non-safety grade controls. However, where the non-safety grade water hammer interlocks interface directly with safety class equipment electrical isolation devices preclude any adverse impact on the safety class equipment caused by potential failure of the non-safety equipment.

1. FEEDWATER ISOLATION VALVE (FIV)

In order to minimize the potential for water hammer, interlocks are provided to prevent the FIVs from being opened (if they are closed) or, to close them if the FIVs are open, and route the feedwater through the auxiliary feedwater nozzle via the preheater bypass line. All of the listed interlocks must be present and the absence of a feedwater isolation signal to allow the FIVs to open.

a. Feedwater Flow

Feedwater flow must be above a low-flow set-point, as measured by a flow switch at the feedwater flow venturi meters. FS-2189, -2190, -2191, and -2192 are provided for loops 1, 2, 3 and 4 respectively. The set-point for these flow switches corresponds to approximately 12 to 15 percent of full feedwater flow.

Once the flow permissives have been cleared allowing the FIV to open, the FIV can remain open irrespective of flow, providing the FW temperature remains high (above the set point as described in item (b) below). Interlocks are provided for this condition.

b. Feedwater Temperature

The feedwater temperature must be above approximately 250°F (as measured by resistance temperature detectors on the main feedwater lines). In addition, the difference in temperature between the RTDs installed outside containment, downstream of the FIVs and RTDs mounted at a piping low point on the feedwater lines inside containment, near the main feedwater nozzle must be within about 10°F of each other. This arrangement of temperature sensors is used to preclude pocketing of cold water at the piping low point during startups and, also, to avoid the possibility of a single RTD open circuit failure causing a false temperature permissive signal to open the FIVs.

Once the temperature permissives have been cleared allowing the FIV to open, the FIV can remain open irrespective of temperature, providing the FW flow remains high (above the low flow set point as described in item (a) above). Interlocks are provided for this condition.

2. FEEDWATER PREHEATER BYPASS VALVE (FPBV)

The FPBV bypasses feedwater from the main feedwater line to the upper auxiliary nozzle. It also serves as a containment isolation valve and is redundantly interlocked with the feedwater isolation signals provided by Westinghouse.

The FPBV is automatically opened whenever the FIV is closed due to the absence of water hammer permissive signals as described above for the FIV. Likewise, it is automatically closed whenever the water hammer permissives are cleared and the FIV is opened. It is also automatically closed on Feedwater isolation actuation signal. The FPBVs are provided with three position close-auto-open control switches on the main board. The switch spring returns to auto; the valves fail closed on loss of control power or air.

3. FEEDWATER SPLIT FLOW BYPASS VALVE (FSBV)

The FSBV is provided to maintain a split bypass flow from the main feedwater line to the upper auxiliary feedwater nozzle. This bypass valve is open at about 30 percent RTP due to system hydraulics when feedwater is admitted to the steam generators through the main feedwater nozzle with the Feedwater Isolation Valve open, and is used to minimize flow induced tube vibration in the preheater section as well as thermal transients at the auxiliary feedwater nozzle.

The FSBV receives an open permissive signal when water hammer interlocks are satisfied. Likewise, the FSBV is automatically closed when the water hammer interlock permissive signal is removed. The FSBV is also automatically closed on an AFW initiation signal to direct AFW to the auxiliary feedwater nozzle, to minimize delays in admitting AFW to the steam generator, and to prevent steam generator blowdown through the auxiliary nozzle during a main feedwater line break.

An annubar flow meter, downstream of the FSBV, measures the flow and provides a signal to a controller to modulate the valve. A control board flow indicator and a manual/auto station is provided for each FSBV. A high-flow alarm is provided to alert the operator when the maximum allowable flow into the preheater is exceeded.

4. FEEDWATER ISOLATION BYPASS VALVE (FIBV)

The FIBV is a containment isolation valve, of the air-operated, fail-closed type. It is redundantly isolated by the Westinghouse feedwater isolation signals from Train A and Train B.

The FIBV stays closed whenever the lack of water hammer permissive signals causes the FIV to close, and is automatically closed after the water hammer permissives are cleared and the FIV has been fully opened manually. The FIBV is manually opened to provide a means of purging the main feedwater line whenever the FIV is closed in order to preclude pockets of cold water from collecting in the main feedwater piping downstream of the FIV. The magnitude of the purge flow is limited by a restriction orifice upstream of the FIBV.

The FIBVs are controlled individually by main control board mounted, three-position, close-auto-open switches which spring-return to auto.

10.4.8 STEAM GENERATOR BLOWDOWN SYSTEM

Under all normal operational conditions, the Steam Generator Blowdown System (SGBS) is used in conjunction with the Chemical Feed and Sampling Systems to maintain optimum secondary-side water chemistry in the recirculating steam generators and to control radioactivity levels associated with nonvolatile radionuclides.

The system components are located in the Auxiliary Building at elevations 778 ft, 790 ft 6 in, and 831 ft 6 in. Each unit has a separate and independent SGBS.

10.4.8.1 Design Bases

10.4.8.1.1 General

Each SGBS is designed to continuously treat 100 percent of the blowdown from each unit. The treated blowdown is returned to the condenser or heater drain tank for reuse as secondary coolant. Typically the blowdown system is operated with the demineralizers in service.

The design primary-to-secondary system leakage is assumed to be 20 gpd, measured at 25°C. The reactor coolant activity is based on 1.00 percent fuel defect. Radioactivity is reduced by the demineralizers until the resins are exhausted. The resin bed capacities are determined by resin saturation with the primary plant chemicals, which greatly exceed the concentration of radioactive dissolved solids. Thus, the resin usage factor depends on neither the amount of radionuclides in the secondary fluid nor the fuel defect levels, but rather on the volume of blowdown processed. Therefore, the magnitude of a primary-to-secondary leak limits operations only to the extent that the leak increases blowdown rates and the quantity of chemicals to secondary water.

Operation of the system is considered normal if a total combined blowdown flow, obtained from the normal blowdown connection on each steam generator, is 0 to 155,620 (Unit 1)/186,000 (Unit 2) pounds per hour; the temperature at the blowdown heat exchanger outlet is 130°F; the condensate polisher is in operation; and the condenser tube leakage is 0 to 2 gpm.

The design bases of the system are as follows: there is a total combined blowdown flow of 308,704 pounds per hour, obtained from the normal and supplemental connections on each steam generator; the temperature at the blowdown heat exchanger outlet is 130°F; and the condensate polisher is in operation.

All materials of construction for the filters are capable of withstanding a total radiation dosage of 4×10^7 rads.

The secondary chemistry water quality will meet the specifications of the EPRI guidelines, except where CPNPP evaluations justify exceptions. Water Chemistry is addressed further in section 10.3.5.

The part of the system from the steam generator to the automatic blowdown isolation valves, outside of the Containment, is an extension of the steam generator boundary. This portion of the system is designated Safety Class 2 because it is necessary for the safe shutdown of the plant. It is also designated as seismic Category I. The portion of the system downstream of the blowdown isolation valves is not essential for safe shutdown of the nuclear plant. Therefore, these various components are classified non-nuclear-safety-related and are designed, fabricated, and tested in accordance with the ASME B&PV Code, Section VIII.

10.4.8.1.2 Seismic and Quality Group Classification

Seismic and quality group classification of the SGBS is in accordance with Section 3.2.

10.4.8.2 System Description and Operation

10.4.8.2.1 System Operation

The SGBS process flow diagram is shown on Figure 10.4-10. The physical layout of the system is shown on Figures 1.2-31 and 33. The system consists of a heat exchanger, a pressure-reducing valve, two inlet filters, two cation demineralizers, two mixed-bed demineralizers, one steam generator blowdown spent resin storage tank, one steam generator blowdown spent resin sluice pump, one steam generator blowdown spent resin sluice pump filter, and the necessary instrumentation and controls. Depending on the mode of operation, all or some of the components are placed in service. The spent resin part of the SGBS is common to both units.

1. Demineralizers in Service

During normal operation, continuous blowdown from each steam generator is routed to a common line outside the Containment. The total blowdown flows through the tube side of the heat exchanger and, after pressure reduction, through the filters and demineralizers. The cooled, demineralized, low-pressure blowdown then mixes with condensate; the combined flow is used as the coolant on the shell side of the heat exchanger. The coolant, which is now at a higher temperature because of the heat picked up from the hot blowdown, flows to the heater drain tank.

During higher levels of condenser tube leakage, supplemental blowdown can be obtained by opening the normally closed supplemental blowdown connection on the steam generator. This supplemental flow and normal blowdown flow from each steam generator are routed together and are processed as described for the normal mode of operation. The routing of the blowdown lines within the Containment is shown on Figure 10.3-1. Blowdown from each steam generator flows into a common header. Each line has a Containment isolation valve (HV-2397, HV-2398, HV-2399, and HV-2400), a steam generator HELB isolation valve (HV-2397A, HV-2398A, HV-2399A, and HV-2400A), a blowdown control valve HV-5175, HV-5176, HV-5177, and HV-5178), and a manual isolation valve (MS-151, MS-153, MS-155, and MS-149). The blowdown header is connected to a high-pressure bulk nitrogen gas supply system. The nitrogen gas is used to provide a nitrogen blanket to the steam generator during wet layups and to mix the layup chemicals.

The blowdown enters the heat exchanger at a temperature of 567°F and a pressure of 1107 psi and leaves the heat exchanger at a temperature of 130°F and a pressure of approximately 1000 psia. Flow element FE-5219 measures the total blowdown flow which, in conjunction with the blowdown water chemistry measurements (see Process Sampling System, Section 9.3.2), guides the operator in adjusting balancing valves HV-5175 through HV-5178 to control individual blowdown. Once the balancing valves are set they are locked in position and do not change. Temperature element TE-5182, located downstream of the flow measuring instruments, measures the exit temperature of the cooled blowdown. Valve TV-5182 is used to regulate the flow of condensate coolant, thus maintaining the design exit temperature.

The cooled blowdown is depressurized from 1000 psia to 300 psia before entering the filters and demineralizers. PV-5180 is a pressure-reducing throttling valve. In case of a

failure in the pressure-reducing system, the fluid pressure is relieved through pressure relief valve SB-020 to the condenser.

The cooled and depressurized fluid is then filtered. There are two filters in the system, which are normally operated in parallel in order to maximize filter life. The filters remove suspended solids. When the pressure drop across a filter reaches a predetermined value that filter is removed from service. The spent filter cartridge assembly is removed as a complete unit and is then disposed of in an appropriate manner. A new filter cartridge assembly is inserted, and the filter is then placed in service. The filtered water temperature is measured by TE-5186 and indicated on the local panel.

The filtered water is then passed through the demineralizers, comprised of four ion exchange units, along with the necessary valves and instrumentation. Manual samples can be taken from each demineralizer outlet. Any resin fines which pass through the ion exchangers are removed in the SGBS resin traps which are Y-type strainers with fine mesh screening designed to arrest resin fines. The effluent flow is measured by flow transmitter FT-5221 and is indicated on the control panel. The processed blowdown is sampled for radioactivity as described in Section 11.5.4 and Table 11.5-4. Process sampling to determine resin exhaustion and performance is described in Section 9.3.2 and Table 9.3-4. If the activity level is within acceptable limits, the fluid is recycled to the heater drain tank or the condenser. If the secondary side radioactivity level is above the acceptable limits, operator action closes valve RV-5179 and valves HV-2397 through HV-2400, thus isolating the system.

2. Demineralizers Bypassed

The demineralizers may be bypassed when the following conditions exist:

a. Activity level of the blowdown, as checked by RE-4200, in the Process Sampling System, is below the detectable level.

b. The processed blowdown is not being routed to the heater drain tank.

c. A transient chemistry condition, such as condenser in leakage, is not in effect.

Under those conditions, the blowdown is cooled, depressurized, filtered, and sent back to either the condenser (normal operation and startup), the Condensate Storage Tank (draining the steam generator for level control), or the vent and drain system of the Turbine Building (during draining of the steam generator for dry layup).

3. Resin Replacement and Removal

When an exchanger vessel is exhausted, as indicated by either high pressure loss across a vessel or high conductivity in the effluent (measured in the Process Sampling System), the spent resin is removed and replaced with fresh resin. Pressure loss across the cation vessels is measured by the difference between the vessel inlet and outlet pressure gauges (PI-5184, PI-5193, and PI-5194). Similarly, pressure differential for the mixed-bed vessels is measured using gauges PI-5195, PI-5198, and PI-5199. Conductivity is measured from the samples routed to the Process Sampling System

downstream of the cation resin trap and upstream from control valve RV-5179. High conductivity also indicates resin exhaustion.

To remove resin, one vessel is valved out of service, and the entire flow is routed to the remaining vessel. By proper alignment of valves, the SGBS spent resin sluice pump pumps water from the SGBS spent resin storage tank to the exhausted vessel through the normal outlet underdrain. Water exits the vessel from the upper screened outlet and flows back to the SGBS spent resin storage tank. This backwash loosens the resin bed.

Next, the SGBS spent resin storage tank sluice pump discharge is routed through the underdrain and the resin exits the vessel bottom. This fluidizes the resin to a 30 to 50 weight percent slurry which is routed to the SGBS spent resin storage tank. The ion exchange vessel is drained after resin is removed, and new resin is added through a flexible connection. The vessel and the fresh resin bed is then rinsed and pressurized. At this point, the vessel is ready for service.

Resin slurry is allowed to settle in the SGBS spent resin storage tank. A lateral distribution of suction screens lead to a common nozzle at the tank wall to provide suction to the SGBS spent resin sluice pump. The tank has a nitrogen gas pressurizing line at the top and six other nitrogen gas lines at the bottom for fluidizing the settled resin. By pressurizing this tank, the fluidized resin slurry is routed to the mobile processing connection.

The water pumped by the SGBS spent resin sluice pump is filtered to remove suspended solids. This filter is identical to the SGBS inlet filters.

10.4.8.2.2 Component Description

1. Steam Generator Blowdown Heat Exchanger

   One heat exchanger cools the blowdown before it is sent to the demineralizers for cleanup. The heat exchanger is of a shell and tube design with blowdown water on the tube side cooled by a combined flow of the cooled blowdown and condensate on the shell side.

2. Steam Generator Blowdown Inlet Filter

   This filter removes particulate matter from the blowdown before it flows to the demineralizers. Two filters are provided.

3. Steam Generator Blowdown Cation Demineralizers

   Two cation-bed demineralizers are provided to remove radioactive materials and other dissolved solids. The demineralizers are connected to the sluice pump and the spent resin storage tank during resin replacement.

4. Steam Generator Blowdown Mixed-Bed Demineralizers

   Two mixed-bed demineralizers, are provided downstream of the cation demineralizers to remove additional radioactive materials and dissolved solids not removed by the cation

demineralizers. The demineralizers are connected to the sluice pump and the spent resin storage tank during resin replacement.

5. Steam Generator Blowdown Spent Resin Sluice Pump

   This canned, centrifugal pump is provided to sluice the spent resin from the demineralizers to the spent resin storage tank. The pump draws water from the spent resin storage tank through a screen and directs it to the demineralizers. A resin slurry is formed and routed to the spent resin storage tank. This pump, which is common to both units, is also used to backwash and rinse the demineralizers.

6. Steam Generator Blowdown Sluice Filter

   This filter removes resin fines from the sluice water which is directed to the demineralizers. This filter is common to both units.

7. Steam Generator Blowdown Spent Resin Storage Tank

   This tank is used to collect and store the spent resin from the SGBS demineralizers until the resin is transferred for disposal. This tank is vented to the plant ventilation system; it is common to both units.

10.4.8.2.3 Equipment Design Criteria

Design criteria for the equipment are listed in Table 10.4-6.

10.4.8.3 Safety Evaluation

Process control and protective instrumentation are provided to ensure functional integrity and the rating efficiency of the system.

The SGBS does not discharge radioactivity to the environment. Whether or not the SGBS is available, plant safety is not affected. The steam generator can be operated without blowdown during the time required for SGBS maintenance. If blowdown cleanup is not available for a period of time, the secondary-side chemistry can exceed the operating limitations.

During abnormally high primary-to-secondary leakage, the operation of demineralizers, will process the blowdown for recycle to the condenser. The demineralizers become exhausted more frequently as the blowdown rate is increased. Because the loss of the SGBS is unlikely, steam generator shell-side radioactivity concentration should not exceed the values given in Table 11.1-7.

Failure of this system has no effect upon safety-related systems. A failure and effects analysis of the system components is shown in Table 10.4-7.

Materials of construction of the steam generators and the blowdown cleanup system are compatible with secondary cycle water chemistry during any primary-to-secondary leaks.

10.4.8.4 Tests and Inspections

1. Inspections

To keep radiation exposures to operating personnel ALARA, the SGBS is designed to control leakage and permit maintenance in accordance with the guidelines of NRC Regulatory Guide 8.8. Planned and carefully prepared periodic visual inspections and preventive maintenance will be conducted by experienced and trained personnel. All components are accessible for internal and external inspection and maintenance.

2. Tests

The SGBS will be tested in place initially and also following any major repair or replacement. Visual inspection of the system and all associated components will be made before each test.

10.4.9 AUXILIARY FEEDWATER SYSTEM

10.4.9.1 Design Bases

The Auxiliary Feedwater System shown on Figure 10.4-11 is designed to provide a supply of high-pressure feedwater to the secondary side of the steam generators for reactor coolant heat removal following a loss of normal feedwater. The system is used in lieu of the main feedwater during cooldown, and startup operations. It also provides a cooling source in the event of a loss-of-coolant accident (LOCA) for small breaks (see Section 15.6). Furthermore, the system is used in the event of a main steam line break, feedwater line break, Control Room evacuation, and steam generator tube rupture.

The system functions over the normal operating pressure range of the steam generators, 100 psia to 1106 psia (Unit 1) or 1107 psia (Unit 2), and is capable of supplying the minimum required flow to at least two of the effective steam generators against a back pressure equivalent to the accumulation pressure of the lowest set safety valve (1236 psia) plus the system frictional and static losses.

The Auxiliary Feedwater System is designed to preclude the effects of hydraulic instability due to water hammer by supplying water to the secondary side of the steam generator through a separate upper auxiliary feedwater nozzle. This permits the cold auxiliary feedwater to be heated as it comes down the side of the steam generator prior to reaching the feedwater preheater. For further discussion of the feedwater system upper nozzle arrangement and considerations see Subsection 10.4.7.

The water level in steam generators is maintained at the proper level to prevent a temperature rise in the RCS, which could result in the release of primary coolant through the pressurizer relief valves.

To ensure that steam does not flow back to the auxiliary feedwater nozzle, the steam generator water level should be above the auxiliary feedwater discharge pipe at all operating levels where steam formation can occur. Therefore, the operator is cautioned to maintain SG level above the auxiliary nozzle internal pipe extension when the temperature is above 212°F. This pipe is

designed with a loop seal immediately upstream of the SG nozzle to prevent steam backleakage. Forward flushing is provided through the auxiliary feedwater nozzle at all loads.

In the event that steam does flow back past the loop seal and the check valves in the feedwater line to the upper nozzle, the temperature of this line will increase causing temperature elements in the line to alarm in the control room. This allows the operator to take action to resolve the problem.

With either only onsite or only offsite power available with an assumed single failure, sufficient auxiliary feedwater flow to two steam generators is provided to permit operation at hot standby for four hours, followed by a cooldown period, at a cooldown rate of 50°F/hr, to reduce the T_{avg} to 350°F, at which time the RHR System can be operated. In the event of a main steam or feedwater line break, sufficient auxiliary feedwater is provided to permit operation at hot standby for two hours followed by the above mentioned cooldown period.

Two motor-driven pumps and one turbine-driven-pump are provided to ensure an adequate supply of auxiliary feedwater to the intact steam generators following a feedwater line break accident, coincident with the assumed single active failure. Subsequent action to provide additional auxiliary feedwater flow is described in Section 15.2.8.

All redundant components are physically separated from each other by an arrangement of concrete barriers designed to preclude coincident damage to equipment in the event of a postulated pipe rupture, equipment failure, or missile generation [6] [7].

The system has been designed to withstand the adverse environmental conditions delineated in Section 3.11, including the effects of flooding which are further discussed in Subsection 10.4.9.2, System Description.

The system is classified as nuclear-safety-related and consists of ANS Safety Class 2 and 3 piping and equipment, except for the non-nuclear-safety condensate transfer pump and associated piping and valves used to provide makeup, recirculation and drainage for the Condensate Storage Tank [3] [4]. Seismic Category I design criteria are considered for all ANS Safety Class 2 or 3 components. Seismic requirements are given in Section 3.2 [5]. The piping is designed to meet the requirements of Branch Technical Positions APCSB 3-1 and MEB 3-1. The system is designed in accordance with 10 CFR Part 50, GDC 2, 4, 5, 19, 44, 45, 46, 54, and 57 [8], [9], [10], [11], [12], [13], [14], [17], [28].

Unit 1

On Unit 1 the steam generators have been replaced with a model Delta 76 that employs a feedring design that eliminates the preheater. Auxiliary feedwater and feedwater entering the steam generator merge with the saturated liquid removed by the moister separators. The combined feed then flows down the annulus formed by the steam generator shell and the tube bundle wrapper where it enters the tube bundle.

To ensure that steam does not flow back through either the main feedwater or auxiliary feedwater nozzles, the main feedwater and auxiliary feedwater lines should be water filled at conditions where steam formation can occur. The operator is cautioned to maintain the SG level above the feedring at normal operating conditions. The normal operating Narrow Range level indication of

60% to 75% would result in sufficient water inventory to maintain the feedring and the AFW vertical perforated spray pipe submerged during the majority of expected transients like +/-10% load changes, turbine synchronizations, and plant loadings/unloadings. The supply piping to the auxiliary nozzle is designed with a loop seal immediately upstream of the SG nozzle to prevent steam backleakage.

In the event that steam does flow back past the loop seal and the check valves in the auxiliary feedwater line to the upper nozzle, the temperature of this line will increase causing temperature elements in the line to alarm in the control room. This allows the operator to take action to resolve the problem.

Unit 2

The Auxiliary Feedwater System is designed to preclude the effects of hydraulic instability due to water hammer by supplying water to the secondary side of the steam generator through a separate upper auxiliary feedwater nozzle. The Unit 2 preheater design steam generators permit the cold auxiliary feedwater to be heated as it comes down the side of the steam generator prior to reaching the feedwater preheater. For further discussion of the feedwater system upper nozzle arrangement and considerations see Subsection 10.4.7.

To ensure that steam does not flow back to the auxiliary feedwater nozzle, the steam generator water level should be above the auxiliary feedwater discharge pipe at all operating levels where steam formation can occur. Therefore, the operator is cautioned to maintain SG level above the auxiliary nozzle internal pipe extension when the temperature is above 212^{o}F. This pipe is designed with a loop seal immediately upstream of the SG nozzle to prevent steam backleakage. Forward flushing is provided through the auxiliary feedwater nozzle virtually at all loads by AFW flow at startup conditions, by preheater bypass flow during initial power ascension and by the split flow of the feedwater at RTPs above 30% full load.

In the event that steam does flow back past the loop seal and the check valves in the feedwater line to the upper nozzle, the temperature of this line will increase causing temperature elements in the line to alarm in the control room. This allows the operator to take action to resolve the problem.

10.4.9.2 System Description

The Auxiliary Feedwater System is comprised of two electric motor-driven auxiliary feedwater pumps and associated valves, piping, and controls and a third turbine-driven auxiliary feedwater pump with associated valves, piping, and controls, which is independent of the electrical power supply to the motor-driven pumps. Three pumps are necessary to ensure an adequate supply of auxiliary feedwater following an accident, coincident with the single failure of a pump. The design parameters of the auxiliary feedwater pumps are given in Table 10.4-8.

All three pumps normally draw suction from the Nuclear Safety Class 3 Condensate Storage Tank. A single line supplies water through a common locked-open valve to the suction of the motor-driven auxiliary feedwater pumps, and a second line supplies water to the suction for the turbine driven auxiliary feedwater pump. Of the approximately 500,000-gal capacity, approximately 282,000 gal remain for auxiliary feedwater. The rest of the tank is used as condensate storage for the Condensate System. The reserved auxiliary feedwater cannot be

drained by the non-nuclear-safety systems because of the elevation of the outlet nozzles. Piping tie-ins are provided to allow connection of equipment to process and chemically treat the inventory of the Condensate Storage Tank.

While the Condensate Storage Tank is the preferred water supply, another ANS Safety Class 3 alternate supply is provided. The Auxiliary Feedwater System has the capability to draw suction from the service water system (SWS) in the event of loss of the Condensate Storage Tank. Two normally closed, key-switch activated, motor-operated butterfly valves in the SWS and three motor-operated gate valves in the auxiliary feedwater system prevent contamination of the auxiliary feedwater by station service water. The three AFW valves are normally de-energized to prevent a fire induced hot-short from causing a mal-operation of the valve(s). This also de-energizes position indication in the Control Room. Therefore, these valves are “Locked Closed” to provide positive indication of correct valve position during normal operation. In addition, high-and low-point leakoff connections are provided between the SWS isolation valves and the motor-operated gate valves to allow detection of any station service water inleakage. An orifice has been installed in the leakoff lines to limit the leakage rate into the safeguard building.

Each motor-driven pump normally feeds two steam generators. A normally closed interconnection between the motor-driven pump discharge lines permits either pump to feed to all four steam generators. This interconnection provides the operator with the means to maintain the water level in all steam generators on a long-term basis following a LOCA by operating either motor driven pump. The motor driven pumps can be manually started or stopped from the Control Room or the hot shutdown panel. The turbine-driven pump discharge line branches into four separate lines each feeding one steam generator. The turbine-driven pump can be manually started from the Control Room or the hot shutdown panel.

Each of the lines that connects the three auxiliary feedwater pumps to the steam generators is provided with: a normally open, pneumatically operated feed regulator control valve; a flow-limiting orifice; a check valve; a motor operated isolation valve; and two manual isolation valves. Remote manual control of the feed regulator control valve is provided from the Control Room with provision for local manual operation on the hot shutdown panel. Air accumulators are provided for the pneumatically operated valves with sufficient capacity to permit remote valve closure in the event of a secondary system break where local valve operation cannot be accomplished within the required time period following the incident. The instrument air system and the air accumulators are described in Section 9.3.1.2. As described in Section 6.2.1.4.4, isolation of a faulted steam generator is an operator action required within 10 minutes after a secondary pipe break. Each air accumulaor is required to have enough air to close its control valve and recirculation valve and maintain them closed for a mission time of 30 minutes (i.e., until local manual control is established). Testing criteria is based on one close stroke of the valves and 30 minutes of air loss/leakage. The control valves are located near each AFW pump to allow isolation or local manual flow control as required. Flow control is not required for the 30 minutes following a secondary line break; however, if isolation is not required and instrument air is lost, the operator may control flow remotely using the air accumulators until instrument air can be restored or local manual flow control can be established.

The flow limiting orifices are provided to limit flow to any one steam generator to a maximum of 1380 gpm, in the event of either a main feed line break or a main steam line break inside containment.

An orifice-type flow measuring device is located in each of the auxiliary feedwater lines to indicate flow to each steam generator and to provide a means of detecting grossly uneven flow to the steam generators. Readout for these flow measuring devices is located in the Control Room and on the hot shutdown panel. To avoid the possibility of a single active failure stopping all auxiliary feedwater flow to a steam generator, there are no valves located in the common main feedwater lines.

The Auxiliary Feedwater System operates over an extended period of time following a LOCA. The two motor-driven pumps start automatically and they provide an additional means for removing core residual heat in the event of a LOCA for small breaks. During large break LOCA conditions, the system is used to maintain an adequate water level above the tubes in the steam generators to prevent primary to secondary leakage. The operator shuts down the pumps at his discretion and manually adjusts feed flow to individual steam generators.

Depending on the severity of the event, either the two motor driven pumps or all three auxiliary feedwater pumps start automatically after either a main steam line break or a feedwater line break (see Subsection 10.4.9.5). At an early stage in the accident, the operator isolates the feedwater to the affected steam generator. The system provides for the cooldown of the unaffected steam generators to prevent the RCS from being repressurized. The operator shuts down the pumps at his discretion.

After a loss of the main feedwater system (condition II event), either the two motor-driven auxiliary feedwater pumps together, the turbine-driven pump alone, or any combination are capable of providing sufficient flow to the steam generators to allow the plant to be taken to a safe shutdown condition (and remain a condition II event). The operator shuts down the pumps at his discretion.

The operation of the Auxiliary Feedwater System following a steam generator tube rupture is manually initiated. The two motor-driven pumps are started manually and are used to maintain the required water level in the steam generators as the plant is shut down. The operator identifies the affected steam generator and isolates it and the operator shuts down the pumps at his discretion.

The operation of the Auxiliary Feedwater System following a Control Room evacuation is manually initiated and is controlled from the hot shutdown panel. The operator maintains water level in the steam generators with either the two motor-driven pumps or the turbine-driven pump. The pumps are used to maintain the required water level in the steam generators as the plant is shut down. Again the operator shuts down the pumps at his discretion.

Each power supply train for the motor-driven pumps, control valves, and instrumentation is supplied from a separate and independent Class 1E bus that is capable of supplying the minimum required power for the safety-related loads required following a LOCA or loss of offsite power (blackout), or both. Each bus can be powered from two independent offsite power sources or by the diesel generator assigned to the bus [15].

The control power for the non-safety related turbine driven pump test panel is from the Train A Class 1E bus and is isolated by an “S” signal.

Unit 1

Downstream of the last isolation valve, each line from the motor-driven pumps joins with a corresponding line from the turbine-driven pump to form a common line that connects with the feedwater line that connects to the auxiliary nozzle on the Unit 1 steam generators.

Unit 2

Downstream of the last isolation valve, each line from the motor-driven pumps joins with a corresponding line from the turbine-driven pump to form a common line that connects with the feedwater preheater bypass line. The preheater bypass line connects to the auxiliary nozzle on the Unit 2 steam generators.

10.4.9.3 Safety Evaluation

The Auxiliary Feedwater System is designed to ANS Safety Class 2 and 3 requirements, and in the event of loss of offsite power, the backup turbine driven auxiliary feedwater pump operates. The turbine drive does not have any auxiliaries requiring electrical power. For redundancy, steam for the turbine driver is supplied from two steam generators. Either supply can meet the turbine driver requirements. The turbine steam supply valves are fail-open air-operated types each with a pilot solenoid valve supplied from a redundant Class 1E power supply. The Train A steam admission valve supplies steam from steam generator 4 and the Train B steam admission valve supplies steam from steam generator 1.

The turbine speed control governor is of the mechanical/hydraulic type, which is capable of maintaining the turbine at the high speed setting without any outside sources of power.

During normal plant operation the turbine speed is controlled by the speed setting signal which is converted to a pneumatic signal for the turbine governor controls. Loss of this remote speed setting signal and/or the air supply will result in the turbine running at the high speed setting.

The power supply for the turbine speed setting signal is from the station inverters which are supplied from the 125 volts dc batteries. The air supply is from the station instrument air system.

Safe shutdown of the unit relies upon the availability of the Auxiliary Feedwater System. Loads which are required for the safe shutdown of the unit are connected to the Class 1E power supply.

The auxiliary feedwater pumps are designed to deliver the minimum required flow within 60 seconds (including sensor/signal response time). The 60 second design response time for the Turbine Driven Auxiliary Feedwater pump is less than that evaluated in the accident analyses. A maximum response time of 85 seconds from the Steam Generator LO-LO setpoint (including sensor/signal response time) for the Turbine Driven Auxiliary Feedwater pump to deliver the minimum required flow is required based on the accident analysis assumptions.

In the event of a LOCA or loss of all offsite power (blackout), or both, the motor-driven auxiliary feedwater pumps and their associated motor-operated valves are automatically sequenced onto their respective emergency buses.

Motor-operated valves stop automatically when valve action is completed while the motor-driven auxiliary feedwater pumps must be manually stopped.

A failure mode analysis (electrical and mechanical) is shown in Table 10.4-9 and on Figure 10.4-12, respectively.

In the event of a feedwater line break inside the Containment, the larger than normal flow is detected by the flow-measuring device in the line.

Sufficient redundancy is provided throughout the Auxiliary Feedwater System and supporting systems to ensure safe plant shutdown with only one motor-driven auxiliary pump by supplying the required flow to a minimum of two steam generators while subject to a single active failure in the short-term or a single active or passive failure in the long-term. This flow is sufficient to maintain the unit in a safe condition.

The Auxiliary Feedwater System, as shown on Figure 10.4-11, is capable of withstanding adverse environmental conditions. It is designed to seismic Category I requirements. Except for the exhaust stack and flash tank atmospheric vents, it is located within tornado resisting structures and is protected from tornado-generated missiles. The exhaust stack and flash tank atmospheric vents have been evaluated for tornado missiles. See Section 3.5.1.4 for details. The Condensate Storage Tank is designed against tornadoes and missile penetration, except as described in Section 9.2.6. The supply lines from the tank to the Safeguards Building are located in seismic Category I structures and the auxiliary feedwater pumps are located in an enclosed bay of the Safeguards Building at a floor elevation of 790 ft 6 in.

All redundant components (including pumps, controls, Class 1E power sources, and electric cable) are separated from each other by a proper arrangement of barriers or suitable physical separation. This barrier separation is provided to preclude coincident damage to redundant equipment in the event of a postulated pipe rupture, equipment failure, or missile generation. Each pump is situated in a separate compartment and is protected by walls constructed to seismic Category I requirements.

Two sources of flooding are considered. One source of flooding is a pipe break in the auxiliary feedwater pump discharge. Separate compartment design, as well as access and drainage, prevents flooding of adjacent equipment. The second source of flooding considered is a crack in a 24-in. diameter 0.375-in. wall, component cooling water pipe that runs adjacent to the bay occupied by the auxiliary feedwater pumps. This piping is considered a piping system containing moderate-energy fluids during reactor operation. Floor drains are provided to accommodate any water leakage as a result of a postulated crack.

Redundancy of cooling water source is ensured by a connection with the SWS, which is of Safety Class 3 design.

This backup source of water, which has lower quality standards than those specified for steam generator feed, would be used only in case of extreme emergency, when safety overrides water quality consideration. The required capacity and design conditions for the use of the SWS and SSI as an alternate feedwater source are described in Section 9.2.5, and show that the SSI is able to supply water to the steam generators for the required time.

For Unit 2, design of the Auxiliary Feedwater System is such that the effects of water hammer are precluded by the use of a separate upper auxiliary feedwater nozzle on the steam generator. (See Subsection 10.4.7 for further discussion.)

10.4.9.4 Inspection and Testing Requirements

All system components are tested and inspected in accordance with the applicable codes. The system is capable of being tested while the plant is in operation. A test line to the Condensate Storage Tank is provided on each pump discharge. This provision allows each pump discharge valve to be closed. Each pump can be started manually and recirculated back to the tank. Only one pump at a time is tested and pressure and flow indications at the pump discharge are used for checking the pump performance.

Refer to Section 6.2.4 and 6.2.6 for containment isolation valve testing.

10.4.9.5 Instrumentation Requirements

1. General

The instrumentation and controls for the Auxiliary Feedwater System provide for automatic or manual and remote or local operation of the system. Controls for manual operation of the system at local stations and at the Hot Shutdown Panel are provided in addition to auto/manual controls in the Control Room. Controls from the Hot Shutdown Panel override all other signals and activate an override alarm in the Control Room.

For a description of the Auxiliary Feedwater System instrumentation and controls, refer to Section 7.3.1.1.4, item 5.

Automatic Initiation of the AFW System will automatically isolate steam generator blowdown and sampling for all steam generators.

2. Feedwater Flow Control

During cooldown, the operator maintains the required steam generator water level by varying the auxiliary feedwater flow. Motor-driven auxiliary feedwater pump flow to each steam generator is remote manually controlled by feed regulator control valves. The valves are located near the pumps to allow manual operation. Control stations on the Main Control Board and Hot Shutdown Panel enable the operator to control the flow manually from the Control Room or from the Hot Shutdown Panel in conjunction with a nearby patch panel for valve control.

An automatic motor driven auxiliary feedwater pump start signal automatically trips the manual flow control to automatic full open to ensure flow to the steam generators. Motor driven auxiliary feedwater pump runout protection is provided by the system piping configuration, and flow restricting orifices. Each auxiliary feedwater regulator control valve is air-operated and is provided with a nuclear safety-related air accumulator to permit valves to close in the event of a secondary system break and an instrument air system failure. The valves fail open on loss of air or electric failure.

All controls for motor-driven pump A are electrical Train A oriented; all controls for motor-driven pump B are electrical Train B oriented; controls for the turbine-driven pump are fed from the Train A, Class 1E 125VDC System.

On loss of electrical power or air supply, the turbine-driven pump accelerates to maximum speed demand. Since the turbine-driven pump is supplied with a fail-closed trip and throttle valve, this valve is latched in the open position. Two redundant steam supply lines, each with an air operated supply valve, provide steam to start and accelerate the turbine-driven pump. These air-operated valves fail open, ensuring that the pump starts on loss of air supply or electrical power. Speed control is accomplished with a Woodward mechanical/hydraulic type governor. A mechanical overspeed trip device is provided to trip the turbine at 116.6-percent rated speed. Manual speed control is from the Control Room or the Hot Shutdown Panel. The manual control from the Hot Shutdown Panel overrides all other signals. There is speed indication on the Control Room Panel and Hot Shutdown Panel and at the local panel. Flow from the turbine-driven pump to each steam generator is regulated by control valves under manual control from the Control Room, the Hot Shutdown Panel, or locally. Each valve has an air accumulator to permit the valves to close in the event of a secondary system break and an air system failure.

3. Feedwater Supply Control

Filling of the Condensate Storage Tank may be performed manually by the operator aligning demineralized water, or automatically by associated level instrumentation when the demineralized water source is continuously aligned. Tank level is indicated locally and remotely and HI-HI, LO, and LO-LO tank level alarms are provided. Redundant level transmitters are used.

The Condensate Storage Tank supplies water to the auxiliary feedwater pumps. The automatic starting of any auxiliary feedwater pump initiates the automatic isolation of the Condensate Storage Tank from all its other users. This maximizes the water supply to the auxiliary feedwater pumps whenever they are started.

The condensate transfer pump is manually started and stopped from a main control board switch. The pump is automatically stopped in the event of an “S” signal or on low pump suction pressure.

4. Emergency Feedwater Supply Control

Inlet motorized control valves are manually controlled by a key lock switch to admit service water to the suction of the auxiliary feedwater pumps.

5. Display Information, Alarms, and Controls

Control switches and position indication lights are provided for all remotely operated valves.

The following display information and alarms, in addition to those already mentioned, are provided in the Control Room:

a. Suction pressure indication and low alarm for each auxiliary feedwater pump

b. Temperature indication for each steam generator auxiliary feedwater line

c. Low pressure alarm for alternate feed supply from service water system (Unit 1 only).

d. Discharge pressure indication for each auxiliary feedwater pump discharge; pressure indication on hot shutdown panel for these pressures; low discharge pressure alarm for each motor driven auxiliary feedwater pump

e. Flow in the discharge line to each steam generator; indication is also on the hot shutdown panel.

f. Flow in the discharge line from each pump

g. Alarms for local override control from the hot shutdown panel; local indicators for temperature, pressure, flow, and level are provided as shown on the flow diagram.

h. Alarms for overspeed trip, lube oil system trouble and high drip pot level for AF pump turbine.

10.4.10 EXTRACTION STEAM SYSTEM

The primary function of the Extraction Steam System is to convey the steam required for regenerative heating of condensate and feedwater by the feedwater heaters. The extraction system also conveys auxiliary steam for process use in the NSSS.

10.4.10.1 Design Bases

The Extraction Steam System design meets the requirements of the following codes and standards:

1. American National Standards Institute (ANSI)
2. American Society of Mechanical Engineers (ASME)
3. American Society for Testing and Materials (ASTM)
4. American Welding Society (AWS)
5. Heat Exchange Institute (HEI)
6. Manufacturer’s Standardization Society (MSS)
7. Occupational Safety and Health Act (OSHA)
8. Steel Structures Painting Council (SSPC)
9. Tubular Exchanger Manufacturers Association (TEMA)

The system design parameters are based on the turbine cycle heat balances shown in Section 10.1.

10.4.10.2 System Description

The Extraction Steam System and interconnected piping is shown on Figure 10.4-13. The extraction steam lines to the different heaters are separated into two groups in order to prevent water induction into the turbine, turbine overspeed, and line drainage. One group comprises extraction lines to two high pressure heaters and the other group includes extraction lines to four low pressure heaters. The heaters are numbered 1 through 6 with No. 1 being the highest pressure heater. The heaters are in two parallel strings and are differentiated as 1A, 1B, 2A, 2B and so forth.

Each extraction line, with the exception of heater Nos. 5 and 6, is provided with drain pots to remove condensate to the condenser. The extraction lines to heaters 1, 2, 3 and 4 are each equipped with a motorized stop valve and power-assisted non-return check valve.

The power-assisted check valve, because of its fast closing time, provides adequate protection for the turbine against the possibilities of overspeed caused by steam which may flow back into the turbine in the event of a turbine trip.

The motor-operated shutoff valve is a slower closing valve, but provides protection to the turbine against the possibility of water induction when there is a high water level in the feedwater heater.

The stop valves can be manually operated from the Control Room. The drain valves can also be operated manually during startup and periods of light load operation. Each of the air-operated drain valves, power-operated check valves, and motorized stop valves has position-indicating lights in the Control Room.

Heaters 5 and 6 are located in the condenser neck with the channel ends extended out of the condenser. Check valve and motorized stop valve combinations are not provided on the extraction lines because of anti-flash baffles to restrict the reverse flow from these heaters to a sufficiently low flow so that it cannot adversely affect turbine overspeed. These lines do not contain enough available energy to overspeed the turbine.

To prevent water induction into the low-pressure turbines from low-pressure heaters 5 and 6 as a result of tube rupture, high-high water level switches on the heaters transmit signals to close the motorized valve located on the discharge line of the condensate pump thereby preventing the flow into the heaters.

In addition, the Extraction Steam System also provides steam for process use.

On turbine trip, pressure transmitters from the turbine provide a signal to close all the power-operated check valves and the motorized stop valves.

Safety relief valves have been provided on the shell side of the high-pressure heaters and the low-pressure heaters, with the exception of condenser neck heaters, to protect against overpressurization. To conform with the standards of the Heat Exchange Institute, the safety relief valves are sized to pass a minimum of 10 percent of the tube side flow.

Extraction steam lines from any particular extraction point are connected by a pressure equalizing header before the lines enter the respective heaters. Pressure equalizing headers are not provided for heaters 5 and 6 because each of the same numbered heaters receives

extraction steam from different low pressure turbines and the extraction steam lines from these turbines are contained in their respective condensers.

All heaters, piping, valves, and miscellaneous components of the Extraction Steam System with the exception of condenser neck piping and heaters 5 and 6 that have a surface temperature of 125°F or more are insulated to minimize heat loss and afford personnel protection.

In order to provide adequate piping flexibility and to keep loads and moments on the turbine extraction nozzles within acceptable limits, all the extraction lines from the low pressure turbine are provided with flexible expansion joints located inside the condenser neck.

10.4.10.3 Safety Evaluation

The system is non-nuclear-safety-related, and seismic design criteria have not been considered.

10.4.10.4 Tests and Inspections

All system components are tested and inspected in accordance with the applicable codes.

10.4.10.5 Instrumentation Requirements

Each turbine side drain pot is provided with an extraction line drain valve. These valves are opened and closed manually with control-board-mounted switches. After a valve opens, it remains open until manually closed. The valves are air operated and fail open on loss of electrical signal or air supply. These valves are automatically opened to drain lines when the extraction stop valve in the extraction line closes.

High water level in the heater side drain pots, which are located between stop valves and heaters, alarms in the Control Room.

Extraction line positive closing check valves for heaters 1, 2, 3 and 4 are automatically closed by high-high level in the associated heater, or a turbine trip (pressure signal from turbine oil control system). After automatic closure, valves must be opened manually by control board switches. They are not capable of being closed manually. In addition, high-high levels in the heaters and drain pots trigger alarm signals.

The extraction line motorized stop valves, like positive closing check valves, close on a high-high heater level or a turbine trip. Manual opening and closing from a control board switch is possible. Each stop valve has its own control board switch with three positions: close, auto, and open. Switches spring return to auto. Extraction line drain valves are interlocked to open when extraction line stop valves close.

Heaters 5 and 6, in the condenser necks, do not have extraction shutoff valves.

The motorized stop valve in the auxiliary steam supply extraction line is automatically closed by a high-high level signal from the extraction line drain pot level switch or by turbine trip.

Heaters 1 thru 4 have a pressure transmitter in their extraction lines to transmit signals to the control board pressure indicator. Heaters 1 through 4 have local pressure indicators on the turbine side of extraction line stop valves. These are used when a heater is put back in service

after being taken out to ensure that heater shell pressure is lower than corresponding turbine stage pressure before the stop valve is opened.

All drain pot level switches alarm in the Control Room on high-high water level. All air-operated drain pot drain valves have open-close indication lights on the control board and they alarm when opened.

Extraction line stop valves and positive closing check valves also have open-close indicator lights on the control board.

10.4.11 HEATER DRAINS SYSTEM

This section discusses the Heater Drains System in accordance with the requirements of NRC Regulatory Guide 1.70 [16].

10.4.11.1 Design Bases

The Heater Drains System functions to aid in regeneratively heating feedwater by cascading the higher energy drains through successively lower energy stages of feedwater heaters. The system also returns the saturated water from the extraction steam to the Feedwater System and the Condensate System in a manner which results in good cycle efficiency.

The Heater Drains System is designed to perform the following functions:

1. To drain the moisture removed from the moisture separator/reheater (MSR) shell and separator sides to the MSR shell and separator drain tanks from where it is continuously drained to the feedwater heater drain tank 01.

2. To collect the steam condensed in the reheater tubes of the MSR in the reheater drain tanks from where the saturated water is continuously drained to the high-pressure feedwater heaters

3. To cascade the higher energy drains through successively lower stages of feedwater heaters to provide regenerative heating for feedwater before its admission to the steam generators

4. To collect drains in the feedwater heater drain tanks from which the heater drain pumps take their suction and inject the condensate into the Feedwater System

5. To provide the alternate drain paths for condensate flow during emergency conditions

The Heater Drains System also functions to remove the noncondensable gases from the system before gas accumulation to improve the heat transfer performance.

The Heater Drains System also incorporates a controlled cold water injection (condensate pump discharge) to the suction side of the heater drain pumps. The Condensate System also supplies seal water injection for the heater drain pumps.

All components of the Heater Drains System are designed in accordance with the applicable requirements of the following codes:

1. American National Standards Institute (ANSI)
2. American Society of Mechanical Engineers (ASME)
3. American Society for Testing and Materials (ASTM)
4. American Welding Society (AWS)
5. Heat Exchange Institute (HEI)
6. Hydraulic Institute (HI)
7. Institute of Electrical and Electronics Engineers (IEEE)
8. Manufacturer's Standardization Society (MSS)
9. National Electrical Code (NEC)
10. National Fire Protection Association (NFPA)
11. Occupational Safety and Health Act (OSHA)
12. Steel Structures Painting Council (SSPC)
13. Tubular Exchanger Manufacturers Association (TEMA)

The Heater Drains System piping is designed in accordance with ANSI B31.1; the piping is insulated where exposed to the weather.

The design bases of the Heater Drains System are as shown on the turbine cycle heat balances in Section 10.1.

10.4.11.2 System Description

10.4.11.2.1 General

The secondary plant steam turbine power cycle uses a regenerative feedwater heating arrangement consisting of two parallel strings of two stages of high-pressure feedwater heaters in series and four stages of low-pressure feedwater heaters. For each heater, except for the heater No. 3 and heater No. 6, an integral drain cooler is provided. Heater No. 6 incorporates an external drain cooler. The feedwater heaters are numbered from 1 through 6 with No. 1 heater being the highest pressure heater. Since the heaters are in two parallel strings, they are further differentiated as 1A and 1B, and so forth. In addition, two MSRs are provided to separate the moisture from steam exhausted from the high-pressure turbine and reheat it before its admission to the low-pressure turbine cylinders.

The system is provided with all the necessary instrumentation, controls, piping, and valves.

The Heater Drains System and interconnected piping are shown on Figure 10.4-14.

All drain flows from Feedwater Heaters 2A, 2B, 3A and 3B are directed to Heater Drain Tank 2. All level control instrumentation is connected to Heater Drain Tank 2 from which the heater drain pumps take suction. All other MSR drains and the Steam Generator Blowdown Heat Exchanger cooling water flow are directed to Heater Drain Tank 1 which is interconnected to Heater Drain Tank 2 through an equalization vent header between the tanks' vapor spaces and an equalization heater between the tanks' liquid spaces.

The heater drain pump recirculation flow is routed to the heater drain tank equalizing liquid space line.

10.4.11.2.2 Moisture Separator/Reheater

For each unit, two MSR assemblies are provided and located in the crossover/crossunder piping system between the high-pressure and the low-pressure turbines. The MSR are housed in one pressure vessel.

The drainage of the MSR is accomplished by the heater drains system using separate piping and drain tanks.

The drain from the MSR is by gravity. To establish a steam water interface, MSR shell, separator, and reheater drain tanks are provided. The drain tanks are of the horizontal, cylindrical with spherical-head type.

10.4.11.2.3 Feedwater Heaters

The feedwater heaters are designed to maintain a drainage flow within the limits required for safe, continuous plant operation under constant or fluctuating load conditions.

All feedwater heaters with the exception of heater No. 3 and No. 6 are equipped with an integral drain cooler. An external drain cooler is provided for heater No. 6. On its way through the drain cooler, the steam condensed is subcooled, providing an additional heating for the feedwater and improving the cycle thermal efficiency.

10.4.11.2.4 Heater Drain Pumps

Two 50-percent-capacity, motor-driven, constant-speed, horizontal centrifugal pumps are provided for the Heater Drains System. Each heater drain pump is equipped in the suction line with a gear operated, wafer-body, butterfly valve for pump isolation, with an expansion joint for thermal and vibration movements, and with an in-line temporary strainer for use during system start-up. The discharge line is equipped with a check and a gear-operated manual valve in series and a warmup line for instant startup service when only one pump is required to operate and the other is on standby.

In addition, each heater drain pump incorporates a minimum recirculation control system and a common cold water injection line.

10.4.11.3 Safety Evaluation

The Heater Drains System is non-nuclear safety-related, and seismic design criteria have not been considered.

Each primary flow path of the MSR drain tanks is provided with a check valve to prevent backflow of flashing steam from interfering with drainage to the condenser when pressure is reduced during reduction of load.

The MSRs and all heaters except Nos. 5 and 6 have steam side safety valves.

Each MSR vessel is provided with a pressure transmitter which lights an indicator in the Control Room to indicate to the operator the high MSR vessel pressure.

10.4.11.4 Tests and Inspections

Each feedwater heater, external drain cooler, pump, drain tank, MSR shell and tube, and valve receives a shop hydrostatic test performed in accordance with applicable codes.

Prior to initial operation, all foreign materials and oxides are removed from the piping, and the system is hydrostatically tested to confirm leaktightness. Visual inspection of pipe weld joints confirms the exterior condition of the weld.

10.4.11.5 Instrumentation and Controls

Each of the MSR drain tanks with the exception of the reheater drain tanks is provided with one level transmitter/ controller to position the control valves in order to maintain the normal water level of the drain tank. Each reheater drain tank is provided with one level/transmitter/controller that uses a split range.

The normal water level of the feedwater heaters is accomplished by their level controller, except heater No. 3, since this heater does not have an integral drain cooler and it drains directly to the heater drain tanks. Heater No. 6 water level is controlled by the level controller located at the discharge piping to its drain cooler.

The normal level control valves in the Heater Drains System have a fail closed position and the high level control valves have a fail-open position.

All primary and alternate drain valves of the feedwater heaters operate on a pneumatic signal from their associated level controller/ transmitters. The valves are of the air-operated globe type except for ball valves on the low-pressure heater drains.

Similar level control valves with controllers are used for the MSR shell and drain tanks. The primary and the alternate drain valves are of the air-operated type, modulated with a 4 to 20 mA electrical signal from an electronic level transmitter on the drain tanks. Each valve can be controlled manually from the manual auto station on the control board.

Heater Nos. 5 and 6 are provided with a high-high level alarm in the Control Room.

The Heater Drains System is equipped with level gauges for local level indication.

Heater Nos. 1, 2, 4 and 5 are provided with level switches actuated by high-and low-level signals from the level transmitters for high and low alarms in the Control Room. The heater No. 6 and heater No. 3 are provided with level switches to actuate a high-level alarm.

Level control signals are originated from heater drain tank 2 only. Each heater drain tank incorporates a high-level alarm actuated by a high-level signal from the drain tank level transmitter. Heater drain tank 2 incorporates a low-level alarm actuated by a low-level signal from the drain tank level transmitter.

The MSR drain tanks are provided with level switches for high alarms in the Control Room.

The Heater Drains System also incorporates the following monitors in the Control Room:

1. Pressure indicator for MSR

2. Drain temperature detector for computer input

3. Open and close position indicator lights for all control valves

4. Differential pressure indicator switch for in line strainers

10.4.12 TURBINE PLANT COOLING WATER SYSTEM

The Turbine Plant Cooling Water (TPCW) System is a closed-loop system designed to cool all turbine-generator-associated equipment and conventional plant equipment and systems which are classified as non-nuclear-safety-related [4]. The system acts to remove residual heat from all turbine plant associated equipment by means of the turbine-plant-cooling water heat exchangers and rejects this heat to the Circulating Water System, which acts as a heat sink.

System components are located in the Turbine Building.

10.4.12.1 Design Bases

The TPCW System is designed to remove heat from the turbine generator auxiliary systems and from various equipment of the secondary plant. The system provides corrosion-inhibited demineralized cooling water to components that can be adversely affected by direct reservoir water cooling because of corrosion or fouling. It is also designed to supply each component with the proper flow rate of cooling water (see Table 10.4-10) at a maximum cooling temperature of 107°F. The cooling water flow rate through each component can be adjusted by means of either butterfly or globe valves on both the inlet and outlet of the individual components. The water exits each component up to 118°F, whereupon it is directed back to the TPCW heat exchanger. The excess heat is removed by circulating water that flows through the tube side of the heat exchanger at a design inlet temperature of 102°F and a design tube velocity of 5 ft/sec. At a temperature approximately 12°F above the inlet temperature, the circulating water is then directed back to the SCR, where the excess heat is dissipated.

The anticipated analyses of the TPCW (through the shell side) and the circulating water (through the tube side) are shown in Tables 10.4-11 and 10.4-12, respectively.

Based on this analysis, calculation of both stability and saturation indices indicates that the water tends to form scale, which will be cleaned during extended outages. The water on the tube side of the heat exchanger has a corrosion control and dispersant chemical added. A solution of sodium hypochlorite or sodium hypochlorite and sodium bromide is added to the water to control organic and biological growth in the cooling system.

10.4.12.2 System Description

The TPCW System, schematically shown on Figure 10.4-15, is provided with one full-capacity heat exchanger and two full-capacity pumps for each unit. The pumps are of the horizontal, centrifugal single-stage type and are designed to deliver 20,000 gpm of cooling water at a design total dynamic head (TDH) of 100 feet. All pumps are connected to a common suction header and discharge manifolds. Either pump may serve as the lead pump.

The full-capacity heat exchanger utilizes circulating water on the tube side as the coolant with turbine plant cooling water on the shell side.

Each component cooled by the TPCW System is provided with either globe valves or butterfly valves on the inlet and outlet of each component. These serve a dual purpose: the valves can be used for the initial regulation of cooling flow (throttling) to provide overall system flow balancing and equalization of pressure drops, and they can be used for isolation and shutoff in the event of an individual component malfunction.

Each system is also provided with one TPCW head tank, which is located at the system's highest point and is connected to the pump suction. The tank contains demineralized water to which a suitable corrosion inhibitor has been added.

The tank is used as an initial system-filling reservoir and as a makeup water reservoir during system operation; it also provides an adequate pump net positive suction head (NPSH).

10.4.12.3 Safety Evaluation

The TPCW System is not required to maintain plant safety.

Seismic design criteria are not applicable.

Safe shutdown of the units does not rely on the TPCW System. Hence, there is no provision to provide power to this system from the emergency diesel generators.

10.4.12.4 Tests and Inspection

The TPCW head tank, heat exchanger, and associated piping are hydrostatically tested by the manufacturer prior to installation. The control system is also tested by the manufacturer to ensure proper valve and pump operation sequences. After installation, the system is again hydrostatically tested.

10.4.12.5 Instrumentation Requirements

10.4.12.5.1 Indication and Monitoring

1. The TPCW pump inlet and outlet pressures and the head tank level are locally observed and monitored. The pump suction and discharge lines are provided with local pressure indicators. Suction low pressure, discharge high pressure, and head tank low and high-high levels are alarmed, and a low-low level signal stops the TPCW pumps; low- and high-level signals control the head tank level.

2. The total cooling water supply to all coolers is measured, indicated on the control board, and high and low flows are alarmed.

   The temperature of the main supply cooling water is indicated on the control board. A resistance temperature detector (RTD) type temperature sensor is applied.

   The main cooling water return header is provided with similar temperature measurement as well as with local pressure measurement. The heat exchanger inlet and outlet are provided with local temperature indicators, all of which are bimetal thermometers.

3. Each return cooling water header of similar cooler group is provided with a local temperature indicator and (except the instrument air compressors) a local cell-type flow indicator. Two additional local differential pressure cell-type flow indicators are placed in the two main return cooling water headers; these measure the overall return cooling water flow from two groups of cooling units.

10.4.12.5.2 Control and Interlocking

1. The lead TPCW pump is started manually and is stopped by the head tank low-low level signal.

   The backup standby pump can be started and stopped by manual overriding or started automatically by a trip signal from its respective lead pump and is stopped by the head tank low-low level signal.

   The pump trip signal or auto stop signal is alarmed to alert an operator of any abnormal pump condition.

2. The cooling water head tank water supply valve is automatically opened and closed by low- and high-level head tank signals, respectively. The valve is of the air-operated, fail close type.

3. The turbine plant cooling water flow from the electrohydraulic control fluid coolers is temperature controlled by throttling an air-operated control valve. The temperature signal is supplied by the turbine manufacturer. An auto-manual bias control station is a part of this control loop.

   A similar control is used for the common cooling water of the three-unit turbine lubrication oil cooler.

4. The return cooling water header from the main generator primary water coolers and the eight-unit hydrogen coolers are provided with electric-motor-operated throttling-type control valves. Both of the valves' control signals are supplied from temperature control circuits designed and furnished by the turbine manufacturer.

10.4.13 AUXILIARY STEAM SYSTEM

This section discusses the Auxiliary Steam System in accordance with the requirements of NRC Regulatory Guide 1.70 [16].

The Auxiliary Steam System shown on Figure 10.4-16 is designed to provide a supply of low pressure steam to, and collect condensate from, miscellaneous warmup and process services in the Turbine and Auxiliary buildings. The system serves to conserve condensed steam for reuse. The system also minimizes possible leakage of radioactive contamination into the Unit 1 Turbine Building Sump #2 in case of steam coil leakage within the evaporators.

The system is non-nuclear safety-related. Seismic design criteria are not considered.

10.4.13.1 Design Bases

The Auxiliary Steam System is designed to meet the requirements listed in Table 10.4-13.

Drip pots are provided to collect the auxiliary steam condensate. Lines connect the drip pots to the main condenser and return the condensate to the condensate system.

Individual NSSS components which require auxiliary steam and which discharge condensate into the auxiliary steam drain tank are shown in Table 10.4-14.

Make-up water for the auxiliary boiler is demineralized deaerated water pumped from the Unit 1 and 2 condensate system.

10.4.13.2 System Description

10.4.13.2.1 General

The Auxiliary Steam System supplies steam at 50 and 150 psig to the distribution points shown in Table 10.4-13. This system also collects condensate from individual components listed in Table 10.4-14.

The Auxiliary Steam System consists of an auxiliary steam header (from which the steam is piped to the different points of application), an auxiliary steam drain tank, two auxiliary steam drain tank pumps, an Auxiliary boiler (provided with two boiler feedwater transfer pumps, a deaerating tank, two boiler feedwater pumps, a chemical feed system, an oil storage tank, and two oil feed pumps), and the associated pipings, valves, and instrumentation and controls. This arrangement is shown on Figure 10.4-16.

10.4.13.2.2 Startup

Prior to the initial startup of Unit 1, the electric auxiliary boiler was the only source of auxiliary steam. The original electric Auxiliary Boiler is not available for use and has been replaced by a fuel oil fired Auxiliary Steam Boiler.

The auxiliary boiler steam is initially utilized to clean the major pipelines. Steam from the auxiliary boiler or the other unit in operation is used for preheating the hot well via the sparger, for preheating the moisture separator/reheater tube sheets, and for sealing the main and feedwater pump turbine glands. Once the plant is in operation, the auxiliary boiler may be shut down and steam may be furnished by the normal supply sources.

10.4.13.2.3 Normal Operation

There are two normal operation supply sources for the Auxiliary Steam System - the main steam and extraction steam systems. These can be furnished by either one or both units.

High-pressure main steam is supplied to the Auxiliary Steam System via the main steam inlet header until it is reduced to 150 psig by means of a pressure-regulating control valve. The extraction steam supply serves as the low-pressure steam supply source and is reduced to 50 psig by means of a pressure control valve. The 50-psig steam serves as the normal supply of process steam for the three evaporator packages and the volume control system boric acid batching tank.

Figure 10.4-16 shows the path for supply of the steam to the floor drain waste evaporator package, waste evaporator package (liquid waste management system), recycle evaporator package (boron recycle system), and boric acid batching tank located in the Auxiliary Building.

The flow rate of steam to each evaporator is varied to maintain temperature or pressure set points in each evaporator by means of control valves.

Individual condensate lines, with differing pressures, are connected to a common header before the condensate flows pass into the condensate cooler, which is designed to cool the maximum flow of condensate down to 150°F at atmospheric pressure. A loop seal is provided between the condensate cooler and the drain tank. A bleed-off connection from the loop seal passes a small percentage of the total flow through a sample line before discharging into the auxiliary steam drain tank. Individual condensate lines are also connected to the sample line for leak detection purposes. A diaphragm valve on each bleed-off line is provided to ensure positive closure of the sample line. In normal operation, the bleed-off valve from the inlet header to the sample line remains open, while the other bleed-off valves remain closed.

Upon detection of radioactivity in the sample, the operator manually closes the bleed-off valve from the inlet header to the radiation monitor and opens the individual bleed-off valves one by one, in order to determine the source of the leakage.

The level-controlled drain tank is provided with an overflow line, a drain line for maintenance, a connection from demineralized water for washing, and a discharge connection to two drain pumps. These pumps are sized for 100-percent-capacity each. One pump serves as standby. Discharge of the pumps is directed to the Unit 1 Turbine Building Sump #2.

10.4.13.2.4 Auxiliary Boiler

The auxiliary boiler is designed to supply steam at a pressure of 150 psig and a rate to supply startup requirements for one unit startup. A fuel oil tank is furnished with the boiler to provide a continuous source of fuel for full load boiler operation. The fuel oil tank is bermed to prevent spillage to the environment. Fuel oil is pumped to the boiler by one of two 100% capacity auxiliary boiler fuel oil pumps. The boiler is also furnished with controls to shut itself down in the event of excess steam pressure or low, or high, water level. Feedwater is pumped to the boiler by one of two 100 percent auxiliary boiler feedwater pumps.

Intermittent boiler blowdown is required to dispose of accumulated feedwater impurities. This is accomplished by the auxiliary boiler blowdown flash tank which operates at atmospheric

pressure. Wastes are drained to the auxiliary boiler sump for final disposal to the low volume waste ponds. An open vent discharges flashed steam to the atmosphere. Blowdown to the flash tank originates from the main blowdown, and surface blowdown lines identified on Figure 10.4-16.

10.4.13.2.5 Abnormal Conditions

In the event of evaporator coil leakage, the sampling and analysis would detect radioactive contamination and the operator would shut off the steam supply control valves. Demineralized water is then introduced into the system to decrease the amount of radioactivity in the lines. All possible contaminated condensate is directed to the Liquid Waste Management System to preclude the entrance of any radioactive contamination to the Unit 1 Turbine Building Sump.

To determine the source of the leakage, individual condensate return test lines from each evaporator discharge are provided. Each test line is opened individually and the condensate is sampled until the source of the leakage is determined. Upon completion of repair, the steam supply valves are manually reopened, and normal operation of the system is resumed. An auxiliary steam sample cooler is provided to cool the condensate to approximately 100°F before it enters the sample line to protect personnel.

10.4.13.2.6 Electrical Requirements

Safe shutdown of the unit does not rely on the availability of the Auxiliary Steam System. Hence, power to this system is not provided from the standby diesel generators. Power is supplied by non-Class 1E sources.

10.4.13.3 Safety Evaluation

The amount of noncondensable gases released by the condensed steam is negligible; therefore, there is no need for the use of radioactivity monitoring throughout the system.

Drain pots are provided at the piping low points to remove any condensate in the steam lines during startup and normal operations.

Relief valves are installed downstream of every pressure regulator to relieve the system pressure in the event the regulator malfunctions. The auxiliary boiler is also equipped with safety valves set to open at maximum boiler working pressure.

10.4.13.4 Tests and Inspection

Normal preoperational tests are performed in accordance with applicable codes and standards. Initial inspection and testing are performed to ensure system integrity and completeness.

10.4.13.5 Instrumentation Requirements

10.4.13.5.1 Indication and Monitoring

Local pressure indications are provided for the auxiliary boiler feedwater transfer pump discharge, 50- and 150-psig steam headers, steam outlet from the auxiliary boiler, and the discharge side of pressure reducing control valves. The auxiliary boiler control panel, which is

part of the package, has sufficient instrumentation to control, supervise, monitor, and alarm all required system functions. A description of the radiation monitoring system can be found in Section 11.5.

10.4.13.5.2 Control

All functions of the auxiliary boiler are controlled from the auxiliary boiler control panel, except control of the two feedwater transfer pumps which are controlled from the local control panel. Auxiliary boiler fuel oil recirculation pump and chemical feed system pumps and mixers are controlled from local push button stations. All other system controls not located on the auxiliary boiler panel are controlled from the main control board in the Control Room. Self-regulated pressure control valves are provided for monitoring a constant pressure in the header.

10.4.13.5.3 Alarm

All alarms related to the auxiliary boiler are located on the auxiliary boiler control panel. A common alarm contact, which functions with the system annunciator, is provided to indicate abnormalities in the auxiliary boiler on the main control board. Alarms are also provided for high- and low-pressure in the steam headers and for high-level in the drain tank.

10.4.14 TURBINE OIL PURIFICATION SYSTEM

1. The Turbine Oil Purification System of the CPNPP is designed to continuously purify lubricating oil for the main and feedwater pump turbines. Provisions for the storage, handling, and heating of turbine oil are also included. System components are located in the Turbine Building and the yard.

2. This system is non-nuclear-safety-related. Seismic design criteria are not considered.

10.4.14.1 Design Bases

The Turbine Oil Purification System is designed to do the following:

1. To provide for the continuous removal of water and particulate contaminants in excess of three microns from the turbine oil at a flow rate of 50 gpm

2. To provide turbine oil purification without affecting the lubricating or sealing properties of the oil and without increasing the acidity or removing additives or inhibitors

3. To furnish makeup oil storage, filtration of all fresh and recoverable oil before transfer to system components, and storage for approximately 25,000 gal of oil whenever drainage of system components is required

4. To provide heating of the main oil feed to its optimum temperature

5. To provide for exhausting extracted moisture and oil vapors from system components

10.4.14.2 System Description

The Turbine Oil Purification System for each unit of the CPNPP as shown on Figure 10.4-17 consists of automatically controlled-skid mounted centrifuge purification units including local control panels (two for Unit 1 and one for Unit 2), one 300-gal sludge holdup tank, one 25,000-gal lubricating oil storage tank, one drain pump, all associated piping, valves, instrumentation, and controls, and one oil conditioner filtration unit. System components are located in the Turbine Building and the yard.

System filling is by oil tank trucks that discharge into the lubricating oil storage tank.

During normal Unit operation, the oil conditioner is primarily used. Oil flows from the main and FWP turbine oil tank discharge line into the oil conditioner. The oil flows through the precipitation compartment and into the filtration compartment before entering the storage compartment. From there the oil flows into the circulating pump which discharges to the polishing filter and then back to the main and FWP turbine oil tanks. If the oil conditioner is not being used the oil will flow through the centrifuge skid as described below.

During alternate system operation, turbine oil flows continuously by gravity from the main turbine and feedwater pump turbine oil tanks to the centrifuge skids. Normally, only one skid will be in operation. Purification is by centrifugal separation, with final polishing through a micronic oil filter. Clean, dry oil is then fed to the main turbine and feedwater pump turbine oil tanks. Each tank is furnished with a discharge line taking suction from an internal standpipe to prevent accidental tank drainage due to siphon effects.

Oil vapor exhausters are provided to remove air from the tanks and prevent moisture condensation. Drains from the oil vapor exhausters and mist eliminators are directed to the sludge holdup tank. All equipment drains not operating continuously are furnished with normally closed valves to prevent accidental drainage.

A 225-gpm drain pump is provided for the discharge of spent oil from the plant.

Electrical loads for this system are supplied from 480-V non-Class 1E power buses.

10.4.14.3 Safety Evaluation

The turbine oil skids and lubricating oil tank are all located in an enclosed room at elevation 778 ft 0 in. in the Turbine Building. As required by the fire codes, this room is sealed to prevent oil leakage outside the room in the event of a major tank rupture. The components supplied by the turbine generator manufacturer, as well as the main turbine oil tank, are similarly enclosed on the 803 ft 0 in. elevation. Safe shutdown of the unit does not rely on the availability of the turbine oil purification system. Therefore, there is no provision to provide power to this system from the standby diesel generators.

10.4.14.4 Test and Inspection

Normal periodic replacement of the micronic oil filters is performed when high differential pressure is noted. Cleaning of the centrifuge components will be performed on a periodic basis.

10.4.14.5 Instrumentation and Controls

The sludge holdup tank, and the lubricating oil storage tank are provided with high-level switches wired to the Control Room for alarm.

Local pressure indicators are provided on the drain pump and centrifuge skids. Local flow instrumentation is also included on the centrifuge skids.

The control and supervisory instrumentation associated with the two feedwater pump turbine oil tanks are wired to the Control Room for remote pressure indication and low-pressure alarms.

Remote high oil cooler discharge temperature alarm and indication for feedwater pump turbine are wired to the Control Room from the temperature indicator.

Two independent temperature switches control heater operation in the 90°F through 120°F design temperature range. A flow switch is also provided and is interlocked with the centrifuge skid feed pump. The interlock trips the heater if the pump stops or fails to operate for any reason, thus reducing the possibility of carbonization of the turbine oil by avoiding stagnation of oil in the heater. Heater controls and supervisory instrumentation are located in the skid mounted control panels. Each skid mounted control panel is wired to a common local control panel.

The 225-gpm drain pump is manually controlled from the common local control panel.

The continuously operating centrifuge skids are started and stopped manually from the controls located on their respective control panels.

10.4.15 NITROGEN AND HYDROGEN SUPPLY SYSTEMS

10.4.15.1 Nitrogen Supply System

10.4.15.1.1 Design Basis

The Nitrogen Supply System is designed to provide storage capacity to ensure regular replenishment of consumed nitrogen. The system is divided into two separate, permanent subsystems: the bulk storage supply and the bottle supply. Each of these subsystems has two supply manifolds, one for normal operation, the other to act as a standby source.

The design storage capacity for each supply package is based on the following:

1. Storage of 30 days is required for gases which are continuously consumed.

2. Storage of one gas charge is required for intermittent uses.

3. A safety margin of 10 to 15 percent of Items 1 and 2 is included.

The Nitrogen Supply System is designed to meet the applicable requirements of the ASME B&PV Code, Section VIII, Pressure Vessels, Part A, ANSI B31.1, Code for Pressure Piping, Power Piping, and ANSI B.31.8, Gas Transmission and Distribution Piping Systems.

Tables 10.4-16 and 10.4-17 list the components which receive nitrogen from this system. Certain components require only a one-time fill prior to reactor startup or shutdown. The capacity for such service is not included in the storage capacity of this system.

10.4.15.1.2 System Description

The Nitrogen Supply System is shown on Figure 10.4-18. Nitrogen is supplied to the various components from one or more of the following three types of storage facilities:

1. Bulk Supply

   The following components are supplied by a 2400 psig pressure bulk storage system:

   a. Low Pressure Supply System (100 psig)

      For intermittent use, nitrogen is supplied to the following:

      1. Volume control tank
      2. Pressurizer relief tank
      3. Catalytic recombiners
      4. Spent resin storage tank
      5. Steam generators
      6. Containment Spray System (CSS) chemical additive tank
      7. Steam Generator Blowdown System (SGBS) spent resin storage tank
      8. Waste Processing System (WPS) gas decay tanks
      9. Turbine generator
      10. Vent and Drain System Atmospheric Drain Tanks
      11. Reactor Makeup Water Storage Tank
      12. Boric Acid Storage Tank

   b. High-Pressure Supply System (700 psig)

      For intermittent use, nitrogen is supplied to the Safety Injection System (SIS) accumulators and Pressurizer PORVs (after pressure reduction)

2. Bottle Supply

The Containment Air and Reactor Coolant PASS Remote Operating Modules are supplied with a permanent source of nitrogen via gas cylinders to allow for continuous operation of Post Accident Sampling System when needed.

Nitrogen bottles are also available for the Secondary Sampling Ion Chromatograph and for supporting leak rate testing of the Electrical Penetration Assemblies (EPAs) on an intermittent basis. The bottles are required to be isolated from the EPAs during normal operation to prevent overpressurization.

In addition, nitrogen bottle supply is provided as a backup source to the Plant Gas Nitrogen system in support of the Generator Gas system. A Bottle rack is also located outside the FW Penetration Isolation Valve rooms to provide high pressure nitrogen for valve maintenance purposes.

Nitrogen bottles provide a continuous flow of purge gas to the hydrogen sensor on the RCS hydrogen/oxygen dissolved gas analyzers in the Process Sample System.

3. Temporary Tube Trailer Supply

The following components are supplied from temporary sources which are furnished only when required:

a. Gas decay tanks (initial plant startup)

b. SIS accumulators (initial plant startup)

c. Volume control tank (initial plant shutdown)

d. Pressurizer PORV accumulators (initial plant startup)

The bulk storage supply is common to the two units. Nitrogen is supplied to two high pressure manifolds with each manifold capable of supplying 100 percent of the plant needs. Safety valves are furnished on each manifold as well as downstream of the high-pressure regulators. These valves are set approximately 10 percent above operating pressure. Excessive pressure in the system can be caused by a regulating valve malfunction or a large increase in the stored gas temperature.

Two 700-psig-pressure nitrogen supply lines (one from each manifold) supply the SIS accumulators and the accumulators for the pressurizer PORV operators. Each of the supply lines branches into two more lines, one for Unit 1 and one for Unit 2. In this way, each unit's accumulators are supplied by redundant nitrogen supply manifolds. Pressure and level switches on the accumulators indicate when nitrogen is required. Isolation and feed valves are opened manually from the Control Room.

A second set of pressure regulators in a supply line delivers high-flow 100-psig nitrogen to a header which is common to the two units. This header in turn supplies local headers at each unit. The services supplied by the local headers are grouped in the following manner. Each unit has a separate header which supplies nitrogen to its four steam

generators. Another separate header for the two units supplies nitrogen to the volume control tank, the CSS chemical additive tank, the RCS pressurizer relief tank, steam generator blowdown, and the vent and drain system atmospheric tank. Each unit has a separate line supplied by the main low-pressure header to the turbine generator. Other supply lines from the main header supply the WPS gas decay tanks and hydrogen recombiners, WPS spent resin storage tank, and the SGBS spent resin storage tank. These systems are common to the two units.

Under normal operation, the lines feeding the components which require only intermittent supplies of gas are valved closed, and these feed valves are manually opened only when the need arises, such as during startups, shutdowns, low gas pressure, and so forth. Pressure control is accomplished by pressure regulating valves for each feedline. In addition, each manifold header has a high pressure reducing valve. All regulating valves to distribution points are process actuated and respond to changes in the downstream pressure.

The piping for the Nitrogen Supply System is non-nuclear safety-class, except for those portions which connect to a safety-related component or penetrate the Containment. Where this piping does connect to a safety-related component requiring an isolation valve, this valve and the piping between the valve and the connection are designed to the same safety class as the component. Lines which do not require isolation valves are non-nuclear-safety-class up to the connection of the safety-related component. The nitrogen supply lines which penetrate the Containment are in accordance with ASME B&PV Code, Section III, Safety Class 2, and are discussed in Section 6.2.4.

10.4.15.1.3 Safety Evaluation

The nitrogen supply is a non-nuclear-safety-class system. Safety related components which are intermittently supplied by the system do not require continuous supply during an accident condition. The SIS accumulators, which require nitrogen for safe shutdown of the unit, have sufficient nitrogen within the system to operate as required. Where nitrogen is supplied to a component containing potentially radioactive material, a check valve is provided to prevent contamination of the Nitrogen Supply System.

The bulk storage of the nitrogen is located outdoors where a tank rupture would not affect any safety-related equipment.

10.4.15.1.4 Inspection and Testing Requirements

All piping and pressure vessels for the nitrogen supply system are leak tested in accordance with ANSI B31.1. The systems are inspected periodically to ensure that no leakage exists.

10.4.15.1.5 Instrumentation Applications

The Nitrogen Supply System is furnished with the following instrumentation:

1. Manifold Components (two manifolds per system)

    a. Local pressure indicator

b. Local temperature indicator

2. Supply Header Components (One header per system)

a. Local pressure indicator

b. Low-pressure switch

All low-pressure switches send signals to the local system panel which in turn alarms a single annunciator (“Bulk Gas System N2 Trouble”) in the Control Room to alert the operator to abnormal pressure conditions. Supply lines to the various components are equipped with a pressure indicator (included in the pressure control package) to set and locally monitor gas supply pressure to the component.

Each main and backup supply to the distribution header is furnished with a high-pressure regulating valve which maintains system supply pressure.

Most component supply lines are equipped with a second stage pressure regulator which controls the pressure according to individual component requirements. Table 10.4-16 shows equipment pressures and other pertinent data.

10.4.15.2 Hydrogen Supply System

10.4.15.2.1 Design Basis

The Hydrogen Supply System is designed to provide bulk gas storage capacity to ensure regular replenishment of consumed gases. The system has two manifolds, one for normal operation, the other for standby reserve.

The design storage capacity for the supply package is based on the following:

1. Storage of 30 days is required for gases which are continuously consumed.

2. Storage of one gas charge is required for intermittent uses

3. A safety margin of 10 to 15 percent of Items 1 and 2 is included.

The Hydrogen Supply System is designed to meet the applicable requirements of the ASME B&PV Code, Section VIII, Pressure Vessels, Part A, ANSI B31.1, Code for Pressure Piping, Power Piping, and ANSI B31.8, Gas Transmission and Distribution Piping Systems. This system also meets the applicable National Fire Protection Association (NFPA) codes.

Tables 10.4-16 and 10.4-17 list the components supplied by this system and their requirements. Components requiring only one-time fill prior to startup or shutdown of the unit are not included in the storage capacity of this system.

10.4.15.2.2 System Description

The Hydrogen Supply System is shown on Figure 10.4-19. Hydrogen is supplied by the following systems:

1. Bulk Gas Supply

   The following components are supplied by the bulk gas storage system:

   a. For Intermittent Use

      Gas decay tanks

   b. For Continuous and Intermittent Use

      1. Turbine generator

      2. Primary water tank

      3. Volume Control Tank

2. Bottle Gas Supply

   For continuous use, the following components are supplied by separate bottle gas storage supplies:

   a. Turbine generator and primary water tank (backup to bulk gas storage system)

   b. Reactor coolant drain tank

   The two independent bulk gas storage systems each have redundant supply manifolds. Each supply manifold is capable of supplying 100 percent of the system requirements.

   Safety valves are furnished on each manifold as well as downstream of the high pressure regulators. The safety valves are set approximately 10 percent above operating pressure. Excessive pressure in the system can be caused by regulating valve malfunction or by a large increase in the stored gas temperature.

   The hydrogen bulk gas storage system supplies a header which is common to the two units. This header divides into the NSSS supply line and the turbine generator supply line. The NSSS supply line supplies each unit's volume control tank and the WPS gas decay tanks which are common to the two units. The turbine generator supply line branches into two lines, one for each unit. This line supplies the turbine generator and the primary water tank.

   In addition to the bulk gas supply, the turbine generator and primary water tank for each unit have a backup bottle gas supply. The reactor coolant drain tank for each unit has an independent bottle gas supply. Each of the drain tank supply systems has redundant supply bottles.

Pressure control is accomplished by pressure regulating valves for each feedline. In addition, each manifold header has a high pressure reducing valve. All regulating valves to distribution points are process actuated and respond to changes in the downstream pressure.

Bulk storage tanks are recharged periodically by one or several tube trailers. Truck fill connections are provided on each system for both refilling the supply and furnishing initial charge for components requiring this.

The piping for the Hydrogen Supply System is non-nuclear-safety class except for those portions which connect to a safety-related component or penetrate the Containment. Where this piping does connect to a safety-related component requiring an isolation valve, this valve and the piping between the valve and the connection are designed to the same safety-class as the component. The hydrogen supply lines which penetrate the Containment are in accordance with the ASME B&PV Code, Section III, Safety Class 2, and are discussed in Section 6.2.4.

10.4.15.2.3 Safety Evaluation

The Hydrogen Supply System is a non-nuclear-safety class system. Continuous supply of hydrogen to safety-related equipment is not required for safe shutdown of the plant. Lines to components containing potentially radioactive material are provided with check valves to prevent contamination of the Hydrogen Supply System.

The bulk hydrogen storage is located outdoors where a tank rupture would not affect any safety-related equipment. The outdoor location of these tanks prevents any explosive concentration of hydrogen from accumulating as a result of leaks.

In the event of a break in a hydrogen supply line, the excess flow control valve automatically closes.

10.4.15.2.4 Inspection and Testing Requirements

All piping and pressure vessels for the Hydrogen Supply System are hydrostatically tested to 1.5 times their design pressure or pneumatically tested to at least 1.2 times their design pressure. The systems are inspected periodically to ensure that no leakage exists.

10.4.15.2.5 Instrumentation Application

Each of the normal and backup gas supply systems is furnished with the following instrumentation.

1. Manifold Components (two manifolds per system)

    a. Local pressure indicator

    b. Local temperature indicator

2. Supply Header Components (One header per system)

   a. Local pressure indicator

   b. High- and low-pressure switches

All high- and low-pressure switches send signals to the local system panel which in turn alarms a single annunciator (“Bulk Gas System H2 Trouble”) in the Control Room to alert the operator to abnormal pressure conditions. Supply lines to the various components are equipped with a pressure indicator (included in the pressure control package) to set and locally monitor gas supply pressure to the component.

The hydrogen bulk gas storage systems are furnished with high pressure regulating valves on both the main and backup supply manifold. These valves maintain the system supply pressure. Component supply lines are equipped with a second stage pressure regulator which controls the pressure according to individual component requirements.

10.4.16 SECONDARY PLANT SAMPLING SYSTEM

The secondary plant sampling (SPS) system is provided to monitor water chemistry in condensate, steam generator feedwater, steam generator blowdown, main and auxiliary condensers, condensate polishing, heater drains, condensate storage tank, vacuum deaerator, main steam, atmospheric drain tank #1, and the demineralized water storage tank. The main panel is a factory assembled, skid mounted package located at elevation 778 ft in the Electrical & Control Building. Additional remote sample stations are located in the plant and send analyzer signals back to the main panel.

10.4.16.1 Design Basis

The SPS system is designed as the central point to which are routed samples and/or signals from various fluid streams in the plant. A flow diagram of the system is shown on Flow Diagram M1(2)-0222. The pressure and temperature conditions at the various sample points are noted in Table 10.4-18.

Specially designed sample pumps are used to pump the main condenser hot well samples to the sample panel.

The design criteria for the hot well sample pumps, the chilled water return pump, and the sample coolers are given in Table 10.4-19.

10.4.16.2 System Description

The SPS system is designed to continuously measure the selected parameters in samples from the secondary cycle, as shown in Table 10.4-20.

Where required, sample lines are cooled and depressurized by cooling coils, an isothermal bath, and pressure reducing and regulating valves. This process allows for uniform and safe operation of the analyzers and the grab sample sink.

Five remote sample stations are located outside the main panel room and transmit analyzer signals back to the main panel.

1. Located in the Safeguards Building pipe tunnel, the Condensate Storage Tank upper and lower volumes are sampled for dissolved oxygen, cation conductivity, and sodium ion (local panel only).

2. Located in the Turbine Building (adjacent to the condensate pumps), the Atmospheric Drain Tank #1 is sampled for cation conductivity.

3. Located in the Turbine Building (adjacent to the vacuum deaerator), the Vacuum Deaerator is sampled for dissolved oxygen and the Demineralized Water Storage Tank is sampled for specific conductivity.

4. Located in the Turbine Building are two analyzers which monitor and display condenser sodium concentration from hotwells A and B. Signals are sent to the main sampling panel recorders. High sodium concentration alarm signals are announced on the main sampling panel.

A Secondary Side Ion Chomatograph is located in its own enclosure in the turbine building (near the circulating water booster pumps). The analyzer, common to both units, takes samples of Unit 1 and 2 hotwell A.

Local Dissolved Oxygen Monitors are located at various locations in the Turbine Building including one at the Feedwater sampling station on elevation 803' and at Common Condensate Discharge Monitoring station on elevation 778'. Data is relayed to the Plant Computer.

Secondary Corrosion Product Monitor panels are provided at Feedwater, Steam Generator Blowdown, Heater Drains and Condensate sampling stations located throughout the Turbine Building and Electrical Control Building.

10.4.16.3 Safety Evaluation

This system is non-nuclear-safety-related. Seismic design criteria are not applicable. Safe shutdown of the unit does not rely on the SPS system. Hence, there is no provision to provide power to this system from emergency diesel generators.

10.4.16.4 Tests and Inspections

All tests and inspections of the SPS system are performed in accordance with applicable codes.

10.4.16.5 Instrumentation and Control Requirements

It is intended that the SPS system perform only advisory and early warning functions (except for amine and hydrazine pump control) by continuously monitoring critical parameters in certain fluid streams; therefore, an extensive system alarm annunciator is provided. Activation of any of the alarms is automatically relayed to the common trouble alarm located on the main control board.

REFERENCES

1. 10 CFR Part 50, Appendix A, General Design Criterion 60, Control of Releases of Radioactive Material to the Environment.

2. 10 CFR Part 50, Appendix A, General Design Criterion 64, Monitoring Radioactivity Releases.

3. NRC Regulatory Guide 1.26, Quality Group Classifications and Standards for Water-, Steam- and Radioactive-Waste-Containing Components of Nuclear Power Plants, Revision 3, February 1976.

4. ANSI N18.2, Nuclear Safety Criteria for the Design of Pressurized Water Reactors.

5. NRC Regulatory Guide 1.29, Seismic Design Classification, Revision 2, February 1976, U.S. Nuclear Regulatory Commission.

6. Branch Technical Position APCSB 3-1, Protection Against Postulated Piping Failures in Fluid Systems Outside Containment, attached to Standard Review Plan 3.6.1.

7. Branch Technical Position MEB 3-1, Postulated Break and Leakage Locations in Fluid System Piping Outside Containment, attached to Standard Review Plan 3.6.2.

8. 10 CFR Part 50, Appendix A, GDC 2, Design Bases for Protection Against Natural Phenomena.

9. 10 CFR Part 50, Appendix A, GDC 4, Environmental and Missile Design Bases.

10. 10 CFR Part 50, Appendix A, GDC 5, Sharing of Structures, Systems, and Components.

11. 10 CFR Part 50, Appendix A, GDC 19, Control Room.

12. 10 CFR Part 50, Appendix A, GDC 44, Cooling Water.

13. 10 CFR Part 50, Appendix A, GDC 45, Inspection of Cooling Water System.

14. 10 CFR Part 50, Appendix A, GDC 46, Testing of Cooling Water System.

15. Branch Technical Position APCSB 10-1, Design Guidelines for Auxiliary Feedwater System Pump Drive and Power Supply Diversity for Pressurized Water Reactor Plants.

16. NRC Regulatory Guide 1.70, Standard Format & Content of Safety Analysis Report for Nuclear Power Plants, LWR edition, Revision 2, September 1975.

17. 10 CFR Part 50, Appendix A, GDC 57, Closed System Isolation Valves.

18. NUREG-0291, An Evaluation of PWR Steam Generator Water Hammer by Creare, Inc, Dec. 31, 1976.

19. WCAP-09364, Vol. 1, High Pressure Water Hammer Test Program for the Counterflow Preheat Steam Generator.

20. Westinghouse Report “Counterflow Preheat Steam Generator Tube Expansion Report,” June 1983.

21. Westinghouse Report “Counterflow Preheat Steam Generator Vibration Summary,” June 1983.

22. NRC Regulatory Guide 1.32, “Criteria for Safety Related Electric Power Systems for Nuclear Power Plants”, Revision 2, February 1977, U. S. Nuclear Regulatory Commission.

23. NRC Regulatory Guide 1.47, Bypassed and Inoperable Status Indication for Nuclear Power Plant Safety Systems, May 1973, U. S. Nuclear Regulatory Commission.

24. NRC Regulatory Guide 1.53, Application of the Single Failure Criterion to Nuclear Power Plant Protection Systems, June 1973, U. S. Nuclear Regulatory Commission.

25. NRC Regulatory Guide 1.75, Physical Independence of Electric Systems, Revision 1, January 1975, U. S. Nuclear Regulatory Commission.

26. Westinghouse Guidelines for Secondary Water Chemistry (SGT-5.1.1-4468), dated February 1985.

27. EPRI PWR Secondary Water Chemistry Guidelines, Revision 2, dated November 1988.

28. 10CFR50, Appendix A, GDC 54, Piping Systems Penetrating Containment.

TABLE 10.4-1
CONDENSER EVACUATION SYSTEM EQUIPMENT DESIGN CRITERIA

Condenser Exhausting Vacuum Pump Package Data Sheets

a. Condenser Exhausting Vacuum Pump

Type of Pump	Rotary, two stage
Design holding capacity, scfm at 1.0 in. Hg abs.	25
Horsepower input to pump at design conditions, bhp	130
Speed, rpm	
Pump	435
Motor	1800
Motor rated hp/service factor	150/1.15
Motor, voltage/hertz/phase	460/60/3
Hogging capacity, dry air, acfm at 10 in. Hg. abs.	2300
Maximum flow rate of air, vapor at discharge, scfm at 15 in. Hg abs.	1250

b. Condenser Exhausting Vacuum Pump Heat Exchanger

Shell-side design temperatures, °F	300
Shell-side design flow rate, gpm	90
Tube-side design temperatures, °F	300
Tube-side design velocity, ft/sec	5.3
Tube-side design flow rate, gpm	700
Shell-side design pressure, psig	150
Tube-side design pressure, psig	75
Tube-side pressure drop (ft)	5.075

TABLE 10.4-2
STEAM DUMP VALVES DESIGN DATA

Safety class	Non-nuclear-safety-related, non-seismic Category I
Design pressure, psia	1300
Design temperature, °F	700
Operating steam flow, lb/hr/valve	530,000
Operating pressure, psia	965 to 1107
Operating temperature, °F	540 to 557
Maximum steam flow, lb/hr/valve (at 1185 psig pressure)	1,109,000
Valve opening, sec	3
Valve closing, sec	5
Actuator	Pneumatic
Size, in.	8

TABLE 10.4-3
ANTICIPATED WATER QUALITY OF SQUAW CREEK RESERVOIR

Substance	Range as CaCo3 (mg/l)
Calcium	360 to 940
Magnesium	80 to 410
Sodium	1810 to 2220
Bicarbonate	200 to 210
Carbonate	0 to 28
Sulfate	620 to 1040
Chloride	1630 to 2310
Silica	8 to 68 (as Si02)
Ammonia	1.2 (as N2)
Phenolphthalein alkalinity	0 to 8
Methyl orange alkalinity	200 to 238
Hardness	400 to 1350 ppm
Total dissolved solids	2450 to 3570 ppm

TABLE 10.4-4
(DELETED)

TABLE 10.4-5
CONDENSATE AND FEEDWATER SYSTEM EQUIPMENT DESIGN PARAMETERS

Steam Generator Feedwater Pumps	
Safety class	NNS
Capacity at design rating, gpm	19,800
Capacity at normal operation, gpm (356°F)	17,090
TDH at design rating, ft	2322
TDH at normal operation, ft	2278
Minimum NPSH required at design rating, ft	232
Pump efficiency at design rating, percent	88.6
Design pressure, psig	2000
Design temperature, °F	400
Condensate Pumps	
Safety class	NSS
Capacity at design rating, gpm	11,460
Capacity at normal operation, gpm (121°F)	9950
TDH at design rating, ft	1095
TDH at normal operation, ft	1120
Minimum recirculation flow, gpm	2900
TDH at minimum recirculation flow, ft	1340
Condensate Storage Tank	
See Table 10.4-8 and Section 10.4.9	

Note:

Total dynamic head (TDH); net positive suction head (NPSH)

TABLE 10.4-6
STEAM GENERATOR BLOWDOWN SYSTEM EQUIPMENT DESIGN CRITERIA

(Sheet 1 of 4)

SGBS Heat Exchanger	
Quantity	One
Type	Horizontal U-tube
TEMA class	R
Tag Nos.	CP1-SBAHSB-01 and CP2-SBAHSB-01
Operating mode	1 x 100 percent
Design heat transfer rate, Btu/hr	150.73×10^6
Effective heat transfer area, ft^2	15,200
Overall length, in.	362
Outside diameter, in.	84

	Tube Side	Shell Side
System fluid	SGB	Condensate and SGB
Design flow, lb/hr	330,000	635,000
Design pressure, psig	1200	300
Design temperature, °F	700	600
Inlet temperature, °F	567	125
Outlet temperature, °F	130	361
Thermal relief valves	One	One
Set pressure, psig	1200	300
Material	SA 249, Type 304	SA 516, Grade 70

TABLE 10.4-6
STEAM GENERATOR BLOWDOWN SYSTEM EQUIPMENT DESIGN CRITERIA

(Sheet 2 of 4)

SGBS Inlet Filters	
Quantity	Two
Tag Nos.	CP1-SBFLSB-01 and CP1-SBFLSB-02
Design temperature, °F	150
Design pressure, psig	300
Design flow, gpm	360
Dirt-holding capacity, grams of ferric oxide	1320
Material	AISI 304 stainless steel
Micron Rating (μM)	0.45 to 40 absolute (100% retention)
SGBS Spent Resin Sluice Pump Filter	
Quantity	One
Tag No.	CP1-SBFLSR-01
Design Temperature, °F	150
Design pressure, psig	150
Design flow, gpm	140
Dirt-holding capacity, grams of ferric oxide (Fe304)	3204
Material	AISI 304 stainless steel
Micron Rating (μM)	0.45 to 100 absolute (100% retention)

TABLE 10.4-6
STEAM GENERATOR BLOWDOWN SYSTEM EQUIPMENT DESIGN CRITERIA

(Sheet 3 of 4)

Steam Generator Blowdown System Demineralizers

Quantity	Four
Tag Nos.	CP1-SBDMCD-01 CP1-SBDMCD-02 CP2-SBDMAD-01 CP2-SBDMAD-02
Type	Flushable
Design pressure, psig	300
Design temperature, °F	150
Design flow, gpm	640
Resin type	Commercial available resins determined to be acceptable for use in the Steam Generator Blowdown system by the Secondary Chemistry Control Program.
Resin volume, ft^3	80
Overall Dimensions	
Outside diameter, in.	60
Shell height, in.	96
Material	AISI 304 stainless steel
Decontamination factor	See Table 11.2-6.

TABLE 10.4-6
STEAM GENERATOR BLOWDOWN SYSTEM EQUIPMENT DESIGN CRITERIA

(Sheet 4 of 4)

Steam Generator Blowdown Spent Resin Sluice Pump

Quantity	One
Tag No.	CPX-SBAPRS-01
Operating mode	1 x 100 percent
Type	Horizontal centrifugal, canned
Designed flow, gpm	150
Design heat, ft	230
Material	AISI 316 stainless steel
Motor	
hp	20
Service factor	1.15
rpm	3450
V/phase/Hz	460/three/60

Steam Generator Blowdown Spent Resin Storage Tank

Quantity	One
Tag No.	CPX-SBATSR-01
Safety class	NNS
Type	Vertical cylindrical
Design capacity, gal	3740
Design pressure, psig	100
Design temperature, °F	200
Material	AISI 304 stainless steel

TABLE 10.4-7
STEAM GENERATOR BLOWDOWN CLEANUP SYSTEM FAILURE ANALYSIS

Components	Malfunction	Remarks
SGBS isolation valves	One valve fails closed.	Blowdown from one steam generator is not available. Plant safety is not affected.
	One valve fails open	These valves are designed to fail closed. Manual valves are provided in series with the automatic valves.
Isolation valves downstream of the SGBS heat exchanger	One valve fails closed. One valve fails open.	Plant safety is not affected.
Demineralizers (cation or mixed bed or both)	One demineralizer fails or is exhausted.	Decontamination capability is reduced temporarily. Plant safety is not affected.

TABLE 10.4-8
EQUIPMENT DESIGN PARAMETERS

(Sheet 1 of 3)

1. Motor-Driven Auxiliary Feedwater Pump

Safety class	3
Seismic class	Category I
Design flow, gpm	570
Design TDH, ft	3160
Shutoff TDH	3650
Normal flow	N/A
Normal TDH	N/A
Casing design temperature, °F and pressure, psig	150/1600
Runout flow, gpm	800
Minimum flow, gpm	100
Efficiency at design point, percent	73
Design point bhp	645
Maximum bhp	690
Motor hp and service factor	700/1.15
Motor electrical requirements, V/Hz/phase	6600/60/3
Rpm	3560
Miscellaneous information	horizontal, split-casing, centrifugal pump, Ingersoll Rand pump type 4HMTA-9 stage
Material, casing	ASME SA-216, WCB
Impeller material	ASME A-296-CA6NM
Suction and discharge nozzle material	ASME SA-216, WCB

TABLE 10.4-8
EQUIPMENT DESIGN PARAMETERS

(Sheet 2 of 3)

2. Turbine-Driven Auxiliary Feedwater Pump	
Safety class	3
Seismic class	Category I
Design flow, gpm	1145
Design TDH, ft	3160
Shutoff TDH	3722
Normal flow	N/A
Normal TDH	N/A
Casing design temperature, °F, and pressure, psig	150/1700
Runout flow, gpm	2118
Minimum flow, gpm	100
Efficiency at design point, percent	76.5
Design point bhp	1194
Maximum bhp	1300
Rpm	4075
Miscellaneous information	horizontal, split-casing, centrifugal pump, Ingersoll Rand pump type 5HMTA-6 stage
Material, casing	ASME SA-216, WCB
Impeller material	ASME A-296-CA6NM
Suction and discharge nozzle material	ASME SA-216, WCB
3. Condensate Storage Tank	
Safety class	3
Seismic class	Category I
Design pressure, psig and temperature, °F	0/150

TABLE 10.4-8
EQUIPMENT DESIGN PARAMETERS

(Sheet 3 of 3)

Total Volume of Tank, gal	500,000
Medium	condensate
Normal operating pressure and temperature, °F	atmospheric/40-120
Material of construction	concrete with stainless steel liner
Type of vessel and orientation	cylindrical/vertical
Diaphragm	Yes
Atmospheric relief	Yes
Miscellaneous	vertical, concrete, site-fabricated, outdoor tank

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 1 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
1	Diesel Generator A (active failure)	Provides power to Auxiliary feedwaterr pump motor 1 and motor controlled valve 17	Diesel Generator A fails to start		Meets single failure criteria:
				a. Auxiliary feedwater pump 1 is inoperative.	a. AF Pumps 2 and 3 are operable
				b. Steam generators 1 and 2 cannot be supplied by pump 1.	b. Steam generators 1 and 2 supplied from AF Pump 3 (Turbine Driven). Steam generator 3 supplied from AF Pumps 2 and 3.
				c. Valve 17, Train A, cannot be closed from Control Room	c. Valves 12 and 13 are closed from Control Room manually. (Isolates SG 4 Break).
2	Diesel Generator B (active failure)	Provides power to Auxiliary feedwater pump motor 2 and motor controlled valve 16	Diesel Generator B fails to start		Meets single failure criteria:
				a. Auxiliary feedwater pump 2 is inoperative.	a. AF Pumps 1 and 3 are operable.
				b. Steam generators 3 and 4 cannot be supplied by pump 2.	b. Steam generators 1 and 2 supplied from AF Pumps 1 and 3 and Steam Generator 3 supplied from AF Pump 3.
				c. Valve 16, on Train B, cannot be closed from Control Room.	c. Valves 12 and 13 are closed from Control Room manually. (Isolates SG 4 Break).

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 2 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
3	Battery A System (passive failure)		Battery A System lost		Meets single failure criteria:
		a. Pump 1 motor circuit breaker		a. Auxiliary feedwater pump 1 is inoperative.	a. AF Pumps 2 and 3 are operable.
		b. Provides control power to valves 6, 8, 11 and 13.		b. Steam generators 1 and 2 cannot be supplied by pump 1.	b. Steam generators 1 and 2 are supplied from AF Pump 3. Steam Generator 3 supplied from AF Pumps 2 and 3.
		c. AF Pump 3 speed indication		c. Valves 6, 8, 11 and 13 fail open.	c. Valves 12 and 16 are closed from Control Room manually. Throttling control of valves 6, 8, 11 and 13 is not available in Control Room; requires manual action locally.
		d. Provides control power to recirc. Valve 14		d. No remote speed indication on AF Pump 3.	d. AF Pump 3 runs at highest speed of speed setter. Pump may be tripped locally if required.
				e. AF Pump 1 recirc. valve open.	e. AF Pump 1 already inoperative, no effect.
3A	118 Vac “A” instrumentation system (passive failure)	a. Speed control for AF Pump 3	118 Vac “A” Bus Lost	Signal from speed set lost.	Same as 3d above.

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 3 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
4	Battery B System (passive failure)	a. Pump 2 motor circuit breaker b. Provides control power to valves 7, 9, 10 and 12 c. Provides control power to recirc. valve 15.	Battery B System lost	a. Auxiliary feedwater pump 2 is inoperative. b. Steam generator 3 cannot be supplied by pump 2. c. Valves 7, 9, 10 and 12 fail open. d. AF Pump 2 recirc. valve opens.	Meets single failure criteria: a. AF Pumps 1 and 3 are operable. b. Steam generators 1 and 2 are supplied from AF Pump 1 and 3. Steam generator 3 is supplied from AF Pump 3. c. Valves 13 and 17 closed manually from Control Room. Throttling control of valves 7, 9, 10 and 12 is not available in Control Room; requires manual action locally. d. AF Pump 2 inoperative, no effect.
5	Train A cable system (passive failure)	See failures 1, 3, and 3A.	See failures 1, 3 and 3A	See failures 3 and 3A	Meets single failure criteria: See failures 3 and 3A.
6	Train B cable system (passive failure)	See failures 2 and 4.	See failures 2 and 4.	See failure 4.	Meets single failure criteria: See failure 4.

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 4 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
7	Motor driven pump No 1	Supplies water to Steam Generators 1 and 2.	Stops pumping.	Steam Generators 1 and 2 cannot be supplied by Pump 1.	Meets single failure criteria: AF Pumps 2 and 3 are operable. 1) Motor-driven pump No. 2 feeds steam generator No. 3 while valve 12 (or 17) is manually closed from Control Room. 2) Turbine driven pump No. 3 feeds Steam Generators No. 1, 2 and 3 while valve 13 (or 16) is manually closed from Control Room.
8	Turbine driven pump (No. 3)	Supplies water to Steam Generators 1, 2, 3 and 4.	Does not operate. Loss of Turbine Lube Oil.	Steam Generators 1, 2, 3 and 4 cannot be supplied by Pump 3.	Meets single failure criteria: AF Pumps 1 and 2 are operable. 1) Motor-driven pump No. 1 feeds Steam Generators No. 1 and 2. 2) Motor-driven pump No. 2 feeds Steam Generator No. 3 while valve 12 (or 17) and 13 (or 16) is manually closed from Control Room.
9	Valve No. 12	Isolates Pump No. 1 flow to Steam Generator No. 4.	Fails to close	Valve No. 12 cannot isolate faulted steam generator.	Meets single failure criteria: Valve 17 closed manually from Control Room.
10	Valve No. 13	Isolates Pump No. 3 flow to Steam Generator No. 4.	Fails to close	Valve No. 13 cannot isolate faulted steam generator.	Meets single failure criteria: Valve 16 closed manually from Control Room.

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 5 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
11	Motor Control	Motor operation.	a) Loss of control power b) Any control circuit fault	Motor will not start	Meets single failure criteria: Similar to Failures 1, 2, 3, 3A,5, 6, and 7
12	Turbine Control	Turbine speed control.	a) Loss of control power b) Loss of instrument air	Speed control goes to rated hi-speed control.	Meets single failure criteria: Steam admission valves fail open on loss of elec. or air (See Failure Nos. 3 and 3A) Turbine speed control goes to hi-speed on loss of elec or air. Shut valves 16 and 17 remotely.
13	AO Feed Regulator Valves (6, 8, 10, 12) (Motor-driven pumps)	Regulate Flow to Steam Generators/Isolate faulted Steam Generator.	a) Loss of air b) Loss of control power c) "Trip to Auto" fails	a) No effect for 30 minutes, the valves fail open. b) On loss of signal valves fails open c) Valve remains in manual control	Meets single failure criteria: a) Air accumulator ensure isolation function for 30 minutes and allows up to 30 minutes of flow control after which operator controls valves locally. Shut valves 16 and 17 remotely b) Valve controlled manually or pump shutdown (See Failure Nos. 3 and 3A). Shut valves 16 & 17 remotely. c) Same as b) above.

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 6 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
14	Pump recirc. control	Isolates miniflow to ensure required flow to Steam Generators.			Meets single failure criteria:
			a) Loss of air	a) No effect for 30 minutes, then valve fails open.	a) Air accumulator ensures 30 minutes of isolation after which operator controls recirculation manually.
			b) Loss of control power or signal	b) Recirc. valve fails open	b) See Failure 3 and 3A.
15	AO Feed Regulator Valves (7, 9, 11, 13) (Turbine-driven pump)	Regulate flow to Steam Generator/Isolate faulted Steam Generator.			Meets single failure criteria:
			a) Loss of air	a) Accumulator supplies 30 minutes of air to allow remote isolation of a faulted steam generator, then valve fails open	a) Remote manual control available to isolate AF flow for 30 minutes. Then resort to local manual manipulation of valves. AF flow assured as all valves fail open.
			b) Loss of control power	b) Remote manual capability lost..	b) Flow control is not required for 30 minutes. Shut valves 16 and 17 remotely.
16	Train A SSPS Fails	Starts Aux Feedwater Pump 1 and opens Steam Admission Valve on MSL 4	Loss of Inverter		Meets single failure criteria:
				a. Aux Feed Pump 1 does not start.	a. Steam generators 1 and 2 supplies from AF Pump 3 (Turbine Driven). Steam Generator 3 supplied from AF Pump 2 and 3.
				b. Steam Admission Valve on MSL 4 does not open.	b. Steam Admission Valve on MSL Loop 1 opened by Train B SSPS.

TABLE 10.4-9
FAILURE MODE ANALYSIS

(Sheet 7 of 7)

BREAK ON FEEDWATER PIPE TO STEAM GENERATOR NO. 4 [NOTE 1], INSIDE CONTAINMENT AND SINGLE FAILURE

Failure	Equipment [Note 2]	Function	Failure Mode	Effect of Failure	Analysis
17	Train B SSPS Fails	Starts Aux Feedwater Pump 2 and opens Steam Admission Valve on MSL 1	Loss of Inverter		Meets single failure criteria
				a. Aux Feedwater Pump 2 does not start.	a. Steam Generators 1 & 2 are fed by AF Pump 1
				b. Steam Admission Valve on MSL 1 does not open.	b. Steam Admission Valve on MSL 1 is manually opened.

Notes:

1 Pipe break assumed on line to steam generator 4 for failure mode analysis. Analysis and results for breaks on lines to steam generators 1, 2 or 3 are similar.

2 The equipment numbers are also shown on Figure 10.4-12.

TABLE 10.4-10
MAXIMUM SYSTEM HEAT SINK REQUIREMENTS

(Sheet 1 of 3)

Component To Be Cooled	Number Provided	Number in Operation	Total Cooling Water Flow Rate (gpm)	Total Operating Heat Load (Btu/hr)
Steam generator feedwater pump turbine drive oil coolers	4	2	480	$.99 \times 10^6$
Electrohydraulic control fluid	2	1	200	0.496×10^6 coolers
Turbine lube oil coolers	3	2	3700	14.77×10^6
Heater drain pump lube oil	2	2	35	0.005×10^6 coolers
Condensate pump motor	2	2	34	0.612×10^6 bearing coolers
Exciter air coolers	2	2	792.5	2.184×10^6
Hydrogen coolers	8	8	4403	18.532×10^6
Isophase forced cooling unit	2	1	220	1.91×10^6
Primary water coolers	6	4	6000	32.696×10^6
Primary water leakage unit	1	1	154	0.83×10^6

TABLE 10.4-10
MAXIMUM SYSTEM HEAT SINK REQUIREMENTS

(Sheet 2 of 3)

Component To Be Cooled	Number Provided	Number in Operation	Total Cooling Water Flow Rate (gpm)	Total Operating Heat Load (Btu/hr)
Generator seal oil unit	2	1	66	0.45×10^6
Rotary-Screw Instrument air compressor CPX-CICACO-01	1	1 (common)	40	0.46×10^6
Rotary-Screw Instrument air compressor package CPX-CICACO-02	1	1 (common)	40	0.46×10^6
Auxiliary steam condensate cooler	1	1 (common)	150	4.65×10^6
Auxiliary steam condensate sample cooler	1	1 (common)	5	0.025×10^6
Secondary sampling system	1	1	50	1.25×10^6
Auxiliary boiler sample cooler and conductivity cell	1	1 (common)	1	$.0225 \times 10^6$
Condensate polishing system Vacuum deaerator vacuum pump	1	1	21	0.158×10^6
seal water coolers	2	2	72	0.14×10^6

TABLE 10.4-10
MAXIMUM SYSTEM HEAT SINK REQUIREMENTS

(Sheet 3 of 3)

Component To Be Cooled	Number Provided	Number in Operation	Total Cooling Water Flow Rate (gpm)	Total Operating Heat Load (Btu/hr)
Generator H_2 Gas Dryer Regen. Cycle Cooler (Thermer)	1	1	1	$.0055 \times 10^6$
TOTAL	46	38	16,194	79.9266×10^6

TABLE 10.4-11
ANTICIPATED ANALYSIS OF THE TURBINE PLANT COOLING WATER

1.	Corrosion Inhibitor A suitable corrosion inhibitor is used.	
2.	pH at 25°C	> 8.0
3.	Chloride, mg/l (max.)	10.0

TABLE 10.4-12
ANTICIPATED ANALYSIS OF THE CIRCULATING WATER

Substance	mg/l, as Calcium Carbonate (except as noted)
Calcium	560 to 940
Magnesium	80 to 410
Sodium	1810 to 2220
Bicarbonate	200 to 210
Carbonate	0 to 28
Sulfate	620 to 1040
Chloride	1630 to 2310
Silica, as silicon dioxide	8 to 68
Ammonia, as N	1 to 2
Phenolphthalein alkalinity	0 to 8
Methyl orange alkalinity	200 to 238
Total hardness	640 to 1350
Total dissolved solids	2450 to 3570
pH at 25 °C	8.0 to 8.2
Turbidity as Jackson Units	5 to 10

Note:

The composition of this water is based on predicted average values for the SCR. The peak values may be somewhat higher than the listed values.

TABLE 10.4-13
AUXILIARY STEAM SYSTEM REQUIREMENTS

Auxiliary Steam Requirement	Point Required	Maximum Flow (lb/hr)	Steam Conditions (psig)
Chemical Pipe cleaning	Prior to startup	As required	50 to 150 psig SAT
Turbine Components			
Turbine Gland Steam System			
a. Main Turbine	During Startup	6,832(Normal) 11,614(Enlarged)	60-150 psig SAT.
b. FWP Turbine	During Startup	1,150(575 Each)	60-150 psig SAT.
	Power Operations	230 (115 Each)	60 - 150 psig SAT.
Main Steam Reheat Steam Dump System (Reheater Tube Sheet Warmup)	During Startup (After Hotwell Sparger Warmup)	As required	150 psig SAT.
Condensate System (Hot Well Sparger)	During Startup	10,000	150 psig SAT.
NSSS Components			
Floor Drain Waste Evaporator Package	During Evaporator Operation	10,600	50 psig SAT.
WPS Waste Evaporator Package	During Evaporator Operation	10,600	50 psig. SAT.
BRS Recycle evaporator package	During Evaporator Operation	10,600	50 psig SAT.
CVCS Boric Acid Batching Tank	During Weekly Batches	500	50 psig SAT.

TABLE 10.4-14
AUXILIARY STEAM SYSTEM REQUIREMENTS - NSSS

Auxiliary Steam Requirement	Steam Flow at 50 psig and 297 °F (lb/hr)	Condensate Return Pressure, Minimum to Maximum (psig)
Evaporator Packages (Each 15 gpm) Floor Drain Waste, WPS Waste, BRS Recycle		
Feed preheater	1400	5 to 15
Evaporator	9100	5 to 45
Evaporator condenser	100	none
Boric acid batching tank heater	500	40 (maximum)

TABLE 10.4-15
HAS BEEN DELETED

TABLE 10.4-16
NITROGEN AND HYDROGEN SUPPLY REQUIREMENTS

(Sheet 1 of 3)

Component	Gas	Operating Pressure (psig)	Operating Temperature (°F)	Flowrate (Normal/Max.)	Service (Continuous or Intermittent)	Normal Consumption	Comments
Volume control	1. Hydrogen	15-60	115	a. 0.7 scfm	Continuous Intermittent	1000 scf per day	
				b. 30 scfm		1000 scf per startup	
	2. Nitrogen	10-15	Ambient	Variable	Required for initial shutdown	4500 scf	Fills one shutdown gas decay tank which is recycled for subsequent shutdown
Reactor Coolant (per tank)	Hydrogen	2-6	120-170	Negligible	Continuous	Normally none	Provides H2 makeup; drain tank processed H2 is mainly from reactor coolant. Separate H2 supply to be provided (388 scf)
Pressurizer relief tank (per tank)	Nitrogen	0-3	Ambient	a. Neglig ble	Intermittent	Neglig ble	Cover gas (not recycled) N2 to permit pressurizer draining during shutdown (not recycled)
				b. Variable	Intermittent	2000 scf	
Gas decay tanks (8 tanks)	Nitrogen	4	Ambient	Variable	Required for initial plant startup	6500 scf	Used to initially inert and to pressurize gas decay tanks to provide sufficient suction for waste gas compressors

TABLE 10.4-16
NITROGEN AND HYDROGEN SUPPLY REQUIREMENTS

(Sheet 2 of 3)

Component	Gas	Operating Pressure (psig)	Operating Temperature (°F)	Flowrate (Normal/Max.)	Service (Continuous or Intermittent)	Normal Consumption	Comments
Catalytic recombiner (per recombiner)	Nitrogen	0	Ambient	Variable	Intermittent	By user	Used for initial inerting and recombiner) maintenance
Spent resin storage tank	Nitrogen	a. 0.5-2.0	Ambient	Negligible	Continuous	Neglig ble	Cover gas (not recycled)
		b. 90 max.	Ambient	Variable	Intermittent	1200 scf	Provides motive force to discharge spent resin to bulk disposal (not recycled)
SIS accumulators (per tank)	Nitrogen	660	Ambient	a. Variable	Continuous	Variable	Usage dependent on containment temperature variations (not recycled)
				b. Variable	Intermittent	22000 scf	Required for initial pressurization
Steam generators (per steam generator)	Nitrogen	2	Ambient	Variable	Intermittent	6000 scf dry layup 1500 scf wet layup	Inerting
CSS chemical additive tank (per tank)	Nitrogen	2	Ambient	Negligible	Intermittent	134 scf	For cover gas
SGBD spent resin storage tank	Nitrogen	a. 0.5-2.0	Ambient	Negligible	Continuous	Variable	Cover gas
		b. 90 max.	Ambient	Variable	Intermittent	1200 scf	To force spent resin to disposal

TABLE 10.4-16
NITROGEN AND HYDROGEN SUPPLY REQUIREMENTS

(Sheet 3 of 3)

Component	Gas	Operating Pressure (psig)	Operating Temperature (°F)	Flowrate (Normal/Max.)	Service (Continuous or Intermittent)	Normal Consumption	Comments
Boron Recycle System Evaporator Condenser	Nitrogen	1-5	250	15-25 scfm	Intermittent	Variable	To provide additional NPSH for the concent rator pump during pump-out
Vent and Drain Atmospheric Drain Tanks (per tank)	Nitrogen	<1 psig	Ambient	1 scfm	Continuous	1 scfm	Required during Heater Drain Pump Operation
Secondary Sampling Ion Chromatograph	Nitrogen	90-100	Ambient				
Reactor Makeup Water Storage Tank	Nitrogen	<1	Ambient	≤40 scfh	Intermittent or Continuous	Variable	For sparging. Nitrogen is supplied into RMWP Recirculation Line
Reactor Makeup Water Storage Tank	Nitrogen	<1/3 psig	Ambient	Negligible	Continuous	Variable	For Blanketing
Demineralized water storage tank DWST Sparge/Purge	Nitrogen	30	Ambient	Variable	Continuous	0-30 scfm	Sparge/Purge gas
		80	Ambient	Variable	Continuous	0-300 scfm	
PASS Purge System	Nitrogen	1800	Ambient		Intermittent		Required for Post Accident Sampling to flush sample lines.
PS RCS H2/O2 Analyzer	Nitrogen	75	Ambient	0.011 scfm	Continuous	0.011 scfm	Purge gas to H2 sensor
Boric Acid Storage Tank	Nitrogen	100	Ambient	≤50 scfh	Intermittent	Variable	Sparge gas

TABLE 10.4-17
TURBINE GENERATOR SYSTEM

Component	Gas	Operating Pressure (psig)	Operating Temperature (°F)	Service Pressure	Service (Continuous or Intermittent)	Normal Consumption	Comments
Nitrogen supply	N2	Bottle pressure, 2844 psig; downstream of pressure reducer, 5.7-21.5 psig	Room temperature	0.2-0.3 bar[(a)] 2.8-4.3 psig	Only on draining and priming of primary water unit	With Allis-Chalmers units. One bottle of a volume of 10$Nm^{3(a)}$ (= 353 ft^3) is provided.	Purging of primary water tank
Nitrogen supply	N2	Bottle pressure, 2844 psig; downstream of pressure reducer, 5.7-21.5 psig	Room temperature	0.428 psig	Only on defect of measuring device	With Allis-Chalmers units. One bottle of a volume of 10$Nm^{3(a)}$ (= 353 ft^3) is provided.	
Hydrogen supply	H2	200-300 psig of central supply	Room temperature	Approximately 3 psig	Continuous	Approximately 200 Nl/h[(a)] =7.06 scf/h	Gas cushion over primary water tank
Hydrogen supply	H2	200-300 psig of central supply	Room temperature	14-70 psig	Only on priming	2.5 x generator volume + 1 x generator volume per 14.7 psi pressure increase	Priming of generator
Hydrogen supply	H2	200-300 psig of central supply	Room temperature	5.1 bar[(a)] 75 psig	Continuous	Losses 500 Nl/h[(a)] =424 scf/day Purity Measurement 450 nl/day =18.75 nl/h[(a)] =0.66 scf/h	Maintaining of H2 concentration in generator casing

Notes:

a) Metric units:

Nl/h = Normal liters per hour (at atmospheric pressure and standard temperature)

Nm^3 = Normal cubic meters (at atmospheric pressure and standard temperature)

bar = 14.7 psia (1 atmosphere)

TABLE 10.4-18
SAMPLE POINTS PRESSURE AND TEMPERATURE CONDITIONS

Source	Design Temperature	Design Pressure
Hot wells	121°F	3.5 in. Hg abs
Condensate pump discharge	150°F	600 psig
Polisher outlet	120°F	600 psig
Heater drain pump discharge and HD Tank	400°F	600 psig
Equalization Header[a]	400°F	200 psig
Steam generator feedwater	500°F	1800 psig
Main steam	600°F	1200 psig
Auxiliary condenser outlet	121°F	3.5 in Hg abs
Main steam after MSR	390°F	220 psia
Condensate storage tank	150°F	14.7 psia
Atmospheric drain tank #1	120°F	14.7 psia
Demineralized water	150°F	150 psig
Steam generator blowdown	130°F	300 psig

a) The Heater Drain Tank Equalization Header sample can be obtained by selective alignment of valves in the two Heater Drain sampling lines. Although either sample may be selected at any time, Chemistry typically will check the Equalization Header sample only during HD pump startup and prior to forward flowing Heater Drains to the Condensate System.

TABLE 10.4-19
EQUIPMENT DESIGN CRITERIA

(Sheet 1 of 2)

1. Hot Well Sample Pumps

Number of units	Two
Type horizontal centrifugal	
Operating mode, per unit	2 x 100%
Design flow, gpm	5
Design TDH, psi	32
NPSH available, psi	2
Casing design pressure, psig at 150°F	150
Material	type 316 stainless steel

2. Chilled Water Return Pump

Number of units	two
Type horizontal centrifugal	
Safety Class	NNS
Operating mode, per unit	1 x 100%
Design flow, gpm	15
Design TDH, psi	45
Normal TDH, psi	50
NPSH available, psi	5
Casing design pressure and temperature, psig at 150°F	125
Material	carbon steel

TABLE 10.4-19
EQUIPMENT DESIGN CRITERIA

(Sheet 2 of 2)

3.	Sample Coolers	
	Safety class	NNS
	Operating mode, percent each	100
	Tube side design flow, ml/min	500 to 2500
	Tube side design pressure and temperature	See Table 10.4-18
	Shell side design flow, gpm each	5
	Shell side design pressure, psig at 110°F	150
	Tube side temperatures	
	Inlet	varies per Table 10.4-18
	Outlet, °F	120
	Shell side temperatures	
	Inlet, °F	110
	Outlet, °F	160
	Heat exchanger type	shell-in-tube
	Material	
	Tube	stainless steel
	Shell	carbon steel

Note:

Total dynamic head (TDH); net positive suction head (NPSH); non-nuclear-safety-related (NNS)

TABLE 10.4-20
SECONDARY PLANT SAMPLING SYSTEM MEASURED PARAMETERS

(Sheet 1 of 3)

Sample Source	Parameter Measured	Receiver and Function
Hotwells (2 samples from 6 locations)	cation conductivity	recorder with high alarm
	sodium ion	recorder with high alarm
	anion/cation	computer database
Condensate pump discharge	pH	Note 1
	sodium ion	recorder with high alarm
	hydrazine	recorder with high and low alarms
	cation conductivity	recorder with high alarm
	specific conductivity	reorder
	anion/cation	computer database
	Secondary Corrosion Products	local sample panel monitored by Chemistry personnel
Heater drain pump discharge and HD Tank Equalization Header[a]	cation conductivity	recorder with high alarm
	anion/cation	computer database
	Secondary Corrosion Products	local sample panel monitored by Chemistry personnel
Polisher outlet	cation conductivity	recorder with high alarm
	sodium ion	recorder with high alarm
	anion/cation	computer database

TABLE 10.4-20
SECONDARY PLANT SAMPLING SYSTEM MEASURED PARAMETERS

(Sheet 2 of 3)

Sample Source	Parameter Measured	Receiver and Function
Final feedwater	hydrazine	recorder with high and low alarms
		recorder
	specific conductivity	recorder with high alarm
	sodium ion	recorder with high alarm
	cation conductivity	Note 1
	pH	
	anion/cation	computer database
	Secondary Corrosion Products	local sample panel monitored by Chemistry personnel
Vacuum deaerator outlet	dissolved oxygen	recorder with high alarm
Condensate storage tank	dissolved oxygen	recorder with high alarm
	cation conductivity	recorder with high alarm
	sodium ion	Local sample panel monitored by Chemistry personnel
Main Steam after MSR	cation conductivity	recorder with high alarm
	dissolved oxygen	recorder
	sodium ion	recorder with high alarm
	anion/cation	computer database
Atmosphere Drain Tank #1	cation conductivity	recorder with high alarm

TABLE 10.4-20
SECONDARY PLANT SAMPLING SYSTEM MEASURED PARAMETERS

(Sheet 3 of 3)

Sample Source	Parameter Measured	Receiver and Function
Demineralized Water Storage Tank	specific conductivity	recorder with high alarm
Steam Generator Blowdown Heat Exchanger outlet	Secondary Corrosion Products Secondary Chemistry Parameters as needed	local sample panel monitored by Chemistry personnel

a) The Heater Drain Tank Equalization Header sample can be obtained by selective alignment of valves in the two Heater Drain sampling lines. Although either sample may be selected at any time, Chemistry typically will check the Equalization Header sample only during HD pump startup and prior to forward flowing heater Drains to the Condensate System.

Note 1: The pH sample uses a common sensor selectable to one of the 2 sample sources. Chemistry typically will select the condensate sample during plant startup and select the feedwater sample during normal operation with indication via a recorder with high and low alarms.

11.0 RADIOACTIVE WASTE MANAGEMENT

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11.1 SOURCE TERMS

Source terms and models used in the design and evaluation of the waste processing systems are compared to operating plant data where available [1].

Source term models are presented which have been used for plant shielding design and effluent release analysis. Source terms for shielding design were based on design basis reactor coolant activity [2] assuming 1 percent fuel defects. Source terms for the effluent release analysis were based on the realistic model for reactor coolant activity as formulated in a standard for the American Nuclear Society (ANS) [3] and calculated per NUREG-0017[6].

Tritium production and fuel operating experience are fully addressed in References [1] and [4], respectively. For conservatism in the shielding design, radioactive waste streams are assumed to be recycled in the plant.

11.1.1 DESIGN BASIS MODEL FOR REACTOR COOLANT ACTIVITY

Except as noted, the discussion on radiation source terms presented in this section represents information used by the original license application to establish plant shielding and radwaste effluent assessments. It is retained for historical purposes to avoid loss of original design basis and is not subject to future update.

The parameters used in calculating the reactor coolant inventory at the SPU power level is presented in [New] Table 11.1-1A and the concentrations presented in [New] Table 12.2-3A. (Comment: These tables should be obtained from Westinghouse.)

Fission Products

The parameters used in the calculation of the reactor coolant fission product concentrations are summarized in Table 11.1-1 Concentrations are presented in Table 12.2-3.

The fission product concentrations are computed using the following differential equations:

For parent nuclides in the coolant:

$$\frac{dN_{wi}}{dt} = D\upsilon_i NC_i - \left(\lambda i + R\eta_i + \frac{B'}{B_o - tB'}\right)N_{wi}$$

For daughter nuclides in the coolant:

$$\frac{dN_{wj}}{dt} = D\upsilon_j NC_j - \left(\lambda j + R\eta_j + \frac{B'}{B_o - tB'}\right)N_{wj} + \lambda_i N_{wi}$$

where

N = population of nuclides (atoms)

t = time (sec)

D = clad defects, as a fraction of rated core thermal power being generated by rods with clad defects

R = purification flow (coolant system volumes per second)

B_o = initial boron concentration (ppm)

B' = boron concentration reduction rate by feed and bleed (ppm per second)

η = removal efficiency of purification cycle for nuclide

λ = radioactive decay constant (sec^{-1})

ν = escape rate coefficient for diffusion into coolant (sec^{-1})

subscripts

C = refers to core

w = refers to coolant

i = refers to parent nuclide

j = refers to daughter nuclide

The fission products are removed by decay, cleanup in the Chemical and Volume Control System and by letdown to the Boron Recycle System. No degassing is assumed in the volume control tank.

Corrosion Products

The corrosion product activities, which are independent of fuel defect level, are based on measurements at operating reactors [1]. The corrosion product concentrations are given in Table 12.2-3.

11.1.2 REALISTIC MODEL FOR REACTOR COOLANT ACTIVITY

The nominal plant design and operating parameters utilized by the referenced standard [3], together with the corresponding specific plant related parameters, are given in Table 11.1-3. The range of plant parameters used by the PWR-GALE computer code [6] is also provided. Corrections are made where a parameter falls outside the given range as noted in Reference [6]. The Gaseous Waste Processing System is assumed to strip fission gases from the volume control tank. The overall Y parameter, as given in Reference [3], is interpreted as being equivalent to the stripping fractions. A separate value of the stripping fraction for isotopes is used as indicated in Table 11.1-3. A stripping efficiency of 0.4 is used. This assumes a less

efficient separation in the volume control tank which results in a conservative estimate of the Reactor Coolant System radioactivity.

The stripping fractions and stripping efficiencies are related as follows:

$$SE = \frac{C_R - C_L}{C_R - C_{L_{eq}}}$$

$$SF = \frac{C_R - C_L}{C_R}$$

where

SE	=	stripping efficiency
SF	=	stripping fraction
C_R	=	gas concentration in the liquid phase entering the volume control tank
C_L	=	gas concentration in the liquid phase leaving the volume control tank
$C_{L_{eq}}$	=	gas concentration in the liquid phase leaving the volume control tank, assuming the ratio of the gas concentration in the vapor and liquid phases in the volume control tank follows Henry's Law.

Specific activities in the primary coolant, based on the realistic model with corrections for plant specific parameters, are given in Table 11.1-4.

Leakage from the primary system to the secondary side in the steam generator is possible; therefore, secondary side activity has been evaluated. The activity levels for the secondary side are presented in Table 11.1-7.

11.1.3 SOURCE TERMS FOR SHIELDING DESIGN

Liquid Waste Processing System

The source terms which were used for plant shielding design are based upon the coolant activity described in Section 11.1.1. However, for shielding purposes, the reactor coolant was assumed to be degassed. Shielding design source terms are presented for components outside the Containment which may contain reactor coolant or higher concentration of radioactivity. Source terms which have been used for shielding design are given in Table 11.1-5.

Gaseous Waste Processing System

The major components are the gas compressors, the hydrogen recombiners and the gas decay tanks. The gases are continuously recirculated; hence, the radiation sources in each component are identical. The total gaseous content is based on 40 years storage and stripping of the reactor coolant during reactor shutdown. The reactor coolant activity in Table 12.2-3 and 100 percent

stripping efficiency were assumed for shielding design. The resulting source terms are given in Table 11.1-5.

11.1.4 SOURCE TERMS FOR COMPONENT FAILURE

Updated source terms obtained from Reference [5] are used for plant shielding design review and to assess consequences of postulated component failure. The updated source terms represent results of improved calculational techniques and nuclear data bases.

Liquid Waste Processing System

The isotopic inventory of the 30,000 gallon floor drain tank no. 3 is given in Table 15.7-3. The tank inventory is based on updated reactor coolant activities obtained from Reference [5]. The capacity of the tank, and type of waste collected, show that analyzing the failure of this component is the most conservative assumption regarding the Liquid Waste Processing System.

Gaseous Waste Processing System

The isotopic inventories in one gas decay tank to be used for gas decay tank rupture are given in Table 15.7-2. The inventories are based on updated reactor coolant activities obtained from Reference [5] and assuming 0.7 scfm volume control tank purge rate to define the maximum activities. The activity is further based on 40 years inventory of the Kr-85 and equilibrium levels of all other isotopes. It is assumed two units are operating simultaneously and decay tanks are switched regularly.

REFERENCES

1. “Source Term Data for Westinghouse Pressurized Water Reactors,” WCAP-8253, Amendment 1, July 1975.

2. “Radiation Analysis Design Manual, 4 Loop Plant,” WCAP-7664, Revision 1, October 1972.

3. American National Standard Radioactive Source Term for Normal Operation of Light Water Reactors, ANSI/AnS-18.1-1984.

4. “Operational Experience with Westinghouse Cores,” WCAP-8183, Revision 6, June 1977.

5. “Radiation Analysis Manual, Standard Plant Model 412,” Revision 3 Westinghouse Electric Corporation, November 1978.

6. “Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Pressurized Water Reactors (PWR-GALE Code),” NUREG-0017, Revision 1, 1985.

TABLE 11.1-1
PARAMETERS USED IN THE CALCULATION OF DESIGN BASIS PRIMARY COOLANT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 3)

1.	Ultimate core thermal power (MWt)	3565
2.	Clad defects, as a percent of rated core thermal power being generated by rods with clad defects	1.0
3.	Reactor coolant liquid volume (ft^3)	12,000
4.	Reactor coolant full power average temperature (°F)	590
5.	Purification flow rate, normal (gpm[b])	75
6.	Effective cation demineralizer flow (gpm[b])	7.5
7.	Volume control tank volumes	
	a. Vapor (ft^3)	240
	b. Liquid (ft^3)	160
8.	Fission product escape rate coefficients[c]	
	a. Noble gas isotopes (sec^{-1})	6.5×10^{-8}
	b. Br, Rb, I and Cs isotopes (sec^{-1})	1.3×10^{-8}
	c. Te isotopes (sec^{-1})	1.0×10^{-9}
	d. Mo isotopes (sec^{-1})	2.0×10^{-9}
	e. Sr and Ba isotopes (sec^{-1})	1.0×10^{-11}
	f. Y, Zr, Nb, La, Ce, Pr isotopes (sec^{-1})	1.6×10^{-12}
9.	Mixed bed demineralizer decontamination factors	
	a. Noble gases and Cs-134, 136, 137, Y-90, 91 and Mo-99	1.0
	b. All other isotopes including corrosion products	10.0

TABLE 11.1-1
PARAMETERS USED IN THE CALCULATION OF DESIGN BASIS PRIMARY COOLANT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

10. Cation bed demineralizer decontamination factor for Cs-134, 137, Y-90, 91 — 10.0

11. Volume control tank noble gas stripping fraction

Isotope	Stripping Fraction: Assuming Operation of GWPS[d]	Stripping Fraction: Plant Without GWPS
Kr-85	2.3×10^{-1}	2.3×10^{-5}
Kr-85m	2.9×10^{-1}	2.7×10^{-1}
Kr-87	6.0×10^{-1}	6.0×10^{-1}
Kr-88	4.3×10^{-1}	4.3×10^{-1}
Xe-131m	2.5×10^{-1}	1.0×10^{-2}
Xe-133	2.5×10^{-1}	1.6×10^{-2}
Xe-133m	2.6×10^{-1}	3.7×10^{-2}
Xe-135	2.8×10^{-1}	1.8×10^{-1}
Xe-135	8.0×10^{-1}	8.0×10^{-1}
Xe-138	1.0	1.0

12. Partition factor[e] for iodine in the volume control tank — 100.0

13. Boron concentration and reduction rates

a. B_o, initial cycle, (ppm) — 805
   B^1, initial cycle, (ppm/day) — 2.06

b. B_o, equilibrium cycle, (ppm) — 1080
   B^1, equilibrium cycle, (ppm/day) — 3.96

TABLE 11.1-1
PARAMETERS USED IN THE CALCULATION OF DESIGN BASIS PRIMARY COOLANT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

14.	Pressurizer volumes	
	a. Vapor (ft^3)	720
	b. Liquid (ft^3)	1080
15.	Spray line flow (gpm[b])	1.0
16.	Pressurizer stripping fractions	
	a. Noble gases	1.0
	b. All other elements	0

a) Historical, not subject to update; has been retained to preserve original design basis. Parameters used to develop the SPV RCS source terms presented in Table 11.1-1A.

b) At reference condition, 130°F and 2300 psig, 1 gpm = 493.6 lb/hr.

c) Escape rate coefficients are based on fuel defect tests performed at the Saxton Reactor. Recent experience at plants operating with fuel rod defects has verified the listed escape rate coefficients

d) Volume control tank purge rate is 0.7 scfm and volume control tank stripping efficiency is 40 percent.

e) Refer to Reference [1].

TABLE 11.1-2
TABLE 11.1-2 HAS BEEN DELETED.

TABLE 11.1-3
PARAMETERS USED IN THE CALCULATION OF REALISTIC PRIMARY COOLANT ACTIVITIES ORIGINAL LICENSING BASIS[(a)]

(Sheet 1 of 3)

Parameter	Symbol	Units	Nominal Value	Range		
				Maximum	Minimum	CPNPP
Thermal Power	P	MWT	3400	3800	3000	3565
Steam flow rate	FS	lb/hr	$1.5x10^7$	$1.7x10^7$	$1.3x10^7$	$1.5x10^7$
Weight of water in Reactor Coolant System	WP	Lb	$5.5x10^5$	$6.0x10^5$	$5.0x10^5$	$5.1x10^5$
Weight of water in all steam generators	WS	Lb	$4.5x10^5$	$5.0x10^5$	$4.0x10^5$	$3.84 x10^5$
Reactor coolant letdown flow (purification)	FD	lb/hr	$3.7x10^4$	$4.2x10^4$	$3.2x10^4$	$3.8x10^4$
Reactor coolant letdown flow (yearly average for boron control)	FB	lb/hr	500	1000	250	100
Steam generator blowdown flow (total)	FBD	lb/hr	$7.5x10^4$	$1.0x10^5$	$5.0x10^4$	$6.96 x10^4$
Fraction of radioactivity in blowdown stream which is not returned to secondary coolant system	NBD	-	1.0	1.0	0.9	1.0
Flow through the purification system cation demineralizer	FA	lb/hr	3700	7500	0.0	$3.8x10^3$
Ratio of condesate demineralizer flow rate to the total steam flow rate	NC	-	0.0	0.01	0.0	0.7

TABLE 11.1-3
PARAMETERS USED IN THE CALCULATION OF REALISTIC PRIMARY COOLANT ACTIVITIES ORIGINAL LICENSING BASIS[(a)]

(Sheet 2 of 3)

Ratio of the total amount of noble gases routed to gaseous radwaste from the purification system to the total amount of noble gases routed to the primary coolant system from the purification system (not including the Boron Recycle System)	Y	-	0.0	0.01	0.0	See below

Isotope	Y Parameter
Kr-83m	0.37
Kr-85m	0.34
Kr-85	0.28
Kr-87	0.38
Kr-88	0.36
Kr-89	0.40
Xe-131m	0.25
Xe-133m	0.26
Xe-133	0.25
Xe-135m	0.39
Xe-135	0.30
Xe-137	0.40
Xe-138	0.39

TABLE 11.1-3
PARAMETERS USED IN THE CALCULATION OF REALISTIC PRIMARY COOLANT ACTIVITIES ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11.1-4
REALISTIC REACTOR COOLANT FISSION AND CORROSION PRODUCT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 3)

Isotope	Reactor Coolant Activity (μCi/gram)	Isotope	Reactor Coolant Activity (μCi/gram)
Group I - Noble Gases		Group III - Cs, Rb	
Kr-83m	2.13×10^{-2}		
Kr-85m	1.03×10^{-1}		
Kr-85	7.12×10^{-3}		
Kr-87	6.20×10^{-2}	Cs-134	7.10×10^{-3}
Kr-88	1.97×10^{-1}	Cs-136	8.70×10^{-4}
Kr-89	5.41×10^{-3}	Cs-137	9.40×10^{-3}
Xe-131m	1.89×10^{-2}		
		Group IV - N-16[b]	
Xe-133m	1.03×10^{-1}		
Xe-133	5.15	N-16	4.00×10^{1}
Xe-135m	1.40×10^{-2}		
Xe-135	2.98×10^{-1}	Group V - Tritium[b]	
Xe-137	9.73×10^{-3}		
Xe-138	4.73×10^{-2}	H-3	1.00
Group II - Halogens			
I-131	4.50×10^{-2}		
I-132	2.10×10^{-1}		
I-133	1.40×10^{-1}		
I-134	3.40×10^{-1}		

TABLE 11.1-4
REALISTIC REACTOR COOLANT FISSION AND CORROSION PRODUCT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

Isotope	Reactor Coolant Activity (μCi/gram)	Isotope	Reactor Coolant Activity (μCi/gram)
I-135	2.60×10^{-1}		
Group VI - Other Isotopes		Group VI - Other Isotopes	
Na-24	4.70×10^{-2}		
Cr-51	3.10×10^{-3}	Te-131m	1.50×10^{-3}
Mn-54	1.60×10^{-3}	Te-131	7.70×10^{-3}
Fe-55	1.20×10^{-3}	Te-132	1.70×10^{-3}
Fe-59	3.00×10^{-4}		
Co-58	4.60×10^{-3}	Ba-140	1.30×10^{-2}
Co-60	5.30×10^{-4}	La-140	2.50×10^{-2}
Zn-65	5.10×10^{-4}		
Sr-89	1.40×10^{-4}	Ce-141	1.50×10^{-4}
Sr-90	1.20×10^{-5}	Ce-143	2.80×10^{-3}
Sr-91	9.60×10^{-4}	Ce-144	3.90×10^{-3}
		W-187	2.50×10^{-3}
Y-91m	4.60×10^{-4}		
Y-91	5.20×10^{-6}	Np-239	2.20×10^{-3}
Y-93	4.20×10^{-3}		
Zr-95	3.90×10^{-4}		
Nb-95	2.80×10^{-4}		
Mo-99	6.40×10^{-3}		
Tc-99m	4.70×10^{-3}		

TABLE 11.1-4
REALISTIC REACTOR COOLANT FISSION AND CORROSION PRODUCT ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

Isotope	Reactor Coolant Activity (μCi/gram)	Isotope	Reactor Coolant Activity (μCi/gram)
Ru-103	7.50×10^{-3}		
Ru-106	9.00×10^{-2}		
Ag-110m	1.30×10^{-3}		
Te-129m	1.90×10^{-4}		
Te-129	2.40×10^{-2}		

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) Values provided from ANSI/ANS-18.1-1984 [3]

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[(a)]

(Sheet 1 of 6)

Waste Evaporator Condensate Demineralizer

($30\ ft^3$ at Resin)

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.4	2.6×10^4
0.8	2.0×10^4
1.3	3.5×10^3
1.7	1.7×10^3
2.2	6.2×10^2

Waste Monitor Tank Demineralizer

($30\ ft^3$ at Resin)

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.4	1.1×10^6
0.8	3.0×10^6
1.3	9.9×10^5
1.7	4.8×10^5
2.2	1.8×10^5

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[(a)]

(Sheet 2 of 6)

Waste Holdup Tank, Floor Drain Tank, and Waste Monitor Tank

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.4	3.6×10^4
0.8	2.0×10^5
1.3	1.3×10^5
1.7	5.0×10^4
2.2	1.9×10^4

Spent Resin Storage Tank

($350\ ft^3$ at Resin)

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.4	1.5×10^8
0.8	3.4×10^8
1.3	3.3×10^7
1.7	1.7×10^7
2.2	4.5×10^6
2.5	1.0×10^5
3.5	2.0×10^5

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 6)

Evaporator Concentrates[b]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.8	1.2×10^{6}

Drumming Station Spent Resin

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.8	1.0×10^{8}
1.3	1.0×10^{7}

Waste Monitor Tank Filter[c]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.8	3.4×10^{7}
1.3	8.9×10^{6}

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 4 of 6)

Waste Evaporator Condensate Filter[c]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.4	1.2×10^5
0.8	2.2×10^5
1.3	2.5×10^4

Waste Evaporator Vent Condenser Vapor[c]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.1	1.1×10^7
0.4	3.5×10^6
0.8	1.5×10^6
1.7	1.0×10^6
2.2	3.8×10^6
2.5	5.2×10^6

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 5 of 6)

Hydrogen Recombiner, Waste Gas Compressor,and Gas Decay Tanks

Gamma Energy (Mev/)	Specific Source Strength (Mev/cc-sec)
0.1	1.8×10^{6}
0.4	3.4×10^{5}
0.8	8.6×10^{4}
1.7	6.7×10^{4}
2.2	1.2×10^{5}
2.5	2.9×10^{5}

Waste Evaporator Feed Filter, and Floor Drain Tank Filter[d]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.8	3.4×10^{7}
1.3	8.9×10^{6}

TABLE 11.1-5
SOURCE TERMS FOR SHIELDING DESIGN SHIELDING SOURCES WASTE PROCESSING SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 6 of 6)

Spent Resin Sluice Filter[d]

Gamma Energy (Mev/γ)	Specific Source Strength (Mev/cc-sec)
0.8	1.1×10^7
1.3	3.0×10^6

a. Historical, not subject to future updating. Has been retained to preserve original design basis.

b. 1.89×10^6 grams - maximum capacity for evaporator concentrate.

c. Homogeneous sources with the following dimensions and compositions

	Source Dimensions(in)		Source Composition
Component	Radius	Length	(Volume Percent)
Waste Monitor Tank, Waste Evaporator Condensate Filters	1.25	19	30% air, 70% water
Waste Evaporator Vent Condenser	4	20	30% s/s, 22% water, 48% air

d. Homogeneous sources with the following dimensions and compositions:

	Source Dimensions(in)		Source Composition
Component	Radius	Length	(Volume Percent)
Waste Evaporator Drain Tank	1.25	19	30% air, 70% water
Spent Resin Sluice	3.375	19	62% air, 38% water

TABLE 11.1-6
TABLE 11.1-6 HAS BEEN DELETED.

TABLE 11.1-7
REALISTIC SECONDARY SIDE (WATER) EQUILIBRIUM ACTIVITIES[a] -
ORIGINAL LICENSING BASIS[b]

(Sheet 1 of 3)

Isotope	Concentration (uCi/gm)
H-3	1.00E-03[c]
N-16	1.00E-06[c]
Na-24	1.55E-06
Cr-51	1.30E-07
Mn-54	6.51E-08
Fe-55	4.91E-08
Fe-59	1.20E-08
Co-58	1.90E-07
Co-60	2.20E-08
Zn-65	2.10E-08
Sr-89	5.71E-09
Sr-90	4.91E-10
Sr-91	2.93E-08
Y-91	2.11E-10
Y-91m	3.65E-09
Y-93	1.25E-07
Zr-95	1.60E-08
Nb-95	1.10E-08
Mo-99	2.53E-07
Tc-99m	1.17E-07

TABLE 11.1-7
REALISTIC SECONDARY SIDE (WATER) EQUILIBRIUM ACTIVITIES[a] -
ORIGINAL LICENSING BASIS[b]

(Sheet 2 of 3)

Isotope	Concentration (uCi/gm)
Ru-103	3.11E-07
Ru-106	3.71E-06
Ag-110M	5.31E-08
Te-129	2.49E-07
Te-129m	7.82E-09
Te-131	3.35E-08
Te-131m	5.51E-08
Te-132	6.66E-08
I-131	1.44E-06
I-132	3.10E-06
I-133	4.02E-06
I-134	2.59E-06
I-135	5.96E-06
Cs-134	3.05E-07
Cs-136	3.71E-08
Cs-137	4.07E-07
Ba-140	5.22E-07
La-140	9.45E-07
Ce-141	6.12E-09
Ce-143	1.02E-07
Ce-144	1.60E-07

TABLE 11.1-7
REALISTIC SECONDARY SIDE (WATER) EQUILIBRIUM ACTIVITIES[a] -
ORIGINAL LICENSING BASIS[b]

(Sheet 3 of 3)

Isotope	Concentration (uCi/gm)
W-187	8.91E-08
Np-239	8.50E-08

a) Equilibrium activity is based on the following:

1. Primary coolant leak - 75 lb/day
2. Primary coolant activity - in accordance with Table 11.1-4
3. Steam generator blowdown rate - 6.96 x 10^4 lb/hr
4. Ratio of condensate demineralizer flow to steam flow rate - 0.70
5. Decontamination Factors

	Blowdown	Condensate Demineralizer
Cs, RB	100	1
Anions	100	1
Others	1000	1

b) Historical, not subject to future updating. Has been retained to preserve original design basis.

c) Values provided from ANSI/ANS - 18.1-1984

11.2 LIQUID WASTE MANAGEMENT SYSTEM

The CPNPP Liquid Waste Processing System (LWPS) services both units with shared components as described in Section 1.2.2.12 of the FSAR. The system is classified as non-nuclear safety-related.

The radioactivity values presented in this section are the design basis values used for the design of the liquid waste management system. As such they are considered historical and are not subject to future updating. The information is retained to avoid loss of original design basis. Actual radioactivity release quantities can be found in the annual radioactive effluent release reports submitted to the NRC.

11.2.1 DESIGN BASES

11.2.1.1 Design Objective

The LWPS is designed to control, collect, process, handle, store, and dispose of liquid radioactive waste generated as the result of normal operation, including anticipated operational occurrences as follows:

1. LWPS equipment malfunction

2. Excessive leakage in Reactor Coolant System (RCS) equipment

3. Excessive leakage in auxiliary system equipment

These modes of operation are considered in combination with fuel cladding defects of 1.0 percent.

Table 11.2-8 shows the design inventory of radionuclides in LWPS components.

The system design considers potential population and occupational exposures and ensures that quantities of radioactive releases to the environment meet the requirements specified in 10 CFR Parts 20 and 50 and the dose design objectives specified in Appendix I of 10 CFR Part 50, during both normal and anticipated operational occurrences.

11.2.1.2 Design Criteria

1. Release limits

   The following summarizes the basic radioactive liquid release limits established by 10 CFR Part 20[a]:

   a. The concentration limit on an unidentified, instantaneous release basis is as defined in 10 CFR Part 20, Appendix B.

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a. Design criteria based on the provisions of 10 CFR 20.1-20.601.

b. The concentration limit on an identified basis is defined in 10 CFR Part 20, Appendix B, Table II, Column 2.

These concentration limits are considered at the point of discharge to Squaw Creek Reservoir. For purposes of calculating doses from liquid effluents, the concentration within the exclusion area boundary may be determined by applying appropriate factors for dilutions, dispersion, and decay between the point of discharge and the exclusion area boundary.

2. Dose to Individuals

Except for occasional variation, as specified in 10 CFR Part 20[a], the permissible dose of exposure for an individual is as follows:

a. Restricted Area

1 1/4 rems per calendar quarter to the whole body

b. Unrestricted Area

0.5 rem per calendar year to the whole body

A cost-benefit analysis for the LWPS was not required because the limitations imposed by the annex Appendix I of 10 CFR Part 50 were met by the LWPS as described in the FSAR during construction.

Evaluations showing that the proposed systems are capable of controlling releases within the numerical design objectives of Appendix I of 10 CFR Part 50 may be found in Appendix 11A.

11.2.1.3 Release Control Capability

The system is capable of controlling release and reducing doses. The subsystems have components such as filters, evaporators and demineralizers, all of which remove radioactivity to various degrees.

The system is capable of recycling water back to plant systems.

The decontamination factors for the processing equipment, which were used to determine isotopic release quantities, are listed in Table 11.2-6.

The surge capacity of the LWPS permits the system to accommodate waste until failures can be fixed and normal plant operation is resumed.

11.2.1.4 Seismic Design Classification and Quality Group Classification

The seismic design classification and quality group classification for the LWPS components and structures are listed in Appendix 17A, Table 17A-1.

11.2.1.5 Special Features

The system has been designed and the equipment chosen to effect an appreciable reduction in maintenance downtime, leakage, gas release, cleaning, etc.

The tanks have sufficient capacity to store abnormal surge loads. They are equipped with high-level alarms and low-level alarms to provide operator notification of a potential overflow condition or leaking tank.

A majority of the pumps are of the canned-rotor design. The canned-rotor design minimizes overall liquid leakage and also minimizes the release of entrained radioactive gases in the leaking fluid to the plant building atmosphere.

Provisions are made to stop radioactive releases by providing a stop valve interlocked with a radiation monitor on the system discharge line. The valve automatically closes when the waste activity reaches the monitor set point. As an additional precaution, the stop valve is also interlocked to block flow if both units have less than two circulating water pumps running.

11.2.2 SYSTEM DESCRIPTIONS

11.2.2.1 General

The LWPS collects and processes potentially radioactive wastes for recycle or disposal during the normal mode of operation. Provisions are made to sample and analyze fluids before they are discharged. Based on this analysis, these wastes are either released under controlled conditions via the circulating water discharge canal or retained for further processing. The circulating waterflow serves to reduce the concentration of radioactivity in the plant effluent by diluting the LWPS discharges. A permanent record of radioactivity releases is provided by analyses of known volumes of effluent.

Normally, radioactive liquids discharged from the RCS are recycled or processed by the Boron Recycle System (BRS), thereby limiting inputs into the LWPS. Water in the Recycle Holdup Tank that needs to be processed is sent to the Filter/Demineralizer System (FDS).

The original system design intent was for Comanche Peak to be a “no release” plant, thereby recycling and reusing all effluent streams. For ALARA reasons, this philosophy has changed and the intent now is to prevent any tritiated or otherwise contaminated effluent from reaching the Reactor Makeup Water Storage Tanks. For this reason the isolation valves between the WP Waste Evaporator Condensate Pumps and the DD/RMW System are presently administratively controlled in a closed position. Also being maintained closed in the BR System is the upstream isolation valve to the radiation controlled three-way diversion valve. In any case, previously designed processing of effluents is possible by simply opening these valves and aligning other valves as appropriate.

11.2.2.2 Subsystems

The LWPS is designed to segregate different effluents from equipment leaks and drains according to their chemical and radiochemical properties. In addition interconnecting piping is available to allow for operating flexibility and provide for efficient utilization of purification equipment. The system is divided into the following subsystems:

1. Reactor Coolant Drain Tank

2. Drain Channel A

3. Drain Channel B

4. Drain Channel C

5. In addition, the LWPS provides capability for handling and storage of spent ion-exchange resins

6. Drain systems are also provided inside the Containment, Fuel, Auxiliary, and Safeguards Buildings to collect drains and leaks and to transfer them to appropriate tanks

7. A filter/demineralizer system (FDS) is typically used to process liquid waste. This filter/demineralizer system is located in Fuel Building Room X-247 on elevation 800'-2".

8. An interim/preprocessing drum dryer system may be used for conditioning or volume reduction of non-routine liquid radioactive water generated from activities like tank cleaning, facility/equipment decontamination, or tank bottoms. This system is a self-contained skid mounted system located in the Fuel Building.

The LWPS subsystems have process capacities as outlined in the following paragraphs.

In the event of equipment faults of moderate frequency, the LWPS is capable of processing up to one-gpm of primary coolant leakage with no change in system operation. During these equipment faults, the load on the waste processing system is increased.

As a practical upper limit of system operation, the LWPS can process 50 gpm.

It is possible to operate the LWPS at these upper limit processing rates for extended periods. However, close operator surveillance is required, and radioactive discharge must be kept within the as low as reasonably achievable (ALARA) limit. Assuming the discharge limits are not exceeded and there are no equipment malfunctions, the LWPS can process all anticipated inputs to the system.

Instrumentation and controls necessary for the operation of the LWPS are located on control boards in the Auxiliary Building and Fuel Building. Any alarm on these control boards is relayed to the main control boards in the Control Room.

1. Reactor Coolant Drain Tank Subsystem

   Recyclable reactor-grade effluents enter this subsystem from equipment leaks and drains, valve leakoffs, pump seal leakoffs, loop drain leakoffs, and from other deaerated tritiated water sources inside the Containment. Connections are provided for various drains and leakoffs and for cooling the Pressurizer Relief Tank. This deaerated tritiated liquid is normally pumped directly to the Recycle Holdup Tanks via the Reactor Coolant Drain Tank Heat Exchanger. This liquid may be processed by the BRS rather than by the LWPS. In addition, refueling canal drains can be routed to the Spent Fuel Purification System using the Reactor Coolant Drain Tank pumps.

The Reactor Coolant Drain Tank is also connected to the Gaseous Waste Processing System vent header. Except in times immediately prior to a planned unit shutdown, hydrogen gas is normally supplied to the Reactor Coolant Drain Tank to maintain a hydrogen blanket and limit hydrogen usage. Provisions for sampling gas are provided. Details of the Reactor Coolant Drain Tank Subsystem are shown on Figure 11.2-2.

2. Drain Channel A Subsystem

Aerated tritiated liquid enters Drain Channel A through lines connected to the Waste Holdup Tank. Sources of this aerated liquid are as follows:

a. Accumulator drainage (to Reactor Coolant Drain Tank Pump suction)

b. Sample room sink drains (to Waste Holdup Tank);

c. Ion exchanger, filter, pump, and other equipment drains (to Waste Holdup Tank)

The Waste Holdup Tank is one of the initial collecting points for liquid waste to be processed through the waste evaporator or the FDS.

Abnormal liquid sources include leaks which may develop in the RCS and its auxiliary systems. Considerable surge and processing capacity is incorporated in the LWPS to accommodate abnormal operations.

The basic composition of the liquid collected in the Waste Holdup Tank is boric acid and water with some radioactivity.

The Waste Evaporator Feed Pump delivers the contents of the Waste Holdup Tank through a filter to the Waste Evaporator Package, the FDS or Floor Drain Tank 3 for removal of radioisotopes and boron prior to reuse or discharge.

If the waste evaporator is processing liquid waste, the condensate leaving the waste evaporator can be routed either directly to a tank, or filtered and then routed to a tank, or demineralized then filtered and then routed to a tank. The final disposition of the condensate is typically dependent on the radioactivity levels found in samples taken from the process stream. Based upon sample results, the condensate in the tanks can then be transferred to a monitor tank and discharged, or be processed through the FDS. The evaporator bottoms may be packaged or processed.

The effluent from the FDS will be directed to one of the six monitor tanks.

The waste evaporator or the FDS may also be utilized as a backup to the Boron Recycle Evaporator to process Recycle Holdup Tank contents.

Details of the Drain Channel A subsystem are shown on Figure 11.2-3.

3. Drain Channel B Subsystem

Drain Channel B collects and processes non-reactor-grade liquid wastes. These include floor drains, equipment drains containing non-reactor-grade water, and other non-reactor-grade sources.

Drain Channel B equipment includes three floor drain tanks, a common filter and evaporator, two waste monitor tanks with a common demineralizer and filter, and two additional monitor tanks.

Three floor drain tanks (numbered 1, 2, and 3) are provided to collect liquid waste from both units. Floor Drain Tanks 1 and 2 each have a capacity of 10,000-gal. Floor Drain Tank 1 normally collects waste water from the Unit 1 and common sources. Floor Drain Tank 2 normally collects waste water from the Unit 2 sources. Floor Drain Tank 3 has a 30,000-gal. capacity. Although any of the Floor Drain Tanks can be directly processed, Floor Drain Tank 3 is normally used as a batch tank for processing Floor Drain Tanks 1 and 2. In addition, Floor Drain Tank 3 is an alternate liquid waste collector for some of the common sources.

Laboratory samples which contain reagent chemicals (and possibly tritiated liquid) are sent to the Chemical Drain Tank. Rinse water from the laboratory is discarded to either the Floor Drain Tanks or the Chemical Drain Tank.

When Floor Drain Tank 3 is sufficiently full, the contents are normally sampled and analyzed to determine the type of processing required. The capability to recycle the waste is dependent on its chemical properties, water inventories and radioactivity level.

The processing flow paths for floor drain wastes are through the floor drain tank pump, strainer, filter, and evaporator; or through the FDS. Evaporator effluents may then be routed to a waste monitor tank. The evaporator bottoms are disposed of as required. Floor drain evaporator or FDS process effluents are collected in either the Waste Monitor Tanks, the Laundry Holdup & Monitor Tanks or the Plant Effluent Holdup and Monitor Tanks. Processed fluid in a monitor tank is mixed and sampled. Depending on analyses, a processed batch of waste may be subject to controlled release or returned to Drain Channel B for additional treatment.

Any releases to the environment have radioactivity levels that are ALARA (within the limits of Appendix I of 10 CFR Part 50) and are administratively controlled by using a downstream stop valve interlocked with a radiation monitor. The monitor initiates automatic closure of the valve when the setpoint is exceeded. The radiation monitor setpoint is established at a value below the required limits to provide a margin of assurance that release limits will not be exceeded. A permanent record of radioactive releases is provided by an analysis of the known volumes of waste effluent released. The stop valve is also interlocked with a minimum of two circulating water pumps running.

Details of the Drain Channel B subsystem are shown on Figure 11.2-4.

4. Drain Channel C Subsystem

Drain Channel C is provided to collect and process waste effluents from onsite vendor laundry, personnel decontamination showers and sinks, and surface decontamination. These liquids may be collected in the Laundry and Hot Shower Tank. Drain Channel C

equipment includes a Laundry and Hot Shower Tank, strainer and filter, two Laundry Holdup and Monitor Tanks, and a Laundry Water Head Tank.

Based on operating plant data, the volume of laundry and hot shower waste is estimated to be approximately 20,000-gal per year per unit. Processing of contaminated plant laundry will be performed by an outside vendor.

The liquid collected in the Laundry and Hot Shower Tank is pumped through the Laundry and Hot Shower Tank Strainer and filter to one of the two 5000-gal Waste Monitor Tanks. The waste water is then sampled to determine if the liquid is to be discharged or reprocessed through the FDS or the waste evaporator. With the use of the FDS the Laundry Holdup Monitor Tanks may also receive effluent.

5. Spent Resin Handling Subsystem

This subsystem collects, handles, and processes spent resins from the primary fluid systems prior to their disposal.

Normally, resin from the primary system demineralizers is transported to and stored in the Spent Resin Storage Tank prior to being packaged for disposal. The Spent Resin Sluice portion of the LWPS consists of a Spent Resin Sluice Filter, Spent Resin Sluice Pump, and the Spent Resin Storage Tank. The resin sluice water, after being directed to an ion exchange vessel by the sluice pump, is returned to the spent resin storage tank for reuse. Thus, sluicing of spent resin from primary plant demineralizers is normally accomplished without generating a large volume of additional liquid waste.

For additional system flexibility, the Spent Resin Storage Tank may be bypassed and the resin sluiced directly from a demineralizer to a mobile system for disposal.

The resin slurry from the Spent Resin Storage Tank is transferred to the waste processing area by pressurizing the tank with nitrogen.

Resin from the Steam Generator Blowdown System (SGBS) demineralizers is handled in a similar manner as that described for the primary system demineralizers. The SGBS resin handling is physically separated from primary demineralizers and is discussed in Section 10.4.8. The primary system resins and the SGBS resins are segregated during all phases of handling so that no cross contamination is possible.

Powdered resin from the Condensate Cleanup System (CCS) is processed as described in Section 10.4.6. When specified by the Radiological Effluents Control Program the powdered resin will be disposed of as radioactive waste using vendor equipment. The actual process for the packaging of resins for disposal is discussed in Subsection 11.4.

Details of the spent resin handling subsystem are shown on Figures 11.2-6 and 11.2-7.

11.2.2.3 Liquids from Sources Other Than Liquid Waste Processing System

1. Steam Generator Blowdown

Blowdown from the steam generators of each unit is cooled, filtered, demineralized, and returned to the condenser or heater drain tank for reuse as secondary coolant. This Blowdown Processing System is described in Section 10.4.8.

2. Turbine Building Sumps

Discharges from the Turbine Building Sumps are routed to the Wastewater Management System (WMS), as discussed in Section 9.2.8. As shown on Figure 9.2-15, these discharges are normally routed to the Low Volume Waste treatment facilities. However, when radioactivity is present above specified levels, the discharges are diverted to the Co-Current Waste treatment facilities. These facilities are also part of the WMS. After batching, these wastes are sampled for radioactivity and if required, treated for conventional pollutants and discharged to the Circulating Water Discharge Canal.

11.2.2.4 Equipment Description and Design Basis

11.2.2.4.1 General

The locations of the LWPS components are shown in Table 11.2-4.

The Waste Holdup Tank, Floor Drain Tanks, and Laundry and Hot Shower Tank are located at the lowest possible levels of the plant so that gravity feed into the tanks is possible from the majority of the components being drained.

Typically parts or components in contact with borated water are fabricated from or clad with austenitic stainless steel. Except for flanged joints which are provided at certain equipment connections to reduce personnel exposure during maintenance or during replacement of that equipment, all-welded construction is used. Compression fittings will be used on small diameter instrument lines. The design parameters of this equipment are listed in Table 11.2-3.

Tank overflow protection is provided as described in item 9 of Section 11.2.2.4.2.

11.2.2.4.2 Equipment

1. Pumps

The two basic types of pumps in the LWPS are a canned-rotor pump design and a mechanical seal design.

Valves are installed in pump discharge lines. Design bases for individual pumps are given below:

a. Reactor Coolant Drain Tank Pump

In accordance with the design basis, this pump may perform as a plant drain. Two pumps are furnished because of the relative inaccessibility of the Containment during plant operation. One pump provides sufficient flow for normal operation of the Reactor Coolant Drain Tank portion of the LWPS. The Reactor Coolant Drain Tank Pumps are of the canned-rotor design.

b. Waste Evaporator Feed Pump

This standard design canned-centrifugal pump supplies feed to the Waste Evaporator or FDS based on the level in the Waste Holdup Tank.

Pump head requirements include elevation, the head loss through the feed filter, and the required waste evaporator feed supply pressure.

c. Waste Evaporator Condensate Pump

The Waste Evaporator Condensate Tank Pump is a canned-rotor design transfer pump. One pump is used to transfer the contents of the Waste Condensate Tank to the BRS holdup tank or to one of the Waste Monitor Tanks.

d. Floor Drain Tank Pumps

Mechanical-seal type pumps are supplied for Floor Drain Tank Pumps 1 and 2. Floor Drain Tank Pump 3 is a canned-rotor design pump. Each pump is capable of supplying feed at the required pressure to the Floor Drain Evaporator, the Waste Monitor Tank Demineralizers and the FDS.

e. Waste Monitor Tank Pumps

One canned-rotor pump is used for each of the two Waste Monitor Tanks. These pumps can discharge water from the plant or recycle it for further processing if required.

f. Laundry and Hot Shower Tank Pump

This pump is of the mechanical-seal type and has performance characteristics identical to the standard LWPS pump. The pump is designed to transfer water from the Laundry and Hot Shower Tank to the Waste Monitor Tanks via the strainer and filter.

g. Reverse Osmosis Concentrates Tank Pump

This pump has been abandoned in place. The Reverse Osmosis System will not be used.

h. Laundry Holdup and Monitor Tank Pump

These pumps are canned-rotor type. These pumps can discharge water from the plant or recycle it for further processing, if required.

i. Chemical Drain Tank Pump

One standard canned-rotor pump is used for recirculation and to transfer liquid from the Chemical Drain Tank to the Waste Conditioning Tank.

j. Spent Resin Sluice Pump

This pump is a canned-rotor design and operates during resin sluicing. The design flow rate is based on a conservative fluid velocity sufficient to sluice resin in a three-inch pipe.

k. Plant Effluent Holdup and Monitor Tank Pumps

Two horizontal centrifugal pumps provided with mechanical seals operate to permit recirculation, sampling, and controlled release or retreatment of processed waste in the floor drain waste system (Drain Channel B).

2. Tanks

a. Reactor Coolant Drain Tank

One tank is provided for each unit. The purpose of the Reactor Coolant Drain Tank is to collect leakoff-type drains inside the Containment at a central collection point for further disposition through a single penetration via the Reactor Coolant Drain Tank Pumps. The tank provides surge volume and net positive suction head (NPSH) to the pumps.

Only water which can be directed to the Boron Recycle Holdup Tanks enters the Reactor Coolant Drain Tank. The water must be compatible with reactor coolant and not contain dissolved air or nitrogen. A hydrogen gas blanket is normally maintained on the tank. Within approximately 24 hours of a scheduled unit shutdown, nitrogen may be introduced into the RCDT to purge the hydrogen in preparation for the shutdown.

It is intended that the tank be maintained at a constant level to minimize the amount of gas sent to the GWPS.

The level is maintained by running one pump continuously and installing a proportional control valve in the discharge line. This valve recirculates a portion of the pump discharge back to the Reactor Coolant Drain Tank to maintain the tank level.

Continuous flow is maintained through the heat exchanger in order to prevent loss of pump NPSH resulting from a sudden inflow of hot liquid into the Reactor Coolant Drain Tank.

b. Waste Holdup Tank

An atmospheric tank is provided to collect equipment drains, valve and pump seal leakoffs (outside the Containment), boron recycle holdup tank overflows, and other water from tritiated aerated sources. The design bases for the Waste Holdup Tank are to provide sufficient surge capacity (3000 gal) to warrant evaporator startup and to provide an additional margin to accept a 10-gpm leak from one unit for eight hours (4800 gal).

c. Waste Evaporator Condensate Tank

There is one Waste Evaporator Condensate Tank, it has a diaphragm to exclude air. This tank collects condensate from the waste evaporator. The tank size is based on providing sufficient condensate tankage to allow a 15-gpm evaporator to operate without interruption for a four-hr period.

d. Floor Drain Tanks

There are three floor drain tanks. These atmospheric tanks are used to collect floor drains from the controlled areas of the primary plant. The tanks provide sufficient surge capacity for the floor drains within their collection areas and in connection with the Waste Holdup Tank, provide surge capacity for abnormal primary system leaks. The design bases for the tanks are the same as for the Waste Holdup Tank.

e. Waste Monitor Tanks

There are two Waste Monitor Tanks. These atmospheric waste monitor tanks are provided to collect floor drain evaporator condensate and/or FDS effluents. The tanks are used for liquid holdup and monitoring prior to reprocessing or discharge from the plant.

f. Laundry and Hot Shower Tank

There is one Laundry and Hot Shower Tank. This atmospheric tank is used to collect hot shower drains within the controlled areas. The tank is sized to furnish a 15-day surge capacity for a twin-unit station during normal operation of both units and a four-day surge capacity during refueling of a single unit.

g. Laundry Holdup and Monitor Tank

There are two Laundry Holdup and Monitor Tanks. These atmospheric tanks are provided to collect processed fluid for sampling and analysis prior to recycle, reprocessing or discharge from the plant.

h. Reverse Osmosis Concentrate Tank

This tank has been abandoned in place. The Reverse Osmosis System will not be used.

i. Not Used

j. Laundry Water Head Tank

The tanks normal discharge has been capped, it can be drained to the Laundry and Hot Shower Tank. Water is fed to this tank from demineralized water storage.

k. Chemical Drain Tank

One tank is provided to collect chemically contaminated and radioactive water from the laboratories. This tankage is sufficient for a two-unit station.

l. Waste Conditioning Tank

One tank is provided to collect evaporator concentrates and chemical drain tank effluents before they are packaged for disposal. By monitoring flow to the tank, the activity and chemistry of the burial packages can be controlled.

m. Spent Resin Storage Tank

The purpose of the Spent Resin Storage Tank is to provide a collection point for spent resin and to allow for decay of short-lived radionuclides before disposal. The tank also serves as a head tank for the Spent Resin Sluice Pump.

One vertical cylindrical tank with sufficient capacity to handle the spent resin storage needs is provided. A vertical cylindrical tank is used because the symmetrical bottom facilitates the removal of resin. The tank is designed so that sufficient pressure can be applied in the gas space of the tank to transfer the resin slurry to the waste processing area.

The spent resin storage tank and associated equipment which can contain radioactive material are shielded to limit the dose to personnel.

n. Evaporator Reagent Tanks

There are two evaporator reagent tanks, one tank is provided with the waste evaporator and the other with the floor drain evaporator. The tanks are used to feed chemicals to their respective evaporators.

o. Plant Effluent Holdup and Monitor Tanks

There are two Plant Effluent Holdup and Monitor Tanks. The tanks are atmospheric waste monitor tanks that collect filter/demineralizer effluent. The tanks are used for liquid holdup and monitoring prior to reprocessing or discharge from the plant. The capacity of each tank is greater than the daily design flow rate so as to provide adequate holding time for sampling and analysis.

3. Demineralizers

The demineralizers are sized for a 35-gpm flow rate and hold 30 ft^3 of ion exchange resin.

a. Waste Evaporator Condensate Demineralizer

One demineralizer is provided to remove trace ionic contaminants from the influent.

b. Waste Monitor Tank Demineralizer

One demineralizer provided to remove trace ionic contaminants from the influent.

4. Filters

The following filters are provided in the LWPS:

a. Waste Evaporator Feed Filter

b. Laundry and Hot Shower Tank Filter

c. Floor Drain Tank Filter

d. Spent Resin Sluice Filter

e. Waste Evaporator Condensate Filter

f. Waste Monitor Tank Filter

The filters are of the disposable-cartridge type and are designed to pass 35 gpm of fluid at a minimal pressure drop. The spent resin sluice filter is of similar design to the other filters, but has a design flow rate of 250 gpm. All but the Laundry and Hot Shower Tank Filter require shielding.

The methods used to change filter elements are dependent on activity levels. If the radiation level of the filter is low enough, the filter elements may be changed manually. If activity levels do not permit manual change, the spent element cartridge assembly is removed remotely with temporary shielding to protect personnel. The spent cartridge assembly is placed in a shielded container for removal to the solid waste disposal area. Filter elements are normally changed because of high differential pressure or high radiation levels. Filter cartridge assembly handling operations are further described in Section 11.4.

5. Strainers

Strainers are provided in the LWPS for the following:

a. Laundry and Hot Shower Tank Strainer

b. Floor Drain Tank strainers

The Laundry and Hot Shower and Floor Drain Tank Strainers are rated at 35 gpm.

Strainers are of the basket type with mesh construction.

The Laundry and Hot Shower and Floor Drain Tank Strainers prevent clogging of filters and lines downstream because of large particles being sluiced through the lines during liquid transfer operations. The basket type laundry and hot shower strainer is not replaced after use but is cleaned and put back into service.

6. Waste and Floor Drain Evaporators

Two 15-gpm forced-recirculation-type evaporators are provided in the LWPS. Evaporator distillate is either recycled, reprocessed, or discharged, depending on its chemistry, radioactivity and operational needs.

The Waste and Floor Drain Evaporators are identical units, and their feed and discharge headers are interconnected so that the Waste and Floor Drain Evaporators or Waste and Boron Recycle Evaporators can be used in place of each other. All evaporators use auxiliary steam from onsite sources as their process heat source. Component cooling water is furnished to the coolers and condensers of the evaporators.

The Waste and Floor Drain Evaporators are maintained as backup equipment for liquid waste processing.

7. Reactor Coolant Drain Tank Heat Exchanger

The Reactor Coolant Drain Tank Heat Exchanger is a U-tube type with one shell pass. The unit is sized to maintain the Reactor Coolant Drain Tank fluid at 170°F or less with a 10-gpm inleakage of reactor coolant at 600°F. The normal temperature is 130°F.

The Reactor Coolant Drain Tank Heat Exchanger can also maintain the Reactor Coolant Drain Tank fluid at 170°F or less with a 25-gpm flow from the Excess Letdown Heat Exchanger during heatup or draining operations and cool the contents of the Pressurizer Relief Tank from 200°F to 120°F in less than eight hours.

8. Reverse Osmosis System

The Reverse Osmosis System will not be utilized as part of the Liquid Waste Processing System. Since an outside vendor is to be utilized to process contaminated plant laundry, the capability to process liquids with high detergent concentrations is not needed. The Reverse Osmosis System has been removed with Reverse Osmosis Concentrates Tank and Reverse Osmosis Concentrates Tank Pump abandoned in place.

9. Tank Overflow Protection

All tanks in the Chemical and Volume Control System (CVCS), Boron Recycle System (BRS), Steam Generator Blowdown Processing System (SGBPS) and WPS that could potentially contain radioactive liquids are designed to provide adequate warning of potential overflow conditions. A summary of the overflow protection features is given in Table 9.3-5. These tanks are provided with level indication instrumentation which has an alarm function on high liquid level in the tank. Alarm annunciation is provided separately on the local system control panel and further relayed to a common annunciator on the main control board in the control room for each system. A description of the level instrumentation provided for these systems is given in Sections 9.3.4.1, 9.3.4.2, Table 9.3-5 and 11.2.2.5 for the CVCS, BRS, SGBPS, and LWPS, respectively.

In addition to tank level monitoring and warning of potential overflow conditions, provisions are made in the systems design to collect and process overflows from tanks containing potentially radioactive liquids.

10. Filter Demineralizer System (FDS)

The Liquid Waste Processing System is equipped with connection to allow for processing of liquid wastes from the following sources through a filter demineralizer system (FDS):

Waste Holdup Tank	Waste Monitor Tanks 1 & 2
Floor Drain Tanks 1, 2, & 3	Laundry & Holdup Monitor
Chemical Drain Tank	Tanks 1 & 2
Laundry & Hot Shower Tank	Radwaste Liner dewatering waste (HIC)
Plant Effluent Holdup and Monitor Tanks 1&2	Recycle Holdup Tanks 1 & 2

Additionally other waste streams may be processed, as necessary, by hose connection to the FDS.

The FDS is a skid mounted system which consists of a number of components which may include:

a. Demineralizer Vessels

b. Booster Pumps

c. Chemical Treatment Systems

d. System Filters

Examples of processing media that may be loaded in the FDS Demineralizer vessels include; activated carbon, organic resin, and/or ion specific media.

Liquid wastes are pumped to the FDS process components. A system booster pump may be utilized to improve system flow rates. The demineralizer vessels may be valved into service or bypassed and sequenced in any order to provide an acceptable cleanup of the process waste stream.

The number and sequence of demineralizer vessels used will depend on vessel media loading and the chemical and radiochemical composition of the waste stream. The vessel configuration and the processing media will be selected to optimize cleanup of waste stream and to minimize solid waste production.

11. Drum Dryer System

Non-routine liquid radioactive water resulting from activities like tank cleaning, facility/ equipment decontamination, or tank bottoms that may not be suited for Filter Demineralizer System or floor drain processing may be preconditioned or volume reduced by using a self-contained, skid-mounted drum dryer system located in the Fuel Building. A vacuum is established in the drum as the liquid is heated. Steam from the

drum is condensed. Liquid effluents are routed to floor drains and the remaining solids are handled as Dry Active Waste or sludge.

11.2.2.5 Instrumentation Design

11.2.2.5.1 General

The instrumentation readout is located mainly on the LWPS panel in the Auxiliary Building. Some instruments are read locally.

Alarms are shown separately on the LWPS panel and further relayed to the common LWPS annunciator on the main control board in the Control Room.

Pumps are protected against loss of suction pressure by a control set point on the level instrumentation for the respective vessels feeding the pumps. The Reactor Coolant Drain Tank Pumps and the Spent Resin Sluice Pump are, in addition, interlocked with flow-rate instrumentation and stop operating when the delivery flows reach minimum set points.

Pressure indicators upstream from and downstream of filters, strainers, and demineralizers provide local indications of pressure drops across each component.

Releases to the environment are monitored for radioactivity by radiation detectors. This instrumentation is further described in Section 11.5.

The system instrumentation for Figures 11.2-2 to 11.2-7 is described in detail in Table 11.2-5.

11.2.2.5.2 Tank Overflow Protection

All tanks in the LWPS that could potentially contain radioactive liquids are designed to provide adequate warning of potential overflow conditions.

These tanks are provided with level indication instrumentation which has an alarm function on high liquid level in the tank. Alarm annunciation is provided separately on the local system control panel and further relayed to a common annunciator on the main control board in the Control Room for each system.

In addition to tank level monitoring and warning of potential overflow conditions, provisions are made in the system design to collect and process overflows from tanks containing potentially radioactive liquids.

The Spent Resin Storage Tank (SRST) level instrumentation is interlocked with the SRST vent line isolation valve. On a high level signal the vent line isolation valve will close.

11.2.2.6 Operating Procedure

The LWPS is manually operated in a batch-type method except for some functions of the Reactor Coolant Drain Tank circuit. The system includes adequate control equipment to protect system components and adequate instrumentation and alarm functions to provide operator information to ensure proper system operation. Pumps in the system have low-level shutoffs, and filters and demineralizers have pressure indication upstream and downstream to signify fouling.

1. Normal Operation

Operation of the LWPS is essentially the same during all phases of normal reactor plant operation; the only differences are in the load on the system. The LWPS is not regarded as an Engineered Safety Features (ESF) System.

2. Reactor Coolant Drain Tank Subsystem Operation

Normal operation of the Reactor Coolant Drain Tank is automatic and requires no operator action. The system can be put in the manual mode if desired. The leakage rate into the tank can be estimated by putting the system in the manual mode, stopping the pump, and watching the rate of level change. The venting system is aligned to the Gaseous Waste Processing System when venting is required.

Operation of the system during refueling is the same as for power operation, although the load on the system can be increased when refueling is complete; the water remaining in the canal following normal drain down is pumped to the discharge of the Refueling Water Purification Pump by the Reactor Coolant Drain Tank Pumps. For this operation, the level control valve is opened and the minimum flow line is valved closed.

3. Drain Channel A Subsystem Operation

Water is accumulated in the Waste Holdup Tank until a sufficient quantity exists to warrant waste processing. If the Waste Evaporator is used the distillate may be passed through the Waste Evaporator Condensate Demineralizer before being transferred to the Waste Condensate Tank or the Waste Monitor or Laundry Holdup Monitor Tanks.

The bottoms from the waste evaporator are concentrated to approximately 4 to 5 percent by weight boric acid and are normally recycled to the boric acid tanks. A reagent tank is included for chemical addition to the evaporator in the event of evaporator tube fouling. The Waste Holdup Tank may also be processed through the vendor supplied FDS with the effluent directed to the Waste Monitor Tanks, Laundry Holdup Monitor Tanks, or Plant Effluent Holdup and Monitor Tanks.

Drain Channel A has been provided with interconnections to Drain Channel B as follows:

a. Waste Evaporator inlet to Floor Drain Evaporator inlet

b. Waste Evaporator Condensate Demineralizer outlet to Waste Monitor Tank

c. Floor Drain Tank Filter outlet to Waste Holdup Tank

These interconnections add flexibility to the LWPS and are provided with double valves and telltale leak-detection connections.

4. Drain Channel B Subsystem Operation

Water is accumulated in the plant's three floor drain tanks. Normally, when Floor Drain Tank 1 or 2 is full it is transferred to Floor Drain Tank 3. When Floor Drain Tank 3 is full, the tank may be sampled prior to processing. Floor Drain Tank 3 is normally processed

through the FDS or the Floor Drain Evaporator. The processed water is then subsequently transferred to the Waste Monitor Tanks, Laundry Holdup Monitor Tanks, or Plant Effluent Holdup and Monitor Tanks. The contents of the monitor tanks are then sampled for acceptability to discharge or recycle.

Water leaving Drain Channel B that is found unacceptable for either discharge from the plant or return to the secondary cycle is reprocessed. The monitor tanks are normally discharged into the circulating water tunnel.

The discharge pipeline to the Circulating Water System is common to both Drain Channel B and Channel C and uses a flowmeter, a radiation monitor, and an automatic isolation valve.

Drain Channel B is provided with interconnections to Drain Channel C as follows:

a. Floor Drain Tank 3 Pump discharge to the FDS

b. Laundry and Hot Shower Tank Filter outlet to the Floor Drain Waste Monitor tank inlet header.

A connection from Floor Drain Tanks 1, 2, and 3 to the Waste Conditioning Tank is also provided for the rare instance when fluid in these tanks would have to be disposed of without processing.

5. Drain Channel C Subsystem Operation

Laundry and hot shower water enters the Laundry and Hot Shower Tank for holdup. When the Laundry and Hot Shower Tank is approximately half full, the water is routed via the Laundry and Hot Shower Tank Strainer and Filter to the Waste Monitor Tank and is discharged or reprocessed as needed.

The Laundry Water Head Tank is provided with high- and low-level alarms and remote level indication. Makeup to the tank can be provided from the demineralized water supply system on low level in the tank.

6. Spent Resin Handling Subsystem

This portion of the system sluices resin from the demineralizers, transports resin from the Spent Resin Storage Tank to the waste processing area, and flushes the line from the Waste Evaporator to the waste processing area.

a. Resin Sluicing

Before resin sluicing begins, the demineralizers are backflushed utilizing a closed recirculation flow loop. This loop uses the resin sluice pump or demineralized water line as a motive force and is between the demineralizer vessel and the spent resin storage tank. Resin transfer is then performed by opening the resin sluice outlet and closing the backflush outlet.

Fresh resin is added via the resin fill line and then subsequently the vessel is backflushed to the Spent Resin Storage Tank which removes the resin fines. The valves are then aligned for standby operation.

b. Resin Disposal

Resin may be disposed of by use of a vendor-supplied mobile system via the bulk disposal connection. When sufficient resin has accumulated to warrant disposal, the Spent Resin Storage Tank is pressurized with nitrogen, and resin is transferred to the bulk disposal connection. Upon completion of transfer, the Spent Resin Storage Tank is vented to the plant vent, and flush water is pumped through all lines to ensure resin removal is complete.

7. Potential Bypass Routes

The potential bypass routes in the LWPS are as follows:

a. Drain Channel A Subsystem

Waste accumulated in the waste holdup tank may be evaporated. The evaporator distillate can go directly to the waste evaporator condensate tank, bypassing the waste evaporator condensate demineralizer, unless waste holdup tank samples indicate a high radioactivity level. The Waste Holdup Tank may also be processed through the FDS by bypassing the evaporator purification train.

b. Drain Channel B Subsystem

The wastewater in the Floor Drain Tanks may be sampled to determine the degree of processing required. Based on the result, the waste can be sent directly to a Waste Monitor Tank, Laundry Holdup and Monitor Tank or Plant Effluent Holdup and Monitor Tank, bypassing the Waste Evaporator, the Waste Monitor Tank Demineralizer, and/or the FDS. The Floor Drain Tanks may be processed through the FDS by bypassing the Floor Drain Evaporator Purification train.

11.2.2.7 Faults Of Moderate Frequency

The system is designed to handle the occurrences of equipment faults of moderate frequency such as:

1. Malfunction in the Liquid Waste Processing System

Malfunction in this system can include pump or valve failures or evaporator failure. There is sufficient surge capacity in the system to accommodate waste until the failures can be remedied and normal plant operation resumed.

2. Excessive Leakage in Reactor Coolant System Equipment

The LWPS is designed to handle a 1-gpm reactor coolant leak in addition to the expected leakage during normal operation. Operation of the system is almost the same as for normal operation except the load on the system is increased. A 1-gpm leak into the

Reactor Coolant Drain Tank is handled automatically but may increase the load factor of the recycle evaporator. If a 1-gpm leak enters the Waste Holdup Tank, operation is the same as normal except for the increased load on the evaporator or FDS. Abnormal liquid volumes of reactor coolant or Auxiliary Building equipment leakage can also be accommodated by the Floor Drain Tank and processed by the LWPS.

3. Excessive Leakage in Auxiliary System Equipment

Leakage of this type can include water from steamside leaks and fan cooler leaks inside the Containment which are collected in the Containment sump and sent to a floor drain tank. Other sources can be Component Cooling Water leaks, Service Water leaks, and secondary side water leaks. This water enters the Floor Drain Tank and is processed and discharged as during normal operation.

11.2.3 RADIOACTIVE RELEASES

11.2.3.1 Expected Releases

Expected release quantities in Table 11.2-2c are based on normal operation, including anticipated operational occurrences of the plant and the LWPS, and a realistic estimation of the potential input sources, based on operating experience. The quantities and isotopic concentrations in the liquid discharge from the LWPS are highly dependent upon the actual operation of the plant.

The input sources assumed in the release determinations are summarized in Table 11.2-1. The specific activities used for these computations are detailed in Section 11.1. It is assumed that the waste entering the floor drain tank is 1300 gpd of reactor coolant and 11,700 gpd of non-reactor grade water. The isotopic composition of reactor grade water is as indicated in Section 11.1. The isotopic concentrations at key locations in the LWPS are given in Table 11.2-2. A comparison between the expected release and the limiting concentrations (in accordance with 10 CFR Part 20) are shown in Table 11.2-9.

The total releases, excluding tritium, from the LWPS are approximately 3.3 Ci/yr. The associated release of tritium is less than 2600 Ci/yr. The expected tritium concentration and its influence on plant water balance is discussed in Section 11.2.1.4 of WCAP-8665.

An excessive reactor coolant leakage of 1 gpm in addition to expected leakage during normal operation can be handled with an increased load on the processing equipment. The releases are approximately the same as during normal operation. Therefore, the design objectives outlined in Subsection 11.2.1.1 are met.

The radioactive releases associated with 1 percent failed fuel are given in Table 11.2-10. The releases determined in accordance with NRC Regulatory Guide 1.112 are given in Table 11.2-9.

11.2.3.2 Release Points

Treated liquid effluent is discharged from the LWPS to the environment via the circulating water discharge canal. The release point is indicated on LWPS flow diagrams in the Flow Diagram Volume.

11.2.3.3 Dilution Factors

Treated liquid effluent is diluted by the circulating water flow prior to discharge to ensure that the concentration of radionuclides at the discharge is well below applicable limits. Details on dilution factors are given in Appendix 11A.

11.2.3.4 Estimated Doses

The estimated doses resulting from anticipated annual releases into the Squaw Creek Reservoir are presented in Appendix 11A. The dose estimates are considered historical and are not subject to future updating. The information is retained to avoid loss of original design basis. Actual radioactivity release quantities can be found in the annual radioactive effluent release reports submitted to the NRC.

TABLE 11.2-1

PARAMETERS USED IN THE CALCULATION OF EXPECTED ACTIVITY IN LIQUID WASTES[a] - ORIGINAL DESIGN BASIS[b]

Collector Tank with Sources	Expected Volume of Liquid Waste	Basis	Collection Period Assumed Before Processing
Floor Drain Tank (Dirty Wastes)	13,000 gpd	Derived from 1991 operating data (2.37×10^6 gal/yr/unit)	1.85 days
Waste Holdup Tank (Clean Wastes)	830 gpd	Derived from 1991 operating data (1.51×10^5 gal/yr/unit)	9.64 days
Recycle Holdup Tank (Chemical Shim)	4200 gpd	Derived from 1991 operating data (7.66×10^5 gal/yr/unit)	23.8 days

a) This table does not represent system capacity. The volume of liquid waste assumes two unit operation.

b) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11.2-2

EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 1 of 9)

a. Tank Inventories

Nuclide	Reactor Coolant Activity (uCi/ml)	Tank Inventory[b] – Ci				
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank	Plant Effluent Holdup and Monitor Tank
Na-24	4.7E-02	5.3E-01	1.5E+00	2.1E+01	1.8E-04	1.1E-03
Cr-51	3.1E-03	3.5E-02	1.0E-01	1.4E+00	6.5E-05	3.9E-04
Mn-54	1.6E-03	1.8E-02	5.1E-02	7.2E-01	3.5E-05	2.2E-04
Fe-55	1.2E-03	1.4E-02	3.9E-02	5.4E-01	2.6E-05	1.6E-05
Fe-59	3.0E-04	3.4E-03	9.7E-03	1.4E-01	6.5E-06	3.9E-05
Co-58	4.6E-03	5.2E-02	1.5E-01	2.1E+00	1.0E-04	6.0E-04
Co-60	5.3E-04	6.0E-03	1.7E-02	2.4E-01	1.2E-05	7.2E-05
Zn-65	5.1E-04	5.8E-03	1.6E-02	2.3E-01	1.1E-05	6.7E-05
W-187	2.5E-03	2.8E-02	8.0E-02	1.1E+00	1.7E-05	1.0E-04
Np-239	2.2E-03	2.5E-02	7.1E-02	9.9E-01	2.6E-05	1.6E-04
Sr-89	1.4E-04	1.6E-03	4.5E-03	6.3E-02	3.1E-06	1.8E-05
Sr-90	2.0E-05	2.3E-04	6.5E-04	9.0E-03	4.3E-07	2.7E-06
Sr-91	9.6E-04	1.1E-02	3.1E-02	4.3E-01	1.8E-06	1.1E-05
Y-91m	4.6E-04	5.2E-03	1.5E-02	2.1E-01	1.2E-06	7.5E-06

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 2 of 9)

Nuclide	Reactor Coolant Activity (uCi/ml)	Tank Inventory[b] – Ci				
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank	Plant Effluent Holdup and Monitor Tank
Y-91	5.2E-06	5.9E-05	1.7E-04	2.3E-03	2.5E-07	1.5E-06
Y-93	4.2E-03	4.8E-02	1.4E-01	1.9E+00	9.2E-06	5.6E-05
Zr-95	3.9E-04	4.4E-03	1.3E-02	1.8E-01	8.5E-06	5.1E-05
Nb-95	2.8E-04	3.2E-03	9.0E-03	1.3E-01	6.3E-06	3.7E-05
Mo-99	6.4E-03	7.3E-02	2.1E-01	2.9E+00	8.3E-05	5.4E-04
Tc-99m	4.7E-03	5.3E-02	1.5E-01	2.1E+00	7.8E-05	4.7E-04
Ru-103	8.2E-03	9.3E-02	2.6E-01	3.7E+00	1.6E-04	1.0E-03
Ru-106	1.7E-01	1.8E+00	5.4E+00	7.7E+01	3.7E-04	2.2E-03
Ag-110m	1.3E-03	1.5E-02	4.2E-02	5.8E-01	2.9E-05	1.7E-04
Te-129m	1.9E-04	2.2E-03	6.1E-03	8.5E-02	4.0E-06	2.4E-05
Te-129	2.4E-02	2.7E-01	7.7E-01	1.1E+01	2.6E-06	1.6E-05
Te-131m	1.5E-03	1.7E-02	4.8E-02	6.7E-01	1.2E-05	7.2E-05
Te-131	7.7E-03	8.7E-02	2.5E-01	3.5E+00	2.2E-06	1.2E-05
I-131	4.5E-02	5.1E-01	1.5E+00	2.0E+01	8.5E-05	4.9E-04
Te-132	1.7E-03	1.9E-02	5.5E-02	7.6E-01	2.3E-05	1.4E-04
I-132	2.1E-01	2.4E+00	6.8E+00	9.4E+01	2.3E-05	1.4E-04

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 3 of 9)

Nuclide	Reactor Coolant Activity (uCi/ml)	Tank Inventory[b] – Ci				
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank	Plant Effluent Holdup and Monitor Tank
I-133	1.4E-01	1.6E+00	4.5E+00	6.2E+01	8.1E-05	4.9E-04
I-134	3.4E-01	3.9E+00	1.1E+01	1.5E+02	4.6E-08	2.7E-07
Cs-134	2.1E-02	2.4E-01	6.8E-01	9.4E+00	5.0E-06	2.9E-05
I-135	2.6E-01	3.0E+00	8.4E+00	1.2E+02	2.6E-05	1.6E-04
Cs-136	8.7E-04	9.9E-03	2.8E-02	3.9E-01	1.8E-07	1.1E-06
Cs-137	1.8E-02	2.1E-01	5.7E-01	8.0E+00	4.4E-06	2.7E-05
Ba-140	1.3E-02	1.5E-01	4.2E-01	5.8E+00	2.5E-04	1.5E-03
La-140	2.5E-02	2.8E-01	8.0E-01	1.1E+01	3.8E-04	2.3E-03
Ce-141	1.5E-04	1.7E-03	4.8E-03	6.7E-02	3.1E-06	1.8E-05
Ce-143	2.8E-03	3.2E-02	9.0E-02	1.3E+00	2.5E-05	1.5E-04
Ce-144	4.4E-03	5.0E-02	1.5E-01	2.0E+00	9.7E-05	5.8E-04
H-3	2.0E+00	2.2E-01	6.5E+01	9.1E+02	4.1E+00	2.2E+01

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 4 of 9)

b. Tank Vents Isotopic Discharge Rate (uCi/min)

Vent No.	Vent Description	Temperature (F)	Pressure (psig)	Flow Rate (cc/min)	Kr-85m	Kr-85
1	Reactor coolant drain tank to GWPS	130	5	8.8	1.07E-01	1.76E+000
2	Waste holdup tank vent to plant vent	amb	atm	431	neg	neg
3	Evaporator package vents to plant vent	150	2	6.0	neg	neg
4	Floor drain tank vent to plant vent	amb	atm	2203	neg	neg
5	Waste monitor tanks local vent	amb	atm	2148	neg	neg
6	Chemical drain tank to plant vent	amb	atm	7.56	neg	neg
7	Laundry and hot shower local vent	Var	atm	866	neg	neg
8	Laundry holdup and monitor tank local vent	amb	atm	866	neg	neg
9	Plant Effluent Holdup and Monitor Tanks[c]					

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 5 of 9)

Vent No.	Kr-87	Kr-88	XE-133m	XE-133	XE-135	I-131	I-133
1	1.76E-02	1.22E-01	1.21E+00	1.31E+02	6.11E-01	2.38E-04	2.64E-04
2	neg	neg	neg	neg	neg	1.77E-03	3.28E-04
3	neg	neg	neg	neg	neg	neg	neg
4	neg	neg	neg	neg	neg	1.56E-03	5.95E-04
5	neg	neg	neg	neg	neg	1.55E-07	5.8E-08
6	neg	neg	neg	neg	neg	4.08E-06	6.2E-07
7	neg	neg	neg	neg	neg	neg	neg
8	neg	neg	neg	neg	neg	neg	neg
9[c]							

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 6 of 9)

c. Processing Lines

Nuclide	Reactor Coolant Activity (uCi/ml)	Activity in Line[d] – uCi/ml					
		Line No. 1	Line No. 2	Line No. 3	Line No. 4	Line No. 5	Line No. 6
Na-24	4.7E-02	4.7E-03	4.7E-05	4.0E-02	4.0E-04	4.5E-02	8.9E-10
Cr-51	3.1E-03	3.1E-04	3.1E-06	2.6E-03	2.6E-05	2.9E-03	5.9E-11
Mn-54	1.6E-03	1.6E-04	1.6E-06	1.4E-03	1.4E-05	1.5E-03	3.0E-11
Fe-55	1.2E-03	1.2E-04	1.2E-06	1.0E-03	1.0E-05	1.1E-03	2.3E-11
Fe-59	3.0E-04	3.0E-05	3.0E-07	2.6E-04	2.6E-06	2.9E-04	5.7E-12
Co-58	4.6E-03	4.6E-04	4.6E-06	3.9E-03	3.9E-05	4.4E-03	8.7E-11
Co-60	5.3E-04	5.3E-05	5.3E-07	4.5E-04	4.5E-06	5.0E-04	1.0E-11
Zn-65	5.1E-04	5.1E-05	5.1E-07	4.3E-04	4.3E-06	4.8E-04	9.7E-12
W-187	2.5E-03	2.5E-04	2.5E-06	2.1E-03	2.1E-05	2.4E-03	4.8E-11
Np-239	2.2E-03	2.2E-04	2.2E-06	1.9E-03	1.9E-05	2.1E-03	4.2E-11
Sr-89	1.4E-04	1.4E-05	1.4E-07	1.2E-04	1.3E-06	1.3E-04	2.7E-12
Sr-90	2.0E-05	2.0E-06	2.0E-08	1.7E-05	1.7E-07	1.8E-05	3.8E-13
Sr-91	9.6E-04	9.6E-05	9.6E-07	8.2E-04	8.2E-06	9.1E-04	1.8E-11
Y-91m	4.6E-04	4.6E-05	4.6E-07	3.9E-04	3.9E-06	4.4E-04	8.7E-12
Y-91	5.2E-06	5.2E-07	5.2E-09	4.4E-06	4.4E-08	4.9E-06	9.9E-14

TABLE 11.2-2

EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 7 of 9)

Nuclide	Reactor Coolant Activity (uCi/ml)	Activity in Line[d] – uCi/ml					
		Line No. 1	Line No. 2	Line No. 3	Line No. 4	Line No. 5	Line No. 6
Y-93	4.2E-03	4.2E-04	4.2E-06	3.6E-03	3.6E-05	4.0E-03	8.0E-11
Zr-95	3.9E-04	3.9E-05	3.9E-07	3.3E-04	3.3E-06	3.7E-04	7.4E-12
Nb-95	2.8E-04	2.8E-05	2.8E-07	2.4E-04	2.4E-06	2.7E-04	5.3E-12
Mo-99	6.4E-03	6.4E-04	6.4E-06	5.4E-03	5.4E-05	6.1E-03	1.2E-10
Tc-99m	4.7E-03	4.7E-04	4.7E-06	4.0E-03	4.0E-05	4.5E-03	8.9E-11
Ru-103	8.2E-03	8.2E-04	8.2E-06	7.0E-03	7.0E-05	7.7E-03	1.5E-10
Ru-106	1.7E-01	1.7E-02	1.7E-04	1.4E-01	1.4E-03	1.6E-01	3.2E-09
Ag-110m	1.3E-03	1.3E-04	1.3E-06	1.1E-03	1.1E-05	1.2E-03	2.5E-11
Te-129m	1.9E-04	1.9E-05	1.9E-07	1.6E-04	1.6E-06	1.8E-04	3.6E-12
Te-129	2.4E-02	2.4E-03	2.4E-05	2.0E-02	2.0E-04	2.3E-02	4.6E-10
Te-131m	1.5E-03	1.5E-04	1.5E-06	1.3E-03	1.3E-05	1.4E-03	2.9E-11
Te-131	7.7E-03	7.7E-04	7.7E-06	6.5E-03	6.5E-05	7.3E-03	1.5E-10
I-131	4.5E-02	4.5E-03	4.5E-06	3.8E-02	3.8E-05	4.3E-02	8.6E-09
Te-132	1.7E-03	1.7E-04	1.7E-06	1.4E-03	1.4E-05	1.6E-03	3.2E-11
I-132	2.1E-01	2.1E-02	2.1E-05	1.8E-01	1.8E-04	2.0E-01	4.0E-08
I-133	1.4E-01	1.4E-02	1.4E-05	1.2E-01	1.2E-04	1.3E-01	2.7E-08

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 8 of 9)

Nuclide	Reactor Coolant Activity (uCi/ml)	Activity in Line[d] – uCi/ml					
		Line No. 1	Line No. 2	Line No. 3	Line No. 4	Line No. 5	Line No. 6
I-134	3.4E-01	3.4E-02	3.4E-05	2.9E-01	2.9E-04	3.2E-01	6.5E-08
Cs-134	2.1E-02	2.1E-03	2.1E-07	1.8E-02	1.8E-06	2.0E-02	3.8E-10
I-135	2.6E-01	2.6E-02	2.6E-05	2.2E-01	2.2E-04	2.5E-01	4.9E-08
Cs-136	8.7E-04	8.7E-05	8.7E-09	7.4E-04	7.4E-08	8.3E-04	1.7E-11
Cs-137	1.8E-02	1.8E-03	1.8E-07	1.5E-02	1.5E-06	1.7E-02	3.4E-10
Ba-140	1.3E-02	1.3E-03	1.3E-05	1.1E-02	1.1E-04	1.2E-02	2.5E-10
La-140	2.5E-02	2.5E-03	2.5E-05	2.1E-02	2.1E-04	2.4E-02	4.8E-10
Ce-141	1.5E-04	1.5E-05	1.5E-07	1.3E-04	1.3E-06	1.4E-04	2.9E-12
Ce-143	2.8E-03	2.8E-04	2.8E-06	2.4E-03	2.4E-05	2.7E-03	5.3E-11
Ce-144	4.4E-03	4.4E-04	4.4E-06	3.7E-03	3.7E-05	4.2E-03	8.3E-11
H-3	2.0E+00	2.0E-01	2.0E-01	1.7E+00	1.7E+00	1.9E+00	1.9E+00

TABLE 11.2-2
EXPECTED PROCESS PARAMETERS FOR THE LIQUID WASTE PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

(Sheet 9 of 9)

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a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) All tanks are at ambient temperature and atmospheric pressure. Tank volume are as follows:

Floor Drain Tank III	30,000 gallons
Waste Holdup Tank	10,000 gallons
Recycle Holdup Tank	112,000 gallons
Waste Monitor Tank	5,000 gallons
Plant Effluent Holdup and Monitor Tank[e]	30,000 gallons

Tank inventories (except for the Waste Monitor Tank) are derived from NUREG-0017 primary coolant activities. The Waste Monitor Tank inventory is derived from NUREG-0017 effluent release concentrations which includes 0.32 Ci/yr (two units) for anticipated operational occurrences.

c) Plant Effluent Holdup and Monitor Tanks serve as an alternate to the Waste Monitor Tanks when they are utilized.

d) Line No. 1 - Output from Floor Drain Tank III to Filter Demineralizer System
Line No. 2 - Treated Floor Drain Tank III fluid output from the Filter Demineralizer System to the Waste Monitor Tank
Line No. 3 - Output from Waste Holdup Tank to Filter Demineralizer System
Line No. 4 - Treated Waste Holdup Tank fluid output from the Filter Demineralizer System to the Waste Monitor Tank
Line No. 5 - Output from Recycle Holdup Tank to Boron Recycle System evaporator
Line No. 6 - Boron Recycle System evaporator output to the Waste Monitor Tank

Processing line concentrations are derived from NUREG-0017 primary coolant activities.

e) Plant Effluent Holdup and Monitor Tanks serve as an alternate to the Waste Monitoring Tanks and therefore process at the same activity concentrations as the Waste Monitor Tanks when they are utilized.

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 1 of 15)

Components	Parameters
Pumps	
1. Reactor Coolant Drain Tank Pumps	
Quantity	Four
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	100
2	140
Design Head, ft	
1	300
2	250
Material	SS
2. Waste Evaporator Feed Pump	
Quantity	One
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	35
2	100
Design Head, ft	
1	250
2	200
Material	SS

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 2 of 15)

Components	Parameters
3. Waste Evaporator Condensate Tank Pump	
Quantity	One
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	35
2	100
Design Head, ft	
1	250
2	200
Material	SS
4. Chemical Drain Tank Pump	
Quantity	One
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	35
2	100
Design Head, ft	
1	250
2	200
Material	SS
5. Spent Resin Sluice Pump	
Quantity	One
Type	Canned

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 3 of 15)

Components	Parameters
Design pressure, psig	150
Design temperature, F	180
Design Flow, gpm	
1	140
Design Head, ft	
1	250
Material	SS
6. Laundry and Hot Shower Tank Pump	
Quantity	One
Type	Mechanical seal
Design pressure, psig	150
Design temperature, F	180
Design Flow, gpm	
1	35
Design Head, ft	
1	250
Material	SS
7. Floor Drain Tank Pumps	
Quantity	Three

	Pump #1 and Pump #2	Pump #3
Type	Mech. Seal Centrifugal	Canned Motor/ Single
Design Pressure, (psig)	150	150
Design Temperature, (F)	200	120
Design Flow, (gpm)	35	5
Design Head, (ft)	250	230
Material	SS	SS

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 4 of 15)

Components	Parameters
8. Waste Monitor Tank Pumps	
Quantity	Two
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	35
2	100
Design Head, ft	
1	250
2	200
Material	SS
9. Reverse Osmosis Concentrates Tank Pump[(a)]	
Quantity	One
Type	Canned
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	
1	10
Design Head, ft	
1	115
Material	SS
10. Laundry Holdup and Monitor Tank Pump	
Quantity	Two
Type	Canned
Design pressure, psig	150

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 5 of 15)

Components	Parameters
Design temperature, F	200
Design Flow, gpm	35
Design Head, ft	230
Material	SS
11. Plant Effluent Holdup and Monitor Tank Pumps	
Quantity	Two
Type	Centrifugal
Design pressure, psig	150
Design temperature, F	200
Design Flow, gpm	100
Design Head, ft	231
Material	SS
Heat Exchangers	
1. Reactor Coolant Drain Tank Heat Exchangers	
Quantity	Two
Type	U-tube
Estimated UA, Btu/hr/F	70,000
Design Pressure, psig	
Shell	165
Tube	240
Design Temperature, F	
Shell	250
Tube	200
Design Flow, lb/hr	
Shell	112,000
Tube	44,600

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 6 of 15)

Components	Parameters
Temperature (In), F	
Shell	105
Tube	180
Temperature (Out), F	
Shell	125
Tube	130
Material	
Shell	CS
Tube	SS
Tanks	
1. Reactor Coolant Drain Tanks	
Quantity	Two
Usable volume, gal	350
Type	Horizontal
Design pressure, psig[(b)]	100
Design temperature, F	250
Material	SS
Diaphragm	No
2. Waste Holdup Tank	
Quantity	One
Usable volume, gal	10,000
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	200
Material	SS
Diaphragm	No

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 7 of 15)

Components	Parameters
3. Waste Evaporator Condensate Tanks	
Quantity	One
Usable volume, gal	5000
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	200
Material	SS
Diaphragm	Yes
4. Chemical Drain Tank	
Quantity	One
Usable volume, gal	600
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	200
Material	SS
Diaphragm	No
5. Spent Resin Storage Tank	
Quantity	One
Usable volume, gal	4100
Type	Vertical
Design pressure, psig	150
Design temperature, F	200
Radiation level inside compartment, rads/hr	1000
Material	SS
Diaphragm	No

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 8 of 15)

Components	Parameters	
6. Laundry and Hot Shower Tank		
Quantity	One	
Usable volume, gal	10,000	
Type	Vertical	
Design pressure, psig	Atmospheric	
Design temperature, F	200	
Material	SS	
Diaphragm	No	
7. Floor Drain Tanks	NNS	VIII
Quantity	Three	
Usable volume, gal	10,000 (Two tanks) 30,000 (One tank)	
Type	Vertical	
Design pressure, psig	Atmospheric	
Design temperature, F	200	
Material	SS	
Diaphragm	No	
8. Waste Monitor Tanks		
Quantity	Two	
Usable volume, gal	5000	
Type	Vertical	
Design pressure, psig	Atmospheric	
Design temperature, F	200	
Material	SS	
Diaphragm	No	

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 9 of 15)

Components	Parameters
9. Laundry Holdup and Monitor Tanks	
Quantity	Two
Usable volume, gal	5000
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	150
Material	SS
Diaphragm	No
10. Reverse Osmosis Concentrates Tank[(a)]	
Quantity	One
Usable volume, gal	1000
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	200
Material	SS
Diaphragm	No
11. Laundry Water Head Tank	
Quantity	One
Usable volume, gal	5000
Type	Vertical
Design pressure, psig	Atmospheric
Design temperature, F	200
Material	SS
Diaphragm	No

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 10 of 15)

Components	Parameters
12. Evaporator Reagent Tanks	
Quantity	Two
Usable volume, gal	5
Type	Vertical
Design pressure, psig	150
Design temperature, F	200
Material	SS
Diaphragm	No
13. Plant Effluent Holdup and Monitor Tanks	
Quantity	Two
Usable Volume, gal.	29,000
Type	Vertical
Design Pressure, psig.	Atmospheric
Design Temperature, F.	200
Material	Reinforced Concrete w/ Stainless Steel Liner
Diaphragm	No
Demineralizers	
1. Waste Evaporator Condensate Demineralizer	
Quantity	One
Type	Flushable
Design pressure, psig	300
Design temperature, F	250
Design flow, gpm	35 (Max. 120)
Resin volume, ft^3	30
Material	SS

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 11 of 15)

Components	Parameters
Resin type	IRN-150(c)
Design process decontamination factor	No
2. Waste Monitor Tank Demineralizer	
Quantity	One
Type	Flushable
Design pressure, psig	300
Design temperature, F	250
Design flow, gpm	35 (Max. 120)
Resin volume, ft^3	30
Material	SS
Resin type	IRN-150(c)
Design process decontamination factor	10
Filters	
1. Waste Evaporator Feed Filter	
Quantity	One
Design pressure, psig	200
Design temperature, F	250
Design flow, gpm	35
Micron rating	20 to 100 absolute (100% retention)
Surface radiation level, rads/hr	100
Materials	
Housing	SS
Filter element	EICF(d)

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 12 of 15)

Components	Parameters
2. Waste Evaporator Condensate Filter	
Quantity	One
Design pressure, psig	200
Design temperature, F	250
Design flow, gpm	35
Micron rating	2 to 40 absolute (100% retention)
Surface radiation level, rads/hr	<1
Materials	
Housing	SS
Filter element	EICF or equivalent
3. Spent Resin Sluice Filter	
Quantity	One
Design pressure, psig	300
Design temperature, F	250
Design flow, gpm	250
Micron rating	2 to 40 absolute (100% retention)
Surface radiation level, rads/hr	<100
Materials	
Housing	SS
Filter element	EICF or equivalent
4. Laundry and Hot Shower Tank Filter	
Quantity	One
Design pressure, psig	200
Design temperature, F	250
Design flow, gpm	35

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 13 of 15)

Components	Parameters
Micron rating	10 to 40 absolute (100% retention)
Surface radiation level, mr/hr	<100
Materials	
Housing	SS
Filter element	EICF or equivalent
5. Floor Drain Tank Filter	
Quantity	One
Design pressure, psig	200
Design temperature, F	250
Design flow, gpm	35
Micron rating	2 to 40 absolute (100% retention)
Surface radiation level, rads/hr	100
Materials	
Housing	SS
Filter element	Resin Bonded Glass Fiber or equivalent
6. Waste Monitor Tank Filter	
Quantity	One
Design pressure, psig	200
Design temperature, F	250
Design flow, gpm	35
Micron rating	10 to 40 absolute (100% retention)
Surface radiation level, rads/hr	90
Materials	
Housing	SS
Filter element	EICF or equivalent

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 14 of 15)

Components	Parameters
Strainers	
1. Laundry and Hot Shower Tank Strainer	
Quantity	One
Type	Basket
Design pressure, psig	150
Design temperature, F	200
Design flow, gpm	35
Pressure drop at design flow, psi	0.2 (Clean)
Nominal rating, in.	0.0625
Surface radiation level	Negligible
Material	SS
2. Floor Drain Tank Strainers	
Quantity	Three
Type	Basket
Design pressure, psig	150
Design temperature, F	200
Design flow, gpm	35
Pressure drop at design flow, psi	0.2 (Clean)
Nominal rating, in.	1/16
Surface radiation level	Negligible
Material	SS
Evaporators	
1. Waste Evaporator Process side Steam side	
Quantity	One
Steam design pressure, psig	50

TABLE 11.2-3
EQUIPMENT PRINCIPAL DESIGN PARAMETERS

(Sheet 15 of 15)

Components	Parameters
Design flow, gpm	15
Feed concentration, mg/l boron	10-2500
Bottoms concentration, mg/l boron	7000-21,000
Design process decontamination factor	1000
2. Floor Drain Evaporator Process side Steam side	
Quantity	One
Steam design pressure, psig	50
Design flow, gpm	15
Feed concentration, mg/l boron	10-2500
Bottoms concentration, mg/l boron	7000-21,000
Design process decontamination factor	1000

a) Refer to subsection 11.2.2.4.2 for additional information on the reverse osmosis system.

b) External design pressure is 60 psig.

c) Rohm and Haas Amberlite or equivalent

d) Epoxy-impregnated cellulose fiber or equivalent.

TABLE 11.2-4
LOCATIONS OF LIQUID WASTE PROCESSING SYSTEM EQUIPMENT

(Sheet 1 of 3)

Equipment	Quantity	Building	Elevation
Reactor Coolant Drain Tanks			
Unit 1	1	Containment	808 ft 0 in
Unit 2	1	Containment	808 ft 0 in
Reactor Coolant Drain Tank Pumps			
Unit 1	2	Containment	808 ft 0 in
Unit 2	2	Containment	808 ft 0 in
Reactor Coolant Drain Tank Heat Exchangers			
Unit 1	1	Containment	808 ft 0 in
Unit 2	1	Containment	808 ft 0 in
Spent resin storage tank	1	Auxiliary	810 ft 6 in
Spent resin sluice pump	1	Auxiliary	810 ft 6 in
Spent resin sluice filter	1	Auxiliary	842 ft
Waste holdup tank	1	Auxiliary	790 ft 6 in
Waste evaporator feed pump	1	Auxiliary	790 ft 6 in
Waste evaporator feed filter	1	Auxiliary	842 ft
Waste evaporator	1	Auxiliary	810 ft 6 in
Waste evaporator condensate demineralizer	1	Auxiliary	832 ft 6 in
Waste evaporator condensate filter	1	Auxiliary	842 ft
Waste evaporator condensate tank	1	Auxiliary	790 ft 6 in
Waste evaporator condensate tank pump	1	Auxiliary	790 ft 6 in
Chemical drain tank	1	Auxiliary	790 ft 6 in

TABLE 11.2-4
LOCATIONS OF LIQUID WASTE PROCESSING SYSTEM EQUIPMENT

(Sheet 2 of 3)

Equipment	Quantity	Building	Elevation
Chemical drain tank pump	1	Auxiliary	790 ft 6 in
Floor drain tank I (10,000 gal)	1	Unit 1 safeguards	773 ft 0 in
Floor drain tank II (10,000 gal)	1	Unit 2 safeguards	773 ft 0 in
Floor drain tank III (30,000 gal)	1	Auxiliary	790 ft 6 in
Floor drain tank pump I with strainer	1	Unit 1 safeguards	773 ft 0 in
Floor drain tank pump II with strainer	1	Unit 2 safeguards	773 ft 0 in
Floor drain tank pump III with strainer	1	Auxiliary	790 ft 6 in
Floor drain tank filter	1	Auxiliary	842 ft
Floor drain evaporator	1	Auxiliary	810 ft 6 in
Waste monitor tank demineralizer	1	Auxiliary	832 ft
Waste monitor tank filter	1	Auxiliary	842 ft 6 in
Waste monitor tank	2	Auxiliary	790 ft 6 in
Waste monitor tank pumps	2	Auxiliary	790 ft 6 in
Laundry and hot shower tank	1	Auxiliary	790 ft 6 in
Laundry and hot shower tank strainer	1	Auxiliary	790 ft 6 in
Laundry and hot shower tank filter	1	Auxiliary	790 ft 6 in
Laundry holdup and monitor tanks	2	Auxiliary	790 ft 6 in
Laundry holdup and monitor tank pump	1	Auxiliary	790 ft 6 in
Laundry head tank	1	Auxiliary	852 ft 6 in

TABLE 11.2-4
LOCATIONS OF LIQUID WASTE PROCESSING SYSTEM EQUIPMENT

(Sheet 3 of 3)

Equipment	Quantity	Building	Elevation
Evaporator reagent tanks	2	Auxiliary	810 ft 6 in
Laundry and hot shower tank pump	1	Auxiliary	790 ft 6 in
Waste Conditioning	1	Fuel	841 ft 0 in
Plant effluent holdup and monitor tanks	2	Yard	813 ft 6 in
Plant effluent holdup and monitor tank pumps	2	Fuel	810 ft 6 in
Filter Demineralizer Skid	1	Fuel	800 ft 2 in
Resin Dewatering Skid	1	Fuel	800 ft 2 in
Rad Vaults	1 or 2	Fuel	800 ft 2 in
Drum Dryer System	1	Fuel	(mobile)

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 1 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
Flow Instrumentation					
X-F-1007	Waste evaporator pump discharge	0 to 30 gpm	N/A	N/A	Local
1/2-F-1008	Reactor coolant drain tank pump discharge	0 to 250 gpm	N/A	Low	LWPS panel
1/2-F-1009	Reactor coolant drain tank recirculation	0 to 100 gpm	Low	N/A	LWPS panel
X-F-1011	Spent resin sluice pump discharge	0 to 100 gpm	Low	Low	LWPS panel
X-F-1085A, C	Waste monitor tank pump No. 1 discharge	0 to 100 gpm	N/A	N/A	LWPS panel and local
X-F-1085B, D	Waste monitor tank pump No. 2 discharge	0 to 100 gpm	N/A	N/A	LWPS panel and local
X-F-5286	Laundry holdup and monitor tank pump discharge	0 to 150 gpm	N/A	N/A	LWPS panel
1/2-Q-1014	Reactor coolant drain tank discharge	N/A	N/A	N/A	Local

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 2 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
Pressure Instrumentation					
1/2-P-1004	Reactor coolant drain tank	0 to 100 psig	High	N/A	LWPS panel
X-P-1006	Spent resin storage tank	0 to 100 psig	High	N/A	LWPS and drumming panels
X-P-1016	Waste evaporator feed pump discharge	0 to 160 psig	N/A	N/A	Local
X-P-1017	Waste evaporator feed header	0 to 160 psig	N/A	N/A	Local
1/2-P-1018A	Reactor coolant drain tank pump No. 1 discharge	0 to 160 psig	N/A	N/A	Local
1/2-P-1018B	Reactor coolant drain tank pump No. 2 discharge	0 to 160 psig	N/A	N/A	Local
X-P-1018C	Laundry and hot shower tank pump discharge	0 to 150 psig	N/A	N/A	Local
X-P-1018D	Chemical drain tank pump discharge	0 to 150 psig	N/A	N/A	Local
X-P-1018G	Waste evaporator condensate pump discharge	0 to 160 psig	N/A	N/A	Local
X-P-1074	Waste evaporator outlet	0 to 160 psig	N/A	N/A	Local
X-P-1075	Waste evaporator condensate demineralizer outlet	0 to 150 psig	N/A	N/A	Local

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 3 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
X-P-1076	Waste evaporator condensate filter	0 to 160 psig	N/A	N/A	Local
X-P-1078	Floor drain tank filter inlet	30" Hg to 150 psig	N/A	N/A	Local
X-P-1079	Floor drain tank filter outlet	0 to 150 psig	N/A	N/A	Local
X-P-1080	Laundry and hot shower tank filter inlet	0 to 150 psig	N/A	N/A	Local
X-P-1081	Laundry and hot shower tank filter outlet	0 to 150 psig	N/A	N/A	Local
1/2-P-1024	Reactor Coolant Drain Tank Vent Pressure	0 to 150 psig	High	N/A	Local
X-P-1084A	Waste monitor tank pump No. 1 discharge	0 to 150 psig	N/A	N/A	Local
X-P-1084B	Waste monitor tank pump No. 2 discharge	0 to 150 psig	N/A	N/A	Local
X-P-5274	Laundry holdup and monitor tank pump discharge	0 to 160 psig	N/A	N/A	Local
X-P-1086	Resin sluice filter inlet	0 to 150 psig	N/A	N/A	Local
X-P-1087	Resin sluice filter outlet	0 to 150 psig	N/A	N/A	Local
X-P-1088	Waste monitor tank filter inlet	0 to 150 psig	N/A	N/A	Local

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[a]

(Sheet 4 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
X-P-1089	Waste monitor tank filter outlet				
X-P-1090A	Floor drain tank pump No. 1 discharge	0 to 150 psig	N/A	N/A	Local
X-P-1090B	Floor drain tank pump No. 2 discharge	0 to 150 psig	N/A	N/A	Local
X-P-5279	Floor drain tank pump No. 3 discharge	0 to 150 psig	N/A	N/A	Local
X-P-5272[b]	Reverse osmosis concentrates tank pump discharge				
X-P-5300A	Plant effluent holdup and monitor tank No. 1	0 to 200 psig	N/A	N/A	Local
X-P-5300B	Plant effluent holdup and monitor tank No. 2	0 to 200 psig	N/A	N/A	Local
Level Instrumentation					
X-L-1001	Waste holdup tank	0 to 100%	High-high High Low	Low	Local and LWPS panel
X-L-1002	Chemical drain tank	0 to 100%	High Low	Low	Local and LWPS drumming panels

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 5 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
1/2-L-1003	Reactor coolant drain tank	0 to 100%	High Low	Low	LWPS panel
X-L-1005	Spent resin storage tank	0 to 100%	High Low	High Low	LWPS and drumming panels
X-L-1010	Laundry and hot shower tank	0 to 100%	High Low	Low	LWPS panel and local
X-L-1012	Waste evaporator condensate tank	0 to 100%	High Low	Low	LWPS panel and local
X-L-1077A	Floor drain tank 1	0 to 100%	High Low	Low	LWPS panel and local
X-L-1077B	Floor drain tank 2	0 to 100%	High Low	Low	LWPS panel and local
X-L-5278	Floor drain tank 3	0 to 100%	High Low	Low	LWPS panel and local
X-L-1082	Waste monitor tank No. 1	0 to 100%	High Low	Low	LWPS panel and local
X-L-1083	Waste monitor tank No. 2	0 to 100%	High Low	Low	LWPS panel and local
X-L-5275	Laundry holdup and monitor tank No. 1	0 to 100% 0' to 10'	High Low	Low	LWPS panel and local

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[a]

(Sheet 6 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
X-L-5276	Laundry holdup and monitor tank No. 2	0 to 100% 0' to 10'	High Low	Low	LWPS panel and local
X-L-5287[c]	Laundry water head tank	0 to 100% 1' to 13'	High Low	Low	LWPS panel and local
X-L-5271A[d]	Reverse osmosis concentrates tank				
X-L-5271B[d]	Reverse osmosis concentrates tank				
X-L-5300A, C, E	Plant effluent holdup and monitor tank No. 1	0 to 100% 0' to 24.5'	High-High Low Low-Low	High Low	Local
X-L-5300B, D, F	Plant effluent holdup and monitor tank No. 2	0 to 100% 0' to 24.5'	High-High Low Low-Low	High Low	Local
Temperature Instrumentation					
1/2-T-1058	Reactor coolant drain tank	250	50 to 250°F	High	LWPS Panel

TABLE 11.2-5
LIQUID WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[a]

(Sheet 7 of 7)

Channel No.	Location Primary Sensor	Range	Alarm Function	Control Function	Location of Readout
Radiation Instrumentation					
X-RE-5253	Laundry holdup and monitor tank discharge line	1E-05 to 5E-02 µCi/cm^3	High	Variable	LWPS and radiation monitor panels
X-RE-5251A	Common discharge line of AB Sump 3,11, DG Sumps and CCW Drain Tanks	1E-05 to 5E-02 µCi/cm^3	High	Variable	CPX-ECPRLV-01

a) The following abbreviations are used in this table only:

F	=	flow	R	=	radiation
Q	=	flow integrator	X	=	Common instrument
P	=	pressure	1/2	=	Unit 1 and Unit 2 instruments
L	=	level			
T	=	temperature			

b) Refer to subsection 11.2.2.4.2 for additional information on the reverse osmosis system.

c) X-LS-5287A has control function; X-LS-5287 indicates 0 to 100% tank level.

d) Refer to subsection 11.2.2.4.2 for additional information on the reverse osmosis system.

TABLE 11.2-6
DECONTAMINATION FACTORS FOR LIQUID WASTE PROCESSING SYSTEM EQUIPMENT

Process Stream	Anion	Cs,Rb	Other Nuclides
Dirty Wastes	10^3	10^4	10^2
(Filter Demineralizer System[(a)])	(Max.)	(Max.)	(Max.)
Clean Wastes	10^3	10^4	10^2
Filter Demineralizer System[(a)])			

	Iodine	All Nuclides
Chemical Shim (Total)	$5x10^6$	$5x10^7$
CVCS Letdown	10^2	50
Evaporator Feed	10^2	10^2
Evaporator Condensate	5	10
Boron Recycle System	10^2	10^3

a) The Filter Demineralizer System (FDS) has multiple demineralizers of various sizes which may be operated in series or parallel. These vessels may be loaded with any combination of organic resin, charcoals, or ion specific media and may be customized according to the isotopes in the process stream. In addition, the waste monitor tank mixed bed demineralizer was included, but is typically not used. The FDS also has chemical injection for waste stabilization and filtration.

TABLE 11.2-7
THIS TABLE HAS BEEN DELETED.

TABLE 11.2-8
LIQUID WASTE PROCESSING SYSTEM COMPONENT DESIGN INVENTORIES - ORIGINAL LICENSING BASIs[(a)]

(Sheet 1 of 3)

Nuclide	Reactor Coolant Activity at 1% Fuel Defects (uCi/gm)	Tank Inventory (Ci)			
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank
Cr-51	5.5E-03	6.2E-02	1.8E-01	2.5E+00	1.2E-04
Mn-54	4.4E-04	5.0E-03	1.4E-02	2.0E-01	9.6E-06
Fe-55	2.0E-03	2.3E-02	6.5E-02	9.0E-01	4.3E-05
Fe-59	5.2E-04	5.9E-03	1.7E-02	2.4E-01	1.1E-05
Co-58	1.5E-02	1.7E-01	4.9E-01	6.8E+00	3.3E-04
Co-60	1.9E-03	2.2E-02	6.1E-02	8.6E-01	4.3E-05
Sr-89	4.3E-03	4.9E-02	1.4E-01	1.9E+00	9.5E-05
Sr-90	2.0E-04	2.3E-03	6.5E-03	9.0E-02	4.3E-06
Sr-91	6.2E-03	7.1E-02	2.0E-01	2.8E+00	1.2E-05
Y-91m	3.3E-03	3.7E-02	1.1E-01	1.5E+00	8.6E-06
Y-91	5.7E-04	6.5E-03	1.9E-02	2.5E-01	2.7E-05
Y-93	3.8E-04	4.3E-03	1.3E-02	1.7E-01	8.3E-07
Zr-95	6.5E-04	7.3E-03	2.2E-02	3.0E-01	1.4E-05

TABLE 11.2-8
LIQUID WASTE PROCESSING SYSTEM COMPONENT DESIGN INVENTORIES - ORIGINAL LICENSING BASIs[(a)]

(Sheet 2 of 3)

Nuclide	Reactor Coolant Activity at 1% Fuel Defects (uCi/gm)	Tank Inventory (Ci)			
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank
Nb-95	6.5E-04	7.4E-03	2.1E-02	3.0E-01	1.5E-05
Mo-99	7.5E-01	8.6E+00	2.5E+01	3.4E+02	9.7E-03
Tc-99m	6.9E-01	7.8E+00	2.2E+01	3.1E+02	1.1E-02
Ru-103	6.2E-04	7.1E-03	2.0E-02	2.8E-01	1.2E-05
Ru-106	2.6E-04	3.0E-03	8.4E-03	1.2E-01	5.8E-07
Ag-110m	1.4E-03	1.6E-02	4.5E-02	6.2E-01	3.1E-05
Te-129m	1.9E-02	2.2E-01	6.1E-01	8.5E+00	4.0E-04
Te-129	1.8E-02	2.0E-01	5.8E-01	8.3E+00	1.9E-06
Te-131m	2.6E-02	2.9E-01	8.3E-01	1.2E+01	2.1E-04
Te-131	1.2E-02	1.4E-01	3.9E-01	5.5E+00	3.4E-06
I-131	2.8E+00	3.2E+01	9.3E+01	1.2E+03	5.0E-03
Te-132	2.9E-01	3.2E+00	9.4E+00	1.3E+02	3.9E-03
I-132	2.8E+00	3.2E+01	9.1E+01	1.3E+03	3.1E-04

TABLE 11.2-8
LIQUID WASTE PROCESSING SYSTEM COMPONENT DESIGN INVENTORIES - ORIGINAL LICENSING BASIs[(a)]

(Sheet 3 of 3)

Nuclide	Reactor Coolant Activity at 1% Fuel Defects (uCi/gm)	Tank Inventory (Ci)			
		Floor Drain Tank III	Waste Holdup Tank	Recycle Holdup Tank	Waste Monitor Tank
I-133	4.2E+00	4.8E+01	1.4E+02	1.9E+03	2.4E-03
I-134	5.7E-01	6.5E+00	1.8E+01	2.5E+02	7.7E-08
Cs-134	6.8E+00	7.6E+01	2.2E+02	2.9E+03	1.6E-03
I-135	2.3E+00	2.7E+01	7.4E+01	1.1E+03	2.3E-04
Cs-136	2.9E+00	3.3E+01	9.3E+01	1.3E+03	6.0E-04
Cs-137	2.9E+00	3.4E+01	9.1E+01	1.3E+03	7.1E-04
Ba-140	4.2E-03	4.8E-02	1.4E-01	1.9E+00	8.1E-05
La-140	1.4E-03	1.6E-02	4.5E-02	6.2E-01	2.1E-05
Ce-141	6.3E-04	7.1E-03	2.0E-02	2.8E-01	1.3E-05
Ce-143	5.0E-04	5.7E-03	1.6E-02	2.3E-01	4.5E-06
Ce-144	4.4E-04	5.0E-03	1.5E-02	2.0E-01	9.7E-06
H-3	7.1E+00 (max)	7.9E+01	2.2E+02	3.2E+03	1.4E+01

TABLE 11.2-8

LIQUID WASTE PROCESSING SYSTEM COMPONENT DESIGN INVENTORIES - ORIGINAL LICENSING BASIS[(a)]

(Sheet 4 of 4)

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11.2-9
EXPECTED RELEASE CONCENTRATION VERSUS LIMITING CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 3)

Isotope	Expected Release Concentration[b] (uCi/gm)	Limiting Concentration[c] (uCi/gm)	
		Soluble	Insoluble
H-3	2.0E-01	3E-03	3E-03
Na-24	9.7E-06	2E-04	3E-05
Cr-51	3.4E-06	2E-03	2E-03
Mn-54	1.9E-06	1E-04	1E-04
Fe-55	1.4E-06	8E-04	2E-03
Fe59	3.4E-07	6E-05	5E-05
Co-58	5.3E-06	1E-04	9E-05
Co-60	6.3E-07	5E-05	3E-05
Ni-63	1.6E-09	3E-05	7E-04
Zn-65	5.9E-07	1E-04	2E-04
Sr-89	1.6E-07	3E-06	3E-05
Sr-90	2.3E-08	3E-07	4E-05
Sr-91	9.7E-08	7E-05	5E-05
Y-90	6.5E-09	2E-05	2E-05
Y-91	1.3E-08	3E-05	3E-05
Y-91m	6.6E-08	3E-03	3E-03
Y-93	4.1E-07	3E-05	3E-05
Zr-95	4.5E-07	6E-05	6E-05
Nb-95	3.3E-07	1E-04	1E-04
Nb-95m	3.2E-09	No Data	No Data

TABLE 11.2-9
EXPECTED RELEASE CONCENTRATION VERSUS LIMITING CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

Isotope	Expected Release Concentration[b] (uCi/gm)	Limiting Concentration[c] (uCi/gm)	
		Soluble	Insoluble
Mo-99	4.4E-06	2E-04	4E-05
Tc-99m	4.1E-06	6E-03	3E-03
Ru-103	8.8E-06	8E-05	8E-05
Ru-106	2.1E-05	1E-05	1E-05
Rh-103m	8.1E-06	1E-02	1E-02
Ag-110m	1.5E-06	3E-05	3E-05
Te-129	1.4E-07	8E-04	8E-04
Te-129m	2.1E-07	3E-05	2E-05
Te-131	1.1E-07	No Data	No Data
Te-131m	6.3E-07	6E-05	4E-05
Te-132	1.2E-06	3E-05	2E-05
I-131	4.3E-06	3E-07	6E-05
I-132	1.2E-06	8E-06	2E-04
I-133	4.3E-06	1E-06	4E-05
I-134	2.4E-09	2E-05	6E-04
I-135	1.4E-06	4E-06	7E-05
Cs-134	2.6E-07	9E-06	4E-05
Cs-136	9.7E-09	9E-05	6E-05
Cs-137	2.3E-07	2E-05	4E-05
Ba-137m	1.1E-07	No Data	No Data

TABLE 11.2-9
EXPECTED RELEASE CONCENTRATION VERSUS LIMITING CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

Isotope	Expected Release Concentration[b] (uCi/gm)	Limiting Concentration[c] (uCi/gm)	
		Soluble	Insoluble
Ba-140	1.3E-05	3E-05	2E-05
La-140	2.0E-05	2E-05	2E-05
Ce-141	1.6E-07	9E-05	9E-05
Ce-143	1.3E-06	4E-05	4E-05
Ce-144	5.1E-06	1E-05	1E-05
Pr-143	1.8E-07	5E-05	5E-05
Pr-144	4.5E-06	No Data	No Data
W-187	8.9E-07	7E-05	6E-05
Np-239	1.4E-06	1E-04	1E-04

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) Derived from Table 11A-1, assuming a discharge flow rate of 17,820 gal/day and no dilution due to Circulating Water.

c) From 10 CFR Part 20, Appendix B, Table II, Column 2. Design assessment made with the provision of 10CFR20.1-20.601.

TABLE 11.2-10
DESIGN RELEASE WITH 1-PERCENT FAILED FUEL - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 2)

Isotope	Reactor Coolant Activity at 1% Fuel Defects (uCi/gm)	Design Release Concentration[b] (uCi/gm)	Limiting Concentration[c] (uCi/gm)	
			Soluble	Insoluble
H-3	7.1E+0 (max)	7.1E-1	3E-03	3E-03
Cr-51	5.5E-3	6.0E-6	2E-03	2E-03
Mn-54	4.4E-4	5.2E-7	1E-04	1E-04
Fe-55	2.0E-3	2.3E-6	8E-04	2E-03
Fe-59	5.2E-4	5.9E-7	6E-05	5E-05
Co-58	1.5E-2	1.7E-5	1E-04	9E-05
Co-60	1.9E-3	2.3E-6	5E-05	3E-05
Sr-89	4.3E-3	4.9E-6	3E-06	3E-05
Sr-90	2.0E-4	2.3E-7	3E-07	4E-05
Sr-91	6.2E-3	6.3E-7	7E-05	5E-05
Y-91	5.7E-4	1.4E-6	3E-05	3E-05
Y-91m	3.3E-3	4.7E-7	3E-03	3E-03
Y-93	3.8E-4	4.4E-8	3E-05	3E-05
Zr-95	6.5E-4	7.5E-7	6E-05	6E-05
Nb-95	6.5E-4	7.7E-7	1E-04	1E-04
Mo-99	7.5E-1	5.2E-4	2E-04	4E-05
Tc-99m	6.9E-1	6.0E-4	6E-03	3E-03
Ru-103	6.2E-4	6.8E-7	8E-05	8E-05
Ru-106	2.6E-4	3.2E-8	1E-05	1E-05
Ag-110m	1.4E-3	1.6E-6	3E-05	3E-05
Te-129	1.8E-2	1.1E-7	8E-04	8E-04

TABLE 11.2-10
DESIGN RELEASE WITH 1-PERCENT FAILED FUEL - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 2)

Isotope	Reactor Coolant Activity at 1% Fuel Defects (uCi/gm)	Design Release Concentration[b] (uCi/gm)	Limiting Concentration[c] (uCi/gm)	
			Soluble	Insoluble
Te-129m	1.9E-2	2.1E-5	3E-05	2E-05
Te-131	1.2E-2	1.7E-7	No Data	No Data
Te-131m	2.6E-2	1.1E-5	6E-05	4E-05
Te-132	2.9E-1	2.0E-4	3E-05	2E-05
I-131	2.8E+0	2.7E-4	3E-07	6E-05
I-132	2.8E+0	1.6E-5	8E-06	2E-04
I-133	4.2E+0	1.3E-4	1E-06	4E-05
I-134	5.7E-1	4.0E-9	2E-05	6E-04
I-135	2.3E+0	1.2E-5	4E-06	7E-05
Cs-134	6.8E+0	8.5E-5	9E-06	4E-05
Cs-136	2.9E+0	3.2E-5	9E-05	6E-05
Cs-137	2.9E+0	3.6E-5	2E-05	4E-05
Ba-140	4.2E-3	4.2E-6	3E-05	2E-05
La-140	1.4E-3	1.1E-6	2E-05	2E-05
Ce-141	6.3E-4	6.7E-7	9E-05	9E-05
Ce-143	5.0E-4	2.3E-7	4E-05	4E-05
Ce-144	4.4E-4	5.1E-7	1E-05	1E-05

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) Derived from expected concentrations in Table 11.2-9. No credit is taken for dilution due to circulating water.

TABLE 11.2-1
DESIGN RELEASE WITH 1-PERCENT FAILED FUEL - ORIGINAL LICENSING BASIS[a]

(SHEET 3 OF 3)

c) From 10 CFR Part 20, Appendix B, Table II, Column 2. Design assessment made with the provisions of 10 CFR20.1-20.601.

11.3 GASEOUS WASTE MANAGEMENT SYSTEMS

This section describes the capabilities of the plant to control, collect, process, store and dispose of gaseous radioactive wastes generated as a result of normal operation including anticipated operational occurrences. The section discusses the design and operating features of the Gaseous Waste Processing System (GWPS) and the performance of other gas treatment and ventilation systems. Total gaseous releases from the plant for normal operation and the resulting offsite doses are also presented.

The radioactivity values presented in this section are the design basis values used for the design of the gaseous waste management system. As such they are considered historical and are not subject to future updating. The information is retained to avoid loss of original design basis. Actual radioactivity release quantities can be found in the annual radioactive effluent release reports submitted to the NRC.

11.3.1 DESIGN BASES

The gaseous waste systems are designed to collect, process, store and release gaseous wastes generated due to plant operations including anticipated operational occurrences. The systems are designed to assure that the release of gaseous effluents from the plant and expected offsite doses are as low as reasonably achievable as defined in the design objectives in Appendix I of 10CFR50. An evaluation of plant conformance to Appendix I is given in Appendix 11A. The gaseous systems have sufficient capacity and redundancy to meet discharge concentration limits of 10CFR20 during periods of design basis fuel leakage, as discussed in Section 11.3.3.

In addition, the GWPS meets the requirements of General Design Criterion 60 by providing long term holdup capacity, thus precluding the release of radioactive effluents during unfavorable environmental conditions. All gaseous effluent discharge paths are monitored for radioactivity, in compliance with General Design Criterion 64. Monitoring of radioactive effluents is discussed in Section 11.5.

The design of the GWPS is based on continuous operation of the Nuclear Steam Supply System assuming that fission products associated with 1 percent of the core power generation are available for leakage from the fuel into the coolant. This condition is assumed to exist over the life of the plant.

A discussion of system design, component design and materials of construction is contained in Sections 11.3.2.1, 11.3.2.1.1, and 11.3.2.1.2. System operation during normal operation and anticipated operational occurrences is described in Section 11.3.2.1.3.

The expected and design inventories of nuclides in the GWPS components are provided in Section 11.3.2.1.4. Design provisions incorporated to control the release of radioactive materials in gaseous effluents are discussed in Sections 11.3.2.1.1 and 11.3.2.1.2. The GWPS is designed to maintain gas composition outside the range of flammable and explosive mixtures as described in Section 11.3.2.1.1 and 11.3.2.1.2.

11.3.2 SYSTEMS DESCRIPTION

This section describes the design and operating features of the GWPS. The performance of the GWPS and other plant gaseous waste management systems with respect to the collection and

control of radioactive gases is also discussed in this section. Detailed descriptions of plant ventilation systems and the condenser vacuum system are presented in Sections 9.4 and 10.4, respectively.

11.3.2.1 Gaseous Waste Processing System Design

The piping and instrumentation diagrams for the GWPS are shown in the Flow Diagram Volume as indexed against Figure 11.3-1. These diagrams indicate safety classes for all components and piping.

The GWPS is shared between Comanche Peak Nuclear Power Plant (CPNPP) Unit 1 and Unit 2. The main flow path in the GWPS is a closed loop comprised of two waste gas compressors, two catalytic hydrogen recombiners, eight gas decay tanks for normal power service and two gas decay tanks for service at shutdown and startup. The eight gas decay tanks used for normal power service can also be used to function as shutdown gas decay tanks at shutdown and startup. The system also includes a gas decay tank drain pump, four gas traps, and a waste gas drain filter. All of the equipment is located in the primary Auxiliary Building.

The GWPS stores fission gases removed from the Reactor Coolant System (RCS). This reduces the escape of fission gases from the RCS during maintenance operations or through equipment leakage.

The primary location from which radioactive gases are removed from the RCS is the volume control tank. Smaller quantities are received via the vent connections, from the reactor coolant drain tank, the pressurizer relief tank, and the recycle holdup tanks. The waste and recycle evaporator gas strippers are normally vented to the Auxiliary Building exhaust.

The largest contributor to the nonradioactive gas accumulation is helium generated by a boron-10 (n,α) lithium-7 reaction in the reactor core. The second largest contributors are impurities in the bulk hydrogen and oxygen supplies. Stable and long-lived isotopes of fission gases also contribute small quantities to the system gas accumulation.

Gases from the above sources should be contained as long as practical, thus the discharges from the GWPS to the environment for normal plant operation should occur infrequently.

Operation of the system is such that fission gases are distributed throughout the eight normal operation gas decay tanks. Separation of the GWPS gaseous inventory in several tanks reduces the amount of fission gases that would be released in the event of a gas decay tank rupture. Radiological consequences of such a postulated rupture are discussed in Chapter 15.

The GWPS also provides capacity for holdup of gases generated during reactor shutdown. A portion of the gas from shutdowns is typically contained in one of the shutdown gas decay tanks. The second shutdown tank is normally at low pressure and is used to accept relief valve discharges from the normal operation gas decay tanks.

11.3.2.1.1 Component Design

Gaseous waste processing equipment design parameters are given in Table 11.3-1. This system performs no function related to the safe shutdown of the plant. Component safety classes,

American Society of Mechanical Engineers (ASME) Code and seismic design are discussed in Chapter 3. Except for flanged joints, all-welded construction is used whenever it is practical.

Pressure retaining components of gaseous waste processing equipment meet the design guidance, including quality assurance requirement, as outlined in Branch Technical Position ETSB 11-1 and FSAR Table 17A-1.

The waste gas compressor is a water sealed centrifugal displacement unit which maintains continuous circulation of nitrogen around the waste gas loop. The compressor is provided with a mechanical shaft seal to minimize water leakage.

The catalytic recombiner recombines hydrogen brought into the GWPS with oxygen to form water. This is accomplished by adding a controlled amount of oxygen to the recombiner which reacts with the hydrogen as the gas flows through a catalyst bed. The catalyst is 0.5% paladium on a Kaolin Base. The recombiner is designed with a low flow rate across the catalyst bed to maintain the reaction within the recombiner. The control system for the recombiner maintains an oxygen lean mixture to preclude the possibility of a hydrogen explosion. This is further discussed in Section 11.3.2.1.2.

The valves in this system are designed to minimize outleakage of radioactive gases.

Relief valves have soft seats and the normal operating pressures are typically two-thirds of the relief valve set pressure. The relief valves of the major components discharge to the shutdown tanks. This permits decay and controlled disposal of all expected discharges. The relief valves are designed to relieve full flow from both waste gas compressors.

The gas decay tank drain pump directs water from the gas decay tanks (due to condensation or maintenance) to the waste holdup tank or Recycle Holdup Tanks. It is used when there is insufficient pressure in the gas system to drive the fluid. All parts of the pump in contact with the drain water are of austenitic stainless steel. The pump is a canned motor type.

The waste gas drain filter is a disposable cartridge filter provided to prevent particulate matter from migrating into other systems. All parts of the filter in contact with the drain water are of austenitic stainless steel.

The four waste gas traps are designed to prevent undissolved gases from leaving the GWPS when water is drained off the system.

11.3.2.1.2 Instrumentation and Control Design

The main system instrumentation is described in Table 11.3-2 and shown on the piping and instrumentation diagrams in the Flow Diagram Volume.

The instrumentation readout is located mainly on the Waste Processing System (WPS) panel in the Auxiliary Building. Some instruments are read where the equipment is located. All alarms are shown separately on the WPS panel and further relayed to one common WPS annunciator on the main control board of the plant. Where suitable, instrument lines are provided with diaphragm seals to prevent fission gas outleakage through the instrument.

The compressors are interlocked with the seal water inventory in the moisture separators and trips off on either high or low moisture separator level. During normal operation the proper seal water inventory is maintained automatically.

The catalytic recombiner system is designed for automatic operation. Each package includes two online gas analyzers, one to measure hydrogen in and oxygen in, and one to measure hydrogen out and oxygen out, which are the primary means of recombiner control.

Process gas flow rate is measured upstream of the recombiner preheater. Local pressure gauges indicate pressure at the recombiner inlet and the oxygen supply pressure.

The following is a summary of the controls and alarms incorporated to maintain the gas composition outside the range of flammable and explosive mixtures. Specific system operational requirements/limitations are specified in the CPNPP Technical Requirements Manual.

1. If the hydrogen concentration in the recombiner feed reaches 9 percent by volume, a hi hydrogen alarm sounds, the oxygen feed is terminated, and the volume control tank hydrogen purge flow is terminated.

2. If the oxygen concentration in the recombiner feed reaches the maximum capacity of the catalytic hydrogen recombiner an alarm sounds and oxygen feed flow is limited so that no further increase in flow is possible. This control maintains the normal system oxygen concentration at a level which is below the flammable limit for hydrogen-oxygen mixtures. Oxygen concentration is always limited to ≤4% in this system.

3. If hydrogen in the recombiner discharge is excessive an alarm sounds. This alarm warns of high hydrogen feed, possible reactor malfunction, or loss of oxygen feed.

4. If oxygen in the recombiner discharge becomes excessive an alarm sounds and oxygen feed is terminated. This control maintains system oxygen lean in case of reactor malfunction.

5. On low flow through the recombiner, oxygen feed is terminated.

6. High discharge temperature from the cooler-condenser (downstream from the recombiner reactor) will terminate oxygen feed.

7. High temperature indication by any one of six thermocouples in the catalyst bed will terminate oxygen feed.

8. High temperature indication at the recombiner reactor discharge will terminate oxygen feed to the recombiner.

11.3.2.1.3 System Operation

Normal Operation

During normal power operation, nitrogen gas and fission gases are typically circulated around the GWPS loop by one of the two compressors. Hydrogen gas is introduced to the volume control tank where it is mixed with fission gases stripped from the reactor coolant by the action of

the volume control tank letdown line nozzle spray. The gas stream may then be vented from the volume control tank into the circulating nitrogen stream in the waste gas system, at the compressor suction.

The resulting mixture of nitrogen, hydrogen and fission gases is pumped by one of the compressors to one of the two catalytic hydrogen recombiners where enough oxygen is added to react with and reduce the hydrogen to a low residual level. Water vapor formed in the recombiner by the hydrogen-oxygen reaction is condensed and removed, and the cooled gas stream (now composed primarily of nitrogen and fission gases) is discharged from the recombiner, routed through a gas decay tank, and sent back to the compressor suction to complete the loop circuit. Depending on gas decay tank pressure the waste gas may be pumped by the compressor to a gas decay tank prior to processing by the hydrogen recombiner.

Only one gas decay tank is valved into the waste gas loop at any time. By switching tanks at regular intervals, the radioactive gas inventory in the system is distributed through all the gas decay tanks.

Startup

At plant startup, the system is first flushed free of air and filled with nitrogen. One compressor, one recombiner and one Shutdown Gas Decay Tank are in service. The reactor is at the cold shutdown condition. In the twin unit CPNPP, the volume control tank purge from the second unit is terminated while the unit of interest is started up. Hydrogen is charged to the volume control tank and the volume control tank vent gas mixes with the circulating nitrogen stream in the GWPS. This circulating mixture enters the compressor suction, passes through the recombiner and Shutdown Gas Decay Tank and returns to the compressor suction. When the RCS hydrogen concentration is within operating specifications, the Shutdown Gas Decay Tank is isolated and the gas flow directed to one of the Gas Decay Tanks provided for normal power operation. Hydrogen purge flow from the second unit is then restarted. Gases accumulated in the Shutdown Gas Decay Tank will be retained for reuse during hydrogen stripping from the RCS.

Shutdown and Degassing of the Reactor Coolant

Plant shutdown operations are essentially startup operations in reverse sequence. The volume control tank hydrogen purge is maintained until after the reactor is shutdown and coolant fission gas concentrations have been reduced to specified level. During this operation hydrogen purge flow may be increased to speed up coolant degassing. Next the hydrogen purge to the unit still at power is stopped and the gas decay tank in service for normal power operation is valved out. A nitrogen purge from the Shutdown Gas Decay Tank to the volume control tank of the shutdown unit is begun. This tank is placed in the process loop at the compressor discharge so that the gas mixture from the volume control tank vents to the compressor suction, passes through the Shutdown Gas Decay Tank and to the recombiner where hydrogen is removed and returned to the compressor suction. The nitrogen purge continues until reactor coolant hydrogen concentration in the volume control tank reaches the required level. At this point the Shutdown Gas Decay Tank may be isolated and the GWPS returned to its previous operating mode for the unit still at power. Depending upon system pressure the shutdown and degassing flow path may be volume control tank to waste gas compressor to hydrogen recombiner to shutdown gas decay tank and returning to the volume control tank.

During the first plant cold shutdown, nitrogen was charged to the volume control tank to strip hydrogen from the reactor coolant. The resulting accumulation of nitrogen in the Shutdown Gas Decay Tank is accommodated by allowing the tank pressure to increase. During subsequent shutdowns, however, there should be minimal additional accumulation since the gas from the first shutdown will be reused.

11.3.2.1.4 Radioactivity in the Gaseous Waste Processing System

Design Basis Case

Table 11.3-3 shows the accumulated activity in the GWPS after 40 years of plant operation using the design basis assumptions given in Table 11.1-1 but assuming 100 percent stripping efficiency. The values in this table are thus the maximum accumulated fission product activity in the GWPS. Figure 11.3-4 shows gaseous fission product accumulation as a function of time over the life of the plant, using design basis assumptions. As is seen in the figure, the increase in activity over the plant life is due to the build up of krypton-85. Other gaseous isotopes reach equilibrium in approximately 30 days.

Realistic Case

Plant activity release for environmental impact evaluations are calculated using realistic assumptions. Table 11.3-4 is the expected 40 year isotopic inventory in the GWPS, using the realistic basis assumptions of Table 11.1-3.

These values are used in calculating the radioactive release from the GWPS (refer to Section 11.3.3). The process flow diagram for the GWPS is shown in the Flow Diagram Volume as indexed against Figure 11.3-1. Table 11.3-5 presents the gas composition, flow rate and isotopic concentration at each point indicated in the process flow diagram. Isotopic inventories of major components and isotopic concentrations of liquid streams in the GWPS are also shown in the table.

For purposes of this table and release calculations, it was assumed that the gas decay tank in the GWPS circulating loop is switched on a daily basis.

11.3.2.2 Ventilation System Design

Details of the plant ventilation system, including the atmospheric cleanup filtration system are presented in Section 9.4.

11.3.2.3 Condenser Vacuum System Design

Details of the Condenser Vacuum System are presented in Section l0.4.2.

11.3.3 ESTIMATED RELEASES

This section describes the estimated gaseous release from the plant for normal operation and anticipated operational occurrences. In general, the following major sources will contribute to the expected releases of radioactive materials in gaseous effluents resulting from normal operations, including anticipated operational occurrences:

1. Leakage from the GWPS.

2. Reactor coolant leakage to the Containment.

3. Reactor coolant leakage to the Auxiliary Building.

4. Secondary side steam and water leakage to the Turbine Building.

5. Discharge from the condenser air ejector.

6. Venting during equipment drainage and sampling.

7. Venting of WGDT for maintenance.

11.3.3.1 Estimated Release from the Gaseous Waste Processing System

The GWPS is designed to provide a holdup of waste gases such that frequent, scheduled releases are not anticipated. When operationally planned releases of radioactive gases from the GWPS become necessary, such releases will be permitted only under controlled conditions subject to the limitations of 10CFR20 and 10CFR50, Appendix I. As shown in Figure 11.3-5, Vistra OpCo has agreed to the stipulation that CPNPP will be operated such that dose commitments from planned gaseous radioactive releases are as low as is reasonably achievable (ALARA). The estimated annual release from the GWPS is calculated based on an assumed leakage from the system. Leakage rates have been estimated for all components in the system and are shown in Table 11.3-6. Based on this leak rate and the expected system activity inventory shown in Table 11.3-4 the annual gaseous release from the system is presented in Table 11.3-7.

11.3.3.2 Estimated Releases from Ventilation Systems

The estimated annual releases from the ventilation system are presented in Appendix 11A, Table 11A.4.

TABLE 11.3-1
GASEOUS WASTE PROCESSING SYSTEM COMPONENT DATA

Waste Gas Compressors	
Type	Centrifugal
Quantity	2
Design pressure (psig)	150
Design temperature (°F)	180
Operating temperature (°F)	70 to 130
Design suction pressure, N_2 at 130°F (psig)	0.5
Design discharge pressure (psig)	110
Design flow, N_2 at 130°F (scfm)	40
Gas Decay Tanks	
Type	Vertical
Quantity	10
Design pressure (psig)	150
Design temperature (°F)	180
Volume, each (ft^3)	600
Material of construction	Carbon Steel
Recombiners	
Type	Catalytic
Quantity	2
Design inlet pressure (psig)	110
Design inlet temperature (°F)	140
Design flow rate (scfm)	50
Design hydrogen recombiner rate (scfm)	3.0
Design discharge pressure (psig)	25
Design discharge temperature (°F)	140
Material of construction	Stainless Steel

TABLE 11.3-2
GASEOUS WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 1 of 3)

Channel Number	Location of Primary Sensor	Design Pressure (psig)	Design Temperature (°F)	Range	Alarm Setpoint	Control Setpoint	Location Of Readout
Flow Instrumentation							
FIA-1094	Volume control tank discharge flow	150	140	0.0-1.2 scfm	1.2 scfm	-	WPS panel
QIA-1091	Gas decay tank water flush	150		0-30 gpm	30 gpm (adjustable)	-	Local
HIC-1094	Volume control tank purge control	150		0-100 pct	None	Manual control (normal flow 0.7 scfm)	WPS panel
Pressure Instrumentation							
PI-1031	Moisture separator	150	180	0-100 psig		-	Local
PI-1033	Moisture separator	150	180	0-100 psig		-	Local
PIA-1036	Gas decay tank no. 1	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1037	Gas decay tank no. 2	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1038	Gas decay tank no. 3	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel

TABLE 11.3-2
GASEOUS WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[(a)]

(Sheet 2 of 3)

Channel Number	Location of Primary Sensor	Design Pressure (psig)	Design Temperature (°F)	Range	Alarm Setpoint	Control Setpoint	Location Of Readout
PIA-1039	Gas decay tank no. 4	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1052	Gas decay tank no. 5	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1053	Gas decay tank no. 6	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1054	Gas decay tank no. 7	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1055	Gas decay tank no. 8	150	180	0-150 psig 0-30 psig	100 psig 20 psig	-	WPS panel
PIA-1056	Gas decay tank no. 9	150	180	0-150 psig 0-30 psig	80 psig	-	WPS panel
PIA-1057	Gas decay tank no. 10	150	180	0-150 psig 0-30 psig	80 psig 20 psig	-	WPS panel
PIA-1065	Hydrogen supply header	150	180	0-150 psig	90 psig	-	WPS panel
PIA-1066	Nitrogen supply header	150	180	0-150 psig	53 psig	-	WPS panel
PICA-1092	Compressor suction header	150	180	2 psi vac. 2 psig	0.5 psi vac	0.5 psi vac.	WPS panel
PI-1093	Gas decay tank makeup water	240	180	0-160 psig	-	Local	

TABLE 11.3-2
GASEOUS WASTE PROCESSING SYSTEM INSTRUMENTATION DESIGN PARAMETERS[a]

(Sheet 3 of 3)

Channel Number	Location of Primary Sensor	Design Pressure (psig)	Design Temperature (°F)	Range	Alarm Setpoint	Control Setpoint	Location Of Readout
PI-1094	Volume control tank discharge pressure	150	250	0-20 psig	-	-	Local
Level Instrumentation							
LICA-1030	Compressor moisture separator	150	180	0-100% (6-33 in wc)	Low-7 inch High - 21.1 inch	21.1" to 16" in wc 16" to 14" in wc 14" to 11" in wc 11" to 7" in wc	WPS panel and local
LICA-1032	Compressor moisture separator	150	180	0-100% (6-33 in wc)	Low-7 inch High - 21.1 inch	21.1 to 16" in wc 16" to 14" in wc 14" to 11" in wc 11" to 7" in wc	WPS panel and local

a)
F - Flow
Q - Water Integrator
P - Pressure
T - Temperature
L - Level
R - Radiation
I - Indication
C - Control
A - Alarm

TABLE 11.3-3
DESIGN BASIS ACCUMULATED RADIOACTIVITY IN THE GASEOUS WASTE PROCESSING SYSTEM AFTER FORTY YEARS OPERATION (BASED ON OPERATION OF TWO UNITS) - ORIGINAL LICENSING BASIS [a]

Isotope	Activity (Curies) At Plant Shutdown
Kr-85	78,000
All other noble gases	
Kr-85m	116
Kr-87	10.8
Kr-88	110
Xe-131m	1240
Xe-133	132,000
Xe-133m	6600
Xe-135	960
Xe-135m	3.8
Xe-138	0.26

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

This table is based on 40 years continuous operation with 1 percent fuel defect. Power assumed to be 3565 MWt. The data are based on the design basis assumptions listed in Table 11.1-1 but assuming 100 percent stripping efficiency.

TABLE 11.3-4
REALISTIC ACCUMULATED RADIOACTIVITY IN THE GASEOUS WASTE PROCESSING SYSTEM AFTER FORTY YEARS OPERATION (BASED ON OPERATION OF TWO UNITS) - ORIGINAL LICENSING BASIS[(a)]

Isotope	Activity (Curies) At Plant Shutdown
Kr-85	15,540
All other noble gases	
Kr-85m	2.86
Kr-87	0.208
Kr-88	2.58
Xe-131m	63.6
Xe-133	7,480
Xe-133m	60.0
Xe-135	21.6
Xe-135m	0.0022
Xe-138	0.00646
Iodine	
I-131	0.1864
I-132	0.0008
I-133	0.0284
I-134	0.000126
I-135	0.0046

Inventories are based on reactor coolant concentrations given in Table 11.1-4. The table is based on 40 years continuous operation. Power assumed to be 3565 MWt. The data are based on a volume control tank purge rate of 0.7 scfm, a 40 percent stripping efficiency and the stripping fractions listed in Table 11.1-3.

(a) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11.3-5
PROCESS PARAMETERS FOR GASEOUS WASTE PROCESSING SYSTEM[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 1 of 4)

Description	Flow (scfm)	N_2 (%)	H_2 (%)	Isotopic Concentration[c] (µc/cc)						
				Kr-85[d]	Kr-85m	Kr-87	Kr-88	Xe-133	Xe-133m	Xe-135
Volume control tank purge	1.4	0	100	2.67×10^{-2}	1.78×10^{-1}	4.52×10^{-2}	2.56×10^{-1}	1.63×10^{1}	3.12×10^{-1}	6.50×10^{-1}
Gas decay tank discharge to compressor	40	99.9	0.1	1.70×10^{1}	1.51×10^{-1}	1.10×10^{-2}	1.37×10^{-1}	$7/48 \times 10^{1}$	3.17	1.14
Compressor suction	41.4	96.6	3.4	1.64×10^{1}	1.52×10^{-1}	1.22×10^{-2}	1.41×10^{-1}	7.28×10^{1}	3.08	1.13
Compressor discharge to Recombiner	41.4	96.6	3.4	1.64×10^{1}	1.52×10^{-1}	1.22×10^{-2}	1.41×10^{-1}	7.28×10^{1}	3.08	1.13
Recombiner discharge to gas decay tanks	40	99.9	0.1	1.70×10^{1}	1.58×10^{-1}	1.26×10^{-2}	1.45×10^{-1}	7.54×10^{1}	3.19	1.17
Miscellaneous vents, evaporators, RCDT, recycle holdup tank educator	Neg.	0	100	0	0	0	0	0	0	0
Recombiner oxygen supply	0.70	0	0	0	0	0	0	0	0	0
Recombiner calibrating gas	0.004	94	6	0	0	0	0	0	0	0
Recombiner calibrating gas	0.004	94	6	0	0	0	0	0	0	0

TABLE 11.3-5
PROCESS PARAMETERS FOR GASEOUS WASTE PROCESSING SYSTEM[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 2 of 4)

Waste gas system nitrogen supply	0	100	0	0	0	0	0	0	0	0

Description	Flow (scfm)	N_2 (%)	H_2 (%)	Isotopic Concentration[c] (µc/cc)						
				Kr-85[d]	Kr-85m	Kr-87	Kr-88	Xe-133	Xe-133m	Xe-135
NSSS nitrogen supply	0	100	0	0	0	0	0	0	0	0
Nitrogen relief to plant vent	0	100	0	0	0	0	0	0	0	0
NSSS hydrogen supply	1.4	0	100	0	0	0	0	0	0	0
Volume control tank hydrogen	1.4	0	100	0	0	0	0	0	0	0
Hydrogen relief to plant vent	0	0	100	0	0	0	0	0	0	0
Waste gas discharge to plant vent	0	100	0	1.70×10^{1}	0	0	0	0	0	0
Recycle gas to volume control tank	0	100	0	0	0	0	0	0	0	0
Pressurizer relief tank vent and return	0	100	0	0	0	0	0	0	0	0
Shutdown tank relief	0	100	0	0	0	0	0	0	0	0

TABLE 11.3-5
PROCESS PARAMETERS FOR GASEOUS WASTE PROCESSING SYSTEM[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 3 of 4)

Description	Flow (scfm)	N_2 (%)	H_2 (%)	Isotopic Concentration[e] (µc/cc) Kr-85[d]	Kr-85m	Kr-87	Kr-88	Xe-133	Xe-133m	Xe-135
Liquid Streams										
Waste gas compressor drain		4.37	4.05×10^{-2}	3.24×10^{-3}	3.74×10^{-2}	1.60×10^{1}	6.77×10^{-1}	2.48×10^{-1}		
Recombiner drain		3.38	3.14×10^{-2}	2.51×10^{-3}	2.90×10^{-2}	1.24×10^{1}	5.25×10^{-1}	1.92×10^{-1}		
Gas decay tank drains		1.19	1.06×10^{-2}	7.72×10^{-4}	9.58×10^{-3}	4.34	1.84×10^{-1}	6.63×10^{-2}		
System drains to volume control tank		1.74	1.58×10^{-2}	1.21×10^{-3}	1.44×10^{-2}	6.36	2.69×10^{-1}	9.77×10^{-2}		
Recombiner reactor makeup water		0	0	0	0	0	0	0		
Compressor makeup water		0	0	0	0	0	0	0		

Description	Flow (scfm)	N_2 (%)	H_2 (%)	Component Inventory (Curies) Kr-85[d]	Kr-85m	Kr-87	Kr-88	Xe-133	Xe-133m	Xe-135
Compressor	4	96.6	3.4	7.80	6.96×10^{-2}	5.06×10^{-3}	6.28×10^{-2}	3.44×10^{1}	1.46	5.26×10^{-1}
Recombiner	4	99.9	0.1	5.84	5.21×10^{-2}	3.79×10^{-3}	4.70×10^{-2}	2.57×10^{1}	1.09	3.93×10^{-1}

TABLE 11.3-5
PROCESS PARAMETERS FOR GASEOUS WASTE PROCESSING SYSTEM[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 4 of 4)

Description	Flow (scfm)	N_2 (%)	H_2 (%)	Component Inventory (Curies)						
				Kr-85[d]	Kr-85m	Kr-87	Kr-88	Xe-133	Xe-133m	Xe-135
Gas decay tank	600	99.9	0.1	1.93×10^{3}	2.74	1.99×10^{-1}	2.47	1.35×10^{3}	5.74×10^{1}	2.07×10^{1}
Total system				1.55×10^{4}	2.86	2.08×10^{-1}	2.58	7.48×10^{3}	6.00×10^{1}	2.16×10^{1}

Basis

Power level - 3565 Mwt per unit

Gas decay tanks[f] - 8

Operating interval - 1 day

Stripping efficiency - 0.4

a) Based on GWPS isotopic inventories from Table 11.3-4 and reactor coolant activities from Table 11.1-4.

b) Historical, not subject to future updating. Has been retained to preserve original design basis.

c) Concentrations in µc/cc of gas at atmospheric pressure and 140°F.

d) Kr-85 concentrations are maximum values, but do not occur simultaneously with other isotope maximum concentrations.

e) Concentrations in µc/cc liquid at room temperature.

f) Includes two shutdown tanks.

TABLE 11.3-6
ESTIMATED LEAKAGE RATES FROM MAJOR GASEOUS WASTE
PROCESSING SYSTEM COMPONENTS

	scf/year
Valves	~45
Compressors	~10
Tanks and Valves	~10
Pipe Flanges	~20
Total	~85[(a)]

a) The above value of 85 scf/year has been rounded off to 100 scf/year which has been used in the calculation of expected releases from the GWPS

TABLE 11.3-7
ESTIMATED GASEOUS WASTE PROCESSING SYSTEM RELEASES DUE TO LEAKAGE - ORIGINAL LICENSING BASIS[a]

Isotopes	Leakage (Curies/year)
Kr-85	92
Kr-85m	0.44
Kr-87	0.032
Kr-88	0.40
Xe-133	218
Xe-133m	2.6
Xe-135	3.32
Xe-135m	0
Xe-138	0
I-131	4.8×10^{-3}
I-133	2.4×10^{-3}
I-135	7.2×10^{-4}

This assumes operation of two units with reactor coolant inventories given in Table 11.1-4 and a leakage rate of 100 standard cubic feet per year from the GWPS. The stripping efficiency used for noble gases in the volume control tank is 40 percent. Separation factor for iodines in the volume control tank is 100.

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

11.4 SOLID WASTE MANAGEMENT SYSTEM

The solid waste management system (SWMS) is designed to control, collect, condition, handle, process, package, and temporarily store, prior to offsite shipment, solid radioactive waste generated as a result of normal operation, including anticipated operational occurrences. The SWMS consists of ATCOR system components (see Section 11.4.2.1) and the waste baling system (WBS). The WBS has a baler located in the Fuel Building. An additional waste baler was abandoned in place in the Unit 1 Containment Building.

11.4.1 DESIGN BASES

11.4.1.1 Design Objectives

The design objectives of the waste baler meet the requirements of 10 CFR Parts 20, 50, 61 and 71 and United States Department of Transportation (DOT) Hazardous Materials Regulation 49 CFR Parts 170 through 178.

Connections have been provided to allow for the bulk disposal of wastes to a truck mounted or mobile waste processing system. These connections supply waste from the Chemical Drain Tank, Waste Conditioning Tank, the NSSS Spent Resin Transfer System, and the Steam Generator Blowdown Spent Resin Transfer System.

The WBS consists of a compactor type-baler which may be used to package low-radiation-level compressible wastes such as paper, disposable clothing, rags, towels, floor coverings, shoe covers, plastics, cloth smears, and respirator filters in 55-gal drums. These wastes are products of plant operation and maintenance.

11.4.1.2 Design Criteria

The input to the SWMS is from various sources. The incompressible solids are packaged in suitable containers while the compressible solid wastes are collected and compressed (baled) into containers suitable for disposal or sent offsite for vendor processing. The various sources, quantities, and activity levels are detailed in Table 11.4-1. Maximum volumes may be produced as a result of excessive equipment leakage, steam generator tube leakage, and so forth.

The principal nuclides shipped from the plant site include the following:

Iodine-131	Iron-59
Cesium-134	Manganese-54
Cesium-136	Manganese-56
Cesium-137	Molybdenum-99
Cobalt-58	Strontium-89
Cobalt-60	Strontium-90
Iron-55	Chromium-51
Hydrogen-3	

The seismic design classification of the Fuel Building housing the SWMS is discussed in Section 3.2. The seismic design and equipment design codes for the SWMS components and piping are in accordance with ETSB Technical position 11-1. See Appendix 17A.

11.4.2 SYSTEM DESCRIPTION

11.4.2.1 Solid Waste System Description

The ATCOR Radwaste Solidification System originally installed at CPNPP is not used and many of the components have been removed from the facility. Only those components necessary for the collection and conditioning of the various waste streams prior to discharge via the bulk disposal connections on elevation 810' of the Fuel Building are utilized.

The major ATCOR system components which are utilized include the Waste Conditioning Tank, Waste Feeder Pumps, Emergency Waste Return Pump, Powdex Transfer Pump, Chemical Addition Tank and their associated valves, piping and controls.

11.4.2.2 Spent Filter Cartridge Assembly Processing

Transfer of expended filter cartridge assemblies from the filter housing to the waste processing area may be accomplished by means of a filter transfer cask. The base of the cask is removed and the cask positioned above the filter housing. The cask hoist grapple is lowered to engage the filter cartridge assembly and the cartridge raised into the shielded cavity. The base of the transfer cask is replaced, then the cask is moved by monorail to the filter drop zone hatchway where it is lowered into the Fuel Building. Other methods for transferring filters may be used predicated on the filter dose rate and ALARA considerations. For example, the filter transfer cask hoist may be used to remove the filter and lower it into the fuel building and into a smaller shielded container, without the use of the filter transfer cask; or the filter may be manually removed and transported to the fuel building.

Filters may be processed on site to comply with state, federal and burial site transportation and disposal regulations or they may be shipped to a vendor for offsite processing.

11.4.2.3 Large Solid Waste Materials and Equipment

Large waste materials and special equipment that have been neutron activated during reactor operation (e.g., core components) are handled and packaged in a safe manner on a case-by-case basis.

11.4.2.4 Waste Baling System Description

Processing of compressible low-radiation-level solid wastes for disposal may be accomplished onsite using the fuel building baler to compact the waste into 55 gallon drums, or waste may be packaged in suitable transportation containers and shipped to an offsite vendor for immediate processing and volume reduction prior to disposal.

The Fuel Building baler’s shroud is ducted to the plant ventilation system to remove dust or particles that may be emitted from the drum during compression of the wastes.

This provision eliminates any potential hazard from airborne radioactivity. In addition, the assembly incorporates a fail-safe switch that does not permit baler operation with the baler door open.

11.4.2.5 Component Description

All components which are located in the Fuel Building are at elevations of 810, 822, and 840 feet.

Associated components of the SWMS are as follows:

1. Containers

   All waste that is processed at CPNPP for disposal shall be packaged in strong, tight containers meeting all applicable DOT, NRC and burial site requirements pertaining to the storage, shipment, and burial of radioactive waste.

   Any waste packaged at CPNPP for shipment to an intermediate processor shall be packaged in strong, tight containers meeting all applicable DOT and NRC requirements pertaining to storage and transportation of waste.

2. Storage Shields

   The storage shields may be used to protect personnel from radiation exposure during onsite storage and other operational handling of the filled containers.

3. Filter Transfer Cask

   One or more filter transfer casks are provided to be used as a shielding and carrier vehicle to protect personnel from radiation exposure while transferring spent filter cartridges from the filter housing to the waste processing area. Movement of a filter transfer cask is by a monorail. Each cask is provided with a removable drip pan to collect any dripping liquid from the filter cartridges. Each cask is designed with a stainless steel interior to facilitate washdown and decontamination.

4. Baler

   The baler is a commercially available assembly used in conjunction with standard 55-gal drums which receive the low radiation level, solid, compressible wastes.

11.4.2.6 Other Design Features

1. Process Control Program (PCP)

   The PCP contains or makes reference to the current formulas, sampling, analyses, tests, and determinations made to ensure that processing and packaging of wet solid radioactive waste based on demonstrated processing of actual or simulated wastes will be accomplished in such a way to ensure compliance with federal and state regulations, burial site criteria, and other requirements governing the disposal of radioactive waste.

Waste processing is performed by a mobile processing vendor. The Vistra OpCo Process Control Program requires that the vendor operate in accordance with a process control program and procedures which have been reviewed and approved by Vistra OpCo Additionally, any vendor selected to provide waste processing services or products used to achieve the 10 CFR 61 stability requirements shall have documentation demonstrating compliance with 10CFR61 stability requirements.

The primary purposes of the Vistra OpCo Control Program are as follows:

a. The PCP defines the expected wet radioactive waste streams to be processed at CPNPP. These waste streams may include bead and powdered resins, cartridge filters, evaporator concentrates, sludges from tanks and sumps, and miscellaneous liquids.

b. The PCP establishes criteria for waste processing that must be addressed in topical reports, process control programs, and/or procedures of vendors selected for waste processing. Such criteria include establishment of bounds for critical parameters and operating limits such as chemical constituents, pH, concentration of radioactive materials, boric acid content, etc.

c. The PCP establishes requirements for prequalification testing of each waste stream.

d. The PCP establishes requirements for sampling of waste streams to provide data necessary for estimating curie content and classifying the waste, and to ensure that the waste stream parameters are within the bounds for critical parameters established in the vendor's PCP.

e. The PCP establishes the testing requirements for each processing method to ensure that the 10CFR61 waste form and stability requirements and the applicable NRC burial site criteria are achieved.

f. The PCP establishes the administrative controls and quality assurance activities for waste processing required to ensure compliance with applicable regulations and requirements.

2. Overflow of Tanks

The Waste Conditioning Tank is provided with an ultrasonic level sensing device which provides level indication over the range of tank operating levels. In addition, level detectors are provided which provide interlock inputs for waste conditioning tank high and low levels. The high level interlock alarms and prevents any restart of waste stream addition to the Waste Conditioning Tank while the low level interlock stops the tank agitator.

3. Tanks Using Compressed Gases

Compressed gas is not directly used in any SWMS tank. However, the Waste Conditioning Tank is vented to the plant ventilation system since bead resin slurry is transported to the waste conditioning tank from the spent resin storage tanks by means of

nitrogen gas pressure. The volume and flow rate of the gas used for transferring one batch is estimated to be 1200 scf at 50 scfm. The expected radionuclide concentration of the vent gases will be negligible. The treatment provided (atmospheric cleanup system) for the vent gases is described in Section 9.4.

11.4.2.7 Packaging, Storage, and Shipment

1. Packaging

The SWMS product is a container of radioactive material packaged in accordance with all applicable NRC, DOT and disposal site requirements.

2. Storage

Capacity for storage is provided in Area 247 in the Fuel Building (see Figure 1.2-38). Wastes (e.g., spent resins and filters, dry active waste, waste oil) may also be stored pending shipment outside the Fuel Building or in a warehouse in a fenced area designated for radioactive materials staging and handling. This fenced area is part of the Radiologically Controlled Area. For higher radiation level waste containers, appropriate shielded storage containers are provided to ensure that radiation levels at and beyond the radioactive materials staging and handling area fence are within 10CFR20 limits for unrestricted areas. Storage of radioactive materials (e.g., contaminated scaffolding and outage equipment, etc.) may also be stored in these areas.

Storage time is a variable and depends on shipment schedules and disposal site availability. Interim storage of low-level radioactive waste may be required due to disposal site unavailability. Interim storage locations include a warehouse area, (2K7) and a fenced area (3L15) shown in Figure 1.2-1. Interim storage areas would be part of a Radiologically Controlled Area.

Prior to container shipment, drum smear samples are taken to determine the surface contamination. If required, the container surface is decontaminated and smear samples are taken again to determine if the desired decontamination has been achieved.

3. Shipment

Radwastes are stored in a designated staging area prior to shipment. Shipment of the radwaste originates from the staging area. All radwaste shipments will be in compliance with the applicable regulatory standards and requirements of the NRC, DOT, Texas Regulations for Radiation Control, and burial site.

11.4.2.8 Instrumentation and Control

The instrumentation and controls of the SWMS & WBS are designed by the equipment supplier.

11.4.2.9 Safety Evaluation

The SWMS & WBS are not safety-related and cannot affect the safe shutdown of the plant or the operation of other systems which are required to safely shut down the plant. Failures of the

SWMS & WBS have been analyzed. The thyroid and whole body doses are below the values set forth in 10 CFR 100.

11.4.2.10 The Old Steam Generator Storage Facility

The Old Steam Generator Storage Facility provides a secure long-term storage facility for the four Unit 1 Old Steam Generators and the Reactor Head Vessel Heads and associated Control Rod Drive Mechanisms for Units 1 and 2. Design features for storage of radioactive materials in the Old Steam Generator Storage facility are described in Section 12.2.1.6.

TABLE 11.4-1
SOURCE INPUTS[a] (2-UNIT OPERATION)

Type of Waste	Expected Annual Disposal Volume (ft^3)	Avg. Container Contact Dose Rate
Dry Active Waste (DAW)	10,000	≤ 50 mR/hr.
Processed Wet Waste		
Evap Conc and Chem Drains	270	≤ 200 mR/hr.
Filter Cartridges	1,000	≤ 5 R/hr.
Resins	1,510	≤ 100 R/hr.
Total Waste	12,780	

a) Based on current industry average for PWRs during the mid 1980's. Actual disposal volumes are dependent on waste processing and volume reduction techniques used.

TABLE 11.4-2
COMPONENT DESIGN PARAMETERS

(DELETED)

11.5 PROCESS AND EFFLUENT RADIOLOGICAL MONITORING AND SAMPLING SYSTEMS

11.5.1 DESIGN BASES

The Process Radiation Monitoring System (PRMS) provides a means for measuring and controlling radioactive process streams and effluents throughout the plant. The system is in compliance with the requirements of 10 CFR Part 50, 10 CFR Part 20, NRC Regulatory Guide 1.21, and GDC 60, 63, and 64. The system aids in ensuring the protection of the general public and plant personnel from exposure to radiation or radioactive materials in excess of those allowed by applicable regulations or governmental agencies. In addition, the system is designed in response to 10 CFR Part 50, Appendix I, to limit radiation levels to ALARA. The design objectives of this system for normal operation are as follows:

1. To provide continuous indication and on demand record of radiation levels, and automatic record of alarms in process and effluent streams over the range from clean-plant background to levels commensurate with Technical Specification limits.

2. To provide data useful in reporting total released activity.

3. To give early warning of increasing levels of radioactivity in process and effluent streams by using an alert-alarm set point.

4. To give an alarm and initiate a control function, where appropriate, upon high radiation levels in plant process and effluent streams using a high-alarm set point.

5. To give an indication that there is a monitor malfunction, equipment failure, or loss of counts by using a channel signal valid data check. If the test is negative, a “fault” alarm is initiated.

The system also performs post-accident functions as described in Sections 6.2.4 and 7.5 to satisfy GDC-13 and 54.

11.5.1.1 Digital Radiation Monitoring System

The Digital Radiation Monitoring System (DRMS) is comprised of the following subsystems:

1. The Area Radiation Monitoring System (ARMS), which continually monitors radiation fields in various representative regions within the plant. The ARMS is described in Section 12.3.4, Area Radiation and Airborne Radioactivity Monitoring Instrumentation.

2. The PRMS, which provides a means for assessing radioactivity levels in plant process and effluent streams, and controls plant process and effluent streams including the handling and processing of radioactive waste.

11.5.1.2 Process Radiation Monitoring System

The PRMS is comprised of the following monitors:

1. Airborne, Steam and Gas Process and Effluent Radiation Monitors (also see Section 12.3.4)

    a. Containment air monitors (1 per unit)

    b. Control Room ventilation intake monitors (4, 2/intake)

    c. Main steam and feedwater area vent duct monitors (1 per unit)

    d. Auxiliary Building vent duct monitor (1, common)

    e. HVAC room vent duct monitor (1, common)

    f. Fuel Building vent duct monitor (1, common)

    g. Safeguard Building vent duct monitors (1 per unit)

    h. Plant vent stack monitors (2/stack, common)

    i. Main Steam Line Monitors (4 per unit)

    j. Steam generator Leak Rate Monitors (4 per unit)

2. Liquid Process and Effluent Radiation Monitors

    a. Deleted

    b. Deleted

    c. Component cooling water monitors (3 per unit)

    d. Service water monitors (2 per unit)

    e. Deleted

    f. Steam generator blowdown sample monitors (1 per unit)

    g. Deleted

    h. Turbine Building sump effluent monitors (1 per unit)

    i. Liquid waste effluent monitor (1, common)

    j. Waste gas monitor (1, common)

    k. Condenser off-gas monitors (1 per unit)

l. Deleted

m. Failed-fuel monitors (1 per unit)

n. Auxiliary Building to LVW Pond monitor (1, common)

These monitors, as described in the following sections, serve in conjunction with varied portable and mobile monitoring equipment as well as with a comprehensive sampling program. Since continuous on-line isotopic analysis and measurement of exceedingly low radionuclide concentrations are not within the practicable state of the art, the sampling program is the primary method for quantitatively and qualitatively evaluating process system and effluent activity levels.

In the event of a radiation release the PRMS and the ARMS can provide immediate information about the concentration and dispersion of radioactivity throughout the plant, enabling operating personnel to evaluate the severity and to mitigate the consequences. For some anticipated operational occurrences resulting primarily from operator error and under certain conditions (where radiation levels exceed set points, the Radiation Monitoring System (RMS) automatically actuates necessary valves and thereby limits the consequences of the release.

Additionally, process monitors are provided upstream of potential radionuclide discharge points between the potential sources and release points as described in this section and as shown on flow diagrams.

The effluent release paths are monitored by the following:

The Containment air monitor provides for monitoring of a continuous sample drawn from the Containment for noble radioactive gas, iodines in elemental and organic-compound forms, and activity in particulate form. During purges or vents, Containment effluent is discharged via the plant vent stacks where it is monitored by off-line particulate, iodine, and gas (PIG) monitors and/or the Wide Range Gas Monitors (WRGM).

The condenser off-gas and waste gas effluent are monitored by beta-sensitive detectors enroute to the primary plant ventilation system and plant vent stack where they are ultimately monitored and discharged.

When the Liquid Waste Processing System is in the proper valve configuration, liquid can be discharged to the Circulating Water System and will be monitored by a gamma-sensitive detector before release to the circulating water discharge tunnel. Turbine Building sump effluent is monitored by gamma-sensitive monitors prior to release to the Low Volume Waste Pond. Upon detection of radioactivity, this effluent is diverted to the Waste Water Holdup Tanks where it is sampled and if appropriate, released from the holdup tanks on a batch basis to the Circulating Water Discharge Tunnel.

11.5.2 SYSTEM DESCRIPTION

11.5.2.1 Design Criteria - Continuous Monitoring

The following criteria govern the design of the PRMS:

1. To facilitate compliance with applicable regulations (e.g., 10 CFR Part 20), monitors and detectors have sensitivities and ranges compatible with radiation levels anticipated at specific detector locations.

2. The alarm setpoints and other important database parameters are controlled by approved plant procedures utilizing designed supervisory control features inherent to the process monitors.

3. The detected radiation levels of the monitored process and effluent systems and the most recent 28 daily, 24 hourly, or 24 10-minute averages are available on the RMS consoles in the Control Room. This information may also be printed on demand on printers located in the Control Room.

4. Control Room alarms annunciate high radiation levels and radiation monitor malfunction, as appropriate.

5. Monitor electronics have independent power supplies.

6. A check source or current is provided for each detector, which is operable from the Control Room, except the failed fuel monitor which is checked during preventive maintenance.

7. Local alarms and indication are provided for monitors where local process control panels are provided. (See Table 11.5-3.)

8. Environmental design conditions for the components are as described in Table 11.5-2.

9. Monitors are placed where they are accessible for maintenance and inspection.

10. Monitors are placed where the background is determined to be the lowest available, in accordance with governing ALARA design considerations, and for detection sensitivity considerations.

11. Detectors, electronics, and electric equipment associated with the containment air and Control Room ventilation intake monitors are seismically qualified.

Sufficient lead shielding against ambient background radiation is provided so that adequate sensitivity is achieved. The count rate (or detector output signal) at the minimum detectable activity (MDA) will be equal to or greater than twice the design background standard deviation. Design background for the sensitivity limit is considered to be the larger of either 1 mR/hr of 1 MeV gamma radiation or a reasonable design maximum for the area in which the monitor is located.

The high-alarm set points are at a value not exceeding plant technical specification limits, assuming the maximum anticipated flow rates.

The proper operation of some detectors are checked with a built-in check source which can be controlled from the Control Room. Check source indications are dependent upon expected background and are engineered to yield a positive indication under all anticipated conditions. The design of air stream sampling considers the guidance of ANSI N13.1-1969. The type of monitors used are off-line, on-line and in- line, as shown in Table 11.5-1. Use of in-line monitors provides complete monitoring of the sampling media and reduces sampling errors from auxiliary devices, pumps, and valves.

The off-line monitors are provided, when required, because of temperature, line size, or other considerations. Where a pressure differential between sample connections is insufficient to ensure proper off-line flow, a pump with flow indicator is provided. All off-line sample lines have isolation valves which allow removing the monitor from service. The off-line monitor sample chamber and auxiliary piping is made of stainless steel of the quality necessary to conform to ASTM standards.

An on-line (or adjacent-to-line) type monitor is provided because of the special considerations of the monitored fluid. The on-line monitor has the advantage of simplicity, freedom from sampling problems, and flow control in that existing piping need not be altered, since flow impediments and sediment collection points are avoided. The use of on-line monitors is based on sensitivity considerations.

11.5.2.2 Detector Location Selection

Detector locations conform with the general criterion of NRC Regulatory Guide 1.21, which states that “all normal and potential paths for release of radioactive material during normal reactor operation including anticipated operational occurrences should be monitored.” In accordance with this guide, the points to be monitored are those that provide data on effluent releases to the plant environs; these monitoring points are indicated schematically on diagrams referenced in Table 11.5-1. Monitors are provided for the following:

1. Process streams that normally discharge low-level activity directly to the environment.

2. Continuous process streams that discharge directly to the environment but do not normally carry radioactive material. Such monitoring indicates if any radioactive leaks into these process lines have occurred.

3. Process lines which contain radioactivity but do not normally discharge to the environment. Such monitoring indicates if process malfunctions have occurred.

11.5.2.3 Expected Radioactivity Concentrations and Quantity Measured

The expected radiation levels in the process and effluent streams are such that concentrations in gaseous effluents at the Exclusion Area Boundary and liquid effluents at the discharge point are a small fraction of 10 CFR Part 20 limits. The calculation of radioactive concentrations and the levels to be monitored or sampled are described in Section 11.1 and Table 11.5-4.

Each channel, except the iodine monitors, measures gross radioactivity.

11.5.2.4 Detector Type, Sensitivity, and Range

The range of each detector, along with other pertinent information such as detector type and reference nuclide are summarized in Table 11.5-1.

Detector location and sample line routes from sampling point to detector are chosen to minimize sample line length, and the number of direction changes are chosen to minimize transport losses. Sample lines are appropriately sized; where applicable, stainless steel tubing is made with long radii elbows in conformance with ANSI N13.1-1969.

Process and effluent stream activity detectors are scintillation crystals, Cd Te (Cl), PM Tubes, GM Tubes, etc. which detect beta or gamma radiation over an energy range appropriate for the process stream. Design sensitivities and ranges have been selected so that the instrument either reads on scale or stays within system limits during normal operation, taking into consideration the limitations of such equipment.

11.5.2.5 Digital System Description

11.5.2.5.1 General

The RMS is a dedicated distributed microprocessor based digital monitoring system. System communication connections require only two twisted-shielded pairs of low-voltage cable. Details are provided in the following text and Figure 11.5-1. Certain channels are seismically qualified to remain operable during and after the SSE. The Control Room Cabinet consist of Liquid Crystal Displays (LCDs) and printers. The monitor detector locations have control, data processing, data storage, and multilevel alarming features. See Table 11.5-3.

The microprocessor-based RMS is comprised of the following components:

1. Display/Operator (SCADA-A & B), color LCDs, alarm printers, and keyboards; two operator console LCDs (View Nodes) and keyboards

2. Dedicated microprocessors for each monitor

3. Remote displays and controls on process system operator panels as applicable

4. Monitors with associated equipment

5. Mirror SCADA-A & B with dedicated LAN to support the LAN-based radiation monitoring clients. Locations outside the control room, including the TSC and EOF, receive RMS data via the Plant Computer System.

6. Control Room equipment racks (two) furnished with display/control modules for selected monitors

The system concept is a distributed data base with each individual monitor processor maintaining its own data base and stored data. A stand-alone configuration includes redundant computers in a Display/Operator Cabinet that handle five loops of up to 31 monitors per loop. Alarm messages are sent to the Control Room from the detector microprocessor when polled by the respective PC-11. The requested data are then returned to the PC-11 where they are displayed,

printed, and announced. Alarms are displayed. Operator-initiated display requests cause requests for data from the monitor processors. The computers contain the communications links to the monitor processor loops and the necessary memory and programming to provide overall system status, monitor group profiles, and individual monitor trend displays. These computers are interconnected to provide alternate communications paths, thus tolerating a single communications cable fault or monitor malfunction without loss of function in the system. In addition to the display requests from the keyboard, other control functions such as check-source readings, pump motor control, and purge control are provided.

Certain radiation monitor channels are seismically qualified, because of their importance to the safe operation of the plant. These channels are capable of withstanding a Safe Shutdown Earthquake (SSE). Dedicated control/display modules on seismically qualified cabinets for the containment air monitors, plant vent stack monitors (WRGMs and PIGs), and the class 1E Control Room ventilation intake monitors are provided in the control room. The individual control/display modules in the cabinets utilize the same microprocessor chip as the monitor processor but on a simpler board. These modules perform basically the same control and display functions as the PC-11A, B LCD Displays, but with limited display capability. The control/display module has lighted alarm indicators, pushbuttons for the check-source test, purge or filter step, and pump motor control. Since the seismic channels are also connected to the PC-11A, B Displays, the normal use of these modules is as a backup to the PC-11A, B Displays which are not seismically qualified.

Digital communication is the basis for component interface for the multiprocessor system. Communication circuits tie the monitor processors and the Control Room Display/Operator Cabinet into four loops. The communications cabling uses two twisted-shielded pairs in a daisy-chain configuration. Each daisy chain forms a loop from Scada A through a group of monitors to Scada B inside the Control Room. It provides two communication paths to and from the Control Room. In case of a failure in a loop section the monitor processors on the first console side of such a failure communicate with that console, and those on the other side communicate with the second console. The consoles share information with each other and lose no information after any single communication failure. Communication consists of request for information via alarm and data polls or commends from the Control Room to the field microprocessor units.

Each monitor processor has a data base which includes count rate, conversion factors, high alarm limit, alert alarm limit, check-source limit, and where monitored, sample and process parameters. Calculations are performed by microprocessor using these parameters and the raw data from the detector. The raw data from the detector are accumulated into two separate computational sections. One section performs averaging calculations and history filing; the other computes the current rate and tests for alarm conditions.

Processed data are averaged and stored in memory for a historical trend. The history file of detector data consists of 24 10-min averages, 24 1-hr averages, and 28-1 day averages.

Alarm determinations are based on having a significant number of counts available to calculate, with a 95 percent confidence level that count alarm rates are being reached. Counts are incremented into an accumulator and tested for significance at time intervals consistent with system alarm response requirements. If the count is significant and the rate is beyond an alarm limit, alarming occurs. The rate computed in this alarm section is displayed as the current rate.

11.5.2.5.2 Monitoring Assemblies

Each monitoring assembly is the equivalent of a digital ratemeter. Each Scada with its color LCD display, keyboard, and printer is a remote digital display and recorder for the monitoring assemblies. All field wiring for communication circuits consists of twisted-shielded pairs that are daisy-chained from monitor to monitor and to the display cabinet. Each monitor and the display cabinet are furnished with safety related or reliable non-safety related power (backed by diesel/ non-safety related station batteries) as applicable.

At each monitor, control, data processing, data storage, and multilevel alarming are all performed by the local microprocessor independently from the rest of the system. If Control Room displays and computers are down, the data stored locally at each individual monitor are available for later communication when the Control Room displays and computers are back on line. The hourly and daily averages continue to be updated and stored 24 hours may pass without loss of hourly averages. Twenty-eight days may pass without loss of daily averages. No data need be lost as a consequence of multiple failures in the equipment common to all monitor channels.

The microprocessor at each radiation monitor assembly controls the monitor, processes and stores data, and communicates messages through intervening monitors to the Control Room Display/Operator Cabinet.

The parameters (such as setpoints, conversion factors and channel identification) that specify how to control each channel and how to process its data are provided in the microprocessor database.

Data processing at each monitor microprocessor is briefly listed as follows:

1. Monitor Microprocessor Controls (on command from the Control Room Display/Operator Cabinet)

    a. Operates the check source upon manual command from the Control Room Display/Operator Cabinet

    b. Steps the filter (where applicable) upon manual command from the Control Room Display/Operator Cabinet

2. Monitor Microprocessor Data Processing

    a. Accumulates gross pulses from the detector up to a maximum pulse rate of 107 cpm

    b. Subtracts the operator entered background value from the gross count

    c. Converts net count rate to engineering units ($\mu Ci/cm^3$) via a specified conversion factor

    d. Where appropriate, via a parameter input point, measures the process fluid flow rate analog signal and uses it to convert $\mu Ci/cm^3$ to $\mu Ci/sec$, using a specified scale factor

e. Differentiates the filter-integrated particulate monitor data from fixed particulate monitor data

f. Tests for alert and high alarms on the basis of a significant quantity of counts accumulated. Actuates applicable audio and visual alarms and control relays. Resets the local horn. Issues a status message to the Control Room Display/ Operator Cabinet when the alarm status changes.

g. At the appropriate stage of processing, tests for the other alarm conditions listed previously

h. Checks for data validity (see Subsection 11.5.2.5.1) and marks each stored value questionable if the majority of its counts are not valid

i. Maintains a rotating file of past values of radiation data: 24 10-min averages, 24 1-hr averages, and 28 1-day averages.

j. Uses a specification table (Database items such as conversion factor, etc.) to control the processing. All entries for a monitor's specifications are sent as messages to that monitor's microprocessor, where they are stored in the specification table.

11.5.2.5.3 Control Room Display/Operator Cabinet

The PC based system is comprised of two operator consoles and a Display/Operator Cabinet. It is a complete, stand-alone system, as shown on Figure 11.5-1. The RMS consoles are dedicated to radiation monitoring and function independently of the plant computer.

The central (RMS) Display/Operator Cabinet (PC-11 SCADA - A & B, see Figure 11.5-1) is divided, with each section having a 18-in. color LCD display, a keyboard, a printer, and a computer.

The operator consoles (PC-11 View Nodes - A & B, see Figure 11.5-1) each have a 18-in. LCD display and keyboard which are wired to a dedicated PC at the Display/Operator Cabinet.

The consoles give the operators consolidated and fully processed information on the RMS throughout the plant. The basic displays on a RMS display are six grids and a control menu display. The characters identify the monitor and the background color indicates monitor status,(i.e. alarm conditions, check source testing, off scan, no response, and normal operation). Additional displays/functions for a radiation monitor/channel are accessed from a standard keyboard.

The next level of displays allow operators to request detailed information on individual monitors. The following are the baseline displays/functions:

1. Monitor Specific Display

   The monitor specific displays contain information such as the current radiation reading, real time trends, alarm setpoints, database item, communication status, etc. For monitors

with flow, a diagram showing pumps, detectors, and other prominent components is also provided.

2. Monitor Trend Display

   The monitor trend display is used to access the trend histories of 24 10-min. averages, 24 1-hour averages, or 28 1-day averages. These trends are displayed in bar chart format scaled to the high alarm limits. The monitor trend display also provides information such as the current activity in engineering units, the data base background value, the high alarm setpoints, and process and sample flow rates, if applicable. In addition to this, a real time trends screen for activity, sample flow, and process flow, as applicable, is provided.

3. Controls Display

   The Control Displays provide cursor selectable control options for pump on/off, purge on/off, filter advance, and check source requests. Supervisory control display allows alteration of database and alarm setpoints.

4. Monitor/Channel Data Base Items Display

   The monitor/channel data base items display provides the current values for the monitor/channel data base items for a particular monitor/channel. The data base changes are password controlled.

5. Alarm History Display

   The alarm history display lists messages regarding the current status of the Radiation Monitoring System.

6. Data Base Compare

   PC-11 provides an operator selectable function to compare the individual Radiation Monitor data base with a resident “master” data base for proper configuration control. This option prints a report identifying any data base discrepancies.

The four computers communicate with each other and with the remote monitor assemblies.

11.5.2.5.4 Control Room Equipment Racks

A seismically qualified control/display module for each Containment Air, plant vent stack (WRGMs and PIGs) and Control Room Ventilation intake monitor is located within seismic equipment racks in the Control Room. Digital displays are provided for each channel, in addition to switches for pump control, filter step, check source operation, and alarm acknowledge. The channel status indicators (operate, alert, high) are indicated by means of green, yellow, and red lamps, respectively, provided on each module.

Each seismic channel has a separate radiation detector with internal power supply. A local microprocessor is provided for each monitor. Each microprocessor has three communication

ports which provide for connection to the central Display/Operator Cabinet and the control/ display modules in the Control Room.

Should communications between local microprocessors and the Control Room fail, the microprocessor at the local detector would continue to monitor radiation levels, and to store daily averages for up to 28 days. Thus, a high degree of reliability and independence is assured. The microprocessor is also equipped with a connector where a portable digital readout device can be connected to display data. The monitoring system has independent power supply electronics for each monitor.

Non-seismic monitors are locally identical to seismic monitors, but are provided with necessary communication ports to communicate with the non-seismic Display/Operator Cabinet computers in the Control Room. Therefore, the non-seismic channels are inherently as reliable as the seismic channels, except for possible loss of Control Room readouts in the event of an SSE.

11.5.2.5.5 Set Points

Set points for appropriate process and effluent monitors are established to meet plant Technical Specifications and Radiological Effluent Controls Program limits, which are governed by 10 CFR Parts 20 and 50, Appendix I, objectives. Set points for process monitors are established to provide timely warning of increased system activity that requires corrective action to maintain safe operating conditions.

Two independently adjustable bi-level radiation set points are provided. The alert set point activates only an alarm, while the high set point not only is alarmed but also initiates control action where appropriate. Alert set points may be set as close to the background as feasible to provide early warning without generating statistically nonsignificant alarms. The bases for alarm set points and the associated flow diagram references are listed in Table 11.5-1. Alert-alarm set points are between background and the high-alarm set point. The system is designed to provide physical security for access to alarm setpoints. These setpoints are administratively controlled by approved station procedures.

The instrumentation and/or control diagrams for safety systems shown/identified in Section 1.7 and 7.2 include radiation monitoring systems.

11.5.2.5.6 Annunciators and Alarms

All process and effluent radiation monitors are annunciated on the RMS Display/Operator Cabinet in the Control Room. The individual detector microprocessors respond to pollings from the respective PC-11 by initiating alarm messages in the control room when set points are exceeded, although all automatic control functions and local alarms are initiated directly from the microprocessor. The routine and automatic polling of monitors is for the purpose of accumulating data for trend history, radiation alarms and monitor failures. The functions of alarming via a LCD Display, sounding a horn, and printing alarm events on a printer are performed by the system.

The RMS Display/Operator Cabinet stores complete alarm status for all radiation monitors. Keyboard access is provided for alarm status and alarm sort screens.

Each new alarm condition is logged at redundant alarm printers. The alarm message, as typed, contains the date, time, alarm code, channel or monitor identification, and brief description. When channels in alarm return to normal, this condition is logged in the same manner.

11.5.2.6 Airborne and Steam Radioactivity Monitoring Systems

Radiation monitors are provided for continuous detection and measurement of airborne radioactivity for gaseous process streams and for plant gaseous effluents as summarized in Table 11.5-1. General design and performance objectives for all elements of the PRMS are covered in Subsections 11.5.1 and 11.5.2. In addition, the following criteria for air sampling are met:

1. Detectors are located as close to sampler intakes as feasible.

2. In the design of sample nozzles and lines, the guidelines of ANSI N13.1-1969 are considered to obtain representative sampling and reduce settling and plateout losses of particulates in transport to the detector.

3. The Containment atmosphere particulate and gaseous radioactivity monitors are part of the reactor coolant pressure boundary (RCPB) leak detection system. Accordingly, all elements of these channels required for radiation indication are designed and qualified to remain functional following an SSE, in compliance with NRC Regulatory Guide 1.45.

11.5.2.6.1 Sampling Devices

For each off-line gaseous monitor, a sample is drawn from a system through a sample line to the monitor skid. The sample is routed through the monitor skid and then returned to the system from which it was extracted. Samples from the plant vent stacks are drawn using isokinetic nozzles.

Sample pumps are utilized to draw an appropriate sample through the monitor. Each monitor has a low sample flow alarm. A local flow indicator is provided for vent stack monitors which have particulate and iodine filters, so that the total volume that has passed through the filters can be determined. The filter papers used to collect particulates have a collection efficiency of at least 99 percent for 0.3-micrometer particulates. The cartridges used to collect iodine have been shown to have a minimum efficiency of 95 percent for elemental and organic iodine.

Each off-line gas monitor has manually operated sample valves. This allows room air to be routed through the gas monitor to check the background radiation level and allows for samples to be taken or for the introduction of calibrated gas to check monitor calibration.

The location of sample probes and off-line monitors has been chosen to minimize sample plateout. Unavoidable bends are made with radii not less than five times the tubing diameter. Stainless steel lines and appropriate sampling valves are used. In addition, a local sample tap is provided for the containment air monitoring systems so that a test can be performed to determine the sample deposition (plateout) after system installation.

11.5.2.6.2 Containment Air Monitoring System

This system is provided to monitor Containment air continuously for particulate, iodine (elemental, particulate, and organic forms), and noble gas activities. The monitoring system takes a continuous air sample from the Containment atmosphere, which is drawn outside the Containment in a closed system. The sample air passes through a particulate filter which is continuously monitored by a beta-sensitive scintillation detector. The filter paper collects 99 percent of all particulate matter greater than 0.3 micrometers in size. Shielding is provided to reduce ambient background radiation to a level that provides adequate detector sensitivity. After the sample leaves the particulate monitor, it passes through a closed system to an adsorber cartridge to collect iodine, which is viewed by a shielded gamma-sensitive scintillation detector. The particulate filter and iodine filter are replaced as required to support operation of the monitor. When the sample air leaves the iodine cartridge it is directed through a closed system to a shielded stainless steel gas sampling chamber viewed by a beta-sensitive scintillation detector. The sampled air is finally returned to the Containment atmosphere. Indication and annunciation are provided in the Control Room. The detection of high radiation levels by the particulate or noble gas channels causes the high-level set point to trip and initiates Containment Ventilation Isolation.

The particulate and gas monitor are also used as part of the RCPB leakage detection system. The sensitivity and response time of this part of the leakage detection system, which is used for monitoring unidentified leakage to the Containment, is discussed in section 5.2.5.3.

The containment atmosphere can also be manually sampled in a number of ways and via a number of outlets if required. Some examples are: by removal of a section of the existing sampling line particulate filter after a known collection period and flow rate and submitting the filter for laboratory assay; by means of an in-line filter and vacuum pump sampling from the test valve in the containment sampling line; and by evacuated flask sampling from the same test valve. Other sampling points potentially exist on the continuous air monitor unit as well as from other containment penetrating equipment, such as hydrogen analyzer equipment.

11.5.2.6.3 Plant Vent Stack Monitoring System

Each plant vent stack is equipped with a gas monitor. They are off-line monitors that sample the plant vent effluent prior to discharge.

The plant vent stack gas monitors draw representative air samples from the plant vent stack via isokinetic nozzles in the stack. Lead shielding is provided to reduce background to levels that allow the detection of the specified instrument ranges given in Table 11.5-1.

For normal range noble gas monitoring when no automatic control functions are required, the Plant Vent Stack Gas Monitor may be used as a backup for the Wide Range Gas Monitor, which is the primary monitor that is used for plant vent stack noble gas detection.

A separate Wide Range Gas Monitor (WRGM) is provided for each stack for extended range noble gas monitoring. Each WRGM has two isokinetic nozzles; one for sampling during normal conditions and one for accident conditions. The stack sample passes first through a sample conditioning unit which filters particulates and iodine and may be used to take grab samples. The WRGM particulate and iodine filters are replaced at least weekly and counted for activity for release accountability. The sample then passes through the detector assembly which uses three

detectors to cover the complete range required. These Wide Range Gas Monitors satisfy the requirements of NUREG-0737, II.F.1 for provisions for sampling plant effluents for iodines and particulates and for noble gas effluent monitors for the plant vent.

The detection of high radiation levels by the WRGM causes the high-level set point alarm to initiate the automatic closure of the gas release valve in the waste gas processing system.

11.5.2.6.4 Control Room Ventilation Intake Monitors

These monitors continuously assess the intake air to the Control Room for an indication of abnormal atmospheric radiation concentration. Two monitor assemblies per intake are provided and are powered from separate electrical supplies. In the event of high radiation, the monitors initiate Control Room ventilation emergency recirculation through filters.

11.5.2.6.5 Condenser Off-Gas Monitor

Gaseous samples are drawn through an off-line system by a vacuum pump from the discharge of the vacuum pump exhaust header of the condenser. This channel monitors the gaseous sample for radioactivity which would be indicative of reactor coolant leakage to the secondary side. The gaseous radioactivity levels are monitored by a beta-sensitive scintillation detector. From the monitor the condenser off-gas is then routed to the Primary Plant Ventilation System. Indication and annunciation are provided in the Control Room. Each of the off-line monitor assemblies comprises a thin plastic scintillator, photomultiplier tube, four pi lead shielding, Cl-36 check source, indicators, and associated electronics. This monitor complements the steam generator blowdown process sample monitors. Monitor range is discussed in Table 11.5-1, and expected concentrations are discussed in Section 11.1.

The COG Radiation Monitors have no control functions. They are installed to aid in detecting a slow propagating steam generator tube leak (detection of tube rupture). Note that argon gas may be injected into the RCS to raise the primary coolant activity. This helps to ensure a reliable COG Monitor response in the absence of other fission and activation product inventories.

11.5.2.6.6 Waste Gas Monitor

This in-line monitor provides an indication of magnitude of the gross gaseous activity in the GWPS by monitoring the input line to the GWPS compressors by means of a CdTe(Cl) scintillation detector. Indication and annunciation are provided in the Control Room, as well as on the local process control panel.

11.5.2.6.7 Auxiliary Building Vent Duct Monitor

This monitor assembly is mounted directly in the ventilation duct with the detector window looking at the duct gas. The beta-gas radiation detector is a thin plastic scintillation phosphor behind a thin aluminum window mounted integrally with a photomultiplier tube. A Cl-36 beta check source is provided for verification of channel operation. This channel monitors the accumulation of airborne radioactivity that is drawn by the Auxiliary Building ventilation system from the various equipment rooms, corridors, and cubicles of the Auxiliary Building. This monitor also initiates the automatic closure of the gas release valve in the waste gas processing system on detection of high radiation. Indication and annunciation are provided in the Control Room.

11.5.2.6.8 HVAC Room Vent Duct Monitor

This monitor is identical to the beta-gas scintillator of the Auxiliary Building vent duct monitor. It is located inside the heating and ventilation room ventilating duct to monitor potential airborne radioactivity in the ventilation system and to give an indication of possible changes in process systems.

11.5.2.6.9 Fuel Building Vent Duct Monitor

The Fuel Building vent duct monitor is identical to the monitor described in Subsection 11.5.2.6.8, except that it provides an indication of Fuel Building airborne activity. Fuel Building radiation monitors are not safety-related; however, an appropriate operational Quality Assurance program will be applied.

11.5.2.6.10 Safeguard Building Vent Duct Monitor

These monitors are similar to the monitor described in Subsection 11.5.2.6.8, except that they provide an indication of Safeguard Building airborne activity.

11.5.2.6.11 Main Steam and Feedwater Area Vent Duct Monitor

These monitors are similar to the monitor described in Subsection 11.5.2.6.8, except that they provide an indication of airborne activity in the main steam and feedwater penetration area.

11.5.2.6.12 Main Steam Line Monitors

These monitors utilize GM tube detectors to monitor increases in secondary side activities. They are mounted adjacent to the main steam lines and are located between the PORV and the first safety valve.

11.5.2.6.13 Steam Generator Leak Rate Monitors (SGLRM)

These monitors are provided to detect N-16 radioactivity in the main steam lines. The SGLRMs are mounted on the main steam lines just upstream of the MSIVs. Detected radioactivity is indicative of a primary-to-secondary system leak and provides an estimate of the range of potential steam generator leak rate. Because these channels monitor the secondary system radioactivity levels, it supplements the main steam line monitors, the steam generator blowdown sample monitor and the condenser off-gas monitor. Gamma-sensitive scintillation detectors are provided. Monitor output indication and alarm annunciation are provided at the Control Room RMS Display/Operator Cabinet.

The SGLRMs have no control function. They are installed to aid in detecting a slow propagating steam generator tube leak (steam generator tube rupture). The primary-to-secondary leak rate program at CPNPP uses the SGLRMs as the primary source for indication and the condenser off gas monitors are used as a back-up/confirmatory indication.

11.5.2.7 Description of Liquid Monitors

Each liquid monitor channel of the system contains an integrated modular assembly; specific details of each monitor are described in Subsections 11.5.2.7.1 through 11.5.2.7.12.

1. General Description - Sampling Devices

There are four basic types of liquid monitors used in the system:

- Off-line without a sample pump; (Sodium Iodide (NaI(Tl) Detector)
- Off-line with a sample pump; (Sodium Iodide (NaI(Tl) Detector)
- Off-line failed fuel; (GM Tube Detector)
- Adjacent-to-line; (Sodium Iodide (NaI(Tl) Detector)

Off-line monitors draw a sample from a process line, through a shielded sample chamber, through the sample pump, where one is required, and then return it to the process system. Each sample pump is capable of drawing at least one gallon per minute of liquid through the monitor. The sample flow rate is controlled by means of a manual valve. All off-line monitors have a low-sample-flow alarm. Adjacent-to-line monitors provide continuous monitoring of the process fluid through the process pipe line. The detector shield configuration is factory designed to accommodate mounting installation on the existing plant piping. A flow switch is not required because the detector is mounted adjacent-to-line with the pipe.

The monitor inlet and outlet lines are flanged and the sample piping has isolation valves so that the monitor can be disassembled for decontamination.

2. Detector-Photomultiplier-Preamplifier Unit

With the exception of the GM Tube, each detector is a NaI gamma-sensitive scintillation detector. A photomultiplier and preamplifier are mounted integrally with the detector. The detectors are designed to remain fully operational over a wide range of temperatures, as shown in Table 11.5-2. If they are exposed to high radiation transients up to 100 times the channel range, the channel maintains its operation and returns to normal functioning when the transients have subsided. Since gamma detectors are used, it is possible to compare the monitor readout with the results of grab samples analyzed in the plant multichannel analyzer as a check of proper monitor operation. Solenoid-operated check sources are provided to check detector response. Monitor characteristics are listed in Tables 11.5-1 and 11.5-3. Monitors with local process control panels have local radiation alarms and/or indication of channel radioactivity levels.

11.5.2.7.1 Steam Generator Blowdown Sample Monitor

This channel monitors the liquid phase of the secondary side of the steam generators for radioactivity to provide an indication of a primary-to-secondary system leak. This provides backup information to the condenser off-gas monitor. The radiation detector is able to monitor radioactivity from blowdown for each steam generator. Blowdown sample lines from each of the steam generators are combined in a common header, and the common sample is continuously monitored by a shielded gamma-sensitive detector located downstream of the sampler assembly. After sampling and monitoring, the blowdown sample is routed to the atmospheric drain tank for return to the Condensate System. Control Room alarm and indication are provided to alert operating personnel. Additionally, an alert alarm is also provided on the local process panel. In

the event of high activity, the monitors high-level set point trips and initiates an alarm and the automatic closure of the isolation valves in the blowdown and sample lines. Prior to this event, the alert alarm set point gives an alarm and indication of increasing activity. The isolation valves for blowdown sampling can be opened by overriding the monitor, as needed, for operation and analysis. Identification of which steam generator is leaking is made by overriding each of the sample line isolation valves and drawing samples from each steam generator blowdown line separately for analysis.

11.5.2.7.2 Steam Generator Blowdown Monitor

“The Steam Generator Blowdown Monitor has been removed”.

11.5.2.7.3 Liquid Waste Monitor

Discharges from the LWPS (see Section 11.2) are continuously monitored by a shielded gamma-sensitive (NaI) scintillation detector. A radiation monitor and a control valve discharges processed waste to the circulating water discharge tunnel. The discharge control valve is administratively controlled; if activity concentrations exceed the discharge monitor high radiation alarm set point, automatic closure of the discharge control valve is initiated. Indication and annunciation are provided on the WPS control panel and in the Control Room.

11.5.2.7.4 Auxiliary Steam Condensate Monitor

This channel has been removed from the plant. Sample fluid is drawn from the WPS waste evaporator, floor drain waste evaporators and the BRS recycle evaporator auxiliary steam condensate return lines prior to delivery to the auxiliary steam drain tank, and analyzed periodically whenever the evaporator packages are operated. Upon detection of predetermined levels of radioactivity, manual closure of isolation valves is initiated to minimize contamination of systems serviced by the auxiliary steam system. A heat exchanger is located upstream from the radiation monitor in the process line to cool the sample.

11.5.2.7.5 Service Water Monitor

These monitors are provided to monitor the Service Water Systems for radiation since in leakage from radioactive fluid systems could cause potential radioactive leakage to the environment. A shielded NaI scintillation detector is located in an off-line sampler assembly downstream of each component cooling water heat exchanger to monitor service water being discharged. Indication and annunciation are provided at the Control Room RMS Display/Operator Cabinet.

11.5.2.7.6 Component Cooling Water Monitor

This off-line channel monitors the CCWS for radiation. Any radiation detected would be indicative of a leak of reactor coolant from the RCS or from the Residual Heat Removal (RHR) System to the CCWS.

A monitor sample line is located in each of the component cooling water return headers upstream from the surge tank. A shielded NaI gamma-sensitive scintillation detector monitors the pump-drawn fluid which is returned to the CCWS after monitoring. If a high radiation level is detected, an alarm signal is initiated in the Control Room.

11.5.2.7.7 Boron Recycle Evaporator Condensate Monitor

This channel and the discharge piping to the Reactor Makeup Water Storage Tank (RMWST), have been removed from the plant.

11.5.2.7.8 Spent Fuel Pool Demineralizer Monitors

These monitors have been removed from the plant. Sampling of the demineralizer discharge is performed in accordance with Table 11.5-4.

11.5.2.7.9 Spent Fuel Pool Cooling Water Monitor

These monitors have been removed from the plant. Sampling of the spent fuel pool is performed in accordance with Table 11.5-4.

11.5.2.7.10 Turbine Building Sump Monitors

These monitors are provided to detect radioactivity in condensate that has drained into the turbine sumps. Detected radioactivity is indicative of a primary-to-secondary system leak and provides control of potential radioactive effluent discharge. Because this channel monitors the secondary system radioactivity levels, it supplements the main steam line monitors, the steam generator blowdown sample monitor, the condenser off-gas monitor, and the steam generator leak rate monitors. Gamma-sensitive scintillation detectors are provided for the outlet header of each unit's Turbine Building sump pumps. Monitor output indication and alarm annunciation are provided at the Control Room RMS Display/Operator Cabinet. Upon detection of high radiation the monitor trips the valve to stop flow to the Low Volume Waste Pond and directs the discharge to the Waster Water Holdup Tanks (see Section 11.2.2.3).

11.5.2.7.11 Failed-Fuel Monitor

A Geiger-Mueller tube is mounted on the reactor coolant letdown line, after the letdown heat exchanger, to monitor fission-product activity. Detection of increased system activity may be indicative of failed fuel; the monitor initiates an alarm in the Control Room.

11.5.2.7.12 Auxiliary Building to LVW Pond Monitor

Discharges from Auxiliary Building Sumps 3 and 11, the Diesel Generator Sumps, and the CCW Drain Tanks are continuously monitored by a shielded gamma-sensitive (NaI) scintillation detector. If activity concentrations exceed the discharge monitor high radiation alarm setpoint, the discharge is automatically diverted to the Cocurrent Waste System Wastewater Holdup Tanks.

11.5.2.8 Monitor Indication and Alarm

All process and effluent radiation monitors are annunciated, indicated, and have a printout (on demand) of monitor output at the Control Room RMS Display/Operator Cabinet. Containment Air, plant vent stack (PIGs and WRGMs), and control room ventilation intake monitors also have control/display modules in the Control Room. Two distinct, visible alarms are provided for all channels for alert and high alarm. A common audible alarm is provided for alert and high set point trip and for loss-of-instrument signal or background. Certain monitors have local indicators

associated with their channels. Some channels have their indication and annunciation on process system control panels, as listed in Table 11.5-3.

11.5.2.9 Accuracy

The overall accuracy of a radiation monitor system is governed by four major areas: (a) factory calibration and alignment, (b) detector characteristics and environment, (c) microprocessor environment, and (d) field alignment. Factory calibration is performed with standards traceable to the National Institute of Standards and Technology (NIST). Detector energy response and linearity are demonstrated by factory calibration and site calibration procedures subsequent to operations. Detector and microprocessor environmental variations are determined from qualification tests. Overall monitor error is the root-mean-square sum of all system errors and for accident conditions is within a factor of 2 over the entire range.

11.5.2.10 Sensitivity

Threshold sensitivity is defined as twice the background standard deviation.

11.5.2.11 Calibration

The initial calibration of each complete monitoring system was performed by the manufacturer at the factory. These calibrations were primary calibrations performed in accordance with ANSI N13.10 on prototype monitors except for several custom designed monitors. Each process monitor was tested at two or more activity levels with calibration sources traceable to the National Institute of Standards and Technology (NIST). In addition, a linearity test was performed on most monitors to verify that the detection system responds linearly when comparing count rate to activity strength for the operating range of one decade above background to the uppermost decade. The source-to-detector geometry during factory calibration was the same as the sample-to-detector geometry in actual use.

Subsequent calibrations are performed onsite at frequencies specified by the Technical Specifications, Radiological Effluent Controls Program or station procedures. These calibrations are performed using one of the following techniques.

1. Onsite calibrations may be performed using a primary calibration technique similar to the factory calibration described above. This type of calibration is performed using NBS traceable sources or by comparing detector response with the results of laboratory analysis of appropriate grab samples. Onsite primary calibrations may be performed in the Radiation Protection Calibration Facility or in the field.

2. Onsite calibrations may also be performed using portable transfer sources related to an onsite primary calibration. This type of calibration consists of exposing the detector to a transfer calibration source and comparing the detector response to a reference value determined during the onsite primary calibration. Transfer calibrations are performed in the field.

3. A third method of performing onsite calibrations is using transfer sources related to the primary calibration performed at the factory. This type of calibration consists of exposing the detector to a transfer calibration source and comparing the detector response to a

reference value determined during the factory primary calibration. Transfer calibrations are performed in the field.

The calibration technique used for a particular monitor is dependent on several factors including monitor function, complexity of performing a primary calibration onsite, availability of suitable NBS traceable sources, and ALARA concerns. Sources used for calibration, whether primary or transfer, are selected to provide energies representative of the expected average energy of the process stream. The calibration techniques used for each monitor type are specified in approved station procedures.

Each monitor is provided with a check source or check current to check detector response, except for the failed fuel monitor which uses a keep-alive source. For gamma detectors a Cs-137 source is used and for beta detectors a Cl-36 source is used. Requirements for check source tests are discussed in Section 11.5.2.12.

11.5.2.12 Tests and Inspections

11.5.2.12.1 General

The system automatically monitors the operability of every channel by monitoring for conditions such as loss of sample flow, loss of counts from detector and filter torn/out of paper for moving filters. If any of these conditions exist, a channel operate failure alarm is initiated. Additionally, certain critical detectors specified in the plant Technical Specifications 3.4.15.1 or the ODCM are response checked by actuation of the associated check source or current at specified intervals. If the detector fails the check source test, the cause of the failure is investigated. If the investigation determines that the detector or monitor electronics are at fault, the channel is declared inoperable and corrective action to restore the channel to an operable status is initiated.

11.5.2.12.2 Detailed Inspection and Test Procedures

1. The tests and inspections required for the following monitors are listed in the plant Technical Specifications or the ODCM.

   a. Plant Vent Wide Range Gas Monitors (WRGM)

   b. Containment Air PIG Monitors

   c. Liquid Waste Effluent Monitor

   d. Turbine Building Sump Monitors

   e. Service Water Monitors

   f. Control Room Ventilation Intake Monitors

   g. Auxiliary Building to LVW Pond Monitor

   These tests and inspections may consist of source checks, channel checks, digital channel operational tests, and/or channel calibrations. The types and frequencies of

tests required for each of the above monitors is specified in the Technical Specifications or the ODCM.

2. The tests and inspections for all other process monitors are as follows:

   a. Channel trends maintained by the system may be used to evaluate channel behavior.

   b. Channel calibrations are performed at a frequency determined by plant procedure(s), based on performance and function.

   c. During channel calibrations, channel alarm functions and initiation of automatic control functions, if appropriate, are tested by simulation of activity above the setpoint, and by simulating loss of channel power (loss of counts).

11.5.2.13 Maintenance

The channel detector and electronics are serviced and maintained to ensure reliable operations. Manufacturer's recommendations on service and maintenance are available for use. Such maintenance includes cleaning and the replacement or adjustment of any components required after performing a test or calibration check. If any work is performed which could affect the calibration, a recalibration is performed at the completion of the work.

11.5.2.14 Sampling

The radiological sampling system is designed to provide representative liquid and gas samples at controlled temperatures and pressures for laboratory analysis to determine discharge activities and equipment performance and to provide information to assist in operator decision making.

The Process Sampling System (primary) is described in Section 9.3.2. Local effluent sampling and process sampling requirements are discussed in Sections 11.5.3 and 11.5.4, respectively.

Local samples can be obtained from the LWPS and the GWPS at the locations shown in Table 11.5-4. Samples are drawn either batchwise or periodically, as appropriate, for radiological examination. The frequencies of sampling, parameters measured, expected flows, analytical procedures, sensitivities, concentrations, and the basis for selecting each location are delineated in Table 11.5-4. As indicated in the table, the sampling program provides a means to evaluate equipment performance, and in some applications the results determine the further routing of each batch of waste.

Samples are drawn from lines or tanks as grab samples and taken to the radiochemistry laboratory for examination. The radiological sampling monitoring connections are not connected as part of the Process Sampling System (see Section 9.3.2), but the radiological sampling system can furnish radiation information for guidance in process operation.

The guidance of Regulatory Guide 1.21 is followed as described below:

1. The gaseous and liquid monitors are designed to detect the most abundant isotopes of an effluent stream and give indication of activity concentrations.

2. The determination of iodine type and its concentration are provided by an appropriate sampling method and detecting systems

3. Sampling programs will be used in conjunction with a continuous monitoring system to determine total quantity and type of radioisotopes released

4. The proper detecting systems will give the effluent concentration and flow rate

5. Gamma Isotopic analysis from the sampling program will determine isotopic content

6. The meteorological monitoring program provides site environmental data for the dose calculations

The data and parameters obtained from the various subsystems and programs are used to report the necessary information according to the Regulatory Guide 1.21 format.

Sampling techniques are discussed further in Section 9.3.2.

11.5.3 EFFLUENT MONITORING AND SAMPLING

The requirement of GDC 64 for the monitoring of effluent discharge paths is implemented by providing continuous radiation detection and/or periodic sampling for all liquid and gaseous effluent paths from which detectable quantities of radioactivity can be released from the plant during normal operation, including anticipated operational occurrences and accidents. The criteria for locating radiation monitors and sampling points are discussed in Sections 11.5.2.2 and 11.5.2.14, respectively. Effluent sampling of all potentially radioactive liquid and gaseous effluent paths is conducted on a regular basis to verify the adequacy of effluent processing to meet the discharge limits. The effluent sampling program provides the information for the effluent measuring and reporting programs required by 10 CFR Part 50.36a in annual reports to the NRC. In general, the effluent sampling program requires that stored wastes be sampled and analyzed prior to release to the environment, and that potentially radioactive, continuously released effluents be periodically sampled and analyzed. Sampling and analysis of each effluent release pathway is performed in accordance with the requirements established in the Radiological Effluent Controls Program required by the Technical Specifications.

Table 11.5-4 summarizes the sampling and analysis requirements for each release pathway. The requirements included in this summary are considered normal. Requirements for increased sampling frequencies and allowances for exemptions to the sampling requirements are included in the Radiological Effluent Controls sampling program required by the Technical Specifications.

The following paragraphs summarize the categories of effluent release types, list the specific pathways associated with each category, and provide a general discussion of sampling and analysis requirements.

11.5.3.1 Radioactive Liquid Effluents Released on a Batch Basis to the Circulating Water Discharge

Release pathways in this category include discharges from the Waste Monitor Tanks, the Laundry Holdup and Monitor Tanks, Waste Water Holdup Tanks and the Plant Effluent Holdup and Monitor Tanks. Gamma isotopic measurements are made on each batch of effluent released

and kept as a record together with the volume of the batch, the average dilution water flow used during discharge, and the date and time of release. Additionally, samples from each batch are composited and periodically analyzed for difficult to measure isotopes (i.e., alpha emitters and beta emitters). Composite samples are samples in which the quantity of liquid added to the composite from each batch released is proportional to the quantity of liquid in that batch. The results of sample analyses are used for effluent accountability and to ensure that release limits are not exceeded.

The Waste Water Holdup Tanks (WWHTs) discussed above are discharged to the Circulating Water Discharge when activity levels exceed limits for discharge to the LVW Pond specified in the Radiological Effluent Controls Program. If secondary wastes are transferred to the WWHTs when such diversion is not required by the Radiological Effluent Controls Program, the tanks may be discharged to the Low Volume Waste Pond. Discharges via this pathway are only allowed when the sample analysis shows that radioactivity concentrations do not exceed specified levels.

11.5.3.2 Potentially Radioactive Liquid Effluents Released Directly to the Environment

1. Continuous Releases

   The release pathways in this category are the discharges from the Turbine Building Sumps, Auxiliary Building Sumps 3 and 11, Diesel Generator Sumps and the Component Cooling Water Drain Tanks to the Low Volume Waste Pond. These waste streams are required to be sampled if detectable quantities of radioactivity are present. They are continuously monitored by process radiation monitors. If a monitor is out of service, then periodic sampling maybe performed. If radioactivity is present above specified levels, the waste stream is diverted to the Waste Water Holdup Tanks where it is isolated, sampled, and released as discussed in Section 11.5.3.1, above. Additionally, Low Volume Waste Pond discharges to Squaw Creek Reservoir are sampled for final accountability of radioactive materials that have been released to the pond.

2. Batch Releases

   The release pathway in this category is the discharge from the Condensate Backwash Recovery Tanks to the Low Volume Waste Pond. These tanks are sampled in accordance with the Radiological Effluent Controls Program. If radioactivity is present above specified levels, the tank discharge is diverted to the Waste Water Holdup Tanks where it is isolated, sampled, and released as discussed in Section 11.5.3.1, above.

11.5.3.3 Radioactive Gaseous Effluents Released on Batch Basis to the Environment

Release pathways in this category include intentional discharges from the containment purge exhaust and the Waste Gas Decay Tanks via the plant vent stacks. These confinements are sampled and analyzed prior to release. Additionally, the releases are continuously monitored by the plant vent monitors. The Plant Vent Stack particulate and iodine filters are changed and analyzed at least weekly to determine the quantities of radionuclides with half-lives greater than 8 days and radioiodine released, respectively. The results of sample analyses are used for effluent accountability and to ensure that release limits are not exceeded.

11.5.3.4 Radioactive Gaseous Effluents Released Continuously to the Environment

The release pathway in this category is the routine continuous ventilation system discharge via the plant vent monitors. Potentially radioactive gases are continuously discharged from the Fuel Building, Safeguards Building, and Auxiliary Building ventilation exhaust systems, and the Condenser Off-Gas System via the plant vent stacks. A representative sample from the plant vent stacks is collected and analyzed for noble gas isotopes at least monthly. A continuous sample from the plant vent is drawn through particulate and iodine filters. These filters are changed and analyzed at least weekly. The particulate filters are also composited and periodically analyzed for more difficult to measure isotopes (i.e., alpha emitters and pure beta emitters). The sampling frequencies are increased during refueling outages and after process changes or other occurrences that could alter the mixture of radionuclides. The results of sample analyses are used for effluent accountability and to ensure that release limits are not exceeded.

11.5.3.5 Analytical Procedures

Samples of process and effluent gases and liquids are analyzed in the laboratory (onsite or offsite) by the following techniques:

1. Gross beta counting

2. Gross alpha counting

3. Gamma spectrometry

4. Liquid scintillation counting

5. Radiochemical separations (by in-house or outside laboratory)

Instrumentation, which is available in the laboratory for the measurement of radioactivity, includes the following:

1. Proportional counter

2. Liquid scintillation counter

3. Gamma Spectrometer

    a. HP(Ge) detector, coupled with

    b. Multichannel analyzer with interfaced computer

Gross alpha analyses of liquid effluent samples and liquid process samples are performed with the proportional counter. These samples are evaporated to dryness on planchets prior to counting. Sample volume, counting geometry, and counting time are chosen to achieve the required measurement sensitivities. Correction factors are applied for sample decay, counter-resolving time, and other parameters as necessary to assure accuracy.

Gross alpha analyses of air particulate samples are performed by direct counting of the filters with the proportional counter.

Gross beta analyses of air particulate or liquid samples may be performed in the laboratory using the capabilities of the proportional counter or by employing counting techniques using beta scintillation detection.

Gamma spectrometry is used extensively for isotopic analyses of gaseous, air particulate, and liquid samples. High-resolution, high-purity Germanium detectors are available for this purpose. All detectors are calibrated against gamma energy for a variety of sample detector geometries.

Liquid samples for tritium analysis are normally purified as necessary prior to analysis by distilling the samples. The liquid scintillation counter is used to count the samples.

Radiochemical separations are performed by an approved outside laboratory or in-house for the purpose of the routine analysis of Sr-89 and Sr-90.

11.5.4 PROCESS MONITORING AND SAMPLING

The requirement of GDC 60 is to provide for control of the releases of radioactivity to the environment. The requirement of GDC 60, with respect to suitably controlling releases from gaseous and liquid effluent discharge paths, is implemented by providing monitors (as described in Section 11.5.2) that detect excessive radiation levels.

If radiation levels in the liquid discharge lines exceed predetermined monitor set points, the monitor trips and initiates closure of the discharge isolation valve to terminate release. If radiation levels in the primary plant ventilation system exceed predetermined monitor setpoints, the monitor trips and initiates closure of the gaseous waste processing system release valve.

GDC 63 requires the provision of instrumentation to detect excessive radiation levels in fuel storage and radioactive waste processing systems and associated handling areas. This requirement is implemented by the use of the ARMS, the PRMS, the portable monitoring and sampling system, and the process sampling program. The process radiation monitoring instrumentation detects and monitors gamma radiation contained in fluids of the Liquid Waste Processing System, and the Gaseous Waste Processing System. Increased fluid activity concentrations in these systems are promptly sensed by the monitors because the monitors are shielded for background levels of radioactivity. The small fluid leakage from the Waste Processing Systems (WPS) provides low airborne radioactivity concentrations. When there are large increases of WPS fluid activities, the leaking fluid provides higher airborne activities which are detected by the ventilation and area monitoring systems.

Process streams of the various subsystems of the Liquid Waste Processing System have monitors that are augmented by a sampling system, and this combination provides radiation monitoring to determine changing levels of system radioactivity and equipment performance.

Proper sampler system design for the medium monitored is provided to assure representative samples from radioactive process streams and tanks.

Provisions for sampling from ducts and stacks are in agreement with ANSI N13.1.

TABLE 11.5-1
PROCESS RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 1 of 5)

Detector Nos.[a]									
Unit 1	Unit 2	Detector Type	Monitor Service	Monitor Locations (El., Figure Numbers)	Principal Isotopes Monitored	Monitored Medium	Measurement Made	Specified Instrument Range (uCi/cm3)	Bases for Alarm Set Points
Auxiliary Building									
XRE-5567A XRE-5567B		Beta scintillator	Plant vent effluent -- noble gases (off-line)	El. 873 feet 6 in. Fig. 12.3-17	Kr-85, Xe-135 Xe-133[b]	Air	Gross beta	1E-06 to 1E-02	Note 2
XRE 5570A XRE 5570B		Beta scintillator CdTe(Cl)	Plant vent effluent -- noble gases (off-line)	El. 873 feet 6 inches Figures: 12.3-8 12.3-9, 12.3-23.6, and 12.3-23.7 Flow Diag. Fig. 9.4-9	Kr-85, Xe-135 Xe-133[b], Kr-88, Xe-138	Air	Gross beta 1E+05	1E-06 to 1E+05	Note 2
XRE 5701	-	Beta scintillator	Auxiliary Building ventilation air exhaust - noble gases (in-line)	Vent duct El. 873 feet 6 in. Fig. 12.3-17 Flow Diag. Fig. 9.4-2	Kr-85, Xe-135 Xe-133[b]	Air	Gross beta	1E-04 to 1E 00	Note 2, 4
1RE 5637	2RE 5637	Beta scintillator	Main steam and feedwater area ventilation air exhaust noble gases (in-line)	Vent duct El. 872 feet 6 in. Figures: 12.3-8 and 12.3-23.6 Flow Diag. Fig. 9.4-4	Kr-85, Xe-135 Xe-133[b]	Air	Gross beta	1E-04 to 1E 00	Note 5
XRE 5250	-	Gamma scintillator	Aux. Bldg. Waste Gas Adjacent to line	GWPS El. 862 feet 6 in. Fig. 12.3-16 Flow Diag. Fig. 11.3-1	Kr-85, Xe-135 Xe-133[b]	Gas	Gross gamma	1E-01 to 1E+04	Note 1
1RE 4269 1RE 4270	2RE 4269 2RE 4270	Gamma scintillator	Aux. Bldg. Service Water (off-line)	El. 790 feet 6 in. Fig. 12.3-12 Flow Diag. Fig. 9.2-1	I-131, I-133 Cs-134, Cs-137, Co-58, Co-60[b]	Water	Gross gamma 5E-02	1E-05 to 5E-02	Note 2
1RE 4509 1RE 4510 1RE 4511	2RE 4509 2RE 4510 2RE 4511	Gamma scintillator	Component cooling water (off-line)	El. 810 feet 6 in. Fig. 12.3-13 Flow Diag. Fig. 9.2-3	1-131, I-133 Cs-134, Cs-137, Co-58, Co-60[b]	Water	Gross gamma	1E-05 to 5E-02	Note 7

TABLE 11.5-1
PROCESS RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 2 of 5)

Detector Nos.[a]									
Unit 1	Unit 2	Detector Type	Monitor Service	Monitor Locations (El., Figure Numbers)	Principal Isotopes Monitored	Monitored Medium	Measurement Made	Specified Instrument Range (uCi/cm3)	Bases for Alarm Set Points
XRE 5253		Gamma scintillator	Effluent Liquid Waste (off-line)	El. 790 feet 6 in. Fig. 12.3-12 Flow Diag. Fig. 11.2-5	I-131, I-133, Cs-134, Cs-137, Co-58, Co-60[b]	Water	Gross gamma	1E-05 to 5E-02	Note 2
XRE 5251A		Gamma scintillator	Auxiliary Bldg. to LVW Pond Liquid Waste Effluent (adjacent-to-line)	El. 778 feet Fig. 12.3-21.2 Flow Diag. Fig. 9.3-8	I-131,I-133 Cs-134, CS-137 Co-58, Co-60[b]	Water	Gross gamma	1E-05 to 5E-02	Note 2
1RE 5698	2RE 5698	Beta scintillator	Safeguards building ventilation air (in-line)	Vent duct El. 873 feet. 6 in. Fig. 12.3-17 Flow Diag. Fig. 9.4-2	Xe-133[b], Xe-135 Kr-85	Air	Gross beta	1E-04 to 1E-00	Note 4
XRE 5700	-	Beta scintillator	Fuel building ventilation air (in-line)	Vent duct. El. 886 feet Fig. 12.3-17 Flow Diag. Fig. 9.4-2	Xe-133[b], Xe-135 Kr-85	Air	Gross beta	1E-04 to 1E-00	Note 4
XRE 5702	-	Beta scintillator	HVAC room ventilation air (in-line)	Vent duct El. 873 feet 6 in. Fig. 12.3-17 Flow Diag. Fig. 9.4-2	Xe-133[b], Xe-135 Kr-85	Air	Gross beta	1E-04 to 1E-00	Note 4
Safeguards Building									
1RE 4200	2RE 4200	Gamma scintillator	Steam generator blowdown sample (off-line)	El. 810 feet 6 in. Figures 12.3-6 and 12.3-23.4 Flow Diag. Fig. 9.3-4	I-131, I-133, Cs-134, Cs-137 Co-58, Co-60[b]	Water	Gross gamma	1E-05 to 5E-02	Note 3
1RE 5502	2RE 5502	Beta scintillator	Containment air-particulate (off-line)	El. 831 feet 6 in. Figures 12.3-7 and 12.3-23.5 Flow Diag. Fig. 9.4-6	Cs-137[b], Rb-88, I-133	Air	Gross beta	5E-11 to 5E-07	Note 7

TABLE 11.5-1
PROCESS RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 3 of 5)

Detector Nos.[a]									
Unit 1	Unit 2	Detector Type	Monitor Service	Monitor Locations (El., Figure Numbers)	Principal Isotopes Monitored	Monitored Medium	Measurement Made	Specified Instrument Range (uCi/cm3)	Bases for Alarm Set Points
1RE 5566	2RE 5566	Gamma scintillator	Containment air iodine (off-line)	El. 831 feet 6 in. Figures 12.3-7 and 12.3-23.5 Flow Diag. Fig. 9.4-6	I-131[b], I-133	Air	Isotopic - I-131	Note 6	Note 4
1RE 5503	2RE 5503	Beta scintillator	Containment air noble gas (off-line)	El. 831 feet 6 in. Figures 12.3-7 and 12.3-23.5 Flow Diag. Fig. 9.4-6	Xe-133[b], Kr-85 Xe-135	Air	Gross beta	1E-06 to 1E-02	Note 2
1RE 406	2RE 406	Geiger-Mueller tube	Reactor coolant letdown line liquid / failed fuel (off-line)	El. 831 feet 6 in. Figures 12.3-7 and 12.3-23.5 Flow Diag. Fig. 9.3-10	Co-60[b], Co-58 Cs-134, Cs-137	Water	Gross gamma	1E-00 to 1E+05	Note 1
1RE 2325	2RE 2325	Geiger-Mueller tube	Main steam noble gas (on-line)	El. 873 feet 6 in. Figures 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	Xe-133[b], Kr-85 Xe-135	Steam	Gross gamma	1E-01 to 1E+03	Note 3
1RE 2326	2RE 2326	Geiger-Mueller tube	Main steam noble gas (on-line)	El.873 feet 6 in. Figures 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	Xe-133[b], Kr-85 Xe-135	Steam	Gross gamma	1E-01 to 1E+03	Note 3
1RE 2327	2RE 2327	Geiger-Mueller tube	Main steam noble gas (on-line)	El. 873 feet 6 inches Figures 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	Xe-133[b], Kr-85 Xe-135	Steam	Gross gamma	1E-01 to 1E+03	Note 3
1RE 2328	2RE 2328	Geiger-Mueller tube	Main steam noble gas (on-line)	El. 873 feet 6 in. Figures 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	Xe-133[b], Kr-85 Xe-135	Steam	Gross gamma	1E-01 to 1E+03	Note 3

TABLE 11.5-1
PROCESS RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 4 of 5)

Detector Nos.[a]									
Unit 1	Unit 2	Detector Type	Monitor Service	Monitor Locations (El., Figure Numbers)	Principal Isotopes Monitored	Monitored Medium	Measurement Made	Specified Instrument Range (uCi/cm3)	Bases for Alarm Set Points
IRE 2325A	2RE2325A	Gamma scintillator	N-16 (on-line)	El. 873 feet 6 in. Figure 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	N-16	Steam	Gamma	1E-03 to 1E+03 Note 9	Note 3
IRE 2326A	2RE2326A	Gamma scintillator	N-16 (on-line)	El. 873 feet 6 in. Figure 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	N-16	Steam	Gamma	1E-03 to 1E+03 Note 9	Note 3
IRE 2327A	2RE2327A	Gamma scintillator	N-16 (on-line)	El. 873 feet 6 in. Figure 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	N-16	Steam	Gamma	1E-03 to 1E+03 Note 9	Note 3
IRE 2328A	2RE2328A	Gamma scintillator	N-16 (on-line)	El. 873 feet 6 in. Figure 12.3-9 and 12.3-23.7 Flow Diag. Fig. 10.3-1	N-16	Steam	Gamma	1E-03 to 1E+03 Note 9	Note 3
Electrical and Control Building									
XRE 5895A XRE 5896A XRE 5895B XRE 5896B		Beta scintillator	Control Room ventilation intake (off-line)	El. 854 feet 4 in. El. 852 feet 6 in. Fig. 12.3-16 Flow Diag. Fig. 9.4-1	Xe-133[b], Xe-135 Kr-85, I-131	Air	Gross beta	1E-06 to 1E-02	Note 4
Turbine Building									
1RE 2959	2RE 2959	Beta scintillator	Condenser off-gas (off-line)	El. 803 feet Figures 12.3-23 and 12.3-23.1 Flow Diag. Fig 10.4-3	Kr-85, Xe-133[b] Xe-135	Gas	Gross beta	1E-05 to 1E-01	Note 3

TABLE 11.5-1
PROCESS RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 5 of 5)

Detector Nos.[a]									
Unit 1	Unit 2	Detector Type	Monitor Service	Monitor Locations (El., Figure Numbers)	Principal Isotopes Monitored	Monitored Medium	Measurement Made	Specified Instrument Range (uCi/cm3)	Bases for Alarm Set Points
1RE 5100	2RE 5100	Gamma scintillator	Turbine Building drains liquid (Adjacent-to-line)	El. 775 feet 3 in. Figures 12.3-21.1 and 12.3-22.1 Flow Diag. Fig 9.3-8	Co-60[b], Cs-134, Co-58, Cs-137	Water	Gross gamma	1E-05 to 5E-02	Note 2

a) Detector numbers preceded by an “X” (i.e., XRE 5701) are common to both units.

b) Reference nuclide

NOTES:

1. Alarm set points are based on process system requirements for normal operation.
2. Alarm set points are based on effluent requirements for normal operation.
3. Alarm set points are based on Steam Generator Tube Rupture (SGTR) detection.
4. Alarm set points are based on airborne radioactivity considerations for personnel protection.
5. Alarm set points are based on indication of containment integrity for accident conditions.
6. Since these iodine detectors monitor the accumulated activity of iodine in the iodine adsorber cartridge, the specified instrument range (uCi/Cm3) and MDC (uCi/Cm3) are dependent on the frequency of the cartridge replacement, flow rate, sensitivity, and detector count rate. The activity concentration may be calculated as follows:

$$\text{Activity } (\mu Ci/cm^3) = \frac{\text{Count Rate (counts/min)}}{\text{Time since cartridge replacement (min)} \times \text{Sample flow rate } (cm^3/min) \times \text{Fractional cartridge efficiency} \times \text{Sensitivity (counts per min/uCi)}}$$

The sensitivity of the detector is > 4E+4 cpm/uCi.

7. Alarm setpoints are based upon RCS pressure boundary leakage detection.
8. GM tube (RD-10) detector activity between 1E-03 and 1E-01 is outside the 2 sigma, 95% confidence level delineated by ANSI 13.10.
9. This indicates sensitivity range of detector. The readout is in gallons per day.

TABLE 11.5-2
PROCESS RADIATION MONITORING SYSTEM ENVIRONMENTAL DESIGN CONDITIONS

(Sheet 1 of 2)

Item	Temperature Range (F)		Relative Humidity (%)		Pressure (in. psig)	
	Normal	Accident	Normal	Accident	Normal	Accident
Airborne, Steam and Gaseous Channels						
a. Control Room electronics, printers and CRT displays	70 to 80 (air)	80 (air)	50	50	0.1	0.1
b. Local electronics and sampling equipment (pumps, motors, and so forth)	60 to 104 (air)	122 (air)	50 to 100	100	-0.1	Atmospheric
c. Detectors (except below) (Note 1)	40 to 122 (air)	122 (air)	50 to 100	100	-0.1	Atmospheric
d. Detector XRE 5250	40 to 180 (air)	180 (air)	50 to 100	100	-0.1	Atmospheric
e. Detectors RE 2325, 2326, 2327, and 2328	40 to 160 (air)	165 (air)	70	95		Atmospheric
f. Detectors RE 2325A, 2326A, 2327A, and 2328A	40 to 120 (air)	140 (air)	70	N/A		Atmospheric
Liquid Channels						
a. Control Room electronics, printer and CRT displays	70 to 80 (air)	80 (air)	50	50	0.1	0.1
b. Local electronics and sampling equipment (pumps, motors, and so forth)	60 to 104 (air)	122 (air)	50 to 100	100	-0.1	Atmospheric
c. Detector for Monitor RE 4200	40 to 122 (Liquid)	122 (Liquid)	N/A	N/A	--	--
d. Deleted						
e. Deleted						
f. Detectors for Monitors RE 4269, RE 4270, RE 4509 RE 4510, RE 4511	32 to 104 (Liquid)	122 (Liquid)	N/A	N/A	--	--
g. Detector for Monitor RE-5251A, RE-5100	40 to 180 (Liquid)	180 (Liquid)	N/A	N/A	--	--
h. Detector for Monitor RE-5253	32 to 122 (Liquid)	122 (Liquid)	N/A	N/A	--	--

TABLE 11.5-2
PROCESS RADIATION MONITORING SYSTEM ENVIRONMENTAL DESIGN CONDITIONS

(Sheet 2 of 2)

Item	Temperature Range (F)		Relative Humidity (%)		Pressure (in. psig)	
	Normal	Accident	Normal	Accident	Normal	Accident
i. Detector for Monitor RE-406	40-140 (Liquid)	140 (Liquid)	N/A	N/A	--	--

Note:

1. For detectors X-RE-5895B, X-RE-5896A, 1-RE-2959, 2-RE-2959, X-RE-5567A, X-RE-5567B, 1-RE-5503 and 2-RE-5503, the extreme maximum operational temperature for the process air entering the detectors is 140 °F.

TABLE 11.5-3
PROCESS RADIATION MONITORING SYSTEM CHARACTERISTICS

(Sheet 1 of 5)

	Channel Nos.		Indicator and Alarm Types		
Monitor	Unit 1	Unit 2	Local (Monitor System)	Remonte (Control Room)	Monitor Automatic Control Actions
Plant vent stack (noble gas)	XRE 5567A	XRE 5567B	None	Console (CRT, printer) with audible and Visible alarm (alert, high, and failure) and rack mounted operator panel with visible alarm and operator console CRT with visible alarm (alert, high and failure)	None
Plant vent stack (noble gas)	XRE 5570A	XRE 5570B	None	"	High radiation signal (Note 1) initiatives closure of valve HCV-014 in the GWPS.
Containment (particulate)	1RE 5502	2RE 5502	None	Console (CRT, printer) with audible and visible alarm (alert, high, and failure) and seismic panel with visible alarm and operator console CRT with visible alarm (alert, high and failure)	High radiation signal initiates Containment ventilation isolation.
Containment (iodine)	1RE 5566	2RE 5566	None	"	None

TABLE 11.5-3
PROCESS RADIATION MONITORING SYSTEM CHARACTERISTICS

(Sheet 2 of 5)

Monitor	Channel Nos.		Indicator and Alarm Types		
	Unit 1	Unit 2	Local (Monitor System)	Remonte (Control Room)	Monitor Automatic Control Actions
Containment (noble gas)	1RE 5503	2RE 5503	None	"	High radiation signal initiates Containment ventilation isolation.
Control Room Ventilation Intake	XRE 5895A XRE 5896A XRE 5895B XRE 5896B		None	Display/Operator Cabinet (LCD Display, printer) with audible and visible alarm (alert, high, and failure) and seismic/ Class 1E panel with visible alarm and operator console CRT with visible alarm (alert, high and failure)	High radiation signal initiates Control Room ventilation emergency recirculation through filters
Condenser off-gas	1RE 2959	2RE 2959	None	Display/Operator Cabinet (LCD Display, printer) with audible and visible alarm (alert, high, and failure) and operator console CRT with visible alarm (alert, high and failure)	None
Waste Gas	XRE 5250		GWPS panel, (ratemeter and alarm)	"	"

TABLE 11.5-3
PROCESS RADIATION MONITORING SYSTEM CHARACTERISTICS

(Sheet 3 of 5)

Monitor	Channel Nos.		Indicator and Alarm Types		Monitor Automatic Control Actions
	Unit 1	Unit 2	Local (Monitor System)	Remonte (Control Room)	
Steam generator blowdown sample	1RE 4200	2RE 4200	Steam Generator Blowdown Sample panel (alert alarm)	Display/Operator Cabinet (LCD Display, printer) with audible and visible alarm (alert, high and failure) and operator console CRT with visible alarm (alert, high and failure)	High radiation signal closes isolation valves in the blowdown and sample lines.
Auxiliary Building to LVW Pond	XRE 5251A		CPX-ECPRLV-01 (Auxiliary Bldg. to pond radiation detector control panel)	"	High radiation signal closes valve to LVW pond and diverts discharge to Cocurrent Wastewater Holdup Tanks
Waste Liquid	XRE 5253		LWPS panel (ratemeter and alert alarm)	"	High radiation signal closes discharge valve to circulating water discharge (Note 1)
Service Water	1RE 4269	2RE 4269	None	"	None
Service Water	1RE 4270	2RE 4270	None	Display/Operator Cabinet (LCD Display, printer) with audible and visible alarm (alert, high, and failure) and operator console CRT with visible alarm (alert, high and failure)	None

TABLE 11.5-3
PROCESS RADIATION MONITORING SYSTEM CHARACTERISTICS

(Sheet 4 of 5)

Monitor	Channel Nos.		Indicator and Alarm Types		
	Unit 1	Unit 2	Local (Monitor System)	Remonte (Control Room)	Monitor Automatic Control Actions
Component cooling water	1RE 4509	2RE 4509	None	"	"
Component cooling water	1RE 4510	2RE 4510	None	"	"
Component cooling water	1RE 4511	2RE 4511	None	"	"
Turbine building drains	1RE 5100	2RE 5100	Turbine Bldg. Sump Pump Control Panel (CP1-EIPRLV-25)	Display/Operator Cabinet (LCD Display, printer) with audible and visible alarm (alert, high, and failure) and operator console CRT with visible and alarm (alert, high and failure)	High radiation signal closes discharge valve to low volume waste pond directs discharge to a wastewater holdup tank. (Note 1)
Auxiliary Building vent duct	XRE 5701		Gaseous Waste Processing System (GWPS) Panel (ratemeter and alert alarm)	"	High radiation signal closes valve HCV-014 in GWPS (Note 1).
HVAC room vent duct	XRE 5702		None	"	None
Fuel Building vent duct	XRE 5700		None	"	None

TABLE 11.5-3
PROCESS RADIATION MONITORING SYSTEM CHARACTERISTICS

(Sheet 5 of 5)

Monitor	Channel Nos.		Indicator and Alarm Types		
	Unit 1	Unit 2	Local (Monitor System)	Remonte (Control Room)	Monitor Automatic Control Actions
Safeguards Building vent	1RE 5698	2RE 5698	None	“	None
Main Steam and FW Area	1RE 5637	2RE 5637	None	“	None
Failed fuel	1RE 406	2RE 406	None	“	None
Main Steam Line	1RE 2325 1RE 2326 1RE 2327 1RE 2328	2RE 2325 2RE 2326 2RE 2327 2RE 2328	None	“	None
Steam Generator Leak Rate Monitor	1RE 2325A 1RE 2326A 1RE 2327A 1RE 2328A	2RE 2325A 2RE 2326A 2RE 2327A 2RE 2328A	None	“	None

Note:

1. In addition to the high alarm signal, circuit failure, loss of counts, loss of flow, or channel out of service will initiate the control function.

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 1 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
Chemical and Volume Control System (CVCS)							
Recycle holdup tanks	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec.	N.A.	To determine purification requirements of waste and evaluate performance of recycle evaporator
Upstream of recycle evaporator feed demineralizers	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec.	N.A.	To evaluate performance of recycle evaporator feed demineralizer
Downstream of recycle evaporator feed demineralizer	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec.	N.A.	To evaluate performance of evaporator feed demineralizer
Recycle evaporator distillate	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec.	N.A.	To evaluate performance of recycle evaporator
Downstream of recycle evaporator condensate demineralizer	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec.	N.A.	To evaluate performance of recycle evaporator condensate demineralizer
Recycle Evaporator concentrates	As Required	Liquid Grab Sample	60	Primary Coolant	Gamma Spec	N.A.	To determine whether to reprocess, recycle or solidify
Refueling water tank	As Required	Liquid Grab Sample	60	Negligible	Gamma Spec.	N.A.	General monitoring

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 2 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
CCWS							
Downstream of the CCW pumps	As Required	Liquid Grab Sample	500	Negligible	Gamma Spec.	N.A.	To check on CCWS radiation monitors
SWS							
Downstream of the CCW heat exchangers	As Required	Liquid Grab Sample	500	Negligible	Gamma Spec.	N.A.	To monitor any in-leakage from the CCWS
Recyclable Waste Treatment System							
Waste holdup tank	As Required	Liquid Grab Sample	60	Table 11.2-2	Gamma Spec.	N.A.	To determine processing requirements, evaluate performance of waste evaporator
Waste evaporator distillate	As Required	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To evaluate performance of waste evaporator
Downstream of waste evaporator distillate demineralizer	As Required	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To evaluate performance of waste evaporator condensate demineralizer
Waste condensate tank	As Required	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To determine disposition of processed condensate
Waste evaporator concentrates	As Required	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To determine whether to recycle, return to BRS or to solid waste management system

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 3 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
Non-Reactor Grade Waste Treatment System							
Laundry and hot shower tank	As Required	Liquid Grab Sample	1000	Negligible	Gamma Spec.	N.A.	To determine whether to process or discharge
Upstream of waste monitor tank demineralizer	As Required	Liquid Grab Sample	1000	Will Vary	Gamma Spec.	N.A.	To evaluate performance of demineralizer
Downstream of waste waste monitor tank demineralizer	As Required	Liquid Grab Sample	1000	Will Vary	Gamma Spec.	N.A.	To evaluate performance of demineralizer
Primary Water Makeup System							
Reactor Makeup Water Storage tank	As Required	Liquid Grab Sample	1000	Will Vary	Gamma Spec.	N.A.	Evaluation of systems for recycling waste water
Spent Fuel Pool Cooling and Demineralizer System							
Spent Fuel Pool	Weekly	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To determine purification requirements; evaluate leakage from spent fuel
Demineralizer Effluent	Weekly	Liquid Grab Sample	60	Will Vary	Gamma Spec.	N.A.	To determine purification performance
Condensate and Feedwater System							

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 4 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
Steam generator blowdown or other secondary sample point	Per Tech Spec	Liquid Grab Sample	1000	Negligible	Dose Equivalent I-131	N.A.	Primary to Secondary leak determination
Pressurizer relief tank vapor space and/ or Pressurize Vapor Space	As Required	Gas Grab Sample	30	Concentrations of individual isotopes relative to one another is proportional to entrained noble gases in primary coolant.	Gamma Spec.	N.A.	To evaluate fuel cladding (PRT) integrity, PRT leakage, and necessity for PRT purging
Auxiliary Steam System							
Condensate	As Required	Liquid Grab Sample	1000	Will Vary	Gamma Spec.	N.A.	To monitor for radwaste evaporator tube leakage
Reactor Coolant Drain Tank	As Required	Gas Grab Sample	30	Concentrations of individual isotopes relative to one another is proportional to entrained noble gases in primary coolant.	Gamma Spec.	N.A.	To determine whether to recycle or process
Recycle Holdup Tanks	As Required	Gas Grab Sample	30	Concentration of individual isotopes relative to one another is proportional to entrained noble gases in primary coolant.	Gamma Spec.	N.A.	To evaluate tank leakage; determine whether to purge to GWPS
Radioactive Effluent Sampling							

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 5 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
Waste gas decay tanks	Per ODCM	Gas Grab Sample	30	Table 11.3-4 Gamma	Principal emitters	1.0E-04	Effluent monitoring
Containment atmosphere	Per ODCM	Gas and H-3 Grab Sample	30 (gas) 100 liters (H-3)	Table 12.2-26	Principal gamma emitters and tritium	1.0E-04 (gas) 1.0E-06 (H-3)	Effluent monitoring
Plant Vent collection header	Per ODCM	Gas, H-3, Particulate, charcoal or silver zeolite grab sample	1000(gas), 100 liters (H-3), The remaining samples are dependent upon operation	Concentrations will vary depending upon effluent releases occurring, other gaseous process considerations and airborne contaminants.	Principal gamma emitters, H-3 I-131, Sr-89 Sr-90, Gross Alpha Noble gas	1.0E-4 (gas) 1.0E-12 (I-131) 1.0E-6 (H-3) 1.0E-11 (Particulate, Gross alpha, Sr-89, Sr-90) 1.0E-6 (Noble gas)	Effluent monitoring
Liquid Process Systems							
Radioactive waste processing system (Waste Monitor Tanks & Laundry Holdup & Monitor Tanks)	Per ODCM	Liquid Grab Sample	500	Will vary Sr-89, Sr-90 Fe-55	Principal gamma emitters, I-131, Dissolved and entrained gasses, H-3, Gross alpha (Sr-89, Sr-90)	5.0E-7 (Principle Gamma Emitters) 1.0E-6 (I-131, Fe-55) 1. 0E-7 (Gross Alpha) 5.0E-8 1.0E-5 (dissolved and entrained noble gases, H-3)	Effluent monitoring or to determine whether to reprocess
Condensate polisher backwash recovery tanks & component cooling water drain tanks and turbine building sumps	Per ODCM	Liquid Grab Sample	1000	Negligible but can vary	Principal Gamma emitters I-131, H-3	1.OE-06 (I-131) 5.OE-07 (Principle Gamma Emitters) 1.OE-05 (H-3)	Effluent monitoring

TABLE 11.5-4
PROCESS AND EFFLUENT RADIOLOGICAL SAMPLING SYSTEM CHARACTERISTICS

(Sheet 6 of 6)

SAMPLING LOCATION	SAMPLING FREQUENCY	SAMPLE TYPE	EXPECTED SAMPLE AMOUNT (ml)	EXPECTED SAMPLE CONCENTRATIONS	ANALYSIS PERFORMED	REQUIRED LOWER LIMIT OF DETECTION (micro Ci/ml)	SAMPLE PROCESS
Low volume Waste Pond discharges	Per ODCM	Composite Sample over discharge period	1000	Negligible but can vary	Principal gamma emitters, I-131, Dissolved and entrained gasses, H-3, Gross alpha Sr-89, Sr-90 Fe-55	5.OE-7 (Principle Gamma Emitters) 1.0E-6 (I-131, Fe-55) 1.0E-7 (Gross Alpha) 5.0E-8 (Sr-89, Sr-90) 1.0E-5 (dissolved and entrained noble gases, H-3)	Secondary effluent monitoring
Waste Water Holdup Tanks (WWHT)	Per ODCM	Liquid Grab Sample	1000	Negligible but can vary	Principal gamma emitters, I-131, Dissolved and entrained gasses, H-3, Gross alpha Sr-89, Sr-90 Fe-55	5.OE-7 (Principle Gamma Emitters) 1.0E-6 (I-131, Fe-55) 1.0E-7 (Gross Alpha) 5.0E-8 (Sr-89, Sr-90) 1.0E-5 (dissolved and entrained noble gases, H-3)	Secondary effluent monitoring
Downstream of SG Blowdown Demineralizers	As Required	Liquid Grab Sample	1000	Will vary	Gamma Spec.	N.A.	To evaluate performance of demineralizers

11A COMPLIANCE WITH APPENDIX I

The dose evaluation presented in this Appendix was developed in support of the original license. These release and dose estimates are considered historical and not subject to future updating. This information is retained to avoid loss of original design basis.

The Offsite Dose Calculation Manual (ODCM) provides guidance requirements for system operation, dose calculations, and monitoring requirements to ensure CPNPP compliance with effluent limits. Actual measured concentrations of radioactivity released and real time dilution and dispersion estimates are utilized to verify compliance with effluent limits. Therefore, CPNPP operation within the requirements of the ODCM ensures compliance within effluent limits, rather than operations within the nominal assumptions utilized in the dose evaluation presented in this Chapter. Actual radioactivity release quantities and associated doses to the public can be found in the annual radioactive effluent release reports submitted to the NRC.

11A.1 CRITICAL PATHWAYS

Pathways of human exposure to plant radioactive emissions likely to account for most of the exposure from plant operation are discussed in this section. These discussions are based upon a knowledge of the characteristics of the site environment and on predicted effluent releases presented in Tables 11A-1 and 11A-4.

These analyses show compliance with 10CFR50 Appendix I design objectives with respect to reactor siting and obtaining an operating license for CPNPP Units 1 and 2. The analyses and discussions are considered complete except for potential future updates that may be required to reflect specific source term changes due to plant design modifications or changes in operating practices. Changes in the local site environment will normally be updated per routine annual land use census surveys according to the methodology and annual reporting requirements of the CPNPP Offsite Dose Calculation Manual (ODCM). Since the CPNPP ODCM effectively monitors Vistra OpCo's continued compliance with 10CFR50, Appendix I radioactive effluent limits, dynamic revision of this FSAR section due to local site environmental changes is not warranted.

A schematic depicting generalized potential pathways of human exposure to radionuclides in plant effluents is presented in Figure 11A-1, which has been taken from Appendix H of Regulatory Guide 4.2[1]. Studies of current and projected land and water use in the area around the Comanche Peak Nuclear Power Plant have revealed that some pathways shown in Figure 11A-1 do not present significant, potential pathways of exposure from plant effluents.

The cultivated land in the area of the site is primarily peanut fields and improved pasture, with some acreage in commercial vegetables such as squash, okra, and some tomatoes and cucumbers. Minor use is made of Lake Granbury waters for the irrigation of croplands [2]. By far the greatest amount of cultivated land is devoted to livestock production, of which beef production is most important. Ingestion of goat milk or crops actually cultivated on irrigated land are not considered significant pathways in evaluating the radiological impact of the plant. As is normally the case with nuclear plants located on or near fresh water bodies, ingestion of invertebrates or aquatic plants harvested from Squaw Creek Reservoir, Lake Granbury, or Lake Whitney does not present a significant exposure pathway to man. Since the water from Squaw Creek Reservoir is not used as a drinking water supply, ingestion of drinking water from the reservoir is not considered to be an exposure pathway.

The relative importance of the remainder of the potential pathways to man has been evaluated by calculating estimated doses from routine operation of the Comanche Peak Nuclear Power Plant for each pathway. The assumptions, methodology and results of the evaluation are presented below along with the summarized dose calculation results. Annual liquid and gaseous releases include consideration of an extended fuel cycle. Expected long term concentrations in Squaw Creek Reservoir and the resultant liquid pathway doses for the total body, liver, and GI-LLI are considered to be primarily affected by the longer fuel cycle and, therefore, have been evaluated numerically.

Examination of Table 11A-2 reveals that, based upon the dose calculation assumptions described above, the liquid pathway of primary importance in individual total-body exposure will probably be ingestion of fish caught in Squaw Creek Reservoir. Exposure from shoreline activity will generally be of less importance, and swimming and boating pathways would contribute only a minor amount to the dose from plant liquid effluents. The crop irrigation pathway is generally minor, except if Squaw Creek Reservoir water actually was to be used. Results of the dose calculations suggest that the relative importance of liquid pathways is about the same for individual organs as for total-body exposure. The relative importance of each liquid pathway is shown in Table 11A-2 for each of the three reservoirs (Squaw Creek Reservoir, Lake Granbury, and Lake Whitney).

A summary of the gaseous pathway dose calculation results which appears in Table 11A-3 indicates that the vegetable pathway produces the greatest total-body doses and the cow milk pathway yields the maximum organ dose to the thyroid gland of an infant. Cloud submersion, air inhalation, and meat consumption produce smaller contributions to the total dose; irradiation from ground-plane disposition of radionuclides in the plume is the least important total-body exposure pathway.

11A.2 DISPERSION OF NORMAL RADIONUCLIDE RELEASES INTO SURFACE WATERS

During normal operation of the plant, some radionuclide elements will be released into the circulated cooling water. The dispersion of these radionuclides in the surface water environment and an estimation of their maximum concentrations are discussed below.

The released radionuclides will enter the Squaw Creek Reservoir and follow the flow pattern shown in Figure 11A-2. Normally, only a small amount (typically 1759 acre-feet/year) of water will be released from Squaw Creek Reservoir into Squaw Creek. For blow-down purposes, water may be pumped from the reservoir to Lake Granbury, which is a lake created by the De Cordova Bend Dam built on the Brazos River. To keep the volume of Squaw Creek Reservoir constant, a typical volume of 27,900 acre-feet of water per year may be pumped from Lake Granbury into the Squaw Creek Reservoir to replace evaporation and blow-down losses. A portion of this water may be diverted into Lower Squaw Creek. The make-up water to Lake Granbury is provided by the Brazos River. Water released from the De Cordova Ben Dam will flow down the Brazos River 80 miles to Whitney Reservoir.

In order to calculate radionuclide concentrations due to normal releases in the surface water bodies a mathematical model was developed based upon mass balances using a modification of models found in NRC Reg. Guide 1.113[3]. The sketch of the model is presented in Figure 11A-3. The flow in Lake Granbury and Whitney Reservoir was assumed to be plug flow

and the flow in Squaw Creek Reservoir was assumed to be completely mixed. All of the reservoir volumes were assumed to be constant.

The terms used in developing the model are defined below:

C_0 = radionuclide concentration in the Brazos River upstream from Lake Granbury

C_1 = radionuclide concentration flowing into Squaw Creek Reservoir

= C_3 (for conservatism)

C_2 = radionuclide concentration in Squaw Creek Reservoir

C_3 = radionuclide concentration in Lake Granbury

C_4 = radionuclide concentration flowing into Lake Granbury

C_5 = radionuclide concentration of Whitney Reservoir

C_6 = inflow radionuclide concentration into Whitney Reservoir

q_b = pumpage rate to Lake Granbury

q_s = flow rate from Squaw Creek Reservoir to Squaw Creek

Q_B = Annual average flow rate in the Brazos River downstream from Lake Granbury

Q_{Bo} = annual average flow rate in the Brazos River downstream from Whitney Reservoir

Q_L = pumpage rate from Lake Granbury

Q_{EG} = evaporation from Lake Granbury

Q_{ES} = evaporation from Squaw Creek Reservoir

Q_{EW} = evaporation from Whitney Reservoir

$t_{1/2}$ = half-life of the radionuclide considered

V_G = volume of Lake Granbury

V_S = volume of Squaw Creek Reservoir

V_W = volume of Whitney Reservoir

W = rate of release of the radionuclide considered

For calculation purposes, the evaporation from Squaw Creek Reservoir is assumed to equal (Q_L-q_b-q_s). Evaporation in Lake Granbury averages 58 inches per year (Climatic Atlas of the United States, U.S. Department of Commerce, June 1969). The constant average volume of Lake Granbury is assumed to be 83,000 acre-feet, which is the average of the top gate capacity of 150,000 acre-feet and crest capacity of 15,000 acre-feet. The corresponding surface area for 83,000 acre-feet is about 5,000 acres. This area yields about 24,000 acre-feet of evaporation loss per year. Evaporation in Whitney Reservoir averages 54 inches and for calculation purposes, is equal to (Q_B-Q_{Bo}+q_s).

Using the complete mixing model, the concentration in Squaw Creek Reservoir can be found as shown below:

$$C_2 = \frac{(C_3 - Q_L) + W}{q_s + q_b + \left(\frac{V_s \cdot 1n2}{t_{1/2}}\right)} \qquad 11A\text{-}1$$

For the case of tritiated water only, an evaporation term for Squaw Creek Reservoir (Q_{ES}) is included:

$$C_2 = \frac{(C_3 - Q_L) + W}{Q_{ES} + q_s + q_b + \left(\frac{V_s - 1n2}{t_{1/2}}\right)} \qquad 11A\text{-}2$$

The concentration balance equation for water entering Lake Granbury is:

$$C_0 (Q_B + Q_L + Q_{Eg}) + C_{2qb} = (Q_B + Q_L + Q_{Eg}) C_4 \qquad 11A\text{-}3$$

$$C_0 = 0 \text{ (fresh makeup water)} \qquad 11A\text{-}3a$$

Using the plug flow formula, the concentrations in Lake Granbury and Whitney Reservoir can be found as shown:

$$C_3 = \exp\left[-\left(\frac{V_G \cdot 1n2}{t_{1/2} \cdot (Q_B + Q_L)}\right)\right] C_4 \qquad 11A\text{-}4$$

$$C_5 = \exp\left[-\left(\frac{V_w \cdot 1n2}{t_{1/2} \cdot Q_{Bo}}\right)\right] C_6 \qquad 11A\text{-}5$$

where $Q_{Bo} = Q_B + q_s$- Evaporation from Whitney Reservoir

Concentration Balance of water into Lake Whitney:

$$C_6(Q_B + q_s) = C_2q_s + C_3Q_B \qquad \text{11A-6}$$

The five unknowns (C_2, C_3, C_4, C_5, C_6) can be solved for from the equations presented above.

In the mathematical model above, the following constants were used:

q_s	=	1,759 ac-ft/yr
q_b	=	24,900 ac-ft/yr (for Lake Granbury and Whitney Reservoir Concentration Calculations)
	=	0 ac-ft/yr (for Squaw Creek Reservoir Concentration Calculation)
V_s	=	151,000 ac-ft
V_G	=	83,000 ac-ft
Q_B	=	1,000,000 ac-ft/yr
V_w	=	627,000 ac-ft
Q_{Bo}	=	994,000 ac-ft/yr

The steady state concentrations for each raionuclide element subject to normal releases were calculated using the mathematical model discussed above. The results for Squaw Creek Reservoir, Lake Granbury, and Whitney Reservoir are presented in Table 11A-1.

11A.3 ESTIMATED DOSES FROM LIQUID EFFLUENTS

Doses to individuals in the environs of the plant were calculated for all potentially significant liquid pathways. Doses to individuals were calculated for drinking water, aquatic food consumption, recreational activity and crop irrigation pathways, as applicable to Squaw Creek Reservoir, Lake Granbury, or Whitney Reservoir. Assumptions, including point of exposure, are described for each pathway in the following paragraphs; the calculated liquid pathway doses are summarized in Table 11A-2. Release values presented in Table 11A-1, which reflect two unit operation, were used. All results were based upon the computational techniques presented in Reg. Guide 1.109[4]. All usage and consumption values, transport times, bioaccumulation factors, dose conversion factors, and other constants utilized were those suggested in Reg. Guide 1.109 and Ref. [9]. The midpoint of plant life is 20 years, in accordance with Ref. [9].

The nearest downsteam municipal use of Brazos River water is at Waco, nearly 140 miles downstream of Lake Granbury. The Brazos River Water Authority has an application for 10,000 acre-feet of water from Lake Granbury for municipal use. At present, however, the city of Marlin (located 170 miles downstream of Lake Granbury) is the nearest municipal water user who has contracted to purchase water from the Brazos River Authority. Nevertheless, the dose

to an individual obtaining his entire annual water requirement from either Whitney Reservoir or Lake Granbury was estimated. It was assumed that 12 hours would elapse from the time of withdrawal to uptake by the individual. The maximum calculated dose to a single organ from this pathway was 7.14×10^{-2} mrem/yr to a child's GI-LLI; the maximum whole-body dose was 6.92×10^{-2} mrem/yr to a child. Drinking water doses at actual points of withdrawal are expected to be lower than those presented due to additional dilution and decay in transit.

Radionuclides released from the plant were assumed to be immediately available for uptake by aquatic food. Radionuclide concentrations in aquatic food were calculated on the assumption that they are in equilibrium with average concentrations in the respective reservoir where they reside after buildup during the assumed plant lifetime of forty years. The maximum predicted dose to a single organ from the fish consumption pathway was due to the fish from Squaw Creek Reservoir, 5.07 mrem/yr to a teen's liver. The maximum whole-body dose was 3.47 mrem/yr to an adult eating fish from Squaw Creek Reservoir.

Exposure to an individual engaging in recreational activity on the shore was evaluated with the 40 year equilibrium concentrations assumed the same at every point about the circumference of the particular reservoir. The doses from shoreline exposure are summarized in Table 11A-2. The maximum predicted total dose to a single organ from recreational activity at Squaw Creek Reservoir was 8.82×10^{-2} mrem/yr to the skin of a teen. The maximum calculated teen total-body dose was 7.52×10^{-2} mrem/yr.

Conservative assumption regarding the contaminated food pathway were made in order to evaluate this pathway in accordance with Appendix A, paragraph 2d of Reg. Guide 1.109. It was assumed that an individual was exposed to vegetation that was irrigated with contaminated water, and exposed to milk and meat from animals which consumed contaminated food and water. The contaminated water was assumed to originate from Lake Granbury or Whitney Reservoir, and the concentration of radionuclides existing in each reservoir was the same concentration used for irrigation and consumption by animals.

An average irrigation rate of 0.1 liter/square-meter/hour was used. The total dose to an individual was based upon ingestion of produce, leafy vegetables, goat-milk, and meat. The largest expected total body dose was 1.91×10^{-1} mrem/yr to a child influenced by water in Whitney Reservoir, and the maximum organ dose was 3.06×10^{-1} mrem/yr to the GI-LLI of an adult based on water from Lake Granbury. Doses for both reservoirs are shown in Table 11A-2. The calculated doses were still quite low considering the conservative assumptions used.

The maximum individuals doses calculated as described above were used to evaluate the status of conformance of predicted liquid effluents from the Comanche Peak Nuclear Power Plant with the requirements of Appendix I of 10 CFR Part 50. The assumptions and results of this evaluation are summarized in Table 11A-5. Note that the calculated doses indicate that liquid effluents from the plant conform to the "as low as reasonably achievable" criteria established in paragraphs IIA, IIB, and IIC of Appendix I [5].

11A.4 ESTIMATED DOSES FROM GASEOUS EFFLUENTS

Potential pathways of exposure of man to radionuclides in gaseous effluents from the CPNPP were identified and discussed previously. Dose to individuals in the environs of the plant were

calculated for all potentially significant pathways. The results of the calculations and the assumptions and methodology are described in the following sections.

All results were obtained based upon the computational techniques presented in Reg. Guide 1.109. Except in the case of an eight-month cow grazing season [7] and use of a 20 year midpoint of plant life [9], all usage and consumption values, transport times, bioaccumulation factors, dose conversion factors, and other constants utilized were those suggested in Reg. Guide 1.109.

Dilution factors for atmospheric pathways were calculated according to methods prescribed in Reg. Guide 1.111[8], as discussed in Chapter 11.

Maximum estimated doses to individuals were calculated for cloud submersion, ground plane contamination, inhalation, and cow's milk, goat's milk, vegetable, and meat ingestion pathways. Assumptions, including point of exposure, are described for each pathway in the following paragraphs; the calculated gaseous pathway doses are summarized in Table 11A-3. All estimates were based upon predicted gaseous releases in Table 11A-4. Each dose was calculated at the location of the highest dose offsite at which the pathway could be assumed to exist.

Exposure to an individual from submersion in a cloud containing radioactive noble gas effluents was evaluated at the exclusion boundary with the largest atmospheric dilution, 1.29 miles in the north-northwest sector, and at two residences, one 1.55 miles west and the other 1.95 miles west-northwest of the plant. The estimated total-body doses for the three locations were 1.52×10^{-1} mrem/yr, 3.59×10^{-2} mrem/yr, and 3.54×10^{-2} mrem/yr, respectively; corresponding skin doses were 3.91×10^{-1} mrem/yr, 9.23×10^{-2} mrem/yr, and 9.11×10^{-2} mrem/yr, respectively.

External irradiation from activity deposited on ground surfaces was also evaluated at the exclusion area boundary point with maximum concentrations as well as at the two above-mentioned most important residences. At the north-northwest boundary, 2.81×10^{-2} mrem/yr to the total-body and 3.29×10^{-2} mrem/yr to the skin can be expected annually. An individual at the residence 1.55 miles west of the plant will experience estimated doses of 3.83×10^{-3} mrem/yr to the total body and 4.49×10^{-3} mrem/yr to the skin, respectively, while at the residence 1.95 miles west-northwest of the plant, 4.11×10^{-3} mrem/yr to the total body and 4.82×10^{-3} to the skin are the estimated levels.

The maximum individual dose from the air inhalation pathway is reported at the same three locations at which plume submersion and ground deposition doses were reported. The maximum organ dose to an individual at these locations is expected to a teen's thyroid gland.

Inhalation of tritium causes about 79 percent of the thyroid dose. The adult thyroid doses are estimated as 4.73×10^{-1} mrem/yr, 1.12×10^{-1} mrem/yr, and 1.10×10^{-1} mrem/yr at the north-northwest site boundary, the residence at 1.55 miles west, and the residence at 1.95 miles west-northwest of the plant, respectively.

The predicted dose to an individual obtaining 76 percent (stipulated in Regulatory Guide 1.109) of his vegetable (including leafy vegetable) consumption from a garden adjacent to the two residences was estimated. Gardens were assumed to be located at 1.55 miles west and

1.89 miles west-northwest and correspond to the two above-mentioned residences. The maximum calculated exposure from this pathway was to a child's bone, 1.41×10^{-1} mrem/yr 1.55 miles west of the plant and 1.48×10^{-1} mrem/yr 1.95 miles west-northwest of the plant. Corresponding total-body doses at these locations were 5.62×10^{-1} and 5.91×10^{-1} mrem/yr. Approximately 99 percent of the bone dose and 50 percent of the total body dose was due to C-14.

For eight months of the year, cows and goats are expected to graze on pasture land exposed to radioparticulates deposited from the CPNPP effluent plume. Since an objective of this study was to determine the maximum point of exposure for each pathway, the meteorological diffusion conditions were used to select the highest dose location. It has been determined that two milk cows are used for home use and feeding calves at 1.55 miles in the west sector. In addition, two Holstein cows are reportedly milked for home use 1.89 miles west- northwest of the plant. The cow-milk pathway was evaluated at both locations since the resultant thyroid doses were deemed to be of the same magnitude. The maximum organ dose from ingestion of milk from a cow grazing eight months of the year at the two locations mentioned above occurred to an infant's thyroid gland and was estimated as 1.51 mrem/yr and 1.70 mrem/yr, respectively. It is postulated that the infant will receive the maximum total-body dose at those locations, 4.44×10^{-1} mrem/yr and 4.67×10^{-1} mrem/yr, respectively. Seventy percent of these thyroid doses are attributed to I-131, as expected, while 62 percent of the total body dose came from C-14.

There are indications that domestic goats may be used locally for milk in the vicinity of the plant. Therefore, consumption of goat's milk is considered an expected pathway to man during the life of the plant, even though there is no indication of the sale of goat's milk commercially. Five goats are milked locally at a site 4.18 miles north of the plant. In addition, about 30 brush goats are located in the same wind sector 3.37 miles from the plant. While it is remote that an individual will catch and milk these brush goats, the possibility cannot be ruled out. Therefore, both the milk goats at 4.18 miles north and the brush goats at 3.37 miles north were evaluated for the goat's milk pathway. It was estimated that the maximum organ dose is again to an infant's thyroid 8.17×10^{-1} mrem/yr from the milk goats at 4.18 miles north and 1.29 mrem/yr from the brush goats. The respective maximum total-body doses were to an infant at 2.14×10^{-1} mrem/yr and 3.10×10^{-1} mrem/yr.

Exposure from consumption of meat was evaluated at a site 2.12 miles east-southeast of the plant where there are yearlings among beef cattle but no milk cows. The maximum organ dose to an individual from ingestion of meat from cattle grazing eight months of the year was to a child's bone at this location, 1.14×10^{-1} mrem/yr. The maximum total body dose from the meat ingestion pathway was estimated as 3.15×10^{-2} mrem/yr.

Maximum individual doses calculated as described above were used to evaluate the status of conformance of predicted gaseous effluents from the CPNPP with the requirements of Appendix I [5]. The assumptions and results of this evaluation are summarized in Table 11A-6. Beta and gamma air doses were calculated in accordance with the methods stated in Reg. Guide 1.109[4]. It will be noted that the calculated doses indicate that the plant design conforms to the "as low as reasonably achievable" criteria established in paragraphs IIA, IIB, and IIC of Appendix I [5].

REFERENCES

1. Regulatory Guide 4.2, Revision 2, Preparation of Environmental Reports for Nuclear Power Stations, July 1976.

2. Texas Water Rights Commission, 1977, Personal Communications.

3. Regulatory Guide 1.113, Revision 1, Estimating Aquatic Dispersion of Effluents from Accidental and Routine Reactor Releases for the Purpose of Implementing Appendix I, April 1977.

4. Regulatory Guide 1.109, Revision 1, Calculation of Annual Doses to Man from Routine Releases of Reactor Effluents for the Purpose of Evaluating Compliance with 10 CFR Part 50, Appendix I, October 1977.

5. Title 10, Code of Federal Regulations, Part 50, Appendix I.

6. Deleted.

7. Final Environmental Statement, Comanche Peak Steam Electric Station Units 1 and 2, Texas Utilities Generating Company, Docket Nos. 50-445 and 50-446, United States Atomic Energy Commission (June 1974) p. 5-29.

8. Regulatory Guide 1.111, Revision 1, methods for Estimating Atmospheric Transport and Dispersion of Gaseous Effluents in Routine Releases from Light-Water-Cooled Reactors, July 1977.

9. NUREG/CR-4013 “LADTAP II - Technical Reference and User Guide,” April 1986.

TABLE 11A-1
EXPECTED CONCENTRATIONS[b] OF RADIOACTIVE MATERIALS IN ENVIRONMENTAL MEDIA FROM LIQUID EFFLUENTS OF THE COMANCHE PEAK NUCLEAR POWER PLANT - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 3)

Isotope	Annual[b] Release (Ci/yr)	Expected Long[c] Term Concentration of Squaw Creek Reservoir uCi/ml	Expected Long[c] Term Concentration of Lake Granbury uCi/ml	Expected Long[c] Term Concentration of Whitney Reservoir uCi/ml
H-3	5.19E+03	9.40E-05	6.39E-07	6.62E-07
CR-51	8.40E-02	4.95E-11	5.34E-13	1.96E-15
MN-54	4.60E-02	2.92E-10	4.90E-12	3.12E-12
FE-55	3.40E-02	6.56E-10	8.69E-12	7.90E-12
FE-59	8.40E-03	8.00E-12	1.10E-13	3.52E-15
CO-58	1.30E-01	1.96E-10	3.07E-12	3.59E-13
CO-60	1.56E-02	5.84E-10	5.83E-12	5.76E-12
ZR-95	1.10E-02	1.51E-11	2.32E-13	2.19E-14
NB-95	8.20E-03	6.08E-12	7.48E-14	8.81E-16
NP-239	3.40E-02	1.69E-12	1.54E-17	[d]
SR-89	4.00E-03	4.40E-12	6.35E-14	3.26E-15
MO-99	1.08E-01	6.39E-12	1.97E-16	[d]
TC-99M	1.02E-01	5.41E-13	[d]	[d]
RU-106	4.86E-01	3.72E-09	3.27E-11	2.28E-11
AG-11OM	3.80E-02	2.02E-10	3.43E-12	1.97E-12
TE-129M	5.20E-03	3.74E-12	4.54E-14	4.68E-16
I -131	1.06E-01	1.81E-11	3.96E-14	1.68E-22
TE-132	3.00E-02	2.07E-12	1.61E-16	[d]
I -132	3.00E-02	6.10E-14	[d]	[d]

TABLE 11A-1
EXPECTED CONCENTRATIONS[b] OF RADIOACTIVE MATERIALS IN ENVIRONMENTAL MEDIA FROM LIQUID EFFLUENTS OF THE COMANCHE PEAK NUCLEAR POWER PLANT - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

Isotope	Annual[b] Release (Ci/yr)	Expected Long[c] Term Concentration of Squaw Creek Reservoir uCi/ml	Expected Long[c] Term Concentration of Lake Granbury uCi/ml	Expected Long[c] Term Concentration of Whitney Reservoir uCi/ml
I- 133	1.06E-01	1.97E-12	3.79E-23	[d]
CS-134	6.46E-03	9.92E-11	4.81E-13	4.18E-13
I -135	3.40E-02	2.01E-13	[d]	[d]
CS-136	2.40E-04	6.61E-14	3.42E-16	2.08E-21
CS-137	5.72E-03	8.86E-10	1.77E-12	1.87E-12
CE-144	1.26E-01	7.51E-10	1.12E-11	6.92E-12
NA- 24	2.40E-01	3.18E-12	1.48E-26	[d]
NI- 63	4.00E-05	1.12E-11	2.57E-14	2.75E-14
ZN- 65	1.46E-02	7.50E-11	1.28E-12	7.21E-13
W -187	2.20E-02	4.65E-13	1.12E-22	[d]
SR- 90	5.66E-04	8.39E-11	1.99E-13	2.10E-13
Y - 90	1.60E-04	9.06E-15	2.08E-19	[d]
SR- 91	2.40E-03	2.05E-14	[d]	[d]
Y - 91M	1.62E-03	1.19E-15	[d]	[d]
Y - 91	3.20E-04	3.98E-13	5.96E-15	4.35E-16
Y - 93	1.20E-02	1.08E-13	[d]	[d]
NB-95M	8.00E-05	6.36E-15	1.04E-18	[d]
RU-103	2.18E-01	1.83E-10	2.19E-12	4.34E-14
RH-103M	2.00E-01	1.68E-13	[d]	[d]

TABLE 11A-1
EXPECTED CONCENTRATIONS[b] OF RADIOACTIVE MATERIALS IN ENVIRONMENTAL MEDIA FROM LIQUID EFFLUENTS OF THE COMANCHE PEAK NUCLEAR POWER PLANT - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

Isotope	Annual[b] Release (Ci/yr)	Expected Long[c] Term Concentration of Squaw Creek Reservoir uCi/ml	Expected Long[c] Term Concentration of Lake Granbury uCi/ml	Expected Long[c] Term Concentration of Whitney Reservoir uCi/ml
TE-129	3.40E-03	3.45E-15	[d]	[d]
TE-131M	1.54E-02	4.08E-13	4.04E-21	[d]
TE-131	2.80E-03	1.03E-15	[d]	[d]
I -134	6.00E-05	4.67E-17	[d]	[d]
BA-137M	2.60E-03	9.76E-17	[d]	[d]
BA-140	3.20E-01	8.68E-11	4.39E-13	2.22E-18
LA-140	5.00E-01	1.78E-11	7.36E-18	[d]
CE-141	4.00E-03	2.75E-12	3.24-14	2.66E-16
CE-143	3.20E-02	9.37E-13	3.67E-20	[d]
PR-143	4.40E-03	1.28E-12	7.09E-15	7.95E-20
PR-144	1.12E-01	2.85E-14	[d]	[d]

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) per two reactor site based on extended fuel cycle

c) 40 year maximum equilibrium concentration (Squaw Creek Reservoir concentration based on extended fuel cycle)

d) negligible concentration after 40 years

TABLE 11A-2
SUMMARY OF CALCULATED LIQUID PATHWAY DOSES[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 1 of 3)

Pathway	Location	Age Group	Organ Receiving Maximum Dose: Organ	Organ Receiving Maximum Dose: Dose (mrem/yr)	Total-Body Dose (mrem/yr)
Drinking Water	Lake Granbury	Adult	GI-LLI	5.51E-02	4.94E-02
		Teen	GI-LLI	3.90E-02	3.48E-02
		Child	GI-LLI	7.05E-02	6.68E-02
		Infant	GI-LLI	6.78E-02	6.54E-02
Drinking Water	Whitney Reservoir	Adult	GI-LLI	5.48E-02	5.12E-02
		Teen	GI-LLI	3.88E-02	3.61E-02
		Child	GI-LLI	7. 14E-02	6.92E-02
		Infant	GI-LLI	6.91E-02	6.77E-02
Fish Ingestion	Squaw Creek Reservoir	Adult	Liver	4.93E+00	3.47E+00
		Teen	Liver	5.07E+00	2.02E+00
		Child	Liver	4.53E+00	9.00E-01
Aquatic Food Ingestion	Lake Granbury	Adult	GI-LLI	5.63E-02	1.36E-02
		Teen	GI-LLI	4.02E-02	9.43E-03
		Child	Liver	2.24E-02	6.48E-03
Aquatic Food Ingestion	Whitney Reservoir	Adult	GI-LLI	3.63E-02	1.26E-02
		Teen	GI-LLI	2.60E-02	8.34E-03
		Child	Liver	1.90E-02	5.30E-03

TABLE 11A-2
SUMMARY OF CALCULATED LIQUID PATHWAY DOSES[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 2 of 3)

Pathway	Location	Age Group	Organ Receiving Maximum Dose: Organ	Organ Receiving Maximum Dose: Dose (mrem/yr)	Total-Body Dose (mrem/yr)
Shoreline Activity	Squaw Creek Reservoir	Adult	Skin	1.58E-02	1.34E-02
		Teen	Skin	8.82E-02	7.52E-02
		Child	Skin	1.84E-02	1.57E-02
Shoreline Activity	Lake Granbury	Adult	Skin	1.09E-04	9.27E-05
		Teen	Skin	6.09E-04	5.17E-04
		Child	Skin	1.27E-04	1.09E-04
Shoreline Activity	Whitney Reservoir	Adult	Skin	1.01E-04	8.59E-05
		Teen	Skin	5.64E-04	4.79E-04
		Child	Skin	1.17E-04	1.00E-04
Crop Irrigation	Lake Granbury	Adult	Liver	3.06E-01	1.02E-01
		Teen	Liver	2.61E-01	1.20E-01
		Child	Liver	2.05E-01	1.85E-01
		Infant	Liver	1.67E-01	1.56E-01
Crop Irrigation	Whitney Reservoir	Adult	GI-LLI	2.45E-01	1.05E-01
		Teen	GI-LLI	2.20E-01	1.24E-01
		Child	GI-LLI	2.51E-01	1.91E-01
		Infant	Liver	1.73E-01	1.61E-01

TABLE 11A-2
SUMMARY OF CALCULATED LIQUID PATHWAY DOSES[a] - ORIGINAL LICENSING BASIS[b]

(Sheet 3 of 3)

Pathway	Location	Age Group	Organ Receiving Maximum Dose: Organ	Organ Receiving Maximum Dose: Dose (mrem/yr)	Total-Body Dose (mrem/yr)
Milk-cow/Meat	Squaw Creek Reservoir	Adult	GI-LLI	3.98E+00	2.52E+00
		Teen	GI-LLI	3.71E+00	2.80E+00
		Child	GI-LLI	4.82E+00	4.28E+00
		Infant	Liver	5.88E+00	5.76E+00

a) Pathways from Squaw Creek Reservoir to total body, liver, and GI-LLI based on extended fuel cycle.

b) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11A-3
SUMMARY OF CALCULATED GASEOUS PATHWAY DOSES - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 2)

Pathway	Location	Age Group	Organ Receiving Maximum Dose: Organ	Organ Receiving Maximum Dose: Dose (mrem/yr)	Total-Body Dose (mrem/yr)
Cloud Submersion	Site Boundary (1.29 miles NNW)	All	Skin	3.91E-01	1.52E-01
Ground Plane	Site Boundary (1.29 miles NNW)	All	Skin	3.29E-02	2.81E-02
Contamination Air Inhalation	Site Boundary (1.29 miles NNW)	Adult	Thyroid	4.50E-01	3.76E-01
		Teen	Thyroid	4.73E-01	3.81E-01
		Child	Thyroid	4.47E-01	3.41E-01
		Infant	Thyroid	2.95E-01	1.98E-01
Vegetable Ingestion	Garden (1.89 Miles WNW)	Adult	Thyroid	4.37E-01	2.43E-01
		Teen	Bone	6.14E-01	3.14E-01
		Child	Bone	1.48E+00	5.91E-01
Cows' Milk Ingestion	1.89 Miles WNW	Adult	Thyroid	2.49E-01	8.79E-02
		Teen	Thyroid	3.87E-01	1.31E-01
		Child	Thyroid	7.63E-01	2.55E-01
		Infant	Thyroid	1.70E-00	4.67E-01

TABLE 11A-3
SUMMARY OF CALCULATED GASEOUS PATHWAY DOSES - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 2)

Pathway	Location	Age Group	Organ Receiving Maximum Dose: Organ	Organ Receiving Maximum Dose: Dose (mrem/yr)	Total-Body Dose (mrem/yr)
Goats' Milk Ingestion	Brush Goats 3.37 Miles N	Adult	Thyroid	1.98E-01	7.13E-02
		Teen	Thyroid	3.01E-01	1.00E-01
		Child	Thyroid	5.81E-01	1.80E-01
		Infant	Thyroid	1.29E-00	3.10E-01
Meat Ingestion	2.12 Miles ESE	Adult	Bone	7.16E-02	2.66E-02
		Teen	Bone	6.05E-02	1.94E-02
		Child	Bone	1.14E-01	3.15E-02

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

TABLE 11A-4
EXPECTED MAXIMUM OFF-SITE CONCENTRATIONS[a] OF RADIOACTIVE MATERIALS IN GASEOUS EFFLUENTS - ORIGINAL LICENSING BASIS[b]

Isotope	Annual [c] Release (Ci/Yr)	(Expected) [d] Exclusion Area Boundary Concentration (Ci/ml)
H^{3}	2.80E+03	2.93E-10
Kr^{83m}	2.00E+00	2.09E-13
Kr^{85m}	3.00E+01	3.14E-12
Kr^{85}	5.00E+02	5.23E-11
Kr^{87}	8.00E+00	8.37E-13
Kr^{88}	4.60E+01	4.81E-12
Xe^{131m}	1.40E+01	1.46E-12
Xe^{133m}	4.20E+01	4.39E-12
Xe^{133}	2.20E+03	2.30E-10
Xe^{135}	1.14E+02	1.19E-11
I^{131}	5.00E-02	5.23E-15
I^{133}	5.20E-02	5.44E-15
Mn^{54}	8.80E-04	9.21E-17
Fe^{59}	3.00E-04	3.14E-17
Co^{58}	3.00E-03	3.14E-16
Co^{60}	1.34E-03	1.40E-16
Sr^{89}	6.60E-05	6.91E-18
Sr^{90}	1.18E-05	1.23E-18
Cs^{134}	8.80E-04	9.21E-17
Cs^{137}	1.48E-03	1.55E-16
C^{14}	1.60E+01	1.67E-12
Ar^{41}	5.00E+01	5.23E-12

a) Per two reactor site

b) Historical, not subject to future updating. Has been retained to preserve original design basis.

c) No radioactive decay assumed.

d) Expected concentration in worst sector averaged over a one-year period.

TABLE 11A-5

APPENDIX I CONFORMANCE SUMMARY TABLE FOR LIQUID EFFLUENTS - ORIGINAL LICENSING BASIS[a]

	Appendix I Criteria		Comanche Peak Nuclear Power Plant[b]	
Type of Dose	Design[c] Objective	Point of Dose Evaluation	Calculated Dose	Point of Dose Evaluation
Liquid Effluents				
Dose to total body from all pathways	6 mrem/yr per site	Location of the highest dose offsite	5.99 mrem/yr[d]	Squaw Creek Reservoir[e]
Dose to any organ from all pathways	20 mrem/yr per site	Same as above	8.87 mrem/yr[f]	Squaw Creek Reservoir[e]

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) Points given correspond to points of dose evaluation under Appendix I heading.

c) Design objectives as specified in 10CFR50, Appendix I.

d) Dose to adult from fish ingestion, recreational exposure, and milk-cow and meat from animals drinking water.

e) 40 year equilibrium concentration maximum in Squaw Creek Reservior.

f) Dose to child liver from fish ingestion, recreational exposure, and milk-cow and meat from animals drinking water.

TABLE 11A-6

APPENDIX I CONFORMANCE SUMMARY TABLE FOR GASEOUS EFFLUENTS - ORIGINAL LICENSING BASIS[a]

Appendix I Criteria		Comanche Peak Nuclear Power Plant[b]		
Type of Dose	Design[c] Objective	Point of Dose Evaluation	Calculated Dose	Point of Dose Evaluation
Gaseous Effluents[d]				
Gamma dose in air	20 mrad/yr per site	Location of the highest dose offsite [e]	0.238 mrad/yr	Location of highest annual average concentration at the site boundary (NNW at 1.29 mile)
Beta dose in air	40 mrad/yr per site	Same as above	0.427 mrad/yr	Same as above
Dose to total body	5 mrem/yr per site	Location of the highest dose offsite [f]	0.152 mrem/yr	Same as above
Dose to skin of an individual	15 mrem/yr per site	Same as above	0.391 mrem/yr	Same as above
Dose to any organ from all pathways	30 mrem/yr per site	Location of the [g] highest dose offsite	2.20 mrem/yr[h]	Worst cow and garden[i] (WNW at 1.89 miles)

a) Historical, not subject to future updating. Has been retained to preserve original design basis.

b) Doses due to tritium and C-14 intake from terrestrial food chains are included in this category.

c) Design objectives as specified in 10CFR50, Appendix I

d) Calculated only for noble gases.

e) Evaluated at a location that could be occupied during the term of plant operation.

f) Evaluated at a location that is anticipated to be occupied during plant lifetime or evaluated with respect to such potential land and water usage and food pathways as could actually exist during the term of plant operation.

g) Evaluated at a location where an exposure pathway actually exists at time of licensing. However, if the applicant determines design objectives with respect to radioactive iodine on the basis of existing conditions and if potential changes in land and water usage and food pathways could result in exposures in excess of the guideline values given above, the applicant should provide reasonable assurance that a monitoring and surveillance program will be performed to determine: 1) the quantities of radioactive iodine actually released to the atmosphere and deposited relative to those estimated in the determination of design objectives; 2) whether changes in land and water usage and food pathways which would result in individual exposures greater than originally estimated have occurred; and 3) the content of radioactive iodine and foods involved in the changes if and when they occur.

h) Dose to a child's bone primarily from air inhalation, vegetation and cow milk ingestion.

i) Two Holstein cows currently milked at this location for home use.

12.0 RADIATION PROTECTION

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12.1 ENSURING THAT OCCUPATIONAL RADIATION EXPOSURES ARE AS LOW AS REASONABLY ACHIEVABLE

12.1.1 POLICY CONSIDERATIONS

The Vistra OpCo policy, through guidance from Regulatory Guides 1.8, 8.8, and 8.10 and 10 CFR Part 20, includes a strong commitment to ensure that occupational radiation exposures (ORE) are As Low As Reasonably Achievable (ALARA). Vistra OpCo will require acceptable radiation protection practices for all applicable station activities, and will provide a qualified and vigilant health physics organization and atmosphere to accomplish this goal. Management also recognizes and will emphasize the importance of each individual's responsibilities to maintain ORE-ALARA.

The Plant Manager (See Section 13.1.1.2.1) has the ultimate responsibility for establishing an ALARA philosophy for all applicable aspects of operating, maintaining, refueling and testing of the station. The operating organizational structure and responsibilities are discussed in Section 13.1.2. The Plant Manager informs station personnel of management's commitment to the ORE-ALARA philosophy and authorizes implementation of ALARA policies by issuing appropriate special orders, policy letters or administrative orders. The Manager, Nuclear Radiation Protection is responsible for implementing the ALARA policy.

The Manager, Nuclear Radiation Protection and his staff prepare or review applicable written procedures, develop dose-saving techniques and methods, provide health physics training and personnel guidance, and vigilantly observe for possible deviations from acceptable radiation protection practices.

12.1.2 DESIGN CONSIDERATIONS

Design considerations to ensure that occupational radiation exposures are as low as reasonably achievable, consistent with benefits received and costs incurred, are the following:

1. Applying experience from past and present engineering designs; cognizant engineers and designers have access to project files of completed projects and can contact personnel assigned to other current projects.

2. Applying experience from operating plants; applicable sources for operating plant experience are the following:

   a. NRC Current Events Reports

   b. NRC Operating Experience Bulletin Information Reports

   An example of "applying experience from operating plants" is the use of a concrete wall with a thickness of 2 ft 6 in. instead of a thin steel shell wall for the Refueling Water Storage Tank (RWST) because it has been found that for an operating plant the surface dose on the steel tank exceeds the upper limit of the radiation zone category for the unrestricted area.

3. Following facility and equipment design practices to reduce radiation level and maintenance; these practices are governed by NRC Regulatory Guide 8.8, Revision 2,

Section C.2, which requires that specific features be incorporated in the engineering design to limit ORE to ALARA. Details are presented in Section 12.3.1.

4. Introducing mechanisms that provide for design review by competent professionals in radiation protection and establishing procedures for continuing facility design review.

5. Providing general and specific guidance to individual designers for implementation of design practices which will reduce radiation levels and time spent for maintenance; this general guidance is provided by the comments generated during the review of design drawings.

6. When the operating experience of the Nuclear Steam Supply System (NSSS) supplier was transmitted to both the utility and the Architect/Engineer, design changes were made that could reduce ORE at a reasonable cost.

The primary method of minimizing and controlling the buildup, transport and deposition of activated corrosion products in the reactor coolant system of the Comanche Peak units is by controlling the materials in contact with the primary coolant system. The cobalt isotopes 58 and 60 are the major contributors to occupational radiation exposure during operation, inspection and maintenance of the plant. These isotopes are produced by activation of nickel-58 and cobalt 59, respectively, deposited on the core during power operation. These materials are deposited on the reactor core surfaces after erosion and/or corrosion of the material from piping, valves, etc. in contact with reactor coolant.

The NSSS materials in contact with the reactor coolant are restricted in the quantity of the trace impurity cobalt-59 in the stainless steel and Inconel. The restrictions are shown below:

Component	Material	Maximum Cobalt Content (Wgt %)
Reactor Internals (non-active region)	SS	0.20
Reactor Internals (active region)	SS	0.12
Reactor Vessel Cladding	SS	0.20
Reactor Coolant Piping	SS	0.20
Reactor Internals Bolting MH	SS	0.25
Reactor Coolant Pumps	SS	0.20
Steam Generator Tubes	Inconel	0.016 (Unit 1) 0.10 (Unit 2)
Fuel (non-active region)	SS	0.12
Fuel (active region)	SS	0.08
Fuel	Inconel	0.10

Fuel	Zircaloy	0.002

The other source of cobalt-59 which can be a major contributor to the total cobalt-59 deposited on the reactor core is Stellite hard surfacing materials. The use of Stellite in the NSSS is limited to the minimum application feasible. It is only used in those cases where other hard-surfacing materials are not determined to be suitable from a reliability standpoint. The Stellite application in the NSSS is given below:

Component	Approximate Surface Area
Reactor Internals	3.23 Ft^2
Reactor Coolant Pump Journals	4.30 Ft^2 per pump
Control Rod Drive Mechanisms	10.76 Ft^2
Reactor Coolant System Valves	2.6 Ft^2

The production of cobalt-58 is limited by minimizing the amount of nickel based alloys in contact with the reactor coolant. The reactor coolant system materials in contact with reactor coolant are stainless steel with the exception of steam generator tubing which is the nickel based alloy Inconel. Inconel is used in this application to increase component reliability thereby reducing inspection and maintenance requirements. This in turn leads to a reduction in total occupational radiation exposures. It is expected that the Model D series steam generators at Comanche Peak will exhibit the expected reliability which will minimize inspection and maintenance requirements.

Other methods of minimizing and controlling the buildup, transport and deposition of actuated corrosion products are under investigation by Westinghouse and others, principally under a significant program in this area funded by the Electric Power Research Institute (EPRI) of Palo Alto, California. One of the promising programs being carried out by Westinghouse under the EPRI program is the control of the deposition and transport by stricter controls on reactor coolant chemistry. This and other areas of Westinghouse investigation was presented to the ACRS Environmental Subcommittee on January 25, 1978. Many of these operational techniques, when proven, can be applied to the Comanche Peak units.

The CPNPP design of the auxiliary systems (BOP systems) incorporates recommendations of WCAP-8872 Section 8.0 with the purpose of maintaining occupational radiation exposures as low as reasonably achievable (ALARA).

12.1.3 OPERATIONAL CONSIDERATIONS

Operational considerations at CPNPP that promote the ALARA philosophy are the determination of the origins of radiation exposures, the proper training of personnel, the preparation of radiation protection procedures, the development of conditions for implementing these procedures, and the formation of a review system to assess the effectiveness of the ALARA philosophy.

The Manager, Nuclear Radiation Protection and his staff working closely with other departments review and study station systems such as the NSSS, the radioactive waste systems, the

Residual Heat Removal System, the Spent Fuel Pool Cooling and Cleanup System, and other systems that collect, store, contain, or transport liquid, gaseous, or solid radioactive material. Objectives are to understand the functional aspects of each system, to identity the origins of radiation exposures in the station, and to know these exposure origins by location, operation, and job category.

The radiation protection procedures, prepared and written by the Manager, Nuclear Radiation Protection and his staff, will emphasize acceptable health physics techniques and methods, and when practical will incorporate guidance from Regulatory Guides 8.8 and 8.10. The information received from other operating nuclear power plants, relating their approach and methods used in solving occupational exposure problems are carefully studied and provide significant assistance in the development of the CPNPP procedures. A description of the radiation protection procedures for CPNPP is located in Section 12.5.3.

The implementation of the ALARA philosophy is directed by the Manager, Nuclear Radiation Protection and is accomplished by two modes. In the review and approval of those plant procedures that include the potential for occupational exposure, appropriate radiation protection procedure may be referenced in the plant procedures to ensure that any occupational exposure is maintained as low as reasonably achievable. Entrance to the restricted areas at CPNPP is controlled by the radiation protection section and requires the issuance of a permit. The description of the permit system is discussed in Section 12.5.3. When the permit is initiated, the work assignment and applicable procedures are listed. As determined on a case-by-case basis, additional radiation protection procedures can be implemented at this time if needed.

The Manager, Nuclear Radiation Protection is responsible for the review of exposure records, investigating not only the individual exposures, but the exposures as classified by job description and job location. Information obtained from this review will be compared with exposure results from past experience and with data obtained from average exposure results from other plants to assess the effectiveness of the ALARA effort at CPNPP.

The Nuclear Oversight organization reviews the radiation protection program for compliance and effectiveness and provides a valuable assistance by this critique in assuring that ORE is ALARA.

The station management will periodically review the exposure records and discuss them with the Manager, Nuclear Radiation Protection This review will seek to identify excessive exposure areas, excessive exposures by job categories, and other exposure trends. They will determine if improvements are needed in plant procedures, radiation protection procedures, or plant equipment; and they will initiate these improvements if they will substantially reduce exposures at a reasonable cost.

In the radiation protection program, Section 12.5.3, there is provided a section for radiation protection training at CPNPP. All occupational workers will be instructed as to the philosophy, why Vistra OpCo has this philosophy, and how the ORE-ALARA philosophy is implemented at CPNPP.

12.2 RADIATION SOURCES

The radiation source term data provided in this section are the design basis values used for the design of plant shielding. As such they are considered historical and not subject to future updating. This information is retained to avoid loss of original licensing basis. As discussed in Chapter 12.0 compliance with occupational exposure limits and controls is ensured and controlled by compliance with 10CFR20 which is implemented at CPNPP via the Radiation Protection Program and the ALARA Policy.

The parameters used in calculating the reactor coolant inventory at the SPU power level is presented in [New] Table 11.1-1A and the concentrations presented in [New] Table 12.2-3A. (Comment: These tables should be obtained from Westinghouse.)

12.2.1 CONTAINED SOURCES

The radiation source terms used as the basis for shield design calculations and dose evaluations are presented for three general conditions of the reactor plant: full-power operation, shutdown, and postulated design basis accident (DBA).

12.2.1.1 Full-Power Operation Design Sources

12.2.1.1.1 Sources of Radiation Inside the Reactor Containment

The primary sources of contained radiation inside the Reactor Containment Building are the core and the Reactor Coolant System (RCS) equipment and pipes, which contain Nitrogen-16 (N-16), fission products from fuel clad defects (reactor coolant activities are calculated based upon fission products from rods generating one percent of the core thermal power), and activated nuclei and corrosion products. Argon-41 (Ar-41) production in the primary coolant is virtually nonexistent.

Tables 12.2-1 and 12.2-2 list the design basis N-16 concentrations around the reactor coolant loop (RCL) and the pressurizer source activities, respectively. Table 12.2-3 details reactor coolant fission and corrosion product activities. These source data represent equilibrium isotropic concentrations based on a 40-year plant life. Table 12.2-4 presents a four-group summary of the gamma and neutron fluxes at the surface of the primary shield concrete.

12.2.1.1.2 Sources of Radiation in Auxiliary Systems

Shielding thicknesses external to the Containment in the Auxiliary Building are determined by the amounts of radiation sources in the reactor plant systems and the adjacent radiation access requirements. The systems which contain radioactivity during operation and are considered in the shield design are the Chemical and Volume Control System (CVCS), Boron Recycle System (BRS), Waste Processing System (WPS), Steam Generator Blowdown Processing System (SGBPS), and the Condensate Polishing System (CPS).

1. Chemical and Volume Control System

   The CVCS functions primarily to provide continuous purification of reactor coolant water. In addition, two subsystems, the boron thermal regeneration system and the seal water

system, provide control of boron concentration in the reactor coolant and supply gland seal water to reactor coolant pumps, respectively.

The following major equipment of the CVCS contain sources of radiation:

a. Heat Exchangers

   1. Regenerative

   2. Excess letdown

   3. Letdown

   4. Boron thermal regeneration

   5. Seal water

b. Filters

   1. Reactor coolant

   2. Seal water injection

   3. Seal water return

c. Demineralizers

   1. Mixed bed (30 ft^3 of resin)

   2. Cation bed (20 ft^3 of resin)

   3. Boron thermal regeneration (70 ft^3 of resin)

d. Other Tanks and Equipment

   1. Volume control tank

   2. Charging pumps

The sources in the volume control tank are considered to correspond to normal operating levels in the tank and are specified as 160 ft^3 of liquid and 240 ft^3 of vapor. The filters are considered homogeneous cylindrical sources with the following dimensions and compositions:

Filter	Source Compositions (in.)	Source Dimensions (Volume %)

Reactor coolant	radius 3.375, length 19.0	62% air, 38% water
Seal water return	radius 3.375, length 19.0	62% air, 38% water
Seal water injection	radius 1.3, length 20.0	90% air, 10% stainless steel

Table 12.2-5 lists the design volumetric or specific source strengths as a function of gamma energies for the equipment in the CVCS.

Additional heat exchanger data are presented in Table 12.2-6.

2. Boron Recycle System

The BRS reclaims boric acid and primary coolant by demineralization, gas stripping, and evaporation. The following equipment in this system contain sources of radiation during normal operation:

a. Demineralizers

1. Recycle evaporator feed (30 ft^3 of resin)

2. Recycle evaporator condensate (20 ft^3 of resin)

b. Filters

1. Recycle evaporator feed

2. Recycle evaporator condensate

3. Recycle evaporator concentrates

c. Evaporator

1. Recycle evaporator

2. Vent condenser

d. Other Tanks and Equipment

1. Recycle holdup tank

2. Recycle evaporator feed pump

The sources in the recycle holdup tank are based on a vapor volume of 500 ft^3 in a holdup tank for a two-unit system. The filters are considered homogeneous cylindrical sources with the following dimensions and compositions:

Filter	Source Compositions (in.)	Source Dimensions (Volume %)
Evaporator feed	radius 3.375, length 19.0	62% air, 38% water
Evaporator condensate and evaporator concentrates	radius 1.25, length 19.0	30% air, 70% water

The vent condenser portion of the recycle evaporator is considered a homogeneous cylindrical source with a diameter of 8 in., a length of 20 in., and the following composition in volume percents: 30 percent stainless steel, 22 percent water, and 48 percent vapor. The evaporator section is considered as a cylinder with a diameter of 3.5 ft and a length of 9.9 feet. The composition in volume percents is 10 percent stainless steel, 77 percent water, and 13 percent air.

Table 12.2-7 lists the design volumetric sources or specific activities as a function of gamma energy for the equipment in the BRS.

3. Waste Processing System

The WPS is designed to store and process various radioactive byproduct materials of the plant. The WPS cleans and recycles the reactor coolant water, collects floor drains, collects radioactive gases, and processes all radioactive waste materials for safe release or transport.

Design basis source strengths at selected gamma energies for the equipment in the WPS are presented in Section 11.1.

4. Steam Generator Blowdown Processing System

The SGBPS provides filtration and demineralization of steam generator secondary coolant to remove radioactivity which can leak from the primary into the secondary coolant.

The filters are considered as homogeneous cylindrical sources 6.625 in. in diameter and 27.5 in. in length, weighing 25 lb (wet). Each demineralizer has 80 ft^3 of resin and the spent resin storage tank has a volume of 3700 gallons.

Table 12.2-8 presents the volumetric source strengths as a function of gamma energies in the filters, demineralizers and spent resin storage tank in the SGBPS.

12.2.1.2 Shutdown Design Sources

During normal shutdown conditions, the following are the primary sources of radiation:

1. The Residual Heat Removal (RHR) system

2. The Spent Fuel Pool Cooling and Purification (SFPCP) System

3. The reactor core fission decay product and activation sources

4. Corrosion product activity in major components

12.2.1.2.1 Residual Heat Removal Sources

The RHR System is placed into operation within four hours after shutdown to reduce the reactor coolant temperature to approximately 140°F. During this time, the RHR pump and heat exchanger are radioactive. Table 12.2-9 lists the maximum design specific activity in the RHR loop for this mode of operation. RHR heat exchanger dimension data are presented in Table 12.2-6.

12.2.1.2.2 Spent Fuel Pool Cooling and Purification Sources

The spent fuel pool is designed to act as a temporary repository for spent reactor fuel. The SFPCP system maintains proper water temperature and water conditions in this pool during and after refueling.

There are two spent fuel pool demineralizers supplied by Hungerford & Terry. Each spent fuel pool demineralizer has 50 ft^3 of resin. Resin used in these demineralizers are as per manufacturer/Westinghouse recommendation. The filter is considered a homogeneous cylindrical source, 10.25 in. in diameter and 20.5 in. in length, weighing 25 lb (wet). Table 12.2-10 lists the volumetric source strengths for the SFPCP filters and demineralizers.

Weekly samples are pulled to determine spent fuel pool demineralizer decontamination factors, and a surveillance will be performed to monitor differential pressures of the demineralizer and filters. Results of the decontamination factor determination which indicates an inability to control ionic contaminants and differential pressure readings which approach the manufacturer's design recommendations will indicate the need for resin and filter replacement. See Section 9.1.3.4 for a brief description of the chemical analyses and operational surveillance performed when the spent fuel pool is in use.

12.2.1.2.3 Core Shutdown Fission Decay and Activation Sources

The core gamma shutdown sources presented in Table 12.2-11 are based upon an average power assembly with an irradiation time of 10^8 sec, or 3.1 years. These source strengths are tabulated per unit volume of homogenized core material as a function of time after shutdown. The homogenization of the core material is in accordance with the volume fractions presented in Table 12.2-12.

The irradiated control rod sources are used in establishing shielding requirements during refueling operations and during shipping of irradiated rods. The neutron absorber material used in the control rods is silver-indium-cadmium (Ag-In-Cd). The source strengths associated with the control rods are presented in Table 12.2-13 as a function of various times after shutdown. The values are per cm of the height of a single rod for an irradiation period of 100,000 hours.

The in-core detector drive wire sources are used in establishing shielding requirements for the wires when the detectors are not in the core and during shipment when the detectors have failed. Table 12.2-14 lists the detector wire sources per cm of length of wire; it is assumed that the detector has been lodged in the core for one year.

12.2.1.2.4 Corrosion Product Activity in Major Components

Deposited crud activity in major components is a significant source of radiation to personnel in the immediate vicinity of these components. Table 12.2-2 presents deposited crud concentrations in the pressurizer.

The activities on the primary side surfaces of the steam generator are used to determine access limitations in and around the steam generators at plant shutdown. Nominal values of deposited activity are listed in Table 12.2-15 for several operating times.

12.2.1.3 Design Basis Accident Sources

The fission product sources considered released to the Containment following an equivalent 100 percent core meltdown are based on the following assumptions stated in TID-14844, Calculation of Distance Factors for Power and Test Reactor Sites [1]:

1. 100 percent of the noble gases

2. 50 percent of the halogens

3. 1 percent of the remaining fission product inventory

The sources from which the postaccident radiation levels outside the Containment are calculated appear in Table 12.2-16.

The loss of coolant accident (LOCA) is the design basis accident (DBA) used to develop the radiation source terms for equipment qualification and vital area dose evaluation. The post-accident radiation sources of importance are listed in Table 12.2-16, along with the fractions of core activity inventory mixed in the respective fluids. The appropriate source concentrations presented below represent those developed to support the original license. The impact of the Stretch Power Uprate on post-accident radiation environments was developed using source term scaling techniques using the core inventory utilized to support the original license and that developed for the Stretch Power Uprate and the fractional mix presented in Table 12.2-16.

12.2.1.3.1 Containment Atmosphere

The airborne activity within the containment building is the source of radiation to areas outside the containment, in the form of direct radiation through the walls, streaming through penetrations,

and skyshine. The containment atmosphere source terms at various times after a LOCA are given in Table 12.2-17 for Unit 1 and in Table 12.2-17A for Unit 2.

12.2.1.3.2 Containment Atmosphere in the Post-Accident Sampling System

The post accident sampling system, which is located outside the containment building, is not used post accident.

12.2.1.3.3 Primary Coolant

The pressurized LOCA is a postulated LOCA event in which the primary coolant system is conservatively assumed not to depressurize. The systems which may include pressurized primary coolant after a pressurized LOCA include the reactor coolant system and the residual heat removal system. The corresponding primary coolant source terms at various times after a LOCA are provided in Table 12.2-19 for Unit 1 and in Table 12.2-19C for Unit 2.

12.2.1.3.4 Containment Sump Water

The systems outside the containment building which may contain sump water after a depressurized LOCA include the residual heat removal system, the containment spray system, and portions of the safety injection and chemical and volume control systems. The sump water source terms at various times after a LOCA are given in Table 12.2- 19A for Unit 1 and in Table 12.2-19D for Unit 2.

12.2.1.3.5 Post-Accident Sampling System

The post-accident sampling system is not used at CPNPP. Post-accident samples can be obtained using the Process Sampling System (PSS) which may contain either containment atmosphere concentrations described in Section 12.2.1.3.2, containment sump water described in Section 12.2.1.3.4, or primary coolant described in Section 12.2.1.3.3 (with appropriate adjustment for fluid density). The source terms for the latter case at various times after a LOCA are presented in Table 12.2-19B for Unit 1 and in Table 12.2-19E for Unit 2.

12.2.1.3.6 Gap Activity Release

The gap activity is the fraction of core activity that diffuses to the gap between the fuel and cladding. For design basis accidents, the noble gas and iodine inventory in the fuel gas is assumed to be 10 percent of the core noble gas and iodine inventory, except for Kr-85 which is assumed to be 30 percent. The design basis gap activities are presented in Table 15.6-8.

12.2.1.4 Byproduct, Source and Special Nuclear Materials

Information on the projected types, quantities, forms and uses of byproduct, source and special nuclear material as defined in 10 CFR Part 30, 10 CFR Part 40 and 10 CFR Part 70 is provided in Table 12.2-20.

These radioactive materials are handled in accordance with the controls described in Section 12.5.3.7, “Radioactive Materials Control”. Special attention is given to the 5000 Ci Cs-137 source used for calibration of portable radiation detection instrumentation, including high range instrumentation available for performing surveys during emergency conditions. This

source is located in the Radiation Protection Calibration Facility and is housed in a Calibration Well Assembly. The calibration facility, well assembly, shield design, physical barriers and administrative and procedural controls associated with this source are discussed in Section 12.5.2.1.

12.2.1.5 Assumptions, Calculation Methods and Design Basis for Contained Radiation Sources

Radiation shielding is designed to provide radiation protection for plant personnel during plant operation at maximum calculated thermal power and to limit the normal operation radiation levels at the exclusion area boundary to below those levels allowed for continuous nonoccupational exposure. The plant is capable of continued safe operation with fuel cladding defects in the equivalent to one percent of the fuel rods.

In addition, the shielding provided ensures that, in the event of a hypothetical accident, the integrated offsite exposure due to the contained activity does not result in any harmful offsite radiation exposures.

The documents 10 CFR Part 20 and 10 CFR Part 100, or applicable portions thereof, are used as the bases for defining acceptable exposures in areas within and beyond the exclusion area boundary. The access requirements for each piece of plant equipment, instrumentation, or control equipment used in the plant are considered in setting the radiation zone for that area. The major radiation sources considered in the Containment Building are the core fission neutrons and gammas and the N-16 activity while at power and the spent fuel sources during refueling operations. The major source of activity in the Auxiliary Building is that activity associated with the assumption of small cladding defects in the equivalent of one percent of the fuel rods.

The calculation model for determining contained radiation sources used for shielding design is based on the design parameters given in Tables 12.2-12 and 12.2-21 through 12.2-23.

The assumptions used in analyzing postulated accident situations are listed in Table 12.2-24. The sources associated with an equivalent core meltdown accident are consistent with those set forth in TID-14844, Calculation of Distance Factors for Power and Test Reactor Sites [1]. The sources associated with a gap activity release accident are based on the assumption that the fission product activity in the space between the fuel rods and the cladding is released as a result of cladding failure.

Additional assumptions, calculation methods and design bases for contained radiation source data are given in Section 11.1, Source Terms.

12.2.1.6 The Old Steam Generator Storage Facility

The Old Steam Generator Storage Facility is designed to store the four Unit 1 Old Steam Generators and the Old Reactor Vessel Heads and associated Control Rod Drive Mechanisms for Units 1 and 2. The facility satisfies all design requirements and criteria for temporary storage of radioactive materials contained within the Old Steam Generators and Reactor Vessel Heads. The radiological design of the Old Steam Generator storage facility provides adequate shielding to satisfy Comanche Peak licensing basis requirements as per 10CFR Part 20 and incorporates ALARA design features as per Regulatory Guide 8.8.

The maximum contact dose rates at the facility surfaces including walls, door shield blocks, and roofs is less than or equal to 0.25 mR/hr. This results in a radiation zone classification of zone 1 as per UFSAR Table 12.3-6 and allows unrestricted access in the adjacent areas.

Personnel access to the Old Steam Generator Storage Facility openings is controlled by Radiation Protection personnel. Access to the facility will be infrequent, and there are no appreciable normal releases of the radiological materials from the facility.

12.2.2 AIRBORNE RADIOACTIVE MATERIAL SOURCES

12.2.2.1 Introduction

This section discusses the models, parameters, and assumptions necessary to evaluate airborne concentrations of radionuclides during normal plant operations in areas where personnel access may be required. The plant conditions that are considered for the calculations of expected concentrations of radioactive airborne material include normal power operations, shutdown, and refueling operations. The final design of each reactor unit ensures that the expected airborne isotopic concentrations in the normally accessible areas during normal operating conditions are below the maximum permissible concentrations. Expected airborne concentrations in areas requiring infrequent or periodic access are also acceptable, considering the expected occupancy in the regions.

12.2.2.2 Areas Containing Sources of Airborne Radioactivity

The major buildings considered to contain sources of airborne radioactivity are as follows:

1. Fuel Building
2. Safeguards Buildings
3. Auxiliary Building
4. Turbine Buildings
5. Reactor Containment Buildings

All other areas are considered to have negligible leak sources for airborne radioactivity.

12.2.2.3 Sources of Airborne Radioactive Material

Plant systems and equipment that handle radioactive fluids are potential sources of airborne radioactive material. The most significant of these potential sources which are included in the analysis are the Reactor Coolant System (RCS), Chemical and Volume Control System (CVCS), Residual Heat Removal System (RHRS), Spent Fuel Pools, and the Main Steam System. The specific leakage activities are dependent upon the concentrations of radionuclides in these systems. Potential leakage points considered for these source activities include the opening of normally closed systems and components, such as reactor vessel head removal and servicing of valves and pumps, transfer of spent fuel, and leakage from components in service, such as leakage from valve stem seals and leak-offs, pump shaft seals, relief valve discharges, vents, and drains.

12.2.2.4 Airborne Concentration Calculation Methodology

The basic methodology of NUREG-0017, Revision 1 and conservative assumptions and parameters are used for the calculation of airborne radioactivity concentrations. These assumptions and parameters including the concentrations of radioisotopes in the primary coolant, secondary water and steam are listed in Table 12.2-25. Equations 1 and 2 are used to calculate the airborne concentrations of the radioisotopes presented in Table 12.2-26. For each case, it is assumed that the initial concentration is zero, and that the leakage has occurred long enough to reach the equilibrium condition in that area.

$$C_i(t) = (LR)_i \, A_i \, (PF)_i \, \frac{(1 - e^{-\lambda Ti^t})}{V^{\lambda Ti}} \qquad \text{(Equation 1)}$$

$(LR)_i$ = leak or evaporation rate of the i^{th} radioisotope in the region (grams/sec).

A_i = activity concentration of the i^{th} leaking or evaporating radioisotope in μ Ci/gram.

$(PF)_i$ = partition factor, or fraction of the leaking activity that is airborne for the i^{th} radioisotope.

$^{\lambda}Ti$ = total removal rate constant for the i^{th} isotope from the region (sec^{-1}). This value is the sum of the radioactive decay constant (ln(2) / half life), and the ventilation rate in the room (air changes per second).

t = time interval between the start of the leak and the time at which the concentration is evaluated (sec).

V = volume of the region in which the leak occurs (cc).

Ci(t) = airborne concentration of the ith isotope at time t in the region (μCi/cc).

With high exhaust rates, the equilibrium concentration will be reached within a few hours. From Equation 1, it is evident that this concentration may be determined as shown in Equation 2, below:

$$C_i,Eq = (LR)_i \, A_i \, (PF)_i \, /(V^{\lambda}Ti) \qquad \text{(Equation 2)}$$

The design of the ventilation systems considers airborne radioactivity. This design provides flow paths from areas of low potential airborne concentration to areas of higher potential concentrations. Also, the ventilation design provides exhaust flow directly into the Primary Plant Ventilation Exhaust System supply plenum for most rooms with the highest expected airborne concentrations.

REFERENCES

1. TID-14844, Calculation of Distance Factors for Power and Test Reactor Sites.

2. NUREG-0017, Revision 1, Calculation of Releases of Radioactive Materials in Gaseous and Liquid Effluents from Pressurized Water Reactors.

TABLE 12.2-1
REACTOR COOLANT N-16 ACTIVITY - ORIGINAL LICENSING BASIS[a]

Position in Loop	Loop Transit Time (sec)	N-16 Activity (uCi/g)
Leaving core	0.0	136
Leaving reactor vessel	1.3	113
Entering steam generator	1.7	109
Leaving steam generator	5.8	74
Entering reactor coolant pump	6.5	69
Entering reactor vessel	7.2	65
Entering core	9.2	54
Leaving core	9.9	136

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-2
PRESSURIZER SOURCE ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 2)

Isotope	Activity (uCi/g)
Liquid Phase	
N-16 (max.)	1.3
Rb-88	1.1×10^{-2}
Mo-99	2.2
I-131	1.6
I-132	2.0×10^{-2}
I-133	7.0×10^{-1}
I-134	5.5×10^{-3}
I-135	1.4×10^{-1}
Cs-134	2.5×10^{-1}
Cs-136	1.1×10^{-1}
Cs-137	1.3
Cs-138	5.5×10^{-3}
Steam Phase	
Kr-85	5.1×10^{1}
Kr-85M	1.0×10^{-1}
Kr-87	1.8×10^{-2}
Kr-88	1.2×10^{-1}
Xe-131M	4.7
Xe-133	3.6×10^{2}

TABLE 12.2-2
PRESSURIZER SOURCE ACTIVITIES - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 2)

Isotope	Activity (uCi/g)
Xe-133M	1.8
Xe-135	6.5×10^{-1}
Xe-135M	5.0×10^{-4}
Xe-138	2.2×10^{-3}
Deposited Crud	
Cr-51	9.8×10^{-2}
Mn-54	1.5×10^{-1}
Mn-56	2.2×10^{-2}
Co-58	3.8
Co-60	1.6×10^{-1}
Fe-59	1.4×10^{-1}

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-3
DESIGN BASIS REACTOR COOLANT FISSION AND CORROSION PRODUCT ACTIVITY - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 2)

Isotope	Activity (uCi/g)	Isotope	Activity (uCi/g)
H-3	3.5 (max.)	Cs-136	0.15
Br-84	4.3×10^{-2}	Cs-137	1.5
Rb-88	3.7	Cs-138	0.98
Rb-89	1.1×10^{-1}	Ba-140	4.3×10^{-3}
Sr-89	3.3×10^{-3}	La-140	1.5×10^{-3}
Sr-90	1.7×10^{-4}	Ce-144	3.4×10^{-4}
Sr-91	1.9×10^{-3}	Pr-144	3.4×10^{-4}
Sr-92	7.4×10^{-4}	Kr-85	8.8 (peak)
Y-90	2.0×10^{-4}	Kr-85M	2.1
Y-91	6.1×10^{-3}	Kr-87	1.2
Y-92	7.2×10^{-4}	Kr-88	3.7
Zr-95	7.0×10^{-4}	Xe-131M	1.9
Nb-95	6.9×10^{-4}	Xe-133	2.81×10^{2}
Mo-99	5.3	Xe-133M	3.1
I-131	2.5	Xe-135	6.3
I-132	0.9	Xe-135M	0.7
I-133	4.0	Xe-138	0.7
I-134	0.6	Cr-51	9.5×10^{-4}
I-135	2.2	Mn-54	7.9×10^{-4}
Te-132	0.26	Mn-56	3.0×10^{-2}
Te-134	2.9×10^{-2}	Co-58	2.6×10^{-2}
Cs-134	0.3	Co-60	7.7×10^{-4}

TABLE 12.2-3
DESIGN BASIS REACTOR COOLANT FISSION AND CORROSION PRODUCT ACTIVITY - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 2)

Fe-59	1.1×10^{-3}

a) Historical. Not subject to future updating. Has been retained to preserve original design basis. Design basis RCS source term for SPU conditions presented in Table 12.2-3A.

TABLE 12.2-4
NEUTRON AND GAMMA FLUXES[a] AT THE SURFACE OF THE PRIMARY CONCRETE - ORIGINAL LICENSING BASIS[b]

(Sheet 1 of 2)

Maximum Neutron Fluxes Incident on the Primary Concrete

E > 1.0 MeV	7.6×10^{8} n/cm^2-sec
5.53 KeV < E < 1.0 MeV	1.2×10^{10} n/cm^2-sec
0.625 eV E < 5.53 keV	7.1×10^{9} n/cm^2-sec
E < 0.625 eV	1.8×10^{9} n/cm^2-sec

Maximum Gamma Ray Energy Fluxes Incident on the Primary Concrete

Group	Flux (MeV/cm^2-sec)	Group Energy (MeV)
1	3.7×10^{9}	7.5
2	3.3×10^{9}	4.0
3	1.7×10^{9}	2.5
4	1.0×10^{9}	0.8

Normalized Axial Variation for Neutron and Gamma Fluxes

Distance Above and Below Core Midplane (ft)	F
0	1.0
1	1.0
2	0.99
3	0.93
4	0.76
5	0.51
6	0.30

TABLE 12.2-4
NEUTRON AND GAMMA FLUXES[a] AT THE SURFACE OF THE PRIMARY CONCRETE - ORIGINAL LICENSING BASIS[b]

(Sheet 2 of 2)

a) Fluxes given at core midplane

b) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-5
RADIATION SOURCE ACITVITIES FOR EQUIPMENT IN THE CHEMICAL AND VOLUME CONTROL SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 3)

Letdown Coolant Fission and Corrosion Product Activity

Gamma Energy (MeV)	Specific Activity (MeV/g-sec)
0.4	4.5×10^5
0.8	2.7×10^5
1.3	1.7×10^5
1.7	1.2×10^5
2.2	1.4×10^5
2.5	1.6×10^5
3.5	1.9×10^4

Filter Activities

Gamma Energy (MeV)	Reactor Coolant (MeV/cm^3-sec)	Seal Water Injection (MeV/cm^3-sec)	Seal Water Return (MeV/cm^3-sec)
0.8	5.7×10^7	4.8×10^7	1.1×10^7
1.3	1.5×10^7	1.2×10^7	3.0×10^6

Demineralizer Activities

Gamma Energy (MeV)	Mixed Bed (MeV/cm^3-sec)	Cation Bed (MeV/cm^3-sec)	Boron Thermal Regeneration[b] (MeV/cm^3-sec)
0.4	1.5×10^8	1.9×10^6	2.5×10^6
0.8	3.4×10^8	3.3×10^8	1.7×10^6
1.3	3.3×10^7	4.8×10^6	6.8×10^5
1.7	1.7×10^7	3.4×10^6	3.3×10^5
2.2	4.5×10^6	-	1.2×10^5
2.5	1.0×10^5	-	-

TABLE 12.2-5
RADIATION SOURCE ACITVITIES FOR EQUIPMENT IN THE CHEMICAL AND VOLUME CONTROL SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

3.5	2.0×10^5	-	-

Volume Control Tank

Gamma Energy (MeV)	Vapor Phase (MeV/cm^3-sec)	Liquid Phase[b] (MeV/gm-sec)
0.1	2.3×10^6	3.1×10^5
0.4	5.0×10^5	7.0×10^4
0.8	1.7×10^5	4.5×10^4
1.3	-	1.6×10^4
1.7	1.3×10^5	2.6×10^4
2.2	3.0×10^5	4.0×10^4
2.5	6.2×10^5	8.3×10^4
3.5	-	1.7×10^3

Regenerative Heat Exchanger and Excess Letdown Heat Exchanger (Specific Source Strength)

Gamma Energy (MeV)	Regenerative-Shell Excess Letdown Tubes (MeV/g-sec)	Regenerative[c] Tubes (MeV/g-sec)
0.4	4.5×10^5	3.8×10^5
0.8	2.7×10^5	4.5×10^4
1.3	1.7×10^5	1.6×10^4
1.7	1.2×10^5	2.6×10^4
2.2	1.4×10^5	4.0×10^4
2.5	1.6×10^5	8.3×10^4
3.5	1.9×10^4	1.7×10^3
6.1	2.2×10^6	-
7.1	1.8×10^5	-

TABLE 12.2-5
RADIATION SOURCE ACITVITIES FOR EQUIPMENT IN THE CHEMICAL AND VOLUME CONTROL SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

Thermal Regeneration

	Letdown Heat Exchanger, Tube Side Seal Water Heat Exchanger, Tube Side	Moderating Heat Exchanger, Tube and Shell Sides, Chiller Heat Exchanger, Tube Side, Letdown Reheat Heat Exchanger Tube and Shell Sides
Gamma Energy (MeV)	Specific Source Strength (MeV/g-sec)	Specific Source Strength (MeV/g-sec)
0.4	4.5×10^5	4.2×10^5
0.8	2.7×10^5	7.6×10^5
1.3	1.7×10^5	1.6×10^4
1.7	1.2×10^5	3.9×10^4
2.2	1.4×10^5	1.2×10^5
2.5	1.6×10^5	1.6×10^5
3.5	1.9×10^4	1.7×10^3

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

b) These data are also applicable to charging pump sources.

c) Modified to account for activity removed by demineralizers and volume control tank

TABLE 12.2-6
HEAT EXCHANGER DESIGN DATA

(Sheet 1 of 2)

Heat Exchanger		Fluid Volume (ft^3)	
		Shell	Tubes
1.	Excess letdown	4.06	1.15
2.	Letdown	195 gal	80 gal
3.	Letdown chiller	23.08	9.50
4.	Letdown reheat	1.12	0.85
5.	Moderating	17.20	9.80
6.	Reactor coolant drain tank	6.72	3.25
7.	Regenerative	7.00	1.33
8.	Residual	174.00	112.00
9.	Seal water	16.86	7.89

	Length (ft)	OD (in.)	Shell Thickness (in.)
1.	12.578	8.625	0.322
2.	16.260	22.00	0.375
3.	15.167	20.00	0.375
4.	5.667	8.625	0.148
5.	16.292	18.00	0.1875
6.	13.760	12.75	0.330
7.	-	8.625	0.75
8.	22.583	44.25	0.50
9.	11.917	20.00	0.375

TABLE 12.2-6
HEAT EXCHANGER DESIGN DATA

(Sheet 2 of 2)

	Tubes			
	Length (ft)	Number ID	ID (in.)	OD (in.)
1.	23.25	25 U's	0.495	0.625
2.	27.9	142 U's	0.652	0.75
3.	26.25	94 U's	0.652	0.75
4.	8.9	15 U's	0.652	0.75
5.	29	101 U's	0.652	0.75
6.	24.75	40 U's	0.652	0.75
7.	33	108	0.245	0.375
8.	46.5	729 U's	0.652	0.75
9.	19.6	94 U's	0.652	0.75

TABLE 12.2-7
RADIATION SOURCE ACTIVITIES FOR EQUIPMENT IN THE BORON RECYCLE SYSTEM - ORIGINAL LICENSING BASIS[(a)]

(Sheet 1 of 3)

Filter Activities

Gamma Energy (MeV)	Recycle Evaporator Feed (MeV/cm^3-sec)	Recycle Evaporator Condensate (MeV/cm^3-sec)	Recycle Evaporator Concentrates (MeV/cm^3-sec)
0.4	-	1.6×10^5	-
0.8	1.1×10^7	8.0×10^4	1.2×10^4
1.3	3.0×10^6	3.3×10^4	-

Demineralizer Activities

Gamma Energy (MeV)	Recycle Evaporator Feed (MeV/cm^3-sec)	Evaporator Condensate (MeV/cm^3-sec)
0.4	1.4×10^7	4.4×10^4
0.8	4.5×10^7	2.2×10^4
1.3	2.2×10^6	5.5×10^3
1.7	1.4×10^6	2.6×10^3
2.2	3.3×10^5	1.0×10^3

TABLE 12.2-7
RADIATION SOURCE ACTIVITIES FOR EQUIPMENT IN THE BORON RECYCLE SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 3)

Recycle Evaporator and Waste Evaporator

Gamma Energy (MeV)	Vent Condenser Vapor Source Strength (MeV/cm3-sec)	Evaporator Concentrates Source Strength (MeV/gm-sec)
0.1	1.1×10^{7}	-
0.4	3.6×10^{6}	-
0.8	1.5×10^{6}	1.2×10^{6}
1.3	-	-
1.7	1.0×10^{6}	-
2.2	3.8×10^{6}	-
2.5	5.2×10^{6}	-

Recycle Holdup Tanks

Gamma Energy (MeV)	Vapor Phase (MeV/cm3-sec)	Liquid Phase[b] (MeV/g-sec)
0.1	2.1×10^{6}	3.1×10^{5}
0.4	4.4×10^{5}	1.1×10^{5}
0.8	1.2×10^{5}	4.9×10^{4}
1.3	-	1.7×10^{3}
1.7	9.3×10^{4}	3.1×10^{4}
2.2	1.9×10^{5}	1.1×10^{5}
2.5	4.3×10^{5}	1.6×10^{5}

TABLE 12.2-7
RADIATION SOURCE ACTIVITIES FOR EQUIPMENT IN THE BORON RECYCLE SYSTEM - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 3)

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

b) These data are also applicable to the recycle evaporator feed pump.

TABLE 12.2-8
RADIATION SOURCE ACTIVITIES FOR EQUIPMENT IN THE STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM - ORIGINAL DESIGN BASIS[a]

Filter Activities

Gamma Energy (MeV)	Inlet (MeV/cm3-sec)	Resin Sluice (MeV/cm3-sec)
0.8	4.02×10^{5}	1.24×10^{4}
1.3	1.10×10^{5}	4.45×10^{2}
1.7	8.07×10^{3}	6.05×10^{1}
2.5	1.62×10^{2}	9.68×10^{-1}
4.0	5.58	4.34×10^{-2}
5.0	-	7.29×10^{-3}

Demineralizer and Spent Resin Storage Tank Activities

Gamma Energy (MeV)	Demineralizers (MeV/cm3-sec)	Spent Resin Storage Tank (MeV/cm3-sec)
0.8	1.24×10^{5}	7.22×10^{4}
1.3	4.45×10^{3}	2.60×10^{3}
1.7	6.05×10^{2}	3.54×10^{2}
2.5	9.68	5.66
4.0	4.34×10^{-1}	2.53×10^{-1}
5.0	7.29×10^{-2}	4.26×10^{-2}

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-9
RADIATION SOURCE ACTIVITY FOR THE RESIDUAL HEAT REMOVAL LOOP - ORIGINAL DESIGN BASIS[a]

Gamma Energy (MeV)	Specific Source Strength (MeV/g-sec)
0.4	3.4×10^5
0.8	1.3×10^5
1.3	4.4×10^4
1.7	2.9×10^4
2.2	2.5×10^4
2.5	4.1×10^4

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-10
RADIATION SOURCE ACTIVITY FOR EQUIPMENT IN THE SPENT FUEL POOL COOLING AND PURIFICATION SYSTEM - ORIGINAL LICENSING BASIS[a]

Gamma Energy (MeV)	Demineralizer (MeV/cm^3-sec)	Filter (MeV/cm^3-sec)
0.4	2.1×10^6	-
0.8	7.4×10^5	1.1×10^7
1.3	4.1×10^3	3.0×10^6
1.7	4.7×10^3	-

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-11
CORE SHUTDOWN SOURCES - ORIGINAL LICENSING BASIS[a]

Core Shutdown Sources (MeV/cm^3-sec)

Photon Energy (MeV)	Time After Shutdown		
	4 Hours	12 Hours	1 Day
0.4	3.1×10^{11}	2.3×10^{11}	1.9×10^{11}
0.8	1.3×10^{12}	9.8×10^{11}	8.0×10^{11}
1.3	3.9×10^{11}	2.9×10^{11}	2.5×10^{11}
1.7	5.1×10^{11}	3.8×10^{11}	3.3×10^{11}
2.2	7.2×10^{10}	2.6×10^{10}	1.5×10^{10}
2.5	8.9×10^{10}	4.7×10^{10}	3.7×10^{10}
3.5	8.2×10^{9}	2.0×10^{9}	1.3×10^{9}

Photon Energy (MeV)	Time After Shutdown		
	1 Week	1 Month	3 Months
0.4	9.2×10^{10}	3.8×10^{10}	1.3×10^{10}
0.8	4.0×10^{11}	2.3×10^{11}	1.2×10^{11}
1.3	1.6×10^{11}	1.2×10^{11}	5.8×10^{10}
1.7	2.3×10^{11}	6.2×10^{10}	2.9×10^{9}
2.2	8.5×10^{9}	6.7×10^{9}	5.0×10^{9}
2.5	2.5×10^{10}	7.9×10^{9}	3.5×10^{8}
3.5	9.6×10^{8}	2.0×10^{8}	1.5×10^{7}

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-12
REACTOR CORE COMPOSITION - ORIGINAL LICENSING BASIS[a]

Material	Volume Fractions
UO_2	0.3052
Zirconium	0.0943
Stainless steel	0.0053
Inconel	0.0043
Water	0.5909

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-13
CONTROL ROD SOURCES - ORIGINAL LICENSING BASIS[a]

Irradiated Ag-In-Cd Control Rod Sources (MeV/cm-sec/rod)

Photon Energy (MeV)	Time After Shutdown	
	0	1 Week
0.8	3.2×10^{12}	3.2×10^{12}
1.3	9.0×10^{11}	8.9×10^{11}

Photon Energy (MeV)	Time After Shutdown		
	1 Month	6 Months	1 Year
0.8	3.0×10^{12}	2.0×10^{12}	1.3×10^{12}
1.3	8.4×10^{11}	5.8×10^{11}	3.8×10^{11}

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-14
INCORE DETECTOR SOURCES - ORIGINAL LICENSING BASIS[a]

Irradiated In-Core Detector Drive Wire Sources (MeV/cm-sec)

Photon Energy (MeV)	Time After Shutdown		
	0	4 Hours	12 Hours
0.8	4.1×10^{10}	1.1×10^{10}	2.6×10^{9}
1.3	1.2×10^{9}	1.2×10^{9}	1.2×10^{9}
2.2	4.0×10^{10}	1.3×10^{10}	1.5×10^{9}

Photon Energy MeV)	Time After Shutdown		
	1 Day	1 Week	1 Month
0.8	1.1×10^{9}	1.0×10^{9}	1.0×10^{9}
1.3	1.2×10^{9}	1.1×10^{9}	8.4×10^{8}
2.2	5.8×10^{7}	-	-

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-15
DEPOSITED CORROSION PRODUCT ACTIVITY ON STEAM GENERATOR SURFACES - ORIGINAL LICENSING BASIS[a]

Deposited Activity (uCi/cm^2)

	Operating Time (Months)				
Isotope	0	6	12	24	36
Mn-54	0.0	0.15	0.60	1.5	2.0
Mn-56	0.0	3.3	3.3	3.3	3.3
Co-58	0.0	4.5	10.2	11.0	11.0
Fe-59	0.0	1.4	3.0	3.0	3.0
Co-60	0.0	0.20	0.80	2.0	3.5

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-16
POST-LOCA RADIATION SOURCES

Radiation Source	Fraction of Core Activity Release
Primary coolant (Pressurized LOCA)	100% noble gases, 50% halogens, 1% remainder
Sump Water (Depressurized LOCA)	50% halogens, 1% remainder
Containment atmosphere (Inside containment)	100% noble gases, 50% halogens

TABLE 12.2-17
SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 1 CONTAINMENT ATMOSPHERE[a] - ORIGINAL LICENSING BASIS[b]

Source Strength at Time After Accident (MeV/cc-sec)

MeV/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	6.41E+07	2.24E+07	1.98E+07	1.81E+07	1.56E+07
0.8	2.40E+08	1.54E+08	1.25E+08	8.81E+07	3.16E+07
1.3	1.43E+08	5.88E+07	5.34E+07	4.16E+07	1.81E+07
1.7	1.39E+08	5.06E+07	4.01E+07	2.15E+07	1.13E.07
2.2	6.37E+07	2.64E+07	1.81E+07	1.23E+07	2.95E+06
2.5	6.71E+07	3.77E+07	3.15E+07	2.36E+07	5.31E+06
3.5	6.43E+07	1.63E+06	1.12E+06	4.47E+05	2.53E+04
6.15	8.67E+06	1.38E+04	6.76E+03	1.79E+03	0.00E+00

MeV/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	1.18E+07	5.06E+06	5.69E+05	1.27E+00	0.00E+00
0.8	1.48E+07	1.27E+06	1.31E+05	6.33E+02	5.90E+02
1.3	4.06E+06	6.33E+04	3.92E+02	0.00E+00	0.00E+00
1.7	1.96E+06	2.99E+04	2.11E+02	0.00E+00	0.00E+00
2.2	1.81E+05	7.17E+03	5.48E+01	0.00E+00	0.00E+00
2.5	2.07E+05	0.00E+00	0.00E+00	0.00E+00	0.00E+00
3.5	3.58E+02	0.00E+00	0.00E+00	0.00E+00	0.00E+00
6.15	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Note:

a. Based on minimum total dry containment volume of 2.985×10^6 ft^3 and release of 100 percent of core noble gases and 50 percent of core halogens to the containment atmosphere.

b. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-17A
SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 2 CONTAINMENT ATMOSPHERE[a][b] - ORIGINAL LICENSING BASIS[c]

Source Strength at Time After Accident (MeV/cc-sec)

Mev/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	6.32E+07	2.20E+07	1.95E+07	1.79E+07	1.54E+07
0.9	2.37E+08	1.52E+08	1.23E+08	8.68E+07	3.11E+07
1.35	1.41E+08	5.79E+07	5.26E+07	4.10E+07	1.79E+07
1.8	1.37E+08	4.99E+07	3.95E+07	2.47E+07	1.12E+07
2.2	6.27E+07	2.60E+07	1.78E+07	1.21E+07	2.91E+06
2.6	6.61E+07	3.71E+07	3.10E+07	2.33E+07	5.24E+06
3.0	3.23E+07	4.07E+05	2.83E+05	1.64E+05	2.08E+04
4.0	2.16E+07	1.43E+06	8.10E+05	2.74E+05	4.10E+03
5.0	1.71E+07	2.72E+04	1.33E+04	3.53E+03	0.0
0.0	0.0	0.0	0.0	0.0	0.0

Mev/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	1.16E+07	4.99E+06	5.61E+05	1.25E+00	0.0
0.9	1.46E+07	1.25E+06	1.29E+05	6.23E+02	5.82E+02
1.35	4.00E+06	6.23E+04	3.86E+02	0.0	0.0
1.8	1.93E+06	2.95E+04	2.08E+02	0.0	0.0
2.2	1.78E+05	7.06E+03	5.41E+01	0.0	0.0
2.6	2.04E+05	0.0	0.0	0.0	0.0
3.0	3.53E+02	0.0	0.0	0.0	0.0
4.0	0.0	0.0	0.0	0.0	0.0
5.0	0.0	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0

Note:

a. Based on a minimum total dry containment volume of 3.03E06 cu.ft. and release of 100 percent of core noble gases and 50 percent of core halogens to the containment atmosphere.

b. Source strength data from Westinghouse Radiation Analysis Manual and normalized to approximate power level and reactor containment building volume.

c. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-18
DELETED

TABLE 12.2-18A
DELETED

TABLE 12.2-19
SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 1 PRESSURIZED PRIMARY COOLANT[a][b] - ORIGINAL LICENSING BASIS[c]

Source Strength at Time After Accident (MeV/cc-sec)

MeV/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	1.85E+10	6.53E+09	5.73E+09	5.18E+09	4.44E+09
0.8	6.91E+10	4.58E+10	3.74E+10	2.64E+10	9.80E+10
1.3	4.15E+10	1.95E+10	1.66E+10	1.21E+10	5.11E+09
1.7	4.02E+10	2.01E+10	1.58E+10	8.88E+09	3.56E+09
2.2	1.180+10	1.01E+10	7.89E+09	5.94E+09	1.39E+09
2.5	1.89E+10	1.26E+10	1.02E+10	7.07E+09	1.49E+09
3.5	1.86E+09	2.32E+09	1.82E+09	9.98E+08	1.45E+08
6.15	2.49E+09	1.92E+07	1.87E+07	1.50E+07	3.37E+06

MeV/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	3.38E+09	1.44E+09	1.71E+08	3.02E+06	1.74E+06
0.8	4.83E+09	8.83E+08	4.20E+08	1.05E+08	3.74E+07
1.3	1.19E+09	5.53E+07	1.41E+07	2.09E+06	1.51E+06
1.7	8.90E+08	2.64E+08	7.68E+07	1.28E+06	9.41E+05
2.2	7.31E+07	1.06E+07	4.32E+06	1.39E+06	8.72E+05
2.5	7.79E+07	1.63E+07	4.65E+06	1.39E+03	0.00E+00
3.5	3.14E+06	3.88E+05	1.12E+05	0.00E+00	0.00E+00
6.15	6.39E+04	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Note:

a. Based on the total coolant volume of 10833 ft^3 which is equal to total RCS liquid volume of 11500 ft^3 at 590°F and 2250 psia minus pressurizer liquid volume of 1080 ft^3 plus RHR Train A total volume of 413 ft^3.

b. 100% Noble gases, 50% halogens, and 1% remainder.

c. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-19A
SOURCE STRENGTH FOR THE UNIT 1 SUMP WATER[a][b] - ORIGINAL LICENSING BASIS[c]

Source Strength at Time After Accident (MeV/cc-sec)

MeV/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	7.97E+08	5.34E+08	4.75E+08	4.20E+08	3.71E+08
0.8	9.07E+09	7.34E+09	6.04E+09	4.24E+09	1.60E+09
1.3	4.63E+09	3.22E+09	2.77E+09	2.04E+09	8.77E+08
1.7	3.38E+09	2.83E+09	2.34E+09	1.33E+09	5.65E+08
2.2	3.00E+08	6.63E+08	6.30E+08	5.53E+08	1.39E+08
2.5	6.65E+08	4.75E+08	3.41E+08	1.59E+08	3.77E+07
3.5	1.70E+09	3.93E+08	3.00E+08	1.63E+08	2.42E+07
6.15	4.05E+08	3.32E+06	3.28E+06	2.64E+06	5.91E+03

MeV/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	3.28E+08	1.91E+08	2.69E+07	5.30E+05	3.06E+05
0.8	8.26E+08	1.55E+08	7.36E+07	1.84E+07	6.53E+06
1.3	2.08E+08	9.18E+06	2.47E+06	3.67E+05	2.65E+05
1.7	1.55E+08	4.63E+07	1.35E+07	2.24E+05	1.65E+05
2.2	1.08E+07	1.86E+06	7.57E+05	2.45E+05	1.53E+05
2.5	9.58E+06	2.85E+06	8.16E+05	2.45E+02	0.00E+00
3.5	5.33E+05	6.81E+04	1.96E+04	0.00E+00	0.00E+00
6.15	1.12E+04	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Note:

a. Based on the total sump water volume of 61740 ft^3.

b. 0% noble gases, 50% halogens, 1% remainder.

c. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-19B

SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 1 PRIMARY COOLANT SAMPLE FLUID[a] - ORIGINAL LICENSING BASIS[b]

Source Strength at Time After Accident (MeV/cc-sec)

MeV/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	2.61E+10	9.22E+09	8.09E+09	7.32E+09	6.27E+09
0.8	9.76E+10	6.46E+10	5.28E+10	3.72E+10	1.38E+10
1.3	5.86E+10	2.75E+10	2.34E+10	1.71E+10	7.22E+09
1.7	5.68E+10	2.84E+10	2.23E+10	1.25E+10	5.02E+09
2.2	2.54E+10	1.42E+10	1.11E+10	8.38E+09	1.97E+09
2.5	2.67E+10	1.78E+10	1.44E+10	9.97E+08	2.11E+09
3.5	2.62E+10	3.28E+09	2.57E+09	1.41E+09	2.05E+08
6.15	3.51E+09	2.71E+07	2.64E+07	2.12E+07	4.76E+06

MeV/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	4.77E+09	2.03E+09	2.41E+08	4.27E+06	2.46E+06
0.8	6.82E+09	1.25E+09	5.92E+08	1.48E+08	5.27E+07
1.3	1.67E+09	7.38E+07	1.98E+07	2.95E+06	2.13E+06
1.7	1.26E+09	3.73E+08	1.08E+08	1.80E+06	1.33E+06
2.2	1.03E+08	1.49E+07	6.09E+06	1.97E+06	1.23E+06
2.5	1.10E+08	2.30E+07	6.56E+06	1.97E+03	0.00E+00
3.5	4.43E+06	5.48E+05	1.57E+05	0.00E+00	0.00E+00
6.15	9.02E+04	0.00E+00	0.00E+00	0.00E+00	0.00E+00

Note:

a. Based on source strengths for the primary coolant given in Table 12.2-19 which are divided by a density correction factor of 0.7084 to obtain the sample water source strengths.

b. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-19C
SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 2 PRESSURIZED PRIMARY COOLANT[a][b][c] - ORIGINAL LICENSING BASIS[d]

Source Strength at Time After Accident (MeV/cc-sec)

Mev/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	1.85E+10	6.53E+09	5.73E+09	5.18E+09	4.44E+09
0.9	6.91E+10	4.58E+10	3.74E+10	2.64E+10	9.80E+09
1.35	4.15E+10	1.95E+10	1.66E+10	1.21E+10	5.11E+09
1.8	4.02E+10	2.01E+10	1.58E+10	8.88E+09	3.56E+09
2.2	1.80E+10	1.01E+10	8.01E+09	5.94E+09	1.39E+09
2.6	1.89E+10	1.26E+10	1.02E+10	7.06E+09	1.49E+09
3.0	9.47E+10	1.51E+09	1.24E+09	6.97E+08	9.99E+07
4.0	6.65E+09	8.63E+07	5.64E+08	2.86E+08	4.18E+07
5.0	4.93E+09	3.78E+07	3.74E+07	3.00E+07	6.74E+06
6.0	2.21E+07	2.79E+05	5.81E+02	0.0	0.0

Mev/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	3.38E+09	1.44E+09	1.71E+08	3.02E+06	1.74E+06
0.9	4.83E+09	8.83E+08	4.20E+08	1.05E+08	3.73E+07
1.35	1.19E+09	5.23E+07	1.41E+07	2.09E+06	1.51E+06
1.8	8.90E+08	2.64E+08	7.67E+07	1.28E+06	9.41E+05
2.2	7.31E+07	1.06E+07	4.31E+06	1.39E+06	8.71E+05
2.6	7.79E+07	1.63E+07	4.65E+06	1.39E+03	0.0
3.0	2.19E+06	2.79E+05	8.02E+04	0.0	0.0
4.0	8.83E+05	1.09E+05	3.14E+04	0.0	0.0
5.0	1.28E+05	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0

Note:

a. Based on the total coolant volume of 10833 cu.ft. which is equal to total RCS liquid volume of 11500 cu.ft. at 590ºF and 2250 psia minus pressurizer volume of 1080 cu.ft. plus RHR train A total volume of 413 cu.ft.

b. 100% noble gases, 50% halogens and 1% remainder.

c. Source strength data from Westinghouse Radiation Analysis Manual and normalized to approximate power level and total coolant volume.

d. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-19D
SOURCE STRENGTH FOR THE UNIT 2 SUMP WATER[a][b][c] - ORIGINAL LICENSING BASIS[d]

Source Strength at Time After Accident (MeV/cc-sec)

Mev/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	7.71E+08	5.16E+08	5.49E+08	4.06E+08	3.59E+08
0.9	8.77E+09	7.10E+09	5.84E+09	4.10E+09	1.55E+09
1.35	4.47E+09	3.11E+09	2.68E+09	1.97E+09	8.48E+08
1.8	3.27E+09	2.74E+09	2.27E+09	1.48E+09	5.46E+08
2.2	2.90E+08	6.41E+08	6.49E+08	5.34E+08	1.34E+08
2.6	6.43E+08	4.59E+08	3.29E+08	1.54E+08	3.65E+07
3.0	8.18E+08	2.43E+08	2.00E+08	1.11E+08	1.60E+07
4.0	4.38E+08	1.34E+08	8.67E+07	4.34E+07	6.90E+06
5.0	7.75E+08	6.33E+06	6.35E+06	5.10E+06	1.14E+06
6.0	3.75E+06	4.73E+04	9.86E+01	0.0	0.0

Mev/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	3.17E+08	1.85E+08	2.60E+07	5.13E+05	2.96E+05
0.9	7.98E+08	1.50E+08	7.12E+07	1.77E+07	6.31E+06
1.35	2.01E+08	8.87E+06	2.38E+06	3.55E+05	2.56E+05
1.8	1.50E+08	4.48E+07	1.30E+07	2.17E+05	1.60E+05
2.2	1.04E+07	1.79E+06	7.32E+05	2.37E+05	1.48E+05
2.6	9.27E+06	2.76E+06	7.89E+05	2.37E+02	0.0
3.0	3.55E+05	4.73E+04	1.36E+04	0.0	0.0
4.0	1.50E+05	1.85E+04	5.32E+03	0.0	0.0
5.0	2.17E+04	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0

Note:

a. Based on total sump water volume from: 388,677 gal from Refueling Water Storage Tank, 3800 cu.ft. from accumulators, and 11500 cu.ft. (@ 590°F) of Reactor Coolant.

b. 0% noble gases, 50% halogens and 1% remainder.

TABLE 12.2-19D
SOURCE STRENGTH FOR THE UNIT 2 SUMP WATER[a][b][c] -
ORIGINAL LICENSING BASIS[d]

c. Source strength data from Westinghouse Radiation Analysis Manual and normalized to approximate power level and total coolant volume.

d. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-19E
SOURCE STRENGTH FOR THE POST-ACCIDENT UNIT 2 PRIMARY COOLANT SAMPLE FLUID[a][b] - ORIGINAL LICENSING BASIS[c]

Source Strength at Time After Accident (MeV/cc-sec)

Mev/Gamma	0 HR	0.5 HR	1 HR	2 HR	8 HR
0.4	2.61E+10	9.22E+09	8.09E+09	7.31E+09	6.27E+09
0.9	9.75E+10	6.47E+10	5.28E+10	3.73E+10	1.38E+10
1.35	5.86E+10	2.75E+10	2.34E+10	1.71E+10	7.21E+09
1.8	5.67E+10	2.84E+10	2.23E+10	1.25E+10	5.03E+09
2.2	2.54E+10	1.43E+10	1.13E+10	8.39E+09	1.96E+09
2.6	2.67E+10	1.78E+10	1.44E+10	9.97E+09	2.10E+09
3.0	1.34E+10	2.13E+09	1.75E+09	9.84E+08	1.41E+08
4.0	9.39E+09	1.22E+09	7.96E+08	4.04E+08	5.90E+07
5.0	6.96E+09	5.34E+07	5.28E+07	4.23E+07	9.51E+06
6.0	3.12E+07	3.94E+05	8.20E+02	0.0	0.0

Mev/Gamma	24 HR	168 HR	720 HR	4380 HR	8760 HR
0.4	4.77E+09	2.03E+09	2.41E+08	4.26E+06	2.46E+06
0.9	6.82E+09	1.25E+09	5.93E+08	1.48E+08	5.27E+07
1.35	1.68E+09	7.38E+07	1.99E+07	2.95E+06	2.13E+06
1.8	1.26E+09	3.73E+08	1.08E+08	1.81E+06	1.33E+06
2.2	1.03E+08	1.50E+07	6.08E+06	1.96E+06	1.23E+06
2.6	1.10E+08	2.30E+07	6.56E+06	1.96E+03	0.0
3.0	3.09E+06	3.94E+05	1.13E+05	0.0	0.0
4.0	1.25E+06	1.54E+05	4.43E+04	0.0	0.0
5.0	1.81E+05	0.0	0.0	0.0	0.0
6.0	0.0	0.0	0.0	0.0	0.0

Note:

a. Based on source strengths for the primary coolant given in Table 12.2-19C which are divided by the density correction factor of 0.7084 to obtain the sample water source strengths.

b. Source strength data from Westinghouse Radiation Analysis Manual and normalized to approximate power level and total coolant volume.

c. Historical. Not subject to future updating. Has been retained to preserve original design basis.

TABLE 12.2-20
RADIOACTIVE SOURCES

(Sheet 1 of 2)

	Byproduct, source, and/or special nuclear material	Form		Maximum quantity	
A.	Any radioactive material with Atomic Numbers 1 through 83, inclusive	A.	Any	A.	Not to exceed 1 curie per radionuclide, except: Hydrogen-3 6 curies Sodium-24 3 curies
B.	Any radioactive material with Atomic Numbers 84, 85, 87, 89, 91, 93, 95, or 96	B.	Any	B.	Not to exceed 500 microcuries per radionuclide
C.	Any radioactive material with Atomic Numbers 4 through 84, inclusive	C.	Sealed or plated sourced or foils	C.	Not to exceed 10 curies per source or foil, except: Cesium-137 6000 curies
D.	Any radioactive material with Atomic Number 85, 87, 89, 91, 93, 95, or 96	D.	Sealed or plated sources or foils	D.	Not to exceed 2 millicuries per source or foil, except: Americium-241 4 curies
E.	Thorium-230	E.	Any	E.	.9 gram
F.	Uranium-233	F.	Any	F.	.9 gram
G.	Uranium-234	G.	Any	G.	.9 gram
H.	Uranium-235	H.	Any	H.	.9 gram
I.	Uranium-238	I.	Any	I.	.9 gram

TABLE 12.2-20
RADIOACTIVE SOURCES

(Sheet 2 of 2)

	Byproduct, source, and/or special nuclear material	Form		Maximum quantity	
J.	Plutonium-238	J.	Any	J.	.9 gram
K.	Plutonium-239	K.	Any	K.	.9 gram
L.	Plutonium-238	L.	Sealed sources	L.	One source of 2.2 curies
M.	Californium-252	M.	Sealed sources	M.	600 millicuries
N.	Uranium enriched in the U-235 isotope	N.	In unirradiated reactor fuel assemblies	N.	See Technical Specification 4.3, “Fuel Storage”
O.	Uranium enriched in the U-235 isotope	O.	Contained in excore neutron detectors	O.	40 grams contained U-235 in uranium enriched to nor more than 93.5% in U-235

Authorized uses:

A. through L.

To be used for systems security checks; equipment standardization and calibration; process control; gauging and quality assurance testing; teaching; and nuclear reactor operations.

M. Primary reactor startup sources.

N. Reactor fuel

O. Excore neutron detectors

TABLE 12.2-21
REACTOR COOLANT SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

(Sheet 1 of 3)

Power Rating	
Total core thermal, MW	3565
Power density, W/cm	109.0
Effective Dimensions, in.	
Height	144.0
Diameter	132.7
Failed Fuel Fraction	
Equivalent fraction of average fuel rods containing cladding defects	0.01
Operation Times, Equivalent Full-Power Hours	
Initial cycle	9264
Equilibrium cycle	6480
Fission Product Escape Rate Coefficients During Full-Power Operation, sec^{-1}	
Noble gas isotopes	6.5×10^{-8}
Br, Rb, I, and Cs isotopes	1.3×10^{-8}
Te isotopes	1.0×10^{-9}
Mo isotopes	2.0×10^{-9}
Sr and Ba isotopes	1.0×10^{-11}
Y, Zr, Nb, La, Ce, and Pr isotopes	1.6×10^{-12}
Fission Product Escape Rate Coefficients During Spent Fuel Pit Storage, sec^{-1}	
Noble gas isotopes	6.5×10^{-13}
Br, Rb, I, and Cs isotopes	1.3×10^{-13}
Te isotopes	1.0×10^{-14}
Mo isotopes	2.0×10^{-14}
Sr and Ba isotopes	1.0×10^{-16}

TABLE 12.2-21
REACTOR COOLANT SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

(Sheet 2 of 3)

Y, Zr, Nb, La, Ce, and Pr isotopes	1.6×10^{-17}
Number of reactor coolant loops	4
System water volume, ft^3	12,000
System normal operating pressure, psia	2250
Average temperature in core, F	590
Transit Times, sec	
In core	0.75
Out of core	9.16
Total	9.91
Core Wetted Areas, Effective, in^2	
Zirconium	9.42×10^6
Stainless steel	6.09×10^6
Inconel	1.01×10^6
Out-of-core wetted area, Inconel, in^2	2.74×10^7
Coolant Velocity, ft/sec	
Core	16.1
Steam generator	19.6
Nominal Base Metal Release Rates, mg/dm^2-mo	
Zirconium	0.0
Stainless steel	0.5
Inconel	1.0
Coolant crud level, ppm	0.1
Permanent Crud Film, Nominal, mg/dm^2	
In core	50
Out of core	50
Transient Crud Layer, Nominal, mg/dm^2	
In core	50

TABLE 12.2-21
REACTOR COOLANT SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

(Sheet 3 of 3)

Out of core	50
Initial Boron Concentrations, ppm	
Initial cycle	805
Equilibrium cycle	1080
Pressurizer	
Liquid volume, nominal, ft^3	1080
Vapor volume, nominal, ft^3	720
Total wetted area, ft^2	1200
Spray line flow, gpm	1.0
Isotopic Stripping Fractions	
Noble gases	1.0
Other isotopes	0.0
10-Percent Step Load Transients	
Maximum surge rate in pressurizer, gpm	1720
Duration of transient, sec	40

TABLE 12.2-22
CHEMICAL AND VOLUME CONTROL SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

(Sheet 1 of 2)

Letdown rate, normal purification, gpm	75
Boron Removal Rates, ppm/day	
Initial cycle	2.06
Equilibrium cycle	3.96
Effective Deborating Demineralizer Cut-In	
Concentration, ppm	100
Mixed-Bed Demineralizer	
Resin volume, ft^3	30.0
Demineralizer Isotopic Decontamination Factors	
Noble gases	1.0
Y-90, Y-91, Cs-134, Cs-136, Cs-137, Mo-99	1.0
Other isotopes	10.0
Cation-Bed Demineralizer	
Effective flow, gpm	7.5
Resin volume, ft^3	30.0
Demineralizer Isotopic Decontamination Factors	
Noble gases	1.0
Y-90, Y-91, Cs-134, Cs-137	10.0
Other isotopes	1.0
Volume Control Tank	
Liquid volume, nominal, ft^3	160
Vapor volume, nominal, ft^3	240

TABLE 12.2-22
CHEMICAL AND VOLUME CONTROL SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

(Sheet 2 of 2)

Isotopic Stripping Fractions	
Kr-85	2.3×10^{-5}
Kr-85M	2.7×10^{-1}
Kr-87	6.0×10^{-1}
Kr-88	4.3×10^{-1}
Xe-131M	1.0×10^{-2}
Xe-133	1.6×10^{-2}
Xe-133M	3.7×10^{-2}
Xe-135	1.8×10^{-1}
Xe-135M	8.0×10^{-1}
Xe-138	1.0

TABLE 12.2-23
STEAM GENERATOR BLOWDOWN PROCESSING SYSTEM DESIGN PARAMETERS - TO BE UPDATED BY WESTINGHOUSE

Primary to secondary side leakage rate, gpd	20
Partition Factors in the Steam Generator	
Iodine	0.01
Others	0.0
Partition Factors in the Condenser and Feedwater Heaters	
Iodine	0.0005
Others	0.0
Decontamination factors for the CPS	
Iodine	10
Cs, Rb	2
Tritium	1
Others	10
Decontamination Factors for the SGBPS Demineralizers	
Iodine	1000
Cs, Rb	2000
Tritium	1
Others	100,000
Resin charge and filter replacement per year	1

TABLE 12.2-24
DESIGN PARAMETERS FOR POST-ACCIDENT DOSE EVALUATIONS -
SHOULD BE UPDATED BY WESTINGHOUSE

Power Level	3,565 MWt
Three region equillbrium cycle core at end of life. The three regions have operated at a specific power of 40.00 MW/MTU for 300, 600, and 900 EFPD, respectively.	
Design Basis Accident	
Reactor coolant volume	11500 ft^3
Refueling water volume	50240 ft^3
Fraction of total core fission product inventory released	
Noble gases	1.0
Halogens	0.5
Remaining fission product inventory	0.01
Gap activity-release	
Fraction of total core fission product inventory released	
I-131	0.08
Kr-85	0.10
Other Iodines and Noble gases	0.05
Fuel handling accident	
Number of fuel rods in discharge region experiencing gap activity release	264
Release fractions are the same as for the total gap acitivity release accident	
Radial peaking factor	1.65
Decay interval between reactor shutdown and commencement of refueling operations (hours)	75

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[(a)]

(Sheet 1 of 10)

1. NUREG-0017, Revision 1 release rates were used for the determination of airborne concentrations, unless otherwise noted.

2. Partition and plate-out factors are assumed to be considered in the source concentrations, which are based upon NUREG-0017, Revision 1.

3. Leakage from the CVCS is assumed to be the primary contributor to the radioactive airborne concentrations in the Safeguards and Auxiliary Buildings. Leakage from the RHRS is also included for shutdown conditions. Both systems are assumed to contain reactor coolant.

4. No purging of the Containment Buildings is assumed. A continuous containment vent for pressure relief equal to 10% of the maximum vent flow rate is assumed to relieve pressure buildup in the building due to thermal transients and air leakage (e.g., Instrument Air System) within the building.

5. Of the total yearly NUREG-0017 tritium activity release from each reactor unit, 1400 Ci is assumed to be released via gaseous pathways, and therefore contributes to the airborne concentrations in each unit. Tritium concentrations in the Containment Buildings during normal power operations are based upon 100% relative humidity and 120°F. The humidity in the containment atmosphere is assumed to contain reactor coolant tritium activity. For all other cases, tritium is conservatively treated as a noble gas, with a partition factor of 1.0 (no removal).

6. The leak rates provided for the auxiliary building in NUREG-0017 apply to both the Auxiliary and Safeguards Buildings, and are assumed to be equally distributed between the two buildings for each plant unit.

7. Building Volumes:

Containment Building (each unit)	2.985×10^6 ft^3
Auxiliary Building	2.0×10^6 ft^3

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 10)

Safeguards Building (each unit)	1.07×10^6 ft^3
Fuel Handling Building	1.3×10^6 ft^3
Turbine Building (each unit)	
Mezzanine	1.77×10^6 ft^3
Basement	1.72×10^6 ft^3
8. Ventilation Flow Rates:	
Containment Building (each unit)	
Purge	26,000 scfm (shutdown)
Pressure Relief	78.4 scfm (normal power operations)
Auxiliary Building	107,600 scfm
Safeguards Building (each unit)	20,200 scfm
Fuel Building	33,500 scfm
Turbine Building (each unit):	
Mezzanine	276,000 scfm (assumed)

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 10)

Basement	276,000 scfm

9. Release Rates for Particulates and Other Isotopes, Per Building, Per Unit, (Ci/year).

Isotope	Auxiliary[b] Building	Safeguards Building	Fuel Handling Building	Containment Building	Waste Gas System
Cs-134	3.03E-04	2.70E-04	1.70E-03	2.50E-03	3.30E-05
Cs-136	2.93E-05	2.40E-05	0.00E+00	3.20E-03	5.30E-06
Cs-137	4.37E-04	3.60E-04	2.70E-03	5.50E-03	7.70E-05
Cr-51	1.74E-04	1.60E-04	1.80E-04	9.20E-03	1.40E-05
Mn-54	4.11E-05	3.90E-05	3.00E-04	5.30E-03	2.10E-06
Fe-59	2.68E-05	2.50E-05	0.00E+00	2.70E-03	1.80E-06
Co-57	0.00E+00	0.00E+00	0.00E+00	8.20E-04	0.00E+00
Co-58	9.59E-04	9.50E-04	2.10E-02	2.50E-02	8.70E-06
Co-60	2.69E-04	2.55E-04	8.20E-03	2.60E-03	1.40E-05
Sr-89	4.19E-04	3.75E-04	2.10E-03	1.30E-02	4.40E-05
Sr-90	1.62E-04	1.45E-04	8.00E-04	5.20E-03	1.70E-05

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 4 of 10)

Sb-125	1.95E-06	1.95E-06	5.70E-05	0.00E+00	0.00E+00
Zr-95	5.05E-04	5.00E-04	3.60E-06	0.00E+00	4.80E-06
Isotope	Auxiliary[b] Building	Safeguards Building	Fuel Handling Building	Containment Building	Waste Gas System
Nb-95	1.87E-05	1.50E-05	2.40E-03	1.80E-03	3.70E-06
Ru-103	1.47E-05	1.15E-05	3.80E-05	1.60E-03	3.20E-06
Ru-106	5.70E-06	3.00E-06	6.90E-05	0.00E+00	2.70E-06
Ba-140	2.23E-04	2.00E-04	0.00E+00	0.00E+00	2.30E-05
Ce-141	1.52E-05	1.30E-05	4.40E-07	1.30E-03	2.20E-06
Other Isotopes, Per Unit					
C-14	3.45E+00	2.25E+00	0.00E+00	1.60E+00	1.20E+00
Ar-41			3.40E+01		

10. Primary and Secondary Coolant Concentrations (based upon NUREG-0017, Revision 1).

		Steam Generator Activity (uCi/g)	
Isotope	RCS Conc. (uCi/g)	Water	Steam

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 5 of 10)

Group I			
Kr-85m	1.39E-01		2.94E-08
Kr-85	1.37E-02		2.84E-09
		Steam Generator Activity (uCi/g)	
Isotope	RCS Conc. (uCi/g)	Water	Steam
Kr-87	1.53E-01		3.06E-08
Kr-88	2.61E-01		5.50E-08
Xe-131m	9.54E-02		1.96E-08
Xe-133m	2.70E-02		5.78E-09
Xe-133	5.90E-01		1.23E-07
Xe-135m	1.43E-01		2.98E-08
Xe-135	6.64E-01		1.41E-07
Xe-137	3.82E-02		7.98E-09
Xe-138	1.33E-01		2.76E-08
Group II			

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 6 of 10)

I-131	5.08E-02	2.19E-06	2.19E-08
I-132	2.37E-01	3.98E-06	3.98E-08
I-133	1.58E-01	5.92E-06	5.92E-08
		Steam Generator Activity (uCi/g)	
Isotope	RCS Conc. (uCi/g)	Water	Steam
I-134	3.84E-01	3.13E-06	3.13E-08
I-135	2.94E-01	8.30E-06	8.30E-08
Group III			
Cs-134	4.28E-02	2.15E-06	1.11E-08
Cs-136	4.23E-03	2.10E-07	1.05E-09
Cs-137	5.70E-02	2.87E-06	1.44E-08
Group IV			
H-3	1.00E+00	1.00E-03	1.00E-03
Group V			
Cr-51	3.45E-03	1.56E-07	7.56E-10

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 7 of 10)

Mn-54	1.78E-03	7.79E-08	3.96E-10
Fe-59	3.34E-04	1.44E-08	7.32E-11
Co-58	5.12E-03	2.28E-07	1.13E-09

Isotope	RCS Conc. (uCi/g)	Steam Generator Activity (uCi/g) Water	Steam
Co-60	5.90E-04	2.64E-08	1.32E-10
Sr-89	1.54E-04	6.77E-09	3.44E-11
Sr-90	1.32E-05	5.82E-10	2.97E-12
Zr-95	4.60E-02	2.03E-06	1.00E-08
Nb-95	2.08E-02	8.79E-07	4.56E-09
Ru-103	6.11E-01	2.72E-05	1.41E-07
Ru-106	2.45E+01	1.09E-03	5.28E-06
Ba-140	4.09E-01	1.76E-05	8.82E-08
Ce-141	1.05E-02	4.59E-07	2.33E-09

11. Containment Building Release Rates, (Each Reactor Unit)

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 8 of 10)

Noble Gases:	3% of RCS/day	(normal power operations)
Iodines:	8.0×10^{-4}%/day	(normal power operations)
	0.32 Ci/yr per uCi/g (RCS)	(shutdown / refueling)
Tritium:	18% of NUREG-0017 release, based upon 100% relative humidity and 120°F during normal power operations, and the remainder of the total yearly containment release during shutdown / refueling conditions.	
Particulates and Other Isotopes:	As noted previously.	

12. Auxiliary / Safeguards Buildings Release Rates (combined) for each reactor unit

Noble Gases:	160 lb/day	(normal power operations)
Iodines:	0.68 Ci/yr per uCi/g (RCS)	(normal power operations)
Iodines:	2.50 Ci/yr per uCi/g (RCS)	(shutdown / refueling)
Tritium:	32% of NUREG-0017 release.	
Particulates and Other Isotopes:	As noted previously.	

13. Worst Potential Auxiliary / Safeguards Room

Release Rates:	10% of the combined Auxiliary / Safeguards Buildings release rates, based upon relative leakage potential in Auxiliary and Safeguards Building rooms (normal power operations).

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[(a)]

(Sheet 9 of 10)

7% of the combined Auxiliary / Safeguards Buildings release rates, based upon relative leakage potential in Auxiliary and Safeguards Building rooms (shutdown and refueling operations).

Volume: 1653 ft^3

Ventilation Flow: 200 scfm

14. Turbine Generator Building Release Rates from each reactor unit

Noble Gases: 1700 lb/hr of secondary steam

Iodines: 3800 Ci/yr per uCi/g of secondary steam (normal power operations)

Iodines: 420 Ci/yr per uCi/g of secondary steam (shutdown and refueling operations)

Tritium: 1700 lb/hr of secondary steam

Particulates and Other Isotopes: 1700 lb/hr of secondary steam

15. Fuel Handling Building Release Rates from each reactor unit

Noble Gases: Produced predominately from decay of iodines. Due to the high ventilation rates in the vicinity of the spent fuel pools, noble gas concentrations due to iodine decay are assumed to be negligible.

Iodines: 0.038 Ci/yr per uCi/g (RCS) (normal power operations)

Iodines: 0.093 Ci/yr per uCi/g (RCS) (shutdown / refueling)

TABLE 12.2-25
ASSUMPTIONS AND PARAMETERS USED IN DETERMINING AIRBORNE ACTIVITY CONCENTRATIONS - ORIGINAL LICENSING BASIS[a]

(Sheet 10 of 10)

Tritium:	32% of NUREG-0017 release.
Particulates and Other Isotopes:	As noted previously.

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

b) The CPNPP Auxiliary Building release rates are determined by adding 50% of the NUREG-0017 auxiliary building release rates to the Waste Gas System release rates from each reactor unit. This adjustment is made since the Waste Gas System at CPNPP is contained with the Auxiliary Building.

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 1 of 10)

(Normal Operations)

Isotope	Auxiliary Building	Each Safeguards Building	Each Containment Building	Fuel Building	Each Turbine Mezzanine	Each Turbine Basement	Worst Room
Kr-85m	2.19E-09	5.38E-09	3.03E-06		4.77E-14	1.19E-14	1.21E-07
Kr-85	2.27E-10	6.05E-10	2.96E-05		4.67E-15	1.17E-15	1.22E-8
Kr-87	2.16E-09	4.55E-09	9.54E-07		4.75E-14	1.19E-14	1.27E-07
Kr-88	4.02E-09	9.45E-09	3.58E-06		8.82E-14	2.21E-14	2.25E-07
Xe-131m	1.58E-09	4.20E-09	8.19E-05		3.23E-14	8.07E-15	8.50E-08
Xe-133m	4.45E-10	1.18E-09	6.35E-06		9.51E-15	2.38E-15	2.40E-08
Xe-133	9.76E-09	2.59E-08	2.87E-04		2.02E-13	5.04E-14	5.25E-07
Xe-135m	1.29E-09	1.86E-09	1.80E-07		3.80E-14	9.56E-15	9.29E-08
Xe-135	1.07E-08	2.75E-08	2.94E-05		2.30E-13	5.74E-14	5.85E-07
Xe-137	1.45E-10	1.59E-10	1.21E-08		6.09E-15	1.54E-15	1.37E-08
Xe-138	1.15E-09	1.63E-09	1.55E-07		3.46E-14	8.72E-15	8.42E-08
Ar-41	N/A	N/A	1.47E-07		N/A	N/A	N/A
I-131	2.62E-11	6.97E-11	8.97E-09	9.40E-12	2.47E-14	6.17E-15	1.41E-09
I-132	1.12E-10	2.58E-10	7.10E-10	3.68E-11	4.34E-14	1.09E-14	6.33E-09

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 2 of 10)

(Normal Operations)

Isotope	Auxiliary Building	Each Safeguards Building	Each Containment Building	Fuel Building	Each Turbine Mezzanine	Each Turbine Basement	Worst Room
I-133	8.09E-11	2.11E-10	4.13E-09	2.87E-11	6.64E-14	1.66E-14	4.38E-09
I-134	1.60E-10	3.12E-10	4.43E-10	4.72E-11	3.25E-14	8.14E-15	9.64E-09
I-135	1.47E-10	3.70E-10	2.51E-09	5.10E-11	9.23E-14	2.31E-14	8.05E-09
Cs-134	4.60E-13	1.09E-12	2.54E-09	8.30E-12	1.82E-14	4.55E-15	2.21E-11
Cs-136	4.45E-14	9.69E-14	1.38E-09		1.73E-15	4.32E-16	1.96E-12
Cs-137	6.64E-13	1.46E-12	5.73E-09	1.32E-11	2.37E-14	5.91E-15	2.94E-11
Cr-51	2.64E-13	6.47E-13	5.77E-09	8.78E-13	1.24E-15	3.11E-16	1.31E-11
Mn-54	6.24E-14	1.58E-13	5.22E-09	1.46E-12	6.51E-16	1.63E-16	3.19E-12
Fe-59	4.07E-14	1.01E-13	2.00E-09		1.20E-16	3.01E-17	2.04E-12
Co-57			8.01E-10				
Co-58	1.46E-12	3.84E-12	2.07E-08	1.02E-10	1.86E-15	4.64E-16	7.77E-11
Co-60	4.09E-13	1.03E-12	2.69E-09	4.00E-11	2.17E-16	5.43E-17	2.08E-11
Sr-89	6.37E-13	1.52E-12	9.94E-09	1.02E-11	5.67E-17	1.42E-17	3.07E-11
Sr-90	2.46E-13	5.87E-13	5.41E-09	3.90E-12	4.89E-18	1.22E-18	1.19E-11

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 3 of 10)

(Normal Operations)

Isotope	Auxiliary Building	Each Safeguards Building	Each Containment Building	Fuel Building	Each Turbine Mezzanine	Each Turbine Basement	Worst Room
Sb-125	2.96E-15	7.89E-15		2.78E-13	1.59E-13		
Zr-95	7.67E-13	2.02E-12		1.76E-14	1.65E-14	4.13E-15	4.09E-11
Nb-95	2.84E-14	6.07E-14	1.23E-09	1.17E-11	7.50E-15	1.87E-15	1.23E-12
Ru-103	2.23E-14	4.65E-14	1.14E-09	1.85E-13	2.31E-13	5.79E-14	9.40E-13
Ru-106	8.66E-15	1.21E-14		3.37E-13	8.70E-12	2.17E-12	2.45E-13
Ba-140	3.39E-13	8.08E-13			1.45E-13	3.63E-14	1.63E-11
Ce-141	2.31E-14	5.26E-14	8.67E-10	2.15E-15	3.84E-15	9.60E-16	1.06E-12
C-14	5.24E-09	9.11E-09	1.67E-06		1.84E-07		
H-3	3.37E-07	8.97E-07	7.85E-05	2.36E-06	1.65E-09	4.11E-10	1.81E-05
Total[b] MPC	9.01E-02	2.32E-01	7.40E+01	4.88E-01	1.80E-03	4.49E-04	4.90E+00

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 4 of 10)

(Shutdown and Refueling Operations)

Isotope	Each Auxiliary Building	Each Safeguards Building	Each Containment Building	Each Fuel Building	Turbine Mezzanine	Turbine Basement	Worst Room
I-131	4.45E-10	1.18E-09	2.34E-10	1.06E-10	1.26E-14	3.15E-15	1.67E-08
I-132	1.90E-09	4.37E-09	6.98E-10	4.15E-10	2.22E-14	5.54E-15	7.50E-08
I-133	1.37E-09	3.59E-09	6.91E-10	3.24E-10	3.39E-14	8.47E-15	5.18E-08
I-134	2.71E-09	5.29E-09	7.11E-10	5.32E-10	1.66E-14	4.15E-15	1.14E-07
I-135	2.49E-09	6.28E-09	1.14E-09	5.75E-10	4.71E-14	1.18E-14	9.53E-08
H-3	4.45E-10		2.53E-06	4.72E-06			
Total MPC[b]	1.35E-01	3.46E-01	5.71E-01	9.75E-01	3.14E-06	7.86E-07	5.14E+00

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 5 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Auxiliary Building	Normal Power Unit Safeguards Building	Shutdown/ Refueling Unit Safeguards Building	Normal Power Unit Reactor Building	Shutdown/ Refueling Unit Reactor Building	Fuel Building
Kr-85m	1.09E-09	5.38E-09		3.03E-06		
Kr-85	1.14E-10	6.05E-10		2.96E-05		
Kr-87	1.08E-09	4.55E-09		9.54E-07		
Kr-88	2.01E-09	9.45E-09		3.58E-06		
Xe-131m	7.89E-10	4.20E-09		8.19E-05		
Xe-133m	2.22E-10	1.18E-09		6.35E-06		
Xe-133	4.88E-09	2.59E-08		2.87E-04		
Xe-135m	6.44E-10	1.86E-09		1.80E-07		
Xe-135	5.37E-09	2.75E-08		2.94E-05		
Xe-137	7.26E-11	1.59E-10		1.21E-08		
Xe-138	5.76E-10	1.63E-09		1.55E-07		
Ar-41	N/A	N/A		1.47E-07		
I-131	2.36E-10	6.97E-11	1.18E-09	8.97E-09	2.34E-10	5.77E-11
I-132	1.01E-09	2.58E-10	4.37E-09	7.10E-10	6.98E-10	2.26E-10

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 6 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Auxiliary Building	Normal Power Unit Safeguards Building	Shutdown/ Refueling Unit Safeguards Building	Normal Power Unit Reactor Building	Shutdown/ Refueling Unit Reactor Building	Fuel Building
I-133	7.27E-10	2.11E-10	3.59E-09	4.13E-09	6.91E-10	1.76E-10
I-134	1.43E-09	3.12E-10	5.29E-09	4.43E-10	7.11E-10	2.90E-10
I-135	1.32E-09	3.70E-10	6.28E-09	2.51E-09	1.14E-09	3.13E-10
Cs-134	2.30E-13	1.09E-12		2.54E-09		4.15E-12
Cs-136	2.22E-14	9.69E-14		1.38E-09		
Cs-137	3.32E-13	1.46E-12		5.73E-09		6.59E-12
H-3	1.68E-07	8.97E-07		7.85E-05	2.53E-06	3.54E-06
Cr-51	1.32E-13	6.47E-13		5.77E-09		4.39E-13
Mn-54	3.12E-14	1.58E-13		5.22E-09		7.32E-13
Fe-59	2.04E-14	1.01E-13		2.00E-09		
Co-57				8.01E-10		
Co-58	7.28E-13	3.84E-12		2.07E-08		5.12E-11
Co-60	2.04E-13	1.03E-12		2.69E-09		2.00E-11
Sr-89	3.18E-13	1.52E-12		9.94E-09		5.12E-12

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 7 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Auxiliary Building	Normal Power Unit Safeguards Building	Shutdown/ Refueling Unit Safeguards Building	Normal Power Unit Reactor Building	Shutdown/ Refueling Unit Reactor Building	Fuel Building
Sr-90	1.23E-13	5.87E-13		5.41E-09		1.95E-12
Sb-125	1.48E-15	7.89E-15				1.39E-13
Zr-95	3.83E-13	2.02E-12				8.78E-15
Nb-95	1.42E-14	6.07E-14		1.23E-09		5.85E-12
Ru-103	1.12E-14	4.65E-14		1.14E-09		9.27E-14
Ru-106	4.33E-15	1.21E-14				1.68E-13
Ba-140	1.69E-13	8.08E-13				
Ce-141	1.15E-14	5.26E-14		8.67E-10		1.07E-15
C-14	2.62E-09	9.11E-09		1.67E-06		0.00E+00
Total[b] MPC:	1.13E-01	2.32E-01	3.46E-01	7.40+01	5.71E-01	7.32E-01

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 8 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Normal Power Unit Turbine Mezzanine	Shutdown/ Refueling Unit Turbine Mezzanine	Normal Power Unit Turbine Basement	Shutdown/ Refueling Unit Turbine Basement	Normal Power Unit Worst Room	Shutdown/ Refueling Unit Worst Room
Kr-85m	4.77E-14		1.19E-14		1.21E-07	
Kr-85	4.67E-15		1.17E-15		1.22E-08	
Kr-87	4.75E-14		1.19E-14		1.27E-07	
Kr-88	8.82E-14		2.21E-14		2.25E-07	
Xe-131m	3.23E-14		8.07E-15		8.50E-08	
Xe-133m	9.51E-15		2.38E-15		2.40E-08	
Xe-133	2.02E-13		5.04E-14		5.25E-07	
Xe-135m	3.80E-14		9.56E-15		9.29E-08	
Xe-135	2.30E-13		5.74E-14		5.85E-07	
Xe-137	6.09E-15		1.54E-15		1.37E-08	
Xe-138	3.46E-14		8.72E-15		8.42E-08	
Ar-41	N/A		N/A		N/A	
I-131	2.47E-14	1.26E-14	6.17E-15	3.15E-15	1.41E-09	1.67E-08
I-132	4.34E-14	2.22E-14	1.09E-14	5.54E-15	6.33E-09	7.50E-08

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 9 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Normal Power Unit Turbine Mezzanine	Shutdown/ Refueling Unit Turbine Mezzanine	Normal Power Unit Turbine Basement	Shutdown/ Refueling Unit Turbine Basement	Normal Power Unit Worst Room	Shutdown/ Refueling Unit Worst Room
I-133	6.64E-14	3.39E-14	1.66E-14	8.47E-15	4.38E-09	5.18E-08
I-134	3.25E-14	1.66E-14	8.14E-15	4.15E-15	9.64E-09	1.14E-07
I-135	9.23E-14	4.71E-14	2.31E-14	1.18E-14	8.05E-09	9.53E-08
Cs-134	1.82E-14		4.55E-15		2.21E-11	
Cs-136	1.73E-15		4.32E-16		1.96E-12	
Cs-137	2.37E-14		5.91E-15		2.94E-11	
H-3	1.65E-09		4.11E-10		1.81E-05	
Cr-51	1.24E-15		3.11E-16		1.31E-11	
Mn-54	6.51E-16		1.63E-16		3.19E-12	
Fe-59	1.20E-16		3.01E-17		2.04E-12	
Co-57					7.77E-11	
Co-58	1.86E-15		4.64E-16		2.08E-11	
Co-60	2.17E-16		5.43E-17		3.07E-11	
Sr-89	5.67E-17		1.42E-17		1.19E-11	

TABLE 12.2-26
RADIOACTIVE AIRBORNE CONCENTRATIONS (uCi/cc) - ORIGINAL LICENSING BASIS[a]

(Sheet 10 of 10)

(1 Unit in Normal Operations, 1 Unit Shutdown or Refueling)

Isotope	Normal Power Unit Turbine Mezzanine	Shutdown/ Refueling Unit Turbine Mezzanine	Normal Power Unit Turbine Basement	Shutdown/ Refueling Unit Turbine Basement	Normal Power Unit Worst Room	Shutdown/ Refueling Unit Worst Room
Sr-90	4.89E-18		1.22E-18		1.59E-13	
Sb-125	4.09E-11					
Zr-95	1.65E-14		4.13E-15		1.23E-12	
Nb-95	7.50E-15		1.87E-15		9.40E-13	
Ru-103	2.31E-13		5.79E-14		2.45E-13	
Ru-106	8.70E-12		2.17E-12		1.63E-11	
Ba-140	1.45E-13		3.63E-14		1.06E-12	
Ce-141	3.84E-15		9.60E-16		1.84E-07	
C-14						
Total[b] MPC	1.80E-03	3.14E-06	4.49E-04	7.86E-07	4.90E+00	5.14E+00

a) Historical. Not subject to future updating. Has been retained to preserve original design basis.

b) Design assessment made with the provision of 10 CFR 20.1-20.601.

12.3 RADIATION PROTECTION DESIGN FEATURES

12.3.1 FACILITY DESIGN FEATURES

The design guidance given in NRC Regulatory Guide 8.8, Revision 3, Section C.2 was applied in the facility and equipment design, as practicable, consistent with benefits received and costs incurred. The following features are provided in the design of equipment and development of plant arrangements.

12.3.1.1 Equipment Design Features

The following design features are provided for the nuclear system equipment that contains sources of radiation to ensure that ORE's are ALARA.

12.3.1.1.1 Evaporators

1. Evaporator packages are designed with consideration given to occupational exposure. A 24-in. thick concrete shield separates the evaporator concentrator tank, gas stripper, and vent condenser from the pumps, valves, and instrumentation. The 24-in. thick concrete provides significant dose reductions. The systems are skid-mounted and arranged so that the evaporator components are easily accessible.

2. Instruments installed in lines which service the evaporators are located outside the evaporator compartments.

3. Shielding is provided for local sample vessels associated with evaporator streams.

12.3.1.1.2 Pumps, Filters, and Demineralizers

1. Canned pumps are used, where appropriate in the WPS to eliminate the potential for leaks. In addition, pumps which carry concentrated boric acid are supplied with high-quality mechanical seals to minimize pump maintenance.

2. Where it is possible, NSSS pumps are provided with flanges to facilitate removal and replacement.

3. Canned pump stators are air-filled, as opposed to silicon oil-filled, to eliminate the potential for process stream contamination and maintenance.

4. In cases where pumps are located in shielded compartments and local start stop or jog switches are required, these switches are located outside the pump compartment.

5. Filters are designed for removal from the top, using a minimum number of tools. Filter bolt lead-in for tool entry is provided to facilitate remote loosening. A detailed description of expended filter cartridge processing is provided in Section 11.4.2.

6. It is possible to remove filter cartridge assemblies into self-shielded transfer containers without the requirements of visual observation; see Section 11.4.2.

7. All demineralizers in the NSSS are considered potentially radioactive. Therefore, each demineralizer is located inside an individually shielded compartment except the steam generator cation blowdown demineralizers and mixed bed blowdown demineralizers, which are grouped and shielded by function and unit.

8. To minimize personnel exposure, special care was taken to ensure that only the required process lines enter into or pass through the demineralizer cubicles.

9. Design changes have been incorporated in the reactor coolant pump seals to improve reliability/maintainability. The No. 2 and No. 3 seal consists of a factory tested and assembled cartridge seal assembly. This unit reduces the time required for seal changeout and maintenance. It is projected the RCP seal life is approximately 4.5 years as opposed to previous 3 year time frame. Thus, radiation exposures will be reduced due to the quicker change time and the reduced frequency of this maintenance.

   An improved reactor coolant pump maintenance system has been developed which allows shaft seal access without removing the pump motor. This reduces manpower, radiation exposure, and pump maintenance downtime.

10. The use of a refueling cavity recirculation cleanup system, equipped with filters, will minimize radioactive particulates in the cavity shield water. This filtration reduces the radiation level at the surface of the water and reduces contamination of the cavity walls upon cavity draining at completion of refueling. Thus, radiation exposure to personnel during cavity cleaning is reduced.

12.3.1.1.3 Tanks and Heat Exchangers

1. To facilitate draining, horizontal and flat-bottom tanks are normally sloped toward the drain, thus allowing radioactive sediment to be more easily flushed from the tank, resulting in lower exposures to plant personnel.

2. To prevent radioactive gases and vapors from escaping to the atmosphere via tank overflows, loop seals are provided, where appropriate.

3. NSSS heat exchangers are designed without the need for special maintenance fixtures, such as blind flanges, which add to maintenance time. For example, by threading every fourth hole in the heat exchanger tube sheet, the shell-to-tube sheet joint need not be broken for maintenance and inspection. The shell and tube assembly can be lifted intact above the channel head to expose the tube ends for inspection or testing, or both, for leaks.

12.3.1.1.4 Valves

1. Special attention is given to valve locations so that operation and maintenance is conveniently performed from standard service equipment such as ladders, platforms, and floors.

2. All modulating and three-way control valves which normally carry radioactive fluid have leakoffs.

3. Bolted body-to-bonnet forgings are used. This design permits use of ultrasonic testing in place of time-consuming radiography during inservice inspection. In addition, this design facilitates assembly/disassembly, and eliminates the need for gaskets as required by the pressure seal valve design. On two in. Pipes and smaller, most valves are designed for zero stem leakage.

12.3.1.1.5 Piping

1. Where possible, piping is arranged so that ample space is available for installing and maintaining electrical heat-tracing equipment, thus reducing jobtime requirements and exposure.

2. Pipe runs carrying radioactive fluid are arranged so that potential crud traps are minimized.

3. Where possible, for lines which are used to transport spent resin, six-diameter bends and continually sloping lines are used.

4. The number of vents and drains on piping are minimized to reduce possible leak paths and likely crud traps.

12.3.1.1.6 Reactor Coolant Piping Insulation

Insulation in the areas of the reactor vessel nozzle welds is fabricated sections with a thin stainless sheet covering, featuring “suitcase” type clasps, to hold the sections in place. By merely unsnapping four, a matter of minutes time, a 29-in.-diameter pipe weld can be exposed for examination. Replacing the insulation after the job is as fast, thereby significantly reducing exposure time.

12.3.1.1.7 Process Systems

Concepts such as minimizing radioactive wastes and thus reducing radiation exposure were realized through process design techniques. Water reuse designs will significantly reduce the quantity of water processed in the liquid radwaste system. By virtue of processing less radioactive liquids, a reduced volume of evaporator bottoms is generated, thereby reducing potential exposure due to handling in the solid waste management system. The design for reuse of resin sluice water is illustrated on Figure 12.3-1. Water pumped from the spent resin storage tank into demineralizers transports the refueling water/resin slurry back to the tank. The sluice water is then reused after passing through a resin retention screen within the tank. Another means of minimizing radioactive effluent handling was to incorporate a boron thermal regeneration system (BTRS) [1]. Effluents generated during load follow operations with BTRS range from 2 to 10 system volumes per year compared to 50 to 60 system volumes per year for plants without BTRS.

Arrangement of unit operations in the system design precludes excess exposure which might result from subsequent maintenance operations. Demineralization and filtration of process streams prior to entry into large storage tanks is one method of equipment arrangement employed. The objective is to retard activity accumulation within the tanks thereby minimizing the source of radiation. In the BRS, the recycle evaporator feed demineralizers are located upstream from the recycle holdup tanks. The prime reason for this location is to minimize the

activity concentration of the effluent entering the tanks and thereby keep the tanks as clean as possible. Table 12.3-1 gives demineralizer decontamination factors (DF) for various isotopes measured in two operating plants [2]. These demineralizers are used for reactor coolant purification. Decontamination factors of similar magnitude are expected for the recycle evaporator feed demineralizers which contain resin in the hydrogen hydroxyl form.

12.3.1.1.8 Refueling

The design of refueling components incorporates such improved features as:

1. An integral missile shield (Unit 1 only) and a roll-away missile shield (Unit 2)

2. Quick disconnect cables for CRDM's

3. Elimination of part-length CRDM's

4. A permanent reactor cavity seal ring

5. A quick reactor vessel head removal system

6. An upgraded manipulator crane control system

7. In-Containment fuel storage

8. A Containment fuel handling bridge crane

9. A reactor vessel head shield.

These refueling features can considerably reduce the time required to perform refueling operations and, consequently, the overall downtime for reactor refueling. These features are designed with consideration given to reducing ORE; however, they cannot be economically justified solely on the basis of reduced ORE.

The pressure vessel head closure system is designed to reduce the time required to tension and detention the reactor vessel studs and to remove and insert the studs into the vessel flange. The system includes quick disconnect/connect stud tensioners which have split, quick acting stud gripper devices to eliminate the need for threading tensioners onto the studs. The system also includes pneumatic tools for removing/inserting studs into the vessel flange.

Connectors on electrical cables permit easy disconnect/connect of the cables when the reactor vessel head is removed, thereby reducing the time spent in making and verifying electrical connections following refueling. A cable tray also permits convenient storage of these cables. These developments significantly reduce personnel exposure during refueling operations.

An integral TV camera that has been designed into the upgraded refueling machine (manipulator crane) or a portable camera may be used for visual confirmation of positioning during fuel handling. This camera or a portable camera is to be used for core-mapping. The speed and accuracy of the refueling machine movement and positioning has also been improved. The features incorporated into the upgrade refueling machine serve to reduce both the refueling time and the radiation exposure associated with refueling operations.

Since the occupational radiation exposures for refueling vary from shutdown to shutdown at a given plant and vary among different plants, no single estimate can be established on the savings in occupational radiation exposure attributable to the improved refueling concept. However, an estimate may be obtained by using data from a typical refueling which usually results in radiation exposures ranging from 30 to 50 man-rem. It is estimated that the improved refueling concept at CPNPP could reduce these values to the range of 10 to 25 man-rem.

12.3.1.2 Facility Design Considerations

The following considerations are incorporated into the arrangement and design of the facility to ensure that occupational radiation exposures are ALARA:

12.3.1.2.1 Maintenance Considerations

The layout of personnel access areas, pipe routing, and equipment arrangement minimize occupational radiation exposure during maintenance activities. Radioactive equipment requiring service is normally located in individually shielded cubicles of a size designed to provide easy accessibility for maintenance. In addition, adequate equipment-handling facilities, such as cranes and overhead rails, are provided for maintenance work.

Accessibility for equipment removal was considered during the plant design so that component removal for servicing is possible. Access control and traffic patterns are considered in the plant layout to minimize the spread of contamination. The following design features are provided to minimize the spread of contamination:

1. Cubicles containing components such as pumps, heat exchangers, and valve stations are provided with vent and drain systems. The equipment vents and drains are piped to a collection system to prevent any contaminated fluid from flowing across the floor of the cubicle.

2. All-welded piping systems are used in radioactive systems to the maximum extent possible to reduce system leakage.

3. Valves designated for radioactive system service are normally provided either with leakoff connections piped directly to a collection system or with diaphragm or bellow seals. Valve designs without stem seals or leakoffs are used when ALARA considerations are accomplished by other means.

4. Tank cubicles are provided with raised entrances to prevent the spread of contamination in the event of a tank rupture.

For maintenance purposes where equipment access is required, various equipment such as pumps and heat exchangers can be remotely flushed before direct maintenance occurs by personnel.

The capability to decontaminate potentially contaminated areas is facilitated by the availability of decontamination water hose stations and by the use of a suitable concrete-finish surface design. Protective coating materials used in radiation areas were originally selected using ANSI N512-1974 and the coating manufacturer's recommendations for guidance to provide decontaminable finishes. See Section 6.1B.2 for a description of the selection and qualification of protective

coatings. Floor drains are provided in all areas of the plant that are subject to potential contamination.

Systems which may become contaminated are normally designed for local decontamination of components prior to servicing. The capability of performing local decontamination is provided by a system design that incorporates proper slopes for drainage, proper vents at high points, drains at low points, and valving for major equipment isolation.

12.3.1.2.2 Penetrations

Because precautions are taken to minimize the contribution of radiation streaming through penetrations, penetrations through shield walls or slabs for pipes, extension stems, ventilation ducts, and cable trays are located where possible so that they are not in a direct line with major radiation sources. In addition, bends and offsets are used when required. Also, filling materials (such as lead wool) are used where applicable to fill voids in penetrations.

Separate shielded areas in the Safeguard Buildings are provided for radioactive and nonradioactive Containment piping penetrations.

12.3.1.2.3 Remote Handling Equipment

In accordance with NRC Regulatory Guide 8.8, Revision 3, remote handling equipment and other special tools are provided, when necessary, to reduce external radiation exposure. Careful design in conjunction with proper use of equipment minimizes radiation exposures to plant personnel. Where necessary in areas of high radioactivity, provisions, such as stem extensions through shielding walls and slabs, are made to reduce personnel exposure. Specifically, remote handling equipment is used for the following:

1. Remote handling equipment is used for waste processing in the Fuel Building. A description of the solid waste management system is given in Section 11.4.

2. Remote handling equipment is used in the in-core instrumentation room where the flux mapping system is used to drive the neutron flux detectors into and out of the reactor core through the flux thimbles. This system consists mainly of drive mechanisms, path transfer devices, and a movable frame assembly.

3. Remote handling equipment is also used for transfer of spent filter cartridges to a solid waste disposal area in the Fuel Building. Filters considered to be potentially radioactive are grouped together and located inside individual shielded compartments. Provisions are made for safe handling of replaced filter cartridges and the spent filter cartridges are transported to the solid waste disposal area via a shielded filter element drop zone. A description of the handling operation is in Section 11.4.

12.3.1.2.4 Ventilation System

The ventilation system is designed to ensure control of airborne contaminants during all modes of operation including maintenance activities. Those aspects of the design that relate to controlling the concentration of airborne radioactivity in equipment cubicles, corridors, and operating areas which are normally occupied by operating personnel are discussed in Section 9.4.

The ventilation system is designed for easy access and service to keep doses ALARA during maintenance and filter changes. The design provisions that are included to achieve ALARA exposures in these situations are discussed in Subsection 12.3.3.

12.3.1.2.5 Tool Design for Maintenance and Inspection

Tools and procedures have been developed which are aimed at keeping ORE's ALARA during maintenance and inservice inspection tasks. The major emphasis is in areas with the potential to prevent large accumulations of radiation exposure, as described in the following paragraphs.

12.3.1.2.5.1 Inservice Inspection

1. Steam Generator Tubing

   Periodic inservice inspection of the steam tubing is required to monitor the integrity of the tubing and to maintain surveillance of potential mechanical degradation or tube deterioration throughout the life of the steam generator. Examination via eddy current testing is the most practical technique. It was developed so that accurate data can be obtained. This technique requires that a long probe be inserted in a tube and then withdrawn at a constant rate of speed, during which time the signal is read and recorded.

   Until recently, this technique required that a man stand at the open steam generator manway with a length of aluminum conduit and hold it in such a position that when an eddy current probe was pushed through the conduit, it would then enter the steam generator tube for the examination. This procedure would take as long as three to five minutes for examination of one tube, with the operator standing in a field of radiation which might range from 150 mrem/hr to 800 mrem/hr. To reduce exposures from this source, a fixture is used which remotely positions the probe in any desired tube and mechanically inserts and withdraws the probe. The control of the total examination is done at a remote location within the Containment, with the operators in a low field of radiation (typically 10 mrem/hr). This fixture does require the entry of individuals into the steam generator for short periods of time for installation and removal. However, exposure is significantly less than that using the hand probing method.

2. Reactor Vessel and Piping Inspection

   Selected welds in the RCS of a NSSS must be examined periodically throughout the life of the plant to ensure the continued integrity of the welds. This examination is done by ultrasonic testing and dye penetrant techniques in close proximity to the component. Of special interest in the periodic examinations are the welds which attach the reactor coolant pipe nozzles to the reactor vessel. The examination is frequently complicated by the fact that most of the welds are covered by thick layers of thermal insulation which must be removed prior to the examination and replaced upon completion of the inspection.

   A dual approach is taken to minimize radiation exposures during these inspections. The first provides an optimum design configuration of the component to readily accommodate the UT sensor. For example, the reactor vessel nozzle area is tapered along the reinforced areas to ensure a smooth transition. The branch pipe locations are selected to

ensure no interference from one branch to the next. All weld-to-pipe interfaces require a smooth, high-quality finish.

The second approach for reducing exposures is the use of remote inspection techniques. An inspection tool has been developed for examining the reactor vessel nozzle welds by positioning the ultrasonic crystals remotely and by providing readout of the signals in a low radiation area. This tool is inserted in the open reactor vessel after removal of the fuel and reactor vessel internals. The refueling cavity is flooded with water so that the operators are shielded by about 30 ft of water.

12.3.1.2.5.2 Maintenance and Repair Operations (Steam Generator Tube Plugging)

Occupational Radiation Exposure reduction is considered in the design of components for the steam generator tube plugging process, through maximizing the use of remote mechanical tube plugging methods.

12.3.1.2.6 Materials Selection and Manufacturing Process

Materials selection and manufacturing processes in the design and manufacture of components in direct contact with the reactor coolant can have a profound effect on radiation levels and subsequent potential radiation exposures. The two major contributors to radiation levels in the pressurized water reactor (PWR) are Co-58 and Co-60. Co-58 is formed from an (n,p) reaction on Ni-58, which is present in many alloys in contact with the RCS, particularly the Inconel steam generator tube material. Co-60 is formed from an (n, γ) reaction on Co-59 which is an impurity in stainless steel and Inconel. An additional potential source of Co-59 is Stellite (Registered trademark of Haynes Stellite Company), which is approximately 50 percent cobalt and is used on wearing surfaces.

The mechanism of accumulating deposits of radioactive materials in components involves the following steps [4]:

1. Release of base metal to the RCS

2. Deposition of base metal in active core region

3. Activation of base metal

4. Release of activated metal from core region

5. Deposition of activated metal in components

Any steps taken during the design and operation of a PWR to reduce any of the aforementioned actions will potentially reduce radiation levels in components.

Four areas related to materials selection and manufacturing processes are as follows:

1. Material selection to minimize maintenance

2. Bright annealing of steam generator tubes

3. Equipment specifications on cobalt content of materials

4. Chemistry control to minimize radiation levels and corrosion

1. Materials Selection to Minimize Maintenance

Reduced occupational exposures in operating PWRs can best be accomplished by the minimization of maintenance, repairs, and inspection.

a. Steam Generator Tubing

For a number of years a program has attempted to evaluate alternate steam generator materials. The program consists of various tests simulating steam generator conditions for a number of materials. For the Unit 1 replacement steam generators (RSGs), experience indicates that thermally-treated Inconel 690 is a superior material and it was selected for this unit. For the Unit 2 steam generators, it was concluded that Inconel 600 was one of the best steam generator materials for maintaining the integrity of the steam generator tubes in steam generators.

An additional consideration related to steam generator material selection is the quantity of base metal that can be activated and can result in subsequent radiation exposure. If the corrosion products released from two materials of different nickel content had the same composition as the base metal, then the material with the lower nickel content would result in the production of less Co-58 than the material with the higher nickel content. However, crud from operation reactor plants with Inconel 600 steam generators has been found to consist of nickel ferrite, Ni_xFe_{3-x} O4, with x less than or equal to 1.0. Thus, it is concluded that the released corrosion products from Inconel 600 have less nickel than base metal. If the crud composition for lower nickel content materials is the same, no benefit in Co-58 radioactivity buildup is found. In addition, there is no evidence of nickel in other compounds in core crud [4].

b. Hard Facing Materials

It has been suggested that consideration be given to Colmonoy and Hastelloys as replacements for Stellite as a means of potentially reducing the cobalt in crud. However, the choice of materials should be based on the best combination of superior wear characteristic or reduced inspection and maintenance requirements, or both, and potential contribution to inplant radiation exposure. With regard to Hastelloys, our experience shows that these materials do not have the required wear characteristics, even in cold worked condition, for our applications. Colmonoy has been investigated for possible application in components with the following conclusions:

1. Experience is with Colmonoy 5 (casting) and Colmonoy 6 (powder) weld deposited by oxyacetylene and plasma transferred arc with the latter being superior in the quality of the finished product.

2. Colmonoy thrust shoes undergo unacceptable corrosion attack in contact with Graphitar during wet layup. Thus, all thrust shoes continue to be made from Stellite.

3. Plasma hardsurfacing with Colmonoy requires wider base metal surface because of the increased fluidity as compared to Stellite.

4. Present Colmonoy hardsurfacing procedures require processing in the sensitization temperature range of austenitic stainless steels (900 to 1200°F).

5. Corrosion performance of the Colmonoy materials has not been demonstrated in borated water systems.

In addition, it should be noted that if these substitute materials have lesser wear characteristics compared to Stellite, the quantity of nickel introduced into the circulating coolant from these substitute materials could increase drastically. This would result in an increase in Co-58 levels in the plant.

Stellite was found to be a major contributor to cobalt in the Shippingport Atomic Station [5]. However, this reactor had check valves and stop valves (among others) in the active RCS, which contributed heavily to the cobalt contribution. Additionally, core crud deposits in the Shippingport reactor showed approximately 0.4 w/o cobalt.

(See the following discussion for significance of the cobalt content of core crud deposits.)

A review of the Stellite inventory in the NSSS indicates that there are four major areas of use: reactor internals, RCS pump journals, RCS valves, and CRDMs. The Stellite application in these areas is limited to load bearing surfaces in which excessive wear might otherwise result. A summary of the surface area wetted by the reactor coolant of these three applications is given in Table 12.3-2.

Figure 12.3-2 provides justification that Stellite is not a major contributor since the cobalt weight percent in crud is generally well below the 0.1-percent level. As a final point on this subject, tungsten is a component material in the Stellite alloy (Stellite has 45 percent tungsten). If significant contributions to the deposited radiation source term were originating from wear of Stellite material, a measurable radioactive tungsten level would be detected in the crud deposits. W-187 has only been detected in one or two reactors, but the level is approximately 40 times that which could be expected based on Co-59 measurements. Thus, the RCS W-187 is due to other sources in the system (i.e., Inconel impurity). Through implementation of a cobalt reduction program, newly developed low cobalt or cobalt-free iron based alloys which have been industry tested to demonstrate similar corrosion and wear characteristics as the Stellite family of hardfacing alloys are considered, primarily for use in RCS valves and other system valves which communicate with the reactor coolant system.

2. Bright Annealing of Steam Generator Tubes

Another technique which demonstrates efforts to reduce the source of occupational exposure is bright annealing of steam generator tubes. Bright annealing involves the

annealing of the Inconel tubes while both the inside and outside tube surfaces are exposed to a hydrogen atmosphere.

This process reduces the corrosion tendency of steam generator tubes and also minimizes the crud retention properties of the tubes. This technique has been used on all steam generators with Inconel tubes dating back to the Inconel 600 tubes used in the Zorita steam generators. The Connecticut-Yankee and San Onofre steam generators are the only ones using Inconel 600 which were not bright annealed. The tubes were grit blasted with $A1_2O_3$ to minimize crud retention properties.

3. Equipment Specifications on Cobalt Content of Materials

The current limits on cobalt content of materials in contact with reactor coolant are given in Table 12.3-3. The cobalt content of materials wetted by reactor coolant is restricted, although there have been infrequent occasions in the past when the restrictions were temporarily relaxed due to unavailability of material with the specified cobalt content. However, this relaxation has been for minor components of minimal surface area. Through implementation of a cobalt reduction program, specifications for valves which potentially contribute to the cobalt inventory reaching the reactor core have been updated to add consideration for the procurement of low cobalt or cobalt-free materials for replacement valves and parts. Such materials should have mechanical and physical characteristics suitable for valve functionality.

To evaluate the adequacy of these equipment specifications, input must be available regarding the actual cost of materials at various cobalt levels and the total cost benefit evaluation of materials at various cobalt levels:

a. Actual Cost of Materials at Various Cobalt Levels

The estimated additional costs for 304 stainless steel plates and forgings for various cobalt levels are given in Table 12.3-4. These values were prepared as part of the CPNPP Operating License application prior to Unit 1 operation and will not be updated. If the entire nuclear industry decided to order low cobalt <0.1-w/o stainless steel for all RCS components, the supply and demand would shift dramatically, thereby resulting in higher costs for reduced cobalt. The potential exists for using carbon steel clad internally with low cobalt stainless steel as an alternative to all low cobalt stainless steel. This potential has been investigated with the conclusion that the cost differential based on using carbon steel is offset by the cladding process costs, special welding costs, and so forth.

The equipment specification for Inconel 600 used in manufacturing steam generator tubes of 0.1-w/o cobalt results in field delivered tubes with cobalt contents ranging from 0.05-w/o to 0.1-w/o cobalt. The average cobalt content of field delivered Inconel 600 tubes is approximately 0.07-w/o cobalt. Within any one steam generator, the same statistical distribution of cobalt content of tubes exists. The manufacturers of Inconel have indicated that Inconel 600 with a guaranteed cobalt content less than 0.1-w/o for all material is unavailable at any price in the quantity needed for steam generator tubes.

b. Materials Availability

The material prices for various cobalt levels are based on the present availability of electrolytic nickel. If the supply situation of virgin nickel were to deteriorate and suppliers began using scrap to a greater extent, it is conceivable that the 0.10-w/o cobalt stainless could be unobtainable. For example, a sharp rise was noted in cobalt levels in unspecified stainless steel in 1973 due to the unobtainability of virgin nickel. The availability of low cobalt stainless also depends on the availability of low cobalt ferro-chrome. Since most of the United States' supply of chromium is from Third World countries, its availability cannot be ensured (nor can the prices).

c. Impact Evaluation

An impact evaluation, pertaining to reduction in cobalt content of materials, cannot be performed without taking into consideration alternate means of reducing occupational exposure. The basic philosophy is that dollars should be used where they can be most effective in overall product quality. The further reduction of cobalt content alone may not be cost effective in reducing radiation exposure. One must determine if there are other methods of reducing exposure at a lesser incremental cost per unit of exposure avoided.

Included in this impact evaluation is the need to consider the specific source of the cobalt that results in radiation exposures. As a first approximation, the radiation contribution might be equated on the basis of the surface area of the components in contact with the reactor coolant, weighted by the cobalt content of that material. The weights and surface areas are given in Table 12.3-2. On a weighted surface area basis (the product of the surface area and the cobalt content limit), it can be seen that the steam generators are predicted to be the largest contributor, as given in Table 12.3-5. In this type of analysis, it might be concluded that the cobalt content of materials should be limited in the steam generators.

The second approximation considers the corrosion rate of the stainless steel and Inconel in addition to the surface area and cobalt content. Using an approximation to the corrosion rate at steady state of 0.5 mg/dm^2-mo for stainless steel and 1.5 mg/dm^2-mo for Inconel 600 [6], the second approximation as given in Table 12.3-5 can be constructed. As shown in the table, the steam generator Inconel is predicted to be the primary contributor to the total cobalt inventory in the plant. This conclusion is further substantiated by comparing the iron to nickel ratio in deposited crud on the core. If the stainless steel has a corrosion release elemental distribution of Ni/Fe = 0.1, and assuming the Inconel has a Ni/Fe ratio = 0.5 (based on releasing Nickel ferrite, $NiFe_2O_4$), then the mixed corrosion products should have a ratio of nearly 0.5 for the corrosion rates given previously. Figure 12.3-3 gives a compilation of Ni/Fe ratios measured at operating plants. From the figure, it can be concluded that the mean Ni/Fe ratio is about 0.35. This is slightly less than that assumed for Inconel corrosion, but nearer to that assumed for Inconel than stainless steel. If the Inconel 600 corrosion is the major source of cobalt, then the percent of cobalt in the crud should be near that for the Inconel base metal.

Figure 12.3-2 shows the measurements of cobalt in core crud deposits from many plants over a number of years. The figure shows that most deposits have a cobalt concentration (in weight percent) less than 0.01 indicating that most of the crud comes from corrosion of materials very low in cobalt content. Figure 12.3-2 also substantiates the claim that Stellite is not a major contributor to the radiation levels due to Co-60 since the cobalt level in core crud deposits is very low.

Therefore, it is concluded that the present equipment specifications on cobalt content of materials wetted by the RCS are adequate.

However, implementation of a CPNPP cobalt reduction program provides additional assurance that efforts to limit and reduce cobalt based source materials are practiced especially for valve replacement and maintenance activities involving valves which communicate with the reactor coolant system.

4. Chemistry Control to Minimize Radiation Levels and Corrosion

a. Radiation Level Minimization

A third area of involvement in attempting to reduce the source of occupational exposure is in methods to reduce radiation levels in the reactor cavity refueling water. Upon transition of the reactor coolant from a reducing environment to an oxidizing environment during the refueling shutdown, a significant quantity of Cobalt-58 (100 to 2000 Ci) and a lesser quantity of Cobalt-60 is solubilized in the reactor coolant. The concept of peroxide (H_2O_2) addition to the reactor coolant after cooldown to 140°F has been considered in an attempt to effect the solubilization under controlled conditions. By the addition of peroxide, the solubilization occurs at such a time as the reactor coolant purification system can cleanup the circulating cobalt. The result of not adding peroxide at this time may be the solubilization when the reactor head is lifted, thereby allowing for mixing of the cobalt in the refueling water. It should be noted that the peroxide addition does not lower radiation exposures due to deposited crud nor is it a decontamination technique. It only allows for the solubilization of some radioactive crud (which would eventually solubilize anyway) under optimum conditions.

b. Corrosion Minimization

The hot hydro (hot) functional is conducted with only LiOH (no boric acid), which according to Riess [2] puts a protective coating on materials wetted by the RCS, thereby reducing corrosion.

Other programs such as the all-volatile steam generator chemistry might be considered exposure reduction programs as they are also aimed at reducing the maintenance, repairs, and inspection of aimed at reducing the maintenance, repairs, and inspection of steam generators.

The various decontaminations at Canadian National Deuterium Uranium reactors (CANDUs), boiling water reactors (BWRs), PWRs, and experimental reactors [7,8]

do not demonstrate that major decontaminations are either practicable or necessary at a PWR at this time.

While major decontaminations have reduced occupational exposures in the first few years after the decontamination, there is no evidence to show that long-term (e.g., greater than three to five years) exposures are reduced significantly in comparison to that expected with no decontamination. Surveys show that levels increase for the first two or three years after which a leveling trend is noted. Thus a system decontamination would reduce radiation levels only during this short time period. In evaluating the practicality of decontamination in PWRs, the extra radiation exposure accrued during the decontamination process must be compared to the projected savings in radiation exposure over the succeeding two year period. After this two year period, no savings should be projected due to the similarity of radiation levels thereafter and the large variability in radiation exposures from year to year. The overall practicality of decontamination must also include such factors as cost of the decontamination, cost of replacement power due to the extended shutdown period required for decontamination, and additional waste disposal cost. Further, decontaminations of CANDUs and BWRs do not show that decontaminations of large scale PWRs are practicable. Those techniques (including on-line decontamination) [8] used at CANDU reactors are not directly applicable to the PWR, since the reactor coolant chemistries are different.

CAN DECON (CANDU on-line decontamination) techniques cannot be used on-line but could be used four days after shutdown since the decon solution decomposes in a high radiation field. Since volume of coolant in an active core region is small in a CANDU, they get very slow decomposition. Since the volume of water in a core region is large (approximately 20-percent of RCS volume), in a PWR, it would decompose rapidly (approximately 15 minutes). At four days after shutdown, the decontamination solution would be effective for approximately 10 hours. CANDU experience indicates most of the decontamination is accomplished in approximately one hour with a DF = 6 for carbon steel and DF = 2 for stainless steel [9].

12.3.1.2.7 Criteria for Routing Radioactive Piping

There are no unshielded radioactive or potentially radioactive pipes in Zone I areas. (See Subsection 12.3.1.3.)

Pipes containing N-16 are routed behind the primary or secondary shield, or behind special shielding to a point where at least one min decay time has elapsed. Radioactive piping with contact dose rates in excess of the maximum design dose rate for a specified zone are not routed unshielded through areas where access is required for reading and adjustment of instruments, inspection, or maintenance and operation of nonradioactive equipment. Pipe shielding is designed to conform to the appropriate adjacent zone designation limitations.

12.3.1.2.8 Equipment Shielding

Equipment for radioactive systems is arranged in accordance with guidelines provided in the description of plant zoning given in Subsection 12.3.1.3. Major components such as tanks,

demineralizers, filters, and standard equipment packages are located in individual, shielded cubicles. Such equipment cubicles contain the minimum necessary piping so that access to these compartments is not needed during equipment operation. In the case of standard equipment packages where valves are not separated from the equipment, operation of these valves is performed from operating panels located in low radiation areas. The connecting piping and valves of small service pumps and major equipment are located in separate, shielded valve rooms. The lengths of pipes in these rooms are minimized to practical limits.

Inspection and maintenance of service pumps and valves can be accomplished when the system is not in operation. Manually operated valves in radioactive systems are operated from shielded valve operating areas located adjacent to valve rooms by the use of extension stems which penetrate the shield walls or slabs. Components of radioactive systems are fully shielded so that general areas such as corridors and stairways can be occupied by radiation workers 40 hr/wk, 50 wk/yr.

The standard equipment packages where valves are not separated from the equipment skid are the waste and recycle evaporators, the waste gas catalytic hydrogen recombiners, and the waste gas compressors.

The equipment packages are designed so that most valves are remotely manipulated. Locally operated valves on the equipment packages do not require manipulation during normal operation.

Valves on other components requiring durable packing are not used in the Westinghouse-supplied standard equipment packages other than the hydrogen recombiner located in the auxiliary building. Westinghouse provides diaphragm valves, control valves with bellows seals, or other valves which do not require durable packings. Graphite foil is used in the hydrogen recombiner, but since the recombiner is a gas processing equipment package it is not contaminated with corrosion products. Thus, maintenance of the recombiner valves does not constitute a major exposure source.

Furthermore, it is expected that the catalytic recombiner is a low-maintenance equipment package.

In addition, the equipment on the Westinghouse evaporator package skids are arranged so that serviceable equipment is separated from those components which accumulate significant amounts of radioactivity. Specifically, the arrangement is such that a 24-inch concrete wall separates the pumps, valves, and instruments from the evaporator concentrator tank, the gas stripper, and the vent condenser. Figure 12.3-13 illustrates this shielding wall in the recycle and waste evaporator cubicles.

Permanent flushing connections are provided on the waste evaporator, recycle evaporator, and catalytic hydrogen recombiner equipment packages. Flushing with water prior to maintenance assists in reducing exposure during maintenance on these units.

A sampling room for each unit is provided for sampling the primary plant systems. Personnel exposure to radioactive fluids is minimized by the use of sample hoods, isolation valves, sample vessel connections, and control of the sample pressure (by proper valve lineup) during all operations. Personnel exposure to radioactive gases is minimized by maintaining airflow through sample hoods, proper valve and sample vessel connection maintenance, and proper valve

operation. Permanent shielding is provided, where practicable, between the operator and the potentially highly contaminated components.

These equipment and facility design features limit occupational radiation exposure to ALARA levels. The equipment arrangements are provided in Figures 12.3-5 to 12.3-23.8.

12.3.1.2.9 Hot Lab and Counting Room

A Chemistry Hot Lab and Counting Room are provided to facilitate the handling, preparation, and analysis of radiochemical samples. The Hot Lab is located to provide easy access to the Primary Sample Rooms and is equipped with a fume hood that exhausts to the Primary Plant Ventilation System. Sink and floor drains in the Hot Lab are directed to the Liquid Waste Processing System. This room is equipped with an alarming area radiation monitor to detect and provide warning of abnormal radiological conditions. Additionally, remote readouts of area radiation monitors located in the Primary Sample Rooms are provided in the Hot Lab.

The Chemistry Counting Room is located adjacent to the Hot Lab and is designed to provide a low background area for performing radiological analyses of chemistry samples. A sample pass through window is provided between the Counting Room and Hot Lab to aid in contamination control and to minimize the handling of samples during transport.

12.3.1.3 Design Objectives and Zoning Guidelines

The facility, equipment, and radiation shielding are designed to provide protection against radiation for personnel and equipment, both inside and outside the plant, and for the general public. The plant radiation shielding is designed to perform the following functions:

1. Keep the radiation doses to plant personnel, construction workers, vendors, and authorized visitors within the limits set forth in 10 CFR Part 20.

2. Prevent excessive neutron activation and limit the accumulated dose to system components to specified limits.

3. Ensure that the dose rate levels in controlled access areas are kept below the maximum dose rate limit of the assigned zone.

4. Reduce radiation dose rates during inspection and maintenance caused by contained sources without unwarranted plant outages.

5. Permit access to and occupancy of the Control Room and Technical Support Center under accident conditions, including loss-of-coolant accidents (LOCA), and to limit the whole body dose to five rem for the duration of the accident as required by criterion 19 of 10 CFR Part 50, Appendix A.

6. Limit the dose rate at the exclusion area boundary caused by contained sources in the unlikely event of a LOCA.

The design of the facility and radiation shielding is greatly influenced by the radiation zoning philosophy used in the development of plant arrangements. Proper zoning of a plant facilitates

the arrangement of radioactive equipment in accordance with the requirements of 10 CFR Part 20 and the guidelines of NRC Regulatory Guide 8.8.

The following design guidelines are used to subdivide the plant into controlled and uncontrolled access areas as shown and described on Figures 12.3-4 to 12.3-23.8: For design purposes, the plant is subdivided into two areas: unrestricted access and restricted access. Unrestricted access areas will be occupied by personnel not involved in radiation work (Zone I).

The restricted access areas for radiation workers are subdivided into four zones. The first two zones are designed for personnel to perform routine work, Zone II for unlimited access time within a normal work week, and Zone III for limited periodic occupancy. The second two zones are designed for locating radioactive equipment and piping. In Zone IV, the amount of radioactive sources are limited to a practical minimum to allow access when needed. In Zone V, the amount of radioactive sources is not limited, so this designation is used for equipment and piping areas to which access is not normally required.

It should be noted that these radiation zone criteria are used for facility design and location of equipment and components based on expected maximum radiation levels; they are design assessments made with the provisions of 10 CFR 20.1-20.601. They are not intended for use as criteria for establishing requirements for radiation posting and access control during plant operations. Such operational requirements will be established in accordance with the requirements of the revised 10 CFR Part 20 based on actual radiation levels as discussed in Sections 12.5.3.2 and 12.5.3.3.

Zone I - Zone I areas are designed for no restriction on occupancy. Zone I areas are designed to limit radiation exposure caused by occupancy on a 40 hr/wk and a 50 wk/yr basis to the whole body dose limit of 0.5 rem/yr. Areas such as the Control Room, portions of the Turbine Building and offices are designated as Zone I areas. The maximum allowable external dose rate is 0.25 mrem/hr and represents a conservative estimate of the maximum values in areas within the station complex where plant personnel, construction workers, or routine visitors are permitted with unrestricted access. This upper limit is based on normal operation with one percent failed fuel and includes the effects of anticipated operational occurrences. Experience at operating pressurized water reactors, however, indicates that the average fraction of failed fuel is actually much less than 0.25 percent. A value of 0.12 percent is considered representative of expected operation. Accordingly, Zone I average external dose rates are expected to be well below the maximum. It is further expected that construction workers or visitors to the site will receive considerably less than 0.5 rem/yr limit because of the relatively short time they will be on the site.

Zone II - Zone II comprises the lowest of four levels of controlled access areas. Zone II areas are designed such that station personnel and authorized visitors can occupy these areas on a 40 hr/wk, 50 wk/yr basis and not exceed the allowable whole body dose of 1 1/4 rems per calendar quarter.

Corridors and areas outside radioactive enclosures where personnel can walk freely are included in this zone. Auxiliary equipment such as valves, dampers, and instrumentation and controls located in this zone can be maintained and operated without draining any equipment.

Average external dose rates and cumulative doses to individuals in Zone II areas are expected to be well below the maximum value cited in Table 12.3-6.

Zone III - Zone III constitutes the next level of areas designed for controlled access and includes areas where only weak radiation sources are expected to be present. Design of Zone III areas allows for personnel occupancy on a periodic basis consistent with cumulative exposure limitations. This zone includes valve-operating areas with stems which extend through shielding decks and auxiliary equipment such as valves, instrumentation, and controls.

Zone IV - Zone IV areas are designed to include high radiation areas with very restricted, controlled access and limited occupancy. Design of these areas considers the need for physical access controls such as lockable doors or gates or other physical barriers as discussed in 10 CFR Part 20.1601.

This zone includes all areas containing radioactive pipes, valves, pumps, heat exchangers, and tanks which need short-time access for operational activities such as reading gauges or manual operation.

Zone V - Zone V areas are designed to include those high radiation areas (10 CFR Part 20.1601) which are normally inaccessible when the reactor is at power or when equipment has large radioactive inventories. Design of these areas considers the need for physical access controls such as lockable doors or gates or other physical barriers as discussed in 10 CFR Part 20.1601.

Zone V is designated for large radioactive equipment which does not require maintenance while the reactor is at power, such as the primary coolant loop including pumps, steam generators, and the pressurizer.

Zone V is also designated for equipment such as the volume control tank, holdup tanks, gas decay tanks, demineralizers, filters, resin and waste storage tanks, pumps and heat exchangers which does not require maintenance when large radioactive inventories are present.

Table 12.3-6 presents a summary of the zoning criteria which was used to zone the facility. Figures 12.3-5 through 12.3-32 provide illustrative examples of equipment and piping layouts for liquid filters, demineralizers, recombiners, tanks, evaporators, pumps, steam generators, valve operating stations, and sampling stations. Instrumentation and control panel and motor control centers (MCCs) locations are provided on Figures 12.3-5 through 12.3-23.8. Sampling port locations are provided on Figures 12.3-24 through 12.3-32.

12.3.1.4 Decommissioning

The design features employed to maintain occupational radiation exposure as low as is reasonably achievable (ALARA) during operation of the plant are also applicable to decommissioning. These radiation protection facility design features are noted in:

Section 12.3.1.1 - Equipment Design Feature

Section 12.3.1.2 - Facility Design Considerations

Section 12.3.1.3 - Design Objectives and Zoning Guidelines

These sections include such features as equipment arrangement for ease of accessibility and maintenance; component design to minimize crud buildup; provisions for decontamination, temporary shielding, and development/use of remote handling equipment; provisions for remote

flushing of equipment; and specifications and limitations on the cobalt content in equipment components to minimize the buildup of activated corrosion products. The specifications and limitations on cobalt content in equipment components will serve to limit radiation doses from crud buildup during both operation and subsequent decommissioning. A summary of the features in Westinghouse PWRs that reduce occupational exposure are given in Reference [13]. Further discussion of design considerations relevant to maintaining personnel exposure ALARA during decommissioning are provided in Section 12.1.2.

With these types of features, the AIF study on decommissioning [12] has estimated occupational radiation exposure for PWRs for the primary decommissioning alternatives as follows:

Mothballing	150 man-rem
Entombing	130 man-rem
Prompt Removal/Dismantling	630 man-rem
Mothballing - Delayed	
Removal/Dismantling	460 man-rem
Entombing - Delayed Removal/Dismantling	440 man-rem

12.3.2 SHIELDING

12.3.2.1 Shield Design Criteria

In general, radiation shielding throughout the plant is designed to meet the specifications established in 10 CFR Part 20 and to keep personnel radiation exposure ALARA. More specifically, the criteria for penetrations presented in Subsection 12.3.1.2.2, the zoning criteria, and the general guidelines stated in Subsection 12.3.1.3 are used as a basis for shielding calculations and engineering decisions affecting potential radiation exposures.

12.3.2.2 Calculation Method

Shield wall and slab thicknesses are determined by using basic shielding data and equations. Nuclear data is taken from Nuclear Data Sheets, National Academy of Sciences [11]. The methods used for calculations are described in The Engineering Compendium of Radiation Shielding [10].

Computer codes used for the original shielding design are as follows:

1. COSACS - Program calculates and tabulates dose rates from cylindrical radiation sources through finite thick slab shields using the point kernal integration method. It also plots results (dose rate vs. shield thickness).

2. ACTIV 4 & 5 - These programs calculate the source term for composite gamma radioactive sources (MeV/cm^3 sec) from nuclide activity inventories (Ci/gm) into source strength composition based on selected energies.

3. HOMES - Program converts heterogeneous equipment sources, consisting of a cylindrical vessel with interior tube bundles into a homogeneous cylindrical volume source.

4. CREED - Program computes environmental and Control Room doses from the radiation sources at given distances and for various wall and roof slab density by means of (a) Geller-Epstein's Source Integral, (b) source supplied by manufacturers, (c) already calculated source.

5. G^3 (G cubed) - Program computes fluence or dose rates from a point radiation source for direct and single scattered gamma radiation.

The geometry used for shielding evaluation is that of a finite shielded cylinder for tanks, demineralizers, filters, heat exchangers, and pipes.

Calculations are performed to determine the gamma dose rate through a laminated shield at the sides and ends of a cylindrical source to obtain exposure values. Source strengths are divided into selected energy bins for different gamma energies. Credit is taken for self-attenuation.

12.3.2.3 Shielding Material

The bulk of shielding material is ordinary concrete with a minimum density of 141 lb/ft^3 after 28 days of curing. Commercial non shrink grout and mortar having a minimum density of 130 lb/ft^3 may be utilized in congested areas when closing temporary construction blockouts. In the Containment wall, however, concrete with a minimum density of 136 lb/ft^3 is used. The amount of reinforcement per cubic foot of ordinary concrete shield walls and slabs is a minimum of 6.26 lb. Where space requirements prohibit the use of ordinary concrete, high density concrete (220 lb/ft^3) with ilmenite and hematite aggregate is used. Observation windows in the drumming area and filter transfer area are of high-density glass with attenuation properties that provide adequate shielding to reduce radiation levels to be within specified dose limits (zones).

12.3.2.4 Shield Descriptions

12.3.2.4.1 Primary Shield

The primary shield is a large mass of reinforced concrete, which is a minimum of six feet thick throughout the full-length of the active reactor core and of irregular geometrical configuration. It surrounds the reactor vessel and extends upward from the Containment mat to form the walls of the refueling cavity. The primary shield is designed to perform the following services:

1. Reduce, in conjunction with the secondary shield, the radiation level from sources within the reactor vessel and allow limited access to the Containment during normal full power operation.

2. Limit the radiation level after shutdown from sources within the reactor vessel and permit limited access to the RCS equipment.

3. Limit neutron flux activation of component and structural materials over the life of the plant.

12.3.2.4.2 Secondary Shielding

The secondary shield is a reinforced concrete wall and slab structure that surrounds the RCS equipment, including pipes, pumps and steam generators. This shield, which is a minimum of 2 ft 9 in. thick, protects personnel from direct gamma radiation emanating from reactor coolant fission and activation products that are carried from the core by the reactor coolant. The secondary shield reduces the radiation from the reactor and doses from the N-16 activity in the primary loops. It supplements the primary shield to attenuate radiation to low levels; this permits limited access to the Containment for inspection during full-power operation and maintenance of equipment in certain areas during hot shutdown.

The secondary shield provides protection for refueling operations, inspection, repair and maintenance during refueling, and shutdown periods. During the transport of spent fuel assemblies, concrete shielding provides protection in the spaces close to the fuel transfer route and the reactor internals temporary storage area. The fuel transfer route is flooded with water which protects the airspace above the fuel transfer path. For a more detailed discussion of fuel transfer tube shielding, see Section 12.3.2.4.7 below.

12.3.2.4.3 Containment Building

The Containment Building is a cylindrical reinforced concrete structure that completely surrounds the NSSS. At full-power operation, this shield attenuates the radiation level outside the primary-secondary shield complex to ensure that radiation levels outside the Containment correspond to zone designations of adjacent areas. The Containment wall and dome are 4-1/2 ft and 2-1/2 ft thick, respectively. These thicknesses ensure that the cumulative dose from sources contained within the plant during the 2-hr period following a DBA would not exceed 10 CFR Part 100 guidelines.

12.3.2.4.4 Auxiliary Shielding

The auxiliary shielding consists of wall and slab structures for various systems which carry process fluids and gases or handle radioactive materials. Certain equipment located in the Safeguards Building are assumed to be radioactive during normal operation, or could become so under special conditions. Such components are analyzed to establish their maximum radiation levels and the shielding is sized accordingly. Equipment compartments are individually shielded from radioactive components in adjacent compartments.

Auxiliary shielding includes all concrete walls, slabs, covers, and removable blocks that shield the numerous radiation sources in systems such as the CVCS, the BRS, and the RHR system. Highly radioactive equipment is isolated individually, or in groups, in shielded compartments. The entrances of such compartments are provided with labyrinths for full protection of adjacent areas. Where concentrations of radioactive equipment and piping occur, the associated manually operated valves are grouped in adjacent valve rooms. Pumps and compressors serving highly contaminated tanks are located in separate shielded compartments.

Auxiliary shielding thicknesses are designed so that dose rates in adjacent areas are below the limitations set forth by the zone designations of such areas.

12.3.2.4.5 Refueling Cavity

The refueling cavity is flooded with water during refueling operations. In addition to providing laydown space for the reactor internals, the cavity serves as the transfer medium in which spent fuel is moved from the reactor to the spent fuel area. Portions of the cavity are sized on the basis of the internals package physical size and moving path; the remainder is sized on the basis of the spent fuel physical size and moving path.

Shielding provides protection during all phases of spent fuel removal and storage. Operations that require shielding of personnel include spent fuel removal from the reactor, temporary spent fuel storage (in the Containment), spent fuel transfer through the refueling canal and transfer tube, spent fuel storage (in the Fuel Building), and spent fuel shipping cask loading prior to offsite transportation. All spent fuel removal and transfer operations are performed under borated water to provide radiation protection.

During movement of fuel assemblies within containment, at least 23 feet of water is maintained over the top of the reactor vessel flange. At least 23 feet of water shall be maintained over the top of irradiated fuel assemblies seated in the storage racks and 10 feet in the transfer canal. This limits the dose at the water surface to less than 2.5 mrem/hr for an assembly in a vertical position.

The 6-ft-thick walls of the fuel transfer canal and 6-ft-thick spent fuel pool walls supplement the water shielding and limit the maximum radiation dose levels in adjacent working areas to less than those specified by their zone designations.

The refueling water and concrete walls also shield personnel from activated control rod clusters and reactor internals that will be removed during refueling. Dose rates generally are expected to be less than 2.5 mrem/hr in working areas. However, certain manipulations of fuel assemblies, rod clusters, or reactor internals may produce short term exposures in excess of 2.5 mrem/hr.

12.3.2.4.6 Control Room Shielding

The Control Room shielding is described in Section 6.4.2.5

12.3.2.4.7 Fuel Transfer Tube Shielding

All the unrestricted areas are shielded from the fuel transfer tube either by stationary concrete or permanently installed removable shielding or by water shielding during spent fuel transfer. Access to the fuel transfer tube is further restricted by removing the inspection access ladder.

Figures 12.3-33 sheets 1 through 3 show the shielding design on the relevant plan, elevation, and section drawings, to assure acceptable levels of radiation in potentially occupied areas. Special attention is given to gaps in the concrete and lead shielding which are required for structural reasons.

Dose rate at contact with the 8" lead shielding is less than 6 Rads/hr where the direct radiation path is through the 8" lead shield. It is provided to shield as much of the 4" structural gap as possible. It supplements the existing permanent shielding concrete structures and water as shown in Figure 12.3-33 (3 sheets). It provides radiation protection for structural gap areas above and below the transfer tube by reducing the streaming and scattering of gamma rays.

The large gap (approx. 18") in the lead shield shown in section 3-3 of Figure 12.3-33 (sheet 3 of 3) is to provide more access space near the fuel transfer tube area in the inspection shaftway. This shaftway extends from elevation 808' up to the fuel transfer tube area, and is also shown in Figure 12.3-33 (sheet 1 of 3).

Within the Fuel Building and the Containment Building, the provision of shielding material above the transfer tube in conjunction with administrative procedures provides adequate protection to assure compliance with 10 CFR 20.

To further reduce dose rates from scattered radiation during spent fuel transfer, additional steel shield plates are installed at strategic locations around the fuel transfer tube (see Fig. 12.3-33, sheet 3 of 3). The shielding provided by these steel plates in conjunction with the lead shielding and administrative procedures reduces maximum expected dose rates in the vicinity of the fuel transfer tube area at 808’ to less than 1.0 rem/hr.

12.3.2.5 Use of Guidance Provided by Regulatory Guides

NRC Regulatory Guide 1.69 states that ANSI N101.6-1972, Concrete Radiation Shields, is considered applicable to shielding structures for nuclear power plants, subject to certain conditions. These conditions are stated in regulatory positions Nos. 1 through 8 in NRC Regulatory Guide 1.69. The guidance provided in this regulatory guide is adhered to as discussed in Appendix 1A(B).

12.3.3 VENTILATION

12.3.3.1 Design Objectives

The plant ventilation systems are designed to provide suitable and safe ambient conditions for personnel and operating equipment. In addition to the conventional function of preventing extreme thermal environmental conditions, the ventilation systems are designed to ensure that the maximum airborne radioactivity levels for normal and anticipated operational occurrences are within the limits of 10 CFR Part 20, Appendix B, for areas within the plant structures and for restricted areas on the plantsite where workers and visitors are permitted. The maximum levels correspond to the design basis reactor coolant isotopic inventory. The plant ventilation systems are also designed to provide an environmentally suitable and radiologically safe environment for continuous personnel occupancy in the Control Room under normal and postaccident conditions in accordance with 10 CFR Part 50, Appendix A, Criterion 19, and to ensure that the maximum radiological exposures to the general public following a DBA are within the limits of the guidelines of 10 CFR Part 100.

In order to meet the design objectives for the plant ventilation systems, the following design guidelines are incorporated into the design basis for the ventilation systems:

1. Air movement patterns are provided from clean areas to potentially contaminated areas, thus preventing the spread of airborne contamination.

2. Negative pressure differentials, with respect to surrounding areas, are maintained, where applicable, inside potentially contaminated areas to prevent uncontrolled spread of airborne contamination.

3. Atmosphere cleanup units are installed to filter the exhaust of potentially radioactive areas in order to maintain the offsite doses ALARA. These offsite doses result from the activity of the ventilation discharge.

12.3.3.2 System Design Criteria

The atmosphere cleanup units remove exhaust air contaminants from the primary plant ventilation exhaust system (which includes the controlled access exhaust, containment purge exhaust, and Fuel Building exhaust ventilation systems) and from the Hydrogen Purge System.

These systems are described in Sections 6.2, 6.5, and 9.4. Figures 6.5-1 and 9.4-10 illustrate typical layouts for the engineered safety features (ESF) and non-ESF atmosphere cleanup units.

In addition to the use of the atmosphere cleanup units referred to previously, the emergency filtration units and emergency pressurization units of the Control Room heating, ventilation, and air-conditioning (HVAC) system remove contaminants from the Control Room environment and prevent contaminants from entering the Control Room atmosphere during emergency modes of operation. The Control Room HVAC system is discussed in Sections 6.4 and 9.4.1. Atmosphere cleanup units are also being used in each Containment Building. These preaccess filtration units are described in Section 9.4.

12.3.3.3 Component Design Criteria

The normal primary plant ventilation exhaust system atmosphere cleanup units and the ESF Control Room emergency filtration units each contain a prefilter, two high-efficiency particulate air (HEPA) filters, a charcoal bed filter for iodine adsorption, and related instrumentation. The HEPA filters are located immediately upstream and downstream from the charcoal bed filter. In addition to the components previously described, the ESF Control Room emergency pressurization and Hydrogen Purge Exhaust System atmosphere cleanup units each contain a demister and electric heater section to ensure that the relative humidity of the incoming air is maintained below 70 percent. The Primary Plant ESF atmosphere cleanup units which normally remain on standby are similar to the Control Room emergency pressurization and Hydrogen Purge Exhaust System atmosphere cleanup units, except that they do not contain a prefilter section.

12.3.3.4 Testing, Isolation, and Decontamination

Each atmosphere cleanup unit housing is furnished with test connections for dioctyl phthalate (DOP) leak testing of the HEPA filters and refrigerant leak testing of the charcoal bed iodine adsorber section. In addition, a water spray system is provided to suppress an adsorber section fire caused by the presence of activated charcoal in the atmosphere cleanup units. The operation of the water spray system is discussed in Section 9.5. Inspection and testing requirements for ESF and non-ESF atmosphere cleanup units are discussed in Section 9.4. The arrangement of these units allows access to the unit and equipment by personnel for maintenance and testing. All housings have multiple drains which are connected to the Equipment and Floor Drainage System. Decontamination water stations are in proximity to each unit to facilitate decontamination. Figures 6.5-1 and 9.4-10 illustrate typical layouts for ESF and non-ESF atmosphere cleanup units, respectively.

12.3.3.5 Maintenance

All atmosphere cleanup unit housings are designed to provide ample room for maintenance of equipment and filters and to minimize radiation exposure to personnel during filter replacement. The criteria for replacement of filters are established on the basis of maximum allowable resistance of the dirty filter or minimum radiation exposure to personnel, or both. Each atmosphere cleanup unit is provided with a pressure indicator. High differential pressure across the entire unit actuates an alarm in the Control Room. Each filter section has provisions to monitor its own status with a local differential pressure indicator. Upon reaching the maximum resistance, the filters are replaced. The maximum allowable resistance for each type of filter is indicated in Table 9.4-4.

The criterion for replacement of the adsorber section is based on the deterioration of adsorber ability to remove radioactive iodine from the exhaust air. This efficiency is determined by laboratory testing of representative samples of the activated charcoal exposed simultaneously to the same service conditions as the adsorber section. Each sample has the same qualification and batch test characteristics as the system adsorber. Samples are tested periodically in accordance with Regulatory Position C.6.b of NRC Regulatory Guide 1.52 or 1.140. The adsorber section will be replaced if one of the samples fails to meet the requirements of NRC Regulatory Guides 1.52 or 1.140, Table 2. See Appendix 1A(B) for a discussion of compliance with NRC Regulatory Guide 1.52 and 1.140.

12.3.3.6 Conformance to NRC Regulatory Guide 1.52

The ESF atmosphere cleanup units conform to the regulatory positions and recommendations of NRC Regulatory Guide 1.52 as indicated in Table 6.5-1 and Appendix 1A(B). Non-ESF atmospheric cleanup units conform to the regulatory positions and recommendations of NRC Regulatory Guide 1.140 as indicated in Appendix 1A(B).

12.3.4 AREA RADIATION AND AIRBORNE RADIOACTIVITY MONITORING INSTRUMENTATION

12.3.4.1 Airborne Radioactivity Monitoring System

Detection and measurement of airborne radioactivity is accomplished by the process radiation monitoring system (PRMS) in conjunction with the area radiation monitoring system (ARMS). Together, the PRMS and ARMS comprise the CPNPP Radiation Monitoring System (RMS). PRMS and ARMS monitor activity levels and dose rates resulting from gaseous and particulate airborne radioactivity in various areas of the plant; the systems provide alarms, indication, and on-demand hard copy printouts in the Control Room.

When an alarm is annunciated in the Control Room, Radiation Protection (RP) will be contacted. RP personnel will establish access control to the identified building(s) to prevent inadvertent entry of personnel. RP personnel will then seek to further identify and locate the specific source or source location within the building(s) identified. This is accomplished by means of a review of area radiation monitor readouts for abnormal levels (utilizing the computer capabilities of the RMS as described in Section 11.5.2.5.3), collecting local air samples in specific areas (using mobile air sampling systems), and communication with Operations personnel to identify abnormal system configurations, component failures or other possible causes. A listing of all local monitoring assemblies and a discussion of their alarm setpoints is provided in

Section 11.5.2.5.6. For further discussion of the role of RP personnel, see Section 12.5. Detailed descriptions of PRMS/ARMS are given in Sections 11.5 and 12.3.4.2. Portable instrumentation and the airborne sampling program are discussed in detail in Sections 12.5.2 and 12.5.3, respectively.

12.3.4.1.1 Design Criteria

The process radiation monitoring system, which performs airborne radioactivity monitoring, is designed to measure, display and store airborne radioactivity levels, to alarm on high airborne radioactivity levels, and, when required, to control the release of radioactive gases and particulates produced in the operation of the plant. It also aids compliance with the requirements of 10 CFR Part 20, 10 CFR Part 50, General Design Criterion (GDC) 60, 63 and 64 and NRC Regulatory Guides 1.21 and 8.8. The process radiation monitoring system aids in the protection of the general public and plant personnel from exposure to airborne radioactivity in excess of that allowed by applicable regulations, by, when appropriate, controlling or terminating releases exceeding discharge limits and warning plant personnel so that they can take appropriate protective measures. The design objectives of the fixed system for normal operation are as follows:

1. To give early warning of increasing radioactivity levels indicative of equipment failure or malfunction or deteriorating system performance.

2. To initiate prompt corrective action, either automatically or through operator response, on high airborne radioactivity level.

3. To provide continuous surveillance of noble gas radioactivity levels within certain buildings by indicating and recording in the main Control Room, exhaust duct activity levels for up to 28 days. Abnormal activity levels are alarmed in the Control Room to aid in preventing a person from inadvertently entering a building where he can inhale noble gas activity in excess of limits defined in 10 CFR Part 20. Also, there are area monitors in these buildings which will alarm locally upon detection of radiation due to increased airborne radioactivity in these areas (see Section 12.3.4.2.2 for monitor location bases).

For some anticipated operational occurrences resulting from accidents or operator error, e.g., if the airborne radioactivity levels exceed alarm set points, the PRMS will activate necessary isolation valves to terminate or reduce releases. Independence of backup monitors is maintained by providing adequate separation of detectors, signal cabling, power supplies, and actuation circuits for isolation valves.

The fixed continuous monitors, which are described in detail in Section 11.5, serve in conjunction with a comprehensive air sampling program using portable airborne sampling equipment. The portable air sampling equipment and the program for performing air sampling are described in detail in Sections 12.5.2 and 12.5.3. The portable airborne sampling equipment is available to complement fixed instrumentation and maybe used in areas where fixed instrumentation is not provided but where there is a need for short term or periodic monitoring or when fixed instrumentation is not operable. Portable radiation and radioactivity monitoring equipment are highly reliable, commercial quality equipment. The operation, maintenance and calibration of this equipment is subject to the Operations Quality Assurance Program. This equipment either collects samples which are then processed and evaluated, or monitors continuously. The system

also performs post-accident functions as described in Section 6.2.4 and 7.5 to satisfy GDC-13 and 54.

12.3.4.1.2 Airborne Monitor Locations

Fixed airborne monitors of the process radiation monitoring system provide information about the airborne activity within the following plant areas (described in Section 11.5):

1. Containment Building

2. Auxiliary Building

3. Heating and ventilation equipment area

4. Fuel Building

5. Safeguards Building

6. Main steam and feedwater area

Locations of airborne monitors that monitor effluent streams and the above areas are listed in Table 11.5-1.

12.3.4.1.3 System Description

The airborne monitors are part of the process radiation monitoring system, which is a dual computer and distributed microprocessor digital monitoring system described in detail in Section 11.5.

Detection of airborne activity in the Containment buildings during accident conditions is provided by the Containment High Range Radiation Monitors (Section 12.3.4.2) located inside Containment, and by grab samples for laboratory analysis.

The Containment particulate, iodine and noble gas (PIG) monitors are located outside the Containment building in the Safeguards buildings at elevation 831 ft, 6 in. The sample collector draws containment air through an open sample line from the Containment to the monitors. A correction factor was determined to account for particulate deposition line losses. The sample pump for the containment PIG monitor is located with the monitor in the Safeguards Building. The guidelines in ANSI N13.1 are considered to ensure that representative samples are obtained.

Various areas of the plant that have potential airborne activity have their ventilation directed to a common header that discharges to the plant vent stacks. Each stack, located on the Auxiliary Building, has off-line noble gas monitors (see Section 11.5) with sample pumps monitoring plant effluents. Each stack is also equipped with an off-line Wide Range Gas Monitor (WRGM) used to monitor both normal and post-accident ranges of noble gases. The WRGM has sample pumps and isokinetic sample nozzles for both normal (high stack flow rate) and accident (low stack flow rate) conditions. Both types of monitors draw representative samples of stack effluent and route them to the detectors. A correction factor was determined to account for particulate deposition line losses.

There are also in-line gaseous monitors in the vent ducts of the Auxiliary, Fuel, and Safeguards building, the heating and ventilation room exhaust duct, and the main steam and feedwater area. They are located at the outlet of each area or building ventilation header. These monitors have no sample lines or pumps associated with them because they are located in each area's ventilation duct. The monitors are beta sensitive scintillation detectors with photomultiplier tubes and associated electronics. They provide continuous service (described in 12.3.4.1.1, Objective 3) and are not designed to detect particulates or iodines. The monitors identify which building has abnormal airborne activity and provide indication that some system within the building or area being ventilated has equipment leaking significant gaseous or liquid activity. Refer to Table 11.5-1 for monitor description and parameters.

The Control Room is provided with four gas monitors (2 per intake) each capable of monitoring Control Room ventilation inlet air. A high radiation signal will be initiated upon detection of sufficient airborne radioactivity in the inlet air and will initiate Control Room emergency ventilation and start recirculation of Control Room air through filters. The monitors are beta sensitive scintillation detectors with photomultiplier tubes and associated electronics. Two of the monitors are located near the Control Room ventilation duct in the heating and ventilation area of the Electrical and Control Building. The other two monitors are located in the Auxiliary Building.

Section 11.5 describes in detail the PRMS and the Digital Radiation Monitoring System (DRMS), monitor calibration methods and frequency, tests and inspections.

The off-line Containment PIG monitors are seismically designed monitors. They have digital indication on the control/display module installed on the seismic equipment rack, located in the Control Room. All airborne monitors have remote indication, alarm, and printout via the peripherals of the microcomputer consoles located in the Control Room.

12.3.4.2 Area Radiation Monitoring System

The objectives of the ARMS are as follows:

1. To provide plant operators and personnel with a system which informs them of radiological conditions is selected plant areas and provide an indication of changing radiological conditions.

2. To indicate, alarm, and record for up to 28 days abnormal radiation levels in areas where radioactive material is present, stored, handled, or inadvertently introduced.

The primary functions of the ARMS are as follows:

1. To provide continuous surveillance of radiation dose rates of representative accessible and restricted areas inside reactor containment.

2. To alert personnel of dose rates above a predetermined set point.

3. To provide a direct reading indication and on demand record of daily radiation dose rate averages for up to 28 days for each monitor location.

4. To indicate in the Control Room the radiation levels at selected locations within the plant building where spent fuel is stored or handled.

5. To warn operating personnel when a monitor is not operational

6. To have sufficient range and sensitivity to monitor normal and anticipated operational occurrences

7. To monitor the Containment radiation level after a DBA (see Section 7.5).

8. To monitor area radiation in locations where access may be required after a DBA (see Section 7.5).

9. To monitor area radiation in areas adjacent to containment which contain electrical or mechanical penetrations (see Section 7.5).

Locations of the area radiation monitors are shown on Figure 12.3-4 through 12.3-23.

12.3.4.2.1 Design Criteria

The following design criteria are used in the design of the ARMS:

1. Detectors and cabling for safety related monitors located in containment are designed to operate during and following the applicable DBA, including LOCA.

2. To facilitate compliance with applicable regulations (e.g., 10 CFR Part 20), monitors and detectors have matching ranges in accordance with dose rate levels anticipated at specific detector locations.

3. The system is designed to provide for physical security for access to alarm setpoints and other important data base parameters.

4. Radiation dose rates are continuously monitored, indicated and annunciated on the Control Room's Display/Operator Cabinets, and recorded on demand at the associated printers.

5. Local audible alarm and analog indication are provided for selected ARMS microprocessor or detector location.

6. Control Room alarms annunciate high radiation dose rates and circuit failures.

7. Monitors electronics have independent internal power supplies.

8. A remotely operable check source or current is provided for each detector.

9. Environmental design conditions for monitoring components are described in Table 12.3-9.

10. Low and high-range area monitors register full-scale if exposed to radiation levels up to 100 times full-scale indication; they do not fold over throughout this range.

11. Monitors are readily accessible for maintenance and inspection wherever possible.

12. The unique function of each monitor is considered in determining alarm set points.

The specified range for each area radiation monitor is provided in Table 12.3-8. The full scale value for each monitor is, as a minimum, in the upper half of the top decade. For example, an ionization chamber detector with a specified range of 10^2 -10^7mR/hr will have a full scale reading of at least 5x10^6mR/hr.

12.3.4.2.2 Monitor Location Bases

Area monitors are located in representative areas (See Table 12.3-8) where the following occurs:

1. Personnel perform infrequent duties, but a significant probability exists for hazardous dose rates (i.e., inside containment).

2. Accident monitoring requirements apply (see Section 7.5)

3. Nuclear material is handled, used, or stored (i.e., R.G. 1.13).

12.3.4.2.3 System Description

The ARMS is part of the microprocessor-based radiation monitoring system. This digital monitoring system is described in detail in Section 11.5. System parameters, nominal instrument range, alarm set point bases, and the location of detectors are described in Table 12.3-8. Location and type of readouts and alarms for the airborne monitor are described in Section 11.5. Location of local readouts and alarms for area monitors are shown on Figures 12.3-4 through 12.3-23.

The basic ARMS (integrated with PRMS) is comprised of four dedicated computers in communication with each other (two in each of two central display consoles), distributed dedicated microprocessors (one for each local detector/monitor assembly), and a report computer. At each monitor, control, data processing, data storage, and multilevel alarming are performed locally by the dedicated microprocessor; also, processed data are communicated to central consoles upon polling requests. These monitor functions are performed at each monitor independently of the rest of the system. This independence is insured by use of optical couplers in monitor input/output circuits and by the distances separating monitors. No area monitor electronics share a microprocessor or internal power supply. Thus, each monitor is a stand-alone system performing independently and locally all functions of a conventional analog ratemeter, but, in addition, providing data storage and self-contained function-checking diagnostic routines in read only memory (ROM).

In addition to a 120-V, single-phase, 60-Hz local power supply for each monitor assembly, field wiring consists of two shielded, twisted-pair cable daisy-chained from monitor to monitor and to display consoles. Each daisy chain is a bidirectional control and data bus looped from one central computer through a group of monitors to the other central computer, operating at logic voltage and power levels. With this arrangement, two paths from each monitor to the central consoles are provided. See Figure 11.5-1.

The central Display/Operator Cabinet includes two microcomputers, an 18-in. color LCD display, a shared color graphics printer, printers, and keyboards for entering data and commands. The

cabinet is located in the common control room area for Units 1 and 2. The cabinet is divided into two sections, each section can display data associated with any monitor or monitor group in the plant.

Each operator console view node includes an 18-in. color LCD display and a keyboard for entering commands. One view node is located at the operator's console (CP1-ECBDCB-21, Operator Console B) in the Unit 1 control room area and the other is located at the operator's console (CP2-ECBDCB-21, Operator Console A) in the Unit 2 control room area. Each view node can display data associated with any monitor or monitor group in the plant.

1. Detectors

   The following two types of ARMS detectors are used: Geiger-Muller tubes for lower-level radiation and ionization chambers for the higher levels. (See Table 12.3-8.) There is an indicator and visual/audible high alarms at selected local microprocessor locations. In addition, certain detectors located remotely from the associated microprocessor are also provided with meters and alarms.

2. Local Electronics

   Associated with the detector is an Intel 8085A microprocessor with sufficient memory to control the monitor, process and store its data, and communicate processed results to the two, identical remote processors in central consoles.

   The monitor microprocessor and all of the generalized digital circuits are included on a board that is designed to fit into a National Electrical Manufacturers Association (NEMA) 12 enclosure with a lock. A printed-circuit backplane board accepts the processor board and the digital preamp, relays, power supply, and an address plug. The address plug gives the monitor a unique address. Also included are the detector high-voltage power supply, analog indicator, and indicator lights.

3. Display/Operator Cabinet

   The RMS Display/Operator Cabinet is a common panel divided into two (2) sections (SCADA A an B). Each section includes two IBM industrial personnel computers with math coprocessor. One computer is a SCADA node and contains a realtime interface coprocessor which allows it to communicate with the local microprocessor of each monitor. The other computer is the operator console view node. An IBM multistation access unit (MAU) interconnects the two computers to an IBM token-ring network (TRN). Each computer stores in memory the complete alarm status for all monitors in the Unit 1 and Unit 2 ARMS. Each section also includes a color LCD, an output printer, a shared graphics printer and an input keyboard. Each operator console view node includes a display color LCD Display and an input keyboard. Selection of the proper keys on the console keyboards can display the complete alarm status on the color LCD Display via a grid presentation and a control menu display, with each monitor represented by a colored rectangle with six superimposed characters. The characters identify the monitor. The color of the rectangle indicates the urgency of the alarm condition. Similar information is provided in Section 11.5.

4. Power Supplies

The radiation monitor processors and the central Display/Operator Cabinet are furnished with safety related or reliable non-safety related power (backed by diesel/non-safety related station batteries) as applicable.

Operating voltages for local electronics and high voltage for detectors are derived from power supplies in their respective local monitor-assembly enclosures.

5. Calibrations and Source Checks

A channel calibration is performed for each safety related area monitor channel at least every 18 months or during refueling outages. The channel calibration for non-safety related area monitors is performed at a frequency determined by plant procedure(s), based on performance and function. These calibrations are performed in accordance with approved station procedures. Electronic calibrations and checks, including alarm initiating actions, for each area monitor channel are performed in the field. Calibration of all area monitor detectors, with the exception of the post accident High Range Radiation Monitors located in containment, is performed as follows. Prior to initial installation in the field, each area monitor detector will receive a primary calibration onsite in the Radiation Protection Calibration Facility using a calibration source to generate radiation fields and a test microprocessor to test detector response. The onsite primary calibration process consists of determining the proper operating voltage for each detector and then testing the detector in radiation fields of at least three dose rate levels to determine detector response and demonstrate detector linearity. Traceability of the calibration to the National Bureau of Standards (NBS) is obtained by the use of a transfer instrument with an NBS traceable calibration to quantify the radiation fields generated by the calibration source. Upon completion of this calibration process, the detector is exposed to a portable transfer source and a reference detector response is determined. The detector is then installed in the field and the same transfer source is used to verify proper detector response. After initial installation in the field, subsequent calibrations of the detector are performed by either removing the detector from the field and repeating the primary calibration procedure described above, or by performing a transfer calibration in the field using the portable transfer source and reference data generated in the original primary calibration.

Calibration of the post accident High Range Radiation Monitors located in containment is performed in place using a portable source to verify detector response at a single point below 10R/hr. The monitor performance for points above 10R/hr is verified electronically by simulating detector input with an electronic signal.

Each low range area monitor GM detector assembly is provided with a built-in radiation check source. The check source is attached to the detector assembly in a shielded container in a fashion such that loss of check source actuation power will cause the source to return to a shielded position. Higher range area monitor ionization detectors have ranges that are too high to check with radiation sources. These are provided with check currents that test the electronic circuitry of the monitor. These monitor checking capabilities are remotely operable from the control room. These checks provide convenient operational checks of these monitors. Additionally, the radiation check source provides a gross response check of the detector.

System operability may be verified by observation of channel behavior or by use of the check source or current. Any detector whose response cannot be verified by these methods may have its response checked with a portable check source.

6. Test Calibration Circuitry

Each monitor microprocessor contains self-checking diagnostics in its ROM, a rotary switch on a printed circuit (PC) card to select each diagnostic program, and a red and green light (also on each PC card) to show the result of each diagnostic test. These allow the microprocessor board to be functionally checked while installed in its enclosure. Typical tests include the following:

a. Communication loop (continuity)

b. ROM

c. Local indicator

d. Local alarms

7. Maintenance

Area monitor channel detectors and electronics are serviced and maintained to ensure reliable operations. Manufacturer's recommendations for Service and Maintenance are available for use. Such maintenance includes cleaning, adjustment, and/or replacement of any components required after performing a test or calibration. If any work is performed that could affect the calibration, a recalibration will be performed at the completion of the work.

8. Control Room Equipment Rack#78#

A dedicated control/display module for each high range radiation monitor (two per containment building) is located within a Class 1E, seismic equipment rack in the Control Room. This equipment rack has the same characteristics and provides the same monitor indications as the control room equipment rack for selected process monitor channels (described in Section 11.5.2.5.5), except that power to the rack is provided directly by class 1E buses.

12.3.4.3 Accuracy

The overall accuracy of a radiation monitor system is governed by four major areas: (a) factory calibration and alignment, (b) detector characteristics and environment, (c) microprocessor environment, and (d) field alignment. Factory calibration is performed with standards traceable to the National Bureau of Standards (NBS). Detector energy response and linearity are demonstrated by factory calibration and site calibration procedures subsequent to operations. Detector and microprocessor environmental variations are determined from qualification tests. Overall monitor error is the root-mean-square sum of all system errors and for all conditions is within a factor of 2 over the entire range. An exception to the above accuracy statement occurs with the High Range Radiation Monitors (HRRMs). With a post Main Steam Line Break (MSLB) high temperature environment, a drop in the insulation resistance of the signal cable could cause

an erroneous bias current that could exceed three times the threshold radiation current. This situation, combined with other possible system inaccuracies, could exceed the stated accuracy limits. This will not affect the intended use of the monitor.

Another exception to the accuracy requirements has been identified with the High Range Area Monitors (HRAMS), at the low end of the monitors scale. When the dose rate from the keep alive source is combined with other system errors the indicated dose rate can exceed the previously specified system accuracy. This inaccuracy is conservative.

The precision of a radiation monitor represents the repeatability of a determination. It signifies the closeness of agreement among a number of consecutive measurements of the output for the same value of input under identical operating conditions. The reference accuracy which defines the limit that errors will not exceed when the system is used includes the combined conformity, hysteresis and repeatability errors discussed previously.

12.3.4.4 Compliance With Regulatory Guides and Standards

When applicable, the guidance provided by Regulatory Guides 8.2, 8.8, 1.21, and ANSI N13.1-1969 has been followed in the design of the area radiation and airborne radioactivity monitoring system.

1. For Regulatory Guide 8.2, the system was designed to provide information to plant operators pertaining to radiological conditions in selected plant areas and in releases of radioactive materials in effluents. Additionally, the system is designed to allow for hard copy printouts of the radiation levels measured by area monitors and radioactivity concentration levels measured by airborne monitors.

2. For Regulatory Guide 8.8, applicable guidance was followed by:

    a. Review of system design using the points outlined in the guide

    b. Proper design of this system to ensure that radiation exposures are minimized

3. Based on Regulatory Guide 1.21, the system was designed to measure and report releases of radioactive materials in gaseous effluents and to classify and report the categories and curie content of wastes.

    a. The airborne monitors detect radioactivity and give indication of effluent stream activity concentrations (detector is designed to monitor the most representative isotope of an effluent stream)

    b. Sampling programs described in Section 11.5 will be used in conjunction with the continuous monitoring system to determine total quantity of radioisotopes and types of isotopes released. Monitors will provide concentrations of sample streams by processing count rates from detectors and, where applicable, flow determinations from flow measuring devices. This information along with multichannel analysis from the sampling program will determine isotopic content. The meteorological program provides data for dose evaluation. The data and information obtained from the various subsystems and programs are used to generate the required reports according to Regulatory Guide 1.21 format.

4. The aerodynamic sampling guidelines for off-line vent stack airborne monitors provided in ANSI N13.1 are considered. Representative samples of the effluent streams are obtained by isokinetic nozzles, sample lines with no obstructions and large radii bends and appropriate sample valves.

5. Based on GDC-54, the system was designed to monitor areas which contain containment penetrations and associated piping including closed systems outside containment. See Section 6.2.5 for details.

6. Compliance with RG1.97 is described in Section 7.5.

REFERENCES:

1. Loose R. A. and Battaglia J. A., Design Considerations in PWR Waste Processing Systems, AIChE 69th National Meeting, May 16, 1971, Cincinnati, Ohio.

2. Source Term Data for Westinghouse Pressurized Water Reactors, WCAP-8253, Westinghouse Electric Corporation, June 1975.

3. Clemons, L., Jr., and Battaglia, J. A., Minimizing Radiation Exposures in PWR Plants, ANS Twenty-First Annual Meeting, June 8, 1975, New Orleans, La.

4. Solomon Y., and Roesmer, J., Measurement of Fuel Element Crud Deposits in Pressurized Water Reactors, Nuclear Technology, May 1976.

5. Bergman C. A., and Bohr, D. P., Radiation Level Buildup on Plant Piping During Second Core Operation of Shippingport Plant, Nuclear Technology, Vol. 13, March 1972.

6. Fletcher, W. D. Primary Coolant Chemistry of PWRs, Proceedings of the 31st International Water Conference, Pittsburgh, Pa. October 1970.

7. Decontamination in Nuclear Reactors, Westinghouse Proprietary Class 2 Report.

8. Decontamination of Nuclear Plants Seminar, Ohio State University, May 1975.

9. Commentary by H. Thexton, AECI, Decontamination of Nuclear Plants Seminar, May 1975, Ohio State University.

10. Engineering Compendium on Radiation Shielding, Vols. 1 and 3. Springer Verlag, Inc., New York, 1968. Sponsored by the International Atomic Energy Agency - Vienna.

11. Nuclear Data Sheets by the National Academy of Sciences. National Research Council (continuously updated).

12. “An Engineering Evaluation of Nuclear Power Reactor Decommissioning Alternative” National Environmental Studies Project (AIF/NESP-009) Industrial Forum, Inc., November 1976.

13. Design, Inspection, Operation and Maintenance Aspects of the Westinghouse NSSS to Maintain Occupational Radiation Exposures As Low As Reasonably Achievable, WCAP-8872, Westinghouse Electric Corporation, April 1977.

TABLE 12.3-1
RANGE OF DF[a] VALUES MEASURED ACROSS PWR PURIFICATION DEMINERALIZER

Isotope	Demineralizer	DF Flange	
		Plant A	Plant B
Mn-54	Mixed bed[b]	5-7	-
Mn-54	Mixed bed + cation bed[c]	150-600	67-1100
Co-58	Mixed bed	140-215	-
Co-58	Mixed bed + cation bed	190-970	100-650
Co-60	Mixed bed	18-20	-
Co-60	Mixed bed + cation bed	170-1000	-
I-131	Mixed bed	300-500	-
I-131	Mixed bed + cation bed	160-380	3.5-3800
Cs-134	Mixed bed	2	-
Cs-134	Mixed bed + cation bed	195-580	470-3000
Cs-137	Mixed bed	1-2	-
Cs-137	Mixed bed + cation bed	180-1000	360-7900

a) Ratio of inlet concentration to outlet concentration

b) Generic name for demineralizer used in W system, normally in the lithium borate form; it essentially behaves as an anion bed for isotopes listed.

c) Normally in the hydrogen form

d) Taken from Reference 3

TABLE 12.3-2
APPROXIMATE RCS WETTED SURFACE AREAS AND WEIGHTS TYPICAL FOUR-LOOP PLANT

	Material	Surface Area (ft^2)	Weight (lb)
Reactor internals	SS	4236	402,000[(a)]
Reactor vessel clad	SS	2190	19,200
RCS piping	SS	2758	299,890
Reactor internal bolting materials	SS	Negligible	Negligible
RCS pumps	SS	Negligible	684,800
Auxiliary heat exchanger surfaces	Inconel	Negligible	Negligible
Steam generators	Inconel	1.90×10^5	96,800
Fuel	SS(0.12)	2000	-
Fuel	SS(0.08)	3600	-
Fuel	Inconel	7.80×10^3	-
Fuel	Zircaloy	7.78×10^4	-
Reactor internals	Stellite	3.23	-
RCS pump journals	Stellite	17.21	-
Control rod drive mechanisms	Stellite	10.76	-
RCS valves	Stellite	2.6	-

SS = Stainless Steel

a) Includes 228,300 lb of large forgings and 173,300 lb of steel plate and small forgings

TABLE 12.3-3
PRESENT SPECIFICATIONS ON COBALT CONTENT OF MATERIALS

	Material	Cobalt (w/o)
Reactor internals	SS	0.12 maximum
Reactor vessel clad	SS	0.2 maximum
RCS piping	SS	0.2 maximum
Reactor internal bolting materials	SS	0.25 maximum
RCS pumps	SS	0.2 maximum
Auxiliary heat exchanger surfaces exposed to RCS	SS	0.2 maximum
Steam generators	Inconel	0.1 maximum
Fuel	SS[(a)]	0.12 maximum
Fuel	SS[(b)]	0.08 maximum
Fuel	Inconel	0.1 maximum
Fuel	Zircaloy	0.002 maximum

SS = Stainless Steel

a) Refers to stainless steel outside active region on zircaloy clad fuel (e.g. top and bottom nozzles)

b) Refers to stainless steel inside active region.

TABLE 12.3-4
COST TO PROVIDE REDUCED COBALT CONTENT DOLLARS/TYPICAL FOUR-LOOP PLANT

	Cobalt Content (w/o)		
	0.25 percent	0.1 percent	0.05 percent
Reactor internals (in core periphery only)	$48,300	$155,000	Unavailable
Reactor vessel clad	300	2000	$4000
RCS piping	5000	31,500	66,000
Reactor internal bolting materials	Negligible	Negligible	Negligible
RCS pumps	10,300	71,900	150,600
Auxiliary heat exchanger surfaces	Negligible	Negligible	Negligible
Steam generator (four) Inconel	5800	40,700	Unavailable

Approximate Adders to Cost for Reduced Cobalt Content in Stainless Steel Plate and Small Forgings (e.g. lower core support plate)

Percent	Cents/lb
>0.25	No charge
0.25-0.15	1.5
0.15-0.10	5.25
0.10-0.05	10.5
<0.05	22

Approximate Adders to Cost for Reduced Cobalt Content in Large Stainless Steel Forgings (e.g. lower-core support plate)

Percent	Cents/lb
>0.25	No charge
0.25-0.10	20
0.10-0.05	60
<0.05	Not available

TABLE 12.3-5[a]
APPROXIMATIONS TO COBALT SOURCES AND CONTRIBUTION TO TOTAL COBALT IN RCS

Component	Surface Area (ft^2) x Cobalt Limit (w/o)	Surface Area (ft^2) x Cobalt Limit (w/o) x Corrosion Rate (mg/dm^2-mo)
Reactor internals (SS)	508	254
Reactor vessel clad (SS)	438	219
RCS piping (SS)	552	276
Steam generator tubes (Inconel)	19,000	2
Fuel (SS)	528	264
Fuel (Inconel)	780	1170
Reactor Internals (Stellite)	1.6	Not available
RCS pump journals (Stellite)	8.6	Not available
CRDM (Stellite)	5.4	Not available
RCS Valves (Stellite)	1.3	Not available
Contribution to Total (Percent)		
Stainless steel	9.28	3.30
Inconel	90.64	96.70
Stellite	0.077	
Total	100.0	100.0

SS = Stainless Steel

a) Source: WCAP-8872, Table 6-4

TABLE 12.3-6
RADIATION ZONES

Zone No.	Maximum Design Dose Rate (mrem/hr)	Design Criteria Description and Explantion
I	Less than 0.25	Uncontrolled, unrestricted access area. Nonradiation personnel and visitor occupancy: 40 hr/wk, 50 wk/yr
II	Less than 2.5	Controlled, occupational access area. Radiation worker and authorized visitor occupancy: maximum 40 hr/wk, 50 wk/yr
III	Less than 25.0	Controlled, limited access area.
IV	More than 25.0	Controlled, restricted access area with limited occupancy. Normally equipped with lockable door/gate.
V	No limitation	Controlled, restricted access area, normally inaccessible. Access is via lockable door.

TABLE 12.3-7
THIS TABLE HAS BEEN DELETED

TABLE 12.3-8
AREA RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 1 of 5)

Channel Nos. Unit 1	Unit 2	Detector Type	Monitor Location	Specified Instrument Range (mR/hr)	Bases for Alarm Set Points
Containment Building					
1RE 6290A	2RE 6290A	Ionization Chamber	Elevation 905 ft. 9 in	10^3 - 10^{10}	Note 5
1RE 6290B	2RE 6290B	Ionization Chamber	Elevation 905 ft. 9 in	10^3 - 10^{10}	Note 5
1RE 6251	2RE 6251	G-M Tube	Elevation 860 ft. 0 in	10^{-1} - 10^4	Note 6
1RE 6253	2RE 6253	G-M Tube	Elevation 860 ft. 0 in.	10^{-1} - 10^4	Note 6
1RE 6255	2RE 6255	Ionization Chamber	Elevation 808 ft. 0 in.	10^2 - 10^7	Note 6
1RE 6256	2RE 6256	Ionization Chamber	In-core instrumentation room Elevation 849 ft. 0 in.	10^2 - 10^7	Note 6
1RE 6285	2RE 6285	Ionization Chamber	Seal table room, Elevation 831 ft. 6 in.	10^2 - 10^7	Note 6

TABLE 12.3-8
AREA RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 2 of 5)

Channel Nos. Unit 1	Unit 2	Detector Type	Monitor Location	Specified Instrument Range (mR/hr)	Bases for Alarm Set Points
Safeguards Building					
1RE 6259	2RE 6259	G-M Tube	Filter Storage Area Elevation 873 ft. 6 in.	10^{-1} - 10^{4}	Note 2
1RE 6259A	2RE 6259A	Ionization Chamber	Mechanical Penetration Area Elevation 810 ft. 6 in.	10^{2} - 10^{7}	Note 3
1RE 6259B	2RE 6259B	Ionization Chamber	Mechanical Penetration Area Elevation 810 ft. 6 in.	10^{2} - 10^{7}	Note 3
1RE 6260A	2RE 6260A	Ionization Chamber	RHR Pump Room Elevation 773 ft. 0 in.	10^{2} - 10^{7}	Note 4
1RE 6260B	2RE 6260B	Ionization Chamber	RHR Pump Room Elevation 773 ft. 0 in.	10^{2} - 10^{7}	Note 4
1RE 6261	2RE 6261	GM Tube	Sampling Room Elevation 810 ft. 6 in.	10^{-1} - 10^{4}	Note 2
1RE 6291A	2RE 6291A	Ionization Chamber	Valve isolation tank room Elevation 790 ft. 6 in.	10^{2} - 10^{7}	Note 3

TABLE 12.3-8
AREA RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 3 of 5)

Channel Nos. Unit 1	Unit 2	Detector Type	Monitor Location	Specified Instrument Range (mR/hr)	Bases for Alarm Set Points
1RE 6291B	2RE 6291B	Ionization Chamber	Valve isolation tank room Elevation 790 ft. 6 in	10^2 - 10^7	Note 3
1RE 6292	2RE 6292	Ionization Chamber	Switchgear Room Elevation 810 ft. 6 in.	10^2 - 10^7	Note 3
1RE 6293	2RE 6293	Ionization Chamber	Piping Penetration Area Elevation 831 ft. 6 in.	10^2 - 10^7	Note 3
1RE 6294	2RE 6294	Ionization Chamber	Electrical Equipment Room Elevation 831 ft. 6 in.	10^2 - 10^7	Note 3
1RE 6295	2RE 6295	Ionization Chamber	Containment Access Hall Elevation 831 ft. 6 in.	10^2 - 10^7	Note 3
1RE 6296	2RE 6296	Ionization Chamber	Switchgear Room Elevation 852 ft. 6 in.	10^2 - 10^7	Note 3
1RE 6297	2RE 6297	Ionization Chamber	Emergency airlock Elevation 896 ft. 6 in.	10^2 - 10^7	Note 3

TABLE 12.3-8
AREA RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 4 of 5)

Channel Nos. Unit 1	Unit 2	Detector Type	Monitor Location	Specified Instrument Range (mR/hr)	Bases for Alarm Set Points
Auxiliary Building					
Fuel Building					
XRE 6272	XRE 6273	G-M Tube	Spent Fuel Pool Operating floor Elevation 860 ft. 0 in.	10^{-1} - 10^{4}	Note 1
XRE 6274	XRE 6275	G-M Tube	Spent Fuel Pool Operating floor Elevation 860 ft. 0 in.	10^{-1} - 10^{4}	Note 1
Turbine Building					
XRE 6283		G-M Tube	Hot lab area Elevation 810 ft. 6 in.	10^{-1} - 10^{4}	Note 2

TABLE 12.3-8
AREA RADIATION MONITORING SYSTEM PARAMETERS

(Sheet 5 of 5)

Channel Nos. Unit 1	Unit 2	Detector Type	Monitor Location	Specified Instrument Range (mR/hr)	Bases for Alarm Set Points
Electrical and Control Building					
XRE 6281	XRE 6282	G-M Tube	Control Room Elevation 830 ft. 0 in.	10^{-1} - 10^{4}	Note 2

Notes:

1. Alarm set points are based on personnel occupancy for normal operation.
2. Alarm set points are based on personnel occupancy for accident conditions.
3. Alarm set points are based on verification of containment integrity for accident conditions.
4. Alarm set points are based on long term equipment surveillance for accident conditions.
5. Alarm set points are based on requirements for emergency planning.
6. Alarm set points are based on surveillance of restricted areas for normal operation.

TABLE 12.3-9
AREA RADIATION MONITORING SYSTEM ENVIRONMENTAL DESIGN CONDITIONS

Item			Normal	Accident	Normal	Accident	Normal	Accident
a.	Control room electronics		80	80	50	50	0.1	0.1
b.	Detectors and Local electronics							
	1)	Outside containment	122	122	95	100	-0.1	ATM
	2)	Inside containment (non-1E)	122	300 (3 hours)	70	Steam, air, and water mixture	ATM	50 psig
	3)	Inside containment (1E)	122	333	70	100	ATM	50 psig

Note:

ATM - atmospheric

12.4 DOSE ASSESSMENT

The following discussion includes historical estimates of personnel occupancy in plant radiation zones and personnel radiation exposures expected annually. The discussion includes a basis for, and an estimate of, the total annual in-plant personnel radiation exposure expected for CPNPP operations. These estimates were prepared as part of the CPNPP Operating License application prior to Unit 1 operation and will not be further maintained in this FSAR Section. Vistra OpCo maintains records of actual in-plant personnel radiation exposures and routinely provides various reports of personnel exposure data, including annual cummulative person-rem totals, as part of the ongoing operational Radiation Protection Program.

12.4.1 ESTIMATED PLANT OCCUPANCY REQUIREMENTS

Table 12.4-1 presents estimated station personnel occupancy requirements of plant radiation areas (radiation zones II, III, IV, and V) during normal operation and anticipated operational occurrences.

12.4.2 DESIGN DOSE RATES

The zone descriptions presented in Section 12.3.1.3 and Table 12.3-6 outline the design maximum dose rate limits for the design of shielding in zones I, II, and III during normal operation, shutdown, and refueling. These limits are 0.25, 2.5, and 25 mrem/hr, respectively. For in-plant areas, these maxima are not expected to occur because plant shielding and access control are based on radionuclide activities corresponding to one percent failed fuel. Because fuel failure and associated radionuclide concentrations are considerably less than those used on a design basis, the maximum expected dose rates in these areas would, likewise, be considerably less. All in-plant areas are designed so that occupational radiation exposure (ORE) is ALARA. During maintenance and inspection in zone IV areas, ORE is kept ALARA by using administrative procedures, drainage, purging, flushing, backwashing, and decontamination. Occupancy in high-radiation areas is limited by administrative control procedures based on routine radiation survey measurements.

12.4.3 ESTIMATED MAN-REM EXPOSURES FOR MAJOR PLANT FUNCTIONS

Total in-plant exposures from 26 operating pressurized water reactor (PWR) units were reported in Reference [1], for 1975, as an average of 309 man-rems with a standard deviation of 275 man-rems. Using the reported average value of 309 man-rems, and normalizing to CPNPP capacity per reactor (1150 MWe, nominal), yields an estimated total in-plant exposure of 421 man-rems per year.

In 1975, an average of 578 persons per reactor received measurable exposures, as reported in Reference [1]. Based on the above value for total dose (421 man-rem), total number of in-plant personnel exposed (578), and distribution data from Table 5 of Reference [1], the distribution of personnel and man-rem, and number of personnel and cummulative man-rem, may be estimated for various in-plant work functions. This data is provided in Tables 12.4-2 and 12.4-3, respectively. Estimates of man-rem doses during refueling are presented in Table 12.4-4. The estimates provided are the expected exposures to an experienced refueling crew; however, refueling operations where trouble is encountered can result in much higher exposures.

In 1975, the average exposure per individual was reported as 0.8 rem [1]. The average exposure per individual for maintenance and refueling activities was 0.77 rem, whereas the average exposure per individual for reactor operations was 0.91 rem. This latter value is a factor of 5.5 below the 5-rem/yr limit presented in 10 CFR Part 20 upon which the 100-mrem/week design basis is established. Therefore, the design approach used on similar plants in which the 100-mrem/week value is used would result in operational levels of individual exposure, a factor of 5.5 below the 10 CFR Part 20[a] limits.

Because most of the anticipated exposures occur under circumstances which are difficult to predict analytically, such as maintenance on radioactive components, shielding design is based on worst case assumptions, and design features are provided which reduce exposures. So, actual exposures should be consistent with the ALARA approach.

The estimated total in-plant exposure of 421 man-rems shown in Table 12.4-3 is within the range of values reported in Reference [1], in 1975, and is not expected to be exceeded by long-term exposures. This value is representative of the average annual exposure during the station lifetime.

12.4.4 ESTIMATED DOSE TO CONSTRUCTION WORKERS

The total dose to construction workers at the site during power operation is the sum of dose contributions from gaseous releases and direct and scattered radiation from contained sources.

12.4.4.1 Construction Worker Dose as a Result of Gaseous Releases

In addition to radiation fields in the general construction area as a result of direct radiation from all Unit 1 sources, there is a contribution to construction worker dose from airborne radionuclides released from the normal operation of Unit 1. These doses, resulting from releases, are calculated assuming the release rates shown in Table 11A-4 of the FSAR. The X/Q values used in estimating exposures in the Unit 2 construction area have been estimated using the straight-line diffusion of NRC Regulatory Guide 1.111, as applied in Appendix 11A of the FSAR, and are presented in Table 12.4-5. All transport times, dose conversion factors and other constants used were those suggested in NRC Regulatory Guide 1.109. Distances and directions from the Unit 1 Containment are plotted as shown on Figure 12.4-1, Building Complex Layout. The corners of the Unit 2 building under construction are chosen arbitrarily because these extremities offer the most vulnerable locations for construction personnel to work.

The location and number of workers during the estimated 16-month Unit 2 construction period are listed in Table 12.4-6. Individual predicted doses are presented at 10 potential worker sites of interest and at 16 site boundary points at the midpoint of each affected wind sector. It is assumed that construction workers spend 2500 hr/yr at the jobsite (50 hr/week for 50 weeks/yr). Also factored into the resultant expected doses shown in Table 12.4-7 is the fact that the breathing rate for occupational exposures is taken to be 10 m^3 per 8-hr period vs. 20 m^3 adult inhalation rate for non-occupational individuals for a 24-hr period. Results are tabulated for total body and skin doses for cloud submersion and total body and thyroid inhalation doses. Estimates are presented on an annual basis (1981 period) such that the 4-man-rem dose

a. Design assessment made with the provisions of 10 CFR 20.1-20.601.

estimates for cloud submersion are derived by exposing the average numbers of workers at each building under construction, shown in Table 12.4-6, to the worst computed individual dose for that particular structure. The results presented in Table 12.4-8 distinguish the 4-month period in 1980 from the 1981 1-yr period.

The estimated doses resulting from airborne radioactivity received by construction workers on Unit 2 as a result of the operation of Unit 1 are well within the limits of 10 CFR Part 20 for exposure of occupationally employed individuals in unrestricted areas.

12.4.4.2 Estimated Dose To Construction Workers Resulting From Contained Sources

The greater part of the construction activity that occurs after the startup of Unit 1 is performed inside buildings. For the purpose of the following calculations, it is assumed that workers spend 15 percent of their workday outdoors. A dose reduction factor of approximately 5.5 (see Section 12.4.3) is applied to the maximum design dose rates of radiation zones (see Section 12.3.1.3) corresponding to the construction activities indoors near contained sources (taken to be Zone II, i.e., 2.5 mrem/hr) and construction activities outdoors and indoors not near contained sources (taken to be Zone I, i.e., 0.25 mrem/hr). The resulting estimated average annual exposures (based on 50 weeks/year and 50 hours/week occupancy) are 1.14 rem/year and 0.11 rem/year, respectively.

Estimates of construction manpower requirements following startup of Unit 1, based on data presented in Table 12.4-6, are shown in Table 12.4-9. Anticipated annual exposures at various construction locations are presented in Table 12.4-10. Total construction worker ORE expressed in man-rems is the product of manpower (in man-yr) and the anticipated annual exposure (in rem/yr). Results of this calculation are presented in Table 12.4-11. These results may be summarized as follows:

1. Over the period considered, the total exposure to the total construction force resulting from the contained sources is 297.71 man-rems.

2. The average annual exposure to an individual is 0.46 rem/yr (297.71 man-rem/ 653.33 man-yr).

3. The average annual exposure to an individual working in the Auxiliary Building, Fuel Building, or Safeguards Building, Unit 2 (after fuel load of Unit 2) is 0.98 rem/yr (0.85 x 1.14 rem/yr + 0.15 x 0.11 rem/yr).

4. The highest exposure received by a construction worker during the entire period considered is 1.31 rems (0.98 rem/yr x 1.33 yr).

12.4.5 ESTIMATED ANNUAL DOSE AT THE EXCLUSION BOUNDARY AND TO THE POPULATION AT LARGE

At the minimum exclusion boundary distance the estimated annual exposure to an individual as a result of contained sources is approximately 3.25×10^{-3} mrems. The estimated additional annual exposure due to the ISFSI with 84 casks is well below 1 mrem. The total exposure to the cumulative population within 50 miles of the site (for census year 2000) as a result of contained sources is approximately 1.53×10^{-3} man-rem.

REFERENCES

1. Murphy, T.D., et al., Occupational Radiation Exposure at Light Water Cooled Power Reactors, 1969-1975, U.S. Nuclear Regulatory Commission, NUREG-0109, August 1976.

TABLE 12.4-1
ESTIMATED STATION PERSONNEL OCCUPANCY OF PLANT RADIATION AREAS (MAN-HOURS/YEAR)

	Radiation Zone					
Personnel	I	II	III	IV	V	Totals
Operators	40,000	25,000	3,600	950	250	120,000
Maintenance	52,000	42,000	4,800	1,080	120	100,000
Technical Support	80,000	35,000	4,000	900	100	120,000
Administrative	54,000	6,000	(a)	(a)	(a)	60,000

a) Denotes insignificant occupancy

TABLE 12.4-2
DISTRIBUTION OF PERSONNEL AND COLLECTIVE DOSE EQUIVALENTS
FOR VARIOUS WORK FUNCTIONS

Work Function	Percentage of Total Personnel	Percentage of Total Man-Rem
Reactor Operations And Surveillance	9.2	10.8
Routine Maintenance	59.5	52.6
Inservice Inspection	4.1	3.0
Special Maintenance	16.3	19.0
Waste Processing	7.6	6.9
Refueling	3.3	7.7

TABLE 12.4-3
ESTIMATES OF PERSONNEL AND COLLECTIVE DOSE EQUIVALENTS FOR VARIOUS WORK FUNCTIONS

(Sheet 1 of 3)

Work Function			Est. Dose Range mrem/hr.	Duration man hrs/yr	Dose Man-Rem
1)	Reactor Operations & Surveillance (estimated number of personnel = 53)				
	a)	Routine Patrols and Operations	.25/300	18,000	8
	b)	Radiation Surveys	.25/2.5	5,500	6
	c)	Radiation Surveys	2.5/100	4,000	13
	d)	Radiation Surveys	>100	50	7
	e)	Periodic Tests, Insp. & Calib.	.25/2.5	10,000	3
	f)	Periodic Tests, Insp. & Calib.	2.5/25	1,500	4
	g)[(a)]	Control Room Operations	<.25	50,000	5
					46
2)	Routine Maintenance (estimated number of personnel = 344)				
	a)	Normal Operations			
		1) Pumps	.25/50	2,500	5
		2) Valves	.25/50	4,000	10
		3) Instrument & Controls	.25/25	4,000	3
		4) Motors	.25/25	1,000	2
		5) Heat Exchangers	.25/1500	1,000	4
		6) Waste Processing Systems	.25/1500	1,000	7
		7) Demineralizer (Resin Change)	2.5/25	1,000	3
		8) Filter Changes	2.5/25	1,000	3
		9) HVAC	.25/25	500	1
		10) Misc.	.25/25	1,500	6

TABLE 12.4-3
ESTIMATES OF PERSONNEL AND COLLECTIVE DOSE EQUIVALENTS FOR VARIOUS WORK FUNCTIONS

(Sheet 2 of 3)

Work Function				Est. Dose Range mrem/hr.	Duration man hrs/yr	Dose Man-Rem
	b)	Refueling Operations				
		1)[b]	Preparation for Head Removal	5/200	1,900	40
		2)	Head Removal	5/200	80	11
		3)	Upper Internals Removal	2.5/150	100	2
		4)	Installation of Upper Internals	2.5/150	100	2
		5)[b]	Preparation for Head Installation	5/1500	1,900	45
		6)	Head Installation	5/200	80	11
		7)	Refueling Cavity Clean-Up	5/100	500	10
		8)	Preparation for Containment Closure	2.5/100	2,800	36
		9)	Steam Generator Test	5/15,000	1,000	10
		10)	Incore Instrumentation	10/50	500	10
						221
3)	Inservice Inspection (estimated number of personnel = 24)			2.5/200	700	13
4)	Special Maintenance (estimated number of personnel = 94)					
	a)	Steam Generator Maintenance		5/15,000	4,000	22
	b)	Reactor Coolant Pump Seal Inspection and Repair		5/500	1,000	8
	c)	Evaporator		5/100	500	5
	d)	Heat Exchanger		5/100	700	10
	e)	Demineralizer		5/100	500	6

TABLE 12.4-3
ESTIMATES OF PERSONNEL AND COLLECTIVE DOSE EQUIVALENTS FOR VARIOUS WORK FUNCTIONS

(Sheet 3 of 3)

Work Function			Est. Dose Range mrem/hr.	Duration man hrs/yr	Dose Man-Rem
	f)	Safety System Modification	5/100	500	5
	g)	Pumps	.25/100	700	6
	h)	Waste Processing System	.25/1500	500	7
	i)	Motors	.25/25	500	5
	j)	Instrument & Control	.25/25	300	3
	k)	HVAC	.25/25	500	3
					80
5)	Waste Processing (estimated number of personnel = 44)		2.5/100	4,400	29
6)	Refueling (estimated number of personnel = 19)				
	a)	Health Physics Operations	.25/1500	7,000	14
	b)	Fuel Handling	.25/25	1,300	12
	c)	Testing & Inspections Specific to Refueling	2.5/100	6,000	6
					32
TOTALS		(estimated number of personnel = 578)			421

a) The control room is a Zone I radiation area, but it is included in this table due to the large duration associated with it.

b) Items (1) and (5) include detensioning/removal of reactor stud bolts and installation/ tensioning of same

TABLE 12.4-4
ESTIMATES OF COLLECTIVE DOSE EQUIVALENTS DURING REFUELING

Operation	Average Radiation Field (mrem/hr)	Man-Rem
Missile shield	25	0.40
Reactor vessel head connections	60	4.60
Cavity seal ring	50	0.00
Reactor vessel stud work	80	10.24
Guide studs and stud hole plugs	80	1.28
Reactor vessel head lift and cavity fill/drain	15	1.05
CRDM drive shafts	15	0.72
Upper internals lift	15	0.72
Fuel shuffle	15	12.00
TOTAL		31.73

TABLE 12.4-5
EXPECTED ATMOSPHERIC RELATIVE CONCENTRATION VALUES AT CALCULATION LOCATIONS

Building/ Location	Distance (m)	Sector	X/Q (sec m^{-3})
Auxiliary Building	105	NW	9.00E-04
" "	81	W	5.60E-04
" "	50	WNW	2.28E-03
" "	66	N	1.92E-03
Containment, Unit 2	89	N	1.08E-03
Safeguards Building, Unit 2	87	NW	1.28E-03
" "	131	NNW	6.40E-04
" "	122	N	5.80E-04
Fuel Building	71	NNE	1.04E-03
" "	37	ENE	2.24E-03

TABLE 12.4-6
LOCATION AND NUMBER OF CONSTRUCTION WORKERS[a]

Building	1980 Average for 4 Months	1981 Average for 12 Months
Containment, Unit 2	249	179
Auxiliary Building	99	96
Fuel Building	25	25
Safeguards Building, Unit 2	201	162
Average for period	574	462

a) Includes preoperational testing and start-up personnel.

TABLE 12.4-7
INDIVIDUAL EXPECTED CONSTRUCTION WORKER DOSE EQUIVALENT[a]

Building/ Location	Distance (m)	Affected Sector	Total Body[b] Cloud Dose (mrem/yr)	Cloud[b] Skin Dose (mrem/yr)	Total Body[b][c] Inhalation Dose (mrem/yr)	Thyroid[b][c] Inhalation Dose (mrem/yr)
Auxiliary Building	105	NW	4.65E+00	1.32E+01	1.68E+01	2.08E+01
" "	81	W	2.09E+00	8.20E+01	1.04E+01	1.30E+01
	50	WNW	1.18E+01	3.34E+01	4.24E+01	5.25E+02
	66	N	9.80E+01	2.78E+01	3.53E+01	4.41E+01
Containment, Unit 2	89	N	5.60E+01	1.58E+01	2.01E+01	2.50E+01
Safeguards Bldg. Unit 2	87	NW	6.60E+01	1.87E+01	2.38E+01	2.98E+01
" "	131	NNW	3.31E+00	9.35E+00	1.19E+01	1.48E+01
	122	N	3.00E+00	8.50E+01	1.08E+01	1.34E+01
Fuel Building	71	NNE	5.40E+00	1.53E+01	1.94E+01	2.42E+01
" "	37	ENE	1.16E+01	3.28E+01	4.18E+01	5.20E+02

a) In the tables "Dose" is used for "Dose Equivalent"

b) Workers spend 50 hr/week, 50 weeks/yr on the job.

c) Occupational inhalation rate 10 m^3/8-hr day or 30 m^3/24-hr period vs. 20 m^3 adult breathing rate assumed for nonoccupation exposure.

TABLE 12.4-8
ESTIMATION OF COLLECTIVE DOSE EQUIVALENT TO CONSTRUCTION WORKERS DURING 1980 TO 1981 PERIOD

	No. of Workers (1980)		No. of Workers (1981)	
Building	Average for 4 Months	Man-Rems	Average for 12 Months	Man-Rems
Containment, Unit 2	249	4.63E-01	179	1.00E+00
Auxiliary Building	99	3.89E-01	96	1.13E+00
Fuel Building	25	9.65E-01	25	2.90E-01
Safeguard Building, Unit 2	201	4.42E-01	162	1.07E+00
Average for period	574	1.39E+00	462	3.49E+00

TABLE 12.4-9
CONSTRUCTION MANPOWER REQUIREMENTS

Location	Manpower (man-yr)	
	Inside	Outside
Containment, Unit 2	222.70	39.30
Auxiliary Building	109.65	19.35
Fuel Building	28.33	5.00
Safeguards Building Unit 2 (prior to fuel load of Unit 2)	114.33	20.18
Safeguards Building Unit 2 (after fuel load of Unit 2)	80.33	14.18
Subtotals	555.33	98.00
TOTAL		653.33

TABLE 12.4-10
ANNUAL DOSE EQUIVALENT TO CONSTRUCTION WORKERS RESULTING FROM CONTAINED SOURCES

Location	Annual Dose Equivalent (rem/yr)	
	Inside	Outside
Containment, Unit 2[(a)]	0.11	0.11
Auxiliary Building	1.14	0.11
Fuel Building	1.14	0.11
Safeguards Building Unit 2 (prior to fuel load of Unit 2)	0.11	0.11
Safeguards Building Unit 2 (after fuel load of Unit 2)	1.14	0.11

a) All work assumed to be completed prior to fuel load of Unit 2.

TABLE 12.4-11
CONSTRUCTION WORKER OCCUPATIONAL COLLECTIVE DOSE EQUIVALENT
RESULTING FROM CONTAINED SOURCES

	ORE (man-rem)	
Location	Inside	Outside
Containment, Unit 2	25.32	4.47
Auxiliary Building	124.69	2.20
Fuel Building	32.21	0.57
Safeguards Building, Unit 2 (prior to fuel load of Unit 2)	13.00	2.29
Safeguards Building, Unit 2 (after fuel load of Unit 2)	91.34	1.61
Subtotals	286.57	11.14
TOTAL		297.71

12.5 RADIATION PROTECTION

12.5.1 ORGANIZATION

The CPNPP radiation protection program is established to provide an effective means of radiation protection for station personnel, visitors, and the general public. To provide this, the radiation protection program incorporates a dedicated philosophy from management (Section 12.1.1), qualified personnel to direct and to implement the radiation protection program, sufficient equipment, facilities, written procedures and instructions based upon acceptable radiation protection practices and guidance. The CPNPP operational radiation protection program is developed and implemented through the applicable guidance of Regulatory Guides; 8.2 (February 1973); 8.8, Revision 3 (June 1978); and 8.10, Revision 1-R (May 1977).

The Radiation Protection Department implements and enforces the CPNPP radiation protection program. The Radiation Protection and Safety Manager (see Section 13.1.1.2.1 for organizational structure) is responsible for oversight and implementation of the radiation protection program. The Radiation Protection and Safety Manager, directs and supervises radiation protection personnel and activities in the plant.

The Radiation Protection and Safety Manager is responsible for insuring compliance with regulatory requirements in the following areas:

1. Shipping and receiving of all radioactive material.

2. Transfer and storage of radioactive waste.

3. Approval of all radioactive waste containers.

4. Collection, sorting, and packaging of radioactive waste.

Qualified Radiation Protection technicians are assigned to the station and are familiar with station lay-out, personnel, procedures, and equipment. Additional information on the qualifications and experience of the Radiation Protection and Safety Manager and technicians can be found in Section 13.1 of the FSAR. Radiation Protection supports the Operations, Maintenance, and Engineering Departments and provides radiation protection coverage for activities that involve exposure to radiation or radioactive material. During major maintenance and refueling outages, additional qualified technicians may be secured, as required, based on the work load. In addition, Radiation Protection is organized to provide the following services:

1. Developing radiation protection procedures and instructions for routine and non-routine activities that may be encountered in operating, maintaining, inspecting, and testing the station.

2. Ensuring that the provisions and standards of 10 CFR Part 20 are implemented.

3. Providing a personnel radiation dosimetry program and maintaining dosimetry records.

4. Performing radiological surveys of station areas and maintaining records of survey results.

5. Assisting in the station training program by assuring that adequate radiation protection training is provided.

6. Maintaining an adequate supply of calibrated radiation detection instruments and equipment for assessing the radiological conditions at CPNPP.

7. Providing, maintaining, and issuing protective clothing and equipment.

8. Preparing and issuing reports dealing with the radiological aspects of CPNPP, as required by regulatory agencies and station procedures.

9. Directing the decontamination of personnel, equipment, and facilities at CPNPP.

10. Developing, implementing and maintaining the CPNPP radiological respiratory protection program.

It is the responsibility of each individual to obey radiological control procedures and to report to Radiation Protection personnel any circumstances where procedures are not being followed or unsafe activities are occurring. Each individual assigned to work in Radiologically Controlled Areas (RCA) is responsible for demonstrating a thorough working knowledge of radiation protection rules and procedures. Personnel requiring access to radiation areas are trained in radiation protection principles and are responsible for demonstrating proficiency in radiation protection skills. New employees required to work in RCAs are required to complete the training program and demonstrate their proficiency prior to beginning work assignments. Visiting or contract individuals will be escorted in the RCA until they demonstrate an adequate knowledge of radiation protection principles through the completion of the training program.

12.5.2 EQUIPMENT, INSTRUMENTATION, AND FACILITIES

Radiation Protection procures and maintains the appropriate monitoring instrumentation and protective equipment and supplies necessary for operating and maintaining the CPNPP radiation protection program. Additionally, adequate facilities are available for the conduct of radiation protection activities such as controlling access to Radiologically Controlled Areas, locating and storing equipment, performing calibrations, etc.

Equipment, facilities and instruments necessary to implement the Radiation Protection Program are highly reliable, commercial grade items. The operation, maintenance and calibration of this equipment is subject to the Operations Quality Assurance Program. Activities necessary to implement the Radiation Protection Program are controlled by approved procedures and are subject to proper Quality Assurance controls and surveillances. The following sections provide information on the types of equipment available, including criteria for selection, and radiation protection facilities.

12.5.2.1 Radiation Detection Instrumentation and Calibration Facilities

Portable, non-portable and laboratory radiation detection equipment are available to provide the appropriate detection capabilities, ranges, sensitivities, and accuracies required for the anticipated types and levels of radiation to be found at CPNPP, whether during normal operations and maintenance, or emergency conditions. The minimum inventories of instruments and equipment is listed in Table 12.5-1. This table inventory includes equipment assigned for

use in both normal plant operations and emergency preparedness. Some equipment assigned for use in plant operations may periodically exceed the designated instrument calibration date depending on circumstances, e.g., period after an outage with no intention of use. However, all equipment is considered available and is calibrated prior to use in accordance with calibration schedules and station procedures. Sufficient inventories of instrumentation are maintained to support radiation protection programs and activities for plant operations and maintenance. Non-portable and laboratory equipment is available to support radiation protection and radiochemistry functions, including a gamma spectroscopy system, a liquid scintillation counter, a low-background proportional counter, whole body counting systems, and personnel contamination monitors. Functional checks are performed on each counting system in accordance with procedures or instructions in order to determine such things as instrument response, background count rates and counting efficiencies. Records are maintained for each instrument or counting system. Repair and maintenance of this equipment is performed by station or contract personnel. Available portable instrumentation includes: low and high range ion chambers; G-M survey meters, smear counters and friskers; alpha scintillation survey meters; neutron dose-equivalent rate meters; continuous air monitors; and airborne grab samplers.

Storage locations for radiation protection instrumentation will depend on the function and frequency of use of the instrument. In general, such instrumentation is stored at locations that offer convenient accessibility to the work areas, such as the access control office.

Normally, calibration of portable and non-portable radiation protection equipment is performed onsite by station personnel. Calibration instructions are written with detailed records of calibration and maintenance of each instrument maintained at the station.

Calibration of portable radiation protection equipment is performed in the Radiation Protection Instrumentation Calibration Facility (or other suitable location) or by a qualified vendor. The Radiation Protection Instrumentation Calibration Facility is located in the Turbine Building Annex adjacent to the Unit 1 Turbine Building. This facility is conveniently located and provides a low background environment, a storage area for calibration sources, and adequate work space. Calibrations are performed using radioactive sources traceable to the National Institute of Standards and Technology (NIST) or using transfer instruments, such as electrometers, which have been calibrated using NIST traceable sources. Each instrument is labeled with a calibration sticker after calibration.

The Radiation Protection Instrumentation Calibration Facility includes a high activity Cs-137 source (initially about 5,000 Ci in 1983) housed in a calibration well assembly. The well assembly provides shielding for the source and consists of a 30 foot (below grade) vertical well containing the calibration source which rests on an elevator assembly.

The well assembly is designed to ensure that radiation exposures to individuals in unrestricted areas are in compliance with 10CFR Part 20. An elevator operating assembly, consisting of a chain and sprocket assembly with braking ability and a device to indicate the position of the source, is used to raise, lower or hold the vertical position of the well source elevator assembly. The calibration well assembly also includes five moveable shields mounted in tracks that are used for establishing desired exposure rates. These shields are locked to shield the active area above the calibration well during periods when the Calibration Well Source is not in use. Each shield is also equipped with a handle which allows the operator to move the shields along their

tracks without extending any portion of the body into or over the active area of the Calibration Well.

During periods when the source is in use, a high radiation area exists on the roof of the calibration facility. This area is fenced to prevent access, posted with appropriate warning signs and equipped with a warning light that is energized when the source is in use. There is no formal or routine access to the roof of the facility.

The following physical and administrative controls are used to prevent inadvertent exposures resulting from the Calibration Well Source:

1. During periods when the Calibration Facility is not attended, the source is lowered to its bottom position, and the physical barriers designed to prevent access to or operation of the Calibration Well Source are locked. These physical barriers include locking the elevator assembly hand wheel, locking the moveable shields, and locking the entrance to the well source room.

2. Warning lights are mounted on the roof of the calibration facility and in the well source room. These lights are activated by any movement of the moveable shields from their stored position or whenever the well source is raised off the bottom of the calibration well.

3. The well source assembly contains an upper limit stop which prevents the source from being raised above a designated height that would result in excessive exposures to the operator and in areas outside of the calibration facility.

4. All operations of the Calibration Well Source are performed in accordance with approved station procedures.

5. All activities involving use of the Calibration Well Source require the issuance of a Radiation Work Permit.

6. Use of the Calibration Well Source is only allowed by personnel specifically trained and qualified in its use.

12.5.2.2 Radiation Protection, Access Control, Decontamination and Laboratory Facilities

Radiation Protection and Access Control facilities are located in the Turbine Building on Elevation 810'-6". These areas provide office space for Radiation Protection personnel and are arranged to maintain positive control for entry and exit of personnel and equipment to and from the Radiologically Controlled Areas (RCA).

Access Control Facilities are used to facilitate access into and out of the RCA. Male and female dressing areas are located in the access control areas. These dressing areas are used for donning modesty clothing prior to RCA entry; therefore, separate change areas within the RCA are not necessary. Information necessary for entry into the RCA is available in the access control areas. Normal information reviewed in this area may include the RWP/GAP and the radiological conditions of the room or building in which personnel may be working. Personnel log into the RCA at the Access Control Point located on the 810'-6" elevation. The individual’s GAP/ RWP authorization, whole body count, current Radiation Worker Training and GAP/RWP requirements are verified prior to access.

Upon completion of RCA entry, all items and personnel must be monitored for contamination prior to exit and release from the RCA.

A Personnel Decontamination room is available near the Access Control Point to decontaminate workers. The decontamination room is supplied with protective clothing, cleaning agents and decontamination facilities (i.e. sink, showers, etc.). A Personnel Decontamination room is normally used after radioactive contamination has been detected at a monitoring location inside the primary RCA or during RCA exit at the Access Control Point. Drains from the decontamination room are routed to the Liquid Waste Processing System.

Personnel exit and log out of the RCA through the Access Control Point after monitoring themselves for contamination. Other exit processing may include collection of dosimetry and any special equipment that may have been issued. After exiting the Access Control Point, personnel may return to the appropriate locker room to change from modesty clothing back to personal clothing.

A low-background Chemistry counting room is available and is used for counting and/or identifying radioactivity in samples. Work activities in the counting room are controlled by station procedures. These procedures are based upon acceptable radiation protection techniques and philosophy. The counting room is equipped with a gamma spectroscopy system, liquid scintillation detector, proportional counter and other assorted counting and ancillary equipment. A hot laboratory is available where primary side samples are processed and prepared for counting. The sink and floor drains in this room are directed to the Liquid Waste Processing System and the fume hood exhaust is directed to the ventilation system. Equipment included in this room consists of such items as survey meters, and miscellaneous laboratory instruments such as pH meters, spectrophotometer, conductivity meter, and various glassware and chemicals.

12.5.2.3 Protective Clothing

Protective clothing is prescribed by Radiation Protection based upon the actual or potential radiological conditions expected for the job assignment. Protective clothing issue stations are established and maintained at inplant locations required to ensure efficient operations and to preclude the spreading of contamination. Used protective clothing is normally laundered by contracted vendor services.

12.5.2.4 Respiratory Protection Equipment

Respiratory protection equipment is available to station personnel and is prescribed and issued to individuals as required by the actual or potential radiological conditions of the work assignment. Selection of respiratory protection equipment such as full-face masks, self-contained breathing apparatus, and chemical cartridge respirators is made following the guidance of applicable regulations contained in 30 CFR Part 11 and the National Institute for Occupational Safety and Health's (NIOSH) Certified Equipment Manual. Respiratory devices available at CPNPP include the following:

1. Full-face masks with high-efficiency particulate and charcoal filters.

2. Full-face masks with airline respirator.

3. Hoods and suits with airline respirator.

4. Full-face masks with self-contained breathing apparatus.

Respiratory protection equipment is stored at the Access Control area, Control Room (SCBAs only), EOF (SCBAs only), and Fuel Building. The Respiratory Protection Program is described in detail in Section 12.5.3.6.

12.5.2.5 Personnel Exposure Monitoring Equipment

Thermoluminescent Dosimeter (TLD) Badges or Electronic Alarming Dosimetry (EAD), and Pocket Ion Chambers (PICs) are available in accordance with plant procedures for monitoring external radiation exposure to station personnel, visitors, and support personnel who enter Radiologically Controlled Areas at CPNPP. Neutron dosimetry devices are also available for those individuals subject to neutron exposure as required by job assignment and location. TLD multi badges and extremity dosimeters will be available for those assignments that require them. The TLD Badges are processed and a definitive dose evaluation is provided by the approved service organization as requested.

Whole body counters and personnel contamination monitors are used for in-vivo radiation measurements of station personnel, visitors, or support personnel to assess internal radiation exposure. Whole body counters and personnel contamination monitors will provide preliminary background information, periodic evaluation, and emergency capability for detecting internal exposure conditions.

The CPNPP personnel exposure monitoring program is described in detail in Section 12.5.3.5.

12.5.3 PROGRAMS AND PROCEDURES

Procedures for personnel radiation protection shall be prepared consistent with the requirements of 10 CFR 20 and shall be approved, maintained, and adhered to for all operations involving personnel radiation exposure. The radiation protection programs, procedures, and instructions developed for Comanche Peak Nuclear Power Plant are an integral part of the ALARA policy as discussed in Section 12.1. These procedures, utilized by well-trained and qualified personnel, should contribute significantly to the overall reduction of the occupational radiation exposures. The provisions and suggestions of Regulatory Guides 8.8, Revision 3 (June 1978), and 8.10, Revision 1-R (May 1977), are closely followed in the development of the radiation protection procedures and instructions. The facilities and equipment necessary to support these programs, procedures, and instructions are discussed in Section 12.5.2.

12.5.3.1 Radiation Surveillance

Radiation Protection routinely surveys selected areas at CPNPP for the assessment of radiation-field, radioactive contamination, and airborne radioactivity levels. Portable instruments, equipment, and techniques used for these surveys are selected based upon the type of survey required and the anticipated types and levels of radioactivity at the location. Survey results are recorded and archived for future reference, reports, and radiation trend studies.

Area radiation surveys are performed on the basis of good ALARA practices and depend upon several factors such as location, actual or potential radiation levels, station status, and

occupancy factor. Surveys are periodically performed in areas subject to change. Examples of these areas inside the RCA include rooms with RHR piping during system operation, and areas outside the RCA include the Turbine Building condensate polishing skid area when there is primary to secondary leakage through one or more of the steam generators. The frequency of surveys may be weekly, monthly, quarterly, semi-annual, annual or more often as dictated by operating conditions and as directed by the Radiation Protection Manager. All survey results are recorded, filed and may be posted locally to assure adequate radiological controls. Caution placards are posted locally to comply with 10 CFR Part 20 requirements.

Contamination surveys are performed at selected locations throughout the station to evaluate the hazard due to removable and non-removable radioactive contamination. The selected locations represent strategic points where there exists the potential for the spread of contamination. Contamination surveys are made using the “smear”, “swipe”, or “large area masslin” technique or by using an appropriate portable instrument. Frequency of surveys depends upon factors such as the actual or potential radioactive concentration, occupancy factor, location, and station status. Routine surveys are performed weekly, monthly, quarterly, semi-annual, annual or more often as required by actual operating conditions and as directed by the Radiation Protection Manager. As required, contamination surveys are performed on personnel, equipment, and in uncontrolled areas to ascertain contamination levels and to ensure that control methods are adequate. Contamination survey results are recorded, filed, and as needed, posted for daily information.

Surveys to assess the airborne radioactivity levels are performed to ensure 10 CFR Part 20 limits are not exceeded, engineering controls are functioning, and respiratory protection techniques are adequate. Areas to be routinely surveyed are determined by a Radiation Protection Supervisor and are based upon the actual or potential hazard, task to be performed, and occupancy factor. Frequency of routine surveys for airborne radioactivity will vary depending upon several factors. For instance, selected areas may be surveyed daily, weekly, monthly or continuously depending upon the location, work to be performed, and actual or potential hazard. Monitoring may be performed with continuous air monitors (CAMs) or with grab samples.

In order to warn personnel of changing airborne conditions, CAM alarm setpoints are normally set at a value corresponding to 25% of the appropriate concentration values given in 10CFR20, Appendix B, Table 1, Column 3. These setpoints may be adjusted as necessary based on actual plant conditions in accordance with station procedures. Appropriate survey results necessary to demonstrate adequate radiological controls are recorded and filed and may be posted locally.

12.5.3.2 Radiological Work Control

Access to Radiologically Controlled Areas at CPNPP is controlled by administrative and physical measures and in accordance with procedural requirements. Station management assures entry control to radiation and high radiation areas through the administration of access permits that stipulate purpose of entry, work location, radiological conditions, surveillance and dosimetry requirements, protective clothing and equipment, and other procedural requirements. The objectives for issuing radiation work permits (RWP)/general access permits (GAP) is to provide the following:

1. Control the access to Radiologically Controlled Areas so that entrance is allowed only to those individuals whose job functions require entry.

2. Provide a detailed assessment of the actual and potential radiological hazards that are associated with the job function and area.

3. Ensure that proper protective measures are taken to safely perform the required duties in the area and to maintain occupational radiation exposure ALARA.

4. Ensure that the individual acknowledges his understanding of the radiological conditions, the protective and safety measures required, and his willingness to follow the GAP/RWP requirements.

5. Provide an avenue for dose-accounting that correlates the individual, the job function, and job for ALARA reports and studies.

6. Provides a communication check which ensures that the appropriate supervisors are aware of the task being performed, the radiological conditions, and the prescribed protective measures.

7. Provide a means for maintaining the accountability of personnel in RCAs.

General Access Permits (GAP) are issued for activities in RCAs where the radiation levels are known and the radiological conditions are not subject to erratic or large changes. The GAP requires administrative authorization and is issued in accordance with procedural requirements.

Radiation Work Permits (RWP) are issued for non-routine, specific activities in areas where radiation levels are significant and may change frequently, rapidly or significantly. Issuance, termination, authorization, protective measures, surveillance requirements and other stipulations for the RWP follow procedural requirements.

12.5.3.3 Posting of Radiological Conditions

Radiation areas, high radiation areas and other radiological zones are identified with the appropriate sign, label, or placard and the control of access into these areas is in accordance with the regulations of 10 CFR 20. Additionally, administrative controls for the posting of and access to high radiation areas are established in the plant Technical Specifications.

Areas that may be contaminated with radioactive material at CPNPP are decontaminated to a level that is reasonably achievable using the most effective methods and techniques. At times certain areas will be designated “contaminated areas” and therefore posted with the proper warning placards and barricaded to isolate the area from routine traffic patterns. Entry to these areas is controlled by Radiation Protection and allowed only through the issuance of a GAP or RWP. Personnel, equipment, and material exiting from contaminated areas are monitored by frisker or other appropriate radiation detecting instruments to prevent the spread of contamination to clean areas. Contaminated equipment is properly packaged and identified before removal from a contaminated area. A final survey will be conducted to ensure that all personnel, material, and equipment are free of significant contamination to provide assurance that no radioactive material is spread to unrestricted areas of the station.

12.5.3.4 Radiation Protection Training Program

The Manager, Nuclear Radiation Protection is responsible for the Radiation Protection Training Program at CPNPP. All permanent station personnel who are required to work in radiologically controlled areas (RCAs) complete the basic training courses, lectures, and exercises and demonstrate their proficiency and competence prior to being allowed to work in RCAs. The Radiation Protection Training Program maintains employee proficiency through periodic retraining lectures and exercises.

The degree of training received by employees is commensurate with the individual's job function and anticipated radiation hazards. The Radiation Protection Training Program includes as a minimum the following topics:

1. Fundamentals of radiation and radioactivity
2. Radiation Protection techniques
3. Biological effects of radiation
4. Measurement of radioactivity and radiation
5. Use of protective clothing and equipment
6. Radiation Protection procedures and instructions
7. Station and federal rules and regulations
8. Emergency planning
9. ALARA program - policy, concepts and methods

Visitors and temporary maintenance and service personnel will be trained in the above topics to the extent necessary to assure safe execution of their duties. As required, all non-permanent personnel and visitors are properly escorted unless they demonstrate an adequate knowledge of the appropriate topics included in the Radiation Protection Training Program. Additional training information is contained in Section 13.2.

12.5.3.5 Personnel Exposure Monitoring

Station and support personnel, who enter Radiologically Controlled Areas at CPNPP are monitored for external radiation exposure using Thermoluminescent Dosimeter (TLD) Badges, alarming electronic dosimeters or pocket ionization chambers (PICs). Issuance of TLD Badges, electronic dosimeters or PICs, is performed in accordance with approved station procedures. TLD Badges processing and dose determination are performed by an approved offsite dosimetry service organization with approved quality manual(s). Additional dosimetry equipment, such as multibadges is available and issued in accordance with procedural and radiological control requirements. The TLD Badge service organization shall maintain accreditation with the National Voluntary Laboratory Accreditation Program (NVLAP).

Electronic alarming dosimeters or direct reading pocket dosimeters are issued as a secondary method of determining gamma exposure for dose control/tracking purposes when TLD Badges are used as the primary means of monitoring exposure. The use, care and testing of pocket dosimeters follows the applicable guidance of Regulatory Guide 8.4 (February 1973).

Exposure records for each individual are maintained in accordance with Regulatory Guide 8.7. Current exposure data for each individual is available and may be reviewed by the individual upon request. Departmental exposure reports are issued to supervisors, upon request, to provide job scheduling assistance. Evaluation of employee exposure records by radiation protection personnel is performed to analyze individual and job exposure trends and to assist in maintaining occupational radiation exposures ALARA.

Internal radiation exposures are identified and assessed by the CPNPP Bioassay Program. This program follows the guidance of Regulatory Guide 8.9 (July 1993). Analysis of internal deposition of radioactive materials is performed using either a whole body counter or excreta samples. The whole body counter and personnel contamination monitor are used for in-vivo radiation measurements of station personnel, support personnel, and visitors. Those individuals who regularly enter areas where the potential exists for inhalation, ingestion, or absorption of radioactive materials are counted prior to initial entry. Special monitoring and evaluation of known or suspected uptakes is performed on a case-by-case basis in accordance with Regulatory Guide 8.9 (July 1993). If necessary, excreta samples are collected and sent to a qualified laboratory for analysis. Significant doses resulting from uptake of radioactive material(s) are calculated in accordance with ICRP 30 and documented on the individual's NRC Form 5 (equivalent) as specified in station procedures.

12.5.3.6 Respiratory Protection

It is management's intent to control airborne radioactivity levels as effectively as practicable by proper preventive measures, engineering controls, and good housekeeping techniques. At times, inadvertent airborne radioactivity problems will exist. Prompt assessment of the airborne activity levels is very important and was discussed previously in this section. Section 12.3.4 provides information on airborne radioactivity monitoring instrumentation.

Control of airborne radioactivity levels is assured through the use of the station's heating, ventilation and air conditioning (HVAC) systems and portable air movers and filters. The HVAC systems provide controlled air movement and filtration for those areas with a high potential for airborne radioactivity problems. As required, special control techniques are used, such as enclosures that isolate and vent airborne radioactivity arising from special work projects. To be used as a final alternative, respiratory protection equipment is available for use in those situations where airborne radioactivity hazards exist and other control measures prove inadequate.

The Manager, Nuclear Radiation Protection is the individual responsible for the development and implementation of the Respiratory Protection Program at CPNPP. The Respiratory Protection Program is developed through the guidance of Regulatory Guide 8.15 (October 1999) and satisfies the requirements of 10 CFR Part 20. A written policy statement exists to insure the provisions of the program are adequately followed and enforced. As a minimum the CPNPP Respiratory Protection Program will provide:

1. Written procedures covering such aspects of the program as description of the equipment, issuance of equipment, selection of equipment based upon hazards, maintenance of equipment and training.

2. Training and fitting sessions for those individuals whose job may entail the wearing of respiratory protection equipment.

3. Proper storage facilities, maintenance, and quality control for equipment used in the respiratory protection program.

4. An adequate medical surveillance program for respirator users.

5. Periodic inspection and surveillance of routine and emergency respiratory equipment.

6. Use of only NIOSH/MSHA-certified or NRC-authorized equipment.

7. A periodic review of the overall Respiratory Protection Program, based upon inspection, maintenance, and quality control reports and bioassay results, to verify the effectiveness of the program.

12.5.3.7 Radioactive Materials Control

Procedures and instructions are developed to control, handle, and store radioactive materials, including radioactive sources and contaminated materials such as tools and equipment. The objectives of these procedures and instructions are to maintain accountability of material, to satisfy regulatory requirements, and to ensure that occupational radiation exposures due to this material are ALARA. Subjects covered by these procedures and instructions include, but are not limited to:

1. Properly identifying radioactive material as required by applicable regulatory and procedural requirements. Warning labels and signs are applied to areas and containers.

2. Utilizing adequate storage facilities for radioactive materials. Storage areas for licensed sealed sources are maintained locked, except when in use. Access to radioactive material is controlled (e.g., by RWP, shielded or locked if appropriate) to minimize exposure to personnel.

3. Using the elements of time, distance, and shielding to work safely with this material and to maintain exposures ALARA. Special extension and remote handling tools are used when applicable.

4. Periodically testing licensed radioactive material sealed sources to verify the integrity of the sealed material.

5. Inventorying non-exempt (reference 10 CFR 30) radioactive sources at regular intervals to maintain accountability.

6. Instructions for proper actions to be taken in the event of leakage and spills.

7. Requirements for the movement and use of radioactive materials including shielding requirements, area control, and accountability requirements.

TABLE 12.5-1
RADIATION MONITORING INSTRUMENTATION AND OTHER RADIATION PROTECTION EQUIPMENT

(Sheet 1 of 2)

The minimum quantity, type of radiation sensitivity, typical range of instruments and equipment are shown below:

	Quantity (Approximate)	Type of Radiation Detected	Range (Typical)
SURVEY INSTRUMENTS:	10 per unit	Beta, Gamma	0-50 R/hr & 0-5R/hr
	15 per unit	Gamma	0-5 R/hr
	4 per unit	Beta, Gamma	0-200 R/hr
	1 per unit	Gamma	0-10,000 R/hr
	1 per unit	Neutron	0-5 Rem/hr
	1 per unit	Alpha	0-20,000 dpm
ELECTRONIC ALARMING DOSIMETERS:	200 per unit	Gamma	1 mR - 100R
POCKET ION CHAMBER DOSIMETERS:	150 per unit	Gamma	0-200 mR or 0-500 mR
(NOTE: Direct reading pocket ion chambers (PICs) in the ranges indicated may be used in lieu of electronic alarming dosimeters)	50 per unit	Gamma	0-1.5 R

TABLE 12.5-1
RADIATION MONITORING INSTRUMENTATION AND OTHER RADIATION PROTECTION EQUIPMENT

(Sheet 2 of 2)

	Quantity (Approximate)	Type of Radiation Detected	Use
OTHER AVAILABLE EQUIPMENT:			
Gamma Spectroscopy System	1	Gamma	Routine counting of liquids, gases and solids
Whole Body Counting System	2	Gamma	Used for in Vivo bioassay
Gas Proportional Counter	1	Beta, Gamma, Alpha	Used for counting smear and air samples
Liquid Scintillation Counter	1	Beta, Gamma	Routine tritium and gross beta analysis
Personnel Contamination/ Portal Monitor	2 per unit	Beta, Gamma	Personnel contamination monitoring at RCA exits. Personnel contamination monitors are also used to perform passive whole body counting (in-vivo bioassay).
Continuous Air Monitor (CAMs)	2 per unit	Beta	Work area air monitoring
Airborne Grab Sampler	15 per unit	Beta Gamma	Work area air monitoring
Count Rate Meter	25 per unit	Beta	Contamination frisking

TABLE 12.5-2
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13.0 CONDUCT OF OPERATIONS

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13.1 ORGANIZATIONAL STRUCTURE OF APPLICANTS

Vistra Operations Company LLC and Comanche Peak Power Company LLC are subsidiaries of Vistra Corp. A simplified organizational diagram is shown in Figure 13.1-1.

Vistra Operations Company LLC has corporate responsibility for the design, construction and operation of CPNPP, which includes the functions of procurement, fuel management and quality assurance. Within Vistra OpCo, the fossil group is responsible for the operation and related activities associated with the fossil fueled plants. The nuclear group, designated as the Nuclear Generation Group, has been delegated to furnish design, engineering, construction, licensing, operation and fuel management support to CPNPP. The Nuclear Generation Group has been split into the following organizations: Nuclear Engineering & Support, Nuclear Operations, Nuclear Oversight, Regulatory Affairs, and Strategic Support.

13.1.1 MANAGEMENT AND TECHNICAL SUPPORT ORGANIZATION

Project management and technical support for CPNPP are the responsibility of Vistra Operations Company LLC.

13.1.1.1 Design and Operating Responsibilities

The design and operating responsibilities are divided into three categories: design and construction activities, pre-operational activities and technical support for operation. The Nuclear Generation organizations are responsible for these activities as discussed below.

13.1.1.1.1 This section deleted in its entirety as it does not apply to an operating plant.

13.1.1.1.2 This section deleted in its entirety as it does not apply to an operating plant.

13.1.1.1.3 Technical Support for Nuclear Operations

Technical services and backup support for Nuclear Operations are furnished by Nuclear Engineering. Personnel are available who are competent in technical matters related to plant safety and other engineering and scientific support aspects. In the event Nuclear Operations needs assistance with specific problems, the services of qualified individuals will be engaged as appropriate.

The special capabilities that are available are:

- Nuclear, mechanical, structural, electrical, thermal-hydraulic, metallurgical and materials, chemistry, and instruments and controls engineering
- Nuclear Safety
- Plant Chemistry
- Health Physics
- Fueling and refueling operations support

- Maintenance support
- Licensing
- Industrial Safety
- Fire Protection

13.1.1.2 Organizational Arrangement

Responsible positions in the organization are described below. Certain executive and management positions may have deputies assigned to the position. Deputies may act with the full authority of the position to which assigned.

Chief Operations Officer, Vistra Operations Company LLC - The Chief Operations Officer, Vistra Operations Company LLC has the overall corporate responsibility for the design, construction and operation of CPNPP. He provides guidance to the Senior Vice President & Chief Nuclear Officer. The Nuclear Generation Group is shown in Figure 13.1-2.

Senior Vice President & Chief Nuclear Officer- The responsibilities of the Senior Vice President & Chief Nuclear Officer are:

- Directing the engineering, construction, start-up, testing, operation and maintenance of CPNPP.
- Providing the Quality Assurance Program and associated audit services applicable to all nuclear activities.
- Providing engineering services, technical services, nuclear fuel services, and licensing services to Nuclear Operations.
- Providing technical direction and guidance to the Nuclear Generation Vice Presidents and other direct reporting managers.
- Establishing and maintaining the Operations Review Committee.
- Providing corporate responsibility for overall plant nuclear safety and taking any measures needed to ensure acceptable performance of the staff in operating, maintaining, and providing technical support to the plant to ensure nuclear safety.
- Maintaining the independence and credibility of the Program.
- Ensuring adequate responses are provided to employees with concerns.

13.1.1.2.1 Organization - Nuclear Operations

The CPNPP operating organization is discussed in Section 13.1.2 and illustrated in Figure 13.1-3. Organization responsibilities which support plant operations as illustrated in Figure 13.1-2 are discussed below.

The Site Vice President is responsible for:

- Ensuring CPNPP operation, maintenance, emergency planning, radiation protection, nuclear training and plant support activities are conducted in compliance with federal, state, and local laws, regulations, licenses, codes, and within established corporate and Nuclear Generation policies, plans and procedures.

- Ensuring the safe and reliable operation of the plant and overall responsibility for security of the plant.

The Plant Manager is responsible for:

- Operational and maintenance support of CPNPP.

- Management of all operations activities at CPNPP.

The Director, Operations is responsible for:

- Operations of CPNPP.

- Management and training of Operations Department personnel.

- Coordinating the generation of power and changes in operating modes.

- Participating in power ascension test program and refueling efforts.

- Fire Protection.

The Director, Maintenance is responsible for:

- Maintenance activities associated with mechanical and electric equipment, instrumentation and controls.

- Maintenance and testing of fire protection equipment.

- Implementing the preventative maintenance activities during routine operation and refueling outages and activities associated with the power ascension test program are conducted in accordance with approved procedures and instructions, regulatory requirements and applicable policies and directives.

The Director, Work Management is responsible for:

- CPNPP work control program.

- Planning, maintenance and work order packages.

- Providing CPNPP an integrated schedule.

- Outage management.

The Manager Chemistry & Environmental is responsible for:

- Monitoring and maintaining the station's fluid systems chemistry.
- Compliance with Non-Radiological environmental requirements of local, state, and federal regulatory agencies.

The Director, Organizational Effectiveness is responsible for:

- Monitoring and improving personnel and plant performance to ensure that CPNPP is operated and maintained in accordance with the provisions of the operating license, Nuclear Regulatory Commission and other federal regulations including the state of Texas regulations and permits, industry codes and standards, and Vistra OpCo requirements.

The Manager Nuclear Emergency Preparedness is responsible for:

- Development of the Emergency Plan and procedures, maintenance of emergency response facilities and equipment, and training of the emergency response organization.
- Interfacing with local, state and federal officials to ensure integrated onsite and offsite plans.

The Manager, Nuclear Security is responsible for:

- Overall development and implementation of the security program at CPNPP as outlined in the Security plans.

The Manager, Corrective Action is responsible for:

- Corrective Action Program - Development and implementation for station problem identification and resolution process. Includes Conditions Adverse to Quality, which require reporting and resolution by 10CFR50 Appendix B or other requirements; also includes evaluation and resolution of other conditions.

The Radiation Protection & Safety Manager is responsible for:

- CPNPP Radiation Protection program.
- Supervision of Radiation Protection Manager and Supervisors.
- Transportation of radioactive material.
- Implementation of station policy of maintaining operational radiation exposures "as low as reasonable achievable".
- Industrial Safety.

The Director, Nuclear Training is responsible for:

- Directing the analysis, design, development, implementation, evaluation and revision of nuclear training programs in order to provide personnel with the requisite skills and knowledge for effectively performing functions important to the operation and maintenance of CPNPP.

13.1.1.2.2 Organization - Nuclear Engineering & Support and Regulatory Affairs

The organization is illustrated in Figure 13.1-2. Organization responsibilities which support plant operations are discussed below.

The Senior Director, Engineering & Regulatory Affairs is responsible for:

- Project Development, Plant Modification, Reactor Engineering, Design Engineering, Strategic Engineering, and Regulatory Affairs.

- Represent Vistra OpCo in committees and task forces established for the nuclear industry.

The Director, Site Engineering is responsible for:

- Implementing a system health program to manage system performance problems and recommend effective improvements; implementing the predictive maintenance programs; implementing the maintenance effectiveness monitoring program (Maintenance Rule Program); and performing assigned surveillance and preventive maintenance test procedures.

- Providing support to Nuclear Generation and to other Vistra OpCo organizations in the areas of Nuclear Fuel Management, Reload Design Engineering, Thermal Hydraulic Analysis, Integrated Risk and Availability Modeling, Nuclear Safety Analysis, Severe Accident Management, and Core Performance.

The Director, Engineering Projects is responsible for:

- Providing for the development and implementation of specific major scope plant modifications and/or engineering projects including interface with involved offsite vendor organizations.

- Providing for the development and implementation of an integrated administrative services program.

The Manager, Regulatory Affairs is responsible for:

- Providing liaison with governmental agencies and implementing the actions necessary to obtain and maintain the permits, licenses, and approvals needed to construct and operate CPNPP.

- Providing technical expertise to evaluate and resolve issues which affect permits and licenses.

- Assisting corporate management in the development and implementation of positions on licensing issues consistent with Nuclear Generation Group policies and procedures.

- Providing formal communication to regulatory agencies (e.g., NRC) with any control over CPNPP, except for that communication assigned to Environmental Services.

The Manager, Design Engineering is responsible for:

- Assuring the consistency of design documentation; providing Operations with timely design engineering services for analyses and technical evaluations; assuring that design activities conducted for Comanche Peak meet the requirements of the design control program; assuring that design outputs are consistent with the design basis of the plant; and providing engineering specialists.

The Manager, Strategic Engineering is responsible for:

- The overall engineering support for operation and maintenance of plant systems.

The Manager, Program and Tactical Engineering is responsible for:

- The overall development and coordination of the Fire Protection Program (duties are further described in Section 13.3B.1.4), Inservice Testing Program, Inservice Inspection Program, core performance, equipment reliability, and prompt engineering support of plant operational issues.

13.1.1.2.3 Organization - Nuclear Oversight

The Director, Nuclear Oversight is responsible for:

- Ensuring that activities at CPNPP are conducted in a manner that is conducive to quality results.

- Keeping the Senior Vice President & Chief Nuclear Officer apprised of the status of Nuclear Oversight activities.

- Recommending actions or methods to improve or correct conditions which are adverse to quality.

- Developing, implementing and managing the vendor control program which includes the review of vendor QA programs, review of procurement documents, evaluation of vendor activities, and source and/or receipt inspection.

- Directing site related Qualify Control activities for CPNPP Operations and Maintenance.

- Performing observations of plant activities to ensure the activities are performed safely, and to ensure plant safety. Make independent safety recommendations to the Senior Vice President & Chief Nuclear Officer on ways to improve the overall quality and safety of operation, maintenance, and engineering activities.

- Ensuring that Quality Assurance Program requirements are met by conducting evaluations (i.e., audits, surveillances, assessments, inspections...).

- Providing for independent verification of critical attributes associated with safety-related equipment or work activities.

- Providing, when necessary, independent review and concurrence for quality-related activities such as procurement, nonconformance and corrective action, stop-work orders, and other activities as designated in the Quality Assurance Program.

- Performing systematic monitoring and assessment of plant operation and maintenance activities, and provide senior management with critical assessments of plant practices and issues that may affect nuclear safety and the operation and maintenance of CPNPP.

- Providing engineering services to support the Operations Review Committee (ORC)

- Providing Health Physics review for those programs necessary for radiological protection of Company personnel and property, the environment and the general public.

- Supporting the development of appropriate emergency plans for Vistra OpCo in accordance with regulatory guidelines and requirements, and performance of the 10CFR50.54(t) independent review of the Emergency Planning Program.

The Employee Concerns Coordinator is responsible for:

- Managing the Employee Concerns Program for the review and investigation of employee safety concerns.

- Ensuring both exiting and continuing employees are provided an opportunity to have a confidential interview to express any concern regarding CPNPP.

13.1.1.3 Qualification of Management and Technical Support Personnel

Resumes of current key Nuclear Generation managerial and technical support personnel available upon request.

13.1.2 OPERATING ORGANIZATION

13.1.2.1 Nuclear Operations Organization

The Plant Manager is responsible for the operation and maintenance of the Comanche Peak Nuclear Power Plant (CPNPP). Reporting interfaces and relationships are shown on Figures 13.1-2 and 13.1-3. Nuclear Oversight is illustrated on Figure 13.1-2 and discussed in Section 13.1.1.2.3 (organization) and Chapter 17 (program).

13.1.2.1.1 Personnel Functions, Responsibilities, and Authorities

Personnel responsibilities relating to membership on the Station Operations Review Committee (SORC) and the Operations Review Committee (ORC) are discussed in Section 13.4.

13.1.2.1.2 Operations Department

The Director, Operations is responsible for the operation of CPNPP and the management and training of Operations Department personnel. He coordinates the generation of power and changes in plant operating modes and participates in the power ascension test program and the refueling efforts.

The Director, Operations provides technical assistance for the development and maintenance of Operations Department procedures to ensure CPNPP is operated as prescribed in Section 13.5. He is also responsible for the operation of the radioactive waste handling systems, for the processing and packaging of radioactive waste.

The Senior Manager Shift Operations is responsible for post-trip reviews, for refueling support and for reactor operator training support. The Senior Manager Shift Operations directs the Shift Managers and is responsible for ensuring that shift operations personnel are trained and qualified (see Section 13.2). The Senior Manager Shift Operations is the position designated to meet ANSI N18.1-1971 qualification requirements for “Operations Manager” and is required to maintain a USNRC Senior Reactor Operator License.

Shift Operations

The Shift Managers are members of management responsible to the Senior Manager Shift Operations for the operation of the CPNPP units. Each Shift Manager is responsible for supervising the evolutions conducted during his shift and ensuring that they are conducted in accordance with the operating license, station procedures, and applicable directives and policies. The Shift Managers are responsible for supervising shift operations personnel and for conducting on-shift training. During periods when senior management personnel are not on site, the Shift Managers assume responsibility for all station activities. The Shift Managers are required to maintain an USNRC Senior Reactor Operator License.

The Unit Supervisors are members of management and assist the Shift Managers in discharging their responsibilities for supervision of the CPNPP units. The Unit Supervisor may assume the duties of the Shift Manager in his absence. The Unit Supervisors are required to maintain an USNRC Senior Reactor Operator License.

The Reactor Operators are responsible for routine evolutions on their assigned unit and for monitoring the status of that unit. The Reactor Operators are supervised by the Shift Manager or a Unit Supervisor and are required to maintain a USNRC Reactor Operator License.

Plant Equipment Operators work under the direction of the Shift Manager, Unit Supervisor, or Radwaste Supervisor. The Plant Equipment Operators’ responsibilities include operating equipment from the Control Room and operating and servicing equipment remote from the Control Room at the direction of Control Room operations personnel.

Shift Technical Advisors will be on each shift unless the Shift Manager or the individual with a Senior Reactor Operator license meets the qualifications described in Option 1 of the Commission Policy Statement on Engineering Expertise (50 FR 43621, October 28, 1985). They will report to the Shift Managers.

A Radwaste Supervisor is responsible to the Manager, Shift Operations Support for the operation of the liquid and gaseous waste systems. He supervises the various liquid and gaseous waste processing activities and serves as an interface between the Operations Department and other CPNPP personnel involved in waste or water handling evolutions.

Chemistry

The Manager, Nuclear Chemistry & Environmental is responsible for the supervision of chemistry and environmental personnel, for monitoring and maintaining the station's fluid systems chemistry. In discharging these responsibilities, he ensures that his personnel are trained and that safety-related activities are conducted in accordance with applicable procedures, instructions, policies and regulations.

13.1.2.1.3 Maintenance Department

The Director, Maintenance is responsible for maintenance activities associated with mechanical and electrical equipment, instrumentation and controls, and for implementing the preventive maintenance program. The Director, Maintenance ensures that maintenance personnel are trained and qualified. He ensures that maintenance activities during routine operation and refueling outages and activities associated with the power ascension test program are conducted in accordance with approved procedures and instructions, regulatory requirements, and applicable policies and directives. The Director, Maintenance is responsible for developing and maintaining procedures and instructions as described in Section 13.5.

Maintenance Teams

The Maintenance Teams are divided into one multi-discipline team (Electrical and Mechanical) and one Instrumentation and Control team. These teams are system oriented established to provide ownership and accountability within the maintenance organization.

The Maintenance Team Managers are responsible to the Director, Maintenance for the maintenance of electrical and mechanical plant systems and their instrumentation and control systems. They ensure that the electricians, mechanics, and I&C technicians are trained and that safety-related activities are conducted in accordance with applicable procedures, instructions, policies and regulations. They are responsible for managing their respective areas/systems through the Maintenance Team Supervisors who direct the day-to-day activities of their personnel.

Maintenance Plant Support Team

The Manager, Maintenance Plant Support is responsible to the Director, Maintenance for maintaining plant valves, welding and machining activities, and oversight to the PROMPT Team.

PROMPT Team

The PROMPT Team is a 24 hour/365 day multi-discipline maintenance team reporting to the Maintenance Plant Support Manager. Specific duties and responsibilities include, but are not limited to, ensuring PROMPT Team activities are performed in accordance with the applicable site procedures. The PROMPT Team is also responsible for providing immediate response to plant emergent maintenance items.

Maintenance Support Team

The Manager, Maintenance Support Group is responsible to the Director, Maintenance for providing technical, administrative and field support for the Maintenance Department.

13.1.2.1.4 Radiation Protection and Safety Services

Radiation Protection

The Radiation Protection & Safety Manager is responsible for the supervision of the Radiation Protection Supervisors, for the transportation of radioactive material, for the CPNPP Radiation Protection program (see Section 12.5), for implementation of the station policy of maintaining operational radiation exposures "as low as reasonably achievable," and industrial safety. In discharging these responsibilities, he ensures that his personnel are trained and that safety-related activities are conducted in accordance with applicable procedures, instructions, policies and regulations.

13.1.2.1.5 Plant Support

The Director, Work Management is responsible for planning and scheduling and outage management.

Planning and Scheduling

The Director, Work Management is responsible for:

- maintaining the work control process, scheduling on-line work and tests, and administrating the risk assessment process.
- preparation and execution of planned outages, scheduling outage activities, ensuring the implementation of the risk assessment process on outage activities, and incorporation of outage lessons learned.

Station Security

The Manager, Nuclear Security is responsible for the overall development and implementation of the security program at CPNPP as outlined in the Security plans.

Emergency Planning

The Manager, Nuclear Emergency Planning is responsible for the development of the Emergency Plan and procedures, maintenance of emergency response facilities and equipment, and training of the emergency response organization. The Manager, Nuclear Emergency Planning is also responsible for interfacing with local, state and federal officials to ensure integrated onsite and offsite plans.

13.1.2.2 Supervisory Succession

The Plant Manager is responsible for the operation of CPNPP. If the Plant Manager is absent or becomes incapacitated, then, unless otherwise designated, the following station staff assume the subject responsibilities in the order listed:

1. Director, Operations

2. Director, Maintenance

During back shift and weekend periods when the station staff is not on site, the Shift Manager is responsible for all activities at CPNPP.

13.1.2.3 Shift Crew Composition

The minimum on-duty shift complement for various modes of single and dual unit operation is shown in Table 13.1-2 and is as follows:

Two USNRC Licensed Operators should be in the Control Room for each reactor while undergoing a start-up, scheduled shutdown or reactor trip recovery.

With two units licensed to operate (both Units in Mode 5 or 6), each shift crew shall have at least six members, including one Shift Manager and two USNRC Licensed Operators.

With two units licensed to operate and one or both operating (Mode 1, 2, 3, 4), each crew shall have at least eight members, including one Shift Manager, one Unit Supervisor and three USNRC Licensed Operators.

An organization chart for CPNPP for one and two unit operation is provided in Figure 13.1-3.

13.1.3 QUALIFICATION REQUIREMENTS FOR PLANT PERSONNEL

13.1.3.1 Minimum Qualification Requirements

The minimum qualification requirements for Licensed Senior Reactor Operators and Reactor Operators are in accordance with Regulatory Guide 1.8 Rev. 3 and are displayed in Table 13.1-1. All other plant personnel will meet or exceed the minimum qualification requirements of Regulatory Guide 1.8, Rev. 2.

13.1.3.2 Qualifications of Plant Personnel

Resumes of current key operating organization personnel are available upon request.

TABLE 13.1-1
MINIMUM QUALIFICATIONS OF PLANT PERSONNEL IN ACCORDANCE WITH REGULATORY GUIDE 1.8, REVISION 2

(Sheet 1 of 4)

Notes: (1) Recommended, but not required (2) Recommended in addition to experience (3) Experience required to independently perform job (4) In accordance with Regulatory Guide 1.8, Rev. 3, May 2000	H. S. DIPLOMA	B.S. in ENGINEERING OR SCIENCE	POWER PLANT EXPERIENCE	NUCLEAR PLANT EXPERIENCE	EXPERIENCE IN AREA OF EXPERTISE	EQUIVALENT EDUCATIONAL EXPERIENCE	REACTOR OPERATOR LICENSE REQUIRED	SENIOR REACTOR OPERATOR LIC/CERT REQUIRED
SITE VICE PRESIDENT OR PLANT MANAGER		X	10	3		4		X(1)
DIRECTOR, OPERATIONS		X(1)	8	3		2		X(1)
DIRECTOR, MAINTENANCE		X(1)	7	1		2		
MAINTENANCE DEPARTMENT MANAGERS		X(1)	5	1		4		
SENIOR MANAGER, SHIFT OPERATIONS		X(1)	8	3		2		X
SHIFT MANAGER	X		3(4)	3(4)		2(2)		X
SENIOR REACTOR OPERATOR	X		3(4)	3(4)		2(2)		X

TABLE 13.1-1
MINIMUM QUALIFICATIONS OF PLANT PERSONNEL IN ACCORDANCE WITH REGULATORY GUIDE 1.8, REVISION 2

(Sheet 2 of 4)

Notes: (1) Recommended, but not required (2) Recommended in addition to experience (3) Experience required to independently perform job (4) In accordance with Regulatory Guide 1.8, Rev. 3, May 2000	H. S. DIPLOMA	B.S. in ENGINEERING OR SCIENCE	POWER PLANT EXPERIENCE	NUCLEAR PLANT EXPERIENCE	EXPERIENCE IN AREA OF EXPERTISE	EQUIVALENT EDUCATIONAL EXPERIENCE	REACTOR OPERATOR LICENSE REQUIRED	SENIOR REACTOR OPERATOR LIC/CERT REQUIRED
SHIFT TECHNICAL ADVISOR		X		1				
REACTOR OPERATOR	X		2(4)	1			X	
NUCLEAR EQUIPMENT OPERATOR	X							
MANAGER, CORE PERFORMANCE ENGINEERING		X		2(3)				
MECHANIC					3(3)			
ELECTRICIAN					3(3)			
INSTRUMENT & CONTROL TECHNICIAN					2(3)	1(2)		

TABLE 13.1-1
MINIMUM QUALIFICATIONS OF PLANT PERSONNEL IN ACCORDANCE WITH REGULATORY GUIDE 1.8, REVISION 2

(Sheet 3 of 4)

Notes: (1) Recommended, but not required (2) Recommended in addition to experience (3) Experience required to independently perform job (4) In accordance with Regulatory Guide 1.8, Rev. 3, May 2000	H. S. DIPLOMA	B.S. in ENGINEERING OR SCIENCE	POWER PLANT EXPERIENCE	NUCLEAR PLANT EXPERIENCE	EXPERIENCE IN AREA OF EXPERTISE	EQUIVALENT EDUCATIONAL EXPERIENCE	REACTOR OPERATOR LICENSE REQUIRED	SENIOR REACTOR OPERATOR LIC/CERT REQUIRED
DIRECTOR, WORK MANAGEMENT		X(1)	5	1		2		
DIRECTOR, SITE ENGINEERING		X	8	1		4		
MANAGER, DESIGN ENGINEERING		X	8	1		4		
DIRECTOR, PROJECT ENGINEERING		X	8	1		4		
RADIATION PROTECTION AND SAFETY MANAGER		X(1)		3	4	4		
MANAGER, CHEMISTRY AND ENVIRONMENTAL		X(1)		1	5	4		
STAFF HEALTH PHYSICIST		X(1)						

TABLE 13.1-1
MINIMUM QUALIFICATIONS OF PLANT PERSONNEL IN ACCORDANCE WITH REGULATORY GUIDE 1.8, REVISION 2

(Sheet 4 of 4)

Notes: (1) Recommended, but not required (2) Recommended in addition to experience (3) Experience required to independently perform job (4) In accordance with Regulatory Guide 1.8, Rev. 3, May 2000	H. S. DIPLOMA	B.S. in ENGINEERING OR SCIENCE	POWER PLANT EXPERIENCE	NUCLEAR PLANT EXPERIENCE	EXPERIENCE IN AREA OF EXPERTISE	EQUIVALENT EDUCATIONAL EXPERIENCE	REACTOR OPERATOR LICENSE REQUIRED	SENIOR REACTOR OPERATOR LIC/CERT REQUIRED
STAFF CHEMIST		X(1)						
RADIATION PROTECTION TECHNICIAN					2(3)	1(2)		
CHEMISTRY TECHNICIAN					2(3)	1(2)		
MANAGER, REGULATORY AFFAIRS	X	X(1)	5	4	6(3)	4		
DIRECTOR, NUCLEAR OVERSIGHT	X			2	6(3)	4		

TABLE 13.1-2
MINIMUM SHIFT CREW COMPOSITION[a]

(Dual Units with common Control Room)

POSITION[c]	MINIMUM CREW COMPOSITION[b]		
	BOTH UNITS in MODES 1,2,3, or 4	ONE UNIT in MODE 1,2,3, or 4; and ONE UNIT in MODE 5,6, or DEFUELED	BOTH UNITS in MODE 5,6, or DEFUELED
SM	1	1	1
SRO	1	1	NONE
STA	1	1	1
RO	3	3	2
PEO	4	4	4
RP Tech	2	2	2
Chem Tech	1	1	1

a) Additional minimum on-shift staffing requirements are contained in the CPNPP Emergency Plan.

b) As allowed by note 1 in 10 CFR 50.54(m)(2)(i), temporary deviations from the numbers of licensed operators required by this table shall be in accordance with the following criteria: The minimum Operations shift crew composition may be one less than shown in Table 13.1-2 for not more than (2) hours to accommodate unexpected absences of on-duty crew members, provided immediate action is taken to restore the crew composition within the minimum shown in Table 13.1-2. This exception does not permit any crew composition to be unmanned upon shift turnover due to an oncoming crew member being late or absent.

c) SM - (Shift Manager) with a dual unit Senior Reactor Operator license assigned to both units when either unit contains fuel. During the absence of the SM from the control room, any currently licensed SRO shall be designated to assume the control room command function.

SRO - Individual with a dual unit Senior Reactor Operator license assigned to each unit containing fuel. During core alterations on either unit, at least one currently licensed SRO (or SRO limited to fuel handling) will be present and responsible for fuel handling activities with no other concurrent duties assigned.

RO - Individual with a dual unit Reactor Operator or Senior Reactor Operator license assigned to each unit containing fuel. At least (1) RO will be assigned as a relief operator when either unit is in MODE 1, 2, 3, or 4.

PEO - A plant equipment operator shall be assigned to each unit containing fuel.

STA - (Shift Technical Advisor) to be assigned to both units in all MODES or when DEFUELED. The STA position may be filled by an on-shift SRO provided the individual meets the dual role requirements described in the Commission Policy Statement on Engineering Expertise on Shift (50 FR 43621) and has dose assessment capability. The SM shall not fulfill the duties as Emergency Coordinator and dose assessor concurrently.

RP/Chem Tech -

At least (2) Radiation Protection Technicians and (1) Chemistry Technician shall be on-site to assume emergency response functions. The Radiation Protection and the Chemistry Technicians may be less than the minimum requirements for a period of 2 hours in order to accommodate unexpected absence, provided immediate action is taken to fill the required position.

13.2 TRAINING

13.2.1 NUCLEAR TRAINING PROGRAM

CPNPP has a training program which is systems approach to training (SAT) based in accordance with the requirements of 10 CFR parts 50 and 55. CPNPP will comply with the SAT-based training requirements by maintaining accreditation as conferred by the National Nuclear Accrediting Board on 10/25/90. Program descriptions are contained in sections 13.2.1.1 and 13.2.1.2.

13.2.1.1 Training Programs for Licensed Personnel

1. Replacement License Training

   Replacement license training will provide SAT-based training to reactor operator (RO) and senior reactor operator (SRO) candidates in accordance with 10 CFR 55. Replacement license training will comply with the SAT-based training requirements by maintaining accreditation.

2. License Operator Requalification Training

   License operator requalification training will provide SAT-based requalification training to ROs and SROs in accordance with 10 CFR 55. License operator requalification training will comply with the SAT-based training requirements by maintaining accreditation.

3. Simulator Training

   ROs and SROs may perform control manipulations on the simulator required as part of the training in section 13.2.1.1.1 and 13.2.1.1.2. Training will maintain a certified simulator in accordance with the provisions of 10 CFR 55.

4. Training of Licensed Supervisors

   The shift supervisor training program will comply with the SAT-based training requirements, and consists of training in those administrative duty areas above and beyond licensed duties.

5. Training for Shift Technical Advisors

   Shift Technical Advisors not meeting the dual-role SRO/STA requirements as described in Section 13.1.2.1.2, will be trained and qualified in accordance with Option 2 of the Commission Policy Statement on Engineering Expertise (50 FR 43621, October 28, 1985). Shift technical advisor training will comply with the SAT-based training requirements.

13.2.1.2 Training Programs for Nonlicensed Personnel

1. Training for Mechanical Maintenance Personnel

   Mechanical maintenance training will provide SAT-based training to mechanical maintenance personnel in accordance with 10 CFR 50.120. Mechanical maintenance training will comply with the SAT-based training requirements by maintaining accreditation.

2. Training for Electrical Maintenance Personnel

   Electrical maintenance training will provide SAT-based training to electrical maintenance personnel in accordance with 10 CFR 50.120. Electrical maintenance training will comply with the SAT-based training requirements by maintaining accreditation.

3. Training for Instrument and Control Personnel

   Instrument and control training will provide SAT-based training to instrument and control personnel in accordance with 10 CFR 50.120. Instrument and control training will comply with the SAT-based training requirements by maintaining accreditation.

4. Training for Radiation Protection Personnel

   Radiation protection training will provide SAT-based training to radiation protection personnel in accordance with 10 CFR 50.120. Radiation protection training will comply with the SAT-based training requirements by maintaining accreditation.

5. Training for Chemistry Personnel

   Chemistry training will provide SAT-based training to chemistry personnel in accordance with 10 CFR 50.120. Chemistry training will comply with the SAT-based training requirements by maintaining accreditation.

6. Training for Non-licensed Operators

   Non-licensed operator training (plant equipment operators) will provide SAT-based training to non-licensed operators in accordance with 10 CFR 50.120. Non-licensed operator training will comply with the SAT-based training requirements by maintaining accreditation.

7. Training for Engineering Support Personnel

   Engineering support training will provide SAT-based training to engineering support staff in accordance with 10 CFR 50.120. Engineering support training will comply with the SAT-based training requirements by maintaining accreditation.

8. Training on the Emergency Diesel Generator

   Operations and maintenance personnel responsible for emergency diesel generator performance will receive training on the EDG commensurate with the level of expertise required.

13.2.1.3 General Employee Training

1. All employees (and others) who require unescorted access to the Protected Area of the station will receive training in the following areas:

   a. General description of plant and facilities

   b. Emergency Plan

   c. Fire Protection Program

   d. Security Requirements and Practices

   e. Safety Program

   f. Quality Assurance Program

   g. Radiological Protection Program

2. All employees (and others) who have unescorted access to Radiation Controlled Areas of the station will receive in-depth instruction in all aspects of radiation protection and, as required, respiratory protection. Subject material will include but will not be limited to the following:

   a. Handling radioactive material

   b. Controls and access requirements

   c. Biological effects of ionizing radiation

3. Some employees receive, as part of their specialty training, a General Plant Information Course. This lecture course consists of reactor theory, primary plant systems, secondary systems, and instrumentation and control.

4. Review of appropriate department and station procedures are specified in department specialty training programs.

13.2.1.4 Fire Protection Training

The CPNPP Fire Protection Training Program is described in section 13.3B.

13.2.2 REPLACEMENT AND RETRAINING

13.2.2.1 Licensed Operators - Requalification Training

NOTE: All replacement and retraining requirements are embedded within the accredited programs as described in sections 13.2.1.1 and 13.2.1.2.

13.2.3 APPLICABLE NRC DOCUMENTS

The NRC Regulations, Regulatory Guides and Reports listed below were used to provide guidance in the area of training for staff personnel. Compliance to these items is indicated below.

1. 10 CFR Part 50, “Licensing of Production and Utilization Facilities.” Full compliance in the area of training.

2. 10 CFR Part 55, “Operators’ Licenses.” Full compliance in the area of training.

3. 10 CFR Part 19, “Notices, Instructions, and Reports to Workers; Inspection.” Full compliance in the area of training.

4. Regulatory Guide 1.8, “Personnel Selection and Training.” Compliance in this area is discussed in Section 12.5.1, “Organization,” and 13.1.3.1, “Minimum Qualification Requirements, and 13.2, “Training”.”

5. Regulatory Guide 8.2, “Guide for Administrative Practices in Radiation Monitoring.” Full compliance in the area of training.

6. Regulatory Guide 8.8, “Information Relevant to Maintaining Occupational Radiation Exposure As Low As Reasonably Achievable (Nuclear Power Reactors).” Full compliance in the area of training except as noted in Item #4 above.

7. Regulatory Guide 8.10, “Operating Philosophy for Maintaining Occupational Radiation Exposures as Low As Is Reasonable Achievable.” Full compliance in the area of training.

8. Regulatory Guide 8.13, “Instruction Concerning Prenatal Radiation Exposure”. Full compliance in the area of training.

9. Generic Letter 87-07, “Information Transmittal of Final Rulemaking for Revisions to Operator Licensing - 10 CFR 55 and Conforming Amendments”. Used as guidance only.

10. NUREG-0737,” Clarification of TMI Action Plan Requirements”, Published November, 1980.

11. 50 FR 43621, “NRC Policy Statement on Engineering Expertise on Shift”. Full compliance with Option 2.

13.3 EMERGENCY PLANNING

Emergency planning for Comanche Peak Nulclear Power Plant is directed toward mitigating the consequences of emergencies and provides reasonable assurance that appropriate measures can and will be taken to protect health and safety and prevent damage to property in the event of an emergency. It is done with the paramount objective to protect the health and safety of the general public, persons temporarily visiting or assigned to the station, and station employees.

The CPNPP Emergency Plan is described in Section 13.3A below.

The CPNPP Fire Protection Program is described in Section 13.3B below.

13.3A EMERGENCY PLAN

The CPNPP Emergency Plan has been upgraded in order to assure that adequate protective measures can and will be taken in the event of a radiological emergency. The guidance of NUREG-0654, “Criteria for Preparation and Evaluation of Radiological Emergency Response Plans and Preparedness in Support of Nuclear Power Plants,” was used to extensively revise the Emergency Plan which is provided in a separate binder titled “Comanche Peak Nulclear Power Plant Emergency Plan.”

13.3B CPNPP FIRE PROTECTION PROGRAM

The senior management position responsible for this program is the Site Vice President. The Site Vice President has delegated to the Operations Review Committee the responsibility to assess the effectiveness of the fire protection program which is accomplished through periodic audits as discussed in FSAR Section 17.2.18. Recommendations and the findings from these audits are reported to the Site Vice President.

The CPNPP Fire Protection Program meets the requirements of the D. B. Vassallo letter of August 1977 entitled “Nuclear Plant Fire Protection Functional Responsibilities, Administrative Controls and Quality Assurance,” as described in the following sections.

The CPNPP Fire Protection Program is established to ensure that a fire will not prevent safe plant shutdown and will not endanger the health and safety of the public. Fire protection at CPNPP is accomplished by using a defense-in-depth approach to include fire detection and extinguishing systems and equipment, (See Section 9.5.1) administrative controls and procedures, and trained personnel. Procedures for implementing the CPNPP Fire Protection Program will be developed prior to Start-Up. CPNPP is committed to meeting the requirements of the Fire Protection Report.

13.3B.1 ORGANIZATION

13.3B.1.1 Plant Manager (See Section 13.1.1.2.1)

The Plant Manager has the responsibility for maintenance, and implementation of the Fire Protection Program for operations at CPNPP. He has delegated this responsibility to the Director, Operations.

13.3B.1.2 Director, Maintenance

The Director, Maintenance is responsible to the Plant Manager (see Section 13.1.1.2.1) for the maintenance and testing of the Fire Protection equipment. The Director, Maintenance has assigned to the Maintenance Team Manager the responsibility of assuring the overall maintenance and testing of the Fire Protection equipment.

13.3B.1.2.1 Director, Operations

The Director Operations is responsible for the implementation of the Fire Protection Program. The Director, Operations has assigned to the Manager Fire / NEIL Program the responsibility of assuring the overall implementation of the Fire Protection Program (see Figure 13.1-3).

13.3B.1.3 Director, Site Engineering

The Director, Site Engineering has the responsibility for the development of the Fire Protection Program at CPNPP. The Director, Site Engineering has assigned to the Manager, Program & Tactical Engineering the overall responsibility for the development of the CPNPP Fire Protection Program.

13.3B.1.4 Manager, Program & Tactical Engineering

The Manager, Program & Tactical Engineering is responsible to the Director, Site Engineering for the overall development and coordination of the CPNPP Fire Protection Program. Duties and responsibilities include:

1. The continued maintenance of the Fire Protection Licensing Documents and evaluation of regulatory requirements for impact.

2. Review and approve Fire Protection Design documents, licensing documents, and Station Procedures including revisions, to verify technical accuracy and regulatory compliance.

3. The continued implementation of the Fire Protection Engineering requirements such that modifications and changes do not impact the ability to safely shutdown the plant in the event of a fire.

The Manager, Strategic Engineering is responsible to the Director, Site Engineering for technical support of the Fire Protection Program in relation to the review and implementation of design modifications and the resolution of system issues.

13.3B.1.5 Manager, Fire / NEIL Program

The Manager, Fire / NEIL Program is responsible to the Director, Operations for the overall implementation of the Fire Protection Program. Duties and responsibilities include:

1. Assures the implementation of periodic inspections to minimize the amount of combustibles in safety-related areas; determine the effectiveness of housekeeping practices; assure the availability and acceptable condition of all fire protection systems/

equipment, and assures that prompt actions are taken to correct conditions adverse to fire protection and preclude their recurrence.

2. Ensures that periodic testing and maintenance of fire protection systems and equipment is being performed and evaluated for availability and acceptability.

3. The development of the Fire Protection Program administrative procedures and the fire protection systems and equipment testing and maintenance requirements.

4. Assures the implementation of the administrative procedures of the Fire Protection Program such that the ability to safely shutdown the plant in the event of a fire is not compromised due to hot work, systems or equipment being impaired, compensatory measures or the control of transient materials and/or flammable/combustible liquids and gases.

5. To assist the Nuclear Training Manager in the development and implementation of fire protection training programs for personnel and the fire brigade at CPNPP.

13.3B.1.6 Director, Nuclear Training

The Director, Nuclear Training reports to the Site Vice President (see Section 13.1.1.2.1) and assists in the development and implementation of fire protection training programs for operating personnel and the fire brigade at CPNPP as requested. The Director, Nuclear Training documents and maintains records of the fire protection training of operations personnel and fire brigade.

13.3B.2 CPNPP FIRE BRIGADE

The Comanche Peak Nulclear Power Plant will have an organized fire brigade to deal with fires and related emergencies when they occur. The size and nature of the CPNPP fire brigade will be based upon an evaluation of the potential magnitude of a fire emergency at CPNPP and the availability of fire fighting assistance from offsite departments. However, CPNPP will be self-sufficient with respect to fire fighting activities to protect critical areas within the plant.

The CPNPP fire brigade organization will be based upon a team concept. The brigade will be organized into fire teams consisting of five (5) men each. This fire team size is consistent with the equipment that must be put into service during a fire emergency (2 1/2" hose station, 1 1/2" hose station, wheeled and hand held portable extinguishers). Each fire team will have a designated fire team leader to direct the action of the fire team. The fire team leader will also maintain close communication with the Shift Manager, keeping him apprised of the situation at the fire site. Two (2) men will perform the primary fire fighting function, i.e. operate the fire suppression equipment. The largest equipment to be used by these two men is the 2 1/2 inch fire hose located on the Turbine Building Operating Deck. The remaining two (2) men will be utilized for:

1. operating additional fire suppression equipment,

2. operating portable smoke ejectors, or

3. resupplying fresh bottles for the breathing apparatus.

A sufficient number of operations personnel will receive fire brigade training and will qualify to be members of the CPNPP fire brigade. A five (5) man fire team consisting of the fire team leader and four (4) additional personnel will be on duty each working shift and at periods when the plant is shutdown. The Fire Brigade shall not include the Shift Manager and the two other members of the minimum shift crew necessary for safe shutdown of the unit and any personnel required for other essential functions during a fire emergency. Other plant personnel will also receive fire brigade training and likewise be qualified members of the CPNPP fire brigade.

The Fire Brigade may be one less than the minimum requirements for a period of time not to exceed 2 hours in order to accommodate unexpected absence of on-duty crew members provided immediate action is taken to restore the brigade composition to within the minimum requirements. This provision does not permit the brigade to be unmanned below the minimum upon shift change due to an oncoming member being late or absent.

The on-duty Shift Manager will assume overall responsibility in the event of a fire emergency. He will be responsible for the following actions based upon his assessment of the magnitude of the fire emergency from reports received from the fire team leader:

1. Safe shutdown of the plant if required.

2. Implementation of the Emergency Plan.

3. Notification of management.

4. Requesting assistance from off duty fire teams if deemed necessary.

5. Contacting local fire departments if required.

If the decision is made to implement the emergency plan as a result of a fire emergency, the Shift Manager is designated the Emergency Coordinator until relieved by a designated alternate. However, the Shift Manager will continue to receive reports from the team leader at the fire site while coordinating other emergency activities with the Emergency Coordinator.

To qualify as a member of the CPNPP fire brigade, individuals must be available to answer alarms and to attend required training sessions.

13.3B.2.1 Fire Brigade Training

A training program will be established to assure that the capability to fight potential fires is developed and documented. The program will consist of a classroom instruction program supplemented with periodic classroom retraining and practice in fire fighting and fire drills. Classroom instruction and training will be conducted by qualified individuals knowledgeable of fighting the types of fires that could occur in the plant and in using available fire fighting equipment.

13.3B.2.1.1 Classroom Instruction

Fire brigade members will receive classroom instruction in fire protection and fire fighting techniques at planned meetings. Instruction will include:

1. Identification of fire hazards and associated types of fires that could occur in the plant and an identification of the location of such hazards.

2. Identification of the location of fire fighting equipment for each fire area, and familiarization with layout of the plant including ingress and egress routes to each area.

3. The proper use of available fire fighting equipment and the correct method of fighting each type of fire, to include electrical fires, cable and cable tray fires, hydrogen fires, flammable liquids, waste/debris fires, fires involving radioactive materials, and record file fires.

4. Review of the CPNPP fire fighting plan with coverage of each individual's responsibilities.

5. The proper use of communication, lighting, ventilation and emergency breathing equipment.

6. The direction and coordination of the fire fighting activities.

7. The toxic and radiological characteristics of expected products of combustions.

8. The proper method of fighting fires inside buildings and tunnels.

9. Review of fire fighting procedures and procedure changes.

10. Review of fire protection-related plant modifications and changes in fire fighting plans.

Instruction will be provided for all employees once a year and repeated on an annual basis. The instruction will include, as appropriate, the fire protection plan, evacuation routes and procedure for reporting a fire.

Instruction will be provided for security personnel including entry procedures for outside fire departments and crowd control for people exiting the station.

Instruction will be provided for construction personnel and temporary employees to include alarm responses, evacuation routes and procedure for reporting fires.

Off-site fire organizations will receive training on basic radiation principles and practices and typical radiation hazards that may be encountered when fighting fires.

13.3B.2.1.2 Practice

Practice sessions will be held on an annual basis for fire brigade members and will include the proper method of fighting various types of fires similar to those which might occur in a nuclear power plant. These sessions will provide brigade members with experience in actual fire extinguishment and the use of emergency breathing apparatus.

Emphasis will be placed on training brigade members to fight fires in safety-related areas.

13.3B.2.1.3 Drills

Fire brigade drills will be conducted on a quarterly (every 3 months) basis at CPNPP. Drills will be of two types: announced and unannounced. Drills will allow fire team individuals to practice together as a team. A sufficient number of drills will be conducted in a period of three months so that all fire teams will participate in at least one drill. Each individual member of a fire team will participate in at least two drills per year. Offsite fire organizations will be included in a fire brigade drill at least annually. Training objectives will be established prior to the drill. Afterwards, to determine how well the training objectives have been met, the drill will be critiqued to include:

1. Assessment of fire alarm effectiveness, time required to notify and assemble fire brigade, and selection, placement and use of equipment.

2. Assessment of brigade leader's effectiveness in directing the fire fighting effort.

3. Assess brigade member's knowledge of fire fighting strategy, procedures, and use of equipment in the area assumed to contain the fire.

Employees who receive plant access training are instructed in fire response including evacuation. Evacuation of employees (site evacuation / accountability) is also exercised as an element of the CPNPP Emergency Plan.

13.3B.2.1.4 Records

Records of training provided for each fire brigade member will be maintained to assure that each member of the fire brigade receives training in all parts of the program.

13.3B.3 FIRE FIGHTING PROCEDURES

Fire fighting procedures will be established to cover such items as notification of a fire, fire emergency procedures, and coordination of fire fighting activities with local fire departments. The procedures will identify:

1. Actions to be taken by the individual discovering the fire, such as notification of Control Room, attempt to extinguish fire, and activation of local fire suppression systems.

2. Actions to be taken by Control Room personnel, such as sounding fire alarms and notifying the Shift Manager/fire brigade leader of type, size and location of fire.

3. Actions to be taken by fire brigade after notification of a fire, including location to assemble, directions given by fire brigade leader, and responsibilities of brigade members such as selection of fire fighting and protective equipment and use of pre-planned strategies for fighting fires in specific areas.

4. Actions to be taken by plant management and Security after notification of fire.

5. Actions to be taken that will coordinate fire fighting activities with offsite fire departments including identification of person responsible for assessing situation and calling in local fire department's assistance if deemed necessary.

6. The strategies established for fighting fires in safety-related areas and areas presenting a hazard to safety-related equipment; strategies such as identification of combustibles in each plant zone covered by a fire fighting procedure type fire extinguishers best suited for controlling the fires with the combustible loadings of the zone, and instructions for plant operators and general plant personnel during fire.

13.3B.4 ADMINISTRATIVE PROCEDURES AND CONTROLS

Administrative procedures and controls will be established to ensure the reliable performance of fire protection personnel, systems and equipment. Effective measures will be established to control the use and storage of combustibles and ignition sources.

13.3B.4.1 Control of Combustibles

Effective administrative controls will be established to minimize the amount of combustibles that safety-related areas may be exposed to during operation or maintenance periods. These controls will govern the following:

1. Proper storage and handling of flammable gases and liquids, HEPA and charcoal filters, dry, unused ion exchange resins or other combustible supplies in safety-related areas.

2. Transient fire loads during maintenance and modifications such as combustibles and flammable liquids, wood and plastic materials in buildings containing safety-related systems or equipment. This control will require an in-plant review of work activities to identify transient fire loads. The supervisor or foreman responsible for reviewing the work activity will specify any required additional fire protection.

3. Waste, debris, scrap, and oil spills resulting from a work activity in a safety-related area will be minimized while work is in progress and removed upon completion of the activity.

4. Periodic inspection for accumulation of combustibles.

13.3B.4.2 Control of Ignition Sources

Effective administrative controls will be established to protect safety-related equipment from fire damage or loss resulting from work involving ignition sources, such as welding, cutting, grinding or open flame work. Administrative controls will prohibit the use of open flame or combustion smoke for leak testing and other ignition sources in certain areas.

Administrative controls will be established to ensure that the following precautions are taken:

1. Welding, cutting, grinding or open flame work will be authorized by the responsible foreman or supervisor through a work permit. The responsible foreman or supervisor will have received sufficient fire fighting and fire prevention training covering anticipated fires, such as electrical fires, fires in cables and cable trays, hydrogen fires, hydrocarbon fires, solvent fires, waste/debris fires, and record file fires.

2. Before work is performed on or near safety-related equipment, combustible materials located in the vicinity of cutting, welding or open flame work will be removed or protected

by curtains, metal guards, or flameproof curtains. Fire extinguishers or other fire fighting equipment will be provided in the vicinity of the work location, if needed.

3. Smoking will not be allowed in safety-related areas. Smoking is only allowed in outside areas as outlined by the Company Smoking Policy.

13.3B.5 QUALITY ASSURANCE PROGRAM

The CPNPP QA Program establishes the quality assurance requirements and controls to be implemented throughout the testing and operation phases of CPNPP. This program defines responsibility and authority and prescribes measures for the control and accomplishment of activities affecting the quality and operation of safety-related structures, systems and components. The structures, systems, and components covered by the Operations Quality Assurance Program are listed in Appendix 17A. These include both nuclear safety-related items subject to 10CFR50, Appendix B QA and non safety-related items which are not subject to Appendix B (e.g. Fire Protection). The provisions of the plan apply to all activities such as operating, maintaining, repairing and modifying which could affect the safety-related functions of safety-related systems and components. The preparation, review and approval of Fire Protection Program procedures will be handled in the same manner as safety-related procedures.

The quality assurance to be applied to the CPNPP Operation's Fire Protection Program is as follows:

1. Design Control and Procurement Document Control.

   Measures are established to assure that all design-related guidelines of the Branch Technical Position are included in design and procurement documents and that any deviations therefrom are controlled. These procedures include provisions, as necessary, to ensure that:

   a. Design and procurement document changes are subject to the same controls that were applicable to the original design.

   b. Design documents specify quality requirements or reference quality standards, as necessary, such as appropriate fire protection codes.

   c. Design and procurement documents are reviewed by qualified personnel to assure inclusion of appropriate fire protection requirements.

2. Instructions, Procedures and Drawings.

   Inspections, tests, administrative controls, fire drills and training, that govern the fire protection program is prescribed by documented instructions, procedures or drawings and are accomplished in accordance with the applicable document.

3. Control of Purchased Material, Equipment and Services.

   Measures will be established to assure that purchased material, equipment and services conform to the procurement documents.

4. Inspection.

   A program for independent inspection of activities affecting fire protection is established and executed by, or for, the organization performing the activity to verify conformance with documented installation drawings and test procedures for accomplishing the activity.

5. Test and Test Control.

   A test program is established to implement the operational surveillance testing required by the CPNPP technical specifications on fire protection. These tests are performed by written and approved procedures and will be subject to the same controls required for surveillance tests for safety related equipment. Test results will be properly evaluated and acted on.

6. Inspection, Test, and Operating Status.

   Measures will be established to provide for the identification of items that have satisfactorily passed required tests and inspections.

7. Nonconforming Items.

   Measures are established to control items that do not conform to specified requirements to prevent inadvertent use or installation.

8. Corrective Action.

   Measures are established to ensure that conditions adverse to fire protection such as failures, malfunctions, deficiencies or deviations, defective components, uncontrolled combustible material and non-conformances are promptly identified, reported, and corrected.

9. Records.

   Records will be prepared and maintained to assure conformance to fire protection requirements. These records will include results of inspections, tests, reviews, and evaluations; nonconformance and corrective action reports; and maintenance and modification records. Record retention requirements will be established.

10. Audits.

    Audits will be conducted and documented to verify compliance with the fire protection program, including design and procurement documents; instructions; procedures and drawings; and inspection and test activities. These audits will be performed by personnel not having direct responsibility for the Fire Protection Program.

13.4 REVIEW AND AUDIT

A program for review and audit of activities affecting station safety during the operational phase has been established. The program provides a system to insure that these activities are performed in accordance with company policy and rules, approved procedures and license provisions. This program will provide review of safety-related plant changes, tests, and procedures.

13.4.1 ONSITE REVIEW

See Section 17.2.1.1.2.1 for Station Operations Review Committee (SORC).

13.4.2 INDEPENDENT REVIEW

Activities affecting station safety occurring during the CPNPP operational phase will be independently reviewed by Nuclear Oversight and/or the Operations Review Committee (ORC).

13.4.2.1 Operations Review Committee (ORC)

See Section 17.2.1.3 for details.

The purpose for ORC review of violations, deviations, and reportable events which require reporting to the NRC in writing is to verify the adequacy of any investigations of such events and the corrective actions taken to prevent or reduce the probability of recurrence of such events.

ORC will also conduct independent reviews of the following:

1. All inspection reports conducted by the NRC and selected reports of audits conducted in accordance with Section 17.2.18.2 to verify compliance with the CPNPP Quality Assurance Manual and CPNPP license requirement.

2. Selected reports and documentation, as specified by the ORC, which pertain to reviews of safety-related activities conducted by the CPNPP Operating Staff. The purpose is to verify the adequacy of the on-site review process and provide an overview of other SORC activities.

3. Other safety-related matters deemed appropriate by ORC members or referred to the ORC by the Station Operations Review Committee.

13.4.3 AUDITS

Audits will be conducted in accordance with Chapter 17 of the FSAR.

13.5 PLANT PROCEDURES AND INSTRUCTIONS

The CPNPP staff is responsible for assuring the safe and efficient operation of the station, under the overall responsibility and direction of the Plant Manager (see Section 13.1.1.2.1). All activities which affect safety-related structures, systems and components will be conducted by detailed, written and approved procedures and instructions. This section identifies the activities that must be conducted by procedures and instructions and provides an appropriate method to develop and approve these procedures and instructions.

13.5.1 ADMINISTRATIVE PROCEDURES

The Plant Manager (see Section 13.1.1.2.1) is responsible for management of station procedures that assign the responsibilities and authorities of the CPNPP staff and provide the control measures for the preparation, review, approval, revision and use of all station procedures and instructions which govern quality-related activities. Station procedures and instructions include plant operating procedures, test procedures, equipment control procedures, calibration procedures and instructions, maintenance or modification procedures and instructions, refueling procedures, procurement procedures, analysis procedures, material control procedures and special orders. Administrative procedures will ensure that station procedures and instructions are reviewed by qualified personnel, approved by authorized personnel and distributed to and used by the personnel performing the prescribed activity.

13.5.1.1 Conformance with Regulatory Guide 1.33

The administrative controls utilized during the operations phase, which are described in this section, will be consistent with the provisions of Regulatory Guide 1.33, “Quality Assurance Program Requirements (Operation),” Revision 2, February 1978.

13.5.1.2 Preparation of Procedures and Instructions

Preparation of plant operating procedures will take place in approximately the same time frame as the preparation of preoperational and initial startup test procedures. Administrative procedures which govern the assignment of responsibilities for preparation, review and approval of other station procedures and instructions will be prepared initially. Other administrative procedures will be prepared as necessary to implement the operational phase of programs such as security and visitor control, housekeeping, and document control and records management, etc. Operation department administrative procedures and operating procedures necessary for operator training and preparation for operator license examinations will be completed six months prior to fuel loading. All other procedures and instructions will be prepared and approved prior to their use for performing the prescribed safety-related activity.

The station management position designated responsible for a given activity, as prescribed in the Quality Assurance Manual, is also responsible for the preparation of procedures and instructions for that activity. The actual preparation of procedures and instructions may be performed by other Vistra OpCo personnel or by outside contractors, but the final responsibility lies with the designated responsible position.

The Plant Manager shall designate approval authority for Station Administrative Procedures. Security Plan implementing procedures and Emergency Plan implementing procedures are approved in accordance with provisions of the CPNPP Security Plans and Emergency Plan,

respectively. Quality-related procedures and instructions shall be reviewed by at least one individual other than the preparer and approved by an appropriate manager.

Changes to approved quality-related procedures and instructions which clearly do not change the intent of the procedure and which require urgent implementation may be approved by two members of the Nuclear Operations staff, at least one of whom is SRO licensed. These changes shall be approved by the original approval authority within 14 days of implementation.

Other changes to procedures and instructions shall be reviewed and approved in the same manner as a permanent revision to that document or as otherwise directed by SORC in writing.

13.5.1.3 Procedures

The Plant Manager (see Section 13.1.1.2.1) is responsible for management of station procedures that provide a clear understanding of operating philosophy and management policies. As stated in 13.5.1.2, administrative procedures will be implemented that provide methods for the preparation, review and approval of all other station procedures including permanent procedures, temporary procedures or any procedures that might be of a transient or self-cancelling nature.

Administrative procedures will be developed by the operations department that will provide operations Shift Managers and shift crews with a clear understanding of how they are to conduct plant operations. Included will be procedures that specifically describe who may manipulate the controls of the reactor and who may operate any apparatus or mechanism that might affect the reactivity of the reactor. Procedures also will be prepared that describe the responsibilities of the senior reactor operators who will direct the licensed activities of licensed operators and who will be present at the facility or on call.

An operator requalification program will be implemented as described in Section 13.2.2.1 of the FSAR.

Examples of Operations Department Administration Procedures that may be included are listed in Table 13.5-9.

The operations department will also implement procedures and instructions as required specifying shift manning requirements which will be in accordance with the Technical Specifications. The responsibilities and authorities of the senior licensed personnel shall be delineated such as:

1. Determine the circumstances, analyze the cause, and determine that operations can proceed safely before the reactor is returned to power after a trip or an unscheduled or unexplained power reduction.

2. Accept and respond conservatively to instrument indications unless they are proven incorrect.

3. Adhere to the plant's Technical Specifications.

4. Review routine operating data to assure safe operation.

Procedures and instructions shall prescribe the conduct of shift operational activities including the following:

1. Definition of the specific area where the reactor operator who is at the controls of the unit must remain. This area is shown on Figure 13.5-1 and defines the area of applicability of position C.1.n of Regulatory Guide 1.29 (See Appendix 1A(B)).

2. Measures to control access to the Control Room.

3. Procedures for proper shift relief and turnover.

4. Procedures and instructions for the control of log and record keeping.

During station operation, the Shift Manager shall be responsible for ensuring that equipment control procedures are followed and properly implemented. These procedures will provide control of equipment, as necessary, to maintain personnel safety and reactor safety, and to avoid unauthorized operation of equipment. To secure and identify equipment in a controlled status, measures such as temporary bypass lines, electrical jumpers, lifted electrical leads, and temporary trip point settings shall be controlled by approved procedures which shall include requirements for independent verification where appropriate. A log will be maintained of the current status of temporary modifications which are not specified and controlled by approved procedures or instructions.

The plant operations and technical support departments shall be responsible for developing and implementing procedures, instructions and schedules to describe and control a surveillance inspection program for those areas for which they are responsible. A master surveillance schedule will be established which reflects the status of all surveillance test and inspections required by the Technical Specifications.

The nuclear operations departments will develop and implement administrative procedures that describe and control a preventive maintenance program. These administrative procedures will be written before initial station startup and will provide the general rules for the development of procedures and instructions under the preventive maintenance program. This program will provide for advance planning and scheduling of required routine preventive repair and maintenance activities.

The plant operations and plant engineering departments will also establish administrative procedures and instructions to control and document major repair and modification. Repairs or modifications which may affect the functioning of safety-related structures, systems or components will be performed in a manner to ensure quality at least equivalent to that specified by the original design specifications, materials specifications and inspection requirements.

Operations shall consider potential flooding when authorizing maintenance on systems which utilize circulating water and which could cause flooding.

Administrative procedures and controls will be established to ensure the reliable performance of fire protection personnel, systems and equipment. FSAR Section 13.3, Appendix B, Section 5.0 describes these procedures in more detail.

13.5.2 STATION OPERATING AND MAINTENANCE PROCEDURES

13.5.2.1 Station Operating Procedures

Operating procedures for all anticipated conditions affecting reactor safety will be written prior to initial fuel loading. The format of the station operating procedures ensures the contents include the applicable elements of ANSI N18.7-1976. These procedures will be grouped into the following classifications:

1. Integrated Plant Operating Procedures

2. System Operating Procedures

3. Alarm Procedures

4. Abnormal Conditions Procedures

5. Emergency Operating Procedures [addressed by TMI Action item I.C. 1.]

6. Radwaste Systems Procedures

13.5.2.1.1 Integrated Plant Operating Procedures

These procedures provide guidance for integrated plant operations and they will include major steps to be performed, references to system operating procedures, valve alignments and verification of instrument calibrations and approvals. Check lists will be included where appropriate.

Examples of procedures that may be included in this category are given in Table 13.5-1.

13.5.2.1.2 System Operating Procedures

These procedures include instructions for energizing, filling, venting, draining, startup, shutdown, changing operational modes and other appropriate instructions for the normal operation of safety-related systems. Valve lineups and check-off sheets will be included where necessary to ensure proper system operation.

Examples of procedures that may be included in this category are listed in Table 13.5-2.

13.5.2.1.3 Alarm Procedures

Each main control board annunciator for safety-related parameters will have a written procedure to identify the proper action to be taken by the operator in response to an alarm. Each of these procedures will include the annunciator identification, alarm setpoints and the proper corrective action to be taken.

13.5.2.1.4 Abnormal Conditions Procedures

These procedures will address abnormal operating conditions of safety- related plant systems. Each procedure will contain corrective action instructions for the malfunction involved.

Examples of procedures that may be included in this category are given in Table 13.5-3.

13.5.2.1.5 Emergency Operating Procedures

The Emergency Operating Procedures will be provided to guide operations in order to prevent or lessen the consequences of emergency conditions. These procedures will include automatic actions that will occur in the event of an emergency, immediate operator actions required to prevent or mitigate the consequences of an emergency and subsequent operator actions necessary to stabilize plant conditions. Immediate operator actions required to be memorized will be identified.

The Emergency Operating Procedures will be written to provide for a conservative course of action on the part of the operator and will be sufficiently flexible to accommodate variations.

Examples of procedures that may be included in this category are listed in Table 13.5-4.

13.5.2.1.6 Radwaste Systems Procedures

These procedures will be instructions for the operation of radioactive liquid, solid and gaseous waste systems and will be written to provide guidance for collection, storage, processing and discharge of these materials. Sound radioactive waste management will be incorporated into these procedures to support the effort to minimize radiation exposure and precisely control the release of radioactive material to the environment.

Examples of procedures that may be included in this category are given in Table 13.5-5.

13.5.2.2 Other Procedures

13.5.2.2.1 Maintenance and Modification Procedures

Maintenance or modification that may affect the functioning of safety- related structures, systems, or components shall be performed in accordance with applicable codes, bases, standards, design requirements, material specifications, and inspection requirements.

Maintenance of safety-related equipment will be properly pre-planned and performed in accordance with written procedures, written instructions, or drawings appropriate to the circumstances (for example, skills normally possessed by qualified maintenance personnel may not require detailed step-by-step delineation in a written procedure). It is the responsibility of the Director, Maintenance to implement a maintenance program for safety-related mechanical and electrical equipment and instruments and controls. It is the responsibility of the Director, Site Engineering to assist in performance testing of safety-related mechanical and electrical equipment to assure adequate levels of performance.

General rules for the control and administration of the maintenance program will be written before fuel loading. These general rules will form the basis for developing the repair or replacement procedures and instructions at the time of failure.

Procedures and instructions will be written early in plant life for maintenance of safety-related equipment expected to require recurring maintenance. The causes of malfunctions shall be determined, evaluated and recorded. The type and depth of the evaluation shall be

commensurate upon the significance of the event or malfunction and shall be performed per procedural and program requirements. Since the probability of failure is usually unknown and the time and mode of failure are usually unpredictable, procedures and instructions will not generally be written for repair of most equipment prior to failure. As experience is gained in operation of the plant, routine maintenance will be altered to improve equipment performance and repair procedures and instructions will be written and improved as required.

A preventive maintenance schedule will be developed which will describe the frequency and type of maintenance to be performed. A preliminary schedule will be developed early in plant life and will be refined and changed as experience with the equipment is gained.

Maintenance will be scheduled so as not to jeopardize the safety of the reactor. Scheduling shall consider the possible safety consequences of concurrent or sequential maintenance, testing, or operating activities. Equipment required to be operable for the mode in which the reactor exists will be available, and maintenance will be performed in a manner such that the license limits are not violated.

A managed maintenance program was developed to provide the plant staff with the maintenance data and information systems necessary to support proper planning and management of the maintenance activities. This is accomplished by a systematic evaluation of each plant component in which all maintenance activities are assessed. Examples of these resources are: manpower, radiation exposure, special tools, spare tools, spare parts, procedure number, and plant condition required for performing the activity.

Once all maintenance activities have been identified, two sets of maintenance plans are generated. The first set is the on-line preventive maintenance plan which includes all of those maintenance activities which can be performed with the plant at power. These activities are scheduled on a computer with various printouts and worksheets for the craft and supervisory personnel.

The second set of plans includes the outage-related work which will normally be performed concurrent with refueling. These activities, along with the refueling sequence are plotted on a computer which is used for managing the outage.

A significant point to note is that, because of inservice inspection requirements, the outage plan is repetitive with a ten-year cycle. The plant staff has completed the outage plans for the first ten-year cycle and, because of the repetitive nature of the work, has a plan for each year of commercial operation throughout the life of the plant. The program is designed to be an active program which will be updated as plant conditions and requirements change.

Off-the-shelf components shall be used only when the proper quality assurance documents are available or when the required quality assurance can be obtained by inspection and testing prior to being placed in service. Major modifications to safety-related equipment shall be designed and performed in accordance with applicable codes, standards, bases, design requirements, materials specifications and inspection requirements.

13.5.2.2.2 Plant Radiation Protection Procedures and Instructions

Detailed written and approved procedures and instructions will be used by the Comanche Peak Station Nuclear Operations Organization to ensure that occupational radiation exposure is

maintained as low as reasonably achievable. It is the responsibility of the Radiation Protection Manager to prepare and maintain the plant radiation protection procedures and instructions. Careful administrative control of the use of these procedures and instructions will ensure that a sound health physics philosophy becomes an integral part of station operation and maintenance. Plant radiation protection procedures and instructions will be developed for activities such as those listed in Table 13.5-6.

13.5.2.2.3 Emergency Preparedness Procedures

The Manager, Nuclear Emergency Planning is responsible for preparing and maintaining procedures that implement the protective measures outlined in the CPNPP Emergency Plan. A list of implementing procedures is maintained in the CPNPP Emergency Plan.

13.5.2.2.4 Chemical and Radiochemical Control Procedures and Instructions

The preparation of detailed, written and approved chemical and radio- chemical procedures and instructions are the responsibility of the Manager, Nuclear Chemistry.

These procedures and instructions will ensure primary and secondary side chemical/ radio-chemical quality, protection of component integrity, and promotion of efficient plant operation. Examples of areas to be covered by written and approved procedures and instructions are listed in Table 13.5-8.

13.5.2.2.5 Instrument Calibration and Test Procedures and Instructions

The Director, Maintenance is responsible for preparing procedures and instructions for proper control and periodic calibration of plant measuring and test equipment to maintain accuracy within necessary limits and to confirm adequacy of calibration frequency.

Specific procedures are prepared for surveillance tests performed on safety-related equipment and instrumentation. These procedures have provisions for assuring measurement accuracies are adequate to keep safety parameters within operational and safety limits. A master surveillance schedule reflecting the status of all planned in-plant surveillance testing is maintained. Control measures exist to assure appropriate documentation, reporting, and evaluation of test results.

13.5.2.2.6 Material Control Procedures and Instructions

Procedures and instructions will be provided for the proper procurement, documentation, and control of safety-related materials and components necessary for plant maintenance and modification. The procedures will be sufficiently detailed to ensure that purchased materials and components associated with safety-related structures or systems are:

1. Purchased to specifications and codes which ensure performance at least equivalent to the original equipment;

2. Produced or fabricated under quality control which ensures performance at least equivalent to that of the original equipment;

3. Properly documented to show compliance with applicable specifications, codes and standards;

4. Properly inspected, identified, and stored to provide protection against damage or misuse;

5. Properly controlled to ensure the identifications, segregation, and disposal of non-conforming material.

13.5.2.2.7 Security Procedures

It is the responsibility of the Manager, Nuclear Security to prepare and maintain detailed, written and approved procedures to implement the security plan. These procedures will supplement the physical barriers and other features designed to control access to the station and, as appropriate, to vital areas within the station. Information concerning specific design features and administrative provisions of the security plan is accorded limited distribution on a need-to-know basis.

13.5.2.2.8 Fuel Handling Procedures

The Director, Engineering Support and Director, Operations are responsible for developing fuel handling procedures to include receipt and receipt inspection, fueling/refueling and fuel handling, storage of fuel, and fuel shipment.

13.5.2.2.9 Fire Protection Procedures

The Manager, Technical Support is responsible for the development of procedures for fire protection. These procedures are described in more detail in FSAR Section 13.3B.

13.5.2.2.10 Test Procedures

The Manager, Startup was responsible for the preparation and implementation of procedures for prerequisite and preoperational acceptance testing. The Director, Site Engineering is responsible for design modification acceptance, station performance, initial startup tests, and those surveillance tests assigned to the System Engineering Department.

TABLE 13.5-1
INTEGRATED PLANT OPERATING PROCEDURES

- Plant Heatup From Cold Shutdown to Hot Standby
- Plant Startup From Hot Standby to Minimum Load
- Power Operations
- Plant Shutdown From Minimum Load to Hot Standby
- Plant Cooldown From Hot Standby to Cold Shutdown
- Maintaining Hot Standby
- Plant Equipment Shutdown Following a Trip
- Reactor Coolant System Mid Loop Operations

TABLE 13.5-2
SYSTEM OPERATING PROCEDURES

(Sheet 1 of 3)

- Reactor Coolant System
- Residual Heat Removal System
- Chemical and Volume Control System
- Reactor Make-up and Chemical Control System
- Boron Thermal Regeneration System
- Concentration Boric Acid System
- Reactor Coolant Pump
- Pressurizer Relief Tank
- Safety Injection System
- Safety Injection Accumulators
- Containment Spray System
- Hydrogen Purge Supply and Exhaust System

- Main Steam System
- Feedwater System
- Condensate System
- Auxiliary Feedwater System
- Steam Generator Blowdown and Cleanup System
- Condensate Polishing System
- Extraction Steam System
- Heater Drains System
- Condenser Vacuum and Waterbox Priming System
- Circulating Water System
- Auxiliary Steam System
- Chemical Feed System
- Turbine Plant Cooling Water System
- Steam Generator Recirculation System
- High Head Chemical Injection System
- Condensate and Feedwater System Long Term Wet Layup and Recirculation
- Turbine Control Fluid System
- Turbine Gland Steam System

TABLE 13.5-2
SYSTEM OPERATING PROCEDURES

(Sheet 2 of 3)

- Turbine Oil Purification System
- Turbine Lube Oil System
- Main Generator System
- Generator Hydrogen System
- Generator Seal Oil System
- Generator Primary Water System
- Station Service Water System
- Component Cooling Water System
- Service Water Chlorination System
- Low Volume Waste
- Spent Fuel Pool Cooling and Cleanup System
- Demineralized and Reactor Make-up Water System
- Service Air System
- Instrument Air System
- Vents and Drains System
- Nitrogen Gas System
- Hydrogen Gas System
- Carbon Dioxide Gas System
- Oxygen Gas System
- 138 KV and 345 KV Startup Transformers
- 6900 v Switchgear
- 480 V Switchgear Motor Control Centers and Power Panels
- 480 v Switchgear and Motor Control Centers
- 125 v DC Switchgear and Distribution System, and Batteries and Battery Chargers (Safety-Related)
- 24/48 v & 125/250 v DC Switchgear and Dist. Systems. Batteries and Battery Chargers (Non-Safety-Related)
- 118 v AC Distribution System and Inverters
- 208 v/120 v AC Distribution System
- Diesel Generator System
- Diesel Generator Fuel Oil Storage and Transfer System

TABLE 13.5-2
SYSTEM OPERATING PROCEDURES

(Sheet 3 of 3)

- Isolated Phase Bus Duct Cooling System
- 345 KV Switchyard Relay House
- Rod Control System
- Excore Instrumentation System
- Area Radiation Monitoring Systems
- Process and Effluent Radiation Monitoring Systems
- Incore Instrumentation System
- Containment Ventilation System
- Control Room Ventilation System
- Switchgear Area Ventilation System
- Battery Room Ventilation Systems
- Auxiliary Building Ventilation System
- Main Steam and Feedwater Area and Electrical Area Ventilation System
- Fuel Building Ventilation System
- Diesel Generator Rooms Ventilation System
- Uncontrolled Access Area Ventilation System
- Office and Service Areas Ventilation System
- Service Water Intake Ventilation System
- Ventilation Chilled Water System
- Safety Chilled Water System
- Primary Plant Ventilation System
- Safeguards Building Ventilation System
- Squaw Creek Reservoir Makeup System
- Squaw Creek Reservoir Return and Service Outlet System
- Halon Fire Protection Systems
- Water Fire Protection System
- Containment Personnel Airlocks

TABLE 13.5-3
ABNORMAL CONDITIONS PROCEDURES

(Sheet 1 of 2)

- REACTOR COOLANT PUMP TRIP/MALFUNCTION
- HIGH REACTOR COOLANT ACTIVITY
- EXCESSIVE REACTOR COOLANT LEAKAGE
- RESIDUAL HEAT REMOVAL SYSTEM MALFUNCTION
- CHEMICAL AND VOLUME CONTROL SYSTEM MALFUNCTIONS
- HIGH SECONDARY ACTIVITY
- EMERGENCY BORATION
- UPS HVAC MALFUNCTION
- LOSS OF NON-SAFEGUARDS VENTILATION SYSTEMS
- INSTRUMENT AIR SYSTEM MALFUNCTION
- FEEDWATER, CONDENSATE, HEATER DRAIN SYSTEM MALFUNCTION
- AUXILIARY FEEDWATER SYSTEM MALFUNCTION
- MAIN TURBINE MALFUNCTION
- MAIN GENERATOR MALFUNCTION
- STATION SERVICE WATER SYSTEM MALFUNCTION
- COMPONENT COOLING WATER SYSTEM MALFUNCTIONS
- RESPONSE TO A 138/345 KV SYSTEM MALFUNCTION
- RESPONSE TO A 6900/480 V SYSTEM MALFUNCTION
- LOSS OF PROTECTION AND/OR INSTRUMENT BUS
- SOURCE RANGE INSTRUMENTATION MALFUNCTION
- INTERMEDIATE RANGE INSTRUMENTATION MALFUNCTION
- POWER RANGE INSTRUMENTATION MALFUNCTION
- Tc/N-16 INSTRUMENTATION MALFUNCTION
- PRESSURIZER PRESSURE INSTRUMENTATION MALFUNCTION
- PRESSURIZER LEVEL INSTRUMENTATION MALFUNCTION
- STEAM FLOW INSTRUMENTATION MALFUNCTION
- FEEDWATER FLOW INSTRUMENTATION MALFUNCTION
- STEAM LINE PRESSURE, STEAM HEADER PRESSURE, AND TURBINE 1st-STAGE PRESSURE INSTRUMENT MALFUNCTION
- STEAM GENERATOR LEVEL INSTRUMENTATION MALFUNCTION
- ROD CONTROL SYSTEM MALFUNCTION

TABLE 13.5-3
ABNORMAL CONDITIONS PROCEDURES

(Sheet 2 of 2)

- RCS FLOW INSTRUMENTATION MALFUNCTION
- WIDE RANGE Tc OR TH INSTRUMENTATION MALFUNCTION
- RCS LOOP PRESSURE INSTRUMENTATION MALFUNCTION
- RESPONSE TO A FIRE IN THE CONTROL ROOM OR CABLE SPREADING ROOM
- RESPONSE TO FIRE IN THE SAFEGUARDS BUILDING
- RESPONSE TO FIRE IN THE AUXILIARY BUILDING OR THE FUEL BUILDING
- RESPONSE TO FIRE IN THE ELECTRICAL AND CONTROL BUILDING
- RESPONSE TO FIRE IN THE CONTAINMENT BUILDING
- RESPONSE TO FIRE IN SERVICE WATER INTAKE STRUCTURE
- RESPONSE TO FIRE IN THE TURBINE BUILDING
- FIRE PROTECTION SYSTEM MALFUNCTION
- ACCIDENTAL RELEASE OF RADIOACTIVE GAS
- ACCIDENTAL RELEASE OF RADIOACTIVE LIQUID
- LOSS OF CONTROL ROOM HABITABILITY
- PLANT PROCESS COMPUTER SYSTEM MALFUNCTION
- ACTS OF NATURE
- FUEL HANDLING ACCIDENT
- SPENT FUEL POOL/REFUELING CAVITY LEAKAGE
- LOOSE PARTS MONITORING
- SHUTDOWN LOSS OF COOLANT
- CONTROL ROOM VENTILATION SYSTEM MALFUNCTION
- MAIN CONDENSER AND CIRCULATING WATER SYSTEM MALFUNCTION
- TURBINE TRIP RESPONSE
- RAPID POWER REDUCTION
- LOSS OF NON 1E INSTRUMENT BUS
- 25 Kv MALFUNCTION
- CONTROL ROOM ANNUNCIATOR SYSTEM AND STATUS LIGHT MALFUNCTION
- COLD WEATHER PREPARATION/HEAT TRACING AND FREEZE PROTECTION SYSTEM MALFUNCTION

TABLE 13.5-4
EMERGENCY OPERATING PROCEDURES

(Sheet 1 of 2)

- Reactor Trip or Safety Injection
- Loss of Reactor Coolant or Secondary Coolant
- Faulted Steam Generator Isolation
- Steam Generator Tube Rupture
- Rediagnosis
- Reactor Trip Response
- Natural Circulation Cooldown
- Natural Circulation cooldown with Steam Void in vessel (with RVLIS)
- Natural circulation cooldown with Steam Void in vessel (without RVLIS)
- Safety Injection Termination
- Post LOCA cooldown and depressurization
- Transfer to cold leg recirculation
- Transfer to hot leg recirculation
- Post-SGTR cooldown using backfill
- Post-SGTR cooldown using blowdown
- Post-SGTR cooldown using Steam Dump
- Loss of all ac power
- Loss of all ac power recovery without SI required
- Loss of all ac power recovery with SI required
- Loss of emergency coolant recirculation
- LOCA outside containment
- Uncontrolled depressurization of all steam generators
- SGTR with loss of reactor coolant - subcooled recovery
- SGTR with loss of reactor coolant saturated recovery desired

TABLE 13.5-4
EMERGENCY OPERATING PROCEDURES

(Sheet 2 of 2)

- SGTR without pressurizer pressure control
- Response to Nuclear Power Generation/ATWT
- Response to Loss of Core Shutdown
- Response to Inadequate Core Cooling
- Response to Degraded Core Cooling
- Response to Saturated Core Cooling
- Response to Loss of Secondary Heat Sink
- Response to steam generator overpressure
- Response to steam generator high level
- Response to Loss of Normal Steam Release Capabilities
- Response to Steam Generator Low Level
- Response to Imminent Pressurized Thermal Shock Condition
- Response to Anticipated Pressurized Thermal Shock Condition
- Response to High Containment Pressure
- Response to Containment Flooding
- Response to High Containment Radiation Level
- Response to High Pressurizer Level
- Response to Low Pressurizer Level
- Response to Voids in Reactor Vessel

TABLE 13.5-5
RADWASTE SYSTEMS PROCEDURES

- Reactor Coolant Drain Tank Subsystem
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13.6 SECURITY

A comprehensive security program has been developed at CPNPP in accordance with the requirements of applicable portions of 10CFR73, “Physical Protection of Plants and Materials.” The security plans that describe the program contain security related information (SRI) or safeguards information (SGI) and are withheld from public disclosure in accordance with 10CFR2.390 or 10CFR73.21. The CPNPP security plans are presented in separate submittals.

14.0 INITIAL TEST PROGRAM

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14.0 INITIAL TEST PROGRAM

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Number	Title
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14.0 INITIAL TEST PROGRAM

LIST OF FIGURES

Number	Title
14.2-1	Test Review Group
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14.2-3	Preoperational Test Schedule (2 Sheets)
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14.1 SPECIFIC INFORMATION TO BE INCLUDED IN PRELIMINARY SAFETY ANALYSIS REPORTS

This section is not applicable; see Section 14.2 for specific information.

14.2 INITIAL TEST PROGRAM

14.2.1 SUMMARY OF TEST PROGRAM AND OBJECTIVES

The Initial Test Program for CPNPP Units 1 and 2 has been completed. The activities described in Section 14.2 for the preoperational and initial startup test programs will be maintained as historical information and will not be revised to reflect updates to organizational structures or position titles and responsibilities. Those types of changes will be reflected in Chapters 13 and 17, as applicable.

The purpose of the test program for Comanche Peak Nuclear Power Plant (CPNPP) is to assure that the installed station structures, systems and components will be subjected to tests as required to verify that the plant has been properly designed and constructed and is ready to operate in a manner that will not endanger the health and safety of the public, and to provide assurance of total plant reliability for operation. The test program will also ensure that the procedures for operating the plant safely have been evaluated and demonstrated, and that the operating organization is knowledgeable about the plant and procedures and is prepared to safely operate the facility.

The necessary procedures to control, implement, and document the test program are established by startup and plant administrative procedures and summarized in the following sections. TU Electric, as the applicant, has responsibility for overall direction and management of the test program.

The testing activities to be performed on safety-related systems at the CPNPP are divided into two major phases: preoperational testing and initial startup testing.

The preoperational testing phase is divided into two overlapping parts: prerequisite testing and preoperational testing. Prerequisite testing will be conducted in order to verify the integrity, proper installation, cleanliness, and functional operability of the system components. Preoperational testing will be performed in order to demonstrate the capability of systems, structures, and components perform their safety functions.

These structures, systems, and components are identified as quality related items in Table 17A-1 of FSAR Section 17A. In addition, systems and components which have applicable testing requirements specified in Regulatory Guide 1.68 and discussed in Appendix 1A(B) will be subjected to preoperational tests. Summaries of individual preoperational tests are provided in Section 14.2.12.

Preoperational testing will be completed prior to fuel load with certain limited exceptions where tests or parts of tests will be deferred to the Initial Startup Test Program. Such an exception is the complete testing of the control rod drive mechanisms with the rods attached. In such cases sufficient tests will be performed prior to fuel loading to provide reasonable assurance that the post-loading tests will be successful.

Initial startup tests will be performed beginning with activities leading to fuel loading and ending with full power operation. The intent of these tests is to assure that tests deferred from the preoperational test program are performed; that fuel loading is effected in a safe manner; that the plant is safely brought to rated capacity; that plant performance is satisfactory in terms of established design criteria; and to demonstrate, where practical, that the plant is capable of

withstanding anticipated transients and postulated accidents. Testing activities related to fuel load and initial criticality are described in Section 14.2.10. Initial startup tests including low power physics testing and power ascension testing are described in the test summaries provided in Section 14.2.12.

For systems and components which have no nuclear safety-related requirements, acceptance testing will be performed to verify proper system performance and to ensure reliable and efficient operation of the plant.

Preoperational and initial startup tests will be conducted in accordance with approved test procedures. Review, approval, and revision of test procedures and the evaluation and disposition of test results will be accomplished by methods specified in the appropriate administrative procedures and summarized in Sections 14.2.3 and 14.2.5.

The startup program will utilize, to the extent practical, operations personnel and operating procedures to provide familiarization with the plant installation and demonstrate the adequacy of operating procedures.

Preoperational testing activities will be coordinated through a Joint Test Group (JTG), as described in Section 14.2.2.6. Initial startup testing activities will be coordinated through a Test Review Group (TRG), as described in Section 14.2.2.7.

14.2.2 ORGANIZATION AND STAFFING

14.2.2.1 General Description

TU Electric is responsible for the overall administration and technical direction of the CPNPP test program. The Performance and Test Manager is responsible for the coordination of the initial startup test program. The Manager, Startup is responsible for the coordination, direction, and implementation of the prerequisite and preoperational test program.

The TRG and JTG organizations are shown in Figures 14.2-1 and 14.2-2. The SORC membership is described in Section 13.4.1.1.1.

14.2.2.2 Performance and Test Organizational Responsibilities

The duties and responsibilities of the Performance and Test organization under the direction of the Performance and Test Manager are as follows:

1. Preparation of testing schedules;

2. Coordination with the appropriate engineering organizations to resolve component/ system design and operating problems;

3. Preparation of test procedures;

4. Conduct tests to demonstrate safe plant operation;

5. Perform the initial startup test sequence to ensure a safe and orderly power ascension program.

Support personnel including craft labor and technicians who will be engaged in testing activities may be supplied by TU Electric or contractors. Such support will vary according to test program requirements.

14.2.2.2.1 Performance and Test Manager

The responsibilities of the Performance and Test Manager include:

1. Development and implementation of the Performance and Test surveillance tests;

2. Development and implementation of the Initial Startup Test Program;

3. Coordination of testing with the activities of operations, maintenance, engineering, and construction organizations;

4. Administration of the development of plans and schedules to support the assigned testing activities;

5. Administration of the development of individual test procedures;

6. Review test procedures and results or assure that such reviews are conducted by qualified personnel within the Performance and Test organization;

7. Assure proper and timely notifications and reports pertaining to assigned testing activities are submitted to the Nuclear Regulatory Commission and other regulatory agencies;

8. Implement the program to utilize testing and operating experience in other similar nuclear plants.

14.2.2.2.2 Performance and Test Engineers

The duties and responsibilities of the Performance and Test engineers include:

1. Preparation of assigned test procedures to direct and guide specific tests in accordance with a standard format;

2. Direction to support personnel and others during performance of tests including appropriate interface with station operators;

3. Ensuring the safety of personnel and plant equipment during testing;

4. Familiarization of support personnel with specific tests;

5. Identification of deficiencies that could adversely affect test performance;

6. Assembly of test data and preparation of test reports for evaluation of test results by others; and

7. Responsibility to disallow or terminate testing due to conditions which could endanger personnel or equipment.

14.2.2.3 Startup Department Organizational Responsibilities

The duties and responsibilities of the Startup Department under the direction of the Manager, Startup are as follows:

1. Preparation of testing schedules;

2. Coordination with appropriate construction organizations for release of completed components/systems to facilitate testing;

3. Coordination with the appropriate engineering organizations to resolve component/ system design and operating problems;

4. Preparation of test procedures;

5. Conduct tests to demonstrate adequate and safe component and system performance.

6. Administration of the preventive and corrective maintenance program for Unit 2 equipment under project jurisdiction.

Support personnel including craft labor and technicians who will be engaged in testing activities may be supplied by TU Electric or contractors. Such support will vary according to test program requirements.

14.2.2.3.1 Manager, Startup

The responsibilities of the Manager, Startup include:

1. Administration of the development of plans and schedules regarding the status of the startup program;

2. Administration of the development of individual test procedures;

3. Continuing analysis of construction and equipment installation schedules for compatibility with testing schedules and implementation of corrective actions to minimize conflicts;

4. Review test procedures and results or assure that such reviews are conducted by qualified personnel within the Startup Department;

5. Assure proper and timely notifications and reports pertaining to testing activities are submitted to the Nuclear Regulatory commission and other regulatory agencies.

14.2.2.3.2 Assistant Startup Managers

The Assistant Startup Managers are responsible for the startup program and will have the following duties and responsibilities:

1. Provide technical direction to system test engineers and others assigned to the preoperational test group;

2. Review of Startup Department administrative and prerequisite test procedures and ensure that the procedures and test results have been approved at the appropriate level;

3. Review and submittal of design change requests originated by the Startup Department in accordance with the appropriate Startup Department administrative procedures;

4. Assure testing activities are conducted in accordance with the Quality Assurance Program and applicable procedures; and

5. Coordination of testing with the activities of the engineering, construction and operations organizations.

14.2.2.3.3 Startup Engineers

The duties and responsibilities of the Startup engineers include:

1. Preparation of assigned test procedures to direct and guide specific tests;

2. Direction to support personnel and others during performance of tests including appropriate interface with station operators;

3. Ensuring the safety of personnel and plant equipment during testing;

4. Familiarization of support personnel with specific test;

5. Identification of deficiencies that could adversely affect test performance;

6. Assembly of test data and preparation of test reports for evaluation of test results by others;

7. Implementation of tagging procedures; and

8. Responsibility to disallow or terminate testing due to conditions which could endanger personnel or equipment.

14.2.2.4 Operating Staff

The CPNPP operating staff will be involved in the test program in several capacities throughout preoperational and initial startup testing. This involvement will include review of test procedures and results and the direct participation of operating personnel in test activities. Operating staff technicians will be assigned to assist test engineers in performing tests. Station operators will assist test engineers in performing tests and will take over the routine operations of systems for which prerequisite-type testing has been completed. The operating staff will direct the fuel loading and will be responsible for the operation of the plant during initial startup testing.

The duties and responsibilities of the operating staff during plant operation are described in Chapter 13. The duties of key operating personnel with regard to the startup program are summarized below.

14.2.2.4.1 Vice President, Nuclear Operations

The Vice President, Nuclear Operations has overall responsibility for station operations and the test program.

14.2.2.4.2 Plant Manager

The Plant Manager will provide a continual analysis of testing schedules, operator training schedules, and plant operating staff workloads in order to minimize conflicts with the test program, and he also will provide coordination between the operating staff and the Startup Department.

14.2.2.4.3 Operations Department

The Manager, Operations will be responsible for the proper operation of all equipment in the custody of the operations department and for ensuring that the conduct of the test program does not place the plant in an unsafe condition. He will provide personnel from the operating staff as required to support the conduct of testing activities. In addition, he will direct the development of station operating procedures.

The Shift Supervisors report to the Manager, Operations and are responsible for the safe operation of the plant during assigned shifts. They also are responsible for the implementation of appropriate safety and custody tagging procedures and have responsibility to disallow or terminate testing due to conditions which could endanger personnel or equipment or violate Technical Specifications.

14.2.2.4.4 Maintenance Department

The Manager, Maintenance will be responsible for performing preventive and corrective maintenance when required on Unit 2 components and systems that have been released to operations from the startup department. He will provide personnel from the maintenance department to support testing activities as required.

14.2.2.4.5 Plant Engineering

The Manager, Plant Engineering will provide personnel from the Plant Engineering organization as required to support testing activities. The Performance and Test Manager reports to the Manager, Plant Engineering and is responsible for the implementation of the Initial Startup program.

14.2.2.4.6 Director, Oversight & Regulatory Affairs

The Director, Oversight & Regulatory Affairs is responsible for approving and verifying implementation of the TU Electric CPNPP QA Program.

14.2.2.5 Major Participating Organizations

14.2.2.5.1 Design Engineering

Design Engineering will be responsible for assisting to the extent required in ensuring that tests sufficiently verify the adequacy of system design and in the formulation of changes required to correct detected design deficiencies. This responsibility includes:

1. Review of preoperational and initial startup test objectives and acceptance criteria to verify implementation of operating license commitments and consistency with the station safety analyses;

2. Review and approval of design change requests originated during the course of the test program.

3. Management of contractors.

14.2.2.5.2 Brown & Root, Inc. (B&R)

B&R, as the constructor for CPNPP, is responsible for the construction completion, performance of associated constructions tests, and orderly release of components and systems to TU Electric consistent with the test program schedules. This responsibility includes:

1. Completion of construction and construction testing activities;

2. Provision of craft technical manpower support as required for performance of the test program.

14.2.2.5.3 Westinghouse Electric Corporation

Westinghouse, as the Nuclear Steam Supply System (NSSS) supplier, is responsible for providing technical direction to TU Electric during preoperational and initial startup testing performed on the NSSS and associated auxiliary equipment. Technical direction is defined as technical guidance, advice and counsel based on current engineering, installation, and testing practices. This responsibility includes:

1. Assignment of personnel to provide advice and assistance to TU Electric for test and operation of all equipment and systems in the Westinghouse area of responsibility.

2. Assignment of an operational physicist to the site organization during fuel loading and low power testing; and

3. Provision of test procedure outlines and technical assistance for tests of Westinghouse furnished components and systems.

14.2.2.5.4 Utility Power Corporation (UPC)

UPC, as supplier of the main turbine generator set, is responsible for providing technical support to TU Electric during preoperational and initial startup testing performed on the turbine generator and related auxiliary equipment.

14.2.2.6 Joint Test Group (JTG)

The JTG is comprised of certain station supervisory and technical personnel, is charged with reviewing Startup activities and advises the Manager, Startup on the disposition of those items reviewed.

14.2.2.6.1 JTG Membership

As a minimum, the JTG shall be composed of managers, supervisors, or engineers from the organizations listed below.

Startup

Operations

Engineering (Design & Technical Support functions)

Westinghouse (for all matters concerning preoperational testing performed on the NSSS and associated auxiliary systems)

Quality Assurance

The members and their alternates, including the Chairman, shall be designated in writing by their cognizant Vice President.

The primary function of the JTG is the review and approval of all preoperational test procedures, procedure revisions, and test results.

In addition to the above, representatives of other organizations will participate, as requested by the Chairman.

14.2.2.7 Test Review Group (TRG)

The TRG is comprised of certain station supervisory and technical personnel. The TRG functions as a subcommittee of the Station Operations Review Committee (SORC) for initial startup testing matters. The TRG is charged with reviewing initial startup testing activities and advising the SORC on the disposition of those items reviewed. The SORC may act in lieu of the TRG for activities described below.

14.2.2.7.1 TRG Membership

As a minimum, the TRG shall be composed of management, supervisory or technical representatives from the areas listed below.

- Operations
- Engineering (design, testing and technical support functions)
- Westinghouse (for all matters concerning initial startup testing performed on the NSSS and associated auxiliary systems)

The members and their alternates, including the Chairman, shall be designated in writing by the Vice President, Nuclear Operations.

The primary function of the TRG is the review and approval of initial startup program test procedures, procedure revisions, and test results.

In addition to the above, representatives of other organizations will participate, as requested by the Chairman.

14.2.2.8 Nuclear Oversight

The Vistra OpCo Nuclear Oversight Department is responsible for assuring the quality of construction, plant testing, and operations activities as described in Chapter 17.

14.2.2.9 Qualifications of Key Personnel

The minimum qualifications of individuals responsible for review and approval of preoperational and initial startup test procedures and results shall be; (1) Bachelor's degree in engineering or the physical sciences and four years of applicable power plant experience; or (2) high school diploma or equivalent and eight years of applicable power plant experience. Credit for up to two years for related technical training may be substituted on a one-for-one time basis. A minimum of two years shall be applicable nuclear power plant experience.

The minimum qualifications of individual that direct or supervise the conduct of preoperational test at the time of performance of the duties, shall be; (1) Bachelor's degree in engineering or the physical sciences and one year of experience in power plant testing or operation. Included in the one year shall be three months of familiarization of system and component operation unique to the design of similar nuclear power plants or (2) a high school diploma or equivalent and four years of experience in power plant testing or operation or credit for up to two years for related technical training may be substituted on a one-for-one time basis. Included in the four years shall be three months of familiarization of system and component operation unique to the design of similar nuclear power plants.

The minimum qualifications of individuals that direct or supervise the conduct of initial startup test at the time of performance of the duties, shall be: (1) Bachelor's degree in engineering or the physical sciences and two years of experience in power plant testing or operation. Included in the two years shall be a minimum of one year of nuclear power plant testing, operation or training on a nuclear facility, or (2) a high school diploma or equivalent and five years of experience in power plant testing of which two years will be nuclear power plant experience. Credit for up to two years for related technical training may be substituted on a one-for-one time basis.

Test personnel will be indoctrinated in the use of applicable administrative procedures, test procedures and familiarized with applicable quality assurance requirements.

14.2.3 TEST PROCEDURES

14.2.3.1 General

Preoperational and initial startup tests will be performed in accordance with written approved test procedures. The following sections describe the general methods employed to control procedure

development and review, and they also describe the responsibilities of the various organizational units participating in this process.

The detailed controls and methods will be prescribed in the startup administrative procedures and station administrative procedures, as applicable.

14.2.3.2 Development of Procedures

Technical information required for the preparation of the test procedures will be provided by the appropriate engineering organizations. This information will consist of system descriptions, technical specifications, design drawings and other technical documents which define the functional requirements and performance objectives for the various systems and components. Additional technical data may also be obtained from the various component vendors and other contractors as required.

The applicable functional requirements provided by the system designers will be incorporated into the acceptance criteria for each test procedure. This information will also be used by the test engineer in developing the detailed test methods which ensure that the capability of systems and components to function properly within design specifications is adequately demonstrated.

14.2.3.3 Review and Approval of Test Procedures

Preoperational and initial test procedures will be reviewed by members of the plant operating staff and appropriate design organizations.

Preoperational test procedures will be forwarded to appropriate members of the Joint Test Group for review and comment.

Initial startup test procedure will be forwarded to the appropriate members of the Test Review Group for review and comment.

The responsibility for final approval of preoperational test procedures rests with the Manager, Startup. The responsibility for final approval of initial startup test procedures rests with the Plant Manager.

14.2.3.4 Format of Test Procedures

Preoperational and initial startup tests will be prepared based on formats specified by administrative procedures. These standard formats will help ensure that each procedure contains the information and instructions required to satisfactorily perform and document the test.

The procedures format and content will reflect the guidance provided in Regulatory Guide 1.68 as discussed in Appendix 1A(B). The standard format will include, as a minimum, the following:

1. Test Objectives:

   A detailed statement of the test objectives and method of system or plant operation to be demonstrated.

2. References:

   References to technical specifications, supporting procedures, vendor's manual, or other technical documents will be included as required.

3. Prerequisites:

   Prerequisites and initial conditions are satisfied and the necessary prerequisite tests and construction activities have been satisfactorily completed.

4. Precautions and Limitations:

   Precautions and limitations relating to personnel safety, equipment integrity, and overall plant safety will be specified.

5. Instructions:

   The instruction section will contain detailed step by step instructions for operating the system in the test configuration, performing actual test manipulations, and for use of off-normal procedures such as jumper cables or mechanical bypasses and restoration of the system to normal status following test. The detailed test instructions will utilize normal and emergency station operating procedures to the extent practical.

6. Acceptance Criteria:

   The performance objectives and functional requirements for system operations will be specified. The criteria used to judge the success or failure of the test may be qualitative or quantitative.

7. Data Collection:

   Provisions will be made for recording all pertinent test data regarding system conditions and performance. Records will identify the observer, type of test instrumentation used, acceptability of results, and any deficiencies, and will become a part of the permanent station records.

14.2.3.5 Revisions to Procedures

Revisions to preoperational test procedures will be reviewed by appropriate members of the JTG and approved by the Manager, Startup. Revisions to initial startup test procedures will be reviewed by the appropriate members of the TRG and approved by the Plant Manager.

Preoperational and Initial Startup Test procedure modifications required during conduct of test will be approved as follows;

1. Preoperational Test Procedures

   a. Test procedure modifications that do not change the intent of the test will be approved by the system test engineer.

b. Test procedure modifications that change the intent of the test will be approved by the test engineer and another member of the startup organization qualified to review preoperational test procedures.

c. All procedure modifications will be included with the completed procedure and be subject to review and approval with the test results as described by paragraph 14.2.5.

2. Initial Startup Test Procedures

a. Modifications to initial startup tests that do not change the intent of the test will have the Shift Supervisor's concurrence and approval of the person in charge of the test.

b. Modifications to initial startup tests that change the intent of the test will be reviewed and approved by the Plant Manager.

14.2.4 CONDUCT OF TEST PROGRAM

14.2.4.1 Administrative Procedures

The conduct of the test program will be controlled by administrative procedures. These procedures will prescribe controls for testing activities such as the following:

1. Preparation, review and approval, of test procedures

2. Turnover of systems

3. Format and content of test procedures

4. Safety and custody tagging procedures

5. Temporary system modifications

6. Design change processing

7. Test deficiency processing

8. Review of Reactor Operating/Startup experiences.

14.2.4.2 Prerequisite Testing

Startup administrative procedures will be established to ensure that applicable prerequisites are met before testing is initiated. Upon completion of construction phase activities, custody of the component/system will be normally transferred to TU Electric for conduct of prerequisite testing and preoperational testing. In select cases prerequisite testing may be conducted prior to custody transfer with agreement of construction and startup. During the turnover process, systems and components will be reviewed for completeness, installation damage and conformance with appropriate installation and/or design documents. Outstanding construction, document and test deficiencies will be identified and controlled prior to fuel load.

14.2.4.3 Preoperational Testing

Technical direction and administration, including test procedure preparation, test execution, and data recording, of the preoperational testing is the responsibility of the Startup Department, with the operating staff retaining responsibility for performing actual equipment operations and maintenance.

The Manager, Startup is responsible for the administration and implementation of all preoperational testing activities during the startup program.

The test engineers will direct support personnel in the performance of tests and will provide appropriate interface with station operators. The Shift Supervisors will be responsible for insuring that the conduct of testing does not place the plant in an unsafe condition at any time. Additionally, the shift supervisors and test engineers have the authority to terminate or disallow testing at any time.

14.2.4.4 Initial Startup Testing

Technical direction and administration, including test procedure preparation, test execution, and data recording, of initial startup testing is the responsibility of the Performance and Test organization, with the operating staff retaining responsibility for performing actual equipment operations and maintenance.

The Performance and Test Manager is responsible for the administration and implementation of all initial startup testing activities during the test program.

The test engineers will direct support personnel in the performance of tests and will provide appropriate interface with station operators. The Shift Supervisors will be responsible for insuring that the conduct of testing does not place the plant in an unsafe condition at any time. Additionally, the Shift Supervisors, other on-shift licensed operators, and test engineers have the authority to terminate or disallow testing at any time.

14.2.4.5 Test Prerequisites

Each test procedure will contain a set of prerequisites or initial conditions as prescribed by administrative procedures. The test engineer will ensure that all specified prerequisites are met prior to performing the test. The format for test procedures is described in Section 14.2.3.4.

14.2.4.6 Phase Evaluation

The test results of all preoperational tests performed in the preoperational testing phase will be reviewed by the JTG prior to commencing the initial startup testing phase of the initial test program.

Between each major phase of the initial startup test program the test results for all tests that have been performed will be reviewed by the TRG.

This review ensures that all required systems have been tested satisfactorily and that test results have been evaluated before proceeding to the next stage of testing. This review is described in Section 14.2.5.

14.2.4.7 Design Modifications

Modifications to the design of safety-related equipment during the test program may be initiated in order to correct deficiencies discovered as a result of testing. Any such modification will be referred to the appropriate engineering organization for approval. Modifications made to safety-related components or systems after completion of preoperational or initial startup testing will be reviewed for retesting requirements on affected portions of the system.

14.2.5 REVIEW, EVALUATION, AND APPROVAL OF TEST RESULTS

Following completion of a particular test, a test engineer will assemble the test data package for evaluation. Preoperational tests will be reviewed by appropriate members of the JTG. Initial startup test results will be reviewed by appropriate members of the TRG.

Each test data package will be reviewed to ensure that the test has been performed in accordance with the written approved procedure and that all required data, checks, and signatures have been properly recorded and that system performance meets the approved acceptance criteria.

Deficiencies identified in the review process will be resolved to the satisfaction of the appropriate review group. If the evaluation indicates that deficiencies in the test method are responsible for unsatisfactory test results, the test procedure will be modified accordingly before retesting is initiated. Whenever an evaluation of test results indicates deficiencies in system performance, the problem will be referred to the appropriate engineering organization for evaluation.

The responsibility for final approval of preoperational test results rests with the Manager, Startup. The responsibility for final approval of initial startup test results and authorization to proceed to higher power level following completion of major power plateau testing (i.e., 0-5%, 50% and 75%) rests with the Plant Manager.

Following each major phase of the test program, test results and/or test status will be reviewed to ensure that all required tests have been performed and that the test results have been evaluated. This evaluation will ensure that all required systems are operating properly and that testing for the next major phase will be conducted in a safe and efficient manner. This type of review will be performed to the extent required before major test phases such as fuel load, initial criticality, and power escalation. During the power escalation phase, review and evaluation of initial startup test procedure results will be completed for each major power plateau prior to proceeding with power ascension testing to the next plateau.

14.2.6 TEST RECORDS

Test procedures and test data relating to preoperational and initial startup testing will be retained in accordance with the measure described in Section 17.2.17, “Quality Assurance Records.”

14.2.7 CONFORMANCE OF TEST PROGRAMS WITH REGULATORY GUIDES

The test program will be conducted in accordance with guidance given in the applicable Nuclear Regulatory Commission Regulatory Guides which are listed below.

1. Regulatory Guide 1.41, “Preoperational Testing of Redundant On- Site Electric Power System to Verify Proper Load Group Assignments”

2. Regulatory Guide 1.52, “Design, Testing, and Maintenance Criteria for Atmosphere Cleanup System Air Filtration and Absorption Units of Light-Water-Cooled Nuclear Power Plants”

3. Regulatory Guide 1.68, “Initial Test Programs for Water-Cooled Nuclear Power Plants”

4. Regulatory Guide 1.68.2, “Initial Startup Test Program to Demonstrate Remote Shutdown Capability for Water Cooled Nuclear Power Plants.”

5. Regulatory Guide 1,69, “Concrete Radiation Shields for Nuclear Power Plants.”

6. Regulatory Guide 1.20, “Comprehensive Vibration Assessment Program for Reactor Internals During Preoperational and Initial Startup Testing.”

7. Regulatory Guide 1.79, “Preoperational Testing of Emergency Core Cooling Systems for Pressurized Water Reactors.”

Exception has been taken to certain provisions of the above Regulatory Guides and are discussed in Appendix 1A(B)

14.2.8 UTILIZATION OF REACTOR OPERATING AND TESTING EXPERIENCE IN DEVELOPMENT OF TEST PROGRAM

The CPNPP test program will utilize information gained from operating and testing experience in other similar nuclear plants. This information will be used to provide guidance in developing test procedures and schedules and to alert personnel to potential problem areas in the testing program.

The development of this program for utilizing testing and operating experience is the responsibility of the Vice President, Nuclear Operations with the direct implementation of the program being the responsibility of the Manager, Startup and the Performance and Test Manager.

Significant event information received from NSAC and INPO will be reviewed by the test organization to identify special testing which should be included in the test program.

The operating experience program will consist of an initial review to be administered prior to conducting preoperational tests and an ongoing review during the remainder of the test program. The initial review will examine all pertinent operating data and abnormal events on similar plants which occurred during a period of two years prior to the review. The review will be conducted so as to allow sufficient time for data to be analyzed and incorporated in test procedures. Any new information will be reviewed on a regular basis during the test program so as to address current testing problems.

Additionally, testing and operating experience gained from Unit 1 will be incorporated into the Unit 2 test program.

14.2.9 TRIAL USE OF PLANT OPERATING AND EMERGENCY PROCEDURES

To the extent practical, the plant operating, emergency, and surveillance procedures will be use-tested during the test program and will also be used in the development of preoperational and initial startup procedures. The trial use of operating procedures serves to familiarize operating personnel with systems and plant operation during the testing phase and also serves to assure the adequacy of the procedures under actual or simulated operating conditions before plant operation begins.

Prior to fuel load, draft operating procedures may be utilized for equipment operation and may be informally altered to meet special test considerations.

In general the development of plant operating procedures will take place in approximately the same time frame as the preparation of preoperational and initial startup test procedures. The operating procedures will be revised as necessary to reflect experience gained during the actual testing.

During the test program, operating procedures employed for periodic tests and inspections may be used directly for certain preoperational and initial startup tests. Operating procedures used directly for preoperational or initial startup tests to obtain quantitative data to be used for evaluation of system performance will be performed in accordance with the CPNPP Station Administrative Manual.

14.2.10 INITIAL FUEL LOADING, CRITICALITY, AND POWER OPERATION

Fuel loading will begin after the required preoperational tests are satisfactorily completed. At the completion of fuel loading, the reactor upper internals and pressure vessel head will be installed and additional mechanical and electrical tests will be performed to prepare the plant for nuclear operation. After final precritical tests, nuclear operation of the reactor will begin. This phase of testing includes initial criticality, low power testing and power level escalation. The purpose of these tests is to establish the operational characteristics of the unit and core, to acquire data for the proper calibration of setpoints, and to ensure that operation is within license requirements. Section 14.2.12.2 summarizes the individual tests which will be performed from core load to rated power. The core loading and post loading tests are described below.

14.2.10.1 Fuel Loading

The overall responsibility for coordination of initial core loading activities will be exercised by the Plant Manager, or his designated representative, with technical assistance provided by Westinghouse. The overall process of initial core loading will be directed from the refueling area of the containment structure by a licensed Senior Reactor Operator.

The core configuration will be specified as part of the core design studies conducted well in advance of fuel loading. In the event mechanical damage is sustained during core loading operations to a fuel assembly, an alternate core loading scheme will be determined. Any such changes will be approved by the appropriate TU Electric and Westinghouse personnel.

Core loading procedures will specify the condition of fluid systems to prevent inadvertent changes in boron concentration of the reactor coolant; the movement of fuel to preclude the

possibility of mechanical damage; the conditions under which loading can proceed; and the responsibility and authority for continuous and complete fuel and core component accountability.

The following conditions will be met prior to core loading:

1. The reactor Containment structure will be complete and Containment integrity established and maintained during fuel loading.

2. Fuel handling tools and equipment will have been checked out and operators familiarized in the use and operation of equipment. Inspections of fuel assemblies, rod cluster control assemblies, and reactor vessel will be satisfactorily completed.

3. The reactor vessel and associated components will be in a state of readiness to receive fuel. Water level will be maintained above the nozzles and recirculation maintained to assure the required boron concentration. Boron concentration can be increased via the recirculation path to the reactor vessel.

Criteria for safe loading require that loading operations stop immediately if any of the following conditions occur.

1. An unanticipated increase in the neutron count rates by a factor of two occurs on all responding nuclear channels during any single loading step after the initial nucleus of eight fuel assemblies is loaded.

2. An unanticipated increase in the count rate by a factor of five on any individual responding nuclear channel during any single loading step after the initial nucleus of eight fuel assemblies is loaded.

3. An unanticipated decrease in boron concentration greater than 20 ppm is determined from two successive samples of the reactor coolant.

Loading operations may not be re-started until the situation is evaluated. An alarm in the Containment and main Control Room will be coupled to the source range channels with a setpoint equal to or less than five times the current count rate. This alarm will automatically alert the loading operation personnel of high count rate and an immediate stop of all operations will be required until the situation is evaluated. In the event the evacuation alarm is actuated during core loading and after it has been determined that no hazards to personnel exist, preselected personnel will be permitted to reenter the Containment to evaluate the cause and determine future action.

The core will be assembled in the reactor vessel, submerged in reactor grade water containing sufficient dissolved boric acid to maintain a calculated core effective multiplication factor of 0.95 or lower. The refueling cavity will be dry during initial core loading with the exception of the refueling canal and upender pits which contain water lubricated equipment. Core moderator chemistry conditions (particularly boron concentration) will be prescribed in the core loading procedure document and verified by chemical analysis of moderator samples taken prior to and during core loading operations.

At least two artificial neutron sources will be introduced into the core at specified points in the core during the loading program to ensure a detector response of at least 1/2 count per second attributable to neutrons.

Core loading instrumentation consists of two permanently installed source range (pulse type) nuclear channels and two temporary incore source range channels. A third temporary channel may also be used as a spare. The permanent channels, when responding, will be monitored in the Main Control Room and the temporary channels will be installed and monitored in the Containment. At least one permanent channel will be equipped with an audible count rate indicator. Both plant channels will have the capability of displaying the neutron flux level on a strip chart recorder. The temporary channels will indicate on scalers. Minimum count rates of 1/2 count per second, attributable to core neutrons will be required on at least two available source range channels at all times following installation of the initial nucleus of eight fuel assemblies. A response check of nuclear instruments to a neutron source will be performed within eight hours prior to loading of the core, or upon resumption of loading if delay is for more than eight hours.

Fuel assemblies together with inserted components (control rod assemblies, burnable poison inserts, source spider, or thimble plugging devices) will be placed in the reactor vessel one at a time according to a previously established and approved sequence developed to provide reliable core monitoring with minimum possibility of core mechanical damage. The core loading procedures document and prescribe the successive movements of each fuel assembly and its specified inserts from its initial position in the storage racks to its final position in the core. Checks will be made of component serial numbers and types at successive transfer points to guard against possible inadvertent exchanges or substitutions of components. Fuel assembly status boards will be maintained throughout the core loading operation.

An initial nucleus of eight fuel assemblies, one containing a neutron source, is the minimum source-fuel nucleus which will permit subsequent meaningful inverse count rate monitoring. This initial nucleus is determined by calculation to be markedly subcritical (Keff≤0.95) under the required conditions of loading.

Each subsequent fuel addition will be accompanied by detailed neutron count rate monitoring to determine that the just-loaded fuel assembly does not excessively increase the count rate and that the extrapolated inverse count rate ratio is behaving as expected. These results for each loading step will be evaluated before the next fuel assembly is placed in the active core region. The final, as loaded, core configuration will be subcritical (Keff≤0.95) under the required loading conditions.

14.2.10.2 Initial Criticality

Prior to initial criticality, the following tests will be performed and the results evaluated.

1. At the completion of core loading, the reactor upper internals and pressure vessel head will be installed. A pressure test will be conducted after filling, venting and heatup is completed to check the leak tightness of the vessel head installation.

2. Mechanical and electrical tests will be performed on the control rod drive mechanisms. These tests include a complete operational checkout of the mechanisms and calibration checks of the individual rod position indicators.

3. A test will be performed on the reactor trip circuits to test manual trip operation, and actual control rod assembly drop times will be measured for each control rod assembly. At all times that the control rod drive mechanisms are being tested, the boron concentration in the coolant will be maintained such that the shutdown margin requirements specified in the Technical Specifications are met. During individual RCCA or RCC bank motion, source range instrumentation will be monitored for unexpected changes in core reactivity.

4. The Reactor Control and Reactor Protection Systems will be checked with simulated inputs to produce trip signals for various trip conditions.

5. A functional electrical and mechanical check will be made of the incore nuclear flux mapping system near normal operating temperature and pressure.

Initial criticality will be achieved by a combination of shutdown and control bank withdrawal and reactor coolant system boron concentration dilution. The plant conditions, precautions and specific instructions for the approach to criticality will be specified by approved procedures.

Initially, the shutdown and control banks of control rods will be withdrawn incrementally in the normal withdrawal sequence leaving the last withdrawn control bank partially inserted in the core to provide effective control when criticality is achieved. The boron concentration in the Reactor Coolant System will then be reduced and criticality achieved by boron dilution or by subsequent rod withdrawal following boron dilution. Throughout this period, samples of the primary coolant will be obtained and analyzed for boron concentration.

Inverse count rate ratio monitoring, using data from the normal plant source range instrumentation, will be used as an indication of the proximity and rate of approach to criticality. Inverse count rate ratio data will be plotted as a function of rod bank position during rod motion and as a function of reactor makeup water addition during reactor coolant system boron concentration reduction.

14.2.10.3 Low Power Testing

Following initial criticality, a program of reactor physics measurements will be undertaken to verify that the basic static and kinetic characteristics of the core are as expected and that the values of the kinetic coefficients assumed in the safeguards analysis are conservative.

Procedures will specify the sequence of tests and measurements to be conducted and the conditions under which each is to be performed in order to ensure both safety of operation and the validity and consistency of the results obtained. If test results deviate significantly from design predictions, if unacceptable behavior is revealed, or if unexplained anomalies develop, the plant will be brought to a safe stable condition and the situation reviewed to determine the course of subsequent plant operation.

These measurements will be made at low power and primarily at or near normal operating temperature and pressure. Measurements will be made in order to verify the calculated values of control rod bank reactivity worths, the isothermal temperature coefficient, differential boron concentration reactivity worth, and critical boron concentrations. In addition, measurements of the relative power distributions will be made. For Unit 1, these measurements are conducted prior to exceeding 5% power. For Unit 2 the corresponding measurements are conducted prior to

exceeding 30% power. For Units 1 and 2, tests will be conducted on the instrumentation including power and intermediate range nuclear channels.

14.2.10.4 Power Level Escalation

After the operating characteristics of the reactor have been verified by low power testing, a program of power level escalation will bring the unit to its full rated power level in successive stages. The minimum test requirements for each successive stage of power escalation will be specified in the initial startup test procedures.

Measurements will be made to determine the relative power distribution in the core as functions of power level and control assembly bank position.

Secondary system heat balance measurements will ensure that the indications of power level are consistent and provide bases for calibration of the power range nuclear channels. The ability of the Reactor Coolant System to respond effectively to signals from primary and secondary instrumentation under a variety of conditions encountered in normal operations will be verified.

At prescribed power levels the dynamic response characteristics of the primary and secondary systems will be evaluated. System response characteristics will be measured for design step load changes, a rapid load reduction, and plant trips.

Adequacy of radiation shielding will be verified by gamma and neutron radiation surveys at selected points inside the Containment Building and throughout the station site at various power levels. Periodic sampling will be performed to verify the chemical and radio-chemical analysis of the reactor coolant.

14.2.11 TEST PROGRAM SCHEDULE

The sequential schedule for the preoperational testing of individual systems and components is shown in Figure 14.2-3. The sequential schedule for initial startup performed following fuel loading is shown in Figures 14.2-4A and B. These schedules show certain milestones at which time the tests, or portions of tests, will be completed and the overall time frame in which the tests will be conducted. The detailed schedules for testing will be prepared, reviewed and revised on a continuing basis as station construction progresses.

Preoperational testing of the various systems and components, which will continue up to fuel loading, is scheduled to commence from 10 to 18 months prior to fuel load. Initial startup tests are scheduled to be conducted over a period of approximately five months subsequent to fuel load. Fuel load, initial criticality, and power ascension tests are scheduled to be accomplished during this time period.

Station structures, systems, and components which are relied upon to prevent or mitigate consequences of postulated accidents will be fully tested to the extent practical prior to exceeding the 50 percent power level. Certain systems will have part of the preoperational testing performed after fuel load due to system configuration (e.g. control rod drive mechanism require fuel in the vessel to be fully tested). Such systems will be sufficiently tested prior to fuel load to provide reasonable assurance of successful testing after fuel load.

The development of test procedures will be an ongoing process, which will consist of preparation, review, and revision. The preparation of preoperational test procedures is scheduled to start no later than 15 months prior to fuel loading.

Preparation of initial startup procedures is scheduled to commence approximately five months prior to fuel loading. Preoperational test procedures will be available for examination by the NRC regional personnel approximately 60 days prior to the scheduled performance of the test. Initial startup test procedures will be available for examination by the NRC regional personnel approximately 60 days prior to the scheduled fuel loading date.

Testing efforts on either unit will not be diluted by activities on the other unit.

14.2.12 INDIVIDUAL TEST DESCRIPTIONS

14.2.12.1 Preoperational Tests

The test summaries for each of the preoperational tests to be performed, along with an index to these summaries, are provided in Table 14.2-2. These summaries describe the various tests which are specified as preoperational tests in Regulatory Guide 1.68 as discussed in Appendix 1A(B). The scope and titles of these summaries may not in all cases correspond directly to the actual test procedures which will be used during the Startup Program. Certain test procedures may include more than one test as described in these summaries, and in some cases tests described in one summary may be covered under more than one procedure. The overall scope and content of the tests described in these summaries will be addressed in final procedures.

There will be certain prerequisites which will apply in general to all preoperational tests. For convenience these general prerequisites are listed here rather than included in each summary.

General Prerequisites:

1. Construction activities have been completed on the system and on the necessary portions of supporting systems and any deficiencies have been properly dispositioned.

2. Functional operability of individual components or subsystems has been demonstrated for items such as valves, pumps, motors, instrumentation, and controls.

3. Test equipment necessary for test performance is available and calibrated.

4. Electrical power and air supplies are available as required for test performance.

14.2.12.2 Initial Startup Tests

The initial startup phase of the test program consists of preoperational tests deferred to the Initial Startup Program, Initial Startup Tests prior to and including fuel loading, precritical tests, initial criticality, low power physics tests, and power ascension tests. Fuel load and initial criticality procedures and tests are described in Section 14.2.10. Test summaries providing descriptions of all other major initial startup tests are provided in Table 14.2-3 along with an index for these test summaries.

TABLE 14.2-1
NOT USED

TABLE 14.2-2
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Vents and Drains System	8
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Spent Fuel Pool Cooling and Cleanup System	11
Residual Heat Removal System	12
CVCS - Charging, Letdown and Seal Water Subsystem	13
CVCS - Chemical Control, Purification and Makeup Subsystem	15
CVCS - Boron Thermal Regeneration Subsystem	17
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TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 4 of 71)

STATION SERVICE WATER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of each train of the Station Service Water System to supply adequate cooling water flows to its required loads, including the CCW heat exchangers, for the subsystem associated with unit being tested.

PREREQUISITES

1. There is sufficient water in the Safe Shutdown Impoundment to provide adequate suction for the service water pumps.

TEST METHOD

1. Verify automatic starting of the standby SSW pump upon tripping of the operating SSW pump.

2. For Unit 1 only, verify proper operation of the recirculation line pressure controllers.

3. Check for proper functioning of instrumentation, alarms, and interlocks.

4. Confirm that the isolation valving systems will properly respond to an isolation signal.

5. Verify proper system flow balancing and SSW pump hydraulic performance.

ACCEPTANCE CRITERIA

The Station Service Water System Pumps meet design requirements. System flow and pressure requirements are satisfied. Instrumentation, controls, annunciators and interlocks function properly in response to simulated or normal input signals. The system meets design requirements when operating from the emergency power source.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 5 of 71)

DEMINERALIZED AND REACTOR MAKEUP WATER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Demineralized and Reactor Makeup Water System to furnish deaerated demineralized water for use as reactor coolant.

PREREQUISITES

1. All necessary supporting equipment is operational.

TEST METHOD

1. Verify that the system is capable of producing demineralized water at the required rate.

2. Verify the proper operation of the vacuum deaerator and transfer pumps and confirm that the deaerated demineralized water is delivered to the Reactor Makeup Water Storage Tanks.

3. Verify proper operation of the reactor makeup water pumps and confirm their capability to supply reactor makeup water to all distribution points.

4. Demonstrate proper functioning of instrumentation, interlocks, and alarms.

ACCEPTANCE CRITERIA

The Demineralized Water System is capable of producing demineralized water which meets or exceeds the design criteria for water quality and capacity and is capable of supplying adequate quantities of demineralized, deaerated water to the system distribution points. Instrumentation, controls, annunciators and interlocks function properly in response to simulated or normal input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 6 of 71)

COMPONENT COOLING WATER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of each train of the Component Cooling Water System to supply adequate cooling water flows to its associated safety-related loads for the subsystem associated with unit being tested.

PREREQUISITES

1. Station Service Water System is operational

2. An acceptable water supply is available to component cooling water surge tanks.

TEST METHOD

1. Demonstrate all manual modes of operation including various pump/loop combinations.

2. Verify automatic starting of the standby CCW pump upon tripping of the operating CCW pump.

3. Demonstrate automatic isolation of the nonsafety-related loop from the rest of the subsystem, and isolation between the two safety related loops.

4. Check all other alarms and interlocks.

5. Verify proper system flow balancing of the safety and non- safety-related loops and CCW pump Hydraulic Performance.

6. Verify proper system flow balancing and CCW pump hydraulic performance.

ACCEPTANCE CRITERIA

The Component Cooling Water System pumps meet design requirements. System flow and pressure requirements are satisfied. The system supplies adequate cooling to both the safety and non-safety-related loops in the normal plant configuration and to the safety-related loops in the LOCA configuration. The system operates properly from the emergency power source. Instrumentation, controls, annunciators and interlocks, including system isolation functions, operate properly in response to normal or simulated input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 7 of 71)

PROCESS SAMPLING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Process Sampling System to provide liquid and gas samples through the correct flow path from all sample points in the primary and secondary systems, and to demonstrate the adequacy of sampling procedures and determine sample line holdup times.

PREREQUISITES

1. Plant conditions are established as necessary to facilitate drawing of liquid and gas samples from the required sampling locations.

2. Necessary tanks and sampling devices are available for receiving sample effluents and relief valve discharge.

TEST METHOD

1. Demonstrate proper system operation with regard to flow paths, flow capacity, and mechanical operability.

2. Verify the operability of the sample coolers and pressure reducing and regulating equipment.

3. Demonstrate the operability of the automatic on-line analyzers and in-line radiation monitors.

4. Check operation of instrumentation, interlocks, and alarms.

5. Demonstrate by flow rate vs line length that the sample line holdup time is within allowable limits.

6. Demonstrate adequacy of sampling procedures

ACCEPTANCE CRITERIA

The Secondary and Process Sampling System provides samples for analysis within the design requirements for flow rates, temperature and pressure. Sample line delay times are within acceptable limits. Analyzing equipment operates within design requirements. Instrumentation, controls, annunciators and interlocks function properly in response to simulated or normal input signals. The sampling procedures are adequate to provide proper samples and protect personnel.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 8 of 71)

VENTS AND DRAINS SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Vents and Drains System to collect and separate potentially radioactive and non-radioactive liquid wastes from valve and pump leakoffs, tank overflows, and tank drains.

PREREQUISITES

1. The Liquid Waste Processing System is capable of receiving liquid wastes and drainage.

TEST METHOD

1. Demonstrate the flow capability of the system.

2. Verify that all sump pumps are started and stopped automatically by the sump level controllers.

3. Check for proper operation of instrumentation, alarms, and interlocks.

ACCEPTANCE CRITERIA

The Vents and Drains System is capable of processing collected liquids at or exceeding design flow rates. Sump pump controls and interlocks function in accordance with design requirements. Potentially radioactive and non-radioactive liquids are separated as per the design criteria.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 9 of 71)

FIRE PROTECTION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Fire Protection System, including its fire detection and fire suppression functions in accordance with design requirements.

PREREQUISITES

1. The Fire Protection System flow element and indicator and associated instrumentation is operational.

2. An acceptable treated water supply is available from the Fire Protection System Water Storage Tanks for fire pump and system testing.

TEST METHOD

1. Verify the proper functioning of the fire detection devices to activate the automatic fire protection system, alert the Control Room operators, initiate fire alarms, and to activate automatic closure of fire dampers, as required.

2. Verify proper operation of the fire suppression system and obtain flow rates through the underground loop.

3. Demonstrate the automatic start feature of the Fire Protection System pumps.

4. Verify proper flows of the Fire Protection System Pumps by use of the system flow element and indicator.

5. Demonstrate flow in wet pipe sprinkler systems.

6. Demonstrate proper operation of system instrumentation, alarms, and interlocks.

ACCEPTANCE CRITERIA

The Fire Detection portion of the Fire Protection System, in response to simulated input signals, provides indication, annunciation and/or fire suppression activation outputs in accordance with system design. The fire pumps meet or exceed the CPNPP Fire Protection Report requirements and provide flow to the water fire suppression system. The Halon Fire Suppression System interlocks, controls, annunciators, instrumentation and active fire isolation devices function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

Halon Fire Suppression Systems will be accepted utilizing either of the two methods below:

1. Testing and certification provided by the equipment supplier, and accepted by TU Electric or

2. Testing performed by TU Electric (or designated vendor), reviewed by JTG and approved by the Manager, Startup.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 11 of 71)

SPENT FUEL POOL COOLING AND CLEANUP SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Spent Fuel Pool Cooling and Cleanup System to provide required cooling water flows to the spent fuel pool and verify proper operation of the purification loops and skimmers.

PREREQUISITES

1. As required for portions of the test, the Spent Fuel pool is filled with demineralized or borated water.

TEST METHOD

1. Verify proper operation and actuation of pumps and valves and verify correct flows in the spent fuel pool cooling loops.

2. Verify correct cooling water flows to the heat exchangers.

3. Verify proper operation and actuation of pumps and valves in the spent fuel pool cleanup loop and verify correct flows to filters and demineralizers.

4. Verify proper actuation and operation of the pump and skimmers in the spent fuel pool skimmer loop and verify correct flows in the loop.

5. Check operation of instrumentation, interlocks, and alarms.

6. Verify correct flows entering and leaving the system.

7. Verify proper operation of the spent fuel pool anti-siphon device.

ACCEPTANCE CRITERIA

The Spent Fuel Pool Cooling and Cleanup System pumps meet design requirements. System flow and pressure requirements are satisfied. Flows to heat exchangers, filters, demineralizers, and to and from the system are in accordance with design. The fuel pool anti-siphon devices and skimmer loop operates properly. Instrumentation, controls, annunciators and interlocks function properly in response to simulated or normal input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

RESIDUAL HEAT REMOVAL SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Residual Heat Removal System and its capability to provide recirculation flows required to remove heat from the Reactor Coolant System.

PREREQUISITES

1. For portions of the test, as required, the RCS is filled with water.

TEST METHOD

1. Verify the Residual Heat Removal System (RHRS) inlet valve interlocks and trips operate properly.

2. Demonstrate the miniflow control valves operate properly.

3. Demonstrate acceptable pump performance while operating on the miniflow bypass line with the discharge isolation valves closed. Record sufficient data to insure pump operation is within acceptable regions of the pump manufacturer's flow curves.

4. Demonstrate recirculation capability within the isolated RHRS.

5. Demonstrate RHRS operation during heatup with letdown through RHRS.

6. Demonstrate RHRS operation during plant cooldown following hot functional testing.

7. Verify proper operation of instrumentation, interlocks and alarms.

Note: The safety injection functions of RHRS will be demonstrated during the testing of the Safety Injection System.

ACCEPTANCE CRITERIA

The Residual Heat Removal System and components function in accordance with design requirements. System flow capability in the various operating modes meets design requirements. Controls, interlocks, instrumentation and annunciators function properly in response to normal or simulated input signals.

TABLE 14.2-2
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CHEMICAL AND VOLUME CONTROL SYSTEM CHARGING, LETDOWN, AND SEAL WATER SUBSYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Chemical and Volume Control System, including capability to maintain charging and letdown flows, to maintain seal-water injection flow to the reactor coolant pumps, and to provide auxiliary spray flow to the pressurizer.

PREREQUISITES

1. Applicable portions of the Reactor Coolant System are capable of operationally interfacing with the Chemical and Volume Control System as required.

2. A cooling water supply is available for the heat exchangers as necessary for test performance.

3. Systems required to supply cover gas to the Volume Control Tank are operational and adequate supplies of gas are available.

TEST METHOD

1. Verify proper functioning of charging and letdown system components, including the hydraulic performance of the charging pumps, and operability of heat exchangers, letdown orifices, and control valves.

2. Demonstrate the capability to maintain seal water flow to the reactor coolant pumps, and the proper operation of seal water return and seal water bypass systems.

3. Verify proper flows and pressure drops for seal injection, seal water return, and reactor coolant filters.

4. Verify proper operation of excess letdown and alternate charging systems and components.

5. Verify proper operation of auxiliary spray to the pressurizer.

6. Verify proper operation of the volume control tank level control system, diversion valves, and cover gas system.

7. Check operation of instrumentation, interlocks, and alarms.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

ACCEPTANCE CRITERIA

The charging pumps meet design requirements. System flow and pressure requirements are satisfied. The charging and letdown normal and alternate flow paths, including heat exchangers, letdown orifices, and control valves, function in accordance with design requirements. The Volume Control Tank Level System, diversion valves and cover gas system function properly. Instrumentation, controls, annunciators and interlocks function properly in response to normal or simulated input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 15 of 71)

CHEMICAL AND VOLUME CONTROL SYSTEM
CHEMICAL CONTROL, PURIFICATION, AND MAKEUP SUBSYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the subsystem of the Chemical and Volume Control System which is used to maintain control of water chemistry conditions, to remove gases from reactor coolant, to provide reactor makeup control, and to adjust reactor coolant boron concentration.

PREREQUISITES

1. Applicable portions of the Reactor Coolant System are capable of operationally interfacing with the CVCS as required.

2. Portions of systems supplying gas to the volume control tank are operational and adequate supplies of gas are available.

TEST METHOD

1. Demonstrate the chemical control function of the CVCS by verifying the capability of the system to introduce chemicals into the charging flow for pH and oxygen control, and that the system is capable of maintaining a gas pressure in the volume control tank as required during the applicable modes of operation.

2. Verify proper flows to mixed bed demineralizers, and determine pressure drops and effectiveness of demineralizers and filters.

3. Demonstrate the CVCS reactor make up control function, including maintaining a fluid inventory in the volume control tank, and testing of the automatic makeup, dilute, alternate dilute, borate, and manual modes of reactor makeup control.

4. Confirm proper functioning of pumps and control valves.

5. Check operation of instrumentation, interlocks, and alarms.

6. Verify proper operation of pumps, mixers and valves in the Boric Acid System.

7. Verify that a solution of boric acid can be mixed, transferred, recirculated and stored.

8. Verify that boric acid can be transferred to other systems as required.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

ACCEPTANCE CRITERIA

The chemical control, purification, and makeup subsystem, the boric acid subsystem, and components of the Chemical and Volume Control System operate in accordance with the design requirements. Reactor makeup control, gas pressure control and chemical control functions are demonstrated to meet design performance requirements. Instrumentation, controls, alarms and interlocks function properly in response to normal or simulated input signals. Boric acid is able to be batched, stored and transferred.

TABLE 14.2-2
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CHEMICAL AND VOLUME CONTROL SYSTEM
BORON THERMAL REGENERATION SUBSYSTEM TEST SUMMARY

OBJECTIVE

Demonstrate the operability of the boron thermal regeneration subsystem.

PREREQUISITES

1. Applicable portions of the Reactor Coolant System are capable of operationally interfacing with the Chemical and Volume Control System.

2. A cooling water supply is available as required for heat exchangers.

TEST METHOD

1. Verify flows to and from thermal regeneration demineralizers, including proper operation of flow control valves, chiller units, heat exchangers, and filters.

2. Verify proper operation of water chillers and temperature control devices.

3. Check operation of instrumentation, interlocks, and alarms.

ACCEPTANCE CRITERIA

The boron thermal regeneration subsystem components of the Chemical and Volume Control System operates in accordance with design specifications. Flows to and from the thermal regeneration demineralizers and control of temperature into the demineralizers are in accordance with design requirements, instrumentation, controls, alarms and interlocks function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

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BORON RECYCLE SYSTEM TEST SUMMARY

OBJECTIVE

Demonstrate the operability of the Boron Recycle System.

PREREQUISITES

1. The recycle evaporator of the Boron Recycle System, and other interrelated or supporting equipment, are operational as required.

2. A steam supply and cooling water supply are available for the evaporator packages as required.

TEST METHOD

1. Demonstrate the capability of the Boron Recycle System to collect water from its various designated sources.

2. Verify proper operation of system components, including the evaporator package, the feed, distillate and concentrates pumps.

3. Verify the proper operation of the piping heat tracing.

4. Check for operability of instrumentation, alarms, and interlocks.

ACCEPTANCE CRITERIA

The Boron Recycle System components function in accordance with design specifications. Flow paths are verified operational. Heat tracing and other system related heating methods maintain system minimum temperatures per design requirements. Interlocks, alarms, controls and instrumentation function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

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SAFETY INJECTION SYSTEM HYDRAULIC PERFORMANCE TEST SUMMARY

OBJECTIVE

Demonstrate the ability of the centrifugal charging pumps, the safety injection pumps, and RHR pumps to deliver required flows to the RCS during injection and recirculation modes of operation.

PREREQUISITES

1. The Refueling Water Storage Tank is filled with demineralized water or boric acid at refueling concentration.

2. The Reactor Coolant System is empty and the reactor vessel head is removed as required for full flow testing.

3. The Reactor Coolant System is closed and in a hot condition as required for portions of the test.

TEST METHOD - INJECTION MODE

1. Verify that all valves and components required for SIS operation are sequenced properly and are actuated within minimum required times. Verify proper operation of all instrumentation, interlocks and alarms.

2. With the RHR, centrifugal charging and SI pumps aligned to take suction from the RWST, sufficient data will be taken with the pump discharge valves shut to ensure satisfactory operation in the miniflow mode.

3. With reactor vessel head and internals removed, and the pumps taking suction from the RWST or the Reactor Coolant System hot legs (RHR pumps only), perform full flow tests for each pump with the miniflow valves closed. Measure flow and discharge pressure and adjust system as necessary to ensure that pumps do not exceed their maximum runout conditions. Measure flows in branch lines and adjust as necessary to ensure that flow distributions for all injection lines are within limits.

4. Align high head pumps to deliver to RCS hot legs. Measure and adjust flows as necessary.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

TEST METHOD - RECIRCULATION MODE

1. Ensure that suction lines from the Containment Building sumps are clean and free from obstruction.

2. Verify auto operation of RHR pump containment sump suction valves upon receipt of RWST lo-lo level signal.

3. Align safety injection and centrifugal charging pumps to take suction from the discharge of the RHR pumps and deliver to the RCS cold legs with RHR taking suction from the RWST or the Reactor Coolant System hot leg. Measure flows and adjust as necessary to ensure that RHR pumps deliver adequate flow to high head pump suction under runout conditions.

4. Verify RHR pump NPSH and anti-vortex capabilities using containment sump model testing.

ACCEPTANCE CRITERIA

The flow rates and flow distribution for all branch lines and the pump runout flows are within design limits as used in FSAR accident analyses with the discharge portion of the SIS aligned for injection and recirculation modes of operation. Instrumentation, controls, interlocks and alarms function properly in response to normal or simulated input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 21 of 71)

SAFETY INJECTION ACCUMULATORS TEST SUMMARY

OBJECTIVE

Verify the discharge characteristics and proper system actuation for each of the SI accumulators.

PREREQUISITES

1. The Reactor Coolant System is drained down and the reactor vessel head and internals are removed for the full flow portion of the test, and is closed and in a hot condition as required for portions of the test relating to check valve hot operability.

2. A supply of nitrogen or compressed air is available for pressurizing the accumulators.

3. The accumulators are filled to their normal operating level with demineralized water or boric acid to refueling concentration.

TEST METHOD

1. Each accumulator is partially pressurized and its discharge valve is then opened allowing discharge into the RCS. Level and pressure measurements are used to calculate line resistance and accumulator injection performance.

2. Verify operability of nitrogen fill, venting and relief valves, accumulator drains, and accumulators makeup systems.

3. Verify ability of the accumulator isolation valves to open automatically on a safety injection signal against a differential pressure equal to the accumulator at maximum operating pressure and the Reactor Coolant System depressurized.

4. Verify operability of check valves in the accumulator discharge lines with the RCS in a hot condition.

5. Verify proper operation of all instrumentation, interlocks, and alarms.

ACCEPTANCE CRITERIA

Accumulator discharge performance is within design limits used for FSAR accident analyses. The accumulator systems function in accordance with design requirements. Controls, interlocks, alarms and instrumentation function properly in response to simulated or normal input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

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CONTAINMENT SPRAY SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the hydraulic performance of the containment spray pumps, proper functioning of spray nozzles and headers, and the proper operation and actuation of the system components including the chemical additive system.

PREREQUISITES

1. A water supply is available for the containment spray pumps.

2. An air supply is available for testing the containment spray nozzles.

3. The containment spray pumps discharge lines are isolated to prevent spraying water into the Containment Building.

4. Verify that the piping between valves ICT-4776 and manual isolation valves ICT-141; and IHV-4777 and manual isolation valve ICT-146 has been satisfactorily flushed.

TEST METHOD

1. Through test connections, pass air under pressure to the spray nozzles and ring headers to ensure they are free of obstructions.

2. Operate the containment spray pumps through the RWST recirculation path to verify pump hydraulic performance.

3. Verify proper sequencing of valves and pumps upon safeguards actuation signals.

4. Demonstrate that the containment spray chemical additive system will automatically actuate and function as required.

5. Verify proper performance of chemical eductors and chemical additive tank vacuum breakers.

ACCEPTANCE CRITERIA

Containment spray pumps meet hydraulic design criteria. Unobstructed flow paths are verified for the containment spray nozzles and ring headers. Sequencing of all valves and pumps upon safeguards action signals occurs properly. Chemical additive system functions in accordance with design requirements. Controls, interlocks, instrumentation and annunciators, in response to normal or simulated signals, function properly.

TABLE 14.2-2
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INTEGRATED ENGINEERED SAFETY FEATURES TEST SUMMARY

OBJECTIVE

Demonstrate proper automatic alignment and operation of all safeguards systems upon a safety injection signal using both normal offsite and emergency power sources.

PREREQUISITES

1. The RCS is cold and drained down and the reactor vessel head and internals are removed.

2. The Refueling Water Storage tank has an adequate supply of demineralized water or boric acid at refueling concentration.

3. The containment spray pump discharge lines are manually isolated to prevent spraying into the Containment Building.

4. Actuation circuitry has been tested and is capable of actuating all equipment upon a manually initiated safety injection signal.

5. All systems required to actuate on a safety injection signal are operational and are aligned for normal operation.

TEST METHOD

1. With the SIS aligned for normal power operation, manually initiate a safety injection signal (“S” signal) and verify proper starting of all safeguards pumps and measure flowrates. Verify proper alignment of valves, including centrifugal charging pump suction and discharge valves, and pump mini-flow lines.

2. Verify proper actuation, alignment, and operation of all systems which are required to operate or change mode of operation upon receipt of a safety injection signal including the Station Service Water System, Component Cooling Water System, and required ventilation systems.

3. Repeat test with no offsite power supply available and verify proper operation of emergency diesel generators in conjunction with safety injection actuation including proper load sequencing, pump start times, and valve sequencing.

ACCEPTANCE CRITERIA

The safeguards systems actuate in the proper sequence, automatically align in the proper manner, and provide required flowrates to the RCS injection points in accordance with design requirements.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

LIQUID WASTE PROCESSING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Liquid Waste Processing System to collect and process potentially radioactive wastes for recycle or for release to the environment.

PREREQUISITES

1. All necessary supporting equipment is operational.

TEST METHOD

1. Demonstrate capability of the Liquid Waste Processing System to process waste water in accordance with design requirements.

2. Demonstrate proper operation of all components in the system, including the waste evaporator, waste holdup tanks, system demineralizers, and laundry reverse osmosis system.

3. Verify the proper operation of the reactor coolant drain tank and associated equipment.

4. Check for proper functioning of instrumentation, interlocks, controls, and alarms.

ACCEPTANCE CRITERIA

The Liquid Waste Processing System collects, segregates and processes liquid wastes in accordance with design requirements. Operation of system components meet design specifications. Instrumentation, controls, annunciators and interlocks function properly in response to simulated or normal input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

SOLID WASTE PROCESSING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the subsystem used to transfer spent resins, evaporator concentrates, and chemical drain tank effluents and of the baling subsystem used to package low-radiation level compressible wastes.

PREREQUISITES

1. All necessary supporting equipment is operational.

2. Equipment, such as pumps, piping, baling units and associated controls and interlocks are operational.

TEST METHOD

1. Demonstrate the operability of the waste baling units.

2. Demonstrate the ability to transfer spent resins evaporator bottoms and chemical drain tank effluents from their source to the bulk disposal outlet.

3. Verify proper operation of associated instrumentation and interlocks.

ACCEPTANCE CRITERIA

The Solid Waste System is capable of transferring waste forms to the bulk disposal outlet and baling low level compressible waste. Controls, interlocks, and instrumentation function properly.

TABLE 14.2-2
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(Sheet 26 of 71)

GASEOUS WASTE PROCESSING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Gaseous Waste Processing System including its capability to remove and process gases from specified sources, including the volume control tank, boron recycle evaporator, reactor coolant drain tank, and waste evaporator.

PREREQUISITES

1. All necessary supporting equipment is operational.

TEST METHOD

1. Demonstrate proper operation of the system and its components, including filling and isolation of the gas decay tanks, waste gas compressor performance, and the catalytic recombiner operation.

2. Check for proper functioning of instrumentation, interlocks, and alarms.

ACCEPTANCE CRITERIA

The Gaseous Waste Processing System components meet design operational requirements. Flow paths and flow rates are in accordance with design criteria. Instrumentation, interlocks, controls and alarms function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

CONTAINMENT ATMOSPHERE MONITORING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Containment Atmosphere Monitoring System.

PREREQUISITES

1. Setpoints for alarms and indicating lights have been set.

TEST METHOD

1. Demonstrate proper operation of the hydrogen analyzer/monitoring equipment by functional testing from the microprocessor keyboard.

2. Demonstrate proper operation of the temperature and humidity instrumentation.

3. Demonstrate proper operation of the containment pressure transmitters.

ACCEPTANCE CRITERIA

The Containment Atmosphere Monitoring System functions in accordance with design requirements.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 28 of 71)

PROCESS RADIATION MONITORING SYSTEM TEST SUMMARY
(UNIT 1)

OBJECTIVE

To demonstrate the capability of the Process Radiation Monitoring System to continuously monitor radiation levels attributed to plant processes and plant effluents, record radiation levels, and initiate radiation alarm signals.

PREREQUISITES

1. A remotely operable check source is available for each radiation detector as required.

TEST METHOD

1. Verify proper operation and response of the radiation monitors.

2. Check for proper setting of the low-level radiation alarm setpoints.

3. Verify proper operation of Control Room high radiation level alarms.

4. Ensure proper operation of indicators and recording devices.

ACCEPTANCE CRITERIA

The radiation monitors respond properly to radiation check sources. Alarm setpoints are in accordance with design criteria. Remote instrumentation and recording devices operate in accordance with design requirements.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 29 of 71)

PROCESS RADIATION MONITORING SYSTEM TEST SUMMARY
(UNIT 2)

OBJECTIVE

To demonstrate the capability of the Process Radiation Monitoring System to continuously monitor radiation levels attributed to plant processes and plant effluents, record radiation levels, and initiate radiation alarm signals.

PREREQUISITES

1. A remotely operable check source is available for each radiation detector as required.

2. Each radiation monitor and detector(s) has been calibrated using a radioactive source of known activity.

TEST METHOD

1. Verify proper operation and response of the radiation monitors.

2. Check for proper setting of the low-level radiation alarm setpoints.

3. Verify proper operation of Control Room high radiation level alarms.

4. Ensure proper operation of indicators and recording devices.

ACCEPTANCE CRITERIA

The radiation monitors successfully pass their calibration tests. Alarm setpoints are in accordance with design criteria. Remote instrumentation and recording devices operate in accordance with design requirements.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

AREA RADIATION MONITORING SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Area Radiation Monitoring System to provide continuous surveillance of radiation dose rates throughout accessible areas of the plant, alert personnel of excessive dose rate levels, and provide direct reading indication and recording of dose rates at each monitor location.

PREREQUISITES

1. Setpoints for radiation alarms and indicating lights have been set.

2. A remotely operable check source is available for each radiation detector as required.

TEST METHOD

1. Verify proper operation of the Area Radiation Monitoring System, including the indicating and recording functions of the system.

2. Check for proper setting of the radiation alarm setpoints.

3. Verify that the Control Room alarms will annunciate both high radiation levels and circuit failures.

4. Ensure proper operation of instrumentation, alarms, and interlocks.

ACCEPTANCE CRITERIA

Radiation monitors respond properly to radiation check sources. Alarm setpoints are in accordance with design criteria. Remote instrumentation, recording devices and interlocks operate in accordance with design requirements.

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CONTROL ROOM VENTILATION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Control Room Ventilation System to maintain suitable and safe ambient conditions for operating equipment and personnel in the Control Room and associated areas.

PREREQUISITES

1. A cooling water supply is available for the Control Room condensing air compressor units.

2. Compressed air is available for system valve and damper operators.

TEST METHOD

1. Demonstrate proper functioning of the Control Room Ventilation System in the normal emergency recirculation and emergency ventilation modes of operation.

2. Verify that the system responds properly to safeguards actuation signals.

3. Verify proper pressure differentials.

4. Demonstrate proper operation of chlorine gas detectors, fans, valves, dampers, and filters.

5. Check operation of system instrumentation, interlocks, and alarms.

ACCEPTANCE CRITERIA

The Control Room Ventilation System operates in accordance with design requirements. Flow and pressure requirements are met. Filtering systems exceed requirements. Alarms, detectors, and interlocks function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

AUXILIARY, FUEL, AND SAFEGUARDS BUILDING VENTILATION TEST SUMMARY

OBJECTIVE

To demonstrate the proper operation of the ventilation systems for the Auxiliary Building, Safeguards Building, and Fuel Building.

PREREQUISITES

1. A cooling water supply is available for the cooling coils.

2. Compressed air is available for valve damper operators.

TEST METHOD

1. For the Auxiliary Building, Safeguards Building, and Fuel Building ventilation systems, demonstrate proper operation of the ventilation fans and filters, and verify adequate air flow to areas served by each system.

2. Demonstrate proper operation of the inlet and exhaust isolation dampers associated with equipment rooms in the Safeguards Building.

3. Demonstrate proper operation of the auxiliary cooling units in the Safeguards Building equipment rooms.

4. Verify the system’s ability to maintain proper building pressure differentials.

5. Ensure proper operation of instrumentation, interlocks, and alarms.

6. Demonstrate adequate performance of each ESF pump room cooler.

ACCEPTANCE CRITERIA

The Auxiliary Building, Safeguards Building, and Fuel Building ventilation systems operate in accordance design requirements. Air flows and pressure differentials are within design specification. The isolation dampers, alarms and interlocks function properly.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 33 of 71)

CONTAINMENT VENTILATION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Containment Ventilation System to provide containment air-recirculation, control rod drive mechanism cooling, neutron detector well and nozzle support cooling, preaccess filtration, containment purging, hydrogen purging, and reactor coolant pipe penetration cooling.

PREREQUISITES

1. A cooling water supply is available for the fan-cooling units of the system.

2. For testing portions of the system as applicable, the control rod drive mechanisms and neutron detector are capable of being energized, and plant conditions are established as required.

3. Compressed air is available for valve and damper operators.

TEST METHOD

1. Demonstrate the capability of the Containment Ventilation System to provide for containment recirculation and heat removal, by testing operation of the axial fans, centrifugal water chillers and the cooling coils, and by ensuring adequate flow is delivered to components and areas inside Containment as required.

2. Verify that the control rod drive mechanism shroud ventilation units are capable of maintaining temperatures within the CRDM shroud within design limits.

3. Demonstrate proper operation of the fans, cooling coils, and inlet and outlet dampers associated with the neutron detector well and nozzle support cooling system, including verification during Hot Functional Testing and that the cooling system will maintain neutron detector temperature within specifications.

4. Confirm the capability of the containment preaccess filtration units to provide air circulation and filtration throughout all containment areas as required.

5. Verify proper operation of the containment purge supply and purge exhaust equipment, including functioning of the heating coils and fans.

6. Demonstrate proper operation of all Containment Ventilation System instrumentation, interlocks and alarms.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

7. Demonstrate during hot functional testing that the reactor coolant primary shield pipe penetration cooling system maintains concrete within design temperature limits.

8. Verify proper operation of the Hydrogen Purge Supply and Purge Exhaust systems.

ACCEPTANCE CRITERIA

The Containment Ventilation System components function in accordance with design requirements. Adequate ventilation flow is provided to containment areas to maintain or limit temperatures to design valves. System interlocks, instrumentation, alarms and controls operate properly. The reactor coolant pipe penetration concrete shall be maintained less than design limits as specified in FSAR Section 9.4A.2.7.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

COMBUSTIBLE GAS CONTROL SYSTEMS TEST SUMMARY

OBJECTIVE

To demonstrate the proper operation of the Hydrogen Recombiners and the Hydrogen Purge System.

PREREQUISITES

1. The hydrogen recombiners are operable and their associated power panel is energized.

2. The Hydrogen Purge System is operational.

TEST METHOD

1. Verify the proper operation of the hydrogen recombiners, including the heaters, controllers and temperature indicators.

2. Verify proper operation of the Hydrogen Purge System, including the system fans and filters.

ACCEPTANCE CRITERIA

The combustible gas control systems operate in accordance with design criteria. The hydrogen recombiners, system fans and filters perform their design function. Instrumentation, controls, annunciators and interlocks function properly in response to normal or simulated signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

ELECTRICAL AREA AND BATTERY ROOM VENTILATION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the proper operation of the Electrical Area and Battery Room Ventilation Systems.

PREREQUISITES

1. Supporting systems are operational as required.

2. Compressed air is available for valve and damper operations.

TEST METHOD

1. For the Electrical Area and Battery Room Ventilation System, demonstrate proper operation of the ventilation supply and exhaust fans, heating and cooling coils, shutoff louvers and filters, and adequate flow to areas served by each portion of the system.

2. Demonstrate operability of the exhaust fans and ducts with each battery room.

3. Verify proper operation of the Uncontrolled Access Area Ventilation System and demonstrate its capability to supply ventilation airflow to each of the battery rooms.

ACCEPTANCE CRITERIA

The Electrical Area and Battery Room Ventilation Systems operate in accordance with design requirements and provide adequate ventilation flows. Components which receive emergency power operate properly when supplied from their emergency power sources.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

DIESEL GENERATOR COMPARTMENT VENTILATION SYSTEMS TEST SUMMARY

OBJECTIVE

To demonstrate the capability of each diesel generator compartment ventilation system to provide adequate ventilation for its associated diesel generator compartment.

PREREQUISITES

1. All necessary supporting equipment is operational.

TEST METHOD

1. Demonstrate proper operation of the system intake and exhaust louvers and exhaust fans.

2. Verify that the ventilation system exhaust fans start automatically on receipt of a diesel generator start signal.

3. Verify adequate ventilation during Emergency Diesel Generator operations.

4. Check operation of instrumentation and alarms.

ACCEPTANCE CRITERIA

The Diesel Generator Compartment Ventilation Systems operate in accordance with design requirements and maintain the diesel generator compartment space temperature at or below design maximum when the diesel generators are operating. Instrumentation, controls, annunciators and interlocks function properly.

TABLE 14.2-2
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DIESEL GENERATORS TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the diesel generator unit and associated auxiliaries, including the proper starting and load sequencing.

PREREQUISITES

1. DC power is available for controls, alarms, protective relays, air starting solenoid valves and generator field flashing.

2. Compressed air is available for air starting of the diesel generator unit.

TEST METHOD

1. Verify proper performance of the diesel generator fuel oil storage and transfer system, cooling and heating systems, and the lubricating oil system.

2. Demonstrate the capability of the unit to start manually and automatically as designed and attain the required voltage and frequency within the acceptable time limitation.

3. Verify the capability of the diesel generator unit to accept the sequenced equipment by utilizing actual plant loads appropriate for existing plant conditions. The loads will be added at the proper sequence and time duration, while maintaining an acceptable voltage and frequency.

4. Verify proper operation of voltage and frequency controllers. Perform load rejection tests as required and verify proper operation of overspeed control devices.

5. Perform at least 35 valid start and load tests per diesel generator with no failures from cold ambient conditions to at least 50 percent continuous rating.

6. Demonstrate loading capability for at least 24 hours, of which 22 hours will be a load equivalent to the diesel generator continuous rating and 2 hours will be a load equivalent to the diesel generator 2-hour rating.

7. Demonstrate the ability to synchronize the diesel generator unit with offsite power while the unit is connected to the emergency load, transfer this load to the offsite power, isolate the diesel generator unit, and restore it to standby status.

8. Demonstrate that the diesel generator capability to supply emergency load is not impaired during periodic testing.

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PREOPERATIONAL TESTS SUMMARIES INDEX

9. Demonstrate functional capability at full-load conditions by reperforming the automatic start and load sections of Steps 2 and 3 immediately after Step 6 is completed.

10. Demonstrate that no common failure modes exist by starting the redundant diesel generators simultaneously.

ACCEPTANCE CRITERIA

The diesel generators supply emergency power to the 6.9KV Class 1E buses on loss of the preferred and alternate offsite power feeds to the emergency buses. The diesels start, attain speed and voltage and accept emergency load or rated load as designed. The fuel oil, lubricating oil, cooling and heating systems function as designed. Interlocks, alarms, controls and tripping devices function properly in response to normal or simulated input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

AC POWER DISTRIBUTION TEST SUMMARY

OBJECTIVE

To demonstrate that the AC Power Distribution System provides a reliable power source to safety-related loads, and that no interaction occurs between redundant trains.

PREREQUISITES

1. Control power is available for necessary switchgear.

2. Diesel generator sets and supporting auxiliaries, including DC power, are operational, where applicable.

3. Metering and relaying circuits are calibrated.

TEST METHOD

1. Confirm that loss of either redundant safety-related train will not impair the ability of the remaining system to supply power to required safety-related loads.

2. Verify performance of alarms, tripping devices, and interlocks.

3. Check local and remote operation and indication of switchgear.

4. Verify proper operation of manual/automatic transfer schemes for;

    a. For Class 1E 6.9KV Buses -

        1. Automatic transfer, slow, from preferred off-site source to alternate off-site source. Automatic transfer, slow, from preferred off-site to on-site, and from alternate off-site to on-site sources.

        2. Manual transfer, live, from any one of the three sources - preferred off-site, alternate off-site and on-site - to the other (six combinations).

    b. For Non-Class 1E 6.9kV Buses

        1. Automatic transfer fast and slow from unit auxiliary transformer to off-site source.

        2. Manual transfer from off-site source to unit auxiliary transformer and vice versa.

TABLE 14.2-2
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5. Verify proper load shedding and/or undervoltage tripping of all 6.9kV and 480V switchgear.

6. Verify load-carrying capability of Class 1E and non-Class 1E transformers, cables, switchgear and feeder breakers.

ACCEPTANCE CRITERIA

The AC Power Distribution System provides power to safety-related loads when supplied from normal and emergency power sources. No interaction occurs between redundant trains. Automatic transfer of power supplies to vital buses occurs properly and meets design requirements. Interlocks, alarms, controls and tripping devices function properly in response to normal or simulated input signals. Emergency bus voltages are maintained within design limits under load.

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DC POWER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the ability of the two redundant 125 VDC systems to supply power to safety-related loads during normal plant conditions and upon loss of all AC power system.

PREREQUISITES

1. Batteries are maintained in a nominally fully charged condition.

2. AC power is available to battery chargers.

3. Protective devices (including ground detection), breakers, and system status sensing devices, relays and meters have been calibrated for desired response to abnormal system conditions.

4. Distribution components, breakers, cables and panels have been tested and verified energized, free of grounds, providing nominal DC power to all Class 1E DC loads.

TEST METHOD

1. Confirm that the batteries have sufficient capacity to carry essential loads continuously without battery chargers for the minimum required time, and that individual cell limits are not exceeded.

2. Confirm that loss of either redundant safety-related train will not impair the ability of the remaining system to supply power to safety-related loads.

3. Check functioning of alarms, local and remote indicators, breaker status indications, and other protective devices including transfer devices and interlocks.

4. Demonstrate the ability of battery chargers to recharge the batteries from the design minimum charge state to the fully charged state within 24 hours while supplying the steady state loads under all modes of plant operation.

ACCEPTANCE CRITERIA

The batteries demonstrate the ability to carry the essential loads continuously for the design time period with the battery chargers out of service. The battery chargers maintain the batteries at full charge and are demonstrated to be able to restore the batteries to full charge within design time limits. Protective devices, controls, interlocks, alarms and instrumentation function properly in response to normal or simulated inputs.

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PREOPERATIONAL TESTS SUMMARIES INDEX

INSTRUMENTATION AND CONTROL POWER SUPPLY SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Class 1E I&C Power Supply System, including the proper functioning of the system inverters.

PREREQUISITES

1. 125 VDC power supplies are available to the inverter units.

2. The common backup source of 120 VAC power is available to the instrument distribution panels.

TEST METHOD

1. Verify that the system inverters maintain the current and voltage to the instrument distribution panels within the required limits.

2. Demonstrate that the loss of either redundant I&C Power Supply System or its associated power supplies will not cause a loss of power to more than one of the redundant vital buses.

3. Check operation of all interlocks and alarms.

4. Upon loss of the normal power supply, demonstrate the capability of the inverters to maintain output by means of the alternate power supply.

ACCEPTANCE CRITERIA

The Instrumentation and Control Power Supply System operates in accordance with design requirements. Inverters transfer to alternate power supplies at the correct setpoint. Redundant trains are verified independent. Alarms and interlocks function properly.

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COMPUTER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate that both the hardware and software present accurate indications of plant parameters and processed information.

PREREQUISITES

1. The computer system has been installed as necessary for test performance

2. Inputs or simulated inputs are available for testing

TEST METHOD

1. Verify inputs are connected correctly

2. Verify that both hardware and software processing of input signals results in accurate information

ACCEPTANCE CRITERIA

The computer system records and displays accurate information from plant parameters using both direct inputs and processed information.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

DC EMERGENCY LIGHTING SYSTEMS TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the DC Emergency Lighting System to provide lighting in those locations where safety-related functions are performed.

PREREQUISITES

1. DC power is available to the Control Room DC Emergency Lighting System.

2. The battery-pack operated lighting units are in service.

TEST METHOD

1. Demonstrate proper operation of the DC Emergency Lighting System in the Control Room.

2. Verify the operability of the 8 hour battery-pack lighting units and integral automatic charger.

ACCEPTANCE CRITERIA

The DC Emergency Lighting System operates in accordance with design requirements, supplying illumination to the required areas. The DC powered lighting initiates when AC powered emergency lighting is lost.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

COMMUNICATION SYSTEMS TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the intraplant and plant-to-offsite communication systems to provide adequate communication coverage and audibility.

PREREQUISITES

1. As required to check the audibility of intraplant communication systems, equipment is operating to create an ambient noise level that would be expected during normal plant operation.

TEST METHOD

1. Demonstrate proper functioning of the Public Address System, including the capability of the voice paging channel output to be clearly audible over the highest expected noise levels.

2. Verify proper operation of the Intraplant Telephone System and Intraplant Sound-Powered Telephone System.

3. Demonstrate operation of the Offsite Communication Systems, including the public telephone and two-way radio systems and the direct telephone line to the system dispatcher.

4. Verify proper functioning of the Emergency Evacuation Alarm System.

ACCEPTANCE CRITERIA

The communication systems function in accordance with design requirements. The Public Address and Emergency Evacuation Alarm Systems are audible during the highest expected ambient noise levels. Intraplant communications channels function properly and communications to offsite can be established by public telephone, two- way radio and direct telephone line to the system dispatcher.

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REACTOR CONTROL SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Reactor Control System to respond properly to simulated input signals and to transmit proper control signals to other plant control systems and components.

PREREQUISITES

The Reactor Control System is energized, calibrated, and aligned in accordance with the test documents.

TEST METHOD

1. Demonstrate proper operation of the Reactor Control System under various simulated plant conditions.

2. Perform tests utilizing signals or simulated signals on each of the Reactor Control System inputs in accordance with the applicable manufacturer's instruction manual.

3. Check each individual control channel from its input to its output signal to other plant control systems and components.

   NOTE: A functional demonstration at approximately 50% power is performed to verify that the Reactor Coolant System automatically maintains the proper reactor coolant average temperature conditions (see Table 14.2-3)

ACCEPTANCE CRITERIA

The Reactor Control System responds properly to simulated or normal input signals to interfacing systems. Rod blocks and permissives are verified correct in accordance with design.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

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REACTOR PROTECTION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Reactor Protection System and Engineered Safety Features Actuation System to respond properly to logic initiation signals prior to initial fuel loading.

PREREQUISITE

1. The instrument and protection systems are energized, calibrated, and aligned.

TEST METHOD

1. Demonstrate proper operation of the Reactor Protection System and Engineered Safety Features Actuation System under various logic conditions.

2. Determine the process fluid variable to sensor input delay time analytically.

3. Perform tests utilizing signals or simulated signals on each of the Reactor Protection System analog inputs in accordance with the applicable specifications.

4. Verify response time of the logic channels. Signals will be input to the primary sensor and timed through to the Reactor Trip Breaker.

5. Demonstrate redundancy and safe failure on loss of power.

6. Verify coincidence for all logic conditions.

ACCEPTANCE CRITERIA

The Reactor Protection System and Engineered Safety Features Actuation System function in accordance with design specifications in response to logic initiation signals. Total response time of the logic channels including process fluid delay time is in accordance with technical specification requirements.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

EXCORE NUCLEAR INSTRUMENTATION TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the Excore Nuclear Instrumentation including its ability to supply signals to the Reactor Protection System for generating appropriate trip signals and alarms, and indicating reactor power levels.

PREREQUISITES

1. Prior to Fuel Load.

TEST METHOD

1. Demonstrate the capability of source, intermediate, and power range circuitry to respond to a simulated test signal.

2. Verify that the source range detectors properly respond to a neutron source.

3. Demonstrate proper operation of the auctioneering circuits and flux channel deviation signals.

4. Check all channels to verify high level trip functions, alarm setpoints, and audible count rates where applicable.

ACCEPTANCE CRITERIA

The Excore Nuclear Instrumentation responds with design specifications to simulated input signals and provides proper output to the Reactor Protection System. Trip and alarm set points are in accordance with predetermined limits and proper neutron flux levels are displayed. The source range responds properly to a neutron source.

NOTE: Refer also to Section 14.2.10.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

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SATURATION MARGIN MONITOR TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Saturation Margin Monitor to accurately calculate and display Saturation Margin and to display hottest sensor temperature. Additionally, to demonstrate accuracy of the in-core thermocouples.

PREREQUISITES

1. Plant conditions are established as required.

2. Reactor Coolant System resistance temperature detectors are operational.

3. RCS Hot Leg and pressurizer pressure channels are operational.

4. The plant/ERF computers are operational.

TEST METHOD

1. Verify the identification of each thermocouple.

2. Measure the resistance and verify the continuity of each thermocouple.

3. Calibrate the incore thermocouples to the average of the RCS resistance temperature detectors.

4. Check the incore thermocouples temperature compensation equipment.

5. Verify that incore thermocouple signals are properly transmitted to the plant/ERF computers.

6. Verify that the Core Cooling Monitor responds properly to simulated input signals and that Saturation Margin and hottest sensor temperature are accurately calculated and displayed on the control board and computers.

ACCEPTANCE CRITERIA

The incore thermocouples achieve satisfactory calibration and provide proper inputs for the core cooling monitor and computers. Saturation margin and hottest sensor temperature are accurately calculated and displayed on the control board and computers. Core exit temperatures are accurately displayed on the computers.

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AUXILIARY STARTUP INSTRUMENTATION TEST SUMMARY

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SEISMIC INSTRUMENTATION TEST SUMMARY

OBJECTIVE

To demonstrate the operability of installed seismic instrumentation and its capability to monitor and record seismic disturbance.

PREREQUISITES

1. A calibrated seismic test signal is available.

TEST METHOD

1. Align the seismic instrumentation to a calibrated test signal.

2. Verify that the seismic trigger will activate the instrumentation as required at an acceptable level.

3. Demonstrate proper operation of seismic instrumentation components and alarms, including the triggering device, recording and playback system, peak acceleration recorders, and seismoscopes.

ACCEPTANCE CRITERIA

The seismic instrumentation is capable of being aligned within specification requirements. The system instruments, including trip settings, demonstrate correct response and outputs in response to simulated input signals.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

STEAM GENERATOR SAFETY & RELIEF VALVES TEST SUMMARY

OBJECTIVE

To demonstrate the proper functioning of the safety valves and power-operated relief valve associated with each steam generator.

PREREQUISITES

1. Plant conditions are established as necessary for test performance.

2. A pressure-assist device is available for use in lifting the relief and safety valves, and equipment is available to measure lifting and reseating pressures and valve stroke time.

TEST METHOD

1. Verify proper actuation and operation of the power-operated relief valves at normal system operating pressure by simulating a pressure signal to the controller.

2. With steam pressure near the safety valve setpoint, lift each safety valve with a pressure-assist device and measure the lifting pressure to ensure setpoints are as required.

3. Determine the full stroke time of the power-operated relief valves at normal system operating pressure by simulating a pressure signal to the controller.

4. With the steam generators at normal operating pressure, verify that safety valve leakage is within acceptable limits, the valves seat properly and do not chatter by visual and audio observation.

ACCEPTANCE CRITERIA

The steam generator safety valves and power-operated relief valves operate in accordance with design requirements. Safety valve lift settings meet technical specification requirements. Valves reset properly and do not chatter. Safety valve leakage at normal operating pressure is within design criteria.

TABLE 14.2-2
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MAIN STEAM AND FEEDWATER ISOLATION VALVES TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the main steam and feedwater isolation valves including their capability to close automatically as required.

PREREQUISITES

1. Plant conditions are established as necessary for test performance.

2. Equipment is available to measure the closure time of the isolation valves.

TEST METHOD

1. Demonstrate remote shutdown panel and Control Room operation of the main steam isolation valves and Control Room operation of the feedwater isolation valves.

2. Verify that the isolation valves will close upon receipt of an isolation signal.

3. Measure the closure time of the main feedwater isolation valves to ensure they close within the maximum and minimum time required and measure the closure time of the main steam isolation valves to ensure they close within the maximum allowed time.

4. Verify that the main steam isolation valves will close upon receipt of an isolation signal after having been isolated from the Instrument Air System for a minimum of 30 minutes.

ACCEPTANCE CRITERIA

The main steam and feedwater isolation valves respond properly to remote shutdown panel and control room operation, as applicable, and close upon receipt of an isolation signal. Valve closure times meet Technical Specification requirements.

TABLE 14.2-2
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AUXILIARY FEEDWATER SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability and reliability of the Auxiliary Feedwater System to supply feedwater to the steam generators and to maintain steam generator water inventory as required.

PREREQUISITES

1. Steam supply is available for the turbine-driven Auxiliary Feedwater pump.

2. An acceptable water supply is available for the Condensate Storage Tank.

TEST METHOD

1. Demonstrate proper manual and automatic operation of suction and discharge valves including steam supply to the turbine-driven pump.

2. Verify proper functioning of local and remote means of component control.

3. Verify that the hydraulic performance of each pump meets design requirements.

4. Demonstrate that the motor driven and turbine-driven pumps are capable of delivering flow to the steam generators within the acceptable time after an initiating signal.

5. Check for proper operation of instrumentation, control interlocks and alarms.

6. Verify the capability of the flow limiter to prevent pump runout.

7. Verify the capability of the Station Service Water System to supply water to the Auxiliary Feedwater Suction Header.

8. Verify that each Auxiliary Feedwater Control Valve can isolate the Auxiliary Feedwater to the Steam Generators for a minimum of 30 minutes while isolated from the Instrument Air System.

9. Verify system reliability with five consecutive, cold quick starts.

10. Demonstrate the ability of the AFWS to restore steam generator level after a low-level transient without causing unacceptable feedwater/steam generator waterhammer.

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PREOPERATIONAL TESTS SUMMARIES INDEX

11. Perform a 48 hour endurance on each of the auxiliary feedwater pumps after achieving the following conditions:

    a. Pump achieves operating speed, rated discharge pressure, and flow. A reduced flow not less than vendor limitations is acceptable.

    b. Steam temperature for the turbine driven AFW pump to be no less than 400°F. The AFW pump turbine is to be operated for a minimum of two hours without forced ventilation.

    c. Upon completion of 48 hour run the pump shall be shut down and allowed to cool to within 20°F of initial conditions, but not less than eight hours. The pump shall then be started and allowed to run for one hour.

ACCEPTANCE CRITERIA

The Auxiliary Feedwater System operates in accordance with technical specification requirements, and its performance is within limits assumed in the appropriate accident analysis. System automatic controls function properly in response to normal or simulated signals. Controls, interlocks, annunciators and instrumentation function in accordance with design requirements. Flow limiters limit runout flow to design values. The control valves function for a minimum of 30 minutes on loss of instrument air. System reliability is demonstrated by five consecutive, cold, quick starts. No damage occurs to the feedwater system from waterhammer.

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THERMAL EXPANSION TEST SUMMARY

OBJECTIVE

To demonstrate that Reactor Coolant System components and Class 1, 2 and 3 systems and piping systems which experience temperatures of greater than 200°F expand and move without obstruction during heatup and return to ambient. Additionally, to verify that loads and clearance gaps of selected piping system snubbers, variable spring hangers and pipe rupture restraints are properly set and allow the piping to move without obstruction or binding.

PREREQUISITES

1. The piping system which is to be subjected to the thermal expansion has been reviewed to ensure that all hangers which allow or restrict thermal growth have been installed and adjusted.

2. Preliminary shimming has been completed and shim crevices and support surfaces have been inspected to ensure debris has been removed.

3. The required heat load is available to subject the system to its designed thermal expansion.

4. Preservice examination of all snubbers verified completed within 6 months.

TEST METHOD

1. Verify that the selected system can expand without obstruction or interference during system heatup from ambient condition to operating condition.

2. Verify that the Reactor Coolant System piping and components are capable of returning to their cold position within specified tolerance limits.

3. Inspect snubbers and spring cans at specified temperature intervals (250°F, 350°F and 450°F) to ensure their expected thermal movements and swing clearances are such that from cold to hot condition and return to ambient condition are within the criteria per applicable design drawings.

   NOTE: For those systems which do not attain operating temperature, verify that the snubbers and spring cans are set at their calculated cold load position and observe that the system supports are free to move without binding or obstruction.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

ACCEPTANCE CRITERIA

The systems and components defined by FSAR Section 3.9 which have been verified by the thermal expansion tests are free to move without obstruction or binding due to thermal loads. The system supports (i.e., snubbers, spring cans) move within the allowable range per applicable design specifications.

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PRESSURIZER SAFETY AND RELIEF VALVES TEST SUMMARY

OBJECTIVE

To demonstrate the proper operation of the pressurizer power-operated relief valves, the proper functioning of the pressurizer relief tank, and the leaktightness of the pressurizer safety valves.

PREREQUISITES

1. Plant conditions are established as necessary for test performance.

2. The pressurizer relief tank is filled to its normal level and a water supply is available for the relief tank sprays.

TEST METHOD

1. Verify proper actuation and operation of the power-operated relief valves by raising pressurizer pressure above the setpoints with the relief valve isolate.

2. With the Reactor Coolant System at normal operating pressure verify that safety valve leakage is within acceptable limits.

3. Check operation of the discharge header leak detection devices.

4. Verify the ability of the pressurizer relief tank to condense a steam discharge from the pressurizer.

5. Verify proper operation of the pressurizer relief tank level control system, instrumentation, interlocks, and alarms.

6. Confirm that cooling spray to the pressurizer relief tank is initiated as required.

7. The pressurizer safety valves will be hydrostatically bench tested to verify their set points.

ACCEPTANCE CRITERIA

The pressurizer safety valves and power-operated relief valves and the pressurizer relief tank function in accordance with design requirements. Safety valve leakage is within acceptable limits. Controls, instrumentation, interlocks and alarms operate properly in response to simulated or normal input signals.

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FUEL HANDLING AND VESSEL SERVICING EQUIPMENT TEST SUMMARY

OBJECTIVE

To demonstrate the operability of the fuel handling and vessel servicing equipment, including the handling equipment components and fuel transfer system.

PREREQUISITES

1. The refueling cavity, refueling canal, and spent fuel pool are clean and areas adjacent to the system equipment are clear.

2. Dummy assembly and test weights are available as required for testing the polar crane, manipulator crane, spent fuel pit bridge crane and fuel building overhead crane.

TEST METHOD

1. With the use of a dummy assembly demonstrate the proper operation of all system components, including the manipulator crane, spent fuel pit bridge and electric hoist, new fuel elevator, fuel transfer system, rod cluster control changing fixtures, various handling tools, and indexing of system.

2. Verify operation of interlocks and proper setting of limit switches.

3. Verify leaktightness of sectionalizing devices as necessary.

4. Perform a 125% static load test and a 100% load full operational test on the polar crane, fuel building overhead crane, manipulator crane and the spent fuel pit bridge crane.

ACCEPTANCE CRITERIA

The fuel handling and vessel servicing system components operate in accordance with design requirements. Limit switches are properly set and interlocks function properly, including, as applicable, conforming to technical specifications requirements. Sectionalizing devices meet design requirements for leaktightness.

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REACTOR COOLANT SYSTEM COLD HYDROSTATIC TEST SUMMARY

OBJECTIVE

To verify the integrity and leak-tightness of the Reactor Coolant System by performing a hydrostatic test of the system, in accordance with Section III of ASME Boiler and Pressure Vessel Code.

PREREQUISITES

1. The reciprocating charging pump is operational.

2. A water supply within acceptable chemistry and temperature limits is available for pressurizing the Reactor Coolant System.

TEST METHOD

1. Pressurize the Reactor Coolant System within the maximum rate and in the prescribed increments until the desired test pressure is obtained, and stabilize at the test pressure for the required time.

2. Perform inspections of the Reactor Coolant System welds, joints, piping, and components.

3. Reduce system pressure to below the test pressure value.

ACCEPTANCE CRITERIA

The cold hydrostatic test satisfactorily verifies the integrity and leak-tightness of the Reactor Coolant System welds.

TABLE 14.2-2
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INTEGRATED HOT FUNCTIONAL TEST SUMMARY

OBJECTIVE

To demonstrate the satisfactory performance of systems and components during Reactor Coolant System heatup, at normal operating pressure and temperature, and during Reactor Coolant System cooldown.

PREREQUISITES

1. The RCS Cold Hydrostatic Test has been completed.

2. All systems, or portions of systems, and components whose adequacy or proper operations are to be verified under hot plant conditions, are operational.

3. Reactor vessel internals have been inspected.

TEST METHOD

1. Heat the RCS to normal operating temperatures and pressure utilizing heat from the reactor coolant pumps and pressurizer heaters.

2. Check the thermal expansion of system components and piping. (See Thermal Expansion Test Summary (Sheet 59 of 31))

3. Perform isothermal calibration of Resistance Temperature Detector and incore thermocouples.

4. Verify capability of the Chemical and Volume Control System to provide charging water at rated flow against normal reactor coolant pressure, check letdown design flow rate for each applicable operating mode, and check response of the system changes in pressurizer level.

5. Demonstrate proper operation of the pressurizer relief valves, and verify proper operation of the Pressurizer Relief Tank.

6. Verify proper operation of steam generator instrumentation to changes in steam generator level.

7. Demonstrate proper functioning of the Main Steam Isolation Valves under normal operating pressure and temperature conditions.

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8. Operate the RC pumps for a minimum of 240 hours at full flow in order to achieve greater than one million cycles on vessel internals. Following hot functional testing, the internals are removed and inspected for vibration effects.

9. Perform periodic vibration measurements on RCS components as required.

10. Verify acceptability of the excess letdown and seal water flows.

11. Perform a controlled plant cooldown by using steam dump from the steam generators and operating the Residual Heat Removal System.

12. Demonstrate that the effectiveness of the pressurizer heaters is within acceptable limits.

ACCEPTANCE CRITERIA

The systems and components checked during Hot Functional Testing function in accordance with design specifications and applicable FSAR requirements. Applicable Technical Specification requirements are satisfied.

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OPERATIONAL VIBRATION TESTING TEST SUMMARY

OBJECTIVE

To verify that the vibration level of selected (1) Class 1, 2, and 3 piping (2) other high-energy piping systems inside Seismic Category I Structures, (3) high-energy portions of systems whose failure could reduce the functioning of any Seismic Category I Plant Feature to an unacceptable level is within acceptable levels, for systems requiring test. See FSAR sections 3.9B.2.1.2 and 3.9B.2.1.3.

PREREQUISITES

1. Systems are operational as required.

2. Instrumentation is in place for testing as required.

TEST METHOD

1. Subject the specified piping systems to various flow modes and transients such as pump trips and valve closures as required.

2. Visually inspect and/or measure the vibration level of the piping and components at the specified locations.

3. Following completion of the system transient test, visually inspect the piping and supports including snubbers for damage, looseness of parts etc.

ACCEPTANCE CRITERIA

The vibration level for piping and components are within acceptable limits. For acceptance criteria basis see FSAR Sections 3.9B.2.1.2 and 3.9B.2.1.3.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

CONTAINMENT LOCAL LEAK RATE TEST SUMMARY

OBJECTIVE

To detect local leaks and measure leakage across each pressure-containing or leakage-limiting boundary for the primary reactor containment penetrations as applicable, including containment isolation valves.

PREREQUISITES

Equipment is available to provide measurement of local leak rate at penetrations, and containment isolation valves.

TEST METHOD

1. Examine the individual containment penetrations and containment isolation valves for leakage per Appendix J of 10 CFR Part 50.

2. Measure the rate of pressure loss across the containment penetration pressure-containing or leakage-limiting boundaries, per Appendix J of 10 CFR Part 50.

ACCEPTANCE CRITERIA

The Containment Local Leak Rate Tests meet the requirement of 10 CFR Part 50, Appendix J.

TABLE 14.2-2
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CONTAINMENT INTEGRATED LEAK RATE TEST SUMMARY

OBJECTIVE

To verify the primary reactor containment overall integrated leakage rate is within acceptable limits.

PREREQUISITES

1. Fluid system conditions are established as applicable to simulate post accident conditions which extend the boundary of the Containment Building.

2. Containment component and isolation valve leak tests have been satisfactorily performed.

3. All containment isolation valves have been closed by normal actuation methods.

TEST METHOD

1. Perform the containment integrated leak rate test per Appendix J of 10 CFR Part 50.

2. Perform the leakage rate calculation by using the mass-point methodology as described by ANSI/ANI 56.8-1987 or the methods of ANSI N45.4-1972.

3. If during the performance of a type A test, excessive leakage occurs through locally testable penetrations or isolation valves, these leakage paths may be isolated and the Type A test continued until completion. The sum of the post repaired minimum pathway local leakage rate values will be added to the UCL per ANSI 56.8- 1981.

ACCEPTANCE CRITERIA

The Containment Integrated Leak Rate Test meets the requirements of Appendix J of 10 CFR Part 50.

Note: The containment structural integrity test described in FSAR Section 3.8 may be performed concurrently with the Integrated Leak Rate Test.

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CONTAINMENT ISOLATION SYSTEM TEST SUMMARY

OBJECTIVE

To verify that the containment isolation valves close as required on a containment isolation signal.

PREREQUISITES

1. Plant conditions are established as necessary for test performance.

TEST METHOD

1. Verify that the containment isolation valves close properly upon receipt of a containment isolation signal.

ACCEPTANCE CRITERIA

The Containment Isolation System functions in accordance with FSAR specifications.

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REACTOR VESSEL WATER LEVEL INDICATION SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Reactor Vessel Water Level Indication system to detect the approach to inadequate core cooling by indication of collapsed water level in the upper head and upper plenum regions of the Reactor Vessel.

PREREQUISITES

1. Reactor Vessel Head is installed and Heated Junction Thermocouple Probes are in place.

2. A temporary external means of determining actual Reactor Vessel water level is installed.

3. The ERF Computer is operational.

4. Plant conditions are established as required to allow control of Reactor Vessel water level.

TEST METHOD

1. Verify that Reactor Vessel water level changes inside the range of the Heated Junction Thermocouple Probes are indicated sequentially on the control board and the ERF computer.

2. Verify that a correlation exists between actual Reactor Vessel water level and indicated level.

ACCEPTANCE CRITERIA

The Reactor Vessel Water Level Indication System responds properly to water levels inside its measurement range. Redundant instrumentation channels indicate agreement with actual water level.

TABLE 14.2-2
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ANTICIPATED TRANSIENT WITHOUT SCRAM MITIGATION SYSTEM ACTUATION CIRCUITRY TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the anticipated transient without scram mitigation system actuation circuitry to respond properly to logic initiation signals.

PREREQUISITE

1. Prior to fuel load.

TEST METHOD

1. Demonstrate proper operation of the anticipated transient without scram mitigation system actuation circuitry under various logic conditions.

2. Perform tests utilizing signals or simulated signals on each of the anticipated transient without scram mitigation system actuation circuitry inputs in accordance with the applicable manufacturer's instruction manual.

ACCEPTANCE CRITERIA

The anticipated transient without scram mitigation system actuation circuitry functions in accordance with design specifications in response to logic initiation signals.

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SAFETY RELATED AIR ACCUMULATOR TEST SUMMARY

OBJECTIVE

To verify the ability of safety related components with backup air accumulators to perform their function on a rapid or gradual loss of the instrument air supply.

PREREQUISITES

The Instrument Air System is operational and aligned in accordance with test documents.

TEST METHOD

1. Subject the air accumulators to a rapid loss of air supply upstream of the check valves to demonstrate the ability of the check valves to seat properly.

2. Subject the air accumulators to a gradual loss of air supply upstream of the check valves to demonstrate the ability of the check valves to seat properly.

3. Perform a test of the air operated components supplied by accumulators to demonstrate their ability to perform their design basis function.

ACCEPTANCE CRITERIA

The check valves seat on both a gradual and rapid loss of instrument air supply. The accumulators have sufficient capacity, including steady state consumption and possible leakage, to supply air to the device so that it can perform its design basis function.

TABLE 14.2-2
PREOPERATIONAL TESTS SUMMARIES INDEX

(Sheet 71 of 71)

INSTRUMENT AIR SYSTEM BRANCH HEADER TEST SUMMARY

OBJECTIVE

To verify that all air operated components supplied by branches of the Instrument Air System which supply safety related accumulators will respond as expected to a loss of air supply.

PREREQUISITES

1. The Instrument Air System is operational and aligned in accordance with test documents.

2. Air operated components of the branch to be tested are in the proper position in accordance with test documents.

TEST METHOD

Isolate the branch header from the air supply and allow the branch header pressure to decay through normal steady state consumption of air operated components.

ACCEPTANCE CRITERIA

All air operated components of the branch respond as expected. Accumulator check valves seat properly.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

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TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

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REACTOR COOLANT SYSTEM FLOW TEST SUMMARY

OBJECTIVE

To verify predicted Reactor Coolant System cold leg volumetric flow rates at normal operating temperature and pressure with all reactor coolant pumps running in hot standby and during power ascension testing and demonstrate that pressurizer spray is within acceptable limits.

PREREQUISITES

1. The reactor is at the specified power level.

2. The RCS is at the specified conditions.

3. All reactor coolant pumps are operational.

TEST METHOD

1. During hot standby operation, measure and record loop elbow differential pressures and determine cold leg volumetric flow rates. At 50% and 75% power for Unit 1 and 75% and 100% power for Unit 2 use the N-16 Transit Time Flow Meter and a precision secondary calorimetric to determine loop cold leg volumetric flow rates.

2. Verify that the reactor coolant system flow transmitters have been aligned for zero flow and 100 percent flow at normal operating conditions.

3. Demonstrate that the effectiveness of the pressurizer spray is within acceptable limits.

ACCEPTANCE CRITERIA

The measured Reactor Coolant System flow is within design flow limits specified in FSAR Chapter 5, and the flow transmitters are satisfactorily aligned for zero flow and full flow conditions. Pressurizer spray is within acceptable limits.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 3 of 28)

REACTOR COOLANT SYSTEM FLOW COASTDOWN TEST SUMMARY

OBJECTIVE

To measure the reactor coolant system flow rate decrease subsequent to a simultaneous trip of all four reactor coolant pumps, and to measure the delay times associated with assumptions of the loss of flow accident analysis.

PREREQUISITES

1. The reactor is in the hot standby condition.

2. Applicable portions of the Reactor Coolant System Flow Test have been satisfactorily completed, and any pressure damping devices, if installed for that test, have been removed.

TEST METHOD

1. With the reactor coolant system in the hot standby condition, simultaneously trip all four reactor coolant pumps and measure the rate at which reactor coolant flow decreases.

2. Determine the delay times associated with the low flow reactor trip circuitry.

ACCEPTANCE CRITERIA

The rate of reactor coolant flow decrease, upon tripping of all four reactor coolant pumps, satisfies the NSSS vendor acceptance criteria which verifies the loss of flow analysis of the FSAR, and the associated delay times are within acceptable limits.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 4 of 28)

CONTROL ROD DRIVE TEST SUMMARY

OBJECTIVE

To verify control rod bank start and stop setpoints, verify proper slave cycler timing and drive mechanism operation, check rod speeds, and demonstrate the capability of the CRDMs to respond to signals from the Reactor Control System.

PREREQUISITES

1. The rod cluster control assemblies, control rod drive mechanisms and Reactor Control System are operational.

2. Plant conditions are established as necessary.

TEST METHOD

1. Verify proper rod bank start and stop positions during rod insertion and withdrawal, and check the Rod Control System bank- overlap setpoints and rod speeds.

2. Demonstrate proper slave cycler timing and drive mechanism operation.

3. Demonstrate proper functioning of the Rod Control System during steady state critical operations, and observe the automatic response of the CRDM’s to signals from the Reactor Control System.

ACCEPTANCE CRITERIA

All mechanical operating features of the rod drive mechanisms and associated rod cluster control assemblies function properly, rod speeds are within acceptable limits, and the CRDM’s satisfactorily respond to signals from the Reactor Control System.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 5 of 28)

ROD POSITION INDICATION TEST SUMMARY

OBJECTIVE

To verify that the Digital Rod Position Indication system provides proper rod position indication and alarms based on simulated and/or actual inputs over the entire length of travel of each rod cluster control assembly.

PREREQUISITES

1. The full-length rod cluster control assemblies are prepared for operation.

2. The Digital Rod Position Indication system is energized and properly aligned.

TEST METHOD

1. Verify proper response of the Digital Rod Position Indication system over the entire length of travel of the rod cluster control assemblies.

2. Verify proper functioning of the indication and alarm features associated with the rod position indication circuitry.

ACCEPTANCE CRITERIA

The rod position indicators perform satisfactorily for each full- length rod cluster control assembly over its entire length of travel. The rod-bottom alarms function in accordance with design specifications. Indication accuracy and alarm setpoints are within design criteria.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 6 of 28)

REACTOR TRIP SYSTEM TEST SUMMARY

OBJECTIVE

To demonstrate the proper functioning of the Reactor Trip System, including the capability to test the operation of the reactor trip breakers using bypass breakers.

PREREQUISITES

1. Fuel loading is complete.

2. Electrical power is available for the Reactor Trip System circuitry.

3. Control rod drive mechanisms are operational (applies only to the test method #1 unlatching test).

TEST METHOD

1. Demonstrate that each control rod drive mechanism will properly unlatch upon opening of the trip breakers.

2. Confirm that providing a simulated trip signal to test a reactor trip breaker, with its associated bypass breaker closed, will open the reactor trip breaker without opening its associated bypass breaker.

3. Demonstrate the capability to manually open the trip breakers.

4. With all control rods fully inserted attempt to close both reactor trip breaker bypass breakers and verify interlocks cause both bypass breakers to trip. With one reactor trip breaker bypass breaker closed, verify that by placing the opposite trip channel in test causes both reactor trip breakers and both bypass breakers to open.

ACCEPTANCE CRITERIA

Each control rod drive mechanism unlatches upon opening of the trip breakers. The associated reactor trip breaker bypass breaker remains closed when each reactor trip breaker is opened for test. Interlocks which prevent closing both reactor trip breaker bypass breakers simultaneously function in accordance with design requirements.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 7 of 28)

AUXILIARY STARTUP INSTRUMENTATION TEST SUMMARY

OBJECTIVE

To demonstrate the proper response of the temporary neutron detectors and proper functioning of their associated indicating and recording functions.

PREREQUISITE

1. A neutron source is available.

TEST METHOD

1. With the use of the neutron source, verify that the auxiliary startup instrumentation responds properly with indication and an optional audible signal.

2. Verify that an increase in the indicated count rate occurs during the initial stages of fuel load.

ACCEPTANCE CRITERIA

The auxiliary startup instrumentation responds properly to a neutron source providing indication and an optional audible signal. The instrumentation indicates increasing neutron levels during early stages of fuel loading.

Note: Refer also to Section 14.2.10.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 8 of 28)

CALIBRATION OF PROCESS TEMPERATURE AND NUCLEAR INSTRUMENTATION
TEST SUMMARY

OBJECTIVE

To calibrate and adjust the operational settings of the source, intermediate, and power range detectors. To verify channel indication overlap and power range detector output linearity. Also, to calibrate and adjust the operational settings of the N-16 power detectors and reactor coolant average temperature instrumentation system.

PREREQUISITES

1. The preoperational tests of the nuclear and temperature instrumentation systems have been satisfactorily completed.

2. Reactor power level is established as necessary.

TEST METHOD

1. Determine the source range channel voltages versus detector output and readjust as necessary prior to initial fuel load.

2. Perform an isothermal alignment of the N-16 power and T_{ave} instrumentation prior to initial criticality.

3. Measure overlap between the source range, intermediate range, and power range channels.

4. At each major power plateau, determine reactor power by secondary calorimetric and calibrate each power range excore NIS and each N-16 power channel to correspond to the measured thermal power. Also, adjust each N-16 T_{ave} channel as necessary.

5. Verify acceptability of each power range channel output by measuring and plotting power range detector currents versus power level. Determine the linearity or each power range channel and the degree of uniformity between channels.

6. Verify proper functioning of the nuclear instrumentation indication and selected alarm features.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

ACCEPTANCE CRITERIA

The source range channel operating voltages are set for optimum neutron response. The source, intermediate, and power range channels exhibit satisfactory overlap. The power range excore detectors display a linear output over the range of normal power operation, and accurately indicate the actual reactor power level as determined by calorimetric measurements. Selected alarm features function properly. The N-16 power and T_{ave} channels accurately indicate the actual reactor power and reactor coolant average temperature as determined by calorimetric measurements.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 10 of 28)

CHEMICAL TEST SUMMARY

OBJECTIVE

To establish the water chemistry of the Reactor Coolant System, and to verify the capability to maintain Reactor Coolant System chemistry at power and during power escalation.

PREREQUISITES

1. The chemical addition portion of the CVCS is operational.

2. Sampling analysis equipment is operational.

TEST METHOD

1. Prior to fuel load, establish the Reactor Coolant System water chemistry within specifications.

2. Verify that the water chemistry can be maintained within specifications during criticality and power escalation.

ACCEPTANCE CRITERIA

Reactor Coolant System water chemistry is satisfactorily established prior to fuel load, and maintained within specifications during criticality and power escalation.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 11 of 28)

RADIATION SURVEYS TEST SUMMARY

OBJECTIVE

To verify radiation shielding effectiveness by measuring radiation dose levels at preselected locations within the plant during low, intermediate and high reactor power level operation.

PREREQUISITES

1. The radiation survey instruments are calibrated against known sources.

2. Reactor power level is established as necessary.

TEST METHOD

1. At specified steady state power levels between zero and 100 percent power, measure radiation levels at preselected locations within the plant to determine effectiveness of the radiation shielding.

2. Upon completion of the radiation surveys, check the calibration of the survey instruments.

ACCEPTANCE CRITERIA

The measured radiation levels are within limits for the zone designation of each area surveyed.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 12 of 28)

PROCESS AND EFFLUENT RADIATION MONITORING TEST SUMMARY
(UNIT 1)

OBJECTIVE

To verify the proper performance of the process and effluent radiation monitoring equipment under actual nuclear operating conditions.

PREREQUISITES

1. The process and effluent radiation monitors have been checked against known sources.

2. The reactor has been operating at power for a time sufficient to provide representative process and effluents samples.

TEST METHOD

1. By radiochemical analysis verify the suitability of liquid effluents for discharge.

2. When conditions permit, perform an actual liquid discharge and observe the response of the effluent monitors.

3. By radiochemical analysis of the process and effluent samples confirm the satisfactory performance of the process and effluent monitors.

ACCEPTANCE CRITERIA

The installed process and effluent monitors properly indicate the radioactive content of the fluid monitored and perform in accordance with design specifications.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 13 of 28)

PROCESS AND EFFLUENT RADIATION MONITORING TEST SUMMARY
(UNIT 2)

This test conducted during preoperational testing

(See Table 14.2-2)

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

MODERATOR TEMPERATURE REACTIVITY COEFFICIENT TEST SUMMARY

OBJECTIVE

To confirm that the actual moderator temperature coefficient of reactivity is within acceptable limits.

PREREQUISITES

1. The reactor is critical, in the hot, zero power condition.

TEST METHOD

1. Initiate a small change in temperature of the Reactor Coolant System.

2. Determine the amount of reactivity inserted or removed by the temperature change, and calculate the value of the moderator temperature coefficient.

ACCEPTANCE CRITERIA

The moderator temperature coefficient of reactivity is determined to be within acceptable limits.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

CONTROL ROD REACTIVITY WORTHS TEST SUMMARY

OBJECTIVE

To verify the design rod worths of the control and shutdown RCCA banks.

PREREQUISITES

1. The reactor is critical, in the hot, zero power condition.

TEST METHOD

1. With the use of boron concentration sampling data, rod position indication and reactivity measurement, verify the worths of the control and shutdown RCCA banks utilizing the method of Bank Exchange (or Boron Concentration Exchange).

ACCEPTANCE CRITERIA

The rod worths are determined to be within design specifications.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 16 of 28)

BORON REACTIVITY WORTH TEST SUMMARY

OBJECTIVE

To measure the reactivity worth of the boron in the reactor coolant.

PREREQUISITES

1. The reactor is critical, in a hot, zero power condition.

TEST METHOD

1. Determine the reactivity worth of the boron in solution with the reactor coolant by changing the boron concentration, and compensating for the reactivity effect by inserting or removing reactivity with control rod motion.

2. Equate the change in reactivity due to rod motion to the worth of the change in boron concentration.

ACCEPTANCE CRITERIA

The value of the boron reactivity worth is determined to be within acceptable limits.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

CORE REACTIVITY BALANCE TEST SUMMARY

OBJECTIVE

To verify that actual core reactivity effects are in agreement with design values.

PREREQUISITES

1. The reactor power is established as necessary.

TEST METHOD

1. Measure the all rods out, hot zero power, xenon free, critical boron concentration.

2. Measure the all rods out, hot full power, equilibrium xenon, critical boron concentration.

3. Perform a core reactivity balance.

ACCEPTANCE CRITERIA

The overall core reactivity balance is within acceptable limits.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

LOSS OF OFFSITE POWER TEST SUMMARY

OBJECTIVE

To demonstrate the proper plant response following a plant trip with no offsite power available.

PREREQUISITES

The Turbine-Generator output is approximately 130 MWe (greater than 10% reactor power) with non-Class 1E buses being supplied from the unit auxiliary transformer and the Class 1E buses being supplied from their offsite power source.

TEST METHOD

1. Manually generate a main turbine trip and isolate offsite power sources.

2. Verify proper starting and load sequencing of diesel generators and transfer of power supplies for all required equipment.

3. Verify the Reactor Coolant System can be maintained in a shutdown condition for a minimum of 30 minutes utilizing the power operated atmospheric relief valves to remove decay heat from the reactor core.

ACCEPTANCE CRITERIA

The on-site power supplies (i.e., diesel generators) shall auto-start and operate the necessary controls, equipment and indication to remove decay heat and maintain the Reactor Coolant System in a shutdown condition for the duration of the test.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 19 of 28)

ROD DROP TEST SUMMARY

OBJECTIVE

To determine the rod drop time of each full-length rod cluster control assembly (RCCA) under hot full flow Reactor Coolant System conditions.

PREREQUISITES

1. The Reactor Coolant System is at nominal, no load, operating temperature and pressure, prior to initial criticality.

2. Rod position indication is functional.

3. All four Reactor Coolant Pumps are operating.

TEST METHOD

1. Withdraw each full-length rod cluster control assembly, interrupt the electrical power to the associated rod drive mechanism, and measure and record the rod drop time. This test is performed with the reactor in hot full flow conditions.

2. Perform at least three additional rod drop tests for each rod whose measured drop time deviates from the mean for all rods by more than two standard deviations.

   NOTE: Additional rod drop tests may be performed at the option of the test engineer. These optional tests are not used to satisfy acceptance criteria and may be performed under hot or cold RCS conditions with any RCS flow condition.

ACCEPTANCE CRITERIA

The hot, full flow rod drop times are acceptable in accordance with plant Technical Specifications.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 20 of 28)

FLUX DISTRIBUTION MEASUREMENTS TEST SUMMARY

OBJECTIVE

To determine the reactor core power distribution.

PREREQUISITES

1. Incore instrumentation and process computer are operable for incore flux mapping.

2. Reactor is critical and power level is established as necessary.

TEST METHOD

Complete an incore flux map for the All Rods Out (ARO) control rod configurations with reactor power stabilized below 5 percent, for Unit 1 and prior to exceeding 30 percent for Unit 2.

Note: ARO is defined for this measurement as Control Bank D above 190 steps withdrawn and all other banks fully withdrawn.

ACCEPTANCE CRITERIA

The core flux distributions indicated by the flux map are acceptable in accordance with plant Technical Specifications where applicable.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 21 of 28)

CORE PERFORMANCE EVALUATION TEST SUMMARY

OBJECTIVE

To verify the operating characteristics of the core and the calibration of the flux and temperature instrumentation during power escalation.

PREREQUISITES

1. Reactor power level is established as necessary.

TEST METHOD

1. At steady state power levels of 30, 50, 75, 90 and 100 percent, record Reactor Coolant System parameters. Incore data for flux maps are recorded at 50, 75 and 100 percent.

2. Analyze the data obtained at each power level to determine core performance margins and verify flux and temperature instrumentation calibration.

ACCEPTANCE CRITERIA

The core performance margins are within design predictions for normal rod configurations and the calibration of the flux and temperature instrumentation has been verified.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 22 of 28)

UNIT LOAD TRANSIENTS TEST SUMMARY

OBJECTIVE

To demonstrate satisfactory plant transient response to various specified load changes and trips, to monitor the behavior of reactor control systems during these transients, and, if necessary, optimize the reactor control system setpoints.

PREREQUISITES

1. Reactor power level is established as necessary for each transient.

2. All reactor control systems are operational and their setpoints have been set to their recommended values.

TEST METHOD

1. Initiate a step change in power level of 10 percent and monitor Reactor Coolant System behavior in response to the transients. For Unit 1, this test will be performed at approximate power levels of 50 percent, 30 percent (following completion of 50 percent testing) and 100 percent. For Unit 2, this test will be performed at approximate power levels of 50 percent and 75 percent.

2. Monitor plant response to a 50 percent load reduction, from power levels of approximately 75 percent (Unit 1 only) and 100 percent.

3. Monitor plant response to a plant trip from power levels up to 100 percent.

4. If necessary, adjust the reactor control system setpoints until optimal response is obtained during subsequent test performance.

ACCEPTANCE CRITERIA

Plant response to the unit load transients is acceptable in accordance with design specifications, and the Reactor Control System parameters reach steady state values without appreciable overshoot or oscillation subsequent to a step change.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 23 of 28)

REMOTE SHUTDOWN TEST SUMMARY

OBJECTIVE

To demonstrate the capability of performing a safe plant shutdown, maintain the plant in a hot standby condition, and to demonstrate the ability to cooldown from hot standby to cold shutdown conditions from outside the control room, using the minimum shift crew. Verify that the Remote Shutdown Panel selector switches properly transfer control from the Control Room to the Remote Shutdown panel.

PREREQUISITES

1. The equipment and instrumentation associated with the Remote Shutdown Panel are available for achieving and maintaining the plant in a hot standby condition.

2. The plant is at a power level greater than 10% generator power but less than 25% reactor power, for the reactor trip portion of the test.

3. For the cooldown portion, the plant is in a stable hot standby condition.

TEST METHOD

1. With the generator at greater than 10 percent power, perform a safe shutdown of the plant from outside the Control Room using the minimum shift crew.

2. Check functioning of instrumentation, controls, interlocks and alarms. Credit may be taken for preop/prereq functional tests.

3. Demonstrate the capability to achieve and maintain the plant in a hot standby condition from the Remote shutdown panel for a minimum of 30 minutes.

4. Demonstrate the potential for cooldown to cold shutdown conditions by placing the residual heat removal system into service and reducing the reactor coolant temperature to approximately 300°F.

ACCEPTANCE CRITERIA

Transfer of control to outside the Control Room can be achieved in accordance with design requirements, remote shutdown instrumentation, controls, alarms and interlocks function properly. The potential ability to perform a safe shutdown, to achieve and maintain hot standby conditions from outside the Control Room has been demonstrated. The potential ability to cool down to cold shutdown conditions from outside the control room has been demonstrated.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

TURBINE TRIP/GENERATOR LOAD REJECTION TEST SUMMARY

OBJECTIVE

To demonstrate the ability of the plant to sustain a full load rejection of the turbine generator at full power, and to evaluate plant response to the transient.

PREREQUISITES

1. Reactor is at a steady state, approximately 100% power condition.

2. Electrical distribution system is aligned for normal power operation.

3. Recording devices are available to record the following parameters; RCS pressure and temperature, pressurizer level, steam pressure, power level and turbine speed.

TEST METHOD

1. With the plant at approximately 100 percent power, manually open the generator main breakers and, observe the effects of the resultant plant trip, and record values of pertinent parameters during the transient.

2. Observe the response of the plant control systems during the transient.

ACCEPTANCE CRITERIA

A Loss of Load trip from full power conditions and resultant plant trip is sustained and the control systems function properly within specified limits to preclude lifting of the pressurizer or main steam safety valves, and subsequent to the trip, the plant can be maintained in a hot standby condition.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 25 of 28)

REACTOR COOLANT LEAK TEST SUMMARY

OBJECTIVE

To demonstrate the leak tightness of Reactor Coolant System pressure boundary and of the reactor vessel flange after the system has been closed following fueling.

To determine the leak rate for primary-to-secondary leakage, Reactor Coolant Pump seal leakage, other identified leakage, and any unidentified leakage.

PREREQUISITES

1. At least one RC pump is operating.

2. Normal operating pressure is being maintained.

3. RC temperature is between 400°F and normal no-load average temperature.

4. The reactor is shutdown prior to initial criticality.

TEST METHOD

1. Verify there is no vessel flange leakoff.

2. Verify Reactor Coolant System integrity by visual inspection.

3. Determine the primary-to-secondary leak rate by sampling the secondary for boron.

4. Determine the Reactor Coolant Pump seal leak rate.

5. Obtain results of leakage rates testing of Reactor Coolant Pressure Isolation Valves.

6. Determine the identified and unidentified leakage rate by conducting a mass balance of the primary system.

ACCEPTANCE CRITERIA

1. There is no indication of vessel flange leakage.

2. Visual inspection of the Reactor Coolant System is satisfactory.

3. The leakage rates are within Technical Specification limitations.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 26 of 28)

ROD CONTROL SYSTEM TEST SUMMARY

OBJECTIVE

Demonstrate that the Full Length Rod Control System performs the required control and indication functions in order to verify it is operational and ready for use prior to initial criticality.

PREREQUISITES

1. All rod drop tests and mechanism timing alignment is complete and verified to be within specifications.

2. Nuclear Instrumentation Source Range channels shall be aligned and operable and shall be monitored at all times during conduct of this test.

3. The plant is at normal no load operating temperature and pressure prior to initial criticality.

4. The digital Rod Position Indication System is operable and in service.

5. The boron concentration is being maintained at refueling shutdown concentration.

6. The bank overlap settings are set to allow overlap operation of all control banks and still maintain a minimum amount of rod withdrawal.

TEST METHOD

1. The manual mode of control is checked for each applicable position of the bank selector switch.

2. Rod speed and direction are verified to be in accordance with manufacturer’s instruction manual.

3. Verify proper operation of status lights, step counters, rod position and speed indications.

4. Verify proper operation of rod bank overlap.

ACCEPTANCE CRITERIA

The Rod Control System responds properly to normal input signals in the manual mode, providing correct bank overlapping, rod speed, direction and indication.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 27 of 28)

AUTOMATIC REACTOR CONTROL SYSTEM TEST SUMMARY

OBJECTIVE

Demonstrate the ability of the Automatic Reactor Control System to return reactor coolant temperature to the programmed setpoint and to maintain it.

PREREQUISITES

1. The reactor is at approximately 50% power.

2. Reactor Coolant System temperature and pressure are stable.

3. Pressurizer pressure and level control are in automatic.

4. Steam Generator level control is in automatic.

TEST METHOD

1. At 50% power, raise Reactor Coolant System temperature and observe the Automatic Reactor Control System response. Then lower Reactor Coolant System temperature and observe the Automatic Reactor Control System response.

ACCEPTANCE CRITERIA

The Reactor Coolant System Temperature (T_{avg}) returns to within 1.5°F of the setpoint (T_{ref}) following the positive and negative temperature transient.

TABLE 14.2-3
INITIAL STARTUP TEST SUMMARIES

(Sheet 28 of 28)

INCORE NUCLEAR INSTRUMENTATION TEST SUMMARY

OBJECTIVE

To demonstrate the capability of the Incore Nuclear Instrumentation to remotely position the incore neutron detectors for the purpose of core flux mapping, and to supply the appropriate digital and analog signals to the plant computer.

PREREQUISITES

1. Fuel has been loaded in the core and all thimbles have been inserted and sealed.

2. Dummy cables and other equipment necessary for the test performance are available.

TEST METHOD

1. Ensure free passage and index the system by driving a dummy cable to all thimble positions and determine preliminary limit switch settings.

2. Verify operation of 5-path and 10-path transfer assemblies in all modes using a dummy cable.

3. Provide a check of the leak detection system.

4. During flux mapping at power, verify detector response to neutron flux and make final limit switch settings.

5. Verify each detector/cable combination can be inserted to the top of core limit switches, automatically stops and can be withdrawn.

ACCEPTANCE CRITERIA

The Incore Nuclear Instrumentation channels function properly and supply appropriate outputs to the plant computer. The indexing system functions in accordance with design requirements. After fuel loading, free passage to all positions is demonstrated. Response to neutron flux at power meets design criteria.

15.0 ACCIDENT ANALYSIS

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15.0 ACCIDENT ANALYSIS

This chapter addresses the representative initiating events listed on pages 15-10, 15-11, and 15-12 of Regulatory Guide 1.70, Revision 2 as they apply to the Comanche Peak Nuclear Power Plant (CPNPP).

Certain items in the guide warrant comment, as follows:

Items 1.3 and 2.1 - There are no pressure regulators in the Nuclear Steam Supply System (NSSS) pressurized water reactor (PWR) design whose malfunction or failure could cause a steam flow transient.

Item 6.2 - No instrument lines from the Reactor Coolant System boundary in the NSSS PWR design penetrate the Containment.[a]

15.0.1 CLASSIFICATION OF PLANT CONDITIONS

Since 1970 the American Nuclear Society (ANS) classification of plant conditions has been used which divides plant conditions into four categories in accordance with anticipated frequency of occurrence and potential radiological consequences to the public. The four categories are as follows:

1. Condition I: Normal Operation and Operational Transients.

2. Condition II: Faults of Moderate Frequency.

3. Condition III: Infrequent Faults.

4. Condition IV: Limiting Faults.

The basic principle applied in relating design requirements to each of the conditions is that the most probable occurrences should yield the least radiological risk to the public and those extreme situations having the potential for the greatest risk to the public shall be those least likely to occur. Where applicable, Reactor Trip System and engineered safeguards functioning is assumed to the extent allowed by considerations such as the single failure criterion in fulfilling this principle (i.e. seismic Category I, Class 1E, and IEEE qualified equipment, instrumentation, and components are used in the ultimate mitigation of the consequences of Condition II, III and IV events; see Section 15.0.8).

Step-by-step sequence-of-events diagrams are provided for each transient in Figures 15.0-8 through 15.0-31. Figure 15.0-7 provides the legend used in these diagrams.

a. For the definition of the Reactor Coolant System boundary, refer to ANSI-N18.2, “Nuclear Safety Criteria for the Design of Stationary PWR Plants,” Section 5, 1973.

15.0.1.1 Condition I - Normal Operation and Operational Transients

Condition I occurrences are those which are expected frequently or regularly in the course of power operation, refueling, maintenance, or maneuvering of the plant. As such, Condition I occurrences are accommodated with margin between any plant parameter and the value of that parameter which would require either automatic or manual protective action. Inasmuch as Condition I occurrences occur frequently or regularly, they must be considered from the point of view of affecting the consequences of fault conditions (Conditions II, III and IV). In this regard, analysis of each fault condition described is generally based on a conservative set of initial conditions corresponding to adverse conditions which can occur during Condition I operation.

A typical list of Condition I events is listed below:

1. Steady state and shutdown operations

   a. Mode 1, Power operation ($K_{eff} \geq 0.99$, > 5 percent of rated thermal power).

   b. Mode 2, Startup ($K_{eff} \geq 0.99$, $\leq$ 5 percent of rated thermal power).

   c. Mode 3, Hot standby ($K_{eff} < 0.99$, $T_{avg} \geq 350°F$).

   d. Mode 4, Hot shutdown ($K_{eff} < 0.99$, $200°F < T_{avg} < 350°F$, all reactor head closure bolts fully tensioned).

   e. Mode 5, Cold shutdown ($K_{eff} < 0.99$, $T_{avg} \leq 200°F$, all reactor head closure bolts fully tensioned).

   f. Mode 6, Refueling (One or more reactor vessel head closure bolts less than fully tensioned).

2. Operation with permissible deviations

   Various deviations which may occur during continued operation as permitted by the plant Technical Specifications must be considered in conjunction with other operational modes. These include:

   a. Operation with components or systems out of service (such as power operation with a reactor coolant pump out of service).

   b. Leakage from fuel with clad defects.

   c. Radioactivity in the reactor coolant

      1. Fission products.

      2. Corrosion products.

      3. Tritium.

d. Operation with steam generator leaks up to the maximum allowed by the Technical Specifications.

e. Testing as allowed by the Technical Specifications.

3. Operational transients

a. Plant heatup and cooldown (up to l00°F/hour for the Reactor Coolant System; 200°F/hour for the pressurizer during cooldown and 100°F/hour for the pressurizer during heatup).

b. Step load changes (up to ± 10 percent).

c. Ramp load changes (up to 5 percent/minute).

d. Load rejection up to and including design full load rejection transient.

15.0.1.2 Condition II - Faults of Moderate Frequency

These faults, at worst, result in the reactor trip with the plant being capable of returning to operation. By definition, these faults (or events) do not propagate to cause a more serious fault, i.e., Condition III or IV events. In addition, Condition II events are not expected to result in fuel rod failures or Reactor Coolant System or secondary system overpressurization. This is demonstrated by assuring that DNBR remains greater than the limit value, and thus, the number of rods calculated to be in DNB corresponds to the criterion in Section 4.4.1.

For the purposes of this report, the following faults are included in this category:

1. Feedwater system malfunctions that result in a decrease in feedwater temperature (Section 15.1.1).

2. Feedwater system malfunctions that result in an increase in feedwater flow (Section 15.1.2).

3. Excessive increase in secondary steam flow (Section 15.1.3).

4. Inadvertent opening of a steam generator relief or safety valve (Section 15.1.4).

5. Loss of external electrical load (Section 15.2.2).

6. Turbine trip (Section 15.2.3).

7. Inadvertent closure of main steam isolation valves (Section 15.2.4).

8. Loss of condenser vacuum and other events resulting in turbine trip (Section 15.2.5).

9. Loss of nonemergency AC power to the station auxiliaries (Section 15.2.6).

10. Loss of normal feedwater flow (Section 15.2.7).

11. Partial loss of forced reactor coolant flow (Section 15.3.1).

12. Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low power startup condition (Section 15.4.1).

13. Uncontrolled rod cluster control assembly bank withdrawal at power (Section 15.4.2).

14. Rod cluster control assembly misalignment (dropped full length assembly, dropped full length assembly bank, or statically misaligned full or part length assembly) (Section 15.4.3).

15. Startup of an inactive reactor coolant pump at an incorrect temperature (Section 15.4.4).

16. Chemical and Volume Control System malfunction that results in a decrease in the boron concentration in the reactor coolant (Section 15.4.6).

17. Inadvertent operation of the Emergency Core Cooling System during power operation (Section 15.5.1).

18. Chemical and Volume Control System malfunction that increases reactor coolant inventory (Section 15.5.2).

19. Inadvertent opening of a pressurizer safety or relief valve (Section 15.6.1).

15.0.1.3 Condition III - Infrequent Faults

By definition Condition III occurrences are faults which may occur very infrequently during the life of the plant. They will be accommodated with the failure of only a small fraction of the fuel rods although sufficient fuel damage might occur to preclude resumption of the operation for a considerable outage time. The release of radioactivity will not be sufficient to interrupt or restrict public use of those areas beyond the exclusion radius. A Condition III fault will not, by itself, generate a Condition IV fault or result in a consequential loss of function of the Reactor Coolant System or Containment barriers. For the purposes of this report the following faults are included in this category:

1. Steam system piping failure (minor) (Section 15.1.5).

2. Complete loss of forced reactor coolant flow (Section 15.3.2).

3. Rod cluster control assembly misalignment (single rod cluster control assembly withdrawal at full power) (Section 15.4.3).

4. Inadvertent loading and operation of a fuel assembly in an improper position (Section 15.4.7).

5. Loss of coolant accidents resulting from a spectrum of postulated piping breaks within the reactor coolant pressure boundary (small break) (Section 15.6.5).

6. Radioactive Gas Waste System leak or failure (Section 15.7.1).

7. Radioactive Liquid Waste System leak or failure (Section 15.7.2).

8. Postulated radioactive releases due to liquid tank failures (Section 15.7.3).

9. Spent fuel cask drop accidents (Section 15.7.5).

10. Pipe breaks which result in the release of reactor coolant outside containment (Section 15.6.2).

15.0.1.4 Condition IV - Limiting Faults

Condition IV occurrences are faults which are not expected to take place, but are postulated because their consequences would include the potential of the release of significant amounts of radioactive material. They are the most drastic which must be designed against and represent limiting design cases. Condition IV faults are not to cause a fission product release to the environment resulting in an undue risk to public health and safety in excess of guideline values of 10CFR100. A single Condition IV fault is not to cause a consequential loss of required functions of systems needed to cope with the fault including those of the Emergency Core Cooling System and the Containment. For the purposes of this report the following faults have been classified in this category:

1. Steam system piping failure (major) (Section 15.1.5).

2. Feedwater system pipe break (Section 15.2.8).

3. Reactor coolant pump shaft seizure (locked rotor) (Section 15.3.3).

4. Reactor coolant pump shaft break (Section 15.3.4).

5. Spectrum of rod cluster control assembly ejection accidents (Section 15.4.8).

6. Steam generator tube failure (Section 15.6.3).

7. Loss of coolant accidents resulting from the spectrum of postulated piping breaks within the reactor coolant pressure boundary (large break) (Section 15.6.5).

8. Design basis fuel handling accidents (Section 15.7.4).

15.0.2 OPTIMIZATION OF CONTROL SYSTEMS

The NSSS Control System Setpoint Study was performed to simulate performance of the reactor control and protection systems. The Study describes setpoints and performance of the Reactor Control System for operation up to the rated power level with chemical shim operation. Emphasis was placed on the development of a control system which safely and automatically maintains prescribed conditions in the plant, even under the most conservative set of reactivity parameters, with respect to both system stability and transient performance.

Using a detailed analog computer simulator coupled with a digital computer as a hybrid system, various design plant disturbances were studied to ensure continued plant operation under these prescribed conditions. The Study presents the sensitivity of the control system performance to

variations in plant system parameter setpoints. For each mode of plant operation, a group of optimum controller setpoints was determined. In areas where the resultant setpoints are different, compromises based on the optimum overall performance were made and verified. A consistent set of control system setpoints was derived to ensure that no setpoint adjustments are required during core lifetime for the full range of automatic control between 15 and 100 percent power. Based on such setpoints, the plant transient response to each specified disturbance was predicted for various times throughout core life to provide an indication of the sensitivity of transient performance to the reactor physics parameters associated with chemical shim operation.

The results show that with these setpoints the control system will provide stable response with adequate margins to protection system setpoints during expected transient operations. The effects of the control system setpoints on the transient analyses are bounded by the assumptions used in accident analyses within this Chapter.

The Study was comprised of analyses of the following control systems: rod control, steam dump, steam generator level, pressurizer pressure, pressurizer level, feedwater pump speed, control rod insertion limits, and rod stops and turbine runbacks.

15.0.3 PLANT CHARACTERISTICS AND INITIAL CONDITIONS ASSUMED IN THE ACCIDENT ANALYSES

15.0.3.1 Design Plant Conditions

Table 15.0-1 lists the principal power rating values which are assumed in analyses performed in this report. The nominal NSSS thermal power output includes the rated thermal power plus the thermal power generated by the reactor coolant pumps.

The initial power operating conditions assumed in the accident analyses include allowance for errors in steady state power determination unless otherwise noted.

15.0.3.2 Initial Conditions

For most accidents which are DNB limited, nominal values of initial conditions are assumed. The allowances on power, temperature, and pressure are determined on a statistical basis and are included in the DNBR limit. This procedure is known as the Revised Thermal Design Procedure (Reference 6) and is discussed more fully in Section 4.4. For accidents which are not DNB limited, or in which the Revised Thermal Design Procedure is not employed, the initial conditions are obtained by adding the maximum steady state errors to rated values.

The following steady state errors were assumed in the analyses:

1.	Core power	±0.6 percent allowance for calorimetric error
2.	Average Reactor Coolant Temperature	± 6°F allowance for controller System temperature deadband and measurement error and steam generator fouling penalty

3.	Pressurizer pressure	±30 pounds per square inch (psi) allowance for steady state fluctuations and measurement error
4.	Pressurizer water level	+5% of normal programmed water level, unless indicated otherwise

Initial values for core power, average Reactor Coolant System temperature and pressurizer pressure are selected to minimize the initial departure from nucleate boiling ratio (DNBR) unless otherwise stated in the sections describing specific accidents.

Pressurizer water level is a controlled parameter, programmed to vary with temperature during normal plant operation. For those events which are sensitive to the initial pressurizer water volume, the value used in the Chapter 15 analyses is chosen to be consistent with the normal level program including an error allowance. No credit has been taken in the Chapter 15 non-LOCA analyses for safety functions initiated by pressurizer water level trips.

15.0.3.3 Power Distribution

The transient response of the reactor system is dependent on the initial power distribution. The nuclear design of the reactor core minimizes adverse power distribution through the placement of control rods and operating instructions. Power distribution may be characterized by the radial peaking factor ($F_{\Delta H}$) and the total peaking factor (F_q). The peaking factor limits are given in the Technical Specifications.

For transients which may be DNB limited the radial peaking factor is of importance. The radial peaking factor increases with decreasing power level due to rod insertion. This increase in $F_{\Delta H}$ is included in the core limits illustrated in Figure 15.0-1. All transients that may be DNB limited are assumed to begin with a $F_{\Delta H}$ consistent with the initial power level defined in the Technical Specifications.

The axial power shapes used in the DNB calculations are discussed in Section 4.4

The radial and axial power distributions described above are input to the DNB analysis as described in Section 4.4.

For transients which may be overpower limited the total peaking factor (F_q) is of importance. All transients that may be overpower limited are assumed to begin with plant conditions including power distributions which are consistent with reactor operation as defined in the Technical Specifications.

For overpower transients which are slow with respect to the fuel rod thermal time constant, for example the Chemical and Volume Control System malfunction that results in a decrease in the boron concentration in the reactor coolant incident which lasts many minutes, and the excessive increase in secondary steam flow incident which may reach equilibrium without causing a reactor trip, the fuel rod thermal evaluations are performed as discussed in Section 4.4. For overpower transients which are fast with respect to the fuel rod thermal time constant, for example the uncontrolled rod cluster control assembly bank withdrawal from subcritical or low power startup

and rod cluster control assembly ejection incidents which result in a large power rise over a few seconds, a detailed fuel heat transfer calculation must be performed.

15.0.4 REACTIVITY COEFFICIENTS ASSUMED IN THE ACCIDENT ANALYSES

The transient response of the reactor system is dependent on reactivity feedback effects, in particular the moderator temperature coefficient and the Doppler power coefficient. These reactivity coefficients and their values are discussed in detail in Chapter 4.

In the analysis of certain events, conservatism requires the use of large reactivity coefficient values whereas in the analysis of other events, conservatism requires the use of small reactivity coefficient values. Some analyses such as loss of reactor coolant from cracks or ruptures in the Reactor Coolant System do not depend on reactivity feedback effects. The justification for use of conservatively large versus small reactivity coefficient values is treated on an event-by-event basis. In some cases conservative combinations of parameters are used to bound the effects of core life. For example, in a load increase transient it is conservative to use a small Doppler defect and a small moderator coefficient.

A moderator temperature coefficient which is positive at reduced power levels has been considered in the analyses for both Units. As shown in Figure 15.0-6, the moderator temperature coefficient would be no greater than +5 pcm/°F below 70% of rated power, ramping down to 0 pcm/°F at 100% power.

15.0.5 ROD CLUSTER CONTROL ASSEMBLY INSERTION CHARACTERISTICS

All accident analysis results contained herein are applicable to all the control rod types defined in Chapter 4. The negative reactivity insertion following a reactor trip is a function of the acceleration of the rod cluster control assemblies and the variation in rod worth as a function of rod position. With respect to accident analyses, the critical parameter is the time of insertion up to the dashpot entry or approximately 85 percent of the rod cluster travel. The rod cluster control assembly position versus time assumed in accident analyses is shown in Figure 15.0-3. The RCCA drop time assumed in the accident analyses is greater than or equal to the value specified in the plant Technical Specifications.

Figure 15.0-4 shows the fraction of total negative reactivity insertion versus normalized rod position for a core where the axial distribution is skewed to the lower region of the core. An axial distribution which is skewed to the lower region of the core can arise from an unbalanced xenon distribution. This curve, or another like it which has been similarly developed, is used to compute the negative reactivity insertion versus time following a reactor trip which is input to all point kinetics core models used in transient analyses. The bottom skewed power distribution itself is not an input into the point kinetics core model.

There is inherent conservatism in the use of a curve such as Figure 15.0-4 in that it is based on a skewed flux distribution which would exist relatively infrequently. For cases other than those associated with unbalanced xenon distributions, significant negative reactivity would have been inserted due to the more favorable axial distribution existing prior to trip.

A representative normalized rod cluster control assembly negative reactivity insertion versus time is shown in Figure 15.0-5. The curve shown in this figure was obtained from Figures 15.0-3 and 15.0-4. A total negative reactivity insertion of 4% $\Delta k/k$ following a trip is assumed in the transient

analyses except where specifically noted otherwise. This assumption is conservative with respect to the calculated trip reactivity worth available as shown in Table 4.3-3. For Figures 15.0-3 and 15.0-4, the rod cluster control assembly drop time is normalized to the insertion time to dashpot entry in order to provide a bounding analysis for all rod cluster control assemblies to be used in the CPNPP cores, as previously stated.

The normalized rod cluster control assembly negative reactivity insertion versus time curve for an axial power distribution skewed to the bottom is used in those transient analyses for which a point kinetics core model is used. Where special analyses require use of three dimensional or axial one dimensional core models, the negative reactivity insertion resulting from the reactor trip is calculated directly by the reactor kinetics code and is not separable from the other reactivity feedback effects. In this case, the rod cluster control assembly position versus time of Figure 15.0-3 is used as code input.

15.0.6 TRIP POINTS AND TIME DELAYS TO TRIP ASSUMED IN ACCIDENT ANALYSES

A reactor trip signal acts to open two trip breakers connected in series feeding power to the control rod drive mechanisms. The loss of power to the mechanism coils causes the mechanisms to release the rod cluster control assemblies which then fall by gravity into the core. There are various instrumentation delays associated with each trip function, including delays in signal actuation, in opening the trip breakers, and in the release of the rods by the mechanisms. Limiting trip setpoints assumed in accident analyses and the time delay which occurs between generation of the reactor trip signal and the point at which the rods are free to fall for each trip function are given in Table 15.0-4. Reference is made in that table to the Overtemperature and Overpower N-16 trip shown in Figure 15.0-1.

The difference between the limiting trip point assumed for the analysis and the nominal trip point represents an allowance for process measurement errors, instrumentation channel error and setpoint error. For additional information, see Section 7.1.2.1.9 and the CPNPP Technical Specifications. Nominal trip setpoints are specified in the plant Technical Specifications. During plant startup tests it will be demonstrated that actual instrument time delays are equal to or less than the assumed values. Additionally, protection system channels are calibrated and instrument response times determined periodically in accordance with the plant Technical Specifications.

15.0.7 INSTRUMENTATION DRIFT AND CALORIMETRIC ERRORS - POWER RANGE NEUTRON FLUX

This section has been deleted. See Section 15.0.6 above.

15.0.8 PLANT SYSTEMS AND COMPONENTS AVAILABLE FOR MITIGATION OF ACCIDENT EFFECTS

The NSSS is designed to afford proper protection against the possible effects of natural phenomena, postulated environmental conditions and dynamic effects of the postulated accidents. In addition, the design incorporates features which minimize the probability and effects of fires and explosions. Chapter 17 discusses the quality assurance program which has been implemented to assure that the NSSS will satisfactorily perform its assigned safety functions. The incorporation of these features in the NSSS, coupled with the reliability of the design, ensures that the normally operating systems and components listed in Table 15.0-6 will be available for mitigation of the events discussed in Chapter 15. In determining which systems

are necessary to mitigate the effects of these postulated events, the classification system of ANSI-N18.2-1973 is utilized. The design of "systems important to safety" (including protection systems) is consistent with IEEE Standard 379-1972 and Regulatory Guide 1.53 in the application of the single failure criterion.

In the analysis of the Chapter 15 events, control system action is considered only if that action results in more severe accident results. No credit is taken for control system operation if that operation mitigates the results of an accident. For some accidents, the analysis is performed both with and without control system operation to determine the worst case.

15.0.9 FISSION PRODUCT INVENTORIES

15.0.9.1 Activities in the Core

The calculation of the core iodine fission product inventory is consistent with the inventories given in TID-14844 [1]. The fission product inventories for other isotopes which are important from a health hazards point of view are calculated using the data from APED- 5398 [2]. These inventories are given in Table 15.0-7. The isotopes included in Table 15.0-7 are the isotopes controlling from considerations of inhalation dose (iodines) and from direct dose due to immersion (noble gases).

The isotopic yields used in the calculations are from the data of APED-5398, utilizing the isotopic yield data for thermal fissioning of uranium-235 as the sole fissioning source. The change in fission product inventory resulting from the fissioning of other fissionable atoms has been reviewed. The results of this review indicated that inclusion of all fission source data would result in a small (less than 10 percent) change in the isotopic inventories.

15.0.9.2 Activities in Coolant

The activities in the reactor coolant as well as in the volume control tank, pressurizer and waste gas decay tanks are given in Chapter 11 along with the data on which these computations are based.

15.0.10 RESIDUAL DECAY HEAT

15.0.10.1 Total Residual Heat

Residual heat in a subcritical core is calculated for the loss of coolant accident per the requirements of Appendix K of 10CFR50.46. (See Section 15.6 for additional details). These requirements include assuming infinite irradiation time before the core goes subcritical to determine fission product decay energy. For all other accidents, conservative core residual heat generation is based upon the American National Standard for Decay Heat Power in Light Water Reactors, ANSI/ANS-5.1-1979, and the assumption of an infinite irradiation time and a two-sigma uncertainty allowance.

15.0.10.2 Distribution of Decay Heat Following Loss of Coolant Accident

During a loss of coolant accident the core is rapidly shut down by void formation or rod cluster control assembly insertion, or both, and a large fraction of the heat generation to be considered comes from fission product decay gamma rays. This heat is not distributed in the same manner

as steady state fission power. Local peaking effects which are important for the neutron dependent part of the heat generation do not apply to the gamma ray contribution. The steady state factor of 97.4 percent which represents the fraction of heat generated within the clad and pellet drops to a cycle-specific value (typically approximately 0.96) for the hot rod in a loss of coolant accident.

For example, consider the transient resulting from the postulated double ended break of the largest Reactor Coolant System pipe; 1/2 second after the rupture about 30 percent of the heat generated in the fuel rods is from gamma ray absorption. The gamma power shape is less peaked than the steady state fission power shape, reducing the energy deposited in the hot rod at the expense of adjacent colder rods. A conservative estimate of this effect is a reduction of 10 percent of the gamma ray contribution or 3 percent of the total. Since the water density is considerably reduced at this time, an average of about 98 percent of the available heat is deposited in the fuel rods, and the remainder is absorbed by water, thimbles, sleeves and grids. The net effect is a cycle-specific factor (typically about 0.96), which is less than 0.974, to be applied to the heat production in the hot rod.

15.0.11 COMPUTER CODES UTILIZED

The computer codes used to analyze each transient are discussed in the methods of analysis section of that transient.

REFERENCES

1. DiNunno, J. J., et al., “Calculation of Distance Factors for Power and Test Reactor Sites,” TID-14844, March 1962.

2. Meek, M. E. and Rider, B. F., “Summary of Fission Product Yields for U-235, U-238, Pu-239, and Pu-241 at Thermal Fission Spectrum and 14 Mev Neutron Energies,” APED-5398, March 1968.

3. Toner, D. F. and Scott, J. S., “Fission-Product Release from UO_2,” Nuc. Safety 3, No. 2, 15-20, December 1961.

4. Belle, J., “Uranium Dioxide Properties and Nuclear Applications,” Naval Reactors, Division of Reactor Development United States Atomic Energy Commission, 1961.

5. Booth, A. H., “A Suggested Method for Calculating the Diffusion of Radioactive Rare Gas Fission Products From UO_2 Fuel Elements,” DCI-27, 1957.

6. Friedland, A. J., and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

TABLE 15.0-1
NUCLEAR STEAM SUPPLY SYSTEM POWER RATINGS

Nominal NSSS thermal power output (MWt)	3628
Assumed net thermal power generated by the reactor coolant pumps (MWt)	16
Rated core thermal power (MWt)	3612

TABLE 15.0-2
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TABLE 15.0-3
HAS BEEN DELETED.

TABLE 15.0-4
TRIP POINTS AND TIME DELAYS TO TRIP ASSUMED IN ACCIDENT ANALYSES

(Sheet 1 of 2)

Trip Function	Limiting Trip Point Assumed In Analysis	Time Delays (Seconds)
Power range high neutron flux, high setting	115%	0.5
Power range high neutron flux, low setting	35%	0.5
High neutron flux, P-8	60%	0.5
Overtemperature N-16	Variable see Figure 15.0-1	(a)
Overpower N-16	118.5%	2.0
High pressurizer pressure	2445 psig (Unit 1) 2422 psig (Unit 2)	1.25 (Units 1 and 2)
Low pressurizer pressure	1845 psig	2.0
Low reactor coolant flow (from loop flow detectors)	87% loop flow	1.0
Undervoltage trip	(b)	1.5(b)

TABLE 15.0-4
TRIP POINTS AND TIME DELAYS TO TRIP ASSUMED IN ACCIDENT ANALYSES

(Sheet 2 of 2)

Trip Function	Limiting Trip Point Assumed In Analysis	Time Delays (Seconds)
Underfrequency trip	57.2 Hz	0.6
Turbine trip	Not applicable	
Low-low steam generator water level	10% of narrow range level span	2.0
High steam generator level trip of the feedwater pumps and closure of feedwater systems valves, and turbine trip	100% of narrow range level span	2.0

a) Total time delay is 9.3 seconds. This total time delay has been analyzed separately to include up to a maximum RTD response time of 6.7 seconds or up to a maximum electronic response time of 3.3 seconds. However, the sum of the RTD response time and electronic response time must not exceed 9.3 seconds.

b) The undervoltage reactor trip setpoint is not explicitly modeled. Instead, the undervoltage delay assumed in the complete loss of flow (CLOF) analysis includes an allowance for the time delay between the loss of voltage to the RCP bus and the time at which the loss of voltage is detected. Therefore, the definition of the of the “reactor trip delay time” for the CLOF event is unique in that the assumed delay begins when the event begins (loss of power to the RCP busses), and not when the RCP bus undervoltage condition is detected.

TABLE 15.0-5
THIS TABLE IS DELETED

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 1 of 6)

	Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
15.1	Increase in Heat Removed by the Secondary System				
	Feedwater system malfunctions that result in a decrease in feedwater temperature	Overpower N-16, Power range high flux, Overtemperature N-16, Manual	-	-	-
	Feedwater system malfunctions that result in an increase in feedwater flow	Power range high flux, high steam generator level, manual	High steam generator level-produced feedwater isolation and turbine trip	Feedwater isolation valves	-
	Excessive increase in secondary steam flow	Power range high flux, Overtemperature N-16, Overpower N-16, manual	-	Pressurizer self-actuated safety valve steam generator safety valves	-
	Inadvertent opening of a steam generator relief or safety valve	Low pressurizer pressure, manual, SIS	Low pressurizer pressure, low compensated steam line pressure, manual	Feedwater isolation valves, steam line stop valves	Auxiliary Feedwater System, Safety Injection System

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 2 of 6)

	Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
	Steam system piping failure	SIS, low pressurizer pressure, manual	Low pressurizer pressure, low compensated steam line pressure, hi-1 Containment pressure, manual	Feedwater isolation valves, steam line stop valves	Auxiliary Feedwater Safety Injection
15.2	Decrease in Heat Removal by the Secondary System				
	Loss of external electrical load/ turbine trip	High pressurizer pressure, Overtemperature N-16, manual	-	Pressurizer safety valves, steam generator safety valves	-
	Loss of non-emergency AC power to the station auxiliaries	Steam generator lo-lo level, manual	Steam generator lo-lo level	Steam generator safety valves	Auxiliary Feedwater System
	Loss of normal feedwater flow	Steam generator lo-lo level, manual	Steam generator lo-lo level	Steam generator safety valves	Auxiliary Feedwater System

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 3 of 6)

	Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
	Feedwater system pipe break	Steam generator lo-lo level, high pressurizer pressure, SIS, manual	High Containment pressure, steam generator lo-lo water level, low compensated steam line pressure	Steam line isolation valves, feedline isolation, pressurizer safety valves, steam generator safety valves	Auxiliary Feedwater System, Safety Injection System
15.3	Decrease in Reactor Coolant System Flow Rate				
	Partial and complete loss of forced reactor coolant flow	Low flow, under-voltage, underfrequency, manual	-	Steam generator safety valves	-
	Reactor coolant pump shaft seizure (locked rotor)	Low flow, manual	-	Pressurizer safety valves, steam generator safety valves	-
15.4	Reactivity and Power Distribution Anomalies				
	Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low power startup condition	Power range high flux (low setpoint), manual	-	-	-

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 4 of 6)

Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
Uncontrolled rod cluster control assembly bank withdrawal at power	Power range high flux, Power range high flux rate. Overtemperature N-16 Overpower N-16, high pressurizer pressure, manual	-	Pressurizer safety valves, steam generator safety valves	-
Rod cluster control assembly misalignment	Low pressurizer pressure, Overtemperature N-16, manual	-	-	-
Chemical and Volume Control System malfunction that results in a decrease in boron concentration in the reactor coolant	Source range high flux, power range high flux, power range low flux, Overtemperature N-16, manual	-	Low insertion limit annunciators for boration, Boron dilution mitigation system VCT water level-high and CVCS/RMWS annunciators	-
Spectrum of rod cluster control assembly ejection accidents	Power range high flux, power range low flux, high positive flux rate, manual	-	-	-

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 5 of 6)

	Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
15.5	Increase in Reactor Coolant Inventory				
	Inadvertent operation of the ECCS during power operation	Low pressurizer pressure, manual, safety injection trip	-	-	Safety Injection System
15.6	Decrease in Reactor Coolant Inventory				
	Inadvertent opening of a pressurizer safety or relief valve	Low Pressurizer pressure, Overtemperature N-16, manual	-	-	-
	Steam generator tube failure	Low pressurizer pressure Overtemperature N-16 manual	Low pressurizer pressure, manual	Service Water System, Component Cooling Water System, steam generator safety and/or relief valves, steam line stop valves, pressurizer relief valves	Emergency Core Cooling System, Auxiliary Feedwater System, Emergency Power System

TABLE 15.0-6
PLANT SYSTEMS AND EQUIPMENT AVAILABLE FOR TRANSIENT AND ACCIDENT CONDITIONS

(Sheet 6 of 6)

Incident	Reactor Trip Functions	ESF Actuation Functions	Other Equipment	ESF Equipment
Loss of coolant accidents resulting from the spectrum of postulated piping breaks within the reactor coolant pressure boundary	Reactor Trip System	Engineered Safety Features Actuation System	Service Water System, Component Cooling Water System, steam generator safety and/or relief	Emergency Core Cooling System, Auxiliary Feedwater System, Containment Heat Removal System, Emergency Power System

TABLE 15.0-7
iCORE ACTIVITIES

Isotope	Curies in Core ($\times 10^7$)
Kr-83m	1.15
Kr-895	2.43
Kr-85m	0.113
Kr-87	4.75
Kr-88	6.36
Xe-131m	0.109
Xe-133	20.6
Xe-133m	0.644
Xe-135	3.65
Xe-138	17.4
I-131	10.2
I-132	14.7
I-133	20.6
I-134	23.0
I-135	19.6

TABLE 15.0-8
POWER-TEMPERATURE DISTRIBUTION FOR FULL CORE

Fuel Temperature Range (°F)	Percent Volume of Core Within Temperature Range	Percent of Power Within Temperature Range (MWt)
3800 - 3600	0.005	0.01
3600 - 3400	0.03	0.08
3400 - 3200	0.15	0.33
3200 - 3000	0.46	0.96
3000 - 2800	1.10	2.16
2800 - 2600	2.23	4.11
2600 - 2400	3.64	6.31
2400 - 2200	4.93	8.18
2200 - 2000	6.03	9.67
<2000	81.43	68.19

15.1 INCREASE IN HEAT REMOVAL BY THE SECONDARY SYSTEM

A number of events have been postulated which could result in an increase in heat removal from the Reactor Coolant System (RCS) by the Secondary System. Detailed analyses are presented for several such events which have been identified as limiting cases.

Discussions of the following RCS cooldown events are presented in this section:

1. Feedwater system malfunctions that result in a decrease in feedwater temperature.

2. Feedwater system malfunctions that result in an increase in feedwater flow.

3. Excessive increase in secondary steam flow.

4. Inadvertent opening of a steam generator relief or safety valve.

5. Steam system piping failure.

The above are considered to be American Nuclear Society (ANS) Condition II events, with the exception of steam system piping failures, which are considered to be ANS Condition III (minor) and Condition IV (major) events. Section 15.0.1 contains a discussion of ANS classifications and applicable acceptance criteria.

15.1.1 FEEDWATER SYSTEM MALFUNCTIONS THAT RESULT IN A DECREASE IN FEEDWATER TEMPERATURE

15.1.1.1 Identification of Causes and Accident Description

Reductions in feedwater temperature will cause an increase in core power by decreasing reactor coolant temperature. Such transients are attenuated by the thermal capacity of the secondary plant and of the RCS. The overpower/overtemperature protection (neutron overpower, Overtemperature and Overpower N-16 trips) prevents any power increase which could lead to a departure from nucleate boiling ratio (DNBR) less than the limit value.

A reduction in feedwater temperature may be caused by the accidental opening of a feedwater bypass valve which diverts flow around a portion of the feedwater heaters. In the event of an accidental opening of the1 bypass valve, there is a sudden reduction in feedwater inlet temperature to the steam generators. At power, this increased subcooling will create a greater load demand on the RCS.

With the plant at no-load conditions the addition of cold feedwater may cause a decrease in RCS temperature and thus a reactivity insertion due to the effects of the negative moderator coefficient of reactivity. However, the rate of energy change is reduced as load and feedwater flow decrease, so the no-load transient is less severe than the full power case.

The net effect on the RCS due to a reduction in feedwater temperature is similar to the effect of increasing secondary steam flow; i.e., the reactor will reach a new equilibrium condition at a power level corresponding to the new steam generator ΔT unless terminated by a reactor trip.

A decrease in normal feedwater temperature is classified as an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

The protection available to mitigate the consequences of a decrease in feedwater temperature is discussed in Section 15.0.8 and listed in Table 15.0-6.

15.1.1.2 Analysis of Effects and Consequences

Method of Analysis

This transient is analyzed by computing conditions at the feedwater pump inlet following opening of the low pressure feedwater heater bypass valve. This gives the new feedwater conditions at the steam generator inlet. The RETRAN computer code (Reference 3) is used to determine the consequences of the reduced feedwater temperature.

The following assumptions are made:

1. Plant initial power level corresponding to guaranteed Nuclear Steam Supply System (NSSS) thermal output.

2. Low pressure heater bypass valve opens, resulting in condensate flow splitting between the bypass line and the low pressure heaters; the flow through each path is proportional to the pressure drops.

3. Heater drain pumps trip; this increases the effect of the cold bypass flow.

4. All high pressure extraction steam is isolated as a direct consequence of the low pressure feedwater heater bypass valve opening.

Opening of a low pressure heater bypass valve, tripping of the heater drain pumps, and isolating all high pressure extraction steam causes a reduction in feedwater temperature which increases the thermal load on the primary system. The calculated reduction in feedwater temperature is about 246°F, resulting in a significant increase in heat load on the primary system.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

Reactor power increases until the reactor protection system functions to trip the reactor due to an overpower N-16 condition. No single active failure will prevent operation of the reactor protection system. Note that Unit 1 bounds Unit 2 because the Unit 1 steam generators have a greater heat transfer area, which is conservative for cooldown events.

15.1.1.3 Conclusions

For Units 1 and 2 the results of the analysis show that the DNBR remains above the limit value; thus the acceptance criterion for the event is satisfied.

15.1.2 FEEDWATER SYSTEM MALFUNCTIONS THAT RESULT IN AN INCREASE IN FEEDWATER FLOW

15.1.2.1 Identification of Causes and Accident Description

Addition of excessive feedwater will cause an increase in core power by decreasing reactor coolant temperature. Such transients are attenuated by the thermal capacity of the secondary plant and of the RCS. The overpower/overtemperature protection (neutron overpower, Overtemperature and Overpower N-16 trips) prevents any power increase which could lead to a DNBR less than the limit value.

An example of excessive feedwater flow would be the full opening of two feedwater control valves due to a feedwater control system malfunction or an operator error. At power this excess flow causes a greater load demand on the RCS due to increased subcooling in the two steam generators. With the plant at no-load conditions, the addition of cold feedwater may cause a decrease in RCS temperature and thus a reactivity insertion due to the effects of the negative moderator coefficient of reactivity.

Continuous addition of excessive feedwater is prevented by the steam generator high-high level trip. When the steam generator water levels in the affected loops reach the high-high level setpoint, all feedwater control valves and feedwater pump discharge valves are automatically closed and the main feedwater pumps are tripped. This prevents continuous addition of the feedwater. In addition, a turbine trip is initiated.

Following turbine trip, the reactor will be automatically tripped, either directly due to turbine trip or due to an Overtemperature N-16 signal. If the reactor were in the automatic control mode, the control rods would be inserted prior to the turbine trip, and the ensuing transient would then be similar to a loss of load (turbine trip event) as analyzed in Section 15.2.3.

An increase in normal feedwater flow is classified as an ANS Condition II event, a fault of moderate frequency. See Section 15.0.1 for a discussion of ANS Condition II events.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6.

15.1.2.2 Analysis of Effects and Consequences

Method of Analysis

The excessive heat removal due to a feedwater system malfunction transient is analyzed using the RETRAN computer code (Reference 3). For cases initiated from zero load conditions, the VIPRE computer code (Reference 4) is also used.

The system is analyzed to demonstrate plant behavior in the event that excessive feedwater addition occurs due to a control system malfunction or operator error which allows two feedwater control valves to open fully. The following cases are analyzed:

1. Accidental opening of one and two feedwater control valves with the reactor just critical at zero load conditions assuming a conservatively large negative moderator temperature coefficient.

2. Accidental opening of one and two feedwater control valves with the reactor at full power with automatic and manual rod control.

Asymmetric feedwater flow may result in asymmetric core inlet fluid conditions. These asymmetries are considered in the DNBR analyses.

The reactivity insertion rate following a feedwater system malfunction is calculated with the following assumptions:

1. For the multiple-loop feedwater control valve accident at full power, two feedwater control valves are assumed to malfunction resulting in a step increase to 168.2 percent of nominal feedwater flow to each of the two affected steam generators. The failure of a single feedwater control valve is also considered.

2. For the single-loop feedwater control valve accident at full power, one feedwater control valve is assumed to malfunction resulting in a step increase to 207.5 percent of nominal feedwater flow to the affected steam generator.

3. For the multiple-loop feedwater control valve accident at zero load condition, two feedwater control valves are assumed to malfunction, which results in a step increase to 158.7 percent of the nominal feedwater flow to the two affected steam generators.

4. For the single-loop feedwater control valve accident at zero load condition, one feedwater control valve is assumed to malfunction, which results in a step increase to 254.3 percent of the nominal feedwater flow to the affected steam generator.

5. For the zero load condition, feedwater temperature is at a conservatively low value of 70°F.

6. No credit is taken for the heat capacity of the RCS and steam generator thick metal in attenuating the resulting plant cooldown.

7. The feedwater flow resulting from the fully open control valves is terminated by a steam generator high-high level trip signal which closes all feedwater control and isolation valves, trips the main feedwater pumps, and trips the turbine.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

Normal reactor control systems and engineered safety systems are not required to function. The Reactor Protection System may function to trip the reactor due to overpower or high-high steam generator water level conditions. No single active failure will prevent operation of the Reactor Protection System. Note that Unit 1 bounds Unit 2 because the Unit 1 steam generators have a greater heat transfer area, which is conservative for cooldown events.

15.1.2.3 Conclusions

For Units 1 and 2 the results of the analysis show that the DNBR encountered for an excessive feedwater addition at power is above the limit value; the DNBR design basis is described in Section 4.4. Additionally, it has been shown that the feedwater malfunction event at no-load is bounded by the feedwater malfunction event at full power.

15.1.3 EXCESSIVE INCREASE IN SECONDARY STEAM FLOW

15.1.3.1 Identification of Causes and Accident Description

An excessive increase in secondary system steam flow (excessive load increase incident) is defined as a rapid increase in steam flow that causes a power mismatch between the reactor core power and the steam generator load demand. The Reactor Control System is designed to accommodate a 10 percent step load increase or a 5 percent per minute ramp load increase in the range of 15 to 100 percent of full power. Any loading rate in excess of these values may cause a reactor trip actuated by the Reactor Protection System. Steam flow increases greater than 10 percent are analyzed in Sections 15.1.4 and 15.1.5.

This accident could result from either an administrative violation such as excessive loading by the operator or an equipment malfunction in the steam dump control or turbine speed control.

During power operation, steam dump to the condenser is controlled by reactor coolant condition signals; i.e., high reactor coolant temperature indicates a need for steam dump. A single controller malfunction does not cause steam dump; an interlock is provided which blocks the opening of the valves unless a large turbine load decrease or a turbine trip has occurred.

For all cases, the plant rapidly reaches a stabilized condition at the higher power level. Normal plant operating procedures would then be followed to reduce power.

The excessive load increase incident is an overpower transient for which the fuel temperatures will rise. Reactor trip may not occur for some of the cases analyzed, and the plant reaches a new equilibrium condition at a higher power level corresponding to the increase in steam flow.

Protection against an excessive load increase accident is provided by the following Reactor Protection System signals:

1. Overpower N-16.

2. Overtemperature N-16.

3. Power range high neutron flux.

An excessive load increase incident is considered to be an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

15.1.3.2 Analysis of Effects and Consequences

Method of Analysis

This event does not lead to a serious challenge to the acceptance criteria and a reactor trip is not typically generated. As such, a detailed reanalysis of this event is not necessary. A simplified statepoint evaluation, assuming a 10% step load increase, was performed and the results confirmed that core DNB limits are not challenged following this event.

Four cases are analyzed to demonstrate the plant behavior following a 10 percent step load increase from rated load. These cases are as follows:

1. Reactor control in manual with minimum moderator reactivity feedback.

2. Reactor control in manual with maximum moderator reactivity feedback.

3. Reactor control in automatic with minimum moderator reactivity feedback.

4. Reactor control in automatic with maximum moderator reactivity feedback.

For the minimum reactivity feedback cases, the core has the least negative moderator temperature coefficient of reactivity and therefore the least inherent transient capability. For the maximum reactivity feedback cases, the moderator temperature coefficient of reactivity has its highest absolute value. This results in the largest amount of reactivity feedback due to changes in coolant temperature. For the cases with automatic rod control, no credit was taken for N-16 trips on overtemperature or overpower in order to demonstrate the inherent transient capability of the plant. Under actual operating conditions, such a trip may occur, after which the plant would quickly stabilize.

A conservative limit on the turbine throttle valve opening is assumed, and all cases are studied without credit being taken for pressurizer heaters. Initial operating conditions are assumed at maximum values consistent with the steady state full power operation, plus uncertainties for calibration and instrument errors. These assumptions result in minimum margin to core DNB at the start of the accident.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

Normal reactor control systems and engineered safety systems are not required to function. The Reactor Protection System is assumed to be operable; however, reactor trip is not encountered for most cases due to the error allowances assumed in the setpoints. No single active failure will prevent the Reactor Protection System from performing its intended function.

The cases which assume automatic rod control are analyzed to ensure that the worst case is considered. The automatic function is not required.

15.1.3.3 Conclusions

The analysis discussed above showed for Units 1 and 2 that for a 10 percent step load increase, the DNBR remained above the limit value; the design basis for DNBR is described in Section 4.4. The plant reaches a stabilized condition rapidly following the load increase. Furthermore, the results of a simplified statepoint evaluation performed for a 10% step load increase with a nominal core power of 3612 MWt confirm that the core thermal limit lines are not challenged, and that the minimum DNBR during this transient will remain above the safety analysis limit value for Units 1 and 2.

15.1.4 INADVERTENT OPENING OF A STEAM GENERATOR RELIEF OR SAFETY VALVE

15.1.4.1 Identification of Causes and Accident Description

The most severe core conditions resulting from an accidental depressurization of the Main Steam System are associated with an inadvertent opening, with failure to close, of the largest of any single relief or safety valve, or of the steam dump system. The analyses performed assuming a rupture of a main steam line are given in Section 15.1.5.

The steam release as a consequence of this accident results in an initial increase in steam flow which decreases during the accident as the steam pressure falls. The energy removal from the RCS causes a reduction of coolant temperature and pressure. In the presence of a negative moderator temperature coefficient, the cooldown results in an insertion of positive reactivity.

Should the reactor be just critical or operating at power at the time of a steam release, the reactor will be tripped by the normal overpower protection when power level reaches a trip point. Following a trip at power, the RCS contains more stored energy than at no-load, the average coolant temperature is higher than at no-load and there is appreciable energy stored in the fuel. Thus, the additional stored energy is removed via the cooldown caused by the steam release before the no-load conditions of RCS temperature and shutdown margin assumed in the analyses are reached. After the additional stored energy has been removed, the cooldown and reactivity insertions proceed in the same manner as in the analysis which assumes no-load condition at time zero. However, since the initial steam generator water inventory is greatest at no-load, the magnitude and duration of the RCS cooldown are less for steam line release occurring at power.

An analysis of the inadvertent opening of a steam generator relief or safety valve must demonstrate that the following criterion is satisfied:

Assuming a stuck rod cluster control assembly, with offsite power available, and assuming a single failure in the Engineered Safety Features System there will be no consequential damage to the core or the RCS, after reactor trip for a steam release equivalent to the spurious opening, with failure to close, of the largest of any single steam dump, relief, or safety valve.

Accidental depressurization of the secondary system is classified as an ANS Condition II event. See Section 15.0.1 for a discussion of Condition II events.

The following systems provide the necessary protection against an accidental depressurization of the Main Steam System.

1. Safety Injection System actuation from any of the following:

   a. Two out of four pressurizer low pressure signals.

   b. Two out of three high-1 Containment pressure signals.

   c. Two out of three low steamline pressure signals in a loop.

2. The overpower reactor trips (neutron flux and N-16) and the reactor trip occurring in conjunction with receipt of the safety injection signal.

3. Redundant isolation of the main feedwater lines

   Sustained high feedwater flow would cause additional cooldown. Therefore, in addition to the normal control action which will close the main feedwater valves following reactor trip, a safety injection signal will rapidly close all feedwater isolation valves and back up feedwater control valves, trip the main feedwater pumps, and close the feedwater pump discharge valves.

4. Trip of the fast-acting steam line stop valves (designed to close in less than 5 seconds) on:

   a. High-2 Containment pressure.

   b. Two out of three low steam line pressure signals in any loop (above Permissive P-11).

   c. High negative steam pressure rate indication from two out of three signals in any loop (below Permissive P-11).

Plant systems and equipment which are available to mitigate the effects of the accident are also discussed in Section 15.0.8 and listed in Table 15.0-6.

Due to the size limitations placed on the steam generator safety and relief valves, the failure of a single valve is capable of increasing steam flow by approximately 5 percent of full plant steam load. At full power, the power increase would be less severe than the excessive increase in secondary system steam flow transient presented in Section 15.1.3. For post reactor trip conditions or hot shutdown conditions, the analysis of this event is bounded by the main steam line rupture analysis presented in Section 15.1.5 due to the large break area modeled, which is equivalent to the maximum steam generator steam nozzle flow area (1.4 ft^2).

Since an analysis of the inadvertent opening of a steam generator relief or safety valve event is always bounded by the main steam line rupture analysis presented in Section 15.1.5, no explicit inadvertent opening of a steam generator relief or safety valve analysis was performed for the power uprate program.

15.1.4.2 Analysis of Effects and Consequences (Historical)

Method of Analysis

The following analyses of a secondary system steam release are performed for this section.

1. A full plant digital computer simulation using the system analyses code to determine RCS temperature and pressure during cooldown, and the effect of safety injection.

2. Analyses to determine that there is no damage to the core or the reactor coolant system.

The following conditions are assumed to exist at the time of a secondary steam system release:

1. End-of-life shutdown margin at no-load, equilibrium xenon conditions, with the most reactive rod cluster control assembly stuck in its fully withdrawn position. Operation of rod cluster control assembly banks during core burnup is restricted in such a way that addition of positive reactivity in a secondary system steam release accident will not lead to a more adverse condition than the case analyzed.

2. A positive moderator density coefficient corresponding to the end-of-life rodded core with the most reactive rod cluster control assembly in the fully withdrawn position.

3. Minimum capability for injection of concentrated boric acid solution corresponding to the most restrictive single failure in the Safety Injection System. This corresponds to the flow delivered by one charging pump, delivering its full contents to the cold leg header. Unborated water must be swept from the safety injection lines downsteam of the Refueling Water Storage Tank (RWST) and the valve realignment from the Volume Control Tank (VCT) to the RWST must be completed prior to the delivery of concentrated boric acid to the reactor coolant loops. This effect has been accounted for in the analysis.

4. The case studied is a steam flow of 308 pounds per second at 1200 pounds per square inch absolute (psia) with offsite power available. This is the maximum capacity of any single steam dump, relief, or safety valve. Because the return-to-power condition is bounded by the analysis of the steam line Break accident presented in Section 15.1.5, the inadvertent opening of a relief valve is analyzed from full power conditions to demonstrate compliance with the event acceptance criteria near the time of reactor trip.

5. Perfect moisture separation in the steam generator is assumed.

6. No credit is taken for the energy stored in the system metal other than that of the fuel elements and other steam generator tubes. Since the transient occurs over a period of about 5 minutes, the neglected stored energy is likely to have a significant effect in slowing the cooldown.

15.1.4.3 Conclusions

For Units 1 and 2, the analysis shows that the criteria stated earlier in this section are satisfied. For an accidental depressurization of the main steam system the DNB design limits are not exceeded. The radiological consequences of this event are not limiting. This case is bounded by the main steam line rupture case described in Section 15.1.5. An additional case has been examined wherein the steam dump system is assumed to inadvertently actuate. No credit is taken for the proper operation of the non-Class 1E steam dump solenoid valves when the low-low T_{avg} (P-12) setpoint was exceeded. If a Main Steam Isolation Valve is assumed to fail to close, a non uniform cooldown of the RCS would ensue; however, the event would remain bounded by the analysis of the Steamline Break event presented in Section 15.1.5, and all event acceptance criteria would continue to be met.

15.1.5 STEAM SYSTEM PIPING FAILURE

15.1.5.1 Identification of Causes and Accident Description

The steam release arising from a rupture of a main steam line would result in an initial increase in steam flow which decreases during the accident as the steam pressure falls. The energy removal from the RCS causes a reduction of coolant temperature and pressure. In the presence of a negative moderator temperature coefficient, the cooldown results in an insertion of positive reactivity. If the most reactive rod cluster control assembly (RCCA) is assumed stuck in its fully withdrawn position after reactor trip, there is an increased possibility that the core will become critical and return to power. A return to power following a steam line rupture is a potential problem mainly because of the high power peaking factors which exist assuming the most reactive RCCA to be stuck in its fully withdrawn position. The core is ultimately shut down by injection of boric acid delivered by the Safety Injection System and the ECCS accumulators.

The analysis of a main steam line rupture is performed to demonstrate that the following criteria are satisfied:

Assuming a stuck RCCA with or without offsite power, and assuming a single failure in the engineered safety features systems, the core remains in place and intact.

The offsite dose limits for a steam pipe rupture with a pre-accident iodine spike are the guideline values of 10CFR100. These guideline values are 300 rem thyroid and 25 rem whole body. For a steam pipe rupture with an accident-initiated iodine spike, the acceptance criterion is a “small fraction” of the 10CFR100 guideline values; or 30 rem thyroid and 2.5 rem whole body.

Although DNB and possible clad perforation following a steam pipe rupture are not necessarily unacceptable, the following analysis, in fact, shows that no DNB occurs for any rupture assuming the most reactive assembly stuck in its fully withdrawn position. The DNBR design basis is discussed in Section 4.4.

A major steam line rupture is classified as an ANS Condition IV event, i.e., a limiting fault. See Section 15.0.1 for a discussion of Condition IV events.

Effects of minor secondary system pipe breaks are bounded by the analysis presented in this section. Minor secondary system pipe breaks are classified as Condition III events, i.e., infrequent faults, as described in Section 15.0.1.3.

The major rupture of a steam line is the most limiting cooldown transient and is analyzed at zero power with no decay heat. Decay heat would retard the cooldown thereby reducing the return to power.

A detailed analysis of this transient with the most limiting break size, a double ended rupture, is presented here.

The following functions provide the protection for a steam line rupture:

1. Safety Injection System actuation from any of the following:

    a. Two out of four low pressurizer pressure signals.

b. Two out of three high-1 Containment pressure signals.

c. Two out of three low steam line pressure signals in any loop.

2. The overpower reactor trips (neutron flux and N-16) and the reactor trip occurring in conjunction with receipt of the safety injection signal.

3. Redundant isolation of the main feedwater lines

   Sustained high feedwater flow would cause additional cooldown. Therefore, in addition to the protective action which will close the main feedwater valves on a reactor trip coincident with low average RCS temperature, a safety injection signal will rapidly close all feedwater control valves and feedwater isolation valves, as well as trip the main feedwater pumps.

4. Trip of the fast acting steam line stop valves (designed to close in less than 5 seconds) on:

   a. High-2 Containment pressure.

   b. Two out of three low steam line pressure signals in any loop (above Permissive P-11).

   c. High negative steam pressure rate indication from two out of three signals in any loop (below Permissive P-11).

Fast-acting isolation valves are provided in each steam line; these valves will fully close within 10 seconds of a large break in the steam line. For breaks downsteam of the isolation valves, closure of all valves would completely terminate the blowdown. For any break, in any location, no more than one steam generator would experience an uncontrolled blowdown even if one of the isolation valves fails to close. For the steam line break accident analysis, credit is taken for a safety injection actuation (“S”) signal which causes all main steam stop valves to close. It also causes all feedwater isolation valves to close. In addition, the “S” signal generates a reactor trip signal which causes closure of other valves as described in Section 10.2.2.7.7. A description of steam line isolation is included in Chapter 10.

The main steam line flow restrictor for each loop is installed in the steam generator outlet nozzle and is an integral part of the shell of the steam generator. The effective throat area of the nozzle is 1.4 square feet, which is considerably less than the main steam pipe area; thus, the nozzle serves to limit the maximum steam flow for a break at any location.

Equipment required to be operational in the event of a main steam line break accident to mitigate the accident consequences and assure a safe shutdown of the plant includes instrumentation, electrical cables, containment electrical penetration assemblies, valves and pumps. Table 15.1-2 lists the equipment required in the recovery from a high energy line rupture. Not all equipment is required for any one particular break, since the requirements will vary depending upon postulated break location and details of balance of plant design and pipe rupture criteria as discussed elsewhere in this application. Design criteria and methods of protection of safety-related equipment from the dynamic effects of postulated piping ruptures are provided in Section 3.6N.

Equipment provided and relied upon for the Post-Accident Monitoring System is discussed in Section 7.5.

15.1.5.2 Analysis of Effects and Consequences

Method of Analysis

The analysis of the steam pipe rupture has been performed to determine:

1. The core heat flux and RCS temperature and pressure resulting from the cooldown following the steam line break. The RETRAN computer code (Reference 3) has been used.

2. The thermal-hydraulic behavior of the core following a steam line break. The detailed thermal-hydraulic digital computer code VIPRE (Reference 4) has been used to determine if the DNB design basis is met for the core conditions computed by RETRAN.

The following conditions were assumed to exist at the time of a main steam line break accident:

1. End-of-life shut down margin at no-load, equilibrium xenon conditions, and the most reactive RCCA stuck in its fully withdrawn position. Operation of the rod cluster control assembly banks during core burnup is restricted in such a way that addition of positive reactivity in a steam line break accident will not lead to a more adverse condition than the case analyzed.

2. A positive moderator density coefficient corresponding to the end-of-life rodded core with the most reactive RCCA in the fully withdrawn position.

   The core properties associated with the sector nearest the affected steam generator and those associated with the remaining sector were conservatively combined to obtain average core properties for reactivity feedback calculations. Further, it was conservatively assumed that the core power distribution was uniform. These two conditions cause underprediction of the reactivity feedback in the high power region near the stuck rod. To verify the conservatism of this method, the reactivity as well as the power distribution was checked for the limiting statepoints for the cases analyzed.

   This core analysis considered the Doppler reactivity from the high fuel temperature near the stuck RCCA, moderator feedback from the high water enthalpy near the stuck RCCA, power redistribution and nonuniform core inlet temperature effects. For cases in which steam generation occurs in the high flux regions of the core, the effect of void formation was also included. It was determined that the reactivity employed in the kinetics analysis sufficiently matched the reactivity calculated including the above local effects for the statepoints.

3. Minimum capability for injection of concentrated boric acid solution from the RWST corresponding to the most restrictive single failure in the Safety Injection System. The Emergency Core Cooling System consists of three systems: 1) the passive accumulators, 2) the Residual Heat Removal System, and 3) the Safety Injection System. Only the Safety Injection System and the accumulators are modeled for the steam line break accident analysis.

The Safety Injection System flow in the system model corresponds to that delivered by one charging pump delivering its full flow to the cold leg header. Unborated water must be swept from the safety injection lines downstream of the RWST and the valve realignment from the VCT to the RWST must be completed prior to the delivery of concentrated boric acid to the reactor coolant loops. This effect has been accounted for in the analysis.

For the cases where offsite power is assumed, a total delay of 27 seconds was assumed to account for SI signal generation, opening of the RWST suction isolation valves, closing of the volume control tank (VCT) outlet isolation valves, startup of the charging pumps, isolation of the normal charging flow path from the VCT and to align suction to the borated water in the RWST. The volume containing the unborated water is swept into the core before the borated water reaches the core. This delay, described above, is inherently included in the modeling.

In cases where offsite power is not available, an additional 10 second delay is assumed to start the diesels and then load the necessary safety injection equipment onto them.

4. Since the steam generators are provided with integral flow restrictors with a 1.4 square foot throat area, any rupture with a break area greater than 1.4 square feet, regardless of location, would have the same effect on the NSSS as the 1.4 square foot break.

5. Offsite power is assumed available so that full reactor coolant flow exists. The transient assumes an uncontrolled steam release from only one steam generator. Should the core be critical at near zero power when the rupture occurs, the initiation of safety injection by low steam line pressure will trip the reactor. During full power operations, a reactor trip will occur due to the generation of either an overpower/overtemperature N-16, or, when the containment temperature is greater than 200°F, a Hi-1 containment pressure trip signal. Steam release from more than one steam generator will be prevented by automatic trip of the fast acting isolation valves in the steam lines, by high containment pressure signals, or by low steam line pressure signals. Even with the failure of one valve, release is limited to a few seconds for the other steam generators while the one generator blows down. The steam line stop valves are designed to be fully closed in less than 5 seconds from receipt of a closure signal.

6. Power peaking factors corresponding to one stuck RCCA and non-uniform core inlet coolant temperatures are determined at end of core life. The coldest core inlet temperatures are assumed to occur in the sector with the stuck rod. The power peaking factors account for the effect of the local void in the region of the stuck control assembly during the return to power phase following the steam line break. This void in conjunction with the large negative moderator coefficient partially offsets the effect of the stuck assembly. The power peaking factors depend upon the core power, temperature, pressure, and flow, and, thus, are different for each case studied.

   The core parameters used for the analysis correspond to values determined from the respective transient analysis.

   The analysis assumes initial hot shutdown conditions at time zero since this represents the most pessimistic initial condition. Should the reactor be just critical or operating at power at the time of a steam line break, the reactor will be tripped by the normal

overpower protection system when power level reaches a trip point. Following a trip at power, the RCS contains more stored energy than at no-load, the average coolant temperature is higher than at no-load and there is appreciable energy stored in the fuel. Thus, the additional stored energy is removed via the cooldown caused by the steam line break before the no-load conditions of RCS temperature and shutdown margin assumed in the analyses are reached. After the additional stored energy has been removed, the cooldown and reactivity insertions proceed in the same manner as in the analysis which assumes no-load conditions at time zero.

15.1.5.3 Environmental Consequences

The postulated accidents involving release of steam from the secondary system do not result in a release of radioactivity unless there is leakage from the RCS to the secondary system in the steam generators. A conservative analysis of the potential offsite doses resulting from this accident is presented considering equilibrium operation based upon Technical Specification limits of primary coolant activity concentration and primary to secondary leakage.

The following assumptions and parameters are used to calculate the activity releases and offsite doses for a postulated main steam line break.

1. Offsite power is lost; thus, the main steam condenser is not available for steam dump.

2. Air ejector release and steam generator blowdown do not occur during the accident.

3. Steam and activity release to the environment from the intact SGs is terminated at 11 hours due to primary and secondary side pressure equalization.

4. The release of activity that leaks into the faulted SG continues until the RCS is cooled below 212 degrees F (25.75 hours).

5. The elemental iodine partition factor in the affected steam generator throughout the accident is 1.0.

6. The elemental iodine partition factor in the unaffected steam generators throughout the accident is 0.01.

7. All releases are assumed to be at ground level.

8. Two separate cases are analyzed:

   a. An iodine spike occurs coincident with the steam line break. The iodine appearance rates from the fuel are increased to 500 times the equilibrium appearance rates associated with a primary coolant concentration of 1.0 uCi/gm Dose Equivalent I-131. The spike appearance rates are given in Table 15.1-3.

   b. A transient has occurred prior to the steam line break. The transient causes an iodine spike that raises the primary coolant activity to 60 uCi/gm Dose Equivalent I-131. The preaccident spike concentrations are given in Table 15.1-3.

9. Other assumptions are in Table 15.1-3.

Based on the foregoing model, the thyroid doses and whole body doses for the pre-accident iodine spike case were calculated to be 1.2 rem and 0.0025 rem, respectively, at the EAB and 0.67 rem and 0.0011 rem, respectively, at the LPZ. These doses are below the applicable limit values of 300 rem thyroid and 25 rem whole body set forth in 10CFR100. The thyroid doses and whole body doses for the accident-initiated iodine spike case were calculated to be 1.5 rem and 0.0049 rem, respectively, at the EAB and 3.1 rem and 0.0054 rem, respectively, at the LPZ. These doses are below a "small fraction" of the 10CFR100 values; or 30 rem thyroid and 2.5 rem whole body.

15.1.5.4 Conclusions

The analysis has shown that the criteria stated earlier in Section 15.1.5.1 are satisfied for Units 1 and 2.

Although DNB and possible clad perforation following a steam line rupture are not necessarily unacceptable and not precluded by the criteria, the analysis shows that no DNB or clad perforation occurs for any main steam line rupture assuming the most reactive RCCA stuck in its fully withdrawn position. The radiological consequences are within the guideline values of 10CFR100 for the pre-accident iodine spike case and within a "small fraction" (10%) of the 10CFR100 guideline values for the accident-initiated iodine spike case.

REFERENCES

1. "Westinghouse Anticipated Transients Without Trip Analysis," WCAP-8330, August 1974.

2. International Commission on Radiological Protection, "Limits for Intakes of Radionuclides by Workers," ICRP Publication 30, Volume 3 No. 1-4, 1979.

3. Huegel, D. S., et al., "RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses," WCAP-14882-P-A, April 1999.

4. Sung, Y. XI, et al., "VIPRE-01 Modeling and Qualification for Pressurized Water Reactor Non-LOCA Thermal-Hydraulic Safety Analysis," WCAP-14565-P-A, October 1999.

TABLE 15.1-1
HAS BEEN DELETED.

TABLE 15.1-2
EQUIPMENT REQUIRED FOLLOWING A RUPTURE OF A MAIN STEAM LINE

(Sheet 1 of 2)

Short Term (Required for Mitigation of Accident)	Hot Standby	Required for Cooldown
Reactor trip and engineered safety feature actuation channels including sensors, circuitry, and processing equipment (the protection circuits used to trip the reactor on undervoltage, underfrequency, and turbine trip may be excluded).	Auxiliary Feedwater System including pumps, water supply, and system valves and piping (this system must be placed in service to supply water to operable steam generators no later than 10 minutes after the incident).	Steam generator power operated relief valves (can be manually operated locally). Control for defeating automatic safety injection actuation during a cooldown and depressurization.
Safety Injection System including the pumps, the refueling water storage tank, and the system valves and piping.	Reactor Containment ventilation cooling units. Capability for obtaining a Reactor Coolant System sample.	Residual Heat Removal System including pumps, heat exchanger, and system valves and piping necessary to cool and maintain the Reactor Coolant System in a cold shutdown condition.
Diesel generators and emergency power distribution equipment.		
Containment engineered safety features cooling equipment.		
Auxiliary Feedwater System including pumps, water supplies, piping and valves.		
Main feedwater control valves (trip closed feature).		
Bypass feedwater control valves (trip closed feature).		
Primary and secondary safety valves.		

TABLE 15.1-2
EQUIPMENT REQUIRED FOLLOWING A RUPTURE OF A MAIN STEAM LINE

(Sheet 2 of 2)

Short Term (Required for Mitigation of Accident)	Hot Standby	Required for Cooldown
Circuits and/or equipment required to trip the main feedwater pumps.		
Main feedwater isolation valves (trip closed feature).		
Main steam line stop valves (trip closed feature).		
Steam generator blowdown isolation valves (automatic closure feature).		
Batteries (Class IE).		
Control Room ventilation.		
Control Room equipment must not be damaged to an extent where any equipment will be spuriously actuated or any of the equipment contained elsewhere in this list cannot be operated.		
Emergency lighting.		
Post Accident Monitoring System[(a)].		

a) See Section 7.5 for a discussion of the Post Accident Monitoring System

TABLE 15.1-3
PARAMETERS FOR POSTULATED MAIN STEAM LINE BREAK ACCIDENT ANALYSIS

(Sheet 1 of 3)

1.	Data and assumptions used to estimate radioactive source from postulated accidents		
	a.	Power level (MWt)	3684
	b.	Primary coolant activity concentration prior to accident	1.0 uCi/gm Dose Equivalent I-131
	c.	Secondary side coolant activity concentration prior to accident	0.1 uCi/gm Dose Equivalent I-131
	d.	Primary to Secondary leakage in affected steam generator (gpm)	0.3472
	e.	Primary to Secondary leakage in unaffected steam generators (gpm)	0.6528
	f.	Offsite power lost	lost
	g.	Failed fuel	none
2.	Data and assumptions used to estimate activity released		
	a.	Iodine partition factor for initial and long term steam release from affected steam generator	1.0
	b.	Iodine partition factor in unaffected steam generators prior to and during accident	0.01
	c.	Initial steam release from affected steam generator (lbm) (0 to 2 min)	176,000
	Note: The entire secondary side liquid inventory in the affected steam generator is assumed to be released.		
	d.	Long term steam release from affected steam generator (lbm) (2 min to 20 hr)	0
	e.	steam release from three unaffected steam generators	
		(lbm) (0 to 2 hr)	434,000
		(2 to 8 hr)	970,000

TABLE 15.1-3
PARAMETERS FOR POSTULATED MAIN STEAM LINE BREAK ACCIDENT ANALYSIS

(Sheet 2 of 3)

	(8 to 11 hr)			265,000
3.	Dispersion data			
	a.	EAB and LPZ distances		2080m and 4 miles
	b.	x/Q		@ EAB (0 – 2 hr) 1.6×10^{-4} sec/m^3 @ LPZ (0 – 8 hr) 2.4×10^{-5} sec/m^3 @ LPZ (8 – 24 hr) 1.6×10^{-5} sec/m^3
4.	Dose data			
	a.	Method of dose calculations		See Appendix 15B
	b.	Dose conversion assumptions		See Appendix 15B
	c.	Primary and secondary side equilibrium activities		See Table 15.1-4
	d.	Iodine spike appearance rates		I-131: 225.5 Ci/min I-132: 766 Ci/min I-133: 439.5 Ci/min I-134: 354 Ci/min I-135: 346.5 Ci/min
	e.	Pre-accident spike iodine concentrations		I-131: 46.1 uCi/gm I-132: 49.2 uCi/gm I-133: 73.8 uCi/gm I-134: 10.7 uCi/gm I-135: 40.6 uCi/gm
	f.	Doses (rem)	thyroid	whole body (gamma dose)
	@EAB (0-2 hr)			
		concurrent iodine spike	1.5	0.0049
		preaccident iodine spike	1.2	0.0025
	@LPZ (0-30 days)			
		concurrent iodine spike	3.1	0.0049

TABLE 15.1-3
PARAMETERS FOR POSTULATED MAIN STEAM LINE BREAK ACCIDENT ANALYSIS

(Sheet 3 of 3)

preaccident iodine spike	0.67	0.0011

TABLE 15.1-4
PRIMARY AND SECONDARY SIDE EQUILIBRIUM ACTIVITIES (ACCIDENT ANALYSIS)

Nuclide	Primary Coolant Concentration (uCi/gm)[a]	Secondary Liquid Concentration (uCi/gm)[b]
Kr-83m	4.77E-01	0.00×10^{-0}
Kr-85m	1.95E+00	0.00×10^{-0}
Kr-85	8.73E+00	0.00×10^{-0}
Kr-87	1.29E+00	0.00×10^{-0}
Kr-88	3.61E+00	0.00×10^{-0}
Xe-131m	3.20E+00	0.00×10^{-0}
Xe-133m	4.81E+00	0.00×10^{-0}
Xe-133	2.80E+02	0.00×10^{-0}
Xe-135m	5.36E-01	0.00×10^{-0}
Xe-135	8.18E+00	0.00×10^{-0}
Xe-138	7.25E-01	0.00×10^{-0}
I-131	7.68E-01	7.68E-02
I-132	8.20E-01	8.20E-02
I-133	1.23E+00	1.23E-01
I-134	1.78E-01	1.78E-02
I-135	6.77E-01	6.77E-02

a) Based on a primary coolant specific activity of 1.0 uCi/gm Dose Equivalent I-131. Noble gases are based on operation with 1% fuel defects.

b) Based on a secondary side specific activity of 0.10 uCi/gm Dose Equivalent I-131.

TABLE 15.1-4A
THIS TABLE IS DELETED

15.2 DECREASE IN HEAT REMOVAL BY THE SECONDARY SYSTEM

A number of transients and accidents have been postulated which could result in a reduction of the capacity of the secondary system to remove heat generated in the Reactor Coolant System (RCS). Detailed analyses are presented in this section for several such events which have been identified as more limiting than the others.

Discussions of the following RCS coolant heatup events are presented in Section 15.2:

1. Steam pressure regulator malfunction or failure that results in decreasing steam flow.
2. Loss of external electrical load.
3. Turbine trip.
4. Inadvertent closure of main steam isolation valves.
5. Loss of condenser vacuum and other events resulting in turbine trip.
6. Loss of nonemergency AC power to the station auxiliaries.
7. Loss of normal feedwater flow.
8. Feedwater system pipe break.

The above items are considered to be American Nuclear Society (ANS) Condition II events, with the exception of a feedwater system pipe break, which is considered to be an ANS Condition IV event. Section 15.0.1 contains a discussion of ANS classifications and applicable acceptance criteria.

15.2.1 STEAM PRESSURE REGULATOR MALFUNCTION OR FAILURE THAT RESULTS IN DECREASING STEAM FLOW

There are no steam pressure regulators in the Comanche Peak Nuclear Power Plant (CPNPP) whose failure or malfunction could cause a steam flow transient.

15.2.2 LOSS OF EXTERNAL ELECTRICAL LOAD

15.2.2.1 Identification of Causes and Accident Description

A major load loss on the plant can result from loss of external electrical load due to some electrical system disturbance. Offsite alternating current (AC) power remains available to operate plant components such as the reactor coolant pumps; as a result, the onsite emergency diesel generators are not required to function for this event. Following the loss of generator load, an immediate fast closure of the turbine control valves will occur. This will cause a sudden reduction in steam flow, resulting in an increase in pressure and temperature in the steam generator shell. As a result, the heat transfer rate in the steam generator is reduced, causing the reactor coolant temperature to rise, which in turn causes coolant expansion, pressurizer insurge, and RCS pressure rise.

For a loss of external electrical load without subsequent turbine trip, no direct reactor trip signal would be generated. The plant would be expected to trip from the Reactor Protection System if a safety limit were approached. A continued steam load of approximately 5 percent would exist after total loss of external electrical load because of the steam demand of plant auxiliaries.

In the event that a safety limit is approached, protection would be provided by the high pressurizer pressure and Overtemperature N-16 trips. Voltage and frequency relays associated with the reactor coolant pump provide no additional safety function for this event. Following a complete loss of load, the maximum turbine overspeed would be approximately 8 to 9 percent, resulting in an overfrequency of less than 6 hertz (Hz). This resulting overfrequency is not expected to damage the sensors (non-Nuclear Steam Supply System) in any way. However, it is noted that frequent testing of this equipment is required by the Technical Specifications. Any degradation in their performance could be ascertained at that time. Any increased frequency to the reactor coolant pump motors will result in slightly increased flow rate and subsequent additional margin to safety limits. For postulated loss of load and subsequent turbine generator overspeed, any overfrequency condition is not seen by other safety-related pump motors, Reactor Protection System equipment, or other engineered safety features (ESF) loads. These ESF loads are supplied from offsite power or, alternatively, from emergency diesels. Reactor Protection System equipment is supplied from the 118 volt AC instrument power supply system, which in turn is supplied from the inverters; the inverters are supplied from a direct current (DC) bus energized from batteries or by a rectified AC voltage from Class 1E buses.

In the event the steam dump valves fail to open following a large loss of load, the steam generator safety valves may lift and the reactor may be tripped by the high pressurizer pressure signal, the high pressurizer water level signal, or the Overtemperature N-16 signal. The steam generator shell side pressure and reactor coolant temperatures will increase rapidly. The pressurizer safety valves and steam generator safety valves are, however, sized to protect the RCS and steam generator against overpressure for all load losses without assuming the operation of the steam dump system, pressurizer spray, pressurizer power operated relief valves, automatic rod cluster control assembly control or direct reactor trip on turbine trip.

The steam generator safety valve capacity is sized to remove the steam flow at the engineered safety features rating (105 percent of steam flow at rated power) from the steam generator without exceeding 110 percent of the steam system design pressure. The pressurizer safety valve capacity is sized based on a complete loss of heat sink with the plant initially operating at the maximum calculated turbine load along with operation of the steam generator safety valves. The pressurizer safety valves are then able to relieve sufficient steam to maintain the RCS pressure within 110 percent of the RCS design pressure.

A more complete discussion of overpressure protection can be found in Reference [1], which also provides the original bases for establishing the capacities of the steam generator and pressurizer safety valves.

A loss of external load is classified as an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

A loss of external load event results in a Nuclear Steam Supply System transient that is less severe than a turbine trip event (see Section 15.2.3). Therefore, a detailed transient analysis is not presented for the loss of external load.

The primary side transient is caused by a decrease in heat transfer capability from primary to secondary due to a rapid termination of steam flow to the turbine, accompanied by an automatic reduction of feedwater flow (should feed flow not be reduced, a larger heat sink would be available and the transient would be less severe). Termination of steam flow to the turbine following a loss of external load occurs due to automatic fast closure of the turbine control valves in less than 0.5 seconds. Following a turbine trip event, termination of steam flow occurs via turbine stop valve closure, which occurs in less than 0.5 seconds. Therefore, the transient in primary pressure, temperature, and water volume will be less severe for the loss of external load than for the turbine trip due to a slightly slower loss of heat transfer capability.

The protection available to mitigate the consequences of a loss of external load is the same as that for a turbine trip, as listed in Table 15.0-6.

15.2.2.2 Analysis of Effects and Consequences

Refer to Section 15.2.3.2 for the method used to analyze the limiting transient (turbine trip) in this grouping of events. The results of the turbine trip event analysis are more severe than those expected for the loss of external load, as discussed in Section 15.2.2.1.

Normal reactor control systems and engineered safety systems are not required to function. The Auxiliary Feedwater System may, however, be automatically actuated following a loss of main feedwater; this will further mitigate the effects of the transient.

The Reactor Protection System may be required to function following a complete loss of external load to terminate core heat input and prevent departure from nucleate boiling (DNB). Depending on the magnitude of the load loss, pressurizer safety valves and/or steam generator safety valves may be required to open to maintain system pressure below allowable limits. No single active failure will prevent operation of any system required to function. Refer to Section 15.8 and Reference [2] for a discussion of anticipated transients without trip (ATWT) considerations.

15.2.2.3 Conclusions

Based on results obtained for the turbine trip event (Section 15.2.3) and considerations described in Section 15.2.2.1, the applicable acceptance criteria for a loss of external load event are met.

15.2.3 TURBINE TRIP

15.2.3.1 Identification of Causes and Accident Description

For a turbine trip event, the reactor would be tripped directly (unless below approximately 50 percent power) from a signal derived from the turbine stop emergency trip fluid pressure and turbine stop valves. The turbine stop valves close rapidly (typically less than 0.5 seconds) on loss of trip fluid pressure actuated by one of a number of possible turbine trip signals. Turbine trip initiation signals include:

1. Generator trip.

2. Low condenser vacuum.

3. Loss of lubricating oil.

4. Turbine thrust bearing failure.

5. Turbine overspeed.

6. Moisture Separator Reheater high level.

7. Manual trip.

Upon initiation of stop valve closure, steam flow to the turbine stops abruptly. Sensors on the stop valves detect the turbine trip and initiate steam dump and, if above 50 percent power, a reactor trip. The loss of steam flow results in an almost immediate rise in secondary system temperature and pressure with a resultant primary system transient as described in Section 15.2.2.1 for the loss of external load event. A slightly more severe transient occurs for the turbine trip event due to the more rapid loss of steam flow caused by the more rapid valve closure.

The automatic steam dump system would normally accommodate the excess steam generation. Reactor coolant temperatures and pressure do not significantly increase if the steam dump system and pressurizer pressure control system are functioning properly. If the turbine condenser were not available, the excess steam generation would be dumped to the atmosphere and main feedwater flow would be lost. For this situation feedwater flow would be maintained by the Auxiliary Feedwater System to ensure adequate residual and decay heat removal capability. Should the steam dump system fail to operate, the steam generator safety valves may lift to provide pressure control. See Section 15.2.2.1 for a further discussion of the transient.

A turbine trip is classified as an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

A turbine trip event is more limiting than loss of external load, loss of condenser vacuum, and other turbine trip events. As such, this event has been analyzed in detail. Results and discussion of the analysis are presented in Section 15.2.3.2.

The plant systems and equipment available to mitigate the consequences of a turbine trip are discussed in Section 15.0.8 and listed in Table 15.0-6.

15.2.3.2 Analysis of Effects and Consequences

Method of Analysis

In this analysis, the behavior of the unit is evaluated for a complete loss of steam load from full power without direct reactor trip. This is done primarily to show the adequacy of the pressure relieving devices and also to demonstrate core protection margins; that is, the turbine is assumed to trip without actuating all the sensors for reactor trip on the turbine stop valves. The assumption delays reactor trip until conditions in the RCS result in a trip due to other signals. Thus, the analysis assumes a worst transient. In addition, no credit is taken for steam dump. Main feedwater flow is terminated at the time of turbine trip, with no credit taken for auxiliary feedwater to mitigate the consequences of the transient.

The turbine trip transients are analyzed using the RETRAN computer code (Reference 3).

Major assumptions are summarized below:

1. Initial operating conditions

   For the peak RCS and main steam system (MSS) pressure analyses, the initial reactor power is assumed at the maximum value consistent with steady state full power operation including allowances for calibration and instrument errors. The initial pressurizer pressure is assumed at a minimum value consistent with steady state full power operation including allowances for calibration and instrument errors. For the peak MSS pressure analysis, the initial RCS temperature is chosen at the highest value plus uncertainties. For the peak RCS pressure analysis, the initial RCS temperature is chosen at the highest value (within uncertainties) such that the MSSVs open after the time of peak RCS pressure. For the DNB analysis, the initial reactor power, pressurizer pressure, and RCS temperature are assumed to be at their nominal values consistent with steady state full power operation; uncertainties in initial conditions are accounted for in the DNBR limit as described in Reference 4. Tube plugging is assumed to be the maximum value for the peak RCS pressure and DNB analyses, and zero for the peak MSS pressure analysis.

2. Moderator and Doppler coefficients of reactivity

   The turbine trip is analyzed with minimum reactivity feedback. The minimum feedback (BOL) cases assume the most positive moderator temperature coefficient and the least negative Doppler coefficient.

3. Reactor control

   From the standpoint of the maximum pressures and minimum DNBRs attained, it is conservative to assume that the reactor is in manual control. If the reactor were in automatic control, the control rod banks would move prior to trip and reduce the severity of the transient.

4. Steam release

   No credit is taken for the operation of the steam dump system or steam generator power operated relief valves. The steam generator pressure rises to the safety valve setpoint where steam release through safety valves limits secondary steam pressure at the setpoint value.

5. Pressurizer spray and power operated relief valves

   Two combinations are modeled:

   a. The pressurizer spray and power operated relief valves were modeled to conservatively reduce or limit the coolant pressure for the DNB and peak MSS pressure analyses. Safety valves were also assumed to be available.

b. No credit was taken for the effect of pressurizer spray and power operated relief valves in reducing or limiting the coolant pressure for the peak RCS pressure analysis. Safety valves were assumed to be operable.

6. Feedwater flow

Main feedwater flow to the steam generators is assumed to be lost at the time of turbine trip. No credit is taken for auxiliary feedwater flow since a stabilized plant condition will be reached before auxiliary feedwater initiation is normally assumed to occur; however, the auxiliary feedwater pumps would be expected to start on a trip of the main feedwater pumps. The auxiliary feedwater flow would remove core decay heat following plant stabilization.

7. Reactor trip is actuated by the first Reactor Protection System trip setpoint reached with no credit taken for the direct reactor trip on the turbine trip. Trip signals are expected due to high pressurizer pressure, Overtemperature N-16, high pressurizer water level, and low-low steam generator water level.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

Except as discussed above, normal Reactor Control System and engineered safety systems are not required to function.

The Reactor Protection System may be required to function following a turbine trip. Pressurizer safety valves and/or steam generator safety valves may be required to open to maintain system pressures below allowable limits. No single active failure will prevent operation of any system required to function. A discussion of ATWT considerations is presented in Section 15.8 and Reference [2].

15.2.3.3 Conclusions

For Units 1 and 2, the results of the analyses show that the plant design is such that a turbine trip without a direct or immediate reactor trip presents no hazard to the integrity of the RCS or the main steam system. Pressure relieving devices incorporated in the two systems are adequate to limit the maximum pressures to within 110 percent of the system design pressures.

The integrity of the core is maintained by operation of the Reactor Protection System, i.e., the DNBR will be maintained above the limit value. The DNBR design basis is described in Section 4.4. Applicable acceptance criteria as listed in Section 15.0.1 have been met. The above analysis results demonstrate the ability of the Nuclear Steam Supply System to safely withstand a full load rejection.

15.2.4 INADVERTENT CLOSURE OF MAIN STEAM ISOLATION VALVES

Inadvertent closure of the main steam isolation valves would result in a turbine trip. Turbine trips are discussed in Section 15.2.3.

15.2.5 LOSS OF CONDENSER VACUUM AND OTHER EVENTS RESULTING IN TURBINE TRIP

Loss of condenser vacuum is one of the events that can cause a turbine trip. Turbine trip initiating events are described in Section 15.2.3. A loss of condenser vacuum would preclude the use of steam dump to the condenser; however, since steam dump is assumed not to be available in the turbine trip analysis, no additional adverse effects would result if the turbine trip were caused by loss of condenser vacuum. Therefore, the analysis results and conclusions contained in Section 15.2.3 apply to loss of condenser vacuum. In addition, analyses for the other possible causes of a turbine trip, as listed in Section 15.2.3.1, are covered by Section 15.2.3. Possible overfrequency effects due to a turbine overspeed condition are discussed in Section 15.2.2.1 and are not a concern for this type of event.

A decreasing condenser vacuum would result in higher steam flow, lower steam pressure, constant or decreasing electrical load, constant or decreasing T_{avg}, and increasing reactor power, depending on initial plant conditions. This is the result of the main turbine control system attempting to maintain a constant electrical output.

The increase in reactor power will be limited by the magnitude of the change in condenser backpressure, the turbine backpressure trip point, initial power level, and/or turbine nozzle sizing. The worst case turbine trip accident analyzed in Section 15.2.3 assumes an initial reactor power level that is consistent with steady state full power operation including allowances for calibration and instrument errors. Should condenser backpressure increase at 100%, the rise in reactor power will be limited to approximately 105% due to turbine nozzle sizing. The limiting criterion for a turbine trip is the resulting pressure excursion; in other words, it is an overpressure accident. The analysis described in Section 15.2.3 verifies the adequacy of the Westinghouse overpressure protection for Units 1 and 2.

15.2.6 LOSS OF NONEMERGENCY AC POWER TO THE STATION AUXILIARIES

15.2.6.1 Identification of Causes and Accident Description

A complete loss of nonemergency AC power may result in the loss of all power to the station auxiliaries, i.e., the reactor coolant pumps, condensate pumps, etc. The loss of power may be caused by a complete loss of the offsite grid accompanied by a turbine generator trip at the station, or by a loss of the onsite AC distribution system.

This transient is more severe than the turbine trip event analyzed in Section 15.2.3. For this case, the decrease in heat removal by the secondary system is accompanied by a flow coastdown which further reduces the capacity of the primary coolant to remove heat from the core. The reactor will trip: 1) due to turbine trip; 2) upon reaching one of the trip setpoints in the primary and secondary systems as a result of the flow coastdown and decrease in secondary heat removal; or 3) due to loss of power to the control rod drive mechanisms as a result of the loss of power to the plant.

Following a loss of AC power with turbine and reactor trips, the sequence described below will occur:

1. Plant vital instruments are supplied from emergency DC power sources.

2. As the steam system pressure rises following the trip, the steam generator power operated relief valves may be automatically opened to the atmosphere. Steam dump to the condenser is assumed not to be available. If the steam flow rate through the power relief valves is not available, the steam generator self actuated safety valves may lift to dissipate the sensible heat of the fuel and coolant plus the residual decay heat produced in the reactor.

3. As the no-load temperature is approached, the steam generator power operated relief valves (or the self actuated safety valves, if the power operated relief valves are not available) are used to dissipate the residual decay heat and to maintain the plant at the hot shutdown condition.

4. The emergency diesel generators, started on loss of voltage on the plant emergency buses, begin to supply plant vital loads.

The Auxiliary Feedwater System is started automatically as follows:

Two motor driven auxiliary feedwater pumps are started on any of the following:

1. Low-low level in any steam generator.

2. Trip of all main feedwater pumps.

3. Any safety injection signal.

4. Loss of offsite power.

5. Manual actuation.

6. AMSAC

One turbine driven auxiliary feedwater pump is started on any of the following:

1. Low-low level in any two steam generators.

2. Loss of offsite power.

3. Manual actuation.

4. AMSAC

Refer to Section 10.4.9 for a description of the Auxiliary Feedwater System.

The motor driven auxiliary feedwater pumps are supplied power by the diesels and the turbine driven pump utilizes steam from the secondary system. The turbine exhausts the secondary steam to the atmosphere. The auxiliary feedwater pumps take suction from the Condensate Storage Tank for delivery to the steam generator.

Upon the loss of power to the reactor coolant pumps, coolant flow necessary for core cooling and the removal of residual heat is maintained by natural circulation in the reactor coolant loops. A

loss of nonemergency AC power to the station auxiliaries is classified as an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

A loss of AC power event, as described above, is a more limiting DNB event than the turbine trip-initiated decrease in secondary heat removal without loss of AC power, which was analyzed in Section 15.2.3. However, the DNB transient for this event is bounded by the complete loss of flow transient analyzed in Section 15.3.2. A loss of AC power to the station auxiliaries as postulated above could also result in a loss of normal feedwater if the condensate pumps lose power to operate. Therefore, detailed analytical results for a for a loss of normal feedwater transient with a loss of AC power are presented here. The results of the analysis in Section 15.2.7 address the loss of feedwater with AC power available.

Following the reactor coolant pump coastdown caused by the loss of AC power, the natural circulation capability of the RCS will remove residual and decay heat from the core, aided by auxiliary feedwater in the secondary system. An analysis is presented here to show that the natural circulation flow in the RCS following a loss of AC power event is sufficient to remove residual heat from the core.

The plant systems and equipment available to mitigate the consequences of a loss of AC power event are discussed in Section 15.0.8 and listed in Table 15.0-6.

15.2.6.2 Analysis of Effects and Consequences

15.2.6.2.1 Method of Analysis

A detailed analysis using the RETRAN computer code (Reference 3) is performed in order to obtain the plant transient following a loss of nonemergency AC power. The simulation describes the plant thermal kinetics, RCS including the natural circulation, pressurizer, steam generators and feedwater system. The digital program computes pertinent variables including the steam generator level, pressurizer water level, and reactor coolant average temperature.

The assumptions used in the analysis are as follows:

1. A 0.6% allowance for uncertainties associated with the plant calorimetric measurement for operation at an NSSS power of 3628 MWt.

2. A conservative core residual heat generation based upon long term operation at the initial power level preceding the trip.

3. A heat transfer coefficient in the steam generator associated with RCS natural circulation.

4. Reactor trip occurs on steam generator low-low level. No credit is taken for immediate release of the control rod drive mechanisms caused by a loss-of-offsite power.

5. Both motor-driven auxiliary feedwater pumps supply a total of 860 gpm to all four steam generators within 60 seconds after the steam generator low-low water level setpoint is exceeded. The turbine-driven auxiliary feedwater pump is the assumed single active failure.

6. Secondary system steam relief is achieved through the steam generator safety valves.

7. Since the plant is allowed to operate over a range of reactor vessel average temperatures (Tavg), and because the pressurizer level program varies over the range, analyses were performed to establish the most limiting initial conditions. For Unit 1, a Tavg of 574.2°F minus 6°F for uncertainty is limiting. For Unit 2, a Tavg of 584.7°F minus 6°F for uncertainty is limiting. An initial pressurizer pressure that is 30 psi higher than nominal is limiting for both Units 1 and 2.

8. The core nuclear physics parameters are chosen to maximize the possibility of water relief from the coolant system by maximizing the coolant system expansion.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

15.2.6.3 Conclusions

For Units 1 and 2, the analysis of the natural circulation capability of the RCS has demonstrated that sufficient heat removal capability exists following reactor coolant pump coastdown to prevent fuel or clad damage.

15.2.7 LOSS OF NORMAL FEEDWATER FLOW

15.2.7.1 Identification of Causes and Accident Description

A loss of normal feedwater (from pump failures, valve malfunctions, or loss of offsite AC power) results in a reduction in capability of the secondary system to remove the heat generated in the reactor core. If an alternative supply of feedwater were not supplied to the plant, core residual heat following reactor trip would heat the primary system water to the point where water relief from the pressurizer would occur, resulting in a substantial loss of water from the RCS. Since the plant is tripped well before the steam generator heat transfer capability is reduced, the primary system variables never approach a DNB condition.

The following events occur upon loss of normal feedwater (assuming main feedwater pump failures or valve malfunctions):

1. As the steam system pressure rises following the trip, the steam generator power operated relief valves are automatically opened to the atmosphere. Steam dump to the condenser is assumed not to be available. If the steam flow rate through the power operated relief valves is not available, the steam generator self actuated safety valves may lift to dissipate the sensible heat of the fuel and coolant plus the residual decay heat produced in the reactor.

2. As the no-load temperature is approached, the steam generator power operated relief valves (or the self actuated safety valves, if the power operated relief valves are not available) are used to dissipate the residual decay heat and to maintain the plant at the hot shutdown condition.

A loss of normal feedwater is classified as an ANS Condition II event, fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

The reactor trip on low-low water level in any steam generator provides the necessary protection against a loss of normal feedwater.

The Auxiliary Feedwater System is started automatically as discussed in Section 15.2.6.1. The steam driven auxiliary feedwater pump utilizes steam from the secondary system and exhausts to the atmosphere. The motor driven auxiliary feedwater pumps are supplied by power from the diesel generators. The pumps take suction directly from the condensate storage tank for delivery to the steam generators.

An analysis of the system transient is presented below to show that following a loss of normal feedwater, the auxiliary feedwater system is capable of removing the stored and residual heat, thus preventing either overpressurization of the RCS or loss of water from the reactor core, and returning the plant to a safe condition.

15.2.7.2 Analysis of Effects and Consequences

15.2.7.2.1 Method of Analysis

A detailed analysis using the RETRAN computer code (Reference 3) is performed in order to obtain the plant transient following a loss of normal feedwater. The simulation describes the plant thermal kinetics, RCS, pressurizer, steam generators and feedwater system. The digital program computes pertinent variables including the steam generator level, pressurizer water level, and reactor coolant average temperature.

Assumptions made in the analysis are:

1. A 0.6% allowance for uncertainties associated with the plant calorimetric measurement for operation at an NSSS power of 3628 MWt.

2. An error of +5% in the full power programmed pressurizer water level is assumed since, although the surge rate is low, the total volumetric expansion of the primary coolant is sufficient to nearly fill the pressurizer. It should be noted for this event that, even if the pressurizer did fill with water, the low surge rate would not cause an excessive pressure rise.

3. A conservative core residual heat generation based upon long term operation at the initial power level preceding the trip.

4. Reactor trip occurs on steam generator low-low level.

5. Both motor-driven auxiliary feedwater pumps supplying a total of 860 gpm to all four steam generators within 60 seconds after the steam generator low-low water level setpoint is exceeded. The turbine-driven auxiliary feedwater pump is the assumed single active failure.

6. Secondary system steam relief is achieved through the steam generator self-actuated safety valves. Note that steam relief will, in fact, be through the power-operated relief valves or condenser dump valves for most cases of loss of normal feedwater. However, for the sake of analysis these have been assumed unavailable.

7. Since the plant is allowed to operate over a range of reactor vessel average temperatures (Tavg), and because the pressurizer level program varies over the range, analyses were performed to establish the most limiting initial conditions. For Unit 1, a Tavg of 574.2°F

minus 6°F for uncertainty is limiting. For Unit 2, a Tavg of 589.2°F plus 6°F for uncertainty is limiting. An initial pressurizer pressure that is 30 psi higher than nominal is limiting for Unit 1 and that is 30 psi lower than nominal is limiting for Unit 2.

8. The core nuclear physics parameters are chosen to maximize the possibility of water relief from the coolant system by maximizing the coolant system expansion.

The loss of normal feedwater analysis is performed to demonstrate the adequacy of the reactor protection and engineered safety features systems (e.g., the Auxiliary Feedwater System) in removing long term decay heat and preventing excessive heatup of the RCS with possible resultant RCS overpressurization or loss of RCS water.

As such, the assumptions used in this analysis are designed to minimize the energy removal capability of the system and to maximize the possibility of water relief from the coolant system by maximizing the coolant system expansion, as noted in the assumptions listed above.

For the Loss of Normal Feedwater transient, the reactor coolant volumetric flow would remain at its normal value and the reactor trips via the low-low steam generator level trip. The reactor coolant pumps would be manually tripped at some later time to reduce heat addition to the RCS. The Auxiliary Feedwater System has sufficient capacity, even assuming the worst single failure, to preclude filling the pressurizer should the pumps not be tripped.

Because the two Comanche Peak units have different steam generators (see Section 5.4.2), the effect of this difference has been considered in the analysis. The initial steam generator level and the trip setpoints assumed in the analysis are conservative for the unit analyzed.

All Unit 2 steam generators are equipped with separate feedwater connections for injection of auxiliary feedwater and main feedwater at low power operation (the Unit 1 steam generators have completely separate connections at the same steam generator level for both main and auxiliary feedwater at all power levels). The major effect of injecting auxiliary feedwater into the upper section of the downcomer is that most of the flow will bypass the preheat region due to the higher resistance to flow in the preheater. This will result in a slight decrease in heat removal capability. However, the auxiliary feedwater injection point is now much closer to the steam generator, resulting in a much smaller volume of hot feedwater which must be purged before the colder auxiliary feed enters the steam generators. This effect has been considered in the analysis for Unit 2 (not applicable for Unit 1).

Plant characteristics and initial conditions are further discussed in Section 15.0.3. Plant systems and equipment which are available to mitigate the effects of a loss of normal feedwater accident are discussed in Section 15.0.8 and listed in Table 15.0-6. Normal reactor control systems are not required to function. The Reactor Protection System is required to function following a loss of normal feedwater as analyzed here. The Auxiliary Feedwater System is required to deliver a minimum auxiliary feedwater flow rate. No single active failure will prevent operation of any system required to function. A discussion of ATWT considerations is presented in Section 15.8 and Reference [2].

15.2.7.3 Conclusions

For Units 1 and 2, results of the analysis show that a loss of normal feedwater does not adversely affect the core, the RCS, or the steam system since the auxiliary feedwater capacity is such that

reactor coolant water is not relieved from the pressurizer relief or safety valves, and the water level in all steam generators is maintained above the tubesheets.

15.2.8 FEEDWATER SYSTEM PIPE BREAK

15.2.8.1 Identification of Causes and Accident Description

A major feedwater line rupture is defined as a break in a feedwater line large enough to prevent the addition of sufficient feedwater to the steam generators to maintain shell side fluid inventory in the steam generators. If the break is postulated in a feedline between the check valve and the steam generator, fluid from the steam generator may also be discharged through the break. Further, a break in this location could preclude the subsequent addition of auxiliary feedwater to the affected steam generator. (A break upstream of the feedline check valve would affect the Nuclear Steam Supply System only as a loss of feedwater. This case is covered by the evaluation in Section 15.2.7).

Depending upon the size of the break and the plant operating conditions at the time of the break, the break could cause either a RCS cooldown (by excessive energy discharge through the break) or a RCS heatup. Potential RCS cooldown resulting from a secondary pipe rupture is evaluated in Section 15.1.5. Therefore, only the RCS heatup effects are evaluated for a feedwater line rupture.

A feedwater line rupture reduces the ability to remove heat generated by the core from the RCS for the following reasons:

1. Feedwater flow to the steam generators is reduced. Since feedwater is subcooled, its loss may cause reactor coolant temperatures to increase prior to reactor trip.

2. Fluid in the steam generator may be discharged through the break, and would then not be available for decay heat removal after trip.

3. The break may be large enough to prevent the addition of any main feedwater after trip.

An Auxiliary Feedwater System is provided to assure that adequate feedwater will be available such that:

1. No substantial overpressurization of the RCS shall occur.

2. Sufficient liquid in the RCS shall be maintained in order to provide adequate decay heat removal.

Refer to Section 10.4.9 for a description of the Auxiliary Feedwater System interfaces.

A major feedwater line rupture is classified as an ANS Condition IV event (i.e., a limiting fault). See Section 15.0.1 for a discussion of Condition IV events.

A main feedwater line rupture is the most limiting event in the decrease in secondary heat removal category. Therefore, a full transient analysis is presented here.

The severity of the feedwater line rupture transient depends on a number of system parameters including break size, initial reactor power, and credit taken for the functioning of various control and safety systems. A number of cases of feedwater line break have been analyzed. Based on these analyses, it has been shown that the most limiting feedwater line ruptures are the double ended rupture of the largest feedwater line, occurring at full power with and without loss of offsite power, with no credit taken for pressurizer pressure control. These cases are analyzed below.

The following provides the necessary protection for a main feedwater line rupture:

1. A reactor trip on any of the following conditions:

    a. High pressurizer pressure.

    b. Overtemperature N-16.

    c. Low-low steam generator water level in any steam generator.

    d. Safety injection signals from any of the following:

        1. Low steam line pressure.

        2. High Containment pressure (hi-1).

    (Refer to Chapter 7 for a description of the actuation system).

2. An Auxiliary Feedwater System to provide an assured source of feedwater to the steam generators for decay heat removal. (Refer to Section 10.4.9 for a description of the Auxiliary Feedwater System).

15.2.8.2 Analysis of Effects and Consequences

15.2.8.2.1 Method of Analysis

The feedline break is analyzed with the RETRAN computer code (Reference 3) to demonstrate the adequacy of the Auxiliary Feedwater System for removing decay heat. The relevant acceptance criteria are that the hot leg and cold leg fluid remains subcooled and the RCS and MSS pressures remain below 110% of their respective design values.

The cases analyzed assume a double ended rupture of the largest feedwater pipe at full power. Major assumptions made in the analyses are as follows:

1. The plant is initially operating at Rated Thermal Power level plus uncertainties (3628 MWt).

2. Initial reactor coolant average temperature is 6.0°F above the nominal full power value, and the initial pressurizer pressure is 30 psi below its nominal value.

3. The pressurizer power operated relief valves are assumed operable as they minimize RCS pressure, which results in a lower saturation temperature. No credit is taken for the pressurizer spray and heaters.

4. For Unit 1, initial steam generator water level is at the nominal value, plus 10 percent in the faulted steam generator and at the nominal value minus 10 percent in the intact steam generators. For Unit 2, the initial steam generator water level is at the nominal value, plus 18 percent in the faulted steam generator and at the nominal value minus 7 percent in the intact steam generators.

5. Main feedwater flow to all steam generators is assumed to be lost at the time the break occurs (all main feedwater spills out through the break).

6. The worst possible break area is assumed. This maximizes the blowdown discharge rate following the time of trip, which maximizes the resultant heatup of the reactor coolant.

7. Reactor trip occurs on steam generator low-low water level.

8. The limiting single active failure in the Auxiliary Feedwater System, the failure of one motor-driven auxiliary feedwater pump, is assumed. The flow from the second motor-driven auxiliary feedwater pump is assumed to spill out the break through the faulted steam generator. No credit is taken for heat removal through the faulted steam generator. The turbine-driven auxiliary feedwater pump is assumed to supply a total of 430 gpm to the three steam generators against a steamline backpressure of 1236 psia, beginning 85 seconds after the low-low steam generator water level setpoint is reached. The remainder of the flow from the turbine-driven auxiliary feedwater pump is assumed to spill our the break through the faulted steam generator.

   Approximately 250 seconds are required to purge the feedwater lines of main feedwater before relatively cold (120°F) auxiliary feedwater can enter the unaffected steam generators (Unit 2 only; Unit 1 has completely separate connections for main and auxiliary feedwater).

9. Thirty minutes after the reactor trip, operator action to isolate the Auxiliary Feedwater System from the break is assumed. In addition, and depending on the specific single failure assumptions, the operators may also have to start an additional MDAFP or open any closed TDAFP Steam Admission valves. At that time, an additional 370 gpm is assumed to be supplied to the three intact steam generators. No credit is taken for the delivery of additional auxiliary feedwater from the second motor-driven auxiliary feedwater pump.

10. No credit is taken for heat energy deposited in RCS metal during the RCS heatup.

11. No credit is taken for charging or letdown.

12. Steam generator heat transfer area is assumed to decrease as the shell side liquid inventory decreases.

13. Conservative core residual heat generation is based upon the American National Standard For Decay Heat Power In Light Water Reactors, ANSI/ANS-5.1-1979 and assumes an infinite irradiation time and a two sigma uncertainty allowance.

14. No credit is taken for the following potential protection logic signals to mitigate the consequences of the accident:

a. High pressurizer pressure.

b. Overtemperature N-16.

c. High pressurizer level.

d. High Containment pressure.

15. Although it is expected that the actuation of the safety injection system would occur during this event, the analysis conservatively does not model safety injection flow.

Receipt of a low-low steam generator water level signal in at least one steam generator starts the motor driven auxiliary feedwater pumps, which then deliver auxiliary feedwater flow to the steam generators. The turbine driven auxiliary feedwater pump is initiated if the low-low steam generator water level signal is reached in at least two steam generators. Similarly, receipt of a low steam line pressure signal in at least one steam line initiates a steam line isolation signal which closes the main steam line isolation valves in all steam lines. This signal also gives a safety injection signal which initiates flow of borated water into the RCS (not credited in this analysis). The amount of safety injection flow is a function of RCS pressure.

Emergency operating procedures following a secondary system line rupture typically will call for the following actions to be taken by the reactor operator:

1. Isolate feedwater flow spilling out the break of ruptured steam generator and align system so level in intact steam generators recovers. In addition, and depending on the specific single assumptions, the operators may also have to start an additional MDAFP or open any closed TDAFP Steam Admission valves.

2. Turn off all reactor coolant pumps (if offsite power is still available). The emergency operating procedure used in the event of secondary system line rupture when offsite power is available calls for reactor coolant pumps to be stopped in the case of either a normal or adverse (harsh) containment environment, if RCS subcooling is below specified limits and at least one CCP or SIP is operating. At this point, natural circulation will provide sufficient heat removal capability.

3. Stop high head safety injection pumps if RCS subcooling is adequate, and there is sufficient inventory in the intact steam generators, and the RCS pressure is stable or increasing, and the pressurizer level is increasing.

Shutting off the reactor coolant pumps (action 2, above) serves to decrease the addition of energy to the RCS. Isolating feedwater flow through the break allows additional auxiliary feedwater flow to be diverted to the intact steam generators (see assumption 9, above).

Subsequent to recovery of level in the intact steam generators, the high head safety injection pumps will be turned off and plant operating procedures will be followed in cooling the plant to hot shutdown conditions. These actions are not modeled in the safety analysis.

Plant characteristics and initial conditions are further discussed in Section 15.0.3.

No reactor control systems are assumed to function. The Reactor Protection System is required to function following a feedwater line rupture as analyzed here. No single active failure will prevent operation of this system.

The engineered safety features system assumed to function is the Auxiliary Feedwater System. The limiting single failure has been used, i.e., a failure of a motor-driven auxiliary feedwater pump.

Following the trip of the reactor coolant pumps, there will be a flow coastdown until flow in the loops reaches the natural circulation value. The natural circulation capability of the RCS has been shown in Section 15.2.6, for the loss of AC power transient, to be sufficient to remove core decay heat following reactor trip. Pump coastdown characteristics are demonstrated in Sections 15.3.1 and 15.3.2, where DNB considerations are addressed for single and multiple reactor coolant pump trips, respectively.

A detailed description and analysis of the Safety Injection System is provided in Section 6.3. The Auxiliary Feedwater System is described in Section 10.4.9.

15.2.8.3 Conclusions

For Units 1 and 2, the results of the analyses show that for the postulated feedwater line rupture, the assumed Auxiliary Feedwater System capacity is adequate to remove decay heat, to prevent overpressurizing the RCS, and to prevent uncovering the reactor core.

Water relief can occur for feedwater line break events. Any water relief, however, would occur through the pressurizer power operated relief valves since they were conservatively assumed in the analysis to minimize RCS pressure, which results in a lower saturation temperature.

15.2.8.4 Analysis of Radiological Effects and Consequences

Radioactivity doses from the postulated feedwater line rupture are less than those previously presented for the postulated steam line break. All applicable acceptance criteria are therefore met.

REFERENCES

1. Mangan, M. A., “Overpressure Protection for Westinghouse Pressurized Water Reactors”, WCAP-7769, Rev. 1, June 1972.

2. “Westinghouse Anticipated Transients Without Trip Analysis”, WCAP-8330, August 1974.

3. Huegel, D. S., et all, “RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses,” WCAP-14882-P-!, April 1999.

4. Friedlant, A. J., and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

TABLE 15.2-1
TABLE HAS BEEN DELETED.

TABLE 15.2-2
NATURAL CIRCULATION FLOW DELETED

15.3 DECREASE IN REACTOR COOLANT SYSTEM FLOW RATE

A number of faults are postulated which could result in a decrease in Reactor Coolant System (RCS) flow rate. These events are discussed in this section. Detailed analyses are presented for the most limiting of these events.

Discussions of the following flow decrease events are presented in Section 15.3:

1. Partial loss of forced reactor coolant flow.

2. Complete loss of forced reactor coolant flow.

3. Reactor coolant pump shaft seizure (locked rotor).

4. Reactor coolant pump shaft break.

Item 1 above is considered to be an American Nuclear Society (ANS) Condition II event, item 2 an ANS Condition III event, and items 3 and 4 ANS Condition IV events. Section 15.0.1 contains a discussion of ANS classifications.

15.3.1 PARTIAL LOSS OF FORCED REACTOR COOLANT FLOW

15.3.1.1 Identification of Causes and Accident Description

A partial loss of coolant flow accident can result from a mechanical or electrical failure in a reactor coolant pump, or from a fault in the power supply to the pump or pumps supplied by a reactor coolant pump bus. If the reactor is at power at the time of the accident, the immediate effect of loss of coolant flow is a rapid increase in the coolant temperature. This increase could result in departure from nucleate boiling (DNB) with subsequent fuel damage if the reactor is not tripped promptly.

Normal power for the pumps is supplied through individual buses connected to the generator. When a generator trip occurs, the buses are automatically transferred to a transformer supplied from external power lines, and the pumps will continue to supply coolant flow to the core. Following any automatic turbine trip where there are no electrical or turbine faults which require tripping the generator from the network, the generator remains connected to the network for approximately 30 seconds. The reactor coolant pump buses remain connected to the network thus ensuring full flow for approximately 30 seconds after the reactor trip before any transfer is made. For a manual turbine trip, or a turbine trip due to certain turbine faults or certain generator protection signals, the generator remains connected to the network for approximately 11.5 seconds. For a complete description of the turbine/generator trip logic, see Figure 7.2-1, Sheet 16 and Figure 10.2-1.

This event is classified as an ANS Condition II incident (a fault of moderate frequency) as defined in Section 15.0.1.

The necessary protection against a partial loss of coolant flow accident is provided by the low primary coolant flow reactor trip signal which is actuated in any reactor coolant loop by two out of three low flow signals. Above Permissive 8, low flow in any loop will actuate a reactor trip. Between approximately 10 percent power (Permissive 7) and the power level corresponding to

Permissive 8, low flow in any two loops will actuate a reactor trip. Above Permissive 7, two or more reactor coolant pump circuit breakers opening will actuate the corresponding undervoltage relays. This results in a reactor trip which serves as a back up to the low flow trip.

15.3.1.2 Analysis of Effects and Consequences

Method of Analysis

This transient (loss of one reactor coolant pump with four loops in operation) is analyzed using two digital computer codes: RETRAN and VIPRE. The RETRAN code (Reference 3) is used to calculate the loop and core flow during the transient, the time of reactor trip based on the calculated flows, the nuclear power transient, and the primary system pressure and temperature transients. The VIPRE code (Reference 4) is then used to calculate the heat flux and DNBR during the transient based on the nuclear power, core inlet enthalpy and flow, and core exit pressure calculated by RETRAN.

Initial Conditions

Initial reactor power, pressurizer pressure, and reactor coolant system (RCS) temperature are assumed to be at their nominal values. Uncertainties in initial conditions are included in the DNBR limit as described in the Revised Thermal Design Procedure (RTDP) (Reference 5). Plant characteristics and initial conditions are further discussed in Section 15.0.3.

Reactivity Coefficients

A conservatively large absolute value of the Doppler coefficient is used. The most positive moderator temperature coefficient is assumed because this results in the maximum core power during the initial part of the transient when the minimum DNBR is reached.

Flow Coastdown

The flow coastdown analysis is based on a momentum balance around each reactor coolant loop and across the reactor core. This momentum balance is combined with the continuity equation, a pump momentum balance and the pump characteristics.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of this accident.

15.3.1.3 Conclusions

For Units 1 and 2 the analysis shows that the DNBR will not decrease below the limit value at any time during the transient. The DNBR design basis is described in Section 4.4. All applicable acceptance criteria are met.

15.3.2 COMPLETE LOSS OF FORCED REACTOR COOLANT FLOW

15.3.2.1 Identification of Causes and Accident Description

A complete loss of forced reactor coolant flow may result from a simultaneous loss of electrical supplies to all reactor coolant pumps. If the reactor is at power at the time of the accident, the immediate effect of loss of coolant flow is a rapid increase in the coolant temperature. This increase could result in DNB with subsequent fuel damage if the reactor were not tripped promptly.

Normal power for the reactor coolant pumps is supplied through buses from a transformer connected to the generator. When a generator trip occurs, the buses are automatically transferred to a transformer supplied from external power lines, and the pumps will continue to supply coolant flow to the core. Following any automatic turbine trip where there are no electrical or turbine faults which require tripping the generator from the network, the generator remains connected to the network for approximately 30 seconds. The reactor coolant pump buses remain connected to the network, thus ensuring full flow for 30 seconds after the reactor trip before any transfer is made. For a manual turbine trip, or a turbine trip due to certain turbine faults or certain generator protection signals, the generator remains connected to the network for approximately 11.5 seconds. For a complete description of the turbine/generator trip logic, see Figure 7.2-1, Sheet 16 and Figure 10.2-1.

This event is classified as an ANS Condition III incident (an infrequent fault) as defined in Section 15.0.1. However, this transient is analyzed to meet Condition II safety acceptance criteria.

The following signals provide the necessary protection against a complete loss of flow accident:

1. Reactor coolant pump power supply undervoltage or underfrequency.

2. Low reactor coolant loop flow.

The reactor trip on reactor coolant pump undervoltage is provided to protect against conditions which can cause a simultaneous opening of all reactor coolant pump breakers. This function is blocked below approximately 10 percent power (Permissive P-7).

The reactor trip on reactor coolant pump underfrequency is provided to trip the reactor for an underfrequency condition, resulting from frequency disturbances on the power grid. Reference [2] provides analyses of grid frequency disturbances and the resulting Nuclear Steam Supply System protection requirements which are applicable to the Comanche Peak Nuclear Power Plant (CPNPP). The maximum expected frequency decay rate for the Vistra OpCo system power grid is 2.4 Hz./sec. (see Section 8.2.2); thus, the nominal underfrequency setpoint of 57 Hz., as recommended in Reference [2], (which examined grid frequency decay rates as high as 5 Hz./sec.) is appropriate for Comanche Peak.

Due to the rapid decrease in the reactor coolant pump bus frequency following a loss of offsite power, the underfrequency reactor trip function will usually initiate a reactor trip prior to the undervoltage reactor trip function.

The reactor trip on low primary coolant loop flow is provided to protect against loss of flow conditions which affect only one reactor coolant loop. This function is generated by two out of three low flow signals per reactor coolant loop. Above Permissive P-8, low flow in any loop will actuate a reactor trip. Between approximately 10 percent power (Permissive P-7) and the power level corresponding to Permissive P-8, low flow in any two loops will actuate a reactor trip. For a decrease in reactor coolant flow during operation below approximately 10 percent power (Permissive P-7), there is no immediate reactor trip function required. Natural circulation flow in the RCS provides adequate core cooling capability. If the maximum grid frequency decay rate is less than approximately 2.5 Hz./sec., this trip function will protect the core from underfrequency events. This effect is fully described in Reference [2]. In addition, the overtemperature N-16 trip function remains active, and will result in a reactor trip if the core protection limits described in Section 15.0 of the FSAR are approached.

15.3.2.2 Analysis of Effects and Consequences

Method of Analysis

This transient (complete loss of forced reactor coolant flow) is analyzed using two digital computer codes: RETRAN and VIPRE. The RETRAN code (Reference 3) is used to calculate the loop and core flow during the transient, the time of reactor trip based on the calculated flows, the nuclear power transient, and the primary system pressure and temperature transients. The VIPRE code (Reference 4) is then used to calculate the heat flux and DNBR during the transient based on the nuclear power, core inlet enthalpy and flow, and core exit pressure calculated by RETRAN.

The method of analysis and the assumptions made regarding initial operating conditions and reactivity coefficients are similar to those discussed in Section 15.3.1, except that following the loss of power supply to all pumps at power, a reactor trip is actuated by either reactor coolant pump power supply undervoltage or underfrequency.

15.3.2.3 Conclusions

For Units 1 and 2, the analysis performed has demonstrated that for the complete loss of forced reactor coolant flow, the DNBR does not decrease below the limit value at any time during the transient. The design basis for DNBR is described in Section 4.4. All applicable acceptance criteria are met.

15.3.3 REACTOR COOLANT PUMP SHAFT SEIZURE (LOCKED ROTOR)

15.3.3.1 Identification of Causes and Accident Description

The accident postulated is an instantaneous seizure of a reactor coolant pump rotor such as is discussed in Section 5.4. Flow through the affected reactor coolant loop is rapidly reduced, leading to an initiation of a reactor trip on a low flow signal.

Following initiation of the reactor trip, heat stored in the fuel rods continues to be transferred to the coolant causing the coolant to expand. At the same time, heat transfer to the shell side of the steam generators is reduced, first because the reduced flow results in a decreased tube side film coefficient and then because the reactor coolant in the tubes cools down while the shell side temperature increases (turbine steam flow is reduced to zero upon plant trip). The rapid

expansion of the coolant in the reactor core, combined with reduced heat transfer in the steam generators causes an insurge into the pressurizer and a pressure increase throughout the RCS. The insurge into the pressurizer compresses the steam volume, actuates the automatic spray system, opens the power operated relief valves, and opens the pressurizer safety valves, in that sequence. The two power operated relief valves are designed for reliable operation and would be expected to function properly during the accident. However, for conservatism, their pressure reducing effect as well as the pressure reducing effect of the spray is not included in the analysis.

This event is classified as an ANS Condition IV incident (a limiting fault) as defined in Section 15.0.1. This classification is used in the Chapter 15 safety analysis. However, the results of the analysis for this event meet the safety criteria for a Condition III event. The peak RCS pressure is maintained below 110% of the design pressure, and the peak clad average temperature is well below 2700°F.

15.3.3.2 Analysis of Effects and Consequences

Method of Analysis

This transient (locked rotor) is analyzed using two digital computer codes: RETRAN and VIPRE. The RETRAN code (Reference 3) is used to calculate the loop and core flow transients following the pump shaft seizure, the time of reactor trip based on the loop flow transients, the nuclear power transient, and the primary system pressure and temperature transients. The thermal behavior of the fuel located at the core hot spot is analyzed using the VIPRE code (Reference 4), which uses the core flow and the nuclear power calculated by RETRAN. The VIPRE code includes a film boiling heat transfer coefficient.

Two locked rotor cases are analyzed. The first case maximizes the peak clad temperature (PCT) and peak RCS pressure. The second case maximizes the percentage of fuel rods that experience DNB conditions.

At the beginning of the postulated locked rotor accident for the PCT/peak RCS pressure case, i.e., at the time the shaft in one of the reactor coolant pumps is assumed to seize, the plant is assumed to be in operation under the most adverse steady state operating conditions, i.e., maximum guaranteed steady state thermal power, maximum steady state pressure, and maximum steady state coolant average temperature, with consideration for uncertainties in the measurement and control channels. Plant characteristics and initial conditions are further discussed in Section 15.0.3. The accident as evaluated accounts for a loss of offsite power by assuming power to the unaffected reactor coolant pumps is lost coincident with the reactor trip.

For the peak pressure evaluation, a positive moderator temperature coefficient is assumed and the initial pressure is conservatively estimated as 30 pounds per square inch (psi) above nominal pressure (2250 psia) to allow for errors in the pressurizer pressure measurement and control channels.

This is done to obtain the highest possible rise in the coolant pressure during the transient.

For the rods-in-DNB case, the initial core power is assumed to be at the nominal value consistent with steady-state, full-power operation. The initial pressurizer pressure and vessel average temperature are assumed to be at the respective nominal values, and the reactor coolant flow is

at the minimum measured flow value. Uncertainties in the initial conditions are accounted for in the DNBR limit value as described in Reference 5.

Evaluation of the Pressure Transient

After pump seizure, the neutron flux is rapidly reduced by control rod insertion. Rod motion is assumed to begin one second after the flow in the affected loop reaches 87 percent of nominal flow. No credit is taken for the pressure reducing effect of the pressurizer relief valves, pressurizer spray, steam dump or controlled feedwater flow after plant trip.

Although these operations are expected to occur and would result in a lower peak pressure, an additional degree of conservatism is provided by ignoring their effect.

The pressurizer safety valves are fully open at 2575 psia and their capacity for steam relief is as described in Section 5.4. No water relief through the pressurizer safety valves occurs for the reactor coolant pump locked rotor (and shaft break) events.

Evaluation of PCT and Zirconium-Water Reaction During the Accident

For this evaluation, DNB is assumed to occur in the core at the initiation of the transient in order to conservatively predict the peak clad temperature. Therefore, an evaluation of the consequences with respect to fuel rod thermal transients is performed. Results obtained from analysis of this “hot spot” condition represent the upper limit with respect to clad temperature and zirconium-water reaction.

In the evaluation, the rod power at the hot spot is assumed to be 2.5 times the average rod power (i.e., $F_Q = 2.5$) at the initial core power level.

Film Boiling Coefficient

The film boiling coefficient is calculated using the Bishop-Sandberg-Tong film boiling correlation. The fluid properties are evaluated at film temperature (average between wall and bulk temperatures). The program calculates the film coefficient at every time step based upon the actual heat transfer conditions at the time. The neutron flux, system pressure, bulk density and mass flow rate as a function of time are used as program input.

Fuel Clad Gap Coefficient

The magnitude and time dependence of the heat transfer coefficient between fuel and clad (gap coefficient) has a pronounced influence on the thermal results. The larger the value of the gap coefficient, the more heat is transferred between pellet and clad. Based on investigations on the effect of the gap coefficient upon the maximum clad temperature during the transient, the gap coefficient was assumed to increase from a steady state value consistent with the initial fuel temperature to a conservatively large value at the initiation of the transient. Thus the large amount of energy stored in the fuel because of the small initial value is released to the clad at the initiation of the transient.

Zirconium-Steam Reaction

The zirconium-steam reaction can become significant above 1800°F (clad temperature). The Baker-Just parabolic rate equation shown below is used to define the rate of the zirconium-steam reaction.

$$\frac{d(w^2)}{dt} = 33.3 \times 10^6 \exp(-45,500/1.986T)$$

where

w = amount reacted (mg/cm^2)

t = time (sec)

T = temperature (°K)

The heat of reaction is 1510 cal/gm.

The effect of the zirconium-steam reaction is included in the calculation of the “hot spot” clad temperature transient.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of the accident.

Evaluation of the Radiological Consequences

An evaluation of the number of rods in DNB was performed in order to assess the radiological consequences of this event. Analysis assumptions were made to maximize the heat flux and thus minimize the DNB, consistent with the current FSAR. The number of rods experiencing DNB is determined from a pin census showing the relative peaking factors of each rod.

Locked Rotor

The doses resulting from a locked rotor accident are based on a conservative fission product release to the reactor coolant of the gap activity shown in sheet 2 of Table 15.0-7 from 15% of the fuel rods in the core. The method follows Regulatory Guide 1.195. Prior to the accident, it is assumed that the plant has been operating with simultaneous fuel defects and steam generator tube leakage for a sufficient amount of time to establish equilibrium levels of activity in the primary and secondary coolants equal to the Table 15.1-4 values.

Following the postulated locked rotor accident, the activity released from the pellet-gap due to the fuel failure is assumed to be instantaneously released to the primary coolant and is assumed to be immediately available for release from the RCS.

The following assumptions are used in the analysis:

1. The initial Reactor Coolant System (RCS) iodine activity is 1.0 uCi/gm DE I-131.

2. The initial RCS noble gas activity is based on 1% fuel defects.

3. The initial secondary coolant iodine activity is 0.1 uCi/gm DE I-131.

4. A radial peaking factor ($F_{\Delta H}$) of 1.65 is applied in the calculation of gap activities released (per Regulatory Guide 1.195).

5. 15% of rods are assumed to undergo clad damage sufficient to release all of their gap activity.

6. The primary to secondary leak rate is 1.0 gpm plant total.

A summary of the parameters used in the analysis is given in Table 15.3-1.

Based on the foregoing model, the thyroid and whole body doses are conservatively calculated to be 2.7 rem and 0.17 rem, respectively, for the exclusion area boundary (EAB), and 5.1 rem and 0.058 rem respectively, for the low population zone (LPZ). These doses are well below the values of 30 rem thyroid and 2.5 rem whole body set forth in Regulatory Guide 1.195.

15.3.3.3 Conclusions

For Units 1 and 2, the following conclusions are made:

1. Because the peak RCS pressure reached during any of the transients is less than that which would cause stresses to exceed the faulted condition stress limits, the integrity of the primary coolant system is not endangered.

2. Because the peak clad surface temperature calculated for the hot spot during the worst transient remains considerably less than 2700°F, the core will remain in place and intact with no loss of core cooling capability.

3. The number of rods experiencing DNB remains less than the number of rods assumed to experience DNB for the radiological analysis (i.e., the radiological doses are below the dose values set forth in 10CFR100).

15.3.4 REACTOR COOLANT PUMP SHAFT BREAK

15.3.4.1 Identification of Causes and Accident Description

The accident is postulated as an instantaneous failure of a reactor coolant pump shaft, such as discussed in Section 5.4. Flow through the affected reactor coolant loop is rapidly reduced, though the initial rate of reduction of coolant flow is greater for the reactor coolant pump rotor seizure event. With a failed shaft, the impeller could conceivably be free to spin in the reverse direction instead of being fixed in position. The effect of such reverse spinning is a slight decrease in the end point (steady state) core flow.

The analysis presented in Section 15.3.3 represents the limiting condition, assuming a locked rotor for forward flow but a free-spinning shaft for reverse flow in the affected loop.

15.3.4.2 Conclusions

The conclusions of Section 15.3.3.3 apply for the Reactor Coolant Pump Shaft Break event.

REFERENCES

1. Not Used.

2. Baldwin, M. S., Merrian, M. M., Schenkel, H. S. and Van De Walle, D. J., “An Evaluation of Loss of Flow Accidents Caused by Power System Frequency Transients in Westinghouse PWRs,” WCAP-8424, Revision 1, June 1975.

3. Huegel, D. S., et al., “RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses,” WCAP-14882-P-A, April 1999.

4. Sung, Y. X. et al., “VIPRE-01 Modeling and Qualification for Pressurized Water Reactor Non-LOCA Thermal-Hydraulic Safety Analysis,” WCAP-14565-P-A, October 1999.

5. Friedland, A. J., and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

TABLE 15.3-1
Page 1 of 2
PARAMETERS FOR POSTULATED LOCKED ROTOR ACCIDENT ANALYSIS

1.	Data and assumptions used to estimate radioactive source from postulated accidents			
	a.	Power level		3684
	b.	Primary coolant activity concentration prior to accident		1.0 uCi/gm Dose Equivalent I-131
	c.	Secondary side coolant activity concentration prior to accident		0.1 uCi/gm Dose Equivalent I-131
	d.	Primary to secondary leakage		1 gpm plant total
	e.	Offsite power		lost
	f.	Failed fuel		15%
	g.	Radial Peaking Factor		1.65
2.	Data assumptions used to estimate activity released			
	a.	Iodine partition factor for steam release from secondary side		0.01
	b.	Steam release from secondary side		450,000 (0 - 2 hours) 1,300,000 (2 - 11 hours)
3.	Dispersion data			
	a.	EAB and LPZ distances		2080 m and 4 miles
	b.	*X*/Q		@ EAB (0 - 2 hours) 1.6×10^{-4} sec/m^3 @ LPZ (0 - 8 hours) 2.4×10^{-5} sec/m^3 @ LPZ (8 - 24 hours) 1.6×10^{-5} sec/m^3
4.	Dose Data			
	a.	Method of dose calculations		See Appendix 15B
	b.	Dose conversion assumptions		See Appendix 15B
	c.	Doses	thyroid	whole body (gamma dose)

TABLE 15.3-1
Page 2 of 2
PARAMETERS FOR POSTULATED LOCKED ROTOR ACCIDENT ANALYSIS

@ EAB (0 - 2 hours)	2.7 rem	0.17 rem
@ LPZ (0 - 11 hours)	5.1 rem	0.058 rem

TABLE 15.3-2
HAS BEEN DELETED.

15.4 REACTIVITY AND POWER DISTRIBUTION ANOMALIES

A number of faults have been postulated which could result in reactivity and power distribution anomalies. Reactivity changes could be caused by control rod motion or ejection, boron concentration changes, or addition of cold water to the Reactor Coolant System (RCS). Power distribution changes could be caused by control rod motion, misalignment, or ejection, or by static means such as fuel assembly mislocation. These events are discussed in this section. Detailed analyses are presented for the most limiting of these events.

Discussions of the following incidents are presented in Section 15.4:

1. Uncontrolled rod cluster control assembly bank withdrawal from a subcritical or low power startup condition.

2. Uncontrolled rod cluster control assembly bank withdrawal at power.

3. Rod cluster control assembly misalignment.

4. Startup of an inactive reactor coolant pump at an incorrect temperature.

5. A malfunction or failure of the flow controller in a BWR loop that results in an increased reactor coolant flow rate. (Not applicable to the Comanche Peak Nuclear Power Plant (CPNPP).)

6. Chemical and Volume Control System malfunction that results in a decrease in the boron concentration in the reactor coolant.

7. Inadvertent loading and operation of a fuel assembly in an improper position.

8. Spectrum of rod cluster control assembly ejection accidents.

Items 1, 2, 4, and 6 above are considered to be American Nuclear Society (ANS) Condition II events, item 7 an ANS Condition III event, and item 8 an ANS Condition IV event. Item 3 entails both Condition II and III events. Section 15.0.1 contains a discussion of ANS classifications.

15.4.1 UNCONTROLLED ROD CLUSTER CONTROL ASSEMBLY BANK WITHDRAWAL FROM A SUBCRITICAL OR LOW POWER STARTUP CONDITION

15.4.1.1 Identification of Causes and Accident Description

A rod cluster control assembly (RCCA) withdrawal accident is defined as an uncontrolled addition of reactivity to the reactor core caused by withdrawal of one or more RCCA banks resulting in a power excursion. Such a transient could be caused by operator action or by a malfunction of the reactor control or rod control systems. This could occur with the reactor subcritical, at hot zero power or at power. The “at power” case is discussed in Section 15.4.2.

The RCCA drive mechanisms are wired into preselected bank configurations which are not altered during reactor life. These circuits prevent the RCCAs from being automatically withdrawn in other than their respective banks. Power supplied to the banks is controlled such that no more than two banks can be withdrawn at the same time and in their proper withdrawal sequence. The

RCCA drive mechanisms are of the magnetic latch type and coil actuation is sequenced to provide variable speed travel. The maximum reactivity insertion rate analyzed in the detailed plant analysis is greater than that occurring with the simultaneous withdrawal of the combination of two sequential control banks having the maximum combined worth at maximum speed.

This event is classified as an ANS Condition II incident (a fault of moderate frequency) as defined in Section 15.0.1.

The neutron flux response to a continuous reactivity insertion is characterized by a very fast rise terminated by the reactivity feedback effect of the negative Doppler coefficient. This self limitation of the power excursion is of primary importance since it limits the power to a tolerable level during the delay time for protective action. Should a continuous RCCA withdrawal accident occur, the transient will be terminated by the following automatic features of the Reactor Protection System:

1. Source range high neutron flux reactor trip

   Actuated when either of two independent source range channels indicates a neutron flux level above a preselected manually adjustable setpoint. This trip function may be manually bypassed only after an intermediate range flux channel indicates a flux level above a specified level. It is automatically reinstated when both intermediate range channels indicate a flux level below a specified value.

2. Intermediate range high neutron flux reactor trip

   Actuated when either of two independent intermediate range channels indicates a flux level above a preselected manually adjustable setpoint. This trip function may be manually bypassed only after two out of the four power range channels are reading above approximately 10 percent of full power and is automatically reinstated when three out of the four channels indicate a power level below this value.

3. Power range high neutron flux reactor trip (low setting)

   Actuated when two out of the four power range channels indicate a power level above approximately 25 percent of full power. This trip function may be manually bypassed when two out of the four power range channels indicate a power level above approximately 10 percent of full power and is automatically reinstated only after three out of the four channels indicate a power level below this value.

4. Power range high neutron flux reactor trip (high setting)

   Actuated when two out of the four power range channels indicate a power level above a preset setpoint. This trip function is always active.

5. High nuclear flux rate reactor trip

   Actuated when the positive rate of change of neutron flux on two out of four nuclear power range channels indicates a rate above the preset setpoint. This trip function is always active.

In addition, control rod stops on high intermediate range flux level (one out of two) and high power range flux level (one out of four) serve to discontinue rod withdrawal and prevent the need to actuate the intermediate range flux level trip and the power range flux level trip, respectively.

15.4.1.2 Analysis of Effects and Consequences

Method of Analysis

The analysis of the uncontrolled RCCA bank withdrawal from subcritical accident is performed in three stages. First, a spatial neutron kinetics computer code, TWINKLE (Reference 4), is used to calculate the core average nuclear power transient, including various core feedback effects, i.e., Doppler and moderator reactivity. Next, the FACTRAN computer code (Reference 5) uses the average nuclear power calculated by TWINKLE and performs a fuel rod transient heat transfer calculation to determine the core average heat flux and hot spot fuel temperature transients. Finally, the core average heat flux calculated by FACTRAN is used in the VIPRE computer code (Reference 6) for DNBR calculations.

Plant characteristics and initial conditions are discussed in Section 15.0.3. In order to give conservative results for a startup accident, the following assumptions are made:

1. The magnitude of the power peak reached during the initial part of the transient for any given rate of reactivity insertion is strongly dependent on the Doppler-only power defect. Therefore, a conservatively low absolute value is used to maximize the nuclear power transient.

2. Contribution of the moderator reactivity coefficient is negligible during the initial part of the transient because the heat transfer time between the fuel and the moderator is much longer than the neutron flux response time. However, after the initial neutron flux peak, the succeeding rate of power increase is affected by the moderator reactivity coefficient. The most positive value of the moderator temperature coefficient is used in the analysis to yield the maximum peak heat flux.

3. The reactor is assumed to be at hot zero power. This assumption is more conservative than that of a lower initial system temperature. The higher initial system temperature yields a larger fuel-water heat transfer coefficient, larger specific heats, and a less negative (smaller absolute magnitude) Doppler coefficient, all of which tend to reduce the Doppler feedback effect thereby increasing the neutron flux peak. The initial effective multiplication factor is assumed to be 1.0 since this results in the worst nuclear power transient.

4. Reactor trip is assumed to be initiated by power range high neutron flux (low setting). The most adverse combination of instrument and setpoint errors, as well as delays for trip signal actuation and RCCA release, is taken into account. A 10 percent increase is assumed for the power range flux trip setpoint raising it from the nominal value of 25 percent to 35 percent. Since the rise in the neutron flux is so rapid, the effect of errors in the trip setpoint on the actual time at which the rods are released is negligible. In addition, the reactor trip insertion characteristic is based on the assumption that the highest worth RCCA is stuck in its fully withdrawn position. See Section 15.0.5 for RCCA insertion characteristics.

5. Reactivity insertion rates are assumed to bound the maximum positive reactivity insertion rate that would result from the simultaneous withdrawal of the combination of the two sequential control banks having the greatest combined worth at maximum speed (45 inches/minute). Control rod drive mechanism design is discussed in Section 4.6.

6. The most limiting axial and radial power shapes, associated with having the two highest combined worth banks in their high worth position, are assumed in the DNB analysis.

7. The initial power level was assumed to be below the power level expected for any shutdown condition (10^{-9} of nominal power). The combination of highest reactivity insertion rate and lowest initial power produces the highest peak heat flux.

8. Two reactor coolant pumps are assumed to be in operation.

9. The RCS pressure is below nominal pressure by 30 psi.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of the accident.

15.4.1.3 Conclusions

For Units 1 and 2, in the event of an RCCA withdrawal accident from the subcritical condition, the core and the RCS are not adversely affected, since the combination of thermal power and the coolant temperature result in a DNBR greater than the limit value. The DNBR design basis as described in Section 4.4; is met.

15.4.2 UNCONTROLLED ROD CLUSTER CONTROL ASSEMBLY BANK WITHDRAWAL AT POWER

15.4.2.1 Identification of Causes and Accident Description

Uncontrolled RCCA bank withdrawal at power results in an increase in the core heat flux. Since the heat extraction from the steam generator lags behind the core power generation until the steam generator pressure reaches the relief or safety valve setpoint, there is a net increase in the reactor coolant temperature. Unless terminated by manual or automatic action, the power mismatch and resultant coolant temperature rise could eventually result in DNB.

Therefore, in order to avert damage to the fuel clad the Reactor Protection System is designed to terminate any such transient before the DNBR falls below the limit value.

This event is classified as an ANS Condition II incident (a fault of moderate frequency) as defined in Section 15.0.1.

The automatic features of the Reactor Protection System which prevent core damage following the postulated accident include the following:

1. Power range neutron flux instrumentation actuates a reactor trip if two out of four channels exceed an overpower setpoint.

2. Reactor trip is actuated if any two out of four N-16 channels exceed an Overtemperature N-16 setpoint. This setpoint is automatically varied with axial power imbalance, coolant temperature and pressure to protect against DNB.

3. Reactor trip is actuated if any two out of four N-16 channels exceed an Overpower N-16 setpoint. This setpoint can be automatically varied with axial power imbalance to ensure that the allowable heat generation rate (kW/ft) is not exceeded.

4. A high pressurizer pressure reactor trip actuated from any two out of four pressure channels, which is set at a fixed point. This set pressure is less than the set pressure for the pressurizer safety valves.

5. A high pressurizer water level reactor trip actuated from any two out of three level channels when the reactor power is above approximately 10 percent (Permissive-7).

6. Reactor trip is actuated if any two of four power range neutron flux high positive rate channels exceed a setpoint to protect against exceeding the Reactor Coolant System pressure limits.

In addition to the above listed reactor trips, there are the following RCCA withdrawal blocks:

1. High neutron flux (one out of four power range).

2. Overpower N-16 (two out of four).

3. Overtemperature N-16 (two out of four).

The manner in which the combination of Overpower and Overtemperature N-16 trips provide protection over the full range of RCS conditions is described in Chapter 7. Figure 15.0-1 presents allowable reactor coolant loop T_{cold} and power for the design power distribution and flow as a function of primary coolant pressure. The boundaries of operation defined by the Overpower N-16 trip and the Overtemperature N-16 trip are represented as “protection lines” on this diagram. The utility of this diagram is in the fact that the limit imposed by any given DNBR can be represented as a line. The DNB lines represent the locus of conditions for which the DNBR equals the limit value. All points below and to the left of a DNB line for a given pressure have a DNBR greater than the limit value. The diagram shows that DNB is prevented for all cases if the area enclosed with the maximum protection lines is not traversed by the applicable DNBR line at any point.

The area of permissible operation (power, pressure and temperature) is bounded by the combination of reactor trips: high neutron flux (fixed setpoint); high pressure (fixed setpoint); low pressure (fixed setpoint); Overpower and Overtemperature N-16 (variable setpoints).

15.4.2.2 Analysis of Effects and Consequences

Method of Analysis

This transient is analyzed for Units 1 and 2 using the RETRAN computer code (Reference 10).

Multiple cases (with varying reactivity insertion rates) at initial power levels of 100%, 60% and 10% of rated thermal were analyzed.

Plant characteristics and initial conditions are discussed in Section 15.0.3. In order to obtain conservative results for an uncontrolled rod withdrawal at power accident, the following assumptions are made:

1. The initial core power, pressurizer pressure and vessel average temperature are assumed to be at the respective nominal values, and the reactor coolant flow is at the minimum measured flow value. Uncertainties in the initial conditions are accounted for in the DNBR limit value as described in Reference 11.

2. Reactivity coefficients - two feedback conditions are analyzed:

   a. Minimum reactivity feedback

      A least negative or positive value of the moderator temperature coefficient of reactivity is assumed corresponding to the beginning of core life. A conservatively small (in absolute magnitude) value of the Doppler coefficient is assumed.

   b. Maximum reactivity feedback

      A conservatively large positive moderator density coefficient and a large (in absolute magnitude) negative Doppler coefficient are assumed.

3. The reactor trip on high neutron flux is assumed to be actuated at a conservative value of 115 percent of nominal full power. The N-16 trips include allowances for all adverse instrumentation and setpoint errors; the delays for trip actuation are assumed to be the maximum values.

4. The RCCA trip insertion characteristic is based on the assumption that the highest worth assembly is stuck in its fully withdrawn position.

5. The maximum positive reactivity insertion rate is greater than that for the simultaneous withdrawal of the combinations of the two control banks having the maximum combined worth at maximum speed.

The effect of RCCA movement on the axial core power distribution is accounted for by causing a decrease in Overtemperature N-16 trip setpoint proportional to a decrease in margin to DNB. For conservatism, this feature is not credited in the analysis of the uncontrolled RCCA bank withdrawal at power event.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of the accident.

15.4.2.3 Conclusions

For Units 1 and 2, all relevant event acceptance criteria are satisfied. The reactor is tripped sufficiently fast during the RCCA withdrawal at power transient to ensure that the ability of the

primary coolant to remove heat from the fuel rods is not reduced. Thus, the fuel cladding temperature does not rise significantly above its initial value during the transient.

The high neutron flux, Overtemperature N-16 and Overpower N-16 trip channels provide adequate protection over the entire range of possible reactivity insertion rates; i.e., the minimum value of DNBR is always larger than the limit value.

15.4.3 ROD CLUSTER CONTROL ASSEMBLY MISALIGNMENT (SYSTEM MALFUNCTION OR OPERATOR ERROR)

15.4.3.1 Identification of Causes and Accident Description

Rod cluster control assembly (RCCA) misoperation accidents include:

1. One or more dropped RCCAs within the same group,

2. A dropped RCCA bank,

3. Statically misaligned RCCA,

4. Withdrawal of a single RCCA.

Each RCCA has a position indicator channel which displays the position of the assembly. The displays of assembly positions are grouped for the operator's convenience. Fully inserted assemblies are further indicated by a rod at bottom signal, which actuates an alarm and a control room annunciator. Group demand position is also indicated.

Full length RCCAs are moved in preselected banks, and the banks are moved in the same preselected sequence. Each control bank of RCCAs is divided into two groups. The rods comprising a group operate in parallel through multiplexing thyristors. The two groups in a bank move sequentially such that the first group is always within one step of the second group in the bank. A definite schedule of actuation (or deactuation of the stationary gripper, movable gripper, and lift coils of a mechanism) is required to withdraw the RCCA attached to the mechanism. Since the stationary gripper, movable gripper, and lift coils associated with the four RCCAs of a rod group are driven in parallel, any single failure which would cause rod withdrawal would affect a minimum of one group. Mechanical failures are in the direction of insertion, or immobility.

The dropped RCCA assemblies, dropped RCCA assembly bank, and statically misaligned RCCA assembly events are classified as ANS Condition II incidents (incidents of moderate frequency) as defined in Section 15.0.1. The single RCCA withdrawal incident is classified as an ANS Condition III event (i.e., an infrequent fault), as discussed below.

No single electrical or mechanical failure in the rod control system could cause the accidental withdrawal of a single RCCA from the inserted bank at full power operation. The operator could withdraw a single RCCA in the control bank since this feature is necessary in order to retrieve an assembly should one be accidentally dropped. The event analyzed must result from multiple wiring failures (probability for single random failure is of the order of 10^{-4}/year; refer to Section 7.7.2.2) or multiple significant operator errors and subsequent and repeated operator

disregard of event indication. The probability of such a combination of conditions is considered sufficiently low that the limiting consequences may include slight fuel damage.

Thus, consistent with the philosophy and format of ANSI N18.2, the event is classified as a Condition III event. By definition “Condition III occurrences include incidents, any one of which may occur during the lifetime of a particular plant”, and “shall not cause more than a small fraction of fuel elements in the reactor to be damaged...”

The selection of this criterion is in accordance with GDC 25 which states, “The protection system shall be designed to assure that specified acceptable fuel design limits are not exceeded for any single malfunction of the reactivity control systems, such as accidental withdrawal (not ejection or dropout) of control rods.” (Emphases have been added). It has been shown that single failures resulting in RCCA bank withdrawals do not violate specified fuel design limits. Moreover, no single malfunction can result in the withdrawal of a single RCCA. Thus, it is concluded that the criterion established for the single rod withdrawal at power accident is appropriate and in accordance with GDC 25.

A dropped RCCA or RCCA bank is detected by:

1. Sudden drop in the core power level as seen by the Nuclear Instrumentation System,

2. Asymmetric power distribution as seen on out-of-core neutron detectors or core exit thermocouples,

3. Rod at bottom signal,

4. Rod deviation alarm, or

5. Rod position indication.

Misaligned RCCAs are detected by:

1. Asymmetric power distribution as seen on out-of-core neutron detectors or core exit thermocouples,

2. Rod deviation alarm, or

3. Rod position indicators.

Rod position is displayed in 6 step increments with an accuracy of ±4 steps. Deviation of any RCCA from its group by twice this distance (12 steps) will not cause power distributions worse than the design limits. The deviation alarm alerts the operator to rod deviation with respect to the group position in excess of 6 steps. If the rod deviation alarm is not operable, the operator is required to take action as required by the Technical Requirements Manual (TRM).

If one or more rod position indicator channels should be out of service, detailed operating instructions shall be followed to assure the alignment of the non-indicated RCCAs. The operator is also required to take action as required by the technical specifications.

In the extremely unlikely event of simultaneous electrical failures which could result in single RCCA withdrawal, rod deviation and rod control urgent failure would both be displayed on the plant annunciator, and the rod position indicators would indicate the relative positions of the assemblies in the bank. The urgent failure alarm also inhibits automatic rod motion in the group in which it occurs. Withdrawal of a single RCCA by operator action, whether deliberate or by a combination of errors, would result in activation of the same alarm and the same visual indications. Withdrawal of a single RCCA results in both positive reactivity insertion tending to increase core power, and an increase in local power density in the core area associated with the RCCA. Automatic protection for this event is provided by the Overtemperature N-16 reactor trip, although due to the increase in local power density it is not possible in all cases to provide assurance that the core safety limits will not be violated.

Plant systems and equipment which are available to mitigate the effects of the various control rod misoperations are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of the accident.

15.4.3.2 Analysis of Effects and Consequences

1. Dropped RCCAs, dropped RCCA bank, and statically misaligned RCCA.

   Method of Analysis

   a. One or more dropped RCCAs from the same group.

      For evaluation of the dropped RCCA event, the transient system response is calculated using the LOFTRAN computer code (Reference 7) with the methodology described in Reference 8.

      Statepoints are calculated and nuclear models are used to obtain a hot channel factor consistent with the primary system conditions and reactor power. By incorporating the primary conditions from the transients and the hot channel factor from the nuclear analysis, the DNB basis can be shown to be met using dropped rod limit lines developed with the VIPRE computer code (Reference 6).

      Single or multiple dropped RCCAs within the same group result in a negative reactivity insertion. If the worth of the dropped rod(s) is sufficiently large, and if the turbine load is constant, a reactor trip may occur on low pressurizer pressure due to the mismatch between the reactor power and the turbine power. The core is not adversely affected during this period since power is decreasing rapidly. Following reactor trip, normal shutdown procedures are followed. The operator may manually retrieve the RCCA by following approved operating procedures.

      For those dropped RCCAs which do not result in a reactor trip, power may be reestablished either by reactivity feedback or control bank withdrawal. Following a dropped rod event in manual rod control, the plant will establish a new equilibrium condition. The equilibrium process without control system interaction is monotonic, thus removing power overshoot as a concern, and establishing the automatic rod control mode of operation as the limiting case.

For a dropped RCCA event in the automatic rod control mode, the Rod Control System detects the drop in power and initiates control bank withdrawal. Power overshoot may occur due to this action by the automatic rod controller after which the control system will insert the control bank to restore nominal power. Uncertainties in the initial conditions are included in the DNB evaluation. In all cases, the minimum DNBR remains above the limit value.

b. Dropped RCCA Bank

A dropped RCCA bank typically results in a negative reactivity insertion greater than 500 pcm. Due to the relatively large worth of the dropped bank, and if the turbine load is constant, a reactor trip may occur on low pressurizer pressure due to the mismatch between the reactor power and the turbine power. The core is not adversely affected during this period, since power is decreasing rapidly. Following reactor trip, normal shutdown procedures are followed to further cool down the plant. Any action required of the operator to maintain the plant in a stabilized condition will be in a time frame in excess of ten minutes following the incident.

c. Statically Misaligned RCCA

No transient analysis is performed, rather steady state power distributions are analyzed using the ANC computer code (Reference 9 and beginning with Unit 2 Cycle 19 and Unit 1 Cycle 22, Reference 9 and Reference 12). The peaking factors are then compared to peaking factor limits developed using the VIPRE computer code (Reference 6), which are based on meeting the DNB design criterion.

The most severe misalignment situations with respect to DNBR at significant power levels arise from cases in which one RCCA is fully inserted, or where bank D is fully inserted with one RCCA fully withdrawn. Multiple independent alarms, including a bank insertion limit alarm, alert the operator well before the postulated conditions are approached. The bank can be inserted to its insertion limit with any one assembly fully withdrawn without the DNBR falling below the limit value.

The insertion limits in the technical specifications may vary from time to time depending on a number of limiting criteria. It is preferable, therefore, to analyze the misaligned RCCA case at full power for a position of the control bank as deeply inserted as the criteria on minimum DNBR and power peaking factor will allow. The full power insertion limits on control bank D must then be chosen to be above that position and will usually be dictated by other criteria. Detailed results will vary from cycle to cycle depending on fuel arrangements.

For this RCCA misalignment, with bank D inserted to its full power insertion limit and one RCCA fully withdrawn, DNBR does not fall below the limit value. This case is analyzed assuming the initial reactor power, and RCS pressure and temperature are at their nominal values including uncertainties but with the increased radial peaking factor associated with the misaligned RCCA.

DNB calculations have not been performed specifically for RCCAs missing from other banks; however, power shape calculations have been done as required for the RCCA ejection analysis. Inspection of the power shapes shows that the DNB and peak kw/ft situation is less severe than the bank D case discussed above assuming insertion limits on the other banks equivalent to a bank D full-in insertion limit.

For RCCA misalignments with one RCCA fully inserted, the DNBR does not fall below the limit value. This case is analyzed assuming the initial reactor power, and RCS pressure and temperature are at their nominal values, including uncertainties but with the increased radial peaking factor associated with the misaligned RCCA.

DNB does not occur for the RCCA misalignment incident and thus the ability of the primary coolant to remove heat from the fuel rod is not reduced. The peak fuel temperature corresponds to a linear heat generation rate based on the radial peaking factor penalty associated with the misaligned RCCA and the design axial power distribution. The resulting linear heat generation is well below that which would cause fuel melting.

Following the identification of a RCCA group misalignment condition by the operator is required to take action as required by the plant technical specifications and operating instructions.

2. Single RCCA Withdrawal

Method of Analysis

Power distributions within the core are calculated using the ANC computer code (Reference 9 and beginning with Unit 2 Cycle 19 and Unit 1 Cycle 22, Reference 9 and Reference 12). The peaking factors are then used by the VIPRE computer code (Reference 6) to calculate the DNBR for the event. The case of the worst rod withdrawn from bank D inserted at the insertion limit, with the reactor initially at full power, is analyzed. This incident is assumed to occur at the initial core power, pressurizer pressure and vessel average temperature are assumed to be at the respective nominal values, and the reactor coolant flow is at the minimum measured flow value. Uncertainties in the initial conditions are accounted for in the DNBR limit value as described in Reference 11.

For the single rod withdrawal event, two cases are considered as follows:

a. If the reactor is in the manual control mode, continuous withdrawal of a single RCCA results in both an increase in core power and coolant temperature, and an increase in the local hot channel factor in the area of the withdrawing RCCA. In terms of the overall system response, this case is similar to those presented in Section 15.4.2; however, the increased local power peaking in the area of the withdrawn RCCA results in lower minimum DNBRs than for the withdrawn bank cases. Depending on initial bank insertion and location of the withdrawn RCCA, automatic reactor trip may not occur sufficiently fast to prevent the minimum DNBR from falling below the limit value. Evaluation of this case at the power and

coolant conditions at which the Overtemperature N-16 trip would be expected to trip the plant shows that an upper limit for the number of fuel rods with a DNBR less than the limit value is 5 percent of the total rods in the core.

b. If the reactor is in the automatic control mode, the multiple failures that result in the withdrawal of a single RCCA will result in the immobility of the other RCCAs in the controlling bank. The transient will then proceed in the same manner as Case a described above.

For such cases as above, a reactor trip will ultimately ensue, although not sufficiently fast in all cases to prevent a minimum DNBR in the core of less than the limit value. Following reactor trip, normal shutdown procedures are followed. No single failure of the Reactor Trip System will negate the protection functions required for the single RCCA withdrawal accident, or adversely affect the consequences of the accident.

15.4.3.3 Conclusions

For Units 1 and 2, it is confirmed that the DNBR remains greater than the limit value for all cases of dropped RCCAs and dropped banks; therefore the DNB design basis is met. It is shown for all cases which do not result in reactor trip that the DNBR remains greater than the limit value and, therefore, the DNB design is met.

For all cases of any RCCA fully inserted, or bank D inserted to its rod insertion limits with any single RCCA in that bank fully withdrawn (static misalignment), the DNBR remains greater than the limit value.

For the case of the accidental withdrawal of a single RCCA, with the reactor in the automatic or manual control mode and initially operating at full power with bank D at the insertion limit, an upper bound of the number of fuel rods experiencing DNB is 5 percent of the total fuel rods in the core.

15.4.4 STARTUP OF AN INACTIVE REACTOR COOLANT PUMP AT AN INCORRECT TEMPERATURE

15.4.4.1 Identification of Causes and Accident Description

If the plant is operating with one pump out of service, there is reverse flow through the inactive loop due to the pressure difference across the reactor vessel. The cold leg temperature in an inactive loop is identical to the cold leg temperature of the active loops (the reactor core inlet temperature). If the reactor is operated at power, and assuming the secondary side of the steam generator in the inactive loop is not isolated, there is a temperature drop across the steam generator in the inactive loop and, with the reverse flow, the hot leg temperature of the inactive loop is lower than the reactor core inlet temperature.

CPNPP technical specifications require all four reactor coolant loops to be in operation while at power or in startup conditions (Modes 1 and 2). Since by definition, the event can only occur during power operations (Mode 1 and 2) when a loop is out of service, a detailed analysis of this event is not presented.

15.4.5 A MALFUNCTION OR FAILURE OF THE FLOW CONTROLLER IN A BWR LOOP THAT RESULTS IN AN INCREASED REACTOR COOLANT FLOW RATE

This section is not applicable to the CPNPP.

15.4.6 CHEMICAL AND VOLUME CONTROL SYSTEM MALFUNCTION THAT RESULTS IN A DECREASE IN THE BORON CONCENTRATION IN THE REACTOR COOLANT

15.4.6.1 Identification of Causes and Accident Description

One of the two principal means of positive reactivity insertion to the core is the addition of unborated, primary grade water from the Demineralized and Reactor Makeup Water System (RMWS) into the RCS through the reactor makeup portion of the Chemical and Volume Control System (CVCS). Boron dilution with these systems is a manually initiated operation under strict administrative controls requiring close operator surveillance with procedures limiting the rate and duration of the dilution. A boric acid blend system is available to allow the operator to match the makeup's boron concentration to that of the RCS during normal charging.

A comprehensive review of the primary system showed that a single failure in the NaOH spray system would not result in a boron dilution of the Reactor Coolant System and that the CVCS malfunction represents the most limiting potential source of dilution. Based on this review, it is clear that the analysis results presented below bounds all potential sources of inadvertent dilution under all modes of operation.

The principal means of causing an inadvertent boron dilution are the opening of the primary water makeup control valve and failure of the blend system, either by controller or mechanical failure. The CVCS and RMWS are designed to limit, even under various postulated failure modes, the potential rate of dilution to values which, with indication by alarms and instrumentation, will allow sufficient time for automatic or operator response (depending on the mode of operation) to terminate the dilution. An inadvertent dilution from the RMWS may be terminated by closing the primary water makeup control valve. All expected sources of dilution may be terminated by closing isolation valves in the CVCS, 1,2-LCV-112B and C. The lost shutdown margin (SDM) may be regained by the opening of isolation valves to the RWST, 1,2-LCV-112D and E, thus allowing the addition of borated water to the RCS.

The rate at which unborated water can be added to the RCS is limited by the design of the CVCS and RMWS. Alarms in the CVCS are available to alert the reactor operators to high flow rates. In addition, the makeup flow is limited to a maximum dilution flow rate of 167 gpm for Cold Shutdown, Hot Shutdown and Hot Standby.

Generally, to dilute, the operator must perform two distinct actions:

1. Switch control of the makeup from the automatic makeup mode to the dilute mode, and

2. Turn the RCS makeup actuation handswitch to the "on" position.

Failure to carry out either of the above actions prevents initiation of dilution. Also during normal operation the operator may add borated water to the RCS by blending boric acid from the boric acid storage tanks with primary grade water. This requires the operator to determine the

concentration of the addition and to set the blended flow rate and the boric acid flow rate. The makeup controller will then limit the sum of the boric acid flow rate and primary grade water flow rate to the blended flow rate after turning the RCS system makeup actuation switch to the “on” position (i.e., the controller determines the primary grade water flow rate).

The status of the RCS makeup is continuously available to the operator by:

1. Indication of the boric acid and blended flow rates,

2. CVCS and RMWS pump status lights,

3. Deviation alarms if the boric acid or blended flow rates deviate by ± 1 gpm or ± 8 gpm, respectively, from the preset values,

4. High charging flow rate alarm,

5. Volume control tank high and high-high water level and high pressure alarms,

6. Letdown divert valve position alarm,

7. Source Range Neutron Flux - when reactor is subcritical;

    a. High Flux at Shutdown Alarm,

    b. Indicated Source Range Neutron Flux count rates,

    c. Audible Source Range Neutron Flux count rate, and

    d. Source Range Neutron Flux - Doubling Alarm.

8. With the reactor critical

    a. Axial Flux Difference Alarm (reactor power ≥ 50% RTP),

    b. Control Rod Insertion Limit Low and Low-Low Alarms,

    c. Overtemperature N-16 Alarm (at power),

    d. Overtemperature N-16 turbine runback (at power),

    e. Overtemperature N-16 Reactor Trip, and

    f. Power Range Neutron Flux - High, both high and low setpoint Reactor Trips.

This event is classified as an ANS Condition II incident (a fault of moderate frequency) as defined in Section 15.0.1.

15.4.6.2 Analysis of Effects and Consequences

To cover all phases of plant operation, boron dilution during Refueling, Cold Shutdown, Hot Shutdown, Hot Standby, Startup, and Power modes of operation are considered in this analysis. Conservative values for necessary parameters were used, i.e., high RCS critical boron concentrations, high boron worths, minimum shutdown margins, and lower than actual RCS volumes. These assumptions result in conservative determinations of the time available for operator or system response after detection of a dilution transient in progress.

Dilution During Refueling

An uncontrolled boron dilution transient cannot occur during this mode of operation. Inadvertent dilution is prevented by administrative controls which isolate the RCS from the potential source of unborated water. Either valve 1,2CS-8455 or valves 1,2CS-8560, 1,2FCV-111B, 1,2CS-8441, 1,2CS-8453 and 1,2CS-8439 in the CVCS will be locked closed during refueling operations. These valves block all flow paths that could allow significant rates of unborated makeup water to reach the RCS. Any makeup which is required during refueling will be borated water supplied from the RWST.

Dilution During Cold Shutdown, Hot Shutdown and Hot Standby

The Technical Specifications require the reactor to be shutdown by the amount specified in the Core Operating Limits Report when the unit is operating in Mode 5 (Cold Shutdown), Mode 4, or 3, (Hot Shutdown or Hot Standby). The following conditions are assumed for the analysis of the inadvertent boron dilution event while in these operating modes:

1. An assumed dilution flow rate of 157.5 gpm is used. This value corresponds to the high charging flow rate alarm plus a 5% allowance for uncertainties.

2. The following minimum RCS water volumes are used while in these operating modes:

Mode(s)	RCS Water Volume (ft^3)	Assumption Used to Minimize RCS Water Volume
3, 4, & 5 (filled)	9903.7 (Unit 1) 8594.1 (Unit 2)	The volumes for the pressurizer, surge line and reactor vessel upper head were excluded.
5 (drained)	4513.0 (Units 1 & 2)	The RCS water level is drained down to the mid-plane of the hot legs.

3. If no reactor coolant pump is in operation, all dilution sources are isolated or under administrative control.

Dilution During Startup

Startup is a transitory mode of operation. In this mode the plant is being taken from one long term mode of operation, Hot Standby, to another, Power. The plant is maintained in the Startup mode only for the purpose of startup testing at the beginning of each cycle. During this mode of operation the plant is in manual control, i.e., T_{avg}/ rod control is in manual. All normal actions required to change power level, either up or down, require operator initiation. The Technical

Specifications require a SDM of at least 1.3% ΔK/K and four reactor coolant pumps operating. Other conditions assumed are:

1. Dilution flow rate is limited by the design of the CVCS and RMWS. The makeup flow rate is limited to a maximum of 157.5 gpm for startup, which corresponds to the high charging flow alarm plus uncertainties.

2. A minimum RCS water volume of 11072 (Unit 1) and 9762 (Unit 2) ft^3. These very conservative estimates of the active RCS volume exclude the pressurizer and surge line volumes.

Dilution During Full Power Operation

The plant may be operated at power two ways, automatic T_{avg}/rod control or under manual (operator) rod control. The Technical Specifications require an available shutdown margin as specified in the Core Operating Limits Report and four reactor coolant pumps operating. With the plant at power and the RCS at pressure, the dilution rate is limited by the capacity of the centrifugal charging pumps (analysis is performed assuming two charging pumps are in operation even though normal operation is with one pump). Conditions assumed for this mode are:

1. Dilution flow rate is limited by the design of the CVCS and RMWS. The makeup flow rate is limited to a maximum of 157.5 gpm, which corresponds to the high charging flow alarm plus uncertainties. When the pressurizer level control is in manual, the maximum assumed dilution flow rate is 157.5 gpm and when in automatic pressurizer level control, the dilution is limited to the maximum letdown flow rate (approximately 125 gpm).

2. A minimum RCS water volume of 11072 (Unit 1) and 9762 (Unit 2) ft^3. These very conservative estimates of the active RCS volume exclude the pressurizer and surge line volumes.

15.4.6.3 Conclusions

Dilution During Refueling

Dilution during this mode has been precluded through administrative control of valves in the possible dilution flow paths (see Section 15.4.6.2).

Dilution During Cold Shutdown, Hot Shutdown, or Hot Standby

In Modes 3, 4, or 5, the reactor operators are relied upon to detect and recover from an inadvertent boron dilution event. Numerous alarms from the CVCS and RMWS and in the Nuclear Instrumentation System are available to provide assistance to the reactor operator in the detection of an inadvertent boron dilution event. In the analyses of the event initiated from Mode 3, 4, or 5, the VCT high water level alarm (at 70% span) is credited as the initial alarm. The time required to perform the VCT/RWST valve swap-over and to purge the charging lines of dilute water are included in the calculation of the time available to the reactor operators to initiate corrective actions. For both units, in Modes 3, 4 and 5, analyses have demonstrated that the

reactor operators have at least 15 minutes in which to initiate actions to terminate the dilution and initiate boration of the RCS.

Dilution During Startup

This mode of operation is a transitory mode to go to power and is the operational mode in which the operator intentionally dilutes and withdraws control rods to take the plant critical. During this mode the reactor is in manual rod control with the operator required to maintain a very high awareness of the plant status. For a normal approach to criticality the operator must manually initiate a limited dilution and subsequently manually withdraw the control rods, a process that takes several hours. The plant Technical Specifications require that the operator determine the estimated critical position of the control rods prior to approaching criticality thus assuring that the reactor does not go critical with the control rods below the insertion limits. Once critical, the power escalation must be sufficiently slow to allow the operator to manually block the Source Range reactor trip after receiving P-6 from the Intermediate Range. Too fast a power escalation (due to an unknown dilution) would result in reaching P-6 unexpectedly, leaving insufficient time to manually block the Source Range reactor trip. Failure to perform this manual action results in a reactor trip and immediate shutdown of the reactor.

However, in the event of an unplanned approach to criticality or dilution during power escalation while in the Startup mode, the plant status is such that minimal impact will result. The plant will slowly escalate in power to a reactor trip on the Power Range Neutron Flux - High, low setpoint (nominally 25% RTP). After reactor trip there is at least 15 minutes for operator action prior to return to criticality. The required operator action is the opening of valves 1,2-LCV-112D and E to initiate boration and the closing of valves 1,2-LCV-112B and C to terminate dilution.

Dilution During Full Power Operation

With the reactor under manual rod control and no operator action taken to terminate the transient, the power and temperature rise will cause the reactor to reach the Overtemperature N-16 trip setpoint resulting in a reactor trip. After reactor trip there is at least 15 minutes for operator action prior to return to criticality. The required operator action is the opening of valves 1,2-LCV-112D and E and the closing of valves 1,2-LCV-112B and C. The boron dilution transient in this case is essentially equivalent to an uncontrolled rod withdrawal at power. A reactor trip occurs when either the Hi Neutron Flux or the Overtemperature N-16 setpoint is reached. The maximum reactivity insertion rate for a boron dilution transient is conservatively estimated to be within the range of insertion rates analyzed for uncontrolled rod withdrawal at power.

It should be noted that prior to reaching the Overtemperature N-16 reactor trip the operator will have received an alarm on Overtemperature N-16 and an Overtemperature N-16 turbine runback. With the reactor in automatic rod control the pressurizer level controller will limit the dilution flow rate to the maximum letdown rate, approximately 125 gpm. If a dilution rate in excess of the letdown rate is present, the pressurizer level controller will throttle charging flow down to match the letdown rate.

Thus, with the reactor in automatic rod control, a boron dilution will result in a power and temperature increase such that the rod controller will attempt to compensate by slow insertion of the control rods. This action by the controller will result in at least three alarms to the operator:

1. rod insertion limit - low level alarm,

2. rod insertion limit - low-low level alarm if insertion continued after (1) above, and

3. axial flux difference alarm (ΔI outside of the target band).

Given the many alarms, indications, and the inherent slow process of dilution at power, the operator has sufficient time for action. For example, the operator has at least 15 minutes from the rod insertion limit low-low alarm until the shutdown margin is completely eroded at beginning-of-life. The time would be significantly longer at end-of-life, due to the low initial boron concentration, when shutdown margin is a concern.

The above results demonstrate that in all modes of operation an inadvertent boron dilution is precluded, or sufficient time is available for operator action to terminate the transient. Following termination of the dilution flow and initiation of boration, the reactor is in a stable condition with the operator regaining the required shutdown margin.

15.4.7 INADVERTENT LOADING AND OPERATION OF A FUEL ASSEMBLY INTO AN IMPROPER POSITION

15.4.7.1 Acceptance Criteria

This event is classified as a Condition III event. Condition III events are defined as those events that do not cause more than a small fraction of fuel rods to fail, although sufficient fuel damage might occur to preclude immediate resumption of operation. The specific acceptance criteria for this even are as follows:

a. To meet the requirements of General Design Criteria 13, plant operating procedures should include a provision requiring that reactor instrumentation be used to search for potential fuel loading errors after fueling operations.

b. In the event the error is not detectable by the instrumentation system and fuel rod failure limits could be exceeded during normal operation, the offsite consequences should be a small fraction of the 10 CFR Part 100 guidelines.

15.4.7.2 Identification of Causes and Accident Description

The Inadvertent Loading Event comprises core misloading scenarios such as the loading of one or more fuel assemblies into improper positions, the loading of a fuel rod during manufacture with one or more pellets of the wrong enrichment, or the loading of a full fuel assembly during manufacture with pellets of the wrong enrichment. In addition to these scenarios, misloading events involving burnable absorbers are theoretically possible, scenarios such as the placement of a cluster of 20 burnable absorbers into a core location slated to have 24 burnable absorbers. All of these misloading scenarios potentially result in a core reactivity distribution that differs from the intended core reactivity distribution. As a result, the core power distribution and peaking factors may differ from predictions. Specifically, misloading errors can lead to increased local power peaking at the location of the misloading if the misloading results in a local reactivity increase relative to the intended pattern. If the misloading results in a local reactivity decrease, power peaking increases away from the location of the misloading are possible due to

unintended power tilts. These kinds of increases, however, are generally distributed over a large core volume and are small relative to those where the local reactivity is increased.

Fuel misloads are prevented by the manufacturing controls employed to build the fuel and the core loading controls used to assemble the core. The manufacturing controls include checks on fuel rod weight to confirm the uranium loading in the fuel rod, active and passive gamma scans of individual fuel rods to confirm fuel enrichments, pellet stack lengths, pellet types, and the absence of pellet gaps during fuel manufacturing, and bar coding of each fuel rod to confirm its proper placement in the fuel assembly.

To reduce the probability of core loading errors during fuel loading, each fuel assembly and core component is marked with an identification number and loaded in accordance with a core loading diagram. During core loading, the fuel identification numbers are checked before each assembly is moved into the core. Identification numbers read during fuel movement are subsequently recorded on the loading diagram as a further check on proper placement after the loading is completed. The correct type of insert is also confirmed for each location. These procedures make the likelihood of core misloadings very small.

The severity and detectability of fuel misloads are influenced by several factors: the local reactivity perturbation relative to the intended core loading pattern, the core position of the misload, the local environment of the misloaded fuel assembly, and the number of operable incore detector locations and their proximity to the misload location. Should misloadings occur, the incore system of movable flux detectors, which is used to verify power distributions during startup and throughout the operating cycle, is capable of revealing enrichment errors or misloadings which would cause the kind of substantial power distribution perturbation that would be necessary to induce large numbers of fuel rod failures. In addition, thermocouples and excore detectors can provide additional indications of power distribution anomalies. This instrumentation, along with the startup testing performed each cycle, make the detection of severe misloadings highly likely.

15.4.7.3 Evaluation

The incore moveable detector system is used to search for potential fuel misloads at the start of each operating cycle. Following fuel loading and low power physics testing, an initial core power distribution measurement is made. The core power level of this initial flux map is typically between ~30% and ~50% of rated thermal power. This initial power distribution measurement is used to confirm that the measured power distribution is consistent with the predicted power distribution. Observed flux map deviations in excess of the flux map review criteria (see Table 1) would prompt an investigation of a possible core anomaly. This satisfies the first acceptance criterion given above.

In Reference 1, a large number of misloads were evaluated for representative core designs employing current fuel types and fuel features. The simulated misloads, involving one or two fuel assemblies, covered a wide range of local reactivity perturbations and core positions. The resulting hot full power (HFP) $F\Delta H$ peaking factors ranged from benign to very severe. Severe misloads with peaking factors that exceed the $F\Delta H$ limit for DNB at normal operation conditions have the potential for fuel failure if they remain undetected. The simulated misloads were assessed with respect to severity and detectability.

The detectability assessments of Reference 1 demonstrated that the incore detector system is very robust with respect to detection of misloads severe enough to fail fuel during normal operation. By examining a large number of moveable detector thimble patterns, the detectability assessments considered the effect of inoperable moveable detector thimbles on misload detectability. Even when the minimum number of operable detector locations allowed per the plant licensing bases was assumed, the incore detector system was capable of reliably detecting misloads severe enough to fail fuel during normal operation.

Fuel misloads involving a single fuel rod or fuel pellet were not evaluated as part of Reference 1. Such misloads, in general, will not be detectable using the incore detector system due to the very small power distribution perturbation. In terms of increased peaking factors and reduced DNBR values, however, the consequences of such misloads will be very small and limited to the affected fuel rod and the immediately adjacent fuel rods.

Detection of fuel misloads is, in part, a function of the number of available incore detector locations. Reference 1 demonstrated that the flux map review criteria of Table 1 are effective in detecting fuel misloads that could lead to fuel failures during normal operation. To enhance the probability that significant misloads will be detected, tighter review criteria are employed when the number of available detector locations is reduced.

The detectability assessments of Reference 1 confirm that the moveable detector system can reliably detect fuel misloads that could fail fuel during normal operation when the Table 1 review criteria are employed. Specifically, Reference 1 demonstrated that only a small fraction of 1% of misloads severe enough to fail fuel during normal operation would be undetected at startup using these limited review criteria. Furthermore, it was judged that even these “undetected” misloads would very likely be detected if other attributes of the startup power distribution measurement (e.g., tilts and reaction rate error contours) were considered along with the results of low power physics testing. Given that detection of >99% of misloads severe enough to fail fuel is expected using these review criteria, a radiological consequences analysis is deemed unnecessary. Failures in fresh fuel during startup would have negligible radiological consequences since there is only a small fission product inventory. Following startup, any fuel rod failures would occur gradually and would be detected by coolant activity monitoring. Since the number of potential fuel rod failures due to a core misload would be extremely small and such failures would occur gradually, any coolant activity releases would initially be well within the cleanup capacity of the plant. Any trend in increased coolant activity would warrant further investigation and evaluation. Therefore, the second acceptance criterion for this event would be satisfied since failures would be gradual, detectable, and the operations would be maintained within Technical Specification coolant activity guidelines.

15.4.7.4 Conclusion

Fuel misloads are prevented by manufacturing controls and core loading controls. In the unlikely event that a fuel misload should occur, the incore moveable detector system is capable of reliably detecting misloads that could fail fuel at normal operation conditions. Exceeding the review criteria herein would initiate an investigation to identify potential core anomalies. Any failures associated with an undetected fuel misload would be gradual, detectable, and the operations would be maintained within Technical Specification coolant activity guidelines.

15.4.8 SPECTRUM OF ROD CLUSTER CONTROL ASSEMBLY EJECTION ACCIDENTS

15.4.8.1 Identification of Causes and Accident Description

This accident is defined as the mechanical failure of a control rod mechanism pressure housing resulting in the ejection of a RCCA and drive shaft. The consequence of this mechanical failure is a rapid positive reactivity insertion together with an adverse core power distribution, possibly leading to localized fuel rod damage.

15.4.8.1.1 Design Precautions and Protection

Certain features in the CPNPP pressurized water reactors are intended to preclude the possibility of a rod ejection accident, or to limit the consequences if the accident were to occur. These include a sound, conservative mechanical design of the rod housings, together with a thorough quality control (testing) program during assembly, and a nuclear design which lessens the potential ejection worth of RCCAs, and minimizes the number of assemblies inserted at high power levels.

Mechanical Design

The mechanical design is discussed in Section 4.6. Mechanical design and quality control procedures intended to preclude the possibility of a RCCA drive mechanism housing failure are listed below:

1. Each full length control rod drive mechanism housing is completely assembled and shop tested at 4100 psi.

2. The mechanism housings are individually hydrotested after they are attached to the head adapters in the reactor vessel head, and are checked during the hydrotest of the completed reactor coolant system.

3. Stress levels in the mechanism are not affected by anticipated system transients at power, or by the thermal movement of the coolant loops. Moments induced by the design earthquake can be accepted within the allowable primary working stress range specified by the American Society of Mechanical Engineers (ASME) Code, Section III, for Class 1 components.

4. The latch mechanism housing and rod travel housing are each a single length of forged Type 304 stainless steel. This material exhibits excellent notch toughness at all temperatures which will be encountered.

A significant margin of strength in the elastic range together with the large energy absorption capability in the plastic range gives additional assurance that gross failure of the housing will not occur. The joints between the latch mechanism housing and head adapter, and between the latch mechanism housing and rod travel housing, are threaded joints reinforced by canopy type rod welds. Administrative regulations require periodic inspections of these (and other) welds.

Nuclear Design

Even if a rupture of a RCCA drive mechanism housing is postulated, the operation of a plant utilizing chemical shim is such that the severity of an ejected RCCA is inherently limited. In general, the reactor is operated with the RCCAs inserted only far enough to permit load follow. Reactivity changes caused by core depletion and xenon transients are compensated by boron changes. Further, the location and grouping of control RCCA banks are selected during the nuclear design to lessen the severity of a RCCA ejection accident. Therefore, should a RCCA be ejected from its normal position during full power operation, only a minor reactivity excursion, at worst, could be expected to occur.

However, it may be occasionally desirable to operate with larger than normal insertions. For this reason, a rod insertion limit is defined as a function of power level. Operation with the RCCAs above this limit guarantees adequate shutdown capability and acceptable power distribution. The position of all RCCAs is continuously indicated in the Control Room. An alarm will occur if a bank of RCCAs approaches its insertion limit or if one RCCA deviates from its bank. Operating instructions when the LO-LO alarm is received requires the shutdown margin to be verified within limits or to take actions to restore the shutdown margin in accordance with the Technical Specifications.

Reactor Protection

The reactor protection in the event of a rod ejection accident is provided by the high neutron flux trip (high and low setting) and the high rate of neutron flux increase trip. These protection functions are described in detail in Section 7.2.

Effects on Adjacent Housings

Disregarding the remote possibility of the occurrence of a RCCA mechanism housing failure, investigations have shown that failure of a housing due to either longitudinal or circumferential cracking would not cause damage to adjacent housings. The full length control rod drive mechanism is described in Section 3.9N.4.

Effects of Rod Travel Housing Longitudinal Failures

If a longitudinal failure of the rod travel housing should occur, the region of the position indicator assembly opposite the break would be stressed by the reactor coolant pressure of 2250 psia. The most probable leakage path would be provided by the radial deformation of the position indicator coil assembly, resulting in the growth of axial flow passages between the rod travel housing and the hollow tube along which the coil assemblies are mounted.

If failure of the position indicator coil assembly should occur, the resulting free radial jet from the failed housing could cause it to bend and contact adjacent rod housings. If the adjacent housings were on the periphery, they might bend outward from their bases. The housing material is quite ductile; plastic hinging without cracking would be expected. Housings adjacent to a failed housing, in locations other than the periphery, would not be bent because of the rigidity of multiple adjacent housings.

Effect of Rod Travel Housing Circumferential Failures

If circumferential failure of a rod travel housing should occur, the broken-off section of the housing would be ejected vertically because the driving force is vertical and the position indicator coil assembly and the drive shaft would tend to guide the broken-off piece upwards during its travel. Travel is limited by the missile shield, thereby limiting the projectile acceleration. When the projectile reached the missile shield it would partially penetrate the shield and dissipate its kinetic energy. The water jet from the break would continue to push the broken-off piece against the missile shield.

If the broken-off piece of the rod travel housing were short enough to clear the break when fully ejected, it would rebound after impact with the missile shield. The top end plates of the position indicator coil assemblies would prevent the broken piece from directly hitting the rod travel housing of a second drive mechanism. Even if a direct hit by the rebounding piece were to occur, the low kinetic energy of the rebounding projectile would not be expected to cause significant damage.

Possible Consequences

From the above discussion, the probability of damage to an adjacent housing must be considered remote. However, even if damage is postulated, it would not be expected to lead to a more severe transient since RCCAs are inserted in the core in symmetric patterns, and control rods immediately adjacent to the worst ejected rods are not in the core when the reactor is critical. Damage to an adjacent housing could, at worst, cause that RCCA not to fall on receiving a trip signal; however this is already taken into account in the analysis by assuming a stuck rod adjacent to the ejected rod.

Summary

The considerations given above lead to the conclusion that failure of a control rod housing, due either to longitudinal or circumferential cracking, would not cause damage to adjacent housings that would increase the severity of the initial accident.

15.4.8.1.2 Limiting Criteria

This event is classified as an ANS Condition IV incident (i.e., a limiting fault). See Section 15.0.1 for a discussion of ANS classifications. Due to the extremely low probability of a RCCA ejection accident, some fuel damage could be considered an acceptable consequence.

In view of the above experimental results, criteria are applied to ensure that there is little or no possibility of fuel dispersal in the coolant, gross lattice distortion, or severe shock waves. These criteria are:

1. Average fuel pellet enthalpy at the hot spot must remain less than 200 cal/gm for irradiated fuel. This bounds non-irradiated fuel, which has a slightly higher enthalpy limit.

2. Peak reactor coolant pressure less than that which could cause stresses to exceed the faulted condition stress limits.

3. Fuel melting will be limited to less than 10 percent of the fuel pellet volume at the hot spot even if the average fuel pellet enthalpy is below the limits of criterion 1 above.

15.4.8.2 Analysis of Effects and Consequences

Method of Analysis

The calculation of the RCCA ejection transient is performed in two stages, first an average core channel calculation and then a hot region calculation. The average core calculation is performed to determine the average power generation with time including the various total core feedback effects, i.e., Doppler reactivity and moderator reactivity. Enthalpy and temperature transients in the hot spot are then determined by multiplying the average core energy generation by the hot channel factor and performing a fuel rod transient heat transfer calculation. The power distribution calculated without feedback is pessimistically assumed to persist throughout the transient.

A detailed discussion of the method of analysis can be found in Reference 2.

Average Core Analysis

The spatial kinetics computer code, TWINKLE (Reference 4), is used for the average core transient analysis. This code solves the two group neutron diffusion theory kinetic equation in one, two or three spatial dimensions (rectangular coordinates) for six delayed neutron groups and up to 2,000 spatial points. The computer code includes a detailed multi-region, transient fuel-clad-coolant heat transfer model for calculations of point-wise Doppler and moderator feedback effects. In this analysis, the code is used as a one dimensional axial kinetics code since it allows a more realistic representation of the spatial effects of axial moderator feedback and RCCA movement. However, since the radial dimension is missing, it is still necessary to employ very conservative methods (described below) of calculating the ejected rod worth and hot channel factor.

Hot Spot Analysis

In the hot spot analysis, the initial heat flux is equal to the nominal times the design hot channel factor. During the transient, the heat flux hot channel factor is linearly increased to the transient value in 0.1 second, the time for full ejection of the rod. Therefore, the assumption is made that the hot spot before and after ejection are coincident. This is very conservative since the peak after ejection will occur in or adjacent to the assembly with the ejected rod, and prior to ejection the power in this region will necessarily be depressed.

The hot spot analysis is performed using the detailed fuel and clad transient heat transfer computer code, FACTRAN (Reference 5). This computer code calculates the transient temperature distribution in a cross section of a metal clad UO2 fuel rod, and the heat flux at the surface of the rod, using as input the nuclear power versus time from the TWINKLE code (as discussed above) and the local coolant conditions. The zirconium-water reaction is explicitly represented, and all material properties are represented as functions of temperature. A parabolic radial power distribution is used within the fuel rod.

FACTRAN uses the Dittus-Boelter or Jens-Lottes correlation to determine the film heat transfer before DNB, and the Bishop-Sandburg-Tong correlation (Reference 3) to determine the film

boiling coefficient after DNB. The Bishop-Sandburg-Tong correlation is conservatively used assuming zero bulk fluid quality. The DNBR is not calculated; instead the code is forced into DNB by specifying a conservative DNB heat flux. The gap heat transfer coefficient is calculated by the code; however, it adjusted in order to force the full power steady state temperature distribution to agree with the fuel heat transfer design codes.

System Overpressure Analysis

Because safety limits for fuel damage specified earlier are not exceeded, there is little likelihood of fuel dispersal into the coolant. The pressure surge may therefore be calculated on the basis of conventional heat transfer from the fuel and prompt heat generation in the coolant. Due to the very conservative method of analysis, the peak surge rate is high enough to cause the reactor coolant pressure to exceed the pressurizer safety valve actuation pressure; however, this condition exists only for a few seconds. The pressurizer water volume does not change significantly. Therefore, these transients are not sensitive to the initial pressurizer water volume.

The pressure surge was calculated generically (Reference 2) by first performing the fuel heat transfer calculation to determine the average and hot spot heat flux versus time. Using this heat flux data, a calculation was performed to determine the volume surge. Finally, the volume surge is simulated in a plant transient computer code. This code calculates the pressure transient taking into account fluid transport in the RCS and heat transfer to the steam generators. No credit is taken for the possible pressure reduction caused by the assumed failure of the control rod pressure housing.

15.4.8.2.1 Calculation of Basic Parameters

Input parameters for the analysis are conservatively selected on the basis of values calculated for this type of core. A summary of parameters used in the analysis is given in Table 15.4-3. The more important parameters are discussed below.

Ejected Rod Worths and Hot Channel Factors

The values for ejected rod worths and hot channel factors are calculated using either three dimensional static methods or by a synthesis method employing one dimensional and two dimensional calculations. Standard nuclear design codes are used in the analysis. No credit is taken for the flux flattening effects of reactivity feedback. The calculation is performed for the maximum allowed bank insertion at a given power level, as determined by the rod insertion limits. Adverse xenon distributions are considered in the calculation.

Appropriate margins are added to the ejected rod worth and hot channel factors to account for any calculational uncertainties, including an allowance for nuclear power peaking due to densification.

Reactivity Feedback Weighting Factors

The largest temperature rises, and hence the largest reactivity feedbacks occur in channels where the power is higher than average. Since the weight of a region is dependent on flux, these regions have high weights. This means that the reactivity feedback is larger than that indicated by a simple channel analysis. Physics calculations have been carried out for temperature

changes with a flat temperature distribution, and with a large number of axial and radial temperature distributions.

Reactivity changes were compared and effective weighting factors determined. These weighting factors take the form of multipliers which, when applied to single channel feedbacks, correct them to effective whole core feedbacks for the appropriate flux shape. No weighting is applied to the moderator feedback.

A conservative radial weighing factor is applied to the transient fuel temperature to obtain an effective fuel temperature as a function of time accounting for the missing spatial dimension. These weighting factors have also been shown to be conservative compared to three dimensional analysis (Reference 2).

Moderator and Doppler Coefficient

The critical boron concentrations and BOL and EOL were adjusted in the nuclear code in order to obtain moderator density coefficient curves that were conservative when compared to the actual design conditions for the plant. As discussed above, no weighting factor is applied to these results. The applicable MTCs modeled were +5 pcm/^{o}F at zero-power nominal Tavg and 0 pcm/^{o}F at full-power Tavg for the BOL cases. For the EOL cases, the applicable zero-power MTC was -16.817 pcm/^{o}F and the full-power MTC was -22.920 pcm/^{o}F.

The Doppler reactivity defect is determined using a one dimensional steady state computer code with a Doppler weighting factor of 1.0. The Doppler weighting factor will increase under accident conditions, as discussed above.

Delayed Neutron Fraction

Calculations of the effective delayed neutron fraction (β_{eff}) typically yield values no less than 0.75 percent at beginning-of-life and 0.50 percent at end-of-life. The accident is sensitive to β_{eff} if the ejected rod worth is equal to or greater than β_{eff} as in zero power transients. In order to allow for future cycles, pessimistic estimates of 0.55 percent at beginning of cycle and 0.44 percent at end of cycle were used in the analysis.

Trip Reactivity Insertion

The trip reactivity insertion includes the effect of one stuck RCCA. These values are reduced by the ejected rod reactivity. The shutdown reactivity was simulated by dropping a rod of the required worth into the core. The start of rod motion is assumed to occur 0.5 seconds after the high neutron flux trip point was reached.

The minimum design shutdown margin available for this plant at hot zero power (HZP) may be reached only at end-of-life in the equilibrium cycle. This value includes an allowance for the worst stuck rod, and adverse xenon distribution, conservative Doppler and moderator defects, and an allowance for calculational uncertainties. Physics calculations for this plant have shown that the effect of two stuck RCCAs (one of which is the worst ejected rod) is to reduce the shutdown margin by about an additional 1 percent Δk. Therefore, following a reactor trip

resulting from an RCCA ejection accident, the reactor will be subcritical when the core returns to HZP.

Depressurization calculations have been performed for a typical four-loop plant assuming the maximum possible size break (2.75 inch diameter) located in the reactor pressure vessel head. The results show a rapid pressure drop and a decrease in system water mass due to the break. The Safety Injection System is actuated on low pressurizer pressure within 1 minute after the break. The RCS pressure continues to drop and reaches saturation (1200 psi) in about 2 to 3 minutes. Due to the large thermal inertia of the primary and secondary system, there has been no significant decrease in the RCS temperature below no-load by this time, and the depressurization itself has caused an increase in shutdown margin by about 0.2 percent Δk due to the pressure coefficient. The cooldown transient could not absorb the available shutdown margin until more than 10 minutes after the break. The addition of borated safety injection flow starting 1 minute after the break is sufficient to ensure that the core remains subcritical during the cooldown.

Reactor Protection

As discussed in Section 15.4.8.1.1, reactor protection for a rod ejection is provided by high neutron flux trip (high and low setting) and high rate of neutron flux increase trip. These protection functions are part of the Reactor Trip System. No single failure of the Reactor Trip System will negate the protection functions required for the rod ejection accident, or adversely affect the consequences of the accident.

15.4.8.3 Environmental Consequences

The ejection of an RCCA constitutes a break in the RCS, located in the reactor pressure vessel head. The effects and consequences of loss of coolant accidents are discussed in Section 15.6.5. Following the RCCA ejection, the operator would follow the same emergency instructions as for any other loss of coolant accident to recover from the event.

Fission Product Release

It is assumed that fission products are released from the gaps of all rods entering DNB. In all cases considered, less than 15 percent of the rods entered DNB.

Pressure Surge

A detailed calculation of the pressure surge for an ejection worth of one dollar at beginning-of-life, hot full power, indicates that the peak pressure does not exceed that which would cause stress to exceed the faulted condition stress limits [2]. Since the severity of the present analysis does not exceed the “worst case” analysis, the accident for this plant will not result in an excessive pressure rise or further damage to the RCS.

Lattice Deformations

A large temperature gradient will exist in the region of the hot spot. Since the fuel rods are free to move in the vertical direction, differential expansion between separate rods cannot produce distortion. However, the temperature gradients across individual rods may produce a differential expansion tending to bow the midpoint of the rods toward the hotter side of the rod. Calculations

have indicated that this bowing would result in a negative reactivity effect at the hot spot since the core is under-moderated, and bowing will tend to increase the under-moderation at the hot spot. In practice, no significant bowing is anticipated, since the structural rigidity of the core is more than sufficient to withstand the forces produced. Boiling in the hot spot region would produce a net flow away from that region. However, the heat from the fuel is released to the water relatively slowly, and it is considered inconceivable that cross flow will be sufficient to produce significant lattice forces. Even if massive and rapid boiling, sufficient to distort the lattice, is hypothetically postulated, the large void fraction in the hot spot region would produce a reduction in the total core moderator to fuel ratio, and a large reduction in this ratio at the hot spot. The net effect would therefore be a negative feedback. It can be concluded that no conceivable mechanism exists for a net positive feedback resulting from lattice deformation. In fact, a small negative feedback may result. The effect is conservatively ignored in the analysis.

Radiological Consequences

The doses resulting from a rod ejection accident are based on a conservative fission product release to the reactor coolant of the gap activity shown in sheet 2 of Table 15.0-7 from 15 percent of the fuel rods in the core plus the core activity shown in sheet 2 of Table 15.0-7 from the assumed 0.375 percent core melt. The method of analysis complies with the requirements of Position 3 of Regulatory Guide 1.195.

Following a postulated rod ejection accident, two alternate activity release paths are considered when determining radiological consequences of the accident. The first release path is via Containment leakage resulting from release of activity from the primary coolant to the Containment. The second path assumes that the Reactor Coolant System remains intact and that the activity enters the secondary system due to tube leakage and then is released to the environment as contaminated steam and is dumped through the relief valves, since offsite power is assumed to be lost.

Prior to the accident, it is assumed that the plant has been operating with simultaneous fuel defects and steam generator tube leakage for a sufficient period of time to establish equilibrium levels of activity in the primary and secondary coolants equal to the Table 15.1-4 values.

Following a postulated rod ejection accident, the activity released from the pellet-clad gap due to failure of a portion of the fuel rods and the melted fuel is assumed to be instantaneously released to the primary coolant. The activity released to the primary coolant is assumed to be uniformly mixed throughout the coolant instantaneously. Thus, the total activity of the primary coolant is assumed to be immediately available for release from the RCS. Of the activity released to the Containment from the coolant through the rupture in the reactor vessel head, 100 percent is assumed to be mixed instantaneously throughout the Containment. Fifty (50) percent of the iodine activity released from the melted fuel is assumed to immediately plate out on Containment surfaces. The remaining activity is available for leakage from the Containment at the design leak rate of 0.10 percent of Containment volume per day for the first 24 hours, and at a rate of 0.05 percent of Containment volume per day for the duration of the accident. The only removal processes considered in the Containment for the activity remaining after the above plate out are radioactive decay and leakage from the Containment.

The model for the activity available for release to the atmosphere from the relief valves assumes that the release consists of the activity in the secondary coolant prior to the accident plus that fraction of the activity leaking from the primary coolant through the steam generator tubes

following the accident. The leakage of primary coolant to the secondary side of the steam generator is assumed to continue at its initial rate, which is assumed to be the same rate as the leakage prior to the accident. With the assumption of coincident loss of offsite power, activity is assumed to be released to the atmosphere from a steam dump through the relief valves until the Reactor Coolant System temperature and pressure are reduced sufficiently to allow operation of the Residual Heat Removal System at 11 hours.

A summary of parameters used in the analysis is given in Table 15.4-4.

Fuel melting, limited to less than the innermost 10 percent of the fuel pellet at the hotspot, is included in the design criteria to ensure that fuel dispersion into the coolant does not occur [1]. Even though centerline melting in a small fraction of the core is not expected, a conservative upper limit of fission product release from the core as a result of a rod ejection accident can be estimated.

The Regulatory Guide 1.195 limit of fission product release from the core for this very conservative case is determined using the following assumptions:

1. 100 percent of the noble gases and iodines in the clad gaps of those fuel rods experiencing clad damage (assumed to be 15 percent of the rods in the core) are released to the reactor coolant. The affected fuel rods are assumed to be operating at 1.65 times the core average and thus have 1.65 times the average fission product inventory. The core activities are presented in Table 15.0-7.

2. 25 percent of the iodines and 100 percent of the noble gases in the fuel that melts are assumed to be released to the reactor coolant and available for release via the containment building. 50 percent of the iodines and 100 percent of the noble gases in the fuel that melts are assumed to be available for release from the secondary systems via primary-to-secondary leakage. The core activities are presented in Table 15.0-7.

3. The fraction of fuel melting is conservatively assumed to be 0.375 percent of the core, as determined by the following method [1]:

    a. A conservative upper limit of 50 percent of the rods experiencing clad damage also may experience centerline melting (a total of 7.5 percent of the rods in the core).

    b. Of rods experiencing centerline melting, only a conservative maximum of the innermost 10 percent of the volume actually melts (equivalent to 0.5 percent of the core that could experience melting).

    c. A conservative maximum of 50 percent of the axial length of the rod experiences melting due to the power distribution (0.5 of the 0.75 percent of the core = 0.375 percent of the core).

4. Prior to the accident, the plant is assumed to be operating at full power with coincident fuel defects and steam generator tube leakage. The steam generator tube leak rate is assumed to be 1.0 gpm. The initial primary coolant and secondary liquid activity concentrations are given in Table 15.1-4.

5. Instantaneous mixing in the Containment of all activity released from the coolant is assumed.

6. Fifty (50) percent of the iodine activity released from the melted fuel to the Containment atmosphere immediately plates out on Containment surfaces.

7. The Residual Heat Removal System is brought into service at 11 hours, thus terminating steam releases.

8. Loss of offsite power is assumed and, 1,750,000 lb of steam are discharged from the secondary system through the relief valves, during the 11 hours following the accident. Steam dump is terminated after 11 hours.

9. All releases to the atmosphere are assumed to be at ground level.

10. Other assumptions are detailed in Table 15.4-4.

Based on the foregoing model, the thyroid and whole body doses from the containment leakage for this accident are conservatively calculated to be 20.0 rem and 0.082 rem, respectively, for the exclusion area boundary (EAB), and 28.0 rem and 0.043 rem, respectively, for the low population zone (LPZ). The thyroid and whole body doses from the secondary side releases for this accident are conservatively calculated to be 4.2 rem and 0.42, respectively, for the EAB, and 7.9 rem and 0.15 rem, respectively, for the LPZ. For each scenario, the doses at these distances are below the dose values of 6.3 rem whole body and 75 rem thyroid set forth in RG 1.195.

15.4.8.4 Conclusions

For Units 1 and 2, even on a pessimistic basis, the analyses indicate that the described fuel and clad limits are not exceeded. It is concluded that there is no danger of sudden fuel dispersal into the coolant. Since the peak pressure does not exceed that which would cause stresses to exceed the faulted condition stress limits, it is concluded that there is no danger of further consequential damage to the RCS. The analyses have demonstrated that upper limit in fission product release as a result of a number of fuel rods entering DNB amounts to 15 percent.

REFERENCES

1. WCAP-16676-NP, R. D. Ankney and J. L. Grover, “Analysis Update for the Inadvertent Loading Event,” March 2009.

2. Risher, D. H., Jr., “An Evaluation of the Rod Ejection Accident in Westinghouse Pressurized Water Reactors Using Spatial Kinetics Methods,” WCAP-7588, Revision 1-A, January 1975.

3. Bishop, A. A., Sandburg, R. O. and Tong, L. S., “Forced Convection Heat Transfer at High Pressure After the Critical Heat Flux,” ASME 65-HT-31, August 1965.

4. Barry, R. F., Jr. and Risher, D. Hl, “TWINKLE - A Multi-dimensional Neutron Kinetics Computer Code,“ WCAP-7979-P-A, January 1975.

5. Hargrove, H. G., “FACTRAN - A FORTRAN-IV Code for Thermal Transients in a UO_2 Fuel Rod,” WCAP-7908-A, December 1989.

6. Sung, Y. X. et al., “VIPRE-01 Modeling and Qualification for Pressurized Water Reactor Non-LOCA Thermal-Hydraulic Safety Analysis,” WCAP-14565-P-A, October 1999.

7. Burnett, T. W. T., et al., “LOFTRAN Code Description,” WCAP-7907-P-A, April 1984.

8. Haessler, R. L., et al., “Methodology for the Analysis of the Dropped Rod Event,” WCAP-11394-P-A, January 1990.

9. Liu, Y. S., et al., “ANC: A Westinghouse Advanced Nodal Computer Code,” WCAP-10965-P-A, September 1986.

10. Huegal, D. S., et al., “RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses,” WCAP-14882-P-A, April 1999.

11. Friedland, A. J. and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

12. Zhang, B., et. al., “Qualification of the New Pin Power Recovery Methodology,” WCAP-10965-P-A, Addendum 2-A, September, 2010.

TABLE 15.4-1
HAS BEEN DELETED

TABLE 15.4-2
HAS BEEN DELETED

TABLE 15.4-3

PARAMETERS USED IN THE ANALYSIS OF THE ROD CLUSTER CONTROL ASSEMBLY EJECTION ACCIDENT

	Beginning of Core Life (HFP)	Beginning of Core Life (HZP)	End of Core Life (HFP)	End of Core Life (HZP)
Power Level (percent)	100.6	0	100.6	0
Ejected Rod Worth (percent Δk)	0.24	0.75	0.25	0.84
Delayed Neutron Fraction (percent)	0.55	0.55	0.44	0.44
Feedback Reactivity Weighting	1.2927	2.0079	1.3549	3.7649
Trip Reactivity (percent Δk)	4.0	2.0	4.0	2.0
F_Q Before Rod Ejection	2.50	-	2.50	-
F_Q After Rod Ejection	5.5	11.0	6.0	26.0
Doppler Defect (pcm)	-1000	-1000	-950	-950
Number of Operational Reactor Coolant Pumps	4	2	4	2

TABLE 15.4-4
PARAMETERS FOR POSTULATED ROD EJECTION ACCIDENT ANALYSIS

(Sheet 1 of 3)

1.	Data and assumptions used to estimate radioactive source from postulated accidents		
	a.	Power level (MWt)	3684
	b.	Primary Side equilibrium activity concentration	1.0 uCi/gm Dose Equivalent I-131
	c.	Secondary Side equilibrium activity concentration	0.1 uCi/gm Dose Equivalent I-131
	d.	Steam generator tube leak rate prior to and during steam dump (gpm)	1
	e.	Failed fuel	15 percent of fuel rods in core
	f.	Radial peaking factor	1.65
	g.	Activity released to reactor coolant from failed fuel and available for release (i.e., gap fraction)	
		Noble gases	10 percent of core inventory
		Iodines	10 percent of core inventory
	h.	Melted fuel	0.375 percent of core
	i.	Activity released to reactor coolant from melted fuel	
		Noble gases inventory	0.375 percent of core
		Iodines	0.1875 percent of core inventory
	j.	Iodine fractions (organic, elemental, and particulate)	Regulatory Guide 1.195
2.	Data and assumptions used to estimate activity released		
	a.	Plate out of iodine activity	50 percent released to Containment (applied only to fuel melt releases)

TABLE 15.4-4
PARAMETERS FOR POSTULATED ROD EJECTION ACCIDENT ANALYSIS

(Sheet 2 of 3)

b.	Containment leak rate		0.1 percent of containment volume per day ($0 \leq t \leq 24$ hr) 0.05 percent of containment volume per day ($24 < t < 720$ hr)
c.	Iodine partition factor in steam generators prior to and during accident		0.01
d.	Offsite power		lost
e.	Steam dump from relief valves (lbm)		450,000 (0 - 2 Hours) 1,300,000 (2 - 11 Hours)
f.	Duration of dump from relief valves (hour)		11
g.	Steam dump to condenser (lbm)		0.0
3. Dispersion data			
a.	EAB and LPZ distances		2080m and 4 miles
b.	X/Q		
		@EAB	1.6×10^{-4} sec/m^3 (0-2 hr)
		@LPZ	2.4×10^{-5} sec/m^3 (0-8 hr)
		@LPZ	1.6E-05 sec/m^3 (8-24 hr)

TABLE 15.4-4
PARAMETERS FOR POSTULATED ROD EJECTION ACCIDENT ANALYSIS

(Sheet 3 of 3)

4.	Dose data		
	a.	Method of dose calculation	See Appendix 15B
	b.	Dose conversion assumptions	See Appendix 15B
	c.	Doses	@EAB (0-2 hr)
			thyroid dose
		Containment leakage	= 20.0 Rem
		Primary to secondary leakage	= 4.2 rem
			Whole body dose (gamma dose)
		Containment leakage	= 0.082 rem
		Primary to secondary leakage	= 0.42 rem
			@LPZ (0 - 11 hours)
			thyroid dose
		Containment leakage	= 28.0 rem
		Primary to secondary leakage	= 7.9 rem
			Whole body dose (gamma dose)
		Containment leakage	= 0.043 rem
		Primary to secondary leakage	= 0.015 rem

TABLE 15.4.7.3-1
FLUX MAP REVIEW CRITERIA FOR A 4-LOOP CORE

Number of Available Detector Locations	Measured vs. Predicted Detector Reaction Rate Comparison[a]	Symmetric Thimble Reaction Rate Comparison[b]
55 to 58	10%	7%
49 to 54	8%	5%
44 to 48	6%	5%

a) The review criterion is the table value (%) or an absolute normalized reaction rate difference equal to the table value divided by 100% (e.g., 10%/100%=0.1), whichever is greater.

b) Applicable to symmetric thimbles with normalized reaction rates above 0.7. The review criterion is relative to the expected reaction rate difference.

15.5 INCREASE IN REACTOR COOLANT INVENTORY

Discussion and analysis of the following events are presented in this section:

1. Inadvertent operation of the Emergency Core Cooling System during power operation

2. Chemical and Volume Control System malfunction that increases reactor coolant inventory

3. A number of BWR transients. (Not applicable to the Comanche Peak Nuclear Power Plant (CPNPP)).

These events, considered to be American Nuclear Society (ANS) Condition II event (i.e., a fault of moderate frequency), cause an increase in reactor coolant inventory. Section 15.0.1 contains a discussion of ANS classifications.

15.5.1 INADVERTENT OPERATION OF THE EMERGENCY CORE COOLING SYSTEM DURING POWER OPERATION

15.5.1.1 Identification of Causes and Accident Description

Spurious Emergency Core Cooling System (ECCS) operation at power could be caused by operator error or a false electrical actuation signal. A spurious signal may originate from any of the safety injection actuation channels as described in Section 7.3.

Following the actuation signal, the suction of the coolant charging pumps is diverted from the volume control tank to the Refueling Water Storage Tank. The charging pumps then force highly concentrated boric acid solution from the Refueling Water Storage Tank, through the header and injection line and into the cold leg of each loop. The safety injection pumps also start automatically but provide no flow when the Reactor Coolant System (RCS) is at normal pressure. The passive injection system and the low head system also provide no flow at normal RCS pressure.

A Safety Injection Actuation System (SIAS) signal normally results in a reactor trip followed by a turbine trip. However, it cannot be assumed that any single fault that actuates the SIAS will also produce a reactor trip. If a reactor trip is generated by the spurious SIAS signal, the operator would consult the appropriate emergency operating procedure to determine if the spurious signal was transient or steady state in nature.

The emergency operating procedures clearly define the diagnostics for a spurious Safety Injection signal. These emergency procedures provide direction to the operator as to the specific actions to be taken based on measured process variables (e.g., pressure, temperature, etc.) so it is unlikely that any judgment errors will be made. Thus, by following the procedures, the operator would be able to diagnose the event and take the appropriate corrective action.

If the operator incorrectly diagnoses an event as a spurious SI signal, this could result in the termination of safety injection for an accident where safety injection is needed (e.g., LOCA). The consequences of this action and the system failures are given in WCAP-9691, “NUREG-0578 2.1.9.c, Transient and Accident Analysis”. If the operator incorrectly diagnoses a spurious safety injection event as some other accident, the emergency operating procedures for

that accident would be followed until it became apparent (when monitoring the process variables called for in the emergency procedure) that the wrong accident had been identified. The operator would then revert to the correct procedure and, based on the results of process variable monitoring, he would take the appropriate actions to bring the plant to a safe shutdown condition.

Following a spurious safety injection signal, the operator cannot reset SI until all the required conditions given in the emergency operating procedures are met. Also, an interlock exists which does not allow manual resetting of SI for at least 60 seconds.

If a loss of offsite power occurs after resetting safety injection, manual action will be required to load the engineered safety features equipment onto the diesel powered emergency busses. The procedures to be followed by the operator in performing this action have been incorporated in the CPNPP Emergency Operating Procedures for the manual reset of safety injection.

If the ECCS actuation instrumentation must be repaired, future plant operation would be in accordance with the Technical Specifications.

If the Reactor Protection System does not produce an immediate trip as a result of the spurious SIAS signal, the reactor experiences a negative reactivity excursion due to the injected boron causing a decrease in reactor power. The power mismatch results in a drop in T_{avg} and consequent coolant shrinkage causing pressurizer pressure and water level to drop. Load will decrease due to the effect of reduced steam pressure on load after the turbine throttle valve is fully open. If automatic rod control is used, these effects will be lessened until the rods have moved out of the core. The transient is eventually terminated by the Reactor Protection System low pressure trip or by manual trip.

The time to trip is affected by initial operating conditions including core burnup history which affects initial boron concentration, rate of change of boron concentration, and Doppler and moderator coefficients.

Recovery from this second case is made in the same manner as described for the case where the SIAS signal results directly in a reactor trip. The only differences are the lower T_{avg} and pressurizer pressure associated with the power mismatch during the transient. The time at which reactor trip occurs is of little concern for this transient. At lower loads coolant contraction will be slower resulting in a longer time to trip.

This event is non-limiting with respect to Departure from Nucleate Boiling (DNB) since the conditions resulting from injecting borated water into the RCS are beneficial with respect to DNB. Depending on the control systems in operation, core power and RCS temperatures either remain near the initial nominal conditions or decrease during this event. The RCS flow remains constant throughout the event. A decrease in RCS pressure is the only condition that may occur which would adversely affect DNB. However, for the decrease in RCS pressure that may occur, the effects are more than offset by beneficial reduction in power and temperature. The net effect is a DNB Ratio (DNBR) that remains near the initial DNBR or increases throughout the event.

The major concern resulting from an Inadvertent ECCS Actuation at Power event is the potential to violate the ANS Condition II acceptance criterion where an incident of moderate frequency should not generate a more serious plant condition without other faults occurring independently.

The pressurizer water volume increase for this event is a result of continuous safety injection flow. Operator action is required to terminate safety injection flow prior to filling the pressurizer, thus demonstrating that this ANS Condition II event does not propagate to a more serious plant condition. Therefore, this event is analyzed to demonstrate that sufficient time is available for the operator to take appropriate actions to preclude a pressurizer water-solid condition.

This event is classified as a Condition II incident (a fault of moderate frequency) as defined in Section 15.0.1.

15.5.1.2 Analysis of Effects and Consequences

Method of Analysis

The Inadvertent Actuation of the ECCS System at Power event is analyzed by employing the detailed digital computer code RETRAN (Reference 1). The code simulates the neutron kinetics, and models the reactor coolant system, pressurizer, pressurizer relief and safety valves, pressurizer spray, steam generator, steam generator safety valves and the safety injection system. The code computes pertinent plant variables including temperature, pressures, and power level.

The assumptions are as follows:

1. Initial operating conditions

   The initial reactor power is assumed at its maximum value consistent with the steady state full power plus 0.6 percent allowance for calibration and instrument errors. The initial pressurizer water level is at the full power value of 60% span plus 5% uncertainty allowance and pressurizer pressure is 30 psi lower than nominal. The initial RCS temperature is assumed at the full power value minus 6^{o}F uncertainty allowance.

2. Moderator and Doppler coefficients of reactivity

   This event is analyzed with the maximum reactivity feedback such that the most negative moderator temperature coefficient and most negative Doppler power coefficient are assumed.

3. Reactor Control System

   The Reactor Control System is assumed to be in the manual mode of operation.

4. Pressurizer Pressure and Level Control System

   Pressurizer sprays are assumed to be operable to increase the rate of pressurizer filling. The pressurizer sprays act to reduce the RCS pressure, thus increasing ECCS injection. The pressurizer heaters are assumed to be de-energized (the pressurizer heaters at CPNPP are automatically de-energized upon receipt of a safety injection signal by safety-related SSPS relays).

5. Boron injection

   At time zero two charging pumps inject borated water into the cold leg of each loop. The RCS pressure does not decrease to the point where the Safety Injection pumps could deliver flow to the RCS.

6. Reactor trip

   The reactor and turbine are assumed to trip upon receipt of the SIAS signal. Assuming a reactor trip on the SIAS signal and a turbine trip derived from the reactor trip minimizes the heat removal capability of the RCS, thereby maximizing the RCS inventory increase through SI flow and thermal expansion of the RCS fluid.

7. Pressurizer PORVs

   No credit is taken for any pressurizer PORV operation.

8. Operator Actions

   The first operator action assumed is the initiation of the opening of at least three of the four steam generator atmospheric relief valves (ARVs) within 8 minutes after the event initiation to control Tavg to the no-load temperature of 557°F.

   The operators are then assumed to secure ECCS within 14 minutes after the event initiation. Operator action is also required to take positive control of the Pressurizer Pressure and Level Control System.

Plant systems and equipment which are available to mitigate the effects of the accident are discussed in Section 15.0.8 and listed in Table 15.0-6. No single active failure in any of these systems or equipment will adversely affect the consequences of the accident.

15.5.1.3 Conclusions

The filling of the pressurizer is the acceptance criterion of concern for this analysis. This criterion is more conservative than preventing water relief through the Pressurizer Safety Valves (PSVs), as it is possible to fill the pressurizer at a pressure below the PSV set pressure.

Results of the analysis show for Units 1 and 2 that the pressurizer does not reach a water-solid condition as a consequence of the inadvertent operation of ECCS during power operation, provided that the operator initiates the opening of the ARVs within 8 minutes to reduce Tavg close to 557°F, and the SI flow is turned off within 14 minutes after the event initiation.

15.5.2 CHEMICAL AND VOLUME CONTROL SYSTEM MALFUNCTION THAT INCREASES REACTOR COOLANT INVENTORY

An increase in reactor coolant inventory which results from the addition of cold, unborated water to the RCS is analyzed in Section 15.4.6, Chemical and Volume Control System malfunction that results in a decrease in boron concentration in the reactor coolant. An increase in reactor coolant inventory which results from the injection of highly borated water into the RCS is

analyzed in Section 15.5.1, inadvertent operation of the Emergency Core Cooling System during power operation.

15.5.3 A NUMBER OF BWR TRANSIENTS

This section is not applicable to the CPNPP.

REFERENCES

1. Huegal, D. S., et al., “RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses,” WCAP-14882-P-A, April 1999.

TABLE 15.5-1
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15.6 DECREASE IN REACTOR COOLANT INVENTORY

Events which result in a decrease in reactor coolant inventory as discussed in this section are as follows:

1. Inadvertent opening of a pressurizer safety or relief valve.

2. Break in instrument line or other lines from reactor coolant pressure boundary that penetrate containment.

3. Steam generator tube failure.

4. Spectrum of boiling water reactor (BWR) steam system piping failures outside of Containment. (Not applicable to the CPNPP.)

5. Loss of coolant accident (LOCA) resulting from a spectrum of postulated piping breaks within the reactor coolant pressure boundary.

6. A number of BWR transients. (Not applicable to the CPNPP.)

15.6.1 INADVERTENT OPENING OF A PRESSURIZER SAFETY OR RELIEF VALVE

15.6.1.1 Identification of Causes and Accident Description

An accidental depressurization of the Reactor Coolant System (RCS) could occur as a result of an inadvertent opening of a pressurizer relief or safety valve. Since a safety valve is sized to relieve approximately twice the steam flow rate of a relief valve (i.e., 420,000 lb/hr vs. 210,000 lb/hr), and will therefore allow a much more rapid depressurization upon opening, the most severe core conditions resulting from an accidental depressurization of the RCS are associated with an inadvertent opening of a pressurizer safety valve. The flow of released coolant due to the inadvertent opening of a pressurizer safety valve is equivalent to approximately a 2.0 in. break in the RCS hot leg. It is shown [13, 14] that a hot leg break transient is much less severe than an equivalent sized break in the cold leg. Thus, the transient due to the inadvertent opening of a pressurizer safety valve would be much less severe than the small break cold leg transients discussed in Section 15.6.5. Initially the event results in a rapidly decreasing RCS pressure until this pressure reaches a value corresponding to the hot leg saturation pressure. At this time, the pressure decrease is slowed considerably. The pressure continues to decrease throughout the transient. The effect of the pressure decrease would be to decrease or increase the neutron flux depending on the moderator density feedback, but the Reactor Control System (if in the automatic mode) functions to maintain the power and average coolant temperature essentially constant until the reactor trip occurs. Initially, the pressurizer level increases due to expansion caused by the depressurization, and then decreases following the reactor trip.

The reactor may be tripped by the following Reactor Protection System signals:

1. Overtemperature N-16.

2. Pressurizer low pressure.

An inadvertent opening of a pressurizer safety valve is classified as an American Nuclear Society (ANS) Condition II event, a fault of moderate frequency. See Section 15.0.1 for a discussion of Condition II events.

15.6.1.2 Analysis of Effects and Consequences

Method of Analysis

This transient is analyzed using the RETRAN computer code (Reference 1).

Plant characteristics and initial conditions are discussed in Section 15.0.3. In order to give conservative results in calculating the departure from nucleate boiling ratio (DNBR) during the transient, the following assumptions are made:

1. The initial core power is assumed to be at the nominal value consistent with steady-state, full-power operation. The initial pressurizer pressure and vessel average temperature are assumed to be at the respective nominal values, and the reactor coolant flow is at the minimum measured flow value. Uncertainties in the initial conditions are accounted for in the DNBR limit value as described in Reference 2.

2. A zero moderator temperature coefficient of reactivity is assumed in order to provide a conservatively low amount of negative reactivity feedback due to changes in the moderator density. The spatial effect of void due to local or subcooled boiling is not considered in the analysis with respect to reactivity feedback or core power shape. These voids would tend to flatten the core power distribution.

3. The least negative Doppler coefficient of reactivity is assumed such that the resultant amount of negative feedback is conservatively low in order to maximize any power increase due to moderator reactivity feedback.

Plant systems and equipment which are necessary to mitigate the effects of RCS depressurization caused by an inadvertent safety valve opening are discussed in Section 15.0.8 and listed in Table 15.0-6.

Normal reactor control systems are not required to function. The Rod Control System is assumed to be in the manual mode in order to maximize the nuclear power increase and to prevent rod insertion prior to reactor trip. The Reactor Protection System functions to trip the reactor on the appropriate signal. No single active failure will prevent the Reactor Protection System from functioning properly.

15.6.1.3 Conclusions

For Units 1 and 2, the results of the analysis show that the pressurizer low pressure and the Overtemperature N-16 Reactor Protection System signals provide adequate protection against the RCS depressurization event.

15.6.2 BREAK IN INSTRUMENT LINE OR OTHER LINES FROM REACTOR COOLANT PRESSURE BOUNDARY THAT PENETRATE CONTAINMENT

There are no instrument lines connected to the RCS that penetrate the Containment. However, the 3 inch CVCS letdown line and grab sample lines from the hot legs of reactor coolant loops 1 and 4, and from the steam and liquid space of the pressurizer, do penetrate the Containment. The grab sample lines are provided with normally closed isolation valves on both sides of the Containment wall and are designed in accordance with the requirements of General Design Criterion 55.

The most severe pipe rupture with regard to radioactivity release during normal plant operation occurs in the Chemical and Volume Control System (CVCS). This Condition III event would be a complete severance of the 3 inch letdown line just outside containment, between the outboard letdown isolation valve and letdown heat exchanger (see Figure 9.3-10), at rated power condition with a maximum letdown rate of 140 gpm. The occurrence of a complete severance of the letdown line would result in a loss-of-reactor coolant through two letdown orifices at the rate of about 190 gpm, which is within the makeup capacity of any two of the three charging pumps (see Table 9.3-7, sheet 1). Two non-safety related pressure switches located in the CVCS letdown line provide a means for detecting a rupture of the line. A low pressure signal, initiated by the rupture, will activate an alarm on the main control board, alerting the operator of the rupture, causing him to manually isolate the system. No credit is taken for Engineered Safety Features System actuation. Frequent operation of the automatic Reactor Makeup System and increased charging flow will also provide the operator indication of the loss-of-reactor coolant. No transient analysis is performed for this accident.

The time required for the operator to identify the accident and initiate the closure of the letdown isolation valve is expected to be within 30 minutes after accident initiation including 10 seconds for the letdown isolation valve closure time. Reactor coolant is assumed to be released until the isolation valve is fully closed. It is conservatively assumed that 20.1 percent of the leaking coolant flashes to steam. All of the iodine in this steam is assumed to become airborne and is available for release to the atmosphere. All noble gases contained in the leaking primary coolant are available for release to the atmosphere. A ground level release was postulated with an atmospheric dilution factor of 1.6×10^{-4} sec/m^3 at the minimum EAB distance. The equilibrium concentration of radioactive nuclides in the reactor coolant is given in Table 15.1-4. Effects of a concurrent iodine spike are included in the analysis.

Based on the foregoing model, the doses at the EAB are conservatively calculated to be 6.2 rem to the thyroid and 0.05 rem to the whole body. The LPZ doses are conservatively calculated to be 1.0 rem to the thyroid and 0.007 rem to the whole body. As expected, the radiological consequences resulting from the failure of the 3 inch CVCS letdown line do not exceed a small fraction of the dose values set forth in 10CFR100.

In Mode 3, below 500 degrees F, with three letdown orifices open (in lieu of two), the doses at the EAB are conservatively calculated to be 9.4 rem to the thyroid and 0.07 rem to the whole body. The LPZ doses are conservatively estimated to be 1.5 rem to the thyroid and 0.01 rem to the whole body. As expected, the radiological consequences resulting from the failure of the 3 inch CVCS letdown line in Mode 3 do not exceed a small fraction of the dose values set forth in 10CFR100.

15.6.3 STEAM GENERATOR TUBE FAILURE

15.6.3.1 Identification of Causes and Accident Description

The accident examined is the complete severance of a single steam generator tube. This event is considered an ANS Condition IV event, a limiting fault (see Section 15.0.1) and is analyzed to demonstrate that the resulting radiological doses are within the guideline values of 10CFR100. Timely operator response is required to terminate the primary-to-secondary break flow and to ensure that the ruptured steam generator does not fill with water and flood the main steamlines. This criterion is important because the main steamlines and relief valves are not designed for liquid flow.

The accident is assumed to take place at power with the reactor coolant contaminated with fission products corresponding to continuous operation with a limited amount of defective fuel rods. The accident leads to an increase in contamination of the secondary system due to leakage of radioactive coolant from the RCS. In the event of a coincident loss of offsite power, or failure of the condenser dump system, discharge of activity to the atmosphere takes place via the steam generator safety and/or power operated relief valves.

In view of the fact that the steam generator tube material is Inconel-600 (for Unit 2) and Inconel-690 (for Unit 1), both of which are a highly ductile material, it is considered that the assumption of a complete severance of a tube is somewhat conservative. The more probable mode of tube failure would be one or more minor leaks of undetermined origin. Activity in the Steam and Power Conversion System is subject to continual surveillance and an accumulation of minor leaks which exceeds the limits established in the Technical Specifications is not permitted during unit operation.

The reactor operators are expected to determine that a steam generator tube rupture has occurred, and to identify and isolate the ruptured steam generator on a restricted time scale in order to minimize the contamination of the secondary system and to ensure the termination of the radioactive release to the atmosphere from the ruptured steam generator. In the following discussions, the steam generator with the ruptured U-tube is referred to as the ruptured steam generator. The operator is then expected to carry out the appropriate recovery procedures on a restricted time scale in order to terminate the primary-to-secondary break flow before the water level in the ruptured steam generator rises into the main steam system piping. Sufficient indications and controls are provided to enable the operator to carry out these functions satisfactorily.

Assuming normal operation of the various plant control systems, the following sequence of events is initiated by a tube rupture:

1. Pressurizer low pressure and low level alarms are actuated and charging pump flow is increased to maintain pressurizer level. On the secondary side, there is a steam flow/ feedwater flow mismatch prior to reactor trip as the feedwater flow to the faulted steam generator is reduced due to the break flow being supplied to that unit.

2. The steam generator blowdown liquid monitor and the condenser off gas radiation monitor will alarm, indicating a sharp increase in radioactivity in the secondary system and the steam generator blowdown liquid monitor will automatically terminate steam generator blowdown. The steam generator leak rate monitor will also provide an alarm.

3. Continued loss of reactor coolant inventory may result in an Overtemperature N-16 initiated turbine runback, eventually followed by a reactor trip signal being generated by either the Overtemperature N-16 or Low Pressurizer Pressure reactor trip signals. The resultant plant cooldown following reactor trip leads to a rapid decrease in the pressurizer level, and the safety injection actuation signal, initiated on low pressurizer pressure, follows soon after the reactor trip. The safety injection actuation signal automatically causes the termination of the normal feedwater supply and the initiation of the Auxiliary Feedwater System.

4. The reactor trip automatically trips the turbine and if offsite power is available, the steam dump valves open, permitting steam release to the condenser. In the event of a coincident station blackout, the steam dump valves would automatically close to protect the condenser. The steam generator pressure would rapidly increase, resulting in steam discharge to the atmosphere through the steam generator safety and/or power operated relief valves.

Operator Responses

The immediately apparent symptoms of a tube rupture accident, such as falling pressurizer pressure and level and increased charging pump flow, are also symptoms of small steamline breaks and loss of coolant accidents. Therefore, it is important for the operator to identify the accident as a steam generator tube rupture in order to execute the correct recovery procedure. The steam generator tube rupture event can be uniquely identified by alarms from the condenser offgas, steam generator blowdown, or main steamline radiation monitors. The steam generator leak rate monitor will also provide an alarm. In addition, following reactor trip, the narrow range water level will rise more rapidly in the faulted steam generator than in the other steam generators due to the primary-to-secondary break flow.

Following identification of the event as a steam generator tube rupture, the following operator responses are required to terminate the primary-to-secondary break flow and to terminate the event.

1. The ruptured steam generator is isolated to minimize the release of radioactivity. Isolation of the ruptured steam generator is accomplished by closing the main steam isolation valve, isolating auxiliary feedwater, and increasing the setpoint pressure of the atmospheric relief valve.

2. The operators then cool the RCS at the maximum rate using the steam dump valves if offsite power is available, or the atmospheric relief valve on the intact steam generators if offsite power is lost. The cooldown is continued until the core exit temperature is less than the saturation temperature of the ruptured steam generator. This action ensures that the RCS can be rapidly depressurized to terminate the break flow.

3. Following the termination of the maximum rate cooldown, the RCS is depressurized using the pressurizer sprays, if available, or a pressurizer power operated relief valve if offsite power has been lost. In accordance with the emergency operating procedures, the depressurization is continued until the RCS pressure is lower than the faulted steam generator pressure. The primary to secondary leakage flow is terminated when the RCS pressure is less than the pressure in the ruptured steam generator.

4. The Emergency Core Cooling System (ECCS) pumps are then secured to prevent the RCS from repressurizing and reestablishing the break flow.

5. Following the termination of the break flow, the operators initiate a controlled cooldown and depressurization of the RCS to maintain the RCS pressure less than the pressure in the ruptured steam generator, thus preventing any further leakage of RCS fluid into the ruptured steam generator.

15.6.3.1.1 Analysis of Effects and Consequences

Determination of Operator Response Times

Because the reactor operators are required to terminate the primary- to-secondary break flow in a restricted period of time, the time needed for the operators to complete the required actions is important in evaluating the effects of the postulated event. Operator responses are separated into four significant actions:

1. Identification and isolation of the faulted steam generator;

2. Initiation of RCS cooldown;

3. Initiation of RCS depressurization; and,

4. Termination of the ECCS flow.

Conservative time requirements for completion of each of the important actions listed above were determined through observation of numerous simulator exercises using licensed reactor operators.

Conservative times required to perform each of the significant actions listed above were determined to be:

1.	Identify and isolate the ruptured steam generator after initiation of the steam generator tube rupture event:	13 minutes
2.	Initiate the RCS cooldown after steam generator isolation:	5 minutes
3.	Initiate the RCS depressurization after the end of the RCS cooldown:	2 minutes
4.	Terminate Safety Injection Flow following the end of the RCS depressurization:	3 minutes

Method of Analysis

The postulated Steam Generator Tube Rupture event is analyzed to demonstrate that the operators can terminate the primary-to-secondary break flow in a time frame compatible with preventing the filling of the ruptured steam generator. The RETRAN02 computer code is used to

calculate the transient response. Additional details concerning the design-basis analysis methodology are contained in References 37, 38 and 39.

Assumptions for initial conditions, control and protection system setpoints, and offsite power availability are selected for conservatism with respect to the maximum liquid volume in the ruptured steam generator.

WCAP-10698-P-A [Ref. 37] concluded that higher decay heat was conservative for the SGTR margin-tooverfill analysis, and therefore 120% of the 1971 ANS decay heat curve should be used in future SGTR margin-to-overfill analyses. This was found to be non-conservative for Comanche Peak. The margin to overfill analysis was revised using a conservative lower decay heat rate. The decreased heat input minimized the ruptured SG mass release due to steaming.

The major analysis assumptions are summarized below:

1. Initial Operating Conditions

   The initial reactor power and RCS temperatures are assumed to be at their nominal operating values consistent with steady-state, full power operation. The initial RCS pressurizer pressure and level are assumed to be consistent with steady-state full power operation, including appropriate allowances for calibration and instrumentation errors. The initial water inventory in the ruptured steam generator is conservatively selected to minimize the margin to overfill.

2. ESF System Capacities

   The capacities of the ECCS components, i.e., the centrifugal charging pumps and the safety injection pumps, are conservatively maximized to increase the primary to secondary leak rate, thereby maximizing the liquid volume in the ruptured steam generator. The total auxiliary feedwater flow to the ruptured steam generator from the turbine-driven and motor-driven auxiliary feedwater pumps is also conservatively maximized.

3. Offsite Power

   Offsite power is assumed to be lost coincident with the reactor trip and turbine trip. Thus, the reactor coolant pumps begin to coast down immediately following the reactor trip and the pressurizer sprays are not available for RCS depressurization. The steam dump valves are also inoperable following the loss of offsite power.

4. Equipment Availability

   To conservatively increase the break flow rate, the pressure in the ruptured steam generator is maintained at or below the setpoint pressure of the atmospheric relief valve rather than that of the main steam safety valves. The atmospheric relief valves on all intact steam generators are available for cooling the RCS.

   The pressurizer power-operated relief valves and their associated block valves are assumed to be operable. Nitrogen accumulators are provided for the pressurizer PORVs should offsite power not be available.

5. Protection and Control Systems

The turbine and rod control systems are assumed to be in the manual mode of operation; the effects of automatic control have been bounded by the selection of conservative initial conditions. The Steam Generator Water Level Control System is assumed to function normally prior to reactor trip. If the steam generator water level controller were in manual, the faulted steam generator would be identified earlier, thus decreasing the maximum liquid volume. To maximize the liquid volume in the ruptured steam generator, the turbine is tripped and auxiliary feedwater actuation is initiated 30 seconds after the reactor trip. No delay time is assumed between the initiation of the auxiliary feedwater and ECCS systems and the delivery of fluid from these pumps.

6. Single Active Failure

The failure of one (Unit 1) or two (Unit 2) intact steam generator ARVs to open is the limiting single active failure with respect to filling the ruptured steam generator. This leaves two ARVs (Unit 1) or one ARV (Unit 2) on intact steam generators available for cooling the RCS. This assumed failure extends the time required to cool the RCS to the point that the RCS can be depressurized and the break flow terminated.

15.6.3.2 Environmental Consequences

Evaluation of the steam generator tube rupture event indicates that DNBR is not reached and, thus, no clad damage would be expected in this transient. This is consistent with the fact that when the reactor is at power, the reactor coolant pumps are operating and, for this event, only a small fraction of the total primary system fluid inventory has leaked to the secondary side. Thus, it is very unlikely that DNB would occur as a result of the reduced RCS flow. The RCS depressurization that results due to flow out of the tube rupture presents another possibility for obtaining a low DNBR. However, the depressurization that occurs in a steam generator tube rupture is much less than considered in the depressurization transient analyzed in Section 15.6.1 for the Inadvertent Opening of a Pressurizer Safety or Relief Valve. In the analysis of that event, it was determined that the DNBR remains above the limit value throughout the transient and, thus, no clad damage is expected. From this, it is concluded that no clad damage is expected in the steam generator tube rupture event.

At the time of event termination, the liquid volume in the ruptured steam generator is less than that required to fill the steam generator to the level of the steam generator outlet nozzle. Thus, the acceptance criterion of preventing the filling of the ruptured steam generator is satisfied.

A conservative analysis of the postulated steam generator tube rupture assumes the loss of offsite power. The scenario involves the release of steam from the secondary system caused by a turbine trip in conjunction with loss of main steam dump capabilities, and subsequent venting to the atmosphere through the atmospheric relief valves.

The limiting single active failure is chosen in order to maximize the amount of radioactivity released directly to the atmosphere. It has been demonstrated that no single active failure scenario results in the flooding of the main steamlines. Thus, the limiting single active failure for the steam generator tube rupture event is the failure to close of the atmospheric relief valve on the loop with the ruptured steam generator. After identifying the stuck open atmospheric relief valve, operations personnel are dispatched to locally close the atmospheric relief valve block

valve. For the environmental consequences analysis, it is assumed that the block valve is closed within 30 minutes after identifying the stuck open atmospheric relief valve, thus terminating the release of radioactive steam from the ruptured steam generator to the atmosphere. Following the closure of the block valve, the primary and secondary system pressures are equalized in a manner similar to that previously described. Equalization of the primary and secondary system pressures terminates the leakage flow into the ruptured steam generator in sufficient time to prevent filling the ruptured steam generator. The volume available prior to filling the ruptured steam generator is greater than for the previous scenario.

The time dependent mass releases used to assess the radiological consequences of the postulated steam generator tube rupture are calculated from the RETRAN02 thermal-hydraulic analysis described above. Time-dependent values of the leakage rate into the ruptured steam generator and the flashing fraction were also used to assess the radiological consequences for the 0-2 hour time period following the event. Following the closure of the atmospheric relief valve block valve, the additional radiological dose is due to the leakage from the primary system into the intact steam generators and the initial concentration of radioactivity contained in the intact steam generators. The steam releases from 2 hours to residual heat removal system (RHRS) cut-in are determined from mass and energy balances using the RCS and intact steam generator conditions. Following termination of break flow, the intact steam generators' ARVs are assumed to cool down the plant at less than the maximum allowable rate to the RHRS in-service temperature. The ruptured steam generator is assumed to be depressurized to the RHRS in-service pressure immediately after the cooldown of the RCS. The amount of steam released is determined from mass and energy balances; no changes in thermodynamic conditions are assumed from termination of tube flow until depressurization is started since the ruptured steam generator is isolated. Steam releases from all steam generators are considered terminated when the RHRS in-service conditions are reached.

Two separate iodine spikes are considered:

Case I — A reactor transient has occurred prior to the tube rupture and raised the primary coolant iodine concentration to 60 uCi/gm Dose Equivalent Iodine-131 (DEQ I-131). The resulting preaccident isotopic iodine concentrations are shown in Table 15.6-2.

Case II — The reactor trip or primary system depressurization associated with the postulated accident creates an iodine spike in the primary system. The spike is assumed to increase the iodine appearance rate (inleakage from the defective fuel rods to the primary coolant) to 335 times the equilibrium appearance rate. The concurrent iodine spike appearance rates are presented in Table 15.6-3.

The assumptions below are used to determine the initial primary and secondary activities and to calculate the activity released and the offsite doses for the postulated steam generator tube rupture accident.

1. The initial primary coolant iodine activity (i.e., prior to any iodine spike considerations) is assumed to be at 1.0 uCi/gm DEQ I-131.

2. The primary coolant activity has been leaking into the secondary side at one gpm for a period of time long enough to establish equilibrium activity concentrations in the steam generators.

3. All noble gas activity transported from the primary system to the secondary system is immediately released to the environment. The initial iodine activity in the water region of the ruptured steam generator increases over time due to the unflashed portion of the leakage.

4. Due to the pressure differential between the primary and secondary sides, a fraction of the primary coolant that is discharged to the defective steam generator flashes to steam. This flashed fraction does not mix with the steam generator water and, therefore, is not subjected to any iodine removal process in the steam generator. However, the flashed fraction experiences iodine removal in the condenser when that path is available.

5. Radioactive decay of parent iodines to noble gas products is not considered during the iodine spiking processes or as unflashed iodine accumulates in the steam generators.

6. A ground level release is assumed. No credit is taken for radioactive decay or cloud depletion due to ground deposition during plume transport.

7. Worst case, five percentile atmospheric dispersion factors are assumed.

8. A breathing rate of 3.5E-04 m^3/sec is assumed.

9. Conservative iodine partition factors of 0.01 and 0.15 are used in the steam generator and condenser, respectively, to account for iodine removal effects within those components.

10. The dose contribution from the intact generator is based upon the effects of iodine spiking and a 1 gpm primary-to-secondary leak rate lasting for 11 hours. The leakage is assumed to mix with the secondary side water.

11. Other assumptions are detailed in Table 15.6-1.

Based on the foregoing model, the thyroid and whole body doses at the EAB are conservatively calculated to be 40 rem and 0.14 rem, respectively, for the preaccident iodine spike case and 26 rem and 0.18 rem, respectively, for the concurrent iodine spike case. The thyroid and whole body doses at the LPZ are conservatively calculated to be 6.0 rem and 0.03 rem, respectively, for the preaccident iodine spike case and 4.0 rem and 0.03 rem, respectively, for the concurrent iodine spike case. As expected, the doses are well below the values of 300 rem to the thyroid and 25 rem to the whole body set forth in 10CFR100.

15.6.4 SPECTRUM OF BWR STEAM SYSTEM PIPING FAILURES OUTSIDE OF CONTAINMENT

This section is not applicable to the CPNPP.

15.6.5 LOSS OF COOLANT ACCIDENT RESULTING FROM A SPECTRUM OF POSTULATED PIPING BREAKS WITHIN THE REACTOR COOLANT PRESSURE BOUNDARY

15.6.5.1 Identification of Causes and Frequency Classification

A LOCA is the result of a pipe rupture of the RCS pressure boundary. For the analyses reported here, a major pipe break (large break) is defined as a rupture with a total cross-sectional area equal to or greater than 1.0 square feet (ft^2). This event is considered an ANS Condition IV event, a limiting fault, in that it is not expected to occur during the lifetime of the plant but is postulated as a conservative design basis (see Section 15.0.1).

A minor pipe break (small break), as considered in this section, is defined as a rupture of the reactor coolant pressure boundary (see Section 5.2) with a total cross-sectional area less than 1.0 ft^2 in which the normally operating charging system flow is not sufficient to sustain pressurizer level and pressure. This is considered a Condition III event, in that it is an infrequent fault which may occur during the life of the plant.

The Acceptance Criteria for the LOCA are described in 10CFR50.46 (Reference 1) as follows:

1. The calculated maximum fuel element cladding temperature shall not exceed 2200 °F.

2. The calculated total amount of hydrogen generated from the chemical reaction of the cladding with water or steam shall not exceed 0.01 times the hypothetical amount that would be generated if all the metal in the cladding cylinders surrounding the fuel, excluding the cladding surrounding the plenum volume, were to react.

3. The calculated total oxidation of the cladding shall nowhere exceed 0.17 times the total cladding thickness before oxidation.

4. Calculated changes in core geometry shall be such that the core remains amenable to cooling.

5. After any successful operation of the ECCS, the calculated core temperature shall be maintained at an acceptably low value and decay heat shall be removed for the extended period of time required by the long-lived radioactivity remaining in the core.

These criteria were established to provide significant margin in Emergency Core Cooling System (ECCS) performance following a LOCA. Reference 2 presents a study in regard to the probability of occurrence of RCS pipe ruptures.

In all cases for Comanche Peak Units 1 and 2, small breaks (less than 1.0 ft^2) yield results with more margin to the Acceptance Criteria limits than large breaks.

15.6.5.1.1 References

1. 10CFR50.46 and Appendix K of 10CFR50. Federal Register, Volume 39, Number 3, January 4, 1974. “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Cooled Nuclear Power Reactors.”

2. WASH-1400, NUREG-75/014, October 1975. “Reactor Safety Study - An Assessment of Accident Risks in U. S. Commercial Nuclear Power Plants.”

15.6.5.2 Best Estimate Large Break Loss of Coolant Analysis (BE-LBLOCA)

15.6.5.2.1 General

When the Final Acceptance Criteria (FAC) governing the loss-of-coolant accident (LOCA) for Light Water Reactors was issued in Appendix K of 10 CFR 50.46, both the Nuclear Regulatory Commission (NRC) and the industry recognized that the stipulations of Appendix K were highly conservative. That is, using the then accepted analysis methods, the performance of the Emergency Core Cooling System (ECCS) would be conservatively underestimated, resulting in predicted Peak Clad Temperatures (PCTs) much higher than expected. At that time, however, the degree of conservatism in the analysis could not be quantified. As a result, the NRC began a large-scale confirmatory research program with the following objectives:

1. Identify, through separate effects and integral effects experiments, the degree of conservatism in those models permitted in the Appendix K rule. In this fashion, those areas in which a purposely prescriptive approach was used in the Appendix K rule could be quantified with additional data so that a less prescriptive future approach might be allowed.

2. Develop improved thermal-hydraulic computer codes and models so that more accurate and realistic accident analysis calculations could be performed. The purpose of this research was to develop an accurate predictive capability so that the uncertainties in the ECCS performance and the degree of conservatism with respect to the Appendix K limits could be quantified.

Since that time, the NRC and the nuclear industry have sponsored reactor safety research programs directed at meeting the above two objectives. The overall results have quantified the conservatism in the Appendix K rule for LOCA analyses and confirmed that some relaxation of the rule can be made without a loss in safety to the public. It was also found that some plants were being restricted in operating flexibility by the overly conservative Appendix K requirements. In recognition of the Appendix K conservatism that was being quantified by the research programs, the NRC adopted an interim approach for evaluation methods. This interim approach is described in SECY-83-472. The SECY-83-472 approach retained those features of Appendix K that were legal requirements, but permitted applicants to use best-estimate thermal-hydraulic models in their ECCS evaluation model. Thus, SECY-83-472 represented an important step in basing licensing decisions on realistic calculations, as opposed to those calculations prescribed by Appendix K.

In 1998, the NRC Staff amended the requirements of 10 CFR 50.46 and Appendix K, “ECCS Evaluation Models”, to permit the use of a realistic evaluation model to analyze the performance of the ECCS during a hypothetical LOCA. This decision was based on an improved understanding of LOCA thermal-hydraulic phenomena gained by extensive research programs. Under the amended rules, best-estimate thermal-hydraulic models may be used in place of models with Appendix K features. The rule change also requires, as part of the LOCA analysis, an assessment of the uncertainty of the best-estimate calculations. It further requires that this analysis uncertainty be included when comparing the results of the calculations to the prescribed

acceptance criteria of 10 CFR 50.46. Further guidance for the use of best-estimate codes is provided in Regulatory Guide 1.157.

To demonstrate use of the revised ECCS rule, the NRC and its consultants developed a method called the Code Scaling, Applicability, and Uncertainty (CSAU) evaluation methodology (NUREG/CR-5249). This method outlined an approach for defining and qualifying a best-estimate thermal-hydraulic code and quantifying the uncertainties in a LOCA analysis.

A LOCA evaluation methodology for three- and four-loop Pressurized Water Reactor (PWR) plants based on the revised 10 CFR 50.46 rules was developed by Westinghouse with the support of EPRI and Consolidated Edison and has been approved by the NRC (WCAP-12945-P-A).

More recently, Westinghouse developed an alternative uncertainty methodology called ASTRUM, which stands for Automated Statistical TReatment of Uncertainty Method (WCAP-16009-P-A). This method is still based on the CQD methodology and follows the steps in the CSAU methodology (NUREG/CR-5249). However, the uncertainty analysis (Element 3 in the CSAU) is replaced by a technique based on order statistics. The ASTRUM methodology replaces the response surface technique with a statistical sampling method where the uncertainty parameters are simultaneously sampled for each case. The ASTRUM methodology has received NRC approval for referencing in licensing calculations in WCAP-16009-P-A.

The three 10 CFR 50.46 criteria (peak clad temperature, maximum local oxidation, and core-wide oxidation) are satisfied by running a sufficient number of WCOBRA/TRAC calculations (sample size). In particular, the statistical theory predicts that 124 calculations are required to simultaneously bound the 95th percentile values of three parameters with a 95-percent confidence level.

This analysis is in accordance with the applicability limits and usage conditions defined in Section 13-3 of WCAP-16009-P-A, as applicable to the ASTRUM methodology. Section 13-3 of WCAP-16009-P-A was found to acceptably disposition each of the identified conditions and limitations related to WCOBRA/TRAC and the CQD uncertainty approach per Section 4.0 of the ASTRUM Final Safety Evaluation Report appended to this topical report.

15.6.5.2.2 Method of Analysis

The methods used in the application of WCOBRA/TRAC to the large break LOCA with ASTRUM are described in WCAP-12945-P-A and WCAP-16009-P-A. A detailed assessment of the computer code WCOBRA/TRAC was made through comparisons to experimental data. These assessments were used to develop quantitative estimates of the code's ability to predict key physical phenomena in a PWR large break LOCA. Modeling of a PWR introduces additional uncertainties which are identified and quantified in the plant-specific analysis. WCOBRA/TRAC MOD7A was used for the execution of ASTRUM for Comanche Peak Units 1 and 2 (WCAP-16009-P-A).

WCOBRA/TRAC combines two-fluid, three-field, multi-dimensional fluid equations used in the vessel with one-dimensional drift-flux equations used in the loops to allow a complete and detailed simulation of a PWR. This best-estimate computer code contains the following features:

1. Ability to model transient three-dimensional flows in different geometries inside the vessel

2. Ability to model thermal and mechanical non-equilibrium between phases

3. Ability to mechanistically represent interfacial heat, mass, and momentum transfer in different flow regimes

4. Ability to represent important reactor components such as fuel rods, steam generators, reactor coolant pumps, etc.

A typical calculation using WCOBRA/TRAC begins with the establishment of a steady-state, initial condition with all loops intact. The input parameters and initial conditions for this steady-state calculation are discussed in the next section.

Following the establishment of an acceptable steady-state condition, the transient calculation is initiated by introducing a break into one of the loops. The evolution of the transient through blowdown, refill, and reflood proceeds continuously, using the same computer code (WCOBRA/TRAC) and the same modeling assumptions. Containment pressure is modeled with the BREAK component using a time dependent pressure table. Containment pressure is calculated using the COCO code (WCAP-8326 and WCAP-8327) and mass and energy releases from the WCOBRA/TRAC calculation.

The final step of the best-estimate methodology, in which all uncertainties of the LOCA parameters are accounted for to estimate a PCT, Local Maximum Oxidation (LMO), and Core-Wide Oxidation (CWO) at 95-percent probability, is described in the following sections.

1. Plant Model Development:

   In this step, a WCOBRA/TRAC model of the plant is developed. A high level of noding detail is used in order to provide an accurate simulation of the transient. However, specific guidelines are followed to ensure that the model is consistent with models used in the code validation. This results in a high level of consistency among plant models, except for specific areas dictated by hardware differences, such as in the upper plenum of the reactor vessel or the ECCS injection configuration.

2. Determination of Plant Operating Conditions:

   In this step, the expected or desired operating range of the plant to which the analysis applies is established. The parameters considered are based on a “key LOCA parameters” list that was developed as part of the methodology. A set of these parameters, at mostly nominal values, is chosen for input as initial conditions to the plant model. A transient is run utilizing these parameters and is known as the “initial transient”. Next, several confirmatory runs are made, which vary a subset of the key LOCA parameters over their expected operating range in one-at-a-time sensitivities. Because certain parameters are not included in the uncertainty analysis, these parameters are set at their bounding condition. This analysis is commonly referred to as the confirmatory analysis. The most limiting input conditions, based on these confirmatory runs, are then combined into the

model that will represent the limiting state for the plant, which is the starting point for the assessment of uncertainties.

3. Assessment of Uncertainty:

   The ASTRUM methodology is based on order statistics. The technical basis of the order statistics is described in Section 11 of WCAP-16009-P-A. The determination of the PCT uncertainty, LMO uncertainty, and CWO uncertainty relies on a statistical sampling technique. According to the statistical theory, 124 WCOBRA/TRAC calculations are necessary to assess against the three 10 CFR 50.46 criteria (PCT, LMO, CWO).

   The uncertainty contributors are sampled randomly from their respective distributions for each of the WCOBRA/TRAC calculations. The list of uncertainty parameters, which are randomly sampled for each time in the cycle, break type (split or double-ended guillotine), and break size for the split break are also sampled as uncertainty contributors within the ASTRUM methodology.

   Results from the 124 calculations are tallied by ranking the PCT from highest to lowest. A similar procedure is repeated for LMO and CWO. The highest rank of PCT, LMO, and CWO will bound 95 percent of their respective populations with 95-percent confidence level.

4. Plant Operating Range:

   The plant operating range over which the uncertainty evaluation applies is defined. Depending on the results obtained in the above uncertainty evaluation, this range may be the desired range or may be narrower for some parameters to gain additional margin.

15.6.5.2.3 Analysis Assumptions

The expected PCT and its uncertainty developed are valid for a range of plant operating conditions. The range of variation of the operating parameters has been accounted for in the uncertainty evaluation. Tables 15.6-4 and 15.6-5 summarize the operating ranges as defined for the proposed operating conditions which are supported by the Best-Estimate LBLOCA analysis for Comanche Peak Unit 1 and Unit 2, respectively. Tables 15.6-6 and 15.6-7 summarize the LBLOCA containment data used for calculating containment pressure (for both units). If operation is maintained within these ranges, the LBLOCA results developed in this report using WCOBRA/TRAC are considered to be valid. Note that some of these parameters vary over their range during normal operation (accumulator temperature) and other ranges are fixed for a given operational condition (T_{avg}).

15.6.5.2.4 Design Basis Accident

The Comanche Peak Units 1 and 2 PCT-limiting transients are both double-ended cold leg guillotine break which analyze conditions that fall within those listed in Table 15.6-4 (Unit 1) and Table 15.6-5 (Unit 2). Traditionally, cold leg breaks have been limiting for large break LOCA. This location is the one where flow stagnation in the core appears most likely to occur. Scoping

studies with WCOBRA/TRAC have confirmed that the cold leg remains the limiting break location (WCAP-12945-P-A).

The large break LOCA transient can be divided into convenient time periods in which specific phenomena occur, such as various hot assembly heatup and cooldown transients. For a typical large break, the blowdown period can be divided into the Critical Heat Flux (CHF) phase, the upward core flow phase, and the downward core flow phase. These are followed by the refill, reflood, and long-term cooling periods. Specific important transient phenomena and heat transfer regimes are discussed below, with the transient results shown in Figures 15.6-1 to 15.6-14 for Unit 1 and Figures 15.6-15 to 15.6-28 for Unit 2. (The PCT-limiting case for each unit was chosen to show a conservative representation of the response to a large break LOCA.)

1. Critical Heat Flux (CHF) Phase:

   Immediately following the cold leg rupture, the break discharge rate is subcooled and high (Figures 15.6-2, 15.6-3 for Unit 1 and 15.6-16, 15.6-17 for Unit 2). The regions of the RCS with the highest initial temperatures (core, upper plenum, upper head, and hot legs) begin to flash to steam, the core flow reverses and the fuel rods begin to go through departure from nucleate boiling (DNB). The fuel cladding rapidly heats up (Figure 15.6-1 for Unit 1 and 15.6-15 for Unit 2) while the core power shuts down due to voiding in the core. This phase is terminated when the water in the lower plenum and downcomer begins to flash (Figures 15.6-7 and 15.6-12 for Unit 1 and 15.6-21 and 15.6-26 for Unit 2, respectively). The mixture swells and intact loop pumps, still rotating in single-phase liquid, push this two-phase mixture into the core.

2. Upward Core Flow Phase:

   Heat transfer is improved as the two-phase mixture is pushed into the core. This phase may be enhanced if the pumps are not degraded, or if the break discharge rate is low due to saturated fluid conditions at the break. If pump degradation is high or the break flow is large, the cooling effect due to upward flow may not be significant. Figures 15.6-4 (Unit 1) and 15.6-18 (Unit 2) show the void fraction for one intact loop pump and the broken loop pump. Each figure shows that the intact loop remains in single-phase liquid flow for several seconds, resulting in enhanced upward core flow cooling. This phase ends as the lower plenum mass is depleted, the loop flow becomes two-phase, and the pump head degrades.

3. Downward Core Flow Phase:

   The loop flow is pushed into the vessel by the intact loop pumps and decreases as the pump flow becomes two-phase. The break flow begins to dominate and pulls flow down through the core, up the downcomer to the broken loop cold leg, and out the break. While liquid and entrained liquid flow provide core cooling, the top third of core vapor flow (Figure 15.6-5 for Unit 1 and Figure 15.6-19 for Unit 2) best illustrates this phase of core cooling. Once the system has depressurized to the accumulator pressure (15.6-6 for Unit 1 and 15.6-20 for Unit 2), the accumulators begin to inject cold borated water into the intact cold legs (Figure 15.6-9 for Unit 1 and 15.6-23 for Unit 2). During this period, due to steam upflow in the downcomer, a portion of the injected ECCS water is calculated to be

bypassed around the downcomer and out the break. As the system pressure continues to fall, the break flow, and consequently the downward core flow, is reduced. The core begins to heat up as the system pressure approaches the containment pressure and the vessel begins to fill with ECCS water (Figure 15.6-8 for Unit 1 and 15.6-22 for Unit 2).

4. Refill Period:

As the refill period begins, the core begins a period of heatup and the vessel begins to fill with ECCS water (Figures 15.6-9, 15.6-10 for Unit 1 and 15.6-23, 15.6-24 for Unit 2). This period is characterized by a rapid increase in cladding temperatures at all elevations due to the lack of liquid and steam flow in the core region. This period continues until the lower plenum is filled and the bottom of the core begins to reflood and entrainment begins.

5. Reflood Period:

During the early reflood phase, the accumulators begin to empty and nitrogen enters the system. This forces water into the core, which then boils, causing system re-pressurization, and the lower core region begins to quench (Figure 15.6-11 for Unit 1 and 15.6-25 for Unit 2). During this time, core cooling may increase due to vapor generation and liquid entrainment. During the reflood period, the core flow is oscillatory as cold water periodically rewets and quenches the hot fuel cladding, which generates steam and causes system re-pressurization. The steam and entrained water must pass through the vessel upper plenum, the hot legs, the steam generators, and the reactor coolant pumps before it is vented out of the break. This flow path resistance is overcome by the downcomer water elevation head, which provides the gravity driven reflood force. From the later stage of blowdown to the beginning of reflood, the accumulators rapidly discharge borated cooling water into the RCS, filling the lower plenum and contributing to the filling of the downcomer. The pumped ECCS water aids in the filling of the downcomer and subsequently supplies water to maintain a full downcomer and complete the reflood period. As the quench front progresses up the core, the PCT location moves higher into the top core region. As the vessel continues to fill, the PCT location is cooled and the early reflood period is terminated.

A second cladding heatup transient may occur due to boiling in the downcomer. The mixing of ECCS water with hot water and steam from the core, in addition to the continued heat transfer from the hot vessel and vessel metal, reduces the subcooling of ECCS water in the lower plenum and downcomer. The saturation temperature is dictated by the containment pressure. If the liquid temperature in the downcomer reaches saturation, subsequent heat transfer from the vessel and other structures will cause boiling and level swell in the downcomer. The downcomer liquid will spill out of the broken cold leg and reduce the driving head, which can reduce the reflood rate, causing a late reflood heatup at the upper core elevations. Figures 15.6-12 (Unit 1) and 15.6-26 (Unit 2) show only a slight reduction in downcomer level and indicates that a late reflood heatup does not occur.

15.6.5.2.5 Post Analysis of Record Evaluations

In addition to the analyses presented in this section, evaluations and reanalyses may be performed as needed to address computer code errors and emergent issues, or to support plant changes. The issues or changes are evaluated, and the impact on the Peak Cladding Temperature (PCT) is determined. The resultant increase or decrease in PCT is applied to the analysis of record PCT. The PCT, including all penalties and benefits is presented in Tables 15.6-8 (Unit 1) and 15.6-10 (Unit 2) for the large break LOCA. The current PCT is demonstrated to be less than the 10 CFR 50.46(b) requirement of 2200 ºF.

In addition, 10 CFR 50.46 requires that licensees assess and report the effect of changes to or errors in the evaluation model used in the large break LOCA analysis. These reports constitute addenda to the analysis of record provided in the UFSAR until the overall changes become significant as defined by 10 CFR 50.46. If the assessed changes or errors in the evaluation model results in significant changes in calculated PCT, a schedule for formal reanalysis or other action as needed to show compliance will be addressed in the report to the NRC.

Finally, the criteria of 10 CFR 50.46 requires that holders and users of the evaluation models establish a number of definitions and processes for assessing changes in the models or their use. Westinghouse, in consultation with the PWR Owner's Group (PWROG), has developed an approach for compliance with the reporting requirements. This approach is documented in WCAP-13451, Westinghouse Methodology for Implementation of 10 CFR 50.46 Reporting. Vistra OpCo provides the NRC with annual and 30-day reports, as applicable, for Comanche Peak Units 1 and 2. Vistra OpCo intends to provide future reports required by 10 CFR 50.46 consistent with the approach described in WCAP-13451.

15.6.5.2.6 Conclusions

It must be demonstrated that there is a high level of probability that the limits set forth in 10 CFR 50.46 are met. The demonstration that these limits are met is as follows:

(b)(1) The limiting PCT corresponds to a bounding estimate of the 95th percentile PCT at the 95-percent confidence level. The analysis confirms that 10 CFR 50.46 acceptance criterion (b)(1), i.e., "Peak Clad Temperature less than 2200 ºF", is demonstrated. The results are shown in Table 15.6-9 for Unit 1 and Table 15.6-11 for Unit 2.

(b)(2) The maximum cladding oxidation corresponds to a bounding estimate of the 95th percentile LMO at the 95-percent confidence level. The analysis confirms that 10 CFR 50.46 acceptance criterion (b)(2), i.e., "Local Maximum Oxidation of the cladding less than 17 percent", is demonstrated. The results are shown in Table 15.6-9 for Unit 1 and Table 15.6-11 for Unit 2.

(b)(3) The limiting core-wide oxidation (CWO) corresponds to a bounding estimate of the 95th percentile CWO at the 95-percent confidence level. The limiting Hot Assembly Rod (HAR) total maximum oxidation is 0.0 percent for both units. A detailed CWO calculation takes advantage of the core power census that includes many lower power assemblies. Because there is significant margin to the regulatory limit, the CWO value can be conservatively chosen as that calculated for the limiting HAR. A detailed CWO calculation is therefore not needed because

the outcome will always be less than the HAR value. The analysis confirms that 10 CFR 50.46 acceptance criterion (b)(3), i.e., “Core-Wide Oxidation less than 1 percent”, is demonstrated. The results are shown in Table 15.6-9 for Unit 1 and Table 15.6-11 for Unit 2.

(b)(4) 10 CFR 50.46 acceptance criterion (b)(4) requires that the calculated changes in core geometry are such that the core remains amenable to cooling. This criterion has historically been satisfied by adherence to criteria (b)(1) and (b)(2), and by assuring that fuel deformation due to combined LOCA and seismic loads is specifically addressed. It has been demonstrated that the PCT and maximum cladding oxidation limits remain in effect for Best-Estimate LOCA applications. The approved methodology (WCAP-12945-P-A) specifies that effects of LOCA and seismic loads on core geometry do not need to be considered unless grid crushing extends beyond the 44 assemblies in the low-power channel. This situation has not been calculated to occur for Comanche Peak Units 1 and 2. Therefore, acceptance criterion (b)(4) is satisfied.

(b)(5) 10 CFR 50.46 acceptance criterion (b)(5) requires that long-term core cooling be provided following the successful initial operation of the ECCS. Long-term cooling is dependent on the demonstration of continued delivery of cooling water to the core. The actions, automatic or manual, that are currently in place at these plants to maintain long-term cooling remain unchanged with the application of the ASTRUM methodology (WCAP-16009-P-A).

Based on the ASTRUM Analysis results (Tables 15.6-9 and 15.6-11), it is concluded that Comanche Peak Units 1 and 2 continue to maintain a margin of safety to the limits prescribed by 10 CFR 50.46.

15.6.5.2.7 References

1. “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Cooled Nuclear Power Reactors”, 10 CFR 50.46 and Appendix K of 10 CFR 50. Federal Register, Volume 39, Number 3, January 4, 1974.

2. SECY-83-472, Information Report from W.J. Dircks to the Commissioners, “Emergency Core Cooling System Analysis Methods”, November 17, 1983.

3. Regulatory Guide 1.157, Best-Estimate Calculations of Emergency Core Cooling System Performance, USNRC, May 1989.

4. NUREG/CR-5249, Qualifying Reactor Safety Margins: Application of Code Scaling Applicability and Uncertainty (CSAU) Evaluation Methodology to a Large Break Loss-of-Coolant-Accident, B. Boyack, et. al., 1989.

5. Bajorek, S.M., et. al., 1998, “Code Qualification Document for Best-Estimate LOCA Analysis”, WCAP-12945-P-A, Volume 1, Revision 2 and Volumes 2 through 5, Revision 1.

6. WCAP-16009-P-A, “Realistic Large-Break LOCA Evaluation Methodology Using the Automated Statistical Treatment of Uncertainty Method (ASTRUM)”, (Proprietary), January 2005.

7. Containment Pressure Analysis Code (COCO), WCAP-8327 (Proprietary) and WCAP-8326 (Non-Proprietary), June 1974.

8. WCAP-13451, “Westinghouse Methodology for Implementation of 10 CFR 50.46 Reporting”, October 1992.

15.6.5.3 Small Break Loss of Coolant Accident

15.6.5.3.1 Sequence of Events and System Operations

A LOCA can result from a rupture of the reactor coolant system (RCS) or of any line connected to that system up to the first isolation valve. Ruptures of small cross section will cause expulsion of the coolant at a rate that can be accommodated by the charging pumps. Breaks of greater size (up to 1 ft^2 area) are defined as small breaks and are analyzed with the NOTRUMP computer code. A rupture in the reactor coolant system results in the discharge to the containment of reactor coolant and associated energy. The result of this discharge is a decrease in coolant pressure in the reactor coolant system and an increase in containment temperature and pressure.

The reactor trip signal subsequently occurs when the pressurizer low pressure trip setpoint is reached. A safety injection system (SIS) signal is actuated when the appropriate pressurizer low pressure setpoint is reached, activating the Emergency Core Cooling System (ECCS) (assuming the most limiting single failure of ECCS equipment).

Before the break occurs, the unit is assumed to be in an equilibrium condition, (i.e., the heat generated in the core is being removed via the secondary system). In the small break LOCA, the blowdown phase of the small break occurs over a long time period. Thus for a small break LOCA, there are three characteristic stages: (1) a gradual blowdown in which the decrease in water level is checked by the inventory replenishment associated with safety injection, (2) core recovery, and (3) long-term recirculation. The heat transfer between the reactor coolant system and the secondary system may be in either direction, depending on the relative temperature. For the case of continued heat addition to the secondary side, the secondary side pressure increases and the main steam safety valves may actuate to reduce the pressure. Makeup to the secondary side is automatically provided by the auxiliary feedwater system. Coincident with the SIS, normal feedwater flow is stopped by closing the main feedwater control valves and tripping the main feedwater pumps. Emergency feedwater flow is initiated by starting the auxiliary feedwater pumps. The secondary side flow aids in the reduction of RCS pressure with the break being the primary means of heat removal. When the reactor coolant system depressurizes to approximately 600 psia, the accumulators begin to inject borated water into the reactor coolant loops. Reflecting the loss of offsite power assumption, the reactor coolant pumps are assumed to be tripped at the time of reactor trip, and the effects of pump coastdown are included in the blowdown analysis.

15.6.5.3.2 Small Break Evaluation Model

The NOTRUMP and LOCTA-IV computer codes are used to perform the analysis of LOCAs due to small breaks in the RCS. The NOTRUMP computer code, approved for this use by the U.S. Nuclear Regulatory Commission in May 1985 (Reference 1), is used to calculate the transient depressurization of the RCS as well as to describe the mass and enthalpy of the flow through the

break. This code is a one-dimensional general network code incorporating a number of features, including the utilization of nonequilibrium thermal calculation in all-fluid volumes, flow regime-dependent drift flux calculations with counter-current flooding limitations, mixture level tracking logic in multiple-stacked fluid nodes, and regime-dependent heat transfer correlations. Also, safety injection into the broken loop is modeled using the COSI condensation model (Reference 2). The NOTRUMP small-break LOCA ECCS evaluation model was developed to determine the RCS response to design basis small-break LOCAs and to address the NRC concerns expressed in NUREG-0611, “Generic Evaluation of Feedwater Transients and Small Break Loss-of-Coolant Accidents in Westinghouse-Designed Operating Plants.”

In NOTRUMP, the RCS is subdivided into fluid-filled control volumes (fluid nodes) and metal nodes interconnected by flow paths and heat transfer links. The transient behavior of the system is determined from the governing conservation equations of mass, energy, and momentum applied to these nodes. The broken loop is modeled explicitly, and the intact loops are lumped into a second loop. A detailed description of the NOTRUMP code is provided in References 1, 2, and 3.

In the NOTRUMP model, the reactor core is represented as a vertical stack of heated control volumes with an associated bubble rise model to permit a transient mixture height calculation. The multinode capability of the program enables an explicit and detailed spatial representation of various system components. In particular, it enables a proper calculation of the behavior of the loop seal during a LOCA.

Clad thermal analyses are performed with the LOCTA-IV code (Reference 4) which uses the RCS pressure, fuel rod power history, steam flow past the uncovered part of the core, and mixture height from the NOTRUMP hydraulic calculations as input. For all computations where core uncovery occurred, the NOTRUMP and LOCTA runs were terminated after the time the core mixture level reached the top of the core following uncovery.

A schematic representation of the computer code interfaces is given in Figure 15.6.5-29.

15.6.5.3.3 Input Parameters and Initial Conditions

Significant input parameters are given in Table 15.6.5-13.

The bases used to select the numerical values that are input parameters to the analysis have been conservatively determined from sensitivity studies (References 5, 6, and 7). In addition, the requirements of 10CFR50 Appendix K regarding specific model features were met by selecting models which provide a significant overall conservatism in the analysis. The assumptions made pertain to the conditions of the reactor and associated safety system equipment at the time that the LOCA occurs and include such items as the core-peaking factors and ECCS performance. Decay heat generated throughout the transient is also conservatively calculated.

For the small break LOCA (SBLOCA) analysis, loss of offsite power (LOOP) is assumed, which results in the limiting single failure assumption of the loss of one Emergency Diesel Generator (EDG) and a subsequent loss of one train of pumped Emergency Core Cooling System (ECCS) flow. The SBLOCA analysis assumes that reactor trip occurs coincident with the LOOP, which results in the following: (a) Reactor Coolant Pump (RCP) trip and coastdown and (b) Steam Dump System being inoperable.

The SBLOCA analysis is performed with a high nominal vessel average temperature (T_{avg}) value of 589.2°F. The high T_{avg} value chosen is evaluated to be applicable over the range of nominal vessel average temperature values (574.2°F to 589.2°F).

The safety injection (SI) system consists of gas pressurized accumulator tanks and pumped injection systems. The accumulators are modeled to inject borated water into the RCS when the RCS depressurizes to 603 psia. Minimum ECCS availability is assumed for the analysis from one centrifugal charging pump (CCP), one SI pump and one RHR pump. Assumed pumped SI characteristics as a function of RCS pressure used as boundary conditions in the analysis are shown in Table 15.6.5-14 and corresponding Figure 15.6.5-31 for break sizes less than 8.75 inches and Table 15.6.5-15 and corresponding Figure 15.6.5-32 for the 8.75 inch break case. For the break sizes less than 8.75 inches the broken loop safety injection flow is assumed to spill to RCS pressure and for the 8.75 inch break case the broken loop safety injection flow is assumed to spill to the containment back pressure of 0 psig. The ECCS was assumed to deliver to the RCS 22 seconds after the generation of the SI signal. This delay includes time required for sensor response, diesel generator startup and emergency power bus loading in case of a loss of offsite power coincident with an accident.

15.6.5.3.4 Small Break LOCA Results

The calculated peak clad temperature (PCT) resulting from the limiting small break LOCA is less than that calculated for the limiting large break for both Units. To insure that the worst possible small break size has been identified, calculations were performed for a spectrum of small breaks (2, 3, 4, 6 inch equivalent diameter cold leg breaks) as well as an 8.75 inch equivalent diameter accumulator line break for each unit. The results of these analyses are summarized in Tables 15.6.5-16 and 15.6.5-17 for Unit 1 and in Tables 15.6.5-18 and 15.6.5-19 for Unit 2.

Based on the results of these analyses, the limiting small break was determined to be a 4-inch diameter rupture of the RCS cold leg for both Unit 1 and Unit 2 with a PCT of 1013ºF and 1210ºF for Unit 1 and Unit 2, respectively. The clad did not rupture in any of the cases simulated.

Figures 15.6.5-33 through 15.6.5-80 present the principal parameters of interest for the small-break ECCS analyses. For all cases analyzed, the following transient parameters, where applicable, are presented (note that since there was no or minimal core uncovery for the 2, 6, and 8.75-inch cases, no LOCTA calculations were performed and no clad temperature or maximum local oxidation figures are provided for these breaks):

- Reactor Coolant System Pressure
- Core Mixture Height
- Core Exit Vapor Temperature
- Clad Temperature at Peak Clad Temperature Elevation
- Maximum Local ZrO2 Thickness

For the limiting (4-inch) breaks analyzed, the following additional transient parameters are presented:

- Total Reactor Coolant System Mass
- Core Exit Steam Flow Rate
- Total Break Flow and Safety Injection Flow
- Clad Surface Heat Transfer Coefficient at Peak Clad Temperature Elevation
- Fluid Temperature at Peak Clad Temperature Elevation

The maximum calculated PCT for all small breaks analyzed is 1210°F. These results are well below all corresponding acceptance criteria limits of 10CFR50.46.

15.6.5.3.5 References

1. WCAP-10079-P-A (Proprietary) and WCAP-10080-A (Non-proprietary), 1985. Meyer, P. E., “NOTRUMP, A Nodal Transient Small Break and General Network Code.”

2. WCAP-10054-P-A, Addendum 2, Revision 1 (Proprietary), 1995. Thompson, C.M., et al, “Addendum to the Westinghouse Small Break ECCS Evaluation Model Using the NOTRUMP Code: Safety Injection into the Broken Loop and COSI Condensation Model.”

3. WCAP-10054-P-A (Proprietary) and WCAP-10081-A (Non-proprietary), 1985. Lee, N. Rupprecht, S. D., Schwarz, W. R, and Tauche, W. D., “Westinghouse Small ECCS Evaluation Model Using the NOTRUMP Code.”

4. WCAP 8301 (Proprietary) and WCAP 8305 (Non-proprietary), 1974. Bordelon, F. M. et al. LOCTA IV Program: Loss of Coolant Transient Analysis.

5. WCAP-8340 (Proprietary) and WCAP-8356 (Non-proprietary), 1974. Salvatori, R. Westinghouse ECCS - Plant Sensitivity Studies.

6. WCAP-8341 (Proprietary) and WCAP-8342 (Non-proprietary), 1974. Westinghouse ECCS Evaluation Model Sensitivity Studies.

7. WCAP-8565-P-A (Proprietary) and WCAP-8566-A (Non-proprietary), 1975. Johnson, W.J.; Massie, H.W.; and Thompson, C.M. Westinghouse ECCS-Four Loop Plant (17 x 17) Sensitivity Studies.

15.6.5.4 Environmental Consequences

To demonstrate in a conservative manner that the operation of a nuclear power station does not present any undue radiological hazard to the general public, a hypothetical accident involving a gross release of fission products is evaluated. No mechanism for such a release has been postulated because it would require a number of simultaneous failures to occur in the engineered safety features. The core fission product inventory is assumed to be released into the containment as described in Regulatory Guide 1.195 [34]. Numerical values for the total core fission product inventory of the isotopes considered in calculating the radiation doses are listed in Table 15.0-7.

The radiological evaluation of this accident is divided into two parts: internal (thyroid) dose from inhalation of iodines in the leakage plume, and external (whole body) exposure as a result of immersion in the leakage plume.

The radiological consequences due to the release of core fission products during a postulated loss-of-coolant accident are evaluated in the following sections:

1. Radiological consequences of containment leakage

   The integrated thyroid doses and the integrated whole body doses are calculated using methods and assumptions in conformance with Regulatory Guide 1.195. The assumptions used in the analysis are listed below.

   a. Twenty-five percent of the equilibrium radioactive iodine inventory developed from maximum full-power operation of the core is immediately available for leakage from the reactor containment. Of this 25 percent, 91 percent is in the form of elemental iodine, 5 percent is in the form of particulate iodine, and 4 percent is in the form of organic iodides.

   b. All (i.e., 100 percent) of the equilibrium radioactive noble gas inventory developed from maximum full-power operation of the core is immediately available for leakage from the reactor containment.

   c. The effects of radiological decay during holdup in the containment are taken into account.

   d. The containment volume is divided into separate regions by concrete floors at different elevations (see Section 6.5.2). A radial gap between the concrete floors and the inner wall of the Containment Building permits a limited amount of convective mixing between these regions. The region not covered by containment spray is treated as a separate unsprayed volume which is assumed to mix with the volumes in the sprayed areas at a mixing rate of two turnovers per hour.

      The Containment Spray System is actuated by a high containment pressure signal. For a discussion of the sequence of events of spray system operation, see Section 6.5.2. A sodium hydroxide spray is used to reduce the amount of fission product iodine available for release during the LOCA. The containment spray solution is assumed to interact with the elemental iodine and particulate iodine. The mathematical models which calculate the iodine spray removal coefficients are presented in Section 6.5.2. For each region the calculated elemental iodine removal coefficients are above 10 hr^{-1}. The removal coefficient for elemental iodine used in the offsite dose calculation is limited to a maximum value of 10 hr^{-1} [22]. The calculated value of 11.4 hr^{-1} is used for the particulate iodine removal coefficient, until a DF of 50 is reached at which time the particulate removal coefficient is reduced by a factor of 10. The elemental iodine removal effectiveness may be expected to diminish after the concentration in the containment atmosphere has been reduced by several orders of magnitude. The elemental iodine removal effectiveness of the spray system is conservatively

assumed to cease after a decontamination of 100 in the containment atmosphere has been achieved.

e. The iodine and noble gases available for release to the environment are assumed to leak from the Containment at a maximum leak rate of 0.10 percent of the containment volume per day for the first 24 hours, and at 0.05 percent of the containment volume per day for the duration of the accident.

f. The duration of the accident is considered to be 30 days.

g. A ground-level release is assumed. Atmospheric dilution factors are discussed in Section 2.3 [23] and listed in Table 15.6-20.

h. No credit is taken for depletion of fission products in the plume due to ground deposition or radioactive decay in transit.

i. For the first 8 hours after the accident, the breathing rate of persons offsite is assumed to be 3.5×10^{-4} cubic meters per second (m3/sec). Eight to 24 hours following the accident, the breathing rate is assumed to be 1.8×10^{-4} m^3/sec. From 24 hours through 30 days after the accident, the rate is assumed to be 2.3×10^{-4} m^3/sec.

j. The mathematical model and dose conversion factors presented in Appendix 15B are used for evaluating the radiological consequences of the LOCA

k. Other assumptions are detailed in Table 15.6-20.

Using the assumptions presented above and the mathematical models presented in Appendix 15B, the doses at the EAB were conservatively calculated to be 46.4 rem to the thyroid and 0.655 rem to the whole body; the doses at the LPZ were conservatively calculated to be 18.7 rem to the thyroid and 0.224 rem to the whole body.

2. Radiological consequences of engineered safety features equipment leakage outside containment.

Following a postulated LOCA, a potential source of fission product release is the leakage of water from engineered safety features (ESF) equipment located outside the containment. Such leakage could occur during the recirculation phase through components such as pump flanges, valves, and heat exchangers. The fission products could then be released from the water into the atmosphere resulting in offsite radiological consequences that contribute to the total dose from the LOCA.

An analysis of the offsite effects attributable to ESF equipment leakage is performed based on the following conservative assumptions:

a. 50 percent of the halogens originally present in the core are intimately mixed with the coolant water and are assumed to be available for release through ESF equipment outside containment (see Table 15.6-21).

b. The leakage from all ESF components is conservatively assumed to start 10 minutes after the LOCA and continue for the duration of the accident at a rate of 2 gallons per minute.

c. An iodine partition factor of 0.1 is assumed. This factor is taken as the fraction of iodine in the leakage that becomes airborne.

d. Gaseous radioactivity released to rooms housing the leaking components is considered to be immediately swept away by the ventilation system and released to the atmosphere. (See Section 9.4.5 and Figures 1.2-16, 1.2-35 and 9.4-9).

e. An iodine adsorber efficiency of 95 percent is applied since the ventilation exhaust passes through HEPA filters and iodine adsorbers prior to release to the atmosphere. The iodine adsorbers are designed to the requirements of NRC Regulatory Guide 1.52 (See Appendix 1A(B)) as discussed in Section 9.4.3.

f. No credit is taken for an elevated release; all meteorological parameters are considered to be identical to those previously defined in this section.

Based upon the foregoing model, the thyroid and whole body dose contributions due to ESF equipment leakage are conservatively calculated to be 12.1 rem and 0.0402 rem, respectively, for the EAB. The LPZ doses are conservatively calculated to be 24.8 rem to the thyroid and 0.0266 rem to the whole body.

3. Environmental consequences of releases through the containment pressure relief line in the event of a LOCA

An analysis of the radioactive effluents escaping the Containment to the environment after a LOCA, via the line through the controlled access area exhaust system, was performed using the following assumptions:

a. The maximum containment air/steam mass release to the environment was conservatively calculated assuming containment pressures for a large break LOCA, and critical flow via the 3-3/8 inch orifice plate at the inlet to the pressure relief system ductwork. No credit was taken for line losses in the ductwork or two butterfly valves.

b. Only reactor coolant activity is assumed to be released. It was conservatively assumed that all iodine and noble gas activity in the primary coolant was instantaneously released to the containment atmosphere.

c. The primary coolant iodine activity is assumed to be 1.0 micro curie per gram dose equivalent I-131. The noble gas activity is based on operation with 1% fuel defects. The activity concentrations are presented in Table 15.1-4.

d. The containment pressure relief line isolation valve closure time including instrumentation delays will not exceed 5 seconds. The analysis did not consider the reduction of mass flow during the valve closure time of five seconds; full flow was assumed until the lapse of five seconds.

e. No credit was taken for radioactive decay.

f. No credit was taken for an elevated release.

Based on the foregoing assumptions, the doses to the thyroid and whole body were both conservatively calculated to be less than 1 rem at the exclusion area boundary. The doses from this accident are well within the values set forth in 10CFR100.

4. Total dose due to a postulated LOCA

The total dose attributed to a postulated LOCA is the combined doses due to containment leakage ESF equipment leakage, and releases through the containment pressure relief line. The combined EAB doses are 59.0 rem to the thyroid and 0.70 rem to the whole body. The combined LPZ doses are 44.0 rem to the thyroid and 0.25 rem to the whole body. As expected, the doses are below the values set forth in 10CFR100.

The dose to personnel engaging in mineral extraction operations within the exclusion area, in the event of a postulated LOCA, would be less than the dose values of 300 rem to the thyroid and 25 rem to the whole body set forth in 10CFR100.

5. Dose to the control room occupants

In the event of a Design Basis Accident (DBA), the safety injection actuation signal or a high radiation signal from the control room air intake monitors will initiate emergency recirculation and pressurization of the Control Room air conditioning system. Later, the emergency ventilation air makeup system can be brought into operation as described in Section 9.4.1.

The control room doses were analyzed for various design basis accidents. It was determined that the LOCA doses represent the limiting case. Therefore the methodology and the doses calculated for the LOCA are reported here.

The following assumptions are applied in the calculations of the dose to the control room occupants following the LOCA:

a. The basic assumptions presented in Items 1 and 2, above, are applied, except a constant breathing rate of 3.5×10^{-4} m^3/sec is assumed throughout the accident.

b. The control room pressurization (air intake) and recirculation iodine adsorbers are assigned a 99 percent decontamination efficiency for both elemental and organic iodines in accordance with Table 2 of Regulatory Guide 1.52 (See Appendix 1A(B)). The pressurization adsorbers are arranged in series with the recirculation adsorbers (see Figure 9.4-1) during the emergency pressurization mode, thus providing an equivalent decontaminating efficiency in excess of 99 percent for both elemental and organic iodines from the pressurization makeup air.

c. The control room air-conditioning system runs either in the emergency recirculation mode or the emergency ventilation mode during a LOCA.

d. During the emergency recirculation mode of operation, a constant air intake flow rate of 800 ft^3/min is assumed. This makes up for losses caused by leaks and maintains the control room atmosphere at a positive pressure of 0.125 inch water gauge relative to adjacent areas.

Since both recirculation trains are actuated by the safety injection signal, the outside air intake flow rate during dual train operation is conservatively estimated to be 1600 cfm. If both trains are assumed to operate for one hour, the calculated thyroid dose would decrease, due to the additional iodine filtration. The calculated whole body gamma and beta skin doses would increase slightly due to the additional intake of outside air. In both cases, the calculated doses remain below the limits specified in 10 CFR 50, Appendix A, "General Design Criteria for Nuclear Power Plants," Criterion 19. The reported control room doses are the higher values of the two cases.

e. During the emergency ventilation mode of operation, 3800 ft^3/min of outdoor air is used to introduce fresh air into the control room.

f. The emergency ventilation mode of operation is under administrative control so that the dose to the control room occupants is minimized, and the need for air change is satisfied.

The operating mode sequence used in this analysis is as follows:

Time Period	Operating Mode	Time in Mode	Air Intake Flow Rate	Filtered Recirculation Flow Rate
0 to 96 hours[(a)]	Emergency recirculation	96 hours	800 ft^3/min	7200 ft^3/min
96 to 117 hours	Emergency ventilation	21 hours	3800 ft^3/min	4200 ft^3/min
117 to 720 hours	Emergency recirculation	603 hours	800 ft^3/min	7200 ft^3/min

a) Since both recirculation trains are actuated by the SI signal, one train must be turned off manually by the operators within one (1) hour.

g. The distance from the Containment to the control room air intake is 94 feet, and the air intake is located 56 feet above ground. The distance from the primary plant vent stack (i.e. the ESF leakage release point) to the closest air intake is 138 ft.

h. Atmospheric dilution factors are determined from the following equation based on Reference [24]:

$$X/Q = [U(\pi\sigma_y \sigma_z + A/(K+2)]^{-1}$$

where:

U = wind speed at an elevation of 10 meters (m/sec)

σ_y, σ_z = standard deviation of the gas concentration in the horizontal and vertical crosswind directions, respectively, both being evaluated at a distance of 94 feet for the containment leakage source and at a distance of 138 feet for the ESF leakage source

$$K = \frac{3}{(S/D)^{1.4}}$$

S = distance between containment surface or primary plant vent and closest control room intake

D = diameter of Containment for the containment leakage source and a combination of portions of the Containment and Auxiliary Building for the ESF leakage source

A = projected area of Containment Building (3265 m^2) for the containment leakage source and 2088 m^2 for the ESF leakage source

Table 15.6-22 summarizes the X/Q values calculated utilizing this expression.

i. The CPNPP CREFS design is zone isolation, with filtered recirculation air, and with a positive pressure. This design maximizes the iodine protection factors and minimizes the dose from iodine. The total unfiltered infiltration rate in the control room is conservatively assumed to be 27 cfm, including 10 cfm due to ingress/egress, 2 cfm leakage from the ductwork passing through the control room pressure boundary, and 15 cfm from other sources. Filtered inleakage through the closed inlet bubble tight dampers due to the pressure differential is conservatively included for the impact on whole body due to noble gases. Though negligible, any inleakage through the bubble tight dampers will be filtered by the recirculation filtration units.

Because the control room door ingress/egress is to a stairwell which is equivalent to a two-door vestibule, backflow will not occur with the CPNPP CREFS design and the 10 cfm is not applicable per SRP 6.4. The ductwork has all welded joints which were leak tested prior to operation. Therefore, the assumed unfiltered inleakage from adjacent areas is conservative with respect to the SRP review criteria.

j. Habitability of the control room is based on the following occupancy factors:

Time Period	Occupancy Factor
0 to 24 hours	1.0
1 to 4 days	0.6
4 to 30 days	0.4

k. The air volume in the control room used to determine exposures to operators is 423,032 ft^3.

l. The models for the major contributors to the control room dose are provided in Appendix 15B.

Using the above assumptions and procedures, the thyroid dose is conservatively calculated to be 40.0 rem in the control room for the duration of the accident. The thyroid dose can be further reduced by the use of the available respiratory protection equipment. The total whole body gamma dose is conservatively calculated to be 1.2 rem. This calculated dose includes whole body dose contributions from containment sources (both direct and scattered radiation), the external passing cloud, control room atmosphere, activity buildup on filters, and streaming through doors and penetrations. These calculated doses are less than the limiting values specified in Regulatory Guide 1.195.

The skin dose received in the control room during the accident period is conservatively calculated to be 13.0 rem. This calculated beta skin dose is less than the 50 rem limit allowed by Regulatory Guide 1.195 if special protective clothing and eye protection is not used. However, special protective clothing and eye protection are provided for use, if required, to reduce the beta skin doses to the operators.

6. Environmental consequences of containment purging to control containment hydrogen concentration after a LOCA

   Purging of the containment atmosphere for controlling potential hydrogen accumulation in the Containment following a postulated LOCA is not required. Thus, an analysis of the radiological consequences of containment purging is not provided.

15.6.6 A NUMBER OF BWR TRANSIENTS

This section is not applicable to the CPNPP.

REFERENCES

1. Huegel, D. S., et al., “RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses,” WCAP-14882-P-A, April 1999.

2. Friedland, A. J., and Ray, S., “Revised Thermal Design Procedure,” WCAP-11397-P-A, April 1989.

3. “Acceptance Criteria for Emergency Core Cooling Systems for Light Water Cooled Nuclear Power Reactors,” 10CFR50.46 and Appendix K of 10CFR50. Federal Register, Volume 39, Number 3, January 4, 1974.

4. “Reactor Safety Study - An Assessment of Accident Risks in U. S. Commercial Nuclear Power Plants,” WASH-1400, NUREG-75/014, October 1975.

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20. Deleted

20a. Deleted

21. DiNunno, J.J., Anderson, F. D., Baker, P.E. and Waterfield, R. L., “Calculation of Distance Factors for Power and Test Reactor Sites,” TID-14844, U. S. Nuclear Regulatory Commission, Division of Licensing and Regulation, March 1962.

22. Postma, A. K., and Pasedag, W. F., “A Review of Mathematical Models for Predicting Spray Removal of Fission Products in Reactor Containment Vessels,” WASH-1329, U. S. Nuclear Regulatory Commission, Accident Analysis Branch.

23. Meteorology and Atomic Energy," U. S. Nuclear Regulatory Commission, Division of Technical Information, 1968.

24. Murphy, K. G. and Campe, Dr. K. M., "Nuclear Power Plant Control Room Ventilation System Design for Meeting General Criterion 19", U. S. Nuclear Regulatory Commission.

25. Deleted

26. Deleted

27. Deleted

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33. Deleted

34. Regulatory Guide 1.195, "Methods and Assumptions For Evaluating Radiological Consequences Of Design Basis Accidents At Light-Water Nuclear Power Reactors," May 2003.

35. Deleted

36. Deleted

37. WCAP-10698, "SGTR Analysis Methodology to Determine the Margin to Steam Generator Overfill," August 1987.

38. Supplement 1 to WCAP-10698-P-A, "Evaluation of Offsite Radiation Doses for a Steam Generator Tube Rupture Accident," March 1986.

39. WCAP-14882-P-A, "RETRAN-02 Modeling and Qualification for Westinghouse Pressurized Water Reactor Non-LOCA Safety Analyses," April 1999.

TABLE 15.6-1
PARAMETERS FOR POSTULATED STEAM GENERATOR TUBE RUPTURE ACCIDENT ANALYSIS

(Sheet 1 of 2)

	Parameter	Value
1.	Data and Assumptions Used to Estimate Radioactive Source Term For the Postulated Steam Generator Tube Rupture Accident	
	a. Steam generator tube leak prior to and during accident (gpm)	1.0
	b. Offsite power	lost on reactor trip
2.	Data and Assumptions Used to Estimate Activity Released to the Atmosphere	
	a. Iodine partition factor in steam generators prior to and during accident	0.01
	b. Iodine partition factor in condensers prior to accident	0.15
	c. Duration of plant cooldown by secondary system after accident (hr)	11
3.	Dispersion Data	
	a. Exclusion area boundary and low population zone distances	2080 m and 4 miles
	b. x/Q (sec/m^3) 0 - 2 hours (EAB) 0 - 8 hours (LPZ) 8 - 24 hours (LPZ)	 1.6×10^{-4} 2.4×10^{-5} 1.6×10^{-5}
4.	Dose Data	
	a. Method of dose calculation	See Appendix 15B
	b. Dose conversion assumptions	See Appendix 15B

TABLE 15.6-1
PARAMETERS FOR POSTULATED STEAM GENERATOR TUBE RUPTURE ACCIDENT ANALYSIS

(Sheet 2 of 2)

Parameter	Value
c. Doses (rem)	
With preaccident iodine spike	
@ exclusion area boundary	
(0 - 2 hours)	
Thyroid inhalation	40.0
Whole body gamma	0.14
@ low population zone	
(0 - 8 hours)	
Thyroid inhalation	6.0
Whole body gamma	0.03
With concurrent iodine spike	
@ exclusion area boundary	
(0 - 2 hours)	
Thyroid inhalation	26.0
Whole body gamma	0.18
@ low population zone	
(0 - 8 hours)	
Thyroid inhalation	4.0
Whole body gamma	0.03

TABLE 15.6-2
PREACCIDENT IODINE SPIKE CONCENTRATION IN THE PRIMARY COOLANT[a]

Spike Isotope	Preaccident Iodine Concentration (μCi/gm)
I-131	46.1
I-132	49.2
I-133	73.8
I-134	10.7
I-135	40.6

a) This concentration corresponds to 60 μCi/gm Dose Equivalent I-131, the maximum allowed at Rated Thermal Power by the Technical Specifications.

TABLE 15.6-3
IODINE APPEARANCE RATE TO THE REACTOR COOLANT AFTER THE ACCIDENT

Isotope	Iodine Appearance Rate[(a)] (Ci/min)
I-131	151.1
I-132	513.2
I-133	294.5
I-134	237.2
I-135	232.2

a) Corresponds to 335 times the equilibrium appearance rate based on operation with reactor coolant activity at 1.0 uCi/gm Dose Equivalent I-131.

TABLE 15.6-4
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 1

(Sheet 1 of 3)

	Parameter	As-Analyzed Value or Range	Operating Range or Target Value
1.0	**Plant Physical Description**		
a)	Dimensions	Nominal	N/A
b)	Pressurizer location	On an intact loop	N/A
c)	Hot assembly location	Anywhere in core[a]	N/A
d)	Hot assembly type[b]	17 x 17 OFA fuel design	17 x 17 OFA fuel design
e)	Steam generator tube plugging level	$\leq$ 10%	0% $\leq$ SGTP $\leq$ 10% (in any one SG)
f)	Fuel assembly type[b]	17 x 17 OFA fuel with ZIRLO cladding, non-IFBA or IFBA	17 x 17 OFA fuel with ZIRLO cladding, non-IFBA or IFBA
2.0	**Plant Initial Operating Conditions**		
2.1	Reactor Power		
a)	Core power	$\leq$ 100.6% of 3612 MWt	$\leq$ 3612 MWt
b)	Peak heat flux hot channel factor (F_Q)[b]	$\leq$ 2.5	$\leq$ 2.5
c)	Peak hot rod enthalpy rise hot channel factor ($F_{\Delta H}$)[b]	$\leq$ 1.60	$\leq$ 1.60
d)	Hot assembly radial peaking factor (F_{HA})[b]	$\leq$ 1.60/1.04	$\leq$ 1.60/1.04
e)	Hot assembly heat flux hot channel factor (F_{QHA})	$\leq$ 2.5/1.04	$\leq$ 2.5/1.04

TABLE 15.6-4
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 1

(Sheet 2 of 3)

	Parameter	As-Analyzed Value or Range	Operating Range or Target Value
f)	Axial power distribution (P_{BOT}, P_{MID})[b]	Figure 15.6-13	Figure 15.6-13
g)	Low power region relative power (P_{LOW})[b]	02. $\leq P_{LOW} \leq$ 0.7	02. $\leq P_{LOW} \leq$ 0.7
h)	Hot assembly burnup	$\leq$ 75,000 MWD/MTU, lead rod[a]	$\leq$ 62,000 MWD/MTU, lead rod[a][c]
i)	MTC	$\leq$ 0 at hot full power (HFP)	$\leq$ 0 at hot full power (HFP)
j)	Typical cycle length	18 months	18 months
k)	Minimum core average burnup[b]	$\geq$ 10,000 MWD/MTU	$\geq$ 10,000 MWD/MTU
l)	Maximum steady state depletion, F_Q[b]	2.0	2.0
2.2	Fluid Conditions		
a)	T_{AVG}	574.2 - 6.5ºF $\leq T_{AVG} \leq$ 589.2 + 6.5ºF	574.2 - 589.2
b)	Pressurizer pressure	2250 - 30 psia $\leq P_{RCS} \leq$ 2250 + 30 psia	2250 psia[d]
c)	Loop flow	TDF $\geq$ 95,700 gpm/loop	mmf $\geq$ 99,100 gpm/loop[e]
d)	Upper head design	T_{COLD}	T_{COLD}
e)	Pressurizer level	60.0% of span (T_{AVG} of 589.2ºF) 43.1% of span (T_{AVG} of 574.2ºF)	60.0% of span (T_{AVG} of 589.2ºF)[d] 43.1% of span (T_{AVG} of 574.2ºF)[d]

TABLE 15.6-4
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 1

(Sheet 3 of 3)

Parameter			As-Analyzed Value or Range	Operating Range or Target Value
	f)	Accumulator temperature	$88^{o}F \leq T_{ACC} \leq 120^{o}F$	$88^{o}F \leq T_{ACC} \leq 120^{o}F$
	g)	Accumulator pressure	603 psia $\leq P_{ACC} \leq$ 693 psia	603 psia $\leq P_{ACC} \leq$ 693 psia
	h)	Accumulator liquid volume	6119 gallon $\leq V_{ACC} \leq$ 6597 gallon	6119 gallon $\leq V_{ACC} \leq$ 6597 gallon
	i)	Accumulator fL/D	4.3736[(f)]	Current line configuration
	j)	Minimum accumulator boron	$\geq$ 2300 ppm	$\geq$ 2300 ppm
3.0	**Accident Boundary Conditions**			
	a)	Minimum safety injection	Table 15.6-12	Table 15.6-12
	b)	Safety injection temperature	$40^{o}F \leq$ SI Temp $\leq 120^{o}F$	$40^{o}F \leq$ SI Temp $\leq 120^{o}F$
	c)	Safety injection delay	17 seconds (with offsite power) 27 seconds (with LOOP)	$\leq$ 17 seconds (with offsite power) $\leq$ 27 seconds (with LOOP)
	d)	Containment modeling	See Figure 15.6-14 and raw data in Tables 15.6-6 and 15.6-7	See Figure 15.6-14 and raw data in Tables 15.6-6 and 15.6-7
	e)	Minimum containment air partial pressure	13.54 psia	$\geq$ 13.54 psia
	f)	Containment spray initiation delay	22 seconds	$\geq$ 22 seconds
	g)	Recirculation spray initiation delay	Not Applicable	Not Applicable
	h)	Single failure	Loss of one ECCS train	Loss of one ECCS train

a) 44 peripheral locations will not physically be lead power assembly.

b) In the Westinghouse Reload Safety Analysis Checklist (RSAC) process, this parameter is identified as a key safety analysis parameter that could be impacted by a fuel reload.

c) Please note that the fuel temperature and rod internal pressure data is only provided up to 62,000 MWD/MTU. In addition, the hot assembly/hot rod will not have a burnup this high in ASTRUM analyses.

d) Plant control systems are designed to control these parameters to the stated values.

e) $\geq$ TDF plus uncertainties.

f) fL/D based on average L/D of 312.4.

TABLE 15.6-5
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 2

(Sheet 1 of 3)

	Parameter	As-Analyzed Value or Range	Operating Range or Target Value
1.0	**Plant Physical Description**		
a)	Dimensions	Nominal	N/A
b)	Pressurizer location	On an intact loop	N/A
c)	Hot assembly location	Anywhere in core[a]	N/A
d)	Hot assembly type[b]	17 x 17 OFA fuel design	17 x 17 OFA fuel design
e)	Steam generator tube plugging level	$\leq$ 10%	0% $\leq$ SGTP $\leq$ 10% (in any one SG)
f)	Fuel assembly type[b]	17 x 17 OFA fuel with ZIRLO cladding, non-IFBA or IFBA	17 x 17 OFA fuel with ZIRLO cladding, non-IFBA or IFBA
2.0	**Plant Initial Operating Conditions**		
2.1	Reactor Power		
a)	Core power	$\leq$ 100.6% of 3612 MWt	$\leq$ 3612 MWt
b)	Peak heat flux hot channel factor (F_Q)[b]	$\leq$ 2.5	$\leq$ 2.5
c)	Peak hot rod enthalpy rise hot channel factor ($F_{\Delta H}$)[b]	$\leq$ 1.60	$\leq$ 1.60
d)	Hot assembly radial peaking factor (F_{HA})[b]	$\leq$ 1.60/1.04	$\leq$ 1.60/1.04
e)	Hot assembly heat flux hot channel factor (F_{QHA})	$\leq$ 2.5/1.04	$\leq$ 2.5/1.04

| | Parameter | As-Analyzed Value or Range | Operating Range or Target Value |

TABLE 15.6-5
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 2

(Sheet 2 of 3)

	f)	Axial power distribution (P_{BOT}, P_{MID})[b]	Figure 15.6-27	Figure 15.6-27
	g)	Low power region relative power (P_{LOW})[b]	02. $\leq P_{LOW} \leq$ 0.7	02. $\leq P_{LOW} \leq$ 0.7
	h)	Hot assembly burnup	$\leq$ 75,000 MWD/MTU, lead rod[a]	$\leq$ 62,000 MWD/MTU, lead rod[a][c]
	i)	MTC	$\leq$ 0 at hot full power (HFP)	$\leq$ 0 at hot full power (HFP)
	j)	Typical cycle length	18 months	18 months
	k)	Minimum core average burnup[b]	$\geq$ 10,000 MWD/MTU	$\geq$ 10,000 MWD/MTU
	l)	Maximum steady state depletion, F_Q[b]	2.0	2.0
2.2	Fluid Conditions			
	a)	T_{AVG}	574.2 - 6.5ºF $\leq T_{AVG} \leq$ 589.2 + 6.5ºF	574.2 - 589.2
	b)	Pressurizer pressure	2250 - 30 psia $\leq P_{RCS} \leq$ 2250 + 30 psia	2250 psia[d]
	c)	Loop flow	TDF $\geq$ 95,700 gpm/loop	mmf $\geq$ 99,100 gpm/loop[e]
	d)	Upper head design	T_{COLD}	T_{COLD}
	e)	Pressurizer level	60.0% of span (T_{AVG} of 589.2ºF) 43.1% of span (T_{AVG} of 574.2ºF)	60.0% of span (T_{AVG} of 589.2ºF)[d] 43.1% of span (T_{AVG} of 574.2ºF)[d]

Parameter	As-Analyzed Value or Range	Operating Range or Target Value

TABLE 15.6-5
PLANT OPERATING RANGE ALLOWED BY THE BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNIT 2

(Sheet 3 of 3)

	f)	Accumulator temperature	88ºF $\leq T_{ACC} \leq$ 120ºF	88ºF $\leq T_{ACC} \leq$ 120ºF
	g)	Accumulator pressure	603 psia $\leq P_{ACC} \leq$ 693 psia	603 psia $\leq P_{ACC} \leq$ 693 psia
	h)	Accumulator liquid volume	6119 gallon $\leq V_{ACC} \leq$ 6597 gallon	6119 gallon $\leq V_{ACC} \leq$ 6597 gallon
	i)	Accumulator fL/D	4.3736[(f)]	Current line configuration
	j)	Minimum accumulator boron	$\geq$ 2300 ppm	$\geq$ 2300 ppm
3.0	**Accident Boundary Conditions**			
	a)	Minimum safety injection	Table 15.6-12	Table 15.6-12
	b)	Safety injection temperature	40ºF $\leq$ SI Temp $\leq$ 120ºF	40ºF $\leq$ SI Temp $\leq$ 120ºF
	c)	Safety injection delay	17 seconds (with offsite power) 27 seconds (with LOOP)	$\leq$ 17 seconds (with offsite power) $\leq$ 27 seconds (with LOOP)
	d)	Containment modeling	See Figure 15.6-28 and raw data in Tables 15.6-6 and 15.6-7	See Figure 15.6-28 and raw data in Tables 15.6-6 and 15.6-7
	e)	Minimum containment air partial pressure	13.54 psia	$\geq$ 13.54 psia
	f)	Containment spray initiation delay	22 seconds	$\geq$ 22 seconds
	g)	Recirculation spray initiation delay	Not Applicable	Not Applicable
	h)	Single failure	Loss of one ECCS train	Loss of one ECCS train

a) 44 peripheral locations will not physically be lead power assembly.

b) In the Westinghouse Reload Safety Analysis Checklist (RSAC) process, this parameter is identified as a key safety analysis parameter that could be impacted by a fuel reload.

c) Please note that the fuel temperature and rod internal pressure data is only provided up to 62,000 MWD/MTU. In addition, the hot assembly/hot rod will not have a burnup this high in ASTRUM analyses.

d) Plant control systems are designed to control these parameters to the stated values.

e) $\geq$ TDF plus uncertainties.

f) fL/D based on average L/D of 355.2.

TABLE 15.6-6
LARGE-BREAK LOCA CONTAINMENT DATA USED FOR CALCULATION OF CONTAINMENT PRESSURE FOR COMANCHE PEAK UNITS 1 AND 2

(Sheet 1 of 2)

Structural Heat Sinks

	Wall (feet)	T_{air} (ºF)	Area (ft^2)	Height (ft)	$T_{initial}$ (ºF)
1.	0.000583 paint, 0.041670 steel, 2.458 concrete	40.0	29,487.0	10.0	88.0
2.	0.000583 paint, 0.031330 steel, 4.467 concrete	40.0	84,091.0	10.0	88.0
3.	0.000583 paint, 0.125 steel	88.0	3371.0	10.0	88.0
4.	0.000583 paint, 4.27 concrete	88.0	198,763.5	10.0	88.0
5.	0.000583 paint, 0.00125 steel	88.0	1,078,360.0	10.0	88.0
6.	0.000583 paint, 0.02630 steel	88.0	93,014.86	10.0	88.0
7.	0.000583 paint, 0.03980 steel	88.0	7477.66	10.0	88.0
8.	0.000583 paint, 0.048580 steel	88.0	2520.82	10.0	88.0
9.	0.000583 paint, 0.06250 steel	88.0	753.28	10.0	88.0
10.	0.000583 paint, 0.08330 steel	88.0	2061.96	10.0	88.0
11.	0.000583 paint, 0.15920 steel	88.0	3239.25	10.0	88.0
12.	0.000583 paint, 0.20830 steel	88.0	6819.36	10.0	88.0
13.	0.000583 paint, 0.250 steel	88.0	963.64	10.0	88.0

TABLE 15.6-6
LARGE-BREAK LOCA CONTAINMENT DATA USED FOR CALCULATION OF CONTAINMENT PRESSURE FOR COMANCHE PEAK UNITS 1 AND 2

(Sheet 2 of 2)

Structural Heat Sinks

	Wall (feet)	T_{air} (°F)	Area (ft^2)	Height (ft)	$T_{initial}$ (°F)
14.	0.000583 paint, 0.500 steel	88.0	7012.0	10.0	88.0
15.	0.000583 paint, 0.750 steel	88.0	730.0	10.0	88.0
16.	0.000583 paint, 1.500 steel	88.0	1287.0	10.0	88.0
17.	0.000583 paint, 0.710 steel	88.0	1152.1	10.0	88.0
18.	0.000583 paint, 6.500 concrete, 0.020830 steel, 12.0 concrete	40.0	3012.0	10.0	88.0
19.	0.000583 paint, 2.500 concrete, 0.020830 steel, 12.0 concrete	40.0	8798.0	10.0	88.0
20.	0.000583 paint, 0.016250 steel	40.0	4086.8	10.0	88.0

TABLE 15.6-7
LARGE-BREAK LOCA CONTAINMENT DATA USED FOR CALCULATION OF CONTAINMENT PRESSURE FOR COMANCHE PEAK UNITS 1 AND 2

Containment Net Free Volume	3,063,000ft^3
Initial Conditions	
Minimum air initial containment partial pressure at full power operation	13.54 psia
Minimum steam initial containment partial pressure at full power operation	0.66 psia
Minimum initial containment temperature at full operation	88ºF
RWST temperature	40.0ºF
Temperature outside containment	40.0ºF
Initial spray temperature	40.0ºF
Spray System	
Number of containment spray pumps operating	4
Post-accident containment spray system initiation delay	22.0 sec(a)
Maximum spray system flow from all containment spray pumps	18,000 gal/min.
Fan Coolers	Not Applicable
Recirculation spray	Not Applicable

a) Assumes offsite power is available.

TABLE 15.6-8
PEAK CLAD TEMPERATURE INCLUDING ALL PENALTIES AND BENEFITS, BEST-ESTIMATE LARGE BREAK LOCA (BE LBLOCA) FOR COMANCHE PEAK UNIT 1

PCT for Analysis-of-Record (AOR)	**1492ºF**
PCT Assessments Allocated to AOR	
1. Revised Heat Transfer Multiplier Distributions	-6ºF
Fuel Pellet Thermal conductivity Degradation and Peaking Factor Burndown	+122ºF
BE LBLOCA PCT for Comparison to 10 CFR 50.46 Requirements	1608ºF

TABLE 15.6-9
COMANCHE PEAK UNIT 1 BEST-ESTIMATE LARGE BREAK LOCA RESULTS

10 CFR 50.46 Requirement	Value	Criteria
95/95 PCT[a] (ºF)	1614	<2,200
95/95 LMO[b] (%)	0.23	<17
95/95 CWO[c] (%)	0.0	<1

a) Peak Cladding Temperature

b) Local Maximum Oxidation

c) Core-Wide Oxidation

TABLE 15.6-10
PEAK CLAD TEMPERATURE INCLUDING ALL PENALTIES AND BENEFITS, BEST-ESTIMATE LARGE BREAK LOCA (BE LBLOCA) FOR COMANCHE PEAK UNIT 2

PCT for Analysis-of-Record (AOR)	**1632ºF**
PCT Assessments Allocated to AOR	
1. Revised Heat Transfer Multiplier Distributions	-17ºF
Fuel Pellet Thermal Conductivity Degradation and Peaking Factor Burndown	+190ºF
BE LBLOCA PCT for Comparison to 10 CFR 50.46 Requirements	1805ºF

TABLE 15.6-11
COMANCHE PEAK UNIT 2 BEST-ESTIMATE LARGE BREAK LOCA RESULTS

10 CFR 50.46 Requirement	Value	Criteria
95/95 PCT[a] (ºF)	1822	<2,200
95/95 LMO[b] (%)	0.71	<17
95/95 CWO[c] (%)	0.0	<1

a) Peak Cladding Temperature

b) Local Maximum Oxidation

c) Core-Wide Oxidation

TABLE 15.6-12
TOTAL MINIMUM INJECTED SAFETY INJECTION FLOW USED IN BEST-ESTIMATE LARGE-BREAK LOCA ANALYSIS FOR COMANCHE PEAK UNITS 1 AND 2

RCS Pressure (psia)	Flow Rate (gpm)
14.7	3864.2
34.7	2538.5
54.7	1963.8
74.7	1324.2
94.7	609.4
114.7	604.7
114.8	0.0

TABLE 15.6-13
INPUT PARAMETERS USED IN THE SMALL BREAK LOCA ANALYSIS

	Unit 1	Unit 2
100% Licensed Core Power[(a)] (MWt)	3612	
Peak Linear Power (kW/ft)	13.897	
Total Peaking Factor [F_Q]	2.50	
Axial Peaking Factor [F_Z]	1.5625	
Hot Channel Enthalpy Rise Factor [$F_{\Delta H}$]	1.60	
Hot Assembly Peaking Factor [P_{HA}]	1.4245	
Power Shape	Figure 15.6.5-30	
Fuel Type	17x17 Vantage 5 with IFMs	
Accumulator Water Volume (ft^3)	850	
Accumulator Gas Pressure, Minimum (including uncertainties), (psia)	603	
Pumped Safety Injection Flow	Tables 15.6.5-11 and 15.6.5-12 Tables 15.6.5-31 and 15.6.5-32	
Thermal Design Flow (gpm/loop)	95,700	
Nominal Vessel Average Temperture (ºF)	574.2 - 589.2	
Reactor Coolant Pressure (including uncertainties) (psia)	2280	
Steam Pressure, (psia)	1006.39	949.24
Steam Generator Tube Plugging Level (%)	10	
AFW Flow (Minimum) to all 4 Steam Generators (gpm)	430 (107.5 gpm/SG * 4)	

a) 0.6 percent is added to this power to account for calorimetirc uncertainty. Reactor coolant pump heat is not modeled in LOCA analysis.

TABLE 15.6-14
SAFETY INJECTION FLOWS FOR 2-INCH TO 6-INCH BREAK SIZES
(SPILLING TO RCS PRESSURE)
(UNIT 1 AND UNIT 2)

(Sheet 1 of 2)

RCS Pressure (psia)	Injected Flow (lbm/s)	Spilled Flow (lbm/s)
14.7	527.37	199.93
34.7	501.77	190.22
114.7	375.32	142.33
134.7	330.65	125.43
154.7	274.99	104.37
174.7	190.59	72.51
194.7	89.88	31.43
214.7	89.88	31.84
314.7	87.24	31.27
414.7	83.89	30.06
514.7	80.43	28.81
614.7	76.81	27.49
714.7	72.89	26.09
814.7	68.79	24.60
914.7	64.45	23.04
1014.7	59.76	21.32
1114.7	54.37	19.36
1214.7	48.08	17.07
1314.7	40.89	14.45
1414.7	29.56	10.31
1514.7	23.02	7.94

TABLE 15.6-14
SAFETY INJECTION FLOWS FOR 2-INCH TO 6-INCH BREAK SIZES
(SPILLING TO RCS PRESSURE)
(UNIT 1 AND UNIT 2)

(Sheet 2 of 2)

RCS Pressure (psia)	Injected Flow (lbm/s)	Spilled Flow (lbm/s)
1614.7	21.69	7.47
1714.7	20.26	6.99
1814.7	18.79	6.48
1914.7	17.29	5.96
2014.7	15.72	5.42
2114.7	13.70	4.72
2214.7	11.28	3.88
2314.7	7.45	2.57
2414.7	0	0

TABLE 15.6-15
SAFETY INJECTION FLOWS FOR 8.75-INCH BREAK SIZE
(SPILLING TO CONTAINMENT PRESSURE)
(UNIT 1 AND UNIT 2)

(Sheet 1 of 2)

RCS Pressure (psia)	Injected Flow (lbm/s)	Spilled Flow (lbm/s)
14.7	527.63	190.65
34.7	346.45	344.63
54.7	267.12	397.75
74.7	180.32	451.99
94.7	83.82	509.29
114.7	83.17	507.64
134.7	82.49	507.78
214.7	79.81	508.30
314.7	76.37	508.98
414.7	72.80	509.64
514.7	68.97	510.28
614.7	65.01	510.94
714.7	60.83	511.60
814.7	56.43	512.28
914.7	51.70	512.97
1014.7	46.47	513.65
1114.7	40.65	514.38
1214.7	32.94	517.74
1314.7	21.01	518.82
1414.7	19.10	519.57
1514.7	17.74	519.44

TABLE 15.6-15
SAFETY INJECTION FLOWS FOR 8.75-INCH BREAK SIZE
(SPILLING TO CONTAINMENT PRESSURE)
(UNIT 1 AND UNIT 2)

(Sheet 2 of 2)

RCS Pressure (psia)	Injected Flow (lbm/s)	Spilled Flow (lbm/s)
1614.7	16.34	519.73
1714.7	14.72	520.02
1814.7	12.89	520.28
1914.7	10.95	520.57
2014.7	8.54	522.85
2114.7	5.89	523.24
2214.7	2.35	523.64
2314.7	0	524.00
2414.7	0	524.00

TABLE 15.6-16
NOTRUMP TRANSIENT RESULTS
(UNIT 1)

Event (sec)	2-inch[a]	3-inch	4-inch	6-inch	8.75-inch[a]
Break Initiated	0.0	0.0	0.0	0.0	0.0
Reactor Trip Signal	95.94	20.92	12.17	7.56	6.28
Safety Injection Signal	105.75	30.60	21.25	14.56	8.61
Safety Injection Begins[b]	127.25	52.60	43.25	36.56	30.61
Loop Seal Clearing Occurs[c]	1370	~535	~315	70, 135	~22
Core Uncovery	N/A	~955	~715	~440	~22
Accumulator Injection Begins	N/A	N/A[d]	~970	~405	~200
Core Recovery	N/A	~3000	~1520	~485	N/A
RWST Low Level[e]	1608.5	1600.19	1586.95	1574.98	1474.99

a) There is no core uncovery for the 2-inch and 8.75-inch cases, and only minimal core uncovery for the 6-inch case.

b) Safety injection begins 22.0 seconds (SI delay time) after the safety injection signal is reached.

c) Loop seal clearing is considered to occur when the broken loop seal vapor flow rate is sustained above 1 lbm/s.

d) Accumulator actually injects at 3485 seconds for Unit 1, but this is after the core has recovered and has no bearing on the LOCA recovery.

e) The analysis assumes minimum usable RWST volume (440,300 gal) before the low-1 RWST water level signal for switchover to cold leg recirculation is reached.

TABLE 15.6-17
SBLOCTA BOL RESULTS
(UNIT 1)

Result	2-inch[a]	3-inch	4-inch	6-inch[a]	8.75-inch[a]
PCT, ^{o}F	N/A	950	1013	N/A	N/A
PCT Time, sec		2026	1058		
PCT Elevation, ft		10.75	11.00		
HR Burst Time[b], sec		N/A	N/A		
HR Burst Elevation[b], ft		N/A	N/A		
Max. Local Transient ZrO_2, %		0.02	0.01		
Max. Local ZrO_2 Elevation, ft		10.75	11.00		
Average ZrO_2, %		0.00	0.00		

a) The core either does not uncover or only uncovers for a very short time and therefore does not warrant SBLOCTA calculations for these break sizes.

b) None of the hot rods nor the hot assembly rods burst during the SBLOCTA calculations.

TABLE 15.6-18
NOTRUMP TRANSIENT RESULTS
(UNIT 2)

Event (sec)	2-inch[a]	3-inch	4-inch	6-inch	8.75-inch[a]
Break Initiated	0.0	0.0	0.0	0.0	0.0
Reactor Trip Signal	101.81	35.60	12.52	7.79	6.19
Safety Injection Signal	111.82	44.53	21.99	14.81	8.58
Safety Injection Begins[b]	133.82	66.53	43.99	36.81	30.58
Loop Seal Clearing Occurs[c]	1140	510	255	40, 60, 135	~17
Core Uncovery	N/A	~670	~575	~420	N/A
Accumulator Injection Begins	N/A	N/A[d]	~850	~380	~170
Core Recovery	N/A	~2800	~1630	~470	N/A
RWST Low Level[e]	1609.1	1599.1	1585.37	1574.83	1474.55

a) There is no core uncovery for the 2-inch and 8.75-inch cases, and only minimal core uncovery for the 6-inch case.

b) Safety injection begins 22.0 seconds (SI delay time) after the safety injection signal is reached.

c) Loop seal clearing is considered to occur when the broken loop loop seal vapor flow rate is sustained above 1 lbm/s.

d) Accumulator actually injects at ~3330 seconds for Unit 2, but this is after the core has recovered and has no bearing on the LOCA recovery.

e) The analysis assumes minimum usable RWST volume (440,300 gal) before the low-1 RWST water level signal for switchover to cold leg recirculation is reached.

TABLE 15.6-19
SBLOCTA BOL RESULTS
(UNIT 2)

Result	2-inch[a]	3-inch	4-inch	6-inch[a]	8.75-inch[a]
PCT, ^{o}F		1069	1210		
PCT Time, sec		1788	920		
PCT Elevation, ft		11.00	11.00		
HR Burst Time[b], sec	N/A	N/A	N/A	N/A	N/A
HR Burst Elevation[b], ft		N/A	N/A		
Max. Local Transient ZrO_2, %		0.04	0.06		
Max. Local ZrO_2 Elevation, ft		11.00	11.00		
Average ZrO_2, %		0.01	0.01		

a) The core either does not uncover or only uncovers for a very short time and therefore does not warrant SBLOCTA calculations for these break sizes.

b) None of the hot rods nor the hot assembly rods burst during the SBLOCTA calculations.

TABLE 15.6-20
PARAMETERS FOR POSTULATED LOCA ANALYSIS[a]

(Sheet 1 of 2)

1.	Data and assumptions used to estimate radioactive source from postulated accidents		
	a.	Power level (MWt)	3684
	b.	Plateout of iodine activity released to containment	50%
	c.	Activity released to containment and available for release to environment	
		Noble gases	100% of core inventory
		Iodines	25% of core inventory
	d.	Iodine fractions (organic, elemental, and particulate)	Regulatory Guide 1.195
2.	Data and assumptions used to estimate activity released		
	a.	Containment volume	
		Directly sprayed volumes	1.706×10^6 ft^3
		Unsprayed volumes affected by spray	1.325×10^6 ft^3
		Total free volume	3.031×10^6 ft^3
	b.	Mixing rate between sprayed and unsprayed volumes	2 turnovers per hour
	c.	Spray removal coefficient	
		Elemental iodine	10 hr^{-1}
		Organic iodines	0 hr^{-1}
			11.4 hr^{-1} for DF less than or equal to 50
		Particulate iodine	1.14 hr^{-1} for DF greater than 50
	d.	Effective decontamination factor of spray for elemental iodine	100
	e.	Containment leak rate	0.1% containment volume per day, ($0 \le t \le 24$ hrs) 0.05% of containment volume per day, ($t > 24$ hrs)

TABLE 15.6-20
PARAMETERS FOR POSTULATED LOCA ANALYSIS[a]

(Sheet 2 of 2)

	f.	Duration of removal effectiveness	
		Elemental iodine	2.518 hr (DF of 100 attained)
		Particulate iodine	2.03 hr (DF of 50 attained)
			4.0 hr sprays assumed to be terminated
3.	Dispersion data		
	a.	Exclusion area boundary and LPZ distances	2080 m and 4 miles
	b.	x/Q's (for time intervals of 2 hours, 8 hours, 24 hours, 4 days, 30 days)	@ EAB, onsite 5 percentile data 1.6×10^{-4} sec/m^3 (0-2 hrs) @LPZ, onsite 5 percentile data 2.4e-05 sec/m^3 (0-8 hrs) 1.6e-05 sec/m^3 (8-24 hrs) 6.2e-06 sec/m^3 (1-4 days) 1.7e-06 sec/m^3 (4-30 days)
4.	Dose due to containment and ESF equipment leakage and the containment pressure relief line.		
	a.	Method of dose calculation	See Appendix 15B
	b.	Dose conversion assumptions	See Appendix 15B
	c.	Doses	@EAB, (0-2 hrs) thyroid = 59.0 rem whole body gamma = 0.7 rem @LPZ (0-30 days) thyroid = 44.0 rem whole body gamma = 0.25 rem

a) Per Regulatory Guide 1.195

TABLE 15.6-21
ACTIVITY AVAILABLE FOR RELEASE VIA ESF COMPONENTS AT T=O
FOLLOWING LOCA

ISOTOPE	ACTIVITY IN CURIES
I-131	5.1×10^{7}
I-132	7.35×10^{7}
I-133	1.03×10^{8}
I-134	1.15×10^{8}
I-135	9.80×10^{7}

TABLE 15.6-22
ATMOSPHERIC DILUTION FACTORS IN CONTROL ROOM DOSE ANALYSIS

A. Containment Leakage[a]

Period	Corresponding Wind Speed Percentile	Representative Wind Speed (m/sec)	Representative Wind Speed Factor	Representative Wind Direction Factor, F=0.352	Control Room Occupancy Factor	Overall Reduction Factor	X/Q (sec/m^3)
0-8 Hours	5	0.75	1	1	1	1	3.04×10^{-3}
8-24 Hours	10	1.05	0.71	0.84	1	0.60	1.82×10^{-3}
1-4 Days	20	1.49	0.50	0.68	0.6	0.20	6.08×10^{-4}
4-30 Days	40	2.25	0.33	0.352	0.4	0.046	1.40×10^{-4}

B. ESF Equipment Leakage Outside Containment[b]

Period	Corresponding Wind Speed Percentile	Representative Wind Speed (m/sec)	Representative Wind Speed Factor	Representative Wind Direction Factor, F=0.401	Control Room Occupancy Factor	Overall Reduction Factor	X/Q (sec/m^3)
0-8 Hours	5	0.83	1	1	1	1	2.96×10^{-3}
8-24 Hours	10	1.12	0.74	0.85	1	0.63	1.86×10^{-3}
1-4 Days	20	1.56	0.53	0.70	0.6	0.22	6.51×10^{-4}
4-30 Days	40	2.32	0.36	0.401	0.4	0.06	1.78×10^{-4}

a) S, SSW, SW, WSW, W, WNW and NW are considered to be affected sectors.

b) SW, WSW, W, WNW, NW and NNW are considered to be affected sectors.

15.7 RADIOACTIVE RELEASE FROM A SUBSYSTEM OR COMPONENT

15.7.1 RADIOACTIVE GAS WASTE SYSTEM LEAK OR FAILURE

15.7.1.1 Identification of Causes and Accident Description

The Gaseous Waste Processing System (GWPS), as discussed in Chapter 11, is designed to remove fission product gases from the reactor coolant. The system consists of a closed loop with waste gas compressors, hydrogen recombiners, waste gas decay tanks for service at power and other waste gas decay tanks for service at shutdown and startup.

The accident is defined as an unexpected and uncontrolled release of radioactive xenon and krypton fission product gases stored in a waste decay tank as a consequence of a failure of a single gas decay tank or associated piping.

Nonvolatile fission product concentrations are greatly reduced as the coolant is passed through the purification demineralizers. An iodine removal factor of 10 is expected in the mixed bed demineralizers, and an iodine partition factor of 100 is expected between the liquid and vapor phases in the volume control tank. Based on the above analysis and operating experience at Yankee-Rowe and Saxton, activity stored in a gas decay tank consists of the noble gases released from the processed coolant and only negligible quantities of less volatile isotopes.

15.7.1.2 Analysis of Effects and Consequences

The maximum activity in a single gas decay tank, the assumptions on which the activity is based, and the radiological consequences resulting from the release of this activity are discussed in Section 15.7.1.3.

15.7.1.3 Environmental Consequences

With the exception of the gas decay tank inventory assumption, the assumptions and equations delineated in Regulatory Guide 1.24 and summarized in Table 15.7-1 are used to evaluate the radiological consequences resulting from the postulated tank rupture. The noble gas inventory in the tank at the time of the rupture is assumed to be the activity in a single waste gas decay tank immediately after it has been isolated from the GWPS (see Table 15.7-2).

The GWPS is designed with sufficient tank capacity to store radioactive gases generated by the plant to allow for holdup of the gases.

The system is also designed to operate so that release of the maximum activity in any one gas decay tank, from a rupture of the tank, will not result in offsite doses that exceed the values set forth in l0CFRl00. The system is equipped with a radiation monitor which is installed so that it indicates the concentration level in the tank on stream. The monitor alarms at the waste panel when the concentration level in the on stream tank reaches a predetermined level. Initiation of the alarm on the waste panel also initiates a general alarm in the Control Room. The gaseous activity in the tanks then decays while the other tanks in the system are being filled with gaseous radioactivity. The maximum activity that can be released as a result of a gas decay tank rupture is the activity stored in one gas decay tank immediately after it has been isolated from the GWPS.

The entire content of a single waste gas decay tank inventory of 200,000 Ci of noble gases (considered as dose equivalent Xe-133) is assumed to be released to the Auxiliary Building, and all of the noble gases are assumed to leak from the building at ground level over a 2 hour period. Based on this model and an onsite 0 to 2 hr atmospheric dilution factor ($1.6x10^{-4}$ sec/m^3), the whole body dose at the nearest point on the exclusion area boundary (EAB) is conservatively calculated to be 0.19 rem. This dose is substantially below the 25 rem whole body value set forth in 10CFR100; it may be concluded that such an incident would not interrupt or restrict public use of areas beyond the EAB.

15.7.2 RADIOACTIVE LIQUID WASTE SYSTEM LEAK OR FAILURE

15.7.2.1 Identification of Causes and Accident Description

The accident is defined as the uncontrolled atmospheric release from the 30,000 gallon floor drain tank due to the postulated rupture of the tank. This tank is the highest potential atmospheric release source term because of its large volume and the fact that it is assumed to be 80 percent full of reactor coolant.

15.7.2.2 Analysis of Effects and Consequences

The activity in the 30,000 gallon floor drain tank, the assumptions on which the activity is based, and the radiological consequences resulting from the release of the activity are discussed in Section 15.7.2.3.

15.7.2.3 Environmental Consequences

This analysis assumes that the plant has been operating with 1 percent failed fuel for an extended period sufficient to achieve equilibrium radioactive concentrations. Floor drain tank III is assumed to contain the inventory as indicated in Table 15.7-3. The entire contents of the tank are assumed to be released to the atmosphere at ground level over a 2 hr period. Other conservative assumptions are detailed in Table 15.7-4.

Based on the foregoing model, the thyroid and whole body doses at the EAB are conservatively calculated to be 2.1 rem and 3.8E-03 rem, respectively. The doses from this accident are substantially below the values set forth in 10CFR100.

15.7.3 POSTULATED RADIOACTIVE RELEASES DUE TO LIQUID TANK FAILURES

The analysis is presented in Section 2.4.12 and 2.4.13.3.

15.7.4 DESIGN BASIS FUEL HANDLING ACCIDENTS

15.7.4.1 Identification of Causes and Accident Description

The accident is defined as dropping of a spent fuel assembly in the Containment Building or spent fuel pool fuel storage area floor resulting in the rupture of the cladding of all the fuel rods in the assembly despite many administrative controls and physical limitations imposed on fuel handling operations. All refueling operations are conducted in accordance with prescribed procedures under the direct surveillance of a supervisor.

15.7.4.2 Analysis of Effects and Consequences

Method of Analysis

The method of analysis used for evaluating the potential radiological consequences of a fuel handling accident is in compliance with Regulatory Guide 1.195.

Fission Product Inventories

The actual fission product gap inventory in the fuel assembly is dependent on the linear heat generation rate of the assembly and the temperature of the fuel. However, the gap inventories assumed in fuel handling accident analyses were based on the conservative guidance contained in Regulatory Guide 1.195. Table 15.7-6 lists the fission product activities released from the damaged fuel rods at the time of the fuel handling accident. These activities are consistent with the assumptions used in analyzing the environmental consequences of postulated fuel handling accidents detailed in Section 15.7.4.3.

Iodine Decontamination Factors

The elemental iodine DF of 400 presented in RG 1.195, is based a fuel depth of 23 ft. and an internal fuel rod pressure of 1200 psig. RG 1.195 states that "If the depth of water is not at least 23 feet, the DF will have to be determined on a case-by-case basis," and "For release pressures greater than 1,200 psig, the iodine DF's will be less than those assumed in this guide and must be calculated on a case-by-case basis using assumptions comparable in conservatism to those of this guide." Since the analysis uses conditions of 1500 psig fuel rod internal pressure and a 21-foot pool depth, a new overall DF that is comparable in conservatism to the RG 1.195 value was calculated. This yields an overall pool decontamination factor of 160 for iodine.

15.7.4.3 Environmental Consequences

15.7.4.3.1 Postulated Fuel Handling Accident Outside Containment

The analysis of a postulated fuel handling accident is performed as follows:

1. The accident is assumed to occur 50 hours after plant shutdown.

2. All of the rods in one fuel assembly are ruptured.

3. The damaged assembly is the highest powered assembly in the core region to be discharged. The values for individual fission product inventories in the damaged assembly are calculated assuming full power operation at the end of core life and a 50 hour shutdown. A radial peaking factor of 1.65 is used.

4. The minimum water depth between the top of the damaged fuel rods and the spent fuel pool surface is 21 ft.

5. All of the gap activity in the rods of the damaged assembly is released to the spent fuel pool. It is assumed that 10% of the fuel in the damaged assembly has an average burnup greater than 54 GWD/MTU and average rod power in excess of 6.3 kW/ft. The gap activity in the 10% of the assembly that exceeds the above limits is 30% for krypton-85,

12% for iodine-131, and 10% for the remaining iodines and noble gases. The gap activity for the remaining 90% of the assembly is 10% for krypton-85, 8% for iodine-131, and 5% for the remaining iodines and noble gases. The gap activities released as a result of this accident are presented in Table 15.7-6.

6. All activity released from the spent fuel pool is released at ground level to the environment over a 2 hour period.

7. The iodine gap inventory is composed of inorganic species (99.75 percent) and organic species (0.25 percent).

8. The overall decontamination factor for the spent fuel pool is 160.

9. No credit is taken for iodine filtration by the primary plant ventilation system.

10. Atmospheric diffusion conditions are assumed to be the 0 to 2 hr ground level case.

The parameters used for this analysis are listed in Table 15.7-7.

Based on the foregoing assumptions, the thyroid and whole body doses at the EAB are conservatively calculated to be 26.0 rem and 0.14 rem, respectively. The corresponding doses at the LPZ are conservatively calculated to be 3.9 rem and 0.021 rem. The calculated doses are within the values set forth in 10CFR100.

15.7.4.3.2 Postulated Fuel Handling Accident Inside Containment

An analysis of the radiological consequences of a fuel handling accident inside the Containment Building would use the same assumptions and yield the same results as those of a fuel handling accident outside the Containment Building. The accident described in the preceding section is considered to represent the limiting case; therefore, no specific analysis of such an accident inside the Containment is provided.

15.7.4.4 Conclusions

The possibility of a fuel handling accident is relatively small due to the many physical, administrative, and safety restrictions imposed on fuel handling operations. However, based on conservative design basis parameters, the calculated doses from a postulated fuel handling accident are well within the values set forth in 10CFR100.

15.7.5 SPENT FUEL CASK DROP ACCIDENTS

The Comanche Peak Nuclear Power Plant Fuel Handling Building overhead crane satisfies NUREG-0554 single-failure-proof requirements and is designed to the requirements of seismic Category I (See Section 3.2). As such it can retain the maximum design load during a Safe Shutdown Earthquake and remain in place under all postulated seismic loadings. The crane is also provided with interlocks which prevent a fuel cask from being lifted more than 29.25 ft above floor elevation or from passing over the new fuel storage area during the spent fuel cask mode of operation (see Section 9.1.2.2). The crane does not pass over the spent fuel pool in any mode of operation. Based on this design approach, the radiological consequences of a spent fuel cask drop accident need not be evaluated.

REFERENCES

1. Bell, M. J., Duhn, E. R., Locante, J. and Malinowski, D. D., “Radiological Consequences of a Fuel Handling Accident,” WCAP- 7518-L (Proprietary) and WCAP-7828 (Non-Proprietary), June 1970.

2. “Radiation Analysis Manual, Standard Plant Model 412, “Rev. 3, Westinghouse Electric Corporation, November 1978.

3. Killough, G. G., Begovich, C. L., Sjoreen, A. L. and Bell, L. W., “A Guide for the TACT III Computer Code,” NUREG/CR-3287, Oak Ridge National Laboratory, May 1983.

TABLE 15.7-1
PARAMETERS FOR POSTULATED WASTE GAS DECAY TANK RUPTURE ACCIDENT

1.	Data and assumptions used to estimate radioactive source from postulated accidents		
	a.	Power level (MWt)	3684
	b.	Percent of fuel defected (%)	1
	c.	Release of activity by nuclide	Table 15.7-2 Values increased to equal 200,000 Ci of dose equivalent Xe-133
2.	All pertinent data and assumptions used to estimate activity released		
	a.	The maximum content of the decay tank is considered for computing noble gas inventory	
	b.	Radiological decay is considered for minimum time required to transfer the gases from the primary system to the decay tank	
	c.	Entire content of one tank is assumed to be released to the building due to failure	
	d.	All of the noble gas is assumed to be released to the atmosphere over a 2 hr period	
3.	Dispersion data		
	a.	Exclusion area boundary (EAB) and low population zone (LPZ) distances	2080 m and 4 miles
	b.	X/Q	1.6×10^{-4} sec/m^3 (0 - 2 hr)
4.	Dose data		
	a.	Method of dose calculations	Appendix 15B
	b.	Doses	dose = 0.19 @EAB, whole body rem (gamma dose)

TABLE 15.7-2
GAS DECAY TANK INVENTORY FOR ACCIDENT ANALYSIS (ONE GAS DECAY TANK)

Isotope	Tank Activity (Ci) Based On Assumptions Below	Tank Activity Corresponding to 200,000 Ci DEQ Xe-133
Kr-85	5.44E + 04	3.28E + 02
Kr-85m	1.66e + 02	1.07E + 05
Kr-87	3.30E +01	6.5E + 01
Kr-88	1.96E + 02	3.87E + 02
Xe-131m	9.12E + 02	1.80E + 03
Xe-133	7.39E + 04	2.19E + 03
Xe-133m	1.11E + 03	1.46E + 05
Xe-135	1.02E + 03	2.01E + 03

Assumptions:

1. 3684 MWT
2. 40 Year continuous operation at 1 percent fuel defects
3. 100 percent stripping efficiency
4. Volume control tank purge rate of 0.7 standard cubic feet per minute (scfm)
5. Tanks switched at regular intervals
6. 8 gas decay tanks shared between two units

TABLE 15.7-3
MAXIMUM RADIOACTIVITY IN A FLOOR DRAIN TANK FOR ACCIDENT ANALYSIS

Isotope	Floor Drain Tank Activity Used To Determine The offsite And Control Room Dose Consequences (Ci)
I-131	2.60E + 02
I-132	2.78E + 02
I-133	4.18E + 02
I-134	6.05E + 01
I-135	2.30E + 02

Assumptions:

a. 3565 MWt

b. Based on 1 percent failed fuel

c. Floor drain tank of 30,000 gal capacity assumed 80 percent full of reactor coolant

TABLE 15.7-4
PARAMETERS FOR POSTULATED RADIOACTIVE LIQUID WASTE SYSTEM LEAK OR FAILURE

1.	Data and assumptions used to estimate radioactive source from postulated accident		
	a.	Power level (MWt)	3684
	b.	Percent of fuel defected (%)	1
	c.	Release of activity by nuclide	Table 15.7-3
2.	Data and assumptions used to estimate activity release		
		a.	Entire content of tank is assumed to be released to the building due to failure
		b.	Activity release to the atmosphere takes place over a two hour period at ground level.
		c.	Iodine Partition factor is 0.1
3.	Dispersion data		
	a.	EAB distance	2080 m
	b.	X/Q	1.6 x 10-4 sec/m^3 (0-2 hr)
4.	Dose data		
	a.	Method of dose calculations	See Appendix 15B
	b.	Doses	@EAB
		thyroid dose =	2.1 rem
		whole body dose =	0.0038 rem (gamma dose)

TABLE 15.7-5
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TABLE 15.7-6
NOBLE GAS AND IODINE ACTIVITIES RELEASED FROM DAMAGED FUEL RODS AS A RESULT OF A FUEL HANDLING ACCIDENT[a]

Nuclide	Curies Released
Kr-83m	1.32E - 2
Kr-85m	5.08E + 0
Kr-88	1.50E - 1
Kr-85	1.16E + 3
Xe-131m	5.08E + 2
Xe-133m	2.18E + 3
Xe-133	8.51E + 4
Xe-135m	7.71E + 1
Xe-135	4.52E + 3
I-131	6.25E + 4
I-132	4.46E + 4
I-133	1.88E + 4
I-135	4.75E + 2

a) Based on the following assumptions:

1. The total gap inventory of one fuel assembly is released to the spent fuel pool.
2. A peaking factor of 1.65 is applied.
3. The accident occurs 50 hours after shutdown.
4. 10% of the rods in the damaged assembly exceed the guidance of RG 1.195, Footnote 7. The Gap activities of the assumed non-compliant 10% of the damaged assembly are 30% for Kr-85, 12% for I-131, and 10% for all other iodines and noble gases. The gap activities for the remaining 90% of the damaged assembly are 10% for Kr-85, 8% for I-131, and 5% for all other iodines and noble gases.

TABLE 15.7-7
PARAMETERS FOR POSTULATED FUEL HANDLING ACCIDENT ANALYSIS

1.	Data and assumptions used to estimate radioactive source from postulated accidents		
	a.	Power level (MWt)	3684
	b.	Release of activity	Regulatory Guide 1.195
	c.	Damage to fuel assembly	All rods ruptured
	d.	Iodine fractions (inorganic and organic)	Regulatory Guide 1.195
2.	Data and assumptions used to estimate activity released		
	a.	Spent fuel pool decontamination factor	160
	b.	All other pertinent data and assumptions	Regulatory Guide 1.195
3.	Dispersion data		
	a.	EAB and LPZ distances	2080 m and 4 miles
	b.	X/Q	@ EAB 1.6 x 10^{-4} sec/m^3 (0 - 2 hr) @ LPZ 2.4 x 10^{-5} sec/m^3 (0 - 8 hr)
4.	Dose data		
	a.	Method of dose calculation	See Appendix 15B
	b.	Dose conversion assumptions	See Appendix 15B
	c.	Doses	@EAB, thyroid dose = 26.0 rem, whole body dose = 0.14 rem (gamma dose) @ LPZ thyroid dose = 3.9 rem, whole body dose = 0.021 rem (gamma dose)

15.8 ANTICIPATED TRANSIENTS WITHOUT SCRAM

A discussion of Anticipated Transients Without Scram (ATWS) is presented in Reference [1]. The information provided in Reference [1] is applicable to the Comanche Peak Nuclear Power Plant.

ATWS procedures, based on procedures generated by the Westinghouse Owner's Group, were reviewed and incorporated into the CPNPP Emergency Operating Procedures prior to fuel load.

The worst common mode failure which is postulated to occur is the failure to trip the reactor after an anticipated transient has occurred. A series of generic studies [1,2] on ATWS showed acceptable consequences would result provided that the turbine trips and auxiliary feedwater flow is initiated in a timely manner. The effects of ATWS events are not considered as part of the design basis for transients analyzed in Chapter 15. The final NRC ATWS rule [3] required that Westinghouse designed plants install an ATWS Mitigation System Actuation Circuitry (AMSAC) to initiate a turbine trip and actuate auxiliary feedwater flow independent of the Reactor Protection System. The CPNPP AMSAC design is described in Section 7.8.

REFERENCES

1. "Westinghouse Anticipated Transients Without Trip Analysis," WCAP-8330, August 1974.

2. Anderson, T. M., "ATWS Submittal," Westinghouse letter NS-TMA-2182 to S. H. Hanaver of the NRC, December 1979.

3. ATWS Final Rule - Code of Federal Regulations, 10 CFR 50.62 and Supplementary Information Package, "Reduction of Risk from Anticipated Transients Without Scram (ATWS) Events for Light- Water-Cooled Nuclear Power Plants."

APPENDIX 15A
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15B DOSE MODELS USED TO EVALUATE THE ENVIRONMENTAL CONSEQUENCES OF ACCIDENTS

15B.1 INTRODUCTION

This section identifies the models used to calculate offsite radiological doses that would result from releases of radioactivity due to various postulated accidents. The postulated accidents are:

1. Steam Line Break
2. Rod Ejection Accident
3. Steam Generator Tube Failure
4. Loss-of-Coolant Accident
5. Radioactive Gas Waste System Leak or Failure
6. Radioactive Liquid Waste System Leak or Failure
7. Design Basis Fuel Handling Accident
8. Primary Coolant Small Line Break Outside Containment
9. Locked Reactor Coolant Pump Rotor

15B.2 ASSUMPTIONS

The following assumptions are basic to the model for the gamma and beta dose due to immersion in a cloud of radioactivity and the model for the thyroid dose due to inhalation of radioactivity.

1. The dose contribution of direct radiation from sources other than the leakage cloud is negligible compared to the dose due to immersion in the leakage cloud.
2. All radioactivity releases are treated as ground level releases regardless of the point of discharge.
3. The dose receptor is a standard man as defined by the International Commission on Radiological Protection (ICRP)[1].
4. Radioactive decay from the point of release to the dose receptor is neglected.
5. Isotopic whole body and beta skin dose conversion factors are taken from references [13] and [14] respectivily.
6. Atmospheric dispersion factors used in the analyses are presented in Section 2.3.

15B.3 GAMMA DOSE

The gamma dose is obtained by considering the dose receptor to be immersed in a radioactive cloud which is infinite in all directions above the ground plane, i.e., a semi-infinite cloud. The concentration of radioactive material within this cloud is taken to be uniform and equal to the maximum centerline ground level concentration that would exist in the cloud at the appropriate distance from the point of release.

The whole body dose is a result of exposure to external gamma radiation. The whole body dose due to immersion in a semi-infinite cloud is given by:

$$D_{\gamma} = K \cdot X/Q \sum_{i}^{A} {}_{Ri} \cdot DCF_{\gamma i} \qquad (15B\text{-}1)$$

where:

D_{γ} is the whole body gamma dose from immersion in a semi-infinite cloud for a given time period, rem.

A_{R_i} is the activity of isotope i released during a given time period, curies

X/Q is the atmospheric dilution factor for a given time period, sec/m3

K is the conversion factor 3.7E12 rem-Bq/Ci-Sv

$DCF_{\gamma i}$ is the whole body gamma dose conversion factor for isotope I, Sv-m^3/Bq-sec

15B.4 THYROID INHALATION DOSE

The thyroid inhalation dose is obtained from the following expression:

$$D_{THY} = K \cdot \chi/Q \cdot B \cdot \sum_{i} Q_{i} \cdot (DCF)_{i} \qquad (15B\text{-}2)$$

where:

D_{THY} = thyroid inhalation dose, rem

X/Q = atmospheric dilution factor for a given time period, sec/m^3

B = breathing rate for a given time period t, m^3/sec

Q_i = total activity of iodine isotope i released in time period t, curies

$(DCF)_i$ = dose conversion factor for iodine isotope i, Sv/Bq inhaled

K is the conversion factor 3.7E12 rem-Bq/Ci-Sv

The thyroid inhalation dose conversion factors and offsite breathing rates used in the above model are given in Table 15B-1.

15B.5 CONTROL ROOM DOSE

The thyroid inhalation, whole body gamma, and beta skin dose models for the major contributors to the Control Room dose are described below. The dose to the Control Room occupants due to a postulated accident is calculated on the basis of source strength, atmospheric transport, dosimetry and Control Room emergency pressurization and filtration considerations as illustrated in the following equations.

The thyroid inhalation dose is obtained from the following expression:

$$D_{THY} = K \cdot \sum_i \sum_j BR \cdot DCF_i \cdot \int_{Tj-1}^{Tj} A_j(t)dt \qquad (15B\text{-}3)$$

where:

i = isotope index

j = time interval index

BR = breathing rate, m^3/sec

DCF_i = dose conversion factor for iodine isotope i, Sv/Bq inhaled

$A_i(t)$ = airborne concentration of iodine isotope i at time t (sec), in the Control Room, curies/m^3

K = is the conversion factor 3.7E12 rem-Bq/Ci-Sv

The thyroid inhalation dose conversion factors and Control Room breathing rate used in the above model are presented in Table 15B-1.

The whole body gamma dose due to inleakage is calculated using the following equation:

$$D_R = K \cdot 1/GF \cdot \sum_i A_i \cdot (DCF)_{\gamma i} \qquad (15B\text{-}4)$$

where:

D_R = Whole body gamma dose, rem

A_i = time integrated concentration of nuclide i, Ci-sec/m^3

K = the conversion factor 3.7E12 rem-Bq/Ci-Sv

GF = Geometry Factor to adjust the dose to reflect the finite cloud in the control room

$(DCF)_{\gamma i}$ = the whole body gamma dose conversion factor for isotope i, Sv-m^3/Bq-sec

$$GF = 1173/V^{0.338} \qquad (15B\text{-}5)$$

V = control room volume, ft^3

The whole body gamma dose conversion factors are presented in Table 15B-1

The beta skin dose due to inleakage is calculated using the following equation:

$$D_B = K \cdot \sum_i A_i \cdot (DCF)_{Bi} \qquad (15B\text{-}6)$$

where:

DB = beta skin dose from immersion in a semi-infinite cloud, rem

A_i = time integrated concentration of nuclide i, Ci-sec/m^3

K = the conversion factor 3.7E12 rem-Bq/Ci-Sv

DCF_{Bi} = the beta skin dose conversion factor for isotope i, Sv-m^3/Bq-sec

The beta skin dose conversion factors are presented in Table 15.B-1.

The whole body gamma dose to Control Room personnel due to a cloud external to the Control Room is calculated using the following equation:

$$D_c = \sum_j \left(X/Q_j \cdot \sum_j A_{ij} \cdot CF \right) \qquad (15B\text{-}7)$$

where:

D_c = whole body gamma dose due to external cloud shine, rem

X/Q_j = atmospheric dispersion factor for the time period j, calculated at the Control Room air intake assuming a ground level release, sec/m3

Ai_j = total activity of nuclide i released during time period j, Ci

CF_i = a dose rate response function for a unit concentration of nuclide i, rem-m^2/Ci-sec.

15B.6 COMPUTER CODES USED IN ACCIDENT ANALYSIS

The computer programs described below may have been used to calculate design-basis source terms and radiological consequences of design basis accidents.

1. RADIOISOTOPE [6]

   Program RADIOISOTOPE calculates the activity of isotopes in a closed system by solving the appropriate simultaneous radioactive decay equations. Based on the activity of any isotope in the system at an initial time, the program calculates the activity of that isotope and its offspring at any later time, provided that the decay scheme is contained in the program library. Furthermore, because gamma activity is important for dose rate and shielding calculations. RADIOISOTOPE also calculates the energy releases in seven gamma energy groups from the decay of an inventory of radionuclides.

2. DRAGON [7]

   Program DRAGON evaluates the activities, dose rates, and time-integrated dose in the reactor building and control room of a nuclear facility or at an adjacent site following release of halogens and noble gases from some control volume. The fission product release to the atmosphere, together with the activities and time-integrated activity concentrations of the halogens which are accumulated in the system, are also computed. Site dose calculations performed by DRAGON employ the semi-infinite cloud models suggested by Regulatory Guide 1.4. The gamma dose in the control room is computed based upon a finite cloud model. Average beta and gamma energies are used in all dose calculations.

3. QADMOD [8]

Program QADMOD calculates dose rates at specified detector locations for a number of different source points representing volumetric sources. The calculational technique employed by this program is commonly referred to as the “point kernel” technique. QADMOD is a modified version of the QAD P-5 program written at Los Alamos Scientific Laboratory by R.E. Malenfant. This modification includes a) the FASTER geometry routines, b) a single point source option, c) a translated cylindrical source volume option, and d) an internal data library which contains flux-to-dose rate conversion factors, buildup factor polynomial coefficients, and mass attenuation coefficients for several materials and compositions.

4. ANISND [9]

   ANISND solves the one-dimensional Boltzmann transport equation for neutrons or gamma rays in rectangular, spherical or cylindrical geometry. The source may be fixed, fission, or a subcritical combination of the two. Criticality search may be performed on any one of several parameters. Cross sections may be weighted using the space and energy dependent flux generated in solving the transport equation. The external cloud dose calculations use Bugle 80, a P-3 cross section library.

5. EFFLMDA [10]

   EFFLMDA calculates the effective spray removal constant as a function of time. The calculation is based on a multi- region model with intermixing among the sprayed and unsprayed regions in a reactor containment.

   The EFFLMDA results are merged with DRAGON input data. This enables the DRAGON program, a single spray region computer code, to simulate a multi-region spray model with step changes in the effective spray constants.

6. RADTRAD [12]

   Program RADTRAD performs similar analysis functions as described above for DRAGON.

REFERENCES

1. ICRP Publication 2, Report of Committee II on Permissible Dose for Internal Radiation (1959). The International Commission on Radiological Protection.

2. Not Used.

3. Not Used.

4. Not Used.

5. Not Used.

6. RADIOISOTOPE, SWEC proprietary computer code NU-007, Version 01, Level 02, April, 1985.

7. DRAGON, SWEC proprietary computer code NU-115, Version 04, Level 02, September 1984 and Version 05 Level 00, April, 1986.

8. QADMOD, SWEC proprietary computer code NU-137U, Version 00, Level 03, November 1985.

9. ANISND, SWEC proprietary computer code NU-146, Version 01, Level 01, January 1985.

10. EFFLMDA, SWEC proprietary computer code NU-159, Version 01, Level 00, December, 1984.

11. International Commission on Radiological Protection, "Limits for Intakes of Radionuclides by Workers," ICRP Publication 30, Volume 3 No. 1-4, 1979.

12. NUREG/CR-6604, "RADTRAD: A Simplified Model for RADionuclide Transport and Removal And Dose Estimation," December 1997.

NUREG/CR-6604, Supplement 1, "RADTRAD: A Simplified Model for RADionuclide Transport and Removal and Dose Estimation," June 1999.

NUREG/CR-6604, Supplement 2, "RADTRAD: A Simplified Model for RADionuclide Transport and Removal And Dose Estimation," October 2002.

13. K. F. Eckerman and J. C. Ryman, "External Exposure to Radionuclides in Air, Water, and Soil," Federal Guidance Report No. 12, EPA 402-R-93-081, Environmental Protection Agency, September 1993.

14. DOE/EH-0070, "External Dose-Rate Conversion Factors for Calculation of Dose to the Public," July 1988.

15. NRC Regulatory Issue Summary 2001-19: "Deficiencies in the Documentation of Design Basis Radiological Analyses Submitted in Conjunction with License Amendment Requests," 10/18/01.

TABLE 15B-1

DOSE CONVERSIONS FACTORS

Isotope	Thyroid DCF[(a)] (Sv/Bg)	Whole Body DCF[(b)] (Sv-m[(c)]/Bg-sec)	Beta-Skin DCF[c] (Sv-m[c]/Bg-sec)
I-131	2.9E-7	1.82E-14	8.66E-15
I-132	1.7E-9	1.12E-13	3.02E-14
I-133	4.9E-8	2.94E-14	2.44E-14
I-134	2.9E-10	1.30E-13	3.87E-14
I-135	8.5E-9	7.98E-14	2.16E-14
Kr-83m	N/A	1.50E-18	0.00E+00
Kr-85m	N/A	7.48E-15	1.38E-14
Kr-85	N/A	1.19E-16	1.35E-14
Kr-87	N/A	4.12E-14	9.08E-14
Kr-88	N/A	1.02E-13	2.13E-14
Xe-131m	N/A	3.89E-16	4.07E-15
Xe-133m	N/A	1.37E-15	8.46E-15
Xe-133	N/A	1.56E-15	2.85E-15
Xe-135m	N/A	2.04E-14	5.85E-15
Xe-135	N/A	1.19E-14	1.75E-14
Xe-138	N/A	5.77E-14	3.99E-14

a) Thyroid DCFs from ICRP-30 (Reference 11).

b) Whole-body DCFs from Federal Guidance Report 12 (Reference 13).

c) Beta-skin DCFs from DOE/EH-0070 (Reference 14). Skin doses are modeled without the contribution from photon emissions as supported by the NRC in Reference 15.

Breathing Rates (m^c/sec)

Time Period (hours)	Offsite	Control Room
0 - 8	3.5E-4	3.5E-4
8 - 24	1.8E-4	3.5E-4
24 - 720	2.3E-4	3.5E-4

16.0 TECHNICAL SPECIFICATIONS

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16.1 PRELIMINARY TECHNICAL SPECIFICATIONS

This section is not applicable; see Section 16.2 for specific information.

16.2 PROPOSED FINAL TECHNICAL SPECIFICATIONS

16.2.1 DISCUSSION

The previous information included in this Section was removed to facilitate compliance with the Standard Technical Specifications (STS) for Westinghouse Pressurized Water Reactors provided to TXU Electric by the NRC (Reference 1). The format of the STS addresses the categories of the Technical Specifications required by 10CFR50.36. Section 16.2.2 briefly describes the submittal and issuance of Final Technical Specifications for CPSES Units 1 and 2.

16.2.2 FINAL TECHNICAL SPECIFICATION

Final Technical Specifications for CPSES Unit 1 based on Westinghouse Standard Technical Specifications as provided by the NRC (Reference 1) were submitted to the NRC for review and approval. The NRC, by NUREG-1399 (Reference 2) approved CPSES Unit 1 Technical Specifications, as Appendix “A” to the CPSES Unit 1 Operating License.

Final Technical Specifications for CPSES Units 1 and 2, based on Westinghouse Standard Technical Specifications provided by the NRC (Reference 1) were submitted to the NRC for review and approval. The NRC, by NUREG-1468 (Reference 3), approved CPSES Units 1 and 2 Technical Specifications, as Appendix “A” to the CPSES Units 1 and 2 Operating Licenses.

These Technical Specifications are updated periodically by the NRC issuance of License Amendments.

Certain information from the Technical Specifications has been relocated to a separate document titled the CPNPP Technical Requirements Manual (TRM).

Changes to the CPNPP TRM are subject to safety evaluations completed in accordance with the provisions of 10CFR50.59.

REFERENCES

1. NRC letter, dated August 14, 1987, “NRC Current Westinghouse Standard Technical Specifications - Comanche Peak - Unit 1”.

2. NUREG-1399, “Technical Specifications - Comanche Peak Steam Electric Station, Unit 1”, April 1990.

3. NUREG-1468, “Technical Specifications - Comanche Peak Steam Electric Station, Unit Nos. 1 and 2”, February, 1993.

17.0 QUALITY ASSURANCE (QA)

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LIST OF FIGURES

Number	Title
17.2-1	Deleted (see Figure 13.1-2)
17.2-2	Deleted (see Figure 13.1-3)
17.2-3	Deleted

Vistra OpCo is submitting this application as Licensee for Comanche Peak Nuclear Power Plant (CPNPP). Vistra OpCo acts as owners agent for construction and operation of CPNPP and is therefore responsible for the design, engineering, procurement, fabrication, and construction technical support of CPNPP. This delegation of authority has been formally established among the Owner. Vistra Operations Company LLC (Vistra OpCo) (operator) and Comanche Peak Power Company LLC (CP PowerCo) (owner) are subsidiaries of Vistra Corp.

To establish and maintain the high quality level required for all quality-related activities for CPNPP, Vistra OpCo has developed a comprehensive Quality Assurance Program (QA Program) as documented in this chapter of the FSAR. Vistra OpCo has implemented those portions of the Quality Assurance Program that are commensurate with the quality activities currently being performed. The program requires, as a minimum, that the quality activities performed by Vistra OpCo and its contractors/vendors comply with the NRC criteria established in 10 CFR Part 50, Appendix B, Licensing of Production and Utilization Facilities, "Quality Assurance Criteria for Nuclear Power Plants". Where appropriate, the requirements of regulatory or safety guides have been incorporated into the program.

The Vistra OpCo Quality Assurance Program requires that a Quality Assurance Manual be established to provide references to the written policies, procedures and instructions used to implement the QA Program for each nuclear power plant project for which it provides service. The combination of the requirements documented in the Quality Assurance Program and the Quality Assurance Manual provides Vistra OpCo with the means of fully executing its assignment.

Appendix 17A identifies all safety-related items for CPNPP within the scope of the Quality Assurance Program.

17.1 QUALITY ASSURANCE DURING DESIGN AND CONSTRUCTION

Section 17.1 has been deleted in its entirety.

17.2 QUALITY ASSURANCE DURING THE OPERATIONS PHASE

17.2.1 ORGANIZATION

17.2.1.1 Organizational Structure

Vistra OpCo, as the licensee, has overall responsibility for the operation of the Comanche Peak Nuclear Power Plant (CPNPP) which includes the Dry Cask Storage System. The Nuclear Generation Group (NG) has been designated by Vistra OpCo to coordinate the design, construction and operation of CPNPP. The organizational structure of Vistra OpCo and the Nuclear Generation Group are described in Section 13.1.

The following paragraphs amplify upon Section 13.1 with regard to establishment and execution of the quality assurance program for the operation of CPNPP. Figure 13.1-2 shows the structure and relationships of those elements of Nuclear Generation Group which function under the control of the QA program. Figure 13.1-3 shows the CPNPP Nuclear Operations organizational structure.

Vistra OpCo may, from time to time, assign responsibility for executing certain portions of the program to qualified consultants and contractors. However, Vistra OpCo retains ultimate responsibility for the CPNPP operations quality assurance program.

The reference to Dry Cask Storage System in this document includes the Independent Spent Fuel Storage Installation (ISFSI) facility, activities supporting spent fuel dry cask loading, and the associated dry cask storage equipment.

17.2.1.1.1 Senior Vice President & Chief Nuclear Officer

The Senior Vice President & Chief Nuclear Officer is responsible for the overall management of Nuclear Generation operations, including operation of CPNPP, and for the establishment of company policies related to Nuclear Generation. He has the overall responsibility for the establishment and execution of the quality assurance program for the operation of CPNPP.

The Senior Vice President & Chief Nuclear Officer has assigned to the Site Vice President the overall responsibility for operation of CPNPP and for implementation of the quality assurance program for the Nuclear Operations function at CPNPP.

The Senior Vice President & Chief Nuclear Officer has assigned to the Senior Director, Engineering & Regulatory Affairs the overall responsibility for engineering and support of CPNPP, and for implementation of the quality assurance program for the Nuclear Engineering function at CPNPP.

17.2.1.1.2 Site Vice President

The Site Vice President is responsible to the Senior Vice President & Chief Nuclear Officer for plant operations at CPNPP.

Specific duties and responsibilities of the Site Vice President include the following:

1. Technical and administrative direction of the Director, Organizational Effectiveness

2. Technical and administrative direction of the Director, Nuclear Training.

3. Operational and technical support of CPNPP.

4. Technical and administrative direction for the implementation of quality assurance requirements and controls at CPNPP.

5. Technical and administrative direction of Plant Manager and assumes duties if the position is not occupied.

17.2.1.1.3 Plant Manager

The Plant Manager is responsible to the Site Vice President for plant operations at CPNPP.

Specific duties and responsibilities of the Plant Manager include the following:

1. Technical and administrative direction of the direct reports shown in Figure 13.1-3.

2. Operational and maintenance support of CPNPP.

3. Manager of all operations activities at CPNPP.

4. Chairmanship of the Station Operations Review Committee (SORC).

5. Membership on the Operations Review Committee upon appointment by the Senior Vice President & Chief Nuclear Officer.

17.2.1.1.3.1 Station Operations Review Committee (SORC)

The SORC shall function to advise the Plant Manager on all matters related to nuclear safety.

The SORC shall as a minimum be composed of the Chairman and six individuals who collectively have experience and expertise in the areas listed below and meet the requirements of Regulatory Guide (RG) 1.8, Rev. 2 for required experience.

Operations

Maintenance

Instrumentation and Controls

Technical Support

Radiation Protection

Quality Assurance

The Plant Manager shall serve as the chairman of SORC. A senior health physicist is acceptable for the Radiation Protection representative on SORC. The SORC members shall be designated, in writing, by the Plant Manager.

All alternate members shall be appointed in writing by the Plant Manager to serve on a temporary basis; however, no more that two alternates shall participate as voting members in SORC activities at any one time.

The SORC shall meet periodically as determined by the SORC Chairman or the designated alternate.

The quorum of the SORC necessary for the performance of the SORC responsibility and authority shall consist of the Chairman or his designated alternate and a majority of the regular members (or their alternates).

The SORC shall be responsible for:

1. Review of applicable administrative procedures recommended in Appendix A of Regulatory Guide 1.33, Revision 2, February 1978.

2. Review of: (1) procedures, (2) change to procedures, equipment, systems or facilities, and (3) tests or experiments where nuclear safety is adversely affected;

3. Review of proposed procedures and changes to procedures, equipment, systems or facilities which require an amendment to the operating license;

4. Review of proposed test or experiments which require an amendment to the operating license;

5. Review of proposed changes to Technical Specifications or the Operating License:

6. Investigation of all violations of the Technical Specifications requiring a written report to the NRC, including the forwarding of reports covering evaluation and recommendations to prevent recurrence to the Senior Vice President & Chief Nuclear Officer and to the ORC:

7. Review of reports or operating abnormalities, deviations from expected performance of plant equipment and of unanticipated deficiencies in the design or operation of structures, systems or components that affect nuclear safety;

8. Review of all events submitted pursuant to 10CFR50.73. Other nonroutine reports may be reviewed at the discretion of the SORC Chairman or the responsible manager.

9. Review of changes to the PROCESS CONTROL PROGRAM, the OFFSITE DOSE CALCULATION MANUAL, and Radwaste Treatment Systems;

10. Review of any accidental, unplanned or uncontrolled radioactive release including the preparation of reports covering evaluation, recommendations, and dispositions of the corrective action to prevent recurrence and the forwarding of these reports to the Senior Vice President & Chief Nuclear Officer and to the ORC;

11. Review of Unit operations to detect potential hazards to nuclear safety;

12. Investigations or analysis of special subjects as requested by the Chairman of the ORC or the Plant Manager.

13. Review of the Fire Protection Report and implementing procedures and submittal of recommended Fire Protection Report changes to the ORC; and

14. Review of the Technical Requirements Manual and revision thereto.

17.2.1.1.3.1.1 The SORC shall:

1. Recommend in writing to the designated manager (see Section 17.2.1.6) approval or disapproval of items considered under items 1 through 5 above. prior to their implementation; and

2. Provide written notification within 24 hours to the Senior Vice President & Chief Nuclear Officer, and the Operations Review Committee of disagreement between the SORC and the designated manager (see Section 17.2.1.6) however, the Plant Manager shall have responsibility for resolution of such disagreements pursuant to Technical Specification 5.1.1.

The SORC shall maintain written minutes of each SORC meeting that, at a minimum, document the results of all SORC activities performed under the responsibility provisions above. Copies shall be provided to the Senior Vice President & Chief Nuclear Officer and the Operations Review Committee.

Any changes in the conduct of operation of the SORC will be made with the approval of the Plant Manager. Changes in the organization of the SORC will be made with the approval of the Plant Manager.

17.2.1.1.4 Director, Nuclear Oversight

The Director, Nuclear Oversight reports directly to the Senior Vice President & Chief Nuclear Officer and is responsible for assuring effective implementation of the Quality Assurance Program. This reporting relationship assures that the Director, Nuclear Oversight has sufficient authority, organizational freedom, and independence from undue influence from, or responsibility for, costs and schedules such that the Director, Nuclear Oversight can effectively assure implementation of and compliance with the CPNPP operations quality assurance requirements and controls.

The Director, Nuclear Oversight communicates directly with the Nuclear Generation Group supervisory and management personnel and with appropriate management levels in consultant and contractor quality assurance organizations to identify quality problems; initiate, recommend or provide solutions; and to verify implementation of solutions to quality problems. The Director, Nuclear Oversight also has authority to “stop work” during the operations phase.

Specific duties and responsibilities of the Director, Nuclear Oversight include the following:

1. Providing technical services and management in areas of Nuclear Oversight including quality assurance/quality control

2. Membership on the Operations Review Committee upon appointment by the Senior Vice President & Chief Nuclear Officer.

Specific duties and responsibilities of the Director, Nuclear Oversight include the following:

1. Technical and administrative direction of Nuclear Oversight personnel.

2. Verification through evaluations that procedures for the control of quality-related activities comply with quality assurance requirements.

3. Verification through evaluations of the implementation of the quality assurance program within the Nuclear Generation Group and evaluation of its effectiveness.

4. Assurance through evaluation and inspection that consultants, contractors and suppliers providing quality-related items or services have established and implemented an adequate quality assurance program.

5. Responsible for submitting the Quality Assurance Manual for concurrence and approval to the Senior Vice President & Chief Nuclear Officer.

17.2.1.1.5 Senior Director, Engineering & Regulatory Affairs

The Senior Director, Engineering & Regulatory Affairs is responsible for providing engineering related technical services in support of CPNPP operations.

Specific duties and responsibilities of the Senior Director, Engineering & Regulatory Affairs include the following:

1. Technical support to Nuclear operations.

2. Technical direction and administrative guidance to engineering staff.

3. Assistance, as required, in the procurement of equipment, materials, and services for the operation, maintenance or modification of CPNPP.

17.2.1.2 Nuclear Oversight

Nuclear Oversight, under the direction of the Director, Nuclear Oversight functions to assure effective implementation of the quality assurance program.

Nuclear Oversight has sufficient authority and organizational freedom at CPNPP to identify quality problems, recommend solutions, verify implementation of solutions, to stop unsatisfactory work and control further processing, delivery or installation of nonconforming material until proper disposition has occurred.

The Director, Nuclear Oversight advises the Senior Vice President & Chief Nuclear Officer of the status of the quality assurance program at CPNPP and of any significant conditions which are adverse to quality.

Specific functions performed by Nuclear Oversight include:

1. Quality assurance evaluations of Nuclear Generation quality-related activities, both offsite and onsite.

2. Evaluation of consultants', contractors', and suppliers' quality assurance programs and implementing procedures.

3. Evaluations and inspections conducted at equipment and material suppliers' facilities.

4. Review of procurement documents to assure incorporation of adequate quality assurance requirements for procured items and services.

5. Evaluations and inspections of site quality related activities to assure compliance with the applicable quality requirements.

6. Engineering services to support the Operations Review Committee (ORC).

7. Health Physics review of programs necessary for radiological protection of Company personnel and property, the environment and the general public.

8. Support the development of appropriate emergency plans for Vistra OpCo in accordance with regulatory guidelines and requirements and performance of the 10CFR50.54(t) independent review of the Emergency Planning Program.

9. Other independent verification/review responsibilities as described in Section 13.1.1.2.4 for Director, Nuclear Oversight.

10. Proposed changes to Technical Specifications or the Operating License;

17.2.1.3 Operations Review Committee

Independent reviews of activities affecting plant safety during the operations phase are performed by the Operations Review Committee.

The ORC shall function to provide independent review of designated activities in the areas of:

1. Nuclear power plant operations,

2. Nuclear engineering,

3. Chemistry and radiochemistry,

4. Metallurgy,

5. Nondestructive testing,

6. Instrumentation and control,

7. Radiological safety,

8. Mechanical and electrical engineering,

9. Administrative controls and quality assurance practices, and

10. Other appropriate fields associated with the unique characteristics of CPNPP.

The ORC shall report to and advise the Senior Vice President & Chief Nuclear Officer on those areas of responsibility specified in Section 17.2.1.3. ORC members should report all matters adversely affecting nuclear safety to the Senior Vice President & Chief Nuclear Officer (via the ORC Chairman) upon identification.

The ORC shall be composed of at least five individuals of whom no more than a minority are members of the CPNPP nuclear operations staff. The Chairman and all members will be appointed by the Senior Vice President & Chief Nuclear Officer.

The ORC Chairman shall hold a bachelor's degree in an engineering or physical science field or equivalent experience and a minimum of 6 years technical managerial experience.

The ORC members shall hold a bachelor's degree in an engineering or physical science field or equivalent experience and a minimum of 5 years technical experience. It is the responsibility of the Chairman to ensure experience and competence is available to review problems in areas listed in items 1 through 9 above. To a large measure, this experience and competence rests with the membership of the ORC. In specialized areas, this experience may be provided by personnel who act as consultants to the ORC.

The alternate for the Chairman and all alternate members shall be appointed in writing by the Senior Vice President & Chief Nuclear Officer to serve on a temporary basis.

Consultants shall be utilized as determined by the Chairman, to provide expert advice to the ORC.

The ORC shall meet not less than twice per year.

The quorum for formal meetings shall consist of not less than a majority of the principals, or duly appointed alternates, and shall be subject to the following constraints: the Chairman or his designated alternate shall be present for all formal meetings and no more than a minority of the quorum shall be members of the CPNPP nuclear operations staff. The Chairman or his alternate (if a member of ORC) will be included in the overall quorum count.

17.2.1.3.1 ORC Reviews

The ORC shall be responsible for the review of:

1. The 10CFR50.59 evaluations for: (1) changes to procedures, equipment, or systems; and (2) tests or experiments.

2. Violations of codes, regulations, orders, Technical Specifications, license requirements, or of internal procedures or instructions having nuclear safety significance;

3. Significant operating abnormalities or deviations from normal and expected performance of unit equipment that affect nuclear safety;

4. All events submitted pursuant to 10CFR50.73;

5. All recognized indications of an unanticipated deficiency in some aspect of design or operation of structures, systems, or components that could affect nuclear safety; and

6. Reports and meeting minutes of the SORC.

7. ORC shall perform periodic reviews of the audit program, including the audits discussed in 17.2.18.

17.2.1.3.2 ORC Records

Records of ORC activities shall be prepared, approved and distributed as indicated below:

1. Minutes of each ORC meeting shall be prepared, approved and forwarded to the Senior Vice President & Chief Nuclear Officer within 30 days following each meeting. Meeting minutes will be promptly distributed to appropriate members of management;

2. Reports of reviews encompassed by Section 17.2.1.3.1 shall be prepared, approved and forwarded to the Senior Vice President & Chief Nuclear Officer within 30 days following completion of the review.

17.2.1.4 Delegation of Nuclear Oversight Functions

Nuclear Generation (NG) periodically retains qualified consultants and contractors to provide safety-related services. All consultants and contractors providing safety-related services and suppliers providing safety-related equipment or materials for CPNPP are required to establish and implement quality assurance programs appropriate for their scope of supply. NG includes specific requirements in procurement documents with which consultants', contractors', or suppliers' quality assurance programs must comply.

17.2.1.5 Personnel Qualifications

17.2.1.5.1 Director, Nuclear Oversight

The following minimum qualification requirements have been established for the Director, Nuclear Oversight:

1. Engineering or related science degree;

2. Minimum of four years of related experience including two years of nuclear power plant experience, one year of supervisory or management experience, and one year of the experience is in performing quality verification activities.

3. Special requirements shall include management and supervisory skills and experience or training in leadership, interpersonal communication, management responsibilities, motivation of personnel, problem analysis and decision making, and administrative policies and procedures.

Individuals who do not possess these formal education and minimum experience requirements should not be eliminated automatically when other factors provide sufficient demonstration of

their abilities. These other factors are evaluated on a case-by-case basis and approved and documented by senior management.

17.2.1.6 Independent Technical Review and Controls

Activities which affect nuclear safety shall be conducted as follows:

1. Procedures required by Technical Specification 5.4 and other procedures which affect plant nuclear safety, and changes thereto, shall be prepared, reviewed and approved. Each such procedure or procedure change shall be reviewed by a qualified individual/ group other than the individual/group which prepared the procedure or procedure change, but who may be from the same organization as the individual/group which prepared the procedure or procedure change. The Plant Manager shall designate approval authority for station procedures. Changes to procedures which do not change the intent of approved procedures may be approved for implementation by two members of the Nuclear Operations Management Staff, at least one of whom holds a Senior Operation License, provided such approval is prior to implementation and is documented. Such changes will be approved by the original approval authority within 14 days of implementation.

2. Proposed tests and experiments which affect plant nuclear safety shall be prepared, reviewed and approved. Each such test or experiment shall be reviewed by a qualified individual/group other than the individual/group which prepared the proposed test or experiment. Proposed test and experiments shall be approved before implementation by the Plant Manager. Individuals responsible for conducting such reviews shall be members of the Nuclear Operations Management Staff previously designated by the Plant Manager.

3. Proposed changes to Technical Specifications or Operating License. Each such change shall be reviewed by a qualified individual/group meeting the experience requirements of Regulatory Guide 1.8 revision 2, 1987 other than the individual/group which proposed the change.

4. Proposed changes, or modifications to plant nuclear safety-related structures, systems and components shall be reviewed as designated by the Senior Director, Engineering & Regulatory Affairs. Each such modification shall be reviewed by a qualified individual/ group meeting the experience requirements of ANSI N18.1-1971 Section 4.6 other than the individual/group which designed the modification, but who may be from the same organization as the individual/group which designed the modifications. Individuals/groups responsible for conducting such reviews shall be previously designed by the Senior Director, Engineering & Regulatory Affairs. Proposed modifications to plant nuclear safety-related structures, systems, and components shall be approved by the Plant Manager prior to implementation.

5. Individuals responsible for reviews performed in accordance with the requirements of items 1 and 2 above, shall be members of the Nuclear Operations Management staff previously designated by the Plant Manager. Each such review shall include a determination of whether or not additional cross-disciplinary review is necessary. If deemed necessary, such review shall be done in accordance with the appropriate qualification requirements; and

6. Each review shall include a determination of whether or not an amendment to the operating license is required. For items requiring an amendment to the operating license, NRC approval shall be obtained prior to the Plant Manager approval for implementation.

Records of the activities described above shall be provided to the Plant Manager, SORC, and/or ORC as necessary for the required reviews.

17.2.1.6.1 Reportable Events

The following action shall be taken for all events submitted pursuant to 10CFR50.73:

Each event shall be reviewed by the SORC, and the results of this review shall be submitted to the ORC and the Senior Vice President & Chief Nuclear Officer.

Written procedures shall be established, implemented and maintained covering the Process Control Program implementation.

Changes to the PCP:

1. Shall be documented and records of reviews performed shall be retained as required by FSAR Section 17.2.17.1. This documentation shall contain:

   a. Sufficient information to support the change together with the appropriate analyses or evaluations justifying the change(s) and

   b. A determination that the change will maintain the overall conformance of the solidified waste product to existing requirements of Federal, State, or other applicable regulations.

2. Shall become effective after review and acceptance by the SORC and the approval of the Plant Manager.

Each procedure and administrative policy of Technical Specification 5.4.1 and changes thereto, shall be reviewed and approved prior to implementation as set forth above.

17.2.2 QUALITY ASSURANCE PROGRAM

The Quality Assurance Program requires a Quality Assurance manual be developed for each nuclear power plant, which prescribe specific measures to assure the quality of safety-related activities, structures, systems and components of that facility. The quality assurance requirements and controls implemented during operations of CPNPP are established by the portion of the CPNPP Quality Assurance Program in this section (17.2) of the FSAR. Quality assurance requirements and controls established by the CPNPP Quality Assurance Program for the Dry Cask Storage System shall be implemented throughout the period during which the ISFSI is licensed or the spent fuel storage cask is certified.

Quality assurance requirements and controls are established and implemented throughout the testing and operation phases at CPNPP. This program shall be implemented at least 90 days prior to fuel loading. Responsibilities and authority, and measures for the control and accomplishment of activities affecting the quality and operation of safety-related structures,

systems, and components of CPNPP are defined. The structures, systems, and components covered by the quality assurance program are discussed in Appendix 17A. These provisions apply to all activities, such as operating, maintaining, repairing, modifying, and refueling which affect the safety-related functions of those structures, systems, and components. These provisions, as necessary, shall ensure conformance with the approved design of the ISFSI or spent fuel storage cask.

A Quality Assurance Program shall be developed and implemented to attain high levels of quality assurance during the operation of CPNPP. This program shall comply with the requirements of Title 10, Code of Federal Regulations, Part 50, Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Processing Plants," and certain NRC Regulatory Guides and ANSI standards as identified in the Final Safety Analysis Report (FSAR).

The Quality Assurance Program also complies with the requirements of Title 10, Code of Federal Regulations, Part 72, Subpart G, "Quality Assurance Requirements for the Independent Storage of Spent Nuclear Fuel, High Level Radioactive Waste, and Reactor-Related Greater Than Class C Waste". Appendix 17A addresses the applicability of the quality assurance program requirements and procedural controls for the Dry Cask Storage System shall be based on the following considerations concerning the complexity and proposed use of the structures, systems, or components:

1. The impact of malfunction or failure of the item on safety;

2. The design and fabrication complexity or uniqueness of the item;

3. The need for special controls and surveillance over processes and equipment;

4. The degree to which functional compliance can be demonstrated by inspection or test; and

5. The quality history and degree of standardization of the item.

Vistra OpCo may delegate to others such as contractors, agents, or consultants the work of establishing and executing the quality assurance program, or any part thereof, but the overall responsibility for the Quality Assurance (QA) Program lies with the Senior Vice President & Chief Nuclear Officer. Specific responsibility for development and administration of the program rests with the Director, Nuclear Oversight. The Senior Vice President & Chief Nuclear Officer will assure that a biennial independent assessment of the evaluation program is performed.

These independent assessments will be conducted in accordance with predetermined schedules, with results documented, and a follow-up system utilized to assure that corrective action is taken and evaluated when it is considered necessary to verify implementation requirements of 10CFR50, Appendix B and 10CFR72 Subpart G, as applicable. The Vice-Presidents shall meet periodically to assess the status and adequacy of the quality assurance program.

The quality assurance requirements and controls applicable to the operations phase, comply with the requirements of 10 CFR Part 50, Appendix B. Table 17.2-1 provides a matrix showing those sections of the QA Manual which satisfy the requirements of each criterion of 10 CFR Part 50, Appendix B. The quality assurance requirements and controls shall be consistent with the

applicable guidance of those Regulatory Guides and industry standards listed in Table 17.2-2 and discussed in Appendix 1A(B).

The quality assurance requirements and controls are designed to assure that activities affecting the quality and operation of safety-related items are accomplished in a planned and controlled manner. Activities affecting quality are accomplished in accordance with written, approved procedures and instructions under suitably controlled conditions. Controlled conditions include, as applicable, appropriate equipment, suitable environmental conditions, and completion of prerequisites. All procedures prescribing activities affecting quality are controlled and distributed in accordance with the measures described in Section 17.2.6.

The Director, Nuclear Oversight is responsible for assuring, through surveillance and audit, implementation of the Quality Assurance Program. He is responsible for regularly assessing the status and adequacy of the Program, within the Nuclear Generation Group, and as implemented by consultants, contractors, and suppliers. The Director, Nuclear Oversight reports the results of these evaluations to the Senior Vice President & Chief Nuclear Officer. Unresolved issues between the Director, Nuclear Oversight and others concerning quality are brought to the Senior Vice President & Chief Nuclear Officer for resolution.

The Director, Nuclear Oversight has overall responsibility for the identification, scheduling, assignment, conduct and reporting of station activities assigned to the Nuclear Oversight. Station activities affecting quality are subject to quality surveillance and audit by Nuclear Oversight personnel.

In addition, the Director, Nuclear Oversight has responsibility for administration and implementation of the CPNPP quality control program.

The Director, Nuclear Oversight reviews procedures during the operations phase for quality assurance requirements either as a normal function of the Station Operations Review Committee, or as described in section 17.2.6. Procedural implementation of quality requirements is periodically reviewed by the Nuclear Oversight organization through evaluation activities.

An indoctrination and training program is established for those personnel performing activities affecting quality. The scope, objectives, and methods for implementing the indoctrination and training program are prescribed by written, approved procedures. These procedures also prescribe methods for documenting the accomplishment of training. The indoctrination and training program includes provisions that personnel performing activities affecting quality are:

1. Instructed as to the purpose, scope, and implementation of the Quality Assurance Program and related procedures and instructions as appropriate to their activities.

2. Qualified in the principles and techniques of activities for which they are responsible.

3. Retrained, re-examined or recertified, when appropriate, to maintain necessary proficiency in those activities for which they are responsible.

17.2.3 DESIGN CONTROL

Requirements for the control of design activities associated with modifications (involving a new design or change in existing design) of safety-related structures, systems, and components are

consistent with the provisions of Regulatory Guide 1.33, and Regulatory Guide 1.64 as discussed in Appendix 1A(B).

The Plant Manager, or his designee, shall have the responsibility for approving and controlling the implementation of station design modifications. The Senior Director, Engineering & Regulatory Affairs shall have the overall responsibility for developing procedures to maintain and control the design control process.

The SORC will submit all proposed station design modifications which involve a change in CPNPP Technical Specifications or other changes to the operating license to the ORC for review. 10CFR50.59 evaluations on station design modifications which do not require a license amendment or a change in CPNPP Technical Specifications will be reviewed by ORC, however, this will not be a prerequisite for implementation. The Plant Manager, or his designee, approves each station design modification for implementation.

10CFR72.48 establishes the conditions under which an Independent Spent Fuel Storage Installation (ISFSI) licensee or a spent fuel storage cask certificate holder may make changes in the ISFSI facility or spent fuel storage cask design without prior NRC approval. Changes in the conditions specified in the license or Certificate of Compliance (CoC) require prior NRC approval. When changes occur that affect the Dry Cask Storage System structures, systems, and components, review and approval of the change must consider the requirements of 10CFR72.48.

All design modification requests made by station personnel shall be submitted to the Senior Director, Engineering & Regulatory Affairs for coordination of the station level engineering review. The actual change of design for those design modification requests approved by the Plant Manager may be done by Nuclear Engineering personnel or approved engineering services contractors. The above organizations will have approved design procedures and/or instructions before any design modifications are performed by the respective organization. These procedures and/or instructions will assure proper design review and verification. These procedures and instructions will also assure that design control is commensurate with the original design. The Senior Director, Engineering & Regulatory Affairs will assure that the designer is provided with the latest revisions to all drawings, specifications, and other design documents which are applicable.

Design changes, including those originating on site, are subject to the same controls which were applicable to the original design. The Senior Director, Engineering & Regulatory Affairs may designate an organization to make design changes other than the one which prepared the original design. In these cases, the Senior Director, Engineering & Regulatory Affairs will assure that organization has access to pertinent background information, including an adequate understanding of the requirements and intent of the original design, and has demonstrated competence in applicable design areas.

The Senior Director, Engineering & Regulatory Affairs shall coordinate necessary revisions to drawings and other design documents. The Plant Manager shall coordinate necessary revisions to plant procedures and instructions as a result of design changes. Changes are promptly distributed to ensure availability to responsible plant personnel prior to commencement of work and replacement; and delineation of acceptance criteria for inspection and tests. In addition, the above controls are applied as necessary to the Dry Cask Storage System for shielding and features that facilitate documentation.

17.2.3.1 Design Changes

Design changes made to the facility are accomplished in a planned and controlled manner in accordance with written, approved procedures. These procedures include provisions, as necessary, to ensure that:

1. Design documents, specifications, drawings, and procedures and instructions reflect applicable regulatory requirements and design bases.

2. Design documents specify quality requirements or reference quality standards as necessary.

3. There is adequate review of the suitability of materials, parts, components, and processes which are essential to the safety- related functions of structures, systems, and components.

4. Materials, parts, and components which are standard commercial (off the shelf) or which have been previously approved for a different application are evaluated for suitability prior to selection.

5. Design documents are revised to reflect design modifications.

6. Internal and external design interfaces between organizations participating in design modifications are adequately controlled, including the review, approval, release, and distribution of design documents and revisions.

The above controls are applied as necessary to such aspects of design as reactor physics; seismic, stress, thermal, hydraulic, radiation, and accident analyses; compatibility of materials; and accessibility for inservice inspection, maintenance, repair and replacement; and delineation of acceptance criteria for inspections and tests. In addition, the above controls are applied as necessary to the Dry Cask Storage System for shielding and features that facilitate decontamination.

17.2.3.1.1 Design Change Verification

The adequacy of design changes shall be verified by the performance of design reviews, alternate calculations, or qualification testing. The control measures specified in the plan for control of design verification activities are as follows:

1. Personnel responsible for design verification do not include the original designer or the designer’s immediate supervisor.

2. Written procedures identify the positions or organizations responsible for design verification and define their authority and responsibility.

3. Qualification tests to verify the adequacy of the design are performed using the most adverse specified design conditions.

4. Design changes are reviewed to assure that design parameters are defined and that inspection and test criteria are identified.

Any errors or deficiencies found in the design process or the design itself are documented and corrective action taken, as described in Section 17.2.16.

Design documents and revisions thereto are controlled and distributed as described in Section 17.2.6. Records of design activities and design changes are collected, stored, and maintained, as described in Section 17.2.17.

17.2.4 PROCUREMENT DOCUMENT CONTROL

Requirements are established for the control of procurement documents prepared by NP, or their designated agents for safety-related components, materials, and services. These requirements are consistent with the provisions of Regulatory Guide 1.33 and Regulatory Guide 1.123 as discussed in Appendix 1A(B) and apply to procurement documents prepared by NP, or their designated agents.

Procurement documents, such as purchase specifications, contain or reference the following:

1. The design basis technical requirements, including the applicable regulatory requirements, material and component identification requirements, drawings, specifications, codes and industrial standards, and test and inspection requirements, including inspection hold points.

2. The applicable requirements of 10 CFR Part 50, Appendix B and of the QA Program, which must be complied with and described in the supplier's QA program; or, identification and verification of characteristics critical to the safety function of the procured item to provide reasonable assurance that the item will perform its intended safety related function.

   The applicable requirements of 10CFR72, Subpart G are imposed in procurement documents for Dry Cask Storage System items and services as required by Appendix 17A.

   When purchasing commercial grade calibration services from a calibration laboratory accredited by National Voluntary Laboratory Accreditation Program (NVLAP) or American Association for Laboratory Accreditation (A2LA), the procurement documents are not required to impose a quality assurance program consistent with the requirements of 10CR50 Appendix B or identification and verification of characteristics critical to the safety function provided each of the following conditions are met:

   a. The procurement documents impose any additional technical and administrative requirements, as necessary, to comply with NP's QA program and technical provisions. The procurement document shall require that the calibration certificate/report include identification of the laboratory equipment/standard used.

   b. The procurement documents require reporting as-found calibration data when calibrated items are found to be out-of-tolerance.

   c. A documented review of the supplier's accreditation has been performed as discussed in Section 17.2.7.

3. Identification of the documentation to be prepared, maintained, or submitted (as applicable) to NP for review and approval. These documents may include, as necessary, inspection and test records, qualification records, or code required documentation.

4. Identification of those records to be retained, controlled, and maintained by the supplier, and those delivered to the purchaser prior to use or installation of the hardware.

5. NP's right of access to supplier's facilities and records for source inspection and evaluation.

6. Requirements for supplier reporting and dispositioning of nonconformances from procurement requirements.

7. Provisions for extending applicable requirements of the procurement documents to lower-tier suppliers.

NP procurement documents are prepared, reviewed, approved, and controlled in accordance with written procedures which clearly delineate the sequence of actions to be accomplished and which identify the individuals or groups responsible for accomplishing those actions. These procedures include provisions for review of procurement documents. This review is performed to insure that necessary quality requirements are incorporated and correct, and that procurement requirements for spare or replacement parts are equivalent to or better than those used for the original equipment. Documentary evidence of that review and approval is retained and available for verification. Changes to purchase documents are subject to the same degree of control as that utilized in the preparation of the original documents.

NP evaluates supplier quality assurance programs prior to award of contracts or issuance of purchase orders, as discussed in Section 17.2.7.

17.2.5 INSTRUCTIONS, PROCEDURES, AND DRAWINGS

Activities affecting the quality of safety-related structures, systems, and components be prescribed by and accomplished in accordance with documented instructions, procedures, and drawings. The manager or supervisor who has cognizance over a specific safety-related activity is responsible for the development and approval of procedures and instructions for prescribing the accomplishment of that activity. Administrative procedures and instructions are reviewed and approved prior to performance of the activity. The cognizant supervisor is responsible for ensuring that the activity is performed in accordance with the procedures and instructions. The development, review, and use of procedures, instructions, and drawings is reviewed on a periodic basis by Nuclear Oversight personnel as part of the station surveillance and audit program. These requirements are consistent with the provisions of Regulatory Guides 1.33, 1.30, and 1.116 as discussed in Appendix 1A(B).

Requirements regarding the content of various types of instructions and procedures are established which provide for the inclusion, as necessary, of items such as prerequisites, precautions, qualitative or quantitative acceptance criteria, inspection points, and checklists, depending upon the nature of the instruction or procedure.

Administrative procedures clearly delineate the sequence of actions to be accomplished in the preparation, review, approval, and control of those instructions and procedures, and they identify the individuals responsible for those actions.

Confirmation that these instructions and procedures meet requirements of the QA Program and are properly implemented is accomplished through audit activities by Nuclear Oversight.

17.2.6 DOCUMENT CONTROL

Requirements are established for the control of documents that prescribe activities affecting quality. The documents which are to be controlled include:

1. Design Specifications

2. Design, manufacturing, construction, and installation drawings

3. Procurement documents

4. The QA Manual and all station procedures and instructions which implement requirements of the QA Program.

5. Maintenance, modification, and operating procedures and instructions

6. Final Safety Analysis Report

7. Inspection and test procedures and instructions

17.2.6.1 Design Control Requirements

These requirements are consistent with the provisions of Regulatory Guide 1.33 as discussed in Appendix 1A(B) and include the following measures:

1. Documents, and changes thereto, are reviewed for adequacy and approved for release by authorized personnel in accordance with written procedures. These procedures identify those individuals or groups responsible for reviewing, approving, and issuing documents and revisions thereto. These reviews include as appropriate a review for QA-related aspects by Quality Assurance or an individual other than the person who generated the documents but qualified in quality assurance.

2. Documents, and changes thereto, are promptly distributed to ensure availability prior to commencement of work.

3. Changes to documents are reviewed and approved by the same organization that performed the original review and approval unless another qualified organization is designated.

4. Master status lists identifying the current revision of documents are periodically updated and utilized to preclude the use of superseded documents.

5. Obsolete or superseded documents are destroyed or identified to prevent their inadvertent use.

6. Documents generated by NP are controlled in accordance with written, approved procedures and instructions. Maintenance, modification and inspection procedures and instructions affecting safety related equipment are reviewed by a person knowledgeable in QA disciplines to determine:

   a. The need for inspection, identification of inspection personnel, and documentation of inspection results.

   b. That the necessary inspection requirements, methods, and acceptance criteria have been identified.

The Director, Nuclear Oversight is responsible for providing the necessary reviews of these procedures and instructions.

17.2.7 CONTROL OF PURCHASED MATERIAL, EQUIPMENT AND SERVICES

Requirements are established for the control of purchased safety-related material, equipment and services, including spare or replacement parts. These requirements are consistent with the provisions of Regulatory Guides 1.33, 1.38, 1.123, and 1.144 as discussed in Appendix 1A(B).

Measures have been established in procedures which determine the level of quality assurance required for the procurement of an item or service. As required, contractor and suppliers are evaluated by Nuclear Oversight personnel prior to award of a purchase order or contract to assure the contractor's or supplier's capability to comply with procurement document requirements. This evaluation is based on one or more of the following:

1. A review of the supplier's quality assurance program description provided with the proposal/bid.

2. A review of historical evidence of the supplier's performance in providing similar items or services.

3. A preaward survey of the supplier's facilities and QA program.

Technical requirements for items and materials to be procured are developed by the design or engineering organization responsible for the modification or maintenance activity. Procurement documents for safety related items and materials are reviewed for inclusion of technical and quality assurance requirements by inspection personnel prior to acceptance of the procured item or material. The results of the Nuclear Oversight review are documented and retained for future reference.

Evaluation and inspection of suppliers and contractors, are conducted where appropriate, to assure compliance with quality requirements. Nuclear Oversight is responsible for evaluation and inspection of offsite suppliers and contractors and for evaluation of contractors providing services onsite. Evaluation of suppliers and contractors is performed by qualified personnel in accordance with written procedures, instructions and checklists.

When purchasing commercial grade calibration services from a calibration laboratory, source evaluation and inspection of these suppliers need not be performed provided each of the following conditions are met:

1. The procurement documents impose any additional technical and administrative requirements, as necessary, to comply with the NP's QA program and technical provisions. The procurement document shall require the calibration certificate/report to include identification of the laboratory equipment/standard used.

2. The procurement documents require reporting as-found calibration data when calibrated items are found to be out-of-tolerance.

3. A documented review of the supplier's accreditation shall be performed and shall include a verification of each of the following:

    - The calibration laboratory holds accreditation by the National Voluntary Laboratory Accreditation Program (NVLAP) or by the American Association for Laboratory Accreditation (A2LA) as recognized by NVLAP through the International Laboratory Accreditation Cooperation (ILAC) Mutual Recognition Arrangement (MRA).

    - The accreditation is based on ANSI/ISO/IEC 17025.

    - The published scope of accreditation for the calibration laboratory covers the necessary measurement parameters, ranges, and uncertainties.

Evaluation, and inspection of suppliers are performed to an extent consistent with the importance, complexity, and quantity of the item(s) being purchased and include measures to periodically confirm the validity of suppliers' certificates of conformance. For annual supplier evaluations, a grace period of up to 90-days may be used for the scheduled commencement date when conditions, such as plant operational considerations or to accommodate supplier activities, make meeting the specified schedule date impractical. For evaluation activities deferred by using this grace period, the next scheduled due date shall be based on the original scheduled date. For triennial supplier audits, the audit schedule shall be based on the month in which the audit starts. An interval extension grace period of up to 25% may be used for the scheduled commencement date when conditions, such as plant operational considerations or to accommodate supplier activities, make meeting the specified schedule date impractical. When an audit interval extension greater that one month is used, the next audit will be scheduled from the original anniversary month rather than from the month of the extended audit. The total combined time interval for any three consecutive audit intervals should not exceed 3.25 times the specified audit interval. Quality verification records are reviewed by Nuclear Oversight personnel to assure their completeness and their compliance with procurement document requirements.

17.2.7.1 Receipt Inspection and Material Testing

Receipt inspection and material testing at CPNPP is performed by qualified quality control inspectors in accordance with written procedures and instructions to assure that:

1. Materials, equipment, or components are properly identified and correspond with associated documentation.

2. Inspection records or certificates of conformance attesting to the acceptance of materials, equipment, and components are completed and are available at CPNPP prior to installation or use.

3. Materials, equipment, and components are inspected and judged acceptable in accordance with predetermined inspection instructions prior to installation or use.

4. Items accepted or released are identified as to their inspection status prior to forwarding them to a controlled storage area or releasing them for installation or further work.

5. Nonconforming items are clearly identified, controlled, and segregated where practical, until proper disposition is made.

17.2.8 IDENTIFICATION AND CONTROL OF MATERIALS, PARTS, AND COMPONENTS

Requirements are established for the identification and control of safety-related materials, parts, and components, including spare or replacement items, as well as expendable and consumable items. These requirements are consistent with the provisions of Regulatory Guides 1.33, and 1.38 as discussed in Appendix 1A(B).

Materials, parts, and components are identified and controlled to prevent the use of incorrect or defective items. Identification of items is maintained either on the item in a manner that does not affect the function or quality of the item, or on records traceable to the item.

Suppliers of safety-related materials, parts, or components are required by procurement documents to establish a system of identification and control which is consistent with the above requirements.

Procedures and instructions implementing these requirements provide for the following:

1. Verification that items received onsite are properly identified and can be traced to the appropriate documentation, such as drawings, specifications, purchase orders, manufacturing and inspection documents, nonconformance reports, or mill test reports.

2. Verification of item identification consistent with the inventory control system and traceable to documentation which identifies the proper uses or applications of the item.

17.2.9 CONTROL OF SPECIAL PROCESSES

Requirements are established for the control of special processes, which are those processes where direct inspection is impossible or disadvantageous such as welding, heat treating, nondestructive testing, and cleaning, which are consistent with the provisions of Regulatory Guides 1.30, 1.33, 1.37, 1.58, and 1.94 as discussed in Appendix 1A(B).

Special processes are performed by qualified personnel using proper equipment and in accordance with written qualified procedures and instructions. These personnel, procedures and instructions are to be qualified in accordance with applicable codes, standards, and specifications. Qualification records of special process procedures and instructions, and personnel performing special processes are filed, maintained, and available for verification.

Qualification of special processes, equipment, and personnel is the responsibility of the cognizant Managers or Supervisors.

Qualified test laboratories and consultants may be used in qualification of special processes. Procedures shall be developed which delineate the requirements for special process. These procedures shall be reviewed by Nuclear Oversight or other qualified personnel.

17.2.10 INSPECTION

Requirements are established for an inspection program to verify conformance of activities affecting quality with requirements specified for those activities. These requirements are consistent with the provisions of Regulatory Guides 1.30, 1.33, 1.39, 1.58, 1.94, 1.116, and 1.123 as discussed in Appendix 1A(B).

The Director, Nuclear Oversight is responsible for administering and implementing an effective inspection program at CPNPP. Quality Control inspections are performed by personnel who are qualified and certified in accordance with ANSI N45.2.6-1978 and who are independent of the individuals performing or directly supervising the activity being inspected. Personnel performing these inspections may be from the same department but are not from the same group that performed the work. Personnel performing inspections may be selected from among any of the Nuclear Generation departments or may be contract personnel. Qualification criteria for inspection personnel are reviewed and approved by a Level III inspector and concurred with by the Director, Nuclear Oversight. Inspection personnel have authority to stop unsatisfactory work and control further processing, delivery, or installation of nonconforming material, parts or components. The inspectors' qualification and certification records/documentations are maintained by the Nuclear Generation training program. The inspectors' qualifications and certifications are maintained current by actual performance of inspections on a periodic basis. Quality independent verifications are identified by designated inspection personnel and performed by personnel who did not perform the work as assigned by line management. These verifications are overviewed by Nuclear Oversight to ensure compliance to requirements.

Inspections at CPNPP are performed in accordance with written procedures, instructions, or checklists, appropriate to the circumstances which provide for the following:

1. Identification of characteristics and activities to be inspected, including inspection hold points.

2. Acceptance and rejection criteria.

3. Method of inspection.

4. Recording the results of the inspection and identification of the quality control inspector.

5. Indirect control by monitoring of processing methods, equipment, and personnel when direct inspection is not possible.

6. Identification of any required procedures, drawings, or specifications.

Station administrative procedures controlling the Measuring and Test Equipment program contain criteria for determining the accuracy of M&TE to be used in performing inspections depending upon the accuracy requirements of the parameters being measured.

Maintenance, repair, and modification procedures and instructions containing inspection criteria shall be reviewed by a Level II or III inspector qualified in accordance with ANSI N45.2.6-1978 to ensure that adequate inspection hold points are included and that the inspection methods are adequate. Criteria contained in appropriate station administrative procedures and in applicable codes and standards shall be used in determining when inspections and tests are required.

In addition, administrative control procedures and instructions are reviewed by Nuclear Oversight to assure that required controls are included and to provide for the necessary reviews for the assignment of inspection hold points.

Inspection results are documented in accordance with procedures and instructions developed and approved for that activity. Inspection results are evaluated and then acceptability determined by individuals qualified to perform that function in accordance with the station training program. Records of the evaluations are documented and retained in the station quality records.

Contractors performing work at CPNPP and equipment and material suppliers are required to work under inspection programs consistent with applicable codes and standards. These contractors and suppliers are required to provide work plans or inspection and fabrication procedures or outlines, which are reviewed for adequacy by NP personnel.

17.2.11 TEST CONTROL

Requirements are established for the control of testing of safety-related systems, equipment, and structures. These requirements are consistent with the provisions of Regulatory Guides 1.30, 1.33, 1.58, 1.68, 1.68.2, 1.94, and 1.116 as discussed in Appendix 1A(B).

17.2.11.1 Test Program

Preoperational and initial startup testing is performed in accordance with Section 14.2 of the FSAR.

Surveillance testing is performed during the operational phase to verify continuing operational readiness and adequacy for those systems and components which are normally in a standby condition and to evaluate whether there has been any degradation of performance, or any departure from the prescribed operating conditions for the systems or components normally in service.

Tests are performed following station modifications or repairs to demonstrate satisfactory performance prior to placing affected items in service. When pressure boundaries are breached functional tests shall be conducted to the extent required to demonstrate acceptability of the repair or maintenance.

17.2.11.2 Test Procedures

Testing is identified, documented, and controlled in accordance with written administrative procedures. Each test is accomplished in accordance with written test procedures by qualified personnel.

The administrative procedures controlling the test program identify the necessary test procedures, the provisions to be included in those procedures, the method of reviewing and approving those procedures, and the methods for documenting and evaluating the results.

Test procedures include the following provisions as appropriate:

1. Prerequisites - those items of work which must be completed prior to establishing initial conditions for the test, including:

    a. Calibrated instrumentation;

    b. Adequate and appropriate equipment;

    c. Initial conditions and completeness of the item to be tested; and

    d. Suitable environmental conditions, if applicable.

2. Special precautions - items needed for safety of personnel or equipment. Special situations where caution or extraordinary attentiveness to operational circumstances is required.

3. Instructions for performing the test - steps required to conduct the test, observations to be made, data to be recorded.

4. Acceptance criteria - criteria against which the success or failure of the test can be determined.

5. Provisions for collecting, documenting, or recording test data and results.

6. Requirements and acceptance limits contained in the ISFSI or spent fuel storage cask license or Certificate of Compliance.

17.2.11.3 Test Results

Records of test results are reviewed by qualified personnel to assure acceptability. These records are retained as quality verification records in accordance with the controls described in Section 17.2.17.

17.2.12 CONTROL OF MEASURING AND TEST EQUIPMENT

Measuring and Test Equipment (M&TE) used for safety-related structures, systems and components are highly reliable, commercial grade items. The Vistra OpCo Quality Assurance Program provides measures for the control of M&TE used as the basis for acceptance in inspection, testing, and measurement activities which affect quality.

The Nuclear Generation organization is responsible for the development of procedures for the M&TE control program. These procedures delineate responsibilities for the implementation of this program and methods for the procurement, handling, storage, control, scheduling, and calibration of M&TE and reference standards. For the operations phase, these methods are consistent with the provisions of Regulatory Guide 1.33 as discussed in Appendix 1A(B).

Vistra OpCo Nuclear Oversight performs pre-award evaluations/surveys as well as periodic evaluations to assure adequacy and effective implementation of the M&TE program.

M&TE and reference standards are traceable to their calibration records. These devices are labeled or tagged to indicate the next calibration due date or otherwise controlled in accordance with approved procedures which ensure only M&TE and reference standards with current calibration are utilized.

Periodic calibration and adjustment of M&TE is performed and controlled to assure accuracy is maintained within limits necessary to verify that design and operating condition requirements have been met. M&TE is normally calibrated against reference standards which have an uncertainty of no more than one fourth (1/4) of the required uncertainty of the M&TE being calibrated. If this 4:1 accuracy requirement is not reasonably achievable, a documented evaluation of the adequacy of the calibration is performed and approved by the responsible Engineering group.

Calibrating standards have accuracy greater than or equal to reference standards being calibrated. If equal accuracy is used, the basis for calibration is documented and approved by the Vistra OpCo manager of the organization which is responsible for the calibration.

M&TE and reference standards are traceable to nationally or internationally recognized standards or natural physical constants. Where no nationally or internationally recognized standard or natural physical constant exists, the basis for calibration is documented and approved by the Vistra OpCo manager of the organization which is responsible for the calibration.

When M&TE or reference standards are lost or found to be out of calibration, a documented review is conducted to determine the validity of all inspection, test and/or measurement results gained through the use of the affected device since its last acceptable calibration. Corrective action is taken for items determined to be suspect.

Contractor and supplier organizations which provide engineering/calibration services to Vistra OpCo are responsible for implementing measures to ensure that M&TE accuracy is maintained within required limits. The M&TE programs of these organizations shall satisfy the Vistra OpCo M&TE program requirements.

17.2.13 HANDLING, STORAGE, AND SHIPPING

Requirements are established for the control, handling, storage, shipping, cleaning, and preservation of material and equipment in accordance with established instructions, procedures, or drawings. These requirements are consistent with the provisions of Regulatory Guides 1.33, 1.38, and 1.39 as discussed in Appendix 1A(B) and include the following provisions, as necessary:

1. For critical, sensitive, perishable, or high value items, specific written procedures and instructions for handling, storing, packing, shipping, and preserving are used. These procedures and instructions reflect design and specification requirements such as inert gas atmosphere, specific moisture content levels, and temperature levels, and reflect manufacturers recommendations in regards to special handling and storage requirements such as shelf life and environmental controls.

2. Personnel responsible for handling these special items are qualified to the extent required by these special handling instructions.

3. Special handling tools and equipment are inspected and tested in accordance with written procedures to verify that they are adequately maintained.

17.2.14 INSPECTION, TEST, AND OPERATION STATUS

Requirements are established for identification and control of the inspection, test, and operating status of safety-related structures, systems, and components. These requirements are consistent with the provisions of Regulatory Guide 1.33 as discussed in Appendix 1A(B).

Written procedures and instructions prescribe the use of tags, labels, and logs to indicate the inspection, test, and operating status of systems and equipment at CPNPP. These procedures and instructions also provide for tagging of nonconforming, inoperative, or malfunctioning equipment to prevent inadvertent use. In addition, these procedures and instructions identify those individuals who are authorized to apply or remove those tags and labels and provide for the use of logs to maintain the status of tags and labels in use at CPNPP.

CPNPP personnel and contractor personnel working onsite are instructed regarding the purpose of, and precautions associated with, the various tags and labels used at CPNPP. Proper use of tags and labels to indicate inspection, test, and operating status is verified through evaluations by the Nuclear Oversight.

17.2.15 NONCONFORMING MATERIALS, PARTS, OR COMPONENTS

Requirements are established for the control of nonconforming materials, parts or components. These requirements are consistent with the provisions of Regulatory Guides 1.33, 1.38, and 1.123 as discussed in Appendix 1A(B).

Material, parts, or components found nonconforming through review, inspection, or testing are controlled by administrative procedures. These procedures provide for the following:

1. Identification of nonconforming items, prior to installation, by use of nonconformance tags, and segregation of those items, if practical, to prevent inadvertent use pending proper disposition and reinspection.

2. Identification of those individuals or organizations responsible for disposition of nonconforming items.

3. Preparation of nonconformance documents which identify nonconforming items and describe the nonconformance, the disposition of the nonconformance, and the

reinspection or testing performed to determine the acceptability of the item after the disposition has been completed.

4. Review of nonconformance documents written on installed plant equipment to determine impact on operability. The administrative controls assure that nonconforming materials do not affect the operability of safety related equipment in violation of Technical Specification requirements.

5. Conditional releases allow issuance of nonconforming items from the warehouse for installation, testing, and operation, pending disposition of the nonconformance. For nonconforming items installed with a conditional release, and affecting Technical Specifications, credit may be taken for Technical Specification operability of the item, provided that: a) resolution of the nonconforming condition is tracked, and b) evaluation of the nonconforming condition supports operability of the component. Each conditional release also describes any limitations or special precautions required. Conditional releases are periodically evaluated as to their status and the results forwarded to management for their review.

6. Verification of the acceptability of rework/repair of items by reinspection or testing of the item as originally performed or by a method which is equivalent to the original inspection and testing method.

7. Nonconformance reports which are dispositioned “use as is” or “repair” are made part of the quality verification records associated with the items.

8. Periodic analysis of these reports to be performed and forwarded to management to show quality trends.

Responsibility for the implementation of activities related to nonconformance control including disposition and closeout is assigned to the cognizant manager of the area of concern. Nonconformances which are resolved by repair or use-as-is dispositions are reviewed and approved by Engineering.

Independent evaluation of activities related to nonconformance control are performed by appropriate Nuclear Oversight personnel or designee.

Marking and segregation of nonconforming items, when required, are addressed in station procedures. Compliance with these administrative requirements is verified through the station evaluation program.

17.2.16 CORRECTIVE ACTION

Requirements are established for the identification and correction of conditions adverse to quality. These requirements are consistent with the provisions of Regulatory Guide 1.33 as discussed in Appendix 1A(B). The Director, Organizational Effectiveness is responsible for administrating and facilitating the corrective action program.

Conditions adverse to quality, such as failures, malfunctions, deficiencies and deviations, identified through review of documents, evaluations, or experience during operation, are documented and dispositioned. Significant conditions adverse to quality are evaluated to

determine the cause of the condition and the corrective action to be taken to preclude recurrence.

Reports of significant conditions adverse to quality are reviewed by the Operation Review Committee and that committee's decisions and/or recommendations regarding corrective action are forwarded to appropriate management personnel. Follow-up reviews to verify proper implementation of corrective action are conducted by Nuclear Oversight personnel.

17.2.17 QUALITY ASSURANCE RECORDS

Requirements are established for the identification, collection, and storage of quality assurance records. These requirements are consistent with the provisions of Regulatory Guides 1.33 and 1.88 as discussed in Appendix 1A(B).

Sufficient records are maintained to provide documentary evidence of the quality of items and of the accomplishment of activities affecting quality. Records to be maintained include such items as drawings, specifications, procurement documents, nonconformance reports, corrective action reports, operating logs, personnel and procedure qualifications, results of inspections and test, material certifications and test results, and audit reports.

Quality assurance records are maintained in accordance with procedures and instructions which assign responsibilities for the collection, maintenance, and protection of records. These procedures and instructions provide a system of record identification to assure retrievability and prescribe retention periods for various types of records.

Senior Director, Engineering & Regulatory Affairs is responsible for development of procedures and instructions to implement the management requirements related to QA records.

Quality assurance records are stored in specially constructed storage facilities at CPNPP to prevent their destruction, deterioration, or theft. These storage facilities are designed, constructed, and maintained in accordance with the applicable requirements of the regulatory guides referenced above. Access to the records facilities is controlled so that only authorized personnel have access to the records areas. As allowed by ANSI N45.2.9-1974, maintenance of duplicate records, stored in a remote location may be used as an alternative to the utilization of these storage facilities, and the appropriate administrative controls for the maintenance of duplicate records are prescribed by procedures and instructions. The program for storage of records on microfilm, dual storage or in electronic format meets the preservation requirement for retention of Quality Assurance records. Electronic records should meet the intent of NIRMA Guidelines TG 11-2011, TG 15-2011, TG 16-2011, and TG 21-2011.

17.2.17.1 Record Retention

In addition to the applicable record retention requirements of Title 10, Code of Federal Regulations, the following records shall be retained for at least the minimum period indicated.

The following records shall be retained for at least 5 years:

1. Records and logs of unit operation covering time interval at each power level;

2. Records and logs of principal maintenance activities, inspections, repair and replacement of principal items of equipment related to nuclear safety;

3. All events reported pursuant to 10CFR50.73;

4. Records of surveillance activities, inspections and calibrations required by the Technical Specifications, Technical Requirements Manual, and Fire Protection Report, except as explicitly covered in Section 17.2.17.1.2;

5. Records of changes made to the procedures required by Technical Specification 5.4.1;

6. Deleted

7. Records of sealed source and fission detector leak tests and results; and

8. Records of annual physical inventory of all sealed source material of record.

17.2.17.2 The following records shall be retained for the duration of the unit Operating License:

1. Records and drawing changes reflecting unit design modifications made to systems and equipment described in the Final Safety Analysis Report;

2. Records of new and irradiated fuel inventory, fuel transfers, and assembly burnup histories;

3. Records of doses received by all individuals for whom monitoring was required by 10CFR Part 20;

4. Records of gaseous and liquid radioactive material released to the environs;

5. Records of transient or operational cycles for those unit components identified in FSAR Table 3.9N-1A;

6. Records of reactor tests and experiments;

7. Records of training and qualification for current members of the unit staff;

8. Records of inservice inspections performed pursuant to the Technical Specifications;

9. Records of Nuclear Oversight activities required by the Quality Assurance Manual;

10. Records of reviews performed for changes made to procedures or equipment or reviews of tests and experiments pursuant to 10CFR50.59 to 10CFR50.59 and 10CFR72.48, as applicable;

11. Records of meetings of the SORC and the ORC;

12. Records of the service lives of all hydraulic and mechanical snubbers required by the Technical Requirements Manual including the date at which the service life commences and associated installation and maintenance records;

13. Records of secondary water sampling and water quality;

14. Records of analyses required by the Radiological Environmental Monitoring Program that would permit evaluation of the accuracy of the analysis at a later date. This should include procedures effective at specified times and QA records showing that these procedures were followed; and

15. Records of reviews performed for changes made to the OFFSITE DOSE CALCULATION MANUAL and the PROCESS CONTROL PROGRAM.

16. Records of radioactive shipments.

17.2.17.3 The following additional record retention requirements apply to the Dry Cask Storage System structures, systems, and components that are classified as important to safety in accordance with 10CFR72 and NUREG/CR-6407, "Classification of Transportation Packaging and Dry Spent Fuel Storage System Components According to Importance to Safety":

1. Documentary evidence that material and equipment conform to procurement specifications shall be retained for the life of the ISFSI or spent fuel storage cask as defined in the applicable appendix to the CPNPP QA Manual.

2. Records pertaining to the design, fabrication, erection, testing, maintenance, and use of structures, systems, and components important to safety shall be retained until the NRC terminates the license or Certificate of Compliance (CoC).

3. Records showing the receipt, inventory (including location), disposal, acquisition, and transfer of special nuclear material including spent fuel must be retained for as long as the material is stored and for a period of 5 years after the material is disposed of or transferred out of the ISFSI. These records must include, as a minimum, the name of the shipper of the material to the ISFSI, the estimated quantity of radioactive material per item (including special nuclear material in spent fuel), item identification and seal number, storage location, onsite movements of each fuel assembly or storage canister, and ultimate disposal. These records must be kept in duplicate at separate locations sufficiently remote from the original records that a single event would not destroy both sets of records.

17.2.18 AUDITS

Requirements are established for an Audit program. The Audit program is consistent with the applicable portions of Regulatory Guides 1.33, 1.144, and 1.146 as discussed in Appendix 1A(B).

Planned and periodic audits are performed in accordance with written procedures to verify compliance with all aspects of the quality assurance program. Audits shall be performed at the intervals designated herein for each audit area. Schedules shall be based on the month in which

the audit starts. Audit reports shall be forwarded to the Senior Vice President & Chief Nuclear Officer and to the management positions responsible for the areas audited within 30 days after completion of the audit by the auditing organization. Responsibility for the evaluation program has been assigned to the Director, Nuclear Oversight. Audits are conducted or coordinated by Nuclear Oversight personnel and shall include evaluation and examination of the following quality-related activities:

1. The conformance of unit operation to provisions contained within the Technical Specifications and applicable license conditions at least once per 24 months;

2. The performance, training and qualifications of the entire unit staff at least once per 24 months;

3. The results of actions taken to correct deficiencies occurring in unit equipment, structures, systems or method of operation that affect nuclear safety, at least once per 24 months;

4. The performance of activities required by the Operational Quality Assurance Program to meet the criteria of Appendix B, 10CFR50, at least once per 24 months;

5. The fire protection programmatic controls, including the implementing procedures, program implementation, and fire protection equipment at least once per 24 months by qualified licensee Nuclear Oversight personnel and qualified offsite fire protection engineers;

6. The fire protection equipment and program implementation at least once per 36 months utilizing an outside independent fire protection consultant.

7. The Radiological Environmental Monitoring Program and the results thereof at least once per 24 months;

8. The OFFSITE DOSE CALCULATION MANUAL and implementing procedures at least once per 24 months;

9. The PROCESS CONTROL PROGRAM and implementing procedures for processing and packaging of radioactive wastes at least once per 24 months;

10. The performance of activities required by the Quality Assurance Program for effluent and environmental monitoring at least once per 24 months.

11. Any other area of unit operation considered appropriate by the ORC or the Senior Vice President & Chief Nuclear Officer; and

12. The performance of activities required by the Technical Requirements Manual at least once per 24 months.

13. The performance of activities required by the Operational Quality Assurance Program to meet the criteria of subpart G, 10CFR72, at least once per 24 months.

17.2.18.1 Audit Organizations

Organizations performing activities affecting quality that are subject to evaluation include the following:

1. The nuclear engineering, operations, regulatory affairs and Nuclear Oversight organizations for CPNPP.

2. Contractors, consultants, and suppliers of quality related items or service. NVLAP and A2LA accredited commercial grade calibration suppliers do not require audit when evaluated in accordance with Section 17.2.7.

17.2.18.2 Nuclear Oversight

As part of the Quality Assurance Program Nuclear Oversight:

1. Utilizes an audit planning document which defines the organizations and activities to be evaluated and the frequency of the audits. The audit schedule is performance-based and additional audits may be scheduled based upon program or functional area performance or other factors that indicate the need for increased assessment. For audits, other than those audit areas with a maximum frequency specified by regulation (Security and Emergency Plan), an interval extension grace period of up to 25% may be used for the scheduled commencement date when conditions, such as plant operational considerations or to accommodate performance-based observations of activities related to the audit area, make meeting the specified schedule date impractical. When an audit interval extension greater than one month is used, the next audit for that particular audit area will be scheduled from the original anniversary month rather than from the month of the extended audit.

2. Requires auditors to be familiar with the type of activities to be evaluated and have no direct responsibilities in the area being evaluated.

3. Provide audit checklists or other objective guidelines to identify those activities which affect quality.

4. Requires examination of the essential characteristics of the quality activity examined.

5. Requires an audit report be prepared that notes the extent of examination and deficiencies found.

6. Requires the audit report be sent to management responsible for the area evaluated for review and corrective action for deficiencies.

7. Requires corrective action taken as result of the audit be reported.

8. Requires re-audit of deficient areas when it is considered necessary to verify implementation of required corrective actions.

9. Requires vendors/subcontractors to comply with items 1-8 above to the extent necessary.

Documentation of audits performed by participating contractors is made available to Nuclear Generation for evaluation.

17.2.18.2.1 Nuclear Generation Verification

In summary, Nuclear Generation verifies conformance of the regulatory audit requirements by three methods:

1. Review of contractor's/vendors' quality assurance methods for auditing.

2. Review of documentation of the audit report performed by these contractors/vendors.

3. Internal and external evaluations performed by Nuclear Oversight personnel.

TABLE 17.2-1
CPNPP QA MANUAL COMPLIANCE MATRIX

(Sheet 1 of 2)

COMANCHE PEAK QUALITY ASSURANCE MANUAL		APPENDIX B QUALITY ASSURANCE CRITERIA																	
		I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII	XVIII
1.0	Organization	X																	
2.0	Quality Assurance Plan		X																
3.0	Design Control			X															
4.0	Procurement Document Control				X														
5.0	Instructions, Procedures and Drawings					X													
6.0	Document Control						X												
7.0	Control of Purchased Items and Services							X											
8.0	Identification and Control Items								X										
9.0	Control of Construction Processes									X									

TABLE 17.2-1
CPNPP QA MANUAL COMPLIANCE MATRIX

(Sheet 2 of 2)

COMANCHE PEAK QUALITY ASSURANCE MANUAL		APPENDIX B QUALITY ASSURANCE CRITERIA																	
		I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV	XVI	XVII	XVIII
10.0	Examinations, Tests and Inspections										X								
11.0	Test Control											X							
12.0	Control of Measuring and Test Equipment												X						
13.0	Handling, Storage, and Preservation													X					
14.0	Examination or Test Status														X				
15.0	Nonconforming Items															X			
16.0	Corrective Action																X		
17.0	Quality Assurance Records																	X	
18.0	Audits																		X

TABLE 17.2-2
REGULATORY GUIDES AND INDUSTRY STANDARDS

(Sheet 1 of 2)

The CPNPP quality assurance program is consistent with the applicable guidance of the NRC Regulatory Guides and industry standards listed below. Vistra OpCo will commit to comply with the respective regulatory positions as discussed in Appendix 1A(B).

Regulatory Guide	Title
1.8	Personnel Selection and Training (endorses ANSI N18.1-1971)
1.26	Quality Group Classifications and Standards for Water-, Steam-, and Radioactive- Waste-Containing Components of Nuclear Power Plants (see Appendix 1A(B) for CPNPP position and compliance)
1.29	Seismic Design Classification
1.30	Quality Assurance Requirements for Installation, Inspection, and Testing of Instrumentation and Electric Equipment (endorses ANSI N45.2.4-1972)
1.33	Quality Assurance Program Requirements (Operations) (endorses ANSI N18.7-1976)
1.37	Quality Assurance Requirements for Cleaning of Fluid Systems and Associated Components of Water-Cooled Nuclear Power Plants (endorses ANSI N45.2.1-1973)
1.38	Quality Assurance Requirements for Packaging, Shipping, Receiving, Storage, and Handling of Items for Water-Cooled Nuclear Power Plants (endorses ANSI N45.2.2-1972)
1.39	Housekeeping Requirements for Watercooled Nuclear Power Plants (endorses ANSI N45.2.3-1973)
1.58	Qualification of Nuclear Power Plant Inspection. Examination, and Testing Personnel (endorses ANSI N45.2.6-1978)
1.64	Quality Assurance Requirements for Design of Nuclear Power Plants (endorses ANSI N45.2.11-1974)
1.74	Quality Assurance Terms and Definitions (endorses ANSI N45.2.10-1973)

TABLE 17.2-2
REGULATORY GUIDES AND INDUSTRY STANDARDS

(Sheet 2 of 2)

Regulatory Guide	Title
1.88	Collection, Storage and Maintenance of Nuclear Power Plant Quality Assurance Records (endorses ANSI N45.2.9-1974)
1.94	Quality Assurance Requirements for Installation, Inspection, and Testing of Structural Concrete and Structural Steel During the Construction Phase of Nuclear Power Plants (endorses ANSI N45.2.5-1974)
1.116	Quality Assurance Requirements for Installation, Inspection, and Testing of Mechanical Equipment and Systems (endorses ANSI N45.2.8-1975)
1.123	Quality Assurance Requirements for Control of Procurement of Items and Services for Nuclear Power Plants (endorses ANSI N45.2.13-1976)
1.144	Auditing of Quality Assurance Programs for Nuclear Power Plants (endorses ANSI N45.2.12 - see Appendix 1A(B) for CPNPP position and compliance)
1.146	Qualification of Quality Assurance Program Audit Personnel for Nuclear Power Plants (endorses ANSI N45.2.23-1978)

17A LIST OF QUALITY ASSURED ITEMS

17A.1 SCOPE OF QA APPLICABILITY

This appendix identifies major safety-related items for Comanche Peak Nuclear Power Plant (CPNPP) within the scope of the nuclear quality assurance (QA) programs established to meet the requirements of Appendix B to 10 CFR Part 50. Safety-related items listed require special consideration during design, purchasing, fabrication, handling, shipment, storage, cleaning, erection, installation, inspection, testing, operation, maintenance, repair, refueling, and modification. These items include structures, systems, and components or portions thereof that prevent or mitigate the consequences of postulated accidents that could cause undue risk to the health and safety of the public.

This appendix also identifies major non-safety related items for CPNPP and the appropriate level of quality assurance where applicable.

Table 17A-1 (the Q-List) identifies the systems and components and their applicable quality assurance requirements for the operation of CPNPP.

The quality assurance applied to the safety related structures, systems, and components as identified in Table 17A-1 are consistent with the safety function that a structure, system, or component is to perform and are determined by engineering as a design function. Table 17A-2 provides the quality assurance notes applicable to Table 17A-1 and the classification of equipment to which they are applied.

Respective descriptions of the quality assurance related activities of these organizations (Criteria I through XVIII of the Appendix B to 10 CFR Part 50) are presented in Sections 17.1 and 17.2. As a minimum, Table 17A-1 includes all safety-related structures, systems, and components discussed in Section 3.2.1.1 (Seismic Category I) and Section 3.2.1.2 (Seismic Category II).

Table 17A-1 also includes other items of the plant not classified as safety-related but for which some degree of quality assurance applies. These items augment reliable performance of safety-related items. In order to ensure that appropriate quality assurance measures are applied to these items and other programmatic activities during the operations phase, Vistra OpCo has established a specifically structured non-Appendix B quality assurance program to augment the Appendix B quality assurance program for safety-related systems and activities described in Section 17.2. This specifically structured program is defined in an Appendix to the CPNPP QA Manual.

The existing quality program controls established to meet 10CFR50 Appendix B may be applied to the Dry Cask Storage System structures, systems and components that are important to safety. The applicability of existing quality assurance program requirements for the Dry Cask Storage System are defined in an Appendix to the CPNPP QA Manual.

Changes to FSAR Table 17A-1 are reviewed and accepted by the Director, Nuclear Oversight. The FSAR amendment promulgating the change is approved and submitted by the Senior Vice President & Chief Nuclear Officer.

17A.2 CRITERIA FOR APPLICATION OF QA

Because the Table 17A-1 serves as a basis for identifying safety- related structures, systems, and components which require consideration in accordance with the QA programs, the following criteria were applied for establishment of the list:

1. Safety Classification

   Systems and components, consistent with their function, were designated as Safety Class 1, 2, 3, 1E, Non-1E, N/A or non-nuclear safety (NNS) (see Section 3.2.2). Safety-related components or their portions designed to the ASME B&PV Code, Section III, are subject to QA programs, and are listed in Table 17A-1.

2. Seismic Category Classification

   a. Seismic Category I

      Structures, systems, and components designated as seismic Category I in accordance with Reg.Guide 1.29 are subject to QA and are listed in the Table 17A-1.

   b. Seismic Category II

      Systems and components which require some seismic design requirements in accordance with Reg.Guide 1.29, but not to the extent of seismic Category I, are listed as seismic Category II.

   c. Seismic Category NONE

      Those systems and components which have no seismic design requirements are listed as seismic Category “NONE”.

3. Additional Classification of Mechanical Components

   All NNS piping and plumbing designated as Class 5 are listed. See Section 3.2.2 for a discussion of the Class 5 categorization.

4. Fire Protection

   Systems and components provided for fire protection for safety- related areas and subject to quality assurance requirements as described in Section 9.5.1 are listed.

5. Radioactive Waste Management Systems (RWMS)

   Systems and components which process radioactive waste are included. The boundaries of RWMS are shown on the flow diagrams.

6. Other

Those systems and components which are given some QA because of criteria not discussed above are also listed. Some components which have no QA requirements are also listed to satisfy information requirements of FSAR Section 3.2 and to provide clarification on QA applicability.

17A.3 INDEX

Table 17A-1 is divided into the following systems:

SYSTEM NO.	SYSTEM
1.	Reactor Coolant System (RCS)
2.	Chemical and Volume Control System (CVCS)
3.	Boron Thermal Regeneration Subsystem (BTRS)
4.	Safety Injection System (SIS)
5.	Residual Heat Removal (RHR) System
6.	Boron Recycle System (BRS)
7.	Containment Spray System (CSS)
8.	Containment Isolation System
9.	Combustible Gas Control System
10.	Component Cooling Water System (CCWS)
11.	Station Service Water System (SSWS)
12.	Main Steam, Reheat and Steam Dump System
13.	Auxiliary Feedwater System
14.	Steam Generator Feedwater System
15.	Diesel Generator, Fuel Oil, and Auxiliary Systems
16.	Spent Fuel Pool Cooling and Cleanup System
17	Liquid Waste Processing System (LWPS)
18.	Gaseous Waste Processing System (GWPS)
19.	Solid Waste Management System (SWMS)
20.	Demineralized and Reactor Makeup Water System
21.	Vents and Drains System

SYSTEM NO.	SYSTEM
22.	Containment Ventilation Systems
23.	Control Room Air-Conditioning System
24.	Safeguards Building HVAC System
25.	Fuel Building Ventilation System
26.	Diesel Generator Building Ventilation System
27.	Uncontrolled Access Area Ventilation System
28.	Primary Plant Ventilation System
29.	Auxiliary Building HVAC System
30.	Service Water Intake Structure Ventilation System
31.	Chilled Water Systems
32.	Process Sampling System
32.a	Post Accident Sampling System
32.b	Secondary Sampling System
33.	Fuel Handling Equipment and Fuel Assemblies
34.	Containment Building Miscellaneous Equipment
35.	Miscellaneous Handling Equipment
36.	Structures
37.	Electrical Equipment
38.	Radiation Monitoring System
39.	Fire Protection System
40.	Plant Gas System
41.	Instrumentation and Control Equipment
42.	Tornado Venting Components
43.	Compressed Air Systems
44.	Protective Coatings
45.	Potable and Sanitary Water System

SYSTEM NO.	SYSTEM
46.	Condensate System
47.	Auxiliary Steam System
48.	Steam Generator Blowdown & Cleanup System
49.	Pipe Whip Restraints
50.	Meteorological Instrumentation
51.	Uninterruptible Power Supply (UPS) Area Air-Conditioning System
52.	Turbine Plant Cooling Water System
53.	Condensate Polishing System
54.	Condenser Vacuum and Waterbox Primary System
55.	Heater Drains System
56.	Chemical Feed System
57.	Security Systems
58.	Chlorination Systems

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

(Sheet 1 of 59)

System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
1. Reactor Coolant System (RCS)							
Reactor vessel	1	ASME III	1	I	Note A	5.3	
CRDM housings	1	ASME III	1	I	Note A		
CRDM head adapter plugs	1	ASME III	1	I	Note A		
Reactor vessel supports	1	ASME III	1	I	Note A	5.4.14	
Steam generator							
Tube side 1	1	ASME III	1	I	Note A	5.4.2	
Shell side 1	2	ASME III	1	I	Note A		Note 8
Steam generators supports	1	ASME III	1	I	Note A	5.4.14	
Steam generator restraints	1	ASME III	1	I	Note A		
Pressurizer	1	ASME III	1	I	Note A	5.4.10, II.E.3.1	Note 1b
Pressurizer support skirt	1	ASME III	1	I	Note A	5.4.14	
Reactor coolant hot- and cold-leg piping and fittings, and fabrication	1	ASME III	1	I	Note A	5.4.3	
Surge pipe & fittings and fabrication	1	ASME III	1	I	Note A	5.4.10	
Piping and Valves (including RCS vents)	2	ASME III	2	I	Note A	II.B.1	Note 79
Piping and valves (including RCS vents)	1	ASME III	1	I	Note A	3.9N, II.B.1	Note 79
Crossover leg piping & fittings and fabrication	1	ASME III	1	I	Note A	5.4.3	
Pressurizer safety valves	1	ASME III	1	I	Note A	5.4.13	Note 79
Power-operated relief valves	1	ASME III	1	I	Note A	5.4.13	Note 79
PORV accumulators	3	ASME III	3	I	Note A		

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

(Sheet 2 of 59)

System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Check valves for PORV Accumulators	3	ASME III	3	I	Note A		Note 79
Tubing between piping and PORV Accumulators (Unit 1 Only)	3	ASME III	-	I	Note A	3.9B	Note 41
Pressurizer PORV Block Valves	1	ASME III	1	I	Note A	5.4.12	Note 79
PORV and Safety Valve Limit Switches	1E	IEEE-323	-	I	Note A	II.D.3	
Valves of Safety Class 1 to Safety Class 2 interface	1	ASME III	1	I	Note A	5.4.12	Note 79
High Point vent valves	2	ASME III	2	I	Note A	5.1	Note 79
Pressurizer relief tank	NNS	ASME VIII	-	NONE	Note E	5.4.11	
Reactor Coolant Pump							
Casing	1	ASME III	1	I	Note A	5.4.1	
Main flange	1	ASME III	1	I	Note A		
Thermal barrier	1	ASME III	1	I	Note A		
Thermal barrier heat exchanger	1	ASME III	1	I	Note A		
No. 1 seal housing	1	ASME III	1	I	Note A		Note 97
No. 2 seal housing	2	ASME III	1	I	Note A		Note 8
Pump Shaft	2	-	-	I	Note A	5.4.1	
Pump Impeller	2	-	-	I	Note A	5.4.1	
Pressure-retaining bolting	1	ASME III	1	I	Note A		
Vertical and lateral supports	1	ASME III	1	I	Note A	5.4.14	
Reactor Coolant Pump Motor							
Motor rotor	2	-	-	I	Note A		Note 9
Motor shaft	2	-	-	I	Note A		Note 9
Shaft coupling	2	-	-	I	Note A		Note 9

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

(Sheet 3 of 59)

System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Spool piece	2	-	-	I	Note A		Note 9
Flywheel	2	-	-	I	Note A		Note 9
Bearing (Motor upper thrust)	2	-	-	I	Note A		Note 9
Motor bolting	2	-	-	I	Note A		Note 9, 30
Motor stand	2	-	-	I	Note A		Note 9
Motor frame	2	-	-	I	Note A		Note 9
Upper oil reservoir	3	Mfrs Stds	-	I	Note A		
Cooling coil	3	ASME III	3	I	Note A		
Lower oil reservoir	3	Mfrs Stds	-	I	Note A		
Cooling coil	3	ASME III	3	I	Note A		
Lube-oil piping	3	Mfrs Stds	-	I	Note A		
Motor-air coolers	3	ASME III	3	I	Note A		Note 2
Reflective insulation assemblies (except for the portions installed on the RCS cold leg and hot leg pipes which are located inside the biological shield tunnels)	N/A	Mfrs Stds	-	II	Note B	6.1B, 6.2.2	
Reactor vessel nozzles non-crushable insulation	N/A	Mfrs Stds	-	II	Note B	3.9N.1.4.6	Note 64
Reflective insulation assemblies on RCS cold leg and hot leg pipes and located in biological shield tunnels	N/A	Mfrs Stds	-	NONE	Note E	6.1B	
Supports for Class 1 piping	1	ASME III	1	I	Note A	3.9N	
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

(Sheet 4 of 59)

System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
2. Chemical and Volume Control System (CVCS)							
Regenerative heat exchanger	2	ASME III	2	I	Note A	9.3.4	
Letdown Heat Exchanger							
Tube side	2	ASME III	2	I	Note A	9.3.4	
Shell side	3	ASME III	3	I	Note A	9.3.4	Note 1c, 2
Mixed-bed demineralizers	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Cation-bed demineralizer	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Reactor coolant filter	2	ASME III	2	I	Note A	9.3.4	
Volume control tank	2	ASME III	2	I	Note A	9.3.4	
Centrifugal charging pump	2	ASME III	2	I	Note A	9.3.4	Note 1a, 1d, 2
Centrifugal charging pump lube oil system piping and valves	3	Mfrs Stds	-	I	Note A	9.3.4	
Positives displacement pump	2	ASME III	2	I	Note A	9.3.4	Note 1a, 1c, 2
Positive displacement pump discharge dampener	2	ASME III	2	I	Note A	9.3.4	
Positive displacement pump suction stabilizer tank	2	ASME III	2	I	Note A	9.3.4	
Seal water injection filter	2	ASME III	2	I	Note A	9.3.4	
Letdown orifices	2	ASME III	2	I	Note A	9.3.4	
Excess letdown heat exchanger							
Tube side	2	ASME III	2	I	Note A	9.3.4	
Shell side	2	ASME III	2	I	Note A	9.3.4	Note 1c
Seal water return filter	2	ASME III	2	I	Note A	9.3.4	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Seal Water Heat Exchanger							
Tube side	2	ASME III	2	I	Note A	9.3.4	
Shell side	3	ASME III	3	I	Note A	9.3.4	Note 1c, 2
Boric acid transfer pump	3	ASME III	3	I	Note A	9.3.4	Note 1b
Boric acid filter	3	ASME III	3	I	Note A	9.3.4	
Boric acid batching tank	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Chemical mixing tank	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Reactor coolant pump seal bypass orifice	1	ASME III	1	I	Note A	9.3.4	
R.C. Pump standpipe	NNS	ASME VIII	-	NONE	Note E		
Boric acid tanks	3	ASME III	3	I	Note A	9.3.4	
Boric acid blender	3	ASME III	3	I	Note A	9.3.4	
Piping and valves	2	ASME III	2	I	Note A	9.3.4	Note 34, 79
Piping and valves	3	ASME III	3	I	Note A	9.3.4	Note 34, 79
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Reflective insulation assemblies	NNS	Mfrs Stds	-	II	Note B	6.1, 6.2.2	
3. Boron Thermal Regeneration Sub-system (BTRS)							
Moderating Heat Exchanger							
Tube side	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Shell side	3	ASME III	3	NONE	Note A	9.3.4	Note 11

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Letdown Chiller Heat Exchanger							
Tube side	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Shell side	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Letdown Reheat Heat Exchanger							
Tube side	2	ASME III	2	I	Note A	9.3.4	
Shell side	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Thermal regeneration demineralizers	3	ASME III	3	NONE	Note A	9.3.4	Note 11
Chiller pump	NNS	Mfrs Stds	-	NONE	Note E	9.3.4	
Chiller surge tank	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Chiller Unit							
Evaporator	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Condenser	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Compressor	NNS	Mfrs Stds	-	NONE	Note E	9.3.4	
Piping and valves	2	ASME III	2	I	Note A	9.3.4	Note 79
Piping and valves	3	ASME III	3	I	Note A	9.3.4	Note 79
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
4. Safety Injection System (SIS)							
Accumulators	2	ASME III	2	I	Note A	6.3	
Intermediate-head SIS pumps	2	ASME III	2	I	Note A	6.3	Note 1a, 1d, 2

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Intermediate-head SIS pumps lube oil system piping and valves	3	Mfrs Stds	-	I	Note A	6.3	
Valve isolation tanks	NNS	ASME VIII	-	II	Note B	6.2, 6.3	
Valve isolation tank expansion joints	NNS	ASME VIII	-	II	Note B	6.2, 6.3	
Valve isolation tank piping	NNS	ANSI B31.1	-	II	Note B	6.3	Note 57
Piping and valves	2	ASME III	2	I	Note A	6.3	Note 79
Piping and valves	1	ASME III	1	I	Note A	6.3	Note 79
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 1 piping	1	ASME III	1	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Reflective insulation assemblies	NNS	Mfrs Stds	-	II	Note B	6.1, 6.2.2	
Piping (Injection Header Alternate Relief Path)	NNS	ANSI B31.1	-	II	Note B	6.3.5	Note 44, 57
5. Residual Heat Removal (RHR) System							
RHR pump	2	ASME III	2	I	Note A	5.4.7	Note 1a, 1c, 2
Residual Heat Exchanger							
Tube side	2	ASME III	2	I	Note A	5.4.7	
Shell side	3	ASME III	3	I	Note A	5.4.7	Note 1c
Piping and valves	1	ASME III	1	I	Note A	5.4.7	Note 79
Piping and valves	2	ASME III	2	I	Note A	5.4.7	Note 79
Process tubing	2	ASME III	2	I	Note A	5.4.7.2.2	
Supports for Class 1 piping	1	ASME III	2	I	Note A	3.9B	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Reflective insulation assemblies	NNS	Mfrs Stds	-	II	Note B	6.1, 6.2.2	
Relief Valves	2	ASME III	2	I	Note A	5.4.7	Note 79
6. Boron Recycle System (BRS)							
Recycle evaporator feed pump	NNS	Mfrs Stds	-	NONE	Note C	9.3.4	Note 34
Recycle evaporator feed demineralizer	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34
Recycle evaporator feed filter	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34
Recycle evaporator condensate demineralizer	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Recycle evaporator condensate filter	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Recycle holdup tank	NNS	API 650 or AWWA D-100	-	NONE	Note C	9.3.4	Note 34
Recycle holdup tank vent ejector	NNS	ANSI B31.1 or ASME VIII	-	NONE	Note C	9.3.4	Note 34
Recycle evaporator concentrate filter	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Recycle evaporator reagent tank	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Recycle evaporator package							
Feed preheater							
a. feed side	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34
b. steam side	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Gas stripper	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34
Submerged tube Evaporator							
a. feed side	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
b. steam side	NNS	ASME VIII	-	NONE	Note E	9.3.4	
Evaporator condenser							
a. Distillate water side	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34, 72
b. Cooling water side	3	ASME III	3	I	Note A	9.3.4	
Distillate Cooler							
a. Distillate side	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34, 72
b. Cooling water side	3	ASME III	3	I	Note A	9.3.4	
Absorption tower Vent Condenser	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34
a. Gas side	NNS	ASME VIII	-	NONE	Note C	9.3.4	Note 34, 72
b. Cooling water side	3	ASME III	3	I	Note A	9.3.4	
Distillate pump	NNS	Mfrs Stds	-	NONE	Note C	9.3.4	Note 34
Concentrate pump	NNS	Mfrs Stds	-	NONE	Note C	9.3.4	Note 34
Piping and valves							
a. Feed	NNS	ANSI B31.1	-	NONE	Note C	9.3.4	Note 34
b. Distillate	NNS	ANSI B31.1	-	NONE	Note C	9.3.4	Note 34
c. Concentrate	NNS	ANSI B31.1	-	NONE	Note C	9.3.4	Note 34
d. Cooling	3	ASME II	3	I	Note A	9.3.4	Note 79
e. Vent	NNS	ANSI B31.1	-	NONE	Note C	9.3.4	Note 34
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Valves	NNS	ANSI B31.1	-	NONE	Note C		Note 34
7. Containment Spray System (CSS)							
Chemical additive tank	3	ASME III	3	I	Note A	6.2.2, 6.5	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Containment spray pumps	2	ASME III	2	I	Note A	6.2.2	Note 1a, 1c, 2
Chemical Eductor	2	ASME III	2	I	Note A	6.5	
Containment Spray Heat Exchanger							
Tube side	2	ASME III	2	I	Note A	6.2.2	
Shell side	3	ASME III	3	I	Note A	6.2.2	Note 1c
Spray nozzles	2	Mfrs Stds	-	I	Note A	6.2.2	
Ring header line orifices	2	ASME III	2	I	Note A	6.2.2	
Emergency Sump Suction Strainers	N/A	AISC	-	I	Note A	6.2.2	
Debris Interceptors	N/A	AISC	-	I	Note A	6.2.2	
Trash Racks	N/A	AISC	-	II	Note B	6.2.2	
Piping and Valves	2	ASME III	2	I	Note A	6.2.2	Note 79
Piping and Valves	3	ASME III	3	I	Note A	6.2.5, 6.5	Note 79
Valve isolation tanks	NNS	ASME VIII	-	II	Note B	6.2	
Valve isolation tank expansion joints	NNS	ANSI B31.1	-	II	Note B	6.2	
Valve isolation tank piping	NNS	ANSI B31.1	-	II	Note B	6.2	Note 57
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Refueling water storage tank	2	ACI 318-71 AISC	-	I	Note A	3.8.4, 6.2.2	Note 55
Refueling water storage tank vents	NNS	B31.1	-	II	Note B	6.3.3.7	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
8. Containment Isolation System							
Piping and valves of all process systems penetrating Containment; from isolation inside Containment to isolation outside Containment	2	ASME III	2	I	Note A	6.2.4, II.E.4.2	Note 79
Penetration assemblies (Mechanical and electrical)	2	ASME III	2 & MC	I	Note A	6.2.4, 8.3.1	
Process Sampling Tubing, Fittings, Valves, and Supports	2	ASME III	-	I	Note A	6.2.4	Note 41
Airlock control, instrument and testing tubing, fittings and components	2	-		I	Note A		Note 41
Airlock instrumentation (pressure gauges)	2	-	-	I	Note A	-	Note 95
Personnel Airlock Hydraulic System (Unit 1)	2	Mfrs Stds/ASME	-	I	Note A	6.2.4	Note 86
Alternate Penetration Insert	NNS	-	-	NONE	Note E	Figure 9.4-6	Note 94
9. Combustible Gas Control System							
a. Hydrogen Recombiner System							
Electric hydrogen recombiner	NA	NA	-	II	Note B	6.2.5	
b. Hydrogen Purge System							
Exhaust filter units	NNS	Mfrs Stds	-	II	Note B		
Exhaust ductwork, supports & dampers (outside Containment)	NNS	Mfrs Stds	-	II	Note B		Note 71
Exhaust Fans	NNS	Mfrs Stds	-	NONE	Note E		Note 70
Supply ductwork, supports & dampers (outside Containment)	NNS	Mfrs Stds	-	II	Note B		Note 71
Supply blowers	NNS	Mfrs Stds	-	NONE	Note E		Note 70
Piping & valves	2	ASME III	2	I	Note A		Note 79

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B		Note 44
Supports for Class 2 piping	2	ASME III	2	I	Note A		
10. Component Cooling Water System (CCWS)							
Heat exchangers	3	ASME III	3	I	Note A	9.2.2	Note 1d
Pumps	3	ASME III	3	I	Note A	9.2.2	Note 1a
Surge tank	3	ASME III	3	I	Note A	9.2.2	
Piping and valves	2	ASME III	2	I	Note A	9.2.2	Note 79
Piping and valves	3	ASME III	3	I	Note A	9.2.2	Note 79
Piping (Rad. Monitor Sample lines, safeguards loops)	NNS	ANSI B31.1	-	II	Note B	9.2.2	Note 44, 57
Recirculation loop orifice	3	ASME III	3	I	Note A	9.2.2	
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Air accumulators (Flow Control Valves)	3	ASME III	-	I	Note A	9.3.1	
Check valves for accumulators	3	ASME III	-	I	Note A	9.3.1	Note 79
Tubing and supports (between check valves upstream of air accumulator and AOV)	3	ASME III	-	I	Note A	3.9B	Note 41
11. Station Service Water System (SSWS)							
Service Water Pumps	3	ASME III	3	I	Note A	9.2.1	Note 1a
Piping and valves	3	ASME III	3	I	Note A	9.2.1	Note 79
Screen wash pumps	NNS	Mfrs Stds	-	NONE	Note E	9.2.1	Note 72

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Traveling screens	N/A	Mfrs Stds	-	I	Note A	9.2.1	Note 23, 46
Recirculation loop orifice	3	ASME III	3	I	Note A	9.2.1	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7b	Note 44
SWIS Debris and Fish Barrier System	**N/A**	**Mfrs Stds**	**-**	**II**	**Note B**		**Note 23 [SWIS: SW Intake Structure]**
12. Main Steam, Reheat and Steam Dump System							
Main steam piping	2	ASME III	2	I	Note A	10.3	
Piping, valves, and drain pots	2	ASME III	2	I	Note A	10.3	Note 79
Piping, valves, and drain pots	3	ASME III	3	I	Note A	10.3	Note 79
Main steam safety valves	2	ASME III	2	I	Note A	10.3	Note 79
Main steam relief valves	2	ASME III	2	I	Note A	10.3	Note 79
Steam generator PORV air accumulator	3	ASME III	-	I	Note A	Appendix 5A, 9.3.1	
Turbine driven auxiliary feedwater pump steam supply isolation valve accumulator tanks	3	ASME III	-	I	Note A	9.3.1	
Check valves for accumulator tanks	3	ASME III	-	I	Note A	9.3.1	Note 79
Tubing and supports (between check valves upstream of air accumulator and AOV)	3	ASME III	-	I	Note A	3.9B	Note 41
Steam generator blowdown system piping	2	ASME III	2	I	Note A	10.3	
Steam flow restrictor (integral to steam generator)	2	ASME III	2	I	Note A	10.3, 5.4.4	
Main steam isolation valves	2	ASME III	2	I	Note A	10.3	Note 8, 79

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Main steam isolation bypass valves and bypass valve piping	2	ASME III	2	I	Note A	10.3	Note 8, 79
Reflective insulation assemblies	NNS	Mfrs Stds	-	II	Note B	6.2.2	
Piping and valves	2	ASME III	2	I	Note A	10.3	Note 79
Piping and valves	3	ASME III	3	I	Note A	10.3	Note 79
Orifices	3	ASME III	3	I	Note A	10.3	
Supports for Class 2 Piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 Piping	3	ASME III	3	I	Note A	3.9B	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44, 81
13. Auxiliary Feedwater System							
Motor-driven auxiliary feedwater pumps	3	ASME III	3	I	Note A	10.4.9	Note 1a
Turbine-driven auxiliary feedwater pump	3	ASME III	3	I	Note A	10.4.9	Note 1e
Auxiliary feedwater pump turbine driver and associated equipment	3	Mfrs Stds	-	I	Note A	10.4.9, 10.3	
Piping and valves	2	ASME III	2	I	Note A	10.4.9	Note 79
Piping and valves	3	ASME III	3	I	Note A	10.4.9	Note 79
Air accumulators (AFW Flow and Miniflow Control Valves)	3	ASME III	-	I	Note A	9.3.1	
Check valves for accumulators	3	ASME III	-	I	Note A	9.3.1	Note 79
Tubing and supports (between check valves upstream of accumulator and AOV)	3	ASME III	-	I	Note A	3.9B	Note 41
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Condensate storage tank	3	ACI 318-71 AISC	-	I	Note A	3.8.4, 10.4.9	Note 55
Condensate storage tank vents	NNS	B31.1	-	II	Note B	9.2.6	
Breakdown orifices	3	ASME III	3	I	Note A	10.4.9	
Orifices	2/3	ASME III	2/3	I	Note A	10.4.9	
14. Steam Generator Feedwater System							
Piping and valves	2	ASME III	2	I	Note A	10.4.7	Note 79
Feedwater isolation valves	2	ASME III	2	I	Note A	10.4.7	Note 79
Orifices	2	ASME III	2	I	Note A	10.4.7	
Reflective insulation assemblies	NNS	Mfrs Stds	-	II	Note B	6.2.2	
Supports for Class 2 Piping	2	ASME III	2	I	Note A	3.9B	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Feedwater Pump Turbine Speed Control	N1E	Mfrs Stds	-	NONE	Note C	10.4.7.5	
15. Diesel Generator, Fuel Oil, and Auxiliary Systems						8.3.1, 9.5.4, 9.5.5, 9.5.6, 9.5.7, 9.5.8	
Diesel Generator Set (including voltage regulator and excitation system)	1E	Mfrs Stds	-	I	Note A		Note 22
Diesel generator Fuel Oil Storage & Transfer System						9.5-52	
Fuel Oil Storage Tank	3	ASME III	3	I	Note A		
Fuel Oil Transfer Pumps	3	Mfrs Stds	-	I	Note A		Note 21

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Y-Strainers	3	ASME III	3	I	Note A		
Fuel Oil Day Tank	3	ASME III	3	I	Note A		
Piping and Valves	3	ASME III	3	I	Note A		Note 79
Supports for piping and components	3	ASME III	3	I	Note A		Note 25
Vacuum Relief Valves	NNS	Mfrs Stds	-	NONE	Note C	9.5.4	
Flame Arrestors	NNS	Mfrs Stds	-	NONE	Note C	9.5.4	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Fuel Oil System on Diesel Engine and on Aux. module							
a. Duplex Strainers w/change valves	N/A	Mfrs Stds	-	I	Note A		Note 22, 79
b. Fuel Oil Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
c. Fuel Oil Booster (Feed) Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
d. Duplex Filter w/change valves	N/A	Mfrs Stds	-	I	Note A		Note 22, 79
e. Injection Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
f. Piping and valves							
On engine	N/A	Mfrs Stds	-	I	Note A		Note 22, 79
On Aux. module	3	ASME III	-	I	Note A		Note 79
g. Supports for piping and components							
On engine	N/A	Mfrs Stds	-	I	Note A		Note 22
On Aux. module	3	ASME III/AISC	-	I	Note A	9.5.4.3	Note 25
h. Drip Waste Tank	N/A	Mfrs Stds	-	I	Note A		Note 22
i. Drip Waste Pump	N/A	Mfrs Stds	-	I	Note A		Note 22

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Diesel Generator Starting Air System						9.5-55	
Compressors	N/A	Mfrs Stds	-	NONE	Note E		Note 22, 72
After Coolers	N/A	Mfrs Stds	-	NONE	Note E		Note 22, 72
Air Dryers	N/A	Mfrs Stds	-	NONE	Note E		Note 22, 72
Starting Air Receivers	3	ASME III	3	I	Note A		
Piping and Valves							
On Engine	N/A	Mfrs Stds	-	I	Note A		Note 22, 79
Off Engine	3	ASME III	3	I	Note A		Note 79
Supports for piping and components							
On engine	N/A	Mfrs Stds	-	I	Note A		Note 22
Off Engine	3	ASME III	3	I	Note A	3.9B	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B		Note 44
Diesel Generator Jacket Water System						9.5-54	
Jacket Water Cooler	3	ASME III	-	I	Note A		
Standpipe	3	ASME III	-	I	Note A		
Jacket Water Heater	N/A	Mfrs Stds	-	I	Note A		Note 22
Engine Jacket Water Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
Auxiliary Jacket Water Pumps	3	ASME III	-	I	Note A		
Keep Warm Pump	3	ASME III	-	I	Note A		
Thermostatically Controlled Valve	3	ASME III	-	I	Note A		Note 79
Piping and Valves							
On Engine	N/A	Mfrs Stds	-	I	Note A		Note 22, 79

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
On Aux. Module	3	ASME III	-	I	Note A		Note 79
Supports for piping and components							
On engine	N/A	Mfrs Stds	-	I	Note A		Note 22
On Aux. module	3	ASME III/AISC	-	I	Note A	9.5.5.3	Note 25
Diesel Generator Lube Oil System						9.5-56	
Lube Oil Sump Tank	3	ASME III	-	I	Note A		
Engine Lube Oil Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
Auxiliary Lube Oil Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
Lube Oil Cooler	3	ASME III	-	I	Note A		
Duplex Filter	3	ASME III	-	I	Note A		
Strainer	N/A	Mfrs Stds	-	I	Note A		Note 22
Lube Oil Heater	N/A	Mfrs Stds	-	I	Note A		Note 22
Prelube Pump	N/A	Mfrs Stds	-	I	Note A		Note 22
Prelube Filter	3	ASME III	-	I	Note A		
Prelube Strainer	N/A	Mfrs Stds	-	I	Note A		Note 22
Piping and Valves							
On Engine	N/A	Mfrs Stds	-	I	Note A		Note 22, 79
On Aux. Module	3	ASME III	-	I	Note A		Note 79
Foot Valves (in tank)	3	Mfrs Stds	-	I	Note A		Note 22, 79
Supports for piping and components							
On engine	N/A	Mfrs Stds	-	I	Note A		Note 22
On Aux. module	3	ASME III/AISC	-	I	Note A	9.5.7.3	Note 25

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Diesel Generator Combustion Air Intake & Exhaust System						9.5-57	
Intake Air Filter	3	Mfrs Stds	-	I	Note A		Note 22
Intake Air Silencers	N/A	Mfrs Stds	-	I	Note A		Note 22
Intake Air Flexible Connectors	3	Mfrs Stds	-	I	Note A		
Intake Air Piping and supports	3	ASME III	3	I	Note A	3.9B	
Exhaust Air Silencers	N/A	Mfrs Stds	-	I	Note A		Note 22
Exhaust Air Flexible Connectors	3	Mfrs Stds	-	I	Note A		
Exhaust Relief Valve	NNS	Mfrs Stds	-	NONE	Note E		Note 75
Exhaust Piping and supports							
In Bldg	3	ASME III	-	I	Note A		Note 82
On Roof	NNS	ANSI B31.1	-	II	Note B	3.7B	Note 44
Flexible Connectors	NNS	Mfrs Stds	-	NONE	Note E		Note 70
Diesel Generator Control Panels (local)	1E	IEEE 323	-	I	Note A	8.3.1	
16. Spent Fuel Pool Cooling and Cleanup System							
Spent fuel pool cooling water pumps	3	ASME III	3	I	Note A	9.1.3	Note 1b
Spent fuel pool heat exchangers	3	ASME III	3	I	Note A	9.1.3	Note 1c
Spent fuel pool demineralizers	3	ASME III	3	I	Note A	9.1.3	
Spent fuel pool suction screens	3	Mfrs Stds	-	I	Note A	9.1.3	Note 21
Resin traps	3	ASME III	3	I	Note A	9.1.3	
Piping and valves	2	ASME III	2	I	Note A	9.1.3	Note 79
Stilling Wells Level	3	ASME III	-	I	Note A	9.1.3	Note 68

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Refueling Cavity Drain Strainers	N/A	AISC	-	I	Note A	6.2.2	
Refueling Cavity Debris Screens	N/A	AISC	-	I	Note A	6.2.2	
Piping and valves	3	ASME III	3	I	Note A		Note 34, 79
Flow restricting orifice	3	ASME III	3	I	Note A	9.1.3	
Supports for Class 2 Piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 Piping	3	ASME III	3	I	Note A	3.9B	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
17. Liquid Waste Processing System (LWPS)							
Reactor coolant drain tank	NNS	ASME VIII	-	NONE	Note E	11.2	
Reactor coolant drain tank pump	NNS	Mfrs Stds	-	NONE	Note E	11.2	
Reactor coolant drain tank heat exchanger							
Tube side	NNS	ASME VIII	-	NONE	Note E	11.2	
Shell side	2	ASME III	2	I	Note A	11.2	
Waste holdup tank	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Waste evaporator feed pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34, 72
Waste evaporator feed filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Waste evaporator package							
a. Feed preheater							
Feed side	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Steam side	NNS	ASME VIII	-	NONE	Note E	11.2	Note 72
b. Gas Stripper	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
c. Submerged Tube evaporator							
Feed side	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Steam side	NNS	ASME VIII	-	NONE	Note E	11.2	Note 72
d. Evaporator Condenser							
Distillate side	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Cooling water side	3	ASME III	3	I	Note A	11.2	
e. Distillate Cooler							
Distillate side	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Cooling water side	3	ASME III	3	I	Note A	11.2	
f. Absorption Tower	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
g. Vent Condenser							
Gas side	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34, 72
Cooling water side	3	ASME III	3	I	Note A	11.2	
h. Distillate Pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34, 72
i. Concentrate Pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34, 72
j. Piping and Valves							
Feed	NNS	ANSI B31.1	-	NONE	Note C	11.2	Note 34, 72
Distillate	NNS	ANSI B31.1	-	NONE	Note C	11.2	Note 34, 72
Concentrate	NNS	ANSI B31.1	-	NONE	Note C	11.2	Note 34, 72
Cooling	3	ASME III	3	I	Note A	11.2	Note 79
Vent	NNS	ANSI B31.1	-	NONE	Note C	11.2	Note 34, 72
Waste evaporator condensate demineralizer	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Waste evaporator condensate filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Waste evaporator condensate tank	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Waste evaporator condensate tank pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Chemical drain tank pump	NNS	Mfrs Stds	-	NONE	Note E		
Chemical drain tank	NNS	ASME VIII	-	NONE	Note E	11.2	
Spent resin storage tank	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Spent resin sluice pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Spent resin sluice filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Laundry and hot shower tank	NNS	ASME VIII	-	NONE	Note E	11.2	
Laundry and hot shower tank pump	NNS	Mfrs Stds	-	NONE	Note E	11.2	
Laundry and hot shower filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Floor drain tanks I and II	NNS	ASME VIII	-	NONE	Note E	11.2	
Floor drain tank III	NNS	API 620	-	NONE	Note E	11.2	
Floor drain tank pumps	NNS	Mfrs Stds	-	NONE	Note E	11.2	
Waste monitor tanks	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Waste Monitor Tank Pumps	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Waste Monitor tank demineralizer	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Waste monitor tank filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Floor drain tank filter	NNS	ASME VIII	-	NONE	Note C	11.2	Note 34
Floor drain tank strainer	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Floor Drain Waste Evaporator Package						11.2	
a. Feed preheater							

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System and Components (45)		Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
	Feed side	NNS	ASME VIII	-	NONE	Note C		Note 34
	Steam side	NNS	ASME VIII	-	NONE	Note E		
b.	Gas stripper	NNS	ASME VIII	-	NONE	Note C		Note 34
c.	Submerged tube evaporator							
	Feed side	NNS	ASME VIII	-	NONE	Note C		Note 34
	Steam side	NNS	ASME VIII	-	NONE	Note E		
d.	Evaporator condenser							
	Distillate side	NNS	ASME VIII	-	NONE	Note C		Note 34, 72
	Cooling water side	3	ASME III	3	I	Note A		
e.	Distillate cooler							
	Distillate side	NNS	ASME VIII	-	NONE	Note C		Note 34, 72
	Cooling water side	3	ASME III	3	I	Note A		
f.	Absorption tower	NNS	ASME VIII	-	NONE	Note C		Note 34
g.	Vent condenser							
	Gas side	NNS	ASME VIII	-	NONE	Note C		Note 34, 72
	Cooling water side	3	ASME III	3	I	Note A		
h.	Distillate pump	NNS	Mfrs Stds	-	NONE	Note C		Note 34
I.	Concentrate pump	NNS	Mfrs Stds	-	NONE	Note C		Note 34
j.	Piping and valves							
	Feed	NNS	ANSI B31.1	-	NONE	Note C		Note 34
	Distillate	NNS	ANSI B31.1	-	NONE	Note C		Note 34
	Concentrate	NNS	ANSI B31.1	-	NONE	Note C		Note 34

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Cooling	3	ASME III	3	I	Note A		Note 79
Vent	NNS	ANSI B31.1	-	NONE	Note C		Note 34
Laundry and hot shower tank strainers	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Waste evaporator reagent tank	NNS	ASME VIII	-	NONE	Note E	11.2	
Laundry holdup and monitor tanks	NNS	API 650	-	NONE	Note C	11.2	Note 34
Laundry holdup and monitor tank pump	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
Laundry water head tank	NNS	API 620	-	NONE	Note E	11.2	
Plant effluent holdup and monitor tanks (2)	NNS	ACI 318-71 AISC, 7th ed.	-	II	Note B, C	11.2	Note 34
Piping and valves	3	ASME III	3	I	Note A	11.2	Note 34, 79
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Piping & valves	NNS	ANSI B31.1	-	NONE	Note C	11.2	Note 34
Pumps	NNS	Mfrs Stds	-	NONE	Note C	11.2	Note 34
18. Gaseous Waste Processing System (GWPS)							
Gas compressors	NNS	Mfrs Stds	-	NONE	Note C	11.3	Note 34
Gas decay tanks (numbers 1-8)	3	ASME III	3	I	Note A	11.3	Note 34
Gas decay tanks (numbers 9 & 10)	NNS	ASME VIII	-	NONE	Note C	11.3	Note 34, 72
Hydrogen recombiner (catalytic)	NNS	ASME VIII	-	NONE	Note C	11.3	Note 34
Waste gas drain filter	NNS	Mfrs Stds	-	NONE	Note C	11.3	Note 34
Gas decay tank drain pump	NNS	Mfrs Stds	-	NONE	Note C	11.3	Note 34
Piping and valves	3	ASME III	3	I	Note A	11.3	Note 34, 79
Valves	NNS	ANSI B31.1	-	NONE	Note E	11.3	Note 34

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Gas Trap	NNS	Mfrs Stds	-	NONE	Note C	11.3	Note 34
19. Solid Waste Management System (SWMS)							
Solidification System	NNS	Mfrs Stds	-	NONE	Note C	11.4	Note 34
Valves	NNS	ANSI B31.1	-	NONE	Note C		Note 34
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Handling Equipment							
Filter transfer cask	NNS	Mfrs Stds	-	NONE	Note E	11.4	
Flat bed trailer	NNS	Mfrs Stds	-	NONE	Note E	11.4	
Remote handling tools	NNS	Mfrs Stds	-	NONE	Note E	11.4	
20. Demineralized and Reactor Makeup Water System							
Reactor makeup water pumps	3	ASME III	3	I	Note A	9.2.3	
Reactor makeup water storage tank	3	ACI 318-71 AISC	-	i	Note A	9.2.3	Note 55
Piping and valves	2	ASME III	2	I	Note A	3.9B	Note 79
Piping and valves	3	ASME III	3	I	Note A	3.9B	Note 79
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
21. Vents and Drains System							
Safeguards building sump pumps	3	ASME III	3	I	Note A	9.3.3	
Piping and valves	2	ASME III	2	I	Note A	9.3.3	Note 79
Piping and valves (pump discharge)	3	ASME III	3	I	Note A	9.3.3	Note 79
Supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Supports for Class 3 piping	3	ASME III	3	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Containment Sump Flow Monitor	1E	IEEE-323	-	I	Note A	5.2.5	
22. Containment Ventilation Systems							
a. Containment recirculation							
cooling units	NNS	Mfrs Stds	-	II	B	9.4A	Note 98
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
b. CRDM cooling system							
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
c. Neutron detector well cooling							
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
d. Reactor coolant sleeve cooling							
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
e. Containment preaccess filtration							
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
filtration unit	NNS	Mfrs Stds	-	II	Note B	9.4A	Note 23
f. Containment purge supply and exhaust							
ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4A	
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4A	
dampers	3	Mfrs Stds	-	I	Note A	9.4A	
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4A	Note 71
containment isolation valves and piping	2	ASME III	2	I	Note A	9.4A	Note 79
g. Containment pressure relief							
ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4A	
containment isolation valves and piping	2	ASME III	2	I	Note A	9.4A	Note 79
Suction pipe	NNS	ANSI B31.1	N/A	II	Note B	9.4A	Note 62
Debris screen	NNS	Mfrs Stds	N/A	II	Note B	9.4A	Note 62
Flow Orifice Plate (Restrictive)	NNS	Mfrs Stds	N/A	II	Note B	9.4A	Note 62
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
23. Control Room Air-Conditioning System							
Air-conditioning units							
Roughing filter, fan, heating and direct-expansion refrigerant type coils	3	Mfrs Stds	-	I	Note A	9.4.1	Note 21
Refrigerant piping & tubes	3	ASME B42 or ASTM B88	-	I	Note A	9.4.1	Note 21
Water side	3	ASME III	3	I	Note A	9.4.1	
Fans	3	Mfrs Stds	-	I	Note A	9.4.1	
Emergency pressurization and filtration units (roughing, charcoal, and HEPA filters and fans)	3	Mfrs Stds	-	I	Note A	9.4.1	Note 21
Deluge and drain piping for filtration units	3	ASME III	-	I	Note A	9.4.1	Note 68
Piping and valves	3	ASME III	3	I	Note A	9.4.1	Note 79
Dampers and supports	3	Mfrs Stds	-	I	Note A	9.4.1	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.1	Note 21
Air accumulators (intake dampers)	3	ASME III	-	I	Note A	9.4.1, 9.3.1	
Tubing and supports (between check valves upstream of accumulators and intake dampers)	3	ASME III	-	I	Note A	3.9B	Note 41
Check valves for accumulators	3	ASME III	-	I	Note A	9.3.1	Note 79
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
24. Safeguard Building HVAC System							
a. Safeguards area							
dampers	3	Mfrs Stds	-	I	Note A	9.4.5	Note 21
ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.5	Note 21
dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4.5	Note 71
ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4.5	
Emergency Fan Coil Units							
fans and housing	3	Mfrs Stds	-	I	Note A	9.4.5	Note 21
cooling coils	3	ASME III	3	I	Note A	9.4.5	
b. Electrical area							
dampers	3	Mfrs Stds	-	I	Note A	9.4C.3	Note 21
dampers	NNS	Mfrs Stds	-	NONE	Note E		Note 71
ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4C.3	Note 21
ductwork and supports	NNS	Mfrs Stds	-	II	Note B		
thermostats	1E	IEEE-344	-	I	Note A	9.4C.3	
supply units/fans	NNS	Mfrs Stds	-	NONE	Note E	9.4C.3	Note 70
Emergency Fan Coil Units							
fans and housing	3	Mfrs Stds	-	I	Note A	9.4C.3	Note 21
cooling coils	3	ASME III	3	I	Note A	9.4C.3	
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
25. Fuel Building Ventilation System							
Dampers	3	Mfrs Stds	-	I	Note A	9.4.2	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.2	Note 21
Dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4.2	Note 71
Ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4.2	
Demister in fuel pool exhaust	NNS	Mfrs Stds	-	NONE	Note E	9.4.2	
Emergency Fan Coil Units							
Cooling coils	3	ASME III	3	I	Note A	9.4.2	
Fans and housing	3	Mfrs Stds	-	I	Note A	9.4.2	
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	
26. Diesel Generator Building Ventilation System							
Fans	3	Mfrs Stds	-	I	Note A	9.4C.1	Note 21
Dampers	3	Mfrs Stds	-	I	Note A	9.4C.1	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4C.1	Note 21
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	
27. Uncontrolled Access Area Ventilation System							
Battery room exhaust system							
Fans	3	Mfrs Stds	-	I	Note A	9.4C.4	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4C.4	Note 21
Dampers	3	Mfrs Stds	-	I	Note A	9.4C.4	Note 21

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Balance of system							
Supply unit/fan	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	Note 70
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.3	
Ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4.3	
Ductwork and supports	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	
Dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	Note 71
Electric Unit Heaters For Train A & B Battery Rooms	1E	IEEE-344	-	I	Note A	9.4C.4	
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	
28. Primary Plant Ventilation System							
a. Exhaust							
Modular exhaust filtration units (ESF)	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Modular exhaust filtration units (non-ESF)	3	Mfrs Stds	-	I	Note A	9.4.3	
Fans	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Dampers	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Supports	3	Mfrs Stds	-	I	Note A	9.4.3	
Deluge and drain piping for filtration units	3	ASME III	-	I	Note A	9.4.3	Note 68
Plant Vent Stack and Supports	NNS	Mfrs Stds	-	II	Note B	9.4D	
b. Supply							
Units/fans	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	Note 70
Ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4.3	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	Note 71
Fans discharge gravity dampers	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	
29. Auxiliary Building HVAC System							
Dampers	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Dampers	NNS	Mfrs Stds	-	NONE	Note E	9.4.3	Note 71
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Ductwork and supports	NNS	Mfrs Stds	-	II	Note B	9.4.3	
Emergency fan coil units							
Fans and housing	3	Mfrs Stds	-	I	Note A	9.4.3	Note 21
Cooling coils	3	ASME III	3	I	Note A	9.4.3	
Emergency Fans for High Pressure Chemical Feed Room Ventilation System	3	Mfrs Stds	-	I	Note A	9.4C	
HVAC panels (control room)	1E	IEEE-344	-	I	Note A	9.4	
30. Service Water Intake Structure Ventilation System							
Fans (pump room exhaust)	3	Mfrs Stds	-	I	Note A	9.4B	
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4B	
Dampers	3	Mfrs Stds	-	I	Note A	9.4B	
Outside Air Intake Screens	NNS	AISC	-	II	Note B	9.4B	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
31. Chilled Water Systems							
a. Plant Ventilation chilled water system							
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	9.4E, 3.6B	Note 44
b. Safety chilled water system							
Chillers	1E	Mfrs Stds	-	I	Note A	9.4F	
Water side of chillers	3	ASME III	3	I	Note A	9.4	
Piping and valves	3	ASME III	3	I	Note A	9.4F	Note 79
Surge Tank	3	ASME III	3	I	Note A	9.4F	
Pumps	3	ASME III	3	I	Note A	9.4F	
Supports for Class 3 Piping	3	ASME III	3	I	Note A	3.9B	
32. Process Sampling System							
Sample Panels	NNS	Mfrs Stds	-	NONE	Note E	9.3.2	
Delay coil	2	ASME III	2	I		9.3.2	
Piping and valves	2	ASME III	2	I	Note A	9.3.2	Note 79
Reflective insulation	NNS	Mfrs Stds	-	II	Note B	6.1B	
Supports for Class 2 Piping	2	ASME III	2	I	Note A	3.9B	
Class 5 Piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Process Sampling Tubing, Fittings, Valves, and Supports	2	ASME III	-	I	Note A	6.2.4	Note 41

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
32a. Post Accident Sampling System (DELETED by License Amendment 91)							
32b. Secondary Sampling System							
Sample Panel	NNS	Mfrs Stds	-	NONE	Note E	10.4.16, 3.7B.2.8	
Class 5 Piping and Supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
33. Fuel Handling Equipment and Fuel Assemblies							
Refueling machine	N/A	Mfrs Stds	-	II	Note B	9.1	Note 16
Refueling machine interlocks	N/A	Mfrs Stds	-	NONE	Note C	7.6	
Containment Fuel Handling Bridge Crane	N/A	CMAA 74	-	II	Note B		Note 23, 24
Fuel Handling Bridge Crane	N/A	ASME NUM - 1 (TYPE B) ASME NOG - 1 (TYPE II)	-	II	Note B		Note 9, 23, 24
Rod cluster control changing fixture	N/A	Mfrs Stds	-	NONE	Note E		Note 16
Reactor vessel stud tensioner	N/A	Mfrs Stds	-	NONE	Note E		Note 15
Spent fuel handling tool	N/A	Mfrs Stds	-	NONE	Note C		Note 10
Fuel Transfer System							
Fuel Transfer Tube & Flange	2	ASME III	MC	I	Note A		(also evaluated as part of containment structure)
Conveyer System & Controls	N/A	Mfrs Stds	-	NONE	Note E		
Remainder of System	N/A	Mfrs Stds	-	NONE	Note E		
Interlocks	N/A	Mfrs Stds	-	NONE	Note C	7.6	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Refueling gates	N/A	AISC	-	I	Note A		
Fuel transfer tube expansion joint	2	ASME III	MC	I	Note A		
Stud hole plug handling fixture	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Stud hole plugs	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Lower internals storage stand	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Upper internals storage stand	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Rod cluster control thimble plug tool	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Source installation guide	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Crane scales	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Control rod drive shaft handling fixture	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Radial Arm Hoist Assembly	N/A	Mfrs Stds	-	NONE	Note E		Note 93
Irradiation tube end plug seat jack	N/A	Mfrs Stds	-	NONE	Note E		Note 10
New fuel elevator	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Portable underwater lights	N/A	Mfrs Stds	-	NONE	Note C		Note 90
Load cells	N/A	Mfrs Stds	-	NONE	Note E		
Burnable Poison Rod handling tool	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Irradiation sample handling tool	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Burnable poison assembly rack inserts	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Control rod drive shaft unlatching tool, full length	N/A	Mfrs Stds	-	NONE	Note E		Note 10
New fuel elevator winch	N/A	Mfrs. Stds.	-	NONE	Note E		Note 10
New fuel storage racks	3	ASME III	-	I	Note A	9.1	Note 14, 68
Spent fuel storage racks	3	ASME III	-	I	Note A	9.1	Note 14, 68

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Refueling cavity seal ring	N/A	Mfrs Stds	-	NONE	Note E		Note 10
New fuel handling tool	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Rod control cluster assembly handling fixture	N/A	Mfrs Stds	-	NONE	Note E		Note 10
Fuel Assemblies	3	N/A	-	I	Note A	4.2	Note 99
34. Containment Building Miscellaneous Equipment							
Containment polar crane	N/A	CMAA 70	-	I	Note A	9.1, 3.8	Note 53
Telescopic Jib Crane	N/A	ASME NOG-1 ASME NUM-1 API Spec 2C	-	II	Note B	3.8	Notes 23,24,89
Reactor vessel head lifting device	1	Mfrs Stds	-	I	Note A	9.1	Note 48
Upper internals lifting device	N/A	Mfrs Stds	-	NONE	Note E	9.1	
Containment dome access rotating platform	N/A	CMAA 70	-	II	Note B		Note 23, 24
Neutron detector positioner	2	Mfrs Stds	-	I	Note A		Note 23
Reactor vessel or core related components							
Reactor vessel shoes and shims	1	Mfrs Stds	-	I	Note A		
Irradiation Sample holder	2	Mfrs Stds	-	I	Note A		Note 18
Irradiation Samples	N/A	Mfrs Stds	-	NONE	Note E		
Full length control rod cluster	2	Mfrs Stds	-	I	Note A		
Control rod drive mechanism (CRDM) dummy can assemblies	NNS	Mfrs Stds	-	NONE	Note E		Note 10
CRDM air cooling shroud assemblies (Unit 1 only)	NNS	Mfrs Stds	-	II	Note E		Note 10
CRDM air cool baffle assemblies (Unit 2 only)	NNS	Mfrs Stds	-	NONE	Note E		Note 10
CRDM seismic support platform	2	Mfrs Stds	-	I	Note A		Note 19

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
CRDM seismic support spacer plate	2	Mfrs Stds	-	I	Note A		Note 19
CRDM seismic support tie rod assemblies	2	Mfrs Stds	-	I	Note A		Note 19
Burnable poison rod assemblies	3	Mfrs Stds	-	NONE	Note A		Note 10
Reactor vessel insulation	NNS	Mfrs Stds	-	NONE	Note E		Note 10
Reactor vessel internals	2	Mfrs Stds	-	I	Note A		
Primary source rods	3	Mfrs Stds	-	NONE	Note A		
Incore Instrumentation							
Seal table assembly	1	Mfrs Stds	-	I	Note A		Provides support for safety class 1 pressure boundary tubing
Flux thimble tubing	2	ASME III	2	I	Note A		Note 20
Flux thimble tubing fittings	2	ASME III	2	I	Note A		Note 20
Roll-away missile shield (Unit 2 only)							
critical parts	2	CMAA 70	-	I	Note A		Note 23
non-critical parts	NNS	CMAA 70	-	NONE	Note E		Note 70
Sump Screen	2	AISC Code	-	I	Note A	6.2.2	Frame structure supports recirculation piping moment restraint
Sump vortex suppression device	2	AISC Code	-	I	Note A		

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
35. Miscellaneous Handling Equipment							
Drumming storage area crane	N/A	CMAA 70	-	II	Note B	9.1	Note 23, 24, 78
Fuel Building Overhead Crane	N/A	CMAA 70	-	I	Note A	9.1, 3.8	Note 53, 78
SWIS Crane	N/A	CMAA 70	-	II	Note B	9.1	Note 23, 24, 78
Auxiliary Filter Hoist	N/A	CMAA 70	-	I	Note A		Note 53, 78
Safety Chiller Hoist	N/A	CMAA 70	-	I	Note A		Note 53, 78
Misc. Cranes and Hoists	N/A	CMAA 70	-	NONE	Note C		Note 78
Wonderhoist	N/A	NOG-1	-	I	Note A	9.1.4	Note 53, 78
36. Structures							
Containment Building	N/A	Note 36	-	I			
Containment internal structure (including containment sump)	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.3	
Safeguards buildings (including diesel generator room and emergency SWGR room)	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.4	
Auxiliary Building	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.4	
Electrical and Control Bldg.	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.4	
Fuel Building	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.4	
Service Water Intake Structure	N/A	ACI 318-71 & AISC Code	-	I	Note A	3.8.4	
Safe Shutdown Impoundment Dam	N/A	-	-	I	Note A	2.4 & 2.5	
Containment Personnel Airlock	2	ASME III	MC	I	Note A	3.8.1.1.6	
Containment Equipment Hatch	2	ASME III	MC	I	Note A	3.8.1.1.6	
Containment Emergency Airlock	2	ASME III	MC	I	Note A	3.8.1.1.6	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Masonry Walls	N/A	-	-	NONE	Note E	3.8.4	Note 52
Removable Precast Block Walls	N/A	-	-	II	Note B	3.8.4	Note 63, 70
Gypsum Walls	N/A	-	-	II	Note B		Note 63, 70
Missile Barriers (concrete)	N/A	-	-	I	Note A	3.3	
Missile Resisting Doors and Hatches	N/A	Mfrs Stds	-	II	Note B		Note 23
Watertight Doors	N/A	Mfrs Stds	-	I	Note A		
Handrails in Seismic Category I Buildings	N/A	ACI 318-71& AISC Specification	-	II	Note B		
Scaffold Pole Framework	N/A	OSHA	-	II	Note B		Note 23
37. Electrical Equipment							
6900-V Switchgear (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
6900-V to 480-VAC transformers (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
480-VAC switchgear (safety related (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
480-VAC motor local control stations (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
Low voltage AC distribution panels (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
118-V uninterrupt ble AC instrument distribution panels and subpanels (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
480 to 208/120-VAC transformers (safety related)	1E	IEEE-344	-	I	Note A	8.3.1	
125-VDC station batteries (safety related)	1E	IEEE-344	-	I	Note A	8.3.2	
125-VDC switchboards and distribution panels (safety related)	1E	IEEE-344	-	I	Note A	8.3.2	
125-VDC Battery Chargers (safety related)	1E	IEEE-344	-	I	Note A	8.3.2	
118-VAC static uninterruptible power systems (BOP safety related)	1E	IEEE-344	-	I	Note A	8.3.1	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
125-V station battery racks and supports (safety related)	1E	IEEE-344	-	I	Note A	8.3.2	
Auxiliary relay racks safety related (BOP)	1E	IEEE-344	-	I	Note A	8.3.1	
Solid-state safeguard sequencer	1E	IEEE-344	-	I	Note A	8.3.1	
Solid-state isolation equipment	1E	IEEE-344	-	I	Note A	8.3.1	
Containment electrical penetration assemblies	1E	IEEE-344	-	I	Note A	8.3.1	
Electrical equipment supports (associated with safety related equipment)	N/A	AISC Code	-	I	Note A	8.3.1	
Motors (safety related)	1E	IEEE-323	-	I	Note A	8.3.1	
Power cables (associated with safety related equipment)	1E	IEEE-323	-	N/A	Note A	8.3.1	Note 80
Instrumentation and control cable (associated with safety related equipment)	1E	IEEE-323	-	N/A	Note A	8.3.1	Note 80
Wire and Cable Raceway System (in Seismic Category I Buildings)							
Cable trays (Containing Class 1E wires or cables)	N/A	-	-	I	Note A	8.3	
Cable trays (Not containing Class 1E wires or cables)	N/A	-	-	II	Note B		
Conduit (Containing Class 1E wires or cables)	N/A	-	-	I	Note A		
Conduit (Not containing Class 1E wires or cables)	N/A	-	-	NONE	Note E		Note 67
Supports (For cable trays or conduit that contain Class 1E wires or cables)	N/A	AISC	-	I	Note A	3.10B.3	
Supports (For cable trays that do not contain Class 1E wires or cables)	N/A	AISC	-	II	Note B		
Supports (For conduit that does not contain Class 1E wires or cables	N/A	AISC	-	II/NONE	Note B		Note 67
480/120V Bypass transformers	1E	IEEE-344	-	I	Note A	8.3.1	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
480/208-120V isolation transformers	1E	IEEE-344	-	I	Note A	8.3.1	
Priority Panels	N/A	-	-	I	Note A	8.3.1	
Prefabricated Cables and Connectors (associated with Safety Related equipment)	1E	IEEE-323	-	I	Note A	8.3.1	
8KV Cable termination and transition joints	1E	IEEE-323	-	N/A	Note A	8.3.1	
Terminal blocks	1E	IEEE-323	-	I	Note A	8.3.1	
Conduit Seals	1E	IEEE-323	-	I	Note A	8.3.1	
Heat Shrinkable cable insulation sleeves	1E	IEEE-323	-	N/A	Note A	8.3.1	Note 80
Spent Fuel Pool and Transfer Canal Underwater lights	N/A	Mfrs Stds	-	NONE	Note C	9.5.3	Note 72, 90
38. Radiation Monitoring System							
Containment High Range Radiation Monitors	1E	IEEE-323	-	I	Note A	12.3, 7.5, II.F.1	
Containment Air Monitors (including Particulate and Gas Channels)	3,N1E	Mfrs Stds, IEEE-344	-	I	Note A	11.5, 5.2.5	Note 88
Plant Vent Stack Gas Monitors (including Particulate and Iodine Samplers)	NNS,N1E	Mfrs Stds	-	NONE	Note C	11.5, 7.5, II.F.1	Note 56
Main Steam Line Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5, 7.5, II.F.1	Note 23, 56, 73
High Range Area Monitors	N1E	Mfrs Stds	-	NONE	Note C	12.3, 7.5	Note 56
Low Range Area Monitors	N1E	Mfrs Stds	-	NONE	Note C	12.3, 7.5	Note 56
Component Cooling Water Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5, 9.2	Note 23, 74, 83
Waste Gas Monitor	N1E	Mfrs Stds	-	NONE	Note C	11.5, 11.3	Note 23, 73
Condenser Off-gas Monitor	N1E	Mfrs Stds	-	NONE	Note C	11.5, 7.5	Note 56
Ventilation Duct Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5, 7.5	Note 56

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Liquid Waste Effluent Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5	Note 34, 83
Turbine Building Drains Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5, 7.5	Note 56, 83
Failed Fuel Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5	Note 83
Service Water Monitors	N1E	Mfrs Stds	-	NONE	Note C	11.5, 9.2	Note 83
Steam Generator Blowdown Sample Monitors	NNS, N1E	Mfrs Stds	-	NONE	Note C	11.5, 9.3, 7.5	Note 56, 83
Control Room Equipment Rack (seismic)	N1E	Mfrs Stds, IEEE-344	-	I	Note A	11.5	Note 88
Control Room Equipment Rack (1E)	1E	IEEE-344	-	I	Note A	11.5, 12.3	
Control Room Ventilation Monitors	1E	IEEE-344	-	I	Note A	11.5, 9.4	
39. Fire Protection System							
Fire Suppression Systems	NNS	NFPA	-	NONE	Note D	9.5	Note 42, 58, 72
Class 5 Piping and supports	NNS	NFPA	-	II/NONE	Note B	3.6B	Note 44
Portable Fire Extinguishers	N/A	NFPA	-	NONE	Note D	9.5	Note 42, 59
Fire Extinguisher Brackets	N/A	NFPA	-	NONE	Note D	9.5	Note 42, 72
Fire Stops and Seals	N/A	ASTM E119	NONE	NONE	Note D	9.5.1.5	Note 42
Fire Rated Coating Systems	N/A	ASTM E119	-	NONE	Note D	9.5	Note 42, 72
RCP Lube Oil Collection System						9.5.1.3.6 (1)	
RCP Cowlings	NNS	Mfrs Stds	-	II	Note B		
Piping	NNS	ANSI B31.1	-	II	Note B		Note 44, 57
Tanks	NNS	ASME III	-	II	Note D,B		Note 42
Fire Detection System	N1E	NFPA-72D	-	NONE	Note D	9.5.1.4.2 (1)	Note 42, 72

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Fire Dampers	N/A	Mfrs Stds	-	II/NONE	Note D,B	9.5	Note 42, 54
Fire Doors	N/A	Mfrs Stds	-	NONE	Note D	9.5	Note 42, 72
Fire Rated Barriers	N/A	ASTM E119	-	NONE	Note D	3.8, 9.5	Note 42, 72
40. Plant Gas System							
a) Nitrogen system							
piping and valves	2	ASME III	2	I	Note A	3.9B	
Valves	NNS	ANSI B31.1	-	NONE	Note C		Note 34
supports for Class 2 piping	2	ASME III	2	I	Note A	3.9B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
b) Hydrogen system							
piping and valves	2	ASME III	2	I	Note A	3.9B	
supports for Class 2 piping	2	ASME III	2	I	Note A	3.7B	
Class 5 piping and supports	NNS	ANSI B31.1	-	II	Note B	3.6B	Note 44, 57
41. Instrumentation and Control							
Electronic transmitters (pressure and differential pressure)	1E	IEEE-323	-	I	Note A	7.0	
Electronic transmitters (pressure, flow)	N1E	IEEE-323	-	NONE	Note C	7.5, II.F.1	Note 56
Electronic transmitters (flow)	1E	IEEE-323	-	I	Note A	7.0	
Pressure switches	1E	IEEE-323	-	I	Note A	7.0	Note 39
Level Switches	2,3	ASME III	2,3	I	Note A	7.0	Note 85a
Level Switches	2,3	Mfrs Stds	-	I	Note A	7.0	Note 85b

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Limit Switches	1E	IEEE-323	-	I	Note A	7.0	
Electronic transmitters (pressure and differential pressure)	1E	IEEE-323	-	NONE	Note A	7.0	Note 77
Pressure switches	1E	IEEE-323	-	NONE	Note A	7.0	Note 77
Limit Switches	1E	IEEE-323	-	NONE	Note A	7.0	Note 77
Level Transmitters	1E	IEEE-323	-	I	Note A	7.0, II.F.1	Functional Integrity only
Thermowells	2,3	ASME III	2,3	I	Note A	3.2	Pressure Integrity only
Resistance Temperature Detectors	1E	IEEE-323	-	I	Note A	7.0	
Temperature Elements	N1E	IEEE-323	-	NONE	Note C	7.5	Note 56
Source and Intermediate Range Neutron Detectors	1E	IEEE-323	-	I	Note A	7.2	
Power Range neutron detectors	1E	IEEE-323	-	I	Note A	7.2	
Thermowells (RWMS)	NNS	ANSI B31.1	-	NONE	Note C		Note 34
Neutron Flux Monitors	1E	IEEE-323	-	I	Note A	7.0	
Flow elements (RWMS)	NNS	ANSI B31.1	-	NONE	Note C		Note 34
Containment hydrogen analyzers	1E	IEEE-323	-	I	Note A	7.5, II.F.1	
Differential Pressure Switches	1E	IEEE-323	-	I	Note A	7.0	
Rotameters	3	ASME III	3	I	Note A	7.0	Pressure Integrity only
Rotameters (RWMS)	NNS	Mfrs Stds	-	NONE	Note C	11.3	Note 34
Acoustic Sensors	N1E	IEEE-323	-	NONE	Note C	7.5	Note 56
Orifice Plates (Flow Metering)	2, 3	Mfrs Stds		I	Note A	7.0	
Impulse tubing, fittings, valves and supports	2, 3	ASME III	-	I	Note A	3.9B.3.4.3	Note 31, 41, 65

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Impulse tubing, fittings, valves and supports	3	Mfrs Stds	-	I	Note A	9.4	Note 92
Supports for impulse tubing, fittings and valves	NNS	-	-	II/NONE	Note B	7.0	Note 51
Instrument supports	N/A	-	-	I	Note A	7.0	Note 38a
Instrument supports	N/A	-	-	II	Note B	7.0	Note 38b
Nuclear instrument racks (NIS)	1E	IEEE-344	-	I	Note A	7.0	
Process instrumentation and control racks (NSSS)	N1E		-	NONE	Note E	7.0	
Rod control equipment	N/A	-	-	NONE	Note C	7.0	
Rod position indication containment cabinets	N/A	-	-	NONE	Note C	7.0	
I & C Power supply inverters (NSSS)	1E	IEEE-344	-	I	Note A	7.0	
Solid-state protection system cabinet	1E	IEEE-344	-	I	Note A	7.0	
Control board demultiplexer	N1E	-	-	NONE	Note C	7.0	
Hot shutdown panel	1E	IEEE-344	-	I	Note A	7.0	
Shutdown Transfer Panel	1E	IEEE-344	-	I	Note A	7.0	
Process instrumentation protection racks	1E	IEEE-344	-	I	Note A	7.0	
Reactor trip switchgear	1E	IEEE-344	-	I	Note A	7.0	
Protective relay rack (6.9-kv bus UV/UF)	1E	IEEE-344	-	I	Note A	7.0	
Cable termination racks	1E	IEEE-344	-	I	Note A	7.0	
Local instrument racks	N/A	IEEE-344	-	I	Note A	7.0	Note 31, 38a
Local instrument racks	N/A	-	-	II	Note B	7.0	Note 31, 38b
Control Room Benchboards, Vertical Panels or Hot Shutdown Panel Mounted Equipment Classified 1E	1E	IEEE-344	-	I	Note A	7.0	
Proportional Integral Derivative (PID) Controller PC-455A	N1E	-	-	NONE	Note C	II.K.3.9	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Control Room Benchboards, Vertical Panels or Hot Shutdown Panel Mounted Equipment Classified as Non-1E	NNS	Mfrs Stds	-	NONE	Note E	7.0	Note 70
Analog Instrumentation Cabinets (BOP Safety Related)	1E	IEEE-344	-	I	Note A	7.0	
Auxiliary Relay Rack (NSSS)	N/A	Mfrs Stds	-	I	Note A	7.0	Note 69
Auxiliary Relay Rack Isolation Relays (NSSS)	1E	IEEE-344	-	I	Note A	7.0	Note 69
Upgrade Protection and Surveillance Cabinet (NSSS)	1E	IEEE-344	-	I	Note A	7.0	
AMSAC Cabinet	N/A	Mfrs Stds	-	I	Note A	7.8	
AMSAC Isolation Relays	1E	IEEE-344	-	I	Note A	7.8	
AMSAC Actuation Circuit (Logic)	N1E	Mfrs Stds	-	NONE	Note C	7.8	Note 66
Core Cooling Monitor	1E	IEEE-344	-	I	Note A	II.F.2	
Reactor Vessel Level Indicating System (RVLIS)	1E	IEEE-344	-	I	Note A	II.F.2	
Safety Parameter Display System (SPDS)/ERF Computer	N1E	Mfrs Stds	-	NONE	Note C	III.A.1.2, I.D.2	
Main Steam Isolation Valve (MSIV) Train C Solenoids	N1E	IEEE 323	-	NONE	Note C	3.6B.2.5.2, Table 3.6B-4, Table 3.6B-5	
Turbine Overspeed Protection System	N1E	Mfrs Stds	-	NONE	Note C	10.2.2	
Explosive Gas Monitoring	N1E	Mfrs Stds	-	NONE	Note C	11.3	
Area Temperature Monitoring	N1E	Mfrs Stds	-	NONE	Note C	3.11B	
Movable Incore Detection Systems	N1E	Mfrs Stds	-	NONE	Note C	7.7	
Seismic Instrumentation	N1E	IEEE 344	-	II	Note C	3.7B.4	Note 100
Loose Parts Monitoring Systems	N1E	Mfrs Stds	-	NONE	Note C	4.4.6.4	
Feedwater Isolation Valve (FWIV) Temperature Monitoring	N1E	Mfrs Stds	-	NONE	Note C		
Feedwater Flow Measurement (LEFM)	N1E	Mfrs Stds	-	NONE	Note C	10.4-9	

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Safety Related Digital System Upgrade	1E	IEEE-344	-	I	Note A	7.0	
Non-Safety Related Digital System Upgrade	N1E	Mfrs Stds	-	II/NONE	Note C	7.0	
Spent Fuel Pool Instrumentation (NRC Order EA-12-051)	N1E	Mfrs Stds	-	NONE	Note F	1.2.5	Note 96
Accident Monitoring:							
Class 1E, Category 1,2,3	1E	IEEE-323	-	I	Note A	7.5	Note 56
Non-1E, Category 2	N1E	IEEE-323	-	NONE	Note C	7.5	Note 56
Non-1E, Category 3	N1E	-	-	NONE	Note C	7.5	Note 45
42. Tornado Venting Components							
Dampers	NNS	Mfrs Stds	-	II/NONE	Note B	3.3.2	Note 24
Blowout Panels	NNS	Mfrs Stds	-	II/NONE	Note B	3.3.2	Note 24, 76
43. Compressed Air Systems							
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	9.3.1	Note 44, 51
SFP Swing Gate Seal Pressures Gauges	3	ASME GRADE 1A	N/A	I	Note A	9.3.1	
SFP Swing Gate Seal Isolation Valves	3	ASME CLASS 2080	N/A	I	Note A	9.3.1	
SFP Swing Gate Seal Check Valves	NNS	Mfrs Stds	-	I	Note A	9.3.1	
44. Protective Coatings						6.1B	
Coatings inside the Containment	N/A	ASTM D 5144-00	-	NONE	Note C		Note 33
Coatings outside the Containment	N/A	NONE	-	NONE	Note C		Note 33

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
45. Potable and Sanitary Water System							
Domestic water storage tank	NNS	ASME VIII	-	II	Note B	9.2.4	Note 23, 91
46. Condensate System							
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	10.4.7	Note 44
47. Auxiliary Steam System							
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	10.4.13	Note 44
48. Steam Generator Blowdown & Cleanup System							
Steam Generator Blowdown Heat Exchanger	NNS	ASME VIII	-	NONE	Note C	10.4.8	Note 34
Blowdown Cation Demineralizers	NNS	ASME VIII	-	NONE	Note C	10.4.8	Note 34
Blowdown Mixed Bed Demineralizers	NNS	ASME VIII	-	NONE	Note C	10.4.8	Note 34
Mixed Bed Resin Trap	NNS	ANSI B31.1	-	NONE	Note C	10.4.8	Note 34
Cation Resin Trap	NNS	ANSI B31.1	-	NONE	Note C	10.4.8	Note 34
Steam Generator Spent Resin Sluice Pump	NNS	Mfrs Stds	-	NONE	Note C	10.4.8	Note 34
Filter	NNS	ANSI B31.1	-	NONE	Note C	10.4.8	Note 34
Steam Generator Blowdown Spent Resin Storage Tank	NNS	ASME VIII	-	NONE	Note C	10.4.8	Note 34
Valves	NNS	ANSI B31.1	-	NONE	Note C	10.4.8	Note 34
Class 5 piping and supports	NNS	ANSI B31.1	-	II/NONE	Note B	3.7B	Note 44
Steam generator blowdown filter	NNS	ANSI B31.1		NONE	Note C	10.4.8	Note 34

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
49. Pipe Whip Restraints	N/A	Mfrs Stds	-	I	Note A	3.6B	Note 43
50. Meteorological Instrumentation							
Meteorological Tower	N/A	Mfrs Stds	-	NONE	Note C	2.3	
Sensors	N/A	Mfrs Stds	-	NONE	Note C	2.3	
Signal Conditioners							
Power Supplies	N/A	Mfrs Stds	-	NONE	Note C	2.3	
Data Logging and Computational Devices	N/A	Mfrs Stds	-	NONE	Note C	2.3	
METSYS Computer	N/A	Mfrs Stds	-	NONE	Note C	2.3	
51. Uninterruptible Power Supply (UPS) and Distribution Rooms System							
Air-conditioning units							
Roughing filter, fan, and direct-expansion refrigerant coils	3	Mfrs Stds	-	I	Note A	9.4C.8	Note 21
Refrigerant piping & tubes	3	ASTM B42 ASTM B88	-	I	Note A	9.4-15	Note 21
Water Side	3	ASME III	3	I	Note A	9.4-15	
Dampers and supports	3	Mfrs Stds	-	I	Note A	9.4-15	Note 21
Ductwork and supports	3	Mfrs Stds	-	I	Note A	9.4-15	Note 21
Booster Return Fans	3	Mfrs Stds	-	I	Note A	9.4-15	Note 21
52. Turbine Plant Cooling Water System							

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Class 5 piping and supports	NNS	ANSI B31.1		II/None	Note B	10.4.12	Note 44
53. Condensate Polishing System							
Class 5 piping and supports	NNS	ANSI B31.1		II/None	Note B	10.4.6	Note 44
54. Condenser Vacuum and Waterbox Priming System	NNS	ANSI B31.1		II/None	Note B	10.4.2	Note 44
Class 5 piping and supports							
55. Heater Drains System							
Class 5 piping and supports	NNS	ANSI B31.1		II/None	Note B	10.4.11	Note 44, 81
56. Chemical Feed System							
Class 5 piping and supports	NNS	ANSI B31.1		II/None	Note B	10.3-1	Note 44
57. Security Systems							
Barriers	N/A	Mfrs Stds	-	II/NONE	Note B, C	13.6	
Doors	N/A	Mfrs Stds	-	NONE	Note C	13.6	
Intrusion Detection/Monitoring	N1E	Mfrs Stds	-	NONE	Note C	13.6	
Lighting	N1E	Mfrs Stds	-	NONE	Note C	13.6	
Access Control System	N1E	Mfrs Stds		NONE	Note C	13.6	
58. Chlorination Systems							

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System and Components (45)	Safety Class (7)	Applicable Code or Standard (12)	Code Class (40)	Seismic Category	Quality Assurance (87)	Reference Section	Remarks
Class 5 Piping and Supports	NSSS	ANSI B31.1	-	II/NONE	Note B	9.2-1	Note 44

Notes

1. Services provided to support a safety or other necessary function:
   a. Emergency power - automatic loading
   b. Emergency power - manual loading
   c. Component cooling water
   d. Service water
   c. Steam
2. Portions of equipment containing component cooling or service water are Safety Class 3, Code Class 3.
3. Deleted.

3a Deleted.

4. Deleted.
5. Deleted.

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6. Deleted.
7. Safety classes for fluid system components are defined by the engineering flow diagrams and are in accordance with ANSI N18.2, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants except as descr bed in Section 3.2.2. Safety classes for electrical, instrumentation and control components are defined by the one-line diagrams, electrical wiring diagrams, and instrumentation and control diagrams. Safety classes for reactor containment pressure boundary components are in accordance with ANSI N18.2.
8. The Main Steam Isolation and Bypass Valves were originally procured to ASME Class 1 as permitted by paragraph NA-2134 of the ASME Code, Section III. Procurement of ASME Code components to classifications higher than their end use is common (and acceptable); however; the application shown is the required classification for system and for ASME XI activities.
9. Parts must meet the structural integrity requirements of the specification and applicable quality assurance requirements of 10 CFR Part 50, Appendix B.
10. Failure can cause no nuclear safety problem, although an economic loss may result.
11. This component is Safety Class 3 under the definition 2.2.3(1), (3), or (4) of ANSI N18.2-1973 and qualifies for no special seismic design by meeting the four following conditions. Portions of systems in which this component is located that perform the same safety function likewise qualify for no special seismic design.

    Conditions to be met for exemption are the following:

    a. Failure would not directly cause a Condition III or IV event (as defined in ANSI N18.2-1973, Nuclear Safety Criteria for the Design of Stationary Pressurized Water Reactor Plants).

    b. There is not safety function to mitigate nor could failure prevent mitigation of the consequences of a Condition III or IV event.

    c. Failure during or following any Condition II event would result in consequences no more severe than allowed for a Condition III event.

    d. Routine post seismic procedures would disclose loss of the safety function.
12. The applicable code or standard provided is the primary ASME code or industry standard which applies to the given component. Additional information about the design requirements is provided in the referenced FSAR Sections. IEEE 323 applies to harsh equipment only. ASME: American Society of Mechanical Engineers. III stands for Section III of the ASME Boiler and Pressure Vessel (B&PV) Code. VIII stands for Section VIII of the ASME B&PV Code. Pressure vessels which are part of the RCPB meet the requirements of 1971 Version, with application of all Addenda through and including the Summer 1972 Addenda. Pumps, valves, and piping which are part of the RCPB meet the requirements of the 1971 Version, with application of all Addenda through and including the Winter 1972 Addenda. Later Code versions may be used optionally.
13. Not Used
14. Must maintain fuel array to prevent criticality under adverse conditions including occurrences of the SSE.
15. To be safety classified, failure of the tool must be directly a nuclear safety problem. If a nuclear safety problem arises from tool failures combined with a procedural failure thereafter, the tool is Non-Nuclear Safety.
16. Failure inside isolable reactor Containment prevents substantial release to the environment of radioactive gases from damaged spent fuel.
17. Deleted.

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18. Any reactor vessel internal, the single failure of which could cause release of a mechanical piece having potential for direct damage (as to the vessel cladding) or flow blockage, shall be classified to a minimum of Safety Class 2.
19. These items are not required as mechanical supports for CRDM housings, but are required to ensure functioning of the control rods.
20. Failure could cause a LOCA, but less than a Condition III loss-of-coolant.
21. This equipment is not commercially available as ASME Code, Section III, Class 3.
22. This equipment is included in the scope of IEEE 387; in that, ANSI N18.2 endorses IEEE-308 which in turn endorses IEEE-387, for diesel generators. IEEE-387 provides principal design criteria, etc., for individual diesel generator units which enable them to meet their functional requirements as part of the standby power supply.
23. Critical parts required to maintain structural integrity during a seismic event are subjected to QA program.
24. These devices will remain in place following a SSE.
25. Component supports are designed in accordance with ASME B&PV Code, Section III, Class 3 but fabricated to AISC -1970.
26. Deleted.
27. Deleted.
28. Deleted.
29. Deleted.
30. Applies only to that bolting involved with coastdown function.
31. Tubing, valves and fittings connected to ASME Class 2 and 3 systems are procured to ASME Section III Class 2 and 3 requirements. Instruments and racks are not covered by ASME Code. Also see Note 41.
32. Deleted.
33. The requirements of FSAR Section 6.1B.2 for Containment and Section 12.3.1.2 for radiation areas outside Containment are implemented by approved procedures. These procedures will establish the requirements necessary to ensure a good quality coating system. The fire protection requirements for protective coatings are in FSAR Section 9.5.1.5.3. The quality assurance requirements for the protective coatings program for Service Level I are in accordance with EPRI Report 1003102, “Guideline on Nuclear Safety-Related Coatings”, Revision 1 (formerly TR-109937) and ASTM D 5144-00, Standard Guide for Use of Protective Coating Standards in Nuclear Power Plants.
34. The quality requirements of Branch Technical Position ETSB 11-1, Rev. 1, are satisfied. Specifically, codes and standards required in the design, fabrication, inspection and testing of pressure vessels, tanks, heat exchangers, piping and valves, and pumps for the liquid and solid radwaste systems, and the steam generator blowdown processing system comply as a minimum with Table 1 of Branch Technical Position ESTB 11-1, Revision 1, except for the instance stated in Note 83 to this Table. Note: This is not within the scope of FSAR Section 17.2. Where “Note 34” is used for safety-related equipment (Safety Class 1, 2, or 3, or Seismic Category I), the QA provisions of Appendix B envelope and satisfy the requirements of ETSB 11-1. The scope of ESTB 11-1 is defined in the engineering flow diagrams by "RWMS" boundary demarcation.
35. ATCOR topical report no. 132A gives a breakdown of system components and applicable design codes.

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36. The applicable code for the Containment is the proposed Standard Code for Concrete Reactor Vessels and Containment (April 1973) issued for trial use and comments. It was developed by the Joint ACI-ASME Technical Committee on Concrete Pressure Components for Nuclear Service (see Section 3.8.1.2.1).

37. Deleted.

38a. This applies to safety-related and non-safety related instruments which are connected to nuclear safety related piping or ducting with seismic Category I tubing and supports.

38b. This applies to non-safety related instruments connected to NNS piping or ducting by NNS tubing and supports.

39. All pressure switches are differential pressure switches with the low side open to atmosphere.

40. Code Class 1, 2 or 3 indicates item within ASME Code Boundary. Code Class "-" indicates item not within ASME Code Boundary.

41. The ASME III Code (excluding NA-4000, NA-5000, and NA-8000) shall apply to all tubing, valves and fittings. Tubing, valves and fittings are procured as ASME material but will not be installed with Third Party Inspection, Code Stamping, and Code Data Reports as specified in ASME Subsection NA-5000 and NA-8000. Site fabrication and installation of tubing, valves and fittings will be in accordance with an NRC approved QA program governing non-ASME work which meets the requirements of Appendix-B to 10 CFR part 50 (excluding ASME III Subsection NA-4000). Piping installed between root valves and instruments is subject to the same requirements as stated previously in this note for instrument tubing, valves and fittings. Tubing and valve supports shall be installed as Seismic Category I but shall not comply with ASME III Subsection NF requirements. See Section 3.9B.3.4.3 for details.

42. The quality assurance requirements of Appendix A of the Branch Technical Position APCSB 9.5-1 of Standard Review Plan, Section 9.5.1, Revision 1, are satisfied.

43. The quality assurance requirements of Branch Technical Position MEB 3-1 and APCSB 3-1 are satisfied.

44. Process piping meets quality assurance requirements as set forth in mechanical specification 2323-MS-46B, "Non-Nuclear Pipe Hanger and Supports" and mechanical specification 2323-MS-100, "Piping Erection". Tubing and tubing supports used in NNS process piping, control piping, and sampling piping meets the qualify assurance requirements as set forth in specification CPES-I-1018, Installation of Piping/Tubing and Instrumentation Class 5 piping and its supports are designated Seismic Category II or None. Seismic Category II is utilized for piping included in ASME Section III stress analyses, high energy lines, and other special cases indicated by engineering evaluation. Seismic Category None is normally utilized for 2 inch and under piping, and 4 inch and under air filled copper tubing/pipe, unless otherwise indicated by engineering evaluation. Seismic Category II piping has Seismic Category II supports. Seismic Category NONE piping has Seismic Category NONE supports. In Category I Buildings, all Class 5 piping has been evaluated to show that seismic failure is not cred ble and therefore, the safety function of nuclear safety-related items is not adversely affected.

45. See system No. 40 for instrument piping (tubing). Piping includes pipe, tubing, fittings and valves.

46. Safety function is achieved by remaining in place. Component safety function is non-active.

47. Applicable codes and standards for concrete radiation shielding are listed in Section 3.8.4.2.

48. Only portions that furnish support to CRDMs, are Safety Class 1.

49. Deleted.

50. Deleted.

51. Supports for NNS instrumentation lines, instrument air lines, and pneumatic signal lines installed in Seismic Category I areas shall be Seismic Category II or NONE. Seismic Category NONE supports shall only be used if engineering evaluation shows that a seismic failure would not adversely affect the safety function of nuclear safety related items.

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52. Masonry walls have been evaluated for seismic interactions (remained non-seismic), removed, or replaced with seismically designed walls of other material.
53. This load handling equipment satisfies NUREG-0554 single-failure-proof requirements. Seismic Category I denotes the capability to retain control of the lifted load and to maintain structural integrity during and after a seismic event. The loading combinations include OBE and SSE with lifted load except for the polar crane due to the low probability of SSE with lifted load (less than 1×10^{-7} per year). The polar crane criteria is discussed in FSAR Section 9.1.4.3.2.
54. Fire dampers are functionally part of fire barriers but are physically located in ventilation ducts and penetrations. The safety class designation on ventilation diagrams does not apply to the fire dampers. Fire dampers which are required to remain open after an SSE have thermal links that are designated seismic Category II and are qualified to keep the damper open.
55. The tank boundary extends to the first weld connecting the penetration nozzles to system piping outside the tank. The tank and associated piping inside the tank is not N-stamped.
56. The application of R.G. 1.97 is applied to those channels identified in FSAR Section 7.5.
57. The Class 5 piping and valves in these 2" and under seismic Category II lines are seismically analyzed and are supported by seismic Category II supports. Installation requirements are the same as other Class 5, seismic Category II pipes.
58. Excluding existing buried yard piping. QA requirements of Appendix A to BTP APCSB 9.5-1 of Standard Review Plan Section 9.5.1 Revision 1 are to be satisfied for future activities associated with buried yard piping that supplies fire protection water to safety related buildings.
59. QA for water extinguishers limited to UL listing.
60. Deleted.
61. Deleted.
62. This component has been qualified as seismic category I by analysis.
63. Applicable where wall is supported by structural steel specified seismically, e.g., all access blockouts that have removable block closures have seismically designed steel supports to prevent collapse of the blocks.
64. Application of modified GDC-4 (leak-before-break) removes the need for non-crushable insulation. The non-crushable insulation may be replaced.
65. The leak testing of the instrument tubing between the instrument isolation valve and the instrument will be accomplished during normal instrument calibration. Tubing systems previously pressure tested as being satisfactory, which were later repaired or replaced, shall not require an additional pressure test prior to initial plant startup but shall be evaluated under an in-service leak test program.
66. For the Non-1E portion of the AMSAC, the requirements of Generic Letter 85-06 will be implemented through application of a quality assurance program that satisfies applicable requirements of Regulatory Guide 1.33, Revision 2.
67. Seismic Category II conduit supports are subject to Operations Quality Assurance Note B.
68. ASME III is the primary code for design and/or fabrication. This item is not within the ASME III, jurisdictional boundary.
69. This applies to auxiliary relay racks 1 and 2 only. The only 1E function of the Auxiliary Relay Racks is to provide isolation.
70. Seismic Category II anchor bolts, structural steel and/or mounting are subject to Operations Quality Assurance Note B.

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

71. NNS HVAC dampers are Seismic Category NONE. Operations Quality Assurance Note B applies to mounting and supports in Seismic Category II ductwork.
72. Seismic Category II mounting, anchors, and/or supports are subject to Operations Quality Assurance Note B.
73. Critical parts of these radiation monitors subject to Operations Quality Assurance Note B are the lead shielding for the detectors and the supports and anchors for those shields.
74. Critical parts of these radiation monitors subject to Operations Quality Assurance Note B are the lead shielding and fluid system pressure boundary and the supports and anchors for these items.
75. Valve required during and after a tornado.
76. Only Control Room Blowout Panels are Seismic Category II (to remain closed) and subject to Operations Quality Assurance Note B.
77. Certain non-safety related (Seismic Category NONE) instrumentation and controls located outside Seismic Category I areas are denoted as Class 1E. These devices are fully Class 1E except for seismic qualification. Seismic qualification and installation are provided to the extent practical considering that the specific location (e.g. building) is not seismically qualified. See Figure 7.1-3 (Sht. 16), Section 7.2.1.1.6, Section 8.3.1.4.1, and Section 10.4.7 for descriptions of these items.
78. Includes the design features and administrative measures to assure the safe handling of heavy loads.

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

79. Valve operators for safety related valves are classified and have QA applicability as follows:

Valve Operator Type	Applicable Safety Class	Code/ Standard	Code Class	Seismic Category	Quality Assurance	Remarks
Self Actuated (e.g. check valve, relief valve, etc.): Included in table listing because "operator" is integral to valve						
Manual: Included in the table listing because the classification of a manual “Operator” is included in the valve classification. Table 3.9B-10 identifies where handwheels are credited for manual operation of active valves.						
Extension Stems (Active)	N/A	Mfrs Stds	-	I	Note A	
Extension Stems (Passive)	N/A	Mfrs Stds	-	NONE	Note C	Note 72
Solenoid: Included in table listing because "operator” is integral to valve; however, Safety Class may be 1E or N1E with applicable corresponding standards in addition to the ANSI safety class.						
Motor (Active Vlv)	1E	IEEE 323	-	I	Note A	
Motor (Passive Vlv)	N1E	Mfrs Stds	-	NONE	Note C	
Air/hydraulic (Active Vlv*)	3, N/A**	Mfrs Std	-	I	Note A	
Air/hydraulic (Passive Vlv)	N/A	Mfrs Stds	-	I#	Note A	
Air/hydraulic (Passive Vlv)	N/A	Mfrs Stds	-	NONE##	Note C	

* Active valves for each system are identified in Tables 3.9N-10 and 3.9B-10.

** Safety Class 3 is applicable to the air/hydraulic valve operator when an air accumulator is required for the valve to perform a safety function. Otherwise, Safety Class is "N/A".

\# Operator with Class 1E Appurtenances.

Operator with non-1E Appurtenances.

80. Cables and heat shrinkable cable insulation sleeves are not seismically qualified. However, the basic standards to which cables and heat shrinkable cable insulation sleeves are procured and their associated installation requirements assure their functioning following a seismic event.

81. High energy line piping segments which are listed in FSAR Section 3.7B.2.8 as located in the Turbine Building are designated class 5 piping and classified as Seismic Category II even though the locations are in a non-seismic building. Additional analyses are performed to demonstrate that unacceptable interactions of these piping segments with Non-Category I structures and components will not occur during a seismic event. These piping segments are seismically analyzed for break postulation and seismic qualification.

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

82. Piping and supports for the Diesel Generator exhaust piping the building, from the flex ble connectors to just upstream of the exhaust relief valve, are designed, analyzed, installed and inspected to the requirements of ASME Section III, including Code Case N-253-2. In some of the pipe support qualifications, part of the material stress allowables are extracted from ANSI B31.1. In all cases, the requirements of Code Case N-253-2 were met or exceeded. The piping and associated supports are exempt from Code Stamping.
83. Radiation monitors are not designed to ASME VIII, Division 1; however, manufacturer's design standards include hydrotesting to 1.5 times design pressure and compliance with ASME IX welding requirements.
84. Steam Generator nozzle dams are used during shutdown and refueling. ANSI N18.2 is not applicable. Seismic and QA requirements apply to critical components of the nozzle dam and support frame and to the diaphragm seal.
85. Non-1E level switches installed in ASME piping are safety related for pressure boundary integrity only. There are two types:
    a. Displacement Type (ASME).
    b. Thermal Dispersion Type (Non-ASME).
86. The Safety Class 2 (containment isolation) portion of the Unit 1 personnel airlock hydraulic system contains both ASME and non-ASME components as denoted on Figure 3.8-22. The ASME code is not applicable to the tubing and hydraulic system components.
87. See Table 17A-2 for the Operations Quality Assurance Notes.
88. The Containment Air Radiation Monitor is classified as N1E because it is not required to function during or following a LOCA and is not environmentally qualified for that event. It is electrically isolated from the 1E bus during and after a LOCA; therefore, IEEE-323 does not apply. The monitor's Nuclear Safety Function is performed in normal operation and after seismic events in accordance with RG 1.45 [FSAR Section 5.2.5]. The monitor is classified as seismic Category I and is seismically qualified in accordance with IEEE-244.
89. Requirements of applicable codes meet or exceed those of CMAA 70 in applicable areas.
90. Underwater lights containing mercury (such as high pressure sodium lights) are subject to inspections and precautions and limitations that assure nuclear safety is not compromised.
91. Abandoned in place.
92. ANSI Safety Class 3 tubing, valves and fittings connected to HVAC systems may be procured to ASME Section III Class 3 requirements; however, the ASME Code is not applicable to the design or installation. ANSI Safety Class 3 tubing, valves, and fittings are designed and installed to seismic Category I requirements.
93. The Radial Arm Hoist Assembly (RAHA) is used during Modes 5 and 6 to aid in reactor vessel head stud removal and other refueling activities. Because the Mode 5 and 6 configuration does not meet RG 1.29, Seismic Category II requirements, the hoists and their trolleys are removed during Modes 1 through 4 and the radial arms are secured in a Seismic Category II configuration that meet the requirements of Reg. Guide 1.29, Rev. 2, Position C.2 and QA Note B.
94. Alternate penetration insert is for use in Modes 5 and 6.
95. The airlock instrumentation is a special case as descr bed in FSAR Section 6.2.4.1.3.
96. The SFP level instrument mounting brackets and anchorages were designed in accordance with the CPNPP requirements as a Seismic Category I component support. Other evaluations provided in letters TXX-15084 and TXX-15122 demonstrate that adequate and reasonable designs have been provided to support the other SFP level instrument channel components for the site's SSE.

TABLE 17A-1
LIST OF QUALITY ASSURED STRUCTURES, SYSTEMS AND COMPONENTS

97. The Westinghouse SHIELD Shutdown Seal is a separate component from the Reactor Coolant Pump No. 1 seal and located within the No. 1 seal insert package. The SHIELD Shutdown Seal is secured in a Seismic Category II configuration that meets the requirements of R.G. 1.29, Rev. 2 Position C.2, Augmented Quality Assurance and FSAR Table 17A-2 Note B.

98. Containment Recirculation Cooling Units at Containment Elevation 905 in Unit 2 only are provided with removable silencers which may be installed in Modes 5 and 6 and with the reactor de-fueled to reduce noise during refueling outages.

99. Fuel Assemblies are not classified by ANSI N18.2. Safety Class 3 id based on ANSI/ANS 51.1 (1983).

100. The portion of seismic instrumentation in the control room is installed seismic category II (i.e., seismic monitoring system cabinet).

TABLE 17A-2
QUALITY ASSURANCE SUMMARY

SAFETY CLASS	SEISMIC CATEGORY	SAFETY RELATED	OPERATIONS QA APPLIED
1	I	YES	YES, A
2	I	YES	YES, A
3	I	YES	YES, A
3, 1E	NONE	NO	YES, A
1E	I, N/A	YES	YES, A
N/A, N1E	I	YES	YES, A
NNS, N/A	II	NO	YES, B
NNS, N/A, N1E	NONE	NO	YES, C or D
NNS, N/A, N1E	NONE	NO	NO, E

Operations Quality Assurance Notes:

A. An Operations QA Program will be implemented which satisfies applicable requirements of Regulatory Guide 1.33, Rev. 2, “Quality Assurance Program Requirements” (Operations).

B. Critical parts required to maintain structural integrity during a seismic event are subjected to an Operations QA Program which satisfies applicable requirements of Regulatory Guide 1.33, Rev. 2.

C. Full QA requirements were not imposed for manufacture and/or installation; however, a specifically structured non-Appendix B QA program will be instituted during operation of equipment.

D. An Operations QA Program will be implemented which satisfies applicable requirements of Appendix A of the Branch Technical Position APCSB 9.5-1 of Standard Review Plan, Section 9.5.1, Rev. 1.

E. No formal quality assurance program is required. Normal applications of administrative controls for non-safety related items is applied.

F. Appropriate quality assurance measures are in accordance with Appendix A-1 of NEI 12-02, Revision 1, “Industry Guidance for Compliance with NRC Order EA-12-051, “To Modify Licenses with Regard to Reliable Spent Fuel Pool Instrumentation.”

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I OPERATIONAL SAFETY

I.A OPERATING PERSONNEL

I.A.1 OPERATING PERSONNEL AND STAFFING

OBJECTIVE:

"Complex transients in nuclear power plants place high demands on the operators in the control room. The objective of the actions described in this task is to increase the capability of the shift crews in the control room to operate the facility in a safe and competent manner by assuring that a proper number of individuals with the proper qualifications and fitness are on shift at all times."

-NUREG 0660, pg. I.A.1-1

I.A.1.1 Shift Technical Advisor

Action Plan Requirements:

"Provide on shift at each nuclear power plant a qualified person (the shift technical advisor) with a bachelor's degree or equivalent in a science or engineering discipline with specific training in the plant response to off-normal events and in accident analysis of the plant."

"Shift technical advisors shall serve in an advisory capacity to shift supervisors. The licensee shall assign normal duties to the shift technical advisor that pertain to the engineering aspects of assuring safe operations of the plant, including the review and evaluation of operating experience."

-NUREG 0578, pg. 13

Also see NUREG 0737

CPNPP Response

Refer to revised Section 13.2.1.1, Item 5 .

I.A.1.2 Shift Supervisor Administrative Duties

Action Plan Requirements:

"The administrative duties of the shift supervisor shall be reviewed by the senior officer of each utility responsible for plant operations. Administrative functions that detract from or are subordinate to the management responsibility for assuring the safe operation of the plant shall be delegated to other operations personnel not on duty in the control room."

- NUREG 0578, pg. A-48

CPNPP Response

A list of administrative duties of the Shift Supervisor will be provided to the Vice President, Nuclear, for his review prior to fuel load.

I.A.1.3 Shift Manning

Action Plan Requirements:

"The minimum shift crew for a unit shall include three operators, plus an additional three operators when the unit is operating. Shift staffing may be adjusted at multi-unit stations to allow credit for operators holding licenses on more than one unit."

"In each control room, including common control rooms for multiple units, there shall be at all times a licensed reactor operator for each reactor loaded with fuel and a senior reactor operator licensed for each reactor that is operating. There shall also be onsite at all times, an additional relief operator licensed for each reactor, a licensed senior reactor operator who is designated as shift supervisor, and any other licensed senior reactor operators required so that their total number is at least one more than the number of control rooms from which a reactor is being operated."

"Administrative procedures shall be established to limit maximum work hours of all personnel performing a safety-related function to no more than 12 hours of continuous duty with at least 12 hours between work periods, no more than 72 hours in any 7 day period, and no more than 14 consecutive days of work without at least 2 consecutive days off."

-NUREG 0694, pg. 11

CPNPP Response

Procedures have been established to address shift manning and an overtime policy. These procedures will be implemented prior to fuel load.

CPNPP currently follows shift manning and overtime policy prescribed by 10 CFR Part 26 Subpart I which supercedes NUREG-0694 criteria.

I.A.2 TRAINING AND QUALIFICATION OF OPERATING PERSONNEL

OBJECTIVE:

"Improve the capability of operators and supervisors to understand and control complex reactor transients and accidents, and improve the general capability of an operations organization to respond rapidly and effectively to upset conditions. Increase the education, experience, and training requirements for operators, senior operators, supervisors, and other personnel in the operations organization to substantially improve their capability to perform their duties."

-NUREG 0660, pg. I.A.2-1

I.A.2.1 Immediate Upgrading of Operator and Senior Operator Training and Qualifications

Action Plan Requirements:

"A. Eligibility Requirements to be Administered an Examination.

"1. Experience*

a. Applicants for senior operator licenses shall have 4 years of responsible power plant experience. Responsible power plant experience should be that obtained as a control room operator (fossil or nuclear) or as a power plant staff engineer involved in the day-to-day activities of the facility, commencing with the final year of construction. A maximum of 2 years power plant experience may be fulfilled by academic or related technical training, on a one-for-one time basis. Two years shall be nuclear power plant experience. At least 6 months of the nuclear power plant experience shall be at the plant for which he seeks a license.

b. Applicants for senior operator licenses shall have held an operator's license for 1 year."

"2. Training

a. Senior operator*: Applicants shall have 3 months of shift training as an extra man on shift.

b. Control room operator*: Applicants shall have 3 months training on shift as an extra person in the control room.

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*Precritical applicants will be required to meet unique qualifications designed to accommodate the fact that their facility has not yet been in operation.

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c. Training programs shall be modified, as necessary, to provide:

1) Training in heat transfer, fluid flow and thermodynamics.

2) Training in the use of installed plant systems to control or mitigate an accident in which the core is severely damaged.

3) Increased emphasis on reactor and plant transients."

"3. Facility Certifications

Certifications completed pursuant to Sections 55.10(a) (6) and 55.33a(4) and (5) of 10 CFR Part 55 shall be signed by the highest level of corporate management for plant operation (for example, Vice President for Operations)."

-NRC Letter dated March 28, 1980, Enclosure 1

Also see NUREG 0737

CPNPP Response

Refer to revised Section 13.2.1.1. The referenced program is consistent with 10 CFR 55 rulemaking issued after the TMI-2 event.

I.A.2.3 Administration of Training Programs

Action Plan Requirements

"Training center and facility instructors who teach systems, integrated responses, transient and simulator courses shall demonstrate their competence to NRC by successful completion of a senior operator examination."

"Instructors shall be enrolled in appropriate requalification programs to assure they are cognizant of current operating history, problems, and changes to procedures and administrative limitations."

-NRC Letter dated March 28, 1980, Enclosure 1

CPNPP Response

Refer to revised Section 13.2.1. The referenced program is consistent with 10 CFR 55 rulemaking issued after the TMI-2 event.

I.A.3 LICENSING AND REQUALIFICATION OF OPERATING PERSONNEL

OBJECTIVE:

"Upgrade the requirements and procedures for nuclear power plants operator and supervisor licensing to assure that safe and competent operators and senior operators are in charge of the

day-to-day operation of nuclear power plants. Increase the requirements for initial issuance of licenses and for license renewals and provide closer NRC monitoring of licensed activities."

-NUREG 0660, Pg. I.A.3-1

I.A.3.1 Revise Scope and Criteria for Licensing Examinations

Action Plan Requirements:

"B. NRC Examinations

1. Increase Scope of Examinations

a. A new category shall be added to the operator written examination entitled, "Principles of Heat Transfer and Fluid Mechanics."

b. A new category shall be added to the senior operator written examination entitled, "Theory of Fluids and Thermodynamics."

c. Time limites shall be imposed for completion of the written examinations:

1. Operator: 9 hours.

2. Senior Operator: 7 hours.

d. The passing grade for the written examination shall be 80% overall and 70% in each category.

e. All applicants for senior operator licenses shall be required to be administered an operating test as well as the written examination.

f. Applicants will grant permission to NRC to inform their facility management regarding the results of the examinations for purposes of enrollment in requalification programs."

-NRC Letter, dated March 28, 1980
Enclosure 1, pg. 4

2. Simulator Examinations

"Simulator examinations will be included as part of the licensee examinations."

"The administration of simulator examinations will be deferred for applicants whose facilities do not have simulators on site as of October 1, 1980. These deferred simulator examinations will be initiated by October 1, 1981."

- NUREG 0737

CPNPP Response

Refer to revised Section 13.2.1.1 . The referenced program is consistent with 10 CFR 55 rulemaking issued after the TMI-2 event.

All CPNPP senior operator cold license candidates will be prepared for an operating test as well as a written examination.

NRC will be provided with a signed release granting permission for NRC to inform CPNPP management regarding the results of the examinations for purposes of enrollment in requalification programs.

CPNPP takes the position that operations on the specified plant should be reinforced by examination at the time of licensing rather than operations on a non-identical plant. Reasoning from this position, CPNPP maintains that simulator examinations on a non-identical plant simulator should not be scheduled consecutively with the balance of the license examination. Furthermore, scheduling examinations at a remote location for an entire operating staff on a fully-booked simulator where the examination target date is not completely predictable cannot be accomplished by a cold license applicant in the narrow window of time prescribed.

I.B SUPPORT PERSONNEL

I.B.1 MANAGEMENT FOR OPERATIONS

OBJECTIVE:

"Improve licensee safety performance and ability to respond to accidents by upgrading the licensee groups responsible for radiation protection and plant operation. The areas to be upgraded include (1) staff size; (2) education and experience of staff members; (3) plant operating and emergency prodecures; (4) management awareness of and attention to safety matters; and (5) numbers and types of personnel available to respond to accidents. Licensee safety performance would further improved if (1) a full-time, dedicated, onsite safety engineering staff were established, and (2) and integrated program for the systematic review of operating experience were provided with the concurrent dissemination of information to plant personnel."

- NUREG 0660, pg. I.B.1-1

I.B.1.2 Evaluation of Organization and Management Improvements of Near – Term Operating License Applicants

Action Plan Requirements:

Each applicant for an operating license shall establish an onsite independent safety engineering group (ISEG) to perform independent reviews of plant operations.

The pricipal function of the ISEG is to examine plant operating characteristics, NRC issuances, Licensing Information Service advisories, and other appropriate sources of plant design and operating experience information that may indicate areas for improving plant safety. The ISEG is to perform independent reviews and audits of plant activities including maintenance, modifications, operating problems, and operational analysis, and aid in the establishment of programmatic requirements for plant activities. Where useful improvements can be achieved, it is expected that this group will develop and present detailed recommendations to corporate management for such things as revised procedures or equipment modifications.

Another functions of the ISEG is to maintain surveillance of plant operations and maintenance activities to provide independent verification that these activities are performed correctly and that human errors are reduced as far as practicable. ISEG will then be in a position to advise utility management on the overall quality and safety of operations. ISEG need not perform detailed audits of plant operations and shall not be responsible for sign-off functions such that it becomes involved in the operating organization.

- NUREG 0737

CPNPP Response

The principal functions and composition of the former ISEG have been distributed throughout CPNPP organizations.

Since the time of NUREG-0737 (1980) and CPNPP licensing and initial operations (1990), the CPNPP organization has continually evolved and improved. Many new plant programs and

processes have been developed and implemented (e.g., improved Corrective Action Program, System Engineering Department and related programs, a Maintenance Rule Program, individual plant department self-assessments, etc.). These improved organizational attributes overlap and/ or complement the original ISEG intended functions. Therefore it was determined that continued description of the ISEG in the FSAR was unnecessary.

I.B.2 INSPECTION OF OPERATING REACTORS

I.B.2.2 Resident Inspector at Operating Reactors

“An NRC resident inspector will be assigned to each site.”

- NUREG 0694, Pg. 26

CPNPP Response

CPNPP will provide facilities for each resident inspector as required.

I.C OPERATING PROCEDURES

OBJECTIVE:

"Improve the quality of procedures to provide greater assurance that operator and staff actions are technically correct, explicit and easily understood for normal, transient, and accident conditions. The overall content, wording, and format of procedures that affect plant operation, administration, maintenance, testing, and surveillance will be included. A principal part of this work is to improve procedures for dealing with abnormal conditions and emergencies by improving the delineation of symbols, events, and plant conditions that identify emergency or off-normal situations that confront the operators and, once identified, to assure consistency with operator training."

- NUREG-0660, pg. I.C-1

I.C.1 SHORT-TERM ACCIDENT ANALYSIS AND PROCEDURES REVISION

Action Plan Requirements:

"a. Provide the analysis, emergency procedures, and training to substantially improve operator performance during a small break loss-of-coolant accident."

"b. Provide the analysis, emergency procedure, and training needed to assure that the reactor operator can recognize and respond to conditions of inadequate core cooling."

"c. Provide the analysis, emergency procedures, and training to substantially improve operator performance during transients and accidents, including events that are caused or worsened by inappropriate operator actions."

- NUREG 0578, Pg. 12

Also see NUREG 0737.

CPNPP Response

Small break loss-of-coolant accident analyses have been performed and submitted to Staff in WCAP-9600. The report presents a comprehensive study of Westinghouse system response to small breaks. On-going efforts aimed at improving emergency operating procedure guidelines have already been discussed by Westinghouse and NRC Staff.

Inadequate core cooling is an item to be addressed, as stated in I.E. Bulletin 79-06C. Further definition of the scope of the item is needed form the NRC Staff, such as system failure and operator error assumptions to be made in the analyses. At present Westinghouse model preparation is in progress to permit response to identified action.

Westinghouse does plan to perform pre-test calculations of the LOFT tests when the necessary input information is available.

The purpose of the recommended transient and accident analysis is to provide an increase in safety by improving the performance of reactor operators during transient and accident

conditions. The primary concern is that the operator training and emergency operating procedures are based on the conservative FSAR Chapter 15 analyses. Chapter 15 should continue to be used for design basis analyses since these show the most limiting initial approach to safety limits, both core thermal and system overpressurization. What is needed to meet the intent of this section of NUREG-0578 is to show the long-term consequences using realistic assumptions (better estimate modeling) incorporating the effects of the following:

1. Operator's failure to act when required.

2. Operator's inappropriate actions during an accident.

3. Additional failures.

4. Selected system operations (e.g., re-starting of RCP's etc.)

The results of these analyses can be used to evaluate information available to the operator and the adequacy of existing procedures. Appropriate changes can then be incorporated into the existing procedures, designs, and training programs. Development of the models to incorporate such effects is in itself a long-term effort before detailed analyses can be run. Significant interaction between industry and the NRC is required to agree on the assumptions, bases, appropriate actions or misactions to be modeled, and best estimate boundary conditions. When completed, the analyses results using the better estimate modeling tools can enhance the current operator training programs by providing additional insight into the course of events the operator will likely encounter during a transient.

The Westinghouse Owners Group (WOG) will submit a detailed description of a program to comply with the requirements of item I.C.1 to the NRC in early 1981. The program will identify WOG's previous submittals to the NRC and the additional effort required to obtain full compliance with this item. After approval of the program by the NRC, CPNPP will submit to the NRC a schedule and program, based on the WOG program, for meeting the requirements of item I.C.1.

NUREG-0737 states that reviews of emergency procedures being conducted under the pilot program, item I.C.8. (NUREG-0660), will be discontinued when the long-term program item I.C.9 (NUREG-0660) is made available. Because it is anticipated that the requirements and guidelines of the long-term program will soon be available, CPNPP will comply with the long-term program rather than submitting procedures for review under the pilot monitoring program.

I.C.2 SHIFT AND RELIEF TURNOVER PROCEDURES

Action Plan Requirements:

"Review and revise plant procedures as necessary to assure that a shift turnover checklist is provided and required to be completed and signed by the on-going and off-going individuals responsible for command of operations in the control room. Supplementary checklists and shift logs should be developed for the entire operations organization, including instrument technicians, auxiliary operators, and maintenance personnel."

- NUREG 0578, Pg. 13

CPNPP Response

A procedure has been established for shift and relief turnover to provide a checklist for reactor operators and shift supervisors. The checklist for auxiliary operators, a system to evaluate the shift turnover, and the implementation of these procedures will be completed prior to fuel load.

I.C.3 SHIFT SUPERVISOR'S RESPONSIBILITIES

Action Plan Requirements:

"Review plant administrative and management procedures. Revise as necessary to assure that reactor operations command and control responsibilities and authority are properly defined. Corporate management shall revise and promptly issue an operations policy directive that emphasizes the duties, responsibilities, and authority and lines of command of the control room operators, the shift technical advisor, and the person responsible for reactor operations command in the control room (i.e., the senior reactor operator)."

- NUREG 0578, Pg. 12

CPNPP Response

Prior to OL and annually thereafter, the Senior Vice President & Principal Nuclear Officer will issue a management directive that emphasizes the primary management responsibility of the Shift Manager for safe operation of the plant under all conditions and that clearly defines his command duties.

Procedures have been established which define the responsibilities and authorities of the Shift Manager and establish lines of succession. These procedures explicitly address the NUREG concerns dealing with retaining breadth of perspective of operational conditions affecting safety and remaining in the control room under accident conditions to direct operational activities. These new provisions will be incorporated into the training program for the Shift Manager.

I.C.4 CONTROL ROOM ACCESS

Action Plan Requirements:

"Review plant emergency procedures, and revise as necessary, to assure that access to the control room under normal and accident conditions is limited to those persons necessary to the safe command and control of operations."

- NUREG 0578, Pg. 13

CPNPP Response

Administrative procedures have been written to formalize policies which allow the Shift Manager, or his designated alternate, to restrict access to the Control Room during both normal operations and emergencies. The procedures establish a clear line of authority, responsibility and succession in the Control Room in the event of an emergency.

I.C.5 PROCEDURES FOR FEEDBACK OF OPERATING EXPERIENCE TO PLANT STAFF

Action Plan Requirements:

"In accordance with Task Action Plan I.C.5, Procedures for Feedback of Operating Experience to Plant Staff (NUREG-0660), each applicant for an operating license shall prepare procedures to assure that operating information pertinent to plant safety originating both within and outside the utility organization is continually supplied to operators and other personnel and is incorporated into training and retraining programs. These procedures shall:

(1) Clearly identify organizational responsibilities for review of operating experience, the feedback of pertinent information to operators and other personnel, and the incorporation of such information into training and retraining programs;

(2) Identify administrative and technical review steps necessary in translating recommendations by the operating experience assessment group into plant actions (e.g., changes to procedures; operating orders);

(3) Identify the recipients of various categories of information from operating experience (i.e., supervisory personnel, shift technical advisors, operators, maintenance personnel, health physics technicians) or otherwise provide means through which such information can be readily related to the job functions of the recipients;

(4) Provide means to assure that affected personnel become aware of and understand information of sufficient importance that should not wait for emphasis through routine training and retraining programs;

(5) Assure that plant personnel do not routinely receive extraneous and unimportant information on operating experience in such volume that it would obscure priority information or otherwise detract from overall job performance and proficiency;

(6) Provide suitable checks to assure that conflicting or contradictory information is not conveyed to operators and other personnel until resolution is reached; and,

(7) Provide periodic internal audit to assure that the feedback program functions effectively at all levels."

- NUREG 0737

CPNPP Response

Procedures have been established to ensure that reports of industry operating experiences distributed either as INPO Event Reports or INPO SEE-IN reports, vendor technical correspondence, or as NRC Correspondence are assessed, and that applicable reports are distributed directly to appropriate station personnel, including, when appropriate, the Shift Operations Manager and the Nuclear Training Manager. For INPO Event Reports, INPO SEE-In reports and NRC Information Notices, these procedures direct recipients to develop appropriate action plans in response to the recommendations developed from assessed reports and to inform the Performance Improvement Department of their plans. The Performance Improvement

Department will monitor recipients' efforts and related developments until the problems identified and/or the concerns raised by the report processing are adequately addressed. Vendor technical correspondence is screened for operating experience by the Vendor Equipment Technical Information Program and distributed to the cognizant organization for action/information. The cognizant organization must acknowledge review of the vendor technical correspondence. NRC Bulletins and other pertinent NRC assessments of operating experience are screened by Performance Improvement and Regulatory Affairs for appropriate distribution and forwarded to the cognizant organization for action/information.

Correspondence which requires a written response or a specific action are tracked to completion by Regulatory Affairs.

Procedures have been established to provide a mechanisim for: reporting problems which occur at CPNPP, correcting the condition, evaluating the reports to identify trends, and determining what changes, if any, are required to prevent recurrence. Procedures assign responsibility for ensuring all required corrective actions are completed. Information on these problems are available to management and plant staff through various methods for dissemination of relevant information to their personnel.

Procedures have been developed in both the Operations department and the Training department to ensure that appropriate information is communicated to operations personnel and incorporated into training programs. These procedures include mechanisms for timely dissemination of information when circumstances warrant. The Shift Operations Manager has the responsibility of screening the available information to ensure both that operations personnel are not burdened with extraneous and unimportant information, and that they do not receive conflicting or contradictory information.

The Performance Improvement Department will distribute operating experience related information to appropriate licensed and other interested personnel. The Performance Improvement Department will ensure that all appropriate personnel remain cognizant of recent industry and station operating experience.

I.C.6 PROCEDURES FOR VERIFYING CORRECT PERFORMANCE OF OPERATING ACTIVITIES

Action Plan Requirements:

"It is required (from NUREG-0660) that licensees' procedures be reviewed and revised, as necessary, to assure that an effective system of verifying and correct performance of operating activities is provided as a means of reducing human errors and improving the quality of normal operations. This will reduce the frequency of occurrence of situations that could result in or contribute to accidents. Such a verification system may include automatic system status monitoring, human verification of operations and maintenance activities independent of the people performing the activity (see NUREG-0585, Recommendation 5), or both.

Implementation of automatic status monitoring if required will reduce the extent of human verification of operations and maintenance activities but will not eliminate the need for such verification in all instances. The procedures adopted by the licensees may consist of two phases – one before and one after installation of automatic status monitoring equipment, if required, in accordance with item I.D.3."

- NUREG 0737

CPNPP Response

Procedures will be established which designate the responsibilities and authorities of personnel associated with the control of equipment. The procedures will address the second, independent verification of proper tagging and the return-to-service of equipment by qualified operators. These procedures will be implemented prior to fuel load.

I.C.7 NSSS VENDOR REVIEW OF PROCEDURES

Action Plan Requirements:

"Obtain nuclear steam supply system (NSSS) vendor review of low-power testing procedures to further verify their adequacy."

- NUREG 0694, Pg. 14

"Obtain NSSS vendor review of power-ascension test and emergency procedures to further verify their adequacy."

- NUREG 0694, Pg. 20

CPNPP Response

CPNPP shall have Westinghouse review the low-power testing procedures prior to fuel loading and the power-ascension and emergency procedures prior to full-power operation.

I.C.8 PILOT MONITORING OF SELECTED EMERGENCY PROCEDURES FOR NTOL APPLICANTS

Action Plan Requirements:

"Correct emergency procedures, as necessary, based on the NRC audit of selected plant emergency operating procedures (e.g., small-break LOCA, loss of feedwater, restart of engineered safety features following a loss of ac power, steam-line break, or steam-generator tube rupture)."

- NUREG 0694, Pg. 21

Also see NUREG-0737.

CPNPP Response

NUREG-0737 states that reviews of emergency procedures being conducted under the pilot monitoring program, item I.C.8. (NUREG-0660), will be discontinued when the long-term program item I.C.9 (NUREG-0660) is made available. Because it is anticipated that the requirements and guidelines of the long-term program will soon be available, CPNPP will comply with the long-term program rather than submitting procedures for review under the pilot monitoring program.

I.D CONTROL ROOM DESIGN

OBJECTIVE:

"Improve the ability of nuclear power plant control room operators to prevent accidents or cope with accidents if they occur by improving the information provided to them."

- NUREG 0660, Pg. I.D-1

I.D.1 CONTROL ROOM DESIGN REVIEW

Action Plan Requirements:

"In accordance with Task Action Plan I.D.1, Control Room Design Reviews (NUREG-0660), all licensees and applicants for operating licenses will be required to conduct a detailed control-room design review to identify and correct design deficiencies. This detailed control-room design review is expected to take about a year. Therefore, the Office of Nuclear Reactor (NRR) requires that those applicants for operating licenses who are unable to complete this review prior to issuance of a license make preliminary assessments of their control rooms to identify significant human factors and instrumentation problems and establish a schedule approved by NRC for correcting deficiencies. These applicants will be required to complete the more detailed control room reviews on the same schedule as licensees with operating plants."

- NUREG 0737

CPNPP Response

The Detailed Control Room Design Review (DCRDR) has been completed in accordance with NUREG-0700 and the reports submitted to the NRC. Onsite audits/reviews have been conducted by the NRC. Implementation of the required design changes has been completed. A final environmental survey was conducted for Unit 1 when construction and modifications were completed. Based on the results of the environmental surveys, Unit 2 control boards were compared to Unit 1 control boards to assess any design differences. Supplements to the DCRDR were submitted prior to fuel loads for Unit 1 and Unit 2 respectively.

Procedures have been developed to formalize the ongoing Human Factors Engineering (HFE) program at CPNPP. The major elements of the HFE program include:

a) HFE review of control room design changes which includes the following:

   1. HFE design review of existing "legacy" control room equipment/systems per guidelines contained in NUREG-0700, Revision 0 [1981].

   2. HFE design review of new and replacement plant computer and plant digital control systems per guidelines contained in NUREG-0700, Revision 2 [2002].

b) Multi-discipline team for the assessment, disposition and verification of Training, Operations and Engineering HFE concerns

c) Site specific HFE design guide

d) Function and task analysis of Emergency Response Guidelines (ERGs) and the comparison with control room capabilities

e) Computer Aided Design (CAD) detailed control board layout drawings

f) Standard abbreviations and acronyms for control room labels and procedures

The DCRDR and the ongoing HFE program were audited by the NRC in January, 1989.

I.D.2 PLANT SAFETY PARAMETER DISPLAY CONSOLE

Action Plan Requirements:

"In accordance with Task Action Plan 1.D.2, Plant Safety Parameter Display Console (NUREG-0660), each applicant and license shall install a safety parameter display system (SPDS) that will display to operating personnel a minimum set of parameters which define the safety status of the plant. This can be attained through continuous indication of direct and derived variables as necessary to assess plant safety status."

- NUREG 0737

Also see NRC Letter dated August 1, 1980.

CPNPP Response

A plant safety parameter display console has provided for CPNPP to assess plant safety status. This system is designed in accordance with NUREG-0696 and NUREG-0737 Supplement 1.

I.G PREOPERATIONAL AND LOW POWER TESTING

OBJECTIVE:

"Increase the capability of the shift crews to operate facilities in a safe and competent manner by assuring that training for plant changes and off-normal events if conducted. Near-term operating license facilities will be required to develop and implement intensified training exercises during the low-power testing programs. This may involve the repetition of startup test on different shifts for training purposes. Based on experiences form the near-term operating license facilities, requirements may be applied to other new facilities or incorporated into the plant drill requirement (Item I.A.2.5). Review comprehensiveness of test programs."

- NUREG 0660, Pg. I.G-1

I.G.1 TRAINING REQUIREMENTS

Action Plan Requirements:

"Define and commit to a special low-power testing program approved by NRG to be conducted at power levels no greater than 5 percent for the purposes of providing meaningful technical information beyond that obtained in the normal startup test program and to provide supplemental training."

- NUREG 0694, Pg. 15

"Supplement operator training by completing the special low-power test program. Test may be observed by other shifts or repeated on other shifts to provide training to the operators."

- NUREG 0694, Pg. 21

CPNPP Response

Refer to revised section 13.2.1.1. The referenced program is consistent with 10 CFR 55 rulemaking issued after the TMI-2 event.

Testing to demonstrate the length of time required to stabilize natural circulation, core flow distribution, ability to establish and maintain natural circulation with or without onsite and offsite power, the ability to uniformly borate and cool down to hot shutdown conditions using natural circulation, and subcooling monitor performance have been performed at the Sequoyah 1 and Salem 2 facilities, which are comparable prototype plants, and will not be performed at CPNPP.

A description of the natural circulation demonstrations that are to be conducted as part of the training program during the low-power test program will be included in the CPNPP FSAR (Chapter 14 - Initial Test Program).

II SITING AND DESIGN

II.B CONSIDERATION OF DEGRADED OR MELTED CORES IN SAFETY REVIEW

OBJECTIVE:

"Enhance public safety and reduce individual and societal risk by developing and implementing a phased program to include in safety reviews, consideration of core degradation and melting beyond the design basis. The program phases are (1) short- and medium-term actions for scoping and implementation; (2) added requirements for high population density sites; (3) research programs and design studies to develop additional needed information; and (4) a rulemaking proceeding to establish long-term policy, goals, and requirements related to accidents involving core damage greater than the present design basis."

- NUREG 0660, Pg. II.B-1

II.B.1 REACTOR COOLANT SYSTEM VENTS

Action Plan Requirements:

"Each applicant and licensee shall install reactor coolant system (RCS) and reactor vessel head point vents remotely operated from the control room. Although the purpose of the system is to vent noncondensible gases from the RCS which may inhibit core cooling during natural circulation, the vents must not lead to an unacceptable increase in the probability of a loss-of-coolant accident (LOCA) or a challenge to containment integrity. Since these vents form a part of the reactor coolant pressure boundary, the design of the vents shall conform to the requirements of Appendix A to 10 CFR Part 50, "General Design Criteria." The vent system shall be designed with sufficient redundancy that assures a low probability of inadvertent or irreversible actuation.

"Each licensee shall provide the following information concerning the design and operation of the high point vent system:*

(1) Submit a description of the design, location, size, and power supply for the vent system along with results of analyses for loss-of-coolant accidents initiated by a break in the vent pipe. The results of the analyses should demonstrate compliance with the acceptance criteria of 10 CFR 50.46.

(2) Submit procedures and supporting analysis for operator use of the vents that also include the information available to the operator for initiating or terminating vent usage."

"*It was the intent of the October 30, 1979 letter to delete the requirement to meet the criteria of 10 CFR 50.44 and SRP 6.2.5 for beyond-design-basis events. The analysis requirements of Position 2 in the September 13, 1979 letter are therefore unnecessary."

CPNPP Response

CPNPP has the capability to remotely vent gases from the reactor vessel head and pressurizer vapor space during a post-accident situation where large quantities of non-condensible gas may collect.

Venting of the reactor vessel head is provided via a line isolated by two one-inch valves in series which vents directly to containment. The venting of the pressurizer vapor space is provided in the same manner. The valves are fail-closed solenoid valves which are environmentally and seismically qualified. The head vent valves are powered by Train A 1E power and the pressurizer vent valves are powered by Train B 1E power.

These vents are operable from the Control Room and positive indication of valve position is also provided in the Control Room.

Procedures will be developed to address the use of the Reactor Coolant System vents. The procedures will address information available to the operator and instructions for initiating or terminating vent usage. This will be implemented upon staff approval.

Details of the design and supporting information required by NUREG- 0737 is provided in Section 1.7, 3.10N, 3.11N, 3A, 5.1 and 17A.

II.B.2 PLANT SHIELDING TO PROVIDE ACCESS TO VITAL AREAS AND PROTECT SAFETY EQUIPMENT FOR POSTACCIDENT OPERATION

Action Plan Requirements:

“With the assumption of a postaccident release of radioactivity equivalent to that described in Regulatory Guides 1.3 and 1.4 (i.e., the equivalent of 50% of the core radioiodine, 100% of the core noble gas inventory, and 1% of the core solids are contained in the primary coolant), each licensee shall perform a radiation and shielding-design review of the spaces around systems that may, as a result of an accident, contain highly radioactive materials. The design review should identify the location of vital areas and equipment, such as the control room, radwaste control stations, emergency power supplies, motor control centers, and instrument areas, in which personnel occupancy may be unduly limited or safety equipment may be unduly degraded by the radiation fields during postaccident operations of these systems.”

“Each licensee shall provide for adequate access to vital areas and protection of safety equipment by design changes, increased permanent or temporary shielding, or postaccident procedural controls. The design review shall determine which types of corrective actions are needed for vital areas throughout the facility.”

CPNPP Response

A design review of CPNPP radiation and shielding for post-accident operations has been performed in accordance with the guidelines of NUREG-0737. The plant shielding review is a non-nuclear safety-related study, and is not subject to Quality Assurance. The review considered the potential radiation exposure to operators in vital areas and to Class 1E equipment. The following section describes the assumptions and methodology employed in the review and a summary of the results. Any design changes resulting from the study are covered by the

QA requirements for the items being changed. Any additions or modifications to shielding during operations are subject to an appropriate Operations QA Program. The results of the review of radiation qualification of Class 1E equipment will be included in FSAR Section 3.11.

II.B.2.1 Source Terms

The source terms used in this evaluation of shielding are based on the postulated post-accident release of radioactivity equivalent to that described in Regulatory Guide 1.4 and TID-14844.

The reactor core inventory of radioisotopes is derived from the assumptions listed in Table 12.2-24.

The source terms and corresponding systems employed in the evaluation are described in Section 12.2.1.3. The specific subsections are 12.2.1.3.3 for nondepressurized primary coolant, 12.2.1.3.4 for depressurized containment sump water, 12.2.1.3.5 for Process Sampling System sources, and 12.2.1.3.1 for containment atmosphere. These source terms are based on the core activity release fractions identified in Table 12.2-16.

II.B.2.2 System Review

After establishing the source terms noted above, the systems which may contain highly radioactive materials in a post-accident situation have been evaluated. The specific systems reviewed are described in the following subsections.

II.B.2.2.1 Containment

Direct radiation penetrating through the containment walls and radiation streaming through penetrations was based on the source term discussed in Section 12.2.1.3.1. Radiation streaming causes a significant dose rate outside the containment equipment hatch in a field within ″ 25° from the hatch centerline. The dose rate changes with time and distance. Immediately following a LOCA the dose rate is calculated to be about 44 rem per hour at a point 30 feet from the containment surface and about 4 rem per hour at 240 feet. These dose rates decrease to less than 2 rem per hour and 0.06 rem per hour respectively at 24 hours after the accident.

II.B.2.2.2 Emergency Core Cooling System

Plant shielding has been evaluated considering the Emergency Core Cooling System (ECCS) to be operating in the recirculation mode. The residual heat removal pumps take suction from the containment sump, and the safety injection pumps and centrifugal charging pumps take suction from the residual heat removal pump discharge.

Prior to the start of recirculation, the ECCS will contain water from the refueling water storage tank. Although recirculation operation is not initiated until approximately 10 minutes to several hours after the start of an accident the shielding review has assumed that at the beginning of the accident the ECCS contains the source terms described in Section 12.2.1.3.4.

II.B.2.2.3 Residual Heat Removal System

Plant shielding has been evaluated considering the Residual Heat Removal System (RHRS) contains the source terms specified in Section 12.2.1.3.3 for nondepressurized LOCA conditions, or the source terms specified in Section 12.2.1.3.4 for depressurized LOCA conditions as appropriate.

II.B.2.2.4 Containment Spray System

Plant shielding has been evaluated considering the Containment Spray System (CSS) to be operating in the recirculation mode. Prior to the start of recirculation, the CSS will contain water from the refueling water storage tank. Although CSS recirculation operation is not initiated until approximately 15 minutes to several hours after an accident (if required), the shielding evaluation has assumed that at the beginning of the accident the CSS contains the source terms described in Section 12.2.1.3.4.

II.B.2.2.5 Chemical And Volume Control System

A portion of the chemical and volume control system (CVCS), namely the charging pumps and the connecting piping to the RHRS and safety injection system, may contain post-accident sources during depressurized LOCA conditions. Source terms in this portion of the CVCS are described in Section 12.2.1.3.4. The remainder of the CVCS is not expected to become highly radioactive in a post-accident situation because the letdown system is automatically isolated and is not required for accident mitigation.

II.B.2.2.6 Gas Waste Processing System

The Gas Waste Processing System (GWPS) is not expected to contain highly radioactive material generated following an accident since the system is automatically isolated from components that do contain the radioactive material, and it is not required to mitigate an accident. Therefore, the system has been excluded as a potential source of highly radioactive material following a LOCA.

II.B.2.2.7 Post-Accident Sampling System

The Process Sampling System (PSS) is used for post-accident sampling.

Plant shielding has been evaluated considering PSS components and sample lines containing liquid sources as described in Section 12.2.1.3.3 or 12.2.1.3.4.

II.B.2.3 Shielding Methods

Evaluation of plant shielding for equipment qualification dose evaluations following an accident includes direct radiation from the containment, and radiation from piping and components of systems discussed in Section II.B.2.2, as appropriate. Direct radiation through shield walls and streaming and scattered radiation through penetrations were considered. Direct radiation and streaming radiation dose rates for the post-accident shielding evaluation were calculated by point-kernel shielding codes. Exponential attenuation through the shields with dose build-up factors corresponding to the number of mean free paths in the shield were considered. The scattered radiation dose rate from surrounding surfaces was determined by the albedo method.

Gamma energy levels of the radiation source are typically divided into eight or ten energy groups for the analysis.

II.B.2.4 Design Review

The plant shielding design review has identified the radiation exposure rates in vital areas requiring personnel access following a LOCA. The vital areas of the plant which may need periodic access after a LOCA for performing manual operations to mitigate the consequences of the accident are described in Table II.B.2-4. This table also presents the operator path or route numbers, the time after LOCA when access to the area is required, and the calculated radiation dose that may be received by the operator during the performance of the assigned task.

Plant drawings have also been developed to illustrate the post LOCA access routes to the vital areas. These drawings are presented in Figures II.B.2-25 through II.B.2-86. Radiation dose rates between various points of the operator path, the time spent between consecutive points and the path or route number are shown in Table II.B.2-5 and II.B.2-6.

The Control Room Complex and the Technical Support Center (TSC) are vital areas requiring full-time occupancy during the course of an accident. The integrated dose for these areas is less than 5 rem whole-body, or equivalent, for the duration of the accident in accordance with GDC 19. Further details pertaining to the Control Room and TSC dose analyses are provided in Section 15.6.5.4.4.

Motor control centers (MCCs) are not considered to be vital areas for purpose of this review, since single failure criteria were assumed; and a single failure would not prohibit the ability of the MCCs' functions to be performed from the control room.

Access to radwaste control panels is not considered to be vital, since the safety injection signal during postulated DBA conditions will isolate reactor coolant letdown. Therefore, no highly radioactive post-accident fluids will be present in the radwaste systems.

II.B.2.5 Radiation Qualification of Class 1E Equipment

The post-accident radiation exposure to Class 1E equipment located outside and inside containment has been evaluated for post-accident sources postulated to exist following a LOCA. The source terms for equipment located outside containment are described in Section 12.2.1.3.3, 12.2.1.3.4, or 12.2.1.3.5, as applicable. For Class 1E equipment inside containment, the larger dose resulting from sources described in either Section 12.2.1.3.3 or Sections 12.2.1.3.1 and 12.2.1.3.4 were utilized. Equipment dose evaluations for Class 1E equipment inside and outside containment are based on the guidelines of NUREG 0588 and Regulatory Guide 1.89. These dose evaluations consider both gamma and beta radiation, as appropriate for post-accident conditions, and include contributions from normal operations. The calculated post-accident radiation dose values for Class 1E equipment are included in FSAR Appendix 3A.

II.B.3 POST-ACCIDENT SAMPLING

Action Plan Requirements:

"A design and operational review of the reactor coolant and containment atmosphere sampling line systems shall be performed to determine the capability of personnel to promptly obtain (less

than 1 hour) a sample under accident conditions without incurring a radiation exposure to any individual in excess of 3 and 18-3/4 rem to the whole body or extremities, respectively. Accident conditions should assume a Regulatory Guide 1.3 or 1.4 release of fission products. If the review indicates that personnel could not promptly and safely obtain the samples, additional design features or shielding should be provided to meet criteria.

"A design and operational review of the radiological spectrum analysis facilities shall be performed to determine the capability to promptly quantify (in less than 2 hours) certain radionuclides that are indicators of the degree of core damage. Such radionuclides are noble gases (which indicate cladding failure), iodines and cesiums (which indicate high fuel temperatures), and nonvolatile isotopes (which indicate fuel melting). The initial reactor coolant spectrum should correspond to a Regulatory Guide 1.3 or 1.4 release. The review should also consider the effects of direct radiation from piping and components in the auxiliary building and possible contamination and direct radiation from airborne effluents. If the review indicates that the analyses required cannot be performed in a prompt manner with existing equipment, then design modifications or equipment procurement shall be undertaken to meet the criteria.

"In addition to the radiological analyses, certain chemical analyses are necessary for monitoring reactor conditions. Procedures shall be provided to perform boron and chloride chemical analyses assuming a highly radioactive initial sample (Regulatory Guide 1.3 or 1.4 source term). Both analyses shall be capable of being completed promptly (i.e., the boron sample analysis within an hour and the chloride sample analysis within a shift)."

CPNPP Response

II.B.3.1 General Description

FSAR Section 9.3.2 describes the process sampling system installed in CPNPP. This system was designed for the capability of obtaining samples during normal operating conditions. During postulated accident conditions, radiation levels would be significantly increased.

The Post-Accident Sampling System (PASS) is comprised of two independent subsystems; the Reactor Coolant PASS (RC PASS) and the Containment Air PASS (CA PASS). Figures II.B.3-1 and Flow Diagram M1(2)-301-A illustrate the basic system flow diagrams for the RC PASS and CA PASS respectively.

II.B.3.2 Design Criteria and Functional Requirements

The requirements of NUREG-0737, II.B.3 have been eliminated for CPNPP in accordance with License Amendment 91.

II.B.3.3 PASS Equipment

As stated above, the CPNPP PASS is comprised of two independent subsystems, the Reactor Coolant PASS (RC PASS) and the Containment Air (CA PASS). A discussion of the major components of each of these subsystems is provided below.

II.B.3.3.1 Reactor Coolant PASS

The RC PASS, as shown in Figure II.B.3-1, includes the following major components:

- RC PASS Sample Module
- RC PASS Remote Operating Module
- RC PASS Flush and Diversion Manifold
- RC PASS Auxiliary Module
- RC PASS Flush Module
- RC PASS Sample Cooler

The RC PASS is in layup.

II.B.3.3.2 Containment Air PASS

The Containment Air PASS (CA PASS) will be used to obtain micro-volume samples of the containment atmosphere. The CA PASS, as shown in Flow Diagram

M1(2)-301-A, includes the following major components:

- CA PASS Sample Module
- CA PASS Remote Operating Module
- CA PASS Sample Diversion Valve

The CA PASS is in layup.

II.B.3.3.3 PASS Isolation Valve Control

II.B.3.3.3.1 PASS Isolation Valve Control Panel

The PASS Isolation Valve Control Panel (PIVCP) is a Class 1E, Seismic Category I electrical control panel. It is located in the cable spreading room, a low radiation area.

The PIVCP is provided to permit control for all isolation valves associated with both the RC PASS and the CA PASS. The bypass function of the PIVCP has been disabled.

The PIVCP has control switches for the operation of twelve (12) isolation valves. Three (3) of these valves can also be controlled during normal operation of the Sample Valve Control Panel (SVCP).

Train oriented selector switches are provided at the PIVCP to select controls at the SVCP or PIVCP. The remaining nine (9) valves are controlled exclusively at the PIVCP.

The PIVCP is under direct administrative control of the control room operator. Train oriented control switches configured “CLOSE-LOCAL” are provided in the control room.

Placing these switches in the “LOCAL” position permits operation of the 12 isolation valves from the PIVCP.

II.B.3.3.3.2 Instrument Air to Containment Isolation Valve Control

Instrument air must be available for operation of the PSS Isolation Valves. Instrument air to Containment valve HV-3487 is isolated by a Containment Isolation Phase A signal. This valve must be opened to admit instrument air to tubing run inside containment in order to obtain a sample from RCS.

Opening valve HV-3487 is accomplished by operating the control switch for this valve on Main Control panel CB-01 to the OPEN position. This is a spring-return position in which the switch must be manually held to over-ride the Containment Isolation signal. Releasing the switch will cause the valve to close if the Containment Isolation signal is still present. This operation is in accordance with FSAR Section 7.3.1.1.4.2 Containment Isolation System.

Operating pressure for Instrument air is greater than the maximum pressure in Containment following an accident. The volume of Containment is so great that the capacity and use of the Instrument Air system will not significantly increase Containment pressure. A check valve inside Containment will close if the Containment pressure is greater than the Instrument Air system pressure. Since it meets containment isolation requirements and does not bring reactor coolant or containment atmosphere out of containment post-accident, instrument air is not an isolated auxiliary system (e.g. CVCS letdown) and may be used for this function.

II.B.4 TRAINING FOR MITIGATING CORE DAMAGE

Action Plan Requirements:

“A program is to be developed to ensure that all operating personnel are training in the use of installed plant systems to control or mitigate an accident in which the core is severely damaged. The training program should include the following topics.”

"1. Incore Instrumentation

a. Use of fixed or movable incore detectors to determine extent of core damage and geometry changes.

b. Use of thermocouples in determining peak temperatures; methods for extended range readings; methods for direct readings at terminal junctions.

c. Methods for calling up (printing) incore data from the plant computer.”

"2. Excore Nuclear Instrumentation (NIS)

a. Use of NIS for determination of void formation; void location basis for NIS response as a function of core temperatures and density changes.”

"3. Vital Instrumentation

a. Instrumentation response in an accident environment; failure sequence (time to failure, method of failure); indication reliability (actual vs indicated level).

b. Alternative methods for measuring flows, pressures, levels, and temperatures.

1) Determination of pressurizer level if all level transmitters fail.

2) Determination of letdown flow with a clogged filter (low flow).

3) Determination of other Reactor Coolant System parameters if the primary method of measurement has failed."

"4. Primary Chemistry

a. Expected chemistry results with severe core damage; consequences of transferring small quantities of liquid outside containment; importance of using leak tight systems.

b. Expected isotopic breakdown for core damage; for clad damage.

c. Corrosion effects of extended immersion in primary water; time to failure."

"5. Radiation Monitoring

a. Response of Process and Area Monitors to severe damages; behavior of detectors when saturated; method for detecting radiation readings by direct measurement at detector output (overranged detector); expected accuracy of detectors at different locations; use of detectors to determine extent of core damage.

b. Methods of determining dose rate inside containment from measurements taken outside containment."

"6. Gas Generation

a. Methods of H_2 generation during an accident; other sources of gas (Xe, Ke); techniques for venting or disposal of non-condensibles.

b. H_2 flammability and explosive limit; sources of 0_2 in containment or Reactor Coolant System.

- NRC Letter dated March 28, 1980, ENCLOSURE 3

"Complete the training of all operating personnel in the use of installed systems to monitor and control accidents in which the core may be severely damaged."

- NUREG 0694, Pg. 22

Also see NUREG 0737

CPNPP Response

Refer to revised sections 13.2.1.1 and 13.2.1.2. The referenced program is consistent with 10 CFR 55 rulemaking issued after the TMI-2 event.

TABLE II.B.2-1
THIS TABLE HAS BEEN DELETED

TABLE II.B.2-2
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TABLE II.B.2-3
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TABLE II.B.2-4
VITAL AREA ACCESS

(Sheet 1 of 5)

	Vital Access Area		Description of Task[1]	Time After LOCA (hrs) When Access is Required	UNIT 1 Route No.	UNIT 1 Operator Dose[7] (rem)	UNIT 2 Route No.	UNIT 2 Operator Dose[7] (rem)
1.	Room No. 78 and adjacent areas, Process Sampling System	A	To lineup the valves in the PSS in the Appropriate Positions so that the required Post Accident samples can be obtained.	0.75	1A Fig. II.B.2-25	2.4	1A Fig. II.B-2.43 Fig. II.B-2-44	2.4
		B	Sample retrieval and transport to the hot lab for analysis.	1.0	1B Fig. II.B.2-26	4.8 (WB) 7.4 (EX)	1B Fig. II.B.2-45 Fig. II.B.2-46	4.7 (WB) 7.4 (EX)
	Room No. 30 and Hot lab	C	To ready the hot lab, the count room and process the samples	0.5	1C Fig. II.B.2-27	0.53 (WB) 1.8 (EX)	1C Fig. II.B.2-47	0.13 (WB) 1.4 (EX)
2.	Room No. 107 Wide Range gas monitoring (WRGM) sample skid		To retrieve sample from the plant Post LOCA effluent being released through the stacks, using the WRGM system X-RE-5570 A&B	Various times are needed starting at t=30 min.	2 Fig. II.B.2-30	0.21[2] (WB) 0.4 (EX)	2 Fig. II.B.2-50 Fig. II.B.2-51	0.21[2](WB) 0.4 (EX)
3.	Room No. 245 Mechanical equipment room CCW surge tank area		To manually adjust the supply valves to fill the CCW surge tank, in case of an instrument air malfunction[3]	Various times as needed starting at t=30	3 Fig. II.B.2-31	2.8[2]	3 Fig. II.B.2-52 Fig. II.B.2-53	2.8[2]

TABLE II.B.2-4
VITAL AREA ACCESS

(Sheet 2 of 5)

	Vital Access Area	Description of Task[1]	Time After LOCA (hrs) When Access is Required	UNIT 1 Route No.	UNIT 1 Operator Dose[7] (rem)	UNIT 2 Route No.	UNIT 2 Operator Dose[7] (rem)
4.	Room No. 245 Mechanical equipment room, Safety chilled water valves	To manually throttle valves to fill the safety chilled water surge tank in case of an instrument air malfunction[3]	Various times as needed starting at t=30	4 Fig. II.B.2-31	2.8[2]	4 Fig. II.B.2-52 Fig. II.B.2-53	2.8[2]
5.	Room No. 115A, or 115B Safety chiller equipment area	To adjust the pressure controller valve on the safety chiller condenser refrigerant in case of an instrument air malfunction.[3]	0.5	5 Fig. II.B.2-32	0.05	5 Fig. II.B.2-54 Fig. II.B.2-55	0.07
6.	Room No. 72, 73, and 74, AFW Pump rooms	To manually throttle the auxiliary feedwater flow or cycle the AFW pumps to control S/G level.	0.5	6 Fig. II.B.2-33	2.2[3]	6 Fig. II.B.2-56 Fig. II.B.2-57	2.2[3]
7.	Room No. 174[4] Laundry holdup area, Reactor makeup water pump area	To switch over the reactor makeup water pumps in case of a low pressure alarm	Between 2 min. to 30 min. after LOCA	7 Fig. II.B.2-34	0.04	7 Fig. II.B.2-58 Fig. II.B.2-59	0.43
8.	Room No. 174[4] Laundry holdup area, Reactor makeup water pump area	To isolate the non-nuclear safety-related portion of the reactor makeup water system	Between 2 to 30 min. after LOCA Fig. II.B.2-59	8 Fig. II.B.2-34	0.04	8 Fig. II.B.2-58	0.43

TABLE II.B.2-4
VITAL AREA ACCESS

(Sheet 3 of 5)

	Vital Access Area	Description of Task[1]	Time After LOCA (hrs) When Access is Required	UNIT 1 Route No.	UNIT 1 Operator Dose[7] (rem)	UNIT 2 Route No.	UNIT 2 Operator Dose[7] (rem)
9.	Room No. 247A, Fuel pool cooling pipe tunnel	To manually maintain water level in the spent fuel pool	17.0	9 Fig. II.B.2-35	0.01	9 Fig. II.B.2-60 Fig. II.B.2-61	0.01
10.	Rooms No. 84 and 85[5] Diesel generator control station	To inspect the fuel oil storage tank level, fill the oil storage tanks when needed	1.0	10 Fig. II.B.2-36	0.86	10 Fig. II.B.2-62 Fig. II.B.2-63 Fig. II.B.2-64	0.84
11.	Rooms No. 84 and 85[5] Diesel generator control station	To inspect the DG lube oil makeup duplex filter and strainers	1.0	11 Fig. II.B.2-36	0.86	11 Fig. II.B.2-62 Fig. II.B.2-63 Fig. II.B.2-64	0.84
12.	Equipment Yard Diesel generator fuel oil tank filling area	To fill main diesel generator oil storage tanks from a delivery tanker truck	Three days after LOCA[8]	12 Fig. II.B.2-37	2.2[9]	12 Fig. II.B.2-65	2.2[9]
13.	Task deleted						
14.	Rooms No. 82, 95, 96, 103, 113, 133, 134 and 241	To deenergize lights, in the case of loss of the non ESF ventilation after a LOCA to prevent temperature excursions	0.5	14 Fig. II.B.2-39	0.61	14 Fig. II.B.2-68 through Fig. II.B.2-73	0.64

TABLE II.B.2-4
VITAL AREA ACCESS

(Sheet 4 of 5)

				UNIT 1		UNIT 2	
	Vital Access Area	Description of Task[1]	Time After LOCA (hrs) When Access is Required	Route No.	Operator Dose[7] (rem)	Route No.	Operator Dose[7] (rem)
15.	Task deleted.						
16.	Room No.79 Safeguards Drain Panel (TAG No. CPI-E1PRLV-24)	Diagnose passive failures in the ESF Systems.	24.0	16 Fig. II.B.2-41	0.11	16 Fig. II.B.2-78	0.07
17.	Rooms No.150 and 150A, Mechanical equipment area	To manually open control room intake dampers.	Within 48	17	0.3[6] Fig. II.B.2-42	17 Fig. II.B.2-79 Fig. II.B.2-80	0.30[6]
18.	Control Room and Technical Support Center	General responsibilities of the Control Room and the TSC personnel	All times	N/A	Refer to Section 15.6.5.4.4.	N/A	Refer to Section 15.6.5.4.4
19.	Rooms 213, 249 249A, 249B, 264 and 268	Manually realign valves to maintain spent fuel pool cooling	0.5	19 Fig. II.B.2-42.1 Fig. II.B.2-42.2 Fig. II.B.2-42.3	1.5	19 Fig. II.B.2-81 Fig. II.B.2-82 Fig. II.B.2-83 Fig. II.B.2-84	1.9

TABLE II.B.2-4
VITAL AREA ACCESS

(Sheet 5 of 5)

				UNIT 1		UNIT 2	
			Time After LOCA (hrs)		Operator Dose[7]		Operator Dose[7]
	Vital Access Area	Description of Task[1]	When Access is Required	Route No.	(rem)	Route No.	(rem)
20.	Rooms 108A and 108B	Manually isolate steam supply to the AFWPT	7.0	20 Fig. II.B.2-42.4	4.0[10]	20 Fig. II.B.2-85 Fig. II.B.2-86	4.1[10]

Notes:

(1) It is assumed that each task is performed by one operator (except for Task 20).

(2) This is the maximum dose which occurs due to buildup on ESF filters at 168 hours after LOCA.

(3) These tasks may have to be repeated several times and more than one operator may be needed.

(4) Tasks No. 7 and 8 may be performed by the same operator.

(5) Task Nos. 10 and 11 may be performed by the same operator.

(6) This is the maximum dose which occurs due to buildup on Control Room pressurization filters and intake air inducting upstream of the filters at 8 hours after LOCA.

(7) Values listed represent total accumulated radiation dose to the whole body and whole body dose = extremity dose, unless separate values are noted by WB (Whole Body) or EX (Extremity). Values include all applicable dose components (e.g., task standby/briefing, en route transit, task performance, handling of radioactive materials.)

(8) Although tank is not required to be filled for one week, operators may choose to begin filling the tank after 3 days.

(9) This is the maximum dose which assumes access plate removal and replacement for each tanker truck delivery. In addition, one operator is assumed to perform the task for all deliveries.

(10) Two operators are required to perform this Task. Access is required to each room, 108A and 108B, (one separate operator per room). Dose reflects operator dose to either room.

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 1 of 12)

A = letters correspond to points on operator access route
t = time (seconds) between consecutive points
dr = dose rate (mrem/hr) between consecutive points
task = vital operator action

Operator Task and Route No.	A		B		C		D		E		F		G	
	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1. Post accident sampling														
Route #1A Initial Valve Lineup	13	27	26	27	11	190	38	2,700	7	39,000	15	16,000	4	36,100
Route #1B sample retrieval	13	27	26	27	11	190	38	2,700	7	39,000	15	16,000	30	36,100
Route #1C preparation of hot lab	13	34	26	34	-	1.1	-	34	- Task 120 min.	230				
2. Sampling of plant gaseous release from stack WRGM Route #2	13	0.3	26	0.3	11	26	3	23	14	2.5	13	5.2	37	2.5
3. Filling the CCW surge tank Route #3	13	0.3	33	0.3	14	2.5	13	5.2	37	2.5	29	5.0	64	49
4. Filling the safety chilled Water surge tank Route #4	13	0.3	33	0.3	14	2.5	13	5.2	37	2.5	29	5.0	64	49
5. Safety chilled water chiller valve adjustment Route #5	13	77	33	34	14	4.1	13	305	37	4.1	15	5.0	23	28
6. Adjusting the AFW flow Route #6	13	77	33	34	14	4.1	13	305	37	4.1	15	5.0	23	28
7. Switch over the reactor makeup water pump Route #7	13	77	33	34	14	4.1	13	305	37	4.1	15	5.0	23	28

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 2 of 12)

Operator Task and Route No.		A		B		C		D		E		F		G	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
8.	Isolation of NNS portion of the reactor makeup water system Route #8	13	77	33	34	14	4.1	13	305	37	4.1	15	5.0	23	28
9.	Maintaining the fuel pool water level Route #9	13	18	33	18	14	3.3	13	170	37	3.3	15	5.0	23	17
10.	Inspection of the diesel generator fuel tanks level Route #10	13	27	33	27	14	3.7	13	253	37	3.7	15	5.0	6	5.0
11.	Inspection of the diesel generator lube oil makeup duplex filters and strainers Route #11	13	27	33	27	14	3.7	13	253	37	3.7	15	5.0	6.0	5.0
12.	Filling the main diesel generator fuel storage tank Route #12	54	0.3	60	0.6	90	0.5	30	125	- Task 3 hrs[2]	125				
13.	Task deleted														
14.	Deenergizing the lights Route #14	46	77	14	4.1	67 Task 60 sec.	3,300	14	4.1	13	305	37	4.1	29	5.0
15.	Task deleted.														
16.	Safeguards Drain Panel Access Route #16	13	1.0	33	1.0	11	1.0	24	108	7	2,000	-	2,000 Task 3 min.	-	-

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 3 of 12)

Operator Task and Route No.	A		B		C		D		E		F		G	
	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
17. Manually open Control Room HVAC Intake Damper Route #17	30	1.0	3	24.8	20	1.66	10	24.8	15 Task 30 min.	565	-	-	-	-
19. Manually realign valves to maintain spent fuel pool cooling	12.5	77	33	34	14.3	4.1	13.3	305	36.7	4.1	16.7	5	47	5
20. Manually isolate steam supply to the AFWPT	12.5	18	33	18	14.3	3.3	13.3	170	36.7	3.3	29.3	5	64.3	24.7

Operator Task and Route No.	H		I		J		K		L		M	
	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1. Route 1A (contd)	6	36,100	- Task 3.7 min.	18,000	6	18,000	10	27,000	20	22,300	20	22,300
Route 1B (contd)	- Task 6 min.	16,200	- Task 2 min.	36,100								
Route 1C (contd)												
2. Sampling of plant (contd)	29	5.0	64	49	15	7.0	10	4.4	15 Task 18 min.	600		
3. Filling the CCW (contd)	15	7.0	12	4.4	6	300	Task 65 min.	2,600				
4. Filling the safety (contd)	15	7.0	12	4.4	6	300	Task 65 min.	2,600				

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 4 of 12)

Operator Task and Route No.		H		I		J		K		L		M	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
5.	Safety chilled water (contd)	17	66	9	125	9	740	26 Task 60 min.	5.0				
6.	Adjusting the AFW (contd)	17	66	9	125	9	740	5	2,600	6	20,000	20	81,000
7.	Switch over (contd)	30 Task 20 min.	4.1										
8.	Isolation of NNS (contd)	30 Task 10 min.	4.1										
9.	Maintaining the fuel (contd)	30	3.3	28	380	7 Task 97 min.	2.5						
10.	Inspection of diesel (contd)	38	3,600	10	280	19	280	22	548	7	890	15	38,000
11.	Inspection of diesel (contd)	38	3,600	10	280	19	280	22	548	7	890	15	38,000
12.	Filling the main diesel (contd)												
13.	Task deleted												
14.	Deenergizing the lights (contd)	97 Task 90 sec.	24.7	15	5.0	64	5.0	31	4,400	13 Task 30 sec.	6,500	8	4,400
15.	Task deleted.												
16.	Safeguards Drain Panel Access (contd)												

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 5 of 12)

Operator Task and Route No.	H t	H dr	I t	I dr	J t	J dr	K t	K dr	L t	L dr	M t	M dr
17. Manually open control room HVAC Intake Damper (contd)												
19. Manually realign valves to maintain spent fuel pool cooling	39.6	3510	126 Task 10 min.	5580	39.6	3510	47	5	31.3	5	23.3	28
20. Manually isolate steam supply to the AFWPT	14.7	89.5	12	42	9.3	250	13	96000	- Task 124 sec.			

Operator Task and Route No.	N t	N dr	O t	O dr	P t	P dr	Q t	Q dr	R t	R dr	S t	S dr	T t	T dr
1. Post accident (contd)														
2. Sampling of plant (contd)														
3. Filling the CCW (contd)														
4. Filling the safety (contd)														
5. Safety chilled water (contd)														
6. Adjusting the AFW (contd)	8	29,000	6	1,240	4	13,300	9	13,000	3 Task 15 min.	1,900				
7. Switch over (contd)														
8. Isolation of NNS (contd)														
9. Maintaining the fuel (contd)														

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 6 of 12)

Operator Task and Route No.		N		O		P		Q		R		S		T	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
10.	Inspection of diesel (contd)	12	38,000	11	6,200	16 Task 5 min.	5.0	19	6,200	18	9,800	16 Task 5 min.	5.0		
11.	Inspection of diesel (cont)	12	38,000	11	6,200	16 Task 10 min.	5.0	19	6,200	18	9,800	16 Task 10 min.	5.0		
12.	Filling the main diesel (contd)														
13.	Task deleted														
14.	Deenergizing the lights (contd)	22 Task 30 sec.	350	24.3	686	7.3	1,100	15	8,500	10 Task 30 sec.	17,700	12	10,800	27 Task 15 sec.	410
15.	Task deleted.														
16.	Safeguards Drain Panel Access (contd)														
17.	Manually open Control Room HVAC Intake Damper (contd)														
19.	Manually realign valves to maintain spent fuel pool cooling	30.3	4.1	27.8	380	25	2.5	13.3	25	77.3 Task 32 min.	25	96.6 Task 28 min.	398	42 Task 5 min.	25
20.	Manually isolate steam supply to the AFWPT														

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 7 of 12)

Operator Task and Route No.	U t	U dr	V t	V dr	W t	W dr	X t	X dr	Y t	Y dr	Z t	Z dr
1. Post accident (contd)												
2. Sampling of plant (contd)												
3. Filling the CCW (contd)												
4. Filling the safety (contd)												
5. Safety chilled water (contd)												
6. Adjusting the AFW (contd)												
7. Switch over (contd)												
8. Isolation of NNS (contd)												
9. Maintaining the fuel (contd)												
10. Inspection of diesel (contd)												
11. Inspection of diesel (contd)												
12. Filling the main diesel (contd)												
13. Task deleted												
14. Deenergizing the lights (contd)	19 Task 30 sec.	1,120	22	410	12	10,800	10	17,700	15	8,500	6	1,100
15. Task deleted.												
16. Safeguards Drain Panel Access (contd)												

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 8 of 12)

Operator Task and Route No.	U t	U dr	V t	V dr	W t	W dr	X t	X dr	Y t	Y dr	Z t	Z dr
17. Manually Open Control Room HVAC Intake Damper (contd)												
19. Manually realign valves to maintain spent fuel pool cooling	13.3	25	25	2.5	27.8	380	30.3	4.1	23.3	28	14.7	5
20. Manually isolate steam supply to the AFWPT												

Operator Task and Route No.	AA t	AA dr	BB t	BB dr	CC t	CC dr	DD t	DD dr	EE t	EE dr	FF t	FF dr	GG t	GG dr
1. Post accident (contd)														
2. Sampling of plant (contd)														
3. Filling the CCW (contd)														
4. Filling the safety (contd)														
5. Safety chilled water (contd)														
6. Adjusting the AFW (contd)														
7. Switch over (contd)														
8. Isolation of NNS (contd)														
9. Maintaining the fuel (contd)														
10. Inspection of diesel (contd)														

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 9 of 12)

Operator Task and Route No.	AA t	AA dr	BB t	BB dr	CC t	CC dr	DD t	DD dr	EE t	EE dr	FF t	FF dr	GG t	GG dr
11. Inspection of diesel (contd)														
12. Filling the main diesel (contd)														
13. Task deleted														
14. Deenergizing the lights (contd)	24.3	686	24.3	350	38	4,400	64	5.0	15	5.0	37	4.1	13	305
15. Task deleted.														
16. Safeguards Drain Panel Access (contd)														
17. Manually Open Control Room HVAC Intake Damper (contd)														
19. Manually realign valves to maintain spent fuel pool cooling	36.7	4.1	13.3	305	14.3	4.1	33	34	12.5	77				
20. Manually isolate steam supply to the AFWPT														

Operator Task and Route No.	HH t	HH dr	II t	II dr	JJ t	JJ dr	KK t	KK dr	LL t	LL dr	MM t	MM dr
1. Post accident (contd)												
2. Sampling of plant (contd)												
3. Filling the CCW (contd)												

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 10 of 12)

Operator Task and Route No.		HH		II		JJ		KK		LL		MM	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
4.	Filling the safety (contd)												
5.	Safety chilled water (contd)												
6.	Adjusting the AFW (contd)												
7.	Switch over (contd)												
8.	Isolation of NNS (contd)												
9.	Maintaining the fuel (contd)												
10.	Inspection of diesel (contd)												
11.	Inspection of diesel (contd)												
12.	Filling the main diesel (contd)												
13.	Task deleted												
14.	Deenergizing the lights (contd)	14	4.1	26	34	56 Task 30 sec.	5.0	6	1.7	45 Task 30 sec.	5.0	6	1.7
15.	Task deleted.												
16.	Safeguards Dring Panel Access (contd)												
17.	Manually Open Control Room HVAC Intake Damper (contd)												

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 11 of 12)

Operator Task and Route No.	HH t	HH dr	II t	II dr	JJ t	JJ dr	KK t	KK dr	LL t	LL dr	MM t	MM dr
19. Manually realign valves to maintain spent fuel pool cooling (contd)												
20. Manually isolate steam supply to the AFWPT												

Operator Task and Route No.	NN t	NN dr	OO t	OO dr	PP t	PP dr	QQ t	QQ dr	RR t	RR dr	SS t	SS dr	TT t	TT d
1. Post accident (contd)														
2. Sampling of plant (contd)														
3. Filling the CCW (contd)														
4. Filling the safety (contd)														
5. Safety chilled water (contd)														
6. Adjusting the AFW (contd)														
7. Switch over (contd)														
8. Isolation of NNS (contd)														
9. Maintaining the fuel (contd)														
10. Inspection of diesel (contd)														
11. Inspection of diesel (cont)														
12. Filling the main diesel (contd)														

TABLE II.B.2-5

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 1 OPERATOR ROUTE[1]

(Sheet 12 of 12)

Operator Task and Route No.	NN		OO		PP		QQ		RR		SS		TT	
	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	d
13. Task deleted														
14. Deenergizing the lights (contd)	33	5.0	20	77										
15. Task deleted.														
16. Safeguards Drain Panel Access (contd)														
17. Manually Open Control Room HVAC Intake Damper (contd)														
19. Manually realign valves to maintain spent fuel pool cooling (contd)														
20. Manually isolate steam supply to the AFWPT														

Notes:

1) Dose rates and times are shown for operator transit and task performance only. The total accumulated radiation doses, which also include task standby/briefing and/or handling of radioactive materials, are listed in Table II.B.2-4.

2) This task is assumed to initiate every 12 hours, three days following a LOCA for each of the nine tanker truck deliveries. The dose rates listed are the maximum for any delivery.

TABLE II.B.2-6
DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 1 of 14)

A = Letters correspond to points on operator access route.
t = Time (seconds) between consecutive points.
dr = Dose rate (mrem/hr) between consecutive points.
task =Vital operator action

		A		B		C		D		E		F		G	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post Accident Sampling														
1A.	Inital valve lineup	12.5	77	26.4	34	10.4	39	5	5	36	4.1	13.1	305	14.1	4.1
1B.	Sample Retrieval	12.5	27	26.4	27	10.4	36	10	5	72	3.7	26.2	253	28.2	3.7
1C.	Hot lab preparation	12.5	77	26.4	34	5	1.1	20	34	5	39	- Task 120 min	39		
2.	Sampling of plant gaseous release from stack WRGM	12.5	0.3	26.4	0.3	10.4	25	29.3	5	64.3	49	14.7	6.7	9.6	4.4
3.	Filling the CCW surge tank	12.5	0.3	33	0.3	5	2.5	29.3	5	64.3	49	14.7	7	12	4.4
4.	Filling the safety chilled water surge tank	12.5	0.3	33	0.3	5	2.5	29.3	5	64.3	49	14.7	7	12	4.4
5.	Safety chilled water system valve adjustment	12.5	77	33	34	5	5	14.7	5	23.3	28	17.3	66	9.3	125
6.	Adjusting the AFW flow rate	12.5	77	33	34	5	5	14.7	5	5	2600	9.3	740	9.3	125
7.	Switchover the reactor makeup water pumps in case of low pressure alarm	12.5	77	33	34	5	5	14.7	5	23.3	28	14	66	30.3	1040

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 2 of 14)

		A		B		C		D		E		F		G	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
8.	Isolation of NNS portion of the reactor makeup water system	12.5	77	33	34	5	5	14.7	5	23.3	28	14	66	30.3	1040
9.	Maintaining the fuel pool water level	12.5	6	33	6	5	5	14.7	5	23.3	8	17.3	17	30.3	195
10.	Inspection of the diesel generator fuel tank level	12.5	27	33	27	5	5	14.7	5	64.3	5	38.3	3600	9.8	280
11.	Inspection of the diesel generator lube oil makeup duplex filters & strainers	12.5	27	33	27	5	5	14.7	5	64.3	5	38.3	3600	9.8	280
12.	Filling the main diesel generator fuel storage tank	97	0.3	60	0.6	90 Task 3 hrs.[2]	0.5	37.5	125	-	125				
13.	Task Deleted														
14.	De-energizing lights	12.5	77	33	34	14.3 Task 1 min.	4.1	13.3	305	36.7	4.1	66.6	3300	36.7	4.1
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access	12.5	1	33	1	14.3	2.5	13.3	17	36.7	2.5	16	110	9	2000

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 3 of 14)

		A		B		C		D		E		F		G	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
17.	Manually open Control Room HVAC intake damper	12.5	18	9	18	39.3	5	31.6	1.7	7	24.7	20	1.7	12	24.7
19.	Manually realign valves to maintain spent fuel pool cooling	12.5	77	33	34	5	5	16.7	5	21.7	5	39.6	3510	126 Task 10 min	5580
20.	Manually isolate steam supply to the AFWPT	12.5	18	33	18	5	33	29.3	5	64.3	24.7	14.7	89.5	12	42

		H		I		J		K		L		M		N	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post-accident sampling														
1A.	Initial valve lineup	38	3300	7	47000	15	19000	4	43700	6	43700	- Task 3.7 min.	21700	6	21700
1B.	Sample retrieval	76	2700	14	39000	15	16000	30	36100	60	16200	60 Task 2 min.	16200	- Task 4 min.	36100
1C.	Hot lab preparation														
2.	Sampling of plant gaseous release from stack WRGM	14.7	600	- Task 18 min.	600										
3.	Filling the CCW surge tank	6.4	300	- Task 65 min.	2600										

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 4 of 14)

		H		I		J		K		L		M		N	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
4.	Filling the safety chilled water surge tank	6.4	300	- Task 65 min.	2600										
5.	Safety chilled water system valve adjustment	9.3	740	26.3	5	- Task 60 min.	5								
6.	Adjustng the AFW flow rate	17.3	66	23.3	28	5.8	20000	20	81000	8	20000	6.3	1240	4.3	13300
7.	Switchover the reactor makeup water pumps in case of low pressure alarm	- Task 20 min.	1040												
8.	Isolation of NNS portion of the reactor makeup water system	- Task 10 min.	1040												
9.	Maintaining the fuel pool water level	27.8	380	6.7	2.5	- Task 97 min.	2.5								
10.	Inspection of the diesel generator fuel tank level	19	280	22	548	7	890	14.7	38000	12	38000	11.2	6200	16.1	5
11.	Inspection of the diesel generator lube oil makeup duplex filters & strainers	19	280	22	548	7	890	14.7	38000	12	38000	11.2	6200	16.1	5
12.	Filling the main diesel generator fuel storage tank														
13.	Task Deleted														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 5 of 14)

		H		I		J		K		L		M		N	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
14.	De-energizing lights	13.3	305	14.3	4.1	29.3	5	96.6 Task 1.5 min.	24.7	14.7	5	64.3	5	30.7	4400
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access	- Task 3 min.	1200												
17.	Manually open Control Room HVAC intake damper	12	561	- Task 30 min.	561										
19.	Manually realign valves to maintain spent fuel pool cooling	39.6	3510	21.7	5	31.3	5	23.3	28	14	66	30.3	1040	27.8	380
20.	Manually isolate steam supply to the AFWPT	9.3	250	13	97000	- Task 124 sec									

		O		P		Q		R		S		T		U	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post-accident Sampling														
1A.	Initial valve lineup	10	21700	20	27000	20	27000								

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 6 of 14)

		O		P		Q		R		S		T		U	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1B.	Sample retrieval														
1C.	Hot lab preparation														
2.	Sampling of plant gaseous release from stack WRGM														
3.	Filling the CCW surge tank														
4.	Filling the safety chilled water surge tank			2.5 2900	- 1900	6 1900									
5.	Safety chilled water system valve adjustment														
6.	Adjusting the AFW flow rate	6	13000	2.5	11500	2.5	1900	- Task 15 min.	1900	7.7	13300	6 Task 15 min.	1900		
7.	Switchover the reactor makeup water pumps in case of low pressure alarm														
8.	Isolation of NNS portion of the reactor makeup water system														
9.	Maintaing the fuel pool water level														
10.	Inspection of the diesel generator fuel tank level	- Task 15 min.	5	2.8	6200	18.3	9800	16.1	5	- Task 15 min.	5				

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 7 of 14)

		O		P		Q		R		S		T		U	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
11.	Inspection of the diesel generator lube oil makeup duplex filters and strainers	- Task 10 min.	5	2.8	6200	18.3	9800	16.1	5	- Task 10 min.	5				
12.	Filling the main diesel generator fuel storage tank														
13.	Task Deleted														
14.	De-energizing lights	12.6 Task 0.5 min.	6500	7.7	4400 Task 0.5 min.	22	350	24.3	686	7.5	1100	14.7	8500	10	17700
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access														
17.	Manually open Control Room HVAC intake damper														
19.	Manually realign valves to maintain spent fuel pool cooling	25	2.5	13.3	25 Task 32 min	77.3	25	96.6 Task 28 min	1150	42 Task 5 min	25	13.3	25	25	2.5
20.	Manually isolate steam supply to the AFWPT														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 8 of 14)

Route	Operator Task	V		W		X		Y		Z		AA		BB	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post-accident Sampling														
1A.	Initial valve lineup														
1B.	Sample Retrieval														
1C.	Hot lab preparation														
2.	Sampling of plant gaseous release from stack WRGM														
3.	Filling the CCW surge tank														
4.	Filling the safety chilled water surge tank														
5.	Safety chilled water system valve adjustment														
6.	Adjusting the AFW flow rate														
7.	Switchover the reactor makeup water pumps in case of low pressure alarm														
8.	Isolation of NNS portion of the reactor makeup water system														
9.	Maintaining the fuel pool water level														
10.	Inspection of the diesel generator fuel tank level														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 9 of 14)

		V		W		X		Y		Z		AA		BB	
Route	Operator Task	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
11.	Inspection of the diesel generator lube oi makeup duplex filters and strainers														
12.	Filling the main diesel generator fuel storage tank														
13.	Task Deleted														
14.	De-energizing lights	12	10800	27.3 Task 0.25 min.	410	18.6 Task 0.5 min.	1120	21.7	410	12	10800	10	17700	14.7	8500
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access														
17.	Manually open Control Room HVAC intake damper														
19.	Manually realsign valves to maintain spent fuel pool cooling	27.8	380	30.3	1040	14	66	23.3	28	14.7	5	5	5	33	34
20.	Manually isolate steam supply to the AFWPT														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 10 of 14)

Route	Operator Task	CC		DD		EE		FF		GG		HH		II	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post-accident Sampling														
1A.	Initial valve lineup														
1B.	Sample Retrieval														
1C.	Hot lab preparation														
2.	Sampling of plant gaseous release from stack WRGM														
3.	Filling the CCW surge tank														
4.	Filling the safety chilled water surge tank														
5.	Safety chilled water system valve adjustment														
6.	Adjusting the AFW flow rate														
7.	Switchover the reactor makeup water pumps in case of low pressure alarm														
8.	Isolation of NNS portion of the reactor makeup water system														
9.	Maintaining the fuel pool water level														
10.	Inspection of the diesel generator fuel tank level														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 11 of 14)

Route	Operator Task	CC		DD		EE		FF		GG		HH		II	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
11.	Inspection of the diesel generator lube oi makeup duplex filters and strainers														
12.	Filling the main diesel generator fuel storage tank														
13.	Task Deleted														
14.	De-energizing lights	8	1100	22	686	28.2	350	38.3	4400	64.3	5	14.7	5	26.4	34
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access														
17.	Manually open Control Room HVAC intake damper														
19.	Manually realign valves to maintain spent fuel pool cooling	12.5	77												
20.	Manually isolate steam supply to the AFWPT														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 12 of 14)

Route	Operator Task	JJ		KK		LL		MM		NN		OO		PP	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
1.	Post-accident Sampling														
1A.	Initial valve lineup														
1B.	Sample Retrieval														
1C.	Hot lab preparation														
2.	Sampling of plant gaseous release from stack WRGM														
3.	Filling the CCW surge tank														
4.	Filling the safety chilled water surge tank														
5.	Safety chilled water system valve adjustment														
6.	Adjusting the AFW flow rate														
7.	Switchover the reactor makeup water pumps in case of low pressure alarm														
8.	Isolation of NNS portion of the reactor makeup water system														
9.	Maintaining the fuel pool water level														
10.	Inspection of the diesel generator fuel tank level														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 13 of 14)

Route	Operator Task	JJ		KK		LL		MM		NN		OO		PP	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
11.	Inspection of the diesel generator lube oi makeup duplex filters and strainers														
12.	Filling the main diesel generator fuel storage tank														
13.	Task Deleted														
14.	De-energizing lights	56.4 Task 0.5 min.	5	9.7	1.7	45.2 Task 1 min.	5	9.7	1.7	33.2	5	14.9	5		
15A.	Task deleted.														
15B.	Task deleted.														
16.	Safeguards drain panel access														
17.	Manually open Control Room HVAC intake damper														
19.	Manually realign valves to maintain spent fuel pool cooling														

TABLE II.B.2-6

DOSE RATES AND TIME SPENT BETWEEN CONSECUTIVE POINTS OF UNIT 2 OPERATOR ROUTE[1]

(Sheet 14 of 14)

Route	Operator Task	JJ		KK		LL		MM		NN		OO		PP	
		t	dr	t	dr	t	dr	t	dr	t	dr	t	dr	t	dr
20.	Manually isolate steam supply to the AFWPT														

Notes:

1) Dose rates and times are shown for operator transit and task performance only. The total accumulated radiation doses, which also include task standby/briefing and/or handling of radioactive materials, are listed in Table II.B.2-4.

2) This task is assumed to initiate every 12 hours, three days following a LOCA for each of the nine tanker truck deliveries. The dose rates listed are the maximum for any delivery.

TABLE II.B.3-1
DELETED

II.D REACTOR COOLANT SYSTEM RELIEF AND SAFETY VALVES

OBJECTIVE:

"Demonstrate by testing and analysis that the relief and safety valves, block valves and associated piping in the reactor coolant system are qualified for the full range of operating and accident conditions. Anticipated transients without scram (ATWS) may be considered in later phases of the test program. In addition, design changes or modifications will be made that are necessary to provide positive indication of valve position."

- NUREG 0660, Pg II.D-1

II.D.1 TESTING REQUIREMENTS

Action Plan Requirements:

"Pressurized-water reactor and boiling-water reactor licensees and applicants shall conduct testing to qualify the reactor coolant system relief and safety valves under expected operating conditions for design-basis transients and accidents."

CPNPP Response

CPNPP participated fully in the EPRI Safety and Relief Valve Test Program to demonstrate acceptable valve operability under expected flow conditions. Table II.D.1-1 key plant parameters pertaining to the pressurizer safety, relief, and PORV block valves. Also see Section 5.4.13 for additional details about the safety and relief valves. Valves identical to those installed at CPNPP were tested in the EPRI test program as documented in Reference 1. Block valves tested are documented in Reference 2. Inlet fluid conditions for the tests adequately envelope those expected at CPNPP as documented in Reference 3 and 8. Justification that the actual valve test conditions adequately represent anticipated plant conditions is provided in Reference 4, Sections 3.7 and 4.7 for relief and safety valves respectively. Test results for the CPNPP-representative safety and block valves are summarized in Reference 5, Sections 3.5 and 4.6 respectively.

A. The CPNPP safety valve inlet piping configuration is shorter than that employed in the tests. However, the test conditions report (Reference 4, Section 4.7), states that only the long inlet piping configuration was tested because, "…the operability of such valves with a shorter inlet is expected to be as good or better than that observed with the test configuration." Safety valve ring settings for CPNPP are comparable to the "reference" ring settings of the following tests: 929, 931a, 932, 1406, 1411, 1415, and 1419. Further discussion of the inlet piping and valve ring settings comparison with the EPRI Tests is provided in Reference 8.

   Expected valve discharge back pressure for CPNPP was modeled by Continuum Dynamics for EPRI and the results are presented in Reference 6, page 3-8. The calculated back pressure of 291 psi is well below that of the referenced 900-series of EPRI tests (650 psi) and only moderately above the 245 psi of the 1400-series of tests.

   The relief valve tests results (NP-2628-LD, Section 4.6), reflect acceptable operation in both the Marshall Steam Station tests and the Wyle Phase III tests.

Based upon the above indicated references to the appropriate reports produced in support of EPRI Project V102, the pressurizer safety and relief valves, as installed at CPNPP, were adequately represented by the previously referenced tests. Westinghouse, in Reference 7, concluded that those same tests adequately demonstrated acceptable performance of the tested safety valves for pressurizer relief at CPNPP.

Reanalysis of the CPNPP Unit 1 pressurizer safety and relief valve piping determined it to be adequate provided a loop seal temperature profile similar to that produced in EPRI Test 917 is maintained. Consequently, the safety valve loop seals will be insulated to maintain the referenced or hotter temperature profile. The calculated piping support loads have been incorporated into a reanalysis of the existing piping supports. Based on the EPRI test temperature profile and Westinghouse recommendation, since the presence of insulation on the upstream piping, which slopes to the valve, would prevent heat transfer from the piping and thus promote flashing of the liquid into steam, no insulation is installed upstream of the PORVs. A water seal is maintained at the PORVs via the sloped piping.

B. Block Valves - A small number of block valves, including the type and model used for CPNPP, were tested at the Marshall Steam Station Test Facility. After modifications to the valve operator, the valve successfully closed under high flow and high differential pressure conditions. Sufficient data was gathered to determine the modifications required to assure correct valve operation under all fluid conditions.

Subsequently, in response to IE Bulletin 81-02, CPNPP has modified the PORV block valves by changing the operator gear ratio to guarantee adequate thrust capability and by rewiring the operator for limit closing control.

Additional block valve tests are not necessary because:

1. The operability of block valves is not a safety issue. Plant procedures provide methods for safely shutting down in the event of a small break LOCA.

2. Results of block valve tests performed at the Marshall Station have provided sufficient information to address valve operability.

3. The CPNPP PORV was demonstrated to perform well over a wide range of operating and accident conditions during the EPRI relief valve testing.

4. The probability of a relief valve failure to perform its intended function is low and on the same magnitude as other [[b]] break LOCA initiators. Operability of the associated block valve does not increase the probability of a small break LOCA.

C. ATWS Testing - Although not specifically addressed by the valve testing program, the results provide most of the information necessary to address ATWS events (i.e., relief capability at high pressure).

D. For additional information on CPNPP response to this item, see References 8 and 9.

References

1) EPRI NP-2292-LD; Valve Selection/Justification Report, March 1982

2) EPRI NP-2514-LD; EPRI-Marshall Electric Motor- Operated Valve (Block Valve) Interim Test Data Report, July 1982

3) EPRI NP-2296-LD; Valve Inlet Fluid Conditions for Pressurizer Safety and Relief Valves in Westinghouse-Designed Plants, March 1982

4) EPRI NP-2460-LD; EPRI PWR Safety and Relief Valve Program Test Condition Justification Report, June 1982

5) EPRI NP-2628-LD; Safety and Relief Valve Test Report, September 1982

6) Continuum Dynamics, Quasi-Steady Backpressure For Pressurized Water Reactor Safety and Relief Valves, Vol. III

7) Westinghouse Nuclear Energy Systems, WCAP-10105, Review of Pressurizer Safety Valve Performance as Observed in the EPRI Safety and Relief Valve Test Program, June 1982

8) TXX-4849 dated June 13, 1986, TU Electric response to NRC Request for Additional Information (RAI) dated July 5, 1985.

9) TXX-6398 dated April 15, 1987, TU Electric response to NRC Request for Additional Information (RAI) dated March 27, 1987.

II.D.3 RELIEF AND SAFETY VALVE POSITION INDICATION

Action Plan Requirements:

"Reactor coolant system relief and safety valves shall be provided with a positive indication in the control room derived from a reliable valve-position detection device or a reliable indication of flow in the discharge pipe."

- NUREG 0737

CPNPP Response

The pressurizer PORVs have both open and close limit switches with control indicating lights mounted at their respective control switches on the main control board. The limit switches are snap acting, positive throw switches mounted on the valve yokes. They are operated by the actual valve stem motion.

The pressurizer safety valves will be equipped with Reed type valve position indicating switches with indicating lights on the main control boards. Opening of either pressurizer PORVs or safety valves will alarm in the Control Room.

Procedures will be developed/revised to address the RCS relief valve and safety valve indications. The use of this equipment will be integrated into the operator training. This will be implemented by fuel load.

These valve position indicators shall be seismically and environmentally qualified in accordance with IEEE Stds 323-1974 and 344-1975. The indicators shall be qualified and installed prior to fuel load.

TABLE II.D.1-1
SAFETY VALVE INFORMATION

(Sheet 1 of 2)

Safety Valve Information	
-Valve Parameters	
Number of Valves	3
Manufacturer	Crosby Valve & Gage Company
Type	HB-BP-86
Size	6M6
Rated Capacity (steam)	420,000 lbm/hr @2500 psia set pressure
Ring Settings	See Ref. 8
-Inlet Piping Parameters	
Diameter	6 in.
Length (water)	61 in.
Volume (water)	1282 in.3 (5.5 gallon)
Type	Loop Seal-Pressurizer 650°F[(a)] -Valve inlet 300°F
-Actuation Transient Parameters	
Fluid Range	Pressurizer-Saturated Steam Valve - Subcooled Water
Maximum Back Pressure	500 psi
Relief Valve Information	
-Valve Parameters	
Number of Valves	2
Manufacturer	Copes-Vulcan
Type	Globe D-100-160
Size	3” NPS, 316 W/Stellite Plug and 17-4PH Cage
Capacity (steam)	210,000 lbm/hr @2350 psia
Operator	Copes-Vulcan, Diaphragm Actuating (160 in^2), Reverse Acting, 100 psig air

TABLE II.D.1-1
SAFETY VALVE INFORMATION

(Sheet 2 of 2)

-Inlet Piping Parameters	
Type	Water Seal
-Actuation Transient Parameters	
Fluid range	Pressurizer-Steam
	Relief Valve-Subcooled Water
Block Valve Information	
Manufacturer	Westinghouse
Type	Gate
Model	3GM88
Size	3 in.
Operator	Limitorque SB-00-15

a) Following installation of loop seal insulation

II.E SYSTEM DESIGN

II.E.1 AUXILIARY FEEDWATER SYSTEM

OBJECTIVE:

"Improve the reliability of the auxiliary feedwater system (AFWS)."

- NUREG 0660, Pg. II.E.1-1

II.E.1.1 Auxiliary Feedwater System Evaluation

Action Plan Requirements:

"The Office of Nuclear Reactor Regulation is requiring reevaluation of the auxiliary feedwater (AFWS) systems for all PWR operating plant licensees and operating license applications. This action includes:

1. Perform a simplified AFW system reliability analysis that uses event-tree and fault-tree logic techniques to determine the potential for AFW system failure under various loss-of-main-feedwater-transient conditions. Particular emphasis is given to determining potential failures that could result from human errors, common causes, single-point vulnerabilities, and test and maintenance outages:

2. Perform a deterministic review of the AFW system using the acceptance criteria of Standard Review Plan

3. Reevaluate the AFW system flowrate design bases and criteria."

- NUREG 0737

4. Based on the analyses performed modify the AFWS, as necessary.

- NUREG 0694, Pg. 22

Also see NRC Letter From D. F. Ross, Jr., dated March 10, 1980.

CPNPP Response

1. The Applicant has performed a simplified Auxiliary Feedwater system reliability analysis similar in method to that described in Appendix III to NUREG-0611. The evaluation is a non-nuclear safety-related study, and is not subject to Quality Assurance. The results of this analysis presented below were prepared in response to the NRC Letter dated March 10, 1980.

   A. System Description

      A complete description of the Auxiliary Feedwater System is in Section 10.4.9. The Auxiliary Feedwater System is designed to provide a supply of high-pressure feedwater to the secondary side of the steam generators for reactor coolant heat

removal following a loss of normal feedwater. The system is comprised of two electric motor-driven pumps and a third turbine-driven pump. All three pumps draw suction from the Nuclear Safety Class 3 Condensate Storage Tanks. A single line supplies water through a common locked-open valve to the suction of the motor-driven pumps, and a second line supplies water to the suction of the turbine-driven pump. The motor-driven pumps normally feed two steam generators each. The turbine-driven pump discharge line branches into four separate lines each feeding one steam generator. The actuation of the Auxiliary Feedwater System following an accident is automatic. A simplified system flow diagram is shown in Figure II.E.1.1-1.

B. Testing

The Auxiliary Feedwater System testing is controlled by the Surveillance Requirements in the CPNPP Technical Specifications.

C. Technical Specification

See Testing above.

D. Reliability Evaluation Results

The results of the reliability evaluations are presented below for the following three events: loss of main feedwater (MFW) with offsite power available, loss of MWF and loss of offsite AC, and loss of feedwater and loss of all AC.

i. Loss of MFW with Offsite Power Available

The simplified fault tree for this transient is shown in Figure II.E.1.1-2 (Sheet 1). The reliability analysis of the Auxiliary Feedwater System based on this event did not identify any single failures that would result in insufficient auxiliary feedwater flow. One double failure was identified. This failure is the result of the closure or blocking of each of the valves in the two supply lines from the condensate storage tank. This double failure is the dominant failure mode.

ii. Loss of MFW and Loss of Offsite AC

The simplified fault tree for this transient is shown in Figure II.E.1.1-2 (Sheet 2). This accident differs from the above only In that the power for the motor driven pumps will come from the diesel generators. To determine what effect this has on overall Auxiliary Feedwater System reliability, the quantified fault tree was expressed as a function of diesel generator reliability. As expected, if the diesels have high reliability, the system reliability is the same as the “loss of feed” event and if the diesels have no reliability, the system reliability is the same as the “loss of feed and the loss of all AC” in the event offsite power is lost. For a diesel generator failure probability as high as .03 the dominant failure mode remains the same as the “loss of feed” event.

iii. Loss of Feedwater and Loss of All AC

The simplified fault tree for this transient is shown in Figure II.E.1.1-2 (Sheet 3). If all AC power is lost, there is only one steam-driven Auxiliary Feedwater pump for this train. The reliability analysis indicates that the dominant failure mode is the steam driven train being out for test or maintenance.

E. Principal Dependencies Identified

The principal dependency found in this analysis was the failure of all trains due to closure of the manual valves in the two supply lines from the condensate storage tank. These valves are locked open and will have special administrative controls-.

There were no design deficiencies noted by this simplified reliability analysis; therefore, no design changes were required.

F. Recommendations

The NRC March 10, 1980 letter instructed applicants to factor the recommendations of Appendix III to NUREG—0611 into plant design. Those recommendations applicable to Comanche Peak are discussed here.

i. GS-1 Tech. Spec. LCO Train Outage Time Limit

CPNPP will have the time limit recommended in the Westinghouse Standard Technical Specifications.

ii. GS—4 Emergency Procedures Backup Water Supply

CPNPP will have an emergency procedure for operators to switch to the backup water supply. See Section 10.4.9.

iii. GS-6 Flow Path Verification

CPNPP will confirm flow path availability of a train that has been out of service to perform periodic testing or maintenance.

iv. Primary AFW Water Source Low Level Alarm

The CPNPP Condensate Storage Tanks have a useable volume of 28,780 gallons of water at the Lo—Low level alarm. Assuming the largest capacity AFW pump is operating, this alarm will allow the operator at least 20 minutes to anticipate the need to make up water or transfer AFW pump suction to the Service Water System.

v. AFW Pump Endurance Test

CPNPP will perform a pump endurance test as part of the pre-startup testing program.

2. For a discussion of how the Auxiliary Feedwater System meets acceptance criteria see FSAR Subsection 10.4.9.

3. Below are listed the AFW System Flowrate design bases and criteria, by question as listed in Enclosure 2 to the March 10, 1980 NRC letter which originally requested the information.

Question 1

a. Identify the plant transient and accident conditions considered in establishing AFWS flow requirements, including the following events:

1) Loss of Main Feed (LMFW)

2) LMFW w/loss of offsite AC power

3) LMFW w/loss of onsite and offsite AC power

4) Plant cooldown

5) Turbine trip with and without bypass

6) Main steam isolation valve closure

7) Main feed line break

8) Main steam line break

9) Small break LOCA

10) Other transient or accident conditions not listed above.

b. Describe the plant protection acceptance criteria and corresponding technical bases used for each initiating event identified above, The acceptance criteria should address plant limits such as:

1) Maximum RCS pressure (PORV or safety valve actuation)

2) Fuel temperature or damage limits (DNB, PCT, maximum fuel central temperature)

3) RCS cooling rate limit to avoid excessive coolant shrinkage

4) Minimum steam generator level to assure sufficient steam generator heat transfer surface to remove decay heat and/or cool down the primary system.

Response to Question 1.a

The reactor plant conditions which impose safety-related performance requirements on the design of the Auxiliary Feedwater System are as follows for Comanche Peak.

- Loss of Main Feedwater Transients
  - Loss of main feedwater with offsite power available
  - Loss of non-emergency AC power (i.e., loss of main 78 feedwater without offsite power available)
- Secondary System Pipe Ruptures
  - Feedline rupture
  - Steamline rupture
- Loss of all AC Power
- Loss of Coolant Accident (LOCA)
- Cool down

Loss of Main Feedwater Transients

The design loss of main feedwater transients are those caused by:

- Interruptions of the Main Feedwater System flow due to a malfunction in the feedwater or condensate system
- Loss of offsite power with the consequential shutdown of the system pumps, auxiliaries, and controls

The transients are discussed in Sections 15.2.7 and 15.2.6.

The Loss of Non-Emergency AC Power (15.2.6), Loss of Normal Feedwater (15.2.7), and Feedline Rupture (15.2.8) events serve as the basis for the minimum flow required by the Auxiliary Feedwater System for CPNPP. The auxiliary feedwater pumps are sized such that sufficient flow against the steam generator safety valve set pressure (with 3% accumulation) will be provided to remove the stored and residual heat even with the worst single failure in the Auxiliary Feedwater System. The Loss of Non-Emergency AC Power and Loss of Normal Feedwater analyses presented in Chapter 15 demonstrate that the Auxiliary Feedwater System is capable of removing the stored and residual heat, thus preventing the overpressurization of the RCS and loss of water through either the pressurizer relief or safety valves. The Feedline Rupture analyses also demonstrate that the Auxiliary Feedwater System capacity is sufficient to prevent RCS overpressurization and uncovering the reactor core.

Secondary System Pipe Ruptures

The feedwater line rupture accident is discussed in Section 15.2.8.

The main steamline rupture accident is discussed in Section 15.1.5. Auxiliary feedwater is not needed during the early phase of the transient but flow to the faulted loop will contribute to an excessive release of mass and energy to containment. Thus, steamline rupture conditions establish the upper limit on auxiliary feedwater flow delivered to a faulted loop. Eventually, however, the Reactor Coolant System will heat up again and auxiliary feedwater flow will be required to be delivered to the non-faulted loops, but at somewhat lower rates than for the loss of feedwater transients described previously. As discussed in Section 10.4.9., provisions have been made in the design of the Auxiliary Feedwater System to limit, control, or terminate the auxiliary feedwater flow to the faulted loop as necessary in order to prevent containment overpressurization following a steamline break inside containment, and to ensure the minimum flow to the remaining unfaulted loops.

Loss of All AC Power

The loss of all AC power is postulated as resulting from accident conditions wherein not only onsite and offsite AC power is lost but also AC emergency power is lost as an assumed common mode failure.

Battery power for operation of protection circuits is assumed available. The Auxiliary Feedwater System provides both an auxiliary feedwater pump power and control source which are not dependent on AC power and which are capable of maintaining the plant at hot shutdown until AC power is restored.

Loss-of-Coolant Accident LOCA

The loss of coolant accidents discussed in Section 15.6.5 do not impose any flow requirements on the AFWS in addition to those required by the other accidents. The following description of the small LOCA is provided here for the sake of completeness to explain the role of the Auxiliary Feedwater System in this transient,

Small LOCA’s are characterized by relatively slow rates of decrease in reactor coolant system pressure and liquid volume. The principal contribution from the Auxiliary Feedwater System following such small LOCA’s is basically the same as the system function during hot shutdown or following a spurious safety injection signal which trips the reactor. Maintaining a water level inventory in the secondary side of the steam generators provides a heat sink for removing decay heat and establishes the capability for providing a buoyancy head for natural circulation. The Auxiliary Feedwater System may be utilized to assist in a system cooldown and depressurization following a small LOCA while bringing the reactor to a cold shutdown condition.

Cooldown

The cooldown function of the Auxiliary Feedwater System is discussed in Section 10.4.9.

Response to Question 1.b

Table II.E.1.1-2 summarizes the criteria which are the general design bases for each event, discussed in the response to Question 1.a, above. Specific assumptions used in the analyses to verify that the design bases are met are discussed in response to Question 2.

The primary function of the Auxiliary Feedwater System is to provide sufficient heat removal capability following reactor trip and to remove the decay heat generated by the core and prevent systam overpressurization. Other plant protection systems are designed to meet short term or pre-trip fuel failure criteria. The effects of excessive coolant shrinkage are evaluated by the analysis of the rupture of a main steam pipe transient. The maximum flow requirement determined by other bases are incorporated into this analysis, resulting in no additional flow requirements.

Question 2

Describe the analyses and assumptions and corresponding technical justification used with plant condition considered in 1.a above including:

a. Maximum reactor power (including instrument error allowance) the time of the initiating transient or accident.

b. Time delay from initiating event to reactor trip.

c. Plant parameter(s) which initiates AFWS flow and time between initiating eveit End introduction AFWS flow steam generator(s)

d. Minimum steam generator water level when initiating event occurs.

e. Initial steam generator water inventory and depletion rate before and after AFWS flow commences -- identify reactor decay heat rate used.

f. Maximum pressure at which steam is released from steam generator(s) and against which the AFW pump must develop sufficient head.

g. Minimum number of steam generators that must receive AFW flow; e.g., 1 out of 2? 2 out of 4?

h. RC flow condition – continued operation of RC pumps or natural circulation.

i. Maximum AFW inlet temperature.

j. Following a postulated steam or feed line break, time delay assumed to isolate temperature and direct AFW flow to intact steam generator(s). AFW pump flow capacity allowance to accommodate the time delay and maintain minimum steam generator water level. Also identify credit taken for primary system heat removal due to blowdown.

k. Volume and maximum temperature of water in main feed lines between steam generator(s) and AFWS connection to main feed line.

l. Operating condition of steam generator normal blowdown following initiating event.

m. Primary and secondary system water and metal sensible heat used for cooldown and AFW flow sizing.

n. Time at hot standby and time to cooldown RCS to RHR system cut in temperature to size AFW water source inventory,

Response to 2

Analyses have been performed for the limiting transients which define the AFWS performance requirements. These analyses have been provided for review. Specifically, they include:

- Loss of main feedwater
- Loss of non-emergency AC power
- Rupture of a main feedwater pipe
- Rupture of a main steam pipe inside containment

In addition to the above analyses, calculations have been performed specifically for Comanche Peak to determine the plant cooldown flow (storage capacity) requirements. The loss of all AC power is evaluated via a comparison to the transient results of a Blackout, assuming an available auxiliary pump having a diverse (non-AC) power supply. The LOCA analysis, as discussed in the response Question 1.b, incorporates the system flow requirements as defined by other transients, and therefore is not performed for the purpose of specifying AFWS flow requirements. Each of the analyses listed above are explained in further detail in the following sections of this response.

Loss of Main Feedwater/Loss of Non-Emergency AC Power

Analyses of a loss of main feedwater, with and without assuming a loss of power to the reactor coolant pumps, were performed in Sections 15.2.6 and 15.2.7 for the purpose of showing that the peak RCS pressure remains below the criterion for Condition II transients and no water relief occurs through the pressurizer relief or safety valves. Sections 15.2.6 and 15.2.7 summarize the assumptions used in these analyses. The analyses assume that the plant is initially operating at 102% (calorimetric error) of the engineered safety features (ESF) design rating, a conservative assumption in defining decay heat and stored energy in the RCS. The reactor is assumed to be tripped on low-low steam generator water level, allowing for level 78 uncertainty. Sections 15.2.6 and 15.2.7 show that there is considerable margin with respect to filling the pressurizer even with the most limiting single active failure in the Auxiliary Feedwater System.

These analyses contribute in establishing the capacity of the auxiliary feedwater pumps and also establish train association of equipment so that these analyses remain valid assuming the most limiting single failure.

Rupture of Main Feedwater Pipe

The double ended rupture of a main feedwater pipe downstream of the main feedwater line check valve is analyzed in Section 15.2.8 which summarizes the assumptions used in this analysis. Reactor trip is assumed to occur when the faulted steam generator is at the low-low level setpoint (adjusted for errors). This conservative assumption maximizes the stored heat prior to reactor trip and minimizes the ability of the steam generator to remove heat from the RCS following reactor trip due to a conservatively small total steam generator inventory. As in the loss of normal feedwater analysis, the initial power rating was assumed to be 102% of the ESF design rating. Auxiliary feedwater flow of 430 gpm was assumed to be delivered to the two non-faulted steam generators 1 minute after reactor trip. At 30 minutes after reactor trip, it is assumed that the operator has isolated the AFWS from the break and a flow of 265 gpm from the second motor-driven auxiliary feed pump is delivered to the third non-faulted steam generator. The criterion that the reactor core remains covered with water is met.

This analysis contributes in establishing the capacity of the auxiliary feedwater pumps, establishes requirements for layout to preclude indefinite loss of auxiliary feedwater to the postulated break, and establishes train association requirements for equipment so that the AFWS can deliver the minimum flow required in 1 minute assuming the worst single failure.

Rupture of a Main Steam Pipe Inside Containment

Because the steamline break transient is a cooldown, the AFWS is not needed to remove heat in the short term. Furthermore, addition of excessive auxiliary feedwater to the faulted steam generator will affect the peak containment pressure following a steamline break inside containment. This transient is performed at four power levels for several break sizes. Auxiliary feedwater is assumed to initiate at the time of the break, independent of system actuation signals. The maximum flow is used for this analysis. Table II.E.1.1-3 summaries the assumptions used in this analysis. At 10 minutes after the break, it is assumed that the operator has isolated the AFWS from the faulted steam generator which subsequently blows down to ambient pressure. The criteria stated in Table II.E.1.1-2 are met.

This transient establishes the maximum allowable auxiliary feedwater flow rate to a single faulted steam generator assuming all pumps operating, establishes the basis for runout protection, if needed, and establishes layout requirements so that the flow requirements may be met considering the worst single failure.

Plant Cooldown

Maximum and minimum flow requirements from the previously discussed transients meet the flow requirements of plant cooldown. This operation, however, defines the basis for tank size, based on the required cooldown duration, maximum decay heat input and maximum stored heat in the system. As previously discussed in the response to Question 1.a, the auxiliary feedwater system partially cools the system to the point where the RHRS may complete the cooldown, i.e., 350°F in the RCS. Table II.E.1.1-3 shows the assumptions used to determine the cooldown heat capacity of the auxiliary feedwater system.

The cooldown is assumed to commence at the maximum rated power, and maximum trip delays and decay heat source terms are assumed when the reactor is tripped. Primary metal, primary water, secondary system metal and secondary system water are all included in the stored heat to

be removed by the AFWS. See Table II.E.1.1-4 for the items constituting the sensible heat stored in the NSSS.

This operation is analyzed to establish minimum tank size requirements for auxiliary feedwater fluid source which are normally aligned.

Question 3

Verify that the AFW pumps in your plant will supply the necessary flow to the steam generator(s) as determined by items 1 and 2 above considering a single failure, Identify the margin in sizing the pump flow to allow for pump recirculation flow, seal leakage and pump wear.

Response to 3

The Auxiliary Feedwater System (AFWS) is designed, and the equipment specified, considering all criteria established by the NSSS supplier.

Each unit of the CPNPP has two motor driven auxiliary feedwater pumps, and one turbine-driven auxiliary feedwater pump. The turbine driven pump can function without AC power.

FSAR Section 10.4.9 provides the auxiliary feedwater pump capacities.

Pump seals are designed to minimize leakage and expected leakage is considered negligible. The pumps are subjected to periodic inspection, testing and maintenance programs. Performance standards will be set for the pumps. If performance degrades below set limits due to wear, the wear rings in the pump will be replaced. The AFWS is equipped with flow-limiting orifices that will limit the flow to a faulted steam generator from the motor-driven pumps to 700 gpm and from the turbine-driven pump to 600 gpm. The total flow to the faulted steam generator will not be more than 1300 gpm, which is less than the 1380 gpm maximum flow stated in Table II.E.1.1-3. The other steam generators will receive more than the minimum required flow.

AFWS flow rates are provided in Sections 15.1.5, 15.2.6, 15.2.7 and 15.2.8. The flow rates assumed in the accident analyses have considered the most limiting single active failure.

4. See Response to 1.E above.

II.E.1.2 Auxiliary Feedwater System Automatic Initiation And Flow Indication

Action Plan Requirements:

"a. Automatic Initiation of the Auxiliary Feedwater System

Provide automatic initiation of all auxiliary feedwater systems, The initiation signals and circuits shall be designed in such a manner that a single failure will not result in the loss of auxiliary feedwater system function. Testability of the initiating signals and circuits shall be a feature of the design. The initiating signals and circuits shall be powered from the emergency buses. Manual capability to initiate the auxiliary feedwater system from the control room must be retained and must be implemented in such a manner that a single failure in the manual circuits will not result in the loss of system function. The a-c motor driven pumps and valves in the auxiliary feedwater system must be included in the

automatic actuation (simultaneous or sequential) of the loads to the emergency buses. The design of the automatic initiating signals and circuits must be such that their failure will not result in the loss of manual capability to initiate the auxiliary feedwater system from the control room."

NUREG 0578, Pg. 10

"b. Auxiliary Feedwater Flow Indication to Steam Generators

Provide safety-grade indication in the control room of auxiliary feedwater flow for each steam generator. The flow instrument channels shall be powered from the emergency buses, consistent with satisfying the power diversity requirements for auxiliary feedwater systems."

- NUREG 0578, Pg. 11

Also see NUREG 0737.

Response to a.

Auxiliary feedwater system initiation as described herein is safety-related. The CPNPP design complies with the stated positions of NUREG-0578 also as described in the detailed responses below; therefore, no design changes were required as a result of related Action Plan requirements. Components which provide this function are listed in Table 17A-1.

NUREG—O578 Positions 1 to 7

"Consistent with satisfying the requirements of General Design Criterion 20 of Appendix A to 10 CFR 50 with respect to the timely initiation of the auxiliary feedwater system, the following requirements shall be implemented in the short term:"

NUREG—O578 Position 1

"The design shall provide for the automatic initiation of the auxiliary feedwater system."

CPNPP Response

The Auxiliary Feedwater System consists of three separate trains. Two of the trains each consist of one 50 percent capacity electric motor driven auxiliary feedwater pump, valves, piping and controls, The third train consists of a 100% capacity turbine driven auxiliary feedwater pump, valves, piping and controls. See FSAR Fig. 10.4-11. The electric motor driven pumps start automatically on:

1. Two out of four Lo-Lo steam generator level signals from any one of four steam generators

2. Trip of both main feedwater pumps (as sensed by hydraulic pressure switches on the feedwater pumps)

3. Station blackout sequence signal

4. Safety injection signal.

The turbine driven pump starts automatically on:

1. Two out of four Lo-Lo steam generator level signals from any two of four steam

2. Station blackout sequence signal.

The pneumatic flow control valve in each of the auxiliary feedwater lines to the steam generators is a normally open, fail open valve. The motor-operated and manual isolation valves in these lines are maintained open by administrative procedure.

NUREG-0578 Position 2

"The automatic initiation signals and circuits shall be designed so that a single failure will not result in the loss of auxiliary feedwater system function."

CPNPP Response

As shown in our response to NUREG-0578, Position 1 above, the auxiliary feedwater initiation circuitry is part of the Engineered Safety Features (ESF) system, and as such, is installed in accordance with IEEE Standard 279. This Standard is referenced in 10 CFR 50.55a(h).

NUREG-0578 Position 3

"Testability of the initiating signals and circuits shall be a feature of the design."

CPNPP Response

The auxiliary feedwater initiation signals and circuitry are testable. Such testability is included as a requirement of the technical specifications.

NUREG-0578 Position 4

"The initiating signals and circuits shall be powered from the emergency buses,"

CPNPP Response

As shown in our responses to Positions 1 and 2, the initiating signals and circuits are a part of the plant engineered safety features. The requirements for the ESF system dictate that the system shall meet the "single failure criteria." To accomplish this, the initiating signals and circuits are powered from separate emergency buses.

The initiating sensors such as steam generator low-low level are powered from separate and redundant nuclear instrumentation and control panels, each of which is supplied by either on-site emergency generators or station emergency batteries. Each of the two redundant Solid State Protection System (SSPS) trains is supplied by separate power sources.

NUREG-0578 Position 5

“Manual capability to initiate the auxiliary feedwater system from the control room shall be retained and shall be implemented so that a single failure in the manual circuits will not result in the loss of system function.”

CPNPP Response

Manual initiation for each train exists in the control room. No single failure in the manual initiation portion of the circuit can result in the loss of auxiliary feedwater system function.

NUREG-0578 Position 6

“The a-c motor-driven pumps and valves in the auxiliary feedwater system shall be included in the automatic actuation (simultaneous and/or sequential) of the loads to the emergency buses.”

CPNPP Response

As with all ESF equipment, the a-c motor-driven pumps and all valves in the system are automatically transferred to, and sequentially loaded on, the emergency buses on loss of offsite power.

NUREG-0578 Position 7

“The automatic initiating signals and circuits shall be designed so that their failures will not result in the loss of manual capability to initiate the AFWS from the control room.”

CPNPP Response

All automatic and manual initiating signals and circuits are installed in accordance with regulatory requirements and are safety grade and redundant. No single failure in the automatic portion of the system will result in loss of the capability to manually initiate the AFWS from the control room.

NUREG-0578 Position

“In the long term, the automatic initiation signals and circuits shall be upgraded in accordance with safety grade requirements.”

CPNPP Response

As described in the preceding responses, the automatic initiating signals presently meet safety grade requirements. No upgrading is required.

CPNPP Response to b.

Auxiliary feedwater system flow as described herein is safety-related. The CPNPP design complies with the stated positions of NUREG-0578 also as described in the responses below; therefore, no design changes were required as a result of related Action Plan requirements. Components which provide this function are listed in Table 17A-1.

NUREG-0578 Positions 1 and 2

Consistent with satisfying the requirements set forth in General Design Criterion 13 to provide the capability in the Control Room to ascertain the actual performance of the AFWS when it is called to perform its intended function, the following requirements shall be implemented:

NUREG-0578 Position 1

Safety-grade indication of auxiliary feedwater flow to each steam generator shall be provided in the Control Room.

CPNPP Response

Two safety-grade indicators of auxiliary feedwater flow to each steam generator are provided in the Control Room. In addition, safety-grade auxiliary feedwater flow indication is provided for each auxiliary feedwater pump.

These flow indicators are included in the human factors analysis of the Control Room which is discussed under item I.D.1.

NUREG-0578 Position 2

The auxiliary feedwater flow instrument channels shall be powered from the emergency buses consistent with satisfying the emergency power diversity requirements of the auxiliary feedwater system set forth in Auxiliary Systems Branch Technical Position 10-1 of Standard Review Plan, Section 10.4.9.

CPNPP Response

Of the two safety-grade indicators of auxiliary feedwater flow to each steam generator, one is powered from a Train A vital instrument ESF bus and the other from a Train B vital instrument ESF bus.

The pump flow indicators are powered from the vital instrument ESF bus of the same train as the pump. The turbine driven pump flow indication is from Train A.

II.E.3 DECAY HEAT REMOVAL

OBJECTIVE:

“Improve the reliability and capability of nuclear power plant systems for removing decay heat and achieving safe shutdown conditions following transients and under post accident conditions.”

- NUREG 0600, Pg. II.E.3-1

II.E.3.1 Reliability of Power Supplies for Natural Circulation

Action Plan Requirements:

Consistent with satisfying the requirements of General Design Criteria 10, 14, 15, 17, and 20 of Appendix A to 10 CFR Part 50 for the event of loss of offsite power, the following positions shall be implemented:

(1) The pressurizer heater power supply design shall provide the capability to supply, from either the offsite power source or the emergency power source (when offsite power is not available), a predetermined number of pressurizer heaters and associated controls necessary to establish and maintain natural circulation at hot standby conditions. The required heaters and their controls shall be connected to the emergency buses in a manner that will provide redundant power supply capability.

(2) Procedures and training shall be established to make the operator aware of when and how the required pressurizer heaters shall be connected to the emergency buses. If required, the procedures shall identify under what conditions selected emergency loads can be shed from the emergency power source to provide sufficient capacity for the connection of the pressurizer heaters.

(3) The time required to accomplish the connection of the preselected pressurizer heater to the emergency buses shall be consistent with the timely initiation and maintenance of natural circulation conditions.

(4) Pressurizer heater motive and control power interfaces with the emergency buses shall be accomplished through devices that have been qualified in accordance with safety-grade requirements.

- NUREG 0737

CPNPP Response

Power is supplied to four pressurizer heater groups from offsite power, when available, and from the onsite emergency diesel generators through ESF buses. Redundancy is provided by supplying two groups of pressurizer heaters from each redundant ESF train, Control power for manual on/off control of each of these four heater groups is supplied from the 125 volt DC ESF bus in the same train as the main power supply.

Procedures are established to control Reactor Coolant System pressure and temperature. Per Westinghouse analysis, pressurizer heaters are not required during natural circulation cooldown to Hot Shutdown or Cold Shutdown.

No loads need to be shed from the emergency power source to provide sufficient capacity for the connection of the pressurizer heaters.

The electrical interface between each of the four pressurizer heater groups and its associated F bus of concern is through a circuit breaker which trips on an “S” signal and is qualified in accordance with safety-grade requirements. The manual controls for these breakers are qualified in accordance with safety-grade requirements.

II.E.4 CONTAINMENT DESIGN

OBJECTIVE:

“Improve the reliability and capability of nuclear power plant containment structures to reduce the radiological consequences and risks to the public from design basis events and degraded—core and core—melt accidents.”

- NUREG 0660, Pg. II.E.4-1

II.E.4.1 Dedicated Penetration

Action Plan Requirements:

Plants using external recombiners or purge systems for postaccident combustible gas control of the containment atmosphere should provide containment penetration systems for external recombiner or purge systems that are dedicated to that service only, that meet the redundancy and single—failure requirements of General Design Criteria 54 and 56 of Appendix A to 10 CFR 50, and that are sized to satisfy the flow requirements of the recombiner or purge system.

The procedures for the use of combustible gas control systems following an accident that results in a degraded core and release of radioactivity to the containment must be reviewed and revised, if necessary.

- NUREG 0737

CPNPP Response

Post accident combustible gas contol vis external hydrogen purge or internal electric hydrogen recombiners is no longer required as described in Section 6.2.5. See Section 6.2.4 for containment penetration isolation.

II.E.4.2 Isolation Dependability

Action Plan Requirements:

“Provide containment isolation on diverse signals in conformance with Section 6.2.4 of the Standard Review Plan, review isolation provisions for non-essential systems and revise as necessary, and modify containment isolation designs as necessary to eliminate the potential for inadvertent reopening upon reset of the isolation signal.”

- NUREG 0578, Pg. 8

"(1) Containment isolation system designs shall comply with the recommendations of Standard Review Plan Section 6.2.4, (i.e., that there be diversity in the parameters sensed for the initiation of containment isolation).

"(2) All plant personnel shall give careful consideration to the definition of essential and nonessential systems, identify each system determined to be nonessential, describe the

basis for selection of each essential system, modify their containment isolation designs accordingly, and report the results of the reevaluation to the NRC.

"(3) All nonessential systems shall be automatically isolated by the containment isolation signal.

"(4) The design of control systems for automatic containment isolation valves shall be such that resetting the isolation signal will not result in the automatic reopening of containment isolation valves. Reopening of containment isolation valves shall require deliberate operator action.

"(5) The containment setpoint pressure that initiates containment isolation for nonessential penetrations must be reduced to the minimum compatible with normal operating conditions.

"(6) Containment purge valves that do not satisfy the operability criteria set forth in Branch Technical Position CSB 6-4 or the Staff Interim Position of October 23, 1979 must be sealed closed, as defined in SRP 6.2.4, Item II.3.f during operational conditions 1, 2, 3, and 4. Furthermore, these valves must be verified to be closed at least every 31 days." (A copy of the Staff Interim Position is enclosed as Attachment 1.)

"(7) Containment purge and vent isolation valves must close on a high radiation signal.

- NUREG 0737

CPNPP Response

1. There are two phases of containment isolation at CPNPP. Phase A isolates all nonessential process lines but does not effect safety injection, containment spray, component cooling water supplied to the reactor coolant pumps, or auxiliary feedwater. Phase B isolates all process lines except safety injection, containment spray, and auxiliary feedwater.

   Phase A isolation is initiated by high containment pressure, low steam line pressure, low pressurizer pressure, or manual initiation. Phase B isolation is initiated by high-high containment pressure or manual initiation.

   In addition, containment ventilation isolation is initiated by Phase A isolation, containment spray actuation, or high containment radioactivity.

   This system fully complies with Section 11.6 of the SRP 6.2.4. Compliance with all remaining sections of SRP 6.2.4 is documented in Section 6.2.4.

2. A reevaluation of the classification assigned to each process penetration has been performed and is described in revised Section 6.2.4.3.

3. All nonessential systems use either normally closed manual valves or the valves are automatically isolated on a containment isolation signal. See revised Section 6.2.4.3.

4. The CPNPP design complies with the position. The operator must reset each device individually as desired. See revised Section 7.3.1.1.4.

5. The CPNPP containment pressure setpoint (Hi-1) that initiates containment isolation for non-essential penetrations is 3.2 psig. This setpoint is the minimum compatible with normal operating conditions and the value used in accident analyses.

6. The containment isolation valves for the containment purge system are automatic isolation valves. Furthermore, these valves will be closed during operational modes 1, 2, 3 and 4 (see response to Question 022.20). These valves will be verified to be closed at least every 31 days.

7. CPNPP complies as described in Section 6.2.4.

TABLE II.E.1.1-1
PRIMARY EVENT PROBABILITIES

Figure II.E.1.1-2

Variables	Event	Value
P_a	Blocked Butterfly Valve 006	1×10^{-4} (a)
P_b	Blocked Butterfly Valve 007 Using Coupling	1×10^{-1}
P_c	Train A Logic Failure Using Coupling	7×10^{-3} (a)
P_d	Train B Logic Failure Using Coupling	.2
P_e	Turbine Train Logic Failure	.5
P_f	No Manual Start	5×10^{-3} (a)
P_g	Train A pump mechanical failure	1×10^{-3} (a)
P_h	Train A pump control circuit failure	4×10^{-3} (a)
P_i	Turbine Pump Mechanical failure	1×10^{-3} (a)
P_j	No Steam to the Turbine Train	1×10^{-3}
P_k	Train Out for test or Maintenance	7.8×10^{-3} (a)
P_l	Diesel Generator Failure is assigned a variable probability	
P_m	Manual Valve Blocked	1×10^{-4} (a)
P_n	Check Valve Blocked	1×10^{-4} (a)
P_o	Air Operated Valve Blocked	3×10^{-4} (a)
P_p	Motor Operated Valve blocked	1×10^{-4} (a)
P_r	Not Used	

a) Values taken from NUREG-0611, Table III-2

TABLE II.E.1.1-2
CRITERIA FOR AUXILIARY FEEDWATER SYSTEM DESIGN BASIS CONDITIONS

Condition or Transient	Classification[a]	Criteria	Additional Design Criteria
Loss of Main Feedwater	Condition II	Peak RCS pressure not to exceed design pressure +10%. No consequential fuel failures	Pressurizer does not fill.
Loss of Non-Emergency AC Power	Condition II	(Same as LMFW)	Pressurizer does not fill.
Feedline Rupture	Condition IV	10CFR100 dose limits. Containment design pressure not exceeded	Core does not uncover.
Loss of all A/C Power	N/A	Note 1	Same as loss of non-emergency A/C power assuming turbine driven pump
Loss of Coolant	Condition IIII	10 CFR 100 dose limits 10 CFR 50 PCT limits	
	Condition IV	10 CFR 100 dose limits 10 CFR 50 PCT limits	
Main Steamline Break	Condition IV	Containment design pressure and temperature not exceeded	
Cooldown	N/A		100°F/hr 557° to 350°F

a) Ref: ANSI N18.2 (This information provided for those transients performed in the FSAR).

Note:

1 Although this transient establishes the basis for AFW pump powered by a diverse power source, this is not evaluated relative to typical criteria since multiple failures must be assumed to postulate this transient.

TABLE II.E.1.1-3
SUMMARY OF ASSUMPTIONS USED IN AFWS DESIGN VERIFICATION ANALYSES

(Sheet 1 of 2)

Transient		Cooldown	Main Steamline Break (containment)
a.	Max reactor power	3644 MWt	0, 30, 70, 102% of rated (percent of 3425 MWt)
b.	Time delay from event to Rx trip	2 sec	variable
c.	AFWS actuation signal/time delay for AFWS flow	N/A	Assumed immediately @ 0 sec (no delay)
d.	SG water level at time of reactor trip	N/A	N/A
e.	Initial SG inventory	70,760 lbm/SG at 544.6°F	consistent with power
	Rate of change before & after AFSW actuation	N/A	N/A
	Decay heat	N/A	ANS + 20%
f.	AFW pump design pressure	1236 psia	N/A
g.	Minimum # of SGs which must receive AFW flow	N/A	N/A
h.	RC pump status	Tripped	All operating
i.	Maximum AFW temperature	100°F	440°F
j.	Operator action	N/A	10 minutes
k.	MFW purge volume/ S/G temperature	450 ft^3/440°F	800 ft^3/loop (for dryout time)

TABLE II.E.1.1-3
SUMMARY OF ASSUMPTIONS USED IN AFWS DESIGN VERIFICATION ANALYSES

(Sheet 2 of 2)

Transient		Cooldown	Main Steamline Break (containment)
l.	Normal blowdown	none assumed	none assumed
m.	Sensible heat	Table II.E.1.1-4	N/A
n.	Time at standby/time to cooldown to RHR	2 hr/4 hrs	N/A
o.	AFW flow rate	variable	1380 gpm (constant) to faulted SG.

Note:

Refer to Sections 15.2.6, 15.2.7 and 15.2.8 for the assumptions used in the Loss of Non-Emergency AC Power, Loss of Main Feedwater and Feedline Break events.

TABLE II.E.1.1-4
SUMMARY OF SENSIBLE HEAT SOURCES

Primary Water Sources (initially at rated power temperature and inventory)

- RCS fluid
- Pressurizer fluid (liquid and vapor)

Primary Metal Sources (initially at rated power temperature)

- Reactor coolant piping, pumps and reactor vessel
- Pressurizer
- Steam generator tube metal and tube sheet
- Steam generator metal below tube sheet
- Reactor vessel internals

Secondary Water Sources (initially at rated power temperature and inventory)

- Steam generator fluid (liquid and vapor)
- Main feedwater purge fluid between steam generator and AFWS piping.

Secondary Metal Sources (initially at rated power temperature)

- All steam generator metal above tube sheet, excluding tubes.

TABLE II.E.1.1-5
THIS TABLE HAS BEEN DELETED

II.F INSTRUMENTATION AND CONTROLS

OBJECTIVE:

"Provide instrumentation to monitor plant variables and systems during and following an accident. Indications of plant variables and status of systems important to safety are required by the plant operator (licensee) during accident situations to (1) provide information needed to permit the operator to take preplanned manual actions to accomplish safe plant shutdown; (2) determine whether the reactor trip, engineered safety features systems, and manually initiated systems are performing their intended functions (i.e., reactivity control, core cooling, maintaining reactor coolant system integrity, and maintaining containment integrity); (3) provide information to the operator that will enable him to determine the potential for a breach of the barriers to radioactivity release (i.e., fuel cladding, reactor coolant pressure boundary, and containment) and if a barrier has been breached; (4) furnish data for deciding on the need to take unplanned action if an automatic or manually initiated safety system is not functioning properly or the plant is not responding properly to the safety systems in operation; (5) allow for early indication of the need to initiate action necessary to protect the public and for an estimate of the magnitude of the impending threat; and (6) improve requirements and guidance for classifying nuclear power plant instrumentation, control, and electrical equipment important to safety."

-NUREG 0660, pg. II.F-1

II.F.1 ADDITIONAL ACCIDENT MONITORING INSTRUMENTATION.

Action Plan Requirements:

"Item II.F.1 of NUREG-0660 contains the following subparts:

(1) Noble gas effluent radiological monitor:

(2) Provisions for continuous sampling of plant effluents for postaccident releases of radioactive iodines and particulates and onsite laboratory capabilities (this requirement was inadvertently omitted from NUREG-0660; see Attachment 2 that follows, for position);

(3) Containment high-range radiation monitor;

(4) Containment pressure monitor;

(5) Containment water level monitor; and

(6) Containment hydrogen concentration monitor.

"NUREG-0578 provided the basic requirements associated with items (1) through (3) above. Letters issued to all operating nuclear power plants dated September 13, 1979 and October‘ 30, 1979 provided clarification of staff requirements associated with items (1) through (6) above. Attachments 1 through 6 present the NRC position on these matters."

-NUREG-0737

CPNPP Response

The use of the additional accident monitoring instrumentation as listed will be integrated into the operating procedures and training programs prior to fuel load:

(1) Noble Gas Monitor

The CPNPP design will include wide range noble gas monitors for the plant vent stack which will detect and measure concentrations of noble gas fission products in plant gaseous effluents during and following an accident.

An adjacent-to-line monitor will be provided for each main steam line to monitor the concentration in steam that may be released to the environment by the safety or relief valves.

For description of these monitors, see Section 11.5.

(2) Iodine/Particulate Sampling

The wide range noble gas monitor discussed above provides the capability to sample the plant vent stacks as required.

For description, see Section 11.5.

(3) Containment High Range Radiation Monitor

The redundant Category 1 monitors will be located in each Containment Building at Elevation 905'-9". Exact location is provided in Figure II.F-1. To ensure valid data, these monitors will be located at least 90E apart and will not be located adjacent to process piping.

Monitor special calibration and environmental qualification will be performed as specified in Table II.F.1-3 of NUREG-0737.

For further discussion, see Section 12.3.

(4) Containment Pressure

The CPNPP design will include redundant wide range pressure indication (0 - 150 psig) on the Main Control Board meeting Regulatory Guide 1.97 Rev. 2 requirements. In addition this parameter will be provided as input to the SPDS high level display.

The present CPNPP design includes four channels of intermediate range pressure indication (-5 to + 60 psig) on the Main Control Board meeting Regulatory Guide 1.97 Rev. 2 requirements. In addition this parameter will be provided as input to the SPDS high level display.

(5) Containment Water Level

CPNPP design will include redundant wide range containment level indication (elevation 808'-3" to 817' - 6) on the Main Control Board meeting the intent of Regulatory Guide 1.97 Rev. 2 requirements. In addition this parameter will be provided as input to the SPDS high level display. These transmitters measure volume in excess of 600,000 gallons. CPNPP design will also include normal sump level indications (0 - 3 feet) on the Main Control Board. The containment wide range level indication covers the maximum expected flooding levels assuming the reactor cavity is blocked and does not receive any flood water to simulate the most limiting post-accident condition (flooding starts at El. 808'). Actual post-accident conditions would include flooding into the reactor cavity sump (781'-2") and reactor cavity (783'-7"). Therefore, containment narrow range level indication is not considered as required for accident monitoring.

(6) Containment Hydrogen

See Sections 6.2.5.2.3 and 7.5 for containment hydrogen monitoring.

II.F.2 IDENTIFICATION OF AND RECOVERY FROM CONDITIONS LEADING TO INADEQUATE CORE COOLING

Action Plan Requirements:

"Licensees shall provide a description of any additional instrumentation or controls (primary or backup) proposed for the plant to supplement existing instrumentation (including primary coolant saturation monitors) in order to provide an unambiguous, easy-to-interpret indication of inadequate core cooling (ICC). A description of the functional design requirements for the system shall also be included. A description of the procedures to be used with the proposed equipment, the analysis used in developing these procedures, and a schedule for installing the equipment shall be provided."

-NUREG-0737

CPNPP Response

The CPNPP design will include redundant instrumentation to monitor the approach to, existence of and recovery from inadequate core cooling. The monitored parameters will be the reactor coolant system (RCS) saturation margin, the collapsed water level above the reactor core and the RCS temperature at the core exit.

An indication of a declining margin to saturation in the RCS will provide the earliest warning that conditions are developing which could lead to ICC. If the event is allowed to progress, saturation conditions will be observed along with an indication of a declining collapsed water level above the reactor core. Collapsed water level alone does not identify the existence of ICC, only the potential for ICC. Maintaining the collapsed water level at a point above the core is not essential for adequate core cooling. A steam/water froth region extending down into the core could equate to a collapsed water level below the top of the core and yet provide adequate core cooling. Only as the top of the froth region drops below the top of the core would ICC tend to occur. RCS pressure and core exit temperatures would indicate this phase of the event by a "saturation" margin in the superheat region. Alternatively, the recovery from ICC and the subsequent stages

of the event can be monitored to verify that corrective actions taken have resulted in the expected plant response.

The instrumentation to be used for monitoring the RCS saturation margin, collapsed water level, and core exit temperatures is described below.

II.F.2.1 RCS Saturation Margin

The core cooling monitor is a microprocessor-based system capable of determining the saturation margin in the RCS. This determination is based on the lower measurement of the hot leg or pressurizer pressure and the highest measured temperature in the RCS. The temperature used for the most current calculation and the saturation margin are displayed together on the main control board. Redundant microprocessor systems and sensor devices will maintain independence between the two separate trains of temperature and pressure data.

II.F.2.2 Collapsed Water Level

To provide the capability for measurement of the reactor coolant inventory in the upper head and plenum regions of the reactor vessel, CPNPP has decided to install a heated junction thermocouple (HJTC) system supplied by Combustion Engineering, Inc. (C-E). This system includes two identical probes located 180° apart, each in the immediate vicinity of a cold leg inlet. Each probe assembly contains a series of eight HJTC sensors with individual splash shields which are axially distributed inside a separator tube. An HJTC sensor consists of two physically separated thermocouple junctions, one of which is electrically heated. The basic principle of the system is to determine whether or not a sensor is covered with water by detecting the temperature difference between adjacent heated and unheated thermocouples.

The HJTC probe has undergone extensive tests by C-E and has demonstrated its ability to indicate the coolant inventory in the reactor vessel above the core.

When the collapsed water level inside the probe falls below a given sensor location during a loss-of coolant event, the heated junction temperature increases due to the relatively poor cooling ability (lower heat transfer coefficient) of steam instead of water. When the relative temperature difference between heated and unheated junctions exceeds a predetermined value, the sensor registers as being uncovered (i.e., surrounded by steam only). Once a sensor becomes uncovered, the potential increase in temperature of the heated junction depends upon the sensor heater power and the pressure. At low pressures, a high heater power can produce excessive junction temperatures and possible sensor failure. Therefore, a heater power control system is used to preclude excessive temperatures. However, should sensor damage occur due to overheating, the loss of a single HJTC sensor within a probe will not seriously compromise the resolution of usefulness of the HJTC probe measurements due to system redundancy and the independence of the sensors within the probe.

The probes for CPNPP will be of the “split-probe” design, having two sensors located in the upper head region and six sensors located in the upper plenum region. Sudden depressurization of the upper plenum could result in localized voiding with no immediate comparable occurrence in the upper head region due to the limited pressure equalization vent area in the upper support plate. The resulting pressure disequilibrium could result in high ambiguous indications without the “split-probe” design. This design will allow an unambiguous indication of collapsed water level in either region regardless of their instantaneous relative pressure. The measurement of

collapsed water level in either region above the core will give immediate indication to the reactor operator that a loss-of-coolant condition exists and that the approach to ICC is imminent unless proper corrective actions are taken.

The control board display will be in the form of two “mimic” probes, each composed of a column of light emitting diodes (LED’s) corresponding to the eight HJTC sensors in each probe. Sensors which are surrounded by water will be indicated by energizing the corresponding LED.

II.F.2.3 Reactor Coolant System Temperatures

The original, vendor supplied system of core exit thermocouples (CET) and resistance temperature detectors (RTD) will be used to supply the necessary temperature data for the core cooling monitoring system. The saturation margin monitor will be the primary data link for the CET’s. The plant process/ERF computers will have on-line access to the data but are secondary and isolated from this primary link. The core cooling monitor will pass a complete set of input and output data in engineering units to these computers following each calculation cycle. The set of fifty CET’s have been divided into two redundant trains, each representative of all four quadrants of the core. Although originally a non-safety class system, necessary modifications have been made to upgrade all of the system components to Class 1E. Maximum achievable train separation is maintained; however, complete separation is impossible in the reactor head region. Due to retrofit limitations, the two trains share reactor vessel head penetrations where a degree of physical separation is achieved with each individual thermocouple in a separate stainless steel sleeve. Above the vessel head, the messenger wires of the two safety trains are bundled separately and are physically separated from each other and from non-safety Train C cables to the greatest extent possible. Complete train separation is achieved above the CRDM Seismic Support Assembly’s platform (Unit 1) and the missile shield (Unit 2). All connectors and cold junction terminals have been upgraded to Class 1E. Thermocouple leads from the thermocouples to the cold junction terminals are mineral-oxide insulated and stainless steel jacketed.

The average bulk coolant temperature in each hot or cold leg of the primary reactor coolant loop is individually monitored by three RTDs in separate thermowells in each leg near the reactor vessel. These are also divided into two trains with either a hot or cold leg from each of the four loops represented in each train.

Computer software for the core cooling monitor is designed to limit the error introduced by the calculations themselves to less than 0.5°F. Thus, the primary sources of system inaccuracy are sensor inaccuracy first and the initial analog to digital conversion second. Core Cooling Monitor system accuracy is ±3.5°F. User modifiable parameter ranges are incorporated in the software to facilitate automatic testing for out-of-range data. The primary thermocouple temperature limits directly result from these ranges and from the potential loss of physical integrity of Type K thermocouples at elevated temperatures. Instrumentation ranges are listed in Table II.F.2-1.

II.F.2.4 Implementation

CPNPP will establish acceptable operation of the upgraded CETs and the core cooling monitors prior to fuel loading in Unit 1. Necessary modifications to the RCS pressure boundary have been incorporated to allow installation of the HJTC probes. Final installation and implementation of the HJTC probes are complete.

CPNPP Emergency Operating Procedures (EOPs) have incorporated the saturation margin monitor instrumentation and the HJTC instrumentation. The CPNPP EOPs are based on the analyses and generic Emergency Response Guideline Procedures developed by the Westinghouse Owners Group.

TABLE II.F.2-1
SATURATION MARGIN MONITOR AND CORE EXIT THERMOCOUPLE
SPECIFICATION SUMMARY

	PARAMETER	RANGE
INPUT:	Coolant Pressure (1)	0-3000 PSIG
	Pressurizer Pressure (1)	1700-2500 PSIG
	Coolant Temperature (4)	0-700°F
	Type K Thermocouple (25)	0-2300°F
	Cold Junction Temperature (3)	32°-500°F
OUTPUT:	Margin (1)	300°F subcooling to 300°F superheat
	Hottest Thermocouple (1)	0-2300°F
	Data (2)	All inputs and outputs in engineering units each cycle.

Core Cooling Monitor System Accuracy: ± 3.5°F

II.G ELECTRICAL POWER

OBJECTIVE:

"Increase the reliability and diversification of the electrical power supplies for certain safety-related equipment."

-NUREG 0660, pg. II.G-1

II.G.1 POWER SUPPLIES FOR PRESSURIZER RELIEF VALVES AND LEVEL INDICATORS

Action Plan Requirements:

"Consistent with satisfying the requirements of General Design Criteria 10, 14, 15, 17 and 20 of Appendix A to 10 CFR Part 50 for the event of loss-of-offsite power, the following positions shall be implemented:

Power Supply for Pressurizer Relief and Block Valves and Pressurizer Level Indicators

(1) Motive and control components of the power-operated relief valves (PORVs) shall be capable of being supplied from either the offsite power source or the emergency power source when the offsite power is not available.

(2) Motive and control components associated with the PORV block valves shall be capable of being supplied from either the offsite power source or the emergency power source when the offsite power is not available.

(3) Motive and control power connections to the emergency buses for the PORVs and their associated block valves shall be through devices that have been qualified in accordance with safety-grade requirements.

(4) The pressurizer level indication instrument channels shall be powered from the vital instrument buses. The buses shall have the capability of being supplied from either the offsite power source or the emergency power source when offsite power is not available."

-NUREG 0737

CPNPP Response

The Pressurizer PORVs are pneumatically operated valves with pilot solenoids.

The pilot solenoids are powered from 125 volt DC emergency buses which can be powered from batteries, offsite power or emergency onsite power.

The Pressurizer PORV controls are powered from the inverter buses which are capable of being fed from offsite and onsite emergency ac power, and the onsite dc power system.

The Pressurizer PORV block valves for CPNPP are motor operated valves. The power for the motors is supplied from class 1E 480 volt buses which may be supplied from either offsite power or onsite emergency diesel generators. Control power is provided from the same buses.

The Pressurizer PORV dc power and control power and the PORV block valve motor power and control power are connected to different emergency ESF buses through devices that are qualified in accordance with safety-grade requirements.

The Pressurizer level indicated circuits are safety-grade and post-accident qualified. AC power for these Class 1E instrument channels is supplied from inverters which are supplied from ESF buses with automatic backup from the emergency batteries.

The power supplies for the PORVs and block valves are listed appropriately in Table 17A-1.

II.K MEASURES TO MITIGATE SMALL-BREAK LOSS-OF-COOLANT ACCIDENT AND LOSS-OF-FEEDWATER ACCIDENTS

OBJECTIVE:

"To perform systems reliability analyses and to effect changes in emergency operating procedures and operator training to improve the capability of plants to mitigate the consequences of the small-break loss-of-coolant accidents (LOCA) and loss-of-feedwater events."

- NUREG 0660, Pg. II.K-1

II.K.1 IE BULLETINS

Action Plan Requirements:

"C.1.5* Review all valve positions, positioning requirements, positive controls and related test and maintenance procedures to assure proper ESF functioning. (See Bulletin 79-06A Item 8, 79-06B Item 7, 79-08 Item 6.)"

"C.1.10 Review and modify, as required, procedures for removing safety-related systems for service (and restoring to service) to assure operability status is known. (See Bulletin 79-05A Item 10, 79-06A Item 10, 79-06B Item 9, 79-08 Item 8.)"

"C.1.17 For Westinghouse-designed reactors, trip the pressurizer low-level coincident signal bistables, so that safety injection would be initiated when the pressurizer low-pressure setpoint is reached regardless of the pressurizer level. (See Bulletin 79-06A and Revision 1, Item 3.)"

-NUREG 0694, pg. 17

CPNPP Response

II.K.1 Procedures will be reviewed and, as necessary, revised to address the removal of safety-related systems from service (and restoring to service) to assure proper operability.

II.K.1.5 CPNPP will review all safety-related valve positions, positioning requirements, and positive controls to assure that valves remain positioned (open or closed) in a manner to ensure the proper operation of engineered safety features.

CPNPP will review and modify as necessary those related maintenance, testing, startup, and supervisory periodic surveillance procedures to ensure that safety-related valves are returned to their correct positions following manipulations and are so maintained during all operational modes.

II.K.1.10 CPNPP will review and modify as necessary the maintenance and test procedures to ensure that they require:

(a) Verification, by test or inspection, of the operability of redundant safety-related systems prior to the removal of any safety-related system from service.

(b) Verification of the operability of all safety-related systems when they are returned to service following maintenance or testing.

(c) Explicit notification of involved reactor operations personnel whenever a safety-related system is removed from and returned to service.

II.K.1.17 FSAR Section 15.6 discusses CPNPP protection from decrease in reactor coolant inventory. Pressurizer low level trips are not utilized at CPNPP.

II.K.2.13 Thermal Mechanical Report – Effect of High-Pressure Injection on Vessel Integrity for Small-Break Loss-of-Coolant Accident With No Auxiliary Feedwater

Action Plan Requirements:

"A detailed analysis shall be performed of the thermal-mechanical conditions in the reactor vessel during recovery from small breaks with an extended loss of all feedwater."

Changes to Previous Requirements and Guidance

"Also, this requirement has been changed to include all operating pressurized-water reactors (PWRs) and applicants."

- NUREG 0737

CPNPP Response

II.K.2.13 To address the requirements of detailed analysis of the thermal-mechanical conditions in the reactor vessel during recovery from small-break LOCAs with loss of all feedwater, the Westinghouse Owner's Group (WOG) has initiated a number of programs applicable to generic Westinghouse plants. Results of these programs are summarized in the following reports.

(1) "Summary Report on Reactor Vessel Integrity for Westinghouse Operating Plants," WCAP-10019, December 1981 (transmitted via Ref. #1).

(2) "Summary of Evaluations Related to Reactor Vessel Integrity performed for the Westinghouse Owner's Group," May 1982 (transmitted via Ref. 2).

Although these reports are primarily directed to demonstrating that there are no near-term safety concerns for operating Westinghouse plants, the

analytical methodologies and general conclusions are applicable to CPNPP.

Pressurized thermal shock is not expected to be a problem at CPNPP because of reactor vessel material properties. Based on the conservative material assumptions described in the Technical Specifications, the limiting material RT_{ndt} changes from an initial value of 40°F to 110°F (1/4 T) after 16 EFPY of fast neutron fluence is well below the proposed NRC screening criterion of 270°F (Ref. 3).

The limiting RT_{ndt} will be benchmarked and updated during CPNPP operation using neutron fluence data provided by the in-situ capsule surveillance program. This updating will include the reevaluation of the prediction of RT_{ndt}. If an extrapolated RT_{ndt} value were to violate the NRC screening criteria, then CPNPP would respond with an appropriate action plan according to NRC requirements.

Reference

1. Letter OG-66, dated December 30, 1981, O. D. Kingsley (WOG) to H. R. Denton (NRC).

2. Letter OG-70, dated May 28, 1982, O. D. Kingsley (WOG) to H. R. Denton (NRC).

3. “NRC Evaluation of Pressurized Thermal Shock” (draft), U.S. NRC, September 13, 1982.

II.K.2.17 Potential for Voiding in the Reactor Coolant System During Transients

Action Plan Requirements:

“Analyze the potential for voiding the reactor coolant system (RCS) during anticipated transients.”

Changes to Previous Requirements and Guidance

“The previous requirement has been to include all PWR operating reactors and applicants.”

- NUREG 0737

CPNPP Response

II.K.2.17 Westinghouse (in support of the Westinghouse Owner’s Group) has performed a study which addresses the potential for void formation in Westinghouse designed Nuclear Steam Supply Systems during natural circulation cooldown/ depressurization transients. The study has been submitted to the NRC by the Westinghouse Owners Group (Letter OG-57, dated April 20, 1981, R. W. Jurgensen to P. S. Check) and is applicable to Comanche Peak Nuclear Power Plant.

In addition, the Westinghouse Owners Group has developed a natural circulation cooldown guideline that takes the results of the study into account so as to preclude void formation in the upper head region during natural circulation cooldown/depressurization transients, and specifies those conditions under which upper head voiding may occur. These Westinghouse Owners Group generic guidelines have been submitted to the NRC (Letter OG-64, dated November 30, 1981, R. W. Jurgensen to D. G. Eisenhut). The generic guidance developed by the Westinghouse Owners Group (augmented as appropriate with plant specific consideration) will be utilized in the implementation of the CPNPP Emergency Operating Procedures.

II.K.2.19 Sequential Auxiliary Feedwater Flow Analysis

Action Plan Requirements:

"Provide a benchmark analysis of sequential auxiliary feedwater (AFW) flow to the steam generators following a loss of main feedwater."

Changes to Previous Requirements and Guidance

"The previous requirement has been changed to include all operating pressurized-water reactors (PWRs) and applicants for operating license."

- NUREG 0737

CPNPP Response

II.K.2.19 This requirement is not applicable to plants with inverted U-tube steam generators such as the Westinghouse steam generators for Comanche Peak.

II.K.3 FINAL RECOMMENDATIONS FOR B&O TASK FORCE

II.K.3.1 Installation and Testing of Automatic Power-Operated Relief Valve Isolation System

Action Plan Requirements:

"All PWR licensees should provide a system which uses the PORV block valve to protect against a small-break loss-of-coolant accident. This system will automatically cause the block valve to close when the reactor coolant system pressure decays after the PORV has opened. Justification should be provided to assure that failure of this system would not decrease overall safety by aggravating plant transients and accidents."

"Each licensee should perform a confirmatory test of the automatic block valve closure system following installation."

- NUREG 0737

CPNPP Response

I.K.3.1 Based on the reduction in PORV LOCA frequency due to post-TMI modifications already implemented, and automatic PORV block valve closure system is unnecessary and therefore not incorporated. See the response to II.K.3.2 below and WCAP-9804 for further discussion.

II.K.3.2 Report on Overall Safety Effect of Power-Operated Relief Valve Isolation System

Action Plan Requirements:

"(1) The licensee should submit a report for staff review documenting the various actions taken to decrease the probability of a small-break loss-of-coolant accident (LOCA) caused by a stuck- open power-operated relief valve (PORV) and show how these actions constitute sufficient improvements in reactor safety."

"(2) Safety-valve failure rates based on past history of the operating plants designed by the specific nuclear steam supply system (NSSS) vendor should be included in the report submitted in response to (1) above."

- NUREG 0737

CPNPP Response

II.K.3.2 The Westinghouse Owners Group submitted WCAP-9804, "Probabilistic Analysis and Operational Data in Response to Item II.K.3.2 for Westinghouse NSSS Plants" to the NRC on March 13, 1981. This report describes various modifications to Westinghouse plants since TMI and, using probabilistic analysis via event trees, estimates the effect of the post-TMI changes, the modifications described in WCAP-9804 have been incorporated by CPNPP. Therefore, the improvements in reactor safety demonstrated by the WCAP are applicable.

II.K.3.3 Reporting of SV & RV Failures & Challenges

Action Plan Requirements:

"Assure that any failure of a PORV or safety valve to close will be reported to the NRC promptly. All challenges to the PORVs or safety valves should be documented in the annual report."

- NUREG 0694, Pg. 24

CPNPP Response

II.K.3.3 Reporting of SV and RV Failures and Challenges

The CPNPP Emergency Plan requires prompt notification to the NRC of the failure of a SV or RV to close. The reporting of SV & RV challenges was moved from the annual report to the monthly operating report (TS 6.9.1.5), then License Amendment 64 deleted this reporting requirement altogether.

II.K.3.5 Automatic Trip of Reactor Coolant Pumps During Loss-Of-Coolant Accident

Action Plan Requirements:

"Tripping of the reactor coolant pumps in case of a loss-of-coolant accident (LOCA) is not an ideal solution. Licensees should consider other solutions to the small-break LOCA problem (for example, an increase in safety injection flow rate). In the meantime, until a better solution is found, the reactor coolant pumps should be tripped automatically in case of a small-break LOCA. The signals designated to initiate the pump trip are discussed in NUREG-0623."

- NUREG 0737

CPNPP Response

II.K.3.5 TMI Action Plan Item II.K.3.5 has been evaluated in accordance with the guidelines set forth in NRC Generic Letter 83-10c. The Westinghouse Owner's Group (WOG) has performed a series of analyses to determine (1) what unique indicators exist that would distinguish between a SBLOCA and a SGTR, and (2) the 10 CFR 50, Appendix K and Best Estimate predictions of the Peak Cladding Temperatures (PCT) associated with the delayed Reactor Coolant Pump (RCP) trip scenario.

Reference 1 provided the basis for calculating the three (3) unique distinguishing identifiers that exist for CPNPP. The generic WOG Emergency Response Guidelines were then used to develop the plant specific Emergency Operating Procedure (EOP) for the scenario. The CPNPP EOPs will require a manual trip of the RCPs when conditions warrant the action. Reference 2 documents that even for the most limiting Westinghouse plant, the PCTs are below the limits established in NRC Generic Letter 83-10c. This result provides the justification for allowing a manual trip of the RCPs.

References:

(1) OG-110, dated 1 December 1983, J. J. Sheppard (WOG) to R. J. Mattson (NRC).

(2) OG-117, dated 12 March 1984, J. J. Sheppard (WOG) to R. J. Mattson (NRC).

II.K.3.9 Proportional Integral Derivative Controller Modification

Action Plan Requirements:

"The Westinghouse-recommended modification to the proportional integral derivative (PID) controller should be implemented by affected licensees."

- NUREG 0737

CPNPP Response

II.K.3.9 CPNPP has reconfigured the pressurizer power operated relief valve (PORV) controller for proportional-integral (PI) control action. Without derivative action the contribution to the error signal from reactor coolant pressure rate of change has been eliminated. This results in fewer challenges to the reactor protection system from spurious pressure signal noise.

II.K.3.10 Proposed Anticipatory Trip Modification

Action Plan Requirements:

"The anticipatory trip modification proposed by some licensees to confine the range of use to high-power levels should not be made until it has been shown on a plant-by-plant basis that the probability of a small-break loss-of-coolant accident (LOCA) resulting from a stuck- open power-operated relief valve (PORV) is substantially unaffected by the modification."

- NUREG 0737

CPNPP Response

II.K.3.10 The CPNPP design includes the P-9 interlock which blocks the direct reactor trip on a turbine trip at or below 50% of rated thermal power. Analyses have been performed which demonstrate that a turbine trip without direct reactor trip at or below 50% of rated thermal power does not pose any undue or additional challenges to the pressurizer PORVs. Assuming all control systems are operable, neither the RCS PORVs nor the steam generator ARVs reached their respective pressure setpoints. Even when single credible failures in the control systems are considered, the pressurizer PORVs do not reach their setpoints. Also see FSAR Section 7.2.

II.K.3.11 Justification of Use of Certain PORVs

Action Plan Requirements:

"Demonstrate that the PORV installed in the plant has a failure rate that is not significantly less than the valves for which there is an operating history."

- NUREG 0694, Pg. 18

CPNPP Response

II.K.3.11 The CPNPP Pressurizer Power Operated Relief Valves are Copes Vulcan Model No. D-100-160 globe valves. The report on failure rates required by II.K.3.2 will establish the operating history for these valves.

II.K.3.12 Confirm Existence of Anticipatory Reactor Trip Upon Turbine Trip

Action Plan Requirements:

"Licensees with Westinghouse-designed operating plants should confirm that their plants have an anticipatory reactor trip upon turbine trip. The licensee of any plant where this trip is not present should provide a conceptual design and evaluation for one installation of this trip."

- NUREG 0737

CPNPP Response

II.K.3.12 CPNPP has the anticipatory reactor trip on turbine trip as described in Section 7.2.1.1.2, item 6.

II.K.3.17 Report on Outages of Emergency Core-Cooling Systems Licensee Report and Proposed Technical Specification Changes

Action Plan Requirements:

"Several components of the emergency core-cooling (ECC) systems are permitted by technical specifications to have substantial outage times (e.g., 72 hours for one diesel-generator; 14 days for the HPCI system). In addition, there are no cumulative outage time limitations for ECC systems. Licensees should submit a report detailing outage dates and lengths of outages for all ECC systems for the last 5 years of operation. The report should also include the causes of the outages (i.e., controller failure, spurious isolation)."

Changes to Previous Requirements and Guidance

"This clarification adds the requirement to propose changes that will improve and control availability."

- NUREG 0737

CPNPP Response

II.K.3.17 ECCS Outages

The NRC has closed this item for CPNPP. The requirements of 10 CFR 50.72 and 50.73 and reporting on the NPRDS is adequate for reporting ECCS outages.

II.K.3.25 Effect of Loss of Alternating-Current Power on Pump Seals

Action Plan Requirements:

"The licensees should determine, on a plant-specific basis, by analysis or experiment, the consequences of a loss of cooling water to the reactor recirculation pump seal coolers. The

pump seals should be designed to withstand a complete loss of alternating-current (ac) power for at least 2 hours. Adequacy of the seal design should be demonstrated."

Changes to Previous Requirements and Guidance

"The evaluation and proposed modifications shall be submitted by July 1, 1981. The May 7, 1980 letter called for modifications by January 1, 1982. This clarification adds a documentation requirement for the evaluation to the submitted by July 1, 1981. The modification date remains unchanged. Additionally, this task has changed to include Westinghouse and Combustion Engineering operating reactors and operating reactor applicants."

- NUREG 0737

CPNPP Response

II.K.3.25 Loss of cooling water to the reactor coolant pump seal coolers due to loss of offsite power is not applicable to CPNPP because the Component Cooling Water supply to the Non-safeguards loop of the CCWS is through motor operated butterfly valves (1-HV-4526 and 4527 at location E-3 and 1-HV-4524 and 4525 at location D-3 on FSAR Figure 9.2-3) which fail-as-is on loss of power. As discussed in Section 5.4.1.3.1 the effect of loss of offsite power is to cause a temporary stoppage of the cooling water. Power is restored to the component cooling water pumps after 30 seconds (see Table 8.3-2). Therefore, cooling of the seal water heat exchanger and the reactor coolant pump packages described in FSAR Section 9.2.2.2 is restored in a timely manner on a loss of offsite power. The components described are covered under the appropriate items in Table 17A-1.

II.K.3.30 Revised Small-Break Loss-of-Coolant-Accident Methods to Show Compliance With 10 CFR Part 50, Appendix K

Action Plan Requirements:

"The analysis methods used by nuclear steam supply system (NSSS) vendors and/or fuel suppliers for small-break loss-of-coolant accident (LOCA) analysis for compliance with Appendix K to 10 CFR Part 50 should be revised, documented, and submitted for NRC approval. The revisions should account for comparisons with experimental data, including data from the LOFT Test and Semiscale Test facilities."

Changes to Previous Requirements and Guidance

"The changed requirement (1) allows for justification of acceptability of present small-break LOCA models by comparison with test date, and (2) requests each licensee to outline scope and schedule for model revision or comparison with test data by late fall, 1980. The original requirement did not allow provision for showing acceptability of present models by comparison with plant data."

- NUREG 0737

CPNPP Response

II.K.3.30 TUGCO has participated, as a member of the Westinghouse Owners Group in the development of the modified SBLOCA code NOTRUMP. A technical description of the code was submitted to the NRC via References 1 and 2. A favorable SER was released on May 21, 1985, Reference 3.

References

(1) Westinghouse Letter, E. P. Rahe to C. O. Thomas, NS-EPR-2694, December 22, 1982. (Transmitting WCAP- 10054)

(2) Westinghouse Letter, E. P. Rahe to C. O. Thomas, NS-2681, November 12, 1982. (Transmitting WCAP-10079)

(3) Safety Evaluation. TMI Action Item II.K.3.30 for Westinghouse Plants attached to NRC letter V. S. Noonan to M. D. Spence dated June 6, 1985.

II.K.3.31 Plant-Specific Calculations to Show Compliance with 10 CFR Part 50.46

Action Plan Requirements:

"Plant-specific calculations using NRC-approved models for small-break loss-of-coolant accidents (LOCAs) as described in item II.K.3.30 to show compliance with 10 CFR 50.46 should be submitted for NRC approval by all licensees."

- NUREG 0737

CPNPP Response

II.K.3.31 In Reference 1, the NRC Staff indicated that the resolution of TMI Action Plan Item II.K.3.31 may be accomplished by generic analyses to demonstrate that the previous NRC approved WFLASH SBLOCA results were conservative when compared with the new NOTRUMP SBLOCA. Such generic studies were undertaken by the Westinghouse Owners Group (WOG) of which Texas Utilities is a participating member. The WOG completed these generic studies and has submitted the results of the analyses to the NRC in the topical report WCAP-11145 (Reference 2). Topical report WCAP-11145 is therefore referenced in order to demonstrate that the requirements to TMI Action Plan II.K.3.31 are satisfied for CPNPP in a generic fashion in accordance with Reference 1.

Topical report WCAP-11145 documents the results of a series of Small Break LOCA (SBLOCA) analyses performed with the NRC approved NOTRUMP SBLOCA Evaluation Model. Cold leg break spectrum analyses were performed for the limiting SBLOCA plant form each of the Westinghouse 4- loop, 4-loop Upper Head Injection (UHI), 3-loop, and 2- loop plant categories. The limiting SBLOCA plant in each category was defined on the basis of previous SBLOCA analyses which were performed with the WFLASH SBLOCA code. In addition to

the cold leg break spectrums, a hot leg and pump suction break were performed as part of the 4- loop plant analyses, confirming that the cold leg was still the worst break location. Comparison of the NOTRUMP cold leg break spectrum results with the previously generated WFLASH results, showed that the WFLASH results were conservative for all plant categories. In particular, for the previous bounding 4-loop peak clad temperature (PCT) plant the NOTRUMP SBLOCA code calculated a PCT which was 537 °F lower than the WFLASH code calculated value. This result is consistent with the other generic studies which indicated that the NOTRUMP SBLOCA calculated PCTs tended to be significantly lower than the previously bounding WFLASH PCTs.

The generic results documented in WCAP-11145, demonstrate that a plant specific reanalysis of CPNPP with the NOTRUMP SBLOCA should result in the calculation of a limiting PCT which would be significantly lower than the 1788°F PCT currently calculated with the WFLASH. Hence, the WFLASH results which currently form the licensing basis for CPNPP are conservative and still valid for demonstrating the adequacy of the Emergency Core Cooling System to mitigate the consequences of a SBLOCA, as required by 10CFR50.46. It is therefore concluded that a plant specific analysis is not needed in order for CPNPP to comply with TMI Action Item II.K.3.31. Rather, Texas Utilities references WCAP-11145 in order to comply with TMI Action Item II.K.3.31 on a generic basis in accordance with Reference 1.

References

1) NRC Generic Letter 83-35 from D. G. Eisenhut, “Clarification of TMI Action Plan Item II.K.3.31”, November 2, 1983.

2) L. D. Butterfield letter to J. Lyons, “Westinghouse Owner’s Group Transmittal of WCAP-11145”, OG-190, June 11, 1986.

III EMERGENCY PREPAREDNESS AND RADIATION EFFECTS

III.A EMERGENCY PREPAREDNESS AND RADIATION EFFECTS

III.A.1 IMPROVE LICENSEE EMERGENCY PREPAREDNESS - SHORT TERM

OBJECTIVE:

"Promptly improve and upgrade licensee emergency preparedness by requiring improvements in facilities, plans, procedures, offsite support, technical assistance, equipment and supplies required to adequately respond to and manage an accident."

- NUREG 0660, Pg. III.A.1-1

III.A.1.1 Upgrade Emergency Preparedness

Action Plan Requirements:

"Licensees will upgrade emergency preparedness in accordance with the requirements described in the NRC "Action Plan for Promptly Improving Emergency Preparedness" (SECY 79-450), which was distributed to all licensees during regional meetings in August 1979, and in accordance with subsequently issued acceptance criteria (NUREG-0654). These actions include:

(1) Preparing and submitting upgraded plans which satisfy the NRR supplemental acceptance criteria provided by the NRC emergency preparedness review teams, with special attention to the establishment of emergency action levels in accordance with NUREG-0610, "Basis for Emergency Action Levels for Nuclear Power Facilities."

(2) Implementing the short-term emergency planning recommendations of NUREG-0578.

(3) Establishing an onsite Technical Support Center, an onsite Operational Support Center, and a near-site Emergency Operations Facility.

(4) Establishing improved offsite radiological monitoring capability in accordance with the NRR/RAB technical position.

(5) Providing planning assistance to appropriate Federal, State, and Local governments to assure that their emergency response roles are properly coordinated with the facility plan and that such plans satisfy the NRC acceptance criteria.

(6) Providing resources as necessary to State and Local governments for implementing the emergency planning zone concept, in accordance with NUREG-0396, "Planning Basis for the Development of State and Local Government Radiological Emergency Response Plans in Support of Light Water Nuclear Power Plants."

(7) Participating in periodic joint exercises involving Federal, State, and Local government emergency response organizations.

- NUREG 0660, Pg. III.A.1-9

"Provide an emergency response plan in substantial compliance with NUREG-0654, "Criteria for Preparedness and Evaluation of Radiological Emergency Response Plans and Preparedness in Support of Nuclear Power Plants" (which may be modified after May 13, 1980 based on public comments except that only a description of and completion schedule for the means for providing prompt notification to the population (App. 3), the staffing for emergencies in addition to that already required (Table B.1), and an upgraded meteorological program (App. 2) need be provided. NRC will give substantial weight to FEMA findings on offsite plans in judging the adequacy against NUREG-0654. Perform an emergency response exercise to test the integrated capability and a major portion of the basic elements existing within emergency preparedness plans and organizations."

- NUREG 0694, Pg. 25

CPNPP Response

1) The CPNPP Emergency Plan has been written in accordance with the draft version of NUREG-0654 and was submitted to the NRC in October 1980 for review.

2) For implementation of NUREG-0578, 2.1.8.a, Post Accident Sampling, see FSAR Section II.B.3.

   For implementation of NUREG-0578, 2.1.8.b, Accident Monitoring Instrumentation, see FSAR Section II.F.1.

   For implementation of NUREG-0578, 2.1.8.c, Improved Inplant Iodine Instrumentation, see FSAR Section III.D.3.3.

3) See FSAR Section III.A.1.2 and CPNPP Emergency Plan, Section 6.0.

4) See CPNPP Emergency Plan, Section 7.0.

5) Planning assistance is provided to State and Federal agencies upon request. Local planning assistance is provided by Vistra OpCo.

6) Resources to implement the State and Local emergency plans will be made available to those agencies.

7) The State and Local emergency plans provide for periodic joint exercises with CPNPP.

III.A.1.2 Upgrade Licensee Emergency Support Facilities

Action Plan Requirements:

"A separate technical support center shall be provided for use by plant management, technical, and engineering support personnel. In an emergency, this center shall be used for assessment of plant status and potential offsite impact in support of the control room command and control function. The center should also be used in conjunction with implementation of onsite and offsite emergency plans, including communications with an offsite emergency response center. Provide at the onsite technical support center the as-built drawings of general plant arrangements and

piping, instrumentation, and electrical systems. Photographs of as-built system layouts and locations may be an acceptable method of satisfying some of these needs."

"Each operating nuclear power plant should establish and maintain a separate onsite operational support center outside the control room. In the event of an emergency, shift support personnel (e.g., auxiliary operators and technicians) other than those required and allowed in the control room shall report to this center for further orders and assignment."

- NUREG 0578, Pg. 13

Also see NRC Letter, dated April 25, 1980; August 1, 1980; and September 5, 1980.

CPNPP Response

I. INTRODUCTION

CPNPP established emergency response facilities (ERFs), consisting of a Technical Support Center (TSC), an Operational Support Center (OSC), and an Emergency Operations Facility (EOF), prior to issuance of an operating license. These facilities meet the guidelines of NUREG 0578, 0654, 0694, and also meet the intent of NUREG 0696 "Functional Criteria for Emergency Response Facilities," as stated in the description below.

The ERFs have been upgraded by incorporation of an integrated data acquistion and display system. This system provides data to the TSC and EOF and also provides the Safety Parameter Display System (SPDS) displays to the Control Room, TSC and EOF. A description of this system is provided in the discussion below.

The CPNPP Emergency Plan has been revised to describe the existence and function of the current CPNPP emergency response facilities; therefore, facility description in Sections IV, V and VI below for the TSC, OSC and EOF, respectively, are not described in FSAR Section III.A.1.2 but are relocated to the CPNPP Emergency Plan.

II. EMERGENCY RESPONSE FACILITY INSTRUMENTATION, DATA SYSTEM EQUIPMENT, AND POWER SUPPLIES

The data display equipment provided in the TSC and EOF is part of the integrated Emergency Response Facility (ERF) Computer System. As shown in Figure III.A.1.2-2, the integrated ERF Computer System gathers, stores and displays data needed in the TSC and EOF to analyze the plant conditions. The system performs its function independent of action in the Control Room and without degradation or interfering with Control Room and plant functions. Two of four TSC displays are powered by a Non-1E battery backed, Uninterruptable Power Supply (UPS) System; two TSC displays are powered from a Non-Class 1E 120 VAC common supply; EOF displays are powered from an AC source (the 25 KV Plant Support Power Loop).

Signals to the integrated ERF Computer System, which are received from sources providing signals to safety-related equipment or displays, are isolated to ensure that the system will not degrade performance of the safety-related equipment or displays.

Power supplied to the ERF computer will be provided from a Non-1E Battery UPS System to insure the continuity of TSC functions if temporary loss of primary TSC Power sources occurs. Upon loss of all A/C power or a seismic event, the ERF computer will not be functional; however, plant process data can be acquired from other sources.

The design goal for the data system reliability is to achieve an operational unavailability goal of 0.01 during all plant operating conditions above cold shutdown.

The SPDS display equipment used in the TSC and EOF is not seismically qualified; it will have the same design goal of meeting the TSC and EOF data system reliability and performance criteria. The design of the TSC and EOF data system equipment incorporates human factors engineering with consideration for both operating and maintenance personnel. The TSC and EOF display equipment discribed herein is included under the appropriate items in Table 17A-1.

III. TECHNICAL DATA AND DATA SYSTEM

The ERF Computer System receives, stores, processes, and displays information acquired from different areas of the plant as needed to perform the TSC and EOF function. The ERF Computer System provides access to accurate and reliable information sufficient to determine:

1. Plant steady-state operating conditions prior to an accident.

2. Transient conditions producing the initiating event.

3. Plant systems dynamic behavior throughout the course of an accident.

The data set available to the data system is complete enough to permit accurate assessment of an accident without interference with the Control Room emergency operation.

The data set available to the ERF Computer System includes Category 1 and applicable Category 2 Type A, B, C and E Accident Monitoring variables (as described in FSAR Section 7.5) and Type D Accident Monitoring variables which are also identified as Type D variables in Regulatory Guide 1.97, Rev. 2, Table 2.

All sensor data and calculated variables used in the data set for SPDS, EOF, or for transmission to offsite locations are available for display. The accuracy of the data displayed is substantially the same as the accuracy of comparable data displayed in the Control Room. The time resolution of data acquisition is sufficient to provide data without loss of information during transient conditions where that data is necessary for operator action. The time resolution of each sensor signal will respond on the potential transient behavior of the variable being measured.

The ERFCS contains a long-term archival system with disk storage for the TSC and EOF data set. At least one fuel cycle of data is stored by the ERFCS. The sample frequency is consistent with use of the data. Archival data storage is provided automatically and retrieval of historical data is incorporated in the standard HMI display package located on every TSC and EOF workstation and does not impact ERFCS data acquisition function.

A sufficient number of display and printout devices is provided in the TSC and EOF to allow designated personnel to perform their assigned tasks. The displays include alphanumeric and graphical representations of:

1. Plant system variables

2. In-plant radiological variables

3. Meteorological information

Trend information and time history display capability are available to personnel in the TSC and EOF. The displays are designed so that call up, manipulation, and presentation of data can easily be performed. The data display presents information in easily understood formats.

The SPDS can also be displayed in the TSC and EOF.

IV. TECHNICAL SUPPORT CENTER (TSC)

See the CPNPP Emergency Plan, Section 6 and specifically 6.2.

V. OPERATIONAL SUPPORT CENTER (OSC)

See the CPNPP Emergency Plan, Section 6 and specifically 6.3.

VI. EMERGENCY OPERATIONS FACILITY (EOF)

See the CPNPP Emergency Plan, Section 6 and specifically 6.4.

VII. SAFETY PARAMETER DISPLAY SYSTEM (SPDS)

A. FUNCTION

The purpose of the Safety Parameters Display System (SPDS) is to assist control room personnel in evaluating the safety status of the plant. The SPDS provides a continuous high level graphical display of plant parameters or derived variables representative of the safety status of the plant.

As an operator aid, SPDS serves to concentrate a minimum set of plant parameters from which the plant safety status can be assessed. More detailed plant information will be provided by several secondary displays.

Human factor engineering is incorporated in the SPDS design to enhance the functional effectiveness of the control room personnel.

The SPDS will be operational during normal, as well as, emergency conditions except for loss of all A/C power or seismic events. However, data acquisition will remain available from other sources. The SPDS will be able to display pertinent information concerning steady-state and transient conditions.

All portions of the SPDS have been designed and specified such that all design basis safety questions have been considered.

Revision 3 to the SPDS Safety Analysis Report contains further supporting information and analysis, and was transmitted via Reference [1].

B. LOCATION

SPDS displays are available in the Control Room, TSC, and EOF. The SPDS location in the control room will be visible to both the control room operator and the Senior Reactor Operator.

C. SIZE

The SPDS will be compatible with the existing space in the control room area and will not interfere with normal movement or with full visual access to other control room operating systems or displays.

D. STAFFING

The SPDS will be of such design that no operating personnel, in addition to the normal control room operating staff, are required for its operation.

E. DISPLAY CONSIDERATIONS

Vistra OpCo was a member of the Safety Assessment System committee to develop and implement the necessary displays to meet the requirements for the SPDS. The committee consisted of Quadrex Corporation and eleven other utilities owning Westinghouse PWRs. The Luminant Power implementation of the generic SAS design has been integrated into the Emergency Response Facility Computer System (ERFCS). The following discussion serves as a general description of the SPDS portion of the ERFCS.

1. General Considerations

The ERFCS satisfies the requirements of the Safety Parameter Display System (SPDS). This section describes the portion of the ERFCS which meets the SPDS requirements of NUREG-0696 and the intent of NUREG-0737 Supplement 1. The ERFCS provides a centralized, flexible, computer-based data and display system to assist control room personnel in evaluating the safety status of the plant. This assistance is accomplished by providing the operator and also Emergency Response Facilities (ERFs) a high-level graphical display containing a minimum set of key plant parameters representative of the plant safety status. More detailed plant information is provided by several secondary displays. All graphical displays are presented to the control room operator on high resolution multiple color monitors.

All data displayed by the SPDS is validated by comparing redundant sensors, checking the value against reasonable limits, calculating rates of change, and/or checking temperature versus pressure curves.

All displays on the SPDS have been carefully designed by persons with plant operating experience and evaluated against human factors design criteria. The intent of the SPDS is to present to the control room personnel a few easily understandable displays which use color coding and pattern recognition techniques to indicate off-normal values.

These displays are updated and validated on an essentially real-time basis.

The SPDS will be operable during normal and abnormal plant operating conditions. The SPDS will operate during all SPDS required modes of plant operation. The normal operation mode will encompass all plant conditions at or above normal operating pressure and temperature. When the reactor coolant system is intentionally cooled below normal operating values, the operator will select the Heatup-Cooldown mode which alters the limit checking algorithm for the key parameters. An additional mode may be provided to address concerns of cold shutdown plant conditions.

2. Display Hardware Locations and Operation

The SPDS portion of the ERFCS has been incorporated on every Plant Computer workstation located in controlled locations of the control room visible to the control room operator and the Senior Reactor Operator. Every Plant Computer workstation contains the high-level display from which the overall safety status of the plant may be assessed. The standard Human-Machine Interface package located on every Plant Computer workstation allows operator selection of several predetermined second level (trend) displays at any time.

The SPDS has been designed such that control room personnel can utilize its features without requiring additional operations personnel.

The SPDS displays have been provided to other ERFs, such as the Technical Support Center and Emergency Operations Facility.

3. Display Contents

The primary display consists of bar graphs of selected parameter values, digital status indicators for important safety system parameters and digital values. The parameters indicated by bar graphs and digital values include: RCS pressure, RCS temperature, pressurizer level, steam generator levels and steam generator pressures. Status indicators are provided for containment environment, secondary system radiation, and reactor vessel water level. Core exit temperature, amount of subcooling and containment radiation are indicated by digital values.

In addition, there is a message area which will be used to indicate that an appropriate secondary display provides further information in case an off-normal value is detected or an event is occurring.

Each of the bar graphs indicate wide-range values. If a parameter's value is outside the normal range, the bar color will turn yellow, then red. Arrows next to

the bar indicate the trend direction (increasing or decreasing) based on data smoothing algorithms.

Secondary displays may be selected by the operator. Trend graph groups of selected parameters, showing the last thirty minutes of plant operation are available. These trend groupings were chosen to keep like parameters or related parameters on one display "page."

4. Human Factor Engineering

Human factors engineering and industrial design techniques have been effectively combined to establish man-machine interface design requirements, maximize system effectiveness, reduce training and skill demands, and minimize operator error.

The monitor color graphic formats and functional keyboard designs have been developed through an interdisciplinary team of senior operational, human factors, industrial design and computer interface personnel.

Minimum use of color combined with simplified format through the monitor presentation have key design features to provide both normal and off-normal pattern recognition. The operator, who is the end user, has been directly involved from the conception to insure that man-machine interface goals of SPDS have been satisfied. Human factor engineering standards and testing verification have been used which are consistent with accepted practices.

5. Validation and Verification

The SPDS is implemented on a digital computer system which includes a peripheral display generator computer for color graphic displays. The software that controls the sensor data validation, key parameter construction, and display formats has been developed under a software quality assurance program based on guidance from ASME NQA-1 Subpart 2.7. Verification and validation activities have been performed on the system using NSAC-39, ANSI/ANS 10.4-1987, and IEEE 1012-1986 as guidelines.

F. DESIGN CRITERIA

The total SPDS will not be Class 1E or meet the single-failure criterion. The sensors and signal conditioners (such as preamplifiers, isolation devices, etc.) will be designed and qualified to meet Class 1E standards for those SPDS parameters that are also used by safety systems.

The control room operating staff will be provided with sufficient information and criteria to allow for performance of an operability evaluation of SPDS if an earthquake should occur.

The SPDS, as used in the control room, will be designed to an operational unavailability goal of 0.01. The cold shutdown unavailability goal for the SPDS during the cold shutdown and refueling modes for the reactor will be 0.2.

The unavailability goal of 0.01 is more stringent than can be reasonably achieved without some redundancy. Therefore, dual processing nodes in a Primary/Backup configuration, data multiplexors, and other critical peripherals have been installed.

The isolation of the SPDS from safety-related devices is described as follows for each type of safety-related device that has input to the SPDS computer system (e.g. multiplexed analog and digital inputs, core cooling monitor inputs, radiation monitoring system inputs, reactor vessel level indicaiton system inputs).

1. Multiplexed analog and digital inputs are obtained from various systems throughout the plant (e.g., analog signals from tank levels, system pressures, flow rates, etc., digital signals from valve postion, breaker position, etc.). All multiplexed inputs from safety-related devices are routed to their respective (Train A or B) qualified multiplexer cabinets (MUX) located in the environmentally protected cable spreading room. Since the signals from safety-related devices are utilized in the non-qualified Plant Computer System (ERFCS, SPDS), an appropriate qualified isolation device was chosen that would electrically isolate signals from qualified SSCs from the Plant Computer System. All credible adverse conditions were considered and tested per specifications given in the qualification test reports. The isolation devices are in compliance with applicable NRC guidelines and IEEE standards for environmental and seismic qualification, and the PCS isolators are designed and qualified for fail-safe operation.

2. Inputs to the SPDS from the core exit thermocouples and the margin to saturation indicator are by way of the two (Train A and B) qualified safety-related Core Cooling Monitors (CCMs). Electrical isolation between the CCMs and the SPDS is accomplished by means of dc to dc converters and optical isolators. DC to dc converters provide isolated dc power to the non-Class 1E CCM circuits by isolating the Class 1E power supply from the non-Class 1E circuits. Optical isolators provide electrical isolation between the Class 1E input signal circuitry and the non-Class 1E output signal circuitry. Testing was performed to verify that the input side of the isolator suffered no degradation as a result of faults applied to the isolator output. The application of the maximum credible fault voltage and current meets this criterion, thus assuring the adequacy of the dc to dc converter and optical isolator as an isolation device for this application. Adequacy was determined by vendor testing, which is documented in Reference [2].

3. All inputs to the SPDS from the Radiation Monitoring System (RMS) are transmitted as ASCII data from the RMS computer system, except for containment radiation inputs which are routed to the multiplexers as described in Item 1. The RMS computers are non-safety-related devices; therefore, no provision for electrical isolation between them and the SPDS is required.

4. All inputs to the SPDS from the qualified Reactor Vessel Level Indication System (RVLIS) are transmitted as ASCII data through an FOC data link from the qualified, safety-related RVLIS cabinet to the SPDS computer system. All credible adverse conditions were considered and tested per specifications given in the qualification test reports. The isolation devices are in compliance with applicable NRC guidelines and IEEE standards for environmental and seismic qualification, and the entire RVLIS system is designed for fail-safe operation.

The power supply will be from Non-1E Battery, UPS supply.

Expandability in both numbers of parameters and processing power will be provided.

Verification and validation are applied as appropriate.

References

1) Letter from W. J. Cahill, Jr. (TUElectric) to the NRC, "Submittal of the SPDS Safety Analysis Report, Revision 3", TXX-89531, July 31, 1989.

2) Letter from W. J. Cahill, Jr. (TUElectric) to the NRC, "Submittal of Core Cooling Monitor (CCM) Isolator Test Report", TXX-89079, February 17, 1989.

VIII. INTEGRATED ERF COMPUTER SYSTEM

A. GENERAL DESCRIPTION

In general, the CPNPP integrated ERF computer system consists of a system configuration as shown in Figure III.A.1.2-2. The overall system principally consists of:

1. Data Acquisition System (DAS) units that isolate, digitize, and transmit field sensor signals to the ERF computers.

2. Redundant processing nodes will provide the data processing/distribution/ and record keeping functions required. The processing nodes will be powered from a highly reliable Non-1E Battery/UPS System.

3. For each unit the Human/Machine Interface (HMI) network will consist of sixteen (16) PCS workstations which implement the ERF/SPDS displays generated by the Westinghouse Ad-hoc Safety Assessment System Group. Nine workstations including four monitors on the Main Control Boards will be located in the control room, four in the TSC, and three in the EOF. All of the workstations will have full functional capability to display ERF/SPDS/Plant parameters and displays. The control room, and two of the TSC workstations will be powered from Non-1E, Battery/UPS power supplies. The rest of the workstations are powered from a Non-Class 1E 120 VAC common supply.

   The integrated ERF Computer System reliability design goal is to achieve 0.01 unavailability during all plant operating modes above cold shutdowns.

B. IMPLEMENTING SCHEDULE

The ERF Computer hardware and software was installed with a sufficient set of input variables to accomplish the SPDS and Regulatory Guide 1.23 functions by fuel load.

III.A.2 IMPROVING LICENSEE EMERGENCY PREPAREDNESS--LONG TERM

OBJECTIVE:

To upgrade the emergency preparedness of nuclear power plants.

- NUREG 0660

Action Plan Requirements:

Each nuclear facility shall upgrade its emergency plans to provide reasonable assurance that adequate protective measures can and will be taken in the event of a radiological emergency. Specific criteria to meet this requirement is delineated in NUREG-0654 (FEMA-REP-1), “Criteria for Preparation and Evaluation of Radiological Emergency Response Plans and Preparation in Support of Nuclear Power Plants.”

- NUREG 0737

CPNPP Response

The CPNPP Emergency Plan is currently being revised to reflect requirements expressed in NUREG-0654, Revision 1. Equipment, facilities and instruments necessary to implement the Emergency Plan are subject to an Operations Quality Assurance Program. Activities necessary to implement the Emergency Plan are controlled by approved procedures and subject to the proper QA controls and surveillances.

Prior to fuel load of Unit One, the meteorological measurements program for CPNPP shall consist of the following:

1) A primary meteorological measurements program;

2) A backup meteorological measurements system;

3) A system for making near real-time predictions of the atmospheric effluent transport and diffusion;

4) A capability for remote interrogation, on demand, of the atmospheric measurements and prediction systems by CPNPP emergency response organizations and the NRC staff.

III.A.3 IMPROVING NRC EMERGENCY PREPAREDNESS

OBJECTIVE:

To enable NRC, in the event of a nuclear accident at a licensed reactor facility, to (1) monitor and evaluate the situation and potential hazards, (2) advise the licensee’s operating staff as needed, and (3) in an extreme case, be able to issue orders governing such operations.

III.A.3.3 Communications

Action Plan Requirements

Install direct dedicated telephone lines between each plant and the NRC Operations Center.

- NUREG 0694, Pg. 27

CPNPP Response

The Federal Telecommunications System (FTS) 2000 network has been installed at CPNPP for communications with the NRC Operations Center.

III.D RADIATION PROTECTION

III.D.1 RADIATION SOURCE CONTROL

OBJECTIVE:

"Perform evaluations to establish additional design features that should be included in the rulemaking proceeding of Item II.B.8. The purpose of these evaluations is to identify design features that will reduce the potential for exposure to workers at nuclear power plants and to offsite populations following an accident."

-NUREG 0660, Pg. III.D.1-1

III.D.1.1 Primary Coolant Sources Outside The Containment Structure

Action Plan Requirements:

"Applicants shall implement a program to reduce leakage from systems outside containment that would or could contain highly radioactive fluids during a serious transient or accident to as-low-as-practical levels. This program shall include the following:

(1) Immediate leak reduction

(a) Implement all practical leak reduction measures for all systems that could carry radioactive fluid outside of containment.

(b) Measure actual leakage rates with system in operation and report them to the NRC.

"(2) Continuing Leak Reduction -- Establish and implement a program of preventive maintenance to reduce leakage to as-low-as-practical levels. This program shall include periodic integrated leak tests as intervals not to exceed each refueling cycle."

- NUREG 0737

CPNPP Response

The residual heat removal (RHR) system; portions of the containment spray (CS) system; portions of the safety injection (SI) system; portions of the chemical and volume control (CVCS) system and the RCS Sampling System (post accident sampling system portion only until such time as a modification eliminates the PASS penetration as a potential leakage path) have been identified as systems which process primary coolant, and could contain high level radioactive materials. (See Section II.B.2.2.) Programs will be implemented to reduce and maintain leakage to as-low-as-practical levels. These programs will include but not be limited to the requirements of the ASME Boiler and Pressure Vessel Code, Section XI.

1. The RHR, CS, SI, and CVCS systems are ASME Code Class 2 and 3, and are subject to the in-service inspection requirements of the ASME Boiler and Pressure Vessel Code, Section XI, including pressure tests.

2. At intervals not exceeding refueling outages, Radioactive System Leakage Inspection (RSLI) tests will be performed on appropriate portions of the RHR, SI, CS, CVCS and RCS Sampling Systems.

3. Excessive leakage into controlled areas will also be indicated by abnormally high airborne radioactivity levels.

4. Plant Initial Leakage Rates

   The initial leakage rates of the system to be tested will be determined from the system hydrostatic (or pneumatic) test which is conducted prior to plant operations and/or the initial Radioactive System Leakage Inspection (RSLI) test.

5. Test Methods

   The primary testing method will be by system walkdown while at normal operating pressure with quantified measurements obtained at all visually observed leakage paths. Systems or subsystems not readily testable in this manner will require either leakage makeup to maintain test pressure testing or pressure drop testing. Actual testing experience may suggest alternate methods which may be incorporated.

6. Limiting Leakage Value

   The effort at CPNPP will be to minimize the overall leakage of fluid from potentially radioactive systems to the environment. Therefore, the limiting leakage value will be based on a cumulative total of the systems within each Unit. The leakage will be limited to a maximum of 1 gpm/Unit.

7. Maintaining Leakage Value

   While shutdown or during operations, the affected systems will receive a periodic inspection for leakage at intervals not to exceed each refueling cycle. The leak tests will be scheduled to coincide with routine system operability tests whenever possible during operations. To minimize the actual leakage rate, packings or seals with provisions for adjustment will be adjusted whenever leakage is noted during the inspection. Replacements of seals, gaskets, o-rings, etc., will be accomplished on a selective basis to prevent exceeding the limiting leakage value.

   Should the limiting leakage value be exceeded, a reinspection and leak test will be conducted on the system(s) with high leakage rates. Excessive leakage paths will be identified and evaluated. Areas of continued high leakage will be considered for modification or improvement in repair methods.

   Modifications or repairs will be accomplished as quickly as possible. Consideration as to the type of leakage, fluid involved, leakage location and collectability will influence repair or modification schedules.

8. Incidence of Leakage at North Anna

A reviews and comparison of the designs of North Anna and Comanche Peak as pertains to the leakage incidence indicates that the design of the Comanche Peak volume control tank relief system is sufficiently different to preclude a similar occurrence. In addition a further review of procedures and as-built configurations will be performed as they become available. Areas or activities which could result in release of radioactive material will be evaluated for modification.

III.D.3 WORKER RADIATION PROTECTION IMPROVEMENT

OBJECTIVES:

“Improve nuclear power plant worker radiation protection to allow workers to take effective action to control the course and consequences of an accident, as well as to keep exposures as low as reasonably achievable (ALARA) during normal operation and accidents, by improving radiation protection plans, health physics, inplant radiation monitoring, control room habitability, and radiation worker exposure data base.”

III.D.3.3 Improved Inplant Iodine Instrumentation Under Accident Conditions

Action Plan Requirements:

"(1) Each licensee shall provide equipment and associated training and procedures for accurately determining the airborne iodine concentration in areas within the facility where plant personnel may be present during an accident.

"(2) Each applicant for a fuel-loading license to be issued prior to January 1, 1981 shall provide the equipment, training, and procedures necessary to accurately determine the presence of airborne radioiodine in areas within the plant where plant personnel may be present during an accident.”

- NUREG 0737

CPNPP Response

An in-plant radiation monitoring program shall be established, implemented and maintained to ensure the capability to accurately determine the airborne/iodine concentration in vital areas under accident conditions. This program shall include provisions for the training of personnel, procedures for monitoring, and provisions for maintenance of sampling and analysis equipment.

During accident conditions portable grab sampling equipment will be used to obtain iodine samples. These samples may be taken in all areas occupied by plant personnel.

The portable grab sampling equipment used consists of a high volume air sampler, a sample cartridge head and a silver zeolite iodine cartridge. This equipment is identical to the equipment normally used for routine iodine sampling. The only difference is the type of collection media used (silver zeolite versus charcoal).

After the iodine sample is obtained, the silver zeolite cartridge will be surveyed and counted in accordance with applicable Emergency Plan Procedures. Specific calculations contained in these procedures will allow personnel taking samples to determine if iodines are present. All used silver zeolite cartidges will be properly labeled, bagged and retained for further analysis, if required. Should additional analysis be required, the selected samples will be transported to the Chemistry Hot Lab. In the Hot Lab, the silver zeolite cartridges will be prepared for counting, then transferred to the Counting Room for isotopic analysis.

In the Counting Room, the silver zeolite cartridge will be counted on a Gamma Spectroscopy System consisting of: a HPGe detector, Multi-Channel Analyzer and Process Computer. The process computer identifies the iodine isotopes present, performs the calculations, and prints out a listing of all iodine isotopes identified along with their respective concentrations.

Several other lab facilities can be made available on short notice to provide back-up radiological analysis of post accident samples. These other facilities are listed in Section 6.8 of the CPNPP Emergency Plan. Availability and location of non-portable radiation detection/measuring equipment is listed in FSAR Table 12.5-1. Both stationary and portable monitors are commercial grade, and subject to an appropriate Operations QA Program.

All procedures and instructions associated with in-plant iodine monitoring under accident conditions shall be prepared, approved, and implemented prior to Unit 1 fuel load.

All training programs and procedures associated with sample acquisition, sample quantification and data evaluation pertaining to in-plant iodine monitoring activities under accident conditions shall be prepared, approved, and implemented prior to Unit 1 fuel load.

III.D.3.4 Control Room Habitability Requirements

Action Plan Requirements:

“In accordance with Task Action Plan item III.D.3.4 and control room habitability, licensees shall assure that control room operators will be adequately protected against the effects of accidental release of toxic and radioactive gases and that the nuclear power plant can be safely operated or shut down under design basis accident conditions (Criterion 19, “Control Room,” of Appendix A, “General Design Criteria for Nuclear Power Plants,” to 10 CFR Part 50).”

- NUREG 0737

CPNPP Response

A reevaluation of control room habitability has been performed.

Sections 2.2.1, 2.2.2, 2.2.3 and 6.4 of the Standard Review Plan (SRP) were reviewed for compliance and are discussed as follows:

1) FSAR Sections 2.2.1 and 2.2.2 is in compliance with SRP Sections 2.2.1 and 2.2.2.

2) FSAR Section 2.2.3 is in compliance with SRP Section 2.2.3.

3) FSAR Section 6.4 shows that the present CPNPP design for the control room is in compliance with SRP Section 6.4. An analysis was performed to evaluate the effect of a postulated steam generator blowdown piping break in the Electrical and Control Building on the Control Room. The results of the analysis indicate that the Control Room is not affected.

Information required by NUREG-0737, III.D.3.4, Attachment 1, is provided below:

(1) The control room ventilation mode of operation for radiological accident isolation is described in FSAR Section 6.4.2.

(2) Control room characteristics

(a) The control room air volume is provided in FSAR Section 15.6.5.4 (item 4(j)).

(b) The control room envelope (i.e., emergency zone) is described in FSAR Section 6.4.

(c) The control room ventilation system schematic is provided by FSAR Figure 9.4-1.

(d) Control room infiltration leakage is discussed in FSAR Sections 6.4.2.3 and 9.4.1.2.

(e) HEPA filter and charcoal adsorber efficiencies are provided in FSAR Section 15.6.5.4 (item 4(b)).

(f) The distance between the containment and air intake is provided in FSAR Section 15.6.5.4 (item 4(g)).

(g) Layouts are provided in FSAR Figures 1.2-1, 1.2-33, 1.2-34 and 6.4-3.

(h) Control room shielding is discussed in FSAR Section 6.4.2.5.

(i) Automatic isolation capability is discussed in FSAR Sections 6.4.2.2 and 9.4.1.2.

(j) Toxic chemicals are discussed in FSAR Section 2.2.3.

(k) Portable self-contained breathing apparatus (SCBA) are provided in the control room. There will be an adequate supply of air to sustain the five-man emergency team for a six-hour period. At least one SCBA will be provided for each member of the emergency team. There will be one additional SCBA to serve as a spare in case one unit fails.

(l) Bottled air supplys are discussed in FSAR Section 6.4.1.3.

(m) Emergency food and water supplies are provided in FSAR Section 6.4.1.1.

(n) The Shift Manager controls the number of personnel allowed into the control room during both normal and emergency conditions. The personnel capacity is

dependent on specific conditions at the time in question. The minimum capacity under design basis accident conditions is five men for a period of five days.

(o) Potassium iodide (KI) will be available to plant emergency personnel. The details of KI use and availability are controlled in accordance with provisions of the CPNPP Emergency Plan.

(3) Onsite Storage of Chlorine and Other Hazardous Chemicals

Toxic chemicals as discussed in FSAR Section 2.2.3.

(4) Offsite manufacturing, storage, or transportation facilities of hazardous chemicals is discussed in FSAR Section 2.2.

(5) Technical Specifications for CPNPP will comply with the Standard Technical Specifications for Westinghouse Pressurized Water Reactors (NUREG-0452) for the control room emergency filtration.

Items described in this response are listed, as appropriate, in Table 17A-1 under Items 23, 36, 37, 38 and 41.

COMANCHE PEAK NUCLEAR POWER PLANT
FINAL SAFETY ANALYSIS REPORT (FSAR)

PREFACE

The Effective Page Listing indicates which pages, tables and figures are current in the CPNPP/ FSAR. All pages/sheets within a FSAR section or FSAR table have an amendment number in the footer. Each sheet of a FSAR figure, which is not an “M” or “E” series station drawing, will be identified with an individual amendment number, date or “original”. The effective revision number and amendment number for “M” or “E” series figures in the FSAR are identified in Table 3.2-3. FSAR Amendment numbers are not shown on “E” and “M” series figures themselves, and the current versions are identified by revision number only.

In the EPL, a table number is preceded by a "T" (e.g., "T1.3-1") and a figure is preceded by an "F" (e.g., "F9.1-3"). Text portions of the FSAR are identified by section number only (e.g., “Section 5.4”).

For text and tables, the amendment number and/or date is generally located at the bottom of the sheet, near the margin. For non “E” and “M” series figures, the amendment number and/or date is generally located in the vicinity of the title block for that figure.

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ORIGINAL	Submitted February 27, 1978
MARCH 21, 1978	Update
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AMENDMENT 2	July 27, 1978
AMENDMENT 3	November 30, 1978
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AMENDMENT 11	July 31, 1980
AMENDMENT 12	October 8, 1980 (Also Original Submittal of Emergency Plan)
AMENDMENT 13	December 15, 1980
AMENDMENT 14	January 30, 1981
AMENDMENT 15	February 20, 1981
AMENDMENT 16	March 31, 1981 (Also Revision 1 to Emergency Plan)
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AMENDMENT 19	April 30, 1981
AMENDMENT 20	May 7, 1981
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AMENDMENT 25	August 7, 1981
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AMENDMENT 29	December 21, 1981
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101a August 09, 2007 (continued)	SA-2006-38 (JCH)	3B, 3.6B, 3.9B, 7.1, 7.3, 10.4, T3.6B-1, T3.9B-1F, T3.9B-10, T6.2.4-2, T6.2.4-3, T6.2.4-6, T7.3-4, T7.5-5, T7.5-7B, T10.3-6, F3.6B-11, F3.6B-12, F3.6B-13, F3.6B-14, F3.6B-19, F3.6B-20, F3.6B-21, F3.6B-22, F3.6B-26, F3.6B-27, F3.6B-28, F3.6B-29, F3.6B-30, F3.6B-34, F3.6B-35, F3.6B-36, F3.6B-37, F3.6B-38, F3.6B-39-1, F3.6B-40, F3.6B-41, F3.6B-55, F3.6B-67-1, F3.6B-82-1, F3.6B-208 sh. 2, F3.6B-208 sh. 3, F6.2.4-1 sh. 9, F7.7-14, F10.4-12, F10.4-22A, F10.4-22B, FII.E.1.1-1
	SA-2006-39 (JDS)	1A(B), Ch. 3 LOT, Ch. 3 LOF, 3.5, T3.5-2A, T3.5-2B, T3.5-6, 3.7N, 3.8, T3.8-1, 3.9N, T3.9N-20, 4.5, 5,2, 5.3, T5.3-2A, T5.3-5, T5.3-15A, T5.3-1, 6.2, 7.7, 8A, 8.3, 9.1, T9.4-2, 12.3, T17A-1, II.F, F3.9N-5A, F3.9N-5B, F3.9N-6A, F3.9N-6B, F3.5-1, F3.5-2, F1.2-8, F1.2-13, F3.8-15, F3.8-2, F5.1-3
	SA-2007-6 (RJK)	T1.6-1, 4A, 15.6
	SA-2003-21 (JCH)	T3.9B-10
	SA-2007-9 (RJK)	6.3
101b February 4, 2008	SA-2006-1 (RJK)	12.3, F12.3-33 sh. 3, F12.3-6, F12.3-23.4
	SA-2007-17 (RJK)	T1.6-1, 5.2, Ch. 5 LOF, Deleted F5.2.2, Deleted F5.2.3
	SA-2004-30 (TJE)	1.2, 1A(N), 1A(B), 3.1, T3.10N-1, Ch. 6 TOC, Ch. 6 LOT, Ch. 6 LOF, 6.2, T6.2.5-1, T6.2.5-2, T6.2.5-4, T6.2.5-5, T6.2.5A-1, T6.2.5A-2, T6.2.5A-3, T6.2.5A-5, T6.2.5A-6, F6.2.5-1, Delete F6.2.5-2, Delete F6.2.5A-1, Delete F6.2.5A-2, Delete F6.2.5A-3, Delete F6.2.5A-5, Delete F6.2.5A-7, Delete F6.2.5A-9, 7.1, T7.1-2.6, 7.2, 7.3, T7.5-3, T7.5-4, T7.5-7A, T7.5-7D, T7.5-7E, 7.6, T8.1-1, T8.3-1, T8.3-2, T13.5-2, 15.6, T17A-1, TII.B.2-4, TII.B.2-5, TII.B.2-6, II.E, II.F

Amendment No.	LDCR No./Licensing Lead	Effective Sections
101b February 4, 2008 (continued)	SA-2007-11 (JDS)	CH. 6 LOT, CH. 6 LOF, 6.2, T6.2.1-2A.1, T6.2.1-2A.2, Delete T6.2.1-2A, T6.2.1-2B.1, T6.2.1-2B.2, Delete T6.2.1-3, Delete T6.2.1-3A, Delete T6.2.1-4, Delete T6.2.1-4A, T6.2.1-5, Delete T6.2.1-9, T6.2.1-9A, T6.2.1-9B, Delete T6.2.1-10, T6.2.1-10A, T6.2.1-10B, Delete T6.2.1-15, Delete T6.2.1-21, Delete T6.2.1-26, Delete T6.2.1-29, Delete T6.2.1-30, Delete T6.2.1-37, Delete T6.2.1-38, Delete T6.2.1-39, Delete T6.2.1-40, Delete T6.2.1-41, Delete T6.2.1-42, Delete T6.2.1-43, Delete T6.2.1-44, Delete T6.2.1-45, Delete T6.2.1-46, Delete T6.2.1-47, Delete T6.2.1-48, Delete T6.2.1-50, Delete T6.2.1-50A, Delete T6.2.1-51, Delete T6.2.1-52, Delete T6.2.1-53, Delete T6.2.1-54, Delete T6.2.1-55, Delete T6.2.1-56, Delete T6.2.1-57, Delete T6.2.1-58, Delete T6.2.1-59, Delete T6.2.1-60, Delete T6.2.1-64, Delete T6.2.1-65, Delete T6.2.1-66, Delete T6.2.1-67, Delete T6.2.1-68, Delete T6.2.1-69, Delete T6.2.1-77, Delete T6.2.1-91, Delete T6.2.1-93, Delete T6.2.1-95, Delete T6.2.1-96, Delete T6.2.1-97, Delete T6.2.1-98, Delete T6.2.1-99, Delete T6.2.1-100, Delete T6.2.1-101, Delete T6.2.1-102, Delete T6.2.1-103, Delete T6.2.1-104, Delete T6.2.1-1, Delete T6.2.1-2, Delete T6.2.1-3, F6.2.1-9, F6.2.1-10, F6.2.1-11, F6.2.1-12, F6.2.1-13, F6.2.1-14, F6.2.1-15, F6.2.1-16, Delete F6.2.1-20, Delete F6.2.1-, Delete F6.2.1-30, Delete F6.2.1-31, Delete F6.2.1-32, Delete F6.2.1-33, Delete F6.2.1-34, Delete F6.2.1-35, Delete F6.2.1-36, Delete F6.2.1-37, Delete F6.2.1-38, Delete F6.2.1-39, Delete F6.2.1-40, Delete F6.2.1-41, Delete F6.2.1-2, Delete F6.2.1-43, Delete F6.2.1-44, Delete F6.2.1-45, Delete F6.2.1-49, Delete F6.2.1-50, Delete F6.2.1-51, Delete F6.2.1-52, Delete F6.2.1-54, Delete F6.2.1-55, Delete F6.2.1-56, Delete F6.2.1-58, Delete F6.2.1-59, Delete F6.2.1-60, Delete F6.2.1-61, Delete F6.2.1-62, Delete F6.2.1-63, Delete F6.2.1-64, Delete F6.2.1-65, Delete F6.2.1-66, Delete F6.2.1-67, Delete F6.2.1-68, Delete F6.2.1-69, Delete F6.2.1-70

Amendment No.	LDCR No./Licensing Lead	Effective Sections
101b February 4, 2008 (continued)	SA-2006-10 (JDS)	1A(B), T3.9B-10, 6.2, F6.2.2-3, F6.2.2-3A, F6.2.2-4, F6.2.2-5, 6.3, 6.5, T6.5-4, T6.3-7, T6.3-11, 7.6, T17A
	SA-2006-31 (JDS)	9.1
	SA-2007-19 (JDS)	6.2
	SA-2007-23 (GLM)	9.1, T1.6-1
	SA-2007-22 (JDS)	1A(B), 6.1B, 6.2, T17A-1
	SA-2006-24 (JCH)	T17A-1, T3.9B-10
	SA-2006-34 (JCH)	10.4
	SA-2007-16 (RJK)	5.3
	SA-2007-5 (JDS)	F4A.2.2-1, F4A.2.2-2, F4A.2.2-3, 4A, T4A.1.2-1, T4A.2.2-1, T4A.2.2-2, 15.0, 15.1, 15.2, 15.4, 15.5, 15,6, T15.6-5
	SA-2007-21 (GLM)	9.2
102 August 1, 2008	SA-2007-18 (RJK)	5.3
	SA-2007-20 (TJEW)	9.1, T9.1-1, T12.2-24, 15.7, T15.7-6, T15.7-7
	SA-2006-41 (CBC)	6.2
	SA-2006-23 (JCH)	9.5
	SA-2005-29 (JDS)	T7.5-7B, F5.4-6, F6.2.2-1
	SA-2008-4 (RJK)	9.3, T9.3-6, T9.3-7, 15.6
	SA-2006-3 (JCH)	10.2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
102 August 1, 2008 (continued)	SA-2007-2 (TJEW)	T1.6-1, 3.5, 10.2,
	SA-2006-25 (JCH)	10.2, T10.4-10, T10.1-1, T10.4-17
	SA-2008-8 (RAS)	Ch. 13 TOC, Ch. 13 LOF, 13.1, T13.1-1, 13.1A-1, 13.3, 13.5, F13.1-1, F13.1-2, F13.1-3, Ch. 17 TOC, 17.1, 17.2
	SA-2008-9 (RJK)	5.2
102a February 2, 2009	SA-2008-14 (TJEW)	13.1
	SA-2008-15 (JDS)	12.3
	SA-2007-15 (JCH)	10.4
	SA-2006-36 (JDS)	6.2, T6.1B-3, T6.2.2-4, 6.3,
	SA-2008-17 (RAS)	6.4
	SA-2008-16 (JCH)	10.2
	SA-2008-1 (JDS)	6.2
102b August 3, 2009	SA-2008-2 (GLM)	9.4E
	SA-2009-13 (GLM)	I.D-1
	SA-2009-8 (JCH)	10.2, F10.2-1
	SA-2008-22 (SCD)	12.5
	SA-2009-2 (TJD)	17.2
	SA-2009-6 (JDS)	5.4, 6.3

Amendment No.	LDCR No./Licensing Lead	Effective Sections
102b August 3, 2009 (continued)	SA-2008-10 (JCH)	5.3, T5.3-4C
	SA-2009-11 (JDS)	T5.2-2,1A(N)
	SA-2008-23 (TJD)	17.2, Ch. 17 TOC
103 February 11, 2010	SA-2004-26 (SCD)	T10.4-10
	SA-2009-5 (CBC)	I.A
	SA-2009-7 (TJD)	17.2
	SA-2009-17 (TJD)	1A(B), 17.2
	SA-2009-3 (SCD)	Ch. 13 TOC, 13.1, T13.1-1, 13.1A, 13.3, 13.5
	SA-2009-10 (TJD)	T2.5.6-7, F2.5.6-56
	SA-2009-16 (TJEW)	T8.3-3
	SA-2009-23 (TJEW)	T8.3-4
	SA-2008-11 (JCH)	10.2, 10.4, T10.4-10
	SA-2008-12 (JCH)	10.4
	SA-2008-20 (JCH)	13.3
	SA-2009-19 (JDS)	4.2, 9.3
	SA-2007-10 (JDS)	1.2, T10.1-1, Ch. 10 LOF, 10,1, 10.2, 10.4, F10.1-1, F10.1-2
	SA-2004-18 (JDS)	5.3

Amendment No.	LDCR No./Licensing Lead	Effective Sections
103 February 11, 2010 (continued)	SA-2008-13 (JDS)	T1.6-1, Ch. 6 TOC, Ch. 6 LOT, Ch. 6 LOF, 6.2, T6.2.1-2, T6.2.1-2A, T6.2.1.3, T6.2.1-3A, T6.2.1-3B, T6.2.1-4, T6.2.1-5, T6.2.1-6, T6.2.1-8, T6.2.1-9, T6.2.1-10, T6.2.1-11, T6.2.1-12, T6.2.1-39, T6.2.1-41, T6.2.1-42, T6.2.1-43, T6.2.1-49, T6.2.1-50, T6.2.1-51, T6.2.1-54, T6.2.1-56, T6.2.1-57, F6.2.1-1, F6.2.1-2, F6.2.1-3, F6.2.1-4, Ch. 7 TOC, 7.7, 9.3, Ch. 15 LOT, Ch. 15 LOF, 15.0, T15.0-1, T15.0-4, T15.0-6, T15.0-7, F15.0-1, F15.0-3, F15.0-4, F15.0-5, 15.1, T15.1-3, T15.1-4, 15.2, T15.3-1, 15.4, T15.4-4, 15.5, 15.6, T15.6-1, T15.6-2, T15.6-3, T15.6-20, T15.6-21, F15.6-1, F15.6-2, F15.6-3, F15.6-4, F15.6-5, F15.6-6, F15.6-7, F15.6-8, F15.6-9, F15.6-10, F15.6-11, F15.6-12, F15.6-13, F15.6-14, F15.6-15, F15.6-16, F15.6-17, F15.6-18, F15.6-19, F15.6-20, F15.6-21, F15.6-22, F15.6-23, F15.6-24, F15.6-25, F15.6-26, F15.6-27, F15.6-28, F15.6-29, F15.6-30, F15.6-31, F15.6-32, F15.6-33, F15.6-34, F15.6-35, F15.6-36, F15.6-37, F15.6-38, F15.6-39, F15.6-40, F15.6-41, F15.6-42, F15.6-43, F15.6-44, F15.6-45, F15.6-46, F15.6-47, F15.6-48, F15.6-49, F15.6-50, F15.6-51, F15.6-52, F15.6-53, F15.6-54, F15.6-55, F15.6-56, F15.6-57, F15.6-58, F15.6-59, F15.6-60, F15.6-61, F15.6-62, F15.6-63, F15.6-64, F15.6-65, F15.6-66, F15.6-67, F15.6-68, F15.6-69, F15.6-70, F15.6-71, F15.6-72, F15.6-73, F15.6-74, F15.6-75, F15.6-76, F15.6-77, F15.6-78, F15.6-79, F15.6-80, 15.7, T15.7-1, T15.7-2, T15.7-3, T15.7-4, T15.7-6, T15.7-7
103a (August 2, 2010)	SA-2008-19 (JCH)	13.3
	SA-2010-2 (JCH)	13.3
	SA-2009-24 (JDS)	6.2
	SA-2009-12 (SCD)	Ch. 6 TOC, 5.2, 6.2, 6.3, F6.2.4-1, Shts. 3 & 12

Amendment No.	LDCR No./Licensing Lead	Effective Sections
103a (August 2, 2010) (continued)	SA-2010-11 (TJD)	1A(B), II.K
	SA-2009-25 (TJEW)	T3.2-3, T3.2-4, Ch. 9 LOF, Deleted M1-0313A (F9.4-15)
	SA-2009-21 (JDS)	Ch. 15 TOC, 15.4
	SA-2010-10 (JDS)	7.2, T7.2-4, T15.0-6, 15.4
103b (February 1, 2011	SA-2010-6 (JCH)	T10.4-13
	SA-2009-9 (RAS)	3.8, 9.1
	SA-1010-7 (SCD)	11.3
	SA-2010-13 (TJD)	2.4
	SA-2010-9 (TJD)	Ch. 17 TOC, 17.2
	SA-2011-1 (TJEW)	T5.1-1B
104 (August 1, 2011)	SA-2010-18 (RAS)	7.4, T7.4-1, T7.4-3
	SA-2009-14 (SCD)	6.3
	SA-2011-8 (SCD)	11.4, F1.2-1
	SA-2010-20 (TJD	T17A-1
	SA-2008-21 (JCH)	T17A-1
	SA-2010-5 (JDS)	4.2, 1A(N)
	SA-2010-8 (JDS)	7.6, CH. 9 LOF, 9.1, F9.1-4, 9.5, T17A-1,
	SA-2011-5 (JDS)	17.2, 17A

Amendment No.	LDCR No./Licensing Lead	Effective Sections
104 (August 1, 2011) (continued)	SA-2010-16 (JCH)	9.5
	SA-2011-3 (SCD)	12.1, 12.5, T13.1, 13.1A
	SA-2010-17 (TJEW)	3.11B
	SA-2009-4 (TJEW)	F1.2-1, Ch. 8 LOF, 8.2, F8.2-1, F8.2-4, F8.2-5 Sh. 1, F8.2-10 Sh. 1 & 2, F8.2-11, 8.3, 9.5, F10.2-1
	SA-2011-6 (JDS)	1.2, F1.2-1, Ch. 9 TOC, 9.1, 9.5, 12.4
	SA-2011-12 (TJEW)	9.5, F9.5-51
	SA-2010-19 (TJEW)	10.4
	SA-2009-20 (TJEW)	1.2, 8.1, 8.2, 8.3, F8.2-1, F8.2-4, F8.2-5 Sht. 1, F8.2-5 Sht. 2, F8.2-12
	SA-2010-1 (SCD)	11.1, T11.1-1, T11.1-2, T11.1-3, T11.1-4, 11.1-5, T11.1-7, 11.2, T11.2-1, T11.2-2, T11.2-8, T11.2-9, T11.2-10, 11.2, T11.3-3, T11.3-4, T11.3-5, T11.3-7, 11A, T11A-1, T11A-2, T11A-3, T11A-4, T11A-5, T11A-6, 121.2, T12.2-1, T12.2-2, T12.2-3, T12.2-4, T12.2-5, T12.2-7, T12.2-8, T12.2-9, T12.2-10, T12.2-11, T12.2-12, T12.2-13, T12.2-14, T12.2-15, T12.2-17, T12.2-17A, T12.2-19, T12.2-19A, T12.2-19B, T12.2-19C, T12.2-19D, T12.2-19E, T12.2-21, T12.2-22, T12.2-23, T12.2-24, T12.2-25, T12.2-26, 6.2, 3.6B, T3.6B-4A, T3.6B-4B, 9.1, 9.2, 10.4, T10.4-10
104a (February 1, 2012)	SA-2010-14 (GLM)	9.2
	SA-2011-13 (JDS)	T1.6-1, 4.1, T4.1-1, 4.2, 4.4, T4.4-1, T15.6-4, T15.6-5
	SA-2011-14 (JDS)	15.5
	SA-2011-2 (RAS)	7.2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
104a (February 1, 2012) (continued)	SA-2009-22 (TJD)	17.2
	SA-2011-15 (TJEW)	T3A.3-2
	SA-2011-18 (TJEW)	9.5
	SA-2009-18 (TJEW)	7.5, T7.5-2, T7.5-7A, T7.5-7C, T7.5-7D, T7.5-7E, II.F-3
	SA-2011-21 (TJEW)	T7.5-7B
	SA-2011-17 (JDS)	T3A.3-1, 6.2
	SA-2011-7 (JDS)	T3.7B(A)-1
	SA-2011-16 (SCD)	12.1, Ch. 13 TOC, 13.1, 13.1A, 13.4, 14.2, Ch. 17 TOC, 17.2
104b (August 1, 2012)	SA-2012-9 (RAS)	7.7
	SA-2012-7 (TJEW)	T3.2-3, T3.2-4, Ch. 6 LOF
	SA-2012-11 (JCH)	T3.9N-10
	SA-2012-10 (SCD)	Ch. 11 TOC, 11.4, Ch. 12 TOC, 12.2
	SA-2012-14 (SCD)	T17A-1
	SA-2012-17 (SCD)	3.4
	SA-2010-3 (TJEW)	1A(B), 9.5
	SA-2012-15 (SCD)	1A(B), T6.5-1

Amendment No.	LDCR No./Licensing Lead	Effective Sections
105 (February 4, 2013)	SA-2011-11 (GLM)	T9.4-2, 9.4C
	SA-2011-19 (GLM)	9.2
	SA-2012-2 (TJEW)	T7.5-7B
	SA-2012-28 (JDS)	15.5
	SA-2012-23 (CBC)	1A(B)
	SA-2010-12 (GLM)	9.4E
	SA-2012-22 (JCH)	3.6B
	SA-2012-6 (JDS)	9.1
	SA-2012-16 (SCD)	11.5, F11.5-1
	SA-2012-26 (TJEW)	T3.9B-10
	SA-2012-4 (TJEW)	9.5, T17A-1
	SA-2012-8 (JDS)	3.5, T7.5-7B, 10.4, 15.1, T17A-1
	SA-2012-19 (TJEW)	T3A.3-2, 5.3, 6.2, 8.3
	SA-2012-24 (JDS)	15.6, T15.6-8, T15.6-9, T15.6-10, T15.6-11
	SA-2013-2 (TJEW)	T3.9B-10
	SA-2012-18 (JDS)	Ch. 6 LOF, 6.2, T6.2.1-2, Deleted F6.2.1-1 and F6.2.1-2
	SA-2012-27 (SCD)	12.5

Amendment No.	LDCR No./Licensing Lead	Effective Sections
105 (February 4, 2013) (continued)	SA-2012-21 (SCD)	13.1, T13.1-1, 13.1A
	SA-2012-20 (SCD)	17.2
	SA-2013-3 (SCD)	F9.2-4A, F9.3-1
105a (August 5, 2013)	SA-2013-5 (TJEW)	F8.2-4, F8.2-5, Sh. 1
	SA-2013-1 (JDS)	4.1, 4.2
	SA-2013-8 (JDS)	T6.3-11
	SA-2012-12 (JDS)	T3.9B-10
	SA-2012-5 (JCH)	10.4
	SA-2012-1 (JCH)	5A, T9.3-9
105b (February 5, 2014)	SA-2013-10 (JCH)	10.2
	SA-2013-9 (JDS)	9.3
	SA-2013-20 (JDS)	15.4
	SA-2013-22 (RAS)	7.7
	SA-2013-23 (RAS)	F7.3-4
	SA-2013-12 (TJEW)	8B-2
	SA-2013-19 (RAS)	3.6B
	SA-2013-24 (SCD)	Ch. 17 TOC, 17.2, 17A, TMI I.B
	SA-2013-15 (GLM)	9.2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
105b (February 5, 2014) (continued)	SA-2013-4 (GLM)	3.5
	SA-2013-17 (JDS)	4.3, 6.2
	SA-2013-21 (GLM)	9.2
106 July 31, 2014	SA-2013-7 (GLM)	9.2
	SA-2013-26 (GLM)	2.4, 9.2
	SA-2014-1 (JDS)	T15.6-8, T15.6-10
	SA-2014-2 (SCD)	6.3
	SA-2014-3 (SCD)	T17A-1
	SA-2014-5 (GLM)	1.2
	SA-2014-7 (GLM)	Ch. 9 LOF, F9.3-1
	SA-2014-6 (SCD)	T12.5-1
	SA-2014-9 (SCD)	6.3, T6.3-1
	SA-2014-11 (TJEW)	8.3
	SA-2012-13 (TJEW)	F1.2-1, 8.2, F8.2-1, F8.2-4, F8.2-7 Sh. 1 & 2, F8.2-9, F8.2-11, F8.2-11A, 9.5
	SA-2014-15 (TJEW)	8.3
	SA-2014-14 (SCD)	Ch. 17 TOC, 17.2, TMI I.B
	SA-2014-16 (JDS)	1.1

Amendment No.	LDCR No./Licensing Lead	Effective Sections
106 July 31, 2014 (continued)	SA-2014-17 (JDS)	9.1
	SA-2014-18 (TJEW)	Ch. 8 LOF
	SA-2014-13 (SCD)	Ch. 13 TOC, 13.1, T13.1-1, F13.1-2, F13.1-3, 13.1A
	SA-2011-4 (JDS)	15.6
106a February 2, 2015	SA-2014-20 (JCH)	T3.9B-10
	SA-2013-6 (JDS)	T4.1-2, 4.3, 9.1 T9.1-4
	SA-2014-21 (SCD)	T9.3-7
	SA-2014-26 (JDS)	T6.2.4-3
	SA-2014-12 (SCD)	T17A-1
	SA-2014-22 (SCD)	T3.2-3, T3.2-4, Ch. 11 LOF, F11.3-1
	SA-2014-19 (SCD)	6.2, T17A-1
	SA-2014-25 (JDS)	T15.0-4
	SA-2015-1 (SCD)	9.3, T17A-1

Amendment No.	LDCR No./Licensing Lead	Effective Sections
106b September 3, 2015	SA-2015-4 (JEB)	13.6
	SA-2014-23 (RAS)	10.4
	SA-2015-7 (SCD)	T6.5-1
	SA-2014-28 (JDS)	T3.7BA-1
	SA-2015-8 (JDS)	9.1
	SA-2015-10 (JDS)	1A(B), 6.2, 6.3, T6.3-7
107 February 1, 2016	SA-2015-12 (SCD)	6.3
	SA-2015-13 (GLM)	9.2
	SA-2015-15 (GLM)	9.2
	SA-2015-18 (SCD)	T6.3-3
	SA-2015-17 (GLM)	Chapter 9 LOF
	SA-2014-4 (CBC)	1.2, F1.2-1, 9.1, T17A-1, T17A-2
	SA-2015-16 (SCD)	17.2
	SA-2014-27 (JDS)	6.2, T6.2.1-2, T6.2.1-2a
	SA-2015-19 (SCD)	Chapter 13 TOC, 13.1, T13.1-1, 13.1A, F13.1-2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
108 August 1, 2017	SA-2012-003 (KJV)	F1.2-1, 8.2
	SA-2013-025 (JCH)	T7.4-1, T7.4-3
	SA-2015-002 (GLM)	9.2, Chapter 9 LOT, T9.2-9, T9.2-10
	SA-2015-11 (CBC)	5.4, 9.3, T17A-1
	SA-2015-014 (KJV)	15.6
	SA-2016-2 (GLM)	TMI I.C
	SA-2016-3 (GWS)	4.3
	SA-2016-4 (CBC)	6.2
	SA-2016-6 (SCD)	12.3
	SA-2016-008 (SCD)	T11.5-2
	SA-2016-009 (KJV)	8.2, F8.2-1, F8.2-4, 8.2-12
	SA-2016-010 (KJV)	5.4, T5.2-3, T5.2-7
	SA-2016-012 (CBC)	F1.2-1
	SA-2016-014 (KJV)	3.5, 10.4
	SA-2016-016 (CBC)	Chapter 1 TOC, Chapter 1 LOT, 1.1, 1.4, T1.4-1, 1A(B), 2.1, 2.3, 3.7B, 3.11B, 3A, T4.1-1, 6.6, 7.1, 7.2, 8.3, 9.5, 11.3, 11.4, 11A, 12.1, 12.4, 13.1, 13.5, 14.2, 15.3, 15.6, 17.1, 17.2, T17.2-2, 17A, TMI III-A
	SA-2016-018 (SCD)	Chapter 13 TOC, Chapter 13 LOF, 13.1, T13.1-1, 13.1A, F13.1-1, F13.1-2, F13.1-3
	SA-2016-019 (JCH)	T3.9B-10

Amendment No.	LDCR No./Licensing Lead	Effective Sections
108 August 1, 2017 (continued)	SA-2017-1 (SCD)	13.2
	SA-2017-003 (KJV)	3.5, 9.2
	SA-2017-004 (KJV)	Appendix 5A
	SA-2017-005 (GWS)	15.1
	SA-2017-006 (CBC)	3.4
	SA-2017-007 (JEB)	1A(B)
	SA-2017-008 (SCD	Chapter 17 TOC, 17.2, 17A
	SA-2017-009 (KJV)	9.4A, T17A-1
	SA-2017-011 (KJV)	Appendix 5A
	SA-2017-012 (CBC)	T5.4-16
	SA-2017-013 (JCH)	10.4
	SA-2017-014 (CBC)	10.2
	SA-2017-015 (SCD)	1A(B)
	SA-2017-016 (GWS)	9.2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
109 February 1, 2019	SA-2011-009 (JCH)	8.3, 9.5
	SA-2013-016 (JCH)	9.5
	SA-2014-010 (KJV)	8.3, T8.3-3, F8.2-4, F8.2-11, F8.2-11A
	SA-2017-002 (JCH)	T10.4-10
	SA-2017-010 (GWS)	9.1
	SA-2017-017 (GWS)	Chapter 10 LOF
	SA-2017-019 (GWS)	3.11B, 3A, T7.5-7A, T7.5-7B, T7.5-7C, T17A-1
	SA-2017-020 (KJV)	T15.1-3
	SA-2017-021 (GWS)	T1.6-2, 3.5, 10.2
	SA-2017-023 (GLM)	9.3
	SA-2017-024 (GWS)	9.5, T9.5-18
	SA-2017-025 (KJV)	5.4
	SA-2017-026 (CBC)	13.1
	SA-2018-001 (KJV)	8.3
	SA-2018-003 (KJV)	17A, T17A-1
	SA-2018-004 (CBC)	17.2
	SA-2018-006 (KJV)	8.3, T8.3-4
	SA-2018-008 (CBC)	F1.2-1

Amendment No.	LDCR No./Licensing Lead	Effective Sections
109 February 1, 2019 (continued)	SA-2018-009 (GWS)	9.5
	SA-2018-010 (KJV)	T1.6-1, 1A(B), 9.1, T17A-1
	SA-2018-012 (GWS)	7.7
	SA-2018-017 (KJV)	1A(B)
	SA-2019-001 (GWS)	T17A-1
	SA-2019-003 (CBC)	1A(N), 1A(B), 17.2
110 August 3, 2020	SA-2016-015 (KJV)	8.3
	SA-2018-007 (KJV)	8B
	SA-2018-011 (GLM)	3.8
	SA-2018-014 (KJV)	8.2
	SA-2018-015 (CBC)	F1.2-1
	SA-2019-002 (GWS)	F8.2-1, F8.2-4
	SA-2019-006 (GLM)	9.4E
	SA-2019-007 (KJV)	9.5
	SA-2019-008 (GLM)	9.2
	SA-2019-009 (GWS)	T8.3-1, T8.3-2
	SA-2019-010 (CBC)	1A(B), 13.5, 17.2

Amendment No.	LDCR No./Licensing Lead	Effective Sections
110 August 3, 2020 (continued)	SA-2019-015 (GWS)	9.1
	SA-2019-017 (CBC)	12.5, Ch. 13 TOC, 13.1, F13.1-2, F13.1-3, 13.3B, 17.2
	SA-2019-018 (CBC)	T13.1-1
	SA-2019-020 (GLM)	1.2
	SA-2019-021 (CBC)	13.3B
	SA-2019-022 (GWS)	1A(B), 9.5
	SA-2019-023 (JEB)	TMI III.D
	SA-2020-001 (CBC)	F7.1-3 (SH 5)
	SA-2020-002 (CBC)	T17A-1
	SA-2020-004 (CBC)	1A(B), 17.2
	SA-2020-006 (GWS)	7.2
	SA-2020-007 (CBC)	2.4
	SA-2020-008 (CBC)	2.4
	SA-2020-009 (CBC)	1.1, 1.4, 13.1, F13.1-1, 17.1
	SA-2020-010 (GWS)	T8.3-1
	SA-2020-011 (CBC)	T1.6-1
	SA-2020-012 (KJV)	9.5

Amendment No.	LDCR No./Licensing Lead	Effective Sections
111 February 1, 2022	SA-2017-022 (NWB)	7.5, 7.7, F7.7-14, 11.5, F11.5-1, TMI III.A, FIII.A.1.2-2
	SA-2019-004 (NWB)	T3.9B-10, 8.3, T8.3-1, T8.3-2, T8.3-11, 9.5, T9.5-13
	SA-2019-012 (CBC)	6.3
	SA-2020-003 (CBC)	T5.1-1B
	SA-2020-013 (NWB)	9.2, 9.4B, T8.3-11
	SA-2021-002 (CBC)	6.2, T6.2.1-2, T6.2.1-3, T6.2.1-3A, T6.2.1-3B, T6.2.1-4, T6.2.1-9, T6.2.1-11, T6.2.1-12, T6.2.1-39, T6.2.1-41, T6.2.1-42, T6.2.1-43, T6.2.1-50, T6.2.1-51, T6.2.1-54
	SA-2021-003 (CBC)	Ch 8 LOF, 8.2, 8.3, F8.2-4, F8.2-5 Deleted, F8.2-7 Deleted
	SA-2021-005 (NWB)	3.2, 3.7B
	SA-2021-006 (CBC)	5.4
	SA-2021-007 (CBC)	3.9N, App. 3B TOC, 3B, 5.4
	SA-2021-010 (NWB)	15.7, T15.7-6
	SA-2021-011 (NWB)	T1.6-1, T4.1-2, 4.3, 15.4
	SA-2022-001 (NWB)	9.5

FSAR Amendment 100, Supplement a

LDCR-SA-2006-12 (RAS):

Administrative change for software conversion only. The changes are editorial in nature and contain no technical changes. The electronic files have been converted from Microsoft WORD to Adobe FrameMaker and published in Adobe Portable Document Format (pdf). The type of changes include changes such (1) correction of spelling errors, (2) correction of inadvertent word processing errors from previous changes, and (3) style guide changes (e.g., changing from a numbered bullet list to an alphabetized bullet list and vice versa, change numbering of footnote naming scheme). The entire FSAR will be reissued as Amendment 100, Supplement a except as noted below. For the text and tables there will be no change bars in the page margins for the editorial changes. The list of effective pages is being replaced with a list of effective sections, tables, and figures. Some Figures and Tables will retain an Amendment number prior to Amendment 100 since the source file is not in Adobe FrameMaker (e.g., scanned page).

Sections Revised: All

Tables Revised: All (except as noted above)

Figures Revised: All (except as noted above)

FSAR Amendment 100, Supplement b

LDCR-SA-2005-8, EVAL-2003-002423-02 (DWS):

Table 17A-1, sheets 45, 53 & 58

Revise the Q-List criteria to add instrument tubing that is connected to ANSI Safety Class 3 ductwork to clarify that criteria appropriate for the application is not the same as for ASME applications covered by FSAR Section 3.9B.3.4.3.

Although not included in commitment CDF-22112, FSAR Table 17A-1 previously applied the same criteria for instruments connected to Safety Class 3 HVAC ductwork as for ASME piping. The change to Table 17A-1 would document that this very conservative practice may be continued; however, it is not required. The tubing should only have to meet the seismic requirements comparable to the equipment to which it is connected.

Section 9.4: FSAR Figure 9.4-9 is revised to show the impulse lines to the non-safety instruments on the Non-ESF PPVES units are being declassified to NNS. This would allow seismic Category II supports rather than seismic Category I supports.

Where the tubing is reclassified as NNS in accordance with ANSI N18.2 [as described in DBD-ME-028], the instrument should only have to meet the requirements of the tubing. FSAR Figure 9.4-9 is revised to show the impulse lines to the non-safety instruments on the Non-ESF PPVES units are being declassified to NNS based on no credible failure which could adversely affect the Nuclear Safety Function of the PPVES.

LDCR-SA-2005-22, EVAL-2005-002874-01 (DWS):

Section 17.2.12

Revise paragraph on M&TE and reference standards to indicate the standards are traceable to nationally and internationally recognized standards. National laboratories no longer support all types of calibration.

NUREG-0800 Standard Review Plan Section 7.1 "Instrumentation and Controls - Introduction" endorses Branch Technical Position HICB-12 "Guidance on Establishing and Maintaining Instrument Setpoints" which endorses ANSI/NCSL Z540-1-1994 "American National Standard for Calibration _ Calibration Laboratories and Measuring and Test Equipment _ General Requirements".

ANSI/NCSL Z540-1-1994 Section 9.2 states in part ".....measurements made by the laboratory are traceable to national, international, or intrinsic standards....

LDCR-SA-2005-22, EVAL-2005-002874-01 (DWS) (continued):

Section 1A(B) - Regulatory Guide 1.30

NUREG-0800 Standard Review Plan Section 7.1 "Instrumentation and Controls - Introduction" endorses Branch Technical Position HICB-12 "Guidance on Establishing and Maintaining Instrument Setpoints" which endorses ANSI/NCSL Z540-1-1994 "American National Standard for Calibration _ Calibration Laboratories and Measuring and Test Equipment _ General Requirements".

ANSI/NCSL Z540-1-1994 Section 9.2 states in part ".....measurements made by the laboratory are traceable to national, international, or intrinsic standards...."

LDCR-SA-2005-33, EVAL-2004-003442-01 (DWS):

Delete from Section FSAR 17.2.1.3.2 and 17.2.18.2 as follows "For audits other that those audit areas with a maximum frequency specified by regulation (Security, Emergency Planning, (Delete "and Radiation Protection") a grace period.....

Historically the periodic review described in 10CFR20.1101(c) has been satisfied by an independent quality assurance audit. This annual review will be satisfied as described below and every 24 months a quality assurance audit will be performed which is consistent with ANSI N18.7 sect 4.5

NOD Audits (which usually include industry peers)

* NOD currently performs annual audits.

NOD Surveillances

Manager and Supervisory Observations (LEADING)

Annual performance reviews

Corrective Action Program (documentation of procedural noncompliance in SmartForms)

RP Self- Assessments (which may include industry peers or contractors)

Third Party Audits (NRC, INPO, NVLAP and ANI)

FSAR Amendment 100, Supplement b

LDCR-SA-2006-11, EVAL-2006-000798-01 (DWS):

Section 17.2: Revise ORC membership composure, definition of a ORC quorum, and audit functions. Move audits listed under Section 17.2.1.3.2 to Section 17.2.18

This is an administrative change only and is in compliance with the requirements of ANSI N18.7-1976.

LDCR-SA-2006-16, EVAL-2006-001294-01 (DWS):

Section 17.2: Add statement "TXU Power may delegate to others such as contractors, agents, or consultants the work of establishing and executing the quality assurance program, or any part thereof". And clarify QA responsibilities on page 21, 23, 28 and 29.

Title 10, Code of Federal Regulations, Part 50, Appendix B, "Quality Assurance Criteria for Nuclear Power Plants and Fuel Processing Plants," allows the use of contractors to perform QA functions as long as the overall responsibility for the Quality Assurance (QA) Program remains with the Senior Vice President & Chief Nuclear Officer.

LDCR-SA-2006-9, EVAL-2005-003061-02 (DWS):

Section 17.2: Revise composure of the Station Operations Review Committee (SORC) and current organizational positions. Replace "Vice President, Nuclear Engineering" with "Vice President, Nuclear Engineering and Support" wherever it appears. Replace "Vice President, Nuclear Operations" with "Site Vice President" wherever it appears. Define the responsibilities of the Site Vice President and Plant Manager. Delete some references to Section 13.1.1.2.1. Insert Figure 13.1-2 and 13.1-3 for Figures 17.2-1 and 17.2-2.

Revised composure of SORC will reflect current organizational changes. This would resolve the discrepancy between current directors and the generic title of managers, clearly indicate the minimum number of SORC members and clarify that the listed individuals are in areas of experience, not organizational or position titles.

LDCR-SA-2006-14, EVAL-2006-001365-01 (DWS):

Section 17.2.1.3.1: Revise FSAR Section 17.2.1.3.1 to delete ORC review of changes to the Technical Specifications or the Operating License. Move this review requirement to Quality Assurance FSAR Section 17.2.1.6

American National Standard N18.7-1976/ANS-3.2, Section 4.3 requires the independent review of operational activities and allows these activities to be performed either by a Standing Committee (ORC) (Section 4.3.2) or by Organizational Units (Section 4.3.3). This change deletes the use of ORC for review of changes to the Technical Specifications or Operating License and incorporates the alternative method of using Organizational Units to perform the Independent Review.

FSAR Amendment 100, Supplement b

LDCR-SA-2006-22, EVAL-2006-002136-01 (DWS):

17.2.1.1.3.1 change reference for required experience from ANSI N18.1-1971 to Regulatory Guide 1.8, Rev.2. The RG is a higher tier document, therefore there is no change in commitment.

17.2.1.3 change reference to items "a through h" above to "1 through 9". This is editorial only to match the new format.

17.2.3 remove requirement for ORC approval before proceeding with implementation of TS changes or other changes to the operating license. This change was approved in LDCR SA-2006-14, however, this section did not reflect the change.

17.2.6.1 "Number second paragraph "6" and reformat bullets under this paragraph from "1 & 2" to "a & b". This is editorial only to match the new format and eliminate redundant numbering under same section.

LDCR-SA-2004-36, EVAL-2001-002847-05 (CBC):

EVAL-2001-002847-05 (LDCR-SA-2004-036) / Sections 1.2.2.3.7, 6.4.1.2, 6.4.2.2, 6.4.2.3, 6.4.2.4, T6.4-1 (Sheet 1), 9.4.1.1, 9.4.1.2, 9.4.1.3, 9.4.1.4, T9.4-10 (Sheet 1 - item 2.a) (Sheet 2 - items 2.c, 2.e) (Sheet 3 -item 2.f), 15.6.5.4: Per FSAR Section 9.4 and 6.5, the basic design functions of the inlet dampers associated with each Train of Control Room (CR) HVAC ventilation system are to: (1) open and let air into the CR via the ventilation system and to (2) close to isolate the CR ventilation system from outside air. The existing dampers do not perform this 2nd function as readily as desired as tracer gas testing has demonstrated some filtered in-leakage occurs when the ESF systems are lined up to operate in emergency modes.

Replacing the existing dampers with bubble tight dampers will minimize the amount of filtered in-leakage into the CRE when operating in emergency modes and one of the Trains of CR HVAC is shutdown. This activity does not change the original design functions of the individual dampers and thus, will not change the design function of the CR HVAC System.

LDCR-SA-2004-036 updates the FSAR providing information for the new type of dampers and their associated design functions of ensuring the assumptions made in the dose analyses remain valid and in accordance with the original design function of the CR HVAC system.

EVAL-2001-002847-05 (LDCR-SA-2004-036) / Section 9.4.1.1: The word "inoperative" is replace with the word "inoperable." This is an editorial clarification.

EVAL-2001-002847-05 (LDCR-SA-2004-036) / Section 9.4.1.2: Replace "resistand with "resistan." This is an editorial correction.

EVAL-2001-002847-05 (LDCR-SA-2004-036) / Section 9.4.1.2: Replace "The main toilet" with "The toilet". This is an editorial clarification.

LDCR-SA-2004-36, EVAL-2001-002847-05 (CBC) (continued):

EVAL-2001-002847-05 (LDCR-SA-2004-036) / Section9.4.1.3: Per FSAR Section 9.4.

LDCR-SA-2005-18, EVAL-2001-002657-03 (CBC):

Section 9.1.4.2.3(14): Revise Section 9.1.4.2.3(14), "Fuel Handling System / System Description / Component Description / Polar Crane," to clarify operation of the polar crane during MODES 1 through 4. Previously the FSAR indicated the crane remained essentially in the parked position during MODES 1 and 2 and the crane may be operated through its full 360 degree travel range during MODES 3 and 4. The revised discussion indicates the crane remains essentially in the parked position during MODES 1 through 4 unless maintenance is being performed on it. The Containment Access Rotating Platform is allowed to be used for maintenance on (including modifications to) the Polar Crane in MODES 1 through 4. During Polar Crane maintenance the polar crane may be operating through its full 360 degree travel range. This change is acceptable since any maintenance activities performed on the polar crane (including its movement) during MODES 1 through 4 are subject to risk assessments performed in accordance with the maintenance rule 10CFR50.65(a)(4).

The phrase "...any safety related SSC (e.g., Hydrogen Recombiner)." is changed to "... any safety related SSC." This excessive level of detail is not required for the FSAR.

Change the phrase "No load will be carried during MODES 1-4." to "No load will be carried by the polar crane during MODES 1-4." This consistent with the changes above.

The phrase "A compensatory measure (operator, in direct communication..." is changed to "A compensatory measure (a dedicated person, ...". An operator is not required for this administrative function.

LDCR-SA-2005-20, EVAL-2005-003243-01 (CBC):

Section 9.1.4.2.3: EVAL-2005-003243-01 / LDCR-SA-2005-20: Changes description in FSAR Section 9.1.4.2.3, "Fuel Handling System / System Description / Component Description / Reactor Vessel Stud Tensioner," by revising the discussion of the stud tensioning process for the reactor vessel head. The method used to pretension the reactor vessel head studs has changed. The requirement that the tensioning method prevent unequal loading of the studs and excessive stress in the flange region remains unchanged, therefore the change is acceptable. The old description contained excessive detail and has been deleted and a statement added that an approved sequence is used to tension the studs.

LDCR-SA-2005-28, EVAL-2004-002831-04 (CBC):

This change to Section 9.1.4.2.3(4) allows the new fuel elevator to also be used in the reconstitution of irradiated fuel. The new fuel elevator will have mechanical stops added that will not adversely affect the elevator's design and its electrical limit switches readjusted to ensure the 10 feet of water shielding is maintained during fuel

LDCR-SA-2005-28, EVAL-2004-002831-04 (CBC) (continued):

reconstitution. During normal operation the mechanical stops are removed and electrical limits will be reset, allowing the new fuel elevator to lower new fuel into the cask pit area. The addition of the mechanical and electrical stops will have no adverse impact on any SSC design function.

LDCR-SA-2005-31, EVAL-2001-001914-14 (CBC):

Section 8.2.1: In Section 8.2.1, revise the stated use of Spare Startup Transformer XST1/2 to be a physical replacement of Startup Transformer XST2 instead of being energized at its storage location west of the Turbine Building from the 345 kV line by closing a normally open motor-operated air-switch. The description of the physical relocation of Spare Transformer XST1/2 and connection to the 138 kV line to XST1 has not changed. Spare Startup Transformer XST1/2 is presently located west of the Turbine Building across the access road under the 345 kV line feeding Station Service Transformer 1ST and Startup Transformer XST2. The Spare Transformer high voltage bushings are connected thorough a motor-operated air-switch to the 345 kV line. The air switch is locked open. The original intent was to use the Spare Transformer from this location in the event of a failure of Startup Transformer XST2 by building a 6.9 kV cable bus across the access road and into the TB for temporary connection into the plant 6.9 kV cable bus system. However, a more practical use of Spare Transformer XST1/2 in the event of a failure of XST2 is to physically move out XST2 and place the Spare Transformer on the XST2 pad.

This allows the 6.9 kV secondary cable bus to be directly connected to the Spare Transformer without modification of the secondary cable bus. The space between the 6.9 kV bushing phases and their height is exactly the same for both transformers. The only modification required is to identify the alternate terminations for power, control and protective relay circuits. FDA-2001-001914-16 is a paper change only contingency modification providing alternate connections for Spare Transformer XST1/2 when physically replacing Startup Transformer XST2. This change to the FSAR more accurately describes the use of Spare Transformer XST1/2 in the physical replacement of XST2.

LDCR-SA-2004-19, EVAL-1999-003133-08 (CBC):

Section 3.3.2.2, 9.4C.2, and 9.4C.3: Revise description of Section 3.3.2.2 , " Tornado Protection Design Features - Tornado Pressure Relief Dampers In Interior Fire Walls", Section 9.4C.2 " Main Steam and Feedwater Piping Area Ventilation System", Section 9.4C.3 " Electrical Area (Safeguards) Ventilation System"

The Safeguards Electrical Area Ventilation System (EAVS) may not be able to maintain the minimum design and licensing limiting temperature in the MS/FW Pipe & Electrical Area HVAC Room rooms 1-104 and 2-104, during extreme wintertime conditions. The Main Steam and Feedwater (MS & FW) Area Ventilation System may not be able to maintain the maximum design and licensing limiting temperature in the Main Steam and

LDCR-SA-2004-19, EVAL-1999-003133-08 (CBC) (continued):

Feedwater Pipe Penetration Areas, during extreme summertime conditions. [Reference SMF-1999-003133]

This modification, FDA-1999-003133-01-02, connects the EAVS and MS &FW Area Ventilation Systems in order to enhance the efficiency of each ventilation system in performing its design function. The modified ventilation systems establish new EAVS exhaust air flow paths through the areas served by EAVS. EAVS tempered exhaust air is drawn through a common stairwell, open doors and floor openings from the electrical areas to the fresh air intake of the MS & FW Area Ventilation System air handling unit located in the Mechanical Equipment Room on Elevation 852'-6".

This modification does not result in an adverse impact on the design function both stated and implicit for those plant components, systems, programs and licensing basis requirements impacted by this activity. The activity does not result in an adverse impact to the design functions of SSC's as described in the FSAR.

LDCR-SA-2005-23, EVAL-2002-000747-01 (CBC):

Table 3.9B-10: EVAL-2002-000747-01 (LDCR-SA-2005-23) Revise Table 3.9B-10, "Active Valves" to clarify the method of actuation for LV-4500, LV-4501, and LV-4500-1. Section 9.2.2.5.3 of the FSAR notes the manual operation of the valves as well as the safety function. This is a trivial administrative clarification of a supporting section.

LDCR-SA-2004-43, EVAL-2004-000038-01 (CBC):

Figures 3.6B-64-1 and 3.6B-64-2: EVAL-2004-000038-01 (LDCR-SA-2004-043): Inactive pipe whip restraint RC-1-007-902-C47W is being deleted and removed from the plant. This LDCR will update the Restraint Location drawing in the FSAR to reflect the facilities configuration. The removal of a non-functioning (i.e., abandoned) pipe whip restraint does not result in an adverse impact on the ability of the plant SSC's to mitigate the consequences of a postulated accident. This restraint presents obstructions to outage maintenance activities that result in increased dose and the potential for injury.

LDCR-SA-2005-15, EVAL-2002-004206-03 (RJK):

In FSAR Section 3.6B.2.5.2 the design pressure for the portion of the Unit 1 Safety Injection (SI) System from and including the SI Pumps up to the outside containment isolation valves is being increased from 1750 psig to 1860 psig to resolve discrepancies identified during an NRC SSD&PC Inspection in 2002. The discrepancies identified were that the SI pump shutoff head identified in the FSAR and DBD was less than the maximum shutoff head shown on the vendor certified pump curves.

The Unit 1 system design pressure has been established at 1860 psig to bound the maximum SI pump discharge pressure that could exist following a LOCA and while in the recirculation mode. The system piping and components have been re-evaluated for this higher design pressure. In Table 6.3-2 the relief valve set points have been changed to

LDCR-SA-2005-15, EVAL-2002-004206-03 (RJK) (continued):

1860 psig, i.e., just above the maximum postulated SI pump discharge pressure, to prevent their opening while in the recirculation mode. Increasing the relief valve set points from 1750 psig to 1860 reduces the probability that the valves will open while the SI system is operating in the recirculation mode and, therefore, provides greater assurance that they can perform their function as a boundary for a closed system outside containment.

In Table 6.3-1 the Unit 1 system design pressure has been established at 1860 psig to bound the maximum SI pump discharge pressure that could exist following a LOCA and while in the recirculation mode. The system piping and components have been re-evaluated for this higher design pressure.

LDCR-SA-2005-2, EVAL-2001-001255-12 (TJE):

Sections 8.3.1.1.11.1 and 8.3.1.1.11.2: The changes in sections 8.3.1.1.11.1 and 8.3.1.1.11.2 provide a description of the new Unit 1 Train B upgraded digital excitation/ voltage regulator that will be consistent with the previous digital excitation/voltage regulator modifications on Unit 1 Train A and Unit 2 Trains A and B. See Markup in EVAL-2001-001255-12.

NOTE: The DG Exciter/Voltage Regulator Upgrade/replacement will replace the obsolete Exciter/Voltage Regulator for the Emergency Diesel Generator with a new digital-controlled exciter/voltage regulator which will be able to be serviced and replaced for the next 20 years.

Section 8.3.1.2.1, Part J: The changes in section 8.3.1.2.1 Part J, dealing with Compliance with the NRC Regulatory Guide 1.75, provides a description of the new Unit 1 Train B upgraded digital multifunction protective device that will be consistent with the previous digital multifunction protective device modifications on Unit 1 Train A and Unit 2 Trains A and B. See Markup in EVAL-2001-001255-12.

NOTE: The DG Exciter/Voltage Regulator Upgrade/replacement will replace the obsolete Exciter/Voltage Regulator for the Emergency Diesel Generator with a new digital-controlled exciter/voltage regulator which will be able to be serviced and replaced for the next 20 years.

LDCR-SA-2005-7, EVAL-2001-002847-10 (TJE):

Section 8.3.1.1.11.2: Currently, in section 8.3.1.1.11.2 under Protection Systems the sentence reads, "The valve motors are protected against sustained fault conditions by circuit breakers if full load currents are equal to or greater than 0.66 amps, and by fused disconnect switches if full load currents are less than 0.66 amps."

Change the sentence to read, "The valve motors are protected against sustained fault conditions by circuit breakers. The valve motors may be protected by fused disconnect switches if full load currents are less than 1.3 amps."

LDCR-SA-2005-7, EVAL-2001-002847-10 (TJE) (continued):

This change does not affect the ability of the MOVs to perform their required safety functions. The alternate design allowance for fuse to be used for MOV circuit protection is limited to only a small set of MOVs, originally provided with fuse protection by the MCC vendor, to further reduce the exposure of fuse blowing. As such the alternate design allowance for fuse to be used for MOV circuit protection is acceptable.

Section 8.3.1.2.1.6, Part c.1: Currently, in section 8.3.1.2.1.6 part c.1 the sentence reads, "For all starter circuits feeding motors having full load currents equal to or greater than 0.66A, and for all feeder circuits, a circuit breaker and starter with a thermal overload relay as primary protection in series with a circuit breaker as backup protection or two circuit breakers in series for primary and backup protection are provided in each circuit."

The sentence will be modified to read, "For all starter circuits feeding motors and for all feeder circuits, primary protection is generally provided with a circuit breaker and/or starter with a thermal overload relay in series with a circuit breaker as backup protection or two circuit breakers in series for primary and backup protection are provided in each circuit."

Section 8.3.1.2.1.6, part c.2: Additionally, section 8.3.1.2.1.6 part c.2 reads, "For all starter circuits feeding motors having full load currents less than 0.66A, primary and backup protection is provided by means of a fused disconnect switch and a circuit breaker respectively."

The sentence in c.2 will be modified to read, "For starter circuits feeding motors having full load currents less than 1.3 A, primary and backup protection may be provided by means of a fused disconnect switch and a circuit breaker respectively."

This change does not affect the ability of the MOVs to perform their required safety functions. The alternate design allowance for fuse to be used for MOV circuit protection is limited to only a small set of MOVs, originally provided with fuse protection by the MCC vendor, to further reduce the exposure of fuse blowing. As such the alternate design allowance for fuse to be used for MOV circuit protection is acceptable.

LDCR-SA-2005-10, EVAL-2000-000526-5 (TJE):

Figure 8.3-16: In FSAR Figure 8.3-16, removed unit differences in lower left hand table under titles "NORMAL" and "TOTAL".

Note: The fiber optic modules are installed to support the Wireless LAN Extension into the Unit 1 Containment Building and to support the new Radiation Protection Camera LAN system. The fiber optic modules will not impact the other circuits in the penetrations. FSAR Sections 8.3.1.4.4, 8.3.1.4.5 and 8.3.1.4.10 all state there is no separation required between Non-1E fiber circuits and 1E circuits. Fiber Optic circuits will not interfere with 125 Vdc or 120 Vac circuits in Type 3 Penetrations, nor will the control circuits interfere with the fiber optic circuits. Therefore, there is no concern that the fiber optic will cause a

LDCR-SA-2005-10, EVAL-2000-000526-5 (TJE) (continued):

fire that could damage the other circuits nor will the fiber impair the operation of any other circuits.

LDCR-SA-2005-19, EVAL-2003-001365-14 (TJE):

Section 8.2.1: In section 8.2.1, 1.) add the following sentence to the end of the 11th paragraph, "The network transmission line terminals of 345-kV switchyard are provided with a third set of relay protection to enhance the reliability of transmission system fault isolation."

In section 8.2.1, 2.) add the following sentence to the end of the 12th paragraph, "A separate 125-VDC system is provided for the third set of relay protection in the 345-kV switchyard."

Note: The third scheme of relay protection can provide this last resort backup protection locally, thereby minimizing the impact on the system. This will increase the availability and reliability of the 345kV offsite source for CPSES safety related buses during 345kV system faults.

LDCR-SA-2005-25, EVAL-2002-003739-1 (TJE):

Section 1A(B): In section 1A(B), under the Discussion of RG 1.75, the proposed changes will change the wording "...upstream circuit breakers..." to "...upstream circuit protective device..." in the first sentence of the second paragraph. Currently, the sentence reads as follows:

"Regulatory Position C.1 Non Class 1E power or control circuits may be isolated from their Class 1E power source by two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit breaker, and the circuit breakers periodically tested. Non safety instrument circuits powered from distribution panels 1PC1, 1PC2, 1PC3, and 1PC4 will have a non safety circuit breaker or fuse connected in series with the panel circuit breaker."

The sentence will be revised to read:

"Regulatory Position C.1 Non Class 1E power or control circuits may be isolated from their Class 1E power source by two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective device, and the circuit breakers periodically tested. Non safety instrument circuits powered from distribution panels 1PC1, 1PC2, 1PC3, and 1PC4 will have a non safety circuit breaker or fuse connected in series with the panel circuit breaker."

Note: The FSAR descriptions of, breakers and starters tripped by SIAS as an isolation device, and a single fuse as an isolation device for NSSS protection system, are being revised to include the coordination requirement with the upstream protective device. This is a clarification change to improve the readers under standing of an isolation device.

LDCR-SA-2005-25, EVAL-2002-003739-1 (TJE) (continued):

Section 8.3.1.2.1.7, part a: Currently, section 8.3.1.2.1.7 under “a. Power Circuits” reads,

"Power Circuits

The following types of devices are used in the CPSES design for isolation of power circuits:

1) Circuit breaker tripped by a safety injection signal.

2) Starter contactor opened by a safety injection signal.

3) Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit breaker, and the circuit breakers periodically tested."

The proposed changes will revise this section to read,

"Power Circuits

The following types of devices are used in the CPSES design for isolation of power circuits:

1) Circuit breaker coordinated with the upstream protective device and tripped by a safety injection signal.

2) Circuit protective device of a starter coordinated with the upstream protective device and starter contactor opened by a safety injection signal.

3) Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective device, and the circuit breakers periodically tested."

Note: The FSAR descriptions of, breakers and starters tripped by SIAS as an isolation device, and a single fuse as an isolation device for NSSS protection system, are being revised to include the coordination requirement with the upstream protective device. This is a clarification change to improve the readers under standing of an isolation device.

Section 8.3.1.2.1.7, part b: Currently, section 8.3.1.2.1.7 and 8.3.1.2.1.7 under “b. Control and Instrumentation Circuits” parts 5 and 6 read,

"5) Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit breaker, and the circuit breakers periodically tested, provide isolation between Class 1E and non Class 1E circuits.

6) Within the NSSS protection system cabinets, a fuse or breaker is an isolation device to prevent malfunctions in Non-Class 1E portions of a circuit from causing unacceptable influences on the Class 1E function of the circuit (Reference 48). This exception to the

LDCR-SA-2005-25, EVAL-2002-003739-1 (TJE) (continued):

isolation devices described in the preceding paragraph (5), is applicable only to the NSSS circuit design by Westinghouse. See Section 7.1.2.2.1."

The proposed changes will revise these two paragraphs to read,

"5) Two circuit breakers, two fuses or a breaker and a fuse in series, both coordinated with an upstream circuit protective devise, and the circuit breakers periodically tested, provide isolation between Class 1E and non Class 1E circuits.

6) Within the NSSS protection system cabinets, a fuse or breaker, coordinated with the upstream protective devise, is an isolation device to prevent malfunctions in Non-Class 1E portions of a circuit from causing unacceptable influences on the Class 1E function of the circuit (Reference 48). This exception to the isolation devices described in the preceding paragraph (5), is applicable only to the NSSS circuit design by Westinghouse. See Section 7.1.2.2.1."

Note: The FSAR descriptions of, breakers and starters tripped by SIAS as an isolation device, and a single fuse as an isolation device for NSSS protection system, are being revised to include the coordination requirement with the upstream protective device. This is a clarification change to improve the readers under standing of an isolation device.

LDCR-SA-2005-32, EVAL-1999-002406-2 (TJE):

Section 1A(B): 1.) In 1A(B) under the Discussion of RG 1.75 after 36th paragraph, insert the following words:

"Physical separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits inside the Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A/02A is not required based on the analysis provided in Section 8.3.1.4.5."

Section 8.3.1.4.5: 2.) Insert the following words after the last paragraph of sections 8.3.1.4.5:

"Separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits in the Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A and 02A, listed in Table 8.3-10, is not required based on the following analysis. These panels are re-classified as Multi-Train cabinets. Both the AC and DC circuits for the Diesel Generator Engine Control Panels are supplied by a Class 1E power supply. The non-Class 1E circuits in the control panels are electrically isolated by an approved design having two circuit breakers in series for electrical isolation to assure that the Class 1E power source is not adversely affected by a fault in the non-Class 1E circuit.

Internal Train Separation Exemption:

Per Regulatory Guide 1.75, non-Class 1E circuits shall be physically separated from Class 1E circuits and non-Class 1E circuits shall be physically separated from associated

LDCR-SA-2005-32, EVAL-1999-002406-2 (TJE) (continued):

Class 1E circuits by the minimum separation requirements. All cables installed in the Diesel Generator Engine Control Panels are flame retardant and meet IEEE-383 flame test requirement. Any non-Class 1E cable fault, short circuit, short to ground, or over load condition will be isolated by an isolation breaker and will not cause the cable temperature to exceed normal operating temperature. The non-Class 1E cables inside the control panel will not be degraded or will not reach the temperature that can affect the Class 1E or associated Class 1E circuits. Open non-Class 1E circuit, have no impact on Class 1E or associated Class 1E circuits. The non-Class 1E cables when maintained within their rated temperature, have no adverse impact on adjacent Class 1E or associated Class 1E cables with no physical separation because the Class 1E and associated Class 1E cables also have the same temperature ratings.

Therefore, the adequacy of the safety-related circuits to perform their safety function is maintained with no separation between non-Class 1E and Class 1E circuits or non-Class 1E and associated Class 1E circuits inside the Diesel Generator Engine Control Panels. Any faults in the non-Class 1E circuits will not compromise the safety function of the Class 1E or associated Class 1E circuits internal to the Diesel Generator Engine Control Panels. Therefore, there is no need for internal electrical separation requirements within Diesel Generator Engine Control Panels CP1/2-MEDGEE-01A and 02A."

Table 8.3-10, sheet 9: 3.) Add the next four lines at the end of TABLE 8.3-10 and before the Notes:

"CP1-MEDGEE-01A DIESEL GENERATOR 1-01 DIESEL ENGINE CONTROL PANEL -- 18

CP1-MEDGEE-02A DIESEL GENERATOR 1-02 DIESEL ENGINE CONTROL PANEL -- 18

CP2-MEDGEE-01A DIESEL GENERATOR 2-01 DIESEL ENGINE CONTROL PANEL -- 18

CP2-MEDGEE-02A DIESEL GENERATOR 2-02 DIESEL ENGINE CONTROL PANEL -- 18"

Table 8.3-10, sheet 10: 4.) Add Note "18" to end of Table 8.3-10.

"18. Equipment supplied by Delaval Engine & Compressor has been analyzed and is exempt from internal separation requirement. (Reference FSAR Section 8.3.1.4.5.)"

LDCR-SA-2006-6, EVAL-2006-000595-1 (TJE):

Section 1.6: 1.) Currently in section 1.6, the first sentence of the first paragraph reads, "Table 1.6-1 lists topical reports and other licensing documents which provide information additional to that provided in this FSAR and have been filed separately with the Nuclear Regulatory Commission (NRC) in support of the CPSES operating licenses."

LDCR-SA-2006-6, EVAL-2006-000595-1 (TJE) (continued):

Delete the words, "topical reports and other licensing" from this sentence.

The sentence will then read, "Table 1.6-1 lists documents which provide information additional to that provided in this FSAR and have been filed separately with the Nuclear Regulatory Commission (NRC) in support of the CPSES operating licenses."

2.) Currently in section 1.6, the last sentence of the first paragraph reads, "Later revisions to these documents are not part of the FSAR unless they are incorporated into this table."

Add the following words to the end of this sentence, "(except for the Technical Requirements Manual (including TRM Bases) and the Technical Specifications Bases)"

The sentence will then read, "Later revisions to these documents are not part of the FSAR unless they are incorporated into this table (except for the Technical Requirements Manual (including TRM Bases) and the Technical Specifications Bases)"

Note:

This change is consistent with Technical Specification 5.5.14 and 5.5.17 which allow changes to the TRM and TSB under 10CFR50.59. Appendix A.4.3 of NEI-98-03, "Guidelines for Updating Final Safety Analysis Reports," Revision 1, allows these type of documents to be incorporated into the FSAR by reference. NEI-98-03, Revision 1, is endorsed by NRC Regulatory Guide 1.181, "Content of the Updated Final Safety Analysis Report in Accordance with 10 CFR 50.71(e)." This is also consistent with FSAR Section 16.2. This will help eliminate confusion and allow certain changes (e.g., reformatting, consistency, and others as defined in NEI-03, Revision 1) to be excluded from 10CFR50.59 review requirements.

Table 1.6-1, sheet 2: Replace the words in the title of Table 1.6-1 from "TOPICAL REPORTS" to "DOCUMENTS" on Sheet 2.

Note:

This change is consistent with Technical Specification 5.5.14 and 5.5.17 which allow changes to the TRM and TSB under 10CFR50.59. Appendix A.4.3 of NEI-98-03, “Guidelines for Updating Final Safety Analysis Reports,” Revision 1, allows these type of documents to be incorporated into the FSAR by reference. NEI-98-03, Revision 1, is endorsed by NRC Regulatory Guide 1.181, “Content of the Updated Final Safety Analysis Report in Accordance with 10 CFR 50.71(e).” This is also consistent with FSAR Section 16.2. This will help eliminate confusion and allow certain changes (e.g., reformatting, consistency, and others as defined in NEI-03, Revision 1) to be excluded from 10CFR50.59 review requirements.

Table 1.6-1, sheet 3: 1.) Replace the words in the title of Table 1.6-1 from "TOPICAL REPORTS" to "DOCUMENTS" on Sheet 3.

LDCR-SA-2006-6, EVAL-2006-000595-1 (TJE) (continued):

2) Add the following two Reports and Reference Sections to the end of Table 1.6-1 on Sheet 3:

Technical Requirements Manual (TRM) for Comanche Peak Steam Electric Station Units 1 and 2	All Sections
Technical Specifications Bases for Comanche Peak Steam Electric Station Units 1 and 2	All Sections

Note:

This change is consistent with Technical Specification 5.5.14 and 5.5.17 which allow changes to the TRM and TSB under 10CFR50.59. Appendix A.4.3 of NEI-98-03, “Guidelines for Updating Final Safety Analysis Reports,” Revision 1, allows these type of documents to be incorporated into the FSAR by reference. NEI-98-03, Revision 1, is endorsed by NRC Regulatory Guide 1.181, “Content of the Updated Final Safety Analysis Report in Accordance with 10 CFR 50.71(e).” This is also consistent with FSAR Section 16.2. This will help eliminate confusion and allow certain changes (e.g., reformatting, consistency, and others as defined in NEI-03, Revision 1) to be excluded from 10CFR50.59 review requirements.

LDCR-SA-2006-7, EVAL-2006-000627-2 (DWS):

Section 13.3B-1: Make organizational change "Senior Vice President and Principal Nuclear Officer" to "Senior Vice President and Chief Nuclear Officer"

TMI - I.C-3: Make organizational change. Change "Vice President, Nuclear Operations" to "Site Vice President"

In 13.1.1.2.1, change "Vice President, Nuclear Operations" to "Site Vice President" wherever it appears. Change "Vice President, Nuclear Engineering" to "Vice President, Nuclear Engineering and Support" wherever it appears. Realign reporting requirements for some managers. Delete obsolete manager titles and include new titles as appropriate.

Section 13.1A: Change "Vice President, Nuclear Operations" to "Site Vice President" wherever it appears. Change "Vice President, Nuclear Engineering" to "Vice President, Nuclear Engineering and Support" wherever it appears. Delete obsolete manager titles and include new titles as appropriate. Add QA manager title and resume.

In Figure 13.1-2, change "Vice President, Nuclear Operations" to "Site Vice President" . Change "Vice President, Nuclear Engineering" to "Vice President, Nuclear Engineering and Support" wherever it appears. Realign reporting requirements for some managers. Reinstate "Plant Manager" position.

LDCR-SA-2006-7, EVAL-2006-000627-2 (DWS) (continued):

In Figure 13.1-3, change "Vice President, Nuclear Operations" to "Site Vice President" and reinstate "Plant Manager" position.

In Table 13.1-1, change "Vice President, Nuclear Operations" to "Site Vice President" and add QA Manager

LDCR-SA-2006-20, EVAL-2006-000798-2 (DWS):

Section 13.3B - Revise ORC responsibility to assess the effectiveness of the fire protection program through review of periodic audits as discussed in FSAR 17.2.18 and corrected management title.

Audits have been moved to Section 17.2.18 and are conducted by the Nuclear Overview organization.

Section 13.4 - Revised to correct wording in FSAR from NRC audits to NRC inspection reports and to reflect changes made in FSAR Section 17.2.

LDCR-SA-2004-47, EVAL-2004-001171-2 (MJR):

FSAR Table 6.2.4-1, "Containment Isolation Valving Application," footnote a) for Table item 129 is revised to clarify that valve SF-0001 associated with the Fuel Transfer Tube penetration is a non-safety related valve that has no direct containment isolation function during MODES 1 through 4. This valve is normally closed in MODES 1-4. The valve may be opened in MODES 1-4 except when the Fuel Transfer Canal is flooded to maintain the Fuel Transfer Flange as leak rate tested. Opening the valve allows for equipment testing and to support preparations for refueling operations. (This change is consistent with clarifications added to DBD-ME-013 and DBD-ME-080 per approved and implemented FDA-2004-001171-01-00.)

LDCR-SA-2003-47 (MJR):

Appendix 1A(B), Regulatory Guide 1.75, is revised to add the Unit 1 and 2 Emergency Diesel Generators to the discussion of the Partial Discharge Monitor Bus Couplers. PDM Bus Couplers were installed on EDGs per FDA-2001-001255-02. The previous reference to Unit 2 only for Component Cooling Water motor feeders is deleted to reflect that PDM Bus Couplers have now been installed on both Unit 1 and 2 Component Cooling Water motors per FDA-2000-000127.

Appendix 1A(B), Regulatory Guide 1.131, is revised to add the Unit 1 and 2 Emergency Diesel Generators to the discussion of the Partial Discharge Monitor Bus Coupler jumpers. PDM Bus Couplers were installed on EDGs per FDA-2001-001255-02.

Section 8.3.1.2.1.7, Onsite Power Systems, Compliance, RG 1.75 and IEEE-354, is revised to add the Unit 1 and 2 Emergency Diesel Generators to the discussion of the Partial Discharge Monitor Bus Coupler jumpers. PDM Bus Couplers were installed on

LDCR-SA-2003-47 (MJR) (continued):

EDGs per FDA-2001-001255-02, and specific component tag numbers and wire size details are deleted as below the level of detail required for the FSAR description. Also, the Component Cooling Water Pump Motors were added to the discussion for completeness. PDM Bus Couplers were installed on CCWP motors per FDA-2000-000127.

Section 9.2.2.2 System Description is revised to delete the previous reference to Unit 2 for the CCW pump motors to reflect that PDM Bus Couplers have now been installed on both Unit 1 and 2 Component Cooling Water pump motors per FDA-2000-000127.

Section 9.5.1.6.2, Fire Protection System, Justification for Items of Noncompliance to Appendix A to Branch Technical Position APCSB 9.5-1, Guideline D.3.f, is revised to add the Unit 1 and 2 Emergency Diesel Generators to the discussion of the Partial Discharge Monitor Bus Coupler jumpers. PDM Bus Couplers were installed on EDGs per FDA-2001-001255-02. Also, specific component tag numbers are deleted as below the level of detail required for the FSAR description.

Table 8.3-3, "Failure Mode and Effect Analysis for Auxiliary AC Power System," is revised to add items 15C, 15D, 16C, and 16D for the Partial Discharge (PD) bus couplers installed on the Emergency Diesel Generators per FDA-2001-001255-05, and to add an abbreviation note to define "partial discharge (PD)."

LDCR-SA-2004-22, EVAL-2004-002167-1 (GLM):

Revise Section 9.2.8 to be consistent with the current approved waste effluent release permit. This includes changes to reflect current terminology as well as changes to bring the FSAR Section in line with the current wastewater discharge permit.

Section 9.2.8 of the FSAR documents the Design Basis and System Description of the Waste Management System (WMS) at CPSES. The design function of the Waste Management System is the collection, retention, treatment and discharge of normally non-radioactive wastewaters. The system does this in accordance with the Texas Commission on Environmental Quality (TCEQ) and the Texas Pollutant Discharge Elimination System (TPDES) permits. The WMS collects wastewaters for processing in either the Low Volume Waste (LVW) or Co-Current Waste (COW) treatment facilities. The goal of most of the changes is to provide the description of the major components in the system and to reiterate that the release of effluents is done within the limitation of our Discharge Permit. Specifically, the wording changes involved in this submittal are administrative in nature. The regulatory agency functions of the agencies in regard to CPSES have remained the same, however, the names of the agencies have changed. The Texas Natural Resource Conservation Commission (TNRCC) is now the Texas Commission on Environmental Quality (TCEQ). The National Pollutant Discharge Elimination System (NPDES) is now the Texas Pollutant Discharge Elimination System (TPDES). While this change request was being processed, it was determined that the description of the systems should be updated to be more consistent with the wording of the current approved discharge permit and some change was necessary to outline some

LDCR-SA-2004-22, EVAL-2004-002167-1 (GLM) (continued):

current operating practices. This submittal also includes corrections in system terminology in the descriptions. The LVW facilities required more change than did the COW facility descriptions. For the most part, the facility descriptions remain unchanged although the wording has been reformatted to align better with the Discharge Permit. The changes requested better describe the use of the facilities and reinforce some detail regarding both use and operation of the facilities which fall within applicable TCEQ permit limitations.

Revise Section 9.2.8 to be consistent with the current approved waste effluent release permit. This includes changes to reflect current terminology as well as changes to bring the FSAR Section in line with the current wastewater discharge permit.

Section 9.2.8 of the FSAR documents the Design Basis and System Description of the Waste Management System (WMS) at CPSES. The design function of the Waste Management System is the collection, retention, treatment and discharge of normally non-radioactive wastewaters. The system does this in accordance with the Texas Commission on Environmental Quality (TCEQ) and the Texas Pollutant Discharge Elimination System (TPDES) permits. The WMS collects wastewaters for processing in either the Low Volume Waste (LVW) or Co-Current Waste (COW) treatment facilities. The goal of most of the changes is to provide the description of the major components in the system and to reiterate that the release of effluents is done within the limitation of our Discharge Permit. Specifically, the wording changes involved in this submittal are administrative in nature. The regulatory agency functions of the agencies in regard to CPSES have remained the same, however, the names of the agencies have changed. The Texas Natural Resource Conservation Commission (TNRCC) is now the Texas Commission on Environmental Quality (TCEQ). The National Pollutant Discharge Elimination System (NPDES) is now the Texas Pollutant Discharge Elimination System (TPDES). While this change request was being processed, it was determined that the description of the systems should be updated to be more consistent with the wording of the current approved discharge permit and some change was necessary to outline some current operating practices. This submittal also includes corrections in system terminology in the descriptions. The LVW facilities required more change than did the COW facility descriptions. For the most part, the facility descriptions remain unchanged although the wording has been reformatted to align better with the Discharge Permit. The changes requested better describe the use of the facilities and reinforce some detail regarding both use and operation of the facilities which fall within applicable TCEQ permit limitations. The second thru fifth paragraphs were deleted since they provide both duplicate information and more detail than is necessary to describe the design volumes in the settling ponds. The sixth paragraph was deleted since it is redundant to information in a new paragraph. This paragraph details periodic testing of Condensate Polishing wastes. This information is accurate but is already covered by the statements in the second paragraph which indicates that releases are controlled via the ODCM program. Typically wastes sent to the Low Volume Waste Management System do not include radioactive wastes. Although there are accident scenarios which postulate such releases, there is no normal release of such wastes to the Low Volume Waste System. Releases

LDCR-SA-2004-22, EVAL-2004-002167-1 (GLM) (continued):

are controlled via the ODCM limits as well as the limits established by the Release Permit.

Revise Section 9.2.8 to be consistent with the current approved waste effluent release permit. This includes changes to reflect current terminology as well as changes to bring the FSAR Section in line with the current wastewater discharge permit.

Section 9.2.8 of the FSAR documents the Design Basis and System Description of the Waste Management System (WMS) at CPSES. The design function of the Waste Management System is the collection, retention, treatment and discharge of normally non-radioactive wastewaters. The system does this in accordance with the Texas Commission on Environmental Quality (TCEQ) and the Texas Pollutant Discharge Elimination System (TPDES) permits. The WMS collects wastewaters for processing in either the Low Volume Waste (LVW) or Co-Current Waste (COW) treatment facilities. The goal of most of the changes is to provide the description of the major components in the system and to reiterate that the release of effluents is done within the limitation of our Discharge Permit. Specifically, the wording changes involved in this submittal are administrative in nature. The regulatory agency functions of the agencies in regard to CPSES have remained the same, however, the names of the agencies have changed. The Texas Natural Resource Conservation Commission (TNRCC) is now the Texas Commission on Environmental Quality (TCEQ). The National Pollutant Discharge Elimination System (NPDES) is now the Texas Pollutant Discharge Elimination System (TPDES). While this change request was being processed, it was determined that the description of the systems should be updated to be more consistent with the wording of the current approved discharge permit and some change was necessary to outline some current operating practices. This submittal also includes corrections in system terminology in the descriptions. The LVW facilities required more change than did the COW facility descriptions. For the most part, the facility descriptions remain unchanged although the wording has been reformatted to align better with the Discharge Permit. The changes requested better describe the use of the facilities and reinforce some detail regarding both use and operation of the facilities which fall within applicable TCEQ permit limitations. The second paragraph regarding Chemical Sump Wastes was also deleted. This paragraph indicates that Chemical Sump Wastes are being discharged to the Equalization Basin and in turn to the Natural Neutralization Basin, Neutralization Clearwell with an ultimate discharge and/or commingling with LVW Pond effluents. These elements and processing within the LVW facilities are no longer in normal practice. Currently, effluents from the Chemical Sumps are processed together with the other plant wastewaters via the Separation Surge Basin, API Oil Separator and Settling Basins. Descriptions regarding the use of the Equalization Basin lineup is no longer relevant or necessary based on the current normal operating lineups or practices.

Revise Section 9.2.8 to be consistent with the current approved waste effluent release permit. This includes changes to reflect current terminology as well as changes to bring the FSAR Section in line with the current wastewater discharge permit.

LDCR-SA-2004-22, EVAL-2004-002167-1 (GLM) (continued):

Section 9.2.8 of the FSAR documents the Design Basis and System Description of the Waste Management System (WMS) at CPSES. The design function of the Waste Management System is the collection, retention, treatment and discharge of normally non-radioactive wastewaters. The system does this in accordance with the Texas Commission on Environmental Quality (TCEQ) and the Texas Pollutant Discharge Elimination System (TPDES) permits. The WMS collects wastewaters for processing in either the Low Volume Waste (LVW) or Co-Current Waste (COW) treatment facilities. The goal of most of the changes is to provide the description of the major components in the system and to reiterate that the release of effluents is done within the limitation of our Discharge Permit. Specifically, the wording changes involved in this submittal are administrative in nature. The regulatory agency functions of the agencies in regard to CPSES have remained the same, however, the names of the agencies have changed. The Texas Natural Resource Conservation Commission (TNRCC) is now the Texas Commission on Environmental Quality (TCEQ). The National Pollutant Discharge Elimination System (NPDES) is now the Texas Pollutant Discharge Elimination System (TPDES). While this change request was being processed, it was determined that the description of the systems should be updated to be more consistent with the wording of the current approved discharge permit and some change was necessary to outline some current operating practices. This submittal also includes corrections in system terminology in the descriptions. The LVW facilities required more change than did the COW facility descriptions. For the most part, the facility descriptions remain unchanged although the wording has been reformatted to align better with the Discharge Permit. The changes requested better describe the use of the facilities and reinforce some detail regarding both use and operation of the facilities which fall within applicable TCEQ permit limitations.

LDCR-SA-2004-44, EVAL-2001-000158-7 (GLM):

Revise Section 9.3.1 to reflect the following: 1) the word "Spare" is no longer applicable as the common compressors may be operated as the lead compressor for either unit. Therefore, spare is changed to "common", 2) each common and Unit 2 instrument air compressor has a dedicated trim cooler cooled by non-safety chilled water, 4) the Unit 1 CCW system remains in a series flow path for the unit 1 compressors. Therefore, a single trim cooler is used, 3) due to the elevated temperatures in the Turbine Building during the Summer months, a non-safety chilled water ventilation cooler is provided to duct cooled air into the compressor air inlet, ensuring intake parameters are within operational limits.

Revise Section 9.3.1 to reflect the following: Each instrument air dryer has two 100% capacity prefilters.

Revise Table 9.3-1 to reflect the following: On sheets 1 and 2, Part "A" is deleted as all 435 SCFM, rotary, two stage, oil free, water cooled air compressors have been removed from the plant.

LDCR-SA-2004-44, EVAL-2001-000158-7 (GLM) (continued):

Revise Table 9.3-1 to reflect the following: On sheet 3, revise the last 2 entries in the table to reflect that the exit cooling water temperature for the Common and Unit 2 compressors is 122 degrees F and the required cooling water flow for the Common and Unit 2 compressors is 40 gpm.

Revise Section 8.3.1.2.1 to reflect the following: The new instrument air compressor control panel CP2-CICACO-01B is Train C and some associated cables are terminated at the control panel without isolation devices.

Revise Table 8.3-10 to reflect the following: Add new instrument air compressor control panel CP2-CICACO-01B to the table.

Revise Table 8.3-11 to reflect the following: The Unit 1 power source and ID numbers are editorially corrected for instrument air compressors 1-01 and 1-02 and instrument air dryer control panel 1-02.

Revise Table 8.3-11 to reflect the following: Add CP2-CICACO-01, CP2-CICACO-02, CP2-CICYIA-01, and CP2-CICYIA-02 to the table.

LDCR-SA-2005-27, EVAL-2004-002128-4 (GLM):

Revise Section 9.2.3.1 to add description of nitrogen sparger that was installed in the Unit 1 RMWST per FDA-2004-002128-01. Adding nitrogen to the RMWST will improve the control of the dissolved oxygen concentration.

LDCR-SA-2006-8, EVAL-1999-000248-8 (GLM):

Revise Table 9.4-2 to add a note indicating that the temperatures in the DG area are maintained with the help of the DG keepwarm system.

The current design basis was established by Calculation 1-EB-302A-1 Rev. 5 "As built HVAC Calc Diesel Generator Area--Space Heat Gains, Space Heat Losses and Maximum and Minimum Temperatures - Unit 1. (dated 9/20/1989). The same design was confirmed in Unit 2 Calc. 2-HV-0009 Rev. 1. 1-EB-302-1 Rev. 5 on page 3 indicates the reason for Rev. 5 "Calculation Rev. 5 is a complete revision which takes credit for the D.G. "Keepwarm" system for space heating as per TU-Electric Letter No. OSE-89026 dated 9-4-89." The justification for this licensing basis change and the Engineering Basis for FDA-1999-000248-03 is Calculation 1-EB-302-1 Rev. 5. issued on 9/20/1989. The calculation demonstrates that the DG Jacket water 52.7 KW heater with the 27.5 KW Lube oil heater and the two installed 5 KW room heaters was more than enough heat to keep all portions of the Unit 1 DG rooms well above 40ºF. 1-EB-302-1 Rev 5 determined that the available safety margin is 56.8%. However, if we have equipment failure, or if the outside ambient temperature falls below the design basis 20ºF temperature or if we have excessive infiltration, supplemental heating and/or other compensatory measures may be required too maintain DG room temperature at or above 40ºF.

LDCR-SA-2006-19, EVAL-2004-003769-19 (GLM):

Revise Section 9.3.1.2 to delete "to regulate flow" since, although it is a function of the valves, it is not a design function of the accumulators. Discussion of the flow regulation is moved to Section 10.4.9.2 which is referenced for clarification.

Clarification suggested by SMF-2004-003769 generated in response to NRC questions. There are no changes to the design function or the method of performing the design function. See FDA-2004-003769-01 for design impact assessments and additional details.

Revise Section 10.4.9.2 to reflect the following: Add clarification to incorporate information from Section 6.2.1, SSER 17, and CPSES-200402640. This places all pertinent information in one location.

Clarification suggested by SMF-2004-003769 generated in response to NRC questions. There are no changes to the design function or the method of performing the design function. See FDA-2004-003769-01 for design impact assessments and additional details.

LDCR-SA-2006-17, EVAL-2006-000809-1 (JCH):

Table 10.4-11 entry for chlorides is incorrect. Chloride value should be "10.0" vs "1.0" mg/l (max)

This appears to be a typo. DBD-ME-212 was changed from 1.0 to 10.0 mg/l by FDA-2001-002476-01 to match existing Chemistry procedures (which had been revised to match EPRI Closed cooling Water Chemistry Guideline 1007820). The FDA failed to identify the FSAR as impacted.

LDCR-SA-2006-21, EVAL-2006-002113-1 (DWS):

Section 13.1A: Added new plant manager's resume.

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LDCR-SA-2005-13, EVAL-1999-003132-05 (JCH):

Section 10.4.10.5: states the following- "Each turbine side drain pot is provided with an extraction line drain valve. These valves are opened and closed manually with control-board-mounted switches. After a valve opens, it remains open until manually closed.

Revise Section 10.4.10.5 to remove extraction line turbine side drain pot level switches and their associated level indication, level control and level alarm functions. Extraction Steam system drip pots (located between the Turbine and Feedwater Heater sides of the supply line stop and check valves) are not required to provide “Level Indication and Control Functions” in order to satisfy any CPSES commitments or applicable codes and standards. This change resolves the unit difference between Unit 1 and Unit 2.

LDCR-SA-2005-30, EVAL-2003-002540-04 (JCH):

Section 10.4.2: To ensure more reliable operation of the vacuum priming pumps year-round, a modification was implemented to provide full required seal water flow rate of 3 gpm per pump from the Demineralized and Reactor Makeup Water System. The water is not cooled and recirculated but is sent to the nearby floor drain as it exits the pump.

Removal of the vacuum primary pump heat exchangers ensures that the setpoint vacuum can be produced by the vacuum priming pumps. Other clarifications were to accurately reflect the system design per FDA-2003-002540-03 and -04.

LDCR-SA-2006-26, EVAL-2004-000897-02 (GLM):

In Section 10.4.12.5.1: Revise the first and second sentence to delete reference to the pitot tube type flow meter.

FDA-2004-00897-01 changes the way the TPCW total flow is determined for Unit 2. The existing Unit 2 total flow indication is erratic and varies depending upon which pump is running. To obtain a more reliable flow indication, the East, West and recirc flows are summed then transmitted to the control room. The control room flow indication and alarms will not be changed. Unit 1 is not modified (still has the pitot tube type flow meter), this change only removes the level of detail regarding the specific type of flow meter.

LDCR-SA-2005-17, EVAL-2004-002063-7 (JDS):

Revised Table 7.5.7E, sheet 4 of 4 to incorporate commitment made during license submittal for TSTF 447 for H2 recombiner removal. This was approved by the NRC in License Amendment 117.

LDCR-SA-2005-24, EVAL-2002-001952-3 (JDS):

Revised Ch. 1A(B) to provide discussion of deviations to Regulatory Guide 1.82 that are being addressed as part of resolution of the generic issue regarding containment emergency sump performance (GSI-191).

LDCR-SA-2005-24, EVAL-2002-001952-3 (JDS) (continued):

Provide for description of the transport of debris through the radiation protection doors as a result of modifications to improve containment emergency sump performance.

LDCR-SA-2006-13, EVAL-2006-000212-02 (JDS):

Section 4.4.2.2.2 - The FSAR description of the calculation of the fuel assembly spring holddown force has been revised. The use of the mechanical design RCS flow has been replaced by the use of the best-estimate RCS flow with margin.

The current Westinghouse methods involve the use of best estimate flow (vs mechanical design flow) to calculate best estimate lift forces. An uncertainty factor (calculated by convoluting uncertainties, including an uncertainty to account for the difference of best estimate flow to design flow) is then applied resulting in the design lift forces. These design lift forces are then used to ensure the functional compliance of the holddown springs. The hold down springs remain capable of performing their function.

Section 5.1, under Best Estimate Flow and Thermal Design Flow - The FSAR description of the calculation of the fuel assembly spring holddown force has been revised. The use of the mechanical design RCS flow has been replaced by the use of the best-estimate RCS flow with margin.

The current Westinghouse methods involve the use of best estimate flow (vs mechanical design flow) to calculate best estimate lift forces. An uncertainty factor (calculated by convoluting uncertainties, including an uncertainty to account for the difference of best estimate flow to design flow) is then applied resulting in the design lift forces. These design lift forces are then used to ensure the functional compliance of the holddown springs. The hold down springs remain capable of performing their function.

LDCR-SA-2006-33, EVAL-2001-001255-17 (TJE):

Section 8.3.1.1.11.1: Replace the words "Unit 1 and Unit 2 Diesel Generators use" with the words "Each EDG uses."

Currently, this paragraph reads,

"Unit 1 and Unit 2 Diesel Generators use an upgraded digital excitation/voltage regulator. This digital exciter/voltage regulator is a Safety Related Siemens THYRIPART generator field excitation unit. This THYRIPART excitation unit utilizes a digital control system consisting of a 32-bit technology module and communications module. This system controls the output of a chopper controller circuit to achieve +0.5 percent Diesel Generator voltage regulation, and has capability of both manual as well as automatic control."

LDCR-SA-2006-33, EVAL-2001-001255-17 (TJE) (continued):

The paragraph will be revised to read,

"Each EDG uses a digital excitation/voltage regulator. This digital exciter/voltage regulator is a Safety Related Siemens THYRIPART generator field excitation unit. This THYRIPART excitation unit utilizes a digital control system consisting of a 32-bit technology module and communications module. This system controls the output of a chopper controller circuit to achieve +0.5 percent Diesel Generator voltage regulation, and has capability of both manual as well as automatic control."

Section 8.3.1.1.11.2: 1.) Add the following item to the reason for the automatic tripping of the DGs

"t. Generator underfrequency"

Delete the following paragraph

"These protective functions on the Unit 1 and the Unit 2 DG units utilize an upgraded digital multifunction protective device (MPR-1) locally which generates a common alarm to the control room."

Delete the following paragraph,

"For Unit 1 and Unit 2, the individual protective alarms except for the differential protection signal target are available locally on device MPR-1."

LDCR-SA-2006-15, EVAL-2006-001324-02 (TJE):

Section 8.3.1.1.11.3: FSAR Section 8.3.1.1.11, sub-section 3. "Testability and Maintenance," is revised to remove portions of the FSAR description of Diesel Generator (DG) periodic testing that are no longer applicable. This section of the FSAR formerly included additional detail that distinguished between DG testing performed during unit at-power operation and that which was previously restricted to performance during unit shutdowns. Surveillance tests for the DGs and the Offsite AC Sources are performed in accordance with the requirements of Technical Specifications (TS) 3.8.1 and are further described in the corresponding TS Bases. This change updates the FSAR description of DG periodic testing to be consistent with the current TS 3.8.1 surveillance mode restrictions as revised by License Amendment 124, effective April 24, 2006.

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LDCR-SA-2006-18, EVAL-1999-003485-02 (NSH):

Changes to Table 10.4-18, Page 1 of 1, corrects the Heater Drain Pump discharge pressure from 200 psig to 600 psig, identify temperature as 400^{O}F and identify the split of the Equalization Header as a sample point. This corrects the current FSAR Table 10.4-18, “Sample Points Pressure and Temperature Conditions” table. The Table in the FSAR currently incorrectly states that the Heater Drain Pump discharge parameters are the same as the Heater Drain Tank Equalization Header.

LDCR-SA-2006-32, EVAL-2005-001011-02 (RJK):

In FSAR table 9.3-4 the RCS Pressurizer Steam Space minimum sample rate is 0.75 GPM; in COP-101A/B (section 5.4.1) the minimum sample rate is 0.6 GPM. Based on a calculation performed, the Reynolds number for flow at 0.75 GPM is 3869. The Reynolds number for flow at 0.6 GPM is 3096. While the Reynolds number is less that the value for 0.75 GPM the number is greater that laminar flow values, therefore the minimum flow of 0.6 GPM in COP-101A/B provides flow sufficiently above laminar values to provide a representative sample.

In FSAR table 9.3-4 the SI Accumulator Tank minimum sample rate is 0.75 GPM; in COP-202A/B (section 5.1) the minimum sample rate is 1 liter/min. Based on a calculation performed, the Reynolds number for flow at 0.75 GPM is 3869. The Reynolds number for flow at 1 liter/min is 1363. A search of several references indicates that a Reynolds number of <2000 - 2300 is considered laminar flow and a Reynolds number of >4000 indicate turbulent flow. Based on the calculation, the lower flow referenced in the COP is laminar in nature. To compensate for the lower flow rate the purge volume is increased to 30 liters to ensure a representative sample of the accumulator is collected for analysis. The only item sampled for in the SI accumulator is boron, the Technical Specification limits for boron is 2300-2600 ppm, boron at the specified concentration will remain soluble well below temperatures seen in containment. Also, no particulate sampling/analysis is required from the SI accumulators. Based on recent testing on the sample system while installing new throttle valves, 1 liter/min is the maximum flow rate that can be expected through the lines as constructed.

LDCR-SA-2006-5, EVAL-2002-004206-04 (RJK):

In FSAR Section 3.6B.2.5.2 the Unit 2 system design pressure has been established at 1860 psig to bound the maximum SI pump discharge pressure that could exist following a LOCA and while in the recirculation mode. The system piping and components have been re-evaluated for this higher design pressure. In Table 6.3-2 the relief valve set points have been changed to 1860 psig, i.e., just above the maximum postulated SI pump discharge pressure, to prevent their opening while in the recirculation mode. Increasing the relief valve set points from 1750 psig to 1860 reduces the probability that the valves will open while the SI system is operating in the recirculation mode and, therefore, provides greater assurance that they can perform their function as a boundary for a closed system outside containment.

LDCR-SA-2006-5, EVAL-2002-004206-04 (RJK) (continued):

The Unit 2 system design pressure has been established at 1860 psig to bound the maximum SI pump discharge pressure that could exist following a LOCA and while in the recirculation mode. The system piping and components have been re-evaluated for this higher design pressure.

In Table 6.3-1 the Unit 2 relief valve set points have been changed to 1860 psig, i.e., just above the maximum postulated SI pump discharge pressure, to prevent their opening while in the recirculation mode. Increasing the relief valve set points from 1750 psig to 1860 reduces the probability that the valves will open while the SI system is operating in the recirculation mode and, therefore, provides greater assurance that they can perform their function as a boundary for a closed system outside containment.

LDCR-SA-2005-26, EVAL-2005-002515-02 (JDS):

Revised sections 4.2, 4.3.2, 4.4.1, Appendix 4A Tables 4A.1.2-1, 4A.2.2-2, and 1.6-1: The Unit 1 Cycle 12 reload core design consists of 88 Westinghouse IFM/OFA fresh fuel assemblies, 89 Westinghouse Revised P+ once-burned fuel assemblies from U1C11(including one reconstituted with 3 stainless steel rods) and 16 twice-burned Westinghouse Revised P+ assemblies from U1C11. The Westinghouse Revised P+ design includes a small hole, debris-filtering bottom nozzle design, longer solid end plugs on the fuel rods, and an alternate "P-grid" located just above the bottom nozzle to trap any debris making it through the bottom nozzle. In addition to the above features, the U1C12 fresh IFM/OFA fuel assemblies include IFM grids for enhanced fuel performance and an oxide coating at the bottom 7 inches of each fuel rod for fuel rod protection. Oxide forms naturally on the fuel cladding during normal operation so this coating has no impact on fuel design function. The pressure drop through the Westinghouse Revised P+ assemblies is slightly lower than that through the co-resident fresh Westinghouse IFM/ OFA assemblies while the heat transfer characteristics are improved by the IFMs. The effects of the mixed core on the thermal-hydraulic response are considered. The impact of the stainless steel, unheated rods in the reconstituted assembly on the mechanical, nuclear and thermal-hydraulic designs was also considered.

Added COLR references from Section 5.6.5 of the Technical Specifications for completeness.

Added Zirlo reference for the specific cladding used by Westinghouse fuel.

Added discussion on ZIROL fuel, reconstitutided fuel, and Westinghouse Optimized fuel with axial blankets or integral fuel burnable absorber. These fuel types and options are loaded into Unit 1 Cycle 12.

Added fuel cladding discussions on fretting, cycling and fatigue, and irradiation stability of the cladding and reference.

Added Zirlo reference for the specific cladding used by Westinghouse fuel.

LDCR-SA-2005-26, EVAL-2005-002515-02 (JDS) (continued):

Added FRA-ANP (framatome) to cover both fuel vendors used.

Added references for to Westinghouse topicals (WCAPs) used to support Unit 1 cycle 12.

Add discussion for reconstituted fuel used in Unit 1 cycle 12.

LDCR-SA-2006-27, EVAL-2006-002148-02 (JCH):

Replace "Engineering Programs Manager" with "Technical Support Manager in paragrahs 13.3B.1.2 and 13.3B.1.4.

Move item 2 under paragraph 13.3B.1.4 to item 3 under paragraph 13.3B.1.2 and renumber subsequent items under 13.3B.1.2 and 13.3B.1.4.

Reassignment of responsibilities is due to reorganization as a result of business process outsourcing.

LDCR-SA-2006-29, EVAL-2001-002001-01 (DWS):

Table 17A-1: Allow permanent installation and erection of Scaffold poles inside Containment Steam Generators Compartments or Loop Rooms and permit storage racks for scaffold material outside the loop rooms at the lower elevation. Reference FDA 2006-002001-01.

LDCR-SA-2006-28, EVAL-2006-001288-03 (JDS):

Update of FSAR for the Unit 2 Cycle 10 core configuration and operation. This update replaces the current FSAR Section 4B in its entirety.

Add reference to section 9.1.4.2.1 for a description of the soluble boron credit.

Add additional description for the soluble boron credit. 9.1.4.2.1 describes the soluble boron credit

Add Step 5 for conservative actions to be performed for operator actions to be performed for primary to secondary break.

LDCR-SA-2007-1, EVAL-2006-002939-02 (DWS):

Section 13.4-1: Change organizational name from "Nuclear Overview" to "Quality Assurance".

Section 14.2.2.4.6 Change title from "Director of Nuclear Overview" to "Director, Oversight & Regulatory Affairs". Section 14.2.2.6 & 14.2.2.8 change organizational name from "Nuclear Overview" to Quality Assurance".

LDCR-SA-2007-1, EVAL-2006-002939-02 (DWS) (continued):

Section 1.B.1.2 CPSES Response - change organizational name from "Nuclear Overview" to "Quality Assurance".

Section 1.C.5 change organizational name from "Nuclear Overview Department" to "Performance Improvement Department" in three places.

Section 17.2: TOC Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs"

Section 17.2.1.1.2 Add additional responsibility for to the Site Vice President. Responsible for "Technical and administrative direction of the Director, Performance Improvement."

Section 17.2.1.1.3 Add responsibility for Plant Manager. Responsible for Tecnical and administrative direction to the Director, Plant Support.

Section 17.2.1.1.4 Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs", change organizational name from "Nuclear Overview" to "Quality Assurance", and revise duties and responsibilities for the Director Nuclear Oversight & Regulatory Affairs and add duties and responsibilities for the Quality Assurance Manager.

Section 17.2.1.2 Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs", change organizational name from "Nuclear Overview" to "Quality Assurance", and remove function number 10 "Administering and facilitating the corrective action program."

Section 17.2.1.5.1 Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs."

Section 17.2.2 Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs", change organizational name from "Nuclear Overview" to "Quality Assurance", add duties for the Quality Assurance Manager, remove evaluations and replace with "surveillance and audits", and remove the responsibility for "Administering and facilitating the corrective action program."

Section 17.2.5 Change organizational name from "Nuclear Overview" to "Quality Assurance", remove evaluations and replace with "surveillance and audits"

Section 17.2.6.1 Change title from "Director Nuclear Overview" to "Quality Assurance Manager", change organizational name from "Nuclear Overview" to "Quality Assurance"

Section 17.2.7 Change organizational name from "Nuclear Overview" to "Quality Assurance"

Section 17.2.9 Change "procedures shall be reviewed by the Nuclear Overview Department or designee" to "review by Quality Assurance or other qualified personnel."

LDCR-SA-2007-1, EVAL-2006-002939-02 (DWS) (continued):

Section 17.2.10 Change title from "Director Nuclear Overview" to "Quality Assurance Manager", change organizational name from "Nuclear Overview" to "Quality Assurance"

Section 17.2.12 Change organizational name from "Nuclear Overview" to "Quality Assurance" and add "TXU Power" in front of manager of the oraganization responsible for the calibration.

Section 17.2.14 & 17.2.15 Change organizational name from “Nuclear Overview” to “Quality Assurance”.

Section 17.2.16 Add "The Director, Performance Improvement is responsible for administrating and faciltating the corrective action program".

Section 17.2.18 & .1 & .2 & .2.1 Replace "evaluation" with audit, change title from "Director Nuclear Overview" to "Quality Assurance Manager", and change organizational name from "Nuclear Overview" to "Quality Assurance"

Section 13.1: TOC Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs", Change organizational name from "Nuclear Overview" to "Quality Assurance".

Section 13.1 Change organizational name from "Nuclear Engineering to Nuclear Engineering & Support; Regulatory Affairs to Oversight & Regulatory Affairs; Nuclear Overview to Quality Assurance.

Section 13.1.1.2.1 added "Senior" to title of "Vice President & Chief Nuclear Officer" and added position of "Director, Performance Improvement" with responsiblities outlined. Add "Supply Chain and Procurement to Director, Plant Support responsibilities. Removed position of "Emergency Planning Manager"

Section 13.1.1.2.2 Revise organization name to Nuclear Engineering & Support. Removed position of "Manager, Administrative Services" and moved responsibility to Manager, Project Engineering.

Section 13.1.1.2.3 Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs" and revise organizational name.

Section 13.1.1.2.4 Change organizational name from "Nuclear Overview" to "Quality Assurance", Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs", remove responsibilities for "Providing for the review and assessment of reports of nuclear industry operating experience.", "Administering and facilitating the corrective action program.", and "Provide for the trending and analysis of conditions adverse to quality." The responsibilities are now under the Director, Performance Improvenment who reports to the Site Vice President.

Section 13.1.2.1.2 Rewrite responsibilities of "Shift Operation Manager"

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LDCR-SA-2007-1, EVAL-2006-002939-02 (DWS) (continued):

Section 13.1.2.1.3 Revise organizational manager titles.

Section 13.1.2.1.4 Editorial changes

Section 13.1.2.1.5 Remove the organization "Central Organization for Reliable Plant Systems (CORPS). This organization has been reassigned and renamed.

Section 13.1.2.1.6 Delete section number (number only).

Table 13.1: Revise titles, remove PLANT MOD TEAM MANAGER and insert Project Engineering Manager, Change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs".

Section 13.1A-1: Insert resumes and titles for new personnel and change title from "Director Nuclear Overview" to Director, Oversight & Regulatory Affairs". Remove old titles and resumes.

Section 13.3 Delete redundant and historical information on Fire Protection.

Section 13.3B Editorial changes to Fire Protection discription.

Section 13.3B.1.2 & .1.3 & .1.4 & .1.5 Revise management titles

Section 13.4-1: Change organizational name from "Nuclear Overview" to "Quality Assurance"

Sections 13.5-1, 5, 7: Correct name of company from TXU Energy to TXU Power and revise management titles.

Figure 13.1-2: Added Director, Performance Improvement and re-alignment of reporting respondibilities of the Corrective Action Program Manager and the Quality Assurance Manager.

Figure 13.1-3: Added Director, Performance Improvement and change various management titles.

LDCR-SA-2004-42, EVAL-2004-000773-01 (JCH):

Revised sections 3.5.1, 5.4.1, 7.1.2, 7.3.2, 7.7.2, 9.5.1, 10.2.2, 10.2.3, 10.2.4 and Table 7.3-4: The requirement for a complete maintenance inspection of the mechanical overspeed protection system every refueling outage has been removed, based on the fact that the mechanical overspeed system has been removed and replaced with a digital electronic protection system. The new digital overspeed protection system is auotmatically tested periodically during each shift and is verified operational every 2 weeks. The turbine trip block is also tested every 2 weeks

LDCR-SA-2004-42, EVAL-2004-000773-01 (JCH) (continued):

Deleted mechanical overspeed trip since the device is removed and added words to describe the new electronic overspeed systems

Revised "3. Turbine" of this section. This section lists equipment that cannot be tested at full power so as not to damage equipment or upset plant operation. The turbine trip solenoids were tested at power every 2 weeks under OPT-217, "Protective devices test". Under the new system, the turbine trip block trip solenoid valves are testable online and will also be tested every 14 days. The Turbine Trip Block will be tested during each startup from refueling outage.

Inserted language to indicate that the slave-relay test results may be observed by the Operator on the OM screen

Remove the statement "A load demand greater than full power is prohibited by the turbine control load limit devices."

There were two such considered ‘load limiting devices.’ One was built into the original EHC control system. The operator could set this to limit the adjusted setpoint. This feature presented certain control problems that resulted in operator confusion and therefore not used. This feature was deleted when the TXP EHC control system was installed because there was no added value.

The second load limiting device is built into the mechanical hydraulic controller. The load limiting portion of the startup and load limiting device mechanically limited the hydraulic governor output which limited the hydraulic output of the electrohydraulic converter. This feature could limit speed since it acting on the hydraulic speed controller but could only limit megawatt output by limiting the amount the control valves could open due to the hydraulic limitation. This feature could not be effectively used to limit full output power because there is no accurate way to set it to perform this function and not interfere with the EHC control system.

The MHC is removed, and therefore, no longer available for this function. In reality, the MHC could never perform this function.

The only limitation to load is the normal load setpoint set by the operator on the operator control screen.

The following component are being removed and replaced by the new system:

- Mechanical Hydraulic Controller (MHC): The new system has enough built-in redundancy and diversity to protect the turbine against an overspeed event.The MHC is no longer needed with the new EHC control system which incorporates a new EH converter design.

Added fire test standard IEC60332-3.

LDCR-SA-2004-42, EVAL-2004-000773-01 (JCH) (continued):

Various pages in Section 10.2

The mechanical turbine trip devices are being deleted and replaced with a fully electronic trip system. The existing electronic turbine protection systems are being upgraded to a modern digital fail-safe system. Changes to the FSAR are necessary to remove references to deleted equipment as well as add section to describe the new system.

The new turbine protection system consists of a redundant automation system with redundant CPUs and a triple redundant fail-safe turbine trip system. The interface between the turbine trip system and the turbine hydraulics is performed by a redundant hydraulic trip block containing 3 trip solenoids. The turbine trip is performed with 2 of 3 trip logic resulting in rapid de-pressurization of the turbine trip fluid. The loss of trip fluid causes all stop and control valves to rapidly close isolating steam from the main turbine.

All of the analog trip signals are connected to the automation system. The redundant input signals are independent from each other by connecting them to different input/ output modules with separate power supplies. The software with the applicable logic (2 of 2, or 2 of 3) is running on the redundant CPUs of the automation system. The automation system provides 3 redundant binary outputs (trip) signals to the fail-safe turbine trip system.

Binary inputs from field devices are directly connected to the fail-safe turbine trip system. Each binary input is connected to each channel of the turbine trip system. The output of each channel of the turbine trip system is connected to each normally energized trip solenoid (3 each) of the hydraulic trip block.

The turbine trip block is the interface between the turbine trip system and the turbine hydraulic system. The three trip solenoids of the trip block are normally energized by the turbine trip system. If any two trip solenoids valves are de-energized, the turbine will trip. This fail safe arrangement provides high degree of availability, reliability and protection of the turbine generator system.

The actual position of the trip piston in each individual channel of the trip block is measured by a position transducer. The position signals are used for status indication and feedback signals.

With the exception of the hydraulic trip devices, the new Automatic Turbine Tester (ATT) performs the same functions as the old ATT, however, the new design is implemented in software rather than hardware. The new ATT Trip Block test will be an operator initiated test.

LDCR-SA-2004-42, EVAL-2004-000773-01 (JCH) (continued):

Specific Changes

The following components are being removed and replaced by the new system:

a) Mechanical Hydraulic Controller (MHC): The new system has enough built-in redundancy and diversity to protect the turbine against an overspeed event.The MHC is no longer needed with the new EHC control system which incorporates a new EH converter design.

b) Load-limiting device: The load limiting device existed but was never used in the old system. Under the old system, the load limit was set at the maximum allowable, 1500 MWe, so that it wouldn't interfere with slight fluctuations that occurred during normal turbine operation. The new system does not have a load limit device. The MHC also had a so called load limiting device also. This was also no used for the same reason. The reference to load limit devices is being removed from the FSAR.

d) Extended Turbine Protection, which housed the automatic vibration trips and electrical low vacuum system. The electrical low vacuum system is incorporated into the new Turbine Protection system, where two out of three trip logic is performed in the software. The vibration trips will no longer be automatic, based on the following:

1) There is no code, standard, or regulatory requirement for these vibration trips.

2) The trips are not required by the plant insurer.

3) Operating experience with large turbine generators (fossil and nuclear) indicates that half of the turbine trips due to this system are spurious, and a vast majority of the remaining trips are also covered by other tripping events.

4) Several large turbine generator manufacturers (including Westinghouse) do not ship their equipment with this trip installed.

5) The high vibration alarm on the control board is retained, which will annunciate whenever any vibration is high to alert the operator. The operator will take the necessary action including turbine trip, if required to protect the turbine.

LDCR-SA-2005-16, EVAL-2004-001338-02 (JCH):

Revise section 3.9B: Delete the following: "After completion of testing , snubbers are disassembled and examined."

Snubbers are not disassembled after being tested, they are installed back in their locations unless they fail. If a snubber fails, then it may be disassembled to determine the cause of the failure.This is consistent with the ASME OM Code 1998 Edition through 2000 addenda and STA-742, "Snubber Surveillance Program."

FSAR Amendment 101, Supplement a

LDCR-SA-2006-2, EVAL-2004-001966-02 (TJE):

Justification: The new Seismic Monitoring System with only free-field ground motion input will not compromise the ability of the system to determine whether or not the OBE was exceeded and subsequent shutdown of both units per Appendix A of 10CFR100. Therefore, the new Seismic Monitoring System does not present any added risk to public health or nuclear safety.

Replace the discussion of RG 1.12 in 1A(B) that says,

"The installation of instrumentation for earthquakes in the CPSES plant is in conformance with the requirements of Revision 1 (4/74) of this regulatory guide."

with the words,

"The installation of instrumentation for earthquakes in the CPSES plant meets the intent of Revision 1 (4/74) of this regulatory guide with respect to the ability to determining exceedance of the OBE in a timely manner. Seismic instrumentation includes only the free-field triaxial accelerometer installed in the Yard. Determination of OBE exceedance will be based on the methods of ANSI/ANS-2.10-2003, "Criteria for the Handling and Initial Evaluation of Records from Nuclear Power Plant Seismic Instrumentation." "

1.) Delete all of section 3.7B.4.1 and insert the following:

"The seismic monitoring system meets the intent of RG 1.12, Rev. 1, with respect to the ability to determining exceedance of the OBE in a timely manner. Seismic instrumentation includes only the free-field triaxial accelerometer installed in the Yard. For the purpose of determining whether or not the OBE has been exceeded, ANSI/ANS-2.10-2003, "Criteria for the Handling and Initial Evaluation of Records from Nuclear Power Plant Seismic Instrumentation," uses the free-field accelerometer data as input to address 10CFR100 Appendix A requirements in paragraph (V)(a)(2). Exceedance above the OBE ground motion level will require structural response analyses to be performed (ANSI/ANS-2.23-2002, "Nuclear Power Plant Response to an Earthquake"). The seismic monitoring system used to record a seismic event is in accordance with the requirements described in ANSI/ANS-2.2-2002, "Earthquake Instrumentation Criteria for Nuclear Power Plants." The seismic monitoring system has the following components:

1. A triaxial accelerometer located in the free-field. The function of the triaxial accelerometer is to provide the acceleration time-history responses where the affects associated with surface features, buildings, and components will be insignificant.

2. A digital recorder that continuously monitors the free-field accelerometers with a solid state archival system. The function of the recorder is to provide a recording of the seismic event for post-event evaluation. The recorder will capture the 3-seconds preceding the exceedance of the seismic trigger threshold, and will continue to record data for a minimum of 5-seconds beyond the last recorded exceedance of the seismic trigger threshold.

LDCR-SA-2006-2, EVAL-2004-001966-02 (TJE) (continued):

3. A controller that will evaluate the recorded data in a timely manner. The function of the controller is to determine whether the OBE has been exceeded.

4. An uninterrupted power supply battery backup. The function of the battery backup is to ensure a minimum of 25-minutes of recorded data will be collected.

5. The free-field accelerometers will also function as the seismic switch. The recorder will continuously monitor the free-field accelerometers. The seismic trigger is a threshold ground acceleration value designed to initiate the event recording process at a value significantly below that of the OBE.

A schematic diagram of the seismic monitoring system is presented on Figure 3.7B-54. The free-field triaxial accelerometer described previously is provided for CPSES Unit 1 as allowed by Section 4.5 of ANSI/ANS-2.2-2002. The seismic monitoring system conforms to the requirements of ANSI/ANS-2.2-2002, as described above, including calibration and channel checks."

2.) Delete all of section 3.7B.4.2 and insert the following words,

"The recorder, controller, computer screen display, printer, and uninterruptible power system are located in an instrument rack in the Control Room. In case of any seismic activity of sufficient intensity to activate the seismic monitoring system, the Control Room operator is alerted by means of the seismic annunciation system, which consists of visual and audible alarms. The first notification will be that an event is in progress by the exceedance of the seismic trigger threshold and activation of the recorder. The second notification will be provided by the controller to indicate that the OBE has been exceeded and that shut down is required. A redundant set of notification indications is provided on the front of the seismic monitoring system in the form of light-emitting diodes (LEDs).

Delete all of section 3.7B.4.3 and insert the following words:

"This equipment computes and displays the free-field response spectra automatically on the computer screen following the event and prints a hardcopy on the system's printer. No Operator action is required to compute and print the response spectra. The seismic monitoring instrumentation facilitates the Operators ability to rapidly determine whether or not the OBE has been exceeded. In the event that the system fails to provide annunciation to one or both Control Rooms, local LED indication is provided on the face of the controller. The Control Room Staff will be alerted to the seismic event through perception of the building motions and external news sources. In addition, the control room operator is provided various screen displays that present the results of the analysis of the ground motion data. Therefore, failure to fully annunciate will not prevent the Control Room Staff from being alerted to the situation. The operator is required to request maintenance and engineering support to evaluate the validity of any alarm or indication of a seismic event. In response to a seismic event, the operator is required to perform focused walkdowns to assess plant damage and the availability of safe shutdown

LDCR-SA-2006-2, EVAL-2004-001966-02 (TJE) (continued):

equipment. The operator is required to shutdown both units if the Seismic Monitoring System determines that the OBE has been exceeded. In the event that the System is not available or functioning, the operator is required to request engineering to determine whether or not the OBE was exceeded."

Delete all of section 3.7B.4.4 and insert the following words,

"Recorded actual time histories from a significant seismic event that has occurred at the site are used to determine whether the OBE has been exceeded. The acceleration response spectrum from the actual time histories will be compared to the design spectra in the FSAR Figures 3.7B-1 and 3.7B-6. Due to large conservatism in the generation of in-structure responses, a detailed comparison of measured to predicted responses is not required."

This figure will be revised to reflect the new seismic instrumentation schematic diagram. See EVAL-2004-001966-01 for the markup.

LDCR-SA-2007-4, EVAL-2006-003263-04 (TJE):

Delete the "F" in front of the word "DISCUSSION" from the last entry in the Table of Contents

1A(B) FDISCUSSION OF REGULATORY GUIDES

In 1A(B), under the discussion of exceptions to compliance with RG 1.137:

1.) add "[49]" after "ANSI N195-1876" in item 1

2.) in item 4.a, add a "*" before and "when tested in accordance with ASTM D1298-1980" after the words "API or Specific Gravity"

3.) in item 4.d, remove the sapce between "D4176-" and "1982" and replace "ASTM-D1796-1968" with "ASTM D1796-1986"

4.) In the paragraph after item 4.d, near the end of the sentence, replace the "or" with "," and add ", or ASTM D4294-2003" at the end of the sentence.

5.) Add the following note after the paragraph that follows section 4.d,

"*If new fuel oil does not meet the diesel generator manufacturer’s requirements for absolute specific gravity at 60/60°F of "greater than or equal to" 0.8348 or for API gravity at 60°F of "less than or equal to" 38°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer’s recommendations."

LDCR-SA-2007-4, EVAL-2006-003263-04 (TJE) (continued):

At the end of the discussion of exceptions to RG 1.137, add exception 9:

"9. C.2.a

If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60°F of "greater than or equal to" 0.8348 or for API gravity at 60°F of "less that or equal to" 38°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations."

At the end of the Reference secion in 9.5, add reference number 49,

"49. ANSI N195-1976, 'Fuel Oil Systems for Standby Diesel Generators'"

In the paragraph before section 9.5.4.2.2, add the following paragraph,

"If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60°F of "greater than or equal to" 0.8348 or for API gravity at 60°F of "less than or equal to" 38°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations."

Before section 9.5.4.2.2.6, add the following paragraph,

"If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60°F of "greater than or equal to" 0.8348 or for API gravity at 60°F of "less than or equal to" 38°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations."

After the second paragraph of section 9.5.4.4, add the following paragraph,

"If new fuel oil does not meet the diesel generator manufacturer's requirements for absolute specific gravity at 60/60°F of "greater than or equal to" 0.8348 or for API gravity at 60°F of "less than or equal to" 38°, it is acceptable to add new fuel oil to the storage tank(s) only if, after being added, the entire storage tank(s) will meet the manufacturer's recommendations."

LDCR-SA-2007-3, EVAL-2002-000162-06 (GLM):

Revise section 9.3.4.1.1.1 to indicate that if the BTRS chiller and associated components are used in the future that the flow and piping vibration response will be verified prior to use.

This change clarifies the description of the BTRS to add the commitment for testing the BTRS chiller and associated components. Since load follow ops are not used at CPSES,

LDCR-SA-2007-3, EVAL-2002-000162-06 (GLM) (continued):

there are no current plans to ever use these components. The addition of this commitment to perform testing prior to use in section 9.3.4.1.1.1 ensures that this testing will be incorporated into operating procedures if this part of the system is ever used in the future.

LDCR-SA-2006-4, EVAL-2005-000657-01 (RJK):

This change reflects construction of a long-term storage facility (OSGSF) for the long-term storage of four Unit-1 Old Steam Generators, one Unit -1 Old Reactor Vessel Storage head and one Unit-2 Old Reactor Vessel Head. The design function of the OSGSF is to limit the on-site and off-site radiation exposure from the old SGs and old RVHs and any attached components that are stored inside the OSGSF. The OSGSF provides sufficient shielding to meet the external dose limits specified in 10CFR20 and 40CFR190. Any postulated failure of the OSGs and ORVHs/CRDMs, due to the collapse of the OSGSF due to seismic event or any other natural phenomenon, or due to the impact of the components with the ground when they are dropped, will not result in an off-site dose greater than 0.5 rem to the whole body or its equivalent to any part of the body in accordance with the 10CFR20 and 40CFR190.

This change reflects construction of a long-term storage facility (OSGSF) for the long-term storage of four Unit-1 Old Steam Generators, one Unit -1 Old Reactor Vessel Storage head and one Unit-2 Old Reactor Vessel Head. This change extends the area shown on the site plot to now include that new facility.

LDCR-SA-2005-12, EVAL-2005-001957-02 (CBC):

FSAR Sections 1, 6 and 15 are revised to reflect new methods an assumptions for evaluating radiological consequences for design basis accidents which are adopted consistent with NRC Regulatory Guide 1.195 and the new Control Room Habitablity Program / Control Room Integrity Program. These methods and assumptions were approved by the NRC in License Amendment 130. See TXX-06151 and TXX-06193 for additional information.

LDCR-SA-2007-12, EVAL-2004-002120-12 (TJE):

LDCR SA-2007-012 removes any ambiguity from the FSAR that the spare Startup Transformer XST1/2 can be energizied promptly, via a motor operated switch, is available to back up XST2.

1.) In the first full paragraph on page 8.2-2, replace the word "The" with "A" such that the sentence reads:

"A spare startup transformer, XST1/2 with dual primary windings (345-kV and 138-kV), is stored in a dedicated location under the 345-kV line to XST2 (refer to Figure 8.2-1)."

LDCR-SA-2007-12, EVAL-2004-002120-12 (TJE) (continued):

2.) In the next sentence of the same paragraph, replace the words "can be used" to the words "must be physically relocated", such that the sentence reads,

"This transformer must be physically relocated to replace XST2 or XST1 if required."

LDCR-SA-2006-42, EVAL-2003-002426-21 (GLM):

Revise Figure 1.2-1 to show deletion of the RCA Yard Entrance Radiation Protection Building.

The RCA Yard Entrance Radiation Protection Building was removed during the Unit 1 Steam Generator Replacement Project. The building has no radiation protection functions and only functions as storage and an access point into the Unit 1 yard area.

Add "Item 8" to Section 3.8.1.1.6 which is titled "Containment Alternate Access for the Steam Generator and Reactor Pressure Vessel Head Replacement (Unit 1)." Item 8 details the codes and specification used for the Unit 1 RSG Containment Alternate Access (CAA) restoration.

Add reference to structural welding code AWS D1.4-1998 in Section 3.8.1.2.4.

This code was used for the Unit 1 Containment Alternate Access during the Steam Generator Replacement Project.

Section 3.8.1.6.2, Item 1, add reference to ASTM A706 for Unit 1. Section 3.8.1.6.2, Item 3, add references to codes and standards used in support of the Unit 1 Steam Generator Replacement Project.

These codes and standards were used in support of the Unit 1 Steam Generator Replacement Project.

Clarification to sections 3.8.1.6.3 and 3.8.1.6.4 related to codes and standards used in support of the Unit 1 Steam Generator Replacement Project.

These codes and standards were used in support of the Unit 1 Steam Generator Replacement Project.

LDCR-SA-2006-40, EVAL-2003-002426-20 (RJK):

Update reference power in Section 1.2.2 for Unit 1 to 3458 MWt to reflect uprate that was implemented in 1RF09.

Justification: this 1.4% power uprate previously occurred during 1RF09 as implementation of License Amendment 89..

Adds five new references to Section 1.6 to reflect RSGs.

LDCR-SA-2006-40, EVAL-2003-002426-20 (RJK) (continued):

Adds new code reference in Section 1A(N) for Unit 1 RSGs (ANSI N45.2.2-1978) in RG 1.38 discussion.

Updates material descriptions in RG 1.43 discussion of Section 1A(N) to reflect RSGs and new RX head.

Adds new code reference in Section 1A(N) for Unit 1 RSGs (ANSI/ASME NQA-1) in RG 1.64 discussion.

Updates Table 1A(N)-1 to reflect Unit 2 only for RG 1.66 applicability.

Updates accident analysis discussions in 3.6B to specify new analyses for Unit 1 RSGs.

Updates text references in 3.6B to identify separate Tables for Unit 1 and Unit 2 parameters.

Adds 8 new references to Section 3.6B to reflect RSGs.

Split Table 3.6B-4 into separate Tables (4A and 4B), one for each unit.

Added NOTE that Table 3.6B-6 is for original SGs only and that Unit 1 RSGs had separate mass and energy release data calculated with RELAP5/MOD3.

Specified that discussion of critical damping values in Section 3.7N.1.3 are for Unit 2 only. Unit 1 is now discussed in Section 3.9N.1.4.3.

add the following statement to Sections 3.8.4.3.2 and 3.8.4.3.3:

"See Section 3.6B.1.2.3 "Structural Design Margins" for additional requirements in the MS/FW penetration areas inside the Safeguards Building."

Update Section 3.9N listing of Computer Programs Used in Analyses to reflect RSG models.

Update Section 3.9N discussion of seismic design considerations in Sections 3.9N.1.4.2, 3.9N.1.4.3, 3.9N.1.4.4, and 3.9N.2.5 to reflect RSGs.

Add eight new references to reflect RSG models to Section 3.9N.

Add the following NOTE to Figure 3.9N-1:

"FOR UNIT 1, THE REPLACEMENT STEAM GENERATOR ACTUALLY HAS 8 MASSES DISTRIBUTED ALONG THE VERTICAL CENTERLINE, AND THE SNUBBERS HAVE BEEN ELIMINATED FROM THE UPPER SUPPORT"

Split Table 3.9N-14 into separate Tables (14A and 14B), one for each unit.

LDCR-SA-2006-40, EVAL-2003-002426-20 (RJK) (continued):

Split Table 3.9N-15 into separate Tables (15A and 15B), one for each unit.

Split Table 3.9N-16 into separate Tables (16A and 16B), one for each unit.

Split Table 3.9N-21 into separate Tables (21A and 21B), one for each unit.

Split Figure 4.4-21 into separate Figures (21A and 21B), one for each unit.

Split Table 5.1-1 into separate Tables (1A and 1B), one for each unit.

Added applicable code cases used for manufacture of RSGs; N-20-4, 2124-1, and 2143-1 to Section 5.2.

Updated Table 5.2-1 to reflect new code addenda for Unit 1 RSGs and RX head.

Updated Table 5.2-2 to reflect new RCS pressure boundary materials for Unit 1 RSGs and RX head.

Split Table 5.2-6 into separate Tables (6A and 6B), one for each unit.

Updated Table 5.2-7 to reflect applicable code cases used for manufacture of RSGs; N-20-4, 2124-1, and 2143-1.

Replace the data in the first 8 lines of Table 5.3-16A to reflect the Fracture Toughness Test report numbers for the RSGs.

Add weld metal fracture toughness data for Unit 1 Replacement Steam Generator in Table 5.3-17A.

Add new section 5.4.2A to discuss Unit 1 RSG materials. Renumber existing Section 5.4.2 to 5.4.2B and make it Unit 2 specific.

Update hotlinks throughout section to either 5.4.2A or 5.4.2B as appropriate.

Update references in Section 5.4.2 to reflect Unit 1 RSGs.

Update SG design data in Table 5.4-3 to reflect Unit differences between Unit 2 original SGs and Unit 1 RSGs.

Split Table 5.4-4 into separate Tables (4A and 4B), one for each unit.

Split Figure 5.4-4 into separate Figures (4A and 4B), one for each unit.

Add the following NOTE to Figure 5.4-13:

"The snubbers exist only on Unit 2. They have been removed from Unit 1."

LDCR-SA-2006-40, EVAL-2003-002426-20 (RJK) (continued):

Updated discussion of Compliance with BTP RSB 5-1 (Diablo Canyon RHR comparison) in Section 5A to reflect separate calculations for each unit.

Update chart in Section 12.1 to reflect differing cobalt Wgt% for Unit 1 RSG tubes.

Update discussion in Section 12.3 of SG tubing materials to reflect RSGs and material selection.

Update Section 1.2.2 nominal performance characteristics of NSSS to reflect RSGs in Unit 1. Includes addition of two new reference documents into new section 1.2.2.13.

Justification:

- Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2005-000659-02, FDA-2005-000659-03 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

editorial change on page 5.3-4 to correct typographic error

Justification:

- Typographic error only, no change in technical content.

LDCR-SA-2006-37, EVAL-2003-002426-17 (JCH):

Insert the following into Section 10.3.2.2, "Atmospheric Relief Valves":

In Unit 1 the Westinghouse steam generators model D-4 have been replaced with a newer model D-76. Two solenoid valves are installed for each ARV. These solenoid valves are supplied from the opposite train power supply. These solenoid valves are operated from a key locked hand switch in the main control room to open the ARVs.

During a Unit 1 steam generator fault, if normal DC power is lost to the Train A or B buses, the ARVs can still be opened if required from the opposite train power to cool and depressurize the reactor and terminate primary to secondary leak.

Justification: The replacement steam generators (RSGs) installed in Unit 1 are physically different than the original SGs and when the SGTR analysis was revised to reflect the thermal hydraulic performance of the RSGs under the accident conditions, one ARV was not sufficient for the primary side cooldown and additional relief capacity is required.

In Section 10.4.8.1.1, change the total combined blowdown flow for Unit 1 from "69, 600" to "155,620".

LDCR-SA-2006-37, EVAL-2003-002426-17 (JCH) (continued):

Justification: RSG have higher SGBD flow.

Added "balancing" before "valves HV-5175 through HV-5178".

Add the following sentence:

"Once the balancing valves are set they are locked in position and do not change.

Justification: clarification on how to set blowdown for each steam generator.

LDCR-SA-2006-38, EVAL-2003-002426-18 (JCH):

DESCRIPTION: Appendix 3B

Page 3B-ii: For computer program 44 delete "*" before "ME215" and insert "(ME101FE)" after "ME215.

Page 3B-iv: Section 3.44 delete "*" before "ME215" and insert "(ME101FE)" after "ME215.

Page 3B-54: Insert "(ME101FE)" after "ME215" in the section title and in the first paragraph and delete "(Unit 2 only)" in the section title.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01.

These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION: Section 3.6B

3.6B.1.2.3.E - Break Exclusion Areas (Feedwater)

Page 3.6B-7 second paragraph last sentence after "..... region" insert "(Unit 2 only)" since the feedwater bypass piping has been eliminated for the Unit 1 RSG.

3.6B.2.2.2 - High - Energy Piping Other Than RCS Main Loop

Page 3.6B-22 second paragraph first sentence insert "or 34" after "[6" for new Reference 34 and in the last sentence after "[29]" insert "or other" and delete the comma and after "programs to" insert "the."

LDCR-SA-2006-38, EVAL-2003-002426-18 (JCH) (continued):

Page 3.6B-23 first paragraph first sentence after "[30" insert "or 35" for new Reference 35.

Page 3.6B-53 add new References 34 and 35. Reference 34 is "RELAP5/MOD 3.2, NUREG/CR-5535, June 1995" and Reference 35 is "ANSYS, Version 8.1."

3.6B.2.5.2.2 - Feedwater System - A. General Description

Page 3.6B-40 after first sentence insert "Due to the Unit 1 RSG installation the 18" x 16" reducing elbow material at the RSG nozzle inlet is now SA-508, Grade 2, Class 1."

3.6B.2.5.2.3 - Auxiliary Feedwater System - D. Environmental Analysis

Page 3.6B-43 after "".....250F maximum" insert "(Unit 2 only)."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

For convenience the main steam system, steam generator blowdown system and miscellaneous FSAR figures pertaining to stress node break point and restraint location have been included in this EVAL.

Fig 3.6B-11 - Main Steam System: Loop 1 Inside Containment Stress Node Break Point and Restraint Location

Changed break 8C stress point to 22 and break 1C stress point to SGB and revised the legend.

Fig 3.6B-12 - Main Steam System: Loop 2 Inside Containment Stress Node Break Point and Restraint Location

Changed break 13C stress point to 28 and break 9C stress point to SGB and revised the legend.

Fig 3.6B-13 - Main Steam System: Loop 3 Inside Containment Stress Node Break Point and Restraint Location

LDCR-SA-2006-38, EVAL-2003-002426-18 (JCH) (continued)

Changed break 14C stress point to N03 and revised the legend.

Fig 3.6B-14 - Main Steam System: Loop 4 Inside Containment Stress Node Break Point and Restraint Location

Changed break 29C stress point to 800 and revised the legend.

Fig 3.6B-19 - Feedwater System: Loop 1 Inside Containment Stress Node Break Point and Restraint Location

Changed break 401C stress point to D04 and break 410C stress point to V09. Revised the notes and legend. Added snubbers and spring hangers to the drawing and deleted split flow bypass line. Revised piping for elevated RSG FW nozzle and deleted chemical feed system piping. Deleted lines capped.

Fig 3.6B-20 - Feedwater System: Loop 2 Inside Containment Stress Node Break Point and Restraint Location

Changed break 411C stress point to 50 and break 418C stress point to B50. Revised notes and legend. Added snubbers, rigid restraints and spring hangers to the drawing and deleted split flow bypass line. Revised piping for elevated RSG FW nozzle and deleted chemical feed system piping. Deleted lines capped.

Fig 3.6B-21 - Feedwater System: 1-3 Inside Containment Stress Node Break Point and Restraint Location

Changed break 414C stress point to H01 and break 434C stress point to V09. Revised notes and legend. Added snubbers, rigid restraints and spring hangers to the drawing and deleted split flow bypass line. Revised piping for elevated RSG FW nozzle and deleted chemical feed system piping. Deleted lines capped.

Fig 3.6B-22 - Feedwater System: Loop 4 Inside Containment Stress Node Break Point and Restraint Location

Changed break 421C stress point to 35 and break 429C stress point to N02. Revised notes and legend. Added snubbers, rigid restraints and spring hangers to the drawing and deleted split flow bypass line. Revised piping for elevated RSG FW nozzle and deleted chemical feed system piping. Deleted lines capped.

Fig 3.6B-26 - Aux. F.W. System Stress Node Break Point and Restraint Location (Units 1 and 2)

Revised to show preheater bypass lines (6-FW-1-91-2003-2 through 6-FW-1-94-2003-2) are for Unit 2 only.

LDCR-SA-2006-38, EVAL-2003-002426-18 (JCH) (continued)

Fig 3.6B-27 - Auxiliary F.W. System: Loop 1 Outside Containment Stress Node Break Point and Restraint Location

Deleted preheater bypass line with a capped pipe section and revised the legend.

Fig 3.6B-28 - Auxiliary F.W. System Outside Containment Stress Node Break Point and Restraint Location

Deleted preheater bypass line with a capped pipe section and revised the legend.

Fig 3.6B-29 - Auxiliary F.W. System: Loop 3 Outside Containment Stress Node Break Point and Restraint Location

Deleted preheater bypass line with a capped pipe section and revised the legend.

Fig 3.6B-30 - Auxiliary F.W. System Outside Containment Stress Node Break Point and Restraint Location

Deleted preheater bypass line with a capped pipe section and changed break 575C stress point to 34J. Revised the legend.

Fig 3.6B-34 - Feedwater System: Loop 1 Inside Containment Stress Node Break Point and Restraint Location

Changed break 597C stress point to 30 and break 598C to stress point B63. Deleted break 599C. Deleted split flow bypass line. Revised piping to the RSG AFW nozzle. Indicated that the chemical feed system and drain piping 2-1303-2 are eliminated. Drain piping 2-1303-2 was previously eliminated. Revised the legend and deleted 1-152A from the title block.

Fig 3.6B-35 - Feedwater System: Loop 2 Inside Containment Stress Node Break Point and Restraint Location

Changed break 628C stress point to 77 and break 629C to stress point 152. Deleted break 630C. Deleted split flow bypass line. Revised piping to the RSG AFW nozzle. Deleted chemical feed system piping. Deleted drain piping 2-1303-2 (previously eliminated). Revised the legend.

Fig 3.6B-36 - Feedwater System: Loop 3 Inside Containment Stress Node Break Point and Restraint Location

Changed break 622C stress point to A10 and break 623C to stress point NZ3. Deleted break 624C. Deleted split flow bypass line. Revised piping to the RSG AFW nozzle. Deleted chemical feed system piping. Deleted drain piping 2-1303-2 (previously eliminated). Revised the legend and deleted 1-154A from the title block.

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Fig 3.6B-37 - Feedwater System: Loop 4 Inside Containment Stress Node Break Point and Restraint Location

Changed break 575C stress point to 34J and break 577C to stress point N03. Deleted break 576C. Deleted split flow bypass line. Revised piping to the RSG AFW nozzle. Deleted chemical feed system piping. Deleted drain piping 2-1303-2 (previously eliminated). Revised the legend.

Fig 3.6B-38 - STM. GEN. Blowdown Sys. Loop 1 Inside Containment Stress Node Break Point and Restraint Location

Changed break 202C stress point to A80, break 203C stress point to W85 and break 201C to stress point Q68. All intermediate breaks are deleted (21C, 22C, 48C, 113C, 116C, 117C, 139C, 205C, 206C, 208C, 1101C, 1160C, 1600C, 2100C and 8100C). Revised piping from the RSG blowdown and supplemental blowdown connections. Revised the legend.

Fig 3.6B-39-1 - STM. GEN. Blowdown Sys. Loop 2 Inside Containment Stress Node Break Point and Restraint Location

Changed break 192C stress point to 260, break 185C stress point to E01 and break 184C to stress point B67. All intermediate breaks are deleted (200C, 1127C/L, 1903C and 9006C). Revised piping from the RSG blowdown and supplemental blowdown connections. Revised the legend.

Fig 3.6B-40 - STM. GEN. Blowdown System Loop 3 Inside Containment Stress Node Break Point and Restraint Location

Changed break 45C stress point to 545. Intermediate break 9624C/L is deleted. Revised piping from the RSG blowdown and supplemental blowdown connections. Revised the legend.

Fig 3.6B-41 - STM. GEN. Blowdown System Loop 4 Inside Containment Stress Node Break Point and Restraint Location

Revised piping from the RSG blowdown and supplemental blowdown connections. Deleted chemical feed system piping. Revised the legend.

Fig 3.6B-55 - Safety Injection System Inside Containment Stress Node Break Point and Restraint Location

Break 8TCA moved to downstream side of check valve 1SI-8819C.

Fig 3.6B-67-1 - Reactor Coolant System Inside Containment Stress Node Break Point and Restraint Location

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Breaks 71C, 72C, 73C, 80C, 81C, 82C, 83C, 96C, 140C and 142C where revised to 4370, 88, 89, 1105, 5210, 105, 106, 82, 102/1115 and 85/82 respectively.

Fig 3.6B-82-1 - CVCS Outside Containment Stress Node Break Point and Restraint Location

Deleted breaks 1421-1C and 2C based on piping analyses (CS-1-046A-PB, Rev. 4). Added pipe break analysis calculation (CS-1-042A-PB, Rev. 2) to provide the source for break 1324C inside the Reactor Building illustrated on Isometric 1-46A.

Figure 3.6B-208 - Safeguard Building Main Steam & Feedwater Lines Arrangement

Sheet 2 of 3 show that the preheater bypass lines 6-FW-1-091-2003-2 through 6-FW-1-094-2003-2 are for Unit 2 only.

Sheet 3 of 3 Sections "53-53" and "54-54" show that the preheater bypass lines 6-FW-1-091-2003-2 through 6-FW-1-094-2003-2 are for Unit 2 only.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION: Table 3.6B-1 - High Energy Line List (See Notes 22 and 23)

Page 11 of 31:

Revise "Line Number" "3FW-1-93-203-2" to "6FW-2-93-203-2" and under "Remarks" revise the note to be "(Notes 1 and 25)." Under "Remarks" for "Line Number" "6FW-1-97-1303-2" insert "(Note 26)" after "bypass line."

Top of page under "Remarks" revise note to "(Notes 1 and 26)" after "bypass." Revise "Line Number" "6FW-1-101-1303-2" to "6FW-2-101-1303-2" and under "Remarks" insert "(Note 25)" after "SG Number 3." Revise "Line Number" "6FW-1-92-2003-2" to "6FW-2-92-2003-2" and under "Remarks" revise note to "(Notes 1 and 25)" after "bypass line."

Page 12 of 31:

Under "Remarks" for "Line Number" "6FW-1-96-1303-2" revise note to "(Notes 1 and 26)" after "bypass line."

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Bottom of page after "bypass line" revise note to "(Notes 1 and 26)."

Revise "Line Number" "6FW-1-100-1303-2" to "6FW-2-100-1303-2" and under "Remarks" insert "(Note 25)" after "SG Number 2." Revise "Line Number" "6FW-1-91-2003-2" to "6FW-2-91-2003-2" and under "Remarks" revise note to "(Notes 1 and 25)" after "bypass line."

Page 13 of 31:

Under "Remarks" for "Line Number" "6FW-1-95-1303-2" revise note to "(Notes 1 and 26)" after "bypass line." Bottom of page after "bypass line" revise note to "(Notes 1 and 26)."

Revise "Line Number" "6FW-1-99-1303-2" to "6FW-2-99-1303-2" and under "Remarks" insert "(Note 25)" after "SG Number 1." Revise "Line Number" "6FW-1-94-2003-2" to "6FW-2-94-2003-2" and under "Remarks" insert "(Note 25)" after "bypass line."

Revise "Line Number" "6F2-1-93-1303-2" to "6FW-2-93-1303-2 and under "Remarks" revise note to "(Notes 1 and 26)."

Page 14 of 31:

Revise "Line Number" "6FW-1-102-1303-2" to "6FW-2-102-1303-2."

Top of page under "Remarks" insert "(Note 25)" after "SG Number 4."

Revise "Line Number" "2FW-1-900-1303-2" through "2FW-1-907-1303-2" to "2FW-2-900-1303-2" through "2FW-2-907-1303-2."

Page 31 of 31: Insert the following notes:

25. Unit 2 only. Line eliminated on Unit 1 with RSG installation.

26. Preheater Bypass Line Unit 2 only. For Unit 1, line partially removed from connection to 18" main feedwater line to connection with 4" auxiliary feedwater line. Remaining portion of line is for auxiliary feedwater flow only."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

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DESCRIPTION:

3.9B.1.2.6 - SAPCAS (ME 215) Stress Analysis for Pipe Containment and Pipe Support Using Finite Element Method - Unit 2 only

Page 3.9B-9: In section title insert "(ME101FE)" after "(ME 215)" and delete "- Unit 2 only." ME 215 is now called ME101FE. Both Unit 1 and Unit 2 have been analyzed using ME215 (ME101FE).

First paragraph after "ME 215" insert "(ME101FE)."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 3.9B-1F - Uses of Code Case N-318

Revise pipe support "FW-1-017-717-S525" to "FW-1-017-717-C52S" to correct typographical error.

Pipe support FW-1-102-700-C62R deleted based on the elimination of the split flow bypass line on Loop 4.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 3.9B-10 - Active Valves

Page 5: Insert "2" before "FV-2193 through 2196."

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Pages 5 and 6: Insert "2" before "FW-0191 through 0194 and FV-2181 and -2182."

Page 6: Insert "2" before "FV-2183 through 2184."

The above mentioned valves are deleted on Unit 1 for the RSG installation.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Figure 6.2.4-1 - Containment Isolation Valving

Sheet 9 of 12, Valve Arrangement 36: Make the "feedwater preheater bypass line" "Unit 2 only" with the piping ending between the check valve and junction with the "auxiliary feedwater" line.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 6.2.4-2 (Sheet 3) Containment Isolation Valving Application (Note 8)

Item "20c" insert "2-" before "FV-2193 and delete "29'-6." Item "22c" insert "2-" before "FV-2194 and delete "29'-6." Item "24c" insert "2-" before "FV-2195 and delete "30'-9." Item "26c" insert "2-" before "FV-2196 and delete "31'-10.

LDCR-SA-2006-38, EVAL-2003-002426-18 (JCH) (continued)

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 6.2.4-3 (Sheets 2 and 3 of 13) Containment Isolation Valving Application (Note 1)

Item "20c", "Remarks" insert after signal "(Unit 2 only, valve deleted Unit 1)." Item "22c", "Remarks" insert after signal "(Unit 2 only, valve deleted Unit 1)." Item "24c", "Remarks" insert after signal "(Unit 2 only, valve deleted Unit 1)." Item "26c", "Remarks" insert after signal "(Unit 2 only, valve deleted Unit 1)."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 6.2.4-6 (Sheets 3 and 4) Classification of Systems Paths Penetration Containment Wall

Item "20a" under "System" insert "/AFW" and item "20c" after "Steam Generator #1 insert "(Unit 2 only)." Item "22a" under "System" insert "/AFW" and item "22c" after "Steam Generator #2 insert "(Unit 2 only)." Item "24a" under "System" insert "/AFW" and item "24c" after "Steam Generator #3 insert "(Unit 2 only)." Item "26a" under "System" insert "/AFW" and item "26c" after "Steam Generator #4 insert "(Unit 2 only)."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

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DESCRIPTION:

7.1.2.5 Conformance to Regulatory Guide 1.22

Page 7.1-18, items 9 and 11 insert "Unit 2" before "Feedwater."

Page 7.1-21, item 9, insert "(Unit 2 only)" after "(Close)."

Page 7.1-22, item 10 insert "Unit 2" before "FWIV" and "FIBVs" and item 11 insert "(Unit 2 only)" after "(close)."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

7.3.1.1.4 - BOP Furnished Engineered Safety Features Systems - (5. Auxiliary Feedwater System,

Pages 7.3-14 and 15: In paragraph starting with "The auto start...." insert "the Unit 2" before "feedwater split flow bypass valves." Under "d. Interlocks" insert "the Unit 2" before "feedwater splitflow bypass valves."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 7.3-4 Safety Injection Actuated Equipment List

Sheet 2 of 21: Delete lines beginning with IFV-2181, 2182, 2183, 2184, 2193, 2194, 2195 and 2196.

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JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 7.5-5 Type D - Variables

Sheet 2 of 6: Insert "Unit 2" before "Feedwater Split-flow."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 7.5-7B instrument Summary Data for Accident Monitoring Variables for CPSES not in Table 2 of Reg. Guide 1.97, Rev. 2

Sheet 4 of 10: Insert "2-" in front of "FV-2193 to 2196" and "FV-2181 to 2184."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

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DESCRIPTION:

Figure 7.7-14 Main Control Board Functional Layout Units 1 and 2

For CB-09 make the "Feedwater FW WTR. HMMR CONT" "Unit 2 only."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Table 10.3-6 - Materials of Main Steam and Feedwater Valves and Piping

Sheet 2 of 2: Under "Fittings", "3 in. to 24 in." insert a superscript "1" for Note 1 after "WPL-6" and insert the following after the "Flanges" section "Notes: 1. Due to the Unit 1 RSG installation the 18" x 16" reducing elbow material at the RSG nozzle inlet is now SA-508, Grade 2, Class 1."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

10.4.7.2. - 2. Feedwater System

Page 10.4-32: Sixth paragraph first sentence revise sentence to include "When originally constructed both Unit 1 and Unit 2 incorporated a Feedwater Bypass System on each main feedwater line to a steam generator."

Insert the following paragraph: "With the installation of Delta 76 steam generators on Unit 1, the Feedwater Bypass System is no longer required by the new feedring steam generators and with the exception of the small bypass line around the main feedwater isolation valve has been totally removed from Unit 1. (Note: The Unit 1 feedwater system is operated in a different manner after RSG installation. The FIV remains open during

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startup and a small flow through the valve and not the FIBV is used for purging purposes.) Unit 2 however still retains the Feedwater Bypass System in its entirety."

Add "The Unit 2" after paragraphs discussing feedwater bypass system, feedwater preheater bypass line, and feedwater split flow bypass line.

10.4.7.2 - 3. Electrical Systems

Page 10.4-33: After "The split flow bypass valves," "feedwater split flow bypass valves" and "preheater bypass valves" insert "(Unit 2 only)."

10.4.7.3 - Safety Evaluation

Page 10.4-34: Insert the following paragraph- "The Unit 1 Delta 76 Feedring RSGs incorporate a feedring design in lieu of a preheater design and this feedring design feature prevents or mitigates the possibility of a steam generator water hammer event. (CP1-RSG-05-184)."

Page 10.4-34 & 35: Insert reference to Unit 2 on the paragraphs discussing waterhammer.

10.4.7.5 - Instrumentation Requirements

Page 10.4-37: Insert the following-"Based on section 10.4.7.3 Unit 1 does not require any instrumentation to mitigate feedline water hammer. The Unit 1 feedwater system is operated in a different manner after RSG installation. The FIV remains open during startup and a small flow through the valve and not the FIBV is used for purging purposes."

10.4.7.5 - 3. Feedwater Split Flow Bypass Valve (FSBV).

Page 10.4-39: Second sentence after "open" insert "at about 30 percent RTP due to system hydraulics" and delete "ever" from "whenever."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

10.4.9.1 - Design Bases

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Page 10.4-45: Second paragraph first sentence insert "1106 psia (Unit 1) or" after "100 psia to" and insert "(Unit 2)" after "1107 psia."

Page 10.4-46: Insert the following at the end of Section 10.4.9.1:

Unit 1

"On Unit 1 the steam generators have been replaced with a model Delta 76 that employs a feedring design that eliminates the preheater. Auxiliary feedwater and feedwater entering the steam generator merge with the saturated liquid removed by the moisture separators. The combined feed then flows down the annulus formed by the steam generator shell and the tube bundle wrapper where it enters the tube bundle.

To ensure that steam does not flow back through either the main and auxiliary feedwater nozzles, the steam generator water level should be above the feedring at all operating levels where steam formation can occur. The operator is cautioned to maintain the normal SG level above the feedring at normal operating conditions. The normal operating Narrow Range level indication of 60% to 75% would result in sufficient water inventory to maintain the feedring and the AFW vertical perforated spray pipe submerged during the majority of expected transients like +/-10% load changes, turbine synchronizations, and plant loadings/unloadings. The supply piping to the auxiliary nozzle is designed with a loop seal immediately upstream of the SG nozzle to prevent steam backleakage.

In the event that steam does flow back past the loop seal and the check valves in the auxiliary feedwater line to the upper nozzle, the temperature of this line will increase causing temperature elements in the line to alarm in the control room. This allows the operator to take action to resolve the problem.

Unit 2

The Auxiliary Feedwater System is designed to preclude the effects of hydraulic instability due to water hammer by supplying water to the secondary side of the steam generator through a separate upper auxiliary feedwater nozzle. The Unit 2 preheater design steam generators permit the cold auxiliary feedwater to be heated as it comes down the side of the steam generator prior to reaching the feedwater preheater. For further discussion of the feedwater system upper nozzle arrangement and considerations see Subsection 10.4.7.

To ensure that steam does not flow back to the auxiliary feedwater nozzle, the steam generator water level should be above the auxiliary feedwater discharge pipe at all operating levels where steam formation can occur. Therefore, the operator is cautioned to maintain SG level above the auxiliary nozzle internal pipe extension when the temperature is above 212oF. This pipe is designed with a loop seal immediately upstream of the SG nozzle to prevent steam backleakage. Forward flushing is provided through the auxiliary feedwater nozzle virtually at all loads by AFW flow at startup

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conditions, by preheater bypass flow during initial power ascension and by the split flow of the feedwater at RTPs above 30% full load.

In the event that steam does flow back past the loop seal and the check valves in the feedwater line to the upper nozzle, the temperature of this line will increase causing temperature elements in the line to alarm in the control room. This allows the operator to take action to resolve the problem.

10.4.9.2 - System Description

Page 10.4-47: Delete the paragraph beginning with "Downstream of the last isolation valve....." t

Page 10.4-48: Insert the following at the end of Section 10.4.9.2-

Unit 1

Downstream of the last isolation valve, each line from the motor-driven pumps joins with a corresponding line from the turbine driven pump to form a common line that connects with the feedwater line that connects to the auxiliary nozzle on the Unit 1 steam generator."

Unit 2

Downstream of the last isolation valve, each line from the motor-driven pumps joins with a corresponding line from the turbine driven pump to form a common line that connects with the feedwater preheater bypass line. The preheater bypass line connects to the Unit 2auxiliary nozzle on the steam generator."

10.4.9.3 - Safety Evaluation

Page 10.4-50: Insert "For Unit 2" at the start of the last paragraph

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Figure 10.4-12 - Auxiliary Feedwater Failure Mode Analysis Flow Diagram

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A dotted line was placed around the main feedwater piping to the SG to the preheater bypass piping and back to the main feed system identification block and text added that is "For Unit 2 Only."

Figure 10.4-22 - Main Feedwater Loop Seal Arrangement

Figure 10.4-22 was revised for Unit 1 and Unit 2. Figure 10.4-22A is for Unit 1 and is a new Figure. Figure 10-4-22B is the current figure with the title block revised to show "Unit 2" only and the a "B" added to Figure 10.4-22.

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

DESCRIPTION:

Figure II.E.1.1-1 - Auxiliary Feedwater System Simplified Flow Diagram

A dotted line was place around the main feedwater piping to the SG to the preheater bypass piping and back to the main feed water identification block and text added that is "For Unit 2 Only."

JUSTIFICATION:

Reflects system design modification made by the Steam Generator Replacement Project as defined in FDA-2005-000224-01, -02, -03, -04, -05, -06 and FDA-2003-002426-01. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and /or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS):

9.1.4.2.2 - Refueling Procedure

Item 2 - Phase II Reactor Disassembly

Insert the following sentence before the current first sentence: "The external air duct is removed from the CRDM Ventilation System's Air Handling Units and plenum (Unit 1 only)."

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Current first sentence after "shield" insert "(Unit 2 only)." Current second sentence after "Air ducts" insert "(Unit 2 only)."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

9.1.4.3.1 - Safe Handling

op paragraph next to last sentence insert "an integral steel missile shield (Unit 1 only) and a" after "employs" and insert "(Unit 2 only)" after current "shield.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.7N.1.3 - Critical Damping Values

Clarification added that the damping values given apply to both Units 1 and 2.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

4.5.1.1 - Materials Specifications

The Unit 1 RSG uses different stainless steel than Unit 2. Therefore, under "Pressure vessel" insert "Type F304LN for Unit 1," before "Type" and insert "for Unit 2" after "304."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

TMI II.F.2.3 - Reactor Coolant System Temperature

This revision indicates the differences between Unit 1 and Unit 2 in the location of the complete core exit thermocouples (CET) cable train separation. Therefore, the following changes are identified: In the next to last sentence on Page II.F-5 insert "CRDM Seismic Support Assembly's platform (Unit 1) and the missile" and insert "(Unit 2)" after "shield."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 17A-1 - List of Quality Assured Structures, Systems and Components

After "Control rod drive mechanism…" insert a new row as follows: "CRDM air cooling shroud assemblies (Unit 1 only), NNS, Mfrd Stds, - , II, Note E, and Note 10" respectively.

After "CRDM air cool baffle assemblies" insert "(Unit 2 only)."

After "Roll-away missile shield" insert "(Unit 2 only)."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.8.3.4.1 General Analysis of Internal Structure

Section revised to indicate that the STRUDL computer program was used for the evaluation of the Unit 1 steam generator compartments.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Table 3.8-1 Summary Table for Category 'I' Structures

Table was revised to include the latest stresses for the Unit 1 Reactor Building and steam generator upper lateral support beam. Revised loads due to the Replacement Steam Generator necessitated that these structures be re-analyzed.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.3.1.2 - Special Processes Used for Manufacturing and Fabrication

Divide Item 3 into a Unit 1 and Unit 2 section as follows: Insert paragraph "A: Unit 1" and the following: "The surfaces of the guide studs are chrome plated to prevent possible galling of the mated parts." Before the existing sentence insert "B: Unit 2."

Divide Item 6 into a Unit 1 and Unit 2 section as follows: Insert paragraph "A: Unit 1" and the following: "The location of full penetration weld seams in the vessel bottom head are restricted to areas that permit accessibility during inservice inspection. There are no full penetration welds in the upper closure head." Before the existing sentence insert "B: Unit 2."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.3.3.1 - Design

First paragraph delete "The reactor vessel closure head" and insert the following the following for Unit 1 and Unit 2: Insert "A: Unit 1" and the following paragraph "The Unit 1 reactor vessel closure head contains head adapters. These head adapters are tubular members, attached by partial penetration welds to the underside of the closure head. The upper end of these allow for welding to the control rod drive mechanisms. There are also two penetrations for the Reactor Vessel Level Measuring System (RVLMS) and four penetrations for the Core Exit Thermocouple Nozzle Assemblies (CETNA)." Insert "B. Unit 2" and "The Unit 2 reactor vessel closure head" before the existing "contains head adapters."

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.3.3.5 - Shipment and Installation

After the first paragraph insert Unit 1 and Unit 2 paragraphs as follows: Insert "A. Unit 1" and the following paragraph: "The Unit 1 replacement closure head is shipped with a protective shipping cover and skid. Two support plates within the shipping cover protect the control rod drive mechanism housings. All head openings are sealed to prevent the entrance of moisture and an adequate quantity of desiccant bags are placed inside the head. In addition, nitrogen purge is maintained to protect the control rod drive mechanism internals. The shipping cover also acts as a lifting frame for handling the replacement closure head." Inset "B. Unit 2" and then insert "Unit 2" before "closure head" of the existing paragraph.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.3.3.7 - Inservice Surveillance

The HAUP design does not include a knuckle transition piece. Therefore revise as follows: Second paragraph third sentence insert "For Unit 2," and delete "T" and insert "t" at the beginning of the sentence.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.3.3.8 - Degradation of Control Rod Drive Mechanism Nozzle and Other Vessel Closure Head Penetrations (GL 97-01)

The Unit 1 RRVCH uses material Alloy 690. Therefore insert the following after the first paragraph making the remainder of the section applicable to Unit 2: "A: Unit 2 (Unit 1 RRVCH replaced Alloy 600 with Alloy 690)."

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.9N.4.1 - Descriptive Information of CRDS

ivide item 1 first paragraph into a paragraph for the Unit 1 RSG and Unit 2.

Insert "A. Unit 1" and the following paragraph" "For Unit 1, the pressure vessel is a one-piece housing which is connected to the reactor vessel head adapters by a full penetration weld. The lower portion of the pressure vessel contains the latch assembly. The upper portion of the pressure vessel provides space for the drive rod during its upward movement as the control rods are withdrawn from the core."

Item 2 second paragraph after "housing" in both sentences insert "/pressure vessel."

Second paragraph after item 4 insert "A. Unit 1" and the following paragraph "The Unit 1 control rod drive mechanism is welded with a full penetration weld to an adaptor on top of the reactor vessel and is coupled to the rod cluster control assembly directly below." Then insert "B. Unit 2" and insert "Unit 2" after "The" in the existing first sentence.

Fifth paragraph after item 4 after "housing" insert "/pressure vessel."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.9N.4.3.3 - Latch Assembly and Coil Stack Assembly

Unit 1 RSG and Unit 2 have different stainless steel materials.

Under "Latch Assembly - Thermal Clearances" first sentence insert "Type F304LN (Unit 1) or" before "Type 304" and insert "(Unit 2)" after "Type 304."

First, second and third paragraph under "Coil Stack Assembly - Thermal Clearances" after "latch housing" insert "/pressure vessel."

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.9N.4.1 - Descriptive Information of CRDS

Divide item 1 first paragraph into a paragraph for the Unit 1 RSG and Unit 2.

Insert "A. Unit 1" and the following paragraph" "For Unit 1, the pressure vessel is a one-piece housing which is connected to the reactor vessel head adapters by a full penetration weld. The lower portion of the pressure vessel contains the latch assembly. The upper portion of the pressure vessel provides space for the drive rod during its upward movement as the control rods are withdrawn from the core."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

6.2.2.3.3 - Recirculation Sump Design

Clarification of location of the Unit 1 RSG and respective piping accelerations. Page 6.2-38, third paragraph, after first sentence insert "As a result of the Steam Generator Replacement Project, the accelerations for the Unit 1 steam generators, the new main feedwater piping and the new auxiliary feedwater piping are location specific.."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Appendix 1A(B) -- Regulatory Guide 1.75

The discussion regarding compliance with Regulatory Position C.9 of Regulatory Guide 1.75 was modified to include the CRDM cable splices being installed in cable trays as part of FDA-2004-002710-02. In addition, a reference was added to Section 8.3, which addresses reduced separation requirements for instrumentation cables.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

8.3.1.4 - Independence of Redundant Systems

A paragraph was added to state that, in plant areas free from potential hazards such as missiles, external fires, and pipe whip and in non-hazard areas, the minimum separation the minimum separation between redundant instrumentation cables/trays is one inch horizontally and three inches vertically.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

3.7N.1.3 - Critical Damping Values

Clarification added that the damping values given apply to both Units 1 and 2.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 3.9N-20 - CRDM Head Adaptor Bending Moments

Actual versus allowable bending moments for the new Unit 1 CRDM Head Adaptors are provided in this Table

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Table 5.3-1 - Reactor Vessel Quality Assurance Program

Clarification added that for Unit 1, the closure head is a one-piece forging.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

7.7.1.9.1 - Thermocouples

Insert sentence after "......assemblies" "The Unit 1 thermocouples are provided with a grafoil seal and swage type seal from conduit to head." After "The" now third sentence insert "Unit 2."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 5.3-5 - Reactor Vessel Design Parameters

Under "Overall height of vessel….." insert the Unit 1 RSG value and identify the Unit 2 value as follows: Insert "43-6.4 (Unit 1)" and after "-10" insert "(Unit 2)."

Under "Closure head thickness (in.)" insert the Unit 1 RSG value and identify the Unit 2 value as follows" Insert "7 (Unit 1)" and after "-1/2" insert "(Unit 2)."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 5.3-15A - Unit 1 Reactor Vessel Non-Beltline Weld Metal Toughness Properties

The HAUP closure head design is not part of this table.

Delete the first four rows of this table for the HAUP. The closure head is a one piece design.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 5.3-2A - Unit 1 Reactor Vessel Fracture Toughness Properties

Delete the first three rows and insert the following row as row one : "Closure Head Forging, A508, C3, - 0.05, -, 0.87, -50, 10, -50, - and 184.0 respectively.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 9.4-2 - Design Conditions - Indoor

The "CRDM Shroud (Air Temperature)" at the inlet and the outlet remain the same for the HAUP/AHU and Unit 2, however Table 9.4-2 is revised to show that the HAUP/AHU inlet temperature is measured at the AHU inlet and not at the CRDM shroud as for Unit 2. Therefore, under "Reactor Containment Bldg. (RCB)" revise item 3 first column as follows: Insert "163 (CRDM Air Handling Unit inlet) (Unit 1)" before existing "163 (outlet)" and insert "(Unit 2)" after "163 (outlet)."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

5.2.3.4.4 - Prevention of Intergranular Attack of Unstabilized Austenitic Stainless Steels

Clarification added that CRDM seal welds only apply to Unit 2.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Figure 3.9N-5 - Full Length Control Rod Drive Mechanism

The Unit 1 CRDM drive mechanism is different than the Unit 2 drive mechanism. A new figure has been developed for Unit 1 with the same title as the existing figure and the figure number has been changed to "Figure 3.9N-5A." For the existing figure delete "S 1 and" and insert "B" after the figure number in the title block for the Unit 2 CRDM.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 3.9N-6 - Full Length Control Rod Drive Mechanism Schematic

The Unit 1 CRDM drive mechanism schematic is different than the Unit 2 drive mechanism schematic. A new figure has been developed for Unit 1 with the same title as the existing figure and the figure number has been changed to "Figure 3.9N-6A." For the existing figure delete "S 1 and" and insert "B" after the figure number in the title block for the Unit 2 CRDM schematic.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Table 3.5-6 - Internally Generated Missiles (Inside Containment)

Due to the new CRDM design on Unit 1, the Unit 1 and Unit 2 missiles and type of protection are different. Therefore, revise as follows:

First row under "Identification of Missiles" insert "Unit 1," "Missile from CRDM drive shaft" and "Unit 2" before the existing text.

First row under "Missile Protection Provided" insert "Unit 1," "Integral steel missile shield above CRDM housing" and "Unit 2" before the existing text.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Figure 3.5-1 - Stationary Storage Scheme Roll-Away Missile Shield

This figure revised to show it is applicable to Unit 2 only. Insert "Unit 2 only" and "Unit 2" as shown on the markup.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 3.5-2 - Roll-Away Missile Shield

This figure revised to show it is applicable to Unit 2 only. Insert "Unit 2 only" and "Unit 2" as shown on the markup.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 1.2-8 - Building Cross-Section D-D

The Missile Shield and CRDM Cooling Unit have been deleted for Unit 1. Also, delete "Rev. No. 6" for drawing "A1-0529" and "Amendment 10, March 31, 1980."

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 1.2-13 - Primary Plant - Unit 1 Containment & Safeguard Buildings - Plans At EL. 852'-6" and 860'-0"

The indication of the position of the Stored Rolling Missile Shield has been deleted for Unit 1. For drawing A1-0503 delete "Rev. No. 8" and insert "Sheet -." Also delete all the "Roof Details" depicted on the left side of the figure for Unit 1. This information is now located on drawing "A1-503, Sheet A."

LDCR-SA-2006-39, EVAL-2003-002426-19 (JDS) (continued):

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 3.8-2 - Structural Plant Arrangement Section A-A

The Missile Shield above the reactor vessel has been deleted for Unit 1. Add note to Figure.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

Figure 3.8-15 - Containment Internal Structure

Notes added to Figure to indicate Missile Shield above reactor vessel is for Unit 2 only.

Reflects system design modification made by the Steam Generator Replacement Project (SGRP) as defined in FDA-2004-002710-02, FDA-2004-002710-03, FDA-2004-0002711-01 and FDA-2003-002426-02. These changes have been reviewed by the 50.59 process and were determined to not involve a change to an SSC that adversely affects an UFSAR described design function and/or that do not require prior NRC approval since the methodology had been previously approved by the NRC for the intended application.

LDCR-SA-2007-6, EVAL-2005-001957-21 (RJK):

Table 1.6-1, Section 4A and Section 15.6:

Update listing of approved topical reports by adding the following:

ERX-04-004-A; “Replacement Steam Generator Supplement To TXU Power's Large and Small Break Loss Of Coolant Accident Analysis Methodologies” Revision 0, March 2007.

ERX-04-005-A; “Application of TXU Power's Non-LOCA Transient Analysis Methodologies to a Feed Ring Steam Generator Design” Revision 0, March 2007.

This is part of implementation of License Amendment 135.

FSAR Amendment 101, Supplement a

LDCR-SA-2003-21, EVAL-1999-000276-07 (JCH):

Table 3.9B-10

Changed valve type for 1DD-0020 and XDD-0103 from "diaphragm" to "ball."

Replacing the diaphragm valves with ball valves provides a more robust design that is unaffected by closure agsinst system pressure thus eliminating the maintenance issue and making the system more reliable.

LDCR-SA-2007-9, EVAL-2006-003910-01 (RJK):

Replace incorrect valve numbers HV-4872 and HV-4873 with valve numbers HV-4782 and HV-4783. Corrects typographic error which transposed the digits of the valve numbers.

LDCR-SA-2006-1, EVAL-2005-003364-01 (RJK):

Section 12.3 - Barriers around the fuel transfer tube area will be removed from the plant. The primary reason to remove these barriers is to increase containment spray drainage flow to the containment sump. These non-essential metal barriers will be removed under FDA-2005-003364-07 (for Unit 2) and FDA-2005-003364-17 (for Unit 1) to support the response to GSI-191 concerns regarding containment sump operation. In order to enhance the flow from containment spray to the sumps during a postulated LOCA, these unnecessary wire mesh cages and doors will be removed.

Engineering analyses (via ME-CA-4000-0950) provided estimated dose rate projections for the RB 808' elev underneath the fuel transfer tube based on worse case source term assemblies and transient time. The results of this calculation define the dose rates at this location to be 187 mR/hr. EVAL-2002-003029-01 provides further assurance via typical field data from past fuel transfer tube activities that potential dose rates are less than that estimated in ME-CA-4000-0950 and less than the dose/dose rate limits prescribed in 10CFR20. As such, this is an acceptable change. Compliance with the requirements of TS 5.7 is provided administratively through the STAs. Therefore, removal of the barriers under the fuel transfer tube will not result in exposures to outage workers that exceed TS 5.7 limits and it is therefore an acceptable change.

LDCR-SA-2007-17, EVAL-2005-001957-20 (RJK):

Replace the reference to Figures 5.2-2 and 5.2-3 with "…the Pressure and Temperature Limits Report." Figures 5.2-2 and 5.2-3 are deleted from the FSAR.

The information contained in Figures 5.2-2 and 5.2-3 is present in the owner controlled Pressure and Temperature Limits Report, developed in accordance with Tech Spec 5.6.6. There is no reduction in control of the information. Maintaining the same information in two places is not necessary, so the information is being relocated to a more relevant document.

Delete Figure 5.2-2 from the FSAR.

The information contained in Figure 5.2-2 is present in the owner controlled Pressure and Temperature Limits Report, developed in accordance with Tech Spec 5.6.6. There is no reduction in control of the information. Maintaining the same information in two places is not necessary, so the information is being relocated to a more relevant document.

Delete Figure 5.2-3 from the FSAR.

The information contained in Figure 5.2-3 is present in the owner controlled Pressure and Temperature Limits Report, developed in accordance with Tech Spec 5.6.6. There is no reduction in control of the information. Maintaining the same information in two places is not necessary, so the information is being relocated to a more relevant document.

Add WCAP-14040 to the list of material incorporated in the FSAR by reference.

LDCR-SA-2007-17, EVAL-2005-001957-20 (RJK) (continued):

WCAP-14040 is the approved methodology as listed in Tech Spec 5.6.6 for development of the RCS H/U and C/D curves for the PTLR.

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE):

In section 1.2.2.3.5:

1. Delete all of 1.2.2.3.5 except of the last paragraph and insert the following before the last paragraph:

"Combustible gas control systems are not required to be Engineered Safety Feature Systems."

2. At the end of the last paragraph insert,

"This system purges the containment atmosphere through filters which reduce radioactive releases. See Section

6.2.5 for details."

Hydrogen recombiners are being removed consistent with the revised 10CFR50.44 and TSTF-447.

Delete the description of the hydrogen recombiners for regulatory guide 1.7 from 1A(N) and insert,

"Based on a revision to 10CFR40.44, Regulatory Guide 1.7 no longer applies to CPSES. See Section 6.2.5." Hydrogen recombiners are being removed consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete description of the hydrogen recombiners for regulatory guide 1.7 in 1A(B) and insert, "Based on a revision to 10CFR40.44, Regulatory Guide 1.7 no longer applies to CPSES. See Section 6.2.5." Hydrogen recombiners are being removed consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 3.1.2.7, delete discussion of the hydrogen removal system. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In sections 3.1.4.13 and 3.1.4.14, delete discussion of the hydrogen recombiners. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 6.2.5 to 6.2.5.3.1, delete discussion of the hydrogen recombiners. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE) (continued):

In section 6.2.5.3.3 to 6.2.5A.6, delete discussion of the hydrogen recombiners and hydrogen gas production. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Tables 6.2.5-1 and -2,

1. Delete Table 6.2.5-1 titled "Applicable Codes, Standards, and Regulatory Guides Used in the Design of the Electric Hydrogen Recombiner"

2.) Delete Table 6.2.5-2 titled "Electric Hydrogen Recombiner Parameters"

Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Table 6.2.5-4 for zinc corrosion rate. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete section 1 and 3 of Table 6.2.5-5 for FMEA for the hydrogen recombiners. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Table 6.2.5A-1 Sheets 1 to 3, Table 6.2.5A-2 Sheets 1 and 2, and Table 6.2.5A-3. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Table 6.2.5A-5 and Table 6.2.5A-6. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 7.1.1.1.2., delete discussion number 4 on combustable gas control. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

On Table 7.1-2.6 sheets 1 to 4, delete the column titled "Hydrogen Monitoring." Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 7.2.1.2.5, delete line item 3. for explosion from hydrogen buildup. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 7.3.1.2.5, delete line item 3. for explosion from hydrogen buildup. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 7.3.2.2, delete the third sentence which makes reference to GDC 41 for Hydrogen recombiners. Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE) (continued):

On Table 7.5-3, sheet 2, revise "Containment Hydrogen" Variable Function from "Key" to "Backup" and Type/Category from "B1" to "B3." On Table 7.5-4 revise "Containment Hydrogen" Variable Function from "Key to Backup" and Type/Category from "C1" to "C3". Removal consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Table 7.5-7A, sheet 3, revise category for containment hydrogen concentration from "B1, C1" to "B3, C3." Revision consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Identify deviation from RG 1.97 for hydrogen monitors.

On Table 7.5-7D, Sheet 7, at the bottom of the page starting from the first column add "Hydrogen Monitor," second column add "Category," third column add "Category 1," forth column add "Category 3," and the last column add "(24)."

Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Incorporate commitment to maintain hydrogen monitors capable of diagnosing beyond design bases accidents.

In Table 7.5-7E, Sheet 1 and 3,

1.) Remove the reference of "Containment Hydrogen Concentration" in numbers (3) on Sheet 1 and (19) on Sheet 3.

2.) Also on Sheet 3, iin number (19), second sentence, replace the word "are" with "is," and add "and License Amendment 117" at the end of number (24)

Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

On Table 8.1-1 sheet 1, delete reference to hydrogen recombiner and containment hydrogen purge system. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

For Table 8.3-1,

On Sheet 7, remove the reference to

1.) Electric H2 Recombiner Power Supply Panel (Post Accident) and

2.) Hydrogen Purge Exhaust Fan Filter Heater

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE) (continued):

On Sheet 8 of the same table above, remove the reference to

1.) Containment Hydrogen Purge Air Exhaust Fan.

On Sheet 9 of the same table above,

1.) Replace Specific Notes 12. and 13 with the words "Not used."

Include hydrogen recombiners to non class 1E equipment connected to safety related bus and footnote to assure remains disconnected during modes where the bus is required to be operable by technical specifications. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Table II.B.2-4 sheet 4, delete reference in number 15 to hydrogen recombiner. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete reference 15 to powering up hydrogen recombiner on Sheets 2, 4, 6, 7, 9, 10, and 12 of Table II.B.2-5. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete references 15A and 15B to powering up the hydrogen recombiner on Sheets 2, 5, 7, 9, 11, and 13 of Table II.B.2.-6. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section II.E.4.1 replace the CPSES Response with "Post accident combustible gas control vis external hydrogen purge or internal electric hydrogen recombiners is no longer required as described in Section 6.2.5. See Section 6.2.4 for containment penetration isolation." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section II.F.1.6, delete reference to hydrogen monitors as required and refer to section 6.2.5.2.3 and 7.5 for required commitment related to the monitors. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete the reference to "Electric Hydrogen Recombiner EQDP-SP-1" from Table 3.10N-1. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Chapter 6 Table of Contents, delete Sections, "6.2.5A.1 to 6.2.5A.6." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Chapter 6 List of Tables, replace the title of numbers, 6.2.5-1, 6.2.5-2, 6.2.5-4, 6.2.5A-1, 6.2.5A-2, 6.2.5A-3, 6.2.5A-5, and 6.2.5A-6 with the words "This Table has been deleted." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE) (continued):

In Chapter 6 List of Figures, replace figure titles of 6.2.5-2, 6.2.5A-1, 6.2.5A-2, 6.2.5A-3, and 6.2.5A-5 with the word "Deleted." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Chapter 6 List of Figures, replace the title of Numbers 6.2.5A-7, and 6.2.5A-9 with the words "Deleted." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Section 6.2.5.3.2.3 , delete the third paragraph which reads,

"After 30 days, the hydrogen generation rate from core radiolysis is 0.68 scfm. The generation rates from sump radiolysis and corrosion are 0.175 scfm, and 2.38 scfm, respectively."

Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

On page 6.2-80, delete the entire References. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Add the words "Abandoned in Place" at the top of Figure 6.2.5-1. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Figure 6.2.5-2. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Figures 6.2.5A-1, -02, and -03. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete the Figures 6.2.5A-5, -7, and -9. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In section 7.6.9,

1. replace the word "DBA" with "severe accident" and delete the word "System" in the first sentence of the first paragraph.

2.) delete the sentences in 7.6.9.1 and 7.6.9.2.

3.) delete the following words from the last paragraph, "System, which includes Hydrogen Monitoring, Hydrogen Recombiners and the Hydrogen Purge System."

Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

On Table 8.3-2, sheet 5, delete the Reference to "Electric H2 Recombiners Power Supply Panel. Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

LDCR-SA-2004-30, EVAL-2004-002063-03 (TJE) (continued):

On Table 8.3-2, sheet 12, replace Notes 12 and 13 with the words "Not used." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

From T 13.5-2 sheet 1, remove the reference to "Electric Hydrogen Recombiner System." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Section 15.6.5.4.5, remove the reference to "hydrogen purge system" and "electric hydrogen recombiners." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Table 17A-1, sheet 11, under number 9.a and in the 2nd and 3rd column, replace with "NA." Under the 5th column, replace the seismic category "I" with "II." Change the 6th column from "Note A" to "Note B." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

In Sections 6.2.5.3.2.3 and 6.2.5.3.2.4,

a. Relocate 6.2.5.3.2.3.b "Reactor Cavity" to 6.2.5.3.2.4.c,

b. Relocate 6.2.5.3.2.4.a "Pressurizer Subcompartment" to 6.2.5.3.2.3.b

c. Delete 6.2.5.3.2.3.c "Pressurizer Relief Tank Subcompartment" and insert the words into 6.2.5.3.2.4.a.

Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

Delete Figure and replace the title with "Deleted." Consistent with the revised 10CFR50.44 and TSTF 447. (SMF-EVAL-2004-002063-03)

LDCR-SA-2006-10, EVAL-2005-003364-03 (JDS):

Appendix 1A(B) RG 1.82 is revised to show conformance for new sump strainers. See EVAL-2005-003364-16 for the comparison of the old and new designs to the Regulatory Guide.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Table 3.9B-10 is revised to show the HV-4758/4759 are now butterfly valves.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

LDCR-SA-2006-10, EVAL-2005-003364-03 (JDS) (continued):

Section 6.2.2.2.1 is revised to describe the new sump strainer design and update the outdated description of emergency sump antivortex design found in FSAR Section 6.2.2.2.1 [SMF-2006-003988 issue].

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 6.2.2.3.3 is revised to describe the new sump strainer design and add the refueling cavity drain modifications.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 6.2.2.3.4 is revised to correct the NPSH description based on ME-CA-0232-5046 R0 which is the basis for Figure 6.2.2-2.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 6.2.2.5 is revised to describe the new RWST setpoint volumes. The containment water level instrument descritption is clarified to match the ref.d figures inlieu of the conrol board indicators.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Figure 6.2.2-3 is revised and Figure 6.2.2-3A added to show the new sump strainer design.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Figure 6.2.2-4 is revised to show the new refueling cavity drain strainers and debris screens.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Figure 6.2.2-5 is revised to show the removal of the pipe reducer in the refueling cavity drains.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

LDCR-SA-2006-10, EVAL-2005-003364-03 (JDS) (continued):

Section 6.3.2.8 is revised to reflect the new RWSTsetpoints and the new emergency strainer design. Typographical errors in valves tag numbers are corrected. Operator reset of SI is also clarified.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Table 6.3-7 is revised to reflect change in EOS-1.3 to start spray switchover after the Empty alarm based on ME(B)-389 R10.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Table 6.3-11 is revised to reflect change in the RWSToutflow analysis and ME(B)-389 R10 based on the above modifications.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 6.5.2.3.2 is revised to reflect the change in pH due to the RWST setpoint changes.

Section 6.5.2.2.1 is revised to clarify CAT isolation valves and vacuum breakers. [ACTN-SMF-2006-004073-01 and ACTN-SMF-2006-004097-01]

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Table 6.5-4 is revised to clarify CAT isolation valves and vacuum breakers. [ACTN-SMF-2006-004073-01 and ACTN-SMF-2006-004097-01]

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 7.6.5 is revised to show the instrument alarm uncertainty for new transmitters (changed from 2.5% to 2.3%).

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

Section 6.5.2.5.3 is revised to reflect some valves are motor and air operated.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

LDCR-SA-2006-10, EVAL-2005-003364-03 (JDS) (continued):

Table 17A-1 is revised to add the new strainers and debris screens.

Updates the FSAR for Modificartions to the plant in response to GSI-191 including License Amendment 129 [see SMF-2005-001869]

LDCR-SA-2006-31, EVAL-2006-001521-01 (JDS):

The sentence "The interlocks are redundant and can withstand a single failure." is being removed. This interlock is not redundant for system operation in normal mode, however, it satisfies the single failure criteria by being fail-safe rather than redundant. In addition, there is a manual override switch located on the fuel transfer system control panel. In case of mechanical failure of the toggle switch, this override feature can be utilized under administrative control. Therefore, redundancy in the upender clear interlocks is not required.

LDCR-SA-2007-19, EVAL-2001-002201-21 (JDS):

This type of insulation was not utilized and was not described in the insulation specification. In accordance with specification CPES-M-2012 Section 3.5.2 M -

Anti-sweat insulation used inside containment shall

1. be low to medium density fiberglass (e.g. < 8 lb/ft3)

2. not employ a vapor barrier, and

3. be stainless steel jacketed.

LDCR-SA-2007-23, EVAL-2007-002718-02 (GLM):

This proposed change is to add Design Basis Document, DBD-ME-006, to Chapter 1, Section 1.6, Table 1.6-1, DOCUMENTS INCORPORATED BY REFEFENCE.

DBD-ME-006, Attachment 1 captures the final response to NUREG-0612 and GL 81-07 as provided by TXX-3659. Documented in DBD-ME-006 are additional correspondence between the licensee and the NRC occurred after TXX-3659 was issued. This correspondence, including that from the NRC, form the basis for DBD-ME-006 content. By adding DBD-ME-006 to Table1.6-1, information that was provided by TXX-3659 to include subsequent correspondence with the NRC for heavy load controls (parameter values and assumptions, such as lift height, load weight and medium present under the load) would result in information describing the heavy load controls program that would normally be required in the FSAR (by the use of the Incorporated by Reference method). Any changes to the described heavy loads control program (changes to the DBD) have been and will be evaluated per the requirements of design change control process, as

LDCR-SA-2007-23, EVAL-2007-002718-02 (GLM) (continued):

well as 10CFR50.59 process. The original responses were provided by TXX-3659 [7.2.8]. DBD-ME-006 is maintained current under 10CFR50.59. By updating Table 1.6-1 to include DBD-ME-006 as incorporated by reference, and changing reference 17 to DBD-ME-006 (with historical listing of TXX-3659), information supporting the necessary details for FSAR content will be complete. This change has not affected on existing commitments. By implementation of this proposed change, the stated Regulatory Expections from RIS 2005-025 Supplement 1 will have been met.

Revise Section 9.1.4, Reference 17, to be DBD-ME-006 (with TXX-3659 being referenced for historical purposes).

DBD-ME-006, Attachment 1 captures the final response to NUREG-0612 and GL 81-07 as provided by TXX-3659. Documented in DBD-ME-006 are additional correspondence between the licensee and the NRC occurred after TXX-3659 was issued. This correspondence, including that from the NRC, form the basis for DBD-ME-006 content. By adding DBD-ME-006 to Table1.6-1, information that was provided by TXX-3659 to include subsequent correspondence with the NRC for heavy load controls (parameter values and assumptions, such as lift height, load weight and medium present under the load) would result in information describing the heavy load controls program that would normally be required in the FSAR (by the use of the Incorporated by Reference method). Any changes to the described heavy loads control program (changes to the DBD) have been and will be evaluated per the requirements of design change control process, as well as 10CFR50.59 process. The original responses were provided by TXX-3659 [7.2.8]. DBD-ME-006 is maintained current under 10CFR50.59. By updating Table 1.6-1 to include DBD-ME-006 as incorporated by reference, and changing reference 17 to DBD-ME-006 (with historical listing of TXX-3659), information supporting the necessary details for FSAR content will be complete. This change has not affected on existing commitments. By implementation of this proposed change, the stated Regulatory Expections from RIS 2005-025 Supplement 1 will have been met.

LDCR-SA-2007-22, EVAL-2004-002882-07 (JDS):

RG 1.54 description is revised to describe the revised program from Non-safety to Service Level I and the applicable standards. This is an administrative update of the FSAR to reflect an upgrade to the Protective Coatings Program for inside containment. Coatings do not have safety function per se and are not Basic Components as defined in 10CFR50.2. They are not subject to 10CFR21. In addition, they are not "Safety Related" Structures, Systems, or Components as defined in 10CFR50.2. However, coatings are subject to applicable portions of 10CFR50, Appendix B via GDC-1 because their failure has the potential to be detrimental to Safety Related Structures, Systems, and Components. This is most similar to non-seismic items (seismic category II equipment). The CPNPP quality assurance program for such items is covered by Appendix D of the QA Manual. See DBD-ME-028 for additional information on classification of SSCs. [Also note that this change is consistent with ONE 97-1016, Applicability of Appendix B to protective coatings applied to the interior surfaces of safety related components]

LDCR-SA-2007-22, EVAL-2004-002882-07 (JDS) (continued):

Updated description of the Protective Coatings Program for containment. This is an administrative update of the FSAR to reflect an upgrade to the Protective Coatings Program for inside containment.

Updated to remove obsolete information. This is an administrative update of the FSAR to reflect an upgrade to the Protective Coatings Program for inside containment.

Update and clarify description of the quality assurance program (under QA Manual Appendix D). This is an administrative update of the FSAR to reflect an upgrade to the Protective Coatings Program for inside containment. Coatings do not have safety function per se and are not Basic Components as defined in 10CFR50.2. They are not subject to 10CFR21. In addition, they are not "Safety Related" Structures, Systems, or Components as defined in 10CFR50.2. However, coatings are subject to applicable portions of 10CFR50, Appendix B via GDC-1 because their failure has the potential to be detrimental to Safety Related Structures, Systems, and Components. This is most similar to non-seismic items (seismic category II equipment). The CPNPP quality assurance program for such items is covered by Appendix D of the QA Manual. See DBD-ME-028 for additional information on classification of SSCs. [Also note that this change is consistent with ONE 97-1016, Applicability of Appendix B to protective coatings applied to the interior surfaces of safety related components]

LDCR-SA-2006-24, EVAL-2003-000188-03 (JCH):

Table 17A-1

Added notes to the table for manual and extension passive and active valves to provide for the installation of Safety Related / Seismic Category I replacement remote valve stem extensions and gear driven valve operators for the Atmospheric Relief Valves (ARV) upstream isolation block valves, and Seismic Category I supports.

The table lists valve operator types. Clarifying UFSAR Table 17A-1 to describe the requirements for Active valve manual and extension stem operator quality attributes and Passive valve manual and extension stem operator quality attributes is acceptable based on the different quality requirement attributes for active and passive components. This clarification also maintains consistency between the table requirements for safety related valves and the referenced Note 79. This clarification is supported by the requirements of DBD-ME-028, Rev. 13, Attachment 19, Table 2.

Table 3.9B-10

Provided editorial clarification of valve/operator type and method of operation descriptions by specifying handwheels or extension stems as appropriate

LDCR-SA-2006-34, EVAL-2003-002117-03 (JCH):

Section 10.4.3.1-Design Basis-2. Feedwater Pump Turbine Seals

Delete "The feedwater pump turbine is self-sealing at loads above approximately half of the rated load on the turbine."

Section 10.4.3.2-System Description-1. Normal Operation Conditions-b. Feedwater Pump Turbine Steam Seals

Delete "during startup of the turbine and during low-power operation."

Justification: The FWPT technical manual shows that the originally supplied design anticipated seal steam would be supplied to the turbine during both start-up and full load operation; therefore, self sealing is not a design attribute of the FWPT's at CPSES. Operations has never seen a self-sealing indication for the feedwater pump turbine in either steam seal supply or dump valve positions. This does not affect the operational capability of the seal system and is congruent with the expectations outlined by the vendor.

LDCR-SA-2007-16, EVAL-2006-002471-02 (RJK):

The values presently listed in Section 5.3.2.2.1 are historic to pre-operational calculations performed IAW Reg Guide 1.99 revision 1 for pre-licensing certification of the reactor vessels. These values were recalculated just prior to Unit 1 and 2 startup using revision 2 of Reg Guide 1.99, but were never updated in the FSAR. These updated values are currently listed in the CPSES PTLR (Table 2-1) and were previously reported to the USNRC via letters TXX-95220 and TXX-97275 in response to Generic Letter 92-01.

LDCR-SA-2007-5, EVAL-2006-003933-05 (JDS):

The Unit 1 Cycle 13 core configuration consists of 88 fresh Westinghouse fuel assemblies (Region 15), 89 once-burned Westinghouse assemblies (88-Region 14 & 1-Region 13), and 16 twice-burned Westinghouse assemblies (Region 13). The single once-burned Region 13 fuel assembly is reloaded from the spent fuel pool. The Region 14/15 Westinghouse fuel assemblies are of the Vantage-Plus with Intermediate Flow Mixing (IFM) grids) and include a thin zirconium oxide coating applied to the bottom seven inches of each fuel rod to provide additional debris fretting protection in this area. Oxide forms naturally on the fuel cladding during normal operation so this coating has no impact on fuel design function. The Region 13 Westinghouse fuel assemblies are also of the Vantage-Plus design, but do not include IFMs or the applied oxide coating but are dimensionally equivalent to the Vantage-Plus design. All fuel assemblies are equipped with the “Small Hole” debris filtering bottom nozzle, an alternate protective grid (P-grid) located just above the bottom nozzle to trap any debris making it through the bottom nozzle, and long solid end plugs. The pressure drop through the Vantage-Plus (without IFMs) assemblies is slightly lower than that through the co-resident fresh Vantage-Plus / IFM assemblies while the heat transfer characteristics are improved by the IFMs. The effects of the mixed core on the thermal hydraulic response are considered.

LDCR-SA-2007-5, EVAL-2006-003933-05 (JDS) (continued):

All the Westinghouse assemblies use ZIRLO (as opposed to Zircaloy-4) for the fuel rod cladding, guide tubes and assembly mid-span grids. In addition, integrated fuel burnable absorbers (IFBAs) and discrete wet annular burnable absorbers (WABAs) are used to shape the power distribution and achieve a desirable moderator temperature coefficient.

The thermal-hydraulic and neutronic responses of the Unit 1 Cycle 13 reload core configuration potentially affect the design functions of many plant components and systems. The acceptability of the Cycle 13 reload core configuration is based on meeting the event acceptance criteria in FSAR Sections 3.6B, 4.2, 4.3, 4.4, 6.2, 6.3, and all of FSAR Chapter 15.

The neutronic design criteria of the Cycle 13 reload core configuration continue to be met. The thermal-hydraulic design criteria of the Cycle 13 reload core configuration continue to be met.

The adequacy of the systems required to mitigate or prevent accidents, shutdown the plant and maintain the plant in a safe shutdown condition is demonstrated in the analyses presented in FSAR Chapter 15 and summarized in the Cycle 13 update of FSAR Appendix 4A. The relevant event acceptance criteria include compliance with the DNBR and peak RCS and steamline pressure acceptance criteria for ANS Condition II transients and acceptable radiological consequences for ANS Condition III and IV accidents. Relevant event acceptance criteria continue to be met with the current cycle-specific reactor physics parameters.

LDCR-SA-2007-21, EVAL-2006-004074-01 (GLM):

Remove the paragraph in FSAR Section 9.2.1.2.1 which states that the Circulating Water Bleed line is used as a continuous makeup for SSI and add a note to Section 9.2.1.2.1 which states that additional details can be found in Section 9.2.5.2.

As stated in FSAR Section 9.2.5.1 the function of the ultimate heat sink is to dissipate heat rejected from the station service water system during post accident shutdown and normal cooldown conditions. The Safe Shutdown Impoundment (SSI) is a part of the Ultimate Heat Sink. The level of water in the SSI is maintained through an equalization canal between the Squaw Creek Reservoir (SCR) and the SSI as described in the FSAR in Sections 9.2.1.2.1 and 9.2.5.2. The equalization canal between the SSI and Squaw Creek Reservoir is current the normal source for makeup to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

A bleed line is provided from the Circulating Water System, at the intake structure, to the Service Water Intake Structure pump suction pit. The FSAR indicates that when operated the circulating water bleed-off ensures continuous makeup to the SSI and provides circulation to prevent excessive dissolved solids concentration in the SSI. Based on the operations procedures (SOP-310A and SOP-310B) above this is not

LDCR-SA-2007-21, EVAL-2006-004074-01 (GLM) (continued):

currently the way that the system is operated. The procedure indicates that the bleed line is normally closed. This is supported by the valve line-up attachment to the procedures which indicates that valve 1CW-0019 is normally closed.

IOR No. 89-64 was written to address environmental regulations which require that the discharge from the Safe Shutdown Impoundment to the Squaw Creek Reservoir be monitored. This includes the flow from the Circulating Water Bleed line. The IOR states that valve 1CW-0019 is intended to remain closed during plant operation until the plasitc is removed from the bleed line. Based on the changes made by IOR No. 89-64 and CPSES Procedure Change Form for SOP-310A, Rev. 5, the method of operation was revised in 1990. The IOR states that the flow Circulating Water System to the Safe Shutdown Impoundment via the bleed line be closed using valve 1CW-0019. In lieu of using the Circulating Water bleed line as the normal makeup, the equalization canal between the SSI and Squaw Creek Reservoir is used as the normal source for make-up to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

Delete the last paragraph in FSAR Section 9.2.5.2 and add a new paragraph describing the Circulating Water Bleed line and its current operation.

As stated in FSAR Section 9.2.5.1 the function of the ultimate heat sink is to dissipate heat rejected from the station service water system during post accident shutdown and normal cooldown conditions. The Safe Shutdown Impoundment (SSI) is a part of the Ultimate Heat Sink. The level of water in the SSI is maintained through an equalization canal between the Squaw Creek Reservoir (SCR) and the SSI as described in the FSAR in Sections 9.2.1.2.1 and 9.2.5.2. The equalization canal between the SSI and Squaw Creek Reservoir is current the normal source for makeup to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

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LDCR-SA-2007-21, EVAL-2006-004074-01 (GLM) (continued):

using the Circulating Water bleed line as the normal makeup, the equalization canal between the SSI and Squaw Creek Reservoir is used as the normal source for make-up to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

In Section 9.2.5.3, under "2. Hydraulic Performance," delete the word "creates" and replace with "can be used if required."

As stated in FSAR Section 9.2.5.1 the function of the ultimate heat sink is to dissipate heat rejected from the station service water system during post accident shutdown and normal cooldown conditions. The Safe Shutdown Impoundment (SSI) is a part of the Ultimate Heat Sink. The level of water in the SSI is maintained through an equalization canal between the Squaw Creek Reservoir (SCR) and the SSI as described in the FSAR in Sections 9.2.1.2.1 and 9.2.5.2. The equalization canal between the SSI and Squaw Creek Reservoir is current the normal source for makeup to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

A bleed line is provided from the Circulating Water System, at the intake structure, to the Service Water Intake Structure pump suction pit. The FSAR indicates that when operated the circulating water bleed-off ensures continuous makeup to the SSI and provides circulation to prevent excessive dissolved solids concentration in the SSI. Based on the operations procedures (SOP-310A and SOP-310B) above this is not currently the way that the system is operated. The procedure indicates that the bleed line is normally closed. This is supported by the valve line-up attachment to the procedures which indicates that valve 1CW-0019 is normally closed.

IOR No. 89-64 was written to address environmental regulations which require that the discharge from the Safe Shutdown Impoundment to the Squaw Creek Reservoir be monitored. This includes the flow from the Circulating Water Bleed line. The IOR states that valve 1CW-0019 is intended to remain closed during plant operation until the plasitc is removed from the bleed line. Based on the changes made by IOR No. 89-64 and CPSES Procedure Change Form for SOP-310A, Rev. 5, the method of operation was revised in 1990. The IOR states that the flow Circulating Water System to the Safe Shutdown Impoundment via the bleed line be closed using valve 1CW-0019. In lieu of using the Circulating Water bleed line as the normal makeup, the equalization canal between the SSI and Squaw Creek Reservoir is used as the normal source for make-up to the SSI as indicated by the Operations procedures SOP-310A and SOP-310B.

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LDCR-SA-2007-11, EVAL-2005-001957-23 (JDS):

Update list of Tables to match revised Tables.

Update list of Figures to Match revised or deleted Figures.

Sections 6.2.1.1.3.1.1 and 6.2.1.1.3.1.2 were modified due to the methodology change and Steam Generator replacement which resulted in differences in the analysis results for the two units.

Section 6.2.1.1.3.2 was modified to reflect the changes in the Figures and the RSGs.

Section 6.2.1.1.3.7 is modified to reflect the changes in the Tables and the RSGs.

Section 6.2.1.2.1 is modified to reflect the changes in the methodology and RSGs. Section 6.2.1.2.2 is modified to reflect the changes in the Figures.Section 6.2.1.2.3 is modified to reflect the changes in the methodology and the addition of the RSGs for Unit 1. Also, many of the Figures and Tables described in this Section were determined to be no longer required, and therefore removed.

Section 6.2.1.3 was re-written due to the methodology update.

Sections 6.2.1.4.1 and 6.2.1.4.2 is modified to reflect the changes or deletions in the Tables

Section 6.2.1.4.9 is deleted due to the deletion of the Tables described in this Section.

Section 6.2.1.5.6 is modified to reflect the changes in the Tables

Section 6.2.2.3.2 is modified to reflect the changes in the Tables

6.2.1-2A was replaced with 6.2.1-2A.1 and 6.2.1-2A.2 due to differences between Units 1 and 2 as a result of Unit 1 Steam Generator Replacement.

6.2.1-2B was replaced with 6.2.1-2B.1 and 6.2.1-2B.2 due to differences between Units 1 and 2 as a result of Unit 1 Steam Generator Replacement.

Tables 6.2.1-3 and -3A were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

LDCR-SA-2007-11, EVAL-2005-001957-23 (JDS) (continued):

Tables 6.2.1-4, -4A, and -4B were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-15 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

6.2.1-9 was replaced with 6.2.1-9A and 6.2.1-9B due to differences between Units 1 and 2 as a result of Unit 1 Steam Generator Replacement.

6.2.1-10 was replaced with 6.2.1-10A and 6.2.1-10B due to differences between Units 1 and 2 as a result of Unit 1 Steam Generator Replacement.

Table 6.2.1-21 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-26 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-29 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

LDCR-SA-2007-11, EVAL-2005-001957-23 (JDS) (continued):

Table 6.2.1-30 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Tables 6.2.1-37 through -49 were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-50 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-50A was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-51 through -56 were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-58 through -60 were deleted as the information contained in ithem is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

LDCR-SA-2007-11, EVAL-2005-001957-23 (JDS) (continued):

Table 6.2.1-64 through -69 were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-77 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-91 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-93 was deleted as the information contained in it is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Table 6.2.1-95 through -104 were deleted as the information contained in them is not required. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-1 through -3 were deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-9 through -16 were modified due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

LDCR-SA-2007-11, EVAL-2005-001957-23 (JDS) (continued):

Figure 6.2.1-20 was deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-30 through -45 were deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-49 through -52 were deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-54 through -56 were deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

Figures 6.2.1-58 through -70 were deleted due to the revised analysis. The Mass and Energy Release Data for both LOCA and SLB for Unit 2 is unchanged. However, for Unit 1, the mass and energy release data for both LOCA and SLB have been modified. For LOCA, the change in the M&E release data is due to both the RSGs and a methodology change. For SLB, the change in the M&E release data is due solely to the RSGs.

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LDCR-SA-2007-18, EVAL-2003-000167-04 (RJK):

Neutron dosimetry sensors used to monitor neutron exposure for continued surveillance of reactor vessel integrity were installed in CPNPP unit 1 reactor cavity annulus per FDA 2003-000167-01 and the FSAR was updated per LDCR SA-2003-059. This LDCR was required to document that the neutron dosimetry sensors are also now installed in the unit 2 reactor cavity annulus and FSAR section 5.3.1.6.2 now applies to both units.

LDCR-SA-2007-20, EVAL-2006-000629-04 (TJEW):

In 9.1.3.1.1.1, the first sentence of the second paragraph reads, "The spent fuel pool bulk water temperatures are maintained at less than 150EF for normal operation based on decay heat generation from a normal full core offload at 7days after shutdown, plus decay heat from the opposite unit’s last refueling discharge plus decay heat from fuel assemblies from a maximum number of previous refuelings in both pools."

1.) In the 150EF, replace the "E" with a degree symbol, and

2.) Change "7 days" to "125 hours".

In 9.1.3.1.1.1, the third and fourth sentence in the fourth paragraph reads, "The start of the second outage is conservatively assumed to begin 45 days after the first. A normal full core offload is conservatively assumed to start at 100 hours after the reactor is subcritical and complete at 168 hours."

1.) Change "45 days" to "4 months",

2.) Change "100" hours to "75" hours, and

3.) Change "168" hours to "125" hours.

4.) In the last paragraph of 9.1.3.1.1.1, for the term "140EF" replace the "E" with a degree sign.

In 9.1.3.1.1.3, the second paragraph reads, "The spent fuel pool water temperatures are maintained at less than 212°F for two loop operation based on an emergency core offload 150 hours after shutdown, plus the most recent refueling discharge 36 days after shutdown, plus the opposite unit’s previous refueling discharge 66 days after shutdown, plus decay heat from a maximum number of previous refuelings in both pools."

1.) Change "150" hours to "125" hours,

2.) Change "36 days after" to "31 days prior to", and

3.) Change "66 days after" to "5 months prior to"

1.) On Table 9.1-1, under Parameters,

LDCR-SA-2007-20, EVAL-2006-000629-04 (TJEW) (continued):

a.) delete the row "Numbr of fuel assemblies stored" and

b.)change the exponent of Decay Heat Producted from "10 to the 5" to 10 to the 6"

2.) On the same row of Decay heat produced, make the following changes:

Under MAX. DESIGN

a.) MAX POOL from 50.7 to 52.0

b.) MIN POOL from 5.55 to 9.09

Under MAX. SUMMER DESIGN

a.) MAX POOL from 21.3 to 23.28

b.) MIN POOL from 5.55 to 6.08

Under ABNORMAL MAX. DESIGN

a.) MAX POOL from 56.5 to 62.92

b.) MIN POOL from 6.0 to 6.6

3.) For the parameter Time to boil, change MAX SUMMER DESIGN, MAX POOL from >7 to >3

4.) Under the "NOTES" section,

a.) Replace the words in the note with “Deleted”, and

b.1) in note 3, replace the words "assumes most resent spent fuel discharges are" with the word “loads”, and

b.2) delete "to" after the words "MAX Pool"

On Table 12.2-24, "DESIGN PARAMETERS FOR POST-ACCIDENT DOSE EVALUATIONS"

1.) Under Gap activity-release

a.) Replace "Noble gases other than Kr-85 0.1" with "I-131 0.08"

b.) Change the value of Kr-85 from "0.3" to "0.10"

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LDCR-SA-2007-20, EVAL-2006-000629-04 (TJEW) (continued):

c.) Replace "Iodines other than I-127 and I-129 0.1" with "Other Iodines and Noble Gases 0.05"

2.) Change the Decay interval between reactor shutdown and commencement of refueling operations (hours) from "100" to "75"

In 15.7.4.2.1, change "100 hr" to "75 hours"

In 15.7.4.3.1.1, change "100 hr" to "75 hours"

1.) In 15.7.4.3.1.3, change "100" hours to "75" hours

2.) In the last paragraph of 15.7.4.3.1, change "0.087" rem to "0.090" rem

Revise Table 15.7-6, NOBLE GAS AND IODINE ACTIVITIES RELEASED FROM DAMAGED FUEL RODS AS A RESULT OF A FUEL HANDLING ACCIDENT as follows:

1.) Revise Kr-85 from "8.62" to "9.66"

2.) Add "Kr-85m" to be "9.62E-02" curries released

3.) Add "Kr-88" to be "3.05E-04" curries released

4.)Revise Xe-131m from "3.01" to "4.57"

5.) Revise Xe-133m from "5.56" to "1.52"

6.) Revise Xe-133 from "6.93" to "6.88"

7.) Revise Xe-135m from "3.61" to "5.04E+00"

8.) Revise Xe-135 from "1.20" to "7.05"

9.) No revision to I-131

10.) Revise I-132 from "2.87" from to 3.25

11.) Revise I-133 from "3.47" to "7.44"

12.) Revise I-135 from "2.36" to "3.07E+01"

13.) Revise a.3 from "100" hours to "75" hours

On Table 15.7-8, PARAMETERS FOR POSTULATED FUEL HANDLING ACCIDENT ANALYSIS, change

LDCR-SA-2007-20, EVAL-2006-000629-04 (TJEW) (continued):

1.) In 1.a, Power Level (MWt) from "3565" to "3694" and

2.) In 4.c, Doses, change "0.087" rem to "0.090" rem

LDCR-SA-2006-41, EVAL-2006-003110-02 (CBC):

In paragraph 6.2.6.4, item 1, revise the first by inserting an "s" at the end of "exception" and deleting the numeral "1" at the end of "5.5.16.a.1". The new sentence will read as follows:

"Containment integrated leakage-rate test (Type A) are performed at least once per 48 months with possible extension to 10 yr. intervals based upon acceptable performance history as determined in accordance with NEI 94-01 and Reg. Guide 1.163, as modified by the exceptions in Technical Specification 5.5.16.a."

FSAR Section 6.2.6.4 is updated by eliminating the concrete containment inspections required by Appendix J, Option B and taking credit for the concrete containment inspections performed under the IWL and IWE of the ASME program (which CPSES is already committed to). The FSAR is updated to be consistent with changes to TS 5.5.16 approved by the NRC in License Amendment 141.

LDCR-SA-2006-23, EVAL-2005-004439-02 (JCH):

Page 9.5-71: add "wireless intercom system" to the types of communication systems listed

Justification: The Refuleing and Emergency sound Powered Loops have been replaced by a wireless intercom with radio backup and emergency communications is supplied by the Inplant Radio System with Gaitronics backup.

Delete references to refueling and emergency sound powered loops

Justification: The Refuleing and Emergency sound Powered Loops have been replaced by a wireless intercom with radio backup and emergency communications is supplied by the Inplant Radio System with Gaitronics backup.

Added new section 9.5.2.2.8 which discusses the wireless intercom system and renumbered subsequent sections.

Justification: The Refuleing and Emergency sound Powered Loops have been replaced by a wireless intercom with radio backup and emergency communications is supplied by the Inplant Radio System with Gaitronics backup.

Added paragraph that power to the wireless intercom system is provided by plant support power and each headset has an individual battery supply.

LDCR-SA-2006-23, EVAL-2005-004439-02 (JCH) (continued):

Justification: The Refuleing and Emergency sound Powered Loops have been replaced by a wireless intercom with radio backup and emergency communications is supplied by the Inplant Radio System with Gaitronics backup.

Deleted reference to testing of refueling and emergency sound powered loops

Justification: The Refuleing and Emergency sound Powered Loops have been replaced by a wireless intercom with radio backup and emergency communications is supplied by the Inplant Radio System with Gaitronics backup.

LDCR-SA-2005-29, EVAL-2005-003364-07 (JDS):

Suction pressure accident monitoring is added for the RHR and Containment Spray pumps. Table 7.5-7B is updated. Flow Diagrams 5.4-6 and 6.2.2-1 are updated as detailed in FDA-2005-003364-08 and -18. These modifications will be implemented at power in 2007.

Suction pressure accident monitoring is added for the RHR and Containment Spray pumps.These components have been identified in Attachment 1 of TXX-05162 as requiring monitoring for system performance. The are utilized as a key variable for emergency sump performance. These instruments are not required by Table 2 of RG 1.97 Rev. 2. CPSES utilization is an alternate to narrow range sump level, Type B Category 2. It is classifed as a Type D Category 2 variable because it provides information to indicate the operation of individual safety systems and other systems important to safety. These variables are to help the operator make appropriate decisions in using the individual systems important to safety in mitigating the consequences of an accident.

LDCR-SA-2008-4, EVAL-2007-001435-01 (RJK):

Changes system description to allow use of both CVCS mixed bed demineralizers during RCS cleanup during outages. FDA-2007-001435-01 provides the basis for allowing increased flow during RCS cleanup based upon less than minimal impact on SSCs and radiological consequences due to higher flow rates.

The FSAR is also administratively updated to clarify the maximum letdown flow. DCN 9970 provides the basis for the normal CVCS flow for a nominal 120 gpm design is a maxiumum of 140 gpm based on accuracies of the orifice construction. This flow rate corresponds to a critical input to the radiological consequences in Section 15.6.2.

The FSAR is updated to clarify the maximum letdown flow. DCN 9970 provides the basis for the normal CVCS flow for a nominal 120 gpm design is a maxiumum of 140 gpm based on accuracies of the orifice construction. This flow rate corresponds to a critical input to the radiological consequences in Section 15.6.2.

LDCR-SA-2008-4, EVAL-2007-001435-01 (RJK) (continued):

Section 15.6.2 is administratively corrected to note that the pressure switches are not safety related and may NOT be credited as "the" method of detection.

LDCR-SA-2006-3, EVAL-2001-001214-01 (JCH):

Section 10.2.2.6 Generator

Sectin 10.2.2.6.2 Fire Prevention and Protection

Page 10.2-6: Change the referenced inert gas from "carbon dioxide" to "argon".

Justification: Change of the inert gas used when purging the generator was changed from carbon dioxide to argon due equipment obsolescence, personnel safety and improved outage purging time.

LDCR-SA-2007-2, EVAL-2005-002580-04 (TJEW):

Revise turbine HP & LP stop and control valves test frequency from 12 weeks to 26 weeks. This recommendation was based on a quantitative evaluation on the probability of failure of the overspeed trip and protection system as a function of the turbine stop and control valve test interval. The basic principles of the methodology are the same as those used by Siemens in previous studies which included the CPSES original designs, Grand Gulf, Connecticut Yankee, and Limerick and which have been previously reviewed and accepted by the NRC.

For Table 1.6-1, sheet 2 of 3, replace the 9th Reference which reads:

"'Missile Probability Analysis Methodology for TXU Generation Company, LP, Comanche Peak Units 1 & 2 with Siemens Retrofit Turbines,' CT-27331 (CONFIDENTIAL), Revision No. 3, December 15, 2003"

with the following words,

"TP-03143, "Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC," July 31, 2003."

In Section 3.5.1.3.2,

1.) In the 1st sentence in the 4th paragraph, replace the word "probabiliy" with "probability"

2.) In the 5th paragraph change the Section number from "3.5.1.3.3.1" to "3.5.1.3.3" and delete the words "Unit 1"

In Section 3.5.1.3.3, add "[17]" to the end of the first paragraph.

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LDCR-SA-2007-2, EVAL-2005-002580-04 (TJEW) (continued):

1.) In section 3.5.1.3.3 on page 3.5-9, replace the words,

"P1 = 3.42E-5 per year for 100,000 hour inspection interval [15]"

with the words,

"P1 = 1.56E-4 at 100,000 hours (inspection interval) for a 26 weeks turbine valve test frequency [15]"

2.) In the last pargrapgh second sentence after "P1" add the following works, "at 100,000 hours inspection interval" and after "the NRC limit of" add the following words, "11.42E-4 when adjusted to 100,000 hours inspection interval verses".

3.) In section 3.5.1.3.4, number 2, change "12" to "26"

4.) In section 3.5.1.3.4 in the last paragraph, change "5x10^-6" to "1.37x10^-5" and

replace References " [5,11,13,][14]" with the following words, "and 26 week turbine valve test interval [15]"

5.) Delete the last sentence of the last paragraph that reads, "CPSES turnine missile analysis is based on a conservative high turbine missile generation probability of 4 x 10^-5 per turbine year."

6.) In section 3.5.1.3.5, replace the following sentence that reads,

"Periodic testing of the stop and control valves is necessary to insure reliability and continuity of service; therefore, valve testing for all nuclear units is recommended with the aid of an automatic turbine tester (ATT) every 12 weeks."

with the following sentence,

"Periodic testing of the stop and control valves is necessary to insure reliability and continuity of service; therefore, valve testing for all nuclear units is recommended with a manual test or with the aid of an automatic turbine tester (ATT) every 26 weeks."

At the end of the first paragraph and at the end of the 17 th of 10.2.3.5, replace Reference [10] with [2].

In the Reference section of 3.5,

1.) replace References 5 with the word "Deleted"

2.) replace References 11, 13 and 14 with the word "Deleted"

LDCR-SA-2007-2, EVAL-2005-002580-04 (TJEW) (continued):

3.) change the revision number of Reference 15 from "4" to "5," change the date from "December 15, 2003" to "January 18, 2007," and delete the VL number.

1.) For Reference 3.5, delete the VL number from Reference 16.

2.) For Reference 3.5, add Reference 17 that will read

"TP-03143, "Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC," July 31, 2003."

In section 10.2.2.1, first paragraph, change "44" to "46."

In section 10.2.2.7.6, number 2, change "12" to "26."

In the 4th paragraph of 10.2.3.6, change "12" to "26."

1.) In the Reference section of 10.2, replace the VL numbers in Reference 1 with "July 1975" and change "46" to "44"

2.) In Reference 2

a.) delete "Allis-Chalmer's Power Systems, Inc.,"

b.) replace "rpm" with "r/min",

c.) replace "46" with "44", and

d.) replace "[VL 04-0051540}" with "October 1975"

3.) In the Reference section of 10.2, replace References 7, 8, and 9 with the word "Deleted"

4.) In the Reference section of 10.2, replace the revision number of Reference 10 from "4" to "5", add "LP" after "Company," change the date from "December 15, 2003" to "January 18, 2007," and delete the VL number.

In the Reference section of 10.2, replace Reference 11 with the words "TP-03143, "Missile Analysis Methodology for GE Nuclear Steam Turbine Rotors by the SWPC," July 31, 2003."

LDCR-SA-2006-25, EVAL-2004-002055-01 (JCH):

Revise the second sentence in paragraph 10.2.2.6 to rep[lace "60 psig" with "psig"

Revise the next to last paragraph in 10.2.2.6.2.1 by replacing "seal oil storage tank" with "Hydrogen Degassing Storage Tank".

LDCR-SA-2006-25, EVAL-2004-002055-01 (JCH) (continued):

Revise the last sentence in the paragraph to clarify that the loop seal is permanently filled.

Revise the last paragraph to state "The seal oil drained from the seal oil tank passes into the Hydrogen Degassing Storage Tank to which the vapor exhauster is also connected. The exhauster creates a slight vacuum in the Hydrogen Degassing StorageTank so that oil saturated with hydrogen is degassed."

In Table 10.4-10, sheet 2 of 3, replace the current a. and b. item entries under "Generator Seal Oil Unit" with the following:

"Air Side Coolers", "2", "1", "113", and "0.65 x 10E6"

Revise the "Maximum Guaranteed Rating" and "Valves Wide Open" values for Hydrogen Pressure from 60 psig to 75 psig.

Revise the "service pressure" values for Hydrogen from "4.2 bar" aand "60 psig" to "5.1 bar" and "75 psig".

LDCR-SA-2008-8, EVAL-2007-002816-02 (RAS):

Revises position titles and organization descriptions to reflect approved titles maintained in EIS and the current CPNPP organizational structure.

LDCR-SA-2008-9, EVAL-2007-002223-01 (RJK):

CP License Amendment 128 changed the limit for identified primary to secondary leakage in TS LCO 3.4.13 to 150 GPD for any SG in each unit. This change removed an existing unit difference and brought the TS LCO into agreement with the standard technical specifications as modified by TSTF-449 R4. The limit of 150 gallons per day per SG is based on the operational LEAKAGE performance criterion in NEI 97-06, Steam Generator Program Guidelines. The Steam Generator Program operational LEAKAGE performance criterion in NEI 97-06 states, "The RCS operational primary to secondary leakage through any one SG shall be limited to 150 gallons per day." The limit is based on operating experience with SG tube degradation mechanisms that result in tube leakage. The operational leakage rate criterion in conjunction with the implementation of the Steam Generator Program is an effective measure for minimizing the frequency of steam generator tube ruptures.

FSAR Amendment 102, Supplement a

LDCR-SA-2008-14, EVAL-2008-002865-01 (TJEW):

Change title from "Director, Oversight & Nuclear Overview" to "Director, Oversight & Regulatory Affairs"

LDCR-SA-2008-15, EVAL-2008-002399-01 (JDS):

ANSI N512-1974 has been replaced with ASTM D 3912 Standard Test Method for Chemical Resistance of Coatings Used in Light-water Nuclear Power Plants which is endorsed by Reg guide 1.54 rev 1. The decontamination test per ANSI 512 was not carried into the replacement standard listed by the ASTM D33 committee that writes these procedures. FSAR Section 6.1B.2 was previously revised in Amendment 102 to remove ANSI N512 and update to current industry standards.

LDCR-SA-2007-15, EVAL-2005-004951-02 (JCH):

Section 10.4.2.2.1 System Operation : page 10.4-8

Deleted reference to priming tank.

Section 10.4.2.5 Instrumentation Requirements: page 10.4-9

Deleted reference to priming tank. Added description of the pressure switch that on low vacuum priming header pressure provides input to a main condenser priming trouble alarm. The main condenser vacuum priming trouble alarm is also activated if the main condenser outlet waterbox vacuum breaker air operated valve is not fully closed.

Justification: The priming tank was removed and a loop seal installed to prevent carry over of the water to the priming pumps due to repetitive reliability problems with the vacuum priming tank level control components.

LDCR-SA-2006-36, EVAL-2005-003364-19 (JDS):

Section 6.2.2 is updated for changes to the sump design and licensing basis to reflect the results of the mechanistic analysis requested in Generic Letter 2004-02. Section 6.2.2 References are administratively updated.

Section 6.2.2 is updated for changes to the sump design and licensing basis to reflect the results of the mechanistic analysis requested in Generic Letter 2004-02. Section 6.2.2.3.3 and 6.2.2.3.4 are revised to reflect the GSI-191 analysis and testing documented in SMF-2001-002201, SMF-2004-003972, and SMF-2005-003364 (see individual references and calculations above) as required by Generic Letter 2004-02. The change to the licensing basis is conservative in that analyses and testing have been performed to the current state of the art.

LDCR-SA-2006-36, EVAL-2005-003364-19 (JDS) (continued):

Section 6.2.2 is updated for changes to the sump design and licensing basis to reflect the results of the mechanistic analysis requested in Generic Letter 2004-02. Table 6.2.2-4 is administratively updated per 10CFR50.71(e) to update and clarify the material description for the sump strainers per SFS-CP-GA-01.

Section 6.3.2.2.10 is updated for changes to the sump design and licensing basis to reflect the results of the mechanistic analysis requested in Generic Letter 2004-02. Section 6.3.2.2.10 is revised to reflect the NPSH analysis for RHR in ME(B)-325 R3.

Table 6.1B-1 is administratively updated per 10CFR50.71(e) to update the quantities of organic materials (cables and oil) from ACTN-MAN-2001-002201-72 and EE-CA-0000-5329.

LDCR-SA-2008-17, EVAL-2008-003064-02 (RAS):

Last sentence in section 6.4.5 incorrectly refers to the "Control Room Envelope Habitability Program" as the "Control Room Integrity Program". This change revises the sentence to use the correct program title.

LDCR-SA-2008-16, EVAL-2008-003163-02 (JCH):

Page 10.2-17 of Section 10.2.3.1, Materials Selection:

Changed compressive residual stress level from " shall not exceed 11,500 psi" to "shall not be lower than 14,504 psi (100 Mpa) and higher than 36,260 psi (250 Mpa) in the middle of the hub area; shall not be lower than 29,007 psi (200 Mpa) and higher than 58, 015 psi (Mpa) in the area of the rim of the hub; shall not be lower than 14, 504 psi (100 Mpa) and higher than 36,260 psi (250 Mpa) in the rim area.

Deleted " For up to 120 percent of rated speed the maximum disk stress at the shrink fit is less than half the burst strength of the material" and inserted " The maximum disk stress at the shrink fit is required to remain less than 50-60% of the yield strength of the material at nominal conditions."

Justification: New advanced design low pressure turbines were installed in Units 1 and 2. Advances in manufacturing techniques and material property selection have resulted in revision to the residual stess criteria. The new advanced design low pressure turbines incorporate disc which have these new advanced manufacturing techniques and new material compositions.

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LDCR-SA-2008-1, EVAL-2007-002946-02 (JDS):

FSAR changed to state, "Due to the sensitive nature of the seals, seal injection flow should be maintained when the Reactor Coolant System is not completely depressurized or when the Reactor Coolant System inventory is moving through the seal package." During outages seal injection flow has been secured and the RCPs are placed on their backseat which fulfills the intent of this section by isolating the RCP seals from the RCS. With RCPs not on a backseat during refueling, the FSAR was clarified to explain under what circumstances seal injection flow needs to be maintained to assure the RCP seal cleanliness.

LDCR-SA-2008-2, EVAL-2006-003828-04 (GLM):

Due to degradation, non-safety vent chiller X-08 (CPX-CHCICE-08) is being replaced. The replacement is a York YK model higher pressure centrifigul liquid type ventilation chiller using R134a refrigerant. Since this chiller is a direct open versus hermetically sealed compressor a change to Section 9.4E.2 of the FSAR is necessary to reflect the modification changes in FDA-2006-3828-06.

Per FSAR Appendix 9.4E.1, the design function of the plant ventilation chilled water system is to remove heat rejected by plant equipment and to maintain ambient temperature below design limits within the areas it serves. The system is shown on Figure 9.4-11. The plant ventilation chilled water system is not required to mitigate the consequences of an accident or to maintain the plant in a safe shutdown condition. Except for the containment penetrations and containment isolation valves, the plant ventilation chilled water system is non-nuclear safety and non-seismic. By replacing chiller X-08 with a similar 660-ton chiller, the degraded system cooling capacity and efficiency is regained. The new chiller essentially provides the 660 ton desired cooling. This activity, therefore, improves the reliability of the ventilation chilled water system and has no adverse impact on the design function of the system.

LDCR-SA-2009-13, EVAL-2008-000762-02 (GLM):

In the 4th sentence from the bottom, change "Revision 1" to "Revision 0" for NUREG-700 (1981).

This is an editorial change. NUREG-0700, Revision 0 was issued in September 1981 and NUREG-0700, Revision 1 was issued in June 1996.

LDCR-SA-2009-8, EVAL-2009-000832-01 (JCH):

Section 10.2.2.10, Turbine Trips

Page 10.2-15: Delete the following: "The primary water pump shaft vibration trip shown in figure 10.2-1 is allowed to be defeated by the use of a time delay."

Justification: These trips are no longer automatic trips. They are manual. The automatic trips were removed with the installation of the new Turbine Protection System via FDA-2004-000773-01 (Unit 1) and FDA-2004-000773-02 (Unit 2).

Figure 10.2-1, "Turbine Trip Logic Diagram", needs to be revised to remove the logic for the following trips:

-Turb Gen Bearing Vibration (ETP)

-P.W. Pump Shaft Vibration

LDCR-SA-2009-8, EVAL-2009-000832-01 (JCH) (continued):

Justification: These trips are no longer automatic trips. They are manual. The automatic trips were removed with the installation of the new Turbine Protection System via FDA-2004-000773-01 (Unit 1) and FDA-2004-000773-02 (Unit 2).

LDCR-SA-2008-22, EVAL-2008-002039-02 (SCD):

Removes references to TLD readers from section 12.5.2.1. Changes Thermoluminescent dosimetry (TLD) to Optically Stimulated Luminescence (OSL) Badges, replaces TLD with OSL and replaces reference to onsite TLD reading with "The OSL Badges are processed and a definitive dose evaluation is provided by the approved service organization as requested." in section 12.5.2.5. Section 12.5.3.5 is changed to replace Thermoluminescent dosimetry (TLD) to Optically Stimulated Luminescence (OSL) Badges, replaces TLD with OSL and remove reference to onsite dosimetry reading. Requirement for offsite dosimetry service organization to maintain accreditation through the National Voluntary Laboratory Accreditation Program (NVLAP) is incoporated. Table 12.5-1 is changed to delete TLD Readers and Beta, Gamma, and Neutron Used for personnel dosimetry program.

LDCR-SA-2009-2, EVAL-2008-002473-02 (TJD):

Description of Change: Delete "the safety evaluations for" from FSAR section 17.2.1.1.3.1(2), which currently reads: "Review of the safety evaluations for: (1) procedures, (2) change to procedures,equipment, systems or facilities, and (3) tests or experiments where nuclear safety is adversely affected;"

Technical Justification: This change was originally approved as a part of LDCR SA-2000-23, but the current FSAR (Amendment 102) does not reflect the change. From a review of the current and previous FSAR versions, it appears that this portion of LDCR SA-2000-23 was inadvertently not incorporated into the FSAR. (Ref. correspondence, CPSES-200101291 and CPSES-200100873, for information on LDCR SA-2000-23.)

LDCR-SA-2009-6, EVAL-2004-001688-02 (JDS):

Document the use of the RHR pumps to fill the SI accumulators. The previously accepted method to fill and add water to the accumulators is by using the SI pump through the SI test header. When the RCS is less than 350 degrees F, TS 3.4.12 requires that the SI pumps be not "capable of injecting into the RCS." To accomplish accumulator fill with the SI pumps and meet the not "capable of injecting into the RCS" requires 40 fuses to be pulled to remove power to the valves on the SI test header. This method adds unnecessary challenges to equipment reliability (fuse clips) and increases the likelihood of human errors. Pulling 40 fuses is also manpower intensive, particularly during outage periods. Therefore, a method to fill and add water to the accumulator using RHR was implemented.

LDCR-SA-2008-10, EVAL-2007-002677-04 (JCH):

Change Section 5.3.1.5 by adding "and 5.3.4-C" to first paragraph on page 5.3-5.

Justification: Changes reflect the use of a hydraulic nut closure system. The HydraNut is a hydraulically actuated, self-tensioning nut. The result of the installation will be a set of HydraNuts that tension the reactor vessel studs in one operation using hydraulic pressure, as opposed to several passes using stud tensioners currently required for the conventional nuts. As a result, the HydraNut will significantly decrease the time required to tension the reactor vessel studs.

Changed section 5.3.1.7 first paragraph to add the following statement: "Conventional nuts and washers were originally supplied with the reactor vessel. Hydraulic nuts were procured to be used with the reactor vessel studs as appropriate."

Added reference to Table 5.3-4C at the end of paragraph 2.

Justification: Changes reflect the use of a hydraulic nut closure system. The HydraNut is a hydraulically actuated, self-tensioning nut. The result of the installation will be a set of HydraNuts that tension the reactor vessel studs in one operation using hydraulic pressure, as opposed to several passes using stud tensioners currently required for the conventional nuts. As a result, the HydraNut will significantly decrease the time required to tension the reactor vessel studs.

Added new Table 5.3-4C, Unit 1 and 2 Reactor Vessel Closure Bolting Material Properties - Hydranuts & Washers.

Justification: Changes reflect the use of a hydraulic nut closure system. The HydraNut is a hydraulically actuated, self-tensioning nut. The result of the installation will be a set of HydraNuts that tension the reactor vessel studs in one operation using hydraulic pressure, as opposed to several passes using stud tensioners currently required for the conventional nuts. As a result, the HydraNut will significantly decrease the time required to tension the reactor vessel studs.

LDCR-SA-2009-11, EVAL-2009-001859-02 (JDS):

Delete inserts and adaptors from under Closure studs, nuts, washers due to implementation of reactor vessel hydraulic nuts (HydraNuts).

Delete as modified by Code Case 1605 from under SA-540 Class-3 B24. Code case 1605 is not applicable to the Comanche Peak Unit 1 and 2 reactor vessel studs, nuts and washers. The chemistry requirements and increased material size limit for SA-540 Grade B24 bolting material referenced in Code Case 1605 does not apply to the Comanche Peak reactor vessel studs, nuts and washers. The Code Case is also not applicable to the new HydraNut reactor vessel modification.

LDCR-SA-2009-11, EVAL-2009-001859-02 (JDS) (continued):

Delete Code Case 1605. Code case,1605 is not applicable to the Comanche Peak Unit 1 and 2 reactor vessel studs, nuts and washers. The chemistry requirements and increased material size limit for SA-540 Grade B24 bolting material referenced in Code Case 1605 does not apply to the Comanche Peak reactor vessel studs, nuts and washers. The Code Case is also not applicable to the new HydraNut reactor vessel modification.

LDCR-SA-2008-23, EVAL-2008-000624-02 (TJD):

The following sentence is being deleted from FSAR, section 17.2.17, at the top of page 17.2-26: "The Nuclear Overview Department reviews and approves the administrative control procedures and instructions and the retention periods assigned for quality assurance records." This change will make section 17.2.17 consistent with FSAR Section 17.2. Also, FSAR section 17.2 pages 17-ii and 17.2-29 are being revised to reflect current organizational titles.

The review of procedures for quality assurance requirements is now accomplished via the SORC process, as identified in FSAR Section 17.2.2. The change to the SORC process was a reduction in commitment which was reviewed and approved by the NRC via LDCR SA-89-243, therefore; LDCR SA-2008-023 does not constitute a reduction in commitment. Commitment 22910 addresses this review process. Additionally, ACTN-MAN-2008-000624-02 has been created to delete commitment 22906, which discusses the pre-SORC review process and should have been deleted when the current review process was approved by the NRC.

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LDCR-SA-2004-26, EVAL-2003-000239-03 (SCD):

EVAL-2003-000239-03 (LDCR-SA-2004-026): Corrects note reference for the Heater Drain Tank Equalization Header from "**" to "*". This is an editorial change.

EVAL-2003-000239-03 (LDCR-SA-2004-026): Adds Note 1 to Table 10.4-20 to describe the use of the 2 pH sample points.

EVAL-2003-000239-03 (LDCR-SA-2004-026): Delete dissolved oxygen and pH monitoring from various sample points as identified in FSAR Table 10.4-20. This activity provides new and enhanced models of the existing oxygen and pH analyzers. The Secondary Chemistry Control Program monitoring capabilities will continue to meet the EPRI "PWR Secondary Water Chemistry Guidelines."

LDCR-SA-2009-5, EVAL-2009-000834-02 (CBC):

TMI Section I.A.1.3, "Shift Manning," is updated to reflect the work hour requirements prescribed by 10 CFR Part 26, Subpart I, effective 10/01/2009. Work hour restrictions are being deleted from Technical Specification 5.2.2.d (License Amendment Request 09-006) to support compliance with 10 CFR Part 26. This change will make the FSAR consistent with the Technical Specifications and the new rule. (EVAL-2009-000834-01).

LDCR-SA-2009-7, EVAL-2008-002474-05 (TJD):

Revise the title of FSAR, section 17.2.7.1, from "Receipt Inspection" to "Receipt Inspection and Material Testing."

This change will clearly define the role of the Procurement Quality Assurance Receipt Inspection as "Receipt Inspection and Material Testing."

Revise the first sentence in FSAR, section 17.2.7.1, from "Receipt inspections at CPSES are . . ." to "Receipt inspection and material testing at CPNPP is . . ."

This change will clearly define the role of the Procurement Quality Assurance Receipt Inspection as "Receipt Inspection and Material Testing."

Change FSAR 17.2.7.1, item 2, from CPSES to CPNPP.

Update plant name.

Change FSAR, section 17.2.10, second sentence, from ". . . and 1.116 as . . ." to ". . . 1.116, and 1.123 as . . ."

This change updates the FSAR to include a Regulatory Guide (1.123) that we are currently commited to follow.

Update two places on FSAR page 17.2-20 from CPSES to CPNPP.

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LDCR-SA-2009-7, EVAL-2008-002474-05 (TJD) (continued):

Update plant name.

Page 17.2-20, Change the wording in the next to last paragraph from ". . . level III inspector . . ." to ". . . Level II or III inspector . . ."

Revise the FSAR to allow a Level II inspector to assign hold points and ensure that inspection methods are adequate.

LDCR-SA-2009-17, EVAL-2007-001302-01 (TJD):

Revise the Discussion section for Regulatory Guide 1.33 to include an additional exception for commercial grade calibrations.

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

Revise the Discussion section for Regulatory Guide 1.123 to include an additional exception for commercial grade calibrations.

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

Revise the Discussion section for Regulatory Guide 1.144 to include an additional exception for NVLAP or A2LA accredited commercial grade calibrations.

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

Revise FSAR section 17.2.4 to describe the procurement document control requirements needed to support the use of NVLAP or A2LA accredited commercial grade calibration services.

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

Revise FSAR section 17.2.7 to describe changes to the control of purchased safety-related material, equipment and services needed to support the use of NVLAP or A2LA accredited commercial grade calibration services.

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

Revise FSAR section 17.2.18.1, "Audit Organizations," to explain the audit implications when using NVLAP or A2LA accredited commercial grade calibration services.

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LDCR-SA-2009-10, EVAL-2005-000768-02 (TJD):

This change supports a new option for NVLAP or A2LA accredidation of commercial grade calibration services in lieu of supplier audit by licensee.

FSAR Table 2.5.6-7, Station 46+17, the Type should be changed from "Pneumatic" to "Pneumatic (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 55+95, the Type should be changed from "Pneumatic" to "Pneumatic (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 56+05, the Type should be changed from "Pneumatic" to "Wellpoint (Replaced)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 56+05, the Piezometer No. should be changed from "P-II-1a" to "P-II-1" The "a" needs to be removed from the piezometer number.

This change corrects the SCR dam piezometer number.

FSAR Table 2.5.6-7, Station 56+02, the Type should be changed from "Wellpoint" to "Wellpoint (Replaced)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 56+22, the Type should be changed from "Wellpoint" to "Wellpoint (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 56+92, the Type should be changed from "Wellpoint" to "Wellpoint (Replaced)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 56+06, the Type should be changed from "Pneumatic" to "Pneumatic (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 69+00, the Type should be changed from "Wellpoint" to "Wellpoint (Replaced)"

LDCR-SA-2009-10, EVAL-2005-000768-02 (TJD) (continued):

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 69+05, the Type should be changed from "Wellpoint" to "Wellpoint (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 69+15, the Type should be changed from "Wellpoint" to "Wellpoint (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 69+09, the Type should be changed from "Pneumatic" to "Pneumatic (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Table 2.5.6-7, Station 69+20, the Type should be changed from "Pneumatic" to "Pneumatic (Abandoned)"

This change updates the table to reflect changes to the SCR dam piezometers.

FSAR Figure 2.5.6-56 is updated to reflect the same piezometer changes that were made to FSAR Table 2.5.6-7, as detailed in the previous 13 records of this database.

This change updates Figure 2.5.6-56 to reflect changes to the SCR dam piezometers.

LDCR-SA-2009-16, EVAL-2009-003424-01 (TJEW):

On Table 8.3-3, sheet 1, for items:

3A, revise the words, "loss of power to buses 1EA1 and 1EA2 from XST1"

to read, "For open circuit: Loss of alternate power to buses 1EA1 and 1EA2 from XST1.

For short circuit: Loss of power to buses 2EA1, 2EA2, and loss of alternate power source for buses 1EA1 and 1EA2."

3B, revise the words, "loss of power to buses 2EA1 and 2EA2 from XST1"

to read, "For open circuit: Loss of power to buses 2EA1 and 2EA2 from XST1.

LDCR-SA-2009-16, EVAL-2009-003424-01 (TJEW) (continued):

For short circuit: Loss of power to buses 2EA1, 2EA2, and loss of alternate power source for buses 1EA1 and 1EA2."

On Table 8.3-3, sheet 2, for items:

4A, revise the words, "loss of power to buses 1EA1 and 1EA2 from XST2"

to read, "For open circuit: Loss of power to buses 1EA1 and 1EA2 from XST2.

For short circuit: Loss of power to buses 1EA1, 1EA2, and loss of alternate power source for buses 2EA1 and 2EA2."

4B, revise the words, "loss of power to buses 2EA1 and 2EA2 from XST2"

to read, "For open circuit: Loss of alternate power to buses 2EA1 and 2EA2 from XST2.

For short circuit: Loss of power to buses 1EA1, 1EA2, and loss of alternate power source for buses 2EA1 and 2EA2."

LDCR-SA-2009-3, CR-EVAL-2009-000413-01 (SCD):

Changes to CPNPP Executive Management assignments and Maintenance Department reorganization.

LDCR-SA-2009-23, EVAL-2006-003201-02 (TJEW):

The revised duty cycle load of the batteries envelops the calculated duty cycle load of the batteries and will maintain the end of the duty cycle battery voltage of 1.75 VPC = battery voltage of 105V, as required by Section 4.1.2.1.B of DBD-EE-044 Rev 24 and FSAR section 8.3.2.1 item "2" Capacity sub item "a" Batteries. Ref calculation EE-CA-0000-5454 Rev 0. The listing of enveloping duty cycle load capability of battery in Table 8.3-4 will minimize unnecessary FSAR changes when a battery load changes but remain bounded by the enveloping load. 59SC-2006-003201-01-00 provide the basis that a 50.59 evaluation is not required for this change.

FSAR Table 8.3-4 is revised to list the duty cycle load of batteries which envelop the batteries calculated load and maintains end of the duty cycle voltage of 105V.

FSAR Amendment 103

LDCR-SA-2008-11, CR-EV-2006-003080-00-150 (JCH):

Table 10.4-10: Maximum System Heat Sink Requirements

1.Exciter air coolers: The turbine generator exciter air coolers currently have a heat load of 2.143x10^6 Btu/hr, this will be increased to 2.184x10^6 Btu/hr for SPU. Additionally, the exciter air coolers currently have a cooling water flow rate of 790 gpm, this will be increased to 792.5 gpm for SPU.

LDCR-SA-2008-11, CR-EV-2006-003080-00-150 (JCH) (continued):

2.Hydrogen coolers: The turbine generator Hydrogen coolers currently have a heat load of 17.02x10^6 Btu/hr, this will be decreased to 16.79x10^6 Btu/hr for SPU. Additionally, the Hydrogen coolers have a cooling water flow rate of 4480 gpm, this will be decreased to 4403 gpm for SPU.

3. Isophase forced cooling unit: The isophase bus forced cooling units currently have a heat load of 1.03x10^6 Btu/hr, this will be increased to 1.808x10^6 Btu/hr for SPU. Additionally, the isophase forced cooling units have a cooling water flow rate of 137 gpm, this will be increased to 220 gpm for SPU.

4. Primary water coolers: The turbine generator primary water coolers currently have a heat load of 31.17x10^6 Btu/hr, this will be increased to 32.70x10^6 Btu/hr for SPU. Additionally, the primary water coolers have a cooling water flow rate of 5720 gpm, this will be increased to 7115 gpm for SPU. To accomplish this five of six cooler will be placed in operation for SPU, currently four of six coolers are in operation.

Justification:

Heat load, flow, and pressure requirements for the exciter air coolers, Hydrogen coolers, and primary water coolers can be found in Vendor Document EC-06153 "Thermal Uprate Study 4.5% for TXU Comanche Peak Units 1 and 2" (Ref ENR-2008-003080-12). Heat load for these coolers is calculated using conservative assumptions, each cooler has its own set of conservative conditions.

Heat load and flow requirements for the isophase bus forced cooling units can be found in Section 3.13.1 of ER-SPU-4.2.6 "Turbine Plant Cooling Water System" (Ref ENR-2008-003080-39).

These changes are required to maintain configuration control.

Section 10.2.2.6.1: Hydrogen Storage and Supply

Hydrogen pressure will be increased to 65 psig for SPU, currently pressure is 60 psig.

LDCR-SA-2008-11, CR-EV-2006-003080-00-150 (JCH) (continued):

Justification:

Heat load, flow, and pressure requirements for the exciter air coolers, Hydrogen coolers, and primary water coolers can be found in Vendor Document EC-06153 "Thermal Uprate Study 4.5% for TXU Comanche Peak Units 1 and 2" (Ref ENR-2008-003080-12). Heat load for these coolers is calculated using conservative assumptions, each cooler has its own set of conservative conditions.

Heat load and flow requirements for the isophase bus forced cooling units can be found in Section 3.13.1 of ER-SPU-4.2.6 "Turbine Plant Cooling Water System" (Ref ENR-2008-003080-39).

These changes are required to maintain configuration control.

Section 10.4.12.1: Design Basis

The TPCW system currently has a maximum operating heat exchanger outlet temperature of 105F, this will be increased to 107F for SPU. Currently the circulating water temperature that flows through the tube side of the TPCW heat exchanger is 95F, this will be increased to 102F post-SPU.

Justification:

Heat load, flow, and pressure requirements for the exciter air coolers, Hydrogen coolers, and primary water coolers can be found in Vendor Document EC-06153 "Thermal Uprate Study 4.5% for TXU Comanche Peak Units 1 and 2" (Ref ENR-2008-003080-12). Heat load for these coolers is calculated using conservative assumptions, each cooler has its own set of conservative conditions.

Heat load and flow requirements for the isophase bus forced cooling units can be found in Section 3.13.1 of ER-SPU-4.2.6 "Turbine Plant Cooling Water System" (Ref ENR-2008-003080-39).

These changes are required to maintain configuration control.

LDCR-SA-2008-12, CR-EV-2006-003080-00-103 (JCH):

Section 10.4.10.5 : remove the following "Heater drain tank control valves to heaters 3A and 3B are interlocked to close when the corresponding extraction steam motorized stop valve is colsed. Note however that Operations may set vent valves in the OPEN position by pulling their fuses. The Heater Drain System Operating Procedures will control the valves configuration."

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LDCR-SA-2008-12, CR-EV-2006-003080-00-103 (JCH) (continued):

Justification: The 3A and 3B heater drain tank control vent valves are being removed and replaced with straight pipe to allow free flow of steam from the heater drain tanks to the 3A and 3B heaters. This will allow the heater drain system to vent with less restriction and better respond to plant trainsients after the power uprate. The valves are currently locked open and act as an oriface or restriction in the line impeeding proper steam vent flow.

Section 10.4.11.5: remove the following "Heater drain tank control vent valves to heater 3A and 3B are interlocked to close when the corresponding extraction steam motorized stop valve is closed. Note the vent valves may be disabled in the open position by removing the control fuses and/or isolating the positioner instrument air supply and restraining the valve handwheels. The vent globe valves are air operated and fail open"

Justification: The 3A and 3B heater drain tank control vent valves are being removed and replaced with straight pipe to allow free flow of steam from the heater drain tanks to the 3A and 3B heaters. This will allow the heater drain system to vent with less restriction and better respond to plant trainsients after the power uprate. The valves are currently locked open and act as an oriface or restriction in the line impeeding proper steam vent flow.

LDCR-SA-2008-20, CR-EV-2007-002129-00-4 (JCH):

Section 13.3B.1.1 Plant Manager:

Revised Plant Manager responsibilities to be maintenance and implementation of the Fire Protection Progrram, not development.

Justification: This change is required due to various personnel and organizational changes as CPNPP. This change is administrarive in nature and does not change the Fire Protection Program requirements at CPNPP.

13.3B.1.2 Director, Maintenance:

Director, Maintenance responsible for maintenance and implementation of the Fire Protection Program, not implementation.

The Manager, Technical Support does not report to him so the reference to him was deleted.

13.B.1.3 Director, Site Engineering:

Added new section for Director Site, Engineering who is responsible for the development of the Fire Protection Program. This responsibility is assigned to the Manager, Technical Suport.

LDCR-SA-2008-20, CR-EV-2007-002129-00-4 (JCH) (continued):

13.B.1.4 Manager, Technical Support:

Manager, Technical Support is responsible to the Director, Site Engineering instead of the Director, Maintenance.

Added the responsibility for "review and apporval of Fire Protection Design documents, licensing documents, and Station Procedures including revisions, to verity technical accuracy and regulatory compliance.

Added the following paragraph- "The Manager, System Engineering is responsible to the Director, Site Engineering for technical support of the Fire Protection Program in relation to the review and implementation of design modifications and the resolution of system issues.

13.3B.1.5 Maintenance Team Manager:

Deleted "Smart" from Maintenance Team Manager title.

Added responsibility "to assist the Nuclear Training Manager in the development and implementation of fire protection training programs for personnel and the fire brigade at CPNPP.

Justification: This change is required due to various personnel and organizational changes as CPNPP. This change is administrarive in nature and does not change the Fire Protection Program requirements at CPNPP.

LDCR-SA-2008-13, CR-EV-2006-003080-94 (JDS):

Incorporate Westinghouse Topical Reports associated with conversion to the Westinghouse transient and accident analysis methodologies.

Include relevant information resulting from the removal of Appendix 4.A and 4.B into this section for consistency with other Westinghouse plants using the Westinghouse transient and accident analysis methodologies.

Remove Appendix 4.A and incorporate the relevant information into chapters 4.1, 4.2, 4.3, and 4.4 for consistency with other Westinghouse plants using the Westinghouse transient and accident analysis methodologies.

Update Chapter 6.2 to reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Insert Tables to reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

LDCR-SA-2008-13, CR-EV-2006-003080-94 (JDS) (continued):

Replace Tables to reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Remove Table as table does not reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Modify Table to reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Delete Tables which does not reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Insert Figures to reflect GOTHIC code analysis provided to the NRC for containment accident conditions resulting from 4.5% power uprate of Units 1 and 2.

Describe Power Distribution Monitoring System Using BEACON.

Update time required for boric acid solution to counteract xenon decay when charging to the RCS is via reactor coolant

pump seal injection at the rate of approximately 5 gpm per pump from approximately 3.5 hours to 6 hours. This is consistent with the Westinghouse Methodology.

Revise this section for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Revise these Tables for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Replace this figure for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Insert these Figures for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Revise these sections for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Insert these Tables for consistency with Westinghouse transient and accident analysis methodologies now used on Units 1 and 2.

Update Table on Reactor Design Comparison for consistency with uprated conditions of Units 1 and 2.

LDCR-SA-2008-13, CR-EV-2006-003080-94 (JDS) (continued):

Update Table on Analytical Techniques in reactor design for consistency with Westinghouse Methodologies used.

Insert Table on Thermal and Hydraulic Comparison for pre- and post- uprate conditions of Units 1 and 2.

LDCR-SA-2009-19, CR-EV-2009-000214-11 (JDS):

Update Chapter 9 to modify the time required to add enough boric acid solution to bring the plant to hot shutdown and counteract xenon decay if normal charging is unavailable.

Update Chapter 4 to include a description of the Westinghouse Integrated Nozzle (WIN) top nozzle design. The mechanical design evaluations idocument that the new fuel assemblies are compatible with existing fuel, reactor vessel internals, spent fuel racks, fuel handling equipment and RCCAs and that all structural integrity and seismic requirements are satisfied.

LDCR-SA-2007-10, CR-EV-2006-003080-202 (JDS):

Revise the Main Generator rating from 1350 MV to 1410 MVA for each Unit in the first sentence of the second paragraph insection 1.2.2.5.

Revised the titles for Figures 10.1-1 and 10.1-2 as follows:

10.1-1 Unit 1, 100% Power Heat Balance (See VL-07-001239)

10.1-2 Unit 2, 100% Power Heat Balance (See VL-07-001240)

The existing figures are replaced with new heat balance drawings supplied by Siemens. These figures contain information proprietary to Siemens. Consequently, no figures will be included but the List of Figures is revised to refer to the vendor control number for the drawings which can be accessed via the plant record system.

Revised the sixth paragraph in section 10.1 as follows:

"Heat balances reflecting the design at an NSSS rated thermal power of 3628 MWt (maximum guaranteed rating) are shown on Figures 10.1-1 and 10.1-2, respectively. The rating of the electric generator is 1,410,000 kVA, 60 Hz, 0.90 power factor."

Replace the 2nd and 3rd paragraphs with the following:

"The Unit 1 turbine generator is designed to produce 1263.4 MWe when operating at 14,774,975 lb/hr saturated steam at 980 psia and 1192.3 Btu/lb with 0.21% moisture at the throttle and 1.414 in Hg absolute at the turbine exhaust. The Unit 2 turbine generator is idesigned to produce 1253.4 MWe when operating at 14,715,715 lb/hr saturated steam

LDCR-SA-2007-10, CR-EV-2006-003080-202 (JDS) (continued):

at 948.9 psia and 1192.1 Btu/lb with 0.40% moisture at the throttle and 1.414 in Hg absolute at the turbine exhaust."

Revise the main generator kVA rating in the first sentence in section 10.2.2.6 by replacing "1,350,000" with "1,410,000" and deleting "and a short circuit ratio of 0.58."

In section 10.2.2.6.1, reevise the maximum pressure in the second sentence from "60" psig to "75" psig.

In section 10.4.1.1.1, revise the value for:

Exhaust steam from "8 x10E6" to "8.44x10E5"

Condenser duty from "7.7x10E9" to "8.06x10E9"

Condenser pressure at turbine exhaust from "3.5" to "3.55"

Outlet temperature form "110" to "110.8".

In section 10.4.5.1, in the third sentence, revise the value for total heat removed from "8.4X10E9" to "8.8x10E9" and amount removed from the main cindenser from "8.0x10E9" to "8.4x10E9".

Delete Figures 10.1-1 and 10.1-2 for Unit 1 100% Power Heat Balance and Unit 2 100% Power Heat Balance

Replace existing Table 10.1-1 with new Table reflecting new values for Maximum Guaranteed Rating for Unit1 and for Unit 2. Delete the column for "Valves Wide Open" values.

LDCR-SA-2004-18, EVAL-2003-000167-02 (JDS):

This Activity installs neutron dosimetry sensors used to monitor neutron exposure for continued surveillance of reactor vessel integrity. The equipment is installed in the reactor cavity annulus in all four quadrants utilizing the neutron detector well ventilation ducts for support.

By installing the Ex-Vessel Dosimetry, an alternative dosimetry program is instituted to monitor reactor vessel neutron exposure through the license renewal period. Therefore, CPSES is following the guidance for reactor vessel surveillance as stated in the NUREG-1801, "Generic Aging Lessons Learned (GALL) Report," April 2001.

FSAR Amendment 103a

LDCR-SA-2009-12, EV-CR-2008-003459-00-11 (SCD):

Amend UFSAR Section 6.2.4.1.3.1 to change the containment isolation valve typical arrangements applied to the high head safety injection lines and to describe that pressure indication is provided to monitor ECCS piping pressure.

Add Section 6.3.5.2.6 and REFERENCES to the Table Of Contents, add Section 6.3.5.2.6 to describe CPNPP response to GL 2008-01, describe the new ECCS piping pressure monitoring function and add Reference section with GL 2008-01 as a reference.

Amend UFSAR Figure 6.2.4-1 Valve Arrangement 9 to depict the locked closed manual containment isolation valve and downstream pressure transmitter.

Amend UFSAR Figure 6.2.4-1 to add new Valve Arrangement 46.

Amend UFSAR Section 5.2.5.2.3 to describe that pressure indication is provided to monitor ECCS piping pressure.

LDCR-SA-2008-19, EV-CR-2007-002125-00-2 (JCH):

13.3.B.5 QUALITY ASSURANCE PROGRAM

1. Design Control and Procurement Document Control

Changed first paragraph to read as follows:

"Measures will be established to assure that NRC applicable guidelines or NRC approved alternatives are included in design and procurement documents and that any deviations therefrom are controlled. There procedures include provisions, as necessary, to ensure that:"

4. Inspection

Changed to read as follows:

"A program for independent inspection of activities affecting fire protection will be established and executed by, or for, the organization performing the activity to verify conformance with documented installation drawings and test procedures for accomplishing the activity."

Justification:

The proposed changes revise the language in FSAR 13.3B.5 to reflect the guidance language in Appendix A to BTP APCSB 9.5-1. CPSES is committed to the meet the requirements of Appendix A to BTP APCSB 9.5-1 as stated in NRC SSERs 1, 12, and 21.

The changes will not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

LDCR-SA-2008-19, EV-CR-2007-002125-00-2 (JCH) (continued):

13.3.B.5 QUALITY ASSURANCE PROGRAM

5. Test and Test Control

Added the following sentence:

"Test results will be properly evaluated and acted on.

7. Nonconforming Items.

Changed to read as follows:

"Measures are established to control items that do not conform to specified requirements to prevent inadvertent use of installation."

8. Corrective Action

Changed to read as follows:

"Measures will be established to ensure that conditions adverse to fire protection such as failures, malfunctions, deficiencies or deviations, defective components, uncontrolled combustible material and non-conformances are promptly identified, reported, and corrected."

10. Evaluations

Changed heading to "Audits".

Changed to read:

"Audits will be conducted and documented to verify compliance with the fire protection program, including design and procurement documents, instructions, procedures and drawings, and inspection and test activities. These audits will be performed by personnel not having direct responsibility for the Fire Protection Program."

Justification:

The proposed changes revise the language in FSAR 13.3B.5 to reflect the guidance language in Appendix A to BTP APCSB 9.5-1. CPSES is committed to the meet the requirements of Appendix A to BTP APCSB 9.5-1 as stated in NRC SSERs 1, 12, and 21.

The changes will not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

FSAR Amendment 103a

LDCR-SA-2010-2, EV-CR-2005-003122-00-4 (JCH):

6. Inspection, Test and Operating Status.

Change to read "Measures will be established to provide for the identification of items that have satisfactorily passed required tests and inspections.

Justification:

FSAR Section 13.3B.5.6 should reflect what is stated in C.6 of Appendix A to BTP APCSB 9-5.1 and should not contain the language related to tagging of non-conformances. Non-conformances are discussed in FSAR 13.3B5.7. As presently worded this section can lead to unintended and overly restrictive requirements placed on the Fire Protection Program. This change will not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

LDCR-SA-2009-24, EV-CR-2006-004121-00-1 (JDS):

Add a paragraph to document the use of the demineralized water storage tank and transfer pump to manually makeup to the containment spray pump suction piping located downstream of the containment sump isolation valves. The change allows for the fill of a section of containment spray pump suction piping with demineralized water, rather than borated water. This method is also used to fill the containment spray risers in containment.

LDCR-SA-2010-11, EV-CR-2010-001826-00-5(TJD):

Delete all the text in the Discussion section of Regulatory Guide 1.16, and replace it with the following: "This regulatory guide was withdrawn on August 11, 2009."

Technical Justification: Regulatory Guide 1.16 was withdrawn by the NRC on August 11, 2009 (Ref. Federal Register, 74 FR 40244, 8/11/2009).

Delete the following sentence from the CPSES Response portion of FSAR section II.K.3.3, "All challenges to SV or RV shall be documented in the monthly operating report." Then replace the deleted sentence with the following: "The reporting of SV & RV challenges was moved from the annual report to the monthly operating report (TS 6.9.1.5), then License Amendment 64 deleted this reporting requirement altogether."

Technical Justification: License Amendment 64 was implemented in July 1999 and deleted TS 6.9.1.5, including the requirement to report challenges to PORVs and safety valves, which was being reported in the annual operating report.

FSAR Amendment 103a

LDCR-SA-2009-25, EV-CR-2007-002527-00-2 (TJEW):

Changes to the AAP are controlled by the site facility program (reference STA-716) and are not required to be tracked as plant equipment. Therefore, the current AAP configuration will not be tracked on plant drawing, and drawing M1-0313 Sht A Sht - will be retired. This activity has no adverse impact on the design function of any SSC because the drawings that are being retired reflect a component was not implemented in the field and have inaccurate depictions of the current AAP status.

Delete the reference to M1-0310-A on Sheet 12 of Table 3.2-3

Changes to the AAP are controlled by the site facility program (reference STA-716) and are not required to be tracked as plant equipment. Therefore, the current AAP configuration will not be tracked on plant drawing, and drawing M1-0313 Sht A Sht - will be retired. This activity has no adverse impact on the design function of any SSC because the drawings that are being retired reflect a component was not implemented in the field and have inaccurate depictions of the current AAP status.

Delete the reference to M1-0310-A on Sheet 10 of Table 3.2-4

Changes to the AAP are controlled by the site facility program (reference STA-716) and are not required to be tracked as plant equipment. Therefore, the current AAP configuration will not be tracked on plant drawing, and drawing M1-0313 Sht A Sht - will be retired. This activity has no adverse impact on the design function of any SSC because the drawings that are being retired reflect a component was not implemented in the field and have inaccurate depictions of the current AAP status.

Delete the reference to M1-0310-A on page 9-xv

LDCR-SA-2009-21, EV-CR-2009-000214-00-8 (JDS):

The misloaded assembly analysis methodology described in the FSAR is being changed from the Luminant Power methods detailed in topical report RXE-91-002 to the original Westinghouse methodology previously reviewed and approved for Comanche Peak by the NRC (NUREG-0797, Section 15.3.3). This change is necessary to fully complete the transition to the Westinghouse accident analysis methodology. An improved description of the original Westinghouse methodology has been recently provided in WCAP-16676-NP which is also referenced as part of the adoption of the misloaded assembly analysis methodology. The improved description (WCAP-16676-NP) provides additional clarifying details regarding the original methodology but does not modify or alter the method relative to the original Westinghouse methodology approved by the NRC for Comanche Peak (NUREG-0797, Section 15.3.3). The specific application of the methodology is to analyze reactor operation with a fuel assembly inadvertently loaded into an improper location. Luminant Power is using the original analysis methodology approved for Comanche Peak in the specific application as approved by the NRC. The SER did not require any additional restrictions or limitations beyond those described in the methodology.

Update table of contents to reflect revised section.

FSAR Amendment 103a

LDCR-SA-2010-10, EV-CR-2009-000465-00-3 (JDS):

Recently, Westinghouse determined that existing Rod Withdrawal at Power (RWAP) RCS overpressure analyses do not cover the full range of reactor power operations for all Westinghouse-designed PWRs (W-PWRs).

Westinghouse performed generic RWAP analyses to demonstrate that the peak RCS pressure criterion was met for most W-PWRs equipped with a positive flux rate reactor trip (PFRT).

Additionally, an evaluation was performed against a generic analysis for Comanche Peak Units 1 and 2. The evaluation against the generic RWAP overpressure analyses concluded that credit for the PFRT was sufficient to demonstrate compliance with the RCS overpressure criterion. Westinghouse has performed analyses for various plants to conservatively calculate Reactor Coolant System (RCS) pressure following a Rod Withdrawal at Power (RWAP) event. Specifically, the Westinghouse analysis for Comanche Peak Units 1 and 2 calculated an acceptable result for RCS peak pressure.

Comanche Peak has reviewed the information and this FSAR update captures the appropriate information from the evaluation regarding the safety analysis by adding the Uncontrolled RCCA Bank Withdrawal to the Power Range high positive neutron flux rate trip discussion.

Comanche Peak has reviewed the information and this FSAR update captures the appropriate information from the evaluation regarding the safety analysis by inserting Power range high flux rate in the uncontrolled rod cluster control assembly bank withdrawal at power.

Recently, Westinghouse determined that existing Rod Withdrawal at Power (RWAP) RCS overpressure analyses do not cover the full range of reactor power operations for all Westinghouse-designed PWRs (W-PWRs).

Comanche Peak has reviewed the information and this FSAR update captures the appropriate information from the evaluation regarding the safety analysis by updationg the reactor protection system automatic features with a discussion of a trip when any two of four power range neutron flux high positive rate channels exceed a setpoint.

FSAR Amendment 103b

LDCR-SA-2010-6, EV-CR-2003-002117-00-2 (JCH):

Section 10.4, Table 10.4-13 (Auxiliary Steam System Requirements)

Description of Change:

Modify the entry stating that the Turbine Gland Steam System is only required by the Feedwater Pump Turbine during startup only to add "Power Operations: 230 (115 Each) 60-150 psig SAT".

Technical Justification:

The Auxiliary Steam System supplies 150 psig steam to the Feedwater Pump Turbine (FWPT) Gland Steam (GS) supply valves (u-PV-3918A). The GS supply valves reduce the 150 psig Auxiliary Steam (AS) supply to 2-4 psig. Each GS supply valve supplies both FWPTs for the associated unit.

This LDCR will change FSAR Table 10.4-13 to agree with the current FSAR description of the AS/GS system operation.

The FSAR Turbine GS System description states the following (10.4.3.2(1.b) :

“The pressure at the high-pressure end is insufficient to seal the low-pressure seals. Auxiliary steam from the Auxiliary Steam System is provided at the inner annular chambers through the steam seal supply valve.”

The FSAR Turbine GS System design bases states the following (10.4.3.1(2) :

“The seal steam supply header has a normal flow rate of 230 lb/hr at 2 to 4 psig. At startup, 1150 lb/hr of auxiliary steam from the Auxiliary Steam System is supplied at 2 to 4 psig.”

This indicates that the FWPT seals are not self-sealing, and the flow rate of the seal steam will vary with loading, from 1150 lb/hr at startup to 230 lb/hr during normal plant operations.

LDCR-SA-2009-9, EV-CR-2007-001888-00-6 (RAS):

Insert a parenthetical reference to Fig. 1.2-8 and 1.2-15 in the 1st sentence of the 3rd paragraph to clarify the physical location of the Jib Crane on the 905 level of the Containment.

In the 4th sentence of the 3rd paragraph, replace "Crane" with "telescopic jib crane" to clarify the particular crane of interest.

LDCR-SA-2009-9, EV-CR-2007-001888-00-6 (RAS) (continued):

In the last sentence of the 3rd paragraph, replace "jib" with "The telescopic jib" to clarify the particular crane of interest

Insert the following immediatley after the 3rd paragraph to describe the administrative controls applicable to Jib Crane operation:

"Both the Polar Crane and the Telescopic Jib Crane operating at the same time constitutes Multi-crane Operation which requires special administrative controls to reduce the likelihood of unplanned crane interactions. These administrative controls include a qualified multi-crane coordinator located in a position of good visibility to opbserve both cranes."

Revise the first four sentences of the last paragraph as follows:

"A Containment Anti-Collision Control System is also installed on the telescopic jib crane to reduce the likelihood of unplanned crane interactions with the polar crane bridge. The function of the Progammable Logic Controller (PLC) and associated components is to keep the jib crane boom below the Polar Crane bridge. In addition, an administratively controlled key, which is normally placed in the Polar Crane allows full operation from the polar crane radio controller. In this mode without the key, the PLC restricts the height of the Jib Crane boom."

Delete the following last sentence from the last paragraph:

"These administrative controls include a multi-crane coordinator located in a position of good visibility to observe both cranes."

Insert a reference to Figure 9.1-14 at the end of the first paragraph.

Insert the following sentence at the end of the second paragraoh:

"See section 9.1.4.3.2 for the Polar Crane seismic considerations. See section 3.8.3.4.10 for the applicable industry standard."

Insert the following paragraph following the 2nd paragraph:

"A telescopic jib crane is also provided inside Containment for refueling and maintenance. It is supported by a structural steel support mounted on the east-west divider wall between S. G. compartments at elevation 905'-9". Both the Polar Cane and Telescopic Jib Crane operating at the same time constitutes Multi-crane Operation which requires special administrative controls to reduce the likelihood of unplanned crane interactions. See section 3.8.3.1.2 for a description of the administrative controls and design features for the Polar Crane and the Telescopic Jib Crane."

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LDCR-SA-2010-7, EV-CR-2010-003216-6 (SCD):

Modify the FSAR to provide the flexibility of using the normal Gas Decay Tanks in a Shutdown Gas Decay Tank capacity in the event that any of the designated Shutdown Gas Decay Tanks become degraded. The designated Shutdown Gas Decay Tanks are normally used during plant startup and shutdown. This system performs no function related to the safe shutdown of the plant. [11.3.2.1.1] This change has no safety impact on the plant.

LDCR-SA-2010-13, EV-CR-2010-002062-19 (TJD):

Add the following sentence to the end of FSAR section 2.4.13.1.5:

A Groundwater Monitoring Program is established and incorporates several sentinel wells in various locations throughout the plant site for prompt identification of potential radiological release source locations.

Justifcation - Update FSAR to include information on groundwater monitoring program from May 2010 hydrogeology study performed by Pastor, Behling, & Wheeler.

Change FSAR section 2.4.13.4 to read as follows (delete end of existing sentence #1, and add new sentence after existing sentence #1):

No planned releases to the ground water environment will take place at the plantsite. However a Groundwater Monitoring Program exists for early detection of inadvertent releases of radioactive material. Pertinent information is provided in Section 6.1 of the Environmental Report.

Justifcation - Update FSAR to include information on groundwater monitoring program from May 2010 hydrogeology study performed by Pastor, Behling, & Wheeler.

LDCR-SA-2010-9, EV-CR-2010-000988-14 (TJD):

Change Executive Vice President & Chief Nuclear Officer title to Senior Vice President & Chief Nuclear Officer.

Justification - Update of VP title.

Also, revise section 17.2.1.3.2, item 3., to read as follows:

Audit program review reports encompassed by Section 17.2.1.3.1(7) shall be forwarded to the Senior Vice President & Chief Nuclear Officer within 30 days following completion of the review.

Justification - Clarify the distribution requirements for audit program review reports.

LDCR-SA-2010-9, EV-CR-2010-000988-14 (TJD) (continued):

Section 17.2.18:

Change the second paragraph in section 17.2.18 to add the following sentence after the existing first sentence: "Audit reports shall be forwarded to the Senior Vice President & Chief Nuclear Officer and to the management positions responsible for the areas audited within 30 days after completion of the audit by the auditing organization."

The new second paragraph will read as follows:

Planned and periodic audits are performed in accordance with written procedures to verify compliance with all aspects of the quality assurance program. Audit reports shall be forwarded to the Senior Vice President & Chief Nuclear Officer and to the management positions responsible for the areas audited within 30 days after completion of the audit by the auditing organization. Responsibility for the evaluation program has been assigned to the Manager, Quality Assurance. Audits are conducted or coordinated by Quality Assurance personnel and shall include evaluation and examination of the following quality-related activities:

Justification - Clarify the distribution requirements for audit reports.

LDCR-SA-2011-1, EV-CR-2010-003005-5 (TJEW):

The FSAR will be revised to reflect the disconnection and abandonment of pressurizer heater elements #19, #20, and #45 and the corresponding reduction in heating capacity from 1800 kW to 1731 kW. This activity has no impact on the pressurizer design function as the minimum heater requirements are not affected.

On Table 5.1-1B (Unit 2) Sheet 1, revise the Pressurizer heater capacity, (kW) from 1,800 to 1,731.

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LDCR-SA-2010-18, EV-CR-2010-006120-13 (RAS):

Added the following sentence to the end of item #5:

"To prevent any adverse impact on controls at the HSP or local control stations when Control Room is evacuated due to a fire, the surveillance test program validates the opening of isolation contacts for transfer switches used to transfer controls in the event of a control room fire."

Clarifies surveillance testing requirements for controls used for shutdown from outside the Control Room.

Added note "(d)" to HS-4524FL and CS-BT-1EB13L

Added note "(d)" to HS-2452C

Added note "(b)" to HS-2452B

LDCR-SA-2009-14, EV-CR-2008-001609-00-11 (SCD):

Revise section 6.3.2.2.9.2 of the FSAR to include description of the new replacement SI to CL check valves (1/2SI-8819D). The replacement check valves are a nozzle style check as opposed to the currently described spring loaded, lift piston types for sizes 2" and smaller in the Emergency Core Cooling System.

LDCR-SA-2011-8, EV-CR-2010-011041-2 (SCD):

Insert the following prior to the last sentence in the second paragraph in section 11.4.2.7:

"Interim storage locations include a warehouse area (2K7) and a fenced area (3L15) shown in Figure 1.2-1."

The existing FSAR description does not reflect locations of interim radioactive material/ waste storage areas at CPNPP.

Revise Figure 1.2-1 to add a label for the Interim Radioactive Waster Storage Area 3L15.

LDCR-SA-2010-20, EV-CR-2010-008660-2 (TJD):

Correct description of PORV Tubing (between piping and PORV) and note that it applies to Unit 1 only. This change is required by regulation to match the as-built plant as shown on FSAR Figure 6.3-1 (M1-0262 sh. – and M2-0263-B). In this case, the Nitrogen Accumulators are inside a Code N-5 Boundary (i.e., are Code Class 3). Note 41 is only applicable to the tubing for Unit 1.

LDCR-SA-2010-20, EV-CR-2010-008660-2 (TJD (continued):

Correct the line item for CCW air accumulators (as shown on FSAR Figure 9.3-3) to be consistent with all the other air accumulators and associated check valves and tubing as shown on FSAR Figure 9.3-1). As shown on the flow diagram, these accumulators are installed in the instrument air tubing system. This tubing was designed and installed in accordance with CPES-I-1018 and Table 17A-1 Note 41. Because they were installed in a tubing system, the check valves and accumulators were not included in an N-5 Code boundary. In accordance with T17A-1, Note 40, the Code Class must be "-".

Revise Reference FSAR Sections for Air Accumulators from Section 10.3 to Section 9.3.1 for consistency.

Revise Reference FSAR Sections for Tubing and supports from Section 9.3.1 to Section 3.9B for consistency.

Add Reference to FSAR Section 9.3.1 for Air Accumulators for consistency. Revise Reference FSAR Section for Tubing and supports from Section 9.3.1 to Section 3.9B for consistency.

Fix typographical error in Note 40. - "_" should be "-".

LDCR-SA-2008-21, EV-CR-2008-001221-00-12 (JCH):

Description of Change: Update FSAR to reflect radial arm hoist assembly (RAHA) QA Requirements.

Table 17A-1 Sheet 35 of 58: Changed Stud tensioner Handling Device to Radial Arm Hoist Assembly.

Table 17A-1 Sheet 58 of 58: Added Not 93.

Technical Justification:

FDA-2008-001221-04 is changing Unit 2 RAHAs to have quick disconnects for the hoists, similar to Unit 1. It also documents the current plan to remove the radial arms in Modes 1 - 4. The FSAR update is required to update the Q List for the pre-OL change from a Stud Tensioner Handling Device, which was a simple hoist, to the Radial Arm Hoist Assembly. The change is administrative and is required to document the applicable quality assurance requirements. Upgrade of the RAHA design to seismic Category II had been found impractical.

LDCR-SA-2010-5, EV-CR-2006-003080-00-320 (JDS):

Include reference in the discussion of Regulatory Guide 1.126 to reflect the vendor model for Westinghouse methods for the analysis of fuel densification used after the transition to the Westinghouse methodologies.

LDCR-SA-2010-5, EV-CR-2006-003080-00-320 (JDS) (continued):

Add reference 12, the Westinghouse method (WCAP-8219-A) for the analysis of fuel densification used after the transition to the Westinghouse methodologies.

Remove reference 18 to Regulatory Guide 1.126, the Westinghouse method (WCAP-8219-A) for the analysis of fuel densification is used after the transition to the Westinghouse methodologies.

LDCR-SA-2010-8, EV-CR-2007-003164-00-3 (JDS):

Update title from "Fuel Handling Maching" to "Fuel Handling Bridge Crane" for consistency with other sections of the FSAR.

The changes are two electric trolleys hoist (one on East side and other one on West side) on an overhead structure instead of one electric trolley hoist. The hoist speeds are limited to 3 feet per minute in the slow zone and 21 feet per minute in the fast zone. For text of the change see attachment below. These changes reflect the current configuration of the new replacement Fuel Handling Bridge Crane.

In addition, the description of the features of this new crane (including speeds and automatic features) are included in the text.

This change identifies the code required ASME NUM-1 (Type B) for hoisting equipment and ASME NOG-1 (Type II) for trolley and crane structure for the design and construction of the new fuel handling bridge crane. For text of change see attachment below. The codes invoked encompass the requirements of CMAA-70 and include requirements typically recognized within the Nuclear Industry. Further, ASME NUM-1 and ASME NOG-1 codes apply to cranes operating in Nuclear Power Plant.

Item 33 of this Table:

Delete non-IEEE-383 cable and its PVC insulation associated with the fuel handling bridge crane anticrabbing control and computer control cables to the fuel handling bridge crane anticrabbing control TCS device. The anticrabbing control for existing crane will be removed along with the crane, since these will not be required for the new replacement crane.

LDCR-SA-2011-5, EV-CR-2009-000859-00-15 (JDS):

Changes to Quality Assurance section of FSAR to support implemtation of the Dry Cask Storage System to include the Independent Spent Fuel Storage Installation (ISFSI) facility, activities supporting spent fuel dry cask loading, and the associated dry cask storage equipment.

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LDCR-SA-2010-16, EV-CR-2010-008336-3(JCH):

Section 9.5.2 Communication Systems

Description: Update the FSAR to replace Private Automatic Branch Exchange(PABX) Telephone with Intraplant(IP) Telephone System in Section 9.5.2.

Page 9.5-71 Section 9.5.2.1 Change private automatic branch exchange (PABX) to Intraplant (IP).

Page 9.5-72 Section 9.5.2.2.2 Change PABX to IP.

Page 9.5-73 Section 9.5.2.2.5 Add IP and delete (PABX telephone system).

Page 9.5-74 Sections 9.5.2.2.7 and 9.5.2.2.10 Change PABX to IP.

Page 9.5-75 Section 9.5.2.2.10 Change PABX to IP.

Justification:

Administrative change by standardizing the name of Intraplant (IP) Telephone System used within FSAR which creates less confusion when telephone technology changes.

LDCR-SA-2011-3, EV-CR-2010-010560-2(SCD):

Update CPNPP management resumes and position descriptions.

LDCR-SA-2010-17, EV-CR-2009-002851-00-4(TJEW):

This activity does not provide a relaxation in the technical or maintenance requirements used when a Mild or Harsh equipment designation is applied. The FSAR is being revised to ensure that components installed within potentially harsh environments are consistently identified as harsh. It has no adverse impact and does not affect the safe shutdown capability of the plant, nuclear safety, industrial safety, plant, system or component reliability. No new test or experiments not described in the LBDs are associated with this activity.

Two CDF were identified as needing revision (22835 and 25701)

On page 3.11B-2,

1.) delete the second paragraph of the note under section 3 that reads,

"If radiation is the only harsh environment criterion exceeded for an area and an evaluation concludes that the component materials for a piece of equipment has a radiation threshold level greater than the postulated radiation environment, then, for purposes of environmental qualification, the equipment will be considered to be located in

LDCR-SA-2010-17, EV-CR-2009-002851-00-4(TJEW) (continued):

a mild environment. The radiation threshold level is the lowest radiation exposure at which property changes in component material is documented. Such documentation can be from materials handbooks, textbooks, government reports, laboratory data, and industry sources."

and

2.)delete the note in section 4 that reads,

"If relative humidity is the only harsh environment parameter exceeded for an area and an evaluation concludes that the subject equipment can perform its safety-related function(s) when exposed to the postulated relative humidity environment, then, for purposes of environmental qualification, the equipment will be considered to be located in a mild environment."

LDCR-SA-2009-4, EV-CR-2010-006971-17 (TJEW):

The installation of transfer panels will facilitate the connection of XST2A or XST2 to the 1E busses within the 72 hour LCO.

Revise transformer name from XST1/2 to XST2A in Figure 1.2-1.

In 8.2-10, revise transformer name from XST1/2 to XST2A on page 8-iv.

Revise transformer name from XST1/2 to XST2A on page 8.2-1.

Replact the paragraph on page 8.2-2 that reads,

"A spare startup transformer, XST1/2 with dual primary windings (345-kV and 138-kV), is stored in a dedicated location under the 345-kV line to XST2 (refer to Figure 8.2-1). This transformer must be physically relocated to replace XST2 or XST1 if required. This transformer is provided to prevent an extended interruption of offsite power in case of failure of any startup transformer."

with a paragraph that will read,

Spare startup transformer XST2A, with dual primary windings (345-kV and 138-kV), is in a dedicated location under the 345 kV line to XST2 (refer to Figure 8.2-1) to serve as a replacement of XST2. Cable buses from secondary X and Y windings of XST2 and XST2A are connected to two 6.9 kV transfer panels to provide 345 kV offsite power to Units 1 or 2 safety related buses. These transfer panels allow transfer of the 345 kV offsite power source for safety related buses from XST2 to XST2A and vice verse. This

LDCR-SA-2009-4, EV-CR-2010-006971-17 (TJEW) (continued):

spare transformer may be physically relocated to a dedicated location near XST1, to serve as a replacement of XST1.

In Figure 8.2-1,

1. revise the 345 transformer storage location name from "XST1/2" to " XST2A"

2. revise the 138 transformer storage location name from "XST1/2" to " XST1A"

3. correct the orientation and delete the extra transformer figure that does not exist next to 2ST

In Figure 8.2-4,

1. revise transformer name from XST1/2 to XST2A

2. swap name tags between XST2A and 1ST.

3. show a connection of 1ST to the non-safeguards buses and shorten the line representing the non-safeguards busses

4. revise the connections of XST2 and XST2A to show connection to the Transfer Panels

5. show the connection from the Transfer Panels to the 6.9 kV buses

6. relocate the "6.9KV SAFEGUARD BUSES (SEE FIGURE 8.3-1)" below the Transfer Panels

In Figure 8.2-5, Sheet 1, revise transformer name from XST1/2 to XST2A.

In Figure 8.2-10, Sheet 1,

1.) revise transformer name from XST1/2 to XST2A for the motor operarted switch, on the spare transformer, and on the Figure title

2.) add the new cable trench and cable bus from XST2A to the turbine building wall,

3.) Deleted "Amendment 100".

In Figure 8.2-10, Sheet 2,

1.) revise transformer name from XST1/2 to XST2A for the motor operated switch, transformer, and Figure title,

2.) Show the new cable trench and cable bus to the turbine building wall, and

LDCR-SA-2009-4, EV-CR-2010-006971-17 (TJEW) (continued):

3.) Deleted "Amendment 100".

In Figure 8.2-11,

1, revise transformer name from XST1/2 to XST2A

2, show the cable trench to from XST2 and XST2A to the Transfer Panels and to the 6.9kV buses.

In Table 8.3-3 Sheet 2,

1.) In "Item 4", under the "Function" column, add the words "through CPX-EPTSST-02Y" after Unit 1 and add the words

"through CPX-EPTSST-02X" after Unit 2.

2.) In "Item 4A", under the "Fuction" column, replace the words that read "buses 1EA1 and 1EA2 of Unit 1" with the words "transfer panel CPX-EPTSST-02Y".

3.) Under the "Effects on System" column, add the following words at the end of "Item 4" and "Item 4A", "Can also realign to spare startup transformer XST2A (Item 4C).

On Table 8.3-3, Sheet 3,

1.) In the "Function" column replace the words that say, "XST2 to buses 2EA1 and 2EA2" with the words "transfer panel CPX-EPTSST-02X from XST2",

2.) Add the words, "Can also realign to spare startup transformer XST2A (Item 4C)"

3.) Add the Failure Mode and Effect Analysis "4C" for the "Spare Startup transformer XST2A", "4D" for the "15-KV Cable" Y-winding connection, and "4E" for the X-winding connection.

In 9.5.1.5.6,

1. In the sentence that reads,

"The unit auxiliary transformer and the Unit 1 startup transformer are separated from the Turbine Buildings by a three-hr rated fire wall."

Delete the words "Unit 1" and after word "transformer" insert the words, "s XST1 and XST2"

2. Delete the sentence that reads,

LDCR-SA-2009-4, EV-CR-2010-006971-17 (TJEW) (continued):

"The Unit 2 startup transformer is also separated from the Turbine Building by a three-hr rated fire wall."

3. In the last sentence that reads,

" The Unit 1 station service transformer is separated from the Turbine Building by a distance greater than 50 feet."

Delete the words "The Unit 1", capitalize the "s" in station, and insert the following words after "transformer"

"s 1ST and 2ST and spare transformer XST2A are" and

delete the word "is"

On Figure 10.2-1 add "\XST2A" after XST2 in three locations.

LDCR-SA-2011-6, EV-CR-2009-000859-00-16 (JDS):

Include a description of the Independent Spent Fuel Storage Installation for the facility desciption for completeness.

Revise drawing to show location of the Independent Spent Fuel Storage Installation.

Update Table of Contents to reflect new section for Spenf Fuel Dry Cask Storage and clarify that section 9.1.2 is Spent fuel storage in the spent fuel pools and containment.

Update title to include description of spent fuel storage in "the spent fuel pools and containments."

Update text to refer to heavy load of the Holtec Hi-Trac tansfer cask for the Fuel building overhead crane.

Insert new section 9.1.5 to describe the independent spent fuel storage installation and the specific Certificate of Compliance (1014) used for dry cask storage.

Add an additional three references for the Cask FSAR, the Holtec Certificate of Compliance (1014) and the NUREG-6407 for Transportation and packaging of spent fuel.

Add reference to the Fire Hazards Analysis for ISFSI and Haul Path to the ISFSI.

Include reference to the CPNPP 10CFR72.212 Evaluation report.

Include statement for a fully loaded (84 Casks) bounding annual exposure for the ISFSI in the estimated annual dose at the exclusion boundary and to the population at large.

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LDCR-SA-2011-12, EV-CR-2010-008829-2 (TJEW):

On Figure 9.5-52 (M1-0215), Sheet F:

1.) add note 10 that says, "10. Alternate fill location connection point. Box with Access Cover Plate is maintained watertight"

2.) On the truck fill watertight valve box add "COVER" in between "REMOVABLE" AND "PLATE"

3.) add an arrow with a note that says "See Note 10" at the quick disconnect fitting

4.) at the quick disconnect fitting add "COVER" in between "REMOVABLE" AND "PLATE"

5.) add the removable access cover plate and add an arrow pointing to the removable access cover plate and the note that says, "REMOVABLE ACCESS COVER PLATE"

On Figure 9.5-52 (M2-0215), Sheet F:

1.) add note 12 that says, "12. Alternate fill location connection point. Box with Access Cover Plate is maintained watertight"

2.) On the truck fill watertight valve box add "COVER" in between "REMOVABLE" AND "PLATE"

3.) add an arrow with a note that says "See Note 12" at the quick disconnect fitting

4.) at the quick disconnect fitting add "COVER" in between "REMOVABLE" AND "PLATE"

5.) add the removable access cover plate and add an arrow pointing to the removable access cover plate and the note that says, "REMOVABLE ACCESS COVER PLATE"

On Figure 9.5-52 (M1-0215), Sheet G:

1.) add note 10 that says, "10. Alternate fill location connection point. Box with Access Cover Plate is maintained watertight"

2.) On the truck fill watertight valve box add "COVER" in between "REMOVABLE" AND "PLATE"

3.) add an arrow with a note that says "See Note 10" at the quick disconnect fitting

4.) at the quick disconnect fitting add "COVER" in between "REMOVABLE" AND "PLATE"

5.) add the removable access cover plate and add an arrow pointing to the removable access cover plate and the note that says, "REMOVABLE ACCESS COVER PLATE"

LDCR-SA-2011-12, EV-CR-2010-008829-2 (TJEW) (continued):

On Figure 9.5-52 (M2-0215), Sheet G:

1.) add note 12 that says, "12. Alternate fill location connection point. Box with Access Cover Plate is maintained watertight"

2.) On the truck fill watertight valve box add "COVER" in between "REMOVABLE" AND "PLATE"

3.) add an arrow with a note that says "See Note 12" at the quick disconnect fitting

4.) at the quick disconnect fitting add "COVER" in between "REMOVABLE" AND "PLATE"

5.) add the removable access cover plate and add an arrow pointing to the removable access cover plate and the note that says, "REMOVABLE ACCESS COVER PLATE"

On page 9.5-81, Section 9.5.4.2.1 add the following words to:

1.) at the end of the first sentence of the forth paragraph add, "or via an alternate fill method which utilizes a special diffuser tool to fill through the Sample Grab Point quick-disconnect fitting connection, which is also in a watertight box." and

2.) After the third sentence of the fourth paragraph add the sentence, " When the alternate fill method is used, the special diffuser tool is also capped at the bottom and perforated with holes."

On figure 9.5-51,

1.) add note 1 that says, "Note 1. Sample Grab Point Connection provides additional truck fill method via use of special diffuser tool. This box is maintained watertight." in the lower left hand corner and

2.) add an arrow pointing at the Sample Gab Point Connection and a text box that says, "See Note 1"

LDCR-SA-2010-19, EV-CR-2010-010870-1 (TJEW):

On page 10.4 page 33, third paragraph, replace the word "each" with "Unit 2" and add a "s" after the first occurrence of the word "generator".

LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW):

Justification: Changes to the description of the 345 kV Switchyard are required for addition of two breakers and their associated lines (Everman & Parker 2) and relocation of the TO's breaker controls/protection to TO's control building. With the new relay control panels a third set of protective relays are not required for line breakers. Therefore, the lines will only have primary or secondary (backup) sets. In addition, the

LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW) (continued):

FSAR and associated Figures are being revised to reflect name changes of TXU Electric Delivery to Transmission Operator (TO) in the affected sections. Further, the approximated length of transmission lines to the substations connected to the CPNPP Switchyards is deleted, because this detail is not required for the FSAR. Also the discussion of transmission towers on pages 8.2-3 and 8.3-4 is being deleted from the FSAR because the information is shown in Figure 8.2-12. Section 8.2.2 analysis is updated to reflect the impact of loss of double circuit towers as a result of addition of new lines. In addition Oncor has reviewed the the section 8.2.2 analysis for its adequacy due to the addition of new lines.

On Page 1.2-9, second paragraph, section 1.2.2.5

1.) change "five" to "seven" and

2.) In the same paragraph, change "owner's" to "Transmission Operator's"

On Page 8.1-1,

1.) Update the Utility Grid Description to reflect the changes in the ERCOT organization.

2.) change all of "TXU Electric Delivery" to "TO"

On Page 8.2-1, change

1.) all of "TXU Electric Delivery" to "TO" and

2.) the reference to "five" 345kV transmission lines to "seven" lines and

3.) In the third paragraph, add the word system before the words "as shown on Figure 8.2-4"

4.) Add the following sentence, from page 8.2-3, to the end of the third paragraph, "The 138-kV switchyard is physically and electrically independent of the 345-kV switchyard."

On Page 8.2-2,

1.) In the 4th paragraph, delete the words that say, "The network transmission line terminals of 345-kV switchyard are provided with a third set of relay protection to enhance the reliability of transmission system fault isolation."

2.) In the 5th paragraph, delete the words that say, "A separate 125-VDC system is provided for the third set of relay protection in the 345-kV switchyard."

3.) In the 6th paragraph,

LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW) (continued):

a.) add the following words, "The 345-
kV switchyard TO circuit breakers may be operated from either the 345-
kV Switchyard Control Building or remotely from TO's System Operations Center."

b.) In the next sentence change the word "all" to "CPNPP", delete the words "the spare", and add the word XST2A after transformer.

c.) In the next sentence capitalize the first letters in the following words, "switchyard relay house", add "138-kV" before the words

"switchyard relay house", and replace the words "through TXU Electric Delivery" with "from TO".

4.) In the 8th paragraph,

a.) delete the words, "and the approximate length of their respective transmission lines", rearrange the list of tranmission lines to match , as shown below, the switchyard Figure 8.2-4, and delete the transmissison line lengths.

"* DeCordova (138-kV)

* Stephenville (138-kV)

* DeCordova (345-kV)

* Wolf Hollow (345-kV)

* Everman (345-kV)

* Johnson Switch (345-kV)

* Comanche Switch (345-kV)

* Parker No. 1 (345-kV)

* Parker No. 2 (345-kV)"

On page 8.2-3,

1.) The order of the CPNPP transmission lines, including the Stephenville line, will be revised as shown on Page 8.2-2 and therefore, the Stephenville line detail on the first line of this page will be deleted.

2.) Insert the following words before the first paragraph,

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LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW)(continued):

"The layout of transmission lines from TO's CPNPP switchyards to other switching stations, in the vicinity of CPNPP switchyards, is shown in Figure 8.2-12.

The following combinations of lines form double-circuits routed on common transmission towers:

345-kV Parker No. 1 and 345-kV Parker No. 2

345-kV Comanche Switch and 138-kV Stephenville

345-kV Everman and 345-kV Johnson Switch

345-kV DeCordova and 345-kV Wolf Hollow

3.) Delete the next 6 paragraphs.

4.) In the 7th paragraph,

a.) delete, the 1st sentence,

b.) in the 2nd paragraph, add "138-kV" after "The".

c.) delete the third sentence.

5.) In the 8th paragraph,

a.) Delete the first sentence,

b.) in the second sentence, replace "The line then" with "The 138-kV line to Stephenville", add "Everman," after "Wolf Hollow", and replace "Venus before making another 90° turn" with "Johnson Switch".

c.) Delete the 3rd sentence.

On page 8.2-4, delete the words from the paragraph that started on page 8.2-3, "Upon making the second turn, the line then shares common towers with the 345-kV line to Comanche Switch."

On Page 8.2-5, change all of "TXU Electric Delivery" and "TDSP" to "TO".

On page 8.2-7,

1.) second paragraph of section 8.2.2,

a.) add a comma in the first sentence after the words "Comanche Switch line"

LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW) (continued):

b.) in the two places where "Parker Line" is written, before the word "line" insert, "No. 1 and Parker No. 2 double-circuit", in the three places where the word "Venus" is written, replace it with "Everman and Johnson Switch double-circuit", add the word "the" before the second occurrence of "Venus"

c.) In the 4th sentence, add a "-" between "double" and "circuit"

2.) In the third paragraph of section 8.2.2

a.) delete the words "Electric Reliability Council of Texas" and the "()" around ERCOT

b.)change "TXU Electric Delivery" to "TO"

On Page 8.2-8,

1.) change "TXU Electric Delivery" and "TXU Delivery" to "TO".

2.) Delete the entire 7th paragraph.

On Page 8.2-9,

a.) change all of "TXU Electric Delivery" to "TO".

B.) Replace the words, "There are no restrictive operating limits (real and reactive power, voltage, frequency and other) established for CPSES by ERCOT. However, CPSES requires that TXU Electric Delivery transmission" with "In order to satisfy offsite power requirements, the TO should"

On Page 8.3-2, second paragraph, change

1.) "TXU" to "TO" and

2.) the reference to "five" 345kV transmission lines to "seven" lines

Revise Figure 8.2-1 to show the new 345 kV Parker and Everman Lines, delineate the Parker No. 1 and No. 2 lines, add the TO Control Building, and change the Venus line to Johnson Switch.

Revise Figure 8.2-4 to show the new 345 kV Parker and Everman Lines and Breakers at both ends of the lines, delineate the Parker No. 1 and No. 2 lines, change TXU Energy Delivery to TO, change the Venus line to Johnson Switch, and revise the breaker arrangement at Johnson Switch and Wolf Hollow.

Revise Figure 8.2-5, sheet 1 to show

1.) to show the new Oncor Control Building

LDCR-SA-2009-20, EV-CR-2009-005301-7 (TJEW) (continued):

2.) the new West Bus, Bay 9 Breaker for the Parker No. 2 transmission line

Revise Figure 8.2-5, sheet 2 to show

1.) change "Venus" to "Johnson Switch"

2.) the addition the new 345 kV Everman Line Breaker and associated equipment on the East Bus of bay 4, change the Venus line to Johnson Switch

Revise Figure 8.2-12 to show

1.) the new 345 kV Parker and Everman Lines,

2.) delineate the Parker No. 1 and No. 2 lines,

3.) change the Venus line to Johnson Switch.

LDCR-SA-2010-1, EV-CR-2006-003080-44 (SCD):

The changes made to FSAR Section 11 reflect the change in isotopic inventory associated with the increase in power as well as the transition from historical TXU calculation methodologies to those of Westinghouse.

The changes made to FSAR section 12 reflect the revised dose estimations associated with normal operations and assuming the revised isotopic inventory. Note dose assessments associated with accident conditions, i.e., FSAR Chapter 15 events, were performed by Westinghouse and are outside of the scope of this LDCR.

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LDCR-SA-2010-14, EV-CR-2007-003115-00-12 (GLM):

The FSAR is being revised because the existing clarifier based pre-treatment subsystem portion of the Water Treatment (WT) System has deteriorated due to the aggressive nature of the lake water and chemicals required for treatment. The pre-treatment subsystem portion of the WT System is being replaced using current microfiltration technology.

With this modification, the clarifier is being taken out of service and replaced with microfiltration units. Therefore, changes to the system description are required, which maintains the design requirements. This modification removes the clarifier, pressure filters, and filtered water forwarding pumps from the WT System and replaces it with microfiltration units. These units are a membrane based filtration system which removes suspended solids and achieves water quality requirements in a one-step filtration process. The system includes filtration tanks, pump skid, module skid, and a clean-in-place (CIP) skid. These changes reflect the new equipment being installed and maintain the design requirements.

LDCR-SA-2011-13, EV-CR-2010-000979-15 (JDS):

Delete TM symbol from Zirlo as this a Registered tradmark. Zirlo is now used to refer to this type of cladding material.

Add rod bow penalty WCAPs (8691 and 8692) to the material incorporated bny reference. Text added to chapter 4 describes this as the accepatable methodology for use to calculate rod bow penalty.

Include the discussion of the use of the Standardized Debris Filter Bottom Nozzle (SDFBN) and the Robust Protective Grid (RPG). These fuel assembly components are now characteristic for reloads at CPNPP.

Update Minimum DNBR for design transients to reflect appropriate values based on WRB correlation used. Update the Average mass velocity for correct description for the values provided. Update maximum overpower value used for Peak linear power resulting from overpower transients/operator errors. Insert RPG for robust p grid usage.

Include the discussion of the use of the Robust Protective Grid (RPG). This fuel assembly components is now characteristic for reloads at CPNPP.

Provide a description of the Standardized Debris Filter Bottom Nozzle (SDFBN) for Westinghouse 17x17 fuel. This design is now standard for reloads for CPNPP units 1 and 2.

Provide a description of the Westinghouse Robust Protective Grid (RPG) design. This design is now standard for reloads for CPNPP units 1 and 2.

Clarify use of nomenclature used for Westinghouse fuel for the Vantage+ fuel design.

LDCR-SA-2011-13, EV-CR-2010-000979-15 (JDS) (continued):

Update discussion on Effects of Rod Bow on DNBR to reflect NRC approved methodology.

Include reference to Effects of Rod Bow on DNBR approval by the NRC (WCAP 8691 and 8692)

Update maximum overpower value used for Peak linear power resulting from overpower transients/operator errors.

LDCR-SA-2011-14, EV-CR-2011-001279-1 (JDS):

Update 7 minutes 30 seconds to 8 minutes and 13 minutes to 14 minutes for consistency with previous statement on this same page and the supporting analysis.

LDCR-SA-2011-2, EV-CR-2009-001766-00-4 (RAS):

Add the following to the end of the sentence for item 'b.' under "Testing of Reactor Trip Breakers":

"...or by a Reactor Trip Test pushbutton via a keylocked test selector switch."

LDCR-SA-2009-22, EV-CR-2007-000293-00-1 (TJD):

FSAR section 17.2.15 is being changed to utilize a conditional release to not only install the nonconforming component in the plant but also to declare the affected SSC Operable per Technical Specifications (TS), provided that resolution of the nonconforming condition is tracked in the Corrective Action Program (CAP), and an evaluation of the nonconforming condition shows that it does not adversely affect Operability of the component. This change was approved by the NRC in a Safety Evaluation dated 9/23/11.

Revise item 5 as follows:

Conditional releases allow issuance of nonconforming items from the warehouse for installation, testing, and operation, pending disposition of the nonconformance. For nonconforming items installed with a conditional release, and affecting Technical Specifications, credit may be taken for Technical Specification operability of the item, provided that: a) resolution of the nonconforming condition is tracked, and b) evaluation of the nonconforming condition supports operability of the component. Each conditional release also describes any limitations or special precautions required. Conditional releases are periodically evaluated as to their status and the results forwarded to management for their review.

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LDCR-SA-2011-15, EV-CR-2008-003340-00-6 (TJEW):

Table 3A.3-2, Sheet 5, section 1.3, is changed via FDA-2008-003340 and in the FSAR via LDCR SA-2011-015 in EV-CR-2008-003340-6 to revise boron concentration from '2200' to '2600' and from "initial" to "final" spray solution to (1) align configuration control and (2) correct the statement so that it is understood that short period exposure to out of specification ph chemical spray will have no effect on the ability of equipment inside containment to perform their safety function.

LDCR-SA-2009-18, EV-CR-2009-001588-00-31 (TJEW):

Technical Justification:

Issue 1 - FSAR Table 7.5-7C note 21 and the response to the TMI Action Plan Section II.F.1 state that the "containment water level covers the entire range of expected water level in the containment for post-accident conditions." This statement regarding the wide range span and the requirement for a separate narrow range instrument was based on a misinterpretation of the maximum flood level calculation 525-00. The calc had conservatively assumed that the reactor cavity was not flooded to maximize the peak flood level. Based on Calculation 525 dated 3-23-1979, the range of containment flooding was calculated from El. 808 to El. 817'-6". Based on the current calculation NU(B)-057 R3, the range of flooding is from the El. 781'-2" rector cavity sump and El 783'-7" reactor cavity [see A1-0529] to the maximum flood El. 816'-10". The FSAR requires update and clarification.

Issue 2 - FSAR table 7.5-7E note 18 states "Containment Water level transmitters are multipoint sensors and failure of any one sensor would not cause ambiguity and hence will not have any adverse effect on the monitoring of containment water level." This statement was based on the assumption that the analog control board indicators for the original level instrumentation would be multipoint LEDs corresponding to the RTDs. However, the original 4-20 ma analog indicators were not replaced. The failure mode of the loop is such that the failure of one sensor can be read as one foot high or one foot low across the scale. This is an ambiguous failure. The FSAR requires update and clarification.

On page 7.5-10, in Section 7.5.2.1, replace "10. Containment Water Level" with the word "Deleted"

In Table 7.5-2, delete "Containment Water Level Key A1"

In Table 7.5-7A, Sheet 2 of 15, for the row containing "CONTAINMENT WATER LEVEL (NOTE 21), delete "A1" and add "-3"" after "808"'.

In Table 7.5-7C, Sheet 2 of 4, number 21, 1.) Insert the words "(from prior to the start of switchover from injection to recirculation to the maximum containment flood level)" between the words "level" and "in" and 2.) After the first sentence, insert the sentence, "The water level in the sump is assured by the RWST Low-Low level setpoint for the start of switchover. Operator action based on containment water level is not required."

LDCR-SA-2009-18, EV-CR-2009-001588-00-31 (TJEW) (continued):

In Table 7.5-7D, Sheet 1 of 7, and in the row of "Containment Water Level" add "-3"" after "808"'.

In Table 7.5-7E, Sheet 3 of 3, number (18), delete all the words after "transmitters" and add the following words, "were intended to be multipoint sensors which would prevent an ambiguous failure from occurring. The original indicators were never replaced, however, and the original analog indicators rely on an additive logic to compute the water level. This results in the potential for a failed indicator to cause a false reading of ± 1 foot. The containment water level indicators are for information only and will not adversely affect the ability to safely mitigate an accident."

In section 11.F.1, CPNPP Response (5) on page II.F-3, 1.) Add "-3"" after "808"' 2.) Add the words "the intent of" between "meeting" and "Regulatory" 3.) replace the words "entire range of expected water level in the Containment for post accident conditions" with the words, "maximum expected flooding levels assuming the reactor cavity is blocked and does not receive any flood water to simulate the most limiting post-accident condition (flooding starts at El. 808'). Actual post-accident conditions would include flooding into the reactor cavity sump (781' -2") and reactor cavity (783' - 7").

LDCR-SA-2011-18, EV-CR-2011-008509-4 (TJEW):

Revise FSAR Section 9.5.5.2 to reflect information in FSAR prior to issuance of LDCR-SA-2004-014 which changed the Jacket Water Heat Exchanger leakage rate. The justification for this change is that during the 2011 PI&R Inspection, a potential Severity Level IV violation of 10CFR50.59 was identified for an incorrect 50.59 screen related to changing the EDG jacket water leakage rates in the FSAR. The screen incorrectly concluded that an evaluation was not required.

On page 87 of sectio 9.5.5.2, 6th paragraph, change 408 gallons to 310, change 17 gallons per hour to 1.5, and change 24 hours to seven (7) days.

LDCR-SA-2011-21, EV-CR-2011-014200-1 (TJEW):

Components FT-6708 and FT-6709 are identified within DBD-EE-004 and FSAR Table 7.5-7B (Sheet 8 of 11) as being required to be seismically qualified. The devices are in fact only required to be seismically mounted. DBD-EE-004 section 4.3.2.1.1 states 'Category 2 instrumentation supplied from Non-1E power need not be seismically qualified unless it is part of a seismic Category I system and its failure may degrade this system.' The flow transmitters are N1E, Process Instruments(Isolable, Offline). DBD-ME-028, attachment 19, page 7 of 23, identifies that these types of transmitters are ANSI class: N/A, IEEE Class: N1E, Seismic Category: None, Safety Related: No, and EQ: No.

DBD-ME-028, Table 1, note 3, is associated with components identified as N1E, Process Instruments(Isolable, Offline). This note states that Seismic Category II mounting applies to the non-1E transmitter or gauge. This mounting requirement is identified within Maximo-MEL. The Location screen contains attribute 'CFC2'. For FT-6708 and FT-6709

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LDCR-SA-2011-21, EV-CR-2011-014200-1 (TJEW) (continued):

the components are identified as SMTQ. The SMTQ designation identifies that the components require 'Seismic Integrity Evaluation and Mounting of Non-safety Related Instruments Connected to ANSI 2 or 3 Fluid Systems'.

Revise the note in Table 7.5-7B, Sheet 8 of 11, Variable - Safety Chilled Water Flow from "Note 5" to "Note 6"

LDCR-SA-2011-17, EV-CR-2011-003925-5 (JDS):

Reference Westinghouse Letter transmitting the GOTHIC containment pressure and temperature analyis for EQ.

Remove curves ad replace with methods. Reflects that the containment temperature and pressure curves used for EQ are no longer the same as those used for the containment integrity analysis.

LDCR-SA-2011-7, EV-CR-2009-000859-00-34 (JDS):

Update Table 3.7B-2 with newer versions of the ANSYS program. These versions of ANSYS were used for ancillary equipment used in the fuel building supporting the Dry Cask Storage project.

LDCR-SA-2011-16, EV-CR-2011-005954-4 (SCD):

Incorporate change of Quality Assurance (QA) to Nuclear Oversight.

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LDCR-SA-2012-9, EV-CR-2011-012481-1 (RAS):

Revise the paragraph at the top of the page as follows to reflect the implementation of the Relaxed Axial Offset Control Methodology:

The axial flux difference deviation alarms are derived from the plant process computer which determines the 1 minute averages of the excore detector outputs to monitor delta flux in the reactor core and alerts the operator where delta flux alarm conditions exist. When power level is 50 percent or greater, an alarm message is output immediately upon determining a delta flux exceeding the Technical Specifications (this alarm requires 2 of the 4 AFD channels to exceed the power dependent limits). The signals from the four section excore detectors are summed and calibrated to a power measurement made using a secondary system calorimetric. The signals from the upper two sections and lower two sections are combined to give an upper flux and lower flux signal as shown in Figure 7.1-2. These upper and lower signals must be calibrated to agree with axial offset measurements derived from flux maps made using the movable incore system. The NIS provides both slope and zero-offset adjustments to achieve this calibration.

LDCR-SA-2012-7, EV-CR-2012-002362-1 (TJEW):

Background: During preparation of FDA-2011-000004-01, an error concerning Flow Diagram M2-0263, Sheet No. C, Safety Injection System Sheet 6 of 6, was found. It is not included in FSAR Figure 6.3-1. This flow diagram is part of the Safety Injection System and Containment Isolation System described in the FSAR Chapter 6 text and tables. Without it, FSAR Figure 6.3-1 is incomplete.

Non-conforming Condition: The Note block designating the drawing as part of FSAR Figure 6.3-1 is missing. Therefore, the Drawing was not included in the FSAR figure as required by RG 1.70 R2 and the FSAR is not current as required by10CFR50.71(e). The FSAR is not being update for changes to the facility as required by regulations (e.g. FDA-2008-003459-04). Corrective Action required: M2-0263 Sheet No. C must be revised and added to FSAR Figure 6.3-1 and to the Chapter 6 TOC.

Additionally, while performing a consistency review of the LDCR, found that FSAR Tables 3.2-3 and 3.2-4 also need to include M2-0263-C.

On Table 3.2-3, Sheet 10 fo 13, there is no entry for Unit 1; however, for Unit 2 add, "M2-0263-C CP-7 104* 6.3-1"

On Table 3.2-4, Sheet 1 of 16, there will be no entry for Unit 1; however, for Unit 2 add "6.3-1 M2-0263-C"

On page 6-xvii, the Table of "List of Figures" add ", M2-0263-C” at the end of the title for 6.3-1.

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LDCR-SA-2012-11, EV-CR-2011-001492-12 (JCH):

Description: Revise Table 3.9N-10, "Unit 1 & 2 Active Valves," to include active relief valves 1-8121 and 2CS-8000.

Justification: RCP seal water return line relief valves, 1-8121 and 2CS-8000, are required to protect the excess letdown/seal water return header from overpressurization in the event that the line exiting containment is isolated while the excess letdown and/or the RCP No. 1 seal leakoff continues in service. The IST plan correctly designates these valves as active for Containment Penetration Overpressure Protection from both seal injection and thermal expansion however DBD-ME-255 Table 4-1 and FSAR Table 3.9N-10 show these valves as passive. This change will align the design and licensing basis documents to the IST plan. There is no change to the valve or its subcomponents and no change to the design function of the valve.

LDCR-SA-2012-10, EV-CR-2012-000999-1 (SCD):

Description of the Old Steam Generator Storage Facility was not adequately incorporated into the proper FSAR chapters.

LDCR-SA-2012-14, EV-CR-2011-008520-14 (SCD):

The Safe Shutdown Impound (SSI) is a thriving dynamic biological system with significant biological populations that can contribute to debris entering the Station Service Water (SSW) system. This debris can result from die-off of small fish (i.e. Threadfin Shad or Silver Sides), algae, live or dead Harris mud crabs and storm water runoff debris. Shad die-offs have occurred in the SSI that caused challenges to the SSW system safety related strainers. The SSI communicates with Squaw Creek Reservoir (SCR), which allows exchange of fish species and complicates measures to reduce or control fish populations. The installed floating booms minimize the ability of floating debris to enter the SSW system while the equalization canal net minimizes the migration of fish from SCR into the SSI.

Clarification is required on the limitation of use of the stop gates as a sole source of flood protection of the Turbine Building when the Circulating Water System is opened for maintenance. Due to the access opening in the operating deck of the circ-water discharge structure, the stop gates only isolate the CW Tunnels and any open CW system when the lake level is below 778 feet. As a result, any open CW system at elevation below the PMF level of 789.7 feet must be reassembled/closed or isolated when the lake level is on a rise and evident that it will reach/exceed a level of 778 feet. This ensures the isolation of the lake levels above 778 from the open pathway that could lead to flooding of the Turbine Building and lower level of the Electrical & Control Building on Elevation 778.

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LDCR-SA-2010-3, EV-CR-2010-000528-8 (TJEW):

On page 1A(B)-55, in note a at the bottom of the page, change "0.8348" to "0.8299" and "38" to "39"

In 1A(B) on page 56, Part 9.C.2.a, change "0.8348" to "0.8299" and "38" to "39"

On page 9.5-82, change "0.8348" to "0.8299" and "38" to "39"

On page 9.5-83, change "0.8348" to "0.8299" and "38" to "39"

On page 9.5-86, change "0.8348" to "0.8299" and "38" to "39"

LDCR-SA-2012-17, EV-CR-2011-008700-17 (SCD):

Clarification is required on the limitation of use of the stop gates as a sole source of flood protection of the Turbine Building when the Circulating Water System is opened for maintenance. Due to the access opening in the operating deck of the circ-water discharge structure, the stop gates only isolate the CW Tunnels and any open CW system when the lake level is below 778 feet. As a result, any open CW system at elevation below the PMF level of 789.7 feet must be reassembled/closed or isolated when the lake level is on a rise and evident that it will reach/exceed a level of 778 feet. This ensures the isolation of the lake levels above 778 from the open pathway that could lead to flooding of the Turbine Building and lower level of the Electrical & Control Building on Elevation 778.

LDCR-SA-2012-15, EV-CR-2010-011195-4 (SCD):

10CFR50.71(e) requires the FSAR to be maintained current. This is an administrative update to the FSAR.

The FSAR 1A(B) exceptions to Revision 1 of RG 1.52 were reconciled with the RG and FSAR Table 6.5-1.

This Appendix 1A(B) discussion of RG 1.197 was not correct. It stated the TSTF incorporates the specific aspects of the RG whereas the TSTF took exception to specific aspects of RG1.197. This error was made at some point after LA 136.

The following are CPNPP exceptions to Sections C and C.1.2 of RG 1.197 documented in TS 5.5.20 which are not reflected in FSAR Appendix 1A(B):

1. C. - Section 4.3.2 “Periodic CRH Assessment” from NEI 99-03 Revision 1 will be used as input to a site specific Self Assessment procedure.

LDCR-SA-2012-15, EV-CR-2010-011195-4 (SCD) (continued):

2. C.1.2 - No peer reviews are required to be performed.

This Appendix 1A(B) discussion of RG 1.197 was not correct. It stated the TSTF incorporates the specific aspects of the RG whereas the TSTF took exception to specific aspects of RG1.197. This error was made at some point after LA 136.

The following are CPNPP exceptions to Sections C and C.1.2 of RG 1.197 documented in TS 5.5.20 which are not reflected in FSAR Appendix 1A(B):

1. C. - Section 4.3.2 “Periodic CRH Assessment” from NEI 99-03 Revision 1 will be used as input to a site specific Self Assessment procedure.

2. C.1.2 - No peer reviews are required to be performed.

The FSAR 1A(B) exceptions to Revision 1 of RG 1.52 were reconciled with the RG and FSAR Table 6.5-1.

LDCR-SA-2011-11, EV-CR-2009-001966-11 (GLM):

In the first sentence of Section 9.4C.6, "OFFICE AND SERVICE AREA HVAC SYSTEM" delete the words "...and relative humidity." In the first sentence, second paragraph of Section 9.4C.6, "OFFICE AND SERVICE AREA HVAC SYSTEM" change "air conditioning units" to "air handling units." Delete the second sentence, second paragraph of Section 9.4C.6, "OFFICE AND SERVICE AREA HVAC SYSTEM." In the third sentence, second paragraph of Section 9.4C.6, "OFFICE AND SERVICE AREA HVAC SYSTEM" change " direct expansion refrigerant cooling coil" to "chilled water cooling coil" and delete the word "humidifier."

The replacement air handling units are rated to provide a level of cooling which exceeds the rated capacity of the previous split air-conditioners (109.4 tons versus 94.5 tons) therefore this change does not adversely impact any design function of the Office and Service Area system, and the resulting changes to the FSAR are required to maintain configuration control. The humidifier does not perform or support a safety-related function for the O&SA HVAC system, therefore this change is not adverse. These changes do not adversely impact the O&SA HVAC system design function which is to provide a comfortable environment for operating personnel by maintaining the temperature and humidity as indicated by UFSAR Table 9.4-2.

For the Office and Service Area A/C, change the Relative Humidity (%) from "35-50" to (b).

The replacement air handling units are rated to provide a level of cooling which exceeds the rated capacity of the previous split air-conditioners (109.4 tons versus 94.5 tons) therefore this change does not adversely impact any design function of the Office and Service Area system, and the resulting changes to the FSAR are required to maintain configuration control. The humidifier does not perform or support a safety-related function for the O&SA HVAC system, therefore this change is not adverse. These changes do not adversely impact the O&SA HVAC system design function which is to provide a comfortable environment for operating personnel by maintaining the temperature and humidity as indicated by UFSAR Table 9.4-2.

LDCR-SA-2011-19, EV-CR-2009-001930-5 (GLM):

Revise the last sentence to state that the potable water supply is from the Somervell Co. Water District public water system and groundwater wells are a backup.

Justification:

The station potable and sanitary water system is designed to provide the following: 1) Water for toilets, sinks, showers, and drinking purposes in all permanent personnel areas of the plant site, as required, 2) Water for emergency eyewash and showers, as required, 3) Water to fire protection hoses for various on-site buildings, 4) Water to fill and to provide normal makeup to the Fire Protection Storage Tanks. The change requested by this LDCR is a change of source which is authorized under SFER-4205839. The changes will not have adverse impact to the design bases described in FSAR Section 4.2.5.1.

LDCR-SA-2011-19, EV-CR-2009-001930-5 (GLM) (continued):

Using SCWD as the primary source, the plant potable water distribution piping will function as it currently does.

As discussed in FSAR Section 9.2.4.3, this system is not connected to any radiological systems. Since the system is physically independent of radiological systems, this change to the potable water system has no impact on the plant radiological systems. Should a radiological spill occur, the conclusions of the FSAR for the potable water system are not changed - i.e., the change to the distribution piping is in the same general vicinity as existing potable water distrubition piping and cannot be directly impacted from a radiological spill outside the plant. Therefore the requested change does not adversely affect any FSAR described design function.

Delete the first sentence and revise the second paragraph to reflect that potable water is chlorinated to conform to TDH standards and the emergency shower in the relay house is chlorinated.

Justification:

The station potable and sanitary water system is designed to provide the following: 1) Water for toilets, sinks, showers, and drinking purposes in all permanent personnel areas of the plant site, as required, 2) Water for emergency eyewash and showers, as required, 3) Water to fire protection hoses for various on-site buildings, 4) Water to fill and to provide normal makeup to the Fire Protection Storage Tanks. The change requested by this LDCR is a change of source which is authorized under SFER-4205839. The changes will not have adverse impact to the design bases described in FSAR Section 4.2.5.1. Using SCWD as the primary source, the plant potable water distribution piping will function as it currently does.

As discussed in FSAR Section 9.2.4.3, this system is not connected to any radiological systems. Since the system is physically independent of radiological systems, this change to the potable water system has no impact on the plant radiological systems. Should a radiological spill occur, the conclusions of the FSAR for the potable water system are not changed - i.e., the change to the distribution piping is in the same general vicinity as existing potable water distrubition piping and cannot be directly impacted from a radiological spill outside the plant. Therefore the requested change does not adversely affect any FSAR described design function.

LDCR-SA-2012-2, EV-CR-2012-000111-3 (TJEW):

Justification:

Components FT-142, FT-143, FT-144, and FT-145 are identified within DBD-EE-004 and FSAR Table 7.5-7B (Sheet 3 of 11) as being required to be seismically qualified. The devices are in fact only required to be seismically mounted. DBD-EE-004 section 4.3.2.1.1 states 'Category 2 instrumentation supplied from Non-1E power need not be

LDCR-SA-2012-2, EV-CR-2012-000111-3 (TJEW)(continued):

seismically qualified unless it is part of a seismic Category I system and its failure may degrade this system.' The flow transmitters are N1E, Process Instruments(Isolable,

Offline). DBD-ME-028, attachment 19, identifies that these types of transmitters are ANSI class: N/A, IEEE Class: N1E, Seismic Category: None, Safety Related: No, and EQ: No.

DBD-ME-028, Attachment 19, Table 1, note 3, is associated with components identified as N1E, Process Instruments(Isolable, Offline). This note states that Seismic Category II mounting applies to the non-1E transmitter or gauge. This mounting requirement is identified within Maximo-MEL. The Location-Qualification screen denotes a Component Function Code 2 ('CFC2') attribute of ‘SMTQ’ for FT-142, FT-143, FT-144, and FT-145. From Table 6 of DBD-ME-028, Attachment 19, the SMTQ designation identifies that the components require "Seismic Integrity Evaluation and Mounting of Non-safety Related Instruments Connected to ANSI 2 or 3 Fluid Systems."

Delete "Note 5" for the Variable RCP Seal Water Injection Flow on Sheet 3 of 11 in Table 7.5-7B

LDCR-SA-2012-28, EV-CR-2012-003858-2 (JDS):

FSAR section 15.5.1.2 states under section 4 of the assumptions that "The pressurizer heaters are assumed to be de-energized (the pressurizer heaters at CPNPP are automatically de-energized upon receipt of a safety injection signal by safety-related SSPS relays)." In section 15.5.1.3 of the FSAR, turning off the pressurizer heaters is incorrectly included in the discussion with SI for actions that must occur within 14 minutes of receiving an inadvertent safety injection. Corrected discussion for consistency.

LDCR-SA-2012-23, EV-CR-2012-009618-2 (CBC):

The discussion for RG 1.108 is updated to indicate the frequency for the integrated test sequence / loss of offsite power testing is revised from "At least once every 18 months" to "At the frequency specified in accordance with Technical Specification 5.5.21, 'Surveillance Frequency Control Program' ".. As documented in EV-CR-2011-002186-21, this change was evaluated and approved in accordance with TS 5.5.21.

LDCR-SA-2010-12, EV-CR-2008-003933-5 (GLM):

In the first paragraph, change direct driven open compressors to hermetically sealed type compressors.

Justification:

Plant ventilation chilled water system, subsystem 2, chiller compressors are no longer hermetically sealed type and are now direct driven open compressors. Changing from a hermetically sealed compressor to a direct drive compressor has no impact on the function of the chillers or the plant ventilation chilled water system.

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LDCR-SA-2012-22, EV-CR-2010-006088-3 (JCH):

Section 3.6B.2.1.1 Reactor Coolant System (RCS) Main Loop Piping

Description: Add the following to the end of Section 3.6B.2.1.1-

"The new Leak Before Break (LBB) leak rate calculation methodology for Alloy 82/182 welds with Structural Weld Overlay (SWOL), has been approved by the Nuclear Regulatory Commission for use in Pressurizer Surge Line (PSL) LBB analysis at another nuclear power plant. This methodology is applicable for CPNPP and has been utilized for the LBB analysis for Alloy 82/182 welds with SWOL for the pressurize surge line nozzle."

Justification:

The revised LBB analysis was performed by Westinghouse and is applicable to both CPNPP U1 & U2 Pressurizer surge lines with SWOL of Alloy 52/52M. The analysis was applied at the Alloy 82/182 Weld locations. The design margin of ten on leak rate, two on flaw size and margin of one on loads remain sufficient to account for changes due to SWOL. Further, the LBBvalidation is not a safety concern since weld overlay has been applied to mitigate the Alloy 82/182 PWSCC issue. Alloy 52/52M is resistant to PWSCC. The revised analysis demonstrated that the required LBB margins are met with no adverse impact on SSCs design function as provided in the referenced UFSAR sections.

The use of LBB analysis was previously approved by the Commission. The current LBB analysis evaluates the piping with the addition of a SWOL on the DMW in order to mitigate PWSCC. The current LBB analysis uses a conservative crack morphology factor that accounts for the PWSCC on Alloy 82/182 weld. The conclusions are that all the recommended margins are satisfied. It is concluded that the previous LBB analyses conclusions remain valid and pressurizer surge line breaks should not be considered in the structural design basis of the Comanche Peak Units 1 and 2 after the SWOL application at the Alloy 82/182 weld locations.

The NRC has approved the current methodology for the Waterford Plant as documented in their safety evaluation. This NRC approved methodology may be utilized by other Licensees so it is not considered a departure from a method of evaluation under 10 CFR 50.59.

The use of the revised LBB analysis with SWOL does not constitute a departure from a method of evaluation described in the UFSAR used in establishing the design basis. Therefore NRC approval and license amendment are not required.

LDCR-SA-2012-6, EV-CR-2012-000748-2 (JDS):

Correct the characterization of the HI-TRAC to show that it is a critical load along with other new heavy loads hoisted over spent fuel in the HI-TRAC/MPC and an MPC. Other paragraph formatting and clarifications were made to reflect the use of the Holtec Dry Cask Storage. The corrected, updated and reformatted FSAR description of critical loads

LDCR-SA-2012-6, EV-CR-2012-000748-2 (JDS) (continued):

moved by the Fuel Building Overhead Crane does not change the description of the crane or its functions to control heavy loads. The correction to the description of the HI-

TRAC as being a critical load is in accordance with the CPNPP heavy loads program and NUREG-0554.

LDCR-SA-2012-16, EV-CR-2011-004569-3 (SCD):

Further detail can be found in the Engineering Basis of FDA-2011-0000133-11 and -12.

Title 10, Part 73, “Physical Protection of Plants and Materials,” Section 73.54, “Protection of Digital Computer and Communication Systems and Networks,” requires licensees provide high assurance that digital computer and communication systems and networks are adequately protected against cyber attacks, up to and including the design basis threat as described in 10 CFR Part 73, Section 73.1.

LDCR-SA-2012-26, EV-CR-2011-012069-5 (TJEW):

FSAR Table 3.9B-10 is being updated to add the active function of “Containment Penetration Protection” for FW check valves FW-0195 through FW-0202. FW check valves FW-0199 through FW-0202 are being revised to indicate that valves 1-FW-0199 through 1-FW-0202 have a Normal Position of "Closed," while valves 2-FW-0199 through 2-FW-0202 still have the Normal Position of "Open." FSAR Table 3.9B-10 is being updated and does not adversely affect or alter the design or functional requirements of the check valves.

On Table 3.9B-10, Sheet 5 of 22,

a.) for valves FW-0195, FW-0196, FW-0197, and FW-0198, add the following words in the last column after the words, "AFW Flow Path"

", Containment Penetration Protection"

b.) add the following four valves after valve FW-0198:

1-FW-0199 FW Check 6 2 Self-Acutated Closed AFW Flow Path, Containment Penetration Protection

1-FW-0200 FW Check 6 2 Self-Acutated Closed AFW Flow Path, Containment Penetration Protection

1-FW-0201 FW Check 6 2 Self-Acutated Closed AFW Flow Path, Containment Penetration Protection

1-FW-0202 FW Check 6 2 Self-Acutated Closed AFW Flow Path, Containment Penetration Protection

LDCR-SA-2012-26, EV-CR-2011-012069-5 (TJEW) (continued):

c.) Also on the same table and sheet, add "2-" to the for the followng valves:

FW-0199

FW-0200

FW-0201

FW-0202

and add the following words in the last column after the words, "AFW Flow Path"

", Containment Penetration Protection"

d.) for valves FW-0195, FW-0196, FW-0197, and FW-0198, add the following words in the last column after the words, "AFW Flow Path"

", Containment Penetration Protection"

LDCR-SA-2012-4, EV-CR-2012-000575-2 (TJEW):

Justification: Liquid mercury is known to promote liquid metal embrittlement (LME) of steels, resulting in premature intergranular cracking. There is the potential for lamp breakage to occur and mercury being released into the pools. Therefore, an evaluation by Westinghouse has been completed to provide a technical analysis of the possible effects of mercury on the integrity of the fuel, stainless steel liner plates, storage racks, and the RCS. After reviewing several possible lamp breakage scenarios and materials compatibility information and using the most limiting scenario, it was determined that the small amount of mercury contamination expected due to lamp breakage will have no significant impact on material integrity in the specified locations. The SPF filters provide removal of fission products and other contaminates by means of filtration and ion exchange. Therefore, any dissolved mercury will be removed via the purification loops.

On page 9.5-79 add the following words in the new section 9.5.3.3:

9.5.3.3 Underwater Lighting

High Pressure Sodium underwater lights are used in the spent fuel pools, transfer canals and wet cask loading pit in the Fuel Building. These lights are subject to inspections and precautions and limitations to control the free mercury contained in the bulbs. These controls minimize the likelihood of the loss of the mercury and requires an evaluation if mercury is lost and not subsequently recovered.

In Table 17A-1, sheet 41 of 58, add the following line at the end of section 37.:

LDCR-SA-2012-4, EV-CR-2012-000575-2 (TJEW) (continued):

"Spent Fuel Pool and Transfer Canal Underwater lights N/A Mfrs Stds - NONE Note C 9.5.3 Note 72, 90 "

In Table 17A-1, sheet 58 of 58, on Note 90:

Delete the word "Portable" and capitalize the "U" in "underwater"

LDCR-SA-2012-8, EV-CR-2007-002309-00-52 (JDS):

Correct descriptions of instruments located in the turbine building (i.e., non-safety related areas) and the feedwater control and bypass valve appurtenances (located in non safety related area mounted on non-safety related valves) Required corrections as noted to be consistent with other FSAR Sections (7.2.1.1.2, 8.3.1.4, and TS Bases 3.7.3). The correction that the instruments located in the turbine building and the feedwater control valve appurtenances are non-safety related is required to be consistent with FSAR Tables 17A-1 and 17A-2 which correctly show that Class 1E, seismic None items are not Nuclear Safety Related. NNS FCVs are backup to the Safety Class 2 FIVs.

These are administrative changes to correct errors made in the FSAR. The changes make these sections consistent with the other FSAR sections.

Corrected Main Feedwater Control and Bypass Valve Status. Seismic Qualification is not applicable. The correction is required to be consistent with FSAR Tables 17A-1 and 17A-2 which correctly show Class 1E, seismic None.

Deleted "safety related" with respect to the turbine trip pressure switches. Section 10.4.7 requires corrections as noted to be consistent with other FSAR Section 8.3.1.4. The correction that the instruments located in the turbine building are non-safety related is required to be consistent with FSAR Tables 17A-1 and 17A-2 which correctly show that Class 1E, seismic None items are not Nuclear Safety Related.

The change makes this section consistent with the other FSAR section.

Changes to correctly show NNS FCVs to be backup to the Safety Class 2 FIVs. Required correction as noted to be consistent with TS Bases 3.7.3.

The changes make this section consistent with the TS Bases.

Separated the Seismic None 1E devices from the Seismic 1E devices to make the application of Note 77 clearer.

LDCR-SA-2012-19, EV-CR-2012-006572-2 (TJEW):

FSAR Table 3A.3-2 references Table 3-1. References to Table 3-1 should read 3A.3-1. This is an administrative issue only and should be corrected in FSAR amendment.

LDCR-SA-2012-19, EV-CR-2012-006572-2 (TJEW) (continued):

On sheet 1 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 2 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 3 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 5 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 6 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 7 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 8 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 9 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 10 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 11 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 13 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 14 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 18 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 19 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 23 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On sheet 30 of Table 3A.3-2, change all references to Table 3-1 to Table 3A.3-1.

On page 5.3-12, last sentence of the first paragraph in section 5.3.1.6.2.1, change "Table 3-1" to "The Table below".

On page 6.2-59, last paragraph of sections 6.2.5.1.4, delete ", Table 3-1".

On page 8.3-47, second paragraph of section 8.3.1.2.4, delete the words "Table 3-1 of"

LDCR-SA-2012-24, EV-CR-2011-013842-8 (JDS):

Section 15.6.5.2.6 is revised to remove the calculated values for Peak Clad Temperature, Local Maximum Oxidation, and Core-Wide Oxidation because the values are reported in Tables 15.6-8, 15.6-9, 15.6-10, and 15.6-11. This change removes redundant information from the FSAR.

LDCR-SA-2012-24, EV-CR-2011-013842-8 (JDS) (continued):

Table is revised to include results for assessment of the thermal conductivity degradation (TCD) issue identified in NRC Information Notice 2011-21 while crediting peaking factor burndown.

LDCR-SA-2013-2, EV-CR-2011-012069-12 (TJEW):

On Table 3.9B-10, Sheet 5 of 22, for valves, FW-0195 to FW-0198, 1-FW-0199 to 1-FW-0202, and 2-FW-0199 to 2-FW-0202,

1.) replace the words under the column "Normal Position" with the words "Open/Closed" and

2.) replace the words under the column "Function (Note 5)" with "See Note 9, below"

On Table 3.9B-10, Sheet 22 of 22, add Note 9 that says,

"9) The feedwater check valves in the auxiliary feedwater lines to the steam generator open to provide the AFW flowpath during design basis events. These valves must close in the event of a line break upstream of the outboard check valve in order to terminate a Condition III loss of feedwater event. In addition, these valves also must close post-LOCA to protect the cold containment penetration during intermittent operation of AFW."

LDCR-SA-2012-18, EV-CR-2011-008432-7 (JDS):

Figures 6.2.1-1 and 6.2.1-2 are removed and the List of Figures for Section 6.0 is updated to reflect their deletion as a result of the impact of NSAL 11-5 "Westinghouse LOCA Mass and Energy Release Calculation Issues" and correct the Time for Peak Temperature for DEPSG - MAXSI.

Figure 6.2.1-1 removed as a result of the impact of NSAL 11-5 "Westinghouse LOCA Mass and Energy Release Calculation Issues" and correct the Time for Peak Temperature for DEPSG - MAXSI.

Figure 6.2.1-2 removed as a result of the impact of NSAL 11-5 "Westinghouse LOCA Mass and Energy Release Calculation Issues" and correct the Time for Peak Temperature for DEPSG - MAXSI.

Section 6.2.1.1.3.2, the reference to WPT-17251 with regards to EQ qualification and the containment pressure and temperature analysis for LOCA is removed since the EQ qualification was revised based to reflect the impact of NSAL 11-5 in RXE-LCA-CPX/0-106.

The values listed in FSAR Table 6.2.1-2 were taken from WPT-17127 "Revised LOCA Containment Response Data", Table 2. The Time for Peak Temperature for the DEPS-MAXSI was incorrectly transcribed into the FSAR from the time for peak sump

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LDCR-SA-2012-18, EV-CR-2011-008432-7 (JDS) (continued):

temperature (1722.0 sec) instead of the time for peak steam temperature (24.0 sec). These values are corrected on Table 6.2.1-2.

LDCR-SA-2012-27, EV-CR-2012-010400-1 (SCD):

Administrative change from OSL (Optically Stimulated Luminescence) to TLD (Thermoluminescent Dosimeter) use at CPNPP.

LDCR-SA-2012-21, EV-CR-2012-008468-1 (SCD):

Update of titles and resumes of CPNPP Key personnel occuring since last FSAR Amendment

LDCR-SA-2012-20, EV-CR-2010-000880-7 (SCD):

Corrective actions for CR-2010-000880. ANSI-N45.2.9-1974 does not discuss electronic records, therefore the statement about duplicate storage of electronic records per the ANSI is incorrect and is being removed.

LDCR-SA-2013-3, EV-CR-2012-009307-2 (SCD):

The ability to pulse the air scrub was identified by the vendor after the system was installed. The omission of the pulsation device was overlooked and not included in the original design. Without a pulsation device the air scrub would be ineffective and this would result in permanent fouling of the Reverse Osmosis membranes.

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LDCR-SA-2013-5, EV-CR-2009-005301-00-29 (TJEW):

Revise Figure 8.2-4 to show breaker E11 (CB 8075) and its air switches.

Revise Figure 8.2-5 Sheet 1 of 2 to show breaker E11 (CB 8075) and its air switches.

LDCR-SA-2013-1, EV-CR-2012-000272-18 (JDS):

The change clarifies that the top and bottom 6 inches of each fuel rod, or blanket region, typically contain fuel of lower enrichment (2.6 w/o U-235) than found in the remaining fuel rod length (central zone). This clarification is necessary because not all fuel rods have a blanket region with different enrichment. For example, the Unit 2 Cycle 14 core contains 4 fuel assemblies with blanket enrichments equivalent to the central zone enrichment of 2.308 w/o U-235.

LDCR-SA-2013-8, EV-CR-2013-001939-1 (JDS):

Updated FSAR Table 6.3-11, RWST Outflow for a Large Break LOCA with a Worst Single Failure, with the correct "Cumulative Change in RWST Volume" and correct the typo "12,46" to "12,464" Previously approved in LDCR SA-2006-010 (EVAL-2005-003364-03) but failed to be incorporated into the FSAR update. Typo is a clarifiction of volume of the RWST (12,464 gallons) or 6% indicated level of 13,450 gallons as discussed in note 11 of Table 6.3.11.

LDCR-SA-2012-12, EV-CR-2012-003165-2 (JCH):

Description: Add the following valves to Table 3.9B-10, "Active Valves"

Valve ID: DO-0111/DO-0187/DO-0211/DO-0287

System: DO

Valve Type: RELIEF

Size In.: 1-1/2

ANS Safety Class: 3

Method of Actuation: SELF-ACTUATED

Normal Position: CLOSED

Function: Pressure Relief

LDCR-SA-2012-12, EV-CR-2012-003165-2 (JCH) (continued):

Justification: These valves are required to protect diesel fuel oil transfer pumps when backpressure from the "Y" strainers requires it. Since this could happen during the Emergency Diesel Generator mission time (30 Days) and no credit had been taken for online maintenance or operator action to protect the pumps, these valves must be classifed as ACTIVE to perform their function as described in the FSAR.

LDCR-SA-2012-5, EV-CR-2012-000036-1 (JCH):

Description: The following changes to FSAR Section 10 are required to document these system changes.

1. Section 10.4.3.1.2 - Feedwater Pump Turbine Seals

Delete "The feed pump turbine seal steam system also incorporates a steam seal dump valve to establish the required flow of seal steam through the seal steam supply header."

2. 10.4.3.2.1.a Normal Operation Conditions

Delete "Excess steam pressure is dumped to the condenser through the steam seal dump valve."

3. 10.4.3.2.2.b Feedwater Pump Turbine Steam Seal System

Revise Para. 1 and 2, "In the event of excessive packing wear or blowout, the packing leakage increases drastically; this causes the steam seal supply valve to modulate close and the steam seal dump valve to open enough to maintain sealing pressure of four psig. As the SSSV controller senses a rising gland steam pressure, its pneumatic output increases to throttle close the supply valve. If the wide open steam seal dump steam seal supply valve cannot maintain pressure at four psig, the pressure increases until the relief valve starts opening at about 20 psig. At 25-psig seal header pressure, the relief valve is wide open together with the steam seal dump valve, and further increase in flow increases header pressure beyond 25 psig.

Loss of air supply to the controller causes the steam seal supply valve and the steam seal dump valve to open wide, and any pressure increase beyond 20 psig again opens the relief valve."

4. 10.4.3.5.2 Feedwater Pump Turbine Steam Seals

Revise Para. 1, "Gland steam header pressure is maintained at a three psig set point by modulation of the steam seal supply valve and the steam seal dump valve. A split range controller sets the air pressure at each the control valve so that sufficient steam is admitted from the auxiliary Steam System (if needed) and excess steam is dumped to the condenser."

LDCR-SA-2012-5, EV-CR-2012-000036-1 (JCH) (continued):

Justification:

As a result of these changes, the FWP Turbine Gland Steam System will operate with a single supply valve (u-PV-3918A) that modulates as needed to provide sealing steam and maintain header pressure at approximately 3 psig. A loss of air supply, loss of controller input or loss of controller output will cause the supply valve to fail open to provide the maximum sealing steam pressure. With the wide open supply valve, the pressure will increase until the system relief valve (uGS-0110) starts opening at about 20 psig. At 25 psig seal header pressure, the relief valve is wide open. The relief valve is sized to handle the full steam flow and pressure from a wide open supply valve. There is no dump valve to in this configuration.

LDCR-SA-2012-1, EV-CR-2010-004331-62 (JCH):

Description: Add the following to Table 9.3-9, "Failure Mode and Effects Analysis Chemical and Volume Control System Active Components - Normal Plant Operation and Load Follow" on sheet 34 of 35:

Component: 55. Motor operated gate valve 8402A

Failure Mode: Fails closed.

CVCS Operation Function: Charging and Volume Control - charging flow.

Effect on System Operation and Shutdown: Failure inhibits use of normal charging line to RCS for boration, dilution, and coolant makeup operations. Seal water injected path remains available for boration of RCS to a hot standby concentration level and makeup of coolant during operations to bring the reactor to hot standby condition.

Failure Detection Mode: Valve position indication (closed to open position chang) and group monitoring light (valve closed) at CB.

Justification:Justification:

This modification will resolve OMA Group 1 "Normal Charging Isolation" with regards to manual operator actions of motor operated valves 1(2)-8105 and 1(2)-8106. The new 3-inch motor operated valve is classified as Nuclear Safety Class 2. The control switch will be used to isolate and bypass 1(2)-HCV-0182 during maintenance and in case of a fire to ensure that the normal charging path is isolated. Replacement of this manual operated valve with a motor operated gate valve will maintain all design and operational functions of the existing valve. Addition of valve 1(2)-8402A will have similar failure modes and actions to those listed for isolation valves 1(2)-8105 and 1(2)-8106. Therefore, there is a minimal increase in the likelihood of occurrence in a malfunction.

The requirement that the valve is a "PASSIVE" open valve.

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LDCR-SA-2012-1, EV-CR-2010-004331-62 (JCH) (continued):

Description: Add the following to Page 5A-10 after Normal Charging Isolatin Valves 8105 and 8106-

"Alternate Charging Isolation Valve 8402- If this normally open motor-oprerated valve closes spuriously, operator action can be used to denergized the valve operator and reopen the valve with its handwheel.

Justification:Justification:

This modification will resolve OMA Group 1 "Normal Charging Isolation" with regards to manual operator actions of motor operated valves 1(2)-8105 and 1(2)-8106. The new 3-inch motor operated valve is classified as Nuclear Safety Class 2. The control switch will be used to isolate and bypass 1(2)-HCV-0182 during maintenance and in case of a fire to ensure that the normal charging path is isolated. Replacement of this manual operated valve with a motor operated gate valve will maintain all design and operational functions of the existing valve. Addition of valve 1(2)-8402A will have similar failure modes and actions to those listed for isolation valves 1(2)-8105 and 1(2)-8106. Therefore, there is a minimal increase in the likelihood of occurrence in a malfunction.

the requirement that the valve is a "PASSIVE" open valve.

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LDCR-SA-2013-15, EV-CR-2013-007598-1 (GLM):

In the 5th paragraph, clarify the description of the location of instrumentation in section 9.2.2.3.

For #7 "Thermal Barrier Cooler", clarify the description of the location of instrumentation in section 9.2.2.5.3.

Technical Justification: Correction is required by CR-2013-007598. The flow diagram (FSAR Figure 9.2-3) shows the instrumentation (temperature and flow) downstream of each thermal barrier. The ICDs attached to the CR show the redundant and diverse design. The importance of redundancy and diversity in the instrumentation and the valves should be clarified because the Thermal Barrier Rupture event is an intersystem LOCA and could challenge the CCW system outside containment. See SSER 17 for additional information.

LDCR-SA-2013-10, EV-CR-2012-012849-3 (JCH):

Section 10.2.2.7.5 Turbine Stop and Control Valves

Description: Page 10.2-12

Correct information on the closure time of the extraction steam non-return check valves from "two (2) seconds" to "one (1) second".

Justification:

The non return check valves of the extraction steam system are credited for prevention of turbine overspeed with a one second closure requirement to isolate the turbine from the moisture separator reheaters (Ref: DBD-ME-205, Rev 12, Section 4.3.1.2). The one second closure time was also specified in specification 2323-MS-021D.2,which the valve was purchased under. Changing FSAR Section 10.2.2.7.5 to identify the correct closure timerequirement will bring it into alignment with the remainder of the FSAR descriptions of the extraction steam check valves, the DBD, and the valve specification.

LDCR-SA-2013-9, EV-CR-2013-003868-1 (JDS):

FSAR 9.3.4.1.2.3, Reactor Makeup Control System, does not explicitly address the increases in RCS boron caused by burnable poison depletion during the first months of operation. Other sections of the FSAR (e.g. 4.2.2.3, 4.3.2.1, and 4.3.2.4.11) address increased in chemical shim (boron acid concentration) in the primary coolant to offset reactivity effects of burnable poison depletion.

Include description of burnable poison depletion for consistency with chapter 4 of the FSAR.

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LDCR-SA-2013-20, EV-CR-2013-003797-2 (JDS):

Update FSAR 15.4.7.2 to clarify the wording related to core loading verification in a manner consistent with the referenced analysis, WCAP-16676-NP, Ref. 1 to FSAR Section 15.4.7. Clarification needed to ensure the FSAR text is consistent with the CPNPP practice and as outlined in WCAP-16676-NP.

LDCR-SA-2013-22, EV-CR-2013-008406-1 (RAS):

Revise the 2nd sentence in the last full paragraph:

"Power relief valves limit system pressure for large positive pressure transients. In the event of a large load reduction, not exceeding the design plant load rejection capability, the pressurizer power operated relief valves might be actuated for the most adverse conditions, e.g., the most negative Doppler coefficient, and the maximum incremental rod worth."

by replacing the phrase "...the most negative Doppler coefficient, and the maximum incremental rod worth." with "…Beginning of Life conditions."

Westinghouse has advised that the Analysis of Record for a large load rejection not exceeding the plant design capability for the most adverse conditions was for beginning of life, not for most negative Doppler coefficient and maximum incremental rod worth.

LDCR-SA-2013-23, EV-CR-2013-003664-1 (RAS):

Corrects an editorial error by revising a label from "HI-4" to HI-1". There is no "HI-4" signal and the indicated signal path is correctly referred to as "HI-1" in plant design documents.

LDCR-SA-2013-12, EV-CR-2013-005620-1 (TJEW):

On page 8B-2,

1.) in the third paragraph, change the word "modificaitons" to "modifications" and

2.) in the forth paragraph, remove the sentence, "In the defuelled mode, TS do not apply."

LDCR-SA-2013-19, EV-CR-2013-010298-1 (RAS):

Revise the last paragraph in section 3.6B.2.1.1 to specify use of Waterford LBB methodology for Alloy 82/182 welds with structural weld overlay.

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LDCR-SA-2013-24, EV-CR-2013-008362-6 (SCD):

Transitioning Nuclear Oversight (NOS) to be a direct reporting relationship to the Vice President Station Support is based on industry experience and supports the INPO Performance Objectives and Criteria for organizational effectiveness. This organizational structure is consistent with industry practices and helps ensure that Nuclear Oversight independence remains strong. This organizational change will ensure that NOS maintains a strong reporting relationship up to the CNO and other external organizations. The responsibilities of the Director, Oversight and Regulatory Affairs have been re-assigned to the Manager, Nuclear Oversight. The personnel qualifications section, 17.2.1.5.1, has been changed to correspond with Table 13.1-1.

Correct an error that was introduced to the TMI section during incorporation of LDCR-SA 2007-001. Wording was changed to specify that the Performance Improvement Department is assigned the functions of the former ISEG organization.

This organizational change will ensure that NOS maintains a strong reporting relationship up to the CNO and other external organizations. The responsibilities of the Director, Oversight and Regulatory Affairs have been re-assigned to the Manager, Nuclear Oversight.

The responsibilities of the Director, Oversight and Regulatory Affairs have been re-assigned to the Manager, Nuclear Oversight. The personnel qualifications section, 17.2.1.5.1, has been changed to correspond with Table 13.1-1.

The responsibilities of the Director, Oversight and Regulatory Affairs have been re-assigned to the Manager, Nuclear Oversight.

LDCR-SA-2013-4, EV-CR-2012-006134-46 (GLM):

Update Section 3.5.1.4 to include the identification and description of the tornado missile impact analysis results on the Auxiliary Feedwater Pump Turbine Exhaust Stack that is not contained within reinforced concrete building or structures.

Add missing details to the FSAR for identification for this safety-related system/ components subjected to tornado-generated missiles. This also resolves the NRC question and identified NCV issue documented in the NRC Inspection Report dated 07/ 24/2012. Calculation CS-CA-0000-5493 was generated to support the impact analysis, while attempts to locate historical design information supporting the asbuilt configuration could not be located or recovered.

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LDCR-SA-2013-17, EV-CR-2013-009990-1 (JDS):

The peak overpower is stated to be 118% which is inconsistent with Tables 4.1-1 and 15.0-4, which lists overpower to be 118.5%. The correct overpower was verified to be 118.5% for both units by Westinghouse LTR-TA-13-8 and LTR-TA-12-124. The overpower value has been updated to 118.5% for consistency between sections of the FSAR and with the assumptions made in the design basis analysis.

Figures 6.2.1-3 and 6.2.1-4 are both related to containment pressure temperature analysis for MSLB and do not include LOCA information. This wording which implies that these graphs contain LOCA related data was to be updated in a previous FSAR change. Reference to LOCA with regards to Figures 6.2.1-3 and 6.2.1-4 has been removed as originally required by a previous change to the FSAR (LDCR 2012-018).

A previous change to the FSAR (LDCR 2012-008) also updated the information concerning the containment analysis that supports environmental qualification evaluations in Section 6.2.1.1.3.2. This update included the phrase "modified GOTHIC run" which is misleading since no change to the actual GOTHIC code was implemented. The "modified" refers to different design inputs to GOTHIC that were used to support EQ analysis. This paragraph already describes the difference in design inputs and so "modified" has been stricken for clarity.

LDCR-SA-2013-21, EV-CR-2013-001520-1 (GLM):

In the 2nd paragraph change "...normal supply is unavailable." to "...Condensate Storage Tank is depleted."

Justification

Clarify SSW/AFW Interface to show that SSW is the long term back-up to the CST. It is used after the Tech Spec usable volume of the CST is depleted.

The Technical Specification BASES for the ST also cites the back up function. SER Sections 9.2.5, 9.2.6, and 10.4.9 are consistent with the FSAR; but are more clear that failure of the CST is not postulated. SER 9.2.5 is the most specific and makes it clear the SSI water is in addition to the CST, not in lieu of the CST.

"Additional water is also available from the service water system."

Calculation ME(B)-088 determined the flow based flow on demand after depletion of the CST rather than on full flow in lieu of the RWST. This is consistent with the design of the SSW backup function.

In the 6th paragraph update the SSI volume that would be used if the CST was unavailable in a beyond the design basis event for the current licensed thermal power of 3612 MWT

LDCR-SA-2013-21, EV-CR-2013-001520-1 (GLM) (continued):

Justification

The change administratively updates the SSI volume that would be used if the CST was unavailable in a beyond the design basis event to be consistent with the current licensed thermal power of 3612 MWT [(3612 MWT X 60 gal/MWT)/325,900 gal/acre-ft = 0.665 acre-ft].

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LDCR-SA-2013-7, EV-CR-2012-011569-2 (GLM):

The description of the source of potable water to the site and references to the Potable Water Storage Tank and well pumps given in FSAR Sections 9.2.4.2, 9.2.4.3, and 9.2.4.4 are changed to reflect that the site receives water solely from the Somervell County Water District and no longer uses the Potable Water Storage Tank or well pumps. The change to the source of site water is facilitated by the modifications described in FDA-2013-000044-1 and in SFER-4606965.

The modifications described in FDA-2013-000044-1 and in SFER-4606965 facilitate a change to the source of potable water to the site. As shown in 10CFR50.59 Screening 59SC-2013-000044, none of the modifications adversely affect the design bases of the Potable and Sanitary Water System described in FSAR Section 9.2.4.1. SFER-4208539 determined that the Somervell County Water District (SCWD) is capable of being the sole provider of both water pressure and capacity sufficient to fulfill the design basis functions of the Potable and Sanitary Water System. Additionally, the adequate operation of the Potable Water System with SCWD as the preferred water source since the implementation of FDA-2009-001930-01 and SFER-4208539 indicates that the Potable and Sanitary Water System can draw water solely from the Somervell County Water District and continue to fulfill its design basis functions. Therefore, all components associated with the onsite production of water can be disconnected from the rest of the system and abandoned in place.

LDCR-SA-2013-26, EV-CR-2013-001054-8 (GLM):

In the last paragraph, change the last sentence to say ".... safe shutdown of the other unit, or simultaneous shutdown of both units and maintaining them both in a safe shutdown condition."

In the 4th paragraph, change 93 acre-feet to 94 acre-feet and change 3.8 ft to 3.9 ft. In the 5th paragraph, update the max water consumption at 30 days and 39 days to 78.4 acre-feet (766'-4") and 99.4 acre-feet (765'-4"), respectively.

In the 2nd paragraph, change 765 ft 8 in to 766 ft 4 in. at 30 days and 765 ft. 4 in. at 39 days.

Technical Justification: The Safe Shutdown Impoundment Hydrothermal Analysis, ME-CA-0000-3264, was updated due to Stretch Power Uprate. It simulated one unit in a two train cooldown due to a Loss Of Coolant Accident (LOCA) and the other unit during a two train cooldown with a Loss Of Offsite Power (LOOP) as required by RG 1.27. The starting elevation is 769 feet 6 inches which is the invert of the equalization channel between the SSI and Squaw Creek Reservoir. The result for a LOCA is summarized in Table 14 of ME-CA-0000-3264 Revision 3, which shows that the maximum SSI loss due to evaporation of 93.7 ac-ft and a minimum surface water elevation after 39 days is 765 feet 7 inches. The difference between 769 feet 6 inches and 765 feet 7 inches is approximately 3.9 feet. The result for a two train cooldown with a LOOP is summarized in Table 16 of ME-CA-0000-3264 Revision 3, which shows the maximum SSI loss due to evaporation of 78.4 ac-ft at 30 days and 99.4 ac-ft at 39 days, corresponding to water

LDCR-SA-2013-26, EV-CR-2013-001054-8 (GLM) (continued):

surface elevations of 766 ft 4 in and 765 ft 4 in, respectively. Therefore, FSAR Section 2.4.11.5 need to be updated to account for both the effects of an accident in one unit, to permit the safe shutdown of the other unit, or simultaneous shutdown of both units and maintaining them both in a safe shutdown condition. FSAR Section 2.4.11.5 will also be updated to state that the minimum water elevation is 766 ft. 4. At 30 days and 765 ft. 4 in. at 39 days. FSAR Section 9.2.5.3.3 needs to be updated to state that the maximum consumption is approximately 94 ac-ft, and the decrease in surface water elevation is 3.9

LDCR-SA-2014-1, EV-CR-2013-008265-5 (JDS):

Update tables to reflect PCT errors due to ECCS model changes in Best Estimate LOCA for Unit 1 and Unit 2

LDCR-SA-2014-2, EV-CR-2012-011620-14 (SCD):

The nozzle check valve is a ASME Section III Class 1513 valve with a pressure rating of 3632 psig at 100F and a design rating of 2485 psig at 650F per drawing MD22754 Revision E (VDRT-3902424). This meets the requirements of 2323-MS-43B Category 2501 component. For flow characteristics, Westinghouse reviewed the new valve and the old valve and concluded the flow changes are insiginificant and acceptable; the Westinghouse conclusions regarding 2SI-8819A are documented in WPT-17762. The replacement valve is designed and manufactured in accordance with the requirements of ASME B&PV Code Section III, Class 1, 1974 Edition through winter 1975 Summer Addenda.

LDCR-SA-2014-3, EV-CR-2013-003295-5 (SCD):

Update table 17A-1 to include alternate penetration insert.

LDCR-SA-2014-5, EV-CR-2013-011618-2 (GLM):

In Section 1.2.2.8.2 add comma after the word "functions."

This is an editorial change only. This comma was supposed to be included via LDCR SA-2003-013, but for some reason it was not.

LDCR-SA-2014-7, EV-CR-2013-011896-1 (GLM):

Add drawing M1-0216-01 to the list of drawings for Figure 9.3-1 on page 9-xiii..

Add drawing M1-0216-01 to Figure 9.3-1.

Technical Justification: M1-0216-01 is correctly denoted as included in Figure 9.3-1; however, it is not included in the table of contents or in the FSAR Figure. This indicates the error was more than likely made when the FSAR special figures were reverted to

LDCR-SA-2014-7, EV-CR-2013-011896-1 (GLM) (continued):

plant drawings in the 1990's to reduce such errors. This is an editorial change because it corrects an error in incoporating previously approved changes.

Technical Justification: M1-0216-01 is correctly denoted as included in Figure 9.3-1; however, it is not included in the table of contents or in the FSAR Figure. This indicates the error was more than likely made when the FSAR special figures were reverted to plant drawings in the 1990's to reduce such errors. This is an editorial change because it corrects an error in incoporating previously approved changes.

LDCR-SA-2014-6, EV-CR-2013-010811-1 (SCD):

The gamma detectors provide the radiation monitoring during routine operation at CPNPP while the additional beta/gamma radiation detectors availability are necessary for less frequent and emergency condition which might warrant them.

LDCR-SA-2014-9, EV-CR-2013-008403-6 (SCD):

Incorporated editorial corrections to the last paragraph in section 6.3.5.4.1 by replacing "taking" with "the" in the first sentence and "less" with "loss" in the second sentence.

“The change from 28 ft to 22 ft for required NPSH is based on the vendor. The change in NPSHa from 32 ft (based on a previous empty alarm setpoint) to 26.7 ft is based on the current alarm setpoint (based on the end of injection mode of operation as described in Section 6.3.2.2.10.2). These changes reflect corrections to the NPSH calculations to reflect the current plant design.”

LDCR-SA-2014-11, EV-CR-2013-008397-4 (TJEW):

The following is an NRC finding documented in the NRC Component Design Bases Inspection Report, IR 2013007:

The inspectors identified a Severity level IV, non-cited violation of 10 CFR 50.71(e)(4), requires the UFSAR be updated, at intervals not exceeding 24 months, and states in part, "the revisions must reflect all changes made in the facility or procedures described in the UFSAR." Specifically, prior to June 20, 2013, the inspectors identified the alternate power diesel generator system was not described in sufficient detail in the FSAR as required. This LDCR corrects the problem.

On page 8.3-1, Section 8.3.1.1. subsection 3, add the following words, "CPNPP has also provided a set of Alternate Power Diesel Generators (APDGs) for each unit with the capability to connect to a Class 1E train at a time to provide defense-in-depth for safe shutdown of a unit during outages or during extended duration of an inoperable offsite circuit on occurrence of a beyond design basis event of concurrent loss of offsite power and failure of EDGs. The APDGs may provide 3450 kVA to provide long term cooling of each unit to respond to a beyond design basis event."

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LDCR-SA-2012-13, EV-CR-2012-005259-3 (TJEW):

Figure 1.2-1 is revised to show the location of XST1A

Page 8.2-1 will add a new paragraph after the 7th paragraph and will state, "Startup transformer XST1 and alternate startup transformer XST1A are connected to a common overhead line from the 138-kV switchyard. Each transformer is provided with a 138-kV motor-operated air switch such that each transformer can be energized independent of the other transformer."

In the last paragraph on the same page, the word "spare" will be replaced with "alternate"

On page 8.2-2, after the first partial paragraph, the following new paragraph will be added, "Alternate startup transformer XST1A is located under the 138-kV line to XST1 (refer to Figure 8.2-1) to serve as a replacement of XST1 after a future plant modification to connect cable buses from secondary X and Y windings of XST1 and XST1A to transfer panels to provide 138-kV offsite power to Units 1 and 2 safety related buses."

In the next paragraph after the new paragraph above, replace the word "Spare" with "Alternate", delete the word "dedicated" and deleted the last sentences that says, "This spare transformer may be physically relocated to a dedicated location near XST1, to serve as a replacement of XST1."

Figure F 8.2-1 is revised to show location of alternate startup transformer XST1A and revise designation of XST2A.

On Figure 8.2-4 show the electrical network interconnection of XST1/ XST1A and revise the detail for XST2/XST2A.

On Figure 8.2-7 Sheet 1, XST1A willl be added to the transmission lines going to XST1 from the 138kV switchyard.

On Figure 8.2-7 Sheet 2, XST1A will be added to the 138kV transmission tower.

Figure 8.2-9 will be updated to show transformer XST1A in the 138kV transformer connections plan and elevations and new air switch configuration for transformers XST1 and XST1A

Figure 8.2-11 will be updated to reflect the as-built cable bus configuration.

New Figure 8.2-11A will show the as built details of the XST1A cable bus configuration to the 6.9kV switchgear.

On page 9.5-17, last sentence of section 9.5.1.5.6, replace the word "spare" with "alternate, and at the end of the paragraph add the following words, "Alternate transformer XST1A is separated from adjacent structures by a three-hour rated fire wall."

LDCR-SA-2014-15, EV-CR-2013-008397-12 (TJEW):

The linkage to beyond design basis event is being deleted as a result of LDCR SA-2014-011 because the FSAR is a design basis document, i.e., a 10CFR50.2 plant, in FSAR section 8.3.1.1.1.

On page 8.3-1, LDCR SA-2014-011 inserted a paragrapgh after section 8.3.1.1.1.3 that said, "CPNPP has also provided a set of Alternate Power Diesel Generators (APDGs) for each unit with the capability to connect to a Class 1E train at a time to provide defense-in-depth for safe shutdown of a unit during outages or during extended duration of an inoperable offsite circuit on occurrence of a beyond design basis event of concurrent loss of offsite power and failure of EDGs. The APDGs may provide 3450 kVA to provide long term cooling of each unit to respond to a beyond design basis event."

This LDCR will modify the above paragraph as follows:

1. In the first sentence, add the words, "non-safety related" before Alternate Power Diesel Generators.

2. In the first sentence delete the words, "a beyond design basis event of."

3. In the second sentence delete the words, "to respond to a beyond design basis event."

4. Add the following sentence to the end of the paragraph, "See Section 8.3.1.2.1.7.h for a discussion of the sizing of APDG cables."

On page 8.3-38,

1. second paragraph and second sentence of section h., relocate the words "during modes 5 and 6" to after the word "APDGs."

2. third paragraph and first sentence of section h, replace the "," between 6,9kV to a "period."

LDCR-SA-2014-14, EV-CR-2013-008362-17 (SCD):

Due to CPNPP organizational and upper management changes, Chapter 17 and the TMI Section of the FSAR is being revised.

LDCR-SA-2014-16, EV-CR-2013-011292-1 (JDS):

Update core thermal power output from 3458 MWt to the currenctly licensed power level of 3612 MWt.

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LDCR-SA-2014-17, EV-CR-2014-001610-1 (JDS):

Include exception for use of the 193 core offload spaces for storage to assure ability to comply with Cask Technical Specifications. This will assure that 32 spaces will remain available for cask offload.

Clarification. The word "hose" should be "hoist".

LDCR-SA-2014-18, EV-CR-2012-005259-93 (TJEW):

Add Figure 8.2-11A to the List of Figures for Chapter 8 of the FSAR on page 8-iv:

8.2-11A Spare Start-up Transformer XST1A Cable Bus Connection to 6.9 KV Switchgear

LDCR-SA-2014-13, EV-CR-2014-006885-1 (SCD):

Due to CPNPP organizational and upper management changes, Chapter 13 of the FSAR is being revised.

LDCR-SA-2011-4, EV-CR-2010-005675-1 (JDS):

Update UFSAR Section 15.6.2 to provide revised post-SPU EAB and LPZ estimated doses for a CVCS letdown break during a system alignment using 3 letdown orifices at Mode 3 operation where the system is below 500 degrees F. The Thyroid and Whole Body doses limits for the EAB and LPZ are based on a small fraction (10%) of the 10CFR100 as directed by the applicable SRP guidelines. 10CFR100 dose limits for the thyroid and whole body doses are 300 rem and 25 rem, respectively. The resulting dose margins derived remain less than a small fraction of the 10CFR100 limits.

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LDCR-SA-2014-20, EV-CR-2013-003649-5 (JCH):

Table 3.9B-10: Active Valves

Description: Correct typo for valve 2DO-0052 (shown as 2DD-0052 on Sheet 15)

Move valves DO-0111, DO-0187, DO-0211 on page 19 and DO-0287 on page 20 to sheet 15 and change the function of these valves from Pressure Relief to Recirculation Flow Path.

Justification: Pressure relief is not the active safety function of the subject valves. These valves provide Recirculation Flow Path (miniflow) for the fuel oil pumps. The subject valves were also inserted in the wrong location in the table.

LDCR-SA-2013-6, EV-CR-2010-009018-22 (JDS):

Clarification by providing a more detailed nominal dmension of the Region 1 spent fuel storage racks in the fuel building.

Update the minimum Boron Concentration for the spent fuel pools to be consistent with License Amendment 162.

Replace Table 9.1-4 with a revised table reflecting the revised spent fuel pool criticality analysis approved in TS amendment 162.

Update criticality codes used to reference SCALE 5.1 and PARAGON to reflect codes used in the spent fuel pool criticality analysis.

Update to include descriptive information for the revised spent fuel pool criticality analysis approved in amendment 162 and revise the references to the WCAP used in the license amendment.

LDCR-SA-2014-21, EV-CR-2014-010150-2 (SCD):

This activity lowers the minimum acceptable filter mesh size of the Reactor Coolant System (RCS) Filter from 0.1 micron to 0.05 micron in DBD-ME-255 to reduce source term dose in the Chemical Volume and Control System (CVCS).

LDCR-SA-2014-26, EV-CR-2013-006623-5 (JDS):

Correct inconsistency between Table 6.2.4-1,Table 6.2.4-2, Figure 6.2.4-1 and Table 6.2.4-3.

Update the function and classification of 1/2-HV-2154 and 1/2-HV-2155 to no longer function as Containment Isolation Valves. The containment isolation function was previously transferred to locked closed manual isolation valves, FW-0113 and FW-0116. FSAR Tables 6.2.4-1 and 6.2.4-2 identify FW-0113 (Item #22) and FW-0116 (Item #20) as the Feedwater Sample line containment isolation valve for penetration MI-5 and MI-6,

LDCR-SA-2014-26, EV-CR-2013-006623-5 (JDS) (continued):

respectively. Table 6.2.4-1 references Figure 6.2.4-1 Arrangement 16 as the containment isolation valve arrangement which depicts a locked closed isolation valve outside of containment. Table 6.2.4-1 also references FSAR Figure 10.4-9 which shows FW-0113/0116 as normally locked closed for containment isolation. Table 6.2.4-2 describes the isolation valves as locally operated manual valves. This is consistent with DBD-ME-203, Rev. 36 Section 4.3.1.2.i and 5.4.1 which identify a locked closed manual valve containment isolation arrangement fulfilled by FW-0113 and FW-0116. However, previous FSAR updates missed the removal of the reference to the properties of the former containment isolation valves in Table 6.2.4-3 and incorrectly describes the valves as having a containment isolation signal, power source, and closure time. The Table also incorrectly shows the normal valve position as open. DBD-ME-013, Attachment 2 (Items 20 and 22) provides the correct properties for FW-0113/0116 that should be reflected in Table 6.2.4-3.

Table 6.2.4-3 is updated to be consistent with DBD-ME-203, DBE-ME-013, FSAR Tables 6.2.4-1 and 6.2.4-2, and FSAR Figure 6.2.4-1.

LDCR-SA-2014-12, EV-CR-2014-005546-1 (SCD):

Spent fuel pool swing gate pressure seals have been re-classified as safety related from not safety related components. Additional information will be added to show the new spent fuel pool swing gate equipment as shown in the following attachments. (Reference FDA-2013-000052-01)

LDCR-SA-2014-22, EV-CR-2014-005228-1 (SCD):

FSAR Tables 3.2-3 and 3.2-4 are updated to reflect the flow diagram split that created drawings M1-0269 Sh. C and D.

The Ch. 11 Listing of Figures is updated to reflect the flow diagram split that created drawings M1-0269 Sh. C and D.

Figure 11.3.-1 is updated to reflect the flow diagram split that created drawings M1-0269 Sh. C and D.

LDCR-SA-2014-19, EV-CR-2013-008947-12 (SCD):

The pressure gauges for the airlocks are special cases due to their Safety Class 2 function as part of the airlocks. Typical pressure gauges on Safety Class piping are classified as NNS because they are isolable and/or minor leakage is acceptable. The subject valves are not accessible and post-LOCA increase in leakage is not acceptable. Therefore, they are being classified as ANSI Safety Class 2, Seismic Category I under FDA-2012-000230-02.

FSAR Amendment 106a

LDCR-SA-2014-25, EV-CR-2014-002884-11 (JDS):

Revise Note (a) to Table 15.0-4 (applicable to Overtemperature N-16 function). The response times of RCS Cold Leg RTD and OTN16 RTD instrumentation is changed from 6.0 and 3.3 seconds to 6.7 and 3.3 seconds respectively. However, this does not change the overall OTN16 RTD response time requirement of 9.3 seconds and therefore has no adverse affect on the operation of the Tcold RTD signal or the existing safety analyses for the OTN16 function since the safety analyses is insensitive to a trade-off between the RTD response time and the instrumentation response time. Therefore, there is no change to overall response time of 9.3 seconds assumed in the applicable safety analyses.

LDCR-SA-2015-1, EV-CR-2015-000856-1 (SCD):

Additional information will be added to show the new spent fuel pool swing gate equipment. LDCR SA-2014-012 added multiple changes to table 17A-1 and it was found that the addition to system 16 was incorrect and needs to be removed.

Paragraph added to section to describe the passive nature of the refueling gate seals.

LDCR-SA-2015-4, EV-CR-2011-013548-1 (JEB):

ISFSI Security Plan and Cyber Security Plan have been added to the CPNPP security program. Update FSAR to include the new plans by simplifying the section 13.6 statement to make it inclusive of all the plans called for in 10CFR73:

Proposed Wording:

A comprehensive security program has been developed at CPNPP in accordance with the requirements of applicable portions of 10CFR73, "Physical Protection of Plants and Material." The security plans that describe the program contain security related information (SRI) or safeguards information (SGI) and are withheld from public disclosure in accordance with 10CFR2.390 or 10CFR73.21. The CPNPP security plans are presented in separate submittals.

LDCR-SA-2014-23, EV-CR-2010-004331-98 (RAS):

Add the following description for the AFW valves: "The three AFW valve are normally de-energized to prevent a fire induced hot-short from causing a mal-operation of the valve(s). This also de-energized position indication in the Control Room. Therefore, these valves are "Locked Closed" to provide positive indication of correct valve position during normal operation."

LDCR-SA-2015-7, EV-CR-2014-012820-1 (SCD):

This LDCR is a clarification of information in Section 6.5.1.1, which states "Efficiencies of the HEPA filters and iodine adsorbers are in accordance with Table 9.4-4." Testing of the HEPA filters following significant painting, a fire or chemical hazard ensure compliance with the efficiencies in Table 9.4-4. FSAR Table 6.5-1 is revised where Regulatory Guide 1.52, Revision 2 is used by CPNPP rather than Revision 1.

LDCR-SA-2014-28, EV-CR-2014-005590-3 (JDS):

The results obtained from ANSYS version 14.0 for 24 of the 26 problems compared were found to have exact numerical matches to those obtained from version 13.0. The results from the 2 problems that did not match were further evaluated for the significance of their mismatches. The detailed evaluation of these 2 problems found that the results from one differed by less than 1-percent which is the ANSYS expected analytical tolerance. A reviewof the input script for the second problem discovered that the version 14.0 input script was different than what had been used in version 13.0. When the script used for version 13.0 was evaluated by ANSYS version 14.0, it was found to differ by less than 1-percent which is the expected ANSYS analytical tolerance. The input script is user provided input that describes the analytical problem to be solved by the program. A change to the input script is not a change to the program. A review of changes made by ANSYS did not find that any significant issues or problems had been corrected in version 14.0. It is therefore the conclusion of this Evaluation that the analysis results obtained from ANSYS version 14.0 are essentially the same as those obtained from ANSYS

LDCR-SA-2014-28, EV-CR-2014-005590-3 (JDS) (continued):

version 13.0 and ANSYS version 14.0 is not considered a departure from a method of evaluation described in the UFSAR.

LDCR-SA-2015-8, EV-CR-2015-000067-6 (JDS):

FSAR section 9.1.3.2 currently states that the SFP Boron concentration is "approximately the same" as the RWST. As described in EV-CR-2015-000067-1, it is desired to maintain the Spent Fuel Pool boron value between 2600-3000 ppm. The RWST is normally maintained 2400-2600 ppm, which is the TS limit applicable in MODES 1-4 of operation, during which SFP and RWST water do not mix. In Emergency Operating procedures, guidance exists to fill the RWST from the SFP during beyond-design-basis events. As described in EV-CR-2015-000067-1, the boron concentration of this emergency water source does not need to satisfy the normal RWST TS limits, but has its own Technical Specification requirement and SFP water at a boron concentration of up to 3000 ppm remains a valid source of water during this potential beyond-design-basis event.

LDCR-SA-2015-10, EV-CR-2012-011869-1 (JDS):

Revise Section 6.2.2.2 to reference a single table 6.2.1-9, as Tables 6.2.1-9A and 6.2.1-9B have been combined into a single table in Amendment 103 (LDCR-2008-013).

Revise 6.3.2.8 to remove obsolete information (references to Appendix 4A and 4B which were previously removed in LDCR-SA-2008-013) and provide an appropriate reference to Tables 6.2.1-9 and 6.2.1-12.

ReviseTable. 6.3-7 to remove obsolete information (references to Appendix 4A and 4B which were previously removed in LDCR-SA-2008-013) and provide an appropriate reference to Tables 6.2.1-9 and 6.2.1-12.

Delete the last sentence on the page since Cycle/Unit specific information for Units 1 and 2 is no longer provided in Appendices 4A and 4B as these Appendeces were previously removed in LDCR-SA-2008-013.

LDCR-SA-2015-12, EV-CR-2015-006885-2 (SCD):

6.3
In the PSAR (RESAR 3), passive failures were considered coincident with hot leg recirculation at 24 hours. The FSAR originally assumed hot leg recirculation at 24 hours; however, that has changed over time without addressing if the passive failure timing. FDA-2015-000095-01 clarifies the DBD based on the conclusion that the passive failure was considered at switchover to hot leg recirculation and that would be the current licensing basis. Therefore, the FSAR should be updated in accordance with 10CFR50.71(e).

LDCR-SA-2015-13, EV-CR-2015-002140-2 (GLM):

9.2-2
In the last paragraph, correct the CCW fouling factor from 0.0017 to 0.0020.

Justification
Errors were made in previous LDCRs (LDCR-SA-2001-021 and LDCR-SA-2010-001) as described in AI-CR-2015-002140-1. LDCR-SA-2001-021 defined the design fouling as 0.0020 with no plugging and 0.0015 with the plugging allowance. Plugging allowances are no longer used nor are they required. LDCR-SA-2010-001 changed the design fouling factor in two places (not all) and did not change the factor with plugging. The 10CFR50.59 basis for the 2010 LDCR was FDA-2006-003080-01 and -02. These FDAs discuss fouling; HOWEVER, they did discuss a change in the design fouling factor. The actual design fouling factor was not changed. Calculation RXE-LA-CPX/0-018 R8 dated 10/28/08 was performed at the design fouling of 0.002. Calc RXE-LA-CPX/0-018 R9 for power uprate did not change the design fouling; however, it noted it would exceed the 135F limit on the operating unit during a two train CCW outage on an outage unit. Per the attached email and RXE-LA-CPX/0-018 R9, the current design fouling remains 0.002. RXE-LA-CPX/0-020 Rev. 9 for power uprate used a fouling factor of 0.0017 for the single train forced cooldown. The reduction was made to maintain the normal operating temperature limit of 122 F used for the piping analysis. The subject cooldown would be an abnormal condition no subject to the 122F limit. The impact review CVC for this calc revision was not in conformance with ECE-5.03 or ECE-5.01. This revision was the basis for LDCR-SA-2010-001. This calculation was subsequently revised to make corrections. The current version of RXE-LA-CPX/0-020 is R12 dated 11/12/2013 and the design fouling of 0.0020 has been restored. STA-734, Service Water System Fouling Monitoring Program, includes the actual number of plugged tubes in the monitoring based on calculation ME-CA-0229-2188. Plugging limits are not used.

Therefore, correction of the design fouling factor and the deletion of the plugging allowance are required to update the FSAR to be current as required by 10CFR50.71€.

9.2-3
In the first paragraph, correct the CCW fouling factor from 0.0017 to 0.0020. Delete the first and third sentences. In the last sentence, insert the word "fouling" between "periodic" and "monitoring."

LDCR-SA-2015-13, EV-CR-2015-002140-2 (GLM) (continued):

Justification
Errors were made in previous LDCRs (LDCR-SA-2001-021 and LDCR-SA-2010-001) as described in AI-CR-2015-002140-1. LDCR-SA-2001-021 defined the design fouling as 0.0020 with no plugging and 0.0015 with the plugging allowance. Plugging allowances are no longer used nor are they required. LDCR-SA-2010-001 changed the design fouling factor in two places (not all) and did not change the factor with plugging. The 10CFR50.59 basis for the 2010 LDCR was FDA-2006-003080-01 and -02. These FDAs discuss fouling; HOWEVER, they did discuss a change in the design fouling factor. The actual design fouling factor was not changed. Calculation RXE-LA-CPX/0-018 R8 dated 10/28/08 was performed at the design fouling of 0.002. Calc RXE-LA-CPX/0-018 R9 for power uprate did not change the design fouling; however, it noted it would exceed the 135F limit on the operating unit during a two train CCW outage on an outage unit. Per the attached email and RXE-LA-CPX/0-018 R9, the current design fouling remains 0.002. RXE-LA-CPX/0-020 Rev. 9 for power uprate used a fouling factor of 0.0017 for the single train forced cooldown. The reduction was made to maintain the normal operating temperature limit of 122 F used for the piping analysis. The subject cooldown would be an abnormal condition no subject to the 122F limit. The 9.2-3 impact review CVC for this calc revision was not in conformance with ECE-5.03 or ECE-5.01. This revision was the basis for LDCR-SA-2010-001. This calculation was subsequently revised to make corrections. The current version of RXE-LA-CPX/0-020 is R12 dated 11/12/2013 and the design fouling of 0.0020 has been restored. STA-734, Service Water System Fouling Monitoring Program, includes the actual number of plugged tubes in the monitoring based on calculation ME-CA-0229-2188. Plugging limits are not used.

Therefore, correction of the design fouling factor and the deletion of the plugging allowance are required to update the FSAR to be current as required by 10CFR50.71€.

LDCR-SA-2015-15, EV-CR-2015-007540-5 (GLM):

9.2-3
After the last sentence in Section 9.2.1.2.2, add clarification that more restrictive limits are imposed on SSI temp and CCW Hx fouling on the operating unit during outages.

Technical Justification: CCW design fouling and fouling monitoring are described in the FSAR for all normal and accident conditions. However, during the abnormal condition where the CCW on an outage unit is unable to provided cooling to common shared loads, the operating unit is capable of cooling the essential loads. However, restrictions on valve alignments, SSI temperature and/or CCW heat exchanger fouling may be imposed to assure operability is not affected. It may be neccesary to limit the number of supplied components to protect the CCW pumps. CR-2014-007235 is an example of such precautions taken. FDA-2014-000175-01 clarified DBD-ME-229 for the alignment of both spent fuel pool heat exchangers to one unit.

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LDCR-SA-2015-15, EV-CR-2015-007540-5 (GLM) (continued):

9.2-7
In the last sentence on the page, add the word "normally" between "is" and "required." After the last sentence on the page, add clarification that during outages it may be necessary to take both safeguards trains and/or the non-safeguards loop of CCW out of service for maintenance. In this case, the common components are cooled by the operating unit. In this abnormal alignment, more restrictive limits may be imposed as described in Section 9.2.1.2.1.

Technical Justification: CCW design fouling and fouling monitoring are described in the FSAR for all normal and accident conditions. However, during the abnormal condition where the CCW on an outage unit is unable to provided cooling to common shared loads, the operating unit is capable of cooling the essential loads. However, restrictions on valve alignments, SSI temperature and/or CCW heat exchanger fouling may be imposed to assure operability is not affected. It may be neccesary to limit the number of supplied components to protect the CCW pumps. CR-2014-007235 is an example of such precautions taken. FDA-2014-000175-01 clarified DBD-ME-229 for the alignment of both spent fuel pool heat exchangers to one unit.

LDCR-SA-2015-18, EV-CR-2014-013507-2 (SCD):

In the FSAR Table 6.3-3 Sheet 2 of 3, in the "Valve Identification" column for the Charging pump miniflow isolation valves it lists the valve numbers as "8810 and 8111". This is incorrrect, it should be "8110 and 8111".

Technical Justification:
Administrative change to correct error in valve identification.

LDCR-SA-2015-16, EV-CR-2015-008796-1 (SCD):

Clarify QA review of procedures in SORC.This is to support removal of ODA's and MDA's from SORC approval IAW EV-CR-2013-001259-1.

LDCR-SA-2015-17, EV-CR-2015-000973-2 (GLM):

Description of Change: Delete reference to Figure 9.3-3 on List Of Figures (Page 9-xiv).

Technical Justification: Flow diagrams M1-0216-01 and M2-0216-B were changed from referencing FSAR Figure 9.3-3 to referencing FSAR Figure 9.3-1 in CR 2014-008461. This change updates the List Of Figures to reflect this.

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LDCR-SA-2014-4, EV-CR-2012-0002652-27 (CBC):

The FSAR is updated to reflect changes to the plant and plant site required by NRC Orders EA-12-049 and EA-12-051. These changes are for “Beyond-Design-Basis External Events” and are not used nor credited by the current plant design basis. Section 1.2 is updated to reflect a general description of NRC Orders EA-12-049 and EA-12-051. Figure 1.2-1 is updated to show the site contains a FLEX Storage Building. The FLEX Storage Building protects equipment credited for NRC Order EA-12-049. Sections 9.1.3.2 (System Description) and 9.1.3.5 (Instrument Requirements) are updated to reflect the addition of the Spent Fuel Pool Level Instrumentation. FSAR Tables 17A-1 and 17A-2 are updated to reflect the quality assurance attributes of the Spent Fuel Pool Level Instrumentation.

LDCR-SA-2014-27, EV-CR-2014-012821-1 (JDS):

Section 6.2-3: Add a sentence that reflects that the temperature and pressure profiles utilized for EQ evaluations as described in section 3.11B bounds the limiting temperature and pressure transients for postulated LOCA and MSLB events. The sentence describes the relationship between containment integrity and environmental qualification. This addition ensures that changes to the containment integrity analysis will consistently cascade to the EQ analysis.

FSAR Table 6.2.1-2 describes CPNPP’s compliance with GDC-16 and -50 that the peak calculated containment pressure should be less than the containment design pressure of 50 psig following a LOCA and is based on the double-ended hot leg (DEHL) and double-ended pump suction leg (DEPS) breaks assuming minimum and maximum ECCS performance (MINSI and MAXSI) cases analyzed by Westinghouse using the EPITOME code. NSAL 14-2 identified errors in the LOCA M&E release analyses input modification program database and the input preprocessor. The analyses are sensitive to energy stored in the RCS metal, including the SG tubes and the program was using the density for stainless steel in determining the mass of the SG tubes and the stored energy based on a stainless steel specific heat. CPNPP steam generators are Alloy 690 (U1) and Alloy 600 (U2) which have different specific heat values than stainless steel. RXE-LA-CPX/0-106 “Impact of M&E Analysis Errors on CPNPP Post LOCA Containment Pressures and Temperatures Meeting Acceptance Criteria” determined that the pressure penalty is 0.2 psi and the temperature penalty is 0.3°F. These penalties were applied to the peak pressure and temperature values in the attached markup of Table 6.2.1-2.

The note added to Tables 6.2.1-2 and 6.2.1-2A as well as Section 6.2.1.1.3.2 describes the relationship between containment integrity and environmental qualification. This addition ensures that changes to the containment integrity analysis will consistently cascade to the EQ analysis.

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LDCR-SA-2015-19, EV-CR-2015-011212-1 (SCD):

Changes to CPNPP organizational structure due to management changes and restructuring.

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LDCR-SA-2016-06, EV-CR-2015-011847-1 (SCD):

Editorial change removing date from RG 8.7 reference on page 12.5-10

LDCR-SA-2015-11, EV-CR-2015-000483-2 (CBC):

FSAR Section 5.4.1.2 Item 2 is updated to provide a functional description of the SHIELD Shutdown Seal (SSDS) component.

FSAR Section 5.4.1.3.1 is updated to describe the test program for the SHIELD Shutdown Seal (SSDS), and discuss the RCS and RCP design functions that have been demonstrated as not impacted by implementation of the SSDS.

FSAR Section 5.4.1.3.2 is updated to state that actuation of the SHIELD Shutdown Seal (SSDS) on a reactor coolant pump that is still rotating will not have any measurable effect on the reactor coolant pump coastdown.

FSAR Section 5.4.1.3.4 is updated to state that actuation of the SHIELD Shutdown Seal (SSDS) on a reactor coolant pump that is still rotating will not effect the cooling flow to the reactor core.

FSAR Section 5.4.1.3.10 is updated to provide a description of the timeline of changes within the RCP after a “loss of all seal cooling” that result in the actuation of the SHIELD Shutdown Seal (SSDS) and how the SSDS actuates.

FSAR Section 9.3.4.1.2.1 is updated to provide a description of SHIELD Shutdown Seal (SSDS) activation due to “loss of all seal cooling” to limit RCP seal leakoff and to describe that should the RCP continue to run, that the SSDS is assumed to fail to limit leakoff.

FSAR Section 9.3.4.1.3.1 is updated, similar to Section 9.3.4.1.2.1, to provide a description of SHIELD Shutdown Seal (SSDS) activation while the RCP continues to run, that the SSDS is assumed to fail to limit leakoff, which is a RCS letdown flowpath.

FSAR Table 17A-1 is updated to provide a quality classification for the SHIELD Shutdown Seal (SSDS) note within the description of the RCP No. 1 Seal Housing and a new note to FSAR Table 17A-1.

LDCR-SA-2016-4, EV-CR-2014-005127-1 (CBC):

References to FSAR Figure 6.2.2-1 and Table 17A-1 are added to the text discussion. FSAR Section 6.2.2.2.1(6) "piping and valves" is clarified. These changes are clarifications to provide consistency within the FSAR.

LDCR-SA-2017-1, EV-TR-2016-008360-22 (SCD):

Wording change to describe General Employee Training after implementation of NEI Bulletin 16-26b.

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LDCR-SA-2016-3, EV-CR-2015-007215-1 (GWS):

Update the outdated description of the PHOENIX-P cross-section library Change "...42 energy group library, which has been derived mainly from the ENDF/B-V files." to "...70 energy group library, which has been derived mainly from the ENDF/B-VI files."

Comanche Peak began using the PHOENIX-P methodology as part of the reload safety analysis transition to Westinghouse that was reviewed and approved by the NRC in License Amendment 144 which was implemented beginning with Unit 2 Cycle 11 in the Spring of 2008. The UFSAR changes made to implement License Amendment 144 were done in conjunction with those to implement the power uprate (License Amendment 146) (UFSAR Amendment 103, LDCR SA-2008-13, EV-CR-2006-003080-00-94) and incorrectly used the outdated description of PHOENIX-P as using the 42 energy group ENDF/B-V libraries. Note that the License Amendment Request for the power uprate (LAR 07-004) included the correct description of the 70 energy group library based on the ENDF/B-VI files (WCAP-16840/enclosure 1 of TXX-07106) which was reviewed and approved by the NRC in License Amendment 146. This LDCR simply corrects an error in originally incorporating the previously approved change (License Amendments 144 and 146).

LDCR-SA-2016-2, EV-TR-2016-000825-1 (GLM):

Update Section I.C.5 to reflect the current CPNPP OE program.

LDCR-SA-2016-019, EV-CR-2010-008411-28 (JCH):

Delete Vacuum Breakers CP1/2-SWVAVB-03/04 from Table 3.9B-10 (Sheet 14 of 22)

LDCR-SA-2016-014, EV-CR-2015-008517-18 (KJV):

3.5.1.4 - Update to Turbine Driven Aux Feedwater (TDAFW) exhaust stack tornado missile analysis. Changes previously made by LDCR-SA-2013-04 which credit the Motor Driven Auxillary Feedwater Pumps providing the auxiliary feedwater design function following a tornado missile strike are deleted. New information which addresses the impact of tornado missiles on the Main Steam System supply and exhaust for the Turbine Driven AFW pump and the effect on the entire AFW system is inserted in its place.

10.4.9.3 - Clarify missile protection and describe the CST vents consistent with Sections 3.5.1.4, 9.2.6 and 10.4.7.1. Add a description of the Turbine Driven AFW Pump steam supply and exhaust and cross reference Section 3.5.1.4.

LDCR-SA-2017-007, EV-TR-2016-010272-1 (CBC):

Revise Section 1A(B) discussion of RG 1.01, "Emergency Planning for Nuclear Power Plants," to document use of NEI 99.01 Revision 6 for Emergency Action Level (EAL) development. CPNPP adopted and implemented EALs based on the NRC-endorsed guidance in NEI 99-01-R6 (NRC License Amendment 166 - ML16137A056 and ML16168A002).

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LDCR-SA-2016-008, EV-CR-2015-007780-6 (SCD):

T 11.5-2 - This change is being made due to new information received from the vendor with design authority, General Atomics (GA). Documentation received from GA (Ref. VDRT-5243381) states that the detectors located downstream of the sample pumps are limited to an extreme maximum operational temperature of 140 °F. A note is being added in lieu of changing the actual temperature value in FSAR Table 11.5-2 because this VDRT is only applicable to the gas detectors, not the whole gaseous channel associated with each detector.

SA-2017-005 EV-CR-2017-007035-1 (GWS):

15.1-9 - Currently the FSAR sates, “No credit is taken for the proper of the non-Class 1E steam dump solenoid valves when the low-low T_{avg} (P-14) setpoint was exceeded.” The actual low-low T_{avg} signal is P-12. P-14 replaced with P-12.

SA-2013-025 EV-CR-2010-004331-95 (JCH):

Description: Added the following valves to Table 7.4-1 on Sheet 6 of 6:
- Identification: LCV-0112B; Function: (d) Motor-driven Charging Pump-Suction Pressure
- Identification: LCV-0112D; Function: (d) Motor-driven Charging Pump-Suction Pressure

Justification: Installed new control switches on the Hot Shutdown Panel (1-LV-01 and 2-LV-01) for LCV-0112B and LCV-0112D to allow control of these valves from the Hot Shutdown Panel. This will isolate the portions of the control circuit that is routed in the Control Room and the Cable Spread Room.

Description: Added the following valves to Table 7.4-3 on Sheet 3 of 3:
- Identification: LCV-0112B; Function: (b) Motor-driven Charging Pump-Suction Pressure
- Identification: LCV-0112D; Function: (b) Motor-driven Charging Pump-Suction Pressure

Justification: Installed new control switches on the Hot Shutdown Panel (1-LV-01 and 2-LV-01) for LCV-0112B and LCV-0112D to allow control of these valves from the Hot Shutdown Panel. This will isolate the portions of the control circuit that is routed in the Control Room and the Cable Spread Room.

SA-2016-016 EV-TR-2015-010416-33 (CBC):

Section 1, 1.1, 1.4, T1.4-1 (including Table of Contents and List of Tables) - This section is updated to be consistent with NRC License Amendments 167 and 169 (transfer of Operating Licenses to Comanche Peak Power Company LLC (owner) and Vistra Operations Company LLC (operator)). Section 1.4 is updated to remove historical information and level of detail not required for this section. Table 1.4-1 is below the level of detail for this section and is deleted.

SA-2016-016 EV-TR-2015-010416-33 (CBC) (continued):

Sections 1A(B), 2.1, 2.3, 3.7B, 3.11B, 3A, T4.1-1, 6.6, 7.1, 7.2, 8.3, 9.5, 11.3, 11.4, 11A, 12.1, 12.4, 13.1, 13.5, 14.2, 15.3, 15.6, 17.1, 17.2, T17.2-2, 17A, III.A - These sections are updated to be consistent with NRC License Amendments 167 and 169 (transfer of Operating Licenses to Comanche Peak Power Company LLC (CP PowerCo) (owner) and Vistra Operations Company LLC (Vistra OpCo) (operator)). Most of the FSAR sections will refer to Vistra OpCo. CP PowerCo is used within the context of financial responsibility (e.g., decommissioning costs) or property ownership (e.g., mineral rights).

SA-2017-012 EV-CR-2016-007216-2 (CBC):

Table 5.4-16 - “Pressurizer Valves Design Parameters,” indicates that each of the two pressurizer power operated relief valves (PORVs) has a steam relieving capacity of 210, 000 lb/hr at 2235 psig. The pressure is changed from “2235” to “2335” for consistency with CPNPP design basis and the background discussion of Technical Specification Bases 3.4.11 “Pressurizer PORVs”. This is an editorial change correcting a typographical error.

SA-2015-002 EV-TR-2014-003190-4 (GLM):

Revise section 9.2.3.1 to reflect current chemistry control methodology and delete Tables 9.2- 9 and 9.2-10

Information in Tables 9.2-9 and 9.2-10 is not current with industry and CPNPP practice. The details are not necessary in the FSAR, and are contained in the Chemistry Data Sheets, CHM- 140-507-1 and CHM-140-507-2. A discrepancy exists between the specific conductivity limit for the Demineralized Water Storage Tank on chemistry form CHM-140-507-1 and the requirement documented in DBD-ME-241 "Demineralized and Reactor Makeup Water System" and FSAR, Section 9.2. This is acceptable because the DBD chemistry parameters are the specifications that the equipment was originally designed to, while the values in chemistry procedures and the Primary Chemistry Strategic Plan are based on recommendations from EPRI and Westinghouse which evolve to improve plant performance. Early in plant life, chemistry control was conducted in accordance with the design values specified in the DBD. When chemistry control methodology changed neither the DBD nor the FSAR was updated. A review was performed of the chemistry restrictions encompassed in DBD-ME-241 and FSAR Section 9.2. It also included CHM-140-507-1 and CHM-140-507-2, the chemistry procedure forms corresponding to DBD-ME-241 Attachment 1 and Attachment 2, respectively. A review of FSAR Section 9.2 found that chemistry restrictions have not been updated to correspond with the plant practice of following EPRI and Westinghouse guidance.

SA-2015-002 EV-TR-2014-003190-4 (GLM) (continued):

FSAR Table 9.2-9 presents chemistry restrictions for Demineralized Water and FSAR Table 9.2-10 presents chemistry restrictions for Reactor Makeup Water. These restrictions are not in conformance with Westinghouse guidance on the location of sampling or requirements for monitoring. For example, the FSAR states that "the quality of water in the Demineralized Water Storage tank is consistent with the water chemistry listed in Table 9.2-9"; however, the Westinghouse guidance specifies that the monitored values should be measured at the outlet of the water treatment plant rather than the tank itself. Also, the guidance does not require plants to monitor this value if, like CPNPP, it has not been incorporated into Technical Specifications.

Removing the outdated tables will ensure that information discrepancies don't occur in the future as data sheets are updated to keep up with changing chemistry guidelines and water conditions.

SA-2016-018 EV-TR-2016-002219-1 (SCD):

Section 13 Table of Contents and List of Figures:
The TOC is updated to reflect deletion of Appendix 13.1A and the LOF is updated for consistency with NRC License Amendments 167 and 169.

Section 13.1:
This section is updated to align with the current CPNPP organizational structure.

Appendix 13.1A:
Appendix 13.1A is deleted. The resumes of current key personnel are available upon request.

Figures 13.1-1, 13.1-2 and 13.1-3:
Updated figures to show current reporting reporting relationships and titles. Figure 13.1-1 is updated to be consistent with the license transfer associated with NRC License Amendments 167 and 169

SA-2017-004 EV-CR-2016-006853-3 (KJV):

Appendix 5A:
FSAR Appendix 5A is clarified to document the full environmental qualification of the PORVs that bound the worst case BTP RSB 5-1 shutdown. Confusion regarding compliance with BTP RSB 5-1 as described TXX-89204 (dated 5/19/89) is reflected in SSER 22 (additional details in FDA-2017-000084-01). Subsequent to the SSER, the PORVs were environmentally qualified for a harsh environment. This new information is added to the FSAR Appendix 5A to effectively supersede TXX-89204 and SSER 22 in accordance with 10CFR50.71(e).

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SA-2017-014 EV-TR-2017-007340-1 (CBC):

Sections 10.2.2.7, 10.2.2.7.1:
FSAR Sections 10.2.2.7 and 10.2.2.7.1 are revised to remove frequency influence turbine control features which have never been used at CPNPP. Frequency Influence is a feature of the turbine controls which has never been turned on at CPNPP. Frequency influence is grid frequency control. If turned on, the CPNPP turbine-generator EHC turbine load control would regulate the turbine steam valves in an attempt to control grid frequency at 60 hz +/- 0.05 hz. This would constantly impact reactivity; therefore, CPNPP has always been a base load unit with the fossil plants controlling grid frequency. The EHC turbine control is speed/load control only with frequency influence turned OFF. Since Frequency Influence must always be turned off, and the existing operation and monitoring (OM) system has never had these functions (no need to have an Operator capability on the operation and monitoring (OM) system to turn it on or impact it in any way).

SA-2016-010 EV-CR-2016-003267-8 (KJV):

Section 5.4.12:
Clarification provided for unit difference for lower instrument taps on pressurizer level instruments lower taps. Specifically, 1RC-8054A, B, and C are upstream of the reducing orifice whereas on Unit 2 the valves are downstream of the orifices. This configuration has existed since initial construction. This change is supported by 10 CFR 50.59 evaluation EV-CR-2016-003267-10.

Tables 5.2-3 and 5.2-7:
Revised to reflect unit difference and associated valve qualifications.

2017-008 EV-TR-2017-007342-1 (SCD):

Chapter 17
Revise FSAR Chapter 17 to address Nuclear Oversight organization, titles, and qualification changes.

2017-009 EV-CR-2017-001654-1 (KJV):

Appendix 9.4A:
Section 9.4A.2.1, “Containment Recirculation and Cooling System,” is revised to reflect a new modification (i.e., FDA-2013-000133-03) allowing use of silencers in the Unit 2 Containment Recirculation Cooling System during Cold Shutdown and Refueling (or when the core is offloaded). The purpose of this modification was to alleviate heat stress and reduce noise during outages.

Table 17A-1:
Table 17A-1 is updated to reflect use of silencers for Unit 2 Containment Recirculation and Cooling System as described in Section 9.4A.2.1.

FSAR Amendment 108

2017-015 EV-TR-2017-007709-1 (SCD):

App. 1A:
Add Reg Guide 8.10 and exception to Appendix 1A(B).

SA-2017-006 EV-CR-2012-009889-5 (CBC):

Section 3.4.2:
FSAR Section 3.4.2 is updated to describe the historical change in design-basis hydrostatic water level used in the design of safety-related structures. The change in hydrostatic water level was reviewed and documented by the NRC in SSER 25. This is an acceptable editorial change required for consistency with FSAR section 2.4.13.5 and SSER section 2.4.6. Previous ground water levels were retained for historical purposes.

SA-2017-013 EV-TR-2012-007391-1 (GWS):

Update of Section 10.4.3 to Feedwater Pump Turbine Gland Seals following FDA-2005-004990-01 (Unit 1 modification) and FDA-2005-004990-02 (Unit 2 modification).

SA-2017-011 EV-TR-2015-002656-2 (KJV):

Section 5A:
Corrects description of BTP requirement I.2 and II.1 for Cold Shutdown in 36 hours. RHR operational conditions is HOT SHUTDOWN, not COLD SHUTDOWN. This is a correction. BTP RSB 5-1 requires cold shutdown conditions in a reasonable period of time which means cold shutdown be expected to be reached in 36 hours.

SA-2016-009 EV-TR-2016-001674-6 (KJV):

8.2-3 & 8.2-7:
Revision to reflect renaming of DeCordova 345 KV line to Mitchell Bend.

Figure 8.2-1:
Revision to reflect renaming of DeCordova 345 KV line to Mitchell Bend.

Figure 8.2-4:
Revision to reflect renaming of DeCordova 345 KV line to Mitchell Bend.

Figure 8.2-12:
Revision to reflect renaming of DeCordova 345 KV line to Mitchell Bend.

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SA-2017-003 EV-CR-2016-004078-7 (KJV):

Section 3.5.1.4:
Clarifies discussion in FSAR Section 3.5.1.4 that the portions of the SSW discharge piping is not protected from a design basis missile impact but has been evaluated to show the SSW would perform it's safety function during and after a design basis tornado.

Section 9.2.1.4:
Is updated to note that the impact of tornado missiles on the SSW intake trash racks, traveling screens, and a small portion of discharge piping are discussed in further detail in Section 3.5.1.4.

Section 9.2.2.1:
Clarifies the operation / alignment of CCW during planned maintenance in outages.

Section 9.2.2.2.1:
Clarifies the operation of CCW during and after LOCA by noting restrictive limits on the nonsafeguards loop from the accident unit.

SA-2012-003 EV-TR-2009-005301-19 (KJV):

Figure 1.2-1:
Existing layout drawing revised to delete excessive and out of date information regarding switchyard configuration. Configuration is described in FSAR Section 8.

Section 8.2:
Revision made to reflect installation of 8075 breaker.

FSAR Amendment 108

SA-2015-014 EV-TR-2014-009974-1 (KJV):

15.6-6,8:
Update FSAR Section 15.6.3 to incorporate impacts of re-analyis of SGTR margin to overfill in response to NSAL 07-11. The CPNPP Steam Generator Tube Rupture (SGTR) margin-tooverfill analysis was revised by Westinghouse to demonstrate that margin to overfill exists (WPT-17834). The analysis was performed by the RETRAN using conservative assumptions to demonstrate the ability of the operator to limit the system transient and establish parameters for providing a bounding radiological consequence assessment. Important operator actions in the CPNPP emergency operating procedures were explicitly modeled in the analysis. the design input for the delivery time of Auxiliary Feedwater (AFW) to the steam generators was revised to more accurately reflect actual plant response. In the initial SGTR margin-to-overill analysis, Auxiliary Feedwater was delivered instantaneously to the steam generators concurrent with reactor trip. However, on a loss of off-site power, the turbine-driven AFW pump start automatically on a blackout signal and the motor-driven AFW pumps are sequenced to their respective power supplies. The start of the TD AFWPs on a blackout signal may be considered instantaneous; however, there is a 40 second delay for the MD AFWPs as the blackout sequencer establishes loads. A review of surveillance data for the TD AFWPs along with the consideration of the power supply alignment for the MD AFWPs determined that a 30 second delay for delivery of AFW to the SGs was conservative. This was modeled in the analysis as a delay in the actuation of AFW. The operator action to terminate Safety Injection thus increased from the current FSAR value of 2 minutes to 3 minutes.

SA-2017-016 EV-TR-2014-009429-1 (GWS):

9.2-1:
Added that during a loss of all Station Service Water, the Fire Protection water system may be aligned to cool the emergency diesel generator jacket water heat exchangers and the safety injection and centrifugal charging pump lube oil coolers.

SA-2016-012 EV-TR-2016-007332-2 (CBC):

F1.2-1:
FSAR Figure 1.2-1, "Plot Plan," is revised to remove the "TEMP SECURITY" box outside of the Southeast corner of the RCA Yard fence inside the Protected Area. This "TEMP SECURITY" box is below the level of detail required in the FSAR.

FSAR Amendment 109

SA-2011-009 EV-CR-2010-004331-93 (JCH):

Sections 8.3.1.4, 8.3.1.4.5, 9.5.1.2.2, and 9.5.1.5.5:
Updates text to reflect use of three-hour Thermo-Lag. Thermo-Lag is an intumescent fireproofing system. It was designed to provide 1 to 3-hour fire protection for various applications. Previously, CPNPP utilized only 1-hour Thermo-Lag. In order to ensure safe reactor shutdown during a fire, CPNPP requires the use of three-hour Thermo-Lag for a fire scenario outside the Control Room. FDA-2010-000172-11-04 provides testing and specifications demonstrating 3-hour ThermoLag's ability to resist a three-hour fire.

SA-2013-016 EV-TR-2012-005259-78 (JCH):

Section 9.5.1.5.6:
Revised to add new startup transformer XST1A to the list of transformers that are separated from the Turbine Buildings by a three hour fire wall.

SA-2014-010 EV-TR-2014-005538-3 (KJV):

Table 8.3-3 and Section 8.3:
Addition of alternate startup transformers XST1A and XST2A for off-site power sources.

SA-2017-002 EV-TR-2016-009440-4 (JCH)

Update Table 10.4-10 Sheet 3 to reflect new TPCW Heat load from new H2 Gas Dryer in support of FDA-16-000081-01. Value changes from .0051 to .0055.

SA-2017-010 EV-TR-2017-009826-1 (GWS):

Section 9.1.4.2.3:
Update description of the Fuel Handling Bridge Crane (FHBC) and load cell to reflect design Modification FDA-2016-000014-03 which eliminates use of an in line load cell.

SA-2017-017 EV-TR-2017-008401-1 (GWS):

Add drawing M2-0211-02 to Title for Figure 10.4-3 on Page 10-viii.

SA-2017-019 EV-TR-2017-009826-2 (GWS):

Sections 3.11B, 3A, Tables 7.5-7A, 7.5-7B, and 7.5-7C:
Clarification of the equipment qualification program scope for harsh and for mild program equipment.

SA-2017-020 EV-CR-2015-009663-2 (KJV)

Table 15.1-3:
Incorporate changes to MSLB dose consequence analysis per WPT-17889.

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SA-2017-021 EV-TR-2017-009826-4 (GWS):

Table 1.6-2, Sections 3.5.1.3 and 10.2.2.7.6:
Remove the design and analyses information for turbine missiles and revise Testing and Trip Devices to Turbine Valve Testing.

SA-2017-023 EV-TR-2017-010561-1 (GLM):

Update FSAR Sections 9.3.3.2.3 and 9.3.3.2.5 to describe that Auxiliary Building Sump 3 receives drainage from the discharge of the SW tunnel sump pumps and to describe the SW tunnel sumps that will be present in the Fuel Building, respectively.

SA-2017-024 EV-TR-2017-009826-12 (GWS):

Remove jacket water information from lube oil Section 9.5 and replace data in Table 9.5-18.

SA-2017-025 EV-TR-2017-012492-1 (KJV):

Add text to Refueling Section 5.4.7.2.3 that describes alternate reactor core cooling via the safety injection pumps through residual heat removal system.

SA-2017-026 EV-TR-2017-012252-1 (CBC):

Revise Chapter 13, Section 13.1.1.24 to align with commitment 03620 (opportunity to have interview with SAFETEAM).

SA-2018-001 EV-CR-2017-013071-2 (KJV):

Section 8.3.1.1.i1:
Update to EDG voltage control description while on magnetics.

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SA-2018-003 EV-CR-2016-008855-3 (KJV):

FSAR Section 17A.3 is clarified by changing the title of system 33 from “Fuel Handling Equipment” to “Fuel Handling Equipment and Fuel Assemblies”.

FSAR Table 17A-1 is clarified by changing the title of system 33 from “Fuel Handling Equipment” to “Fuel Handling Equipment and Fuel Assemblies”. The Safety Classification for Fuel Assemblies is corrected from Safety Class 2 to 3. During resolution of CR-2016-008855 (Corrective Action Program document to resolve discrepancy between FSAR and equipment list), Westinghouse advised that Fuel Assemblies are Safety Class 3, not 2, based on ANSI 51.1 (ANSI/ANS 51.1 (1983)). Since Table 17A-1 is based on ANSI N18.2 (see Note 7 of FSAR Table 17A-1) which does not provide a classification for Fuel Assemblies, Note 99 is added to Table 17A-1 to clarify that ANSI 51.1 is used as the basis for Safety Class 3. Since the Fuel Assemblies are not subject to the ASME Code by any regulation, by ANSI N18.2, or by ANSI 51.1 and the assemblies are classified as Safety Related and Seismic Category I independent of this change, the change is merely an administrative correction. However, the change is included in the screening for FDA-2018-000015-01 which corrects the Master Equipment List in Maximo.

SA-2018-004 EV-TR-2017-001152-2 (CBC):

Update (revise) the UFSAR Section 17.2 regarding ORC requirements. Specifically, added NDE and administrative controls to designated review activities, decreased ORC composition from six to five members, specified requirements for ORC meetings being conducted twice per year rather that once per six months, removed two outside member minimum for quorum requirement, and removed redundant requirement to forward audit program reviews to SVP & CNO.

SA-2018-006 EV-CR-2013-008394-74 (KJV):

Section 8.3.2.1:
Remove statement “Non-Class IE loads are not connected to these systems.” This was removed as a clarification to remove the inconsistency with FSAR Section 8.3.2.1 Item (7) (i.e., FSAR Figure 8.3-14 Sheet 6 circuit 10)

Section 8.3.2.1 Item 2.a “Capacity / Batteries”:
The battery capacity for BT1ED2 and BT1ED2 is changed from “1950” to “1944.” The amp-hour rating is revised per manufacture data.

Section 8.3.2.1 Item 2.b “Capacity / Battery Chargers”:
Added “+/-” in front of the tolerance range for float and equalize voltage values. This is an editorial clarification.

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SA-2018-006 EV-CR-2013-008394-74 (KJV) (continued):

Section 8.3.2.1 Item 7 “Identification of Safety Loads”:
Changes the reference to figures from “Figures 8.3-14 (Sheets 1 & 2) and 8.3-15B (Sheets 1 & 2)” to “Figures 8.3-14 and 8.3-15B”. This is an editorial correction since are no designated “Sheets 1 & 2”. Figures 8.3-14 and 8.3-15B are a compilation of several plant drawings as noted in Table 3.2-4.

Section 8.3.2.1 Item 7 “Identification of Safety Loads”:
Revised the sentence referencing the maximum length of operating time by changing “DC bus” to “battery”. This is an editorial clarification, Table 8.3-4 refers to battery load.

Table 8.3-4, “125-Vdc CLASS IE BATTERY LOAD REQUIREMENTS”:
The table is revised by updating the amperes, clarifying Note 1, and removing Note 2 to reflect approved engineering calculations.

SA-2018-008 EV-TR-2018-002691-1 (CBC):

Revise the Figure 1.2-1 to show the addition of the Stator Assembly Building, X-YD-3K34, south of Warehouse E.

SA-2018-009 EV-TR-2017-009826-3 (GWS):

Sections 9.5.4.2 and 9.5.6.2:
Add clarifying information for fuel oil transfer pumps and starting air systems with regard to emergency diesel generator operability for Technical Specifications 3.8.1, 3.8.2, and 3.8.3.

SA-2018-010 EV-TR-2018-001537-8 (KJV)

Table 1.6-1, Sections 1A(B) and 9.1.4, and Table 17A-1:
Add Wonderhoist to Miscellaneous Handling Equipment.

SA-2018-012 EV-TR-2018-004370-1 (GWS):

Section 7.7.1.3.1:
Add BEACON Axial Offset as a method to measure core power distribution in addition to flux maps made using the movable incore system.

SA-2018-017 EV-CR-2017-010633-29 (KJV)

Revise the discussion regarding ASME Code Case N-31-1 in FSAR Section 1A(B) Regulatory Guide 1.84.

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SA-2019-001 EV-TR-2017-012914-1 (GWS):

Remove Post Accident Sampling System from FSAR Table 17A-1, List of Quality Assured Structures, Systems, and Components.

SA-2019-003 EV-TR-2017-009826-16 (CBC):

Sections 1A(N), 1A(B), and 17.2.17:
The guidance for electronic storage requirements for Quality Assurance records is updated to reflect the best practices, evolved technology, improvements in electronic record keeping, and more specific details.

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SA-2016-015 EV-CR-2013-000105-3 (KJV):

Section 8.3.1.1.11.2:
Update FSAR Section 8.3.1.1.11.2 to add additional EDG trip functions described in other design documents. This is an administrative change to the facility documents and the FSAR. This activity will ensure consistency throughout design documents.

SA-2018-007 EV-CR-2017-011125-2 (KJV):

Section 8B:
Revise to allow a two train EDG outage when Technical Specifications allow.

SA-2018-011 EV-TR-2017-001993-5 (GLM):

Update section 3.8.3.1.2 to reflect that 2 keys, instead of 1, are used to satisfy the crane anti-collision interlock function.

SA-2018-014 EV-TR-2016-005840-53 (KJV):

Section 8.2.1:
The text discussion is updated to reflect the open phase condition equipment. This change is consistent with design documentation Final Design Authorization (FDA) numbers 2016-000005-02, 2016-00005-04, 2016-000005-05, and 2016-000005-06.

SA-2018-015 EV-TR-2018-004664-10 (CBC):

Figure 1.2-1:
"Plot Plan," is revised to show the addition of the crane column supports in both Turbine Building crane bay. The crane column supports added to the Turbine Building crane bays for both units were installed solely to support an engineering temporary lift assembly used to replace the main generator stators. They are located within the Protected Area but do not adversely affect any plant SSCs and serve no UFSAR-described function.

SA-2019-002 EV-TR-2017-004085-5 (GWS):

Figures 8.2-1 and 8.2-4:
Multiple new breakers added to switchyard for reliability.

SA-2019-006 EV-CR-2016-009517-2 (GLM):

Section 9.4E.2.1.b:
Clarifies the operation of the NNS chillers during a LOOP. The four NNS chillers are loaded on each of the four diesels by the blackout sequencer and are available to cool both containments, both PDP pumps and the secondary plant sampling package. There is no requirement for all four chillers to operate during a LOOP. This change is consistent with statements FSAR 9.4E.1.

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SA-2019-007 EV-CR-2019-003268-1 (KJV):

Section 9.5.5.2:
Adds specific reference to pre-licensing Question and Response (Q&R) regarding post-accident leakage from Emergency Diesel Generator jacket water. For consistency, this is an administrative change which adds explicit reference in FSAR section 9.5.5.2 to the questions and responses (Q&R) between the NRC and CPNPP with regard to allowable leakage from the EDG jacket water standpipe.

SA-2019-008 EV-TR-2018-009176-2 (GLM):

FSAR Section 9.2.1.1:
(Station Service Water System / Design Basis) and Section 9.2.5.3(1) (Ultimate Heat Sink / Safety Evaluation) are updated to reflect the results of the revised hydrothermal analysis of the Safe Shutdown Impoundment to include the most current heat loads from engineering calculations.

SA-2019-009 EV-CR-2019-002068-4 (GWS):

Tables 8.3-1 and 8.3-2:
Replace "AUTO SEQUENCER START" with "AUTO SEQUENCER SIGNAL" to be consistent with Technical Specification Bases for SR 3.8.1.11.

SA-2019-010 EV-TR-2018-007152-4 (CBC):

Section 1A(B) RG 1.33 Rev 2:
Removes biennial reviews for Emergency Response Guidelines (ERGs), Functional Restoration Guidelines (FRGs), and Abnormal Plant Operating Procedures (ABNs). The requirement for biennial reviews is replaced with reliance on current programs and processes. The changes are consistent with NRC RG 1.33 Revision 2. This change was approved by the NRC (i.e., CP-201900476 / ML19228A086, CP-202000024 / ML20016A318, and ML20054A276).

Sections 13.5.1, 17.2.1.6, and 17.2.2:
Updates programmatic discussion of procedures. The discussion of procedures is clarified by removing prescriptive, unnecessary, and inconsistent information. The changes are consistent with NRC RG 1.33 Revision 2. The change to the SORC meeting frequency (from once per month to periodically as determined by the SORC chairman or the designated alternate) was approved by the NRC (i.e., CP-201900476 / ML19228A086, CP-202000024 / ML20016A318, and ML20054A276).

SA-2019-015 EV-TR-2019-006425-1 (GWS):

Section 9.1.4.2.3.17:
Added component description for new mechanical (Framatome) RCCA tool to existing component description of the mechanical with an electro-pneumatic winch RCCA tool.

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SA-2019-017 EV-CR-2019-005757-1 (CBC):

FSAR Sections 12.5, 13.1, 13.3B, and 17.2 and FSAR Figures 13.1-2 and 13.1-3 are updated to reflect the current CPNPP organization relationships and functional responsibility descriptions. This an administrative update.

In accordance with Operating License paragraph 2.G, the changes to CPNPP Fire Protection Program (i.e., FSAR Section 13.3B) do not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

In accordance with 10 CFR 50.54, the changes to the Quality Assurance Program (i.e., FSAR Section 17.2) do not reduce the commitments in the QA Program previously accepted by the NRC.

SA-2019-018 EV-CR-2019-005757-9 (CBC):

FSAR Table 13.1-1 is updated to be consistent with approved LDCR-SA-2019-017 / EV-CR-2019-005757-1. This is an administrative update.

SA-2019-020 EV-TR-2019-001318-1 (GLM):

Add note to page 1-vii to state that the Chapter 1 Figure title blocks will not be updated to change “Comanche Peak S.E.S.” to CPNPP. Change CPSES to CPNPP in Section 1.2.4. This is an administrative update.

SA-2019-021 EV-CR-2019-008333-1 (CBC):

Section 13.3B:
(CPNPP Fire Protection Program) included a reference to periodic audits as discussed in FSAR Section 17.2.1.3.2. This reference is incorrect and is changed from "17.2.1.3.2" to "17.2.18". This is an editorial update and does not adversely affect the ability to achieve and maintain safe shutdown in the event of a fire.

SA-2019-022 EV-TR-2019-007125-1 (GWS):

Sections 1A(B), 9.5.4.4:
Update FSAR Appendix 1A(B) to add exception to C.2.f of RG 1.137. Remove detailed description of fuel oil sampling and refer to sampling programs.

SA-2019-023 EV-TR-2020-001500-1 (JEB):

Revise FSAR TMI III.D.3.3 to remove description of EOF-NOSF laboratory facilities.

The description of the EOF-NOSF laboratory facilities in Section 6.7 of the Emergency Plan was deleted. The EOF-NOSF laboratory facilities are no longer maintained as a back-up for iodine analysis. The available back-up lab facilities are described in Section 6.8 of the Emergency Plan.

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SA-2020-001 EV-TR-2020-001413-1 (CBC):

Figure 7.1-3 (sh 05 of 36):
Figure 7.1-3 (sheet 5 of 33) shows 1-LT-0501, SG Loop 1 Wide Range level transmitter twice and does not show 1-LT-0502, SG Loop 2 Wide Range. The depiction of 1-LT-0501 at the approximate one-o'clock position on the drawing is in error and is changed to 1-LT-0502. The depiction of 1-LT-0501 slightly below the three-o'clock position is correct. This is an editorial correction.

SA-2020-002 EV-CR-2019-004086-2 (CBC):

Table 17A-1:
The seismic classification of Item 41 / Instrumentation & Controls / Seismic Instrumentation of Table 17A-1 is changed from “NONE” to “II” and “Note 100” is added in the Remarks column. Note 100 states “The portion of seismic instrumentation installed in the control room is seismic category II (i.e., seismic monitoring system cabinet).” This change is a clarification and is consistent with the CPNPP design basis.

SA-2020-004 EV-TR-2020-002116-1 (CBC):

Sections 1A(B) (RG 1.33 and RG 1.144), 17.2.7, 17.2.18:
This change updates interval periods for certain QA audits as discussed in Sections 1A(B) (RG 1.33 and RB 1.144), 17.2.7, and 17.2.18. The change adds details regarding grace periods based on conditions (e.g., plant operational considerations, to accommodate supplier activities). This change conforms to the requirements of 10 CFR 50 Appendix B and is consistent with the Section 17.2 of NUREG-0800. This change has been reviewed per 10 CFR 50.54 and does not reduce the commitments or effectiveness of the QA Program (FSAR Chap 17) previously accepted by the NRC.

SA-2020-006 EV-CR-2020-003010-1 (GWS):

Section 7.2.2.3.3:
Correct Pressurizer Pressure coincidence from 2 of 3 to 2 of 4.

SA-2020-007 EV-CR-2017-004389-1 (CBC):

Section 2.4.1.1:
This change updates the discussion in Section 2.4.1.1 which refers to Figures 2.4-4, 2.4-40, and 2.4-41. Site plan drainage configuration changes (placement of new or modified design features, buildings, Security related features and other structures) have been implemented and the design basis updated to reflect these changes and impacts. This change clarifies that although this level of detail is not shown on Figures 2.4-4, 2.4-40, and 2.4-41, the drainage system requirements have been addressed.

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SA-2020-008 EV-TR-2017-006309-5 (CBC):

Section 2.4.13.2.2:
Updates text discussion to describe the plugging and capping of four production wells (two of which were specifically listed in the FSAR). Also described operational pumpage while the two listed productions wells were used. Four production wells were plugged and capped in 2013 as documented in CR-2012-011569 (LDCR-SA-2013-007). Since 2012 Somervell County Water District became the sole supplier of potable water for CPNPP.

SA-2020-009 EV-TR-2020-005363-1 (CBC):

Sections: 1.1, 1.4.1, 13.1, 17 and Figure 13.1-1
Vistra Operations Company LLC (Vistra OpCo) (operator) and Comanche Peak Power Company LLC (CP PowerCo) (owner) are subsidiaries of Vistra Energy Corp. As part of corporate re-branding, "Vistra Energy Corp." has been changed to "Vistra Corp." This is an editorial change only and has no impact to Comanche Peak Nuclear Power Plant.

SA-2020-010 EV-TR-2019-009288-1 (GWS):

Table 8.3-1:
Revise Note 7 last sentence to read; “See DBD-ME-011, Diesel Generator Sets, Attachment 3 for other DBAs where loss of offsite power is mitigated by the BOS.”

SA-2020-011 EV-TR-2017-004708-6 (CBC):

Updates Table 1.6-1 to be consistent with Reference 17 of Section 9.1.

This change is an administrative update to clarify that only the specific response to heavy loads (TXX-3659 / ML20072A945) is considered part of the FSAR and that future revisions to Design Basis Document DBD-ME-006 "Control of Heavy Loads at Nuclear Plants" are not considered part of the FSAR.

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SA-2020-012 EV-TR-2019-000901-4 (KJV):

Section 9.5.6.2:
This change reflects the ongoing staged replacement of the non-safety-related EDG starting air compressors - one compressor for each of the redundant starting air subsystems provided each of the two EDGs per Unit (total of eight). It is anticipated the overall installation period will be several years. Critical characteristics of the safety-related starting air receivers are unaffected by this modification. Each starting air receiver will continue to maintain the ability to support five (5) diesel starts and the minimum allowable receiver pressure Technical Specification limit remain unchanged throughout the installation period. The replacement air compressor design allows for the elimination of existing motor-driven air aftercoolers, unloader solenoid valves and associated circuitry and tubing, and the existing condensate drip traps. The FSAR will be updated in full upon completion of the modification when all EDG starting subsystems will have been modified. This activity is being done per Final Design Authorization Number 2017-000092-01.

SA-2017-022 EV-TR-2009-002875-40 (NWB):

Sections 7.5, 7.7, 11.5, TMI III.A, and Figures 7.7-14, 11.5-1, and III.A.1.2-2:
Update plant computer description following Plant computer replacement project.

SA-2019-004 EV-TR-2019-000901-1 (NWB):

Sections 8.3 and 9.5 and Tables 3.9B-10, 8.3-1, 8.3-2, 8.3-11, and 9.5-13:
UFSAR revisions to reflect the replacement of the Diesel Generator Starting Air Compressors, and also removal of the Starting Air Aftercoolers, Drip Trips, and Starting Air 3-Way Solenoid Valves.

SA-2019-012 EV-CR-2015-011094-7 (CBC):

Section 6.3.2.2.9.2:
The text is updated to reflect that the Safety Injection (ECCS to Cold Legs) check valves 2 inches and smaller are either spring loaded, lift piston types or nozzle check valve types. This change is supported by FDA-2019-000080-01. Both types of valve perform the design function and are subject to the same testing requirements.

SA-2020-003 EV-CR-2019-009290-8 (CBC):

Table 5.1-1B (Unit 2) Sheet 1 of 2:
The Unit 2 Pressurizer heater capacity (kW), has changed from 1731 to 1662 as heater #61 is defective and has been abandoned. This activity is being done per Final Design Authorization Number 2020-000040-01. This activity has no impact on the pressurizer design function as the minimum heater requirements are not affected.

SA-2020-013 EV-TR-2020-008801-1 (CBC):

Sections 9.2.1.10, 9.4B.2 and Table 8.3-11:
Removing obsolete information and designating two of three fire pumps as abandoned in place in the SWIS.

SA-2021-002 EV-CR-2018-002388-10 (CBC):

Sections 6.2.1.1.3.2, 6.2.1.1.3.11, and Tables 6.2.1-2, 6.2.1-3, 6.2.1-3A, 6.2.1-3B, 6.2.1-4, 6.2.1-9, 6.2.1-11, 6.2.1-12, 6.2.1-39, 6.2.1-41, 6.2.1-42, 6.2.1-43, 6.2.1-50, 6.2.1-51, 6.2.1-54:
The FSAR is updated to reflect the results of the re-analysis of the LOCA containment Pressure and Temperature in order to address Westinghouse NSALs (NSAL-06-6-R6, NSAL-11-5-R0, NSAL-14-2-R0) and concerns identified in CR-2018-002388 and CR-2019-002407. This change is supported by FDA-2021-000003-01. No physical changes to the plant resulted from updated analysis. This update does not adversely affect the design function of containment since the revised values do not violate the containment GDCs or equipment qualification requirements.

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SA-2021-003 EV-TR-2020-007054-1 (CBC):

Sections: 8.2, 8.3, F8.2-4, F8.2-5, F8.2-7
The description of the alternate startup transformers (XST1A and XST2A) and the switchyards in FSAR Section 8.2.1 is updated for consistency with FSAR Figure 8.2-1. FSAR Figure 8.2-4 is updated to correct an editorial error (change switch "17441" to "14741"). FSAR Figures 8.2-5 and 8.2-7 are deleted. A Text reference to the deleted figures is replaced with a reference to FSAR Figures 8.2-1 and 8.2-4. A reference to FSAR Figure 8.2-2 in FSAR Section 8.3.1 is replace with a reference to FSAR Figure 8.2-1 (Figure 8.2-2 was deleted in a previous FSAR Amendment).
These changes are clarifications and provide consistency within the FSAR. The deleted Figures are below the level of detail required by NRC RG 1.70-R2 (FSAR format and content)

SA-2021-005 EV-TR-2021-001647-2 (NWB):

Sections 3.2.2.2.d, 3.7B.2.8:
The UFSAR is updated to include the approval for the use of LOKRING fittings for Class 5 piping.

SA-2021-006 EV-TR-2021-002266-2 (CBC):

Section 5.4.1.4:
FSAR Section 5.4.1.4, "Tests and Inspections," is updated and simplified by removing excessive detail regarding reactor coolant pump casings. This detail is not required by NRC RG 1.70-R2 (FSAR format and content).

SA-2021-007 EV-TR-2018-005809-2 (NWB):

Sections 3.9N, 3B and 5.4:
Incorporate description of Unit 1 MSIP application to Reactor Vessel Nozzles performed in 1RF20.

SA-2021-010 EV-CR-2018-003408-2 (NWB):

Section 15.7.4.2 and Table 15.7.6:
Administrative removal of excessively detailed statements which are inconsistent with the Safety Analysis description of the fuel handling accident.

SA-2021-011 EV-TR-2020-002570-2 (NWB):

Sections 4.3.3 and 15.4.3.2 and Tables 1.6-1 and 4.1-2:
Update the UFSAR for the transition to the NEXUS/PARAGON/ANC9 nuclear design software.

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SA-2022-001 EV-TR-2019-000901-7 (NWB):

Section 9.5:
Removal of temporary tracking information statement after completion of EDG starting air compressor changes.